

Mathematical Analysis

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September 19, 2026

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Chapter 1

Elements of set theory

1.1 Basic concepts about classes and sets

Every book about mathematical subjects must be based on one form of set theory. Because the focus of this book is on mathematical analysis instead of the foundations of mathematics, I have decided to use Von Neumann's set theory instead of the set theory of Fraenkel, Skolem and Zermelo. The benefit of Von Neumann's theory is that it is nearer to the naive set theory of Cantor. This book assumes that the basics of mathematical logic are understood, more specifically that the reader knows the meaning of the following terms:

$\wedge$	meaning	and
$\vee$	meaning	or
$\neg$	meaning	not
$\Rightarrow$	meaning	implies
$\Leftrightarrow$	meaning	is equivalent with
$\vdash, \models$	meaning	with, where
$\forall$	meaning	for all
$\exists$	meaning	there exists
$\exists!$	meaning	there exists a unique

and how to use them. Axiomatic set theory is based on two undefined concepts: **class** and the **membership** relation between classes (noted as $\in$). Intuitively you can think of a class as a collection and $x \in A$ to mean that x is part of the collection where A stands for. We introduce then axioms that state what are the true statements about these undefined concepts. Further we introduce different definitions that helps us to simplify our notation. To start with, we define the concept of $\notin$ [not member of]

Definition 1.1. *Let A be a class then $x \notin A$ is equivalent with saying $\neg(x \in A)$.*

Next we introduce **sets** and **elements**, which are two concepts for the same thing. We use **set** if we want to stress that it is a kind of collection and **element** that is also a member of a class in contrast to a class that does not have to be a member. A **set** or **element** is something that is a member of a class.

Definition 1.2. *We say that a **class** x is a **element** if $x \in A$ where A is a class. Another name for a **element** is a **set**. A **proper** class is a class that is not a element [in other words not a set].*

From here on we use the following convention: elements are noted in **lower-case** and sets and classes are noted in **upper-case**. Next we define equality of classes.

Definition 1.3. *Let A, B classes then we say that $A = B$ if and only if*

$$\forall X \text{ we have } A \in X \Rightarrow B \in X \wedge B \in X \Rightarrow A \in X$$

In other words two classes are equal if they are members of the same classes.

Once we have defined equality we can also define inequality.

Definition 1.4. *Let A and B classes then $A \neq B$ is equivalent with $\neg(A = B)$*

If two classes are equal, we expect them to contain the same elements, this is stated in the first set axiom.

Axiom 1.5. (Axiom of extent)

$$A = B \Leftrightarrow [x \in A \Rightarrow x \in B \wedge x \in B \Rightarrow x \in A]$$

Less formally A is equal to B if and only if every element of A is a element of B and every element of B is a element of A , in other words A and B have the same elements.

Definition 1.6. Let A and B classes then A is a sub-class of B noted by $A \subseteq B$ iff

$$x \in A \Rightarrow x \in B$$

So A is a sub-class of B iff every element of A is also a element of B .

Definition 1.7. Let A and B classes then A is a proper sub-class of B noted by $A \subset B$ iff

$$x \in A \Rightarrow x \in B \wedge A \neq B$$

So A is a proper sub-class of B iff A is different from B and every element of A is also a element of B .

Given a set [or equivalently a element] A then we require that all sub-classes of A are also sets [or equivalently elements] by the axiom of subsets.

Axiom 1.8. (Axiom of Subsets) Every sub-class of a set is a set

We can now proof the following relations between classes.

Theorem 1.9. Let A, B, C be classes then the following holds:

1. $A = A$
2. $A = B \Rightarrow B = A$
3. $A = B \wedge B = C \Rightarrow A = C$
4. $A \subseteq B \wedge B \subseteq A \Rightarrow A = B$
5. $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$
6. $A = B \Rightarrow A \subseteq B$

Proof.

1. $x \in A \Rightarrow x \in A$ and $x \in A \Rightarrow x \in A$ are obviously true, hence using the Axiom of Extent [axiom: 1.5] it follows that $A = A$.
2. As $A = B$ we have using the Axiom of Extent [axiom: 1.5] that $x \in A \Rightarrow x \in B \wedge x \in B \Rightarrow x \in A$ which is equivalent with $x \in B \Rightarrow x \in A \wedge x \in A \Rightarrow x \in B$. Using the Axiom of Extent [axiom: 1.5] it follows that $B = A$
3. As $A = B \wedge B = C$ we have by the Axiom of Extent [axiom: 1.5] that

$$x \in A \Rightarrow x \in B \tag{1.1}$$

$$x \in B \Rightarrow x \in A \tag{1.2}$$

$$x \in B \Rightarrow x \in C \tag{1.3}$$

$$x \in C \Rightarrow x \in B \tag{1.4}$$

From [eq: 1.1] and [eq: 1.3] it follows that $x \in A \Rightarrow x \in C$ and from [eq: 1.4] and [eq: 1.2] it follows that $x \in C \Rightarrow x \in A$. Using the Axiom of Extent [axiom: 1.5] it follows then that $A = C$.

4. From $A \subseteq B \wedge B \subseteq A$ it follows that $x \in A \Rightarrow x \in B \wedge x \in B \Rightarrow x \in A$, so by the Axiom of Extent [axiom: 1.5] we have $A = B$
5. As $A \subseteq B \wedge B \subseteq C$ it follows that $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in C$ proving that $x \in A \Rightarrow x \in C$ or $A \subseteq C$

6. If $x \in A$ then as $A = B$ we have by the axiom of extension [axiom: 1.5] that $x \in B$, hence $A \subseteq B$. $\square$

One way to create a new class is to specify a predicate that a object must satisfies and then take the class of all objects that satisfies this predicate. The problem with this construction is that it can lead to paradoxes like the famous Russell paradox. Consider the predicate $R(x) = x \notin x$, this predicate is true for x if x is not a member of itself. Consider now the class that contains all classes that has not them self as member. Does this class contain itself yes or no? If the class contains itself then by definition $R(x)$ should be true so the class should not contain itself leading to a contradiction. If the class does not contain itself then it satisfies $R(x)$, hence it is a member of itself again leading to a contradiction. So we can not test the predicate $R(x)$ for all classes and thus can not define the class of all classes for which $R(x)$ is true. The axiom of class construction allows us to create a new class in a safe way.

Axiom 1.10. (Axiom of Construction) *Let $P(x)$ be a statement about x [using mathematical logic] then there exists a class C such that $x \in C$ iff x is a element and $P(x)$ is true.*

Notation 1.11. *This class C is noted as $C = \{x | P(x)\}$, note the use of lower cases for x , which is a visual indicator that x is a element.*

Note that that C consists of **elements** for which $P(x)$ is true, it is not enough that $P(x)$ is true to belong to C . A object must belong to a class [be a element or equivalently be a set] and $P(x)$ must be true to be a member of C . The following theorem shows how this solves Russell's paradox.

Theorem 1.12. *Russell's class defined by $R = \{x | x \notin x\}$ is not a element [or equivalently a set] so that $R \notin R$ is true [for if $R \in R$ then R is a by definition a element [or equivalently a set].*

Proof. We have the following two possibilities to check:

$R \in R$. Then R is a element and $R \notin R$ giving the contradiction $R \in R \wedge R \notin R$

$R \notin R$. Then R is not a element or $R \in R$ which as $R \notin R$ proves that R is not a element

hence we must have that R is not a element. $\square$

So we have that R is not a element and indeed because of this that $R \notin R$. You can ask yourself if there actually exists elements, none of the axioms up to now can be used to construct elements [or equivalent sets], for this we will need extra axioms.

Definition 1.13. *Let A be a class and $P(x)$ a predicate then $\{x \in A | P(x)\} = \{x | x \in A \wedge P(x)\}$*

Theorem 1.14. *Let A be a class and $P(x)$ a predicate then:*

1. $\{x \in A | P(x)\} \subseteq A$
2. *If A is a set then $\{x \in A | P(x)\}$ is a set*

Proof.

1. If $x \in \{x \in A | P(x)\} \stackrel{\text{definition}}{=} \{x | x \in A \wedge P(x)\}$ then $x \in A$ so that $\{x \in A | P(x)\} \subseteq A$.
2. If A is a set then as by (1) $\{x \in A | P(x)\} \subseteq A$ it follows from the Axiom of subsets [see axiom: 1.8] that $\{x \in A | P(x)\}$ is a set. $\square$

Using the axiom of construction [axiom: 1.10] we can define the universal class $\mathcal{U}$.

Definition 1.15. (Universal class) *The universal class $\mathcal{U}$ is defined by $\mathcal{U} = \{x | x = x\}$.*

The universal class contains all the elements , as is expressed in the following theorem.

Theorem 1.16. *If x is a element then $x \in \mathcal{U}$*

Proof. Let x be a element then, as $x = x$ [see theorem: 1.9] we have that $x \in \mathcal{U}$ $\square$

The universal class is however a proper class not a set as is show in the following theorem.

Theorem 1.17. *$\mathcal{U}$ is a proper class [in other words $\mathcal{U}$ is not a set or element].*

Proof. Take Russell's class $R = \{x | x \notin x\}$. If $x \in R$ then by definition x is a element and $x \stackrel{=}{=}_{[\text{theorem: 1.9}]} x$ so that $x \in \{x | x = x\} = \mathcal{U}$ which proves that

$$R \subseteq \mathcal{U}$$

Assume that $\mathcal{U}$ is a set then by the above and the axiom of subsets [see axiom: 1.8] R is a set which contradicts the fact that by [theorem: 1.12] R is not a set. Hence the assumption must be wrong and $\mathcal{U}$ is not a set [or equivalently a element]. $\square$

1.2 Class operations

We use now the axiom of construction to define the union and intersection of two classes.

Definition 1.18. *Let A, B be two classes then the union of A and B , noted as $A \cup B$ is defined by*

$$A \cup B = \{x | x \in A \vee x \in B\}$$

Definition 1.19. *Let A, B be two classes then the intersection of A and B , noted as $A \cap B$ is defined by*

$$A \cap B = \{x | x \in A \wedge x \in B\}$$

Next we define the empty class, the class that does not contains a element.

Definition 1.20. *The empty class $\emptyset$ is defined by*

$$\emptyset = \{x | x \neq x\}$$

Theorem 1.21. *$\emptyset$ does not contains elements, meaning if x is a element then $x \notin \emptyset$*

Proof. We proof this by contradiction, so assume that there exists a element $x \in \emptyset$ then $x \neq x$, contradicting $x = x$ [see theorem: 1.9]. $\square$

Theorem 1.22. *If A is a class then*

1. $\emptyset \subseteq A$
2. $A \subseteq \mathcal{U}$
3. *If $A \subseteq \emptyset$ then $A = \emptyset$*

Proof.

1. We proof this by contra-position, as $\emptyset \subseteq A$ is equivalent with $x \in \emptyset \Rightarrow x \in A$, we must proof that $x \notin A \Rightarrow x \notin \emptyset$. Well if $x \notin A$ then certainly $x \notin \emptyset$ [by theorem: 1.21] so that $x \notin A \Rightarrow x \notin \emptyset$.
2. If $x \in A$ then x is a element, hence $x \in \mathcal{U}$ by [Theorem: 1.16].
3. By (1) we have $\emptyset \subseteq A$ which together with $A \subseteq \emptyset$ proves by [theorem: 1.9] that $A = \emptyset$. $\square$

We also have that every class with no elements is equal to the empty set [there is only one empty set]

Theorem 1.23. *If A is a a class such that $x \in A$ yields a contradiction then $A = \emptyset$*

Proof. Let $x \in A$ then we have a contradiction, so $x \in A$ must be false and thus $x \in A \Rightarrow x \in \emptyset$ is vacuously true which proves that $A \subseteq \emptyset$, combining this with [theorem: 1.22, 1.9] proves that $A = \emptyset$ $\square$

Corollary 1.24. *Let A be a class such that $A \neq \emptyset$ then $\exists x$ such that $x \in A$*

Proof. We proof this by contradiction. Assume that $\forall x$ we have $x \notin A$ then $x \in A$ yields the contradiction $x \in A \wedge x \notin A$, hence by [theorem: 1.23] $A = \emptyset$ which contradicts $A \neq \emptyset$. $\square$

Definition 1.25. Two classes A, B are disjoint iff $A \cap B = \emptyset$

We define now the complement of a class

Definition 1.26. Let A be a class then the complement of A noted by A^c is defined by

$$A^c = \{x | x \notin A\}$$

Something similar to the complement of a class is the difference between two classes

Definition 1.27. Let A, B be classes then the difference between A and B noted by $A \setminus B$ is defined by

$$A \setminus B = \{x | x \in A \wedge x \notin B\} \underset{\text{shorter notation}}{=} \{x \in A | x \notin B\}$$

We can express the difference of two classes using the intersection and the complement.

Theorem 1.28. Let A, B be classes then

$$A \setminus B = A \cap B^c$$

Proof. Let $x \in A \setminus B$ then $x \in A \wedge x \notin B$ so that $x \in A \wedge x \in B^c$, further if $x \in A \cap B^c$ then $x \in A \wedge x \notin B$. Using then the axiom of extent [axiom: 1.5] proves the theorem. $\square$

Theorem 1.29. Let A, B, C be classes then we have:

commutativity.

1. $A \cup B = B \cup A$
2. $A \cap B = B \cap A$

idem potency.

1. $A \cup A = A$
2. $A \cap A = A$

associativity.

1. $A \cup (B \cup C) = (A \cup B) \cup C$
2. $A \cap (B \cap C) = (A \cap B) \cap C$

Distributivity.

1. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
2. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof.

commutativity.

1. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cup B &\Leftrightarrow x \in A \vee x \in B \\ &\Leftrightarrow x \in B \vee x \in A \\ &\Leftrightarrow x \in B \cup A \end{aligned}$$

2. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cap B &\Leftrightarrow x \in A \wedge x \in B \\ &\Leftrightarrow x \in B \wedge x \in A \\ &\Leftrightarrow x \in B \cap A \end{aligned}$$

idem potency.

1. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cup A &\Leftrightarrow x \in A \vee x \in A \\ &\Leftrightarrow x \in A \end{aligned}$$

2. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cap A &\Leftrightarrow x \in A \wedge x \in A \\ &\Leftrightarrow x \in A \end{aligned}$$

associativity.

1. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cup (B \cup C) &\Leftrightarrow x \in A \vee x \in B \cup C \\ &\Leftrightarrow x \in A \vee (x \in B \vee x \in C) \\ &\Leftrightarrow (x \in A \vee x \in B) \vee x \in C \\ &\Leftrightarrow x \in A \cup B \vee x \in C \\ &\Leftrightarrow x \in (A \cup B) \cup C \end{aligned}$$

2. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cap (B \cap C) &\Leftrightarrow x \in A \wedge x \in B \cap C \\ &\Leftrightarrow x \in A \wedge (x \in B \wedge x \in C) \\ &\Leftrightarrow (x \in A \wedge x \in B) \wedge x \in C \\ &\Leftrightarrow x \in A \cap B \wedge x \in C \\ &\Leftrightarrow x \in (A \cap B) \cap C \end{aligned}$$

distributivity.

1. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cup (B \cap C) &\Leftrightarrow x \in A \vee x \in B \cap C \\ &\Leftrightarrow x \in A \vee (x \in B \wedge x \in C) \\ &\Leftrightarrow (x \in A \vee x \in B) \wedge (x \in A \vee x \in C) \\ &\Leftrightarrow x \in A \cup B \wedge x \in A \cup C \\ &\Leftrightarrow x \in (A \cup B) \cap (A \cup C) \end{aligned}$$

2. This follows from [axiom: 1.5] and

$$\begin{aligned} x \in A \cap (B \cup C) &\Leftrightarrow x \in A \wedge x \in B \cup C \\ &\Leftrightarrow x \in A \wedge (x \in B \vee x \in C) \\ &\Leftrightarrow (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\ &\Leftrightarrow x \in A \cap B \vee x \in A \cap C \\ &\Leftrightarrow x \in (A \cap B) \cup (A \cap C) \end{aligned}$$

□

Theorem 1.30. *Let A, B, C are classes then we have*

1. $A \subseteq A \cup B$
2. $B \subseteq A \cup B$
3. $A \cap B \subseteq A$
4. $A \cap B \subseteq B$
5. $A \setminus B \subseteq A$
6. $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$

7. $(A \cap B) \setminus C = (A \setminus C) \cap B = A \cap (B \setminus C)$
8. If C is a class such that $A \subseteq C$ and $B \subseteq C$ then $A \cup B \subseteq C$
9. If C is a class such that $A \subseteq C$ and D a class such that $B \subseteq D$ then $A \cup B \subseteq C \cup D$
10. If C is a class such that $C \subseteq A$ and $C \subseteq B$ then $C \subseteq A \cap B$
11. If C is a class such that $A \subseteq C$ and D a class such that $B \subseteq D$ then $A \cap B \subseteq C \cap D$

Proof.

1. If $x \in A$ then $x \in A \vee x \in B$ proving that $x \in A \cup B$, hence $A \subseteq A \cup B$.
2. If $x \in B$ then $x \in A \vee x \in B$ proving that $x \in A \cup B$, hence $B \subseteq A \cup B$.
3. If $x \in A \cap B$ then $x \in A \wedge x \in B$, hence $x \in A$ so that $A \cap B \subseteq A$.
4. If $x \in A \cap B$ then $x \in A \wedge x \in B$, hence $x \in B$ so that $A \cap B \subseteq B$.
5. If $x \in A \setminus B$ then $x \in A \wedge x \notin B$, hence $x \in A$ so that $A \setminus B \subseteq A$.
6. We have

$$\begin{aligned}
 (A \cup B) \setminus C &\stackrel{[\text{theorem: 1.28}]}{=} (A \cup B) \cap C^c \\
 &\stackrel{[\text{theorem: 1.29}]}{=} (A \cap C^c) \cup (B \cap C^c) \\
 &\stackrel{[\text{theorem: 1.28}]}{=} (A \setminus C) \cup (B \setminus C)
 \end{aligned}$$

7.

- a. $(A \cap B) \setminus C = (A \cap B) \cap C^c \stackrel{[\text{theorem: 1.29}]}{=} (A \cap C^c) \cap B = (A \setminus C) \cap B$
- b. $(A \cap B) \setminus C = (A \cap B) \cap C^c \stackrel{[\text{theorem: 1.29}]}{=} A \cap (B \cap C^c) = A \cap (B \setminus C)$

8. If $x \in A \cup B$ then $x \in A \xRightarrow{A \subseteq C} x \in C$ or $x \in B \xRightarrow{B \subseteq C} x \in C$ proving that $x \in C$.
9. Using (1) $A \subseteq C \cup D$ and $B \subseteq C \cup D$, so using (6) we have $A \cup B \subseteq C \cup D$.
10. If $x \in C$ then $x \in A$ and $x \in B$ so that $x \in A \cap B$.
11. If $x \in A \cap B$ then $x \in A \xRightarrow{A \subseteq C} x \in C$ and $x \in B \xRightarrow{B \subseteq D} x \in D$ hence $x \in C \cap D$.

□

Theorem 1.31. Let A be a class then $(A^c)^c = A$

Proof. If $x \in (A^c)^c$ then x is a element and $x \notin A^c$ then $x \in A$ [for if $x \notin A$ we have $x \in A^c$]. If $x \in A$ then $x \notin A^c$ [for if $x \in A^c$ then $x \notin A$] so that $x \in (A^c)^c$. □

Theorem 1.32. (DeMorgan's Law) For all classes A, B we have

1. $(A \cup B)^c = A^c \cap B^c$
2. $(A \cap B)^c = A^c \cup B^c$

Proof.

1. If $x \in (A \cup B)^c$ then $x \notin A \cup B$, so that $\neg(x \in A \vee x \in B) = x \notin A \wedge x \notin B$ proving that $x \in A^c \cap B^c$. If $x \in A^c \cap B^c$ then $x \notin A \wedge x \notin B = \neg(x \in A \vee x \in B)$, so that $x \notin A \cup B$ or $x \in (A \cup B)^c$. The proof follows then from the axiom of extent [axiom: 1.5]
2. If $x \in (A \cap B)^c$ then $x \notin A \cap B$, so that $\neg(x \in A \wedge x \in B) = x \notin A \vee x \notin B$ proving that $x \in A^c \cup B^c$. If $x \in A^c \cup B^c$ then $x \notin A \vee x \notin B = \neg(x \in A \wedge x \in B)$, so that $x \in (A \cap B)^c$. The proof follows then from axiom of extent [axiom: 1.5] □

Theorem 1.33. Let A, B be classes such that $A \subseteq B$ then $A \cap B^c = \emptyset$

Proof. Assume that $x \in A \cap B^c$ then $x \in A \wedge x \notin B$, as $A \subseteq B$ we have $x \in B$ so we reach the contradiction $x \in B \wedge x \notin B$. Hence $A \cap B^c = \emptyset$. □

Theorem 1.34. *Let A, B, C be classes then we have*

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C) = (A \setminus B) \setminus C$
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Proof.

1.

$$\begin{aligned}
 A \setminus (B \cup C) & \stackrel{\text{theorem:1.28}}{=} A \cap (B \cup C)^c \\
 & \stackrel{\text{theorem: 1.32}}{=} A \cap (B^c \cap C^c) \\
 & \stackrel{\text{associativity}}{=} (A \cap B^c) \cap C^c \\
 & \stackrel{\text{idem potency}}{=} ((A \cap A) \cap B^c) \cap C^c \\
 & \stackrel{\text{associativity}}{=} (A \cap (A \cap B^c)) \cap C^c \\
 & \stackrel{\text{commutativity}}{=} ((A \cap B^c) \cap A) \cap C^c \\
 & \stackrel{\text{associativity}}{=} (A \cap B^c) \cap (A \cap C^c) \\
 & \stackrel{\text{theorem:1.28}}{=} (A \setminus B) \cap (A \setminus C) \\
 A \setminus (B \cup C) & \stackrel{\text{theorem:1.28}}{=} A \cap (B \cup C)^c \\
 & \stackrel{\text{theorem: 1.32}}{=} A \cap (B^c \cap C^c) \\
 & \stackrel{\text{associativity}}{=} (A \cap B^c) \cap C^c \\
 & \stackrel{\text{theorem:1.28}}{=} (A \setminus B) \setminus C
 \end{aligned}$$

2.

$$\begin{aligned}
 A \setminus (B \cap C) & \stackrel{\text{theorem:1.28}}{=} A \cap (B \cap C)^c \\
 & \stackrel{\text{theorem: 1.32}}{=} A \cap (B^c \cup C^c) \\
 & \stackrel{\text{Distributivity}}{=} (A \cap B^c) \cup (A \cap C^c) \\
 & \stackrel{\text{theorem:1.28}}{=} (A \setminus B) \cup (A \setminus C)
 \end{aligned}$$

□

Theorem 1.35. *Let A be a class then we have:*

1. $\emptyset \cup A = A$
2. $\emptyset \cap A = \emptyset$
3. $A \cup \mathcal{U} = \mathcal{U}$
4. $A \cap \mathcal{U} = A$
5. $A \setminus A = \emptyset$

Proof.

1. If $x \in \emptyset \cup A$ then $x \in \emptyset \wedge x \in A \xRightarrow{x \notin \emptyset \text{ [see theorem: 1.21]}} x \in A$, further if $x \in A$ then $x \in A \vee x \in \emptyset$ so that $x \in \emptyset \cup A$. Hence using [axiom: 1.5] $\emptyset \cup A = A$.
2. If $x \in \emptyset \cap A$ then $x \in \emptyset \wedge x \in A$ leading to $x \in \emptyset$ which is a contradiction by [theorem: 1.21]. Hence using [theorem: 1.23] $\emptyset \cap A = \emptyset$.
3. If $x \in A \cup \mathcal{U}$ then we have either:
 - $x \in A$. Then x is a element so that by [theorem: 1.16] $x \in \mathcal{U}$
 - $x \in \mathcal{U}$. Then trivially $x \in \mathcal{U}$

hence we have in all cases that $x \in \mathcal{U}$. Further if $x \in \mathcal{U}$ then clearly $x \in \mathcal{U} \vee x \in A \Rightarrow x \in \mathcal{U} \cup A$. Hence using [axiom: 1.5] we have that

$$A \cup \mathcal{U} = \mathcal{U}$$

4. If $x \in A \cap \mathcal{U}$ then $x \in A \wedge x \in \mathcal{U} \Rightarrow x \in A$, further if $x \in A$ then x is a element so that by [theorem: 1.16] $x \in \mathcal{U}$, hence $x \in A \cap \mathcal{U}$. Using [axiom: 1.5] it follows that

$$A \cap \mathcal{U} = A$$

5. Let $x \in A \setminus A$ then $x \in A \wedge x \notin A$ a contradiction, so by [theorem: 1.23] we have that $A \setminus A = \emptyset$ $\square$

Theorem 1.36. *If A, B, C are classes then we have*

1. $A \subseteq B$ if and only if $A \cup B = B$
2. $A \subseteq B$ if and only if $A \cap B = A$
3. If $A \subseteq B$ then $C \setminus B \subseteq C \setminus A$
4. $A \setminus (A \setminus B) = A \cap B$
5. If $B \subseteq A$ then $A \setminus (A \setminus B) = B$
6. If $B \subseteq A$ then $B \setminus (A \setminus C) = B \cap C$
7. $A \cap (B \setminus C) = (A \cap B) \setminus C$
8. If $B \subseteq A$ then $B \cap (A \setminus C) = B \setminus C$
9. $(A \setminus B) \cup A = A$
10. $(A \setminus B) \cap A = A \setminus B$
11. If $C \subseteq A$ then $(A \setminus B) \setminus (A \setminus C) = C \setminus B$

Proof.

1.

$\Rightarrow$. If $x \in A \cup B \Rightarrow x \in A \xRightarrow{A \subseteq B} x \in B$ and thus $A \cup B \subseteq B$. From theorem [theorem: 1.30] we have $B \subseteq A \cup B$ so by [theorem: 1.9] we have $A \cup B = B$

$\Leftarrow$. If $x \in A \Rightarrow x \in A \cup B \xRightarrow{A \cup B = B} x \in B$ so that $A \subseteq B$
2.

$\Rightarrow$. If $x \in A \xRightarrow{A \subseteq B} x \in B \Rightarrow x \in A \wedge x \in B \Rightarrow x \in A \cap B$ proving that $A \subseteq A \cap B$. From [theorem: 1.30] we have $A \cap B \subseteq A$ so that by [theorem: 1.9] we have $A \cap B = A$.

$\Leftarrow$. If $A \cap B = A$ we have $x \in A \Rightarrow x \in A \cap B \Rightarrow (x \in A \wedge x \in B) \Rightarrow x \in B$ so $A \subseteq B$.
3. If $x \in C \setminus B$ then $x \in C$ and $x \notin B$. If $x \in A$ then as $A \subseteq B$ we would have $x \in B$ contradicting $x \notin B$ hence $x \notin A$ proving that $x \in C \setminus A$.
4. We have

$$\begin{aligned}
 A \setminus (A \setminus B) & \stackrel{[\text{theorem: 1.28}]}{=} A \cap (A \setminus B)^c \\
 & \stackrel{[\text{theorem: 1.28}]}{=} A \cap (A \cap B^c)^c \\
 & \stackrel{[\text{theorem: 1.32}]}{=} A \cap (A^c \cup (B^c)^c) \\
 & \stackrel{[\text{theorem: 1.31}]}{=} A \cap (A^c \cup B) \\
 & \stackrel{[\text{theorem: 1.29}]}{=} (A \cap A^c) \cup (A \cap B) \\
 & \stackrel{[\text{theorem: 1.28}]}{=} (A \setminus A) \cup (A \cap B) \\
 & \stackrel{[\text{theorem: 1.35}]}{=} \emptyset \cup (A \cap B) \\
 & \stackrel{[\text{theorem: 1.35}]}{=} A \cap B
 \end{aligned}$$

5. We have

$$A \setminus (A \setminus B) \stackrel{(4)}{=} A \cap B \stackrel{[\text{theorem: 1.36}]}{=} B$$

6. If $B \subseteq A$ then we have

$$\begin{aligned} B \setminus (A \setminus C) &\stackrel{[\text{theorem: 1.28}]}{=} B \cap (A \setminus C)^c \\ &\stackrel{[\text{theorem: 1.28}]}{=} B \cap (A \cap C^c)^c \\ &\stackrel{[\text{theorem: 1.32}]}{=} B \cap (A^c \cup (C^c)^c) \\ &\stackrel{[\text{theorem: 1.31}]}{=} B \cap (A^c \cup C) \\ &\stackrel{[\text{theorem: 1.29}]}{=} (B \cap A^c) \cup (B \cap C) \\ &\stackrel{[\text{theorem: 1.33}]}{=} \emptyset \cup (B \cap C) \\ &\stackrel{[\text{theorem: 1.35}]}{=} B \cap C \end{aligned}$$

7. We have

$$\begin{aligned} A \cap (B \setminus C) &\stackrel{[\text{theorem: 1.28}]}{=} A \cap (B \cap C^c) \\ &\stackrel{[\text{theorem: 1.29}]}{=} (A \cap B) \cap C^c \\ &\stackrel{[\text{theorem: 1.28}]}{=} (A \cap B) \setminus C \end{aligned}$$

8. We have

$$\begin{aligned} B \cap (A \setminus C) &\stackrel{(7)}{=} (B \cap A) \setminus C \\ &\stackrel{B \subseteq A \wedge (2)}{=} B \setminus C \end{aligned}$$

9. Using [theorem: 1.30] we have $(A \setminus B) \subseteq A$ so that by (1) $(A \setminus B) \cup A = A$

10. Using [theorem: 1.30] we have $(A \setminus B) \subseteq A$ so that by (2) $(A \setminus B) \cap A = A \setminus B$

11. We have

$$\begin{aligned} (A \setminus B) \setminus (A \setminus C) &\stackrel{[\text{theorem: 1.28}]}{=} (A \setminus B) \cap (A \setminus C)^c \\ &\stackrel{[\text{theorem: 1.28}]}{=} (A \cap B^c) \cap (A \cap C^c)^c \\ &\stackrel{[\text{theorem: 1.32}]}{=} (A \cap B^c) \cap (A^c \cup (C^c)^c) \\ &\stackrel{[\text{theorem: 1.31}]}{=} (A \cap B^c) \cap (A^c \cup C) \\ &\stackrel{[\text{theorem: 1.29}]}{=} ((A \cap B^c) \cap A^c) \cup ((A \cap B^c) \cap C) \\ &\stackrel{[\text{theorem: 1.29}]}{=} (B^c \cap (A \cap A^c)) \cup ((A \cap C) \cap B^c) \\ &\stackrel{[\text{theorem: 1.33}]}{=} (B^c \cap \emptyset) \cup ((A \cap C) \cap B^c) \\ &\stackrel{[\text{theorem: 1.35}]}{=} \emptyset \cup ((A \cap C) \cap B^c) \\ &\stackrel{[\text{theorem: 1.35}]}{=} (A \cap C) \cap B^c \\ &\stackrel{C \subseteq A \wedge (2)}{=} C \cap B^c \\ &\stackrel{[\text{theorem: 1.28}]}{=} C \setminus B \end{aligned}$$

□

Theorem 1.37. (Absorption Laws) *If A, B are classes then*

1. $A \cup (A \cap B) = A$

$$2. A \cap (A \cup B) = A$$

Proof.

1. By [theorem: 1.30] we have $A \cap B \subseteq A$, hence using [theorem: 1.36] we have that $A \cup (A \cap B) = A$
2. By [theorem: 1.30] we have $A \subseteq A \cup B$, hence using [theorem: 1.36] we have that $A \cap (A \cup B) = A$ $\square$

If A, B are classes then $x \in A \cup B$ if and only if $x \in A \vee x \in B$ where the boolean operator satisfies the following truth table

P	Q	$P \vee Q$
True	True	True
False	True	True
True	False	True
False	False	False

So if $x \in A \wedge x \in B$ we have that $x \in A \cup B$ essential because $\vee$ is the inclusive 'or', sometimes we want to use the exclusive 'or' operator so that if $x \in A \cup B$ we have either $x \in A$ or $x \in B$ but not $x \in A \wedge x \in B$. Essential we want to have

$x \in A$	$x \in B$	$x \in A \cup B$
True	True	False
False	True	True
True	False	True
False	False	False

To indicate this we use the following definition

Definition 1.38. Let A, B be classes then if $A \cap B = \emptyset$ we use the notation $A \sqcup B$ instead of $A \cup B$. In other words if we use $A \sqcup B$ for the union then we implicitly assume that $A \cap B = \emptyset$.

Using this notation we have the following usefull set relations

Theorem 1.39. Let A, B be classes then we have

1. $A \cap (B \setminus A) = \emptyset$
2. $(A \cap B) \cap (B \setminus A) = \emptyset$
3. $A \sqcup (B \setminus A) = A \cup B$
4. $A \cup B = (A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A)$
5. $B = (A \cap B) \sqcup (B \setminus A)$
6. If $A \subseteq B$ then $B = A \sqcup (B \setminus A)$

Proof.

1. If $x \in A \cap (B \setminus A)$ then $x \in A$ and $x \in B \setminus A \Rightarrow x \notin A$ leading to the contradiction $x \in A \wedge x \notin A$, hence we have that $A \cap (B \setminus A) = \emptyset$.
2. As $(A \cap B) \subseteq A$ and $B \setminus A \subseteq B \setminus A$ it follows from [theorem: 1.30] that

$$(A \cap B) \cap (B \setminus A) \subseteq A \cap (B \setminus A) \stackrel{(1)}{=} \emptyset$$

so that by [theorem: 1.22]

$$(A \cap B) \cap (B \setminus A) = \emptyset$$

3. If $x \in A \cup (B \setminus A)$ then $x \in A$ or $x \in B \setminus A \Rightarrow x \in B$ so that $x \in A \cup B$, hence

$$A \cup (B \setminus A) \subseteq A \cup B$$

On the other hand if $x \in A \cup B$ we have either $x \in A$ or $x \in B$. If $x \notin A$ then we must have $x \in B$ [otherwise $x \notin A \wedge x \notin B = \neg(x \in A \vee x \in B) = \neg(x \in A \cup B)$ contradicting $x \in A \cup B$] so we must have $x \in A \cup (B \setminus A)$. Hence

$$A \cup B \subseteq A \cup (B \setminus A)$$

So that $A \cup (B \setminus A) = A \cup B$ which combined with (1) gives

$$A \sqcup (B \setminus A) = A \cup B$$

4. By [theorem: 1.30] we have $A \setminus B \subseteq A \subseteq A \cup B$, $A \cap B \subseteq A \subseteq A \cup B$, $B \setminus A \subseteq B \subseteq A \cup B$ so that by [theorem: 1.9]

$$(A \setminus B) \cup (A \cap B) \cup (B \setminus A) \subseteq A \cup B$$

On the other hand if $x \in A \cup B$ then we have either:

$x \in A$. Then we have either

$x \in B$. Then $x \in A \cap B$ so that $x \in (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$

$x \notin B$. Then $x \in A \setminus B$ so that $x \in (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$

$x \in B$. Then we have either

$x \in A$. Then $x \in A \cap B$ so that $x \in (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$

$x \notin A$. Then $x \in B \setminus A$ so that $x \in (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$

hence we have that

$$A \cup B \subseteq (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$$

So that

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$$

which, as by (2) $(A \setminus B) \cap (A \cap B) = \emptyset = (B \setminus A) \cap (A \cap B)$, proves that

$$A \cup B = (A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A)$$

5. If $x \in B$ then we have either $x \in A \Rightarrow x \in A \cap B$ or $x \notin A \Rightarrow x \in B \setminus A$ so that $x \in (A \cap B) \cup (B \setminus A)$ proving

$$B \subseteq (A \cap B) \cup (B \setminus A)$$

If $x \in (A \cap B) \cup (A \setminus B)$ then we have either $x \in A \cap B \Rightarrow x \in B$ or $x \in A \setminus B \Rightarrow x \in B$ so that in all cases we have $x \in B$. Hence we have

$$(A \cap B) \cup (B \setminus A) \subseteq B$$

so that

$$B = (A \cap B) \cup (B \setminus A)$$

which using (2) leads to

$$B = (A \cap B) \sqcup (B \setminus A)$$

6. If $A \subseteq B$ then by [theorem: 1.36] then $A = A \cap B$ so that

$$B \stackrel{(5)}{=} (A \cap B) \sqcup (B \setminus A) = A \sqcup (B \setminus A)$$

□

1.3 Cartesian products

If a is a element we can use the axiom of construction [axiom: 1.10] to define the class $\{x | x = a\}$, this leads to the following definition.

Definition 1.40. If a is a element then $\{a\} = \{x|x=a\}$ is a class containing only one element. The class $\{a\}$ is called a **singleton**.

Lemma 1.41. If a, b are elements such that $a = b$ then $\{a\} = \{b\}$

Proof. If $z \in \{a\}$ then $z = a$ which by $a = b$ and [theorem: 1.9] proves that $z = b$ hence $z \in \{b\}$. Likewise if $z \in \{b\}$ then $z = b$ which by $a = b$ and [theorem: 1.9] proves that $z = a$ hence $z \in \{a\}$. Using the axiom of extent [axiom: 1.5] it follows then that $\{a\} = \{b\}$ $\square$

If a, b are elements then we can use the axiom of construction [axiom: 1.10] to define the class $\{x|x = a \vee x = b\}$ consisting of two elements. This leads to the following definition.

Definition 1.42. If a, b are elements then $\{a, b\} = \{x|x = a \vee x = b\}$ is called a **unordered pair**.

The next axiom ensures we can construct new elements from given elements.. It allows us to create classes that has as members pairs of elements.

Axiom 1.43. (Axiom of Pairing) If a, b are elements then $\{a, b\}$ is a element

Lemma 1.44. If a is a element then $\{a, a\} = \{a\}$

Proof.

$$\begin{aligned} x \in \{a, a\} &\Leftrightarrow x = a \vee x = a \\ &\Leftrightarrow x = a \\ &\Leftrightarrow x \in \{a\} \end{aligned}$$

$\square$

Theorem 1.45. If a is a element then $\{a\}$ is a element

Proof. As a is a element we have by the axiom of pairing [axiom: 1.43] that $\{a, a\}$ is a element, which as $\{a\} \stackrel{\text{lemma: 1.44}}{=} \{a, a\}$ proves that $\{a\}$ is a element. $\square$

The following lemma characterize equality of unordered pairs and will be used later to characterize equality of ordered pairs.

Lemma 1.46. If x, y, x', y' are elements then

$$\{x, y\} = \{x', y'\} \text{ implies } (x = x' \wedge y = y') \vee (x = y' \wedge y = x')$$

Proof. Lets's consider the following possible cases x, y :

$x = y$. Then $\{x\} \stackrel{\text{lemma: 1.44}}{=} \{x, y\} = \{x', y'\}$. From $x' \in \{x', y'\} = \{x\}$ it follows that $x = x'$ and from $y' \in \{x', y'\} = \{x\}$ it follows that $y = x$. As $x = x'$ it follows from [theorem: 1.9] that $y = x'$. So we have that $(x = x' \wedge y = y')$ from which it follows that

$$(x = x' \wedge y = y') \vee (x = y' \wedge y = x')$$

$x \neq y$. Then as $x \in \{x, y\} = \{x', y'\}$ we have by [axiom: 1.5] that $x \in \{x', y'\}$, so by definition we have for x either

$x = x'$. Then as $y \in \{x, y\} = \{x', y'\}$ we have by [axiom: 1.5] that $y \in \{x', y'\}$, so by definition we have for y either:

$y = x'$. As $x = x' \stackrel{\text{theorem: 1.9}}{\Rightarrow} x = y$ we contradict $x \neq y$ so this case does not apply

$y = y'$. Then $(x = x' \wedge y = y')$ hence $(x = x' \wedge y = y') \vee (x = y' \wedge y = x')$

$x = y'$. Then as $y \in \{x, y\} = \{x', y'\}$ we have by [axiom: 1.5] that $y \in \{x', y'\}$, so by definition we have for y either:

$y = x'$. Then $(x = y' \wedge y = x')$ hence $(x = x' \wedge y = y') \vee (x = y' \wedge y = x')$

$y = y'$. As $x = y' \xRightarrow{\text{theorem: 1.9}} x = y$ we contradict $x \neq y$ so this case does not apply

So in all cases that apply we have

$$(x = x' \wedge y = y') \vee (x = y' \wedge y = x') \quad \square$$

TODO check next

Lemma 1.47. *If x, y, x', y' are elements such that $(x = x' \wedge y = y') \vee (x = y' \wedge y = x')$ then $\{x, y\} = \{x', y'\}$*

Proof. Let $z \in \{x, y\}$ then either:

$z = x$. then if $x = x' \wedge y = y'$ we have using [theorem: 1.9] that $z = x'$, hence by definition $z \in \{x', y'\}$ and if $x = y' \wedge y = x'$ we have using [theorem: 1.9] that $z = y'$, hence by definition $x \in \{x', y'\}$

$z = y$. then if $x = x' \wedge y = y'$ we have using [theorem: 1.9] that $z = y'$, hence by definition $z \in \{x', y'\}$ and if $x = y' \wedge y = x'$ we have using [theorem: 1.9] that $z = x'$, hence by definition $x \in \{x', y'\}$

which proves that

$$\{x, y\} \subseteq \{x', y'\} \quad (1.5)$$

Let $z \in \{x', y'\}$ then either:

$z = x'$. then if $x = x' \wedge y = y'$ we have using [theorem: 1.9] that $z = x$, hence by definition $z \in \{x, y\}$ and if $x = y' \wedge y = x'$ we have using [theorem: 1.9] that $z = y$, hence by definition $x \in \{x, y\}$

$z = y'$. then if $x = x' \wedge y = y'$ we have using [theorem: 1.9] that $z = y$, hence by definition $z \in \{x, y\}$ and if $x = y' \wedge y = x'$ we have using [theorem: 1.9] that $z = x$, hence by definition $x \in \{x, y\}$

which proves that

$$\{x', y'\} \subseteq \{x, y\} \quad (1.6)$$

Using [theorem: 1.9] on [eq: 1.5, 1.6] proves that

$$\{x = y\} = \{x' = y'\} \quad \square$$

The above lemma actually shows that the order of the elements in unordered pairs do not matter, to remedy this we construct a ordered pair.

Definition 1.48. *If a, b are elements then*

$$(a, b) = \{\{a\}, \{a, b\}\}$$

Note 1.49. *As $\{a\}, \{a, b\}$ are elements we have again that $\{\{a\}, \{a, b\}\}$ is a element, hence (a, b) is also a element.*

Next we show that the order of elements is important for a tuple

Theorem 1.50. *Let x, y, x', y' are elements then*

$$(x, y) = (x', y') \Leftrightarrow x = x' \wedge y = y'$$

Proof.

$\Rightarrow$. If $(x, y) = (x', y')$ then by definition

$$\{\{x\}, \{x, y\}\} = \{\{x'\}, \{x', y'\}\}$$

By [lemma: 1.46] we have either:

$\{x\} = \{x'\} \wedge \{x, y\} = \{x', y'\}$. then, as $x \in \{x\}$, we have by definition $x = x'$, using [lemma: 1.46] again we have either:

$$x = x' \wedge y = y'. \text{ Then } x = x' \wedge y = y'$$

$x = y' \wedge y = x'$. Then by [theorem: 1.9] and $x = x'$ we have $y' = x'$ so that by [theorem: 1.9] again $y = y'$. Hence we have $x = x' \wedge y = y'$

$\{x\} = \{x', y'\} \wedge \{x, y\} = \{x'\}$. Then as $x', y' \in \{x', y'\} = \{x\}$ we have $x' = x \wedge y' = x$, as $x, y \in \{x, y\} = \{x'\}$ we have $x = x' \wedge y = x'$. Using [theorem: 1.9] on $y' = x \wedge x = x' \wedge y = x'$ we have $y = y'$. Hence $x = x' \wedge y = y'$.

So in all cases we have

$$x = x' \wedge y = y'$$

$\Leftarrow$. As $x = x'$ it follows from [lemma: 1.41] that $\{x\} = \{x'\}$, from $x = x' \wedge y = y'$ we have by [lemma: 1.47] that $\{x, y\} = \{x', y'\}$. Using [lemma: 1.47] gives then that $\{\{x\}, \{x, y\}\} = \{\{x'\}, \{x', y'\}\}$ which by definition gives

$$(x, y) = (x', y') \quad \square$$

We are now ready to define the Cartesian product of two classes, using the axiom of construction [axiom: 1.10].

Definition 1.51. If A, B are classes then the **Cartesian product** of A and B noted by $A \times B$ is defined as

$$A \times B = \{z | z = (a, b) \wedge a \in A \wedge b \in B\}$$

Notation 1.52. Instead of writing $\{z | z = (a, b) \wedge a \in A \wedge b \in B\}$ we use in the future the shorter notation $\{(a, b) | a \in A \wedge b \in B\}$

A special case of the Cartesian product is the Cartesian product of empty sets.

Example 1.53. $\emptyset = \emptyset \times \emptyset$

Proof. If $z \in \emptyset \times \emptyset$ then there exists a $x, y \in \emptyset$ such that $z = (x, y)$ which contradict $x, y \notin \emptyset$ [theorem: 1.21] hence by 1.23 we have $\emptyset \times \emptyset = \emptyset$. $\square$

Theorem 1.54. Let A be a class then $A \times \emptyset = \emptyset$ and $\emptyset \times A = \emptyset$

Proof. If $z \in A \times \emptyset$ then $z = (x, y)$ where $y \in \emptyset$, which contradicts $y \notin \emptyset$ [theorem: 1.21], so using [theorem: 1.23] we have that

$$A \times \emptyset = \emptyset$$

Likewise if $x \in \emptyset \times A$ then $z = (x, y)$ where $x \in \emptyset$, which contradicts $x \notin \emptyset$ [theorem: 1.21], so using [theorem: 1.23] we have that

$$\emptyset \times A = \emptyset$$

$\square$

Theorem 1.55. If A, B, C, D are classes then we have:

1. If $A \subseteq B \wedge C \subseteq D$ then $A \times C \subseteq B \times D$
2. Let $A \neq \emptyset \wedge C \neq \emptyset$ then if $A \times C \subseteq B \times D$ it follows that $A \subseteq B \wedge C \subseteq D$
3. Let $A \neq \emptyset \wedge B \neq \emptyset \wedge C \neq \emptyset$ then $A \times C = B \times D \Leftrightarrow A = B \wedge C = D$

Proof.

1. Let $z \in A \times C$ then there exists a $x \in A$ and $y \in C$ such that $z = (x, y)$. As $A \subseteq B \wedge C \subseteq D$ it follows that $x \in B \wedge y \in D$ so that $z = (x, y) \in B \times D$ Hence

$$A \times C \subseteq B \times D$$

2. Let $x \in A$ then, as $C \neq \emptyset$, we have by [corollary: 1.24] the existence of a $y \in C$, then $(x, y) \in A \times C$ which as $A \times C \subseteq B \times D$ proves that $(x, y) \in B \times D$. By definition we have then that $x \in B$ proving

$$A \subseteq B$$

Likewise, let $y \in C$ then, as $A \neq \emptyset$ we have by [corollary: 1.24] the existence of a $x \in A$, hence $(x, y) \in A \times C$, which as $A \times C \subseteq B \times D$, proves $(x, y) \in B \times D$ and by definition $y \in D$. Hence

$$C \subseteq D$$

3.

$\Rightarrow$. First as $A \times C = B \times D$ we have by [theorem: 1.9] that $A \times C \subseteq B \times D$, using (2) proves then that

$$A \subseteq B \wedge C \subseteq D \quad (1.7)$$

Next as $A \times C = B \times D$ we have by [theorem: 1.9] that $B \times D \subseteq A \times C$, using (2) proves then that

$$B \subseteq A \wedge C \subseteq D \quad (1.8)$$

Combining then [eq 1.7, 1.8] with [theorem: 1.9] proves

$$A = B \wedge C = D$$

$\Leftarrow$. As $A = B \wedge C = D$ we have by [theorem: 1.9] that $A \subseteq B, C \subseteq D, B \subseteq A, D \subseteq C$ which using (1) gives that $A \times C \subseteq B \times D \wedge B \times D \subseteq A \times C$. Using [theorem: 1.9] it follows then that

$$A \times C = B \times D \quad \square$$

Theorem 1.56. Let A, B, C and D be classes then we have

1. $A \times (B \cap C) = (A \times B) \cap (A \times C)$
2. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
3. $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$
4. $(B \cap C) \times A = (B \times A) \cap (C \times A)$
5. $(B \cup C) \times A = (B \times A) \cup (C \times A)$
6. $(A \times B) \setminus (C \times D) = ((A \setminus C) \times B) \cup (A \times (B \setminus D))$
7. $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$
8. $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

Proof.

1. We have

$$\begin{aligned} z \in A \times (B \cap C) &\Leftrightarrow z = (x, y) \wedge x \in A \wedge y \in (B \cap C) \\ &\Leftrightarrow z = (x, y) \wedge x \in A \wedge (y \in B \wedge y \in C) \\ &\Leftrightarrow (z = (x, y) \wedge x \in A \wedge y \in B) \wedge (z = (x, y) \wedge x \in A \wedge y \in C) \\ &\Leftrightarrow z \in A \times B \wedge z \in A \times C \\ &\Leftrightarrow z \in (A \times B) \cap (A \times C) \end{aligned}$$

2. We have

$$\begin{aligned} z \in A \times (B \cup C) &\Leftrightarrow z = (x, y) \wedge x \in A \wedge y \in (B \cup C) \\ &\Leftrightarrow z = (x, y) \wedge x \in A \wedge (y \in B \vee y \in C) \\ &\Leftrightarrow (z = (x, y) \wedge x \in A \wedge y \in B) \vee (z = (x, y) \wedge x \in A \wedge y \in C) \\ &\Leftrightarrow z \in A \times B \vee z \in A \times C \\ &\Leftrightarrow z \in (A \times B) \cup (A \times C) \end{aligned}$$

3. We have

$$\begin{aligned}
 z \in (A \times B) \cap (C \times D) &\Leftrightarrow z \in A \times B \wedge z \in C \times D \\
 &\Leftrightarrow (z = (x, y) \wedge x \in A \wedge y \in B) \wedge (z = (x', y') \wedge x' \in C \wedge y' \in D) \\
 &\stackrel{(x, y) = z = (x', y') \Rightarrow x = x', y = y'}{\Leftrightarrow} z = (x, y) \wedge x \in A \wedge y \in B \wedge x \in C \wedge y \in D \\
 &\Leftrightarrow z = (x, y) \wedge (x \in A \wedge x \in C) \wedge (y \in B \wedge y \in D) \\
 &\Leftrightarrow z = (x, y) \wedge (x \in A \cap C) \wedge (y \in B \cap D) \\
 &\Leftrightarrow z \in (A \cap C) \times (B \cap D)
 \end{aligned}$$

4. We have

$$\begin{aligned}
 z \in (B \cap C) \times A &\Leftrightarrow z = (x, y) \wedge x \in B \cap C \wedge y \in A \\
 &\Leftrightarrow z = (x, y) \wedge x \in B \wedge x \in C \wedge y \in A \\
 &\Leftrightarrow (z = (x, y) \wedge x \in B \wedge y \in A) \wedge (z = (x, y) \wedge x \in C \wedge y \in A) \\
 &\Leftrightarrow z \in B \times A \wedge z \in C \times A \\
 &\Leftrightarrow z \in (B \times A) \cap (C \times A)
 \end{aligned}$$

5. We have

$$\begin{aligned}
 z \in (B \cup C) \times A &\Leftrightarrow z = (x, y) \wedge x \in B \cup C \wedge y \in A \\
 &\Leftrightarrow z = (x, y) \wedge (x \in B \vee x \in C) \wedge y \in A \\
 &\Leftrightarrow (z = (x, y) \wedge x \in B \wedge y \in A) \vee (z = (x, y) \wedge x \in C \wedge y \in A) \\
 &\Leftrightarrow (z \in B \times A) \vee (z \in C \times A) \\
 &\Leftrightarrow z \in (B \times A) \cup (C \times A)
 \end{aligned}$$

6. We have

$$\begin{aligned}
 z \in (A \times B) \setminus (C \times D) &\Leftrightarrow \\
 (z = (x, y) \wedge x \in A \wedge y \in B) \wedge (x, y) \notin C \times D &\Leftrightarrow \\
 (z = (x, y) \wedge x \in A \wedge y \in B) \wedge \neg(x \in C \wedge y \in D) &\Leftrightarrow \\
 (z = (x, y) \wedge x \in A \wedge y \in B) \wedge (x \notin C \vee y \notin D) &\Leftrightarrow \\
 (z = (x, y) \wedge x \in A \wedge y \in B \wedge x \notin C) \vee (z = (x, y) \wedge x \in A \wedge y \in B \wedge y \notin D) &\Leftrightarrow \\
 z = (x, y) \wedge [(x, y) \in (A \setminus C) \times B \vee (x, y) \in A \times (B \setminus D)] &\Leftrightarrow \\
 z \in ((A \setminus C) \times B) \cup (A \times (B \setminus D)) &\Leftrightarrow
 \end{aligned}$$

7. We have

$$\begin{aligned}
 (A \times C) \setminus (B \times C) &\stackrel{(6)}{=} ((A \setminus C) \times B) \cup (A \times (C \setminus C)) \\
 &\stackrel{[theorem: 1.35]}{=} ((A \setminus C) \times B) \cup (A \times \emptyset) \\
 &\stackrel{[theorem: 1.54]}{=} ((A \setminus C) \times B) \cup \emptyset \\
 &\stackrel{[theorem: 1.35]}{=} (A \setminus C) \times B
 \end{aligned}$$

8. We have

$$\begin{aligned}
 (A \times B) \setminus (A \times C) &\stackrel{(6)}{=} ((A \setminus A) \times B) \cup (A \times (B \setminus C)) \\
 &\stackrel{[theorem: 1.35]}{=} (\emptyset \times B) \cup (A \times (B \setminus C)) \\
 &\stackrel{[theorem: 1.54]}{=} \emptyset \cup (A \times (B \setminus C)) \\
 &\stackrel{[theorem: 1.35]}{=} A \times (B \setminus C)
 \end{aligned}$$

□

1.4 Sets

Remember that that another name for **element** is **set** [definition: 1.2]. Up to now we have used the name **element**, because we want to think of a element as a member of a class. However a element is also a class and can contain other elements. If we want to stress the collection aspect then we use the word **set** instead of **element**. The convention is to use uppercase to represent a set and lower cases for a element. Of course set and element are the same thing, we just want to stress different aspects of the same thing. Note that we have two kinds of classes classes that are a member of another class and classes that are not a member of a class. This leads to the following definition.

Definition 1.57. *A class A is a **set** [or **element**] if there exists a class B such that $A \in B$. A class that is never a member of another class is called a **proper class**.*

Up to know we had axioms that given a element/set create a new element/set, but we have not ensured the existence of a element/set. To this we must first define the concept of a successor set.

Definition 1.58. *A set S is a **successor set** iff*

1. $\emptyset \in S$
2. If $X \in S$ then $X \cup \{X\} \in S$

Of course nothing proves that successor set's exists, to ensure the existence of a successor set we have the axiom of infinity.

Axiom 1.59. (Axiom of Infinity) *There exists a **successor set***

This axiom ensures that we have at least one set. We can then use the other axioms about elements/sets to create new elements. Later we will use the Axiom of Infinity to create the Natural Numbers, form which we build all the other numbers (integers, rationals, reals, complex numbers). The Axiom of Infinity ensures also that the empty class is actually a set.

Theorem 1.60. *$\emptyset$ is a set*

Proof. The Axiom of Infinity [axiom: 1.59] ensures the existence of a successor set S . By definition we have then that $\emptyset \in S$ which proves that $\emptyset$ is a set. $\square$

So now we have two sets to start with, the successor set and the empty set. We can use the Axiom of Pairing [axiom: 1.43] to create new sets like singletons, unordered pairs and pairs. We introduce now extra axioms to create new sets given existing sets.

As a application we proof that the intersection of two sets is a set

Theorem 1.61. *Let A, B be sets then $A \cap B$ is a set*

Proof. By [theorem: 1.30] we have that $A \cap B \subseteq A$, so by the axiom of infinity [axiom: 1.59] it follows that $A \cap B$ is a set. $\square$

We define now a more general concept of union and intersection

Definition 1.62. *Let $\mathcal{A}$ be a class then using the Axiom of Construction [axiom: 1.10] we define $\bigcup \mathcal{A} = \{x | \exists y \in \mathcal{A} \text{ such that } x \in y\}$*

Definition 1.63. *Let $\mathcal{A}$ be a class then using the Axiom of Construction [axiom: 1.10] we define $\bigcap \mathcal{A} = \{x | \forall y \in \mathcal{A} \text{ we have } x \in y\}$*

Example 1.64. Let A be a class then

1. $\bigcup \{A\} = A$

2. $\bigcap \{A\} = A$
3. $\bigcup \emptyset = \emptyset$

Proof.

1.

$$x \in \bigcup \{A\} \Leftrightarrow \begin{array}{l} \exists y \in \{A\} \text{ with } x \in y \\ \Leftrightarrow_{y \in \{A\} \Leftrightarrow y=A} x \in A \end{array}$$

proving that

$$\bigcup \{A\} = A$$

2.

$$x \in \bigcap \{A\} \Leftrightarrow \begin{array}{l} \forall y \in \{A\} \text{ we have } x \in y \\ \Leftrightarrow_{y \in \{A\} \Leftrightarrow y=A} x \in A \end{array}$$

proving that

$$\bigcap \{A\} = A$$

3. Assume that $x \in \emptyset$ then $\exists y \in \emptyset$ such that $x \in y$ which lead by the definition of $\emptyset$ [definition: 1.20] to the contradiction that $y \neq y$. $\square$

Example 1.65. Let A and B classes then

1. $\bigcup \{A, B\} = A \bigcup B$
2. $\bigcap \{A, B\} = A \bigcap B$

Proof.

1.

$$x \in \bigcup \{A, B\} \Leftrightarrow \begin{array}{l} \exists y \in \{A, B\} \text{ with } x \in y \\ \Leftrightarrow_{y \in \{A, B\} \Leftrightarrow y=A \vee y=B} x \in A \vee x \in B \\ \Leftrightarrow x \in A \bigcup B \end{array}$$

proving that

$$\bigcup \{A, B\} = A \bigcup B$$

2.

$$x \in \bigcap \{A, B\} \Leftrightarrow \begin{array}{l} \forall y \in \{A, B\} \text{ with } x \in y \\ \Leftrightarrow_{y \in \{A, B\} \Leftrightarrow y=A \vee y=B} x \in A \wedge x \in B \\ \Leftrightarrow x \in A \bigcap B \end{array}$$

proving that

$$\bigcap \{A, B\} = A \bigcap B$$

$\square$

Theorem 1.66. If $\mathcal{A}$ is a class

1. If $A \in \mathcal{A}$ then $\bigcap \mathcal{A} \subseteq A$
2. If $A \in \mathcal{A}$ then $A \subseteq \bigcup \mathcal{A}$
3. If $\forall A \in \mathcal{A}$ we have $C \subseteq A$ then $C \subseteq \bigcap \mathcal{A}$
4. If $\forall A \in \mathcal{A}$ we have $A \subseteq C$ then $\bigcup \mathcal{A} \subseteq C$
5. If $\mathcal{A} \neq \emptyset$ then $\bigcap \mathcal{A}$ is a set

Proof.

1. Let $A \in \mathcal{A}$ then if $x \in \bigcap \mathcal{A}$ we have by definition of $\bigcap \mathcal{A}$ that $x \in A$. Hence $\bigcap \mathcal{A} \subseteq A$
2. If $x \in A$ then $\exists y \in \mathcal{A}$ such that $x \in y$ [take $y = A$] so that $x \in \bigcup \mathcal{A}$
3. If $x \in C$ then $\forall A \in \mathcal{A}$ we have as $C \in \mathcal{A}$ that $x \in A$ so that $x \in \bigcap \mathcal{A}$
4. If $x \in \bigcup \mathcal{A}$ then $\exists A \in \mathcal{A}$ such that $x \in A$ which as $A \subseteq C$ proves that $x \in C$
5. As $\mathcal{A} \neq \emptyset$ there exists a $A \in \mathcal{A}$, which by definition means that A is a set. Using (1) we have $\bigcap \mathcal{A} \subseteq A$, applying then the Axiom of Subsets [axiom: 1.8] it follows that $\bigcap \mathcal{A}$ is a set. $\square$

The above is not applicable for unions, however we state the Axiom of Unions that will ensure that $\bigcup \mathcal{A}$ is a set if $\mathcal{A}$ is a set

Axiom 1.67. (Axiom of Unions) *If $\mathcal{A}$ is a set then $\bigcup \mathcal{A}$ is a set*

A consequence of the above axiom is that the union of two sets is a set

Theorem 1.68. *Let A, B be two sets then $A \cup B$ is a set*

Proof. Using the Axiom of Pairing [axiom: 1.43] we have that $\{A, B\}$ is a set. Further

$$\begin{aligned} x \in A \cup B &\Leftrightarrow x \in A \vee x \in B \\ &\Leftrightarrow \exists C \in \{A, B\} \text{ with } x \in C \\ &\Leftrightarrow \bigcup \{A, B\} \end{aligned}$$

proving by the Axiom of Union [axiom: 1.67] we have that $A \cup B$ is a set. $\square$

Definition 1.69. *Let A be a set then we use the Axiom of Construction to define $\mathcal{P}(A)$ by*

$$\mathcal{P}(A) = \{B \mid B \subseteq A\}$$

We introduce now the Axiom of Power Sets to ensure that $\mathcal{P}(A)$ is a set, called the **power set** of A .

Axiom 1.70. (Axiom of Power Sets) *If A is a set then $\mathcal{P}(A)$ is a set*

Theorem 1.71. *If A is a set and $P(X)$ a predicate then $\{X \mid X \subseteq A \wedge P(X)\}$ is a set.*

Proof. If $B \in \{X \mid X \subseteq A \wedge P(X)\}$ then $B \subseteq A$ so that $B \in \mathcal{P}(A)$, proving that

$$\{X \mid X \subseteq A \wedge P(X)\} \subseteq \mathcal{P}(A)$$

Using the Axiom of Power Sets [axiom: 1.70] $\mathcal{P}(A)$ is a set, so we can use the Axiom of Subsets to prove that $\{X \mid X \subseteq A \wedge P(X)\}$ is a set. $\square$

Lemma 1.72. *If A, B are classes then $A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B))$*

Proof. Let $z \in A \times B$ then there exists a $x \in A$ and a $y \in B$ so that $z = (x, y)$. Now if $e \in \{x\}$ then $e = x$ proving that $e \in A$, hence we have, by definition of the union, that $\{x\} \subseteq A \cup B$. By definition of the $\mathcal{P}(A \cup B)$ set it follows then that

$$\{x\} \in \mathcal{P}(A \cup B)$$

Likewise if $e \in \{x, y\}$ then either $e = x \Rightarrow e \in A$ or $e = y \Rightarrow e \in B$, hence, by definition of the union, we have $\{x, y\} \subseteq A \cup B$. Using the definition $\mathcal{P}(A \cup B)$ we have then

$$\{x, y\} \in \mathcal{P}(A \cup B)$$

Now if $e \in \{\{x\}, \{x, y\}\}$ then either $e = \{x\} \in \mathcal{P}(A \cup B)$ or $e = \{x, y\} \in \mathcal{P}(A \cup B)$ which proves that $\{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(A \cup B)$ or

$$z \in \{\{x\}, \{x, y\}\} \in \mathcal{P}(\mathcal{P}(A \cup B))$$

giving finally

$$A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B)) \quad \square$$

Theorem 1.73. *If A and B are sets then $A \times B$ is a set*

Proof. As A, B are sets we have by [theorem: 1.68] that $A \cup B$ is a set, using the Axiom of Power sets [axiom: 1.70] it follows that $\mathcal{P}(A \cup B)$ is a set, using the Axiom of Power sets [axiom: 1.70] again proves that $\mathcal{P}(\mathcal{P}(A \cup B))$ is a set. Finally by [lemma: 1.72] we have that $A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B))$, which using the Axiom of Subsets [axiom: 1.8] proves that

$$A \times B \text{ is a set} \quad \square$$

We show now that given a set we can find a element that is not in the set.

Theorem 1.74. *Let S be a set then there exists a element x such that $x \notin S$*

Proof. Use the axiom of construction [axiom: 1.10] to define $T = \{x \in S \mid x \notin x\}$ which as $T \subseteq S$ is a set by [axiom: 1.8] is a set. Assume now that $T \in S$ then we have either:

$T \in T$. Then we have $T \notin T$ contradicting $T \in T$.

$T \notin T$. Then we have by definition of T that $T \in T$ contradicting $T \notin T$

As in all cases we have a contradiction the assumption is wrong and we must have $T \notin S$. $\square$

Chapter 2

Partial Functions and Functions

2.1 Pairs and Triples

Although we have already defined the concept of a pair, we can not simple extend this to pairs (and later triples) of classes. If A, B are pure classes (classes that are not elements) then we can not just form $(A, B) = \{A, \{B\}\}$ because this would mean that A, B are elements and not pure classes. So we need another way of forming pairs, triples and so on.

Definition 2.1. *If A, B are classes then $\langle A, B \rangle$ is defined by $\langle A, B \rangle = (A \times \{\emptyset\}) \cup (B \times \{\{\emptyset\}\})$*

We show now that from $\langle A, B \rangle = \langle A', B' \rangle$ it follows that $A = A' \wedge B = B'$, first we need some lemma's

Lemma 2.2. *We have $\emptyset \neq \{\emptyset\}$*

Proof. Assume that $\{\emptyset\} = \emptyset$ then, as $\emptyset \in \{\emptyset\}$ it follows that $\emptyset$ which is a contradiction, hence

$$\emptyset \neq \{\emptyset\}$$

□

Lemma 2.3. *If A, B, C, D are classes then $\langle A, B \rangle = \langle C, D \rangle \Leftrightarrow A = C \wedge B = D$*

Proof.

$\Rightarrow$. Assume that $\langle A, B \rangle = \langle C, D \rangle$ then by definition

$$(A \times \{\emptyset\}) \cup (B \times \{\{\emptyset\}\}) = (C \times \{\emptyset\}) \cup (D \times \{\{\emptyset\}\}) \quad (2.1)$$

Let now $x \in A$ then $(x, \emptyset) \in (A \times \{\emptyset\})$ so that by the axiom of extent [axiom: 1.5] and [eq: 2.1]

$$(x, \emptyset) \in (C \times \{\emptyset\}) \cup (D \times \{\{\emptyset\}\})$$

which by the definition of the union gives

$$(x, \emptyset) \in C \times \{\emptyset\} \vee (x, \emptyset) \in D \times \{\{\emptyset\}\} \quad (2.2)$$

Now if $(x, \emptyset) \in D \times \{\{\emptyset\}\}$ then $\emptyset \in \{\{\emptyset\}\}$ or $\emptyset = \{\emptyset\}$ which is impossible by [lemma: 2.2] so that by [eq: 2.2] we have $(x, \emptyset) \in C \times \{\emptyset\}$, hence $x \in C$. This proves that

$$A \subseteq C \quad (2.3)$$

Likewise, let $x \in C$ then $(x, \emptyset) \in (C \times \{\emptyset\})$ so that by the axiom of extent [axiom: 1.5] and [eq: 2.1]

$$(x, \emptyset) \in (A \times \{\emptyset\}) \cup (B \times \{\{\emptyset\}\})$$

which by the definition of the union gives

$$(x, \emptyset) \in A \times \{\emptyset\} \vee (x, \emptyset) \in B \times \{\{\emptyset\}\} \quad (2.4)$$

Now if $(x, \emptyset) \in B \times \{\{\emptyset\}\}$ then $\emptyset \in \{\{\emptyset\}\}$ or $\emptyset = \{\emptyset\}$ which is impossible by [lemma: 2.2] so that by [eq: 2.4] we have $(x, \emptyset) \in A \times \{\emptyset\}$, hence $x \in A$. This proves that

$$C \subseteq A \quad (2.5)$$

Combining [eq: 2.3,2.5] with [theorem: 1.9] proves

$$A = C$$

Further if $x \in B$ then $(x, \{\emptyset\}) \in B \times \{\{\emptyset\}\}$ so that by the axiom of extent [axiom: 1.5] and [eq: 2.1]

$$(x, \{\emptyset\}) \in (C \times \{\emptyset\}) \cup (D \times \{\{\emptyset\}\})$$

or using the definition of the union that

$$(x, \{\emptyset\}) \in C \times \{\emptyset\} \vee (x, \{\emptyset\}) \in D \times \{\{\emptyset\}\} \quad (2.6)$$

If $(x, \{\emptyset\}) \in C \times \{\emptyset\}$ then $\{\emptyset\} \in \{\emptyset\}$ or $\{\emptyset\} = \emptyset$ which is impossible by [lemma: 2.2], so by [eq: 2.6] we have that $(x, \{\emptyset\}) \in D \times \{\{\emptyset\}\}$, hence $x \in D$. This proves that

$$B \subseteq D \quad (2.7)$$

Likewise, if $x \in D$ then $(x, \{\emptyset\}) \in D \times \{\{\emptyset\}\}$ so that by the axiom of extent [axiom: 1.5] and [eq: 2.1]

$$(x, \{\emptyset\}) \in (A \times \{\emptyset\}) \cup (B \times \{\{\emptyset\}\})$$

or using the definition of the union that

$$(x, \{\emptyset\}) \in A \times \{\emptyset\} \vee (x, \{\emptyset\}) \in B \times \{\{\emptyset\}\} \quad (2.8)$$

If $(x, \{\emptyset\}) \in A \times \{\emptyset\}$ then $\{\emptyset\} \in \{\emptyset\}$ or $\{\emptyset\} = \emptyset$ which is impossible by [lemma: 2.2], so by [eq: 2.8] we have that $(x, \{\emptyset\}) \in B \times \{\{\emptyset\}\}$, hence $x \in B$. This proves that

$$D \subseteq B \quad (2.9)$$

Combining [eq: 2.7,2.9] with [theorem: 1.9] proves

$$B = D$$

$\Leftarrow$. Assume that $A = C \wedge B = D$ then

$$\begin{aligned} x \in \langle A, B \rangle & \Leftrightarrow x \in (A \times \{\emptyset\}) \cup (B \times \{\{\emptyset\}\}) \\ & \Leftrightarrow x \in A \times \{\emptyset\} \vee x \in B \times \{\{\emptyset\}\} \\ & \Leftrightarrow (x = (a, \emptyset) \wedge a \in A) \vee (x = (b, \{\emptyset\}) \wedge b \in B) \\ & \stackrel{[\text{axiom: 1.5}]}{\Leftrightarrow} (x = (a, \emptyset) \wedge a \in C) \vee (x = (b, \{\emptyset\}) \wedge b \in D) \\ & \Leftrightarrow x \in (C \times \{\emptyset\}) \cup (D \times \{\{\emptyset\}\}) \\ & \Leftrightarrow e \in \langle C, D \rangle \end{aligned}$$

so that by the Axiom of Extent [axiom: 1.5]

$$\langle A, B \rangle = \langle C, D \rangle \quad \square$$

We can now easily extend $\langle A, B \rangle$ to a triple $\langle A, B, C \rangle$.

Definition 2.4. Let A, B, C be classes then $\langle A, B, C \rangle$ is defined by

$$\langle A, B, C \rangle = \langle \langle A, B \rangle, C \rangle$$

Lemma 2.5. Let A, B, C, D, E, F be classes then

$$\langle A, B, C \rangle = \langle D, E, F \rangle \Leftrightarrow A = D \wedge B = E \wedge C = F$$

Proof.

$\Rightarrow$. Assume that $\langle A, B, C \rangle = \langle D, E, F \rangle$ then by definition $\langle \langle A, B \rangle, C \rangle = \langle \langle D, E \rangle, F \rangle$, by [lemma: 2.3] then $C = F \wedge \langle A, B \rangle = \langle D, E \rangle$, using [lemma: 2.3] again proves then $A = D \wedge B = E$.

←. Assume that $A = D \wedge B = E \wedge C = F$ then by [lemma: 2.3] $\langle A, B \rangle = \langle D, E \rangle$, using [lemma: 2.3] again we have $\langle \langle A, B \rangle, C \rangle = \langle \langle D, E \rangle, F \rangle$ which by definition proves that

$$\langle A, B, C \rangle = \langle D, E, F \rangle \quad \square$$

2.2 Partial functions and Functions

The concept of a function as a mapping of one value to a unique value is used throughout mathematics, especially in analysis, which is essentially a theory of functions. Note that a function maps a value x to a **unique** value y which in the context of a set theory defines a pair (x, y) . This leads to the following definition of a graph.

2.2.1 Partial function

Definition 2.6. (Graph) A graph is a sub class of $\mathcal{U} \times \mathcal{U}$, or in other words a graph is a collection of pairs.

Definition 2.7. A triple $\langle A, B, f \rangle$ where A, B are classes and f a graph is a **partial function between A and B** if

1. $f \subseteq A \times B$
2. If $(x, y) \in f \wedge (x, y') \in f$ then $y = y'$

We call A the **source** of the partial function, B the **destination** of the partial function and f the **graph** of the partial function.

Remark 2.8. Instead of writing $\langle A, B, f \rangle$ for a partial function between A and B we use the notation $f: A \rightarrow B$ or $A \xrightarrow{f} B$. Further the condition (2) ensures that only one value can be associated with x . So it is useful to use a special notation for this unique value, especially if we have an expression to calculate this unique value.

Definition 2.9. Let $f: A \rightarrow B$ be a partial function then $(x, y) \in f$ is equivalent with $y = f(x)$

From now on we will use the Axiom of Construction [axiom: 1.10] to define different classes related to partial functions without explicitly mentioning this. It is assumed that the reader understands when to use this axiom.

Definition 2.10. Let $f: A \rightarrow B$ be a partial function then its domain noted as $\text{dom}(f)$ and range noted as $\text{range}(f)$ is defined by

$$\begin{aligned} \text{dom}(f) &= \{x \mid \exists y \text{ such that } (x, y) \in f\} \\ \text{range}(f) &= \{y \mid \exists x \text{ such that } (x, y) \in f\} \end{aligned}$$

Theorem 2.11. If $f: A \rightarrow B$ is a partial function then $\text{dom}(f) \subseteq A$ and $\text{range}(f) \subseteq B$

Proof. If $x \in \text{dom}(f)$ then $\exists y$ such that $(x, y) \in f \xrightarrow{f \subseteq A \times B} (x, y) \in A \times B$ proving that $x \in A$, hence

$$\text{dom}(f) \subseteq A$$

Further if $y \in \text{range}(f)$ then $\exists x$ such that $(x, y) \in f \xrightarrow{f \subseteq A \times B} (x, y) \in A \times B$ proving that $y \in B$, hence

$$\text{range}(f) \subseteq B \quad \square$$

Corollary 2.12. If A, B are sets and $f: A \rightarrow B$ a partial function then $\text{dom}(f)$ and $\text{range}(f)$ are sets

Proof. Using [theorem: 2.11] we have that $\text{dom}(f) \subseteq A$ and $\text{range}(f) \subseteq B$, so applying the Axiom of Subsets [axiom: 1.8] proves that $\text{dom}(f)$ and $\text{range}(f)$ are sets. $\square$

Definition 2.13. Let $f: A \rightarrow B$ be a partial function and C a class such that $C \subseteq A$ then **the image of C by f** noted as $f(C)$ is defined by

$$f(C) = \{y | \exists x \in C \text{ such that } (x, y) \in f\}$$

Remark 2.14. Note that we use a conflicting notation here. On one hand $y = f(x)$ can be interpreted as $(x, y) \in f$, on the other hand it can also mean that y is the image of x by f . We adopt the following convention. If lower cases are used as in $y = f(x)$ we interpret this as $(x, y) \in f$ and if we use uppercase like in $f(C)$ we are talking about images. In case of doubt $(f)(C)$ always refers to the image.

Definition 2.15. Let $f: A \rightarrow B$ be a partial function and C a class then **the preimage of C by f** noted as $f^{-1}(C)$ is defined by

$$f^{-1}(C) = \{x | \exists y \in C \text{ such that } (x, y) \in f\}$$

Note 2.16. In contrast with most text books we do not require that $C \subseteq B$, this will give us more flexibility if we compose partial functions.

Theorem 2.17. Let $f: A \rightarrow B$ be a partial function, $C \subseteq A$ and D a class then we have:

1. $f(C) \subseteq \text{range}(f)$
2. $f^{-1}(D) \subseteq \text{dom}(f)$
3. $f(A) = \text{range}(f)$
4. $f^{-1}(B) = \text{dom}(f)$
5. If $E \subseteq C$ then $f(E) \subseteq f(C)$
6. If $E \subseteq D$ then $f^{-1}(E) \subseteq f^{-1}(D)$

and if in addition A, B are sets then $f(C)$ and $f^{-1}(D)$ are sets

Proof.

1. If $y \in f(C)$ then there exists a $x \in C$ such that $(x, y) \in f$, so $y \in \text{range}(f)$. Hence

$$f(C) \subseteq \text{range}(f)$$

2. If $x \in f^{-1}(D)$ then there exists a $y \in D$ such that $(x, y) \in f$, which proves that $x \in \text{dom}(f)$, hence

$$f^{-1}(D) \subseteq \text{dom}(f)$$

3. If $y \in \text{range}(f)$ then $\exists x$ such that $(x, y) \in f$, which as $f \subseteq A \times B$ proves that $x \in A$, hence $y \in f(A)$, or $\text{range}(f) \subseteq f(A)$. From (1) we have $f(A) \subseteq \text{range}(f)$, so using [theorem: 1.9]

$$f(A) = \text{range}(f)$$

4. If $x \in \text{dom}(f)$ then $\exists y$ such that $(x, y) \in f$, which as $f \subseteq A \times B$ proves that $y \in B$, giving $x \in f^{-1}(B)$, hence $\text{dom}(f) \subseteq f^{-1}(B)$. From (2) we have $f^{-1}(B) \subseteq \text{dom}(f)$, so using [theorem: 1.9]

$$f^{-1}(B) = \text{dom}(f)$$

5. If $y \in f(E)$ then $\exists x \in E$ such that $(x, y) \in f$, as $E \subseteq C$ we have $x \in C$ and still $(x, y) \in f$ so that $y \in f(C)$ proving

$$f(E) \subseteq f(C)$$

6. If $x \in f^{-1}(E)$ then $\exists y \in E$ such that $(x, y) \in f$, as $E \subseteq D$ we have $y \in D$ and still $(x, y) \in f$ so that $x \in f^{-1}(D)$ proving

$$f^{-1}(E) \subseteq f^{-1}(D)$$

Finally if A, B are sets then using [theorem: 2.12] $\text{range}(f)$ and $\text{dom}(f)$ are sets, applying then the Axiom of Subsets [axiom: 1.8] proves that $f(C)$ and $f^{-1}(D)$ are sets. $\square$

Next we define the composition of two partial functions.

Definition 2.18. (Composition of graphs) Let f, g be two graphs then $g \circ f$ is defined by

$$g \circ f = \{z \mid z = (x, y) \text{ such that } \exists u \text{ with } (x, u) \in f \wedge (u, y) \in g\}$$

Theorem 2.19. Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be partial functions, then

$$g \circ f: A \rightarrow D$$

is a partial function. We call $g \circ f: A \rightarrow D$ the **composition** of $f: A \rightarrow B$ and $g: C \rightarrow D$

Proof. If $(x, y) \in g \circ f$ then there exist a u such that $(x, u) \in f$ and $(u, y) \in g$, as f, g are partial functions we have that $f \subseteq A \times B$ and $g \subseteq C \times D$. So $(x, u) \in A \times B$ and $(u, y) \in C \times D$. So $x \in A$ and $y \in D$ proving that $(x, y) \in A \times D$. Hence

$$g \circ f \subseteq A \times D$$

Further if $(x, y) \in g \circ f \wedge (x, y') \in g \circ f$ then there exists u, v such that $(x, u) \in f \wedge (u, y) \in g \wedge (x, v) \in f \wedge (v, y') \in g$. From $(x, u) \in f \wedge (x, v) \in f$ it follows [as f is a partial function] that $u = v$. So $(u, y) \stackrel{u=v}{=} (u, y') \in g$. Hence as g is a partial function it follows that $y = y'$. To summarize

$$\text{If } (x, y) \in g \circ f \wedge (x, y') \in g \circ f \text{ then } y = y'$$

So all the requirements for $g \circ f: A \rightarrow D$ to be a partial function are satisfied. $\square$

Note 2.20. In contrast with most textbooks we do not require that $B = C$ in this theorem, there is no need for this because for partial functions $\text{dom}(f \circ g)$ can be different from A . Later we will compose functions and then we will need an extra condition for C .

Theorem 2.21. (Associativity of Composition) Let $f: A \rightarrow B, g: C \rightarrow D$ and $h: E \rightarrow F$ be partial functions then $h \circ (g \circ f) = (h \circ g) \circ f$

Proof. If $(x, z) \in h \circ (g \circ f)$ then $\exists u$ such that $(x, u) \in g \circ f$ and $(u, z) \in h$. As $(x, u) \in g \circ f$ there exists a v such that $(x, v) \in f$ and $(v, u) \in g$. As $(v, u) \in g \wedge (u, z) \in h$ we have that $(v, z) \in h \circ g$, as $(x, v) \in f$ it follows $(x, z) \in (h \circ g) \circ f$.

If $(x, z) \in (h \circ g) \circ f$ there $\exists u$ such that $(x, u) \in f$ and $(u, z) \in h \circ g$. As $(u, z) \in h \circ g$ there $\exists v$ such that $(u, v) \in g$ and $(v, z) \in h$. From $(x, u) \in f$ and $(u, v) \in g$ we have that $(x, v) \in g \circ f$. As $(v, z) \in h$ we have that $(x, z) \in h \circ (g \circ f)$.

Using the Axiom of Extension [axiom: 1.5] it follows that

$$h \circ (g \circ f) = (h \circ g) \circ f \quad \square$$

Let's look now at the domain and range of the composition of two partial functions.

Theorem 2.22. Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be partial functions then for $g \circ f: A \rightarrow D$ we have

1. $\text{dom}(g \circ f) = \text{dom}(f) \cap f^{-1}(\text{dom}(g))$
2. $\text{range}(g \circ f) = g(\text{range}(f) \cap \text{dom}(g))$
3. $\text{range}(g \circ f) \subseteq \text{range}(g)$

Proof.

1. If $x \in \text{dom}(g \circ f)$ then there exist a z such that $(x, z) \in g \circ f$. So there exist a y such that $(x, y) \in f$ and $(y, z) \in g$, hence $x \in \text{dom}(f)$ and $y \in \text{dom}(g) \stackrel{(x, y) \in f}{\Rightarrow} x \in f^{-1}(\text{dom}(g))$. So $x \in \text{dom}(f) \cap f^{-1}(\text{dom}(g))$. Hence

$$\text{dom}(g \circ f) \subseteq \text{dom}(f) \cap f^{-1}(\text{dom}(g)) \quad (2.10)$$

If $x \in \text{dom}(f) \cap f^{-1}(\text{dom}(g))$ then $x \in \text{dom}(f)$ so that $\exists y$ such that $(x, y) \in f$ and $x \in f^{-1}(\text{dom}(g))$ so that $\exists y' \in \text{dom}(g)$ such that $(x, y') \in f$. As f is a partial function it follows that $y = y'$. So $y \in \text{dom}(g)$, from which it follows that $\exists z$ such that $(y, z) \in g$. As we have $(x, y) \in f$ and $(y, z) \in g$ it follows that $(x, z) \in g \circ f$ or $x \in \text{dom}(g \circ f)$. This proves that $\text{dom}(f) \cap f^{-1}(\text{dom}(g)) \subseteq \text{dom}(g \circ f)$, combining this with [eq: 2.10] allows us to use [theorem: 1.9] to get

$$\text{dom}(g \circ f) = \text{dom}(f) \cap f^{-1}(\text{dom}(g))$$

2. If $z \in \text{range}(g \circ f)$ then there exists a $x \in A$ such that $(x, z) \in g \circ f$, so there exist a y such that $(x, y) \in f \wedge (y, z) \in g$. Then $y \in \text{range}(f)$ and $y \in \text{dom}(g)$ or $y \in \text{range}(f) \cap \text{dom}(g)$, which as $(y, z) \in g$ proves that $z \in g(\text{range}(f) \cap \text{dom}(g))$. Hence

$$\text{range}(g \circ f) \subseteq g(\text{range}(f) \cap \text{dom}(g)) \quad (2.11)$$

If $z \in g(\text{range}(f) \cap \text{dom}(g))$ then $\exists y \in \text{range}(f) \cap \text{dom}(g)$ such that $(y, z) \in g$. From $y \in \text{range}(f)$ it follows that there exist a x such that $(x, y) \in f$. So $(x, z) \in g \circ f$ proving that $x \in \text{range}(g \circ f)$, hence $g(\text{range}(f) \cap \text{dom}(g)) \subseteq \text{range}(g \circ f)$. Combining this with [eq: 2.11] allows us to use [theorem: 1.9] to get

$$\text{range}(g \circ f) = g(\text{range}(f) \cap \text{dom}(g))$$

3. If $z \in \text{range}(g \circ f)$ then there exists a x such that $(x, z) \in g \circ f$, so there exists a y such that $(x, y) \in f \wedge (y, z) \in g$. Hence $z \in \text{range}(g)$. $\square$

Theorem 2.23. *If $f: A \rightarrow B$ and $g: C \rightarrow D$ are partial functions then we have*

1. *If $E \subseteq A$ then $(g \circ f)(E) = g(f(E))$*
2. *If $E \subseteq D$ then $(g \circ f)^{-1}(E) = f^{-1}(g^{-1}(E))$*

Proof.

1. If $z \in (g \circ f)(E)$ then there exists a $x \in E$ such that $(x, z) \in g \circ f$. So by definition there exist a y such that $(x, y) \in f \wedge (y, z) \in g$. From $(x, y) \in f$ it follows that $y \in f(E)$ and as $(y, z) \in g$ it follows that $z \in g(f(E))$. Hence

$$(g \circ f)(E) \subseteq g(f(E)) \quad (2.12)$$

On the other hand if $z \in g(f(E))$ there exist a $y \in f(E)$ such that $(y, z) \in g$. As $y \in f(E)$ there exists a $x \in E$ such that $(x, y) \in f$. From $(x, y) \in f \wedge (y, z) \in g$ it follows that $(x, z) \in g \circ f$ so that [as $x \in E$] $z \in (g \circ f)(E)$. Proving $g(f(E)) \subseteq (g \circ f)(E)$, combining this with [eq 2.12] and [theorem: 1.9] gives

$$(g \circ f)(E) = g(f(E))$$

2. If $x \in (g \circ f)^{-1}(E)$ then there exist a $z \in E$ such that $(x, z) \in g \circ f$, hence $\exists y$ such that $(x, y) \in f \wedge (y, z) \in g$. So by definition $y \in g^{-1}(E)$ and as $(x, y) \in f$ it follows that $x \in f^{-1}(g^{-1}(E))$. Hence

$$(g \circ f)^{-1}(E) \subseteq f^{-1}(g^{-1}(E)) \quad (2.13)$$

If $x \in f^{-1}(g^{-1}(E))$ then there exist a $y \in g^{-1}(E)$ such that $(x, y) \in f$, as $y \in g^{-1}(E)$ then there exist a $z \in E$ such that $(y, z) \in g$. From $z \in E \wedge (x, y) \in f \wedge (y, z) \in g$ it follows that $x \in (g \circ f)^{-1}(E)$ proving that $f^{-1}(g^{-1}(E)) \subseteq (g \circ f)^{-1}(E)$. Combining this with [eq: 2.13] and [theorem: 1.9] gives

$$(g \circ f)^{-1}(E) = f^{-1}(g^{-1}(E)) \quad \square$$

2.2.2 Functions

Definition 2.24. *A partial function $f: A \rightarrow B$ is a **function** iff $\text{dom}(f) = A$*

Theorem 2.25. *If $f: A \rightarrow B$ is a partial function then $f: \text{dom}(f) \rightarrow B$ is a partial function*

Proof. As $f: A \rightarrow B$ is a partial function we have that

$$f \subseteq A \times B \text{ and } \forall (x, y), (x, y') \in f \text{ we have } y = y'$$

Let $(x, y) \in f$ then $x \in \text{dom}(f)$ and $y \in B$ so that $(x, y) \in \text{dom}(f) \times B$ which together with the above proves that

$$f: \text{dom}(f) \rightarrow B \text{ is a partial function}$$

and thus by definition that

$$f: \text{dom}(f) \rightarrow B \text{ is a function} \quad \square$$

So every function is also a partial function, hence statements about partial functions applies also for functions. One special benefit of functions is the following.

Theorem 2.26. *If $f: A \rightarrow B$ is a function then for $C \subseteq A$ we have*

1. $C \subseteq f^{-1}(f(C))$
2. $f(f^{-1}(C)) \subseteq C$

Proof.

1. If $x \in C \subseteq A$ then as $A = \text{dom}(f)$ there exist a y such that $(x, y) \in f$ so that $y \in f(C)$, which as $(x, y) \in f$ proves that $x \in f^{-1}(f(C))$. Hence we have $C \subseteq f^{-1}(f(C))$
2. If $y \in f(f^{-1}(C))$ then there exist a $x \in f^{-1}(C)$ such that $(x, y) \in f$. As $x \in f^{-1}(C)$ there exist a $c \in C$ such that $(x, c) \in f$. Hence as $(x, y), (x, c) \in f$ we must have that $y = c \in C$ proving that $f(f^{-1}(C)) \subseteq C$. $\square$

Proposition 2.27. *A partial function $f: A \rightarrow B$ is a function iff $A \subseteq \text{dom}(f)$*

Proof. As $A \subseteq \text{dom}(f)$ and $\text{dom}(f) \subseteq A$ [theorem: 2.11] we have by [theorem: 1.9] that

$$\text{dom}(f) = A \quad \square$$

Example 2.28. Let A, B be elements and define $f = \{(0, A), (1, B)\}$ then $f: \{0, 1\} \rightarrow \{A, B\}$ is a function

Proof. If $(x, y) \in f$ then

$$(x, y) = (0, A) \Rightarrow x = 0 \in \{0, 1\} \wedge y = A \in \{A, B\} \text{ so that } (x, y) \in \{0, 1\} \times \{A, B\}$$

or

$$(x, y) = (1, B) \Rightarrow x = 1 \in \{0, 1\} \wedge y = B \in \{A, B\} \text{ so that } (x, y) \in \{0, 1\} \times \{A, B\}$$

proving that

$$f \subseteq \{0, 1\} \times \{A, B\}$$

If $(x, y), (x, y') \in f$ then for (x, y) we have either:

$$(x, y) = (0, A). \text{ Then } x = 0 \text{ and } y = A \text{ so that } (x', y') = (0, y') \in f \Rightarrow y' = A \text{ hence } y = y'$$

$$(x, y) = (1, B). \text{ Then } x = 1 \text{ and } y = B \text{ so that } (x', y') = (1, y') \in f \Rightarrow y' = B \text{ hence } y = y'$$

which proves that

$$f: \{0, 1\} \rightarrow \{A, B\} \text{ is a partial function}$$

If $x \in \{0, 1\}$ then either $x = 0$ so that $(0, A) \in f$ or $x = 1$ so that $(1, B) \in f$, so $\{0, 1\} \subseteq \text{dom}(f)$. Using [proposition: 2.27] it follows that

$$f: \{0, 1\} \rightarrow \{A, B\} \text{ is a function} \quad \square$$

Although the composition of functions $f: A \rightarrow B$ and $g: C \rightarrow D$ is a partial function [see theorem: 2.19], it does not have to be a function as we need the extra requirement that $\text{dom}(g \circ f) = A$. So we must have a extra condition on A . This is expressed in the following theorem

Theorem 2.29. *Let $f: A \rightarrow B$ and $g: C \rightarrow D$ functions then $g \circ f: A \cap f^{-1}(C) \rightarrow D$ is also a function with $\text{range}(g \circ f) = g(\text{range}(f))$*

Proof. As $f: A \rightarrow B$ and $g: C \rightarrow D$ are functions then we have that

$$f \subseteq A \times B \wedge (x, y), (x, y') \in f \Rightarrow y = y'$$

$$g \subseteq C \times D \wedge (x, y), (x, y') \in g \Rightarrow y = y'$$

$$A = \text{dom}(f) \wedge B = \text{dom}(g)$$

Let $(x, y) \in g \circ f$ then $\exists z$ with $(x, z) \in f$ and $(z, y) \in g$. As $f \subseteq A \times B$ and $g \subseteq C \times D$ it follows that $x \in A \wedge z \in C \wedge y \in D$. As $z \in C$ and $(x, z) \in f$ it follows that $x \in f^{-1}(C)$. So we have $(x, y) \in (A \cap f^{-1}(C)) \times D$. Hence we have that

$$g \circ f \subseteq (A \cap f^{-1}(C)) \times D$$

Further we have if $(x, y), (x, y') \in g \circ f$ that there exist z, z' such that $(x, z) \in f, (z, y) \in g, (x, z') \in f$ and $(z', y') \in g$. As $(x, z), (x, z') \in f$ it follows that $z = z'$ so that $(z, y), (z, y') \in g$ hence $y = y'$. So we conclude that

$$g \circ f: A \cap f^{-1}(C) \rightarrow D \text{ is a partial function} \quad (2.14)$$

Now if $x \in \text{dom}(g \circ f)$ then there exist a z such that $(x, z) \in g \circ f \subseteq (A \cap f^{-1}(C)) \times D$ so that $x \in A \cap f^{-1}(C)$ proving that

$$\text{dom}(g \circ f) \subseteq A \cap f^{-1}(C) \quad (2.15)$$

Further if $x \in A \cap f^{-1}(C)$ then $x \in A$ and there exist a $z \in C$ such that $(x, z) \in f$. As $z \in C = \text{dom}(g)$ there exist a z' such that $(z, z') \in g$. Hence by definition $(x, z') \in g \circ f$ so that $x \in \text{dom}(g \circ f)$. So we conclude that $A \cap f^{-1}(C) \subseteq \text{dom}(g \circ f)$ or using [eq: 2.15]

$$\text{dom}(g \circ f) = A \cap f^{-1}(C)$$

Combining the above with [eqs: 2.14, 2.15] proves that

$$g \circ f: A \cap f^{-1}(C) \rightarrow D \text{ is a function}$$

If $y \in \text{range}(g \circ f)$ then there exist a x such that $(x, y) \in g \circ f$, hence there exist a z such that $(x, z) \in f$ and $(z, y) \in g$. As $(x, z) \in f$ it follows that $z \in \text{range}(f)$, so as $(z, y) \in g$ it follows that $y \in g(\text{range}(f))$ which proves that

$$\text{range}(g \circ f) \subseteq g(\text{range}(f)) \quad (2.16)$$

On the other hand if $y \in g(\text{range}(f))$ then there exist a $x \in \text{range}(f)$ such that $(x, y) \in g$. As $x \in \text{range}(f)$ there exist a z such that $(z, x) \in f$. Hence $(z, y) \in g \circ f$ so that $y \in \text{range}(g \circ f)$ proving that $g(\text{range}(f)) \subseteq \text{range}(g \circ f)$. Combining this with [eq: 2.16] proves that

$$\text{range}(g \circ f) = g(\text{range}(f))$$

□

Theorem 2.30. *Let $f: A \rightarrow B$ and $g: C \rightarrow D$ functions with $f(A) \subseteq C$ then $g \circ f: A \rightarrow D$ is also a function with $\text{range}(g \circ f) = g(\text{range}(f))$*

Proof. Using the previous theorem we have that

$$g \circ f: A \cap f^{-1}(C) \rightarrow D \text{ is a function and } \text{range}(g \circ f) = g(\text{range}(f)) \quad (2.17)$$

Further as $f(A) \subseteq C$ we have by [theorem: 2.17] that $f^{-1}(f(A)) \subseteq f^{-1}(C)$ which as $A \subseteq_{[\text{theorem: 2.26}]} f^{-1}(f(A))$ proves that $A \subseteq f^{-1}(C)$. But then $A \cap f^{-1}(C) = A$ and substituting this in [eq: 2.17] proves that

$$g \circ f: A \rightarrow D \text{ is a function with } \text{range}(g \circ f) = g(\text{range}(f))$$

□

Next we define the class of all the graphs of functions between two classes

Note 2.31. Be aware that some books call partial functions functions and functions mappings.

Definition 2.32. Let A, B be two classes then we define the class B^A [using the Axiom of Construction] as

$$B^A = \{f \mid f: A \rightarrow B \text{ is a function}\}$$

Note 2.33. B^A is not the class of functions between A and B , but the class of graphs of functions between A and B . This distinction is important because it makes the following theorem possible.

Example 2.34. Let A be a class then $A^\emptyset = \{\emptyset\}$

Proof. Let $f \in A^\emptyset$ then $f: \emptyset \Rightarrow A$ is a function, so that $f \subseteq \emptyset \times A = \emptyset$ or $f = \emptyset$ □

Lemma 2.35. If $f: A \rightarrow B$ is a function and $B \subseteq C$ then $f: A \rightarrow C$ is a function

Proof. As $f: A \rightarrow B$ is a function we have $f \subseteq A \times B$ which as by [theorem: 1.55] $A \times B \subseteq A \times C$ means that $f \subseteq A \times C$. Further as $f: A \rightarrow B$ is a function we have also $\text{dom}(f) = A$ and if $(x, y), (x, y') \in f$ then $y = y'$. So by definition $f: A \rightarrow C$ is a function. □

Theorem 2.36. Let A, B, C be classes such that $B \subseteq C$ then $B^A \subseteq C^A$

Proof. Let $f \in B^A$ then $f: A \rightarrow B$ is a function, using the above lemma [lemma: 2.35] we have that $f: A \rightarrow C$ is a function, hence $f \in C^A$ proving that

$$B^A \subseteq C^A \quad \square$$

We have also the following relation between $A \times B$ and B^C

Theorem 2.37. Let A, B be two classes then we have:

1. $B^A \subseteq A \times B$
2. If A, B are sets then B^A is a set

Proof.

1. If $f \in B^A$ then $f: A \rightarrow B$ is a function so that $f \subseteq A \times B$ proving that $B^A \subseteq A \times B$
2. If A, B are sets then by [theorem: 1.73] we have that $A \times B$ is a set. So using the Axiom of Subsets [axiom: 1.8] we have that f is a set, □

Theorem 2.38. Let A, B, C be classes then $A^C \cap B^C = (A \cap B)^C$

Proof. First by [theorem: 1.30] we have $A \cap B \subseteq A$ and $A \cap B \subseteq B$ it follows from the above theorem [theorem: 2.36] that $(A \cap B)^C \subseteq A^C$ and $(A \cap B)^C \subseteq B^C$. Applying then [theorem: 1.36] gives

$$(A \cap B)^C \subseteq A^C \cap B^C \quad (2.18)$$

For the opposite inclusion, let $f \in A^C \cap B^C$ then $f \in A^C \wedge f \in B^C$ so that $f: C \rightarrow A$ and $f: C \rightarrow B$ are functions. Then we have that $f \subseteq C \times A$ and $f \subseteq C \times B$ so that

$$f \subseteq (C \times A) \cap (C \times B) \stackrel{1.56}{=} (C \cap C) \times (A \cap B) \stackrel{[theorem: 1.29]}{=} C \times (A \cap B)$$

Further as $f: A \rightarrow C$ is a function we have $(x, y), (x, y') \in f$ and $\text{dom}(f) = C$ so that

$$f: C \rightarrow (A \cap B) \text{ is a function}$$

proving that $f \in (A \cap B)^C$. So $A^C \cap B^C \subseteq (A \cap B)^C$ which combined with [eq: 2.18] gives

$$A^C \cap B^C = (A \cap B)^C \quad \square$$

We have the follow trivial fact about a function

Proposition 2.39. *Let $f: A \rightarrow B$ be a function then if $f(A) \stackrel{[theorem: 2.17]}{=} \text{range}(f) \subseteq C$ we have that $f: A \rightarrow C$ is a function.*

Proof. If $(x, y) \in f$ then $y \in \text{range}(f)$ hence as $\text{range}(f) \subseteq C$ $y \in C$. As $f \subseteq A \times B$ we have also $x \in A$ so that $(x, y) \in C \times B$. Hence $f \subseteq A \times C$, further if $(x, y), (x, y') \in f$ we have as $f: A \rightarrow B$ is a function that $y = y'$. So

$$f: A \rightarrow C \text{ is a partial function}$$

As $\text{range}(f) = A$ (because $f: A \rightarrow B$ is a function] we have that $f: A \rightarrow C$ a function $\square$

We have the following trivial proposition about the equality of two functions

Proposition 2.40. *Two functions $f: A \rightarrow B$ and $g: A \rightarrow B$ are equal if*

$$[(x, y) \in f \Rightarrow (x, y) \in g \wedge (x, y) \in g \Rightarrow (x, y) \in f]$$

Proof. Note that the statement $f: A \rightarrow B$ and $g: A \rightarrow B$ are equal is equivalent with $\langle A, B, f \rangle = \langle A, B, g \rangle$, which by 2.5 is equivalent with $A = A \wedge B = B \wedge f = g$, As $A = A$ and $B = B$ are true this is equivalent with $f = g$. Now by the Axiom of Extent [axiom: 1.5] we have that

$$f = g \Leftrightarrow [(x, y) \in f \Rightarrow (x, y) \in g \wedge (x, y) \in g \Rightarrow (x, y) \in f]$$

$\square$

If $f: A \rightarrow B$ is a function then for every $x \in A$ we have a unique $y \in B$ such that $(x, y) \in f$. Furthermore in many cases we have actually a expression valid for every $x \in A$ to calculate this unique value. To express this we use the following notation.

Definition 2.41. *If $f: A \rightarrow B$ is a function then*

$$y = f(x) \text{ or } f(x) = y \text{ is equivalent with } (x, y) \in f$$

and

$$f(x) = E(x) \text{ where } E(x) \text{ is a expression depending on } x \text{ is equivalent with } (x, E(x)) \in f$$

Further if $D \subseteq B$ then $f(x) \in D$ is the same as $\exists y \in D$ such that $y = f(x)$ or $(x, y) \in f$

Example 2.42. Let $3 \cdot x + 1$ be the value associated with x , so $f = \{z | z = (x, 3 \cdot x + 1) \in f \wedge x \in A\}$, then we can use the following equivalent notations to define our function

$$f: A \rightarrow B \text{ is defined by } x \rightarrow 3 \cdot x + 1$$

If we have defined a function $f: A \rightarrow B$ using a expression and we want to refer to the expression of the function we use the notation $f(x)$. Hence we define a function also as

$$f: A \rightarrow B \text{ is defined by } x \rightarrow f(x) = 3 \cdot x + 1$$

or

$$f: A \rightarrow B \text{ is defined by } x \rightarrow f(x) \text{ where } f(x) = 3 \cdot x + 1$$

or

$$f: A \rightarrow B \text{ is defined by } f(x) = 3 \cdot x + 1$$

In all of the above cases we actually means that $\langle f, A, B \rangle$ is a function with $f = \{z | z = (x, 3 \cdot x + 1) \wedge x \in A\}$.

Using the above notation we can reformulate [proposition: 2.40] in a form that is easier to work with if we use expressions to define a function.

Proposition 2.43. *Two functions $f: A \rightarrow B$ and $g: A \rightarrow B$ are equal if and only if*

$$\forall x \in A \ f(x) = g(x)$$

Proof. Assume that $f: A \rightarrow B$ and $g: A \rightarrow B$ are equal then if $x \in A$ we have $\exists y \in B$ such that $(x, y) \in f$ or $y = f(x)$, using [proposition: 2.40] we have also $(x, y) \in g$ hence $y = g(x)$ which proves that $f(x) = g(x)$.

On the other hand assume that $\forall x \in A f(x) = g(x)$ then if $(x, y) \in f$ we have $y = f(x) = g(x)$ so that $(x, y) \in g$. If $(x, y) \in g$ then $y = g(x) = f(x)$ or $(x, y) \in f$. Using [proposition: 2.40] we have then that $f: A \rightarrow B$ and $g: A \rightarrow B$ are equal. $\square$

Using the new notation, composition of function is written as

Theorem 2.44. *If $f: A \rightarrow B$ and $g: C \rightarrow D$ are two functions with $f(A) \subseteq C$ then*

$$(g \circ f)(x) = g(f(x))$$

Proof. Take $z = (g \circ f)(x)$ then $(x, z) \in g \circ f$ so that $\exists y$ such that $(x, y) \in f$ and $(y, z) \in g$. Hence $y = f(x)$ and $z = g(y)$ so that $z = g(f(x))$, proving $(g \circ f)(x) = g(f(x))$. $\square$

Image and pre-image can also be expressed in the new notation.

Proposition 2.45. *Let $f: A \rightarrow B$ a function, $C \subseteq A$ and $D \subseteq B$ then*

1. $y \in f(C) \Leftrightarrow \exists x \in A$ such that $y = f(x)$
2. $x \in f^{-1}(D) \Leftrightarrow f(x) \in D$

Proof.

1.

$$\begin{aligned} y \in f(C) &\Leftrightarrow \exists x \in C \text{ such that } (x, y) \in f \\ &\Leftrightarrow \exists x \in C \text{ such that } y = f(x) \end{aligned}$$

2.

$$\begin{aligned} x \in f^{-1}(C) &\Leftrightarrow \exists y \in D \text{ such that } (x, y) \in f \\ &\Leftrightarrow \exists y \in D \text{ such that } y = f(x) \\ &\Leftrightarrow f(x) \in D \end{aligned}$$

$\square$

Let's now look at some example of functions:

Example 2.46. (Empty Function) $\emptyset: \emptyset \rightarrow B$

Proof. First $\emptyset \subseteq \emptyset \times B$ by [theorem: 1.22], if $x \in \text{dom}(\emptyset)$ then $\exists y \in \emptyset$ such that $(x, y) \in \emptyset$ which is a contradiction, so by [theorem: 1.23] we have that $\text{dom}(\emptyset) = \emptyset$. And finally $(x, y) \in \emptyset \wedge (x, y') \in \emptyset \Rightarrow y = y'$ is satisfied vacuously as $(x, y) \in \emptyset \wedge (x, y') \in \emptyset$ is never true. $\square$

Example 2.47. (Constant Function) Let A, B classes and $c \in B$ then $C_c: A \rightarrow B$ is defined by $C_c(x) = c$ or formally $C_c = \{z | z = (x, c) | x \in A\} = A \times \{c\}$

Proof. If $(x, y) \in C_c$ then $x \in A$ and $y = c \in B$ so that $C_c \subseteq A \times B$. If $(x, y) \in C_c \wedge (x, y') \in C_c$ then $y = c \wedge y' = c$ so that $y = y'$. So

$$C_c: A \rightarrow B \text{ is a partial function}$$

Finally if $x \in A$ then $(x, c) \in C_c$ so that $A \subseteq \text{dom}(C_c)$ which by [proposition: 2.27] proves that

$$C_c: A \rightarrow B \text{ is a function}$$

$\square$

Example 2.48. (Characteristics Function) Let A be a class and $B \subseteq A$ then $\mathcal{X}_{A,B}: A \rightarrow \{0, 1\}$ is defined by $\mathcal{X}_{A,B} = (B \times \{1\}) \cup ((A \setminus B) \times \{0\})$ [so that $\mathcal{X}_{A,B}(x) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \in A \setminus B \end{cases}$]

Proof. If $(x, y) \in \mathcal{X}_{A,B}$ then either $(x, y) \in (B \times \{1\}) \Rightarrow x \in B \xrightarrow{B \subseteq A} x \in A$ and $y = 1 \in \{0, 1\}$ or $(x, y) \in ((A \setminus B), \{0\}) \Rightarrow x \in A \setminus B \Rightarrow x \in A$ and $y = 0 \in \{0, 1\}$ so that

$$\mathcal{X}_{A,B} \subseteq A \times \{0, 1\}$$

Also if $(x, y), (x, y') \in \mathcal{X}_{A,B}$ then for (x, y) we have either:

$(x, y) \in B \times \{1\}$. then $x \in B$ so that $(x, y') \in B \times \{1\}$ hence $y = 1 = y'$

$(x, y) \in (A \setminus B) \times \{0\}$. then $x \in A \setminus B$ so that $(x, y') \in (A \setminus B) \times \{0\}$ hence $y = 0 = y'$

or in all cases $y = y'$ and $x \in B \cup (A \setminus B) = A$. Hence $\mathcal{X}_{A,B}: A \rightarrow \{0, 1\}$ is a function. $\square$

Example 2.49. (Identity Function) Let A be a class then $\text{Id}_A: A \rightarrow B$ is defined by

$$\text{Id}_A = \{z | z = (x, x) \wedge x \in A\}$$

is a function,

Proof. Trivially we have $\text{Id}_A \subseteq A \times A$. If $(x, y), (x, y') \in \text{Id}_A$ then $(x, y) = (x, x) = (x, y')$ proving that $y = x = y'$. Hence $\text{Id}_A: A \rightarrow A$ is a partial function. Further if $x \in A$ then $(x, x) \in \text{Id}_A$ so that $x \in \text{dom}(\text{Id}_A)$ or $\text{dom}(\text{Id}_A) \subseteq A$ which by [proposition: 2.27] proves that

$$\text{Id}_A: A \rightarrow A \text{ is a function}$$

$\square$

Proposition 2.50. Let $f: A \rightarrow B$ be a partial function then $f = f \circ \text{Id}_A$ and $f = \text{Id}_B \circ f$

Proof.

1. If $(x, y) \in f$ then as $f \subseteq A \times B$ we have $x \in A \wedge x \in B$, by the definition of Id_A we have $(x, x) \in \text{Id}_A$, as $(x, y) \in f$ we have $(x, y) \in \text{Id}_A \circ f$. If $(x, y) \in f \circ \text{Id}_A$ then $\exists x'$ such that $(x, x') \in \text{Id}_A \wedge (x', y) \in f$. By definition of Id_A we have that $\exists z \in A$ such that $(x, x') = (z, z)$ hence $x = x'$ so that $(x, y) \in f$. Using the Axiom of Extent [axiom: 1.5] we have then that

$$f = f \circ \text{Id}_A$$

2. If $(x, y) \in f$ then as $f \subseteq A \times B$ we have $x \in A \wedge x \in B$, by the definition of Id_B we have $(y, y) \in \text{Id}_B$, so $(x, y) \in \text{Id}_B \circ f$. If $(x, y) \in \text{Id}_B \circ f$ then $\exists y'$ such that $(x, y') \in f \wedge (y, y') \in \text{Id}_B$, from the definition of Id_B we have that $y = y'$ so that $(x, y) \in f$. Using the Axiom of Extent [axiom: 1.5] we have then that

$$f = \text{Id}_B \circ f$$

$\square$

As a function $f: A \rightarrow B$ is a partial function with $\text{dom}(f) = A$ we can refine [theorem: 2.17].

Theorem 2.51. If $f: A \rightarrow B$ is a function $C \subseteq B$ and $D \subseteq B$ then we have

1. $f(C) \subseteq B$
2. $f^{-1}(D) \subseteq A$
3. $f(A) = \text{range}(f)$
4. $f^{-1}(B) = A$

Proof. This follows from 2.17 taking in account that $A = \text{dom}(f)$ $\square$

2.2.3 Injectivity, Surjectivity and bijectivity

First we define injectivity and surjectivity of partial functions.

Definition 2.52. Let $f: A \rightarrow B$ be a partial function then we say that:

1. f is **injective** iff if $(x, y) \in f \wedge (x', y) \in f$ implies $x = x'$
2. f is **surjective** iff $\text{range}(f) = B$

Proposition 2.53. *A partial function $f: A \rightarrow B$ is surjective if $B \subseteq \text{range}(f)$*

Proof. By [theorem: 2.11] $\text{range}(f) \subseteq B$, so if $B \subseteq \text{range}(f)$ it follows from [theorem: 1.9] that $B = \text{range}(f)$, proving surjectivity. $\square$

Using the notation $y = f(x)$ is the same as $(x, y) \in f$ we have

Theorem 2.54. *Let $f: A \rightarrow B$ be a function then*

1. *f is injective if and only if $\forall x, x' \in A$ with $f(x) = f(x')$ we have $x = x'$*
2. *If $B \subseteq C$ and $f: A \rightarrow B$ is injective then $f: A \rightarrow C$ is injective*
3. *f is surjective if and only if $\forall y \in B$ there exists a $x \in A$ such that $y = f(x)$*

Proof.

1.

$\Rightarrow$. Let $x, x' \in A$ then if $y = f(x) = f(x')$ we have $(x, y) \in f$ and (x', y) so that $x = x'$
 $\Leftarrow$. If $(x, y) \in f$ and $(x', y) \in f$ then $y = f(x) \wedge y = f(x')$ so that $f(x) = f(x')$ so that $x = x'$

2. This is trivial because injectivity is a property of the graph of a function.

3.

$\Rightarrow$. As $B = \text{range}(f)$ we have $y \in B$ then $\exists x$ such that $(x, y) \in f \Rightarrow y = f(x)$ which as $f \subseteq A \times B$ proves that $x \in A$. So $\forall y \in B \exists x \in A$ such that $y = f(x)$
 $\Leftarrow$. Let $y \in B$ then $\exists x \in A$ such that $y = f(x)$ or $(x, y) \in f$ proving that $B \subseteq \text{range}(f)$, using [proposition: 2.53] we have that f is surjective $\square$

Example 2.55. Let A, B be classes, $B \subseteq A$ then $i_B: B \rightarrow A$ defined by $i_B = \{(x, x) | x \in B\}$ is a injective function. This function is called the **inclusion** function.

Proof. First if $(x, y) \in i_B$ then $\exists b \in B$ such that $(x, y) = (b, b)$ so that $x = b \in B \wedge y = b \in B \subseteq A$ proving that

$$i_B \subseteq B \times A$$

Further if $(x, y), (x', y') \in i_B$ then $\exists b, b' \in B$ such that $(x, y) = (b, b) \wedge (x', y') = (b', b')$, so that $x = b \wedge y = b \wedge x' = b' \wedge y' = b'$, hence $y = y'$. So

$$i_B: B \rightarrow A \text{ is a partial function}$$

If $x \in B$ then $(x, x) \in i_B$ proving that $A \subseteq \text{dom}(i_b)$ so using [proposition: 2.27] it follows that

$$i_B: B \rightarrow A \text{ is a function}$$

Finally if $(x, y), (x', y') \in i_B$ then there exists $b, b' \in B$ such that $(x, y) = (b, b) \wedge (x', y') = (b', b')$, so that $x = b \wedge y = b \wedge x' = b' \wedge y' = b'$, hence $x = x'$, proving injectivity. $\square$

The following axiom ensures that the image of a set by a surjection is a set.

Axiom 2.56. (Axiom of Replacement) *If A is a set and $f: A \rightarrow B$ a surjection then B is a set.*

Proposition 2.57. *If $f: A \rightarrow B$ is a function and $C \subseteq A, D \subseteq B$ then*

1. $C \subseteq f^{-1}(f(C))$
2. *If f is injective then $C = f^{-1}(f(C))$*
3. *If f is surjective then $D = f(f^{-1}(D))$*

Proof.

1. This is stated in [theorem: 2.26]

2. If $x \in f^{-1}(f(C))$ then $\exists y \in f(C)$ such that $(y, x) \in f^{-1}$, hence $(x, y) \in f$. As $y \in f(C)$ there exists a $x' \in C$ such that $(x', y) \in f$. Given that f is injective it follows from $(x, y), (x', y) \in f$ that $x = x'$, so as $x' \in C$ it follows that $x \in C$. Hence $f^{-1}(f(C)) \subseteq C$ which combined with (1) proves

$$C = f^{-1}(f(C))$$

3. If $y \in f(f^{-1}(D))$ then $\exists x \in f^{-1}(D)$ such that $(x, y) \in f$, hence $\exists z \in D$ such that $(z, x) \in f^{-1} \Rightarrow (x, z) \in f$. As f is a function we have $y = z$ so that $y \in D$. Hence

$$f(f^{-1}(D)) \subseteq D \quad (2.19)$$

If $y \in D$ then as f is a surjection there exist a $x \in A$ such that $(x, y) \in f$, hence $x \in f^{-1}(D)$ proving that $y \in f(f^{-1}(D))$. So $D \subseteq f(f^{-1}(D))$ which together with [eq: 2.19] proves

$$D = f(f^{-1}(D)) \quad \square$$

The importance of injectivity is that it allows us to define the inverse of a partial function. First we define the inverse graph of the graph of a partial function.

Definition 2.58. Let $f: A \rightarrow B$ be a partial function then the **inverse of the graph** f noted as f^{-1} is defined by

$$f^{-1} = \{z : z = (z, y) \text{ where } (y, x) \in f\}$$

Theorem 2.59. Let $f: A \rightarrow B$ be a **injective** partial function then $f^{-1}: B \rightarrow A$ is a partial function

Proof. If $(x, y) \in f^{-1}$ then $(y, x) \in f$ which, as $f \subseteq A \times B$, gives $(y, x) \in A \times B$, so $x \in B \wedge y \in A$, proving $(x, y) \in B \times Y$. Hence

$$f^{-1} \subseteq B \times A$$

Further if $(x, y) \in f^{-1}$ and $(x, y') \in f^{-1}$ then $(y, x) \in f \wedge (y, x') \in f$ which, as f is injective proves that $y = y'$. So all the conditions are satisfied to make $f^{-1}: B \rightarrow A$ a partial function. $\square$

Note 2.60. The requirement that f is injective is needed to make f^{-1} is a partial function. For example assume that $A = \{1, 2, 3\}$, $B = \{10, 20\}$ and $f = \{(1, 10), (2, 10), (3, 20)\}$ then $f^{-1} = \{(10, 1), (10, 2), (20, 3)\}$ which is not the graph of a partial function.

If f is a injective function then the above theorem ensures that f^{-1} is a partial function however f^{-1} can be a graph of a function if we restrict the source of the inverse function.

Theorem 2.61. If $f: A \rightarrow B$ is a injective function then $f^{-1}: f(A) \rightarrow A$ is a function

Proof. First if $(x, y) \in f^{-1}$ then $(y, x) \in f \subseteq A \times B$ so that $y \in A \wedge x \in B$, as $(y, x) \in f$ we have that $x \in f(A)$, hence $(x, y) \in f(A) \times A$. So $f^{-1} \subseteq f(A) \times B$. Further if $(x, y), (x, y') \in f^{-1}$ then $(y, x), (y', x) \in f$ which as f is injective proves $y = y'$. Hence

$$f^{-1}: f(A) \rightarrow A \text{ is a partial function}$$

Further if $x \in f(A)$ then there exists a $y \in A$ such that $(y, x) \in f$, hence $(x, y) \in f^{-1}$ so that $x \in \text{dom}(f^{-1})$, proving that $f(A) \subseteq \text{dom}(f^{-1})$. Hence

$$f^{-1}: f(A) \rightarrow A \text{ is a function} \quad \square$$

Corollary 2.62. If $f: A \rightarrow B$ is a function, $A \neq \emptyset$ then $f: A \rightarrow B$ is injective if and only if there exist a function $g: B \rightarrow A$ such that $g \circ f = \text{Id}_A$

Proof.

$\Rightarrow$. Using the above [theorem: 2.61] we have that $f^{-1}: f(A) \rightarrow A$ is a function. As $A \neq \emptyset$ there exist a $a \in A$ so we can consider the constant function $C_a: B \setminus f(A) \rightarrow A$ [see example: 2.47]. As $f(A) \cap (B \setminus f(A)) = \emptyset$ and $B = f(A) \cup (B \setminus f(A))$ we have by [theorem: 2.84] that

$$g = C_a \cup f^{-1}: B \rightarrow A$$

is a function. If $(x, y) \in g \circ f$ then $\exists z$ such that $(x, z) \in f \wedge (z, y) \in g$. As $(x, z) \in f$ we have that $(z, x) \in f^{-1} \subseteq C_a \cup f^{-1} = g$, as also $(z, y) \in g$ and g is function, we have that $y = x$ so that $(x, y) = (x, x) \in \text{Id}_A$ hence

$$g \circ f \subseteq \text{Id}_A$$

Further if $(x, y) \in \text{Id}_A$ then $x = y$, as $x \in A = \text{dom}(f)$ there exist a $z \in B$ such that $(x, z) \in f \Rightarrow (z, x) \in f^{-1} \subseteq C_a \cup f^{-1} = g$ proving that $(x, y) = (x, x) \in g \circ f$. Hence

$$\text{Id}_A \subseteq g \circ f$$

proving that

$$g \circ f = \text{Id}_A$$

$\Leftarrow$. Assume that there exists a function $g: B \rightarrow A$ such that $g \circ f = \text{Id}_A$ then

$$\begin{aligned} (x, y), (x', y) \in f \subseteq A \times B & \xRightarrow{y \in B, \text{dom}(g)=B} \exists z \vdash (y, z) \in g \\ & \Rightarrow (x, z), (x', z) \in g \circ f = \text{Id}_A \\ & \Rightarrow x = z = x' \\ & \Rightarrow x = x' \\ & \square \end{aligned}$$

Definition 2.63. A function $f: A \rightarrow B$ is a **bijection** iff the function is **injective** and **surjective**.

Definition 2.64. Two classes A and B are **bijjective** iff there exists a bijection between A and B

Example 2.65. The function $\emptyset: \emptyset \rightarrow \emptyset$ is a bijection.

Proof. By [example: 2.46] $\emptyset: \emptyset \rightarrow \emptyset$ is a function. To prove that is a bijection we have:

injectivity. $\forall (x, y), (x', y) \in \emptyset$ we have $x = x'$ is satisfied vacuously.

surjectivity. $\forall y \in \emptyset$ there exist a $x \in \emptyset$ such that $(x, y) \in \emptyset$ is satisfied vacuously. $\square$

Example 2.66. Let A be a class then $\text{Id}_A: A \rightarrow A$ [example: 2.49] is a bijection

Proof. Let $(x, y) \in \text{Id}_A \wedge (x', y) \in \text{Id}_A$ then $\exists z, z' \in A$ such that $(x, y) = (z, z) \wedge (x', y) = (z', z')$. So using [theorem: 1.50] $x = z \wedge y = z \wedge x = z' \wedge y = z'$. Using [theorem: 1.9] repeatedly gives then $x = x'$ proving that

Id_A is injective

If $y \in A$ then by definition $(y, y) \in \text{Id}_A$ so that $\text{range}(\text{Id}_A) \subseteq A$. Using [theorem: 2.53] it follows that

Id_A is surjective $\square$

Example 2.67. Let $I = \{0\}$ B a class and take $f: I \rightarrow \{B\}$ defined by $f = \{(0, B)\}$ is a bijection

Proof. As $0 \in \{0\}$ and $B \in \{B\}$ it follows that $(0, B) \in \{0\} \times \{B\}$, hence $f = \{(0, B)\} \subseteq \{0\} \times \{B\}$. If $(x, y), (x, y') \in f = \{0\} \times \{B\}$ then $y = B = y'$, further $\text{dom}(f) = \{0\} = I$. So we conclude that $f: \{0\} \rightarrow \{B\}$ is indeed a function. Further if $y \in \{B\}$ then $y = B$ and as $(0, B) \in f$ it follows that $y \in \text{range}(f)$ or $\{B\} \subseteq \text{range}(f)$, which by [theorem: 2.53] proves that f is surjective. Finally if $(x, y), (x', y) \in f = \{(0, B)\}$ then $x = 0 = x'$ proving that $f: \{0\} \rightarrow \{B\}$ is a bijection. $\square$

Proposition 2.68. If $f: A \rightarrow B$ is a injective function then $f: A \rightarrow f(A)$ is a bijection

Proof. As injectivity is a property of the graph of a function, the function $f: A \rightarrow B$ is still injective. Further $\text{range}(f) \underset{[\text{theorem: 2.17}]}{=} f(A)$ which proves surjectivity. $\square$

Theorem 2.69. If $f: A \rightarrow B$ is a bijection then $f^{-1}: B \rightarrow A$ is a function

Proof. As $f: A \rightarrow B$ is injective and surjective we have that $f(A) = B$ and by [theorem: 2.61] that $f^{-1}: f(A) \rightarrow A$ is a function. Hence $f^{-1}: B \rightarrow A$ is a function. $\square$

Theorem 2.70. *If $f: A \rightarrow B$ is a bijection then for $f^{-1}: B \rightarrow A$ we have*

1. *If $C \subseteq A$ then $(f^{-1})(C) = f^{-1}(C)$.*
2. *If $C \subseteq B$ then $(f^{-1})^{-1}(C) = f(C)$*

Proof.

1. Let $y \in (f^{-1})(C)$ then there exist a $x \in C$ such that $(x, y) \in f^{-1}$, hence $(y, x) \in f$ so that $y \in f^{-1}(C)$ proving

$$(f^{-1})(C) \subseteq f^{-1}(C)$$

If $x \in f^{-1}(C)$ then there exist a $y \in C$ such that $(x, y) \in f$ hence $(y, x) \in f^{-1}$ proving that $x \in (f^{-1})(C)$, hence $f^{-1}(C) \subseteq (f^{-1})(C)$. Combining this with the above gives

$$(f^{-1})(C) = f^{-1}(C)$$

2. Let $x \in (f^{-1})^{-1}(C)$ then there exist a $y \in C$ such that $(x, y) \in f^{-1}$ hence $(y, x) \in f$ so that $x \in f(C)$ proving

$$(f^{-1})^{-1}(C) \subseteq f(C)$$

If $x \in f(C)$ then there exist a $y \in C$ such that $(y, x) \in f$ hence $(x, y) \in f^{-1}$ proving that $x \in (f^{-1})^{-1}(C)$, hence $f(C) \subseteq (f^{-1})^{-1}(C)$. Combining this with the above gives

$$(f^{-1})^{-1}(C) = f(C) \quad \square$$

Theorem 2.71. *If $f: A \rightarrow B$ is bijective then*

1. $f \circ f^{-1} = \text{Id}_B$
2. $f^{-1} \circ f = \text{Id}_A$

Proof. First $f^{-1}: B \rightarrow A$ is a function by [theorem: 2.69].

1. Let $(x, y) \in f \circ f^{-1}$ then $\exists z$ such that $(x, z) \in f^{-1} \Rightarrow (z, x) \in f$ and $(z, y) \in f$. As f^{-1} is the graph of a function we have that $x = y$. Further from $(x, z) \in f^{-1} \subseteq B \times A$ it follow that $x \in B$. Hence $(x, y) = (x, x) \in \text{Id}_B$, proving that

$$f \circ f^{-1} \subseteq \text{Id}_B \quad (2.20)$$

If $(x, y) \in \text{Id}_B$ then $\exists z \in B$ such that $(x, y) = (z, z)$ so that $x = y \in B$. As $B = \text{dom}(f^{-1})$ there exists a u such that $(y, u) \in f^{-1} \Rightarrow (u, y) \in f$ so that $(y, y) \in f \circ f^{-1} \xRightarrow{x=y} (x, y) \in f \circ f^{-1}$. So $\text{Id}_B \subseteq f \circ f^{-1}$. Combining this with [eq: 2.20] proves that

$$f \circ f^{-1} = \text{Id}_B$$

2. Let $(x, y) \in f^{-1} \circ f$ then $\exists z$ such that $(x, z) \in f \Rightarrow (z, x) \in f^{-1}$ and $(z, y) \in f^{-1}$. As f^{-1} is the graph of a function we have that $x = y$. Further from $(x, z) \in f \subseteq A \times B$ it follows that $x \in A$. Hence $(x, y) = (x, x) \in \text{Id}_A$, proving that

$$f^{-1} \circ f \subseteq \text{Id}_A \quad (2.21)$$

If $(x, y) \in \text{Id}_A$ then $\exists z \in A$ such that $(x, y) = (z, z)$ so that $x = y \in A$. As $A = \text{dom}(f)$ there exists a u such that $(x, u) \in f \Rightarrow (u, x) \in f^{-1}$ so that $(x, x) \in f^{-1} \circ f \xRightarrow{x=y} (x, y) \in f^{-1} \circ f$. So $\text{Id}_A \subseteq f^{-1} \circ f$. Combining this with [eq: 2.21] proves that

$$f^{-1} \circ f = \text{Id}_A \quad \square$$

Corollary 2.72. *If $f: A \rightarrow B$ is bijective then*

1. $\forall x \in A$ we have $(f^{-1})(f(x)) = x$
2. $\forall y \in B$ we have $f((f^{-1})(y)) = y$

Proof.

1. If $x \in A$ then $(f^{-1})(f(x)) = ((f^{-1}) \circ f)(x) \underset{[\text{theorem: 2.71}]}{=} \text{Id}_A(x) = x$
2. If $y \in B$ then $f((f^{-1})(y)) \underset{[\text{theorem: 2.71}]}{=} \text{Id}_B(y) = y$ □

Corollary 2.73. *Let $f: A \rightarrow B$ a function then the following are equivalent:*

1. $f: A \rightarrow B$ is a bijection
2. There exists a function $g: B \rightarrow A$ such that $f \circ g = \text{id}_B$ and $g \circ f = \text{Id}_A$

Proof.

- 1 $\Rightarrow$ 2.** This follows from [theorem: 2.71] by taking $g = f^{-1}$
- 2 $\Rightarrow$ 1.** Let $(x, y), (x', y) \in f \subseteq A \times B$, as $y = \text{dom}(g)$ there exists a z such that $(y, z) \in g$, hence $(x, z), (x', z) \in g \circ f = \text{Id}_A$ so that $x = z = x'$ proving that

$f: A \rightarrow B$ is injective

Further if $y \in B$ then $(y, y) \in \text{Id}_B = f \circ g$ so there exists a $z \in A$ such that $(y, z) \in g$ and $(z, y) \in f$. Proving that $B \subseteq \text{range}(f)$ so by [proposition: 2.53]

$f: A \rightarrow B$ is a surjection □

The inverse of a bijection is again a bijection

Corollary 2.74. *If $f: A \rightarrow B$ is a bijection then $f^{-1}: B \rightarrow A$ is a bijection*

Proof. If $f: A \rightarrow B$ is a bijection then by [theorem: 2.71] $f \circ f^{-1} = \text{Id}_B$ and $f^{-1} \circ f = \text{Id}_A$ which by [theorem: 2.73] proves that $f^{-1}: B \rightarrow A$ is a bijection. □

Proposition 2.75. *If $f: A \rightarrow B$ is a bijection then we have:*

1. If $g: B \rightarrow A$ is such that $f \circ g = \text{Id}_B$ and $g \circ f = \text{Id}_A$ then $g = f^{-1}$
2. $(f^{-1})^{-1} = f$

Proof.

1. We have

$$\begin{aligned}
 f \circ g = \text{Id}_B & \Rightarrow f^{-1} \circ (f \circ g) = f^{-1} \circ \text{Id}_B \\
 & \xRightarrow{[\text{proposition: 2.50}]} f^{-1} \circ (f \circ g) = f^{-1} \\
 & \xRightarrow{[\text{theorem: 2.21}]} (f^{-1} \circ f) \circ g = f^{-1} \\
 & \xRightarrow{[\text{function: 2.71}]} \text{Id}_B \circ g = f^{-1} \\
 & \xRightarrow{[\text{proposition: 2.50}]} g = f^{-1}
 \end{aligned}$$

2. We have

$$\begin{aligned}
 (x, y) \in (f^{-1})^{-1} & \Leftrightarrow (y, x) \in f^{-1} \\
 & \Leftrightarrow (x, y) \in f
 \end{aligned}$$

which by the Axiom of Extent [axiom: 1.5] proves

$$(f^{-1})^{-1} = f \quad \square$$

Corollary 2.76. *If A is a set then we have for the bijection $\text{Id}_U: A \rightarrow A$ [see theorem: 2.66] that $(\text{Id}_U)^{-1} = \text{Id}_U$.*

Proof. As we have for $\text{Id}_U: A \rightarrow A$ that $\text{Id}_A \circ \text{Id}_A \underset{[\text{theorem: 2.50}]}{=} \text{Id}_A$ it follows from [proposition: 2.75] that $(\text{Id}_U)^{-1} = \text{Id}_U$. □

Composition preserves injectivity, surjectivity and bijectivity.

Theorem 2.77. *We have*

1. If $f: A \rightarrow B$ and $g: C \rightarrow D$ are injective functions with $f(A) \subseteq C$ then $g \circ f: A \rightarrow D$ is a injective function.
2. If $f: A \rightarrow B$ and $g: C \rightarrow D$ are injective functions with $f(A) \subseteq C$ then $g \circ f: A \rightarrow g(f(A))$ is a bijective function.
3. If $f: A \rightarrow B$ is a function and $g: C \rightarrow D$ a surjective function so that $f(A) = C$ then $g \circ f: A \rightarrow D$ is a surjective function.
4. If $f: A \rightarrow B$ is a injective function and $g: C \rightarrow D$ a bijective function so that $f(A) = C$ then $g \circ f: A \rightarrow D$ is a bijective function.
5. If $f: A \rightarrow B$ is a injective function and $g: C \rightarrow D$ a bijective function so that $f(A) = C$ then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof.

1. Let $(x, z), (x', z) \in g \circ f$ then $\exists u, v$ such that

$$(x, u) \in f \wedge (x', v) \in f \wedge (u, y) \in g \wedge (v, y) \in g$$

As g is injective we have $u = v$, but that means from the above that $(x, u) \in f \wedge (x', u) \in f$, which as f is injective proves

$$x = x'$$

2. Using (1) we have that $g \circ f: A \rightarrow D$ is injective so that by [theorem: 2.68] $g \circ f: A \rightarrow (g \circ f)(A)$ is a bijection. Further by [theorem: 2.23] $(g \circ f)(A) = g(f(A))$ so that $g \circ f: A \rightarrow g(f(A))$ is a bijection.
3. Let $z \in D$ then as g is surjective there $\exists y \in C$ such that $(y, z) \in g$. As $f(A) = C$ there exists a $x \in A$ such that $(x, y) \in f$. But then $(x, z) \in g \circ f$ proving that $g \circ f$ is surjective.
4. Using (1) and (2) proves that $g \circ f: A \rightarrow D$ is injective and surjective and thus by definition bijective.
5. By (3) $g \circ f$ is a bijection, so by [theorem: 2.71] we have that

$$\begin{aligned}
 (g \circ f)^{-1} \circ (g \circ f) &= \text{Id}_A && \xRightarrow{\text{[associativity: 2.21]}} && ((g \circ f)^{-1} \circ g) \circ f = \text{Id}_A \\
 &&& \Rightarrow && (((g \circ f)^{-1} \circ g) \circ f) \circ f^{-1} = \text{Id}_A \circ f^{-1} \\
 &&& \xRightarrow{\text{[proposition: 2.50]}} && (((g \circ f)^{-1} \circ g) \circ f) \circ f^{-1} = f^{-1} \\
 &&& \xRightarrow{\text{[associativity: 2.21]}} && ((g \circ f)^{-1} \circ g) \circ (f \circ f^{-1}) = f^{-1} \\
 &&& \xRightarrow{\text{[theorem: 2.71]}} && ((g \circ f)^{-1} \circ g) \circ \text{Id}_B = f^{-1} \\
 &&& \xRightarrow{\text{[proposition: 2.50]}} && (g \circ f)^{-1} \circ g = f^{-1} \\
 &&& \Rightarrow && ((g \circ f)^{-1} \circ g) \circ g^{-1} = f^{-1} \circ g^{-1} \\
 &&& \xRightarrow{\text{[associativity: 2.21]}} && (g \circ f^{-1}) \circ (g \circ g^{-1}) = f^{-1} \circ g^{-1} \\
 &&& \xRightarrow{\text{[theorem: 2.71]}} && (g \circ f)^{-1} \circ \text{Id}_A = f^{-1} \circ g^{-1} \\
 &&& \xRightarrow{\text{[proposition: 2.50]}} && (g \circ f)^{-1} = f^{-1} \circ g^{-1}
 \end{aligned}$$

□

In the special case that $B = C$ we have

Corollary 2.78. *We have*

1. If $f: A \rightarrow B$ and $g: B \rightarrow C$ are injective functions then $g \circ f: A \rightarrow C$ is a injective function.

2. If $f: A \rightarrow B$ and $g: B \rightarrow C$ are surjective functions then $g \circ f: A \rightarrow C$ is a surjective function.
3. If $f: A \rightarrow B$ and $g: B \rightarrow C$ are bijective function then $g \circ f: A \rightarrow C$ is a bijective function.
4. If $f: A \rightarrow B$ and $g: B \rightarrow C$ are bijective function then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof.

1. This follows from [theorem: 2.77 (1)] because $f(A) \subseteq B$.
2. This follows from [theorem: 2.77 (2)] because if f is surjective we have $f(A) = B$.
3. This follows from (1) and (2)
4. This follows from [theorem: 2.77 (4)] because if f is bijective, hence surjective, we have $f(A) = B$ □

The following is a example of a bijection between a class and the class of functions in this set.

Theorem 2.79. *Let A be a class then there exists a bijection between A and $A^{\{0\}}$*

Proof. Given $x \in A$ define the function $f_x: \{0\} \rightarrow \{x\}$ where $f_x = \{(0, x)\}$ [see [example: 2.67] to prove that this is a function (even a bijection)]. So $f_x \in \{x\}^{\{0\}}$, which as $\{x\} \subseteq A$ proves by [theorem: 2.36] that $f_x \in A^{\{0\}}$. Define now $f = \{z | z = (x, f_x) \text{ where } x \in A\}$. If $(x, y) \in f$ we have $x \in A$ and thus $y = f_x \in A^{\{0\}}$ hence $(x, y) \in A \times A^{\{0\}}$. Also if $(x, y), (x, y') \in f$ then $y = f_x$ and $y' = f_x$ so that $y = y'$. Further for every $x \in A$ we have by the definition of f that $(x, f_x) \in f$. So we conclude that

$$f: A \rightarrow A^{\{0\}} \text{ is a function}$$

Assume now that $(x, y), (x', y) \in f$ then $f_x = y = f_{x'}$, so that $\{(0, x)\} = \{(0, x')\}$, hence $(0, x) = (0, x')$, from which it follows that $x = x'$. this proves that

$$f: A \rightarrow A^{\{0\}} \text{ is an injective function}$$

If $y \in A^{\{0\}}$ then $y: \{0\} \rightarrow A$ is a function, hence $0 \in \{0\} = \text{dom}(y)$, so there exists a z such that $(0, z) \in y \subseteq \{0\} \times A$ proving that $z \in A$. Hence

$$\{(0, z)\} \subseteq y \wedge z \in A \tag{2.22}$$

If $(u, v) \in y \subseteq \{0\} \times A$ then $u = 0$ so that $(0, u) \in y$, which, as $(0, z) \in y$ and y is a function, proves that $u = z$ or $(u, v) = (0, z) \in \{(0, z)\}$. So $y \subseteq \{(0, z)\}$ which combined with [eq: 2.22] proves that $\{(0, z)\} = y$. As $f_z = \{(0, z)\} = y$ we have that $(z, y) \in f$ which proves that

$$f \text{ is a surjection} \tag{2.22} \quad \square$$

Theorem 2.80. *If A is a class then there is a bijection between $\mathcal{P}(A)$ and $\{0, 1\}^A$ where $0 = \emptyset$ and $1 = \{\emptyset\}$ are different elements.*

Proof. Define $\gamma: \mathcal{P}(A) \rightarrow \{0, 1\}^A$ by $\gamma = \{z | z = (B, \mathcal{X}_{A,B}) \text{ where } B \in \mathcal{P}(A)\}$ where $\mathcal{X}_{A,B} = (B \times \{1\}) \cup ((A \setminus B) \times \{0\})$ is the graph of the Characteristic function [example: 2.48]. If $(B, f) \in \gamma$ then $B \in \mathcal{P}(A)$ and $f = \mathcal{X}_{A,B}$, as $B \in \mathcal{P}(A) \Rightarrow B \subseteq A$ it follow using [example: 2.48] that $\mathcal{X}_{A,B}: A \rightarrow \{0, 1\}$ is a function. So $(B, f) \in \{0, 1\}^A$ giving

$$\gamma \subseteq \mathcal{P}(A) \times (\{0, 1\}^A)$$

If $(B, f), (B, g) \in \gamma$ then $f = \mathcal{X}_{A,B}$ and $g = \mathcal{X}_{A,B}$ so that $f = g$, also by the definition of γ we have that $\text{dom}(\gamma) = \mathcal{P}(A)$, hence

$$\gamma: \mathcal{P}(A) \rightarrow \{0, 1\}^A \text{ is a function}$$

If $(B, f), (B', f) \in \gamma$ then $\mathcal{X}_{A,B} = \mathcal{X}_{A,B'}$ so that

$$\begin{aligned} x \in B & \Leftrightarrow \mathcal{X}_{A,B}(x) = 1 \\ & \Leftrightarrow \mathcal{X}_{A,B'}(x) = 1 \\ & \Leftrightarrow x \in B' \end{aligned}$$

proving that $B = B'$. Hence

$$\gamma: \mathcal{P}(A) \rightarrow \{0, 1\}^A \text{ is injective}$$

Let $f \in \{0, 1\}^A$, define $B = \{x \in A \mid (x, 1) \in f\} \subseteq A$, then $B \in \mathcal{P}(A)$.

If $(x, y) \in f$ then we have for x either:

$x \in B$. Then $(x, 1) \in f$ and as $(x, y) \in f$ we have that $y = 1$ so that $(x, y) = (x, 1) \in \mathcal{X}_{A,B}$

$x \notin B$. Then $(x, 0) \in f$ and as $(x, y) \in f$ we have that $y = 0$ so that $(x, y) = (x, 0) \in \mathcal{X}_{A,B}$ [as $x \in A \setminus B$]

proving that

$$f \subseteq \mathcal{X}_{A,B} \quad (2.23)$$

If $(x, y) \in \mathcal{X}_{A,B}$ then we have for x either:

$x \in B$. Then as $(x, 1) \in \mathcal{X}_{A,B}$ we must have that $y = 1$, using the definition of B we have also $(x, 1) \in f \Rightarrow (x, y) \in f$

$x \notin B$. Then $x \in A \setminus B$ so that $(x, 0) \in \mathcal{X}_{A,B}$ hence we must have that $y = 0$. As $(x, 0) \in f$ [if $(x, 1) \in f$ then $x \in B$ a contradiction] it follows that $(x, y) = (x, 0) \in f$

proving that $\mathcal{X}_{A,B} \subseteq f$, which combined with 2.23 gives

$$\mathcal{X}_{A,B} = f \quad (2.24)$$

So given $f \in \{0, 1\}^A$ we have found a $B \in \mathcal{P}(A)$ such that $\mathcal{X}_{A,B} \stackrel{[\text{eq: 2.24}]}{=} f$, hence $(B, f) \in \gamma$ proving that

$$\gamma: \mathcal{P}(A) \rightarrow \{0, 1\}^A \text{ is a surjective} \quad \square$$

2.2.4 Restriction of a Function/Partial Function

Sometimes we only want to work with functions whose graphs satisfies certain conditions. It could be that the graph of a function does not satisfies these, but that the restriction of this graph to a sub-class satisfies the conditions. For example, the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} 1 & \text{if } x < 1 \\ 0 & \text{if } 1 \leq x \end{cases}$ is not continuous, as it is discontinuous at 1. However restricting this function to $\mathbb{R} \setminus \{1\}$ produces a continuous function. This is the idea of the next definition

Definition 2.81. Let $f: A \rightarrow B$ be a partial function and $C \subseteq A$ a sub-class of A then the restriction of f to C noted by $f|_C$ is defined by

$$f|_C = \{z \mid z = (x, y) \in f \wedge x \in C\} = f \cap (C \times B)$$

which defines the partial function

$$f|_C: C \rightarrow B$$

Proof. We must of course proof that $\{z \mid z = (x, y) \in f \wedge x \in C\} = f \cap (C \times B)$ and that $f|_C: C \rightarrow B$ is indeed a partial function. If $(x, y) \in \{z \mid z = (x, y) \in f \wedge x \in C\}$ then $(x, y) \in f \subseteq A \times B \Rightarrow y \in B$ and $x \in C$, so that $(x, y) \in f \wedge (x, y) \in C \times B$, hence $(x, y) \in f \cap (C \times B)$. If $(x, y) \in f \cap (C \times B)$ then $(x, y) \in f \wedge (x, y) \in C \times B \Rightarrow x \in C$, proving that $(x, y) \in \{z \mid z = (x, y) \in f \wedge x \in C\}$. So we have that

$$f|_C = \{z \mid z = (x, y) \in f \wedge x \in C\} = f \cap (C \times B)$$

From the above it follows, using [theorem: 1.30], that

$$f|_C \subseteq C \times B$$

Finally, if $(x, y), (x, y') \in f|_C$ then $(x, y), (x, y') \in f$ so that $y = y'$. Hence we have that $f|_C: C \rightarrow B$ is a partial function. $\square$

We show now the relation between composition and restriction.

Theorem 2.82. *If $f: A \rightarrow B$ and $g: C \rightarrow D$ are functions [so that by [theorem: 2.29] $g \circ f: A \cap f^{-1}(C) \rightarrow D$ is a function] then*

$$g \circ f = g \circ f|_{A \cap f^{-1}(C)}$$

so that

$$g \circ f|_{A \cap f^{-1}(C)}: A \cap f^{-1}(C) \rightarrow D$$

is also a function that is equal to the function

$$g \circ f: A \cap f^{-1}(C) \rightarrow D$$

Proof. As by [theorem: 2.29] $g \circ f: A \cap f^{-1}(C) \rightarrow D$ is a function we have that

$$g \circ f \subseteq (A \cap f^{-1}(C)) \times D$$

Let $(x, y) \in g \circ f$ then there exist a z such that $(x, z) \in f$ and $(z, y) \in g$. Further, as $(x, y) \in g \circ f$ and $g \circ f \subseteq (A \cap f^{-1}(C)) \times D$, it follows that $x \in A \cap f^{-1}(C)$, so that $(x, z) \in f|_{A \cap f^{-1}(C)}$ and as $(z, y) \in g$ that $(x, y) \in g \circ f|_{A \cap f^{-1}(C)}$. Hence we must conclude that

$$g \circ f \subseteq g \circ f|_{A \cap f^{-1}(C)} \quad (2.25)$$

On the other hand if $(x, y) \in g \circ f|_{A \cap f^{-1}(C)}$ then there exist a z such that $(x, z) \in f|_{A \cap f^{-1}(C)}$ and $(z, y) \in g$. Hence by the definition of $f|_{A \cap f^{-1}(C)}$ it follows that $(x, z) \in f$ so that, as $(z, y) \in g$ it follows that $(x, y) \in g \circ f$. Hence $g \circ f|_{A \cap f^{-1}(C)} \subseteq g \circ f$ which combined with [eq: 2.25] proves finally that

$$g \circ f = g \circ f|_{A \cap f^{-1}(C)} \quad \square$$

Corollary 2.83. *If $f: A \rightarrow B$ and $g: C \rightarrow D$ are functions such that $C \subseteq B$ then*

$$g \circ f: f^{-1}(C) \rightarrow D \text{ is a function}$$

and

$$g \circ f = g \circ f|_{f^{-1}(C)}$$

so that

$$g \circ f|_{f^{-1}(C)}: f^{-1}(C) \rightarrow D$$

is also a function that is equal to the function

$$g \circ f: f^{-1}(C) \rightarrow D$$

Proof. As $f: A \rightarrow B$ is a function it follows from [theorem: 2.51(4)] that

$$A = f^{-1}(B)$$

Further as $C \subseteq B$ it follows from [theorem: 2.17(6)] that $f^{-1}(C) \subseteq f^{-1}(B)$ which by the above proves that

$$f^{-1}(C) \subseteq A$$

Hence using [theorem: 1.36(2)] it follows that

$$A \cap f^{-1}(C) = f^{-1}(C) \quad (2.26)$$

Now using [theorem: 2.29] $g \circ f: A \cap f^{-1}(C) \rightarrow D$ is a function so that by the above

$$g \circ f: f^{-1}(C) \rightarrow D \text{ is a function}$$

Further by the previous theorem [see theorem: 2.82] we have that $g \circ f = g \circ f|_{A \cap f^{-1}(C)}$ so that by [eq: 2.26]

$$g \circ f = g \circ f|_{f^{-1}(C)} \quad \square$$

Theorem 2.84. Let $f: A \rightarrow C$ and $g: B \rightarrow C$ be two partial functions such that $A \cap B = \emptyset$ then

1. $f \cup g: A \cup B \rightarrow C$ is a partial function
2. $f = (f \cup g)|_A$ and $g = (f \cup g)|_B$
3. $\text{dom}(f \cup g) = \text{dom}(f) \cup \text{dom}(g)$
4. If $f: A \rightarrow C$ and $g: B \rightarrow C$ are functions then $f \cup g: A \cup B \rightarrow C$ are functions

Note 2.85. If $x \in A$ then as f is a function there exist a $y \in C$ such that $(x, y) \in f \subseteq f \cup g$ which is written as $y = f(x) \wedge y = (f \cup g)(x)$. If $x \in B$ then as g is a function there exist a $y \in C$ such that $(x, y) \in g \subseteq f \cup g$ which is written as $y = g(x) \wedge y = (f \cup g)(x)$. Hence we can say that

$$f \cup g: A \cup B \rightarrow C \text{ is defined by } (f \cup g)(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B \end{cases}$$

Proof.

1. As $f: A \rightarrow C$ and $g: B \rightarrow C$ are functions we have that $f \subseteq A \times C$ and $g \subseteq B \times C$ so that by [theorem: 1.30]

$$f \cup g \subseteq (A \times C) \cup (B \times C) \stackrel{[\text{theorem: 1.56}]}{=} (A \cup B) \times C$$

Let $(x, y), (x, y') \in f \cup g$. Assume that $y \neq y'$ then we can not have that $(x, y), (x, y') \in f$ for then, as f is a function, we would have $y = y'$, likewise we can not have that $(x, y), (x, y') \in g$, for then, as g is a function, we would have that $y = y'$. So we must that either $(x, y) \in f \wedge (x, y') \in g$ or $(x, y) \in g \wedge (x, y') \in f$, but then we would have $x \in A \cap B$ which contradicts $A \cap B = \emptyset$. So we must have that $y = y'$. Summarized

$$\text{If } (x, y), (x, y') \in f \cup g \text{ then we have } y = y'$$

2. As $f \subseteq A \times C$ we have by [theorem :1.30] that

$$f \cap (B \times C) \subseteq (A \times C) \cap (B \times C) \stackrel{[\text{theorem: 1.56}]}{=} (A \cap B) \times C = \emptyset \times C \stackrel{[\text{theorem: 1.54}]}{=} \emptyset$$

proving using [theorem: 1.22] that

$$f \cap (B \times C) = \emptyset \tag{2.27}$$

As $g \subseteq B \times C$ we have by [theorem :1.30] that

$$g \cap (A \times C) \subseteq (B \times C) \cap (A \times C) \stackrel{[\text{theorem: 1.56}]}{=} (A \cap B) \times C = \emptyset \times C \stackrel{[\text{theorem: 1.54}]}{=} \emptyset$$

proving using [theorem: 1.22] that

$$g \cap (A \times C) = \emptyset \tag{2.28}$$

Further we have

$$\begin{aligned} (f \cup g)|_A &= (f \cup g) \cap (A \times C) \\ &\stackrel{[\text{theorem: 1.29}]}{=} (f \cap (A \times C)) \cup (g \cap (A \times C)) \\ &\stackrel{[\text{eq: 2.28}]}{=} (f \cap (A \times C)) \cup \emptyset \\ &\stackrel{[\text{theorem: 1.35}]}{=} f \cap (A \times C) \\ &\stackrel{f \subseteq A \times C \text{ and } [\text{theorem: 1.36}]}{=} f \\ (f \cup g)|_B &= (f \cup g) \cap (B \times C) \\ &\stackrel{[\text{theorem: 1.29}]}{=} (f \cap (B \times C)) \cup (g \cap (B \times C)) \\ &\stackrel{[\text{eq: 2.27}]}{=} \emptyset \cup (g \cap (B \times C)) \\ &\stackrel{[\text{theorem: 1.35}]}{=} g \cap (B \times C) \\ &\stackrel{g \subseteq B \times C \text{ and } [\text{theorem: 1.36}]}{=} g \end{aligned}$$

3.

$$\begin{aligned}
x \in \text{dom}(f \cup g) &\Leftrightarrow \exists y \text{ such that } (x, y) \in f \cup g \\
&\Leftrightarrow \exists y \text{ such that } (x, y) \in f \vee (x, y) \in g \\
&\Rightarrow x \in \text{dom}(f) \vee x \in \text{dom}(g) \\
&\Rightarrow x \in \text{dom}(f) \cup \text{dom}(g) \\
x \in \text{dom}(f) \cup \text{dom}(g) &\Rightarrow x \in \text{dom}(f) \vee x \in \text{dom}(g) \\
&\Rightarrow (\exists y \text{ such that } (x, y) \in f) \vee (\exists y' \text{ such that } (x, y) \in g) \\
&\Rightarrow (\exists y \text{ such that } (x, y) \in f \cup g) \vee (\exists y' \text{ such that } (x, y) \in f \cup g) \\
&\Rightarrow x \in \text{dom}(f \cup g)
\end{aligned}$$

so

$$\text{dom}(f \cup g) = \text{dom}(f) \cup \text{dom}(g)$$

4. As $f: A \rightarrow C$ and $g: B \rightarrow C$ are functions we have that $A = \text{dom}(f)$, $B = \text{dom}(g)$. So that

$$\text{dom}(f \cup g) \stackrel{(3)}{=} \text{dom}(f) \cup \text{dom}(g) = A \cup B$$

proving that

$$f \cup g: A \cup B \rightarrow C \text{ is a function}$$

□

Corollary 2.86. Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be functions such that $A \cap C = \emptyset$ then

$$f \cup g: A \cup C \rightarrow B \cup D$$

is a function.

Proof. Using [theorem: 2.35] we have that $f: A \rightarrow B \cup D$ and $g: C \rightarrow B \cup D$ are functions. Applying then the previous theorem [theorem: 2.84] proves that $f \cup g: A \cup C \rightarrow B \cup D$ is a function. □**Corollary 2.87.** Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be bijections with $A \cap C = \emptyset$ and $B \cap D = \emptyset$ then

$$f \cup g: A \cup C \rightarrow B \cup D$$

is a bijection.

Proof. Using the previous theorem [theorem: 2.86] we have that $f \cup g: A \cup C \rightarrow B \cup D$ is a function. Now we have:**injectivity.** If $(x, y), (x', y) \in f \cup g \subseteq (A \cup C) \times (B \cup D)$ we have the following possibilities for y : **$y \in B$.** As $f \subseteq A \times B$ and $g \subseteq C \times D$ we can not have $(x, y), (x', y) \in g$ [for then $y \in D \Rightarrow y \in B \cap D = \emptyset$], as g is injective we have $x = x'$. **$y \in D$.** As $f \subseteq A \times B$ and $g \subseteq C \times D$ we can not have $(x, y), (x', y) \in f$ [for then $y \in B \Rightarrow y \in B \cap D = \emptyset$], as f is injective we have $x = x'$.so in all cases we have $x = x'$ proving injectivity of $f \cup g: A \cup C \rightarrow B \cup D$.**surjectivity.** If $y \in B \cup D$ then we have either: **$y \in B$.** Then as f is surjective there exist a $x \in A \subseteq A \cup C$ such that $(x, y) \in f \subseteq f \cup g$. **$y \in D$.** Then as g is surjective there exist a $x \in C \subseteq A \cup C$ such that $(x, y) \in g \subseteq f \cup g$.proving that in all cases there exist a $x \in A \cup C$ such that $(x, y) \in f \cup g$. □**Corollary 2.88.** Let $f: A \rightarrow B$ be a function a, b elements such that $a \notin A$ then

$$g: A \cup \{a\} \rightarrow B \cup \{b\} \text{ defined by } g = \{(a, b)\} \cup f$$

is a function.

Note 2.89. A alternative definition of g is $g(x) = \begin{cases} b & \text{if } x = a \\ f(x) & \text{if } x \in A \end{cases}$

Proof. Using [example: 2.47] we have that $C_b: \{a\} \rightarrow \{b\}$ where $C_b = \{(x, b) | x \in \{a\}\} = \{(a, b)\}$ is a function. As $A \cap \{a\}$ we can use the previous corollary [corollary: 2.86] so that

$$h: A \cup \{a\} \rightarrow B \cup \{b\} \text{ where } h = \{(a, b)\} \cup f \text{ is a function} \quad \square$$

Theorem 2.90. Let $f: A \rightarrow C$ and $g: B \rightarrow C$ functions such that $\forall x \in A \cap B$ we have $(x, y) \in f \wedge (x, y') \in g$ implies $y = y'$ [which is equivalent with saying that $\forall x \in A \cap B$ $f(x) = g(x)$] then we have that:

1. $f \cup g: A \cup B \rightarrow C$ is a function.
2. If $D \subseteq C$ then $(f \cup g)^{-1}(D) = f^{-1}(D) \cup g^{-1}(D)$.

Note 2.91. If $x \in A$ then as f is a function there exist a $y \in C$ such that $(x, y) \in f \subseteq f \cup g$ which is written as $y = f(x) \wedge y = (f \cup g)(x)$. If $x \in B$ then as g is a function there exist a $y \in C$ such that $(x, y) \in g \subseteq f \cup g$ which is written as $y = g(x) \wedge y = (f \cup g)(x)$. Hence we can say that

$$f \cup g: A \cup B \rightarrow C \text{ is defined by } (f \cup g)(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B \end{cases}$$

Proof.

1. First as $f \subseteq A \times C$ and $g \subseteq B \times C$ then we have that

$$f \cup g \subseteq (A \times C) \cup (B \times C) \stackrel{[\text{theorem: 1.56}]}{=} (A \cup B) \times C$$

Second if $(x, y), (x, y') \in f \cup g$ then we have either:

$x \in A \setminus B$. Then $(x, y), (x, y') \notin B \times C \Rightarrow (x, y), (x, y') \notin g$ so that $(x, y), (x, y') \in f$ which as f is a function proves that $y = y'$.

$x \in B \setminus A$. Then $(x, y), (x, y') \notin A \times C \Rightarrow (x, y), (x, y') \notin f$ so that $(x, y), (x, y') \in g$ which as g is a function proves that $y = y'$.

$x \in B \setminus A$. As $(x, y), (x, y') \in f \cup g$ we have either:

$(x, y) \in f$. Then we have either:

$(x, y') \in f$. As f is a function we have that $y = y'$

$(x, y') \in g$. Then we have by the extra condition on $x \in A \cap B$ that $y = y'$.

$(x, y) \in g$. Then we have either:

$(x, y') \in f$. Then we have by the extra condition on $x \in A \cap B$ that $y = y'$.

$(x, y') \in g$. As g is a function we have that $y = y'$

So in all cases we have that $y = y'$ which proves that

$$f \cup g \text{ is a partial function}$$

Let $x \in A \cup B$ then we have either:

$x \in A$. As f is a function there exist a $y \in C$ such that $(x, y) \in f \subseteq f \cup g$.

$x \in B$. As g is a function there exist a $y \in C$ such that $(x, y) \in g \subseteq f \cup g$.

So in all cases we found a $y \in C$ such that $(x, y) \in f \cup g$ which proves that

$$f \cup g \text{ is a function}$$

2. Let $x \in (f \cup g)^{-1}(D)$ then there exist a $y \in D$ such that $(x, y) \in f \cup g$. Hence we have either $(x, y) \in f \Rightarrow x \in f^{-1}(D)$ or $(x, y) \in g \Rightarrow x \in g^{-1}(D)$ so that $x \in f^{-1}(D)$ or $x \in g^{-1}(D)$. Hence we have proved that

$$(f \cup g)^{-1}(D) \subseteq f^{-1}(D) \cup g^{-1}(D) \quad (2.29)$$

On the other hand if $x \in f^{-1}(D) \cup g^{-1}(D)$ then we have either:

$x \in f^{-1}(D)$. Then there exist a $y \in D$ such that $(x, y) \in f \subseteq f \cup g \Rightarrow x \in (f \cup g)^{-1}(D)$

$x \in g^{-1}(D)$. Then there exist a $y \in D$ such that $(x, y) \in g \subseteq f \cup g \Rightarrow x \in (f \cup g)^{-1}(D)$

hence in all possible cases we have $x \in (f \cup g)^{-1}(D)$ which proves that

$$f^{-1}(D) \cup g^{-1}(D) \subseteq (f \cup g)^{-1}(D)$$

which, combined with [eq: 2.29], proves that

$$(f \cup g)^{-1}(D) = f^{-1}(D) \cup g^{-1}(D) \quad \square$$

Theorem 2.92. *Let $f: A \rightarrow B$ be a partial function and $C \subseteq A$ a sub-class of A then we have:*

1. $\text{dom}(f|_C) = C \cap \text{dom}(f)$
2. $\text{range}(f|_C) = f(C)$
3. *If $D \subseteq C$ then $f|_C(D) = f(D)$ and $(f|_C)|_D = f|_D$. Further as $C \subseteq C$ we have automatically that $f|_C(C) = f(C)$.*
4. *If $E \subseteq B$ then $(f|_C)^{-1}(E) = C \cap f^{-1}(E)$*
5. *If $f: A \rightarrow B$ is injective then $f|_C: C \rightarrow B$ is injective*

Proof.

1. If $x \in \text{dom}(f|_C)$ then there exists a y such that $(x, y) \in f|_C$, hence $x \in C$ and $(x, y) \in f$ or $x \in C$ and $x \in \text{dom}(f)$, so that $x \in C \cap \text{dom}(f)$. Hence

$$\text{dom}(f|_C) \subseteq C \cap \text{dom}(f) \quad (2.30)$$

Further if $x \in C \cap \text{dom}(f)$ then $x \in C$ and $x \in \text{dom}(f)$, so there exists a y such that $(x, y) \in f$, hence $(x, y) \in f|_C$ or $x \in \text{dom}(f|_C)$. So $C \cap \text{dom}(f) \subseteq \text{dom}(f|_C)$ which together with [eq: 2.30] gives

$$\text{dom}(f|_C) = C \cap \text{dom}(f)$$

2. If $y \in \text{range}(f|_C)$ then $\exists x$ such that $(x, y) \in f|_C$, hence $(x, y) \in f$ and $x \in C$, so that $y \in f(C)$. On the other hand if $y \in f(C)$ there exists a $x \in C$ such that $(x, y) \in f$, hence $(x, y) \in f|_C$ so that $y \in \text{range}(f|_C)$. Hence using the Axiom of Extent [axiom: 1.5] we have

$$\text{range}(f|_C) = f(C)$$

3. If $y \in f|_C(D)$ then $\exists x \in D$ such that $(x, y) \in f|_C$, hence $(x, y) \in f$ so that $y \in f(D)$. On the other hand if $y \in f(D)$ then $\exists x \in D$ such that $(x, y) \in f$, which as $x \in D \subseteq C \Rightarrow x \in C$ proves that $(x, y) \in f|_C$, so $y \in f|_C(D)$. Hence using the Axiom of Extent [axiom: 1.5] we have

$$f|_C(D) = f(D)$$

Further

$$(f|_C)|_D = (f \cap (C \times B)) \cap (D \times B) \underset{D \times B \subseteq C \times B}{=} f \cap (D \times B) = f|_D$$

4. If $x \in (f|_C)^{-1}(E)$ then there exist a $y \in E$ such that $(x, y) \in f|_C$, hence $x \in C$ and $(x, y) \in f \Rightarrow x \in f^{-1}(E)$, so that $x \in C \cap f^{-1}(E)$. Further if $x \in C \cap f^{-1}(E)$ then $x \in C$ and $x \in f^{-1}(E)$, so there exist a $y \in E$ such that $(x, y) \in f \xrightarrow{x \in C} (x, y) \in f|_C$, hence $x \in (f|_C)^{-1}(E)$. Hence using the Axiom of Extent [axiom: 1.5] we have

$$(f|_C)^{-1}(E) = C \cap f^{-1}(E)$$

5. If $(x, y), (x', y) \in f|_C$ then as $f|_C \subseteq f$ we have $(x, y), (x', y) \in f$ which as f is injective proves $y = y'$ $\square$

Theorem 2.93. Let $f: A \rightarrow B$ be a **partial** function and $\text{dom}(f) \subseteq C$ then $f|_C = f$

Note 2.94. If $f: A \rightarrow B$ is a function then $\text{dom}(f) = A$. Hence if $A \subseteq C$ then $f|_C = f$ in this case.

Proof. If $(x, y) \in f$ then by definition $x \in \text{dom}(f)$ which as $\text{dom}(f) \subseteq C$ proves that $x \in C$ hence $(x, y) \in \{z | z = (x, y) \in f | x \in C\} = f|_C$. Further if $(x, y) \in f|_C$ then $(x, y) \in f$ and $x \in C$, so evidently $(x, y) \in f$. Hence using the Axiom of Extent [axiom: 1.5] we have

$$f|_C = f \quad \square$$

Theorem 2.95. Let $f: A \rightarrow B$ be an injective **partial** function and $C \subseteq A$ then $(f^{-1})|_{f(C)} = (f|_C)^{-1}$

Proof. Let $(x, y) \in (f^{-1})|_{f(C)}$ then $x \in f(C)$ and $(x, y) \in f^{-1} \Rightarrow (y, x) \in f$, as $x \in f(C)$ there exists a $z \in C$ such that $(z, x) \in f$. As f is injective we have that $z = y$, proving that $y \in C$, which as $(y, x) \in f$ gives $(y, x) \in f|_C$ so that $(x, y) \in (f|_C)^{-1}$. Hence

$$(f^{-1})|_{f(C)} \subseteq (f|_C)^{-1} \quad (2.31)$$

If $(x, y) \in (f|_C)^{-1}$ then $(y, x) \in f|_C$ so that $y \in C$ and $(y, x) \in f$. Hence $x \in f(C)$ and as $(y, x) \in f$ gives $(x, y) \in f^{-1}$ we have $(x, y) \in (f^{-1})|_{f(C)}$. This proves that $(f|_C)^{-1} \subseteq (f^{-1})|_{f(C)}$, combining this with [eq: 2.31] gives:

$$(f^{-1})|_{f(C)} = (f|_C)^{-1} \quad \square$$

Theorem 2.96. Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be **partial** functions, $E \subseteq A$ and $f(E) \subseteq F$ then

$$(g \circ f)|_E = g|_{f(E)} \circ f|_E$$

Note 2.97. If $f(E) \subseteq \text{dom}(g)$ then, as by [theorem: 2.93] $g|_{\text{dom}(g)} = g$ we have in this special cases that

$$(g \circ f)|_E = g \circ f|_E$$

Proof. Let $(x, y) \in (g \circ f)|_E$ then $(x, y) \in g \circ f$ and $x \in E$, hence $\exists y$ such that $(x, y) \in f \wedge (y, z) \in g$. As $x \in E$ it follows that $(x, y) \in f|_E$, further from $x \in E$ and $(x, y) \in f$ it follows that $y \in f(E)$ or, as $f(E) \subseteq F$, it follows that $y \in F$ so that $(y, z) \in g|_F$. Hence we have that $(x, z) \in g|_F \circ f|_E$ which proves that

$$(g \circ f)|_E \subseteq g|_F \circ f|_E \quad (2.32)$$

If $(x, y) \in g|_F \circ f|_E$ then there exist a y such that $(x, y) \in f|_E$ and $(y, z) \in g|_F$. Hence $(x, y) \in f \wedge x \in E$ and $(y, z) \in g$ so that $(x, y) \in g \circ f$ and $x \in E$ which proves that $(x, y) \in (g \circ f)|_E$. So $g|_F \circ f|_E \subseteq (g \circ f)|_E$ which combined with [eq: 2.32] proves that

$$(g \circ f)|_E \subseteq g|_F \circ f|_E \quad \square$$

Theorem 2.98. Let $f: A \rightarrow B$ be a function and $C \subseteq A$ a sub-class of A then

1. $f|_C: C \rightarrow B$ is a function
2. $f|_C: C \rightarrow f(C)$ is a surjective function
3. If $f: A \rightarrow B$ is injective then $f|_C: C \rightarrow f(C)$ is a bijection
4. If $f: A \rightarrow B$ is a bijection then $f|_C: C \rightarrow f(C)$ is a bijection

Proof.

1. Using [definition: 2.81] we have that $f|_C: C \rightarrow B$ is a partial function, as by [theorem: 2.92] $\text{dom}(f|_C) = C \cap \text{dom}(f) \underset{f \text{ is a function}}{=} C \cap A \underset{C \subseteq A}{=} C$, it follows that $f|_C: C \rightarrow B$ is a function.
2. Using [theorem: 2.39] we have that $f|_C: C \rightarrow f(C)$ is a function. Further if $y \in f(C)$ then there exist a $x \in C$ such that $(x, y) \in f$. As $x \in C$ we have that $(x, y) \in f \cap (C \times B) = f|_C$. Hence $f|_C: C \rightarrow f(C)$ is a surjective function.

3. Let $(x, y), (x', y) \in f|_C$ then as $f|_C \subseteq f$ we have $(x, y), (x', y) \in f$ so that $x = x'$ proving that $f|_C: C \rightarrow f(C)$ is injective. As by (2) $f|_C: C \rightarrow f(C)$ is surjective it follows that

$$f|_C: C \rightarrow f(C) \text{ is a bijection}$$

4. As a bijection is injective this follows from (3). $\square$

Theorem 2.99. *Let $f: A \rightarrow B$ be a bijection then $f|_{A \setminus \{a\}}: A \setminus \{a\} \rightarrow B \setminus \{b\}$ is a bijection [where $(a, b) \in f$] or in other words $f|_{A \setminus \{a\}}: A \setminus \{a\} \rightarrow B \setminus f(a)$ is a bijection.*

Proof. Let $(x, y) \in f|_{A \setminus \{a\}} = f \cap ((A \setminus \{a\}) \times B)$ then $(x, y) \in f$ and $x \neq a$. Assume that $y = b$ then $(x, b) \in f$ and as $(a, b) \in f$ we must have by injectivity of f that $x = a$ contradicting $x \neq a$, hence we have that $y \in B \setminus \{b\}$. So we have that $f|_{A \setminus \{a\}} \subseteq (A \setminus \{a\}) \times (B \setminus \{b\})$ and

$$f|_{A \setminus \{a\}}: A \setminus \{a\} \rightarrow B \setminus \{b\} \text{ is a function}$$

If $(x, y), (x', y) \in f|_{A \setminus \{a\}}$ then $(x, y), (x', y) \in f$ $\xRightarrow{f \text{ is injective}}$ $x = x'$ proving that $f|_{A \setminus \{a\}}$ is injective. So

$$f|_{A \setminus \{a\}} \text{ is injective}$$

Finally if $y \in B \setminus \{b\}$ then as $f: A \rightarrow B$ is a bijection hence surjective, there exist a $x \in A$ such that $(x, y) \in f$. Assume that $x = a$ then $(a, y) \in f$ which as $(a, b) \in f$ gives $y = b$ contradicting $y \in B \setminus \{b\}$. Hence we must have $x \in A \setminus \{a\}$ so that $(x, y) \in f \cap ((A \setminus \{a\}) \times B) = f|_{A \setminus \{a\}}$ proving that

$$f|_{A \setminus \{a\}}: A \setminus \{a\} \rightarrow B \setminus \{b\} \text{ is surjective} \quad \square$$

Theorem 2.100. *If $f: A \rightarrow B$ is a bijection and $C \subseteq A$ then*

1. $(f^{-1})|_{f(C)} = (f|_C)^{-1}$
2. $(f^{-1})|_{f(C)}: f(C) \rightarrow C$ is a bijection

Proof.

1. If $(y, x) \in (f^{-1})|_{f(C)}$ then $y \in f(C)$ and $(y, x) \in f^{-1} \Rightarrow (x, y) \in f$. As $y \in f(C)$ there exist a $x' \in C$ such that $(x', y) \in f$. As f is injective it follow that $x = x'$ so that $x \in C$, hence $(x, y) \in f|_C$ or $(y, x) \in (f|_C)^{-1}$. From this it follows that

$$(f^{-1})|_{f(C)} \subseteq (f|_C)^{-1} \quad (2.33)$$

If $(y, x) \in (f|_C)^{-1}$ then $(x, y) \in f|_C$ so that $x \in C$ and $(x, y) \in f$ proving that $y \in f(C)$. Further as $(x, y) \in f$ it follows that $(y, x) \in f^{-1}$. Hence $(y, x) \in (f^{-1})|_{f(C)}$ so that $(f|_C)^{-1} \subseteq (f^{-1})|_{f(C)}$. Combining this result with [eq: 2.33] proves

$$(f^{-1})|_{f(C)} = (f|_C)^{-1}$$

2. We have

injectivity. Let $(y, x), (y', x) \in (f^{-1})|_{f(C)}$ then $(y, x), (y', x) \in f^{-1} \Rightarrow (x, y), (x, y') \in f$. Hence as f is a function we have that $y = y'$ which proves that $(f^{-1})|_{f(C)}$ is injective.

surjectivity. Let $x \in C$ then as $C \subseteq A$ $x \in A$. As $f^{-1}: B \rightarrow A$ is a bijection by [theorem: 2.74] there exist a $y \in B$ such that $(y, x) \in f^{-1}$ or $(x, y) \in f$ so that $y \in f(C)$. Hence $(y, x) \in (f^{-1})|_{f(C)}$ proving surjectivity. $\square$

The following theorem will be useful for manifolds later:

Theorem 2.101. *Let $f: A \rightarrow B$ be a function and $g: C \rightarrow D$ be an injective function then we have*

1. $g: C \rightarrow g(C)$ is a bijection
2. $\text{dom}(f \circ g^{-1}) = g(A \cap C)$
3. $f \circ g^{-1}: g(A \cap C) \rightarrow B$ is a function

4. If f is a injection then $f \circ g^{-1}: g(A \cap C) \rightarrow f(A \cap C)$ is a bijection
5. $f \circ g^{-1} = (f \circ g^{-1})|_{g(A \cap C)} = f|_{A \cap C} \circ (g^{-1})|_{g(A \cap C)} = f|_{A \cap C} \circ (g|_{A \cap C})^{-1}$

Proof.

1. This follows from [proposition: 2.68]
2. If $z \in \text{dom}(f \circ g^{-1})$ then $\exists x$ such that $(z, x) \in f \circ g^{-1}$, hence $\exists y$ such that $(z, y) \in g^{-1}$ and $(y, x) \in f$, from which it follows that $(y, z) \in g$ and $(y, x) \in f$. As $g \subseteq C \times B$ and $f \subseteq A \times B$ it follows that $y \in A$ and $y \in C$ so that $y \in A \cap C$, as $(y, z) \in g$ we have $z \in g(A \cap C)$. This proves

$$\text{dom}(g \circ f^{-1}) \subseteq g(A \cap C) \quad (2.34)$$

If $z \in g(A \cap C)$ then $\exists y \in A \cap C$ such that $(y, z) \in g$, hence $(z, y) \in g^{-1}$. As f is a function we have that $A = \text{dom}(f)$, hence as $y \in A \cap C \Rightarrow y \in A$, there exists a x such that $(y, x) \in f$. As $(z, y) \in g^{-1}$ we have $(z, x) \in f \circ g^{-1}$ proving that $z \in \text{dom}(f \circ g^{-1})$. Hence $g(A \cap C) \subseteq \text{dom}(g \circ f^{-1})$ which combined with [eq: 2.34] proves that

$$\text{dom}(g \circ f^{-1}) = g(A \cap C)$$

3. As g is injective it follows from [theorem: 2.59] that $g^{-1}: D \rightarrow C$ is a partial function, hence, using [theorem: 2.19], $f \circ g^{-1}: D \rightarrow B$ is a partial function. Using [theorem: 2.25] it follows that $f \circ g^{-1}: \text{dom}(f \circ g^{-1}) \rightarrow B$ is a function, further as by (2) $\text{dom}(f \circ g^{-1}) = g(A \cap C)$ we must conclude that

$$f \circ g^{-1}: g(A \cap C) \rightarrow B \text{ is a function}$$

4.

injectivity. If $(x, y), (x', y) \in f \circ g^{-1}$ then $\exists z, z'$ such that $(x, z), (x', z') \in f$ and $(z, y), (z', y) \in g^{-1}$. Hence $(y, z), (y, z') \in g$ so that $z = z'$ [as $g^{-1}: D \rightarrow C$ is a partial function by [theorem: 2.59]] hence $(x, z), (x', z) \in f$ giving $x = x'$ [as f is injective]

surjectivity. If $y \in f(A \cap C)$ then $\exists x \in A \cap C$ such that $(x, y) \in f$. As $A \cap C \subseteq C$ we have that $x \in C$, so as $g: C \rightarrow B$ is a function there exist a z such that $(x, z) \in g$, hence $(z, x) \in g^{-1}$. As $(x, y) \in f$ it follows that $(z, y) \in f \circ g^{-1}$.

5. We have

$$\begin{aligned} (f \circ g^{-1}) & \stackrel{[\text{theorem: 2.93}]}{=} (f \circ g^{-1})|_{\text{dom}(f \circ g^{-1})} \\ & \stackrel{(2)}{=} (f \circ g^{-1})|_{g(A \cap C)} \\ & \stackrel{[\text{theorem: 2.96}]}{=} f|_{g^{-1}(g(A \cap C))} \circ (g^{-1})|_{g(A \cap C)} \\ & \stackrel{[\text{proposition: 2.57}]}{=} f|_{A \cap C} \circ (g^{-1})|_{g(A \cap C)} \\ & \stackrel{[\text{theorem: 2.95}] \wedge g \text{ is injective}}{=} f|_{A \cap C} \circ (g|_{A \cap C})^{-1} \end{aligned}$$

□

2.2.5 Set operations and (Partial) Functions

Theorem 2.102. Let $f: A \rightarrow B$ be a function then we have

1. If $C, D \subseteq A$ with $C \subseteq D$ then $f(C) \subseteq f(D)$
2. If $C, D \subseteq B$ with $C \subseteq D$ then $f^{-1}(C) \subseteq f^{-1}(D)$
3. If $C, D \subseteq B$ then $f^{-1}(C \setminus D) = f^{-1}(C) \setminus f^{-1}(D)$
4. If $D \subseteq B$ then $f^{-1}(B \setminus D) = A \setminus f^{-1}(D)$
5. If $C, D \subseteq A$ then $f(C) \setminus f(D) \subseteq f(C \setminus D)$
6. If $C, D \subseteq A$ and f is **injective** then $f(C) \setminus f(D) = f(C \setminus D)$

Proof.

1. Let $y \in f(C)$ then there exist a $x \in C$ such that $(x, y) \in f$, as $C \subseteq D$ we have $x \in D$ so that $y \in f(D)$
2. If $x \in f^{-1}(C)$ there exists a $y \in C$ such that $(x, y) \in f$, as $C \subseteq D$ then $y \in D$ so that $x \in f^{-1}(D)$
3. If $x \in f^{-1}(C \setminus D)$ then $\exists y \in C \setminus D$ such that $(x, y) \in f$. As $y \in C \setminus D$ we have that $y \in C$ and $y \notin D$, from $y \in C$ it follows that $x \in f^{-1}(C)$. Assume that also $x \in f^{-1}(D)$ then $\exists y' \in D$ such that $(x, y') \in f$ which, as f is a function and $(x, y) \in f$, proves that $y = y'$, hence $y \in D$ contradicting $y \notin D$, so we must have $x \notin f^{-1}(D)$, hence $x \in f^{-1}(C) \setminus f^{-1}(D)$ proving

$$f^{-1}(C \setminus D) \subseteq f^{-1}(C) \setminus f^{-1}(D) \quad (2.35)$$

If $x \in f^{-1}(C) \setminus f^{-1}(D)$ then $x \in f^{-1}(C)$ and $x \notin f^{-1}(D)$. As $x \in f^{-1}(C)$ there exists a $y \in C$ such that $(x, y) \in f$. Assume that $y \in D$, then as $(x, y) \in f$ we have $x \in f^{-1}(D)$ contradicting $x \notin f^{-1}(D)$, so we must have $y \notin D$. Hence $y \in C \setminus D$ which proves that $x \in f^{-1}(C \setminus D)$ or $f^{-1}(C) \setminus f^{-1}(D) \subseteq f^{-1}(C \setminus D)$. Combining this with [eq: 2.35] proves

$$f^{-1}(C \setminus D) = f^{-1}(C) \setminus f^{-1}(D)$$

4. As $D \subseteq B \subseteq B$ we have by (3) that

$$\begin{aligned} f^{-1}(B \setminus D) &= f^{-1}(B) \setminus f^{-1}(D) \\ &\stackrel{[\text{theorem: 2.51}]}{=} A \setminus f^{-1}(D) \end{aligned}$$

5. If $y \in f(C) \setminus f(D)$ then $y \in f(C)$ and $y \notin f(D)$. From $y \in f(C)$ it follows that $\exists x \in C$ such that $(x, y) \in f$. Assume that $x \in D$ then as $(x, y) \in f$ we have $y \in f(D)$ contradicting $y \notin f(D)$, so we must have $x \notin D$, proving that $x \in C \setminus D$. Hence $y \in f(C \setminus D)$ or

$$f(C) \setminus f(D) \subseteq f(C \setminus D)$$

6. If $y \in f(C \setminus D)$ then $\exists x \in C \setminus D$ such that $x \in C$, $x \notin D$ and $(x, y) \in f$. From $x \in C$ it follows that $y \in f(C)$. Assume that $y \in f(D)$ then there exist a $x' \in D$ such that $(x', y) \in f$, as f is **injective** we have $x = x'$ so that $x \in D$ contradicting $x \notin D$, hence $y \notin f(D)$. This proves that $y \in f(C) \setminus f(D)$ or $f(C \setminus D) \subseteq f(C) \setminus f(D)$ which combined with (3) gives

$$f(C) \setminus f(D) = f(C \setminus D) \quad \square$$

Theorem 2.103. If $f: A \rightarrow B$ is a function, $E, F \subseteq A$ and $C, D \subseteq B$ then we have

1. $f(E \cup F) = f(E) \cup f(F)$
2. $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$
3. $f(E \cap F) \subseteq f(E) \cap f(F)$
4. If f is injective then $f(E \cap F) = f(E) \cap f(F)$
5. $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$

Proof.

1. Let $y \in f(E \cup F)$ then there exist a $x \in E \cup F$ with $(x, y) \in f$. So $x \in E$ proving that $y \in f(E)$ or $x \in F$ proving $y \in f(F)$. So it follows that $y \in f(E) \cup f(F)$ or

$$f(E \cup F) \subseteq f(E) \cup f(F) \quad (2.36)$$

If $y \in f(E) \cup f(F)$ then we have the following possibilities

$y \in f(E)$. Then $\exists x \in E$ such that $(x, y) \in f$. As by the definition of a union $x \in E \cup F$, it follows that $y \in f(E \cup F)$

$y \in f(F)$. Then $\exists x \in F$ such that $(x, y) \in f$. As by the definition of a union $x \in E \cup F$, it follows that $y \in f(E \cup F)$

So in all cases we have $y \in f(E \cup F)$. Hence $f(E) \cup f(F) \subseteq f(E \cup F)$ which combined with [eq: 2.36] proves

$$f(E \cup F) = f(E) \cup f(F)$$

2. If $x \in f^{-1}(C \cup D)$ there exists a $y \in C \cup D$ such that $(x, y) \in f$. From $y \in C \cup D$ we have $y \in C$ hence $x \in f^{-1}(C)$ or $y \in D$ hence $x \in f^{-1}(D)$. So $x \in f^{-1}(C) \cup f^{-1}(D)$ proving

$$f^{-1}(C \cup D) \subseteq f^{-1}(C) \cup f^{-1}(D) \quad (2.37)$$

If $x \in f^{-1}(C) \cup f^{-1}(D)$ then we have the following possibilities to consider:

$x \in f^{-1}(C)$. Then $\exists y \in C$ such that $(x, y) \in f$. As by the definition of a union $y \in C \cup D$ it follows that $x \in f^{-1}(C \cup D)$

$x \in f^{-1}(D)$. Then $\exists y \in D$ such that $(x, y) \in f$. As by the definition of a union $y \in C \cup D$ it follows that $x \in f^{-1}(C \cup D)$

So in all cases we have $x \in f^{-1}(C \cup D)$, proving $f^{-1}(C) \cup f^{-1}(D) \subseteq f^{-1}(C \cup D)$ which combined with [eq 2.37] proves

$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$$

3. If $y \in f(E \cap F)$ then $\exists x \in E \cap F$ such that $(x, y) \in f$. From $x \in E \cap F$ we have that $x \in E$ hence $y \in f(E)$ and $x \in F$, so that $y \in f(F)$. Hence $y \in f(E) \cap f(F)$ or

$$f(E \cap F) \subseteq f(E) \cap f(F)$$

4. Using (3) we have that

$$f(E \cap F) \subseteq f(E) \cap f(F) \quad (2.38)$$

Let $y \in f(E) \cap f(F)$ then we have $y \in f(E)$ so that $\exists x \in E$ such that $(x, y) \in f$ and $y \in f(F)$ so that $\exists x' \in F$ such that $(x', y) \in f$. As f is injective and $(x, y), (x', y) \in f$ we have $x = x'$ so that $x \in E \cap F$, proving that $f(E) \cap f(F) \subseteq f(E \cap F)$. Combining this result with [eq: 2.38] gives

$$f(E \cap F) = f(E) \cap f(F)$$

5. If $x \in f^{-1}(C \cap D)$ then $\exists y \in C \cap D$ such that $y \in C$, so that $x \in f^{-1}(C)$ and $y \in D$, so that $x \in f^{-1}(D)$. Hence $x \in f^{-1}(C) \cap f^{-1}(D)$ proving

$$f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D) \quad (2.39)$$

If $x \in f^{-1}(C) \cap f^{-1}(D)$ then $x \in f^{-1}(C)$ so there exists a $y \in C$ such that $(x, y) \in f$ and $x \in f^{-1}(D)$ so $\exists y' \in D$ such that $(x, y') \in f$. As f is a function $y = y'$ proving $y \in C \cap D$, hence $x \in f^{-1}(C \cap D)$. So $f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$, combining this with [eq: 2.39] gives

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D) \quad \square$$

Up to now we define a function $f: A \rightarrow B$ by specifying what the classes f, A, B are. However in many cases we have a parameterized expression [based on function calls and operators) to define f . Then we have the following

Proposition 2.104. *Let A, B be classes and suppose that there exists a parameterized expression $F(x)$ that calculates a **unique** value for **every** $x \in A$ then we can define the function $f: A \rightarrow B$ by $f = \{z | z = (x, F(x)) \wedge x \in A\}$*

Proof. If $(x, y), (x, y') \in f$ then there exists $a, a' \in A$ such that $(x, y) = (a, F(a)) \wedge (x, y') = (a', F(a'))$, hence $x = a \wedge x = a' \wedge y = F(a) \wedge y' = F(a') \Rightarrow a = a' \wedge y = F(a) \wedge y = F(a)$ proving that $y = y'$. So

$f: A \rightarrow B$ is a partial function

If $x \in A$ then as $F(x)$ is defined on every $x \in A$ we have that $(x, F(x)) \in f$ so that $x \in \text{dom}(f)$. So $A \subseteq \text{dom}(f)$ we have by 2.27 that

$$f: A \rightarrow B \text{ is a function} \quad \square$$

This leads to a notation that we will gradually start to use

Notation 2.105. The function definition $f: A \rightarrow B$ defined by $f(x) = F(x)$ [where $E(x)$ is a parameterized expression that calculates a unique value for every $x \in A$] is equivalent with

$$f = \{z \mid z = (x, E(x)) \wedge x \in E\} = \{(x, E(x)) \mid x \in X\}$$

Example 2.106. $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \cos(4 \cdot x)$

2.2.6 Indexed sets

In many cases we have to deal with sets indexed by a index, which is actually a function in another form. We will use this in topology and vector spaces.

Definition 2.107. (indexed set) Let $f: I \rightarrow A$ be a surjection then A is called a **indexed set** and noted as

$$A = \{f(i) \mid i \in I\} \text{ or } A = \{f_i \mid i \in I\}$$

So

$$x \in A \Leftrightarrow \exists i \in I \text{ such that } x = f(i) \text{ or } x = f_i$$

I is called the index of the indexed set $\{f_i \mid i \in I\}$.

Definition 2.108. (unique indexed set) $A = \{f_i \mid i \in I\}$ is a **unique indexed set** if $f: I \rightarrow A$ is a bijection. So

$$x \in A \Leftrightarrow \exists i \in I \text{ such that } x = f(i) \text{ or } x = f_i$$

and

$$\text{If } x_i = x_j \text{ then } i = j$$

Example 2.109. Every set can be written as a unique indexed set indexed by itself. So if A is a set then $A = \{\text{Id}_X(i) \mid i \in I\}$.

2.3 Families

2.3.1 Family

We introduce now the idea of a indexed family which is essential a function of a class to another class. It is essential another notation for a function where the emphasis is on the objects in a collection and a way of indexing these objects and less on the function itself

Definition 2.110. Let I, B be classes then a family

$$\{x_i\}_{i \in I} \subseteq B$$

is actually a function

$$f: I \rightarrow B$$

Further x_i is another notation for $f(i)$ so that $y = f_i$ is equivalent with $y = f(i)$ or $(i, y) \in f$

Note 2.111. In the above definition $\{x_i\}_{i \in I}$ only make sense if you specify what the defining function is. To avoid excessive notation, we assume that if we write $\{x\}_{i \in I} \subseteq B$ that the defining function is $x: I \rightarrow B$. However this is sometimes not feasible and in that case we state what the defining function of $\{x_i\}_{i \in I}$ is.

Note 2.112. In most cases we use the notation $\{x_i\}_{i \in I} \subseteq B$ to indicate three parts of a function namely $x: I \rightarrow B$ where x is the graph of a function, I is the domain and B is the target of the function. In some cases we are not interested in the target of the function [we can always assume then that the target is the universal class $\mathcal{U}$] in this case we just use the notation $\{x_i\}_{i \in I}$ for the family.

Example 2.113. The empty function $\emptyset: \emptyset \rightarrow V$ [see example: 2.46] defines a family that is noted as $\{\emptyset_i\}_{i \in \emptyset} \subseteq V$. Further if $\{x_i\}_{i \in \emptyset} \subseteq V$ is a family where the index set is empty then $\{x_i\}_{i \in \emptyset} \subseteq V = \{\emptyset_i\}_{i \in \emptyset} \subseteq V$

Proof. $\{x_i\}_{i \in \emptyset} \subseteq V$ is defined by the function $x: \emptyset \rightarrow V$, as $x \subseteq \emptyset \times V = \emptyset$ we have that $x = \emptyset$ so that $\{x_i\}_{i \in \emptyset} \subseteq V = \{\emptyset_i\}_{i \in \emptyset} \subseteq V$. $\square$

Proposition 2.114. For the family $\{x_i\}_{i \in I} \subseteq \emptyset$ we have $I = \emptyset$ so that $\{x_i\}_{i \in I} \subseteq \emptyset = \{\emptyset_i\}_{i \in \emptyset} \subseteq \emptyset$

Proof. Let $f: I \rightarrow \emptyset$ be the function that defines the family then as f is a function we have that $f(I) = \emptyset$. So if $x \in I$ then $\exists y \in \emptyset$ such that $(x, y) \in f \subseteq I \times \emptyset = \emptyset$ a contradiction, hence we must have $I = \emptyset$. $\square$

Example 2.115. Let A, B be classes then family $\{(\text{Id}_A)_a\}_{a \in A} \subseteq B$ defined by the function $\text{Id}_A: A \rightarrow B$ is noted as $\{x\}_{x \in A}$.

We can now define the concept of a sub family

Definition 2.116. Let $\{A_i\}_{i \in I} \subseteq B$ be a family of objects in B defined by the function $f: I \rightarrow B$ and $J \subseteq I$ then $\{A_i\}_{i \in J} \subseteq B$ is the family defined by the function $f|_J: J \rightarrow B$ [see: theorem: 2.98 for the proof that $f|_J: J \rightarrow B$ is a function]

Definition 2.117. Let I, J, A, B be classes such that $I \cap J = \emptyset$ and

$$\{x\}_{i \in I} \subseteq A \text{ defined by the function } f: I \rightarrow A$$

$$\{y_i\}_{i \in J} \subseteq B \text{ defined by the function } g: J \rightarrow B$$

then $\{z_i\}_{i \in I \cup J} \subseteq A \cup B$ defined by $z_i = \begin{cases} A_i & \text{if } i \in I \\ B_i & \text{if } i \in J \end{cases}$ is the family defined by the function

$$f \cup g: I \cup J \rightarrow A \cup B$$

[see theorem: 2.86 for the proof that $f \cup g: I \cup J \rightarrow A \cup B$ is indeed a function]

Definition 2.118. If I, J are classes then $\{x_{i,j}\}_{(i,j) \in I \times J} \subseteq A$ is defined by a function $x: I \times J \rightarrow A$, based on this we can define the following families:

1. $\forall i \in I \{x_{i,j}\}_{j \in J}$ is defined by the function $x_{i,*}: J \rightarrow A$ where $x_{i,*}(j) = x(i, j) = x_{i,j}$
2. $\forall j \in J \{x_{i,j}\}_{i \in I}$ is defined by the function $x_{*,j}: I \rightarrow A$ where $x_{*,j}(i) = x(i, j) = x_{i,j}$

Composition of functions can also be represented via the above family notation,

Definition 2.119. If you have a function $f: I \rightarrow J$ and a family $\{x_j\}_{j \in J} \subseteq A$ [defined by the function $g: J \rightarrow A$] then

$$\{x_{f(i)}\}_{i \in I}$$

is the family represented by the function

$$g \circ f: I \rightarrow A$$

So a family is just another notation for a function. We introduce also a new notation for the range of this function.

Definition 2.120. If $\{x_i\}_{i \in I}$ is a family of objects in B [standing for the function $f: I \rightarrow B$] then we define $\{x_i | i \in I\}$ by

$$\{x_i | i \in I\} = \text{range}(f) = f(I)$$

The motivation for this definition is the following theorem

Theorem 2.121. *If $\{x_i\}_{i \in I} \subseteq B$ is a family of objects in B with associated function f then*

$$x \in \{x_i | i \in I\} \Leftrightarrow \exists i \in I \text{ such that } x = x_i$$

Proof. As $\{x_i\}_{i \in I} \subseteq B$ is equivalent with $f: I \rightarrow B$ we have

$$\begin{aligned} z \in \{x_i | i \in I\} &\stackrel{\text{define}}{\Leftrightarrow} z \in \text{range}(x) \\ &\Leftrightarrow \exists i \text{ with } (i, z) \in f \\ &\stackrel{f \subseteq I \times B}{\Leftrightarrow} \exists i \text{ with } i \in I \wedge (i, z) \in f \\ &\Leftrightarrow \exists i \in I \text{ with } (i, z) \in f \\ &\Leftrightarrow \exists i \in I \text{ with } z = f(i) \\ &\Leftrightarrow \exists i \in I \text{ with } z = x_i \\ &\quad \square \end{aligned}$$

Theorem 2.122. *If $\{x_i\}_{i \in I} \subseteq B$ is a family such that I and B are sets then $\{x_i | i \in I\}$ is a set*

Proof. $\{x_i\}_{i \in I} \subseteq B$ is actually the function $x: I \rightarrow B$ where $\text{range}(x) = \{x_i | i \in I\}$. As I and B are sets, it follows from [theorem: 2.12] that $\text{range}(x)$ is a set, hence $\{x_i | i \in I\}$ is a set. $\square$

Up to now we consider a family as a indexed collection of objects. What is actually a object, in set theory it is a class which can be either a set or a proper class. A class is a collection so we can talk about the union of these collection. The convention is then to use upper case instead of lower case. If we want to deal with the union and intersection of the objects [considered as collections] in the family we use also a different notation.

Notation 2.123. *If $\{A_i\}_{i \in I} \subseteq B$ is a family of objects in B [standing for the function $A: I \rightarrow B$] then $\bigcup_{i \in I} A_i$ is defined by*

$$\bigcup_{i \in I} A_i = \bigcup \{\text{range}(A)\} \text{ [definition: 1.62]}$$

Definition 2.124. *A family $\{A_i\}_{i \in I} \subseteq B$ is **pairwise disjoint** iff $\forall i, j \in I$ with $i \neq j$ we have $A_i \cap A_j = \emptyset$.*

Notation 2.125. *If $\{A_i\}_{i \in I} \subseteq B$ is pairwise disjoint and we want to indicate this fact when we write the union of the family then we use the notation $\bigsqcup_{i \in I} A_i$. So $\bigsqcup_{i \in I} A_i$ is actually the same as $\bigcup_{i \in I} A_i$, but also relating the information that $\{A_i\}_{i \in I}$ is pairwise disjoint.*

Using this new notation we have the following characterization of the union

Theorem 2.126. *If $\{A_i\}_{i \in I} \subseteq B$ is a family of objects in B then*

$$x \in \bigcup_{i \in I} A_i \Leftrightarrow \exists i \in I \text{ such that } x \in A_i$$

Proof. As $\{A_i\}_{i \in I} \subseteq B$ is actually the function $A: I \rightarrow B$ where $\bigcup_{i \in I} A_i = \bigcup \text{range}(A)$. Then we have

$$\begin{aligned} x \in \bigcup_{i \in I} A_i &\stackrel{\text{definition}}{\Leftrightarrow} x \in \bigcup \text{range}(A) \\ &\stackrel{[\text{definition: 1.62}]}{\Leftrightarrow} \exists y \in \text{range}(A) \text{ such that } x \in y \\ &\Leftrightarrow \exists i \text{ such that } (i, y) \in A \text{ and } x \in y \\ &\stackrel{A \subseteq I \times B}{\Leftrightarrow} \exists i \in I \text{ such that } (x, y) \in A \text{ and } x \in y \\ &\Leftrightarrow \exists i \in I \text{ such that } y = A_i \text{ and } x \in y \\ &\Leftrightarrow \exists i \in I \text{ such that } x \in A_i \\ &\quad \square \end{aligned}$$

Corollary 2.127. *If $\{A_j\}_{j \in J} \subseteq B$ is a family and $f: I \rightarrow J$ is a surjection then*

$$\bigcup_{j \in J} A_j = \bigcup_{i \in I} A_{f(i)}$$

Proof. If $x \in \bigcup_{j \in J} A_j$ then by [theorem: 2.126] there exist a $j \in J$ such that $x \in A_j = A(f(j))$. As f is surjective we have by [theorem: 2.54] that there exist a $i \in I$ such that $j = f(i)$. Hence $x \in A(f(i)) = (A \circ f)(i)$. So by [theorem: 2.126] and the definition of $\bigcup_{i \in I} A_{f(i)}$ we have $x \in \bigcup_{i \in I} A_{f(i)}$. Hence

$$\bigcup_{j \in J} A_j \subseteq \bigcup_{i \in I} A_{f(i)} \quad (2.40)$$

If $x \in \bigcup_{i \in I} A_{f(i)}$ then there exist a $i \in I$ such that $x \in (A \circ f)(i)$, which, as using [theorem: 2.22] $(A \circ f)(i) \in \text{range}(A)$, means that there exists a $j \in J$ such that $A_j = (A \circ f)(i)$. Hence $x \in A_j$ proving by [theorem: 2.126] that $x \in \bigcup_{j \in J} A_j$. So $\bigcup_{i \in I} A_{f(i)} \subseteq \bigcup_{j \in J} A_j$ which combined with [eq: 2.40] gives

$$\bigcup_{j \in J} A_j = \bigcup_{i \in I} A_{f(i)}$$

□

Theorem 2.128. *If $\{A_i\}_{i \in I} \subseteq B$ is a family of objects in B where I and B are sets then $\bigcup_{i \in I} A_i$ is a set.*

Proof. As $\{A_i\}_{i \in I} \subseteq B$ is another way of saying $A: I \rightarrow B$ and I and B are sets, it follows from [theorem: 2.12] that $\text{range}(A)$ is a set. Using the Axiom of Union [axiom: 1.67] $\bigcup \text{range}(A)$ is a set, so by definition $\bigcup_{i \in I} A_i$ is a set. □

Example 2.129. Let $\{A_i\}_{i \in \emptyset} \subseteq B$ be the family defined by $A = \emptyset$ [the empty function $\emptyset: \emptyset \rightarrow B$ [see example: 2.46]] then $\bigcup_{i \in \emptyset} A_i = \emptyset$

Proof. Let $y \in \text{range}(A) = \text{range}(\emptyset)$ then x such that $(x, y) \in \emptyset$, a contradiction. Hence $\text{range}(A) = \emptyset$. So

$$\bigcup_{i \in \emptyset} A_i = \bigcup \text{range}(A) = \bigcup \emptyset \stackrel{1.64}{=} \emptyset \quad \square$$

Definition 2.130. *If $\{A_i\}_{i \in I} \subseteq B$ is a family of objects in B [standing for the function $A: I \rightarrow B$] then $\bigcap_{i \in I} A_i$ is defined by*

$$\bigcap_{i \in I} A_i = \bigcap \text{range}(A) \text{ [definition: 1.63]}$$

Theorem 2.131. *If $\{A_i\}_{i \in I} \subseteq B$ then $x \in \bigcap_{i \in I} A_i \Leftrightarrow \forall i \in I$ we have $x \in A_i$*

Proof. $\{A_i\}_{i \in I} \subseteq B$ is actually the function $A: I \rightarrow B$ where $\bigcap_{i \in I} A_i = \bigcap \text{range}(A)$.

$$\begin{aligned} x \in \bigcap_{i \in I} A_i & \stackrel{\text{definition}}{\Leftrightarrow} x \in \bigcap \text{range}(A) \\ & \stackrel{[\text{definition: 1.63}]}{\Leftrightarrow} \forall y \in \text{range}(A) \text{ we have } x \in y \\ & \stackrel{y \in \text{range}(A) \Leftrightarrow \exists i \text{ with } (i, y) \in A}{\Leftrightarrow} \forall i \in I \text{ with } (i, y) \in A \text{ we have } x \in y \\ & \Leftrightarrow \forall i \in I \text{ with } y = A_i \text{ we have } x \in y \\ & \Leftrightarrow \forall i \in I \text{ we have } x \in A_i \end{aligned}$$

□

Theorem 2.132. *If $\{A_i\}_{i \in I} \subseteq B$ is a family of objects in B such that $I \neq \emptyset$ then $\bigcap_{i \in I} A_i$ is a set.*

Proof. $\{A_i\}_{i \in I} \subseteq B$ is actually the function $A: I \rightarrow B$ where $\bigcap_{i \in I} A_i = \bigcap \text{range}(A)$. As $I \neq \emptyset$ there exists a $i \in I$. Given that A is a function it follows that $\text{dom}(A) = I$, so there exists a y such that $(i, y) \in A$ or $y \in \text{range}(A)$. So $\emptyset \neq \text{range}(A)$ which by [theorem: 1.63] proves that $\bigcap \text{range}(A)$ is a set, hence $\bigcap_{i \in I} A_i$ is a set. □

Example 2.133. Let $I = \{0\}$, B a class and take $A: I \rightarrow \{B\}$ defined by $A = \{(0, B)\}$, defining the family $\{A_i\}_{i \in \{0\}} \subseteq \{B\}$ where $A_0 = B$. For this family we have $\bigcap_{i \in \{0\}} A_i = B$ and $\bigcup_{i \in \{0\}} A_i = B$

Proof. Using [example: 2.67] it follows that $A: I \rightarrow \{B\}$ is bijection, hence a function, so that $\{A_i\}_{i \in \{0\}} \subseteq \{B\}$ is a well defined family. Further as A is a bijection we have that

$$\text{range}(A) = \{B\}$$

Finally

$$\bigcup_{i \in \{0\}} A_i = \bigcup \text{range}(A) = \bigcup \{B\} \stackrel{[\text{example: 1.64}]}{=} A$$

and

$$\bigcap_{i \in \{0\}} A_i = \bigcap \text{range}(A) = \bigcap \{B\} \stackrel{[\text{example: 1.64}]}{=} A \quad \square$$

Example 2.134. Let C, D classes, $I = \{0, 1\}$ and take $A: I \rightarrow \{C, D\}$ defined by $A = \{(0, C), (1, D)\}$ [see example: 2.28], defining the family $\{A_i\}_{i \in \{0, 1\}} \subseteq \{C, D\}$ where $A_0 = C$ and $A_1 = D$. For this family we have $\bigcup_{i \in \{0, 1\}} A_i = C \cup D$ and $\bigcap_{i \in \{0, 1\}} A_i = C \cap D$.

Proof. If $y \in \text{range}(A)$ then $\exists x$ such that $(x, y) \in A = \{(0, C), (1, D)\}$, so that $(x, y) = (0, C) \Rightarrow y = C$ or $(x, y) = (1, D) \Rightarrow y = D$, proving that $x \in \{C, D\}$. Further if $y \in \{C, D\}$ then $y = C \Rightarrow (0, C) \in A \Rightarrow y \in \text{range}(A)$ or $y = D \Rightarrow (1, D) \in A \Rightarrow y \in \text{range}(A)$. So we have

$$\text{range}(A) = \{C, D\}$$

Finally

$$\bigcup_{i \in \{0, 1\}} A_i = \bigcup \text{range}(A) = \bigcup \{C, D\} \stackrel{[\text{example: 1.65}]}{=} C \cup D$$

and

$$\bigcap_{i \in \{0, 1\}} A_i = \bigcap \text{range}(A) = \bigcap \{C, D\} \stackrel{[\text{example: 1.65}]}{=} C \cap D \quad \square$$

2.3.2 Properties of the union and intersection of families

To save space, from now on we use [theorem: 2.126] and [theorem: 2.131] about union and intersection of families without explicit referring to these theorems.

Theorem 2.135. If $\{A_i\}_{i \in I} \subseteq B$ is a family then we have:

1. $\forall i \in I$ we have $A_i \subseteq \bigcup_{i \in I} A_i$
2. $\forall i \in I$ we have $\bigcap_{i \in I} A_i \subseteq A_i$
3. If $\forall i \in I$ we have that $A_i \subseteq C$ then $\bigcup_{i \in I} A_i \subseteq C$
4. If $\forall i \in I$ we have $C \subseteq A_i$ then $C \subseteq \bigcap_{i \in I} A_i$

Proof.

1. Let $i \in I$ and assume that $x \in A_i$ then $\exists i \in I$ such that $x \in A_i$, so $x \in \bigcup_{i \in I} A_i$, proving that $A_i \subseteq \bigcup_{i \in I} A_i$.
2. Let $i \in I$ then if $x \in \bigcap_{i \in I} A_i$ we have $\forall j \in I$ that $x \in A_j \Rightarrow x \in A_i$, proving that $\bigcap_{i \in I} A_i \subseteq A_i$
- 3.

$$\begin{aligned} x \in \bigcup_{i \in I} A_i &\Rightarrow \exists i \in I \vdash x \in A_i \\ &\stackrel{[\text{theorem: 2.126}]}{\Rightarrow} x \in C \\ &\Rightarrow \bigcup_{i \in I} A_i \subseteq C \end{aligned}$$

4.

$$\begin{aligned}
x \in C &\Rightarrow \forall i \in I \models x \in A_i \\
&\Rightarrow x \in \bigcap_{i \in I} A_i \\
&\Rightarrow C \subseteq \bigcap_{i \in I} A_i \\
&\quad \square
\end{aligned}$$

Theorem 2.136. *If $\{A_i\}_{i \in I} \subseteq B$ is a family then*

1. *If $J \subseteq I$ then*

$$a. \bigcup_{i \in J} A_i \subseteq \bigcup_{i \in I} A_i$$

$$b. \bigcap_{i \in I} A_i \subseteq \bigcap_{i \in J} A_i$$

2. *If $I = J \cup K$ then*

$$a. \bigcup_{i \in I} A_i = (\bigcup_{i \in J} A_i) \cup (\bigcup_{i \in K} A_i)$$

$$b. \bigcap_{i \in I} A_i = (\bigcap_{i \in J} A_i) \cap (\bigcap_{i \in K} A_i)$$

Proof.

1.

a. If $x \in \bigcup_{i \in J} A_i$ then $\exists i \in J$ such that $x \in A_i$, as $J \subseteq I$ we have $i \in I$ with $x \in A_i$, so that $x \in \bigcup_{i \in I} A_i$.

b. If $x \in \bigcap_{i \in I} A_i$ then $\forall i \in I$ we have $x \in A_i$, as $J \subseteq I$ we have also $\forall i \in J$ that $x \in A_i$, hence $x \in \bigcap_{i \in J} A_i$.

2.

a. As by [theorem: 1.30] $J, K \subseteq I$ we have using (1) that $\bigcup_{i \in J} A_i \subseteq \bigcup_{i \in I} A_i$ and $\bigcup_{i \in K} A_i \subseteq \bigcup_{i \in I} A_i$. Using [theorem: 1.30] it follows that

$$\left(\bigcup_{i \in J} A_i \right) \cup \left(\bigcup_{i \in K} A_i \right) \subseteq \bigcup_{i \in I} A_i \quad (2.41)$$

If $x \in \bigcup_{i \in I} A_i$ then $\exists i \in I$ such that $x \in A_i$, as $I = J \cup K$ we have $i \in J \Rightarrow x \in \bigcup_{i \in J} A_i$ or $i \in K \Rightarrow x \in \bigcup_{i \in K} A_i$, which proves that $x \in (\bigcup_{i \in J} A_i) \cup (\bigcup_{i \in K} A_i)$. Hence

$$\bigcup_{i \in I} A_i \subseteq \left(\bigcup_{i \in J} A_i \right) \cup \left(\bigcup_{i \in K} A_i \right)$$

which combined with [eq: 2.41] proves

$$\bigcup_{i \in I} A_i = \left(\bigcup_{i \in J} A_i \right) \cup \left(\bigcup_{i \in K} A_i \right)$$

b. As by [theorem: 1.30] $J, K \subseteq I$ we have using (1) that $\bigcap_{i \in I} A_i \subseteq \bigcap_{i \in J} A_i$ and $\bigcap_{i \in I} A_i \subseteq \bigcap_{i \in K} A_i$. Using [theorem: 1.30] it follows that

$$\bigcap_{i \in I} A_i \subseteq \left(\bigcap_{i \in J} A_i \right) \cap \left(\bigcap_{i \in K} A_i \right) \quad (2.42)$$

If $x \in (\bigcap_{i \in J} A_i) \cap (\bigcap_{i \in K} A_i)$ then $x \in \bigcap_{i \in J} A_i$ and $x \in \bigcap_{i \in K} A_i$. So $\forall i \in J$ we have $x \in A_i$ and $\forall i \in K$ we have $x \in A_i$. Hence as $\forall i \in I$ we have $i \in J \Rightarrow x \in A_i$ or $i \in K \Rightarrow x \in A_i$ it follows that $x \in \bigcap_{i \in I} A_i$. So $(\bigcap_{i \in J} A_i) \cap (\bigcap_{i \in K} A_i) \subseteq \bigcap_{i \in I} A_i$ which combined with [eq: 2.42] proves

$$\bigcap_{i \in I} A_i = \left(\bigcap_{i \in J} A_i \right) \cap \left(\bigcap_{i \in K} A_i \right)$$

□

Theorem 2.137. Let $\{A_i\}_{i \in I} \subseteq C$ and $\{B_i\}_{i \in I} \subseteq D$ be two families such that $\forall i \in I$ we have $A_i \subseteq B_i$ then

1. $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} B_i$
2. $\bigcap_{i \in I} A_i \subseteq \bigcap_{i \in I} B_i$

Proof.

1. If $x \in \bigcup_{i \in I} A_i$ there exist a $i \in I$ such that $x \in A_i \xRightarrow{A_i \subseteq B_i} x \in B_i$, hence $x \in \bigcup_{i \in I} B_i$
2. If $x \in \bigcap_{i \in I} A_i$ then $\forall i \in I$ we have $x \in A_i \xRightarrow{A_i \subseteq B_i} x \in B_i$ proving $x \in \bigcap_{i \in I} B_i$ □

We have also the distributive laws for union and intersection [theorem: 1.29]

Theorem 2.138. (Distributivity) Let $\{A_i\}_{i \in I} \subseteq B$ be a family and C a class then

1. $C \cap (\bigcup_{i \in I} A_i) = \bigcup_{i \in I} (C \cap A_i)$
2. $C \cup (\bigcap_{i \in I} A_i) = \bigcap_{i \in I} (C \cup A_i)$
3. $C \cap (\bigcap_{i \in I} A_i) = \bigcap_{i \in I} (C \cap A_i)$
4. $C \cup (\bigcup_{i \in I} A_i) = \bigcup_{i \in I} (C \cup A_i)$

Proof.

1. If $x \in C \cap (\bigcup_{i \in I} A_i)$ then $x \in C$ and $x \in \bigcup_{i \in I} A_i \Rightarrow \exists i \in I$ such that $x \in A_i$. Hence $x \in C \cap A_i$, proving by [theorem: 2.135] that $x \in \bigcup_{i \in I} (C \cap A_i)$. So

$$C \cap \left(\bigcup_{i \in I} A_i \right) \subseteq \bigcup_{i \in I} (C \cap A_i) \quad (2.43)$$

If $x \in \bigcup_{i \in I} (C \cap A_i)$ then there exist a $i \in I$ such that $x \in C$ and $x \in A_i \Rightarrow x \in \bigcup_{i \in I} A_i$, so $x \in C \cap (\bigcup_{i \in I} A_i)$, proving that $\bigcup_{i \in I} (C \cap A_i) \subseteq C \cap (\bigcup_{i \in I} A_i)$. Combining this with [eq: 2.43] proves

$$C \cap \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (C \cap A_i)$$

2. If $x \in C \cup (\bigcap_{i \in I} A_i)$ then we have the following cases to consider:

$x \in C$. then $\forall i \in I$ we have $x \in C \cup A_i$ hence $x \in \bigcap_{i \in I} (C \cup A_i)$

$x \in \bigcap_{i \in I} A_i$. then $\forall i \in I$ we have $x \in A_i$ hence $x \in \bigcap_{i \in I} (C \cup A_i)$

which proves that

$$C \cup \left(\bigcap_{i \in I} A_i \right) \subseteq \bigcap_{i \in I} (C \cup A_i) \quad (2.44)$$

If $x \in \bigcap_{i \in I} (C \cup A_i)$ then we have two cases to consider:

$x \in C$. then $x \in C \cup (\bigcap_{i \in I} A_i)$

$x \notin C$. then, as $\forall i \in I$ we have $x \in C \cup A_i \xRightarrow{x \notin C} x \in A_i$, it follows that $x \in \bigcap_{i \in I} A_i$ hence $x \in C \cup (\bigcap_{i \in I} A_i)$

In all cases we have $x \in C \cup (\bigcap_{i \in I} A_i)$ proving that $\bigcap_{i \in I} (C \cup A_i) \subseteq C \cup (\bigcap_{i \in I} A_i)$, combining this with [eq: 2.44] gives

$$C \cup \left(\bigcap_{i \in I} A_i \right) = \bigcap_{i \in I} (C \cup A_i)$$

3. We have

$$\begin{aligned} x \in C \cap \left(\bigcap_{i \in I} A_i \right) &\Leftrightarrow x \in C \wedge \forall i \in I \text{ we have } x \in A_i \\ &\Leftrightarrow \forall i \in I \text{ we have } x \in C \cap A_i \\ &\Leftrightarrow x \in \bigcap_{i \in I} (C \cap A_i) \end{aligned}$$

Proving

$$C \cap \left(\bigcap_{i \in I} A_i \right) = \bigcap_{i \in I} (C \cap A_i)$$

4. We have

$$\begin{aligned} x \in C \cup \left(\bigcup_{i \in I} A_i \right) &\Leftrightarrow x \in C \vee x \in \bigcup_{i \in I} A_i \\ &\Leftrightarrow x \in C \vee \exists i \in I \text{ with } x \in A_i \\ &\Leftrightarrow \exists i \in I \text{ with } (x \in C \vee x \in A_i) \\ &\Leftrightarrow \exists i \in I \text{ we have } x \in C \cup A_i \end{aligned}$$

proving that

$$C \cup \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (C \cup A_i)$$

□

Theorem 2.139. Let $\{A_i\}_{i \in I} \subseteq C$ and $\{B_i\}_{i \in I} \subseteq D$ be two families then

1. $(\bigcup_{i \in I} A_i) \cup (\bigcup_{i \in I} B_i) = \bigcup_{i \in I} (A_i \cup B_i)$
2. $\bigcup_{i \in I} (A_i \cap B_i) \subseteq (\bigcup_{i \in I} A_i) \cap (\bigcup_{i \in I} B_i)$

Proof.

1. First as $\forall i \in I$ we have by [theorem: 1.30] that $A_i \subseteq A_i \cup B_i$ and $B_i \subseteq A_i \cup B_i$ so it follows using [theorem: 2.137] that $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} (A_i \cup B_i)$ and $\bigcup_{i \in I} B_i \subseteq \bigcup_{i \in I} (A_i \cup B_i)$. Applying then [theorem: 1.30] gives

$$\left(\bigcup_{i \in I} A_i \right) \cup \left(\bigcup_{i \in I} B_i \right) \subseteq \bigcup_{i \in I} (A_i \cup B_i) \quad (2.45)$$

If now $x \in \bigcup_{i \in I} A_i \cup B_i$ then $\exists i \in I$ such that $x \in A_i \cup B_i$, then we have $x \in A_i \Rightarrow x \in \bigcup_{i \in I} A_i$ or $x \in B_i \Rightarrow x \in \bigcup_{i \in I} B_i$. So $x \in (\bigcup_{i \in I} A_i) \cup (\bigcup_{i \in I} B_i)$ proving that $\bigcup_{i \in I} (A_i \cup B_i) \subseteq (\bigcup_{i \in I} A_i) \cup (\bigcup_{i \in I} B_i)$ which combined with 2.45 gives

$$\left(\bigcup_{i \in I} A_i \right) \cup \left(\bigcup_{i \in I} B_i \right) = \bigcup_{i \in I} (A_i \cup B_i)$$

2. As $\forall i \in I$ we have by [theorem: 1.30] that $A_i \cap B_i \subseteq A_i$ and $A_i \cap B_i \subseteq B_i$, $B_i \subseteq A_i \cup B_i$ it follows using [theorem: 2.137] that $\bigcup_{i \in I} (A_i \cap B_i) \subseteq \bigcup_{i \in I} A_i$ and $\bigcup_{i \in I} (A_i \cap B_i) \subseteq \bigcup_{i \in I} B_i$. Using then [theorem: 1.30] we have

$$\bigcup_{i \in I} (A_i \cap B_i) \subseteq \left(\bigcup_{i \in I} A_i \right) \cap \left(\bigcup_{i \in I} B_i \right)$$

□

We have also a variant of the deMorgan's laws [theorem: 1.32]

Theorem 2.140. (deMorgan's Law) *Let $\{A_i\}_{i \in I} \subseteq B$ be a family then we have*

1. $(\bigcup_{i \in I} A_i)^c = \bigcap_{i \in I} (A_i)^c$
2. $(\bigcap_{i \in I} A_i)^c = \bigcup_{i \in I} (A_i)^c$
3. *If C is a class then $C \setminus (\bigcup_{i \in I} A_i) = \bigcap_{i \in I} (C \setminus A_i)$*
4. *If C is a class then $C \setminus (\bigcap_{i \in I} A_i) = \bigcup_{i \in I} (C \setminus A_i)$*

Proof.

1.

$$\begin{aligned}
 x \in \left(\bigcup_{i \in I} A_i \right)^c &\Leftrightarrow x \notin \left(\bigcup_{i \in I} A_i \right) \\
 &\Leftrightarrow \neg \left(x \in \bigcup_{i \in I} A_i \right) \\
 &\Leftrightarrow \neg (\exists i \in I \text{ with } x \in A_i) \\
 &\Leftrightarrow \forall i \in I \text{ we have } \neg(x \in A_i) \\
 &\Leftrightarrow \forall i \in I \text{ we have } x \notin A_i \\
 &\Leftrightarrow \forall i \in I \text{ we have } x \in (A_i)^c \\
 &\Leftrightarrow x \in \bigcap_{i \in I} (A_i)^c
 \end{aligned}$$

proving that

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} (A_i)^c$$

2.

$$\begin{aligned}
 x \in \left(\bigcap_{i \in I} A_i \right)^c &\Leftrightarrow x \notin \left(\bigcap_{i \in I} A_i \right) \\
 &\Leftrightarrow \neg \left(x \in \left(\bigcap_{i \in I} A_i \right) \right) \\
 &\Leftrightarrow \neg (\forall i \in I \text{ we have } x \in A_i) \\
 &\Leftrightarrow \exists i \in I \text{ we have } \neg(x \in A_i) \\
 &\Leftrightarrow \exists i \in I \text{ we have } x \notin A_i \\
 &\Leftrightarrow \exists i \in I \text{ we have } x \in (A_i)^c \\
 &\Leftrightarrow x \in \bigcup_{i \in I} (A_i)^c
 \end{aligned}$$

proving that

$$\left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} (A_i)^c$$

3. We have

$$\begin{aligned}
 C \setminus \left(\bigcup_{i \in I} A_i \right) &\stackrel{[\text{theorem: 1.28}]}{=} C \cap \left(\bigcup_{i \in I} A_i \right)^c \\
 &\stackrel{(1)}{=} C \cap \left(\bigcap_{i \in I} (A_i)^c \right) \\
 &\stackrel{[\text{theorem: 2.138}]}{=} \bigcap_{i \in I} (C \cap (A_i)^c) \\
 &\stackrel{[\text{theorem: 1.28}]}{=} \bigcap_{i \in I} (C \setminus A_i)
 \end{aligned}$$

4. We have

$$\begin{aligned}
 C \setminus \left(\bigcap_{i \in I} A_i \right) &\stackrel{[\text{theorem: 1.28}]}{=} C \cap \left(\bigcap_{i \in I} A_i \right)^c \\
 &\stackrel{(\overline{2})}{=} C \cap \left(\bigcup_{i \in I} (A_i)^c \right) \\
 &\stackrel{[\text{theorem: 2.138}]}{=} \bigcup_{i \in I} (C \cap (A_i)^c) \\
 &= \bigcup_{i \in I} (C \setminus A_i) \\
 &\quad \square
 \end{aligned}$$

Theorem 2.141. *If $\{A_i\}_{i \in I} \subseteq B$ is a family and A a class then we have*

1. $(\bigcup_{i \in I} A_i) \setminus A = \bigcup_{i \in I} (A_i \setminus A)$
2. $(\bigcap_{i \in I} A_i) \setminus A = \bigcap_{i \in I} (A_i \setminus A)$
3. $(\bigcup_{i \in I} A_i) \times A = \bigcup_{i \in I} (A_i \times A)$
4. $A \times (\bigcup_{i \in I} A_i) = \bigcup_{i \in I} (A \times A_i)$
5. $(\bigcap_{i \in I} A_i) \times A = \bigcap_{i \in I} (A_i \times A)$
6. $A \times (\bigcap_{i \in I} A_i) = \bigcap_{i \in I} (A \times A_i)$

Proof.

1.

$$\begin{aligned}
 \left(\bigcup_{i \in I} A_i \right) \setminus A &\stackrel{[\text{theorem: 1.28}]}{=} \left(\bigcup_{i \in I} A_i \right) \cap A^c \\
 &\stackrel{[\text{theorem: 1.29}]}{=} A^c \cap \left(\bigcup_{i \in I} A_i \right) \\
 &\stackrel{[\text{theorem: 2.138}]}{=} \bigcup_{i \in I} (A^c \cap A_i) \\
 &\stackrel{[\text{theorem: 1.29}]}{=} \bigcup_{i \in I} (A_i \cap A^c) \\
 &\stackrel{[\text{theorem: 1.28}]}{=} \bigcup_{i \in I} (A_i \setminus A)
 \end{aligned}$$

2.

$$\begin{aligned}
 \left(\bigcap_{i \in I} A_i \right) \setminus A &\stackrel{[\text{theorem: 1.28}]}{=} \left(\bigcap_{i \in I} A_i \right) \cap A^c \\
 &\stackrel{[\text{theorem: 1.29}]}{=} A^c \cap \left(\bigcap_{i \in I} A_i \right) \\
 &\stackrel{[\text{theorem: 2.138}]}{=} \bigcap_{i \in I} (A^c \cap A_i) \\
 &\stackrel{[\text{theorem: 1.29}]}{=} \bigcap_{i \in I} (A_i \cap A^c) \\
 &\stackrel{[\text{theorem: 1.28}]}{=} \bigcap_{i \in I} (A_i \setminus A)
 \end{aligned}$$

3.

$$\begin{aligned}
(x, y) \in \left(\bigcup_{i \in I} A_i \right) \times A &\Leftrightarrow x \in \bigcup_{i \in I} A_i \wedge y \in A \\
&\Leftrightarrow y \in A \wedge \exists i \in I \text{ with } x \in A_i \\
&\Leftrightarrow \exists i \in I \text{ with } (x \in A_i \wedge y \in A) \\
&\Leftrightarrow \exists i \in I \text{ with } (x, y) \in A_i \times A \\
&\Leftrightarrow (x, y) \in \bigcup_{i \in I} (A_i \times A)
\end{aligned}$$

4.

$$\begin{aligned}
(x, y) \in A \times \left(\bigcup_{i \in I} A_i \right) &\Leftrightarrow x \in A \wedge y \in \bigcup_{i \in I} A_i \\
&\Leftrightarrow x \in A \wedge \exists i \in I \text{ with } y \in A_i \\
&\Leftrightarrow \exists i \in I \text{ with } (x \in A \wedge y \in A_i) \\
&\Leftrightarrow \exists i \in I \text{ with } (x, y) \in A \times A_i \\
&\Leftrightarrow (x, y) \in \bigcup_{i \in I} (A \times A_i)
\end{aligned}$$

5.

$$\begin{aligned}
(x, y) \in \left(\bigcap_{i \in I} A_i \right) \times A &\Leftrightarrow x \in \bigcap_{i \in I} A_i \wedge y \in A \\
&\Leftrightarrow (\forall i \in I \text{ we have } x \in A_i) \wedge y \in A \\
&\Leftrightarrow \forall i \in I \text{ we have } (x \in A_i \wedge y \in A) \\
&\Leftrightarrow \forall i \in I \text{ we have } (x, y) \in A_i \times A \\
&\Leftrightarrow (x, y) \in \bigcap_{i \in I} (A_i \times A)
\end{aligned}$$

6.

$$\begin{aligned}
(x, y) \in A \times \left(\bigcap_{i \in I} A_i \right) &\Leftrightarrow x \in A \wedge y \in \bigcap_{i \in I} A_i \\
&\Leftrightarrow (\forall i \in I \text{ we have } y \in A_i) \wedge x \in A \\
&\Leftrightarrow \forall i \in I \text{ we have } (y \in A_i \wedge x \in A) \\
&\Leftrightarrow \forall i \in I \text{ we have } (x, y) \in A \times A_i \\
&\Leftrightarrow (x, y) \in \bigcap_{i \in I} (A \times A_i)
\end{aligned}$$

□

Theorem 2.142. Let $\{A_i\}_{i \in I} \subseteq B$ a family then

1. If $j \in I$ then $(\bigcup_{i \in I \setminus \{j\}} A_i) \cup A_j = \bigcup_{i \in I} A_i$
2. $\bigcup_{i \in I} A_i = \bigcup_{i \in \{j \in I \mid A_j \neq \emptyset\}} A_i$
3. If $\exists i \in I$ such that $A_i = \emptyset$ then $\bigcap_{i \in I} A_i = \emptyset$

Proof.

1. If $x \in (\bigcup_{i \in I \setminus \{j\}} A_i) \cup A_j$ then either $x \in A_j \subseteq \bigcup_{i \in I} A_i$ [see: 2.135], so that $x \in \bigcup_{i \in I} A_i$ or $x \in \bigcup_{i \in I \setminus \{j\}} A_i \Rightarrow \exists k \in I \setminus \{j\}$ with $x \in A_k$ which as $k \in I$ proves $x \in \bigcup_{i \in I} A_i$. If $x \in \bigcup_{i \in I} A_i$ then $\exists i \in I$ such that $x \in A_i$, we have then for i either $i \in I \setminus \{j\}$ so that $x \in \bigcup_{i \in I \setminus \{j\}} A_i$ or $i = j$ giving $x \in A_j$, proving that $x \in (\bigcup_{i \in I \setminus \{j\}} A_i) \cup A_j$.

2. As $\{j \in I \mid A_j \neq \emptyset\} \subseteq I$ we have by [theorem: 2.136] that

$$\bigcup_{i \in \{j \in I \mid A_j \neq \emptyset\}} A_i \subseteq \bigcup_{i \in I} A_i \quad (2.46)$$

Further if $x \in \bigcup_{i \in I} A_i$ then there exist a $i \in I$ such that $x \in A_i$. As $x \in A_i$ we must have that $A_i \neq \emptyset$ or $i \in \{j \in I \mid A_j \neq \emptyset\}$, proving that $x \in \bigcup_{i \in \{j \in I \mid A_j \neq \emptyset\}} A_i$. So

$$\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in \{j \in I \mid A_j \neq \emptyset\}} A_i$$

combining this with [eq: 2.46] proves

$$\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in \{j \in I \mid A_j \neq \emptyset\}} A_i$$

3. Assume that $i \in I$ such that $A_i = \emptyset$. If $x \in \bigcap_{j \in I} A_j$ we have $\forall j \in I$ that $x \in A_j$, so for sure $x \in A_i$ which contradicts $A_i = \emptyset$. Hence we have that $\bigcap_{j \in I} A_j = \emptyset$. □

Theorem 2.143. If $\{A_i\}_{i \in I} \subseteq C$ a family and $\forall i \in I \{B_{i,j}\}_{j \in J} \subseteq C$ a family such that $A_i = \bigcup_{j \in J} B_{i,j}$ then

$$\bigcup_{i \in I} A_i = \bigcup_{(i,j) \in I \times J} B_{i,j}$$

in other words

Proof. If $x \in \bigcup_{i \in I} A_i$ then $\exists i \in I$ such that $x \in A_i = \bigcup_{j \in J} B_{i,j}$, hence $\exists j \in J$ such that $x \in B_{i,j}$. So as $(i,j) \in I \times J$ we have that $x \in \bigcup_{(i,j) \in I \times J} B_{i,j}$. Further if $x \in \bigcup_{(i,j) \in I \times J} B_{i,j}$ then $\exists (i,j) \in I \times J$ such that $x \in B_{i,j}$, which, as $A_i = \bigcup_{j \in J} B_{i,j}$, proves that $x \in A_i$, hence $x \in \bigcup_{i \in I} A_i$. So we conclude that

$$\bigcup_{i \in I} A_i = \bigcup_{(i,j) \in I \times J} B_{i,j}$$

□

Theorem 2.144. If $f: A \rightarrow B$ is a function, $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ and $\{B_i\}_{i \in I} \subseteq \mathcal{P}(B)$ are families of sub-classes of A and B then

1. $f(\bigcup_{i \in I} A_i) = \bigcup_{i \in I} f(A_i)$
2. $f^{-1}(\bigcup_{i \in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i)$
3. $f(\bigcap_{i \in I} A_i) \subseteq \bigcap_{i \in I} f(A_i)$
4. If f is injective and $I \neq \emptyset$ then $f(\bigcap_{i \in I} A_i) = \bigcap_{i \in I} f(A_i)$
5. $f^{-1}(\bigcap_{i \in I} B_i) = \bigcap_{i \in I} f^{-1}(B_i)$

Proof.

1. If $y \in f(\bigcup_{i \in I} A_i)$ then $\exists x \in \bigcup_{i \in I} A_i$ such that $(x, y) \in f$, hence $\exists i \in I$ such that $x \in A_i$, which as $(x, y) \in f$ proves that $y \in f(A_i)$. So $y \in \bigcup_{i \in I} f(A_i)$ giving

$$f\left(\bigcup_{i \in I} A_i\right) \subseteq \bigcup_{i \in I} f(A_i) \quad (2.47)$$

If $y \in \bigcup_{i \in I} f(A_i)$ then there exists a $i \in I$ such that $y \in f(A_i)$, hence $\exists x \in A_i$ such that $(x, y) \in f$, as $x \in A_i$ this implies $x \in \bigcup_{i \in I} A_i$, so we have that $y \in f(\bigcup_{i \in I} A_i)$. Hence $\bigcup_{i \in I} f(A_i) \subseteq f(\bigcup_{i \in I} A_i)$, which combined with [eq: 2.47] gives

$$f\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f(A_i)$$

2. If $x \in f^{-1}(\bigcup_{i \in I} B_i)$ then there exists a $y \in \bigcup_{i \in I} B_i$ such that $(x, y) \in f$, hence $\exists i \in I$ such that $y \in B_i$. So $x \in f^{-1}(B_i)$ which as $i \in I$ implies that $x \in \bigcup_{i \in I} f^{-1}(B_i)$ or

$$f^{-1}\left(\bigcup_{i \in I} B_i\right) \subseteq \bigcup_{i \in I} f^{-1}(B_i) \quad (2.48)$$

If $x \in \bigcup_{i \in I} f^{-1}(A_i)$ then there exists a $i \in I$ such that $x \in f^{-1}(A_i)$, so $\exists y \in A_i$ with $(x, y) \in f$. As from $y \in A_i$ we have $y \in \bigcup_{i \in I} A_i$ it follows that $x \in f^{-1}(\bigcup_{i \in I} A_i)$. This proves that $\bigcup_{i \in I} f^{-1}(A_i) \subseteq f^{-1}(\bigcup_{i \in I} A_i)$ which combined with [eq: 2.48] gives

$$f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i)$$

3. If $y \in f(\bigcap_{i \in I} A_i)$ then there exists a $x \in \bigcap_{i \in I} A_i$ such that $(x, y) \in f$. From $x \in \bigcap_{i \in I} A_i$ it follows that $\forall i \in I$ $x \in A_i$, which as $(x, y) \in f$ proves that $\forall i \in I$ $x \in f(A_i)$ or $x \in \bigcap_{i \in I} f(A_i)$. So

$$f\left(\bigcap_{i \in I} A_i\right) \subseteq \bigcap_{i \in I} f(A_i)$$

4. Let $y \in \bigcap_{i \in I} f(A_i)$ then $\forall i \in I$ we have $y \in f(A_i)$. As $I \neq \emptyset$ there exists a $i \in I$ and we must thus have that $y \in f(A_i)$. So there exists a $x \in A_i$ such that $(x, y) \in f$. Assume that $x \notin \bigcap_{i \in I} A_i$ then $\exists j \in I$ such that $x \notin A_j$. However as $j \in I$ we must have that $y \in f(A_j)$, so there exists a $x' \in A_j$ such that $(x', y) \in f$. As f is injective and $(x, y), (x', y) \in f$ we must have $x = x'$, but this means that $x \in A_j$ contradicting $x \notin A_j$. So the assumption that $x \notin \bigcap_{i \in I} A_i$ is wrong, hence $x \in \bigcap_{i \in I} A_i$. As $(x, y) \in f$ we have $y \in f(\bigcap_{i \in I} A_i)$, proving that $\bigcap_{i \in I} f(A_i) \subseteq f(\bigcap_{i \in I} A_i)$, which combined with (3) proves

$$f\left(\bigcap_{i \in I} A_i\right) = \bigcap_{i \in I} f(A_i)$$

5. If $x \in f^{-1}(\bigcap_{i \in I} B_i)$ then there exists a $y \in \bigcap_{i \in I} B_i$ such that $(x, y) \in f$. Hence $\forall i \in I$ we have that $y \in B_i \Rightarrow_{(x, y) \in f} x \in f^{-1}(B_i)$ proving that $x \in \bigcap_{i \in I} f^{-1}(B_i)$. So

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) \subseteq \bigcap_{i \in I} f^{-1}(B_i) \quad (2.49)$$

If $x \in \bigcap_{i \in I} f^{-1}(B_i)$ then $\forall i \in I$ we have $x \in f^{-1}(B_i)$ or $\exists y \in B_i$ with $(x, y) \in f$. So $y \in \bigcap_{i \in I} B_i$ which as $(x, y) \in f$ proves that $x \in f^{-1}(\bigcap_{i \in I} B_i)$. So $\bigcap_{i \in I} f^{-1}(B_i) \subseteq f^{-1}(\bigcap_{i \in I} B_i)$ which combined with 2.49 gives

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) = \bigcap_{i \in I} f^{-1}(B_i) \quad \square$$

2.4 Product of a family of sets

The Cartesian product $A \times B$ consists of all the possible pairs that you can form, where the first element is a element of A and the second element is a element of B . We want now to construct a generalized product of a family of classes consisting of tuples whose elements are indexed by the index of the family.

Definition 2.145. (Product of a family of sets) Let $\{A_i\}_{i \in I} \subseteq B$ a family then the **product** of $\{A_i\}_{i \in I}$ noted as $\prod_{i \in I} A_i$ is defined by

$$\prod_{i \in I} A_i = \left\{ f: f \in \left(\bigcup_{i \in I} A_i \right)^I \text{ where } \forall i \in I \text{ we have } f(i) \in A_i \right\}$$

If $x \in \prod_{i \in I} A_i$ then x_i is defined as

$$x_i = x(i)$$

Here $(\bigcup_{i \in I} A_i)^I$ is the class of function graphs of functions between I and $\bigcup_{i \in I} A_i$ [definition: 2.32] and $f(i)$ is the unique y such that $(i, y) \in f$. So $\prod_{i \in I} A_i$ is the class of graphs of functions from I to $\bigcup_{i \in I} A_i$ such that $\forall i \in I f_i = f(i) \in A_i$.

Theorem 2.146. Let $\{A_i\}_{i \in I} \subseteq B$ be such that $\exists i_0 \in I$ with $A_{i_0} = \emptyset$ then $\prod_{i \in I} A_i = \emptyset$

Proof. Let $x \in \prod_{i \in I} A_i$ then we have that $x: I \rightarrow \bigcup_{i \in I} A_i$ is a function such that $x(i_0) \in A_{i_0}$, contradicting $A_{i_0} = \emptyset$. Hence we must have that $\prod_{i \in I} A_i = \emptyset$. $\square$

The following shows that the product of a family of only one class is ‘almost’ that class itself.

Example 2.147. Let $\{A_i\}_{i \in \{0\}} \subseteq \{B\}$ be the family in [example: 2.133] defined by $A: \{0\} \rightarrow \{B\}$ where $A = \{(0, B)\}$ then there exists a bijection between B and $\prod_{i \in \{0\}} A_i$ or as $A_0 = B$ there exists a bijection between A_0 and $\prod_{i \in \{0\}} A_i$.

Proof. First using [example: 2.133] we have

$$B = \bigcup_{i \in \{0\}} A_i \quad (2.50)$$

hence

$$\left(\bigcup_{i \in \{0\}} A_i \right)^{\{0\}} = B^{\{0\}} \quad (2.51)$$

Let $f \in B^{\{0\}} \stackrel{[\text{eq: 2.51}]}{=} (\bigcup_{i \in \{0\}} A_i)$ then if $i \in \{0\}$ we must have $i = 1$ hence $f(i) = f(0) \in B = A(0) = A_0$ proving that $\forall i \in \{0\}$ we have $f(i) \in A_i$. Hence $f \in \prod_{i \in \{0\}} A_i$ from which it follows that $B^{\{0\}} \subseteq \prod_{i \in \{0\}} A_i$. As clearly $\prod_{i \in \{0\}} A_i \subseteq (\bigcup_{i \in \{0\}} A_i)^{\{0\}} \stackrel{[\text{eq: 2.51}]}{=} B^{\{0\}}$ we have that

$$\prod_{i \in \{0\}} A_i = B^{\{0\}}$$

Now by [theorem: 2.79] there exists a bijection between B and $B^{\{0\}}$ which by the above proves the example. $\square$

The next theorem shows that the product of a family of two classes is ‘almost’ the Cartesian product of these classes.

Theorem 2.148. Let $\{A_i\}_{i \in \{0,1\}} \subseteq \{C, D\}$ be the family in [example: 2.134] defined by $A: \{0, 1\} \rightarrow \{C, D\}$ where $A = \{(0, C), (1, D)\}$ then there exists a bijection between $A \times B$ and $\prod_{i \in \{0,1\}} A_i$

Proof. First using [example: 2.134]: we have that

$$\bigcup_{i \in \{0,1\}} A_i = C \cup D \quad (2.52)$$

so that

$$\left(\bigcup_{i \in \{0,1\}} A_i \right)^{\{0,1\}} = (C \cup D)^{\{0,1\}} \quad (2.53)$$

So

$$\prod_{i \in \{0,1\}} A_i = \{f \mid f \in (C \cup D)^{\{0,1\}} \text{ where } f(0) \in C \wedge f(1) \in D\} \quad (2.54)$$

Given $(c, d) \in C \times D \Rightarrow c \in C \wedge d \in D$, define $f_{c,d} = \{(0, c), (1, d)\}$. If $(x, y) \in f_{c,d}$ we have either

$$(x, y) = (0, c) \Rightarrow x = 0 \in \{0, 1\} \wedge y = c \in C \subseteq C \cup D \Rightarrow (x, y) \in \{0, 1\} \times (C \cup D)$$

or

$$(x, y) = (1, d) \Rightarrow x = 1 \in \{0, 1\} \wedge y = d \in D \subseteq C \cup D \Rightarrow (x, y) \in \{0, 1\} \times (C \cup D)$$

proving that

$$f_{a,b} \subseteq \{0,1\} \times (C \cup D) \wedge f_{a,b}(0) \in C \wedge f_{a,b}(1) \in D \quad (2.55)$$

If $(x, y), (x, y') \in f_{c,d}$ then either

$$(x, y) = (0, c) \Rightarrow x = 0 \Rightarrow (0, y') \in f_{c,d} \Rightarrow (0, y') = (0, c) \Rightarrow y' = c \Rightarrow y = y'$$

or

$$(x, y) = (1, d) \Rightarrow x = 1 \Rightarrow (1, y') \in f_{c,d} \Rightarrow (1, y') = (1, d) \Rightarrow y' = d \Rightarrow y = y'.$$

Together with [eq: 2.55] this proves that

$$f_{a,b}: \{0,1\} \rightarrow C \cup D \text{ is a partial function} \quad (2.56)$$

If $x \in \{0,1\}$ then either $x = 0 \Rightarrow (0, c) \in f_{c,d}$ or $x = 1 \Rightarrow (1, d) \in f_{c,d}$ proving that $\{0,1\} \subseteq \text{dom}(f_{c,d})$ which by [theorem: 2.27] proves that

$$f_{c,d}: \{0,1\} \rightarrow C \cup D \text{ is a function} \quad (2.57)$$

As by [eq: 2.55] $f_{c,d}(0) \in C \wedge f_{c,d}(1) \in D$ proving that

$$f_{c,d} \in \prod_{i \in \{0,1\}} A_i \quad (2.58)$$

Define now γ by $\gamma = \{((c, d), f_{c,d}) \mid (c, d) \in C \times D\}$. If $(x, y) \in \gamma$ then $x = (c, d) \in C \times D$ and $y = f_{c,d} \xrightarrow{[\text{eq: 2.58}]}$, hence $y \in (C \cup D)^{\{0,1\}}$. This proves that $(x, y) \in (C \times D) \times (\prod_{i \in \{0,1\}} A_i)$ or

$$\gamma \subseteq (C \times D) \times \left(\prod_{i \in \{0,1\}} A_i \right) \quad (2.59)$$

If $(x, y), (x, y') \in \gamma$ then $\exists (c, d) \in C \times D$ such that $(x, y) = ((c, d), f_{c,d})$ and $(x, y') = ((c, d), f_{c,d})$ so that $y = f_{c,d} = y'$ hence $y = y'$. Combining this with [eq: 2.59] proves that

$$\gamma: C \times D \rightarrow \left(\prod_{i \in \{0,1\}} A_i \right) \text{ is a partial function} \quad (2.60)$$

If $(c, d) \in C \times D$ then by definition of γ we have $((c, d), f_{c,d}) \in \gamma$ so that $(c, d) \in \text{dom}(\gamma)$ proving that $C \times D \subseteq \text{dom}(\gamma)$. By [theorem: 2.27] and [eq: 2.60] we have

$$\gamma: C \times D \rightarrow \left(\prod_{i \in \{0,1\}} A_i \right) \text{ is a function} \quad (2.61)$$

If $(x, y), (x', y) \in \gamma$ then there exists $(c, d), (c', d') \in C \times D$ such that $x = (c, d) \wedge x' = (c', d')$ and $f_{c,d} = y = f_{c',d'}$. As $(0, c) \in f_{c,d} = f_{c',d'}$ we have $(0, c) = (0, c')$ giving $c = c'$ and from $(1, d) \in f_{c,d} = f_{c',d'}$ we have $(1, d) = (1, d')$ giving $d = d'$. So $(c, d) = (c', d')$ proving that

$$\gamma: C \times D \rightarrow \left(\prod_{i \in \{0,1\}} A_i \right) \text{ is an injection}$$

If $g \in \prod_{i \in \{0,1\}} A_i$ then $g: \{0,1\} \rightarrow C \cup D$ is a function and $g(0) \in C \wedge g(1) \in D$. So there exists a $c \in C$ such that $(0, c) \in g$ and there exists a $d \in D$ such that $(1, d) \in g$. So $g = \{(0, c), (1, d)\} = f_{c,d}$ which proves that

$$\gamma: C \times D \rightarrow \left(\prod_{i \in \{0,1\}} A_i \right) \text{ is a surjection} \quad \square$$

Theorem 2.149. Let $\{A_i\}_{i \in I} \subseteq A$ and $\{B_i\}_{i \in I} \subseteq B$ classes such that $\forall i \in I \ A_i \subseteq B_i$ then

$$\prod_{i \in I} A_i \subseteq \prod_{i \in I} B_i$$

Proof. Let $x \in \prod_{i \in I} A_i$ then $x \in (\bigcup_{i \in I} A_i)^I$ and $\forall i \in I \ x(i) \in A_i$. Using [theorem: 2.137] it follows that $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} B_i$, applying [theorem: 2.36] proves then $(\bigcup_{i \in I} A_i)^I \subseteq (\bigcup_{i \in I} B_i)^I$, so that

$$x \in \left(\bigcup_{i \in I} B_i \right)^I$$

If $i \in I$ then $x(i) \in A_i$, which as $A_i \subseteq B_i$ gives $x(i) \in B_i$, combining this with the above proves that $x \in \prod_{i \in I} B_i$. Hence we have

$$\prod_{i \in I} A_i \subseteq \prod_{i \in I} B_i \quad \square$$

Theorem 2.150. Let $\{A_i\}_{i \in I} \subseteq C$ and $\{B_i\}_{i \in I} \subseteq D$ are two families then

$$\left(\prod_{i \in I} A_i \right) \cap \left(\prod_{i \in I} B_i \right) = \prod_{i \in I} (A_i \cap B_i)$$

Proof. First, as $\forall i \in I$ we have by [theorem: 1.30] $A_i \cap B_i \subseteq A_i$ and $A_i \cap B_i \subseteq B_i$ it follows that by [theorem: 2.149]

$$\prod_{i \in I} (A_i \cap B_i) \subseteq \prod_{i \in I} A_i \text{ and } \prod_{i \in I} (A_i \cap B_i) \subseteq \prod_{i \in I} B_i$$

so that by [theorem: 1.30]

$$\prod_{i \in I} (A_i \cap B_i) \subseteq \left(\prod_{i \in I} A_i \right) \cap \left(\prod_{i \in I} B_i \right) \quad (2.62)$$

Now for the opposite inclusion Let $x \in \left(\prod_{i \in I} A_i \right) \cap \left(\prod_{i \in I} B_i \right)$ then $x \in \prod_{i \in I} A_i$ and $x \in \prod_{i \in I} B_i$. So $x \in \left(\bigcup_{i \in I} A_i \right)^I \wedge \forall i \in I \models x(i) \in A_i$ and $x \in \left(\bigcup_{i \in I} B_i \right)^I \wedge \forall i \in I \models x(i) \in B_i$. Hence

$$\begin{aligned} x: I &\rightarrow \bigcup_{i \in I} A_i \text{ is a function} \\ x: I &\rightarrow \bigcup_{i \in I} B_i \text{ is a function} \\ \forall i \in I &\quad \text{we have } x(i) \in A_i \cap B_i \end{aligned}$$

Now if $(i, y) \in x$ we have $i \in I$ [as $x \subseteq I \times (\bigcup_{i \in I} A_i)$] and $y = x(i) \in A_i \cap B_i \subseteq \bigcup_{i \in I} (A_i \cap B_i)$ so that $(i, y) \in I \times (\bigcup_{i \in I} (A_i \cap B_i))$ giving

$$x \subseteq I \times \left(\bigcup_{i \in I} (A_i \cap B_i) \right) \text{ and } \forall i \in I \text{ we have } x(i) \in A_i \cap B_i \quad (2.63)$$

Further as $x: I \rightarrow \bigcup_{i \in I} A_i$ is a function we have $\forall (i, y), (i, y')$ that $y = y'$ and that $\text{dom}(x) = I$. Combining this with [eq: 2.63] proves that $f: I \rightarrow \bigcup_{i \in I} (A_i \times B_i)$ is a function and $\forall i \in I$ we have $x(i) \in A_i \cap B_i$. This proves that $x \in \prod_{i \in I} (A_i \cap B_i)$ giving $\left(\prod_{i \in I} A_i \right) \cap \left(\prod_{i \in I} B_i \right) \subseteq \prod_{i \in I} (A_i \cap B_i)$ which combined with 2.62 gives finally

$$\prod_{i \in I} (A_i \cap B_i) \subseteq \left(\prod_{i \in I} A_i \right) \cap \left(\prod_{i \in I} B_i \right) \quad \square$$

We can easily generalize the above theorem.

Theorem 2.151. Let I, J be sets and $\{\{A_{i,j}\}_{j \in J}\}_{i \in I}$ be a family of sets then

$$\bigcap_{i \in I} \left(\prod_{j \in J} A_{i,j} \right) = \prod_{j \in J} \left(\bigcap_{i \in I} A_{i,j} \right)$$

Proof. As $\forall i \in I, j \in I$ we have $\bigcap_{i \in I} A_{i,j} \subseteq A_{i,j}$ it follows from [theorem: 2.149] that $\prod_{j \in J} (\bigcap_{i \in I} A_{i,j}) \subseteq \prod_{j \in J} A_{i,j}$. Hence by [theorem: 2.135]

$$\prod_{j \in J} \left(\bigcap_{i \in I} A_{i,j} \right) \subseteq \bigcap_{i \in I} \left(\prod_{j \in J} A_{i,j} \right) \quad (2.64)$$

If $x \in \bigcap_{i \in I} (\prod_{j \in J} A_{i,j})$ then $\forall i \in I, x \in \prod_{j \in J} A_{i,j}$ hence $x: J \rightarrow A_{i,j}$ is a function such that $\forall j \in J$ we have $x(j) \in A_{i,j}$. Hence $x(j) \in \bigcap_{i \in I} A_{i,j} \forall j \in J$ so that $x \in \prod_{j \in J} (\bigcap_{i \in I} A_{i,j})$, so

$$\bigcap_{i \in I} \left(\prod_{j \in J} A_{i,j} \right) \subseteq \prod_{j \in J} \left(\bigcap_{i \in I} A_{i,j} \right)$$

which combined with [eq: 2.64] gives

$$\bigcap_{i \in I} \left(\prod_{j \in J} A_{i,j} \right) = \prod_{j \in J} \left(\bigcap_{i \in I} A_{i,j} \right)$$

□

The following theorem is a motivation for the notation A^B for the graphs of functions from B to A .

Theorem 2.152. Let I, B be classes and consider the family $\{A_i\}_{i \in I} \subseteq \{B\}$ based on the constant function $A: I \rightarrow \{B\}$ where $A = C_B = I \times \{B\}$ so that $\forall i \in I, A(i) = B$ [see example: 2.47] then $\prod_{i \in I} A_i = A^I$

Proof. For I we have the following cases to consider:

$I = \emptyset$. Using [example: 2.34] we have that

$$\left(\bigcup_{i \in \emptyset} A_i \right)^\emptyset = \{\emptyset\}$$

Further $\forall i \in \emptyset$ we have $\emptyset(i) \in A_i$ is satisfied vacuously proving that $\emptyset \in \prod_{i \in \emptyset} A_i$ so that $\{\emptyset\} \subseteq \prod_{i \in \emptyset} A_i \subseteq (\bigcup_{i \in \emptyset} A_i)^\emptyset = \{\emptyset\}$ or taking $I = \emptyset$

$$\prod_{i \in I} A_i = A^I$$

$I \neq \emptyset$. If $y \in \text{range}(A)$ then $\exists x$ such that $(x, y) \in C_B = I \times \{B\}$, so that $y \in \{B\}$. Hence

$$\text{range}(A) \subseteq \{B\} \quad (2.65)$$

As $I \neq \emptyset$ there exists a $i \in I$, which by the definition of C_B means that $(i, B) \in C_B$, hence $B \in \text{range}(A)$. So if $y \in \{B\}$ then $y = B \in \text{range}(A)$ proving that $\{B\} \subseteq \text{range}(A)$ which combined with [eq: 2.65] gives

$$\text{range}(A) = \{B\}$$

hence

$$\bigcup_{i \in I} A_i = \bigcup (\text{range}(A)) = \bigcup \{B\} \stackrel{[\text{example: 1.64}]}{=} B$$

so that

$$\left(\bigcup_{i \in I} A_i \right)^I = B^I \quad (2.66)$$

Now if $f \in B^I$ then $\forall i \in I$ we have $f(i) \in B = A(i) = A_i$ proving that

$$f \in \{f \mid f \in B^I \wedge \forall i \in I, f(i) \in A_i\} \stackrel{[\text{eq: 2.66}]}{=} \left\{ f \mid f \in \left(\prod_{i \in I} A_i \right)^I \wedge \forall i \in I, f(i) \in A_i \right\} = \prod_{i \in I} A_i$$

proving that

$$B^I \subseteq \prod_{i \in I} A_i \quad (2.67)$$

Further

$$\prod_{i \in I} A_i = \left\{ f \mid f \in \left(\prod_{i \in I} A_i \right)^I \wedge \forall i \in I f(i) \in A_i \right\} \subseteq \left\{ f \mid f \in \left(\prod_{i \in I} A_i \right)^I \right\} = \{ f \mid f \in B^I \} = B^I$$

which combined with [eq: 2.67] proves that

$$B^I = \prod_{i \in I} A_i$$

□

Theorem 2.153. Let I, J, B be classes, $f: I \rightarrow J$ a bijection and $\{A_j\}_{j \in J}$ then

$$\beta: \prod_{j \in J} A_j \rightarrow \prod_{i \in I} A_{f(i)} \text{ where } \beta(x) = x \circ f$$

is a bijection.

Proof. First as $f: I \rightarrow J$ is a bijection, hence surjective, we have by [theorem: 2.127] that

$$\bigcup_{j \in J} A_j = \bigcup_{i \in I} A_{f(i)} \quad (2.68)$$

Let $x \in \prod_{j \in J} A_j$ then $x \in (\bigcup_{j \in J} A_j)^J$, which is equivalent with $x: J \rightarrow \bigcup_{j \in J} A_j$ is a function, and $\forall j \in J$ we have $x(j) \in A_j$. So $x \circ f: I \rightarrow \bigcup_{j \in J} A_j \stackrel{[\text{eq: 2.68}]}{=} \bigcup_{i \in I} A_{f(i)}$ is a function, proving that $x \circ f \in (\bigcup_{i \in I} A_{f(i)})^I$, further if $i \in I$ then $(x \circ f)(i) = x(f(i)) \in A_{f(i)}$, hence

$$x \circ f \in \prod_{i \in I} A_{f(i)} \quad (2.69)$$

So

$$\beta: \prod_{j \in J} A_j \rightarrow \prod_{i \in I} A_{f(i)}$$

is indeed a function. To prove that it is a bijection note:

injectivity. Assume that $\beta(x) = \beta(y)$ then

$$\begin{aligned} x \circ f = y \circ f & \quad f \text{ is bijective} \Rightarrow (x \circ f) \circ f^{-1} = (y \circ f) \circ f^{-1} \\ & \Rightarrow x \circ (f \circ f^{-1}) = y \circ (f \circ f^{-1}) \\ & \Rightarrow x \circ \text{Id}_J = y \circ \text{Id}_J \\ & \Rightarrow x = y \end{aligned}$$

surjectivity. If $y \in \prod_{i \in I} A_{f(i)}$ then $y: I \rightarrow \bigcup_{i \in I} A_{f(i)} \stackrel{[\text{eq: 2.68}]}{=} \bigcup_{j \in J} A_j$ is a function and $\forall i \in I$ we have $y(i) \in A_{f(i)}$. As $f^{-1}: J \rightarrow I$ is a bijection we have that $y \circ f^{-1}: J \rightarrow \bigcup_{j \in J} A_j$ is a function, so that $y \circ f^{-1} \in (\bigcup_{j \in J} A_j)^J$, and $(y \circ f^{-1})(j) = y(f^{-1}(j)) \in A_{f(f^{-1}(j))} = A_j$. So that

$$y \circ f^{-1} \in \prod_{j \in J} A_j$$

Finally $\beta(y \circ f^{-1}) = (y \circ f^{-1}) \circ f = y \circ (f^{-1} \circ f) = y \circ \text{Id}_I = y$ proving surjectivity. □

Definition 2.154. Let $\{A_i\}_{i \in I} \subseteq B$ be a family and $J \subseteq I$ then $\prod_{i \in J} A_i$ is the product based on the sub-family $\{A_i\}_{i \in J} \subseteq B$ [see definition: 2.116] or equivalently

$$\prod_{i \in J} A_i = \left\{ f: f \in \left(\bigcup_{i \in J} A_i \right)^J \text{ where } \forall i \in J \text{ we have } f(i) \in A_i \right\}$$

The following theorem will be used later in induction arguments.

Theorem 2.155. Let $\{A_i\}_{i \in I} \subseteq B$, $i \in I$ and $b \in A_i$ then

$$\text{if } x \in \prod_{j \in I \setminus \{i\}} A_j \text{ we have } y \in \prod_{i \in I} A_j$$

where y is defined by

$$y_j = y(j) = \begin{cases} b & \text{if } j = i \\ x_j & \text{if } j \in I \setminus \{i\} \end{cases} \stackrel{\text{def}}{=} \begin{cases} b & \text{if } j = i \\ x(j) & \text{if } j \in I \setminus \{i\} \end{cases}$$

Proof. If $x \in \prod_{j \in I \setminus \{i\}} A_j$ then $x \in (\bigcup_{j \in I \setminus \{i\}} A_i)^{I \setminus \{i\}}$ so that $x: I \setminus \{i\} \rightarrow \bigcup_{j \in I \setminus \{i\}} A_j$ is a function. As $i \notin (I \setminus \{i\})$, $I = (I \setminus \{i\}) \cup \{i\}$ and $\bigcup_{j \in I} A_j \stackrel{[\text{theorem: 2.142}]}{=} A_i \cup (\bigcup_{j \in I \setminus \{i\}} A_j)$ we have by [theorem: 2.88] that

$$y: I \rightarrow \bigcup_{i \in I} A_i \text{ where } y(j) = \begin{cases} b & \text{if } j = i \\ x(j) & \text{if } j \in I \setminus \{i\} \end{cases}$$

is a function, so

$$y \in \left(\bigcup_{i \in I} A_i \right)^I \quad (2.70)$$

Further if $j \in I$ then either $j = i$ so that $y_j = y(i) = b \in A_i = A_j$ or $j \in I \setminus \{i\}$ then $y_j = y(j) = x(j) = x_j \in A_j$. Hence

$$\forall j \in I \text{ we have } y_j \in A_j \quad (2.71)$$

From [eq: 2.70] and [eq: 2.71] it follows by

$$y \in \prod_{i \in I} A_i$$

□

We introduce now the projection operator

Definition 2.156. Let $\{A_i\}_{i \in I} \subseteq B$ be family then for $i \in I$ we define the projection function

$$\pi_i: \prod_{j \in I} A_j \rightarrow A_i$$

where

$$\pi_i = \left\{ z \mid z = (x, x(i)) \mid x \in \prod_{j \in I} A_j \right\}$$

In other words $(x, y) \in \pi_i \Leftrightarrow x \in \prod_{j \in I} A_j$ and $y = x(i) \Leftrightarrow (i, y) \in x$

Proof. This definition only make sense if $\forall i \in I$ that $\pi_i: \prod_{j \in I} A_j \rightarrow A_i$ is a function. First if $(x, y) \in \pi_i$ we have that $x \in \prod_{j \in I} A_j$ and $y = x(i)$ giving $y \in A_i$, so $(x, y) \in (\prod_{i \in I} A_i) \times A_i$. Hence

$$\pi_i \subseteq \left(\prod_{i \in I} A_i \right) \times A_i \quad (2.72)$$

If $(x, y), (x, y') \in \pi_i$ then $y = x(i) \wedge y' = x(i)$ proving that $y = y'$ or

$$\pi_i: \prod_{j \in I} A_j \rightarrow A_i \text{ is a partial function}$$

If $x \in \prod_{j \in I} A_j$ then by definition $(x, x(i)) \in \pi_i$ proving that $x \in \text{dom}(\pi_i)$ proving that $\prod_{j \in I} A_i \subseteq \text{dom}(\pi_i)$, which by [theorem: 2.27] gives

$$\pi_i: \prod_{j \in I} A_j \rightarrow A_i \text{ is a function} \quad \square$$

We are not yet finished with the product of a family of classes, however for some of the theorems we need the Axiom of Choice. For example to prove that the projection function is a surjection we need the Axiom of Choice.

Chapter 3

Relations

3.1 Relation

The idea of a relation is that we can specify which elements of a class are related to each other. You do this by specifying a class of pairs.

Definition 3.1. Let A be a class then a relation in A is a sub-class of $A \times A$

Notation 3.2. So a relation is a set of pairs from elements of the same class, to avoid confusion with the graph of a function we use the following notation:

If $R \subseteq A \times A$ is relation then instead of writing $(x, y) \in R$ we write xRy

Example 3.3. Let A be a class then $A \times A$ is a relation [as $A \times A \subseteq A \times A$]

We define now the following properties that a relation can have

Definition 3.4. If A is a class and $R \subseteq A \times A$ a relation then we say that R is

reflexive. iff $\forall x \in A$ we have

$$xRx$$

in other words every element is related to itself.

symmetric. iff

$$xRy \Rightarrow yRx$$

in other words if one element is related to a second element then the second element is related to the first element.

anti symmetric. iff

$$xRy \wedge yRx \Rightarrow x = y$$

in other words if one element is related to a second element and the second element is related to the first element then the two elements are the same.

transitive. iff

$$xRy \wedge yRz \Rightarrow xRz$$

in other words if one element is related to a second element and the second element is related to the third element then the first element is also related to the third element.

3.2 Equivalence relations

3.2.1 Equivalence relations and equivalence classes

Note that for classes and equality we have by [theorem: 1.9] that

- $A = A$
- $A = B \Rightarrow B = A$
- $A = B \wedge B = C \Rightarrow A = C$

If we want to create a relation that defines a kind of equality then it must behave in the same way as the equality for classes. This is the idea behind the following definition.

Definition 3.5. (Equivalence Relation) *If A is a class then a relation R is a **equivalence relation** iff it is reflexive, symmetric and transitive or in other words if*

reflexivity. $\forall x \in A \ x R x$

symetricity. $x R y \Rightarrow y R x$

transitivity. $x R y \wedge y R z \Rightarrow x R z$

Given a set A and a equivalence relation in A then it is useful to partition the set in subsets containing all the elements that are equivalent with each other. To do this we must first define what a partition of a set is.

Definition 3.6. *Let A be a set then a **partition** of A is a family $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ of non empty subsets of A $[\forall i \in I \text{ we have } A_i \neq \emptyset]$ such that:*

1. $\bigcup_{i \in I} A_i = A$
2. $\forall i, j \in I \text{ we have } A_i \cap A_j = \emptyset \vee A_i = A_j$

Note 3.7. Condition (2) in the above definition is a weaker condition than pairwise disjointness. For example if we define the family $(A_i)_{i \in \{1,2,3\}}$ by $A_1 = \{1\}$, $A_2 = \{1\}$ and $A_3 = \{2\}$ then this family is not pairwise disjoint as $1 \neq 2$ and $A_1 \cap A_2 \neq \emptyset$, however (2) is clearly satisfied.

We can also reformulate the definition of a partition of A in the following way

Theorem 3.8. *Let A be a set and $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ a family of non empty subsets of A then we have the following equivalences*

1. $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ is a partition of A
2. $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ satisfies
 - a. $\forall x \in A$ there exists a $i \in I$ such that $x \in A_i$
 - b. $\forall i, j \in I$ with $A_i \cap A_j \neq \emptyset$ we have $A_i = A_j$
3. $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ satisfies
 - a. $\forall x \in A$ there exists a $i \in I$ such that $x \in A_i$
 - b. $\forall i, j \in I$ with $A_i \neq A_j$ we have $A_i \cap A_j = \emptyset$

Proof.

1 $\Rightarrow$ 2.

- a) If $x \in A$ then as $A = \bigcup_{i \in I} A_i$ there exists a $i \in I$ such that $x \in A_i$
- b) Let $i, j \in I$ with $A_i \cap A_j \neq \emptyset$. As by definition of a partition $A_i \cap A_j = \emptyset \vee A_i = A_j$ we must have that $A_i = A_j$.

2 $\Rightarrow$ 3.

- a) This is trivial
- b) Let $i, j \in I$ with $A_i \neq A_j$. Assume that $A_i \cap A_j \neq \emptyset$ then by (2.b) we have $A_i = A_j$ contradicting $A_i \neq A_j$, so we must have that $A_i \cap A_j = \emptyset$

3 $\Rightarrow$ 1.

- a) Using (3.a) it follows that $A \subseteq \bigcup_{i \in I} A_i$. If $z \in \bigcup_{i \in I} A_i$ then there exists a $i \in I$ such that $z \in A_i$ [theorem: 2.126], hence as $A_i \in \mathcal{P}(A) \Rightarrow A_i \subseteq A$ it follows that $z \in A$, proving that $\bigcup_{i \in I} A_i \subseteq A$. So we have that

$$\bigcup_{i \in I} A_i = A$$

- b) Let $i, j \in I$ then if $A_i \neq A_j$ we have by (3b) that $A_i \cap A_j = \emptyset$, so we have that $A_i = A_j \vee A_i \cap A_j = \emptyset$. $\square$

We show now how a equivalence relation can be used to partition a set.

Definition 3.9. Let A be a set and R a equivalence relation in A then given x we define the *equivalence class* of x noted by $R[x]$ by

$$R[x] = \{y \in A \mid x R y\} \subseteq A$$

Note 3.10. Because $R[x] \subseteq A$ and A is a set we have by the axiom of subset 1.8 that $R[x]$ is a set.

We have the following important property for equivalence classes

Theorem 3.11. Let A be a set with a equivalence relation R in A then

1. $\forall x \in A$ we have $x \in R[x]$
2. $\forall x, y \in A$ we have

$$x R y \Leftrightarrow R[x] = R[y]$$

3. $\forall x \in A$ we have

$$y \in R[x] \Leftrightarrow R[x] = R[y]$$

Proof.

1. If $x \in A$ then using reflexivity we have $x R x$ so that $x \in R[x]$
- 2.

$\Rightarrow$. Let $z \in R[x]$ then $x R z$, further from $x R y$ we have $y R x$, so using transitivity it follows that $y R z$ or $z \in R[y]$. If $z \in R[y]$ then $y R z$ so as $x R y$ we have by transitivity that $x R z$ or that $z \in R$.

$\Leftarrow$. Using (1) $x \in R[x] \xRightarrow{R[x]=R[y]} x \in R[y]$ proving that $x R y$

- 3.

$\Rightarrow$. If $y \in R[x]$ then $y R x$ hence by (2) $R[x] = R[y]$

$\Leftarrow$. If $R[x] = R[y]$ then $y R x$ proving that $y \in R[x]$ $\square$

We define now a function that maps a element of as set on its equivalence class and use it to define a family of equivalence classes indexed by the elements of the set.

Definition 3.12. Let A be a set and R a equivalence relation in A then $\{R[x]\}_{x \in A} \subseteq \mathcal{P}(X)$ is the family defined by the function $R[]: A \rightarrow \mathcal{P}(A)$ where $R[] (x) = R[x]$

Note 3.13. As $x \in R[x]$ we have that $\{R[x]\}_{x \in A}$ is a non empty family of subsets of A

Proof. We must of course prove that this a function. First $R[x]$ is defined for every $x \in A$ and calculates a unique set, further $R[x] \subseteq A \Rightarrow R[x] \in \mathcal{P}(A)$. So by [proposition: 2.104] $R[]: A \rightarrow \mathcal{P}[A]$ is a function. $\square$

Theorem 3.14. Let A be a set and R a equivalence relation in A then $\{R[x]\}_{x \in A}$ is a partition of A

Proof. We use [theorem: 3.8] to prove this

1. If $x \in A$ then by [theorem: 3.11] we have that $x \in R[x]$ so that $x \in \bigcup_{x \in A} R[x]$
2. Let $x, y \in A$ such that $R[x] \cap R[y] \neq \emptyset$ then there exists a

$$z \in R[x] \cap R[y] \Rightarrow z R x \wedge z R y \xRightarrow{\text{symmetry}} x R z \wedge z R y \xRightarrow{\text{transitivity}} x R y$$

Using the above together with [theorem: 3.11] we have then that $R[x] = R[y]$

So by [theorem: 3.8] it follows that $\{R[x]\}_{x \in A} \subseteq \mathcal{P}(A)$ is a partition of A $\square$

We have also the opposite of the above theorem in that a partition defines a equivalence relation that generates the same partition.

Theorem 3.15. *Let A be a set and $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$ a partition of A . Define $R \subseteq A \times A$ by*

$$R = \{(x, y) | \exists i \in I \text{ such that } x \in A_i \wedge y \in A_i\}$$

then we have:

1. R is a equivalence relation
2. $\forall i \in I$ there exists a $x \in A$ such that $R[x] = A_i$
3. $\forall x \in A$ there exists a $i \in I$ such that $R[x] = A_i$

*we call R is the called the **equivalence relation associated with the partition** $\{A_i\}_{i \in I} \subseteq \mathcal{P}(A)$*

Proof.

1. We have:

- a. If $x \in A = \bigcup_{i \in I} A_i$ then $\exists i \in I$ such that $x \in A_i$ so that $(x, x) \in R$ or xRx
- b. If xRy or $(x, y) \in R$ then $\exists i \in I$ such that $x \in A_i \wedge y \in A_i \Rightarrow y \in A_i \wedge x \in A_i$. Hence $(y, x) \in R$ or yRx .
- c. If $xRy \wedge yRz$ then $\exists i \in I$ such that $x, y \in A_i$ and $\exists j \in I$ such that $y, z \in A_j$. So $y \in A_i \cap A_j$ or $A_i \cap A_j \neq \emptyset$, by [theorem: 3.8] we have that $A_i = A_j$, hence $x, z \in A_i$ proving that $(x, z) \in R$ or xRz .

2. If $i \in I$ then as $A_i \neq \emptyset$ [a partition is a family of non empty subsets] there exists a $x \in A_i$. Take $y \in A_i$ then $x, y \in A_i$ or yRx proving that $y \in R[x]$. So

$$A_i \subseteq R[x]$$

Take $y \in R[x]$ then yRx so there exist a $j \in I$ such that $x, y \in A_j$, hence $A_i \cap A_j \neq \emptyset$ which by [theorem: 3.8] proves that $A_i = A_j$, so that $y \in A_i$. So $R[x] \subseteq A_i$ giving

$$A_i = R[x]$$

3. If $x \in A$ then $\exists i \in I$ such that $x \in A_i$. Take $y \in A_i$ then $x, y \in A_i$ or yRx proving that $y \in R[x]$, hence

$$A_i \subseteq R[x]$$

Take $y \in R[x]$ then yRx so there exist a $j \in I$ such that $x, y \in A_j$, hence $A_i \cap A_j \neq \emptyset$ which by [theorem: 3.8] proves that $A_i = A_j$, so that $y \in A_i$. So $R[x] \subseteq A_i$ giving

$$A_i = R[x]$$

□

Definition 3.16. *Let A be a set and R a equivalence relation then A/R is defined by*

$$A/R = \{R[x] | x \in A\}$$

Note 3.17. As $\forall x \in X$ $R[x] \in \mathcal{P}(A)$ it follows that

$$R/X \in \mathcal{P}(A).$$

As A is a set it follows from the Axiom Power [axiom: 1.70] that $\mathcal{P}(A)$ is a set, applying the Axiom of Subsets [axiom: 1.8] we have

$$R/X \text{ is a set}$$

Theorem 3.18. *Let A be a set and R a equivalence relation then*

$$A = \bigcup_{I \in A/R} I \text{ and } \forall I, J \in A/R \text{ with } I \neq J \text{ we have } I \cap J = \emptyset$$

Proof. If $x \in A$ then by [theorem: 3.11] $x \in R[x] \in A/R$ so that $x \in \bigcup_{I \in A/R} I$ hence

$$A \subseteq \bigcup_{I \in A/R} I$$

Further, as $\forall I \in A/R$ we have that $\exists x \in A$ such that $I = R[x] \subseteq A$, it follows that $\bigcup_{I \in A/R} I \subseteq A$ which combined with the above gives

$$A = \bigcup_{I \in A/R} I$$

Further if $I, J \in A/R$ with $I \neq J$ then $\exists x, y \in A$ such that $I = R[x]$ and $J = R[y]$. If $z \in R[x] \cap R[y]$ then by [theorem: 3.11] we have $R[x] = R[z] = R[y]$ so that $I = J$ contradicting $I \neq J$. Hence we must have that

$$I \cap J = \emptyset \quad \square$$

3.2.2 Functions and equivalence relations

In this section we show how a function can be decomposed as the composition of a surjection, a bijection and injection. First we examine the relation between functions and equivalence relations. We can use functions to generate a equivalence relation on the domain of the function based on a equivalence relation on the target of the function.

Theorem 3.19. *$f: A \rightarrow B$ a function and R a equivalence relation in A then*

$$f \langle R \rangle = \{(x, y) | f(x) R f(y)\} \subseteq A \times A$$

is a equivalence relation in A

Proof.

reflectivity. If $x \in A$ then $f(x) \in B$ so that $f(x) R f(x)$ hence by definition $x R x$

symmetric. If $x R y$ then $f(x) R f(y)$ so that $f(y) R f(x)$ proving $y R x$

transitivity. If $x R y \wedge y R z$ then $f(x) R f(y) \wedge f(y) R f(z)$ so that $f(x) R f(z)$ proving that $x R z$. $\square$

A equivalence relation on a set induce a equivalence relation on a subset

Theorem 3.20. *Let A be a class, $B \subseteq A$ a sub-class and R a equivalence relation in R then $R|_B$ defined by*

$$R|_B = \{(x, y) | x \in B \wedge y \in B \wedge x R y\} = R \cap (B \times B)$$

is a equivalence relation.

Proof.

reflexivity. If $x \in B$ then $x R x$ so that $x R|_B x$

symmetric. If $x R|_B y \Rightarrow x \in B \wedge y \in B \wedge x R y \Rightarrow y R x \Rightarrow y R|_B x$

transitivity. If $x R|_B y \wedge y R|_B z$ then $x, y, z \in B$ and $x R y \wedge y R z$ so that $x, z \in B$ and $x R z$ proving $x R|_B z$ $\square$

Theorem 3.21. *If $f: A \rightarrow B$ is a function then R_f defined by*

$$R_f = \{(x, y) \in A \times A | f(x) = f(y)\}$$

*is a relation. R_f is called the **equivalence relation determined by f***

Proof.

reflexivity. If $x \in A$ then $f(x) = f(x)$ proving that $x R_f x$

symmetric. If $x R_f y$ then $f(x) = f(y) \Rightarrow f(y) = f(x)$ proving that $y R_f x$

transitivity. If $x R_f y \wedge y R_f x$ then $f(x) = f(y)$ and $f(y) = f(z)$ so that $f(x) = f(x)$ hence $x R_f z$ $\square$

We can also do the opposite and associate a function with a equivalence relation

Theorem 3.22. (Canonical Function) *Let A be a set and R a equivalence relation in A then:*

1. $f_R: A \rightarrow A/R$ defined by $f_R(x) = R[x]$ is a surjective function.
2. $R_{R_f} = R$

$f_R: A \rightarrow A/R$ is called the **Canonical function associated with R**

Proof.

1. As for every $x \in A$ we have the unique $R[x] \in R/X$ it follows from [proposition: 2.104] that

$$f_R: A \rightarrow A/R \text{ is a function}$$

Let $y \in R/X$ then $\exists x \in A$ such that $y = R[x]$ so that $(x, y) = (x, R[x]) \in f_R$ proving that $y \in \text{range}(f_R)$. So $R/X \subseteq \text{range}(f_R)$ which by [theorem: 2.53] proves that

$$f_R: A \rightarrow A/R \text{ is surjective}$$

2. We have

$$\begin{aligned} (x, y) \in R & \Leftrightarrow x R y \\ & \stackrel{[\text{theorem: 3.11}]}{\Leftrightarrow} R[x] = R[y] \\ & \Leftrightarrow f_R(x) = f_R(y) \\ & \Leftrightarrow (x, y) \in R_{f_R} \end{aligned} \quad \square$$

We use the above to decompose every function as the composition of a surjection, bijection and injection.

Theorem 3.23. *Let A, B be sets and $f: A \rightarrow B$ a function and define the following functions:*

- a) $s_f: A/R_f \rightarrow f(A)$ where $s_f = \{(R_f[x], f(x)) | x \in A\}$
- b) $i_{f(A)}: f(A) \rightarrow B$ where $i_{f(A)} = \{(x, x) | x \in f(A)\}$ [the inclusion function see [example: 2.55]]
- c) $f_{R_f}: A \rightarrow A/R_f$ where $f_{R_f}(x) = R_f[x]$ [theorem: 3.22]

then

1. $s_f: A/R_f \rightarrow f(A)$ is a bijection
2. $i_{f(A)}: f(A) \rightarrow B$ is a injective function
3. $f_{R_f}: A \rightarrow A/R_f$ is a surjective function
4. $f = i_{f(A)} \circ (s_f \circ f_{R_f}) \stackrel{[\text{theorem: 2.21}]}{=} (i_{f(A)} \circ s_f) \circ f_{R_f}$

Proof. Using [example: 2.55] and [theorem: 3.22] we have that

$$i_{f(A)}: f(A) \rightarrow B \text{ is a injective function}$$

and

$$f_{R_f}: A \rightarrow A/R_f \text{ is surjective function}$$

We proceed now to prove that s_f is a bijection. If $(x, y) \in s_f$ then there exists a $a \in A$ such that $(x, y) = (R_f[a], f(a))$ hence $x = R_f[a] \in A/R_f$ and $y = f(a) \Rightarrow (a, y) \in f \Rightarrow y \in f(A)$. So that $(x, y) \in (A/R_f) \times f(A)$ or

$$s_f \subseteq (A/R_f) \times f(A)$$

If $(x, y), (x, y') \in s_f$ then there exists $a, a' \in A$ such that

$$(x, y) = (R_f[a], f(a)) \wedge (x, y') = (R_f[a'], f(a'))$$

or

$$x = R_f[a] \wedge y = f(a) \wedge x = R_f[a'] \wedge y' = f(a') \quad (3.1)$$

From the above $R_f[a] = x = R_f[a']$, which using [theorem: 3.11] means that $a R_f a'$, so by the definition of R_f [theorem: 3.21] we have $f(a) = f(a')$. As by [eq: 3.1] $y = f(a) \wedge y' = f(a')$ it follows that $y = y'$. So

$s_f: A/R_f \rightarrow f(A)$ is a partial function

If $x \in A/R_f$ then $\exists a \in A$ such that $x = [a]$, hence if we take $y = f(A)$ we have that $(x, y) = ([a], f(a)) \in s_f$ proving that $x \in \text{dom}(s_f)$. So $A/R_f \subseteq \text{dom}(s_f)$ which by [proposition: 2.27] proves that

$s_f: A/R_f \rightarrow f(A)$ is a function

Let $(x, y), (x', y) \in s_f$ then $\exists a, a' \in A$ such that $(x, y) = (R_f[a], f(a))$ and $(x', y) = (R_f[a'], f(a'))$, hence

$$x = R_f[a] \wedge x' = R_f[a'] \wedge y = f(a) \wedge y = f(a') \quad (3.2)$$

From $f(a) = y = f(a')$ it follows that $f(a) = f(a')$, which by the definition of R_f [theorem: 3.21] proves that $a R_f a'$. Using [theorem: 3.11] it follows that $R_f[a] = R_f[a']$ or using [eq: 3.2] that $x = x'$. So we have proved that

$s_f: A/R_f \rightarrow f(A)$ is injective (3.3)

Let $y \in f(A)$ then there exist a $a \in A$ such that $(a, y) \in f \Rightarrow y = f(a)$. But then $(R_f[a], y) = (R_f[a], f(a)) \in s_f$ proving that $y \in \text{range}(s_f)$. So $A/R_f \subseteq \text{range}(s_f)$ which by [proposition: 2.53] proves that

$s_f: A/R_f \rightarrow f(A)$ is surjective (3.4)

Combining [eq: 3.3] and [eq: 3.4] it follows that

$s_f: A/R_f \rightarrow f(A)$ is a bijection

Now we proceed to prove that $f = (i_{f(A)} \circ s_f) \circ f_{R_f}$. Let $(x, u) \in (i_{f(A)} \circ s_f) \circ f_{R_f}$ then $\exists y$ such that $(x, y) \in f_{R_f} \wedge (y, u) \in i_{f(A)} \circ s_f$, from $(y, u) \in i_{f(A)} \circ s_f \exists z$ such that $(y, z) \in s_f \wedge (z, u) \in i_{f(A)}$, summarized

$$(x, y) \in f_{R_f} \wedge (y, z) \in s_f \wedge (z, u) \in i_{f(A)} \quad (3.5)$$

From $(x, y) \in f_{R_f}$ it follows that $\exists a \in A$ such that $(x, y) = (a, R_f[a])$ or

$$x = a \wedge y = R_f[a] \quad (3.6)$$

From $(y, z) \in s_f$ it follows that $\exists a' \in A$ such that $(y, z) = (R_f[a'], f(a'))$ or $y = R_f[a'] \wedge z = f(a')$. As $y \stackrel{[eq: 3.6]}{=} R_f[a]$ we have that $R_f[a] = R_f[a']$, which by [theorem: 3.11] proves that $a R_f a'$, so by the definition of R_f we have $f(a) = f(a')$ hence $z = f(a)$. From $(z, u) \in i_{f(A)}$ it follows that $z = u$ hence $u = f(a)$. As $x \stackrel{[eq: 3.6]}{=} a$ it follows that $(x, u) = (a, f(a)) \in f$. Hence

$$(i_{f(A)} \circ s_f) \circ f_{R_f} \subseteq f \quad (3.7)$$

Finally if $(x, y) \in f$ then as $f \subseteq A \times B$ proves that $x \in A$ and $f(x) = y \in f(A)$. Hence $(R_f[x], f(x)) \in s_f$, $(x, R_f[x]) \in f_{R_f}$ and $(f(x), y) = (f(x), f(x)) \in i_{f(A)}$. So that $(R_f[x], y) \in i_{f(A)} \circ s_f$ and $(x, R_f[x]) \in f_{R_f}$ proving that $(x, y) \in (i_{f(A)} \circ s_f) \circ f_{R_f}$. So $f \subseteq (i_{f(A)} \circ s_f) \circ f_{R_f}$ which combined with [eq: 3.7] gives

$$f = (i_{f(A)} \circ s_f) \circ f_{R_f} \quad \square$$

Notation 3.24. For the rest of this book we use the standard convention of noting a equivalence relation as $\sim$, The definition of $\sim$ should then be clear from the context. If many equivalence relations are used in the same context we use superscripts like $\sim_{\mathbb{R}}$ and $\sim_{\mathbb{Z}}$ to avoid conflicts.

3.3 Partial ordered classes

3.3.1 Order relation

First we define a partial order relation that allows us to compare two elements and specify which element 'lies before' another element.

Definition 3.25. (Pre-order) Let A be a class then a relation $R \subseteq A \times A$ in A is a *pre-order* if it is **reflective** and **transitive** or in other words:

reflectivity. $\forall x \in A$ we have xRx

transitivity. If $xRy \wedge yRz$ then xRz

Definition 3.26. $\langle A, R \rangle$ is a *pre-ordered class* iff A is a class and R is a pre-order in A

A order relation is a pre-order with one extra condition

Definition 3.27. (Order relation) If A is a class then a relation $R \subseteq A \times A$ in A is a **order** if it is a pre-order that is *anti-symmetric* or in other words:

reflectivity. $\forall x \in A$ we have xRx

anti-symmetry. If $xRy \wedge yRx$ then $x = y$

transitive. If $xRy \wedge yRz$ then xRz

Definition 3.28. (Partial ordered class) $\langle A, R \rangle$ is a **partial ordered class** if A is a class and R is a order.

Notation 3.29. We use the standard convention of noting a pre-order relation as $\leq$, The definition of $\leq$ should then be clear from the context. If many pre-order relations are used in the same context we use superscripts like $\leq_{\mathbb{R}}$ and $\leq_{\mathbb{Z}}$ or $\preceq$ to avoid conflicts.

Definition 3.30. If $\langle A, \leq \rangle$ is a pre-ordered or partial class and $x, y, z \in A$ then we define:

$$\begin{aligned} x \leq y \leq z & \text{ is the same as } x \leq y \wedge y \leq z \\ x \leq y < z & \text{ is the same as } x \leq y \wedge y < z \\ x < y \leq z & \text{ is the same as } x < y \wedge y \leq z \\ x < y < z & \text{ is the same as } x < y \wedge y < z \end{aligned}$$

Definition 3.31. If $\langle A, \leq \rangle$ is a pre-ordered class [or partial ordered class] then $x < y$ is equivalent with $x \leq y \wedge x \neq y$

Theorem 3.32. If $\langle A, \leq \rangle$ is a partially ordered set then

1. $x \leq y \wedge y < z \Rightarrow x < z$
2. $x < y \wedge y \leq z \Rightarrow x < z$
3. $x < y \wedge y < z \Rightarrow x < z$
4. $(x < y \vee x = y) \Leftrightarrow (x \leq y)$

or in other words

1. $x \leq y < z \Rightarrow x < z$
2. $x < y \leq z \Rightarrow x < z$
3. $x < y < z \Rightarrow x < z$
4. $(x < y \vee x = y) \Leftrightarrow x \leq y$

Proof.

1. If $x \leq y \wedge y < z$ then $x \leq y \wedge y \leq z \wedge y \neq z$, so that $x \leq z$ and $y \neq z$. Assume that $x = z$ then $z \leq y \stackrel{y \leq z}{=} z = y$ contradicting $y \neq z$, so we must have $x \neq z$, which together with $x \leq z$ gives

$$x < z$$

2. If $x < y \wedge y \leq z$ then $x \leq y \wedge y \leq z \wedge x \neq y$, so that $x \leq z$ and $x \neq y$. Assume that $x = z$ then $y \leq x \stackrel{x \leq y}{\Rightarrow} y = x$ contradicting $x \neq y$, so we must have $x \neq z$, which together with $x \leq z$ gives

$$x < z$$

3. If $x < y \wedge y < z$ then $x \neq y \wedge x \leq y \wedge y < z$ so that by (1) we have $x < z$
4. We have

$$\begin{aligned} (x < y \vee x = y) &\Leftrightarrow ((x \leq y \wedge x \neq y) \vee x = y) \\ &\Leftrightarrow ((x \leq y \vee x = y) \wedge (x \neq y \vee x = y)) \\ &\Leftrightarrow x \leq y \vee x = y \\ &\Leftrightarrow x \leq y \end{aligned}$$

□

Example 3.33. Let A be a class of classes and $\leq$ defined by $\leq = \{(x, y) \in A \times A \mid x \subseteq y\}$ then $\langle A, \leq \rangle$ is a partial ordered class

Proof.

reflectivity. If $A \in \mathcal{C}$ then by [theorem: 1.9] $A \subseteq A$ so that $A \leq A$

anti-symmetric. If $A \leq B$ and $B \leq A$ then $A \subseteq B \wedge B \subseteq A$ so that by [theorem: 1.9] $A = B$

transitivity. If $A \leq B \wedge B \leq C$ then $A \subseteq B \wedge B \subseteq C$ so that by [theorem: 1.9] $A \subseteq C$ or $A \leq C$ □

Every pre-order can be used as the base to create an order relation as is expressed in the following theorem. The basic idea is that $x \leq y \wedge y \leq x \Rightarrow x = y$ is missing from a pre-order. By defining an equivalence relation $\sim$ such that $x \sim y$ if $x \leq y \wedge y \leq x$ we turn this into equality of equivalence classes. This is a typical example about the use of equivalence relations, they allow you to define a new type of equality, so that objects that are not equal have associated equivalence classes that are equal.

Theorem 3.34. Let $\langle A, \leq \rangle$ be a pre-ordered set then we have

1. $\sim \subseteq A \times A$ defined by $\sim = \{(x, y) \in A \mid x \leq y \wedge y \leq x\}$ is an equivalence relation
2. Define $\preceq \subseteq (A/\sim) \times (A/\sim)$ by

$$\preceq = \{(x, y) \in (A/\sim) \times (A/\sim) \mid \exists x' \in \sim[x] \text{ and } \exists y' \in \sim[y] \text{ such that } x' \leq y'\}$$

then $\preceq$ is an order relation in $A/\sim$. So $\langle A/\sim, \preceq \rangle$ is a partial ordered set

3. $\forall x, y \in A$ we have $x \leq y \Leftrightarrow \sim[x] \preceq \sim[y]$

Proof.

1. To prove that $\sim$ is an equivalence relation note:

reflectivity. If $x \in A$ then $x \leq x$ proving that $x \sim x$

symmetric. If $x \sim y$ then $x \leq y \wedge y \leq x \Rightarrow y \leq x \wedge x \leq y$ so that $y \sim x$

transitive. If $x \sim y$ and $y \sim z$ then $x \leq y \wedge y \leq x \wedge y \leq z \wedge z \leq y$ so that $x \leq z$ and $z \leq x$ or $x \sim z$

2. To prove that $\preceq$ is an order relation we must prove reflectivity, symmetry and transitivity:

reflectivity. Take $\sim[x]$ then as $x \leq x$ there exists a $u \in \sim[x]$ and $v \in \sim[x]$ such that $u \leq v$ [just take $u = x = v$] so that

$$\sim[x] \preceq \sim[x]$$

symmetry. Let $\sim[x] \leq \sim[y]$ and $\sim[y] \leq \sim[x]$ then $\exists x', x'' \in \sim[x], \exists y', y'' \in \sim[y]$ such that

$$x' \leq y' \wedge y'' \leq x''$$

From $\exists x', x'' \in \sim[x], \exists y', y'' \in \sim[y]$ we have

$$x' \leq x \wedge x \leq x' \wedge x'' \leq x \wedge x \leq x'' \wedge y' \leq y \wedge y \leq y' \wedge y'' \leq y \wedge y \leq y''$$

From $x \leq x'$ and $x' \leq y'$ we have $x \leq y'$, as $y' \leq y$ we have

$$x \leq y$$

From $y \leq y''$ and $y'' \leq x''$ we have $y \leq x''$, as $x'' \leq x$ it follows that

$$y \leq x$$

Finally from $x \leq y$ and $y \leq x$ we have that $x \sim y$ which by [theorem: 3.11] gives

$$\sim[x] = \sim[y]$$

transitivity. Assume that $\sim[x] \leq \sim[y]$ and $\sim[y] \leq \sim[z]$ then we have the existence of $x' \in \sim[x], y', y'' \in \sim[y]$ and $z' \in \sim[z]$ such that

$$x' \leq y' \wedge y'' \leq z'$$

From $x' \in \sim[x], y', y'' \in \sim[y]$ and $z' \in \sim[z]$ it follows that

$$x' \leq x \wedge x \leq x' \wedge y' \leq y \wedge y \leq y' \wedge y'' \leq y \wedge y \leq y'' \wedge z' \leq z \wedge z \leq z'$$

From $x \leq x'$ and $x' \leq y'$ we have $x \leq y'$, as $y' \leq y$ we have $x \leq y$, as $y \leq y''$ it follows that $x \leq y''$, from $y'' \leq z'$ we have that $x \leq z'$ and finally from $z' \leq z$ it follows that $x \leq z$. Hence

$$\sim[x] \leq \sim[z]$$

3.

$\Rightarrow$. If $x \leq y$ then as $x \in \sim[x]$ and $y \in \sim[y]$ we have $\sim[x] \leq \sim[y]$

$\Leftarrow$. If $\sim[x] \leq \sim[y]$ then $\exists x' \in \sim[x]$ and $\exists y' \in \sim[y]$ such that

$$x' \leq y'$$

From $x' \in \sim[x]$ and $y' \in \sim[y]$ we have that

$$x' \leq x \wedge x \leq x' \wedge y' \leq y \wedge y \leq y'$$

From $x \leq x'$ and $x' \leq y'$ it follows that $x \leq y'$ and as $y' \leq y$ it follows that

$$x \leq y$$

□

Given a partial ordered class then we can induce the order on a sub-class making the sub-class also a partial ordered class.

Theorem 3.35. If $\langle A, \leq \rangle$ is a partial ordered sets and $B \subseteq A$ then $\leq|_B$ defined by

$$\leq|_B = \bigcap_{x \in B} B \times B = B$$

is a order relation in B making $\langle B, \leq|_B \rangle$ a partial ordered set.

Proof.

reflectivity. If $x \in B$ then $x \leq x$ or $(x, x) \in \leq|_B \Rightarrow (x, x) \in \bigcap (B \times B)$ hence $x \leq|_B y$

symmetry. If $x \leq|_B y \wedge y \leq|_B x \Rightarrow x \leq y \wedge y \leq x \Rightarrow x = y$

transitivity. If $x \leq|_B y \wedge y \leq|_B z \Rightarrow x \leq y \wedge y \leq z \Rightarrow x \leq z \xRightarrow{x, z \in B} x \leq|_B z$

□

Convention 3.36. To avoid excessive usage notation we write $\langle B, \leq \rangle$ instead of $\langle B, \leq|_B \rangle$

The following shows a technique of defining a partial order on the Cartesian product of partial ordered set.

Theorem 3.37. (Lexical ordering) Let $\langle A, \leq_A \rangle$ and $\langle B, \leq_B \rangle$ be partial ordered classes then $\leq_{A \times B}$ defined by

$$\leq_{A \times B} = \{((x, y), (u, v)) \in (A \times B) \times (A \times B) \mid (x \neq u \wedge x \leq_A u) \vee (x = y \wedge y \leq_B v)\}$$

is a order in $A \times B$ making $\langle (A \times B) \times (A \times B), \leq_{A \times B} \rangle$ a partial ordered set

Proof.

reflexivity. If $(x, y) \in A \times B$ then $x \leq_A x \wedge y \leq_B y$ proving that $(x, y) \leq_{A \times B} (x, y)$

symmetry. Let $(x, y) \leq_{A \times B} (u, v) \wedge (u, v) \leq_{A \times B} (x, y)$. If $x \neq u$ we would have $x \leq_A u \wedge u \leq_A x \Rightarrow x = u$ a contradiction. So we must have that $x = u$ but then $y \leq_B v \wedge v \leq_B y \Rightarrow y = v$ proving that

$$(x, y) = (u, v)$$

transitivity. Let $(x, y) \leq_{A \times B} (u, v) \wedge (u, v) \leq_{A \times B} (r, s)$ then we have to consider the following cases:

$x = u$. Then $y \leq_B v$ and we have the following possibilities

$u = r$. Then $v \leq_B s$ so that $y \leq_B s$ which as $x = r$ proves that

$$(x, y) \leq_{A \times B} (r, s)$$

$u \neq r$. Then $u \leq_A r \Rightarrow_{x=u} x \leq_A r$ which as $x \neq r$ proves that

$$(x, y) \leq_{A \times B} (r, s)$$

$x \neq u$. Then $x \leq_A u$ and we have the following possibilities

$u = r$. Then $x \leq_A u \Rightarrow_{u=r} x \leq_A r$ and $x \neq r$ so that

$$(x, y) \leq_{A \times B} (r, s)$$

$u \neq r$. Then $u \leq_A r$ so that $x \leq_A r$. If $x = r$ then we would have $x \leq_A u \wedge u \leq_A x$ giving $x = u$ contradicting $x \neq u$. So we must have $x \neq r$ which as $x \leq_A r$ gives

$$(x, y) \leq_{A \times B} (r, s) \quad \square$$

Definition 3.38. Let $\langle A, \leq \rangle$ be a partial ordered class then $x, y \in A$ are **comparable** if $x \leq y$ or $y \leq x$

Theorem 3.39. Let $\langle A, \leq \rangle$ be a partial ordered class and $x, y \in A$ comparable elements then we have either $x \leq y$ or $y < x$

Proof. As x, y are comparable then we have $x \leq y \vee y \leq x$, consider the following cases:

$x \leq y$. then $x \leq y$

$\neg(x \leq y)$. then we must have $y \leq x$. If $x = y$ then as $x \leq x$ we have $x \leq y$ contradicting $\neg(x \leq y)$ so that $x \neq y$ proving $y < x$.

Hence we have

$$x \leq y \vee y < x \quad \square$$

Definition 3.40. A partial ordered class $\langle A, \leq \rangle$ is a **totally ordered class** iff

$$\forall x, y \in A \text{ we have } x \leq y \vee y \leq x$$

In other words $\langle A, \leq \rangle$ is a **totally ordered class** if every pair of elements are comparable. Other names used in the literature are **fully ordered class** or **linear ordered class**.

Definition 3.41. (chain) Let $\langle A, \leq \rangle$ be a partial ordered class and $C \subseteq A$ then C is called a **chain** if $\forall x, y \in C$ we have that $x \leq y$ or $y \leq x$.

Example 3.42. Let $\langle A, \leq \rangle$ be a partial ordered class then $\emptyset$ is a chain

Proof. The condition $\forall x, y \in \emptyset$ we have that x, y are comparable is satisfied vacuously. $\square$

Theorem 3.43. Let $\langle A, \leq \rangle$ be a partial ordered class and $B \subseteq A$ a chain then $\langle B, \leq|_B \rangle$ is a totally ordered class

Proof. Using [theorem: 3.35] we have that $\langle B, \leq|_B \rangle$ is a partial ordered class. Let $x, y \in B$ then as B is a chain we have that $\forall x, y \in B \ x \leq y \vee y \leq x$ or using the definition of $\leq|_B$ that $x \leq|_B y \vee y \leq|_B x$. $\square$

Theorem 3.44. Let $\langle A, \leq \rangle$ be a totally ordered class and $B \subseteq A$ then B is a chain [hence by [theorem: 3.43] $\langle B, \leq|_B \rangle$ is a totally ordered class]

Proof. If $x, y \in B$ then $x, y \in A$ and as A is totally ordered we have $x \leq y \vee y \leq x$ so B is a chain $\square$

Theorem 3.45. Let $\langle A, \leq_A \rangle$ and $\langle B, \leq_B \rangle$ be totally ordered classes then $\langle A \times B, \leq_{A \times B} \rangle$ is a totally ordered class.

Proof. First $\langle A \times B, \leq_{A \times B} \rangle$ is a partially ordered class by [theorem: 3.37]. If $(x, y), (x', y') \in A \times B$ then we have for x, x' either

$x = x'$. As $\langle B, \leq_B \rangle$ is fully ordered we have either

$y \leq y'$. then $(x, y) \leq (x', y')$

$y' \leq y$. then $(x', y') \leq (x, y)$

$x \neq x'$. As $\langle A, \leq_A \rangle$ is fully ordered we have either

$x \leq x'$. then $(x, y) \leq (x', y')$

$x' \leq x$. then $(x', y') \leq (x, y)$ $\square$

Definition 3.46. (Initial Segment) If $\langle A, \leq \rangle$ is a partial ordered class, $a \in A$ then a **initial segment of A determined by a** noted as $S_{A,a}$ is defined by

$$S_{A,a} = \{x \in A \mid x < a\}$$

We have the following trivial result for initial segments.

Proposition 3.47. If $\langle A, \leq \rangle$ is a partial ordered class and $a, b \in A$ such that $a \leq b$ then $S_{A,a} \subseteq S_{A,b}$

Proof. If $x \in S_{A,a}$ then $x < a \xRightarrow{a \leq b} x < b$ proving that $x \in S_{A,b}$ $\square$

Theorem 3.48. If $\langle A, \leq \rangle$ is a partial ordered class and P is a initial segment of A and Q is a initial segment of P [using the induced order $\leq|_P$] then A is a initial segment of A

Proof. Using the hypothesis there exists $a \in A$ such that $P = \{x \in A \mid x < a\}$ and a $b \in P$ such that $Q = \{x \in P \mid x < b\}$. Consider then the initial segment $S_{A,b} = \{x \in A \mid x < b\}$ of A determined by a then we have

$$\begin{array}{ll}
 x \in S_{A,b} & \Rightarrow x \in A \wedge x < b \\
 & \xRightarrow{b < a \Rightarrow x < b \Rightarrow x < a} x \in A \wedge x < a \wedge x < b \\
 & \Rightarrow x \in P \wedge x < b \\
 & \Rightarrow x \in P \wedge x <|_P b \\
 & \Rightarrow x \in Q \\
 x \in Q & \Rightarrow x \in P \wedge x <|_P b \\
 & \Rightarrow x \in P \wedge x < b \\
 & \xRightarrow{P \subseteq A} x \in A \wedge x < b \\
 & \Rightarrow x \in S_{A,b}
 \end{array}$$

Hence $Q = S_{A,b}$ a initial segment of A □

3.3.2 Order relations and functions

Functions between two partial ordered classes can be classified based on the fact that they preserve or not preserve the order relation. This is expressed in the next definition.

Definition 3.49. Let $\langle A, \leq_A \rangle, \langle B, \leq_B \rangle$ be partial ordered classes and $f: A \rightarrow B$ a function then:

1. $f: \langle A, \leq_A \rangle \rightarrow B$ is **increasing** if $\forall x, y \in A$ with $x \leq y$ we have $f(x) \leq f(y)$. Another name that is used is **a order homeomorphism** [a homeomorphism is a function that preserves a certain operation, in this case the order relation]
2. $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is **strictly increasing** if $\forall x, y \in A$ with $x < y$ we have $f(x) < f(y)$
3. $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is **decreasing** if $\forall x, y \in A$ with $x \leq y$ we have $f(y) \leq f(x)$
4. $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is **strictly decreasing** if $\forall x, y \in A$ with $x < y$ we have $f(y) < f(x)$
5. $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is a **order isomorphism** if $\forall x, y \in A$ with $x \leq y \Leftrightarrow f(x) \leq f(y)$

Definition 3.50. Two partial ordered classes $\langle A, \leq_A \rangle$ and $\langle B, \leq_B \rangle$ are **order isomorphic** noted as $A \cong B$ if there exists order isomorphism between A and B .

Theorem 3.51. Let $\langle A, \leq_A \rangle, \langle B, \leq_B \rangle$ be two partial ordered classes, $D \subseteq B$ and

$f: \langle A, \leq_A \rangle \rightarrow \langle D, (\leq_B)|_D \rangle$ be a order homeomorphism [see theorem: 3.35 for $\langle D, (\leq_B)|_D \rangle$

then

$f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is a order homeomorphism

Proof. The proof is trivial for if $x, y \in A$ with $x \leq_A y$ then $f(x)(\leq_B)|_D f(y) \xrightarrow{3.35} f(x) \leq_B f(y)$ □

Theorem 3.52. Let $\langle A, \leq_A \rangle, \langle B, \leq_B \rangle, \langle C, \leq_C \rangle$ be partial ordered classes, $D \subseteq B$

1. If $D \subseteq B$ is equipped with the induced order from $\langle B, \leq_B \rangle$ [see theorem: 3.35] and

$f: \langle A, \leq_A \rangle \rightarrow \langle D, \leq_B \rangle$ and $g: \langle B, \leq_B \rangle \rightarrow \langle C, \leq_C \rangle$ are order homeomorphisms

then

$g \circ f: \langle A, \leq_A \rangle \rightarrow \langle C, \leq_C \rangle$ is a order homeomorphism

2. If $D \subseteq B$ is equipped with the induced order from $\langle B, \leq_B \rangle$ [see theorem: 3.35] and

$f: \langle A, \leq_A \rangle \rightarrow \langle D, \leq_B \rangle$ and $g: \langle B, \leq_B \rangle \rightarrow \langle C, \leq_C \rangle$ are strictly increasing

then

$g \circ f: \langle A, \leq_A \rangle \rightarrow \langle C, \leq_C \rangle$ is strictly increasing

3. If $D \subseteq B$ is equipped with the induced order from $\langle B, \leq_B \rangle$ [see theorem: 3.35] and

$f: \langle A, \leq_A \rangle \rightarrow \langle D, \leq_B \rangle$ and $g: \langle B, \leq_B \rangle \rightarrow \langle C, \leq_C \rangle$ are order isomorphism

then

$g \circ f: \langle A, \leq_A \rangle \rightarrow \langle g(f(A)), \leq_C \rangle$ is a order isomorphism

or as $D \xrightarrow{f: a \rightarrow D} \overset{=}{=} f(A)$ is bijective

$g \circ f: \langle A, \leq_A \rangle \rightarrow \langle g(D), \leq_C \rangle$ is a order isomorphism

Proof.

1. Let $x, y \in A$ with $x \leq_A y$ then $f(x) \leq_B f(y)$ hence $(g \circ f)(x) = g(f(x)) \leq_C g(f(y)) = (g \circ f)(y)$.
2. Let $x, y \in A$ with $x <_A y$ then $f(x) <_B f(y)$ hence $(g \circ f)(x) = g(f(x)) <_C g(f(y)) = (g \circ f)(y)$.

3. Using [theorem: 2.77] we have that $g \circ f: A \rightarrow g(D) = g(f(A))$ is a bijection. Let $x, y \in A$. If $x \leq_A y$ then $f(x) \leq_B f(y)$ hence $(g \circ f)(x) = g(f(x)) \leq_C g(f(y)) = (g \circ f)(y)$. Also if $(g \circ f)(x) \leq_C (g \circ f)(y)$ then $g(f(x)) \leq_C g(f(y))$ so that $f(x) \leq_B f(y)$, giving $x \leq_A y$. $\square$

Theorem 3.53. *If $\langle A, \leq_A \rangle$ and $\langle B, \leq_B \rangle$ are partially ordered classes and*

$$f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle \text{ a order isomorphism}$$

then $\forall x, y \in A$ we have

$$x <_A y \Leftrightarrow f(x) <_B f(y)$$

Proof.

$\Rightarrow$. If $x <_A y$ then $x \neq y$ and $x \leq_A y \Rightarrow f(x) \leq_B f(y)$. Assume that $f(x) = f(y)$ then as f is a bijection we would have $x = y$ contradicting $x \neq y$. So we must have that $f(x) \neq f(y)$ hence

$$f(x) <_B f(y)$$

$\Leftarrow$. As $f(x) <_B f(y)$ we have that $f(x) \neq f(y)$ so that we must have $x \neq y$. Further as f is a isomorphism we have $x \leq_A y$. So

$$x <_A y \quad \square$$

Theorem 3.54. *If $\langle A, \leq_A \rangle$ and $\langle B, \leq_B \rangle$ are partially ordered classes and $f: A \rightarrow B$ a bijection then $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is a order isomorphism $\Leftrightarrow f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ and $f^{-1}: \langle B, \leq_B \rangle \rightarrow \langle A, \leq_A \rangle$ are increasing functions*

Proof. As $f: A \rightarrow B$ is a bijection we have by [theorems: 2.69, 2.74] that $f^{-1}: B \rightarrow A$ is a bijection.

$\Rightarrow$. As $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is a isomorphism we have that $\forall x, y \in A$ with $x \leq_A y \Rightarrow f(x) \leq f(y)$ hence $f: A \rightarrow B$ is increasing. If $x, y \in B$ with $x \leq_B y$ then

$$f(f^{-1}(x)) = (f \circ f^{-1})(x) \stackrel{[\text{theorem: 2.71}]}{=} x \leq_B y = (f \circ f^{-1})(y) = f(f^{-1}(y))$$

which as f is a isomorphism proves that $f^{-1}(x) \leq_A f^{-1}(y)$, hence f^{-1} is increasing.

$\Leftarrow$. Suppose that f, f^{-1} are increasing functions then if $x \leq_A y \xRightarrow{f \text{ is increasing}} f(x) \leq_B f(y)$. Further if $f(x) \leq_B f(y) \xRightarrow{f^{-1} \text{ is increasing}} f^{-1}(f(x)) \leq_A f^{-1}(f(y)) \Rightarrow x \leq y$ $\square$

Theorem 3.55. *If $\langle A, \leq_A \rangle, \langle C, \leq_C \rangle$ and $\langle B, \leq_B \rangle$ are partially ordered classes then*

1. $1_A: \langle A, \leq_A \rangle \rightarrow \langle A, \leq_A \rangle$ is a order isomorphism
2. If $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is a order isomorphism then $f^{-1}: \langle B, \leq_B \rangle \rightarrow \langle A, \leq_A \rangle$ is a order isomorphism
3. If $f: \langle A, \leq_A \rangle \rightarrow B$ and $g: \langle B, \leq_B \rangle \rightarrow \langle C, \leq_C \rangle$ are order isomorphism's then

$$g \circ f: \langle A, \leq_A \rangle \rightarrow \langle C, \leq_C \rangle \text{ is a order isomorphism}$$

Proof.

1. By 2.49 we have that $\text{Id}_A: A \rightarrow A$ is a bijection then, as $x = \text{Id}_A(x)$ and $y = \text{Id}_A(y)$, we have $x \leq y \Leftrightarrow \text{Id}_A(x) \leq \text{Id}_A(y)$.
2. If $f: A \rightarrow B$ is a isomorphism then by [theorem: 2.74] we have that $f^{-1}: B \rightarrow A$ is a bijection. By the previous theorem [theorem: 3.54] we have that f^{-1} is increasing. Further as by 2.75 $f = (f^{-1})^{-1}$ and by [theorem: 3.54] f is increasing it follows that $(f^{-1})^{-1}$ is increasing. Using then [theorem: 3.54] it follows that f^{-1} is a isomorphism.
3. This follows from [theorem: 3.52] $\square$

Theorem 3.56. *If $\langle A, \leq_A \rangle, \langle B, \leq_B \rangle$ and $\langle C, \leq_C \rangle$ are partially ordered classes then we have*

1. $A \cong A$

2. If $A \cong B$ then $B \cong A$
3. If $A \cong B$ and $B \cong D$ then $A \cong D$

Proof. This follows easily from the previous theorem [theorem: 3.55] $\square$

Theorem 3.57. Let $\langle A, \leq_A \rangle$ be a totally ordered class and $\langle B, \leq_B \rangle$ is a partially ordered class then a bijective and increasing function $f: \langle A, \leq_A \rangle \rightarrow \langle B, \leq_B \rangle$ is a isomorphism

Proof. Suppose that $f(x) \leq_B f(y)$ then since A is fully ordered we have that x, y are comparable therefore by [theorem: 3.38] we have the following exclusive cases

1. $x \leq_A y$ in this case our theorem is proved
2. $y <_A x$ in this case we would have $f(y) \leq_B f(x) \Rightarrow f(y) = f(x) \xRightarrow{f \text{ is injective}} x = y$ a contradiction. So this case does not occurs. $\square$

3.3.3 Min, max, supremum and infimum

Definition 3.58. Let $\langle X, \leq \rangle$ be a pre-ordered class and $A \subseteq X$ then

1. m is a **maximal element** of A iff $m \in A$ and if $\forall x \in A$ with $m \leq x$ we have $x = m$
2. m is a **minimal element** of A iff $m \in A$ and if $\forall x \in A$ with $x \leq m$ we have $x = m$

Definition 3.59. If $\langle X, \leq \rangle$ is a partial ordered class and $A \subseteq X$ then

1. m is the **greatest element** of A iff $m \in A$ and $\forall x \in A$ we have $x \leq m$
2. m is the **least element** of A iff $m \in A$ and $\forall x \in A$ we have $m \leq x$

Note 3.60. There is a subtle difference between the definition of a maximal (minimal) element and the greatest (least) element. If m is the greatest (least) element of A then every element in A is comparable with m , which is not the case if m is a maximal (minimal) element of A .

Note 3.61. The empty set $\emptyset$ can not have a maximal, minimal element, greatest element or least element.

Theorem 3.62. If $\langle X, \leq \rangle$ is a partial ordered class and $A \subseteq X$ then

1. If m, m' are greatest elements of A then $m = m'$
2. If m, m' are least elements of A then $m = m'$

The unique greatest element of A (if it exist) is called the maximum of A and noted as $\max(A)$, the unique least element of A (if it exist) is called the minimum of A and noted as $\min(A)$

Proof.

1. If m, m' are greatest elements of A then as $m, m' \in A$ we have $m \leq m' \wedge m' \leq m$ so that $m = m'$.
2. If m, m' are least elements of A then as $m, m' \in A$ we have $m \leq m' \wedge m' \leq m$ so that $m = m'$. $\square$

Example 3.63. Let $\langle X, \leq \rangle$ be totally ordered set and $x \in X$ then $\max(\{x\}), \min(\{x\})$ exists and $\max(\{x\}) = x = \min(\{x\})$

Proof. As $x \in \{x\}$ and $\forall y \in \{x\} \ y = x \Rightarrow y \leq x \wedge x \leq y$ it follows that $\max(\{x\}), \min(\{x\})$ exists and $\max(\{x\}) = x = \min(\{x\})$. $\square$

Example 3.64. Let $\langle X, \leq \rangle$ be a totally ordered set, $x, y \in X$ then $\max(\{x, y\})$ and $\min(\{x, y\})$ exist.

Proof. As $\langle X, \leq \rangle$ is a totally ordered set we have either:

$x \leq y$. Then as $x, y \in X$, $x \leq x$, $y \leq y$ and $x \leq y$ we have $\min(\{x, y\}) = x$ and $\max(\{x, y\}) = y$.

$y \leq x$. Then as $x, y \in X$, $x \leq x$, $y \leq y$ and $y \leq x$ we have $\min(\{x, y\}) = y$ and $\max(\{x, y\}) = x$. $\square$

Notation 3.65. The above notation suggest the following notation. If $\langle X, \leq \rangle$ is a totally ordered set and $x, y \in X$ then $\max(x, y) = \max(\{x, y\})$ and $\min(x, y) = \min(\{x, y\})$.

Theorem 3.66. Let $\langle X, \leq \rangle$ be a totally ordered set and $A \subseteq X$, $B \subseteq X$ then we have:

1. If $\max(A)$ and $\max(B)$ exist then $\max(A \cup B)$ exist and

$$\max(A \cup B) = \max(\{\max(A), \max(B)\})$$

2. If $\min(A)$ and $\min(B)$ exist then $\min(A \cup B)$ exist and

$$\min(A \cup B) = \min(\{\min(A), \min(B)\})$$

Proof.

1. We have the following possibilities for $\max(A)$ and $\max(B)$:

$\max(A) \leq \max(B)$. Then $\max(B) \in B \subseteq A \cup B$ and $\forall x \in A \cup B$ we have either:

$x \in A$. Then $x \leq \max(A) \leq \max(B)$ so that $x \leq \max(B)$

$x \in B$. Then $x \leq \max(B)$

Hence $\forall x \in A \cup B$ we have that $x \leq \max(B)$ proving that

$$\max(A \cup B) = \max(B) \stackrel{[\text{example: 3.64}]}{=} \max(\{\max(A), \max(B)\})$$

$\max(B) \leq \max(A)$. Then $\max(A) \in A \subseteq A \cup B$ and $\forall x \in A \cup B$ we have either:

$x \in A$. Then $x \leq \max(A)$

$x \in B$. Then $x \leq \max(B) \leq \max(A)$

Hence $\forall x \in A \cup B$ we have that $x \leq \max(A)$ proving that

$$\max(A \cup B) = \max(A) \stackrel{[\text{example: 3.64}]}{=} \max(\{\max(A), \max(B)\})$$

2. We have the following possibilities for $\min(A)$ and $\min(B)$:

$\min(A) \leq \min(B)$. Then $\min(A) \in A \subseteq A \cup B$ and $\forall x \in A \cup B$ we have either:

$x \in A$. Then $\min(A) \leq x$

$x \in B$. Then $\min(A) \leq \min(B) \leq x$ so that $\min(A) \leq x$

Hence $\forall x \in A \cup B$ we have that $\min(A) \leq x$ proving that

$$\min(A \cup B) = \min(A) \stackrel{[\text{example: 3.64}]}{=} \min(\{\min(A), \min(B)\})$$

$\min(B) \leq \min(A)$. Then $\min(B) \in B \subseteq A \cup B$ and $\forall x \in A \cup B$ we have either:

$x \in A$. Then $\min(B) \leq \min(A) \leq x$ so that $\min(B) \leq x$

$x \in B$. Then $\min(B) \leq x$

Hence $\forall x \in A \cup B$ we have that $\min(B) \leq x$ proving that

$$\min(A \cup B) = \min(B) \stackrel{[\text{example: 3.64}]}{=} \min(\{\min(A), \min(B)\}) \quad \square$$

Corollary 3.67. Let $\langle X, \leq \rangle$ be a totally ordered set, $A \subseteq X$ and $x \in X$ then we have

1. If $\max(A)$ exist then $\max(A \cup \{x\})$ exists and $\max(A \cup \{x\}) = \max(x, \max(A))$

2. If $\min(A)$ exist then $\min(A \cup \{x\})$ exists and $\min(A \cup \{x\}) = \min(x, \min(A))$

Proof.

1. Using [example: 3.63] $\max(\{x\})$ exist and $\max(\{x\}) = x$ which, as $\max(A)$ exists, proves, using [theorem: 3.66], that $\max(A \cup \{x\})$ exists and

$$\max(A \cup \{x\}) = \max(\{\max(A), \max(\{x\})\}) = \max(\{\max(A), x\}) \underset{\text{notation}}{=} \max(\max(A), x)$$

2. Using [example: 3.63] $\min(\{x\})$ exist and $\min(\{x\}) = x$ which, as $\min(A)$ exists, proves, using [theorem: 3.66], that $\min(A \cup \{x\})$ exists and

$$\min(A \cup \{x\}) = \min(\{\min(A), \min(\{x\})\}) = \min(\{\min(A), x\}) \underset{\text{notation}}{=} \min(\min(A), x) \quad \square$$

Theorem 3.68. If $\langle X, \leq \rangle$ is a partial ordered class and $A \subseteq X$ such that $\min(A)$ and $\max(A)$ exist then $\min(A) \leq \max(A)$

Proof. As $\min(A) \in A$ we have by definition that $\min(A) \leq \max(A)$. $\square$

Theorem 3.69. Let $\langle X, \leq \rangle$ be a partial ordered class, $A \subseteq X$, $B \subseteq X$ then

1. If $\max(A)$ and $\max(B)$ exist and $\forall x \in A \exists y \in B$ such that $x \leq y$ then $\max(A) \leq \max(B)$
2. If $\min(A)$ and $\min(B)$ exist $\forall x \in B \exists y \in A$ such that $y \leq x$ then $\min(A) \leq \min(B)$

Proof.

1. As $\max(A) \in A$ there exist a $y \in B$ such that $\max(A) \leq y$, as $y \leq \max(B)$ we have

$$\max(A) \leq \max(B)$$

2. As $\min(B) \in B$ there exist a $y \in A$ such that $y \leq \min(B)$, as $\min(A) \leq y$ we have

$$\min(A) \leq \min(B) \quad \square$$

Definition 3.70. If $\langle X, \leq \rangle$ is a partial ordered class and $A \subseteq X$ then

1. $u \in X$ is a **upper bound** of A if $\forall a \in A a \leq u$.
2. A is **bounded above** if it has a upper bound.
3. $l \in X$ is a **lower bound** of A if $\forall x \in A l \leq x$
4. A is **bounded below** if it has a lower bound.
5. $v(A) = \{x \in X \mid x \text{ is a upper bound of } A\}$ [the class of upper bound of A]
6. $\lambda(A) = \{x \in X \mid x \text{ is a lower bound of } A\}$ [the class of lower bounds of A]

Example 3.71. If $\langle X, \leq \rangle$ then $v(\emptyset) = X$ and $\lambda(\emptyset) = X$

Proof. Let $x \in X$ then as $\forall a \in \emptyset a \leq x$ [or $x \leq a$] is vacuously satisfied $X \subseteq v(A)$ and $X \subseteq \lambda(A)$, which as $v(X) \subseteq X$ and $\lambda(X) \subseteq X$ proves $v(A) = X = \lambda(A)$. $\square$

Definition 3.72. If $\langle X, \leq \rangle$ is a partial ordered class and $A \subseteq X$ then

1. If $\min(v(A))$ exists then $\min(v(A))$ is called the **supremum** of A and noted as $\sup(A)$.
2. If $\max(\lambda(A))$ exists then $\max(\lambda(A))$ is called the **infimum** of A and noted as $\inf(A)$

In other words if $v(A)$ has a least element then the supremum of A is this unique, by [theorem: 3.62], element. So $\sup(A)$ is the least upper bound of A [if it exist] and it is itself a upper bound. If $\lambda(A)$ has a least element then the infimum of A is this unique, by [theorem: 3.62], element. So $\inf(A)$ is the greatest lower bound [if it exist] and it is itself a lower bound.

Example 3.73. Let $\mathcal{A}$ be a class of classes and $\langle \mathcal{A}, \leq \rangle$ the partial class where

$$\leq = \{(x, y) \in \mathcal{A} \times \mathcal{A} \mid x \subseteq y\}$$

[see example: 3.33] and $\mathcal{B} \subseteq \mathcal{A}$ we have that

1. If $\bigcap \mathcal{B} \in \mathcal{A}$ then $\inf(\mathcal{B})$ exist and $\inf(\mathcal{B}) = \bigcap \mathcal{B}$
2. If $\bigcup \mathcal{B} \in \mathcal{A}$ then $\sup(\mathcal{B})$ exist and $\sup(\mathcal{B}) = \bigcup \mathcal{B}$

Proof.

1. If $B \in \mathcal{B}$ then by [theorem: 1.66] $\bigcap \mathcal{B} \subseteq B \Rightarrow \bigcap \mathcal{B} \leq B$ so that $\bigcap \mathcal{B} \in \lambda(\mathcal{B})$. Now if $C \in \lambda(\mathcal{B})$ then $\forall B \in \mathcal{B}$ we have that $C \leq B \Rightarrow C \subseteq B$, so that by [theorem: 1.66] we have $C \subseteq \bigcap \mathcal{B} \Rightarrow C \leq \bigcap \mathcal{B}$ so that $\bigcap \mathcal{B}$ is the greatest element of $\lambda(\mathcal{B})$ proving that $\inf(\mathcal{B})$ exists and $\inf(\mathcal{B}) = \bigcap \mathcal{B}$.
2. If $B \in \mathcal{B}$ then by [theorem: 1.66] $B \subseteq \bigcup \mathcal{B} \Rightarrow B \leq \bigcup \mathcal{B}$ so that $\bigcup \mathcal{B} \in v(\mathcal{B})$. Now if $C \in v(\mathcal{B})$ then $\forall B \in \mathcal{B}$ we have that $B \leq C \Rightarrow B \subseteq C$, so that by [theorem: 1.66] we have $\bigcup \mathcal{B} \subseteq C \Rightarrow \bigcup \mathcal{B} \leq C$ so that $\bigcup \mathcal{B}$ is the lowest element of $v(\mathcal{B})$ proving that $\sup(\mathcal{B})$ exists and $\sup(\mathcal{B}) = \bigcup \mathcal{B}$. $\square$

The following theorem will be used a lot of time when dealing with supremum and infimum.

Theorem 3.74. Let $\langle X, \leq \rangle$ be a totally ordered set and $A \subseteq X$ then

1. If $\sup(A)$ exists then $\forall x \in X$ with $x < \sup(A)$ there $\exists a \in A$ such that $x < a \wedge a \leq \sup(A)$
2. If $\inf(A)$ exist then $\forall x \in X$ with $\inf(A) < x$ there $\exists a \in A$ such that $\inf(A) \leq a \wedge a < x$

Proof. First as $\langle X, \leq \rangle$ is totally ordered we have $\forall x, y \in X$ that x, y are comparable, hence by [theorem: 3.39], we have $x \leq y \wedge y < x /$

1. Let $x \in X$ such that $x < \sup(A)$. Assume that $\forall a \in A$ we have $\neg(x < a)$ so that $a \leq x$, so x is a upper bound of A , hence $x \in v(A)$, so that $\sup(A) = \min(v(A)) \leq x$, which, as $x < \sup(A)$, leads to the contradiction $x < x$. So we must have that $\exists a \in A$ such that $x < a$, further as $\sup(A)$ is a upper bound we have that $a \leq \sup(A)$. So

$$\exists a \in A \quad x < a \wedge a \leq \sup(A)$$

2. Let $x \in X$ such that $\inf(A) < x$. Assume that $\forall a \in A$ we have $\neg(a < x)$ so that $x \leq a$, so x is a lower bound of A , hence $x \in \lambda(A)$, so that $x \leq \max(\lambda(A)) = \inf(A)$, which, as $\inf(A) < x$, leads to the contradiction $x < x$. So we must have that $\exists a \in A$ such that $a \leq x$, further as $\inf(A)$ is a lower bound we have that $\inf(A) \leq a$. So

$$\exists a \in A \quad \inf(A) \leq a \wedge a < x$$

$\square$

Lemma 3.75. If $\langle X, \leq \rangle$ is a partially ordered class and $A \subseteq X, B \subseteq X$ with $A \subseteq B$ then

1. If $\max(A)$ and $\max(B)$ exist then $\max(A) \leq \max(B)$
2. If $\min(A)$ and $\min(B)$ exist then $\min(B) \leq \min(A)$

Proof.

1. As $\max(A) \in A$ and $A \subseteq B$ we have that $\max(A) \in B$ so that $\max(A) \leq \max(B)$
2. As $\min(A) \in A$ and $A \subseteq B$ we have that $\min(A) \in B$ so that $\min(B) \leq \min(A)$ $\square$

Lemma 3.76. If $\langle X, \leq \rangle$ is a partially ordered class and $A \subseteq X, B \subseteq X$ with $A \subseteq B$ then

1. $v(B) \subseteq v(A)$
2. $\lambda(B) \subseteq \lambda(A)$

Proof.

1. Let $x \in v(B)$ then $\forall a \in A$ we have, as $A \subseteq B$ that $a \in B$ hence $a \leq x$ proving that x is a upper bound of A or $x \in v(A)$.

2. Let $x \in \lambda(B)$ then $\forall a \in A$ we have as $A \subseteq B$ that $a \in B$ hence $x \leq a$ proving that x is a lower bound of A or $x \in \lambda(A)$. $\square$

Theorem 3.77. Let $\langle X, \leq \rangle$ be a partial ordered class and $A \subseteq X, B \subseteq X$ such that $A \subseteq B$ then

1. If $\sup(A)$ and $\sup(B)$ exist then $\sup(A) \leq \sup(B)$
2. If $\inf(A)$ and $\inf(B)$ exist then $\inf(B) \leq \inf(A)$

Proof.

1. Using [lemma: 3.76] we have that $v(B) \subseteq v(A)$ so that by [lemma: 3.75]

$$\sup(A) = \min(v(A)) \leq \min(v(B)) = \sup(B)$$

2. Using [lemma: 3.76] we have that $\lambda(B) \subseteq \lambda(A)$ so that by [lemma: 3.75]

$$\inf(B) = \max(\lambda(B)) \leq \max(\lambda(A)) = \inf(A)$$

$\square$

Theorem 3.78. Let $\langle X, \leq \rangle$ be a partial ordered class and $A \subseteq X, B \subseteq X$ then

1. If $\sup(A)$, $\sup(B)$ exists and $\forall a \in A \exists b \in B$ such that $a \leq b$ then $\sup(A) \leq \sup(B)$
2. If $\inf(A)$ and $\inf(B)$ exist and $\forall a \in A \exists b \in B$ such that $b \leq a$ then $\inf(B) \leq \inf(A)$

Proof.

1. Let $a \in A$ then $\exists b \in B$ such that $a \leq b$, as $b \leq \sup(B)$ it follows that $a \leq \sup(B)$. Hence $\sup(B) \in v(A)$. So $\sup(A) = \min(v(A)) \leq \sup(B)$, hence

$$\sup(A) \leq \sup(B)$$

2. Let $a \in A$ then $\exists b \in B$ such that $b \leq a$, as $\inf(B) \leq b$ it follows that $\inf(B) \leq a$. Hence $\inf(B) \in \lambda(A)$, So $\inf(B) \leq \max(\lambda(A)) = \inf(A)$, hence

$$\inf(B) \leq \inf(A)$$

$\square$

We have by definition that $\sup(A)$ exists if $\min(v(A))$ exists and $\inf(A)$ exist if $\max(\lambda(A))$ exist. The following theorem shows that there is a weaker condition for the existence of $\sup(A)$ and $\inf(A)$.

Theorem 3.79. Let $\langle X, \leq \rangle$ be a partial ordered class and $A \subseteq X$ then

1. If $\lambda(A)$ has a supremum then A has a infimum and $\sup(\lambda(A)) = \inf(A)$
2. If $v(A)$ has a infimum then A has a supremum and $\inf(v(A)) = \sup(A)$

Proof.

1. If $a \in A$ then $\forall y \in \lambda(A)$ we have $y \leq a$ so that $a \in v(\lambda(A))$. As $\sup(\lambda(A)) = \min(v(\lambda(A)))$ we have that $\sup(\lambda(A)) \leq a$. As $a \in A$ was arbitrary chosen we have that

$$\sup(\lambda(A)) \in \lambda(A) \tag{3.8}$$

If $x \in \lambda(A)$, then, as $\sup(\lambda(A))$ is a upper bound of $\lambda(A)$, we have $x \leq \sup(\lambda(A))$. So

$$\forall x \in \lambda(A) \text{ we have } x \leq \sup(\lambda(A)) \tag{3.9}$$

Using [eq: 3.8] and [eq: 3.9] it follows that $\sup(\lambda(A)) = \max(\lambda(A)) = \inf(A)$ or

$$\sup(\lambda(A)) = \inf(A)$$

2. If $a \in A$ then $\forall y \in v(A)$ we have $a \leq y$ so that $a \in \lambda(v(A))$. As $\inf(v(A)) = \max(\lambda(v(A)))$ we have that $a \leq \inf(v(A))$. As $a \in A$ was arbitrary chosen we have that

$$\inf(v(A)) \in v(A) \tag{3.10}$$

If $x \in v(A)$, then, as $\inf(v(A))$ is a lower bound of $v(A)$, we have $\inf(v(A)) \leq x$. So we have that

$$\forall x \in v(A) \text{ we have that } \inf(v(A)) \leq x \quad (3.11)$$

Using [eq: 3.10] and [eq: 3.11] it follows that $\inf(v(A)) = \min(v(A)) = \sup(A)$ or

$$\inf(v(A)) = \sup(A) \quad \square$$

In general it is not guaranteed that $\sup(A)$ or $\inf(A)$ exists. However there exists partial order classes that guarantees the existence of a supremum for non empty sub-classes that are bounded above.

Definition 3.80. (Conditional Completeness) A partial ordered class $\langle X, \leq \rangle$ is **conditional complete** if every non empty sub-class of A that is bounded above has a supremum.

The next theorem shows that conditional completeness can also be defined based on bounded below and infimum.

Theorem 3.81. If $\langle A, \leq \rangle$ is a partial ordered class then the following are equivalent

1. Every non empty sub-class of X that is bounded above has a supremum [$\langle X, \leq \rangle$ is conditional complete]
2. Every non empty sub-class of X that is bounded below has a infimum

Proof.

1 $\Rightarrow$ 2. Let $A \subseteq X$ a non empty sub-class that is bounded below. As $A \neq \emptyset$ there exists a $a \in A$, further by definition of $\lambda(A)$ we have $\forall y \in \lambda(A)$ that $y \leq a$ so $\lambda(A)$ is bounded above. As A is bounded below we have that $\lambda(A) \neq \emptyset$. So by the hypothesis $\sup(\lambda(A))$ exist. Applying then [theorem: 3.79] proves

$$\inf(A) \text{ exist}$$

2 $\Rightarrow$ 1. Let $A \subseteq X$ a non empty sub-class that is bounded above. As $A \neq \emptyset$ there exists a $a \in A$, further by definition of $v(A)$ we have $\forall y \in v(A)$ that $a \leq y$ so $v(A)$ is bounded below. As A is bounded above we have that $v(A) \neq \emptyset$. So by the hypothesis $\inf(v(A))$ exist. Applying then [theorem: 3.79] proves

$$\sup(A) \text{ exist} \quad \square$$

Next we show that a order isomorphism preserves the concepts of greatest element, least element, upper bound, lower bound, supremum and infimum.

Lemma 3.82. Let $\langle X, \leq_X \rangle, \langle Y, \leq_Y \rangle$ be partial ordered classes, $f: \langle X, \leq_X \rangle \rightarrow \langle Y, \leq_Y \rangle$ is a order isomorphism, $A \subseteq X$ and $B \subseteq Y$ then

1. If u is a upper bound of B then $(f^{-1})(u)$ is a upper bound of $f^{-1}(B)$
2. If l is a lower bound of B then $(f^{-1})(l)$ is a lower bound of $f^{-1}(B)$
3. If u is a upper bound of A then $f(u)$ is a upper bound of $f(A)$
4. If l is a lower bound of A then $f(l)$ is a lower bound of $f(A)$
5. $f(v(A)) = v(f(A))$
6. $f(\lambda(A)) = \lambda(f(A))$
7. If $\max(A)$ exist then $\max(f(A))$ exist and $\max(f(A)) = f(\max(A))$
8. If $\min(A)$ exist then $\min(f(A))$ exist and $\min(f(A)) = f(\min(A))$
9. If $\sup(A)$ exist then $\sup(f(A))$ exist and $\sup(f(A)) = f(\sup(A))$
10. If $\inf(A)$ exist then $\inf(f(A))$ exist and $\inf(f(A)) = f(\inf(A))$

Proof. First using [theorem: 3.54] we have that $f: \langle X, \leq_X \rangle \rightarrow \langle Y, \leq_Y \rangle$ and $f^{-1}: \langle Y, \leq_Y \rangle \rightarrow \langle X, \leq_X \rangle$ are increasing.

1. Let $x \in f^{-1}(B)$ then $\exists y \in B$ such that $y = f(x)$, as u is an upper bound of B , we have that $y \leq_B u$. So $x \stackrel{[\text{theorem: 2.72}]}{=} (f^{-1})(f(x)) = (f^{-1})(y) \leq_A (f^{-1})(u)$, proving that $(f^{-1})(u)$ is an upper bound of $f^{-1}(B)$.
2. Let $x \in f^{-1}(B)$ then $\exists y \in B$ such that $y = f(x)$, as l is a lower bound of B we have that $l \leq_B y$. So $(f^{-1})(l) \leq_A (f^{-1})(y) = (f^{-1})(f(x)) \stackrel{[\text{theorem: 2.72}]}{=} x$, proving that $(f^{-1})(l)$ is a lower bound of $f^{-1}(B)$.
3. If $y \in f(A)$ then $\exists x \in A$ such that $y = f(x)$. As u is an upper bound of A we have that $x \leq_A u$, so $y = f(x) \leq_B f(u)$ proving that $f(u)$ is an upper bound of $f(A)$.
4. If $y \in f(A)$ then $\exists x \in A$ such that $y = f(x)$. As l is a lower bound of A we have that $l \leq_A x$, so $f(l) \leq_B f(x) = y$ proving that $f(l)$ is a lower bound of $f(A)$.
5. If $y \in f(v(A))$ then there $\exists x \in v(A)$ such that $y = f(x)$. As $x \in v(A)$, x is an upper bound of B , so that by (3) $y = f(x)$ is an upper bound of $f(A)$. Hence

$$f(v(A)) \subseteq v(f(A)) \quad (3.12)$$

If $y \in v(f(A))$ then by (1) $(f^{-1})(y)$ is an upper bound of $f^{-1}(f(A)) \stackrel{[\text{theorem: 2.57}]}{=} A$ so that $(f^{-1})(y) \in v(A)$. So $y \stackrel{[\text{theorem: 2.72}]}{=} f((f^{-1})(y)) = y \in f(v(A))$. Hence $v(f(A)) \subseteq f(v(A))$ which combined with [eq: 3.12] proves

$$f(v(A)) = v(f(A))$$

6. If $y \in f(\lambda(A))$ then there $\exists x \in \lambda(A)$ such that $y = f(x)$. As $x \in \lambda(A)$, x is a lower bound of A , so that by (4) $y = f(x)$ is a lower bound of $f(A)$. Hence

$$f(\lambda(A)) \subseteq \lambda(f(A)) \quad (3.13)$$

If $y \in \lambda(f(A))$ then by (2) $(f^{-1})(y)$ is a lower bound of $f^{-1}(f(A)) \stackrel{[\text{theorem: 2.57}]}{=} A$ so that $(f^{-1})(y) \in \lambda(A)$. So $y \stackrel{[\text{theorem: 2.72}]}{=} f((f^{-1})(y)) = y \in f(\lambda(A))$. Hence $\lambda(f(A)) \subseteq f(\lambda(A))$ which combined with [eq: 3.12] proves

$$f(\lambda(A)) = \lambda(f(A))$$

7. If $\max(A)$ exist then $\max(A) \in A$ giving $f(\max(A)) \in f(A)$. Let $y \in f(A)$ then $\exists x \in A$ such that $y = f(x)$, as $\max(A)$ exist we have $x \leq_A \max(A)$ so that $y = f(x) \leq_B f(\max(A))$. So

$$\max(f(A)) \text{ exist and } \max(f(A)) = f(\max(A))$$

8. If $\min(A)$ exist then $\min(A) \in A$ giving $f(\min(A)) \in f(A)$. Let $y \in f(A)$ then $\exists x \in A$ such that $y = f(x)$, as $\min(A)$ exist we have $\min(A) \leq_A x$ so that $f(\min(A)) \leq_B f(x) = y$. So

$$\min(f(A)) \text{ exist and } \min(f(A)) = f(\min(A))$$

9. If $\sup(A)$ exists then $\min(v(A))$ exists and $\sup(A) = \min(v(A))$. Using (8) $\min(f(v(A)))$ exist, As $f(v(A)) \stackrel{(\text{5})}{=} v(f(A))$ we have that $\min(v(f(A)))$ exist and

$$\sup(f(A)) = \min(v(f(A))) \stackrel{(\text{5})}{=} \min(f(v(A))) \stackrel{(\text{8})}{=} f(\min(v(A))) = f(\sup(A))$$

10. If $\inf(A)$ exists then $\max(\lambda(A))$ exists and $\inf(A) = \max(\lambda(A))$. Using (7) $\max(f(\lambda(A)))$ exist, As $f(\lambda(A)) \stackrel{(\text{6})}{=} \lambda(f(A))$ we have that $\max(\lambda(f(A)))$ exist and

$$\inf(f(A)) = \max(\lambda(f(A))) \stackrel{(\text{6})}{=} \max(f(\lambda(A))) \stackrel{(\text{7})}{=} f(\max(\lambda(A))) = f(\inf(A)) \quad \square$$

Theorem 3.83. Let $\langle X, \leq_X \rangle$ be a conditional complete partial ordered set, $\langle Y, \leq_Y \rangle$ a partial ordered class and $f: \langle X, \leq_X \rangle \rightarrow \langle Y, \leq_Y \rangle$ a order isomorphism then $\langle Y, \leq_Y \rangle$ is conditionally complete.

Proof. Let $A \subseteq Y$ be such that A is bounded above and non empty. Let u be an upper bound of A then by [lemma: 3.82] we have that $(f^{-1})(u)$ is an upper bound of $f^{-1}(A)$. As $A \neq \emptyset$ there exists a $a \in A$ which as f is surjective means that $\exists x$ such that $a = f(x)$ hence $x \in f^{-1}(A)$ proving that $f^{-1}(A) \neq \emptyset$. As $\langle X, \leq_X \rangle$ is conditional complete $\sup(f^{-1}(A))$ exist. Using [lemma: 3.82] $\sup(f(f^{-1}(A)))$ exist which as $A \stackrel{[\text{theorem: 2.57}]}{=} f(f^{-1}(A))$ proves that $\sup(A)$ exist. So $\langle X, \leq_Y \rangle$ is conditional complete. $\square$

3.3.4 Well ordering

Definition 3.84. A partial ordered class $\langle X, \leq \rangle$ is **well ordered** is every non empty sub-class of X has a least element. In other words if $\forall A \in \mathcal{P}(X) \min(A)$ exist.

Theorem 3.85. If $\langle X, \leq_X \rangle, \langle Y, \leq_Y \rangle$ are partial ordered sets, $f: \langle X, \leq_X \rangle \rightarrow \langle Y, \leq_Y \rangle$ a order isomorphism then if $\langle X, \leq_X \rangle$ is well ordered $\langle Y, \leq_Y \rangle$ is well ordered.

Proof. Let $A \subseteq Y$ be a non empty subclass of Y . Then $\exists a \in A$ and as f is a bijection there exist a $x \in X$ such that $y = f(x)$, from which it follows that $x \in f^{-1}(A)$. So

$$f^{-1}(A) \neq \emptyset$$

As $\langle X, \leq_X \rangle$ is well ordered we have that $f^{-1}(A)$ has a least element, hence

$$\exists m' \in f^{-1}(A) \text{ such that } \forall a \in f^{-1}(A) \text{ we have } m' \leq_X a$$

Take now $m = f(m')$ then as $m' \in f^{-1}(A)$ we have that

$$m \in A \tag{3.14}$$

Further if $a \in A$ then as f is surjective there exists a $b \in X$ such that $a = f(b)$ or $b \in f^{-1}(A)$, so that $m' \leq_X b$. As f is a order isomorphism we have $m = f(m') \leq_Y f(b) = a$. Hence we have proved that

$$\forall a \in A \text{ we have } m \leq a \tag{3.15}$$

From [eq: 3.14] and [eq: 3.15] we conclude finally that $\langle Y, \leq_Y \rangle$ is well ordered. $\square$

Theorem 3.86. If $\langle X, \leq \rangle$ is a partial ordered class, $B \subseteq X$ then for $\langle B, \leq|_B \rangle$ [see theorem: 3.35] we have

1. If $\langle X, \leq \rangle$ is totally ordered then $\langle B, \leq|_B \rangle$ is totally ordered
2. If $\langle X, \leq \rangle$ is well ordered then $\langle B, \leq|_B \rangle$ is totally ordered

Proof.

1. If $x, y \in B \Rightarrow x, y \in X$ hence $x \leq y \vee y \leq x$ so that $x \leq|_B y \vee y \leq|_B x$.
2. If $C \subseteq B$ is a non empty class then as $B \subseteq X$ we have $\emptyset \neq C \subseteq X$. So there exists a least element c of C . So $c \in C$ and $\forall x \in C$ we have $c \leq x \xRightarrow{x \in B} c \leq|_B x$ proving that c is a least element of C using the order relation $\leq|_B$. $\square$

Well ordering is a stronger condition then conditional completeness and totally ordering

Theorem 3.87. Let $\langle X, \leq \rangle$ is a well ordered class then

1. $\langle X, \leq \rangle$ is totally ordered
2. $\langle X, \leq \rangle$ is conditional complete
3. $\forall x, y \in X$ we have $x \leq y$ or $y < x$

Proof.

1. If $x, y \in X$ then $\{x, y\}$ is a non empty sub-class of X and must have a least element. If x is the least element then $x \leq y$ and if y is the least element then $y \leq x$, so $\langle X, \leq \rangle$ is totally ordered.

2. If A is a non empty sub-class of X that is bounded above then $v(A) \neq \emptyset$. Using well ordering we have that $\sup(A) = \min(v(A))$ exist.
3. As by (1) $\langle X, \leq \rangle$ is totally ordered we have that x and y are comparable, hence by [theorem: 3.39] we have $x \leq y \vee y < x$. $\square$

One difference between the order relation on the set of whole numbers $\mathbb{Z}$ and the set of real numbers $\mathbb{R}$ is that there does not exist a whole number between 1 and 2 while for the real numbers there is the real number 1.5 between 1 and 2. This leads to the following definition.

Definition 3.88. (Immediate successor) Let $\langle X, \leq \rangle$ be a partial ordered set and $x, y \in X$ then y is the **immediate successor** of x iff

1. $x < y$
2. $\neg(\exists z \in X \text{ such that } x < z \wedge z < y)$ [in words there does not exists a $x \in X$ such that $x < z < y$]

Theorem 3.89. Let $\langle X, \leq \rangle$ be a well ordered class then every element that is not a greatest element of X has a immediate successor.

Proof. Using [theorem: 3.87] we have that $\langle X, \leq \rangle$ is totally ordered. Let $x \in X$ such that x is not a greatest element in X . Take $B = \{y \in X \mid x < y\}$ then if $B = \emptyset$ we have that $X \setminus B = X$ so $\forall r \in X$ we have $r \notin B$ or $\neg(x < r)$, by [theorem: 3.87] we have that $r \leq x$, proving that x is a greatest element of X which contradicts our hypothesis.. So we must have that $B \neq \emptyset$, by well ordering there exist a least element b of B , which as $b \in B$ gives $x < b$. Assume that there exist a $a \in X$ such that $x < a \wedge a < b$, then we must have that $a \in B$ and $a < b$. As b is the least element of B and $a \in B$ we have $b < a$ leading to the contradiction $a < a$. So b is a immediate successor of x . $\square$

Definition 3.90. Let $\langle X, \leq \rangle$ be a partial ordered class then $B \subseteq X$ is a **section** of X if

$$\forall x \in X \text{ we have } \forall y \in B \text{ with } x \leq y \text{ that } x \in B$$

Lemma 3.91. Let $\langle X, \leq \rangle$ be a well ordered class and $B \subseteq X$ then

$$B \text{ is a section} \Leftrightarrow B = X \text{ or } B \text{ is a initial segment of } X \text{ [definition: 3.46]}$$

Proof.

$\Rightarrow$. Let B be a section of X then if $B = X$ we are done. So we must prove the theorem for $B \neq X$ or equivalently $X \setminus B \neq \emptyset$. Because X is well ordered, there exists a least element $l \in X \setminus B$. Consider the initial segment $S_{X,l} = \{x \in X \mid x < l\}$ [see definition: 3.46]. Let $x \in S_{X,l}$ so that $x < l$. Assume that $x \notin B$ then $x \in X \setminus B$ so, as l is a least element of $X \setminus B$, we have $l \leq x$ which combined with $x < l$ leads to the contradiction $l < l$. So we must have that $x \in B$ which proves that

$$S_{X,l} \subseteq B \tag{3.16}$$

Let $x \in B$, as X is well ordered we have by [theorem: 3.87] that $l \leq x \vee x < l$. Assume that $l \leq x$ then, as B is a section, we have $l \in B$ contradicting $l \in X \setminus B$ [as l is least element of $X \setminus B$]. So we must have $x < l$ or $x \in S_{X,l}$ so $B \subseteq S_{X,l}$. Combining this result with [eq: 3.16] proves

$$S_{X,l} = B$$

$\Leftarrow$. If $X = B$ then $\forall x \in X$ we have $\forall y \in B = X$ with $x \leq y$ that trivially $x \in X = B$, so B is a section. If B is initial segment then there exist a $l \in X$ such that $B = \{y \in X \mid y < l\}$. Take $x \in X$ then if $y \in B$ with $x \leq y$ we have $y < l$ so that $x < l$ hence $x \in B$, proving that B is a section. $\square$

A application of the above lemma is Transfinite Induction.

Theorem 3.92. (Transfinite Induction) Let $\langle X, \leq \rangle$ be a well ordered class and let $P(x)$ a proposition about x [a statement about x that can be true or false] such that

$$\forall x \in X \text{ such that, if } P(y) \text{ is true for every } y < x \text{ then } P(x) \text{ is true} \tag{3.17}$$

then

$$\forall x \in X \ P(x) \text{ is true}$$

Proof. We prove this by contradiction. Assume that $\exists x \in X$ such that $P(x)$ is false, then $B = \{x \in X \mid P(x) \text{ is false}\}$ is non empty. As X is well ordered there exist a least element $l \in B$. Take $x \in X$ with $x < l$ then $x \notin B$ [for if $x \in B$ then $l \leq x$, which combined with $x < l$ gives the contradiction $l < l$] so that $P(x)$ is true. By the hypothesis [eq: 3.17] we have that $P(l)$ is true, which means that $l \notin B$ contradicting $l \in B$. So we must have that $\forall x \in X \ P(x)$ is true. $\square$

Lemma 3.93. Let $\langle X, \leq \rangle$ be a well ordered class, $B \subseteq X$ and $f: \langle X, \leq \rangle \rightarrow \langle B, \leq \rangle$ a order isomorphism then $\forall x \in X$ we have $x \leq f(x)$

Proof. We prove this by contradiction. Assume that that $\exists x \in X$ such that $\neg(x \leq f(x))$. As $\langle X, \leq \rangle$ if well ordered we have by [theorem: 3.87] that $f(x) < x$, hence $C = \{x \in X \mid f(x) < x\} \neq \emptyset$. By well ordering there exists a least element c of C . As $c \in C$ we have that $f(c) < c$, hence by [theorem: 3.53] $f(f(c)) < f(c)$ so that $f(c) \in C$. As c is the least element of C we have $c \leq f(c)$, which combined with $f(c) < c$ gives the contradiction $c < c$. So we must have $\forall x \in X$ that $x \leq f(x)$. $\square$

Theorem 3.94. Let $\langle X, \leq \rangle$ be a well ordered class then there does not exist a order isomorphism from X to a sub-class of an initial segment of X .

Proof. We prove this by contradiction. So assume that there exists a initial segment $S_{X,a} = \{y \in X \mid y < a\}$ of X , a $B \subseteq S_{X,a}$ and a order isomorphism $f: \langle X, \leq \rangle \rightarrow \langle B, \leq \rangle$. Using the previous lemma [lemma: 3.93] we have that $a \leq f(a)$, so $f(a) \notin S_{X,a}$ [for if $f(a) \in S_{X,a}$ then $f(a) < a$ leading to the contradiction $a < a$]. However as $\text{range}(f) = B \subseteq S_{X,a}$ we must have that $f(a) \in S_{X,a}$ and we reach a contradiction. $\square$

Corollary 3.95. Let $\langle X, \leq \rangle$ be a well ordered class then there does not exist a order isomorphism between X and initial segment of X

Proof. As a initial segment is a sub-class of itself this follows from the previous theorem [theorem: 3.94] $\square$

Theorem 3.96. If $\langle X, \leq_X \rangle, \langle Y, \leq_Y \rangle$ are well ordered classes then if X is order isomorphic with an initial segment of Y we have that Y is not order isomorphic with any sub-class of X .

Proof. Let $S_{Y,y}$ be a initial segment of Y and $f: \langle X, \leq_X \rangle \rightarrow \langle S_{Y,y}, \leq_Y \rangle$ a order isomorphism. Assume that there exist a $A \subseteq X$ and a order isomorphism $g: \langle Y, \leq_Y \rangle \rightarrow \langle A, \leq_A \rangle$. As by [lemma: 2.35], [theorem: 2.54] and the fact that 'increasing' is a property of the graph of a function, we have that $g: \langle Y, \leq_Y \rangle \rightarrow \langle X, \leq_X \rangle$ is a injective increasing function. Using [theorem: 2.77], [theorem: 3.52] we have that $f \circ g: \langle Y, \leq_Y \rangle \rightarrow \langle S_{Y,y}, \leq_Y \rangle$ is a injective increasing function, hence $f \circ g: Y \rightarrow (f \circ g)(Y)$ is a bijective function [see theorem: 2.68] which is increasing, hence by [theorem: 3.57] we have that $f \circ g: \langle Y, \leq_Y \rangle \rightarrow \langle (f \circ g)(Y), \leq_Y \rangle$ is a order isomorphism. As $(f \circ g)(Y) \subseteq \text{range}(f)$ [see theorem: 2.22] and $\text{range}(f) \subseteq S_{Y,y}$ we have a order isomorphism between Y and a sub-class of a initial segment of Y . By [theorem: 3.94] this is impossible so the assumption is false, hence Y is not order isomorphic to a an initial segment of Y . $\square$

Corollary 3.97. If $\langle X, \leq_X \rangle, \langle Y, \leq_Y \rangle$ are well ordered classes such that X is order isomorphic with Y then

1. X can not be order isomorphic with a initial segment of Y
2. Y can not be order isomorphic with a initial segment of X

Proof. We prove this by contradiction. First by the hypothesis we have $X \cong Y$ and by [theorem: 3.56] $Y \cong X$.

1. If X is order isomorphic with a initial segment of Y then as $Y \cong X$ we have that Y is order isomorphic with a sub-class of X , which by [theorem: 3.96] is not allowed.

2. If Y is order isomorphic with a initial segment of X then as $X \cong Y$ we have that X is order isomorphic with a sub-class of Y , which by [theorem: 3.96] is not allowed. $\square$

Lemma 3.98. *Let $\langle X, \leq \rangle$ be a well ordered class and $a, b \in X$ with $a < b$ then $S_{X,a}$ is a initial segment of $S_{X,b}$ [using the order $\leq_{|S_{X,y}}$]*

Proof. First if $x \in S_{X,a}$ then $x < a \xRightarrow{a < b} x < b$ so that $x \in S_{X,b}$, hence

$$S_{X,a} \subseteq S_{X,b}$$

Now if $x \in S_{X,b}$ and $y \in S_{X,a}$ is such that $x \leq_{|S_{X,b}} y$ then $x \leq y \xRightarrow{y \in S_{X,a} \Rightarrow y < a} x < a$ hence $x \in S_{X,a}$. So $S_{X,a}$ is a section of $S_{X,b}$, as $a \notin S_{X,a} \wedge a \in S_{X,b}$ [for $a < b$] we have $S_{X,a} \neq S_{X,b}$ so that, using [theorem: 3.91], $S_{X,a}$ is a initial segment of $S_{X,b}$. $\square$

Theorem 3.99. *Let $\langle X, \leq_X \rangle$ and $\langle Y, \leq_Y \rangle$ be well ordered classes then exactly one of the following cases hold*

1. X is order isomorphic with Y
2. X is order isomorphic with an initial segment of Y
3. Y is order isomorphic with an initial segment of X

Proof. Define

$$C = \{x \in X \mid \exists y \in Y \text{ such that } S_{X,x} \cong S_{Y,y}\} \quad (3.18)$$

and

$$F = \{(x, y) \in C \times Y \mid S_{X,x} \cong S_{Y,y}\} \quad (3.19)$$

We prove now that F is the graph of a order isomorphism between C and $F(C)$. We have trivially from the definition of F that

$$F \subseteq C \times Y \quad (3.20)$$

Let $(x, y), (x', y') \in F$, then $S_{X,x} \cong S_{Y,y}$ and $S_{X,x'} \cong S_{Y,y'}$ so by [theorem: 3.56]

$$S_{Y,y} \cong S_{Y,y'} \quad (3.21)$$

Assume that $y \neq y'$ then, as $\langle Y, \leq_Y \rangle$ is well ordered we have by [theorem: 3.87] either:

$y \leq y'$. then $y < y'$ so that by the previous lemma [lemma: 3.98] we have that $S_{Y,y}$ is a initial segment of $S_{Y,y'}$. Using [corollary: 3.95] we have then that $S_{Y,y}$ is not order isomorphic with $S_{Y,y'}$ contradicting [eq: 3.21].

$y' < y$. then by the previous lemma [lemma: 3.98] we have that $S_{Y,y'}$ is a initial segment of $S_{Y,y}$. Using [corollary: 3.95] we have then that $S_{Y,y}$ is not order isomorphic with $S_{Y,y'}$ contradicting [eq: 3.21].

as in all cases we have a contradiction, the assumption must be wrong. Hence

$$\text{If } (x, y), (x', y') \in F \text{ then } y = y' \quad (3.22)$$

Further if $x \in C$ then by definition of C there exists a $y \in Y$ such that $S_{X,x} \cong S_{Y,y}$ hence $(x, y) \in F$ proving that

$$C \subseteq \text{dom}(F) \quad (3.23)$$

If $(x, y), (x', y) \in F$ then $S_{X,x} \cong S_{Y,y}$ and $S_{X,x'} \cong S_{Y,y}$ so by [theorem: 3.56] we have that

$$S_{X,x} \cong S_{X,x'} \quad (3.24)$$

Assume that $x \neq x'$ then, as $\langle X, \leq_X \rangle$ is well ordered we have by [theorem: 3.87] either:

$x \leq x'$. then $x < x'$ so that by the previous lemma [lemma: 3.98] we have that $S_{X,x}$ is a initial segment of $S_{X,x'}$. Using [corollary: 3.95] we have then that $S_{X,x}$ is not order isomorphic with $S_{X,x'}$ contradicting [eq: 3.24].

$x' \leq x$. then by the previous lemma [lemma: 3.98] we have that $S_{X,x'}$ is a initial segment of $S_{X,x}$. Using [corollary: 3.95] we have then that $S_{X,x}$ is not order isomorphic with $S_{X,x'}$ contradicting [eq: 3.24].

as in all cases we have a contradiction, the assumption must be wrong. Hence

$$\text{If } (x, y), (x', y) \in F \text{ we have } x = x' \quad (3.25)$$

Combining [eq: 3.20], [eq: 3.22], [eq: 3.23] and [eq: 3.25] it follows that $F: C \rightarrow Y$ is a injective function. Applying then [proposition: 2.68] gives if we define $D = F(C)$

$$F: C \rightarrow D \text{ is a bijection} \quad (3.26)$$

Take $x, y \in C$ such that $x \leq_X y$ then by definition of F we have

$$S_{X,x} \cong S_{Y,F(x)} \text{ and } S_{X,y} \cong S_{Y,F(y)} \quad (3.27)$$

Assume now that $\neg(F(x) \leq_Y F(y))$ then as $\langle Y, \leq_Y \rangle$ is well ordered we have by [theorem: 3.87] that $F(y) <_Y F(x)$. So using [theorem: 3.98] we have that $S_{Y,F(y)}$ is a initial segment of $S_{Y,F(x)}$. As $x \leq_X y$ it follows that $S_{X,x} \subseteq S_{X,y}$ [see proposition: 3.47]. So we have using [eq: 3.27]

- a) $S_{X,y}$ is order isomorphic with $S_{Y,F(y)}$ a initial segment of $S_{Y,F(x)}$
- b) $S_{F(x)}$ is order isomorphic with $S_{X,x}$ a sub-class of $S_{X,y}$

Using [theorem: 3.96] we see that (a) and (b) can not be all true, hence our assumption is false so that $F(x) \leq F(y)$. Hence we have that $F: C \rightarrow D$ is a increasing bijection which by [theorem: 3.57] proves that

$$F: \langle C, \leq_X \rangle \rightarrow \langle D, \leq_Y \rangle \text{ is a order isomorphism or } C \cong D \quad (3.28)$$

Next we prove that

$$C \text{ is a section of } X \quad (3.29)$$

Proof. Let $x \in X$ and take $c \in C$ such that $x \leq_X c$. As $S_{X,c} \cong S_{Y,F(c)}$ there exist a order isomorphism

$$g: S_{X,c} \rightarrow S_{Y,F(c)} \quad (3.30)$$

Now as $x \leq_X c$ we have by [proposition: 3.47] that $S_{X,x} \subseteq S_{X,c}$. Hence by 2.98 we have that

$$g|_{S_{X,x}}: S_{X,x} \rightarrow S_{X,c} \text{ is a function} \quad (3.31)$$

Further if $y \in S_{X,x}$ we have that $y <_X x$, so as g is a order isomorphism we have $g(y) <_Y g(x)$ proving that $g|_{S_{X,x}}(y) = g(y) \in S_{Y,g(x)}$ or $\text{range}(g|_{S_{X,x}}) \subseteq S_{Y,g(x)}$. So by [theorem: 2.39] it follows that

$$g|_{S_{X,x}}: S_{X,x} \rightarrow S_{Y,g(x)} \text{ is a function} \quad (3.32)$$

As g is a isomorphism and thus injective it follows from [theorem: 2.92] that

$$g|_{S_{X,x}}: S_{X,x} \rightarrow S_{Y,g(x)} \text{ is injective} \quad (3.33)$$

Further if $y \in S_{Y,g(x)}$ then $y <_Y g(x)$, as $g(x) \in S_{Y,F(c)}$ [see eq: 3.30] we have $g(x) <_Y F(c)$ so that $y <_Y F(c)$ proving $y \in S_{Y,F(c)}$. As g is surjective there exist a $u \in S_{X,c}$ such that $y = g(u)$. Assume that $x \leq_X u$ then $g(x) \leq_Y g(u) = y$, as $y <_Y g(x)$ this gives the contradiction $g(x) < g(x)$. So we have $\neg(x \leq_X u)$ which, as $\langle X, \leq_X \rangle$ is well ordered, gives by [theorem: 3.87] that $u <_X x$ so that $u \in S_{X,x}$. So for $y \in S_{Y,g(x)}$ we found a $u \in S_{X,x}$ such that $g|_{S_{X,x}}(u) = g(u) = y$ proving that

$$g|_{S_{X,x}}: S_{X,x} \rightarrow S_{Y,g(x)} \text{ is surjective} \quad (3.34)$$

Further if $u, v \in S_{X,x}$ are such that $u \leq_X v$ so that $g|_{S_{X,x}}(u) = g(u) \leq_X g(v) = g|_{S_{X,x}}(v)$ proving that

$$g|_{S_{X,x}}: S_{X,x} \rightarrow S_{Y,g(x)} \text{ is increasing} \quad (3.35)$$

Combining [eq: 3.31], [eq: 3.32], [eq: 3.34], [eq: 3.35] we have that $g|_{S_{X,x}}: \langle S_{X,x}, \leq_X \rangle \rightarrow \langle S_{Y,g(x)}, \leq_Y \rangle$ is a order isomorphism so that $S_{X,x} \cong S_{Y,g(x)}$ hence $x \in C$. Proving that C is as section of X . $\square$

Next we prove that

$$D \text{ is a section of } Y \quad (3.36)$$

Proof. Let $y \in Y$ and take $d \in D$ such that $y \leq_Y d$. As $d \in D = \text{range}(F)$ there exist a $c \in C$ such that $F(c) = d$, so $S_{X,c} \cong S_{Y,d} \xrightarrow{[\text{theorem: 3.56}]} S_{Y,d} \cong S_{X,c}$. So there exist a order isomorphism

$$f: S_{Y,d} \rightarrow S_{X,c} \quad (3.37)$$

Now from $y \leq_D d$ we have by [theorem: 3.47] $S_{Y,y} \subseteq S_{Y,d}$. Hence by 2.98 we have that

$$f|_{S_{Y,y}}: S_{Y,y} \rightarrow S_{X,c} \text{ is a function} \quad (3.38)$$

If $x \in S_{Y,y}$ then $x <_Y y$ so, as f is a order isomorphism, $f|_{S_{Y,y}}(x) = f(x) <_X f(y)$, we have that $f|_{S_{Y,y}}(x) \in S_{Y,f(y)}$, so $\text{range}(f|_{S_{Y,y}}) \subseteq S_{X,f(y)}$. By [theorem: 2.39] it follows that

$$f|_{S_{Y,y}}: S_{Y,y} \rightarrow S_{X,f(y)} \text{ is a function} \quad (3.39)$$

As f is a isomorphism and injective it follows from [theorem: 2.92] that

$$f|_{S_{Y,y}}: S_{Y,y} \rightarrow S_{X,f(y)} \text{ is injective} \quad (3.40)$$

If $x \in S_{X,f(y)}$ then $x <_X f(y)$, as by [eq: 3.37] $f(y) \in S_{X,c}$, we have $f(y) < c$, so that $x <_X c$ or $x \in S_{X,c}$. As f is surjective there exists a $u \in S_{Y,d}$ such that $f(u) = x$. As $u \in S_{Y,d}$ we have that $u <_Y d$. Assume now that $y \leq_Y u$ then, as f is a order isomorphism, $f(y) \leq_X f(u) = x$, which as $x <_X f(y)$ gives the contradiction $x <_X x$. So we must have that $\neg(y \leq_Y u)$, which, as $\langle Y, \leq_Y \rangle$ is well ordered, gives by [theorem: 3.87] that $u <_Y y$ or $u \in S_{Y,y}$. So for $x \in S_{X,f(y)}$ there exist a $u \in S_{Y,y}$ such that $f(u) = x$, proving that

$$f|_{S_{Y,y}}: S_{Y,y} \rightarrow S_{X,f(y)} \text{ is surjective} \quad (3.41)$$

Further if $u, v \in S_{Y,y}$ is such that $u \leq v$ then $f|_{S_{Y,y}}(u) = f(u) \leq f(v) = f|_{S_{Y,y}}(v)$ proving that

$$f|_{S_{Y,y}}: S_{Y,y} \rightarrow S_{X,f(y)} \text{ is increasing} \quad (3.42)$$

Combining [eq: 3.39], [eq: 3.40], [eq: 3.41] and [eq: 3.42] we have that

$$f|_{S_{Y,y}}: \langle S_{Y,y}, \leq \rangle \rightarrow \langle S_{X,f(y)}, \leq_X \rangle \text{ is a order isomorphism,}$$

hence $S_{Y,y} \cong S_{X,f(y)}$. As $f(y) \in S_{X,c} \subseteq X$ and $y \in Y$ it follows from the definition of C that $f(y) \in C$, hence by definition of F $(f(y), y) \in F$ or $y = F(f(y)) \in F(C) = D$, giving $y \in D$. Proving that D is a section of Y . $\square$

To summarize [eq: 3.28], [eq: 3.29] and [eq: 3.36] we have

$$C \cong D \wedge C \text{ is a segment of } X \wedge D \text{ is a segment of } Y \quad (3.43)$$

Assume now that C is a initial segment of X and D is a initial segment of Y then there exist a $r \in X$ and a $s \in Y$ such that $C = S_{X,r}$ and $D = S_{Y,s}$. By 3.43 we have that $S_{X,r} \cong S_{Y,s}$ which by definition of C means that $r \in C$ or as $C = S_{X,r}$ that $r < r$ a contradiction. So we have that

$$\neg(C \text{ is a initial segment of } X \wedge D \text{ is a initial segment of } Y) \quad (3.44)$$

As C is a section of X we have by [theorem: 3.91] that

$$X = C \text{ or } C \text{ is a initial segment of } X \quad (3.45)$$

Like wise, as D is a section of Y we have by [theorem: 3.91] that

$$Y = D \text{ or } D \text{ is a initial segment of } Y \quad (3.46)$$

We have taking [eq: 3.45] and [eq: 3.46] in account that either:

$$X = C \wedge Y = D. \text{ then by [eq: 3.43]}$$

$$X \cong Y$$

Using theorem [theorem: 3.97] and the above we have that

X is not order isomorphic with a sub-class of Y

Y is not order isomorphic with a sub-class of X

$X = C \wedge Y \neq D$. then by [eq: 3.46] we have that D is a initial segment of Y , which as by [eq: 3.43] $X = C \cong D$ prove that

X is order isomorphic with a initial segment of Y

If Y is order isomorphic with a initial segment of X then by [theorem: 3.96] we have that X is not order isomorphic to a subset of Y contradicting $X \cong D$ and $X \cong Y$. So

Y is not order isomorphic to a initial segment of X

X is not order isomorphic to Y

$X \neq C \wedge Y = D$. then by [eq: 3.45] we have that C is a initial segment of X , which as by [eq: 3.43] $C \cong D \xRightarrow{[\text{theorem: 3.56}]} Y = D \cong C$ proves that

Y is order isomorphic with a initial segment of X

If X is order isomorphic with a initial segment of Y then by [theorem: 3.96] we have that Y is not order isomorphic to a subset of X contradicting $Y \cong C$ and $Y \cong X$. So

X is not order isomorphic to a initial segment of Y

X is not order isomorphic to Y

$X \neq C \wedge Y \neq D$. Using [eq: 3.45] and [eq: 3.46] we have that C is a initial segment of X and D is a initial segment of Y which contradicts [eq: 3.44]. Hence this case does not apply. $\square$

Corollary 3.100. Let $\langle X, \leq \rangle$ be a well ordered class and $Y \subseteq X$ then we have either (but not both):

1. Y is order isomorphic with X
2. X is order isomorphic with a initial segment of X

Proof. If $Y \subseteq X$ then $\langle Y, \leq|_Y \rangle$ is a well ordered class [see theorem: 3.86], so using the previous [theorem: 3.99] we have either:

1. Y is order isomorphic with X
2. Y is order isomorphic with a initial segment of X
3. X is order isomorphic with a initial segment of Y . By [theorem: 3.96] we may not have that Y is order isomorphic with a sub-class of X . As by [theorem: 3.56] $Y \cong Y$ and Y is a sub-class of X we reach a contradiction, so this case never applies. $\square$

3.4 Axiom of choice

The axiom of choice in it's many equivalent forms like

Hausdorff's Maximal Principle

Zorn's Lemma

Well – Ordering Theorem

plays a major role in some fundamental theorems about the product of sets, the existence of a basis for a vector space, etc.

Definition 3.101. Let A be a class then $\mathcal{P}'(A)$ is defined as

$$\mathcal{P}'(A) = \mathcal{P}(A) \setminus \{\emptyset\}$$

In other words it is the collection of all non empty sub sets of a set

It turns out that if A is a set then $\mathcal{P}'(A)$ is also a set.

Theorem 3.102. *If A is a set then $\mathcal{P}'(A)$ is a set*

Proof. Using the Axiom of Power [axiom 1.70] we have that $\mathcal{P}(A)$ is a set. As $\mathcal{P}'(A) \subseteq \mathcal{P}(A)$ [see [theorem: 1.30]] it follow from the Axiom of Subsets [axiom: 1.8] that $\mathcal{P}'(A)$ is a set. $\square$

Definition 3.103. (Choice Function) *Let A be a set then a **choice function for A** is a function $f: \mathcal{P}'(A) \rightarrow A$ such that $\forall B \in \mathcal{P}'(A)$ we have $f(B) \in B$*

So a choice function picks out one element out of each subset of A and the axiom of choice ensures the existence of a choice function for a set.

Axiom 3.104. (Axiom of Choice) *If A is a set then there exist a choice function for A*

As a application of the axiom of choice we have the following theorem

Theorem 3.105. *If $f: A \rightarrow B$ is a surjective function then there exists a injective function $g: B \rightarrow A$ such that $f \circ g = \text{Id}_B$*

Proof. By the axiom of choice there exists a choice function

$$c: \mathcal{P}'(A) \rightarrow A \text{ such that } \forall A \in \mathcal{P}'(A) \text{ we have } c(A) \in A$$

If $f: A \rightarrow B$ is surjective. Then $\forall y \in B$ we have that $f^{-1}(\{y\})$ is a non empty subset of $A \Rightarrow f^{-1}(\{y\}) \in \mathcal{P}'(A)$. Define then the function

$$g: B \rightarrow Y \text{ by } g(y) = c(f^{-1}(\{y\}))$$

Now if $y \in Y$ then, as c is a choice function, $c(f^{-1}(\{y\})) \in f^{-1}(\{y\})$ so that $f(c(f^{-1}(\{y\}))) = y$. Hence we have that $(f \circ g)(y) = f(g(y)) = f(c(f^{-1}(\{y\}))) = y$ or

$$f \circ g = \text{Id}_B$$

If $g(y) = g(y')$ then we have $f(g(y)) = f(g(y')) \xrightarrow{f \circ g = \text{Id}_B} \text{Id}_B(y) = \text{Id}_B(y') \Rightarrow y = y'$ proving that

$$g: B \rightarrow Y \text{ is injective}$$

$\square$

The important thing to remember in the above is that the axiom of choice ensures the existence of $g: B \rightarrow A$ but does not give a way to construct the function g itself.

We have the following equivalent statements of the axiom of choice

Theorem 3.106. *The following are equivalent*

1. *The Axiom of Choice*
2. *Let $\mathcal{A}$ be a set of sets such that:*

- a. $\forall A \in \mathcal{A}$ we have $A \neq \emptyset$
- b. $\forall A, B \in \mathcal{A}$ with $A \neq B$ we have $A \cap B = \emptyset$

*then there exist a set C called the **choice set for $\mathcal{A}$** such that*

- a. $C \subseteq \bigcup \mathcal{A}$
- b. $\forall A \in \mathcal{A}$ we have $A \cap C \neq \emptyset$ and if $y, y' \in A \cap C$ then $y = y'$

In other words C consists of exactly one element from each $A \in \mathcal{A}$.

3. *If $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ is a family of non empty sets $[\forall i \in I \text{ we have } A_i \neq \emptyset]$ where $I, \mathcal{A}$ are sets then there exists a function $f: I \rightarrow \bigcup_{i \in I} A_i$ such that $\forall i \in I$ we have $f(i) \in A_i$.*

Note 3.107. *Using the definition of a family [see definition: 2.110] this is equivalent with saying that there exist a family $\{f_i\}_{i \in I} \subseteq \bigcup_{i \in I} A_i$ such that $\forall i \in I$ we have $f_i \in A_i$.*

Proof.

1 $\Rightarrow$ 2. Take $U = \bigcup \mathcal{A}$ [see definition: 1.62]. As $\mathcal{A}$ is a set we have by the Axiom of Union [axiom: 1.67] that U is a set. So we can apply the Axiom of Choice [axiom: 3.104] to get a function

$$c: \mathcal{P}'(U) \rightarrow U \text{ such that } \forall A \in \mathcal{P}'(U) \text{ we have } c(A) \in A$$

If $A \in \mathcal{A}$ then $A \neq \emptyset$ and using [theorem: 1.66] we have $A \subseteq U$ proving that $A \in \mathcal{P}'(U)$ hence

$$\mathcal{A} \subseteq \mathcal{P}'(U)$$

so we can take the **image** of $\mathcal{A}$ by c

$$C = c(\mathcal{A})$$

We have now:

- a) If $x \in C$ then $\exists A \in \mathcal{A}$ such that $x = (c)(A)$, which as c is a choice function means that $x \in A$ hence, by [theorem: 1.66], we have that $x \in \bigcup \mathcal{A}$ proving that

$$C \subseteq \bigcup \mathcal{A}$$

- b) Let $A \in \mathcal{A}$ then $(c)(A) \in c(\mathcal{A}) = C$ and, as c is a choice function, $(c)(A) \in A$ [note: $(c)(A)$ is function application and $c(\mathcal{A})$ is the image of $\mathcal{A}$ by c]. Hence

$$A \cap C \neq \emptyset$$

If $y, y' \in A \cap C$ then as $y, y' \in C = c(\mathcal{A})$ there exist $Y, Y' \in \mathcal{A}$ such that $y = (c)(Y)$ and $y' = (c)(Y')$, as c is a choice function we have $y = (c)(Y) \in Y$ and $y' = (c)(Y') \in Y'$. Assume that $Y \neq Y'$ then we have the contradiction $y, y' \in Y \cap Y' = \emptyset$, so we have that $Y = Y'$ but then $y = c(Y) = c(Y') = y'$ proving that $y = y'$. So

$$y, y' \in A \cap C \Rightarrow y = y'$$

so (2.a) and (2.b) is proved.

2 $\Rightarrow$ 1. Let A be a set and let $B \in \mathcal{P}'(A)$ then $\emptyset \neq B \subseteq A$. Define now

$$P_B = \{(B, x) | x \in B\} \quad (3.47)$$

If $(B, x) \in P_B$ then as $B \in \mathcal{P}'(A)$ and $x \in B \subseteq A$ we have $(B, x) \in \mathcal{P}'(A) \times A$ or

$$P_B \subseteq \mathcal{P}'(A) \times A \text{ or } P_B \in \mathcal{P}(\mathcal{P}'(A) \times A) \quad (3.48)$$

As $B \neq \emptyset$ we have that $\exists b \in B$ so that $(B, b) \in P_B$ proving that

$$\forall B \in \mathcal{P}'(A) \text{ we have } P_B \neq \emptyset \quad (3.49)$$

If $x \in P_B \cap P_{B'}$ then $\exists b \in B$ and $b' \in B'$ such that $(B, b) = x = (B', b')$ proving that $B = B'$, hence $P_B = P_{B'}$. From this it follows that

$$\forall B, B' \in \mathcal{P}'(A) \text{ we have If } P_B \neq P_{B'} \text{ then } P_B \cap P_{B'} = \emptyset \quad (3.50)$$

Define

$$\mathcal{A} = \{P_B | B \in \mathcal{P}'(A)\} \subseteq \mathcal{P}(\mathcal{P}'(A) \times A) \quad (3.51)$$

As A is a set we have by [theorem: 3.102] that $\mathcal{P}'(A)$ is a set, using [theorem: 1.73] it follow that $\mathcal{P}'(A) \times A$ is a set, applying the Axiom of Power sets [axiom: 1.70] proves that $\mathcal{P}(\mathcal{P}'(A) \times A)$ is a set. As by [eq: 3.51] we have that $\mathcal{A} \subseteq \mathcal{P}(\mathcal{P}'(A) \times A)$ we can use the Axiom of Sub Sets [axiom: 1.8] giving

$$\mathcal{A} \text{ is a set} \quad (3.52)$$

So the conditions for the hypothesis (2) are satisfied by [eq: 3.52], [eq: 3.49] and [eq: 3.50] hence there exist a choice set C for $\mathcal{A}$ such that:

$$C \subseteq \bigcup \mathcal{A} \text{ and } \forall B \in \mathcal{A} \text{ we have } B \cap C \neq \emptyset \text{ and if } y, y' \in B \cap C \text{ then } y = y' \quad (3.53)$$

If $x \in C$ then $\exists y \in \mathcal{A}$ such that $x \in y$. As $y \in \mathcal{A}$ there exists a $B \in \mathcal{P}'(A)$ such that $y = P_B = \{(B, x) | x \in B\}$, hence there exist a $b \in B$ such that $x = (B, b) \in P_B \subseteq \mathcal{P}'(A) \times A$ [see eq: 3.48] proving that

$$C \subseteq \mathcal{P}'(A) \times A \quad (3.54)$$

If $(B, y), (B, y') \in C$ then $(B, y), (B, y') \in P_B \cap C \xrightarrow[\text{eq: 3.53}]{\Rightarrow} (B, y) = (B, y')$ proving that $y = y'$, so

$$\text{If } (B, y), (B, y') \in C \text{ then } y = y' \quad (3.55)$$

Let $B \in \mathcal{P}'(A)$ then $P_B \in \mathcal{A}$ so that by [eq: 3.53] $P_B \cap C \neq \emptyset$ hence there exist a $y \in B$ such that $(B, y) \in C$ proving that

$$\mathcal{P}'(A) \subseteq \text{dom}(C) \quad (3.56)$$

From [eq: 3.54], [eq: 3.55] and [eq: 3.56] it follows that

$$C: \mathcal{P}'(A) \rightarrow A \text{ is a function} \quad (3.57)$$

Let $B \in \mathcal{P}'(A)$ then $(B, C(B)) \in C \subseteq \bigcup \mathcal{A}$ so that $\exists B' \in \mathcal{P}'(A)$ such that $(B, C(B)) \in P_{B'}$, hence $B = B'$ and $C(B) \in B' = B$ proving that $\forall B \in \mathcal{P}'(A)$ we have $C(B) \in B$, so that

$$C: \mathcal{P}'(A) \rightarrow A \text{ is a choice function}$$

proving (1)

1 $\Rightarrow$ 3. Let $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ be a family of non empty sets where $I, \mathcal{A}$ are sets. Then using [theorem: 2.128] it follows that $\bigcup_{i \in I} A_i$ is a set. Using the Axiom of Choice [axiom: 3.104] there exist a choice function

$$c: \mathcal{P}'\left(\bigcup_{i \in I} A_i\right) \rightarrow \bigcup_{i \in I} A_i \text{ where } \forall A \in \mathcal{P}'\left(\bigcup_{i \in I} A_i\right) c(A) \in A$$

Let $A: I \rightarrow \mathcal{A}$ be the function that defines $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ then $\forall i \in I$ we have that $A(i) = A_i \subseteq \bigcup_{i \in I} A_i$ [see: theorem: 2.135] or $A(i) \in \mathcal{P}(\bigcup_{i \in I} A_i)$, further as $A_i \neq \emptyset$ we have that $A_i \in \mathcal{P}'(\bigcup_{i \in I} A_i)$, hence $\text{range}(A) \subseteq \mathcal{P}'(\bigcup_{i \in I} A_i)$. Using [theorem: 2.39] it follows that $A: I \rightarrow \mathcal{P}'(\bigcup_{i \in I} A_i)$ is also a function. If we take $f = c \circ A$ then

$$f: I \rightarrow \bigcup_{i \in I} A_i \text{ is a function and } \forall i \in I \text{ we have } f(i) = c(A(i)) = c(A_i) \in A_i$$

proving (3).

3 $\Rightarrow$ 1. Let A be a set and define the family $\{B_C\}_{C \in \mathcal{P}'(A)} \subseteq \mathcal{P}'(A)$ by $B = \text{Id}_{\mathcal{P}'(A)}: \mathcal{P}'(A) \rightarrow \mathcal{P}'(A)$ [see example: 2.49]. For every $C \in \mathcal{P}'(A)$ we have $B_C = \text{Id}(C) = C \neq \emptyset$, further as A is a set we have by [theorem: 3.102] that $\mathcal{P}'(A)$ is a set. So the conditions for (3) are satisfied and by (3) there exist a function

$$f: \mathcal{P}'(A) \rightarrow \bigcup_{C \in \mathcal{P}'(A)} B_C \text{ such that } \forall C \in \mathcal{P}'(A) \text{ we have } f(C) \in B_C = \text{Id}(C) = C \quad (3.58)$$

Let $x \in \bigcup_{C \in \mathcal{P}'(A)} B_C$ then $\exists C \in \mathcal{P}'(A)$ such that $x \in B_C = \text{Id}_{\mathcal{P}'(A)}(C) = C \subseteq A \Rightarrow x \in A$. So $\bigcup_{C \in \mathcal{P}'(A)} B_C \subseteq A$. Using then [theorem: 2.35] we have

$$f: \mathcal{P}'(A) \rightarrow A \text{ is a function with } \forall C \in \mathcal{P}'(A) \text{ we have } f(C) \in C$$

which proves that $f: \mathcal{P}'(A) \rightarrow A$ is a choice function for A , proving (1). $\square$

Theorem 3.108. Let A, B be sets such that $\forall a \in A$ there exist a $b \in B$ satisfying $P(a, b)$ [where $P(a, b)$ is a predicate] then there exist a function $f: A \rightarrow B$ such that $\forall a \in A$ $P(a, f(a))$. Furthermore as a function defines a family you can also say that there is a family $\{f_a\}_{a \in A} \subseteq B$ such that $\forall a \in A$ $P(a, f_a)$.

Proof. Let $a \in A$ then by the hypothesis the set $\{b \in B | P(a, b)\}$ is non empty. This allows us to define the function

$$\mathcal{A}: A \rightarrow \mathcal{P}'(B) \text{ by } \mathcal{A}(a) = \{b \in B | P(a, b)\} \subseteq B$$

defining the family

$$\{\mathcal{A}_a\}_{a \in A} \subseteq \mathcal{P}'(B)$$

Applying then [theorem: 3.106 (3)] there exist a function

$$f: A \rightarrow \bigcup_{a \in A} \mathcal{A}_a$$

such that $\forall a \in A$ we have $f(a) \in \mathcal{A}_a$ so that $f(a)$ satisfies $P(a, f(a))$. Further as $\forall a \in A$ $\mathcal{A}_a \subseteq B$ we have that $\bigcup_{a \in A} \mathcal{A}_a \subseteq B$, hence we have that

$$f: A \rightarrow B$$

is a function such that $\forall a \in A$ $P(a, f(a))$. □

As a application of the Axiom of Choice we have the following theorems about the product of a family of sets. First we prove that the projection function is surjective.

Theorem 3.109. *Let $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ be a family of **non empty sets** where $I, \mathcal{A}$ are sets then $\forall i \in I$ we have that the projection function*

$$\pi_i: \prod_{j \in I} A_j \rightarrow A_i \text{ defined by } \pi_j(x) = x(j) \text{ [see definition: 2.156]}$$

is a surjection.

Proof. Let $i \in I$ and take $x \in A_i$. Consider the family $\{A_j\}_{j \in I \setminus \{i\}}$ [see definition: 2.116] then $\forall j \in I \setminus \{i\}$ we have $A_j \neq \emptyset$. So we can use [theorem: 3.106 (3)] to find a function

$$f: I \setminus \{i\} \rightarrow \bigcup_{j \in I \setminus \{i\}} A_j \text{ such that } \forall j \in I \setminus \{i\} \text{ we have } f(j) \in A_j$$

By the definition of the product of a family of sets we have that

$$f \in \prod_{j \in I \setminus \{i\}} A_j$$

Define now $g: I \rightarrow \bigcup_{j \in I} A_j$ by $g(j) = \begin{cases} x & \text{if } j = i \\ f(j) & \text{if } j \in I \setminus \{i\} \end{cases}$ then by [theorem: 2.155] we have that $g \in \prod_{i \in I} A_i$. Finally by $\pi_i(g) = g(i) = x$ proving surjectivity. □

Second we prove that the product of a family of sets is not empty if and only if every set in the family is non empty.

Theorem 3.110. *Let $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ be a family of sets where $I, \mathcal{A}$ are sets then we have*

$$\prod_{i \in I} A_i \neq \emptyset \Leftrightarrow \forall i \in I \text{ we have } A_i \neq \emptyset$$

Proof.

$\Rightarrow$. We prove this by contradiction, so assume that $\exists i \in I$ such that $A_i = \emptyset$. As $\prod_{i \in I} A_i \neq \emptyset$ there exists a $x \in \prod_{i \in I} A_i$ such that $\forall j \in I$ $x_j \in A_j$, in particular we would have $x_i \in A_i$ contradicting $A_i = \emptyset$. So we must have that $\forall i \in I$ we have $A_i \neq \emptyset$.

$\Leftarrow$. If $\forall i \in I$ we have $A_i \neq \emptyset$ we have by [theorem: 3.106 (3)] that there exist a function

$$f: I \rightarrow \bigcup_{i \in I} A_i \text{ such that } \forall i \in I \text{ we have } f(i) \in A_i$$

which by definition of the product means that $f \in \prod_{i \in I} A_i$ proving that

$$\prod_{i \in I} A_i \neq \emptyset \quad \square$$

We can rephrase the above theorem in another way.

Theorem 3.111. Let $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ be a family of sets where $I, \mathcal{A}$ are sets then we have

$$\prod_{i \in I} A_i = \emptyset \Leftrightarrow \exists k \in I \text{ with } A_k = \emptyset$$

Proof.

$\Rightarrow$. Assume that $\forall k \in I A_k \neq \emptyset$ it follows from the previous theorem [theorem: 3.110] it follows that $\prod_{i \in I} A_i \neq \emptyset$ contradicting the hypothesis $\prod_{i \in I} A_i = \emptyset$. So the assumption is wrong and we must have that $\exists k \in I$ with $A_k = \emptyset$.

$\Leftarrow$. Assume that $\prod_{i \in I} A_i \neq \emptyset$ then by the previous theorem [theorem: 3.110] $\exists k \in I$ with $A_k \neq \emptyset$ contradicting the hypothesis. Hence the assumption is wrong and we must have $\prod_{i \in I} A_i = \emptyset$. $\square$

Theorem 3.112. Let $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ and $\{B_i\}_{i \in I} \subseteq \mathcal{B}$ be two families and $\forall i \in I A_i \neq \emptyset$ then

$$\prod_{i \in I} A_i \subseteq \prod_{i \in I} B_i \Leftrightarrow \forall i \in I \text{ we have } A_i \subseteq B_i$$

Note 3.113. The condition $\forall i \in I A_i \neq \emptyset$ is essential, for example if we define $\{A_i\}_{i \in \{1,2\}}$ by $A_1 = \{1\}$, $A_2 = \emptyset$ and $\{B_i\}_{i \in \{1,2\}}$ by $B_1 = \{2\}$ and $B_2 = \{3\}$ then $\prod_{i \in \{1,2\}} A_i = \emptyset \subseteq \prod_{i \in \{1,2\}} B_i$ but $A_1 \not\subseteq B_1$.

Proof.

$\Rightarrow$. Let $i \in I$ then by [theorem: 3.109] $\pi_i: \prod_{j \in I} A_j \rightarrow A_i$ defined by $\pi_i(x) = x(i)$ is a surjection. Given $a \in A_i$ then as π_i is a surjection there exist a $x \in \prod_{i \in I} A_i$ such that $a = \pi_i(x) = x(i)$. As $\prod_{i \in I} A_i \subseteq \prod_{i \in I} B_i$ we have that $x \in \prod_{i \in I} B_i$ hence $a = x(i) \in B_i$ proving that $A_i \subseteq B_i$.

$\Leftarrow$. This follows from [theorem: 2.149]. $\square$

Theorem 3.114. Let $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ and $\{B_i\}_{i \in I} \subseteq \mathcal{B}$ be two families and $\forall i \in I A_i \neq \emptyset$ then we have

$$\prod_{i \in I} A_i = \prod_{i \in I} B_i \Leftrightarrow \forall i \in I A_i = B_i \quad (3.59)$$

Note 3.115. The condition $\forall i \in I A_i \neq \emptyset$ is essential, for example if we define $\{A_i\}_{i \in \{1,2\}}$ by $A_1 = \{1\}$, $A_2 = \emptyset$ and $\{B_i\}_{i \in \{1,2\}}$ by $B_1 = \emptyset$ and $B_2 = \{3\}$ then $\prod_{i \in \{1,2\}} A_i = \emptyset = \prod_{i \in \{1,2\}} B_i$ but $A_1 \neq B_1$.

Proof.

$\Rightarrow$. As $\prod_{i \in I} A_i = \prod_{i \in I} B_i$ we have $\prod_{i \in I} A_i \subseteq \prod_{i \in I} B_i$ so that by [theorem: 3.112] we have

$$\forall i \in I \text{ we have } A_i \subseteq B_i$$

By [theorem: 3.110] it follows that $\prod_{i \in I} A_i = \emptyset$ so that $\prod_{i \in I} B_i = \emptyset$ and using [theorem: 3.110] again that $\forall i \in I B_i \neq \emptyset$. Hence as $\prod_{i \in I} B_i \subseteq \prod_{i \in I} A_i$ we have by [theorem: 3.112] that

$$\forall i \in I \text{ we have } B_i \subseteq A_i$$

Combining this with [eq: 3.59] proves that

$$\forall i \in I \text{ we have } A_i = B_i$$

$\Leftarrow$. This is trivial. $\square$

The Axiom of Choice has also important consequences for partial ordered sets.

Theorem 3.116. Let $\langle X, \leq \rangle$ be a partial ordered **set** such that:

1. X has a least element p

2. Every chain [see definition: 3.41] of X has a supremum

then there is a element $x \in X$ which has no immediate successor [see definition: 3.88]

Proof. We prove this by contradiction, so assume that $\forall x \in X$ there exist a immediate successor. Given $x \in X$ define $T_x = \{y \mid y \text{ is a immediate successor of } x\}$ then $T_x \neq \emptyset$ so that $T_x \in \mathcal{P}'(X)$. Using the Axiom of Choice [axiom: 3.104] there exist a choice function

$$c: \mathcal{P}'(A) \rightarrow A \text{ such that } \forall A \in \mathcal{P}'(X) \text{ we have } c(A) \in A \quad (3.60)$$

As $\forall x \in X$ we have $T_x \in \mathcal{P}'(X)$ so that $c(T_x)$ is well defined we can use [proposition: 2.104] to define the function

$$\text{succ}: X \rightarrow X \text{ by } \text{succ}(x) = c(T_x).$$

If $x \in X$ then $\text{succ}(x) = c(T_x) \in T_x$ so that $\text{succ}(x)$ is a immediate successor of x , to summarize

$$\text{succ}: X \rightarrow X \text{ is a function such that } \forall x \in X \text{ succ}(x) \text{ is a immediate successor of } x \quad (3.61)$$

Before we can reach the contradiction we need to have some definitions and sub lemmas.

Definition 3.117. $A \subseteq X$ is a **p-sequence** iff

1. $p \in A$
2. If $x \in A$ then $\text{succ}(x) \in A$
3. If $C \subseteq A$ is a chain then $\sup(C) \in A$ [note that by hypothesis (2) $\sup(C)$ exist]

Note 3.118. X is a p-sequence so there exist p-sequences.

Proof. First $p \in X$ by the hypothesis (1), second if $x \in X$ then by [eq: 3.61] $\text{succ}(x) \in X$ and finally if C is chain then by definition of the supremum $\sup(C) \in X$ $\square$

Lemma 3.119. Every intersection of a set of p-sequences is a p-sequence

Proof. Let $\mathcal{A}$ be a set of p-sequences then

1. $\forall A \in \mathcal{A}$ A is a p-sequence hence $p \in A$ so that $p \in \bigcap \mathcal{A}$
2. If $x \in \bigcap \mathcal{A}$ then $\forall A \in \mathcal{A}$ we have $x \in A$ which as A is a p-sequence gives that $\text{succ}(x) \in A$ hence $\text{succ}(x) \in \bigcap \mathcal{A}$
3. If $C \subseteq \bigcap \mathcal{A}$ is a chain then $\forall A \in \mathcal{A}$ we have $C \subseteq A$ and as A is a p-sequence we have that $\sup(C) \in A$ so that $\sup(C) \in \bigcap \mathcal{A}$

so by definition of a p-sequence we have that

$$\bigcap \mathcal{A} \text{ is a p-sequence} \quad \square$$

From the above lemma [lemma: 3.119] we have that $\bigcap \{A \in \mathcal{P}(X) \mid A \text{ is a p-sequence}\}$ is a p-sequence and by definition $p \in \bigcap \{A \in \mathcal{P}(X) \mid A \text{ is a p-sequence}\}$. Further if A is a p-sequence then $\bigcap \{A \in \mathcal{P}(X) \mid A \text{ is a p-sequence}\} \subseteq A$. Summarized

$$P = \bigcap \{B \in \mathcal{P}(X) \mid B \text{ is a p-sequence}\} \text{ is a p-sequence} \wedge p \in P \wedge \text{If } A \text{ is a p-sequence} \Rightarrow P \subseteq A \quad (3.62)$$

Definition 3.120. A element $x \in P$ is **select** if x is comparable with every element in P .

Lemma 3.121. If $x \in P$ is select then $\forall y \in P$ with $y < x$ have $\text{succ}(y) \leq x$

Proof. If $y \in P$ with $y < x$ then as P is a p-sequence we have by [definition: 3.117 (2)] that $\text{succ}(y) \in P$. Now as x is select we have that $x, \text{succ}(y)$ are comparable, hence by [theorem: 3.39] we have either $\text{succ}(y) \leq x$ or $x < \text{succ}(y)$. If $x < \text{succ}(y)$ then from $y < x$ it follows that $y < x \wedge x < \text{succ}(y)$ contradicting the fact that by [eq: 3.61] $\text{succ}(y)$ is the immediate successor of y . Hence we must have that

$$\text{succ}(y) \leq x \quad \square$$

Lemma 3.122. *If x is select then $A_x = \{y \in P \mid y \leq x \vee \text{succ}(x) \leq y\}$ is a p -sequence*

Proof.

1. As p is a least element of X we have that $p \leq x$ so that $p \in A_x$
2. Let $y \in A_x$. Then we have either:
 - $y = x$. Then $\text{succ}(x) = \text{succ}(y) \Rightarrow \text{succ}(x) \leq \text{succ}(y)$ so that $\text{succ}(y) \in A_x$.
 - $y < x$. Then as $y \in A_x \subseteq P$ we have by the previous lemma [lemma: 3.121] that $\text{succ}(y) < x \Rightarrow \text{succ}(y) \leq x$ so that $\text{succ}(y) \in A_x$.
 - $\text{succ}(x) \leq y$. As $\text{succ}(y)$ is the immediate successor of y we have $y < \text{succ}(y)$ so that $\text{succ}(x) < \text{succ}(y) \Rightarrow \text{succ}(x) \leq \text{succ}(y)$ proving that $\text{succ}(y) \in A_x$.

so in all cases we have

$$\text{succ}(y) \in A_x$$

3. If $C \subseteq A_x$ is a chain then we have the following excluding cases:

$\exists y \in C$ with $\text{succ}(x) \leq y$. Then as $y \leq \sup(C)$ we have that $\text{succ}(x) \leq \sup(C)$ so that $\sup(C) \in A_x$.

$\forall y \in C$ we have $\neg(\text{succ}(x) \leq y)$. Now $\forall y \in C$ as $y \in C \subseteq A_x$ we have either $y \leq x$ or $\text{succ}(y) \leq y$. As $\neg(\text{succ}(x) \leq y)$ is true we must have $y \leq x$ and thus x is an upper bound of C . So by definition of the supremum as the least upper bound of C we must have that $\sup(C) \leq x$, hence $\sup(C) \in A_x$.

So in all cases we have

$$\sup(C) \in A_x$$

From (1),(2) and (3) it follows then that

$$A_x \text{ is a } p\text{-sequence}$$

□

Corollary 3.123. *If x is select then $\forall y \in P$ we have $y \leq x$ or $\text{succ}(x) \leq y$*

Proof. As A_x is a p -sequence by the previous lemma [lemma: 3.122] we have by [eq: 3.62] that $P \subseteq A_x$ and as by definition of A_x $A_x \subseteq P$ it follows that

$$P = A_x$$

□

Lemma 3.124. *The set $\{x \in X \mid x \text{ is select}\}$ is a p -sequence.*

Proof.

1. As p is a least element of X we have $\forall x \in P$ that $p \leq x$ so it is comparable with every element of p , hence p is select, so $p \in \{x \in X \mid x \text{ is select}\}$.
2. If $x \in \{x \in X \mid x \text{ is select}\}$ then x is select and by [corollary: 3.123] we have $\forall y \in P$ either:

$y \leq x$. Then as $\text{succ}(x)$ is the immediate successor of x we have $x < \text{succ}(x)$ so that $y < \text{succ}(x) \Rightarrow y \leq \text{succ}(x)$ proving that $\text{succ}(x)$ is comparable with y

$\text{succ}(x) \leq y$. Then $\text{succ}(x)$ is comparable with y

from the above it follows that $\text{succ}(x)$ is comparable with every $y \in P$ hence

$$\text{succ}(x) \in \{x \in X \mid x \text{ is selected}\}$$

3. Let $C \subseteq \{x \in X \mid x \text{ is select}\}$ be a chain. Then as $C \subseteq X$ we have the hypothesis (3) that $\sup(C)$ exist. Then $\forall y \in P$ we have the following possibilities for C :

$\exists x \in C$ with $y \leq x$. Then $x \leq \sup(C)$ so that $y \leq \sup(C)$ so that $\sup(C)$ is comparable with y

$\forall x \in C$ we have $\neg(y \leq x)$. Then given $x \in C$ we have as $C \subseteq \{x \in X \mid x \text{ is select}\}$ that x is select. By [corollary: 3.123] we have either $y \leq x$ which is not allowed or $\text{succ}(x) \leq y$. As $\text{succ}(x)$ is a immediate successor of x we have $x < \text{succ}(x)$ so that $x < y$ proving that y is a upper bound of C . Hence $\text{sup}(C) \leq y$ proving that $\text{sup}(C)$ is comparable with y

So in all cases we have that $\text{sup}(C)$ is comparable with y proving that $\text{sup}(C)$ is select and thus that $\text{sup}(C) \in \{x \in X \mid x \text{ is select}\}$

From (1),(2),(3) it follows then that $\{x \in X \mid x \text{ is select}\}$ is a p-sequence. $\square$

Now for the last corollary in the proof.

Corollary 3.125. *P is a chain*

Proof. As by the previous lemma [lemma: 3.124] $\{x \in X \mid x \text{ is select}\}$ is a p-sequence it follows from [eq: 3.62] that $P \subseteq \{x \in X \mid x \text{ is select}\}$. So if $x, y \in P$ then x is select and as $y \in P$ comparable with x , proving that P is a chain. $\square$

We are now finally able to reach a contradiction and prove the theorem. As P is a chain we have by hypothesis (2) that $\text{sup}(P)$ exist. Now as P is a p-sequence [see eq: 3.62] we have by [definition: 3.117 (3)] that $\text{sup}(P) \in P$ and by [definition: 3.117 (2)] that $\text{succ}(\text{sup}(P)) \in P$ so that $\text{succ}(\text{sup}(P)) \leq \text{sup}(P)$. As $\text{succ}(\text{sup}(P))$ is the immediate successor of $\text{sup}(P)$ we have that $\text{sup}(P) < \text{succ}(\text{sup}(P))$. Hence $\text{sup}(P) < \text{sup}(P)$ which is a contradiction. $\square$

This was a long proof but it will be used in the following important theorem.

Definition 3.126. *A partial ordered set $\langle X, \leq \rangle$ is **Hausdorff maximal** if there exist a chain C such that if D is a chain with $C \subseteq D$ then $C = D$. In other words C is maximal when using the order relation defined by $\subseteq$.*

We show now that as a consequence of the Axiom of choice every partial ordered set is Hausdorff maximal.

Theorem 3.127. (Hausdorff's Maximal Theorem) *Let $\langle X, \leq \rangle$ be a partial ordered set then it is Hausdorff maximal. In other words there exists a chain C such that if D is a chain such that $C \subseteq D$ then $C = D$.*

Proof. Define the set of all chain of X

$$\mathcal{C} = \{A \in \mathcal{P}(X) \mid A \text{ is a chain in } \langle X, \leq \rangle\}$$

Using the fact $\mathcal{P}(X)$ is a set by the Axiom of Power Sets [axiom: 1.70] we have by the Axiom of Subsets [axiom: 1.8] and the fact that $\mathcal{C} \subseteq \mathcal{P}(X)$ it follows that

$$\mathcal{C} \text{ is a set} \tag{3.63}$$

Using [example: 3.33] we have that

$$\langle \mathcal{C}, \preceq \rangle \text{ where } \preceq = \{(x, y) \in \mathcal{C} \times \mathcal{C} \mid x \subseteq y\} \text{ is a partial ordered set}$$

As $\forall A \in \mathcal{C}$ we have $\emptyset \subseteq A \Rightarrow \emptyset \preceq A$ and $\emptyset$ is a chain [see example: 3.42] in $\langle X, \leq \rangle$ it follows that

$$\mathcal{C} \text{ has a least element [using } \preceq \text{]} \tag{3.64}$$

Let $\mathcal{D}$ a chain in $\langle \mathcal{C}, \preceq \rangle$ then if $x, y \in \bigcup \mathcal{D}$ there exists $A, B \in \mathcal{D} \subseteq \mathcal{C}$ such that $x \in A \wedge y \in B$ where A, B are chains in $\langle X, \leq \rangle$. As $\mathcal{D}$ is a chain we have either:

$A \subseteq B$. Then $x, y \in B$ which as B is a chain [using $\leq$] means that x, y are comparable [using the order $\leq$]

$B \subseteq A$. Then $x, y \in A$ which as A is a chain [using $\leq$] means that x, y are comparable [using the order $\leq$]

From the above it follows that $\bigcup \mathcal{D}$ is a chain in $\langle X, \leq \rangle$ hence $\bigcup \mathcal{D} \in \mathcal{C}$. Hence by [example: 3.73] it follows that $\bigcup \mathcal{D} = \sup(\mathcal{D})$ [using $\preceq$]. So we have proved that

$$\text{Every chain of } \langle \mathcal{C}, \preceq \rangle \text{ has a supremum} \quad (3.65)$$

Now the conditions for [theorem: 3.116] are satisfied by [eq: 3.63], [eq: 3.64] and [eq: 3.65] so we have

$$\exists C \in \mathcal{C} \text{ [so } C \text{ is a chain in } \langle X, \leq \rangle \text{] which has no immediate successor [using } \preceq \text{]} \quad (3.66)$$

Let now D be a chain in $\langle X, \leq \rangle$ [so that $D \in \mathcal{C}$] such that $C \subseteq D$. Take $d \in D$ and assume that $d \notin C$ then $C \subset C \cup \{d\}$ [as $C \cup \{d\} \notin \mathcal{C} \Rightarrow C \neq C \cup \{d\}$] so that $C \prec C \cup \{d\}$. As C has no immediate successor [using $\prec$] there must be a $H \in \mathcal{C}$ such that $C \prec H \wedge H \prec C \cup \{d\}$ or $C \subset H \wedge H \subset C \cup \{d\}$. As $C \subset H$ there exists a $h \in H$ such that $h \notin C$, but then as $H \subset C \cup \{d\}$ we must have $h \in \{d\}$ or $h = d$, so $d \in H$. Now as $H \subset C \cup \{d\}$ there exists a $y \in C \cup \{d\}$ such that $y \notin H$, we can not have $y = d$ [as $d \in H$] so we must have $y \in C$ but then as $C \subset H$ we have $y \in H$ contradicting $y \notin H$. So we must have $d \in C$. As $d \in D$ was chosen arbitrary we have that $D \subseteq C$ or $C = D$ which proves maximality. $\square$

We state now Zorn's lemma but not prove it yet, it will be show to be directly dependent on the Hausdorff maximal principle, which in turn depends on the Axiom of Choice. So if we accept the Axiom of Choice [which we do as it is expressed as a Axiom] then Zorn's lemma applies.

Lemma 3.128. (Zorn's Lemma) *Let $\langle X, \leq \rangle$ be a partial ordered set such that every chain has a upper bound then X has a maximal element.*

We prove now that the Hausdorff Maximal principle implies Zorn's lemma.

Theorem 3.129. *Let $\langle X, \leq \rangle$ be Hausdorff Maximal then Zorn's lemma follows.*

Proof. Let $\langle X, \leq \rangle$ be a partial ordered set such that every chain in X has a upper bound. As $\langle X, \leq \rangle$ is Hausdorff maximal [definition: 3.126] there exist a chain C such that for every chain D with $C \subseteq D$ we have $C = D$. As C is a chain it has by the hypothesis a upper bound u for C . Assume now that u is not a maximal element of X , then by the definition of a maximal element [definition: 3.58] there exist a $x \in X$ with $u \leq x$ and $u \neq x$ so that $u < x$. If $x \in C$ then as u is a upper bound of C we have $x \leq u$ so that $u < u$ a contradiction. So we must have that $x \notin C$. Consider now $r, s \in C \cup \{x\}$ then we have to consider the following possibilities:

$r = x \wedge s = x$. Then by reflectivity we have $r \leq s$, so r, s are comparable.

$r = x \wedge s \neq x$. Then $s \in C$ so that $s \leq u$, which as $u \leq x$ proves that $s \leq x \xRightarrow[r=x]{} s \leq r$, so r, s are comparable.

$r \neq x \wedge s = x$. Then $r \in C$ so that $r \leq u$, which as $u \leq x$ proves that $r \leq x \xRightarrow[s=x]{} r \leq s$, so r, s are comparable.

$r \neq x \wedge s \neq x$. Then $r, s \in C$, which as C is a chain proves that r, s are comparable

From the above it follows that $C \cup \{x\}$ is a chain such that $C \subseteq C \cup \{x\}$ giving by maximality of C that $C = C \cup \{x\}$ contradicting $x \notin C$. Hence the assumption that u is not a maximal element of X is false. So u is a maximal element of X . $\square$

We show now that Zorn's lemma implies well ordering.

Theorem 3.130. *Zorn's lemma implies that given a set X there exist a order relation $\leq$ on X such that $\langle X, \leq \rangle$ is well ordered [see 3.84]*

Proof. Just like the proof of [theorem: 3.116] this proof will consist of many sub lemma's. Let X be a set and define the class

$$\mathcal{A} = \{(B, R) | B \in \mathcal{P}(X) \wedge R \text{ a order relation on } B \text{ so that } \langle B, R \rangle \text{ is well ordered}\}$$

Define now $\preceq \in \mathcal{A} \times \mathcal{A}$ by

$$\preceq = \{((B, R), (B', R')) | B \subseteq B' \wedge R \subseteq R' \wedge \text{If } x \in B \wedge y \in B' \setminus B \text{ then } (x, y) \in R'\}$$

then we have that

$$\langle \mathcal{A}, \preccurlyeq \rangle \text{ is a order relation} \quad (3.67)$$

Proof. We have to prove reflexivity, anti-symmetry and transitivity:

reflectivity. If $(B, R) \in \mathcal{A}$ then we have

1. $B \subseteq B$
2. $R \subseteq R$
3. If $x \in B \wedge y \in B \setminus B \stackrel{[\text{theorem: 1.35}]}{=} \emptyset$ which can not occur so that $(x, y) \in R$ is satisfied vacuously

proving that $(B, R) \preccurlyeq (B, R)$

anti-symmetry. If $(B, R) \preccurlyeq (B', R') \wedge (B', R') \preccurlyeq (B, R)$ then $B \subseteq B' \wedge R \subseteq R' \wedge B' \subseteq B \wedge R' \subseteq R$ proving that $B = B'$ and $R = R'$ so that $(B, R) = (B', R')$

transitivity. Let $(B, R) \preccurlyeq (B', R')$ and $(B', R') \preccurlyeq (B'', R'')$ then we have

1. $B \subseteq B' \wedge B' \subseteq B'' \Rightarrow B \subseteq B''$
2. $R \subseteq R' \wedge R' \subseteq R'' \Rightarrow R \subseteq R''$
3. If $x \in B \wedge y \in B'' \setminus B$ we have for y to consider the following possibilities

$y \in B'$. Then $y \in B' \setminus B$ so that $(x, y) \in R' \stackrel{R' \subseteq R''}{\Rightarrow} (x, y) \in R''$

$y \notin B'$. Then $y \in B'' \setminus B'$ so that $(x, y) \in R''$

so in all cases we have $(x, y) \in R''$.

proving $(B, R) \preccurlyeq (B'', R'')$. □

We now have the following sub lemma:

Lemma 3.131. *If $\mathcal{C} \subseteq \mathcal{A}$ is a chain in $\langle \mathcal{A}, \preccurlyeq \rangle$ then if*

$$B_{\mathcal{C}} = \bigcup \{B | \exists R \text{ such that } (B, R) \in \mathcal{C}\}$$

$$R_{\mathcal{C}} = \bigcup \{R | \exists B \text{ such that } (B, R) \in \mathcal{C}\}$$

then

$$(B_{\mathcal{C}}, R_{\mathcal{C}}) \in \mathcal{A}$$

Proof. First note that if $(B, R) \in \mathcal{C}$ then

$$B \in \{B | \exists R \text{ such that } (B, R) \in \mathcal{C}\}$$

and

$$R \in \{R | \exists B \text{ such that } (B, R) \in \mathcal{C}\}$$

or

$$\forall (B, R) \in \mathcal{C} \text{ we have } B \subseteq B_{\mathcal{C}} \wedge R \subseteq R_{\mathcal{C}} \quad (3.68)$$

1. If $x \in B_{\mathcal{C}}$ then $\exists (B, R) \in \mathcal{C}$ such that $x \in B$, as $\mathcal{C} \subseteq \mathcal{A}$ we have $(B, R) \in \mathcal{C}$, so that $B \in \mathcal{P}(A)$, hence $B \subseteq A$, proving that $x \in A$. In other words $B_{\mathcal{C}} \subseteq A$ or $B \in \mathcal{P}(A)$.
2. We must prove that $R_{\mathcal{C}}$ is a a order relation on $B_{\mathcal{C}}$:

reflectivity. If $x \in B_{\mathcal{C}}$ then $\exists (B, R) \in \mathcal{C}$ such that $x \in B$, as R is a order relation we have that $(x, x) \in R$ so that by [eq: 3.68] $(x, x) \in R_{\mathcal{C}}$

anti-symmetry. If $(x, y) \in R_{\mathcal{C}} \wedge (y, x) \in R_{\mathcal{C}}$ then $\exists (B, R), (B', R') \in \mathcal{C}$ such that $(x, y) \in R$ and $(y, x) \in R'$. As $\mathcal{C}$ is a chain we have either:

$(B, R) \preccurlyeq (B', R')$. Then $R \subseteq R'$ so that $(x, y) \in R' \wedge (y, x) \in R'$, which as R' is a order relation proves that $x = y$.

$(B', R') \preceq (B, R)$. Then $R' \subseteq R$ so that $(x, y) \in R \wedge (y, x) \in R$, which as R is a order relation proves that $x = y$.

transitivity. If $(x, y) \in R_C \wedge (y, z) \in R_C$ then $\exists (B, R), (B', R') \in \mathcal{C}$ such that $(x, y) \in R$ and $(y, z) \in R'$. As $\mathcal{C}$ is a chain we have either:

$(B, R) \preceq (B', R')$. Then $R \subseteq R'$ so that $(x, y) \in R \wedge (y, z) \in R'$, which as R' is a order relation proves that $(x, z) \in R'$, hence $(x, z) \in R_C$ [see eq: 3.68].

$(B', R') \preceq (B, R)$. Then $R' \subseteq R$ so that $(x, y) \in R \wedge (y, z) \in R$, which as R is a order relation proves that $(x, z) \in R$, hence $(x, z) \in R_C$ [see eq: 3.68].

3. Next we have to prove well ordering of $\langle B_C, R_C \rangle$. Let $D \subseteq B_C$ and $D \neq \emptyset$. Then there exist a $x \in D$ so that $x \in B_C$, hence there exist a $(B, R) \in \mathcal{C}$ such that $x \in B$ or $x \in D \cap B$ proving that $D \cap B \neq \emptyset$. As $\mathcal{C} \subseteq \mathcal{A}$ we have by the definition of $\mathcal{A}$ that $\langle B, R \rangle$ is well ordered, hence there exist a least element $b \in B$. So

$$\forall y \in B \text{ we have } (b, y) \in R \quad (3.69)$$

We prove now that

b is a least element of D

Proof. If $x \in D$ then $\exists (B', R')$ such that $x \in B'$. For x and B we the following possible cases:

$x \in B$. Then by [eq: 3.69] we have that $(b, x) \in R$ so that by [eq: 3.68] $(b, x) \in R_C$.

$x \notin B$. Then $x \in B' \setminus B \wedge b \in B$. As $\mathcal{C}$ is a chain we have the following cases:

$(B, R) \preceq (B', R')$. Then by definition of $\preceq$ we have $(b, x) \in R'$ so that by [eq: 3.68] $(b, x) \in R_C$

$(B', R') \preceq (B, R)$. Then $B' \subseteq B$ and as $x \in B'$ we have $x \in B$ contradicting $x \notin B$. So this case never occurs.

So in all cases that apply we have $(b, x) \in R_C$ proving that b is a least element of D . $\square$

As we have proved that every non empty $D \subseteq B_C$ has a least element [using the order R_C it follows that $\langle B_C, R_C \rangle$ is well ordered.

From (1),(2) and (3) it follows that

$$(B_C, R_C) \in \mathcal{A} \quad \square$$

Lemma 3.132. *If $\mathcal{C}$ is a chain in $\langle \mathcal{A}, \preceq \rangle$ then (B_C, R_C) is a upper bound of $\mathcal{C}$*

Proof. Let $(B, R) \in \mathcal{C}$ then

1. $B \subseteq B_C$ [see eq: 3.68]
2. $R \subseteq R_C$ [see eq: 3.68]
3. Let $x \in B$ and $y \in B_C \setminus B$ then $\exists (B', R') \in \mathcal{C}$ such that $y \in B'$ or as $y \in B_C \setminus B$ that

$$y \in B' \setminus B$$

As $\mathcal{C}$ is a chain we have either $(B, R) \preceq (B', R')$ or $(B', R') \preceq (B, R)$. If $(B', R') \preceq (B, R)$ then $B' \subseteq B$, as $y \in B'$ we would have $y \in B$ contradiction $y \in B_C \setminus B$. So we have

$$(B, R) \preceq (B', R')$$

As $x \in B$ and $y \in B' \setminus B$ we have by definition of $\preceq$ and the above that $(x, y) \in R'$ which as $R' \subseteq R_C$ [see eq: 3.68] proves that $(x, y) \in R_C$

So by the definition of $\preceq$ we have by (1),(2) and (3) that

$$(B, R) \preceq (B_C, R_C) \quad \square$$

Using Zorn's [lemma: 3.128] together with the above lemma [lemma: 3.132] we have

$$\exists (B_m, R_m) \in \mathcal{A} \text{ such that } (B_m, R_m) \text{ is a maximum element of } \mathcal{A} \quad (3.70)$$

We prove now by contradiction that

$$B_m = X$$

Proof. Assume that $X \neq B_m$. Then as $B_m \in \mathcal{P}(X) \Rightarrow B_m \subseteq X$ there exist a

$$x \in X \setminus B_m \Rightarrow x \notin B_m.$$

Define

$$R^* = R_m \cup \{(b, x) | b \in B_m\} \cup \{(x, x)\} \quad (3.71)$$

Then if $(r, s) \in R_m \cap \{(b, x) | b \in B_m\}$ we have as $R_m \subseteq B_m \times B_m$ that $s \in B_m \wedge s = x \notin B_m$ a contradiction, if $(r, s) \in R_m \cap \{(x, x)\}$ then $r \in B_m \wedge r = x \notin B_m$ a contradiction and finally if $(r, s) \in \{(b, x) | b \in B_m\} \cap \{(x, x)\}$ then $r \in B_m \wedge r = x \notin B_m$ a contradiction. So we have

$$R_m \cap \{(b, x) | b \in B_m\} = \emptyset \wedge R_m \cap \{(x, x)\} = \emptyset \wedge \{(b, x) | b \in B_m\} \cap \{(x, x)\} = \emptyset \quad (3.72)$$

Further if $(x, r) \in R^*$ then we have either $(x, r) \in R_m \Rightarrow x \in B_m$ contradicting $x \notin B_m$, $(x, r) \in \{(b, x) | b \in B_m\} \Rightarrow x \in B_m$ contradicting $x \notin B_m$ or $(x, r) \in \{(x, x)\} \Rightarrow r = x$. To summarize we have

$$\text{If } (x, r) \in R^* \text{ then } r = x \quad (3.73)$$

We prove now that $\langle B_m \cup \{x\}, R^* \rangle$ is well ordered.

Proof. First we have:

reflexivity. If $r \in B_m \cup \{x\}$ then we have either:

$r \in B_m$. Then as $\langle B_m, R_m \rangle$ is a partial order we have $(r, r) \in R_m \subseteq R^*$.

$r \notin B_m$. Then $r \in \{x\}$ so that $r = x$ hence $(r, r) = (x, x) \in \{(x, x)\} \subseteq R^*$

proving that $(r, r) \in R^*$.

anti-symmetry. If $(r, s) \in R^*$ and $(s, r) \in R^*$ then we have by [eq: 3.71] for (r, s) either:

$(r, s) \in R_m$. Then as $R_m \subseteq B_m \times B_m$ we have $r, s \in B_m$ so that $r \neq x \neq s$ so that $(s, r) \in R_m$ [if $(s, r) \in \{(b, x) | b \in B_m\} \cup \{(x, x)\}$ then $r = x$ contradicting $r \neq x$], which as $\langle B_m, R_m \rangle$ is a partial order gives that $r = s$.

$(r, s) \in \{(b, x) | b \in B_m\}$. Then $s = x$ so that $(x, r) = (s, r) \in R^* \xRightarrow{[\text{eq: 3.73}]} r = x = s$ hence $s = r$.

$(r, s) \in \{(x, x)\}$. Then $r = x = s \Rightarrow r = s$.

proving $r = s$

transitivity. If $(r, s) \in R^* \wedge (s, t) \in R^*$ then we have by [eq: 3.71] that:

$(r, s) \in R_m$. We have the following case for (s, t) :

$(s, t) \in R_m$. Then as $\langle B_m, R_m \rangle$ is a partial ordered we have $(r, t) \in R_m \subseteq R^*$.

$(s, t) \in \{(b, x) | b \in B_m\}$. Then $t = x$ and $r \in B_m$ so that $(r, t) \in \{(b, x) | b \in B_m\} \subseteq R^*$.

$(s, t) \in \{(x, x)\}$. Then $t = x$ and $r \in B_m$ so that $(r, t) \in \{(b, x) | b \in B_m\} \subseteq R^*$.

$(r, s) \in \{(b, x) | b \in B_m\}$. Then $s = x$ so that $(s, t) = (x, t) \in R^* \xRightarrow{[\text{eq: 3.73}]} t = x$. As $r \in B_m$ we have $(r, t) \in \{(b, x) | b \in B_m\} \subseteq R^*$.

$(r, s) \in \{(x, x)\}$. Then $r = x \wedge t = x$ so that $(x, t) = (s, t) \in R^* \xRightarrow{[\text{eq: 3.73}]} t = x$ hence $(r, t) = (x, x) \in \{(x, x)\} \subseteq R^*$.

proving $(r, t) \in R^*$.

Hence

$$\langle B_m \cup \{x\}, R^* \rangle \text{ is partial ordered}$$

If $\emptyset \neq C \subseteq B_m \cup \{x\}$ is non empty then we have for $C \cap B_m$ the following possibilities:

$C \cap B_m \neq \emptyset$. Then as $\emptyset \neq C \cap B_m \subseteq B_m$ and $\langle B_m, R_m \rangle$ is well ordered [see definition of $\mathcal{A}$] there exist a least element $l \in C \cap B_m$ so

$$\forall r \in C \cap B_m \text{ we have } (l, r) \in R_m \quad (3.74)$$

Now if $r \in C$ we have either:

$r \in B_m$. then $r \in C \cap B_m$ so that by the above [eq: 3.74] $(l, r) \in R_m \subseteq R^*$

$r \notin B_m$. then as $C \subseteq B_m \cup \{x\}$ we have $r = x$ so $(l, r) \in \{(b, x) | b \in B_m\} \cup \{(x, x)\} \subseteq R^*$ proving that $(l, r) \in R^*$. Hence

C has a least element [using $\langle B \cup \{x\}, R^* \rangle$]

$C \cap B_m = \emptyset$. Then $C = \{x\}$ so that $\forall r \in C$ we have $r = x$ so that $(r, x) = (x, x) \in \{(x, x)\} \subseteq R^*$ proving that x is a least element of C .

So in all cases we have that C has a least element, hence

$$\langle B_m \cup \{x\}, R^* \rangle \text{ is well ordered} \quad \square$$

Now as $B_m \cup \{x\} \subseteq X$, we have by the definition of $\mathcal{A}$ and the above that

$$(B_m \cup \{x\}, R^*) \in \mathcal{A}$$

Next we have:

$$1. B_m \subseteq B_m \cup \{x\}$$

$$2. R_m \subseteq R^*$$

$$3. \text{ If } r \in B_m \text{ and } s \in (B_m \cup \{x\}) \setminus B_m \text{ then } s = x \text{ so that } (r, s) = (r, x) \in \{(b, x) | b \in B_m\} \subseteq R^*$$

proving that $(B_m, R_m) \preceq (B_m \cup \{x\}, R^*)$. As (B_m, R_m) is a maximal element of $\langle \mathcal{A}, \preceq \rangle$ we must have $(B_m, R_m) = (B_m \cup \{x\}, R^*)$ so that $B = B \cup \{x\}$ which as $x \notin B_m$ leads to a contradiction. Hence the assumption that $X \neq B_m$ is wrong and we must have that

$$X = B_m \quad \square$$

As $\langle B_m, R_m \rangle$ is a well ordered the above proves that there exists a partial order R_m such that

$$\langle X, R_m \rangle = \langle B_m, R_m \rangle \text{ is well-ordered [by definition of } \mathcal{A} \text{ } B_m \text{ is well ordered]} \quad \square$$

We show now that Well Ordering implies the Axiom of Choice.

Theorem 3.133. Assume that for every X there exist a order relation such that $\langle X, \leq \rangle$ is well ordered then there exists a function $c: \mathcal{P}'(X) \rightarrow X$ such that $\forall A \in \mathcal{P}'(X)$ we have $c(A) \in A$ (Axiom of Choice).

Proof. Let X be a set then by the hypothesis there exist a order $\leq$ on X such that $\langle X, \leq \rangle$ is well ordered. Define now $c = \{(A, x) | A \in \mathcal{P}'(X) \wedge x \text{ is a least element of } A\}$. If $(A, x) \in c$ then $A \in \mathcal{P}'(X)$ and x is a least element of A , so that $x \in A \subseteq X$ proving that $(A, x) \in \mathcal{P}'(X) \times X$. So $c \subseteq \mathcal{P}'(X) \times X$. If $(A, x), (A, x') \in c$ then x and x' are least elements of A , which are unique by [theorem: 3.62] so that $x = x'$. Hence we have that

$$c: \mathcal{P}'(X) \rightarrow X \text{ is a partial function}$$

If $A \in \mathcal{P}'(X)$ then $A \neq \emptyset$ so by well ordering A has a least element l so that $(A, l) \in c$, so $\mathcal{P}'(A) \subseteq \text{dom}(c)$. Hence by [proposition: 2.27] we have that

$$c: \mathcal{P}'(X) \rightarrow X \text{ is a function}$$

If $(A, x) \in c$ then x is the least element of A so that $c(A) = x \in A$ proving that

$$c: \mathcal{P}'(X) \rightarrow X \text{ is a choice function for } X \quad \square$$

We are now ready to specify the different equivalent statements of the Axiom of Choice

Theorem 3.134. *The following statements are equivalent*

1. *Axiom of Choice*
2. *Hausdorff's Maximal Principle*
3. *Zorn's Lemma*
4. *Every set can be well ordered*

Proof.

1 $\Rightarrow$ 2. This follows from [theorem: 3.127]

2 $\Rightarrow$ 3. This follows from [theorem: 3.129]

3 $\Rightarrow$ 4. This follows from [theorem: 3.130]

4 $\Rightarrow$ 1. This follows from [theorem: 3.133] □

As in most of works about mathematics we assume the Axiom of Choice. To summarize the consequences of the Axiom of Choice we have [taking in account [theorem: 3.106] that the following statements are true.

Theorem 3.135.

Axiom of Choice. *Let X be a set then there exist a function $c: \mathcal{P}'(X) \rightarrow X$ such that $\forall A \in \mathcal{P}'(X)$ we have $c(A) \in A$.*

Existence of Choice set. *Let $\mathcal{A}$ be a set of sets such that*

- a) $\forall A \in \mathcal{A}$ we have $A \neq \emptyset$
- b) $\forall A, B \in \mathcal{A}$ with $A \neq B$ we have $A \cap B = \emptyset$

*then there exist a set C [called the **choice set of $\mathcal{A}$**] such that*

- a) $C \subseteq \bigcup \mathcal{A}$
- b) $\forall A \in \mathcal{A}$ we have $A \cap C \neq \emptyset$ and if $y, y' \in A \cap C$ then $y = y'$

Axiom of Choice alternative. *If $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ is a family of non empty sets $[\forall i \in I$ we have $A_i \neq \emptyset]$ where $I, \mathcal{A}$ are sets then there exists a function $f: I \rightarrow \bigcup_{i \in I} A_i$ such that $\forall i \in I$ we have $f(i) \in A_i$*

Hausdorff's Maximal Theorem. *If $\langle X, \leq \rangle$ is a partial ordered set then there exists a chain $C \subseteq X$ such that for every chain $D \subseteq X$ with $C \subseteq D$ we have $C = D$*

Zorn's Lemma. *If $\langle X, \leq \rangle$ is a partial ordered set such that every chain has a upper bound then X has a maximal element.*

Well-Ordering Theorem. *For every set there exists a order relation making $\langle X, \leq \rangle$ well-ordered.*

There is a kind of extension of Zorn's lemma to pre-ordered sets if we change the definition of maximal element slightly.

Theorem 3.136. *Let $\langle X, \leq \rangle$ be a pre-ordered set [see definitions: 3.26, 3.25] such that every chain has a upper bound then there exists a $m \in X$ such that $\forall x \in X$ with $m \leq x$ we have $x \leq m$*

Proof. Using [theorem: 3.34] we have the following

1. $\sim \subseteq X \times X$ defined by $\sim = \{(x, y) \in X \mid x \leq y \wedge y \leq x\}$ is a equivalence relation

2. Define $\preceq \subseteq (X/\sim) \times (X/\sim)$ by

$$\preceq = \{(x, y) \in (X/\sim) \times (X/\sim) \mid \exists x' \in \sim[x] \text{ and } \exists y' \in \sim[y] \text{ such that } x' \leq y'\}$$

then $\preceq$ is a order relation in $X/\sim$. So $\langle X/\sim, \preceq \rangle$ is a partial ordered set

3. $\forall x, y \in A$ we have $x \leq y \Leftrightarrow \sim[x] \preceq \sim[y]$

Let $C \subseteq X/\sim$ be a chain [using the order $\preceq$] and construct $C' = \bigcup C$. If $x, y \in C'$ then $\exists \sim[x'], \sim[y']$ such that $x \in \sim[x']$ and $y \in \sim[y']$, so $x \sim x'$ and $y \sim y'$ or $x \leq x' \wedge x' \leq x$ and $y \leq y' \wedge y' \leq y$. As C is a chain [using $\preceq$] we have the following possibilities:

$\sim[x'] \preceq \sim[y']$. then $x' \leq y'$ and as $x \leq x'$ and $y' \leq y$ we have $x \leq y$

$\sim[y'] \preceq \sim[x']$. then $y' \leq x'$ and as $y \leq y'$ and $x' \leq x$ we have $y \leq x$

proving that x, y are comparable. Hence

C' is a chain [using $\leq$]

By the hypothesis we have that there exist a upper bound u of C' [using $\leq$], in other words

$$\exists u \in X \text{ such that } \forall x \in C' \text{ we have } x \leq u$$

Take now $\sim[z] \in C$ then $z \in \sim[z] \subseteq C'$ so that $z \leq u$ and thus by (3) $\sim[z] \preceq \sim[u]$. So $\sim[u]$ is a upper bound of C . As we just have proved that every chain in $X/\sim$ has a upper bound and $\langle X/\sim, \preceq \rangle$ is a partial order, it follows from Zorn's lemma that there exist a maximal element $\sim[m]$ in $X/\sim$. So by [definition: 3.58] we have

$$\forall \sim[x] \in X/\sim \text{ with } \sim[m] \preceq \sim[x] \text{ we have } \sim[x] = \sim[m]$$

If now $x \in X$ such that $x \leq m$ then by (3) we have $\sim[x] \preceq \sim[m]$ hence by the above we have $\sim[x] = \sim[m]$ so that $x \sim m$ hence $x \leq m$. $\square$

From this point on we will gradually start to use the simpler notations for functions and families that are mentioned in the references [definition: 2.41], [theorem: 2.43], [theorem: 2.44], [theorem: 2.54], [theorem: 2.104], [notation: 2.105], [theorem: 2.126] and [theorem: 2.131] without explicit referring to them. This to avoid excessive notation and difference of notation between this text and standard mathematical practice.

As a interesting application of the Axiom of Choice we prove that every function can be restricted to a injection or bijection.

Theorem 3.137. *Let X, Y be sets, $f: X \rightarrow Y$ a function then there exist a $Z \subseteq X$ such that:*

1. $f|_Z: Z \rightarrow Y$ is a injection
2. $f|_Z(Z) = f(X)$
3. $f|_Z: Z \rightarrow f(X)$ is a bijection

Proof.

1. Define

$$\mathcal{A} = \{f^{-1}(\{y\}) \mid y \in f(X)\}.$$

If $A \in \mathcal{A}$ then $\exists y \in f(X)$ such that $A = f^{-1}(\{y\}) \subseteq X$ and as $y \in f(X)$ there exists a $x \in X$ such that $f(x) = y \in \{y\} \Rightarrow x \in f^{-1}(\{y\}) = A$, proving that $A \neq \emptyset$. So we have proved that

$$\mathcal{A} \subseteq \mathcal{P}'(X)$$

By the Axiom of Choice [axiom: 3.104] there exist a function

$$c: \mathcal{P}'(X) \rightarrow X \text{ such that } \forall A \in \mathcal{P}'(X) \ (c)(A) \in A$$

Take

$$Z = c(\mathcal{A}) \subseteq X$$

and consider the restriction of f to Z

$$f|_Z: Z \rightarrow Y$$

Let $x, y \in Z$ such that $f|_Z(x) = f|_Z(y) \xRightarrow{x, y \in Z} f(x) = f(y)$. As $x, y \in Z = c(\mathcal{A})$ there exists $A_x \in \mathcal{A} \wedge A_y \in \mathcal{A}$ such that $x = (c)(A_x) \in A_x$ and $y = (c)(A_y) \in A_y$. As $A_x, A_y \in \mathcal{A}$ there exist $x', y' \in f(X)$ such that $A_x = f^{-1}(\{x'\})$ and $A_y = f^{-1}(\{y'\})$. Then $f(x) \stackrel{x \in A_x}{=} x'$ and $f(y) \stackrel{y \in A_y}{=} y'$. As $f(x) = f(y)$ we have $x' = y'$ so that $A_x = f^{-1}(\{x'\}) = f^{-1}(\{y'\}) = A_y$. So $x = (c)(A_x) = (c)(A_y) = y$, proving that $x = y$.

2. If $y \in f(X)$ then $f^{-1}(\{y\}) \in \mathcal{A}$ so that $x = (c)(f^{-1}(\{y\})) \in c(\mathcal{A}) = Z$. Further as $(c)(f^{-1}(\{y\})) \in f^{-1}(\{y\})$ we have that $f(x) = f((c)(f^{-1}(\{y\}))) \in \{y\}$ so that $y = f(x) \in f(Z)$, proving that $f(X) \subseteq f(Z)$. As $Z \subseteq X$ we have by [theorem: 2.17] that $f(Z) \subseteq f(X)$ so that

$$f(X) = f(Z)$$

3. From (2) we have that $f|_Z: Z \rightarrow f(X)$ is surjective which together with (1) proves bijectivity. $\square$

Another application of the Axiom of Choice is the following.

Theorem 3.138. *Let I, X, Y be sets, $f: X \rightarrow Y$ a function and $\{y_i\}_{i \in I} \subseteq f(X)$ then $\exists \{x_i\}_{i \in I} \subseteq X$ such that $\forall i \in I$ we have $f(x_i) = y_i$*

Proof. Define

$$\{A_i\}_{i \in I} \text{ by } A_i = f^{-1}(\{y_i\}) \subseteq X$$

Let $i \in I$ then, as $y_i \in f(X)$, there exist a $x \in X$ such that $f(x) = y_i$, hence $x \in f^{-1}(\{y_i\}) = A_i$. This proves that

$$\forall i \in I \ A_i \neq \emptyset$$

Using the Axiom of Choice [axiom: 3.106(3)] there exist a function

$$x: I \rightarrow \bigcup_{i \in I} A_i \text{ such that } \forall i \in I \text{ we have } x(i) \in A_i$$

If $i \in I$ we have that $x(i) \in A_i$ so that $f(x(i)) \in \{y_i\} \Rightarrow f(x_i) = y_i$, this together with the fact that we can extend the function $x: I \rightarrow \bigcup_{i \in I} A_i$ to $f: I \rightarrow X$ proves that we have found a function

$$x: I \rightarrow X \text{ such that } \forall i \in I \text{ we have } f(x_i) = y_i$$

This function defines then the family $\{x_i\}_{i \in I} \subseteq X$ satisfying $\forall i \in I \ f(x_i) = y_i$. $\square$

3.5 Generalized Intervals

Definition 3.139. *Let $\langle A, \leq \rangle$ be a partial ordered class then if $a, b \in A$ we define*

1. $[a, b] = \{x \in A \mid a \leq x \wedge x \leq b\}$
2. $]a, b] = \{x \in A \mid a < x \wedge x \leq b\}$
3. $[a, b[= \{x \in A \mid a \leq x \wedge x < b\}$
4. $]a, b[= \{x \in A \mid a < x \wedge x < b\}$
5. $] -\infty, a] = \{x \in A \mid x \leq a\}$
6. $] -\infty, a[= \{x \in A \mid x < a\}$
7. $[a, \infty[= \{x \in A \mid a \leq x\}$
8. $]a, \infty[= \{x \in A \mid a < x\}$
9. $] -\infty, \infty[= A$

these are the intervals in A

Theorem 3.140. Let $\langle A, \leq \rangle$ be a partial ordered class and $a, b \in A$ then we have:

1. If $b < a$ then $[a, b] = [a, b[=]a, b] =]a, b[= \emptyset$
2. If $b \leq a$ then $]a, b] = [a, b[=]a, b[= \emptyset$

Proof.

1. Let $b < a$ then we have
 - a. If $z \in [a, b]$ then $b < a \wedge a \leq z \wedge z \leq b$ so that $b < z \wedge z \leq b$ a contradiction.
 - b. If $z \in]a, b]$ then $b < a \wedge a < z \wedge z \leq b$ so that $b < z \wedge z \leq b$ a contradiction.
 - c. If $z \in [a, b[$ then $b < a \wedge a \leq z \wedge z < b$ so that $b < z \wedge z < b$ a contradiction.
 - d. If $z \in]a, b[$ then $b < a \wedge a < z \wedge z \leq b$ so that $b < z \wedge z < b$ a contradiction.
2. Let $b \leq a$ then we have
 - a. If $z \in [a, b[$ then $b \leq a \wedge a \leq z \wedge z < b$ so that $b \leq z \wedge z < b$ a contradiction.
 - b. If $z \in]a, b]$ then $b \leq a \wedge a < z \wedge z \leq b$ so that $b < z \wedge z \leq b$ a contradiction.
 - c. If $z \in]a, b[$ then $b \leq a \wedge a < z \wedge z < b$ so that $b < z \wedge z < b$ a contradiction. $\square$

Definition 3.141. (Generalized Interval) Let $\langle A, \leq \rangle$ be a partial ordered class then $I \subseteq A$ is a **generalized interval** if $\forall x, y \in I$ we have $[x, y] \subseteq I$.

We have the following equivalent conditions for a generalized interval.

Theorem 3.142. Let $\langle A, \leq \rangle$ be a partial ordered class that is conditional complete and $I \subseteq A$ then then

$$I \text{ is a generalized interval} \Leftrightarrow \forall x, y \in I \text{ we have }]x, y[\subseteq I$$

Proof.

- $\Rightarrow$. Let $x, y \in I$ then if $z \in]x, y[$ we have $x < z \wedge z < y \Rightarrow x \leq z \wedge z \leq y \Rightarrow z \in [x, y] \subseteq I$ proving that $]x, y[\subseteq I$
- $\Leftarrow$. Let $x, y \in I$ then if $z \in [x, y]$ we have either
- $\mathbf{x = y.}$ Then $x \leq z \wedge z \leq x \Rightarrow z = x \in I$
 - $\mathbf{x < y.}$ Then we have either
 - $\mathbf{z = x.}$ Then $z \in I$
 - $\mathbf{z = y.}$ Then $z \in I$
 - $\mathbf{z \neq x \wedge z \neq y.}$ Then $x < z \wedge z < y$ proving that $z \in]x, y[\subseteq I$

So in all cases we have that $z \in I$ proving that $[x, y] \subseteq I$ $\square$

Theorem 3.143. Let $\langle A, \leq \rangle$ be a fully ordered class that is conditional complete and $\emptyset \neq I \subseteq A$ a generalized interval then we have either

1. If I is bounded below and above [so that by conditional completeness $\inf(I)$ and $\sup(I)$ exists] we have:
 - a. If $\inf(I) \in I \wedge \sup(I) \in I$ then $I = [\inf(I), \sup(I)]$
 - b. If $\inf(I) \notin I \wedge \sup(I) \in I$ then $I =]\inf(I), \sup(I)]$
 - c. If $\inf(I) \in I \wedge \sup(I) \notin I$ then $I = [\inf(I), \sup(I)[$
 - d. If $\inf(I) \notin I \wedge \sup(I) \notin I$ then $I =]\inf(I), \sup(I)[$
2. If I is bounded below and not bounded above [so that by conditional completeness $\inf(I)$ exists] we have
 - a. If $\inf(I) \in I$ then $I = [\inf(I), \infty[$

- b. If $\inf(I) \notin I$ then $I =]\inf(I), \infty[$
3. If I is bounded above and not bounded below [so that by conditional completeness $\sup(I)$ exists] we have
- a. If $\sup(I) \in I$ then $I =]-\infty, \sup(I)]$
- b. If $\sup(I) \notin I$ then $I =]-\infty, \sup(I)[$
4. If I is not bounded above and not bounded below we have $I = A$

In other words a generalized interval I has either one of the following forms

$$\begin{aligned}
 [a, b] & \quad a, b \in A \text{ and } a \leq b \\
]a, b] & \quad a, b \in A \text{ and } a < b \\
 [a, b[& \quad a, b \in A \text{ and } a < b \\
]a, b[& \quad a, b \in A \text{ and } a < b \\
 [a, \infty[& \\
]a, \infty[& \\
]-\infty, a] & \\
]-\infty, a[& \\
]-\infty, \infty[&
 \end{aligned}$$

Proof.

1.

- a. Let $z \in I$ then using the definition of a supremum and infimum as a upper and lower bound that $\inf(I) \leq z \wedge z \leq \sup(I)$ so that

$$I \subseteq [\inf(I), \sup(I)] \quad (3.75)$$

As I is a generalized interval and $\sup(I) \in I \wedge \inf(I) \in I$ we have further that $[\inf(I), \sup(I)] \subseteq I$ which using 3.75 proves

$$I = [\inf(I), \sup(I)]$$

- b. Let $z \in I$ then using the definition of a supremum and infimum as upper and lower bound and the fact that $\inf(I) \notin I$ we have that $\inf(I) < z \wedge z \leq \sup(I)$ so that

$$I \subseteq]\inf(I), \sup(I)] \quad (3.76)$$

If $z \in]\inf(I), \sup(I)]$ we have $\inf(I) < z \wedge z \leq \sup(I)$. Using [theorem: 3.74] there exists a $x \in I$ such that $\inf(I) \leq x < z$ or $z \in [x, \sup(I)] \subseteq I$ [as I is a generalized interval and $x, \sup(I) \in I$]. This proves that $] \inf(I), \sup(I)] \subseteq I$. Combining this with [eq: 3.76] proves

$$I =]\inf(I), \sup(I)]$$

- c. Let $z \in I$ then using the definition of a supremum and infimum as upper and lower bound and the fact that $\sup(I) \notin I$ we have that $\inf(I) \leq z \wedge z < \sup(I)$ so that

$$I \subseteq [\inf(I), \sup(I)[\quad (3.77)$$

If $z \in [\inf(I), \sup(I)[$ we have $\inf(I) \leq z \wedge z < \sup(I)$. Using [theorem: 3.74] there exists a $y \in I$ such that $z < y \leq \sup(I)$ or $z \in [\inf(I), y] \subseteq I$ [as I is a generalized interval and $\inf(I), y \in I$]. This proves that $[\inf(I), \sup(I)[\subseteq I$. Combining this with [eq: 3.77] proves

$$I = [\inf(I), \sup(I)[$$

- d. Let $z \in I$ then using the definition of a supremum and infimum as upper and lower bound and the fact that $\inf(I) \notin I \wedge \sup(I) \notin I$ we have that $\inf(I) < z \wedge z < \sup(I)$ so that

$$I \subseteq]\inf(I), \sup(I)[\quad (3.78)$$

If $z \in]\inf(I), \sup(I)[$ we have $\inf(I) < z \wedge z < \sup(I)$. Using [theorem: 3.74] there exists $x, y \in I$ such that $\inf(I) \leq x < z \wedge z < y \leq \sup(I)$ or $z \in]x, y[\subseteq I$ [as I is a generalized interval and $x, y \in I$]. This proves that $] \inf(I), \sup(I)[\subseteq I$. Combining this with [eq: 3.78] gives

$$I =]\inf(I), \sup(I)[$$

2.

- a. Let $z \in I$ then using the definition of the infimum as a lower bound we have that $\inf(I) \leq z$ proving that

$$I \subseteq [\inf(I), \infty[\quad (3.79)$$

If $z \in [\inf(I), \infty[$ then $\inf(I) \leq z$. As I is not bounded above there exists a $x \in I$ such that $\neg(x \leq z) \xRightarrow{A \text{ is fully ordered}} z < x$. Hence $z \in [\inf(I), x] \subseteq I$ [as I is a generalized interval and $\inf(I), x \in I$]. So $[\inf(I), \infty[\subseteq I$ and combining this with [eq: 3.79] proves

$$I = [\inf(I), \infty[$$

- b. Let $z \in I$ then using the definition of the infimum as a lower bound and the fact that $\inf(I) \notin I$ we have that $\inf(I) < z$ proving that

$$I \subseteq]\inf(I), \infty[\quad (3.80)$$

If $z \in]\inf(I), \infty[$ then $\inf(I) < z$. Using [theorem: 3.74] there exists a $x \in I$ such that $\inf(I) \leq x < z$. As I is not bounded above there exists a $y \in I$ such that $\neg(y \leq z) \xRightarrow{A \text{ is fully ordered}} z < y$. Hence $z \in]x, y[\subseteq I$ [as I is a generalized interval and $x, y \in I$]. So $] \inf(I), \infty[\subseteq I$ and combining this with [eq: 3.80] gives

$$I =]\inf(I), \infty[$$

3.

- a. Let $z \in I$ then using the definition of the supremum as an upper bound we have $z \leq \sup(I)$ proving that

$$I \subseteq]-\infty, \sup(I)] \quad (3.81)$$

If $z \in]-\infty, \sup(I)]$ then $z \leq \sup(I)$. As I is not bounded below there exists a $x \in I$ such that $\neg(z \leq x) \xRightarrow{A \text{ is fully ordered}} x < z$. Hence $z \in [x, \sup(I)] \subseteq I$ [as I is a generalized interval and $x, \sup(I) \in I$]. So $]-\infty, \sup(I)] \subseteq I$ which combined with [eq: 3.81] gives

$$I =]-\infty, \sup(I)]$$

- b. Let $z \in I$ then using the definition of the supremum as an upper bound and the fact that $\sup(I) \notin I$ we have that $z < \sup(I)$ proving that

$$I \subseteq]-\infty, \sup(I)[\quad (3.82)$$

If $z \in]-\infty, \sup(I)[$ then $z < \sup(I)$. Using [theorem: 3.74] there exists a $y \in I$ such that $z < y \leq \sup(I)$. As I is not bounded below there exists a $x \in I$ such that $\neg(z \leq x) \xRightarrow{A \text{ is fully ordered}} x < z$. Hence $z \in]x, y[\subseteq I$ [as I is a generalized interval and $x, y \in I$]. So $] -\infty, \sup(I)[\subseteq I$ which combined with [eq: 3.82] proves

$$I =]-\infty, \sup(I)[$$

4. If $z \in A$ then as I is not bounded below and not bounded above there exists $x, y \in I$ such that $x < z \wedge z < y$ giving $z \in]x, y[\subseteq I$ [as I is a generalized interval and $x, y \in I$]. This proves that $A \subseteq I$ and as trivially we have $I \subseteq A$ it follows that

$$I = A \quad \square$$

We have also the opposite of the above theorem

Theorem 3.144. *Let $\langle A, \leq \rangle$ be a fully ordered class that is conditional complete then we have given $a, b \in A$ that*

1. $[a, b]$ is a generalized interval
2. $[a, b[$ is a generalized interval
3. $]a, b]$ is a generalized interval
4. $]a, b[$ is a generalized interval
5. $[a, \infty[$ is a generalized interval
6. $]a, \infty[$ is a generalized interval
7. $]-\infty, a]$ is a generalized interval
8. $]-\infty, a[$ is a generalized interval
9. A is a generalized interval

Proof.

1. If $x, y \in [a, b]$ and $z \in [x, y]$ then we have $x \leq z \Rightarrow_{a \leq x} a \leq z$ and $z \leq y \Rightarrow_{y \leq b} z \leq b$ proving that $z \in [a, b]$. Hence $[x, y] \subseteq [a, b]$ from which it follows that

$[a, b]$ is a generalized interval

2. If $x, y \in [a, b[$ and $z \in [x, y]$ then we have $x \leq z \Rightarrow_{a \leq x} a \leq z$ and $z \leq y \Rightarrow_{y < b} z < b$ proving that $z \in [a, b[$. Hence $[x, y] \subseteq [a, b[$ from which it follows that

$[a, b[$ is a generalized interval

3. If $x, y \in]a, b]$ and $z \in [x, y]$ then we have $x \leq z \Rightarrow_{a < x} a < z$ and $z \leq y \Rightarrow_{y \leq b} z \leq b$ proving that $z \in]a, b]$. Hence $[x, y] \subseteq]a, b]$ from which it follows that

$]a, b]$ is a generalized interval

4. If $x, y \in]a, b[$ and $z \in [x, y]$ then we have $x \leq z \Rightarrow_{a < x} a < z$ and $z \leq y \Rightarrow_{y < b} z < b$ proving that $z \in]a, b[$. Hence $[x, y] \subseteq]a, b[$ from which it follows that

$]a, b[$ is a generalized interval

5. If $x, y \in [a, \infty[$ and $z \in [x, y]$ we have $x \leq z \Rightarrow_{a \leq x} a \leq z$ proving that $z \in [a, \infty[$. Hence $[x, y] \subseteq [a, \infty[$ from which it follows that

$[a, \infty[$ is a generalized interval

6. If $x, y \in]a, \infty[$ and $z \in [x, y]$ we have $x \leq z \Rightarrow_{a < x} a < z$ proving that $z \in]a, \infty[$. Hence $[x, y] \subseteq]a, \infty[$ from which it follows that

$]a, \infty[$ is a generalized interval

7. If $x, y \in]-\infty, a]$ and $z \in [x, y]$ we have $z \leq y \Rightarrow_{y \leq a} z \leq a$ proving that $z \in]-\infty, a]$. Hence $[x, y] \subseteq]-\infty, a]$ from which it follows that

$]-\infty, a]$ is a generalized interval

8. If $x, y \in]-\infty, a[$ and $z \in [x, y]$ we have $z \leq y \Rightarrow_{y < a} z < a$ proving that $z \in]-\infty, a[$. Hence $[x, y] \subseteq]-\infty, a[$ from which it follows that

$]-\infty, a[$ is a generalized interval

9. If $x, y \in A$ then trivially $[x, y] \subseteq A$ hence A is a generalized interval. □

Theorem 3.145. *Let $\langle A, \leq \rangle$ be a totally ordered class then*

1. $\forall a, b \in A$ we have if $]-\infty, a] =]-\infty, b]$ then $a = b$

2. $\forall a, b \in A$ we have if $] -\infty, a[=] -\infty, b[$ then $a = b$
3. $\forall a, b \in A$ we have if $[a, \infty[= [b, \infty[$ then $a = b$
4. $\forall a, b \in A$ we have if $]a, \infty[=]b, \infty[$ then $a = b$
5. $\forall a, b, c, d$ with $[a, b] \neq \emptyset$ we have if $[a, b] = [c, d]$ then $a = c \wedge b = d$
6. $\forall a, b, c, d$ with $]a, b[\neq \emptyset$ we have if $]a, b[=]c, d[$ then $a = c \wedge b = d$
7. $\forall a, b, c, d$ with $[a, b[\neq \emptyset$ we have if $[a, b[= [c, d[$ then $a = c \wedge b = d$
8. $\forall a, b, c, d$ with $]a, b[\neq \emptyset$ we have if $]a, b[=]c, d[$ then $a = c \wedge b = d$
9. $\forall a, b, c, d$ with $[a, b[\neq \emptyset$ we have if $[a, b[\subseteq [c, d[$ then $c \leq a \wedge b \leq d$

Proof.

1. As $a \in] -\infty, a[=] -\infty, b[$ we have $a \leq b$, further from $b \in] -\infty, b[=] -\infty, a[$ we have $b \leq a$, hence $a = b$.
2. As A is totally ordered we have for a, b either
 - $a = b$. then (2) is proved
 - $a < b$. then $a \in] -\infty, b[=] -\infty, a[$ giving the contradiction $a < a$
 - $b < a$. then $b \in] -\infty, a[=] -\infty, b[$ giving the contradiction $b < b$
 so the only valid conclusion is that $a = b$
3. As $a \in [a, \infty[= [b, \infty[$ we have $b \leq a$, further from $b \in [b, \infty[= [a, \infty[$ we have $a \leq b$, hence $a = b$
4. As A is totally ordered we have for a, b either
 - $a = b$. then (4) is proved
 - $a < b$. then $b \in]a, \infty[=]b, \infty[$ giving the contradiction $b < b$
 - $b < a$. then $a \in]b, \infty[=]a, \infty[$ giving the contradiction $a < a$
 so the only valid conclusion is that $a = b$
5. As $[a, b] = [c, d] \neq \emptyset$ it follows from [theorem: 3.140] that $a \leq b$ and $c \leq d$ so that $a, b \in [a, b] = [c, d] \Rightarrow c \leq a \wedge b \leq d$ and $c, d \in [c, d] = [a, b] \Rightarrow a \leq c \wedge d \leq b$ proving that $a = c \wedge b = d$.
6. As $]a, b[=]c, d[\neq \emptyset$ it follows from [theorem: 3.140] that $a < b \wedge c < d$ hence $b \in]a, b[\wedge d \in]c, d[$. From $b \in]a, b[=]c, d[$ we have $b \leq d$ and from $d \in]c, d[=]a, b[$ we have $d \leq b$ hence we have

$$b = d$$

Assume that $a < c$ then as $c < d = b$ we have $c \in]a, b[=]c, d[$ so that $c < c$ a contradiction, hence we must have

$$c \leq a \tag{3.83}$$

Assume that $c < a$ then as $a < b = d$ we have $a \in]c, d[=]a, b[$ so that $a < a$ a contradiction hence we must have $a \leq c$ which combined with [eq: 3.83] proves that

$$a = c$$

7. As $[a, b[= [c, d[\neq \emptyset$ we have by [theorem: 3.140] that $a < b \wedge c < d$ so that $a \in [a, b[\wedge c \in [c, d[$. From $a \in [a, b[= [c, d[$ it follows that $c \leq a$ and from $c \in [c, d[= [a, b[$ we have $a \leq c$ hence we have

$$a = c$$

Assume that $b < d$ then as $c = a < b$ we have $b \in [c, d[= [a, b[$ so that $b < b$ a contradiction, hence

$$b \leq d \tag{3.84}$$

Assume that $d < b$ then as $a = c < d < b$ we have $d \in [a, b[= [c, d[$ so that $d < d$ a contradiction, hence $d \leq b$ which combined with [eq: 3.84] proves

$$b = d$$

8. As $\emptyset \neq]a, b[=]c, d[$ there exists a $x \in A$ such that $x \in]a, b[=]c, d[$ giving

$$a < x < b \wedge c < x < d \quad (3.85)$$

Assume that $a < c$ then by [eq: 3.85] $a < c < x < b$ so that $c \in]a, b[=]c, d[$ leading to the contradiction $c < c$ hence

$$c \leq a \quad (3.86)$$

Assume that $c < a$ then by [eq: 3.85] $c < a < x < d$ so that $a \in]c, d[=]a, b[$ leading to the contradiction $a < a$ hence $a \leq c$ which combined with [eq: 3.86] gives

$$a = c$$

Assume that $b < d$ then by [eq: 3.85] $c < x < b < d$ so that $b \in]c, d[=]a, b[$ leading to the contradiction $b < b$, hence

$$d \leq b \quad (3.87)$$

Assume that $d < b$ then by [eq: 3.85] $a < x < d < b$ so that $d \in]a, b[=]c, d[$ leading to the contradiction $d < d$ proving that $b \leq d$. Combining this with [eq: 3.87] results in

$$b = d$$

9. As $[a, b[\neq \emptyset$ $a < b$ so that $a \in [a, b[\subseteq [c, d[$ hence

$$c \leq a$$

Assume that $d < a$ then, as $a \in [a, b[\subseteq [c, d[$, we have $a < d$ leading to the contradiction $d < d$ hence we must have

$$a \leq d$$

Assume that $d < b$ then by the above we have $a \leq d < b$ so that $d \in [a, b[\subseteq [c, d[$ leading to the contradiction $d < d$. Hence we must have

$$b \leq d \quad \square$$

Chapter 4

Algebraic constructs

Before we define the different number systems, like the natural numbers, whole numbers, rational numbers, real numbers and complex numbers, we define the algebraic operations and structures that we can define on them. In this way we abstract away the algebraic operations and algebraic structures. First we define the concept of a operator which is short notation for the application of a function with two arguments between a set and itself.

Definition 4.1. (Operator) Let X be a set then a **operator** is function

$$f: X \times X \rightarrow X$$

To avoid using excessive notation we use infix notation instead of the classic function call notation, so

$$f(x, y) \text{ is noted as } xfy$$

4.1 Groups

Definition 4.2. A semi-group is a pair $\langle G, \odot \rangle$ where G is a set and $\odot$ a operator $\odot: G \times G \rightarrow G$ such that:

neutral element. $\exists e \in G$ such that $\forall x \in G$ we have $x \odot e = x = e \odot x$

associativity. $\forall x, y, z \in G$ we have $(x \odot y) \odot z = x \odot (y \odot z)$

Theorem 4.3. If $\langle G, \odot \rangle$ is a semi-group then

1. $G \neq \emptyset$
2. G has only one neutral element

Proof.

1. As G is a group there exist a neutral element $e \in G$ so that $G \neq \emptyset$
2. Assume that there exists two neutral elements e, e' then we have

$$e \underset{e' \text{ is neutral element}}{=} e \odot e' \underset{e \text{ is neutral element}}{=} e' \quad \square$$

Example 4.4. Let X be a set then $\langle X^X, \circ \rangle$ is a semi group [see definition: 2.32]. Here X^X is the set of function graphs between X and X and $\circ$ is the composition between functions.

Proof. As X is a set we have by [theorem: 2.37] that X^X is a set. Further if $f, g \in X^X$ then $f: X \rightarrow$ and $g: X \rightarrow X$ are functions, so that by [theorem: 2.29] $f \circ g: X \rightarrow X$ is a function, hence $f \circ g \in X^X$. So

$$\circ: X^X \times X^X \rightarrow X^X \text{ defined by } \circ(f, g) = f \circ g$$

is a function. The neutral element is Id_X because $\forall f \in X^X$ we have

$$f \circ \text{Id}_X \underset{[\text{theorem: 2.50}]}{=} f \underset{[\text{theorem: 2.50}]}{=} \text{Id}_X \circ f \quad \square$$

A group is a semi-group with the extra condition that is has a inverse element.

Definition 4.5. A **group** $\langle X, \odot \rangle$ is a semi-group with the extra condition

Inverse Element. $\forall x \in G$ there $\exists y \in G$ such that

$$x \odot y = e = y \odot x$$

where e is the neutral element of the group.

One benefit that a group has is the canceling property

Theorem 4.6. If $x, y, z \in \langle G, \odot \rangle$ then $x \odot z = y \odot z$ then $x = y$

Proof. We have

$$\begin{aligned} x \odot z = y \odot z &\Rightarrow (x \odot z) \odot z^{-1} = (y \odot z) \odot z^{-1} \\ &\Rightarrow_{\text{associativity}} x \odot (z \odot z^{-1}) = y \odot (z \odot z^{-1}) \\ &\Rightarrow_{\text{inverse element}} x \odot e = y \odot e \\ &\Rightarrow x = y \\ &\quad \square \end{aligned}$$

Theorem 4.7. If $\langle G, \odot \rangle$ is group then every element has a unique inverse element. So

$$\forall x \in G \exists! y \in G \text{ such that } x \odot y = x = y \odot x$$

this unique element is noted as x^{-1} [or sometimes as $-x$]

Proof. Let $x \in G$ and assume that y, y' are inverse elements for x then we have

$$x \odot y = e = y \odot x \text{ and } x \odot y' = e = y' \odot x$$

So that

$$y = y \odot e = y \odot (x \odot y') = (y \odot x) \odot y' = e \odot y' = y'$$

□

Theorem 4.8. If $\langle G, \odot \rangle$ is a group then $\forall x, y \in G$ we have $(x \odot y)^{-1} = y^{-1} \odot x^{-1}$

Proof. We have

$$\begin{aligned} (x \odot y) \odot (y^{-1} \odot x^{-1}) &= x \odot (y \odot (y^{-1} \odot x^{-1})) \\ &= x \odot ((y \odot y^{-1}) \odot x^{-1}) \\ &= x \odot (e \odot x^{-1}) \\ &= x \odot x^{-1} \\ &= e \\ (y^{-1} \odot x^{-1}) \odot (x \odot y) &= y^{-1} \odot (x^{-1} \odot (x \odot y)) \\ &= y^{-1} \odot ((x^{-1} \odot x) \odot y) \\ &= y^{-1} \odot (e \odot y) \\ &= y^{-1} \odot y \\ &= e \end{aligned}$$

□

Theorem 4.9. If $\langle G, \odot \rangle$ is a group then $\forall x \in G$ we have $(x^{-1})^{-1} = x$ and $e^{-1} = e$ where e is the neutral element.

Proof. If $x \in G$ then $x \odot x^{-1} = e = x^{-1} \odot x$ and $(x^{-1})^{-1} \odot x^{-1} = e = x^{-1} \odot (x^{-1})^{-1}$. So

$$\begin{aligned} x &= x \odot e \\ &= x \odot (x^{-1} \odot (x^{-1})^{-1}) \\ &= (x \odot x^{-1}) \odot (x^{-1})^{-1} \\ &= e \odot (x^{-1})^{-1} \\ &= (x^{-1})^{-1} \end{aligned}$$

Further

$$e^{-1} = e \cdot e^{-1} = e$$

□

Theorem 4.10. *If $\langle G, \odot \rangle$ then $\forall x, y \in X$ we have $x = y \Leftrightarrow x^{-1} = y^{-1}$ [and by contra position $x \neq y \Leftrightarrow x^{-1} \neq y^{-1}$]*

Proof.

$\Rightarrow$. $e = x^{-1} \cdot x = x^{-1} \cdot y$ and $e = x \cdot x^{-1} = y \cdot x^{-1}$ proving by uniqueness of the inverse [see theorem: 4.7] that $y^{-1} = x^{-1}$

$\Leftarrow$. If $x^{-1} = y^{-1}$ then by the above we have $(x^{-1})^{-1} = (y^{-1})^{-1}$ it follows from [theorem: 4.9] that $x = y$.

□

Definition 4.11. *A semi-group or group $\langle G, \odot \rangle$ is Abelian or **commutative** iff*

$$\forall x, y \in G \text{ we have } x \odot y = y \odot x$$

Definition 4.12. *Let $\langle G, \odot \rangle$ be a semi-group then $F \subseteq G$ is a sub-semi-group iff*

1. $\forall x, y \in F$ we have $x \odot y \in F$
2. $e \in F$ [e is the neutral element of G]

Definition 4.13. *Let $\langle G, \odot \rangle$ be groups then $F \subseteq G$ is a sub-group iff*

1. $\forall x, y \in F$ we have $x \odot y \in F$
2. $e \in F$ [e is the neutral element of G]
3. $\forall x \in F$ we have $x^{-1} \in F$

The following show how sub-semi-groups and sub-groups can be used to reduce the work for proving the group axioms.

Theorem 4.14. *Let $\langle G, \odot \rangle$ be a semi-group and $F \subseteq G$ a sub-semi-group then*

1. $\langle F, \odot|_{F \times F} \rangle$ is a semi group with the same neutral element as $\langle G, \odot \rangle$
2. If $\langle G, \odot \rangle$ is Abelian then $\langle F, \odot|_{F \times F} \rangle$ is Abelian

To avoid excessive notation we use $\odot$ instead of $\odot|_{F \times F}$ if it is clear from the context which operation should be used.

Proof. First as G is a set we have by the Axiom of Subsets [axiom: 1.8] that G is a set.

1. For $\langle F, \odot|_{F \times F} \rangle$

neutral element. By definition of a subgroup $e \in F$. Let $x \in F$ then

$$e \odot|_{F \times F} x \underset{e, x \in F}{=} e \odot x = x = x \odot e = x \odot|_{F \times F} e$$

associativity. Let $x, y, z \in F$ then

$$(x \odot|_{F \times F} y) \odot|_{F \times F} z = (x \circ y) \circ z = x \circ (y \circ z) = x \odot|_{F \times F} (y \odot|_{F \times F} z)$$

2. Let $x, y \in F$ then

$$x \odot|_{F \times F} y = x \odot y = y \odot x = y \odot|_{F \times F} x$$

□

Theorem 4.15. *Let $\langle G, \odot \rangle$ be a semi-group $F \subseteq G$ a sub semi-group of $\langle G, \odot \rangle$ and $H \subseteq F$ a sub semi-group of $\langle G, \odot|_{F \times F} \rangle$ then H is a sub semi-group of $\langle G, \odot \rangle$*

Proof.

1. $\forall x, y \in H \subseteq F$ we have $x \odot_{|F \times F} y \in H$ which as $(x, y) \in F \times F$ proves that $x \odot y = x \odot_{|F \times F} y \in H$
2. if e is the neutral element of $\langle G, \odot \rangle$ then by [theorem: 4.14] e is also the neutral element of F , hence $e \in H$. $\square$

Theorem 4.16. *Let $\langle G, \odot \rangle$ be a group and $F \subseteq G$ a sub-group then*

1. $\langle F, \odot_{|F \times F} \rangle$ is a group with same neutral element as $\langle G, \odot \rangle$ and for every $x \in F$ it's inverse element in $\langle G, \odot \rangle$ is also the inverse element in $\langle F, \odot_{|F \times F} \rangle$.
2. If $\langle G, \odot \rangle$ is Abelian then $\langle F, \odot_{|F \times F} \rangle$ is Abelian

To avoid excessive notation we use $\odot$ instead of $\odot_{|F \times F}$ if it is clear from the context which operation should be used.

Proof.

1. For $\langle F, \odot_{|F \times F} \rangle$ we have

neutral element. Let $x \in F$ then $e \odot_{|F \times F} x \stackrel{e, x \in F}{=} e \odot x = x = x \odot e = x \odot_{|F \times F} e$

associativity. Let $x, y, z \in F$ then

$$(x \odot_{|F \times F} y) \odot_{|F \times F} z = (x \circ y) \circ z = x \circ (y \circ z) = x \odot_{|F \times F} (y \odot_{|F \times F} z)$$

inverse element. Let $x \in F$ then also $x^{-1} \in F$ then

$$(x \odot_{|F \times F} x^{-1}) = x \odot x^{-1} = e = x^{-1} \odot c = x^{-1} \odot_{|F \times F} x$$

2. Let $x, y \in F$ then

$$x \odot_{|F \times F} y = x \odot y = y \odot x = y \odot_{|F \times F} x$$

$\square$

Theorem 4.17. *Let $\langle G, \odot \rangle$ be a [semi-]group $F \subseteq G$ a sub [semi-]group of $\langle G, \odot \rangle$ and $H \subseteq F$ a sub [semi-]group of $\langle G, \odot_{|F \times F} \rangle$ then H is a sub [semi-]group of $\langle G, \odot \rangle$*

Proof.

1. $\forall x, y \in H \subseteq F$ we have $x \odot_{|F \times F} y \in H$ which as $(x, y) \in F \times F$ proves that $x \odot y = x \odot_{|F \times F} y \in H$.
2. If e is the neutral element of $\langle G, \odot \rangle$ then by [theorem: 4.16] e is also the neutral element of F , hence $e \in H$.
3. If $x \in H$ then $x \in G$ and by [theorem: 4.16] it's inverse element x^{-1} in $\langle G, \odot \rangle$ is also its inverse element in $\langle F, \odot_{|F \times F} \rangle$ hence $x^{-1} \in H$. $\square$

Example 4.18. Let X be a set, $\langle X^X, \circ \rangle$ the semi-group used in [example: 4.4] then $\langle \mathcal{B}[X], \circ \rangle$ is a group where $\mathcal{B}[X] = \{f \in X^X \mid f: X \rightarrow X \text{ is a bijection}\}$.

Proof. First we prove that $\mathcal{B}[X]$ is a sub-semi-group

1. $\forall f, g \in \mathcal{B}[X]$ we have that $f: X \rightarrow X$ and $g: X \rightarrow X$ are bijections so that by [theorem: 2.77] $f \circ g$ is a bijection so that $f \circ g \in \mathcal{B}[X]$
2. $\text{Id}_X: X \rightarrow X$ is by [theorem: 2.66] a bijection so that $\text{Id}_X \in \mathcal{B}[X]$

Applying then [theorem: 4.14] proves that

$$\langle \mathcal{B}[X], \circ \rangle \text{ is a semi-group}$$

Let $f \in \mathcal{B}[X]$ then $f: X \rightarrow X$ is a bijection and by [theorems: 2.71, 2.74] we have that $f^{-1}: X \rightarrow X$ is a bijection, so that $f^{-1} \in \mathcal{B}[X]$ and $f \circ \text{Id}_X = f = \text{Id}_X \circ f$. $\square$

Definition 4.19. (Group Homeomorphism) *If $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$ be semi-groups then a function $f: F \rightarrow G$ is a **group homeomorphism between $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$** iff $\forall x, y \in F$ we have $f(x \odot y) = f(x) \oplus f(y)$.*

Notation 4.20. We use the following notation for a group homeomorphism between $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$

$$f: \langle F, \odot \rangle \rightarrow \langle G, \oplus \rangle \text{ is a group homeomorphism}$$

Theorem 4.21. If $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$ are semi-groups, $H \subseteq G$ a sub-semi-group of $\langle G, \oplus \rangle$ and

$$f: \langle F, \odot \rangle \rightarrow \langle H, \oplus|_{H \times H} \rangle \text{ is a group homeomorphism}$$

then

$$f: \langle F, \odot \rangle \rightarrow \langle G, \oplus \rangle \text{ is a group homeomorphism}$$

Proof. Let $x, y \in F$ then we have

$$f(x \odot y) = f(x) \oplus_{|H \times H} f(y) = f(x) \oplus f(y)$$

□

Theorem 4.22. If $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$ be semi groups with neutral elements e_F, e_G and $f: F \rightarrow G$ a **group homeomorphism** then:

1. $f(e_F) = e_G$
2. If $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$ are groups then $\forall x \in F$ we have $f(x^{-1}) = f(x)^{-1}$
3. $f(F)$ is a sub-[semi-]group of $\langle G, \oplus \rangle$ if $\langle F, \odot \rangle$ is a [semi-]group

Proof.

1.

$$\begin{aligned} e_G &= f(e_F)^{-1} \oplus f(e_F) \\ &= f(e_F)^{-1} \oplus f(e_F \odot e_F) \\ &= f(e_F)^{-1} \oplus (f(e_F) \oplus f(e_F)) \\ &= (f(e_F)^{-1} \oplus f(e_F)) \oplus f(e_F) \\ &= e_G \oplus f(e_F) \\ &= f(e_F) \end{aligned}$$

2. If $x \in F$ then

$$f(x^{-1}) \oplus f(x) = f(x^{-1} \odot x) = f(e_F) \stackrel{(1)}{=} e_G$$

and

$$f(x) \oplus f(x^{-1}) = f(x \odot x^{-1}) = f(e_F) \stackrel{(1)}{=} e_G$$

so that $f(x)^{-1} = f(x^{-1})$

3. If $x, y \in f(F)$ then there exists $u, v \in F$ such that $x = f(u)$ and $y = f(v)$, then we have

$$x \oplus y = f(u) \oplus f(v) = f(u \odot v) \in f(F)$$

Also

$$e_G \stackrel{(1)}{=} f(e_F) \in f(F)$$

Finally if $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$ are groups and $x \in f(F)$ then there exists a $u \in F$ such that $x = f(u)$, then we have

$$x^{-1} \stackrel{(2)}{=} f(u)^{-1} = f(u^{-1}) \in f(F)$$

□

Definition 4.23. (Group Isomorphism) If $\langle F, \odot \rangle$ and $\langle G, \oplus \rangle$ are semi-groups then a **group isomorphism** is a **bijection** $f: F \rightarrow G$ such that

$$f: \langle F, \odot \rangle \rightarrow \langle G, \oplus \rangle \text{ is a group homeomorphism}$$

Theorem 4.24. Let $\langle F, \odot \rangle, \langle G, \oplus \rangle$ be semi groups and

$$f: \langle F, \odot \rangle \rightarrow \langle G, \oplus \rangle \text{ is a group isomorphism}$$

then

$$f^{-1}: \langle G, \oplus \rangle \rightarrow \langle F, \odot \rangle \text{ is a group isomorphism}$$

Proof. As $f: F \rightarrow G$ is a bijection we have by [theorem: 2.74] that $f^{-1}: G \rightarrow F$ is a bijection. Take $x, y \in G$ then we have

$$\begin{aligned} f^{-1}(x \oplus y) &= f^{-1}(\text{Id}_G(x) \oplus \text{Id}_G(y)) \\ &\stackrel{[\text{theorem: 2.71}]}{=} f^{-1}((f \circ f^{-1})(x) \oplus (f \circ f^{-1})(y)) \\ &\stackrel{[\text{theorem: 2.44}]}{=} f^{-1}(f(f^{-1}(x)) \oplus f(f^{-1}(y))) \\ &\stackrel{f \text{ is homeomorphism}}{=} f^{-1}(f(f^{-1}(x) \odot f^{-1}(y))) \\ &\stackrel{[\text{theorem: 2.44}]}{=} (f^{-1} \circ f)(f^{-1}(x) \odot f^{-1}(y)) \\ &\stackrel{[\text{theorem: 2.71}]}{=} \text{Id}_F(f^{-1}(x) \odot f^{-1}(y)) \\ &= f^{-1}(x) \odot f^{-1}(y) \end{aligned}$$

Further if e_F, e_G are the neutral elements of $\langle F, \oplus_F \rangle, \langle G, \oplus_G \rangle$ then

$$\begin{aligned} e_F &= \text{Id}_F(e_f) \\ &= (f^{-1} \circ f)(e_F) \\ &= f^{-1}(f(e_F)) \\ &= f^{-1}(e_G) \end{aligned}$$

proving that

$$f^{-1}: F \rightarrow G \text{ is a group isomorphism}$$

□

Theorem 4.25. If $\langle A, \oplus_A \rangle, \langle B, \oplus_B \rangle$ and $\langle C, \oplus_C \rangle$ are [semi]-groups then

1. If D is a sub [semi]-group of $\langle B, \oplus_B \rangle$ and

$$f: \langle A, \oplus_A \rangle \rightarrow \langle D, \oplus_B \rangle \text{ and } g: \langle B, \oplus_B \rangle \rightarrow \langle C, \oplus_C \rangle \text{ are group homeomorphism}$$

then

$$g \circ f: \langle A, \oplus_A \rangle \rightarrow \langle C, \oplus_C \rangle \text{ is a group homeomorphism}$$

and

$$g(f(A)) \text{ is a sub [semi]-group of } \langle C, \oplus_C \rangle$$

2. If D is a sub group of $\langle B, \oplus_B \rangle$ and

$$f: \langle A, \oplus_A \rangle \rightarrow \langle D, \oplus_B \rangle \text{ and } g: \langle B, \oplus_B \rangle \rightarrow \langle C, \oplus_C \rangle \text{ are group isomorphisms}$$

then

$$g \circ f: \langle A, \oplus_A \rangle \rightarrow \langle g(f(A)), \oplus_C \rangle \text{ is a group isomorphism}$$

or as $g(D) \stackrel{f: A \rightarrow D \text{ is injective}}{=} g(f(A))$ that

$$g \circ f: \langle A, \oplus_A \rangle \rightarrow \langle g(D), \oplus_C \rangle \text{ is a group isomorphism}$$

Proof.

1. Let $x, y \in A$ then we have

$$\begin{aligned} (g \circ f)(x \oplus_A y) &= g(f(x \oplus_A y)) \\ &\stackrel{f \text{ is a homeomorphism}}{=} g(f(x) \oplus_B f(y)) \\ &\stackrel{g \text{ is a homeomorphism}}{=} g(f(x)) \oplus_C g(f(y)) \\ &= (g \circ f)(x) \oplus_C (g \circ f)(y) \end{aligned}$$

proving that $g \circ f$ is a group homeomorphism. Finally using [theorem: 4.22] we have then that

$g(f(A))$ is a sub group

2. Using [theorem: 2.77] we have that $g \circ f: A \rightarrow g(f(A))$ is a bijection which combined with (1) proves that $g \circ f: \langle A, \oplus_A \rangle \rightarrow \langle C, \oplus_C \rangle$ is a group isomorphism. $\square$

The following theorem show how we can define a group on the product of a family of groups.

Theorem 4.26. *Let $\{\langle A_i, \odot_i \rangle\}_{i \in I}$ be a family of semi-groups then we have*

1. *If $x, y \in \prod_{i \in I} A_i$ then $(x \odot y) \in \prod_{i \in I} A_i$ where $x \odot y$ is defined by $(x \odot y)_i = x_i \odot_i y_i$*
2. *If we define $\odot: (\prod_{i \in I} A_i) \times (\prod_{i \in I} A_i) \rightarrow \prod_{i \in I} A_i$ by $\odot(x, y) = x \odot y$ then*

$$\left\langle \prod_{i \in I} A_i, \odot \right\rangle$$

is a semi-group with neutral element e defined by $(e)_i = e_i$ where e_i is the neutral element of $\langle A_i, \odot_i \rangle$

3. *If $\forall i \in I$ we have that $\langle A_i, \odot_i \rangle$ is Abelian then $\langle \prod_{i \in I} A_i, \odot \rangle$ is Abelian.*
4. *If $\forall i \in I$ we have that $\langle A_i, \odot_i \rangle$ is a group then $\langle \prod_{i \in I} A_i, \odot \rangle$ is a group where the inverse x^{-1} for each $x \in \prod_{i \in I} A_i$ is defined by $(x^{-1})_i = (x_i)^{-1}$ [here $(x_i)^{-1}$ is the inverse of x_i in the group $\langle A_i, \odot_i \rangle$]*

Proof.

1. If $x, y \in \prod_{i \in I} A_i$ then x is a function $x: I \rightarrow \bigcup_{i \in I} A_i$ such that $\forall i \in I$ $x_i = x(i) \in A_i$ and y is a function $y: I \rightarrow \bigcup_{i \in I} A_i$ such that $\forall i \in I$ $y_i = y(i) \in A_i$. So if we define $x \odot y$ by $(x \odot y)(i) = (x \odot y)_i = x_i \odot_i y_i = x(i) \odot_i y(i)$ then $x \odot y: I \rightarrow \bigcup_{i \in I} A_i$ is a function and $\forall i \in I$ we have $(x \odot y)(i) = x(i) \odot_i y(i) \in A_i$ [as $\langle A_i, \odot_i \rangle$ is a semi-group]. Hence $x \odot y \in \prod_{i \in I} A_i$
2. We have

associativity. Let $x, y, z \in \prod_{i \in I} A_i$ then we have for $i \in I$

$$\begin{aligned} (x \odot (y \odot z))(i) &= x(i) \odot_i (y \odot z)(i) \\ &= x(i) \odot_i (y(i) \odot_i z(i)) \\ &\stackrel{\langle A_i, \odot_i \rangle \text{ is a semi group}}{=} (x(i) \odot_i y(i)) \odot_i z(i) \\ &= (x \odot y)(i) \odot_i z(i) \\ &= ((x \odot y) \odot z)(i) \end{aligned}$$

so that

$$x \odot (y \odot z) = (x \odot y) \odot z$$

neutral element. Let $x \in \prod_{i \in I} A_i$ then $\forall i \in I$

$$\begin{aligned} (x \odot e)(i) &= x(i) \odot_i e(i) \\ &= x(i) \odot_i e_i \\ &\stackrel{\langle A_i, \odot_i \rangle \text{ is a semi group}}{=} x(i) \\ (e \odot x)(i) &= e(i) \odot_i x(i) \\ &= e_i \odot_i x(i) \\ &\stackrel{\langle A_i, \odot_i \rangle \text{ is a semi group}}{=} x(i) \end{aligned}$$

so that

$$x \odot e = x = e \odot x$$

3. Let $x, y \in \prod_{i \in I} A_i$ then $\forall i \in I$ we have

$$(x \circ y)(i) = x(i) \odot_i y(i) \stackrel{\langle A_i, \odot_i \rangle \text{ is Abelian}}{=} y(i) \odot_i x(i) = (y \odot x)(i)$$

so that $x \odot y = y \odot x$

4. Let $x \in \prod_{i \in I} A_i$ then we have $\forall i \in I$ that

$$\begin{aligned} (x \odot x^{-1})(i) &= x(i) \odot_i (x^{-1})(i) \\ &= x(i) \odot_i (x_i)^{-1} \\ &= x_i \odot_i (x_i)^{-1} \\ &\stackrel{\langle A_i, \odot_i \rangle \text{ is a group}}{=} e_i \\ &= e(i) \\ (x^{-1} \odot x)(i) &= (x^{-1})(i) \odot_i x(i) \\ &= (x_i)^{-1} \odot_i x(i) \\ &= (x_i)^{-1} \odot_i x_i \\ &= e_i \\ &= e(i) \end{aligned}$$

So that $x \odot x^{-1} = e = x^{-1} \odot x$. Which as by (2) $\langle \prod_{i \in I} A_i, \odot \rangle$ is a semi group proves that $\langle \prod_{i \in I} A_i, \odot \rangle$ is a group. $\square$

The following five definitions will be later used in Linear Algebra.

Definition 4.27. Let $\langle G, \odot \rangle$ be a group with neutral element e and let X be a set then we have the following definitions:

1. A **left group action** is a function $\triangleright: G \times X \rightarrow X$ where $\triangleright(g, x) \stackrel{\text{notation}}{=} g \triangleright x$ such that
 - a. $\forall x \in X$ we have $e \triangleright x = x$
 - b. $\forall g, g' \in G$ and $\forall x \in X$ we have $(g \odot g') \triangleright x = g \triangleright (g' \triangleright x)$
2. A **right group action** is a function $\triangleleft: X \times G \rightarrow X$ where $\triangleleft(x, g) \stackrel{\text{notation}}{=} x \triangleleft g$ such that
 - a. $\forall x \in X$ we have $x \triangleleft e = x$
 - b. $\forall g, g' \in G$ and $\forall x \in X$ we have $x \triangleleft (g \odot g') = (x \triangleleft g) \triangleleft g'$

Definition 4.28. Let $\langle G, \odot \rangle$ be a group, X a set, $\triangleright$ a left group action and $g \in G$ then we define

$$g_{\triangleright}: X \rightarrow X \text{ by } g_{\triangleright}(x) = g \triangleright x$$

Definition 4.29. Let $\langle G, \odot \rangle$ be a group, X , $\triangleleft$ a right group action and $g \in G$ then we define

$$g_{\triangleleft}: X \rightarrow X \text{ by } g_{\triangleleft}(x) = x \triangleleft g$$

Definition 4.30. Let $\langle G, \odot \rangle$ be a group with neutral element e and let X be a set then we have the following definitions for a left group action $\triangleright$

1. $\triangleright$ is **faithful** if

$$g_{\triangleright} = \text{Id}_X \text{ if and only if } g = e$$

or equivalently

$$\{g \in G \mid \forall x \in X \text{ we have } g \triangleright x = x\} = \{e\}$$

2. $\triangleright$ is **transitive** iff $\forall x_1, x_2$ there exist a $g \in G$ such that $g \triangleright x_1 = x_2$
3. $\triangleright$ is **free** iff $\forall x \in X$ we have $\{g \in G \mid g \triangleright x = x\} = \{e\}$

Definition 4.31. Let $\langle G, \odot \rangle$ be a group with neutral element e and let X be a set then we have the following definitions for a right group action $\triangleleft$

1. $\triangleleft$ is **faithful** if

$$g_{\triangleleft} = \text{Id}_X \text{ if and only if } g = e$$

or equivalently

$$\{g \in G \mid \forall x \in X \text{ we have } g \triangleleft x = x\} = \{e\}$$

2. $\triangleright$ is **transitive** iff $\forall x_1, x_2$ there exists a $g \in G$ such that $g \triangleleft x_1 = x_2$
3. $\triangleright$ is **free** iff $\forall x \in X$ we have $\{g \in G \mid g \triangleleft x = x\} = \{e\}$

4.2 Rings

Definition 4.32. (Ring) A triple $\langle R, \oplus, \odot \rangle$ is a ring iff

1. R is a set
2. $\langle R, \oplus \rangle$ is a Abelian group or $\oplus: R \times R \rightarrow R$ is a operator such that
 - associativity.** $\forall x, y, z \in R$ we have $x \oplus (y \oplus z) = (x \oplus y) \oplus z$
 - neutral element.** $\exists 0 \in R$ such that $\forall x \in R$ we have $0 \oplus x = x = x \oplus 0$
 - inverse element.** $\forall x \in R$ there exist a $-x$ such that $x \oplus (-x) = 0 = (-x) \oplus x$
 - commutativity.** $\forall x, y \in R$ we have $x \oplus y = y \oplus x$ $\oplus$ is called the sum operator of the ring.
3. $\odot: R \times R \rightarrow R$ is a operator so that
 - distributivity.** $\forall x, y, z \in R$ we have $x \odot (y \oplus z) = (x \odot y) \oplus (x \odot z)$
 - neutral element.** $\exists 1 \in R$ such that $\forall x \in R$ we have $1 \odot x = x = x \odot 1$
 - commutativity.** $\forall x, y \in R$ we have $x \odot y = y \odot x$
 - associativity.** $\forall x, y, z \in R$ we have $x \odot (y \odot z) = (x \odot y) \odot z$ $\odot$ is called the multiplication operator of the ring.

Definition 4.33. If $\langle R, \oplus, \odot \rangle$ is a ring then a **zero divisor of R** is a $x \in R \setminus \{0\}$ so that $\exists y \in R \setminus \{0\}$ such that $x \odot y = 0$

Definition 4.34. A ring $\langle R, \oplus, \odot \rangle$ is a **integral domain** if it does not contains a zero divisor

Definition 4.35. (Sub-ring) If $\langle R, \oplus, \odot \rangle$ is a ring then a subset $S \subseteq R$ is a sub ring iff

1. $\forall x, y \in S$ we have $x \oplus y \in S$ and $x \odot y \in S$
2. $\forall x \in S$ we have $-x \in S$ [the inverse element for $\oplus$]
3. $1 \in S$ [the neutral element for $\odot$]
4. $0 \in S$ [the neutral element for $\oplus$]

Theorem 4.36. If $\langle R, \oplus, \odot \rangle$ is a ring $F \subseteq R$ a sub ring of $\langle R, \oplus, \odot \rangle$ then

$$F \text{ is a sub group of } \langle R, \oplus \rangle \text{ and } F \text{ is a sub semi-group of } \langle R, \odot \rangle$$

Proof. This follows directly from [definitions: 4.12, 4.13 and 4.35] □

Theorem 4.37. If $\langle R, \oplus, \odot \rangle$ is a ring and $S \subseteq R$ a sub ring then $\langle S, \oplus_{|S \times S}, \odot_{|S \times S} \rangle$ is a ring with the same neutral elements for addition and multiplications and for each $x \in S$ its inverse element is also the inverse element in $\langle S, \oplus_{|S \times S}, \odot_{|S \times S} \rangle$. For simplicity we note this ring as $\langle S, \oplus, \odot \rangle$

Proof.

1. S is a set as R is a set by the Axiom of Subsets [axiom: 1.8].
2. $\langle S, \oplus_{|S \times S} \rangle$ is a Abelian group by [theorem: 4.14]

3. $\odot: R \times R \rightarrow R$ is a operator so that

Distributivity. $\forall x, y, z \in S$ we have

$$x \odot_{|S \times S} (y \oplus_{|S \times S} z) = x \odot (y \oplus z) = (x \odot y) \oplus (x \odot z) = (x \odot_{|S \times S} y) \oplus_{|S \times S} (x \odot_{|S \times S} z)$$

neutral element. For $1 \in R$ we have $\forall x \in S$ that

$$1 \odot_{|S \times S} x = 1 \odot x = x = x \odot 1 = x \odot_{|S \times S} 1$$

commutativity. $\forall x, y \in S$ we have

$$x \odot_{|S \times S} y = x \odot y = y \odot x = y \odot_{|S \times S} x$$

associativity. $\forall x, y, z \in S$ we have

$$x \odot_{|S \times S} (y \odot_{|S \times S} z) = x \odot (y \odot z) = (x \odot y) \odot z = (x \odot_{|S \times S} y) \odot_{|S \times S} z \quad \square$$

Theorem 4.38. Let $\langle R, \oplus, \odot \rangle$ be a ring $F \subseteq R$ a sub-ring of $\langle R, \oplus, \odot \rangle$ and $H \subseteq F$ a sub-ring of $\langle R, \oplus_{|F \times F}, \odot_{|F \times F} \rangle$ then H is a sub-ring of $\langle R, \oplus, \odot \rangle$

Proof.

1. $\forall x, y \in H \subseteq F$ we have $x \oplus_{|F \times F} y \in H$ which as $(x, y) \in F \times F$ proves that

$$x \oplus y = x \oplus_{|F \times F} y \in H$$

and $x \odot_{|F \times F} y \in H$ which as $(x, y) \in F \times F$ proves that

$$x \odot y = x \odot_{|F \times F} y \in H$$

2. If 0 is the additive neutral element of $\langle R, \oplus, \odot \rangle$ then by [theorem: 4.37] 0 is also the additive neutral element of $\langle F, \oplus_{|F \times F}, \odot_{|F \times F} \rangle$ hence $0 \in H$.

3. If 1 is the multiplicative neutral element of $\langle R, \oplus, \odot \rangle$ then by [theorem: 4.37] 1 is also the multiplicative neutral element of $\langle F, \oplus_{|F \times F}, \odot_{|F \times F} \rangle$ hence $1 \in H$.

4. If $x \in H$ then by [theorem: 4.37] its inverse element $-x$ is also the inverse element in $\langle F, \oplus_{|F \times F}, \odot_{|F \times F} \rangle$ hence $x^{-1} \in H$. $\square$

The following theorem shows that the neutral element for the sum in a ring is actual a absorbing element.

Theorem 4.39. Let $\langle X, \oplus, \odot \rangle$ be a ring with 0 the neutral element for $\oplus$ then $\forall x \in R$ we have

$$x \odot 0 = 0 = 0 \odot x$$

Proof. If $x \in R$ then

$$\begin{aligned} 0 & \stackrel{\text{inverse element}}{=} (0 \odot x) \oplus -(0 \odot x) \\ & \stackrel{0 \oplus 0 = 0}{=} ((0 \oplus 0) \odot x) \oplus -(0 \odot x) \\ & \stackrel{\text{distributivity}}{=} [(0 \odot x) \oplus (0 \odot x)] \oplus -(0 \odot x) \\ & \stackrel{\text{associativity}}{=} (0 \odot x) \oplus [(0 \odot x) + (-(0 \odot x))] \\ & \stackrel{\text{inverse element}}{=} (0 \odot x) \oplus 0 \\ & \stackrel{\text{inverse element}}{=} 0 \odot x \\ & \stackrel{\text{commutativity}}{=} x \odot 0 \end{aligned}$$

$\square$

Theorem 4.40. Let $\langle X, \oplus, \odot \rangle$ be a ring with 0 the neutral element for $\oplus$ and 1 the neutral element for $\odot$ then we have:

1. $\forall x \in X$ we have $-x = (-1) \odot x$

2. $\forall x, y \in X$ we have $-(x \odot y) = (-x) \odot y = x \odot (-y)$
3. $\forall x, y \in X$ we have $x \odot y = (-x) \odot (-y)$
4. $(-1) \odot (-1) = 1$

Proof.

1. Let $x \in X$ then

$$x + (-1) \odot x \stackrel{\text{commutative}}{=} (-1) \odot x + x = (-1) \odot x + 1 \odot x = ((-1) + 1) \odot x = 0 \odot x \stackrel{[\text{theorem: 4.39}]}{=} 0$$

which as the inverse element is unique by [theorem: 4.7] proves that $-x = (-1) \odot x$.

2. If $x, y \in X$ then

$$-(x \odot y) \stackrel{(1)}{=} (-1) \odot (x \odot y) = ((-1) \odot x) \odot y \stackrel{(1)}{=} (-x) \odot y$$

and

$$-(x \odot y) = -(y \odot x) \stackrel{(1)}{=} (-1) \odot (y \odot x) = ((-1) \odot y) \odot x \stackrel{(1)}{=} (-y) \odot x = x \odot (-y)$$

3. If $x, y \in X$ then we have

$$(-x) \cdot (-y) \stackrel{(2)}{=} -(x \odot (-y)) \stackrel{(2)}{=} -(-(x \odot y)) \stackrel{[\text{theorem: 4.9}]}{=} x \odot y$$

4. We have $(-1) \odot (-1) \stackrel{(2)}{=} 1 \odot 1 = 1$

□

Definition 4.41. Let $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ be rings then a function $f: A \rightarrow B$ is a **ring homeomorphism** iff

1. $\forall x, y \in A$ we have $f(x \oplus_A y) = f(x) \oplus_B f(y)$
2. $\forall x, y \in A$ we have $f(x \odot_A y) = f(x) \odot_B f(y)$
3. $f(1_A) = 1_B$ where 1_A is the multiplicative neutral element in A and 1_B is the multiplicative neutral element in B .

Notation 4.42. As ring homeomorphism between $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ is noted as

$$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle \text{ is a ring homeomorphism}$$

Note 4.43. Note that a ring homeomorphism $f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle$ is automatically a

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ and } f: \langle A, \odot_A \rangle \rightarrow \langle B, \odot_B \rangle \text{ group homeomorphism}$$

Theorem 4.44. If $\langle F, \oplus_F, \odot_F \rangle$ and $\langle G, \oplus_G, \odot_G \rangle$ are rings, $H \subseteq G$ a sub-ring of $\langle G, \oplus \rangle$ and

$$f: \langle F, \oplus_F, \odot_F \rangle \rightarrow \langle H, (\oplus_G)|_{H \times H}, (\odot_G)|_{H \times H} \rangle \text{ is a ring homeomorphism}$$

then

$$f: \langle F, \oplus_F, \odot_F \rangle \rightarrow \langle G, \oplus_G, \odot_G \rangle \text{ is a ring homeomorphism}$$

Proof. Let $x, y \in F$ then we have

$$f(x \oplus_F y) = f(x)(\oplus_G)|_{H \times H} f(y) = f(x) \oplus_G f(y)$$

and

$$f(x \odot_F y) = f(x)(\odot_G)|_{H \times H} f(y) = f(x) \odot_G f(y)$$

and finally as the neutral element in $\langle H, (\oplus_G)|_{H \times H}, (\odot_G)|_{H \times H} \rangle$ is $1_B \in B$

$$f(1_A) = 1_B$$

□

Using 4.22 we have then the following theorem.

Theorem 4.45. *If $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ are rings with additive units $0_A, 0_B$, multiplicative units $1_A, 1_B$ and*

$$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle \text{ a ring homeomorphism}$$

1. $f(0_A) = 0_B$
2. $\forall a \in A$ we have $f(-a) = -f(a)$
3. $f(A)$ is a sub-ring of $\langle B, \oplus_B, \odot_B \rangle$

Proof. Be careful the same symbol will be used in the context of $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$.

1. We have

$$\begin{aligned} 0_B &= (-f(0_A)) \oplus_B f(0_A) \\ &= (-f(0_A)) \oplus_B f(0_A \oplus_A 0_A) \\ &= (-f(0_A)) \oplus_B (f(0_A) \oplus_B f(0_A)) \\ &= ((-f(0_A)) \oplus_B f(0_A)) \oplus_B f(0_A) \\ &= 0_B \oplus_B f(0_A) \\ &= f(0_A) \end{aligned}$$

2. We have

$$\begin{aligned} f(-x) \oplus_B f(x) &= f(x \oplus_A (-x)) \\ &= f(0_A) \\ &\stackrel{(1)}{=} 0_B \\ f(x) \oplus_B f(-x) &= f(x \oplus_A (-x)) \\ &= f(0_A) \\ &\stackrel{(1)}{=} 0_A \end{aligned}$$

so that

$$f(-x) = -f(x)$$

3. Let $x, y \in f(A)$ then $\exists u, v \in A$ such that $x = f(u)$ and $y = f(v)$ then we have

$$x \oplus_B y = f(u) \oplus_B f(v) = f(u \oplus_A v) \in f(A)$$

and

$$x \odot_B y = f(u) \odot_B f(v) = f(u \odot_A v) \in f(A)$$

and

$$-x = -f(x) \stackrel{(2)}{=} f(-x) \in f(A)$$

and

$$0_B \stackrel{(1)}{=} f(0_A) \in f(A)$$

and

$$1_B = f(1_A)$$

□

Definition 4.46. *If $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ are rings then a function $f: A \rightarrow B$ is a ring isomorphism if it is a bijection and*

$$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle \text{ is a ring homeomorphism}$$

Theorem 4.47. *If $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ are rings and*

$$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle \text{ a ring homeomorphism [or ring isomorphism]}$$

then

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ is a group homeomorphism [or group isomorphism]}$$

and

$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle$ is a group homeomorphism [or group isomorphism]

Proof. This follows directly from [definitions: 4.19, 4.23, 4.41 and 4.46]. $\square$

Theorem 4.48. If $\langle A, \oplus_A, \odot_A \rangle$, $\langle B, \oplus_B, \odot_B \rangle$ and $\langle C, \oplus_C, \odot_C \rangle$ are rings then

1. If D is a sub-ring of $\langle B, \oplus_B, \odot_B \rangle$ and

$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle D, \oplus_B, \odot_B \rangle$ and $g: \langle B, \oplus_B, \odot_B \rangle \rightarrow \langle C, \oplus_C, \odot_C \rangle$ are ring homeomorphisms then

$g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle C, \oplus_C, \odot_C \rangle$ is a ring homeomorphism

and

$g(f(A))$ is a sub-ring of $\langle C, \oplus_C, \odot_C \rangle$

2. If D is a sub-ring of $\langle B, \oplus_B, \odot_B \rangle$ and

$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle D, \oplus_B, \odot_B \rangle$ and $g: \langle B, \oplus_B, \odot_B \rangle \rightarrow \langle C, \oplus_C, \odot_C \rangle$ are ring isomorphisms then

$g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle g(f(A)), \oplus_C, \odot_C \rangle$ is a ring isomorphism

or as $g(f(A)) \xrightarrow{f: A \rightarrow D \text{ is a bijection}} g(D)$

$g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle g(f(A)), \oplus_C, \odot_C \rangle$ is a ring isomorphism

Proof.

1. Let $x, y \in A$ then

$$\begin{aligned}
 (g \circ f)(x \oplus_A y) &= g(f(x \oplus_A y)) \\
 &\stackrel{f \text{ is a homeomorphism}}{=} g(f(x) \oplus_B f(y)) \\
 &\stackrel{g \text{ is a homeomorphism}}{=} g(f(x)) \oplus_C g(f(y)) \\
 &= (g \circ f)(x) \oplus_C (g \circ f)(y) \\
 (g \circ f)(x \odot_A y) &= g(f(x \odot_A y)) \\
 &\stackrel{f \text{ is a homeomorphism}}{=} g(f(x) \odot_B f(y)) \\
 &\stackrel{g \text{ is a homeomorphism}}{=} g(f(x)) \odot_C g(f(y)) \\
 &= (g \circ f)(x) \odot_C (g \circ f)(y) \\
 (g \circ f)(1_A) &= g(f(1_A)) \\
 &\stackrel{f \text{ is a homeomorphism}}{=} g(1_B) \\
 &\stackrel{g \text{ is a homeomorphism}}{=} 1_C
 \end{aligned}$$

proving that $g \circ f$ is a ring homeomorphism. Finally using [theorem: 4.45] we have then that

$g(f(A))$ is a sub-ring of $\langle C, \oplus_C, \odot_C \rangle$

2. Using [theorem: 2.78] we have that $g \circ f: A \rightarrow g(f(A))$ is a bijection which combined with (1) proves that $g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle g(f(A)), \oplus_C, \odot_C \rangle$ is a ring isomorphism. $\square$

Definition 4.49. $\langle R, \oplus, \odot, \leq \rangle$ is a ordered ring if

1. $\langle R, \oplus, \odot \rangle$ is a ring
2. $\langle R, \leq \rangle$ is totally ordered
3. $\forall x, y, z \in R$ with $x < y$ we have $x \oplus z < y \oplus z$

4. $\forall x, y \in R$ with $0 < x$ and $0 < y$ we have $0 < x \odot y$

If in addition $\langle R, \oplus, \odot \rangle$ is a integral domain then we say that $\langle R, \oplus, \odot, \leq \rangle$ is a ordered integral domain.

Theorem 4.50. If $\langle R, \oplus, \odot, \leq \rangle$ is a ordered ring then we have :

1. $\forall x, y, z \in R$ we have $x < y \Leftrightarrow x \oplus z < y \oplus z$
2. $\forall x, y, z \in R$ we have $x \leq y \Leftrightarrow x \oplus z \leq y \oplus z$
3. $\forall x, y \in R$ we have $x < y \Leftrightarrow 0 < y \oplus (-x)$
4. $\forall x, y \in R$ we have $x \leq y \Leftrightarrow 0 \leq y \oplus (-x)$
5. $\forall x, y \in R$ we have $x < y \Leftrightarrow -y < -x$
6. $\forall x, y \in R$ we have $x \leq y \Leftrightarrow -y \leq -x$
7. $\forall x, y, z \in R$ with $0 < z$ we have $x < y \Leftrightarrow x \odot z < y \odot z$
8. $\forall x, y, z \in R$ with $0 < z$ we have $x \leq y \Leftrightarrow x \odot z \leq y \odot z$
9. $\forall x, y, z \in \mathbb{R}$ with $0 \leq z$ and $x \leq y$ we have $x \odot z \leq y \odot z$
10. $\forall x, y, z \in R$ with $z < 0$ we have $x < y \Leftrightarrow y \odot z < x \odot z$
11. $\forall x, y, z \in R$ with $z < 0$ we have $x \leq y \Leftrightarrow y \odot z \leq x \odot z$
12. $\forall x, y, z \in R$ with $z \leq 0$ and $x \leq y$ we have $y \odot z \leq x \odot z$
13. $\forall x \in R$ we have $0 \leq x \cdot x \stackrel{\text{def}}{=} x^2$, further if $x \neq 0$ then $0 < x \odot x \stackrel{\text{def}}{=} x^2$
14. $0 \leq 1$
15. $\forall x, y \in R$ with $0 < x < y$ we have that $x^2 < y^2$ where $x^2 = x \odot x$ and $y^2 = y \odot y$
16. $\forall x, y \in R$ with $0 \leq x \leq y$ we have that $x^2 \leq y^2$ where $x^2 = x \odot x$ and $y^2 = y \odot y$

Proof.

1.

$\Rightarrow$. This follows directly from the definition of a ordered ring.

$\Leftarrow$. If $x \oplus z \leq y \oplus z$ then from the definition of a ordered ring we have

$$x = x + (z \oplus (-z)) = (x \oplus z) \oplus (-z) < (y \oplus z) + (-z) = y \oplus (z \oplus (-z)) = y$$

so that $x < y$

2.

$$\begin{aligned} x \leq y & \Leftrightarrow x = y \vee x < y \\ & \stackrel{(1)}{\Leftrightarrow} x = y \vee x \oplus z < y \oplus z \\ & \stackrel{x=y \Leftrightarrow x \oplus z = y \oplus z}{\Leftrightarrow} x \oplus z = y \oplus z \vee x \oplus z < y \oplus z \\ & \Leftrightarrow x \oplus z \leq y \oplus z \end{aligned}$$

3.

$$\begin{aligned} x < y & \stackrel{(1)}{\Leftrightarrow} x \oplus (-x) < y \oplus (-x) \\ & \Leftrightarrow 0 < y \oplus (-x) \end{aligned}$$

4.

$$\begin{aligned} x \leq y & \stackrel{(2)}{\Leftrightarrow} x \oplus (-x) \leq y \oplus (-x) \\ & \Leftrightarrow 0 \leq y \oplus (-x) \end{aligned}$$

5.

$$\begin{aligned}
x < y & \stackrel{(3)}{\Leftrightarrow} 0 < y \oplus (-x) \\
& \stackrel{\text{commutativity}}{\Leftrightarrow} 0 < (-x) \oplus y \\
& \stackrel{(1)}{\Leftrightarrow} 0 + (-y) < ((-x) \oplus y) \oplus (-y) \\
& \Leftrightarrow -y < -x
\end{aligned}$$

6.

$$\begin{aligned}
x \leq y & \stackrel{(4)}{\Leftrightarrow} 0 < y \oplus (-x) \\
& \stackrel{\text{commutativity}}{\Leftrightarrow} 0 < (-x) \oplus y \\
& \stackrel{(1)}{\Leftrightarrow} 0 + (-y) < ((-x) \oplus y) \oplus (-y) \\
& \Leftrightarrow -y < -x
\end{aligned}$$

7. Let $0 < z$ then we have $\Rightarrow$. As $x < y$ we have by (3) that $0 < y \oplus (-x)$ so by the definition of a ordered ring that

$$\begin{aligned}
0 & < (y \oplus (-x)) \odot z \\
& \stackrel{\text{distributivity}}{=} y \odot z \oplus (-x) \odot z \\
& \stackrel{[\text{theorem: 4.40}]}{=} y \odot z \oplus -(x \odot z)
\end{aligned}$$

which using (3) prove that $x \odot z < y \odot z$. $\Leftarrow$. Then $x \odot z < y \odot z$ and using the totally ordering we have for x, y either: $\mathbf{x = y}$. Then $x \odot z = y \odot z$ contradicting $x \odot z < y \odot z$ so this case never occurs. $\mathbf{y < x}$. Then by (3) we have $0 < x \oplus (-y)$ so that by the definition of a ordered field we have

$$\begin{aligned}
0 & < (x \oplus (-y)) \odot z \\
& \stackrel{\text{distributivity}}{=} x \odot z \oplus (-y) \odot z \\
& \stackrel{[\text{theorem: 4.40}]}{=} x \odot z \oplus -(y \odot z)
\end{aligned}$$

which using (3) proves that $y \odot z < x \odot z$ contradicting $x \odot z < y \odot z$. So this case does not occur. $\mathbf{x < y}$. the remaining case.So the only remaining case is $x < y$

8.

 $\Rightarrow$. As $0 < z$ and $x \leq y$ we have for x, y either: $\mathbf{x = y}$. Then $x \odot z = y \odot z$ hence $x \odot z \leq y \odot z$ $\mathbf{x < y}$. Then by (7) we have $x \odot z < y \odot z$ hence $x \odot z \leq y \odot z$ So in all cases we have $x \odot z \leq y \odot z$. $\Leftarrow$. Assume that $x \odot z \leq y \odot z$. By totally ordering we have for x, y either $x \leq y$ or $y < x$.If $y < x$ then by (7) we have $y \odot z < x \odot z$ contradicting $x \odot z \leq y \odot z$ so we must have that $x \leq y$ 9. Let $0 \leq z$ and $x \leq y$ then we have for z either: $\mathbf{z = 0}$. Then $x \odot z = x \odot 0 \stackrel{[\text{theorem: 4.39}]}{=} 0 \stackrel{[\text{theorem: 4.39}]}{=} y \odot 0 = y \odot z$ so that $x \odot z \leq y \odot z$ $\mathbf{0 < z}$. Then by (8) we have $x \odot z < y \odot z$ So in all cases we have $x \odot z \leq y \odot z$

10. As $z < 0$ we have by (5) that $0 < (-z)$ so that

$$\begin{aligned} x < y & \xLeftrightarrow{(7)} x \odot (-z) < y \odot (-z) \\ & \xLeftrightarrow{[\text{theorem: 4.40}]} -(x \odot z) < -(y \odot z) \\ & \xLeftrightarrow{(5)} y \odot z < z \odot z \end{aligned}$$

11. As $z < 0$ we have by (5) that $0 < (-z)$ so that

$$\begin{aligned} x \leq y & \xLeftrightarrow{(8)} x \odot (-z) \leq y \odot (-z) \\ & \xLeftrightarrow{[\text{theorem: 4.40}]} -(x \odot z) \leq -(y \odot z) \\ & \xLeftrightarrow{(6)} y \odot z \leq z \odot z \end{aligned}$$

12. As $z \leq 0$ we have by (5) that $0 \leq -z$ so that by (9) we have

$$\begin{aligned} x \leq y & \xRightarrow{(9)} x \odot (-z) \leq y \odot (-z) \\ & \xLeftrightarrow{[\text{theorem: 4.40}]} -(x \odot z) \leq -(y \odot z) \\ & \xLeftrightarrow{(6)} y \odot z \leq z \odot z \end{aligned}$$

13. If $x \in \mathbb{R}$ then we have either:

$0 < x$. Then we have by (7) that $0 \xrightarrow{[\text{theorem: 4.39}]} 0 \odot x < x \odot x$ hence $0 < x \odot x$

$0 = x$. Then we have $0 = 0 \odot 0 = x \odot x$ so that $0 \leq x \odot x$

$x < 0$. Then we have by (7) $0 \xrightarrow{[\text{theorem: 4.39}]} 0 \odot (-x) \leq (-x) \odot (-x) \xrightarrow{[\text{theorem: 4.40}]} x \odot x$
hence $0 < x \odot x$

14. Using (13) we have $0 \leq 1 \odot 1 = 1$

15. Let $0 < x < y$ then by (7) we have $x \odot x < y \odot x$ and $x \odot y < y \odot y$ so that $x \odot x < y \odot y$.

16. Let $0 \leq x < y$ then by (8) we have $x \odot x \leq y \odot x$ and $x \odot y \leq y \odot y$ so that $x \odot x < y \odot y$. □

4.3 Fields

A ring has no inverse for a multiplicative element, one of the reasons for this is that is difficult to say what the inverse of 0 is, as expressed in the following computation

$$1 = 0 \odot 0^{-1} \xrightarrow{[\text{theorem: 4.39}]} 0$$

so that we have

$$\forall x \in R \text{ that } x = 1 \odot x = 0 \odot x \xrightarrow{[\text{theorem: 4.39}]} 0$$

and we end up with $R = \{0\}$, which is not a useful ring. However we can avoid this problem if we exclude the 0 of the list of elements that has a inverse element. This is the idea behind a field.

Definition 4.51. A triple $\langle F, \oplus, \odot \rangle$ is a field if $\langle F, \oplus, \odot \rangle$ is a ring and additional

$$\begin{aligned} \forall x \in F \setminus \{0\} \exists b \in F \text{ such that } x \odot b = 1 = b \odot x \\ 0 \neq 1 \end{aligned}$$

where 1 is the neutral element for $\odot$. In other words $\langle F, \oplus, \odot \rangle$ is a field iff

1. F is a set

2. $\langle F, \oplus \rangle$ is a Abelian group or $\oplus: F \times F \rightarrow F$ is a operator such that

associativity. $\forall x, y, z \in F$ we have $x \oplus (y \oplus z) = (x \oplus y) \oplus z$

neutral element. $\exists 0 \in F$ such that $\forall x \in F$ we have $0 \oplus x = x = x \oplus 0$

inverse element. $\forall x \in F$ there exist a $-x$ such that $x \oplus (-x) = 0 = (-x) \oplus x$

commutativity. $\forall x, y \in F$ we have $x \oplus y = y \oplus x$

$\oplus$ is called the sum operator of the field.

3. $\odot: F \times F \rightarrow F$ is a operator so that

Distributivity. $\forall x, y, z \in F$ we have $x \odot (y \oplus z) = (x \odot y) \oplus (x \odot z)$

neutral element. $\exists 1 \in F$ such that $\forall x \in F$ we have $1 \odot x = x = x \odot 1$

commutativity. $\forall x, y \in F$ we have $x \odot y = y \odot x$

associativity. $\forall x, y, z \in F$ we have $x \odot (y \odot z) = (x \odot y) \odot z$

inverse element. $\forall x \in F \setminus \{0\} \exists b \in F$ such that $x \odot b = 1 = b \odot x$

4. $1 \neq 0$

The inverse if it exist is unique

Theorem 4.52. If $\langle F, \oplus, \odot \rangle$ is field then $\forall x \in F \setminus \{0\}$ there exist a **unique** inverse element for $\odot$. We note this element as x^{-1} .

Proof. Let $x \in F \setminus \{0\}$ and assume that $y, y' \in F$ such that $y \odot x = 1 = x \odot y$ and $y' \odot x = 1 = x \odot y'$ then we have

$$y = y \odot 1 = y \odot (x \odot y') = (y \odot x) \odot y' = 1 \odot y' = y' \quad \square$$

Theorem 4.53. If $\langle F, \oplus, \odot \rangle$ is field then $\forall x \in F \setminus \{0\}$ we have $(x^{-1})^{-1} = x$

Proof. First if $x^{-1} = 0$ then $x = x \odot 1 = x \odot (x \odot x^{-1}) = (x \odot x) \odot x^{-1} = (x \odot x) \odot 0 \stackrel{[\text{theorem: 4.39}]}{=} 0$ contradicting $x \in F \setminus \{0\}$. So we must have that $x^{-1} \neq 0$ hence $(x^{-1})^{-1}$ is defined. Further

$$\begin{aligned} 1 = x^{-1} \odot (x^{-1})^{-1} &\Rightarrow x \odot 1 = x \odot (x^{-1} \odot (x^{-1})^{-1}) \\ &\stackrel{\text{neutral element}}{\Rightarrow} x = x \odot (x^{-1} \odot (x^{-1})^{-1}) \\ &\stackrel{\text{associativity}}{\Rightarrow} x = (x \odot x^{-1}) \odot (x^{-1})^{-1} \\ &\stackrel{\text{inverse element}}{\Rightarrow} x = 1 \odot (x^{-1})^{-1} \\ &\stackrel{\text{neutral element}}{\Rightarrow} x = (x^{-1})^{-1} \end{aligned} \quad \square$$

Theorem 4.54. If $\langle F, \oplus, \odot \rangle$ is a field then $\forall x, y \in F \setminus \{0\}$ we have $x^{-1} = y^{-1} \Leftrightarrow x = y$

Proof. We have

$$\Rightarrow. \text{ If } x = y \text{ then trivially } x^{-1} = y^{-1}$$

$$\Leftarrow. \text{ Then we have}$$

$$\begin{aligned} x^{-1} = y^{-1} &\Rightarrow x^{-1} \odot y = y^{-1} \odot y \\ &\stackrel{\text{inverse element}}{\Rightarrow} x^{-1} \odot y = 1 \\ &\Rightarrow x \odot (x^{-1} \odot y) = x \odot 1 \\ &\stackrel{\text{neutral element}}{\Rightarrow} x \odot (x^{-1} \odot y) = x \\ &\stackrel{\text{associativity}}{\Rightarrow} (x \odot x^{-1}) \odot y = x \\ &\stackrel{\text{inverse element}}{\Rightarrow} 1 \odot y = x \\ &\stackrel{\text{neutral element}}{\Rightarrow} y = x \end{aligned} \quad \square$$

Theorem 4.55. *If $\langle F, \oplus, \odot \rangle$ is a field then $\forall x, y \in F$ we have $(x \odot y)^{-1} = x^{-1} \odot y^{-1}$*

Proof.

$$\begin{aligned}
 (x^{-1} \odot y^{-1}) \odot (x \odot y) &\stackrel{\text{commutativity}}{=} (x \odot y) \odot (x^{-1} \odot y^{-1}) \\
 &\stackrel{\text{commutativity}}{=} (x \odot y) \odot (y^{-1} \odot x^{-1}) \\
 &\stackrel{\text{associativity}}{=} ((x \odot y) \odot y^{-1}) \odot x^{-1} \\
 &\stackrel{\text{associativity}}{=} (x \odot (y \odot y^{-1})) \odot x^{-1} \\
 &\stackrel{\text{inverse element}}{=} (x \odot 1) \odot x^{-1} \\
 &\stackrel{\text{neutral element}}{=} x \odot x^{-1} \\
 &\stackrel{\text{inverse element}}{=} 1
 \end{aligned}$$

proving by the uniqueness of the inverse [theorem: 4.52] that

$$(x \odot y)^{-1} = x^{-1} \odot y^{-1}$$

□

Theorem 4.56. *Let $\langle F, \oplus, \odot \rangle$ be a field and $x, y \in F$ and $z \in F \setminus \{0\}$ then $x = y \Leftrightarrow x \odot z = y \odot z$*

Proof. As $z \neq 0$ we have that z^{-1} exist.

$\Rightarrow$. If $x = y$ then clearly $x \odot z = y \odot z$

$\Leftarrow$. If $x \odot z \oplus y \odot z$ then

$$x = x \odot 1 = x \odot (z \odot z^{-1}) = (x \odot z) \odot z^{-1} \stackrel{x \odot z = y \odot z}{=} (y \odot z) \odot z^{-1} = y \odot (z \odot z^{-1}) = y \odot 1 = y \quad \square$$

Theorem 4.57. *If $\langle F, \oplus, \odot \rangle$ is a field and $x, y \in F$ then $x \odot y = 0 \Leftrightarrow x = 0 \vee y = 0$*

Proof.

$\Rightarrow$. If $x \odot y = 0$ then for x we have either

$x = 0$. Then clearly $0 \vee y = 0$ is true.

$x \neq 0$. Then $0 = x^{-1} \odot 0 = x^{-1} \odot (x \odot y) = (x^{-1} \odot x) \odot y = 1 \odot y = y$ so that $y = 0$ or $x = 0 \vee y = 0$ is true.

$\Leftarrow$. If $x = 0$ or $y = 0$ we have by [theorem: 4.39] and the fact that a field is a ring that $x \odot y = 0$ □

Corollary 4.58. *If $\langle F, \oplus, \odot \rangle$ is a field then $\langle F, \oplus, \odot \rangle$ is a integral domain*

Proof. From the definition of a field [definition: 4.51] it follows that $\langle F, \oplus, \odot \rangle$ is a ring. Assume that $x \in F$ is a zero divisor [see 4.33 then $x \neq 0$ and $\exists y \in F \setminus \{0\}$ such that $x \cdot y = 0$. However by [theorem: 4.57] we have $x = 0 \vee y = 0$ contradicting $x \neq 0 \neq y$. So F does not have zero divisors proving that $\langle F, \oplus, \odot \rangle$ is a integral domain. □

Definition 4.59. *If $\langle F, \oplus, \odot \rangle$ is a field then a subset $S \subseteq F$ is a sub-field iff the following is satisfied*

1. $\forall x, y \in F$ we have $x \oplus y \in F$ and $x \odot y \in F$
2. $\forall x \in F$ we have $-x \in F$ [the inverse element for $\oplus$]
3. $1 \in F$ [the neutral element for $\odot$]
4. $0 \in F$ [the neutral element for $\oplus$]
5. $\forall x \in F \setminus \{0\}$ we have $x^{-1} \in F$

Theorem 4.60. *If $\langle F, \oplus, \odot \rangle$ is a field $G \subseteq F$ a sub ring of $\langle F, \oplus, \odot \rangle$ then*

G is a subring of $\langle F, \oplus, \odot \rangle$, G is a sub group of $\langle F, \oplus \rangle$ and G is a sub semi-group of $\langle F, \odot \rangle$

Proof. This follows directly from [definitions: 4.35, 4.59] and [theorem: 4.36]. $\square$

Theorem 4.61. *If $\langle F, \oplus, \odot \rangle$ is a field and $S \subseteq F$ is a sub-field then $\langle S, \oplus|_{S \times S}, \odot|_{S \times S} \rangle$ is a field with the same additive and multiplicative neutral element as $\langle F, \oplus, \odot \rangle$. Further if $x \in S$ then the additive inverse element in $\langle F, \oplus, \odot \rangle$ is also the inverse in $\langle S, \oplus|_{S \times S}, \odot|_{S \times S} \rangle$ and if $x \in S \setminus \{0\}$ then the multiplicative inverse element in $\langle F, \oplus, \odot \rangle$ is also the multiplicative element in $\langle S, \oplus|_{S \times S}, \odot|_{S \times S} \rangle$.*

Proof. Using [theorem: 4.37] it follows that $\langle S, \oplus|_{S \times S}, \odot|_{S \times S} \rangle$ is a ring. Further if $x \in F \setminus \{0\}$ then $1 \in S$ and $x^{-1} \in S$, further $x \odot_S x^{-1} = x \odot x^{-1} = 1 = x^{-1} \odot x = x^{-1} \odot_S x$ proving that $\langle S, \oplus|_{S \times S}, \odot|_{S \times S} \rangle$ is a field. $\square$

Theorem 4.62. *Let $\langle F, \oplus, \odot \rangle$ be a field $G \subseteq F$ a sub-field of $\langle F, \oplus, \odot \rangle$ and $H \subseteq G$ a sub-field of $\langle G, \oplus|_{G \times G}, \odot|_{G \times G} \rangle$ then H is a sub-field of $\langle G, \odot \rangle$*

Proof.

1. $\forall x, y \in H \subseteq G$ we have $x \oplus_{|G \times G} y \in H$ which as $(x, y) \in G \times G$ proves that

$$x \oplus y = x \oplus_{|G \times G} y \in H$$

and $x \odot_{|G \times G} y \in H$ which as $(x, y) \in G \times G$ proves that

$$x \odot y = x \odot_{|G \times G} y \in H$$

2. If 0 is the additive neutral element of $\langle F, \oplus, \odot \rangle$ then by [theorem: 4.61] 0 is also the additive neutral element of $\langle G, \oplus|_{G \times G}, \odot|_{G \times G} \rangle$ hence $0 \in H$.
3. If 1 is the multiplicative neutral element of $\langle F, \oplus, \odot \rangle$ then by [theorem: 4.61] 1 is also the multiplicative neutral element of $\langle G, \oplus|_{G \times G}, \odot|_{G \times G} \rangle$ hence $1 \in H$.
4. If $x \in H$ then by [theorem: 4.61] its additive inverse element $-x$ is also the additive inverse element in $\langle G, \oplus|_{G \times G}, \odot|_{G \times G} \rangle$ hence $x^{-1} \in H$.
5. If $x \in H \setminus \{0\}$ then by [theorem: 4.61] its multiplicative inverse element x^{-1} is also the multiplicative inverse element in $\langle G, \oplus|_{G \times G}, \odot|_{G \times G} \rangle$ hence $x^{-1} \in H$. $\square$

Definition 4.63. *If $\langle A, \odot_A, \oplus_A \rangle$ and $\langle B, \odot_B, \oplus_B \rangle$ are fields with multiplicative units $1_A, 1_B$ then a function $f: A \rightarrow B$ is a field homeomorphism iff*

1. $\forall x, y \in A$ we have $f(x \odot_A y) = f(x) \odot_B f(y)$
2. $\forall x, y \in A$ we have $f(x \oplus_A y) = f(x) \oplus_B f(y)$
3. $f(1_A) = 1_B$

Notation 4.64. *A field homeomorphism between $\langle A, \odot_A, \oplus_A \rangle$ and $\langle B, \odot_B, \oplus_B \rangle$ is noted as*

$$f: \langle A, \odot_A, \oplus_A \rangle \rightarrow \langle B, \odot_B, \oplus_B \rangle \text{ is a field homeomorphism}$$

Note 4.65. If $\langle A, \odot_A, \oplus_A \rangle$ and $\langle B, \odot_B, \oplus_B \rangle$ are fields and

$$f: \langle A, \odot_A, \oplus_A \rangle \rightarrow \langle B, \odot_B, \oplus_B \rangle \text{ a field homeomorphism}$$

then

$$f: \langle A, \odot_A, \oplus_A \rangle \rightarrow \langle B, \odot_B, \oplus_B \rangle \text{ is a ring homeomorphism}$$

and

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ and } f: \langle A, \odot_A \rangle \rightarrow \langle B, \odot_B \rangle \text{ are group homeomorphisms}$$

Proof. As a field is also a ring [see definitions: 4.32 and 4.51] we have by [definitions: 4.41 and 4.63] that

$$f: \langle A, \odot_A, \oplus_A \rangle \rightarrow \langle B, \odot_B, \oplus_B \rangle \text{ is a ring homeomorphism}$$

Finally using [note: 4.43] we have that

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ and } f: \langle A, \odot_A \rangle \rightarrow \langle B, \odot_B \rangle \text{ are group homeomorphisms} \quad \square$$

Definition 4.66. Let $\langle A, \odot_A, \oplus_A \rangle$ and $\langle B, \odot_B, \oplus_B \rangle$ be fields then a field homeomorphism

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ and } f: \langle A, \odot_A \rangle \rightarrow \langle B, \odot_B \rangle$$

is a field isomorphism if it is also a bijection.

Theorem 4.67. If $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ are rings and

$$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle \text{ a field homeomorphism [or field isomorphism]}$$

then

$$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle \text{ a ring homeomorphism [or ring isomorphism]}$$

and

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ is a group homeomorphism [or group isomorphism]}$$

and

$$f: \langle A, \odot_A \rangle \rightarrow \langle B, \odot_B \rangle \text{ is a group homeomorphism [or group isomorphism]}$$

Proof. This follows directly from [definitions: 4.41, 4.46, 4.63 and 4.66] and [theorem: 4.47] $\square$

Theorem 4.68. If $\langle F, \odot_F, \oplus_F \rangle$ and $\langle G, \odot_G, \oplus_G \rangle$ are fields, $H \subseteq G$ a sub-field of $\langle G, \oplus \rangle$ and

$$f: \langle F, \oplus_F, \odot_F \rangle \rightarrow \langle H, (\oplus_G)|_{H \times H}, (\odot_G)|_{H \times H} \rangle \text{ is a field homeomorphism}$$

then

$$f: \langle F, \oplus_F, \odot_F \rangle \rightarrow \langle G, \oplus_G, \odot_G \rangle \text{ is a field homeomorphism}$$

Proof. Let $x, y \in F$ then we have

$$f(x \oplus_F y) = f(x)(\oplus_G)|_{H \times H} f(y) = f(x) \oplus_G f(y)$$

and

$$f(x \odot_F y) = f(x)(\odot_G)|_{H \times H} f(y) = f(x) \odot_G f(y)$$

and finally as the neutral element in $\langle H, (\oplus_G)|_{H \times H}, (\odot_G)|_{H \times H} \rangle$ is $1_B \in B$

$$f(1_A) = 1_B \quad \square$$

Theorem 4.69. If $\langle A, \odot_A, \oplus_A \rangle$ and $\langle B, \odot_B, \oplus_B \rangle$ are fields with multiplicative units $1_A, 1_B$ and

$$f: \langle A, \oplus_A \rangle \rightarrow \langle B, \oplus_B \rangle \text{ and } f: \langle A, \odot_A \rangle \rightarrow \langle B, \odot_B \rangle \text{ is a field isomorphism}$$

then

$$f^{-1}: \langle B, \odot_B \rangle \rightarrow \langle A, \odot_A \rangle \text{ is a field isomorphism}$$

Proof. First using [theorem: 2.74] we have that $f^{-1}: B \rightarrow A$ is a bijection. Further we have:

1. Take $x, y \in B$ then we have

$$\begin{aligned} f^{-1}(x \oplus_B y) &= f^{-1}(\text{Id}_B(x) \oplus_B \text{Id}_B(y)) \\ &\stackrel{[\text{theorem: 2.71}]}{=} f^{-1}((f \circ f^{-1})(x) \oplus_B (f \circ f^{-1})(y)) \\ &\stackrel{[\text{theorem: 2.44}]}{=} f^{-1}(f(f^{-1}(x)) \oplus_B f(f^{-1}(y))) \\ &\stackrel{f \text{ is homeomorphism}}{=} f^{-1}(f(f^{-1}(x) \oplus_A f^{-1}(y))) \\ &\stackrel{[\text{theorem: 2.44}]}{=} (f^{-1} \circ f)(f^{-1}(x) \oplus_A f^{-1}(y)) \\ &\stackrel{[\text{theorem: 2.71}]}{=} \text{Id}_A(f^{-1}(x) \oplus_A f^{-1}(y)) \\ &= f^{-1}(x) \oplus_A f^{-1}(y) \end{aligned}$$

2. Take $x, y \in B$ then we have

$$\begin{aligned}
 f^{-1}(x \odot_B y) &= f^{-1}(\text{Id}_B(x) \odot_B \text{Id}_B(y)) \\
 &\stackrel{[\text{theorem: 2.71}]}{=} f^{-1}((f \circ f^{-1})(x) \odot_B (f \circ f^{-1})(y)) \\
 &\stackrel{[\text{theorem: 2.44}]}{=} f^{-1}(f(f^{-1}(x)) \odot_B f(f^{-1}(y))) \\
 &\stackrel{f \text{ is homeomorphism}}{=} f^{-1}(f(f^{-1}(x) \odot_A f^{-1}(y))) \\
 &\stackrel{[\text{theorem: 2.44}]}{=} (f^{-1} \circ f)(f^{-1}(x) \odot_A f^{-1}(y)) \\
 &\stackrel{[\text{theorem: 2.71}]}{=} \text{Id}_A(f^{-1}(x) \odot_A f^{-1}(y)) \\
 &= f^{-1}(x) \odot f^{-1}(y)
 \end{aligned}$$

3. From $f(1_A) = 1_B$ it follows that

$$f^{-1}(1_B) = f^{-1}(f(1_B)) = (f^{-1} \circ f)(1_B) \stackrel{[\text{theorem: 2.71}]}{=} \text{Id}_A(1_B) = 1_B \quad \square$$

Theorem 4.70. *If $\langle A, \oplus_A, \odot_A \rangle$ and $\langle B, \oplus_B, \odot_B \rangle$ are fields with additive units $0_A, 0_B$ and multiplicative units $1_A, 1_B$ and $f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle$ a field homeomorphism then we have*

1. $f(0_A) = 0_B$
2. $\forall a \in A$ we have $f(-a) = -f(a)$
3. $\forall a \in A$ with $a \neq 0_A$ we have $f(a^{-1}) = (f(a))^{-1}$
4. $f(A)$ is a sub-field of $\langle B, \oplus_B, \odot_B \rangle$

Proof.

1. We have

$$\begin{aligned}
 0_B &= (-f(0_A)) \oplus_B f(0_A) \\
 &= (-f(0_A)) \oplus_B f(0_A \oplus_A 0_A) \\
 &= (-f(0_A)) \oplus_B (f(0_A) \oplus_B f(0_A)) \\
 &= ((-f(0_A)) \oplus_B f(0_A)) \oplus_B f(0_A) \\
 &= 0_B \oplus f(0_A) \\
 &= f(0_A)
 \end{aligned}$$

2. We have

$$\begin{aligned}
 f(-x) \oplus_B f(x) &= f((-x) \oplus_A x) \\
 &= f(0_A) \\
 &\stackrel{(1)}{=} 0_B \\
 f(x) \oplus_B f(-x) &= f(x \oplus_A (-x)) \\
 &= f(0_A) \\
 &\stackrel{(1)}{=} 0_A
 \end{aligned}$$

so that

$$f(-x) = -f(x)$$

3. If $a \in A$ with $a \neq 0_A$ then

$$\begin{aligned}
 f(a^{-1}) \odot_B f(a) &= f(a^{-1} \odot_A a) \\
 &= f(1_A) \\
 &= 1_B \\
 f(a) \odot_B f(a^{-1}) &= f(a \odot_A a^{-1}) \\
 &= f(1_A) \\
 &= 1_B
 \end{aligned}$$

so that

$$f(x^{-1}) = f(x)^{-1}$$

4. Let $x, y \in f(A)$ then $\exists u, v \in A$ such that $x = f(u)$ and $y = f(v)$ so we have

$$x \oplus_B y = f(u) \oplus_B f(v) = f(u \oplus_A v) \in f(A)$$

and

$$x \odot_B y = f(u) \odot_B f(v) = f(u \odot_A v) \in f(A)$$

and

$$-x = -f(x) \stackrel{(2)}{=} f(-x) \in f(A)$$

and if $x \neq 0_B$ then

$$x^{-1} = f(u)^{-1} \stackrel{(3)}{=} f(u^{-1}) \in f(A)$$

and

$$0_B \stackrel{(1)}{=} f(0_A) \in f(A)$$

and by definition of a field homeomorphism

$$1_B = f(1_A)$$

□

Theorem 4.71. *If $\langle A, \oplus_A, \odot_A \rangle$, $\langle B, \oplus_B, \odot_B \rangle$ and $\langle C, \oplus_C, \odot_C \rangle$ are fields then*

1. *If $D \subseteq B$ is a sub-field of $\langle B, \oplus_B, \odot_B \rangle$ and*

$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle D, \oplus_B, \odot_B \rangle$ and $g: \langle B, \oplus_B, \odot_B \rangle \rightarrow \langle C, \oplus_C, \odot_C \rangle$ are field homeomorphism then

$g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle C, \oplus_C, \odot_C \rangle$ is a field homeomorphism

and

$g(f(A))$ is a sub-field of $\langle C, \oplus_C, \odot_C \rangle$

2. *If $D \subseteq B$ is a sub-field of $\langle B, \oplus_B, \odot_B \rangle$ and*

$f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle B, \oplus_B, \odot_B \rangle$ and $g: \langle B, \oplus_B, \odot_B \rangle \rightarrow \langle C, \oplus_C, \odot_C \rangle$ are field isomorphisms then

$g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle g(f(A)), \oplus_C, \odot_C \rangle$ is a field isomorphism

or as $g(f(A)) \stackrel{f: A \rightarrow D \text{ is a bijection}}{=} g(D)$

$g \circ f: \langle A, \oplus_A, \odot_A \rangle \rightarrow \langle g(D), \oplus_C, \odot_C \rangle$ is a field isomorphism

Proof.

1. Let $x, y \in A$ then

$$\begin{aligned} (g \circ f)(x \oplus_A y) &= g(f(x \oplus_A y)) \\ &\stackrel{f \text{ is a homeomorphism}}{=} g(f(x) \oplus_B f(y)) \\ &\stackrel{g \text{ is a homeomorphism}}{=} g(f(x)) \oplus_C g(f(y)) \\ &= (g \circ f)(x) \oplus_C (g \circ f)(y) \\ (g \circ f)(x \odot_A y) &= g(f(x \odot_A y)) \\ &\stackrel{f \text{ is a homeomorphism}}{=} g(f(x) \odot_B f(y)) \\ &\stackrel{g \text{ is a homeomorphism}}{=} g(f(x)) \odot_C g(f(y)) \\ &= (g \circ f)(x) \odot_C (g \circ f)(y) \\ (g \circ f)(1_A) &= g(f(1_A)) \\ &\stackrel{f \text{ is a homeomorphism}}{=} g(1_B) \\ &\stackrel{g \text{ is a homeomorphism}}{=} 1_C \end{aligned}$$

Finally using [theorem: 4.70] we have then that

$$g(f(A)) \text{ is a sub-field of } \langle C, \oplus_C, \odot_C \rangle$$

2. Using [theorem: 2.78] we have that $g \circ f: A \rightarrow g(f(A))$ is a bijection which combined with (1) proves that $g \circ f$ is a field isomorphism. $\square$

Definition 4.72. $\langle F, \oplus, \odot, \leq \rangle$ is a ordered field if

1. $\langle F, \oplus, \odot \rangle$ is a field
2. $\langle F, \leq \rangle$ is totally ordered
3. $\forall x, y, z \in F$ with $x \leq y$ we have $x + z \leq y + z$
4. $\forall x, y \in F$ with $0 < x$ and $0 < y$ we have $0 < x \odot y$

Theorem 4.73. If $\langle F, \oplus, \odot, \leq \rangle$ is a ordered ring then we have :

1. $\forall x, y, z \in F$ we have $x < y \Leftrightarrow x \oplus z < y \oplus z$
2. $\forall x, y, z \in F$ we have $x \leq y \Leftrightarrow x \oplus z \leq y \oplus z$
3. $\forall x, y \in F$ we have $x < y \Leftrightarrow 0 < y \oplus (-x)$
4. $\forall x, y \in F$ we have $x \leq y \Leftrightarrow 0 \leq y \oplus (-x)$
5. $\forall x, y \in F$ we have $x < y \Leftrightarrow -y < -x$
6. $\forall x, y \in F$ we have $x \leq y \Leftrightarrow -y \leq -x$
7. $\forall x, y, z \in F$ with $0 < z$ we have $x < y \Leftrightarrow x \odot z < y \odot z$
8. $\forall x, y, z \in F$ with $0 < z$ we have $x \leq y \Leftrightarrow x \odot z \leq y \odot z$
9. $\forall x, y, z \in F$ with $0 \leq z$ and $x \leq y$ we have $x \odot z \leq y \odot z$
10. $\forall x, y, z \in F$ with $z < 0$ we have $x < y \Leftrightarrow y \odot z < x \odot z$
11. $\forall x, y, z \in F$ with $z < 0$ we have $x \leq y \Leftrightarrow y \odot z \leq x \odot z$
12. $\forall x, y, z \in F$ with $z \leq 0$ and $x \leq y$ we have $y \odot z \leq x \odot z$
13. $\forall x \in F$ we have $0 \leq x \cdot x \stackrel{\text{def}}{=} x^2$, further if $0 \neq x$ then $0 < x \odot x \stackrel{\text{def}}{=} x^2$
14. $0 \leq 1$
15. $\forall x, y \in F$ with $0 < x < y$ we have that $x^2 < y^2$ where $x^2 = x \odot x$ and $y^2 = y \odot y$
16. $\forall x, y \in F$ with $0 \leq x \leq y$ we have that $x^2 \leq y^2$ where $x^2 = x \odot x$ and $y^2 = y \odot y$
17. $\forall x \in F$ with $0 < x$ we have $0 < x^{-1}$
18. $\forall x, y \in F$ we have $0 < x < y \Leftrightarrow 0 < y^{-1} < x^{-1}$
19. $\forall x, y \in F$ we have $0 < x \leq y \Leftrightarrow 0 < y^{-1} \leq x^{-1}$
20. $\forall x \in F$ with $0 < x < 1$ we have $1 < x^{-1}$
21. $\forall x \in F$ with $0 < x \leq 1$ we have $1 \leq x^{-1}$

Proof. Using [definitions: 4.49, 4.72] and the fact that a field is automatically a ring we have that (1 – 14) follows from [theorem: 4.50] so that we only have to proof (15-17). So

1. This follows from [theorem: 4.50 (1)].
2. This follows from [theorem: 4.50 (2)].
3. This follows from [theorem: 4.50 (3)].
4. This follows from [theorem: 4.50 (4)].
5. This follows from [theorem: 4.50 (5)].

6. This follows from [theorem: 4.50 (6)].
7. This follows from [theorem: 4.50 (7)].
8. This follows from [theorem: 4.50 (8)].
9. This follows from [theorem: 4.50 (9)].
10. This follows from [theorem: 4.50 (10)].
11. This follows from [theorem: 4.50 (11)].
12. This follows from [theorem: 4.50 (12)].
13. This follows from [theorem: 4.50 (13)].
14. This follows from [theorem: 4.50 (14)].
15. This follows from [theorem: 4.50 (15)].
16. This follows from [theorem: 4.50 (16)].
17. Let $x \in R$ with $0 < x$. By the totally ordering we have for x^{-1} :

$x^{-1} = \mathbf{0}$. Then $1 = x^{-1} \odot x = \mathbf{0} \odot x \stackrel{[\text{theorem: 4.39}]}{=} \mathbf{0}$ so that $x = x \odot 1 = x \odot \mathbf{0} \stackrel{[\text{theorem: 4.39}]}{=} \mathbf{0}$ contradicting $0 < x$. So this case does not occur.

$x^{-1} < \mathbf{0}$. Then by (7) we have that $1 = x^{-1} \odot x < \mathbf{0} \odot x \stackrel{[\text{theorem: 4.39}]}{=} \mathbf{0}$ so that $1 < \mathbf{0}$ contradicting $0 \leq 1$, so this case will never occur.

$\mathbf{0} < x^{-1}$. This is the remaining case proving (15).

18.

$\Rightarrow$. If $0 < x < y$ then by (15) we have that $0 < x^{-1}$ and $0 < y^{-1}$. Hence

$$\begin{aligned}
 x < y &\stackrel{(7)}{\Rightarrow} x \odot x^{-1} < y \odot x^{-1} \\
 &\Rightarrow 1 < y \odot x^{-1} \\
 &\Rightarrow 1 < x^{-1} \odot y \\
 &\stackrel{(7)}{\Rightarrow} 1 \odot y^{-1} < (x^{-1} \odot y) \odot y^{-1} \\
 &\Rightarrow y^{-1} < x^{-1}
 \end{aligned}$$

$\Leftarrow$. If $0 < y^{-1} < x^{-1}$ then by (15) we have $0 < (x^{-1})^{-1}$ and $0 < (y^{-1})^{-1}$ which using [theorem: 4.53] proves that $0 < x$ and $0 < y$ so that

$$\begin{aligned}
 y^{-1} < x^{-1} &\stackrel{(7)}{\Rightarrow} y^{-1} \odot x < x^{-1} \odot x \\
 &\Rightarrow y^{-1} \odot x < 1 \\
 &\Rightarrow x \odot y^{-1} < 1 \\
 &\stackrel{(7)}{\Rightarrow} (x \odot y^{-1}) \odot y < 1 \odot y \\
 &\Rightarrow x < y
 \end{aligned}$$

19.

$\Rightarrow$. If $0 < x \leq y$ then by (15) we have that $0 < x^{-1}$ and $0 < y^{-1}$. Hence

$$\begin{aligned}
 x \leq y &\stackrel{(7)}{\Rightarrow} x \odot x^{-1} \leq y \odot x^{-1} \\
 &\Rightarrow 1 \leq y \odot x^{-1} \\
 &\Rightarrow 1 \leq x^{-1} \odot y \\
 &\stackrel{(7)}{\Rightarrow} 1 \odot y^{-1} \leq (x^{-1} \odot y) \odot y^{-1} \\
 &\Rightarrow y^{-1} \leq x^{-1}
 \end{aligned}$$

$\Leftarrow$. If $0 < y^{-1} \leq x^{-1}$ then by (15) we have $0 < (x^{-1})^{-1}$ and $0 < (y^{-1})^{-1}$ which using [theorem: 4.53] proves that $0 < x$ and $0 < y$ so that

$$\begin{aligned}
 y^{-1} \leq x^{-1} &\stackrel{(7)}{\Rightarrow} y^{-1} \odot x \leq x^{-1} \odot x \\
 &\Rightarrow y^{-1} \odot x \leq 1 \\
 &\Rightarrow x \odot y^{-1} \leq 1 \\
 &\stackrel{(7)}{\Rightarrow} (x \odot y^{-1}) \odot y \leq 1 \odot y \\
 &\Rightarrow x \leq y
 \end{aligned}$$

20. As $0 < x < 1$ we have by (16) that $0 < 1^{-1} < x^{-1}$ hence $1 < x^{-1}$.

21. As $0 < x \leq 1$ we have by (16) that $0 < 1^{-1} \leq x^{-1}$ hence $1 \leq x^{-1}$.

□

Chapter 5

Natural Numbers

Now we build a tower of different sets of numbers each one based on a previous one.

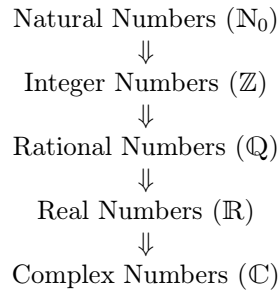


Table 5.1.

We start with the the Natural Numbers. Using the set of Natural numbers we can introduce the concept of finite sets, infinite sets, denumerable sets, countable sets, mathematical induction and recursion. Further we will introduce a total ordering on $\mathbb{N}_0$ and prove that $\langle \mathbb{N}_0, \leq \rangle$ is well ordered. Finally two arithmetic operators sum (+) and product ($\cdot$) are introduced, unfortunately it will turn out that $\langle \mathbb{N}_0, + \rangle$ is not a group but only a semi-group so that for example $x + 1 = 0$ has no solution. To solve this we introduce a new set of numbers, the integer numbers ($\mathbb{Z}$) and embed $\mathbb{N}_0$ in $\mathbb{Z}$ by creating a group and order isomorphism $i_{\mathbb{N}_0}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+$ where $\mathbb{Z}_0^+ \subseteq \mathbb{Z}$ is the embedding of $\mathbb{N}_0$ in $\mathbb{Z}$. Although we will have succeeded in making $\langle \mathbb{Z}, +, \cdot \rangle$ a ring a equation like $2 \cdot x = 1$ has no solution, for this we must have a field. Hence the next step is to construct a new set of numbers, the rational numbers ($\mathbb{Q}$), so that $\langle \mathbb{Q}, +, \cdot \rangle$ forms a field. We will embed then $\mathbb{Z}, \mathbb{N}_0$ in $\mathbb{Q}$ by creating ring, group and order isomorphisms $i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{Q}}, i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \mathbb{N}_0 \rightarrow \mathbb{N}_{0,\mathbb{Q}}$ where $\mathbb{Z}_{\mathbb{Q}} \subseteq \mathbb{Q}$ is the embedding of $\mathbb{Z}$ in $\mathbb{Q}$ and $\mathbb{N}_{0,\mathbb{Q}}$ is the embedding of $\mathbb{N}_0$ in $\mathbb{Q}$. However the totally ordered set $\langle \mathbb{Q}, \leq \rangle$ is not conditional complete which is important for analysis. So we create the set of real numbers $\mathbb{R}$ so that $\langle \mathbb{R}, \leq \rangle$ is conditionally complete. Next we create embeddings $\mathbb{N}_{0,\mathbb{R}}, \mathbb{Z}_{\mathbb{R}}, \mathbb{Q}_{\mathbb{R}}$ in $\mathbb{R}$ for $\mathbb{N}_0, \mathbb{Z}$ and $\mathbb{Q}$. It turns out that a equation like $x^2 = -1$ has no solution, so we create a new set of numbers, the complex numbers ($\mathbb{C}$) to solve this defect. Then we embed $\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ in $\mathbb{C}$ in the form $\mathbb{N}_{0,\mathbb{C}}, \mathbb{Z}_{\mathbb{C}}, \mathbb{Q}_{\mathbb{C}}$ and $\mathbb{R}_{\mathbb{C}}$ such that $\mathbb{N}_{0,\mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}} \subseteq \mathbb{Q}_{\mathbb{C}} \subseteq \mathbb{R}_{\mathbb{C}} \subseteq \mathbb{C}$. From then on, for the rest of this book we work with $\mathbb{C}$ and these embeddings, to avoid excessive notation we use the symbols $\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$ for these embeddings.

5.1 Definition of the Natural Numbers

We are now ready to define the first set of numbers namely the natural numbers which forms the basic of the other number systems but also of the important concepts of finite, infinite sets, countable sets, recursion and mathematical induction. To define the set of natural numbers recall the following definitions and axiom.

Definition 5.1. (Successor Set) *A set A is a **successor set** iff*

1. $\emptyset \in A$
2. *If $X \in A \Rightarrow X \cup \{X\} \in A$*

[see definition: 1.58]

Axiom 5.2. (Axiom of Infinity) *There exists a successor set [see axiom: 1.59].*

Definition 5.3. (Natural numbers) *The set of **natural numbers** $\mathbb{N}_0$ is defined by*

$$\mathbb{N}_0 = \bigcap \{S \mid S \text{ is a successor set}\}$$

Theorem 5.4. $\mathbb{N}_0$ is a set

Proof. By the axiom of infinity it follows that $\{S \mid S \text{ is a successor set}\} \neq \emptyset$ so that by [theorem: 1.66 (5)] $\bigcap \{S \mid S \text{ is a successor set}\}$ is a set. $\square$

Theorem 5.5. *If $n \in \mathbb{N}_0$ then $n \cup \{n\} \in \mathbb{N}_0$*

Proof. If $n \in \mathbb{N}_0$ then for $\forall A \in \{S \mid S \text{ is a successor set}\}$ we have $n \in A$ so that by definition of a successor set we have $n \cup \{n\} \in A$ so that $n \cup \{n\} \in \bigcap \{S \mid S \text{ is a successor set}\} = \mathbb{N}_0$. $\square$

The above theorem allows us to define the successor function

Definition 5.6. (Successor Function) *The function defined by*

$$s: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \text{ where } s(n) = n \cup \{n\}$$

*is called the **successor function**.*

The set $\mathbb{N}_0$ is not empty as is shown in the next theorem.

Theorem 5.7. $\emptyset \in \mathbb{N}_0$

Proof. If A is a successor set then by definition $\emptyset \in A$ so that $\emptyset \in \bigcap \{A \mid A \text{ is a successor set}\}$ $\square$

Further using the successor function we have that $s(\emptyset)$, $s(s(\emptyset))$ etc. are all elements of $\mathbb{N}_0$. we introduce a special notation for this elements that corresponds with the notation used for counting.

Notation 5.8. *We define the numbers $0, 1, 2, 3, \dots$ as follows*

1. $0 = \emptyset$
2. $1 = s(0) = s(\emptyset) = \emptyset \cup \{\emptyset\} = \{\emptyset\} = \{0\}$
3. $2 = s(1) = s(\emptyset) \cup \{s(\emptyset)\} = \{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\} = \{0, 1\}$
4. $3 = s(2) = \{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} = \{0, 1, 2\}$
5. $\dots$

The notation $\mathbb{N}_0$ may seem a little bit strange, the fact is that many mathematicians don't consider 0 a natural number. To express that $0 \in \mathbb{N}_0$ we add the 0 subscript. If we want to indicate that $0 \notin \mathbb{N}_0$ we use the following definition.

Definition 5.9. $\mathbb{N} = \mathbb{N}_0 \setminus \{0\}$

Theorem 5.10. *If $n \in \mathbb{N}_0$ then $s(n) \neq 0$ in particular $0 \neq 1$*

Proof. By definition we have $s(n) = n \cup \{n\}$ so that $n \in s(n)$ proving that $s(n) \neq \emptyset = 0$ $\square$

We introduce now the very important principle of **Mathematical Induction**.

Theorem 5.11. (Mathematical Induction) *If $X \subseteq \mathbb{N}_0$ such that*

1. $0 \in X$

2. If $n \in X$ then $s(n) \in X$

then $X = \mathbb{N}_0$

Proof. By (1), (2) it follows that X is a successor set so that $X \in \{A \mid A \text{ is a successor set}\}$ hence by [theorem: 1.66] $\mathbb{N}_0 = \bigcap \{A \mid A \text{ is a successor set}\} \subseteq X$, which together with $X \subseteq \mathbb{N}$ proves that $X = \mathbb{N}$. $\square$

Theorem 5.12. Let $n, m \in \mathbb{N}_0$ then if $m \in s(n)$ we have $m \in n \vee m = n$

Proof. If $m \in s(n) = n \cup \{n\}$ then we have either $m \in n$ or $m \in \{n\} \Rightarrow m = n$ $\square$

Definition 5.13. A set A is **transitive** if $\forall x \in A$ we have $x \subseteq A$.

As a application of mathematical induction we prove that every natural number is transitive, this fact will be used later, when we define a order relation on $\mathbb{N}_0$ to prove transitivity, hence the name for this property.

Theorem 5.14. $\forall n \in \mathbb{N}_0$ we have that n is transitive [in other words $\forall x \in n$ we have $x \subseteq n$]

Proof. We prove this by mathematical induction, let $S = \{n \in \mathbb{N}_0 \mid n \text{ is transitive}\}$ then clearly $S \subseteq \mathbb{N}_0$. Further we have

$0 \in S$. Because $\forall x \in \emptyset \vdash x \subseteq \emptyset$ is satisfied vacuously.

$n \in S \Rightarrow s(n) \in S$. If $n \in S$ then we have for $m \in s(n)$ by the previous theorem [theorem: 5.12] the following cases:

$m \in n$. Then as $n \in S$, n is transitive so that $m \subseteq n \subseteq n \cup \{n\} = s(n)$

$m = n$. Then $m = n \subseteq n \cup \{n\} = s(n)$

So $\forall m \in s(n)$ we have $m \subseteq s(n)$ which proves that $s(n)$ is transitive, hence $s(n) \in S$

Using mathematical induction [see theorem: 5.11] it follows then that $S = \mathbb{N}_0$. So if $n \in \mathbb{N}_0$ then $n \in S$ or n is transitive. $\square$

Another application of transitivity and mathematical induction is the following theorem.

Theorem 5.15. If $n \in \mathbb{N}_0$ then $n \neq s(n)$

Proof. Let $S = \{n \in \mathbb{N}_0 \mid n \neq s(n)\}$ then we have

$0 \in S$. By [theorem: 5.10] $0 \neq s(0)$.

$n \in S \Rightarrow s(n) \in S$. Assume that $s(s(n)) = s(n)$. As $s(s(n)) = s(n) \cup \{s(n)\}$ we have that $s(n) \in s(s(n)) = s(n)$, so $s(n) \in n \cup \{n\}$. As $n \in S$ we have that $n \neq s(n)$ so we must have that $s(n) \in n$. As by [theorem: 5.14] $s(n)$ is transitive it follows that $s(n) \subseteq n$, further we have that $n \subseteq n \cup \{n\} = s(n)$. So we conclude that $n = s(n)$ proving $n \notin S$ which contradicts $n \in S$. So we must have that $s(s(n)) \neq s(n)$ proving that $s(n) \in S$.

Using mathematical induction it follows then that $\mathbb{N}_0 = S$ so if $n \in \mathbb{N}_0$ then $n \in S$ and thus $n \neq s(n)$. $\square$

The next theorem shows that the successor function is a injection.

Theorem 5.16. If $n, m \in \mathbb{N}_0$ is such that $s(n) = s(m)$ then $n = m$. In other words

$s: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is injective

Proof. As $n \in n \cup \{n\} = s(n) = s(m)$ and $m \in m \cup \{m\} = s(m) = s(n)$ we have that $n \in s(m) \wedge m \in s(n)$. Using [theorem: 5.12] this becomes

$$(n \in m \vee n = m) \wedge (m \in n \vee m = n) \Rightarrow (n \in m \wedge m \in n) \vee n = m$$

If $n = m$ we are done. So we must look at the case that $m \in n \wedge n \in m$. By transitivity [theorem: 5.14] we have then $n \subseteq m$ and $m \subseteq n$ proving that $n = m$. $\square$

The above theorems are part of what is in number theory the Peano Axioms.

Theorem 5.17. (Peano Axioms) $\mathbb{N}_0$ satisfies the following so called Peano Axioms

1. $0 \in \mathbb{N}_0$
2. If $n \in \mathbb{N}_0$ then $s(n) \in \mathbb{N}_0$
3. $\forall n \in \mathbb{N}_0$ we have that $s(n) \neq 0$
4. If $X \subseteq \mathbb{N}_0$ is such that
 - a. $0 \in X$
 - b. $n \in X \Rightarrow s(n) \in X$
 then $X = \mathbb{N}_0$
5. If $n, m \in \mathbb{N}_0$ is such that $s(n) = s(m)$ then $n = m$

Proof.

1. See [theorem: 5.7]
2. See [definition: 5.6]
3. See [theorem: 5.10]
4. See [theorem: 5.11]
5. See [theorem: 5.16] $\square$

Theorem 5.18. If $n \in \mathbb{N}_0 \wedge n \neq 0$ then $\exists! m \in \mathbb{N}_0$ such that $n = s(m)$

Proof. We use mathematical induction to prove this. So let

$$S = \{n \in \mathbb{N}_0 \mid (n = 0) \vee (\exists! m \in \mathbb{N}_0 \text{ such that } n = s(m))\} \subseteq \mathbb{N}_0$$

then we have:

$0 \in S$. As $0 = 0$ we have that $0 \in S$.

$n \in S \Rightarrow s(n) \in S$. Consider $s(n)$ then by [theorem: 5.10] $s(n) \neq 0$, further we have that $m = n$ satisfies $s(n) = s(m)$ proving the existence part. Assume that there is another $m' \in \mathbb{N}_0$ such that $s(n) = s(m')$, then by [theorem: 5.16] we have $n = m'$, proving uniqueness. So $s(n) \in S$.

Mathematical induction [see: 5.11] proves then that $\mathbb{N}_0 = S$. So if $n \in \mathbb{N}_0$ with $n \neq 0$ we have as $n \in S$ that $\exists! m \in \mathbb{N}_0$ such that $n = s(m)$. $\square$

5.2 Recursion

Recursion will be used to essential define things in terms of itself. It is the mathematical equivalent of iteration in many programming languages. Actually, functional languages that are mathematical oriented, like Haskell, have no iteration and loop constructs at all and rely fully on recursion. Recursion is based on the definition of a recursive function that takes the role of iterating. The following theorem ensures the existence of such a function.

Theorem 5.19. (Recursion) Let A be a set, $a \in A$ and $f: A \rightarrow A$ a function then there exists a **unique** function

$$\lambda: \mathbb{N}_0 \rightarrow A$$

such that

1. $\lambda(0) = a$

2. $\forall n \in \mathbb{N}_0$ we have $\lambda(s(n)) = f(\lambda(n))$

Proof. Define

$$\mathcal{G} = \{G \mid G \subseteq \mathbb{N}_0 \times A \text{ such that } (0, a) \in G \text{ and } \forall n \in \mathbb{N}_0 \text{ that } (n, x) \in G \Rightarrow (s(n), f(x)) \in G\}$$

Define $G = \mathbb{N}_0 \times A$ then as $0 \in \mathbb{N}_0$ and $a \in A$ we have $(0, a) \in \mathbb{N}_0 \times A$. Further if $(n, x) \in \mathbb{N}_0 \times A$ then $n \in \mathbb{N}_0$ and $x \in A$ so that $s(n) \in \mathbb{N}_0$ and $f(x) \in A$, hence $(s(n), f(x)) \in \mathbb{N}_0 \times A$. So

$$\mathbb{N}_0 \times A \in \mathcal{G} \quad (5.1)$$

We prove now that

$$\text{If } \lambda = \bigcap \mathcal{G} \text{ then } \lambda \in \mathcal{G}, \lambda \subseteq \mathbb{N}_0 \times A \text{ and } (0, a) \in \lambda \quad (5.2)$$

Proof.

1. By [eq: 5.1] we have $\mathbb{N}_0 \times A \in \mathcal{G}$ so that by [theorem: 1.66] $\bigcap \mathcal{G} \subseteq \mathbb{N}_0 \times A$ hence $\lambda \subseteq \mathbb{N}_0 \times A$

2. $\forall G \in \mathcal{G}$ we have by definition that $(0, a) \in G$ hence $(0, a) \in \bigcap \mathcal{G}$ or $(0, a) \in \lambda$

3. If $(n, x) \in \bigcap \mathcal{G}$ then $\forall G \in \mathcal{G}$ we have $(n, x) \in G \Rightarrow (s(n), f(x)) \in G$, so that $(s(n), f(x)) \in \bigcap \mathcal{G}$.

Using (1), (2) and (3) it follows that $\bigcap \mathcal{G} \in \mathcal{G}$. $\square$

If $x \in \text{dom}(\lambda)$ then $\exists y$ such that $(x, y) \in \lambda \subseteq \mathbb{N}_0 \times A$ [see eq: 5.2] so that $x \in \mathbb{N}_0$, hence

$$\text{dom}(\lambda) \subseteq \mathbb{N}_0 \quad (5.3)$$

As by [eq: 5.2] $(0, a) \in \lambda$ we have that

$$0 \in \text{dom}(\lambda) \quad (5.4)$$

If $n \in \text{dom}(\lambda)$ then $\exists x$ such that $(n, x) \in \lambda$, as by [eq: 5.2] $\lambda \in \mathcal{G}$, we have $(s(n), f(x)) \in \lambda$ so that $s(n) \in \text{dom}(\lambda)$. In other words we have

$$\text{if } n \in \text{dom}(\lambda) \text{ then } s(n) \in \text{dom}(\lambda) \quad (5.5)$$

Now [eq: 5.3], [eq: 5.4] and [eq: 5.5] are the conditions for mathematical induction [theorem: 5.11], so we have proved that

$$\text{dom}(\lambda) = \mathbb{N}_0 \quad (5.6)$$

We use now mathematical induction to prove that λ is the graph of a function. Let

$$S = \{n \in \mathbb{N}_0 \mid \exists! x \text{ such that } (n, x) \in \lambda\} \subseteq \mathbb{N}_0$$

then we have:

0 $\in$ S. By [eq: 5.2] we have $(0, a) \in \lambda$. Assume that $\exists x \in A$ with $x \neq a$ such that $(0, x) \in \lambda$, then $(0, a) \neq (0, x)$. Define now $\beta = \lambda \setminus \{(0, x)\}$ then we have

1. $\beta \subseteq \lambda \subseteq \mathbb{N}_0 \times A$

2. As $(0, a) \neq (0, x)$ and $(0, a) \in \lambda$ we have $(0, a) \in \beta$

3. If $(n, y) \in \beta \xRightarrow{\beta \subseteq \lambda} (n, y) \in \lambda$ so that $(s(n), f(y)) \in \lambda$, as by [theorem: 5.10] $s(n) \neq 0$ we have that $(s(n), f(x)) \neq (0, x)$, hence $(s(n), f(y)) \in \beta$

From (1), (2) and (3) it follows that $\beta \in \mathcal{G}$ so that by [theorem: 1.66] $\lambda = \bigcap \mathcal{G} \subseteq \beta$ which as $(0, x) \in \lambda$ would give $(0, x) \in \beta = \lambda \setminus \{(0, x)\}$ a contradiction. So the assumption is wrong and we must have that $x = a$, proving uniqueness, hence that $0 \in S$.

$n \in S \Rightarrow s(n) \in S$. As $n \in S$ there exist a **unique** $x \in S$ such that $(n, x) \in \lambda$. As $(n, x) \in \lambda$ we have as $\lambda \in \mathcal{G}$ that $(s(n), f(x)) \in \lambda$. Assume now that $\exists y$ such that $(s(n), y) \in \lambda$ and $f(x) \neq y$. Define then $\beta = \lambda \setminus \{(s(n), y)\}$ then we have:

1. $\beta \subseteq \lambda \subseteq \mathbb{N}_0 \times A$

2. As by [theorem: 5.15] $s(n) \neq 0$ we have that $(0, a) \neq (s(n), y)$, as further $(0, a) \in \lambda$ it follows that $(0, a) \in \beta$

3. If $(m, z) \in \beta$ then $(m, z) \in \lambda$ so that $(s(m), f(z)) \in \lambda$ we must now consider two cases for $s(n), s(m)$:

$s(m) = s(n)$. Then by [theorem: 5.16] we have $n = m$ so that $(n, z) = (m, z) \in \lambda$.
 As $n \in S$ and we have $(n, x) \in \lambda$ it follows that $z = x$. So that $(s(m), f(z)) = (s(n), f(x)) \neq (s(n), y)$ [as we assumed that $y \neq f(x)$] hence we have that $(s(m), f(z)) \in \beta$.

$s(m) \neq s(n)$. then $(s(m), f(z)) \neq (s(n), y)$ so that $(s(m), f(z)) \in \beta$

So we have prove that if $(m, z) \in \beta$ then $(s(m), f(z)) \in \beta$

From (1),(2) and (3) it follows that $\beta \in \mathcal{G}$ but then using [theorem: 1.66] we have that $\lambda = \bigcap \mathcal{G} \subseteq \beta$ which as $(s(n), y) \in \lambda$ leads to $(s(n), y) \in \beta = \lambda \setminus \{(s(n), y)\}$ a contradiction. So the assumption is wrong and we must have that $y = f(x)$ proving **uniqueness**, hence we have that $s(n) \in S$.

Using mathematical induction [theorem: 5.11] it follows that $\mathbb{N}_0 = S$. So if $(n, x), (n, x') \in \lambda$ then $n \in \mathbb{N}_0 = S$ so that $y = y'$ giving

$$\text{If } (n, x), (n, x') \in \lambda \text{ then } x = x' \quad (5.7)$$

From [eq: 5.2], [eq: 5.6] and [eq: 5.7] it follows that

$$\lambda: \mathbb{N}_0 \rightarrow A \text{ is a function} \quad (5.8)$$

As $\lambda \in \mathcal{G}$ we have that $(0, a) \in \lambda \Rightarrow a = \lambda(0)$, further if $n \in \mathbb{N}_0 = \text{dom}(\lambda)$ then $\exists x$ such that $(n, x) \in \lambda$ and $(s(n), f(x)) \in \lambda$. Now $(n, x) \in \lambda$ is equivalent with $\lambda(n) = x$ and $(s(n), f(x)) \in \lambda$ is equivalent with $\lambda(s(n)) = f(x) = f(\lambda(n))$. So we have for λ that

$$\lambda(0) = a \text{ and } \forall n \in \mathbb{N}_0 \text{ we have } \lambda(s(n)) = f(\lambda(n)) \quad (5.9)$$

So we have proved the existence of our function, next we must prove that this function is unique. Assume that there exist another function

$$\beta: \mathbb{N}_0 \rightarrow A \text{ such that } \beta(0) = a \text{ and } \forall n \in \mathbb{N}_0 \text{ we have } \lambda(s(n)) = f(\lambda(n))$$

We proceed by mathematical induction, so define $T = \{n \in \mathbb{N}_0 \mid \lambda(n) = \beta(n)\}$ then we have

$0 \in T$. As $\lambda(0) = a = \beta(0)$ we have that $0 \in T$.

$n \in T \Rightarrow s(n) \in T$. As $n \in T$ we have $\lambda(n) = \beta(n)$ but then $\lambda(s(n)) = f(\lambda(n)) = f(\beta(n)) = \beta(s(n))$ so that $s(n) \in T$

Using mathematical induction [theorem: 5.11] we have then $T = \mathbb{N}_0$. So $\forall n \in \mathbb{N}_0$ we have $n \in T$ hence $\lambda(n) = \beta(n)$ which by [theorem: 2.43] proves that

$$\lambda = \beta \quad \square$$

Corollary 5.20. *If A is a set, $a \in A$ and $f: A \rightarrow A$ a function then there exists a unique function*

$$\lambda: \mathbb{N}_0 \rightarrow A$$

such that

1. $\lambda(0) = a$
2. $\forall n \in \mathbb{N}_0$ we have $\lambda(s(n)) = f(\lambda(n))$
3. *If $a \notin f(A)$ and $f: A \rightarrow A$ is injective then λ is injective*

Proof. The first part is easy. Using recursion [theorem: 5.19] there exists a function

$$\lambda: \mathbb{N}_0 \rightarrow A$$

such that

$$\lambda(0) = a \text{ and } \forall n \in \mathbb{N}_0 \text{ we have } \lambda(s(n)) = f(\lambda(n))$$

We use now mathematical induction to prove (3). Assume that $a \notin f(A)$ and take

$$S = \{n \in \mathbb{N}_0 \mid \forall m \in \mathbb{N}_0 \text{ with } \lambda(n) = \lambda(m) \text{ we have } n = m\}$$

then we have:

$0 \in S$. If $\lambda(m) = \lambda(0)$ then as $\lambda(0) = a$ we have that $\lambda(m) = a$. Assume that $m \neq 0$ then by [theorem: 5.18] there exists a $k \in \mathbb{N}_0$ such that $m = s(k)$ so that $a = \lambda(m) = \lambda(s(k)) = f(\lambda(k))$, which proves that $a \in f(A)$ contradicting $a \notin f(A)$. Hence we must have $m = 0$ so that $0 \in S$.

$n \in S \Rightarrow s(n) \in S$. Let $m \in \mathbb{N}_0$ such that $\lambda(s(n)) = \lambda(m)$. Assume that $m = 0$ then $\lambda(s(n)) = \lambda(m) = \lambda(0) = a$ so that $f(\lambda(n)) = \lambda(s(n)) = a$, resulting in $a \in f(A)$ contradicting $a \notin f(A)$. Hence we must have that $m \neq 0$. Using [theorem: 5.18] there exists a $k \in \mathbb{N}_0$ such that $m = s(k)$, from $\lambda(s(n)) = \lambda(m)$ it follows then that $\lambda(s(n)) = \lambda(s(k))$ so that $f(\lambda(n)) = \lambda(s(n)) = \lambda(s(k)) = f(\lambda(k))$. As f is injective we have $\lambda(n) = \lambda(k)$. Now as $n \in S$ we must have $n = k$ or $s(n) = s(k) = m$. This proves that $\forall m \in \mathbb{N}_0$ with $\lambda(s(n)) = \lambda(m)$ we have $s(n) = m$, hence $s(n) \in S$.

Using mathematical induction [theorem: 5.11] it follows that $\mathbb{N}_0 = S$. So if $n, m \in \mathbb{N}_0$ is such that $\lambda(n) = \lambda(m)$ then $n \in S$ and as $m \in \mathbb{N}_0$ we have $n = m$, proving that

λ is injective □

Remark 5.21. To understand how recursion works in the above theorem consider the following, Let $f: A \rightarrow A$ a function, $a \in A$ and $\lambda: \mathbb{N}_0 \rightarrow A$ such that $\lambda(0) = a$ and $\lambda(s(n)) = f(\lambda(n))$

$$\begin{aligned}\lambda(0) &= a \\ \lambda(1) = \lambda(s(0)) &= f(\lambda(0)) = f(a) \\ \lambda(2) = \lambda(s(1)) &= f(\lambda(1)) = f(f(a)) \\ \lambda(3) = \lambda(s(2)) &= f(\lambda(2)) = f(f(f(a))) \\ &\dots \\ \lambda(n) &= \overbrace{f(f(\dots(f(a))))}^{n \text{ times}}\end{aligned}$$

so $\lambda(n)$ is the result of applying f n -times on a value a . If $a \notin f(A)$ and f is injective then λ is injective and we would have that $f(a), f(f(a)), f(f(f(a))), \dots, \overbrace{f(f(\dots(f(a))))}^{n \text{ times}}$ are all different numbers.

To see the conditions for injectivity of λ consider the following two examples:

Example 5.22. Define $f: \{1, 2, 3\} \rightarrow f(\{1, 2, 3\})$ by $f(i) = \begin{cases} 2 & \text{if } i = 1 \\ 3 & \text{if } i = 2 \\ 2 & \text{if } i = 3 \end{cases}$ (so f is not injective) and $a = 3$

then we have

$$\begin{aligned}\lambda(0) &= 3 \\ \lambda(1) &= f(3) = 2 \\ \lambda(2) &= f(f(3)) = f(2) = 1 \\ \lambda(3) &= f(f(f(3))) = f(1) = 2 \\ \lambda(4) &= f(f(f(f(3)))) = f(2) = 1 \\ &\dots\end{aligned}$$

So that $\lambda: \mathbb{N}_0 \rightarrow A$ is clearly not injective.

Example 5.23. Take $f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ by $f(i) = \begin{cases} 2 & \text{if } i = 1 \\ 3 & \text{if } i = 2 \\ 1 & \text{if } i = 3 \end{cases}$ so that f is injective and $a = 2$ so that $a \in f(\{1, 2, 3\})$ then we have

$$\begin{aligned}\lambda(0) &= 2 \\ \lambda(1) &= f(2) = 1 \\ \lambda(2) &= f(f(2)) = f(1) = 2 \\ \lambda(3) &= f(f(f(2))) = f(2) = 1 \\ &\dots\end{aligned}$$

So that $\lambda: \mathbb{N}_0 \rightarrow \{1, 2, 3\}$ is not injective.

We can rephrase the above remark in the iteration principle that is useful in proofs using mathematical induction.

Theorem 5.24. (Iteration) *Let A be a non empty set and $f: A \rightarrow A$ a function. Then $\forall n \in \mathbb{N}_0$ there exist a function*

$$(f)^n: A \rightarrow A$$

such that

1. $(f)^0 = \text{Id}_A$
2. $(f)^{s(n)} = f \circ (f)^n$

Proof. Let $a \in A$ and use the recursion [theorem: 5.19] to find a **unique** function

$$\lambda_a: \mathbb{N}_0 \rightarrow A \text{ such that } \lambda_a(0) = a \text{ and } \forall n \in \mathbb{N}_0 \lambda_a(s(n)) = f(\lambda_a(n))$$

Define now

$$(f)^n: A \rightarrow A \text{ where } (f)^n(a) = \lambda_a(n)$$

Then we have

1. $\forall a \in A$ we have that $(f)^0(a) = \lambda_a(0) = a$ so that

$$(f)^0 = \text{Id}_A$$

2. $\forall a \in A$ we have that $(f)^{s(n)}(a) = \lambda_a(s(n)) = f(\lambda_a(n)) = f((f)^n(a)) = (f \circ (f)^n)(a)$ so that

$$(f)^{s(n)} = f \circ (f)^n \quad \square$$

As illustration of iteration let $f: A \rightarrow A$ then we have

$$\begin{aligned} (f)^0 &= \text{Id}_A \\ (f)^1 &= (f)^{s(0)} = f \circ (f)^0 = f \circ \text{Id}_A = f \\ (f)^2 &= (f)^{s(1)} = f \circ (f)^1 = f \circ f \\ (f)^3 &= (f)^{s(2)} = f \circ (f)^2 = f \circ f \circ f \\ &\dots \\ (f)^n &= \overbrace{f \circ \dots \circ f}^{n \text{ times}} \end{aligned}$$

We can apply the above on a group to define new operations on the group.

Example 5.25. Let $\langle A, \oplus \rangle$ be a group and $a \in A$ define then $\oplus_a: A \rightarrow A$ by $x \rightarrow \oplus_a(x) = x \oplus a$ we define then given $n \in \mathbb{N}$ $a \langle \oplus \rangle n = (\oplus_a)^n(e)$ where e is the neutral element in the group. So

$$\begin{aligned} a \langle \oplus \rangle 0 &= (\oplus_a)^0(e) = \text{Id}_A(e) = e \\ a \langle \oplus \rangle 1 &= (\oplus_a)^1(e) = \oplus_a(e) = a \oplus e = e \\ a \langle \oplus \rangle 2 &= (\oplus_a)^2(e) = \oplus_a(\oplus_a(e)) = a \oplus (a \oplus e) = a \oplus (a \oplus e) = a \oplus a \\ a \langle \oplus \rangle 3 &= (\oplus_a)^3(e) = (\oplus_a(\oplus_a(\oplus_a(e)))) = a \oplus (a \oplus (a \oplus e)) = a \oplus a \oplus a \\ &\dots \\ a \langle \oplus \rangle n &= \overbrace{a \oplus \dots \oplus a}^{n \text{ times}} \end{aligned}$$

Sometimes we consider a group to be additive or multiplicative, this is either noted as $\langle A, + \rangle$ with neutral element 0 or $\langle A, \cdot \rangle$ with neutral element 1. Then we note $a \langle + \rangle n$ as $a \cdot n$ as and $a \langle \cdot \rangle n$ as a^n hence we have

1. Additive group $\langle A, + \rangle$ with neutral element 0 gives

$$\begin{aligned} a \cdot 0 &= 0 \\ a \cdot 1 &= a \\ a \cdot 2 &= a + a \\ a \cdot 3 &= a + a + a \\ &\dots \\ a \cdot n &= \overbrace{a + \dots + a}^{n \text{ times}} \end{aligned}$$

2. Multiplicative group $\langle A, \cdot \rangle$ with neutral element 1 gives

$$\begin{aligned} a^0 &= 1 \\ a^1 &= a \\ a^2 &= a \cdot a \\ a^3 &= a \cdot a \cdot a \\ &\dots \\ a^n &= \overbrace{a \cdot \dots \cdot a}^{n \text{ times}} \end{aligned}$$

Recursion is mostly used in it's step form to define recursive functions.

Theorem 5.26. (Recursion on $\mathbb{N}_0$ Step Form) *Let A be a set, $a \in A$ and $g: \mathbb{N} \times A \rightarrow A$ a function then there exist a **unique** function $\lambda: \mathbb{N}_0 \rightarrow A$ such that*

1. $\lambda(0) = a$
2. $\forall n \in \mathbb{N}_0$ we have $\lambda(s(n)) = g(n, \lambda(n))$

Proof. First we define the projection functions

$$\pi_1: \mathbb{N}_0 \times A \rightarrow \mathbb{N}_0 \text{ where } \pi_1(n, x) = n$$

$$\pi_2: \mathbb{N}_0 \times A \rightarrow A \text{ where } \pi_2(n, x) = x$$

Define now

$$\gamma: \mathbb{N}_0 \times A \rightarrow \mathbb{N}_0 \times A \text{ where } \gamma(x) = (s(\pi_1(x)), g(\pi_1(x), \pi_2(x))) \quad (5.10)$$

Using the iteration [theorem: 5.24] on the above functions gives $\forall n \in \mathbb{N}_0$ the existence of the function

$$(\gamma)^n: \mathbb{N}_0 \times A \rightarrow \mathbb{N}_0 \times A \text{ such that } (\gamma)^0 = \text{Id}_{\mathbb{N}_0 \times A} \text{ and } \forall n \in \mathbb{N}_0 \text{ we have } (\gamma)^{s(n)} = \gamma \circ (\gamma)^n \quad (5.11)$$

We prove now by mathematical induction that $\forall n \in \mathbb{N}_0$ $\pi_1((\gamma)^n(0, a)) = n$. So let

$$S = \{n \in \mathbb{N}_0 \mid \pi_1((\gamma)^n(0, a)) = n\}$$

then we have:

$$0 \in S. \text{ As } \pi_1((\gamma)^0(0, a)) \stackrel{[\text{eq: 5.11}]}{=} \pi_1(\text{Id}_{\mathbb{N}_0 \times A}(0, a)) = \pi_1(0, a) = 0 \text{ we have that } 0 \in S$$

$n \in S \Rightarrow s(n) \in S$. We have

$$\begin{aligned} \pi_1((\gamma)^{s(n)}(0, a)) &\stackrel{[\text{eq: 5.11}]}{=} \pi_1((\gamma \circ (\gamma)^n)(0, a)) \\ &= \pi_1(\gamma((\gamma)^n(0, a))) \\ &\stackrel{[\text{eq: 5.10}]}{=} \pi_1(\pi_1((\gamma)^n(0, a)), g(\pi_1((\gamma)^n(0, a)), \pi_2((\gamma)^n(0, a)))) \\ &\stackrel{n \in S \Rightarrow \pi_1((\gamma)^n(0, a)) = n}{=} \pi_1(n, g(n, \pi_2((\gamma)^n(0, a)))) \\ &= n \end{aligned}$$

proving that $s(n) \in S$

Using mathematical induction [theorem: 5.11] we have $\mathbb{N}_0 = S$, hence

$$\forall n \in \mathbb{N}_0 \text{ we have } \pi_1((\gamma)^n(0, a)) = n \quad (5.12)$$

Define now

$$\lambda: \mathbb{N}_0 \rightarrow A \text{ by } \gamma(n) = \pi_2((\gamma)^n(0, a)) \quad (5.13)$$

then we have:

1. $\lambda(0) = \pi_2((\gamma)^0(0, a)) = \pi_2(\text{Id}_{\mathbb{N}_0 \times A}(0, a)) = \pi_2(0, a) = a$
2. If $n \in \mathbb{N}_0$ then

$$\begin{aligned} \lambda(s(n)) &= \pi_2((\gamma)^{s(n)}(0, a)) \\ &\stackrel{[\text{eq: 5.11}]}{=} \pi_2((\gamma \circ (\gamma)^n)(0, a)) \\ &= \pi_2(\gamma((\gamma)^n(0, a))) \\ &\stackrel{[\text{eq: 5.10}]}{=} \pi_2(\pi_1((\gamma)^n(0, a)), g(\pi_1((\gamma)^n(0, a)), \pi_2((\gamma)^n(0, a)))) \\ &= g(\pi_1((\gamma)^n(0, a)), \pi_2((\gamma)^n(0, a))) \\ &\stackrel{[\text{eq: 5.12}]}{=} g(n, \pi_2((\gamma)^n(0, a))) \\ &\stackrel{[\text{eq: 5.13}]}{=} g(n, \lambda(n)) \end{aligned}$$

This proves the existence of the function we are searching for. Now for uniqueness assume that there is a

$$\beta: \mathbb{N}_0 \rightarrow A \text{ such that } \beta(0) = a \text{ and } \forall n \in \mathbb{N}_0 \text{ that } \beta(s(n)) = g(n, \beta(n))$$

Define now $B = \{n \in \mathbb{N}_0 \mid \lambda(n) = \beta(n)\}$ then we have:

$0 \in B$. As $\beta(0) = a = \lambda(0)$ it follows that $0 \in B$.

$n \in B \Rightarrow s(n) \in B$. As

$$\beta(s(n)) = g(n, \beta(n)) \stackrel{n \in B}{=} g(n, \lambda(n)) = \lambda(s(n))$$

we have that $s(n) \in B$

Using mathematical induction we have $B = \mathbb{N}_0$, so $\forall n \in \mathbb{N}_0$ we have $n \in B$ hence $\beta(n) = \lambda(n)$ proving that

$$\beta = \lambda \quad \square$$

Up to now we have used the successor function $s: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ in the recursion and induction theorems. Once we have introduced the arithmetic of the natural numbers, we will rewrite these theorems by a version where $s(n)$ is replaced by $n + 1$.

5.3 Arithmetic of the Natural numbers

We use recursion to define the sum of two natural numbers.

Definition 5.27. Let $m, n \in \mathbb{N}_0$ then the addition operator $+$ is defined by

$$+: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \mathbb{N}_0 \text{ where } n + m \stackrel{\text{notation}}{=} +(n, m) = (s)^m(n)$$

Here $s: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is the successor function [definition: 5.6] and we use the iteration principle from [theorem: 5.24] to define $(s)^n$.

Example 5.28. Using this definition we can easily calculate that $1 + 1 = 2$

Proof. $1 + 1 = (s)^1(1) = (s \circ (s)^0)(s) = s((s)^0(1)) = s(\text{Id}_{\mathbb{N}_0}(1)) = s(1) = 2 \quad \square$

We will show now that $\langle \mathbb{N}_0, + \rangle$ forms a Abelian semi-group.

Theorem 5.29. (Neutral Element) *Let $n \in \mathbb{N}_0$ then $n + 0 = n = 0 + n$*

Proof.

1. $n + 0 = (s)^0(n) = \text{Id}_{\mathbb{N}_0}(n) = n$
2. For the $0 + n = n$ we use mathematical induction. So let $S = \{n \in \mathbb{N}_0 | 0 + n = n\}$ then we have:

$$0 \in S. \text{ As } 0 + 0 \underset{(1)}{=} 0 \text{ proving } 0 \in S$$

$$n \in S \Rightarrow s(n) \in S. \text{ We have } 0 + s(n) = (s)^{s(n)}(0) = (s \circ (s)^n)(0) = s((s)^n(0)) \underset{n \in S}{=} s(n) \text{ proving that } s(n) \in S$$

Using mathematical induction 5.11 we have $S = \mathbb{N}_0$. So if $n \in \mathbb{N}_0 \Rightarrow n \in S$ then $0 + n = n$. $\square$

Theorem 5.30. $\forall n \in \mathbb{N}_0$ we have $n + 1 = s(n) = 1 + n$

Proof.

1. $n + 1 = (s)^1(n) = (s \circ (s)^0)(n) = s((s)^0(n)) = s(\text{Id}_{\mathbb{N}_0}(n)) = s(n)$
2. For $1 + n = s(n)$ we use induction, so define $S = \{n \in \mathbb{N}_0 | 1 + n = s(n)\}$ then we have:

$$0 \in S. \text{ } 1 + 0 \underset{[\text{theorem: 5.29}]}{=} 1 = s(0)$$

$$n \in S \Rightarrow n + 1 \in S.$$

$$1 + s(n) = (s)^{s(n)}(1) = (s \circ (s)^n)(1) = s((s)^n(1)) = s(1 + n) \underset{n \in S}{=} s(s(n))$$

proving that $s(n) \in S$.

By mathematical induction [theorem: 5.11] we have $S = \mathbb{N}_0$ completing the proof. $\square$

Lemma 5.31. *If $n, m \in \mathbb{N}$ then $n + s(m) = s(n + m)$*

Proof. $n + s(m) = (s)^{s(m)}(n) = (s \circ (s)^m)(n) = s((s)^m(n)) = s(n + m)$ $\square$

Theorem 5.32. (Associativity) *If $n, m, k \in \mathbb{N}$ then $(n + m) + k = n + (m + k)$*

Proof. The proof is by mathematical induction, so given $n, m \in \mathbb{N}_0$ define

$$S_{n,m} = \{k \in \mathbb{N} | (n + m) + k = n + (m + k)\}$$

then we have:

$$0 \in S_{n,m}. \text{ } (n + m) + 0 \underset{[\text{theorem: 5.29}]}{=} n + m \underset{[\text{theorem: 5.29}]}{=} n + (m + 0) \Rightarrow 0 \in S_{n,m}$$

$$k \in S_{n,m} \Rightarrow s(k) \in S_{n,m}. \text{ We have}$$

$$\begin{aligned} (n + m) + s(k) &\underset{[\text{lemma: 5.31}]}{=} s((n + m) + k) \\ &\underset{k \in S}{=} s(n + (m + k)) \\ &\underset{[\text{lemma: 5.31}]}{=} (n + s(m + k)) \\ &\underset{[\text{lemma: 5.31}]}{=} (n + (m + s(k))) \end{aligned}$$

proving that $s(k) \in S_{n,m}$.

By mathematical induction [theorem: 5.11] we have $\mathbb{N}_0 = S_{n,m}$. So if $n, m, k \in \mathbb{N}_0$ then $k \in S_{n,m} \Rightarrow (n + m) + k = n + (m + k)$ $\square$

Theorem 5.33. (Commutativity) *If $n, m \in \mathbb{N}$ then $n + m = m + n$*

Proof. This is done again by induction. Let $n \in \mathbb{N}_0$ and define

$$S_n = \{k \in \mathbb{N}_0 | n + k = k + n\}$$

then we have:

$$0 \in S_n. \text{ Using [theorem: 5.29] it follows that } n + 0 = 0 + n \text{ proving that } 0 \in S_n$$

$k \in S_n \Rightarrow s(k) \in S_n$. We have

$$\begin{aligned}
 n + s(k) &\stackrel{[\text{lemma: 5.31}]}{=} s(n+k) \\
 &\stackrel{k \in S_{n,m}}{=} s(k+n) \\
 &\stackrel{[\text{theorem: 5.30}]}{=} 1 + (k+n) \\
 &\stackrel{[\text{theorem: 5.32}]}{=} (1+k) + n \\
 &\stackrel{[\text{theorem: 5.30}]}{=} s(k) + n
 \end{aligned}$$

Using mathematical induction [theorem: 5.11] we have that $S_n = \mathbb{N}_0$. So if $n, m \in \mathbb{N} \Rightarrow m \in S_n \Rightarrow n+m = m+n$. $\square$

We can summarize the above theorems as follows.

Theorem 5.34. $\langle \mathbb{N}_0, + \rangle$ forms a Abelian semi-group with neutral element 0

Proof.

neutral element. This follows from [theorem: 5.29].

associativity. This follows from [theorem: 5.32].

commutativity. This follows from [theorem: 5.33] $\square$

Next we use recursion to define multiplication in $\mathbb{N}_0$ and prove that $\langle \mathbb{N}_0, \cdot \rangle$ is a Abelian group.

Definition 5.35. (Multiplication) Given $n \in \mathbb{N}_0$ define

$$\alpha_n: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \text{ by } \alpha_n(m) = n + m$$

Then we define the multiplication operator as follows

$$\cdot: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \mathbb{N}_0 \text{ by } n \cdot m = \cdot(n, m) = (\alpha_n)^m(0)$$

Using the above definition we have

We have the following examples to see how multiplication works by repeating summation

$$\begin{aligned}
 2 \cdot 0 &= (\alpha_2)^0(0) = \text{Id}_{\mathbb{N}}(0) = 0 \\
 2 \cdot 1 &= (\alpha_2)^1(0) = (\alpha_2)^{s(0)}(0) = (\alpha_2 \circ (\alpha_2)^0)(0) = \alpha_2(0) = 2+0 = 2 \\
 2 \cdot 2 &= (\alpha_2)^2(0) = (\alpha_2)^{s(1)}(0) = (\alpha_2((\alpha_2)^1(0))) = \alpha_2(2) = 2+2 = 4 \\
 &\dots
 \end{aligned}$$

Theorem 5.36. (Absorbing Element) If $n \in \mathbb{N}_0$ then $n \cdot 0 = 0 = 0 \cdot n$

Proof.

1. $n \cdot 0 = (\alpha_n)^0(0) = \text{Id}_{\mathbb{N}_0}(0) = 0$
2. We prove by induction that $0 \cdot n = 0$, so let $S = \{n \in \mathbb{N}_0 | 0 \cdot n = 0\}$ then we have:

$$0 \in S. \text{ This follows from } 0 \cdot 0 \stackrel{(1)}{=} 0$$

$n \in S \Rightarrow s(n) \in S$. We have

$$\begin{aligned}
 0 \cdot s(n) &= (\alpha_0)^{s(n)}(0) \\
 &= (\alpha_0 \circ (\alpha_0)^n)(0) \\
 &= \alpha_0((\alpha_0)^n(0)) \\
 &= \alpha_0(0 \cdot n) \\
 &\stackrel{n \in S}{=} \alpha_0(0) \\
 &= 0 + 0 \\
 &\stackrel{[\text{theorem: 5.29}]}{=} 0
 \end{aligned}$$

proving that $s(n) \in S$.

By induction [theorem: 5.11] we have that $S = \mathbb{N}_0$ hence the theorem follows. $\square$

Theorem 5.37. (Neutral Element) *If $n \in \mathbb{N}_0$ then $n \cdot 1 = n = 1 \cdot n$*

Proof.

1.

$$\begin{aligned}
 n \cdot 1 &= (\alpha_n)^1(0) \\
 &= (\alpha_n)^{s(0)}(0) \\
 &= (\alpha_n \circ (\alpha_n)^0)(0) \\
 &= \alpha_n((\alpha_n)^0(0)) \\
 &= \alpha_n(\text{Id}(0)) \\
 &= \alpha_n(0) \\
 &= n + 0 \\
 &\stackrel{[\text{theorem: 5.29}]}{=} n
 \end{aligned}$$

2. We prove $1 \cdot n$ by induction, so let $S = \{n \in \mathbb{N}_0 \mid 1 \cdot n = n\}$ then we have:

$$0 \in S. \text{ This follows from } 1 \cdot 0 \stackrel{[\text{theorem: 5.36}]}{=} 0$$

$n \in S \Rightarrow s(n) \in S$. We have

$$\begin{aligned}
 1 \cdot s(n) &= (\alpha_1)^{s(n)}(n) \\
 &= (\alpha_1 \circ (\alpha_1)^n)(0) \\
 &= \alpha_1((\alpha_1)^n(0)) \\
 &= \alpha_1(1 \cdot n) \\
 &\stackrel{n \in S}{=} \alpha_1(n) \\
 &= 1 + n \\
 &\stackrel{[\text{theorem: 5.30}]}{=} s(n)
 \end{aligned}$$

proving that $s(n) \in S$.

By induction [theorem: 5.11] it follows that $S = \mathbb{N}_0$ completing the proof. □

Lemma 5.38. *If $n, m \in \mathbb{N}_0$ then $n \cdot s(m) = n + n \cdot m \stackrel{[\text{theorem: 5.33}]}{=} n \cdot m + n$.*

Proof. $n \cdot s(m) = (\alpha_n)^{s(m)}(0) = (\alpha_n \circ (\alpha_n)^m)(0) = \alpha_n((\alpha_n)^m(0)) = \alpha_n(n \cdot m) = n + n \cdot m$. □

Theorem 5.39. (Distributivity) $\forall n, m, k \in \mathbb{N}_0$ we have $(n + m) \cdot k = n \cdot k + m \cdot k$.

Proof. We use induction to prove this. So given $n, m \in \mathbb{N}_0$ let

$$S_{n,m} = \{k \in \mathbb{N}_0 \mid (n + m) \cdot k = n \cdot k + m \cdot k\}$$

then we have:

$$0 \in S_{n,m}. \quad (n + m) \cdot 0 \stackrel{[\text{theorem: 5.36}]}{=} 0 \stackrel{[\text{theorem: 5.29}]}{=} 0 + 0 \stackrel{[\text{theorem: 5.36}]}{=} n \cdot 0 + m \cdot 0$$

$n \in S_{n,m} \Rightarrow s(n) \in S_{n,m}$. We have

$$\begin{aligned}
 (n + m) \cdot s(k) &\stackrel{[\text{lemma: 5.38}]}{=} (n + m) \cdot k + (n + m) \\
 &\stackrel{k \in S_{n,m}}{=} (n \cdot k + m \cdot k) + (n + m) \\
 &\stackrel{[\text{theorem: 5.32}]}{=} n \cdot k + (m \cdot k + (n + m)) \\
 &\stackrel{[\text{theorem: 5.33}]}{=} n \cdot k + (m \cdot k + (m + n)) \\
 &\stackrel{[\text{theorem: 5.32}]}{=} n \cdot k + ((m \cdot k + m) + n) \\
 &\stackrel{[\text{theorem: 5.33}]}{=} n \cdot k + (n + (m \cdot k + m)) \\
 &\stackrel{[\text{theorem: 5.32}]}{=} (n \cdot k + n) + (m \cdot k + m) \\
 &\stackrel{[\text{lemma: 5.38}]}{=} n \cdot s(k) + m \cdot s(k)
 \end{aligned}$$

proving that $s(k) \in S_{n,m}$.

By induction [theorem: 5.11] it follows that $\mathbb{N}_0 = S_{n,m}$. So if $n, m, k \in \mathbb{N}_0$ then $k \in S_{n,m}$ giving $(n+m) \cdot k = n \cdot k + m \cdot k$. $\square$

Theorem 5.40. (Commutativity) *If $n, m \in \mathbb{N}_0$ then $n \cdot m = m \cdot n$.*

Proof. We prove this by induction so given $n \in \mathbb{N}_0$ let $S_n = \{m \in \mathbb{N}_0 | n \cdot m = m \cdot n\}$ then we have:

$0 \in S_n$. Using [theorem: 5.36] we have $n \cdot 0 = 0 = 0 \cdot n$ proving that $0 \in S_n$.

$m \in S_n \Rightarrow s(m) \in S_n$. We have

$$\begin{aligned} n \cdot s(m) & \stackrel{[\text{lemma: 5.38}]}{=} n + n \cdot m \\ & \stackrel{m \in S_n}{=} n + m \cdot n \\ & \stackrel{[\text{theorem: 5.37}]}{=} 1 \cdot n + m \cdot n \\ & \stackrel{[\text{theorem 5.39}]}{=} (1 + n) \cdot n \\ & \stackrel{[\text{theorem: 5.30}]}{=} s(m) \cdot n \end{aligned}$$

proving that $s(m) \in S_n$.

Using induction [theorem: 5.11] it follows that $\mathbb{N}_0 = S_n$. So if $n, m \in \mathbb{N}_0$ then $m \in S_n$ hence $n \cdot m = m \cdot n$. $\square$

Theorem 5.41. (Associativity) *If $n, m, k \in \mathbb{N}_0$ then $(n \cdot m) \cdot k = n \cdot (m \cdot k)$*

Proof. We prove this by induction. So given $n, m \in \mathbb{N}_0$ define

$$S_{n,m} = \{k \in \mathbb{N}_0 | (n \cdot m) \cdot k = n \cdot (m \cdot k)\}$$

then we have:

$$0 \in S_{n,m}. \text{ This follows from } (n \cdot m) \cdot 0 \stackrel{[\text{theorem: 5.36}]}{=} 0 \stackrel{[\text{theorem: 5.36}]}{=} n \cdot 0 \stackrel{[\text{theorem: 5.36}]}{=} n \cdot (m \cdot 0)$$

$k \in S_{n,m} \Rightarrow s(k) \in S_{n,m}$. We have

$$\begin{aligned} (n \cdot m) \cdot s(k) & \stackrel{[\text{theorem: 5.38}]}{=} (n \cdot m) \cdot k + n \cdot m \\ & \stackrel{k \in S_{n,m}}{=} n \cdot (m \cdot k) + n \cdot m \\ & \stackrel{[\text{theorem: 5.40}]}{=} (m \cdot k) \cdot n + m \cdot n \\ & \stackrel{[\text{theorem: 5.39}]}{=} ((m \cdot k) + m) \cdot n \\ & \stackrel{[\text{theorem: 5.40}]}{=} n \cdot ((m \cdot k) + m) \\ & \stackrel{[\text{theorem: 5.38}]}{=} n \cdot (m \cdot s(k)) \end{aligned}$$

proving that $s(k) \in S_{n,m}$.

Using induction we have then that $\mathbb{N}_0 = S_{n,m}$. So if $n, m, k \in \mathbb{N}_0$ we have $k \in S_{n,m}$ giving $(n \cdot m) \cdot k = n \cdot (m \cdot k)$. $\square$

To summarize the above we have the following;

Theorem 5.42. $\langle \mathbb{N}_0, \cdot \rangle$ is a Abelian semi-group with neutral element 1.

Proof.

neutral element. This follows from [theorem: 5.37]

associativity. This follows from [theorem: 5.41]

commutativity. This follows from [theorem: 5.40] $\square$

Although there is no inverse element for addition in $\mathbb{N}_0$ [this will be solved by the set of whole numbers], we can still solve equations as is expressed in the next theorem.

Theorem 5.43. *If $n, m, k \in \mathbb{N}_0$ then if $n + k = m + k$ it follows that $n = m$*

Proof. We prove this by induction. So given $n, m \in \mathbb{N}_0$ define $S = \{k \in \mathbb{N}_0 \mid \forall n, m \in \mathbb{N}_0 \text{ with } n + k = m + k \text{ we have } n = m\}$ then we have:

$0 \in S$. If $n, m \in \mathbb{N}_0$ are such that $n + 0 = m + 0$ then we have $n \stackrel{5.29}{=} n + 0 = m + 0 \stackrel{5.29}{=} m$ or $n = m$ which proves that $0 \in S$

$k \in S \Rightarrow s(k) \in S$. If $n, m \in \mathbb{N}_0$ are such that $n + s(k) = m + s(k)$ then we have by [theorem: 5.30] that $n + (1 + k) = m + (1 + k)$ or using [theorem: 5.32] that $(n + 1) + k = (m + 1) + k$. As $k \in S$ it follows that $n + 1 = m + 1$ or using [theorem: 5.30] that $s(n) = s(m)$. Finally using [theorem: 5.16] we have $n = m$. So $s(k) \in S$.

Using induction we have then that $\mathbb{N}_0 = S$. So if $n, m, k \in \mathbb{N}_0$ then as $k \in S$ we have if $n + k = m + k$ that $n = m$. $\square$

Note 5.44. We do not have a equivalent theorem for the product of two natural numbers, for example $0 \cdot 0 = 1 \cdot 0$ but we don't have that $1 = 0$.

5.4 Order relation on the natural numbers

Theorem 5.45. *If we define the relation $\leq$ by*

$$\leq = \{(n, m) \in \mathbb{N}_0 \times \mathbb{N}_0 \mid n \in m \vee n = m\}$$

then

$$\langle \mathbb{N}_0, \leq \rangle \text{ is a partial ordered set}$$

Proof.

reflectivity. If $n \in \mathbb{N}_0$ then $n = n \Rightarrow n \in n \vee n = n$ so that $n \leq n$.

anti-symmetry. If $n \leq m \wedge m \leq n$ then we have

$$\begin{aligned} (n \in m \vee n = m) \wedge (m \in n \vee m = n) &\Rightarrow (n \in m \vee n = m) \wedge (m \in n \vee m = n) \\ &\Rightarrow (n \in m \wedge m \in n) \vee n = m \\ &\stackrel{[\text{theorem: 5.14}]}{\Rightarrow} (n \subseteq m \wedge m \subseteq n) \vee n = m \\ &\Rightarrow n = m \vee n = m \\ &\Rightarrow n = m \end{aligned}$$

transitivity. If $n \leq m \wedge m \leq k$ then we have the following possibilities to consider

1. $n \in m \wedge m \in k$ then by [theorem: 5.14] $n \in m \wedge m \subseteq k \Rightarrow n \in k \Rightarrow n \leq k$
2. $n \in m \wedge m = k$ then $n \in k$ so that $n \leq k$
3. $n = m \wedge m \in k$ then $n \in k$ so that $n \leq k$
4. $n = m \wedge m = k$ then $n = k \Rightarrow n \leq k$

So in all cases we have $n \leq k$ proving transitivity. $\square$

Theorem 5.46. $\forall n \in \mathbb{N}_0$ we have $0 \leq n$

Proof. We prove this by induction, so let $S = \{n \in \mathbb{N}_0 \mid 0 \leq n\}$ then we have:

$0 \in S$. $0 = 0$ so that $0 \leq 0$ proving that $0 \in S$.

$n \in S \Rightarrow s(n) \in S$. As $s(n) = n \cup \{n\}$ we have that $n \in s(n)$ so that $n \leq s(n)$, as $n \in S$ $0 \leq n$, so by transitivity we have that $0 \leq s(n)$. Hence we have $s(n) \in S$.

Using induction [theorem: 5.11] it follows that $\mathbb{N}_0 = S$ proving the theorem. $\square$

Theorem 5.47. $\forall n \in \mathbb{N}_0$ we have $n < s(n)$ [in other words using [theorem: 5.30] we have $n < n+1$]

Proof. From $n \in n \cup \{n\} = s(n)$ we have that $n \leq s(n)$ and by [theorem: 5.15] $n \neq s(n)$ so that $n < s(n)$. $\square$

Theorem 5.48. If $n \in \mathbb{N}_0$ then $k \in n \Leftrightarrow k < n$.

Proof.

$\Rightarrow$. We proceed by induction, so let $S = \{n \in \mathbb{N}_0 \mid \text{If } k \in n \Rightarrow k < n\}$ then we have:

$0 \in S$. As $0 = \emptyset$ so that $k \in 0$ is never true hence $k \in 0 \Rightarrow k < 0$ is true, proving that $0 \in S$.

$n \in S \Rightarrow s(n) \in S$. If $k \in s(n) = n \cup \{n\}$ then we have the following cases to consider:

$k \in n$. As $n \in S$ we have $k < n$, further from [theorem: 5.47] we have $n < s(n)$ so that $k < s(n)$.

$k = n$. By [theorem: 5.47] we have $n < s(n)$ so that $k < s(n)$.

So in all cases we have $k < s(n)$ proving that $s(n) \in S$.

By the induction [theorem: 5.11] it follows that $\mathbb{N}_0 = S$, proving the theorem.

$\Leftarrow$. If $k < n$ then $k \neq n$ and $k \leq n \Rightarrow k \in n \vee k = n$ so that $k \in n$. $\square$

Theorem 5.49. If $n, m \in \mathbb{N}_0$ then we have that

1. $n < 0$ is false.
2. If $n \leq 0$ then $n = 0$.
3. $n < m \wedge m < n$ is false.
4. $n \leq m \wedge m < n$ is false.
5. $n < m \wedge m \leq n$ is false.

Proof.

1. If $n < 0$ then by [theorem: 5.48] we have $n \in 0 = \emptyset$ which is false.

2. If $n \leq 0$ then we have either $n < 0$ [which by (1) is false] or $n = 0$.

3. If $n < m \wedge m < n$ then $n \leq m \wedge m \leq n \Rightarrow n = m$ and $n \neq m$ which is a contradiction.

4. If $n \leq m \wedge m < n$ then $n \leq m \wedge m \leq n \Rightarrow n = m$ and $n \neq m$ which is a contradiction.

5. If $n < m \wedge m \leq n$ then $n \leq m \wedge m \leq n \Rightarrow n = m$ and $n \neq m$ which is a contradiction. $\square$

Theorem 5.50. $\forall n, m \in \mathbb{N}_0$ with $n < m$ we have $s(n) \leq m$ [in other words using [theorem: 5.30] $n < m$ implies $n+1 \leq m$].

Proof. We proof this by induction, so given $n \in \mathbb{N}_0$, define $S_n = \{m \in \mathbb{N}_0 \mid n < m \Rightarrow s(n) \leq m\}$ then we have:

$0 \in S_n$. By [theorem: 5.49] $n < 0$ is false, so $n < 0 \Rightarrow s(n) \leq m$ is true, proving that $0 \in S_n$.

$m \in S_n \rightarrow s(m) \in S_n$. Let $n < s(m)$ then we have $n \neq s(m)$ and $n \leq s(m)$ so that $n \in s(m) = m \cup \{m\}$, hence we have to look at:

$n \in m$. By [theorem: 5.48] we have $n < m$, as $m \in S_n$ we have $s(n) \leq m$, as by [theorem: 5.47] $m < s(m)$ it follows by transitivity that $s(n) \leq s(m)$ [actually even $s(n) < s(m)$].

$n = m$. Then $s(n) = s(m)$ so that $s(n) \leq s(m)$.

So we have $s(m) \in S_n$

Using induction [theorem: 5.11] it follows that $\forall n, m \in \mathbb{N}_0$ with $n < m$ we have $s(m) \in S_n$ such that $s(n) \leq m$. $\square$

Theorem 5.51. $\langle \mathbb{N}_0, \leq \rangle$ is a well ordered set.

Proof. We prove this by contradiction. Assume that there exist a A such that $\emptyset \neq A \subseteq \mathbb{N}_0$ with no least element. Define then

$$S_A = \{n \in \mathbb{N}_0 \mid \forall m \in A \text{ we have } n \leq m\}$$

then as A has no least element we must have that $S_A \cap A = \emptyset$ [for if $l \in S_A \cap A$ then $l \in A$ and $\forall m \in A$ we have $l \leq m$ so that l is a least element of A]. For S_A we have

$0 \in S_A$. If $m \in A$ we have by [theorem: 5.46] that $0 \leq m$ so that $0 \in S_A$.

$n \in S_A \Rightarrow s(n) \in S_A$. As $n \in S_A$ we have $\forall m \in A$ that $n \leq m$, $S_A \cap A = \emptyset$ so we have $n \neq m$ so that $n < m$, using then [theorem: 5.50] proves $s(n) \leq m$. Hence $s(n) \in S_A$

Using mathematical induction we have $S_A = \mathbb{N}_0$, so that $S_A \cap A = \mathbb{N}_0 \cap A = A \neq \emptyset$ contradicting $S_A \cap A = \emptyset$. As the assumption gives a contradiction every non empty subset of $\mathbb{N}_0$ has a least element and $\langle \mathbb{N}_0, \leq \rangle$ must be well ordered. $\square$

As a consequence of the above we have:

Corollary 5.52. $\langle \mathbb{N}_0, \leq \rangle$ is totally ordered and conditional complete.

Proof. As $\langle \mathbb{N}_0, \leq \rangle$ is well ordered by [theorem: 5.51] we have by [theorem: 3.87] that $\langle \mathbb{N}_0, \leq \rangle$ is totally ordered and conditional complete. $\square$

Corollary 5.53. If $x, y \in \mathbb{N}_0$ then we have either $x \leq y$ or $y < x$

Proof. As $\langle \mathbb{N}_0, \leq \rangle$ is well ordered the corollary follows from [theorem: 3.87]. $\square$

Theorem 5.54. $\forall n, m \in \mathbb{N}$ then $n < m \Leftrightarrow s(n) < s(m)$

Proof.

$\Rightarrow$. From [theorem: 5.50] we have $s(n) \leq m$, as by [theorem: 5.47] $m < s(m)$ it follows that $s(n) < s(m)$.

$\Leftarrow$. Assume that $m \leq n$ then by [theorem: 5.47] we have $n < s(n)$ so that $n < s(m)$, using [theorem: 5.50] we have $s(n) \leq s(m)$, combining this with $s(n) < s(m) \Rightarrow s(n) \neq s(m) \wedge s(n) \leq s(m)$ gives the contradiction $s(n) = s(m) \wedge s(n) \neq s(m)$, so we have

$$\neg(m \leq n)$$

Using [corollary: 5.53] we have $m \leq n$ or $n < m$ so that we must have

$$n < m$$

$\square$

Theorem 5.55. If $n, m, k \in \mathbb{N}_0$ then we have

$$n < m \Leftrightarrow n + k < m + k$$

which, using [theorem: 5.43], implies that

$$n \leq m \Leftrightarrow n + k \leq m + k$$

Proof. We use induction, so let $S = \{k \in \mathbb{N}_0 \mid \text{If } m, n \in \mathbb{N}_0 \text{ then } n < m \Leftrightarrow n + k < m + k\}$ then we have:

$0 \in S$. If $k = 0$ then for $n, m \in \mathbb{N}_0$ we have, as by [theorem: 5.29] $n = n + 0 \wedge m = m + 0$ that $n < m \Leftrightarrow n + 0 < m + 0$. So $0 \in S$.

$k \in S \Rightarrow s(k) \in S$. then we have

$$\begin{aligned} n < m & \xLeftrightarrow[k \in S] n + k < m + k \\ & \xLeftrightarrow[\text{theorem: 5.50}] s(n + k) < s(m + k) \\ & \xLeftrightarrow[\text{theorem: 5.31}] n + s(k) < m + s(k) \end{aligned}$$

proving that $s(k) \in S$

Induction [theorem: 5.11] proves then $\mathbb{N}_0 = S$ completing the proof. $\square$

Corollary 5.56. *If $n \in \mathbb{N}_0$ then we have:*

1. *If $k \in \mathbb{N}_0 \setminus \{0\}$ then $n < n + k$*
2. *If $k \in \mathbb{N}_0$ then $n \leq n + k$*

Proof.

1. If $k \neq 0$ then $0 < k$ so that by the above theorem [theorem: 5.55] we have

$$n \underset{[\text{theorem: 5.29}]}{=} 0 + n < n + k$$

2. As $0 \leq 0$ it follows from the above theorem [theorem: 5.55] we have that

$$n \underset{[\text{theorem: 5.29}]}{=} 0 + n \leq n + k \quad \square$$

Theorem 5.57. *If $n, k \in \mathbb{N}_0$ then $n + k = 0$ implies $n = k = 0$.*

Proof. Suppose that $k \neq 0$ then as $0 \leq n \underset{[\text{theorem: 5.56}]}{\Rightarrow} 0 \leq n < n + k = 0$ so that $0 \neq 0$ a contradiction, so $k = 0$. But then $n = n + 0 = n + k = 0$. $\square$

Theorem 5.58. *If $n, m \in \mathbb{N}_0$ with $n < s(m)$ then $n \leq m$.*

Note 5.59. As by [theorem: 5.30] $s(m) = m + 1$ this is equivalent with $n < m + 1 \Rightarrow n \leq m$

Proof. Using [corollary: 5.53] we have that either $n \leq m$ or $m < n$. If $m < n$ then by [theorem: 5.50] $s(m) \leq n$, which combined with the hypothesis $n < s(m)$ gives the contradiction $n < m$. Hence we must have $n \leq m$. $\square$

Theorem 5.60. *If $n, m \in \mathbb{N}_0$ with $n < m$ then $\exists! k \in \mathbb{N}_0 \setminus \{0\}$ such that $m = n + k$.*

Proof. First we prove existence by induction, so let

$$S_n = \{m \in \mathbb{N}_0 \mid \text{If } n < m \text{ then there exist a } k \in \mathbb{N}_0 \text{ such that } k \neq 0 \text{ and } m = n + k\}$$

then we have:

$0 \in S_n$. As $n < 0$ is false by [theorem: 5.49], the condition is satisfied vacuously, proving that $0 \in S_n$.

$m \in S_n \Rightarrow s(m) \in S_n$. If $n < s(m)$ then we have by [theorem: 5.58] that $n \leq m$ so that we have the following possibilities to consider:

$n = m$. Then $n + 1 \underset{[\text{theorem: 5.30}]}{=} s(n) = s(m)$, as $1 = s(0) \neq 0$ we have if we take $k = 1$ that $k \neq 0$ and $n + k = s(m)$, proving that $s(m) \in S_n$

$n < m$. Then as $m \in S_n$ there exist a $l \in \mathbb{N}_0$ such that $l \neq 0$ and $n + l = m$. Now

$$s(m) = s(n + l) \underset{[\text{theorem: 5.31}]}{=} n + s(l)$$

Take $k = s(l)$ then $n + k = s(m)$, further by [theorems: 5.46, 5.47] we have $0 \leq l \wedge l < s(l) = k$ so that $0 < k$ hence $k \neq 0$. This proves that in this case we also have $s(m) \in S_n$.

Induction [see theorem: 5.11] proves then that $\mathbb{N}_0 = S_n$. Hence if $n, m \in \mathbb{N}_0$ we have $m \in S_n$ so that if $n < m$ there exist a $k \in \mathbb{N}_0$ such that $k \neq 0$ and $m = n + k$.

Now for uniqueness assume that $n < m$ and there exists $k, l \in \mathbb{N}_0$ such that

$$k + n \underset{[\text{theorem: 5.33}]}{=} n + k = m = n + l \underset{[\text{theorem: 5.33}]}{=} l + n$$

then by [theorem: 5.43] $k = l$. □

Corollary 5.61. *If $n, m \in \mathbb{N}_0$ then $n < m \Leftrightarrow \exists! k \in \mathbb{N}_0 \setminus \{0\}$ such that $n + k = m$*

Proof.

$\Rightarrow$. This follows from the previous theorem [theorem: 5.60].

$\Leftarrow$. Let $k \in \mathbb{N}_0 \setminus \{0\}$ such that $n + k = m$. As $k \in \mathbb{N}_0 \setminus \{0\}$ we have $0 < k$ so that by [theorem: 5.55] $0 + n < k + n \underset{[\text{theorems: 5.29, 5.33}]}{\Rightarrow} n < n + k = m$. □

Corollary 5.62. *If $n, m \in \mathbb{N}_0$ then $n \leq m \Leftrightarrow \exists! k \in \mathbb{N}_0$ such that $m = n + k$*

Proof.

$\Rightarrow$. If $n \leq m$ then we have either:

$n = m$. Then $m \underset{[\text{theorem: 5.29}]}{=} n + 0$ where $0 \in \mathbb{N}_0$.

$n < m$. Then by the previous corollary [corollary: 5.61] there exists a $k \in \mathbb{N}_0 \setminus \{0\} \subseteq \mathbb{N}_0$ such that $m = n + k$.

proving existence. For uniqueness assume that $n + k = m = n + l$ then

$$k + n \underset{[\text{theorem: 5.33}]}{=} n + k = m = n + l \underset{[\text{theorem: 5.33}]}{=} l + n$$

proving by [theorem: 5.43] that $k = l$.

$\Leftarrow$. As $k \in \mathbb{N}_0$ we have either:

$k = 0$. Then $m = n + 0 \underset{[\text{theorem: 5.29}]}{=} n$ so that $n \leq m$.

$0 < k$. Then by the previous corollary [corollary: 5.61] we have $n < m$ so that $n \leq m$. □

The above corollary ensures that the following definition is well defined.

Definition 5.63. *If $n, m \in \mathbb{N}_0$ with $n \leq m$ then the **unique** $k \in \mathbb{N}_0$ such that $m = n + k$ is noted as $m - n$. So we have that $n + (m - n) \underset{[\text{theorem: 5.33}]}{=} (m - n) + n = m$ and using [theorem: 5.29] that $n - n = 0$.*

Note 5.64. The condition $n \leq m$ is essential for the existence of $n - m$ as this is needed for [corollary: 5.62]. Later when we define the set $\mathbb{Z}$ of integers we will relax this condition.

Theorem 5.65. *If $n, m, k \in \mathbb{N}_0$ is such that $n \leq k$ then*

$$(k + m) - n = (k - n) + m = (m + k) - n$$

Proof. As $n \leq k$ we have by [theorem: 5.56] $n \leq k + m$ so that $(k + m) - n$ and $k - n$ are well defined. Now

$$\begin{aligned} ((k - n) + m) + n &\underset{[\text{theorem: 5.32}]}{=} (k - n) + (m + n) \\ &\underset{[\text{theorem: 5.33}]}{=} (k - n) + (n + m) \\ &\underset{[\text{theorem: 5.33}]}{=} ((k - n) + n) + m \\ &\underset{\text{definition}}{=} k + m \end{aligned}$$

So we have that

$$(k + m) - n = (k - n) + m$$

Further using commutativity [theorem: 5.33] we have that $(m + k) - n = (k + m) - n$ so that

$$(m + k) - n = (k - n) + m$$

□

Theorem 5.66. *If $n, k \in \mathbb{N}_0$ then $(n + k) - n = k = (k + n) - n$*

Proof. As $n \leq n$ we can use the previous theorem [see theorem: 5.65] so that

$$(k + n) - n = (n + k) - n = (n - n) + k = 0 + k = k$$

□

Theorem 5.67. *Let $n \in \mathbb{N}_0$ then we have Let $n, m \in \mathbb{N}_0$ such that $n < m$ then $n \leq m - 1$*

Proof. As $n < m$ we have by [theorem: 5.60] a $k \in \mathbb{N}_0 \setminus \{0\}$ such that $m = n + k$. As $0 \neq k$ we have by [theorem: 5.18] that there exist a $l \in \mathbb{N}_0$ such that $k = s(l) = l + 1$, so $m = (n + l) + 1$ which by [definition 5.63] means that $n + l = m - 1$. Further by [theorem: 5.56] we have $n \leq n + l$ so that $n \leq m - 1$. □

Corollary 5.68. *Let $n \in \mathbb{N}_0$ and $m \in \mathbb{N} = \mathbb{N}_0 \setminus \{0\}$ then $n < m \Leftrightarrow n \leq m - 1$*

Proof.

$\Rightarrow$. This follows from the previous theorem [theorem: 5.67]

$\Leftarrow$. By [theorem: 5.47] we have $(m - 1) < (m - 1) + 1 = m$ we have from $n \leq m - 1$ that $n < m$. □

Theorem 5.69. *Let $n \in \mathbb{N}_0$ and $m \in \mathbb{N}_0 \setminus \{0\}$ then $(m - 1) \cdot n = n \cdot (m - 1) = n \cdot m - n$*

Proof. As $0 < m$ we have by [theorem: 5.50] that $1 = s(0) \leq m$ so that $m - 1$ is well defined. Now

$$n + (m - 1) \cdot n \underset{\text{commutativity}}{=} (m - 1) \cdot n + n = (m - 1) \cdot n + 1 \cdot n = ((m - 1) + 1) \cdot n = m \cdot n = n \cdot m$$

so that $(m - 1) \cdot n = n \cdot m - n$ and by commutativity [see theorem: 5.33] $n \cdot (m - 1) = n \cdot m - n$ □

Theorem 5.70. *If $n, m, i \in \mathbb{N}_0$ then*

1. *If $n \leq i < m$ then $0 \leq i - n < m - n$*
2. *If $n \leq i \leq m$ then $0 \leq i - n \leq m - n$*

Proof.

1. As $n \leq i < m$ we have $n < m$. From [corollary: 5.53] it follows that $0 \leq i - n \vee i - n < 0$ and $i - n < m - n \vee m - n \leq i - n$. Now by [theorem: 5.49] we have that $i - n < 0$ is false so we must have that $0 \leq i - n$. If $m - n \leq i - n$ then by [theorem: 5.55] $m = (m - n) + n \leq (i - n) + n = n$ proving that $m \leq n$ which by [theorem: 5.49] contradicts with $n < m$, so we must have $i - n < m - n$.
2. As $n \leq i \leq m$ we have $n \leq m$. From [corollary: 5.53] it follows that $0 \leq i - n \vee i - n < 0$ and $i - n \leq m - n \vee m - n < i - n$. Now by [theorem: 5.49] we have that $i - n < 0$ is false so we must have that $0 \leq i - n$. If $m - n < i - n$ then by [theorem: 5.55] $m = (m - n) + n < (i - n) + n = n$ proving that $m < n$ which by [theorem: 5.49] contradicts with $n \leq m$, so we must have $i - n \leq m - n$. □

Theorem 5.71. *If $k, n, m \in \mathbb{N}_0$ such that $k \leq n \wedge k \leq m$ then we have*

$$n \leq m \Leftrightarrow n - k \leq m - k$$

Proof.

$\Rightarrow$. Using [theorem: 5.53] we have either $m - k < n - k$ or $n - k \leq m - k$, if $m - k < n - k$ we have by [theorem: 5.55] that $(m - k) + k < (n - k) + k$ so that $m < n$ which as $n \leq m$ gives the contradiction $m < m$, so we have $n - k \leq m - k$.

$\Leftarrow$. Using [theorem: 5.55] we have that $(n - k) + k \leq (m - k) + k$ so that $n \leq m$. □

Theorem 5.72. *If $n \in \mathbb{N}_0$ then there does not exist a $k \in \mathbb{N}_0$ such that $n < k < s(n)$*

Proof. Assume that $\exists k \in \mathbb{N}_0$ such that $n < k < s(n)$. As $n < k$ we have by [theorem: 5.50] that $s(n) \leq k$ which combined with $k < s(n)$ gives $s(n) < s(n)$ a contradiction. $\square$

Theorem 5.73. *If $\emptyset \neq A \subseteq \mathbb{N}_0$ is a set such that $\sup(A)$ exist then $\sup(A) \in A$*

Proof. We have the following cases for $\sup(A)$ to consider:

$\sup(A) = 0$. As $A \neq \emptyset$ there exist a $x \in A$, further as the $\sup(A)$ is a upper bound of A we have that $x \leq 0$, which by [theorem: 5.49] proves that $x = 0 = \sup(A)$, giving that $\sup(A) = x \in A$.

$\sup(A) \neq 0$. Using [theorem: 5.18] there exist a $k \in \mathbb{N}_0$ such that $s(k) = \sup(A)$. As $(\mathbb{N}_0, \leq)$ is totally ordered [see theorem: 5.52] and $k < s(k) = \sup(A)$, it follows from the properties of the supremum [theorem: 3.74] that there exist a $a \in A$ such that $k < a \leq \sup(A) = s(k)$. As we can not have $k < a < s(k)$ [see theorem: 5.72], it follows that $a = \sup(A)$ so that $\sup(A) \in A$. $\square$

Theorem 5.74. *If $n, m, r, s \in \mathbb{N}_0$ then*

1. *If $n < m \wedge r < s$ then $n + r < m + s$*
2. *If $n \leq m \wedge r \leq s$ then $n + r \leq m + s$*
3. *If $n < m \wedge r \leq s$ then $n + r < m + s$*
4. *If $n \leq m \wedge r < s$ then $n + m < m + r$*

Proof.

1. Using [theorem: 5.55] to follows that $n + r < m + r$ and $r + m < s + m \xRightarrow{[\text{theorem: 5.33}]} n + r < m + s$ proving, using transitivity, that $n + r < m + s$.
2. Using [theorem: 5.55] to follows that $n + r \leq m + r$ and $r + m \leq s + m \xRightarrow{[\text{theorem: 5.33}]} n + r \leq m + s$ proving, using transitivity, that $n + r \leq m + s$.
3. Using [theorem: 5.55] to follows that $n + r \leq m + r$ and $r + m < s + m \xRightarrow{[\text{theorem: 5.33}]} n + r < m + s$ proving, using transitivity, that $n + r < m + s$.
4. Using [theorem: 5.55] to follows that $n + r < m + r$ and $r + m \leq s + m \xRightarrow{[\text{theorem: 5.33}]} n + r < m + s$ proving, using transitivity, that $n + r < m + s$. $\square$

Theorem 5.75. *Let $n, m \in \mathbb{N}_0 \setminus \{0\}$ then $n \cdot m \in \mathbb{N}_0 \setminus \{0\}$.*

Proof. As $m \neq 0$ it follows from [theorem: 5.18] that $\exists k \in \mathbb{N}_0$ such that $m = s(k)$. So $n \cdot m = n \cdot s(k) \xRightarrow{[\text{theorem: 5.38}]} n + n \cdot k$. Further as $n \neq 0$ we have that $0 < n$, so that by [theorem: 5.55] $n \xRightarrow{[\text{theorem: 5.29}]} n + 0 \leq n + n \cdot k = n \cdot m$, using transitivity gives then finally $0 < n \cdot m$. $\square$

Theorem 5.76. *If $n, m \in \mathbb{N}_0$ such that $n < m$ then*

1. *If $k \in \mathbb{N}_0 \setminus \{0\}$ then $n \cdot k < m \cdot k$*
2. *If $k \in \mathbb{N}_0$ then $n \cdot k \leq m \cdot k$*

Proof.

1. As $n < m$ we have by [theorem: 5.60] that there exist a $l \in \mathbb{N}_0 \setminus \{0\}$ such that $m = n + l$. So

$$m \cdot k = (n + l) \cdot k \xRightarrow{[\text{theorem: 5.39}]} n \cdot k + l \cdot k.$$

As $l, k \in \mathbb{N}_0 \setminus \{0\}$ we have by [theorem: 5.75] that $l \cdot k \neq 0$ so that $0 < l \cdot k$, hence using [theorem: 5.55] we have that

$$n \cdot k \xRightarrow{[\text{theorem: 5.29}]} 0 + n \cdot k < l \cdot k + n \cdot k \xRightarrow{[\text{theorem: 5.33}]} n \cdot k + l \cdot k = m \cdot k$$

so that

$$n \cdot k < m \cdot k$$

2. If $k \in \mathbb{N}_0$ then we have either:

$k = 0$. Then by [theorem: 5.36] we have $n \cdot k = 0 = m \cdot k$ so that $n \cdot k \leq m \cdot k$.

$k \neq 0$. Then by (1) $n \cdot k < m \cdot k \Rightarrow n \cdot k \leq m \cdot k$. $\square$

Theorem 5.77. *If $n, m \in \mathbb{N}_0$ such that $\exists k \in \mathbb{N}_0 \setminus \{n\}$ such that $n \cdot k = m \cdot k$ then $n = m$.*

Proof. Using [corollary: 5.53] we have that $n < m$, $m < n$ or $n = m$. If $n < m$ then by [theorem: 5.76] $n \cdot k < m \cdot k$ contradicting $n \cdot k = m \cdot k$, likewise if $m < n$ then by [theorem: 5.76] $m \cdot k < n \cdot k$ contradicting $n \cdot k = m \cdot k$. So we must have $n = m$. $\square$

Theorem 5.78. (Archimedean Property) *If $x, y \in \mathbb{N}_0$ and $x \neq 0$ then there exists a $z \in \mathbb{N}_0 \setminus \{0\}$ such that $y < z \cdot x$*

Proof. For y we have two possibilities:

$y = 0$. As $x \neq 0$ we have $y = 0 < x \stackrel{[\text{theorem: 5.37}]}{=} 1 \cdot x$, so using $z = 1$ proves the theorem.

$y \neq 0$. Using [corollary: 5.53] we have for $x, y \in \mathbb{N}_0$ either:

$y \leq x$. Then as $1 < s(1) = 2$ [see theorem: 5.47] we have $x \stackrel{[\text{theorem: 5.37}]}{=} 1 \cdot x < 2 \cdot x$ [see: theorem: 5.76], hence $y < 2 \cdot x$, so using $z = 2$ proves the theorem.

$x < y$. Using [theorem: 5.60] there exist $k \in \mathbb{N}_0 \setminus \{0\}$ such that

$$y = x + k \quad (5.14)$$

As $0 < x$ we have by [theorem: 5.50] $1 = s(0) \leq x$ so that by multiplication with k we have [see theorem: 5.76] that

$$k = 1 \cdot k \leq x \cdot k \quad (5.15)$$

As $0 \neq k < s(k)$ and $x \neq 0$ we have by [see theorem: 5.76] that $k \cdot x < s(k) \cdot x \Rightarrow x \cdot k < x \cdot s(k)$ combining this with [eq: 5.15] gives that

$$k < x \cdot s(k) \quad (5.16)$$

Using [theorem: 5.55] we have

$$x + k = k + x < s(k) \cdot x = x + x \cdot s(k) = x \cdot 1 + x \cdot s(k) \stackrel{\text{distributivity}}{=} x \cdot (1 + s(k))$$

or using [eq: 5.14] that $y < x \cdot (s + s(k))$. So if we take $z = 1 + s(k)$ we have that $y < x \cdot z$ which as also $0 < 1 < 1 + s(k)$ proves the theorem. $\square$

Theorem 5.79. (Division) *If $m \in \mathbb{N}_0$ and $n \in \mathbb{N}_0 \setminus \{0\}$ then there exists a **unique** $r \in \mathbb{N}_0$ and a unique $q \in \mathbb{N}_0$ such that*

$$m = n \cdot q + r \text{ and } 0 \leq r < n$$

Proof. First we prove existence of q and r . As $n \in \mathbb{N}_0 \setminus \{0\}$ $n \neq 0$ so that $0 < n$. For m we have the following cases to consider:

$m = 0$. In this case taking $q = 0$ and $r = 0$ gives $n \cdot 0 + 0 \stackrel{[\text{theorem: 5.36}]}{=} 0 + 0 \stackrel{[\text{theorem: 5.29}]}{=} 0 \cdot m + 0$ and $0 \leq 0 < n$, so $q = 0 = r$ satisfies $m = n \cdot q + r$ and $0 \leq r < n$.

$0 < m$. Then we have the following cases for n to consider:

$n = 1$. Take $q = m$ and $r = 0$ then $n \cdot q + r = 1 \cdot m + 0 \stackrel{[\text{theorem: 5.29, 5.37}]}{=} m$ and $0 \leq 0 < n$, so q, r satisfies $m = n \cdot q + r$ and $0 \leq r < n$.

$n \neq 1$. Then as $0 < n \stackrel{[\text{theorem: 5.50}]}{\Rightarrow} 1 = s(0) \leq n$ we have $1 < n$. By [theorem: 5.76] it follows that $m = 1 \cdot m < n \cdot m$, so if we define

$$A_{n,m} = \{x \in \mathbb{N}_0 \mid m < n \cdot x \wedge x \leq m\}$$

then $m \in A_{n,m}$ proving that

$$A_{n,m} \neq \emptyset$$

As $(\mathbb{N}_0, \leq)$ is well ordered [see theorem: 5.51] there exist a least element

$$q' = \min(A_{n,m})$$

If $q' = 0$ then as $q' \in A_{n,m}$ we would have $m < n \cdot 0 \stackrel{[\text{theorem: 5.36}]}{=} 0$ a contradiction, hence we must have that $0 < q'$. So by [theorem: 5.18] there exist a $q \in \mathbb{N}_0$ such that $s(q) = q'$. As $q < s(q) = q'$ [see theorem: 5.47] we must have that $q \notin A_{n,m}$, which, as $q < q' \leq m$, means that $n \cdot q \leq m$. From this we have by [theorem: 5.76] the existence of a $r \in \mathbb{N}_0$ such that

$$m = n \cdot q + r$$

Using [corollary: 5.53] we have either $n \leq r$ or $r < n$. If $n \leq r$ then by [theorem: 5.55] we have $n + n \cdot q \leq r + n \cdot q = n \cdot q + r = m$, hence

$$n \cdot q' = n \cdot s(q) \stackrel{[\text{theorem: 5.38}]}{=} n + n \cdot q \leq m$$

As $q' \in A_{n,m}$ we have by definition that $m < n \cdot q'$ which combined with the above yields the contradiction $m < m$. So we must have

$$0 \leq r < n$$

To summarize we have found q, r such that $m = n \cdot q + r$ and $0 \leq r < n$ proving existence.

Now to prove uniqueness. Assume that $n \cdot q + r = m = n \cdot q'' + r''$ and $0 \leq q < n, 0 \leq q'' < n$ with $q \neq q''$ then by [corollary: 5.53] we have either $q < q'', q'' < q$ or $q = q'$. For the cases $q < q''$ or $q'' < q$ we have

$q < q''$. Then by [theorem: 5.50]

$$\begin{aligned} s(q) \leq q'' &\stackrel{[\text{theorem: 5.76}]}{\Rightarrow} s(q) \cdot n \leq q'' \cdot n \\ &\stackrel{[\text{theorem 5.40}]}{\Rightarrow} n \cdot s(q) \leq q'' \cdot n \\ &\stackrel{[\text{theorem: 5.38}]}{\Rightarrow} n \cdot q + n \leq q'' \cdot n \\ &\stackrel{[\text{theorem: 5.55}]}{\Rightarrow} n \cdot q + n + r + r'' \leq q'' \cdot n + r + r'' \\ &\Rightarrow m + n + r'' \leq m + r \\ &\stackrel{[\text{theorem: 5.55}]}{\Rightarrow} n + r'' \leq r \\ &\stackrel{[\text{theorem: 5.56}]}{\Rightarrow} n \leq n + r'' \leq r \end{aligned}$$

contradicting $r < n$.

$q'' < q$. Then by [theorem: 5.50]

$$\begin{aligned} s(q'') \leq q &\stackrel{[\text{theorem: 5.76}]}{\Rightarrow} s(q'') \cdot n \leq q \cdot n \\ &\stackrel{[\text{theorem 5.40}]}{\Rightarrow} n \cdot s(q'') \leq q \cdot n \\ &\stackrel{[\text{theorem: 5.38}]}{\Rightarrow} n \cdot q'' + n \leq q \cdot n \\ &\stackrel{[\text{theorem: 5.55}]}{\Rightarrow} n \cdot q'' + n + r + r'' \leq q \cdot n + r + r'' \\ &\Rightarrow m + n + r \leq m + r'' \\ &\stackrel{[\text{theorem: 5.55}]}{\Rightarrow} n + r \leq r'' \\ &\stackrel{[\text{theorem: 5.56}]}{\Rightarrow} n \leq n + r \leq r'' \end{aligned}$$

contradicting $r < n$.

So we must have that $q = q''$ but then $r + n \cdot q = n \cdot q + r = m = n \cdot q + r'' = r'' + n \cdot q$ proving by [theorem: 5.55] that $r = r''$. $\square$

5.5 Other forms of Mathematical Induction and Recursion

In this section we rewrite the theorem of induction and recursion using $n + 1$ instead of $s(n)$ [see theorem: 5.30]. First we introduce some definitions.

Definition 5.80. Let $n \in \mathbb{N}_0$ then $\{n, \dots, \infty\}$ is defined as

$$\{n, \dots, \infty\} = \{i \in \mathbb{N}_0 \mid n \leq i\}$$

Note 5.81. $\{0, \dots, \infty\} = \{x \in \mathbb{N}_0 \mid 0 \leq x\} \stackrel{[\text{theorem: 5.46}]}{=} \mathbb{N}_0$

Definition 5.82. Let $n, m \in \mathbb{N}_0$ then $\{n, \dots, m\}$ is defined as

$$\{n, \dots, m\} = \{i \in \mathbb{N}_0 \mid n \leq i \wedge i \leq m\}$$

We have now the following variation on mathematical induction.

Theorem 5.83. (Mathematical Induction) If $n \in \mathbb{N}_0$ and $X \subseteq \{n, \dots, \infty\}$ is such that

1. $n \in X$
2. If $i \in X$ then $i + 1 \in X$

then $X = \{n, \dots, \infty\}$.

Proof. Take $S = \{i \in \mathbb{N}_0 \mid i + n \in X\}$ then we have:

$0 \in S$. As $0 + n \stackrel{[\text{theorem: 5.29}]}{=} n \in X$ we have $0 \in S$.

$i \in S \Rightarrow s(i) \in S$. As $i \in S$ we have $i + n \in X$ so that by the hypothesis $(i + n) + 1 \in X$. Now

$$\begin{aligned} (i + n) + 1 &\stackrel{[\text{theorem: 5.32}]}{=} i + (n + 1) \\ &\stackrel{[\text{theorem: 5.33}]}{=} i + (1 + n) \\ &\stackrel{[\text{theorem: 5.32}]}{=} (i + 1) + n \\ &\stackrel{[\text{theorem: 5.30}]}{=} s(i) + n \end{aligned}$$

so that $s(i) + n \in X$, proving $s(i) \in S$.

By mathematical induction we have that $S = \mathbb{N}_0$. If $i \in \{n, \dots, \infty\}$ then $n \leq i$ so by [theorem: 5.62] $\exists k \in \mathbb{N}_0$ such that $i = n + k \stackrel{[\text{theorem: 5.33}]}{=} k + n \stackrel{k \in \mathbb{N}_0 = S}{\Rightarrow} i \in X$. Hence $\{n, \dots, \infty\} \subseteq X$ which together with $X \subseteq \{1, \dots, n\}$ proves that

$$X = \{1, \dots, \infty\} \quad \square$$

For recursion we have the following theorems that follows from [theorem: 5.20], [theorem: 5.24] and [theorem: 5.26] by replacing $s(n)$ by its equivalent form $n + 1$.

Theorem 5.84. Let A be a set, $a \in A$ and $f: A \rightarrow A$ a function then there exist a **unique** function

$$\lambda: \mathbb{N}_0 \rightarrow A$$

such that:

1. $\lambda(0) = a$
2. $\forall n \in \mathbb{N}_0$ we have $\lambda(n + 1) = f(\lambda(n))$

Further if $f: A \rightarrow A$ is injective and $a \notin f(A)$ then $\lambda: \mathbb{N}_0 \rightarrow A$ is injective.

Theorem 5.85. Let A be a set, $f: A \rightarrow A$ a function then $\forall n \in \mathbb{N}_0$ there exist a function

$$(f)^n: A \rightarrow A$$

such that:

1. $(f)^0 = \text{Id}_A$
2. $(f)^{n+1} = f \circ (f)^n$

Theorem 5.86. Let A be a set, $a \in A$ and $g: \mathbb{N}_0 \times A \rightarrow A$ then there exist a **unique** function

$$\lambda: \mathbb{N}_0 \rightarrow A$$

such that:

1. $\lambda(0) = a$
2. $\forall n \in \mathbb{N}_0 \lambda(n+1) = g(n, \lambda(n))$

Corollary 5.87. Let A be a set, $a \in A$ and $g: \mathbb{N}_0 \times A \rightarrow A$ then there exist a **unique** function

$$\lambda: \mathbb{N}_0 \rightarrow A$$

such that:

1. $\lambda(0) = a$
2. $\forall n \in \{1, \dots, \infty\} \lambda(n) = g(n-1, \lambda(n-1))$

Proof. Using [theorem: 5.86] there exists a **unique** $\lambda: \mathbb{N}_0 \rightarrow A$ such that

$$\lambda(0) = a \text{ and } \forall n \in \mathbb{N}_0 \lambda(n+1) = g(n, \lambda(n)) \quad (5.17)$$

Let $n \in \{1, \dots, \infty\}$ then $1 \leq n$ so by [definition: 5.63] we have that $n-1 \in \mathbb{N}_0$ such that $n = (n-1) + 1$, hence $\lambda(n) = \lambda((n-1) + 1) = g(n-1, \lambda(n-1))$. $\square$

Theorem 5.88. Let A be a set, $a \in A$, $n \in \mathbb{N}_0$ and $g: \{0, \dots, n-1\} \times A \rightarrow A$ a function then there exists a **unique** function $\lambda: \{0, \dots, n\} \rightarrow A$ satisfying

$$\begin{aligned} \lambda(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \lambda(i+1) &= g(i, \lambda(i)) \end{aligned}$$

Proof. Define

$$g': \mathbb{N}_0 \times A \rightarrow A \text{ by } g'(i, x) = \begin{cases} g(i, x) & \text{if } i \in \{0, \dots, n-1\} \\ x & \text{if } i \in \{n, \dots, \infty\} \end{cases}$$

then by [corollary: 5.87] there exists a $\beta: \mathbb{N}_0 \rightarrow A$ such that

$$\beta(0) = a \quad (5.18)$$

$$\forall i \in \mathbb{N}_0 \text{ we have } \beta(i+1) = g'(i, \beta(i)) \quad (5.19)$$

Define now $\lambda: \{0, \dots, n\} \rightarrow A$ by $\lambda = \beta|_{\{0, \dots, n\}}$ then we have

$$\lambda(0) = \beta|_{\{0, \dots, n\}}(0) \underset{0 \in \{0, \dots, n\}}{=} \beta(0) \underset{[\text{theorem: 5.18}]}{=} a$$

and $\forall i \in \{0, \dots, n-1\}$ we have

$$\begin{aligned} \lambda(i+1) &= \beta|_{\{0, \dots, n\}}(i+1) \\ &\underset{i+1 \in \{0, \dots, n\}}{=} \beta(i+1) \\ &\underset{[\text{theorem: 5.19}]}{=} g'(i, \beta(i)) \\ &\underset{i \in \{0, \dots, n-1\}}{=} g'(i, \lambda(i)) \\ &\underset{i \in \{0, \dots, n-1\}}{=} g(i, \lambda(i)) \end{aligned}$$

so that we found a function $\lambda: \{0, \dots, n\} \rightarrow A$ such that

$$\begin{aligned} \lambda(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \lambda(i+1) &= g(i, \lambda(i)) \end{aligned}$$

Next we must prove uniqueness so let $\gamma: \{0, \dots, n\} \rightarrow A$ be such that

$$\begin{aligned}\gamma(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \gamma(i+1) &= g(i, \gamma(i))\end{aligned}$$

and define $S = \{i \in \mathbb{N}_0 \mid i \notin \{0, \dots, n\} \vee \lambda(i) = \gamma(i)\}$ then we have:

$0 \in S$. As $\lambda(0) = a = \gamma(0)$ we have $0 \in S$

$i \in S \Rightarrow i+1 \in S$. then for $i+1$ we have either:

$i+1 \in \{0, \dots, n\}$. Then $i+1 \leq n$ so that $i < n$ and as $i \in S$ we have $0 \leq i$, so it follows that $i \in \{0, \dots, m\}$. Further $\lambda(i+1) = g(i, \lambda(i)) \stackrel{i \in \{0, \dots, m\}}{=} g(i, \gamma(i)) = \gamma(i+1)$ proving that $i+1 \in S$

$i+1 \notin \{0, \dots, n\}$. Then $i+1 \in S$

so in all cases we have $i+1 \in S$.

By mathematical induction [theorem: 5.83] we have that $S = \mathbb{N}_0$. If $i \in \{0, \dots, n\} \subseteq \mathbb{N}_0$ we have $i \in S$ which as $i \in \{0, \dots, n\}$ gives $\lambda(i) = \gamma(i)$ so that $\lambda = \gamma$. $\square$

In the above the function $\lambda: \mathbb{N}_0 \rightarrow A$ is specified by saying what $a \in A$ is and what the function $g: \mathbb{N}_0 \times A \rightarrow A$ is. There exist a more intuitive way of specifying these requirement as is expressed in the following definitions.

Definition 5.89. Let A be a set, $a \in A$ then we can define a function as follows:

$$f: \mathbb{N}_0 \rightarrow A$$

is defined by:

1. $f(0) = a$
2. $f(n+1) = G(n, \lambda(n))$

where $G(n, \lambda(n))$ is a expression of two parameters. The above is equivalent with the function defined by [theorem: 5.86] where $a \in A$ and $g: \mathbb{N}_0 \times A \rightarrow A$ is defined by $g(n, x) = G(n, x)$.

Another way to define a recursive function is based on [corollary: 5.87]

Definition 5.90. Let A be a set, $a \in A$ then we define $f: \mathbb{N}_0 \rightarrow A$ as follows

$$f(n) = \begin{cases} a & \text{if } n=0 \\ G(n-1, f(n-1)) & \text{if } n \in \{1, \dots, \infty\} \end{cases}$$

Which is equivalent with the function defined by [theorem: 5.87] where $a \in A$ and $g: \mathbb{N}_0 \times A \rightarrow A$ is defined by $g(n, x) = G(n, x)$.

We can use the above to define functions by recursion.

Definition 5.91. Let A be a set, $a \in A$, $n \in \mathbb{N}_0$ then we define the function

$$\lambda: \{0, \dots, n\} \rightarrow A$$

by

$$\begin{aligned}\lambda(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \lambda(i+1) &= G(i, \lambda(i))\end{aligned}$$

where $G(n, \lambda(n))$ is a expression of two parameters. The above is equivalent with the function defined by [theorem: 5.88] where $a \in A$ and $g: \{0, \dots, n-1\} \times A \rightarrow A$ is defined by $g(n, x) = G(n, x)$.

Example 5.92. (Faculty) $\text{fac}: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is defined by

$$\text{fac}(n) = \begin{cases} 1 & \text{if } n=0 \\ n \cdot \text{fac}(n-1) & \text{if } n \in \{1, \dots, \infty\} \end{cases} = ((n-1)+1) \cdot \text{fac}(n-1)$$

this is the function defined by [corollary: 5.87] where $a = 1$ and $g: \mathbb{N}_0 \times A \rightarrow A$ is define by $g(n, x) = (n + 1) \cdot x$ then we have

$$\begin{aligned}
 \text{fac}(0) &= 1 \\
 \text{fac}(1) &= g(0, \text{fac}(0)) = (0 + 1) \cdot \text{fac}(0) = 1 \cdot \text{fac}(0) = 1 \cdot 1 = 1 \\
 \text{fac}(2) &= g(1, \text{fac}(1)) = (1 + 1) \cdot \text{fac}(1) = 2 \cdot \text{fac}(1) = 2 \cdot 1 = 2 \\
 \text{fac}(3) &= g(2, \text{fac}(2)) = (2 + 1) \cdot \text{fac}(2) = 3 \cdot \text{fac}(2) = 3 \cdot 2 = 6 \\
 &\dots \\
 \text{fac}(n) &= g(n - 1, \text{fac}(n - 1)) = ((n - 1) + 1) \cdot \text{fac}(n - 1) = n \cdot \text{fac}(n - 1) \\
 &\dots
 \end{aligned}$$

or in other words without using g

$$\begin{aligned}
 \text{fac}(0) &= 1 \\
 \text{fac}(1) &= 1 \cdot \text{fac}(0) = 1 \cdot 1 = 1 \\
 \text{fac}(2) &= 2 \cdot \text{fac}(1) = 2 \cdot 1 = 2 \\
 \text{fac}(3) &= 3 \cdot \text{fac}(2) = 3 \cdot 2 = 6 \\
 &\dots \\
 \text{fac}(n) &= n \cdot \text{fac}(n - 1) \\
 &\dots
 \end{aligned}$$

which is exactly what we mean by the definition

$$\text{fac}(n) = \begin{cases} 1 & \text{if } n=0 \\ n \cdot \text{fac}(n - 1) & \text{if } n \in \{1, \dots, \infty\} \end{cases}$$

Chapter 6

Finite and Infinite Sets

6.1 Equipotence

First we define the concept of equipotency which allows us to state that two sets have the same size without actually counting the number of elements. The latter will turn out to be impossible for every set.

Definition 6.1. Two sets A and B are **equipotent** if there exist a bijection $f: A \rightarrow B$. We note this as $A \approx B$.

Theorem 6.2. Let A, B, C be sets then

1. $A \approx A$
2. If $A \approx B$ then $B \approx A$
3. If $A \approx B \wedge B \approx C$ then $A \approx C$

Proof.

1. $\text{Id}: A \rightarrow A$ is a bijection [see example: 2.66] proving that $A \approx A$
2. As $A \approx B$ there exist a bijection $f: A \rightarrow B$ but then by [theorem: 2.74] $f^{-1}|B \rightarrow A|$ is also a bijection, so that $B \approx A$.
3. If $A \approx B$ and $B \approx C$ then there exists bijections $f: A \rightarrow B$ and $g: B \rightarrow C$, using [theorem: 2.77] we have that $g \circ f: A \rightarrow C$ is a bijection, so $A \approx C$. $\square$

Next we define a relation that says one set is smaller or equal to another set.

Definition 6.3. Let A, B be sets then $A \preceq B$ if there exist a $C \subseteq B$ such that $A \approx C$.

The following relation expresses that one set is smaller then another set.

Definition 6.4. Let A, B be sets then $A \prec B$ if $A \preceq B$ and $\neg(A \approx B)$

Clearly we have the following:

Theorem 6.5. If A is a set then $\mathcal{P}(A) \approx 2^A$

Proof. As $2 = s(1) = s(s(0)) = s(\{\emptyset\}) = \{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\} = \{0, 1\}$ we have that $2^A = \{0, 1\}^A$, finally using [theorem: 2.80] there exist a bijection $\mathcal{P}(A)$ and $\{0, 1\}^A$. $\square$

Theorem 6.6. Let A, B be sets then $A \preceq B$ if and only if there exist a injection $f: A \rightarrow B$

Proof.

- $\Rightarrow$. If $A \preceq B$ then there exist a set $C \subseteq B$ and a bijection $f: A \rightarrow C$, as a bijection is injective we have that $f: A \rightarrow C$ is injective and finally by [theorem: 2.54] $f: A \rightarrow B$ is a injection.
- $\Leftarrow$. If $f: A \rightarrow B$ is a injection then by [theorem: 2.68] $f: A \rightarrow f(A)$ is a bijection where $f(A) \subseteq B$ proving that $A \preceq B$. $\square$

Theorem 6.7. If A is a set then there exist no surjection between A and $\mathcal{P}(A)$

Proof. We prove this by contradiction. So assume that there exists a surjective function

$$f: A \rightarrow \mathcal{P}(A)$$

Define

$$B = \{x \in A \mid x \notin f(x)\}$$

As $B \subseteq A$ we have that $B \in \mathcal{P}(A)$ and by surjectivity there exists a $a \in A$ such that $f(a) = B$. If $a \in B$ then $a \notin f(a) = B$ leading to the contradiction $a \in B \wedge a \notin B$, so we must have $a \notin B = f(a)$ giving the contradiction $a \in B \wedge a \notin B$. So the assumption must be wrong hence there is no surjection between A and $\mathcal{P}(A)$. $\square$

Corollary 6.8. *If A is a set then no subset of A can be equipotent with $\mathcal{P}(A)$ or 2^A*

Proof. First we prove that no subset of A can be equipotent with $\mathcal{P}(A)$. If $B \subseteq A$ then we have the following possible cases to consider:

$B = A$. Then by [theorem: 6.7] we can not have a surjection between B and $\mathcal{P}(A)$, which as a bijection is surjection, proves that there is no bijection between B and $\mathcal{P}(A)$. So B is not equipotent with $\mathcal{P}(A)$.

$B \subset A$. Then $A \setminus B \neq \emptyset, B \cap (A \setminus B) = \emptyset$ and $A = (A \setminus B) \cup B$. Assume now that B is equipotent with $\mathcal{P}(A)$ then a bijection $g: B \rightarrow \mathcal{P}(A)$ exist, take the constant function $C_\emptyset: A \setminus B \rightarrow \mathcal{P}(A)$ where $C_\emptyset(x) = \emptyset$ and form then using [theorem: 2.84] the function

$$f = g \bigcup C_\emptyset: A \rightarrow \mathcal{P}(A)$$

If $C \in \mathcal{P}(A)$ then, as g is bijective $\exists x \in B$ such that $(x, C) \in g \subseteq f$ or $f(x) = C$, hence f is a surjection which is not allowed by [theorem: 6.7]. So B is not equipotent with $\mathcal{P}(A)$.

If $B \approx 2^A$ then, as by [theorem: 6.5] $2^A \approx \mathcal{P}(A)$, we have by [theorem: 6.2] that $B \approx \mathcal{P}(A)$ which we have just shown to be impossible. So B can not be equipotency with 2^A . $\square$

Theorem 6.9. *If A, B are sets, $A \neq \emptyset$ then there exists a injection $f: A \rightarrow B$ if and only there exist a surjection $g: B \rightarrow A$*

Proof.

$\Rightarrow$. Let $f: A \rightarrow B$ be a injection then by [theorem: 2.62] there exist a $g: B \rightarrow A$ such that $g \circ f = \text{Id}_A$. If $x \in A$ then $y = \text{Id}_A(y) = (g \circ f)(y) = g(f(y))$ so that g is surjective.

$\Leftarrow$. Let $g: B \rightarrow A$ be a surjection then by [theorem: 3.105] there exist a injective function $f: A \rightarrow B$. $\square$

Corollary 6.10. *If A, B are sets then $A \preccurlyeq B$ if and only if there exist a surjection $f: B \rightarrow A$*

Proof. This follows from [theorem: 6.6] and the above [theorem: 6.9]. $\square$

Theorem 6.11. *Let A, B, C, D classes with $A \cap C = \emptyset = B \cap D$, $A \approx B$ and $C \approx D$ then*

$$(A \bigcup C) \approx (B \bigcup D)$$

Proof. As $A \approx B$ and $C \approx D$ then there exists bijections $f: A \rightarrow B$ and $g: C \rightarrow D$ then by [theorem: 2.87] there exists a bijection $f \bigcup g: A \bigcup C \rightarrow B \bigcup D$. Hence $A \bigcup C \approx B \bigcup D$. $\square$

Theorem 6.12. *If A, B, C, D are sets such that $A \approx B$ and $C \approx D$ then $A \times C \approx B \times D$*

Proof. As $A \approx B$ and $C \approx D$ there exist bijections $f: A \rightarrow B$ and $g: C \rightarrow D$. Define

$$h: A \times C \rightarrow B \times D \text{ by } h(x, y) = (f(x), g(y))$$

then we have:

injectivity. If $h(x, y) = h(x', y')$ then $(f(x), g(y)) = (f(x'), g(y'))$ so that $f(x) = f(x')$ and $g(y) = g(y')$, as f and g are injective we have $x = x'$ and $y = y'$ so that $(x, y) = (x', y')$.

surjectivity. If $(r, s) \in B \times D$ then as f, g are surjective there exists $x \in A, y \in C$ such that $r = f(x)$ and $s = g(y)$ so that $h(x, y) = (f(x), g(y)) = (r, s)$. $\square$

Theorem 6.13. *If A, B, C, D are sets such that $A \approx B$ and $C \approx D$ then $A^C \approx B^D$*

Proof. As $A \approx B$ and $C \approx D$ then there exists bijections $f: A \rightarrow B$ and $g: D \rightarrow C$. If $x \in A^C$ then $x: C \rightarrow A$ is a function, so $x \circ g: D \rightarrow A$ is a function, hence $f \circ (x \circ g): D \rightarrow B$ is a function, proving that $f \circ (x \circ g) \in B^D$. Define now $h: A^C \rightarrow B^D$ by $h(x) = f \circ (x \circ g)$ then we have:

injectivity. If $x, y \in A^C$ satisfies $h(x) = h(y)$ then

$$\begin{aligned} f \circ (x \circ g) = f \circ (y \circ g) &\Rightarrow f^{-1} \circ (f \circ (x \circ g)) = f^{-1} \circ (f \circ (y \circ g)) \\ &\Rightarrow (f^{-1} \circ f) \circ (x \circ g) = (f^{-1} \circ f) \circ (y \circ g) \\ &\Rightarrow \text{Id}_A \circ (x \circ g) = \text{Id}_A \circ (y \circ g) \\ &\Rightarrow x \circ g = y \circ g \\ &\Rightarrow (x \circ g) \circ g^{-1} = (y \circ g) \circ g^{-1} \\ &\Rightarrow x \circ (g \circ g^{-1}) = y \circ (g \circ g^{-1}) \\ &\Rightarrow x \circ \text{Id}_C = y \circ \text{Id}_C \\ &\Rightarrow x = y \end{aligned}$$

surjectivity. If $y \in B^D$ then $y: D \rightarrow B$ is a function so that $y \circ g^{-1}: C \rightarrow B$ is a function, hence $f^{-1} \circ (y \circ g^{-1}): C \rightarrow A$ is a function or $f^{-1} \circ (y \circ g^{-1}) \in A^C$. Further

$$\begin{aligned} h(f^{-1} \circ (y \circ g^{-1})) &= f \circ ((f^{-1} \circ (y \circ g^{-1})) \circ g) \\ &= f \circ ((f^{-1} \circ y) \circ (g^{-1} \circ g)) \\ &= f \circ ((f^{-1} \circ y) \circ \text{Id}_D) \\ &= f \circ (f^{-1} \circ y) \\ &= (f \circ f^{-1}) \circ y \\ &= \text{Id}_B \circ y \\ &= y \end{aligned} \quad \square$$

Theorem 6.14. *If A, B are sets such that $A \approx B$ then $\mathcal{P}(A) \approx \mathcal{P}(B)$ and $2^A \approx 2^B$*

Proof. As $A \approx B$ and $2 \approx 2$ [see theorem: 6.2], we have by [theorem: 6.13] that $2^A \approx 2^B$. Further by [theorem: 6.8] $\mathcal{P}[A] \approx 2^A$ and $\mathcal{P}(B) \approx 2^B$, so by [theorem: 6.2] it follows that $\mathcal{P}(A) = \mathcal{P}(B)$. $\square$

6.2 Finite, Infinite and Denumerable sets

6.2.1 Finite and Infinite sets

Applying the concept of initial segments [see definition: 3.46] on $\langle \mathbb{N}_0, \leq \rangle$ we have the following definition.

Definition 6.15. *Let $n \in \mathbb{N}_0$ then S_n is defined by*

$$S_n = \{m \in \mathbb{N}_0 | m < n\}$$

Actual we have already encountered the initial segments for $\langle \mathbb{N}_0, \leq \rangle$ because they are actual the natural numbers as is proved in the following theorem.

Theorem 6.16. $\forall n \in \mathbb{N}_0$ we have $n = S_n$.

Proof. We prove this by induction. So let $S = \{n \in \mathbb{N}_0 | n = S_n\}$ then we have:

$0 \in S$. If $x \in S_0$ then $x < 0$ which by [theorem: 5.49] is false, so $S_0 = \emptyset = 0$ proving that $0 \in S$.

$n \in S \Rightarrow s(n) \in S$. As $n \in S$ we have that $n = S_n$ so that

$$s(n) = n \bigcup \{n\} = S_n \bigcup \{n\}$$

If $m \in s(n)$ then we have the following possibilities to consider:

$m = n$. Then by [theorem: 5.47] we have that $m < s(m) = s(n)$ so that $m \in S_{s(n)}$

$m \in S_n$. Then $m < n$ which as by [theorem: 5.47] $n < s(n)$ proves that $m < s(n)$ hence $m \in S_{s(n)}$

this proves that

$$s(n) \subseteq S_{s(n)} \quad (6.1)$$

If $m \in S_{s(n)}$ then $m < s(n)$, now by [theorem: 5.53] we have either $n < m$ or $m \leq n$. If $n < m$ then by [theorem: 5.50] we have $s(n) \leq m$ so that by transitivity we have $m < m$ a contradiction. So we must have that $m \leq n$, if $m = n$ then $m \in n \bigcup \{n\} = s(n)$ and if $m < n$ then $m \in S_n \subseteq S_n \bigcup \{n\} = s(n)$. So in all cases we have $m \in s(n)$ proving that $S_{s(n)} \subseteq s(n)$, combining this with [eq: 6.1] gives

$$s(n) = S_{s(n)}$$

proving that $s(n) \in S$.

Using induction [theorem: 5.11] it follows that $S = \mathbb{N}_0$ proving the theorem. $\square$

Theorem 6.17. Let $n, m \in \mathbb{N}_0$ then

$$n \leq m \Leftrightarrow S_n \subseteq S_m.$$

In other words as $n = S_n$ and $m = S_m$ we have

$$n \leq m \Leftrightarrow n \subseteq m$$

Proof.

$\Rightarrow$. If $x \in S_n$ then $x < n$ which as $n \leq m$ proves that $x < m$ so that $x \in S_m$, hence $S_n \subseteq S_m$.

$\Leftarrow$. By definition if $n \leq m$ then either $n = m \xrightarrow{n=S_n, m=S_m} S_n = S_m \Rightarrow S_n \subseteq S_m$ or $n \in m$ which by [theorem: 5.14] we have that $n \subseteq m \xrightarrow{n=S_n, m=S_m} S_n \subseteq S_m$. $\square$

Theorem 6.18. Let $n, m \in \mathbb{N}_0$ with $n \leq m$ then

$$\beta: \{n, \dots, m\} \rightarrow S_{(m-n)+1} \text{ where } \beta(i) = i - n$$

is a bijection with inverse

$$\beta^{-1}: S_{(m-n)+1} \rightarrow \{n, \dots, m\} \text{ where } \beta^{-1}(i) = i + n$$

Proof. We have for the function $\beta: \{n, \dots, m\} \rightarrow S_{(m-n)+1}$ where $\beta(i) = i - n$ the following:

injectivity. If $k, l \in \{n, \dots, m\}$ such that $\beta(k) = \beta(l)$ then $k - n = l - n$, so by [theorem: 5.43] $k = (k - n) + n = (l - n) + n = l$ proving that $k = l$.

surjectivity. If $k \in S_{(m-n)+1}$ then $0 \leq k < (m - n) + 1$ so that by [theorem: 5.58] $0 \leq k \leq m - n$, then by [theorem: 5.55] we have that $n = 0 + n \leq k + n \leq (m - n) + n = m$. If we take $i = k + n$ then we have $0 \leq i \leq m$ and further $i - n = (k + n) - n \stackrel{[\text{theorem: 5.66}]}{=} k$ proving that $\beta(i) = k$.

So $\beta: \{n, \dots, m\} \rightarrow S_{(m-n)+1}$ is a bijection. Further we have if $k \in S_{(m-n)+1}$ that $k = \beta(\beta^{-1}(k)) = \beta^{-1}(k) - n$ so that by [theorem: 5.43] $k + n = (\beta^{-1}(k) - n) + n = \beta^{-1}(k)$ proving that

$$\beta^{-1}: S_{(m-n)+1} \rightarrow \{n, \dots, m\} \text{ is defined by } \beta^{-1}(k) = k + n \quad \square$$

We define now the concept of a finite set.

Definition 6.19. (Finite Set) A set A is *finite* if $\exists n \in \mathbb{N}_0$ such that $n \approx A$

Example 6.20. $\emptyset$ is finite.

Proof. $\emptyset: \emptyset \rightarrow \emptyset$ is a bijection by [example: 2.65], so as $0 = \emptyset$ we have that $0 \approx \emptyset$. $\square$

Lemma 6.21. *If $n \in \mathbb{N}_0$ then $n \approx \{1, \dots, n\}$*

Proof. If $n \in \mathbb{N}_0$ then we have either:

$n = 0$. Then $n = 0 = \emptyset$ and $\{1, \dots, 0\} = \emptyset$ so that $n \approx \{1, \dots, n\}$

$n \neq 0$. Then $0 < n$ and we have for $\beta: \{1, \dots, n\} \rightarrow S_n$ defined by $\beta(i) = i - 1$ that it satisfies:

injectivity. If $\beta(i) = \beta(j)$ then $i - 1 = j - 1$ so that $i = j$

surjectivity. If $j \in S_n$ then $0 \leq j < n$ so that $0 < j + 1 \leq n \Rightarrow 1 \leq j + 1 \leq n$, so if we take $i = j + 1$ we have that $i \in \{1, \dots, n\}$ and $\beta(i) = (j + 1) - 1 = j$

proving β is a bijection. This proves that

$$\{1, \dots, n\} \approx S_n \stackrel{[\text{theorem: 6.16}]}{=} n \quad \square$$

Theorem 6.22. *As set A is finite if and only if there exist a $n \in \mathbb{N}_0$ such that $\{1, \dots, n\} \approx A$.*

Proof. We have

$$\begin{aligned} A \text{ is finite} & \Leftrightarrow \exists n \in \mathbb{N}_0 \text{ such that } A \approx n \\ & \Leftrightarrow \sim_{[\text{theorem: 6.21}]} \exists n \in \mathbb{N}_0 \text{ such that } A \approx \{1, \dots, n\} \end{aligned}$$

$\square$

Proof. First, if $n \in \mathbb{N}_0$ with $0 < n$ we have for $\beta: \{1, \dots, n\} \rightarrow S_n$ defined by $\beta(i) = i - 1$ that it satisfies:

injectivity. If $\beta(i) = \beta(j)$ then $i - 1 = j - 1$ so that $i = j$

surjectivity. If $j \in S_n$ then $0 \leq j < n$ so that $0 < j + 1 \leq n \Rightarrow 1 \leq j + 1 \leq n$, so if we take $i = j + 1$ we have that $i \in \{1, \dots, n\}$ and $\beta(i) = (j + 1) - 1 = j$

proving β is a bijection. This proves that

$$\forall n \in \mathbb{N}_0 \setminus \{0\} \quad \{1, \dots, n\} \approx S_n \stackrel{[\text{theorem: 6.16}]}{=} n \quad (6.2)$$

Now for the final proof:

$\Rightarrow$. Then for A we have either:

$A = \emptyset$. Then we have as $\{1, \dots, 0\} = \emptyset$ that $\{1, \dots, 0\} \approx A$

$A \neq \emptyset$. Then there exists a $n \in \mathbb{N}_0$ such that $n \approx A$, as $0 \stackrel{\text{def}}{=} \emptyset$ we must have that $0 < n$.
Combining this with [eq: 6.2] gives

$$\{1, \dots, n\} \approx A$$

$\Leftarrow$. As $\exists n \in \mathbb{N}_0$ such that $A \approx \{1, \dots, n\}$ we have for n either:

$n = 0$. Then $\{1, \dots, n\} = \emptyset$ so that $A \approx \emptyset \Rightarrow A = \emptyset$ hence A is finite.

$n \neq 0$. Then by [eq: 6.2] $\{1, \dots, n\} \approx n$ and as $A \approx \{1, \dots, n\}$ we have that $A \approx n$ proving that A is finite. $\square$

Definition 6.23. (Infinite Set) *A set A is **infinite** if A is not **finite**.*

Definition 6.24. (Denumerable Set) *A set A is **denumerable** or **infinite countable** if*

$$\mathbb{N}_0 \approx A.$$

Definition 6.25. (Countable Set) *A set A is **countable** if it is **finite** or **denumerable**.*

Theorem 6.26. *If A, B are sets such that $A \approx B$ then we have*

1. *If A is finite then B is finite*
2. *If A is denumerable then B is denumerable*

3. If A is countable then B is countable.

Proof.

1. As A is finite there exists a $n \in \mathbb{N}_0$ such that $n \approx A$ which as $A \approx B$ proves by [theorem: 6.2] that $n \approx B$ hence B is finite.
2. As A is denumerable $\mathbb{N}_0 \approx A$ which as $A \approx B$ proves by [theorem: 6.2] that $\mathbb{N}_0 \approx B$ hence B is finite.
3. As A is countable it is either finite or denumerable, (1) and (2) ensures then that B is either finite or denumerable. $\square$

Lemma 6.27. If A is a **denumerable** set and $a \in A$ then $A \setminus \{a\}$ is a denumerable set.

Proof. As A is denumerable there exist a bijection $f: \mathbb{N}_0 \rightarrow A$. As $a \in A$ we have by surjectivity that $\exists n \in \mathbb{N}_0$ such that $f(n) = a$. Define now

$$g: \mathbb{N}_0 \rightarrow A \text{ where } g(i) = \begin{cases} f(i) & \text{if } i < n \\ f(i+1) & \text{if } n \leq i \end{cases}$$

which, as $\{x \in \mathbb{N}_0 | x < n\} \cap \{x \in \mathbb{N}_0 | n \leq x\} = \emptyset$ and $\mathbb{N}_0 = \{x \in \mathbb{N}_0 | x < n\} \cup \{x \in \mathbb{N}_0 | n \leq x\}$, is a function. As for bijectivity we have:

injectivity. If $g(i) = g(i')$ then for i, i' we have either:

$i < n \wedge i' < n$. Then $f(i) = g(i) = g(i') = f(i')$ which as f is injective proves that $i = i'$.

$i < n \wedge n \leq i'$. Then $f(i) = g(i) = g(i') = f(i' + 1)$ which as f is injective proves that $i = i' + 1$. Now as $n \leq i' < i' + 1 = i$ and $i < n$ we reach the contradiction $n < n$, so this case is not possible.

$n \leq i \wedge i' < n$. Then $f(i + 1) = g(i) = g(i') = f(i')$ which as f is injective proves that $i + 1 = i'$. Now as $n \leq i < i + 1 = i'$ and $i' < n$ we reach the contradiction $n < n$, so this case is not possible.

$n \leq i \wedge n \leq i'$. Then $f(i + 1) = g(i) = g(i') = f(i' + 1)$, hence, as f is injective, we have $i + 1 = i' + 1$ or by [theorem: 5.43] $i = i'$.

So in all valid cases we have $i = i'$ proving injectivity.

surjectivity. If $y \in A \setminus \{x\}$ then there exists by surjectivity of f a $i \in \mathbb{N}_0$ such that $f(i) = y$. We can not have $i = n$, because we would then have $f(i) = f(n) = y \notin A \setminus \{y\}$. So we have either

$i < n$. Then $g(i) = f(i) = y$

$n < i$. Then by [theorem: 5.67] $n \leq i - 1$, so $g(i - 1) = f((i - 1) + 1) = f(i) = y$

proving surjectivity. $\square$

Lemma 6.28. Let $n \in \mathbb{N}_0$ then n has no denumerable subset. In particular, as $n \subseteq n$, n is not denumerable.

Proof. We prove this by induction, so define

$$S = \{n \in \mathbb{N}_0 | n \text{ does not contain a denumerable subset}\}$$

then we have:

$0 \in S$. As $0 = \emptyset$ we have if $A \subseteq 0$ that $A = \emptyset$. If now $\mathbb{N}_0 \approx A$ then there exists a bijection $f: \mathbb{N}_0 \rightarrow A$ so that $f(0) \in A = \emptyset$ which is a contradiction. So 0 does not contain a denumerable subset.

$n \in S \Rightarrow n + 1 \in S$. We proceed by contradiction, so assume that there exist a $A \subseteq n + 1 = s(n) = n \cup \{n\}$ which is denumerable. If $n \notin A$ then $A \subseteq n$ which is impossible because $n \in S$, so we must have that $n \in A$. Let $a \in A \setminus \{n\} \subseteq n \cup \{n\}$ then, as $a \neq n$, $a \in n$ proving that $A \setminus \{n\} \subseteq n$. Now by the previous lemma [lemma: 6.27] we have, as A is denumerable, that $A \setminus \{n\}$ is denumerable which is forbidden as $n \in S$. So the assumption is wrong, hence every subset of $s(n)$ is not denumerable, proving that $n + 1 \in S$.

Using induction [see theorem: 5.83] it follows that $S = \{0, \dots, \infty\} = \mathbb{N}_0$ proving the lemma. $\square$

Theorem 6.29. *Let A be a set then A is infinite if and only if A contains a denumerable subset.*

Proof.

$\Rightarrow$. Let A be a infinite set. Using the well ordering theorem [see theorem: 3.135] there exists a order relation $\leq_A$ such that $\langle A, \leq_A \rangle$ is a well ordered set. Using [theorem: 3.99] and the fact that $\langle \mathbb{N}, \leq \rangle$ is well ordered [see theorem: 5.51] we have exactly one of the following cases:

$\langle \mathbb{N}_0, \leq \rangle$ is order isomorphic with $\langle A, \leq_A \rangle$. This implies that $A \approx \mathbb{N}_0$ so that A is a denumerable subset of itself.

$\langle \mathbb{N}_0, \leq \rangle$ is order isomorphic with an initial segment of $\langle A, \leq_A \rangle$. This implies that A has a denumerable subset [the initial segment].

$\langle A, \leq_A \rangle$ is order isomorphic with an initial segment of $\langle \mathbb{N}_0, \leq \rangle$. So there exists a $n \in \mathbb{N}_0$ such that $A \approx S_n \stackrel{[\text{theorem: 6.16}]}{=} n$ so that A is finite, contradicting the fact that A is infinite. Hence this case does not apply.

So in all applicable cases we have that A contains a denumerable subset.

$\Leftarrow$. Let $B \subseteq A$ be a denumerable subset of A . Assume that A is finite then there exists a $n \in \mathbb{N}_0$ such that $n \approx A$, hence there exist a bijection $f: A \rightarrow n$. As $B \subseteq A$ we have that $f|_B: B \rightarrow f(B)$ is a bijection [see theorems: 2.92, 2.68] so that $B \approx f(B)$, as B is denumerable $\mathbb{N}_0 \approx B$, so by [theorem: 6.2] it follows that $\mathbb{N}_0 \approx f(B) \subseteq n$. So there exists a denumerable subset of n which by [theorem: 6.28] is impossible. Hence A is not finite which by definition means that A is infinite. $\square$

Corollary 6.30. *Every denumerable set is infinite*

Proof. Let A be a denumerable set then as $A \subseteq A$ it follows from the previous theorem [theorem: 6.29] that A is infinite. $\square$

Corollary 6.31. $\mathbb{N}_0$ is infinite.

Proof. As $\mathbb{N}_0 \approx \mathbb{N}_0$ $\mathbb{N}_0$ is denumerable, clearly $\mathbb{N}_0 \subseteq \mathbb{N}_0$ so by the previous theorem [theorem: 6.29] we have that $\mathbb{N}_0$ is infinite. $\square$

Corollary 6.32. *Every set with a infinite subset is infinite.*

Proof. If A is a set such that there exists a infinite set B with $B \subseteq A$ then, as B is infinite, we have by [theorem: 6.29] the existence of a denumerable set $C \subseteq B$, but then $C \subseteq A$ and thus A has a denumerable subset. Using [theorem: 6.29] it follows that A is infinite. $\square$

Corollary 6.33. *Every subset of a finite set is finite*

Proof. If a finite set would contain a infinite subset then by the previous theorem the finite set would be infinite. $\square$

Theorem 6.34. *If A and B are finite sets then $A \cup B$ is a finite set.*

Proof. As A is finite we have by [theorem: 6.33] that $A \setminus B$ is finite. So there exists $n, m \in \mathbb{N}_0$ such that $n \approx A \setminus B$ and $m \approx B$, hence we have two bijections

$$f: A \setminus B \rightarrow n \stackrel{[\text{theorem: 6.16}]}{=} S_n \text{ and } g': B \rightarrow m \stackrel{[\text{theorem: 6.16}]}{=} S_m \quad (6.3)$$

Define

$$C = \{i \in \mathbb{N}_0 | n \leq i \wedge i < n + m\}$$

If $b \in B$ then $g'(b) \in S_m$, hence $0 \leq g'(b) < m$ so that by [theorem: 5.55] $n = 0 + n \leq g'(b) + n < m + n$ or $g'(b) + n \in C$. So

$$g: B \rightarrow C \text{ where } g(i) = g'(i) + n \quad (6.4)$$

defines a function. Further we have:

injectivity. If $g(b) = g(b')$ then $g'(b) + n = g'(b') + n$, so using [theorem: 5.43] $g'(b) = g'(b')$, hence, as g' is injective, we have $b = b'$.

surjectivity. If $i \in C$ then $n \leq i < n + m$, using [theorem: 5.60] there exist a $k \in \mathbb{N}_0$ such that $n + k = i$. If $m \leq k$ then by [theorem: 5.55] $n + m \leq n + k = i < n + m$ a contradiction. So $k < m$ and thus $k \in S_m$. As g' is surjective there exists a $b \in B$ such that $g'(b) = k$ and thus $g(b) = g'(b) = k + n = i$.

proving that

$$g: B \rightarrow C \text{ is a bijection} \quad (6.5)$$

Further if $i \in n \cap C = S_n \cap C$ then $i < n \wedge n \leq i$ yielding the contradiction $i < i$ so we have that

$$n \cap C \neq \emptyset \quad (6.6)$$

If $i \in n \cup C$ then either

$i \in n$. Then, as $n = S_n$, we have $i < n$ which as $n \leq n + m$ proves that $i < n + m$ hence $i \in S_{n+m}$.

$i \in C$. Then $i < n + m$ so that $i \in S_{n+m}$

proving

$$n \cup C \subseteq S_{n+m} \quad (6.7)$$

If $i \in S_{n+m}$ then $i < n + m$, further we have either $i < n$ so that $i \in S_n = n$ or $n \leq i$ giving $i \in C$, hence $i \in n \cup C$ or $S_{n+m} \subseteq n \cup C$ which by [eq: 6.7] proves that

$$n \cup C = S_{n+m} \quad (6.8)$$

Using [eq: 6.3], [eq: 6.5], [eq: 6.6], [eq: 6.8], $A \cup B = (A \setminus B) \cup B$ and $(A \setminus B) \cap B = \emptyset$ allows use to use [theorem: 2.87] to get the bijection

$$f \cup g: A \cup B \rightarrow S_{n+m}$$

proving that

$$A \cup B \approx S_{n+m}$$

□

Lemma 6.35. If $\{A_i\}_{i \in S_n}$ is such that $\forall i \in S_n$ A_i is finite then $\bigcup_{i \in S_n} A_i$ is finite.

Proof. We use induction to prove this, so define

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{A_i\}_{i \in S_n} \text{ satisfies } \forall i \in S_n \text{ } A_i \text{ is finite then } \bigcup_{i \in S_n} A_i \text{ is finite} \right\}$$

then we have:

$0 \in S$. If $n = 0$ then $S_0 = 0 = \emptyset$ so that $\bigcup_{i \in S_0} A_i = \bigcup_{i \in \emptyset} A_i \stackrel{\text{example: 2.129}}{=} \emptyset$ which is finite, hence $0 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $\{A_i\}_{i \in n+1}$ a family of finite sets. As $S_{n+1} = n + 1 = s(n) = n \cup \{n\} = S_n \cup \{n\}$ and $n \notin S_n$ we have that $S_{n+1} \setminus \{n\} = S_n$. So

$$\bigcup_{i \in S_{n+1}} A_i \stackrel{\text{theorem: 2.142}}{=} \left(\bigcup_{i \in S_{n+1} \setminus \{n\}} A_i \right) \cup A_n = \left(\bigcup_{i \in S_n} A_i \right) \cup A_n$$

As $n \in S$ we have that $\bigcup_{i \in S_n} A_i$ is finite which, as A_n is also finite, proves, using [theorem: 6.34] that $(\bigcup_{i \in S_n} A_i) \cup A_n$ is finite. So $\bigcup_{i \in S_{n+1}} A_i$ is finite proving that $n + 1 \in S$.

Mathematical induction [see theorem: 5.83] proves then the lemma. □

Theorem 6.36. If $\{A_i\}_{i \in I}$ is such that I is finite and $\forall i \in I$ A_i is finite then $\bigcup_{i \in I} A_i$ is finite.

Proof. As I is finite there exists a $n \in \mathbb{N}_0$ and a bijection $f: S_n \rightarrow I$ so that by [theorem: 2.127] we have that

$$\bigcup_{i \in I} A_i = \bigcup_{i \in S_n} A_{f(i)} \quad (6.9)$$

Using the previous lemma [lemma: 6.35] it follows that $\bigcup_{i \in S_n} A_{f(i)}$ is finite, hence using [eq: 6.9] we have

$$\bigcup_{i \in I} A_i \text{ is finite} \quad \square$$

Theorem 6.37. *A set A is infinite if and only if $\exists B \subset A$ such that $B \approx A$. In other words A is infinite if and only if A is equipotent with a proper subset of itself.*

Proof.

$\Rightarrow$. If A is infinite then by [theorem: 6.29] there exist a denumerable $B \subseteq A$. So there exists a bijection $f: \mathbb{N}_0 \rightarrow B$. Define now the function [taking in account that $(A \setminus B) \cap B = \emptyset$ and $A = (A \setminus B) \cup B$]

$$g: A \rightarrow A \text{ where } g(x) = \begin{cases} x & \text{if } x \in A \setminus B \\ f(f^{-1}(x) + 1) & \text{if } x \in B \end{cases}$$

where $f^{-1}: B \rightarrow \mathbb{N}_0$ is the inverse of f .

Then we have:

$$g(A) = A \setminus \{f(0)\} \quad (6.10)$$

Proof. If $y \in g(A)$ then there exists a $x \in A$ such that $y = g(x)$, we have for x either:

$x \in A \setminus B$. Then $y = g(x) = x$ so that $y \in A \setminus B$ or as $f(0) \in B$ that $y \in A \setminus \{f(0)\}$.

$x \in B$. If $f(0) = f(f^{-1}(x) + 1)$ we have, as f is a bijection hence injective, that $0 = f^{-1}(x) + 1$ which contradicts $0 < f^{-1}(x) + 1$. So we must have that

$$f(0) \neq f(f^{-1}(x) + 1) = y.$$

proving $y \in A \setminus \{f(0)\}$.

So we conclude that

$$g(A) \subseteq A \setminus \{f(0)\} \quad (6.11)$$

If $y \in A \setminus \{f(0)\}$ then we have either:

$y \in B$. If $f^{-1}(y) = 0$ we would have that $y = f(f^{-1}(y)) = f(0)$ contradicting $y \in A \setminus \{f(0)\}$. So we have that $f^{-1}(y) \neq 0$ or $0 < f^{-1}(y)$, using [theorem: 5.67] we have then that $0 \leq f^{-1}(y) - 1$. Take then $x = f(f^{-1}(y) - 1) \in B \subseteq A$ then we have:

$$\begin{aligned} g(x) &= f(f^{-1}(x) + 1) \\ &= f(f^{-1}(f(f^{-1}(y) - 1)) + 1) \\ &= f((f^{-1}(y) - 1) + 1) \\ &= f(f^{-1}(y)) \\ &= y \end{aligned}$$

so that $y \in g(A)$.

$y \notin B$. Then $y \in A \setminus B$ so that $g(y) = y$ proving that $y \in g(A)$

So we conclude that $A \setminus \{f(0)\} \subseteq g(A)$ which combined with [eq: 6.11] proves $g(A) = A \setminus \{f(0)\}$. $\square$

Next we proof that $g: A \rightarrow A$ is injective

Proof. Let $x, x' \in A$ such that $g(x) = g(x')$ then for x, x' we have to consider the following possible cases:

$x \in B \wedge x' \in B$. then $f(f^{-1}(x) + 1) = g(x) = g(x') = f(f^{-1}(x') + 1)$ so that

$$\begin{aligned} f(f^{-1}(x) + 1) = f(f^{-1}(x') + 1) &\xRightarrow{f \text{ is injective}} f^{-1}(x) + 1 = f^{-1}(x') + 1 \\ &\Rightarrow f^{-1}(x) = f^{-1}(x') \\ &\xRightarrow{f^{-1} \text{ is injective}} x = x' \end{aligned}$$

$x \in B \wedge x' \notin B$. Then $f(f^{-1}(x) + 1) = g(x) = g(x') = x'$ so that $f(f^{-1}(x) + 1) \notin B$ contradicting $f: \mathbb{N}_0 \rightarrow B$. So this case does not apply.

$x \notin B \wedge x' \in B$. Then $x = g(x) = g(x') = f(f^{-1}(x') + 1)$ so that $f(f^{-1}(x') + 1) \notin B$ contradicting $f: \mathbb{N}_0 \rightarrow B$, So this case never occurs.

$x \notin B \wedge x' \notin B$. Then $x = g(x) = g(x') = x'$. □

So we have proved that

$$g: A \rightarrow A \text{ is injective} \tag{6.12}$$

Using [eq: 6.10] and [eq: 6.12] proves that $g: A \rightarrow A \setminus \{f(0)\}$ is a bijection or

$$A \approx A \setminus \{f(0)\}$$

Further as $f(0) \in B \subseteq A$ we have that $A \neq A \setminus \{f(0)\}$ giving $A \setminus \{f(0)\} \subset A$. Hence we have proved that A is equipotent with a proper subset of itself.

$\Leftarrow$. Assume that there exists a proper subset $B \subset A$ such that $A \approx B$ then there exists a bijection $f: A \rightarrow B$, resulting in the injection [see theorem: 2.54]

$$f: A \rightarrow A \text{ with } f(A) = B \subset A$$

As $f(A) \subset A$ there exists a $a \in A$ such that $a \notin f(A)$. Using recursion [theorem: 5.84] there exist a injection $\lambda: \mathbb{N}_0 \rightarrow A$ such that $\lambda(0) = a$ and $\forall n \in \mathbb{N}_0 \lambda(n+1) = f(\lambda(n))$. Hence we have a bijection $\lambda: \mathbb{N}_0 \rightarrow \lambda(A)$ proving that $\lambda(A)$ is denumerable, as $\lambda(A) \subseteq A$ it follows from [theorem: 6.29] that A is infinite. □

The following theorem allows you to quantify the number of elements in a finite set.

Theorem 6.38. *If $n, m \in \mathbb{N}_0$ such that $n \approx m$ then $n = m$.*

Proof. Assume that $n \approx m$ then by [theorem: 5.53] we have either $n < m$, $m < n$ or $n = m$. If

$n < m$. Then $\forall i \in n = S_n$ we have $i < n < m \Rightarrow i < m$ so that $i \in S_m = m$ which as $n \neq m$ proves that $n \subset m$. So m is equipotent to a proper subset of itself which by [theorem: 6.37] would mean that m is infinite contradicting the fact that m is finite [as $m \approx m$].

$m < n$. Then $\forall i \in m = S_m$ we have $i < m < n \Rightarrow i < n$ so that $i \in S_n = n$ which as $n \neq m$ proves that $m \subset n$. So n is equipotent to a proper subset of itself which by [theorem: 6.37] would mean that n is infinite contradicting the fact that n is finite [as $n \approx n$].

So the only option left is

$$n = m \tag{6.13}$$

□

The previous theorem leads to the following observation: If A is a finite set then there exists a $n \in \mathbb{N}_0$ such that $n \approx A$, if there was also a $n' \in \mathbb{N}_0$ such that $n' \approx A$ then $n \approx n'$, hence $n = n'$. This leads to the following definition.

Definition 6.39. *If A is a **finite** set then $\exists! n \in \mathbb{N}_0$ such that $n \approx A$. This unique number is noted as $\#A$, so $\#A \approx A$. $\#A$ can be interpreted as the number of elements in A .*

Theorem 6.40. *If A is a set then $A = \emptyset \Leftrightarrow \#A = 0$*

Proof.

$\Rightarrow$. If $A = \emptyset$ then by [example: 2.65] $\emptyset: \emptyset \rightarrow \emptyset$ is a bijection, so as $0 = \emptyset$ we have $\# \emptyset = 0$.

$\Leftarrow$. If $\#A = 0$ then as $0 = \emptyset$ there exists a bijection $f: \emptyset \rightarrow A$. Assume that $A \neq \emptyset$ then there exist a $y \in A$ and as f is a bijection we would have a $x \in \emptyset$ such that $f(x) = y$ contradicting the fact that $\forall x \ x \notin \emptyset$. $\square$

Theorem 6.41. *If A, B are finite sets then $A \times B$ is finite and $\#(A \times B) = \#A \cdot \#B$*

Proof. We have for A, B to consider the following possibilities:

$A = \emptyset \vee B = \emptyset$. Then $0 = \emptyset \approx A$ and $0 = \emptyset \approx B$ so that $\#A = 0 = \#B$, further by [theorem: 1.54] $0 = \emptyset = A \times B$ hence $\#(A \times B) = 0 = \#A \cdot \#B$.

$A \neq \emptyset \wedge B \neq \emptyset$. Take $n = \#A \neq 0$ and $m = \#B \neq 0$ then there exist bijections $f: B \rightarrow n = S_n$ and $g: A \rightarrow m = S_m$. Now $\forall x \in A, \forall y \in B$ we have $f(x) < n$ and $g(y) < m$, using [theorem: 5.67] we have $g(y) \leq m - 1$. So by [theorem: 5.76]

$$n \cdot g(x) = g(x) \cdot n \leq (m - 1) \cdot n \stackrel{[\text{theorem: 5.69}]}{=} m \cdot n - n,$$

further by [theorem: 5.74] we have

$$(m \cdot n - n) + f(x) < (m \cdot n - n) + n = m \cdot n = n \cdot m$$

This allows us to define the function

$$h: A \times B \rightarrow S_{n \cdot m} \text{ where } h(x, y) = n \cdot g(x) + f(x)$$

then we have:

injectivity. If $h(x, y) = h(x', y')$ then $n \cdot g(x) + f(x) = n \cdot g(x') + f(x')$. As $0 \leq f(x) < n$ and $0 \leq f(x') < n$ it follows from [theorem: 5.79] that $g(x) = g(x')$ and $f(x) = f(x')$ which as f, g are bijections gives $x = x'$ and $y = y'$ so that $(x, y) = (x', y')$.

surjectivity. If $z \in S_{n \cdot m}$ then $0 \leq z < n \cdot m$, using [theorem: 5.79] there exist a q, r such that $z = q \cdot n + r$ and $0 \leq r < n$. If $m \leq q \stackrel{[\text{theorem: 5.76}]}{\Rightarrow} m \cdot n \leq q \cdot n \stackrel{[\text{theorem: 5.55}]}{\Rightarrow} m \cdot n + r \leq q \cdot n + r = z < n \cdot m$ so that $n \cdot m + r < n \cdot m$ or $r + n \cdot m < 0 + n \cdot m \stackrel{[\text{theorem: 5.55}]}{\Rightarrow} r < 0$ a contradiction, hence $q < m$. So we have proved that $r \in S_n$ and $q \in S_m$, as f, g are bijections there exists $x \in A, y \in B$ such that $f(x) = r$ and $g(y) = q$. So $h(x, y) = n \cdot g(x) + f(x) = n \cdot q + r = z$.

Hence we have $A \times B \approx S_{n \cdot m}$ proving that $A \times B$ is finite and $\#(A \times B) = n \cdot m = \#A \cdot \#B$. $\square$

Theorem 6.42. *If A, B are finite sets such that $A \cap B = \emptyset$ then $\#(A \cup B) = \#A + \#B$*

Proof. Let $n = \#A, m = \#B$ then there exist bijections $f: A \rightarrow S_n$ and $g: B \rightarrow S_m$. If $x \in A$ then $f(x) < n < n + m$ and if $x \in B$ then $g(x) < m \Rightarrow n + g(x) < n + m$, as further $A \cap B = \emptyset$ we can define the function

$$h: A \cup B \rightarrow S_{n+m} \text{ where } h(x) = \begin{cases} f(x) & \text{if } x \in A \\ n + g(x) & \text{if } x \in B \end{cases}$$

We prove now that this is a bijection.

injectivity. If $h(x) = h(x')$ then we have the following cases to consider for $x, x' \in A \cup B$:

$x \in A \wedge x' \in A$. Then $f(x) = h(x) = h(x') = f(x')$ which as f is a bijection gives $x = x'$.

$x \in A \wedge x' \in B$. Then $f(x) = h(x) = h(x') = n + g(x')$, now as $f(x) < n$ we have

$$n + g(x') = f(x) < n + 0$$

so that by [theorem: 5.55] $g(x') < 0$, a contradiction. So this case will never occur.

$x \in B \wedge x' \in A$. Then $n + g(x) = h(x) = h(x') = f(x')$, now as $f(x') < n$ we have

$$n + g(x) = f(x') < n + 0$$

so that by [theorem: 5.55] $g(x) < 0$, a contradiction. So this case will never occur.

$x \in B \wedge x \in B$. Then $g(x) + n = n + g(x) = h(x) = h(x') = n + g(x') = g(x') + n$ so that by [theorem: 5.55] $g(x) = g(x')$, which as g is a bijection proves that $x = x'$.

surjectivity. If $y \in S_{n+m}$ then $y < n + m$ and we have the following cases for y to consider:

$y < n$. Then $y \in S_n$ so that by surjectivity of f we have a $x \in A$ such that $f(x) = y$, hence $h(x) = f(x) = y$

$n \leq y$. Then $n \leq y < n + m$, by [theorem: 5.70] we have then that $0 \leq y - n < (n + m) - n$ [theorem: 5.66] m , proving that $y - n \in S_m$. As g is a surjection there exists a $x \in B$ such that $g(x) = y - n$, hence $h(n) = n + g(x) = n + (y - n) = y$. $\square$

Theorem 6.43. *If A is a finite set and $B \subseteq A$ then:*

1. B is finite
2. $A \setminus B$ is finite
3. $\#B \leq \#A$
4. If $B \subset A$ then $\#B < \#A$
5. $\#A = \#B + \#(A \setminus B)$

Proof. As A is finite there exist $n \in \mathbb{N}_0$ and a bijection $f: n = S_n \rightarrow A$. We have then to consider the following possibilities:

$B = A$. Then obviously B is finite, $A \setminus B = \emptyset$ is also finite, $\#B = \#A \Rightarrow \#B \leq \#A$ and $\#B + \#(A \setminus B) = \#A + \#\emptyset = \#A + 0 = \#A$, So (1), (2), (3), (4) and (5) are satisfied.

$B = \emptyset$. Then clearly B is finite, $A \setminus B = A$ is finite, $\#B = 0 \leq \#A$ and $\#B + \#(A \setminus B) = 0 + \#A = \#A$, further if $B \subset A$ then $A \neq \emptyset$ so that $\#B = 0 < \#A$.

$\emptyset \neq B \subset A$. As every subset of a finite set is finite [see theorem: 6.33] we have that B and $A \setminus B$ are finite, further as $B \subset A$ we have that $A \setminus B \neq \emptyset$ so that

$$0 < \#(A \setminus B).$$

As $B \cap (A \setminus B) = \emptyset$ and $A \cup B = (A \setminus B) \cup B$ it follows from [theorem: 6.42] that

$$\#A = \#B + \#(A \setminus B)$$

Now if $\#A \leq \#B$ then as $0 < \#(A \setminus B)$ it follows from [theorem: 5.74] that

$$\#A = \#A + 0 < \#B + \#(A \setminus B) = \#A$$

a contradiction, so we must have that

$$\#B < \#A$$

So (1), (2), (3), (4) and (5) are satisfied. $\square$

Corollary 6.44. *If A, B are sets, A is finite and $f: A \rightarrow B$ is a surjection then B is finite and $\#B \leq \#A$.*

Proof. If $B = \emptyset$ then B is finite and $\#B = 0 \leq \#A$ proving the corollary in this case. If $B \neq \emptyset$ then by [theorem: 6.9] there exist an injection $g: B \rightarrow A$, leading by [theorem: 2.68] to a bijection $g: B \rightarrow g(B)$, hence $B \approx g(B)$. As $g(B) \subseteq A$ we have by [theorem: 6.43] that $g(B)$ is finite and $\#g(B) \leq \#A$. Finally as $\#g(B) \approx B$ and $B \approx g(B)$ it follows that $\#B = \#g(B) \leq \#A$. $\square$

Corollary 6.45. *If A, B are sets, A finite and $f: A \rightarrow B$ a function then $f(A)$ is finite and $\#(f(A)) \leq \#A$.*

Proof. As $f: A \rightarrow f(A)$ is a surjection it follows from the previous theorem [theorem: 6.44]. $\square$

Theorem 6.46. *Let I be a finite set and $\{x_i\}_{i \in I} \subseteq X$ a finite family of elements in X then $\{x_i | i \in I\}$ is finite and $\#(\{x_i | i \in I\}) \leq \#I$*

Proof. Define the function $f: I \rightarrow \{x_i | i \in I\}$ by $f(i) = x_i$ then if $y \in \{x_i | i \in I\}$ there exist a $i \in I$ such that $y = x_i$, hence $y = f(i)$. This proves that $f: I \rightarrow \{x_i | i \in I\}$ is a surjection, hence using [corollary: 6.44] we have as I is finite that $\{x_i | i \in I\}$ is finite and $\#(\{x_i | i \in I\}) \leq \#I$. $\square$

Theorem 6.47. Let A, B be sets, A infinite and $f: A \rightarrow B$ a injection then B is infinite.

Proof. Assume that B is finite then $f(A) \subseteq B$ is finite and there is a bijection $g: n \rightarrow f(A)$, as $f: A \rightarrow f(A)$ is a bijection we have that $f^{-1}: f(A) \rightarrow A$ is a bijection so that $f^{-1} \circ g: n \rightarrow A$ is a bijection, hence A is finite, contradicting the fact that A is infinite. So the assumption is wrong hence B is infinite. $\square$

Corollary 6.48. Let I be a infinite set and $\{A_i\}_{i \in I}$ a family such that $\forall i \in I$ we have $A_i \neq \emptyset$ and $\forall i, j \in I$ with $i \neq j$ we have $A_i \cap A_j \neq \emptyset$ then $\bigcup_{i \in I} A_i$ is infinite.

Proof. Using the Axiom of Choice [see 3.106] there exist a function $f: I \rightarrow \bigcup_{i \in I} A_i$ such that $\forall i \in I$ $f(i) \in A_i$. Let $i, j \in I$ such that $f(i) = f(j)$ then $f(i) \in A_i$ and $f(j) = f(i) \in A_j$ so that $A_i \cap A_j \neq \emptyset$, hence $i = j$ [for if $i \neq j$ then $A_i \cap A_j = \emptyset$]. Hence $f: I \rightarrow \bigcup_{i \in I} A_i$ is a injection which by [theorem: 6.47] proves that $\bigcup_{i \in I} A_i$ is infinite. $\square$

Theorem 6.49. Let $\langle X, \leq \rangle$ be a totally ordered set, $\emptyset \neq A \subseteq X$ a finite set then $\max(A)$ and $\min(A)$ exists.

Proof. We prove this by induction on $\#A$, so let

$$S = \{n \in \{1, \dots, \infty\} | \text{If } A \subseteq X \text{ with } \#A = n \text{ then } \max(A) \text{ and } \min(A) \text{ exists}\}$$

then we have:

$1 \in S$. As $\#A = 1 = \{0\}$ there exists a bijection $f: \{0\} \rightarrow A$ so that $A = \{f(0)\}$ and $\max(A) = f(0) = \min(A)$.

$n \in S \Rightarrow n + 1 \in S$. Let $A \subseteq X$ with $\#A = n + 1$ then $n + 1 = s(n) = n \cup \{n\}$, so that there exists a bijection $f: n \cup \{n\} \rightarrow A$. If $n \in n$ then $n < n$ a contradiction so we have $n \notin n$. Take now

$$f|_n: n \rightarrow A \setminus \{f(n)\}$$

then by [theorem: 2.92] $f|_n$ is injective. Further if $y \in A \setminus \{f(n)\}$ then, as f is a bijection, there exists a $i \in n + 1$ such that $f(i) = y$, we can not have $i = n$ [because then $f(i) = f(n)$], so $i \neq n \Rightarrow i \in n$, proving that $f|_n(i) = f(i) = y$. Hence $f|_n: n \rightarrow A \setminus \{f(n)\}$ is a surjection, which together with injectivity proving that

$$f|_n: n \rightarrow A \setminus \{f(n)\} \text{ is a bijection hence } \#(A \setminus \{f(n)\}) = n$$

As $n \in S$ we have that $M = \max(A \setminus \{f(n)\})$ and $m = \min(A \setminus \{f(n)\})$ exists. We have now for $M, f(n)$ to consider the following possibilities::

$M \leq f(n)$. Then $\forall x \in A \setminus \{f(n)\}$ we have $x \leq M \leq f(n) \Rightarrow x \leq f(n)$ and for $x = f(n)$ $x \leq f(n)$. So $\forall x \in A$ we have $x \leq f(n)$, proving that $\max(A)$ exist and $\max(A) = f(n)$.

$f(n) < M$. Then $\forall x \in A$ we have $x \leq M$ so that $\max(A)$ exist and $\max(A) = M$

For $m, f(n)$ we need to consider:

$m \leq f(n)$. Then $\forall x \in A$ we have $m \leq x$ so that $\min(A)$ exist and $\min(A) = m$.

$f(n) < m$. Then $\forall x \in A \setminus \{f(n)\}$ we have $m \leq x$ so that $f(n) < m$ and for $x = f(n)$ $x \leq f(n)$. So $\forall x \in A$ we have $f(n) \leq x$ proving that $\min(A)$ exist and that $f(n) = \min(A)$.

As $\min(A)$ and $\max(A)$ exist it follows that $n + 1 \in S$

Using induction [see theorem: 5.83] it follows that $\{1, \dots, \infty\} = S$. Assume now that $\emptyset \neq A \subseteq X$ such that A is finite we must have that $\#A \in \{1, \dots, \infty\}$ [for if $\#A = 0$ then $A = \emptyset$], so that $\min(A)$ and $\max(A)$ exist. $\square$

Theorem 6.50. *If A is a finite set and $f: \mathbb{N}_0 \rightarrow A$ a function then $\exists a \in A$ such that $f^{-1}(\{a\})$ is infinite.*

Proof. Assume that $\forall a \in A$ $f^{-1}(\{a\})$ is finite. As A is finite we have for the family $\{f^{-1}(\{a\})\}_{a \in A}$ by [theorem: 6.36] that $\bigcup_{a \in A} f^{-1}(\{a\})$ is finite. Now

$$\begin{aligned} x \in \bigcup_{a \in A} f^{-1}(\{a\}) &\Leftrightarrow \exists a \in A \text{ such that } x \in f^{-1}(\{a\}) \\ &\Leftrightarrow \exists a \in A \text{ such that } f(x) \in \{a\} \\ &\Leftrightarrow \exists a \in A \text{ such that } f(x) = a \\ &\Leftrightarrow x \in f^{-1}(A) \end{aligned}$$

So that $\mathbb{N}_0 = f^{-1}(A) = \bigcup_{a \in A} f^{-1}(\{a\})$ from which it follows that $\mathbb{N}_0$ is finite contradicting the fact that $\mathbb{N}_0$ is infinite [by theorem: 6.31]. So the assumption is wrong, hence $\exists a \in A$ such that $f^{-1}(\{a\})$ is infinite. $\square$

Corollary 6.51. *If A is finite and $f: \mathbb{N}_0 \rightarrow A$ a function then $\exists a \in A$ such that $\forall n \in \mathbb{N}_0$ there exist a $m \in \{n, \dots, \infty\}$ so that $f(m) = a$.*

Proof. By the preceding theorem [theorem: 6.50] there exist a $a \in A$ such that $f^{-1}(\{a\})$ is infinite. Assume now that $\exists n \in \mathbb{N}_0$ such that $\forall m \in \{n, \dots, \infty\}$ we have $f(m) \neq a$. If $m \in f^{-1}(\{a\})$ then $f(m) \in \{a\} \Rightarrow f(m) = a$, so we must have that $m \notin \{n, \dots, \infty\}$, hence $m < n$ or $m \in S_n$. So we have proved that $f^{-1}(\{a\}) \subseteq S_n$ a finite set, giving by [theorem: 6.43] that $f^{-1}(\{a\})$ is finite contradicting the fact that $f^{-1}(\{a\})$ is infinite. So the assumption must be wrong, hence $\forall n \in \mathbb{N}_0$ there exists a $m \in \{n, \dots, \infty\}$ such that $f(m) = a$. $\square$

6.2.2 Finite families

We show now that every finite family of elements of a totally ordered set can be sorted.

Theorem 6.52. *Let $\langle X, \leq \rangle$ be a totally ordered set, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in S_{n+1}} \subseteq X$ then there exists a bijection $\beta: S_{n+1} \rightarrow S_{n+1}$ such that $\forall i \in S_n$ we have $x_{\beta(i)} \leq x_{\beta(n)}$.*

Proof. We prove this by induction, so let

$S = \{n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in S_{n+1}} \subseteq X \text{ there exist a bijection } \beta: S_{n+1} \rightarrow S_{n+1} \text{ such that } \forall i \in S_n \ x_{\beta(i)} \leq x_{\beta(n)}\}$ then we have:

0 $\in S$. If $\{x_i\}_{i \in S_1 = \{0\}} \subseteq X$ then for the bijection $\beta = \text{Id}_{S_1}: S_1 \rightarrow S_1$ we have $\forall i \in S_0 = \emptyset$ that $x_{\beta(i)} \leq x_{\beta(0)}$ is satisfied vacuously, proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in S_{(n+1)+1}} \subseteq X$ then for $\{x_i\}_{i \in S_{n+1}}$ we have, as $n \in S$, the existence of a bijection $\alpha: S_{n+1} \rightarrow S_{n+1}$ such that $\forall i \in S_n \ x_{\alpha(i)} \leq x_{\alpha(n)}$. For x_{n+1} we have now two cases to consider:

$x_{\alpha(n)} \leq x_{n+1}$. Define

$$\beta: S_{(n+1)+1} \rightarrow S_{(n+1)+1} \text{ by } \beta(i) = \begin{cases} \alpha(i) & \text{if } i \in S_{n+1} \\ n+1 & \text{if } i = n+1 \end{cases}$$

then we have:

injectivity. Let $i, j \in S_{(n+1)+1}$ be such that $\beta(i) = \beta(j)$ then we have the following possibilities:

$i \in S_{n+1} \wedge j \in S_{n+1}$. Then $\alpha(i) = \beta(i) = \beta(j) = \alpha(j)$ which as α is a bijection proves that $i = j$.

$i \in S_{n+1} \wedge j = n+1$. Then $\alpha(i) = \beta(i) = \beta(j) = n+1$ from which it follows that $n+1 = \alpha(i) \in S_{n+1}$ giving the contradiction $n+1 < n+1$. So this case never occurs.

$i = n + 1 \wedge j \in S_{n+1}$. Then $n + 1 = \beta(i) = \beta(j) = \alpha(j)$ from which it follows that $n + 1 = \alpha(j) \in S_{n+1}$ giving the contradiction $n + 1 < n + 1$. So this case never occurs.

$i = n + 1 \wedge j = n + 1$. Then $i = j$

surjectivity. If $j \in S_{(n+1)+1}$ then we have the following possibilities:

$j = n + 1$. Then $n + 1 = \beta(n + 1)$.

$j \in S_n$. Then as α is a bijection there exist a $i \in S_n$ such that $j = \alpha(i) \xRightarrow{i \in S_n} j = \beta(i)$.

So $\beta: S_{(n+1)+1} \rightarrow S_{(n+1)+1}$ is a bijection. Let now $i \in S_{n+1}$ then we have the following possibilities:

$i = n$. Then $x_{\beta(i)} = x_{\alpha(i)} = x_{\alpha(n)} \leq x_{n+1} = x_{\beta(n+1)}$.

$i \in S_n$. Then $x_{\beta(i)} = x_{\alpha(i)} \leq x_{\alpha(n)} \leq x_{n+1} = x_{\beta(n+1)}$.

which proves that in this case we have $n + 1 \in S$.

$x_{n+1} < x_{\alpha(n)}$. Define

$$\beta: S_{(n+1)+1} \rightarrow S_{(n+1)+1} \text{ by } \beta(i) = \begin{cases} \alpha(i) & \text{if } i \in S_n \\ n + 1 & \text{if } i = n \\ \alpha(n) & \text{if } i = n + 1 \end{cases}$$

then we have:

injectivity. Let $i, j \in S_{(n+1)+1}$ such that $\beta(i) = \beta(j)$ then we have the following possibilities:

$i \in S_n \wedge j \in S_n$. Then $\alpha(i) = \beta(i) = \beta(j) = \alpha(j)$ which as β is a bijection gives $i = j$.

$i \in S_n \wedge j = n$. Then $\alpha(i) = \beta(i) = \beta(j) = n + 1$ so that $n + 1 = \alpha(i) \in S_{n+1}$ giving the contradiction $n + 1 < n + 1$, so this case never occurs.

$i \in S_n \wedge j = n + 1$. Then $\alpha(i) = \beta(i) = \beta(j) = \alpha(n)$, which as α is a bijection, gives $i = n$ contradicting $i \in S_n \Rightarrow i < n$, so this case never occurs.

$i = n \wedge j \in S_n$. Then $n + 1 = \beta(i) = \beta(j) = \alpha(j)$ so that $n + 1 = \alpha(j) \in S_{n+1}$ giving the contradiction $n + 1 < n + 1$, so this case never occurs.

$i = n \wedge j = n$. Then $i = j$.

$i = n \wedge j = n + 1$. Then $n + 1 = \beta(i) = \beta(j) = \alpha(n)$ so that $n + 1 = \alpha(n) \in S_{n+1}$ giving the contradiction $n + 1 < n + 1$, so this case never occurs.

$i = n + 1 \wedge j \in S_n$. Then $\alpha(n) = \beta(i) = \beta(j) = \alpha(j)$, which as α is a bijection gives $n = j \in S_n$ resulting in the contradiction $n < n$, so this case never occurs.

$i = n + 1 \wedge j = n$. Then $\alpha(n) = \beta(i) = \beta(j) = n + 1$ so that $n + 1 = \alpha(n) \in S_{n+1}$ leading to the contradiction $n + 1 < n + 1$, so this case never occurs.

$i = n + 1 \wedge j = n + 1$. Then $i = j$.

surjectivity. Let $j \in S_{(n+1)+1}$ then we have the following possibilities to check:

$j = n + 1$. then $\beta(n) = j$

$j \in S_{n+1}$. then as α is a bijection there exist a $i \in S_{n+1}$ so that $\alpha(i) = j$. If $i = n$ then $\beta(n + 1) = \alpha(n) = j$ and if $i \in S_n$ then $\beta(i) = \alpha(i) = j$.

So $\beta: S_{(n+1)+1} \rightarrow S_{(n+1)+1}$ is a bijection. Let now $i \in S_{n+1}$ then we have to consider the following possibilities:

$i = n$. Then $x_{\beta(i)} = x_{n+1} \leq x_{\alpha(n)} = x_{\beta(n+1)}$.

$i \in S_n$. Then $x_{\beta(i)} = x_{\alpha(i)} \leq x_{\alpha(n)} = x_{\beta(n+1)}$,

which proves that in this case $n+1 \in S$.

Mathematical induction [see theorem: 5.83] proves then that $S = \mathbb{N}_0$. $\square$

Theorem 6.53. Let $\langle X, \leq \rangle$ be a totally ordered set, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in S_{n+1}} \subseteq X$ then there exists a bijection $\beta: S_{n+1} \rightarrow S_{n+1}$ such that

$$\forall i \in S_n \text{ we have } x_{\beta(i)} \leq x_{\beta(i+1)}$$

Proof. We proof this by induction, so let

$$S = \{n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in n+1} \subseteq X \text{ there exist a bijection } \beta: S_{n+1} \rightarrow S_{n+1} \text{ such that } \forall i \in S_n \ x_{\beta(i)} \leq x_{\beta(i+1)}\}$$

then we have:

$0 \in S$. Then $S_0 = \emptyset$ and $S_1 = \{0\}$. Let $\{x_i\}_{i \in S_1 = \{0\}} \subseteq X$ then, for the bijection $\beta: S_1 \rightarrow S_1$ where $\beta = \text{Id}_{S_1}$, we have that, $\forall i \in S_0 = \emptyset \ x_{\beta(i)} \leq x_{\beta(i+1)}$, is satisfied vacuously.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in S_{(n+1)+1}} \subseteq X$ then by the previous theorem [theorem: 6.52] there exists a bijection

$$\alpha: S_{(n+1)+1} \rightarrow S_{(n+1)+1} \text{ such that } \forall i \in S_{n+1} \ x_{\alpha(i)} \leq x_{\alpha(n+1)} \quad (6.13)$$

Consider now $\{x_{\alpha(i)}\}_{i \in S_{n+1}}$ then as $n \in S$ we have the existence of a bijection

$$\gamma: S_{n+1} \rightarrow S_{n+1} \text{ such that } \forall i \in S_n \text{ we have } x_{\alpha(\gamma(i))} \leq x_{\alpha(\gamma(i+1))} \quad (6.14)$$

Define now

$$\beta: S_{(n+1)+1} \rightarrow S_{(n+1)+1} \text{ by } \beta(i) = \begin{cases} \alpha(\gamma(i)) & \text{if } i \in S_{n+1} \\ \alpha(n+1) & \text{if } i = n+1 \end{cases}$$

then we have:

injectivity. Let $k, l \in S$ be such that $\beta(k) = \beta(l)$ then we must consider the following possibilities:

$k \in S_{n+1} \wedge l \in S_{n+1}$. Then

$$(\alpha \circ \gamma)(k) = \alpha(\gamma(k)) = \beta(k) = \beta(l) = \alpha(\gamma(l)) = (\alpha \circ \gamma)(l)$$

which as $\alpha \circ \gamma$ is a bijection proves that $k = l$.

$k \in S_{n+1} \wedge l = n+1$. Then $\alpha(n+1) = \beta(l) = \beta(k) = \alpha(\gamma(k))$ which, as α is a bijection, gives $n+1 = \gamma(k)$, as $\gamma(k) \in S_{n+1} \Rightarrow \gamma(k) < n+1$ we reach the contradiction $n+1 < n+1$, so this case never occurs.

$k = n+1 \wedge l \in S_{n+1}$. Then $\alpha(n+1) = \beta(k) = \beta(l) = \alpha(\gamma(l))$ which, as α is a bijection, gives $n+1 = \gamma(l)$, as $\gamma(l) \in S_{n+1} \Rightarrow \gamma(l) < n+1$, we reach the contradiction $n+1 < n+1$, so this case never occurs.

$k = n+1 \wedge l = n+1$. then $k = l$

surjectivity. If $k \in S_{(n+1)+1}$ we have, as α is a bijection, that there exist a $l \in S_{(n+1)+1}$ such that $\alpha(l) = k$, for l we have then the following possibilities:

$l = n+1$. Then $\beta(n+1) = \alpha(n+1) = k$

$l \in S_{n+1}$. Then as γ is a bijection there exist a $i \in S_{n+1}$ such that $l = \gamma(i)$, hence $\beta(i) = \alpha(\gamma(i)) = \alpha(l) = k$.

Further if $i \in S_{n+1}$ we have the following possibilities to consider:

$i = n$. Then $\gamma(n) \in S_{n+1}$ so that by [eq: 6.13] $x_{\alpha(\gamma(i))} \leq x_{\alpha(n+1)} = x_{\beta(n+1)}$ hence

$$x_{\beta(i)} = x_{\alpha(\gamma(i))} \leq x_{\beta(n+1)} = x_{\beta(i+1)}$$

$i \in S_n$. Then by [eq: 6.14] we have $x_{\alpha(\gamma(i))} \leq x_{\alpha(\gamma(i+1))}$ so that

$$x_{\beta(i)} = x_{\alpha(\gamma(i))} \leq x_{\alpha(\gamma(i+1))} = x_{\beta(i+1)}$$

Hence $\forall i \in S_{n+1}$ we have $x_{\beta(i)} \leq x_{\beta(i+1)}$ proving that $n+1 \in S$.

Mathematical induction [see theorem: 5.83] proves that $S = \mathbb{N}_0$ and thus the theorem. $\square$

Corollary 6.54. *Let $\langle X, \leq \rangle$ be a totally ordered set, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq X$ then there exist a bijection $\alpha: \{n, \dots, m\} \rightarrow \{n, \dots, m\}$ such that $\forall i \in \{n, \dots, m-1\} \ x_{\alpha(i)} \leq x_{\alpha(i+1)}$*

Proof. Using [theorem: 6.18] there exists bijections

$$\beta: \{n, \dots, m\} \rightarrow S_{(m-n)+1} \text{ where } \beta(i) = i - n \quad (6.15)$$

and

$$\beta^{-1}: S_{(m-n)+1} \rightarrow \{n, \dots, m\} \text{ where } \beta(i) = i + n \quad (6.16)$$

Let $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq X$ then for $\{x_{\beta^{-1}(i)}\}_{i \in S_{(m-n)+1}}$ we have by [theorem: 6.53] a bijection

$$\gamma: S_{(m-n)+1} \rightarrow S_{(m-n)+1} \text{ such that } \forall i \in S_{m-n} \text{ we have } x_{\beta^{-1}(\gamma(i))} \leq x_{\beta^{-1}(\gamma(i+1))} \quad (6.17)$$

Define now the bijection

$$\alpha = \beta^{-1} \circ \gamma \circ \beta: \{n, \dots, m\} \rightarrow \{n, \dots, m\}$$

If $k \in \{n, \dots, m-1\}$ then $n \leq k \leq m-1 < m$ so that by [theorem: 5.70] we have $0 \leq k - n < m - n$ or $0 \leq \beta(k) < m - n$. So $\beta(k) \in S_{m-n}$ and thus by [eq: 6.17] we have that

$$x_{\beta^{-1}(\gamma(\beta(k)))} \leq x_{\beta^{-1}(\gamma(\beta(k)+1))}$$

Now $\beta(k+1) = (k+1) - n \stackrel{[\text{theorem: 5.65}]}{=} (k-n) + 1 = \beta(k) + 1$ so that by the above we have

$$x_{\beta^{-1}(\gamma(\beta(k)))} \leq x_{\beta^{-1}(\gamma(\beta(k)+1))} \quad (6.18)$$

Hence

$$\begin{aligned} x_{\alpha(k)} &= x_{(\beta^{-1} \circ \gamma \circ \beta)(k)} \\ &= x_{\beta^{-1}(\gamma(\beta(k)))} \\ &\stackrel{[\text{eq: 6.18}]}{\leq} x_{\beta^{-1}(\gamma(\beta(k)+1))} \\ &= x_{(\beta^{-1} \circ \gamma \circ \beta)(k+1)} \\ &= x_{\alpha(k+1)} \end{aligned}$$

So we have found a bijection $\alpha: \{n, \dots, m\} \rightarrow \{n, \dots, m\}$ such that $\forall k \in \{n, \dots, m-1\} \ x_{\alpha(k)} \leq x_{\alpha(k+1)}$ $\square$

Corollary 6.55. *Let $\langle X, \leq \rangle$ be a totally ordered set, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq X$ then there exist a bijection $\alpha: \{n, \dots, m\} \rightarrow \{n, \dots, m\}$ such that $\forall i, j \in \{n, \dots, m\}$ with $i \leq j$ we have $x_{\alpha(i)} \leq x_{\alpha(i+1)}$*

Proof. Using [corollary: 6.54] there exist a bijection $\alpha: \{n, \dots, m\} \rightarrow \{n, \dots, m\}$ such that

$$\forall i \in \{n, \dots, m-1\} \text{ we have } x_{\alpha(i)} \leq x_{\alpha(i+1)}$$

We proceed now by induction so let

$$S = \{k \in \{n, \dots, \infty\} | \text{If } k \in \{n, \dots, m\} \text{ then } \forall i \in \{n, \dots, k\} \text{ we have } x_{\sigma(i)} \leq x_{\sigma(k)}\}$$

then we have:

$n \in S$. As $\forall i \in \{n, \dots, k\} = \{n\}$ we have $i = n$, it follows that $x_{\sigma(i)} = x_{\sigma(n)} = x_{\sigma(k)}$ so that $n \in S$.

$k \in S \Rightarrow k+1 \in S$. If $k+1 \in \{n, \dots, m\}$ then $\{n, \dots, k+1\} \subseteq \{n, \dots, m\}$. Let $i \in \{n, \dots, k+1\}$ then for i we have either:

$i = k+1$. Then trivially $x_{\sigma(i)} = x_{\sigma(k+1)} \leq x_{\sigma(k+1)}$

$i \in \{n, \dots, k\}$. Then as $k \in S$ and $\{n, \dots, k\} \subseteq \{n, \dots, k+1\} \subseteq \{n, \dots, m\}$ it follows that $x_{\sigma(i)} < x_{\sigma(k)}$, further as $k \in \{n, \dots, m\}$ we have $x_{\sigma(k)} \leq x_{\sigma(k+1)}$, hence we have $x_{\sigma(i)} \leq x_{\sigma(k+1)}$.

so in all cases we have $x_{\sigma(i)} \leq x_{\sigma(k+1)}$, proving that $k+1 \in S$.

Mathematical induction proves that $S = \{n, \dots, \infty\}$. So if $i, j \in \{n, \dots, m\}$ with $i \leq j$ then $j \in S$, as $i \in \{n, \dots, j\}$ it follows from the definition of S that $x_{\sigma(i)} \leq x_{\sigma(j)}$. $\square$

The next theorem allows use later to apply induction on the product of a finite family of sets.

Theorem 6.56. *Let $n \in \mathbb{N}_0$ and let $\{A_i\}_{i \in S_{n+1}}$ a family of sets then*

$$\prod_{i \in S_{n+1}} A_i \approx \left(\prod_{i \in S_n} A_i \right) \times A_n$$

Proof. If $x \in \prod_{i \in S_{n+1}} A_i$ then $x \in (\bigcup_{i \in S_{n+1}} A_i)^{S_{n+1}}$ such that $\forall i \in S_{n+1}$ we have $x(i) \in A_i$ or equivalently $x: S_{n+1} \rightarrow \bigcup_{i \in S_{n+1}} A_i$ is a function so that $\forall i \in S_{n+1}$ we have $x(i) \in A_i$. As $\forall i \in S_n$ we have $x(i) \in A_i \subseteq \bigcup_{i \in S_n} A_i$, it follows that $x|_{S_n}: S_n \rightarrow \bigcup_{i \in S_n} A_i$ is a function, so $x|_{S_n} \in \prod_{i \in S_n} A_i$. Hence we can define the following function

$$\beta: \left(\prod_{i \in S_{n+1}} A_i \right) \rightarrow \left(\prod_{i \in S_n} A_i \right) \times A_n \text{ by } \beta(x) \rightarrow (x|_{S_n}, x(n))$$

Then we have:

injectivity. If $\beta(x) = \beta(y)$ then $(x|_{S_n}, x(n)) = (y|_{S_n}, y(n))$ or $x|_{S_n} = y|_{S_n}$ and $x(n) = y(n)$. So if $i \in S_{n+1}$ we have either $i \in S_n$ then $x(i) = x|_{S_n}(i) = y|_{S_n}(i) = y(i)$ or $i = n$ and then $x(i) = x(n) = y(n) = y(i)$, proving that $x = y$.

surjectivity. Let $(y, a) \in (\prod_{i \in S_n} A_i) \times A_n$ then $y \in \prod_{i \in S_n} A_i$ and $a \in A_n$. Define then the function:

$$x: S_{n+1} \rightarrow \bigcup_{i \in S_{n+1}} A_i \text{ by } x(i) = \begin{cases} y(i) & \text{if } i \in S_n \\ a & \text{if } i = n \end{cases}$$

Then $\forall i \in S_{n+1}$ we have either $i \in S_n$ giving $x(i) = y(i) \in A_i$ or $i = n$ giving $x(i) = x(n) = a \in A_n$, proving that $x \in \prod_{i \in S_{n+1}} A_i$. Further as clearly $x|_{S_n} = y$ and $x(n) = a$ we have that $\beta(x) = y$. $\square$

We use the above theorem to prove that the product of a finite family of finite sets is finite.

Theorem 6.57. *Let $n \in \mathbb{N}_0 \setminus \{0\}$ and $\{A_i\}_{i \in S_n}$ be such that $\forall i \in S_n$ A_i is finite then $\prod_{i \in S_n} A_i$ is finite.*

Proof. we proof this by induction so define

$$S = \left\{ n \in \{1, \dots, \infty\} \mid \text{If } \{A_i\}_{i \in S_n} \text{ satisfies } \forall i \in S_n \text{ } A_i \text{ is finite then } \prod_{i \in S_n} A_i \text{ is finite} \right\}$$

then we have:

$1 \in S$. Using [example: 2.147] there exist a bijection $\beta: A_0 \rightarrow \prod_{i \in \{0\}} A_i$, hence as $S_1 = \{0\}$ $A_0 \approx \prod_{i \in S_1} A_i$. As A_0 is finite there exist a $k \in \mathbb{N}_0$ such that $k \approx A_0$ proving that $k \approx \prod_{i \in S_0} A_i$ or that $\prod_{i \in S_1} A_i$ is finite. So $1 \in S$.

$n \in S$ then $n+1 \in S$. Let $\{A_i\}_{i \in S_{n+1}}$ be such that that $\forall i \in S_{n+1}$ we have that A_i is finite. As $n \in S$ we have that $\prod_{i \in S_n} A_i$ is finite so using [theorem: 6.41] it follows that $(\prod_{i \in S_n} A_i) \times A_n$ is finite. Hence $\exists k \in \mathbb{N}_0$ such that $k \approx (\prod_{i \in S_n} A_i) \times A_n$. Using [theorem: 6.56] we have $(\prod_{i \in S_n} A_i) \times A_n \approx \prod_{i \in S_{n+1}} A_i$ proving that $k \approx \prod_{i \in S_{n+1}} A_i$. Hence $\prod_{i \in S_{n+1}} A_i$ is finite proving that $n+1 \in S$.

Using mathematical induction it follows that $S = \{1, \dots, \infty\}$ proving the theorem. $\square$

Corollary 6.58. *Let I be a non empty finite set and $\{A_i\}_{i \in I}$ is such that $\forall i \in I$ we have A_i is finite then $\prod_{i \in I} A_i$ is finite.*

Proof. As I is finite and $I \neq \emptyset$ there exists a $n \in \mathbb{N}_0 \setminus \{0\}$ such that $k \approx I$, so there exist a bijection $f: S_k \rightarrow I$. Using [theorem: 2.153] we have that there exists a bijection $\beta: \prod_{i \in I} A_i \rightarrow \prod_{i \in S_k} A_{f(i)}$ hence $\prod_{i \in I} A_i \approx \prod_{i \in S_k} A_{f(i)}$. By [theorem: 6.57] we have that $\prod_{i \in S_k} A_{f(i)}$ is finite so there exists a $m \in \mathbb{N}_0$ such that $m \approx \prod_{i \in S_k} A_{f(i)}$, hence $m \approx \prod_{i \in I} A_i$, proving that $\prod_{i \in I} A_i$ is finite. $\square$

6.2.3 Denumerable sets

Lemma 6.59. *Every subset of $\mathbb{N}_0$ is either finite or denumerable*

Proof. By [theorem: 5.51] $\langle \mathbb{N}_0, \leq \rangle$ is a well ordered set, hence by [theorem: 3.100] we have for $N \subseteq \mathbb{N}_0$ either:

1. N is order isomorphic with $\mathbb{N}_0$ hence $N \approx \mathbb{N}_0$ proving that N is denumerable.
2. N is order isomorphic with a initial segment of $\mathbb{N}_0$ so there exists a $n \in \mathbb{N}_0$ such that $N \approx S_n$ proving that N is finite. $\square$

Theorem 6.60. *Every subset of a denumerable set is finite or denumerable.*

Proof. Let A be a denumerable set and $B \subseteq A$. As A is denumerable there exists a bijection

$$\beta: A \rightarrow \mathbb{N}_0$$

Using [theorem: 2.92] and [theorem: 2.68] we have that $\beta|_B: B \rightarrow \beta(B)$ is a bijection so that

$$\beta(B) \approx B$$

as $\beta(B) \subseteq \mathbb{N}$ we have by the previous lemma [lemma: 6.59] that either:

- $\beta(B) \approx \mathbb{N}_0$. Then by [theorem: 6.2] $B \approx \mathbb{N}_0$ proving that B is denumerable.
- $\beta(B)$ is finite. Then there exists a $n \in \mathbb{N}_0$ such that $\beta(B) \approx n$, by [theorem: 6.2] $B \approx n$ proving that B is finite. $\square$

Theorem 6.61. $\mathbb{N}_0 \times \mathbb{N}_0 \approx \mathbb{N}_0$, in other words $\mathbb{N}_0 \times \mathbb{N}_0$ is denumerable.

Proof. First define the function

$$f: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0 \text{ where } f(k, m) = \begin{cases} (0, k+1) & \text{if } m=0 \\ (k+1, m-1) & \text{if } m \in \mathbb{N}_0 \setminus \{0\} \end{cases}$$

If $f(k, m) = f(k', m')$ we have the following cases for m, m'

- $m = 0 \wedge m' = 0$. Then $m = m'$ and $(0, k+1) = f(k, m) = f(k', m') = (0, k'+1)$ so that $k+1 = k'+1 \xRightarrow{[\text{theorem: 5.43}]} k = k'$ hence $(k, m) = (k', m')$.
- $m = 0 \wedge m' \in \mathbb{N}_0 \setminus \{0\}$. Then $(0, k+1) = f(k, m) = f(k', m') = (k'+1, m'-1)$ so that $0 = k'+1$ which as $0 < s(k') = k'+1$ is a contradiction, so this case does not occur.
- $m \in \mathbb{N}_0 \setminus \{0\} \wedge m' = 0$. Then $(k+1, m-1) = f(k, m) = f(k', m') = (0, k'+1)$ so that $0 = k+1$ which as $< s(k) = k+1$ is a contradiction, so this case does not occur.
- $m \in \mathbb{N}_0 \setminus \{0\} \wedge m' \in \mathbb{N}_0 \setminus \{0\}$. Then $(k+1, m-1) = f(k, m) = f(k', m') = (k'+1, m'-1)$ so that $k+1 = k'+1 \xRightarrow{[\text{theorem: 5.43}]} k = k'$ and $m-1 = m'-1 \xRightarrow{[\text{theorem: 5.43}]} m = (m-1)+1 = (m'-1)+1 = m'$ so that $(k, m) = (k', m')$

The above proves that

$$f: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0 \text{ is injective} \tag{6.19}$$

Assume that $f(k, m) = (0, 0)$ then if $m = 0$ we have $(0, 0) = (0, k+1)$ giving the contradiction $0 = k+1$ and if $m \neq 0$ we have $(k+1, m-1)$ giving the contradiction $0 = k+1$. So the assumption is incorrect hence

$$(0, 0) \notin f(\mathbb{N}_0 \times \mathbb{N}_0) \tag{6.20}$$

Using [eq: 6.19] and [eq: 6.20] we can use recursion [see theorem: 5.84] to get a **injective** function

$$\lambda: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0 \text{ such that } \lambda(0) = (0, 0) \text{ and } \forall n \in \mathbb{N}_0 \text{ we have } \lambda(n+1) = f(\lambda(n))$$

We prove now the following proposition about λ :

Proposition 6.62. *If there exist a $n, m \in \mathbb{N}_0$ such that $\lambda(n) = (0, m)$ then if $k, l \in \mathbb{N}_0$ is such that $k + l = m$ we have $\lambda(n + k) = (k, l)$.*

Proof. We proof this by induction so let

$$S_{n,m} = \{k \in \mathbb{N}_0 | \text{For } l \in \mathbb{N}_0 \text{ with } k + l = m \text{ we have } \lambda(n + k) = (k, l)\}$$

then we have:

0 $\in S_{n,m}$. If $l \in \mathbb{N}_0$ such that $k + l = m$ then $l = m$ and $\lambda(n + k) = \lambda(n) = (0, m) \stackrel{k=0 \wedge l=m}{=} (k, l)$ proving that $0 \in S_{n,m}$.

$k \in S_{n,m} \Rightarrow k + 1 \in S_{n,m}$. If $l \in \mathbb{N}_0$ such that $(k + 1) + l = m$ then we have $k + (l + 1) = m$ and as $k \in S_{n,m}$ it follows that

$$\lambda(n + k) = (k, l + 1) \tag{6.21}$$

Further

$$\begin{aligned} \lambda(n + (k + 1)) &= \lambda((n + k) + 1) \\ &= f(\lambda(n + k)) \\ &\stackrel{[\text{eq: 6.21}]}{=} f(k, l + 1) \\ &\stackrel{l+1 \neq 0}{=} (k + 1, (l + 1) - 1) \\ &\stackrel{[\text{theorem: 5.66}]}{=} (k + 1, l) \end{aligned}$$

proving that $k + 1 \in S_{n,m}$.

Using induction [theorem: 5.83] it follows that $S_{n,m} = \mathbb{N}_0$ proving the proposition. $\square$

We prove now using induction that $\lambda: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0$ is surjective. So let

$$S = \{n \in \mathbb{N}_0 | \text{For } (k, m) \in \mathbb{N}_0 \times \mathbb{N}_0 \text{ with } k + m = n \text{ there exists a } l \in \mathbb{N}_0 \text{ such that } \lambda(l) = (k, m)\}$$

0 $\in S$. If $(k, m) \in \mathbb{N}_0 \times \mathbb{N}_0$ is such that $k + m = 0$ then we must have $k = m = 0$, as $\lambda(0) = (0, 0) = (k, l)$ we have $0 \in S$.

$n \in S$ then $n + 1 \in S$. Let $(k, m) \in \mathbb{N}_0$ be such that $k + m = n + 1$, then for k we have to consider the following cases:

$k = 0$. Then $m = k + m = n + 1$ so that $(k, m) = (0, m) = (0, n + 1) = f(n, 0)$. As $n \in S$ and $n = n + 0$ there exist a $l \in \mathbb{N}_0$ such that $\lambda(l) = (n, 0)$. So

$$\lambda(l + 1) = f(\lambda(l)) = f(n, 0) = (0, n + 1) \stackrel{k=0}{=} (k, m)$$

$k \neq 0$. Then $0 < k$ so that $0 \leq k - 1$, further as $0 \neq m + 1$ we have that

$$f(k - 1, m + 1) = ((k - 1) + 1, (m + 1) - 1) = (k, m)$$

Let $k' = (k + m) - 1 \stackrel{k+m=n+1}{=} (n + 1) - 1 = n$ and $l' = 0$ then $k' + l' = n$ so that, as $n \in S$, there exist a $l \in \mathbb{N}_0$ such that

$$\lambda(l) = (k', l') = ((k + m) - 1, 0) \tag{6.22}$$

Hence

$$\begin{aligned} \lambda(l + 1) &= f(\lambda(l)) \\ &\stackrel{[\text{eq: 6.22}]}{=} f((k + m) - 1, 0) \\ &= (0, k + m) \end{aligned}$$

Combining the above with [proposition: 6.62] we have that $\lambda((l+1)+k) = (k, m)$, so that $n+1 \in S$.

By mathematical induction [theorem: 5.83] it follows that $S = \mathbb{N}_0$. So if $(k, m) \in \mathbb{N}_0 \times \mathbb{N}_0$ we have that $k+m \in \mathbb{N}_0 = S$ so that $\exists n \in \mathbb{N}_0 \lambda(n) = (k, m)$ which proves that λ is a surjection. Hence as λ is also injective it follows that $\lambda: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0$ is a bijection, proving that $\mathbb{N}_0 \times \mathbb{N}_0$ is denumerable. $\square$

Corollary 6.63. *If A, B are denumerable then $A \times B$ is denumerable*

Proof. As A, B are denumerable we have $\mathbb{N}_0 \approx A$ and $\mathbb{N}_0 \approx B$, proving by [theorem: 6.12] that $\mathbb{N}_0 \times \mathbb{N}_0 \approx A \times B$. Finally as $\mathbb{N}_0 \approx \mathbb{N}_0 \times \mathbb{N}_0$ it follows that $\mathbb{N}_0 \approx A \times B$. $\square$

Corollary 6.64. *If $n \in \mathbb{N}_0 \setminus \{0\}$ then $n \times \mathbb{N}_0$ is denumerable*

Proof. As $n = S_n \subseteq \mathbb{N}_0$ we have by [theorem: 1.55] that $n \times \mathbb{N}_0 \subseteq \mathbb{N}_0 \times \mathbb{N}_0$ so that by [theorem: 6.60]

$$n \times \mathbb{N}_0 \text{ is either finite or denumerable}$$

As $n \neq 0$ we have that $n \neq \emptyset$ so there exist a $m \in n$, define then

$$\beta: \mathbb{N}_0 \rightarrow \{m\} \times \mathbb{N}_0 \text{ by } \beta(i) = (m, i)$$

then we have:

injectivity. If $\beta(i) = \beta(i')$ then $(m, i) = (m, i')$ giving $i = i'$

surjectivity. If $(x, y) \in \{m\} \times \mathbb{N}_0$ then $x = m$ and $y \in \mathbb{N}_0$ so that $\beta(y) = (m, y) = (x, y)$

So $\beta: \mathbb{N}_0 \rightarrow \{m\} \times \mathbb{N}_0$ is a bijection proving that $\{m\} \times \mathbb{N}_0$ is denumerable. As $\{m\} \times \mathbb{N}_0 \subseteq n \times \mathbb{N}_0$ it follows by [theorem: 6.29] that $n \times \mathbb{N}_0$ is not finite so $n \times \mathbb{N}_0$ must be denumerable. $\square$

Corollary 6.65. *If A is a non empty finite set and B a denumerable set then $A \times B$ and $B \times A$ are denumerable sets.*

Proof. As $A \neq \emptyset$ and finite there exist a $n \in \mathbb{N}_0 \setminus \{0\}$ such that $n \approx A$, as B is denumerable $\mathbb{N}_0 \times B$ we have by [theorem: 6.12] that

$$n \times \mathbb{N}_0 \approx A \times B$$

which as $\mathbb{N}_0 \approx \mathbb{N}_0 \times \mathbb{N}_0$ [see corollary: 6.64] proves that $\mathbb{N}_0 \approx A \times B$, hence

$$A \times B \text{ is denumerable}$$

Define the function

$$\beta: A \times B \rightarrow B \times A \text{ by } \beta(x, y) = (y, x)$$

then we have

injectivity. If $\beta(x, y) = \beta(x', y')$ then $(y, x) = \beta(x, y) = \beta(x', y') = (y', x')$ so that $x = x' \wedge y = y'$ proving that $(x, y) = (x', y')$.

surjectivity. If $(x, y) \in B \times A$ we have that $(y, x) \in A \times B$ so that $\beta(y, x) = (x, y)$.

proving that

$$\beta: A \times B \rightarrow B \times A \text{ is a bijection}$$

hence $A \times B \approx B \times A$, which as $A \times B \approx \mathbb{N}_0$ proves that

$$B \times A \text{ is denumerable.} \quad \square$$

Theorem 6.66. *Let $n \in \mathbb{N}_0 \setminus \{0\}$ and $\{A_i\}_{i \in S_n}$ such that $\forall i \in S_n$ A_i is denumerable then $\prod_{i \in S_n} A_i$ is denumerable.*

Proof. We proof this by induction, so define

$$S = \left\{ n \in \{1, \dots, \infty\} \mid \text{If } \{A_i\}_{i \in S_n} \text{ satisfies } \forall i \in S_n \text{ } A_i \text{ is denumerable then } \prod_{i \in S_n} A_i \text{ is denumerable} \right\}$$

then we have:

$1 \in S$. As $S_1 = \{0\}$ we can use [example: 2.147] to find a bijection $\beta: A_0 \rightarrow \prod_{i \in \{0\}} A_i = \prod_{i \in S_1} A_i$ proving that $A_0 \approx \prod_{i \in S_1} A_i$, hence $\prod_{i \in S_1} A_i$ is denumerable [see theorem: 6.26].

$n \in S \Rightarrow n+1 \in S$. Let $\{A_i\}_{i \in S_{n+1}}$ be such that $\forall i \in S_{n+1}$ A_i is denumerable. As $n \in S$ we have that $\prod_{i \in S_n} A_i$ is denumerable, so by [theorem: 6.63] we have that $(\prod_{i \in S_n} A_i) \times A_n$ is denumerable. Finally by [theorem: 6.56] we have $\prod_{i \in S_{n+1}} A_i \approx (\prod_{i \in S_n} A_i) \times A_n$ so that $\prod_{i \in S_{n+1}} A_i$ is denumerable [see theorem: 6.26]. Hence $n+1 \in S$.

Mathematical induction proves then that $S = \{1, \dots, \infty\}$ proving the theorem. $\square$

Corollary 6.67. *If I is non empty and finite and $\{A_i\}_{i \in I}$ such that $\forall i \in I$ A_i is denumerable then $\prod_{i \in I} A_i$ is denumerable.*

Proof. As I is finite and non empty there exists a $n \in \mathbb{N}_0 \setminus \{0\}$ such that $n \approx I$ hence there exist a bijection $f: n = S_n \rightarrow I$, Using [theorem: 2.153] there exists a bijection $\beta: \prod_{i \in I} A_i \rightarrow \prod_{i \in S_n} A_{f(i)}$ so that $\prod_{i \in S_n} A_{f(i)} \approx \prod_{i \in I} A_i$. Using the previous theorem [theorem: 6.66] $\prod_{i \in S_n} A_{f(i)}$ is denumerable, hence by [theorem: 6.26] $\prod_{i \in I} A_i$ is denumerable. $\square$

Theorem 6.68. *If $\{A_i\}_{i \in I}$ is such that $I \neq \emptyset \wedge I$ is finite and $\forall i \in I$ A_i is denumerable then $\bigcup_{i \in I} A_i$ is denumerable. In other words the union of a finite family of denumerable sets is denumerable.*

Proof. As I is finite and non empty there exist $n_0 \in \mathbb{N}_0 \setminus \{0\}$ and a bijection $\beta: n_0 \rightarrow I$. Further as $\forall i \in I$ A_i is denumerable there exist a bijection $\alpha_i: \mathbb{N}_0 \rightarrow A_i$. Define now the function

$$g: n_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ by } g(n, m) = \alpha_{\beta(n)}(m)$$

Now if $y \in \bigcup_{i \in I} A_i$ there exist a $l \in I$ such that $y \in A_l$, as β is bijective there exists a $n \in n_0$ such that $\beta(n) = l$. As $\alpha_l: \mathbb{N}_0 \rightarrow A_l$ is a bijection there exist a $m \in \mathbb{N}_0$ such that $\alpha_l(m) = y$. So

$$g(n, m) = \alpha_{\beta(n)}(m) = \alpha_l(m) = y$$

proving that

$$g: n_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Now by [theorem: 6.64] there exist a bijection $\gamma: \mathbb{N}_0 \rightarrow n_0 \times \mathbb{N}_0$ so that

$$g \circ \gamma: \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Using [theorem: 6.10] we have that $\bigcup_{i \in I} A_i \preceq \mathbb{N}_0$ which by [definition: 6.3] gives that $\exists E \subseteq \mathbb{N}_0$ such that $\bigcup_{i \in I} A_i \approx E$. Using [theorem: 6.60] we have that E is either finite or E is denumerable so that $\bigcup_{i \in I} A_i$ is either finite or denumerable. As $n_0 \neq 0 \Rightarrow 0 < n_0$ we have $0 \in S_{n_0} = n_0$, so that $\beta(0) \in I$, hence $A_{\beta(0)} \subseteq \bigcup_{i \in I} A_i$, which, as $A_{\beta(0)}$ is denumerable, proves by [theorem: 6.29] that $\bigcup_{i \in I} A_i$ is not finite. So we must have that $\bigcup_{i \in I} A_i$ is denumerable. $\square$

Theorem 6.69. *If $\{A_i\}_{i \in I}$ is such that I is denumerable and $\forall i \in I$ A_i is denumerable then $\bigcup_{i \in I} A_i$ is denumerable. In other words every union of a denumerable family of denumerable sets is denumerable.*

Proof. As I is denumerable there exist a bijection $\beta: \mathbb{N}_0 \rightarrow I$. Further as $\forall i \in I$ A_i is denumerable there exist a bijection $\alpha_i: \mathbb{N}_0 \rightarrow A_i$. Define now the function

$$g: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ by } g(n, m) = \alpha_{\beta(n)}(m)$$

Now if $y \in \bigcup_{i \in I} A_i$ there exist a $l \in I$ such that $y \in A_l$, as β is bijective there exists a $n \in \mathbb{N}_0$ such that $\beta(n) = l$. As $\alpha_l: \mathbb{N}_0 \rightarrow A_l$ is a bijection there exist a $m \in \mathbb{N}_0$ such that $\alpha_l(m) = y$. So

$$g(n, m) = \alpha_{\beta(n)}(m) = \alpha_l(m) = y$$

proving that

$$g: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Now by [theorem: 6.61] there exist a bijection $\gamma: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0$ so that

$$g \circ \gamma: \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Using [theorem: 6.10] we have that $\bigcup_{i \in I} A_i \preceq \mathbb{N}_0$ which by [definition: 6.3] gives that $\exists E \subseteq \mathbb{N}_0$ such that $\bigcup_{i \in I} A_i \approx \mathbb{N}_0$. Using [theorem: 6.60] we have that E is either finite or E is denumerable so that $\bigcup_{i \in I} A_i$ is either finite or denumerable. As $A_{\beta(0)} \subseteq \bigcup_{i \in I} A_i$ and $A_{\beta(0)}$ is denumerable it follows from [theorem: 6.29] that $\bigcup_{i \in I} A_i$ is not finite. So we must have that $\bigcup_{i \in I} A_i$ is enumerable. $\square$

6.2.4 Countable Sets

Remember that a countable set is a set that is either finite or denumerable.

Theorem 6.70. *Every subset of a denumerable set is countable*

Proof. This follows from [theorem: 6.60] and the definition of countable sets. $\square$

Theorem 6.71. *Every subset of a countable set is countable*

Proof. If A is countable then A is either denumerable or finite. If A is finite then by [theorem: 6.43] every subset of A is finite hence countable. If A is denumerable then by [theorem: 6.70] every subset of A is countable. $\square$

Corollary 6.72. *If A is a countable set, B a set and $f: A \rightarrow B$ a surjection then B is countable.*

Proof. As A is countable we have either:

A is finite. Then using [theorem: 6.44] B is finite hence countable.

A is denumerable. If $B = \emptyset$ then B is finite hence countable proving the corollary in this case. If $B \neq \emptyset$ then by [theorem: 6.9] there exist an injection $g: B \rightarrow A$, leading by [theorem: 2.68] to a bijection $g: B \rightarrow g(B)$, hence $B \approx g(B)$. As $g(B) \subseteq A$ it follows from [theorem: 6.71] that $g(B)$ is countable. Finally as $g(B) \approx B$ it follows from [theorem: 6.26] that B is countable. $\square$

Corollary 6.73. *If A, B are sets, A countable and $f: A \rightarrow B$ a function then $f(A)$ is countable.*

Proof. As $f: A \rightarrow f(A)$ is a surjection it follows from the previous corollary [corollary: 6.72] that $f(A)$ is countable. $\square$

Corollary 6.74. *Let I be a countable set and $\{x_i\}_{i \in I} \subseteq X$ a countable family of elements in X then $\{x_i | i \in I\}$ is countable.*

Proof. Define the function $f: I \rightarrow \{x_i | i \in I\}$ by $f(i) = x_i$ then if $y \in \{x_i | i \in I\}$ there exist a $i \in I$ such that $y = x_i$, hence $y = f(i)$. This proves that $f: I \rightarrow \{x_i | i \in I\}$ is a surjection, hence using [corollary: 6.72] we have as I is finite that $\{x_i | i \in I\}$ is finite $\square$

Theorem 6.75. *Let A be a non empty set then the following are equivalent:*

1. A is countable

2. There exists a surjection $\beta: \mathbb{N}_0 \rightarrow A$
3. There exists an injection $\alpha: A \rightarrow \mathbb{N}_0$
4. There exist a denumerable set B and an injection $\alpha: A \rightarrow B$

Proof.

1 $\Rightarrow$ 2. If A is countable then we have either:

A is finite. Then $\exists n \in \mathbb{N}_0$ and a bijection $\alpha: n = S_n \rightarrow A$. As $A \neq \emptyset$ there exist a $a \in A$, this allows us to define the function

$$\beta: \mathbb{N}_0 \rightarrow A \text{ where } \beta(i) = \begin{cases} \alpha(i) & \text{if } i < n \\ a & \text{if } n \leq i \end{cases}$$

If $y \in A$ then as α is surjective we have that $\exists i \in S_n = n$ such that $\alpha(i) = y$ so that $\beta(i) = \alpha(i) = y$ proving that $\beta: \mathbb{N}_0 \rightarrow A$ is surjective.

A is denumerable. Then $\mathbb{N}_0 \approx A$ so there exist a bijection, hence surjection, $\beta: \mathbb{N}_0 \rightarrow A$.

2 $\Rightarrow$ 3. Given that there exists a surjection $\beta: \mathbb{N}_0 \rightarrow A$ and $A \neq \emptyset$ we have by [theorem: 6.9] the existence of an injection $\alpha: A \rightarrow \mathbb{N}_0$.

3 $\Rightarrow$ 4. As B is denumerable we have $\mathbb{N}_0 \approx B$ so there exist a bijection $\beta: \mathbb{N}_0 \rightarrow B$. by (3) there exist an injection $\alpha: A \rightarrow \mathbb{N}_0$, hence we have the injection $\beta \circ \alpha: A \rightarrow B$.

4 $\Rightarrow$ 1. As B is denumerable there exist a bijection $\beta: B \rightarrow \mathbb{N}_0$ so that we have an injection $\beta \circ \alpha: A \rightarrow \mathbb{N}_0$. Using [theorem: 2.68] it follows that $\beta \circ \alpha: A \rightarrow (\beta \circ \alpha)(A) \subseteq \mathbb{N}_0$ is a bijection hence

$$A \approx (\beta \circ \alpha)(A) \subseteq \mathbb{N}_0$$

Using [theorem: 6.59] we have that $(\beta \circ \alpha)(A)$ is either finite or denumerable. If $(\beta \circ \alpha)(A)$ is finite then there exist a $n \in \mathbb{N}_0$ such that $n \approx (\beta \circ \alpha)(A)$, hence $n \approx A$ proving that A is finite, hence countable. If $(\beta \circ \alpha)(A)$ is denumerable then $\mathbb{N}_0 \approx (\beta \circ \alpha)(A)$ so that $\mathbb{N}_0 \approx A$ proving that A is denumerable hence countable. So we reach the conclusion that A is countable. $\square$

Theorem 6.76. If $\{A_i\}_{i \in I}$ is such that I is finite and $\forall i \in I$ A_i is countable then $\bigcup_{i \in I} A_i$ is countable.

Proof. As I is finite we have either:

$I = \emptyset$. Then by [example: 2.129] $\bigcup_{i \in I} A_i = \emptyset$ which is finite hence countable.

$I \neq \emptyset$. As I is finite and non empty there exist a $n_0 \in \mathbb{N}_0 \setminus \{0\}$ and a bijection $\beta: n_0 \rightarrow I$. Further as $\forall i \in I$ A_i is countable there exist a surjection $\alpha_i: \mathbb{N}_0 \rightarrow A_i$ [see theorem: 6.75]. Define now the function

$$g: n_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ by } g(n, m) = \alpha_{\beta(n)}(m)$$

Now if $y \in \bigcup_{i \in I} A_i$ there exist a $l \in I$ such that $y \in A_l$, as β is bijective there exists a $n \in \mathbb{N}_0$ such that $\beta(n) = l$. As $\alpha_l: \mathbb{N}_0 \rightarrow A_l$ is a surjection there exist a $m \in \mathbb{N}_0$ such that $\alpha_l(m) = y$. So

$$g(n, m) = \alpha_{\beta(n)}(m) = \alpha_l(m) = y$$

which proves that

$$g: n_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Now by [theorem: 6.64] there exist a bijection $\gamma: \mathbb{N}_0 \rightarrow n_0 \times \mathbb{N}_0$ so that

$$g \circ \gamma: \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Using [theorem: 6.75] it follows that $\bigcup_{i \in I} A_i$ is countable.

So in all cases we have that

$$\bigcup_{i \in I} A_i \text{ is countable}$$

□

Theorem 6.77. *If $\{A_i\}_{i \in I}$ is such that I is denumerable and $\forall i \in I$ A_i is countable then $\bigcup_{i \in I} A_i$ is countable. In other words every union of a denumerable family of countable sets is countable.*

Proof. As I is denumerable there exist a bijection $\beta: \mathbb{N}_0 \rightarrow I$. Further as $\forall i \in I$ A_i is countable there exist a surjection $\alpha_i: \mathbb{N}_0 \rightarrow A_i$ [see theorem: 6.75]. Define now the function

$$g: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ by } g(n, m) = \alpha_{\beta(n)}(m)$$

Now if $y \in \bigcup_{i \in I} A_i$ there exist a $l \in I$ such that $y \in A_l$, as β is bijective there exists a $n \in \mathbb{N}_0$ such that $\beta(n) = l$. As $\alpha_l: \mathbb{N}_0 \rightarrow A_l$ is a surjection there exist a $m \in \mathbb{N}_0$ such that $\alpha_l(m) = y$. So

$$g(n, m) = \alpha_{\beta(n)}(m) = \alpha_l(m) = y$$

which proves that

$$g: \mathbb{N}_0 \times \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Now by [theorem: 6.61] there exist a bijection $\gamma: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \times \mathbb{N}_0$ so that

$$g \circ \gamma: \mathbb{N}_0 \rightarrow \bigcup_{i \in I} A_i \text{ is surjective}$$

Using [theorem: 6.75] it follows that $\bigcup_{i \in I} A_i$ is countable. □

We can summarize the two previous theorems as follows

Theorem 6.78. *If $\{A_i\}_{i \in I}$ is such that I is a countable set and $\forall i \in I$ A_i is countable then $\bigcup_{i \in I} A_i$ is countable.*

Proof. As I is countable we have either:

I is finite. Then by [theorem: 6.76] we have that $\bigcup_{i \in I} A_i$ is countable.

I is denumerable. Then by [theorem: 6.77] we have that $\bigcup_{i \in I} A_i$ is countable. □

The following theorem shows when a union of a denumerable family of countable sets is denumerable.

Theorem 6.79. *If $\{A_i\}_{i \in I}$ is such that I is denumerable and*

1. $\forall i \in I$ A_i is countable and $A_i \neq \emptyset$
2. $\forall i, j \in I$ with $i \neq j$ we have $A_i \cap A_j = \emptyset$

then

$$\bigcup_{i \in I} A_i \text{ is denumerable}$$

Proof. Using [theorem: 6.77] it follows that $\bigcup_{i \in I} A_i$ is countable. As I is denumerable I is infinite [see corollary: 6.30] hence using [theorem: 6.48] $\bigcup_{i \in I} A_i$ is infinite. As a set is countable if it is finite or denumerable we must conclude that $\bigcup_{i \in I} A_i$ is denumerable. □

Corollary 6.80. *If $\{n_i\}_{i \in \mathbb{N}} \subseteq \mathbb{N}$ is a sequence of natural numbers then*

$$\bigcup_{i \in \mathbb{N}} \{i\} \times \{1, \dots, n_i\}$$

is denumerable.

Proof. Let $i \in \mathbb{N}$ then by [theorem: 6.41] $\{i\} \times \{1, \dots, n_i\}$ is finite and non empty. Further if $i, j \in \mathbb{N}$ with $i \neq j$ then we must have that $(\{i\} \times \{1, \dots, n_i\}) \cap (\{j\} \times \{1, \dots, n_j\}) = \emptyset$ [for if $x \in (\{i\} \times \{1, \dots, n_i\}) \cap (\{j\} \times \{1, \dots, n_j\})$ we have that $x = (i, r) = (j, s) \Rightarrow i = j$ contradicting $i \neq j$]. Hence

$$\{\{i\} \times \{1, \dots, n_i\}\}_{i \in \mathbb{N}} \text{ is pairwise disjoint}$$

So using the previous theorem [theorem: 6.79] it follows that

$$\bigcup_{i \in \mathbb{N}} \{i\} \times \{1, \dots, n_i\} \text{ is denumerable} \quad \square$$

Theorem 6.81. *If A, B are countable sets then we have $A \times B$ is countable.*

Proof. For A, B we have the following possibilities:

A is finite and B is finite. Then by [theorem: 6.41] $A \times B$ is finite hence countable.

A is finite and B is denumerable. Then by [theorem: 6.65] $A \times B$ is denumerable hence countable.

A is denumerable and B is finite. Then by [theorem: 6.65] $A \times B$ is denumerable hence countable.

A is denumerable and B is denumerable. Then by [theorem: 6.63] $A \times B$ is denumerable hence countable. $\square$

Lemma 6.82. *Let $n \in \mathbb{N}$ and $\{A_i\}_{i \in S_n}$ such that $\forall i \in S_n$ A_i is countable then $\prod_{i \in S_n} A_i$ is countable.*

Proof. We proof this by induction, so define

$$S = \left\{ n \in \{1, \dots, \infty\} \mid \text{If } \{A_i\}_{i \in S_n} \text{ satisfies } \forall i \in S_n \text{ } A_i \text{ is countable then } \prod_{i \in S_n} A_i \text{ is countable} \right\}$$

then we have:

$1 \in S$. As $S_1 = \{0\}$ we can use [example: 2.147] to find a bijection $\beta: A_0 \rightarrow \prod_{i \in \{0\}} A_i = \prod_{i \in S_1} A_i$ proving that $A_0 \approx \prod_{i \in S_1} A_i$, hence $\prod_{i \in S_1} A_i$ is countable [see theorem: 6.26].

$n \in S \Rightarrow n+1 \in S$. Let $\{A_i\}_{i \in S_{n+1}}$ be such that $\forall i \in S_{n+1}$ A_i is countable. As $n \in S$ we have that $\prod_{i \in S_n} A_i$ is countable, so by [theorem: 6.81] we have that $(\prod_{i \in S_n} A_i) \times A_n$ is countable. Finally by [theorem: 6.56] we have $\prod_{i \in S_{n+1}} A_i \approx (\prod_{i \in S_n} A_i) \times A_n$ so that $\prod_{i \in S_{n+1}} A_i$ is countable [see theorem: 6.26]. Hence $n+1 \in S$

Mathematical induction proves then that $S = \{1, \dots, \infty\}$ proving the theorem. $\square$

Theorem 6.83. *If I is non empty and finite and $\{A_i\}_{i \in I}$ such that $\forall i \in I$ A_i is countable then $\prod_{i \in I} A_i$ is countable.*

Proof. As I is finite and non empty there exists a $n \in \mathbb{N}_0 \setminus \{0\}$ such that $n \approx I$ hence there exist a bijection $f: n = S_n \rightarrow I$, Using [theorem: 2.153] there exists a bijection $\beta: \prod_{i \in I} A_i \rightarrow \prod_{i \in S_n} A_{f(i)}$ so that $\prod_{i \in S_n} A_{f(i)} \approx \prod_{i \in I} A_i$. Using the previous lemma [lemma: 6.82] $\prod_{i \in S_n} A_{f(i)}$ is countable, hence by [theorem: 6.26] $\prod_{i \in I} A_i$ is countable. $\square$

6.3 Finite product of sets

We turn now our attention to the finite product of sets. Using the general definition of a product of a family of sets as is discussed in [definition: 2.145] we can define the finite product of sets.

Definition 6.84. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq B$ a finite family of sets then $\prod_{i=1}^n A_i$ is defined as

$$\prod_{i=1}^n A_i = \prod_{i \in \{1, \dots, n\}} A_i \text{ [see definition: 2.145]}$$

In other words

$$\prod_{i=1}^n A_i \text{ [definition: 2.145]} = \left\{ f \mid f \in \left(\bigcup_{i \in \{1, \dots, n\}} A_i \right)^{\{1, \dots, n\}} \text{ where } \forall i \in \{1, \dots, n\} \text{ we have } f(i) \in A_i \right\}$$

So if $x \in \prod_{i=1}^n A_i$ then $x: \{1, \dots, n\} \rightarrow \bigcup_{i \in \{1, \dots, n\}} A_i$ is a function with $\forall i \in \{1, \dots, n\} x(i) \in A_i$. As x_i is another notation for $x(i)$ we can introduce a new notation for $x \in \prod_{i=1}^n A_i$.

Notation 6.85. $x \in \prod_{i=1}^n A_i$ is noted as $(x_1, \dots, x_n)$ which is equivalent with saying that $x: \{1, \dots, n\} \rightarrow \bigcup_{i \in \{1, \dots, n\}} A_i$ is a function with $\forall i \in \{1, \dots, n\} x_i = x(i) \in A_i$. Using this new notation we have the much shorter specification of $\prod_{i=1}^n A_i$.

$$x \in \prod_{i=1}^n A_i \Leftrightarrow x = (x_1, \dots, x_n) \text{ and } \forall i \in \{1, \dots, n\} \text{ we have } x_i \in A_i$$

Using the above notation and definition we can also rephrase the projection operators [see definition: 2.156]

Definition 6.86. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq B$ a finite family of sets then $\forall i \in \{1, \dots, n\}$

$$\pi_i: \prod_{i=1}^n A_i \rightarrow A_i \text{ is defined by } \pi_i(x) = \pi_i(x_1, \dots, x_i) = x_i$$

Theorem 6.87. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq B$ a finite family of sets then $\forall i \in \{1, \dots, n\}$ we have that $\pi_i: \prod_{i=1}^n A_i \rightarrow A_i$ is a surjection

Proof. This was proved in [theorem: 3.109] □

If we have only two sets involved we use the following notation to distinguish the product of sets from the Cartesian product.

Definition 6.88. Let A, B two sets then for the family $\{X_i\}_{i \in \{1, 2\}} \subseteq A \cup B$ where $X_1 = A \wedge X_2 = B$ we have

$$A \cdot B = \prod_{i \in \{1, \dots, 2\}} X_i = \prod_{i \in \{1, 2\}} X_i$$

so

$$A \cdot B = \{f \mid f \in (A \cup B)^{\{1, 2\}} \text{ where } f(1) \in A \wedge f(2) \in B\}$$

We use the notation $A \cdot B$ to make a distinction with the Cartesian product $A \times B$, of course by [theorem: 2.148] we have that $A \cdot B$ and $A \times B$ are bijective.

Theorem 6.89. Let A, B, C, D be sets then we have

1. $(A \cdot B) \cap (C \cdot D) = (A \cap C) \cdot (B \cap D)$
2. If $A \subseteq B$ then $A \cdot C \subseteq B \cdot C$
3. If $B \subseteq C$ then $A \cdot B \subseteq A \cdot C$
4. $A \cdot (B \cup C) = (A \cdot B) \cup (A \cdot C)$
5. $(A \cup B) \cdot C = (A \cdot C) \cup (B \cdot C)$
6. $A \cdot \emptyset = \emptyset = \emptyset \cdot A$

Proof.

1. This follows from the definition of $\cdot$ and [theorem: 2.150]

2. This follows from the definition of $\cdot$ $C \subseteq C$ and [theorem: 2.149]
3. This follows from the definition of $\cdot$ $A \subseteq A$ and [theorem: 2.149]
4. As $B, C \subseteq B \cup C$ we have by (3) that $A \cdot B \subseteq A \cdot (B \cup C)$ and $A \cdot C \subseteq A \cdot (B \cup C)$ so that

$$(A \cdot B) \cup (A \cdot C) \subseteq A \cdot (B \cup C) \quad (6.23)$$

For the opposite inclusion let $x \in A \cdot (B \cup C)$ then $x: \{1, 2\} \rightarrow A \cup (B \cup C)$ is a function so that $x(1) \in A \wedge x(2) \in B \cup C$, so that either $x(1) \in A \wedge x(2) \in B$ or $x(1) \in A \wedge x(2) \in C$ proving that $x \in A \cdot B$ or $x \in A \cdot C$, hence $x \in (A \cdot B) \cup (A \cdot C)$, This proves that $A \cdot (B \cup C) \subseteq (A \cdot B) \cup (A \cdot C)$ which combined with [eq: 6.23] results in

$$A \cdot (B \cup C) = (A \cdot B) \cup (A \cdot C)$$

5. As $A, B \subseteq A \cup B$ we have by (2) that $A \cdot C \subseteq (A \cup B) \cdot C$ and $B \cdot C \subseteq (A \cup B) \cdot C$ so that

$$(A \cdot C) \cup (B \cdot C) \subseteq (A \cup B) \cdot C \quad (6.24)$$

For the opposite inclusion let $x \in (A \cup B) \cdot C$ then $x: \{1, 2\} \rightarrow (A \cup B) \cup C$ is a function so that $x(1) \in (A \cup B) \wedge x(2) \in C$, so that either $x(1) \in A \wedge x(2) \in C$ or $x(1) \in B \wedge x(2) \in C$ proving that $x \in A \cdot C$ or $x \in B \cdot C$, hence $x \in (A \cdot C) \cup (B \cdot C)$, This proves that $(A \cup B) \cdot C \subseteq (A \cdot C) \cup (B \cdot C)$ which combined with [eq: 6.24] results in

$$(A \cup B) \cdot C = (A \cdot C) \cup (B \cdot C)$$

6. This follows from the definition of $\cdot$ and [theorem: 2.146]. □

We have the following extension for [theorem: 6.89(4),(5)].

Theorem 6.90. *Let $\{A_i\}_{i \in I}$ be a family of sets then we have:*

1. *If B is a set then*

$$B \cdot \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (B \cdot A_i)$$

and

$$\left(\bigcup_{i \in I} A_i \right) \cdot B = \bigcup_{i \in I} (A_i \cdot B)$$

2. *If $\{B_j\}_{j \in J}$ be another family of sets then*

$$\left(\bigcup_{i \in I} A_i \right) \cdot \left(\bigcup_{j \in J} B_j \right) = \bigcup_{i \in I} \left(\bigcup_{j \in J} (A_i \cdot B_j) \right) = \bigcup_{(i,j) \in I \times J} (A_i \cdot B_j)$$

Proof.

1. Let $x \in B \cdot \left(\bigcup_{i \in I} A_i \right)$ then $x: \{1, 2\} \rightarrow B \cup \left(\bigcup_{i \in I} A_i \right)$ is a function such that $x(1) \in B \wedge x(2) \in \bigcup_{i \in I} A_i$, hence there exist a $i \in I$ such that $x(2) \in A_i$, so $x: \{1, 2\} \rightarrow B \cup A_i$ is a function such that $x(1) \in B \wedge x(2) \in A_i$. Hence $x \in B \cdot A_i \subseteq \bigcup_{i \in I} B \cdot A_i$ which proves that

$$B \cdot \left(\bigcup_{i \in I} A_i \right) \subseteq \bigcup_{i \in I} (B \cdot A_i) \quad (6.25)$$

If $x \in \bigcup_{i \in I} (B \cdot A_i)$ then there exist a $i \in I$ such that $x \in B \cdot A_i$. Hence $x: \{1, 2\} \rightarrow B \cup A_i$ is a function with $x(1) \in B \wedge x(2) \in A_i \subseteq B \cdot \left(\bigcup_{i \in I} A_i \right)$, which proves that $x \in B \cdot \left(\bigcup_{i \in I} A_i \right)$. Hence $\bigcup_{i \in I} (B \cdot A_i) \subseteq B \cdot \left(\bigcup_{i \in I} A_i \right)$, which combined with [eq: 6.25] results in

$$B \cdot \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (B \cdot A_i) \quad (6.26)$$

Let $x \in (\bigcup_{i \in I} A_i) \cdot B$ then $x: \{1, 2\} \rightarrow (\bigcup_{i \in I} A_i) \cup B$ is a function such that $x(1) \in \bigcup_{i \in I} A_i \wedge x(2) \in B$, hence there exist a $i \in I$ such that $x(1) \in A_i$, so $x: \{1, 2\} \rightarrow A_i \cup B$ is a function such that $x(1) \in A_i \wedge x(2) \in B$. Hence $x \in A_i \cdot B \subseteq \bigcup_{i \in I} A_i \cdot B$ which proves that

$$\left(\bigcup_{i \in I} A_i \right) \cdot B \subseteq \bigcup_{i \in I} (A_i \cdot B) \quad (6.27)$$

If $x \in \bigcup_{i \in I} (A_i \cdot B)$ then there exist a $i \in I$ such that $x \in A_i \cdot B$. Hence $x: \{1, 2\} \rightarrow A_i \cup B$ is a function with $x(1) \in A_i \wedge x(2) \in B \subseteq (\bigcup_{i \in I} A_i) \cdot B$, which proves that $x \in (\bigcup_{i \in I} A_i) \cdot B$. Hence $\bigcup_{i \in I} (A_i \cdot B) \subseteq (\bigcup_{i \in I} A_i) \cdot B$, which combined with [eq: 6.27] results in

$$\left(\bigcup_{i \in I} A_i \right) \cdot B = \bigcup_{i \in I} (A_i \cdot B) \quad (6.28)$$

2. We have

$$\begin{aligned} \left(\bigcup_{i \in I} A_i \right) \cdot \left(\bigcup_{j \in J} B_j \right) &\stackrel{(1)}{=} \bigcup_{i \in I} \left(A_i \cdot \left(\bigcup_{j \in J} B_j \right) \right) \\ &\stackrel{(1)}{=} \bigcup_{i \in I} \left(\bigcup_{j \in J} (A_i \cdot B_j) \right) \\ &\stackrel{[\text{theorem: 2.143}]}{=} \bigcup_{(i,j) \in I \times J} (A_i \cdot B_j) \end{aligned}$$

□

The following theorem will be useful in induction arguments.

Theorem 6.91. *Let $n \in \mathbb{N}$ and $\{X_i\}_{i \in \{1, \dots, n\}} \subseteq B$ a finite family of sets then $\prod_{i \in \{1, \dots, n+1\}} X_i$ is bijective with $(\prod_{i \in \{1, \dots, n\}} X_i) \times X_{n+1}$*

Proof. Define

$$\beta: \prod_{i \in \{1, \dots, n+1\}} X_i \rightarrow \left(\prod_{i \in \{1, \dots, n\}} X_i \right) \times X_{n+1} \text{ by } \beta(x) = (x|_{\{1, \dots, n\}}, x(n+1))$$

then we have:

injectivity. If $\beta(x) = \beta(y)$ then $(x|_{\{1, \dots, n\}}, x(n+1)) = (y|_{\{1, \dots, n\}}, y(n+1))$, hence $x|_{\{1, \dots, n\}} = y|_{\{1, \dots, n\}}$ and $x(n+1) = y(n+1)$ or $\forall i \in \{1, \dots, n\}$

$$x(i) = x|_{\{1, \dots, n\}}(i) = y|_{\{1, \dots, n\}}(i) = y(i) \wedge x(n+1) = y(n+1)$$

proving that $x = y$.

surjectivity. If $z \in (\prod_{i \in \{1, \dots, n\}} X_i) \times X_{n+1}$ then $z = (z_1, z_2)$ where $z_1 \in \prod_{i \in \{1, \dots, n\}} X_i$ and $z_2 \in X_{n+1}$. Define the function

$$x: \{1, \dots, n+1\} \rightarrow \bigcup_{i \in \{1, \dots, n\}} X_i \text{ by } x(i) = \begin{cases} z_2 \in X_{n+1} & \text{if } i = n+1 \\ z_1(i) \in X_i & \text{if } i \in \{1, \dots, n\} \end{cases} \in X_i$$

then $x \in \prod_{i \in \{1, \dots, n+1\}} X_i$. Further $\forall i \in \{1, \dots, n\}$ $x|_{\{1, \dots, n\}}(i) = x(i) = z_1(i)$ proving that $x|_{\{1, \dots, n\}} = z_1$ so that

$$\beta(x) = (x|_{\{1, \dots, n\}}, x(n+1)) = (z_1, z_2) = z$$

proving surjectivity. □

Next we consider the special case where for $\forall i \in \{1, \dots, n\}$ $A_i = A$ [see also theorem: 2.152]

Definition 6.92. Let $n \in \mathbb{N}$, A a set then A^n is defined by

$$A^n = \prod_{i=1}^n A_i \text{ where } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \{A\} \text{ is defined by } C_A: \{1, \dots, n\} \rightarrow \{A\}$$

so that $\forall i \in \{1, \dots, n\} \ A_i = C_A(i) = A$. So as $A = \bigcup_{i \in \{1, \dots, n\}} A_i$ we have that

$$A^n = \{f \mid f \in A^{\{1, \dots, n\}} \text{ where } \forall i \in \{1, \dots, n\} \text{ we have } f(i) \in A\} = A^{\{1, \dots, n\}}$$

Using [notation: 6.85] we can write also

$$x \in A^n \Leftrightarrow x = (x_1, \dots, x_n) \text{ and } \forall i \in \{1, \dots, n\} \text{ we have } x_i \in A$$

Theorem 6.93. Let $n \in \mathbb{N}$, A a set then $A^n = A^{\{1, \dots, n\}} \stackrel{\text{def}}{=} \{f \mid f: \{1, \dots, n\} \rightarrow A \text{ is a function}\}$

Proof. This follows from [theorem: 2.152] □

Theorem 6.94. If A is a set then

$$\beta: A \rightarrow A^1 \text{ defined by } \beta(a) = \{(1, a)\}$$

is a bijection.

Proof. If $a \in A$ then $(1, a) \in \{1\} \times A = \{1, \dots, 1\} \times A$ so that $\beta(a): \{1, \dots, 1\} \rightarrow A$ is indeed a function, proving that $\beta(A) \subseteq A^1$. Further we have

injectivity. If $\beta(a) = \beta(b)$ then $\{(1, a)\} = \{(1, b)\} \Rightarrow (1, a) = (1, b) \Rightarrow a = b$

surjectivity. If $f \in A^1 = A^{\{1\}}$ then $f: \{1\} \rightarrow A$ is a function, so that $f \subseteq \{1\} \times A$, hence $\exists a \in A$ such that $f = \{(1, a)\} = \beta(a)$ □

Once we have defined the finite power of a set we can consider the finite power of semi-groups

Theorem 6.95. Let $\langle G, + \rangle$ be a semi-group with neutral element 0, $n \in \mathbb{N}$ then if we define

$$+^n: G^n \times G^n \rightarrow G^n \text{ by } x +^n y: \{1, \dots, n\} \rightarrow G \text{ where } (x +^n y)(i) = x(i) + y(i)$$

$$0^n: \{1, \dots, n\} \rightarrow G \text{ by } 0^n(i) = 0$$

or using [notation: 6.85]

$$\forall x, y \in G^n \text{ we have } x +^n y = (x_1 + y_1, \dots, x_n + y_n) \text{ where } (x +^n y)_i = x_i + y_i$$

$$0^n \in G^n \text{ is } \left(\underbrace{0, \dots, 0}_n \right) \text{ so that } (0^n)_i = 0$$

then we have:

1. $\langle G^n, +^n \rangle$ is a semi-group with neutral element 0^n .
2. If $\langle G, + \rangle$ is Abelian semi-group then $\langle G^n, +^n \rangle$ is Abelian.
3. If $\langle G, + \rangle$ is a group where $-x$ is the inverse element of $x \in G$ then $\langle G^n, +^n \rangle$ is a group where for $x \in G^n$ the inverse element

$$\sim x: \{1, \dots, n\} \rightarrow G \text{ is defined by } (\sim x)(i) = -x(i)$$

or using [notation: 6.85]

$$\sim x = \sim(x_1, \dots, x_n) = (-x_1, \dots, -x_n) \text{ so that } (\sim x)_i = -x_i$$

Proof. This was actually proved for a more general case in [theorem: 4.26]. □

Note 6.96. As usual, for the rest of this text we reduce the number of symbols by using $+$, $-$, 0 instead of $+^n$, $\sim$, 0^n .

Chapter 7

The integer numbers

In this chapter we will introduce the set of integers and embed the natural numbers in it. Just as with $\mathbb{N}_0$ we will introduce a order relation, a sum operator, a product operator, neutral elements for addition and multiplication as well as inverse elements for the integers. If we would use different symbols to note these, we introduce a lot of excessive notation clutter. So we use the same symbols for the natural numbers and the integers, and use context to determine the meaning of the symbols involved. A practice also used in programming languages [where it is called 'overloading', the following table should help you in determining the meaning of the different symbols based on the context of there usage.

Context	Expression	Operator
$n, m \in \mathbb{N}_0$	$n+m$	sum in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{N}_0$	$n \cdot m$	product in $\langle \mathbb{N}_0, \cdot \rangle$
$n, m \in \mathbb{N}_0$	$n \leq m$	order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n < m$	strict order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n - m$	subtraction in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{N}_0, \cdot \rangle$
$n \in \mathbb{N}_0$	$-n$	inverse element in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{Z}$	$n+m$	sum in $\langle \mathbb{Z}, + \rangle$
$n, m \in \mathbb{Z}$	$n \cdot m$	product in $\langle \mathbb{Z}, \cdot \rangle$
$n, m \in \mathbb{Z}$	$n \leq m$	order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n < m$	strict order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n - m$	subtraction in $\langle \mathbb{Z}, - \rangle$
$n \in \mathbb{Z}$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{Z}, + \rangle$
$n \in \mathbb{Z}$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{Z}, \cdot \rangle$
$n \in \mathbb{Z}$	$-n$	inverse element in $\langle \mathbb{Z}, + \rangle$

7.1 Definition and arithmetic

One major defect of $\mathbb{N}_0$ is that $n - m$, defined to be the unique natural number such that $(n - m) + m = n$, is only defined for $m \leq n$. If this limitation did not exist then we can easily find a inverse for a number n , just take $-n = 0 - n$, then $(-n) + n = (0 - n) + n = 0$. The purpose of this chapter is to define a new set of numbers, the set of integers, that does not have this defect. One strategy is adding to the set of natural numbers the set of numbers of the form $n - m$ where $n < m$. The numbers of the form $n - m$ where $m \leq n$ is then the set of non negative integers and represent the set of natural numbers and the numbers $n - m$ where $n < m$ forms the set of negative numbers. Of course the expression $n - m$ is only a formal expression and is not defined in $\mathbb{N}_0$ if $n < m$ hence we use pairs to of natural numbers to define integers. So a integer is a pair (n, m) where $n, m \in \mathbb{N}_0$ that must be interpreted as the **formal** expression $n - m$ if $n < m$ and the **real** expression $n - m$ if $m \leq n$. However we encounter then another problem, the representations are not **unique**. For example we know that for the natural number 3 we have that $3 = 3 - 0 = 4 - 1 = 5 - 2 = 6 - 3, \dots$, so that $(3, 0), (4, 1), (5, 2), (6, 3), \dots$, must all represent the same number 3. How can we see if

two representations of a natural number are the same? If (n, m) and (n', m') are representations of the same **natural** number then $m \leq n$ and $m' \leq n'$ and we must have

$$\begin{aligned} n - m = n' - m' &\Rightarrow (n - m) + m = (n' - m') + m \\ &\Rightarrow n = (n' - m') + m \\ &\Rightarrow n = m + (n' - m') \\ &\Rightarrow n + m' = (m + (n' - m')) + m' \\ &\Rightarrow n + m' = m + ((n - m') + m') \\ &\Rightarrow n + m' = m + n' \end{aligned}$$

So (n, m) and (n', m') with $m \leq n$ and $m' \leq n'$ represent the same **natural** number if $n + m' = m + n'$. As we don't use subtraction anymore we can extend this test also to the cases where $n < m$ or $n' < m'$. So we say that two representations (n, m) and (n', m') represent the same **integer** if $n + m' = m + n'$. Hence if we define the relation $(n, m) \sim (n', m')$ iff $n + m' = m + n'$ and prove that is a equivalence relation then the equivalence classes will be our integers.

Theorem 7.1. *The relation $\sim \subseteq (\mathbb{N}_0 \times \mathbb{N}_0) \times (\mathbb{N}_0 \times \mathbb{N}_0)$ defined by*

$$\sim = \{((n, m), (n', m')) \mid n + m' = m + n'\}$$

is a equivalence relation.

Proof.

reflexivity. If $(n, m) \in \mathbb{N}_0 \times \mathbb{N}_0$ then $n + m \stackrel{[\text{theorem: 5.33}]}{=} m + n$ so that $(n, m) \sim (n, m)$.

symmetry. If $(n, m) \sim (n', m')$ then $n + m' = m + n' \stackrel{[\text{theorem: 5.33}]}{\Rightarrow} n' + m = m' + n$ so that $(n', m') \sim (n, m)$.

transitivity. We have

$$\begin{aligned} (n, m) \sim (n', m') \wedge (n', m') \sim (n'', m'') &\Rightarrow n + m' = m + n' \wedge n' + m'' = m' + n'' \\ &\Rightarrow (n + m') + (n' + m'') = (m + n') + (m' + n'') \\ &\Rightarrow (n + m'') + (m' + n') = (m + n'') + (n' + m') \\ &\Rightarrow (n + m'') + (n' + m') = (m + n'') + (n' + m') \\ &\Rightarrow (n + m'') = (m + n'') \end{aligned}$$

so that $(n, m) \sim (n'', m'')$. □

Next we define the set of integers.

Definition 7.2. *The set of integers $\mathbb{Z}$ is defined by $(\mathbb{N}_0 \times \mathbb{N}_0) / \sim$ or in other words*

$$\mathbb{Z} = \{ \sim[(n, m)] \mid (n, m) \in \mathbb{N}_0 \times \mathbb{N}_0 \}$$

Theorem 7.3. *If $\sim[(n, m)] \in \mathbb{Z}$ then if $k \in \mathbb{N}_0$ we have $\sim[(n, m)] = \sim[(n + k, m + k)]$*

Proof. $n + (m + k) = (n + m) + k = (m + n) + k = m + (n + k)$ so that $(n, m) \sim (n + k, m + k)$. Hence by [theorem: 3.11] $\sim[(n, m)] = \sim[(n + k, m + k)]$. □

Theorem 7.4. *If $\sim[(n, m)], \sim[(r, s)], \sim[(n', m')]$ and $\sim[(r', s')]$ are elements of $\mathbb{Z}$ such that $\sim[(n, m)] = \sim[(n', m')]$ and $\sim[(r, s)] = \sim[(r', s')]$ then $\sim[(n + r, m + s)] = \sim[(n' + r', m' + s')]$*

Proof. As $\sim[(n, m)] = \sim[(n', m')]$ and $\sim[(r, s)] = \sim[(r', s')]$ we have

$$n + m' = m + n' \wedge r + s' = s + r' \tag{7.1}$$

then

$$\begin{aligned} (n + r) + (m' + s') &= (n + m') + (r + s') \\ &\stackrel{[\text{eq: 7.1}]}{=} (m + n') + (s + r') \\ &= (m + s) + (n' + r') \end{aligned}$$

so that $(n+r, m+s) \sim (n'+r', m'+s')$ proving that

$$\sim[(n+r, m+s)] = \sim[(n'+r', m'+s')] \quad \square$$

The above theorem ensure that the following definition is well defined:

Definition 7.5. The sum operator $+: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is defined by

$$\sim[(n, m)] + \sim[(r, s)] = \sim[(n+r, m+s)]$$

Lemma 7.6. If $n \in \mathbb{N}_0$ then $\sim[(n, n)] = \sim[(0, 0)]$

Proof. As $n+0 = n+0$ we have $(n, n) \sim (0, 0)$ so that $\sim[(n, n)] = \sim[(0, 0)]$. $\square$

Theorem 7.7. $(\langle \mathbb{Z}, + \rangle$ is a Abelian group) so

Associativity. $\forall n, m, k \in \mathbb{Z}$ we have $(n+m) + k = n + (m+k)$.

Neutral element. $\forall n \in \mathbb{Z}$ we have that $n+0 = 0+n$ where $0 = \sim[(0, 0)]$.

Inverse element. $\forall n \in \mathbb{Z}$ there exist a inverse element $-n$ such that $(-n) + n = 0 = n + (-n)$.
More specifically if $x = \sim[(n, m)]$ then $-x = \sim[(m, n)]$.

Commutativity. $\forall n, m \in \mathbb{N}_0$ we have $n+m = m+n$.

Proof.

Associativity. If $n = \sim[(n_1, m_1)]$, $m = \sim[(n_2, m_2)]$ and $k = \sim[(n_3, m_3)]$ then we have

$$\begin{aligned} (n+m) + k &= (\sim[(n_1, m_1)] + \sim[(n_2, m_2)]) + \sim[(n_3, m_3)] \\ &= \sim[(n_1+n_2, m_1+m_2)] + \sim[(n_3, m_3)] \\ &= \sim[(n_1+n_2)+n_3, (m_1+m_2)+m_3] \\ &= \sim[(n_1+(n_2+n_3), m_1+(m_2+m_3))] \\ &= \sim[(n_1, m_1)] + \sim[(n_2+n_3, m_2+m_3)] \\ &= \sim[(n_1, m_1)] + (\sim[(n_2, m_2)] + \sim[(n_3, m_3)]) \\ &= n + (m+k) \end{aligned}$$

Commutativity. If $n = \sim[(n_1, m_1)]$ and $m = \sim[(n_2, m_2)]$ then

$$\begin{aligned} \sim[(n_1, m_1)] + \sim[(n_2, m_2)] &= \sim[(n_1+n_2, m_1+m_2)] \\ &= \sim[(n_2+n_1, m_2+m_1)] \\ &= \sim[(n_2, m_2)] + \sim[(n_1, m_1)] \end{aligned}$$

Neutral element. If $k = \sim[(n, m)] \in \mathbb{Z}$ then

$$\begin{aligned} 0 + k &\stackrel{\text{commutativity}}{=} k + 0 \\ &= \sim[(n, m)] + \sim[(0, 0)] \\ &= \sim[(n+0, m+0)] \\ &= \sim[(n, m)] \\ &= k \end{aligned}$$

Inverse element. If $k = \sim[(n, m)]$

$$\begin{aligned} k + (-k) &\stackrel{\text{commutativity}}{=} (-k) + k \\ &= \sim[(m, n)] + \sim[(n, m)] \\ &= \sim[(m+n, n+m)] \\ &= \sim[(n+m, n+m)] \\ &\stackrel{[\text{theorem: 7.6}]}{=} \sim[(0, 0)] \end{aligned}$$

$\square$

The following introduce the difference operator that is now defined for all integers.

Definition 7.8. Let $n, m \in \mathbb{Z}_0^+$ then we have $n - m = n + (-m)$

Now to define multiplication in $\mathbb{Z}$, note that (n, m) is to be interpreted as $n - m$. So if $x = (n, m)$ and $y = (r, s)$ are two integers then $x \cdot y = (n, m) \cdot (r, s)$ is to be interpreted as the formal expression $(n - m) \cdot (r - s)$. Which if we formally evaluate it gives

$$\begin{aligned} (n - m) \cdot (r - s) &= n \cdot r - n \cdot s - m \cdot r + m \cdot s \\ &= n \cdot r + m \cdot s - (m \cdot r + n \cdot s) \end{aligned}$$

which suggest us that $(n, m) \cdot (r, s)$ should be equal to $(n \cdot r + m \cdot s, m \cdot r + n \cdot s)$, of course this is based on the representation of x and y . The next theorem proves that this product is independent of the representation, allowing us to define the product.

Theorem 7.9. If $\sim[(n, m)], \sim[(r, s)], \sim[(n', m')]$ and $\sim[(r', s')]$ are elements of $\mathbb{Z}$ such that $\sim[(n, m)] = \sim[(n', m')]$ and $\sim[(r, s)] = \sim[(r', s')]$ then

$$\sim[(n \cdot r + m \cdot s, m \cdot r + n \cdot s)] = \sim[(n' \cdot r' + m' \cdot s', m' \cdot r' + n' \cdot s')]$$

Proof. As $\sim[(n, m)] = \sim[(n', m')] \wedge \sim[(r, s)] = \sim[(r', s')]$ we have

$$n + m' = m + n' \wedge r + s' = s + r' \quad (7.2)$$

So we have

$$\begin{aligned} n \cdot r + m' \cdot r &= (n + m') \cdot r \\ &\stackrel{[\text{eq: 7.2}]}{=} (m + n') \cdot r \\ &= m \cdot r + n' \cdot r \\ m \cdot s + n' \cdot s &= (m + n') \cdot s \\ &\stackrel{[\text{eq: 7.2}]}{=} (n + m') \cdot s \\ &= n \cdot s + m' \cdot s \\ m' \cdot s + m' \cdot r' &= m' \cdot (s + r') \\ &\stackrel{[\text{eq: 7.2}]}{=} m' \cdot (r + s') \\ &= m' \cdot r + m' \cdot s' \\ n' \cdot r + n' \cdot s' &= n' \cdot (r + s') \\ &\stackrel{[\text{eq: 7.2}]}{=} n' \cdot (s + r') \\ &= n' \cdot s + n' \cdot r' \end{aligned}$$

so after summing (underlining common terms).

$$\begin{aligned} n \cdot r + \underbrace{m' \cdot r}_1 + m \cdot s + \underbrace{n' \cdot s}_2 + \underbrace{m' \cdot s}_3 + m' \cdot r' + \underbrace{n' \cdot r}_4 + n' \cdot s' &= \\ m \cdot r + \underbrace{n' \cdot r}_4 + n \cdot s + \underbrace{m' \cdot s}_3 + \underbrace{m' \cdot r}_1 + m' \cdot s' + \underbrace{n' \cdot s}_2 + n' \cdot r' & \end{aligned}$$

Using [theorem: 5.43] to eliminate common terms in the above gives:

$$n \cdot r + m \cdot s + m' \cdot r' + n' \cdot s' = m \cdot r + n \cdot s + m' \cdot s' + n' \cdot r'$$

So that

$$(n \cdot r + m \cdot s, m \cdot r + n \cdot s) \sim (n' \cdot r' + m' \cdot s', m' \cdot r' + n' \cdot s')$$

Hence

$$\sim[(n \cdot r + m \cdot s, m \cdot r + n \cdot s)] = \sim[(n' \cdot r' + m' \cdot s', m' \cdot r' + n' \cdot s')]$$

□

The above theorem ensures that the following definition is sensible.

Definition 7.10. The multiplication operator $\cdot : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is defined by

$$\sim[(n, m)] \cdot \sim[(r, s)] = \sim[(n \cdot r + m \cdot s, m \cdot r + n \cdot s)]$$

Theorem 7.11. $\langle \mathbb{Z}, +, \cdot \rangle$ is a *integral domain* [definition: 4.34], more specific:

1. $\langle \mathbb{Z}, + \rangle$ is a **Abelian** group [see: 7.7]
2. $\langle \mathbb{Z}, \cdot \rangle$ is a **Abelian semi-group**.

Associativity. $\forall n, m, k \in \mathbb{Z}$ we have $n \cdot (m \cdot k) = (n \cdot m) \cdot k$

Neutral Element. There exist a $1 = \sim[(1, 0)]$ such that $\forall n \in \mathbb{N}_0$ we have $n \cdot 1 = n = 1 \cdot n$.

Commutativity. $\forall n, m \in \mathbb{Z}$ we have $n \cdot m = m \cdot n$.

3. Further we have:

Distributivity. $\forall n, m, k \in \mathbb{Z}$ we have $n \cdot (m + k) = n \cdot m + n \cdot k$

There does not exist a zero divisor. If $n, m \in \mathbb{Z}$ is such that $n \cdot m = 0 \Rightarrow n = 0 \vee m = 0$

Proof.

1. This is already proved in [theorem: 7.7].
- 2.

Commutativity. If $\sim[(n, m)], \sim[(r, s)] \in \mathbb{Z}$ we have

$$\begin{aligned} \sim[(n, m)] \cdot \sim[(r, s)] &= \sim[(n \cdot r + m \cdot s, m \cdot r + n \cdot s)] \\ &= \sim[(r \cdot n + s \cdot m, s \cdot n + r \cdot m)] \\ &= \sim[(r, s)] \cdot \sim[(n, m)] \end{aligned}$$

Associativity. Let $\sim[(a, b)], \sim[(c, d)], \sim[(e, f)] \in \mathbb{Z}$ then

$$\begin{aligned} &\sim[(a, b)] \cdot (\sim[(c, d)] \cdot \sim[(e, f)]) = \\ &\sim[(a, b)] \cdot \sim[(c \cdot e + d \cdot f, d \cdot e + c \cdot f)] = \\ &\sim[(a \cdot (c \cdot e + d \cdot f) + b \cdot (d \cdot e + c \cdot f), b \cdot (c \cdot e + d \cdot f) + a \cdot (d \cdot e + c \cdot f))] = \\ &\sim\left[\left(\underbrace{a \cdot (c \cdot e)}_1 + \underbrace{a \cdot (d \cdot f)}_2 + \underbrace{b \cdot (d \cdot e)}_3 + \underbrace{b \cdot (c \cdot f)}_4, \underbrace{b \cdot (c \cdot e)}_5 + \underbrace{b \cdot (d \cdot f)}_6 + \underbrace{a \cdot (d \cdot e)}_7 + \underbrace{a \cdot (c \cdot f)}_8\right)\right] = \\ &\sim\left[\left(\underbrace{(a \cdot c) \cdot e}_1 + \underbrace{(b \cdot d) \cdot e}_3 + \underbrace{(b \cdot c) \cdot f}_4 + \underbrace{(a \cdot d) \cdot f}_2, \underbrace{(b \cdot c) \cdot e}_5 + \underbrace{(a \cdot d) \cdot e}_7 + \underbrace{(a \cdot c) \cdot f}_8 + \underbrace{(b \cdot d) \cdot f}_6\right)\right] \\ &\sim[(a \cdot c + b \cdot d) \cdot e + (b \cdot c + a \cdot d) \cdot f, (b \cdot c + a \cdot d) \cdot e + (a \cdot c + b \cdot d) \cdot f] = \\ &\sim[(a \cdot c + b \cdot d, b \cdot c + a \cdot d)] \cdot \sim[(e, f)] = \\ &(\sim[(a, b)] \cdot \sim[(c, d)]) \cdot \sim[(e, f)] \end{aligned}$$

Neutral element. If $n = \sim[(n, m)] \in \mathbb{Z}$ then we have

$$\begin{aligned} n \cdot 1 &\stackrel{\text{commutativity}}{=} 1 \cdot n \\ &= \sim[(1, 0)] \cdot \sim[(n, m)] \\ &= \sim[(1 \cdot n + 0 \cdot m, 0 \cdot n + 1 \cdot m)] \\ &= \sim(n, m) \end{aligned}$$

3. Further we have:

Distributivity. If $\sim[(a, b)], \sim[(c, d)], \sim[(e, f)] \in \mathbb{Z}$ then

$$\begin{aligned}
 & \sim[(a, b)] \cdot (\sim[(c, d)] + \sim[(e, f)]) = \\
 & \sim[(a, b)] \cdot \sim[(c+e, d+f)] = \\
 & \sim[(a \cdot (c+e) + b \cdot (d+f), b \cdot (c+e) + a \cdot (d+f))] = \\
 & \sim\left[\left(\underbrace{a \cdot c}_1 + \underbrace{a \cdot e}_2 + \underbrace{b \cdot d}_3 + \underbrace{b \cdot f}_4, \underbrace{b \cdot c}_5 + \underbrace{b \cdot e}_6 + \underbrace{a \cdot d}_7 + \underbrace{a \cdot f}_8\right)\right] = \\
 & \sim\left[\left(\underbrace{a \cdot c}_1 + \underbrace{b \cdot d}_3 + \underbrace{a \cdot e}_2 + \underbrace{b \cdot f}_4, \underbrace{b \cdot c}_5 + \underbrace{a \cdot d}_7 + \underbrace{b \cdot e}_6 + \underbrace{a \cdot f}_8\right)\right] = \\
 & \sim[(a \cdot c + b \cdot d, b \cdot c + a \cdot d)] + \sim[(a \cdot e + b \cdot f, b \cdot e + a \cdot f)] = \\
 & \sim[(a, b)] \cdot \sim[(c, d)] + \sim[(a, b)] \cdot \sim[(e, f)] =
 \end{aligned}$$

There does not exist a zero divisor. Let $n = \sim\{(a, b)\}$, $m = \sim[(c, d)]$ such that $n \cdot m = 0$ then

$$\sim[(a, b)] \cdot \sim[(c, d)] = \sim[(a \cdot c + b \cdot d, b \cdot c + a \cdot d)] = \sim[(0, 0)]$$

so we have that $(a \cdot c + b \cdot d) + 0 = (b \cdot c + a \cdot d) + 0$ giving

$$a \cdot c + b \cdot d = b \cdot c + a \cdot d \quad (7.3)$$

Assume that $n \neq 0$ then $\sim[(a, b)] \neq \sim[(0, 0)]$ so that $a + 0 \neq b + 0$ so that $a \neq b$, hence we have the following cases to consider:

$a < b$. Then using [theorem: 5.60] there exists a $k \in \mathbb{N}_0 \setminus \{0\}$ such that $a + k = b$, so substituting this in [eq: 7.3] gives

$$\begin{aligned}
 a \cdot c + (a + k) \cdot d &= (a + k) \cdot c + a \cdot d & \Rightarrow \\
 \underbrace{a \cdot c}_1 + \underbrace{a \cdot d}_2 + k \cdot d &= \underbrace{a \cdot c}_1 + k \cdot c + \underbrace{a \cdot d}_2 & \Rightarrow \\
 k \cdot d &= k \cdot c & \xRightarrow{k \neq 0 \wedge [\text{theorem: 5.77}]} \\
 d &= c
 \end{aligned}$$

$$\text{So } m = \sim[(c, d)] = \sim[(d, d)] \stackrel{[\text{theorem: 7.6}]}{=} \sim[(0, 0)] = 0.$$

$b < a$. Then using [theorem: 5.60] there exists a $k \in \mathbb{N}_0 \setminus \{0\}$ such that $b + k = a$, so substituting this in [eq: 7.3] gives

$$\begin{aligned}
 (b + k) \cdot c + b \cdot d &= b \cdot c + (b + k) \cdot d & \Rightarrow \\
 \underbrace{b \cdot c}_1 + k \cdot c + \underbrace{b \cdot d}_2 &= \underbrace{b \cdot c}_1 + \underbrace{b \cdot d}_2 + k \cdot d & \Rightarrow \\
 k \cdot c &= k \cdot d & \xRightarrow{k \neq 0 \wedge [\text{theorem: 5.77}]} \\
 c &= d
 \end{aligned}$$

$$\text{So } m = \sim[(c, d)] = \sim[(d, d)] \stackrel{[\text{theorem: 7.6}]}{=} \sim[(0, 0)] = 0.$$

So if $n \cdot m = 0$ then we have either $n \neq 0$ but then $m = 0$ or $n = 0$ proving that $n \cdot m = 0 \Rightarrow n = 0 \vee m = 0$. $\square$

Example 7.12. $1 + 1 = 2$ where $2 = \sim[(2, 0)]$

Proof. $1 + 1 = \sim[(1, 0)] + \sim[(1, 0)] = \sim[(1 + 1, 0 + 0)] = \sim[(1 + 1, 0)] \stackrel{[\text{example: 5.28}]}{=} \sim[(2, 0)] = 2$ $\square$

Lemma 7.13. $\forall n \in \mathbb{N}_0 \setminus \{0\}$ we have that $\sim[(n, 0)] \neq 0$

Proof. We prove this by contradiction so assume that $\sim[(n, 0)] = 0 = \sim[(0, 0)]$ then $n + 0 = 0 \Rightarrow n = 0$ contradicting $n \in \mathbb{N}_0 \setminus \{0\}$. So $\sim[(n, 0)] \neq 0$. $\square$

Corollary 7.14. $\forall z \in \mathbb{Z}$ such that $z = -z$ we have $z = 0$

Proof. If $z = -z$ we have that $z + z = (-z) + z = 0$. So $(1 + 1) \cdot z = z \cdot 1 + z \cdot 1 = z + z = 0$, hence

$$(1 + 1) \cdot z = 0$$

As $1 + 1 = \sim[(1, 0)] + \sim[(1, 0)] = \sim[(2, 0)]$ and $2 \neq 0$ we have by [lemma: 7.13] that $1 + 1 \neq 0$, using [theorem: 7.11] on the above proves then that $z = 0$. $\square$

Theorem 7.15. Let $n, k, r \in \mathbb{Z}$ with $r \neq 0$ then $n \cdot r = k \cdot r$ implies $n = k$.

Proof.

$$\begin{aligned} n \cdot r = k \cdot r &\Rightarrow n \cdot r + (-(k \cdot r)) = (k \cdot r) + (-(k \cdot r)) \\ &\Rightarrow n \cdot r + (-(k \cdot r)) = 0 \\ &\stackrel{[\text{theorem: 4.40}]}{\Rightarrow} n \cdot r + (-k) \cdot r = 0 \\ &\Rightarrow (n + (-k)) \cdot r = 0 \end{aligned}$$

As by [theorem: 7.11] $\langle \mathbb{Z}, +, \cdot \rangle$ is a integral domain and $r \neq 0$ we have $n + (-k) = 0$ so that $(n + (-k)) + k = 0 + k$ or $n + ((-k) + k) = k$ proving $n = k$. $\square$

7.2 Order relation on the set of integers

First we define the set of non negative integers.

Definition 7.16. $\mathbb{Z}_0^+ = \{\sim[(n, 0)] \mid n \in \mathbb{N}_0\} \subseteq \mathbb{Z}$

We have the following properties for the set on non negative integers.

Theorem 7.17. The set of non negative integers satisfies

1. $\mathbb{Z}_0^+$ is a sub semi-group of $\langle \mathbb{Z}, + \rangle$ [hence by [theorem: 4.14] $\langle \mathbb{Z}_0^+, + \rangle$ is a Abelian semi-group].
2. $\mathbb{Z}_0^+$ is a sub semi-group of $\langle \mathbb{Z}, \cdot \rangle$ [hence by [theorem: 4.14] $\langle \mathbb{Z}_0^+, \cdot \rangle$ is a Abelian semi-group].
3. The function $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+$ defined by $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) = \sim[(n, 0)]$ is a bijection and
 - a. $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{Z}_0^+, + \rangle$ is a group isomorphism
 - b. $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{Z}_0^+, \cdot \rangle$ is a group isomorphism
4. For every $z \in \mathbb{Z} \exists x, y \in \mathbb{Z}_0^+$ such that $z = x - y$

Proof.

1. Let $z, z' \in \mathbb{Z}$ then $z = \sim[(n, 0)]$ and $z' = \sim[(n', 0)]$ so that

$$z + z' = \sim[(n, 0)] + \sim[(n', 0)] = \sim[(n + n', 0 + 0)] = \sim[(n + n', 0)] \in \mathbb{Z}_0^+$$

further

$$0 = \sim[(0, 0)] \in \mathbb{Z}_0^+.$$

Using [definition: 4.12] it follows that $\mathbb{Z}_0^+$ is a sub semi-group of $\langle \mathbb{Z}, + \rangle$.

2. Let $z, z' \in \mathbb{Z}$ then $z = \sim[(n, 0)]$ and $z' = \sim[(n', 0)]$ so that

$$z \cdot z' = \sim[(n, 0)] \cdot \sim[(n', 0)] = \sim[(n \cdot n' + 0 \cdot 0, 0 \cdot n' + n \cdot 0)] = \sim[(n \cdot n', 0)] \in \mathbb{Z}_0^+$$

further

$$1 = \sim[(1, 0)] \in \mathbb{Z}_0^+$$

Using [definition: 4.12] it follows that $\mathbb{Z}_0^+$ is a sub semi-group of $\langle \mathbb{Z}, \cdot \rangle$.

3. First we show that $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}$ is a bijection:

injectivity. If $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(m)$ then $\sim[(n, 0)] = \sim[(m, 0)]$ so that $n + 0 = 0 + m \Rightarrow n = m$.

surjectivity. If $z \in \mathbb{Z}_0^+$ there exist a $n \in \mathbb{N}_0$ such that $z = \sim[(n, 0)] = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n)$.

Next we have:

- a. First $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n + m) = \sim[(n + m, 0)] = \sim[(n, 0)] + \sim[(m, 0)] = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) + i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(m)$.
Secondly $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(0) = \sim[(0, 0)] = 0 \in \mathbb{Z}_0^+$.
- b. First

$$\begin{aligned} i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) \cdot i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(m) &= \sim[(n, 0)] \cdot \sim[(m, 0)] \\ &= \sim[(n \cdot m + 0 \cdot m, 0 \cdot n + n \cdot 0)] \\ &= \sim[(n \cdot m, 0)] \\ &= i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n \cdot m) \end{aligned}$$

Second $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(1) = \sim[(1, 0)] = 1 \in \mathbb{Z}_0^+$.

4. Let $z \in \mathbb{Z}$ then $z = \sim[(n, m)]$, take $x = \sim[(n, 0)] \in \mathbb{Z}_0^+$ and $y = \sim[(m, 0)] \in \mathbb{Z}_0^+$ then we have

$$x - y = x + (-y) = \sim[(n, 0)] + \sim[(0, m)] = \sim[(n, m)] = z \quad \square$$

Next we define the set of non positive numbers.

Definition 7.18. $\mathbb{Z}_0^- = \{-n | n \in \mathbb{Z}_0^+\} = \{(0, n) | n \in \mathbb{N}_0\} \subseteq \mathbb{Z}$

Definition 7.19. $\mathbb{Z}^+ = \mathbb{Z}_0^+ \setminus \{0\}$ and $\mathbb{Z}^- = \mathbb{Z}_0^- \setminus \{0\}$

The following theorem shows the relation between $\mathbb{Z}_0^+$ and $\mathbb{Z}_0^-$.

Theorem 7.20. $\mathbb{Z} = \mathbb{Z}_0^+ \cup \mathbb{Z}_0^-$ and $\{0\} = \mathbb{Z}_0^+ \cap \mathbb{Z}_0^-$

Proof. As $\mathbb{Z}_0^+ \subseteq \mathbb{Z}$ and $\mathbb{Z}_0^- \subseteq \mathbb{Z}$ it follows that

$$\mathbb{Z}_0^+ \cup \mathbb{Z}_0^- \subseteq \mathbb{Z} \quad (7.4)$$

Let $z \in \mathbb{Z}$ then $\exists n, m \in \mathbb{N}_0$ such that $z = \sim[(n, m)]$, for n, m we have either:

$n \leq m$. then using [theorem: 5.62] there exist a $k \in \mathbb{N}_0$ such that $m = n + k$ so that

$$z = \sim[(n, n + k)] \quad (7.5)$$

Now for $(0, k)$ and $(n, n + k)$ we have $0 + (n + k) = n + k$ so that $(0, k) \sim (n, n + k)$ proving that $\sim[(0, k)] = \sim[(n, n + k)] \stackrel{\text{eq: 7.5}}{=} z$, proving that $z \in \mathbb{Z}_0^- \subseteq \mathbb{Z}_0^+ \cup \mathbb{Z}_0^-$.

$m < n$. Then using [theorem: 5.62] there exist a $k \in \mathbb{N}_0$ such that $n = m + k$ so that

$$z = \sim[(m + k, m)] \quad (7.6)$$

Now for $(k, 0)$ and $(m + k, m)$ we have $k + m = 0 + m + k$ so that $(k, 0) \sim (m + k, m)$ proving that $\sim[(k, 0)] = \sim[(m + k, m)] \stackrel{\text{eq: 7.6}}{=} z$, proving that $z \in \mathbb{Z}_0^+ \subseteq \mathbb{Z}_0^+ \cup \mathbb{Z}_0^-$.

From the above we have $\mathbb{Z} \subseteq \mathbb{Z}_0^+ \cup \mathbb{Z}_0^-$ which by [eq: 7.4] proves that

$$\mathbb{Z} = \mathbb{Z}_0^+ \cup \mathbb{Z}_0^-$$

As $0 = \sim[(0, 0)] \in \mathbb{Z}_0^+$ and $0 = \sim[(0, 0)] \in \mathbb{Z}_0^-$ we have that $\{0\} \in \mathbb{Z}_0^+ \cap \mathbb{Z}_0^-$. Let $z \in \mathbb{Z}_0^+ \cap \mathbb{Z}_0^-$ then there exists $n, m \in \mathbb{N}_0$ such that $z = \sim[(n, 0)] = \sim[(0, m)]$ hence $n + 0 = 0 + m \Rightarrow n = m$. So $z = \sim[(n, 0)] = \sim[(0, n)] = -z$. Applying then [theorem: 7.14] it follows that $z = 0$ or $\mathbb{Z}_0^+ \cap \mathbb{Z}_0^- \subseteq \{0\}$. Hence

$$\mathbb{Z}_0^+ \cap \mathbb{Z}_0^- = \{0\} \quad \square$$

We can now define a order relation on $\mathbb{Z}$.

Theorem 7.21. $\langle \mathbb{Z}, \leq \rangle$ where

$$\leq = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid y + (-x) \in \mathbb{Z}_0^+\}$$

is a partial ordered set that is fully [or total] ordered.

Proof.

reflexivity. If $x \in \mathbb{Z}$ then $x + (-x) = 0 \in \mathbb{Z}_0^+$ so that $x \leq x$.

anti symmetry. Let $x, y \in \mathbb{Z}$ with $x \leq y$ and $y \leq x$ then

$$y + (-x) \in \mathbb{Z}_0^+ \quad \wedge \quad x + (-y) \in \mathbb{Z}_0^+$$

then $\exists n, m \in \mathbb{N}_0$ such that

$$y + (-x) = \sim[(n, 0)] \wedge x + (-y) = \sim[(m, 0)]$$

so taking the sum we have

$$\begin{aligned} \sim[0, 0] &= 0 \\ &= y + (-x) + x + (-y) \\ &= \sim[(n, 0)] + \sim[(m, 0)] \\ &= \sim[(n + m, 0)] \end{aligned}$$

Hence $0 + 0 = 0 + n + m$ so that $n + m = 0$ which by [theorem: 5.57] proves that $n = m = 0$ so that $y + (-x) = \sim[(n, 0)] = \sim[(0, 0)] = 0$. Hence $x = 0 + x = (y + (-x)) + x = y + ((-x) + x) = y$ from which it follows that $x = y$.

transitivity. If $x \leq y$ and $y \leq z$ then $y + (-x) \in \mathbb{Z}_0^+$ and $z + (-y) \in \mathbb{Z}_0^+$. Then we have

$$\begin{aligned} z + (-x) &= (z + (-y)) + 0 \\ &= (z + (-y)) + (y + (-x)) \\ &= (y + (-x)) + (z + (-y)) \end{aligned}$$

which as $y + (-x), z + (-y) \in \mathbb{Z}_0^+$ proves by [theorem: 7.17] that $z + (-x) \in \mathbb{Z}_0^+$ proving that $x \leq z$.

total ordering. If $x, y \in \mathbb{N}_0$ then we have for $x + (-y) \in \mathbb{Z}_{[\text{theorem: 7.20}]}^+ \cup \mathbb{Z}_0^-$ either:

$x + (-y) \in \mathbb{Z}_0^+$. Then $y \leq x$

$x + (-y) \in \mathbb{Z}_0^-$. Then $-(x + (-y)) \in \mathbb{Z}_0^+$, further

$$\begin{aligned} -(x + (-y)) &\stackrel{[\text{theorem: 4.8}]}{=} -x + (-(-y)) \\ &\stackrel{[\text{theorem: 4.9}]}{=} -x + y \\ &= y + (-x) \end{aligned}$$

proving that $x \leq y$. □

Theorem 7.22. $\mathbb{Z}_0^+ = \{x \in \mathbb{Z} \mid 0 \leq x\}$ and $\mathbb{Z}_0^- = \{x \in \mathbb{Z} \mid x \leq 0\}$

Proof. First we have

$$x + (-0) \stackrel{[\text{theorem: 4.9}]}{=} x + 0 = x \tag{7.7}$$

Now

$$\begin{aligned} x \in \mathbb{Z}_0^+ &\stackrel{[\text{eq: 7.7}]}{\Leftrightarrow} x + (-0) \in \mathbb{Z}_0^+ \\ &\Leftrightarrow 0 \leq x \\ &\Leftrightarrow x \in \{x \in \mathbb{Z} \mid 0 \leq x\} \end{aligned}$$

proving

$$\mathbb{Z}_0^+ = \{x \in \mathbb{Z} \mid 0 \leq x\}$$

Further

$$\begin{aligned}
 x \in \mathbb{Z}_0^- &\Leftrightarrow -x \in \mathbb{Z}_0^+ \\
 &\Leftrightarrow 0 + (-x) \in \mathbb{Z}_0^+ \\
 &\Leftrightarrow x \leq 0 \\
 &\Leftrightarrow x \in \{x \in \mathbb{Z} \mid x \leq 0\}
 \end{aligned}$$

proving

$$\mathbb{Z}_0^- = \{x \in \mathbb{Z} \mid x \leq 0\}$$

□

Theorem 7.23. *If $x = \sim[n, m] \in \mathbb{Z}$ then we have*

1. $0 \leq x \Leftrightarrow m \leq n$
2. $0 < x \Leftrightarrow m < n$
3. *If $0 < x$ then $1 \leq x$*

Proof.

1.

$$\begin{aligned}
 0 \leq x &\stackrel{[\text{theorem: 7.22}]}{\Leftrightarrow} x \in \mathbb{Z}_0^+ \\
 &\Leftrightarrow \exists k \in \mathbb{N}_0 \text{ such that } x = \sim[(k, 0)] \\
 &\Leftrightarrow \exists k \in \mathbb{N}_0 \text{ such that } n + 0 = m + k \Leftrightarrow n = m + k \\
 &\stackrel{[\text{theorem: 5.62}]}{\Leftrightarrow} m \leq n
 \end{aligned}$$

2. First

$$\begin{aligned}
 x \neq 0 &\Leftrightarrow \sim[(n, m)] \neq \sim[(0, 0)] \\
 &\Leftrightarrow n + 0 \neq m + 0 \\
 &\Leftrightarrow n \neq m
 \end{aligned}$$

then

$$\begin{aligned}
 0 < x &\Leftrightarrow x \neq 0 \wedge 0 \leq x \\
 &\Leftrightarrow n \neq m \wedge 0 \leq x \\
 &\stackrel{(1)}{\Leftrightarrow} n \neq m \wedge m \leq n \\
 &\Leftrightarrow m < n
 \end{aligned}$$

3. If $0 < x$ then by (2) $m < n$ so that by [theorem: 5.50]

$$m + 1 \leq n.$$

Now

$$x + (-1) = \sim[(n, m)] + \sim[(0, 1)] = \sim[(n, m + 1)]$$

so that $0 \leq x + (-1)$, hence $x + (-1) \in \mathbb{Z}_0^+$ from which we conclude that

$$1 \leq x$$

□

Corollary 7.24. $\forall n \in \mathbb{N}_0$ we have $0 \leq \sim[(n, 0)]$ further if $n \neq 0$ then $0 < \sim[(n, 0)]$

Proof. By [theorem: 5.46] we have $\forall n \in \mathbb{N}_0$ that $0 \leq n$ so that by [theorem: 7.23] [$0 \leq \sim[(n, 0)]$], further if $n \neq 0$ then $0 < n$, hence by [theorem: 7.23] we have that $0 < \sim[(n, 0)]$ □

Example 7.25. $0 < 1$ and $0 < 2$ where $1, 2 \in \mathbb{Z}$

Proof. This follows directly from [corollary: 7.24] and the fact that for $1, 2 \in \mathbb{N}_0$ we have $0 < 1$ and $0 < 2$. □

Theorem 7.26. *If $x, y \in \mathbb{Z}$ and $0 < x \wedge 0 < y$ then $0 < x \cdot y$.*

Proof. $x = \sim[(n.m)]$ and $y = \sim[(r.s)]$ then by [theorem: 7.23] we have $m < n$ and $s < r$, so by [theorem: 5.60] there exists $k, l \in \mathbb{N}_0 \setminus \{0\}$ such that $n = m + k$ and $r = s + l$. Hence

$$\begin{aligned} n \cdot r + m \cdot s &= (m + k) \cdot (s + l) + m \cdot s \\ &= \underbrace{m \cdot s}_1 + \underbrace{m \cdot l}_2 + \underbrace{k \cdot s}_3 + k \cdot l + \underbrace{m \cdot s}_4 \\ m \cdot r + n \cdot s &= m \cdot (s + l) + (m + k) \cdot s \\ &= \underbrace{m \cdot s}_1 + \underbrace{m \cdot l}_2 + \underbrace{m \cdot s}_4 + \underbrace{k \cdot s}_3 \end{aligned}$$

so that

$$n \cdot r + m \cdot s = m \cdot r + n \cdot s + k \cdot l$$

As $0 \neq k \Rightarrow 0 < k$ and $0 \neq l \Rightarrow 0 < l$ it follows from [theorem: 5.76] that $0 < k \cdot l$ so that $k, l \in \mathbb{N}_0 \setminus \{0\}$, using the above together with [theorem: 5.61] proves that

$$m \cdot r + n \cdot s < n \cdot r + m \cdot s \quad (7.8)$$

now

$$x \cdot y = \sim[(n \cdot r + m \cdot s, m \cdot r + n \cdot s)]$$

Combining the above with [eq: 7.8] and [theorem: 7.23] proves finally:

$$0 < x \cdot y \quad \square$$

Theorem 7.27. $\langle \mathbb{Z}, +, \cdot, \leq \rangle$ is a ordered integral domain [definition: 4.49]

Proof. Using [theorem: 7.11] $\langle \mathbb{Z}, +, \cdot \rangle$ is a integral domain and using [theorem: 7.21] we have that $\langle \mathbb{Z}, \leq \rangle$ is totally ordered. Next

1. For $n, m, k \in \mathbb{Z}$ with $n < m$ we have

$$m + (-n) \in \mathbb{Z}_0^+ \text{ and } n \neq m \Rightarrow n + k \neq m + k \quad (7.9)$$

Further

$$\begin{aligned} (m + k) + (-(n + k)) &\stackrel{[\text{theorem: 4.8}]}{=} (m + k) + ((-n) + (-k)) \\ &\stackrel{\text{commutativity}}{=} (m + k) + ((-k) + (-n)) \\ &\stackrel{\text{associativity}}{=} m + (k + ((-k) + (-n))) \\ &\stackrel{\text{associativity}}{=} m + ((k + (-k)) + (-n)) \\ &= m + (0 + (-n)) \\ &= m + (-n) \end{aligned}$$

which by [eq: 7.9] proves that $(m + k) + (-(n + k)) \in \mathbb{Z}_0^+$. Hence $n + k \leq m + k$ and $n + k \neq m + k$ proving that

$$n + k < m + k$$

2. Let $n, m \in \mathbb{Z}$ with $0 < m$ and $0 < m$ then by [theorem: 7.26] we have $0 < n \cdot m$. $\square$

Theorem 7.28. *Let $x, y \in \mathbb{Z}$ with $x < y$ then*

1. $x + 1 \leq y$
2. $x \leq y + (-1)$

Proof.

1. If $x < y$ then by [theorems 7.27, 4.50] $0 < y + (-x)$, using [theorem: 7.23] we have $1 \leq y + (-x)$ so that using [theorem: 7.23] $x + 1 \leq (y + (-x)) + x = y$.
2. By (1) $x + 1 \leq y$ so that by [theorem: 7.23] $x = (x + 1) + (-1) \leq y + (-1)$ $\square$

Theorem 7.29. Define $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+$ by $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) = \sim[(n, 0)]$ then

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{Z}_0^+, \leq \rangle \text{ is a order isomorphism}$$

Proof. Using [theorem: 7.17 (3)] it follows that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+ \text{ is a bijection}$$

Further we have:

$$\begin{aligned} i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(x) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(y) &\Leftrightarrow i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(y) + (-i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(x)) \in \mathbb{Z}_0^+ \\ &\Leftrightarrow \sim[(y, 0)] + (-\sim[(x, 0)]) \in \mathbb{Z}_0^+ \\ &\Leftrightarrow \sim[(y, 0)] + \sim[(0, x)] \in \mathbb{Z}_0^+ \\ &\Leftrightarrow \sim[(y, x)] \in \mathbb{Z}_0^+ \\ &\stackrel{[\text{theorem: 7.23}]}{\Leftrightarrow} x \leq y \end{aligned}$$

□

The above theorem allows us to transfer properties of $\mathbb{N}_0$ to $\mathbb{Z}_0^+$ as is expressed in the following theorems.

Theorem 7.30. (Archimedean property) If $x, y \in \mathbb{Z}$ with $0 < x$ then there exist a $k \in \mathbb{Z}_0^+$ such that $y < k \cdot x$.

Proof. We have the following cases for y :

$y \leq 0$. Take $k = 1 \in \mathbb{Z}_0^+$ then as $y \leq 0 < x = 1 \cdot x = k \cdot x$ proving that $y < k \cdot x$

$0 < y$. Then $y \in \mathbb{Z}_0^+$. Using [theorem: 7.17] $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{Z}_0^+, \cdot \rangle$ defined by $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) = \sim[(n, 0)]$ is a group isomorphism. Take $n = (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(x)$ and $m = (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(y)$ then $x = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n)$ and $n \neq 0$ [otherwise $x = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(0) = 0$]. Using the Archimedean property of the natural numbers [see theorem: 5.78] there exists a $l \in \mathbb{N}_0$ such that $m < l \cdot n$. So by [theorem: 7.29] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(m) < i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(l \cdot n) \stackrel{[\text{theorem: 7.17}]}{=} i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(l) \cdot i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) \quad (7.10)$$

Take $k = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(l) \in \mathbb{Z}_0^+$ then as $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(x)) = x$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(m) = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(y)) = y$ we have by [eq: 7.10] that

$$y < k \cdot x \quad \square$$

Theorem 7.31. $\langle \mathbb{Z}_0^+, \leq \rangle$ is a well-ordered set

Proof. Using [theorem: 7.29] we have that $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+$ is a order isomorphism, further by [theorem: 5.51] $\langle \mathbb{N}_0, \leq \rangle$ is well ordered. so using [theorem: 3.93] we conclude that

$$\langle \mathbb{Z}_0^+, \leq \rangle \text{ is well ordered} \quad \square$$

Theorem 7.32. $\langle \mathbb{Z}_0^+, \leq \rangle$ is conditional complete [see definition: 3.80].

Proof. As by [theorem: 7.31] $\langle \mathbb{Z}_0^+, \leq \rangle$ is well-ordered it follows from [theorem: 3.87] it follows that $\langle \mathbb{Z}_0^+, \leq \rangle$ is conditional complete. □

Theorem 7.33. If $A \subseteq \mathbb{Z}_0^+$ is such that $A \neq \emptyset$ and $\sup(A)$ exists then $\sup(A) \in A$.

Proof. By [theorem: 7.29]

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{Z}_0^+, \leq \rangle \text{ is a order isomorphism}$$

which by [theorem: 3.55] means that

$$(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}: \langle \mathbb{Z}_0^+, \leq \rangle \rightarrow \langle \mathbb{N}_0, \leq \rangle \text{ is a order isomorphism}$$

Assume that $M = \sup(A)$ exists then by [theorem: 3.82] $\sup((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(A))$ exist and $\sup((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(A)) = (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(M)$. By [theorem: 5.73] we have that $\sup((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(A)) \in (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(A)$ so that $(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(M) \in (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(A)$, hence $M = i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(M)) \in (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})((i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})^{-1}(A)) = A$ or

$$\sup(A) \in A \quad \square$$

Definition 7.34. (Absolute Value) If $x \in \mathbb{Z}$ then $|x|$ is defined by

$$|x| = \begin{cases} x & \text{if } 0 \leq x \\ -x & \text{if } x < 0 \end{cases}$$

Theorem 7.35. If $x, y \in \mathbb{Z}$ then $|x \cdot y| = |x| \cdot |y|$

Proof. We have the following possibilities for x, y :

$0 \leq x \wedge 0 \leq y$. Then $|x| = x$ and $|y| = y$. Further by [theorems: 7.27, 4.50] $0 = 0 \cdot y \leq x \cdot y$, hence $x \cdot y = |x \cdot y|$. So we have that $|x \cdot y| = |x| \cdot |y|$.

$0 \leq x \wedge y < 0$. Then $x = |x|$ and $-y = |y|$, further by [theorems: 7.27, 4.50] $x \cdot y \leq 0 \cdot y = 0$, hence $|x \cdot y| = -(x \cdot y)$. So

$$|x| \cdot |y| = x \cdot (-y) \stackrel{[\text{theorem: 4.40}]}{=} -(x \cdot y) = |x \cdot y|.$$

$x < 0 \wedge 0 \leq y$. Then $-x = |x|$ and $y = |y|$, further by [theorems: 7.27, 4.50] $x \cdot y \leq 0 \cdot y = 0$, hence $|x \cdot y| = -(x \cdot y)$. So

$$|x| \cdot |y| = (-x) \cdot y \stackrel{[\text{theorem: 4.40}]}{=} -(x \cdot y) = |x \cdot y|$$

$x < 0 \wedge y < 0$. Then $-x = |x|$, $-y = |y|$, further by [theorems: 7.27, 4.50] $0 = 0 \cdot y < x \cdot y$, hence $|x \cdot y| = x \cdot y$. So

$$|x| \cdot |y| = (-x) \cdot (-y) \stackrel{[\text{theorem: 4.40}]}{=} -(-(x \cdot y)) \stackrel{[\text{theorem: 4.9}]}{=} x \cdot y = |x \cdot y| \quad \square$$

Theorem 7.36. If $x \in \mathbb{Z}$ then $x \leq |x|$

Proof. If $0 \leq x$ then $x = |x|$ so that trivially $x \leq |x|$, if $x < 0$ then by [theorems: 7.27, 4.50] $0 < -x = |x|$ so that by transitivity $x < |x|$ or $x \leq |x|$. $\square$

Theorem 7.37. $\forall x \in \mathbb{Z}$ we have $|x| = 0 \Leftrightarrow x = 0$

Proof.

$\Rightarrow$. If $x = 0$ then $0 \leq x$ so that $|x| = x = 0$ hence $|x| = 0$

$\Leftarrow$. If $|x| = 0$ then if $x < 0$ we would have $-x = |x| = 0$ so that $-x = 0 \Rightarrow x = 0$ contradicting $x < 0$. So we must have $0 \leq x$, hence $x = |x| = 0$ proving that $x = 0$. $\square$

We introduce now division, just as it was done for the natural numbers.

Theorem 7.38. (Division Algorithm) If $n, m \in \mathbb{Z}$ and $0 < n$ then there exists **unique** $r \in \mathbb{Z}_0^+$, $q \in \mathbb{Z}$ such that $0 \leq r < n$ and $m = n \cdot q + r$

Proof. First we prove existence, let $n, m \in \mathbb{Z}$ with $0 < n$. Define

$$A_{n,m} = \{m + n \cdot q \mid q \in \mathbb{Z} \wedge 0 \leq m + n \cdot q\} \subseteq \mathbb{Z}_0^+.$$

Using $0 < n$ and the Archimedean property of $\mathbb{Z}$ [see theorem: 7.30] there exist a $k \in \mathbb{Z}_0^+$ such that $-m < n \cdot k$, using [theorems: 7.27, 4.50] it follows that $0 < n \cdot k + (-(-m)) = n \cdot k + m = m + n \cdot k$ proving that $m + n \cdot k \in A_{n,m}$, hence $A_{n,m} \neq \emptyset$. As $\langle \mathbb{Z}_0^+, \leq \rangle$ is well-ordered [see theorem: 7.31] $A_{n,m}$ has a least element, hence

$$\exists r' \in A_{n,m} \text{ such that } \forall a \in A_{n,m} \text{ we have } r' \leq a \quad (7.11)$$

As $r' \in A_{n,m}$ there exist a $q' \in \mathbb{Z}$ such that

$$r' = m + n \cdot q' \text{ and } 0 \leq r' \quad (7.12)$$

Assume that $n < r'$ then by [theorems: 7.27, 4.50] $k = r' + (-n) \in \mathbb{Z}_0^+ \setminus \{0\}$ so that $r' = n + k$. Hence $m + n \cdot q' = n + k$, so that $0 < k = m + n \cdot q' + (-n) = m + (q' - 1) \cdot n$ proving that $k \in A_{n,m}$. Now $0 < n \xRightarrow{\text{[theorems: 7.27, 4.50]}} k < n + k = r' \Rightarrow k < r'$, as $k \in A_{n,m}$ we have by [eq: 7.11] $r' \leq k$, giving the contradiction $k < k$. So we must have that $r' \leq n$ or

$r' = n$. In this case we have that $m + n \cdot q' = r' = n$, hence

$$m = n + (-(n \cdot q')) = n \cdot 1 + n \cdot (-q') = n \cdot (1 + (-q'))$$

So by taking $q = (1 + (-q'))$ and $r = 0 < n$ we have

$$m = n \cdot q + r \text{ and } 0 \leq r < n$$

$r' < n$. Then as $r' = m + n \cdot q'$ we have $m = r' + (-(n \cdot q')) = r' + n \cdot (-q')$, so taking $q = -q'$ and $r = r'$ then

$$m = n \cdot q + r \text{ and } 0 \leq r' < n$$

Now for uniqueness assume that there exists $q_1, q_2 \in \mathbb{Z}$ and $r_1, r_2 \in \mathbb{Z}_0^+$ such that

$$m = n \cdot q_1 + r_1 \wedge m = n \cdot q_2 + r_2 \wedge 0 \leq r_1 < n \wedge 0 \leq r_2 < n$$

Then

$$\begin{aligned} n \cdot q_1 + r_1 &= n \cdot q_2 + r_2 \Rightarrow n \cdot q_1 + (-(n \cdot q_2)) = r_2 + (-r_1) \\ &\Rightarrow n \cdot (q_1 + (-q_2)) = r_2 + (-r_1) \end{aligned} \quad (7.13)$$

$$\begin{aligned} n \cdot q_1 + r_1 &= n \cdot q_2 + r_2 \Rightarrow n \cdot q_2 + (-(n \cdot q_1)) = r_1 + (-r_2) \\ &\Rightarrow n \cdot (q_2 + (-q_1)) = r_1 + (-r_2) \end{aligned} \quad (7.14)$$

Assume now that $r_1 \neq r_2$ then we have either:

$r_1 < r_2$. Then by [theorems: 7.27, 4.50] $0 < r_2 + (-r_1) \stackrel{\text{[eq: 7.13]}}{=} n \cdot (q_1 + (-q_2))$, hence $0 \cdot n < (q_1 + (-q_2)) \cdot n$, as $0 < n$ we must have by [theorems: 7.27, 4.50] that $0 < q_1 + (-q_2)$. Using [theorem: 7.23] we have

$$1 \leq q_1 + (-q_2) \quad (7.15)$$

As $r_2 < n$ we have by [theorems: 7.27, 4.50] that $r_2 + (-r_1) < n + (-r_1)$, further as $(-r_1) \leq 0$ we have by [theorems: 7.27, 4.50] that $n + (-r_1) \leq n$ so that $r_2 + (-r_1) < n$. Using this with [eq: 7.13] gives $n \cdot (q_1 + (-q_2)) < n = 1 \cdot n$, hence using [theorems: 7.27, 4.50] we have that $q_2 + (-q_1) < 1$, contradicting [eq: 7.15]. So this case never occurs.

$r_2 < r_1$. Then by [theorems: 7.27, 4.50] $0 < r_1 + (-r_2) \stackrel{\text{[eq: 7.14]}}{=} n \cdot (q_2 + (-q_1))$, hence $0 \cdot n < (q_2 + (-q_1)) \cdot n$ as $0 < n$ we must have by [theorems: 7.27, 4.50] that $0 < q_2 + (-q_1)$. Using [theorem: 7.23] we have

$$1 \leq q_2 + (-q_1) \quad (7.16)$$

As $r_1 < n$ we have by [theorems: 7.27, 4.50] that $r_1 + (-r_2) < n + (-r_2)$, further as $(-r_2) \leq 0$ we have by [theorems: 7.27, 4.50] that $n + (-r_2) \leq n$ so that $r_1 + (-r_2) < n$. Using this with [eq: 7.14] gives $n \cdot (q_2 + (-q_1)) < n = 1 \cdot n$, hence using [theorems: 7.27, 4.50] we have that $q_1 + (-q_2) < 1$, contradicting [eq: 7.16]. So this case never occurs.

As all the cases lead to a contradiction the assumption $r_1 \neq r_2$ is wrong. Hence

$$r_1 = r_2$$

So $n \cdot q_1 + r_1 \stackrel{r_1=r_2}{=} n \cdot q_2 + r_1$ giving, by adding $-r_1$ to both sides, that $n \cdot q_1 = n \cdot q_2$. Applying [theorem: 7.15] proves then

$$q_1 = q_2$$

□

Definition 7.39. If $n, m \in \mathbb{Z}$ then we say that n divides m noted as $n|m$ if there exist a $q \in \mathbb{Z}$ such that $q \cdot n = m$, we call n a **divisor** of m .

Example 7.40. Every integer is a divisor of 0.

Proof. If $n \in \mathbb{Z}$ then $n \cdot 0 = 0$ □

Example 7.41. If $n \in \mathbb{Z}$ then $1|n$

Proof. As $1 \cdot n = n$ we have by definition $1|n$. □

Theorem 7.42. Let $m \in \mathbb{Z}$ then if $n|m$ we have that $(-n)|m$. In other words if n is a divisor of m then $-n$ is a divisor of m . So as $|n| = \begin{cases} -n & \text{if } n < 0 \\ n & \text{if } 0 \leq n \end{cases}$ we have also that $n|m \Rightarrow |n||m$.

Proof. If $n|m$ then there exist a q such that $n \cdot q = m$, then $(-n) \cdot (-q) = n \cdot q = m$ so that $(-n)|m$. □

Theorem 7.43. If $m \in \mathbb{Z}$ and $n \in \mathbb{Z} \setminus \{0\}$ a divisor of m then there exists a **unique** q such that $n \cdot q = m$

Proof. Existence follows from the definition of divisor. Now for uniqueness assume that $q_1, q_2 \in \mathbb{Z}$ such that $n \cdot q_1 = m = n \cdot q_2$ then by [theorem: 7.15] $q_1 = q_2$. □

Definition 7.44. If $m \in \mathbb{Z}$ and $n \in \mathbb{Z} \setminus \{0\}$ then the unique number q such that $m = n \cdot q$ is called the **quotient** of n and m and is noted as $m:n$. So $n \cdot (m:n) = m$.

Definition 7.45. (Common Divisor) If $n, m \in \mathbb{Z}$ then d is a **common divisor** of n and m if $d|n$ and $d|m$.

Lemma 7.46. If $n, m \in \mathbb{Z}$ such that $m \neq 0$ and $n|m$ then $n \leq |m|$

Proof. As $n|m$ there exist a $q \in \mathbb{Z}$ such that $n \cdot q = m$, as $m \neq 0$ we must have $q \neq 0$ [otherwise $m = n \cdot q = 0$]. For n, m we have now the following possibilities to consider:

$0 < m \wedge n \leq 0$. In this case we have $n \leq 0 < m \leq |m|$ so that $n \leq |m|$

$0 < m \wedge 0 < n$. If $q \leq 0 \xRightarrow{q \neq 0} q < 0 \xRightarrow{0 < n \wedge [\text{theorems: 7.27, 4.50}]} q \cdot n < 0 \cdot n = 0$ so that $m = q \cdot n < 0$ contradicting $0 < m$, hence we must have that $0 < q$. Using [theorem: 7.23] we have $1 \leq q$ so that by [theorems: 7.27, 4.50] $n = 1 \cdot n \leq q \cdot n = m = |m|$, hence $n \leq |m|$.

$m < 0 \wedge n \leq 0$. Then $0 < -m = |m|$ so that $n \leq 0 < |m|$ giving $n \leq |m|$.

$m < 0 \wedge 0 < n$. If $0 \leq q \xRightarrow{q \neq 0} 0 < q \xRightarrow{0 < n \wedge [\text{theorems: 7.27, 4.50}]} 0 = 0 \cdot n < q \cdot n = m$ contradicting $m < 0$, hence $q < 0$, so that $0 < -q$. Using [theorem: 7.23] we have then

$$1 \leq -q \xRightarrow{[\text{theorems: 7.27, 4.50}]} n = 1 \cdot n \leq (-q) \cdot n = -(q \cdot n) = |m|$$

proving that $n \leq |m|$.

So in all cases we have

$$n \leq |m|$$

□

Theorem 7.47. Let $n, m \in \mathbb{Z}$ with $n \neq 0$ then $\max(\{d \in \mathbb{Z}_0^+ | d \text{ is a common divisor of } n \text{ and } m\})$ exist and $0 < 1 \leq \max(\{d \in \mathbb{Z}_0^+ | d \text{ is a common divisor of } n \text{ and } m\})$

Proof. Let $n, m \in \mathbb{Z}$ and define $D_{n,m} = \{d \in \mathbb{Z}_0^+ | d \text{ is a common divisor of } n \text{ and } m\}$. By [example: 7.41] 1 is a common divisor of n and m , which as $0 < 1$ means that $1 \in D_{n,m}$ so that $D_{n,m} \neq \emptyset$. Let $d \in D_{n,m}$ then as $d|n$ and $n \neq 0$ we have by [lemma: 7.46] that $d \leq |n|$ so that $D_{n,m}$ has a upper bound. As $(\mathbb{Z}_0^+, \leq)$ is conditional complete [see theorem: 7.32] it follows that $\sup(D_{n,m})$ exist, using [theorem: 7.33] $\sup(D_{n,m}) \in D_{n,m}$. Hence $\max(D_{n,m})$ exists. □

The above theorem ensures that the following definition is well defined,

Definition 7.48. Let $n, m \in \mathbb{Z}_0^+$ with $n \neq 0$ then

$$\gcd(n, m) = \max(\{d \in \mathbb{Z}_0^+ \mid d \text{ is a common divisor of } n \text{ and } m\}) \geq 1 > 0$$

$\gcd(n, m)$ is called the **greatest common divisor** of n and m .

Theorem 7.49. If $n, m \in \mathbb{Z}$ with $m \neq 0$ then we have

1. $\{d \in \mathbb{Z} \mid d \mid (n : \gcd(n, m)) \wedge d \mid (m : \gcd(n, m))\} = \{1, -1\}$
2. $\gcd(n : \gcd(n, m), m : \gcd(n, m)) = 1$

Proof. As $\gcd(n, m) \mid n$ and $\gcd(n, m) \mid m$ the quotients $n : \gcd(n, m)$ and $m : \gcd(n, m)$ are well defined.

1. Take $n' = n : \gcd(n, m)$ and $m' = m : \gcd(n, m)$ then $n = n' \cdot \gcd(n, m)$ and $m = m' \cdot \gcd(n, m)$. If $d \mid n'$ and $d \mid m'$ there exists $n'', m'' \in \mathbb{Z}$ such that $n'' \cdot d = n'$ and $m'' \cdot d = m'$. Multiplying both sides by $\gcd(n, m)$ gives

$$(d \cdot \gcd(n, m)) \cdot n'' = (n'' \cdot d) \cdot \gcd(n, m) = n' \cdot \gcd(n, m) = n \quad (7.17)$$

and

$$(d \cdot \gcd(n, m)) \cdot m'' = (m'' \cdot d) \cdot \gcd(n, m) = m' \cdot \gcd(n, m) = m \quad (7.18)$$

proving that $d \cdot \gcd(n, m) \mid n$ and $d \cdot \gcd(n, m) \mid m$ Using [theorem: 7.42] and $0 < \gcd(n, m)$ we have that

$$|d| \cdot \gcd(n, m) \mid n \text{ and } |d| \cdot \gcd(n, m) \mid m$$

So by the definition of $\gcd(n, m)$ we have then that

$$|d| \cdot \gcd(n, m) \leq \gcd(n, m) = 1 \cdot \gcd(n, m)$$

As $0 < \gcd(n, m)$ we have by [theorems: 7.27, 4.50] and the above that

$$|d| \leq 1$$

If $d = 0$ then by [eq: 7.18] $m = 0$ contradicting $m \neq 0$ so we have $d \neq 0$, proving by [theorem: 7.37] that $|d| \neq 0$ which as $0 \leq |d|$ implies that $0 < |d|$ or using [theorem: 7.23] $1 \leq |d|$, which by the above proves that $|d| = 1$ hence $d = 1$ or $d = -1$. So

$$\{d \in \mathbb{Z} \mid d \mid (n : \gcd(n, m)) \wedge d \mid (m : \gcd(n, m))\} = \{1, -1\}$$

2. We have

$$\begin{aligned} \gcd(n : \gcd(n, m), m : \gcd(n, m)) &= \max(\{d \in \mathbb{Z}_0^+ \mid d \mid (n : \gcd(n, m)) \wedge d \mid (m : \gcd(n, m))\}) \\ &\stackrel{(1)}{=} \max(\{1, -1\}) \\ &= 1 \end{aligned}$$

□

Definition 7.50. A $z \in \mathbb{Z}$ is **even** if $2 \mid z$ and **odd** if z is not even.

Theorem 7.51. Let $z \in \mathbb{Z}$ then we have

1. z is even $\Leftrightarrow \exists m \in \mathbb{Z}$ such that $z = 2 \cdot m$
2. z is odd $\Leftrightarrow \exists m \in \mathbb{Z}$ such that $z = 2 \cdot m + 1$

Proof.

- 1.

$$\begin{aligned} z \text{ is even} &\Leftrightarrow 2 \mid z \\ &\Leftrightarrow \exists m \in \mathbb{Z} \text{ such that } z = 2 \cdot m \end{aligned}$$

2. Using the Division Algorithm [see: theorem: 7.38] there exists unique $q, r \in \mathbb{Z}$ such that $z = 2 \cdot q + r$ and $0 \leq r < 2$ proving that $r \in \{0, 1\}$. So

$$\begin{aligned} z \text{ is odd} &\Leftrightarrow z \text{ is not even} \\ &\Leftrightarrow r=0 \Rightarrow z \text{ is even} \\ &\quad \square \\ &\quad z = 2 \cdot q + 1 \end{aligned}$$

Theorem 7.52. *If $z \in \mathbb{Z}$ then we have*

1. $z \text{ is even} \Leftrightarrow z^2 = z \cdot z \text{ is even}$
2. $z \text{ is odd} \Leftrightarrow z^2 = z \cdot z \text{ is odd}$

Proof.

1. If z is even then $z = 2 \cdot m$ so that $z \cdot z = (2 \cdot m) \cdot (2 \cdot m) = 2 \cdot (2 \cdot (m \cdot m))$ proving that $z \cdot z$ is even. If $z \cdot z$ is even then if z is odd we have $z = 2 \cdot m + 1$ so that

$$\begin{aligned} z \cdot z &= (2 \cdot m + 1) \cdot (2 \cdot m + 1) \\ &= 2 \cdot (m \cdot (2 \cdot m + 1)) + 2 \cdot m + 1 \\ &= 2 \cdot (m \cdot (2 \cdot m + 1) + m) + 1 \end{aligned}$$

proving that $z \cdot z$ is odd contradiction the fact that $z \cdot z$ is even, hence z should be even.

2. This follows from (1) by contra position. □

7.3 Denumerability of the Integers

Theorem 7.53. $\mathbb{Z}_0^+$, $\mathbb{Z}_0^-$ and $\mathbb{Z}$ are all denumerable

Proof. Using [theorem: 7.17 (3)] there exists a bijection $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+$ so that $\mathbb{N}_0 \approx \mathbb{Z}_0^+$, hence

$\mathbb{Z}_0^+$ is denumerable

Define now $\beta: \mathbb{Z}_0^+ \rightarrow \mathbb{Z}_0^-$ by $\beta(n) = -n$ then we have

injectivity. If $\beta(n) = \beta(n')$ then $-n = -n' \Rightarrow n = -(-n) = -(-n') = n'$

surjectivity. If $n \in \mathbb{Z}_0^- = \{-n | n \in \mathbb{Z}_0^+\}$ there exists $m \in \mathbb{Z}^+$ such that $n = -m = \beta(m)$

Hence $\beta: \mathbb{Z}_0^+ \rightarrow \mathbb{Z}_0^-$ is a bijection proving that $\mathbb{Z}_0^+ \approx \mathbb{Z}_0^-$. So using [theorem: 6.26] it follows that

$\mathbb{Z}_0^-$ is denumerable

Finally as $\mathbb{Z} \stackrel{=}{\underset{[\text{theorem: 7.20}]}{\mathbb{Z}_0^+ \cup \mathbb{Z}_0^-}}$ it follows by [theorem: 6.68] that

$\mathbb{Z}$ is denumerable □

Chapter 8

The Rational Numbers

In this chapter we will introduce the set of rational numbers and embed the integer numbers in it. Just as with $\mathbb{Z}$ and $\mathbb{N}_0$ we will introduce a order relation, a sum operator, a product operator, neutral elements for addition and multiplication as well as inverse elements. To avoid excessive notation we use the same symbols for the natural numbers, integers and rational numbers and use context to determine the meaning of the symbols involved. The following table should help you in determining the meaning of the different symbols based on the context of their usage.

Context	Expression	Operator
$n, m \in \mathbb{N}_0$	$n+m$	sum in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{N}_0$	$n \cdot m$	product in $\langle \mathbb{N}_0, \cdot \rangle$
$n, m \in \mathbb{N}_0$	$n \leq m$	order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n < m$	strict order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n - m$	subtraction in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{N}_0, \cdot \rangle$
$n \in \mathbb{N}_0$	$-n$	inverse element in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{Z}$	$n+m$	sum in $\langle \mathbb{Z}, + \rangle$
$n, m \in \mathbb{Z}$	$n \cdot m$	product in $\langle \mathbb{Z}, \cdot \rangle$
$n, m \in \mathbb{Z}$	$n \leq m$	order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n < m$	strict order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n - m$	subtraction in $\langle \mathbb{Z}, - \rangle$
$n \in \mathbb{Z}$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{Z}, + \rangle$
$n \in \mathbb{Z}$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{Z}, \cdot \rangle$
$n \in \mathbb{Z}$	$-n$	inverse element in $\langle \mathbb{Z}, + \rangle$
$q, r \in \mathbb{Q}$	$q+r$	sum in $\langle \mathbb{Q}, + \rangle$
$q, r \in \mathbb{Q}$	$q \cdot r$	product in $\langle \mathbb{Q}, \cdot \rangle$
$q, r \in \mathbb{Q}$	$q \leq r$	order in $\langle \mathbb{Q}, \leq \rangle$
$q, r \in \mathbb{Q}$	$q < r$	strict order in $\langle \mathbb{Q}, \leq \rangle$
$q, r \in \mathbb{Q}$	$q - r$	subtraction in $\langle \mathbb{Q}, - \rangle$
$q, r \in \mathbb{Q}$	q/r	division in $\langle \mathbb{Q}, \cdot \rangle$
$q \in \mathbb{Q}$	$q + 0$ or $0 + q$	neutral element in $\langle \mathbb{Q}, + \rangle$
$q \in \mathbb{Q}$	$q \cdot 1$ or $1 \cdot q$	neutral element in $\langle \mathbb{Q}, \cdot \rangle$
$q \in \mathbb{Q}$	$-q$	inverse element in $\langle \mathbb{Q}, + \rangle$

8.1 Definition and arithmetic

One of the problems that the integer numbers have is that the quotient of two numbers n and m is only defined if n divides m . The following example shows this issue.

Example 8.1. If x is a even number and y is a odd number then x can not divide y .

Proof. As x is even there exists a $n \in \mathbb{Z}$ such that $x = 2 \cdot n$ and as y is odd y is not even. Assume that $x|y$ then there exists a $q \in \mathbb{Z}$ such that $y = x \cdot q$ but then $y = (2 \cdot n) \cdot q = 2 \cdot (n \cdot q)$ proving that y is even, contradicting the fact that y is odd. $\square$

The rational numbers will resolve this defect. Just as we have done with set of integers we work with pairs of integers (n, m) that will be interpreted as the quotient $\frac{n}{m}$. We have to be careful however for if $m = 0$ then the quotient only exist if $n = 0$ and then every integer is a quotient. So we should only consider pairs (n, m) where $n \in \mathbb{Z}$ and $m \in \mathbb{Z} \setminus \{0\}$. Further we have that $\frac{8}{4} = \frac{6}{3} = \frac{4}{2} = \frac{2}{1} = 2$ so we have to define a equivalence relation and work with equivalence classes.

Definition 8.2. $\mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$

Theorem 8.3. The relation $\simeq \subseteq (\mathbb{Z} \times \mathbb{Z}^*) \times (\mathbb{Z} \times \mathbb{Z}^*)$ defined by

$$\simeq = \{((n, m), (r, s)) \in (\mathbb{Z} \times \mathbb{Z}^*) \times (\mathbb{Z} \times \mathbb{Z}^*) \mid n \cdot s = m \cdot r\}$$

is a equivalence relation in $\mathbb{Z} \times \mathbb{Z}^*$.

Proof.

reflexivity. If $\{n, m\} \in \mathbb{Z} \times \mathbb{Z}^*$ then $n \cdot m \stackrel{[\text{theorem: 7.7}]}{=} m \cdot n$ so that $(n, m) \simeq (n, m)$

symmetry. If $(n, m) \simeq (r, s)$ then $n \cdot s = m \cdot r \stackrel{[\text{theorem: 7.7}]}{\Rightarrow} r \cdot m = s \cdot n$ proving that $(r, s) \simeq (n, m)$

transitivity. If $(n, m) \simeq (k, l)$ and $(k, l) \simeq (r, s)$ then $n \cdot l = m \cdot k$ and $k \cdot s = l \cdot r$, further

$$\begin{aligned} (n \cdot l) \cdot s &= (m \cdot k) \cdot s && \stackrel{[\text{theorem: 7.11}]}{\Rightarrow} && (n \cdot s) \cdot l = m \cdot (k \cdot s) \\ &&& \Rightarrow && (n \cdot s) \cdot l = m \cdot (l \cdot r) \\ &&& \stackrel{[\text{theorem: 7.11}]}{\Rightarrow} && (n \cdot s) \cdot l = (m \cdot r) \cdot l \\ &&& \stackrel{l \neq 0 \wedge [\text{theorem: 7.15}]}{\Rightarrow} && n \cdot s = m \cdot r \\ &&& \Rightarrow && (n, m) \simeq (r, s) \\ &&& \square \end{aligned}$$

Definition 8.4. The set of rational numbers noted as $\mathbb{Q}$ is defined as

$$\mathbb{Q} = (\mathbb{Z} \times \mathbb{Z}^*) / \simeq$$

or using the definition of $(\mathbb{Z} \times \mathbb{Z}^*) / \simeq$

$$\mathbb{Q} = \{ \simeq[(n, m)] \mid (n, m) \in \mathbb{Z} \times \mathbb{Z}^* \}$$

We note $\simeq[(n, m)] \in \mathbb{Q}$ as $\frac{n}{m}$, n is called the **numerator** and m is called the **denominator**. Using this notation we have that $\frac{n}{m} = \frac{n'}{m'} \Leftrightarrow n \cdot m' = m \cdot n'$. In this new notation we have

$$\mathbb{Q} = \left\{ \frac{n}{m} \mid (n, m) \in \mathbb{Z} \times \mathbb{Z}^* \right\}$$

Theorem 8.5. If $k \in \mathbb{Z}^*$ and $(n, m) \in \mathbb{Z} \times \mathbb{Z}^*$ then

1. $\frac{n}{m} = \frac{n \cdot k}{m \cdot k}$
2. $\frac{0}{n} = \frac{0}{1}$
3. $\frac{n}{m} = \frac{0}{1} \Leftrightarrow n = 0$
4. $\frac{n}{m} \neq \frac{0}{1} \Leftrightarrow n \neq 0$

Proof.

1. First as $k \neq 0$ and $m \neq 0$ we have that $m \cdot k \neq 0$ so that $\frac{n \cdot k}{m \cdot k} \in \mathbb{Q}$. Further

$$n \cdot (m \cdot k) \stackrel{[\text{theorem: 7.11}]}{=} m \cdot (n \cdot k)$$

proving that

$$\frac{n}{m} = \frac{n \cdot k}{m \cdot k}$$

2. As $0 \cdot 1 = 0 = n \cdot 0$ we have $\frac{0}{n} = \frac{0}{1}$
3. $\frac{n}{m} = \frac{0}{1} \Leftrightarrow n \cdot 1 = m \cdot 0 \Leftrightarrow n = 0$
4. This follows from (3) by contra-position. □

Theorem 8.6. Let $\frac{n}{m}, \frac{n'}{m'}, \frac{r}{s}, \frac{r'}{s'} \in \mathbb{Q}$ are such that $\frac{n}{m} = \frac{n'}{m'}$ and $\frac{r}{s} = \frac{r'}{s'}$ then

$$\frac{n \cdot s + r \cdot m}{m \cdot s}, \frac{n' \cdot s' + r' \cdot m'}{m' \cdot s'} \in \mathbb{Q} \text{ and } \frac{n \cdot s + r \cdot m}{m \cdot s} = \frac{n' \cdot s' + r' \cdot m'}{m' \cdot s'}$$

Proof. First as $m \neq 0, m' \neq 0, s \neq 0$ and $s' \neq 0$ we have $m \cdot s \neq 0, m' \cdot s' \neq 0$ so that

$$\frac{n \cdot s + r \cdot m}{m \cdot s}, \frac{n' \cdot s' + r' \cdot m'}{m' \cdot s'} \in \mathbb{Q}$$

As $\frac{n}{m} = \frac{n'}{m'}$ and $\frac{r}{s} = \frac{r'}{s'}$ we have

$$n \cdot m' = m \cdot n' \wedge r \cdot s' = s \cdot r' \quad (8.1)$$

$$\begin{aligned} (n \cdot s + r \cdot m) \cdot (m' \cdot s') &= (n \cdot s) \cdot (m' \cdot s') + (r \cdot m) \cdot (m' \cdot s') \\ &\stackrel{[\text{theorem: 7.11}]}{=} (n \cdot m') \cdot (s \cdot s') + (r \cdot s') \cdot (m \cdot m') \\ &\stackrel{[\text{eq: 8.1}]}{=} (m \cdot n') \cdot (s \cdot s') + (s \cdot r') \cdot (m \cdot m') \\ &\stackrel{[\text{theorem: 7.11}]}{=} (n' \cdot s) \cdot (m \cdot s) + (r' \cdot m') \cdot (m \cdot s) \\ &\stackrel{[\text{theorem: 7.11}]}{=} (n' \cdot s + r' \cdot m') \cdot (m \cdot s) \\ &\stackrel{[\text{theorem: 7.11}]}{=} (m \cdot s) \cdot (n' \cdot s + r' \cdot m') \end{aligned}$$

proving that

$$\frac{n \cdot s + r \cdot m}{m \cdot s} = \frac{n' \cdot s' + r' \cdot m'}{m' \cdot s'}$$

□

The above theorem ensures that the following is well-defined, independent of the representation.

Definition 8.7. The sum operator $+: \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ is defined by

$$\frac{n}{m} + \frac{r}{s} = \frac{n \cdot s + m \cdot r}{m \cdot s}$$

Theorem 8.8. $\langle \mathbb{Q}, + \rangle$ is a **Abelian group** with neutral element $0 = \frac{0}{1}$ and for every $\frac{n}{m} \in \mathbb{Q}$ the inverse element $\frac{-n}{m}$.

Proof.

associativity. Let $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \in \mathbb{Q}$ then

$$\begin{aligned} \frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f} \right) &= \frac{a}{b} + \frac{c \cdot f + d \cdot e}{d \cdot f} \\ &= \frac{a \cdot (d \cdot f) + b \cdot (c \cdot f + d \cdot e)}{b \cdot (d \cdot f)} \\ &\stackrel{[\text{theorem: 7.11}]}{=} \frac{(a \cdot d) \cdot f + (c \cdot b) \cdot f + (b \cdot d) \cdot e}{(b \cdot d) \cdot f} \\ &\stackrel{[\text{theorem: 7.11}]}{=} \frac{(a \cdot d + c \cdot b) \cdot f + (b \cdot d) \cdot e}{(b \cdot d) \cdot f} \\ &= \frac{a \cdot d + c \cdot b}{b \cdot d} + \frac{e}{f} \\ &= \left(\frac{a}{b} + \frac{c}{d} \right) + \frac{e}{f} \end{aligned}$$

commutativity. Let $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ then

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{a \cdot d + b \cdot c}{b \cdot d} \\ &\stackrel{[\text{theorem: 7.11}]}{=} \frac{c \cdot b + d \cdot a}{d \cdot b} \\ &= \frac{c}{d} + \frac{a}{b} \end{aligned}$$

neutral element. Let $\frac{a}{b} \in \mathbb{Q}$ then

$$\begin{aligned} \frac{a}{b} + \frac{0}{1} & \stackrel{\text{commutativity}}{=} \frac{a \cdot 1 + b \cdot 0}{b \cdot 1} \\ & \stackrel{[\text{theorem: 7.11}] \text{ and } [\text{theorem: 4.39}]}{=} \frac{a}{b} \end{aligned}$$

inverse element. Let $\frac{a}{b} \in \mathbb{Q}$ then we have

$$\begin{aligned} \frac{a}{b} + \frac{-a}{b} & = \frac{-a}{b} + \frac{a}{b} \\ & = \frac{(-a) \cdot b + b \cdot a}{b \cdot b} \\ & \stackrel{[\text{theorem: 7.11}]}{=} \frac{b \cdot ((-a) + a)}{b \cdot b} \\ & = \frac{b \cdot 0}{b \cdot b} \\ & \stackrel{[\text{theorem: 4.39}]}{=} \frac{0}{b \cdot b} \\ & \stackrel{[\text{theorem: 8.5}]}{=} \frac{0}{1} \\ & = 0 \end{aligned}$$

□

Definition 8.9. If $x, y \in \mathbb{Q}$ then $x - y = x + (-y)$

Next we define multiplication.

Theorem 8.10. If $\frac{n}{m}, \frac{n'}{m'}, \frac{r}{s}, \frac{r'}{s'} \in \mathbb{Q}$ such that $\frac{n}{m} = \frac{n'}{m'}$ and $\frac{r}{s} = \frac{r'}{s'}$ then

$$\frac{n \cdot r}{m \cdot s}, \frac{n' \cdot r'}{m' \cdot s'} \in \mathbb{Q} \text{ and } \frac{n \cdot r}{m \cdot s} = \frac{n' \cdot r'}{m' \cdot s'}$$

Proof. First as $m \neq 0, m' \neq 0, s \neq 0$ and $s' \neq 0$ we have that $m \cdot s \neq 0$ and $m' \cdot s' \neq 0$ so that $\frac{n \cdot r}{m \cdot s}, \frac{n' \cdot r'}{m' \cdot s'} \in \mathbb{Q}$. As $\frac{n}{m} = \frac{n'}{m'}$ and $\frac{r}{s} = \frac{r'}{s'}$ we have also that

$$n \cdot m' = m \cdot n' \wedge r \cdot s' = s \cdot r' \tag{8.2}$$

$$\begin{aligned} (n \cdot r) \cdot (m' \cdot s') & \stackrel{[\text{theorem: 7.11}]}{=} (n \cdot m') \cdot (r \cdot s') \\ & \stackrel{[\text{eq: 8.2}]}{=} (m \cdot n') \cdot (s \cdot r') \\ & = (m \cdot s) \cdot (n' \cdot r') \end{aligned}$$

so that

$$\frac{n \cdot r}{m \cdot s} = \frac{n' \cdot r'}{m' \cdot s'} \quad \square$$

The above theorem ensures that the next definition is well defined.

Definition 8.11. The product operator $\cdot : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ is defined by

$$\frac{n}{m} \cdot \frac{r}{s} = \frac{n \cdot r}{m \cdot s}$$

Theorem 8.12. $\langle \mathbb{Q}, +, \cdot \rangle$ is a field [see definition: 4.51] more specifically:

1. $\langle \mathbb{Q}, + \rangle$ is a Abelian group [see theorem: 8.8]

2. $\cdot : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ satisfies

distributivity. $\forall x, y, z \in \mathbb{Q}$ we have $x \cdot (y + z) = x \cdot y + x \cdot z$

commutativity. $\forall x, y \in \mathbb{Q}$ we have $x \cdot y = y \cdot x$

neutral element. $\forall x \in \mathbb{Q} \frac{1}{1} \cdot x = 1 = x \cdot \frac{1}{1}$, so $1 \stackrel{\text{definition}}{=} \frac{1}{1}$ is the neutral element.

associativity. $\forall x, y, z \in \mathbb{Q} (x \cdot y) \cdot z = x \cdot (y \cdot z)$

inverse element. $\forall x \in \mathbb{Q} \setminus \{0\} = \mathbb{Q}^*$ there exists a $x^{-1} \cdot x = x \cdot x^{-1}$. More specific if $x = \frac{a}{b} \neq 0$ then $x^{-1} = \frac{b}{a}$.

3. $0 \neq 1$

Proof.

1. This follows from [theorem: 8.8].

2. We have:

distributivity. Let $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \in \mathbb{Q}$ then

$$\begin{aligned}
 \frac{a}{b} \cdot \frac{c}{d} + \frac{a}{b} \cdot \frac{e}{f} &= \frac{\frac{a \cdot c}{b \cdot d} + \frac{a \cdot e}{b \cdot f}}{\frac{(a \cdot c) \cdot (b \cdot f) + (b \cdot d) \cdot (a \cdot e)}{(b \cdot d) \cdot (b \cdot f)}} \\
 &\stackrel{[\text{theorem: 7.11}]}{=} \frac{b \cdot (a \cdot (c \cdot f)) + b \cdot (a \cdot (d \cdot e))}{b \cdot (b \cdot (d \cdot f))} \\
 &\stackrel{[\text{theorem: 7.11}]}{=} \frac{b \cdot (a \cdot (c \cdot f) + a \cdot (d \cdot e))}{b \cdot (b \cdot (d \cdot f))} \\
 &\stackrel{b \neq 0 \wedge [\text{theorem: 8.5}]}{=} \frac{a \cdot (c \cdot f) + a \cdot (d \cdot e)}{b \cdot (d \cdot f)} \\
 &= \frac{a}{b} \cdot \left(\frac{c \cdot f + d \cdot e}{d \cdot f} \right) \\
 &= \frac{a}{b} \cdot \left(\frac{c}{d} + \frac{e}{f} \right)
 \end{aligned}$$

commutativity. Let $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ then

$$\begin{aligned}
 \frac{a}{b} \cdot \frac{c}{d} &= \frac{\frac{a \cdot c}{b \cdot d}}{\frac{c \cdot a}{d \cdot b}} \\
 &\stackrel{[\text{theorem: 7.11}]}{=} \frac{c}{d} \cdot \frac{a}{b} \\
 &= \frac{c}{d} \cdot \frac{a}{b}
 \end{aligned}$$

neutral element. Let $\frac{a}{b} \in \mathbb{Q}$ then

$$\begin{aligned}
 \frac{1}{1} \cdot \frac{a}{b} &\stackrel{\text{commutativity}}{=} \frac{\frac{a}{b} \cdot 1}{\frac{1 \cdot 1}{1}} \\
 &= \frac{a \cdot 1}{b \cdot 1} \\
 &\stackrel{[\text{theorem: 7.11}]}{=} \frac{a}{b}
 \end{aligned}$$

associativity. Let $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \in \mathbb{Q}$ then

$$\begin{aligned}
 \frac{a}{b} \cdot \left(\frac{c}{d} \cdot \frac{e}{f} \right) &= \frac{\frac{a}{b} \cdot \frac{c \cdot e}{d \cdot f}}{\frac{a \cdot (c \cdot e)}{b \cdot (d \cdot f)}} \\
 &\stackrel{[\text{theorem: 7.11}]}{=} \frac{(a \cdot c) \cdot e}{(b \cdot d) \cdot f} \\
 &= \frac{a \cdot c}{b \cdot d} \cdot \frac{e}{f} \\
 &= \left(\frac{a}{b} \cdot \frac{c}{d} \right) \cdot \frac{e}{f}
 \end{aligned}$$

inverse element. Let $\frac{a}{b} \in \mathbb{Q} \setminus \{0\}$ then $\frac{a}{b} \neq \frac{0}{1} \xRightarrow{[\text{theorem: 8.5}]} a \neq 0$ so that $\frac{b}{a} \in \mathbb{Q}$, then

$$\begin{aligned} \frac{a}{b} \cdot \frac{b}{a} & \stackrel{\text{commutativity}}{=} \frac{b}{a} \cdot \frac{a}{b} \\ & = \frac{b \cdot a}{a \cdot b} \\ & \stackrel{[\text{theorem: 7.11}]}{=} \frac{1 \cdot (a \cdot b)}{1 \cdot (a \cdot b)} \\ & \stackrel{a \cdot b \neq 0 \wedge [\text{theorem: 8.5}]}{=} \frac{1}{1} \end{aligned}$$

3. If $\frac{0}{1} = \frac{1}{1}$ then $0 \cdot 1 = 1 \cdot 1$ or that $0 = 1$ which is impossible by [theorem: 5.10]. □

Example 8.13. $1 + 1 = 2$ and $2^{-1} = \frac{1}{2}$ where $2 = \frac{2}{1}$.

Proof. $\frac{1}{1} + \frac{1}{1} = \frac{1 \cdot 1 + 1 \cdot 1}{1 \cdot 1} = \frac{1+1}{1} \stackrel{[\text{theorem: 7.12}]}{=} \frac{2}{1} = 2$, so $2^{-1} = \left(\frac{2}{1}\right)^{-1} = \frac{1}{2}$ □

Theorem 8.14. Let $q, r \in \mathbb{Q}$ and $s \neq 0$ then

1. $q = r \Leftrightarrow q \cdot s = r \cdot s$
2. $q \neq r \Leftrightarrow q \cdot s \neq r \cdot s$

Proof.

1.

$\Rightarrow$. If $q = r$ then $q \cdot s = r \cdot s$

$\Leftarrow$. We have

$$\begin{aligned} q \cdot s = r \cdot s & \xRightarrow{s \neq 0} (q \cdot s) \cdot s^{-1} = (r \cdot s) \cdot s^{-1} \\ & \stackrel{[\text{theorem: 8.12}]}{\Rightarrow} q \cdot (s \cdot s^{-1}) = r \cdot (s \cdot s^{-1}) \\ & \stackrel{[\text{theorem: 8.12}]}{\Rightarrow} q \cdot 1 = r \cdot 1 \\ & \stackrel{[\text{theorem: 8.12}]}{\Rightarrow} q = r \end{aligned}$$

2. This follows by contra-position. □

8.2 Order Relation

Definition 8.15. The set of non negative rational numbers $\mathbb{Q}_0^+$ and the set of non positive numbers $\mathbb{Q}_0^-$ is defined by:

$$\begin{aligned} \mathbb{Q}_0^+ &= \left\{ \frac{a}{b} \mid (a, b) \in \mathbb{Z} \times \mathbb{Z}^* \wedge a \cdot b \in \mathbb{Z}_0^+ \right\} \stackrel{[\text{theorem: 7.22}]}{=} \left\{ \frac{a}{b} \mid (a, b) \in \mathbb{Z} \times \mathbb{Z}^* \wedge 0 \leq a \cdot b \right\} \\ \mathbb{Q}_0^- &= \left\{ \frac{a}{b} \mid (a, b) \in \mathbb{Z} \times \mathbb{Z}^* \wedge a \cdot b \in \mathbb{Z}_0^- \right\} \stackrel{[\text{theorem: 7.22}]}{=} \left\{ \frac{a}{b} \mid (a, b) \in \mathbb{Z} \times \mathbb{Z}^* \wedge a \cdot b \leq 0 \right\} \end{aligned}$$

Theorem 8.16. $\mathbb{Q} = \mathbb{Q}_0^+ \cup \mathbb{Q}_0^-$ and $\{0\} = \mathbb{Q}_0^+ \cap \mathbb{Q}_0^-$

Proof. If $q \in \mathbb{Q}$ then $\exists (a, b) \in \mathbb{Z} \times \mathbb{Z}^*$ such that $q = \frac{a}{b}$, as $a \cdot b \in \mathbb{Z} \stackrel{[\text{theorem: 7.20}]}{=} \mathbb{Z}_0^+ \cup \mathbb{Z}_0^-$ we have either:

$a \cdot b \in \mathbb{Z}_0^+$. Then $q = \frac{a}{b} \in \mathbb{Q}_0^+$

$a \cdot b \in \mathbb{Z}_0^-$. Then $q = \frac{a}{b} \in \mathbb{Q}_0^-$

proving that

$$\mathbb{Q} \subseteq \mathbb{Q}_0^+ \cup \mathbb{Q}_0^-$$

As trivially $\mathbb{Q}_0^+ \subseteq \mathbb{Q}$ and $\mathbb{Q}_0^- \subseteq \mathbb{Q}$ we have that $\mathbb{Q}_0^+ \cup \mathbb{Q}_0^- \subseteq \mathbb{Q}$, which by the above proves that

$$\mathbb{Q} = \mathbb{Q}_0^+ \cup \mathbb{Q}_0^-$$

If $q \in \{0\}$ then $q = \frac{0}{1}$ so that $0 \cdot 1 = 0 \in \mathbb{Z}_0^+$ and $0 \cdot 1 = 0 \in \mathbb{Z}_0^-$ so that $q \in \mathbb{Q}_0^+ \cap \mathbb{Q}_0^-$ proving that

$$\{0\} \subseteq \mathbb{Q}_0^+ \cap \mathbb{Q}_0^- \quad (8.3)$$

If $q \in \mathbb{Q}_0^+ \cap \mathbb{Q}_0^-$ then there exist $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}^* \Rightarrow b \neq 0 \neq d$ such that $a \cdot b \in \mathbb{Z}_0^+ \Rightarrow 0 \leq a \cdot b$, $c \cdot d \in \mathbb{Z}_0^- \Rightarrow c \cdot d \leq 0$ and $\frac{a}{b} = \frac{c}{d} \Rightarrow a \cdot d = b \cdot c$. Assume that $a \neq 0$ then we have either:

0 < a. Assume that $b < 0$ then by [theorems: 7.27, 4.50] $a \cdot b < 0$ contradicting $0 \leq a \cdot b$, so we must have that $0 \leq b$ which as $b \neq 0$ gives

$$0 < b \quad (8.4)$$

As $d \neq 0$ we have by [theorems: 7.27, 4.50] that $0 < d \cdot d$ so that by [theorems: 7.27, 4.50]

$$0 < a \cdot (d \cdot d) = (a \cdot d) \cdot d \stackrel{a \cdot d = b \cdot c}{=} (b \cdot c) \cdot d = (c \cdot d) \cdot b \quad (8.5)$$

Using [theorems: 7.27, 4.50] on [eq: 8.4] and [eq: 8.5] we have that $0 < c \cdot d$ contradicting $c \cdot d \leq 0$.

a < 0. Assume that $0 < b$ then by [theorems: 7.27, 4.50] $a \cdot b < 0$ contradicting $0 \leq a \cdot b$, so we must have that $0 \leq b$ which as $b \neq 0$ gives

$$b < 0 \quad (8.6)$$

As $d \neq 0$ we have by [theorems: 7.27, 4.50] that $0 < d \cdot d$ so that by [theorems: 7.27, 4.50] $a \cdot (d \cdot d) < 0$, hence as $a \cdot d = b \cdot c$

$$(c \cdot d) \cdot b = (b \cdot c) \cdot d = (a \cdot d) \cdot d = a \cdot (d \cdot d) < 0 \quad (8.7)$$

Using [theorems: 7.27, 4.50] on [eq: 8.6] and [eq: 8.7] we have that $0 < c \cdot d$ contradicting $c \cdot d \leq 0$.

As in all cases we reach a contradiction the assumption $a \neq 0$ is wrong, so $a = 0$ or $q = \frac{0}{b} \stackrel{[\text{theorem: 8.5}]}{=} \frac{0}{1} = 0$. Hence $\mathbb{Q}_0^+ \cap \mathbb{Q}_0^- \subseteq \{0\}$ which combined with [eq: 8.3] proves that

$$\mathbb{Q}_0^+ \cap \mathbb{Q}_0^- = \{0\} \quad \square$$

Theorem 8.17. $\mathbb{Q}_0^- = \{-q \mid q \in \mathbb{Q}_0^+\}$

Proof. If $q \in \mathbb{Q}_0^-$ then $\exists (n, m) \in \mathbb{Z} \times \mathbb{Z}^*$ with $n \cdot m \in \mathbb{Z}_0^-$ such that $q = \frac{n}{m}$. By the definition of $\mathbb{Z}_0^-$ it follows that $\exists k \in \mathbb{Z}_0^+$ such that $n \cdot m = -k$. Hence

$$(-n) \cdot m \stackrel{[\text{theorem: 4.40}]}{=} -(n \cdot m) = -(-(k)) \stackrel{[\text{theorem: 4.9}]}{=} k \in \mathbb{Z}_0^+,$$

proving that $-q = \frac{-n}{m} \in \mathbb{Q}_0^+$. Using [theorem: 4.9] we have $q = -(-q)$ so that $q \in \{-q \mid q \in \mathbb{Q}_0^+\}$ or that

$$\mathbb{Q}_0^- \subseteq \{-q \mid q \in \mathbb{Q}_0^+\} \quad (8.8)$$

If $q \in \{-q \mid q \in \mathbb{Q}_0^+\}$ then $\exists (n, m) \in \mathbb{Z} \times \mathbb{Z}^*$ with $n \cdot m \in \mathbb{Z}_0^+$ such that $q = -\frac{n}{m} = \frac{-n}{m}$, as $(-n) \cdot m = -(n \cdot m) \in \mathbb{Z}_0^-$, it follows that $q \in \mathbb{Q}_0^-$. Hence $\{-q \mid q \in \mathbb{Q}_0^+\} \subseteq \mathbb{Q}_0^-$ which together with [eq: 8.8] gives

$$\mathbb{Q}_0^- = \{-q \mid q \in \mathbb{Q}_0^+\} \quad \square$$

Theorem 8.18. $\langle \mathbb{Q}_0^+, + \rangle$ is a sub semi-group of $\langle \mathbb{Q}, + \rangle$ [hence $\langle \mathbb{Q}_0^+, + \rangle$ is a semi-group]

Proof. By [theorem: 8.16]

$$0 \in \mathbb{Q}_0^+ \quad (8.9)$$

If $q, r \in \mathbb{Q}_0^+$ then there exists $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}^*$ with $a \cdot b, c \cdot d \in \mathbb{Z}_0^+$ such that $q = \frac{a}{b}$ and $r = \frac{c}{d}$. Then we have

$$q + r = \frac{a \cdot d + b \cdot c}{b \cdot d}$$

so we have to check that $(a \cdot d + b \cdot c) \cdot (b \cdot d) \in \mathbb{Z}_0^+$. Hence

$$\begin{aligned} (a \cdot d + b \cdot c) \cdot (b \cdot d) &\stackrel{[\text{theorem: 7.11}]}{=} (a \cdot d) \cdot (b \cdot d) + (b \cdot c) \cdot (b \cdot d) \\ &\stackrel{[\text{theorem: 7.11}]}{=} (a \cdot b) \cdot (d \cdot d) + (c \cdot d) \cdot (b \cdot b) \end{aligned} \quad (8.10)$$

Now using [theorems: 7.27, 4.50] we have that $0 \leq d \cdot d \wedge 0 \leq b \cdot b$, as $0 \leq a \cdot b \wedge 0 \leq c \cdot d$, we have by [theorems: 7.27, 4.50] that $0 \leq (a \cdot b) \cdot (d \cdot d) \wedge 0 \leq (c \cdot d) \cdot (b \cdot b)$ or $(a \cdot b) \cdot (d \cdot d), (c \cdot d) \cdot (b \cdot b) \in \mathbb{Z}_0^+$. Using [theorem: 7.17] it follows that $(a \cdot b) \cdot (d \cdot d) + (c \cdot d) \cdot (b \cdot b) \in \mathbb{Z}_0^+$, hence by [eq: 8.10] $(a \cdot d + b \cdot c) \cdot (b \cdot d) \in \mathbb{Z}_0^+$ so that $q + r \in \mathbb{Q}_0^+$. So

$$\forall q, r \in \mathbb{Q}_0^+ \text{ we have } q + r \in \mathbb{Q}_0^+ \quad (8.11)$$

Finally [eq: 8.9] and [eq: 8.11] proves that $\langle \mathbb{Q}_0^+, + \rangle$ is a semi-group. $\square$

Next we define the relation that will later become a order relation on $\mathbb{Q}$.

Definition 8.19. (Order Relation) $\leq \subseteq \mathbb{Q} \times \mathbb{Q}$ is defined as

$$\leq = \{(q, r) \in \mathbb{Q} \times \mathbb{Q} \mid r + (-q) \in \mathbb{Q}_0^+\}$$

So $q \leq r$ if and only if $r + (-q) \in \mathbb{Q}_0^+$

Theorem 8.20. $\mathbb{Q}_0^+ = \{q \in \mathbb{Q} \mid 0 \leq q\}$ and $\mathbb{Q}_0^- = \{q \in \mathbb{Q} \mid q \leq 0\}$.

Proof.

$$\begin{aligned} q \in \mathbb{Q}_0^+ &\stackrel{q = q + (-0)}{\Leftrightarrow} q + (-0) \in \mathbb{Q}_0^+ \\ &\stackrel{\mathbb{Q}_0^+ \subseteq \mathbb{Q}}{\Leftrightarrow} q \in \mathbb{Q} \wedge 0 \leq q \\ &\Leftrightarrow q \in \{q \in \mathbb{Q} \mid 0 \leq q\} \end{aligned}$$

proving that

$$\mathbb{Q}_0^+ = \{q \in \mathbb{Q} \mid 0 \leq q\}$$

Further

$$\begin{aligned} q \in \mathbb{Q}_0^- &\stackrel{[\text{theorem: 8.17}]}{\Leftrightarrow} -q \in \mathbb{Q}_0^+ \\ &\stackrel{0 + (-q) = -q}{\Leftrightarrow} 0 + (-q) \in \mathbb{Q}_0^+ \\ &\stackrel{\mathbb{Q}_0^+ \subseteq \mathbb{Q}}{\Leftrightarrow} q \in \mathbb{Q} \wedge q \leq 0 \\ &\Leftrightarrow q \in \{q \in \mathbb{Q} \mid q \leq 0\} \end{aligned}$$

proving that

$$\mathbb{Q}_0^- = \{q \in \mathbb{Q} \mid q \leq 0\} \quad \square$$

Theorem 8.21. If $q, r \in \mathbb{Q}$ then

1. $q \leq r \Leftrightarrow 0 \leq r + (-q)$
2. $q < r \Leftrightarrow 0 < r + (-q)$

Proof.

1. We have

$$\begin{aligned} q \leq r &\Leftrightarrow r + (-q) \in \mathbb{Q}_0^+ \\ &\stackrel{[\text{theorem: 8.20}]}{\Leftrightarrow} 0 \leq r + (-q) \end{aligned}$$

2. We have

$$\begin{aligned}
 q < r &\Leftrightarrow q \neq r \wedge q \leq r \\
 &\Leftrightarrow q + (-q) \neq r + (-q) \wedge q \leq r \\
 &\Leftrightarrow 0 \neq r + (-q) \wedge q \leq r \\
 &\stackrel{(1)}{\Leftrightarrow} 0 \neq r + (-q) \wedge 0 \leq r + (-q) \\
 &\Leftrightarrow 0 < r + (-q)
 \end{aligned}$$

□

Theorem 8.22. *If $q \in \mathbb{Q}$ satisfies $0 \leq q \wedge q \leq 0$ then $q = 0$*

Proof. As $0 \leq q$ and $q \leq 0$ we have by [theorem: 8.20] that $q \in \mathbb{Q}_0^+ \wedge \mathbb{Q}_0^-$, so $q \in \mathbb{Q}_0^+ \cap \mathbb{Q}_0^- \stackrel{[\text{theorem: 8.16}]}{=} \{0\}$. Proving $q = 0$. □

Theorem 8.23. *$\langle \mathbb{Q}, \leq \rangle$ is a totally ordered set.*

Proof.

reflectivity. If $q \in \mathbb{Q}_0^+$ then $q + (-q) = 0 \in \{0\} \stackrel{[\text{theorem: 8.16}]}{=} \mathbb{Q}_0^+ \cap \mathbb{Q}_0^- \subseteq \mathbb{Q}_0^+$ so that $q \leq q$.

anti symmetry. If $q \leq r$ and $r \leq q$ then $r + (-q) \in \mathbb{Q}_0^+$, $q + (-r) \in \mathbb{Q}_0^+ \Rightarrow -(q + (-r)) \in \mathbb{Q}_0^-$ so that by [theorem: 8.20] $0 \leq r + (-q)$ and $r + (-q) = -(q + (-r)) \leq 0$. Using [theorem: 8.22] it follows that $r + (-q) = 0$ so that $r = q$.

transitivity. If $q \leq r$ and $r \leq s$ then $r + (-q), s + (-r) \in \mathbb{Q}_0^+$ so that by [theorem: 8.18] we have that

$$(r + (-q)) + (s + (-r)) \in \mathbb{Q}_0^+ \quad (8.12)$$

As $(r + (-q)) + (s + (-r)) \stackrel{[\text{theorem: 8.8}]}{=} s + (-q)$ we have by [eq: 8.12] that $s + (-q) \in \mathbb{Q}_0^+$ proving that

$$q \leq s$$

totally order. If $q, r \in \mathbb{Q}$ then as $r + (-q) \in \mathbb{Q} \stackrel{[\text{theorem: 8.16}]}{=} \mathbb{Q}_0^+ \cup \mathbb{Q}_0^-$ we have the following possibilities:

$r + (-q) \in \mathbb{Q}_0^+$. Then $q \leq r$

$r + (-q) \in \mathbb{Q}_0^-$. Then by [theorem: 8.17] we have that $-(r + (-q)) \in \mathbb{Q}_0^+$. Further

$$-(r + (-q)) \stackrel{[\text{theorem: 4.40}]}{=} (-r) + (-(-q)) = q + (-r)$$

so that $q + (-r) \in \mathbb{Q}_0^+$ or $r \leq q$. □

Lemma 8.24. *Let $q \in \mathbb{Q}$ then $0 < q \Leftrightarrow \exists n, m \in \mathbb{Z}$ with $0 < n \wedge 0 < m$ such that $q = \frac{n}{m}$*

Proof.

$\Rightarrow$. As $0 < q$ we have $0 \neq q$ and $0 \leq q \stackrel{[\text{theorem: 8.20}]}{\Rightarrow} q \in \mathbb{Q}_0^+$, so there exists $(n', m') \in \mathbb{Z} \times \mathbb{Z}^*$ with $0 \leq n' \cdot m'$ such that $q = \frac{n'}{m'}$, as $m' \in \mathbb{Z}^*$ we have $m' \neq 0$, further by [theorem: 8.5] $n' \neq 0$. So we have the following resting cases to consider for n', m' :

$0 < n' \wedge 0 < m'$. Then $q = \frac{n'}{m'}$ so if we take $n = n'$ and $m = m'$ we have $0 < n \wedge 0 < m$ such that $q = \frac{n}{m}$.

$0 < n' \wedge m' < 0$. Then by [theorems: 7.27, 4.50] we have $n' \cdot m' < 0$ contradicting $0 \leq n' \cdot m'$ so this is not a valid case.

$n' < 0 \wedge 0 < m'$. Then by [theorems: 7.27, 4.50] we have $n' \cdot m' < 0$ contradicting $0 \leq n' \cdot m'$ so this is not a valid case.

$n' < 0 \wedge m' < 0$. Then by [theorem: 7.27, 4.50] we have $0 < -n' \wedge 0 < -m'$ so that $\frac{-n'}{-m'} = \frac{n' \cdot (-1)}{m' \cdot (-1)} \stackrel{[\text{theorem: 8.5}]}{=} \frac{n'}{m'} = q$. So if we take $n = -n'$ and $m = -m'$ then $0 < n \wedge 0 < m$ and $q = \frac{n}{m}$.

So in all valid cases we found a $n, m \in \mathbb{Z}$ with $0 < n \wedge 0 < m$ and $q = \frac{n}{m}$.

$\Leftarrow$. If $\exists n, m \in \mathbb{Z}$ with $0 < n \wedge 0 < m$ such that $q = \frac{n}{m}$ then by [theorem: 7.26] we have that $0 < n \cdot m$ so that $0 \leq q$, further by [theorem: 8.5] and $n \neq 0$ we have $q \neq 0$, hence $0 < q$. $\square$

Example 8.25. $0 < 1$ where $0, 1 \in \mathbb{Q}$

Proof. As $1 = \frac{1}{1} \in \mathbb{Q}$ and $1 = 1 \cdot 1 \in \mathbb{Z}$ and $0 < 1$ [see example: 7.25] it follows from [lemma: 8.24] that $0 < 1$. $\square$

Corollary 8.26. If $q \in \mathbb{Q}$ then

1. $q < q + 1$
2. $q - 1 = q + (-1) < q$

Proof.

1. $(q + 1) + (-q) = (q + (-q)) + 1 = 1$ and by [example: 8.25] $0 < 1$ so that $q \leq q + 1$. If $q = q + 1$ we have $0 = q + (-q) = (q + 1) + (-q) = 1$ contradicting $0 < 1$ so we must have that

$$q < q + 1$$

2. $q + (-(q + (-1))) \stackrel{[\text{theorem: 4.8}]}{=} q + ((-q) + (-(-1))) = (q + (-q)) + (-(-1)) = 0 + 1 = 1$ and by [example: 8.25] $0 < 1$ so that $q + (-1) \leq q$. If $q + (-1) = q$ then $q = q + 1$ contradicting (1) so $q - 1 \neq q$ and we have

$$q - 1 = q + (-1) < q$$

$\square$

Theorem 8.27. Let $n \in \mathbb{Z}$ and $m \in \mathbb{Z}^*$ then we have

1. $n = m \Leftrightarrow \frac{n}{m} = 1$
2. If $0 < m$ then
 - a. $n < m \Leftrightarrow \frac{n}{m} < 1$
 - b. $m < n \Leftrightarrow 1 < \frac{n}{m}$
 - c. $n \leq m \Leftrightarrow \frac{n}{m} \leq 1$
 - d. $m \leq n \Leftrightarrow 1 \leq \frac{n}{m}$
3. If $m < 0$ then
 - a. $n < m \Leftrightarrow 1 < \frac{n}{m}$
 - b. $m < n \Leftrightarrow \frac{n}{m} < 1$
 - c. $n \leq m \Leftrightarrow 1 \leq \frac{n}{m}$
 - d. $m \leq n \Leftrightarrow \frac{n}{m} \leq 1$

Proof.

1.

$\Rightarrow$. If $n = m$ then $\frac{n}{m} = \frac{n}{n} = \frac{1 \cdot n}{1 \cdot n} \stackrel{[\text{theorem: 8.5}]}{=} \frac{1}{1} = 1$.

$\Leftarrow$. If $\frac{n}{m} = 1$ then $\frac{n}{m} = \frac{1}{1}$ so that $n \cdot 1 = m \cdot 1$ proving that $n = m$.

2.

a. Note that

$$1 + \left(-\frac{n}{m}\right) = \frac{1}{1} + \frac{-n}{m} = \frac{m + (-n)}{m}$$

So

$$\begin{aligned}
 n < m &\Leftrightarrow 0 < m + (-n) \\
 &\stackrel{[\text{theorem: 8.24}]}{\Leftrightarrow} 0 < \frac{m + (-n)}{m} \\
 &\Leftrightarrow 1 + \left(-\frac{n}{m}\right) \\
 &\Leftrightarrow \frac{n}{m} < 1
 \end{aligned}$$

b. Note that

$$\frac{n}{m} + (-1) = \frac{n}{m} + \frac{-1}{1} = \frac{n + (-m)}{m}$$

So

$$\begin{aligned}
 m < n &\Leftrightarrow 0 < m + (-n) \\
 &\stackrel{[\text{theorem: 8.24}]}{\Leftrightarrow} 0 < \frac{m + (-n)}{m} \\
 &\Leftrightarrow 0 < \frac{n}{m} + (-1) \\
 &\Leftrightarrow 1 < \frac{n}{m}
 \end{aligned}$$

c. This follows from (1) and (2.a)

d. This follows from (1) and (2.b)

3. As $m < 0$ we have by [theorem: 7.27, 4.50] that $0 < -m$ so we have

a. Then

$$\begin{aligned}
 n < m &\stackrel{[\text{theorem: 7.27, 4.50}]}{\Leftrightarrow} -m < -n \\
 &\stackrel{(2.b)}{\Leftrightarrow} 1 < \frac{-n}{-m} \\
 &\stackrel{[\text{theorem: 7.27, 4.40}]}{\Leftrightarrow} 1 < \frac{(-1) \cdot n}{(-1) \cdot m} \\
 &\stackrel{[\text{theorem: 8.5}]}{=} 1 < \frac{n}{m}
 \end{aligned}$$

b. Then

$$\begin{aligned}
 m < n &\stackrel{[\text{theorem: 7.27, 4.50}]}{\Leftrightarrow} -n < -m \\
 &\stackrel{(2.a)}{\Leftrightarrow} \frac{-n}{-m} < 1 \\
 &\stackrel{[\text{theorem: 7.27, 4.40}]}{\Leftrightarrow} \frac{(-1) \cdot n}{(-1) \cdot m} < 1 \\
 &\stackrel{[\text{theorem: 8.5}]}{=} \frac{n}{m} < 1
 \end{aligned}$$

c. This follows from (1) and (3.a)

d. This follows from (1) and (3.b) □

Theorem 8.28. *If $q, r \in \mathbb{Q}$ such that $0 < q$ and $0 < r$ then $0 < q \cdot r$.*

Proof. As $0 < q \wedge 0 < r$ we have by [lemma: 8.24] the existence of $a, b, c, d \in \mathbb{Z}$ with $0 < a, 0 < b, 0 < c, 0 < d$ such that $q = \frac{a}{b}$ and $r = \frac{c}{d}$. So by applying [theorems: 7.27, 4.50] we have $0 < a \cdot c \wedge 0 < b \cdot d$, hence $0 < (a \cdot c) \cdot (b \cdot d)$, so that $q \cdot r = \frac{a \cdot c}{b \cdot d} \in \mathbb{Q}_0^+$ or $0 \leq q \cdot r$. As $0 < a \cdot c$ we have by [theorem: 8.5] that $q \cdot r \neq 0$, so

$$0 < q \cdot r \quad \square$$

Theorem 8.29. $\langle \mathbb{Q}, +, \cdot, \leq \rangle$ is a ordered field

Proof. First using [theorem: 8.12] $\langle \mathbb{Q}, +, \cdot \rangle$ is a field. Next

1. For $r, q, s \in \mathbb{Q}$ with $r < q$ we have

$$q + (-r) \in \mathbb{Q}_0^+ \text{ and } r \neq q \Rightarrow r + s \neq q + s \quad (8.13)$$

Further

$$\begin{aligned}
 (q+s) + (-(r+s)) & \stackrel{[\text{theorem: 4.8}]}{=} (q+s) + ((-r) + (-s)) \\
 & \stackrel{\text{commutativity}}{=} (q+s + ((-s) + (-r))) \\
 & \stackrel{\text{associativity}}{=} q + (s + ((-s) + (-r))) \\
 & \stackrel{\text{associativity}}{=} q + ((s + (-s)) + (-r)) \\
 & = q + (0 + (-r)) \\
 & = q + (-r)
 \end{aligned}$$

which by [eq: 8.13] proves that $(q+s) + (-(r+s)) \in \mathbb{Q}_0^+$. Hence $r+s \leq q+s$ and $r+s \neq q+s$ proving that

$$r+s < q+s$$

2. Let $q, r \in \mathbb{Q}$ with $0 < q$ and $0 < r$ then by [theorem: 8.28] $0 < q \cdot r$. □

Theorem 8.30. $\forall q \in \mathbb{Q}$ with $q \neq 0$ we have $-(q^{-1}) = (-q)^{-1}$

Proof. If $q \neq 0$ then $-q \neq 0$ so q^{-1} and $(-q)^{-1}$ exists, further $q = \frac{a}{b}$ where $a, b \neq 0$. Now

$$-(q^{-1}) = -\left(\frac{b}{a}\right) = \left(\frac{-b}{a}\right) = \left(\frac{a}{-b}\right)^{-1} = \left(\frac{-1}{-1} \cdot \frac{a}{-b}\right)^{-1} = \left(\frac{-a}{b}\right)^{-1} = (-q)^{-1} \quad \square$$

Next we embed the set of integer numbers in the set of rational numbers.

Definition 8.31. $\mathbb{Z}_{\mathbb{Q}} = \{\frac{z}{1} | z \in \mathbb{Z}\} \subseteq \mathbb{Q}$

Theorem 8.32. $\mathbb{Z}_{\mathbb{Q}}$ is a sub-ring [see definition: 4.35] of $\langle \mathbb{Q}, +, \cdot \rangle$ and for

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{Q}} \text{ defined by } i_{\mathbb{Z} \rightarrow \mathbb{Q}}(z) = \frac{z}{1}$$

we have

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, \leq \rangle \text{ is a order isomorphism}$$

and

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, +, \cdot \rangle \text{ is a ring isomorphism}$$

Proof. If $q, r \in \mathbb{Z}_{\mathbb{Q}}$ then $\exists n, m \in \mathbb{Z}$ such that $q = \frac{n}{1}$ and $r = \frac{m}{1}$ so we have

$$q+r = \frac{n}{1} + \frac{m}{1} = \frac{n \cdot 1 + 1 \cdot m}{1 \cdot 1} = \frac{n+m}{1}$$

proving that $q+r \in \mathbb{Z}_{\mathbb{Q}}$. Further we have

$$q \cdot r = \frac{n}{1} \cdot \frac{m}{1} = \frac{n \cdot m}{1 \cdot 1} = \frac{n \cdot m}{1}$$

proving that $q \cdot r \in \mathbb{Z}_{\mathbb{Q}}$. Also $-q = \frac{-n}{1} \in \mathbb{Z}_{\mathbb{Q}}$. So we have

$$\forall q, r \in \mathbb{Z}_{\mathbb{Q}} \text{ we have } q+r \in \mathbb{Z}_{\mathbb{Q}}, q \cdot r \in \mathbb{Z}_{\mathbb{Q}} \text{ and } -q \in \mathbb{Z}_{\mathbb{Q}}$$

Further we have $0 = \frac{0}{1}$ and $1 = \frac{1}{1}$ so that

$$0, 1 \in \mathbb{Z}_{\mathbb{Q}}$$

Hence we have that

$$\mathbb{Z}_{\mathbb{Q}} \text{ is a sub-ring of } \langle \mathbb{Q}, +, \cdot \rangle$$

Now for $i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{Q}}$ we have:

injectivity. If $i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x) = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y)$ then $\frac{x}{1} = \frac{y}{1}$ so that $x \cdot 1 = 1 \cdot y \Rightarrow x = y$.

surjectivity. This follows from the definition of $\mathbb{Z}_{\mathbb{Q}}$.

proving that

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{Q}} \text{ is a bijection} \quad (8.14)$$

Further if $x, y \in \mathbb{Z}$ then $i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y) + (-i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x)) = \frac{y}{1} + \left(-\frac{x}{1}\right) = \frac{y}{1} + \frac{-x}{1} = \frac{y \cdot 1 + 1 \cdot (-x)}{1 \cdot 1} = \frac{y + (-x)}{1}$ so

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y) + (-i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x)) = \frac{y + (-x)}{1} \quad (8.15)$$

Hence we have

$$\begin{aligned} i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x) \leq i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y) &\Leftrightarrow i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y) + (-i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x)) \in \mathbb{Q}_0^+ \\ &\stackrel{[\text{eq: 8.15}]}{\Leftrightarrow} \frac{y + (-x)}{1} \in \mathbb{Q}_0^+ \\ &\stackrel{\text{def } \mathbb{Q}_0^+}{\Leftrightarrow} (y + (-x)) \cdot 1 \in \mathbb{Z}_0^+ \\ &\Leftrightarrow y + (-x) \in \mathbb{Z}_0^+ \\ &\Leftrightarrow x \leq y \end{aligned}$$

which combined with [eq: 8.14] proves that

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, \leq \rangle \text{ is a order isomorphism}$$

Now for the proof that $i_{\mathbb{Z} \rightarrow \mathbb{Q}}$ is a ring isomorphism.

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x + y) = \frac{x + y}{1} = \frac{x \cdot 1 + 1 \cdot y}{1 \cdot 1} = \frac{x}{1} + \frac{y}{1} = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x) + i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y)$$

and

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x \cdot y) = \frac{x \cdot y}{1} = \frac{x}{1} \cdot \frac{y}{1} = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x) \cdot i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y)$$

and

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}(1) = \frac{1}{1} = 1$$

proving with [eq: 8.14] that

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, +, \cdot \rangle \text{ is a ring isomorphism} \quad \square$$

Theorem 8.33. $\mathbb{Q} = \{x \cdot y^{-1} | x \in \mathbb{Z}_{\mathbb{Q}} \wedge y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}\}$

Proof. If $q \in \mathbb{Q}$ there exists $x' \in \mathbb{Z}$ and $y' \in \mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$ such that $q = \frac{x'}{y'}$. Define now $x = \frac{x'}{1} = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x')$ and $y = \frac{y'}{1} = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y')$ then by the previous theorems [theorem: 8.32] we have that $x \in \mathbb{Z}_{\mathbb{Q}}$ and $y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}$. Further

$$q = \frac{x'}{y'} = \frac{x'}{1} \cdot \frac{1}{y'} = \frac{x'}{1} \cdot \left(\frac{y'}{1}\right)^{-1} = x \cdot y^{-1} \in \{x \cdot y^{-1} | x \in \mathbb{Z}_{\mathbb{Q}} \wedge y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}\}$$

proving

$$\mathbb{Q} \subseteq \{x \cdot y^{-1} | x \in \mathbb{Z}_{\mathbb{Q}} \wedge y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}\} \quad (8.16)$$

If $q \in \{x \cdot y^{-1} | x \in \mathbb{Z}_{\mathbb{Q}} \wedge y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}\}$ there exists a $x \in \mathbb{Z}_{\mathbb{Q}}$ and $y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}$ such that $q = x \cdot y^{-1}$. Using [theorem: 8.32] again there exists $x' \in \mathbb{Z}$ and $y' \in \mathbb{Z} \setminus \{0\}$ such that $x = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x') = \frac{x'}{1}$ and $y = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y') = \frac{y'}{1}$ so that

$$q = x \cdot y^{-1} = \frac{x'}{1} \cdot \left(\frac{1}{y'}\right)^{-1} = \frac{x'}{1} \cdot \frac{1}{y'} = \frac{x'}{y'} \in \mathbb{Q}$$

proving that $\{x \cdot y^{-1} | x \in \mathbb{Z}_{\mathbb{Q}} \wedge y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}\} \subseteq \mathbb{Q}$ which combined with [eq: 8.16] results in

$$\mathbb{Q} = \{x \cdot y^{-1} | x \in \mathbb{Z}_{\mathbb{Q}} \wedge y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}\} \quad \square$$

Definition 8.34. $\mathbb{N}_{0, \mathbb{Q}} = (i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Q}})(\mathbb{N}_0) \subseteq \mathbb{Q}$ where

$$\begin{aligned} i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{Z} \text{ is defined by } i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n) &= \sim[n, 0] \\ i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Z} \rightarrow \mathbb{Q} \text{ is defined by } i_{\mathbb{Z} \rightarrow \mathbb{Q}}(z) &= \frac{z}{1} \end{aligned}$$

Theorem 8.35. *We have that*

1. $\mathbb{N}_{0,\mathbb{Q}} = \left\{ \frac{n}{1} \mid n \in \mathbb{Z}_0^+ \right\}$

2. *If we define $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \mathbb{N}_0 \rightarrow \mathbb{N}_{0,\mathbb{Q}}$ by $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}} = i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}$ then we have:*

- a. $\langle \mathbb{N}_{0,\mathbb{Q}}, + \rangle$ is a sub semi-group of $\langle \mathbb{Q}, + \rangle$ and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{Q}}, + \rangle \text{ is a group isomorphism}$$

- b. $\langle \mathbb{N}_{0,\mathbb{Q}}, \cdot \rangle$ is a sub semi-group of $\langle \mathbb{Q}, \cdot \rangle$ and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{Q}}, \cdot \rangle \text{ is a group isomorphism}$$

- c.

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{Q}}, \leq \rangle \text{ is a group isomorphism}$$

3. $\forall n \in \mathbb{N}_{0,\mathbb{Q}}$ we have that $0 \leq n$, if $n \neq 0$ then $0 < 1 \leq n$

Proof.

1. We have

$$\begin{aligned} x \in \mathbb{N}_{0,\mathbb{Q}} & \Leftrightarrow \exists n \in \mathbb{N}_0 \text{ such that } x = (i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(n) \\ & \Rightarrow x = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n)) \\ & \Rightarrow x = \frac{\sim[(n, 0)]}{1} \\ & \stackrel{\sim[(n, 0)] \in \mathbb{Z}_0^+}{\Rightarrow} x \in \left\{ \frac{n}{1} \mid n \in \mathbb{Z}_0^+ \right\} \end{aligned}$$

proving that

$$\mathbb{N}_{0,\mathbb{Q}} \subseteq \left\{ \frac{n}{1} \mid n \in \mathbb{Z}_0^+ \right\} \quad (8.17)$$

Further

$$\begin{aligned} x \in \left\{ \frac{n}{1} \mid n \in \mathbb{Z}_0^+ \right\} & \Rightarrow \exists n \in \mathbb{Z}_0^+ \text{ such that } x = \frac{n}{1} \\ & \stackrel{\text{definition of } \mathbb{Z}_0^+}{\Rightarrow} \exists n' \in \mathbb{N}_0 \text{ such that } n = \sim[(n', 0)] \\ & \Rightarrow x = \frac{\sim[(n, 0)]}{1} \\ & \Rightarrow x = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n')) \\ & \Rightarrow (i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(n') \\ & \Rightarrow x \in \mathbb{N}_{0,\mathbb{Q}} \end{aligned}$$

proving that $\left\{ \frac{n}{1} \mid n \in \mathbb{Z}_0^+ \right\} \subseteq \mathbb{N}_{0,\mathbb{Q}}$ which combined with [eq: 8.17] gives

$$\mathbb{N}_{0,\mathbb{Q}} = \left\{ \frac{n}{1} \mid n \in \mathbb{Z}_0^+ \right\}$$

- 2.

- a. Using [theorem: 7.17] and [theorem: 8.32] we have that $\mathbb{Z}_0^+$ is a sub semi group of $\langle \mathbb{Z}, + \rangle$ and that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{Z}_0^+, + \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, + \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, + \rangle \text{ are group isomorphisms}$$

Applying then [theorems: 4.25, 4.17] we find that $(i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(\mathbb{N}_0)$ is a sub semi-group of $\langle \mathbb{Q}, + \rangle$ and that $i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle (i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(\mathbb{N}_0), + \rangle$ is a group isomorphism. Using then the definition of $\mathbb{N}_{0,\mathbb{Q}}$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}$ we get then finally

$\mathbb{N}_{0,\mathbb{Q}}$ is a subgroup of $\langle \mathbb{Q}, + \rangle$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{Q}}, + \rangle$ is a group isomorphism.

- b. Using [theorem: 7.17] and [theorem: 8.32] we have that $\mathbb{Z}_0^+$ is a sub semi group of $\langle \mathbb{Z}, \cdot \rangle$ and that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{Z}_0^+, \cdot \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, \cdot \rangle \text{ are group isomorphisms}$$

Applying then [theorems: 4.25, 4.17] we find that $(i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(\mathbb{N}_0)$ is a sub semi-group of $\langle \mathbb{Q}, \cdot \rangle$ and that $i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle (i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(\mathbb{N}_0), \cdot \rangle$ is a group isomorphism. Using then the definition of $\mathbb{N}_{0, \mathbb{Q}}$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}$ we get then finally

$\mathbb{N}_{0, \mathbb{Q}}$ is a subgroup of $\langle \mathbb{Q}, \cdot \rangle$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{Q}}, \cdot \rangle$ is a group isomorphism

3. Using [theorem: 7.29] and [theorem: 8.32] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{Z}_0^+, \leq \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, \leq \rangle \text{ are order isomorphisms}$$

So using [theorem: 3.52] we have that $i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle (i_{\mathbb{N}_0 \rightarrow \mathbb{Z}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}})(\mathbb{N}_0), \leq \rangle$ is an order isomorphism. Using then the definition of $\mathbb{N}_{0, \mathbb{Q}}$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}$ we get then finally that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{Q}}, \leq \rangle \text{ is an order isomorphism}$$

4. If $n \in \mathbb{N}_{0, \mathbb{Q}}$ then there exist a $n' \in \mathbb{N}_0$ such that

$$n = i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}(n') = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n')) = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(\sim[(n', 0)]) = \frac{\sim[(n', 0)]}{1} \quad (8.18)$$

now $\sim[(n', 0)] \cdot 1 = \sim[(n', 0)] \in \mathbb{Z}_0^+$ so that $n \in \mathbb{Q}_0^+$ or

$$0 \leq n$$

If $n \neq 0$ then by [eq: 8.18] $n' \neq 0 \Rightarrow 0 < n'$ so by [theorem: 5.50] $1 = s(0) \leq n'$. Using (3) we have that

$$1 = i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}(1) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}(n') = n$$

as by [example: 8.25] $0 < 1$ we have

$$0 < 1 \leq n$$

□

Theorem 8.36. (Archimedean Property) *If $x, y \in \mathbb{Q}$ with $0 < x$ then there exist a $n \in \mathbb{N}_{0, \mathbb{Q}}$ such that $y < n \cdot x$*

Proof. For $y \in \mathbb{Q}$ we have the following possibilities to consider:

$y \leq 0$. Take $1 \in \mathbb{Q}$ then by [theorem: 8.35] $1 \in \mathbb{N}_{0, \mathbb{Q}}$ so if we take $n = 1$ then $y \leq 0 < x = 1 \cdot x = n \cdot x$, hence $y < n \cdot x$.

$0 < y$. As also $0 < x$ we have by [theorem: 8.24] the existence of $p, q, r, s \in \mathbb{Z}$ with $0 < p, 0 < q, 0 < r, 0 < s$ such that $x = \frac{p}{q}$ and $y = \frac{r}{s}$. As $0 < p \wedge 0 < s$ we have by [theorem: 7.26] that $0 < p \cdot s$. Using the Archimedean property of $\mathbb{Z}$ [see theorem: 7.30] there exist a $n' \in \mathbb{Z}_0^+$ such that $q \cdot r < n' \cdot (p \cdot s)$ or

$$0 < n' \cdot (p \cdot s) + (-(q \cdot r)) \quad (8.19)$$

As $n' \in \mathbb{Z}_0^+$ there exists a $n'' \in \mathbb{N}_0$ such that $n' = \sim[(n'', 0)]$ so that if we take $n = \frac{n'}{1} = \frac{\sim[(n'', 0)]}{1}$ we have $n = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(n''))$ so that

$$n \in \mathbb{N}_{0, \mathbb{Q}} \quad (8.20)$$

Now

$$\begin{aligned} n \cdot x - y &= \frac{n'}{1} \cdot x + (-y) \\ &= \frac{n'}{1} \cdot \frac{p}{q} + \frac{-r}{s} \\ &= \frac{n' \cdot p}{1 \cdot q} + \frac{-r}{s} \\ &= \frac{n' \cdot p}{q} + \frac{-r}{s} \\ &= \frac{(n' \cdot p) \cdot s + q \cdot (-r)}{q \cdot s} \\ &= \frac{n' \cdot (p \cdot s) + (-(q \cdot r))}{q \cdot s} \end{aligned} \quad (8.21)$$

As $0 < q \wedge 0 < s \xRightarrow{[\text{theorem: 7.26}]} 0 < q \cdot s$ and $0 < n' \cdot (p \cdot s) + (-(q \cdot r))$ [see eq: 8.19] it follows using [theorem: 8.24] that $0 < n \cdot x - y$ hence

$$y < n \cdot x \text{ where } n \in \mathbb{N}_{0, \mathbb{Q}} \quad \square$$

Theorem 8.37. ($\mathbb{Q}$ is dense) *If $x, y \in \mathbb{Q}$ with $x < y$ then there exist a $q \in \mathbb{Q}$ such that $x < q < y$.*

Proof. As $x < y$ we have by [theorems 8.29, 4.73] that $x + x < y + x = x + y$ and $x + y < y + y$. Further $x + x = 1 \cdot x + 1 \cdot x = (1 + 1) \cdot x \stackrel{[\text{example: 8.13}]}{=} \frac{2}{1} \cdot x$ and $y + y = 1 \cdot y + 1 \cdot y = (1 + 1) \cdot y \stackrel{[\text{example: 8.13}]}{=} \frac{2}{1} \cdot y$. So

$$\frac{2}{1} \cdot x < x + y \text{ and } x + y < \frac{2}{1} \cdot y \quad (8.22)$$

As $0 < 1 < 1 + 1 = 2 = \frac{2}{1}$ we have by [theorems: 8.29, 4.73] $0 < \left(\frac{2}{1}\right)^{-1} = \frac{1}{2}$, so using [theorems: 8.29, 4.73] on [eq: 8.22] gives $x < \frac{1}{2} \cdot (x + y)$ and $\frac{1}{2} \cdot (x + y) < y$. So if $q = \frac{1}{2} \cdot (x + y)$ we have that

$$x < q < y \quad \square$$

Theorem 8.38. $\langle \mathbb{N}_{0, \mathbb{Q}}, \leq \rangle$ *is well ordered*

Proof. Using [theorem: 8.35] we have that $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \mathbb{N}_0 \rightarrow \mathbb{N}_{0, \mathbb{Q}}$ is a order isomorphism, further by [theorem: 5.51] $\langle \mathbb{N}_0, \leq \rangle$ is well ordered. so using [theorem: 3.85] we conclude that

$$\langle \mathbb{N}_{0, \mathbb{Q}}, \leq \rangle \text{ is well ordered} \quad \square$$

Theorem 8.39. $\langle \mathbb{N}_{0, \mathbb{Q}}, \leq \rangle$ *is conditional complete*

Proof. As by [theorem: 8.38] $\langle \mathbb{N}_{0, \mathbb{Q}}, \leq \rangle$ is well-ordered it follows from [theorem: 3.87] that $\langle \mathbb{Z}_0^+, \leq \rangle$ is conditional complete. $\square$

Although $\langle \mathbb{N}_{0, \mathbb{Q}}, \leq \rangle$ is conditional complete we have that $\langle \mathbb{Q}, \leq \rangle$ is not conditional complete as we will prove now. First we need a lemma that essentially says that $\sqrt{2}$ is not a rational number.

Lemma 8.40. $\forall q \in \mathbb{Q}$ we have $q^2 \stackrel{\text{def}}{=} q \cdot q \neq 2 = \frac{2}{1}$.

Proof. We prove this by contradiction. Assume that $\exists q' \in \mathbb{Q}$ such that $q' \cdot q' = 2$, then $q' \neq 0$ [if $q' = 0 \Rightarrow q' \cdot q' = 0 = 2$]. Take $q = \begin{cases} -q' & \text{if } q' < 0 \\ q' & \text{if } q' > 0 \end{cases}$ then

$$0 < q \text{ and } q \cdot q = 2$$

Using [theorem: 8.24] there exist $n, m \in \mathbb{Z}$ with $0 < n \wedge 0 < m$ such that $q = \frac{n}{m}$. Take now $n' = n: \gcd(n, m)$ and $m' = m: \gcd(n, m)$ then as $m \neq 0 \Rightarrow m' \neq 0$ we have by [theorem: 7.49] that

$$\{d \in \mathbb{Z} \mid d \mid n' \wedge d \mid m'\} = \{1, -1\} \quad (8.23)$$

Now

$$\frac{n'}{m'} \stackrel{0 < \gcd(n, m) \wedge [\text{theorem: 8.5}]}{=} \frac{n' \cdot \gcd(n, m)}{m' \cdot \gcd(n, m)} = \frac{n}{m} = q$$

so that

$$\frac{n' \cdot n'}{m' \cdot m'} = \frac{n'}{m'} \cdot \frac{n'}{m'} = q \cdot q = \frac{2}{1}$$

Hence $(n' \cdot n') \cdot 1 = (m' \cdot m') \cdot 2$ or $n' \cdot n' = 2 \cdot (m' \cdot m')$ proving that $n' \cdot n'$ is even, using [theorem: 7.52] it follows then that n' is even. So there exist a $k \in \mathbb{Z}$ such that $n' = 2 \cdot k$. Then $2 \cdot (m' \cdot m') = (2 \cdot k) \cdot (2 \cdot k) = 2 \cdot (2 \cdot (k \cdot k))$, hence by [theorem: 7.15] $m' \cdot m' = 2 \cdot (k \cdot k)$ proving that $m' \cdot m'$ is even, by [theorem: 7.52] m' is even hence $\exists l \in \mathbb{Z}$ such that $m' = 2 \cdot l$. So $2 \mid n'$ and $2 \mid m'$ which by [eq: 8.23] means that $2 = 1$ or $2 = -1$ both of which are false, so we reach a contradiction. $\square$

Theorem 8.41. $\langle \mathbb{Q}, \leq \rangle$ *is not conditional complete, so there exist a non empty subset of $\mathbb{Q}$ that is bounded above but does not have a least upper bound.*

Proof. In this prove we make use of the fact that there does not exist a $q \in \mathbb{Q}$ such that $q \cdot q = 2$. So define

$$A = \left\{ q \in \mathbb{Q} \mid 0 < q \wedge q \cdot q < \frac{2}{1} \right\} \subseteq \mathbb{Q}$$

As $0 < \frac{4}{3}$ and $\frac{2}{1} + \left(-\left(\frac{4}{3}\right) \cdot \left(\frac{4}{3}\right)\right) = \frac{18-16}{9} = \frac{2}{9} > 0$ so that $\frac{4}{3} \cdot \frac{4}{3} < 2$ we have that

$$\frac{4}{3} \in A \Rightarrow \emptyset \neq A \quad (8.24)$$

Let $x \in A$ then $0 < x$ and $x \cdot x < \frac{2}{1}$. Assume that $\frac{2}{1} < x$ then by multiplying both sides by x we have by [theorems: 8.29, 4.73] that $\frac{2}{1} \cdot x < x \cdot x < \frac{2}{1} = 1 \cdot \frac{2}{1}$, we have by [theorems: 8.29, 4.73] that $x < 1 < \frac{2}{1}$ contradicting $\frac{2}{1} < x$. So we must have that $x \leq \frac{2}{1}$ hence

$$\frac{2}{1} \text{ is an upper bound of } A \quad (8.25)$$

Assume now that $u = \sup(A)$ exist. As $\frac{4}{3} + (-1) = \frac{4}{3} + \frac{-1}{1} = \frac{4+(-3)}{3} = \frac{1}{3} > 0$ it follows that $1 < \frac{4}{3} \in A$ so that $0 < 1 < u$ and as $\frac{2}{1}$ is an upper bound of A we have

$$0 < 1 < u \leq \frac{2}{1} \quad (8.26)$$

Now for $u \cdot u$ we have by [theorem: 8.40] that $u \cdot u \neq \frac{2}{1}$ so we have only to consider the following possibilities:

$u \cdot u < \frac{2}{1}$. So $0 < \frac{2}{1} + (-u \cdot u)$ and by the Archimedean property [see theorem: 8.36] there exist a $n' \in \mathbb{N}_{0, \mathbb{Q}}$ such that

$$\frac{5}{1} < n' \cdot \left(\frac{2}{1} - u \cdot u \right)$$

Using [theorem 8.35] we have that $\exists n \in \mathbb{Z}_0^+$ such that $n' = \frac{n}{1}$, further $n \neq 0$ [otherwise $\frac{5}{1} < \frac{0}{1} \cdot \left(\frac{2}{1} - u \cdot u \right) = 0$] so there exist a $n \in \mathbb{Z}_0^+ \setminus \{0\}$ such that

$$\frac{5}{1} < \frac{n}{1} \cdot \left(\frac{2}{1} - u \cdot u \right)$$

multiplying both sides by $\frac{1}{n} = \left(\frac{n}{1}\right)^{-1}$ gives

$$\frac{5}{n} < \frac{2}{1} - u \cdot u \quad (8.27)$$

Now

$$\begin{aligned} \left(u + \frac{1}{n}\right) \cdot \left(u + \frac{1}{n}\right) &= u \cdot u + u \cdot \frac{1}{n} + u \cdot \frac{1}{n} + \frac{1}{n} \cdot \frac{1}{n} \\ &= u \cdot u + \frac{2}{1} \cdot u \cdot \frac{1}{n} + \frac{1}{n} \cdot \frac{1}{n} \end{aligned}$$

and thus

$$\begin{aligned} \left(u + \frac{1}{n}\right) \cdot \left(u + \frac{1}{n}\right) < \frac{2}{1} &\Leftrightarrow u \cdot u + \frac{2}{1} \cdot u \cdot \frac{1}{n} + \frac{1}{n} \cdot \frac{1}{n} < \frac{2}{1} \\ &\Leftrightarrow \frac{2}{1} \cdot u \cdot \frac{1}{n} + \frac{1}{n} \cdot \frac{1}{n} < \frac{2}{1} - u \cdot u \\ &\Leftrightarrow \frac{2}{n} \cdot u + \frac{1}{n} \cdot \frac{1}{n} < \frac{2}{1} - u \cdot u \quad (8.28) \end{aligned}$$

As $0 < n$, so that by [theorem: 7.23] $1 \leq n$, hence $0 \leq n - 1$. Now

$$\frac{1}{n} - \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n} - \frac{1}{n \cdot n} = \frac{n \cdot n - n \cdot 1}{n \cdot n} = \frac{n \cdot (n - 1)}{n \cdot n} = \frac{n - 1}{n} \geq 0$$

giving

$$\frac{1}{n} \cdot \frac{1}{n} \leq \frac{1}{n} \quad (8.29)$$

Further as $u \leq \frac{2}{1}$ [see eq: 8.26] we have by [theorems: 8.29, 4.73] and $0 < \frac{2}{n}$ [as $0 < n \wedge 0 < 2$]

$$u \cdot \frac{2}{n} \leq \frac{2}{n} \cdot \frac{2}{1} = \frac{4}{n} \quad (8.30)$$

So

$$\begin{aligned} \frac{2}{n} \cdot u + \frac{1}{n} \cdot \frac{1}{n} &\leq_{[\text{eq: 8.29}]} \frac{2}{n} \cdot u + \frac{1}{n} \\ 1 &\leq_{[\text{eq: 8.30}]} \frac{4}{n} + \frac{1}{n} \\ &= \left(\frac{4}{1} + \frac{1}{1} \right) \cdot \frac{1}{n} \\ &= \frac{5}{1} \cdot \frac{1}{n} \\ &= \frac{5}{n} \\ &<_{[\text{eq: 8.27}]} \frac{2}{1} - u \cdot u \end{aligned}$$

So by [eq: 8.28] we have

$$\left(u + \frac{1}{n} \right) \cdot \left(u + \frac{1}{n} \right) < \frac{2}{1}$$

By [eq: 8.26] $0 < u \Rightarrow 0 < u + \frac{1}{n}$ which together with the above proves that $u + \frac{1}{n} \in A$, so $u + \frac{1}{n} \leq \sup(A) = u$, which as $u < u + \frac{1}{n}$ leads to the contradiction $u < u$. So this case is impossible.

$\frac{2}{1} < \mathbf{u} \cdot \mathbf{u}$. So $0 < u \cdot u + \frac{-2}{1}$ and using the Archimedean property there exist a $n' \in \mathbb{N}_0, \mathbb{Q}$ such that

$$\frac{2}{1} \cdot u < n' \cdot \left(u \cdot u + \frac{-2}{1} \right) \quad (8.31)$$

Using [theorem: 8.35] there exist a $n \in \mathbb{Z}_0^+$ such that $n' = \frac{n}{1}$. If $n = 0 \Rightarrow n' = 0$ so that $\frac{2}{1} \cdot u < 0 \cdot \left(u \cdot u + \frac{-2}{1} \right) = 0 \Rightarrow u < 0$ contradicting $0 < u$ [see eq: 8.26], hence we must have that $n \neq 0$ or $0 < n$, so $(n')^{-1} = \frac{1}{n}$ exist and $0 < \frac{1}{n}$. Next

$$\begin{aligned} \frac{2}{1} \cdot u < n' \cdot \left(u \cdot u + \frac{-2}{1} \right) &\Rightarrow \frac{2}{1} \cdot u < \frac{n}{1} \cdot \left(u \cdot u + \frac{-2}{1} \right) \\ &\Rightarrow_{0 < \frac{1}{n} \wedge [\text{theorems: 8.29, 4.73}]} \left(\frac{2}{1} \cdot u \right) \cdot \frac{1}{n} < \left(\frac{n}{1} \cdot \left(u \cdot u + \frac{-2}{1} \right) \right) \cdot \frac{1}{n} \\ &\Rightarrow \frac{2}{n} \cdot u < u \cdot u + \frac{-2}{1} \\ &\Rightarrow \frac{2}{1} < u \cdot u + \frac{-2}{n} \cdot u \end{aligned}$$

which as $0 < \frac{1}{n} \Rightarrow 0 < \frac{1}{n} \cdot \frac{1}{n}$ proves that

$$\frac{2}{1} < u \cdot u + \frac{-2}{n} \cdot u + \frac{1}{n} \cdot \frac{1}{n} \quad (8.32)$$

As $0 < n$ we have that $1 \leq n$ which by [theorem: 8.27] gives $\frac{1}{n} \leq 1$ and as $1 < u$ we have $\frac{1}{n} < u$, hence

$$0 < u + \left(\frac{-1}{n} \right) \Rightarrow_{[\text{theorems: 8.29, 4.73}]} 0 < \left(u + \frac{-1}{n} \right) \cdot \left(u + \frac{-1}{n} \right) \quad (8.33)$$

As $0 < \frac{1}{n} \Rightarrow_{[\text{theorems: 8.29, 4.73}]} \frac{-1}{n} < 0$ so that $u + \frac{-1}{n} < u$, as $u = \sup(A)$ and $\langle \mathbb{Q}, \leq \rangle$ is totally ordered we have by [theorem: 3.74] that there exist a $q \in A$ such that

$$u + \frac{-1}{n} < q \leq u \quad (8.34)$$

Multiplying both sides of [eq: 8.34] by $u + \frac{-1}{n}$ we have by [theorems: 8.29, 4.73] that

$$\left(u + \frac{-1}{n}\right) \cdot \left(u + \frac{-1}{n}\right) < q \cdot \left(u + \frac{-1}{n}\right)$$

Further as by [eq: 8.33] $0 < u + \frac{-1}{n} < q \Rightarrow 0 < q$ we have, by multiplying both sides of [eq: 8.34] by q , that

$$\left(u + \frac{-1}{n}\right) \cdot q < q \cdot q.$$

Hence $\left(u + \frac{-1}{n}\right) \cdot \left(u + \frac{-1}{n}\right) < q \cdot q$ and as $q \in A$ we have also $q \cdot q < \frac{2}{1}$ so that

$$\left(u + \frac{-1}{n}\right) \cdot \left(u + \frac{-1}{n}\right) < \frac{2}{1} \quad (8.35)$$

Next

$$\begin{aligned} \left(u + \frac{-1}{n}\right) \cdot \left(u + \frac{-1}{n}\right) &= u \cdot u + \frac{-1}{n} \cdot u + \frac{-1}{n} \cdot n + \frac{-1}{n} \cdot \frac{-1}{n} \\ &= u \cdot u + \frac{-2}{n} + \frac{1}{n} \cdot \frac{1}{n} \end{aligned}$$

which by [eq: 8.35] proves that $u \cdot u + \frac{-2}{n} + \frac{1}{n} \cdot \frac{1}{n} < \frac{2}{1}$, combining this with [eq: 8.32] result in $\frac{2}{1} < \frac{2}{1}$ a contradiction. So this case is impossible.

As all possible cases are impossible, the assumption is wrong hence A has no supremum and $\langle \mathbb{Q}, \leq \rangle$ is not conditional complete. $\square$

So we have that $\langle \mathbb{N}_0, \mathbb{Q}, \leq \rangle$ is conditional complete but $\langle \mathbb{Q}, \leq \rangle$ is not. This defect will be resolved by introducing the set of real numbers that will extend the set of rationals.

8.3 Denumerability of the rationals

Theorem 8.42. $\mathbb{N}_0, \mathbb{Q}$ is denumerable.

Proof. Using [theorem: 8.35] $i_{\mathbb{N}_0 \rightarrow \mathbb{Q}}: \mathbb{N}_0 \rightarrow \mathbb{N}_0, \mathbb{Q}$ is a bijection, hence $\mathbb{N}_0 \approx \mathbb{Q}$ proving that $\mathbb{Q}$ is denumerable. $\square$

Theorem 8.43. $\mathbb{Z}_{\mathbb{Q}}$ is denumerable

Proof. Using [theorem: 7.53] we have that $\mathbb{Z}$ is denumerable, further by [theorem: 8.32]

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Q} \rightarrow \mathbb{Z}_{\mathbb{Q}}$$

is a bijection, hence $\mathbb{N}_0 \approx \mathbb{Q} \approx \mathbb{Z}_{\mathbb{Q}}$, proving that $\mathbb{Z}_{\mathbb{Q}}$ is denumerable. $\square$

Theorem 8.44. $\mathbb{Q}$ is denumerable

Proof. Define the mapping $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q}$ by

$$f(x, y) = \begin{cases} \frac{x}{y} & \text{if } (x, y) \in \mathbb{Z} \times \mathbb{Z}^* \\ 0 & \text{if } (x, y) \in \mathbb{Z} \times \{0\} \end{cases}$$

If $q \in \mathbb{Q}$ then there exist a $(x, y) \in \mathbb{Z} \times \mathbb{Z}^* \subseteq \mathbb{Z} \times \mathbb{Z}$ such that $q = \frac{x}{y} = f(x, y)$ proving that

$$f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q} \text{ is a surjection}$$

As $\mathbb{Z}$ is denumerable [see theorem: 7.53] we have by [theorem: 6.63] that $\mathbb{Z} \times \mathbb{Z}$ is denumerable, hence there exist a bijection $g: \mathbb{N}_0 \rightarrow \mathbb{Z} \times \mathbb{Z}$, so $f \circ g: \mathbb{N}_0 \rightarrow \mathbb{Q}$ is a surjection. By [theorem: 6.75] $\mathbb{Q}$ is countable, hence either finite or denumerable. As $\mathbb{N}_0, \mathbb{Q} \subseteq \mathbb{Q}$ and $\mathbb{N}_0, \mathbb{Q}$ is denumerable, it follow from [theorem: 6.29] that $\mathbb{Q}$ is not finite, hence we must have that $\mathbb{Q}$ is denumerable. $\square$

Chapter 9

The real numbers

In this chapter we will introduce the set of real numbers and embed the natural, integer and rational numbers in it. Just as with $\mathbb{Q}$, $\mathbb{Z}$ and $\mathbb{N}_0$ we will introduce a order relation, a sum operator, a product operator, neutral elements for addition and multiplication as well as inverse elements. If we would use different symbols for these we introduce a lot of excessive notation clutter. So we use the same symbols for the natural numbers, integers, rational numbers and real numbers and use context to determine the meaning of the symbols involved. The following table should help you in determining the meaning of the different symbols based on the context of there usage.

Context	Expression	Operator
$n, m \in \mathbb{N}_0$	$n+m$	sum in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{N}_0$	$n \cdot m$	product in $\langle \mathbb{N}_0, \cdot \rangle$
$n, m \in \mathbb{N}_0$	$n \leq m$	order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n < m$	strict order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n - m$	subtraction in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{N}_0, \cdot \rangle$
$n \in \mathbb{N}_0$	$-n$	inverse element in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{Z}$	$n+m$	sum in $\langle \mathbb{Z}, + \rangle$
$n, m \in \mathbb{Z}$	$n \cdot m$	product in $\langle \mathbb{Z}, \cdot \rangle$
$n, m \in \mathbb{Z}$	$n \leq m$	order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n < m$	strict order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n - m$	subtraction in $\langle \mathbb{Z}, - \rangle$
$n \in \mathbb{Z}$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{Z}, + \rangle$
$n \in \mathbb{Z}$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{Z}, \cdot \rangle$
$n \in \mathbb{Z}$	$-n$	inverse element in $\langle \mathbb{Z}, + \rangle$
$q, r \in \mathbb{Q}$	$q+r$	sum in $\langle \mathbb{Q}, + \rangle$
$q, r \in \mathbb{Q}$	$q \cdot r$	product in $\langle \mathbb{Q}, \cdot \rangle$
$q, r \in \mathbb{Q}$	$q \leq r$	order in $\langle \mathbb{Q}, \leq \rangle$
$q, r \in \mathbb{Q}$	$q < r$	strict order in $\langle \mathbb{Q}, \leq \rangle$
$q, e \in \mathbb{Q}$	$q - r$	subtraction in $\langle \mathbb{Q}, - \rangle$
$q, r \in \mathbb{Q}$	q/r	division in $\langle \mathbb{Q}, \cdot \rangle$
$q \in \mathbb{Q}$	$q + 0$ or $0 + q$	neutral element in $\langle \mathbb{Q}, + \rangle$
$q \in \mathbb{Q}$	$q \cdot 1$ or $1 \cdot q$	neutral element in $\langle \mathbb{Q}, \cdot \rangle$
$q \in \mathbb{Q}$	$-q$	inverse element in $\langle \mathbb{Q}, + \rangle$
$q, r \in \mathbb{R}$	$q+r$	sum in $\langle \mathbb{R}, + \rangle$
$q, r \in \mathbb{R}$	$q \cdot r$	product in $\langle \mathbb{R}, \cdot \rangle$
$q, r \in \mathbb{R}$	$q \leq r$	order in $\langle \mathbb{R}, \leq \rangle$
$q, r \in \mathbb{R}$	$q < r$	strict order in $\langle \mathbb{R}, \leq \rangle$
$q, e \in \mathbb{R}$	$q - r$	subtraction in $\langle \mathbb{R}, - \rangle$
$q, r \in \mathbb{R}$	q/r	division in $\langle \mathbb{R}, \cdot \rangle$
$q \in \mathbb{R}$	$q + 0$ or $0 + q$	neutral element in $\langle \mathbb{R}, + \rangle$
$q \in \mathbb{R}$	$q \cdot 1$ or $1 \cdot q$	neutral element in $\langle \mathbb{R}, \cdot \rangle$
$q \in \mathbb{R}$	$-q$	inverse element in $\langle \mathbb{R}, + \rangle$

9.1 Definition and Arithmetic on $\mathbb{R}$

9.1.1 Definition of the real numbers

Definition 9.1. (Dedekind Cut) A subset $\alpha \subseteq \mathbb{Q}$ is a Dedekind's cut of $\mathbb{Q}$ if the following properties hold

1. $\alpha \neq \emptyset$
2. $\alpha \neq \mathbb{Q}$ [which as $\alpha \subseteq \mathbb{Q}$ implies that $\mathbb{Q} \setminus \alpha \neq \emptyset$]
3. $\forall q \in \alpha \wedge \forall r \in \mathbb{Q} \setminus \alpha$ we have $q < r$
4. α does not have a greatest element [or maximum]

So a Dedekind cut α divides $\mathbb{Q}$ in two disjoint pieces so that $\alpha \neq \emptyset \neq \mathbb{Q} \setminus \alpha$ where every element in α is strict lower of elements in $\mathbb{Q} \setminus \alpha$ and α has not a greatest element. The collection of Dedekind cuts will form the set of real numbers.

Definition 9.2. The set of real numbers, noted as $\mathbb{R}$ is the set of Dedekind cuts of $\mathbb{Q}$ hence

$$\mathbb{R} = \{\alpha \subseteq \mathbb{Q} \mid \alpha \text{ is a Dedekind cut}\}$$

Lemma 9.3. $\forall \alpha \in \mathbb{R}$ we have $\forall q \in \alpha$ and $\forall r \in \mathbb{Q}$ with $r \leq q$ that $r \in \alpha$.

Proof. Let $\alpha \in \mathbb{R}$, $q \in \alpha$ and $r \in \mathbb{Q}$ with $r \leq q$. Assume that $r \notin \alpha$ then $r \in \mathbb{Q} \setminus \alpha$ and by [definition: 9.1] we have $q < r$ contradicting $r \leq q$. Hence we must have that $r \in \alpha$. $\square$

We prove now that every rational number can be associated with a Dedekind cut of $\mathbb{Q}$.

Theorem 9.4. (Rational cuts) If $q \in \mathbb{Q}$ then $\alpha_q = \{r \in \mathbb{Q} \mid r < q\}$ is a Dedekind cut. Dedekind cuts of this form are called rational cuts. Furthermore we have:

1. $\alpha_q = \alpha_r \Leftrightarrow q = r$
2. α is a rational cut $\Leftrightarrow q = \min(\mathbb{Q} \setminus \alpha)$ exist and in that case $\alpha = \alpha_q$

Proof. First we prove that given $q \in \mathbb{Q}$ $\alpha_q = \{r \in \mathbb{Q} \mid r < q\}$ is a cut.

1. By [theorem: 8.26] $q - 1 < q$ hence $q - 1 \in \alpha_q$ proving that $\alpha_q \neq \emptyset$.
2. As $q < q$ is false we have that $q \in \mathbb{Q} \setminus \alpha_q$ so that $\mathbb{Q} \setminus \alpha_q \neq \emptyset$.
3. If $r \in \alpha_q$ and $s \in \mathbb{Q} \setminus \alpha_q$ then $r < q$ and $\neg(s < q) \Rightarrow q \leq s$ so that $r < s$.
4. Assume that m is a greatest element of α_q then $m \in \alpha_q$ and $\forall r \in \alpha_q$ we have $r \leq m$. As $m \in \alpha_q$ we have that $m < q$, using the density of $\mathbb{Q}$ [see theorem: 8.37] there exist a $r \in \mathbb{Q}$ such that $m < r < q$. As $r < q$ we have that $r \in \alpha_q$ so that $r \leq m$ contradicting $m < r$. So the assumption is false proving that α_q has no greatest element.

Next we prove (1) and (2)

1.

$\Rightarrow$. If $\alpha_q = \alpha_r$ then if $q \neq r$ we have either

$q < r$. then $q \in \alpha_r$ and so that $q \in \alpha_q$ resulting in the contradiction $q < q$.

$r < q$. then $r \in \alpha_q$ and so that $r \in \alpha_r$ resulting in the contradiction $r < r$.

so we must have $q = r$.

$\Leftarrow$. $s \in \alpha_q \Leftrightarrow s \in \mathbb{Q} \wedge s < q \Leftrightarrow_{q=r} s \in \mathbb{Q} \wedge s < r \Leftrightarrow s \in \alpha_r$ hence $\alpha_q = \alpha_r$

2.

$\Rightarrow$. If α is a rational cut then there exist a $q \in \mathbb{Q}$ such that $\alpha = \{r \in \mathbb{Q} | r < q\}$. So

$$\begin{aligned} s \in \mathbb{Q} \setminus \alpha &\Leftrightarrow s \in \mathbb{Q} \wedge \neg(s < q) \\ &\Leftrightarrow s \in \mathbb{Q} \wedge q \leq s \\ &\Leftrightarrow s \in \{s \in \mathbb{Q} | q \leq s\} \end{aligned}$$

proving that $\mathbb{Q} \setminus \alpha = \{s \in \mathbb{Q} | q \leq s\}$. So $q \in \{s \in \mathbb{Q} | q \leq s\}$ and $\forall s \in \mathbb{Q} \setminus \alpha$ we have $q \leq s$
proving that $q = \min(\mathbb{Q} \setminus \alpha)$ and $\alpha = \{r \in \mathbb{Q} | r < q\} = \alpha_q$

$\Leftarrow$. If $q = \min(\mathbb{Q} \setminus \alpha)$ exists then $q \in \mathbb{Q} \setminus \alpha$ and $\forall r \in \mathbb{Q} \setminus \alpha$ we have $q \leq r$. If now $r \in \alpha$ then by the definition of a cut we have $r < q$, hence $r \in \{r \in \mathbb{Q} | r < q\} = \alpha_q$. Further if $r \in \alpha_q$ then $r < q$, assume that $r \notin \alpha$ then we have $q \leq r$ contradicting $r < q$, so we must have that $r \in \alpha$. Hence we have that

$$\alpha = \alpha_q \text{ where } q = \min(\mathbb{Q} \setminus \alpha) \quad \square$$

Corollary 9.5. $\mathbb{R} \neq \emptyset$

Proof. As $0, 1 \in \mathbb{Q}$ we have that $\alpha_0, \alpha_1 \in \mathbb{R}$ proving that $\mathbb{R} \neq \emptyset$ $\square$

We embed now the rational numbers in the set of reals.

Definition 9.6. The set $\mathbb{Q}_{\mathbb{R}}$ is defined by

$$\mathbb{Q}_{\mathbb{R}} = \{\alpha_q | q \in \mathbb{Q}\} \subseteq \mathbb{R}$$

where $\alpha_q = \{r \in \mathbb{Q} | r < q\}$

To make the above a embedding we need a bijection between $\mathbb{Q}$ and $\mathbb{Q}_{\mathbb{R}}$ and once we have defined sum, product and order prove that it is a field and order isomorphism. We start with providing a bijection.

Theorem 9.7. $i_{\mathbb{Q} \rightarrow \mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{R}}$ defined by $i_{\mathbb{Q}}(q) = \alpha_q$ is a bijection.

Proof. We have

reflexivity. If $i_{\mathbb{Q} \rightarrow \mathbb{R}}(q) = i_{\mathbb{Q} \rightarrow \mathbb{R}}(r)$ then $\alpha_q = \alpha_r$ so that by [theorem: 9.4] $q = r$.

surjective. If $\alpha \in \mathbb{Q}_{\mathbb{R}}$ we have a $q \in \mathbb{Q}$ such that $\alpha = \alpha_q = i_{\mathbb{Q} \rightarrow \mathbb{R}}(q)$ $\square$

Corollary 9.8. The set $\mathbb{Q}_{\mathbb{R}}$ is denumerable.

Proof. As $\mathbb{Q}$ is denumerable we have that $\mathbb{N}_0 \approx \mathbb{Q}$, further from the previous theorem [theorem: 9.7] we have $\mathbb{Q} \approx \mathbb{Q}_{\mathbb{R}}$ so that $\mathbb{N}_0 \approx \mathbb{Q}_{\mathbb{R}}$. Hence $\mathbb{Q}_{\mathbb{R}}$ is denumerable. $\square$

Theorem 9.9. (Gap theorem) If $\alpha \in \mathbb{R}$ then $\forall \varepsilon \in \mathbb{Q}$ with $0 < \varepsilon$ there $\exists q \in \alpha$ and $\exists r \in \mathbb{Q} \setminus \alpha$ such that

$$r - q = r + (-q) < \varepsilon$$

Proof. Let $\alpha \in \mathbb{R}$ and $\varepsilon \in \mathbb{Q} \setminus \{0\}$. By the definition of a cut there exist a $q' \in \alpha$ and a $r' \in \mathbb{Q} \setminus \alpha$ such that $q' < r'$, so $0 < r' + (-q') = r' - q'$ and we have by the Archimedean property [see theorem: 8.36] the existence of a $k \in \mathbb{N}_{0, \mathbb{Q}}$ such that $r' - q' < k \cdot \varepsilon$. If $k = 0$ then we would have that $0 < r' - k' < 0$ a contradiction, so $k \neq 0$ which by [theorem: 8.35] proves that $0 < k$. Applying [theorems: 8.29, 4.73] we have that $0 < k^{-1}$, so multiplying both sides of $r' - q' < k \cdot \varepsilon$ gives

$$k^{-1} \cdot (r' - q') < \varepsilon \quad (9.1)$$

Define now

$$A = \{n \in \mathbb{N}_{0, \mathbb{Q}} \mid q' + (n \cdot k^{-1}) \cdot (r' - q') \notin \alpha\} \subseteq \mathbb{N}_{0, \mathbb{Q}}$$

As $q' + (k \cdot k^{-1}) \cdot (r' - q') = q' + (r' - q') = r' \in \mathbb{Q} \setminus \alpha$ it follows that $k \in A$ so that $A \neq \emptyset$, as $\mathbb{N}_{0, \mathbb{Q}}$ is well ordered [see theorem: 8.38] it follows that $k' = \min(A)$ exist. If $k' = 0$ then as $k' \in A$ we would have $q' = q' + (0 \cdot k^{-1}) \cdot (r' - q') \notin \alpha$ contradicting $q' \in \alpha$, so we must have that $k' \neq 0$ and using [theorem: 8.35] it follows that $1 \leq k'$, hence $0 \leq k' - 1$, giving by [theorem: 8.35] that $k' - 1 \in \mathbb{N}_{0, \mathbb{Q}}$. As by [theorem: 8.26] $k' - 1 < k'$ we have, as $k' = \min(A)$, that $k' - 1 \notin A$ so that

$$q' + ((k' - 1) \cdot k^{-1}) \cdot (r' - q') \in \alpha$$

Define now $q = q' + ((k' - 1) \cdot k^{-1}) \cdot (r' - q')$ and $r = q' + (k' \cdot k^{-1}) \cdot (r' - q')$ then we have

$$q \in \alpha \text{ and } r \in \mathbb{Q} \setminus \alpha$$

Next

$$\begin{aligned} r - q &= (q' + (k' \cdot k^{-1}) \cdot (r' - q')) - (q' + ((k' - 1) \cdot k^{-1}) \cdot (r' - q')) \\ &= (q' + k' - (q' + (k' - 1))) \cdot k^{-1} \cdot (r' - q') \\ &= k^{-1} \cdot (r' - q') \\ &< \varepsilon \quad [\text{see eq: 9.1}] \end{aligned}$$

□

Theorem 9.10. (Negative cut) *If $\alpha \in \mathbb{R}$ then $-\alpha$ defined by*

$$-\alpha = \{r \mid -r \in \mathbb{Q} \setminus \alpha \text{ such that } \exists t \in \mathbb{Q} \setminus \alpha \models t < -r\}$$

*is a Dedekind cut called the **negative cut**.*

Proof.

1. As α is a Dedekind cut we have by [definition: 9.1 (2)] that $\mathbb{Q} \setminus \alpha \neq \emptyset$ so there exist a $q \in \mathbb{Q} \setminus \alpha$. Assume that $q + 1 \in \alpha$ then by [definition: 9.1 (3)] we have $q + 1 < q$ a contradiction, so we must have that $q + 1 \notin \alpha$ or $q + 1 \in \mathbb{Q} \setminus \alpha$. Hence we have $-(-(q + 1)) = q + 1 \in \mathbb{Q} \setminus \alpha$ and $q \in \mathbb{Q} \setminus \alpha$ with $q < q + 1 = -(-(q + 1))$ proving that $-(q + 1) \in -\alpha$ or that

$$-\alpha \neq \emptyset$$

2. As α is a Dedekind cut we have by [definition: 9.1 (1)] that $\alpha \neq \emptyset$ so there exist a $q \in \alpha$ hence $q \notin \mathbb{Q} \setminus \alpha$. If $-q \in -\alpha$ then $q = -(-q) \in \mathbb{Q} \setminus \alpha$ contradicting $q \notin \mathbb{Q} \setminus \alpha$ hence we must have that $-q \notin -\alpha$ proving that

$$-\alpha \neq \mathbb{Q}$$

3. Let $q \in -\alpha$ and $s \in \mathbb{Q} \setminus -\alpha$. Assume that $s \leq q$ then by [theorems: 8.29, 4.73]

$$-q \leq -s \tag{9.2}$$

As $q \in -\alpha$ we have that

$$-q \in \mathbb{Q} \setminus \alpha \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \models t < -q \tag{9.3}$$

If $-s \in \alpha$ then as $-q \in \mathbb{Q} \setminus \alpha$ we have by [definition: 9.1 (3)] that $-s < -q$ contradicting [eq: 9.2] hence we must have that $-s \notin \alpha$ so that $-s \in \mathbb{Q} \setminus \alpha$. Using [eq: 9.2] and [eq: 9.3] we have $\exists t \in \mathbb{Q} \setminus \alpha$ such that $t < -q \leq -s$ so we have that $s \in -\alpha$ contradicting $s \in \mathbb{Q} \setminus -\alpha$. So the assumption is wrong and we have

$$q < s$$

4. Assume that $-\alpha$ has a greatest element m then

$$m \in -\alpha \text{ and } \forall r \in -\alpha \text{ we have } r \leq m \tag{9.4}$$

As $m \in -\alpha$ we have that

$$-m \in \mathbb{Q} \setminus \alpha \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \models t < -m \xRightarrow{[\text{theorems: 8.29, 4.73}]} m < -t \quad (9.5)$$

For $\mathbb{Q} \setminus \alpha$ we have now two cases to consider:

$\min(\mathbb{Q} \setminus \alpha)$ does not exist. As $t \in \mathbb{Q} \setminus \alpha$ and $\min(\mathbb{Q} \setminus \alpha)$ does not exist there exist a $s \in \mathbb{Q} \setminus \alpha$ such that $s < t$, hence we have $-(-t) \in \mathbb{Q} \setminus \alpha \wedge s < -(-t)$ proving that $-t \in -\alpha$ hence by [eq: 9.4] that $-t \leq m$ contradicting [eq 9.5].

$\min(\mathbb{Q} \setminus \alpha)$ exist. As $-m \in \mathbb{Q} \setminus \alpha$ we have $\min(\mathbb{Q} \setminus \alpha) \leq -m$, further as $t \in \mathbb{Q} \setminus \alpha \wedge t < -m$ we have $-m \neq \min(\mathbb{Q} \setminus \alpha)$ so that $\min(\mathbb{Q} \setminus \alpha) < -m$. Using the density of $\mathbb{Q}$ [see 8.37] there exist a $s \in \mathbb{Q}$ such that

$$\min(\mathbb{Q}) < s < -m \quad (9.6)$$

If $s \in \alpha$ then as $\min(\mathbb{Q} \setminus \alpha) \in \mathbb{Q} \setminus \alpha$ we have by [definition: 9.1 (3)] that $s < \min(\mathbb{Q} \setminus \alpha)$ contradicting $\min(\mathbb{Q} \setminus \alpha) < s$, so we must have that $s \in \mathbb{Q} \setminus \alpha$. Hence $s = -(-s) \in \mathbb{Q} \setminus \alpha$, $\min(\mathbb{Q} \setminus \alpha) \in \mathbb{Q} \setminus \alpha$ and $\min(\mathbb{Q}) < s = -(-s)$ proving that $-s \in -\alpha$. Using [eq: 9.4] it follows that $-s \leq m$ or $-m \leq s$ contradicting [eq: 9.6]

So in all cases we reach a contradiction so that the assumption is wrong. Hence

$$-\alpha \text{ has no greatest element} \quad \square$$

For rational cuts there is a simple expression for negative cuts.

Theorem 9.11. *If $q \in \mathbb{Q}$ then $-\alpha_q = \alpha_{-q}$*

Proof. Using [theorem: 9.4] we have that

$$\min(\mathbb{Q} \setminus \alpha_q) \text{ exist and } q = \min(\mathbb{Q} \setminus \alpha_q)$$

If $x \in -\alpha_q$ then $-x \in \mathbb{Q} \setminus \alpha_q \exists t \in \mathbb{Q} \setminus \alpha$ such that $t < -x$ so that $-x \neq \min(\mathbb{Q} \setminus \alpha) = q$. As $\alpha_q = \{r \in \mathbb{Q} \mid r < q\}$ and $-x \in \mathbb{Q} \setminus \alpha_q$ we have $q \leq -x$ or $x \leq -q$ which as $-x \neq q \Rightarrow x \neq -q$, gives $x < -q$. Hence $x \in \{r \in \mathbb{Q} \mid r < -q\} = \alpha_{-q}$ proving that

$$-\alpha_q \subseteq \alpha_{-q} \quad (9.7)$$

If $x \in \alpha_{-q}$ then $x < -q$, so that $q < -x$ hence $-x \notin \{x \in \mathbb{Q} \mid x < q\} = \alpha_q$ and $q < -x$ where $q = \min(\mathbb{Q} \setminus \alpha_q) \in \mathbb{Q} \setminus \alpha_q$ proving that $x \in -\alpha_q$. So $\alpha_{-q} \subseteq \alpha_q$, combining this with [eq: 9.7] gives

$$-\alpha_q = \alpha_{-q} \quad \square$$

9.1.2 Arithmetic in $\mathbb{R}$

9.1.2.1 Addition in $\mathbb{R}$

Definition 9.12. *If $\alpha, \beta \in \mathbb{R}$ then we define $\alpha + \beta$ by*

$$\alpha + \beta = \{q + r \mid q \in \alpha \wedge r \in \beta\}$$

Before we can use the above definition to define the addition operator in $\mathbb{R}$ we must prove that $\alpha + \beta$ is a Dedekind cut, hence a element of $\mathbb{R}$. First we need a little lemma.

Lemma 9.13. *$\forall \alpha \in \mathbb{R}$ and $\forall \varepsilon \in \mathbb{Q}$ with $0 < \varepsilon$ there exist a $r \in \alpha$ such that $r + \varepsilon \in \mathbb{Q} \setminus \alpha$*

Proof. Let $\alpha \in \mathbb{R}$ and $\varepsilon \in \mathbb{Q}$ such that $0 < \varepsilon$. Using [theorem: 9.9] there exist a $q \in \alpha$ and a $r \in \mathbb{Q} \setminus \alpha$ such that $r - q < \varepsilon$. Assume that $q + \varepsilon \in \alpha$ then we have by the definition of a cut that $q + \varepsilon < r$ so that $\varepsilon < r - q$ contradicting $r - q < \varepsilon$. Hence we must have that $q + \varepsilon \notin \alpha$ or $q + \varepsilon \in \mathbb{Q} \setminus \alpha$. $\square$

Theorem 9.14. *$\forall \alpha, \beta \in \mathbb{R}$ we have that $\alpha + \beta \in \mathbb{R}$, hence $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ where $+(\alpha, \beta) = \alpha + \beta$ is a operator on $\mathbb{R}$.*

Proof. Given Dedekind cuts α and β we must prove that $\alpha + \beta$ is a Dedekind cut.

1. As $\alpha \neq \emptyset$ and $\beta \neq \emptyset$ it follows that $\exists a \in \alpha$ and $\exists b \in \beta$ so that $a + b \in \{q + r \mid q \in \alpha \wedge r \in \beta\} = \alpha + \beta$, proving that

$$\alpha + \beta \neq \emptyset$$

2. Given $\varepsilon = \frac{1}{2} \in \mathbb{Q}$ we can as $0 < \varepsilon$ use [lemma: 9.13] to find a $r' \in \alpha$ and a $s' \in \beta$ such that $q' + \varepsilon \in \mathbb{Q} \setminus \alpha$ and $r' + \varepsilon \in \mathbb{Q} \setminus \beta$. Assume that $q' + y' + 1 \in \alpha + \beta$ then there exists a $q \in \alpha$ and $r \in \beta$ such that

$$q' + y' + 1 = q + r \quad (9.8)$$

As $q \in \alpha \wedge q' + \varepsilon \in \mathbb{Q} \setminus \alpha$ and $r \in \beta \wedge r' + \varepsilon \in \mathbb{Q} \setminus \beta$ it follows from the definition of Dedekind cuts that $q < q' + \varepsilon$ and $r < r' + \varepsilon$ so that $q + r < q' + r' + 2 \cdot \varepsilon = q' + r' + 1 \stackrel{[\text{eq: 9.8}]}{=} q + r$ giving the contradiction that $q + r < q + r$. So we must have that $q' + r' + 1 \notin \alpha + \beta$ proving that

$$\alpha + \beta \neq \mathbb{Q}$$

3. Let $s \in \alpha + \beta$ and $t \in \mathbb{Q} \setminus (\alpha + \beta)$ then there exists a $q \in \alpha$ and a $r \in \beta$ such that $s = q + r$. Assume now that that $t \leq s$, then $t \leq q + r$, so that $t - r \leq q$, by [theorem: 9.3] it follows then that $t - r \in \alpha$. From this and the fact that $r \in \beta$ it follows that $t = (t - r) + r \in \alpha + \beta$ contradicting $t \in \mathbb{Q} \setminus (\alpha + \beta)$, hence we must have that $s < t$.
4. Assume that $\alpha + \beta$ has a greatest element m then we have

$$m \in \alpha + \beta \text{ and } \forall q \in \alpha + \beta \text{ we have } q \leq m \quad (9.9)$$

As $m \in \alpha + \beta$ there exists a $q \in \alpha$ and a $r \in \beta$ such that $m = q + r$. As α has no greatest element there exist a $q' \in \alpha$ such that $q < q'$, hence $m = q + r < q' + r$ which as $q' + r \in \alpha + \beta$ contradicts [eq: 9.9]. So the assumption is wrong, hence $\alpha + \beta$ has no greatest element. $\square$

Theorem 9.15. $\langle \mathbb{R}, + \rangle$ is a Abelian group with neutral element $0 = \alpha_0 = \{q \in \mathbb{Q} \mid q < 0\}$ and if $\alpha \in \mathbb{R}$ then $-\alpha$ [the negative cut of α] is the inverse element of α .

Proof. We make use of the fact that $\langle \mathbb{Q}, + \rangle$ is a Abelian group [see theorem: 8.8]. So we have

associativity. If $\alpha, \beta, \gamma \in \mathbb{R}$ then

$$\begin{aligned} z \in (\alpha + \beta) + \gamma &\Leftrightarrow z = r + s \wedge r \in (\alpha + \beta) \wedge s \in \gamma \\ &\Leftrightarrow z = (q + t) + s \wedge q \in \alpha \wedge t \in \beta \wedge s \in \gamma \\ &\Leftrightarrow z = q + (t + s) \wedge q \in \alpha \wedge t \in \beta \wedge s \in \gamma \\ &\Leftrightarrow z = q + r \wedge q \in \alpha \wedge r \in \beta + \gamma \\ &\Leftrightarrow z \in \alpha + (\beta + \gamma) \end{aligned}$$

proving that $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.

commutativity. If $\alpha, \beta \in \mathbb{R}$ then

$$\begin{aligned} z \in \alpha + \beta &\Leftrightarrow z = r + s \wedge r \in \alpha \wedge s \in \beta \\ &\Leftrightarrow z = s + r \wedge r \in \alpha \wedge s \in \beta \\ &\Leftrightarrow z \in \beta + \alpha \end{aligned}$$

neutral element. Let $\alpha \in \mathbb{R}$ and take $\alpha_0 = \{q \in \mathbb{Q} \mid q < 0\}$. If $q \in \alpha + \alpha_0$ then there exists $r \in \alpha$ and $s \in \alpha_0$ such that $q = r + s$, as $s \in \alpha_0$ we have that $s < 0$ so that $q = r + s < r$, using [theorem: 9.3] it follows then that $q \in \alpha$. Hence we have that

$$\alpha + \alpha_0 \subseteq \alpha \quad (9.10)$$

If $q \in \alpha$ then as α has no maximum there exist a $r \in \alpha$ such that $q < r$, so $q - r < 0$ so that $q - r \in \alpha_0$, hence $q = (q - r) + r \in \alpha + \alpha_0$. So $\alpha \subseteq \alpha + \alpha_0$ which together with [eq: 9.10] proves that

$$\alpha = \alpha + \alpha_0 \stackrel{\text{commutativity}}{=} \alpha_0 + \alpha$$

inverse element. Let $\alpha \in \mathbb{R}$ and take

$$-\alpha \stackrel{[\text{theorem: 9.10}]}{=} \{r | -r \in \mathbb{Q} \setminus \alpha \text{ such that } \exists t \in \mathbb{Q} \setminus \alpha \models t < -r\}$$

then we have the following cases to consider:

$\min(\mathbb{Q} \setminus \alpha)$ does not exist. If $q \in \alpha_0$ then $q < 0$ so that by [theorems: 8.29, 4.73] $0 < -q$, by [theorem: 9.13] there exist a $r \in \alpha$ such that $-(q + (-r)) = r + (-q) \in \mathbb{Q} \setminus \alpha$, as $\min(\mathbb{Q} \setminus \alpha)$ does not exist there exist a $s \in \mathbb{Q} \setminus \alpha$ such that $s < r + (-q) = -(q + (-r))$. So we conclude that $q + (-r) \in -\alpha$, hence $q = (q + (-r)) + r \in (-\alpha) + \alpha$ giving

$$\alpha_0 \subseteq (-\alpha) + \alpha \quad (9.11)$$

If $q \in (-\alpha) + \alpha$ there exist a $r \in -\alpha$ and $s \in \alpha$ such that $q = r + s$. As $r \in -\alpha$ we have that $-r \in \mathbb{Q} \setminus \alpha$, using [definition: 9.1 (3)] we have then $s < -r$ so that $q = s + r < 0$ proving that $q \in \alpha_0$. Hence $(-\alpha) + \alpha \subseteq \alpha_0$ which by [eq: 9.11] proves that

$$\alpha_0 = (-\alpha) + \alpha$$

$\min(\mathbb{Q} \setminus \alpha)$ exist. Let $m = \min(\mathbb{Q} \setminus \alpha)$ then by [theorem: 9.4]

$$\alpha = \alpha_m = \{q \in \mathbb{Q} | q < m\}$$

Further by [theorem: 9.11] we have then that

$$-\alpha = -\alpha_m = \alpha_{-m}$$

so that

$$\alpha + (-\alpha) = \alpha_m + \alpha_{-m}$$

If $q \in \alpha + (-\alpha)$ there exist a $r \in \alpha_m$ and a $s \in \alpha_{-m}$ such that $q = r + s$. As $r \in \alpha_m$ we have $r < m$ and as $s \in \alpha_{-m}$ $s < -m$ so that $q = r + s < m + (-m) = 0$ proving that $q \in \alpha_0$. Hence

$$\alpha + (-\alpha) \subseteq \alpha_0 \quad (9.12)$$

Further if $q \in \alpha_0$ then $q < 0$ then as $0 < \frac{1}{2}$ we have that $\frac{1}{2} \cdot q < 0$ so that $m + \frac{1}{2} \cdot q < m$ and $-m + \frac{1}{2} \cdot q < -m$ so that $m + \frac{1}{2} \cdot q \in \alpha_m$ and $-m + \frac{1}{2} \cdot q \in \alpha_{-m}$ hence

$$\left(m + \frac{1}{2} \cdot q\right) + \left(-m + \frac{1}{2} \cdot q\right) \in \alpha_m + \alpha_{-m} = \alpha + (-\alpha)$$

which as $\left(m + \frac{1}{2} \cdot q\right) + \left(-m + \frac{1}{2} \cdot q\right) = \frac{1}{2} \cdot q + \frac{1}{2} \cdot q = \left(\frac{1}{2} + \frac{1}{2}\right) \cdot q = q$ proves that $q \in \alpha + (-\alpha)$. So $\alpha_0 \subseteq \alpha + (-\alpha)$ which combined with [eq: 9.12] gives

$$\alpha_0 = \alpha + (-\alpha) \stackrel{\text{commutativity}}{=} (-\alpha) + \alpha$$

□

9.1.2.2 Multiplication

Before we can define multiplication we have to divide the set of real numbers in the positive real numbers, the negative real numbers and the 0 element. We will then define multiplication of positive real numbers and extend it to all the real numbers.

Definition 9.16. The set of positive real numbers noted by $\mathbb{R}^+$ and negative real numbers noted by $\mathbb{R}^-$ is defined by

$$\begin{aligned} \mathbb{R}^+ &= \{\alpha \in \mathbb{R} | 0 \in \alpha\} \subseteq \mathbb{R} \\ \mathbb{R}^- &= \{\alpha | -\alpha \in \mathbb{R}^+\} \subseteq \mathbb{R} \end{aligned}$$

Further we define the set $\mathbb{R}_0^+$ of non negative numbers and $\mathbb{R}_0^-$ of non positive numbers by

$$\begin{aligned} \mathbb{R}_0^+ &= \mathbb{R}^+ \cup \{0\} \\ \mathbb{R}_0^- &= \mathbb{R}^- \cup \{0\} \end{aligned}$$

The following theorem shows that $\mathbb{R}_0^+ \neq \mathbb{R}^+$ and $\mathbb{R}_0^- \neq \mathbb{R}^-$

Theorem 9.17. $\mathbb{R} = \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ where $\mathbb{R}^+ \cap \mathbb{R}^- = \emptyset$, $\mathbb{R}^+ \cap \{0\} = \emptyset$ and $\mathbb{R}^- \cap \{0\} = \emptyset$

Note 9.18. Be careful here, 0 can mean either $0 \in \mathbb{Z}$ or $0 \in \mathbb{R}$ in which case $0 = \alpha_0$

Proof. As $\{0\} \subseteq \mathbb{R}$, $\mathbb{R}^+ \subseteq \mathbb{R}$ and $\mathbb{R}^- \subseteq \mathbb{R}$ we have

$$\mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\} \subseteq \mathbb{R} \quad (9.13)$$

If $\alpha \in \mathbb{R}$ then we have either:

$0 \in \alpha$. then $\alpha \in \mathbb{R}^+$ so that $\alpha \in \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$

$0 \notin \alpha$. then we have either:

min $(\mathbb{Q} \setminus \alpha)$ **does not exist.** As $0 \notin \alpha$ we have $-0 = 0 \in \mathbb{Q} \setminus \alpha$ and as **min** $(\mathbb{Q} \setminus \alpha)$ does not exist there exist a $s \in \mathbb{Q} \setminus \alpha$ such that $s < 0 = -0$ so that $0 \in -\alpha$. Hence $-\alpha \in \mathbb{R}^+$ proving that $\alpha \in \mathbb{R}^- \subseteq \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$.

min $(\mathbb{Q} \setminus \alpha)$ **exist.** Then by [theorem: 9.4] $\alpha = \alpha_m$ where $m = \min(\mathbb{Q} \setminus \alpha)$

$0 = m$. Then $\alpha = \alpha_0 = 0$ so that $\alpha \in \{0\} \subseteq \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$.

$0 < m$. Then $0 \in \alpha_m = \alpha$ so that $\alpha \in \mathbb{R}^+ \subseteq \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$.

$m < 0$. Then $0 \notin \alpha_m = \alpha$ so that $-0 = 0 \in \mathbb{Q} \setminus \alpha$ and as $m < 0 = -0$, $m \in \mathbb{Q} \setminus \alpha$ it follows that $0 \in -\alpha$, proving that $-\alpha \in \mathbb{R}^+$, hence $\alpha \in \mathbb{R}^- \subseteq \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$

So in all cases we have $\alpha \in \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ proving $\mathbb{R} \subseteq \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ which combined with [eq: 9.13] proves

$$\mathbb{R} = \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$$

Now as $0 = \alpha_0 = \{q \in \mathbb{Q} | q < 0\}$ we have that $0 \notin \alpha_0$ hence $0 = \alpha_0 \notin \mathbb{R}^+$ proving that

$$\mathbb{R}^+ \cap \{0\} = \emptyset$$

Using [theorem: 9.11] it follows that $-\alpha_0 = \alpha_{-0} = \alpha_0$ so that $-0 = -\alpha_0 \notin \mathbb{R}^+$ hence $0 \notin \mathbb{R}^-$ proving that

$$\mathbb{R}^- \cap \{0\} = \emptyset$$

Finally if $\alpha \in \mathbb{R}^+ \cap \mathbb{R}^-$ then $0 \in \alpha$ and $0 \in -\alpha$, as $0 \in -\alpha$ then at least $-0 \in \mathbb{Q} \setminus \alpha$ so that $0 = -0 \notin \alpha$ contradicting $0 \in \alpha$. So we have

$$\mathbb{R}^+ \cap \mathbb{R}^- = \emptyset \quad \square$$

Defining multiplication in $\mathbb{R}$ is difficult. we first define multiplication for $\mathbb{R}^+$ and extend it later to $\mathbb{R}$.

Definition 9.19. Given $\alpha, \beta \in \mathbb{R}^+$ we define $A = \alpha \odot \beta$ by

$$\begin{aligned} \alpha \odot \beta &= \mathbb{Q}_0^- \cup \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\} \\ &= \{r \in \mathbb{Q} | r \leq 0\} \cup \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\} \end{aligned}$$

Theorem 9.20. $\forall \alpha, \beta \in \mathbb{R}^+$ we have that $\alpha \odot \beta \in \mathbb{R}^+$

Proof. First we prove that $\alpha \odot \beta$ is a Dedekind cut.

1. As $0 \in \{r \in \mathbb{Q} | r \leq 0\}$ it follows that

$$\alpha \odot \beta \neq \emptyset$$

2. As $\alpha, \beta \in \mathbb{R}^+$ it follows that $0 \in \alpha \wedge 0 \in \beta$ and as α, β do not have a greatest element we have

$$\exists s_1 \in \alpha, \exists t_1 \in \beta \text{ such that } 0 < s_1 \wedge 0 < t_1 \quad (9.14)$$

As $0 < 1$ we have by [theorem: 9.13] that

$$\exists s_2 \in \alpha, \exists t_2 \in \beta \text{ such that } s_2 + 1 \in \mathbb{Q} \setminus \alpha \wedge t_2 + 1 \in \mathbb{Q} \setminus \beta \quad (9.15)$$

Take now

$$s = \max(\{s_1, s_2\}) \text{ and } t = \max(\{t_1, t_2\}) \quad (9.16)$$

If $s \notin \alpha$ then $s \in \mathbb{Q} \setminus \alpha$ so that by [definition: 9.1 (3)] and [eqs: 9.14, 9.15] we have $s_1 < s$ and $s_2 < s$ contradicting the fact that $s \in \{s_1, s_2\}$, so we must have that $s \in \alpha$. Likewise if $t \notin \beta$ then $t \in \mathbb{Q} \setminus \beta$ so that by [definition: 9.1 (3)] and [eq: 9.14, 9.15] we have $t_1 < t$ and $t_2 < t$ contradicting the fact that $t \in \{t_1, t_2\}$, so we must have that $t \in \beta$. So we have

$$s \in \alpha \wedge t \in \beta \text{ and by [eqs: 9.14, 9.16] that } 0 < s \wedge 0 < t \quad (9.17)$$

If $s + 1 \in \alpha$ then by [definition: 9.1 (3)] and [eq: 9.15] we have $s + 1 < s_2 + 1 \Rightarrow s < s_2$ contradicting $s = \max(s_1, s_2)$. Likewise if $t + 1 \in \beta$ then by [definition: 9.1 (3)] and [eq: 9.15] we have $t + 1 < t_2 + 1 \Rightarrow t < t_2$ contradicting $t = \max(t_1, t_2)$. So we must have

$$s + 1 \in \mathbb{Q} \setminus \alpha \text{ and } t + 1 \in \mathbb{Q} \setminus \beta \quad (9.18)$$

Assume now that $s \cdot t + s + t + 1 \in \alpha \odot \beta$. As $0 < s \wedge 0 < t$ we have that $0 < s \cdot t$ giving $0 < s \cdot t + s + t + 1$ so that $s \cdot t + s + t + 1 \notin \{q \in \mathbb{Q} | q \leq 0\}$ so we must have that

$$s \cdot t + s + t + 1 \in \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\}$$

hence there exists $s' \in \alpha$ and $t' \in \beta$ with $0 < s' \wedge 0 < t'$ such that $s \cdot t + s + t + 1 = s' \cdot t'$. Using [definition: 9.1 (3)] and [eq: 9.18] we have that $s' < s + 1$ and $t' < t + 1$ so $s' \cdot t' < (s + 1) \cdot (t + 1)$ and $t' \cdot (s + 1) < (t + 1) \cdot (s + 1)$, hence $s' \cdot t' < (s + 1) \cdot (t + 1) = s \cdot t + s + t + 1 = s' \cdot t'$ giving the contradiction $s' \cdot t' < s' \cdot t'$. Hence the assumption is false so that $s \cdot t + s + t + 1 \notin \alpha \odot \beta$ proving that

$$\alpha \odot \beta \neq \mathbb{Q}$$

3. Let $q \in \alpha \odot \beta$ and $r \in \mathbb{Q} \setminus \alpha \odot \beta$ then for q we have either:

$q \in \{r \in \mathbb{Q} | r \leq 0\}$. Then $q \leq 0$ further as $r \in \mathbb{Q} \setminus \alpha \odot \beta$ we have that $r \notin \{r \in \mathbb{Q} | r \leq 0\}$ so that $0 < r$ from which it follows that $q < r$.

$q \notin \{r \in \mathbb{Q} | r \leq 0\}$. Then $q \in \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\}$ so that

$$\exists s' \in \alpha, \exists t' \in \beta \text{ with } 0 < s' \wedge 0 < t' \text{ such that } q = s' \cdot t' \quad (9.19)$$

Assume now that $r \leq q$. As $r \in \mathbb{Q} \setminus \alpha \odot \beta$ we have that $r \neq q$ [as $q \in \alpha \odot \beta$] and $r \notin \{r \in \mathbb{Q} | r \leq 0\}$ so that $0 < r$. Hence we have $0 < r < q$ or multiplying by r^{-1} [which by [theorems: 8.29, 4.73] exists and $0 < r^{-1}$] we have $1 = r \cdot r^{-1} < q \cdot r^{-1}$ or if we define $t = q \cdot r^{-1}$, it follows that

$$1 < t \text{ and } t \cdot r = q \quad (9.20)$$

Using the above, we have by [theorems: 8.29, 4.73] that $0 < t^{-1} < 1$ so that by multiplying by s' we have, as $0 < s'$, that

$$t^{-1} \cdot s' < s' \quad (9.21)$$

If now $t^{-1} \cdot s' \notin \alpha$ then $t^{-1} \cdot s' \in \mathbb{Q} \setminus \alpha$ which, as $s' \in \alpha$, means by [definition: 9.1 (3)] that $s' < t^{-1} \cdot s'$ contradicting [eq: 9.21]. Hence we must have that

$$t^{-1} \cdot s' \in \alpha \quad (9.22)$$

As by [eq: 9.19] $t' \in \beta$ we have using the above that $(t^{-1} \cdot s') \cdot t' \in \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\}$ so that

$$(t^{-1} \cdot s') \cdot t' \in \alpha \odot \beta \quad (9.23)$$

Now

$$\begin{aligned}
 (t^{-1} \cdot s') \cdot t' &= t^{-1} \cdot (s' \cdot t') \\
 &\stackrel{[\text{eq: 9.19}]}{=} t^{-1} \cdot q \\
 &\stackrel{[\text{eq: 9.20}]}{=} t^{-1} \cdot (t \cdot r) \\
 &= r
 \end{aligned}$$

which combined with [eq: 9.23] proves that $r \in \alpha \odot \beta$ contradicting the fact $r \in \mathbb{Q} \setminus \alpha \odot \beta$, hence the assumption is wrong and we must have

$$q < r$$

4. Assume now that $\alpha \odot \beta$ has a greatest element m then we have

$$m \in \alpha \odot \beta \text{ and } \forall r \in \alpha \odot \beta \text{ we have } r \leq m \quad (9.24)$$

As $m \in \alpha \odot \beta$ we have the following cases to consider:

$m \in \{r \in \mathbb{Q} | r \leq 0\}$. Then $m \leq 0$. As $\alpha, \beta \in \mathbb{R}^+$ we have that $0 \in \alpha$ and $0 \in \beta$ which, as α, β have no greatest element, there exists $s \in \alpha$ and $t \in \beta$ such that $0 < s$ and $0 < t$, hence $s \cdot t \in \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\}$ proving that

$$s \cdot t \in \alpha \odot \beta \text{ and thus } s \cdot t \leq m$$

As $0 < s \wedge 0 < t$ we have that $0 < s \cdot t$ so, as $m \leq 0$, we have $m < s \cdot t$ contradicting the above.

$m \notin \{r \in \mathbb{Q} | r \leq 0\}$. Then $0 < m$ and $m \in \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\}$ hence there exists $s \in \alpha$ and $t \in \beta$ with $0 < s$ and $0 < t$ such that

$$m = s \cdot t \quad (9.25)$$

As α, β has no greatest element there exists $s' \in \alpha$ and $t' \in \beta$ such that $0 < s < s'$ and $0 < t < t'$. As $0 < s \wedge 0 < t$ we have $s \cdot t < s' \cdot t$ and $t \cdot s' < s' \cdot t'$ so that $s \cdot t < s' \cdot t'$ or using [eq: 9.25]

$$m < s' \cdot t' \quad (9.26)$$

Further as $s' \cdot t' \in \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\}$ we have that $s' \cdot t' \in \alpha \odot \beta$ so that by [eq: 9.24] $s' \cdot t' \leq m$ contradicting [eq: 9.26].

As in all cases we have a contradiction the assumption must be wrong, so $\alpha \odot \beta$ has no greatest element.

By (1),(2),(3) and (4) we have that $\alpha \odot \beta$ is a Dedekind cut, hence

$$\alpha \odot \beta \in \mathbb{R}$$

Finally as $0 \in \{r \in \mathbb{Q} | r \leq 0\}$ we have $0 \in \alpha \odot \beta$ proving that

$$\alpha \odot \beta \in \mathbb{R}^+ \quad \square$$

After we have defined multiplication in $\mathbb{R}^+$ we want to specify the neutral element for $\odot$

Theorem 9.21. $\forall \alpha \in \mathbb{R}^+ \text{ we have } \alpha_1 \odot \alpha = \alpha$

Proof. Let $x \in \alpha_1 \odot \alpha$ then we have either:

$x \leq 0$. As $\alpha \in \mathbb{R}^+$ we have that $0 \in \alpha$ and as $x \leq 0$ it follows from [theorem: 9.3] that $x \in \alpha$.

$0 < x$. Then $x \notin \mathbb{Q}_0^-$ so there exists a $s \in \alpha_1$ and a $t \in \alpha$ with $0 < s \wedge 0 < t$ such that $x = s \cdot t$. From $s \in \alpha_1$ it follows that $s < 1$ so, as $0 < t$ we have that $x = s \cdot t < t \Rightarrow x < t$. As $t \in \alpha$ it follows from [theorem: 9.3] that $x \in \alpha$.

As in all cases $x \in \alpha$ it follows that

$$\alpha_1 \odot \alpha \subseteq \alpha \quad (9.27)$$

If $x \in \alpha$ then we have either:

$x \leq 0$. Then $x \in \{q \in \mathbb{Q} | q \leq 0\}$ so that $x \in \alpha_1 \odot \alpha$

$0 < x$. As α has no greatest element [see definition: 9.1 (4)] there exist a $t \in \alpha$ such that $0 < x < t$. Then as by [theorems: 8.29, 4.73] $0 < t^{-1}$ we have that $0 < x \cdot t^{-1} < t \cdot t^{-1} = 1$ so that $x \cdot t^{-1} \in \alpha_1$. Now $0 < t$, $0 < x \cdot t^{-1}$ so that $x = (x \cdot t^{-1}) \cdot t \in \alpha_1 \odot \alpha$. Hence $\alpha \subseteq \alpha_1 \odot \alpha$ which combined with [eq: 9.27] results in

$$\alpha_1 \odot \alpha = \alpha$$

□

Theorem 9.22. $\forall \alpha, \beta \in \mathbb{R}^+$ we have $\alpha \odot \beta = \beta \odot \alpha$

Proof. Then we have

$$\begin{aligned} q \in \alpha \odot \beta &\Leftrightarrow q \in \{r \in \mathbb{Q} | r \leq 0\} \cup \{s \cdot t | (s, t) \in \alpha \times \beta \wedge 0 < s \wedge 0 < t\} \\ &\Leftrightarrow q \leq 0 \vee \exists (s, t) \in \alpha \times \beta \text{ with } 0 < s \wedge 0 < t \text{ such that } q = s \cdot t \\ &\stackrel{[\text{theorem: 8.12}]}{\Leftrightarrow} q \leq 0 \vee \exists (s, t) \in \alpha \times \beta \text{ with } 0 < s \wedge 0 < t \text{ such that } q = t \cdot s \\ &\Leftrightarrow q \leq 0 \vee \exists (t, s) \in \beta \times \alpha \text{ with } 0 < t \wedge 0 < s \text{ such that } q = s \cdot t \\ &\Leftrightarrow q \in \beta \odot \alpha \end{aligned}$$

proving that

$$\alpha \odot \beta = \beta \odot \alpha$$

□

Theorem 9.23. Let $\alpha, \beta, \gamma \in \mathbb{R}^+$ then we have that $\alpha \odot (\beta \odot \gamma) = (\alpha \odot \beta) \odot \gamma$

Proof. Using the definition of $\odot$ we have

$$\alpha \odot \beta = \mathbb{Q}_0^- \cup \{s \cdot t | (s, t) \in \alpha \times \beta \text{ with } 0 < s \wedge 0 < t\} \quad (9.28)$$

$$\beta \odot \gamma = \mathbb{Q}_0^- \cup \{s \cdot t | (s, t) \in \beta \times \gamma \text{ with } 0 < s \wedge 0 < t\} \quad (9.29)$$

$$\alpha \odot (\beta \odot \gamma) = \mathbb{Q}_0^- \cup \{s \cdot t | (s, t) \in \alpha \times (\beta \odot \gamma) \text{ with } 0 < s \wedge 0 < t\} \quad (9.30)$$

$$(\alpha \odot \beta) \odot \gamma = \mathbb{Q}_0^- \cup \{s \cdot t | (s, t) \in (\alpha \odot \beta) \times \gamma \text{ with } 0 < s \wedge 0 < t\} \quad (9.31)$$

Let $x \in \alpha \odot (\beta \odot \gamma)$ then we have either

$x \leq 0$. Then we have $x \in \mathbb{Q}_0^-$ proving by [eq: 9.31] that $x \in (\alpha \odot \beta) \odot \gamma$

$0 < x$. Then by [eq: 9.30] there exist $q \in \alpha$ and $r \in \beta \odot \gamma$ with $0 < q \wedge 0 < r$ such that

$$x = q \cdot r \quad (9.32)$$

As $0 < r$ and $r \in \beta \odot \gamma$ it follows from [eq: 9.29] that there exists a $s \in \beta$ and $t \in \gamma$ with $0 < s$ and $0 < t$ such that $r = s \cdot t$ hence $x = q \cdot r = q \cdot (s \cdot t) = (q \cdot s) \cdot t$ proving that

$$x = (q \cdot s) \cdot t \quad (9.33)$$

As $q \in \alpha \wedge s \in \beta \wedge 0 < q \wedge 0 < s$ we have by [eq: 9.28] that $q \cdot s \in \alpha \odot \beta$. Further as $0 < q \wedge 0 < s$ we have $0 < q \cdot s$ which together with $t \in \gamma \wedge 0 < t$ proves that $(q \cdot s) \cdot t \in (\alpha \odot \beta) \odot \gamma$ or using [eq: 9.33] that

$$x \in (\alpha \odot \beta) \odot \gamma$$

So we have proved that

$$\alpha \odot (\beta \odot \gamma) \subseteq (\alpha \odot \beta) \odot \gamma \quad (9.34)$$

Let $x \in (\alpha \odot \beta) \odot \gamma$ then we have either:

$x \leq 0$. Then we have $x \in \mathbb{Q}_0^-$ proving by [eq: 9.31] that $x \in \alpha \odot (\beta \odot \gamma)$.

$0 < x$. Then by [eq: 9.31] we have that there exists a $q \in \alpha \odot \beta$ and a $r \in \gamma$ with $0 < q \wedge 0 < r$ such that

$$x = q \cdot r \quad (9.35)$$

As $0 < q$ and $q \in \alpha \odot \beta$ it follows from [eq: 9.28] that there exists a $s \in \alpha$ and $t \in \beta$ with $0 < s \wedge 0 < t$ such that $q = s \cdot t$. Hence $x = q \cdot r = (s \cdot t) \cdot r = s \cdot (t \cdot r)$ giving

$$x = s \cdot (t \cdot r) \quad (9.36)$$

As $t \in \beta \wedge r \in \gamma \wedge 0 < \beta \wedge 0 < \gamma$ we have by [eq: 9.29] that $t \cdot r \in \beta \odot \gamma$. Further as $0 < t \wedge 0 < r$ we have $0 < t \cdot r$ which together with $s \in \alpha \wedge 0 < s$ proves that $s \cdot (t \cdot r) \in \alpha \odot (\beta \odot \gamma)$ or using [eq: 9.36] we have that

$$x \in \alpha \odot (\beta \odot \gamma)$$

So we have proved that $(\alpha \odot \beta) \odot \gamma \subseteq \alpha \odot (\beta \odot \gamma)$ which combined with [eq: 9.34] gives

$$(\alpha \odot \beta) \odot \gamma = \alpha \odot (\beta \odot \gamma) \quad \square$$

Theorem 9.24. $\forall \alpha, \beta, \gamma \in \mathbb{R}^+$ we have that $\alpha \odot (\beta + \gamma) = \alpha \odot \beta + \alpha \odot \gamma$

Proof. Let $x \in \alpha \odot (\beta + \gamma)$ then we have either:

$x \leq 0$. Then $0, x \in \mathbb{Q}_0^-$ so that $x \in \alpha \odot \beta$ and $0 \in \alpha \odot \gamma$ hence $x = x + 0 \in \alpha \odot \beta + \alpha \odot \gamma$.

$0 < x$. Then $x = s \cdot t$ where $s \in \alpha \wedge 0 < s$ and $t \in \beta + \gamma \wedge 0 < t$. As $t \in \beta + \gamma$ there exists $u \in \beta$ and $v \in \gamma$ such that $t = u + v$. Using [theorem: 8.12] we have that

$$x = s \cdot t = s \cdot (u + v) = s \cdot u + s \cdot v \quad (9.37)$$

We have now the following possibilities for u and v :

$u \leq 0 \wedge v \leq 0$. Then $t = u + v \leq 0$ giving the contradiction $0 < t \leq 0$ so this case will not occur.

$u \leq 0 \wedge 0 < v$. Then as $0 < s$ we have $s \cdot u \leq 0 \Rightarrow s \cdot u \in \mathbb{Q}_0^- \Rightarrow s \cdot u \in \alpha \odot \beta$, further as $0 < s \wedge 0 < v$ we have that $s \cdot v \in \alpha \odot \gamma$. Hence $x \stackrel{[eq: 9.37]}{=} s \cdot u + s \cdot v \in \alpha \odot \beta + \alpha \odot \gamma$.

$0 < u \wedge v \leq 0$. Then as $0 < s \wedge 0 < u$ we have that $s \cdot u \in \alpha \odot \beta$, further $s \cdot v \leq 0 \Rightarrow s \cdot v \in \mathbb{Q}_0^- \Rightarrow s \cdot v \in \alpha \odot \gamma$. Hence $x \stackrel{[eq: 9.37]}{=} s \cdot u + s \cdot v \in \alpha \odot \beta + \alpha \odot \gamma$.

$0 < u \wedge 0 < v$. Then as $0 < s \wedge 0 < u \wedge 0 < v$ we have that $s \cdot u \in \alpha \odot \beta$ and $s \cdot v \in \alpha \odot \gamma$. Hence $x \stackrel{[eq: 9.37]}{=} s \cdot u + s \cdot v \in \alpha \odot \beta + \alpha \odot \gamma$.

So in all cases we have that $x \in \alpha \odot \beta + \alpha \odot \gamma$ proving that

$$\alpha \odot (\beta + \gamma) \subseteq \alpha \odot \beta + \alpha \odot \gamma \quad (9.38)$$

For the opposite inclusion let $x \in \alpha \odot \beta + \alpha \odot \gamma$. Then

$$x = r + t \text{ where } r \in \alpha \odot \beta \text{ and } t \in \alpha \odot \gamma \quad (9.39)$$

We must now consider the following cases for x :

$x \leq 0$. Then $x \in \mathbb{Q}_0^-$ so that $x \in \alpha \odot (\beta + \gamma)$

$0 < x$. Then we have to look at the following sub cases:

$r \leq 0 \wedge t \leq 0$. Then $x \stackrel{[eq: 9.39]}{=} r + t \leq 0$ a contradicting $0 < x$, so this case does not occur.

$r \leq 0 \wedge 0 < t$. Then as $t \notin \mathbb{Q}_0^+$ there exists $u \in \alpha$ and $v \in \gamma$ with $0 < u \wedge 0 < v$ such that $t = u \cdot v$. As $\beta \in \mathbb{R}^+ \Rightarrow 0 \in \beta$ and $v \in \gamma$ we have that $(0 + v) \in \beta + \gamma$, further as $0 < u$ and $0 < 0 + v$ it follows that $t = u \cdot v = u \cdot (0 + v) \in \alpha \odot (\beta + \gamma)$. Since $r \leq 0$ we have $x \stackrel{[eq: 9.39]}{=} r + t \leq 0 + t = t$, which, as $t \in \alpha \odot (\beta + \gamma)$, proves by [theorem: 9.3] that

$$x \in \alpha \odot (\beta + \gamma)$$

$0 < r \wedge t \leq 0$. Then as $r \notin \mathbb{Q}_0^-$ there exists $u \in \alpha$ and $v \in \beta$ with $0 < u \wedge 0 < v$ such that $r = u \cdot v$. As $\gamma \in \mathbb{R}^+ \Rightarrow 0 \in \gamma$ and $v \in \beta$ we have that $v + 0 \in \beta + \gamma$, further as $0 < u$ and $0 < v + 0$ it follows that $r = u \cdot v = u \cdot (v + 0) \in \alpha \odot (\beta + \gamma)$. Since $t \leq 0$ we have $x \stackrel{[eq: 9.39]}{=} r + t \leq r + 0 = r$, which, as $r \in \alpha \odot (\beta + \gamma)$, proves by [theorem: 9.3] that

$$x \in \alpha \odot (\beta + \gamma)$$

$0 < r \wedge 0 < t$. Then as $r, t \notin \mathbb{Q}_0^-$ there exists $u, u' \in \alpha$, $v \in \beta$ and $v' \in \gamma$ such that

$$r = u \cdot v \wedge t = u' \cdot v' \wedge 0 < u \wedge 0 < v \wedge 0 < u' \wedge 0 < v' \quad (9.40)$$

For u, u' we must now examine the following possibilities:

$u = u'$. Then

$$x \stackrel{[\text{eqs: 9.39, 9.40}]}{=} u \cdot v + u' \cdot v' = u \cdot v + u \cdot v' = u \cdot (v + v') \quad (9.41)$$

so as $0 < u \wedge 0 < v + v'$ we have that $u \cdot (v + v') \in \alpha \odot (\beta + \gamma)$ hence

$$x \in \alpha \odot (\beta + \gamma)$$

$u < u'$. Then as $0 < u' \wedge 0 < v + v' \wedge u' \in \alpha \wedge v + v' \in \beta + \gamma$ we have

$$u' \cdot (v + v') \in \alpha \odot (\beta + \gamma)$$

Further from $u < u'$, $0 < v$ we have that $u \cdot v < u' \cdot v$, hence

$$x \stackrel{[\text{eq: 9.39, 9.40}]}{=} u \cdot v + u' \cdot v' < u' \cdot v + u' \cdot v' = u' \cdot (v + v') \in \alpha \odot (\beta + \gamma)$$

which by [theorem: 9.3] proves that

$$x \in \alpha \odot (\beta + \gamma)$$

$u' < u$. Then as $0 < u \wedge 0 < v + v' \wedge u \in \alpha \wedge v + v' \in \beta + \gamma$ we have

$$u \cdot (v + v') \in \alpha \odot (\beta + \gamma)$$

Further from $u' < u$, $0 < v'$ it follows that $u' \cdot v' < u \cdot v'$, hence

$$x \stackrel{[\text{eq: 9.39, 9.40}]}{=} u \cdot v + u' \cdot v' < u \cdot v + u \cdot v' = u \cdot (v + v') \in \alpha \odot (\beta + \gamma)$$

which by [theorem: 9.3] proves that

$$x \in \alpha \odot (\beta + \gamma)$$

So in all cases we have $x \in \alpha \odot (\beta + \gamma)$ proving that $\alpha \odot \beta + \alpha \odot \gamma \subseteq \alpha \odot (\beta + \gamma)$ which combined with [eq: 9.38] gives

$$\alpha \odot (\beta + \gamma) = \alpha \odot \beta + \alpha \odot \gamma \quad \square$$

Theorem 9.25. Let $\alpha \in \mathbb{R}^+$ then $\text{inv}(\alpha)$ defined by

$$\begin{aligned} \text{inv}(\alpha) &= \{r \in \mathbb{Q} | r \leq 0\} \cup \{s^{-1} | s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\} \\ &\stackrel{\mathbb{Q}_0^- = \{r \in \mathbb{Q} | r \leq 0\}}{=} \mathbb{Q}_0^- \cup \{s^{-1} | s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\} \end{aligned}$$

is a Dedekind cut such that $\text{inv}(\alpha) \in \mathbb{R}^+$.

Proof. We have

1. As $0 \in \mathbb{Q}_0^-$ it follow that $0 \in \text{inv}(\alpha)$ proving that

$$\text{inv}(\alpha) \neq \emptyset$$

2. As $\alpha \in \mathbb{R}^+$ we have $0 \in \alpha$ and as α has no greatest element there exist a $s \in \alpha$ such that $0 < s$. Hence s^{-1} exist and by [theorems: 8.29, 4.73] $0 < s^{-1}$ so that $s^{-1} \notin \mathbb{Q}_0^-$. Assume that $s^{-1} \in \text{inv}(\alpha)$ then as $s^{-1} \notin \mathbb{Q}_0^-$ we must have that

$$s^{-1} \in \{s^{-1} | s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\}$$

so that $\exists t \in \mathbb{Q} \setminus \alpha$ such that $s^{-1} = t^{-1} \stackrel{[\text{theorem: 4.54}]}{\Rightarrow} s = t$. So $s \in \mathbb{Q} \setminus \alpha$ contradicting $s \in \alpha$ hence $s^{-1} \notin \text{inv}(\alpha)$ proving that

$$\text{inv}(\alpha) \neq \mathbb{Q}$$

3. Let $q \in \text{inv}(\alpha)$ and $r \in \mathbb{Q} \setminus \text{inv}(\alpha)$. For q we have the following possibilities:

$q \leq 0$. Then as $r \in \mathbb{Q} \setminus \text{inv}(\alpha)$ we have $r \notin \text{inv}(\alpha)$ hence $r \notin \mathbb{Q}_0^-$ so that $0 < r$ giving

$$q < r$$

$0 < q$. Then $q \notin \mathbb{Q}_0^-$ hence, as $q \in \text{inv}(\alpha)$, we have:

$$q \in \{s^{-1} \mid s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\}$$

so there exist a $s \in \mathbb{Q} \setminus \alpha$ with $0 < s$ and $\exists t \in \mathbb{Q} \setminus \alpha \vdash t < s$ such that $q = s^{-1}$, as $q^{-1} = (s^{-1})^{-1} \stackrel{[\text{theorem: 4.53}]}{=} s$ we have that

$$q^{-1} \in \mathbb{Q} \setminus \alpha, 0 < q^{-1} \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < q^{-1} \quad (9.42)$$

Further as $r \in \mathbb{Q} \setminus \text{inv}(\alpha)$ we have that $r \notin \mathbb{Q}_0^-$ giving $0 < r$ so that by [theorems: 8.29, 4.73]

$$0 < r \text{ and } 0 < r^{-1} \quad (9.43)$$

For r^{-1} we have the following possibilities:

$r^{-1} \in \alpha$. Then as $q^{-1} \in \mathbb{Q} \setminus \alpha$ [see eq: 9.42] we have by [definition: 9.1 (3)] that $r^{-1} < q^{-1}$, so as $0 < r^{-1}$ we have by [theorems: 8.29, 4.73] that

$$q < r$$

$r^{-1} \notin \alpha$. Then $r^{-1} \in \mathbb{Q} \setminus \alpha$ and we have to look at the following possibilities

$\forall t \in \mathbb{Q} \models r^{-1} \leq t$. Then as $q^{-1} \in \mathbb{Q} \setminus \alpha$ [see eq: 9.42] we have that $r^{-1} \leq q^{-1}$. If $r^{-1} = q^{-1}$ we have by [eq: 9.42] a $t \in \mathbb{Q} \setminus \alpha$ such that $t < r^{-1}$ contradicting $\forall t \in \mathbb{Q} \models r^{-1} \leq t$, hence $r^{-1} \neq q^{-1}$. So $0 < r^{-1} < q^{-1}$ and by [theorems: 8.29, 4.73]

$$q < r$$

$\exists t \in \mathbb{Q}$ such that $t < r^{-1}$. Then as $r^{-1} \in \mathbb{Q} \setminus \alpha$ and $0 < r^{-1}$ we have that $r = (r^{-1})^{-1} \in \{s^{-1} \mid s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\}$ so that $r \in \text{inv}(\alpha)$ contradicting $r \in \mathbb{Q} \setminus \text{inv}(\alpha)$ so this case does not occur.

Hence in all valid cases we have

$$q < r$$

4. Assume that $\text{inv}(\alpha)$ has a greatest element m then we have

$$m \in \text{inv}(\alpha) \text{ and } \forall s \in \text{inv}(\alpha) \text{ we have } s \leq m \quad (9.44)$$

For m we have to look at the following possibilities:

$m \leq 0$. Using [definition: 9.1 (2)] $\emptyset \neq \mathbb{Q} \setminus \alpha$ so there exist a $r \in \mathbb{Q} \setminus \alpha$. As $\alpha \in \mathbb{R}^+$ we have that $0 \in \alpha$ so that by [definition: 9.1 (3)] that

$$0 < r \stackrel{\Rightarrow}{r < r+1} 0 < r+1 \text{ and by [theorems: 8.29, 4.73]} 0 < (r+1)^{-1} \quad (9.45)$$

If $r+1 \in \alpha$ then as $r \in \mathbb{Q} \setminus \alpha$ we have by [definition: 9.1 (3)] that $r+1 < r$ a contradiction, so we must have that $r+1 \notin \alpha$ or $r+1 \in \mathbb{Q} \setminus \alpha$. As further $r < r+1$ and $0 < r+1$ it follows that $(r+1)^{-1} \in \{s^{-1} \mid s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\}$ so that $(r+1)^{-1} \in \text{inv}(\alpha)$ hence by [eq: 9.44] $(r+1)^{-1} \leq m \leq 0$ contradicting $0 < (r+1)^{-1}$ [see eq: 9.45]. So we end with a contradiction.

$0 < m$. Then $m \notin \mathbb{Q}_0^-$ so that, as $m \in \text{inv}(\alpha)$, we have $m \in \{s^{-1} \mid s \in \mathbb{Q} \setminus \alpha \text{ with } 0 < s \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \vdash t < s\}$ so there exist a $s \in \mathbb{Q} \setminus \alpha$ with $0 < s$ and $\exists t \in \mathbb{Q} \setminus \alpha \vdash t < s$ such that $m = s^{-1}$, hence $m^{-1} = s$ so that:

$$m^{-1} \in \mathbb{Q} \setminus \alpha, 0 < m^{-1} \text{ and } \exists t \in \mathbb{Q} \setminus \alpha \text{ such that } t < m^{-1} \quad (9.46)$$

As $t \in \mathbb{Q} \setminus \alpha$ we have that $t \notin \mathbb{Q}_0^-$ so that $0 < t$, further as $t < m^{-1}$ we have by the density of $\mathbb{Q}$ [see theorem: 8.37] that there exist a $s \in \mathbb{Q}$ such that $t < s < m^{-1}$, hence

$$0 < t < s < m^{-1} \text{ which by [theorems: 8.29, 4.73] gives also } m < s^{-1} \quad (9.47)$$

If $s \in \alpha$ then as $t \in \mathbb{Q} \setminus \alpha$ [see eq: 9.46] we have by [definition: 9.1 (3)] that $s < t$ contradicting $t < s < m^{-1}$, so we must have $s \notin \alpha$, hence $s \in \mathbb{Q} \setminus \alpha$, so as $s \notin \alpha$ we have $s \notin \mathbb{Q}_0^-$, so that $0 < s$, which together with $t \in \mathbb{Q} \setminus \alpha$ and $t < s$ proves that $s^{-1} \in \text{inv}(\alpha)$. Using [eq: 9.44] it follows that $s^{-1} \leq m$ which contradicts [eq: 9.47]. So this case ends also in a contradiction.

As all possible cases ends in a contradiction the assumption must be false resulting in

$\text{inv}(\alpha)$ has no greatest element

(1),(2),(3),(4) proves that

$\text{inv}(\alpha)$ is a Dedekind cut

Further as $0 \in \mathbb{Q}_0^-$ we have that $0 \in \text{inv}(\alpha)$ hence

$$\text{inv}(\alpha) \in \mathbb{R}^+ \quad \square$$

We prove now that $\text{inv}(\alpha)$ is the multiplicative inverse for $\mathbb{R}^+$.

Theorem 9.26. *If $\alpha \in \mathbb{R}^+$ then $\alpha \odot \text{inv}(\alpha) = \alpha_1$*

Proof. If $x \in \alpha \odot \text{inv}(\alpha)$ then we have for x either:

$x \leq 0$. Then as $0 < 1$ we have $x < 1$ hence $x \in \alpha_1$

$0 < x$. Then $x \notin \mathbb{Q}_0^-$ we have as $x \in \alpha \odot \text{inv}(\alpha)$ that $\exists s \in \alpha \wedge \exists t \in \text{inv}(\alpha)$ with $0 < s$ and $0 < t$ such that $x = s \cdot t$. For t we have the following cases:

$t \leq 0$. Then from $0 < s$ we have that $x = s \cdot t \leq 0 < 1$ hence $x \in \alpha_1$.

$0 < t$. Then $t \notin \mathbb{Q}_0^-$ which as $t \in \text{inv}(\alpha)$ means that there exist a $s \in \mathbb{Q} \setminus \alpha$ such that $0 < s$ and $\exists r \in \mathbb{Q} \setminus \alpha$ $r < s$ such that $t = s^{-1}$. As $t^{-1} = (s^{-1})^{-1} = s$ we have that $t^{-1} \in \mathbb{Q} \setminus \alpha$.

Using [definition: 9.1 (3)] we have $s < t^{-1}$ so that $x = s \cdot t < 1$ proving that $x \in \alpha_1$.

As in all cases $x \in \alpha_1$ we have that

$$\alpha \odot \text{inv}(\alpha) \subseteq \alpha_1 \quad (9.48)$$

Now for the opposite inclusion, let $x \in \alpha_1$ then $x < 1$ and we have either:

$x \leq 0$. Then $x \in \mathbb{Q}_0^-$ so that $x \in \alpha \odot \text{inv}(\alpha)$.

$0 < x$. As $\alpha \in \mathbb{R}^+$ we have that $0 \in \alpha$, further as α is a Dedekind cut, α has no greatest element [see definition: 9.1 (4)] so

$$\exists s_1 \in \alpha \text{ such that } 0 < s_1 \quad (9.49)$$

As $0 < x < 1$ we have $0 < 1 - x$, and by [theorems: 8.29, 4.73] $0 < x^{-1}$ so that by applying [theorems: 8.29, 4.73] repeatedly we have that $0 < s_1 \cdot (1 - x) \cdot x^{-1}$ or

$$\text{If } \varepsilon = s_1 \cdot (1 - x) \cdot x^{-1} \text{ then } 0 < \varepsilon \quad (9.50)$$

We have by [theorem: 9.13] that there exist a $s_2 \in \alpha$ such that $s_2 + \varepsilon \in \mathbb{Q} \setminus \alpha$. As α has no maximal element and $s_2 \in \alpha$ there exist a $s_3 \in \alpha$ such that $s_2 < s_3$ then $s_2 + \varepsilon < s_3 + \varepsilon$. If $s_3 + \varepsilon \in \alpha$ then by [definition: 9.1 (3)] we have as $s_2 + \varepsilon \in \mathbb{Q} \setminus \alpha$ that $s_3 + \varepsilon < s_2 + s_3$ contradicting $s_2 + \varepsilon < s_3 + \varepsilon$ hence we must have that

$$s_3 + \varepsilon \in \mathbb{Q} \setminus \alpha \text{ and } s_2 + \varepsilon < s_3 + \varepsilon \wedge s_2 + \varepsilon \in \mathbb{Q} \setminus \alpha \quad (9.51)$$

For s_1, s_2 we have either:

$s_3 < s_1$. Then $s_3 + \varepsilon < s_1 + \varepsilon$. If $s_1 + \varepsilon \in \alpha$ then as $s_3 + \varepsilon \in \mathbb{Q} \setminus \alpha$ we have by [definition: 9.1 (3)] that $s_1 + \varepsilon < s_3 + \varepsilon$ contradicting $s_3 + \varepsilon < s_1 + \varepsilon$ so we must have that

$$s_1 + \varepsilon \in \mathbb{Q} \setminus \alpha \quad (9.52)$$

As $0 \in \alpha$ and $x_1 + \varepsilon \in \mathbb{Q} \setminus \alpha$ we have by [definition: 9.1 (3)] that

$$0 < s_1 + \varepsilon \text{ and by [theorems: 8.29, 4.73]} 0 < (s_1 + \varepsilon)^{-1} \quad (9.53)$$

By [eq: 9.52], [eq: 9.53] and the fact that $s_3 + \varepsilon < s_1 + \varepsilon$, $s_3 + \varepsilon \in \mathbb{Q} \setminus \alpha$ we have by the definition of $\text{inv}(\alpha)$ we have that

$$(s_1 + \varepsilon)^{-1} \in \text{inv}(\alpha)$$

As $0 < s_1 \in \alpha$, $0 < (s_1 + \varepsilon)^{-1} \in \text{inv}(\alpha)$ [see eqs: 9.49, 9.52, 9.53] it follows from the definition of $\odot$ that

$$s_1 \cdot (s_1 + \varepsilon)^{-1} \in \alpha \odot \text{inv}(\alpha) \quad (9.54)$$

Now

$$\begin{aligned} s_1 \cdot (s_1 + \varepsilon)^{-1} & \stackrel{[\text{eq: 9.50}]}{=} s_1 \cdot (s_1 + s_1 \cdot (1 - x) \cdot x^{-1})^{-1} \\ & = s_1 \cdot (s_1 + s_1 \cdot x^{-1} - s_1 \cdot x^{-1} \cdot x)^{-1} \\ & = s_1 \cdot (s_1 + s_1 \cdot x^{-1} - s_1)^{-1} \\ & = s_1 \cdot (s_1 \cdot x^{-1})^{-1} \\ & \stackrel{[\text{theorem: 4.55}]}{=} s_1 \cdot (s_1^{-1} \cdot (x^{-1})^{-1}) \\ & = (x^{-1})^{-1} \\ & = x \end{aligned}$$

proving using [eq: 9.54] that

$$x \in \alpha \odot \text{inv}(\alpha)$$

$s_1 \leq s_3$. Then as $0 \in \alpha$ and $s_3 + \varepsilon \in \mathbb{Q} \setminus \alpha$ [see eq: 9.51] it follows from [definition: 9.1] that

$$0 < s_3 + \varepsilon \text{ and by [theorems: 8.29, 4.73]} 0 < (s_3 + \varepsilon)^{-1} \quad (9.55)$$

So using the definition of $\text{inv}(\alpha)$ together with [eqs: 9.51, 9.55] that

$$(s_3 + \varepsilon)^{-1} \in \text{inv}(\alpha) \quad (9.56)$$

Now by [eq: 9.49] $0 < s_1 \leq s_3 \in \alpha$ and $0 < (s_3 + \varepsilon)^{-1} \in \text{inv}(\alpha)$ [see eq: 9.55, 9.51] so that

$$s_3 \cdot (s_3 + \varepsilon)^{-1} \in \alpha \odot \text{inv}(\alpha) \quad (9.57)$$

Now

$$\begin{aligned} s_1 \leq s_3 & \stackrel{0 < 1-x}{\Rightarrow} s_1 \cdot (1-x) \leq s_3 \cdot (1-x) \\ & \stackrel{0 < x^{-1}}{\Rightarrow} s_1 \cdot (1-x) \cdot x^{-1} \leq s_3 \cdot (1-x) \cdot x^{-1} \\ & \Rightarrow s_3 + s_1 \cdot (1-x) \cdot x^{-1} \leq s_3 + s_3 \cdot (1-x) \cdot x^{-1} \\ & \stackrel{[\text{theorems: 8.29, 4.73}]}{\Rightarrow} (s_3 + s_1 \cdot (1-x) \cdot x^{-1})^{-1} \geq (s_3 + s_3 \cdot (1-x) \cdot x^{-1})^{-1} \\ & \stackrel{[\text{eq: 9.50}]}{\Rightarrow} (s_3 + \varepsilon)^{-1} \geq (s_3 + s_3 \cdot (1-x) \cdot x^{-1})^{-1} \\ & \stackrel{0 < s_1 \leq s_3}{\Rightarrow} s_3 \cdot (s_3 + \varepsilon)^{-1} \geq s_3 \cdot (s_3 + s_3 \cdot (1-x) \cdot x^{-1})^{-1} \end{aligned} \quad (9.58)$$

Further

$$\begin{aligned} s_3 \cdot (s_3 + \varepsilon)^{-1} & \stackrel{[\text{eq: 9.58}]}{\geq} s_3 \cdot (s_3 + s_3 \cdot (1-x) \cdot x^{-1})^{-1} \\ & = s_3 \cdot (s_3 + s_3 \cdot x^{-1} - s_3 \cdot x \cdot x^{-1})^{-1} \\ & = s_3 \cdot (s_3 + s_3 \cdot x^{-1} - s_3)^{-1} \\ & = s_3 \cdot (s_3 \cdot x^{-1})^{-1} \\ & \stackrel{[\text{theorem: 4.55}]}{=} s_3 \cdot s_3^{-1} \cdot (x^{-1})^{-1} \\ & = x \end{aligned}$$

which by [theorem: 9.3] and [eq: 9.57] proves that

$$x \in \alpha \odot \text{inv}(\alpha)$$

As in all cases $x \in \alpha \odot \text{in}(\alpha)$ it follows that $\alpha_1 \subseteq \alpha \odot \text{inv}(\alpha)$ which combined with [eq: 9.48] proves finally that

$$\alpha_1 = \alpha \odot \text{inv}(\alpha) \quad \square$$

We prove now that $\mathbb{R} \times \mathbb{R}$ is the disjoint union of sets of the form $A \times B$ where $A, B \in \{\mathbb{R}^+, \mathbb{R}^-, \{0\}\}$

Theorem 9.27. $\mathbb{R} \times \mathbb{R}$ can be expressed as follows

$$\mathbb{R} \times \mathbb{R} = (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

where

$$\begin{aligned} (\mathbb{R}^+ \times \mathbb{R}^+) \cap (\mathbb{R}^+ \times \mathbb{R}^-) &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^+) \cap (\mathbb{R}^- \times \mathbb{R}^+) &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^+) \cap (\mathbb{R}^- \times \mathbb{R}^-) &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^+) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^-) \cap (\mathbb{R}^- \times \mathbb{R}^+) &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^-) \cap (\mathbb{R}^- \times \mathbb{R}^-) &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^-) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) &= \emptyset \\ (\mathbb{R}^- \times \mathbb{R}^+) \cap (\mathbb{R}^- \times \mathbb{R}^-) &= \emptyset \\ (\mathbb{R}^- \times \mathbb{R}^+) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) &= \emptyset \\ (\mathbb{R}^- \times \mathbb{R}^-) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) &= \emptyset \end{aligned}$$

Proof. First note that by [theorem: 9.17]

$$\mathbb{R} = \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\} = \bigcup_{A \in \{\mathbb{R}^+, \mathbb{R}^-, \{0\}\}} A \quad (9.59)$$

and

$$\mathbb{R}^+ \cap \mathbb{R}^- = \emptyset \text{ and } \mathbb{R}^+ \cap \mathbb{R}^+ \cap \{0\} = \emptyset \text{ and } \mathbb{R}^- \cap \{0\} = \emptyset \quad (9.60)$$

First as $\mathbb{R}^+ \subseteq \mathbb{R}$, $\mathbb{R}^- \subseteq \mathbb{R}$, $\{0\} \subseteq \mathbb{R}$ and $\mathbb{R} \subseteq \mathbb{R}$ we have by [theorem: 1.55] that $\mathbb{R}^+ \times \mathbb{R}^+ \subseteq \mathbb{R} \times \mathbb{R}$, $\mathbb{R}^+ \times \mathbb{R}^- \subseteq \mathbb{R} \times \mathbb{R}$, $\mathbb{R}^- \times \mathbb{R}^+ \subseteq \mathbb{R} \times \mathbb{R}$, $\mathbb{R}^- \times \mathbb{R}^- \subseteq \mathbb{R} \times \mathbb{R}$, $((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) \subseteq \mathbb{R} \times \mathbb{R}$ so that

$$(\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) \subseteq \mathbb{R} \times \mathbb{R} \quad (9.61)$$

Let $(x, y) \in \mathbb{R} \times \mathbb{R}$ then $x \in \mathbb{R}_{[\text{eq: 9.59}]} = \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ and $y \in \mathbb{R}_{[\text{eq: 9.59}]} = \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ so that for x we have either:

$x \in \mathbb{R}^+$. Then for y we have either:

$y \in \mathbb{R}^+$. Then $(x, y) \in \mathbb{R}^+ \times \mathbb{R}^+$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

$y \in \mathbb{R}^-$. Then $(x, y) \in \mathbb{R}^+ \times \mathbb{R}^-$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

$y \in \{0\}$. Then $(x, y) \in \mathbb{R} \times \{0\}$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

$x \in \mathbb{R}^-$. Then for y we have either:

$y \in \mathbb{R}^+$. Then $(x, y) \in \mathbb{R}^- \times \mathbb{R}^+$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

$y \in \mathbb{R}^-$. Then $(x, y) \in \mathbb{R}^- \times \mathbb{R}^-$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

$y \in \{0\}$. Then $(x, y) \in \mathbb{R} \times \{0\}$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

$x \in \{0\}$. Then $(x, y) \in \{0\} \times \mathbb{R}$ so that

$$(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$$

So $(x, y) \in (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$ proving that $\mathbb{R} \times \mathbb{R} \subseteq (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}))$ which combined with [eq: 9.61] gives

$$\mathbb{R} \times \mathbb{R} = (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^+ \times \mathbb{R}^-) \cup (\mathbb{R}^- \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-) \cup ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) \quad (9.62)$$

Next we have by [theorem: 1.56] and [theorem: 1.54] that

$$\begin{aligned} (\mathbb{R}^+ \times \mathbb{R}^+) \cap (\mathbb{R}^+ \times \mathbb{R}^-) &= (\mathbb{R}^+ \cap \mathbb{R}^+) \times (\mathbb{R}^+ \cap \mathbb{R}^-) \\ &\stackrel{[\text{eq: 9.60}]}{=} (\mathbb{R}^+ \cap \mathbb{R}^+) \times \emptyset \\ &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^+) \cap (\mathbb{R}^- \times \mathbb{R}^+) &= (\mathbb{R}^+ \cap \mathbb{R}^-) \times (\mathbb{R}^+ \cap \mathbb{R}^+) \\ &\stackrel{[\text{eq: 9.60}]}{=} \emptyset \times (\mathbb{R}^+ \cap \mathbb{R}^+) \\ &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^+) \cap (\mathbb{R}^- \times \mathbb{R}^-) &= (\mathbb{R}^+ \cap \mathbb{R}^-) \times (\mathbb{R}^+ \cap \mathbb{R}^-) \\ &\stackrel{[\text{eq: 9.60}]}{=} \emptyset \times \emptyset \\ &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^+) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) &= ((\mathbb{R}^+ \times \mathbb{R}^-) \cap (\mathbb{R} \times \{0\})) \cup ((\mathbb{R}^+ \times \mathbb{R}^+) \cap (\{0\} \times \mathbb{R})) \\ &= ((\mathbb{R}^+ \cap \mathbb{R}) \times (\mathbb{R}^- \cap \{0\})) \cup ((\mathbb{R}^+ \cap \{0\}) \times (\mathbb{R}^+ \cap \mathbb{R})) \\ &\stackrel{[\text{eq: 9.60}]}{=} ((\mathbb{R}^+ \cap \mathbb{R}) \times \emptyset) \cup (\emptyset \times (\mathbb{R}^+ \cap \mathbb{R})) \\ &= \emptyset \cup \emptyset \\ &= \emptyset \\ (\mathbb{R}^+ \times \mathbb{R}^-) \cap (\mathbb{R}^- \times \mathbb{R}^+) &= (\mathbb{R}^+ \cap \mathbb{R}^-) \times (\mathbb{R}^- \cap \mathbb{R}^+) \end{aligned}$$

$$\begin{aligned}
& \stackrel{=}{=} \emptyset \times \emptyset \\
& \stackrel{[\text{eq: 9.60}]}{=} \emptyset \\
& (\mathbb{R}^+ \times \mathbb{R}^-) \cap (\mathbb{R}^- \times \mathbb{R}^-) = (\mathbb{R}^+ \cap \mathbb{R}^-) \times (\mathbb{R}^- \cap \mathbb{R}^-) \\
& \stackrel{=}{=} \emptyset \times (\mathbb{R}^+ \cap \mathbb{R}^-) \\
& \stackrel{[\text{eq: 9.60}]}{=} \emptyset \\
& (\mathbb{R}^+ \times \mathbb{R}^-) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) = ((\mathbb{R}^+ \times \mathbb{R}^-) \cap (\mathbb{R} \times \{0\})) \cup ((\mathbb{R}^+ \times \mathbb{R}^-) \cap (\{0\} \times \mathbb{R})) \\
& = ((\mathbb{R}^+ \cap \mathbb{R}) \times (\mathbb{R}^- \cap \{0\})) \cup ((\mathbb{R}^+ \cap \{0\}) \times (\mathbb{R}^- \cap \mathbb{R})) \\
& \stackrel{=}{=} ((\mathbb{R}^+ \cap \mathbb{R}) \times \emptyset) \cup (\emptyset \times (\mathbb{R}^- \cap \mathbb{R})) \\
& \stackrel{[\text{eq: 9.60}]}{=} \emptyset \cup \emptyset \\
& \stackrel{=}{=} \emptyset \\
& (\mathbb{R}^- \times \mathbb{R}^+) \cap (\mathbb{R}^- \times \mathbb{R}^-) = (\mathbb{R}^- \cap \mathbb{R}^-) \times (\mathbb{R}^+ \cap \mathbb{R}^-) \\
& = (\mathbb{R}^- \cap \mathbb{R}^-) \times \emptyset \\
& = \emptyset \\
& (\mathbb{R}^- \times \mathbb{R}^+) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) = ((\mathbb{R}^- \times \mathbb{R}^+) \cap (\mathbb{R} \times \{0\})) \cup ((\mathbb{R}^- \times \mathbb{R}^+) \cap (\{0\} \times \mathbb{R})) \\
& = ((\mathbb{R}^- \cap \mathbb{R}) \times (\mathbb{R}^+ \cap \{0\})) \cup ((\mathbb{R}^- \cap \{0\}) \times (\mathbb{R}^+ \cap \mathbb{R})) \\
& \stackrel{=}{=} ((\mathbb{R}^- \cap \mathbb{R}) \times \emptyset) \cup (\emptyset \times (\mathbb{R}^+ \cap \mathbb{R})) \\
& \stackrel{[\text{eq: 9.60}]}{=} \emptyset \cup \emptyset \\
& \stackrel{=}{=} \emptyset \\
& (\mathbb{R}^- \times \mathbb{R}^-) \cap ((\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})) = ((\mathbb{R}^- \times \mathbb{R}^-) \cap (\mathbb{R} \times \{0\})) \cup ((\mathbb{R}^- \times \mathbb{R}^-) \cap (\{0\} \times \mathbb{R})) \\
& = ((\mathbb{R}^- \cap \mathbb{R}) \times (\mathbb{R}^- \cap \{0\})) \cup ((\mathbb{R}^- \cap \{0\}) \times (\mathbb{R}^- \cap \mathbb{R})) \\
& \stackrel{=}{=} ((\mathbb{R}^- \cap \mathbb{R}) \times \emptyset) \cup (\emptyset \times (\mathbb{R}^- \cap \mathbb{R})) \\
& \stackrel{[\text{eq: 9.60}]}{=} \emptyset \cup \emptyset \\
& \stackrel{=}{=} \emptyset
\end{aligned}$$

□

We use now [theorem: 9.20] and [theorem: 9.27] to define the multiplication operator on $\mathbb{R}$.

Definition 9.28. *The multiplication operator $\cdot : \mathbb{R} \times \mathbb{R} \Rightarrow \mathbb{R}$ is defined as*

$$\alpha \cdot \beta = \begin{cases} \alpha \odot \beta & \text{if } (\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^+ \\ -((- \alpha) \odot \beta) & \text{if } (\alpha, \beta) \in \mathbb{R}^- \times \mathbb{R}^+ \\ -(\alpha \odot (- \beta)) & \text{if } (\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^- \\ (- \alpha) \odot (- \beta) & \text{if } (\alpha, \beta) \in \mathbb{R}^- \times \mathbb{R}^- \\ 0 & \text{if } (\alpha, \beta) \text{ if } (\alpha, \beta) \in (\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}) \end{cases}$$

If we want to prove something about multiplication then we have 5 cases to consider for the definition of the multiplication operator. The following lemma allows to reduce the amount work.

Lemma 9.29. $\forall \alpha, \beta \in \mathbb{R} \times \mathbb{R}$ we have $-(\alpha \cdot \beta) = (-\alpha) \cdot \beta = \alpha \cdot (-\beta)$

Proof. We have to consider the following exclusive 5 cases [see theorem: 9.27]:

$(\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^+$. Then

$$\begin{aligned}
 (-\alpha) \cdot \beta & \stackrel{\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+}{=} -((-(-\alpha)) \odot \beta) \\
 & \stackrel{-(-\alpha) = \alpha}{=} -(\alpha \odot \beta) \\
 & \stackrel{\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+}{=} -(\alpha \cdot \beta) \\
 \alpha \cdot (-\beta) & \stackrel{\alpha \in \mathbb{R}^+ \wedge -\beta \in \mathbb{R}^-}{=} -(\alpha \odot (-(-\beta))) \\
 & \stackrel{=}{=} -(\alpha \odot \beta) \\
 & \stackrel{\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+}{=} -(\alpha \cdot \beta)
 \end{aligned}$$

$(\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^-$. Then

$$\begin{aligned}
 -(\alpha \cdot \beta) & \stackrel{\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^-}{=} -(-(\alpha \odot (-\beta))) \\
 & \stackrel{=}{=} \alpha \odot (-\beta) \\
 & \stackrel{\alpha \in \mathbb{R}^+ \wedge -\beta \in \mathbb{R}^-}{=} \alpha \cdot (-\beta) \\
 (-\alpha) \cdot \beta & \stackrel{-\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^-}{=} -((-(-\alpha)) \odot (-\beta)) \\
 & \stackrel{=}{=} \alpha \odot (-\beta) \\
 & \stackrel{\alpha \in \mathbb{R}^+ \wedge -\beta \in \mathbb{R}^+}{=} \alpha \cdot (-\beta) \\
 & \stackrel{[\text{eq: 9.63}]}{=} -(\alpha \cdot \beta)
 \end{aligned} \tag{9.63}$$

$(\alpha, \beta) \in \mathbb{R}^- \times \mathbb{R}^+$. Then

$$\begin{aligned}
 -(\alpha \cdot \beta) & \stackrel{\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+}{=} -(-((-(-\alpha)) \odot \beta)) \\
 & \stackrel{=}{=} (-\alpha) \odot \beta \\
 & \stackrel{-\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+}{=} (-\alpha) \cdot \beta \\
 \alpha \cdot (-\beta) & \stackrel{\alpha \in \mathbb{R}^- \wedge -\beta \in \mathbb{R}^-}{=} (-\alpha) \odot (-(-\beta)) \\
 & \stackrel{=}{=} (-\alpha) \odot \beta \\
 & \stackrel{-\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+}{=} (-\alpha) \cdot \beta \\
 & \stackrel{[\text{eq: 9.64}]}{=} -(\alpha \cdot \beta)
 \end{aligned} \tag{9.64}$$

$(\alpha, \beta) \in \mathbb{R}^- \times \mathbb{R}^-$. Then

$$\begin{aligned}
 -(\alpha \cdot \beta) & \stackrel{\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^-}{=} -((-(-\alpha)) \odot (-\beta)) \\
 (-\alpha) \cdot \beta & \stackrel{-\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^-}{=} -((-(-\alpha)) \odot (-\beta)) \\
 & \stackrel{[\text{eq: 9.65}]}{=} -(\alpha \cdot \beta) \\
 \alpha \cdot (-\beta) & \stackrel{\alpha \in \mathbb{R}^- \wedge -\beta \in \mathbb{R}^+}{=} -((-(-\alpha)) \odot (-\beta)) \\
 & \stackrel{[\text{eq: 9.65}]}{=} -(\alpha \cdot \beta)
 \end{aligned} \tag{9.65}$$

$(\alpha, \beta) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})$. Then

$$\begin{aligned}
 -(\alpha \cdot \beta) & \stackrel{(\alpha, \beta) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})}{=} 0 \\
 & \stackrel{(-\alpha, \beta) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})}{=} (-\alpha) \cdot \beta \\
 -(\alpha \cdot \beta) & \stackrel{(\alpha, \beta) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})}{=} 0 \\
 & \stackrel{(\alpha, -\beta) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})}{=} \alpha \cdot (-\beta)
 \end{aligned}$$

□

Lemma 9.30. *If $\alpha, \beta \in \mathbb{R}^+$ and $\gamma \in \mathbb{R}^-$ then $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$.*

Proof. First we proof that

$$\forall \alpha, \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^- \text{ such that } \beta + \gamma \in \mathbb{R}^+ \text{ we have } \alpha \cdot (\beta + \gamma) \quad (9.66)$$

Proof. As $\beta, -\gamma \in \mathbb{R}^+$ we have $0 \in \beta \wedge 0 \in -\gamma$ so that $0 = 0 + 0 \in \beta + (-\gamma)$ proving that

$$\beta + (-\gamma) \in \mathbb{R}^+ \wedge \beta + \gamma \in \mathbb{R}^+ \quad (9.67)$$

Now

$$\begin{aligned} \alpha \cdot \beta + \alpha \cdot \beta & \stackrel{\alpha, \beta \in \mathbb{R}^+}{=} \alpha \odot \beta + \alpha \odot \beta \\ & \stackrel{\alpha, \beta \in \mathbb{R}^+ \text{ and [theorem: 9.24]}}{=} \alpha \odot (\beta + \beta) \\ & = \alpha \odot ((\beta + (-\gamma)) + (\beta + \gamma)) \\ & \stackrel{[\text{eq: 9.67} + \text{theorem: 9.24}]}{=} (\alpha \odot (\beta + (-\gamma))) + \alpha \odot (\beta + \gamma) \\ & \stackrel{[\beta, -\gamma \in \mathbb{R}^+ + \text{theorem: 9.24}]}{=} \alpha \odot \beta + \alpha \odot (-\gamma) + \alpha \odot (\beta + \gamma) \\ & = \alpha \odot \beta + (-(-(\alpha \odot (-\gamma)))) + \alpha \odot (\beta + \gamma) \\ & \stackrel{\alpha, \beta \in \mathbb{R}^+, \gamma \in \mathbb{R}^-, \beta + \gamma \in \mathbb{R}^+}{=} \alpha \cdot \beta + (-(\alpha \cdot \gamma)) + \alpha \cdot (\beta + \gamma) \end{aligned}$$

so after adding $-(\alpha \cdot \beta) + \alpha \cdot \gamma$ to both sides gives

$$\begin{aligned} \alpha \cdot (\beta + \gamma) &= \alpha \cdot \beta + \alpha \cdot \beta + (-(\alpha \cdot \beta)) + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma \end{aligned} \quad \square$$

For $\beta + \gamma \in \mathbb{R}$ we have three cases to consider:

$$\beta + \gamma \in \mathbb{R}^+. \text{ Then } \alpha \cdot (\beta + \gamma) \stackrel{[\text{eq: 9.66}]}{=} \alpha \cdot \beta + \alpha \cdot \gamma$$

$\beta + \gamma \in \mathbb{R}^-$. Then $(-\beta) + (-\gamma) \stackrel{[\text{theorem: 4.8}]}{=} -(\beta + \gamma) \in \mathbb{R}^+$. So if we take $\gamma' = -\beta \in \mathbb{R}^-$ and $\beta' = -\gamma \in \mathbb{R}^+$ we have that $\beta' + \gamma' = -(\beta + \gamma) \in \mathbb{R}^+$, so we can apply [eq: 9.66] resulting in

$$\alpha \cdot (\beta' + \gamma') = \alpha \cdot \beta' + \alpha \cdot \gamma'$$

which after substituting the formulas for β', γ' gives

$$\alpha \cdot ((-\gamma) + (-\beta)) = \alpha \cdot (-\gamma) + \alpha \cdot (-\beta) \quad (9.68)$$

Now we have

$$\begin{aligned} \alpha \cdot (\beta + \gamma) &= \alpha \cdot (-(-(\beta + \gamma))) \\ & \stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot (-(\beta + \gamma))) \\ & = -(\alpha \cdot ((-\gamma) + (-\beta))) \\ & \stackrel{[\text{eq: 9.68}]}{=} -(\alpha \cdot (-\gamma) + \alpha \cdot (-\beta)) \\ & = -(\alpha \cdot (-\beta) + \alpha \cdot (-\gamma)) \\ & \stackrel{[\text{theorem: 4.8}]}{=} (-(\alpha \cdot (-\beta))) + (-(\alpha \cdot (-\gamma))) \\ & \stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot (-(-\beta)) + \alpha \cdot (-(-\gamma)) \\ & = \alpha \cdot \beta + \alpha \cdot \gamma \end{aligned}$$

$\beta + \gamma = 0$. Then $\gamma = -\beta$ and we have

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= \alpha \cdot 0 \\
 &= 0 \\
 &= \alpha \cdot \beta + (-(\alpha \cdot \beta)) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot \beta + \alpha \cdot (-\beta) \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma \\
 &\quad \square
 \end{aligned}$$

We are finally ready to prove that $\langle \mathbb{R}, +, \cdot \rangle$ is a field

Definition 9.31. If $\alpha \in \mathbb{R} \setminus \{0\}$ then we define α^{-1} by $\alpha^{-1} = \begin{cases} \text{inv}(\alpha) & \text{if } \alpha \in \mathbb{R}^+ \\ -\text{inv}(-\alpha) & \text{if } \alpha \in \mathbb{R}^- \end{cases}$

Proof. As by [theorem: 9.17] $\mathbb{R} \setminus \{0\} = \mathbb{R}^+ \cup \mathbb{R}^-$ and $\mathbb{R}^+ \cap \mathbb{R}^- = \emptyset$ α^{-1} is well defined. $\square$

Theorem 9.32. $\langle \mathbb{R}, +, \cdot \rangle$ is a field where

1. $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined in [theorem: 9.15]
2. $\cdot: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined in [definition: 9.28]
3. $0 = \alpha_0$ is the additive neutral element [see theorem: 9.15]
4. $1 = \alpha_1$ is the multiplicative neutral element.
5. $\forall \alpha \in \mathbb{R}$ the additive inverse is the negative cut of α [see theorem: 9.15]
6. $\forall \alpha \in \mathbb{R} \setminus \{0\}$ we have the multiplicative inverse is defined by [definition: 9.31]

Proof.

1. Using [theorem: 9.15] $\langle \mathbb{R}, + \rangle$ is a Abelian group with neutral element $0 = \alpha_0$ and $\forall \alpha \in \mathbb{R}$ the negative cut $-\alpha$ as inverse.
2. For the multiplication operator $\cdot: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ we have:

commutativity. Let $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ then using [theorem: 9.27] we have to consider the following cases:

$(\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^+$. Then

$$\begin{aligned}
 \alpha \cdot \beta &\stackrel{\alpha, \beta \in \mathbb{R}^+}{=} \alpha \odot \beta \\
 &\stackrel{[\text{theorem: 9.22}]}{=} \beta \odot \alpha \\
 &= \beta \cdot \alpha
 \end{aligned}$$

$(\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^-$. Then

$$\begin{aligned}
 \alpha \cdot \beta &\stackrel{\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^-}{=} -(\alpha \odot (-\beta)) \\
 &\stackrel{[\text{theorem: 9.22}]}{=} -((- \beta) \odot \alpha) \\
 &\stackrel{\alpha \in \mathbb{R}^+ \wedge -\beta \in \mathbb{R}^+}{=} -((- \beta) \cdot \alpha) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-(\beta \cdot \alpha)) \\
 &= \beta \cdot \alpha
 \end{aligned}$$

$(\alpha, \beta) \in \mathbb{R}^- \times \mathbb{R}^+$. Then

$$\begin{aligned}
 \alpha \cdot \beta & \stackrel{\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+}{=} -((- \alpha) \odot \beta) \\
 & \stackrel{[\text{theorem: 9.22}]}{=} -(\beta \odot (- \alpha)) \\
 & \stackrel{- \alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+}{=} -(\beta \cdot (- \alpha)) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} -(-(\beta \cdot \alpha)) \\
 & = \beta \cdot \alpha
 \end{aligned}$$

$(\alpha, \beta) \in \mathbb{R}^- \times \mathbb{R}^-$. Then

$$\begin{aligned}
 \alpha \cdot \beta & \stackrel{\alpha, \beta \in \mathbb{R}^-}{=} (- \alpha) \odot (- \beta) \\
 & \stackrel{[\text{theorem: 9.22}]}{=} (- \beta) \odot (- \alpha) \\
 & \stackrel{\alpha, \beta \in \mathbb{R}^-}{=} \beta \cdot \alpha
 \end{aligned}$$

$(\alpha, \beta) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})$. Then

$$\begin{aligned}
 \alpha \cdot \beta & = 0 \\
 & = \beta \cdot \alpha
 \end{aligned}$$

neutral element. First note that as $0 \in \alpha_1$ we have $\alpha_1 \in \mathbb{R}^+$. Let $\alpha \in \mathbb{R} \stackrel{[\text{theorem: 9.17}]}{=} \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ then we have either:

$\alpha \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot \alpha_1 & \stackrel{\text{commutativity}}{=} \alpha_1 \cdot \alpha \\
 & \stackrel{\alpha_1, \alpha \in \mathbb{R}^+}{=} \alpha_1 \odot \alpha \\
 & \stackrel{[\text{theorem: 9.21}]}{=} \alpha
 \end{aligned}$$

$\alpha \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot \alpha_1 & \stackrel{\text{commutativity}}{=} \alpha_1 \cdot \alpha \\
 & \stackrel{\alpha_1 \in \mathbb{R}^+ \wedge \alpha \in \mathbb{R}^-}{=} -(\alpha_1 \odot (- \alpha)) \\
 & \stackrel{[\text{theorem: 9.21}]}{=} -(- \alpha) \\
 & = \alpha
 \end{aligned}$$

$\alpha = 0$. Then we have

$$\begin{aligned}
 \alpha \cdot \alpha_1 & = \alpha_1 \cdot \alpha \\
 & = 0 \\
 & = \alpha
 \end{aligned}$$

inverse element. Let $\alpha \in \mathbb{R} \setminus \{0\}$ then by [theorem: 9.17] we have to consider:

$\alpha \in \mathbb{R}^+$. Then $\text{inv}(\alpha) \in \mathbb{R}^+$ [see theorem: 9.25] and

$$\begin{aligned}
 \alpha^{-1} \cdot \alpha & \stackrel{\text{commutativity}}{=} \alpha \cdot \alpha^{-1} \\
 & \stackrel{\alpha \in \mathbb{R}^+}{=} \alpha \cdot \text{inv}(\alpha) \\
 & \stackrel{\text{inv}(\alpha) \in \mathbb{R}^+ \wedge \alpha \in \mathbb{R}^+}{=} \alpha \odot \text{inv}(\alpha) \\
 & \stackrel{[\text{theorem: 9.26}]}{=} \alpha_1
 \end{aligned}$$

$\alpha \in \mathbb{R}^-$. Then $-\alpha \in \mathbb{R}^+$ and by [theorem: 9.17] $\text{inv}(-\alpha) \in \mathbb{R}^+$, further

$$\begin{aligned}
 \alpha^{-1} \cdot \alpha & \stackrel{\text{commutativity}}{=} \alpha \cdot \alpha^{-1} \\
 & \stackrel{\alpha \in \mathbb{R}^-}{=} \alpha \cdot (-(\text{inv}(-\alpha))) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot \text{inv}(-\alpha)) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} (-\alpha) \cdot \text{inv}(-\alpha) \\
 & \stackrel{[\text{theorem: 9.26}]}{=} \alpha_1
 \end{aligned}$$

associativity. As $\mathbb{R} \stackrel{[\text{theorem: 9.17}]}{=} \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ we have for $\alpha, \beta, \gamma \in \mathbb{R}$ the following 27 cases to consider:

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) & = \alpha \odot (\beta \odot \gamma) \\
 & \stackrel{[\text{theorem: 9.23}]}{=} (\alpha \odot \beta) \odot \gamma \\
 & = (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) & = \alpha \cdot (-(-(\beta \cdot \gamma))) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot (-(\beta \cdot (-\gamma))) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot (\beta \cdot (-\gamma))) \\
 & = -(\alpha \odot (\beta \odot (-\gamma))) \\
 & \stackrel{[\text{theorem: 9.23}]}{=} -((\alpha \odot \beta) \odot (-\gamma)) \\
 & = -((\alpha \cdot \beta) \cdot (-\gamma)) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} (\alpha \cdot \beta) \cdot (-(-\gamma)) \\
 & = (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+ \wedge \gamma = 0$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) & = \alpha \cdot 0 \\
 & = 0 \\
 & = (\alpha \cdot \beta) \cdot 0 \\
 & = (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) & = \alpha \cdot ((-(-\beta)) \cdot \gamma) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot (-((- \beta) \cdot \gamma)) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot ((- \beta) \cdot \gamma)) \\
 & = -(\alpha \odot ((- \beta) \odot \gamma)) \\
 & \stackrel{[\text{theorem: 9.23}]}{=} -((\alpha \odot (-\beta)) \odot \gamma) \\
 & = -((\alpha \cdot (-\beta)) \cdot \gamma) \\
 & \stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot (-\beta)) \cdot \gamma \\
 & \stackrel{[\text{theorem: 9.29}]}{=} (\alpha \cdot (-(-\beta))) \cdot \gamma \\
 & = (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot ((-(-\beta)) \cdot (-(-\gamma))) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot (-((- \beta) \cdot (-(-\gamma)))) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot (-(-((- \beta) \cdot (-\gamma)))) \\
 &= \alpha \cdot ((-\beta) \cdot (-\gamma)) \\
 &= \alpha \odot ((-\beta) \odot (-\gamma)) \\
 &\stackrel{[\text{theorem: 9.23}]}{=} (\alpha \odot (-\beta)) \odot (-\gamma) \\
 &= (\alpha \cdot (-\beta)) \cdot (-\gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((\alpha \cdot (-\beta)) \cdot \gamma) \\
 &= -((- (\alpha \cdot \beta)) \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-((\alpha \cdot \beta) \cdot \gamma)) \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^- \wedge \gamma = \mathbf{0}$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\beta \cdot \mathbf{0}) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= (\alpha \cdot \beta) \cdot \mathbf{0} \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta = \mathbf{0} \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\mathbf{0} \cdot \gamma) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\alpha \cdot \mathbf{0}) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta = \mathbf{0} \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\mathbf{0} \cdot \gamma) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\alpha \cdot \mathbf{0}) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta = \mathbf{0} \wedge \gamma = \mathbf{0}$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\mathbf{0} \cdot \gamma) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\alpha \cdot \mathbf{0}) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= (-(-\alpha)) \cdot (\beta \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (\beta \cdot \gamma)) \\
 &= -((- \alpha) \odot (\beta \odot \gamma)) \\
 &\stackrel{[\text{theorem: 9.23}]}{=} -(((- \alpha) \odot \beta) \odot \gamma) \\
 &= -(((- \alpha) \cdot \beta) \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- (\alpha \cdot \beta)) \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-((\alpha \cdot \beta) \cdot \gamma)) \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= (-(-\alpha)) \cdot (\beta \cdot (-(-\gamma))) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (\beta \cdot (-(-\gamma)))) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (-(\beta \cdot (-\gamma)))) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-((- \alpha) \cdot (\beta \cdot (-\gamma)))) \\
 &= (-\alpha) \cdot (\beta \cdot (-\gamma)) \\
 &= (-\alpha) \odot (\beta \odot (-\gamma)) \\
 &\stackrel{[\text{theorem: 9.23}]}{=} ((-\alpha) \odot \beta) \odot (-\gamma) \\
 &= ((-\alpha) \cdot \beta) \cdot (-\gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(((- \alpha) \cdot \beta) \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- (\alpha \cdot \beta)) \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-((\alpha \cdot \beta) \cdot \gamma)) \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+ \wedge \gamma = \mathbf{0}$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\beta \cdot \mathbf{0}) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= (\alpha \cdot \beta) \cdot \mathbf{0} \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= (-(-\alpha)) \cdot ((-(-\beta)) \cdot \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot ((-(-\beta)) \cdot \gamma)) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (-((- \beta) \cdot \gamma))) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-((- \alpha) \cdot ((- \beta) \cdot \gamma))) \\
 &= (-\alpha) \cdot ((- \beta) \cdot \gamma) \\
 &= (-\alpha) \odot ((- \beta) \odot \gamma) \\
 &\stackrel{[\text{theorem: 9.23}]}{=} ((-\alpha) \odot (-\beta)) \odot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot ((-\beta) \odot (-\gamma)) \\
 &= -((- \alpha) \odot ((-\beta) \odot (-\gamma))) \\
 &\stackrel{[\text{theorem: 9.23}]}{=} -(((- \alpha) \odot (-\beta)) \odot (-\gamma)) \\
 &= -((\alpha \cdot \beta) \cdot (-\gamma)) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-((\alpha \cdot \beta) \cdot \gamma)) \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^- \wedge \gamma = \mathbf{0}$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\beta \cdot \mathbf{0}) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= (\alpha \cdot \beta) \cdot \mathbf{0} \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta = \mathbf{0} \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\mathbf{0} \cdot \gamma) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\alpha \cdot \mathbf{0}) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta = \mathbf{0} \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\mathbf{0} \cdot \gamma) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\alpha \cdot \mathbf{0}) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta = \mathbf{0} \wedge \gamma = \mathbf{0}$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \alpha \cdot (\mathbf{0} \cdot \gamma) \\
 &= \alpha \cdot \mathbf{0} \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\alpha \cdot \mathbf{0}) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha = \mathbf{0} \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}
 \alpha \cdot (\beta \cdot \gamma) &= \mathbf{0} \cdot (\beta \cdot \gamma) \\
 &= \mathbf{0} \\
 &= \mathbf{0} \cdot \gamma \\
 &= (\mathbf{0} \cdot \beta) \cdot \gamma \\
 &= (\alpha \cdot \beta) \cdot \gamma
 \end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^+ \wedge \gamma = 0$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^- \wedge \gamma = 0$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta = 0 \wedge \gamma \in \mathbb{R}^+$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta = 0 \wedge \gamma \in \mathbb{R}^-$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta = 0 \wedge \gamma = 0$. Then we have

$$\begin{aligned}\alpha \cdot (\beta \cdot \gamma) &= 0 \cdot (\beta \cdot \gamma) \\ &= 0 \\ &= 0 \cdot \gamma \\ &= (0 \cdot \beta) \cdot \gamma \\ &= (\alpha \cdot \beta) \cdot \gamma\end{aligned}$$

distributivity. As $\mathbb{R} \stackrel{[\text{theorem: 9.17}]}{=} \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$ we have for $\alpha, \beta, \gamma \in \mathbb{R}$ the following 27 cases to consider:

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \odot (\beta + \gamma) \\ &\stackrel{[\text{theorem: 9.24}]}{=} \alpha \odot \beta + \alpha \odot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^-$. Then

$$\alpha \cdot (\beta + \gamma) \stackrel{[\text{lemma: 9.30}]}{=} \alpha \cdot \beta + \alpha \cdot \gamma$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^+ \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (\beta + 0) \\ &= \alpha \cdot \beta \\ &= \alpha \cdot \beta + 0 \\ &= \alpha \cdot \beta + \alpha \cdot 0 \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (\gamma + \beta) \\ &\stackrel{[\text{lemma: 9.30}]}{=} \alpha \cdot \gamma + \alpha \cdot \beta \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (-(-(\beta + \gamma))) \\ &\stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot (-(\beta + \gamma))) \\ &\stackrel{[\text{theorem: 4.8}]}{=} -(\alpha \cdot ((-\beta) + (-\gamma))) \\ &= -(\alpha \odot ((-\beta) + (-\gamma))) \\ &\stackrel{[\text{theorem: 9.24}]}{=} -(\alpha \odot (-\beta) + \alpha \odot (-\gamma)) \\ &= -(\alpha \cdot (-\beta) + \alpha \cdot (-\gamma)) \\ &\stackrel{[\text{theorem: 4.8}]}{=} (-(\alpha \cdot (-\beta))) + (-(\alpha \cdot (-\gamma))) \\ &\stackrel{[\text{theorem: 9.29}]}{=} \alpha \cdot (-(-\beta)) + \alpha \cdot (-(-\gamma)) \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta \in \mathbb{R}^- \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (\beta + 0) \\ &= \alpha \cdot \beta \\ &= \alpha \cdot \beta + 0 \\ &= \alpha \cdot \beta + \alpha \cdot 0 \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta = 0 \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (0 + \gamma) \\ &= \alpha \cdot \gamma \\ &= 0 + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta = 0 \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (0 + \gamma) \\ &= \alpha \cdot \gamma \\ &= 0 + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^+ \wedge \beta = 0 \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (0 + \gamma) \\ &= \alpha \cdot \gamma \\ &= 0 + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= -((- \alpha) \odot (\beta + \gamma)) \\ &\stackrel{[\text{theorem: 9.24}]}{=} -((- \alpha) \odot \beta + (- \alpha) \odot \gamma) \\ &\stackrel{[\text{theorem: 4.8}]}{=} (-((- \alpha) \odot \beta)) + (-((- \alpha) \odot \gamma)) \\ &= (-((- \alpha) \cdot \beta)) + (-((- \alpha) \cdot \gamma)) \\ &\stackrel{[\text{theorem: 9.29}]}{=} (-(- \alpha)) \cdot \beta + (-(- \alpha)) \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= (-(- \alpha)) \cdot (\beta + \gamma) \\ &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (\beta + \gamma)) \\ &\stackrel{[\text{lemma: 9.30}]}{=} -((- \alpha) \cdot \beta + (- \alpha) \cdot \gamma) \\ &\stackrel{[\text{theorem: 4.8}]}{=} (-((- \alpha) \cdot \beta)) + (-((- \alpha) \cdot \gamma)) \\ &\stackrel{[\text{theorem: 9.29}]}{=} (-(- \alpha)) \cdot \beta + (-(- \alpha)) \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^+ \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (\beta + 0) \\ &= \alpha \cdot \beta \\ &= \alpha \cdot \beta + 0 \\ &= \alpha \cdot \beta + \alpha \cdot 0 \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= \alpha \cdot (\gamma + \beta) \\
 &= (-(-\alpha)) \cdot (\gamma + \beta) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (\gamma + \beta)) \\
 &\stackrel{[\text{lemma: 9.30}]}{=} -((- \alpha) \cdot \gamma + (- \alpha) \cdot \beta) \\
 &\stackrel{[\text{theorem: 4.8}]}{=} -((- \alpha) \cdot \gamma) + -((- \alpha) \cdot \beta) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} (-(-\alpha)) \cdot \gamma + (-(-\alpha)) \cdot \beta \\
 &= \alpha \cdot \gamma + \alpha \cdot \beta \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= (-(-\alpha)) \cdot (\beta + \gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -((- \alpha) \cdot (\beta + \gamma)) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} (-\alpha) \cdot (-(\beta + \gamma)) \\
 &\stackrel{[\text{theorem: 4.8}]}{=} (-\alpha) \cdot ((-\beta) + (-\gamma)) \\
 &= (-\alpha) \odot ((-\beta) + (-\gamma)) \\
 &\stackrel{[\text{theorem: 9.24}]}{=} (-\alpha) \odot (-\beta) + (-\alpha) \odot (-\gamma) \\
 &= (-\alpha) \cdot (-\beta) + (-\alpha) \cdot (-\gamma) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(\alpha \cdot (-\beta)) + -(\alpha \cdot (-\gamma)) \\
 &\stackrel{[\text{theorem: 9.29}]}{=} -(-(\alpha \cdot \beta)) + -(-(\alpha \cdot \gamma)) \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta \in \mathbb{R}^- \wedge \gamma = \mathbf{0}$. Then

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= \alpha \cdot (\beta + \mathbf{0}) \\
 &= \alpha \cdot \beta \\
 &= \alpha \cdot \beta + \mathbf{0} \\
 &= \alpha \cdot \beta + \alpha \cdot \mathbf{0} \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta = \mathbf{0} \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= \alpha \cdot (\mathbf{0} + \gamma) \\
 &= \alpha \cdot \gamma \\
 &= \mathbf{0} + \alpha \cdot \gamma \\
 &= \alpha \cdot \mathbf{0} + \alpha \cdot \gamma \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta = \mathbf{0} \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= \alpha \cdot (\mathbf{0} + \gamma) \\
 &= \alpha \cdot \gamma \\
 &= \mathbf{0} + \alpha \cdot \gamma \\
 &= \alpha \cdot \mathbf{0} + \alpha \cdot \gamma \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma
 \end{aligned}$$

$\alpha \in \mathbb{R}^- \wedge \beta = 0 \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= \alpha \cdot (0 + \gamma) \\ &= \alpha \cdot \gamma \\ &= 0 + \alpha \cdot \gamma \\ &= \alpha \cdot 0 + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^+ \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^+ \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^- \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta \in \mathbb{R}^- \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta = 0 \wedge \gamma \in \mathbb{R}^+$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta = 0 \wedge \gamma \in \mathbb{R}^-$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

$\alpha = 0 \wedge \beta = 0 \wedge \gamma = 0$. Then

$$\begin{aligned}\alpha \cdot (\beta + \gamma) &= 0 \cdot (\beta + \gamma) \\ &= 0 \\ &= 0 + 0 \\ &= 0 \cdot \beta + 0 \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma\end{aligned}$$

3. Assume that $\alpha_0 = \alpha_1$ which by [theorem: 9.7] proves that in $\mathbb{Q}$ we have $0 = 1$. However by [theorem: 8.12] $\langle \mathbb{Q}, +, \cdot \rangle$ is a field so that $0 \neq 1$ and we reach a contradiction. Hence we must have that

$$0 = \alpha_0 \neq \alpha_1 = 1$$

□

Remember that $x + (-y)$ has a shorthand notation $x - y$, in the same way we have some shorthand notations for multiplication with a inverse element.

Notation 9.33. If $x, y \in \mathbb{R}$ $x \neq 0$ then we use the following shorthand notation

1. x^{-1} is noted as $1/x$
2. $y \cdot x^{-1}$ is noted as y/x

We show now how the rational numbers as a field can be embedded in the field of the real numbers. The primary candidate for this are the rational cuts, so lets review some of the properties of the rational cuts. First we need two little lemmas.

Lemma 9.34. If $r \in \mathbb{Q}$ such that $\alpha_r \in \mathbb{R}^+$ then $\text{inv}(\alpha_r) = \alpha_{r^{-1}}$

Proof. As $\alpha_r \in \mathbb{R}^+$ we have $0 \in \alpha_r$ hence $0 < r \Rightarrow 0 < r^{-1}$. Let $x \in \text{inv}(\alpha_r)$ then we have for x the following to consider:

$x \leq 0$. Then $x \leq 0 < r^{-1}$ proving that $x \in \alpha_{r^{-1}}$.

$0 < x$. Then $x \notin \mathbb{Q}_0^-$ so there exists a $s \in \mathbb{Q} \setminus \alpha_r$ such that $0 < s$ and $\exists t \in \mathbb{Q} \setminus \alpha_r$ such that $x = s^{-1}$. Hence $s = x^{-1}$ and $x^{-1} \in \mathbb{Q} \setminus \alpha_r$ and $\exists t \in \mathbb{Q} \setminus \alpha_r$ such that $t < x^{-1}$. As α_r is a rational cut we have by [theorem: 9.4] that $r = \min(\mathbb{Q} \setminus \alpha_r)$ hence $\forall t \in \mathbb{Q} \setminus \alpha_r$ we have $r \leq t$, from which we conclude that $x^{-1} \neq r$. As $x^{-1} \in \mathbb{Q} \setminus \alpha_r \Rightarrow r \leq x^{-1}$ we conclude that $0 < r < x^{-1}$ or using [theorems: 8.29, 4.73] that $x < r^{-1}$. Hence we have $x \in \alpha_{r^{-1}}$

So it follows that

$$\text{inv}(\alpha_r) \subseteq \alpha_{r^{-1}} \tag{9.69}$$

Let $x \in \alpha_{r^{-1}}$ then for x we have either:

$x \leq 0$. Then $x \in \mathbb{Q}_0^-$ so that $x \in \text{inv}(\alpha_r)$.

$0 < x$. Then as $x \in \alpha_{r^{-1}}$ we have that $0 < x < r^{-1}$ so that by [theorems: 8.29, 4.73] $0 < r < x^{-1}$, hence $x^{-1} \notin \alpha_r$ or $x^{-1} \in \mathbb{Q} \setminus \alpha_r$, further $\mathbb{Q} \setminus \alpha \ni \min(\mathbb{Q} \setminus \alpha_r) \stackrel{[\text{theorem: 9.4}]}{=} r < x^{-1}$. Summarized we have $x^{-1} \in \mathbb{Q} \setminus \alpha_r \wedge 0 < x^{-1} \wedge \exists t \in \mathbb{Q} \setminus \alpha_r \vdash t < x^{-1}$ proving that $x = (x^{-1})^{-1} \in \text{inv}(\alpha_r)$.

So in all cases $x \in \text{inv}(\alpha_r)$, hence $\alpha_{r^{-1}} \subseteq \text{inv}(\alpha_r)$, combining this with [eq: 9.69] gives

$$\text{inv}(\alpha_r) = \alpha_{r^{-1}} \quad \square$$

Lemma 9.35. *Let $\alpha_r, \alpha_s \in \mathbb{R}^+$ then $\alpha_r \odot \alpha_s = \alpha_{r \cdot s}$*

Proof. As $\alpha_r, \alpha_s \in \mathbb{R}^+$ we have $0 \in \alpha_r \wedge 0 \in \alpha_s$ so that

$$0 < r \wedge 0 < s \quad (9.70)$$

Let $x \in \alpha_r \odot \alpha_s$ then we have the following possibilities:

$x \leq 0$. Then as $0 < r \wedge 0 < s$ we have that $0 < r \cdot s$ so that $x < r \cdot s$ proving that $x \in \alpha_{r \cdot s}$.

$0 < x$. Then we have $x \notin \mathbb{Q}_0^-$ so there exists $u \in \alpha_r$ and $v \in \alpha_s$ so that $x = u \cdot v$ with $0 < u$ and $0 < v$. As $u \in \alpha_r$ and $v \in \alpha_s$ we have $u < r$ and $v < s$. So $u \cdot v < r \cdot v$ and $r \cdot v < r \cdot s$ hence $x = u \cdot v < r \cdot s$.

So we conclude that

$$\alpha_r \odot \alpha_s \subseteq \alpha_{r \cdot s} \quad (9.71)$$

Let $x \in \alpha_{r \cdot s}$ then $x < r \cdot s$. For x we have now the following cases to consider:

$x \leq 0$. Then $x \in \mathbb{Q}_0^-$ so that $x \in \alpha_r \cdot \alpha_s$.

$0 < x$. As $0 < r$ we have by the density of $\mathbb{Q}$ [see theorem: 8.37] the existence of $\varepsilon_1 \in \mathbb{Q}$ such that $0 < \varepsilon_1 < r$. From $x < r \cdot s$ it follows that $0 < r \cdot s - x$ hence, as $0 < s \Rightarrow 0 < s^{-1}$ we have that $0 < (r \cdot s - x) \cdot s^{-1} = r - x \cdot s^{-1}$. Using density of $\mathbb{Q}$ again there exist a $\varepsilon_2 \in \mathbb{Q}$ such that $0 < \varepsilon_2 < r - x \cdot s^{-1}$. Take now $\varepsilon = \min(\varepsilon_1, \varepsilon_2)$ then we have

$$0 < \varepsilon \leq \varepsilon_1 < r \text{ and } 0 < \varepsilon \leq \varepsilon_2 < r - x \cdot s^{-1} \quad (9.72)$$

From the above we have $x \cdot s^{-1} < r - \varepsilon$ or as $0 < x \wedge 0 < s^{-1} \Rightarrow 0 < x \cdot s^{-1}$ that $0 < x \cdot s^{-1} < r - \varepsilon$ allowing us to apply [theorems: 8.29, 4.73] giving

$$0 < (r - \varepsilon)^{-1} < (x \cdot s^{-1})^{-1} \stackrel{[\text{theorem: 4.55}]}{=} (s^{-1})^{-1} \cdot x^{-1} = s \cdot x^{-1}$$

multiplying both sides x we get by [theorems: 8.29, 4.73] and $0 < x$ that $0 < x \cdot (r - \varepsilon)^{-1} < s$ so that

$$0 < x \cdot (r - \varepsilon)^{-1} \in \alpha_s \quad (9.73)$$

As $0 < \varepsilon < r$ [see eq: 9.72] we have $0 < r - \varepsilon < r$ so that

$$0 < r - s \in \alpha_r \quad (9.74)$$

Now $(x \cdot (r - \varepsilon)^{-1}) \cdot (r - s) = x$ which combined with [eqs: 9.73, 9.74] proves that $x \in \alpha_r \cdot \alpha_s$.

So in all cases we have $x \in \alpha_r \cdot \alpha_s$ hence it follows that $\alpha_{r \cdot s} \subseteq \alpha_r \cdot \alpha_s$. Combining this with [eq: 9.71] proves that

$$\alpha_{r \cdot s} = \alpha_r \cdot \alpha_s \quad \square$$

Theorem 9.36. *Let $r, s \in \mathbb{Q}$ then we have*

1. $\alpha_r + \alpha_s = \alpha_{r+s}$
2. $-\alpha_r = \alpha_{-r}$
3. $\alpha_r \cdot \alpha_s = \alpha_{r \cdot s}$
4. If $\alpha_r \neq 0$ then $1/\alpha_r \stackrel{\text{notation}}{=} (\alpha_r)^{-1} = \alpha_{r^{-1}}$

Proof.

1. Let $x \in \alpha_r + \alpha_s$ then there exists $u \in \alpha_r$ and $v \in \alpha_s$ such that $x = u + v$. As $u \in \alpha_r$ and $v \in \alpha_s$ we have that $u < r$ and $v < s$ so that $u + v < r + s$ and $v + r < s + r$ giving $x = u + v < r + s$ proving that $x \in \alpha_r + \alpha_s$. Hence we have

$$\alpha_r + \alpha_s \subseteq \alpha_{r+s} \quad (9.75)$$

Let $x \in \alpha_{r+s}$ then $x < r + s$ hence $x - r < s$. Using the density of $\mathbb{Q}$ [see theorem: 8.37] there exist a $z \in \mathbb{Q}$ such that $x - r < z < s$. Then $z \in \alpha_s$ and if we define $\varepsilon = z - (x - r)$ we have $0 < \varepsilon \Rightarrow -\varepsilon < 0$. So $r - \varepsilon = r + (-\varepsilon) < r$ proving that $r - \varepsilon \in \alpha_r$. Hence

$$(r - \varepsilon) + z \in \alpha_r + \alpha_s \quad (9.76)$$

Now

$$\begin{aligned} (r - \varepsilon) + z &= r - (z - (x - r)) + z \\ &= r - z + x - r + z \\ &= x \end{aligned}$$

so that by [eq: 9.76] $x \in \alpha_r + \alpha_s$. Hence $\alpha_{r+s} \subseteq \alpha_r + \alpha_s$ which together with [eq: 9.75] proves

$$\alpha_{r+s} = \alpha_r + \alpha_s$$

2. This is stated in [theorem: 9.11]
3. Using [theorem: 9.27] we have to look at the following five cases:

$\alpha_r \in \mathbb{R}^+ \wedge \alpha_s \in \mathbb{R}^+$. Then

$$\begin{aligned} \alpha_r \cdot \alpha_s &= \alpha_r \odot \alpha_s \\ &\stackrel{[\text{lemma: 9.35}]}{=} \alpha_{r \cdot s} \end{aligned}$$

$\alpha_r \in \mathbb{R}^+ \wedge \alpha_s \in \mathbb{R}^-$. Then

$$\begin{aligned} \alpha_r \cdot \alpha_s &= -(\alpha_r \odot (-\alpha_s)) \\ &\stackrel{(2)}{=} -(\alpha_r \odot \alpha_{-s}) \\ &\stackrel{[\text{lemma: 9.35}]}{=} -\alpha_{r \cdot (-s)} \\ &\stackrel{[\text{theorem: 4.40}]}{=} -\alpha_{-(r \cdot s)} \\ &\stackrel{(2)}{=} \alpha_{-(-(r \cdot s))} \\ &= \alpha_{r \cdot s} \end{aligned}$$

$\alpha_r \in \mathbb{R}^- \wedge \alpha_s \in \mathbb{R}^+$. Then

$$\begin{aligned} \alpha_r \cdot \alpha_s &= -((- \alpha_r) \odot \alpha_s) \\ &= -(\alpha_{-r} \odot \alpha_s) \\ &\stackrel{[\text{lemma: 9.35}]}{=} -(\alpha_{(-r) \cdot s}) \\ &\stackrel{[\text{theorem: 4.40}]}{=} -(\alpha_{-(r \cdot s)}) \\ &\stackrel{(2)}{=} \alpha_{-(-(r \cdot s))} \\ &= \alpha_{r \cdot s} \end{aligned}$$

$\alpha_r \in \mathbb{R}^- \wedge \alpha_s \in \mathbb{R}^-$. Then

$$\begin{aligned} \alpha_r \cdot \alpha_s &= (-\alpha_r) \odot (-\alpha_s) \\ &\stackrel{(2)}{=} \alpha_{-r} \odot \alpha_{-s} \\ &\stackrel{[\text{lemma: 9.35}]}{=} \alpha_{(-r) \cdot (-s)} \\ &\stackrel{[\text{theorem: 4.40}]}{=} \alpha_{r \cdot s} \end{aligned}$$

$(\alpha_r, \alpha_s) \in (\{0\} \times \mathbb{R}) \cup (\mathbb{R} \times \{0\})$. Then we have two sub cases:

$(\alpha_r, \alpha_s) \in \{0\} \times \mathbb{R}$. Then $\alpha_r = 0 = \alpha_0 \Rightarrow_{[\text{theorem: 9.7}]} r = 0$ and

$$\begin{aligned} \alpha_r \cdot \alpha_s &= 0 \cdot \alpha_s \\ &= 0 \\ &= \alpha_0 \\ &= \alpha_{0 \cdot s} \\ &= \alpha_{r \cdot s} \end{aligned}$$

$(\alpha_r, \alpha_s) \in \mathbb{R} \times \{0\}$. Then $\alpha_s = 0 = \alpha_0 \Rightarrow_{[\text{theorem: 9.7}]} s = 0$ and

$$\begin{aligned} \alpha_r \cdot \alpha_s &= \alpha_r \cdot 0 \\ &= 0 \\ &= \alpha_0 \\ &= \alpha_{r \cdot 0} \\ &= \alpha_{r \cdot s} \end{aligned}$$

So in all cases we have

$$\alpha_r \cdot \alpha_s = \alpha_{r \cdot s}$$

4. Let $\alpha_r \in \mathbb{R} \setminus \{0\}$ then we have the following possibilities:

$\alpha_r \in \mathbb{R}^+$. Then $(\alpha_r)^{-1} = \text{inv}(\alpha_r) \stackrel{[\text{lemma: 9.34}]}{=} \alpha_{r^{-1}}$

$\alpha_r \in \mathbb{R}^-$. Then

$$\begin{aligned} (\alpha_r)^{-1} &= -\text{inv}(-\alpha_r) \\ &\stackrel{(2)}{=} -\text{inv}(\alpha_{-r}) \\ &\stackrel{[\text{lemma: 9.34}]}{=} -\alpha_{(-r)^{-1}} \\ &\stackrel{[\text{theorems: 8.30}]}{=} -\alpha_{-(r^{-1})} \\ &\stackrel{(2)}{=} \alpha_{-(-(r^{-1}))} \\ &= \alpha_{r^{-1}} \\ &\quad \square \end{aligned}$$

We show now that $\mathbb{Q}_{\mathbb{R}}$ is a embedding of $\mathbb{Q}$ in $\mathbb{R}$ that conserves the field structure.

Theorem 9.37. For $\mathbb{Q}_{\mathbb{R}} = \{\alpha_r | r \in \mathbb{Q}\}$ [definition: 9.6] we have:

1. $\mathbb{Q}_{\mathbb{R}}$ is a sub-field of $\langle \mathbb{R}, +, \cdot \rangle$
2. The function $i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{R}, +, \cdot \rangle$ defined by $i_{\mathbb{Q} \rightarrow \mathbb{R}}(q) = \alpha_q$ is a field isomorphism

Proof.

1. Let $x, y \in \mathbb{Q}_{\mathbb{R}}$ then we have that $\exists r, s \in \mathbb{Q}$ such that $x = \alpha_r$ and $y = \alpha_s$. Then we have:

- a. $x + y = \alpha_r + \alpha_s \stackrel{[\text{theorem: 9.36}]}{=} \alpha_{r+s} \in \mathbb{Q}_{\mathbb{R}}$
- b. $x \cdot y = \alpha_r \cdot \alpha_s \stackrel{[\text{theorem: 9.36}]}{=} \alpha_{r \cdot s} \in \mathbb{Q}_{\mathbb{R}}$
- c. If $x \neq 0$ then $x^{-1} = (\alpha_r)^{-1} \stackrel{[\text{theorem: 9.36}]}{=} \alpha_{r^{-1}} \in \mathbb{Q}_{\mathbb{R}}$
- d. $0 = \alpha_0 \in \mathbb{Q}_{\mathbb{R}}$

$$\text{e. } 1 = \alpha_1 \in \mathbb{Q}_{\mathbb{R}}$$

which proves that $\mathbb{Q}_{\mathbb{R}}$ is a sub-field of $\langle \mathbb{R}, +, \cdot \rangle$.

2. Using [theorem: 9.7] it follows that

$$i_{\mathbb{Q} \rightarrow \mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R} \text{ is a bijection}$$

Next we have to prove the homeomorphism properties:

- a. If $r, s \in \mathbb{Q}$ then $i_{\mathbb{Q} \rightarrow \mathbb{R}}(r + s) = \alpha_{r+s} \stackrel{[\text{theorem: 9.36}]}{=} \alpha_r + \alpha_s = i_{\mathbb{Q} \rightarrow \mathbb{R}}(r) + i_{\mathbb{Q} \rightarrow \mathbb{R}}(s)$
- b. If $r, s \in \mathbb{Q}$ then $i_{\mathbb{Q} \rightarrow \mathbb{R}}(r \cdot s) = \alpha_r \cdot \alpha_s \stackrel{[\text{theorem: 9.36}]}{=} \alpha_{r \cdot s} = i_{\mathbb{Q} \rightarrow \mathbb{R}}(r) \cdot i_{\mathbb{Q} \rightarrow \mathbb{R}}(s)$
- c. $i_{\mathbb{Q} \rightarrow \mathbb{R}}(1) = \alpha_1 = 1$

□

9.2 Order relation on $\mathbb{R}$

Theorem 9.38. *If $\alpha, \beta \in \mathbb{R}^+$ then*

- 1. $\alpha + \beta \in \mathbb{R}^+$
- 2. $\alpha \cdot \beta \in \mathbb{R}^+$
- 3. $\alpha^{-1} \in \mathbb{R}^+$

Proof.

- 1. As $\alpha, \beta \in \mathbb{R}^+$ we have that $0 \in \alpha$ and $0 \in \beta$ then $0 = 0 + 0 \in \alpha + \beta$ proving that

$$\alpha + \beta \in \mathbb{R}^+$$

- 2. As $0 \in \mathbb{Q}_0^-$ we have $\alpha \odot \beta \in \mathbb{R}^+$ so that

$$\alpha \cdot \beta \stackrel{[\alpha, \beta \in \mathbb{R}^+]}{=} \alpha \odot \beta \in \mathbb{R}^+$$

- 3. We have $\alpha^{-1} \stackrel{[\alpha \in \mathbb{R}^+]}{=} \text{inv}(\alpha) \in \mathbb{R}^+$

□

We define now the relation on $\mathbb{R}$ that later will be proved to be a order relation, this definition mirrors the definition of order in $\mathbb{Z}$ and $\mathbb{Q}$ and is the reason why we have defined $\mathbb{R}^+$. One problem is that $0 \notin \mathbb{R}^+$, so we have first to define $<$ and base $\leq$ on $<$.

Definition 9.39. $< \subseteq \mathbb{R} \times \mathbb{R}$ is defined by

$$< = \{(\alpha, \beta) \in \mathbb{R} \times \mathbb{R} \mid \beta + (-\alpha) \in \mathbb{R}^+\}$$

or in other words for $\alpha, \beta \in \mathbb{R}$ we have

$$\alpha < \beta \Leftrightarrow \beta - \alpha = \beta + (-\alpha) \in \mathbb{R}^+$$

Definition 9.40. $\leq \subseteq \mathbb{R} \times \mathbb{R}$ is defined by

$$\leq = \{(\alpha, \beta) \in \mathbb{R} \times \mathbb{R} \mid \alpha = \beta \vee \beta + (-\alpha) \in \mathbb{R}^+\} = \{(\alpha, \beta) \in \mathbb{R} \times \mathbb{R} \mid \alpha = \beta \vee \alpha < \beta\}$$

or in other words for $\alpha, \beta \in \mathbb{R}$ we have

$$\alpha \leq \beta \Leftrightarrow \alpha = \beta \vee \alpha < \beta \Leftrightarrow \alpha = \beta \vee \beta - \alpha = \beta + (-\alpha) \in \mathbb{R}^+$$

The following theorem shows a simpler way to decide if $\alpha < \beta$ or $\alpha \leq \beta$.

Theorem 9.41. $\forall \alpha, \beta \in \mathbb{R}$ we have

- 1. $\alpha < \beta \Leftrightarrow \alpha \subset \beta$ [strict inclusion]

$$2. \alpha \leq \beta \Leftrightarrow \alpha \subseteq \beta$$

Proof.

1.

$\Rightarrow$. As $\alpha < \beta$ we have that $\beta + (-\alpha) \in \mathbb{R}^+$ so that

$$0 \in \beta + (-\alpha) \quad (9.77)$$

Let $r \in \alpha$. If $-r \in -\alpha$ then by the definition of a negative cut [see definition: 9.10] we have $r = -(-r) \in \mathbb{Q} \setminus \alpha$ contradicting $r \in \alpha$. Hence we must have that $-r \notin -\alpha$ or

$$-r \in \mathbb{Q} \setminus -\alpha \quad (9.78)$$

As $0 \in \beta + (-\alpha)$ [see eq: 9.77] there exists $s \in \beta$ and a $t \in -\alpha$ such that $0 = s + t$ or $s = -t$. As $t \in -\alpha$ and $-r \in \mathbb{Q} \setminus -\alpha$ [see eq: 9.78] it follows from [definition: 9.1 (3)] that $t < -r$ or $r < -t = s \in \beta$. So $r < s \in \beta$ which by [theorem: 9.3] proves that $r \in \beta$. Hence we have

$$\alpha \subseteq \beta \quad (9.79)$$

If now $\alpha = \beta$ then $\beta + (-\alpha) = \beta + (-\beta) = 0 = a_0$, as by [eq: 9.77] $0 \in \beta + (-\alpha)$ we find that $0 \in \alpha_0 = \{q \in \mathbb{Q} | q < 0\}$ a contradiction, so $\alpha \neq \beta$, which combined with [eq: 9.79] gives

$$\alpha \subset \beta$$

$\Leftarrow$. As $\alpha \subset \beta$ there exist a $r \in \beta$ such that $r \notin \alpha$ or

$$r \in \mathbb{Q} \setminus \alpha \quad (9.80)$$

As by [definition: 9.1 (4)] $\max(\beta)$ does not exist we have

$$\exists r' \in \beta \text{ such that } r < r'$$

If $r' \in \alpha$ then as $r \in \mathbb{Q} \setminus \alpha$ we have by [definition: 9.1 (3)] that $r' < r$ contradicting $r < r'$, hence $r' \notin \alpha$ or

$$r' \in \mathbb{Q} \setminus \alpha$$

So we have that $-(-r') \in \mathbb{Q} \setminus \alpha$ and $r < r' = -(-r')$ where $r \in \mathbb{Q} \setminus \alpha$ which by the definition of a negative cut [see definition: 9.10] proves that $-r' \in -\alpha$. As $r' \in \beta$ we have that $0 = r' + (-r') \in \beta + (-\alpha)$ proving that $\beta + (-\alpha) \in \mathbb{R}^+$ or that

$$\alpha < \beta$$

2.

$\Rightarrow$. As $\alpha \leq \beta$ we have by (1) that $\alpha = \beta \vee \alpha \subset \beta$ so that $\alpha \subseteq \beta$

$\Leftarrow$. If $\alpha \subseteq \beta$ then $\alpha = \beta \wedge \alpha \subset \beta$ so that by (1) $\alpha = \beta \vee \alpha < \beta$ proving $\alpha \leq \beta$ $\square$

Corollary 9.42. For $\langle \mathbb{R}, \leq \rangle$ we have that $0 < 1$

Proof. Note that $1 = \alpha_1 = \{q \in \mathbb{Q} | q < 1\}$ and $0 = \alpha_0 = \{q \in \mathbb{Q} | q < 0\}$. So if $q \in \alpha_0$ we have $q < 0 < 1 \Rightarrow q \in \alpha_1$ proving that

$$\alpha_0 \subseteq \alpha_1$$

As in $\mathbb{Q}$ we have $0 < 1$ [see example: 8.25] we have by the density of $\mathbb{Q}$ [see theorem: 8.37] that there exist a $q \in \mathbb{Q}$ such that $0 < q < 1$ hence $q \in \alpha_1$ but $q \notin \alpha_0$ proving that

$$\alpha_0 \subset \alpha_1$$

which by the previous theorem [theorem: 9.41] proves that

$$0 < 1 \quad \square$$

Theorem 9.43. $\langle \mathbb{R}, \leq \rangle$ is a totally ordered set

Proof.

reflexivity. If $\alpha \in \mathbb{R}$ then $\alpha \subseteq \alpha$ so that by [theorem: 9.41] $\alpha \leq \alpha$

anti symmetry. If $\alpha \leq \beta$ and $\beta \leq \alpha$ then by [theorem: 9.41] we have $\alpha \subseteq \beta$ and $\beta \subseteq \alpha$ hence $\alpha = \beta$.

transitivity. If $\alpha \leq \beta$ and $\beta \leq \gamma$ then by [theorem: 9.41] we have $\alpha \subseteq \beta \wedge \beta \subseteq \gamma$ so that $\alpha \subseteq \gamma$ which by [theorem: 9.41] proves that $\alpha \leq \gamma$

totally ordering. Let $\alpha, \beta \in \mathbb{R}$ then for $\alpha + (-\beta)$ we have, as $\mathbb{R} \stackrel{[\text{theorem: 9.17}]}{=} \mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\}$, either:

$\alpha + (-\beta) \in \mathbb{R}^+$. Then $\alpha < \beta \Rightarrow \alpha \leq \beta$

$\alpha + (-\beta) \in \mathbb{R}^-$. Then $\beta + (-\alpha) = -(\alpha + (-\beta)) \in \mathbb{R}^+$ so that $\beta < \alpha \Rightarrow \beta \leq \alpha$

$\alpha + (-\beta) = 0$. Then $\alpha = \beta$ hence $\alpha \leq \beta$ □

Theorem 9.44. *We have the following for $\mathbb{R}$*

1. $\mathbb{R}^+ = \{\alpha \in \mathbb{R} | 0 < \alpha\}$
2. $\mathbb{R}_0^+ = \{\alpha \in \mathbb{R} | 0 \leq \alpha\}$
3. $\mathbb{R}^- = \{\alpha \in \mathbb{R} | \alpha < 0\}$
4. $\mathbb{R}_0^- = \{\alpha \in \mathbb{R} | \alpha \leq 0\}$

Proof.

1. $\alpha \in \mathbb{R}^+ \Leftrightarrow \alpha + (-0) \in \mathbb{R}^+ \Leftrightarrow 0 < \alpha \Leftrightarrow \alpha \in \{\alpha \in \mathbb{R} | 0 < \alpha\}$
2. $\alpha \in \mathbb{R}_0^+ \Leftrightarrow \alpha = 0 \vee \alpha + (-0) \in \mathbb{R}^+ \Leftrightarrow \alpha = 0 \vee 0 < \alpha \Leftrightarrow 0 \leq \alpha \Leftrightarrow \alpha \in \{\alpha \in \mathbb{R} | 0 \leq \alpha\}$.
3. $\alpha \in \mathbb{R}^- \Leftrightarrow -\alpha \in \mathbb{R}^+ \Leftrightarrow 0 + (-\alpha) \in \mathbb{R}^+ \Leftrightarrow \alpha < 0 \Leftrightarrow \alpha \in \{\alpha \in \mathbb{R} | \alpha < 0\}$
4. $\alpha \in \mathbb{R}_0^- \Leftrightarrow \alpha = 0 \vee -\alpha \in \mathbb{R}^- \Leftrightarrow \alpha = 0 \vee 0 + (-\alpha) \in \mathbb{R}^+ \Leftrightarrow \alpha = 0 \vee \alpha < 0 \Leftrightarrow \alpha \leq 0 \Leftrightarrow \alpha \in \{\alpha \in \mathbb{R} | \alpha \leq 0\}$ □

Theorem 9.45. $\langle \mathbb{R}, +, \cdot, \leq \rangle$ is a ordered field

Proof. First using [theorem: 9.32] we have that $\langle \mathbb{R}, +, \cdot \rangle$ is a field. Second we have

1. If $x, y, z \in \mathbb{R}$ with $x < y$ then

$$y + (-x) \in \mathbb{R}^+, \tag{9.81}$$

further

$$\begin{aligned} (y + z) + (-(x + z)) &\stackrel{[\text{theorem: 4.8}]}{=} (y + z) + ((-x) + (-z)) \\ &= (y + z) + ((-z) + (-x)) \\ &= y + (z + ((-z) + (-x))) \\ &= y + ((z + (-z)) + (-x)) \\ &= y + (0 + (-x)) \\ &= y + (-x) \end{aligned}$$

which by [eq: 9.81] proves that $(y + z) + (-(x + z)) \in \mathbb{R}^+$ or

$$x + z < y + z$$

2. If $x, y \in \mathbb{R}$ with $0 < x$ and $0 < y$ then by [theorem: 9.44] $x, y \in \mathbb{R}^+$. So by [theorem: 9.38] we have that $x \cdot y \in \mathbb{R}^+$ which by [theorem: 9.44] proves that

$$0 < x \cdot y \tag{9.82}$$

□

Corollary 9.46. *Let $\alpha \in \mathbb{R}$*

1. $\forall \alpha \in \mathbb{R}$ we have $\alpha < \alpha + 1$

$$2. \alpha - 1 < \alpha$$

Proof.

1. As $0 < 1$ [see corollary: 9.42] we have by [theorems: 9.45, 4.73] that $\alpha = 0 + \alpha < 1 + \alpha = \alpha + 1$
2. As $0 < 1$ we have by [theorems: 9.45, 4.73] that $-1 < 0$ hence using [theorems: 9.45, 4.73] again we have $\alpha - 1 = (-1) + \alpha < 0 + \alpha = \alpha$. $\square$

Theorem 9.47. If $\alpha, \beta \in \mathbb{R}_0^+$ [so that $0 \leq \alpha \wedge 0 \leq \beta$] such that $\alpha + \beta = 0$ then $\alpha = 0 = \beta$

Proof. As $0 \leq \alpha, 0 \leq \beta$ then we have either

$0 < \alpha$. Then $\beta = 0 + \beta < \alpha + \beta = 0$ hence $\beta < 0$ contradicting $0 \leq \beta$, so this case does not occur.

$0 < \beta$. Then $\alpha = \alpha + 0 < \alpha + \beta = 0$ hence $\alpha < 0$ contradicting $0 \leq \alpha$, so this case does not occur.

$\alpha = \beta = 0$. This is the only resting case proving that $\alpha = \beta = 0$ $\square$

Lemma 9.48. Let $r, s \in \mathbb{Q}$ then we have

1. $r < s \Leftrightarrow \alpha_r < \alpha_s$
2. $r \leq s \Leftrightarrow \alpha_r < \alpha_s$

Proof.

1.

$\Rightarrow$. If $x \in \alpha_r$ then $x < r$ which as $r < s$ proves that $x < s$ hence $x \in \alpha_s$, so $\alpha_r \subseteq \alpha_s$. Further as $r < s$ we have by the density of $\mathbb{Q}$ [see theorem: 8.37] that there exists a $q \in \mathbb{Q}$ such that $r < q < s$ hence $q \in \alpha_s$ and $q \notin \alpha_r$, proving that $\alpha_r \subset \alpha_s$. By [theorem: 9.41] it follows then that

$$\alpha_r < \alpha_s$$

$\Leftarrow$. If $\alpha_r < \alpha_s$ then we have by [theorem: 9.41] that $\alpha_r \subset \alpha_s$. Assume that $s \leq r$ then $\forall t \in \alpha_s$ we have $t < s \leq r \Rightarrow t < r \Rightarrow t \in \alpha_r$ proving that $\alpha_s \subseteq \alpha_r$ contradicting $\alpha_r \subset \alpha_s$. As the assumption $s \leq r$ leads to a contradiction we must have that $r < s$.

2.

$$\begin{aligned} r \leq s &\Leftrightarrow r = s \vee r < s \\ &\Leftrightarrow \alpha_r = \alpha_s \vee r < s \\ &\stackrel{(1)}{\Leftrightarrow} \alpha_r = \alpha_s \vee \alpha_r < \alpha_s \\ &\Leftrightarrow \alpha_r \leq \alpha_s \end{aligned}$$

$\square$

The above lemma allows us to show that the embedding of $\mathbb{Q}$ in $\mathbb{R}$ by $i_{\mathbb{Q} \rightarrow \mathbb{R}}$ is not only preserving the field structure but also the order.

Theorem 9.49. The field isomorphism $i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{R}}, +, \cdot \rangle$ defined by $i_{\mathbb{Q} \rightarrow \mathbb{R}}(r) = \alpha_r$ [see theorem: 9.37] is a order isomorphism between $\langle \mathbb{Q}, \leq \rangle$ and $\langle \mathbb{Q}_{\mathbb{R}}, \leq \rangle$

Proof. Using [theorem: 9.37] we have that $i_{\mathbb{Q} \rightarrow \mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{R}}$ is a bijection. Further for $r, s \in \mathbb{Q}$ we have

$$\begin{aligned} r \leq s &\stackrel{[theorem: 9.48]}{\Leftrightarrow} \alpha_r \leq \alpha_s \\ &\Leftrightarrow i_{\mathbb{Q} \rightarrow \mathbb{R}}(r) \leq i_{\mathbb{Q} \rightarrow \mathbb{R}}(s) \end{aligned}$$

$\square$

We have seen in [theorem: 8.41] that the rational numbers are not conditional complete [causing $\langle \mathbb{Q}_{\mathbb{R}}, \leq \rangle$ to be not conditional complete], the prime reason that we construct the real numbers is that the real numbers are conditional complete.

Theorem 9.50. $\langle \mathbb{R}, \leq \rangle$ is conditional complete [definition: 3.80] in other words

$\forall S \subseteq \mathbb{R}$ with $S \neq \emptyset$ such that $\exists v \in \mathbb{R}$ such that $\forall \alpha \in S$ we have $\alpha \leq v$ we have that $\sup(S)$ exist
Using [theorem: 3.81] this is equivalent with

$\forall S \subseteq \mathbb{R}$ with $S \neq \emptyset$ such that $\exists \lambda \in \mathbb{R}$ such that $\forall \alpha \in S$ we have $\lambda \leq \alpha$ we have that $\inf(S)$ exist

Proof. Let $S \subseteq \mathbb{R}$ with $S \neq \emptyset$ such that there exists a $v \in \mathbb{R}$ such that $\forall \alpha \in S$ we have $\alpha \leq v$. Define γ by

$$\gamma = \{q \in \mathbb{Q} \mid \exists \alpha \in S \mid q \in \alpha\}$$

First we prove that $\gamma \in \mathbb{R}$ [or γ is a Dedekind cut]

1. As $S \neq \emptyset$ there exist a $\alpha \in S \subseteq \mathbb{R}$. As α is a Dedekind cut we have by [definition: 9.1 (1)] that $\alpha \neq \emptyset$. Hence $\exists q \in \alpha \subseteq \mathbb{Q}$ so that $q \in \gamma$, proving that

$$\gamma \neq \emptyset$$

2. If $r \in \gamma$ then $\exists \alpha \in S$ such that $r \in \alpha$, as v is an upper bound of S we have that $\alpha \leq v$, so using [9.41] $\alpha \subseteq v$ proving that $r \in v$ so that $\gamma \subseteq v$. As $v \in \mathbb{R}$ we have by [definition: 9.1 (2)] that $v \neq \mathbb{Q}$ so that $\exists q \in \mathbb{Q}$ such that $q \notin v$, which as $\gamma \subseteq v$ proves that $q \notin \gamma$. Hence

$$\gamma \neq \mathbb{Q}$$

3. Let $r \in \gamma$ and $s \in \mathbb{Q} \setminus \gamma$. As $r \in \gamma$ there exists a $\alpha \in S$ such that $r \in \alpha$ and as $s \in \mathbb{Q} \setminus \gamma$ we have that $\forall \zeta \in S$ we have $s \notin \zeta$, so in particular $s \notin \alpha$ hence $s \in \mathbb{Q} \setminus \alpha$. Using [definition: 9.1 (3)] we have that $r < s$. So

$$\text{If } r \in \gamma \wedge s \in \mathbb{Q} \setminus \gamma \text{ then } r < s$$

4. Assume that γ has a greatest element m then

$$m \in \gamma \text{ and } \forall r \in \gamma \text{ we have } r \leq m \quad (9.82)$$

Now as $m \in \gamma$ there exist a $\alpha \in S$ such that $m \in \alpha$. As by [definition: 9.1 (4)] α has no greatest element there exist a $s \in \alpha$ such that $m < s$. As $s \in \alpha \in S$ it follows that $s \in \gamma$ so by [eq: 9.82] we must have that $s \leq m$ contradicting $m < s$. So the assumption is wrong and we have

γ has no greatest element

From (1),(2),(3) and (4) we conclude that γ is a Dedekind cut, hence

$$\gamma \in \mathbb{R}$$

Next we proof that γ is an upper bound of S . So let $\alpha \in S$ then if $q \in \alpha$ we have by definition that $q \in \gamma$ proving that $\alpha \subseteq \gamma$ which by [theorem: 9.41] results in $\alpha \leq \gamma$. Hence

γ is an upper bound of S

Finally let $\lambda \in v(S) = \{\alpha \in \mathbb{R} \mid \alpha \text{ is an upper bound of } S\}$. If $q \in \gamma$ there exist a $\alpha \in S$ such that $q \in \alpha$, as λ is an upper bound of S we have $\alpha \leq \lambda \xRightarrow{\text{[theorem: 9.41]}} \alpha \subseteq \lambda$, so $q \in \lambda$, proving that $\gamma \subseteq \lambda$ or by [theorem: 9.41] that $\gamma \leq \lambda$. Hence γ is the least element of $v(S)$ which by definition proves that

$$\sup(S) \text{ exist} \quad \square$$

9.3 Embeddings in $\mathbb{R}$

First remember that by [theorems: 9.37,9.49] we have an embedding of $\mathbb{Q}$ in $\mathbb{R}$ by the order and field isomorphism $i_{\mathbb{Q} \rightarrow \mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{R}}$ defined by $i_{\mathbb{Q} \rightarrow \mathbb{R}}(r) = \alpha_r$. We show now that there exist also embeddings of $\mathbb{N}_0$ and $\mathbb{Z}$ in $\mathbb{R}$.

Definition 9.51. $\mathbb{Z}_{\mathbb{R}} = (i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}})(\mathbb{Z})$ where

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{Q}} \subseteq \mathbb{Q} \text{ is defined by } i_{\mathbb{Z} \rightarrow \mathbb{Q}}(z) = \frac{z}{1} \text{ [theorem: 8.32]}$$

and

$i_{\mathbb{Q} \rightarrow \mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{R}}$ is defined by $i_{\mathbb{Q} \rightarrow \mathbb{R}}(r) = \alpha_r$ [theorem: 9.49]

so that

$$\mathbb{Z}_{\mathbb{R}} \subseteq \mathbb{Q}_{\mathbb{R}}$$

Theorem 9.52. For $\langle \mathbb{Z}_{\mathbb{R}}, +, \cdot \rangle$ and $i_{\mathbb{Z} \rightarrow \mathbb{R}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{R}}$ defined by $i_{\mathbb{Z} \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{Q}} \circ i_{\mathbb{Q} \rightarrow \mathbb{R}}$ we have

1. $\mathbb{Z}_{\mathbb{R}}$ is a sub ring of $\langle \mathbb{R}, +, \cdot \rangle$ and $i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, +, \cdot \rangle$ is a ring isomorphism.
2. $\mathbb{Z}_{\mathbb{R}}$ is a sub group of $\langle \mathbb{R}, + \rangle$ and $i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, + \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, + \rangle$ is a group isomorphism.
3. $\mathbb{Z}_{\mathbb{R}}$ is a sub semi-group of $\langle \mathbb{R}, \cdot \rangle$ and $i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, \cdot \rangle$ is a group isomorphism.
4. $i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, \leq \rangle$ is a order isomorphism
5. $\mathbb{Z}_{\mathbb{R}}$ is denumerable

Proof. First note that by the definition of $\mathbb{Z}_{\mathbb{R}}$ we have that

$$\mathbb{Z}_{\mathbb{R}} = i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(\mathbb{Z})) \quad (9.83)$$

Second we have

1. Using [theorems: 8.32, 9.49] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, + \rangle \rightarrow \langle \mathbb{Q}, + \rangle \text{ and } i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, + \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{R}}, + \rangle \text{ are ring isomorphism}$$

So using [theorem: 4.48] and [theorem: 8.32] we have that

$$i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(\mathbb{Z})) \text{ is a sub-ring of } \langle \mathbb{Q}_{\mathbb{R}}, +, \cdot \rangle$$

and

$$i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(\mathbb{Z})), +, \cdot \rangle \text{ is a ring isomorphism}$$

Using [eq: 9.83] and [theorem: 4.38, 9.37] we have that

$$\mathbb{Z}_{\mathbb{R}} \text{ is a sub-ring of } \langle \mathbb{R}, +, \cdot \rangle$$

and

$$i_{\mathbb{Z} \rightarrow \mathbb{R}} = i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, +, \cdot \rangle \text{ is a ring isomorphism}$$

2. This follows from (1) and [theorems: 4.36 and 4.47]
3. This follows from (1) and [theorems: 4.36 and 4.47]
4. Using [theorems: 8.32, 9.49] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, \leq \rangle \text{ and } i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, \leq \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{R}}, \leq \rangle \text{ are order isomorphisms}$$

So using [theorem: 3.52] we have that

$$i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(\mathbb{Z})), \leq \rangle \text{ is a order isomorphism}$$

hence using [eq: 9.83] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{R}} = i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, \leq \rangle \text{ is a order isomorphism}$$

5. Using (4) we have that $\mathbb{Z} \approx \mathbb{Z}_{\mathbb{R}}$ which as by [theorem: 7.53] $\mathbb{N}_0 \approx \mathbb{Z}$ proves that $\mathbb{N}_0 \approx \mathbb{Z}_{\mathbb{R}}$ proving that $\mathbb{Z}_{\mathbb{R}}$ is denumerable. $\square$

We can use the same technique to embed the set of natural numbers in $\mathbb{R}$.

Definition 9.53. $\mathbb{N}_{0, \mathbb{R}} = (i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}})(\mathbb{N}_0)$ where

$i_{\mathbb{N}_0}: \mathbb{N}_0 \rightarrow \mathbb{Z}_0^+ \subseteq \mathbb{Z}$ is defined by $i_{\mathbb{N}_0}(z) = \sim[(z, 0)]$ [theorem: 7.17] and

$$i_{\mathbb{Z} \rightarrow \mathbb{R}}: \mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{R}} \text{ [theorem: 9.52]}$$

so that

$$\mathbb{N}_{0,\mathbb{R}} \subseteq \mathbb{Z}_{\mathbb{R}}$$

In many cases we want to exclude 0 from the embedded real numbers, hence the following definition.

Definition 9.54. $\mathbb{N}_{\mathbb{R}} = \mathbb{N}_{0,\mathbb{R}} \setminus \{0\}$

Theorem 9.55. For $\langle \mathbb{N}_{0,\mathbb{R}}, +, \cdot \rangle$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \mathbb{N}_0 \rightarrow \mathbb{N}_{0,\mathbb{R}}$ defined by $i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}$ we have

1. $\mathbb{N}_{0,\mathbb{R}}$ is a sub-semi-group of $\langle \mathbb{R}, + \rangle$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, + \rangle$ is a group isomorphism.
2. $\mathbb{N}_{0,\mathbb{R}}$ is a sub-semi-group of $\langle \mathbb{R}, \cdot \rangle$ and $i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, \cdot \rangle$ is a group isomorphism.
3. $i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, \leq \rangle$ is a order isomorphism.
4. $\mathbb{N}_{0,\mathbb{R}}$ is denumerable.
5. $\langle \mathbb{N}_{0,\mathbb{R}}, \leq \rangle$ is well ordered.

Proof. First note that by the definition of $\mathbb{N}_{0,\mathbb{R}}$ we have that

$$\mathbb{N}_{0,\mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}}(i_{\mathbb{N}_0}(\mathbb{N}_0)) \quad (9.84)$$

1. Using [theorems: 7.17 and 9.52] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{Z}_0^+, + \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, + \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, + \rangle \text{ are a group isomorphisms}$$

So using [theorem: 4.25] and [theorem: 7.17] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{R}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(\mathbb{N}_0)) \text{ is a sub group of } \langle \mathbb{Z}_{\mathbb{R}}, + \rangle$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle i_{\mathbb{Z} \rightarrow \mathbb{R}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(\mathbb{N}_0)), + \rangle \text{ is a group isomorphism}$$

Using [eq: 9.84] and [theorems: 4.17, 9.52] it follows that

$$\mathbb{N}_{0,\mathbb{R}} \text{ is a sub semi-group of } \langle \mathbb{R}, + \rangle$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, + \rangle \text{ is a group isomorphism}$$

2. Using [theorems: 7.17 and 9.52] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{Z}_0^+, \cdot \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, \cdot \rangle \text{ are group isomorphisms}$$

So using [theorem: 4.25] and [theorem: 7.17] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{R}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(\mathbb{N}_0)) \text{ is a sub semi-group of } \langle \mathbb{Z}_{\mathbb{R}}, \cdot \rangle$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle i_{\mathbb{Z} \rightarrow \mathbb{R}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(\mathbb{N}_0)), \cdot \rangle \text{ is a group isomorphism}$$

Using [eq: 9.84] and [theorems: 4.17, 9.52] it follows that

$$\mathbb{N}_{0,\mathbb{R}} \text{ is a sub semi-group of } \langle \mathbb{R}, + \rangle$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, \cdot \rangle \text{ is a group isomorphism}$$

3. Using [theorems: 7.29 and 9.52] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{Z}_0^+, \leq \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, \leq \rangle$$

so using [theorem: 3.52]

$$i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle i_{\mathbb{Z} \rightarrow \mathbb{R}}(i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}(\mathbb{N}_0)), \leq \rangle \text{ is a order isomorphism}$$

Using [eq: 9.84] it follows that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{R}} = i_{\mathbb{Z} \rightarrow \mathbb{R}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, \leq \rangle \text{ is a order isomorphism}$$

4. Using (3) we have that $\mathbb{N}_0 \approx \mathbb{N}_{0,\mathbb{R}}$ proving that $\mathbb{N}_{0,\mathbb{R}}$ is denumerable.
5. By [theorem: 5.51] $\langle \mathbb{N}_0, \leq \rangle$ is well ordered, further by (3) $i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, \leq \rangle$ is a order isomorphism, hence by [theorem: 3.85] it follows that

$$\langle \mathbb{N}_{0,\mathbb{R}}, \leq \rangle \text{ is well ordered} \quad \square$$

For the relation between $\mathbb{N}_{0,\mathbb{R}}$, $\mathbb{Z}_{\mathbb{R}}$, $\mathbb{Q}_{\mathbb{R}}$ and $\mathbb{R}$ we have

Theorem 9.56. *We have the following relation between the embeddings of $\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}$ in $\mathbb{R}$*

$$\mathbb{N}_{0,\mathbb{R}} \subseteq \mathbb{Z}_{\mathbb{R}} \subseteq \mathbb{Q}_{\mathbb{R}} \subseteq \mathbb{R}$$

Proof. Using [definition: 9.6] we have that $\mathbb{Q}_{\mathbb{R}} \subseteq \mathbb{R}$, using [definition: 9.51] we have that $\mathbb{Z}_{\mathbb{R}} \subseteq \mathbb{Q}_{\mathbb{R}}$ and finally by [definition: 9.53] it follows that $\mathbb{N}_{0,\mathbb{R}} \subseteq \mathbb{Z}_{\mathbb{R}}$ $\square$

Finally we can define $\mathbb{Q}_{\mathbb{R}}$ in terms of $\mathbb{Z}$.

Theorem 9.57. $\mathbb{Q}_{\mathbb{R}} = \{x/y | x \in \mathbb{Z}_{\mathbb{R}} \wedge y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}\}$ where q/r is a shorthand for $q \cdot r^{-1}$ [see notation: 9.33]

Proof. Using [theorems: 8.32, 9.37, 9.52] we have the following

$$i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{Q}}, +, \cdot \rangle \text{ is a ring isomorphism} \quad (9.85)$$

$$i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{R}}, +, \cdot \rangle \text{ is a field isomorphism} \quad (9.86)$$

$$i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, +, \cdot \rangle \text{ is a ring isomorphism} \quad (9.87)$$

Let $q \in \mathbb{Q}_{\mathbb{R}}$ then by [eq: 9.86] there exist a $q' \in \mathbb{Q}$ such that $q = i_{\mathbb{Q} \rightarrow \mathbb{R}}(q')$. As $q' \in \mathbb{Q}$ we have by [theorem: 8.33] that there exists $x \in \mathbb{Z}_{\mathbb{Q}}$ and $y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}$ such that $q' = x \cdot y^{-1}$, hence we have

$$\begin{aligned} q &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(q') \\ &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(x \cdot y^{-1}) \\ &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(x) \cdot i_{\mathbb{Q} \rightarrow \mathbb{R}}(y^{-1}) \\ &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(x) \cdot (i_{\mathbb{Q} \rightarrow \mathbb{R}}(y))^{-1} \\ &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(x) / i_{\mathbb{Q} \rightarrow \mathbb{R}}(y) \end{aligned} \quad (9.88)$$

As $x \in \mathbb{Z}_{\mathbb{Q}}$ and $y \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}$ there exists by [eq: 9.85] a $x' \in \mathbb{Z}$ and $y' \in \mathbb{Z}$ such that $x = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x')$ and $y = i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y')$. So by [eq: 9.87] we have that $i_{\mathbb{Q} \rightarrow \mathbb{R}}(x) = (i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}})(x') \in \mathbb{Z}_{\mathbb{R}}$ and $i_{\mathbb{Q} \rightarrow \mathbb{R}}(y) = (i_{\mathbb{Q} \rightarrow \mathbb{R}} \circ i_{\mathbb{Z} \rightarrow \mathbb{Q}})(y') \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}$. Combining this with [eq: 9.88] we have that $q \in \{x/y | x \in \mathbb{Z}_{\mathbb{R}} \wedge y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}\}$ proving that

$$\mathbb{Q}_{\mathbb{R}} \subseteq \{x/y | x \in \mathbb{Z}_{\mathbb{R}} \wedge y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}\} \quad (9.89)$$

If $q \in \{x/y | x \in \mathbb{Z}_{\mathbb{R}} \wedge y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}\}$ then there exists $x \in \mathbb{Z}_{\mathbb{R}}$ and $y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}$ such that $q = x/y$. Using [eq: 9.87] there exists $x' \in \mathbb{Z}$ and $y' \in \mathbb{Z} \setminus \{0\}$ such that $x = i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x'))$ and $y = i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y'))$. From [eq: 9.85] it follows that $i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x') \in \mathbb{Z}_{\mathbb{Q}}$ and $i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y') \in \mathbb{Z}_{\mathbb{Q}} \setminus \{0\}$ which by [theorem: 8.33] gives $i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x') \cdot (i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y'))^{-1} \in \mathbb{Q}$. So by [eq: 9.86]

$$i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x') \cdot (i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y'))^{-1}) \in \mathbb{Q}_{\mathbb{R}} \quad (9.90)$$

Further

$$\begin{aligned} i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x') \cdot (i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y'))^{-1}) &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x')) \cdot i_{\mathbb{Q} \rightarrow \mathbb{R}}((i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y'))^{-1}) \\ &= i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(x')) \cdot (i_{\mathbb{Q} \rightarrow \mathbb{R}}(i_{\mathbb{Z} \rightarrow \mathbb{Q}}(y')))^{-1} \\ &= x \cdot y^{-1} \\ &= q \end{aligned}$$

proving by [eq: 9.90] that $q \in \mathbb{Q}_{\mathbb{R}}$. Hence $\{x/y | x \in \mathbb{Z}_{\mathbb{R}} \wedge y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}\} \subseteq \mathbb{Q}_{\mathbb{R}}$ which combined with [eq: 9.89] proves finally that

$$\mathbb{Q}_{\mathbb{R}} = \{x/y | x \in \mathbb{Z}_{\mathbb{R}} \wedge y \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}\} \quad \square$$

Chapter 10

The complex numbers

One problem that exist in the set of real numbers is that the equation $x^2 = -1$ has no solution because $0 < 1 \Rightarrow -1 < 0$ and by [theorems: 9.45, 4.73] $0 \leq x^2$. This problem will be solved by introducing the set of complex numbers. Note that to avoid having to use different symbols for neutral elements, inverse elements, sum, product etc. we use context to derive the meaning of the different symbols.

Context	Expression	Operator
$n, m \in \mathbb{N}_0$	$n+m$	sum in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{N}_0$	$n \cdot m$	product in $\langle \mathbb{N}_0, \cdot \rangle$
$n, m \in \mathbb{N}_0$	$n \leq m$	order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n < m$	strict order in $\langle \mathbb{N}_0, \leq \rangle$
$n, m \in \mathbb{N}_0$	$n - m$	subtraction in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{N}_0, + \rangle$
$n \in \mathbb{N}_0$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{N}_0, \cdot \rangle$
$n \in \mathbb{N}_0$	$-n$	inverse element in $\langle \mathbb{N}_0, + \rangle$
$n, m \in \mathbb{Z}$	$n+m$	sum in $\langle \mathbb{Z}, + \rangle$
$n, m \in \mathbb{Z}$	$n \cdot m$	product in $\langle \mathbb{Z}, \cdot \rangle$
$n, m \in \mathbb{Z}$	$n \leq m$	order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n < m$	strict order in $\langle \mathbb{Z}, \leq \rangle$
$n, m \in \mathbb{Z}$	$n - m$	subtraction in $\langle \mathbb{Z}, - \rangle$
$n \in \mathbb{Z}$	$n + 0$ or $0 + n$	neutral element in $\langle \mathbb{Z}, + \rangle$
$n \in \mathbb{Z}$	$n \cdot 1$ or $1 \cdot n$	neutral element in $\langle \mathbb{Z}, \cdot \rangle$
$n \in \mathbb{Z}$	$-n$	inverse element in $\langle \mathbb{Z}, + \rangle$
$q, r \in \mathbb{Q}$	$q+r$	sum in $\langle \mathbb{Q}, + \rangle$
$q, r \in \mathbb{Q}$	$q \cdot r$	product in $\langle \mathbb{Q}, \cdot \rangle$
$q, r \in \mathbb{Q}$	$q \leq r$	order in $\langle \mathbb{Q}, \leq \rangle$
$q, r \in \mathbb{Q}$	$q < r$	strict order in $\langle \mathbb{Q}, \leq \rangle$
$q, e \in \mathbb{Q}$	$q - r$	subtraction in $\langle \mathbb{Q}, - \rangle$
$q, r \in \mathbb{Q}$	q/r	division in $\langle \mathbb{Q}, \cdot \rangle$
$q \in \mathbb{Q}$	$q + 0$ or $0 + q$	neutral element in $\langle \mathbb{Q}, + \rangle$
$q \in \mathbb{Q}$	$q \cdot 1$ or $1 \cdot q$	neutral element in $\langle \mathbb{Q}, \cdot \rangle$
$q \in \mathbb{Q}$	$-q$	inverse element in $\langle \mathbb{Q}, + \rangle$
$q, r \in \mathbb{R}$	$q+r$	sum in $\langle \mathbb{R}, + \rangle$
$q, r \in \mathbb{R}$	$q \cdot r$	product in $\langle \mathbb{R}, \cdot \rangle$
$q, r \in \mathbb{R}$	$q \leq r$	order in $\langle \mathbb{R}, \leq \rangle$
$q, r \in \mathbb{R}$	$q < r$	strict order in $\langle \mathbb{R}, \leq \rangle$
$q, e \in \mathbb{R}$	$q - r$	subtraction in $\langle \mathbb{R}, - \rangle$
$q, r \in \mathbb{R}$	q/r	division in $\langle \mathbb{R}, \cdot \rangle$
$q \in \mathbb{R}$	$q + 0$ or $0 + q$	neutral element in $\langle \mathbb{R}, + \rangle$
$q \in \mathbb{R}$	$q \cdot 1$ or $1 \cdot q$	neutral element in $\langle \mathbb{R}, \cdot \rangle$
$q \in \mathbb{R}$	$-q$	inverse element in $\langle \mathbb{R}, + \rangle$
$q, r \in \mathbb{C}$	$q+r$	sum in $\langle \mathbb{C}, + \rangle$
$q, r \in \mathbb{C}$	$q \cdot r$	product in $\langle \mathbb{C}, \cdot \rangle$
$q, r \in \mathbb{C}$	$q \leq r$	order in $\langle \mathbb{C}, \leq \rangle$
$q, r \in \mathbb{C}$	$q < r$	strict order in $\langle \mathbb{C}, \leq \rangle$
$q, e \in \mathbb{C}$	$q - r$	subtraction in $\langle \mathbb{C}, - \rangle$
$q, r \in \mathbb{C}$	q/r	division in $\langle \mathbb{C}, \cdot \rangle$
$q \in \mathbb{C}$	$q + 0$ or $0 + q$	neutral element in $\langle \mathbb{C}, + \rangle$
$q \in \mathbb{C}$	$q \cdot 1$ or $1 \cdot q$	neutral element in $\langle \mathbb{C}, \cdot \rangle$
$q \in \mathbb{C}$	$-q$	inverse element in $\langle \mathbb{C}, + \rangle$

10.1 Definition and arithmetic's

Definition 10.1. The space $\mathbb{C}$ of complex numbers together with two operators $+: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ and $\cdot: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ is defined as follows

1. $\mathbb{C} = \mathbb{R}^2 = \mathbb{R} \cdot \mathbb{R} = \{(x, y) | x, y \in \mathbb{R}\}$
2. $+: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ where $(x, y) + (x', y') = (x + x', y + y')$
3. $\cdot: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ where $(x, y) \cdot (x', y') = (x \cdot x' - y \cdot y', x \cdot y' + y \cdot x')$

Just as $\langle \mathbb{R}, +, \cdot \rangle$ is a field we have that $\langle \mathbb{C}, +, \cdot \rangle$ is a field.

Theorem 10.2. $\langle \mathbb{C}, +, \cdot \rangle$ is a field where

1. The additive neutral element is $(0, 0)$
2. Then multiplicative neutral element is $(1, 0)$
3. The additive inverse element of $(x, y) = (-x, -y)$
4. The multiplicative inverse element for $(x, y) \neq (0, 0)$ is

$$(x/(x^2 + y^2), -y/(x^2 + y^2))$$

As usual we use the following notation convention based on context:

1. The additive neutral element is noted as 0.
2. The multiplicative neutral element is noted as 1.
3. The additive inverse of $z \in \mathbb{C}$ is noted as $-z$.
4. The multiplicative inverse of $x \in \mathbb{C} \setminus \{0\}$ is noted as x^{-1} .

Proof. First we prove that $\langle \mathbb{C}, + \rangle$ is a Abelian group:

associativity. If $(x, y), (x', y'), (x'', y'') \in \mathbb{C}$ then we have

$$\begin{aligned} (x, y) + ((x', y') + (x'', y'')) &= (x, y) + (x' + x'', y' + y'') \\ &= (x + (x' + x''), y + (y' + y'')) \\ &= ((x + x') + x'', (y + y') + y'') \\ &= (x + x', y + y') + (x'', y'') \\ &= ((x, y) + (x', y')) + (x'', y'') \end{aligned}$$

commutativity. If $(x, y), (x', y') \in \mathbb{C}$ then

$$(x, y) + (x', y') = (x + x', y + y') = (x' + x, y' + y) = (x', y') + (x, y)$$

neutral element. If $(x, y) \in \mathbb{C}$ then

$$\begin{aligned} (x, y) + (0, 0) &\stackrel{\text{commutativity}}{=} (0, 0) + (x, y) \\ &= (0 + x, 0 + y) \\ &= (x, y) \end{aligned}$$

inverse element. If $(x, y) \in \mathbb{C}$ then

$$\begin{aligned} (x, y) + (-x, -y) &= (-x, -y) + (x, y) \\ &= ((-x) + x, (-y) + y) \\ &= (0, 0) \end{aligned}$$

Next we prove the rest of the axioms for a field for the multiplication

associativity. If $(x, y), (x', y'), (x'', y'') \in \mathbb{C}$ then

$$\begin{aligned} (x, y) \cdot ((x', y') \cdot (x'', y'')) &= \\ (x, y) \cdot (x' \cdot x'' - y' \cdot y'', x' \cdot y'' + y' \cdot x'') &= \\ (x \cdot (x' \cdot x'' - y' \cdot y'') - y \cdot (x' \cdot y'' + y' \cdot x''), x \cdot (x' \cdot y'' + y' \cdot x'') + y \cdot (x' \cdot x'' - y' \cdot y'')) &= \\ (x \cdot (x' \cdot x'') - x \cdot (y' \cdot y'') - y \cdot (x' \cdot y'') - y \cdot (y' \cdot x''), x \cdot (x' \cdot y'') + x \cdot (y' \cdot x'') + y \cdot (x' \cdot x'') - &= \\ y \cdot (y' \cdot y'')) &= \\ \left(\underbrace{(x \cdot x')}_1 \cdot x'' - \underbrace{(x \cdot y') \cdot y''}_2 - \underbrace{(y \cdot x') \cdot y''}_2 - \underbrace{(y \cdot y') \cdot x''}_1, \underbrace{(x \cdot x') \cdot y''}_3 + \right. &= \\ \left. \underbrace{(x \cdot y') \cdot x'' + (y \cdot x') \cdot x''}_4 - \underbrace{(y \cdot y') \cdot y''}_3 \right) &= \end{aligned}$$

$$\begin{aligned}
& \left(\underbrace{(x \cdot x' - y \cdot y') \cdot x''}_1 - \underbrace{(x \cdot y' + y \cdot y') \cdot y''}_2, \underbrace{(x \cdot x' - y \cdot y') \cdot y''}_3 + \underbrace{(x \cdot y' + y \cdot x') \cdot x''}_4 \right) = \\
& \qquad \qquad \qquad (x \cdot x' - y \cdot y', x \cdot y' + y \cdot y') \cdot (x'', y'') = \\
& \qquad \qquad \qquad ((x, y) \cdot (x', y')) \cdot (x'', y'') =
\end{aligned}$$

commutativity. If $(x, y), (x', y') \in \mathbb{C}$ then

$$(x, y) \cdot (x', y') = (x \cdot x' - y \cdot y', x \cdot y' + y \cdot x') = (x' \cdot x - y' \cdot y, y' \cdot x + x' \cdot y) = (x', y') \cdot (x, y)$$

neutral element. If $(x, y) \in \mathbb{C}$ then

$$\begin{aligned}
(x, y) \cdot (1, 0) & \stackrel{\text{commutativity}}{=} (1, 0) \cdot (x, y) \\
& = (1 \cdot x - 0 \cdot y, 1 \cdot y + 0 \cdot x) \\
& = (x, y)
\end{aligned}$$

inverse element. Let $(x, y) \in \mathbb{C} \setminus \{(0, 0)\}$ then $x \neq 0 \vee y \neq 0$, by [theorems: 9.45, 4.73] it follows that $0 < x^2 \vee 0 < y^2$ giving $0 < x^2 + y^2$, so that $(x^2 + y^2) \neq 0$ hence $(x/(x^2 + y^2), -y/(x^2 + y^2))$ is well defined. Now

$$\begin{aligned}
& (x, y) \cdot (x/(x^2 + y^2), -y/(x^2 + y^2)) \stackrel{\text{commutativity}}{=} \\
& (x/(x^2 + y^2), -y/(x^2 + y^2)) \cdot (x, y) = \\
& (x^2/(x^2 + y^2) + y^2/(x^2 + y^2), x \cdot y/(x^2 + y^2) + (-y) \cdot x/(x^2 + y^2)) = \\
& ((x^2 + y^2)/(x^2 + y^2), (x \cdot y - y \cdot x)/(x^2 + y^2)) = \\
& (1, 0) =
\end{aligned}$$

distributivity. If $(x, y), (x', y'), (x'', y'') \in \mathbb{C}$ then

$$\begin{aligned}
(x, y) \cdot ((x', y') + (x'', y'')) & = (x, y) \cdot (x' + x'', y' + y'') \\
& = (x \cdot (x' + x'') - y \cdot (y' + y''), x \cdot (y' + y'') + y \cdot (x' + x'')) \\
& = (x \cdot x' + x \cdot x'' - y \cdot y' - y \cdot y'', x \cdot y' + x \cdot y'' + y \cdot x' + y \cdot x'') \\
& = ((x \cdot x' - y \cdot y') + (x \cdot x'' - y \cdot y''), (x \cdot y' + y \cdot x') + (x \cdot y'' + y \cdot x'')) \\
& = (x \cdot x' - y \cdot y', x \cdot y' + y \cdot x') + (x \cdot x'' - y \cdot y'', x \cdot y'' + y \cdot x'') \\
& = (x, y) \cdot (x', y') + (x, y) \cdot (x'', y'')
\end{aligned}$$

Finally as in $\mathbb{R}$ we have that $1 \neq 0$ [see theorem: 9.32] we have that

$$0 = (0, 0) \neq (1, 0) = 1$$

□

Just as with the integers, rationals and real numbers we introduce the following shorthand notation.

Notation 10.3. If $x, y \in \mathbb{C}$ then we have the following notation conventions:

1. $x + (-y)$ is noted as $x - y$
2. If $y \in \mathbb{C} \setminus \{0\}$ then $x \cdot y^{-1}$ is noted as x/y and y^{-1} is noted as $1/y$

10.2 Embedding of $\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$ in $\mathbb{C}$

10.2.1 Embeddings

Definition 10.4. We define $\mathbb{R}_{\mathbb{C}}, \mathbb{Q}_{\mathbb{C}}, \mathbb{Z}_{\mathbb{C}}, \mathbb{N}_{0, \mathbb{C}}$ as follows

$$\begin{aligned}
\mathbb{R}_{\mathbb{C}} & = \{(x, 0) | x \in \mathbb{R}\} \\
\mathbb{Q}_{\mathbb{C}} & = \{(x, 0) | x \in \mathbb{Q}_{\mathbb{R}}\} \\
\mathbb{Z}_{\mathbb{C}} & = \{(x, 0) | x \in \mathbb{Z}_{\mathbb{R}}\} \\
\mathbb{N}_{0, \mathbb{C}} & = \{(x, 0) | x \in \mathbb{N}_{0, \mathbb{R}}\}
\end{aligned}$$

It turns out that $\mathbb{R}_{\mathbb{C}}, \mathbb{Q}_{\mathbb{C}}, \mathbb{Z}_{\mathbb{C}}, \mathbb{N}_{\mathbb{C}}$ are embeddings of $\mathbb{R}, \mathbb{Q}, \mathbb{Z}$ and $\mathbb{N}_0$ in $\mathbb{C}$.

Theorem 10.5. *We have that*

1. $\mathbb{R}_{\mathbb{C}}$ is a sub-field of $\langle \mathbb{C}, +, \cdot \rangle$ and

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, +, \cdot \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle \text{ defined by } i_{\mathbb{R} \rightarrow \mathbb{C}}(r) = (r, 0)$$

is a field isomorphism.

2. $\mathbb{Q}_{\mathbb{C}}$ is a sub-field of $\langle \mathbb{C}, +, \cdot \rangle$ and if we define $i_{\mathbb{Q} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Q} \rightarrow \mathbb{R}}$ then

$$i_{\mathbb{Q} \rightarrow \mathbb{C}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field isomorphism}$$

3. $\mathbb{Z}_{\mathbb{C}}$ is a sub-ring of $\langle \mathbb{C}, +, \cdot \rangle$ and if we define $i_{\mathbb{Z} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Z} \rightarrow \mathbb{R}}$ then

$$i_{\mathbb{Z} \rightarrow \mathbb{C}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field isomorphism}$$

4. $\mathbb{N}_{0, \mathbb{C}}$ is a sub semi-group of $\langle \mathbb{C}, + \rangle$ and if we define $i_{\mathbb{N}_0 \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}$ then

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, + \rangle \text{ is a group isomorphism}$$

5. $\mathbb{N}_{0, \mathbb{C}}$ is a sub semi-group of $\langle \mathbb{C}, \cdot \rangle$ and if we define $i_{\mathbb{N}_0 \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}$ then

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, \cdot \rangle \text{ is a group isomorphism}$$

6. $\mathbb{N}_{0, \mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}} \subseteq \mathbb{Q}_{\mathbb{C}} \subseteq \mathbb{R}_{\mathbb{C}} \subseteq \mathbb{C}$

7. $\mathbb{N}_{0, \mathbb{C}}, \mathbb{Z}_{\mathbb{C}}$ and $\mathbb{Q}_{\mathbb{C}}$ are denumerable

Proof.

1. If $x \in \mathbb{R}_{\mathbb{C}}$ then $\exists x' \in \mathbb{R}$ such that $x = (x', 0) = i_{\mathbb{R} \rightarrow \mathbb{C}}(x') \in i_{\mathbb{R} \rightarrow \mathbb{C}}(\mathbb{R})$. Also if $x \in i_{\mathbb{R} \rightarrow \mathbb{C}}(\mathbb{R})$ then $\exists x' \in \mathbb{R}$ such that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(x') = (x', 0) \in \mathbb{R}_{\mathbb{C}}$. So we have that

$$\mathbb{R}_{\mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}}(\mathbb{R}) \tag{10.1}$$

Let $x, y \in \mathbb{R}$ then we have

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(x + y) = (x + y, 0) = (x, 0) + (y, 0) = i_{\mathbb{R} \rightarrow \mathbb{C}}(x) + i_{\mathbb{R} \rightarrow \mathbb{C}}(y)$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(x \cdot y) = (x \cdot y, 0) = (x \cdot y - 0 \cdot 0, x \cdot 0 + 0 \cdot y) = (x, 0) \cdot (y, 0) = i_{\mathbb{R} \rightarrow \mathbb{C}}(x) \cdot i_{\mathbb{R} \rightarrow \mathbb{C}}(y)$$

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(1) = (1, 0) = 1$$

proving that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, +, \cdot \rangle \rightarrow \langle \mathbb{C}, +, \cdot \rangle \text{ is a field homeomorphism} \tag{10.2}$$

As $\langle \mathbb{R}, +, \cdot \rangle$ is a field we can use [theorem: 4.70] together with the above proving that $i_{\mathbb{R} \rightarrow \mathbb{C}}(\mathbb{R})$ is a sub-field of $\langle \mathbb{C}, +, \cdot \rangle$. Combining this with [eq: 10.1] gives

$$\mathbb{R}_{\mathbb{C}} \text{ is a sub-field of } \langle \mathbb{C}, +, \cdot \rangle$$

Further if $i_{\mathbb{R} \rightarrow \mathbb{C}}(x) = i_{\mathbb{R} \rightarrow \mathbb{C}}(y)$ then $(x, 0) = (y, 0)$ proving that $x = y$ hence

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \mathbb{R} \rightarrow \mathbb{C} \text{ is injective}$$

so that $i_{\mathbb{R} \rightarrow \mathbb{C}}: \mathbb{R} \rightarrow i_{\mathbb{R} \rightarrow \mathbb{C}}(\mathbb{R}) \stackrel{[\text{eq: 10.1}]}{=} \mathbb{R}_{\mathbb{C}}$ is a bijection. Hence we have that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, +, \cdot \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field isomorphism} \tag{10.3}$$

2. By [theorem: 9.37] we have that

$$\langle \mathbb{Q}_{\mathbb{R}}, +, \cdot \rangle \text{ is a sub-field of } \langle \mathbb{R}, +, \cdot \rangle \text{ and } i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{R}}, +, \cdot \rangle \text{ is a field isomorphism} \tag{10.4}$$

Using then the above together with [eq: 10.3] we have by [theorem: 4.71] that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q})) \text{ is a sub-field of } \langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle \quad (10.5)$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q})), +, \cdot \rangle \text{ is a field isomorphism} \quad (10.6)$$

Now if $x \in i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q})) \exists x' \in \mathbb{Q}$ such that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(x')) = (i_{\mathbb{Q} \rightarrow \mathbb{R}}(x'), 0)$, using [eq: 10.4] we have that $i_{\mathbb{Q} \rightarrow \mathbb{R}}(x') \in \mathbb{Q}_{\mathbb{R}}$ so that $x \in \mathbb{Q}_{\mathbb{C}}$. Also if $x \in \mathbb{Q}_{\mathbb{C}}$ then there exist a $x' \in \mathbb{Q}_{\mathbb{R}}$ such that $x = (x', 0) = i_{\mathbb{R} \rightarrow \mathbb{C}}(x')$, using [eq: 10.4] there exists a $x'' \in \mathbb{Q}$ such that $x' = i_{\mathbb{Q} \rightarrow \mathbb{R}}(x'')$ so that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(x'')) \in i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q}))$. Hence we have that

$$\mathbb{Q}_{\mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q}))$$

which combined with [eqs: 10.5, 10.6] and [theorem: 4.62] proves that

$$\mathbb{Q}_{\mathbb{C}} \text{ is a sub-field of } \langle \mathbb{C}, +, \cdot \rangle$$

and

$$i_{\mathbb{Q} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field isomorphism}$$

3. By [theorem: 9.52] we have that

$$\langle \mathbb{Z}_{\mathbb{R}}, +, \cdot \rangle \text{ is a sub-ring of } \langle \mathbb{R}, +, \cdot \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, +, \cdot \rangle \text{ is a field isomorphism} \quad (10.7)$$

Using then the above together with [eq: 10.3] we have by [theorem: 4.48] that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z})) \text{ is a sub-ring of } \langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle \quad (10.8)$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z})), +, \cdot \rangle \text{ is a ring isomorphism} \quad (10.9)$$

Now if $x \in i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z})) \exists x' \in \mathbb{Z}$ such that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(x')) = (i_{\mathbb{Z} \rightarrow \mathbb{R}}(x'), 0)$, using [eq: 10.7] we have that $i_{\mathbb{Z} \rightarrow \mathbb{R}}(x') \in \mathbb{Z}_{\mathbb{R}}$ so that $x \in \mathbb{Z}_{\mathbb{C}}$. Also if $x \in \mathbb{Z}_{\mathbb{C}}$ then there exist a $x' \in \mathbb{Z}_{\mathbb{R}}$ such that $x = (x', 0) = i_{\mathbb{R} \rightarrow \mathbb{C}}(x')$, using [eq: 10.7] there exists a $x'' \in \mathbb{Z}$ such that $x' = i_{\mathbb{Z} \rightarrow \mathbb{R}}(x'')$ so that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(x'')) \in i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z}))$. Hence we have that

$$\mathbb{Z}_{\mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z}))$$

which combined with [eqs: 10.8, 10.9] and [theorem: 4.17] proves that

$$\mathbb{Z}_{\mathbb{C}} \text{ is a sub-ring of } \langle \mathbb{C}, +, \cdot \rangle$$

and

$$i_{\mathbb{Z} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, +, \cdot \rangle \text{ is a ring isomorphism}$$

4. By [theorem: 9.55] we have that

$$\langle \mathbb{N}_{0, \mathbb{R}}, + \rangle \text{ is a sub semi-group of } \langle \mathbb{R}, + \rangle \text{ and } i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{R}}, + \rangle \text{ is a group isomorphism} \quad (10.10)$$

Using then the above together with [eq: 10.3] we have by [theorem: 4.48] that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)) \text{ is a sub semi-group of } \langle \mathbb{R}_{\mathbb{C}}, + \rangle \quad (10.11)$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)), + \rangle \text{ is a group isomorphism} \quad (10.12)$$

Now if $x \in i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)) \exists x' \in \mathbb{N}_0$ such that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(x')) = (i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(x'), 0)$, using [eq: 10.10] we have that $i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(x') \in \mathbb{N}_{0, \mathbb{R}}$ so that $x \in \mathbb{N}_{0, \mathbb{C}}$. Also if $x \in \mathbb{N}_{0, \mathbb{C}}$ then there exist a $x' \in \mathbb{C}_{\mathbb{R}}$ such that $x = (x', 0) = i_{\mathbb{R} \rightarrow \mathbb{C}}(x')$, using [eq: 10.10] there exists a $x'' \in \mathbb{N}_0$ such that $x' = i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(x'')$ so that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(x'')) \in i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0))$. Hence we have that

$$\mathbb{N}_{0, \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)) \quad (10.13)$$

which combined with [eqs: 10.11,10.12] and [theorem: 4.17] proves that

$$\mathbb{N}_{0,\mathbb{C}} \text{ is a sub-semi-group of } \langle \mathbb{C}, + \rangle$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{C}}, + \rangle \text{ is a group isomorphism}$$

5. By [theorem: 9.55] we have that

$$\langle \mathbb{N}_{0,\mathbb{R}}, \cdot \rangle \text{ is a sub semi-group of } \langle \mathbb{R}, \cdot \rangle \text{ and } i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{R}}, \cdot \rangle \text{ is a group isomorphism} \quad (10.14)$$

Using then the above together with [eq: 10.3] we have by [theorem: 4.48] that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)) \text{ is a sub semi-group of } \langle \mathbb{R}_{\mathbb{C}}, \cdot \rangle \quad (10.15)$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{C} \rightarrow \mathbb{R}}(\mathbb{N}_0)), \cdot \rangle \text{ is a group isomorphism} \quad (10.16)$$

Combining [eq: 10.13] with [eqs: 10.15,10.16] proves that

$$\mathbb{N}_{0,\mathbb{C}} \text{ is a sub-field of } \langle \mathbb{C}, \cdot \rangle$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \cdot \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{C}}, \cdot \rangle \text{ is a group isomorphism}$$

6. By [theorem: 9.56] $\mathbb{N}_{0,\mathbb{R}} \subseteq \mathbb{Z}_{\mathbb{R}} \subseteq \mathbb{Q}_{\mathbb{R}} \subseteq \mathbb{R}$ so that

$$\{(x, 0) | x \in \mathbb{N}_{0,\mathbb{R}}\} \subseteq \{(x, 0) | x \in \mathbb{Z}_{\mathbb{R}}\} \subseteq \{(x, 0) | x \in \mathbb{Q}_{\mathbb{R}}\} \subseteq \{(x, 0) | x \in \mathbb{R}\} \subseteq \mathbb{C}$$

proving that

$$\mathbb{N}_{0,\mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}} \subseteq \mathbb{Q}_{\mathbb{C}} \subseteq \mathbb{R}_{\mathbb{C}} \subseteq \mathbb{C}$$

7. Using (2),(3) and (4) we have that $\mathbb{Q} \approx \mathbb{Q}_{\mathbb{C}}$, $\mathbb{Z} \approx \mathbb{Z}_{\mathbb{C}}$ and $\mathbb{N}_0 \approx \mathbb{N}_{0,\mathbb{C}}$, further by [theorems: 7.53 and 8.44] we have that $\mathbb{N}_0 \approx \mathbb{Z}$ and $\mathbb{N}_0 \approx \mathbb{Q}$. So $\mathbb{N}_0 \approx \mathbb{N}_{0,\mathbb{C}}$, $\mathbb{N}_0 \approx \mathbb{Z}_{\mathbb{C}}$ and $\mathbb{N}_0 \approx \mathbb{Q}_{\mathbb{C}}$. $\square$

Theorem 10.6. $\mathbb{Q}_{\mathbb{C}} = \{n/m | n \in \mathbb{Z}_{\mathbb{C}} \wedge m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}\}$ where n/m is a shorthand for $n \cdot m^{-1}$ [see notation: 10.3]

Proof. Let $x \in \mathbb{Q}_{\mathbb{C}}$ then by definition there exists a $q \in \mathbb{Q}_{\mathbb{R}}$ such that $x = (q, 0) \stackrel{[\text{theorem: 10.5}]}{=} i_{\mathbb{R} \rightarrow \mathbb{C}}(q)$. Using then [theorem: 9.57] we have $q = n \cdot m^{-1}$ where $n \in \mathbb{Z}_{\mathbb{R}}$ and $m \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}$. So we have

$$\begin{aligned} x &= i_{\mathbb{R} \rightarrow \mathbb{C}}(q) \\ &= i_{\mathbb{R} \rightarrow \mathbb{C}}(n \cdot m^{-1}) \\ &= i_{\mathbb{R} \rightarrow \mathbb{C}}(n) \cdot i_{\mathbb{R} \rightarrow \mathbb{C}}(m^{-1}) \\ &\stackrel{[\text{theorem: 4.70}]}{=} i_{\mathbb{R} \rightarrow \mathbb{C}}(n) \cdot (i_{\mathbb{R} \rightarrow \mathbb{C}}(m))^{-1} \\ &\stackrel{[\text{theorem: 10.5}]}{=} (n, 0) \cdot (m, 0)^{-1} \end{aligned}$$

which, as by definition of $\mathbb{Z}_{\mathbb{C}}$ $(n, 0) \in \mathbb{Z}_{\mathbb{C}}$ and $(m, 0) \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}$, proves that $x \in \{n/m | n \in \mathbb{Z}_{\mathbb{C}} \wedge m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}\}$. Hence

$$\mathbb{Q}_{\mathbb{C}} \subseteq \{n/m | n \in \mathbb{Z}_{\mathbb{C}} \wedge m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}\} \quad (10.17)$$

If $x \in \{n/m | n \in \mathbb{Z}_{\mathbb{C}} \wedge m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}\}$ then there exists $n \in \mathbb{Z}_{\mathbb{C}}$ and $m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}$ such that $x = n \cdot m^{-1}$. So there exists $n' \in \mathbb{Z}_{\mathbb{R}}$ and $m' \in \mathbb{Z}_{\mathbb{R}} \setminus \{0\}$ such that $n = (n', 0)$ and $m = (m', 0)$. Now by [theorem: 9.57] $n' \cdot m'^{-1} \in \mathbb{Q}_{\mathbb{R}}$ so that $(n' \cdot m'^{-1}, 0) \in \mathbb{Q}_{\mathbb{C}}$. As

$$\begin{aligned} (n' \cdot m'^{-1}, 0) &\stackrel{[\text{theorem: 10.5}]}{=} i_{\mathbb{R} \rightarrow \mathbb{C}}(n' \cdot m'^{-1}) \\ &= i_{\mathbb{R} \rightarrow \mathbb{C}}(n') \cdot i_{\mathbb{R} \rightarrow \mathbb{C}}(m'^{-1}) \\ &\stackrel{[\text{theorem: 4.70}]}{=} i_{\mathbb{R} \rightarrow \mathbb{C}}(n') \cdot (i_{\mathbb{R} \rightarrow \mathbb{C}}(m'))^{-1} \\ &= n \cdot m^{-1} \\ &= x \end{aligned}$$

proving that $x \in \mathbb{Q}_{\mathbb{C}}$. Hence $\{n/m \mid n \in \mathbb{Z}_{\mathbb{C}} \wedge m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}\} \subseteq \mathbb{Q}_{\mathbb{C}}$ which combined with [eq: 10.17]

$$\mathbb{Q}_{\mathbb{C}} = \{n/m \mid n \in \mathbb{Z}_{\mathbb{C}} \wedge m \in \mathbb{Z}_{\mathbb{C}} \setminus \{0\}\} \quad \square$$

10.2.2 Order relation

As $\langle \mathbb{R}, \leq \rangle$ is totally ordered by [theorem: 9.43] we could use [theorem: 3.37] to define a lexical order on $\mathbb{C}$. However we can't guarantee $\langle \mathbb{C}, +, \cdot \rangle$ is a ordered field. The proof is by contradiction, so assume that $\langle \mathbb{C}, +, \cdot, \leq \rangle$ is a ordered field then by [theorem: 4.73]

1. If $x < y$ then $x + z < y + z$
2. If $x < y$ and $0 < z$ then $x \cdot z < y \cdot z$

Now for $i = (0, 1)$ we have that $i \cdot i = (0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1$ and as $\langle \mathbb{C}, \leq \rangle$ must be totally ordered we have for i either:

$0 < i$. Then by (2) $0 < i \cdot i = -1$ so that $0 < -1$, hence by (1) we have $1 = 0 + 1 < 1 + (-1) = 0$ or $1 < 0$. But then by (2) we have $i = 1 \cdot i < 0 \cdot i = i$ or $i < 0$ contradicting $0 > i$.

$i < 0$. Then by (1) we have that $0 = i + (-i) < 0 + (-i) = -i$ so that $0 < -i$. Hence using (2) we have that $0 < (-i) \cdot (-i) \stackrel{[\text{theorem: 4.40}]}{=} i \cdot i = -1$ or $0 < -1$ but then by (1) $1 = 0 + 1 < (-1) + 1 < 0$ or $1 < 0$. But then by (2) we have $-i = 1 \cdot (-i) < 0 \cdot (-i) = 0$ or $-i < 0$ contradicting that we found that $0 < -i$.

However we can still have a order relation on the sub-field $\mathbb{R}_{\mathbb{C}}$ that satisfies (1) and (2) as will be showed in the following.

Definition 10.7. The relation $\leq_{\mathbb{C}} \subseteq \mathbb{R}_{\mathbb{C}} \times \mathbb{R}_{\mathbb{C}}$ is defined by

$$\leq_{\mathbb{C}} = \{((x, 0), (y, 0)) \in \mathbb{R}_{\mathbb{C}} \mid x \leq y\}$$

Note: that in $x \leq y$ we use the order of $\langle \mathbb{R}, \leq \rangle$

Theorem 10.8. Using the above order relation we have that

1. $\langle \mathbb{R}_{\mathbb{C}}, \leq_{\mathbb{C}} \rangle$ is a totally ordered set
2. $i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq_{\mathbb{C}} \rangle$ is a order isomorphism

Proof.

1. We have

reflexivity. If $(x, 0) \in \mathbb{R}_{\mathbb{C}}$ then as $\langle \mathbb{R}, \leq \rangle$ is totally ordered we have $x \leq x$ so that $(x, 0) \leq_{\mathbb{C}} (x, 0)$

anti symmetry. If $(x, 0) \leq_{\mathbb{C}} (y, 0) \wedge (y, 0) \leq_{\mathbb{C}} (x, 0)$ then $x \leq y \wedge y \leq x$ which, as $\langle \mathbb{R}, \leq \rangle$ is totally ordered, proves that $x = y$ hence $(x, 0) = (y, 0)$.

transitivity. If $(x, 0) \leq_{\mathbb{C}} (y, 0) \wedge (y, 0) \leq_{\mathbb{C}} (z, 0)$ then $x \leq y \wedge y \leq z$ which, as $\langle \mathbb{R}, \leq \rangle$ is totally ordered, proves that $x \leq z$, hence $(x, 0) \leq_{\mathbb{C}} (z, 0)$

totally order. If $(x, 0), (y, 0) \in \mathbb{R}_{\mathbb{C}}$ then we have as $\langle \mathbb{R}, \leq \rangle$ is totally ordered that either $x \leq y \Rightarrow (x, 0) \leq_{\mathbb{C}} (y, 0)$ or $y \leq x \Rightarrow (y, 0) \leq_{\mathbb{C}} (x, 0)$.

2. First by [theorem: 10.5] $i_{\mathbb{R} \rightarrow \mathbb{C}}: \mathbb{R} \rightarrow \mathbb{R}_{\mathbb{C}}$ is a bijection and second

$$\begin{aligned} i_{\mathbb{R} \rightarrow \mathbb{R}_{\mathbb{C}}}(x) \leq_{\mathbb{C}} i_{\mathbb{R} \rightarrow \mathbb{C}}(y) &\Leftrightarrow (x, 0) \leq_{\mathbb{C}} (y, 0) \\ &\stackrel{\text{definition}}{\Leftrightarrow} x \leq y \end{aligned} \quad \square$$

Theorem 10.9. We have that

1. $i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq_{\mathbb{C}} \rangle$ is a order isomorphism

2. $i_{\mathbb{Q} \rightarrow \mathbb{C}}: \langle \mathbb{Q}, \leq \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{C}}, \leq \rangle$ is a order isomorphism
3. $i_{\mathbb{Z} \rightarrow \mathbb{C}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, \leq \rangle$ is a order isomorphism
4. $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, \leq \rangle$ is a order isomorphism

Proof.

1. This is expressed by [theorem: 10.8].
2. Using (1) and [theorem: 9.49] we have that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq \rangle \text{ and } i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, \leq \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{R}}, \leq \rangle \text{ are order isomorphisms}$$

so using [theorem 3.52]

$$i_{\mathbb{Q} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, \leq \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q})), \leq \rangle \text{ is a order isomorphism}$$

As by [theorem: 10.5] $i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Q} \rightarrow \mathbb{R}}(\mathbb{Q})) = \mathbb{Q}_{\mathbb{C}}$ we have that

$$i_{\mathbb{Q} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Q} \rightarrow \mathbb{R}}: \langle \mathbb{Q}, \leq \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{C}}, \leq \rangle \text{ is a order isomorphism}$$

3. Using (1) and [theorem: 9.52] we have that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq \rangle \text{ and } i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{R}}, \leq \rangle \text{ are order isomorphisms}$$

so using [theorem 3.52]

$$i_{\mathbb{Z} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z})), \leq \rangle \text{ is a order isomorphism}$$

As by [theorem: 10.5] $i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{Z} \rightarrow \mathbb{R}}(\mathbb{Z})) = \mathbb{Z}_{\mathbb{C}}$ we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{Z} \rightarrow \mathbb{R}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, \leq \rangle \text{ is a order isomorphism}$$

4. Using (1) and [theorem: 9.55] we have that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq \rangle \text{ and } i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{R}}, \leq \rangle \text{ are order isomorphisms}$$

so using [theorem 3.52]

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)), \leq \rangle \text{ is a order isomorphism}$$

As by [theorem: 10.5] $i_{\mathbb{R} \rightarrow \mathbb{C}}(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(\mathbb{N}_0)) = \mathbb{N}_{0, \mathbb{C}}$ we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}} = i_{\mathbb{R} \rightarrow \mathbb{C}} \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, \leq \rangle \text{ is a order isomorphism} \quad \square$$

Corollary 10.10. We have $0, 1 \in \mathbb{R}_{\mathbb{C}}$ with $0 < 1$

Proof. Using [theorem: 10.5] we have that $i_{\mathbb{R} \rightarrow \mathbb{C}}(0) = 0$ and $i_{\mathbb{R} \rightarrow \mathbb{C}}(1) = 1$ which as in $\mathbb{R}$ $0 < 1$ proves that $0 = i_{\mathbb{R} \rightarrow \mathbb{C}}(0) < i_{\mathbb{R} \rightarrow \mathbb{C}}(1) = 1$ $\square$

Theorem 10.11. $\forall x \in \mathbb{N}_{0, \mathbb{C}}$ we have $0 \leq x$

Proof. By [theorems: 10.5, 10.9] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, + \rangle \text{ is a group isomorphism hence } i_{\mathbb{N}_0}(0) = 0 \quad (10.18)$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, \leq \rangle \text{ is a order isomorphism} \quad (10.19)$$

If $x \in \mathbb{N}_{0, \mathbb{C}}$ we have by [eq: 10.18] that $\exists x' \in \mathbb{N}_0$ such that $x = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x')$. Using [theorem: 5.46] we have that $0 \leq x'$. So we have $0 \underset{[eq: 10.18]}{=} i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(0) \underset{[theorem: 10.18]}{\leq} i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x') = x$ proving that $0 \leq x$. $\square$

We use now the order isomorphism and field isomorphism to transfer the properties of $\langle \mathbb{R}, \leq \rangle$ and $\langle \mathbb{R}, +, \cdot \rangle$ to $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ and $\langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle$.

Definition 10.12. Using the above order relation we can define $\mathbb{R}_{\mathbb{C}}^+$, $\mathbb{R}_{0,\mathbb{C}}^+$, $\mathbb{R}_{\mathbb{C}}^-$ and $\mathbb{R}_{0,\mathbb{C}}^-$ by

1. $\mathbb{R}_{\mathbb{C}}^+ = \{x \in \mathbb{R}_{\mathbb{C}} \mid 0 < x\}$
2. $\mathbb{R}_{0,\mathbb{C}}^+ = \{x \in \mathbb{R}_{\mathbb{C}} \mid 0 \leq x\}$
3. $\mathbb{R}_{\mathbb{C}}^- = \{x \in \mathbb{R}_{\mathbb{C}} \mid x < 0\}$
4. $\mathbb{R}_{0,\mathbb{C}}^- = \{x \in \mathbb{R}_{\mathbb{C}} \mid x \leq 0\}$

Theorem 10.13. $\langle \mathbb{R}_{\mathbb{C}}, +, \cdot, \leq \rangle$ is a ordered field

Proof. Using [theorems: 10.5, 10.9] we have that

$$\langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field}$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq \rangle \text{ is a order isomorphism} \quad (10.20)$$

and

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, +, \cdot \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field isomorphism} \quad (10.21)$$

Then we have

1. If $x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $x < y$ then $\exists \alpha, \beta, \gamma \in \mathbb{R}$ such that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha)$, $y = i_{\mathbb{R} \rightarrow \mathbb{C}}(\beta)$ and $z = i_{\mathbb{R} \rightarrow \mathbb{C}}(\gamma)$. As $x < y \Rightarrow i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha) < i_{\mathbb{R} \rightarrow \mathbb{C}}(\beta)$ we have by [eq: 10.20] that $\alpha < \beta$. By [theorem: 9.45] we have that $\alpha + \gamma < \beta + \gamma$. Using then [eqs: 10.20, 10.21] we have that

$$\begin{aligned} x + z &= i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha) + i_{\mathbb{R} \rightarrow \mathbb{C}}(\gamma) \\ &= i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha + \gamma) \\ &< i_{\mathbb{R} \rightarrow \mathbb{C}}(\beta + \gamma) \\ &= i_{\mathbb{R} \rightarrow \mathbb{C}}(\beta) + i_{\mathbb{R} \rightarrow \mathbb{C}}(\gamma) \\ &= y + z \end{aligned}$$

proving that

$$x + x < y + z$$

2. If $x, y \in \mathbb{R}_{\mathbb{C}}$ with $0 < x$ and $0 < y$ then there exist $\alpha, \beta \in \mathbb{R}$ such that $x = i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha)$ and $y = i_{\mathbb{R} \rightarrow \mathbb{C}}(\beta)$. As $0 < x$ and $0 < y$ we have by [eq: 10.20] that $0 < \alpha$ and $0 < \beta$, using [theorem: 9.45] it follows that $0 < \alpha \cdot \beta$. Using then [eqs: 10.20, 10.21] we have that

$$0 = i_{\mathbb{R} \rightarrow \mathbb{C}}(0) < i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha \cdot \beta) = i_{\mathbb{R} \rightarrow \mathbb{C}}(\alpha) \cdot i_{\mathbb{R} \rightarrow \mathbb{C}}(\beta) = x \cdot y$$

proving that

$$0 < x \cdot y$$

□

As $\langle \mathbb{R}_{\mathbb{C}}, +, \cdot, \leq \rangle$ is a ordered field we have automatically the following properties:

Theorem 10.14. For $\langle \mathbb{R}_{\mathbb{C}}, +, \cdot, \leq \rangle$ we have

1. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ we have $x < y \Leftrightarrow x + z < y + z$
2. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ we have $x \leq y \Leftrightarrow x + z \leq y + z$
3. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ we have $x < y \Leftrightarrow 0 < y + (-x)$
4. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ we have $x \leq y \Leftrightarrow 0 \leq y + (-x)$
5. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ we have $x < y \Leftrightarrow -y < -x$
6. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ we have $x \leq y \Leftrightarrow -y \leq -x$
7. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $0 < z$ we have $x < y \Leftrightarrow x \cdot z < y \cdot z$
8. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $0 < z$ we have $x \leq y \Leftrightarrow x \cdot z \leq y \cdot z$
9. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $0 \leq z$ and $x \leq y$ we have $x \cdot z \leq y \cdot z$

10. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $z < 0$ we have $x < y \Leftrightarrow y \cdot z < x \cdot z$
11. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $z < 0$ we have $x \leq y \Leftrightarrow y \cdot z \leq x \cdot z$
12. $\forall x, y, z \in \mathbb{R}_{\mathbb{C}}$ with $z \leq 0$ and $x \leq y$ we have $y \cdot z \leq x \cdot z$
13. $\forall x \in \mathbb{R}_{\mathbb{C}}$ we have $0 \leq x \cdot x \stackrel{\text{def}}{=} x^2$, further if $0 \neq x$ then $0 < x \cdot x = x^2$
14. $0 \leq 1$ [actually by [corollary: 10.10] we have $0 < 1$]
15. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ with $0 < x < y$ we have that $x^2 < y^2$ where $x^2 = x \cdot x$ and $y^2 = y \cdot y$
16. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ with $0 \leq x \leq y$ we have that $x^2 \leq y^2$ where $x^2 = x \cdot x$ and $y^2 = y \cdot y$
17. $\forall x \in \mathbb{R}_{\mathbb{C}}$ with $0 < x$ we have $0 < x^{-1}$
18. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ we have $0 < x < y \Leftrightarrow 0 < y^{-1} < x^{-1}$
19. $\forall x, y \in \mathbb{R}_{\mathbb{C}}$ we have $0 < x \leq y \Leftrightarrow 0 < y^{-1} \leq x^{-1}$
20. $\forall x \in \mathbb{R}_{\mathbb{C}}$ with $0 < x < 1$ we have $1 < x^{-1}$
21. $\forall x \in \mathbb{R}_{\mathbb{C}}$ with $0 < x \leq 1$ we have $1 \leq x^{-1}$

Proof. This follows from [theorem: 4.73] □

Corollary 10.15. If $x \in \mathbb{R}_{\mathbb{C}}$ then we have

1. $x < x + 1$
2. $x - 1 < x$

Proof.

1. As $0 < 1$ [see corollary: 10.10] we have $x = 0 + x < 1 + x = x + 1$ [see theorem: 10.14].
2. From (1) we have $x < x + 1$ so that $x - 1 = x + (-1) < (x + 1) + (-1) = x$ [see theorem: 10.14]. □

As for conditional completeness we have the following theorems

Theorem 10.16. $\langle \mathbb{Z}_{\mathbb{C}}, \leq \rangle$ is conditional complete [definition: 3.80] in other words

$\forall S \subseteq \mathbb{Z}_{\mathbb{C}}$ with $S \neq \emptyset$ such that $\exists v \in \mathbb{Z}_{\mathbb{C}}$ such that $\forall \alpha \in S$ we have $\alpha \leq v$ we have that $\sup(S)$ exist

Using [theorem: 3.81] this is equivalent with

$\forall S \subseteq \mathbb{Z}_{\mathbb{C}}$ with $S \neq \emptyset$ such that $\exists \lambda \in \mathbb{Z}_{\mathbb{C}}$ such that $\forall \alpha \in S$ we have $\lambda \leq \alpha$ we have that $\inf(S)$ exist

Proof. Using [theorem: 10.9] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{C}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, \leq \rangle \text{ is an order isomorphism}$$

and by [theorem: 7.32] $\langle \mathbb{Z}, \leq \rangle$ is conditional complete. Hence by [theorem: 3.83]

$$\langle \mathbb{Z}_{\mathbb{C}}, \leq \rangle \text{ is conditional complete} \quad \square$$

Theorem 10.17. $\langle \mathbb{Q}_{\mathbb{C}}, \leq \rangle$ is not conditional complete.

Proof. Using [theorems: 10.9 and 3.55] we have that

$$(i_{\mathbb{Q} \rightarrow \mathbb{C}})^{-1}: \langle \mathbb{Q}_{\mathbb{C}}, \leq \rangle \rightarrow \langle \mathbb{Q}, \leq \rangle \text{ is an order isomorphism}$$

Assume that $\langle \mathbb{Q}_{\mathbb{C}}, \leq \rangle$ is conditional complete then by the above and [theorem: 3.83] $\langle \mathbb{Q}, \leq \rangle$ is conditional complete, contradiction the fact that by [theorem: 8.41] $\langle \mathbb{Q}, \leq \rangle$ is not conditional complete. So the assumption is wrong and $\langle \mathbb{Q}_{\mathbb{R}}, \leq \rangle$ is not conditional complete. □

Theorem 10.18. $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is conditional complete [definition: 3.80] in other words

$\forall S \subseteq \mathbb{R}_{\mathbb{C}}$ with $S \neq \emptyset$ such that $\exists v \in \mathbb{R}_{\mathbb{C}}$ such that $\forall \alpha \in S$ we have $\alpha \leq v$ we have that $\sup(S)$ exist

Using [theorem: 3.81] this is equivalent with

$\forall S \subseteq \mathbb{R}_{\mathbb{C}}$ with $S \neq \emptyset$ such that $\exists \lambda \in \mathbb{R}_{\mathbb{C}}$ such that $\forall \alpha \in S$ we have $\lambda \leq \alpha$ we have that $\inf(S)$ exist

Proof. Using [theorem: 10.9] we have that

$$i_{\mathbb{R} \rightarrow \mathbb{C}}: \langle \mathbb{R}, \leq \rangle \rightarrow \langle \mathbb{R}_{\mathbb{C}}, \leq \rangle \text{ is a order isomorphism}$$

and by [theorem: 9.50] $\langle \mathbb{R}, \leq \rangle$ is conditional complete. Hence by [theorem: 3.83]

$$\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle \text{ is conditional complete} \quad \square$$

Theorem 10.19. $\langle \mathbb{N}_{0, \mathbb{C}}, \leq \rangle$ is well ordered [definition: 3.84]

Proof. Using [theorem: 10.9] we have that

$$i_{\mathbb{N}_0 \rightarrow \cdot}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0, \mathbb{C}}, \leq \rangle \text{ is a order isomorphism}$$

As by [theorems: 5.51] $\langle \mathbb{N}_0, \leq \rangle$ is well ordered we have by the above and [theorem: 3.85] that $\langle \mathbb{N}_{0, \mathbb{C}}, \leq \rangle$ is well ordered. $\square$

Corollary 10.20. (Irrational numbers) $\mathbb{Q}_{\mathbb{C}} \subset \mathbb{R}_{\mathbb{C}}$ so that $\mathbb{R}_{\mathbb{C}} \setminus \mathbb{Q}_{\mathbb{C}} \neq \emptyset$. The set $\mathbb{R}_{\mathbb{C}} \setminus \mathbb{Q}$ is called the set of **irrational numbers**.

Proof. By [theorem: 10.17] $\langle \mathbb{Q}_{\mathbb{C}}, \leq \rangle$ is not conditional complete. Hence there exists a non empty $S \subseteq \mathbb{Q}_{\mathbb{C}}$ with a upper bound such that $\{u \in \mathbb{Q}_{\mathbb{C}} | u \text{ is a upper bound of } S\}$ has no least element. As $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is conditional complete we have that $s = \min(\{u \in \mathbb{R}_{\mathbb{C}} | u \text{ is a upper bound of } S\})$ exist. Assume now that $s \in \mathbb{Q}_{\mathbb{C}}$ then if $u \in \{u \in \mathbb{Q}_{\mathbb{C}} | u \text{ is a upper bound of } S\}$ we have, as $\mathbb{Q}_{\mathbb{C}} \subseteq \mathbb{R}_{\mathbb{C}}$, that $u \in \{u \in \mathbb{R}_{\mathbb{C}} | u \text{ is a upper bound of } S\}$ so that $u \leq s$. Hence s is a least element of $\{u \in \mathbb{Q}_{\mathbb{C}} | u \text{ is a upper bound of } S\}$ contradicting the fact that $\{u \in \mathbb{Q}_{\mathbb{C}} | u \text{ is a upper bound of } S\}$ has no least element. So $s \notin \mathbb{Q}_{\mathbb{C}}$ or $\mathbb{R}_{\mathbb{C}} \setminus \mathbb{Q}_{\mathbb{C}} \neq \emptyset$. $\square$

Theorem 10.21. Let $\emptyset \neq S \subseteq \mathbb{R}_{\mathbb{C}}$ then for $-S = \{-s | s \in S\}$ we have

1. If $\sup(S)$ exist then $\inf(-S)$ exist and $\inf(-S) = -\sup(S)$.
2. If $\inf(S)$ exist then $\sup(-S)$ exist and $\sup(-S) = -\inf(S)$

Proof. First

$$\begin{aligned} s \in -S &\Leftrightarrow \exists s' \in S \text{ such that } s = -s' \\ &\Leftrightarrow \exists s' \in S \text{ such that } -s = s' \\ &\Leftrightarrow -s \in S \end{aligned}$$

proving that

$$-S = \{s | -s \in S\}$$

1. Let $s \in -S$ then $-s \in S$ so that $-s \leq \sup(S) \Rightarrow -\sup(S) \leq s$. So $-S$ is bounded below by $-\sup(S)$, as $\mathbb{R}_{\mathbb{C}}$ is conditional complete [see theorem: 10.18] we have that

$$\inf(-S) \text{ exist and } -\sup(S) \leq \inf(-S) \quad (10.22)$$

Assume that $-\sup(S) < \inf(-S)$ then $-\inf(-S) < \sup(S)$ so that by the definition of the supremum there exist a $s \in S$ such that $-\inf(-S) < s \Rightarrow -s < \inf(-S)$ which, as $-s \in -S \Rightarrow \inf(-S) \leq -s$, results in the contradiction $-s < -s$. Hence we must have that $\inf(-S) \leq -\sup(S)$ which combined with [eq: 10.22] gives

$$\inf(-S) = -\sup(S)$$

2. Let $s \in -S$ then $-s \in S$ so that $\inf(S) \leq -s \Rightarrow s \leq -\inf(S)$. So $-S$ is bounded above by $-\inf(S)$, as $\mathbb{R}_{\mathbb{C}}$ is conditional complete [see theorem: 10.18] we have that

$$\sup(-S) \text{ exist and } \sup(-S) \leq -\inf(S) \quad (10.23)$$

Assume that $\sup(-S) < -\inf(S)$ then $\inf(S) < -\sup(-S)$ so that by the definition of the infimum there exist a $s \in S$ such that $s < -\sup(-S) \Rightarrow \sup(-S) < -s$ which, as $-s \in -S \Rightarrow -s \leq \sup(-S)$, results in the contradiction $-s < -s$. Hence we must have that $-\inf(S) \leq \sup(-S)$ which combined with [eq: 10.23] gives

$$\sup(-S) = -\inf(S) \quad \square$$

Theorem 10.22. Let $\emptyset \neq S \subseteq \mathbb{R}_{\mathbb{C}}$ and $\alpha \in \mathbb{R}_{\mathbb{C}}$ with $0 \leq \alpha$ then we have for $\alpha \cdot S = \{\alpha \cdot s \mid s \in S\}$ that

1. If $\sup(S)$ exist then $\sup(\alpha \cdot S)$ exist and $\sup(\alpha \cdot S) = \alpha \cdot \sup(S)$
2. If $\inf(S)$ exist then $\inf(\alpha \cdot S)$ exist and $\inf(\alpha \cdot S) = \alpha \cdot \inf(S)$

Proof.

1. For α we have two possible cases:

$\alpha = 0$. Then $\alpha \cdot S = \{0\}$ so that $\sup(\alpha \cdot S) = 0 = 0 \cdot \sup(S) = \alpha \cdot \sup(S)$.

$0 < \alpha$. As $\sup(S)$ is a upper bound of S we have that $\forall s \in S \ s \leq \sup(S)$. If $x \in \alpha \cdot S$ then there exist a $s \in S$ such that $x = \alpha \cdot s \leq_{0 \leq \alpha} \alpha \cdot \sup(S)$ proving that

$$\alpha \cdot \sup(S) \text{ is a upper bound of } \alpha \cdot S$$

Hence as $\mathbb{R}_{\mathbb{C}}$ is conditional complete [see theorem: 10.18] we have that

$$\sup(\alpha \cdot S) \text{ exist}$$

and as the supremum is the lowest upper bound that

$$\sup(\alpha \cdot S) \leq \alpha \cdot \sup(S) \quad (10.24)$$

Assume that $\sup(\alpha \cdot S) < \alpha \cdot \sup(S)$ then $\frac{1}{\alpha} \cdot \sup(\alpha \cdot S) < \sup(S)$, hence there exist a $s \in S$ such that $\frac{1}{\alpha} \cdot \sup(\alpha \cdot S) < s \Rightarrow \sup(\alpha \cdot S) < \alpha \cdot s$, as $\alpha \cdot s \in \alpha \cdot S$ we have $\alpha \cdot s \leq \sup(\alpha \cdot S)$ giving the contradiction $\alpha \cdot s < \alpha \cdot s$. So we must have that $\alpha \cdot \sup(S) \leq \sup(\alpha \cdot S)$ which combined with [eq: 10.24] proves that

$$\sup(\alpha \cdot S) = \alpha \cdot \sup(S)$$

So in all cases $\sup(\alpha \cdot S)$ exists and $\sup(\alpha \cdot S) = \alpha \cdot \sup(S)$.

2. For α we have two possible cases:

$\alpha = 0$. Then $\alpha \cdot S = \{0\}$ so that $\inf(\alpha \cdot S) = 0 = 0 \cdot \inf(S) = \alpha \cdot \inf(S)$.

$0 < \alpha$. As $\inf(S)$ is a lower bound of S we have that $\forall s \in S \ \inf(S) \leq s$. If $x \in \alpha \cdot S$ then there exist a $s \in S$ such that $\alpha \cdot \inf(S) \leq_{0 \leq \alpha} \alpha \cdot s = x$ proving that

$$\alpha \cdot \inf(S) \text{ is a lower bound of } \alpha \cdot S$$

Hence as $\mathbb{R}_{\mathbb{C}}$ is conditional complete [see theorem: 10.18] we have that

$$\inf(\alpha \cdot S) \text{ exist}$$

and as the infimum is the highest lower bound that

$$\alpha \cdot \inf(S) \leq \inf(\alpha \cdot S) \quad (10.25)$$

Assume that $\alpha \cdot \inf(S) < \inf(\alpha \cdot S)$ then $\inf(S) < \frac{1}{\alpha} \cdot \inf(\alpha \cdot S)$, hence there exist a $s \in S$ such that $s < \frac{1}{\alpha} \cdot \inf(\alpha \cdot S) \Rightarrow \alpha \cdot s < \inf(\alpha \cdot S)$, as $\alpha \cdot s \in \alpha \cdot S$ we have $\inf(\alpha \cdot S) \leq \alpha \cdot s$ giving the contradiction $\alpha \cdot s < \alpha \cdot s$. So we must have that $\inf(\alpha \cdot S) \leq \alpha \cdot \inf(S)$ which combined with [eq: 10.25] proves that

$$\inf(\alpha \cdot S) = \alpha \cdot \inf(S)$$

So in all cases $\inf(\alpha \cdot S)$ exists and $\inf(\alpha \cdot S) = \alpha \cdot \inf(S)$.

$\square$

Theorem 10.23. Let $S, T \subseteq \mathbb{R}_{\mathbb{C}}$ with $S \neq \emptyset \neq T$ then for

$$S + T = \{\alpha + \beta \mid \alpha \in S \wedge \beta \in T\}$$

we have

1. If $\sup(S), \sup(T)$ exists then $\sup(S + T)$ exist and $\sup(S + T) = \sup(S) + \sup(T)$
2. If $\inf(S), \inf(T)$ exists then $\inf(S + T)$ exist and $\inf(S + T) = \inf(S) + \inf(T)$

Proof. First as $S \neq \emptyset \neq T$ there exists $s \in S$ and $t \in T$ so that $s + t \in S + T$ hence

$$S + T \neq \emptyset$$

1. Let $q \in S + T$ then $\exists s \in S$ and $\exists t \in T$ such that $q = s + t$, as $s \leq \sup(S)$ we have $q = s + t \leq \sup(S) + t$, further as $t \leq \sup(T)$ it follows that $\sup(S) + t \leq \sup(S) + \sup(T)$ giving $q \leq \sup(S) + \sup(T)$. So $\sup(S) + \sup(T)$ is a upper bound of $S + T$ which as $S + T \neq \emptyset$ and $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is conditional complete [see theorem: 10.18] proves that

$$\sup(S + T) \text{ exist and } \sup(S + T) \leq \sup(S) + \sup(T) \quad (10.26)$$

Assume now that $\sup(S + T) < \sup(S) + \sup(T)$ then for $\varepsilon = \sup(S) + \sup(T) - \sup(S + T)$ we have $0 < \varepsilon$. So $-\varepsilon < 0$ and as $0 < 2 \Rightarrow 0 < 2^{-1}$ we have that $-(\varepsilon/2) < 0$. So $\sup(S) - \varepsilon/2 < \sup(S)$ and $\sup(T) - \varepsilon/2 < \sup(T)$. As $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is totally ordered we have by [theorem: 3.74] that there exists $s \in S$ and $t \in T$ such that $\sup(S) - \varepsilon/2 < s$ and $\sup(T) - \varepsilon/2 < t$. So

$$\begin{aligned} s + t &> \sup(S) - \varepsilon/2 + \sup(T) - \varepsilon/2 \\ &= \sup(S) + \sup(T) - (\varepsilon + \varepsilon)/2 \\ &= \sup(S) + \sup(T) - \varepsilon \\ &= \sup(S) + \sup(T) - \sup(S) - \sup(T) + \sup(S + T) \\ &= \sup(s + t) \end{aligned} \quad (10.27)$$

As $s + t \in S + T$ we have that $s + t \leq \sup(S + T)$ contradicting [eq: 10.27], so the assumption is wrong and we must have $\sup(S) + \sup(T) \leq \sup(S + T)$ which combined with [eq: 10.26] proves that

$$\sup(S + T) = \sup(S) + \sup(T)$$

2. Let $q \in S + T$ then $\exists s \in S$ and $\exists t \in T$ such that $q = s + t$, as $\inf(S) \leq s$ we have $\inf(S) + t \leq s + t = q$, further as $\inf(T) \leq t$ it follows that $\inf(S) + \inf(T) \leq \inf(S) + t$ giving $\inf(S) + \inf(T) \leq q$. So $\inf(S) + \inf(T)$ is a lower bound of $S + T$ which as $S + T \neq \emptyset$ and $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is conditional complete [see theorem: 10.18] proves that

$$\inf(S + T) \text{ exist and } \inf(S) + \inf(T) \leq \inf(S + T) \quad (10.28)$$

Assume now that $\inf(S) + \inf(T) < \inf(S + T)$ then for $\varepsilon = \inf(S + T) - \inf(S) - \inf(T)$ we have $0 < \varepsilon$. As $0 < 2 \Rightarrow 0 < 2^{-1}$ we have that $0 < \varepsilon/2$. So $\inf(S) < \inf(S) + \varepsilon/2$ and $\inf(T) < \inf(T) + \varepsilon/2$. As $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is totally ordered we have by [theorem: 3.74] that there exists $s \in S$ and $t \in T$ such that $s < \inf(S) + \varepsilon/2$ and $t < \inf(T) + \varepsilon/2$. So

$$\begin{aligned} s + t &< \inf(S) + \varepsilon/2 + \inf(T) + \varepsilon/2 \\ &= \inf(S) + \inf(T) + \varepsilon \\ &= \inf(S) + \inf(T) + \inf(S + T) - \inf(S) - \inf(T) \\ &= \inf(S + T) \end{aligned} \quad (10.29)$$

As $s + t \in S + T$ we have that $\inf(S + T) \leq s + t$ contradicting [eq: 10.29], so the assumption is wrong and we must have $\inf(S + T) \leq \inf(S) + \inf(T)$ which combined with [eq: 10.28] proves that

$$\inf(S + T) = \inf(S) + \inf(T) \quad \square$$

Corollary 10.24. Let $S \subseteq \mathbb{R}$ with $S \neq \emptyset$ and $\alpha \in \mathbb{R}$ then for $S + \alpha = \{s + \alpha \mid s \in S\}$ we have that

1. If $\sup(S)$ exists then $\sup(S + \alpha)$ exists and $\sup(S + \alpha) = \sup(S) + \alpha$

2. If $\inf(S)$ exists then $\inf(S + \alpha)$ exists and $\inf(S + \alpha) = \inf(S) + \alpha$

Proof. First

$$\begin{aligned} x \in S + \alpha &\Leftrightarrow \exists s \in S \text{ such that } x = s + \alpha \\ &\Leftrightarrow \exists s \in S \wedge \exists t \in \{\alpha\} \text{ such that } x = s + t \\ &\Leftrightarrow x \in S + \{\alpha\} \end{aligned}$$

hence we have

$$S + \alpha = S + \{\alpha\}$$

Now we have

1. If $\sup(S)$ exists then by [theorem 10.23] that $\sup(S + \{\alpha\})$ exist and $\sup(S + \{\alpha\}) = \sup(S) + \sup(\{\alpha\})$ which as $S + \{\alpha\} = S + \alpha$ and $\sup(\{\alpha\}) = \alpha$ proves that

$$\sup(S + \alpha) \text{ exist and } \sup(S + \alpha) = \sup(S) + \alpha$$

2. If $\inf$ exists then by [theorem 10.23] that $\inf(S + \{\alpha\})$ exist and $\inf(S + \{\alpha\}) = \inf(S) + \inf(\{\alpha\})$ which as $S + \{\alpha\} = S + \alpha$ and $\inf(\{\alpha\}) = \alpha$ proves that

$$\inf(S + \alpha) \text{ exist and } \inf(S + \alpha) = \inf(S) + \alpha \quad \square$$

We have similar theorems for the minimum and maximum.

Theorem 10.25. Let $S, T \subseteq \mathbb{R}_{\mathbb{C}}$ with $S \neq \emptyset \neq T$ then for

$$S + T = \{\alpha + \beta \mid \alpha \in S \wedge \beta \in T\}$$

we have

1. If $\max(S), \max(T)$ exists then $\max(S + T)$ exist and $\max(S + T) = \max(S) + \max(T)$
2. If $\min(S), \min(T)$ exists then $\min(S + T)$ exist and $\min(S + T) = \min(S) + \min(T)$

Proof.

1. As $\max(S), \max(T)$ exist we have

$$\max(S) \in S \wedge \max(T) \in T \wedge \forall s \in S \ s \leq \max(S) \wedge \forall t \in T \ t \leq \max(T)$$

As $\max(S) \in S$ and $\max(T) \in T$ it follows by definition that

$$\max(S) + \max(T) \in S + T \quad (10.30)$$

Further if $x \in S + T$ then there exists a $s \in S$ and a $t \in T$ such that $x = s + t$, hence, as $s \leq \max(S) \wedge t \leq \max(T)$, we have $x = s + t \leq \max(S) + \max(T)$. This proves that

$$\forall x \in S + T \text{ we have } x \leq \max(S) + \max(T) \quad (10.31)$$

Finally using the definition of the maximum together with [eqs: 10.30, 10.31] we have that

$$\max(S + T) \text{ exists and } \max(S + T) = \max(S) + \max(T)$$

2. As $\min(S), \min(T)$ exist we have

$$\min(S) \in S \wedge \min(T) \in T \wedge \forall s \in S \ \min(S) \leq s \wedge \forall t \in T \ \min(T) \leq t$$

As $\min(S) \in S$ and $\min(T) \in T$ it follows by definition that

$$\min(S) + \min(T) \in S + T \quad (10.32)$$

Further if $x \in S + T$ then there exists a $s \in S$ and a $t \in T$ such that $x = s + t$, hence, as $\min(S) \leq s \wedge \min(T) \leq t$, we have $\min(S) + \min(T) \leq s + t = x$. This proves that

$$\forall x \in S + T \text{ we have } \min(S) + \min(T) \leq x \quad (10.33)$$

Finally using the definition of the minimum together with [eqs: 10.32, 10.33] we have that

$$\min(S + T) \text{ exists and } \min(S + T) = \min(S) + \min(T) \quad \square$$

Corollary 10.26. *Let $\emptyset \neq S \subseteq \mathbb{R}_{\mathbb{C}}$ and $\alpha \in \mathbb{R}_{\mathbb{C}}$ then we have*

1. *If $\max(S)$ exist then $\max(S + \alpha)$ exist and $\max(S + \alpha) = \max(S) + \alpha$*
2. *If $\min(S)$ exist then $\min(S + \alpha)$ exist and $\min(S + \alpha) = \min(S) + \alpha$*

Proof.

1. Define $T = \{\alpha\}$ then $S + T = S + \alpha$ and $\max(T)$ exist and $\max(T) = \alpha$. Hence by [theorem: 10.25] $\max(S + \alpha) = \max(S + T)$ exist and

$$\max(S + \alpha) = \max(S) + \max(\{\alpha\}) = \max(S) + \alpha$$

2. Define $T = \{\alpha\}$ then $S + T = S + \alpha$ and $\min(T)$ exist and $\min(T) = \alpha$. Hence by [theorem: 10.25] $\min(S + \alpha) = \min(S + T)$ exist and

$$\min(S + \alpha) = \min(S) + \min(\{\alpha\}) = \min(S) + \alpha \quad \square$$

Theorem 10.27. *Let $\emptyset \neq S \subseteq \mathbb{R}$ and $\alpha \in \mathbb{R}$ such that $0 \leq \alpha$ then for $\alpha \cdot S = \{\alpha \cdot x \mid x \in S\}$ we have:*

1. $\max(\alpha \cdot S) = \alpha \cdot \max(S)$
2. $\min(\alpha \cdot S) = \alpha \cdot \min(S)$

Proof.

1. First as $\max(S) \in S$ we have that $\alpha \cdot \max(S) \in \alpha \cdot S$. If $x \in \alpha \cdot S$ then there exist a $s \in S$ such that $x = \alpha \cdot s$, as $s \leq \max(S)$ and $0 \leq \alpha$ we have by [theorem: 10.14] that $x = \alpha \cdot s \leq \alpha \cdot \max(S)$ hence $\max(\alpha \cdot S) = \alpha \cdot \max(S)$.
2. First as $\min(S) \in S$ we have that $\alpha \cdot \min(S) \in \alpha \cdot S$. If $x \in \alpha \cdot S$ then there exist a $s \in S$ such that $x = \alpha \cdot s$, as $\min(S) \leq s$ and $0 \leq \alpha$ we have by [theorem: 10.14] that $\alpha \cdot \min(S) \leq \alpha \cdot s = x$ hence $\min(\alpha \cdot S) = \alpha \cdot \min(S)$. $\square$

Theorem 10.28. *Let $x, y \in \mathbb{Z}_{\mathbb{C}}$ with $x < y$ then*

1. $x + 1 \leq y$
2. $x \leq y - 1$

Proof. By [theorems: 10.5 and 10.9] we have that

$$i_{\mathbb{Z} \rightarrow \mathbb{C}}: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, \leq \rangle \text{ is a order isomorphism} \quad (10.34)$$

$$i_{\mathbb{Z} \rightarrow \mathbb{Z}_{\mathbb{C}}}: \langle \mathbb{Z}, +, \cdot \rangle \rightarrow \langle \mathbb{Z}_{\mathbb{C}}, +, \cdot \rangle \text{ is a field isomorphism} \quad (10.35)$$

Let $x, y \in \mathbb{Z}_{\mathbb{C}}$ then by [eq: 10.34] there exists $x', y' \in \mathbb{Z}$ such that

$$x = i_{\mathbb{Z} \rightarrow \mathbb{C}}(x') \wedge y = i_{\mathbb{Z} \rightarrow \mathbb{C}}(y') \wedge x' < y' \quad (10.36)$$

1. Using [theorem: 7.28] we have that $x' + 1 \leq y'$ so that

$$x + 1 \underset{[\text{eq: 10.35}]}{=} i_{\mathbb{Z} \rightarrow \mathbb{C}}(x') + i_{\mathbb{Z} \rightarrow \mathbb{C}}(1) \underset{[\text{eq: 10.34}]}{=} i_{\mathbb{Z} \rightarrow \mathbb{C}}(x' + 1) \leq_{[\text{eq: 10.34}]} i_{\mathbb{Z} \rightarrow \mathbb{C}}(y') = y$$

proving that

$$x + 1 \leq y$$

2. As by (1) we have that $x + 1 \leq y$ hence $x = (x + 1) + (-1) \leq y + (-1) = y - 1$ $\square$

Theorem 10.29. (Archimedean Property) *If $x, y \in \mathbb{R}_{\mathbb{C}}$ with $0 < x$ then $\exists n \in \mathbb{N}_{0, \mathbb{C}}$ such that $y < n \cdot x$*

Proof. For y we have either

$y \leq 0$. Then for $n = 1$ we have $y \leq 0 < x = 1 \cdot x = n \cdot x$ proving that $y < n \cdot x$

$0 < y$. We prove this by contradiction. Assume that $\forall n \in \mathbb{N}_{0,\mathbb{C}}$ we have $n \cdot x \leq y$. Define

$$A = \{n \cdot x \mid n \in \mathbb{N}_{0,\mathbb{C}}\}$$

then $\forall t \in A$ we have $t \leq y$ so that y is an upper bound of A and as $x = 1 \cdot x \in A$ $A \neq \emptyset$. By [theorem: 10.18] $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is conditional complete so that $\sup(A)$ exists. As $0 < x$ we have $-x < 0$ so that $\sup(A) - x < \sup(A)$, given that by [theorem: 10.8] $\langle \mathbb{R}, \leq \rangle$ is totally ordered, we have by [theorem: 3.74] that $\exists t \in A$ such that $\sup(A) - x < t$. Using the definition of A we have then that $\exists n \in \mathbb{N}_{0,\mathbb{C}}$ such that $t = n \cdot x$ hence $\sup(A) - x < n \cdot x$, so that

$$\sup(A) < n \cdot x + x = (n+1) \cdot x \quad (10.37)$$

As $n+1 \in \mathbb{N}_{0,\mathbb{C}}$ we have that $(n+1) \cdot x \in A$ so that $(n+1) \cdot x \leq \sup(A)$ contradicting [eq: 10.37]. So our assumption is wrong hence

$$\exists n \in \mathbb{N}_{0,\mathbb{R}} \text{ such that } y < n \cdot x \quad \square$$

Corollary 10.30. *Let $x \in \mathbb{R}_{\mathbb{C}}$ then we have*

1. $\exists n \in \mathbb{N}_{0,\mathbb{C}}$ such that $x < n$
2. $\exists n \in \mathbb{N}_{0,\mathbb{C}}$ such that $-n < x$
3. $\exists n \in \mathbb{Z}_{\mathbb{C}}$ such that $n \leq x < n+1$
4. $\exists n \in \mathbb{Z}_{\mathbb{C}}$ such that $n < x \leq n+1$
5. $\exists n \in \mathbb{Z}_{\mathbb{C}}$ such that $n-1 \leq x < n$
6. $\exists n \in \mathbb{Z}_{\mathbb{C}}$ such that $n-1 < x \leq n$
7. If $0 < x$ then $\exists n \in \mathbb{N}_{\mathbb{C}}$ such that $0 < 1/n < x$
8. For $y \in \mathbb{R}_{\mathbb{C}}$ with $x < y$ we have that $\exists n \in \mathbb{N}_{\mathbb{C}}$ such that $x < x + \frac{1}{n} < y$
9. $\mathbb{R}_{\mathbb{C}} = \bigcup_{n \in \mathbb{N}_{\mathbb{C}}}]-n, n[$

Proof.

1. As $0 < 1$ [see corollary: 10.10] we have by the Archimedean property [see theorem: 10.29] that there exist a $n \in \mathbb{N}_{0,\mathbb{C}}$ such that $x < n \cdot 1 = n$
2. By (1) there exists a $n \in \mathbb{N}_0$ such that $-x < n$ hence $-n < -(-x) = x$
3. By (1) $A = \{n \in \mathbb{N}_{0,\mathbb{C}} \mid x < n\} \neq \emptyset$ and by the well ordering of $\langle \mathbb{N}_{0,\mathbb{C}}, \leq \rangle$ [see theorem: 10.19] there exists a least element $m \in A$. As $m-1 < m$ [see corollary: 10.15] we have that $m-1 \notin A$ hence $m-1 \leq x$ and as $m \in A$ we have also $x < m$. Take $n = m-1$ then $n \leq x < m = n+1$.
4. Using (2) there exist a $m \in \mathbb{N}_{0,\mathbb{C}}$ such that $m \leq x < m+1$. As $m \leq x$ we have the following possibilities to consider:

$m = x$. Take then $n = m-1$ so that $n+1 = m$ then we have $n < x \leq n+1$

$m < x$. Take then $n = m$ so that $n < x \leq n+1$

5. Using (2) there exist a $m \in \mathbb{N}_{0,\mathbb{C}}$ such that $m \leq x < m+1$, take then $n = m+1$ so that $m = n-1$, hence $n-1 \leq x < n$.
6. Using (3) there exist a $m \in \mathbb{N}_{0,\mathbb{C}}$ such that $m < x \leq m+1$, take then $n = m+1$ so that $m = n-1$, hence $n-1 < x \leq n$.
7. Using the Archimedean Property [see theorem: 10.29] there exists a $n \in \mathbb{N}_{0,\mathbb{C}}$ such that $1 < n \cdot x$. If $n = 0$ we would have $1 < 0$ a contradiction so $0 \neq n$. Using [theorem: 10.11] we have that $0 \leq n$, so that $0 < n$. Applying then [theorem: 10.14] we have $0 < n^{-1} = 1/n$ which using [theorems: 10.14] on $1 < n \cdot x$ gives $0 < 1/n = 1 \cdot n^{-1} < (n \cdot x) \cdot n^{-1} = x$.
8. As $x < y$ we have that $0 < y-x$ hence using (7) there exist a $n \in \mathbb{N}_{\mathbb{C}}$ such that $0 < \frac{1}{n} < y-x$ hence $x = 0 + x < \frac{1}{n} + x < (y-x) + x = y$.

9. Let $x \in \mathbb{R}_{\mathbb{C}}$ then by (1),(2) there exists $n_1, n_2 \in \mathbb{N}_{0,\mathbb{C}}$ such that $x \leq n_1$ and $-n_2 \leq x$. Take $n = \max(n_1, n_2) + 1$ then we have $0 \leq n_1, n_2 \leq \max(n_1, n_2) < \max(n_1, n_2) + 1 = n$ so that $n \in \mathbb{N}_{\mathbb{C}}$ and $-n < -n_1 \leq x \leq n_2 < n \Rightarrow x \in]-n, n[$. Hence $\mathbb{R} \subseteq \bigcup_{n \in \mathbb{N}_{\mathbb{C}}}]-n, n[$ which as $\forall n \in \mathbb{N}_{\mathbb{C}}$ we $] -n, n[\subseteq \mathbb{R}$ proves that

$$\mathbb{R} = \bigcup_{n \in \mathbb{N}}]-n, n[\quad \square$$

Corollary 10.31. *Let $x, y \in \mathbb{R}_{\mathbb{C}}$ then we have*

1. *If $\forall n \in \mathbb{N}_{\mathbb{C}}$ we have that $x \leq y + 1/n$ then $x \leq y$*
2. *If $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ we have that $x \leq y + \varepsilon$ then $x \leq y$*
3. *Let $a \in \mathbb{R}_{0,\mathbb{C}}^+$ then if $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ we have that $x \leq y + \varepsilon \cdot a$ then $x \leq y$*
4. *Let $b \in \mathbb{R}_{\mathbb{C}}^+$ then if $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ with $\varepsilon < b$ we have that $x \leq y + \varepsilon$ then $x \leq y$*
5. *Let $a \in \mathbb{R}_{0,\mathbb{C}}^+, b \in \mathbb{R}_{0,\mathbb{C}}^+$ then if $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ with $\varepsilon < b$ we have that $x \leq y + \varepsilon \cdot a$ then $x \leq y$*
6. *Let $a \in \mathbb{R}_{0,\mathbb{C}}^+$ then if $\forall n \in \mathbb{N}_{\mathbb{C}}$ we have that $x \leq y + a/n$ then $x \leq y$*

Proof.

1. Assume that $y < x$ then we have $0 < x - y$ so by [corollary: 10.30] there exist a $n \in \mathbb{N}_{\mathbb{C}}$ such that $1/n < x - y$. As we also have that $x \leq y + 1/n \Rightarrow x - y \leq 1/n$ we reach the contradiction $1/n < 1/n$. So the assumption is wrong and we must have that

$$x \leq y$$

2. Assume that $y < x$ then we have $0 < x - y$ so by [corollary: 10.30] there exist a $n \in \mathbb{N}_{\mathbb{C}}$ such that $1/n < x - y$. Take $\varepsilon = 1/n$ then we have also $x \leq y + \varepsilon \Rightarrow x - y \leq \varepsilon = \frac{1}{n}$ so we reach the contradiction $1/n < 1/n$. So the assumption is wrong and we must have that

$$x \leq y$$

3. As $a \in \mathbb{R}_{0,\mathbb{C}}^+$ we have two possibilities to consider:

$a = 0$. Then if we take $\varepsilon = 1 \in \mathbb{R}_{\mathbb{C}}^+$ we have from $x \leq y + a \cdot \varepsilon = y + 0 \cdot 1 = y$ that $x \leq y$

$0 < a$. Then $0 < a^{-1} = 1/a$, take $\varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ then $0 < \varepsilon/a$, hence $\delta = \varepsilon/a \in \mathbb{R}_{\mathbb{C}}^+$, by the assumption we have $x \leq y + a \cdot \delta = y + (\varepsilon/a) \cdot a = y + \varepsilon$. So we have $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ that $x \leq y + \varepsilon$ which by (2) proves that

$$x \leq y$$

4. Let $\varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ then we have either:

$b \leq \varepsilon$. Then as $0 < b$ it follow that $0 < \frac{b}{2} < b \leq \varepsilon$ hence we have by the hypothesis that $x \leq y + \frac{b}{2}$ so that $x \leq y + \varepsilon$.

$\varepsilon < b$. Then by the hypothesis $x \leq y + \varepsilon$

So we have $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ that $x \leq y + \varepsilon$ which by (2) proves that $x \leq y$

5. Let $\varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ then we have either:

$b \leq \varepsilon$. Then as $0 < b$ it follows that $0 < \frac{b}{2} < b \leq \varepsilon$ hence we have by the hypothesis that $x \leq y + \frac{b}{2} \cdot a$ so that $x \leq y + \varepsilon \cdot a$.

$\varepsilon < b$. Then by the hypothesis $x \leq y + \varepsilon \cdot a$

So we have $\forall \varepsilon \in \mathbb{R}_{\mathbb{C}}^+$ that $x \leq y + \varepsilon \cdot a$ which by (3) proves that $x \leq y$

6. As $a \in \mathbb{R}_{0,\mathbb{C}}^+$ we have two possibilities to consider:

$a = 0$. Then if we take $n = 1 \in \mathbb{N}_{\mathbb{C}}$ we have from $x \leq y + a/n = y + (1/1) \cdot 0 = y$ that $x \leq y$

$0 < a$. Take $n \in \mathbb{N}_{\mathbb{C}}$ then

$$0 < n \quad \Rightarrow \quad 0 < a \cdot n \quad \Rightarrow \quad 0 < (a \cdot n)^{-1} = a^{-1} \cdot n^{-1}$$

[theorems: 10.14] [theorems: 10.14]

so that by [corollary: 10.30] there exists a $m \in \mathbb{N}_{\mathbb{R}}$ such that $1/m < a^{-1} \cdot n^{-1}$. By assumption we have now $x \leq y + a/m <_{[\text{theorem: 10.14}]} y + a \cdot (a^{-1} \cdot n^{-1}) = y + 1/n$. So we have $\forall n \in \mathbb{N}_{\mathbb{R}}$ that $x \leq y + 1/n$ which by (1) implies that

$$x \leq y \quad \square$$

The next theorem shows how the embedded rational numbers are dense in the set of real numbers.

Theorem 10.32. (Density Theorem) *If $x \in y \in \mathbb{R}_{\mathbb{C}}$ such that $x < y$ then we have*

1. $\exists q \in \mathbb{Q}_{\mathbb{C}}$ such that $x < q < y$
2. $\exists r \in \mathbb{R}_{\mathbb{C}} \setminus \mathbb{Q}_{\mathbb{C}}$ such that $x < r < y$

In other words if we have two different real numbers then we can always find a rational number and a irrational number that lies between the two real numbers.

Proof.

1. We first prove the case for $0 \leq x$, then we have either

0 < x. From $x < y$ we have that $0 < y - x$ so that by [theorem: 10.14] $0 < (y - x)^{-1}$. Using [corollary: 10.30] there exists a $n \in \mathbb{N}_{0, \mathbb{C}}$ such that $0 < (y - x)^{-1} < n$. As $0 < y - x$ we have that $1 = (y - x)^{-1} \cdot (y - x) < n \cdot (y - x) = n \cdot y - n \cdot x$ so that

$$1 + n \cdot x < n \cdot y \quad (10.38)$$

and from $0 < (y - x)^{-1} < n$ that

$$0 < n \stackrel{[\text{theorem: 10.14}]}{\Rightarrow} 0 < n^{-1} = 1/n \quad (10.39)$$

Using [corollary: 10.30] there exist a $m \in \mathbb{N}_{0, \mathbb{R}}$ such that

$$m - 1 \leq n \cdot x < m \quad (10.40)$$

Multiplying by n^{-1} gives then by [theorem: 10.14] that $x = (n \cdot x) \cdot n^{-1} < m \cdot n^{-1} = m/n$. Take $q = m/n$ then, as $n, m \in \mathbb{N}_{0, \mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}}$ and $n \neq 0$ we have by [theorem: 10.6] that $q \in \mathbb{Q}_{\mathbb{C}}$. So we have

$$x < q \in \mathbb{Q}_{\mathbb{C}} \quad (10.41)$$

From [eq: 10.40] we have that $m \leq n \cdot x + 1 < n \cdot y$ so that $m < n \cdot y$ and by multiplying both sides by n^{-1} that $q = m/n < y$. Combining this with [eq: 10.41] gives finally

$$x < q < y \text{ where } q \in \mathbb{Q}_{\mathbb{C}}$$

x = 0. Then $0 < y$ and by [corollary: 10.30] there exists a $n \in \mathbb{N}_{\mathbb{C}}$ such that $0 < 1/n < y$. As $1, n \in \mathbb{N}_{0, \mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}}$ and $n \neq 0$ we have by [theorem: 10.6] that $1/n \in \mathbb{Q}_{\mathbb{C}}$ hence if we take $q = 1/n$ we have that

$$x < q < y \text{ where } q \in \mathbb{Q}_{\mathbb{C}}$$

So we have proved that

$$\forall x, y \in \mathbb{R}_{\mathbb{C}} \text{ with } 0 \leq x \wedge x < y \text{ there exist a } q \in \mathbb{Q}_{\mathbb{C}} \text{ such that } x < q < y \quad (10.42)$$

So the only case left to prove is where $x < 0$. Then for y we have either

0 < y. Then if we take $q = 0 \in \mathbb{N}_{0, \mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}} \subseteq \mathbb{Q}_{\mathbb{C}}$ so that

$$x < q < y \text{ with } q \in \mathbb{Q}_{\mathbb{C}}$$

y ≤ 0. Then $x < y \leq 0$ or using [theorem: 10.14] $0 \leq -y < -x$. Take $x' = -y$ and $y' = -x$ then $0 \leq x' < y'$ and by [eq: 10.42] there exists a $q' \in \mathbb{Q}_{\mathbb{C}}$ such that $x' < q' < y'$. So $-y < q' < -x$ or $x < -q' < y$. As $\langle \mathbb{Q}_{\mathbb{C}}, + \rangle$ is a sub group of $\langle \mathbb{C}, + \rangle$ it follows that $-q' \in \mathbb{Q}_{\mathbb{C}}$ hence if we take $q = -q'$ then we have

$$x < q < y \text{ where } q \in \mathbb{Q}_{\mathbb{C}}$$

2. Finally we prove the case for the irrational numbers. By [theorem: 10.20] $\mathbb{R}_{\mathbb{C}} \setminus \mathbb{Q}_{\mathbb{C}} \neq \emptyset$ so there exist a $z \in \mathbb{R} \setminus \mathbb{Q}_{\mathbb{R}}$, then from $x < y$ we have that $x + z < y + z$. Using (1) there exist a $r \in \mathbb{Q}_{\mathbb{R}}$ such that $x + z < r < y + z$ or $x < r - z < y$. Take $q = r - z$ then we have $x < q < y$. Assume that $q \in \mathbb{Q}_{\mathbb{R}}$ then as $q, r \in \mathbb{Q}_{\mathbb{R}}$ and $\langle \mathbb{Q}_{\mathbb{R}}, + \rangle$ is a sub group of $\langle \mathbb{R}, + \rangle$ we have that $z = r - q \in \mathbb{Q}_{\mathbb{R}}$ contradicting $z \in \mathbb{R} \setminus \mathbb{Q}_{\mathbb{R}}$. So we must have that $q \in \mathbb{R} \setminus \mathbb{Q}_{\mathbb{R}}$ and we conclude that

$$x < q < y \text{ where } x \in \mathbb{Q}_{\mathbb{R}} \quad \square$$

10.2.3 Recursion and mathematical induction in $\mathbb{C}$

The embedding $\mathbb{N}_{0,\mathbb{C}}$ of $\mathbb{N}_0$ in $\mathbb{C}$ is important because it allows use to extend recursion and induction using $\mathbb{N}_{0,\mathbb{C}}$ instead of $\mathbb{N}_0$.

Definition 10.33. Let $n, m \in \mathbb{N}_{0,\mathbb{C}}$ then we define

$$\{n, \dots, m\} = \{i \in \mathbb{N}_{0,\mathbb{C}} \mid n \leq i \leq m\} \subseteq \mathbb{N}_{0,\mathbb{C}}$$

$$\{n, \dots, \infty\} = \{i \in \mathbb{N}_{0,\mathbb{C}} \mid n \leq i\} \subseteq \mathbb{N}_{0,\mathbb{C}}$$

Lemma 10.34. Let $n, m \in \mathbb{N}_{0,\mathbb{C}}$ then we have

1. $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, m\}) = \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\}$
2. $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{m, \dots, \infty\}) = \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m), \dots, \infty\}$
3. $\mathbb{N}_{0,\mathbb{C}} = \{0, \dots, \infty\}$

Proof. Using [theorems: 10.9] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, \leq \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{C}}, \leq \rangle \text{ is a order isomorphism} \quad (10.43)$$

then we have:

1. If $y \in i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, m\})$ then $\exists x \in \{n, \dots, m\}$ such that $y = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x)$. As $x \leq n \wedge x \leq m$ we have by [eq: 10.43] that $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x) \wedge i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)$ proving that $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq y \wedge y \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)$ hence $y \in \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\}$, so we have

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, m\}) \subseteq \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\} \quad (10.44)$$

Further if $y \in \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\} \subseteq \mathbb{N}_{0,\mathbb{C}}$ then $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq y \wedge y \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)$ and by [eq: 10.43] $\exists x \in \mathbb{N}_0$ such that $y = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x)$, hence

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x) \wedge i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m).$$

Using [eq: 10.43] again we have then that $n \leq x \wedge x \leq m$ so that $x \in \{n, \dots, m\}$ proving that $y \in i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, m\})$. So we have $\{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\} \subseteq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, m\})$ which combined with [eq: 10.44] gives

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, m\}) = \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\}$$

2. If $y \in i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, \infty\})$ then $\exists x \in \{n, \dots, \infty\}$ such that $y = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x)$. As $n \leq x$ we have by [eq: 10.43] that $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x) = y$ hence $y \in \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, \infty\}$, so we have

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, \infty\}) \subseteq \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, \infty\} \quad (10.45)$$

Further if $y \in \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, \infty\} \subseteq \mathbb{N}_{0,\mathbb{C}}$ then $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq y$ and by [eq: 10.43] $\exists x \in \mathbb{N}_0$ such that $y = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x)$, hence $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(x)$ Using [eq: 10.43] again we have then that $n \leq x$ so that $x \in \{n, \dots, \infty\}$ proving that $y \in i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, \infty\})$. So we have $\{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, \infty\} \subseteq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, \infty\})$ which combined with [eq: 10.45] gives

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{n, \dots, \infty\}) = \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n), \dots, \infty\}$$

3. Using [note: 5.81] we have that $\mathbb{N}_0 = \{0, \dots, \infty\}$ so that

$$\mathbb{N}_{0,\mathbb{C}} \underset{[\text{eq: 10.43}]}{=} i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\mathbb{N}_0) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{0, \dots, \infty\}) \underset{(2)}{=} \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(0), \dots, \infty\} = \{0, \dots, \infty\} \quad \square$$

Next we state Mathematical Induction for $\mathbb{N}_0, \mathbb{C}$ using Mathematical Induction using $\mathbb{N}_0$.

Theorem 10.35. *Let $k \in \mathbb{N}_0, \mathbb{C}$ and $S \subseteq \{k, \dots, \infty\}$ such that*

1. $k \in S$
2. *If $n \in S$ then $n + 1 \in S$*

then $S = \{k, \dots, \infty\}$

Proof. Using [theorems: 10.5, 3.55] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_0, \mathbb{C}, + \rangle \text{ is a group isomorphisms and } i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(1) = 1 \quad (10.46)$$

Define $T = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(S)$ and take $k' = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(k)$ then we have

$k' \in T$. As $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(k') = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(k)) = k \in S$ it follows that $k' \in T$

$n \in T \Rightarrow n + 1 \in T$. As $n \in T = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(S)$ we have that $n' = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \in S$, by the hypothesis we have then that $n' + 1 \in S$. Hence

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n + 1) \stackrel{[\text{eq: 10.46}]}{=} i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) + i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(1) \stackrel{[\text{eq: 10.46}]}{=} n' + 1 \in S$$

so that $n + 1 \in (i_{\mathbb{N}_0 \rightarrow \mathbb{R}})^{-1}(S) = T$.

Using [theorem: 5.83] it follows that $T = \{k', \dots, \infty\}$ so

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{k', \dots, \infty\}) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(T) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(S)) \stackrel{[\text{eq: 10.46}]}{=} \stackrel{[\text{theorem: 2.57}]}{=} S \quad (10.47)$$

Next

$$\begin{aligned} i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{k', \dots, \infty\}) &\stackrel{[\text{lemma: 10.34}]}{=} \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(k'), \dots, \infty\} \\ &= \{k, \dots, \infty\} \end{aligned}$$

which combined with [eq: 10.47] gives finally

$$S = \{k, \dots, \infty\} \quad \square$$

As a example of using induction we have the following theorem.

Theorem 10.36. *Let $\langle A, \leq \rangle$ be a partial ordered set then we have*

1. *If $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{n, \dots, \infty\}} \subseteq A$ a sequence such that $\forall i \in \{n, \dots, \infty\} \ x_i \leq x_{i+1}$ then $\forall k \in \{n, \dots, \infty\}$ we have $\forall i \in \{0, \dots, k\}$ that $x_i \leq x_k$*
2. *If $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ a finite family such that $\forall i \in \{n, \dots, m-1\}$ $x_i \leq x_{i+1}$ then $\forall k \in \{n, \dots, m\}$ we have $\forall i \in \{n, \dots, k\}$ we have $x_i \leq x_k$.*
3. *If $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{n, \dots, \infty\}} \subseteq A$ a sequence such that $\forall i \in \{n, \dots, \infty\} \ x_{i+1} \leq x_i$ then $\forall k \in \{n, \dots, \infty\}$ we have $\forall i \in \{n, \dots, k\}$ that $x_k \leq x_i$*
4. *If $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ a finite family such that $\forall i \in \{n, \dots, m-1\}$ $x_{i+1} \leq x_i$ then $\forall k \in \{n, \dots, m\}$ we have $\forall i \in \{n, \dots, k\}$ we have $x_k \leq x_i$.*

Proof.

1. We prove this by induction, take

$$S = \{k \in \{n, \dots, \infty\} \mid \forall i \in \{n, \dots, k\} \text{ we have } x_i \leq x_k\}$$

then we have:

$n \in S$. If $i \in \{n, \dots, n\}$ then $i = n$ so that $x_i = x_n \leq x_n$ proving that $n \in S$.

$k \in S \Rightarrow k + 1 \in S$. Let $i \in \{n, \dots, k + 1\}$ then for i we have either:

$i = k + 1$. Then $x_i = x_{k+1} \leq x_{k+1}$

$i \in \{n, \dots, k\}$. Then as $k \in S$ we have $x_i \leq x_k$ and by the hypothesis $x_k \leq x_{k+1}$ it follows that $x_i \leq x_{k+1}$.

so in all cases we have $x_i \leq x_{k+1}$ which proves that $k+1 \in S$

Mathematical induction [theorem: 10.35] proves then that $S = \{n, \dots, \infty\}$ so $\forall k \in \{n, \dots, \infty\}$ we have $\forall i \in \{n, \dots, k\}$ that $x_i \leq x_k$.

2. This is also proved by induction, take

$$S = \{k \in \{n, \dots, \infty\} | k \in \{m+1, \dots, \infty\} \vee \forall i \in \{n, \dots, k\} \text{ we have } x_i \leq x_k\}$$

then we have:

$n \in S$. If $i \in \{n, \dots, n\}$ then $i = n$ and $x_i = x_n \leq x_n$ proving that $n \in S$.

$k \in S \Rightarrow k+1 \in S$. For $k+1$ we have either:

$k+1 \in \{m+1, \dots, \infty\}$. Then $k+1 \in S$

$k+1 \notin \{m+1, \dots, \infty\}$. Then $k < k+1 < m+1$ so that $k \notin \{m+1, \dots, \infty\}$.

For $i \in \{n, \dots, k+1\}$ we have either:

$i = k+1$. Then $x_i = x_{k+1} \leq x_{k+1}$

$i \in \{n, \dots, k\}$. Then as $k \in S$ and $k \notin \{m+1, \dots, \infty\}$ we must have that $x_i \leq x_k$ which as by the hypothesis $x_k \leq x_{k+1}$ proves that $x_i \leq x_{k+1}$.

so in all cases we have $x_i \leq x_{k+1}$ proving that $k+1 \in S$.

Using mathematical induction it follows then that $S = \{n, \dots, \infty\}$. So if $k \in \{n, \dots, m\}$ then $k \in S$ and $k \notin \{m+1, \dots, \infty\}$ so that $\forall i \in \{n, \dots, k\}$ we have $x_i \leq x_k$.

3. We prove this by induction, take

$$S = \{k \in \{n, \dots, \infty\} | \forall i \in \{n, \dots, k\} \text{ we have } x_k \leq x_i\}$$

then we have:

$n \in S$. If $i \in \{n, \dots, n\}$ then $i = n$ so that $x_n \leq x_n = x_i$ proving that $n \in S$.

$k \in S \Rightarrow k+1 \in S$. Let $i \in \{n, \dots, k+1\}$ then for i we have either:

$i = k+1$. Then $x_{k+1} \leq x_{k+1} = x_i$

$i \in \{n, \dots, k\}$. Then as $k \in S$ we have $x_k \leq x_i$ and as by the hypothesis $x_{k+1} \leq x_k$ it follows that $x_{k+1} \leq x_i$.

so in all cases we have $x_{k+1} \leq x_i$ which proves that $k+1 \in S$

Mathematical induction [theorem: 10.35] proves then that $S = \{n, \dots, \infty\}$ so $\forall k \in \{n, \dots, \infty\}$ we have $\forall i \in \{n, \dots, k\}$ that $x_k \leq x_i$.

4. This is also proved by induction, take

$$S = \{k \in \{n, \dots, \infty\} | k \in \{m+1, \dots, \infty\} \vee \forall i \in \{n, \dots, k\} \text{ we have } x_k \leq x_i\}$$

then we have:

$n \in S$. If $i \in \{n, \dots, n\}$ then $i = n$ and $x_n \leq x_n = x_i$ proving that $n \in S$.

$k \in S \Rightarrow k+1 \in S$. For $k+1$ we have either:

$k+1 \in \{m+1, \dots, \infty\}$. Then $k+1 \in S$

$k+1 \notin \{m+1, \dots, \infty\}$. Then $k < k+1 < m+1$ so that $k \notin \{m+1, \dots, \infty\}$.

For $i \in \{n, \dots, k+1\}$ we have either:

$i = k+1$. Then $x_{k+1} \leq x_{k+1} = x_i$

$i \in \{n, \dots, k\}$. Then as $k \in S$ and $k \notin \{m+1, \dots, \infty\}$ we must have that $x_k \leq x_i$ which as by the hypothesis $x_{k+1} \leq x_k$ proves that $x_{k+1} \leq x_i$.

so in all cases we have $x_{k+1} \leq x_i$ proving that $k+1 \in S$.

Using mathematical induction it follows then that $S = \{n, \dots, \infty\}$. So if $k \in \{n, \dots, m\}$ then $k \in S$ and $k \notin \{m+1, \dots, \infty\}$ so that $\forall i \in \{n, \dots, k\}$ we have $x_k \leq x_i$. $\square$

We turn now to recursion.

Theorem 10.37. *Let A be a set, $a \in A$ and $f: A \rightarrow A$ a function then there exist a **unique** function*

$$\lambda: \mathbb{N}_{0,\mathbb{C}} \rightarrow A$$

such that:

1. $\lambda(0) = a$
2. $\forall n \in \mathbb{N}_{0,\mathbb{R}}$ we have $\lambda(n+1) = f(\lambda(n))$

Proof. Using [theorems: 10.5,4.24] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{C}}, + \rangle \text{ and } (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}: \langle \mathbb{N}_{0,\mathbb{C}}, + \rangle \rightarrow \langle \mathbb{N}_0, + \rangle \text{ are group isomorphisms} \quad (10.48)$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(1) = 1 \quad (10.49)$$

Now by [theorem: 5.84] there exist a **unique** function $\beta: \mathbb{N}_0 \rightarrow A$ such that

1. $\beta(0) = a$
2. $\forall n \in \mathbb{N}_0$ we have $\beta(n+1) = f(\beta(n))$

Define $\lambda: \mathbb{N}_{0,\mathbb{C}} \rightarrow A$ by $\lambda = \beta \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}$ then we have:

1. $\lambda(0) = (\beta \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1})(0) = \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(0)) = \beta(0) = a$
2. Let $n \in \mathbb{N}_{0,\mathbb{C}}$ and take $n' = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n) \in \mathbb{N}_0$ then

$$(i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n+1) = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n) + (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(1) \stackrel{[\text{eq: 10.49}]}{=} (i_{\mathbb{N}_0 \rightarrow \mathbb{R}})^{-1}(n) + 1 = n' + 1$$

So that

$$\begin{aligned} \lambda(n+1) &= (\beta \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1})(n+1) \\ &= \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n+1)) \\ &= \beta(n'+1) \\ &= f(\beta(n')) \\ &= f(\beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n))) \\ &= f(\lambda(n)) \end{aligned}$$

Hence we have proved the existence of $\lambda: \mathbb{N}_{0,\mathbb{C}} \rightarrow A$.

Next we prove uniqueness, so assume that there is another $\gamma: \mathbb{N}_{0,\mathbb{C}} \rightarrow A$ such that $\gamma(0) = a$ and $\gamma(n+1) = f(\gamma(n))$. Then $(\gamma \circ i_{\mathbb{N}_0 \rightarrow \mathbb{C}}): \mathbb{N}_0 \rightarrow A$ is such that

$$(\gamma \circ i_{\mathbb{N}_0 \rightarrow \mathbb{C}})(0) = \gamma(i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(0)) = \gamma(0) = a$$

and

$$\begin{aligned} (\gamma \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}})(n+1) &= \gamma(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(n+1)) \\ &= \gamma(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(n) + i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(1)) \\ &= \gamma(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(n) + 1) \\ &= f(\gamma(i_{\mathbb{N}_0 \rightarrow \mathbb{R}}(n))) \\ &= f((\gamma \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}})(n)) \end{aligned}$$

As β is unique we have by the above that $\beta = \gamma \circ i_{\mathbb{N}_0 \rightarrow \mathbb{R}}$, so that

$$\lambda = \beta \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1} = (\gamma \circ i_{\mathbb{N}_0 \rightarrow \mathbb{C}}) \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1} = \gamma \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}} \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}) = \gamma \circ \text{id}_{\mathbb{N}_0,\mathbb{C}} = \gamma$$

proving uniqueness. □

Theorem 10.38. (Iteration) *Let A be a set and $f: A \rightarrow A$ a function then $\forall n \in \mathbb{N}_{0,\mathbb{C}}$ there exist a function*

$$(f)^n: A \rightarrow A$$

such that:

1. $(f)^0 = \text{Id}_A$
2. $(f)^{n+1} = f \circ (f)^n$

Proof. Let $a \in A$ and use the recursion [theorem: 10.37] to find a **unique** function

$$\lambda_a: \mathbb{N}_{0,\mathbb{R}} \rightarrow A \text{ such that } \lambda_a(0) = a \text{ and } \forall n \in \mathbb{N}_{0,\mathbb{R}} \lambda_a(n+1) = f(\lambda_a(n))$$

Define now

$$(f)^n: A \rightarrow A \text{ where } (f)^n(a) = \lambda_a(n)$$

Then we have

1. $\forall a \in A$ we have that $(f)^0(a) = \lambda_a(0) = a$ so that

$$(f)^0 = \text{Id}_A$$

2. $\forall a \in A$ we have that $(f)^{n+1}(a) = \lambda_a(n+1) = f(\lambda_a(n)) = f((f)^n(a)) = (f \circ (f)^n)(a)$ so that

$$(f)^{n+1} = f \circ (f)^n \quad \square$$

Theorem 10.39. Let A be a set, $a \in A$ and $g: \mathbb{N}_{0,\mathbb{R}} \times A \rightarrow A$ then there exist a **unique** function

$$\gamma: \mathbb{N}_{0,\mathbb{R}} \rightarrow A$$

such that:

1. $\lambda(0) = a$
2. $\forall n \in \mathbb{N}_{0,\mathbb{R}} \lambda(n+1) = g(n, \lambda(n))$

Proof. By [theorems: 10.5, 4.24] we have that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}: \langle \mathbb{N}_0, + \rangle \rightarrow \langle \mathbb{N}_{0,\mathbb{C}}, + \rangle \text{ and } (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}: \langle \mathbb{N}_{0,\mathbb{C}}, + \rangle \rightarrow \langle \mathbb{N}_0, + \rangle \text{ are group isomorphisms} \quad (10.50)$$

and

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(1) = 1 \quad (10.51)$$

Define now

$$h: \mathbb{N}_0 \times A \rightarrow A \text{ by } h(n, a) = g((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})(n), a)$$

Using [theorem: 5.86] there exist a $\beta: \mathbb{N}_0 \rightarrow A$ such that

1. $\beta(0) = a$
2. $\forall n \in \mathbb{N}_{0,\mathbb{R}}$ we have $\beta(n+1) = h(n, \beta(n))$

Define now

$$\lambda: \mathbb{N}_{0,\mathbb{C}} \rightarrow A \text{ by } \lambda = \beta \circ (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}$$

then we have:

1. $\lambda(0) = \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(0)) \stackrel{[\text{eq: 10.50}]}{=} \beta(0) = a$
2. If $n \in \mathbb{N}_{0,\mathbb{C}}$ then

$$\begin{aligned} \lambda(n+1) &= \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n+1)) \\ &\stackrel{[\text{eq: 10.50}]}{=} \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n) + (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(1)) \\ &\stackrel{[\text{eq: 10.51}]}{=} \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n) + 1) \\ &= h((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n), \beta((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n))) \\ &= h((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n), \lambda(n)) \\ &= g(i_{\mathbb{N}_0 \rightarrow \mathbb{C}}((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(n)), \lambda(n)) \\ &= g(n, \lambda(n)) \end{aligned}$$

which proves existence. Now for uniqueness, assume that there is a

$$\gamma: \mathbb{N}_{0,\mathbb{C}} \rightarrow A \text{ such that } \beta(0) = a \text{ and } \forall n \in \mathbb{N}_{0,\mathbb{C}} \text{ that } \beta(n+1) = g(n, \beta(n))$$

Define now $S = \{n \in \mathbb{N}_{0,\mathbb{C}} \mid \lambda(n) = \gamma(n)\}$ then we have:

0 $\in S$. As $\lambda(0) = a = \gamma(0)$ it follows that $0 \in S$.

n $\in S \Rightarrow n+1 \in S$. As

$$\lambda(n+1) = g(n, \lambda(n)) \stackrel{n \in S}{=} g(n, \gamma(n)) = \gamma(n+1)$$

we have that $n+1 \in S$

Using mathematical induction [theorem: 10.35] we have $S = \mathbb{N}_{0,\mathbb{C}}$, so $\forall n \in \mathbb{N}_{0,\mathbb{C}}$ we have $n \in S$ hence $\lambda(n) = \gamma(n)$ proving that

$$\lambda = \gamma \quad \square$$

Corollary 10.40. Let A be a set, $a \in A$ and $g: \mathbb{N}_{0,\mathbb{C}} \times A \rightarrow A$ then there exist a **unique** function

$$\lambda: \mathbb{N}_{0,\mathbb{C}} \rightarrow A$$

such that:

1. $\lambda(0) = a$
2. $\forall n \in \{1, \dots, \infty\} \lambda(n) = g(n-1, \lambda(n-1))$

Proof. Using [theorem: 10.39] there exists a **unique** $\lambda: \mathbb{N}_{0,\mathbb{C}} \rightarrow A$ such that

$$\lambda(0) = a \text{ and } \forall n \in \mathbb{N}_0 \lambda(n+1) = g(n, \lambda(n))$$

Let $n \in \{1, \dots, \infty\}$ then $1 \leq n$ so that $n-1 \in \mathbb{N}_{0,\mathbb{C}}$ such that $n = (n-1) + 1$, hence $\lambda(n) = \lambda((n-1) + 1) = g(n-1, \lambda(n-1))$. $\square$

Theorem 10.41. Let A be a set, $a \in A$, $n \in \mathbb{N}_0$ and $g: \{0, \dots, n-1\} \times A \rightarrow A$ a function then there exists a **unique** function $\lambda: \{0, \dots, n\} \rightarrow A$ satisfying

$$\begin{aligned} \lambda(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \lambda(i+1) &= g(i, \lambda(i)) \end{aligned}$$

Proof. Define

$$g': \mathbb{N}_0 \times A \rightarrow A \text{ by } g'(i, x) = \begin{cases} g(i, x) & \text{if } i \in \{0, \dots, n-1\} \\ x & \text{if } i \in \{n, \dots, \infty\} \end{cases}$$

then by [corollary: 5.87] there exists a $\beta: \mathbb{N}_0 \rightarrow A$ such that

$$\beta(0) = a \quad (10.52)$$

$$\forall i \in \mathbb{N}_0 \text{ we have } \beta(i+1) = g'(i, \beta(i)) \quad (10.53)$$

Define now $\lambda: \{0, \dots, n\} \rightarrow A$ by $\lambda = \beta|_{\{0, \dots, n\}}$ then we have

$$\lambda(0) = \beta|_{\{0, \dots, n\}}(0) \stackrel{0 \in \{0, \dots, n\}}{=} \beta(0) \stackrel{[\text{theorem: 10.52}]}{=} a$$

and $\forall i \in \{0, \dots, n-1\}$ we have

$$\begin{aligned} \lambda(i+1) &= \beta|_{\{0, \dots, n\}}(i+1) \\ &\stackrel{i+1 \in \{0, \dots, n\}}{=} \beta(i+1) \\ &\stackrel{[\text{theorem: 10.53}]}{=} g'(i, \beta(i)) \\ &\stackrel{i \in \{0, \dots, n-1\}}{=} g'(i, \lambda(i)) \\ &\stackrel{i \in \{0, \dots, n-1\}}{=} g(i, \lambda(i)) \end{aligned}$$

so that we found a function $\lambda: \{0, \dots, n\} \rightarrow A$ such that

$$\begin{aligned}\lambda(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \lambda(i+1) &= g(i, \lambda(i))\end{aligned}$$

Next we must prove uniqueness so let $\gamma: \{0, \dots, n\} \rightarrow A$ be such that

$$\begin{aligned}\gamma(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \gamma(i+1) &= g(i, \gamma(i))\end{aligned}$$

and define $S = \{i \in \mathbb{N}_0 \mid i \notin \{0, \dots, n\} \vee \lambda(i) = \gamma(i)\}$ then we have:

$0 \in S$. As $\lambda(0) = a = \gamma(0)$ we have $0 \in S$

$i \in S \Rightarrow i+1 \in S$. then for $i+1$ we have either:

$i+1 \in \{0, \dots, n\}$. $i+1 \in \{0, \dots, n\}$. Then $i+1 \leq n$ so that $i < n$ and as $i \in S$ we have $0 \leq i$, so it follows that $i \in \{0, \dots, n\}$. Further

$$\lambda(i+1) = g(i, \lambda(i)) \quad \text{for } i \in \{0, \dots, n\} \text{ and } i \in S \quad g(i, \gamma(i)) = \gamma(i+1)$$

proving that $i+1 \in S$

$i+1 \notin \{0, \dots, n\}$. Then $i+1 \in S$

so in all cases we have $i+1 \in S$.

By mathematical induction [theorem: 5.83] we have that $S = \mathbb{N}_0$. If $i \in \{0, \dots, n\} \subseteq \mathbb{N}_0$ we have $i \in S$ which as $i \in \{0, \dots, n\}$ gives $\lambda(i) = \gamma(i)$ so that $\lambda = \gamma$. $\square$

The three previous theorems gives a way of defining functions by recursions as is expressed in the following two definitions.

Definition 10.42. Let A be a set, $a \in A$ then we can define a function as follows:

$$f: \mathbb{N}_{0, \mathbb{C}} \rightarrow A$$

is defined by:

1. $f(0) = a$
2. $f(n+1) = G(n, \lambda(n))$

where $G(n, \lambda(n))$ is a expression of two parameters. The above is equivalent with the function defined by [theorem: 10.39] where $a \in A$ and $g: \mathbb{N}_{0, \mathbb{C}} \times A \rightarrow A$ is defined by $g(n, x) = G(n, x)$.

Another way to define a recursive function is based on [corollary: 10.40]

Definition 10.43. Let A be a set, $a \in A$ then we define $f: \mathbb{N}_{0, \mathbb{C}} \rightarrow A$ as follows

$$f(n) = \begin{cases} a & \text{if } n=0 \\ G(n-1, f(n-1)) & \text{if } n \in \{1, \dots, \infty\} \end{cases}$$

Which is equivalent with the function defined by [theorem: 10.40] where $a \in A$ and $g: \mathbb{N}_{0, \mathbb{C}} \times A \rightarrow A$ is defined by $g(n, x) = G(n, x)$.

Definition 10.44. Let A be a set, $a \in A$, $n \in \mathbb{N}_0$ then we define the function

$$\lambda: \{0, \dots, n\} \rightarrow A$$

by

$$\begin{aligned}\lambda(0) &= a \\ \forall i \in \{0, \dots, n-1\} \text{ we have } \lambda(i+1) &= G(i, \lambda(i))\end{aligned}$$

where $G(n, \lambda(n))$ is a expression of two parameters. The above is equivalent with the function defined by [theorem: 10.41] where $a \in A$ and $g: \{0, \dots, n-1\} \times A \rightarrow A$ is defined by $g(n, x) = G(n, x)$.

As a application of recursion we show how we can define a product of a natural number and a element of a field by repeating addition.

Definition 10.45. Let $\langle F, \oplus, \odot \rangle$ be a field with additive neutral element e then we define

$$*: \mathbb{N}_{0, \mathbb{C}} \times F \rightarrow F \text{ where } n * f = f_*(n)$$

where $f_*: F \rightarrow F$ is defined by

$$\begin{aligned} f_*(0) &= e \\ f_*(n+1) &= f \oplus f_*(n) \end{aligned}$$

Example 10.46. Let $\langle F, \oplus, \odot \rangle$ be a field with additive neutral element e then

$$\begin{aligned} 0 * f &= f_*(0) = e \\ 1 * f &= f_*(1) = f \oplus f_*(0) = f \oplus e = f \\ 2 * f &= f_*(2) = f \oplus f_*(1) = f \oplus f \\ 3 * f &= f_*(3) = f \oplus f_*(2) = f \oplus f \oplus f \\ &\dots \\ n * f &= \underbrace{f \oplus \dots \oplus f}_n \end{aligned}$$

The above allows us to define a field of characteristics zero, which we will use if talk about determinant functions.

Definition 10.47. Let $\langle F, \oplus, \odot \rangle$ be a field with additive neutral element e and multiplicative unit u then $\langle F, \oplus, \odot \rangle$ is of **characteristics zero** if $\forall n \in \mathbb{N}_{\mathbb{C}}$ we have $n * u \neq e$

10.3 Power in $\mathbb{C}$

We are ready now to define power in $\mathbb{C}$

Definition 10.48. Let $x \in \mathbb{C}$ then $x^{(\cdot)}: \mathbb{N}_{0, \mathbb{C}} \rightarrow \mathbb{C}$ is defined by $n \rightarrow x^{(\cdot)}(n) = x^n$ where

$$\begin{aligned} x^0 &= 1 \\ x^{n+1} &= x \cdot x^n \end{aligned}$$

Note 10.49. Let $x \in \mathbb{C}$ then $x^0 = 1$, $x^1 = x$, $x^2 = x \cdot x$

Proof. $x^0 \stackrel{\text{def}}{=} 1$, $x^1 = x \cdot x^0 = x \cdot 1 = x$ and $x^2 = x \cdot x^1 = x \cdot x$ □

Theorem 10.50. Let $x, y \in \mathbb{R}_{\mathbb{C}}$ then we have

1. If $0 \leq x < y$ then $x^2 < y^2$
2. If $0 \leq x \leq y$ then $x^2 \leq y^2$

Proof. 10.14

1. As $0 \leq x < y$ we have $0 < y$. For x we have as $0 \leq x$ either

$x = 0$. Then by [theorem: 10.14] $0 = 0 \cdot y < y \cdot y = y^2$. Hence $x^2 = 0 \cdot 0 = 0 < y^2$.

$0 < x$. Then multiplying both sides of $x < y$ by x we have by [theorem: 10.14] that $x^2 = x \cdot x < y \cdot x$. Further multiplying both sides of $x < y$ by y we have by [theorem: 10.14] that $x \cdot y < y \cdot y = y^2$, Hence $x^2 < y \cdot x < y^2$.

2. For x, y we have as $0 \leq x \leq y$ either

$x = y$. Then $x^2 = x \cdot x = y \cdot y = y^2$ hence $x^2 \leq y^2$.

$x < y$. Then by (1) we have $x^2 < y^2$ hence $x^2 \leq y^2$. $\square$

Theorem 10.51. *Let $n \in \mathbb{N}_{0,\mathbb{C}}$ then we have*

1. *If $x \in \mathbb{N}_{0,\mathbb{C}}$ then $x^n \in \mathbb{N}_{0,\mathbb{C}}$*
2. *If $x \in \mathbb{Z}_{\mathbb{C}}$ then $x^n \in \mathbb{Z}_{\mathbb{C}}$*
3. *If $x \in \mathbb{Q}_{\mathbb{C}}$ then $x^n \in \mathbb{Q}_{\mathbb{C}}$*
4. *if $x \in \mathbb{R}_{\mathbb{C}}$ then $x^n \in \mathbb{R}_{\mathbb{C}}$*
5. *If $x \in \mathbb{R}^+$ then $x^n \in \mathbb{R}^+$ [in other words if $0 < x$ then $0 < x^n$]*

Proof. This is easily proved by induction [see: 10.35]

1. Take $S_x = \{n \in \mathbb{N}_{0,\mathbb{C}} | x^n \in \mathbb{N}_{0,\mathbb{C}}\}$ then we have

$0 \in S_x$. As $x^0 = 1 \in \mathbb{N}_{0,\mathbb{C}}$ we have that $0 \in S_x$

$n \in S_x \Rightarrow n+1 \in S_x$. As $n \in S$ we have $x^n \in \mathbb{N}_{0,\mathbb{C}}$ and by the hypothesis $x \in \mathbb{N}_{0,\mathbb{C}}$ so using [theorem: 10.5] $x^{n+1} = x \cdot x^n \in \mathbb{N}_{0,\mathbb{C}}$ proving that $n+1 \in S_x$.

2. Take $S_x = \{n \in \mathbb{N}_{0,\mathbb{C}} | x^n \in \mathbb{Z}_{\mathbb{C}}\}$ then we have

$0 \in S_x$. As $x^0 = 1 \in \mathbb{N}_{0,\mathbb{C}} \subseteq \mathbb{Z}_{\mathbb{C}}$ we have that $0 \in S_x$

$n \in S_x \Rightarrow n+1 \in S_x$. As $n \in S$ we have $x^n \in \mathbb{Z}_{\mathbb{C}}$ and by the hypothesis $x \in \mathbb{Z}_{\mathbb{C}}$ so using [theorem: 10.5] $x^{n+1} = x \cdot x^n \in \mathbb{Z}_{\mathbb{C}}$ proving that $n+1 \in S_x$.

3. Take $S_x = \{n \in \mathbb{N}_{0,\mathbb{C}} | x^n \in \mathbb{Q}_{\mathbb{C}}\}$ then we have

$0 \in S_x$. As $x^0 = 1 \in \mathbb{N}_{0,\mathbb{C}} \subseteq \mathbb{Q}_{\mathbb{C}}$ we have that $0 \in S_x$

$n \in S_x \Rightarrow n+1 \in S_x$. As $n \in S$ we have $x^n \in \mathbb{Q}_{\mathbb{C}}$ and by the hypothesis $x \in \mathbb{Q}_{\mathbb{C}}$ so using [theorem: 10.5] $x^{n+1} = x \cdot x^n \in \mathbb{Q}_{\mathbb{C}}$ proving that $n+1 \in S_x$.

4. Take $S_x = \{n \in \mathbb{N}_{0,\mathbb{C}} | x^n \in \mathbb{R}_{\mathbb{C}}\}$ then we have

$0 \in S_x$. As $x^0 = 1 \in \mathbb{N}_{0,\mathbb{C}} \subseteq \mathbb{R}_{\mathbb{C}}$ we have that $0 \in S_x$

$n \in S_x \Rightarrow n+1 \in S_x$. As $n \in S$ we have $x^n \in \mathbb{R}_{\mathbb{C}}$ and by the hypothesis $x \in \mathbb{R}_{\mathbb{C}}$ so using [theorem: 10.5] $x^{n+1} = x \cdot x^n \in \mathbb{R}_{\mathbb{C}}$ proving that $n+1 \in S_x$.

5. Take $S_x = \{n \in \mathbb{N}_{0,\mathbb{C}} | 0 < x^n\}$ then we have

$0 \in S_x$. Using [corollary: 10.10] $0 < 1 = x^0$ so that $0 \in S_x$.

$n \in S_x \Rightarrow n+1 \in S_x$. As $0 < x$ and $n \in S \Rightarrow 0 < x^n$ we have by [theorem: 10.14] that $0 < x \cdot x^n = x^{n+1}$ proving that $n+1 \in S_x$ $\square$

Theorem 10.52. *If $n, m \in \mathbb{N}_{0,\mathbb{C}}$ then $\forall x \in \mathbb{C}$ we have $x^{n+m} = x^n \cdot x^m$*

Proof. This is proved by induction, so let $x \in \mathbb{C}, n \in \mathbb{N}_{0,\mathbb{C}}$ and define

$$S_{n,x} = \{m \in \mathbb{N}_{0,\mathbb{C}} | x^{n+m} = x^n \cdot x^m\}$$

then we have:

$0 \in S_{n,x}$. We have $x^{n+0} = x^n = x^n \cdot 1 = x^n \cdot x^0$ proving that $0 \in S_{n,x}$.

$m \in S_{n,x} \Rightarrow m+1 \in S_{n,x}$. Then

$$\begin{aligned} x^{n+(m+1)} &= x^{(n+m)+1} \\ &= x \cdot x^{(n+m)} \\ &= x^{n+m} \cdot x \\ &\stackrel{m \in S_{n,x}}{=} (x^n \cdot x^m) \cdot x \\ &= x^n \cdot (x^m \cdot x) \\ &= x^n \cdot (x \cdot x^m) \\ &= x^n \cdot x^{m+1} \end{aligned}$$

proving that $m+1 \in S_{n,x}$

So $\forall n \in \mathbb{N}_{0,\mathbb{C}}$ we have by mathematical induction [theorem: 10.35] that $S_{n,x} = \mathbb{N}_{0,\mathbb{C}}$. So if $n, m \in \mathbb{N}_{0,\mathbb{C}}$ then $m \in S_{n,x}$ so that $x^{n+m} = x^n \cdot x^m$. $\square$

Theorem 10.53. *If $n, m \in \mathbb{N}_{\mathbb{C}}$ then $\forall x \in \mathbb{C}$ we have $(x^n)^m = x^{n \cdot m}$*

Proof. This is proved by induction, let $S_{n,x} = \{m \in \mathbb{N} | (x^n)^m = x^{n \cdot m}\}$ then we have:

$1 \in S_{n,x}$. As $(x^n)^1 = x^n = x^{n \cdot 1}$ we have $1 \in S$.

$m \in S_{n,x} \Rightarrow m+1 \in S_{n,x}$. We have

$$(x^n)^{m+1} = (x^n)^m \cdot x^n \stackrel{m \in S}{=} x^{n \cdot m} \cdot x^n \stackrel{[\text{theorem: 10.52}]}{=} x^{n \cdot m + n} = x^{n \cdot (m+1)}$$

proving that $m+1 \in S_{n,x}$. $\square$

Theorem 10.54. *Let $x \in \mathbb{C} \setminus \{0\}$ then $(x^{-1})^n = (x^n)^{-1}$ or in other words $(1/x)^n = 1/x^n$*

Proof. Theorem we prove this by induction, take $S = \{n \in \mathbb{N}_{\mathbb{C}} | (x^{-1})^n = (x^n)^{-1}\}$ then we have:

$0 \in S$. As $(1/x)^0 = 1 = (1)^{-1} = (x^0)^{-1}$ proving that $0 \in S$.

$n \in S \rightarrow n+1 \in S$. We have

$$\begin{aligned} (x^{-1})^{n+1} &= (x^{-1}) \cdot (x^{-1})^n \\ &\stackrel{n \in S}{=} (x^{-1}) \cdot (x^n)^{-1} \\ &\stackrel{[\text{theorem: 4.55}]}{=} (x \cdot x^n)^{-1} \\ &= (x^{n+1})^{-1} \end{aligned}$$

proving that $n+1 \in S$. $\square$

Theorem 10.55. *Let $n \in \mathbb{N}_{0,\mathbb{C}}$ then we have*

1. *If $n \neq 0$ then $x^n = 0 \Leftrightarrow 0$ [note that by definition $0^0 = 1$ so we must have $n \neq 0$]*
2. $1^n = 1$
3. $(-1)^n = 1 \vee (-1)^n = -1$
4. $(-1)^{2 \cdot n} = 1$
5. $(-1)^{2 \cdot n + 1} = -1$

Proof.

1. We have to prove that $\forall n \in \mathbb{N}_{\mathbb{C}}$ we have $x^n = 0 \Leftrightarrow x = 0$

$\Rightarrow$. We prove this by induction so let $S = \{n \in \mathbb{N}_{\mathbb{C}} | \text{If } x^n = 0 \text{ then } x = 0\}$ then we have:

$1 \in S$. This is trivial because $x = x^1 = 0 \Rightarrow x = 0$

$n \in S \rightarrow n+1 \in S$. If $x^{n+1} = 0$ then as $x^{n+1} = x \cdot x^n$ it follows that $0 = x \cdot x^n$.

Assume that $x \neq 0$ then as $\mathbb{C}$ is a field x^{-1} exist so that

$$0 = x^{-1} \cdot 0 = x^{-1} \cdot (x \cdot x^n) = (x^{-1} \cdot x) \cdot x^n = 1 \cdot x^n = x^n \Rightarrow x^n = 0$$

As $m \in S$ it follows that $x = 0$ contradicting the assumption $x \neq 0$, hence the assumption is wrong and we must have $x = 0$. This proves that $n+1 \in S$.

$\Leftarrow$. As $n \in \mathbb{N}_{\mathbb{C}}$ we have that $0^n = 0 \cdot 0^{n-1} = 0$.

so that $0^n = 0^{m+1} = 0 \cdot 0^m = 0$

2. We proceed by induction, so let

$$S = \{n \in \mathbb{N}_{0,\mathbb{C}} | 1^n = 1\}$$

then we have:

$0 \in S$. $1^0 = 1$ by definition, proving that $0 \in S$

$n \in S \Rightarrow n+1 \in S$. $1^{n+1} = 1 \cdot 1^n \underset{n \in S}{=} 1 \cdot 1 = 1$ proving that $n+1 \in S$

3. Again we use induction, so let

$$S = \{n \in \mathbb{N}_{0,\mathbb{C}} \mid (-1)^n = 1 \vee (-1)^n = -1\}$$

then we have:

$0 \in S$. $(-1)^0 = 1$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ we have either:

$(-1)^n = 1$. Then $(-1)^{n+1} = (-1) \cdot (-1)^n = (-1) \cdot 1 = -1$ so the $n+1 \in S$

$(-1)^n = -1$. Then $(-1)^{n+1} = (-1) \cdot (-1)^n = (-1) \cdot (-1) \underset{[\text{theorem: 4.40}]}{=} 1 \cdot 1 = 1$ so that $n+1 \in S$

4. $(-1)^{2 \cdot n} = (-1)^{(1+1) \cdot n} = (-1)^{n+n} \underset{[\text{theorem: 10.52}]}{=} (-1)^n \cdot (-1)^n \underset{[\text{theorem: 4.40}]}{=} 1$ and (3)

5. $(-1)^{2 \cdot n+1} = (-1) \cdot (-1)^{2 \cdot n} \underset{(4)}{=} (-1) \cdot 1 = -1$ □

Theorem 10.56. For $\mathbb{R}_{\mathbb{C}}$ we have

1. $\forall \alpha \in \mathbb{R}_{\mathbb{C}}$ with $0 < \alpha < 1$ and $n \in \{1, \dots, \infty\}$ that $0 < \alpha^n < 1$
2. $\forall \alpha \in \mathbb{R}_{\mathbb{C}}$ with $0 < \alpha < 1$ and $n \in \{2, \dots, \infty\}$ that $0 < \alpha^n < \alpha$
3. $\forall \alpha, \beta \in \mathbb{R}_{\mathbb{C}}$ such that $1 \leq \alpha$ and $n \in \mathbb{N}_{\mathbb{C}}$ we have:
 - a. If $\alpha < \beta$ then $\alpha < \beta^n$
 - b. If $\alpha \leq \beta$ then $\alpha \leq \beta^n$

Proof.

1. We proof this by induction on n . So let $S = \{n \in \{1, \dots, \infty\} \mid 0 < \alpha^n < 1\}$ then we have:

$1 \in S$. As $0 < \alpha < 1$ we have $0 < \alpha^1 < 1$ so that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ we have $0 < \alpha^n < 1$, so using [theorem: 10.14] we have that $0 = 0 \cdot \alpha < \alpha^n \cdot \alpha < 1 \cdot \alpha = \alpha < 1$ or $0 < \alpha^{n+1} < 1$. Hence $n+1 \in S$

proving that $S = \{1, \dots, \infty\}$ or $\forall n \in \{1, \dots, \infty\}$ we have $0 < \alpha^n < 1$

2. As $n \in \{2, \dots, \infty\}$ we have $2 \leq n \Rightarrow 1 = 2 + (-1) \leq n + (-1) = n-1$ so that $(n-1) \in \{1, \dots, \infty\}$. Using (1) we have $0 < \alpha^{n-1} < 1$ which as $0 < \alpha$ gives by [theorem: 10.14]

$$0 < 0 \cdot \alpha < \alpha^{n-1} \cdot \alpha < 1 \cdot \alpha = \alpha.$$

or as $\alpha^{n-1} \cdot \alpha = \alpha^n$ that $0 < \alpha^n < \alpha$.

3. Let $\alpha, \beta \in \mathbb{R}_{\mathbb{C}}$ such that $1 \leq \alpha$ and $n \in \mathbb{N}_{\mathbb{C}}$

a. If $\alpha < \beta$ we have to prove that $\alpha < \beta^n$. Let $S = \{n \in \{1, \dots, \infty\} \mid \alpha < \beta^n\}$ then we have:

$1 \in S$. As $\alpha < \beta = \beta^1$ we have that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. As $1 \leq \alpha < \beta \Rightarrow 1 < \beta$ we have by [theorem: 10.14] that

$$\alpha = 1 \cdot \alpha < \beta \cdot \alpha = \alpha \cdot \beta \tag{10.54}$$

As $n \in S$ we have that $\alpha < \beta^n$ which by [theorem: 10.14] gives $\alpha \cdot \beta < \beta^n \cdot \beta = \beta^{n+1}$, combining this with [eq: 10.54] proves $\alpha < \beta^{n+1}$. So $n+1 \in S$.

Hence $S = \{1, \dots, \infty\} = \mathbb{N}_{\mathbb{C}}$ hence $\forall n \in \mathbb{N}_{\mathbb{C}}$ we have $\alpha < \beta^n$.

b. If $\alpha \leq \beta$ we have to prove that $\alpha \leq \beta^n$. Let $S = \{n \in \{1, \dots, \infty\} \mid \alpha \leq \beta^n\}$ then we have:

$1 \in S$. As $\alpha \leq \beta = \beta^1$ we have that $1 \in S$

$n \in S \Rightarrow n + 1 \in S$. As $1 \leq \alpha \leq \beta \Rightarrow 1 \leq \beta$ we have by [theorem: 10.14] that

$$\alpha = 1 \cdot \alpha \leq \beta \cdot \alpha = \alpha \cdot \beta \quad (10.55)$$

As $n \in S$ we have that $\alpha \leq \beta^n$ which by [theorem: 10.14] gives $\alpha \cdot \beta \leq \beta^n \cdot \beta = \beta^{n+1}$, combining this with [eq: 10.55] proves $\alpha \leq \beta^{n+1}$. So $n + 1 \in S$.

So $S = \{1, \dots, n\} = \mathbb{N}$ hence $\forall n \in \mathbb{N}$ we have $\alpha \leq \beta^n$. $\square$

Theorem 10.57. $\forall n \in \mathbb{N}_{0,\mathbb{C}}$ we have that $n < 2^n$

Proof. This is proved by induction so let $S = \{n \in \mathbb{N}_{0,\mathbb{C}} | n < 2^n\}$ then we have

$0 \in S$. As $n = 0 < 1 = 2^0$ we have that $0 \in S$

$n \in S \Rightarrow n + 1 \in S$. For $n + 1$ we have the following cases to consider:

$n + 1 = 1$. Then $n + 1 = 1 < 2 = 2^1 = 2^{n+1}$ proving that in this case $n + 1 \in S$

$1 < n + 1$. Then by [theorem: 10.28] we have $1 + 1 \leq n + 1$ so that $1 \leq n$, adding n to both sides gives then $n + 1 \leq n + n = (1 + 1) \cdot n = 2 \cdot n$. Further as $n \in S$ we have $n < 2^n$ so that $2 \cdot n < 2 \cdot 2^n = 2^{n+1}$, hence we have $n + 1 < 2^{n+1}$, proving that $n + 1 \in S$. $\square$

Corollary 10.58. $\forall x \in \mathbb{R}_{\mathbb{C}}$ there exists a $n \in \mathbb{N}_{0,\mathbb{C}}$ such that $x < 2^n$

Proof. Let $x \in \mathbb{R}_{\mathbb{C}}$ then by [corollary: 10.30] there exist a $n \in \mathbb{N}_{0,\mathbb{C}}$ such that $x < n$, using [theorem: 10.57] $n < 2^n$ so we have $x < 2^n$. $\square$

Lemma 10.59. Let $x \in \mathbb{R}_{\mathbb{C}}$ with $1 < x$ then $\forall n \in \mathbb{N}_{0,\mathbb{C}}$ we have

$$n \cdot (x - 1) \leq x^n - 1$$

Proof. We prove this by induction, so let $S = \{n \in \mathbb{N}_{0,\mathbb{C}} | n \cdot (x - 1) \leq x^n - 1\}$ then we have:

$0 \in S$. If $n = 0$ then $n \cdot (x - 1) = 0 \cdot (x - 1) = 0 \leq 0 = 1 - 1 = x^0 - 1 = x^n - 1$ proving that $0 \in S$.

$n \in S \Rightarrow n + 1 \in S$. As $n \in S$ we have $n \cdot (x - 1) \leq x^n - 1$ which as $0 < x$ prove that

$$x \cdot n \cdot (x - 1) \leq x \cdot (x^n - 1) \quad (10.56)$$

As $1 < x$ we have that $0 < x - 1 \xRightarrow{0 \leq n} 0 < n \cdot (x - 1)$ so that

$$n \cdot (x - 1) = 1 \cdot n \cdot (x - 1) < x \cdot n \cdot (x - 1)$$

which combined with [eq: 10.56] proves that

$$n \cdot (x - 1) \leq x \cdot (x^n - 1) \quad (10.57)$$

Further

$$\begin{aligned} x^{n+1} - 1 &= x \cdot x^n - 1 \\ &= x \cdot x^n - x + x - 1 \\ &= x \cdot (x^n - 1) + (x - 1) \\ &\geq_{[\text{eq: 10.57}]} n \cdot (x - 1) + (x - 1) \\ &= (n + 1) \cdot (x - 1) \end{aligned}$$

proving that $n + 1 \in S$. $\square$

Theorem 10.60. If $m \in \mathbb{N}_{0,\mathbb{C}}$ and $x \in \mathbb{R}_{\mathbb{C}}$ with $1 < x$ then $\exists n \in \mathbb{N}_{0,\mathbb{C}}$ such that $m < x^n$.

Proof. For $m \in \mathbb{N}_{0,\mathbb{C}}$ we have either:

$m = 0$. Then $m = 0 < 1 = x^0$ proving the theorem in this case.

$m = 1$. Then $m = 1 < x = x^1$ proving the theorem in this case.

$1 < m$. Then $0 < m - 1$ and $0 < x - 1 \Rightarrow 0 < (x - 1)^{-1}$ so that $(m - 1)/(x - 1)$ is defined. By [corollary: 10.30] there exist a $n \in \mathbb{N}_{0,\mathbb{C}}$ such that $(m - 1)/(x - 1) < n$. As $0 < x - 1$ we have $m - 1 < n \cdot (x - 1)$, by [lemma: 10.59] $n \cdot (x - 1) \leq (x^n - 1)$, hence $m - 1 \leq x^n - 1$ or $m < x^n$, proving the theorem in this case. $\square$

Corollary 10.61. *Let $x, y \in \mathbb{R}_{\mathbb{C}}$ with $1 < x$ then there exist a $n \in \mathbb{N}_{\mathbb{C}}$ such that $y < x^n$*

Proof. Using [theorem: 10.30] there exist a $m \in \mathbb{N}_{\mathbb{C}}$ such that $y < m$ and by the previous theorem [theorem: 10.60] there exist a $n \in \mathbb{N}_{0,\mathbb{C}}$ such that $m < x^n$, hence $y < x^n$. $\square$

Theorem 10.62. *If $x \in \mathbb{R}_{\mathbb{C}}$ with $0 < x < 1$ and $n, m \in \mathbb{N}_{\mathbb{C}}$ such that $n < m$ then $x^m < x^n$.*

Proof. We prove this by induction, let $m \in \mathbb{N}_{0,\mathbb{C}}$ and take $S_m = \{n \in \{1, \dots, \infty\} | x^{m+n} < x^m\}$ then we have:

$1 \in S_m$. As $0 < x$ we have by [theorem: 10.51] that $0 < x^m$, hence from $x < 1$ we have that $x^{m+1} = x \cdot x^m < 1 \cdot x^m = x^m$, proving that $1 \in S_m$.

$n \in S_m \Rightarrow n + 1 \in S_m$. As $x < 1$ and by [theorem: 10.51] $0 < x^{m+n}$ we have that

$$x^{m+(n+1)} = x \cdot x^{m+n} < x^{m+n} \xrightarrow{n \in S_m \Rightarrow x^{m+n} < x^m} x^{m+(n+1)} < x^m$$

proving that $n + 1 \in S_m$.

Using mathematical induction [see theorem: 10.35] we have that $S_m = \{1, \dots, \infty\}$. So take $n, m \in \mathbb{N}_{0,\mathbb{R}}$ with $n < m$ then $k = m - n > 0 \Rightarrow k \geq 1$ so that $k \in \{1, \dots, \infty\} = S_n$ hence $x^m = x^{n+k} < x^n$ completing the proof. $\square$

Theorem 10.63. *If $x \in \mathbb{R}_{\mathbb{C}}$ with $1 \leq x$ then $\forall n, m \in \mathbb{N}_{\mathbb{C},0}$ with $n \leq m$ we have $x^n \leq x^m$*

Proof. We prove this by induction so take $S_n = \{m \in \{0, \dots, \infty\} | x^n \leq x^{n+m}\}$ then we have:

$0 \in S_n$. As $x^n = x^{n+0} \Rightarrow x^n \leq x^{n+0}$ proving that $0 \in S$.

$m \in S \Rightarrow m + 1 \in S$. As $1 \leq x$ and $0 < x$ it follows from [theorem: 10.56] that $0 < x^{n+m}$, hence as $1 \leq x$ it follows from [theorem: 10.14] that $x^{n+m} = 1 \cdot x^{n+m} \leq x \cdot x^{n+m} = x^{n+(m+1)}$. Given that $x^n \leq x^{n+m}$ [because $m \in S$] it follows that $x^n \leq x^{n+(m+1)}$, hence $m + 1 \in S$.

Using mathematical induction [see theorem: 10.35] we have that $S_n = \{0, \dots, \infty\}$. So if $n, m \in \mathbb{N}_{\mathbb{C},0}$ with $n \leq m$ then $k = m - n \in \{0, \dots, \infty\} = S_n$ and $x^n \leq x^{n+k} = x^{n+(m-n)} = x^m$ completing the proof. $\square$

Theorem 10.64. *If $\varepsilon \in \mathbb{R}_{\mathbb{C}}^+$, $x \in \mathbb{R}_{\mathbb{C}}$ with $0 < x < 1$ then $\exists N \in \mathbb{N}_{\mathbb{C}}$ such that $\forall n \in \mathbb{N}_{\mathbb{C}}$ $N \leq n$*

$$0 < x^n < \varepsilon$$

Note 10.65. *As $0 < \frac{1}{2} < 1$ it follows from the above that $\exists N \in \mathbb{N}_{\mathbb{C}}$ such that $\forall n \in \mathbb{N}_{\mathbb{C}}$ with $N \leq n$ we have*

$$0 < \frac{1}{2^n} < \varepsilon$$

Proof. As $0 < \varepsilon \Rightarrow \varepsilon \neq 0$ so that $1/\varepsilon$ is defined. By a consequence of the Archimedean property [theorem: 10.30] there exist a $M \in \mathbb{N}_{\mathbb{C}}$ such that

$$0 < 1/\varepsilon < M$$

As $0 < x < 1$ we have $1 < \frac{1}{x}$ so that by [theorem: 10.60] there exist a $N' \in \mathbb{N}_{\mathbb{C},0}$ such that $M < (\frac{1}{x})^{N'}$, hence $0 < 1/\varepsilon < (\frac{1}{x})^{N'} \underset{[\text{theorem: 10.54}]}{=} 1/x^{N'}$ so that

$$0 < x^{N'} < \varepsilon$$

Take $N = N' + 1$ then if $n \in \mathbb{N}_{0,\mathbb{C}}$ with $N \leq n$ then $N' < n$. So we have by [theorem: 10.62] and $0 < x < 1$ that $x^n < x^{N'} < \varepsilon$. As $0 < x$ we have by [theorem: 10.51] that

$$0 < x^n < \varepsilon$$

$\square$

10.4 The square root in $\mathbb{R}_{\mathbb{C}}$

Theorem 10.66. *The function*

$$(\cdot)^2: \mathbb{R}_{0,\mathbb{C}}^+ \rightarrow \mathbb{R}_{0,\mathbb{C}}^+ \text{ defined by } (\cdot)^2(x) = x^2 = x \cdot x$$

is a bijection.

Proof. We have

injectivity. Let $x, y \in \mathbb{R}_{0,\mathbb{C}}^+ = \{x \in \mathbb{R}_{\mathbb{C}} \mid 0 \leq x\}$ be such that $x^2 = y^2$ then we have the following possibilities:

$x = 0$. Assume that $y \neq 0$ then $0 < y$ so that by [theorems: 10.14] $0 < y \cdot y = y^2 = x^2 = x \cdot x = 0$ leading to the contradiction $0 < 0$. Hence we have $y = 0$ so that $x = y$.

$y = 0$. Assume that $x \neq 0$ then $0 < x$ so that by [theorems: 10.14] $0 < x \cdot x = x^2 = y^2 = y \cdot y = 0$ leading to the contradiction $0 < 0$. Hence we have $x = 0$ so that $x = y$.

$0 < x \wedge 0 < y$. Assume that $x \neq y$ then we have either:

$x < y$. Then by [theorems: 10.14] we have $x \cdot y < y \cdot y = y^2$ and $x^2 = x \cdot x < y \cdot x = x \cdot y$ so that $x^2 < y^2$ contradicting $x^2 = y^2$.

$y < x$. Then by [theorems: 10.14] we have $y \cdot x < x \cdot x = x^2$ and $y^2 = y \cdot y < x \cdot y = y \cdot x$ so that $y^2 < x^2$ contradicting $x^2 = y^2$.

As the assumption $x \neq y$ leads to a contradiction in all cases we must have that $x = y$.

So in all cases we have $x = y$ proving injectivity.

surjectivity. If $y \in \mathbb{R}_{0,\mathbb{C}}^+ = \{x \in \mathbb{R}_{\mathbb{C}} \mid 0 \leq x\}$ then $0 \leq y$ and we have the following possibilities to consider:

$y = 0$. Then $0^0 = 0 \cdot 0 = 0 = y$, hence if $x = 0$ we have $y = x^2$

$y = 1$. Then $1^2 = 1 \cdot 1 = 1 = y$, hence if $x = 1$ we have $y = x^2$

$0 < y \wedge y \neq 1$. Take then $S_y = \{t \in \mathbb{R}_{\mathbb{C}} \mid 0 \leq t \wedge t^2 \leq y\}$. As $0^2 = 0 < y$ we have $0 \in S_y$ hence

$$S_y \neq \emptyset \quad (10.58)$$

As $y \neq 1$ we have either:

$y < 1$. Assume that $\exists t \in S_y$ such that $1 < t$ then by [theorems: 10.14] we have $t < t \cdot t = t^2$, as $t \in S_y$ we have $t^2 \leq y < 1$ so that $t < 1$ contradicting $1 < t$. Hence we have $\forall t \in S_y$ that $t \leq 1$ proving that S_y is bounded above.

$1 < y$. Assume that $\exists t \in S_y$ such that $y < t$ then as $1 < y$ we have $1 < t$ so that $t < t \cdot t = t^2 \leq y$ contradicting the assumption $y < t$. So the assumption is wrong and we have $\forall t \in S_y$ that $t \leq y$ proving that S_y is bounded above.

So in all cases we have that

$$S_y \text{ is bounded above} \quad (10.59)$$

As $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is conditionally complete [see theorem: 10.18] we have thanks to [eqs: 10.58, 10.59] that

$$s_y = \sup(S_y) \text{ exist} \quad (10.60)$$

For $y \neq 1$ we consider again the following possibilities:

$y < 1$. As $0 < y$ we have by [theorems: 10.14] that $y^2 = y \cdot y < 1 \cdot y = y$ so that $y \in S_y$ so $y \leq s_y \xrightarrow{0 < y} 0 < s_y$.

$1 < y$. Then $1^2 = 1 < y$ so as $0 < 1$ we have $1 \in S_y$, hence $1 \leq s_y$ proving that $0 < s_y$.

So in all cases we have

$$0 < s_y \quad (10.61)$$

Let $\varepsilon \in \mathbb{R}_{\mathbb{C}}$ such that $0 < \varepsilon < s_y$. Then $0 < s_y - \varepsilon < s_y < s_y + \varepsilon$, so that by [theorems: 10.14]

$$\begin{aligned}(s_y - \varepsilon)^2 &= (s_y - \varepsilon) \cdot (s_y - \varepsilon) < s_y \cdot (s_y - \varepsilon) \\ &= (s_y - \varepsilon) \cdot s_y < s_y \cdot s_y = s_y^2 \\ s_y \cdot (s_y + \varepsilon) &< (s_y + \varepsilon) \cdot (s_y + \varepsilon) = (s_y + \varepsilon)^2 \\ s_y^2 &= s_y \cdot s_y < (s_y + \varepsilon) \cdot s_y\end{aligned}$$

So that

$$(s_y - \varepsilon)^2 < s_y^2 < (s_y + \varepsilon)^2 \quad (10.62)$$

As s_y is an upper bound of S_y and $s_y < s_y + \varepsilon$ we must have that $s_y + \varepsilon \notin S_y$, which, as $0 < s_y < s_y + \varepsilon$ proves that

$$y < (s_y + \varepsilon)^2 \quad (10.63)$$

As $\langle \mathbb{R}_{\mathbb{C}}, \leq \rangle$ is totally ordered and $s_y - \varepsilon < s_y$ we have by [theorem: 3.74] that $\exists f \in S_y \Rightarrow 0 \leq f$ such that $s_y - \varepsilon < f$. As $0 < s_y - \varepsilon$ we have by [theorems: 10.14] that $(s_y - \varepsilon) \cdot f < f \cdot f = f^2$ and $(s_y - \varepsilon)^2 = (s_y - \varepsilon) \cdot (s_y - \varepsilon) < f \cdot (s_y - \varepsilon)$ so that $(s_y - \varepsilon)^2 < f^2$. As $f \in S_y$ it follows that $f^2 \leq y$ so that

$$(s_y - \varepsilon)^2 < y \quad (10.64)$$

Using [theorems: 10.14] on [eqs: 10.63, 10.64] we have that

$$-(s_y + \varepsilon)^2 < -y < -(s_y - \varepsilon)^2 \quad (10.65)$$

Adding [eq: 10.62] to [eq: 10.65] gives

$$(s_y - \varepsilon)^2 - (s_y + \varepsilon)^2 < s_y^2 - y < (s_y + \varepsilon)^2 - (s_y - \varepsilon)^2 \quad (10.66)$$

Now

$$\begin{aligned}(s_y - \varepsilon)^2 - (s_y + \varepsilon)^2 &= s_y^2 - 2 \cdot \varepsilon \cdot s_y + \varepsilon^2 - (s_y^2 + 2 \cdot \varepsilon \cdot s_y + \varepsilon^2) \\ &= -4 \cdot \varepsilon \cdot s_y\end{aligned}$$

which combined with [eq: 10.66] gives

$$\forall \varepsilon \in \mathbb{R}_{\mathbb{C}} \text{ with } 0 < \varepsilon < s_y \text{ we have } -4 \cdot \varepsilon \cdot s_y^2 < s_y^2 - y < 4 \cdot \varepsilon \cdot s_y \quad (10.67)$$

or using [theorems: 10.14]

$$\forall \varepsilon \in \mathbb{R}_{\mathbb{C}} \text{ with } 0 < \varepsilon < s_y \text{ we have } -4 \cdot \varepsilon \cdot s_y^2 < y - s_y^2 < 4 \cdot \varepsilon \cdot s_y \quad (10.68)$$

Now as $0 < s_y$ we have by [theorem: 10.32] a $\varepsilon_0 \in \mathbb{R}_{\mathbb{C}}$ such that

$$0 < \varepsilon_0 < s_y \quad (10.69)$$

For $s_y^2 - y$ we can have now the following possibilities:

$s_y^2 - y < 0$. Take then $\delta = y - s_y^2$ then $0 < \delta$. Take $\varepsilon = \min(\delta / (4 \cdot s_y), \varepsilon_0)$ then we have as $0 < 4 \cdot s_y, \varepsilon_0$ by [theorems: 10.14] that $0 < \varepsilon$ and $\varepsilon \leq \varepsilon_0 < s_y$ so we have by [eq: 10.68] that $\delta = y - s_y^2 < 4 \cdot \varepsilon \cdot s_y$. As $4 \cdot \varepsilon \cdot s_y \leq (\delta / (4 \cdot s_y)) \cdot 4 \cdot s_y = \delta$ we have the contradiction $\delta < \delta$. So this case does not occur.

$0 < s_y^2 - y$. Take then $\delta = s_y^2 - y$ then $0 < \delta$. Take $\varepsilon = \min(\delta / (4 \cdot s_y), \varepsilon_0)$ then we have as $0 < 4 \cdot s_y, \varepsilon_0$ by [theorems: 10.14] that $0 < \varepsilon$, further $\varepsilon \leq \varepsilon_0 < s_y$, so we have by [eq: 10.67] that $\delta = s_y^2 - y < 4 \cdot \varepsilon \cdot s_y$. As $4 \cdot \varepsilon \cdot s_y \leq (\delta / (4 \cdot s_y)) \cdot 4 \cdot s_y = \delta$ we have the contradiction $\delta < \delta$. So this case does not occur.

$s_y^2 - y = 0$. Then $y = s_y^2$

So the only valid case is where $y = s_y^2$, hence if we take $x = s_y$ then $x^2 = y$.

In all cases we have found a $x \in \mathbb{R}_{\mathbb{C}}$ such that $y = x^2 = (.)^2(x)$ which proves surjectivity. $\square$

Definition 10.67. (Square Root) Using the previous theorem [theorem: 10.66] we have that

$$(.)^2: \mathbb{R}_{0,\mathbb{C}}^+ \rightarrow \mathbb{R}_{0,\mathbb{C}}^+ \text{ defined by } (.)^2(x) = x \cdot x$$

is a bijection so that we have a inverse bijection

$$((.)^2)^{-1}: \mathbb{R}_{0,\mathbb{C}}^+ \rightarrow \mathbb{R}_{0,\mathbb{C}}^+$$

this inverse bijection is called the square root mapping and noted by

$$\sqrt{\cdot}: \mathbb{R}_{0,\mathbb{C}}^+ \rightarrow \mathbb{R}_{0,\mathbb{C}}^+ \text{ where } \sqrt{\cdot} = ((.)^2)^{-1}$$

Hence if $x \in \mathbb{R}_0^+$ then $\sqrt{(x^2)} = (\sqrt{\cdot} \circ (.)^2)(x) = i_{\mathbb{R}_{0,\mathbb{C}}^+}(x) = x$ and $(\sqrt{x})^2 = ((.)^2 \circ \sqrt{\cdot})(x) = i_{\mathbb{R}_{0,\mathbb{C}}^+}(x)$

Note 10.68. The requirement that $x \in \mathbb{R}_{0,\mathbb{C}}^+$ in the above is required because $(.)^2: \mathbb{R}_{\mathbb{C}} \rightarrow \mathbb{R}_{\mathbb{C}}$ is not injective [for example $(1)^2 = 1 = (-1)^2$]

Example 10.69. $\sqrt{0} = 0$ and $\sqrt{1} = 1$

Proof. First, as by [corollary: 10.10] $0 < 1$ so that $\sqrt{0}$ and $\sqrt{1}$ are well defined. As $0^2 = 0 \cdot 0 = 0$ we have $\sqrt{0} = ((.)^2)^{-1}(0) = 0$. Further from $1^2 = 1 \cdot 1 = 1$ we have that $\sqrt{1} = ((.)^2)^{-1}(1) = 1$. $\square$

Note 10.70. $2 \in \mathbb{R}_0^+$ so $\sqrt{2}$ exist but $\sqrt{2} \notin \mathbb{Q}_{\mathbb{C}}$ so that $\sqrt{2} \in \mathbb{R}_{\mathbb{C}} \setminus \mathbb{Q}_{\mathbb{C}}$ or $\sqrt{2}$ is a irrational number.

Proof. Assume that $\sqrt{2} \in \mathbb{Q}_{\mathbb{C}}$ then as by [theorem: 10.5] $i_{\mathbb{Q} \rightarrow \mathbb{C}}: \langle \mathbb{Q}, +, \cdot \rangle \rightarrow \langle \mathbb{Q}_{\mathbb{C}}, +, \cdot \rangle$ is a ring isomorphism there exist a $q \in \mathbb{Q}$ such that $i_{\mathbb{Q} \rightarrow \mathbb{C}}(q) = \sqrt{2}$, hence:

$$\begin{aligned} i_{\mathbb{Q} \rightarrow \mathbb{C}}(q \cdot q) &= i_{\mathbb{Q} \rightarrow \mathbb{C}}(q) \cdot i_{\mathbb{Q} \rightarrow \mathbb{C}}(q) \\ &= \sqrt{2} \cdot \sqrt{2} \\ &= 2 \\ &= 1 + 1 \\ &= i_{\mathbb{Q} \rightarrow \mathbb{C}}(1) + i_{\mathbb{Q} \rightarrow \mathbb{C}}(1) \\ &= i_{\mathbb{Q} \rightarrow \mathbb{C}}(1 + 1) \\ &= i_{\mathbb{Q} \rightarrow \mathbb{C}}(2) \end{aligned}$$

so that by injectivity we have that $q \cdot q = 2$ which by [theorem: 8.40] is impossible. $\square$

Theorem 10.71. $\sqrt{\cdot}: \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$ is strictly increasing.

Proof. We prove this by contradiction. Let $x, y \in \mathbb{R}_{0,\mathbb{C}}^+$ be such that $x < y$ and assume that $\sqrt{y} \leq \sqrt{x}$, then as $\sqrt{x}, \sqrt{y} \in \mathbb{R}_{0,\mathbb{C}}^+$ we can use [theorems: 10.14] getting $y = \sqrt{y} \cdot \sqrt{y} \leq \sqrt{x} \cdot \sqrt{y}$ and $\sqrt{y} \cdot \sqrt{x} \leq \sqrt{x} \cdot \sqrt{x} = x$, so $y \leq x$ contradicting $x < y$. Hence we must have that $\sqrt{x} < \sqrt{y}$. $\square$

Theorem 10.72. If $x \in \mathbb{R}_{\mathbb{C}}$ and $a \in \mathbb{R}_{0,\mathbb{C}}^+$ then we have

1. $x^2 = a \Leftrightarrow x = \sqrt{a} \vee x = -\sqrt{a}$
2. $x^2 \leq a \Leftrightarrow -\sqrt{a} \leq x \leq \sqrt{a}$
3. $x^2 < a \Leftrightarrow -\sqrt{a} < x < \sqrt{a}$

Proof.

1.

$\Rightarrow$. For x we have either:

$$0 \leq x. \text{ Then by [definition: 10.67] we have } x = \sqrt{x^2} = \sqrt{a}$$

$x < 0$. Then $0 < -x$ and $(-x)^2 = x^2 = a$ so that by [definition: 10.67] that $-x = \sqrt{(-x)^2} = \sqrt{a}$ giving $x = -\sqrt{a}$

$\Leftarrow$. If $x = \sqrt{a}$ then $x^2 = (\sqrt{a})^2 = a$ and if $x = -\sqrt{a}$ we have $x^2 = (-\sqrt{a})^2 = (\sqrt{a})^2 = a$

2.

$\Rightarrow$. For x we have either

$0 \leq x$. As $0 \leq \sqrt{a}$ it follows that $-\sqrt{a} \leq 0 \leq x$. Further assume that $\sqrt{a} < x$ then by [theorems: 10.14] and $0 \leq \sqrt{a}, x$ we have $a = (\sqrt{a})^2 < x^2$ contradicting $x^2 \leq a$, so we must have $x \leq \sqrt{a}$. So $-\sqrt{a} \leq x \leq \sqrt{a}$.

$x < 0$. Then as $0 \leq \sqrt{a} \Rightarrow -\sqrt{a} \leq 0$ we have $x < \sqrt{a}$. Further assume that $\sqrt{a} < -x$ we have by [theorems: 10.14] and $0 \leq \sqrt{a}, -x$ that $a = (\sqrt{a})^2 < (-x)^2 = x^2$ contradicting $x^2 \leq a$, hence $-x \leq \sqrt{a}$ or $-\sqrt{a} \leq x$. So $-\sqrt{a} \leq x < \sqrt{a} \Rightarrow -\sqrt{a} \leq x \leq \sqrt{a}$.

$\Leftarrow$. For x we have either:

$0 \leq x$. Then from $x \leq \sqrt{a}$ and $0 \leq \sqrt{a}$ we have by [theorems: 10.14] that $x^2 \leq (\sqrt{a})^2 = a$.

$x < 0$. Then from $-\sqrt{a} \leq x \Rightarrow -x \leq \sqrt{a}$ and $0 \leq \sqrt{a}, -x$ we have by [theorems: 10.14] that $x^2 = (-x)^2 \leq (\sqrt{a})^2 = a$.

3.

$\Rightarrow$. If $x = \sqrt{a}$ or $x = -\sqrt{a}$ then by (1) $x^2 = a$ contradicting $x^2 < a$ hence we must have that $x \neq \sqrt{a}$ and $x \neq -\sqrt{a}$. Using (2) we have that $-\sqrt{a} \leq x \leq \sqrt{a}$ so that $-\sqrt{a} < x < \sqrt{a}$.

$\Leftarrow$. As $-\sqrt{a} < x < \sqrt{a}$ we have $x \neq \sqrt{a}$ and $x \neq -\sqrt{a}$ so that by (1) $x^2 \neq a$, further by (2) we have $x^2 \leq a$ so that $x^2 < a$. $\square$

Corollary 10.73. If $x \in \mathbb{R}_{\mathbb{C}}$ then $x \leq |x| = \sqrt{x^2}$

Proof. For $x \in \mathbb{R}_{\mathbb{C}}$ we have either:

$0 \leq x$. Then $|x| = x$ so that $|x|^2 = x^2$

$x < 0$. Then $|x| = -x$ so that $|x|^2 = (-x) \cdot (-x) = (-1) \cdot (-1) \cdot x \cdot x = x^2$

so that

$$|x^2| = x^2$$

As $|x| \in \mathbb{R}_{0,C}^+$ we have that

$$|x| = \sqrt{|x|^2} = \sqrt{x^2}$$

$\square$

Theorem 10.74. If $x, y \in \mathbb{R}_{0,\mathbb{C}}^+$ then $\sqrt{x \cdot y} = \sqrt{x} \cdot \sqrt{y}$

Proof. As $(\sqrt{x \cdot y})^2 = x \cdot y = (\sqrt{x})^2 \cdot (\sqrt{y})^2 = (\sqrt{x} \cdot \sqrt{y})^2$ we have by the fact that $(\cdot)^2: \mathbb{R}_{0,\mathbb{C}}^+ \rightarrow \mathbb{R}_{0,\mathbb{C}}^+$ is a bijection and thus injective that $\sqrt{x \cdot y} = \sqrt{x} \cdot \sqrt{y}$. $\square$

Theorem 10.75. Let $x, y \in \mathbb{R}_{0,\mathbb{C}}^+$ then $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$

Proof. We prove this by contradiction, so assume that $\sqrt{x} + \sqrt{y} < \sqrt{x+y}$. Then by [theorem: 10.14] we have that $(\sqrt{x} + \sqrt{y})^2 < (\sqrt{x+y})^2 = x+y$. Now

$$(\sqrt{x} + \sqrt{y})^2 = (\sqrt{x})^2 + (\sqrt{y})^2 + 2 \cdot \sqrt{x} \cdot \sqrt{y} = x + y + 2 \cdot \sqrt{x} \cdot \sqrt{y}$$

so that we have that $x + y + 2 \cdot \sqrt{x} \cdot \sqrt{y} < x + y$ or $2 \cdot \sqrt{x} \cdot \sqrt{y} < 0$ giving

$$\sqrt{x} \cdot \sqrt{y} < 0$$

As $0 \leq \sqrt{x}$ and $0 \leq \sqrt{y}$ we have by [theorem: 10.14] that $0 \leq \sqrt{x} \cdot \sqrt{y}$ contradicting the above. So we must have that $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$. $\square$

10.5 Operations on Complex numbers

10.5.1 Notation of complex numbers

Note that the additive neutral element $(0, 0)$ is noted as 0 and the multiplicative unit element $(1, 0)$ is noted as 1. In the following definition we have also a special notation for $(0, 1)$.

Definition 10.76. $i \in \mathbb{C}$ is defined as $i = (0, 1)$ so that as $0 \neq 1$ we have that $i \in \mathbb{C} \setminus \mathbb{R}_{\mathbb{C}}$.

Theorem 10.77. For i we have

1. $i \cdot i = -1$
2. If $z \in \mathbb{C}$ then there exists **unique** $x, y \in \mathbb{R}_{\mathbb{C}}$ such that $z = x + i \cdot y$

Proof.

1. $i \cdot i = (0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -(1, 0) = -1$
2. If $z \in \mathbb{C}$ then $\exists x', y' \in \mathbb{R}$ such that $z = (x', y')$, define then $x = (x', 0) \in \mathbb{R}_{\mathbb{C}}$ and $y = (y', 0) \in \mathbb{R}_{\mathbb{C}}$ then we have

$$\begin{aligned} x + i \cdot y &= (x', 0) + (0, 1) \cdot (y', 0) \\ &= (x', 0) + (0 \cdot y' - 1 \cdot 0, 0 \cdot 0 + 1 \cdot y') \\ &= (x', 0) + (0, y') \\ &= (x', y') \\ &= z \end{aligned}$$

Further if $u, v \in \mathbb{R}_{\mathbb{C}}$ such that $z = u + i \cdot v$ then $\exists u', v' \in \mathbb{R}$ such that $u = (u', 0) \wedge v = (v', 0)$ then we have

$$\begin{aligned} (z', y') &= z \\ &= u + i \cdot v \\ &= (u', 0) + (0, 1) \cdot (v', 0) \\ &= (u', 0) + (0 \cdot v' - 1 \cdot 0, 0 \cdot 0 + 1 \cdot v') \\ &= (u', 0) + (0, v') \\ &= (u', v') \end{aligned}$$

so that $x' = u'$ and $y' = v'$ hence $x = (x', 0) = (u', 0) = u$ and $y = (y', 0) = (v', 0) = v$ proving uniqueness. $\square$

Note 10.78. We have now two ways to **uniquely** represent $z \in \mathbb{C}$ either $z = x + i \cdot y$ where $x, y \in \mathbb{R}_{\mathbb{C}}$ or $z = (x, y)$ where $x, y \in \mathbb{R}$. In this book we follow the common practice of using the first representation.

Definition 10.79. (Real & Imaginary part) The above theorem allows us to define the following functions

$$\text{Re}: \mathbb{C} \rightarrow \mathbb{R}_{\mathbb{C}} \text{ is defined by } \text{Re}(x + i \cdot y) = x$$

and

$$\text{Im}: \mathbb{C} \rightarrow \mathbb{R}_{\mathbb{C}} \text{ by } \text{Im}(x + i \cdot y) = y$$

So $\forall z \in \mathbb{C}$ we have that

$$z = \text{Re}(z) + i \cdot \text{Im}(z)$$

Theorem 10.80. *We have the following properties for Re and Im*

1. $z \in \mathbb{R}_{\mathbb{C}} \Leftrightarrow z = \text{Re}(z)$
2. $z \in \mathbb{R}_{\mathbb{C}} \Leftrightarrow \text{Im}(z) = 0$
3. $\forall z_1, z_2 \in \mathbb{C}$ we have $\text{Re}(z_1 + z_2) = \text{Re}(z_1) + \text{Re}(z_2)$
4. $\forall z_1, z_2 \in \mathbb{C}$ we have $\text{Im}(z_1 + z_2) = \text{Im}(z_1) + \text{Im}(z_2)$
5. $\forall z \in \mathbb{C}$ and $\alpha \in \mathbb{R}$ then $\text{Re}(\alpha \cdot z) = \alpha \cdot \text{Re}(z)$.
6. $\forall z \in \mathbb{C}$ and $\alpha \in \mathbb{R}$ then $\text{Im}(\alpha \cdot z) = \alpha \cdot \text{Im}(z)$

Proof.

1.

$\Rightarrow$. If $z \in \mathbb{R}_{\mathbb{C}}$ then $z = z + i \cdot 0$ where $z, 0 \in \mathbb{R}_{\mathbb{C}}$ so that $\text{Re}(z) = z$

$\Leftarrow$. If $z = \text{Re}(z)$ then as $\text{Re}(z) \in \mathbb{R}_{\mathbb{C}}$ we have $z \in \mathbb{R}_{\mathbb{C}}$

2.

$\Rightarrow$. If $z \in \mathbb{R}_{\mathbb{C}}$ then $z = z + i \cdot 0$ where $z, 0 \in \mathbb{R}_{\mathbb{C}}$ so that $\text{Im}(z) = 0$

$\Leftarrow$. If $\text{Im}(z) = 0$ then $z = \text{Re}(z) + \text{Im}(z) = \text{Re}(z) \in \mathbb{R}_{\mathbb{C}}$

3.

$$\begin{aligned} \text{Re}(z_1 + z_2) &= \text{Re}((\text{Re}(z_1) + i \cdot \text{Im}(z_1)) + (\text{Re}(z_2) + i \cdot \text{Im}(z_2))) \\ &= \text{Re}((\text{Re}(z_1) + \text{Re}(z_2)) + i \cdot (\text{Im}(z_1) + \text{Im}(z_2))) \\ &= \text{Re}(z_1) + \text{Re}(z_2) \end{aligned}$$

4.

$$\begin{aligned} \text{Im}(z_1 + z_2) &= \text{Im}((\text{Re}(z_1) + i \cdot \text{Im}(z_1)) + (\text{Re}(z_2) + i \cdot \text{Im}(z_2))) \\ &= \text{Im}((\text{Re}(z_1) + \text{Re}(z_2)) + i \cdot (\text{Im}(z_1) + \text{Im}(z_2))) \\ &= \text{Im}(z_1) + \text{Im}(z_2) \end{aligned}$$

5.

$$\begin{aligned} \text{Re}(\alpha \cdot z) &\stackrel{(1)}{=} \text{Re}(\text{Re}(\alpha) \cdot (\text{Re}(z) + i \cdot \text{Im}(z))) \\ &= \text{Re}(\text{Re}(\alpha) \cdot \text{Re}(z) + i \cdot (\text{Re}(\alpha) \cdot \text{Im}(z))) \\ &= \text{Re}(\alpha) \cdot \text{Re}(z) \\ &\stackrel{(1)}{=} \alpha \cdot \text{Re}(z) \end{aligned}$$

6.

$$\begin{aligned} \text{Im}(\alpha \cdot z) &\stackrel{(1)}{=} \text{Im}(\text{Re}(\alpha) \cdot (\text{Re}(z) + i \cdot \text{Im}(z))) \\ &= \text{Im}(\text{Re}(\alpha) \cdot \text{Re}(z) + i \cdot (\text{Re}(\alpha) \cdot \text{Im}(z))) \\ &= \text{Re}(\alpha) \cdot \text{Im}(z) \\ &= \alpha \cdot \text{Im}(z) \end{aligned}$$

□

Theorem 10.81. *For i we have*

1. $i^2 = -1$
2. $\forall n \in \mathbb{N}_0$ we have
 - a. $i^{4 \cdot n} = 1$
 - b. $i^{4 \cdot n + 1} = i$
 - c. $i^{4 \cdot n + 2} = -1$

$$d. i^{4 \cdot n + 3} = -i$$

Proof.

$$1. i^2 = i \cdot i \stackrel{[\text{theorem: 10.77}]}{=} -1$$

2. For $n \in \mathbb{N}$ we have:

a. We use induction to prove this, so let $S = \{n \in \mathbb{N} | i^{4 \cdot n} = 1\}$ then we have

$$0 \in S. \text{ As } i^{4 \cdot 0} = i^0 = 1 \text{ we have that } 1 \in S.$$

$$n \in S \Rightarrow n + 1 \in S. \text{ We have}$$

$$\begin{aligned} i^{4 \cdot (n+1)} &= i^{4 \cdot n + 4} \\ &\stackrel{[\text{theorem: 10.52}]}{=} i^{4 \cdot n} \cdot i^4 \\ &\stackrel{n \in S}{=} 1 \cdot i^4 \\ &= i^{2+2} \\ &\stackrel{[\text{theorem: 10.52}]}{=} i^2 \cdot i^2 \\ &\stackrel{(1)}{=} (-1) \cdot (-1) \\ &= 1 \end{aligned}$$

proving that $n \in S$.

$$b. i^{4 \cdot n + 1} \stackrel{[\text{theorem: 10.52}]}{=} i^{4 \cdot n} \cdot i \stackrel{(2.a)}{=} i$$

$$c. i^{4 \cdot n + 2} \stackrel{[\text{theorem: 10.52}]}{=} i^{4 \cdot n} \cdot i^2 \stackrel{(2.a)}{=} i^2 \stackrel{(1)}{=} -1$$

$$d. i^{4 \cdot n + 3} \stackrel{[\text{theorem: 10.52}]}{=} i^{4 \cdot n} \cdot i^3 \stackrel{(2.a)}{=} i^3 \stackrel{[\text{theorem: 10.52}]}{=} i^2 \cdot i \stackrel{(2)}{=} (-1) \cdot i = -i$$

□

10.5.2 Norm on $\mathbb{C}$

Definition 10.82. (Conjugate) Let $z \in \mathbb{C}$ then the *conjugate* of z noted as $\bar{z}$ is defined as

$$\bar{z} = \text{Re}(z) - i \cdot \text{Im}(z)$$

Theorem 10.83. The conjugate has the following properties

1. $\bar{i} = -i$
2. $\forall z \in \mathbb{C}$ we have $\overline{(\bar{z})} = z$
3. $\forall z_1, z_2 \in \mathbb{C}$ we have $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
4. $\forall z_1, z_2 \in \mathbb{C}$ we have $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$
5. $\forall z \in \mathbb{C} \setminus \{0\}$ we have $\overline{z^{-1}} = (\bar{z})^{-1}$
6. $\forall z_1 \in \mathbb{C}, \forall z_2 \in \mathbb{C} \setminus \{0\}$ we have $\overline{z_1 / z_2} = \bar{z}_1 / \bar{z}_2$
7. $z \in \mathbb{R}_{\mathbb{C}} \Leftrightarrow z = \bar{z}$
8. $\forall z \in \mathbb{C}$ we have that $z \cdot \bar{z} = \text{Re}(z)^2 + \text{Im}(z)^2 \in \mathbb{R}_0^+$
9. $\forall z \in \mathbb{C}$ we have $z + \bar{z} = 2 \cdot \text{Re}(z)$
10. $\forall z \in \mathbb{C}$ we have $(z - \bar{z}) = 2 \cdot i \cdot \text{Im}(z)$
11. $\forall z \in \mathbb{C}$ we have $\overline{-z} = -\bar{z}$
12. $\forall z \in \mathbb{C}$ we have $\bar{\bar{z}} = z$
13. Let $n \in \mathbb{N}_{\mathbb{C}}$ and $\{z_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ then $\overline{\sum_{i=1}^n z_i} = \sum_{i=1}^n \bar{z}_i$ [see definition: 11.15]

14. Let $n \in \mathbb{N}_{\mathbb{C}}$ and $z \in \mathbb{C}$ then $\bar{z}^n = (\bar{z})^n$

Proof.

1. As $i = 0 + i \cdot 1$ we have that $\operatorname{Re}(i) = 0$ and $\operatorname{Im}(i) = 1$ so that $\bar{i} = \operatorname{Re}(i) - i \cdot \operatorname{Im}(i) = 0 - i = -i$
- 2.

$$\begin{aligned} \overline{(\bar{z})} &= \overline{\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)} \\ &= \overline{(\operatorname{Re}(z) + i \cdot (-\operatorname{Im}(z)))} \\ &= \operatorname{Re}(z) - i \cdot (-\operatorname{Im}(z)) \\ &= \operatorname{Re}(z) + i \cdot \operatorname{Im}(z) \\ &= z \end{aligned}$$

3.

$$\begin{aligned} \overline{z_1 + z_2} &= \operatorname{Re}(z_1 + z_2) - i \cdot \operatorname{Im}(z_1 + z_2) \\ &\stackrel{[\text{theorem: 10.80}]}{=} \operatorname{Re}(z_1) + \operatorname{Re}(z_2) - i \cdot (\operatorname{Im}(z_1) + \operatorname{Im}(z_2)) \\ &= (\operatorname{Re}(z_1) - i \cdot \operatorname{Im}(z_1)) + (\operatorname{Re}(z_2) - i \cdot \operatorname{Im}(z_2)) \\ &= \bar{z}_1 + \bar{z}_2 \end{aligned}$$

4.

$$\begin{aligned} \overline{z_1 \cdot z_2} &= \overline{(\operatorname{Re}(z_1) + i \cdot \operatorname{Im}(z_1)) \cdot (\operatorname{Re}(z_2) + i \cdot \operatorname{Im}(z_2))} \\ &= \overline{\operatorname{Re}(z_1) \cdot \operatorname{Re}(z_2) - \operatorname{Im}(z_1) \cdot \operatorname{Im}(z_2) + i \cdot (\operatorname{Re}(z_1) \cdot \operatorname{Im}(z_2) + \operatorname{Im}(z_1) \cdot \operatorname{Re}(z_2))} \\ &= \operatorname{Re}(z_1) \cdot \operatorname{Re}(z_2) - \operatorname{Im}(z_1) \cdot \operatorname{Im}(z_2) - i \cdot (\operatorname{Re}(z_1) \cdot \operatorname{Im}(z_2) + \operatorname{Im}(z_1) \cdot \operatorname{Re}(z_2)) \\ &= (\operatorname{Re}(z_1) - i \cdot \operatorname{Im}(z_1)) \cdot (\operatorname{Re}(z_2) - i \cdot \operatorname{Im}(z_2)) \\ &= \bar{z}_1 \cdot \bar{z}_2 \end{aligned}$$

5. We have $\overline{z^{-1}} \cdot \bar{z} = \bar{z} \cdot \overline{z^{-1}} \stackrel{(4)}{=} \overline{z \cdot z^{-1}} = \overline{1} = \overline{1 + i \cdot 0} = 1 - i \cdot 0 = 1$ proving that $(\bar{z})^{-1} = \overline{z^{-1}}$

6. $\overline{z_1/z_2} = \overline{z_1 \cdot z_2^{-1}} \stackrel{(4)}{=} \bar{z}_1 \cdot \overline{z_2^{-1}} \stackrel{(5)}{=} \bar{z}_1 \cdot (\bar{z}_2)^{-1} = \bar{z}_1 / \bar{z}_2$

7.

$$\begin{aligned} \Rightarrow. \text{ If } z \in \mathbb{R}_{\mathbb{C}} \text{ then } \bar{z} &= \operatorname{Re}(z) - i \cdot \operatorname{Im}(z) \stackrel{[\text{theorem: 10.80}]}{=} \operatorname{Re}(z) - i \cdot 0 = \operatorname{Re}(z) \stackrel{[\text{theorem: 10.80}]}{=} z \\ \Leftarrow. \text{ If } z = \bar{z} \text{ then } \operatorname{Re}(z) + i \cdot \operatorname{Im}(z) &= \operatorname{Re}(z) - i \cdot \operatorname{Im}(z) \Rightarrow 2 \cdot \operatorname{Im}(z) = 0 \Rightarrow \operatorname{Im}(z) = 0 \text{ so} \\ &\text{that } z = \operatorname{Re}(z) \in \mathbb{R}_{\mathbb{C}} \text{ [see theorem: 10.80]} \end{aligned}$$

8.

$$\begin{aligned} z \cdot \bar{z} &= (\operatorname{Re}(z) + i \cdot \operatorname{Im}(z)) \cdot (\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)) \\ &= \operatorname{Re}(z) \cdot \operatorname{Re}(z) - i \cdot \operatorname{Re}(z) \cdot \operatorname{Im}(z) + i \cdot \operatorname{Im}(z) \cdot \operatorname{Re}(z) + i \cdot (-i) \cdot \operatorname{Im}(z) \cdot \operatorname{Im}(z) \\ &= \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2 \end{aligned}$$

As $\operatorname{Re}(z), \operatorname{Im}(z) \in \mathbb{R}_{\mathbb{C}}$ we have by [theorem: 10.14] that $0 \leq \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$ so that $z \cdot \bar{z} \in \mathbb{R}_{0, \mathbb{C}}^+$.

9. $z + \bar{z} = (\operatorname{Re}(z) + i \cdot \operatorname{Im}(z)) + (\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)) = 2 \cdot \operatorname{Re}(z)$

10. $z - \bar{z} = (\operatorname{Re}(z) + i \cdot \operatorname{Im}(z)) - (\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)) = 2 \cdot i \cdot \operatorname{Im}(z)$

11.

$$\begin{aligned} -\bar{z} &= \overline{-(\operatorname{Re}(z) + i \cdot \operatorname{Im}(z))} \\ &= \overline{-\operatorname{Re}(z) + i \cdot (-\operatorname{Im}(z))} \\ &= -\operatorname{Re}(z) - i \cdot (-\operatorname{Im}(z)) \\ &= -(\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)) \\ &= -\bar{z} \end{aligned}$$

12. $\bar{\bar{z}} = \overline{\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)} = \operatorname{Re}(z) + i \cdot \operatorname{Im}(z)$

13. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \forall \{z_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C} \text{ we have } \overline{\sum_{i=1}^n z_i} = \sum_{i=1}^n \bar{z}_i \right\}$$

then we have:

$1 \in S$. this follows from $\overline{\sum_{i=1}^1 z_i} = \bar{z}_1 = \sum_{i=1}^1 \bar{z}_i$ for $\{z_i\}_{i \in \{1, \dots, 1\}} \subseteq \mathbb{C}$

$n \in S \Rightarrow n+1 \in S$. we have for $\{z_i\}_{i \in \{1, \dots, n+1\}}$ that

$$\begin{aligned} \overline{\sum_{i=1}^{n+1} z_i} &= \overline{z_{n+1} + \sum_{i=1}^n z_i} \\ &\stackrel{(3)}{=} \overline{z_{n+1}} + \overline{\sum_{i=1}^n z_i} \\ &\stackrel{n \in S}{=} \bar{z}_{n+1} + \sum_{i=1}^n \bar{z}_i \\ &= \sum_{i=1}^{n+1} \bar{z}_i \end{aligned}$$

proving that $n+1 \in S$.

14. We prove this by induction, so let $S = \{n \in \mathbb{N}_{\mathbb{C},0} \mid \bar{z}^n = (\bar{z})^n\}$ then we have:

$0 \in S$. As $\bar{z}^0 = \bar{1} \stackrel{(7)}{=} 1 = (\bar{z})^0$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Then we have

$$\bar{z}^{n+1} = \overline{z^n \cdot z} \stackrel{(4)}{=} \bar{z}^n \cdot \bar{z} \stackrel{n \in S}{=} (\bar{z})^n \cdot \bar{z} = (\bar{z})^{n+1}$$

□

Using the above theorem we have that $\forall z \in \mathbb{C} \ z \cdot \bar{z} \in \mathbb{R}_{0,\mathbb{C}}^+$ so that $\sqrt{z \cdot \bar{z}} \in \mathbb{R}_{0,\mathbb{C}}^+$ is well defined, hence the following definition makes sense.

Definition 10.84. (complex norm)

$$||: \mathbb{C} \rightarrow \mathbb{R}_{0,\mathbb{C}}^+ \text{ is defined by } |z| = \sqrt{z \cdot \bar{z}} \stackrel{[theorem: 10.83]}{=} \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$$

Theorem 10.85. $\forall z, z' \in \mathbb{C}$ we have

1. $|z \cdot z'| = |z| \cdot |z'|$
2. $|\bar{z}| = |z|$
3. $\operatorname{Re}(z) \leq |\operatorname{Re}(z)| \leq_{\mathbb{R}} |z|$ and $\operatorname{Im}(z) \leq |\operatorname{Im}(z)| \leq_{\mathbb{R}} |z|$
4. $z \cdot \bar{z} = |z|^2$
5. $|z + z'| \leq |z| + |z'|$
6. $|-z| = |z|$
7. $|1| = |-1| = |i| = 1$
8. $|z| \leq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$
9. $|z| = 0 \Leftrightarrow z = 0$
10. If $z \neq 0$ then $|z^{-1}| = |z|^{-1}$ or in other words $|1/z| = 1/|z|$
11. If $z \in \mathbb{R}_{\mathbb{C}}$ then

$$|z| = \begin{cases} z & \text{if } 0 \leq z \\ -z & \text{if } z < 0 \end{cases}$$

12. If $n \in \mathbb{N}_{\mathbb{C},0}$ then $|z^n| = |z|^n$

Proof.

1.

$$\begin{aligned} |z \cdot z'| &= \sqrt{(z \cdot z') \cdot \overline{(z \cdot z')}} \\ &\stackrel{[\text{theorem: 10.83}]}{=} \sqrt{(z \cdot z') \cdot (\bar{z} \cdot \bar{z}')} \\ &= \sqrt{(z \cdot \bar{z}) \cdot (z' \cdot \bar{z}')} \\ &\stackrel{[\text{theorem: 10.74}]}{=} \sqrt{z \cdot \bar{z}} \cdot \sqrt{z' \cdot \bar{z}'} \\ &= |z| \cdot |z'| \end{aligned}$$

$$2. |\bar{z}| = \sqrt{\bar{z} \cdot \overline{(\bar{z})}} \stackrel{[\text{theorem: 10.83}]}{=} \sqrt{\bar{z} \cdot z} = \sqrt{z \cdot \bar{z}} = |z|$$

3. Using [corollary: 10.73] we have that

$$\operatorname{Re}(z) \leq \sqrt{\operatorname{Re}(z)^2} = \sqrt{\operatorname{Re}(z)^2 + 0^2} = |\operatorname{Re}(z)|$$

and

$$\operatorname{Im}(z) \leq \sqrt{(\operatorname{Im}(z))^2} = \sqrt{0 + \operatorname{Im}(z)^2} = |\operatorname{Im}(z)|$$

Further as $\operatorname{Re}(z)^2 \leq \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$ and $\operatorname{Im}(z)^2 \leq \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$ and [theorem: 10.71] $\sqrt{\cdot}$ is increasing it follows that

$$\sqrt{\operatorname{Re}(z)^2} \leq \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2} = |z| \text{ and } \sqrt{\operatorname{Im}(z)^2} \leq \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2} = |z|$$

so that $\operatorname{Re}(z) \leq |\operatorname{Re}(z)| \leq_{\mathbb{R}} |z|$ and $\operatorname{Im}(z) \leq |\operatorname{Im}(z)| \leq_{\mathbb{R}} |z|$.

$$4. |z|^2 = (\sqrt{z \cdot \bar{z}})^2 \stackrel{0 \leq z \cdot \bar{z}}{=} z \cdot \bar{z}$$

5. We have

$$\begin{aligned} |z + z'|^2 &\stackrel{(5)}{=} (z + z') \cdot \overline{(z + z')} \\ &\stackrel{[\text{theorem: 10.83}]}{=} (z + z') \cdot (\bar{z} + \bar{z}') \\ &= z \cdot \bar{z} + z \cdot \bar{z}' + z' \cdot \bar{z} + z' \cdot \bar{z}' \\ &\stackrel{(4)}{=} |z|^2 + |z'|^2 + z \cdot \bar{z}' + z' \cdot \bar{z} \\ &\stackrel{[\text{theorem: 10.83}]}{=} |z|^2 + |z'|^2 + z \cdot \bar{z}' + \overline{z \cdot \bar{z}'} \\ &\stackrel{[\text{theorem: 10.83}]}{=} |z|^2 + |z'|^2 + 2 \cdot \operatorname{Re}(z \cdot \bar{z}') \\ &\stackrel{(3)}{\leq} |z|^2 + |z'|^2 + 2 \cdot |z \cdot \bar{z}'| \\ &\stackrel{(1)}{=} |z|^2 + |z'|^2 + 2 \cdot |z| \cdot |z'| \\ &= (|z| + |z'|)^2 \end{aligned}$$

which as $\sqrt{\cdot}$ is increasing [see theorem: 10.71] and $0 \leq |z + z'|, |z| + |z'|$ we have that

$$|z + z'| = \sqrt{|z + z'|^2} \leq \sqrt{(|z| + |z'|)^2} = |z| + |z'|$$

$$6. |-z| = \sqrt{(-z) \cdot \overline{(-z)}} \stackrel{[\text{theorem: 10.83}]}{=} \sqrt{(-z) \cdot (-\bar{z})} = \sqrt{z \cdot \bar{z}} = |z|$$

7.

$$|-1| \stackrel{(6)}{=} |1| = \sqrt{1 \cdot \bar{1}} \stackrel{1 \in \mathbb{R}_{\mathbb{C}} \text{ and } [\text{theorem: 10.83}]}{=} \sqrt{1 \cdot 1} = \sqrt{1} \stackrel{[\text{example: 10.69}]}{=} 1$$

and

$$|i| = \sqrt{i \cdot \bar{i}} \stackrel{[\text{theorem: 10.83}]}{=} \sqrt{-(i \cdot i)} = \sqrt{1} = 1$$

$$8. |z| = |\operatorname{Re}(z) + i \cdot \operatorname{Im}(z)| \leq_{(5)} |\operatorname{Re}(z)| + |i \cdot \operatorname{Im}(z)| \stackrel{(4)}{=} |\operatorname{Re}(z)| + |i| \cdot |\operatorname{Im}(z)| \stackrel{(7)}{=} |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$$

9.

$\Rightarrow$. If $z = 0$ then $z = 0 + i \cdot 0$ so that $\operatorname{Re}(z) = \operatorname{Im}(z) = 0$ hence

$$|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2} = \sqrt{0^2 + 0^2} \stackrel{[\text{example: 10.69}]}{=} 0$$

$\Leftarrow$. If $|z| = 0$ then $\sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2} = 0 \stackrel{[\text{example: 10.69}]}{=} \sqrt{0}$ so as $\sqrt{\cdot}$ is injective it follows that $\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2 = 0$. Using [theorem: 10.14] we have that $0 \leq \operatorname{Re}(z)^2, \operatorname{Im}(z)^2$ so that by [theorem: 10.14] again we have that

$$\operatorname{Re}(z)^2, \operatorname{Im}(z)^2 \leq \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$$

If now either $\operatorname{Re}(z) \neq 0$ or $\operatorname{Im}(z) \neq 0$ then by [theorem: 10.14] $0 < \operatorname{Re}(z)$ or $0 < \operatorname{Im}(z)$ which would give $0 < \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$ contradicting $\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2 = 0$. Hence we have that $\operatorname{Re}(z) = 0 = \operatorname{Im}(z)$ so that $z = \operatorname{Re}(z) + i \cdot \operatorname{Im}(z) = 0$.

$$10. \text{ If } z \in \mathbb{C} \setminus \{0\} \text{ then as } 1 = z \cdot z^{-1} \text{ we have } 1 \stackrel{(7)}{=} |1| = |z \cdot z^{-1}| \stackrel{(1)}{=} |z| \cdot |z^{-1}| \text{ so that } |z^{-1}| = |z|^{-1}$$

$$11. \text{ If } z \in \mathbb{R}_{0,\mathbb{C}}^+ \text{ then we have } \operatorname{Re}(z) \stackrel{[\text{theorem: 10.80}]}{=} z \in \mathbb{R}_{0,\mathbb{C}}^+ \text{ and } \operatorname{Im}(z) = 0 \text{ so that}$$

$$z = \operatorname{Re}(z) = \sqrt{\operatorname{Re}(z)^2} = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2} = |z|$$

proving that

$$\forall z \in \mathbb{R}_{0,\mathbb{C}}^+ \text{ we have } z = |z| \quad (10.70)$$

Now if $z \in \mathbb{R}_{\mathbb{C}}$ we have either

$$\mathbf{0} \leq z. \text{ Then } z \in \mathbb{R}_{0,\mathbb{C}}^+ \text{ and by [eq: 10.70] we have } z = |z|$$

$$z < \mathbf{0}. \text{ Then } 0 < -z \text{ so that by [eq: 10.70] we have } -z = |-z| \stackrel{(6)}{=} |z| \text{ proving } -z = |z|$$

$$12. \text{ We prove this by induction so let } S = \{i \in \{0, \dots, \infty\} \mid |z^n| = |z|^n\} \text{ then we have:}$$

$$\mathbf{0} \in S. \text{ As } |z^0| = |1| \stackrel{(11)}{=} 1 = |1|^0 \text{ proving that } 0 \in S.$$

$$n \in S \Rightarrow n+1 \in S. \text{ We have}$$

$$\begin{aligned} |z^{n+1}| &= |z \cdot z^n| \\ &\stackrel{(1)}{=} |z| \cdot |z^n| \\ &\stackrel{n \in S}{=} |z|^{n+1} \end{aligned}$$

proving that $n+1 \in S$.

□

10.5.3 Finite sets

We define now a characterization of finite sets in terms of $\mathbb{N}_{0,\mathbb{C}}$.

Lemma 10.86. *If $n \in \mathbb{N}_0$ then $\{1, \dots, n\} \approx \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\}$*

Proof. First we have by [theorem: 10.5] that $i_{\mathbb{N}_0 \rightarrow \mathbb{Z}}: \mathbb{N}_0 \rightarrow \mathbb{N}_{0,\mathbb{C}}$ is a bijection, hence $i_{\mathbb{N}_0 \rightarrow \mathbb{C}}$ is injective. Further we have by [theorem: 10.34] that

$$i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\{1, \dots, n\}) = \{i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(1), \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\} \stackrel{[\text{theorem: 10.5}]}{=} \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\}$$

so that

$$(i_{\mathbb{N}_0 \rightarrow \mathbb{C}})|_{\{1, \dots, n\}}: \{1, \dots, n\} \rightarrow \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\}$$

is a bijection.

□

Theorem 10.87. *We have the following characterization of finite sets then*

$$I \text{ is finite} \Leftrightarrow \text{there exists a unique } k \in \mathbb{N}_{\mathbb{C},0} \text{ such that } \{1, \dots, k\} \approx I$$

Proof.

$\Rightarrow$. As I is finite we have by [theorem: 6.22] that $\exists n \in \mathbb{N}_0$ such that $\{1, \dots, n\} \approx I$. Hence if we take $k = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n) \in \mathbb{N}_{\mathbb{C},0}$ we have by [lemma: 10.86] that

$$\{1, \dots, n\} \approx \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\} = \{1, \dots, k\}$$

proving that

$$\{1, \dots, k\} \approx I$$

Now for uniqueness, assume that there is another $l \in \mathbb{N}_{\mathbb{C},0}$ such that $\{1, \dots, l\} \approx I$. Take then $m = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(l)$ we have by [lemma: 10.86] that

$$\{1, \dots, m\} \approx \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(m)\} = \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(l))\} = \{1, \dots, l\} \approx I$$

so that $\{1, \dots, n\} \approx \{1, \dots, m\}$, which, as by [theorem: 6.21] $\{1, \dots, n\} \approx n \wedge \{1, \dots, m\} \approx m$, gives that $n \approx m$. So that $(i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(k) = n = m = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(l)$ giving as $(i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}$ is a bijection that $k = l$.

$\Leftarrow$. Let $k \in \mathbb{N}_{\mathbb{C},0}$ be such that $I \approx \{1, \dots, k\}$ then if we define $n \in \mathbb{N}_0$ by $n = (i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(k)$ we have by [lemma: 10.86] that

$$\{1, \dots, n\} \approx \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\} = \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}((i_{\mathbb{N}_0 \rightarrow \mathbb{C}})^{-1}(k))\} = \{1, \dots, k\} \approx I$$

proving that $\{1, \dots, n\} \approx I$, hence by [theorem: 6.22] I is finite. $\square$

Corollary 10.88. *Let I be a non empty set then*

$$I \text{ is finite} \Leftrightarrow \text{there exists a **unique** } k \in \mathbb{N}_{\mathbb{C}} \text{ such that } \{0, \dots, k-1\} \approx I$$

Proof. Let $k \in \mathbb{N}_{\mathbb{C}} \Rightarrow k-1 \in \mathbb{N}_{\mathbb{C},0}$. Define

$$\tau: \{0, \dots, k-1\} \rightarrow \{1, \dots, k\} \text{ by } \tau(i) = i+1$$

then we have:

injectivity. If $\gamma(i) = \gamma(j)$ then $i+1 = j+1$ hence $i = j$

surjectivity. If $j \in \{1, \dots, k\}$ then $1 \leq j \leq k \Rightarrow 0 \leq j-1 \leq k-1$ proving that $j-1 \in \{0, \dots, k-1\}$. As $\gamma(j-1) = (j-1)+1 = j$ surjectivity is proved.

So we have that

$$\text{If } k \in \mathbb{N}_{\mathbb{C}} \text{ then } \{0, \dots, k-1\} \approx \{1, \dots, k\}$$

$\Rightarrow$. If I is finite there exists by [theorem: 10.87] a $k \in \mathbb{N}_{\mathbb{C},0}$ such that $\{1, \dots, k\} \approx I$ hence there exists a bijection

$$\alpha: \{1, \dots, k\} \rightarrow I.$$

If $k=0$ then $\{1, \dots, k\} = \emptyset$ so that $I = \alpha(\emptyset) = \emptyset$ contradicting $I \neq \emptyset$, hence $k \in \mathbb{N}_{\mathbb{C}}$. As $\{0, \dots, k-1\} \approx \{1, \dots, k\}$ and $\{1, \dots, k\} \approx I$ it follows that

$$\{0, \dots, k-1\} \approx I$$

As for uniqueness. If $l \in \mathbb{N}_{\mathbb{C}}$ such that $\{0, \dots, l-1\} \approx I$ then as $\{0, \dots, l-1\} \approx \{1, \dots, l\}$ we have also $\{1, \dots, l\} \approx I$. Using the previous theorem [theorem: 10.87] it follows then that $k=l$.

$\Leftarrow$. If $k \in \mathbb{N}_{\mathbb{C}}$ such that $\{0, \dots, k-1\} \approx I$ then as $\{0, \dots, k-1\} \approx \{1, \dots, k\}$ it follows that $\{1, \dots, k\} \approx I$. Using the previous theorem [theorem: 10.87] it follows then that I is finite. $\square$

Definition 10.89. *Let I be a finite set then $\text{card}(I)$ is defined as follows*

$$\text{card}(I) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(I))$$

where $\#(I)$ is defined in [definition: 6.39]

Theorem 10.90. *If I is a finite set then we have that*

$$\text{card}(I) \text{ is the unique } k \in \mathbb{N}_{0,\mathbb{C}} \text{ such that } \{1, \dots, k\} \approx I$$

In particular

$$I = \emptyset \Leftrightarrow \text{card}(I) = 0$$

Proof. By definition we have that $\#(I) = n$ where $n \approx I$ then

$$\text{card}(I) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(I)) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)$$

As by [theorem: 6.21] we have that $\{1, \dots, n\} \approx n$, further by [lemma: 10.86]

$$\{1, \dots, n\} \approx \{1, \dots, i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(n)\} = \{1, \dots, \text{card}(I)\}$$

so that $I \approx \{1, \dots, \text{card}(I)\}$. Finally we have

$$I = \emptyset \underset{[\text{theorem: 6.40}]}{\Leftrightarrow} \#(I) = 0 \Leftrightarrow i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(I)) = 0 \Leftrightarrow \text{card}(I) = 0 \quad \square$$

Theorem 10.91. *If A, B are sets such that A is finite then*

1. *If $B \subseteq A$ then B is finite and $\text{card}(B) \leq \text{card}(A)$*
2. *If $B \subset A$ then B is finite and $\text{card}(B) < \text{card}(A)$*

Proof.

1. Using [theorem: 6.43] it follows that B is finite and $\#(B) \leq \#(A)$, so, as by [theorem: 10.9] is a order isomorphism, we have $\text{card}(B) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(B)) \leq i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A)) = \text{card}(A)$.
2. Using [theorem: 6.43] it follows that B is finite and $\#(B) < \#(A)$, so, as by [theorem: 10.9] is a order isomorphism, we have $\text{card}(B) = i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(B)) < i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A)) = \text{card}(A)$ $\square$

Theorem 10.92. *If A, B are finite sets then*

1. *$A \times B$ is finite and $\text{card}(A \times B) = \text{card}(A) \cdot \text{card}(B)$*
2. *$A \cup B$ is finite and $A \cap B = \emptyset$ then $\text{card}(A \cup B) = \text{card}(A) + \text{card}(B)$*

Proof.

1. By [theorem: 6.41] we have that $A \times B$ is finite and $\#(A \times B) = \#(A) \cdot \#(B)$. Further we have that

$$\begin{aligned} \text{card}(A \times B) &= i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A \times B)) \\ &= i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A) \cdot \#(B)) \\ &= i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A)) \cdot i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(B)) \\ &= \text{card}(A) \cdot \text{card}(B) \end{aligned}$$

2. By [theorems: 6.34, 6.42] we have that $A \cup B$ is finite and $\#(A \cup B) = \#(A) + \#(B)$. Further we have that

$$\begin{aligned} \text{card}(A \cup B) &= i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A \cup B)) \\ &= i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A) + \#(B)) \\ &= i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(A)) + i_{\mathbb{N}_0 \rightarrow \mathbb{C}}(\#(B)) \\ &= \text{card}(A) + \text{card}(B) \end{aligned}$$

$\square$

Corollary 10.93. *If A is a finite set and $a \notin A$ then $\text{card}(A \cup \{a\}) = \text{card}(A) + 1$*

Proof. As $A \not\subseteq A$ we have that $A \cap \{a\} = \emptyset$ so by [theorem: 10.92] we have

$$\text{card}(A \cup \{a\}) = \text{card}(A) + \text{card}(\{a\}) = \text{card}(A) + 1 \quad \square$$

Corollary 10.94. *If A is a finite set and $a \in A$ then $\text{card}(A \setminus \{a\}) = \text{card}(A) - 1$*

Proof. As $a \notin A \setminus \{a\}$ and $A = (A \setminus \{a\}) \cup \{a\} = A$ we have that

$$\text{card}(A) \stackrel{[\text{corollary: 10.93}]}{=} \text{card}(A \setminus \{a\}) + 1$$

so that

$$\text{card}(A \setminus \{a\}) = \text{card}(A) - 1 \quad \square$$

Corollary 10.95. *If A is a finite set and $B \subseteq A$ such that $\text{card}(B) = \text{card}(A)$ then $B = A$*

Proof. Assume that $B \neq A$ then as $B \subseteq A$ there exist a $a \in A$ such that $a \notin B$. Hence $B \cup \{a\} \subseteq A$ so that $\text{card}(B \cup \{a\}) \leq \text{card}(A)$, further we have by [corollary: 10.93] that $\text{card}(B \cup \{a\}) = \text{card}(B) + 1$ so that $\text{card}(B) + 1 \leq \text{card}(A)$ or $\text{card}(B) < \text{card}(A)$ contradicting $\text{card}(A) = \text{card}(B)$. Hence we must have that $A = B$. $\square$

Theorem 10.96. *Let I be a finite set, $\{x_i\}_{i \in I} \subseteq X$ then $\{x_i | i \in I\}$ is finite and*

$$\text{card}(\{x_i | i \in I\}) \leq \text{card}(I)$$

Proof. This follows from [theorem: 6.46]. $\square$

The next theorem shows that every finite set can be sorted by a bijection.

Theorem 10.97. *Let $\langle X, \leq \rangle$ be a totally ordered set, A be a finite subset of X with $\text{card}(A) = n \in \mathbb{N}$ then there exist a bijection $\beta: \{1, \dots, n\} \rightarrow A$ such that $\forall i \in \{1, \dots, n-1\}$ we have $\beta(i) < \beta(i+1)$*

Proof. We prove this by induction on $\text{card}(A)$ so define

$S = \{n \in \mathbb{N} | \text{If } A \subseteq X \text{ with } \text{card}(A) = n \text{ then there exist a bijection } \beta: \{1, \dots, n\} \rightarrow A \text{ such that } \forall i \in \{1, \dots, n-1\} \text{ we have } \beta(i) \leq \beta(i+1)\}$

then we have:

$1 \in S$. As $\text{card}(A) = 1$ there exist a bijection $\beta: \{1\} \rightarrow A$. As $\{1, \dots, 1-1\} = \emptyset$ $\beta(i) \leq \beta(i+1)$ is satisfied vacuously. Hence $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $A \subseteq X$ with $\text{card}(A) = n+1$. As $n \in S \subseteq \mathbb{N}$ we have that $n \in \mathbb{N} \Rightarrow n+1 \in \{2, \dots, \infty\}$. As A is finite

As A is finite and $A \neq \emptyset$ [because $n+1 > 0 \Rightarrow \alpha(1) \in A$] $m = \max(A)$ exist. Take $B = A \setminus \{m\}$ then by [theorem: 10.94] $\text{card}(B) = (n+1) - 1 = n$. Hence as $n \in A$ there exist a bijection

$$\alpha: \{1, \dots, n\} \rightarrow B \text{ with } \forall i \in \{1, \dots, n-1\} \alpha(i) < \alpha(i+1)$$

Define then

$$\beta: \{1, \dots, n+1\} \rightarrow A \text{ by } \beta(i) = \begin{cases} \alpha(i) & \text{if } i \in \{1, \dots, n\} \\ m & \text{if } i = n+1 \end{cases}$$

then we have:

injectivity. If $i, j \in \{1, \dots, n+1\}$ with $\beta(i) = \beta(j)$ then we have for i, j either:

$i = n+1 \wedge j = n+1$. Then trivially $i = j$

$i \in \{1, \dots, n\} \wedge j = n+1$. Then $m = \beta(j) = \beta(i) = \alpha(i) \in B = A \setminus \{m\}$ giving the contradiction $m \in A \setminus \{m\}$, hence this case never occurs.

$i = n+1 \wedge j \in \{1, \dots, n\}$. Then $m = \beta(i) = \beta(j) = \alpha(j) \in B = A \setminus \{m\}$ giving the contradiction $m \in A \setminus \{m\}$, hence this case never occurs.

$i \in \{1, \dots, n\} \wedge j \in \{1, \dots, n\}$. Then $\alpha(i) = \beta(i) = \beta(j) = \alpha(j)$ so that, as α is a bijection, $i = j$.

Hence in all valid cases we have $i = j$ which proves injectivity.

surjectivity. Let $y \in A$ then we have either $y = m$ then $y = \beta(n+1)$ or $y \in A \setminus \{m\}$ then, as α is surjective, $\exists i \in \{1, \dots, n\} \subseteq \{1, \dots, n+1\}$ such that $y = \alpha(i) = \beta(i)$.

So we have proved that

$$\beta: \{1, \dots, n+1\} \rightarrow A \text{ is a bijection}$$

Finally let $i \in \{1, \dots, n\}$ then we have for i either:

$i \in \{1, \dots, n-1\}$. Then $\beta(i) = \alpha(i) < \alpha(i+1) = \beta(i+1)$

$i = n$. Then $\beta(n+1) = m$, as $m = \max(A)$ and $\beta(i) \in A$ it follows that $\beta(i) \leq \beta(i+1)$, further, as β is a bijection, $\beta(i) \neq \beta(i+1)$ so that $\beta(i) < \beta(i+1)$

hence

$$\forall i \in \{1, \dots, n\} \text{ we have } \beta(i) < \beta(i+1)$$

proving that $n+1 \in S$. □

The following lemma will later be used to define the extended real numbers.

Lemma 10.98. *There exists at least two different elements that are not element of $\mathbb{R}_{\mathbb{C}}$*

Proof. Using [definitions: 9.1, 9.2] it follows that $\emptyset \notin \mathbb{R}$ and $\mathbb{Q} \notin \mathbb{R}$ and as $0 \in \mathbb{Q}$ we have $\emptyset \neq \mathbb{Q}$ So that $(\emptyset, 0) \notin \mathbb{R}_{\mathbb{C}}$ and $(\mathbb{Q}, 0) \notin \mathbb{R}_{\mathbb{C}}$ and $(\emptyset, 0) \neq (\mathbb{Q}, 0)$ □

10.5.4 Conventions

Now we are finished with the tower of different types of numbers. From now on for the rest of this book we work only with $\mathbb{N}_{0,\mathbb{C}}$, $\mathbb{Z}_{\mathbb{C}}$, $\mathbb{Q}_{\mathbb{C}}$, $\mathbb{R}_{\mathbb{C}}$, $\mathbb{C}$ and $\overline{\mathbb{R}}$. To avoid excessive use of subscripts we leave out the subscripts. Hence for the rest of this book we use the following conventions:

Name of set	Symbol	Meaning
Natural Number	$\mathbb{N}_0$	$\mathbb{N}_{0,\mathbb{C}}$
Positive Natural Numbers	$\mathbb{N}$	$\mathbb{N}_{0,\mathbb{C}} \setminus \{0\}$
Integers	$\mathbb{Z}$	$\mathbb{Z}_{\mathbb{C}}$
Rational Numbers	$\mathbb{Q}$	$\mathbb{Q}_{\mathbb{C}}$
Real Numbers	$\mathbb{R}$	$\mathbb{R}_{\mathbb{C}}$
Extended Real Numbers	$\overline{\mathbb{R}}$	$\overline{\mathbb{R}}$
Non negative real numbers	$\mathbb{R}_0^+$	$\{x \in \mathbb{R} 0 \leq x\}$
Positive real numbers	$\mathbb{R}^+$	$\{x \in \mathbb{R} 0 < x\}$
Non positive numbers	$\mathbb{R}_0^-$	$\{x \in \mathbb{R} x \leq 0\}$
Negative real numbers	$\mathbb{R}^-$	$\{x \in \mathbb{R} x < 0\}$
Complex numbers	$\mathbb{C}$	$\mathbb{C}$

Using this notation we have that

$\mathbb{N}_0 \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$
$\mathbb{N}_0 \subseteq \mathbb{R}_0^+ \subseteq \mathbb{R} \subseteq \mathbb{C}$
$\mathbb{N} \subseteq \mathbb{R}^+ \subseteq \mathbb{R}_0^+ \subseteq \mathbb{R} \subseteq \mathbb{C}$
$\mathbb{R}^- \subseteq \mathbb{R}_0^- \subseteq \mathbb{R} \subseteq \mathbb{C}$
$\forall z \in \mathbb{C}$ there exists unique $x, y \in \mathbb{R}$ such that $z = x + i \cdot y$

Another implicit convention that will be used in this book concerns the finite power of sets where we extend the notation suggested in [theorem: 2.152]

Definition 10.99. *Let A be a set, $n \in \mathbb{N}$ then*

$$A^n = \prod_{i \in \{1, \dots, n\}} A_i \text{ where } \forall i \in \{1, \dots, n\} \ A_i = A$$

Chapter 11

Linear Algebra

11.1 Sums and products

11.1.1 Definition and properties

First we define the concept of a the sum of a finite family in a semi-group.

Definition 11.1. (Finite Sum) Let $\langle A, + \rangle$ be a semi-group, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$ a finite family of elements of A then

$$\sum_{i=0}^{(.)} x_i: \{0, \dots, n\} \rightarrow A$$

is recursively defined [see definition: 10.41] by

$$\begin{aligned} \sum_{i=0}^0 x_i &= x_0 \\ \forall i \in \{0, \dots, n-1\} \sum_{i=0}^{i+1} x_i &= \left(\sum_{i=0}^i x_i \right) + x_{i+1} \end{aligned}$$

Remark 11.2. We typical use the $\sum$ symbol in this text to represent a finite sum in a semi-group. However on the same set we can have many operators that makes the set a semi-group (for example in a ring, field or vector space we have a operator that represent addition and a operator that represent multiplication. To express these differences we use $\sum$ for addition and $\prod$ for multiplication. The following table gives a overview of the use of these symbols.

Structure	Operator	Symbol
additive semi-group	+	$\sum$
multiplicative semi-group	$\cdot$	$\prod$
ring	+	$\sum$
	$\cdot$	$\prod$
field	+	$\sum$
	$\cdot$	$\prod$
vector space	+	$\sum$
	$\cdot$	$\prod$
X^X	$\circ$	$\prod$

All the theorems, propositions, lemma's. corollaries and definitions in this section using the $\sum$ symbol also apply when we use the $\prod$ symbol. The meaning of $\sum$, $\prod$ symbol is always derived from the context where the operator is specified.

Example 11.3. Let $\{2^i\}_{i \in \{0, \dots, 3\}} \subseteq \mathbb{R}$ then for the field $\langle \mathbb{R}, +, \cdot \rangle$ $\sum_{i=0}^3 2^i = 15$

Proof.

$$\begin{aligned}
 \sum_{i=0}^3 2^i &= \left(\sum_{i=0}^2 2^i \right) + 2^3 \\
 &= \left(\left(\sum_{i=0}^1 2^i \right) + 2^2 \right) + 2^3 \\
 &= \left(\left(\left(\sum_{i=0}^0 2^i \right) + 2^1 \right) + 2^2 \right) + 2^3 \\
 &= ((2^0 + 2^1) + 2^2) + 2^3 \\
 &= ((1 + 2) + 4) + 8 \\
 &= 15
 \end{aligned}$$

□

Theorem 11.4. If $\langle A, + \rangle$ is a semi-group, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}}$ then we have $\forall k \in \{0, \dots, n\}$ that

$$\sum_{i=0}^k x_i = \begin{cases} x_0 & \text{if } k=0 \\ (\sum_{i=0}^{k-1} x_i) + x_k & \text{if } k \in \{1, \dots, n\} \end{cases}$$

Proof. For $k \in \{0, \dots, n\}$ we have either:

$k=0$. Then $\sum_{i=0}^k x_i = \sum_{i=0}^0 x_i = x_0$

$k \in \{1, \dots, n\}$. Then $l = k - 1 \in \{0, \dots, n - 1\}$ so that

$$\sum_{i=0}^k x_i = \sum_{i=0}^{l+1} x_i = \left(\sum_{i=0}^l x_i \right) + x_{l+1} = \left(\sum_{i=0}^{k-1} x_i \right) + x_k$$

□

Theorem 11.5. Let $\langle A, + \rangle$ be a semi-group, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$ is such that $\forall i \in \{0, \dots, n\} \ x_i = 0$ then

$$\sum_{i=0}^n x_i = 0$$

Proof. We prove this by mathematical induction so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ with } \forall i \in \{0, \dots, n\} \ x_i = 0 \text{ we have } \sum_{i=0}^n x_i = 0 \right\}$$

then we have:

$0 \in S$. If $\{x_i\}_{i \in \{0, \dots, 0\}} \subseteq A$ with $\forall i \in \{0, \dots, 0\} = \{0\} \ x_i = 0$ we have that $\sum_{i=0}^0 x_i = x_0 = 0$ proving that $0 \in S$.

$n \in S \Rightarrow n + 1 \in S$. If $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ with $\forall i \in \{0, \dots, n+1\} \ x_i = 0$ then we have

$$\begin{aligned}
 \sum_{i=0}^{n+1} x_i &= \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\
 &= \left(\sum_{i=0}^n x_i \right) + 0 \\
 &= \sum_{i=0}^n x_i \\
 &\stackrel{n \in S}{=} 0
 \end{aligned}$$

proving that $n + 1 \in S$.

□

Theorem 11.6. Let $\langle A, + \rangle$ be a **Abelian** semi-group, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$, $\{y_i\}_{i \in \{0, \dots, n\}} \subseteq A$ then

$$\sum_{i=0}^n (x_i + y_i) = \sum_{i=0}^n x_i + \sum_{i=0}^n y_i$$

Proof. We prove this by mathematical induction so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}}, \{y_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ we have } \sum_{i=0}^n (x_i + y_i) = \sum_{i=0}^n x_i + \sum_{i=0}^n y_i \right\}$$

then we have:

$0 \in S$. If $\{x_i\}_{i \in \{0\}}, \{y_i\}_{i \in \{0\}} \subseteq A$ then $\sum_{i=0}^0 (x_i + y_i) = x_0 + y_0 = \sum_{i=0}^0 x_i + \sum_{i=0}^0 y_i$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $\{x_i\}_{i \in \{0, \dots, n+1\}}, \{y_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ then

$$\begin{aligned} \sum_{i=0}^{n+1} (x_i + y_i) &= \left(\sum_{i=0}^n (x_i + y_i) \right) + (x_{n+1} + y_{n+1}) \\ &\stackrel{n \in S}{=} \left(\sum_{i=0}^n x_i + \sum_{i=0}^n y_i \right) + (x_{n+1} + y_{n+1}) \\ &\stackrel{\text{associativity and commutativity}}{=} \left(\sum_{i=0}^n x_i + x_{n+1} \right) + \left(\sum_{i=0}^n y_i + y_{n+1} \right) \\ &= \sum_{i=0}^{n+1} x_i + \sum_{i=0}^{n+1} y_i \end{aligned}$$

proving that $n+1 \in S$. □

Theorem 11.7. Let $\langle A, + \rangle$ is a group, such that $\forall x \in A$ $-x$ is the inverse of x , $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$ then we have

$$\sum_{i=0}^n (-x_i) = - \sum_{i=0}^n x_i$$

Proof. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ we have } \sum_{i=0}^n (-x_i) = - \sum_{i=0}^n x_i \right\}$$

then we have:

$0 \in S$. If $\{x_i\}_{i \in \{0\}} \subseteq A$ then $\sum_{i=0}^0 (-x_i) = -x_0 = - \sum_{i=0}^0 x_i$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ then

$$\begin{aligned} \sum_{i=0}^{n+1} (-x_i) &= \left(\sum_{i=0}^n (-x_i) \right) + (-x_{n+1}) \\ &\stackrel{n \in S}{=} - \sum_{i=0}^n x_i + (-x_{n+1}) \\ &= - \left(\left(\sum_{i=0}^n x_i \right) + x_{n+1} \right) \\ &= - \sum_{i=0}^{n+1} x_i \end{aligned}$$

hence we have $n+1 \in S$ □

Theorem 11.8. Let $\langle R, +, \cdot \rangle$ be a ring, $\alpha \in R$, $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$ then we have

$$\sum_{i=0}^n \alpha \cdot x_i = \alpha \cdot \sum_{i=0}^n x_i$$

Proof. As usually the prove is by induction. So let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq R \text{ we have } \sum_{i=0}^n (\alpha_i \cdot x_i) = \alpha \cdot \sum_{i=0}^n x_i \right\}$$

then we have:

$0 \in S$. Let $\{x_i\}_{i \in \{0, \dots, 0\}} \subseteq R$ then $\sum_{i=0}^0 (\alpha \cdot x_i) = \alpha \cdot x_0 = \alpha \cdot \sum_{i=0}^0 x_i$, proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ then

$$\begin{aligned} \sum_{i=0}^{n+1} (\alpha \cdot x_i) &= \left(\sum_{i=0}^n (\alpha \cdot x_i) \right) + \alpha \cdot x_{n+1} \\ &\stackrel{n \in S}{=} \alpha \cdot \sum_{i=0}^n x_i + \alpha \cdot x_{n+1} \\ &= \alpha \cdot \left(\left(\sum_{i=0}^n x_i \right) + x_{n+1} \right) \\ &= \alpha \cdot \sum_{i=0}^{n+1} x_i \end{aligned}$$

proving that $n+1 \in S$. □

Theorem 11.9. Let $\langle A, + \rangle$ be a **Abelian** semi-group, $n_1, n_2 \in \mathbb{N}_0$ and

$$\{x_{i,j}\}_{(i,j) \in \{0, \dots, n_1\} \times \{0, \dots, n_2\}} \subseteq A$$

then

$$\sum_{i=0}^{n_1} \left(\sum_{j=0}^{n_2} x_{i,j} \right) = \sum_{j=0}^{n_2} \left(\sum_{i=0}^{n_1} x_{i,j} \right)$$

See [definition: 2.118] for the definition of $\{x_{i,k}\}_{k \in \{0, \dots, n_2\}}$ and $\{x_{k,j}\}_{k \in \{0, \dots, n_1\}}$ used in $\sum_{j=0}^{n_2} x_{i,j}$ and $\sum_{i=0}^{n_1} x_{i,j}$.

Proof. We prove this by induction so let $n \in \{0, \dots, n_1\}$ and take

$$S_n = \left\{ m \in \{0, \dots, n_2\} \mid \forall \{x_{i,j}\}_{(i,j) \in \{0, \dots, n\} \times \{0, \dots, m\}} \subseteq A \text{ then } \sum_{i=0}^n \left(\sum_{j=0}^m x_{i,j} \right) = \sum_{j=0}^m \left(\sum_{i=0}^n x_{i,j} \right) \right\}$$

then we have:

$0 \in S_n$. Take $\{x_{i,j}\}_{(i,j) \in \{0, \dots, n\} \times \{0\}}$ then

$$\sum_{i=0}^n \left(\sum_{j=0}^0 x_{i,j} \right) = \sum_{i=0}^n x_{i,0} = \sum_{j=0}^0 \left(\sum_{i=0}^n x_{i,j} \right)$$

proving that $0 \in S_n$.

$m \in S_n \Rightarrow m+1 \in S_n$. Take $\{x_{i,j}\}_{(i,j) \in \{0, \dots, n\} \times \{0, \dots, m+1\}}$ then we have

$$\begin{aligned} \sum_{i=0}^n \left(\sum_{j=0}^{m+1} x_{i,j} \right) &= \sum_{i=0}^n \left(\left(\sum_{j=0}^m x_{i,j} \right) + x_{i,m+1} \right) \\ &\stackrel{[\text{theorem: 11.6}]}{=} \sum_{i=0}^n \left(\sum_{j=0}^m x_{i,j} \right) + \sum_{i=0}^n x_{i,m+1} \\ &\stackrel{m \in S_n}{=} \sum_{j=0}^m \left(\sum_{i=0}^n x_{i,j} \right) + \sum_{i=0}^n x_{i,m+1} \\ &= \sum_{j=0}^{m+1} \left(\sum_{i=0}^n x_{i,j} \right) \end{aligned}$$

proving that $m+1 \in S_n$.

Using mathematical induction we have then that $S_n = \mathbb{N}_0$. So if $n_1, n_2 \in \mathbb{N}_0$ and $\{x_{i,j}\}_{(i,j) \in \{0, \dots, n_1\} \times \{0, \dots, n_2\}} \subseteq A$ then $n_1 \in \mathbb{N}_0$ and $n_2 \in \mathbb{N}_0 = S_{n_1}$ so that

$$\sum_{i=0}^{n_1} \left(\sum_{j=0}^{n_2} x_{i,j} \right) = \sum_{j=0}^{n_2} \left(\sum_{i=0}^{n_1} x_{i,j} \right)$$

□

Theorem 11.10. Let $\langle A, + \rangle$ be a Abelian group, $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$ then for $\{x_{i+1} - x_i\}_{i \in \{0, \dots, n-1\}} \subseteq A$ we have

$$\sum_{i=0}^{n-1} (x_{i+1} - x_i) = x_n - x_0$$

Proof. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ then } \sum_{i=0}^{n-1} (x_{i+1} - x_i) = x_n - x_0 \right\}$$

then we have:

$1 \in S$. If $\{x_i\}_{i \in \{0,1\}} \subseteq A$ then $\sum_{i=0}^{1-1} x_i = \sum_{i=0}^0 (x_{i+1} - x_i) = x_1 - x_0$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ then

$$\begin{aligned} \sum_{i=0}^{(n+1)-1} (x_{i+1} - x_i) &= \sum_{i=0}^n (x_{i+1} - x_i) \\ &= \left(\sum_{i=0}^{n-1} (x_{i+1} - x_i) \right) + (x_{n+1} - x_n) \\ &\stackrel{n \in S}{=} (x_n - x_0) + (x_{n+1} - x_n) \\ &\stackrel{\text{associativity and commutativity}}{=} x_{n+1} - x_0 \end{aligned}$$

proving that $n+1 \in S$. □

Definition 11.11. Let A be a set, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ a family then then the family $\{x_{i+n}\}_{i \in \{0, \dots, n\}}$ is defined as $x \circ \beta$ where $\beta: \{0, \dots, m-n\} \rightarrow \{n, \dots, m\}$ is the function defined by

$$\beta(i) = n + i$$

[see definition: 2.119]

Up to now we have only defined the finite sum of a family of elements indexed by $\{0, \dots, n\}$, we extend now this definition to a more general index set.

Definition 11.12. If $\langle A, + \rangle$ is a semi-group, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then we define $\sum_{i=n}^m x_i$ by

$$\sum_{i=n}^m x_i = \sum_{i=0}^{m-n} x_{n+i}$$

Note 11.13. If $n=0$ then $m-n=m$ and $\{x_i\}_{i \in \{n, \dots, m\}} = \{x_{0+i}\}_{i \in \{0, \dots, m\}} = \{x_i\}_{i \in \{0, \dots, m\}}$ and $\sum_{i=n}^m x_i = \sum_{i=0}^m x_{0+i} = \sum_{i=0}^m x_i$ as expected.

Theorem 11.14. If $\langle A, + \rangle$ is a semi-group, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then for $k \in \{0, \dots, n\}$ we have

$$\sum_{i=n}^m x_i = \sum_{i=k}^{m+k-n} x_{n-k+i}$$

Proof. We have as $k \in \{0, \dots, n\} \subseteq \mathbb{N}_0$ that $k \leq n \leq m$ that

$$\begin{aligned} \sum_{i=k}^{m+k-n} x_{(n-k)+i} &\stackrel{\text{def}}{=} \sum_{i=0}^{(m+k-n)-k} x_{(n-k)+i+k} \\ &= \sum_{i=0}^{m-n} x_{n+i} \\ &= \sum_{i=n}^m x_i \\ &\square \end{aligned}$$

Using the above definition we can rephrase [theorems: 11.4, 11.5, 11.6, 11.9, 11.7, 11.10]

Theorem 11.15. If $\langle A, + \rangle$ is a semi-group, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then we have for $k \in \{n, \dots, m\}$ that

$$\sum_{i=n}^k x_i = \begin{cases} x_n & \text{if } k=n \\ (\sum_{i=n}^{k-1} x_i) + x^k & \text{if } k \in \{n+1, \dots, m\} \end{cases}$$

Proof. Let $k \in \{n, \dots, m\}$ then we have either:

$k = n$. Then $\sum_{i=n}^k x_i = \sum_{i=n}^n x_i = \sum_{i=0}^{n-n} x_{n+i} = \sum_{i=0}^0 x_{n+i} = x_{n+0} = x_n$

$k \in \{n+1, \dots, m\}$. Then as $n < m$ we have $m-n \in \mathbb{N}$ and for $l \in \{1, \dots, m-n\}$

$$\sum_{i=0}^l x_{n+i} \stackrel{[\text{theorem: 11.4}]}{=} \left(\sum_{i=0}^{l-1} x_{n+i} \right) + x_{n+l} \quad (11.1)$$

Further

$$\begin{aligned} \sum_{i=n}^k x_i &= \sum_{i=0}^{k-n} x_{n+i} \\ &\stackrel{k-n \in \{1, \dots, m-n\} \text{ and [eq: 11.1]}}{=} \left(\sum_{i=0}^{(k-n)-1} x_{n+i} \right) + x_{n+(k-n)} \\ &= \left(\sum_{i=n}^{k-1} x_i \right) + x_k \end{aligned}$$

□

Theorem 11.16. If $\langle A, + \rangle$ is a semi-group, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ is such that $\forall i \in \{n, \dots, m\} x_i = 0$ then

$$\sum_{i=n}^m x_i = 0$$

Proof. Note that $\forall i \in \{0, \dots, m-n\} n+i \in \{n, \dots, m\}$, so that $x_{n+i} = 0$, hence

$$\sum_{i=n}^m x_i \stackrel{\text{def}}{=} \sum_{i=0}^{m-n} x_{n+i} \stackrel{[\text{theorem: 11.5}]}{=} 0 \quad \square$$

Theorem 11.17. Let $\langle A, + \rangle$ be a Abelian semi-group, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}}, \{y_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then

$$\sum_{i=n}^m (x_i + y_i) = \sum_{i=n}^m x_i + \sum_{i=n}^m y_i$$

Proof. We have

$$\sum_{i=n}^m (x_i + y_i) = \sum_{i=0}^{m-n} (x_{n+i} + y_{n+i}) \stackrel{[\text{theorem: 11.6}]}{=} \sum_{i=0}^{m-n} x_i + \sum_{i=0}^{m-n} y_i = \sum_{i=n}^m x_i + \sum_{i=n}^m y_i \quad \square$$

Theorem 11.18. Let $\langle R, +, \cdot \rangle$ be a ring, $\alpha \in R$, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then we have

$$\sum_{i=n}^m \alpha \cdot x_i = \alpha \cdot \sum_{i=n}^m x_i$$

Proof.

$$\sum_{i=n}^m \alpha \cdot x_i = \sum_{i=0}^{m-n} \alpha \cdot x_{n+i} \stackrel{[\text{theorem: 11.8}]}{=} \alpha \cdot \sum_{i=0}^{m-n} x_{n+i} = \alpha \cdot \sum_{i=n}^m x_i \quad \square$$

Theorem 11.19. Let $\langle A, + \rangle$ be a Abelian semi-group, $n_1, n_2, m_1, m_2 \in \mathbb{N}_0$ with $n_1 \leq m_1$, $n_2 \leq m_2$ and $\{x_{i,j}\}_{(i,j) \in \{n_1, \dots, m_1\} \times \{n_2, \dots, m_2\}} \subseteq A$ then

$$\sum_{i=n_1}^{m_1} \left(\sum_{j=n_2}^{m_2} x_{i,j} \right) = \sum_{j=n_2}^{m_2} \left(\sum_{i=n_1}^{m_1} x_{i,j} \right)$$

See [definition: 2.118] for $\{x_{i,k}\}_{k \in \{n_2, \dots, m_2\}}$ and $\{x_{k,j}\}_{k \in \{n_1, \dots, m_1\}}$

Proof. We have

$$\begin{aligned} \sum_{i=n_1}^{m_1} \left(\sum_{j=n_2}^{m_2} x_{i,j} \right) &= \sum_{i=0}^{m_1-n_1} \left(\sum_{j=n_2}^{m_2} x_{n_1+i,j} \right) \\ &= \sum_{i=0}^{m_1-n_1} \left(\sum_{j=0}^{m_2-n_2} x_{n_1+i, m_2+j} \right) \\ &\stackrel{[\text{theorem: 11.9}]}{=} \sum_{j=0}^{m_2-n_2} \left(\sum_{i=0}^{m_1-n_1} x_{n_1+i, m_2+j} \right) \\ &= \sum_{j=0}^{m_2-n_2} \left(\sum_{i=n_1}^{m_1} x_{i, m_2+j} \right) \\ &= \sum_{j=n_2}^{m_2} \left(\sum_{i=n_1}^{m_1} x_{i,j} \right) \end{aligned}$$

$\square$

Theorem 11.20. Let $\langle A, + \rangle$ is a group, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then we have

$$\sum_{i=n}^m (-x_i) = - \sum_{i=n}^m x_i$$

Proof. We have

$$\sum_{i=n}^m (-x_i) = \sum_{i=0}^{m-n} (-x_{n+i}) \stackrel{[\text{theorem: 11.7}]}{=} - \sum_{i=0}^{m-n} x_i = - \sum_{i=n}^m x_i \quad \square$$

Theorem 11.21. Let $\langle A, + \rangle$ be a Abelian group, $n, m \in \mathbb{N}_0$ with $n < m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ then for $\{x_{i+1} - x_i\}_{i \in \{n, \dots, m-n-1\}}$ we have

$$\sum_{i=n}^{m-1} (x_{i+1} - x_i) = x_m - x_n$$

Proof. As $n < m$ we have that $m - n \in \mathbb{N}$, so using [theorem: 11.10] we have

$$\sum_{i=0}^{m-n-1} (x_{(n+i)+1} - x_{n+i}) = x_{n+(m-n)} - x_{n+0} = x_m - x_n$$

so that

$$\sum_{i=n}^{m-1} (x_{i+1} - x_i) = \sum_{i=0}^{m-n-1} (x_{(n+i)+1} - x_{n+i}) = x_m - x_n \quad \square$$

Theorem 11.22. If $\langle A, + \rangle$ is a semi-group, $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{0, \dots, n\}}$ then we have

$$\sum_{i=0}^n x_i = x_0 + \sum_{i=1}^n x_i$$

Proof. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N} \mid \text{if } \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ then } \sum_{i=0}^n x_i = x_0 + \sum_{i=1}^n x_i \right\}$$

then we have:

$1 \in S$. If $\{x_i\}_{i \in \{0, \dots, 1\}}$ then $\sum_{i=0}^1 x_i = (\sum_{i=0}^0 x_i) + x_1 = x_0 + x_1 \stackrel{[\text{theorem: 11.15}]}{=} x_0 + \sum_{i=1}^1 x_i$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ then we have

$$\begin{aligned} \sum_{i=0}^{n+1} x_i &= \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\ &\stackrel{n \in S}{=} \left(x_0 + \sum_{i=1}^n x_i \right) + x_{n+1} \\ &\stackrel{\text{associativity}}{=} x_0 + \left(\left(\sum_{i=1}^n x_i \right) + x_{n+1} \right) \\ &\stackrel{[\text{theorem: 11.15}]}{=} x_0 + \sum_{i=1}^{n+1} x_i \end{aligned}$$

proving that $n+1 \in S$ $\square$

11.1.2 Associativity

Theorem 11.23. Let $\langle A, + \rangle$ be a semi-group $n, m \in \mathbb{N}_0$, with $n < m$, $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq A$ and $k \in \{n, \dots, m-1\}$ then

$$\sum_{i=n}^m x_i = \sum_{i=n}^k x_i + \sum_{i=k+1}^m x_i$$

or taking $k = n$ that

$$\sum_{i=n}^m x_i = \sum_{i=n}^n x_i + \sum_{i=n+1}^m x_i = x_n + \sum_{i=n+1}^m x_i$$

Proof. Let $n \in \mathbb{N}$ and define

$$S = \left\{ n \in \mathbb{N} \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ we have } \forall k \in \{0, \dots, n-1\} \sum_{i=0}^n x_i = \sum_{i=0}^k x_i + \sum_{i=k+1}^n x_i \right\}$$

then we have:

$1 \in S$. Let $\{x_i\}_{i \in \{0, \dots, 1\}} = \{0, 1\} \subseteq A$ and let $k \in \{0, \dots, 1-1\} = \{0\}$ then

$$\sum_{i=0}^1 x_i = \left(\sum_{i=0}^0 x_i \right) + x_1 = \left(\sum_{i=0}^0 x_i \right) + x_1 = \sum_{i=0}^0 x_i + \sum_{i=0}^0 x_{1+i} = \sum_{i=0}^0 x_i + \sum_{i=1}^1 x_i$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq A$ and take $k \in \{0, \dots, (n+1)-1\} = \{0, \dots, n\}$ then we have for k either:

$k = n$. Then

$$\begin{aligned} \sum_{i=0}^{n+1} x_i &= \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\ &= \sum_{i=0}^n x_i + \sum_{i=0}^0 x_{(n+1)+i} \\ &= \sum_{i=0}^n x_i + \sum_{i=n+1}^{n+1} x_i \\ &\stackrel{k=n}{=} \sum_{i=0}^k x_i + \sum_{i=k+1}^{n+1} x_i \end{aligned}$$

proving that $n+1 \in S$

$k \in \{0, \dots, n-1\}$. Then we have

$$\begin{aligned} \sum_{i=0}^{n+1} x_i &= \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\ &\stackrel{n \in S \wedge k \in \{0, \dots, n-1\}}{=} \left(\sum_{i=0}^k x_i + \sum_{i=k+1}^n x_i \right) + x_{n+1} \\ &= \sum_{i=0}^k x_i + \left(\left(\sum_{i=k+1}^n x_i \right) + x_{n+1} \right) \\ &= \sum_{i=0}^k x_i + \left(\left(\sum_{i=0}^{n-(k+1)} x_{(k+1)+i} \right) + x_{(k+1)+(n-(k+1))+1} \right) \\ &= \sum_{i=0}^k x_i + \sum_{i=0}^{(n-(k+1))+1} x_{(k+1)+i} \\ &= \sum_{i=0}^k x_i + \sum_{i=k+1}^{n+1} x_i \end{aligned}$$

proving that $n+1 \in S$.

So by mathematical induction we have

$$\forall n \in \mathbb{N}, \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ we have } \forall k \in \{0, \dots, n-1\} \text{ that } \sum_{i=0}^n x_i = \sum_{i=0}^k x_i + \sum_{i=k+1}^n x_i \quad (11.2)$$

Take now $n, m \in \mathbb{N}_0$ with $n < m \Rightarrow m-n \in \mathbb{N}$ then for $\{x_i\}_{i \in \{n, \dots, m\}}$ and $k \in \{n, \dots, m-1\} \Rightarrow k-n \in \{0, \dots, m-n\}$ we have

$$\begin{aligned} \sum_{i=n}^m x_i &= \sum_{i=0}^{m-n} x_{n+i} \\ &\stackrel{k-n \in \{0, \dots, m-n\} \text{ and [eq: 11.2]}}{=} \sum_{i=0}^{k-n} x_{n+i} + \sum_{i=(k-n)+1}^{m-n} x_{n+i} \\ &= \sum_{i=n}^k x_i + \sum_{i=k+1}^m x_i \end{aligned}$$

proving the theorem. $\square$

Theorem 11.24. (Associativity) Let $\langle A, + \rangle$ be a semi-group, $n \in \mathbb{N}$, let $\{(b_i, e_i)\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{N}_0 \times \mathbb{N}_0$ so that

$$1. \forall i \in \{0, \dots, n\} \quad b_i \leq e_i$$

$$2. \forall i \in \{0, \dots, n-1\} \quad e_i + 1 = b_{i+1}$$

then for $\{x_i\}_{i \in \{b_0, \dots, e_n\}} \subseteq A$ we have

$$\sum_{i=b_0}^{e_n} x_i = \sum_{i=0}^n \left(\sum_{j=b_i}^{e_i} x_j \right)$$

Proof. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N} \mid \forall \{(b_i, e_i)\}_{i \in \{0, \dots, n\}} \text{ satisfying (1),(2)}, \forall \{x_i\}_{i \in \{b_0, \dots, e_n\}} \subset A \models \sum_{i=b_0}^{e_n} x_i = \sum_{i=0}^n \left(\sum_{j=b_i}^{e_i} x_j \right) \right\}$$

then we have:

$1 \in S$. Then for $\{(b_i, e_i)\}_{i \in \{0,1\}}$ we have $b_0 \leq e_0 \wedge b_1 \leq e_1 \wedge e_0 + 1 = b_1 \Rightarrow e_0 \in \{b_0, \dots, e_1 - 1\}$, for $\{x_i\}_{i \in \{b_0, \dots, e_1\}} \subseteq A$ we have then

$$\begin{aligned} \sum_{i=b_0}^{e_1} x_i &\stackrel{[\text{theorem: 11.23}]}{=} \sum_{j=b_0}^{e_0} x_j + \sum_{j=e_0+1}^{e_1} x_j \\ &= \sum_{j=b_0}^{e_0} x_j + \sum_{j=b_1}^{e_1} x_j \\ &= \sum_{i=0}^0 \left(\sum_{j=b_i}^{e_i} x_j \right) + \sum_{j=b_1}^{e_1} x_j \\ &= \sum_{i=0}^1 \left(\sum_{j=b_i}^{e_i} x_j \right) \end{aligned}$$

proving that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{(b_i, e_i)\}_{i \in \{0, \dots, n+1\}} \subseteq \mathbb{N}_0 \times \mathbb{N}_0$ with $\forall i \in \{1, \dots, n+1\} \quad b_i \leq e_i, \forall i \in \{0, \dots, (n+1) - 1\} = \{0, \dots, n\} \quad e_i + 1 = b_{i+1}$ and $\{x_i\}_{i \in \{b_0, \dots, e_{n+1}\}} \subseteq A$. In particular we have $b_n \leq e_n = b_{n+1} - 1 \leq e_{n+1} - 1$. Further $\forall i \in \{0, \dots, n\}$ we have $b_i \leq e_i < e_i + 1 = b_{i+1}$ proving by [theorem: 10.36] that $b_0 \leq b_n \leq e_n$ so that $e_n \in \{b_0, \dots, e_{n+1} - 1\}$. Next

$$\begin{aligned} \sum_{j=b_0}^{e_{n+1}} x_j &\stackrel{e_n \in \{b_0, \dots, e_{n+1}-1\}}{=} \sum_{j=b_0}^{e_n} x_j + \sum_{j=e_n+1}^{e_{n+1}} x_j \\ &\stackrel{e_n+1=b_{n+1}}{=} \sum_{j=b_0}^{e_n} x_j + \sum_{j=b_{n+1}}^{e_{n+1}} x_j \\ &\stackrel{n \in S}{=} \sum_{i=0}^n \left(\sum_{j=b_i}^{e_i} x_j \right) + \sum_{j=b_{n+1}}^{e_{n+1}} x_j \\ &= \sum_{i=0}^{n+1} \left(\sum_{j=b_i}^{e_i} x_j \right) \end{aligned}$$

proving that $n+1 \in S$. □

11.1.3 Commutativity

We will now generalize commutativity to finite sum. First we must introduce the concept of permutations.

Definition 11.25. (Permutation) If I is a set then a bijection $\sigma: I \rightarrow I$ is called a **permutation of I** . The set of all the permutation graphs of I is noted as S_I hence

$$S_I = \{\sigma \in I^I \mid \sigma: I \rightarrow I \text{ is a bijection}\}$$

Theorem 11.26. Let I be a set then $\langle S_I, \circ \rangle$ is a group with neutral element Id_I and $\forall \sigma \in S_I$ σ^{-1} as inverse element.

Proof. As the composition of two bijections is a bijection [see theorem: 2.78] we have that

$$\circ: S_I \times S_I \rightarrow S_I \text{ is a operator}$$

Further we have:

associativity. If $\sigma, \beta, \gamma \in S_I$ then $(\sigma \circ \beta) \circ \gamma \stackrel{[\text{function: 2.21}]}{=} \sigma \circ (\beta \circ \gamma)$

neutral element. As $\text{Id}_I: I \rightarrow I$ is a bijection [see: 2.66] we have that $\text{Id}_I \in S_I$. Further by [theorem: 2.50] we have

$$\sigma = \sigma \circ \text{Id}_I = \text{Id}_I \circ \sigma$$

inverse element. If $\sigma \in S_I$ then $\sigma: I \rightarrow I$ is a bijection, hence by [theorem: 2.74] $\sigma^{-1}: I \rightarrow I$ is a bijection so that $\sigma^{-1} \in S_I$. Hence $\sigma \circ \sigma^{-1} \stackrel{[\text{theorem: 2.71}]}{=} \text{Id}_I \stackrel{[\text{theorem: 2.71}]}{=} \sigma^{-1} \circ \sigma$ $\square$

Theorem 11.27. Let I be a set and $\sigma \in S_I$ then if $i \in I$ with $\sigma(i) = i$ then $\sigma|_{I \setminus \{i\}} \in S_{I \setminus \{i\}}$

Proof. As $\sigma: I \rightarrow I$ is a bijection we have by [theorem: 2.99] that

$$\sigma|_{I \setminus \{i\}}: I \setminus \{i\} \rightarrow I \setminus \{\sigma(i)\} \stackrel{[\sigma(i)=i]}{=} I \setminus \{i\}$$

is a bijection. $\square$

We define now a special type of permutation a transposition

Theorem 11.28. (transposition) Let I be a set, $i, j \in I$ then for $(i \leftrightarrow j)$ defined by

$$(i \leftrightarrow j)(k) = \begin{cases} k & \text{if } k \in I \setminus \{i, j\} \\ i & \text{if } k = j \\ j & \text{if } k = i \end{cases}$$

we have that

$$(i \leftrightarrow j) \in S_I$$

The permutation $(i \leftrightarrow j)$ is called a **transposition** of i and j .

Proof. We have

injectivity. If $(i \leftrightarrow j)(k) = (i \leftrightarrow j)(l)$ then for k, l we have either:

$k = i \wedge l = i$. Then trivially $k = l$

$k = i \wedge l = j$. Then $l = j = (i \leftrightarrow j)(k) = (i \leftrightarrow j)(l) = i = k$ so that $k = l$

$k = i \wedge l \neq i, j$. Then $j = (i \leftrightarrow j)(k) = (i \leftrightarrow j)(l) = l \neq i, j$ a contradiction so this case never occurs.

$k = j \wedge l = i$. Then $l = i = (i \leftrightarrow j)(k) = (i \leftrightarrow j)(l) = j = k$ so that $k = l$.

$k = j \wedge l = j$. Then trivially $k = l$.

$k = j \wedge l \neq i, j$. Then $i = (i \leftrightarrow j)(k) = (i \leftrightarrow j)(l) = l \neq i, j$ a contradiction so this case never occurs.

$k \neq i, j \wedge l = i$. Then $i, j \neq k = (i \leftrightarrow_I j)(k) = (i \leftrightarrow_I j)(l) = j$ a contradiction so this case never occurs.

$k \neq i, j \wedge l = j$. Then $i, j \neq k = (i \leftrightarrow_I j)(k) = (i \leftrightarrow_I j)(l) = i$ a contradiction so this case never occurs.

$k \neq i, j \wedge l \neq i, j$. Then $k = (i \leftrightarrow_I j)(k) = (i \leftrightarrow_I j)(l) = l$ so that $k = l$

Hence in all valid cases we have $k = l$ proving injectivity.

surjectivity. If $l \in I$ then we have either:

$l = i$. Then for $k = j$ we have $(i \leftrightarrow_I j)(k) = (i \leftrightarrow_I j)(j) = i = l$

$l = j$. Then for $k = i$ we have $(i \leftrightarrow_I j)(k) = (i \leftrightarrow_I j)(i) = j = l$

$l \neq i, j$. Then for $k = l$ we have $(i \leftrightarrow_I j)(k) = (i \leftrightarrow_I j)(l) = l$

so in all cases we found a $k \in I$ such that $(i \leftrightarrow_I j)(k) = l$. □

Theorem 11.29. Let I be a set then we have

1. If $(i \leftrightarrow_I i) = \text{Id}_I(i)$
2. If $i, j \in I$ then $(i \leftrightarrow_I j) \circ (i \leftrightarrow_I j) = \text{Id}_I$
3. $(i \leftrightarrow_I j) = (j \leftrightarrow_I i)$
4. If $J \subseteq I$ and $i, j \in J$ then $(i \leftrightarrow_J j) = (i \leftrightarrow_I j)|_J$

Proof.

1. If $k \in I$ then we have either:

$k = i$. Then $(i \leftrightarrow_I j)(k) = j \stackrel{i=j}{=} i = k = \text{Id}_I(k)$

$k = j$. Then $(i \leftrightarrow_I j)(k) = j \stackrel{i=j}{=} i = k = \text{Id}_I(k)$

$k \neq i, j$. Then $(i \leftrightarrow_I j)(k) = j \stackrel{i=j}{=} i = k = \text{Id}_I(k)$

proving that $(i \leftrightarrow_I j) = \text{Id}_I$

2. If $k \in I$ then we have either:

$k = i$. Then $(i \leftrightarrow_I j)((i \leftrightarrow_I j)(k)) = (i \leftrightarrow_I j)(j) = i = k$

$k = j$. Then $(i \leftrightarrow_I j)((i \leftrightarrow_I j)(k)) = (i \leftrightarrow_I j)(i) = j = k$

$k \neq i, j$. Then $(i \leftrightarrow_I j)((i \leftrightarrow_I j)(k)) = k$

proving that $(i \leftrightarrow_I j) \circ (i \leftrightarrow_I j) = \text{Id}_I$.

3. If $k \in I$ then we have either:

$k = i$. Then $(i \leftrightarrow_I j)(k) = j = (j \leftrightarrow_I i)(k)$

$k = j$. Then $(i \leftrightarrow_I j)(k) = i = (j \leftrightarrow_I i)(k)$

$k \neq i, j$. $(i \leftrightarrow_I j)(k) = k = (j \leftrightarrow_I i)(k)$

proving that $(i \leftrightarrow_I j) = (j \leftrightarrow_I i)$.

4. If $k \in J$ then we have either:

$k = i$. Then $(i \leftrightarrow_J j)(k) = j = (i \leftrightarrow_I j)(k)$

$k = j$. Then $(i \leftrightarrow_J j)(k) = i = (i \leftrightarrow_I j)(k)$

$k \neq i, j$. Then $(i \leftrightarrow j)(k) = k = (i \leftrightarrow j)(k)$

proving that $(i \leftrightarrow j) = (i \leftrightarrow j)|_J$. $\square$

Lemma 11.30. Let $n \in \mathbb{N}_0$ and $\sigma \in S_{\{0, \dots, n+1\}}$ is such that $\sigma(n+1) \neq n+1$ then for $k = \sigma^{-1}(n+1)$ we have for γ_σ defined by

$$\gamma_\sigma: \{0, \dots, n\} \rightarrow \{0, \dots, n\} \text{ defined by } \gamma_\sigma(i) = \begin{cases} \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right) & \text{if } i \in \{0, \dots, n-1\} \\ \sigma(n+1) & \text{if } i = n \end{cases}$$

that

$$\gamma_\sigma \in S_{\{0, \dots, n\}}$$

Proof. First for $i \in [0, \dots, n]$ we have either:

$i \in \{0, \dots, n-1\}$. Assume that $\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right) = n+1$ then as $\sigma(k) = n+1$ we have as σ is injective that $\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i) = k = \left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(n)$ so that $i = n$ contradicting $i \in \{0, \dots, n-1\}$. So we must have that $\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right) \neq n+1$ or $\gamma_\sigma(i) \neq n+1 \Rightarrow \gamma_\sigma(i) \in \{0, \dots, n\}$.

$i = n$. Then as $\sigma(n+1) \neq n+1$ we have that $\sigma(n+1) \in \{0, \dots, n\}$ or $\gamma_\sigma(i) \in \{0, \dots, n\}$

So we have that $\forall i \in \{1, \dots, n\}$ that $\gamma_\sigma(i) \in \{0, \dots, n\}$ hence we have indeed a function

$$\gamma_\sigma: \{0, \dots, n\} \rightarrow \{0, \dots, n\}$$

Next we have to prove that it is a bijection.

injectivity. Let $r, s \in \{0, \dots, n\}$ such that $\gamma_\sigma(r) = \gamma_\sigma(s)$ then we have for r, s either:

$r = n \wedge s = n$. Then trivially $r = s$.

$r = n \wedge s \neq n$. Then we have for s either:

$s = k$. Then $\sigma(n+1) = \gamma_\sigma(r) = \gamma_\sigma(s) = \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(s)\right) = \sigma(n)$ leading to the contradiction $n+1 = n$, hence this case never occurs.

$s \neq k$. Then $\sigma(n+1) = \gamma_\sigma(r) = \gamma_\sigma(s) = \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(s)\right) = \sigma(s)$ so that $s = n+1$ contradicting $s \in \{1, \dots, n\}$, hence this case does not occur.

$r \neq n \wedge s = n$. Then we have for r either:

$r = k$. Then $\sigma(n+1) = \gamma_\sigma(s) = \gamma_\sigma(r) = \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(r)\right) = \sigma(n)$ leading to the contradiction $n+1 = n$, so this case never occurs.

$r \neq k$. Then $\sigma(n+1) = \gamma_\sigma(s) = \gamma_\sigma(r) = \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(r)\right) = \sigma(r)$ so that $n+1 = r$ contradicting $r \in \{1, \dots, n\}$, hence this case does not occur.

$r \neq n \wedge s \neq n$. Then $\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(r)\right) = \gamma_\sigma(r) = \gamma_\sigma(s) = \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(s)\right)$ so that $\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(r) = \left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(s)$ or $r = s$ [as σ and $\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)$ are injections].

So in all valid cases we have $r = s$ proving injectivity.

surjectivity. Let $r \in \{0, \dots, n\}$ then by the surjectivity of σ there exist a $s \in \{0, \dots, n+1\}$ such that $\sigma(s) = r$. For s we have then either:

$s = n+1$. Then for $n \in \{0, \dots, n\}$ we have $\gamma_\sigma(n) = \sigma(n+1) = \sigma(s) = r$ so that $\gamma_\sigma(n) = r$.

$s = n$. If $s = k$ then $r = \sigma(s) = \sigma(k) = n+1$ contradicting $r \in \{0, \dots, n\}$. So we have that $n = s \neq k$, hence $k \in \{0, \dots, n-1\}$, so that

$$\gamma_\sigma(k) = \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(k)\right) = \sigma(n) = \sigma(s) = r$$

proving that $\gamma_\sigma(k) = r$.

$s \in \{0, \dots, n-1\}$. If $s = k$ then $r = \sigma(s) = \sigma(k) = n+1$ so that $r = n+1$ contradicting $r \in \{0, \dots, n\}$ so we must have that $s \neq k, n$. Hence

$$\gamma_\sigma(s) = \sigma\left(\left(n \begin{smallmatrix} \leftrightarrow \\ \{0, \dots, n\} \end{smallmatrix} k \right)(s)\right) = \sigma(s) = r$$

So in all cases we found a $l \in \{0, \dots, n\}$ such that $\gamma_\sigma(l) = r$ proving surjectivity. $\square$

Theorem 11.31. (Commutativity) Let $\langle A, + \rangle$ be a **Abelian** semi-group, $n \in \mathbb{N}_0$, $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq A$ and $\sigma \in S_{\{0, \dots, n\}}$ a permutation then

$$\sum_{i=0}^n x_i = \sum_{i=0}^n x_{\sigma(i)}$$

Proof. We prove this by induction so take

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{x_i\}_{i \in \{0, \dots, n\}} \subseteq A \text{ and } \sigma \in S_{\{0, \dots, n\}} \text{ then } \sum_{i=0}^n x_i = \sum_{i=0}^n x_{\sigma(i)} \right\}$$

then we have:

$0 \in S$. Let $\{x_i\}_{i \in \{0\}} \subseteq A$ and $\sigma \in S_{\{0\}}$ then $\sigma = \text{Id}_{\{0\}}$ so that

$$\sum_{i=0}^0 x_i = x_0 = x_{\text{Id}_{\{0\}}(0)} = x_{\sigma(0)} = \sum_{i=0}^0 x_{\sigma(i)}$$

proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{0, \dots, n+1\}}$ and $\sigma \in S_{\{0, \dots, n+1\}}$ then for $n+1$ we have either

$n+1 = 1$. Then $S_{\{0, \dots, n+1\}} = S_{\{0, 1\}}$ so that for $\sigma(1)$ we have either:

$\sigma(1) = 0$. Then as $\sigma: \{0, 1\} \rightarrow \{0, 1\}$ is a bijection we must have $\sigma(0) = 1$, so

$$\sum_{i=0}^1 x_i = x_0 + x_1 = x_1 + x_0 = x_{\sigma(0)} + x_{\sigma(1)} = \sum_{i=0}^1 x_{\sigma(i)}$$

$\sigma(1) = 1$. Then as $\sigma: \{0, 1\} \rightarrow \{0, 1\}$ is a bijection we must have $\sigma(0) = 0$, so

$$\sum_{i=0}^1 x_i = x_0 + x_1 = x_{\sigma(0)} + x_{\sigma(1)} = \sum_{i=0}^1 x_{\sigma(i)}$$

so in all cases $\sum_{i=0}^1 x_i = \sum_{i=0}^1 x_{\sigma(i)}$ proving that $n+1 \in S$.

$1 < n+1$. Now for $\sigma(n+1)$ we have either:

$\sigma(n+1) = n+1$. Then by [theorem: 11.27] we have that $\sigma|_{\{0, \dots, n\}} \in S_{\{0, \dots, n\}}$ which as $n \in S$ proves that

$$\sum_{i=0}^n x_i \stackrel{n \in S}{=} \sum_{i=0}^n x_{\sigma|_{\{0, \dots, n\}}(i)} = \sum_{i=0}^n x_{\sigma(i)} \quad (11.3)$$

So

$$\begin{aligned} \sum_{i=0}^{n+1} x_i &= \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\ &\stackrel{[eq: 11.3]}{=} \left(\sum_{i=0}^n x_{\sigma(i)} \right) + x_{n+1} \\ &= \left(\sum_{i=0}^n x_{\sigma(i)} \right) + x_{\sigma(n+1)} \\ &= \sum_{i=0}^{n+1} x_{\sigma(i)} \end{aligned}$$

proving that $n+1 \in S$.

$\sigma(n+1) \in \{0, \dots, n\}$. Take then $k = \sigma^{-1}(n+1) \in \{0, \dots, n\}$ then we have by [lemma: 11.30] that $\gamma_\sigma \in S_{\{0, \dots, n\}}$ where

$$\gamma_\sigma(i) = \begin{cases} \sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right) & \text{if } i \in \{0, \dots, n-1\} \\ \sigma(n+1) & \text{if } i = n \end{cases} \quad (11.4)$$

so as $n \in S$ we have that:

$$\begin{aligned} \sum_{i=0}^n x_i &\stackrel{n \in S}{=} \sum_{i=0}^n x_{\gamma_\sigma(i)} \\ &= \left(\sum_{i=0}^{n-1} x_{\gamma_\sigma(i)} \right) + x_{\gamma_\sigma(n+1)} \\ &\stackrel{11.4}{=} \left(\sum_{i=0}^{n-1} x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right)} \right) + x_{\sigma(n+1)} \end{aligned} \quad (11.5)$$

Further we have

$$\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right) \in S_{\{0, \dots, n\}} \quad (11.6)$$

we have

$$\begin{aligned} \sum_{i=0}^{n+1} x_{\sigma(i)} &= \\ \left(\sum_{i=0}^n x_{\sigma(i)} \right) + x_{\sigma(n+1)} &\stackrel{n \in S \wedge \text{eq: 11.6}}{=} \\ \left(\sum_{i=0}^n x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right)} \right) + x_{\sigma(n+1)} &= \\ \left(\left(\sum_{i=0}^{n-1} x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right)} \right) + x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(n)\right)} \right) + x_{\sigma(n+1)} &= \\ \left(\sum_{i=0}^{n-1} x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right)} \right) + \left(x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(n)\right)} + x_{\sigma(n+1)} \right) &= \\ \left(\sum_{i=0}^{n-1} x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right)} \right) + \left(x_{\sigma(n+1)} + x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(n)\right)} \right) &= \\ \left(\left(\sum_{i=0}^{n-1} x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(i)\right)} \right) + x_{\sigma(n+1)} \right) + x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(n)\right)} &\stackrel{\text{eq: 11.5}}{=} \\ \left(\sum_{i=0}^n x_i \right) + x_{\sigma\left(\left(n_{\{0, \dots, n\}}^{\leftrightarrow k}\right)(n)\right)} &= \\ \left(\sum_{i=0}^n x_i \right) + x_{\sigma(k)} &\stackrel{k = \sigma^{-1}(n+1)}{=} \\ \left(\sum_{i=0}^n x_i \right) + x_{n+1} &= \\ \sum_{i=0}^{n+1} x_i &= \end{aligned}$$

proving that $n+1 \in S$.

proving that $n+1 \in S$.

So in all cases we have $n+1 \in S$.

Mathematical induction proves then that $S = \mathbb{N}_0$ and the theorem. $\square$

11.1.4 Generalized sum

Next we define the sum of elements over a finite index set.

Definition 11.32. Let $\langle A, + \rangle$ be a Abelian semi-group with neutral element 0, I a finite set and $\{x_i\}_{i \in I}$ then we define $\sum_{i \in I} x_i$ as follows:

1. If $I = \emptyset$ then $\sum_{i \in I} x_i = 0$
2. If $I \neq \emptyset$ then as I is finite there exist by [corollary: 10.88] a **unique** $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$, $\sum_{i \in I} x_i$ is then defined by

$$\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)}$$

Note 11.33. If $\langle A, \cdot \rangle$ is a multiplicative Abelian group [for example for a ring or field we want to make distinction between summation and multiplication] with neutral element 1, I a finite set and $\{x_i\}_{i \in I}$ then we use $\prod$ instead of $\sum$ and have then that $\prod_{i \in I} x_i$ is defined by

1. If $I = \emptyset$ then $\prod_{i \in I} x_i = 1$
2. If $I \neq \emptyset$ then I is finite there exist by [corollary: 10.88] a **unique** $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$, $\prod_{i \in I} x_i$ is then defined by

$$\prod_{i \in I} x_i = \prod_{i=0}^{n-1} x_{\beta(i)}$$

Of course all statements about summation applies also to multiplication if you replace $\sum$ by $\prod$ and 0 by 1.

Proof. We must for (2) prove that $\sum_{i \in I} x_i$ is unique. So assume that $\gamma: \{0, \dots, n-1\} \rightarrow I$ is another bijection then $\beta^{-1} \circ \gamma: \{0, \dots, n-1\} \rightarrow \{0, \dots, n-1\}$ is a bijection, hence $\beta^{-1} \circ \gamma \in S_{\{0, \dots, n-1\}}$ so that

$$\sum_{i=0}^{n-1} x_{\beta(i)} \stackrel{[\text{theorem: 11.31}]}{=} \sum_{i=0}^{n-1} x_{\beta(\beta^{-1}(\gamma(i)))} = \sum_{i=0}^{n-1} x_{\gamma(i)} \quad \square$$

Example 11.34. Let $\langle A, + \rangle$ be a Abelian semi-group and $\{x_i\}_{i \in \{k\}} \subseteq A$ then $\sum_{i \in \{k\}} x_i = x_k$

Proof. As $\beta: \{0\} = \{0, \dots, (1-1)\} \rightarrow \{k\}$ by $\beta(0) = k$ we have that $\sum_{i \in \{k\}} x_i = \sum_{i=0}^0 x_{\beta(i)} = x_{\beta(0)} = x_k$ $\square$

This new definition is equivalent with the previous definition as the following theorem shows.

Theorem 11.35. Let $\langle A, + \rangle$ be a Abelian semi-group with neutral element 0, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}}$ then $\sum_{i=n}^m x_i = \sum_{i \in \{n, \dots, m\}} x_i$

Proof. Define $\sigma: \{0, \dots, m-n\} \rightarrow \{n, \dots, m\}$ by $\sigma(i) = n+i$ then we have

injectivity. If $\sigma(i) = \sigma(j)$ then $n+i = n+j$ hence $i = j$.

surjectivity. If $j \in \{n, \dots, m\} \Rightarrow n \leq j \leq m \Rightarrow 0 \leq j-n \leq m-n$ then for $i = j-n$ we have $j = i+n = \sigma(i)$ and $0 \leq i \leq m-n$

so that $\sigma: \{0, \dots, m-n\} \rightarrow \{n, \dots, m\}$ is a bijection and by definition

$$\sum_{i \in \{n, \dots, m\}} x_i = \sum_{i=0}^{m-n} x_{\sigma(i)} = \sum_{i=0}^{m-n} x_{n+i} \stackrel{\text{def}}{=} \sum_{i=n}^m x_i$$

□

Theorem 11.36. Let $\langle A, + \rangle$ be a Abelian semi-group with neutral element 0, I a finite set, $\{x_i\}_{i \in I}$ and $\sigma: J \rightarrow I$ a bijection then

$$\sum_{i \in I} x_i = \sum_{j \in J} x_{\sigma(j)}$$

Proof. For I we have either:

$I = \emptyset$. Then as $\sigma: J \rightarrow I$ is a bijection we have $J = \emptyset$ so that $\sum_{i \in I} x_i = 0 = \sum_{j \in J} x_{\sigma(j)}$

$I \neq \emptyset$. Then there exists a $n \in \mathbb{N}$ such that $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)}$$

As $\sigma^{-1} \circ \beta: \{0, \dots, n-1\} \rightarrow J$ is a bijection we have

$$\sum_{j \in J} x_{\sigma(j)} = \sum_{i=0}^{n-1} x_{\sigma(\sigma^{-1}(\beta(i)))} = \sum_{i=0}^{n-1} x_{\beta(i)} = \sum_{i \in I} x_i$$

□

Using this more general definition of a finite sum we can rewrite [theorems: 11.16, 11.6, 11.8, 11.9 and 11.7].

Theorem 11.37. Let $\langle A, + \rangle$ be a Abelian semi-group with neutral element 0, I a finite set and $\{x_i\}_{i \in I} \subseteq A$ with $\forall i \in I$ $x_i = 0$ then $\sum_{i \in I} x_i = 0$.

Proof. For I we have either

$I = \emptyset$. Then $\sum_{i \in I} x_i = 0$

$I \neq \emptyset$. Then there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)}$$

As $\forall i \in \{0, \dots, n-1\}$ $\beta(i) \in I$ we have $x_{\beta(i)} = 0$ so that by [theorem: 11.5]

$$\sum_{i=0}^{n-1} x_{\beta(i)} = 0,$$

hence we conclude that

$$\sum_{i \in I} x_i = 0$$

□

Theorem 11.38. Let $\langle A, + \rangle$ be a Abelian semi-group, I a finite set and $\{x_i\}_{i \in I} \subseteq A$, $\{y_i\}_{i \in I} \subseteq A$ then

$$\sum_{i \in I} (x_i + y_i) = \sum_{i \in I} x_i + \sum_{i \in I} y_i$$

Proof. For I we have either:

$I = \emptyset$. Then we have

$$\sum_{i \in I} (x_i + y_i) = 0 = 0 + 0 = \sum_{i \in I} x_i + \sum_{i \in I} y_i$$

$I \neq \emptyset$. Then there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} (x_i + y_i) = \sum_{i=0}^{n-1} (x_{\beta(i)} + y_{\beta(i)}), \sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)} \text{ and } \sum_{i \in I} y_i = \sum_{i=0}^{n-1} y_{\beta(i)}$$

By [theorem: 11.6] we have that $\sum_{i=0}^{n-1} (x_{\beta(i)} + y_{\beta(i)}) = \sum_{i=0}^{n-1} x_{\beta(i)} + \sum_{i=0}^{n-1} y_{\beta(i)}$ proving that

$$\sum_{i \in I} (x_i + y_i) = \sum_{i \in I} x_i + \sum_{i \in I} y_i \quad \square$$

Theorem 11.39. Let $\langle R, +, \cdot \rangle$ be a ring, $\alpha \in R$, I a finite set and $\{x_i\}_{i \in I} \subseteq R$ then

$$\sum_{i \in I} \alpha \cdot x_i = \alpha \cdot \sum_{i \in I} x_i$$

Proof. For I we have either:

$I = \emptyset$. Then we have

$$\sum_{i \in I} \alpha \cdot x_i = 0 = \alpha \cdot 0 = \alpha \cdot \sum_{i \in I} x_i$$

$I \neq \emptyset$. Then there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} \alpha \cdot x_i = \sum_{i=0}^{n-1} \alpha \cdot x_{\beta(i)} \stackrel{[\text{theorem: 11.8}]}{=} \alpha \cdot \sum_{i=0}^{n-1} x_{\beta(i)} = \alpha \cdot \sum_{i \in I} x_i \quad \square$$

Theorem 11.40. Let $\langle A, +, \cdot \rangle$ be an Abelian semi-group, I a finite set and $\{x_i\}_{i \in I} \subseteq A$ then

$$\sum_{i \in I} (-x_i) = - \sum_{i \in I} x_i$$

Proof. For I we have either:

$I = \emptyset$. Then we have

$$\sum_{i \in I} (-x_i) = 0 = -0 = - \sum_{i \in I} x_i$$

$I \neq \emptyset$. Then there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} (-x_i) = \sum_{i=0}^{n-1} (-x_{\beta(i)}) \stackrel{[\text{theorem: 11.7}]}{=} - \sum_{i=0}^{n-1} x_{\beta(i)} = - \sum_{i \in I} x_i \quad \square$$

Theorem 11.41. Let $\langle A, + \rangle$ be an Abelian semi-group, I, J finite sets and $\{x_{i,j}\}_{(i,j) \in I \times J} \subseteq A$ then

$$\sum_{i \in I} \left(\sum_{j \in J} x_{i,j} \right) = \sum_{j \in J} \left(\sum_{i \in I} x_{i,j} \right)$$

Proof. For I, J we have either:

$I = \emptyset$. Then we have

$$\sum_{i \in I} \left(\sum_{j \in J} x_{i,j} \right) = 0 \stackrel{[\text{theorem: 11.37}]}{=} \sum_{j \in J} 0 = \sum_{j \in J} \left(\sum_{i \in I} x_{i,j} \right)$$

$J = \emptyset$. Then we have

$$\sum_{i \in I} \left(\sum_{j \in J} x_{i,j} \right) = \sum_{i \in I} 0 \stackrel{[\text{theorem: 11.37}]}{=} 0 = \sum_{j \in J} \left(\sum_{i \in I} x_{i,j} \right)$$

$I \neq \emptyset \wedge J \neq \emptyset$. Then there exists $n, m \in \mathbb{N}$ and bijections

$$\alpha: \{0, \dots, n-1\} \rightarrow I, \beta: \{0, \dots, m-1\} \rightarrow J$$

such that

$$\forall j \in J \text{ we have } \sum_{i \in I} x_{i,j} = \sum_{i=0}^{n-1} x_{\alpha(i),j} \text{ and } \forall i \in I \text{ we have } \sum_{j \in J} x_{i,j} = \sum_{j=0}^{m-1} x_{i,\beta(j)} \quad (11.7)$$

so that

$$\begin{aligned}
 \sum_{i \in I} \left(\sum_{j \in J} x_{i,j} \right) & \stackrel{[\text{eq: 11.7}]}{=} \sum_{i \in I} \left(\sum_{j=0}^{m-1} x_{i,\beta(j)} \right) \\
 & = \sum_{i=0}^{n-1} \left(\sum_{j=0}^{m-1} x_{\alpha(i),\beta(j)} \right) \\
 & \stackrel{[\text{theorem: 11.9}]}{=} \sum_{j=0}^{m-1} \left(\sum_{i=0}^{n-1} x_{\alpha(i),\beta(j)} \right) \\
 & = \sum_{j \in J} \left(\sum_{i=0}^{n-1} x_{\alpha(i),j} \right) \\
 & \stackrel{[\text{eq: 11.7}]}{=} \sum_{j \in J} \left(\sum_{i \in I} x_{i,j} \right)
 \end{aligned}$$

□

We have the equivalent of [theorem: 11.23]

Lemma 11.42. *Let $\langle A, + \rangle$ be a Abelian semi-group with neutral element 0, I a finite set such that $I = M \cup N$ and $M \cap N = \emptyset$ and $\{x_i\}_{i \in I} \subseteq A$ then we have*

$$\sum_{i \in I} x_i = \sum_{i \in M} x_i + \sum_{i \in N} x_i$$

Proof. As I is finite we have by [theorem: 6.33] that M, N are finite and we have either:

$M = \emptyset$. Then $N = I$ and

$$\sum_{i \in I} x_i = \sum_{i \in N} x_i = \sum_{i \in N} x_i + 0 = \sum_{i \in N} x_i + \sum_{i \in M} x_i$$

$N = \emptyset$. Then $M = I$ and

$$\sum_{i \in I} x_i = \sum_{i \in M} x_i = 0 + \sum_{i \in M} x_i = \sum_{i \in N} x_i + \sum_{i \in M} x_i$$

$M, N \neq \emptyset$. Then there exists $n, m \in \mathbb{N}$ such $\alpha: \{0, \dots, n-1\} \rightarrow N$ and $\beta: \{0, \dots, m-1\} \rightarrow M$ are bijections and

$$\sum_{i \in N} x_i = \sum_{i=0}^{n-1} x_{\alpha(i)} \text{ and } \sum_{i \in M} x_i = \sum_{i=0}^{m-1} x_{\beta(i)} \quad (11.8)$$

Define now

$$\gamma: \{0, \dots, n+m-1\} \rightarrow N \cup M \text{ by } \gamma(i) = \begin{cases} \alpha(i) & \text{where } i \in \{0, \dots, n-1\} \\ \beta(i-n) & \text{where } i \in \{n, \dots, n+m-1\} \end{cases}$$

then we have:

injectivity. If $\gamma(i) = \gamma(j)$ then for i, j we have either:

$i \in \{0, \dots, n-1\} \wedge j \in \{0, \dots, n-1\}$. Then $\alpha(i) = \gamma(i) = \gamma(j) = \alpha(j)$ which as α is injective proves $i = j$.

$i \in \{0, \dots, n-1\} \wedge j \in \{n, \dots, n+m-1\}$. Then $\alpha(i) = \gamma(i) = \gamma(j) = \beta(j-n)$ so that $\alpha(i) \in N \cap M$ contradicting $N \cap M = \emptyset$, so this case never occurs.

$i \in \{n, \dots, n+m-1\} \wedge j \in \{0, \dots, n-1\}$. Then $\beta(i-n) = \gamma(i) = \gamma(j) = \alpha(j)$ so that $\alpha(j) \in N \cap M$ contradicting $N \cap M = \emptyset$, so this case never occurs.

$i \in \{n, \dots, n+m-1\} \wedge j \in \{n, \dots, n+m-1\}$. Then $\beta(i-n) = \gamma(i) = \gamma(j) = \beta(j-n)$ which as β is injective gives $i-n = j-n$ proving $i = j$.

So in all valid cases we have $i = j$ proving injectivity.

surjectivity. If $y \in I = N \cup M$ then we have either:

$y \in N$. Then as α is surjective there exist a $i \in \{0, \dots, n-1\}$ such that $y = \alpha(i)$ which as $\gamma(i) = \alpha(i)$ proves that $\gamma(i) = y$.

$y \in M$. Then as β is surjective there exist a $i \in \{0, \dots, m-1\}$ such that $y = \beta(i)$. For $j = i + n$ we have $n \leq i \leq n + m - 1$ and $\gamma(j) = \beta(j - n) = \beta(i) = y$

So

$$\gamma: \{0, \dots, n + m - 1\} \rightarrow N \cup M = I \text{ is a bijection}$$

Hence

$$\begin{aligned} \sum_{i \in I} x_i &= \sum_{i=0}^{n+m-1} x_{\gamma(i)} \\ &\stackrel{[\text{theorem: 11.23}]}{=} \sum_{i=0}^{n-1} x_{\gamma(i)} + \sum_{i=n}^{n+m-1} x_{\gamma(i)} \\ &= \sum_{i=0}^{n-1} x_{\alpha(i)} + \sum_{i=n}^{n+m-1} x_{\beta(i-n)} \\ &= \sum_{i=0}^{n-1} x_{\alpha(i)} + \sum_{i=0}^{m-1} x_{\beta((i+n)-n)} \\ &= \sum_{i=0}^{n-1} x_{\alpha(i)} + \sum_{i=0}^{m-1} x_{\beta(i)} \\ &\stackrel{[\text{theorem: 11.8}]}{=} \sum_{i \in N} x_i + \sum_{i \in M} x_i \\ &\quad \square \end{aligned}$$

We have now the equivalence of [theorem: 11.24]

Theorem 11.43. Let $\langle A, + \rangle$ be a Abelian semi-group, $n \in \mathbb{N}_0$, $\{I_i\}_{i \in \{0, \dots, n\}}$ a family of finite sets such that $\forall i, j \in I$ with $i \neq j$ $I_i \cap I_j = \emptyset$ then for $\{x_i\}_{i \in \bigcup_{j \in \{0, \dots, n\}} I_j} \subseteq A$ we have

$$\sum_{i \in \bigcup_{j \in \{0, \dots, n\}} I_j} x_i = \sum_{i=0}^n \left(\sum_{j \in I_i} x_j \right)$$

Proof. We prove this by induction, so take

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{I_i\}_{i \in \{0, \dots, n\}} \text{ pairwise disjoint family of sets, } \forall \{x_i\}_{i \in \bigcup_{j \in \{0, \dots, n\}} I_j} \subseteq A \text{ we have } \sum_{i \in \bigcup_{j \in \{0, \dots, n\}} I_j} x_i = \sum_{i=0}^n \left(\sum_{j \in I_i} x_j \right) \right\}$$

then we have:

$0 \in S$. Let $\{I_i\}_{i \in \{0\}}$ be a pairwise disjoint family of sets then $\bigcup_{i \in \{0\}} I_i = I_0$, let $\{x_i\}_{i \in \bigcup_{j \in \{0\}} I_j} \subseteq A$ then $\sum_{i \in \bigcup_{j \in \{0\}} I_j} x_i = \sum_{i \in I_0} x_i = \sum_{i=0}^0 \left(\sum_{j \in I_i} x_j \right)$ proving that $0 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $\{I_i\}_{i \in \{0, \dots, n+1\}}$ be a pairwise disjoint family of finite sets and $\{x_i\}_{i \in \bigcup_{j \in \{0, \dots, n+1\}} I_j}$. Take $J = \bigcup_{i \in \{0, \dots, n\}} I_i$ then by [theorem: 6.36] J is finite and

$$J \cap I_{n+1} = \left(\bigcup_{i \in \{0, \dots, n\}} I_i \right) \cap I_{n+1} = \bigcup_{i \in \{0, \dots, n\}} (I_i \cap I_{n+1}) = \bigcup_{i \in \{0, \dots, n\}} \emptyset = \emptyset$$

proving

$$\bigcup_{j \in \{0, \dots, n+1\}} I_j = J \bigcup I_{n+1}, \quad J, I_{n+1} \text{ are finite and } J \cap I_{n+1} = \emptyset \quad (11.9)$$

So we have

$$\begin{aligned} \sum_{i \in \bigcup_{j \in \{0, \dots, n\}} I_j} x_i &\stackrel{[\text{eq: 11.9}] \text{ and } [\text{lemma: 11.42}]}{=} \sum_{i \in J} x_i + \sum_{i \in I_{n+1}} x_i \\ &= \sum_{i \in \bigcup_{j \in \{0, \dots, n\}} I_j} x_i + \sum_{i \in I_{n+1}} x_i \\ &\stackrel{[n \in S]}{=} \sum_{i=0}^n \left(\sum_{j \in I_i} x_j \right) + \sum_{i \in I_{n+1}} x_i \\ &= \sum_{i=0}^{n+1} \left(\sum_{j \in I_i} x_j \right) \end{aligned}$$

proving that $n+1 \in S$. □

Corollary 11.44. *Let $\langle A, + \rangle$ be a Abelian group, I a finite set, $\{I_i\}_{i \in I}$ a family of finite sets such that $\forall i, j \in I$ with $i \neq j$ we have $I_i \cap I_j = \emptyset$ and $\{x_i\}_{i \in \bigcup_{j \in I} I_j} \subseteq A$ then*

$$\sum_{i \in \bigcup_{j \in I} I_j} x_i = \sum_{i \in I} \left(\sum_{j \in I_i} x_j \right)$$

Proof. For I we have either:

$I = \emptyset$. Then $\bigcup_{j \in I} I_j = \emptyset$ so that

$$\sum_{i \in \bigcup_{j \in I} I_j} x_i = \sum_{i \in \emptyset} x_i = 0 = \sum_{i \in \emptyset} \left(\sum_{j \in I_i} x_j \right)$$

$I \neq \emptyset$. Then there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$. Using [theorem: 2.127] we have that

$$\bigcup_{j \in I} I_j = \bigcup_{j \in \{0, \dots, n-1\}} I_{\beta(j)}$$

so that

$$\begin{aligned} \sum_{i \in \bigcup_{j \in I} I_j} x_i &= \sum_{i \in \bigcup_{j \in \{0, \dots, n-1\}} I_{\beta(j)}} x_i \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i=0}^{n-1} \left(\sum_{j \in I_{\beta(i)}} x_j \right) \\ &= \sum_{i \in I} \left(\sum_{j \in I_i} x_j \right) \end{aligned}$$

□

Corollary 11.45. *Let $\langle A, + \rangle$ be a Abelian semi-group, I, J finite sets and $\{x_{i,j}\}_{(i,j) \in I \times J}$ then*

$$\sum_{(i,j) \in I \times J} x_{i,j} = \sum_{i \in I} \left(\sum_{j \in J} x_{i,j} \right) \stackrel{[\text{theorem: 11.41}]}{=} \sum_{j \in J} \left(\sum_{i \in I} x_{i,j} \right)$$

Proof. Given $i \in I$ let $J_i = \{i\} \times J$ then as $I = \bigcup_{i \in I} \{i\}$ we have by [theorem: 2.141] that

$$I \times J = \bigcup_{i \in I} J_i$$

Further if $i, j \in I$ with $i \neq j$ then if $x \in J_i \cap J_j = (\{i\} \times J) \cap (\{j\} \times J)$ then $\exists k, l \in J$ such that $(i, k) = x = (j, l)$ giving $i = j$ contradicting $i \neq j$. So

$$\forall i, j \in I \text{ with } i \neq j \text{ we have } \{i\} \times J \cap \{j\} \times J = \emptyset \text{ or } J_i \neq J_j$$

So we can apply [theorem: 11.44] giving

$$\begin{aligned} \sum_{(i,j) \in I \times J} x_{i,j} &= \sum_{(i,j) \in \bigcup_{k \in I} J_k} x_{i,j} \\ &\stackrel{[\text{theorem: 11.44}]}{=} \sum_{k \in I} \left(\sum_{(i,j) \in J_k} x_{i,j} \right) \\ &\stackrel{=}{=} \sum_{k \in I} \left(\sum_{(k,j) \in J_k} x_{k,j} \right) \end{aligned} \quad (11.10)$$

Given $k \in I$ define $\beta_k: J \rightarrow J_k$ by $\beta_k(j) = (k, j) \in \{k\} \times J = J_k$ then we have

injectivity. If $\beta_k(i) = \beta_k(j)$ then $(k, i) = (k, j)$ giving $i = j$

surjectivity. If $(k, j) \in \{k\} \times J$ then $\beta_k(j) = (k, j)$

so that $\beta_k: J \rightarrow J_k$ is a bijection, hence by [theorem: 11.36]

$$\sum_{j \in J} x_{k,j} = \sum_{j \in J} x_{\beta_k(j)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{(k,j) \in J_k} x_{k,j}$$

which combined with [eq: 11.10] gives

$$\sum_{(i,j) \in I \times J} x_{i,j} = \sum_{k \in I} \left(\sum_{j \in J} x_{k,j} \right)$$

□

The following corollary will be useful in induction proofs.

Corollary 11.46. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle A, + \rangle$ be a Abelian semi group and $\{I_i\}_{i \in \{1, \dots, n\}}$ a family of finite sets and $\{x_{\sigma_1, \dots, \sigma_n}\}_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \subseteq A$ then

$$\sum_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} x_{\sigma_1, \dots, \sigma_n} = \sum_{i \in I_1} \left(\sum_{\sigma \in \prod_{i \in \{2, \dots, n\}} I_i} x_{i, \sigma_1, \dots, \sigma_{n-1}} \right)$$

Note 11.47. As $\{I_i\}_{i \in \{1, \dots, n\}}$ is a finite family of finite sets it follows from [theorem: 6.57] that

$\prod_{i \in \{1, \dots, n\}} I_i$ and $\prod_{i \in \{2, \dots, n\}} I_i$ are finite so that the sums $\sum_{\sigma \in \prod_{i \in \{2, \dots, n\}} I_i} x_{i, \sigma_1, \dots, \sigma_n}$ and $\sum_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} x_{\sigma_1, \dots, \sigma_n}$ are well defined.

Proof. Given $i \in I_1$ define by $J_i = \{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i \mid \sigma_1 = i\}$ then we have, if $\sigma \in \prod_{j \in \{1, \dots, n\}} I_j$, that $\sigma_1 \in I_1$ so that $\sigma \in J_{\sigma_1}$ hence $\prod_{j \in \{1, \dots, n\}} I_j \subseteq \bigcup_{i \in I_1} J_i$. As clearly $J_i \subseteq \prod_{i \in \{1, \dots, n\}} I_i$ it follows that

$$\prod_{i \in \{1, \dots, n\}} I_i = \bigcup_{i \in I_1} J_i$$

Take $i, j \in I_1$ with $i \neq j$. Assume that $\sigma \in J_i \cap J_j$ then $\sigma_1 = i \neq j = \sigma_1$ leading to the contradiction $\sigma_1 \neq \sigma_1$. Hence we must have that $J_i \cap J_j = \emptyset$. So we can apply [theorem: 11.44] to get

$$\sum_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} x_{\sigma_1, \dots, \sigma_n} = \sum_{i \in I_1} \left(\sum_{\sigma \in J_i} x_{\sigma_1, \dots, \sigma_n} \right) \quad (11.11)$$

Given $i \in I_1$ define

$$\beta_i: \prod_{(2, \dots, n)} I_i \rightarrow J_i \text{ by } \beta_i(\sigma) = (i, \sigma_1, \dots, \sigma_{n-1})$$

then we have:

injectivity. If $\beta_i(\sigma) = \beta_i(\tau)$ then $(i, \sigma_1, \dots, \sigma_{n-1}) = (i, \tau_1, \dots, \tau_{n-1})$ so that $\sigma = \tau$ proving injectivity.

surjectivity. If $\sigma \in J_i$ then $\sigma_1 = i \in I_1$ and $\forall j \in \{2, \dots, n\} \sigma_j \in I_j$ so that $\tau = (\sigma_2, \dots, \sigma_n) \in \prod_{j \in \{2, \dots, n\}} I_j$. As

$$\beta(\tau) = (i, \tau_1, \dots, \tau_{n-1}) = (i, \sigma_2, \dots, \sigma_n) \stackrel{i=\sigma_1}{=} (\sigma_1, \dots, \sigma_n)$$

surjectivity is proved.

so that β_i is a bijection. Further if $\sigma \in \prod_{j \in (2, \dots, n)} I_j$ then $\beta_i(\sigma) = (i, \sigma_1, \dots, \sigma_{n-1})$ so that

$$(\beta_i(\sigma))_j = \begin{cases} i & \text{if } j = 1 \\ \sigma_{j-1} & \text{if } j \in \{2, \dots, n\} \end{cases}$$

so that

$$x_{i, \sigma_1, \dots, \sigma_{n-1}} = x_{\beta(\sigma)_1, \dots, \beta(\sigma)_n}$$

Next we have

$$\begin{aligned} \sum_{\sigma \in \prod_{j \in (2, \dots, n)} I_j} x_{i, \sigma_1, \dots, \sigma_{n-1}} &= \sum_{\sigma \in \prod_{j \in (2, \dots, n)} I_j} x_{\beta(\sigma)_1, \dots, \beta(\sigma)_n} \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{\sigma \in J_i} x_{\sigma_1, \dots, \sigma_n} \end{aligned}$$

Substituting the above in [eq: 11.11] gives finally

$$\sum_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} x_{\sigma_1, \dots, \sigma_n} = \sum_{i \in I_1} \left(\sum_{\sigma \in \prod_{j \in (2, \dots, n)} I_j} x_{i, \sigma_1, \dots, \sigma_{n-1}} \right)$$

□

Corollary 11.48. Let $\langle R, +, \cdot \rangle$ be a ring, I, J finite sets and $\{x_i\}_{i \in I} \subseteq R$, $\{y_i\}_{i \in J} \subseteq R$ then

$$\sum_{(i, j) \in I \times J} (x_i \cdot y_j) = \left(\sum_{i \in I} x_i \right) \cdot \left(\sum_{j \in J} y_j \right)$$

Proof. As $\langle R, + \rangle$ is a Abelian group we have

$$\begin{aligned} \sum_{(i, j) \in I \times J} (x_i \cdot y_j) &\stackrel{[\text{corollary: 11.45}]}{=} \sum_{i \in I} \left(\sum_{j \in J} (x_i \cdot y_j) \right) \\ &\stackrel{[\text{theorem: 11.39}]}{=} \sum_{i \in I} \left(x_i \cdot \sum_{j \in J} y_j \right) \\ &\stackrel{[\text{theorem: 11.39}]}{=} \left(\sum_{i \in I} x_i \right) \cdot \left(\sum_{j \in J} y_j \right) \\ &= \left(\sum_{i \in I} x_i \right) \cdot \left(\sum_{j \in J} y_j \right) \end{aligned}$$

□

Theorem 11.49. Let $\langle R, +, \cdot \rangle$ be a ring, $n \in \mathbb{N}$ and $\{\{a_{i,j}\}_{j \in I_i}\}_{i \in \{1, \dots, n\}}$ is a finite family of finite families in R [in other words $\forall i \in \{1, \dots, n\}$ I_i is finite and $\{a_{i,j}\}_{j \in I_i} \subseteq R$] then

$$\prod_{i \in \{1, \dots, n\}} \left(\sum_{j \in I_i} a_{i,j} \right) = \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\prod_{i \in \{1, \dots, n\}} a_{i, \rho_i} \right)$$

Proof. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{ \{ a_{i,j} \}_{j \in \{n_i, \dots, m_i\}} \}_{i \in \{1, \dots, n\}} \text{ is a finite family of finite families in } R \right.$$

$$\text{then } \prod_{i \in \{1, \dots, n\}} \left(\sum_{j \in I_i} a_{i,j} \right) = \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\prod_{i \in \{1, \dots, n\}} a_{i, \rho_i} \right) \Bigg\}$$

then we have:

$1 \in S$. If $\{ \{ a_{i,j} \}_{j \in I_i} \}_{i \in \{1\}}$ is a finite family of finite families. Define

$$\beta: \prod_{i \in \{1\}} I_i \rightarrow I_1 \text{ by } \beta(\rho) = \rho_1$$

then we have:

injectivity. If $\beta(\rho) = \beta(\sigma)$ then $\rho_1 = \sigma_1$ so that $\sigma = (\sigma_1) = (\rho_1) = \rho$.

surjectivity. If $y \in I_1$ then for $\sigma = (y) \in \prod_{i \in \{1\}} I_i$ we have $\beta(\sigma) = \sigma_1 = y$

So β turns out to be a bijection, hence using [theorem: 11.36] it follows that

$$\sum_{j \in I_1} a_{1,j} = \sum_{\rho \in \prod_{i \in \{1\}} I_i} a_{1, \rho_1} \quad (11.12)$$

Hence

$$\begin{aligned} \prod_{i \in \{1\}} \left(\sum_{j \in I_i} a_{i,j} \right) &= \sum_{j \in I_1} a_{1,j} \\ &\stackrel{[\text{eq: 11.12}]}{=} \sum_{\rho \in \prod_{i \in \{1\}} I_i} a_{1, \rho_1} \\ &= \sum_{\rho \in \prod_{i \in \{1\}} I_i} \left(\prod_{i \in \{1\}} a_{i, \rho_i} \right) \end{aligned}$$

which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{ \{ a_{i,j} \}_{j \in I_i} \}_{i \in \{1, \dots, n+1\}}$ be a finite family of finite families with $\forall i \in \{1, \dots, n+1\}$ $n_i \leq m_i$ then we have

$$\begin{aligned} &\prod_{i \in \{1, \dots, n+1\}} \left(\sum_{j \in I_i} a_{i,j} \right) = \\ &\left(\prod_{i \in \{1, \dots, n\}} \left(\sum_{j \in I_i} a_{i,j} \right) \right) \cdot \left(\sum_{j \in I_{n+1}} a_{n+1,j} \right) \stackrel{n \in S}{=} \\ &\left(\sum_{\rho \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\prod_{i \in \{1, \dots, n\}} a_{i, \rho_i} \right) \right) \cdot \left(\sum_{j \in I_{n+1}} a_{n+1,j} \right) = \\ &\sum_{\rho \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, \rho_i} \right) \cdot \sum_{j \in I_{n+1}} a_{n+1,j} \right) = \\ &\sum_{\rho \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\sum_{j \in I_{n+1}} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, \rho_i} \right) \cdot a_{n+1,j} \right) \right) \stackrel{[\text{corollary: 11.45}]}{=} \\ &\sum_{\delta = (\rho, j) \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, \rho_i} \right) \cdot a_{n+1,j} \right) = \\ &\sum_{\delta \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, (\delta_1)_i} \right) \cdot a_{n+1, \delta_2} \right) \end{aligned}$$

so that

$$\prod_{i \in \{1, \dots, n+1\}} \left(\sum_{j \in I_i} a_{i,j} \right) = \sum_{\delta \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, (\delta_1)_i} \right) \cdot a_{n+1, \delta_2} \right) \quad (11.13)$$

Define

$$\gamma: \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1} \rightarrow \prod_{i \in \{1, \dots, n+1\}} I_i \quad \text{where } \gamma(\delta) \text{ is defined by}$$

$$\gamma(\delta)_i = \begin{cases} (\delta_1)_i & \text{if } i \in \{1, \dots, n\} \\ \delta_2 & \text{if } i = n+1 \end{cases}$$

then we have:

injectivity. Let $\delta, \sigma \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}$ with $\gamma(\delta, \sigma)$. Then $\delta_1, \sigma_1 \in \prod_{i \in \{1, \dots, n\}} I_i$ and $\forall i \in \{1, \dots, n\}$ we have $(\delta_1)_i = \gamma(\delta)_i = \gamma(\sigma)_i = (\sigma_1)_i$ so that $\delta_1 = \sigma_1$, further $\delta_2 = \gamma(\delta)_{n+1} = \gamma(\sigma)_{n+1} = \sigma_2$. Hence $\delta = (\delta_1, \delta_2) = (\sigma_1, \sigma_2) = \sigma$.

surjectivity. If $\rho \in \prod_{i \in \{1, \dots, n+1\}} I_i$ then $((\rho_1, \dots, \rho_n), \rho_{n+1}) \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}$ and

$$\gamma((\rho_1, \dots, \rho_n), \rho_{n+1})_i = \begin{cases} (\rho_1, \dots, \rho_n)_i = \rho_i & \text{if } i \in \{1, \dots, n\} \\ \rho_{n+1} & \text{if } i = n+1 \end{cases} = \rho_i$$

so that $\gamma((\rho_1, \dots, \rho_n), \rho_{n+1}) = \rho$

hence

$$\gamma: \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1} \rightarrow \prod_{i \in \{1, \dots, n+1\}} I_i \text{ is a bijection}$$

So that

$$\begin{aligned} & \sum_{\delta \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, (\delta_1)_i} \right) \cdot a_{n+1, \delta_2} \right) = \\ & \sum_{\delta \in \left(\prod_{i \in \{1, \dots, n\}} I_i \right) \times I_{n+1}} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, \gamma(\delta)_i} \right) \cdot a_{n+1, \gamma(\delta)_{n+1}} \right) \stackrel{[\text{theorem: 11.36}]}{=} \\ & \sum_{\delta \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\left(\prod_{i \in \{1, \dots, n\}} a_{i, \delta_i} \right) \cdot a_{n+1, \delta_{n+1}} \right) = \\ & \sum_{\delta \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\prod_{i \in \{1, \dots, n+1\}} a_{i, \delta_i} \right) \end{aligned}$$

Substituting the above in [eq: 11.13] gives

$$\prod_{i \in \{1, \dots, n+1\}} \left(\sum_{j \in I_i} a_{i,j} \right) = \sum_{\delta \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\prod_{i \in \{1, \dots, n+1\}} a_{i, \delta_i} \right)$$

which proves that $n+1 \in \mathcal{S}$.

Mathematical induction finished then the proof of this theorem. $\square$

Theorem 11.50. Let $\langle A, + \rangle$ be a Abelian semi-group, $n \in \mathbb{N}$, $\langle A^n, + \rangle$ the semi-group based on $\langle A, + \rangle$ [see theorem: 6.95] then we have:

1. Let $k, l \in \mathbb{N}_0$ with $k \leq l$ and $\{x_i\}_{i \in \{k, \dots, l\}} \subseteq A^n$ then $\forall i \in \{1, \dots, n\}$ we have

$$\left(\sum_{j=k}^l x_j \right)_i = \sum_{j=k}^l (x_j)_i$$

2. Let J be a finite set and $\{x_j\}_{j \in J} \subseteq A^n$ then for $i \in \{1, \dots, n\}$ we have $(\sum_{j \in J} x_j)_i = \sum_{j \in J} (x_j)_i$.

Proof.

1. We use induction to prove this for the case $\sum_{j=0}^l x_j$, so define

$$S = \left\{ l \in \mathbb{N}_0 \mid \text{If } \{x_i\}_{i \in \{0, \dots, l\}} \subseteq A^n \text{ then } \left(\sum_{j=0}^l x_j \right)_i = \sum_{j=0}^l (x_j)_i \right\}$$

then we have:

$0 \in S$. If $\{x_j\}_{j \in \{0\}} \subseteq A^n$ then for $i \in \{1, \dots, n\}$ we have $(\sum_{j=0}^0 x_j)_i = (x_0)_i = \sum_{j=0}^0 (x_j)_i$ proving that $0 \in S$.

$l \in S \Rightarrow l+1 \in S$. Let $\{x_j\}_{j \in \{0, \dots, l+1\}} \subseteq A^n$ then we have

$$\begin{aligned} \left(\sum_{j=0}^{l+1} x_j \right)_i &= \left(\left(\sum_{j=0}^l x_j \right) + x_{l+1} \right)_i \\ &\stackrel{[\text{theorem: 6.95}]}{=} \left(\sum_{j=0}^l x_j \right)_i + (x_{l+1})_i \\ &\stackrel{l \in S}{=} \left(\sum_{j=0}^l (x_j)_i \right) + (x_{l+1})_i \\ &= \sum_{j=0}^{l+1} (x_j)_i \end{aligned}$$

proving that $l+1 \in S$.

By mathematical induction it follows then that

$$\forall l \in \mathbb{N}_0 \text{ we have for } \{x_j\}_{j \in \{1, \dots, l\}} \subseteq A^n \text{ and } i \in \{1, \dots, n\} \text{ that } \left(\sum_{j=0}^l x_j \right)_i = \sum_{j=0}^l (x_j)_i \quad (11.14)$$

Let now $k, l \in \mathbb{N}_0$ with $k \leq l$ and $\{x_j\}_{j \in \{k, \dots, l\}} \subseteq A^n$ then we have for $i \in \{1, \dots, n\}$

$$\left(\sum_{j=k}^l x_j \right)_i = \left(\sum_{j=0}^{l-k} x_{j+k} \right)_i \stackrel{[\text{eq: 11.14}]}{=} \sum_{j=0}^{l-k} (x_{j+k})_i = \sum_{j=k}^l (x_j)_i$$

2. If I is a finite set, $i \in \{1, \dots, n\}$ and $\{x_j\}_{j \in I} \subseteq A^n$ then we have for I either:

$I = \emptyset$. Then

$$\left(\sum_{j \in I} x_j \right)_i = \left(\sum_{j \in \emptyset} x_j \right)_i = 0_i = \sum_{j \in \emptyset} (x_j)_i$$

$I \neq \emptyset$. Then there exists a $l \in \mathbb{N}_0$ and bijection $\beta: \{0, \dots, l\} \rightarrow I$ then

$$\left(\sum_{j \in I} x_j \right)_i \stackrel{\text{def}}{=} \left(\sum_{j=0}^l x_{\beta(j)} \right)_i \stackrel{[\text{theorem: 11.14}]}{=} \sum_{j=0}^l (x_{\beta(j)})_i = \sum_{j \in I} (x_j)_i \quad \square$$

Theorem 11.51. For the field $\langle \mathbb{K}, \cdot, + \rangle$ where $\mathbb{K} = \mathbb{C}$ or $\mathbb{K} = \mathbb{R}$ we have

1. If $n \in \mathbb{N}_0$ and $\{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{K}$ is defined by $\alpha_i = \alpha$ then

$$a. \sum_{i=0}^n \alpha_i = (n+1) \cdot \alpha$$

- b. $\prod_{i=0}^n \alpha_i = \alpha^{(n+1)}$
2. If $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{\alpha_i\}_{i \in \{n, \dots, m\}} \subseteq \mathbb{K}$ is defined by $\alpha_i = \alpha$ then
- a. $\sum_{i=n}^m \alpha_i = (m - n + 1) \cdot \alpha$
- b. $\prod_{i=n}^m \alpha_i = \alpha^{m-n+1}$
3. If I is a finite set then for $\{\alpha_i\}_{i \in I} \subseteq \mathbb{K}$ defined by $\alpha_i = \alpha$ we have
- a. $\sum_{i \in I} \alpha_i = \text{card}(I) \cdot \alpha$
- b. $\prod_{i \in I} \alpha_i = \alpha^{\text{card}(I)}$

Proof.

1.

a. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{K} \text{ is defined by } \alpha_i = \alpha \text{ then } \sum_{i=0}^n \alpha_i = (n+1) \cdot \alpha \right\}$$

then we have:

$0 \in S$. Then $\sum_{i=0}^0 \alpha_i = \alpha_0 = \alpha = (0+1) \cdot \alpha$ proving that $0 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{\alpha_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{K}$ be defined by $\alpha_i = \alpha$. Then

$$\begin{aligned} \sum_{i=0}^{n+1} \alpha_i &= \sum_{i=0}^n \alpha_i + \alpha_{n+1} \\ &= \sum_{i=0}^n \alpha_i + \alpha \\ &\stackrel{n \in S}{=} (n+1) \cdot \alpha + \alpha \\ &= ((n+1) + 1) \cdot \alpha \end{aligned}$$

proving that $n+1 \in S$

b. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{K} \text{ is defined by } \alpha_i = \alpha \text{ then } \prod_{i=0}^n \alpha_i = (n+1) \cdot \alpha \right\}$$

then we have:

$0 \in S$. Then $\prod_{i=0}^0 \alpha_i \stackrel{[\text{theorem: 11.34}]}{=} \alpha_0 = \alpha = \alpha^1 = \alpha^{(0+1)}$ proving that $0 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{\alpha_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{R}$ be defined by $\alpha_i = \alpha$. Then

$$\begin{aligned} \prod_{i=0}^{n+1} \alpha_i &= \left(\prod_{i=0}^n \alpha_i \right) \cdot \alpha_{n+1} \\ &= \left(\prod_{i=0}^n \alpha_i \right) \cdot \alpha \\ &\stackrel{n \in S}{=} \alpha^{n+1} \cdot \alpha \\ &= \alpha^{(n+1)+1} \end{aligned}$$

proving that $n+1 \in S$

2.

a. $\sum_{i=n}^m \alpha_i = \sum_{i=0}^{m-n} \alpha_{i+n} \stackrel{(1.a)}{=} (m-n+1) \cdot \alpha$

$$\text{b. } \prod_{i=n}^m \alpha_i = \prod_{i=0}^{m-n} \alpha_{i+n} \stackrel{(1.a)}{=} \alpha^{m-n+1}$$

3.

a. For I we have either:

$$I = \emptyset. \text{ Then } \sum_{i \in I} \alpha_i = \sum_{i \in \emptyset} \alpha_i = 0$$

$I \neq \emptyset$. Then $\{0, \dots, \text{card}(I) - 1\} \approx \{1, \dots, \text{card}(I)\}$ so that there exists a bijection $\beta: \{0, \dots, \text{card}(I) - 1\} \rightarrow I$ so that

$$\sum_{i \in I} \alpha_i = \sum_{i=0}^{\text{card}(I)-1} \alpha_{\beta(i)} \stackrel{(1.a)}{=} (\text{card}(I) - 1 + 1) \cdot \alpha = \text{card}(I) \cdot \alpha$$

b. For I we have either:

$$I = \emptyset. \text{ Then } \prod_{i \in I} \alpha_i = \prod_{i \in \emptyset} \alpha_i = 1$$

$I \neq \emptyset$. Then $\{0, \dots, \text{card}(I) - 1\} \approx \{1, \dots, \text{card}(I)\}$ so that there exists a bijection $\beta: \{0, \dots, \text{card}(I) - 1\} \rightarrow I$ so that

$$\prod_{i \in I} \alpha_i = \prod_{i=0}^{\text{card}(I)-1} \alpha_{\beta(i)} \stackrel{(2.a)}{=} \alpha^{(\text{card}(I)-1+1)} = \alpha^{\text{card}(I)} \quad \square$$

Theorem 11.52. *If $\langle F, +, \cdot \rangle$ is a field then we have*

1. *Let I be a finite set, $i \in I$, $x, y \in F$ and $\{\alpha_j\}_{j \in I} \subseteq F$ such that $\alpha_i = x + y$ then*

$$\prod_{j \in I} \alpha_j = \prod_{j \in I} \beta_j + \prod_{j \in I} \gamma_j$$

where $\{\beta_j\}_{j \in I}$ and $\{\gamma_j\}_{j \in I}$ are defined by

$$\beta_j = \begin{cases} x & \text{if } j = i \\ \alpha_j & \text{if } j \in I \setminus \{i\} \end{cases} \quad \text{and} \quad \gamma_j = \begin{cases} y & \text{if } j = i \\ \alpha_j & \text{if } j \in I \setminus \{i\} \end{cases}$$

2. *Let I be a finite set, $i \in I$, $x, \beta \in F$, and $\{\alpha_j\}_{j \in I} \subseteq F$ such that $\alpha_i = \beta \cdot x$ then*

$$\prod_{j \in I} \alpha_j = \beta \cdot \prod_{j \in I} \beta_j$$

where $\{\beta_j\}_{j \in I}$ is defined by $\beta_j = \begin{cases} x & \text{if } j = i \\ \alpha_j & \text{if } j \in I \setminus \{i\} \end{cases}$.

3. *Let $n, m \in \mathbb{N}_0$ with $n \leq m$, $i \in \{n, \dots, m\}$, $x, y \in F$ and $\{\alpha_j\}_{j \in \{n, \dots, m\}} \subseteq F$ such that $\alpha_i = x + y$ then*

$$\prod_{j=n}^m \alpha_j = \prod_{j=n}^m \beta_j + \prod_{j=n}^m \gamma_j$$

where $\{\beta_j\}_{j \in \{n, \dots, m\}}$ and $\{\gamma_j\}_{j \in \{n, \dots, m\}}$ are defined by

$$\beta_j = \begin{cases} x & \text{if } j = i \\ \alpha_j & \text{if } j \in \{n, \dots, m\} \setminus \{i\} \end{cases} \quad \text{and} \quad \gamma_j = \begin{cases} y & \text{if } j = i \\ \alpha_j & \text{if } j \in \{n, \dots, m\} \setminus \{i\} \end{cases}$$

4. *Let $n, m \in \mathbb{N}_0$ with $n \leq m$, $i \in \{n, \dots, m\}$, $x, \beta \in F$, and $\{\alpha_j\}_{j \in \{n, \dots, m\}} \subseteq F$ such that $\alpha_i = \beta \cdot x$ then*

$$\prod_{j=n}^m \alpha_j = \beta \cdot \prod_{j=n}^m \beta_j$$

where $\{\beta_j\}_{j \in \{n, \dots, m\}}$ is defined by $\beta_j = \begin{cases} x & \text{if } j = i \\ \alpha_j & \text{if } j \in \{n, \dots, m\} \setminus \{i\} \end{cases}$.

Proof.

1. We have

$$\begin{aligned}
 \prod_{j \in I} \alpha_j &\stackrel{[\text{theorem: 11.43}]}{=} \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot \left(\prod_{j \in \{i\}} \alpha_j \right) \\
 &\stackrel{[\text{example: 11.34}]}{=} \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot \alpha_i \\
 &= \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot (x + y) \\
 &= \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot x + \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot y \\
 &= \left(\prod_{j \in I \setminus \{i\}} \beta_j \right) \cdot \beta_i + \left(\prod_{j \in I \setminus \{i\}} \gamma_j \right) \cdot \gamma_i \\
 &\stackrel{[\text{theorem: 11.34}]}{=} \left(\prod_{j \in I \setminus \{i\}} \beta_j \right) \cdot \left(\prod_{j \in \{i\}} \beta_j \right) + \left(\prod_{j \in I \setminus \{i\}} \gamma_j \right) \cdot \left(\prod_{j \in \{i\}} \gamma_j \right) \\
 &\stackrel{[\text{theorem: 11.43}]}{=} \prod_{j \in I} \beta_j + \prod_{j \in I} \gamma_j
 \end{aligned}$$

2. We have

$$\begin{aligned}
 \prod_{j \in I} \alpha_j &\stackrel{[\text{theorem: 11.43}]}{=} \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot \left(\prod_{j \in \{i\}} \alpha_j \right) \\
 &\stackrel{[\text{theorem: 11.34}]}{=} \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot \alpha_i \\
 &= \left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot (\beta \cdot x) \\
 &= \beta \cdot \left(\left(\prod_{j \in I \setminus \{i\}} \alpha_j \right) \cdot x \right) \\
 &= \beta \cdot \left(\left(\prod_{j \in I \setminus \{i\}} \beta_j \right) \cdot \beta_i \right) \\
 &\stackrel{[\text{theorem: 11.34}]}{=} \beta \cdot \left(\left(\prod_{j \in I \setminus \{i\}} \beta_j \right) \cdot \left(\prod_{j \in \{i\}} \beta_j \right) \right) \\
 &\stackrel{[\text{theorem: 11.43}]}{=} \beta \cdot \prod_{j \in I} \beta_j
 \end{aligned}$$

3. We have

$$\begin{aligned}
 \prod_{i=n}^m \alpha_i &\stackrel{[\text{theorem: 11.35}]}{=} \prod_{i \in \{n, \dots, m\}} \alpha_i \\
 &\stackrel{(1)}{=} \prod_{i \in \{n, \dots, m\}} \beta_i + \prod_{i \in \{n, \dots, m\}} \gamma_i \\
 &\stackrel{[\text{theorem: 11.35}]}{=} \prod_{i=n}^m \beta_i + \prod_{i=n}^m \gamma_i
 \end{aligned}$$

4. We have

$$\begin{aligned}
 \prod_{i=n}^m \alpha_i &\stackrel{[\text{theorem: 11.35}]}{=} \prod_{i \in \{n, \dots, m\}} \alpha_i \\
 &\stackrel{(2)}{=} \beta \cdot \prod_{i \in \{n, \dots, m\}} \beta_i \\
 &\stackrel{[\text{theorem: 11.35}]}{=} \beta \cdot \prod_{i=n}^m \beta_i
 \end{aligned}$$

□

Theorem 11.53. *If $\langle F, \cdot, + \rangle$ is a field then*

1. *Let $n \in \mathbb{N}_0$ and $\{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq F \setminus \{0\}$ then we have $\prod_{i=0}^n \alpha_i \neq 0$*
2. *Let $n, m \in \mathbb{N}_0$ and $\{\alpha_i\}_{i \in \{n, \dots, m\}} \subseteq F \setminus \{0\}$ then we have $\prod_{i=n}^m \alpha_i \neq 0$*
3. *If I is a finite set and $\{\alpha_i\}_{i \in I} \subseteq F \setminus \{0\}$ then we have $\prod_{i \in I} \alpha_i \neq 0$*

Proof.

1. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq F \setminus \{0\} \text{ then } \prod_{i=0}^n \alpha_i \neq 0 \right\}$$

then we have:

$0 \in S$. Then $\prod_{i=0}^0 \alpha_i \stackrel{[\text{theorem: 11.34}]}{=} \alpha_0 \neq 0$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{\alpha_i\}_{i \in \{0, \dots, n+1\}} \subseteq F \setminus \{0\}$ then we have

$$\prod_{i=0}^{n+1} \alpha_i = \left(\prod_{i=0}^n \alpha_i \right) \cdot \alpha_{n+1}$$

As $n \in S$ we have that $\prod_{i=0}^n \alpha_i \neq 0$ so as, $\alpha_{n+1} \neq 0$, so as a field is a integral domain [see: 4.58], we have that $\prod_{i=0}^{n+1} \alpha_i \neq 0$. So $n+1 \in S$.

2. $\prod_{i=n}^m \alpha_i = \prod_{i=0}^{m-n} \alpha_i \neq 0$ [using (1)]
3. If I is finite then we have either:

$I = \emptyset$. Then $\prod_{i \in S} \alpha_i = \prod_{i \in \emptyset} \alpha_i = 1 \neq 0$.

$I \neq \emptyset$. Then there exists a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} \alpha_i = \sum_{i=0}^{n-1} \alpha_{\beta(i)} \neq 0 \text{ (by (1))}$$

□

Theorem 11.54. *If $\langle R, +, \cdot \rangle$ is a commutative ring with neutral element 0, I a finite set, $\{x_i\}_{i \in I} \subseteq R$ such that $\exists i \in I$ with $x_i = 0$ then*

$$\prod_{j \in I} x_j = 0$$

Proof. This follows from

$$\prod_{j \in I} x_j \stackrel{[\text{theorem: 11.41}]}{=} \left(\prod_{j \in I \setminus \{i\}} x_j \right) \cdot \prod_{j \in \{i\}} x_j \stackrel{[\text{theorem: 11.34}]}{=} \left(\prod_{j \in I \setminus \{i\}} x_j \right) \cdot x_i = \left(\prod_{j \in I \setminus \{i\}} x_j \right) \cdot 0 = 0$$

□

Theorem 11.55. *Let I be a finite set and $\{x_i\}_{i \in I} \subseteq \mathbb{R}$ then we have*

1. *If $\forall i \in I$ we have $0 \leq x_i$ then*

$$0 \leq \sum_{i \in I} x_i$$

Further if additional there exist a $i \in I$ such that $0 < x_i$ then

$$0 < \sum_{i \in I} x_i$$

2. *If $\{y_i\}_{i \in I} \subseteq \mathbb{R}$ is a finite family such that $\forall i \in I$ $x_i \leq y_i$ then*

$$\sum_{i \in I} x_i \leq \sum_{i \in I} y_i$$

Further if additional there exist a $i \in I$ such that $x_i < y_i$ then

$$\sum_{i \in I} x_i < \sum_{i \in I} y_i$$

Proof.

1. We prove this by induction on the size of I so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \text{card}(I) = n \text{ then } \forall \{x_i\}_{i \in I} \subseteq \mathbb{R} \text{ with } \forall i \in I \ 0 \leq x_i \text{ we have } 0 \leq \sum_{i \in I} x_i \right\}$$

then we have:

0 $\in S$. If $\text{card}(I) = 0$ then $I = \emptyset$ such that $\sum_{i \in I} x_i = \sum_{i \in \emptyset} x_i \stackrel{[\text{definition: 11.32}]}{=} 0$ so that $0 \leq \sum_{i \in I} x_i$.

$n \in S \Rightarrow n+1 \in S$. Let I be such that $\text{card}(I) = n+1$ then $I \neq \emptyset$ so that $\exists i_0 \in I$ and by [theorem: 10.94] $\text{card}(I \setminus \{i_0\}) = n$. Let $\{x_i\}_{i \in I} \subseteq \mathbb{R}$ then

$$\begin{aligned} \sum_{i \in I} x_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus \{i_0\}} x_i + \sum_{i \in \{i_0\}} x_i \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in I \setminus \{i_0\}} x_i + x_{i_0} \end{aligned}$$

As $n \in S$ we have that $0 \leq \sum_{i \in I \setminus \{i_0\}} x_i$, further by the hypothesis $0 \leq x_{i_0}$ hence it follows that $n+1 \in S$.

Hence using mathematical induction we have proved that if I is finite and $\{x_i\}_{i \in I} \subseteq \mathbb{R}$ satisfies $\forall i \in I \ 0 \leq x_i$ then

$$0 \leq \sum_{i \in I} x_i$$

Further if there exist a $k \in I$ such that $0 < x_k$ then we have

$$0 < \sum_{i \in I \setminus \{k\}} x_i + x_k \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in I \setminus \{k\}} x_i + \sum_{i \in \{k\}} x_i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I} x_i$$

2. As $\forall i \in I$ we have $x_i \leq y_i$ it follows that $0 \leq y_i - x_i$ so that by (1)

$$\begin{aligned} 0 &\leq \sum_{i \in I} (y_i - x_i) \\ &\stackrel{[\text{theorems: 11.40, 11.38}]}{=} \sum_{i \in I} y_i - \sum_{i \in I} x_i \end{aligned}$$

proving that

$$\sum_{i \in I} x_i \leq \sum_{i \in I} y_i$$

Further if in addition there exist a $k \in I$ such that $x_i < y_i$ then

$$\begin{aligned} 0 &< \sum_{i \in I} (y_i - x_i) \\ &\stackrel{[\text{theorems: 11.40, 11.38}]}{=} \sum_{i \in I} y_i - \sum_{i \in I} x_i \end{aligned}$$

proving that

$$\sum_{i \in I} x_i < \sum_{i \in I} y_i \quad \square$$

11.2 Vector spaces

11.2.1 Definition

Definition 11.56. A vector space $\langle V, \oplus, \odot \rangle$ over a field $\langle F, +, \cdot \rangle$ is a Abelian group $\langle V, \oplus \rangle$ together with a map $\odot: F \times V \rightarrow V$ satisfying

1. $\forall \alpha \in F$ and $\forall x, y \in V$ we have $\alpha \odot (x \oplus y) = \alpha \odot x \oplus \alpha \odot y$
2. $\forall \alpha, \beta \in F$ and $x \in V$ we have $(\alpha + \beta) \odot x = \alpha \odot x \oplus \beta \odot x$
3. $\forall \alpha, \beta \in F$ and $x \in V$ we have $(\alpha \cdot \beta) \odot x = \alpha \odot (\beta \odot x)$
4. If 1 is the multiplicative neutral element of $\langle F, +, \cdot \rangle$ then $1 \odot x = x$

The map $\odot$ is called the scalar product and elements of V are called vectors.

Note 11.57. Some books call a vector space a linear space, which is maybe clearer as we later will introduce linear (in)dependent sets and linear combinations. In this book we follow the convention of most books about mathematics and physics and use the term vector space.

Theorem 11.58. Let $\langle V, \oplus, \odot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ where

- a) 0_v is the neutral element of the Abelian group $\langle V, \oplus \rangle$
- b) 0_f is the additive neutral element of $\langle F, +, \cdot \rangle$
- c) 1 is the multiplicative neutral element of $\langle F, +, \cdot \rangle$
- d) For $x \in V$ $-x$ is the inverse of x in $\langle V, \oplus \rangle$

then we have

1. $\forall x \in V$ we have $0_f \odot x = 0_v$
2. $\forall x \in V$ we have $(-1) \odot x = -x$
3. $\forall \alpha \in F$ we have $\alpha \odot 0_v = 0_v$
4. If $\alpha \in F \setminus \{0_f\}$ then for $x \in V$ with $\alpha \odot x = 0_v$ we have $x = 0_v$
5. If $x \in V \setminus \{0_v\}$ then for $\alpha \in F$ with $\alpha \odot x = 0_v$ we have $\alpha = 0_f$.

Proof.

1. If $x \in V$ then

$$\begin{aligned} 0_v &= (0_f \odot x) \oplus (-(0_f \odot x)) \\ &\stackrel{0_f = 0_f + 0_f}{=} ((0_f + 0_f) \odot x) \oplus (-(0_f \odot x)) \\ &= ((0_f \odot x) \oplus (0_f \odot x)) \oplus (-(0_f \odot x)) \\ &= (0_f \odot x) \oplus ((0_f \odot x) \oplus (-(0_f \odot x))) \\ &= (0_f \odot x) \oplus 0_v \\ &= 0_f \odot x \end{aligned}$$

2. If $x \in V$ then

$$\begin{aligned} x \oplus ((-1) \odot x) &= (1 \odot x) \oplus ((-1) \odot x) \\ &= (1 + (-1)) \odot x \\ &= 0_f \odot x \\ &\stackrel{(1)}{=} 0_v \end{aligned}$$

so that by the definition of a inverse element we have

$$(-1) \odot x = -x$$

3. If $\alpha \in F$ then

$$\begin{aligned} \alpha \odot 0_v &\stackrel{(1)}{=} \alpha \odot (0_f \odot 0_v) \\ &= (\alpha \cdot 0_f) \odot 0_v \\ &\stackrel{[\text{theorem: 4.39}]}{=} 0_f \odot 0_v \\ &\stackrel{(1)}{=} 0_v \end{aligned}$$

4. Let $\alpha \in F \setminus \{0_f\}$ and take $x \in V$ such that $\alpha \odot x = 0_v$. As $\alpha \neq 0_f$ we have that α^{-1} exist. So that

$$0_v \stackrel{(3)}{=} \alpha^{-1} \odot 0_v = \alpha^{-1} \odot (\alpha \odot x) = (\alpha^{-1} \cdot \alpha) \odot x = 1 \odot x = x$$

5. Let $x \in V \setminus \{0_v\}$ and $\alpha \in F$ such that $\alpha \odot x = 0_v$. Assume that $\alpha \neq 0_f$ then by (4) we have that $x = 0_v$ contradicting $x \in V \setminus \{0_v\}$, so we must have that $\alpha = 0_f$. $\square$

Just as we have sub-groups, sub-rings and sub-fields we have sub-spaces of a vector space.

Definition 11.59. Let $\langle V, \oplus, \odot \rangle$ be a vector space over $\langle F, +, \cdot \rangle$ then $W \subseteq V$ is a sub-space of $\langle V, \oplus, \odot \rangle$ if

1. $W \neq \emptyset$
2. $\forall \alpha \in F$ and $x, y \in W$ we have that $(\alpha \odot x) \oplus y \in W$

Example 11.60. Let $\langle V, \oplus, \odot \rangle$ be a vector space over $\langle F, +, \cdot \rangle$, $x \in V$ then $W = \{t \cdot x | t \in F\}$ is a sub-space of V

Proof. Let 1 be the multiplicative neutral element in V then $x = 1 \odot x$ which proves that $x \in W$ hence

$$w \neq \emptyset$$

Let $\alpha \in F$, $y, z \in W$ then there exist $\lambda_y, \lambda_z \in F$ such that $y = \lambda_y \odot x$, $z = \lambda_z \odot x$ so that

$$\begin{aligned} y + \alpha \odot z &= \lambda_y \odot x + \alpha \odot (\lambda_z \odot x) \\ &= \lambda_y \odot x + (\alpha \cdot \lambda_z) \odot x \\ &= \lambda_y \odot x + (\lambda_z \cdot \alpha) \odot x \\ &= (\lambda_y + \lambda_z \cdot \alpha) \odot x \end{aligned}$$

which proves, as $\lambda_y + \lambda_z \cdot \alpha \in F$, that

$$y + \alpha \odot z \in W$$

$\square$

Theorem 11.61. Let $\langle V, \oplus, \odot \rangle$ be a vector space over $\langle F, +, \cdot \rangle$ and $W \subseteq V$ a sub-space then $\langle W, \oplus|_{W \times W}, \odot|_{F \odot W} \rangle$ is a vector space over $\langle F, +, \cdot \rangle$ where the neutral element of W is the neutral element of V .

Proof. First we prove that $\langle W, \oplus|_{W \times W} \rangle$ is a Abelian group:.

$\oplus|_{W \times W}$ is a operator on W . If $x, y \in W$ then $x \oplus y = (1 \odot x) \oplus y \in W$ so that $\oplus|_{W \times W}$ is indeed a function between $W \times W$ and W .

associativity. If $x, y, z \in W$ then

$$x \oplus_{|W \times W|} (y \oplus_{|W \times W|} z) \underset{x, y, z \in W}{=} x \oplus (y \oplus z) = (x \oplus y) \oplus z = (x \oplus_{|W \times W|} y) \oplus_{|W \times W|} z$$

commutativity. If $x, y \in W$ then $x \oplus_{|W \times W|} y = x \oplus y = y \oplus x = y \oplus_{|W \times W|} x$

neutral element. Then as $W \neq \emptyset$ there exist a $w \in W$ so that

$$0_v = (-w) \oplus w \underset{[\text{theorem: 11.58 (2)}]}{=} (-1) \odot w \oplus w \in W$$

proving that $0_v \in W$. Further $\forall x \in W$ we have

$$x \oplus_{|W \times W|} 0_v \underset{\text{commutativity}}{=} 0_v \oplus_{|W \times W|} x = 0_v \oplus x = x$$

inverse element. If $x \in W$ then $-x \underset{[\text{theorem: 11.58 (2)}]}{=} ((-1) \odot x) = ((-1) \odot x) \oplus 0_v \in W$ so that

$$-x \in W. \text{ Further } x \oplus_{|W \times W|} (-x) \underset{\text{commutativity}}{=} (-x) \oplus_{|W \times W|} x = (-x) \oplus x = 0_v$$

Further we have:

1. If $\alpha \in F$ and $x \in W$ then $\alpha \odot x = \alpha \odot x \oplus 0_v \in W$ so that $\odot_{|F \times W|}$ is a function between $F \times W$ and W .
2. If $\alpha \in F$ and $x, y \in W$ then $\alpha \odot_{|F \times W|} (x \oplus_{|W \times W|} y) = \alpha \odot (x \oplus y) = \alpha \odot x \oplus \alpha \odot y = \alpha \odot_{|F \times W|} x \oplus_{|W \times W|} \alpha \odot_{|F \times W|} y$
3. If $\alpha, \beta \in F$ and $x \in W$ then $(\alpha + \beta) \odot_{|F \times W|} x = (\alpha + \beta) \odot x = \alpha \odot x \oplus \beta \odot x = \alpha \odot_{|F \times W|} x \oplus_{|W \times W|} \beta \odot_{|F \times W|} x$
4. If $\alpha, \beta \in F$ and $x \in W$ then $(\alpha \cdot \beta) \odot_{|F \times W|} x = (\alpha \cdot \beta) \odot x = \alpha \odot (\beta \odot x) = \alpha \odot_{|F \times W|} (\beta \odot_{|F \times W|} x)$
5. If $x \in W$ then $1 \odot_{|F \times W|} x = 1 \odot x = x$ □

Note 11.62. To avoid excessive use of subscripts we follow for the rest of this book the convention that if $\langle V, \oplus, \odot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$ and $W \in V$ is a sub-space of $\langle V, \oplus, \odot \rangle$ we use $\oplus$ instead of $\oplus_{|W \times W|}$ and $\odot$ instead of $\odot_{|F \times W|}$. Using this convention we have then that $\langle W, \oplus, \odot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$.

11.2.2 Examples of vector spaces

Example 11.63. Let $\langle F, +, \cdot \rangle$ be a field, e a element and

1. $\oplus: \{e\} \times \{e\} \rightarrow \{e\}$ defined by $e \oplus e = e$
2. $\odot: F \times \{e\} \rightarrow \{e\}$ defined by $\alpha \odot e = e$

then $\langle \{e\}, \oplus, \odot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$ with neutral element e and the inverse of e is e . This vector space is called the **trivial vector space**.

Proof. First we prove that $\langle \{e\}, \oplus \rangle$ is a Abelian group:

associativity.

$$\forall x, y, z \in \{e\} \text{ we have } x \oplus (y \oplus z) = e \oplus (e \oplus e) = e \oplus e = e = e \oplus e = (e \oplus e) \oplus e = (x \oplus y) \oplus z$$

neutral element. $\forall x \in \{e\}$ we have $x = e$ so that $x \oplus e = e \oplus e = e = e \oplus e = e \oplus x$ proving that e is the neutral element.

inverse element. $\forall x \in \{e\}$ we have $x = e$ so that $x \oplus e = e \oplus e = e$ proving that e is the inverse element of x .

commutativity. $\forall x, y \in \{e\}$ we have $x = e = y$ so that $x \oplus y = e \oplus e = e = e \oplus e = y \oplus x$

For the remaining axioms we have

1. $\forall \alpha \in F$ and $x, y \in \{e\}$ we have as $x = e = y$ that

$$\alpha \odot (x + y) = \alpha \odot (e \oplus e) = \alpha \odot e = e = e \oplus e = \alpha \odot e \oplus \alpha \odot e = \alpha \odot x \oplus \alpha \odot y$$

2. $\forall \alpha, \beta \in F$ and $x \in \{e\}$ we have as $x = e$ that

$$(\alpha + \beta) \odot x = (\alpha + \beta) \odot e = e = e \oplus e = \alpha \odot e + \beta \odot e = \alpha \odot x + \beta \odot x$$

3. $\forall \alpha, \beta \in F$ and $x \in \{e\}$ we have as $x = e$ that

$$(\alpha \cdot \beta) \odot x = (\alpha \cdot \beta) \odot e = e = \alpha \odot e = \alpha \odot (\beta \odot e)$$

4. $\forall x \in \{e\}$ we have $1 \odot x = 1 \odot e = e = x$ □

Every field is a vector spaces over itself.

Theorem 11.64. *If $\langle F, +, \cdot \rangle$ is a field then $\langle F, +, \cdot \rangle$ is a vector space over itself*

Proof. As $\langle F, +, \cdot \rangle$ is a field we have by definition of a field [see definition: 4.51] that $\langle F, + \rangle$ is a Abelian group. Further for the rest of the vector axioms we have:

1. $\forall \alpha, x, y \in F$ we have $\alpha \cdot (x + y) \stackrel{\text{distributivity}}{=} \alpha \cdot x + \alpha \cdot y$

2. $\forall \alpha, \beta, x \in F$ we have $(\alpha + \beta) \cdot x \stackrel{\text{distributivity}}{=} \alpha \cdot x + \beta \cdot x$

3. $\forall \alpha, \beta, x \in F$ we have $(\alpha \cdot \beta) \cdot x \stackrel{\text{associativity}}{=} \alpha \cdot (\beta \cdot x)$

4. $\forall x \in F$ we have for the multiplicative neutral element of $\langle F, +, \cdot \rangle$ that $1 \cdot x = x$ □

Using the above and the fact that $\langle \mathbb{Q}, +, \cdot \rangle$, $\langle \mathbb{R}, +, \cdot \rangle$ and $\langle \mathbb{C}, +, \cdot \rangle$ are fields [see theorem: 10.5] we have:

Example 11.65. $\langle \mathbb{Q}, +, \cdot \rangle$ is a vector space over $\langle \mathbb{Q}, +, \cdot \rangle$, $\langle \mathbb{R}, +, \cdot \rangle$ is a vector space over $\langle \mathbb{R}, +, \cdot \rangle$ and $\langle \mathbb{C}, +, \cdot \rangle$ is a vector space over $\langle \mathbb{C}, +, \cdot \rangle$.

Be aware that a vector space depends also on the field used, so for example $\mathbb{C}$ can be used to define another vector space, as is shown in the next example.

Example 11.66. $\langle \mathbb{C}, +, \cdot \rangle$ is a vector space over $\langle \mathbb{R}, +, \cdot \rangle$

Proof. As $\langle \mathbb{C}, +, \cdot \rangle$ is a field we have by definition of a field [see definition: 4.51] that $\langle \mathbb{C}, + \rangle$ is a Abelian group. Further for the rest of the vector axioms we have as $\mathbb{R} \subseteq \mathbb{C}$ that

1. $\forall \alpha \in \mathbb{R}, \forall x, y \in \mathbb{C}$ we have $\alpha \cdot (x + y) \stackrel{\text{distributivity}}{=} \alpha \cdot x + \alpha \cdot y$

2. $\forall \alpha, \beta \in \mathbb{R}, \forall x \in \mathbb{C}$ we have $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$

3. $\forall \alpha, \beta \in \mathbb{R}, \forall x \in \mathbb{C}$ we have $(\alpha \cdot \beta) \cdot x = \alpha \cdot (\beta \cdot x)$

4. As $1 \in \mathbb{R} \subseteq \mathbb{C}$ we have $\forall x \in \mathbb{C}$ $1 \cdot x$ □

Theorem 11.67. *Let $\langle F, +, \cdot \rangle$ be a field [so that $\langle F, +, \cdot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$] then if V is a sub-space of F then $V = \{0\}$ or $V = F$*

Proof. Let V be a sub-space of F then we have either

$V = \{0\}$. Then clearly $V = \{0\}$

$V \neq \{0\}$. Then there exist a $v \in F$ with $v \neq 0$. As F is a field there exist a $v^{-1} \in F$ such that $1 = v^{-1} \cdot v$ which proves that $1 \in V$. Let $x \in F$ then as $x = x \cdot 1 \in V$ so that $F \subseteq V$ which as for a sub-space we have $V \subseteq F$ proves that $V = F$.

so V is either $\{0\}$ or F , □

Definition 11.68. A vector space $\langle V, \oplus, \odot \rangle$ over $\langle \mathbb{R}, +, \cdot \rangle$ is called a **real** vector space and a vector space $\langle V, \oplus, \odot \rangle$ over $\langle \mathbb{C}, +, \cdot \rangle$ is called a **complex** vector space. If we want to refer to a vector space $\langle V, \oplus, \odot \rangle$ over either $\langle \mathbb{R}, +, \cdot \rangle$ or $\langle \mathbb{C}, +, \cdot \rangle$ then we say that $\langle V, \oplus, \odot \rangle$ is a vector space over $\langle \mathbb{K}, +, \cdot \rangle$.

Next we use an existing vector space and pairwise addition and scalar multiplication to define a function space. Later will use this to define more special function spaces that are sub-spaces of this function space.

Theorem 11.69. (function space) Let $\langle V, \oplus, \odot \rangle$ be a vector space over $\langle F, +, \cdot \rangle$, X a set then and define for $V^X = \{f \mid f: X \rightarrow V \text{ is a function}\}$ the following operations:

$\boxplus: V^X \times V^X \rightarrow V^X$ defined by $f \boxplus g$ where $(f \boxplus g)(x) = f(x) \oplus g(x)$ [pairwise addition]

$\boxdot: F \times V^X \rightarrow V^X$ defined by $\alpha \boxdot f$ where $(\alpha \boxdot f)(x) = \alpha \odot f(x)$

then

$\langle V^X, \boxplus, \boxdot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$

where:

- a) $C_0: X \rightarrow V$ is defined by $C_0(x) = 0_V$ [the constant function [see example: 2.47].
- b) If $f \in V^X$ then $-f$ defined by $(-f)(x) = -f(x)$ is the inverse element of f .

Proof. First we prove that $\langle V^X, \boxplus, \boxdot \rangle$ is a Abelian group:

associativity. Let $f, g, h \in V^X$ then $\forall x \in X$ we have

$$\begin{aligned} (f \boxplus (g \boxplus h))(x) &= f(x) \oplus (g \boxplus h)(x) \\ &= f(x) \oplus (g(x) \oplus h(x)) \\ &= (f(x) \oplus g(x)) \oplus h(x) \\ &= (f \boxplus g)(x) \oplus h(x) \\ &= ((f \boxplus g) \boxplus h)(x) \end{aligned}$$

proving that $f \boxplus (g \boxplus h) = (f \boxplus g) \boxplus h$

commutativity. Let $f, g \in V^X$ then $\forall x \in X$ we have

$$\begin{aligned} (f \boxplus g)(x) &= f(x) \oplus g(x) \\ &= g(x) \oplus f(x) \\ &= (g \boxplus f)(x) \end{aligned}$$

so that

$$f \boxplus g = g \boxplus f$$

neutral element. Let $f \in V^X$ then $\forall x \in X$ we have

$$(f \boxplus C_0)(x) = f(x) \oplus C_0(x) = f(x) \oplus 0 = f(x)$$

so that

$$C_0 \boxplus f \underset{\text{commutativity}}{=} f \boxplus C_0 = f$$

inverse element. Let $f \in V^X$ then $\forall x \in X$ we have

$$(f \boxplus (-f))(x) = f(x) \oplus (-f)(x) = f(x) \oplus (-f(x)) = 0 = C_0(x)$$

so that $(-f) \boxplus f \underset{\text{commutativity}}{=} f \boxplus (-f) = C_0$

For the remaining axioms of a vector space we have:

1. If $\alpha \in F$ and $f, g \in V^X$ then $\forall x \in X$ we have

$$\begin{aligned} (\alpha \boxtimes (f \boxplus g))(x) &= \alpha \odot (f \boxplus g)(x) \\ &= \alpha \odot (f(x) \oplus g(x)) \\ &= \alpha \odot f(x) \oplus \alpha \odot g(x) \\ &= (\alpha \boxtimes f)(x) \oplus (\alpha \boxtimes g)(x) \\ &= (\alpha \boxtimes f \boxplus \alpha \boxtimes g)(x) \end{aligned}$$

proving that $\alpha \boxtimes (f \boxplus g) = \alpha \boxtimes f \boxplus \alpha \boxtimes g$.

2. If $\alpha, \beta \in F$ and $f \in V^X$ then $\forall x \in X$ we have

$$\begin{aligned} ((\alpha + \beta) \boxtimes f)(x) &= (\alpha + \beta) \odot f(x) \\ &= \alpha \odot f(x) + \beta \odot f(x) \\ &= (\alpha \boxtimes f)(x) \oplus (\beta \boxtimes f)(x) \\ &= (\alpha \boxtimes f \boxplus \beta \boxtimes f)(x) \end{aligned}$$

so that $(\alpha + \beta) \boxtimes f = \alpha \boxtimes f \boxplus \beta \boxtimes f$.

3. If $\alpha, \beta \in F$ and $f \in V^X$ then $\forall x \in X$ we have

$$\begin{aligned} ((\alpha \cdot \beta) \boxtimes f)(x) &= (\alpha \cdot \beta) \odot f(x) \\ &= \alpha \cdot (\beta \odot f(x)) \\ &= \alpha \cdot (\beta \boxtimes f)(x) \\ &= (\alpha \boxtimes (\beta \boxtimes f))(x) \end{aligned}$$

proving that $(\alpha \cdot \beta) \boxtimes f = \alpha \boxtimes (\beta \boxtimes f)$

4. Let $f \in V^X$ then we have $\forall x \in X$ that $(1 \boxtimes f)(x) = 1 \odot f(x) = f(x)$ so that $1 \boxtimes f = f$. $\square$

Up to now we have used different operator symbols for addition and multiplication in the different structures. To simplify notation we start using the same symbol $+$ for addition in the vector space and field and the same symbol $\cdot$ for the scalar product and multiplication in the field. We also use 0 for the additive neutral element in the vector space and the field and note the inverse of x in the vector space and the field as $-x$.

Referring to the power of a set [see definition: 6.92] we can construct a vector space on the power of a vector space.

Theorem 11.70. *Let $n \in \mathbb{N}$, $\langle V, +, \cdot \rangle$ be a vector space over $\langle F, +, \cdot \rangle$ then $\langle V^n, +, \cdot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$ where*

$$+: V^n \times V^n \rightarrow V^n \text{ is defined by } (x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

$$\cdot: F \times V^n \rightarrow V^n \text{ is defined by } \alpha \cdot (x_1, \dots, x_n) = (\alpha \cdot x_1, \dots, \alpha \cdot x_n)$$

$$0 = (\underbrace{0, \dots, 0}_n) \in V^n \text{ is the additive neutral element in } V^n$$

$$\forall (x_1, \dots, x_n) \in V^n \text{ the additive negative is } (-x_1, \dots, -x_n)$$

Proof. Note that using [theorem: 6.93] we have that

$$V^n = V^{\{1, \dots, n\}} = \{f \mid f: \{1, \dots, n\} \rightarrow V \text{ is a function}\}$$

and $x = (x_1, \dots, x_n)$ is equivalent with $x: \{1, \dots, n\} \rightarrow V$ is a function where $\forall i \in \{1, \dots, n\}$ we have $x(i) = x_i$. So

$$+: V^n \times V^n \rightarrow V^n \text{ is defined by } (x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

is equivalent with:

$$+: V^n \times V^n \rightarrow V^n \text{ is defined by } (x + y) \text{ where } (x + y)(i) = x(i) + y(i)$$

and

$$\cdot: F \times V^n \rightarrow V^n \text{ is defined by } \alpha \cdot (x_1, \dots, x_n) = (\alpha \cdot x_1, \dots, \alpha \cdot x_n)$$

is equivalent with:

$$\cdot: F \times V^n \rightarrow V^n \text{ is defined by } \alpha \cdot x \text{ where } (\alpha \cdot x)(i) = \alpha \cdot x(i)$$

and

$$(0, \dots, 0) \in V^n \text{ is } C_0 \text{ where } C_0(i) = 0$$

and finally

$$\forall (x_1, \dots, x_n) \in V^n \text{ the additive negative is } (-x_1, \dots, -x_n)$$

is equivalent with

$$\forall x \in V^n \text{ the additive negative is } -x \text{ where } (-x)(i) = -x(i)$$

Combining these equivalent definitions with [theorem: 11.69] proves then that $\langle V^n, +, \cdot \rangle$ is a vector space. $\square$

As a application of the above theorem we have that

Corollary 11.71. *Let $n \in \mathbb{N}$ and $\langle F, +, \cdot \rangle$ a field then $\langle F^n, +, \cdot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$*

Proof. This follows from [theorems: 11.64, 11.70] $\square$

Using [examples: 11.65, 11.66] and the above theorem [theorem: 11.70] we have then the following examples of vector spaces:

Example 11.72. Let $n \in \mathbb{N}$ then

1. $\langle \mathbb{Q}^n, +, \cdot \rangle$ is a vector space over $\langle \mathbb{Q}, +, \cdot \rangle$
2. $\langle \mathbb{R}^n, +, \cdot \rangle$ is a vector space over $\langle \mathbb{R}, +, \cdot \rangle$
3. $\langle \mathbb{C}, +, \cdot \rangle$ is a vector space over $\langle \mathbb{C}, +, \cdot \rangle$
4. $\langle \mathbb{C}, +, \cdot \rangle$ is a vector space over $\langle \mathbb{R}, +, \cdot \rangle$

Remark 11.73. Note that in the proof of the previous theorem we use the fact that $V^n = \{f | f: \{1, \dots, n\} \rightarrow V \text{ is a function}\}$, however this is not standard practice. Most books prefers to work with the notation $x \in V^n \Leftrightarrow x = (x_1, \dots, x_n)$ such that $x_i \in V \forall i \in \{1, \dots, n\}$. Likewise for $\prod_{i \in \{1, \dots, n\}} V_i$ most books use the notation $x \in \prod_{i \in \{1, \dots, n\}} V_i \Leftrightarrow x = (x_1, \dots, x_n)$ such that $x_i \in V_i \forall i \in \{1, \dots, n\}$. This is a standard that we will follow for the rest of this book. If needed you can use [definition: 6.84] and [theorem: 6.93] to fallback to the original definitions.

Definition 11.74. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $A, B \subseteq V$, $G \subseteq F$, $x \in V$ and $\alpha \in F$ then we have the following definitions:

1. $x + A \stackrel{\text{def}}{=} \{x + y | y \in A\} \stackrel{\text{commutativity}}{=} \{y + x | y \in A\} \stackrel{\text{def}}{=} A + x$
2. $A + B = \{x + y | x \in A \wedge y \in B\}$
3. $\alpha \cdot A = \{\alpha \cdot x | x \in A\}$
4. $G \cdot A = \{\gamma \cdot x | \gamma \in G \wedge x \in A\}$
5. $A - B = \{x - y | x \in A \wedge y \in B\}$

Theorem 11.75. If $\langle V, +, \cdot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$ then we have

1. $\forall A \subseteq V, x \in V$ we have $y \in x + A \Leftrightarrow y - x \in A$

2. $\forall A, B \subseteq V, x \in V$ we have $x + (A \cup B) = (x + A) \cup (x + B)$
3. $\forall A, B \subseteq V, x \in V$ we have $x + (A \cap B) = (x + A) \cap (x + B)$
4. $\forall A \subseteq V, x, y \in V$ we have $x + (y + A) = (x + y) + A$

Proof.

1. If $y \in x + A$ there exists a $z \in A$ such that $y = x + z \Rightarrow y - x = z \in A$. If $y - x \in A \Rightarrow y = x + (y - x) \in x + A$.
- 2.

$$\begin{aligned}
 y \in x + (A \cup B) &\stackrel{(1)}{\Leftrightarrow} y - x \in A \cup B \\
 &\Leftrightarrow (y - x \in A) \vee (y - x \in B) \\
 &\Leftrightarrow (y \in x + A) \vee (y \in x + B) \\
 &\Leftrightarrow y \in (x + A) \cup (x + B)
 \end{aligned}$$

3.

$$\begin{aligned}
 y \in x + (A \cap B) &\stackrel{(1)}{\Leftrightarrow} y - x \in A \cap B \\
 &\Leftrightarrow (y - x \in A) \wedge (y - x \in B) \\
 &\Leftrightarrow (y \in x + A) \wedge (y \in x + B) \\
 &\Leftrightarrow y \in (x + A) \cap (x + B)
 \end{aligned}$$

4.

$$\begin{aligned}
 z \in (x + (y + A)) &\stackrel{(1)}{\Leftrightarrow} z - x \in y + A \\
 &\stackrel{(1)}{\Leftrightarrow} (z - x) - y \in A \\
 &\Leftrightarrow (z - (x + y)) \in A \\
 &\Leftrightarrow z \in (x + y) + A
 \end{aligned}$$

□

Theorem 11.76. If $\langle V_i, +_i, \cdot_i \rangle_{i \in I}$ is a family of vector spaces over a field $\langle F, +, \cdot \rangle$ then if we define:

1. $+$: $\prod_{i \in I} V_i \times \prod_{i \in I} V_i \rightarrow \prod_{i \in I} V_i$ by $(x, y) \rightarrow x + y$ where $x + y: I \rightarrow \bigcup_{i \in I} V_i$ is defined by $(x + y)(i) = x(i) +_i y(i) = x_i +_i y_i$ [see theorem 4.26]
2. $\cdot$: $F \times \prod_{i \in I} V_i \rightarrow \prod_{i \in I} V_i$ is defined by $(\alpha, x) \rightarrow \alpha \cdot x$ where $\alpha \cdot x: I \rightarrow \bigcup_{i \in I} V_i$ is defined by $(\alpha \cdot x) = \alpha \cdot_i x(i) = \alpha \cdot_i x_i$

then we have that $\langle \prod_{i \in I} V_i, +, \cdot \rangle$ is a vector space over $\langle F, +, \cdot \rangle$

Proof. From [theorem 4.26] it follows that $\langle \prod_{i \in I} V_i, + \rangle$ is an Abelian group. Next if $\alpha \in F$ and $x \in \prod_{i \in I} V_i$ we have by the fact that $\langle V_i, +_i, \cdot_i \rangle$ is a vector space that $\alpha \cdot_i x(i) \in V_i$ so that $\alpha \cdot x: I \rightarrow \bigcup_{i \in I} V_i$ is an element of $\prod_{i \in I} V_i$ and thus that $\cdot: F \times \prod_{i \in I} V_i \rightarrow \prod_{i \in I} V_i$ is indeed a function. Now that we have proved that (1) and (2) are well defined we prove the rest of the vector space axioms.

1. If $\alpha \in F$ and $x, y \in \prod_{i \in I} V_i$ then $\forall i \in I$ we have

$$\begin{aligned}
 (\alpha \cdot (x + y))(i) &= \alpha \cdot_i (x + y)(i) \\
 &= \alpha \cdot_i (x(i) +_i y(i)) \\
 &\stackrel{\langle V_i, +_i, \cdot_i \rangle \text{ is a vector space}}{=} \alpha \cdot_i x(i) +_i \alpha \cdot_i y(i) \\
 &= (\alpha \cdot x)(i) +_i (\alpha \cdot y)(i) \\
 &= (\alpha \cdot x + \alpha \cdot y)(i)
 \end{aligned}$$

so that $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$

2. If $\alpha, \beta \in F$ and $x \in \prod_{i \in I} V_i$ then $\forall i \in I$ we have

$$\begin{aligned} ((\alpha + \beta) \cdot x)(i) &= (\alpha + \beta) \cdot_i x(i) \\ &\stackrel{\langle V_i, +_i, \cdot_i \rangle \text{ is a vector space}}{=} \alpha \cdot_i x(i) +_i \beta \cdot_i x(i) \\ &= (\alpha \cdot x)(i) +_i (\beta \cdot x)(i) \\ &= (\alpha \cdot x + \beta \cdot x)(i) \end{aligned}$$

so that $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$.

3. If $\alpha, \beta \in F$ and $x \in \prod_{i \in I} V_i$ then $\forall i \in I$ we have

$$\begin{aligned} ((\alpha \cdot \beta) \cdot x)(i) &= (\alpha \cdot \beta) \cdot_i x(i) \\ &\stackrel{\langle V_i, +_i, \cdot_i \rangle \text{ is a vector space}}{=} \alpha \cdot_i (\beta \cdot_i x(i)) \\ &= \alpha \cdot_i (\beta \cdot x)(i) \\ &= (\alpha \cdot (\beta \cdot x))(i) \\ &= (\alpha \cdot (\beta \cdot x))(i) \end{aligned}$$

So that $(\alpha \cdot \beta) \cdot x = \alpha \cdot (\beta \cdot x)$.

4. If 1 is the unit in F and $x \in \prod_{i \in I} V_i$ then $\forall i \in I$ we have

$$(1 \cdot x)(i) = 1 \cdot_i x(i) \stackrel{\langle V_i, +_i, \cdot_i \rangle \text{ is a vector space}}{=} x(i)$$

and thus $1 \cdot x = x$ □

11.2.3 Factor spaces of a vector space

A factor space allows us to treat a sub-space of a vector space as a neutral element in higher level of vector space. To create a factor space we first define an equivalence relation, the factor space is then the set of equivalence classes. We add then operators to the factor space based on the operators for addition and scalar product of the original vector space.

Theorem 11.77. Let $\langle X, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and Y a sub-space of X then $\sim_Y$ defined by $\sim_Y = \{(x, y) \in X \times X \mid x - y \in Y\}$ is an equivalence relation.

Proof.

reflectivity. As Y is a sub-space of x we have that $0 \in Y$ [see theorem: 11.61], hence if $x \in X$ we have that $x - x = 0 \in Y$ proving that $\forall x \in X \ x \sim_Y x$.

symmetry. If $x \sim_Y y$ then $x - y \in Y$, so as Y is a vector space [as a sub-space of X] we have that $y - x = -(x - y) \in Y$, hence $y \sim_Y x$.

transitivity. If $x \sim_Y y$ and $y \sim_Y z$ then $x - y \in Y$ and $y - z \in Y$ so that $x - z = (x - y) + (y - z) \in Y$ [as Y is a vector space], hence $x \sim_Y z$. □

We form now the set of equivalence classes and endow it with an addition and scalar product operator based on the operators of the original vector space.

Theorem 11.78. Let X be a vector space over a field F and Y a subspace of X then we use [definition: 3.16] to define X/Y as follows:

$$X/Y = X / \sim_Y \stackrel{\text{definition: 3.16}}{=} \{\sim_Y[x] \mid x \in X\}$$

where

$$\sim_Y[x] = \{y \in X \mid y \sim_Y x\} = \{y \in X \mid y - x \in Y\} \stackrel{y-x \in Y \Leftrightarrow y \in x+Y}{=} x + Y$$

Using this definition we have that the following are well defined functions

$$+: X/Y \times X/Y \rightarrow X/Y \text{ defined by } \sim_Y[x] + \sim_Y[y] = \sim_Y[x + y]$$

$$\cdot: F \times X/Y \rightarrow X/Y \text{ defined by } \alpha \cdot \sim_Y[x] = \sim_Y[\alpha \cdot x]$$

Using the above operators we have that

$$\langle X/Y, +, \cdot \rangle \text{ is a vector space over } \langle F, +, \cdot \rangle$$

where the neutral element is $\sim[0] = Y$ and for every $\sim[x] \sim[-x]$ is the inverse element

Proof. First we must prove that $+$ and $\cdot$ are well defined.

1. Assume that $\sim_Y[x] = \sim_Y[x']$ and $\sim_Y[y] = \sim_Y[y']$ then we have by [theorem: 3.11] that $x \sim_Y x'$ and $y \sim_Y y'$. Hence $x - x' \in Y$ and $y - y' \in Y$ so that $(x + y) - (x' + y') = (x - x') + (y - y') \in Y$ or $x + y \sim_Y x' + y'$. Using then [theorem: 3.11] it follows that $\sim_Y[x + y] = \sim_Y[x' + y']$. So

$$+: X/Y \times X/Y \rightarrow X/Y \text{ defined by } \sim_Y[x] + \sim_Y[y] = \sim[x + y] \text{ is a function}$$

2. Let $\alpha \in F$. Assume that $\sim_Y[x] = \sim_Y[x']$ then we have by [theorem: 3.11] that $x \sim_Y x'$ or $x - x' \in Y$. Hence $\alpha \cdot x - \alpha \cdot x' = \alpha \cdot (x - x') \in Y$ so that $\sim[\alpha \cdot x] = \sim_Y[\alpha \cdot x']$. Using then [theorem: 3.11] it follows that $\sim_Y[\alpha \cdot x] = \sim_Y[\alpha \cdot x']$. So

$$\cdot: F \times X/Y \rightarrow X/Y \text{ defined by } \alpha \cdot \sim[x] = \sim[\alpha \cdot x] \text{ is a function}$$

Next note that $x \in \sim_Y[0] \Leftrightarrow x - 0 \in Y \Leftrightarrow x \in Y$ which proves that

$$\sim_Y[0] = Y$$

Next we have to prove the vector space axioms:

1. $\langle X/Y, + \rangle$ is a Abelian group:

associativity. Let $\sim_Y[x], \sim_Y[y], \sim_Y[z] \in X/Y$ then

$$\begin{aligned} \sim_Y[x] + (\sim_Y[y] + \sim_Y[z]) &= \sim_Y[x] + \sim_Y[y + z] \\ &= \sim_Y[x + (y + z)] \\ &= \sim_Y[(x + y) + z] \\ &= \sim_Y[x + y] + \sim_Y[z] \\ &= (\sim_Y[x] + \sim_Y[y]) + \sim_Y[z] \end{aligned}$$

commutativity. Let $\sim_Y[x], \sim_Y[y] \in X/Y$ then

$$\sim_Y[x] + \sim_Y[y] = \sim_Y[x + y] = \sim_Y[y + x] = \sim_Y[y] + \sim_Y[x]$$

neutral element. Let $\sim_Y[x] \in X/Y$ then

$$\sim_Y[x] + \sim_Y[0] \underset{\text{commutativity}}{=} \sim_Y[0] + \sim_Y[x] = \sim_Y[0 + x] = \sim_Y[x]$$

inverse element. Let $\sim_Y[x] \in X/Y$ then

$$\sim_Y[x] + \sim_Y[-x] \underset{\text{commutativity}}{=} \sim_Y[-x] + \sim_Y[x] = \sim_Y[(-x) + x] = \sim_Y[0]$$

2. Scalar product axioms

- a. Let $\alpha \in F$ and $\sim_Y[x], \sim_Y[y] \in X/Y$ then

$$\begin{aligned} \alpha \cdot (\sim_Y[x] + \sim_Y[y]) &= \alpha \cdot \sim_Y[x + y] \\ &= \sim_Y[\alpha \cdot (x + y)] \\ &= \sim_Y[\alpha \cdot x + \alpha \cdot y] \\ &= \sim_Y[\alpha \cdot x] + \sim_Y[\alpha \cdot y] \\ &= \alpha \cdot \sim_Y[x] + \alpha \cdot \sim_Y[y] \end{aligned}$$

- b. Let $\alpha, \beta \in F$ and $\sim[x] \in X/Y$ then

$$\begin{aligned} (\alpha + \beta) \cdot \sim_Y[x] &= \sim_Y[(\alpha + \beta) \cdot x] \\ &= \sim_Y[\alpha \cdot x + \beta \cdot x] \\ &= \sim_Y[\alpha \cdot x] + \sim_Y[\beta \cdot x] \\ &= \alpha \cdot \sim_Y[x] + \beta \cdot \sim_Y[x] \end{aligned}$$

c. Let $\alpha, \beta \in F$ and $\sim[x] \in X/Y$ then

$$\begin{aligned} (\alpha \cdot \beta) \cdot \sim_Y[x] &= \sim_Y[(\alpha \cdot \beta) \cdot x] \\ &= \sim_Y[\alpha \cdot (\beta \cdot x)] \\ &= \alpha \cdot \sim_Y[\beta \cdot x] \\ &= \alpha \cdot (\beta \cdot \sim_Y[x]) \end{aligned}$$

d. Let $\sim_Y[x] \in X/Y$ then

$$1 \cdot \sim_Y[x] = \sim_Y[1 \cdot x] = \sim_Y[x]$$

□

Theorem 11.79. (Projection Function) Let X be a vector space over a field F and Y a subspace of X then the **projection function** is defined by

$$\pi_Y: X \rightarrow X/Y \text{ where } \pi_Y(x) = \sim_Y[x]$$

This function has the following properties:

1. $\forall x, y \in X$ and $\forall \alpha \in F$ we have $\pi_Y(x+y) = \pi_Y(x) + \pi_Y(y)$ and $\pi_Y(\alpha \cdot x) = \alpha \cdot \pi_Y(x)$ [in other words π_Y is linear]
2. $\pi_Y(X) = X/Y$ [surjectivity]
3. $\forall x \in Y$ we have $\pi_Y(x) = \sim_Y[0] = 0$ the neutral element of X/Y

Proof.

1. Let $x, y \in X$ and $\alpha \in F$ then we have

$$\pi_Y(x+y) = \sim_Y[x+y] = \sim_Y[x] + \sim_Y[y] = \pi_Y(x) + \pi_Y(y)$$

and

$$\pi_Y[\alpha \cdot x] = \sim_Y[\alpha \cdot x] = \alpha \cdot \sim_Y[x] = \alpha \cdot \pi_Y(x)$$

2. $\pi_Y(X) = \{\pi_Y(x) | x \in X\} = \{\sim_Y[x] | x \in X\} = X/Y$
3. If $x \in Y$ then we have $x - 0 = x \in Y$ so that $x \sim_Y 0$, hence by [theorem: 3.11] we have that $\sim_Y[x] = \sim_Y[0]$. Hence

$$\forall x \in Y \text{ we } \pi_Y(x) = \sim_Y[0] = 0$$

□

11.3 Basis of a vector space

11.3.1 Finite sums on a vector space

As a vector space is also a Abelian group we can talk about finite sums of vectors in the vector space.

Theorem 11.80. If $\langle V, +, \cdot \rangle$ is a vector space over a field $\langle F, +, \cdot \rangle$ then we have:

1. If $\alpha \in F$, $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq V$ then $\sum_{i=n}^m \alpha \cdot x_i = \alpha \cdot \sum_{i=n}^m x_i$
2. If $\alpha \in F$, I a finite set, $\{x_i\}_{i \in I} \subseteq V$ then $\sum_{i \in I} \alpha \cdot x_i = \alpha \cdot \sum_{i \in I} x_i$

Proof.

1. We use induction, so take:

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \text{ we have } \sum_{i=0}^n \alpha \cdot x_i = \alpha \cdot \sum_{i=0}^n x_i \right\}$$

then we have:

$$0 \in S. \text{ If } \{x_i\}_{i \in \{0\}} \text{ then } \sum_{i=0}^0 \alpha \cdot x_i = \alpha \cdot x_0 = \alpha \cdot \sum_{i=0}^0 x_i \text{ proving that } 0 \in S$$

$n \in S \Rightarrow n+1 \in S$. If $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq V$ then

$$\begin{aligned} \sum_{i=0}^{n+1} \alpha \cdot x_i &= \left(\sum_{i=0}^n \alpha \cdot x_i \right) + \alpha \cdot x_{n+1} \\ &\stackrel{n \in S}{=} \alpha \cdot \left(\sum_{i=0}^n x_i \right) + \alpha \cdot x_{n+1} \\ &= \alpha \cdot \left(\left(\sum_{i=0}^n x_i \right) + x_{n+1} \right) \\ &= \alpha \cdot \sum_{i=0}^{n+1} x_i \end{aligned}$$

proving that $n+1 \in S$.

So by mathematical induction

$$\forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq V \text{ we have } \sum_{i=0}^n \alpha \cdot x_i = \alpha \cdot \sum_{i=0}^n x_i \quad (11.15)$$

Let now $n, m \in \mathbb{N}_0$ with $n \leq m$ then we have

$$\sum_{i=n}^m \alpha \cdot x_i = \sum_{i=0}^{m-n} \alpha \cdot x_{n+i} \stackrel{[\text{eq: 11.15}]}{=} \alpha \cdot \sum_{i=0}^{m-n} x_{n+i} = \alpha \cdot \sum_{i=n}^m x_i$$

2. If I is finite and $\{x_i\}_{i \in I}$ with $\forall i \in I \ x_i = x$ then we have either:

$I = \emptyset$. Then

$$\sum_{i \in I} \alpha \cdot x_i = 0 = \alpha \cdot 0 = \alpha \cdot \sum_{i \in I} x_i$$

$I \neq \emptyset$. Then there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} \alpha \cdot x_i = \sum_{i=0}^{n-1} \alpha \cdot x_{\beta(i)} \stackrel{[\text{eq: 11.15}]}{=} \alpha \cdot \sum_{i=0}^{n-1} x_{\beta(i)} = \alpha \cdot \sum_{i \in I} x_i \quad \square$$

Theorem 11.81. If $\langle V, +, \cdot \rangle$ is a vector space over a field $\langle F, +, \cdot \rangle$ and $x \in V$ then we have:

1. If $n, m \in \mathbb{N}_0$ with $n \leq m$ $\{\alpha_i\}_{i \in \{n, \dots, m\}} \subseteq F$ then

$$\sum_{i=n}^m \alpha_i \cdot x = \left(\sum_{i=n}^m \alpha_i \right) \cdot x$$

2. If I is a finite set, $\{\alpha_i\}_{i \in I} \subseteq F$ then

$$\sum_{i \in I} \alpha_i \cdot x = \left(\sum_{i \in I} \alpha_i \right) \cdot x$$

Proof.

1. We will use induction in this proof, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq F \text{ we have } \sum_{i=0}^n (\alpha_i \cdot x) = \left(\sum_{i=0}^n \alpha_i \right) \cdot x \right\}$$

then we have:

$0 \in S$. If $\{\alpha_i\}_{i \in \{0\}}$ then we have $\sum_{i=0}^0 (\alpha_i \cdot x) = \alpha_0 \cdot x = \left(\sum_{i=0}^0 \alpha_i \right) \cdot x$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{\alpha_i\}_{i \in \{0, \dots, n+1\}} \subseteq F$ then we have

$$\begin{aligned} \sum_{i=0}^{n+1} (\alpha_i \cdot x) &= \left(\sum_{i=0}^n (\alpha_i \cdot x) \right) + \alpha_{n+1} \cdot x \\ &\stackrel{n \in S}{=} \left(\sum_{i=0}^n \alpha_i \right) \cdot x + \alpha_{n+1} \cdot x \\ &= \left(\left(\sum_{i=0}^n \alpha_i \right) + \alpha_{n+1} \right) \cdot x \\ &= \left(\sum_{i=0}^{n+1} \alpha_i \right) \cdot x \end{aligned}$$

proving that $n+1 \in S$.

By mathematical induction it follows then that

$$\forall n \in \mathbb{N}_0 \text{ and } \{\alpha_i\}_{i \in \{0, \dots, n\}} \text{ we have } \sum_{i=0}^n \alpha_i \cdot x = \left(\sum_{i=0}^n \alpha_i \right) \cdot x \quad (11.16)$$

Let now $n, m \in \mathbb{N}_0$ with $n \leq m$ then for $\{\alpha_i\}_{i \in \{n, \dots, m\}} \subseteq F$ then we have

$$\sum_{i=n}^m \alpha_i \cdot x = \sum_{i=0}^{m-n} \alpha_{n+i} \cdot x \stackrel{[eq: 11.16]}{=} \left(\sum_{i=0}^{m-n} \alpha_i \right) \cdot x = \left(\sum_{i=n}^m \alpha_i \right) \cdot x$$

2. I is a finite set and $\{\alpha_i\}_{i \in I} \subseteq F$ then we have either:

$I = \emptyset$. Then

$$\sum_{i \in I} \alpha_i \cdot x = 0 = 0 \cdot x = \left(\sum_{i \in I} \alpha_i \right) \cdot x$$

$I \neq \emptyset$. Then there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} \alpha_i \cdot x = \sum_{i=0}^{n-1} \alpha_{\beta(i)} \cdot x \stackrel{[eq: 11.16]}{=} \left(\sum_{i=0}^{n-1} \alpha_i \right) \cdot x = \left(\sum_{i \in I} \alpha_i \right) \cdot x$$

□

Theorem 11.82. If $\langle V, +, \cdot \rangle$ is a vector space over $\langle \mathbb{K}, +, \cdot \rangle$ where $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$ then we have:

1. If $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq V$ is such that $\forall i \in \{n, \dots, m\} x_i = x$ then

$$\sum_{i=n}^m x_i = (m - n + 1) \cdot x$$

2. If I is a finite set and $\{x_i\}_{i \in I} \subseteq V$ is such that $\forall i \in I x_i = x$ then

$$\sum_{i \in I} x_i = \text{card}(I) \cdot x$$

Proof.

1. We use induction for the proof, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{K} \text{ with } \forall i \in \{0, \dots, n\} x_i = x \text{ we have } \sum_{i=0}^n x_i = (n+1) \cdot x \right\}$$

then we have:

$0 \in S$. If $\{x_i\}_{i \in \{0\}} \subseteq V$ is such that $x_0 = x$ then $\sum_{i=0}^0 x_i = x_0 = x = 1 \cdot x = (0+1) \cdot x$ proving that $0 \in S$.

$\mathbf{n} \in \mathcal{S} \Rightarrow \mathbf{n} + 1 \in \mathcal{S}$. If $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq V$ is such that $\forall i \in \{0, \dots, n+1\} \ x_i = x$ then we have

$$\sum_{i=0}^{n+1} x_i = \left(\sum_{i=0}^n x_i \right) + x_{n+1} = \left(\sum_{i=0}^n x_i \right) + x \stackrel{n \in \mathcal{S}}{=} n \cdot x + x = (n+1) \cdot x$$

proving that $n+1 \in \mathcal{S}$.

By mathematical induction we have then that

$$\forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq V \text{ with } \forall i \in \{0, \dots, n\} \ x_i = x \text{ we have } \sum_{i=0}^n x_i = (n+1) \cdot x \quad (11.17)$$

If now $n, m \in \mathbb{N}_0$ with $n \leq m$ then

$$\sum_{i=n}^m x_i = \sum_{i=0}^{m-n} x_{n+i} \stackrel{[eq: 11.17]}{=} (m-n+1) \cdot x$$

2. For I is finite and $\{x_i\}_{i \in I}$ with $\forall i \in I \ x_i = x$ then we have either:

$I = \emptyset$. Then

$$\sum_{i \in I} x_i = 0 = 0 \cdot x = \text{card}(I) \cdot x$$

$I \neq \emptyset$. Then there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I \Rightarrow \{0, \dots, n-1\} \approx I$ such that

$$\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)} = (n-1+1) \cdot x = n \cdot x \stackrel{[\text{theorem: 10.90}]}{=} \text{card}(I) \cdot x \quad \square$$

Theorem 11.83. Let $\langle V, +, \cdot \rangle$ be a vector space over $\langle F, +, \cdot \rangle$ and $W \subseteq V$ a sub-space of $\langle V, +, \cdot \rangle$ then:

1. $\forall n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq W$ we have $\sum_{i=n}^m x_i \in W$
2. If I is a finite set and $\{x_i\}_{i \in I} \subseteq W$ then $\sum_{i \in I} x_i \in W$

Proof.

1. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq W \text{ we have } \sum_{i=0}^n x_i \in W \right\}$$

then we have:

$0 \in \mathcal{S}$. If $\{x_i\}_{i \in \{0\}} \subseteq W$ then $x_0 \in W$ so that $\sum_{i=0}^0 x_i = x_0 \in W$ proving that $0 \in W$.

$\mathbf{n} \in \mathcal{S} \Rightarrow \mathbf{n} + 1 \in \mathcal{S}$. Let $\{x_i\}_{i \in \{0, \dots, n+1\}}$ then we have

$$\begin{aligned} \sum_{i=0}^{n+1} x_i &= \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\ &= 1 \cdot \left(\sum_{i=0}^n x_i \right) + x_{n+1} \\ &\in W \text{ [as } n \in \mathcal{S} \Rightarrow \sum_{i=0}^n x_i \in S \wedge x_{n+1} \in S] \end{aligned}$$

proving that $n+1 \in \mathcal{S}$.

using induction we have then that

$$\forall n \in \mathbb{N}_0 \text{ and } \{x_i\}_{i \in \{0, \dots, n\}} \subseteq W \text{ then } \sum_{i=0}^n x_i \in W \quad (11.18)$$

Let now $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq W$ then we have

$$\sum_{i=n}^m x_i = \sum_{i=0}^{m-n} x_{n+i} \in_{[\text{eq: 11.18}]} W$$

2. Let I be finite then we have either:

$I = \emptyset$. By [theorem: 11.61] that $\langle W, +, \cdot \rangle$ is a vector space with $0 \in W$. Hence $\sum_{i \in I} x_i = 0 \in W$.

$I \neq \emptyset$. As $I \neq \emptyset$ there exists a $a \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that

$$\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)} \in W \quad [\text{see eq: 11.18}] \quad \square$$

11.3.2 Linear (in)dependency

11.3.2.1 Finite distinct set

In the previous section we have defined sums and products over finite sets by using the fact that for a finite set there exists a $n \in \mathbb{N}_0$ and a bijection between $\{1, \dots, n\}$ and the finite set. To further develop the theory of dependent, independent sets in a vector space we have to work with these finite sums/products and it will be usefully to introduce a notation that expresses this relation between the finite set, $\{1, \dots, n\}$ and the bijection.

Definition 11.84. (Finite distinct set) Let $n \in \mathbb{N}_0$ and $x: \{1, \dots, n\} \rightarrow A$ a bijection then we note this as $A = \{x_1, \dots, x_n\}$, further if $i \in \{1, \dots, n\}$ then x_i is a shorthand for $x(i)$. In other words if we write $A = \{x_1, \dots, x_n\}$ then we actually means that $n \in \mathbb{N}_0$ and there exist a bijection $x: \{1, \dots, n\} \rightarrow A$.

The notation $A = \{x_1, \dots, x_n\}$ is a way of specifying how the content of the set A must be calculated. In essence this is a extension of the notation $A = \{1.3.5\}$ which defines the set A by specifying that $x \in A \Leftrightarrow x = 1 \vee x = 2 \vee x = 3$. Using the above definition we have then the following theorem.

Theorem 11.85. If $A = \{x_1, \dots, x_n\}$ then we have:

1. $A = \{x(i) | i \in \{1, \dots, n\}\} = \{x_i | i \in \{1, \dots, n\}\}$
2. $\forall i \in \{1, \dots, n\} \ x_i = x(i) \in A$
3. $y \in A \Leftrightarrow \exists i \in \{1, \dots, n\}$ such that $x_i = x(i) = y$
4. $\forall i, j \in I$ with $x_i = x_j$ we have $i = j$
5. $\forall i, j \in I$ with $i \neq j$ we have $x_i \neq x_j$

Proof. As $A = \{x_1, \dots, x_n\}$ $n \in \mathbb{N}_0$ and $x: \{1, \dots, n\} \rightarrow A$ is a bijection. Hence we have:

1. $A = x(\{1, \dots, n\}) = \{x(i) | i \in \{1, \dots, n\}\} \stackrel{x_i \text{ notation}}{=} \{x_i | i \in \{1, \dots, n\}\}$.
2. $\forall i \in I$ we have $x_i = x(i) \in A$.
3. This follows from the fact that a bijection is surjective and (1)
4. This follows from the fact that a bijection is injective.
5. Let $i, j \in \{1, \dots, n\}$ with $i \neq j$. If $x_i = x_j \Rightarrow x(i) = x(j)$ we have by injectivity that $i = j$ contradicting $i \neq j$, so we must have $x_i \neq x_j$.

Remark 11.86. If in a context we have defined $A = \{x_1, \dots, x_n\}$ or equivalently the bijection $x: \{1, \dots, n\} \rightarrow A$ then we will identify the set A with $\{x_1, \dots, x_n\}$. We will do this also if we are not really interested in the exact details of either A or x . Rephrasing the above theorem using this remark we have for $n \in \mathbb{N}_0$ and $\{x_1, \dots, x_n\}$ that:

1. $\{x_1, \dots, x_n\} = \{x(i) | i \in \{1, \dots, n\}\} = \{x_i | i \in \{1, \dots, n\}\}$

2. $\forall i \in \{1, \dots, n\} \ x_i = x(i) \in A$
3. $y \in \{x_1, \dots, x_n\} \Leftrightarrow \exists i \in \{1, \dots, n\}$ such that $x_i = x(i) = y$
4. $\forall i, j \in I$ with $x_i = x_j$ we have $i = j$
5. $\forall i, j \in I$ with $i \neq j$ we have $x_i \neq x_j$

Definition 11.87. If $A = \{x_1, \dots, x_n\}$ and $m \in \{1, \dots, n\}$ then $B = \{x_1, \dots, x_m\}$ represents the bijection $x|_{\{1, \dots, m\}}: \{1, \dots, m\} \rightarrow x(\{1, \dots, m\}) \subseteq A$.

Remark 11.88. If we use the previous remark on the previous definition we have that $\forall n \in \mathbb{N}_0$ and $m \in \{1, \dots, n\}$ that

$$\{x_1, \dots, x_m\} \subseteq \{x_1, \dots, x_n\}$$

Actually every finite set can be written as a finite distinct set.

Theorem 11.89. We have

$$A \text{ is finite} \Leftrightarrow A = \{x_1, \dots, x_n\} \text{ where } n = \text{card}(A)$$

Note that the representation $A = \{x_1, \dots, x_n\}$ is not unique as there could be many bijections between $\{1, \dots, n\}$ and A □

Proof.

$\Rightarrow$. If A is finite then by [theorem: 10.90] there exists a bijection $x: \{1, \dots, n\} \rightarrow A$ where $n = \text{card}(A)$. Hence $A = \{x_1, \dots, x_n\}$ where $n = \text{card}(A)$.

$\Leftarrow$. As $A = \{x_1, \dots, x_n\}$ $x: \{1, \dots, n\} \rightarrow A$ is a bijection so that A is finite. □

Example 11.90. $A = \emptyset \Leftrightarrow A = \{x_1, \dots, x_0\}$

Proof.

$\Rightarrow$. Let $A = \emptyset$ then as $\{1, \dots, 0\} = \emptyset$ we have for $x = \emptyset$ by [example: 2.65] the empty bijection $x = \emptyset: \{1, \dots, 0\} \rightarrow \emptyset$ so that $A = \{x_1, \dots, x_0\}$.

$\Leftarrow$. If $A = \{x_1, \dots, x_0\}$ then we have the bijection $x: \{1, \dots, 0\} = \emptyset \rightarrow A$ so that $A = x(\emptyset) = \emptyset$. □

The following lemma will be used in the Steinitz Lemma later.

Lemma 11.91. $\forall n \in \mathbb{N}_0$ we have $\{x_1, \dots, x_n\} \subseteq \{x_1, \dots, x_{n+1}\} \setminus \{x_{n+1}\}$

Proof. Let $A = \{x_1, \dots, x_{n+1}\}$ and $B = \{x_1, \dots, x_n\}$ then there exists a bijection $x: \{1, \dots, n\} \rightarrow A$ and $x|_{\{1, \dots, m\}}: \{1, \dots, m\} \rightarrow x(\{1, \dots, m\}) = B$. If $y \in A \setminus \{x_{n+1}\}$ then $\exists i \in \{1, \dots, n+1\}$ such that $y = x_i$, we can not have $i = n+1$ because then $y = x_{n+1}$, hence $i \in \{1, \dots, n\}$ and $y \in x(\{1, \dots, n\}) = B$. Hence we have

$$A \setminus \{x_{n+1}\} \subseteq x(\{1, \dots, n\}) = B$$

On the other hand if $y \in x(\{1, \dots, n\}) = B$ there exists a $i \in \{1, \dots, n\}$ such that $y = x_i = x(i) \in A$. Assume that $y = x_{n+1} = x(n+1)$ then by injectivity we have from $x(i) = y = x(n+1)$ that $n+1$ contradicting $i \in \{1, \dots, n\}$ hence $y \neq x_{n+1}$ proving that $y \in A \setminus \{x_{n+1}\}$. So

$$x(\{1, \dots, x_n\}) \subseteq A \setminus \{x_{n+1}\} \quad \square$$

If $x: \{1, \dots, n\} \rightarrow A$ is a bijection and $\beta: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ another bijection then we can consider the bijection $x \circ \beta: \{1, \dots, n\} \rightarrow A$ so that we have the finite distinct set

$$A = \{(x \circ \beta)_1, \dots, (x \circ \beta)_n\}$$

and given $i \in \{1, \dots, n\}$ we have $(x \circ \beta)_i \stackrel{\text{def}}{=} (x \circ \beta)(i) = x(\beta(i)) \stackrel{\text{def}}{=} x_{\beta(i)}$. So the following definition seems logical

Definition 11.92. Let $A = \{x_1, \dots, x_n\}$ be a finite disjoint set and $\beta: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ a bijection then

$$A = \{x_{\beta(1)}, \dots, x_{\beta(n)}\}$$

is the finite distinct set defined by the bijection

$$x \circ \beta: \{1, \dots, n\} \rightarrow A.$$

Definition 11.93. (Ordered Family) A family of the form $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ where $n \in \mathbb{N}_0$ is called a **ordered family**.

The set of all members of a ordered family is always finite as is expressed in the following theorem.

Theorem 11.94. If $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$, $n \in \mathbb{N}_0$ is a ordered family then $\{x_i | i \in \{1, \dots, n\}\}$ is finite. with $\text{card}(\{x_i | i \in \{1, \dots, n\}\}) \leq n$.

Proof. As $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ is actually the function $x: \{1, \dots, n\} \rightarrow X$ we have that

$$x: \{1, \dots, n\} \rightarrow x(\{1, \dots, n\}) = \{x(i) | i \in I\} = \{x_i | i \in I\} \text{ is a surjection}$$

So by [theorem: 6.44] $\{x_i | i \in \{1, \dots, n\}\}$ is finite and $\text{card}(\{x_i | i \in \{1, \dots, n\}\}) \leq n$. $\square$

Theorem 11.95. Let X be a set, $n \in \mathbb{N}_0$, $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ and $Y \subseteq \{x_i | i \in \{1, \dots, n\}\}$ then there exist a bijection

$$\beta: \{1, \dots, \text{card}(Y)\} \rightarrow J \subseteq \{1, \dots, n\}$$

such that

$$Y = \{x_{\beta(1)}, \dots, x_{\beta(\text{card}(Y))}\}$$

Proof. Let $x: \{1, \dots, n\} \rightarrow X$ be the function defining the family $\{x_i\}_{i \in \{1, \dots, n\}}$. Using a consequence of the Axiom of Choice [see theorem: 3.137] there exists a $I \subseteq \{1, \dots, n\}$ such that

$$x|_I: I \rightarrow x(\{1, \dots, n\}) = \{x | i \in \{1, \dots, n\}\} \text{ is a bijection}$$

Let $J = (x|_I)^{-1}(Y) \subseteq I$ then for

$$x|_J \stackrel{=}{\underset{[\text{theorem: 2.92}]}{=}} (x|_I)|_J: J \rightarrow Y$$

we have:

injectivity. Let $i, j \in J$ such that $x|_J(i) = x|_J(j)$ then $x(i) = x(j)$ and as $J \subseteq I$ we have $x|_I(i) = x|_I(j)$. So by injectivity of $x|_I$ it follows that $i = j$.

surjectivity. Let $y \in Y$ then, as $x|_I: I \rightarrow \{x_i | i \in \{1, \dots, n\}\}$ is a bijection and $Y \subseteq \{x_i | i \in \{1, \dots, n\}\}$, there exists a $i \in I$ such that $y = x|_I(i) = x(i)$. As $y \in Y$ we have $i \in (x|_I)^{-1}(Y) = J$ hence $y = x(i) \stackrel{=}{\underset{i \in J}{=}} x|_J(i)$.

From the above we conclude that

$$x|_J: J \rightarrow Y \text{ is a bijection}$$

As Y is finite [using theorems: 10.91, 11.94] we have that J is finite with $\text{card}(J) = \text{card}(Y)$. Hence there exists a bijection

$$\beta: \{1, \dots, \text{card}(Y)\} \rightarrow J$$

As the composition of two bijections is again a bijection we have that

$$x|_J \circ \beta: \{1, \dots, \text{card}(Y)\} \rightarrow Y \text{ is a bijection}$$

so that $Y = \{(x|_J \circ \beta)_1, \dots, (x|_J \circ \beta)_{\text{card}(Y)}\}$. As $\forall i \in \{1, \dots, \text{card}(Y)\}$ we have $(x|_J \circ \beta)_i = x(\beta(i)) = x_{\beta(i)}$ we have

$$Y = \{x_{\beta(1)}, \dots, x_{\beta(\text{card}(Y))}\} \quad \square$$

11.3.2.2 Span of a set

Definition 11.96. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then $v \in V$ is a linear combination of W if there exists a finite set $I \subseteq W$ and a $\{\alpha_i\}_{i \in I} \subseteq F$ such that

$$v = \sum_{i \in I} \alpha_i \cdot i$$

For finite sets (that can always be representation as a finite distinct sets) we have a simpler definition of a linear combination in terms of the finite distinct representation of a finite set.

Theorem 11.97. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}_0$, $W = \{w_1, \dots, w_n\} \subseteq V$ a finite set and $v \in V$ then we have:

1. If $\exists \{\alpha_u\}_{u \in W} \subseteq F$ such that $v = \sum_{u \in W} \alpha_u \cdot w$ then there exists a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $v = \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i$ and $\{\alpha_u | u \in W\} = \{\beta_i | i \in \{1, \dots, n\}\}$.
2. If $\exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i$ then there exists a $\{\beta_w\}_{w \in W} \subseteq F$ such that $v = \sum_{u \in W} \beta_u \cdot u$ and $\{\alpha_i | i \in \{1, \dots, n\}\} = \{\beta_w | w \in W\}$.

Proof. First as $W = \{w_1, \dots, w_n\}$ we have the following two bijections

$$w: \{1, \dots, n\} \rightarrow W \text{ and } w^{-1}: W \rightarrow \{1, \dots, n\} \quad (11.19)$$

Now for the rest of the proof:

1. Define $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ by the function $\beta = \alpha \circ w: \{1, \dots, n\} \rightarrow F$ [so that $\forall i \in \{1, \dots, n\}$ $\beta_i = (\alpha \circ w)(i) = \alpha(w(i)) = \alpha_{w(i)}$] then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i &= \sum_{i \in \{1, \dots, n\}} \alpha_{w(i)} \cdot w(i) \\ &\stackrel{[\text{eq:11.19}] \wedge [\text{theorem: 11.36}]}{=} \sum_{u \in W} \alpha_u \cdot u \\ &= v \end{aligned}$$

Further we have:

$$\begin{aligned} \{\alpha_u | u \in W\} &= \alpha(W) \\ &\stackrel{w: \{1, \dots, n\} \rightarrow W \text{ is bijective}}{=} \alpha(w(\{1, \dots, n\})) \\ &= (\alpha \circ w)(\{1, \dots, n\}) \\ &= \beta(\{1, \dots, n\}) \\ &= \{\beta_i | i \in \{1, \dots, n\}\} \end{aligned}$$

2. Define $\{\beta_u\}_{u \in W} \subseteq F$ by the function $\beta = \alpha \circ w^{-1}: W \rightarrow F$ [so that $\forall i \in \{1, \dots, n\}$ $\beta_u = (\alpha \circ w^{-1})(u) = \alpha(w^{-1}(u)) = \alpha_{w^{-1}(u)}$] then we have

$$\begin{aligned} \sum_{u \in W} \beta_u \cdot u &= \sum_{u \in W} \alpha_{w^{-1}(u)} \cdot w(w^{-1}(u)) \\ &\stackrel{[\text{eq:11.19}] \wedge [\text{theorem: 11.36}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w(i) \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i \end{aligned}$$

Further we have:

$$\begin{aligned} \{\alpha_i | i \in \{1, \dots, n\}\} &= \alpha(\{1, \dots, n\}) \\ &\stackrel{w^{-1}: W \rightarrow \{1, \dots, n\} \text{ is bijective}}{=} \alpha((w^{-1})(W)) \\ &= (\alpha \circ w^{-1})(W) \\ &= \beta(W) \\ &= \{\beta_u | u \in W\} \end{aligned}$$

□

We have a more general version of the above theorem.

Theorem 11.98. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}_0$ and $\{w_i\}_{i \in \{1, \dots, n\}} \subseteq V$ then we have:*

1. *If $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ then there exist a $\{\beta_v\}_{v \in \{w_i | i \in \{1, \dots, n\}\}} \subseteq F$ such that*

$$\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = \sum_{v \in \{w_i | i \in \{1, \dots, n\}\}} \beta_v \cdot v$$

2. *If $\{\alpha_w\}_{w \in \{w_i | i \in \{1, \dots, n\}\}} \subseteq F$ then there exist a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that*

$$\sum_{v \in \{w_i | i \in \{1, \dots, n\}\}} \alpha_v \cdot v = \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i$$

Note 11.99. *Using [theorem: 11.94] $\{w_i | i \in \{1, \dots, n\}\}$ is finite so that the sums over $\{w_i | i \in \{1, \dots, n\}\}$ are well defined.*

Proof.

1. By [theorem: 11.94] we have that

$$\{w_i | i \in \{1, \dots, n\}\} \text{ is finite} \quad (11.20)$$

Let $w \in \{w_i | i \in \{1, \dots, n\}\}$ and define $I_w = \{i \in \{1, \dots, n\} | w_i = w\}$. If $j \in \{1, \dots, n\}$ we have $w_j \in \{w_i | i \in \{1, \dots, n\}\}$ so that $j \in I_{w_j}$ proving that $\{1, \dots, n\} \subseteq \bigcup_{w \in \{w_i | i \in \{1, \dots, n\}\}} I_w$. Further as $I_w \subseteq \{1, \dots, n\}$ it follows that $\bigcup_{w \in \{w_i | i \in \{1, \dots, n\}\}} I_w \subseteq \{1, \dots, n\}$. Hence we have

$$\{1, \dots, n\} = \bigcup_{w \in \{w_i | i \in \{1, \dots, n\}\}} I_w \quad (11.21)$$

Let $w, u \in \{w_i | i \in \{1, \dots, n\}\}$ with $w \neq u$, if $k \in I_w \cap I_u$ then $w = w_k = u$ contradicting $w \neq u$, hence we must have

$$\forall w, u \in \{w_i | i \in \{1, \dots, n\}\} \text{ with } w \neq u \text{ we have } I_w \cap I_u = \emptyset \quad (11.22)$$

Define now

$$\{\beta_w\}_{w \in \{w_i | i \in \{1, \dots, n\}\}} \subseteq F \text{ by } \beta_w = \sum_{i \in I_w} \alpha_i \quad (11.23)$$

Then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i & \stackrel{[\text{theorem: 11.44}] \text{ and } [\text{eqs: 11.20, 11.21, 11.22}]}{=} \sum_{w \in \{w_i | i \in \{1, \dots, n\}\}} \left(\sum_{i \in I_w} \alpha_i \cdot w_i \right) \\ & \stackrel{[i \in I_w \Rightarrow w_i = w]}{=} \sum_{w \in \{w_i | i \in \{1, \dots, n\}\}} \left(\sum_{i \in I_w} \alpha_i \cdot w \right) \\ & \stackrel{[\text{theorem: 11.81}]}{=} \sum_{w \in \{w_i | i \in \{1, \dots, n\}\}} \left(\left(\sum_{i \in I_w} \alpha_i \right) \cdot w \right) \\ & \stackrel{[\text{theorem: 11.23}]}{=} \sum_{w \in \{w_i | i \in \{1, \dots, n\}\}} \beta_w \cdot w \end{aligned}$$

2. Let $w: \{1, \dots, n\} \rightarrow V$ be the function defining $\{w_i\}_{i \in \{1, \dots, n\}}$ then by a consequence of the Axiom of Choice [see theorem: 3.137] there exists a $I \subseteq \{1, \dots, n\}$ such that

$$w|_I: I \rightarrow w(\{1, \dots, n\}) = \{w_i | i \in \{1, \dots, n\}\} \text{ is a bijection}$$

Define now

$$\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \beta_i = \begin{cases} \alpha_{w(i)} & \text{if } i \in I \\ 0 & \text{if } i \in \{1, \dots, n\} \setminus I \end{cases}$$

then we have

$$\begin{aligned}
 \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus I} \beta_i \cdot w_i + \sum_{i \in I} \beta_i \cdot w_i \\
 & = \sum_{i \in \{1, \dots, n\} \setminus I} 0 \cdot w_i + \sum_{i \in I} \alpha_{w(i)} \cdot w(i) \\
 & = \sum_{i \in I} \alpha_{w|_I(i)} \cdot w|_I(i) \\
 & \stackrel{[\text{theorem: 11.36}]}{=} \sum_{v \in \{w_i | i \in \{1, \dots, n\}\}} \alpha_w \cdot w \\
 & \quad \square
 \end{aligned}$$

The set of all linear combinations of a set is called the span of the set.

Definition 11.100. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$\text{span}(W) = \left\{ v \in V \mid \exists \{\alpha_w\}_{w \in I} \subseteq F, I \text{ finite and } I \subseteq W \text{ such that } v = \sum_{w \in I} \alpha_w \cdot w \right\}$$

We can reformulate this definition using ordered families of vectors.

Theorem 11.101. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$\text{span}(W) = \left\{ v \in V \mid \exists n \in \mathbb{N}_0, \{u_i\}_{i \in \{1, \dots, n\}} \subseteq W \text{ and } \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ with } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i \right\}$$

Proof. Let

$$S = \left\{ v \in V \mid \exists n \in \mathbb{N}_0, \{u_i\}_{i \in \{1, \dots, n\}} \subseteq W \text{ and } \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ with } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i \right\}$$

If $v \in \text{span}(W)$ then there exist a finite $I \subseteq W$ and $\{\beta_u\}_{u \in I} \subseteq F$ so that $v = \sum_{u \in I} \beta_u \cdot u$. As I is finite we have by [theorem: 11.89] that $I = \{u_1, \dots, u_n\} \subseteq W$ hence $\{u_i\}_{i \in \{1, \dots, n\}} \subseteq W$. Using [theorem: 11.97] there exists a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i$ proving that $v \in S$. Hence

$$\text{span}(W) \subseteq S \quad (11.24)$$

Let $v \in S$ then there exists a $n \in \mathbb{N}_0$, a $\{u_i\}_{i \in \{1, \dots, n\}} \subseteq W$ and $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i \quad (11.25)$$

Let

$$U = \{u_i \mid i \in \{1, \dots, n\}\} \subseteq W \quad (11.26)$$

then by [theorem: 11.94]

$$U \text{ is finite} \quad (11.27)$$

For $u \in U$ define $I_u = \{i \in \{1, \dots, n\} \mid u_i = u\} \subseteq \{1, \dots, n\}$ then we have

$$\bigcup_{u \in U} I_u \subseteq \{1, \dots, n\} \quad (11.28)$$

Further if $i \in \{1, \dots, n\}$ then $u_i \in U$ and as $u_i \in \{u_i \mid i \in \{1, \dots, n\}\} \stackrel{[\text{eq: 11.26}]}{=} U$ we have $i \in \bigcup_{u \in U} I_u$, so $\{1, \dots, n\} \subseteq \bigcup_{u \in U} I_u$. Combining this with [eq: 11.28] gives

$$\bigcup_{u \in U} I_u = \{1, \dots, n\} \quad (11.29)$$

Further if $u, v \in U$ with $u \neq v$ then if $i \in I_u \cap I_v$ we have that $u = u_i = v$ contradicting $u \neq v$ hence

$$\forall u, v \in U \text{ with } u \neq v \text{ we have } I_u \cap I_v = \emptyset \quad (11.30)$$

As $\forall u \in U \ I_u \subseteq \{1, \dots, n\}$ we have by [theorem: 10.91] that I_u is finite and we can define

$$\{\beta_u\}_{u \in U} \text{ by } \beta_u = \sum_{i \in I_u} \alpha_i \quad (11.31)$$

Then

$$\begin{aligned} v & \stackrel{[\text{eq: 11.25}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i \\ & \stackrel{[\text{theorem: 11.44}] \wedge [\text{eqs: 11.29, 11.30}]}{=} \sum_{u \in U} \left(\sum_{i \in I_u} \alpha_i \cdot u_i \right) \\ & \stackrel{i \in I_u \Rightarrow u_i = u}{=} \sum_{u \in U} \left(\sum_{i \in I_u} \alpha_i \cdot u \right) \\ & \stackrel{[\text{theorem: 11.81}]}{=} \sum_{u \in U} \left(\left(\sum_{i \in I_u} \alpha_i \right) \cdot u \right) \\ & \stackrel{[\text{eq: 11.31}]}{=} \sum_{u \in U} \beta_u \cdot u \end{aligned}$$

which as $U \subseteq W$ is finite [see eqs: 11.26, 11.27] proves that $v \in \text{span}(W)$. Hence $S \subseteq \text{span}(W)$. Combining this with [eq: 11.24] results in $S = \text{span}(W)$ or

$$\text{span}(W) = \left\{ v \in V \mid \exists n \in \mathbb{N}_0, \{u_i\}_{i \in \{1, \dots, n\}} \subseteq W \text{ and } \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ with } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i \right\} \quad \square$$

This definition for the span of a set has to use a finite $I \subseteq W$ so that the the sum is well defined as a finite sum (for infinite sums we have to introduce the concept of limits and topology). For finite sets we have a simpler version of the span.

Theorem 11.102. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $\{w_i\}_{i \in \{1, \dots, n\}} \subseteq V$ then

$$\text{span}(\{w_i \mid i \in \{1, \dots, n\}\}) = \left\{ v \in V \mid \exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i \right\}$$

Note 11.103. If $W \subseteq V$ is a finite set then by [theorem: 11.89] $W = \{w_1, \dots, w_n\}$ where $n = \text{card}(W)$ so that the above becomes

$$\text{span}(W) = \left\{ v \in V \mid \exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i \right\}$$

Proof. Let $x \in \text{span}(\{w_i \mid i \in \{1, \dots, n\}\})$ then by definition there exists a finite $I \subseteq \{w_i \mid i \in \{1, \dots, n\}\}$ and a $\{\alpha_v\}_{v \in I}$ such that

$$x = \sum_{v \in I} \alpha_v \cdot v \quad (11.32)$$

Define now

$$\{\beta_v\}_{v \in \{w_i \mid i \in \{1, \dots, n\}\}} \subseteq F \text{ by } \beta_v = \begin{cases} \alpha_v & \text{if } v \in I \\ 0 & \text{if } v \in \{w_i \mid i \in \{1, \dots, n\}\} \setminus I \end{cases}$$

then using [theorem: 11.98] there exists a $\{\gamma_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$\sum_{v \in \{w_i \mid i \in \{1, \dots, n\}\}} \beta_v \cdot v = \sum_{i \in \{1, \dots, n\}} \gamma_i \cdot w_i \quad (11.33)$$

Hence

$$\begin{aligned}
\sum_{i \in \{1, \dots, n\}} \gamma_i \cdot w_i &\stackrel{[\text{eq: 11.33}]}{=} \sum_{v \in \{w_i | i \in \{1, \dots, n\}\}} \beta_v \cdot v \\
&\stackrel{[\text{theorem: 11.43}]}{=} \sum_{v \in \{w_i | i \in \{1, \dots, n\}\} \setminus I} \beta_v \cdot v + \sum_{v \in I} \beta_v \cdot v \\
&= \sum_{v \in \{w_i | i \in \{1, \dots, n\}\} \setminus I} 0 \cdot v + \sum_{v \in I} \alpha_v \cdot v \\
&= \sum_{v \in I} \alpha_v \cdot v \\
&\stackrel{[\text{eq: 11.32}]}{=} x
\end{aligned}$$

proving that $x \in \{v \in V | \exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i\}$. Hence we have

$$\text{span}(\{w_i | i \in \{1, \dots, n\}\}) \subseteq \left\{ v \in V | \exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i \right\}$$

For the opposite inclusion, let $x \in \{v \in V | \exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i\}$ then there exist a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $x = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i$. Using [theorem: 11.98] there exists a $\{\beta_v\}_{v \in \{w_i | i \in \{1, \dots, n\}\}} \subseteq F$ such that $x = \sum_{v \in \{w_i | i \in \{1, \dots, n\}\}} \beta_v \cdot v$, which, as $\{w_i | i \in \{1, \dots, n\}\}$ is finite and a subset of itself, proves that $x \in \text{span}(\{w_i | i \in \{1, \dots, n\}\})$. Hence we have also

$$\left\{ v \in V | \exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i \right\} \subseteq \text{span}(\{w_i | i \in \{1, \dots, n\}\}) \quad \square$$

Example 11.104. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then $\text{span}(\emptyset) = \{0\}$

Proof. Let $v \in \text{span}(\emptyset)$ then there exists a finite $I \subseteq \emptyset \Rightarrow I = \emptyset$ and a $\{\alpha_w\}_{w \in \emptyset} \subseteq F$ such that

$$v = \sum_{w \in \emptyset} \alpha_w \cdot w \stackrel{[\text{definition: 11.32}]}{=} 0$$

proving that

$$\text{span}(\emptyset) = \{0\} \quad \square$$

Example 11.105. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then $\text{span}(\{0\}) = \{0\}$

Proof. Let $v \in \text{span}(\{0\})$ then there exists a finite $I \subseteq \{0\}$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that $v = \sum_{u \in I} \alpha_u \cdot u$. For I we have either:

$$I = \emptyset. \text{ Then } v = \sum_{u \in I} \alpha_u \cdot u = \sum_{u \in \emptyset} \alpha_u \cdot u \stackrel{[\text{definition: 11.32}]}{=} 0$$

$$I = \{0\}. \text{ Then } v = \sum_{u \in I} \alpha_u \cdot u = \sum_{u \in \{0\}} \alpha_u \cdot u \stackrel{[\text{theorem: 11.37}]}{=} \alpha_0 \cdot 0 = 0$$

proving that

$$\text{span}(\{0\}) = \{0\} \quad \square$$

The coefficients in a span of a set can be assumed to be different from the neutral element as is show in the next theorem.

Theorem 11.106. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$v \in \text{span}(W)$$

$$\Updownarrow$$

$$\text{There exists a finite } I \subseteq W \text{ and a } \{\alpha_w\}_{w \in I} \subseteq F \setminus \{0\} \text{ such that } v = \sum_{w \in I} \alpha_w \cdot w$$

Proof.

$\Rightarrow$. As $v \in \text{span}(W)$ there exist by definition a $J \subseteq W$ and a $\{\beta_w\}_{w \in J} \subseteq F$ such that

$$v = \sum_{w \in J} \alpha_w \cdot w$$

Let $I = \{w \in J \mid \alpha_w \neq 0\} \subseteq J$ and define $\{\alpha_w\}_{w \in I} \subseteq F \setminus \{0\}$ by $\alpha_w = \beta_w$ then we have

$$\begin{aligned} v &= \sum_{w \in J} \beta_w \cdot w \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{w \in J \setminus I} \beta_w \cdot w + \sum_{w \in I} \beta_w \cdot w \\ &= \sum_{w \in J \setminus I} 0 \cdot w + \sum_{w \in I} \beta_w \cdot w \\ &\stackrel{[\text{theorem: 11.37}]}{=} \sum_{w \in I} \beta_w \cdot w \\ &= \sum_{w \in I} \alpha_w \cdot w \end{aligned}$$

$\Leftarrow$. As $F \setminus \{0\} \subseteq F$ this follows from the definition of $\text{span}(W)$. $\square$

Theorem 11.107. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$v \in \text{span}(W)$$

$$\Updownarrow$$

$$\exists U = \{u_1, \dots, u_n\} \text{ with } U \subseteq W \text{ and } \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\} \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i$$

Proof.

$\Rightarrow$. As $v \in \text{span}(W)$ there exists by the previous theorem [theorem: 11.106] a finite $U \subseteq W$ and a $\{\beta_u\}_{u \in U} \subseteq F \setminus \{0\}$ such that $v = \sum_{u \in U} \beta_u \cdot u$. Using [theorem: 11.89] we have that

$$U = \{u_1, \dots, u_n\}$$

where $n = \text{card}(U)$. Applying then [theorem: 11.97] gives a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i$$

and

$$\{\beta_u \mid u \in U\} = \{\alpha_i \mid i \in \{1, \dots, n\}\}$$

As $\{\beta_u\}_{u \in U} \subseteq F \setminus \{0\}$ it follows that

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\}$$

$\Leftarrow$. As $U = \{u_1, \dots, u_n\}$, $U \subseteq W$, $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\} \subseteq F$ and $v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i$ it follows by the definition of the span of a set that $v \in \text{span}(W)$. $\square$

Theorem 11.108. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$W \subseteq \text{span}(W)$$

Proof. Let $w \in W$ then $\{w\}$ is finite and $\{w\} \subseteq W$. Define $\{\alpha_u\}_{u \in \{w\}}$ by $\alpha_w = 1$ then

$$\sum_{u \in \{w\}} \alpha_u \cdot u \stackrel{[\text{theorem: 11.34}]}{=} \alpha_w \cdot w = 1 \cdot w = w$$

proving that $w \in \text{span}(W)$. Hence $W \subseteq \text{span}(W)$. $\square$

Theorem 11.109. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $U, W \subseteq V$ with $U \subseteq W$ then

$$\text{span}(U) \subseteq \text{span}(W)$$

Proof. Let $v \in \text{span}(U)$ then there exists a finite $I \subseteq U$ and a $\{\alpha_i\}_{i \in I} \subseteq F$ such that $v = \sum_{w \in I} \alpha_w \cdot w$. As $U \subseteq W$ we have $I \subseteq W$ so that $v \in \text{span}(W)$. $\square$

Theorem 11.110. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$\text{span}(W) \text{ is a sub-space of } \langle V, +, \cdot \rangle$$

Proof. As $\emptyset \subseteq W$ we have $\{0\} \stackrel{[\text{example: 11.104}]}{=} \text{span}(\emptyset) \subseteq \stackrel{[\text{theorem: 11.109}]}{\subseteq} \text{span}(W)$ proving that

$$0 \in W \text{ hence } W \neq \emptyset$$

Let $x, y \in \text{span}(W)$ and $\alpha \in F$ then there exists finite $I, J \subseteq W$ and $\{\alpha_w\}_{w \in I} \subseteq F, \{\beta_w\}_{w \in J} \subseteq F$ such that

$$x = \sum_{w \in I} \alpha_w \cdot w \text{ and } y = \sum_{w \in J} \beta_w \cdot w$$

As I, J are finite we have by [theorem: 6.34] that $K = I \cup J = (I \setminus J) \cup (I \cap J) \cup (J \setminus I)$ is finite. Define

$$\{\gamma_w\}_{w \in K} \subseteq F \text{ by } \gamma_w = \begin{cases} \alpha \cdot \alpha_w & \text{if } w \in I \setminus J \\ \alpha \cdot \alpha_w + \beta_w & \text{if } w \in I \cap J \\ \beta_w & \text{if } w \in J \setminus I \end{cases}$$

then we have

$$\begin{aligned} & \sum_{w \in K} \gamma_w \cdot w \stackrel{[\text{theorem: 11.43}]}{=} \\ & \sum_{w \in I \setminus J} \gamma_w \cdot w + \sum_{w \in I \cap J} \gamma_w \cdot w + \sum_{w \in J \setminus I} \gamma_w \cdot w = \\ & \sum_{w \in I \setminus J} (\alpha \cdot \alpha_w) \cdot w + \sum_{w \in I \cap J} (\alpha \cdot \alpha_w + \beta_w) \cdot w + \sum_{w \in J \setminus I} \beta_w \cdot w = \\ & \sum_{w \in I \setminus J} \alpha \cdot (\alpha_w \cdot w) + \sum_{w \in I \cap J} (\alpha \cdot (\alpha_w \cdot w) + \beta_w \cdot w) + \sum_{w \in J \setminus I} \beta_w \cdot w \stackrel{[\text{theorem: 11.80}]}{=} \\ & \alpha \cdot \sum_{w \in I \setminus J} \alpha_w \cdot w + \sum_{w \in I \cap J} (\alpha \cdot (\alpha_w \cdot w) + \beta_w \cdot w) + \sum_{w \in J \setminus I} \beta_w \cdot w \stackrel{[\text{theorem: 11.38}]}{=} \\ & \alpha \cdot \sum_{w \in I \setminus J} \alpha_w \cdot w + \sum_{w \in I \cap J} \alpha \cdot (\alpha_w \cdot w) + \sum_{i \in I \cap J} \beta_w \cdot w + \sum_{w \in J \setminus I} \beta_w \cdot w \stackrel{[\text{theorem: 11.80}]}{=} \\ & \alpha \cdot \sum_{w \in I \setminus J} \alpha_w \cdot w + \alpha \cdot \sum_{w \in I \cap J} \alpha_w \cdot w + \sum_{i \in I \cap J} \beta_w \cdot w + \sum_{w \in J \setminus I} \beta_w \cdot w = \\ & \alpha \cdot \left(\sum_{w \in I \setminus J} \alpha_w \cdot w + \sum_{w \in I \cap J} \alpha_w \cdot w \right) + \left(\sum_{w \in J \setminus I} \beta_w \cdot w + \sum_{i \in I \cap J} \beta_w \cdot w \right) \stackrel{[\text{theorem: 11.43}]}{=} \\ & \alpha \cdot \sum_{w \in I} \alpha_w \cdot w + \sum_{w \in J} \beta_w \cdot w = \\ & \alpha \cdot x + y \end{aligned}$$

proving that

$$\alpha \cdot x + y \in \text{span}(W)$$

Hence by the definition of a sub-space [see definition: 11.59] it follows that $\text{span}(W)$ is a sub-space of $\langle V, +, \cdot \rangle$. $\square$

Theorem 11.111. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ a sub-space of V then

$$W = \text{span}(W)$$

Proof. Let $v \in \text{span}(W)$ then there exists a finite $I \subseteq W$ and $\{\alpha_w\}_{w \in I} \subseteq F$ such that

$$v = \sum_{w \in I} \alpha_w \cdot w$$

As W is a sub-space we have $\forall w \in I$ that $\alpha_w \cdot w \in W$, by [theorem: 11.83] it follows then that $v = \sum_{w \in I} \alpha_w \cdot w \in W$. Hence $\text{span}(W) \subseteq W$ which together with [theorem: 11.108] proves

$$W = \text{span}(W) \quad \square$$

Corollary 11.112. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then

$$\text{span}(W) = \text{span}(\text{span}(W))$$

Proof. By [theorem: 11.110] $\text{span}(W)$ is a sub-space of $\langle V, +, \cdot \rangle$, hence using [theorem: 11.111] we have

$$\text{span}(W) = \text{span}(\text{span}(W)) \quad \square$$

Theorem 11.113. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $W \subseteq V$ and $w \in \text{span}(W)$ then

$$\text{span}(W) = \text{span}(W \cup \{w\})$$

Proof. For $w \in \text{span}(W)$ we have either:

$w \in W$. Then $W = W \cup \{w\}$ so that $\text{span}(W) = \text{span}(W \cup \{w\})$

$w \notin W$. Then as $w \in \text{span}(W)$ there exists a finite $I \subseteq W$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that

$$w = \sum_{u \in I} \alpha_u \cdot u \quad (11.34)$$

Let $v \in \text{span}(W \cup \{w\})$ then there exists a finite $J \subseteq W \cup \{w\}$ and $\{\beta_u\}_{u \in J} \subseteq F$ such that

$$v = \sum_{u \in J} \beta_u \cdot u \quad (11.35)$$

For J we have either:

$w \notin J$. Then $J \subseteq W$ so that $v \in \text{span}(W)$

$w \in J$. Take the finite set $K = (I \cup J) \setminus \{w\} \stackrel{[\text{theorem: 1.30}]}{=} (I \setminus \{w\}) \cup (J \setminus \{w\})$ then, as

$I \subseteq W, J \subseteq W \cup \{w\} \Rightarrow I \setminus \{w\}, J \setminus \{w\} \subseteq W$, we have $K \subseteq W$. Further

$$\begin{aligned} K &= (I \cup J) \setminus \{w\} \\ &= ((I \setminus J) \cup (I \cap J) \cup (J \setminus I)) \setminus \{w\} \\ &= ((I \setminus J) \setminus \{w\}) \cup ((I \cap J) \setminus \{w\}) \cup ((J \setminus I) \setminus \{w\}) \\ &\stackrel{w \notin W \Rightarrow w \notin I \Rightarrow w \in I \cap J}{=} (I \setminus J) \cup (I \cap J) \cup ((J \setminus I) \setminus \{w\}) \end{aligned} \quad (11.36)$$

and we have

$$(I \setminus J) \cap (I \cap J) = \emptyset \wedge (I \setminus J) \cap ((J \setminus I) \setminus \{w\}) = \emptyset \wedge (I \cap J) \cap ((J \setminus I) \setminus \{w\}) = \emptyset \quad (11.37)$$

Define now

$$\{\gamma_u\}_{u \in K} \subseteq F \text{ by } \gamma_u = \begin{cases} \beta_u & \text{if } u \in ((J \setminus I) \setminus \{w\}) \\ \beta_w \cdot \alpha_u + \beta_u & \text{if } u \in I \cap J \\ \beta_w \cdot \alpha_u & \text{if } u \in I \setminus J \end{cases}$$

then we have

$$\begin{aligned}
& \sum_{u \in K} \gamma_u \cdot u \stackrel{[\text{theorem: 11.43}]}{=} \\
& \sum_{u \in I \setminus J} \gamma_u \cdot u + \sum_{u \in I \cap J} \gamma_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \gamma_u \cdot u = \\
& \sum_{u \in I \setminus J} (\beta_w \cdot \alpha_u) \cdot u + \sum_{u \in I \cap J} (\beta_w \cdot \alpha_u + \beta_u) \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u = \\
& \sum_{u \in I \setminus J} \beta_w \cdot (\alpha_u \cdot u) + \sum_{u \in I \cap J} (\beta_w \cdot (\alpha_u \cdot u) + \beta_u \cdot u) + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.80}]}{=} \\
& \beta_w \cdot \sum_{u \in I \setminus J} \alpha_u \cdot u + \sum_{u \in I \cap J} (\beta_w \cdot (\alpha_u \cdot u) + \beta_u \cdot u) + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.38}]}{=} \\
& \beta_w \cdot \sum_{u \in I \setminus J} \alpha_u \cdot u + \sum_{u \in I \cap J} \beta_w \cdot (\alpha_u \cdot u) + \sum_{i \in I \cap J} \beta_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.80}]}{=} \\
& \beta_w \cdot \sum_{u \in I \setminus J} \alpha_u \cdot u + \beta_w \cdot \sum_{u \in I \cap J} \alpha_u \cdot u + \sum_{i \in I \cap J} \beta_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u = \\
& \beta_w \cdot \left(\sum_{u \in I \setminus J} \alpha_u \cdot u + \sum_{u \in I \cap J} \alpha_u \cdot u \right) + \sum_{i \in I \cap J} \beta_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.43}]}{=} \\
& \beta_w \cdot \sum_{u \in I} \alpha_u \cdot u + \sum_{i \in I \cap J} \beta_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{eq: 11.34}]}{=} \\
& \beta_w \cdot w + \sum_{i \in I \cap J} \beta_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{w \notin I \cap J}{=} \\
& \beta_w \cdot w + \sum_{i \in (I \cap J) \setminus \{w\}} \beta_u \cdot u + \sum_{u \in (J \setminus I) \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.43}]}{=} \\
& \beta_w \cdot w + \sum_{i \in J \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.34}]}{=} \\
& \sum_{i \in \{w\}} \beta_u \cdot u + \sum_{i \in J \setminus \{w\}} \beta_u \cdot u \stackrel{[\text{theorem: 11.43}]}{=} \\
& \sum_{i \in J} \beta_u \cdot u \stackrel{[\text{eq: 11.35}]}{=} v
\end{aligned}$$

proving, as K is finite and $K \subseteq W$, that $v \in \text{span}(W)$.

So in all cases we have $v \in \text{span}(W)$ proving that $\text{span}(W \cup \{w\}) \subseteq \text{span}(W)$. As further $W \subseteq W \cup \{w\}$ we have by [theorem: 11.109] that $\text{span}(W) \subseteq \text{span}(W \cup \{w\})$ proving that $\text{span}(W) = \text{span}(W \cup \{w\})$.

So in all cases we have

$$\text{span}(W) = \text{span}(W \cup \{w\})$$

completing the proof. $\square$

11.3.2.3 Linear (in)dependent sets

Definition 11.114. (Linear Dependency) Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then W is **linear dependent** if there exists a finite $I \subseteq W$ and a $\{\alpha_w\}_{w \in I} \subseteq F$ satisfying $\exists u \in I$ with $\alpha_u \neq 0$ such that $\sum_{u \in I} \alpha_u \cdot u = 0$.

Example 11.115. Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ such that $0 \in W$ then W is linear dependent.

Proof. Take $I = \{0\}$ and $\{\alpha_u\}_{u \in \{0\}}$ by $\alpha_0 = 1$ then we have $\sum_{u \in \{0\}} \alpha_u \cdot u \stackrel{[\text{theorem: 11.34}]}{=} \alpha_0 \cdot 0 = 0$ proving that W is linear dependent. $\square$

Linear dependent sets can also be described as sets where one of the vector can be written as a linear combination of some other vectors, this is proved in the next theorem.

Theorem 11.116. Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then
 W is linear dependent

$\Updownarrow$

$\exists w \in W$ so that there exists a finite $I \subseteq W \setminus \{w\}$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that $w = \sum_{u \in I} \alpha_u \cdot u$

Proof.

$\Rightarrow$. As W is a linear dependent there exists a finite $J \subseteq W$ and a $\{\beta_u\}_{u \in J} \subseteq F$ satisfying $\exists w \in J$ with $\beta_w \neq 0$ such that $\sum_{u \in J} \beta_u \cdot u = 0$. So we have

$$\begin{aligned} 0 &= \sum_{u \in J} \beta_u \cdot u \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in J \setminus \{w\}} \beta_u \cdot u + \sum_{u \in \{w\}} \beta_u \cdot u \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in J \setminus \{w\}} \beta_u \cdot u + \beta_w \cdot w \end{aligned}$$

proving as $\beta_w \neq 0$ that

$$\begin{aligned} w &= (\beta_w)^{-1} \cdot \left(- \sum_{u \in J \setminus \{w\}} \beta_u \cdot u \right) \\ &\stackrel{[\text{theorem: 11.40}]}{=} (\beta_w)^{-1} \cdot \sum_{u \in J \setminus \{w\}} -(\beta_u \cdot u) \\ &= (\beta_w)^{-1} \cdot \sum_{u \in J \setminus \{w\}} (-\beta_u) \cdot u \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{u \in J \setminus \{w\}} ((\beta_w)^{-1}(-\beta_u) \cdot u) \\ &= \sum_{u \in J \setminus \{w\}} ((\beta_w)^{-1}(-\beta_u)) \cdot u \end{aligned}$$

So if we define $I = J \setminus \{w\}$ and $\{\alpha_u\}_{u \in I}$ by $\beta_u = (\beta_w)^{-1}(-\beta_u)$ then $I \subseteq W \setminus \{w\}$ and

$$\sum_{u \in I} \alpha_u \cdot u = w$$

$\Leftarrow$. By the hypothesis there exists a $w \in W$, a finite $I \subseteq W \setminus \{w\}$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that

$$w = \sum_{u \in I} \alpha_u \cdot u$$

Let $J = I \cup \{w\}$ then J is finite and $J \subseteq W$ and define

$$\{\beta_u\}_{u \in J} \text{ by } \beta_u = \begin{cases} -1 & \text{if } u = w \\ \alpha_u & \text{if } u \in I \end{cases}$$

then

$$\begin{aligned} \sum_{u \in J} \beta_u \cdot u &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in I} \beta_u \cdot u + \sum_{u \in \{w\}} \beta_u \cdot u \\ &= \sum_{u \in I} \alpha_u \cdot u + \sum_{u \in \{w\}} \beta_u \cdot u \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in I} \alpha_u \cdot u + \beta_w \cdot w \\ &= \sum_{u \in I} \alpha_u \cdot u + (-1) \cdot w \\ &= w + (-w) \\ &= 0 \end{aligned}$$

which as $\beta_w = -1 \neq 0$ proves that W is linear dependent. □

For finite sets we have a easier condition for linear dependency.

Theorem 11.117. *Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $W = \{w_1, \dots, w_n\} \subseteq V$ a finite set then we have the following equivalences:*

1. W is linear dependent
2. $\exists \{\alpha_u\}_{u \in W} \subseteq F$ satisfying $\exists u \in W$ with $\alpha_u \neq 0$ such that $\sum_{w \in W} \alpha_w \cdot w = 0$
3. $\exists u \in W$ and a $\{\alpha_w\}_{w \in W \setminus \{u\}} \subseteq F$ such that $u = \sum_{w \in W \setminus \{u\}} \alpha_w \cdot w$
4. $\exists \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists i \in \{1, \dots, n\}$ with $\alpha_i \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = 0$.
5. $\exists k \in \{1, \dots, n\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, n\} \setminus \{k\}} \subseteq F$ such that $w_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot w_i$
6. $\exists k \in \{1, \dots, n\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, k-1\}} \subseteq F$ such that $w_k = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot w_i$

Proof.

1 $\Rightarrow$ 2. As W is linear dependent there exist by [definition: 11.114] a finite set $I \subseteq W$ and a $\{\beta_u\}_{u \in I} \subseteq F$ satisfying $\exists u_0 \in I$ with $\beta_{u_0} \neq 0$ such that $\sum_{u \in I} \beta_u \cdot u = 0$. Define for the finite set W

$$\{\alpha_u\}_{u \in W} \subseteq F \text{ by } \alpha_w = \begin{cases} 0 & \text{if } w \in W \setminus I \\ \beta_w & \text{if } w \in I \end{cases}$$

then we have for $u_0 \in I \subseteq W$ that $\alpha_{u_0} = \beta_{u_0} \neq 0$ and

$$\begin{aligned} \sum_{u \in W} \alpha_u \cdot u &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in W \setminus I} \alpha_u \cdot u + \sum_{u \in I} \alpha_u \cdot u \\ &= \sum_{u \in W \setminus I} 0 \cdot u + \sum_{u \in I} \beta_u \cdot u \\ &\stackrel{[\text{theorem: 11.37}]}{=} 0 + \sum_{u \in I} \beta_u \cdot u \\ &= 0 \end{aligned}$$

2 $\Rightarrow$ 1. As W is finite and $W \subseteq W$ this follows from the definition of linear dependency [see definition: 11.114]

2 $\Rightarrow$ 3. By the hypothesis there exist a $\{\beta_w\}_{w \in W} \subseteq F$ satisfying $\exists u \in W$ with $\beta_u \neq 0$ such that $\sum_{w \in W} \beta_w \cdot w = 0$. Hence

$$\begin{aligned} 0 &= \sum_{w \in W} \beta_w \cdot w \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{w \in W \setminus \{u\}} \beta_w \cdot w + \sum_{w \in \{u\}} \beta_w \cdot w \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{w \in W \setminus \{u\}} \beta_w \cdot w + \beta_u \cdot u \end{aligned}$$

which, as $\beta_u \neq 0$, gives that

$$\begin{aligned} u &= (\beta_u)^{-1} \cdot \left(- \sum_{w \in W \setminus \{u\}} \beta_w \cdot w \right) \\ &\stackrel{[\text{theorem: 11.40}]}{=} (\beta_u)^{-1} \cdot \left(\sum_{w \in W \setminus \{u\}} -(\beta_w \cdot w) \right) \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{w \in W \setminus \{u\}} (\beta_u)^{-1} \cdot (-\beta_w) \cdot w \end{aligned}$$

So if we define $\{\alpha_w\}_{w \in W \setminus \{u\}} \subseteq F$ by $\alpha_w = (\beta_u)^{-1} \cdot (-\beta_w)$ then

$$u = \sum_{w \in W \setminus \{u\}} \alpha_w \cdot w$$

proving (3).

3 $\Rightarrow$ 2. By the hypothesis $\exists u \in W$ and a $\{\beta_w\}_{w \in W \setminus \{u\}} \subseteq F$ such that $u = \sum_{w \in W \setminus \{u\}} \beta_w \cdot w$.

Define $\{\alpha_w\}_{w \in W} \subseteq F$ by $\alpha_w = \begin{cases} -1 & \text{if } w = u \\ \beta_w & \text{if } w \in W \setminus \{u\} \end{cases}$ then

$$\begin{aligned} \sum_{w \in W} \alpha_w \cdot w &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{w \in W \setminus \{u\}} \alpha_w \cdot w + \sum_{w \in \{u\}} \alpha_w \cdot w \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{w \in W \setminus \{u\}} \alpha_w \cdot w + \alpha_u \cdot u \\ &= u + (-1) \cdot u \\ &= 0 \end{aligned}$$

which as $\alpha_u = -1 \neq 0$ proves (2).

2 $\Rightarrow$ 4. By the hypothesis $\exists \{\beta_u\}_{u \in W} \subseteq F$ satisfying $\exists u_0 \in W$ with $\beta_{u_0} \neq 0$ such that $\sum_{u \in W} \beta_u \cdot u = 0$. Using [theorem: 11.97] there exist a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = \sum_{u \in W} \beta_u \cdot u = 0$$

and

$$\{\beta_u | u \in W\} = \{\alpha_i | i \in \{1, \dots, n\}\}$$

As $\beta_{u_0} \in \{\beta_u | u \in W\} = \{\alpha_i | i \in \{1, \dots, n\}\}$ there exists a $i_0 \in \{1, \dots, n\}$ such that $\alpha_{i_0} = \beta_{u_0} \neq 0$. Hence we found a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists i_0 \in \{1, \dots, n\}$ with $\alpha_{i_0} \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = 0$.

4 $\Rightarrow$ 2. By the hypothesis there exist a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists i_0 \in \{1, \dots, n\}$ with $\beta_{i_0} \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i = 0$. Using [theorem: 11.97] there exist a $\{\alpha_u\}_{u \in W} \subseteq F$ such that

$$\sum_{u \in W} \alpha_u \cdot u = \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i = 0$$

and

$$\{\beta_i | i \in \{1, \dots, n\}\} = \{\alpha_u | u \in W\}.$$

As $\beta_{i_0} \in \{\beta_i | i \in \{1, \dots, n\}\} = \{\alpha_u | u \in W\}$ there exist a $u_0 \in W$ such that $\alpha_{u_0} = \beta_{i_0} \neq 0$. Hence we found a $\{\alpha_u\}_{u \in W} \subseteq F$ satisfying $\exists u_0 \in W$ with $\alpha_{u_0} \neq 0$ such that $\sum_{u \in W} \alpha_u \cdot u = 0$.

4 $\Rightarrow$ 5. By the hypothesis there exist a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists k \in \{1, \dots, n\}$ with $\alpha_k \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = 0$. Then we have

$$\begin{aligned} 0 &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot w_i + \sum_{i \in \{k\}} \alpha_i \cdot w_i \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot w_i + \alpha_k \cdot w_k \end{aligned}$$

which as $\alpha_k \neq 0$ gives

$$\begin{aligned} w_k &= (\alpha_k)^{-1} \cdot \left(- \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot w_i \right) \\ &\stackrel{[\text{theorem: 11.40}]}{=} (\alpha_k)^{-1} \cdot \sum_{i \in \{1, \dots, n\} \setminus \{k\}} -(\alpha_i \cdot w_i) \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} (\alpha_k)^{-1} \cdot (-\alpha_i \cdot w_i) \\ &= \sum_{i \in \{1, \dots, n\} \setminus \{k\}} ((\alpha_k)^{-1} \cdot (-\alpha_i)) \cdot w_i \end{aligned}$$

Hence if $\{\beta_i\}_{i \in \{1, \dots, n\} \setminus \{k\}}$ is defined by $\beta_i = (\alpha_k)^{-1} \cdot (-\alpha_i)$ we have

$$w_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot w_i$$

5 $\Rightarrow$ 4. By the hypothesis there $\exists k \in \{1, \dots, n\}$ and a $\{\beta_i\}_{i \in \{1, \dots, n\} \setminus \{k\}} \subseteq F$ such that

$$w_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot w_i$$

Define then

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \alpha_i = \begin{cases} -1 & \text{if } i = k \\ \beta_i & \text{if } i \in \{1, \dots, n\} \setminus \{k\} \end{cases}$$

then $\alpha_k = -1 \neq 0$ and

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot w_i + \sum_{i \in \{k\}} \alpha_i \cdot w_i \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot w_i + \alpha_k \cdot w_k \\ &= w_k + (-1) \cdot w_k \\ &= 0 \end{aligned}$$

4 $\Rightarrow$ 6. By the hypothesis there exist a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists j \in \{1, \dots, n\}$ with $\beta_j \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i = 0$. Hence the set $A = \{j \in \{1, \dots, n\} \mid \alpha_j \neq 0\}$ is non empty and finite so $k = \max(A)$ exist. So $\forall i \in \{k+1, \dots, n\}$ we have $\alpha_i = 0$ and we have

$$\begin{aligned} 0 &= \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i \\ &= \sum_{i \in \{1, \dots, m-1\}} \beta_i \cdot w_i + \sum_{i \in \{k\}} \alpha_i \cdot w_i + \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot w_i \\ &= \sum_{i \in \{1, \dots, m-1\}} \beta_i \cdot w_i + \beta_k \cdot w_k + 0 \end{aligned}$$

which as $\beta_k \neq 0$ proves that

$$w_k = \sum_{i \in \{1, \dots, k-1\}} \frac{-\beta_i}{\beta_k} \cdot w_i = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot w_i$$

where

$$\{\alpha_i\}_{i \in \{1, \dots, m-1\}} \subseteq F \text{ is defined by } \alpha_i = \frac{-\beta_i}{\beta_m}$$

6 $\Rightarrow$ 4. By the hypothesis there exist a $k \in \{1, \dots, n\}$ and a $\{\beta_i\}_{i \in \{1, \dots, k-1\}} \subseteq F$ such that $w_k = \sum_{i \in \{1, \dots, k-1\}} \beta_i \cdot w_i$. Define then

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \alpha_i = \begin{cases} 0 & \text{if } i \in \{k+1, \dots, n\} \\ -1 & \text{if } i = k \\ \beta_i & \text{if } i \in \{1, \dots, k-1\} \end{cases}$$

then $\alpha_k \neq 0$ and

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i &= \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot w_i + \sum_{i \in \{k\}} \alpha_i \cdot w_i + \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot w_i \\ &= \sum_{i \in \{1, \dots, k-1\}} \beta_i \cdot w_i + \alpha_k \cdot w_k + 0 \\ &= \sum_{i \in \{1, \dots, k-1\}} \beta_i \cdot w_i - w_k \\ &= 0 \end{aligned}$$

□

The opposite of a linear dependent set is a linear independent set.

Definition 11.118. (Linear Independence) Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then W is **linear independent** if W is **not linear dependent**.

We have the following equivalent definition of a linear independent set.

Theorem 11.119. *Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ then*

W is linear independent

$\Updownarrow$

If $\{\alpha_u\}_{u \in I} \subseteq F$, I a finite subset of W satisfies $\sum_{u \in I} \alpha_u \cdot u = 0$ then $\forall u \in I$ we have $\alpha_u = 0$

Proof.

$\Rightarrow$. As W is independent we have by definition that W is not dependent. Let $\{\alpha_u\}_{u \in I} \subseteq W$, I a finite subset of W that satisfies $\sum_{u \in I} \alpha_u \cdot u = 0$. Assume that $\exists u \in W$ with $\alpha_u \neq 0$ then by definition W is linear dependent and we have a contradiction, so we must have that $\forall u \in I$ $\alpha_u = 0$.

$\Leftarrow$. Assume that W is linear dependent, then there exists a finite $I \subseteq W$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ satisfying $\exists u \in I$ with $\alpha_u \neq 0$ such that $\sum_{u \in I} \alpha_u \cdot u = 0$ which contradict the hypothesis which says that $\forall w \in I$ $\alpha_w = 0$. $\square$

For finite sets we can simplify the conditions for linear independency.

Theorem 11.120. *Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W = \{w_1, \dots, w_n\} \subseteq V$ a **finite** set then we have the following equivalences:*

1. *W is linear independent*
2. *$\forall \{\alpha_u\}_{u \in W} \subseteq F$ such that $\sum_{u \in W} \alpha_u \cdot u = 0$ we have $\forall u \in W$ that $\alpha_u = 0$*
3. *$\forall \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = 0$ we have $\forall i \in \{1, \dots, n\}$ $\alpha_i = 0$.*

Proof.

1 $\Rightarrow$ 2. As W is finite and $W \subseteq W$ this follows from [theorem: 11.119].

2 $\Rightarrow$ 1. Let $I \subseteq W$ be a finite set and $\{\alpha_u\}_{u \in I} \subseteq F$ satisfying $\sum_{u \in I} \alpha_u \cdot u = 0$. Define

$$\{\beta_u\}_{u \in W} \subseteq F \text{ by } \beta_u = \begin{cases} 0 & \text{if } u \in W \setminus I \\ \alpha_u & \text{if } u \in I \end{cases}$$

then we have as W is finite

$$\begin{aligned} \sum_{u \in W} \beta_u \cdot u &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in W \setminus I} \beta_u \cdot u + \sum_{u \in I} \beta_u \cdot u \\ &= \sum_{u \in W \setminus I} 0 \cdot u + \sum_{u \in I} \alpha_u \cdot u \\ &\stackrel{[\text{theorem: 11.37}]}{=} \sum_{u \in I} \alpha_u \cdot u \\ &= 0 \end{aligned}$$

By the hypothesis we have that $\forall u \in W$ $\beta_u = 0$, so if $u \in I \subseteq W$ we have $\alpha_u = \beta_u = 0$. Hence we have that W is linear independent.

2 $\Rightarrow$ 3. Let $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = 0$. Using [theorem: 11.97] there exist a $\{\beta_u\}_{u \in W} \subseteq F$ such that

$$\sum_{u \in W} \beta_u \cdot u = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i = 0$$

and $\{\alpha_i | i \in \{1, \dots, n\}\} = \{\beta_u | u \in W\}$. By the hypothesis we have $\forall u \in W$ that $\beta_u = 0$. If $i \in \{1, \dots, n\}$ then $\alpha_i \in \{\beta_u | u \in W\}$ so that $\exists u \in W$ such that $\alpha_i = \beta_u = 0$. Hence $\forall i \in \{1, \dots, n\}$ we have that $\alpha_i = 0$.

3 $\Rightarrow$ 2. Let $\{\alpha_u\}_{u \in W} \subseteq F$ such that $\sum_{u \in W} \alpha_u \cdot u = 0$. Using [theorem: 11.97] there exist a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$\sum_{i \in \{1, \dots, n\}} \beta_i = \sum_{u \in W} \alpha_u \cdot u = 0$$

and $\{\alpha_u | u \in W\} = \{\beta_i | i \in \{1, \dots, n\}\}$. By the hypothesis we have that $\forall i \in \{1, \dots, n\} \beta_i = 0$. If $u \in W$ then $\alpha_u \in \{\beta_i | i \in \{1, \dots, n\}\}$ so there exists a $i \in \{1, \dots, n\}$ such that $\alpha_u = \beta_i = 0$. Hence $\forall u \in W$ we have $\alpha_u = 0$. $\square$

Example 11.121. Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ then $\emptyset$ is linear independent.

Proof. Let $I \subseteq \emptyset$, I a finite set then $I = \emptyset$ so that if $\{\alpha_u\}_{u \in I} \subseteq F$ with $\sum_{u \in \emptyset} \alpha_u \cdot u = 0$ we have that $\forall u \in \emptyset \alpha_u = 0$ is satisfied vacuously. $\square$

Example 11.122. Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $v \in V$ such that $v \neq 0$ then $\{v\}$ is a linear independent set.

Proof. Let $\{\alpha_u\}_{u \in \{v\}} \subseteq F$ be such that $\sum_{u \in \{v\}} \alpha_u \cdot u = 0$ then by [theorem: 11.34] $\alpha_v \cdot v = 0$. Assume that $\alpha_v \neq 0$ then $v = (\alpha_v)^{-1} \cdot (\alpha_v \cdot v) = 0$ contradicting $v \neq 0$. Hence $\alpha_v = 0$ or $\forall u \in \{v\} \alpha_u = 0$ proving by [theorem: 11.120] that the finite set $\{v\}$ is linear dependent. $\square$

Theorem 11.123. Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $U, W \subseteq V$ with $U \subseteq W$ then we have

1. If W is linear independent then U is linear independent, in other words every subset of a linear independent set is linear independent.
2. If U is linear dependent then W is linear dependent, in other words every super set of a linear dependent set is linear dependent.

Proof.

1. Let $I \subseteq U$ be a finite set then as $U \subseteq V$ we have $I \subseteq V$. So if $\{\alpha_u\}_{u \in I} \subseteq F$ is such that $\sum_{u \in I} \alpha_u \cdot u = 0$ we have as V is linear independent that $\forall u \in I$ we have $\alpha_u = 0$. Hence U is linear independent.
2. As U is linear dependent there exists a finite $I \subseteq U$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ not all zeroes such that $\sum_{u \in I} \alpha_u \cdot u = 0$. As $U \subseteq V$ we have $I \subseteq V$ proving that V is linear dependent. $\square$

The following lemma shows, how given a finite linear independent set $\{e_1, \dots, e_n\}$ and a set that spans the vector space we can create a new spanning set by replacing n vectors in the spanning set by the vectors in $\{e_1, \dots, e_n\}$. This lemma will be crucial in introducing the concept of the dimension of a vector space.

Lemma 11.124. (Steinitz Lemma) Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}_0$, $E = \{e_1, \dots, e_n\} \subseteq V$ a linear independent finite set and $T \subseteq V$ such that $\text{span}(T) = V$ then there exist a $T_n \subseteq T$ with $\text{card}(T_n) = n$ such that

$$\text{span}((T \setminus T_n) \cup \{e_1, \dots, e_n\}) = V$$

Proof. We prove this by induction so let

$$S = \{k \in \mathbb{N}_0 | \text{If } 0 \leq k \leq n \text{ then } \exists T'_k \subseteq T \text{ with } \text{card}(T'_k) = k \text{ such that } \text{span}((T \setminus T'_k) \cup \{e_1, \dots, e_k\}) = V\}$$

then we have:

0 $\in S$. Take $T'_0 = \emptyset$ then $\text{card}(T'_0) = 0$ and

$$V = \text{span}(T) = \text{span}((T \setminus \emptyset) \cup \emptyset) \stackrel{[\text{example: 11.90}]}{=} \text{span}((T \setminus T'_0) \cup \{e_1, \dots, e_0\})$$

proving that $0 \in S$.

$k \in S \Rightarrow k+1 \in S$. If $0 \leq k+1 \leq n$ then $k < n$ and, as $k \in S \subseteq \mathbb{N}_0$, $0 \leq k < n$. So using the fact that $k \in S$ there exist a $T'_k \subseteq T$ with $\text{card}(T'_k) = k$ such that $\text{span}((T \setminus T'_k) \cup \{e_1, \dots, e_k\}) = V$. Define

$$T_k = (T \setminus T'_k) \cup \{e_1, \dots, e_k\} \quad (11.38)$$

then

$$\text{span}(T_k) = V \quad (11.39)$$

As $\{e_1, \dots, e_{k+1}\} \subseteq \{e_1, \dots, e_n\}$ a linear independent set it follows from [theorem: 11.123] that

$$\{e_1, \dots, e_{k+1}\} \text{ is linear independent} \quad (11.40)$$

As $e_{k+1} \in V = \text{span}(T_k)$ we have by [theorem: 11.106] that

$$\text{there exists a finite } I \subseteq T_k \text{ and } \{\lambda_u\}_{u \in I} \text{ such that } e_{k+1} = \sum_{u \in I} \lambda_u \cdot u \quad (11.41)$$

Assume that $I \subseteq \{e_1, \dots, e_k\}$ $\stackrel{[\text{theorem: 11.91}]}{=} \{e_1, \dots, e_{k+1}\} \setminus \{e_{k+1}\}$ then we have by the above and [theorem: 11.116] that $\{e_1, \dots, e_{k+1}\}$ is linear dependent contradicting [eq: 11.40]. Hence $I \not\subseteq \{e_1, \dots, e_{k+1}\}$ so that

$$\exists t \in I \text{ with } t \notin \{e_1, \dots, e_{k+1}\} \quad (11.42)$$

As $t \in I \subseteq T_k \stackrel{[\text{eq: 11.38}]}{=} (T \setminus T'_k) \cup \{e_1, \dots, e_k\} \subseteq (T \setminus T'_k) \cup \{e_1, \dots, e_{k+1}\}$ we must have by the above that $t \in T \setminus T'_k$ hence $t \notin T'_k$. Using [theorem: 10.93] it follows then that $\text{card}(T'_k \cup \{t\}) = \text{card}(T'_k) + 1 = k+1$. To summarize we have

$$t \in T \text{ and } \text{card}(T'_{k+1}) = k+1 \text{ where } T'_{k+1} = T'_k \cup \{t\} \quad (11.43)$$

Further

$$\begin{aligned} T_k \setminus \{t\} &\stackrel{[\text{eq: 11.38}]}{=} ((T \setminus T'_k) \cup \{e_1, \dots, e_k\}) \setminus \{t\} \\ &= ((T \setminus T'_k) \setminus \{t\}) \cup (\{e_1, \dots, e_k\} \setminus \{t\}) \\ &\stackrel{[\text{eq: 11.42}]}{=} ((T \setminus T'_k) \setminus \{t\}) \cup \{e_1, \dots, e_k\} \\ &\stackrel{[\text{theorem: 1.34}]}{=} (T \setminus (T'_k \cup \{t\})) \cup \{e_1, \dots, e_k\} \end{aligned} \quad (11.44)$$

Let

$$T_{k+1} = (T \setminus T'_{k+1}) \cup \{e_1, \dots, e_{k+1}\} \quad (11.45)$$

Then we have

$$\begin{aligned} T_{k+1} &= (T \setminus T'_{k+1}) \cup \{e_1, \dots, e_{k+1}\} \\ &= (T \setminus T'_{k+1}) \cup (\{e_1, \dots, e_k\} \cup \{e_{k+1}\}) \\ &= ((T \setminus T'_{k+1}) \cup \{e_1, \dots, e_k\}) \cup \{e_{k+1}\} \\ &\stackrel{[\text{eq: 11.43}]}{=} ((T \setminus (T'_k \cup \{t\})) \cup \{e_1, \dots, e_k\}) \cup \{e_{k+1}\} \\ &\stackrel{[\text{eq: 11.44}]}{=} (T_k \setminus \{t\}) \cup \{e_{k+1}\} \end{aligned}$$

proving that

$$T_{k+1} = (T_k \setminus \{t\}) \cup \{e_{k+1}\} \quad (11.46)$$

Further

$$\begin{aligned} e_{k+1} &\stackrel{[\text{eq: 11.41}]}{=} \sum_{u \in I} \lambda_u \cdot u \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in I \setminus \{t\}} \lambda_u \cdot u + \sum_{u \in \{t\}} \lambda_u \cdot u \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in I \setminus \{t\}} \lambda_u \cdot u + \lambda_t \cdot t \end{aligned} \quad (11.47)$$

As $(I \setminus \{t\}) \cup \{e_{k+1}\} \subseteq_{[\text{eq: 11.41}]} (T_k \setminus \{t\}) \cup \{e_{k+1}\} \stackrel{[\text{eq: 11.46}]}{=} T_{k+1}$ we have, taking in account that I is finite that

$$J \subseteq T_{k+1} \text{ and } J \text{ is finite where } J = (I \setminus \{t\}) \cup \{e_{k+1}\} \quad (11.48)$$

For e_{k+1} we have either:

$e_{k+1} \in I \setminus \{t\}$. Then by [eq: 11.47]

$$\begin{aligned} e_{k+1} &= \sum_{u \in I \setminus \{t\}} \lambda_u \cdot u + \lambda_t \cdot t \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in I \setminus \{t, e_{k+1}\}} \lambda_u \cdot u + \sum_{u \in \{e_{k+1}\}} \lambda_u \cdot u + \lambda_t \cdot t \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in I \setminus \{t, e_{k+1}\}} \lambda_u \cdot u + \lambda_{e_{k+1}} \cdot e_{k+1} + \lambda_t \cdot t \end{aligned}$$

as $\{\lambda_u\}_{u \in I} \subseteq F \setminus \{0\}$ we have $\lambda_t \neq 0$, so

$$t = ((\lambda_t)^{-1} \cdot (1 - \lambda_{k+1})) \cdot e_{k+1} - (\lambda_t)^{-1} \cdot \sum_{u \in I \setminus \{t, e_{k+1}\}} \lambda_u \cdot u \quad (11.49)$$

Then for $J \stackrel{[\text{eq: 11.48}]}{=} (I \setminus \{t\}) \cup e_{k+1} = (I \setminus \{t, e_{k+1}\}) \cup \{e_{k+1}\}$ we can define

$$\{\alpha_u\}_{u \in J} \subseteq F \text{ by } \alpha_u = \begin{cases} (\lambda_t)^{-1} \cdot (1 - \lambda_{k+1}) & \text{if } u = e_{k+1} \\ -(\lambda_t)^{-1} \cdot \lambda_u & \text{if } u \in I \setminus \{t, e_{k+1}\} = J \setminus \{e_{k+1}\} \end{cases}$$

giving:

$$\begin{aligned} \sum_{u \in J} \alpha_u \cdot u &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in J \setminus \{e_{k+1}\}} \alpha_u \cdot u + \sum_{u \in \{e_{k+1}\}} \alpha_u \cdot u \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in J \setminus \{e_{k+1}\}} \alpha_u \cdot u + \alpha_{e_{k+1}} \cdot e_{k+1} \\ &= \sum_{u \in J \setminus \{e_{k+1}\}} ((-\lambda_t)^{-1} \cdot \lambda_u) \cdot u + (\lambda_t)^{-1} \cdot (1 - \lambda_{k+1}) \cdot e_{k+1} \\ &\stackrel{[\text{theorem: 11.80}]}{=} (-\lambda_t)^{-1} \cdot \sum_{u \in J \setminus \{e_{k+1}\}} \lambda_u \cdot u + (\lambda_t)^{-1} \cdot (1 - \lambda_{k+1}) \cdot e_{k+1} \\ &\stackrel{[\text{eq: 11.49}]}{=} t \end{aligned}$$

which proves, as by [eq: 11.48] $J \subseteq T_{k+1}$, that

$$t \in \text{span}(T_{k+1})$$

$e_{k+1} \notin I \setminus \{t\}$. Using [eq: 11.47] we have, as $\{\lambda_u\}_{u \in I} \subseteq F \setminus \{0\}$ implies $\lambda_t \neq 0$, that

$$t = (\lambda_t)^{-1} \cdot e_{k+1} - (\lambda_t)^{-1} \cdot \sum_{u \in I \setminus \{t\}} \lambda_u \cdot u \quad (11.50)$$

Further from $e_{k+1} \notin I \setminus \{t\}$ it follows that

$$J \setminus \{e_{k+1}\} \stackrel{[\text{eq: 11.48}]}{=} ((I \setminus \{t\}) \cup \{e_{k+1}\}) \setminus \{e_{k+1}\} = (I \setminus \{t\}) \setminus \{e_{k+1}\} = I \setminus \{t\}$$

giving

$$J \setminus \{e_{k+1}\} = I \setminus \{t\} \quad (11.51)$$

As $e_{k+1} \in (I \setminus \{t\}) \cup \{e_{k+1}\} \stackrel{[\text{eq: 11.48}]}{=} J$ we can define

$$\{\alpha_u\}_{u \in J} \subseteq F \text{ by } \alpha_u = \begin{cases} (\lambda_t)^{-1} & \text{if } u = e_{k+1} \\ -(\lambda_t)^{-1} \cdot \lambda_u & \text{if } u \in J \setminus \{e_{k+1}\} \end{cases}$$

then we have:

$$\begin{aligned}
\sum_{u \in J} \alpha_u \cdot u &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in J \setminus \{e_{k+1}\}} \alpha_u \cdot u + \sum_{u \in \{e_{k+1}\}} \alpha_u \cdot u \\
&\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in J \setminus \{e_{k+1}\}} \alpha_u \cdot u + \alpha_{e_{k+1}} \cdot e_{k+1} \\
&= \sum_{u \in J \setminus \{e_{k+1}\}} ((-\lambda_t)^{-1} \cdot \lambda_u) \cdot u + (\lambda_t)^{-1} \cdot e_{k+1} \\
&\stackrel{[\text{theorem: 11.80}]}{=} (-\lambda_t)^{-1} \cdot \sum_{u \in J \setminus \{e_{k+1}\}} \lambda_u \cdot u + (\lambda_t)^{-1} \cdot e_{k+1} \\
&\stackrel{[\text{eq: 11.51}]}{=} (-\lambda_t)^{-1} \cdot \sum_{u \in I \setminus \{t\}} \lambda_u \cdot u + (\lambda_t)^{-1} \cdot e_{k+1} \\
&\stackrel{[\text{eq: 11.50}]}{=} t
\end{aligned}$$

proving, as by [eq: 11.48] $J \subseteq T_{k+1}$, that

$$t \in \text{span}(T_{k+1})$$

So in all cases we have $t \in \text{span}(T_{k+1})$ which using [theorem: 11.113] proves that

$$\text{span}(T_{k+1} \cup \{t\}) = \text{span}(T_{k+1}) \quad (11.52)$$

As

$$\begin{aligned}
T_{k+1} \cup \{t\} &\stackrel{[\text{eq: 11.46}]}{=} (T_k \setminus \{t\}) \cup \{e_{k+1}\} \cup \{t\} \\
&= ((T_k \setminus \{t\}) \cup \{t\}) \cup \{e_{k+1}\} \cup \{t\} \\
&= T_k \cup \{e_{k+1}\} \\
&\supseteq T_k
\end{aligned}$$

we have by [theorem: 11.109]

$$V \stackrel{[\text{eq: 11.39}]}{=} \text{span}(T_k) \leq \text{span}(T_{k+1} \cup \{t\}) \stackrel{[\text{eq: 11.52}]}{=} \text{span}(T_{k+1}) \subseteq V$$

proving that

$$\text{span}((T \setminus T'_{k+1}) \cup \{e_1, \dots, e_{k+1}\}) \stackrel{[\text{eq: 11.45}]}{=} \text{span}(T_{k+1}) = V$$

Using [eq: 11.43] we have $\text{card}(T'_{k+1}) = k+1$, $t \in T$ and $T'_{k+1} = T'_k \cup \{t\} \subseteq T$ [as $t \in T \wedge T'_k \subseteq T$].

Hence by the definition of S proves that $k+1 \in S$ finishing the induction step.

Mathematical induction proves then that $S = \mathbb{N}_0$. So as $n \in \{1, \dots, n\} \subseteq \mathbb{N}_0 = S$ we have that there exists a $T_n \subseteq T$ with $\text{card}(T_n) = n$ such that $\text{span}((T \setminus T_n) \cup \{e_1, \dots, e_n\}) = V$. $\square$

Corollary 11.125. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$. Suppose that $E = \{e_1, \dots, e_n\} \subseteq V$ is a finite linear independent set and there exist a finite $T \subseteq V$ such that $\text{span}(T) = V$ then $n \leq \text{card}(T)$.*

Proof. As T is finite we have by [theorem: 11.89] that $T = \{t_1, \dots, t_m\}$ where $m = \text{card}(T)$. Assume that $m < n$ then $\{e_1, \dots, e_m\} \subseteq \{e_1, \dots, e_n\}$ a linear independent set. Hence we have by [theorem: 11.123] that

$$\{e_1, \dots, e_m\} \text{ is linear independent}$$

Using the Steinitz lemma [lemma: 11.124] there exists a $T_m \subseteq T$ with $\text{card}(T_m) = m$ such that

$$\text{span}((T \setminus T_m) \cup \{e_1, \dots, e_m\}) = V \quad (11.53)$$

As $T_m \subseteq T$ and $\text{card}(T_m) = m = \text{card}(T)$ we have by [theorem: 10.95] that $T_m = T$ or $T \setminus T_m = \emptyset$ so that $(T \setminus T_m) \cup \{e_1, \dots, e_m\} = \{e_1, \dots, e_m\}$ hence by [eq: 11.53]

$$\text{span}(\{e_1, \dots, e_m\}) = V$$

As $e_{m+1} \in V$ there exists by the above a finite

$$I \subseteq \{e_1, \dots, e_m\} \stackrel{[\text{theorem: 11.91}]}{=} \{e_1, \dots, e_{m+1}\} \setminus \{e_{m+1}\}$$

and a $\{\alpha_u\}_{u \in I}$ such that

$$e_{m+1} = \sum_{u \in I} \alpha_u \cdot u$$

Using [theorem:11.116] it follows then that $\{e_1, \dots, e_{m+1}\}$ is linear dependent. By [theorem: 11.123] and the fact that $\{e_1, \dots, e_{m+1}\} \subseteq \{e_1, \dots, e_n\}$ [because $m+1 \leq n$] we reach the conclusion that $\{e_1, \dots, e_n\}$ is linear dependent which contradicts the hypothesis that $\{e_1, \dots, e_n\}$ is linear independent. So the assumption $m < n$ leads to a contradiction, hence $n \leq m = \text{card}(T)$. $\square$

Corollary 11.126. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$. Suppose that $E \subseteq V$ is a linear independent set and $T \subseteq F$ is a finite set such that $\text{span}(T) = V$ then E is finite.*

Proof. Assume that E is infinite then by [theorem: 6.29] there exists a denumerable set $B \subseteq E$. So there exists a bijection

$$\beta: \mathbb{N}_0 \rightarrow B \subseteq E$$

As $\{1, \dots, \text{card}(T) + 1\} \subseteq \mathbb{N}_0$ we have the bijection

$$e = \beta_{\{1, \dots, \text{card}(T) + 1\}}: \{1, \dots, \text{card}(T) + 1\} \rightarrow \beta(\{1, \dots, \text{card}(T) + 1\}) \subseteq B \subseteq E$$

defining

$$E' = \{e_1, \dots, e_{\text{card}(T) + 1}\} \subseteq E$$

As E is linear independent we have by [theorem: 11.123] that E' is linear independent. Using then the previous corollary [corollary: 11.125] we have that $\text{card}(T) + 1 \leq \text{card}(T)$ leading to the contradiction $\text{card}(T) \neq \text{card}(T)$. Hence E must be finite. $\square$

11.3.2.4 Linear (in)dependent families

Up to now we have talked about linear (in)dependency of sets. However if want to work later with skew-symmetric mappings (needed for the theory of determinant functions) we need the concept of linear (in)dependency of ordered families.

Definition 11.127. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ then a family $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is **linear dependent** if $\exists \{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists i \in \{1, \dots, n\}$ with $\alpha_i \neq 0$ such that*

$$\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i = 0$$

Just as for linear dependent sets there exist a equivalent definition for linear dependent families.

Theorem 11.128. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ then*

$$\begin{aligned} & \{x_i\}_{i \in \{1, \dots, n\}} \text{ is linear dependent} \\ & \Updownarrow \\ & \exists k \in \{1, \dots, n\} \text{ and } \exists \{\alpha_i\}_{i \in \{1, \dots, n\} \setminus \{k\}} \subseteq F \text{ such that } x_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i \end{aligned}$$

Proof.

$\Rightarrow$. As $\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent there exist a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists k \in \{1, \dots, n\}$ such that $\beta_k \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \beta_i \cdot x_i = 0$. Hence

$$\begin{aligned} 0 &= \sum_{i \in \{1, \dots, n\}} \beta_i \cdot x_i \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot x_i + \sum_{i \in \{k\}} \beta_i \cdot x_i \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot x_i + \beta_k \cdot x_k \end{aligned}$$

so that

$$\begin{aligned}
 x_k &= -(\beta_k)^{-1} \cdot \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot x_i \\
 &= \sum_{i \in \{1, \dots, n\} \setminus \{k\}} ((\beta_k)^{-1} \cdot (-\beta_i)) \cdot x_i \\
 &= \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i
 \end{aligned}$$

where

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ is defined by } \alpha_i = ((\beta_k)^{-1} \cdot (-\beta_i))$$

$\Leftarrow$. Assume that $x_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot x_i$ where $\{\beta_i\}_{i \in \{1, \dots, n\} \setminus \{k\}} \subseteq F$. Define

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \alpha_i = \begin{cases} -1 & \text{if } i = k \\ \beta_i & \text{if } i \in \{1, \dots, n\} \setminus \{k\} \end{cases}$$

then we have

$$\begin{aligned}
 \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i + \sum_{i \in \{k\}} \alpha_i \cdot x_i \\
 &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i + \alpha_k \cdot x_k \\
 &= \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \beta_i \cdot x_i + (-1) \cdot x_k \\
 &= x_k = x_k \\
 &= 0
 \end{aligned}$$

proving, as $\alpha_k = -1 \neq 0$, linear dependency. $\square$

Theorem 11.129. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ a ordered family then we have

$\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent

$\Updownarrow$

There exist a $k \in \{1, \dots, n\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, k-1\}} \subseteq F$ such that $x_k = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i$

Proof.

$\Rightarrow$. As $\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent there exist a $\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists j \in \{1, \dots, n\}$ with $\beta_j \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i = 0$. Hence the finite set $A = \{j \in \{1, \dots, n\} | \alpha_j \neq 0\}$ is non empty so that $k = \max(A)$ exist. Then we must have $\forall i \in \{k+1, \dots, n\}$ that $\alpha_i = 0$. So

$$\begin{aligned}
 0 &= \sum_{i \in \{1, \dots, n\}} \beta_i \cdot w_i \\
 &= \sum_{i \in \{1, \dots, m-1\}} \beta_i \cdot w_i + \sum_{i \in \{k\}} \alpha_i \cdot w_i + \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot w_i \\
 &= \sum_{i \in \{1, \dots, m-1\}} \beta_i \cdot w_i + \beta_k \cdot w_k + 0
 \end{aligned}$$

which as $\beta_k \neq 0$ proves that

$$w_k = \sum_{i \in \{1, \dots, k-1\}} \frac{-\beta_i}{\beta_k} \cdot w_i = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot w_i$$

where

$$\{\alpha_i\}_{i \in \{1, \dots, m-1\}} \subseteq F \text{ is defined by } \alpha_i = \frac{-\beta_i}{\beta_m}$$

$\Leftarrow$. By the hypothesis there exist a $k \in \{1, \dots, n\}$ and a $\{\beta_i\}_{i \in \{1, \dots, k-1\}} \subseteq F$ such that $w_k = \sum_{i \in \{1, \dots, k-1\}} \beta_i \cdot w_i$. Define then

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \alpha_i = \begin{cases} 0 & \text{if } i \in \{k+1, \dots, n\} \\ -1 & \text{if } i = k \\ \beta_i & \text{if } i \in \{1, \dots, k-1\} \end{cases}$$

then $\alpha_k \neq 0$ and

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i &= \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot w_i + \sum_{i \in \{k\}} \alpha_i \cdot w_i + \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot w_i \\ &= \sum_{i \in \{1, \dots, k-1\}} \beta_i \cdot w_i + \alpha_k \cdot w_k + 0 \\ &= \sum_{i \in \{1, \dots, k-1\}} \beta_i \cdot w_i - w_k \\ &= 0 \end{aligned}$$

□

Theorem 11.130. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{x_k\}_{k \in \{1, \dots, n\}} \subseteq X$ a ordered family then:

1. If there exists $i, j \in \{1, \dots, n\}$ with $i \neq j$ and $x_i = x_j$ then $\{x_k\}_{k \in \{1, \dots, n\}}$ is linear dependent.
2. If there exists a $i \in \{1, \dots, n\}$ such that $x_i = 0$ then $\{x_k\}_{k \in \{1, \dots, n\}}$ is linear dependent.

Proof.

1. Define

$$\{\alpha_k\}_{k \in \{1, \dots, n\}} \subseteq F \text{ by } \alpha_k = \begin{cases} 1 & \text{if } k = i \\ -1 & \text{if } k = j \\ 0 & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases}$$

then we have

$$\begin{aligned} \sum_{k \in \{1, \dots, n\}} \alpha_k \cdot x_k &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} \alpha_k \cdot x_k + \sum_{k \in \{i\}} \alpha_k \cdot x_k + \sum_{k \in \{j\}} \alpha_k \cdot x_k \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} \alpha_k \cdot x_k + \alpha_i \cdot x_i + \alpha_j \cdot x_j \\ &= \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} 0 \cdot x_k + 1 \cdot x_i + (\alpha_j - 1) \cdot x_j \\ &= x_i - x_j \\ &\stackrel{x_i = x_j}{=} 0 \end{aligned}$$

which as $\alpha_i = 1 \neq 0$ proves linear dependency.

2. Define

$$\{a_k\}_{k \in \{1, \dots, n\}} \subseteq F \text{ by } a_k = \begin{cases} 1 & \text{if } k = i \\ 0 & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

then we have

$$\begin{aligned} \sum_{k \in \{1, \dots, n\}} a_k \cdot x_k &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i\}} a_k \cdot x_k + \sum_{k \in \{i\}} a_k \cdot x_k \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i\}} a_k \cdot x_k + \alpha_i \cdot x_i \\ &= \sum_{k \in \{1, \dots, n\} \setminus \{i\}} 0 \cdot x_k + 1 \cdot 0 \\ &\stackrel{[\text{theorem: 11.37}]}{=} 0 \end{aligned}$$

which as $\alpha_i = 1 \neq 0$ proves linear dependency.

□

The following theorem shows the relation between linear dependency of a family $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ and the linear dependency of its associated set $\{x_i | i \in \{1, \dots, n\}\}$.

Theorem 11.131. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ a ordered family such that $\{x_i | i \in \{1, \dots, n\}\}$ is linear dependent then $\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent.*

Proof. Let $W = \{x_i | i \in \{1, \dots, n\}\}$ then by [theorem: 10.96] W is finite and

$$\text{card}(W) \leq \text{card}(\{1, \dots, n\}) = n.$$

As $\{x_i | i \in \{1, \dots, n\}\}$ is linear dependent there exist a $I \subseteq W$ and $\{\alpha_i\}_{i \in I} \subseteq F$, satisfying $\exists l \in I$ with $\alpha_l \neq 0$, such that

$$\sum_{i \in I} \alpha_i \cdot i = 0$$

By definition of a family, $\{x_i\}_{i \in \{1, \dots, n\}}$ is defined by a function $x: \{1, \dots, n\} \rightarrow x(\{1, \dots, n\}) = W$. Using a consequence of the Axiom of Choice [see theorem: 3.137] there exists a $J \subseteq \{1, \dots, n\}$ so that

$$x|_J: J \rightarrow W \text{ is a bijection}$$

Take $K = (x|_J)^{-1}(I) \subseteq J \subseteq \{1, \dots, n\}$ then we have the bijection

$$x|_K \stackrel{[\text{theorem: 2.92}]}{=} (x|_J)|_K: K \rightarrow x|_K(K) = x(K) = I$$

Define now

$$\{\beta_k\}_{k \in \{1, \dots, n\}} \text{ by } \beta_k = \begin{cases} \alpha_{(x|_K)(i)} & \text{if } k \in K \\ 0 & \text{if } k \in \{1, \dots, n\} \setminus K \end{cases}$$

then

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \beta_i \cdot x_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus K} \beta_i \cdot x_i + \sum_{i \in K} \beta_i \cdot x_i \\ &= \sum_{i \in \{1, \dots, n\} \setminus K} 0 \cdot x_i + \sum_{i \in K} \alpha_{|x|_K(i)} \cdot x|_K(i) \\ &= \sum_{i \in K} \alpha_{|x|_K(i)} \cdot x|_K(i) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in I} \alpha_i \cdot i \\ &= 0 \end{aligned}$$

which as $\beta_{(x|_K)^{-1}(l)} = \alpha_{x|_K((x|_K)^{-1}(l))} = \alpha_l \neq 0$ proves that

$$\{x_i\}_{i \in \{1, \dots, n\}} \text{ is linear dependent} \quad \square$$

The opposite of the above theorem is not true as the following example shows.

Example 11.132. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $v \in V$ with $v \neq 0$ then $\{x_i\}_{i \in \{1, 2\}} \subseteq X$ defined by $x_i = v$ is linear dependent, however $\{x_i | i \in \{1, 2\}\}$ linear independent.

Proof. Define $\{\alpha_i\}_{i \in \{1, 2\}}$ by $\alpha_1 = 1$ and $\alpha_2 = -1$ then $\alpha_1 \neq 0$ and $\sum_{i \in \{1, 2\}} \alpha_i \cdot x_i = (-1) \cdot x + 1 \cdot x = 0$ proving that linear dependency. Further $\{x_i | i \in \{1, 2\}\} = \{v\}$ is by [example: 11.122] linear independent. $\square$

Definition 11.133. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ then $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent if $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is **not** linear dependent.

Just as with linear independent set we have a equivalent definition of linear dependency.

Theorem 11.134. Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and

$\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ then

$\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent

$\Updownarrow$

$\forall \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i = 0$ we have $\forall i \in \{1, \dots, n\} \alpha_i = 0$

Proof.

$\Rightarrow$. As $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent we have by definition that $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ is not dependent. Hence for every $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists i \in I$ with $\alpha_i \neq 0$ we must have $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i \neq 0$. So if $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i = 0$ we must have $\forall i \in \{1, \dots, n\}$ that $\alpha_i = 0$ otherwise we reach the contradiction that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i \neq 0$.

$\Leftarrow$. Assume that $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear dependent then there exists a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ satisfying $\exists k \in \{1, \dots, n\}$ with $\alpha_k \neq 0$ such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i = 0$. From the hypothesis it follows then that $\forall i \in \{1, \dots, n\} \alpha_i = 0$ contradicting $\alpha_k \neq 0$. So we must have that $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent. $\square$

The following theorem shows the relation between linear independency of a family $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ and the linear independency of its associated set $\{x_i | i \in \{1, \dots, n\}\}$.

Theorem 11.135. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ a linear independent family then

1. $\{x_i | i \in \{1, \dots, n\}\}$ is a linear independent set
2. $\{x_i | i \in \{1, \dots, n\}\} = \{x_1, \dots, x_n\}$ [or by definition: 11.84] that $x: \{1, \dots, n\} \rightarrow \{x_i | i \in \{1, \dots, n\}\}$ is a bijection].

Proof.

1. We prove this by contradiction, so assume that $\{x_i | i \in \{1, \dots, n\}\}$ is linear dependent then by [theorem: 11.131] $\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent, contradicting the linear independency of $\{x_i\}_{i \in \{1, \dots, n\}}$. Hence we must have that $\{x_i | i \in \{1, \dots, n\}\}$ is linear independent.
2. Let $x: \{1, \dots, n\} \rightarrow X$ be the function that defines $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ [where $x_i \stackrel{\text{notation}}{=} x(i)$]. Let $i, j \in \{1, \dots, n\}$ such that $x(i) = x(j)$ and assume that $i \neq j$. Define

$$\{\alpha_k\}_{k \in \{1, \dots, n\}} \subseteq F \text{ by } \alpha_k = \begin{cases} 1 & \text{if } k = i \\ -1 & \text{if } k = j \\ 0 & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases}$$

then we have

$$\begin{aligned} \sum_{k \in \{1, \dots, n\}} \alpha_k \cdot x_k &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} \alpha_k \cdot x_k + \sum_{k \in \{i\}} \alpha_k \cdot x_k + \sum_{k \in \{j\}} \alpha_k \cdot x_k \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} \alpha_k \cdot x_k + \alpha_i \cdot x_i + \alpha_j \cdot x_j \\ &= \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} 0 \cdot x_k + 1 \cdot x_i + (-1) \cdot x_j \\ &\stackrel{[\text{theorem: 11.37}]}{=} x_i - x_j \\ &= x(i) - x(j) \\ &= 0 \end{aligned}$$

which, as $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent, results in the contradiction $0 = \alpha_i = -1$. Hence $\forall i, j \in \{1, \dots, n\}$ with $x(i) = x(j)$ we must have $i = j$ proving that x is injective. So

$x: \{1, \dots, n\} \rightarrow x(\{1, \dots, n\}) = \{x_i | i \in \{1, \dots, n\}\}$ is a bijection

or by [definition: 11.84] that $\{x_i | i \in \{1, \dots, n\}\} = \{x_1, \dots, x_n\}$. $\square$

11.3.3 Basis of a vector space

Definition 11.136. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then $B \subseteq V$ is a basis of $\langle V, +, \cdot \rangle$ if

1. B is linear independent
2. $\text{span}(B) = V$

Note 11.137. The basis that we define here is called a Hamel basis in books about vector spaces. Every vector in a vector space can be written as a linear combination of the Hamel basis. So we don't have to bother with infinite sums and convergence of these sums, although the basis itself does not have to be finite. There are other concepts of basis in a vector space that work with infinite sums (for example Fourier series) but these are not covered in this section.

Example 11.138. Let $\langle \{0\}, +, \cdot \rangle$ be the trivial space over a field $\langle F, +, \cdot \rangle$ then $\emptyset$ is the only basis for $\langle \{0\}, +, \cdot \rangle$

Proof. By [example: 11.104] we have $\text{span}(\emptyset) = \{0\}$ and by [example: 11.121] $\emptyset$ is linear independent, proving that

$$\emptyset \text{ is a basis for } \langle \{0\}, +, \cdot \rangle$$

If B is another basis of $\langle \{0\}, +, \cdot \rangle$ then $B \subseteq \{0\}$. If $B = \{0\}$ then $0 \in B$ so that by [theorem: 11.115] B is linear dependent contradicting linear Independence, hence we must have that $B = \emptyset$. $\square$

A basis of a vector space allows us to write every vector of the vector space as a finite linear combination of vectors in the basis.

Theorem 11.139. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $B \subseteq V$ then

$$B \text{ is a basis of } \langle V, +, \cdot \rangle$$

$$\Updownarrow$$

$$\forall v \in V \text{ there exists a finite } I \subseteq B \text{ and a **unique** (1) } \{\alpha_u\}_{u \in I} \subseteq F \text{ such that } v = \sum_{u \in I} \alpha_u \cdot u$$

(1) If $\{\beta_u\}_{u \in I} \subseteq F$ satisfies $v = \sum_{u \in I} \beta_u \cdot u$ then $\{\alpha_u\}_{u \in I} = \{\beta_u\}_{u \in I}$

Proof.

$\Rightarrow$. As $\text{span}(B) = V$ and $v \in V$ there exists a finite $I \subseteq B$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that

$$v = \sum_{u \in I} \alpha_u \cdot u$$

proving existence. For uniqueness let $\{\beta_u\}_{u \in I} \subseteq F$ such that

$$v = \sum_{u \in I} \beta_u \cdot u$$

Define

$$\{\gamma_u\}_{u \in I} \subseteq F \text{ by } \gamma_u = \alpha_u - \beta_u$$

then we have

$$\begin{aligned} \sum_{u \in I} \gamma_u \cdot u &= \sum_{u \in I} (\alpha_u - \beta_u) \cdot u \\ &\stackrel{[\text{theorems: 11.6, 11.40}]}{=} \sum_{u \in I} \alpha_u \cdot u - \sum_{u \in I} \beta_u \cdot u \\ &= v - v \\ &= 0 \end{aligned}$$

As B is linear independent we have $\forall u \in I$ that $\alpha_u - \beta_u = \gamma_u = 0$ proving that

$$\{\alpha_u\}_{u \in I} \subseteq F = \{\beta_u\}_{u \in I} \subseteq F$$

$\Leftarrow$. Let $v \in V$ then by the hypothesis there exists a finite $I \subseteq B$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that $v = \sum_{u \in I} \alpha_u \cdot u$ proving that $v \in \text{span}(B)$. So $V \subseteq \text{span}(B) \subseteq V$ hence

$$\text{span}(B) = V$$

As for linear independence let $I \subseteq B$ a finite set and $\{\alpha_u\}_{u \in I} \subseteq F$ such that $\sum_{u \in I} \alpha_u \cdot u = 0$. Define $\{\beta_u\}_{u \in I} \subseteq F$ by $\beta_u = 0$ then $\sum_{u \in I} \beta_u \cdot u = \sum_{u \in I} 0 \cdot u \stackrel{[\text{theorem: 11.37}]}{=} 0$, by uniqueness we have $\forall u \in I \alpha_u = \beta_u = 0$. So by definition

B is linear independent □

The uniqueness in the above is only true if the index sets are equal, however we can ensure uniqueness also for the index set if we require that the coefficients $\{\alpha_i\}_{i \in I}$ are non zero.

Theorem 11.140. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and B a basis of $\langle V, +, \cdot \rangle$ then $\forall v \in V$ there exist a finite $I \subseteq B$ and $\{\alpha_u\}_{u \in I} \subseteq F$ such that $v = \sum_{u \in I} \alpha_u \cdot u$. Further if there exists another $J \subseteq B$ and a $\{\beta_u\}_{u \in J} \subseteq F$ such that $v = \sum_{u \in J} \beta_u \cdot u$ then*

$$\begin{aligned} \forall u \in I \setminus J & \quad \text{we have} \quad \alpha_u = 0 \\ \forall u \in J \setminus I & \quad \text{we have} \quad \beta_u = 0 \\ \forall u \in I \cap J & \quad \text{we have} \quad \alpha_u = \beta_u \end{aligned}$$

Proof. Let $v \in V$ then using the previous theorem [theorem: 11.139] there exist a finite $I \subseteq B$ and a $\{\alpha_i\}_{i \in I} \subseteq F$ such that $v = \sum_{i \in I} \alpha_i \cdot i$. Assume that there exists another finite $J \subseteq B$ and a $\{\beta_i\}_{i \in J} \subseteq F \setminus \{0\}$ such that $v = \sum_{i \in J} \beta_i \cdot i$. Define

$$\{\gamma_i\}_{i \in I \cup J} \subseteq F \text{ by } \gamma_i = \begin{cases} \alpha_i & \text{if } i \in I \\ 0 & \text{if } i \in (I \cup J) \setminus I \end{cases} \text{ and } \{\delta_i\}_{i \in I \cup J} \text{ by } \delta_i = \begin{cases} \beta_i & \text{if } i \in J \\ 0 & \text{if } i \in (I \cup J) \setminus J \end{cases}$$

then we have

$$\begin{aligned} \sum_{i \in I \cup J} \alpha_i \cdot i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in (I \cup J) \setminus I} \alpha_i \cdot i + \sum_{i \in I} \alpha_i \cdot i \\ & = \sum_{i \in (I \cup J) \setminus I} 0 \cdot i + v \\ & \stackrel{[\text{theorem: 11.37}]}{=} 0 + v \\ & = v \\ & \stackrel{[\text{theorem: 11.37}]}{=} \sum_{i \in (I \cup J) \setminus J} 0 \cdot i + v \\ & = \sum_{i \in (I \cup J) \setminus J} \beta_i \cdot i + \sum_{i \in J} \beta_i \cdot i \\ & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \cup J} \beta_i \cdot i \end{aligned}$$

Applying then the previous theorem [theorem: 11.139] on the above we have

$$\forall i \in I \cup J \text{ that } \gamma_i = \delta_i$$

If $i \in I \setminus J$ then $i \in (I \cup J) \setminus J$ hence $0 = \delta_i = \gamma_i = \alpha_i$ so that

$$\forall i \in I \setminus J \text{ we have } \alpha_i = 0$$

If $i \in J \setminus I$ then $i \in (I \cup J) \setminus I$ hence $0 = \gamma_i = \delta_i = \beta_i$ so that

$$\forall i \in J \setminus I \text{ we have } \beta_i = 0$$

Finally if $i \in I \cap J$ then $\alpha_i = \gamma_i = \delta_i = \beta_i$ hence

$$\forall i \in I \cap J \text{ we have } \alpha_i = \beta_i \quad \square$$

Theorem 11.141. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and B a basis of $\langle V, +, \cdot \rangle$ then $\forall v \in V$ there exists a **unique pair** $(I, \{\alpha_u\}_{u \in I} \subseteq F \setminus \{0\})$ such that I is finite, $I \subseteq B$ and

$$v = \sum_{u \in I} \alpha_u \cdot u$$

Proof. Let $v \in V$. By [theorem: 11.139] there exist a finite $K \subseteq B$ and a $\{\alpha_i\}_{i \in K} \subseteq F$ such that $v = \sum_{i \in K} \alpha_i \cdot i$. Define now $I \subseteq K$ by $I = \{i \in K \mid \alpha_i \neq 0\}$ so that $\{\alpha_i\}_{i \in I} \subseteq F \setminus \{0\}$ then we have

$$\begin{aligned} v &= \sum_{i \in K} \alpha_i \cdot i \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in K \setminus I} \alpha_i \cdot i + \sum_{i \in I} \alpha_i \cdot i \\ &= \sum_{i \in K \setminus I} 0 \cdot i + \sum_{i \in I} \alpha_i \cdot i \\ &\stackrel{[\text{theorem: 11.37}]}{=} 0 + \sum_{i \in I} \alpha_i \cdot i \\ &= \sum_{i \in I} \alpha_i \cdot i \end{aligned}$$

which proves existence. As for uniqueness, assume that there exists a finite $J \subseteq B$ and a $\{\beta_i\}_{i \in J} \subseteq F \setminus \{0\}$ such that $v = \sum_{i \in J} \beta_i \cdot i$. Then by the previous theorem [theorem: 11.140] we have that

$$\forall i \in I \setminus J \alpha_i = 0 \wedge \forall i \in J \setminus I \beta_i = 0 \wedge \forall i \in I \cap J \alpha_i = \beta_i$$

As by the hypothesis $\forall i \in I \alpha_i \neq 0$ and $\forall i \in J \beta_i \neq 0$ we must have that $I \setminus J = \emptyset = J \setminus I$ hence

$$I = J$$

and $\forall i \in I = J = I \cap J$ that $\alpha_i = \beta_i$ or

$$\{\alpha_i\}_{i \in I} = \{\beta_i\}_{i \in J}$$

□

If B is a finite we have a simpler equivalence.

Theorem 11.142. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $B \subseteq V$ where B is **finite** then

$$B \text{ is a basis of } \langle V, +, \cdot \rangle$$

$$\Updownarrow$$

$$\forall v \in V \text{ there exists a **unique** } \{\alpha_u\}_{u \in B} \subseteq F \text{ such that } v = \sum_{u \in B} \alpha_u \cdot e_u$$

Proof.

$\Rightarrow$. If B is a basis we have by [theorem: 11.139] that there exists a finite $I \subseteq B$ and a $\{\beta_u\}_{u \in I} \subseteq F$ such that $v = \sum_{u \in I} \beta_u \cdot u$. Define

$$\{\alpha_u\}_{u \in B} \subseteq F \text{ by } \alpha_u = \begin{cases} 0 & \text{if } u \in B \setminus I \\ \beta_u & \text{if } u \in I \end{cases}$$

then we have

$$\begin{aligned} \sum_{u \in B} \alpha_u \cdot u &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in B \setminus I} \alpha_u \cdot u + \sum_{u \in I} \alpha_u \cdot u \\ &= \sum_{u \in B \setminus I} 0 \cdot u + \sum_{u \in I} \beta_u \cdot u \\ &\stackrel{[\text{theorem: 11.37}]}{=} \sum_{u \in I} \beta_u \cdot u \\ &= v \end{aligned}$$

proving existence. For uniqueness let $\{\lambda_u\}_{u \in B} \subseteq F$ such that

$$v = \sum_{u \in B} \lambda_u \cdot u$$

Define

$$\{\gamma_u\}_{u \in B} \subseteq F \text{ by } \gamma_u = \alpha_u - \lambda_u$$

then we have

$$\begin{aligned} \sum_{u \in B} \gamma_u \cdot u &= \sum_{u \in B} (\alpha_u - \lambda_u) \cdot u \\ &\stackrel{[\text{theorems: 11.6, 11.40}]}{=} \sum_{u \in B} \alpha_u \cdot u - \sum_{u \in B} \lambda_u \cdot u \\ &= v - v \\ &= 0 \end{aligned}$$

As B is linear independent we have $\forall u \in B$ that $\alpha_u - \lambda_u = \gamma_u = 0$ proving that

$$\{\alpha_u\}_{u \in B} \subseteq F = \{\lambda_u\}_{u \in B} \subseteq F$$

$\Leftarrow$. If $v \in V$ then by the hypothesis there exist a $\{\alpha_u\}_{u \in B} \subseteq F$ such that $v = \sum_{u \in B} \alpha_u \cdot u$ proving that $v \in \text{span}(B)$. So $V \subseteq \text{span}(B) \subseteq V$ hence

$$\text{span}(B) = V$$

As for linear independence let $I \subseteq B$ a finite set and $\{\alpha_u\}_{u \in I} \subseteq F$ such that $\sum_{u \in I} \alpha_u \cdot u = 0$. Define $\{\beta_u\}_{u \in I} \subseteq F$ by $\beta_u = 0$ then $\sum_{u \in I} \beta_u \cdot u = \sum_{u \in I} 0 \cdot u \stackrel{[\text{theorem: 11.37}]}{=} 0$. By uniqueness we have $\forall u \in I$ $\alpha_u = \beta_u = 0$. Using [theorem: 11.120] it follows then that

B is linear independent

□

A other usefully alternative characterization of a finite basis is the following

Theorem 11.143. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $E = \{e_1, \dots, e_n\} \subseteq V$ a finite set then

E is a basis of $\langle V, +, \cdot \rangle$

$$\Updownarrow$$

$$\forall v \in V \text{ there exists a **unique** } \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ such that } v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i$$

Proof. $E = \{e_1, \dots, e_n\}$ is defined by the bijection

$$e: \{1, \dots, n\} \rightarrow E \text{ where } n = \text{card}(E)$$

Further we have

$\Rightarrow$. Let $v \in V$ then by [theorem: 11.142] there exists a $\{\alpha_u\}_{u \in B} \subseteq F$ such that

$$v = \sum_{u \in W} \alpha_u \cdot u$$

Define

$$\{\beta_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \beta_i = \alpha_{e(i)}$$

then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \beta_i \cdot e_i &= \sum_{i \in \{1, \dots, n\}} \alpha_{e(i)} \cdot e(i) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{u \in B} \alpha_u \cdot u \\ &= v \end{aligned}$$

proving existence. As for uniqueness let $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq F$ be another family such that $v = \sum_{i \in \{1, \dots, n\}} \lambda_i \cdot e_i$ and define

$$\{\gamma_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ by } \gamma_i = \beta_i - \lambda_i$$

then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \gamma_i \cdot e_i &= \sum_{i \in \{1, \dots, n\}} (\beta_i - \lambda_i) \cdot e_i \\ &= \sum_{i \in \{1, \dots, n\}} (\beta_i \cdot e_i - \lambda_i \cdot e_i) \\ &\stackrel{[\text{theorems: 11.6, 11.40}]}{=} \sum_{i \in \{1, \dots, n\}} \beta_i \cdot e_i - \sum_{i \in \{1, \dots, n\}} \lambda_i \cdot e_i \\ &= v - v \\ &= 0 \end{aligned}$$

As E is linear independent it follows then from the above and [theorem: 11.120] that $\forall i \in \{1, \dots, n\} \gamma_i = 0 \Rightarrow \beta_i = \lambda_i$. Hence $\{\beta_i\}_{i \in \{1, \dots, n\}} = \{\lambda_i\}_{i \in \{1, \dots, n\}}$.

$\Leftarrow$. Let $v \in V$ then by the hypothesis there exists a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$v = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i$$

Define then

$$\{\beta_u\}_{u \in B} \subseteq F \text{ by } \beta_u = \alpha_{e^{-1}(u)}$$

then we have

$$\begin{aligned} \sum_{u \in B} \beta_u \cdot u &= \sum_{u \in B} \alpha_{e^{-1}(u)} \cdot e(e^{-1}(u)) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e(i) \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i \\ &= v \end{aligned}$$

proving that $v \in \text{span}(B)$. Hence $V \subseteq \text{span}(B) \subseteq V$ so that

$$V = \text{span}(V)$$

Let $\{\gamma_i\}_{i \in \{1, \dots, n\}} \subseteq F$ be such that $0 = \sum_{i \in \{1, \dots, n\}} \gamma_i \cdot e_i$ and define $\{\zeta_i\}_{i \in \{1, \dots, n\}}$ by $\gamma_i = 0$ then we have also $\sum_{i \in \{1, \dots, n\}} \zeta_i \cdot e_i = 0$. Hence by uniqueness we have $\forall i \in \{1, \dots, n\}$ that $\gamma_i = \zeta_i = 0$. Using then [theorem: 11.120] it follows that E is linear independent. $\square$

Scaling a finite basis of a vector space results in a finite basis of the same vector space as is show in the next theorem.

Theorem 11.144. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\}$ and a finite basis $X = \{x_1, \dots, x_n\} \subseteq V$ of V then

$$y: \{1, \dots, n\} \rightarrow \{\alpha_i \cdot x_i | i \in \{1, \dots, n\}\} \text{ defined by } y(i) = \alpha_i \cdot x_i \text{ is a bijection}$$

defining the finite set

$$Y = \{y_1, \dots, y_n\} = \{\alpha_1 \cdot x_1, \dots, \alpha_n \cdot x_n\}$$

that is a basis of V . In other words if $\{x_1, \dots, x_n\}$ is a basis for V then if $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\}$ we have that $\{\alpha_1 \cdot x_1, \dots, \alpha_n \cdot x_n\}$ is a basis of V .

Proof. Define

$$y: \{1, \dots, n\} \rightarrow V \text{ by } y(i) = \alpha_i \cdot x_i$$

Assume that $\exists i, j \in \{1, \dots, n\}$ with $y(i) = y(j) \Rightarrow \alpha_i \cdot x_i = \alpha_j \cdot x_j$ such that $i \neq j$. Then as $\{\alpha_k\}_{k \in \{1, \dots, n\}} \subseteq F \setminus \{0\}$ we have $x_j = (\alpha_i / \alpha_j) \cdot x_i$. Define then

$$\{\beta_k\}_{k \in \{1, \dots, n\}} \text{ by } \beta_k = \begin{cases} \alpha_i / \alpha_j & \text{if } k = i \\ -1 & \text{if } k = j \\ 0 & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases}$$

then we have

$$\begin{aligned} \sum_{k \in \{1, \dots, n\}} \alpha_k \cdot x_k &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} \alpha_k \cdot x_k + \sum_{k \in \{i\}} \alpha_k \cdot x_k + \sum_{k \in \{j\}} \alpha_k \cdot x_k \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} \alpha_k \cdot x_k + \alpha_i \cdot x_i + \alpha_j \cdot x_j \\ &= \sum_{k \in \{1, \dots, n\} \setminus \{i, j\}} 0 \cdot x_k + (\alpha_i / \alpha_j) \cdot x_i + (-1) \cdot x_j \\ &= x_j + (-x_j) \\ &= 0 \end{aligned}$$

As $\{x_1, \dots, x_n\}$ is linear independent [because it is a basis] we have by [theorem: 11.120] that $\alpha_i / \alpha_j = 0 \Rightarrow \alpha_i = 0$ contradicting $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\}$. So the assumption must be wrong, hence $\forall i, j \in \{1, \dots, n\}$ with $y(i) = y(j)$ we have $i = j$, proving that y is injective. So

$$y: \{1, \dots, n\} \rightarrow y(\{1, \dots, n\}) = \{y_i | i \in \{1, \dots, n\}\} = \{\alpha_i \cdot x_i | i \in \{1, \dots, n\}\} \text{ is a bijection} \quad (11.54)$$

defining the finite set

$$Y = \{y_1, \dots, y_n\} = \{\alpha_1 \cdot x_1, \dots, \alpha_n \cdot x_n\} \quad (11.55)$$

For linear independency of $\{y_1, \dots, y_n\}$, let $\{\gamma_k\}_{k \in \{1, \dots, n\}} \subseteq F$ be such that $\sum_{k \in \{1, \dots, n\}} \alpha_k \cdot y_k = 0$ and define

$$\{\rho_k\}_{k \in \{1, \dots, n\}} \subseteq F \text{ by } \rho_k = \alpha_k \cdot \gamma_k$$

then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \rho_k \cdot x_k &= \sum_{i \in \{1, \dots, n\}} (\alpha_k \cdot \gamma_k) \cdot x_k \\ &= \sum_{i \in \{1, \dots, n\}} \gamma_k \cdot y_k \\ &= 0 \end{aligned}$$

As $\{x_1, \dots, x_n\}$ is linear independent it follows, using [theorem: 11.120] that $\forall k \in \{1, \dots, n\}$ $\rho_k = 0 \Rightarrow \alpha_k \cdot \gamma_k = 0 \xRightarrow{\alpha_k \neq 0} \gamma_k = 0$, proving by [theorem: 11.120] that

$$\{y_1, \dots, y_n\} \text{ is linear independent} \quad (11.56)$$

Let $v \in V$ then, as $\text{span}(\{x_1, \dots, x_n\}) = V$, there exists by [theorem: 11.102] a $\{\omega_k\}_{k \in \{1, \dots, n\}} \subseteq F$ such that $v = \sum_{k \in \{1, \dots, n\}} \omega_k \cdot x_k$. As $\{\alpha_k\}_{k \in \{1, \dots, n\}} \subseteq F \setminus \{0\}$ we can define

$$\{\xi_k\}_{k \in \{1, \dots, n\}} \subseteq F \text{ by } \xi_k = \omega_k \cdot (\alpha_k)^{-1}$$

then

$$\begin{aligned} \sum_{k \in \{1, \dots, n\}} \xi_k \cdot y_k &= \sum_{k \in \{1, \dots, n\}} (\omega_k \cdot (\alpha_k)^{-1}) \cdot (\alpha_k \cdot x_k) \\ &= \sum_{k \in \{1, \dots, n\}} \omega_k \cdot x_k \\ &= v \end{aligned}$$

Which proves that $V \subseteq \text{span}(\{y_1, \dots, y_n\}) \subseteq V$ so that

$$V = \text{span}(\{y_1, \dots, y_n\}) = \text{span}(\{\alpha_1 \cdot x_1, \dots, \alpha_n \cdot x_n\})$$

combining the above with [eq: 11.56] proves finally that

$$\{y_1, \dots, y_n\} \text{ is a basis of } V \quad \square$$

Up to now we are not certain that every vector space has a basis. We will now use Zorn's lemma, a consequence of the Axiom of Choice, to prove that every vector space has a basis. We start with proving that every linear independent set that is a subset of a spanning set can be extended to a basis.

Theorem 11.145. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $R \subseteq W \subseteq V$ such that*

1. *R is linear independent*

2. *$\text{span}(W) = V$*

then there exists a basis B of $\langle V, +, \cdot \rangle$ such that $R \subseteq B \subseteq W$

Proof. Define the set of linear independent sets between R and B

$$\mathcal{A} = \{X \subseteq V \mid R \subseteq X \subseteq W \text{ and } X \text{ is linear independent}\} \quad (11.57)$$

then we have, as R is linear independent and $R \subseteq W \subseteq V$, that

$$R \in \mathcal{A} \quad (11.58)$$

Using [example: 3.33] we can partial order $\mathcal{A}$ by the inclusion operator, in other words

$$\langle \mathcal{A}, \subseteq \rangle \text{ is a partial ordered set} \quad (11.59)$$

Let $\mathcal{C} \subseteq \mathcal{A}$ be a **non empty** chain [see definition: 3.41]. If $X \in \mathcal{C}$ then $X \in \mathcal{A}$ hence $R \subseteq X \subseteq W$, so that $R \subseteq \bigcup_{X \in \mathcal{C}} X \subseteq W$ giving

$$R \subseteq B_{\mathcal{C}} \subseteq W \text{ and } \forall X \in \mathcal{C} \text{ we have } X \subseteq B_{\mathcal{C}} \text{ where } B_{\mathcal{C}} = \bigcup_{X \in \mathcal{C}} X \quad (11.60)$$

We use now mathematical induction on the size of finite sets to proof that

$$\forall A \subseteq B_{\mathcal{C}} \text{ such that } A \text{ is finite there } \exists X \in \mathcal{C} \text{ such that } A \subseteq X \quad (11.61)$$

Proof. Define

$$\mathcal{S} = \{n \in \mathbb{N}_0 \mid \text{If } A \subseteq B_{\mathcal{C}} \text{ with } \{1, \dots, n\} \approx A \text{ then } \exists X \in \mathcal{C} \text{ such that } A \subseteq X\}$$

then we have:

0 $\in \mathcal{S}$. If $A \subseteq B_{\mathcal{C}}$ with $A \approx \{1, \dots, 0\} = \emptyset$ then by [theorem: 10.87] there exist a bijection

$$\beta: \{1, \dots, 0\} = \emptyset \rightarrow A$$

hence $A = \beta(\emptyset)$ proving that $A = \emptyset$. As $\mathcal{C} \neq \emptyset$ there exists a $X \in \mathcal{C}$ and trivially $A = \emptyset \subseteq X$, proving **0** $\in \mathcal{S}$.

n $\in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $A \subseteq B_{\mathcal{C}}$ with $\{1, \dots, n+1\} \approx A$ then there exist a bijection

$$\beta: \{1, \dots, n+1\} \rightarrow A.$$

Take $A' = A \setminus \{\beta(n+1)\}$ and consider $\beta|_{\{1, \dots, n\}}: \{1, \dots, n\} \rightarrow A'$ then we have:

injectivity. If $k, l \in \{1, \dots, n\}$ and $\beta|_{\{1, \dots, n\}}(k) = \beta|_{\{1, \dots, n\}}(l)$ then $\beta(k) = \beta(l)$ which, as β is a bijection, proves that $k = l$

surjectivity. If $y \in A'$ then $y \in A$ and $y \neq \beta(n+1)$. As β is a bijection there exists a $i \in \{1, \dots, n+1\}$ such that $\beta(i) = y$. If $i = n+1$ then $y = \beta(i) = \beta(n+1)$ contradicting $y \neq \beta(n+1)$, so $i \in \{1, \dots, n\}$, hence $y = \beta(i) = \beta|_{\{1, \dots, n\}}(i)$.

proving that $\beta|_{\{1, \dots, n\}}: \{1, \dots, n\} \rightarrow A'$ is a bijection hence $\{1, \dots, n\} \approx A'$. As **n** $\in \mathcal{S}$ there exist a $X' \in \mathcal{C}$ such that $A' \subseteq X'$. Further as $\beta(n+1) \in A \subseteq B_{\mathcal{C}} = \bigcup_{X \in \mathcal{C}} X$ there exists a $X \in \mathcal{C}$ such that $\beta(n+1) \in X$. Now as $\mathcal{C}$ is a chain we have either:

X' $\subseteq$ **X**. Then as $A \setminus \{\beta(n+1)\} = A' \subseteq X' \subseteq X$ and $\beta(n+1) \in X$ we have that $A \subseteq X$ proving that **n+1** $\in \mathcal{S}$

$X \subseteq X'$. Then as $A \setminus \{\beta(n+1)\} = A' \subseteq X'$ and $\beta(n+1) \in X \subseteq X'$ we have that $A \subseteq X'$ proving that $n+1 \in S$

so in both cases we have that $n+1 \in S$.

By mathematical induction it follows that $S = \mathbb{N}_0$. So for a finite $A \subseteq B_C$ we have by [theorem: 10.87] a $n \in \mathbb{N}_0$ such that $\{1, \dots, n\} \approx A$. Hence as $n \in \mathbb{N}_0 = S$ there exists a $X \in \mathcal{C}$ such that $A \subseteq X$. $\square$

We prove now that B_C is linear independent. Let $I \subseteq B_C$ be a finite set then by [eq: 11.61] we have

$$\exists Y \in \mathcal{C} \text{ such that } I \subseteq Y \quad (11.62)$$

Further as $Y \in \mathcal{C} \subseteq \mathcal{A}$ we have by [eq: 11.57] that

$$R \subseteq Y \subseteq W \text{ and } Y \text{ is linear independent} \quad (11.63)$$

So if $\{\alpha_u\}_{u \in I} \subseteq F$ is such that $\sum_{u \in I} \alpha_u \cdot u = 0$ then as $I \subseteq Y$ and Y is linear independent we have by the definition of linear independence that $\forall u \in I \alpha_u = 0$, which, as $I \subseteq B_C$, I finite was chosen arbitrary, proves that

$$B_C \text{ is linear independent} \quad (11.64)$$

Combining this with [eqs: 11.57, 11.60] it follows that

$$B_C \in \mathcal{A}$$

So for every **non empty** chain $\mathcal{C} \subseteq \mathcal{A}$ we have found a $B_C \in \mathcal{A}$ such that $\forall X \in \mathcal{C}$ we have $X \subseteq \bigcup_{X \in \mathcal{C}} X = B_C$ hence $\mathcal{C}$ has a upper bound B_C . Further as by [eq: 11.58] $R \in \mathcal{A}$ we have for the empty chain $[\mathcal{C} = \emptyset]$ that $\forall X \in \emptyset X \subseteq R$ is satisfied variously, hence R is a upper bound of the empty chain.. So we have that

every chain $\mathcal{C}$ in $\mathcal{A}$ has a upper bound in $\mathcal{A}$

By Zorn's lemma [see theorem: 3.128] there exists a maximal element in $\mathcal{A}$, in other words

$$\exists B \in \mathcal{A} \text{ such that } \forall X \in \mathcal{A} \text{ we have } X \subseteq B \quad (11.65)$$

and as $B \in \mathcal{A}$ we have

$$R \subseteq B \subseteq W \text{ and } B \text{ is linear independent} \quad (11.66)$$

Let now $w \in W$ then we have either:

$w \in B$. As by [theorem: 11.108] $B \subseteq \text{span}(B)$ it follows that $w \in \text{span}(B)$.

$w \notin B$. Assume that $B \cup \{w\}$ is linear independent then as $R \subseteq B \subseteq B \cup \{w\} \subseteq W$ it follows that $B \cup \{w\} \in \mathcal{A}$, using [eq: 11.65] it follows that $B \cup \{w\} \subseteq B$ so that $w \in B$ contradicting $w \notin B$. Hence we must have that

$$B \cup \{w\} \text{ is linear dependent}$$

Hence there exists a finite $I \subseteq B \cup \{w\}$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ satisfying $\exists u_0 \in I$ with $\alpha_{u_0} \neq 0$ such that $\sum_{u \in I} \alpha_u \cdot u = 0$. If $w \notin I$ then $I \subseteq B$ which as B is linear independent would mean that $\forall u \in I$ we have $\alpha_u = 0$ contradicting $\alpha_{u_0} \neq 0$. So we must have

$$w \in I \quad (11.67)$$

Further

$$\begin{aligned} 0 &= \sum_{u \in I} \alpha_u \cdot u \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in I \setminus \{w\}} \alpha_u \cdot u + \sum_{u \in \{w\}} \alpha_u \cdot u \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{u \in I \setminus \{w\}} \alpha_u \cdot u + \alpha_w \cdot w \end{aligned} \quad (11.68)$$

Note as $I \subseteq B \cup \{w\}$ we have $I \setminus \{w\} \subseteq (B \cup \{w\}) \setminus \{w\} \subseteq B$ so that

$$I \setminus \{w\} \subseteq B \quad (11.69)$$

Assume $\alpha_w = 0$ then from [eq: 11.68] we have $0 = \sum_{u \in I \setminus \{w\}} \alpha_u \cdot u$ which, as $I \setminus \{w\} \subseteq B$ [see eq: 11.69] and B is linear independent, gives $\forall u \in I \setminus \{w\} \alpha_u = 0$. So, as we assumed $\alpha_w = 0$, we have $\forall u \in I$ that $\alpha_u = 0$ contradicting $\alpha_{u_0} \neq 0$. Hence we must have that $\alpha_w \neq 0$, and applying this on [eq: 11.68] proves that

$$w = (\alpha_w)^{-1} \cdot \left(- \sum_{u \in I \setminus \{w\}} \alpha_u \cdot u \right) \stackrel{[\text{theorems: 11.40, 11.80}]}{=} \sum_{u \in I} ((\alpha_u)^{-1} \cdot (-\alpha_u)) \cdot u = \sum_{u \in I} \lambda_u \cdot u$$

where $\{\lambda_u\}_{u \in I \setminus \{w\}} \subseteq F$. So as $I \setminus \{w\} \subseteq B$ it follows that

$$w \in \text{span}(B)$$

So, as in all cases we have $w \in \text{span}(B)$, it follows that $W \subseteq \text{span}(B)$, hence

$$V = \text{span}(W) \subseteq \stackrel{[\text{theorem: 11.109}]}{\text{span}(\text{span}(B))} \stackrel{[\text{theorem: 11.112}]}{=} \text{span}(B) \subseteq V$$

proving that

$$\text{span}(B) = V \tag{11.70}$$

By [eqs: 11.66, 11.70] we have that B is a basis of $\langle V, +, \cdot \rangle$ and $R \subseteq B \subseteq W$ proving the theorem. $\square$

Corollary 11.146. *Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and R a linear independent set then there exists a basis B of $\langle V, +, \cdot \rangle$ such that $R \subseteq B$. In other words a linear independent set of a vector space can be extended to a basis of the vector space.*

Proof. As R is linear independent, $R \subseteq V$ and $\text{span}(V) \stackrel{[\text{theorem: 11.111}]}{=} V$ we have by [theorem: 11.145] that there exist a basis B of $\langle V, +, \cdot \rangle$ with $R \subseteq B \subseteq V$. $\square$

Corollary 11.147. *Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ and $W \subseteq V$ such that $\text{span}(W) = V$ then there exist a basis B of $\langle V, +, \cdot \rangle$ such that $B \subseteq W$. In other words every spanning set of a vector space can be reduced to a basis of the vector space.*

Proof. For V we have either:

$V = \{0\}$. By [example: 11.138] $\emptyset$ is a basis for V and trivially $\emptyset \subseteq W$.

$V \neq \{0\}$. As $V = \text{span}(W)$ and $\text{span}(\emptyset) \stackrel{[\text{example: 11.104}]}{=} \{0\}$ we must have $W \neq \emptyset$, also if $W = \{0\}$ then by [example: 11.105] $\text{span}(W) = \{0\}$. Hence there exist a $w \in W$ such that $w \neq 0$. Using [example: 11.122] we have that $R = \{w\}$ is linear independent. As further $R = \{w\} \subseteq W$ we have by [theorem: 11.145] that there exist a basis B of $\langle V, +, \cdot \rangle$ such that $R \subseteq B \subseteq W$. $\square$

Corollary 11.148. *Let $\langle V, +, \cdot \rangle$ be a vector space over the field $\langle F, +, \cdot \rangle$ then there exist a basis $B \subseteq V$ of $\langle V, +, \cdot \rangle$.*

Proof. As $\text{span}(V) \stackrel{[\text{theorem: 11.111}]}{=} V$ we have by [corollary: 11.147] that there exist a basis B of $\langle V, +, \cdot \rangle$. $\square$

11.3.4 Dimension of a vector space

Lemma 11.149. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ that has a **finite** basis $B \subseteq V$ then every basis of $\langle V, +, \cdot \rangle$ is **finite**. Further for every basis A of $\langle V, +, \cdot \rangle$ we have $\text{card}(B) = \text{card}(A)$*

Proof. As B is a basis we have that

$$\text{span}(B) = V, B \text{ is linear independent and } B \text{ is finite}$$

Let A be another basis of $\langle V, +, \cdot \rangle$ then

$$\text{span}(A) = V \text{ and } A \text{ is linear independent}$$

By [corollary: 11.126] it follows that A is finite. Further as A, B are finite we have by [theorem: 11.89] that

$$A = \{a_1, \dots, a_n\}, B = \{b_1, \dots, b_m\} \text{ where } n = \text{card}(A) \text{ and } m = \text{card}(B)$$

From $V = \text{span}(B)$ and the linear independence of A it follows from [corollary: 11.125] that

$$n \leq \text{card}(B) = m$$

From $V = \text{span}(A)$ and the linear independency of B it follows from [corollary: 11.125] that

$$m \leq \text{card}(A) = n$$

So we have that $n = m$ proving that

$$\text{card}(A) = \text{card}(B)$$

□

Corollary 11.150. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ that has a *infinite* basis $B \subseteq V$ then every basis of $\langle V, +, \cdot \rangle$ is *infinite*.

Proof. Assume that there exists a finite basis of $\langle V, +, \cdot \rangle$ then by [lemma: 11.149] B is finite a contradiction. So every basis of $\langle V, +, \cdot \rangle$ must be infinite. □

Definition 11.151. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then V is *finite dimensional* if V has a finite basis and V is *infinite dimensional* if V has infinite basis. Using [corollary: 11.148], [lemma: 11.149] and [corollary: 11.150] we have that:

1. V is either finite dimensional or infinite dimensional but not both.
2. If V is finite dimensional then every basis of V is finite and have the same cardinality.
3. If V is infinite dimensional then every basis of V is infinite.

The above definition allows us to define $\dim(V)$.

Definition 11.152. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then $\dim(V)$ is defined by

$$\dim(V) = \begin{cases} \infty & \text{if } B \text{ is infinite} \\ \text{card}(B) & \text{if } B \text{ is finite} \end{cases}$$

where B is basis of V . By [corollary: 11.148], [lemma: 11.149] and [corollary: 11.150] this is well defined.

Example 11.153. Let $\langle \{0\}, +, \cdot \rangle$ be the trivial vector space over a field $\langle F, +, \cdot \rangle$ [see: 11.63] then

$$\dim(\{0\}) = 0$$

Further if $\langle V, +, \cdot \rangle$ is a vector space over the field $\langle F, +, \cdot \rangle$ with $\dim(X) = 0$ then $V = \{0\}$

Proof. By [example: 11.138] $\emptyset$ is a basis of $\langle \{0\}, +, \cdot \rangle$ so that $\dim(\{0\}) = \text{card}(\emptyset) = 0$. Further if $\langle V, +, \cdot \rangle$ is a vector space over the field $\langle F, +, \cdot \rangle$ with $\dim(X) = 0$ then if B is a basis for $\langle V, +, \cdot \rangle$ we must have that $\text{card}(B) = \dim(X) = 0$. Hence $B = \emptyset$, so by [theorem: 11.104] $V = \text{span}(B) = \text{span}(\emptyset) = \{0\}$. □

Theorem 11.154. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{e_i\}_{i \in \{1, \dots, n\}} \subseteq V$ then $\text{span}(\{e_i | i \in \{1, \dots, n\}\})$ is finite dimensional and $\dim(\text{span}(\{e_i | i \in \{1, \dots, n\}\})) \leq n$

Proof. First using [theorem: 11.94] we have that

$$\{e_i | i \in \{1, \dots, n\}\} \text{ is finite and } \text{card}(\{e_i | i \in \{1, \dots, n\}\}) \leq n$$

Using [theorem: 11.111, 11.61] we have that $\text{span}(\{e_i | i \in \{1, \dots, n\}\})$ is a vector space and trivially equal to itself, hence using [theorem: 11.147] there exists a basis $B \subseteq \{e_i | i \in \{1, \dots, n\}\}$ of $\text{span}(\{e_i | i \in \{1, \dots, n\}\})$. Further as $B \subseteq \{e_i | i \in \{1, \dots, n\}\}$ we have by [theorem: 10.91] that B is finite and

$$\dim(\text{span}(\{e_i | i \in \{1, \dots, n\}\})) = \text{card}(B) \leq \text{card}(\{e_i | i \in \{1, \dots, n\}\}) \leq n$$

□

In a finite dimensional vector space we have that every linear independent set with the same number of elements as the dimension of the vector space is automatically a basis of the vector space.

Theorem 11.155. *Let $\langle V, +, \cdot \rangle$ be a finite dimensional vector space over a field $\langle F, +, \cdot \rangle$ with $\dim(V) = n$ and $B \subseteq V$ a finite linear independent set with $\text{card}(B) = n$ then B is a basis for $\langle V, +, \cdot \rangle$.*

Proof. As $\dim(V) = n$ there exist a basis B' with $\text{card}(B') = n$. Using [theorem: 11.146] there exist a basis B'' such that $B \subseteq B''$. From [theorem: 11.149] we have that $\text{card}(B'') = \text{card}(B') = n$. So we have that $B \subseteq B''$ and $\text{card}(B) = \text{card}(B'')$, applying then [theorem: 10.95] proves that $B = B''$, hence B is a basis. $\square$

Corollary 11.156. *Let $\langle V, +, \cdot \rangle$ be a finite dimensional vector space over a field $\langle F, +, \cdot \rangle$ with $\dim(V) = n$ then if $\{v_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent we have that*

1. $\{v_i | i \in \{1, \dots, n\}\} = \{v_1, \dots, v_n\}$ [or by [definition: 11.84] that $v: \{1, \dots, n\} \rightarrow \{v_i | i \in \{1, \dots, n\}\}$ is a bijection].
2. $\{v_i | i \in \{1, \dots, n\}\}$ is a basis of V

Proof.

1. This follows from [theorem: 11.135 (2)]
2. As $\{v_i\}_{i \in \{1, \dots, n\}} \subseteq V$ is linear independent we have by [theorem: 11.135] that

$$\{v_i | i \in \{1, \dots, n\}\} \text{ is linear independent}$$

From (1) it follows that $\text{card}(\{v_i | i \in \{1, \dots, n\}\}) = n$, so applying the previous theorem [theorem: 11.155], we have that

$$\{v_i | i \in \{1, \dots, n\}\} \text{ is a basis for } V$$

$\square$

Theorem 11.157. *Let $\langle V, +, \cdot \rangle$ be a finite dimensional vector space over a field $\langle F, +, \cdot \rangle$ with $\dim(V) = n$ and $R = \{v_1, \dots, v_m\}$ a linear independent set with $m \in \{1, \dots, n\}$ then there exist a basis $B = \{b_1, \dots, b_n\}$ of V such that $\forall i \in \{1, \dots, m\}$ we have $b_i = v_i$.*

Proof. Let $R = \{v_1, \dots, v_m\}$ then by definition there exist a bijection $v: \{1, \dots, m\} \rightarrow R$. As R is linear independent it follows from [theorem: 11.146] that there exist a basis B of V such that $R \subseteq B$. As $B = R \sqcup (B \setminus R)$ we have that $n = \text{card}(B) = \text{card}(R) + \text{card}(B \setminus R) = m + \text{card}(B \setminus R)$ hence $\text{card}(B \setminus R) = n - m$. We have now two cases to consider:

$n = m$. Then $\text{card}(B \setminus R) = 0$ so that $B \setminus R = \emptyset$ hence $R = B$ and $B = \{v_1, \dots, v_n\}$.

$m < n$. Then there exist a bijection $w: \{1, \dots, n - m\} \rightarrow B \setminus R$ define then

$$b: \{1, \dots, n\} \rightarrow B \text{ by } b(i) = \begin{cases} v(i) & \text{if } i \in \{1, \dots, m\} \\ w(i - m) & \text{if } i \in \{m + 1, \dots, n\} \end{cases}$$

then we have:

injectivity. If $i, j \in \{1, \dots, n\}$ where $b(i) = b(j)$ then we have either:

$i, j \in \{1, \dots, m\}$. Then as $v(i) = b(i) = b(j) = v(j)$ so that, as v is bijective, $i = j$.

$i, j \in \{m + 1, \dots, n\}$. Then as $w(i - m) = b(i) = b(j) = w(j - m)$ so that, as w is bijective, $i - m = j - m$ hence $i = j$.

$i \in \{1, \dots, m\} \wedge j \in \{m + 1, \dots, n\}$. Then $b(i) = v(i) \in R$ and

$$b(i) = b(j) = w(j - m) \in B \setminus R$$

so that $b(i) \in R \cap (B \setminus R) = \emptyset$ a contradiction. Hence this case never occurs.

$i \in \{m + 1, \dots, n\} \wedge j \in \{1, \dots, m\}$. Then $b(i) = w(i - m) \in B \setminus R$ and

$$b(i) = b(j) = v(j) \in R$$

so that $b(i) \in R \cap (B \setminus R) = \emptyset$ a contradiction. Hence this case never occurs.

surjectivity. If $y \in B = R \cup (B \setminus R)$ then we have either:

$y \in R$. Then there exist a $i \in \{1, \dots, m\}$ such that $v(i) = y$

$y \in B \setminus R$. Then there exist a $j \in \{1, \dots, n - m\}$ such that $w(j) = y$, hence for $j + m \in \{m, \dots, n\}$ we have that $b(j + m) = w(j) = y$.

So we have that b is a bijection hence we have

$$B = \{b_1, \dots, b_n\}$$

and by construction $\forall i \in \{1, \dots, n\} \ b_i = b(i) = v(i) = v_i$. $\square$

Theorem 11.158. Let $\langle V, +, \cdot \rangle$ be a finite dimensional vector space over a field $\langle F, +, \cdot \rangle$ with $\dim(V) = n$ then for every $\{v_i\}_{i \in \{1, \dots, n\}} \subseteq V$ with $\text{span}(\{v_i\}_{i \in \{1, \dots, n\}}) = V$ we have that

$$\{v_i\}_{i \in \{1, \dots, n\}} \text{ is a basis of } V$$

Proof. As $\text{span}(\{v_i\}_{i \in \{1, \dots, n\}}) = V$ there exists by [theorem: 11.147] a basis $B \subseteq \{v_i\}_{i \in \{1, \dots, n\}}$. As $\dim(V) = n$ we must have that $\text{card}(B) = n$, further by [theorem: 10.96] $\text{card}(\{v_i\}_{i \in \{1, \dots, n\}}) \leq n$ and as $B \subseteq \{v_i\}_{i \in \{1, \dots, n\}}$ we have by [theorem: 10.91] $n = \text{card}(B) \leq \text{card}(\{v_i\}_{i \in \{1, \dots, n\}})$. So we have that $\text{card}(\{v_i\}_{i \in \{1, \dots, n\}}) = \text{card}(B)$. Finally from [theorem: 10.95] it follows that $B = \{v_i\}_{i \in \{1, \dots, n\}}$ hence $\{v_i\}_{i \in \{1, \dots, n\}}$ is a basis for V . $\square$

Theorem 11.159. Let $\langle V, +, \cdot \rangle$ be a finite dimensional vector space over a field $\langle F, +, \cdot \rangle$ and W a sub-space of $\langle V, +, \cdot \rangle$ then $\dim(W) \leq \dim(V)$ and if V is finite dimensional W is finite dimensional.

Proof. For $\dim(V)$ we have for V either:

V is infinite dimensional. Then V has a infinite basis so that by definition $\dim(V) = \infty$, using the definition of the order on the extended real numbers it follows that $\dim(W) \leq \dim(V)$.

V is finite dimensional. Let $B_W \subseteq W$ a basis of W then B_W is linear independent and $B_W \subseteq V = \text{span}(V)$. Using [theorem: 11.145] there exist then a basis B_V of $\langle V, +, \cdot \rangle$ such that $B_W \subseteq B_V \subseteq V$. As V is finite dimensional B_V is finite and using [theorem: 10.91] it follows that B_W is finite and $\text{card}(B_W) \leq \text{card}(B_V)$. Hence we have that W is finite dimensional with

$$\dim(W) = \text{card}(B_W) \leq \text{card}(B_V) = \dim(V) \quad \square$$

Theorem 11.160. Let $\langle F, +, \cdot \rangle$ a field and consider the vector space of $\langle F, +, \cdot \rangle$ over $\langle F, +, \cdot \rangle$ [see theorem: 11.64]. Then if $1 \in F$ is the multiplicative neutral element we have that $\{1\}$ is a basis of $\langle F, +, \cdot \rangle$, so $\dim(F) = 1$

Proof. First if $\{\alpha_i\}_{i \in \{1\}} \subseteq F$ is such that $\sum_{i \in \{1\}} \alpha_i \cdot i = 0$ then as

$$0 = \sum_{i \in \{1\}} \alpha_i \cdot i \stackrel{[\text{theorem: 11.34}]}{=} \alpha_1 \cdot 1 = \alpha_1$$

hence $\forall i \in \{1\} \ \alpha_i = 0$ proving that

$$\{1\} \text{ is linear independent}$$

Further if $v \in F$ then, if we take $\{\alpha_i\}_{i \in \{1\}} \subseteq F$ defined by $\alpha_1 = v$, we have

$$v = v \cdot 1 \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1\}} \alpha_i \cdot e_i$$

proving that

$$\text{span}(\{1\}) = F$$

Hence we have that

$$\{1\} \text{ is a basis for } \langle F, +, \cdot \rangle \text{ and } \dim(F) = \text{card}(\{1\}) = 1 \quad \square$$

Theorem 11.161. Let $\langle V, +, \cdot \rangle$ be a finite dimensional space over a field $\langle F, +, \cdot \rangle$ then we have:

1. If S is a finite set with $\dim(V) < \text{card}(S)$ then S is linear dependent.
2. If $\{v_i\}_{i \in \{1, \dots, m\}} \subseteq V$ is a finite family with $\dim(V) < m$ then $\{v_i\}_{i \in \{1, \dots, m\}}$ is linear dependent

Proof. As V is finite dimensional there exist a basis E for V with $\dim(V) = \text{card}(E)$.

1. Let $\text{card}(S)$ then we can write S as $S = \{s_1, \dots, s_{\text{card}(S)}\}$. Assume that S is linear independent then as $\text{span}(E) = V$ it follows from a consequence of Steinitz theorem [see corollary: 11.125] that $\text{card}(S) \leq \text{card}(E) = \dim(V)$, so, as $\dim(V) < \text{card}(S)$, we reach the contradiction $\text{card}(S) < \text{card}(S)$. Hence the assumption is wrong and S must be dependent.
2. Assume that $\{v_i\}_{i \in \{1, \dots, m\}} \subseteq V$ is linear independent then by [theorem: 11.135] $S = \{v_i | i \in \{1, \dots, m\}\}$ is a linear independent set with $\text{card}(S) = m > \dim(V)$ which is impossible by (1). Hence $\{v_i\}_{i \in \{1, \dots, m\}}$ is a linear independent family. $\square$

The above theorem proves automatically the following .

Corollary 11.162. We have [see: 11.65]

1. The vector space $\langle \mathbb{Q}, +, \cdot \rangle$ over $\langle \mathbb{Q}, +, \cdot \rangle$ has as basis $\{1\}$ and $\dim(\mathbb{Q}) = 1$
2. The vector space $\langle \mathbb{R}, +, \cdot \rangle$ over $\langle \mathbb{R}, +, \cdot \rangle$ has as basis $\{1\}$ and $\dim(\mathbb{R}) = 1$
3. The vector space $\langle \mathbb{C}, +, \cdot \rangle$ over $\langle \mathbb{C}, +, \cdot \rangle$ has as basis $\{1\}$ and $\dim(\mathbb{C}) = 1$

Be aware that that the field for a vector space is important as the following shows.

Theorem 11.163. The vector space $\langle \mathbb{C}, +, \cdot \rangle$ over $\langle \mathbb{R}, +, \cdot \rangle$ [see example: 11.66] has as basis $\{1, i\}$ so that $\dim(\mathbb{C}) = 2$

Proof. Let $\{\alpha_u\}_{u \in \{1, i\}} \subseteq \mathbb{R}$ be such that $\sum_{u \in \{1, i\}} \alpha_u \cdot u = 0$ then we have

$$\begin{aligned} 0 + 0 \cdot i &= \sum_{u \in \{1, i\}} \alpha_u \cdot u \\ &= \alpha_1 \cdot 1 + \alpha_i \cdot i \\ &= \alpha_1 + \alpha_i \cdot i \end{aligned}$$

so that $\alpha_1 = 0 = \alpha_i$ proving that

$$\{1, i\} \text{ is linear independent}$$

Further if $v \in \mathbb{C}$ then there exists $x, y \in \mathbb{R}$ so that $v = x + i \cdot y$. Hence if we define $\{\alpha_u\}_{u \in \{1, i\}}$ by $\alpha_1 = x$ and $\alpha_i = y$ we have

$$\sum_{u \in \{1, i\}} \alpha_u \cdot u = \alpha_1 \cdot 1 + \alpha_i \cdot i = x \cdot 1 + y \cdot i = x + i \cdot y = v$$

proving that

$$\text{span}(\{1, i\}) = \mathbb{C}$$

$\square$

Now for a basis of higher dimensional space we introduce the Kronecker delta.

Definition 11.164. (Kronecker delta) Let $n \in \mathbb{N}_0$ and $\langle F, +, \cdot \rangle$ a field with additive neutral element 0 and multiplicative neutral element 1 then $\{\delta_{i,j}^{[n]}\}_{(i,j) \in \{1, \dots, n\} \times \{1, \dots, n\}}$ is defined by

$$\delta_{i,j}^{[n]} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

If n is know from the context then we write $\delta_{i,j}$ instead of $\delta_{i,j}^{[n]}$

Theorem 11.165. If $n \in \mathbb{N}$ and $\langle F, +, \cdot \rangle$ a field, $j \in \{1, \dots, n\}$ and $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ then

$$\sum_{i \in \{1, \dots, n\}} \delta_{i,j} \cdot \alpha_i = \alpha_j$$

Proof. We have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \delta_{i,j} \cdot \alpha_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{j\}} \delta_{i,j} \cdot \alpha_i + \sum_{i \in \{j\}} \delta_{i,j} \cdot \alpha_i \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{j\}} \delta_{i,j} \cdot \alpha_i + \delta_{j,j} \cdot \alpha_j \\ &= \sum_{i \in \{1, \dots, n\} \setminus \{j\}} (0 \cdot \alpha_i) + 1 \cdot \alpha_j \\ &\stackrel{[\text{theorem: 11.37}]}{=} \alpha_j \\ &\quad \square \end{aligned}$$

We have a similar theorem for vector spaces.

Theorem 11.166. If $n \in \mathbb{N}$, $\langle X, +, \cdot \rangle$ a vector space over a field $\langle F, +, \cdot \rangle$ a field, $j \in \{1, \dots, n\}$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ then

$$\sum_{i \in \{1, \dots, n\}} \delta_{i,j} \cdot x_i = x_j$$

Proof. We have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \delta_{i,j} \cdot x_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{j\}} \delta_{i,j} \cdot x_i + \sum_{i \in \{j\}} \delta_{i,j} \cdot x_i \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{j\}} \delta_{i,j} \cdot x_i + \delta_{j,j} \cdot x_j \\ &= \sum_{i \in \{1, \dots, n\} \setminus \{j\}} (0 \cdot x_i) + 1 \cdot x_j \\ &\stackrel{[\text{theorem: 11.37}]}{=} x_j \\ &\quad \square \end{aligned}$$

Theorem 11.167. Let $n \in \mathbb{N}$, $\langle F, +, \cdot \rangle$ a field and $\langle F^n, +, \cdot \rangle$ the vector space over $\langle F, +, \cdot \rangle$ [see theorem: 11.71] define

$$e: \{1, \dots, n\} \rightarrow F^n \text{ where } e(i) \in F^n \text{ is defined by } (e(i))_j = \delta_{i,j}$$

then e is injective so that

$$e: \{1, \dots, n\} \rightarrow e(\{1, \dots, n\}) = E \text{ is a bijection}$$

defining $E = \{e_1, \dots, e_n\}$. Then $E = \{e_1, \dots, e_n\}$ is a basis for F^n so that $\dim(F^n) = n$. Further

$$\forall x = (x_1, \dots, x_n) \in F^n \text{ we have } x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$$

$E = \{e_1, \dots, e_n\}$ is called the **canonical basis for F^n** .

Proof. Let $i, j \in \{1, \dots, n\}$ such that $e_i = e_j$. Assume that $i \neq j$ then we have

$$1 = \delta_{i,i} = (e_i)(i) = (e_j)(i) = \delta_{i,j} = 0$$

leading to the contradiction $1 = 0$ [see definition: 4.51], hence we must have that $e: \{1, \dots, n\} \rightarrow F^n$ is an injection and thus that $e: \{1, \dots, n\} \rightarrow F^n$ is a bijection defining the finite distinct set

$$E = \{e_1, \dots, e_n\}$$

Let now $x = (x_1, \dots, x_n) \in F^n$ then for $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ we have that $\forall j \in \{1, \dots, n\}$

$$\begin{aligned} \left(\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right)_j &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{i \in \{1, \dots, n\}} (x_i \cdot e_i)_j \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot (e_i)_j \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot \delta_{i,j} \\ &\stackrel{[\text{theorem: 11.165}]}{=} x_j \end{aligned}$$

proving that

$$\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i = x$$

so that $x \in \text{span}(\{e_1, \dots, e_n\})$, proving $F^n \subseteq \text{span}(\{e_1, \dots, e_n\}) \subseteq F^n$. Hence we have

$$\text{span}(E) = F^n \quad (11.71)$$

Further if $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ is such that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i = 0$ then for every $j \in \{1, \dots, n\}$ we have:

$$\begin{aligned} 0 &= 0_j \\ &= \left(\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i \right)_j \\ &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot (e_i)_j \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot \delta_{i,j} \\ &\stackrel{[\text{theorem: 11.165}]}{=} \alpha_j \end{aligned}$$

proving by [theorem: 11.120] that

$$E \text{ is linear independent} \quad (11.72)$$

From [eqs: 11.71, 11.72] it follows that

$$E \text{ is a basis of } F^n \quad \square$$

11.4 Linear mappings

11.4.1 Linear mappings

Similar to the concepts of group, ring, field homeomorphisms we have also mappings that preserves the structure of vector spaces, these are called linear mappings. Linear mappings are very important in Banach spaces, Hilbert spaces, differential analysis and so on.

Definition 11.168. (Linear Mapping) If $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ are vector spaces over a field $\langle F, +, \cdot \rangle$ then a function $L: X \rightarrow Y$ is a **linear mapping** if

1. $\forall x, y \in X$ we have $L(x + y) = L(x) + L(y)$
2. $\forall x \in X, \forall \alpha \in \mathcal{F}$ we have that $L(\alpha \cdot x) = \alpha \cdot L(x)$

The set of graphs of linear mappings is $\text{Hom}(X, Y)$ so that

$$\text{Hom}(X, Y) = \{L \in Y^X \mid L: X \rightarrow Y \text{ is a linear mapping}\}$$

If $L \in \text{Hom}(X, X)$ then $L: X \rightarrow Y$ is called a **linear transformation** and we note $\text{Hom}(X, X)$ as $\text{Hom}(X)$.

Restrictions of a linear mapping to a sub-space is again a linear mapping.

Theorem 11.169. *If $\langle X, +, \cdot \rangle, \langle Y, +, \cdot \rangle$ be vector spaces over a field $\mathcal{F}$, $W \subseteq X$ a sub-space of X and $L \in \text{Hom}(X, Y)$ then $L|_W \in \text{Hom}(W, Y)$*

Proof. First using [theorem: 11.61] it follows that W is a vector space over the field $\mathcal{F}$ with the induced operators from $\langle X, +, \cdot \rangle$. Further we have:

1. $\forall x, y \in W$ we have $L|_W(x + y) = L(x + y) = L(x) + L(y) = L|_W(x) + L|_W(y)$
2. $\forall x \in W$ and $\alpha \in \mathcal{F}$ we have $L|_W(\alpha \cdot x) = L(\alpha \cdot x) = \alpha \cdot L(x) = \alpha \cdot L|_W(x)$ □

Theorem 11.170. *Let $\langle X, +, \cdot \rangle, \langle Y, +, \cdot \rangle$ are vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then*

1. *If $x \in X$ and $A \subseteq X$ then $L(x + A) = L(x) + L(A)$*
2. *If $A, B \subseteq X$ then $L(A + B) = L(A) + L(B)$*
3. *If $A \subseteq X$ and $\alpha \in F$ then $L(\alpha \cdot A) = \alpha \cdot L(A)$*

Proof.

1. If $y \in L(x + A)$ then there exist a $z \in x + A$ such that $y = L(z)$. As $z \in x + A$ there exist a $a \in A$ such that $z = x + a$ hence $y = L(z) = L(x + a) = L(x) + L(a) \in L(x) + L(A)$. Likewise if $y \in L(x) + L(A)$ then there exist a $z \in L(A)$ such that $y = L(x) + z$. As $z \in L(A)$ there exist a $a \in A$ such that $z = L(a)$. Hence $y = L(x) + z = L(x) + L(a) = L(x + a) \in L(x + A)$.
2. If $y \in L(A + B)$ then there exists a $a \in A, b \in B$ such that $y = L(a + b) = L(a) + L(b)$ so that $y \in L(a) + L(b)$. Likewise if $y \in L(A) + L(B)$ there exist a $a \in A, b \in B$ such that $y = L(a) + L(b) = L(a + b)$ proving, as $a + b \in A + B$, that $y \in L(A + B)$.
3. If $y \in L(\alpha \cdot A)$ then there exist a $a \in A$ such that $y = L(\alpha \cdot a) = \alpha \cdot L(a)$ so that $y \in \alpha \cdot L(A)$. Likewise if $y \in \alpha \cdot L(A)$ then there a $a \in A$ such that $y = \alpha \cdot L(a) = L(\alpha \cdot a)$ proving that $y \in L(\alpha \cdot A)$. □

Theorem 11.171. *Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ are vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ is a linear mapping then $L(0) = 0$.*

Proof. $L(0) = L(0 \cdot 0) = 0 \cdot L(0) = 0$ □

A equivalent definition for a linear mapping is the following theorem:

Theorem 11.172. *If $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ are vector spaces over a field $\langle F, +, \cdot \rangle$ then we have for a function $L: X \rightarrow Y$ that*

$$\begin{aligned} L: X \rightarrow Y \text{ is a linear mapping} \\ \Updownarrow \\ \forall x, y \in X \text{ and } \alpha \in F \text{ we have } L(x + \alpha \cdot y) = L(x) + \alpha \cdot L(y) \end{aligned}$$

Proof.

$\Rightarrow$. Let $x, y \in X$ and $\alpha \in F$ we have as L is linear that

$$L(x + \alpha \cdot y) = L(x) + L(\alpha \cdot y) = L(x) + \alpha \cdot L(y)$$

$\Leftarrow$. Let $x, y \in X$ and $\alpha \in F$ then we have

$$L(x + y) = L(x + 1 \cdot y) = L(x) + 1 \cdot L(y) = L(x) + L(y)$$

and

$$L(\alpha \cdot x) = L(0 + \alpha \cdot x) = L(0) + \alpha \cdot L(x) \stackrel{[\text{theorem: 11.171}]}{=} 0 + \alpha \cdot L(x) = \alpha \cdot L(x)$$

proving by definition that $L: X \rightarrow Y$ is linear. □

Example 11.173. Consider the vector space $\langle \mathbb{C}, +, \cdot \rangle$ and $\langle \mathbb{R}, +, \cdot \rangle$ over $\langle \mathbb{R}, +, \cdot \rangle$ then we have that

1. $\text{Img}: \mathbb{C} \rightarrow \mathbb{R}$ defined by $\text{Img}(x + i \cdot y) = y$ is linear
2. $\text{Re}: \mathbb{C} \rightarrow \mathbb{R}$ defined by $\text{Re}(x + i \cdot y) = x$ is linear
3. If $\{x_i\}_{i \in I} \subseteq \mathbb{C}$ is family with finite support then
 - a. $\text{Img}(\sum_{i \in I} x_i) = \sum_{i \in I} \text{Img}(x_i)$
 - b. $\text{Re}(\sum_{i \in I} x_i) = \sum_{i \in I} \text{Re}(x_i)$

Proof.

1. This follows from [theorem: 10.80].
2. This follows from [theorem: 10.80].
3. This follows from (1), (2) and [theorem: 11.190] □

Example 11.174. Let $n \in \mathbb{N}$, $\langle F, +, \cdot \rangle$ be a field and $\langle F^n, +, \cdot \rangle$ be the vector space over $\langle F, +, \cdot \rangle$ defined by [theorem: 11.71] then $\forall i \in \{1, \dots, n\}$ we have that the projection mapping

$$\pi_i: F^n \rightarrow F$$

is a linear mapping.

Proof. If $x, y \in F^n$ and $\alpha \in F$ then we have

$$\pi_i(x + y) = (x + y)_i = x_i + y_i = \pi_i(x) + \pi_i(y)$$

and

$$\pi_i(\alpha \cdot x) = \alpha \cdot x_i = \alpha \cdot \pi_i(x) \quad \square$$

We can extend the above example to a more general case.

Example 11.175. Let $\{\langle X_i, +_i, \cdot_i \rangle\}_{i \in I}$ be a family of vector spaces over a field $\langle F, +, \cdot \rangle$ and $\langle \prod_{i \in I} X_i, +, \cdot \rangle$ the vector space defined in [theorem: 11.76] then $\forall i \in I$

$$\pi_i: \prod_{i \in I} X_i \rightarrow X_i \text{ defined by } \pi_i(x) = x_i \text{ is a linear mapping}$$

Proof. Let $x, y \in \prod_{i \in I} X_i$ and $\alpha \in F$ then we have

$$\pi_i(x + y) = (x + y)_i \stackrel{[\text{theorem: 11.76}]}{=} x_i +_i y_i = \pi_i(x) +_i \pi_i(y)$$

and

$$\pi_i(\alpha \cdot x) = (\alpha \cdot x)_i = \alpha \cdot_i x_i = \alpha \cdot_i \pi_i(x) \quad \square$$

Definition 11.176. (Linear Isomorphism) If $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ are vector spaces over a field $\langle F, +, \cdot \rangle$ then a mapping $L: X \rightarrow Y$ is a **linear isomorphism** iff

1. $L: X \rightarrow Y$ is a bijection
2. $L: X \rightarrow Y$ is a linear mapping

If between two vector spaces a linear isomorphism exists then we say that the **vector spaces are isomorphic**.

Example 11.177. Let $\langle \mathbb{R}^2, +, \cdot \rangle$ and $\langle \mathbb{C}, +, \cdot \rangle$ be vector spaces over the field $\langle \mathbb{R}, +, \cdot \rangle$ [see theorems: 11.167, 11.163] then $\mathcal{C}: \mathbb{R}^2 \rightarrow \mathbb{C}$ defined by $\mathcal{C}((x, y)) = x + i \cdot y$ is a linear isomorphism so that $\langle \mathbb{R}^2, +, \cdot \rangle$ is isomorphic with $\langle \mathbb{C}, +, \cdot \rangle$.

Proof. If $\mathcal{C}((x, y)) = \mathcal{C}((x', y'))$ then $x + i \cdot y = x' + i \cdot y'$ then by [theorem: 10.77] $x = x'$ and $y = y'$ so that $(x, y) = (x', y')$, hence

$$\mathcal{C}: \mathbb{R}^2 \rightarrow \mathbb{C} \text{ is injective}$$

Further if $z \in \mathbb{C}$ then $z = x + i \cdot y = \mathcal{C}((x, y))$ proving surjectivity. combining this with the above proves that

$\mathcal{C}: \mathbb{R}^2 \rightarrow \mathbb{C}$ is a bijection

Let $(x, y), (x', y') \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$

$$\begin{aligned} \mathcal{C}((x, y) + (x', y')) &= \mathcal{C}((x + x', y + y')) \\ &= (x + x') + i \cdot (y + y') \\ &= (x + i \cdot y) + (x' + i \cdot y') \\ &= \mathcal{C}((x, y)) + \mathcal{C}((x', y')) \\ \mathcal{C}(\alpha \cdot (x, y)) &= \mathcal{C}((\alpha \cdot x, \alpha \cdot y)) \\ &= (\alpha \cdot x) + i \cdot (\alpha \cdot y) \\ &= \alpha \cdot (x + i \cdot y) \\ &= \alpha \cdot \mathcal{C}((x, y)) \end{aligned}$$

proving that

$\mathcal{C}: \mathbb{R}^2 \rightarrow \mathbb{C}$ is a isomorphism □

Theorem 11.178. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over the field $\langle F, +, \cdot \rangle$ and

$L: X \rightarrow Y$ a linear isomorphism

then

$L^{-1}: Y \rightarrow X$ is a linear isomorphism

Proof. Using [theorem: 2.74] we have that

$L^{-1}: Y \rightarrow X$ is a bijection

Let $x, y \in Y$ and $\alpha \in F$ then we have

$$\begin{aligned} L^{-1}(x) + L^{-1}(y) &= L^{-1}(L(L^{-1}(x)) + L^{-1}(y)) \\ &= L^{-1}(L(L^{-1}(x)) + L(L^{-1}(y))) \\ &= L^{-1}(x + y) \end{aligned}$$

Likewise

$$\begin{aligned} \alpha \cdot L^{-1}(x) &= L^{-1}(L(\alpha \cdot L^{-1}(x))) \\ &= L^{-1}(\alpha \cdot L(L^{-1}(x))) \\ &= L^{-1}(\alpha \cdot x) \end{aligned}$$

□

Example 11.179. Let $\langle F, +, \cdot \rangle$ be the vector space over the field $\langle F, +, \cdot \rangle$ [see example: 11.64] and $L: F \rightarrow F$ a function then:

1. L is linear $\Leftrightarrow L = L(1) \cdot \text{Id}_F$
2. L is a linear isomorphism $\Leftrightarrow L(1) \neq 0$ and $L = L(1) \cdot \text{Id}_F$

Proof.

1.

$\Rightarrow$. $\forall x \in F$ that

$$L(x) = L(x \cdot 1) = x \cdot L(1) = L(1) \cdot x = L(1) \cdot \text{Id}_F(x) = (L(1) \cdot \text{Id}_F)(x)$$

which proves that $L = L(1) \cdot \text{Id}_F$.

$\Leftarrow$. If $L = \alpha \cdot \text{Id}_F$ then we have:

$$\text{a. } \forall x, y \in F \quad L(x + y) = L(1) \cdot (x + y) = L(1) \cdot x + L(1) \cdot y = L(x) + L(y).$$

$$\text{b. } \forall x \in F, \forall \alpha \in F \text{ we have } L(\alpha \cdot x) = L(1) \cdot (\alpha \cdot x) = \alpha \cdot (L(1) \cdot x) = \alpha \cdot L(x)$$

which proves that L is a linear mapping.

2.

$\Rightarrow$. As L is a linear isomorphism it is by definition linear so that by (1) we have that $L = L(1) \cdot \text{Id}_F$. Assume that $L(1) = 0$ then we have as $L(0) = 0$ and L is injective that $1 = 0$ which contradicts the definition of a field [see definition: 4.51] where $1 \neq 0$. Hence we must have that $L(1) \neq 0$.

$\Leftarrow$. As $L = L(1) \cdot \text{Id}_F$ we have by (1) that L is a linear mapping. Further we have:

injectivity. If $L(x) = L(y)$ then $L(1) \cdot x = L(1) \cdot y$ we have as $L(1) \neq 0$ that $x = y$ proving injectivity.

surjectivity. Given $y \in F$ take $x = L(1)^{-1} \cdot y$ then

$$L(x) = L(L(1)^{-1} \cdot y) = L(1) \cdot (L(1)^{-1} \cdot y) = y$$

which proves surjectivity.

proving that L is also bijective and thus by definition that L is a linear isomorphism. $\square$

We look now at the composition of linear mappings.

Theorem 11.180. Let $\langle X, +, \cdot \rangle$, $\langle Y, +, \cdot \rangle$ and $\langle Z, +, \cdot \rangle$ be vector spaces over the field $\langle F, +, \cdot \rangle$ and $L_1 \in \text{Hom}(X, Y)$, $L_2 \in \text{Hom}(Y, Z)$ then $L_2 \circ L_1 \in \text{Hom}(X, Z)$

Proof. Let $x, y \in X$ and $\alpha \in f$ then we have for the function

$$L_2 \circ L_1 : X \rightarrow Z$$

$$\begin{aligned} (L_2 \circ L_1)(x + y) &= L_2(L_1(x + y)) \\ &= L_2(L_1(x) + L_1(y)) \\ &= L_2(L_1(x)) + L_2(L_1(y)) \\ &= (L_2 \circ L_1)(x) + (L_2 \circ L_1)(y) \\ (L_2 \circ L_1)(\alpha \cdot x) &= L_2(L_1(\alpha \cdot x)) \\ &= L_2(\alpha \cdot L_1(x)) \\ &= \alpha \cdot L_2(L_1(x)) \\ &= \alpha \cdot (L_2 \circ L_1)(x) \end{aligned}$$

$\square$

Theorem 11.181. Let $\langle X, +, \cdot \rangle$, $\langle Y, +, \cdot \rangle$ and $\langle Z, +, \cdot \rangle$ be vector spaces over the field $\langle F, +, \cdot \rangle$ then we have:

1. If $L \in \text{Hom}(Y, Z)$ and $L_1, L_2 \in \text{Hom}(X, Y)$ then $L \circ (L_1 + L_2) = L \circ L_1 + L \circ L_2$
2. If $\alpha \in F$, $L_1 \in \text{Hom}(X, Y)$ and $L_2 \in \text{Hom}(Y, Z)$ then $L_2 \circ (\alpha \cdot L_1) = \alpha \cdot (L_2 \circ L_1)$

Proof.

1. Let $x \in X$ then we have

$$\begin{aligned} (L \circ (L_1 + L_2))(x) &= L((L_1 + L_2)(x)) \\ &= L(L_1(x) + L_2(x)) \\ &= L(L_1(x)) + L(L_2(x)) \\ &= (L \circ L_1)(x) + (L \circ L_2)(x) \\ &= (L \circ L_1 + L \circ L_2)(x) \end{aligned}$$

proving that

$$L \circ (L_1 + L_2) = L \circ L_1 + L \circ L_2$$

2. Let $x \in X$ then we have:

$$\begin{aligned} (L_2 \circ (\alpha \cdot L_1))(x) &= L_2((\alpha \cdot L_1)(x)) \\ &= L_2(\alpha \cdot L_1(x)) \\ &= \alpha \cdot L_2(L_1(x)) \\ &= \alpha \cdot (L_2 \circ L_1)(x) \\ &= (\alpha \cdot (L_2 \circ L_1))(x) \end{aligned}$$

proving that

$$L_2 \circ (\alpha \cdot L_1) = \alpha \cdot (L_2 \circ L_1) \quad \square$$

It turns out that $\text{Hom}(X, X)$ together with the composition operator $\circ$ forms a semi-group allowing us to define the composition of more than one linear transformations.

Corollary 11.182. *Let $\langle X, +, \cdot \rangle$, $\langle Y, +, \cdot \rangle$ and $\langle Z, +, \cdot \rangle$ be vector spaces over the field $\langle F, +, \cdot \rangle$ and $L_1: X \rightarrow Y$ and $L_2: Y \rightarrow Z$ linear isomorphism then $L_2 \circ L_1: X \rightarrow Z$ is a linear isomorphism.*

Proof. This follows from [theorems: 2.77 and 11.180] $\square$

Theorem 11.183. *Let $\langle X, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then*

$$\langle \text{Hom}(X, X), \circ \rangle$$

is a semi-group with neutral element Id_X .

Proof. First using the previous theorem [theorem: 11.180] we have that the following mapping is well defined

$$\circ: \text{Hom}(X, X) \times \text{Hom}(X, X) \rightarrow \text{Hom}(X, X) \text{ where } \circ(L, K) = L \circ K$$

Further we have for the operator $\circ$ that

associativity. $\forall L_1, L_2, L_3 \in \text{Hom}(X, X)$ we have

$$L_1 \circ (L_2 \circ L_3) \stackrel{[\text{theorem: 2.21}]}{=} (L_1 \circ L_2) \circ L_3$$

neutral element. $\forall L \in \text{Hom}(X, X)$ we have

$$L \circ \text{Id}_X \stackrel{[\text{theorem: 2.49}]}{=} L \stackrel{[\text{theorem: 2.49}]}{=} \text{Id}_X \circ L \quad \square$$

Using $\circ$ as the product operator in the semi-group we can define the finite product (composition) of linear transformations.

Definition 11.184. *Let $n \in \mathbb{N}$, X a finite dimensional vector space with $\dim(X) = n$ and $\{L_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Hom}(X, X)$ then*

$$L_1 \circ \dots \circ L_n = \prod_{i=1}^m L_i$$

[see remark: 11.2]

In [theorem: 11.69] it is proved that $\langle Y^X, +, \cdot \rangle$ is a vector space where for $f, g \in Y^X$, $\alpha \in F$:

1. $f + g: X \rightarrow Y$ is defined by $(f + g)(x) = f(x) + g(x)$
2. $\alpha \cdot f: X \rightarrow Y$ is defined by $(\alpha \cdot f)(x) = \alpha \cdot f(x)$
3. $C_0: X \rightarrow Y$ is defined by $0(x) = 0$
4. $(-f): X \rightarrow Y$ is defined by $(-f)(x) = -f(x)$

As $\text{Hom}(X, Y) \subseteq Y^X$, a natural question to ask is: Is $\text{Hom}(X, Y)$ a sub-space of $\langle Y^X, +, \cdot \rangle$. It turns out that the answer is yes, as is shown in the next theorem.

Theorem 11.185. *Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over the field $\langle F, +, \cdot \rangle$ then*

$$\text{Hom}(X, Y) \text{ is a sub-space of } \langle Y^X, +, \cdot \rangle$$

Applying then [theorem: 11.61] proves that $\langle \text{Hom}(X, Y), +, \cdot \rangle$ is vector space over the field $\langle F, +, \cdot \rangle$.

Proof. Let $L_1, L_2 \in \text{Hom}(X, Y)$, $\alpha \in F$ then for $x, y \in X$ and $\gamma \in F$ we have

$$\begin{aligned}
 (\alpha \cdot L_1 + L_2)(x + y) &= (\alpha \cdot L_1)(x + y) + L_2(x + y) \\
 &= \alpha \cdot L_1(x + y) + L_2(x + y) \\
 &= \alpha \cdot (L_1(x) + L_1(y)) + L_2(x) + L_2(y) \\
 &= \alpha \cdot L_1(x) + \alpha \cdot L_1(y) + L_2(x) + L_2(y) \\
 &= \alpha \cdot L_1(x) + L_2(x) + \alpha \cdot L_1(y) + L_2(y) \\
 &= (\alpha \cdot L_1)(x) + L_2(x) + (\alpha \cdot L_1)(y) + L_2(y) \\
 &= (\alpha \cdot L_1 + L_2)(x) + (\alpha \cdot L_1 + L_2)(y) \\
 (\alpha \cdot L_1 + L_2)(\gamma \cdot x) &= (\alpha \cdot L_1)(\gamma \cdot x) + L_2(\gamma \cdot x) \\
 &= \alpha \cdot L_1(\gamma \cdot x) + L_2(\gamma \cdot x) \\
 &= \alpha \cdot (\gamma \cdot L_1(x)) + \gamma \cdot L_2(x) \\
 &= \gamma \cdot (\alpha \cdot L_1(x)) + \gamma \cdot L_2(x) \\
 &= \gamma \cdot ((\alpha \cdot L_1)(x) + L_2(x)) \\
 &= \gamma \cdot (\alpha \cdot L_1 + L_2)(x) \\
 C_0(x + y) &= 0 \\
 &= 0 + 0 \\
 &= 0(x) + 0(y) \\
 C_0(\gamma \cdot x) &= 0 \\
 &= \gamma \cdot 0 \\
 &= \gamma \cdot 0(x)
 \end{aligned}$$

proving that

$$\alpha \cdot L_1 + L_2 \in \text{Hom}(X, Y) \text{ and } C_0 \in \text{Hom}(X, Y) \Rightarrow \text{Hom}(X, Y) \neq \emptyset$$

hence by [definition: 11.59] we have that $\text{Hom}(X, Y)$ is a sub-space of $\langle Y^X, +, \cdot \rangle$ □

Definition 11.186. (Dual Space) Let $\langle X, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ [which is a vector space over itself] then $\text{Hom}(X, F)$ is called the **dual space** of X and noted as X^* .

Theorem 11.187. Let $\langle X, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ [which is a vector space over itself] and $x \in X$ with $x \neq 0$ then there exists a $L \in \text{Hom}(X, F) = X^*$ such that $L(x) = 1$

Proof. As $x \neq 0$ we have by [theorem: 11.122] that $\{x\}$ is a linear independent, using [theorem: 11.146] there exists a basis B of $\langle V, +, \cdot \rangle$ such that $\{x\} \subseteq B$. Let $y \in X$ then by [theorem: 11.139] there exists a **unique** $\{\alpha_w^{(y)}\}_{w \in B} \subseteq F$ such that $y = \sum_{w \in B} \alpha_w^{(y)} \cdot w$. This allows us to define

$$L: X \rightarrow F \text{ by } L(y) = \alpha_x^{(y)}$$

As $x \in B$ we have that for $\{\alpha_w\}_{w \in B}$ defined by $\alpha_w = \begin{cases} 1 & \text{if } w = x \\ 0 & \text{if } w \in B \setminus \{x\} \end{cases}$ that

$$\begin{aligned}
 \sum_{w \in B} \alpha_w \cdot w &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{w \in B \setminus \{x\}} \alpha_w \cdot w + \sum_{w \in \{x\}} \alpha_w \cdot w \\
 &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{w \in B \setminus \{x\}} \alpha_w \cdot w + \alpha_x \cdot x \\
 &= \sum_{w \in B \setminus \{x\}} 0 \cdot w + 1 \cdot x \\
 &\stackrel{[\text{theorem: 11.37}]}{=} 1 \cdot x \\
 &= x \\
 &= \sum_{w \in B} \alpha_w^{(x)} \cdot w
 \end{aligned}$$

so that by uniqueness of the extension in a basis we have $\{\alpha_w\}_{w \in B} = \{\alpha_w^{(x)}\}_{w \in B}$ then $L(x) = \alpha_x^{(x)} = \alpha_x = 1$ or

$$L(x) = 1 \quad (11.73)$$

Further we have if $y_1, y_2 \in X$ then

$$\begin{aligned} y_1 + y_2 &= \sum_{w \in B} \alpha_w^{(y_1)} \cdot w + \sum_{w \in B} \alpha_w^{(y_2)} \cdot w \\ &\stackrel{[\text{theorem: 11.38}]}{=} \sum_{w \in B} (\alpha_w^{(y_1)} + \alpha_w^{(y_2)}) \cdot w \\ &= \sum_{w \in B} \alpha_w^{(y_1 + y_2)} \cdot w \end{aligned}$$

so that by uniqueness we have $L(y_1 + y_2) = \alpha_x^{(y_1 + y_2)} = \alpha_x^{(y_1)} + \alpha_x^{(y_2)} = L(y_1) + L(y_2)$ so that

$$L(y_1 + y_2) = L(y_1) + L(y_2) \quad (11.74)$$

Finally if $y \in X$ and $\alpha \in F$ we have

$$\begin{aligned} \alpha \cdot w &= \alpha \cdot \sum_{w \in B} \alpha_w^{(y)} \cdot w \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{w \in B} \alpha \cdot (\alpha_w^{(y)} \cdot w) \\ &= \sum_{w \in B} (\alpha \cdot \alpha_w^{(y)}) \cdot w \\ &= \sum_{w \in B} \alpha_w^{(\alpha \cdot y)} \cdot w \end{aligned}$$

so that by uniqueness we have $L(\alpha \cdot y) = \alpha_x^{(\alpha \cdot y)} = \alpha \cdot \alpha_x^{(y)} = \alpha \cdot L(y)$ proving

$$L(\alpha \cdot y) = \alpha \cdot L(y) \quad (11.75)$$

The theorem is then proved by [eqs: 11.73, 11.74, 11.75]. $\square$

Theorem 11.188. Let $\langle X, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ [which is a vector space over itself]. Let $x, y \in X$ such that $\forall L \in \text{Hom}(X, F)$ we have $L(x) = L(y)$ then $x = y$.

Proof. Let $x, y \in X$ such that $\forall L \in \text{Hom}(X, F)$ $L(x) = L(y)$. Assume that $x \neq y$ then $x - y \neq 0$ so by the previous theorem [theorem: 11.187] there exists a $L \in \text{Hom}(X, F)$ such that $L(x - y) = 1$, hence we have $0 \stackrel{L(x)=L(y)}{=} L(x) - L(y) = L(x - y) = 1$ leading to the contradiction $1 = 0$, hence we must have that $x = y$. $\square$

Theorem 11.189. Let $\langle X, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, $n \in \mathbb{N}$ and $\{e_i\}_{i \in \{1, \dots, n\}} \subseteq X$ a linear independent family then there exist a $L \in \text{Hom}(X, F)$ such that

$$L(e_1) = 1 \text{ and } \forall i \in \{2, \dots, n\} \ L(e_i) = 0$$

Proof. As $\{e_i\}_{i \in \{1, \dots, n\}} \subseteq X$ is linear independent we have by [theorem: 11.135] that $E = \{e_1, \dots, e_n\}$ is a linear independent set. Using [theorem: 11.146] then there exist a basis B of X such that $E \subseteq B$. Let $y \in X$ then there exist by [theorem: 11.140] a finite $I \subseteq B$ and a $\{\alpha_i\}_{i \in I} \subseteq F$ such that

$$y = \sum_{i \in I} \alpha_i \cdot e_i$$

Let

$$L_y^{\{\alpha_i\}_{i \in I}} = \begin{cases} 0 & \text{if } e_1 \notin I \\ \alpha_{e_1} & \text{if } e_1 \in I \end{cases}$$

then if there is another $J \subseteq B$ and $\{\beta_i\}_{i \in J} \subseteq F$ such that

$$y = \sum_{i \in J} \beta_i \cdot e_i$$

and let

$$L_y^{\{\beta_i\}_{i \in I \cup J}} = \begin{cases} 0 & \text{if } e_1 \notin J \\ \beta_{e_1} & \text{if } e_1 \in J \end{cases}$$

then by [theorem: 11.140] we have that

$$\begin{aligned} \forall u \in I \setminus J & \quad \alpha_u = 0 \\ \forall u \in J \setminus I & \quad \beta_u = 0 \\ \forall u \in I \cap J & \quad \alpha_u = \beta_u \end{aligned}$$

For e_1 we have then either

$$e_1 \in I \wedge e_1 \in J. \text{ Then } e_1 \in I \cap J \text{ so that } L_y^{\{\beta_i\}_{i \in J}} = \beta_{e_1} = \alpha_{e_1} = L_y^{\{\alpha_i\}_{i \in I}}$$

$$e_1 \notin I \wedge e_1 \in J. \text{ Then } L_y^{\{\beta_i\}_{i \in J}} = \beta_{e_1} = 0 = L_y^{\{\alpha_i\}_{i \in I}}$$

$$e_1 \in I \wedge e_1 \notin J. \text{ Then } L_y^{\{\beta_i\}_{i \in J}} = 0 = \alpha_{e_1} = L_y^{\{\beta_i\}_{i \in J}}$$

$$e_1 \notin I \wedge e_1 \notin J. \text{ Then } L_y^{\{\beta_i\}_{i \in J}} = 0 = L_y^{\{\beta_i\}_{i \in J}}$$

proving that in all cases $L_y^{\{\alpha_i\}_{i \in I}} = L_y^{\{\beta_i\}_{i \in J}}$, hence the following function is well defined:

$$L: X \rightarrow F \text{ by } L(y) = L_y^{\{\alpha_i\}_{i \in I}}$$

Let now $\alpha \in F$, $x, y \in X$ then there exists finite $I, J \subseteq B$ and $\{\alpha_i\}_{i \in I} \subseteq F$, $\{\beta_i\}_{i \in J} \subseteq F$ such that

$$x = \sum_{i \in I} \alpha_i \cdot i \text{ and } y = \sum_{i \in I} \beta_i \cdot i$$

Define

$$\{\gamma_i\}_{i \in I \cup J} \subseteq F \text{ by } \gamma_i = \begin{cases} \alpha_i & \text{if } i \in I \setminus J \\ \alpha \cdot \beta_i & \text{if } i \in J \setminus I \\ \alpha_i + \alpha \cdot \beta_i & \text{if } i \in I \cap J \end{cases}$$

then we have:

$$\begin{aligned} \sum_{i \in I \cup J} \gamma_i \cdot i & \stackrel{[\text{theorem: 11.44}]}{=} \sum_{i \in I \cap J} \gamma_i \cdot i + \sum_{i \in I \setminus J} \gamma_i \cdot i + \sum_{i \in J \setminus I} \gamma_i \cdot i \\ & = \sum_{i \in I \cap J} (\alpha_i + \alpha \cdot \beta_i) \cdot i + \sum_{i \in I \setminus J} \alpha_i \cdot i + \sum_{i \in J \setminus I} \alpha \cdot \beta_i \cdot i \\ & \stackrel{[\text{theorem: 11.38, 11.80}]}{=} \underbrace{\sum_{i \in I \cap J} \alpha_i \cdot i}_1 + \underbrace{\alpha \cdot \sum_{i \in I \cap J} \beta_i \cdot i}_2 + \underbrace{\sum_{i \in I \setminus J} \alpha_i \cdot i}_1 + \underbrace{\alpha \cdot \sum_{i \in J \setminus I} \beta_i \cdot i}_2 \\ & \stackrel{[\text{theorem: 11.44}]}{=} \underbrace{\sum_{i \in I} \alpha_i \cdot i}_1 + \underbrace{\alpha \cdot \sum_{i \in J} \beta_i \cdot i}_2 \\ & = x + \alpha \cdot y \end{aligned}$$

from the above we have that

$$\begin{aligned} L(x + \alpha \cdot y) & = L_{x + \alpha \cdot y}^{\{\gamma_i\}_{i \in I \cup J}} \\ & = \begin{cases} 0 & \text{if } e_1 \notin I \cup J \\ \gamma_{e_1} & \text{if } e_1 \in I \cup J \end{cases} \\ & = \begin{cases} 0 & \text{if } e_1 \notin I \cup J \\ \alpha_{e_1} + \alpha \cdot \beta_{e_1} & \text{if } e_1 \in I \cup J \end{cases} \\ & \stackrel{e_1 \notin I \cup J \Leftrightarrow e_1 \notin I \wedge e_1 \notin J}{=} \begin{cases} 0 & \text{if } e_1 \notin I \\ \alpha_{e_1} & \text{if } e_1 \in I \cup J \end{cases} + \alpha \cdot \begin{cases} 0 & \text{if } e_1 \notin J \\ \beta_{e_1} & \text{if } e_1 \in I \cup J \end{cases} \\ & = L_x^{\{\alpha_i\}_{i \in I}} + \alpha \cdot L_y^{\{\beta_i\}_{i \in J}} \\ & = L(x) + \alpha \cdot L(y) \end{aligned}$$

proving by [theorem: 11.172] that

$$L \in \text{Hom}(X, F)$$

Define now $\{\zeta_i\}_{i \in \{e_1\}} \subseteq B$ by $\zeta_{e_1} = 1$ then $\sum_{i \in \{e_1\}} \zeta_i \cdot i \stackrel{[\text{theorem: 11.34}]}{=} e_1$ so that

$$L(e_1) = \begin{cases} 0 & \text{if } e_1 \notin \{e_1\} \\ 1 & \text{if } e_1 \in \{e_1\} \end{cases} = 1$$

If $i \in \{2, \dots, n\}$ then as $\{e_i | i \in \{1, \dots, n\}\} = \{e_1, \dots, e_n\}$ $e_1 \neq e_i \Rightarrow e_1 \notin \{e_i\}$, define $\{\eta_k\}_{k \in \{e_i\}} \subseteq F$ by $\eta_k = 1$ then $\sum_{k \in \{e_i\}} \eta_k \cdot k \stackrel{[\text{theorem: 11.34}]}{=} e_i$ so that

$$L(e_i) = \begin{cases} 0 & \text{if } e_1 \notin \{e_i\} \\ 1 & \text{if } e_1 \in \{e_i\} \end{cases} = 0$$

Hence we have

$$L(e_1) = 1 \text{ and } \forall i \in \{2, \dots, n\} \ L(e_i) = 0$$

□

Theorem 11.190. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ a linear mapping between X and Y then:

1. If $n \in \mathbb{N}_0$ and $\{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq F$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq X$ then

$$L\left(\sum_{i=0}^n \alpha_i \cdot x\right) = \sum_{i=0}^n \alpha_i \cdot L(x_i)$$

2. If $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{\alpha_i\}_{i \in \{n, \dots, m\}} \subseteq F$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq X$ then

$$L\left(\sum_{i=n}^m \alpha_i \cdot x\right) = \sum_{i=n}^m \alpha_i \cdot L(x_i)$$

3. If $W \subseteq X$ is a finite set and $\{\alpha_w\}_{w \in W} \subseteq F$ then

$$L\left(\sum_{w \in W} \alpha_w \cdot w\right) = \sum_{w \in W} \alpha_w \cdot L(w)$$

Proof.

1. This is proved by induction, let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq F \text{ and } \{x_i\}_{i \in \{0, \dots, n\}} \subseteq X \text{ are families then } L\left(\sum_{i=0}^n \alpha_i \cdot x\right) = \sum_{i=0}^n \alpha_i \cdot L(x_i) \right\}$$

then we have:

$0 \in S$. If $\{\alpha_i\}_{i \in \{0, \dots, 0\}} \subseteq F$ and $\{x_i\}_{i \in \{0, \dots, 0\}} \subseteq X$ are families then we have

$$\begin{aligned} L\left(\sum_{i=0}^0 \alpha_i \cdot x_i\right) &\stackrel{\text{def}}{=} L(a_0 \cdot x_0) \\ &= \alpha_0 \cdot L(x_0) \\ &\stackrel{\text{def}}{=} \sum_{i=0}^0 \alpha_i \cdot L(x_i) \end{aligned}$$

proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{\alpha_i\}_{i \in \{0, \dots, n+1\}} \subseteq F$ and $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq X$ be families then we have

$$\begin{aligned} L\left(\sum_{i=0}^{n+1} \alpha_i \cdot x_i\right) &\stackrel{\text{def}}{=} L\left(\sum_{i=0}^n \alpha_i \cdot x_i + \alpha_{n+1} \cdot x_{n+1}\right) \\ &= L\left(\sum_{i=0}^n \alpha_i \cdot x_i\right) + \alpha_{n+1} \cdot L(x_{n+1}) \\ &\stackrel{n \in S}{=} \sum_{i=0}^n \alpha_i \cdot L(x_i) + \alpha_{n+1} \cdot L(x_{n+1}) \\ &\stackrel{\text{def}}{=} \sum_{i=0}^{n+1} \alpha_i \cdot L(x_i) \end{aligned}$$

proving that $n+1 \in S$

2. Let $n, m \in \mathbb{N}_0$ with $n \leq m$ and $\{\alpha_i\}_{i \in \{n, \dots, m\}} \subseteq F$ and $\{x_i\}_{i \in \{n, \dots, m\}} \subseteq X$ then we have

$$\begin{aligned} L\left(\sum_{i=n}^m \alpha_i \cdot x_i\right) &\stackrel{\text{def}}{=} L\left(\sum_{i=0}^{m-n} \alpha_{i+n} \cdot x_{i+n}\right) \\ &\stackrel{(1)}{=} \sum_{i=0}^{m-n} \alpha_{i+n} \cdot L(x_{i+n}) \\ &\stackrel{\text{def}}{=} \sum_{i=n}^m \alpha_i \cdot L(x_i) \end{aligned}$$

3. If W is finite we have either:

$W = \emptyset$. Then $\sum_{w \in W} \alpha_w \cdot w = 0$ so that

$$L\left(\sum_{w \in W} \alpha_w \cdot w\right) = L(0) \stackrel{[\text{theorem: 11.171}]}{=} 0 = \sum_{w \in W} \alpha_w \cdot L(w)$$

$W \neq \emptyset$. Then there exist by definition a $n \in \mathbb{N}$ and a $\beta: \{0, \dots, n-1\} \rightarrow W$ such that

$$\sum_{w \in W} \alpha_w \cdot w = \sum_{i=0}^{n-1} \alpha_{w(i)} \cdot w(i)$$

so that

$$\begin{aligned} L\left(\sum_{w \in W} \alpha_w \cdot w\right) &= L\left(\sum_{i=0}^{n-1} \alpha_{w(i)} \cdot w(i)\right) \\ &\stackrel{(1)}{=} \sum_{i=0}^{n-1} \alpha_{w(i)} \cdot L(w(i)) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{w \in W} \alpha_w \cdot L(w) \end{aligned}$$

□

11.4.2 Kernel and image of a linear mapping

Definition 11.191. (Kernel of a Linear Mapping) Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then the **kernel** of L noted as $\ker(L)$ is defined by

$$\ker(L) = \{x \in X \mid L(x) = 0\} = L^{-1}(\{0\})$$

Definition 11.192. (Range of a Linear Mapping) Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then the range of L noted as **range**(L) is defined by

$$\text{range}(L) = L(X)$$

Theorem 11.193. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then

1. $\ker(L)$ is a sub-space of X
2. $\text{range}(L)$ is a sub-space of Y

Proof.

1. First as $L(0) \stackrel{[\text{theorem: 11.171}]}{=} 0$ we have that

$$0 \in \ker(L) \Rightarrow 0 \neq \ker(L)$$

Second if $x, y \in \ker(L)$ and $\alpha \in F$ then we have

$$L(x + y) = L(x) + L(y) \stackrel{[\text{theorem: 11.171}]}{=} 0 + 0 = 0$$

and

$$L(\alpha \cdot x) = \alpha \cdot L(x) \stackrel{[\text{theorem: 11.171}]}{=} \alpha \cdot 0 = 0$$

so that

$$x + y \in \ker(L) \text{ and } \alpha \cdot x \in \ker(L)$$

So we have that $\ker(L)$ is a sub-space of X .

2. If $x, y \in \text{range}(L) = L(X)$ and $\alpha \in F$ then we have a $x', y' \in X$ such that $x = L(x')$ and $y = L(y')$. For $x' + y' \in X$ we have then that

$$L(x' + y') = L(x') + L(y') = x + y$$

so that

$$x + y \in L(X) = \text{range}(L)$$

Likewise for $\alpha \cdot x' \in X$ we have

$$L(\alpha \cdot x') = \alpha \cdot L(x') = \alpha \cdot x$$

proving that

$$\alpha \cdot x \in L(X) = \text{range}(L)$$

□

As $\text{range}(L)$ is a vector space the following definition make sense.

Definition 11.194. (Rank of a Linear Mapping) Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then the **rank of L** noted as $\text{rank}(L)$ is defined by

$$\text{rank}(L) = \dim(\text{range}(L)) = \dim(L(X))$$

Theorem 11.195. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then

$$L \text{ is injective} \Leftrightarrow \ker(L) = \{0\}$$

Proof.

$\Rightarrow$. As $L(0) \stackrel{[\text{theorem: 11.171}]}{=} 0$ we have $0 \in \ker(L)$ so that $\{0\} \subseteq \ker(L)$. Let $x \in \ker(L)$ then $L(x) = 0 \stackrel{[\text{theorem: 11.171}]}{=} L(0)$, as L is injective we have $x = 0$ proving that $\ker(L) \subseteq \{0\}$. Hence we have

$$\ker(L) = \{0\}$$

$\Leftarrow$. Let $x, y \in X$ such that $L(x) = L(y)$ then $L(x - y) = L(x) - L(y) = 0$ so that $x - y \in \ker(L) = \{0\}$. Hence $x - y = 0$ or $x = y$ proving that L is injective. □

Theorem 11.196. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$, $W \subseteq X$ and $L \in \text{Hom}(X, Y)$ then

$$L(\text{span}(W)) \subseteq \text{span}(L(W))$$

Further if L is injective [or equivalently $\ker(L) = \{0\}$] then

$$L(\text{span}(W)) = \text{span}(L(W))$$

Proof. For the first part. If $y \in L(\text{span}(W))$ then $\exists x \in \text{span}(W)$ such that $y = L(x)$. As $x \in \text{span}(W)$ there exists by [theorem: 11.101] a finite $\{v_i\}_{i \in \{1, \dots, n\}} \subseteq W$ and $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$x = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot v_i$$

Define

$$\{u_i\}_{i \in \{1, \dots, n\}} \subseteq L(W) \text{ by } u_i = L(v_i)$$

then we have

$$\begin{aligned} L(x) &= L\left(\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot w_i\right) \\ &\stackrel{[\text{theorem: 11.190}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot L(w_i) \end{aligned}$$

By [theorem: 11.101] it follows that $y = L(x) \in \text{span}(W)$ proving that

$$L(\text{span}(W)) \subseteq \text{span}(L(W)) \quad (11.76)$$

For the second part. Let $y \in \text{span}(L(W))$ then there exists a finite $J \subseteq L(W)$ and a $\{\alpha_i\}_{i \in J} \subseteq F$ such that

$$y = \sum_{u \in J} \alpha_u \cdot u \quad (11.77)$$

Let $I = L^{-1}(J)$ then if $i \in I$ we have that $L(i) \in J$, as $J \subseteq L(W)$ there exist a $j \in W$ such that $L(i) = L(j)$ which, as L is injective, proves that $i = j \in W$. Hence

$$I \subseteq W \quad (11.78)$$

Define

$$\beta: I \rightarrow J \text{ by } \beta(i) = L(i)$$

then we have

injectivity. If $i, j \in I$ such that $\beta(i) = \beta(j)$ then $L(i) = L(j)$ which, as L is injective, proves that $i = j$.

surjectivity. If $j \in J$ then as $J \subseteq L(W)$ we have that $\exists i \in W$ such that $j = L(i)$. As $L(i) = j \in J$ we have that $i \in L^{-1}(J) = I$, so there exists a $i \in L^{-1}(J)$ such that $j = L(i) = \beta(i)$.

Hence we have

$$\beta: I \rightarrow J \text{ defined by } \beta(i) = L(i) \text{ is a bijection} \quad (11.79)$$

As J is finite we have also that

$$I \text{ is finite} \quad (11.80)$$

Define

$$\{\gamma_i\}_{i \in I} \subseteq F \text{ by } \gamma_i = \alpha_{\beta(i)} \quad (11.81)$$

By [eqs: 11.78, 11.80] and the definition of a span we have that

$$\sum_{i \in I} \gamma_i \cdot i \in \text{span}(W)$$

Further

$$\begin{aligned} L\left(\sum_{i \in I} \gamma_i \cdot i\right) &\stackrel{[\text{theorem: 11.190}]}{=} \sum_{i \in I} \gamma_i \cdot L(i) \\ &= \sum_{i \in I} \alpha_{\beta(i)} \cdot \beta(i) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in J} \alpha_i \cdot i \\ &= y \end{aligned}$$

proving that $y \in L(\text{span}(W))$. Hence $\text{span}(L(W)) \subseteq L(\text{span}(W))$. Combining this with [eq: 11.76] proves that

$$L(\text{span}(W)) = \text{span}(L(W)) \quad \square$$

Theorem 11.197. Let $\langle X, +, \cdot \rangle$ be a **finite dimensional** vector space over a field F , $\langle Y, +, \cdot \rangle$ a vector space over the same field and $L \in \text{Hom}(X, Y)$ then $L(X)$ is finite dimensional and $\text{rank}(L) \stackrel{\text{def}}{=} \dim(L(X)) \leq \dim(X)$.

Proof. For X we have either:

$X = \{0\}$. Then $L(X) = L(\{0\}) = \{0\}$ so that $L(X)$ is finite dimensional and $\dim(L(X)) = \dim(\{0\}) \stackrel{\text{[example: 11.153]}}{=} 0 = \dim(X)$.

$X \neq \{0\}$. Let $n = \dim(X)$ then there exist a basis B of X . As $\text{span}(L(B))$ is a vector space [see theorem: 11.110] it follows from [theorem: 11.147] that there exist a basis E of $\text{span}(L(B))$ with $E \subseteq L(B)$, further, as B is finite, it follows from [theorem: 6.45] that $L(B)$ is finite with $\text{card}(E) \leq \text{card}(L(B)) \leq \text{card}(B) = \dim(X)$, proving that $\text{span}(L(B))$ is finite dimensional with $\dim(\text{span}(L(B))) = \text{card}(E) \leq \dim(X)$. Using [theorem: 11.196] we have that $L(X) = L(\text{span}(B)) \subseteq \text{span}(L(B))$ so that by [theorem: 11.159] $L(X)$ is finite dimensional and $\text{rank}(L) \stackrel{\text{def}}{=} \dim(L(X)) \leq \dim(\text{span}(L(B))) \leq \dim(X)$. $\square$

Theorem 11.198. Let $\langle X, +, \cdot \rangle$ be a **finite dimensional** vector space over a field F , $\langle Y, +, \cdot \rangle$ a vector space over the same field F and $L \in \text{Hom}(X, Y)$ then we have

L is surjective

$\Updownarrow$

Y is finite dimensional and $\text{rank}(L) = \dim(Y)$

Proof.

$\Rightarrow$. As L is surjective it follows that $Y = L(X)$, further using the previous theorem [theorem: 11.197] it follows that $L(X)$ is finite dimensional, hence Y is finite dimensional. Finally $\text{rank}(L) = \dim(L(X)) = \dim(Y)$.

$\Leftarrow$. For Y we have either:

$\dim(Y) = 0$. Then $Y = \{0\}$ and $L(X) \subseteq Y$, further, as $L(0) = 0 \in Y$ we have $Y \subseteq L(X)$ proving that $Y = L(X)$. Hence L is surjective.

$\dim(Y) \neq 0$. As $L(X) \subseteq Y$, Y is finite dimensional and $L(X)$ is a vector space it follows from [theorem: 11.159] that $L(X)$ is finite dimensional. Further, $\text{rank}(L) = \dim(Y)$ and by definition $\dim(L(X)) = \text{rank}(L)$ so that $\dim(L(X)) = \dim(Y)$. Hence there exist a basis B of $L(X)$ with $\text{card}(B) = \dim(L(X)) = \dim(Y)$. Hence, as $B \subseteq L(X) \subseteq Y$ it follows from [theorem: 11.155] that B is also a basis for Y . So we have

$$L(X) \stackrel{B \text{ is basis of } L(X)}{=} \text{span}(B) \stackrel{B \text{ is basis of } Y}{=} Y$$

proving that L is surjective. $\square$

Theorem 11.199. Let $\langle X, +, \cdot \rangle$, $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$, $W \subseteq X$ and $L \in \text{Hom}(X, Y)$ a injective linear mapping. If W is linear independent then $L(W)$ is linear independent.

Proof. Let $J \subseteq L(W)$ be a finite subset and $\{\alpha_i\}_{i \in J} \subseteq F$ such that

$$\sum_{i \in J} \alpha_i \cdot i = 0 \quad (11.82)$$

Let $I = L^{-1}(J)$ then if $i \in I$ we have that $L(i) \in J$, as $J \subseteq L(W)$ there exist a $j \in W$ such that $L(i) = L(j)$ which, as L is injective, proves that $i = j \in W$. Hence

$$I \subseteq W \quad (11.83)$$

Define

$$\beta: I \rightarrow J \text{ by } \beta(i) = L(i)$$

then we have

injectivity. If $i, j \in I$ such that $\beta(i) = \beta(j)$ then $L(i) = L(j)$ which, as L is injective, proves that $i = j$.

surjectivity. If $j \in J$ then as $J \subseteq L(W)$ we have that $\exists i \in W$ such that $j = L(i)$. As $L(i) = j \in J$ we have that $i \in L^{-1}(J) = I$, so there exists a $i \in L^{-1}(J)$ such that $j = L(i) = \beta(i)$.

$$\beta: I \rightarrow J \text{ defined by } \beta(i) = L(i) \text{ is a bijection} \quad (11.84)$$

Hence as J is finite we have also that

$$I \text{ is finite} \quad (11.85)$$

Define now

$$\{\gamma_i\}_{i \in I} \subseteq F \text{ by } \gamma_i = \alpha_{\beta(i)}$$

Then we have

$$\begin{aligned} L\left(\sum_{i \in I} \gamma_i \cdot i\right) & \stackrel{[\text{theorem: 11.190}]}{=} \sum_{i \in I} \gamma_i \cdot L(i) \\ & = \sum_{i \in I} \alpha_{\beta(i)} \cdot \beta(i) \\ & \stackrel{[\text{theorem: 11.36}] \text{ and } [\text{eq: 11.84}]}{=} \sum_{i \in J} \alpha_i \cdot i \\ & \stackrel{[\text{theorem: 11.82}]}{=} 0 \end{aligned}$$

So $\sum_{i \in I} \gamma_i \cdot i \in \ker(L)$, as L is injective we have by [theorem: 11.195] that $\ker(L) = \{0\}$, so that

$$\sum_{i \in I} \gamma_i \cdot i = 0$$

As I is finite, $I \subseteq W$ and W is linear independent we have by [theorem: 11.119] that $\forall i \in I \ \gamma_i = 0$. Hence, if $i \in J$ then $\beta^{-1}(i) \in I$ so that $\alpha_i = \alpha_{\beta(\beta^{-1}(i))} = \gamma_{\beta^{-1}(i)} = 0$, proving that $\forall i \in J$ we have $\alpha_i = 0$. Applying then [theorem: 11.119] proves that $L(W)$ is linear independent. $\square$

Corollary 11.200. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$, $B \subseteq X$ a basis of $\langle X, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ a injective linear mapping then $L(B)$ is a basis of $\text{range}(L)$.

Proof. As $B \subseteq X$ we have

$$L(B) \subseteq L(X) = \text{range}(L)$$

As B is a basis of X we have that B is linear independent and $\text{span}(B) = X$. As B is linear independent we have by [theorem: 11.199] that

$$L(B) \text{ is linear independent}$$

As B is a basis we have $\text{span}(B) = X$ so that

$$\begin{aligned} \text{range}(L) & = L(X) \\ & = L(\text{span}(B)) \\ & \stackrel{[\text{theorem: 11.196}]}{=} \text{span}(L(B)) \end{aligned}$$

So $L(B)$ is a basis of $\text{range}(L)$. $\square$

Corollary 11.201. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be finite dimensional vector spaces over a field $\langle F, +, \cdot \rangle$ such that $\dim(X) = \dim(Y)$ and $L \in \text{Hom}(X, Y)$ a injective linear mapping then $L: X \rightarrow Y$ is a linear isomorphism

Proof. As X is finite dimensional there exists a basis $B \subseteq X$ with $\text{card}(B) = \dim(X)$. Further as L is injective we have that $L|_B: B \rightarrow L(B)$ is a bijection so that $\text{card}(L(B)) = \text{card}(B) = \dim(X)$. By [theorem: 11.199] we have that $L(B) \subseteq Y$ is linear independent. Using this, the fact that $\text{card}(L(B)) = \dim(X) = \dim(Y)$ it follows from [theorem: 11.155] that $L(B)$ is a basis of Y . Hence

$$L(X) = L(\text{span}(B)) \stackrel{[\text{theorem: 11.196}]}{=} \text{span}(L(B)) \stackrel{L(B) \text{ is a basis of } Y}{=} Y$$

proving that L is surjective, hence as L is also injective it follows that $L: X \rightarrow Y$ is a bijection and by definition that

$$L \text{ is a linear isomorphism} \quad \square$$

Corollary 11.202. Let $\langle X, +, \cdot \rangle$ be a finite dimensional vector space over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, X)$ a injective linear transformation then $L: X \rightarrow X$ is a linear isomorphism.

Proof. As clearly $\dim(X) = \dim(X)$ this follows from the previous theorem. $\square$

Corollary 11.203. Let $\langle X, +, \cdot \rangle$ and $\langle Y, +, \cdot \rangle$ be vector spaces over a field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ a linear isomorphism then we have:

1. If X is infinite dimensional then Y is infinite dimensional
2. If X is finite dimensional then Y is finite dimensional
3. $\dim(X) = \dim(Y)$

Proof. If L is a linear isomorphism then

$$L: X \rightarrow Y \text{ is a bijection so that } \text{range}(L) = L(X) = Y \quad (11.86)$$

Let B be a basis for X then

$$L|_B: B \rightarrow L(B) \text{ is a bijection or } B \approx L(B) \quad (11.87)$$

Further by the previous theorem [theorem: 11.200]

$$L(B) \text{ is a basis for } \text{range}(L) \stackrel{[\text{eq: 11.86}]}{=} Y \quad (11.88)$$

So we have

1. If X is infinite dimensional then B is infinite so that by [theorem: 6.47] and [eq: 11.87] it follows that $L(B)$ is infinite. Hence Y is infinite dimensional.
2. If X is finite dimensional then B is finite so that by [theorem: 6.44] and [eq: 11.87] it follows that $L(B)$ is finite. Hence Y is finite dimensional.
3. For B we have either:

B is infinite. Then $L(B)$ is infinite and $\dim(X) = \infty = \dim(Y)$

B is finite. Then there exists a $n \in \mathbb{N}_0$ such that $\{1, \dots, n\} \approx B \approx L(B)$ so that

$$\dim(X) = n = \dim(Y) \quad \square$$

The above theorem is a special case of a more general theorem.

Theorem 11.204. Let $\langle X, +, \cdot \rangle$ be a finite dimensional vector space over the field $\langle F, +, \cdot \rangle$, $\langle Y, +, \cdot \rangle$ a vector space over $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Y)$ then

$$\dim(X) = \dim(\ker(L)) + \text{rank}(L)$$

Proof. We divide the proof in two cases:

$\ker(L) = \{0\}$. Then by [theorem: 11.195] L is injective so that

$$L: X \rightarrow L(X) = \text{range}(L)$$

is a isomorphism, hence by [theorem: 11.203] $\langle \text{range}(L), +, \cdot \rangle$ is finite dimensional and $\dim(X) = \dim(\text{range}(L)) = \text{rank}(L)$. Further $\dim(\ker(L)) = \dim(\{0\}) \stackrel{\text{example: 11.153}}{=} 0$ so that

$$\dim(X) = \text{rank}(L) = 0 + \text{rank}(L) = \dim(\ker(L)) + \text{rank}(L)$$

$\ker(L) \neq \{0\}$. Let B_k be a basis [hence a linear independent set] of $\langle \ker(L), +, \cdot \rangle$ and use [theorem: 11.146] to find a basis B_x of $\langle X, +, \cdot \rangle$ such that $B_k \subseteq B_x$. As B_x is finite [for X is finite dimensional] we have that B_k is finite. Further, as B_x is a disjoint union of $B_x \setminus B_k$ and B_k , we have by [theorem: 10.92] that

$$\dim(X) = \text{card}(B_x) = \text{card}(B_x \setminus B_k) + \text{card}(B_k) = \text{card}(B_x \setminus B_k) + \dim(\ker(L)) \quad (11.89)$$

Consider the function

$$L|_{B_x \setminus B_k}: B_x \setminus B_k \rightarrow L(B_x \setminus B)$$

then we have

injectivity. Let $x, y \in B_x \setminus B_k$ such that $L|_{B_x \setminus B_k}(x) = L|_{B_x \setminus B_k}(y)$ then $L(x) = L(y)$ so that $L(x - y) = L(x) - L(y) = 0$, hence $x - y \in \ker(L)$. As B_k is a basis for $\ker(L)$ and B_k is finite we have by [theorem: 11.142] that there exists a $\{\alpha_u\}_{u \in B_k}$ such that

$$x - y = \sum_{u \in B_k} \alpha_u \cdot u$$

so that

$$0 = \sum_{u \in B_k} \alpha_u \cdot u + 1 \cdot y + (-1) \cdot x \quad (11.90)$$

Now

$$\begin{aligned} B_x &\stackrel{B_k \subseteq B_x}{=} (B_x \setminus B_k) \cup B_k \\ &= ((B_x \setminus B_k) \setminus \{x, y\}) \cup \{x, y\} \cup B_k \\ &= ((B_x \setminus B_k) \setminus \{x, y\}) \cup \{x\} \cup \{y\} \cup B_k \end{aligned}$$

Assume that $x \neq y$ then $\{x\} \cap \{y\} = \emptyset$, further $((B_x \setminus B_k) \setminus \{x, y\}) \cap \{x\} = \emptyset$, $((B_x \setminus B_k) \setminus \{x, y\}) \cap \{y\} = \emptyset$, $((B_x \setminus B_k) \setminus \{x, y\}) \cap B_k = \emptyset$, $\{x\} \cap B_k \stackrel{x, y \in B_x \setminus B_k}{=} \emptyset$ and $\{y\} \cap B_k \stackrel{x, y \in B_x \setminus B_k}{=} \emptyset$ so that

$$B_x \text{ is the disjoint union of } ((B_x \setminus B_k) \setminus \{x, y\}), \{x\}, \{y\} \text{ and } B_k \quad (11.91)$$

The above allows us to define

$$\{\gamma_u\}_{u \in B_x} \subseteq F \text{ by } \gamma_u = \begin{cases} \alpha_u & \text{if } u \in B_k \\ 1 & \text{if } u \in \{x\} \\ -1 & \text{if } u \in \{y\} \\ 0 & \text{if } u \in ((B_x \setminus B_k) \setminus \{x, y\}) \end{cases}$$

then we have

$$\begin{aligned} &\sum_{u \in B_x} \gamma_u \cdot u \stackrel{\text{theorem: 11.43}}{=} \\ &\sum_{u \in B_k} \gamma_u \cdot u + \sum_{u \in \{x\}} \gamma_u \cdot u + \sum_{u \in \{y\}} \gamma_u \cdot u + \sum_{u \in ((B_x \setminus B_k) \setminus \{x, y\})} \gamma_u \cdot u \stackrel{\text{theorem: 11.34}}{=} \\ &\sum_{u \in B_k} \gamma_u \cdot u + \gamma_x \cdot x + \gamma_y \cdot y + \sum_{u \in ((B_x \setminus B_k) \setminus \{x, y\})} \gamma_u \cdot u = \\ &\sum_{u \in B_k} \alpha_u \cdot u + 1 \cdot x + (-1) \cdot y + \sum_{u \in ((B_x \setminus B_k) \setminus \{x, y\})} 0 \cdot u \stackrel{\text{theorem: 11.37}}{=} \\ &\sum_{u \in B_k} \alpha_u \cdot u + 1 \cdot x + (-1) \cdot y \stackrel{\text{eq: 11.90}}{=} \\ &0 \end{aligned}$$

As B_x is linear independent we have that $\forall u \in B_k \gamma_u = 0$ contradicting the fact that $\gamma_x = 1$. Hence we must have that $x = y$ proving injectivity.

surjectivity. This is trivial.

So we have proved that

$$L|_{B_x \setminus B_k}: B_x \setminus B_k \rightarrow L(B_x \setminus B_k) \text{ is a isomorphism} \quad (11.92)$$

Hence, as $L(B_x \setminus B_k) \approx B_x \setminus B_k \subseteq B_x$ a finite set, we have that

$$L(B_x \setminus B_k) \text{ is finite} \quad (11.93)$$

If $y \in L(X)$ then there exists a $x \in X$ such that $y = L(x)$. As B_x is a basis there exists a $\{\alpha_u\}_{u \in B_x} \subseteq F$ such that $x = \sum_{u \in B_x} \alpha_u \cdot u$. So that

$$\begin{aligned} y &= L(x) \\ &= L\left(\sum_{u \in B_x} \alpha_u \cdot u\right) \\ &\stackrel{[\text{theorem: 11.190}]}{=} \sum_{u \in B_x} \alpha_u \cdot L(u) \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{u \in B_x \setminus B_k} \alpha_u \cdot L(u) + \sum_{u \in B_k} \alpha_u \cdot L(u) \\ &= \sum_{u \in B_x \setminus B_k} \alpha_u \cdot L(u) + \sum_{u \in B_k} \alpha_u \cdot 0 \\ &\stackrel{[\text{theorem: 11.37}]}{=} \sum_{u \in B_x \setminus B_k} \alpha_u \cdot L(u) \end{aligned} \quad (11.94)$$

Define now

$$\{\beta_u\}_{u \in L(B_x \setminus B_k)} \text{ by } \beta_u = \alpha_{(L|_{B_x \setminus B_k})^{-1}(u)}$$

Then we have

$$\begin{aligned} \sum_{u \in L(B_x \setminus B_k)} \beta_u \cdot u &= \sum_{u \in L(B_x \setminus B_k)} \alpha_{(L|_{B_x \setminus B_k})^{-1}(u)} \cdot (L|_{B_x \setminus B_k}((L|_{B_x \setminus B_k})^{-1}(u))) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{u \in B_x \setminus B_k} \alpha_u \cdot L|_{B_x \setminus B_k}(u) \\ &= \sum_{u \in B_x \setminus B_k} \alpha_u \cdot L(u) \\ &\stackrel{[\text{eq: 11.94}]}{=} y \end{aligned}$$

proving, as $L(B_x \setminus B_k)$ is finite [see eq: 11.93], that $y \in \text{span}(L(B_x \setminus B_k))$. Hence

$$L(X) \subseteq \text{span}(L(B_x \setminus B_k)) \quad (11.95)$$

For the opposite inclusion:

$$\begin{aligned} B_x \setminus B_k \subseteq X &\Rightarrow L(B_x \setminus B_k) \subseteq L(X) \\ &\stackrel{[\text{theorem: 11.109}]}{\Rightarrow} \text{span}(L(B_x \setminus B_k)) \subseteq \text{span}(L(X)) \\ &\stackrel{[\text{theorems: 11.193, 11.111}]}{\Rightarrow} \text{span}(L(B_x \setminus B_k)) \subseteq L(X) \end{aligned}$$

which combined with [eq: 11.95] gives

$$L(X) = \text{span}(L(B_x \setminus B_k)) \quad (11.96)$$

As $B_x \setminus B_k \subseteq B_x$ a linear independent set, it follows from [theorem: 11.123] that $B_x \setminus B_k$ is linear independent. As $L|_{B_x \setminus B_k}: B_x \setminus B_k \rightarrow L(B_x \setminus B_k)$ is a isomorphism [see eq: 11.92], we have by [theorem: 11.199]

$$L(B_x \setminus B_k) \text{ is linear independent}$$

The above together with [eq: 11.96] prove that

$$L(B_x \setminus B_k) \text{ is a basis of } L(X)$$

Using [eq: 11.92] we have that $\text{card}(B_x \setminus B_k) = \text{card}(L(B_x \setminus B_k)) = \dim(L(x)) = \text{rank}(L)$, Substituting this in [eq: 11.89] gives finally

$$\dim(X) = \text{rank}(L) + \dim(\ker(L))$$

So in all cases we have proved that

$$\dim(X) = \text{rank}(L) + \dim(\ker(L)) \quad \square$$

Corollary 11.205. *Let $\langle X, +, \cdot \rangle, \langle Y, +, \cdot \rangle$ be finite dimensional vector space over the field $\langle F, +, \cdot \rangle$ with $\dim(X) = \dim(Y)$ with $L \in \text{Hom}(X, Y)$ a **surjective** linear mapping then L is a linear isomorphism.*

Proof. As L is surjective it follows from [theorem: 11.198] that $\text{rank}(L) = \dim(Y)$. Using the above theorem [theorem: 11.204] we have that $\dim(\ker(L)) = \dim(X) - \text{rank}(L) = \dim(Y) - \dim(Y) = 0$ which proves that $\dim(\ker(L)) = 0$. Hence using [example: 11.153] it follows that $\ker(L) = \{0\}$. Hence using [theorem: 11.195] L is injective. This proves that L is a bijection and thus by definition a linear isomorphism. $\square$

11.4.3 Internal Direct Sum

Definition 11.206. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and $X \subseteq V, Y \subseteq V$ then*

$$X + Y = \{v | \exists (x, y) \in X \times Y \text{ such that } v = x + y\} \subseteq V$$

Note 11.207. *As*

$$\begin{aligned} X + Y &= \{v | \exists (x, y) \in X \times Y \text{ such that } v = x + y\} \\ &= \{v | \exists (x, y) \in X \times Y \text{ such that } v = y + x\} \\ &= \{v | \exists (y, x) \in Y \times X \text{ such that } v = y + x\} \\ &= Y + X \end{aligned}$$

we have

$$X + Y = Y + X$$

Theorem 11.208. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and X, Y sub-spaces of V the $X + Y$ is a sub-space of V .*

Proof. Let $x, y \in V_1 + V_2$ and $\alpha \in F$ then there exists $v_1, w_1 \in V_1$ and $v_2, w_2 \in V_2$ such that $x = v_1 + w_1$ and $y = v_2 + w_2$. Then as V_1, V_2 are sub-spaces we have that $\alpha \cdot v_1 + w_1 \in V_1$ and $\alpha \cdot v_2 + w_2 \in V_2$ so that

$$\alpha \cdot x + y = \alpha \cdot (v_1 + w_1) + (v_2 + w_2) = (\alpha \cdot v_1 + w_1) + (\alpha \cdot v_2 + w_2)$$

hence we have

$$\alpha \cdot x + y \in V_1 + V_2.$$

Further as V_1, V_2 are sub-spaces we have that $0 \in V_1$ and $0 \in V_2$ so that $0 = 0 + 0 \in V_1 + V_2$ proving that

$$V_1 + V_2 \neq \emptyset \quad \square$$

Definition 11.209. (Internal Direct Sum) *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then*

$$V = X \oplus Y / V \text{ is the internal direct sum of } X \text{ and } y$$

iff

1. X and Y are sub-spaces of V
2. $\forall v \in V$ there exist a **unique** $x \in X \wedge y \in Y$ such that $v = x + y$

We have the following relation between $X + Y$ and $X \oplus Y$.

Theorem 11.210. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, X, Y sub spaces of V then

$$V = X \oplus Y \Leftrightarrow V = X + Y \text{ and } X \cap Y = \{0\}$$

Note 11.211. As $X + Y = Y + X$ and $X \cap Y = Y \cap X$ it follows that

$$V = X \oplus Y \Leftrightarrow V = Y \oplus X$$

Proof.

$\Rightarrow$. As X, Y are sub-spaces of V we have by [theorem: 11.61] that $0 \in X \wedge 0 \in Y$ so that $0 \in X \cap Y$ proving that

$$\{0\} \in X \cap Y \quad (11.97)$$

On the other hand if $v \in X \cap Y$ then $v \in X \wedge v \in Y$ and $v = v + 0 = 0 + v$. As $V = X \oplus Y$ we must have that $v = 0 \wedge 0 = v$ or $v = 0$. Hence $X \cap Y \subseteq \{0\}$ which combined with [eq: 11.97] proves that

$$X \cap Y = \{0\}$$

$\Leftarrow$. If $v \in V$ then as $V = X + Y$ there exists a $x \in X$ and $y \in Y$ such that $v = x + y$. Assume that there exists also a $x' \in X, y' \in Y$ such that $v = x' + y'$. Then $0 = v - v = (x + y) - (x' + y')$ so that $x - x' = y' - y$. As X, Y are sub-spaces, hence vector spaces [see theorem: 11.61] we have $x - x' \in X \wedge y' - y \in Y$ so that $(x - x'), (y' - y) \in X \cap Y = \{0\}$ proving that $x = x' \wedge y = y'$. Hence we found a **unique** $x \in X, y \in Y$ such that $v = x + y$ which proves that

$$V = X \oplus Y \quad \square$$

Theorem 11.212. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and V_1, V_2 finite dimensional sub spaces of V so that $V_1 + V_2 = V_1 \oplus V_2$ [or equivalently that $V_1 \cap V_2 = \{0\}$] then $V_1 + V_2$ is finite dimensional and

$$\dim(V_1 + V_2) = \dim(V_1 \oplus V_2) = \dim(V_1) + \dim(V_2)$$

Further if $E_1 = \{e_1^1, \dots, e_{n_1}^1\}$ and $E_2 = \{e_1^2, \dots, e_{n_2}^2\}$ are bases for V_1 and V_2 then

$$E_1 \cap E_2 = \emptyset \text{ and } E_1 \cup E_2 \text{ is a basis for } V_1 \oplus V_2$$

Proof. Let $n_1 = \dim(V_1), n_2 = \dim(V_2)$ and $E_1 = \{e_1^1, \dots, e_{n_1}^1\}, E_2 = \{e_1^2, \dots, e_{n_2}^2\}$ bases for V_1, V_2 respectively. Using [theorem: 11.115] it follows that $0 \notin E_1 \wedge 0 \notin E_2$ so that

$$0 \notin E_1 \cap E_2 \subseteq V_1 \cap V_2 = \{0\}$$

which proves that

$$E_1 \cap E_2 = \emptyset \quad (11.98)$$

Let $\{\alpha_i\}_{i \in E_1 \cup E_2} \subseteq F$ be such that

$$\sum_{i \in E_1 \cup E_2} \alpha_i \cdot i = 0 \quad (11.99)$$

As $V_1 = \text{span}(E_1)$ and $V_2 = \text{span}(E_2)$ we have that

$$\sum_{i \in E_1} \alpha_i \cdot i \in V_1 \wedge \sum_{i \in E_2} \alpha_i \cdot i \in V_2$$

Further we have

$$\begin{aligned}
 \underbrace{0}_{\in V_1} + \underbrace{0}_{\in V_2} &= \underbrace{0}_{\in V_1 \oplus V_2} \\
 &= \sum_{i \in E_1 \cup E_2} \alpha_i \cdot i \\
 &\stackrel{[\text{theorem: 11.43}]}{=} \underbrace{\sum_{i \in E_1} \alpha_i \cdot i}_{\in V_1} + \underbrace{\sum_{i \in E_2} \alpha_i \cdot i}_{\in V_2}
 \end{aligned}$$

so by [definition: 11.209] we have that

$$\sum_{i \in E_1} \alpha_i \cdot i = 0 \wedge \sum_{i \in E_2} \alpha_i \cdot i = 0$$

By the linear independency of E_1, E_2 it follows then that $\forall i \in E_1 \alpha_i = 0$ and $\forall i \in E_2 \alpha_i = 0$ or $\forall i \in E_1 \cup E_2$ we have $\alpha_i = 0$. This proves that

$$E_1 \cup E_2 \text{ is linear independent} \quad (11.100)$$

Further if $v \in V_1 \oplus V_2$ there exist a $v_1 \in V_1$ and $v_2 \in V_2$ such that $v = v_1 + v_2$, as E_1, E_2 are bases for V_1, V_2 there exists $\{\alpha_i\}_{i \in E_1} \subseteq V, \{\beta_i\}_{i \in E_2} \subseteq V_2$ such that $v_1 = \sum_{i \in E_1} \alpha_i \cdot i$ and $v_2 = \sum_{i \in E_2} \beta_i \cdot i$. Define

$$\{\gamma_i\}_{i \in E_1 \cup E_2} \subseteq F \text{ by } \gamma_i = \begin{cases} \alpha_i & \text{if } i \in E_1 \\ \beta_i & \text{if } i \in E_2 \end{cases}$$

then

$$\begin{aligned}
 \sum_{i \in E_1 \cup E_2} \gamma_i \cdot i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in E_1} \gamma_i \cdot i + \sum_{i \in E_2} \gamma_i \cdot i \\
 &= \sum_{i \in E_1} \alpha_i \cdot i + \sum_{i \in E_2} \beta_i \cdot i \\
 &= v_1 + v_2 \\
 &= v
 \end{aligned}$$

Hence we have that

$$V_1 \oplus V_2 \subseteq \text{span}(E_1 \cup E_2) \quad (11.101)$$

If $x \in E_1$ [or $x \in E_2$] then $x = x + 0 \in V_1 \oplus V_2$ [or $x = 0 + x \in V_1 \oplus V_2$] so that $E_1 \cup E_2 \subseteq V_1 \oplus V_2$, hence by [theorem: 11.109] we have that $\text{span}(E_1 \cup E_2) \subseteq \text{span}(V_1 \oplus V_2) \stackrel{[\text{theorem: 11.111}]}{=} V_1 \oplus V_2$. Combining this with [eq: 11.101] gives $V_1 \oplus V_2 = \text{span}(E_1 \cup E_2)$, hence we have, taking in account [eq: 11.100] that

$$E_1 \cup E_2 \text{ is a basis for } V_1 \oplus V_2$$

Finally

$$\dim(V_1 \oplus V_2) = \text{card}(E_1 \cup E_2) \stackrel{[\text{eq: 11.98}] \text{ and } [\text{theorem: 10.92}]}{=} \text{card}(V_1) + \text{card}(V_2) = \dim(V_1) + \dim(V_2) \quad \square$$

The following is a trivial example of the decomposition of V in two sub-spaces.

Example 11.213. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ then $V = V \oplus \{0\} = \{0\} \oplus V$

Proof. Clearly $V, \{0\}$ are sub-spaces of V and if $v \in V$ we have that $v = 0 + v$ so that

$$V = \{x + y \mid x \in \{0\} \wedge y \in V\}. \text{ Finally } \{0\} \cap V = \{0\} \quad \square$$

So every vector space can be written as a trivial direct sum, however every vector space containing a sub-space can be written as a direct sum of this subspace and another sub-space even if the sub-space is different from $\{0\}$.

Theorem 11.214. (Fundamental theorem of sub-spaces) Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$ and X a sub-space of $\langle V, +, \cdot \rangle$ then there exists a subspace Y of V such that $V = X \oplus Y$.

Proof. As X is a sub-space, hence a vector space there exists by [theorem: 11.148] a basis $B_X \subseteq X$ for $\langle X, +, \cdot \rangle$. As B_X is linear independent there exists by [theorem: 11.146] a basis B for $\langle V, +, \cdot \rangle$ such that $B_X \subseteq B$. We have now either:

$B_X = B$. Then $X = \text{span}(B_X) = \text{span}(B) = V$ so that $X = V$ and thus by [example: 11.213] we have that for $Y = \{0\}$ that

$$V = X \oplus Y$$

$B_X \neq B$. Take $Y = \text{span}(B \setminus B_X)$ then by [theorem: 11.110] Y is a sub-space of $\langle V, +, \cdot \rangle$. Let $v \in V$ then as $\text{span}(B) = V$ there exists a finite set $I \subseteq B$ and $\{\alpha_i\}_{i \in I} \subseteq F$ such that

$$v = \sum_{i \in I} \alpha_i \cdot i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \cap B_X} \alpha_i \cdot i + \sum_{i \in I \setminus B_X} \alpha_i \cdot i = x + y$$

where $x = \sum_{i \in I \cap B_X} \alpha_i \cdot i \in \text{span}(I \cap B_X)$ and $y = \sum_{i \in I \setminus B_X} \alpha_i \cdot i \in \text{span}(I \setminus B_X)$. As $I \setminus B_X \subseteq B \setminus B_X$ and $I \cap B_X \subseteq B_X$ we have by [theorem: 11.109] that $\text{span}(I \setminus B_X) \subseteq \text{span}(B \setminus B_X) = Y$ and $\text{span}(I \cap B_X) \subseteq \text{span}(B_X) = X$ so that $x \in X$ and $y \in Y$, proving that $V \subseteq X + Y \subseteq V$, hence

$$V = X + Y \quad (11.102)$$

As X, Y are vector spaces we have that $0 \in X \cap Y$ so that $\{0\} \subseteq X \cap Y$. Let $z \in X \cap Y$ then $z \in X = \text{span}(B_X)$ and $z \in Y = \text{span}(B \setminus B_X)$, so there exists finite sets $I \subseteq B_X$, $J \subseteq B \setminus B_X$ and families $\{\alpha_i\}_{i \in I} \subseteq F$, $\{\beta_i\}_{i \in J} \subseteq F$ such that

$$\sum_{i \in I} \alpha_i \cdot i = z = \sum_{i \in J} \beta_i \cdot i \quad (11.103)$$

As $I \cap J \subseteq B_X \cap (B \setminus B_X) = \emptyset$ we can define

$$\{\gamma_i\}_{i \in I \cup J} \text{ by } \gamma_i = \begin{cases} \alpha_i & \text{if } i \in I \\ -\beta_i & \text{if } i \in J \end{cases}$$

then

$$\begin{aligned} \sum_{i \in I \cup J} \gamma_i \cdot i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I} \gamma_i \cdot i + \sum_{i \in J} \gamma_i \cdot i \\ &= \sum_{i \in I} \alpha_i \cdot i + \sum_{i \in J} (-\beta_i) \cdot i \\ &\stackrel{[\text{theorem: 11.40}]}{=} \sum_{i \in I} \alpha_i \cdot i - \sum_{i \in J} \beta_i \cdot i \\ &\stackrel{[\text{eq: 11.103}]}{=} z - z \\ &= 0 \end{aligned}$$

As $I \cup J \subseteq B_X \cup (B \setminus B_X) = B$ a linear independent set we have that $\forall i \in I \cup J \gamma_i = 0$, hence $\forall i \in I$ we have $\alpha_i = \gamma_i = 0$, so $z = \sum_{i \in I} \alpha_i \cdot i = \sum_{i \in I} 0 \cdot i \stackrel{[\text{theorem: 11.37}]}{=} 0$. So it follows that

$X \cap Y \subseteq \{0\}$ proving as $\{0\} \subseteq X \cap Y$ that $X \cap Y = \{0\}$. Combining this with [eq: 11.102] gives finally

$$V = X \oplus Y \quad \square$$

Theorem 11.215. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, U a sub-space of V and $x \in V \setminus U$ then for the sub-space $W = \text{span}(U \cup \{x\})$ of V [see theorem: 11.110] we have that

$$W = U \oplus \text{span}(\{x\})$$

In other words $\forall w \in W$ there exist **unique** $y \in U$ and $\lambda \in F$ such that $w = y + \lambda \cdot x$

Proof. Let $y \in U \cap \text{span}(\{x\})$ then $y \in U$ and there exist a $\lambda \in F$ such that $y = \lambda \cdot x$. Assume that $y \neq 0$ then we must have that $\lambda \neq 0$ so that $x = \frac{1}{\lambda} \cdot y \in U$ which contradicts $x \in V \setminus U$, hence we must have that $y = 0$ which proves that $\text{span}(\{x\}) \cap U \subseteq \{0\}$. As $0 \in \text{span}(\{x\}) \wedge 0 \in U \Rightarrow 0 \in \text{span}(\{x\}) \cap U$ it follows that

$$U \cap \text{span}(\{x\}) = \{0\} \quad (11.104)$$

Let $y \in \text{span}(U \cup \{x\})$ then by [theorem: 11.101] there exists a $n \in \mathbb{N}$, a $\{u_i\}_{i \in \{1, \dots, n\}} \subseteq U \cup \{x\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $y = \sum_{i=1}^n \alpha_i \cdot u_i$. Let $I = \{i \in \{1, \dots, n\} | u_i = x\}$ then we have $\{u_i\}_{i \in I} \subseteq \{x\}$ so that $\sum_{i \in I} \alpha_i \cdot u_i \in \text{span}(\{x\})$, further $\{u_i\}_{i \in \{1, \dots, n\} \setminus I} \subseteq U$ so that $\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i \in \text{span}(U) \stackrel{[\text{theorem: 11.111}]}{=} U$. Hence we have that

$$y = \sum_{i=1}^n \alpha_i \cdot u_i = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot u_i = \sum_{i \in \{1, \dots, n\} \setminus I} \alpha_i \cdot u_i + \sum_{i \in I} \alpha_i \cdot u_i \in U + \text{span}(\{x\})$$

which proves that

$$\text{span}(U \cup \{x\}) \subseteq U + \text{span}(\{x\}) \quad (11.105)$$

Let $y \in U + \text{span}(\{x\})$ then there exist a $u \in U$ and a $\lambda \in F$ such that $y = u + \lambda \cdot x$. Hence if we define $\{u_i\}_{i \in \{1, 2\}} \subseteq U \cup \{x\}$ and $\{\alpha_i\}_{i \in \{1, 2\}} \subseteq F$ by $u_1 = u$, $u_2 = x$, $\alpha_1 = 1$ and $\alpha_2 = \lambda$ then we have

$$y = u + \lambda \cdot x = \sum_{i=1}^2 \alpha_i \cdot u_i \in \text{span}(U \cup \{x\})$$

which proves that $U + \text{span}(\{x\}) \subseteq \text{span}(U \cup \{x\})$. Combining this with [eq: 11.105] proves

$$W = \text{span}(U \cup \{x\}) = U + \text{span}(\{x\})$$

The above together with [eq: 11.104] allows us to use [theorem: 11.210] to get finally that

$$W = U \oplus \text{span}(\{x\}) \quad \square$$

Theorem 11.216. Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, X, Y a sub-spaces of $\langle V, +, \cdot \rangle$ then there exists mappings

$$\pi_X: V \rightarrow X \text{ and } \pi_Y: V \rightarrow Y$$

such that:

1. $\pi_X \in \text{Hom}(V, X)$ and $\pi_Y \in \text{Hom}(V, Y)$
2. $\forall v \in V \ v = \pi_X(v) + \pi_Y(v)$
3. $(\pi_X)|_X = \text{Id}_X$, $(\pi_X)|_Y = C_0$, $(\pi_Y)|_X = C_0$ and $(\pi_Y)|_Y = \text{Id}_Y$

Proof. Let $v \in V = X \oplus Y$ then by [theorem: 11.210] there exists **unique** $v_X \in X$ and $v_Y \in Y$ such that $v = v_X + v_Y$. This allows us to define $\pi_X: V \rightarrow X$ and $\pi_Y: V \rightarrow Y$ by $\pi_X(v) = v_X$ and $\pi_Y(v) = v_Y$ so that

$$v = \pi_X(v) + \pi_Y(v)$$

Let $x, y \in V$ and $\alpha \in F$ then $x = \pi_X(x) + \pi_Y(x)$, $y = \pi_X(y) + \pi_Y(y)$. So

$$x + y = \pi_X(x) + \pi_Y(x) + \pi_X(y) + \pi_Y(y) = \underbrace{(\pi_X(x) + \pi_X(y))}_{\in X} + \underbrace{(\pi_Y(x) + \pi_Y(y))}_{\in Y}$$

which by definition proves

$$\pi_X(x + y) = \pi_X(x) + \pi_X(y) \text{ and } \pi_Y(x + y) = \pi_Y(x) + \pi_Y(y)$$

Further

$$\alpha \cdot x = \alpha \cdot (\pi_X(x) + \pi_Y(y)) = \underbrace{\alpha \cdot \pi_X(x)}_{\in X} + \underbrace{\alpha \cdot \pi_Y(y)}_{\in Y}$$

which by definition proves

$$\pi_X(\alpha \cdot x) = \alpha \cdot \pi_X(x) \text{ and } \pi_Y(\alpha \cdot x) = \alpha \cdot \pi_Y(x)$$

Hence we have

$$\pi_X \in \text{Hom}(V, X) \text{ and } \pi_Y \in \text{Hom}(V, Y)$$

Finally if $v \in X$ then $v = v + 0 = \pi_X(v) + \pi_Y(v)$ and if $v \in Y$ then $v = 0 + v = \pi_X(v) + \pi_Y(v)$ so that

$$\forall v \in X \ v_X(v) = v = \text{Id}_X(v), \ v_Y(v) = 0 = C_0(v) \text{ and } \forall v \in Y \ v_X(v) = 0 = C_0(v), \ v_Y(v) = v = \text{Id}_Y(v)$$

proving that

$$(v_X)_{|X} = \text{id}_X \wedge (v_X)_{|Y} = C_0 \wedge (v_Y)_{|Y} = \text{Id}_Y \wedge (v_Y)_{|X} = C_0 \quad \square$$

Theorem 11.217. *Let $\langle V, +, \cdot \rangle$ be a vector space over a field $\langle F, +, \cdot \rangle$, X a sub-space of $\langle V, +, \cdot \rangle$, $\langle Z, +, \cdot \rangle$ a vector space over the same field $\langle F, +, \cdot \rangle$ and $L \in \text{Hom}(X, Z)$ then there exists a $K \in \text{Hom}(V, Z)$ such that $K_{|X} = L$. In other words K is a extension of L .*

Proof. Using [theorem: 11.214] there exists a sub-space Y of $\langle V, +, \cdot \rangle$ such that

$$V = X \oplus Y$$

Using [theorem: 11.216] there exists mappings

$$\pi_X: V \rightarrow X \text{ and } \pi_Y: V \rightarrow Y$$

such that:

1. $\pi_X \in \text{Hom}(V, X)$ and $\pi_Y \in \text{Hom}(V, Y)$
2. $\forall v \in V \ v = \pi_X(v) + \pi_Y(v)$
3. $(\pi_X)_{|X} = \text{Id}_X$, $(\pi_X)_{|Y} = C_0$, $(\pi_Y)_{|X} = C_0$ and $(\pi_Y)_{|Y} = \text{Id}_Y$

Define then

$$K: V \rightarrow Z \text{ by } K = L \circ \pi_X$$

then if $x, y \in V$ and α we have

$$K(x + y) = L(\pi_X(x + y)) \stackrel{(1)}{=} L(\pi_X(x) + \pi_X(y)) = L(\pi_X(x)) + L(\pi_X(y)) = K(x) + K(y)$$

and

$$K(\alpha \cdot x) = L(\pi_X(\alpha \cdot x)) \stackrel{(1)}{=} L(\alpha \cdot \pi_X(x)) = \alpha \cdot L(\pi_X(x)) = \alpha \cdot K(x)$$

proving that

$$K \in \text{Hom}(V, Z)$$

Finally if $x \in X$ then $K(x) = L(\pi_X(x)) \stackrel{(3)}{=} L(x)$ proving that

$$K_{|X} = L \quad \square$$

11.5 Permutations

In [definition: 11.25] we have introduced the idea of permutations, bijections of a set on itself, that forms a group under function compositions. We consider now the special case of permutations of sets of the form $\{1, \dots, n\}$ where $n \in \mathbb{N}$.

Definition 11.218. *Let $n \in \mathbb{N}$ then*

$$P_n = \{\sigma \in \{1, \dots, n\}^{\{1, \dots, n\}} \text{ such that } \sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\} \text{ is a bijection}\}$$

In other words using [definition: 11.25] $P_n = S_{\{1, \dots, n\}}$

Example 11.219. Let $n \in \mathbb{N}$ then for

$$\rho_n: \{1, \dots, n\} \rightarrow \{1, \dots, n\} \text{ defined by } \rho_n(i) = n - i + 1$$

we have $\rho_n \in P_n$.

ρ_n is the reversion operation, for example:

- $\rho_3(1) = 3 - 1 + 1 = 3$
- $\rho_3(2) = 3 - 2 + 1 = 2$
- $\rho_3(3) = 3 - 3 + 1 = 1$

Proof. We have:

injectivity. If $\rho_n(i) = \rho_n(j)$ then $n - i + 1 = n - j + 1$ so that $-i = -j$ hence $i = j$.

surjectivity. If $k \in \{1, \dots, n\}$ take then $j = n - k + 1 \in \{1, \dots, n\}$ so that $\rho_n(n - k + 1) = n - (n - k + 1) + 1 = k$

proving bijectivity, hence $\rho_n \in P_n$. $\square$

Next we will prove that P_n is a finite set, first we show how we can extend a permutation in P_n to a permutation in P_{n+1} . Then we will show that $\{1, \dots, n+1\} \times P_n \approx P_{n+1}$ and finally we will use mathematical induction to prove that $\forall n \in \mathbb{N}$ P_n is finite and $\text{card}(P_n) = \text{fac}(n) \underset{\text{noted}}{=} n!$. First we define $\text{fac}(n)$ using recursion.

Definition 11.220. Let $\text{fac}: \mathbb{N}_0 \rightarrow \mathbb{N}$ is defined by

$$\text{fac}(n) = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot \text{fac}(n-1) & \text{if } n \in \mathbb{N} \end{cases}$$

We will use the common convention and write $n!$ for $\text{fac}(n)$.

Next we extend a permutation $\sigma \in P_n$ to a permutation $\sigma^{[i]}$ of P_{n+1} .

Lemma 11.221. Let $n \in \mathbb{N}$, $i \in \{1, \dots, n+1\}$ and $\sigma \in P_n$ then define

$$\sigma^{[i]}: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$$

by

1. If $i = n+1$ then

$$\sigma^{[n+1]}(j) = \begin{cases} n+1 & \text{if } j = n+1 \\ \sigma(j) & \text{if } j \in \{1, \dots, n\} \end{cases}$$

2. If $i \in \{1, \dots, n\}$ take $k = \sigma^{-1}(i)$ [hence $\sigma(k) = i$]

$$\sigma^{[i]}(j) = \begin{cases} i & \text{if } j = n+1 \\ n+1 & \text{if } j = k \\ \sigma(j) & \text{if } j \in \{1, \dots, n\} \setminus \{k\} \end{cases}$$

then we have that

$$\sigma^{[i]} \in P_{n+1} \text{ and } \sigma^{[i]}(n+1) = i$$

Proof. For $i \in \{1, \dots, n+1\}$ we have either:

$i = n+1$. Then for $\sigma^{[n+1]}: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$ we have by definition

$$\sigma^{[n+1]}(n+1) = n+1$$

Further we have

injectivity. If $\sigma^{[n+1]}(k) = \sigma^{[n+1]}(l)$ then we have for k either:

$k \in \{1, \dots, n\}$. Then $\sigma^{[n+1]}(k) = \sigma(k) \in \{1, \dots, n\}$ so that $n+1 \neq \sigma^{[n+1]}(k) = \sigma^{[n+1]}(l)$, hence $\sigma^{[n+1]}(l) \in \{1, \dots, n\}$ so that $l \in \{1, \dots, n\}$ [as $\sigma^{[n+1]}(n+1) = n+1 \notin \{1, \dots, n\}$]. So $\sigma(k) = \sigma^{[n+1]}(k) = \sigma^{[n+1]}(l) = \sigma(l)$ which as σ is a bijection proves that $k = l$.

$k = n+1$. If $l \in \{1, \dots, n\}$ then $\sigma^{[n+1]}(l) = \sigma(l) \in \{1, \dots, n\}$ so that $\sigma^{[n+1]}(l) \neq n+1 = \sigma^{[n+1]}(k)$ contradicting $\sigma^{[n+1]}(k) = \sigma^{[n+1]}(l)$, hence $k = n+1 = l$.

surjectivity. If $j \in \{1, \dots, n+1\}$ then we have either:

$j = n+1$. Then $n+1 \in \{1, \dots, n+1\}$ and $\sigma^{[n+1]}(n+1) = n+1 = j$

$j \in \{1, \dots, n\}$. Then, as $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is a bijection, we have a $k \in \{1, \dots, n\} \subseteq \{1, \dots, n+1\}$ such that $j = \sigma(k) = \sigma^{[i]}(k)$

Hence $\sigma^{[n+1]}: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$ is a bijection or

$$\sigma^{[n+1]} \in P_{n+1}$$

$i \in \{1, \dots, n\}$. Let

$$k = \sigma^{-1}[i] \Rightarrow k \in \{1, \dots, n\} \wedge \sigma(k) = i \quad (11.106)$$

then we have for $\sigma^{[i]}: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$ that

injectivity. If $\sigma^{[i]}(r) = \sigma^{[i]}(s)$ then we have to consider the following cases for r, s :

$r = n+1 \wedge s = n+1$. Then $r = s$

$r = k \wedge s = n+1$. Then

$$n+1 = \sigma^{[i]}(k) = \sigma^{[i]}(r) = \sigma^{[i]}(s) = \sigma^{[i]}(n+1) = i \in \{1, \dots, n\}$$

leading to the contradiction $n+1 \leq n$, so this case does not occur.

$r \in \{1, \dots, n\} \setminus \{k\} \wedge s = n+1$. Then

$$\sigma(k) \underset{\text{eq: 11.106}}{=} i = \sigma^{[i]}(n+1) = \sigma^{[i]}(s) = \sigma^{[i]}(r) = \sigma[r]$$

so that, as σ is injective, $r = k$ contradicting $r \in \{1, \dots, n\} \setminus \{k\}$. So this case does not occur.

$r = n+1 \wedge s = k$. Then

$$n+1 = \sigma^{[i]}(k) = \sigma^{[i]}(s) = \sigma^{[i]}(r) = \sigma^{[i]}(n+1) = i \in \{1, \dots, n\}$$

leading to the contradiction $n+1 \leq n$. So this case does not occur.

$r = k \wedge s = k$. Then $r = s$

$r \in \{1, \dots, n\} \setminus \{k\} \wedge s = k$. Then

$$n+1 = \sigma^{[i]}(k) = \sigma^{[i]}(s) = \sigma^{[i]}(r) = \sigma(r) \in \{1, \dots, n\} \setminus \{k\}$$

leading to the contradiction $n+1 \leq n$. Hence this case does not occur.

$r = n+1 \wedge s \in \{1, \dots, n\} \setminus \{k\}$. Then

$$\sigma(k) = i = \sigma^{[i]}(n+1) = \sigma^{[i]}(r) = \sigma^{[i]}(s) = \sigma(s),$$

which, as σ is injective, proves that $k = s$ contradicting $s \in \{1, \dots, n\} \setminus \{k\}$. Hence this case does not occur.

$r = k \wedge s \in \{1, \dots, n\} \setminus \{k\}$. Then

$$n+1 = \sigma^{[i]}(k) = \sigma^{[i]}(r) = \sigma^{[i]}(s) = \sigma(s) \in \{1, \dots, n\}$$

leading to the contradiction $n+1 \leq n$, hence this case does not apply.

$r \in \{1, \dots, n\} \setminus \{k\} \wedge s \in \{1, \dots, n\} \setminus \{k\}$. Then $\sigma(r) = \sigma^{[i]}(r) = \sigma^{[i]}(s) = \sigma(s)$, which as σ is injective proves that $r = s$.

So in all valid cases we have $r = s$ proving injectivity.

surjectivity. Let $j \in \{1, \dots, n+1\}$ then we have either:

$j = n+1$. Then $\sigma^{[i]}(k) = n+1 = j$

$j = i$. Then $\sigma^{[i]}(n+1) = i = j$

$j \in \{1, \dots, n\} \setminus \{i\}$. Then as $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is a bijection there exists $l \in \{1, \dots, n\}$ such that $\sigma(l) = j$. If $l = k$ then $j = \sigma(l) = \sigma(k) = i$ contradicting $j \in \{1, \dots, n\} \setminus \{i\}$. So we must have that $l \in \{1, \dots, n\} \setminus \{k\}$, hence $\sigma^{[i]}(l) = \sigma(l) = j$

proving surjectivity.

From the above it follows that $\sigma^{[i]}: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$ is bijective and thus that

$$\sigma^{[i]} \in P_{n+1} \quad \square$$

Lemma 11.222. *Let $n \in \mathbb{N}$ then $\{1, \dots, n+1\} \times P_n \approx P_{n+1}$*

Proof. Let $n \in \mathbb{N}$ and define the function

$$\beta: \{1, \dots, n+1\} \times P_n \approx P_{n+1} \text{ by } \beta(i, \sigma) = \sigma^{[i]}$$

then we have:

injectivity. Assume that $\beta(k_1, \sigma_1) = \beta(k_2, \sigma_2)$ then $\sigma_1^{[k_1]} = \sigma_2^{[k_2]}$. So that

$$k_1 \underset{[\text{theorem: 11.221}]}{=} \sigma_1^{[k_1]}(n+1) = \sigma_2^{[k_2]}(n+1) \underset{[\text{theorem: 11.221}]}{=} k_2$$

proving that

$$k_1 = k_2 \quad (11.107)$$

We have now to look at two cases for k_1, k_2 corresponding with the definitions of $\sigma_1^{[k_1]}, \sigma_2^{[k_2]}$

$k_1 = k_2 = n+1$. Then if $i \in \{1, \dots, n\}$ we have

$$\sigma_1(i) \underset{[\text{theorem: 11.221}]}{=} \sigma_1^{[k_1]}(i) = \sigma_2^{[k_2]}(i) \underset{[\text{theorem: 11.221}]}{=} \sigma_2(i)$$

proving that

$$\sigma_1 = \sigma_2$$

$k_1 = k_2 \neq n+1$. Let $l_1 = (\sigma_1)^{-1}(k_1)$ and $l_2 = (\sigma_2)^{-1}(k_2)$ then we have

$$\sigma_1^{[k_1]}(l_1) \underset{[\text{theorem: 11.221}]}{=} n+1 \underset{[\text{theorem: 11.221}]}{=} \sigma_2^{[k_2]}(l_2)$$

proving as $\sigma_1^{[k_1]}, \sigma_2^{[k_2]}$ are bijections that $l_1 = l_2$. Now if $i \in \{1, \dots, n\}$ then we have either:

$$i = l_1 = l_2. \text{ Then } \sigma_1(i) = \sigma_1(l_1) \underset{[\text{theorem: 11.221}]}{=} k_1 = k_2 \underset{[\text{theorem: 11.221}]}{=} \sigma_2(l_2) = \sigma_2(i)$$

$$i \neq l_1 = l_2. \text{ Then } \sigma_1(i) \underset{[\text{theorem: 11.221}]}{=} \sigma_1^{[k_1]}(i) = \sigma_2^{[k_2]}(i) \underset{[\text{theorem: 11.221}]}{=} \sigma_2(i)$$

proving that

$$\sigma_1 = \sigma_2$$

So in all cases we have $\sigma_1 = \sigma_2$ which together with [eq: 11.107] gives $(k_1, \sigma_1) = (k_2, \sigma_2)$, proving injectivity.

surjectivity. Let $\rho \in P_{n+1}$ then for ρ we have two cases to consider:

$\rho(n+1) = n+1$. Take $\sigma = \rho|_{\{1, \dots, n\}}$ then by [theorem: 11.27] $\rho|_{\{1, \dots, n\}} \in P_n$. Further $\forall i \in \{1, \dots, n+1\}$ we have

$$\begin{aligned} \beta(n+1, \rho|_{\{1, \dots, n\}})(i) &= (\rho|_{\{1, \dots, n\}})^{[n+1]}(j) \\ &\underset{[\text{theorem: 11.221}]}{=} \begin{cases} n+1 & \text{if } i=n+1 \\ \rho|_{\{1, \dots, n+1\}}(i) & \text{if } i \in \{1, \dots, n\} \end{cases} \\ &= \begin{cases} n+1 & \text{if } i=n+1 \\ \rho(i) & \text{if } i \in \{1, \dots, n\} \end{cases} \\ &= \begin{cases} \rho(n+1) & \text{if } i=n+1 \\ \rho(i) & \text{if } i \in \{1, \dots, n\} \end{cases} \\ &= \rho(i) \end{aligned}$$

proving

$$\beta(n+1, \rho|_{\{1, \dots, n\}}) = \rho$$

$\rho(n+1) \neq n+1$. Then $\rho(n+1) \in \{1, \dots, n\}$. Define

$$k = \rho(n+1) \text{ and } l = \rho^{-1}(n+1) \text{ so that } \rho(l) = n+1 \quad (11.108)$$

Then we have $k \in \{1, \dots, n\}$ and $l \in \{1, \dots, n\}$ [for if $l = n+1$ we have $\rho(n+1) = \rho(l) \stackrel{[\text{eq: 11.108}]}{=} n+1$ contradicting $\rho(n+1) \neq n+1$]. Further if $i \in \{1, \dots, n\} \setminus \{l\}$ we have that $\rho(i) \in \{1, \dots, n\}$ [for if $\rho(i) = n+1 \stackrel{[\text{eq: 11.108}]}{=} \rho(l)$ then, as ρ is injective, $i = l$ contradicting $i \in \{1, \dots, n\} \setminus \{l\}$]. To summarize we have

$$k, l \in \{1, \dots, n\} \text{ and } \forall i \in \{1, \dots, n\} \setminus \{l\} \quad \rho(i) \in \{1, \dots, n\} \quad (11.109)$$

Define now

$$\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\} \text{ by } \sigma(i) = \begin{cases} k & \text{if } i = l \\ \rho(i) & \text{if } i \in \{1, \dots, n\} \setminus \{l\} \end{cases}$$

Then we have:

injectivity. Let $i, j \in \{1, \dots, n\}$ such that $\sigma(i) = \sigma(j)$ then we have either:

$\sigma(i) = \sigma(j) = k$. If $i \neq l$ then $i \in \{1, \dots, n\} \setminus \{l\}$ so that $\sigma(i) = \rho(i)$, hence $\rho(i) = k \stackrel{[\text{eq: 11.108}]}{=} \rho(n+1)$ which as ρ is injective proves that $i = n+1$ contradicting $i \in \{1, \dots, n\} \setminus \{l\}$, hence we must have that $i = l$. Likewise if $j \neq l$ then $j \in \{1, \dots, n\} \setminus \{l\}$ so that $\sigma(j) = \rho(j)$, hence $\rho(j) = k \stackrel{[\text{eq: 11.108}]}{=} \rho(n+1)$ which as ρ is injective proves that $j = n+1$ contradicting $j \in \{1, \dots, n\} \setminus \{l\}$. Hence we must have that $j = l$. So we have $i = l = j$ proving injectivity in this case.

$\sigma(i) = \sigma(j) \neq k$. If $i = l$ then $\sigma(i) = \sigma(l) = k$ contradicting $\sigma(i) \neq k$, likewise if $j = l$ then $\sigma(j) = \sigma(l) = k$ contradicting $\sigma(j) \neq k$. So we have $i, j \in \{1, \dots, n\} \setminus \{l\}$ so that $\rho(i) = \sigma(i) = \sigma(j) = \rho(j)$. As ρ is injective we have that $i = j$, proving injectivity in this case.

surjectivity. Let $j \in \{1, \dots, n\}$ then we have either:

$j = k$. Then $\sigma(l) = k = j$ where $l \in \{1, \dots, n\}$ proving surjectivity in this case.

$j \neq k$. Then $j \in \{1, \dots, n\} \setminus \{k\} \subseteq \{1, \dots, n+1\}$. By surjectivity of $\rho: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$ there exists a $i \in \{1, \dots, n+1\}$ such that $\rho(i) = j$. Assume that $i = n+1$ then

$$j = \rho(i) = \rho(n+1) \stackrel{[\text{eq: 11.108}]}{=} k$$

contradicting $j \neq k$. So we must have that $i \in \{1, \dots, n\}$. Also if $i = l$ then $j = \rho(i) = \rho(l) \stackrel{[\text{eq: 11.108}]}{=} n+1$ contradicting $j \in \{1, \dots, n\}$, hence $i \neq l$ or $i \in \{1, \dots, n\} \setminus \{l\}$, so that $\sigma(i) = \rho(i) = j$, proving surjectivity in this case.

So we have proved that $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is a bijection or

$$\sigma \in P_n$$

Consider now $\sigma^{[k]} \in P_{n+1}$ and take $i \in \{1, \dots, n+1\}$ then for i we have either:

$i = n+1$. Then we have

$$\begin{aligned} \sigma^{[k]}(i) &= \sigma^{[k]}(n+1) \\ &\stackrel{[\text{theorem: 11.221}]}{=} k \\ &\stackrel{[\text{eq: 11.108}]}{=} \rho(n+1) \\ &= \rho(i) \end{aligned}$$

$i = l$. Then as $\sigma(l) = k$ we have $l = \sigma^{-1}(k)$ and

$$\begin{aligned} \sigma^{[k]}(i) &= \sigma^{[k]}(l) \\ &\stackrel{[\text{theorem: 11.221}]\wedge \rho(n+1) \neq n+1 \wedge l = \sigma^{-1}(k)}{=} n+1 \\ &\stackrel{[\text{eq: 11.108}]}{=} \rho(l) \\ &= \rho(i) \end{aligned}$$

$i \in \{1, \dots, n\} \setminus \{l\}$. As $\sigma(l) = k$ we have

$$i \neq l = \sigma^{-1}(k) \text{ and } i \neq n+1 \quad (11.110)$$

so that

$$\begin{aligned} \sigma^{[k]}(i) &\stackrel{[\text{theorem: 11.221}] \text{ and } [\text{eq: 11.110}]}{=} \sigma(i) \\ &= \rho(i) \end{aligned}$$

So in all cases we have $\rho(i) = \sigma^{[k]}(i) = \beta(k, \sigma)(i)$ for every $i \in \{1, \dots, n+1\}$. Hence $\rho = \beta(k, \sigma)$ proving surjectivity of $\beta: \{1, \dots, n+1\} \times P_n \rightarrow P_{n+1}$.

So we have proved that $\beta: \{1, \dots, n+1\} \times P_n \rightarrow P_{n+1}$ is a bijection hence

$$\{1, \dots, n+1\} \times P_n \approx P_{n+1} \quad \square$$

Finally we can use mathematical induction to prove that P_n is finite and calculate its cardinality.

Theorem 11.223. *Let $n \in \mathbb{N}$ then P_n is finite and $\text{card}(P_n) = n!$*

Proof. Define

$$S = \{n \in \mathbb{N} \mid P_n \text{ is finite and } \text{card}(P_n) = n!\}$$

then we have:

$1 \in S$. Let $\sigma \in P_1$ then, as $\sigma: \{1\} \rightarrow \{1\}$ is a bijection, we must have that $\sigma = \text{Id}_{\{1\}}$, so that $P_1 = \{\text{Id}_{\{1\}}\}$. Hence P is finite and $\text{card}(P_1) = 1 = 1!$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. First $\{1, \dots, n+1\}$ is finite with $\text{card}(\{1, \dots, n+1\}) \stackrel{[\text{theorem: 10.90}]}{=} n+1$, second, as $n \in S$, P_n is finite with $\text{card}(P_n) = n!$. By [theorem: 10.92] it follows then that $\{1, \dots, n+1\} \times P_n$ is finite with

$$\text{card}(\{1, \dots, n+1\} \times P_n) = \text{card}(\{1, \dots, n+1\}) \cdot \text{card}(P_n) = (n+1) \cdot n! = (n+1)!$$

Using [lemma: 11.222] we have

$$\{1, \dots, n+1\} \times P_n \approx P_{n+1}$$

so that P_{n+1} is finite and $\text{card}(P_{n+1}) = (n+1)!$, proving that $n+1 \in S$. $\square$

Theorem 11.224. *Let $n \in \mathbb{N}$ then $\langle P_n, \circ \rangle$ is a group called the permutation group*

Proof. Using [definition: 11.25] we have that $P_n = S_{\{1, \dots, n\}}$, the rest follows from [theorem: 11.26]. $\square$

Note 11.225. P_n is not commutative

Proof. For example if $f_1 = \begin{pmatrix} 1 \rightarrow 3 \\ 2 \rightarrow 2 \\ 3 \rightarrow 1 \end{pmatrix}$ and $f_2 = \begin{pmatrix} 1 \rightarrow 2 \\ 2 \rightarrow 1 \\ 3 \rightarrow 3 \end{pmatrix}$ then $f_1 \circ f_2 = \begin{pmatrix} 1 \rightarrow 2 \\ 2 \rightarrow 3 \\ 3 \rightarrow 1 \end{pmatrix}$ and $f_2 \circ f_1 = \begin{pmatrix} 1 \rightarrow 3 \\ 2 \rightarrow 1 \\ 3 \rightarrow 2 \end{pmatrix}$ so that $f_1 \circ f_2 \neq f_2 \circ f_1$ $\square$

Definition 11.226. *Let $n \in \mathbb{N}$. As $\langle P_n, \circ \rangle$ forms a group we can define the finite product of permutations [see remark: 11.2]. Given a family $\{\sigma_i\}_{i \in \{1, \dots, n\}} \subseteq P_n$ we define $\sigma_1 \circ \dots \circ \sigma_n$ by*

$$\sigma_1 \circ \dots \circ \sigma_n = \prod_{i=1}^n \sigma_i$$

(see [definition: 11.12] using $\circ$ for product).

Theorem 11.227. Let $m, n \in \mathbb{N}$ and $\{\sigma_i\}_{i \in \{1, \dots, m\}} \subseteq P_n$ then we have

1. If $m = 1$ then $\sigma_1 \circ \dots \circ \sigma_1 = \sigma_1$
2. If $1 < m$ then

$$\begin{aligned}\sigma_1 \circ \dots \circ \sigma_m &= (\sigma_1 \circ \dots \circ \sigma_{m-1}) \circ \sigma_m \\ &= \sigma_1 \circ (\sigma_2 \circ \dots \circ \sigma_m)\end{aligned}$$

Proof.

1. We have

$$\sigma_1 \circ \dots \circ \sigma_1 = \prod_{i=1}^1 \sigma_1 \stackrel{[\text{theorem: 11.15}]}{=} \sigma_1$$

2. If $1 < m$ then we have

$$\begin{aligned}\sigma_1 \circ \dots \circ \sigma_m &= \prod_{i=1}^m \sigma_i \\ &\stackrel{[\text{theorem: 11.15}]}{=} \left(\prod_{i=1}^{m-1} \sigma_i \right) \circ \sigma_m \\ &= (\sigma_1 \circ \dots \circ \sigma_{m-1}) \circ \sigma_m \\ \sigma_1 \circ \dots \circ \sigma_m &= \prod_{i=1}^m \sigma_i \\ &\stackrel{\text{def}}{=} \prod_{i=0}^{m-1} \sigma_{i+1} \\ &\stackrel{[\text{theorem: 11.22}]}{=} \sigma_{0+1} \circ \prod_{i=1}^{m-1} \sigma_{i+1} \\ &\stackrel{\text{def}}{=} \sigma_1 \circ \prod_{i=2}^m \sigma_i \\ &= \sigma_1 \circ (\sigma_2 \circ \dots \circ \sigma_m)\end{aligned}$$

□

11.5.1 Transpositions

Definition 11.228. (Transposition) Let $n \in \mathbb{N}$ and $i, j \in \{1, \dots, n\}$ then a **transposition** $(i \leftrightarrow j)_n \in P_n$ is defined by

$$(i \leftrightarrow j)_n \stackrel{[\text{definition: 11.28}]}{=} \left(i \begin{smallmatrix} \leftrightarrow \\ \{1, \dots, n\} \end{smallmatrix} j \right)$$

or using [definition: 11.28]

$$(i \leftrightarrow j)_n(k) = \begin{cases} k & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \\ j & \text{if } k = i \\ i & \text{if } k = j \end{cases}$$

If $i \neq j$ then $(i \leftrightarrow j)_n$ is a **strict transposition** [note that for a transposition to be strict we must have that $n \in \mathbb{N} \setminus \{1\}$].

Theorem 11.229. Let $n \in \mathbb{N}$ then we have for $i, j \in \{1, \dots, n\}$

1. If $i = j$ then $(i \leftrightarrow j)_n = \text{Id}_{\{1, \dots, n\}}$
2. $(i \leftrightarrow j)_n = (j \leftrightarrow i)_n$

3. $(i \leftrightarrow_n j) \circ (i \leftrightarrow_n j) = \text{Id}_{\{1, \dots, n\}}$
4. $(i \leftrightarrow_n j) = (i \leftrightarrow_{n+1} j)|_{\{1, \dots, n\}}$

Proof. This is proved in [theorem: 11.29] □

We can always extend a permutation on $\{1, \dots, n\}$ to a permutation on $\{1, \dots, n+1\}$ as the following theorem shows.

Theorem 11.230. *Let $n, m \in \mathbb{N}$ with $n < m$ then for every $\sigma \in P_n$ define*

$$\sigma|_{\{1, \dots, m\}}: \{1, \dots, m\} \Rightarrow \{1, \dots, m\}$$

by

$$\sigma|_{\{1, \dots, m\}}(i) = \begin{cases} i & \text{if } i \in \{n+1, \dots, m\} \\ \sigma(i) & \text{if } i \in \{1, \dots, n\} \end{cases} \in \{1, \dots, m\}$$

then we have

1. $\sigma|_{\{1, \dots, m\}} \in P_m$
2. $\forall i, j \in \{1, \dots, n\}$ we have $(i \leftrightarrow_n j)|_{\{1, \dots, m\}} = (i \leftrightarrow_m j)$
3. If $\sigma, \rho \in P_n$ then $(\sigma \circ \rho)|_{\{1, \dots, m\}} = \sigma|_{\{1, \dots, m\}} \circ \rho|_{\{1, \dots, m\}}$
4. If $k \in \mathbb{N}$ and $\{\sigma_i\}_{i \in \{1, \dots, k\}} \subseteq P_n$ then

$$(\sigma_1 \circ \dots \circ \sigma_n)|_{\{1, \dots, m\}} = (\sigma_1)|_{\{1, \dots, m\}} \circ \dots \circ (\sigma_n)|_{\{1, \dots, m\}}$$

Proof.

1. We have:

injectivity. For $i, j \in \{1, \dots, m\}$ with $\sigma|_{\{1, \dots, m\}}(i) = \sigma|_{\{1, \dots, m\}}(j)$ we have either:

$i, j \in \{1, \dots, n\}$. Then we have $\sigma(i) = \sigma|_{\{1, \dots, m\}}(i) = \sigma|_{\{1, \dots, m\}}(j) = \sigma(j)$ proving, as σ is injective, that $i = j$.

$i \in \{1, \dots, n\} \wedge j \in \{n+1, \dots, m\}$. Then $\sigma(i) = \sigma|_{\{1, \dots, m\}}(i) = \sigma|_{\{1, \dots, m\}}(j) = j$, hence as $\sigma(i) \in \{1, \dots, n\}$ we have that $j \in \{1, \dots, n\}$ contradicting $j \in \{n+1, \dots, m\}$. So this case does not apply.

$i \in \{n+1, \dots, m\} \wedge j \in \{1, \dots, n\}$. Then $\sigma(j) = \sigma|_{\{1, \dots, m\}}(j) = \sigma|_{\{1, \dots, m\}}(i) = i$, hence as $\sigma(j) \in \{1, \dots, n\}$ we have that $i \in \{1, \dots, n\}$ contradicting $i \in \{n+1, \dots, m\}$. So this case does not apply.

$i, j \in \{n+1, \dots, m\}$. Then we have $i = \sigma|_{\{1, \dots, m\}}(i) = \sigma|_{\{1, \dots, m\}}(j) = j$ or $i = j$.

So in all cases $i = j$.

surjectivity. If $j \in \{1, \dots, m\}$ then either $j \in \{n+1, \dots, m\}$ so that $j = \sigma|_{\{1, \dots, m\}}(j)$ or $j \in \{1, \dots, n\}$ and then there exists a $i \in \{1, \dots, n\}$ so that $j = \sigma(i) = \sigma|_{\{1, \dots, m\}}(i)$.

So $\sigma|_{\{1, \dots, m\}}: \{1, \dots, m\} \Rightarrow \{1, \dots, m\}$ is a bijection or $\sigma|_{\{1, \dots, m\}} \in P_m$

2. Let $k \in \{1, \dots, m\}$ then we have either:

$k \in \{1, \dots, n\}$. Then as $i, j \in \{1, \dots, n\}$ we have

$$(i \leftrightarrow_n j)|_{\{1, \dots, m\}}(k) = (i \leftrightarrow_n j)(k) = \begin{cases} i & \text{if } k = j \\ j & \text{if } k = i \\ k & \text{if } i \in \{1, \dots, n\} \setminus \{i, j\} \end{cases} = (i \leftrightarrow_m j)(k)$$

$k \in \{n+1, \dots, m\}$. Then as $i, j \in \{1, \dots, n\}$ we have that $i, j \neq k$ so that

$$(i \leftrightarrow_n j)^{|\{1, \dots, m\}}(k) = k = (i \leftrightarrow_m j)(k)$$

So in all cases $(i \leftrightarrow_n j)^{|\{1, \dots, m\}}(k) = (i \leftrightarrow_m j)(k)$ proving that

$$(i \leftrightarrow_n j)^{|\{1, \dots, m\}} = (i \leftrightarrow_m j)$$

3. Let $i \in \{1, \dots, m\}$ then we have either:

$i \in \{1, \dots, n\}$. Then

$$\begin{aligned} (\sigma \circ \rho)^{|\{1, \dots, m\}}(i) &= (\sigma \circ \rho)(i) \\ &= \sigma(\rho(i)) \\ &= \sigma^{|\{1, \dots, m\}}(\rho^{|\{1, \dots, m\}}(i)) \\ &= (\sigma^{|\{1, \dots, m\}} \circ \rho^{|\{1, \dots, m\}})(i) \end{aligned}$$

$i \in \{n+1, \dots, m\}$. Then

$$\begin{aligned} (\sigma \circ \rho)^{|\{1, \dots, m\}}(i) &= i \\ &= \sigma^{|\{1, \dots, m\}}(i) \\ &= \sigma^{|\{1, \dots, m\}}(\rho^{|\{1, \dots, m\}}(i)) \\ &= (\sigma^{|\{1, \dots, m\}} \circ \rho^{|\{1, \dots, m\}})(i) \end{aligned}$$

So $\forall i \in \{1, \dots, m\}$ we have $(\sigma \circ \rho)^{|\{1, \dots, m\}}(i) = (\sigma^{|\{1, \dots, m\}} \circ \rho^{|\{1, \dots, m\}})(i)$ proving that

$$(\sigma \circ \rho)^{|\{1, \dots, m\}} = \sigma^{|\{1, \dots, m\}} \circ \rho^{|\{1, \dots, m\}}$$

4. We prove this by induction, so let

$$S = \{k \in \mathbb{N} \mid \text{If } \{\sigma_i\}_{i \in \{1, \dots, k\}} \subseteq P_n \text{ then } (\sigma_1 \circ \dots \circ \sigma_k)^{|\{1, \dots, m\}} = (\sigma_1)^{|\{1, \dots, m\}} \circ \dots \circ (\sigma_k)^{|\{1, \dots, m\}}\}$$

then we have:

$1 \in S$. Let $\{\sigma_i\}_{i \in \{1, \dots, 1\}} \subseteq P_n$ then

$$\begin{aligned} (\sigma_1 \circ \dots \circ \sigma_n)^{|\{1, \dots, m\}} &\stackrel{[\text{theorem: 11.227}]}{=} (\sigma_1)^{|\{1, \dots, m\}} \\ &\stackrel{[\text{theorem: 11.227}]}{=} ((\sigma_1)^{|\{1, \dots, m\}}) \circ \dots \circ (\sigma_k)^{|\{1, \dots, m\}} \end{aligned}$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{\sigma_i\}_{i \in \{1, \dots, n+1\}} \subseteq P_n$ then we have

$$\begin{aligned} (\sigma_1 \circ \dots \circ \sigma_{n+1})^{|\{1, \dots, m\}} &\stackrel{[\text{theorem: 11.227}]}{=} \\ ((\sigma_1 \circ \dots \circ \sigma_n) \circ \sigma_{n+1})^{|\{1, \dots, m\}} &\stackrel{(3)}{=} \\ (\sigma_1 \circ \dots \circ \sigma_n)^{|\{1, \dots, m\}} \circ (\sigma_{n+1})^{|\{1, \dots, m\}} &\stackrel{n \in S}{=} \\ ((\sigma_1)^{|\{1, \dots, m\}} \circ \dots \circ (\sigma_n)^{|\{1, \dots, m\}}) \circ (\sigma_{n+1})^{|\{1, \dots, m\}} &\stackrel{[\text{theorem: 11.227}]}{=} \\ ((\sigma_1)^{|\{1, \dots, m\}} \circ \dots \circ (\sigma_{n+1})^{|\{1, \dots, m\}}) & \end{aligned}$$

proving that $n+1 \in S$

□

Corollary 11.231. Let $n \in \mathbb{N}$ and $\sigma \in P_{n+1}$ such that $\sigma(n+1) = n+1$ [so that by [theorem: 11.27] $\sigma|_{\{1, \dots, n\}} \in P_n$] then

$$(\sigma|_{\{1, \dots, n\}})^{|\{1, \dots, n+1\}} = \sigma$$

Proof. Let $k \in \{1, \dots, n+1\}$ then we have either:

$k \in \{1, \dots, n\}$. Then $(\sigma|_{\{1, \dots, n\}})^{|\{1, \dots, n+1\}}(k) = \sigma|_{\{1, \dots, n\}}(k) = \sigma(k)$

$k = n+1$. Then $(\sigma|_{\{1, \dots, n\}})^{|\{1, \dots, n+1\}}(k) = n+1 = \sigma(n+1)$ $\square$

So $\forall k \in \{1, \dots, n+1\}$ we have $(\sigma|_{\{1, \dots, n\}})^{|\{1, \dots, n+1\}}(k) = \sigma(k)$ proving that

$$(\sigma|_{\{1, \dots, n\}})^{|\{1, \dots, n+1\}} = \sigma$$

We show now that every permutation of more then one element can be written as the composition of disjoint transpositions.

Theorem 11.232. Let $n \in \mathbb{N} \setminus \{1\}$ and $\sigma \in P_n$ then there exists a family $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n$ with $\forall k \in \{1, \dots, l\} \ i_k \neq j_k$ such that

$$\sigma = (i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)$$

Proof. We use mathematical induction to prove this, so let

$S = \{n \in \{2, \dots, \infty\} | \text{If } \sigma \in P_n \text{ then } \sigma = (i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l) \text{ where } \{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n \text{ satisfies } i_k \neq j_k \forall k \in \{1, \dots, l\}\}$

then we have:

$2 \in S$. Then for $\sigma \in P_2$ we have for $\sigma(1)$ either:

$\sigma(1) = 1$. Then we must have, as σ is injective, that $\sigma(2) = 2$ hence $\sigma = \text{Id}_{\{1, \dots, 2\}}$. Define $\{(i_k \leftrightarrow j_k)\}_{k \in \{1, \dots, 2\}}$ by $(i_1 \leftrightarrow_2 j_1) = (i_2 \leftrightarrow_2 j_2) = (1 \leftrightarrow_2 2)$, a family of **strict** transpositions. Then

$$(i_1 \leftrightarrow_2 j_1) \circ \dots \circ (i_2 \leftrightarrow_2 j_2) = (1 \leftrightarrow_2 2) \circ (1 \leftrightarrow_2 2) \stackrel{[\text{theorem: 11.229}]}{=} \text{Id}_{\{1, \dots, 2\}} = \sigma$$

proving that in this cases $2 \in S$.

$\sigma(1) = 2$. Then we must have, as σ is injective, that $\sigma(2) = 1$ hence $\sigma = (1 \leftrightarrow_2 2)$. Define $\{(i_k \leftrightarrow j_k)\}_{k \in \{1, \dots, 1\}}$ by $(i_1 \leftrightarrow_2 j_1) = (1 \leftrightarrow_2 2)$, a family of strict transpositions. Then

$$(i_1 \leftrightarrow_2 j_1) \circ \dots \circ (i_1 \leftrightarrow_2 j_1) = (1 \leftrightarrow_2 2) = \sigma$$

$n \in S \Rightarrow n+1 \in S$. Let $\sigma \in P_{n+1}$ then for $\sigma(n+1)$ we have either:

$\sigma(n+1) = n+1$. Using [theorem: 11.27] we have that $\sigma|_{\{1, \dots, n\}} \in P_n$. As $n \in S$ there exists a family of strict transpositions $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n$ such that

$$\sigma|_{\{1, \dots, n\}} = (i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)$$

So that

$$\begin{aligned} \sigma &\stackrel{[\text{theorem: 11.231}]}{=} (\sigma|_{\{1, \dots, n\}})^{|\{1, \dots, n+1\}} \\ &= ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l))^{|\{1, \dots, n+1\}} \\ &\stackrel{[\text{theorem: 11.230}]}{=} ((i_1 \leftrightarrow_n j_1)^{|\{1, \dots, n+1\}} \circ \dots \circ (i_l \leftrightarrow_n j_l)^{|\{1, \dots, n+1\}}) \\ &\stackrel{[\text{theorem: 11.230}]}{=} (i_1 \leftrightarrow_{n+1} j_1) \circ \dots \circ (i_l \leftrightarrow_{n+1} j_l) \end{aligned}$$

proving that in this case $n+1 \in S$

$\sigma(n+1) \neq n+1$. Then as σ is a bijection there exists a $i \in \{1, \dots, n+1\} \setminus \{n+1\} = \{1, \dots, n\}$ such that $\sigma(i) = n+1$. Define then

$$\rho = \sigma \circ (i \leftrightarrow_{n+1} n+1) \tag{11.111}$$

where

$$\left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \text{ is a strict transposition as } i \neq n+1$$

then we have $\rho(n+1) = \left(\sigma \left(\left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) (i) \right) \right) = \sigma(i) = n+1$, so using [theorem: 11.27] we have that $\rho|_{\{1, \dots, n\}} \in P_n$. As $n \in S$ there exists a $\{(r_k \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n$ such that

$$\rho|_{\{1, \dots, n\}} = \left((r_1 \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_1) \circ \dots \circ (r_l \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_l) \right)$$

So that

$$\begin{aligned} \rho &\stackrel{[\text{theorem: 11.231}]}{=} (\rho|_{\{1, \dots, n\}})^{| \{1, \dots, n+1\} } \\ &= \left((r_1 \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_1) \circ \dots \circ (r_l \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_l) \right)^{| \{1, \dots, n+1\} } \\ &\stackrel{[\text{theorem: 11.230}]}{=} \left((r_1 \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_1) \right)^{| \{1, \dots, n+1\} } \circ \dots \circ \left((r_l \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} s_l) \right)^{| \{1, \dots, n+1\} } \\ &\stackrel{[\text{theorem: 11.230}]}{=} \left(r_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} s_1 \right) \circ \dots \circ \left(r_l \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} s_l \right) \end{aligned} \quad (11.112)$$

Define now

$$\left\{ \left(i_k \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j_k \right) \right\}_{k \in \{1, \dots, l+1\}} \text{ by } (i_k \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} j_k) = \begin{cases} \left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) & \text{if } k = l+1 \\ \left(r_k \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} s_k \right) & \text{if } k \in \{1, \dots, l\} \end{cases}$$

then we have

$$\begin{aligned} \sigma &\stackrel{[\text{theorem: 11.229}]}{=} \sigma \circ \left(\left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \circ \left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \right) \\ &\stackrel{\text{associativity}}{=} \left(\sigma \circ \left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \right) \circ \left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \\ &\stackrel{[\text{eq: 11.111}]}{=} \rho \circ \left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \\ &\stackrel{[\text{theorem: 11.112}]}{=} \left((r_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} s_1) \circ \dots \circ (r_l \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} s_l) \right) \circ \left(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} n+1 \right) \\ &= \left(\left(i_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j_1 \right) \circ \dots \circ \left(i_l \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j_l \right) \right) \circ \left(i_{n+1} \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j_{n+1} \right) \\ &\stackrel{[\text{theorem: 11.227}]}{=} \left(i_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j_1 \right) \circ \dots \circ \left(i_{l+1} \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j_{l+1} \right) \end{aligned}$$

proving that $n+1 \in S$

So in all cases we have $n+1 \in S$.

Mathematical induction proves then that $S = \mathbb{N} \setminus \{1\} = \{2, \dots, \infty\}$ completing the proof. $\square$

Lemma 11.233. Let $n \in \mathbb{N}$, $i, j \in \{1, \dots, n\}$, $m \in \mathbb{N}$ and $\{(k_r, l_r)\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n\}$ then if

$$(i \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} j) = (k_1 \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} l_1) \circ \dots \circ (k_m \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} l_m)$$

then

$$(i \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} j) = (k_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} l_1) \circ \dots \circ (k_m \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} l_m)$$

Proof. First we prove by induction that

$$(k_1 \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} l_1) \circ \dots \circ (k_m \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} l_m) = \left((k_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} l_1) \circ \dots \circ (k_m \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} l_m) \right)_{| \{1, \dots, n\}} \quad (11.113)$$

So let

$$S = \left\{ m \in \mathbb{N} \mid \text{If } \{(k_r, l_r)\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n\} \text{ then } (k_1 \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} l_1) \circ \dots \circ (k_m \begin{smallmatrix} \leftrightarrow \\ n \end{smallmatrix} l_m) = \left((k_1 \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} l_1) \circ \dots \circ (k_m \begin{smallmatrix} \leftrightarrow \\ n+1 \end{smallmatrix} l_m) \right)_{| \{1, \dots, n\}} \right\}$$

then we have:

$1 \in \mathbf{S}$. As

$$\begin{aligned} (k_1 \leftrightarrow_n l_1) \circ \dots \circ (k_1 \leftrightarrow_n l_1) &= (k_1 \leftrightarrow_n l_1) \\ &\stackrel{[\text{theorem: 11.229}]}{=} (k_1 \leftrightarrow_{n+1} l_1)_{|\{1, \dots, n\}} \\ &= \left((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_1 \leftrightarrow_{n+1} l_1) \right)_{|\{1, \dots, n\}} \end{aligned}$$

proving that $1 \in S$.

$m \in \mathbf{S} \Rightarrow m+1 \in \mathbf{S}$. Consider $(k_1 \leftrightarrow_n l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_n l_{m+1})$. Let $k \in \{1, \dots, n\}$ then we have

$$\begin{aligned} &((k_1 \leftrightarrow_n l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_n l_{m+1}))(k) = \\ &(((k_1 \leftrightarrow_n l_1) \circ \dots \circ (k_m \leftrightarrow_n l_m)) \circ (k_{m+1} \leftrightarrow_n l_{m+1}))(k) \stackrel{m \in S}{=} \\ &\left(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m))_{|\{1, \dots, n\}} \circ (k_{m+1} \leftrightarrow_n l_{m+1}) \right)(k) = \\ &\left(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m))_{|\{1, \dots, n\}} \right) ((k_{m+1} \leftrightarrow_n l_{m+1})(k)) \stackrel{(k_{m+1} \leftrightarrow_n l_{m+1})(k) \in \{1, \dots, n\}}{=} \\ &\left(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m)) \right) ((k_{m+1} \leftrightarrow_n l_{m+1})(k)) = \\ &\left(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m)) \circ (k_{m+1} \leftrightarrow_n l_{m+1}) \right)(k) = \\ &\left((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_{n+1} l_{m+1}) \right)(k) \stackrel{k \in \{1, \dots, n\}}{=} \\ &\left((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_{n+1} l_{m+1}) \right)_{|\{1, \dots, n\}}(k) \end{aligned}$$

proving that

$$(k_1 \leftrightarrow_n l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_n l_{m+1}) = \left((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_{n+1} l_{m+1}) \right)_{|\{1, \dots, n\}}$$

so that

$$n+1 \in S$$

Next we use induction to prove that

$$((k_1 \leftrightarrow_n l_1) \circ \dots \circ (k_m \leftrightarrow_n l_m))(n+1) = n+1 \quad (11.114)$$

So let

$$S = \left\{ m \in \mathbb{N} \mid \text{If } \{(k_r, l_r)\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n\} \text{ then } ((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m))(n+1) = n+1 \right\}$$

then we have:

$1 \in \mathbf{S}$. Then $((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_1 \leftrightarrow_{n+1} l_1))(n+1) = (k_1 \leftrightarrow_{n+1} l_1)(n+1) \stackrel{n+1 \neq k_1, l_1}{=} n+1$ proving that $1 \in S$.

$m \in \mathbf{S} \Rightarrow m+1 \in \mathbf{S}$. As we have

$$\begin{aligned} &((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_{m+1} \leftrightarrow_{n+1} l_{m+1}))(n+1) = \\ &(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m)) \circ (k_{m+1} \leftrightarrow_{n+1} l_{m+1}))(n+1) = \\ &\left(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m)) \right) ((k_{m+1} \leftrightarrow_{n+1} l_{m+1})(n+1)) \stackrel{n+1 \neq k_{m+1}, l_{m+1}}{=} \\ &\left(((k_1 \leftrightarrow_{n+1} l_1) \circ \dots \circ (k_m \leftrightarrow_{n+1} l_m)) \right) (n+1) \stackrel{m \in S}{=} \\ &n+1 \end{aligned}$$

it is proved that

$$m+1 \in S$$

Now we are ready for the rest of the proof. Let $k \in \{1, \dots, n+1\}$ then we have either:

$k = n+1$. Then we have

$$\begin{aligned} \left(i \leftrightarrow_{n+1} j\right)(k) &= \left(i \leftrightarrow_{n+1} j\right)(n+1) \\ &\stackrel{n+1 \neq i, j}{=} n+1 \\ &\stackrel{[\text{eq: 11.114}]}{=} \left(\left(k_1 \leftrightarrow_{n+1} l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} l_m\right)\right)(n+1) \\ &= \left(\left(k_1 \leftrightarrow_{n+1} l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} l_m\right)\right)(k) \end{aligned}$$

$k \in \{1, \dots, n\}$. Then we have

$$\begin{aligned} \left(i \leftrightarrow_{n+1} j\right)(k) &= \left(i \leftrightarrow_{n+1} j\right)_{|\{1, \dots, n\}}(k) \\ &\stackrel{[\text{theorem: 11.29}]}{=} \left(i \leftrightarrow_n j\right)(k) \\ &= \left(\left(k_1 \leftrightarrow_n l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_n l_m\right)\right)(k) \\ &\stackrel{[\text{theorem: 11.113}]}{=} \left(\left(k_1 \leftrightarrow_{n+1} l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} l_m\right)\right)_{|\{1, \dots, n\}}(k) \\ &= \left(\left(k_1 \leftrightarrow_{n+1} l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} l_m\right)\right)(k) \end{aligned}$$

So in all cases we have $\left(i \leftrightarrow_{n+1} j\right)(k) = \left(\left(k_1 \leftrightarrow_{n+1} l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} l_m\right)\right)(k)$ proving that

$$\left(i \leftrightarrow_{n+1} j\right) = \left(k_1 \leftrightarrow_{n+1} l_1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} l_m\right) \quad \square$$

Theorem 11.234. Let $n \in \mathbb{N} \setminus \{1\}$, $i, j \in \{1, \dots, n\}$ with $i \neq j$ then there exists a $m \in \mathbb{N}$ and $\{k_r\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n-1\}$ such that $\left(i \leftrightarrow_n j\right) = \left(k_1 \leftrightarrow_n k_1 + 1\right) \circ \dots \circ \left(k_m \leftrightarrow_n k_m + 1\right)$

Proof. We prove this by recursion on n so define

$$S = \{n \in \{2, \dots, \infty\} \mid \text{If } (i, j) \in \{1, \dots, n\} \text{ with } i \neq j \text{ then there exist a } \{k_r\}_{r \in \{1, \dots, m\}} \text{ such that } (i \leftrightarrow_n j) = (k_1 \leftrightarrow_n k_1 + 1) \circ \dots \circ (k_m \leftrightarrow_n k_m + 1)\}$$

then we have:

$2 \in S$. As $\left(i \leftrightarrow_n j\right) \stackrel{[\text{theorem: 11.229}]}{=} \left(j \leftrightarrow_n i\right)$ we may assume that $i < j$ so we have that $i = 1$ and $j = 2 = i + 1$ so that $\left(i \leftrightarrow_2 j\right) = \left(1 \leftrightarrow_2 2\right)$ proving that $2 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $(i, j) \in \{1, \dots, n+1\}$ with $i \neq j$. As $\left(i \leftrightarrow_{n+1} j\right) \stackrel{[\text{theorem: 11.229}]}{=} \left(j \leftrightarrow_{n+1} i\right)$ we may assume that $i < j$ so we have the following cases to consider:

$i = n \wedge j = n+1$. Then $\left(i \leftrightarrow_{n+1} j\right) = \left(i \leftrightarrow_{n+1} 1\right)$ proving that $n+1 \in S$ in this case.

$i < n \wedge j = n+1$. As $n \in S$ there exists $\{k_r\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n-1\}$ such that

$$\left(i \leftrightarrow_n n\right) = \left(k_1 \leftrightarrow_n k_1 + 1\right) \circ \dots \circ \left(k_m \leftrightarrow_n k_m + 1\right)$$

Given that $i, n \in \{1, \dots, n\}$ and $\forall r \in \{1, \dots, m\}$ we have $k_r, k_r + 1 \in \{1, \dots, n\}$ it follows from [lemma: 11.233] that

$$\left(i \leftrightarrow_{n+1} n\right) = \left(k_1 \leftrightarrow_{n+1} k_1 + 1\right) \circ \dots \circ \left(k_m \leftrightarrow_{n+1} k_m + 1\right) \quad (11.115)$$

Consider now

$$\rho = \left(n \leftrightarrow_{n+1} n+1\right) \circ \left(i \leftrightarrow_{n+1} n\right) \circ \left(n \leftrightarrow_{n+1} n+1\right)$$

then given $k \in \{1, \dots, n+1\}$ we have either:

$k = n+1$. Then we have

$$\begin{aligned}
 \rho(k) &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) \left(\left(n \overset{\leftrightarrow}{n+1} n+1 \right) (n+1) \right) \right) \\
 &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) (n) \right) \\
 &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) (i) \\
 &\stackrel{i \neq n, n+1}{=} i \\
 &= \left(i \overset{\leftrightarrow}{n+1} n+1 \right) (n+1) \\
 &\stackrel{j=n+1 \wedge k=n+1}{=} \left(i \overset{\leftrightarrow}{n+1} j \right) (k)
 \end{aligned}$$

$k = n$. Then we have

$$\begin{aligned}
 \rho(k) &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) \left(\left(n \overset{\leftrightarrow}{n+1} n+1 \right) (n) \right) \right) \\
 &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) (n+1) \right) \\
 &\stackrel{n+1 \neq i, n}{=} \left(n \overset{\leftrightarrow}{n+1} n+1 \right) (n+1) \\
 &= n \\
 &\stackrel{n \neq i, n+1}{=} \left(i \overset{\leftrightarrow}{n+1} n+1 \right) (n) \\
 &\stackrel{n=k \wedge j=n+1}{=} \left(i \overset{\leftrightarrow}{n+1} j \right) (k)
 \end{aligned}$$

$k = i$. Then

$$\begin{aligned}
 \rho(k) &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) \left(\left(n \overset{\leftrightarrow}{n+1} n+1 \right) (i) \right) \right) \\
 &\stackrel{i \neq n, n+1}{=} \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) (i) \right) \\
 &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) (n) \\
 &= n+1 \\
 &= \left(i \overset{\leftrightarrow}{n+1} n+1 \right) (i) \\
 &\stackrel{i=k \wedge j=n+1}{=} \left(i \overset{\leftrightarrow}{n+1} j \right) (k)
 \end{aligned}$$

$k \in \{1, \dots, n-1\} \setminus \{i\}$. Then

$$\begin{aligned}
 \rho(k) &= \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) \left(\left(n \overset{\leftrightarrow}{n+1} n+1 \right) (k) \right) \right) \\
 &\stackrel{k \neq n, n+1}{=} \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \left(\left(i \overset{\leftrightarrow}{n+1} n \right) (k) \right) \\
 &\stackrel{k \neq i, n}{=} \left(n \overset{\leftrightarrow}{n+1} n+1 \right) (k) \\
 &\stackrel{k \neq n, n+1}{=} k \\
 &\stackrel{k \neq i, n+1}{=} \left(i \overset{\leftrightarrow}{n+1} n+1 \right) (k) \\
 &\stackrel{j=n+1}{=} \left(i \overset{\leftrightarrow}{n+1} j \right) (k)
 \end{aligned}$$

So we have that $\forall k \in \{1, \dots, n+1\}$ that

$$\left(i \overset{\leftrightarrow}{n+1} j \right) (k) = \rho(k) = \left(\left(n \overset{\leftrightarrow}{n+1} n+1 \right) \circ \left(i \overset{\leftrightarrow}{n+1} n \right) \circ \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \right) (k)$$

proving that

$$\left(i \overset{\leftrightarrow}{n+1} j \right) = \left(n \overset{\leftrightarrow}{n+1} n+1 \right) \circ \left(i \overset{\leftrightarrow}{n+1} n \right) \circ \left(n \overset{\leftrightarrow}{n+1} n+1 \right)$$

or combining the above with [eq: 11.115] proves that

$$\left(i \xleftrightarrow{n+1} j\right) = \left(n \xleftrightarrow{n+1} n+1\right) \circ \left(k_1 \xleftrightarrow{n+1} k_1+1\right) \circ \dots \circ \left(k_m \xleftrightarrow{n+1} k_m+1\right) \circ \left(n \xleftrightarrow{n+1} n+1\right)$$

so that $n+1 \in S$ in this case.

$i < j \wedge j \in \{1, \dots, n\}$. Then as $i, j \in \{1, \dots, n\}$ we have by [theorem: 11.229] that $\left(i \xleftrightarrow{n} j\right) = \left(i \xleftrightarrow{n+1} j\right)_{|\{1, \dots, n\}}$. As $n \in S$ it follows that there exist a $\{k_r\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n-1\}$ such that

$$\left(i \xleftrightarrow{n} j\right) = \left(k_1 \xleftrightarrow{n} k_1+1\right) \circ \dots \circ \left(k_m \xleftrightarrow{n} k_m+1\right)$$

Given that $i, j \in \{1, \dots, n\}$ and $\forall r \in \{1, \dots, m\}$ we have $k_r, k_r+1 \in \{1, \dots, n\}$ it follows from [lemma: 11.233] that

$$\left(i \xleftrightarrow{n+1} j\right) = \left(k_1 \xleftrightarrow{n+1} k_1+1\right) \circ \dots \circ \left(k_m \xleftrightarrow{n+1} k_m+1\right)$$

which proves that $n+1 \in S$ in this case.

So in all possible cases we have $n+1 \in S$ which completes the induction argument. $\square$

Permutations are typically used to permute the arguments of functions with several arguments. Remember that X^n is defined as $X^n \stackrel{[\text{definition: 6.92}]}{=} A^{\{1, \dots, n\}}$ so that the following definition makes sense.

Definition 11.235. Let X, Y be sets, $n \in \mathbb{N}$ and $f: X^n \rightarrow Y$ a function and $\sigma \in P_n$ then

$$\sigma f: X^n \rightarrow Y$$

is defined by

$$(\sigma f)(x) = f(x \circ \sigma)$$

or using [notation: 6.85]

$$\sigma f(x) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)})$$

Theorem 11.236. Let X be a set, $\langle Y, +, \cdot \rangle$ a vector space over a field $\langle F, +, \cdot \rangle$, $\sigma \in P_n$, then we have for the vector space $\langle Y^X, +, \cdot \rangle$ [see example: 11.69] that:

1. $\forall f, g \in Y^X$ we have $\sigma(f+g) = \sigma f + \sigma g$
2. $\forall f \in Y^X, \alpha \in F$ we have $\sigma(\alpha \cdot f) = \alpha \cdot \sigma f$

Proof.

1. Let $x \in X$ then

$$(\sigma(f+g))(x) = (f+g)(x \circ \sigma) = f(x \circ \sigma) + g(x \circ \sigma) = (\sigma f)(x) + (\sigma g)(x) = (\sigma f + \sigma g)(x)$$

proving that

$$\sigma(f+g) = \sigma f + \sigma g$$

2. Let $x \in X$ then

$$(\sigma(\alpha \cdot f))(x) = (\alpha \cdot f)(x \circ \sigma) = \alpha \cdot f(x \circ \sigma) = \alpha \cdot ((\sigma f)(x)) = (\alpha \cdot \sigma f)(x)$$

proving that

$$\sigma(\alpha \cdot f) = \alpha \cdot \sigma f$$

$\square$

Theorem 11.237. Let X, Y be sets, $n \in \mathbb{N}$ and $f: X^n \rightarrow Y$ a function then

$$\text{Id}_{\{1, \dots, n\}} f = f$$

Proof. $\forall x \in X^n$ we have $(\text{Id}_{\{1, \dots, n\}} f)(x) = f(x \circ \text{Id}_{\{1, \dots, n\}}) = f(x)$ proving that

$$\text{Id}_{\{1, \dots, n\}} f = f. \quad \square$$

Theorem 11.238. Let X, Y be sets, $n \in \mathbb{N}$ and $f: X^n \rightarrow Y$ a function and $\sigma, \rho \in P_n$ then

$$\sigma(\rho f) = (\sigma \circ \rho) f$$

Proof. $\forall x \in X^n$ we have

$$\sigma(\rho f)(x) = (\rho f)(x \circ \sigma) = f((x \circ \sigma) \circ \rho) = f(x \circ (\sigma \circ \rho)) = (\sigma \circ \rho) f(x)$$

so that $\sigma(\rho f) = (\sigma \circ \rho) f$. $\square$

Theorem 11.239. Let $n \in \mathbb{N}$, $\sigma \in P_n$, X a set, $\langle Y, +, \cdot \rangle$ a ring and $f, g: X^n \rightarrow Y$ then we have using point wise product and adding of functions that

1. $\forall \alpha \in Y \sigma(\alpha \cdot f) = \alpha \cdot (\sigma f)$
2. $\sigma(f + g) = \sigma f + \sigma g$

Proof.

1. Let $x \in X^n$ then $(\sigma(\alpha \cdot f))(x) = (\alpha \cdot f)(x \circ \sigma) \stackrel{\text{def}}{=} \alpha \cdot (f(x \circ \sigma)) = \alpha \cdot (\sigma f)(x)$ proving that

$$\sigma(\alpha \cdot f) = \alpha \cdot (\sigma f)$$

2. Let $x \in X^n$ then

$$(\sigma(f + g))(x) = (f + g)(x \circ \sigma) \stackrel{\text{def}}{=} f(x \circ \sigma) + g(x \circ \sigma) = (\sigma f)(x) + (\sigma g)(x) = (\sigma f + \sigma g)(x)$$

proving that

$$\sigma(f + g) = \sigma f + \sigma g \quad \square$$

Theorem 11.240. Let X, Y be sets, $n \in \mathbb{N}$ and $f: X^n \rightarrow Y$ a function such that $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have $(i \leftrightarrow_n j) f = f$ then if $l \in \mathbb{N}$ and $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n$ we have

$$((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)) f = f$$

Proof. We prove this by induction on l , so take:

$S = \{l \in \mathbb{N} \mid \text{If } f: X^n \rightarrow Y \text{ satisfies } \forall i, j \in \{1, \dots, n\} \text{ with } i \neq j \text{ we have } (i \leftrightarrow_n j) f = f \text{ then for every}$

$\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n \text{ we have } ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)) f = f\}$

then we have:

$1 \in S$. Let $f: X^n \rightarrow Y$ satisfies $\forall i, j \in \{1, \dots, n\} (i \leftrightarrow_n j) f = f$ and $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, 1\}} \subseteq P_n$ then we have

$$((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)) f \stackrel{[\text{theorem: 11.227}]}{=} (i_1 \leftrightarrow_n j_1) f \quad (11.116)$$

For i_1, j_1 we have either:

$i_1 = j_1$. Then

$$(i_1 \leftrightarrow_n j_1) f \stackrel{[\text{theorem: 11.227}]}{=} \text{Id}_{\{1, \dots, n\}} f \stackrel{[\text{theorem: 11.237}]}{=} f$$

$i_1 \neq j_1$. Then by the hypothesis we have $(i_1 \leftrightarrow_n j_1) f = f$

combining this with [eq: 11.116] proves that $((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)) f = f$ hence $1 \in S$.

$l \in S \Rightarrow l + 1 \in S$. If $f: X^n \rightarrow Y$ satisfies $\forall i, j \in \{1, \dots, n\} (i \leftrightarrow_n j) f = f$ and $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l+1\}} \subseteq P_n$. For i_{l+1}, j_{l+1} we have either:

$i_{l+1} = j_{l+1}$. Then

$$(i_{l+1} \leftrightarrow_n j_{l+1})f \stackrel{[\text{theorem: 11.227}]}{=} \text{Id}_{\{1, \dots, n\}}f \stackrel{[\text{theorem: 11.237}]}{=} f$$

$i_{l+1} \neq j_{l+1}$. Then by the hypothesis we have $(i_{l+1} \leftrightarrow_n j_{l+1})f = f$

so in all cases we have

$$(i_{l+1} \leftrightarrow_n j_{l+1})f = f \quad (11.117)$$

Further

$$\begin{aligned} & ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_{l+1} \leftrightarrow_n j_{l+1}))f \stackrel{[\text{eq: 11.227}]}{=} \\ & (((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l)) \circ (i_{l+1} \leftrightarrow_n j_{l+1}))f \stackrel{[\text{theorem: 11.238}]}{=} \\ & ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l))((i_{l+1} \leftrightarrow_n j_{l+1})f) \stackrel{[\text{eq: 11.117}]}{=} \\ & ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l))f \stackrel{[n \in S]}{=} \\ & f \end{aligned}$$

proving that $l+1 \in S$. $\square$

We can extend the above theorem to general permutation's.

Corollary 11.241. *Let X, Y be sets, $n \in \mathbb{N}$ and $f: X^n \rightarrow Y$ a function such that $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have $(i \leftrightarrow_n j)f = f$ then $\forall \sigma \in P_n$ we have $\sigma f = f$.*

Proof. If $n=1$ then $\sigma = \text{Id}_{\{1\}}$ so that $\sigma f = \text{Id}_{\{1\}}f \stackrel{[\text{theorem: 11.237}]}{=} f$. To complete the proof we have to look at the case where $n \in \mathbb{N} \setminus \{1\}$. Using [theorem: 11.232] there exists a $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, l\}} \subseteq P_n$ such that $\sigma = ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l))$. Further

$$\sigma f = ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_l \leftrightarrow_n j_l))f \stackrel{[\text{theorem: 11.240}]}{=} f$$

proving the corollary. $\square$

The above shows that in order to prove that a multi parameter function is stable under a permutation we only have to show that it is stable under arbitrary transpositions. Actually we can weaken this condition. A multi parameter function is stable under a permutation if it is stable under transpositions of the form $(i \leftrightarrow_n i+1)$.

Corollary 11.242. *Let X, Y be sets, $n \in \mathbb{N}$ and $f: X^n \rightarrow Y$ a function such that $\forall i \in \{1, \dots, n-1\}$ we have $(i \leftrightarrow_n i+1)f = f$ then $\forall \sigma \in P_n$ we have $\sigma f = f$.*

Proof. If $n=1$ then $\sigma = \text{Id}_{\{1\}}$ so that $\sigma f = \text{Id}_{\{1\}}f \stackrel{[\text{theorem: 11.237}]}{=} f$. To complete the proof we have to look at the case where $n \in \mathbb{N} \setminus \{1\}$. Let $i, j \in \{1, \dots, n\}$ with $i \neq j$ then by [theorem: 11.234] there exist a $\{k_r\}_{r \in \{1, \dots, m\}} \subseteq \{1, \dots, n-1\}$ such that $(i \leftrightarrow_n j) = (k_1 \leftrightarrow_n k_1+1) \circ \dots \circ (k_m \leftrightarrow_n k_m+1)$. By the hypothesis we have $\forall r \in \{1, \dots, m\} (k_r \leftrightarrow_n k_r+1)f = f$, hence using [theorem: 11.240] it follows that $((k_1 \leftrightarrow_n k_1+1) \circ \dots \circ (k_m \leftrightarrow_n k_m+1))f = f$. So as $(i \leftrightarrow_n j) = (k_1 \leftrightarrow_n k_1+1) \circ \dots \circ (k_m \leftrightarrow_n k_m+1)$ it follows that $(i \leftrightarrow_n j)f = f$. As $i, j \in \{1, \dots, n\}$ was chosen arbitrary we can use the previous corollary [corollary: 11.241] to prove $\sigma f = f$ for every $\sigma \in P_n$. $\square$

11.5.2 Sign of a Permutation

In the following definition we will use the multiplicative Abelian semi-group $\langle \mathbb{Z}, \cdot \rangle$, so we use the symbol $\prod$ instead of $\sum$ [see remark: 11.2]. Further let $n \in \mathbb{N}_0$ then

$\{(i, j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\} \subseteq \{1, \dots, n\} \times \{1, \dots, n\}$ a finite set [see theorems: 10.87, 10.92]

so that by [theorem: 6.43] $\{(i, j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\}$ is a finite set. Hence the following definition makes sense.

Definition 11.243. Let $n \in \mathbb{N}$ then we define

$$\Phi_n: \mathbb{Z}^n \rightarrow \mathbb{Z} \text{ by } \Phi_n(x) = \prod_{(i,j) \in \{(i,j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\}} (x_i - x_j)$$

Example 11.244.

1. $\Phi_1(x) = 1$, as $\{(i,j) \in \{1\} \times \{1\} | i < j\} = \emptyset$ we have $\Phi_1(x) = \prod_{(i,j) \in \emptyset} (x_i - x_j) = 1$ [the neutral element of $\langle \mathbb{Z}, \cdot \rangle$].
2. If $x = (x_1, x_2) \in \mathbb{Z}^2$ then $\Phi_2(x) = x_2 - x_1$ because $\{(i,j) \in \{1, 2\} \times \{1, 2\} | i < j\} = \{(1, 2)\}$ so that

$$\Phi_2(x) = \prod_{(i,j) \in \{(1,2)\}} (x_i - x_j) = x_1 - x_2$$

3. If $x = (x_1, x_2, x_3)$ then $\Phi_3(x) = (x_1 - x_2) \cdot (x_1 - x_3) \cdot (x_2 - x_3)$ because

$$\{(i,j) \in \{1, 2, 3\} \times \{1, 2, 3\} | i < j\} = \{(1, 2), (1, 3), (2, 3)\}$$

so that

$$\Phi_3(x) = \prod_{(i,j) \in \{(1,2), (1,3), (2,3)\}} (x_i - x_j) = (x_1 - x_2) \cdot (x_1 - x_3) \cdot (x_2 - x_3)$$

4. ...

Lemma 11.245. Let $n \in \mathbb{N}$ then for $(x_1, \dots, x_n) \in \mathbb{Z}^n$ defined by $x_i = i$ we have $\Phi_n(x_1, \dots, x_n) \neq 0$, in other words $\Phi_n(1, \dots, n) \neq 0$.

Proof. If we define $(x_1, \dots, x_n)$ by $x_i = i$ then $\forall (i,j) \in \{(i,j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\}$ we have $x_i - x_j = i - j < 0$, so that $\{x_i - x_j\}_{(i,j) \in \{(i,j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\}} \subseteq \mathbb{R} \setminus \{0\}$. Using [theorem: 11.53] it follows then that $\Phi_n(x_1, \dots, x_n) = \prod_{(i,j) \in \{(i,j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\}} (x_i - x_j) \neq 0$. $\square$

Lemma 11.246. Let $n \in \mathbb{N}_0 \setminus \{1\} = \{2, \dots, \infty\}$ then if $k, l \in \{1, \dots, n\}$ with $k \neq l$ we have for the strict transposition $(k \leftrightarrow_n l)$ that

$$(k \leftrightarrow_n l) \Phi_n = (-1) \cdot \Phi_n$$

Proof. First we may always assume that $k < l$ [otherwise exchange k and l]. Define

$$I = \{(i,j) \in \{1, \dots, n\} \times \{1, \dots, n\} | i < j\}$$

Then we have the following excluding possibilities for $(i,j) \in I$

$i = k$. Then as $k < l$ and $i < j$ we have for j either:

$j < l$. Then $(i,j) \in J_0$ where $J_0 = \{(k,j) \in I | k < j < l\} \subseteq I$

$j = l$. Then $(i,j) \in J_1$ where $J_1 = \{(k,l)\} \subseteq I$

$l < j$. Then $(i,j) \in J_2$ where $J_2 = \{(k,j) \in I | l < j\} \subseteq I$

$i \neq k$. Then we have for i either:

$i = l$. Then as $i < j$ we have $(i,j) \in J_3 = \{(l,j) \in I | l < j\} \subseteq I$

$i \neq l$. Then we have for j either:

$j = k$. Then as $i < j \Rightarrow i < k$ we have $(i,j) \in J_4$ where $J_4 = \{(i,k) \in I | i < k\} \subseteq I$

$j = l$. Then as $i = k$ we have either:

$i < k$. Then $(i,j) \in J_5$ where $J_5 = \{(i,l) \in I | i < k\} \subseteq I$

$k < i$. Then $(i,j) \in J_6$ where $J_6 = \{(i,l) \in I | k < i < l\} \subseteq I$

$j \neq l, k$. Then $(i,j) \in J_7$ where $J_7 = \{(i,j) \in I | i \neq k, l \wedge j \neq k, l\} \subseteq I$

So

$$I = J_0 \cup J_1 \cup J_2 \cup J_3 \cup J_4 \cup J_5 \cup J_6 \cup J_7$$

Further for the possible intersections of $J_0, J_1, J_2, J_3, J_4, J_5, J_6$ and J_7 [using commutativity] we have

- $(i, j) \in J_0 \cap J_1$. Then $i = k < j < l \wedge j = l$ a contradiction.
- $(i, j) \in J_0 \cap J_2$. Then $i = k < j < l \wedge l < j$ a contradiction.
- $(i, j) \in J_0 \cap J_3$. Then $i = k < j < l \wedge l < j$ a contradiction.
- $(i, j) \in J_0 \cap J_4$. Then $i = k < j < l \wedge j = k$ a contradiction.
- $(i, j) \in J_0 \cap J_5$. Then $i = k < j < l \wedge j = l$ a contradiction.
- $(i, j) \in J_0 \cap J_6$. Then $i = k < j < l \wedge j = l$ a contradiction.
- $(i, j) \in J_0 \cap J_7$. Then $i = k < j < l \wedge i \neq k$ a contradiction.
- $(i, j) \in J_1 \cap J_2$. Then $i = k \wedge j = l \wedge l < j$ a contradiction.
- $(i, j) \in J_1 \cap J_3$. Then $i = k \wedge j = l \wedge l < j$ a contradiction.
- $(i, j) \in J_1 \cap J_4$. Then $i = k \wedge j = l \wedge i < k$ a contradiction.
- $(i, j) \in J_1 \cap J_5$. Then $i = k \wedge j = l \wedge i < k$ a contradiction.
- $(i, j) \in J_1 \cap J_6$. Then $i = k \wedge j = l \wedge k < i$ a contradiction.
- $(i, j) \in J_1 \cap J_7$. Then $i = k \wedge j = l \wedge i \neq k$ a contradiction.
- $(i, j) \in J_2 \cap J_3$. Then $i = k \wedge l < j \wedge i = l$ contradicting $k < l$.
- $(i, j) \in J_2 \cap J_4$. Then $i = k \wedge l < j \wedge j = k$ contradicting $i < j$.
- $(i, j) \in J_2 \cap J_5$. Then $i = k \wedge l < j \wedge i < k$ a contradiction.
- $(i, j) \in J_2 \cap J_6$. Then $i = k \wedge l < j \wedge k < i$ a contradiction.
- $(i, j) \in J_2 \cap J_7$. Then $i = k \wedge l < j \wedge i \neq k$ a contradiction.
- $(i, j) \in J_3 \cap J_4$. Then $i = l \wedge l < j \wedge i < k$ giving $l < k$ contradicting $k < l$.
- $(i, j) \in J_3 \cap J_5$. Then $i = l \wedge l < j \wedge i < k$ giving $l < k$ contradicting $k < l$.
- $(i, j) \in J_3 \cap J_6$. Then $i = l \wedge l < j \wedge l = j$ a contradicting.
- $(i, j) \in J_3 \cap J_7$. Then $i = l \wedge l < j \wedge i \neq l$ a contradiction.
- $(i, j) \in J_4 \cap J_5$. Then $i < k = j \wedge j = l$ contradicting $k < l$.
- $(i, j) \in J_4 \cap J_6$. Then $i < k = j \wedge j = l$ contradicting $k < l$.
- $(i, j) \in J_4 \cap J_7$. Then $i < k = j \wedge j \neq k$ a contradiction.
- $(i, j) \in J_5 \cap J_6$. Then $i < k \wedge j = l \wedge k < i$ a contradiction.
- $(i, j) \in J_5 \cap J_7$. Then $i < k < i \wedge j = l \wedge j \neq l$ a contradiction.
- $(i, j) \in J_6 \cap J_7$. Then $k < i \wedge j = l \wedge j \neq l$ a contradiction.

So

$$I = \bigcup_{i \in \{1, \dots, 7\}} J_i \text{ and } \forall i, j \in \{1, \dots, 7\} \text{ with } i \neq j \text{ we have } J_i \cap J_j = \emptyset$$

So that

$$\begin{aligned} \Phi_n(x_1, \dots, x_n) &= \prod_{(i,j) \in I} (x_i - x_j) \\ &\stackrel{[\text{theorem: 11.43}]}{=} \prod_{m=0}^7 \left(\prod_{(i,j) \in J_m} (x_i - x_j) \right) \\ &= \prod_{m=0}^7 Q_m \end{aligned} \tag{11.118}$$

where

$$Q_m = \prod_{(i,j) \in J_m} (x_i - x_j) \tag{11.119}$$

and

$$\begin{aligned}
 ((k \leftrightarrow_n l) \Phi_n)(x_1, \dots, x_n) &= \prod_{(i,j) \in I} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\
 &\stackrel{[\text{theorem: 11.43}]}{=} \prod_{m=0}^7 \left(\prod_{(i,j) \in J_m} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \right) \\
 &= \prod_{m=0}^7 R_m
 \end{aligned} \tag{11.120}$$

where

$$R_m = \prod_{(i,j) \in J_m} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \tag{11.121}$$

Now we have:

R_0 .

$$\begin{aligned}
 R_0 &= \prod_{(i,j) \in J_0} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\
 &= \prod_{(i,j) \in \{(k,j) \in I \mid k < j < l\}} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\
 &= \prod_{(i,j) \in \{(k,j) \in I \mid k < j < l\}} (x_l - x_j) \\
 &= \prod_{(i,j) \in J_0} (x_l - x_j)
 \end{aligned} \tag{11.122}$$

Define now

$$\beta: J_6 = \{(i, l) \in I \mid k < i < l\} \rightarrow J_0 = \{(k, j) \in I \mid k < j < l\} \text{ by } \beta(i, j) = (k, i)$$

then we have:

injectivity. Let $(i, j), (i', j') \in J_6$ with $\beta(i, j) = \beta(i', j')$ then $j = l = j'$ and $(k, i) = (k, i') \Rightarrow i = i'$ proving that $(i, j) = (i', j')$.

surjectivity. Let $(r, s) \in J_0$ then $k = r$ and $k < s < l$ so that $(s, s) \in J_7$ which, as $\beta(s, s) = (k, s) = (r, s)$, proves surjectivity.

proving that

$$\beta: J_6 \rightarrow J_0 \text{ is a bijection}$$

Next

$$\begin{aligned}
 \prod_{(i,j) \in J_0} (x_l - x_j) &= \prod_{(i,j) \in J_0} (x_l - x_{(i,j)_2}) \\
 &\stackrel{[\text{theorem: 11.36}]}{=} \prod_{(i,j) \in J_6} (x_l - x_{(\beta(i,j))_2}) \\
 &= \prod_{(i,j) \in J_6} (x_l - x_i) \\
 &= \prod_{(i,j) \in J_6} ((-1) \cdot (x_i - x_l)) \\
 &\stackrel{[\text{theorem: 11.38}]}{=} \left(\prod_{(i,j) \in J_6} (-1) \right) \cdot \prod_{(i,j) \in J_6} (x_i - x_l) \\
 &\stackrel{[\text{theorem: 11.51}]}{=} (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in \{(i,l) \in I \mid k < i < l\}} (x_i - x_l) \\
 &= (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in \{(i,l) \in I \mid k < i < l\}} (x_i - x_j) \\
 &= (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in J_6} (x_i - x_j) \\
 &= (-1)^{\text{card}(J_6)} \cdot Q_6
 \end{aligned}$$

combining the above with [eq: 11.122] gives

$$R_0 = (-1)^{\text{card}(j_6)} \cdot Q_6 \quad (11.123)$$

R₁. Then

$$\begin{aligned} R_1 &= \prod_{(i,j) \in J_1} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\ &= \prod_{(i,j) \in \{(k,l)\}} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\ &\stackrel{[\text{theorem: 11.34}]}{=} x_{(k \leftrightarrow l)_n(k)} - x_{(k \leftrightarrow l)_n(l)} \\ &= x_l - x_k \\ &= (-1) \cdot (x_k - x_l) \\ &\stackrel{[\text{theorem: 11.34}]}{=} (-1) \cdot \prod_{(i,j) \in \{(k,l)\}} (x_i - x_j) \\ &= (-1) \cdot \prod_{(i,j) \in J_1} (x_i - x_j) \\ &= (-1) \cdot Q_1 \end{aligned}$$

proving that

$$R_1 = (-1) \cdot Q_1 \quad (11.124)$$

R₂. Then

$$\begin{aligned} R_3 &= \prod_{(i,j) \in J_2} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\ &= \prod_{(i,j) \in \{(k,j) \in I \mid l < j\}} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\ &= \prod_{(i,j) \in \{(k,j) \in I \mid l < j\}} (x_l - x_j) \\ &= \prod_{(i,j) \in J_2} (x_l - x_j) \end{aligned} \quad (11.125)$$

Define

$$\beta: J_3 = \{(l, j) \in I \mid l < j\} \rightarrow J_2 = \{(k, j) \in I \mid l < j\} \text{ by } \beta(i, j) = (k, j)$$

then we have:

injectivity. Let $(i, j), (i', j') \in J_3$ with $\beta(i, j) = \beta(i', j')$ then $i = l = i'$ and $(k, j) = (k, j') \Rightarrow j = j'$ so that $(i, j) = (i', j')$.

surjectivity. Let $(r, s) \in J_2$ then $r = k$ and $l < s$ so that $(l, s) \in J_3$ which, as $\beta(l, s) = (k, s) = (r, s)$, proves surjectivity.

Hence

$$\beta: J_3 \rightarrow J_2 \text{ is a bijection}$$

Next

$$\begin{aligned} \prod_{(i,j) \in J_2} (x_l - x_j) &= \prod_{(i,j) \in J_2} (x_l - x_{(i,j)_2}) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \prod_{(i,j) \in J_3} (x_l - x_{(\beta(i,j))_2}) \\ &= \prod_{(i,j) \in J_3} (x_l - x_j) \\ &= \prod_{(i,j) \in \{(l,j) \in I \mid l < j\}} (x_l - x_j) \\ &= \prod_{(i,j) \in \{(l,j) \in I \mid l < j\}} (x_i - x_j) \end{aligned}$$

$$\begin{aligned}
&= \prod_{(i,j) \in J_3} (x_i - x_j) \\
&= Q_3
\end{aligned}$$

which combined with [eq: 11.125] proves that

$$R_2 = Q_3 \quad (11.126)$$

R₃. We have

$$\begin{aligned}
R_3 &= \prod_{(i,j) \in J_3} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\
&= \prod_{(i,j) \in \{(l,j) \in I \mid l < j\}} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\
&= \prod_{(i,j) \in \{(k,j) \in I \mid l < j\}} (x_k - x_j) \\
&= \prod_{(i,j) \in J_3} (x_k - x_j) \quad (11.127)
\end{aligned}$$

Define

$$\beta: J_2 = \{(k,j) \in I \mid l < j\} \rightarrow J_3 = \{(l,j) \in I \mid l < j\} \text{ by } \beta(i,j) = (l,j)$$

then we have

injectivity. Let $(i,j), (i',j') \in J_2$ with $\beta(i,j) = \beta(i',j')$ then $i = k = i'$ and $(l,j) = (l,j') \Rightarrow j = j'$ proving $(i,j) = (i',j')$.

surjectivity. Let $(r,s) \in J_3$ then $r = l$ and $l < s$, then $(k,s) \in J_2$ and $\beta(k,s) = (l,s) = (r,s)$ proving surjectivity.

hence we have

$$\beta: J_2 \rightarrow J_3 \text{ is a bijection}$$

Next

$$\begin{aligned}
\prod_{(i,j) \in J_3} (x_k - x_j) &= \prod_{(i,j) \in J_3} (x_k - x_{(i,j)_2}) \\
&\stackrel{[\text{theorem: 11.36}]}{=} \prod_{(i,j) \in J_2} (x_k - x_{(\beta(i,j))_2}) \\
&= \prod_{(i,j) \in J_2} (x_k - x_{(l,j)_2}) \\
&= \prod_{(i,j) \in J_2} (x_k - x_j) \\
&= \prod_{(i,j) \in \{(k,j) \in I \mid l < j\}} (x_k - x_j) \\
&= \prod_{(i,j) \in \{(k,j) \in I \mid l < j\}} (x_i - x_j) \\
&= \prod_{(i,j) \in J_2} (x_i - x_j) \\
&= Q_2
\end{aligned}$$

proving together with [eq: 11.127] that

$$R_3 = Q_2 \quad (11.128)$$

R₄. We have

$$\begin{aligned}
 R_4 &= \prod_{(i,j) \in J_4} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\
 &= \prod_{(i,j) \in \{(i,k) \in I \mid i < k\}} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\
 &= \prod_{(i,j) \in \{(i,k) \in I \mid i < k\}} (x_i - x_l) \\
 &= \prod_{(i,j) \in J_4} (x_i - x_l) \tag{11.129}
 \end{aligned}$$

Define

$$\beta: J_5 = \{(i, l) \in I \mid i < k\} \rightarrow J_4 = \{(i, k) \in I \mid i < k\} \text{ by } \beta(i, j) = (i, k)$$

then we have:

injectivity. Let $(i, j), (i', j') \in J_5$ with $\beta(i, j) = \beta(i', j')$ then $j = l = j'$ and $(i, k) = (i', k) \Rightarrow i = i'$ so that $(i, j) = (i', j')$.

surjectivity. Let $(r, s) \in J_4$ then $s = k$ and $r < k$ then for (r, l) we have $(r, l) \in J_5$ and $\beta(r, l) = (r, k) = (r, s)$ proving surjectivity.

So

$$\begin{aligned}
 \prod_{(i,j) \in J_4} (x_i - x_l) &= \prod_{(i,j) \in J_4} (x_{(i,j)_1} - x_l) \\
 &\stackrel{[\text{theorem: 11.36}]}{=} \prod_{(i,j) \in J_5} (x_{(\beta(i,j))_1} - x_l) \\
 &= \prod_{(i,j) \in J_5} (x_{(i,k)_1} - x_l) \\
 &= \prod_{(i,j) \in J_5} (x_i - x_l) \\
 &= \prod_{(i,j) \in \{(i,l) \in I \mid i < k\}} (x_i - x_l) \\
 &= \prod_{(i,j) \in \{(i,l) \in I \mid i < k\}} (x_i - x_j) \\
 &= \prod_{(i,j) \in J_5} (x_i - x_j) \\
 &= Q_5
 \end{aligned}$$

combining this with [eq: 11.129] gives

$$R_4 = Q_5 \tag{11.130}$$

R₅. We have

$$\begin{aligned}
 R_5 &= \prod_{(i,j) \in J_5} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\
 &= \prod_{(i,j) \in \{(i,l) \in I \mid i < k\}} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\
 &= \prod_{(i,j) \in \{(i,l) \in I \mid i < k\}} (x_i - x_k) \\
 &= \prod_{(i,j) \in J_5} (x_i - x_k) \tag{11.131}
 \end{aligned}$$

Define

$$\beta: J_4 = \{(i, k) \in I \mid i < k\} \rightarrow J_5 = \{(i, l) \in I \mid i < k\} \text{ by } \beta(i, j) = (i, l)$$

then we have:

injectivity. Let $(i, j), (i', j') \in J_4$ with $\beta(i, j) = \beta(i', j')$ then $j = k = j'$ and $(i, l) = (i', l) \Rightarrow i = i'$ proving that $(i, j) = (i', j')$.

surjectivity. Let $(r, s) \in J_5$ then $s = l$ and $r < k$ then for (r, k) we have $(r, k) \in J_4$ and $\beta(r, k) = (r, l) = (r, s)$.

proving that

$$\beta: J_4 \rightarrow J_5 \text{ is a bijection}$$

Next

$$\begin{aligned} \prod_{(i,j) \in J_5} (x_i - x_k) &= \prod_{(i,j) \in J_5} (x_{(i,j)_1} - x_k) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \prod_{(i,j) \in J_4} (x_{(\beta(i,j))_1} - x_k) \\ &= \prod_{(i,j) \in J_4} (x_{(i,l)_1} - x_k) \\ &= \prod_{(i,j) \in J_4} (x_i - x_k) \\ &= \prod_{(i,j) \in \{(i,k) \in I \mid i < k\}} (x_i - x_k) \\ &= \prod_{(i,j) \in \{(i,k) \in I \mid i < k\}} (x_i - x_j) \\ &= \prod_{(i,j) \in J_4} (x_i - x_j) \\ &= Q_4 \end{aligned}$$

combining the above with [eq: 11.131] proves that

$$R_5 = Q_4 \tag{11.132}$$

R₆. We have

$$\begin{aligned} R_6 &= \prod_{(i,j) \in J_6} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\ &= \prod_{(i,j) \in \{(i,l) \in I \mid k < i < l\}} (x_{(k \leftrightarrow_n l)(i)} - x_{(k \leftrightarrow_n l)(j)}) \\ &= \prod_{(i,j) \in \{(i,l) \in I \mid k < i < l\}} (x_i - x_k) \\ &= \prod_{(i,j) \in J_6} (x_i - x_k) \end{aligned} \tag{11.133}$$

Define

$$\beta: J_0 = \{(k, j) \in I \mid k < j < l\} \rightarrow J_6 = \{(i, l) \in I \mid k < i < l\} \text{ by } \beta(i, j) = (j, l)$$

then we have:

injectivity. Let $(i, j), (i', j') \in J_0$ with $\beta(i, j) = \beta(i', j')$ then $i = k = i'$ and $(j, l) = (j', l) \Rightarrow j = j'$ hence $(i, j) = (i', j')$.

surjectivity. Let $(r, s) \in J_6$ then $s = l$ and $k < r < l$ so that for (k, r) we have $(k, r) \in J_0$ and $\beta(k, r) = (r, l) = (r, s)$.

hence

$$\beta: J_0 \rightarrow J_6 \text{ is a bijection so that } \text{card}(J_0) = \text{card}(J_6) \tag{11.134}$$

Further

$$\begin{aligned}
\prod_{(i,j) \in J_6} (x_i - x_k) &= \prod_{(i,j) \in J_6} (x_{(i,j)_1} - x_k) \\
&\stackrel{[\text{theorem: 11.36}]}{=} \prod_{(i,j) \in J_0} (x_{(\beta(i,j))_1} - x_k) \\
&= \prod_{(i,j) \in J_0} (x_{(j,l)_1} - x_k) \\
&= \prod_{(i,j) \in J_0} (x_j - x_k) \\
&= \prod_{(i,j) \in J_0} ((-1) \cdot (x_k - x_j)) \\
&\stackrel{[\text{theorem: 11.38}]}{=} \left(\prod_{(i,j) \in J_0} (-1) \right) \cdot \prod_{(i,j) \in J_0} (x_k - x_j) \\
&\stackrel{[\text{theorem: 11.51}]}{=} (-1)^{\text{card}(J_0)} \cdot \prod_{(i,j) \in J_0} (x_k - x_j) \\
&\stackrel{[\text{eq: 11.134}]}{=} (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in J_0} (x_k - x_j) \\
&= (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in \{(k,j) \in I \mid k < j < l\}} (x_k - x_j) \\
&= (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in \{(k,j) \in I \mid k < j < l\}} (x_i - x_j) \\
&= (-1)^{\text{card}(J_6)} \cdot \prod_{(i,j) \in J_0} (x_i - x_j) \\
&= (-1)^{\text{card}(J_6)} \cdot Q_0
\end{aligned}$$

which together with [eq: 11.133] proves that

$$R_6 = (-1)^{\text{card}(J_6)} \cdot Q_0 \quad (11.135)$$

R₇. We have

$$\begin{aligned}
R_7 &= \prod_{(i,j) \in J_7} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\
&= \prod_{(i,j) \in \{(i,j) \in I \mid i \neq k, l \wedge j \neq k, l\}} (x_{(k \leftrightarrow l)_n(i)} - x_{(k \leftrightarrow l)_n(j)}) \\
&= \prod_{(i,j) \in \{(i,j) \in I \mid i \neq k, l \wedge j \neq k, l\}} (x_i - x_j) \\
&= \prod_{(i,j) \in J_7} (x_i - x_j) \\
&= Q_7
\end{aligned}$$

then

$$R_7 = Q_7 \quad (11.136)$$

Finally we have

$$\begin{aligned}
&((k \leftrightarrow l)_n \Phi_n)(x_1, \dots, x_n) \stackrel{[\text{eq: 11.120}]}{=} \\
&R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot R_5 \cdot R_6 \cdot R_7 \stackrel{[\text{eqs: 11.123, 11.124, 11.126, 11.128, 11.130, 11.132, 11.135, 11.136}]}{=} \\
&(-1)^{\text{card}(J_6)} \cdot Q_6 \cdot (-1) \cdot Q_1 \cdot Q_3 \cdot Q_2 \cdot Q_5 \cdot \\
&Q_4 \cdot (-1)^{\text{card}(J_6)} \cdot Q_0 \cdot Q_7 = \\
&-(Q_1 \cdot Q_2 \cdot Q_3 \cdot Q_4 \cdot Q_5 \cdot Q_6 \cdot Q_7) = \\
&-\Phi_n(x_1, \dots, x_n)
\end{aligned}$$

proving that

$$(k \leftrightarrow_n l)\Phi_n = -\Phi_n \quad \square$$

Corollary 11.247. Let $n, m \in \mathbb{N}$ and $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, m\}} \subseteq P_n$ such that $\forall k \in \{1, \dots, m\} \ i_k \neq j_k$ then

$$((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m))\Phi_n = (-1)^m \cdot \Phi_n$$

Proof. We prove this by induction, so let

$$S = \{m \in \mathbb{N} \mid ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m))\Phi_n = (-1)^m \cdot \Phi_n\}$$

then we have:

$$1 \in S. \quad ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_1 \leftrightarrow_n j_1))\Phi_n = (i_1 \leftrightarrow_n j_1)\Phi_n \stackrel{[\text{theorem: 11.246}]}{=} (-1) \cdot \Phi_n = (-1)^1 \cdot \Phi_n \text{ proving that } 1 \in S.$$

$m \in S \Rightarrow m+1 \in S$. We have

$$\begin{aligned} & ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_{m+1} \leftrightarrow_n j_{m+1}))\Phi_n \stackrel{[\text{theorem: 11.227}]}{=} \\ & (((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m)) \circ (i_{m+1} \leftrightarrow_n j_{m+1}))\Phi_n \stackrel{[\text{theorem: 11.238}]}{=} \\ & ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m))((i_{m+1} \leftrightarrow_n j_{m+1})\Phi_n) \stackrel{[\text{theorem: 11.246}]}{=} \\ & ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m))((-1) \cdot \Phi_n) \stackrel{[\text{theorem: 11.239}]}{=} \\ & (-1) \cdot ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m))(\Phi_n) \stackrel{n \in S}{=} \\ & (-1) \cdot ((-1)^m \cdot \Phi_n) = \\ & (-1)^{m+1} \cdot \Phi_n \end{aligned}$$

proving that $m+1 \in S$ $\square$

Theorem 11.248. Let $n \in \mathbb{N}$ and $\sigma \in P_n$ then there exist a **unique** $\varepsilon_\sigma \in \{-1, 1\}$ such that

$$\sigma\Phi_n = \varepsilon_\sigma \cdot \Phi_n$$

Proof. If $n \in \mathbb{N}$ then we have either

$n = 1$. Then if $\sigma \in P_1$ we have that $P_1 = \text{Id}_{\{1\}}$ so that $\sigma\Phi_1 = \sigma\text{Id}_{\{1\}} \stackrel{[\text{theorem: 11.237}]}{=} \Phi_1$ so that if we take $\varepsilon_\sigma = 1$ then $\sigma\Phi_1 = \varepsilon_\sigma \Phi_1$.

$n \in \mathbb{N} \setminus \{1\}$. Using [theorem: 11.232] there exist a $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, m\}} \subseteq P_n$ with $\forall k \in \{1, \dots, m\} \ i_k \neq j_k$ such that

$$\sigma = (i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m)$$

Hence by [theorem: 11.247] we have that $((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m))\Phi_n = (-1)^m \cdot \Phi_n$. So if we take $\varepsilon_\sigma = (-1)^m$ we have $\sigma\Phi_n = \varepsilon_\sigma \cdot \Phi_n$

So in all cases there exists a ε_σ

$$\sigma\Phi_n = \varepsilon_\sigma \cdot \Phi_n$$

proving existence. Now for uniqueness assume that there exists a $\delta_\sigma \in \{1, -1\}$ such that $\sigma\Phi_n = \delta_\sigma \cdot \Phi_n$ then $\varepsilon_\sigma \cdot \Phi_n = \delta_\sigma \cdot \Phi_n$. hence $\varepsilon_\sigma \cdot \Phi_n(1, \dots, n) = \delta_\sigma \cdot \Phi_n(1, \dots, n)$. By [theorem: 11.245] $\Phi_n(1, \dots, n) \neq 0$ so, after multiplying both sides by $\Phi_n(1, \dots, n)^{-1}$, we have that $\varepsilon_\sigma = \delta_\sigma$. $\square$

The above theorem ensures that the following definition makes sense.

Definition 11.249. Let $n \in \mathbb{N}$ and $\sigma \in P_n$ then $\text{sign}(\sigma) \in \{-1, 1\}$ is the **unique** number such that

$$\sigma\Phi_n = \text{sign}(\sigma) \cdot \Phi_n$$

A permutation σ is called **even** if $\text{sign}(\sigma) = 1$ and **odd** if $\text{sign}(\sigma) = -1$

The concept of **even** or even permutation follows from the following.

Remark 11.250. Let $n \in \mathbb{N} \setminus \{1\}$ and $\sigma \in P_n$ then by [theorem: 11.232] there exist a family $\{(i_k \leftrightarrow_n j_k)\}_{k \in \{1, \dots, m\}} \subseteq P_n$ of **strict** transpositions such that

$$\sigma = (i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m)$$

so that

$$\sigma \Phi_n = ((i_1 \leftrightarrow_n j_1) \circ \dots \circ (i_m \leftrightarrow_n j_m)) \Phi_n \stackrel{[\text{theorem: 11.247}]}{=} (-1)^m \Phi_n$$

proving that

$$\text{sign}(\sigma) = (-1)^m$$

In other words if σ can be written as a odd number of strict transpositions σ is odd and if σ can be written as a even number of transpositions σ is even.

Theorem 11.251. If $n \in \mathbb{N}$ then we have

1. $\forall \sigma, \tau \in P_n$ that $\text{sign}(\sigma \circ \tau) = \text{sign}(\sigma) \cdot \text{sign}(\tau)$
2. If $i, j \in \{1, \dots, n\}$ with $i \neq j$ then $\text{sign}((i \leftrightarrow_n j)) = -1$
3. $\text{sign}(\text{Id}_{\{1, \dots, n\}}) = 1$
4. $\forall \sigma \in P_n$ $\text{sign}(\sigma) = \text{sign}(\sigma^{-1})$

Proof.

1.

$$\begin{aligned} (\sigma \circ \tau) \Phi_n &\stackrel{[\text{theorem: 11.238}]}{=} \sigma(\tau \Phi_n) \\ &= \sigma(\text{sign}(\tau) \cdot \Phi_n) \\ &\stackrel{[\text{theorem: 11.239}]}{=} \text{sign}(\tau) \cdot (\tau \Phi_n) \\ &= \text{sign}(\tau) \cdot \text{sign}(\sigma) \cdot \Phi_n \\ &= \text{sign}(\sigma) \cdot \text{sign}(\tau) \cdot \Phi_n \end{aligned}$$

proving that

$$\text{sign}(\sigma \circ \tau) = \text{sign}(\sigma) \cdot \text{sign}(\tau)$$

2. As $(i \leftrightarrow_n j) \Phi_n \stackrel{[\text{theorem: 11.246}]}{=} (-1) \cdot \Phi_n$ we have

$$\text{sign}((i \leftrightarrow_n j)) = -1$$

3. As $\text{Id}_{\{1, \dots, n\}} \Phi_n \stackrel{[\text{theorem: 11.237}]}{=} \Phi_n$ we have

$$\text{sign}(\text{Id}_{\{1, \dots, n\}}) = 1$$

4. We have

$$1 \stackrel{(2)}{=} \text{sign}(\text{Id}_{\{1, \dots, n\}}) = \text{sign}(\sigma \circ \sigma^{-1}) \stackrel{(1)}{=} \text{sign}(\sigma) \cdot \text{sign}(\sigma^{-1})$$

$$\text{so that } \text{sign}(\sigma) = \text{sign}(\sigma) \cdot 1 = (\text{sign}(\sigma) \cdot \text{sign}(\sigma)) \cdot \text{sign}(\sigma^{-1}) \stackrel{\text{sing}(\sigma) \in \{-1, 1\}}{=} \text{sign}(\sigma^{-1}) \quad \square$$

The following permutation will be used in determinant functions later.

Definition 11.252. Let $n \in \mathbb{N}$ and $i, j \in \{1, \dots, n\}$ then we define

$$(i \rightsquigarrow_n j): \{1, \dots, n\} \rightarrow \{1, \dots, n\}$$

as follows:

1. If $i = j$ then $(i \rightsquigarrow_n j) = \text{Id}_{\{1, \dots, n\}}$
2. If $i < j$ then for $k \in \{1, \dots, n\}$ we have

$$(i \rightsquigarrow_n j)(k) = \begin{cases} k & \text{if } 1 \leq k < i \\ k+1 & \text{if } i \leq k < j \\ i & \text{if } k = j \\ k & \text{if } j < k \leq n \end{cases}$$

3. If $j < i$ then for $k \in \{1, \dots, n\}$ we have

$$(i \rightsquigarrow_n j)(k) = \begin{cases} k & \text{if } 1 \leq k < j \\ i & \text{if } k = j \\ k-1 & \text{if } j < k \leq i \\ k & \text{if } i < k \leq n \end{cases}$$

Example 11.253. To get a idea of the mapping $(i \rightsquigarrow_n j)$ we can look at some examples:

1. $n=6$

$$(2 \rightsquigarrow_6 5) \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \\ 2 \\ 6 \end{pmatrix}$$

2. $n=6$

$$(5 \rightsquigarrow_6 2) \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 2 \\ 3 \\ 4 \\ 6 \end{pmatrix}$$

So we interpret $(i \rightsquigarrow_n j)$ as removing i from its position and inserting it after (before) j if $i < j$ ($j < i$).

Next we prove that $(i \rightsquigarrow_n j)$ is actual a permutation and decompose it in a composition of transpositions so that we can calculate its sign. First we need a little lemma:

Definition 11.254. Let $n \in \mathbb{N}$, $i, j \in \{1, \dots, n\}$ then we define

1. If $i < j$ then $\theta^{i < j}: \{1, \dots, j-i\} \rightarrow \{i+1, \dots, j\}$ by $\theta^{i < j}(l) = j - (l-1)$
2. If $j < i$ then $\theta^{i > j}: \{1, \dots, i-j\} \rightarrow \{j, \dots, i-1\}$ by $\theta^{i > j}(l) = j + (l-1)$

Proof. Of course we must prove that the range is correct.

1. We have

$$\begin{aligned} l \in \{1, \dots, j-i\} &\Leftrightarrow 1 \leq l \leq j-i \\ &\Leftrightarrow 0 \leq l-1 \leq j-i-1 \\ &\Leftrightarrow i-j+1 \leq -(l-1) \leq 0 \\ &\Leftrightarrow i+1 \leq j-(l-1) \leq j \\ &\Leftrightarrow i+1 \leq \theta^{i < j}(l) \leq j \\ &\Leftrightarrow \theta^{i < j}(l) \in \{i+1, \dots, j\} \end{aligned}$$

2. We have

$$\begin{aligned} l \in \{1, \dots, i-j\} &\Leftrightarrow 1 \leq l \leq i-j \\ &\Leftrightarrow 0 \leq l-1 \leq i-j-1 \\ &\Leftrightarrow j \leq j+(l-1) \leq i-1 \\ &\Leftrightarrow j \leq \theta^{i > j}(l) \leq i-1 \\ &\Leftrightarrow \theta^{i > j}(l) \in \{j, \dots, i-1\} \end{aligned}$$

□

Theorem 11.255. *If $n \in \mathbb{N}$ and $i, j \in \{1, \dots, n\}$ then we have:*

1. *If $i < j$ then*

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n^{\theta^{i < j}(1)}) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i < j}(j-i)})$$

2. *If $j < i$ then*

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n^{\theta^{i > j}(1)}) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i > j}(i-j)})$$

3. $(i \rightsquigarrow_n j) \in P_n$

4. $\text{sign}((i \rightsquigarrow_n j)) = (-1)^{|i-j|}$

Proof.

1. Let $n \in \mathbb{N}$. We proceed now by induction on $k = j - i$, so let

$$S_n = \{k \in \mathbb{N} \mid \text{If } i, j \in \{1, \dots, n\} \text{ with } i < j \wedge j - i = k \text{ then } (i \rightsquigarrow_n j) = (i \leftrightarrow_n^{\theta^{i < j}(1)}) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i < j}(j-i)})\}$$

then we have:

1 $\in S$. Let $i, j \in \{1, \dots, n\}$ with $i < j \wedge j - i = 1 \Rightarrow i < j \wedge j = i + 1$. Then

$$\theta^{i < j}(1) = j - (1 - 1) = j = i + 1 \quad (11.137)$$

and we have for $l \in \{1, \dots, n\}$ either:

1 $\leq l < i$. Then

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= l \\ &\stackrel{l < i < i+1 \Rightarrow l \neq i, i+1}{=} (i \leftrightarrow_n^{i+1})(l) \\ &\stackrel{[\text{eq: 11.137}]}{=} (i \leftrightarrow_n^{\theta^{i < j}(1)})(l) \end{aligned}$$

$i \leq l < j$. Then $i \leq l < i + 1$ so that $l = i$ and

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= l + 1 \\ &\stackrel{l=i}{=} i + 1 \\ &= (i \leftrightarrow_n^{i+1})(i) \\ &\stackrel{l=i}{=} (i \leftrightarrow_n^{i+1})(l) \\ &\stackrel{[\text{eq: 11.137}]}{=} (i \leftrightarrow_n^{\theta^{i < j}(1)})(l) \end{aligned}$$

$l = j$. Then $l = j = i + 1$ and

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= i \\ &= (i \leftrightarrow_n^{i+1})(i+1) \\ &\stackrel{l=i+1}{=} (i \leftrightarrow_n^{i+1})(l) \\ &\stackrel{[\text{eq: 11.137}]}{=} (i \leftrightarrow_n^{\theta^{i < j}(1)})(l) \end{aligned}$$

$j < l \leq n$. Then

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= l \\ &\stackrel{i < j < l \Rightarrow i+1 \leq j < l \Rightarrow l \neq i, i+1}{=} (i \leftrightarrow_n^{i+1})(l) \\ &\stackrel{[\text{eq: 11.137}]}{=} (i \leftrightarrow_n^{\theta^{i < j}(1)})(l) \end{aligned}$$

So we have in all cases $(i \rightsquigarrow_n j)(l) = (i \leftrightarrow_n^{\theta^{i < j}(1)})(l)$ hence

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n^{\theta^{i < j}(1)}) \underset{j-i=1}{=} (i \leftrightarrow_n^{\theta^{i < j}(1)}) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i < j}(j-i)})$$

proving that $1 \in S_n$.

$k \in S \Rightarrow k + 1 \in S$. Let $i, j \in \{1, \dots, n\}$ with $i < j$ and $j - i = k + 1$. First we prove that

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n^j) \circ (i \rightsquigarrow_n j - 1) \quad (11.138)$$

Proof. For $l \in \{1, \dots, n\}$ we have the following cases to consider:

$1 \leq l < i$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j - 1)(l)) &= (i \leftrightarrow_n j)(l) \\ &\stackrel{l < i < j \Rightarrow l \neq i, j}{=} \stackrel{l}{=} (i \rightsquigarrow_n j)(l) \\ &\stackrel{1 \leq l < i}{=} \end{aligned}$$

$i \leq l < j - 1$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j - 1)(l)) &= (i \leftrightarrow_n j)(l + 1) \\ &\stackrel{i < i + 1 \leq l + 1 < j \Rightarrow l + 1 \neq i, j}{=} \stackrel{l + 1}{=} (i \rightsquigarrow_n j)(l) \\ &\stackrel{i \leq l + 1 < j}{=} \end{aligned}$$

$l = j - 1$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j - 1)(l)) &= (i \leftrightarrow_n j)(i) \\ &= \stackrel{j}{=} (i \rightsquigarrow_n j)(j - 1) \\ &\stackrel{i < j \Rightarrow i \leq j - 1 = l < j}{=} \stackrel{l = j - 1}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

$j - 1 < l \leq n$. Then $j \leq l$ so we have either:

$j = l$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j - 1)(l)) &= (i \leftrightarrow_n j)(l) \\ &\stackrel{l = j}{=} \stackrel{i}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

$j < l$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j - 1)(l)) &\stackrel{j - 1 < j < l}{=} (i \leftrightarrow_n j)(l) \\ &\stackrel{i < j < l \Rightarrow i, j \neq l}{=} \stackrel{l}{=} (i \rightsquigarrow_n j)(l) \\ &\stackrel{j < l \leq n}{=} \end{aligned}$$

So in all cases $(i \leftrightarrow_n j)((i \rightsquigarrow_n j - 1)(l)) = (i \rightsquigarrow_n j)(l)$ or $(i \rightsquigarrow_n j) = (i \leftrightarrow_n j) \circ (i \rightsquigarrow_n j - 1)$ proving [eq: 11.138]. $\square$

As $\theta^{i < j}(1) = j + (1 - 1) = j$ we have by [eq: 11.138] that

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n \theta^{i < j}(1)) \circ (i \rightsquigarrow_n j - 1) \quad (11.139)$$

Further as $(j - 1) - i \stackrel{j - i = k + 1}{=} k$ we have as $k \in S_n$ that

$$(i \rightsquigarrow_n j - 1) = (i \leftrightarrow_n \theta^{i < j - 1}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i < j - 1}(j - 1 - i))$$

Let $l \in \{1, \dots, j - i\}$ then $\theta^{i < j - 1}(l) = j - 1 - (l - 1) = j - ((l + 1) - 1) = \theta^{i < j}(l + 1)$ so we have that

$$(i \rightsquigarrow_n j - 1) = (i \leftrightarrow_n \theta^{i < j}(2)) \circ \dots \circ (i \leftrightarrow_n \theta^{i < j - 1}(j - i)) \quad (11.140)$$

Further

$$\begin{aligned} (i \rightsquigarrow_n j) &\stackrel{[\text{eq: 11.139}]}{=} (i \leftrightarrow_n \theta^{i < j}(1)) \circ (i \rightsquigarrow_n j - 1) \\ &\stackrel{[\text{eq: 11.140}]}{=} (i \leftrightarrow_n \theta^{i < j}(1)) \circ ((i \leftrightarrow_n \theta^{i < j}(2)) \circ \dots \circ (i \leftrightarrow_n \theta^{i < j - 1}(j - i))) \\ &\stackrel{[\text{theorem: 11.227}]}{=} (i \leftrightarrow_n \theta^{i < j}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i < j - 1}(j - i)) \end{aligned}$$

proving that $k + 1 \in S_n$.

2. Let $n \in \mathbb{N}$. We proceed now by induction on $k = i - j$, so let

$$S_n = \{k \in \mathbb{N} \mid \text{If } i, j \in \{1, \dots, n\} \text{ with } j < i \wedge i - j = k \text{ then } (i \rightsquigarrow_n j) = (i \leftrightarrow_n \theta^{i>j}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i>j}(i - j))\}$$

then we have:

1 $\in S$. Let $i, j \in \{1, \dots, n\}$ with $j < i$ and $i - j = 1$ so that $j = i - 1$. Then

$$\theta^{i>j}(1) = j + (1 - 1) = j = i - 1 \quad (11.141)$$

For $l \in \{1, \dots, n\}$ we have either:

1 $\leq l < j$. Then we have

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= l \\ &\stackrel{l < j < i \Rightarrow l \neq j, i-1}{=} (i \leftrightarrow_n i - 1)(l) \\ &\stackrel{[\text{eq: 11.141}]}{=} (i \leftrightarrow_n \theta^{i>j}(1))(l) \end{aligned}$$

l = **j**. Then we have $l = j = i - 1$

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= i \\ &= (i \leftrightarrow_n i - 1)(i - 1) \\ &\stackrel{l = i-1}{=} (i \leftrightarrow_n i - 1)(l) \\ &\stackrel{[\text{eq: 11.141}]}{=} (i \leftrightarrow_n \theta^{i>j}(1))(l) \end{aligned}$$

j < **l** $\leq$ **i**. Then we have $i - 1 < l \leq i$ so that $l = i$ and

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= l - 1 \\ &\stackrel{l = i}{=} i - 1 \\ &= (i \leftrightarrow_n i - 1)(i) \\ &\stackrel{l = i}{=} (i \leftrightarrow_n i - 1)(l) \\ &\stackrel{[\text{eq: 11.141}]}{=} (i \leftrightarrow_n \theta^{i>j}(1))(l) \end{aligned}$$

i < **l** $\leq$ **n**. Then we have

$$\begin{aligned} (i \rightsquigarrow_n j)(l) &= l \\ &\stackrel{j < i < l}{=} (i \leftrightarrow_n i - 1)(l) \\ &\stackrel{[\text{eq: 11.141}]}{=} (i \leftrightarrow_n \theta^{i>j}(1))(l) \end{aligned}$$

So in all cases we have $(i \rightsquigarrow_n j)(l) = (i \leftrightarrow_n \theta^{i>j}(1))(l)$, hence

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n \theta^{i>j}(1)) \stackrel{i-j=1}{=} (i \leftrightarrow_n \theta^{i>j}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i>j}(i - j))$$

proving that $1 \in S_n$.

k $\in S \Rightarrow$ **k** + **1** $\in S$. Let $i, j \in \{1, \dots, n\}$ with $j < i$ and $i - j = k + 1$. First we prove that

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n j) \circ (i \rightsquigarrow_n j + 1) \quad (11.142)$$

Proof. For $l \in \{1, \dots, n\}$ we have either:

1 $\leq l < j$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j + 1)(l)) &\stackrel{1 \leq l < j < j+1}{=} (i \leftrightarrow_n j)(l) \\ &\stackrel{l < j < i}{=} l \\ &\stackrel{l < j}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

$l = j$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j + 1)(l)) &\stackrel{l=j < j+1}{=} (i \leftrightarrow_n j)(l) \\ &\stackrel{l=j}{=} (i \leftrightarrow_n j)(j) \\ &= i \\ &\stackrel{l=j}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

$j < l \leq i$. Then we have $j + 1 \leq l$ so that either:

$l = j + 1$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j + 1)(l)) &= (i \leftrightarrow_n j)(i) \\ &= j \\ &\stackrel{l=j+1}{=} l - 1 \\ &\stackrel{j < l \leq i}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

$j + 1 < l$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j + 1)(l)) &\stackrel{j+1 < l \leq i}{=} (i \leftrightarrow_n j)(l - 1) \\ &\stackrel{j < j+1 \leq l-1 < l \leq i}{=} l - 1 \\ &\stackrel{j < l \leq i}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

$i < l \leq n$. Then

$$\begin{aligned} (i \leftrightarrow_n j)((i \rightsquigarrow_n j + 1)(l)) &\stackrel{i < l \leq n}{=} (i \leftrightarrow_n j)(l) \\ &\stackrel{j < i < l}{=} l \\ &\stackrel{j < i < l}{=} (i \rightsquigarrow_n j)(l) \end{aligned}$$

So in all cases we have $(i \leftrightarrow_n j)((i \rightsquigarrow_n j + 1)(l)) = (i \rightsquigarrow_n j)(l)$, hence $(i \rightsquigarrow_n j) = (i \leftrightarrow_n j) \circ (i \rightsquigarrow_n j + 1)$, proving [eq: 11.142]. $\square$

Now $\theta^{i>j}(1) = j + (1 - 1) = j$ which combined with [eq: 11.142] gives

$$(i \rightsquigarrow_n j) = (i \leftrightarrow_n \theta^{i>j}(1)) \circ (i \rightsquigarrow_n j + 1) \quad (11.143)$$

Further as $i - (j + 1) = i - j - 1 = k + 1 - 1 = k \in \mathcal{S}_n$ we have that

$$(i \rightsquigarrow_n j + 1) = (i \leftrightarrow_n \theta^{i>j+1}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i>j+1}(i - (j + 1)))$$

Now if $l \in \{1, \dots, i - (j + 1)\}$ then

$$\theta^{i>j+1}(l) = j + 1 + (l - 1) = j + ((l + 1) - 1) = \theta^{i>j}(l + 1)$$

so that

$$(i \rightsquigarrow_n j + 1) = (i \leftrightarrow_n \theta^{i>j}(2)) \circ \dots \circ (i \leftrightarrow_n \theta^{i>j}(i - j)) \quad (11.144)$$

Next

$$\begin{aligned} (i \rightsquigarrow_n j) &\stackrel{[\text{eq: 11.143}]}{=} (i \leftrightarrow_n \theta^{i>j}(1)) \circ (i \rightsquigarrow_n j + 1) \\ &\stackrel{[\text{eq: 11.144}]}{=} (i \leftrightarrow_n \theta^{i>j}(1)) \circ ((i \leftrightarrow_n \theta^{i>j}(2)) \circ \dots \circ (i \leftrightarrow_n \theta^{i>j}(i - j))) \\ &\stackrel{[\text{theorem: 11.227}]}{=} (i \leftrightarrow_n \theta^{i>j}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i>j}(i - j)) \end{aligned}$$

proving that $k + 1 \in S_n$.

3. Let $i, j \in \{1, \dots, n\}$ then we have either:

$i = j$. Then $(i \rightsquigarrow_n j) \stackrel{\text{def}}{=} \text{Id}_{1, \dots, n} \in P_n$

$i < j$. Then by (1) $(i \rightsquigarrow_n j) = (i \leftrightarrow_n \theta^{i < j}(1)) \circ \dots \circ (i \leftrightarrow_n \theta^{i < j}(j - i)) \in P_n$

$j < i$. Then by (2) $(i \rightsquigarrow_n j) = (i \leftrightarrow_n^{\theta^{i>j}}(1)) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i>j}}(i-j)) \in P_n$

4. Let $i, j \in \{1, \dots, n\}$ then we have either:

$i = j$. Then $|i - j| = 0$ and

$$\text{sign}((i \rightsquigarrow_n j)) = \text{sign}(\text{Id}_{\{1, \dots, n\}}) \stackrel{[\text{theorem: 11.251}]}{=} 1 = (-1)^0 = (-1)^{|i-j|}$$

$i < j$. Then as $\theta^{i<j}(\{1, \dots, j-i\}) \stackrel{[\text{definition: 11.254}]}{=} \{i+1, \dots, j\}$ so that $\forall l \in \{1, \dots, j-i\}$ we have that $i \neq \theta^{i<j}(l)$, so $(i \leftrightarrow_n^{\theta^{i<j}}(l))$ is a strict transposition, hence as $i-j = |i-j|$

$$\text{sign}((i \rightsquigarrow_n j)) \stackrel{(1)}{=} \text{sign}((i \leftrightarrow_n^{\theta^{i<j}}(1)) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i<j}}(j-i))) \stackrel{[\text{theorem: 11.250}]}{=} (-1)^{|i-j|}$$

$j < i$. Then as $\theta^{i>j}(\{1, \dots, i-j\}) \stackrel{[\text{definition: 11.254}]}{=} \{j, \dots, i-1\}$ so that $\forall l \in \{1, \dots, i-j\}$ we have that $i \neq \theta^{i>j}(l)$, so $(i \leftrightarrow_n^{\theta^{i>j}}(l))$ is a strict transposition, hence as $j-i = |i-j|$

$$\text{sign}((i \rightsquigarrow_n j)) \stackrel{(1)}{=} \text{sign}((i \leftrightarrow_n^{\theta^{i>j}}(1)) \circ \dots \circ (i \leftrightarrow_n^{\theta^{i>j}}(j-i))) \stackrel{[\text{theorem: 11.250}]}{=} (-1)^{|i-j|} \quad \square$$

11.6 Multilinear mappings

From now on, unless specified otherwise, instead of saying that $\langle X, +, \cdot \rangle, \langle Y, +, \cdot \rangle$ are vector spaces over a field $\langle F, +, \cdot \rangle$ we just say that X and Y are vector spaces over a field F , the addition operators and multiplication operators follows then from the context.

We use the following notation to specify what the i -th element of a tuple is.

Definition 11.256. Let $n \in \mathbb{N}$ and $\{X_i\}_{i \in \{1, \dots, n\}}$ a family of sets, $i \in \{1, \dots, n\}$ and $x \in \prod_{i \in \{1, \dots, n\}} X_i$ then

$$\left(\dots \underbrace{a}_i \dots \right) \text{ is equivalent with } x_i = a$$

using this notation we have

$$y = \left(x_1, \dots, \underbrace{a}_i, \dots, x_n \right) \text{ is equivalent with saying that } y_j = \begin{cases} x_i & \text{if } j = i \\ x_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

Another notation that we use is

$$y = (x_1, x_2, \dots, x_n) \text{ is equivalent with } y_j = \begin{cases} x_1 & \text{if } j = 1 \\ x_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

Likewise

$$y = (x_1, \dots, x_{n-1}, a) \text{ is equivalent with } y_j = \begin{cases} x_n & \text{if } j = n \\ x_j & \text{if } j \in \{1, \dots, n-1\} \end{cases}$$

The notation

$$z = (x_1, \dots, x_n, y_1, \dots, y_m) \text{ is equivalent with } z_j = \begin{cases} x_j & \text{if } j \in \{1, \dots, n\} \\ y_{j-n} & \text{if } j \in \{n+1, \dots, n+m\} \end{cases}$$

Finally the notation

$$z = (x_1, \dots, x_n, y_{n+1}, \dots, y_m) \text{ is equivalent with } z_j = \begin{cases} x_j & \text{if } j \in \{1, \dots, n\} \\ y_j & \text{if } j \in \{n+1, \dots, m\} \end{cases}$$

Proposition 11.257. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ a family of sets, $\sigma \in P_n$, $i \in \{1, \dots, n\}$ and $x \in \prod_{i \in \{1, \dots, n\}} X_i$ such that $x = \left(x_1, \dots, \underbrace{a}_i, \dots, x_n \right)$ then

$$x \circ \sigma = \left((x \circ \sigma)_1, \dots, \underbrace{a}_{\sigma^{-1}(i)}, \dots, (x \circ \sigma)_n \right)$$

Proof. Let $i \in \{1, \dots, n\}$ and $x = \left(x_1, \dots, \underbrace{}_i, \dots, x_n\right)$ then $x(i) = x_i = a$. As

$$(x \circ \sigma)_{\sigma^{-1}(i)} = x(\sigma(\sigma^{-1}(i))) = x(i) = a$$

we have by definition that

$$x \circ \sigma = \left((x \circ \sigma)_1, \dots, \underbrace{}_{\sigma^{-1}(i)}, \dots, (x \circ \sigma)_n\right) \quad \square$$

Definition 11.258. Let $n \in \mathbb{N}$, $i \in \{1, \dots, n\}$, $\{X_j\}_{j \in \{1, \dots, n\}}$ a family of vector spaces over a field F and Y a vector space over the same field then a function $L: \prod_{j \in \{1, \dots, n\}} X_j \rightarrow Y$ is a multilinear mapping if $\forall i \in \{1, \dots, n\}$ we have

1. $\forall u, v \in X_i$ then $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ such that $x_i = u + v$ we have

$$L(x) = L(y) + L(z)$$

where y, z are defined by

$$y_k = \begin{cases} u & \text{if } k = i \\ x_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \end{cases} \quad \text{and } z_k = \begin{cases} v & \text{if } k = i \\ x_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

2. $\forall u \in X_i, \forall \alpha \in F$ then $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ such that $x_i = \alpha \cdot u$ we have

$$L(x) = \alpha \cdot L(y)$$

where y is defined by

$$y_k = \begin{cases} u & \text{if } k = i \\ x_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

Using [definition: 11.256] this can be written more clearly as

1. $\forall u, v \in X_i$ and $\forall \left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n\right) \in \prod_{j \in \{1, \dots, n\}} X_j$

$$L\left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n\right) = L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)$$

2. $\forall \alpha \in F, \forall u \in X_i$ and $\forall \left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n\right) \in \prod_{j \in \{1, \dots, n\}} X_j$

$$L\left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n\right) = \alpha \cdot L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right)$$

The set of graphs of multilinear mappings is noted as $\text{Hom}(X_1 \dots X_n; Y)$, more specific:

$$\text{Hom}(X \dots X_n; Y) = \left\{ L \in Y^{\prod_{i \in \{1, \dots, n\}} X_i} \mid L: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Y \text{ is multilinear} \right\}$$

Theorem 11.259. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ be family of vector spaces over a field F , Y a vector space over the same field F and $L \in \text{Hom}(X \dots X_n; Y)$ then if $x \in \prod_{i \in \{1, \dots, n\}} X_i$ such that $\exists i \in \{1, \dots, n\}$ with $x_i = 0$ we have that $L(x) = 0$.

Proof. If $x \in \prod_{i \in \{1, \dots, n\}} X_i$ such that $\exists i \in \{1, \dots, n\}$ with $x_i = 0$ then

$$x = \left(x_1, \dots, \underbrace{0}_i, \dots, x_n\right)$$

so that

$$\begin{aligned}
 L(x) &= L\left(x_1, \dots, \underbrace{0}_i, \dots, x_n\right) \\
 &= L\left(x_1, \dots, \underbrace{0 \cdot 0}_i, \dots, x_n\right) \\
 &= 0 \cdot \left(x_1, \dots, \underbrace{0}_i, \dots, x_n\right) \\
 &= 0
 \end{aligned}$$

□

Just as with linear functions we have an alternative and simpler condition for multilinearity.

Theorem 11.260. *Let $n \in \mathbb{N}$, $\{X_j\}_{j \in \{1, \dots, n\}}$ a family of vector spaces over a field F and Y a vector space over the same field then for $L: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Y$ we have*

$$L: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Y \text{ is a multilinear mapping}$$

$$\Updownarrow$$

$$\forall i \in \{1, \dots, n\} \text{ then } \forall \left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n\right) \in \prod_{j \in \{1, \dots, n\}} X_j \text{ where } \alpha \in F, u, v \in X_i$$

we have

$$L\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n\right) = L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + \alpha \cdot L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)$$

Proof. We have:

$\Rightarrow$. Let $i \in \{1, \dots, n\}$, $u, v \in X_i$ and $\alpha \in F$ then we have

$$\begin{aligned}
 &L\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n\right) \stackrel{\text{multilinearity}}{=} \\
 &L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + L\left(x_1, \dots, \underbrace{\alpha \cdot v}_i, \dots, x_n\right) \stackrel{\text{multilinearity}}{=} \\
 &L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + \alpha \cdot L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)
 \end{aligned}$$

$\Leftarrow$. Let $i \in \{1, \dots, n\}$, $u, v \in X_i$ and $\alpha \in F$ then we have:

$$\begin{aligned}
 &L\left(x_1, \dots, \underbrace{u + v}_i, \dots, x_n\right) = \\
 &L\left(x_1, \dots, \underbrace{u + 1 \cdot v}_i, \dots, x_n\right) = \\
 &L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + 1 \cdot L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) = \\
 &L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)
 \end{aligned}$$

and

$$\begin{aligned}
 &L\left(x_1, \dots, \underbrace{\alpha \cdot v}_i, \dots, x_n\right) = \\
 &L\left(x_1, \dots, \underbrace{0 + \alpha \cdot v}_i, \dots, x_n\right) = \\
 &L\left(x_1, \dots, \underbrace{0}_i, \dots, x_n\right) + L\left(x_1, \dots, \underbrace{\alpha \cdot v}_i, \dots, x_n\right) \stackrel{[\text{theorem: 11.259}]}{=}
 \end{aligned}$$

$$\alpha \cdot L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right)$$

□

Given a family of linear mappings we can construct a multilinear mapping.

Theorem 11.261. *Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F and $\{L_i\}_{i \in \{1, \dots, n\}}$ a family satisfying $\forall i \in \{1, \dots, n\} L_i \in \text{Hom}(X_i, F)$ then if we define*

$$\prod_{i \in \{1, \dots, n\}} L_i: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow F \text{ by } \left(\prod_{i \in \{1, \dots, n\}} L_i \right)(x_1, \dots, x_n) = \prod_{i \in \{1, \dots, n\}} L_i(x_i)$$

we have

$$\prod_{i \in \{1, \dots, n\}} L_i \in \text{Hom}(X_1 \dots X_n; F)$$

Proof. Let $i \in \{1, \dots, n\}$, $x \in \prod_{j \in \{1, \dots, n\} \setminus \{i\}} X_j$, $u, v \in X_i$ and $\alpha \in F$ then we have for

$$\begin{aligned} r &= \left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n \right) \\ s &= \left(x_1, \dots, \underbrace{u}_i, \dots, x_n \right) \\ t &= \left(x_1, \dots, \underbrace{v}_i, \dots, x_n \right) \\ \left(\prod_{j \in \{1, \dots, n\}} L_j \right) \left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n \right) &= \\ \left(\prod_{j \in \{1, \dots, n\}} L_j \right) (r) &= \\ \prod_{j \in \{1, \dots, n\}} L_j(r_j) & \quad [\text{theorem: 11.43}] \\ \left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(r_j) \right) \cdot L_i(r_i) &= \\ \left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(r_j) \right) \cdot L_i(u + \alpha \cdot v) &= \\ \left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(r_j) \right) \cdot (L_i(u) + \alpha \cdot L_i(v)) &= \\ \left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(r_j) \right) \cdot L_i(u) + \alpha \cdot \left(\left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(r_j) \right) \cdot L_i(v) \right) &= \\ \left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(s_j) \right) \cdot L_i(s_i) + \alpha \cdot \left(\left(\prod_{j \in \{1, \dots, n\} \setminus \{i\}} L_j(t_j) \right) \cdot L_i(t_i) \right) & \quad [\text{theorem: 11.43}] \\ \prod_{j \in \{1, \dots, n\}} L_j(s_j) + \alpha \cdot \prod_{j \in \{1, \dots, n\}} L_j(t_j) &= \\ \left(\prod_{j \in \{1, \dots, n\}} L_j \right) (s) + \alpha \cdot \left(\prod_{j \in \{1, \dots, n\}} L_j \right) (t) &= \\ \left(\prod_{j \in \{1, \dots, n\}} L_j \right) \left(x_1, \dots, \underbrace{u}_i, \dots, x_n \right) + \alpha \cdot \left(\prod_{j \in \{1, \dots, n\}} L_j \right) \left(x_1, \dots, \underbrace{v}_i, \dots, x_n \right) \end{aligned}$$

proving by [theorem: 11.260] that $\prod_{j \in \{1, \dots, n\}} L_j$ is a multilinear mapping. $\square$

A special case of multilinear mappings is where $\{X_i\}_{i \in \{1, \dots, n\}}$ satisfies $\forall i \in \{1, \dots, n\} X_i = X$, so that $\prod_{i \in \{1, \dots, n\}} X_i = X^n$. This leads to the definition of n -linear functions.

Definition 11.262. Let $n \in \mathbb{N}$ and X, Y vector spaces over a field F then a multilinear mapping

$$L: X^n \rightarrow Y$$

is called a **n -linear mapping**. The set of graphs of n -linear mappings is noted as $\text{Hom}^n(X; Y)$ so

$$\text{Hom}^n(X; Y) = \text{Hom}\left(\underbrace{X \dots X}_n; Y\right)$$

Example 11.263. Let $n \in \mathbb{N}$ and F a field then $L_\otimes: F^n \rightarrow F$ defined by

$$L_\otimes(x) = \prod_{i \in \{1, \dots, n\}} x_i$$

is multilinear. So

$$L_\otimes \in \text{Hom}(F^n; F)$$

Proof. This follows from [theorem: 11.52] $\square$

To be able to use induction in proofs about multilinear mappings we have the following theorem.

Theorem 11.264. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n+1\}}$ be a family of vector spaces over a field F , Y a vector space over the same field F , $a \in X_{n+1}$ and $L \in \text{Hom}(X_1 \dots X_{n+1}; Y)$ then if we define

$$L_{\{\dots a\}}: \prod_{j \in \{1, \dots, n\}} X_j \rightarrow Y \text{ by } L_{\{\dots a\}}(x_1, \dots, x_n) = L(x_1, \dots, x_n, a)$$

then

$$L_{\{\dots a\}} \in \text{Hom}(X_1 \dots X_n; Y)$$

Proof. Let $i \in \{1, \dots, n\}$, $\alpha \in F$, $u, v \in X_i$. If $(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then

$$\begin{aligned} L_{\{\dots a\}}\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n\right) &= \\ L\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n, a\right) &= \\ L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n, a\right) + \alpha \cdot L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n, a\right) &= \\ L_{\{\dots a\}}\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + \alpha \cdot L_{\{\dots a\}}\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) \end{aligned}$$

$\square$

Definition 11.265. Let $n \in \mathbb{N}$, $i \in \{1, \dots, n\}$, $\{X_i\}_{j \in \{1, \dots, n\}}$ a family of sets, Y a set, $x \in \prod_{j \in \{1, \dots, n\} \setminus \{i\}} X_j$ and

$$f: \prod_{j \in \{1, \dots, n\}} X_j \rightarrow Y$$

then $f\left(x_1, \dots, \underbrace{*}_i, \dots, x_n\right): \prod_{i \in \{1, \dots, n\} \setminus \{i\}} X_i \rightarrow Y$ is defined by

$$f\left(x_1, \dots, \underbrace{*}_i, \dots, x_n\right)(x) = f\left(x_1, \dots, \underbrace{x}_i, \dots, x_n\right)$$

Theorem 11.266. Let $n \in \mathbb{N}$, $i \in \{1, \dots, n\}$, $\{X_j\}_{j \in \{1, \dots, n\}}$ a family of vector spaces over a field F and Y a vector space over the same field F and $L \in \text{Hom}(X_1 \dots X_n; Y)$ and $x \in \prod_{j \in \{1, \dots, n\} \setminus \{i\}} X_j$ then

$$L\left(x_1, \dots, \underbrace{*}_i, \dots, x_n\right) \in \text{Hom}(X_i, Y)$$

Proof. Let $\alpha \in F$, $u, v \in X_i$ then

$$\begin{aligned} L\left(x_1, \dots, \underbrace{\quad}_i \dots, x_n\right)(u + \alpha \cdot v) &= \\ L\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i \dots, x_n\right) &= \\ L\left(x_1, \dots, \underbrace{u}_i \dots, x_n\right) + \alpha \cdot L\left(x_1, \dots, \underbrace{v}_i \dots, x_n\right) &= \\ L\left(x_1, \dots, \underbrace{\quad}_i \dots, x_n\right)(u) + \alpha \cdot L\left(x_1, \dots, \underbrace{\quad}_i \dots, x_n\right)(v) & \end{aligned}$$

□

Example 11.267. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F and Y a vector space over the same field F then $C_0 \in \text{Hom}(X_1 \dots X_n; Y)$

Proof. Let $x \in \prod_{i=1}^n X_i$ $i \in \{1, \dots, n\}$, $\alpha \in F$ and $u, v \in X_i$ then

$$C_0\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i \dots, x_n\right) = 0 = 0 + 0 = C_0\left(x_1, \dots, \underbrace{u}_i \dots, x_n\right) + \alpha \cdot C_0\left(x_1, \dots, \underbrace{v}_i \dots, x_n\right) \quad \square$$

Theorem 11.268. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ be a family of vector spaces over a field F and Y a vector space over the same field then $\text{Hom}(X_1 \dots X_n; Y)$ is a sub-space of $\langle Y^{\prod_{i=1}^n X_i}, +, \cdot \rangle$. So by [theorem: 11.61] $\langle \text{Hom}(X_1 \dots X_n; Y), +, \cdot \rangle$ is a vector space.

Proof. First using [exercise: 11.267] we have that

$$C_0 \in \text{Hom}(X_1 \dots X_n; Y) \Rightarrow \text{Hom}(X_1 \dots X_n; Y) \neq \emptyset$$

Second if $\alpha \in F$ and $L_1, L_2 \in \text{Hom}(X_1 \dots X_n; Y)$ and $i \in \{1, \dots, n\}$ we have

$$\begin{aligned} (\alpha \cdot L_1 + L_2)(x_1, \dots, \underbrace{u + v}_i \dots, x_n) &= \\ \alpha \cdot L_1(x_1, \dots, \underbrace{u + v}_i \dots, x_n) + L_2(x_1, \dots, \underbrace{u + v}_i \dots, x_n) &= \\ \underbrace{\alpha \cdot L_1(x_1, \dots, \underbrace{u}_i \dots, x_n)}_1 + \underbrace{\alpha \cdot L_1(x_1, \dots, \underbrace{v}_i \dots, x_n)}_2 + \underbrace{L_2(x_1, \dots, \underbrace{u}_i \dots, x_n)}_1 + \\ \underbrace{L_2(x_1, \dots, \underbrace{v}_i \dots, x_n)}_2 &= \\ \underbrace{(\alpha \cdot L_1 + L_2)(x_1, \dots, \underbrace{u}_i \dots, x_n)}_1 + \underbrace{(\alpha \cdot L_1 + L_2)(x_1, \dots, \underbrace{v}_i \dots, x_n)}_2 &= \end{aligned}$$

and

$$\begin{aligned} (\alpha \cdot L_1 + L_2)(x_1, \dots, x_1, \dots, \underbrace{\gamma \cdot u}_i \dots, x_n, \dots, x_n) &= \\ \alpha \cdot L_1(x_1, \dots, \underbrace{\gamma \cdot u}_i \dots, x_n) + L_2(x_1, \dots, \underbrace{\gamma \cdot u}_i \dots, x_n) &= \\ \alpha \cdot \gamma \cdot L_1(x_1, \dots, \underbrace{u}_i \dots, x_n) + \gamma \cdot L_2(x_1, \dots, \underbrace{u}_i \dots, x_n) &= \\ \gamma \cdot \left(\alpha \cdot L_1(x_1, \dots, \underbrace{u}_i \dots, x_n) + L_2(x_1, \dots, \underbrace{u}_i \dots, x_n) \right) &= \\ \gamma \cdot (\alpha \cdot L_1 + L_2)(x_1, \dots, \underbrace{u}_i \dots, x_n) & \end{aligned}$$

proving that

$$\alpha \cdot L_1 + L_2 \in \text{Hom}(X_1 \dots X_n; Y) \quad \square$$

Theorem 11.269. Let $n \in \mathbb{N}$ and $\{X_i\}_{i \in \{1, \dots, n\}}$ be a family of vector spaces over a field F , Y and Z vector spaces over the same field F , $L \in \text{Hom}(X_1 \dots X_n; Y)$ and $S \in \text{Hom}(Y, Z)$ then $S \circ L \in \text{Hom}(X_1 \dots X_n; Z)$

Proof. Let $i \in \{1, \dots, n\}$, $x = (x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n) \in \prod_{j \in \{1, \dots, n\} \setminus \{i\}} X_j$, $\alpha \in F$ and $u, v \in X_i$ then

$$\begin{aligned} (S \circ L)\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n\right) &= \\ S\left(L\left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n\right)\right) &= \\ S\left(L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + \alpha \cdot L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)\right) &= \\ S\left(L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right)\right) + \alpha \cdot S\left(L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)\right) &= \\ (S \circ L)\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + \alpha \cdot (S \circ L)\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) &= \end{aligned}$$

□

Theorem 11.270. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F , Y a vector space over the same field F , $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ and $L \in \text{Hom}(X_1 \dots X_n; Y)$ then if $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$L(\alpha_1 \cdot x_1, \dots, \alpha_n \cdot x_n) = \left(\prod_{i=1}^n \alpha_i\right) \cdot L(x_1, \dots, x_n)$$

Proof. We prove this by induction, so let

$$S = \left\{ k \in \mathbb{N} \mid \text{If } k \leq n \text{ and } L \in \text{Hom}(X_1, \dots, X_k; Y) \text{ then } L(\alpha_1 \cdot x_1, \dots, \alpha_k \cdot x_k) = \left(\prod_{i=1}^k \alpha_i\right) \cdot L(x_1, \dots, x_k) \right\}$$

then we have:

$1 \in S$. Then $L(\alpha_1 \cdot x_1, \dots, \alpha_1 \cdot x_1) = L(\alpha_1 \cdot x_1) = \alpha_1 \cdot L(x_1) = (\prod_i^1 \alpha_i) \cdot L(x_1) = (\prod_i^1 \alpha_i) \cdot L(x_1, \dots, x_1)$ proving that $1 \in S$.

$k \in S \Rightarrow k+1 \in S$. If $k+1 \leq n$ and $L \in \text{Hom}(X_1 \dots X_{k+1}; Y)$ then by [theorem: 11.264] $L_{\{\dots x_{k+1}\}} \in \text{Hom}(X_1 \dots X_k; Y)$ so that as $k \in S$ we have

$$L_{\{\dots x_k\}}(\alpha_1 \cdot x_1, \dots, \alpha_k \cdot x_k) = \left(\prod_{i=1}^k \alpha_i\right) \cdot L_{\{\dots x_k\}}(x_1, \dots, x_k) \quad (11.145)$$

Then we have

$$\begin{aligned} L(\alpha_1 \cdot x_1, \dots, \alpha_{k+1} \cdot x_{k+1}) &= \alpha_{k+1} \cdot L(\alpha_1 \cdot x_1, \dots, \alpha_k \cdot x_k, x_{k+1}) \\ &= \alpha_{k+1} \cdot L_{\{\dots, x_{k+1}\}}(\alpha_1 \cdot x_1, \dots, \alpha_k \cdot x_k) \\ &\stackrel{[\text{eq.: 11.145}]}{=} \alpha_{k+1} \cdot \left(\prod_{i=1}^k \alpha_i\right) \cdot L_{\{\dots x_k\}}(x_1, \dots, x_k) \\ &= \left(\prod_{i=1}^{k+1} \alpha_i\right) \cdot L_{\{\dots x_k\}}(x_1, \dots, x_k) \\ &= \left(\prod_{i=1}^{k+1} \alpha_i\right) \cdot L(x_1, \dots, x_{k+1}) \end{aligned}$$

proving that $k+1 \in S$.

Mathematical induction proves then that $S = \mathbb{N}$, so as $n \leq n$ we have that

$$L(\alpha_1 \cdot x, \dots, \alpha_n \cdot x_n) = \left(\prod_{i=1}^n \alpha_i \right) \cdot L(x_1, \dots, x_n) \quad \square$$

Theorem 11.271. Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ be family of vector spaces over a field F , Z a vector space over the same field F , $\{Y_i\}_{i \in \{1, \dots, n\}}$ a family such that $\forall i \in \{1, \dots, n\}$ Y_i is a sub-space of X_i and $L \in \text{Hom}(Y_1 \dots Y_n, Z)$ then there exist a $K \in \text{Hom}(X_1 \dots X_n, Z)$ such that $K|_{\prod_{i \in \{1, \dots, n\}} Y_i} = L$. In other words K is a multilinear mapping extending L .

Proof. Take $i \in \{1, \dots, n\}$. As Y_i is a sub-space of X_i there exists by [theorem: 11.214] a sub-space Z_i of X_i such that $X_i = Y_i \oplus Z_i$. Using [theorem: 11.216] there exists $\pi_{Y_i}: X_i \rightarrow Y_i$ such that $\pi_{Y_i} \in \text{Hom}(X_i, Y_i)$ and $(\pi_{Y_i})|_{Y_i} = \text{Id}_{Y_i}$. Define

$$K: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Z \text{ by } K(x_1, \dots, x_n) = L(\pi_{Y_1}(x_1), \dots, \pi_{Y_n}(x_n))$$

Let $i \in \{1, \dots, n\}$ $u, v \in X_i$ and $\alpha \in F$ and $(x_1, \dots, \underbrace{\alpha \cdot u + v}_i, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then we have:

$$\begin{aligned} K(x_1, \dots, \underbrace{\alpha \cdot u + v}_i, \dots, x_n) &= \\ L\left(\left(\pi_{Y_1}(x_1), \dots, \underbrace{\pi_{Y_i}(u + \alpha \cdot v)}_i, \dots, \pi_{Y_n}(x_n)\right)\right) &= \\ L\left(\left(\pi_{Y_1}(x_1), \dots, \underbrace{\pi_{Y_i}(u) + \alpha \cdot \pi_{Y_i}(v)}_i, \dots, \pi_{Y_n}(x_n)\right)\right) &= \\ L\left(\left(\pi_{Y_1}(x_1), \dots, \underbrace{\pi_{Y_i}(u)}_i, \dots, \pi_{Y_n}(x_n)\right)\right) + \alpha \cdot L\left(\left(\pi_{Y_1}(x_1), \dots, \underbrace{\pi_{Y_i}(v)}_i, \dots, \pi_{Y_n}(x_n)\right)\right) &= \\ K(x_1, \dots, \underbrace{u}_i, \dots, x_n) + \alpha \cdot K(x_1, \dots, \underbrace{v}_i, \dots, x_n) \end{aligned}$$

proving that

$$K \in \text{Hom}(X_1, \dots, X_n; Y)$$

Finally we if $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} Y_i$ so that $\forall i \in \{1, \dots, n\}$ $x_i \in Y_i$ so that

$$\pi_i(x_i) = (\pi_i)|_{Y_i} = \text{Id}_{Y_i}(x_i) = x_i$$

hence

$$K(x_1, \dots, x_n) = L(\pi_{Y_1}(x_1), \dots, \pi_{Y_n}(x_n)) = L(x_1, \dots, x_n)$$

proving that

$$K|_{\prod_{i \in \{1, \dots, n\}} Y_i} = L \quad \square$$

Theorem 11.272. Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with basis $E = \{e_1, \dots, e_n\}$, Y a vector space over F and $L \in \text{Hom}^n(X; Y)$ then $\forall x \in X$ we have

$$L(x) = L(x_1, \dots, x_n) = \sum_{k \in \{1, \dots, n\}^n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{k_j} \right) \cdot L(e_{k_1}, \dots, e_{k_n})$$

where $\forall i \in \{1, \dots, n\}$ $\{\alpha_j^i\}_{j \in \{1, \dots, n\}}$ is the **unique family** satisfying $x_i = \sum_{j \in \{1, \dots, n\}} \alpha_j^i \cdot e_j$ [which exists by theorem: 11.143]

Proof. We prove this by induction

$$S_n = \left\{ m \in \mathbb{N} \mid \text{If } m \leq n \text{ then } L(x_1, \dots, x_n) = \sum_{k \in \{1, \dots, n\}^m} \left(\prod_{j \in \{1, \dots, m\}} \alpha_{k_j}^j \right) \cdot L(e_{k_1}, \dots, e_{k_m}, x_{m+1}, \dots, x_n) \right\}$$

where

$$(e_{\rho_1}, \dots, e_{\rho_m}, x_{m+1}, \dots, x_n) \text{ is defined by } (e_1, \dots, e_m, x_{m+1}, \dots, x_n)_k = \begin{cases} e_{\rho_k} & \text{if } k \in \{1, \dots, m\} \\ x_k & \text{if } k \in \{m+1, \dots, n\} \end{cases}$$

then we have

$1 \in S_n$. Then we have

$$\begin{aligned} L(x_1, \dots, x_n) &= L\left(\sum_{i \in \{1, \dots, n\}} \alpha_1^i \cdot e_i, x_2, \dots, x_n\right) \\ &\stackrel{[\text{theorems: 11.266, 11.80}]}{=} \sum_{i \in \{1, \dots, n\}} L(\alpha_1^i \cdot e_i, x_2, \dots, x_n) \\ &\stackrel{[\text{theorem: 11.266}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_1^i \cdot L(e_i, x_2, \dots, x_n) \\ &= \sum_{i \in \{1, \dots, n\}} \left(\prod_{j \in \{1\}} \alpha_j^i \right) \cdot L(e_i, x_2, \dots, x_n) \quad (11.146) \end{aligned}$$

Define $\beta: \{1, \dots, n\}^1 \rightarrow \{1, \dots, n\}$ by $\beta(\rho) = \rho_1$ then we have

injectivity. If $\beta(\rho) = \beta(\sigma)$ then $\forall k \in \{1\}$ we have $\rho_k = \rho_1 = \beta(\rho) = \beta(\sigma) = \sigma_1 = \sigma_k$ so that $\rho = \sigma$.

surjectivity. If $i \in \{1, \dots, n\}$ then define $\rho \in \{1, \dots, n\}^1$ by $\rho_1 = i$ then $\beta(\rho) = \rho_1 = i$

so that

$$\beta: \{1, \dots, n\}^1 \rightarrow \{1, \dots, n\} \text{ is a bijection}$$

then we have

$$\begin{aligned} \sum_{\rho \in \{1, \dots, n\}^1} \left(\prod_{j \in \{1\}} \alpha_j^{\rho_j} \right) \cdot L(e_{\rho_j}, x_2, \dots, x_n) &= \\ \sum_{\rho \in \{1, \dots, n\}^1} \left(\prod_{j \in \{1\}} \alpha_j^{\rho_1} \right) \cdot L(e_{\rho_1}, x_2, \dots, x_n) &= \\ \sum_{i \in \{1, \dots, n\}^1} \left(\prod_{j \in \{1\}} \alpha_j^{\beta(\rho)} \right) \cdot L(e_{\beta(\rho)}, x_2, \dots, x_n) &\stackrel{[\text{theorem: 11.36}]}{=} \\ \sum_{i \in \{1, \dots, n\}} \left(\prod_{j \in \{1\}} \alpha_j^i \right) \cdot L(e_i, x_2, \dots, x_n) &\stackrel{[\text{eq: 11.146}]}{=} \\ &L(x_1, \dots, x_n) \end{aligned}$$

proving that $1 \in S_n$.

$m \in S_n \Rightarrow m+1 \in S_n$. If $m+1 \leq n$ then we have

$$\begin{aligned}
 L(x_1, \dots, x_n) &\stackrel{m \in S_n}{=} \\
 &\sum_{\rho \in \{1, \dots, n\}^m} \left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{\rho_j} \right) \cdot L(e_{\rho_1}, \dots, e_{\rho_m}, x_{m+1}, \dots, x_n) = \\
 &\sum_{\rho \in \{1, \dots, n\}^m} \left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{\rho_j} \right) \cdot L(e_{\rho_1}, \dots, e_{\rho_m}, \sum_{i \in \{1, \dots, n\}} \alpha_{m+1}^i \cdot x_i, x_{m+2}, \dots, x_n) = \\
 &\sum_{\rho \in \{1, \dots, n\}^m} \left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{\rho_j} \right) \cdot \sum_{i \in \{1, \dots, n\}} \alpha_{m+1}^i \cdot L(e_{\rho_1}, \dots, e_{\rho_m}, e_i, x_{m+2}, \dots, x_n) = \\
 &\sum_{\rho \in \{1, \dots, n\}^m} \left(\sum_{i \in \{1, \dots, n\}} \left(\left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{\rho_j} \right) \cdot \alpha_{m+1}^i \right) \cdot L(e_{\rho_1}, \dots, e_{\rho_m}, e_i, x_{m+2}, \dots, x_n) \right) =
 \end{aligned}$$

so that

$$L(x_1, \dots, x_n) = \sum_{\rho \in \{1, \dots, n\}^m} \left(\sum_{i \in \{1, \dots, n\}} A_{\rho, i} \right) \quad (11.147)$$

where

$$A_{\rho, i} = \left(\left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{\rho_j} \right) \cdot \alpha_{m+1}^i \right) \cdot L(e_{\rho_1}, \dots, e_{\rho_m}, e_i, x_{m+2}, \dots, x_n) \quad (11.148)$$

Given $\rho \in \{1, \dots, n\}^m$ define

$$I_\rho = \{\rho\} \times \{1, \dots, n\} \text{ and } \pi_2: I_\rho \rightarrow \{1, \dots, n\} \text{ by } \pi_2(\rho, k) = k \quad (11.149)$$

then we have for π_2

injectivity. If $\pi_2(\omega, k) = \pi_2(\sigma, l)$ then as $(\omega, k), (\sigma, l) \in I_\rho = \{\rho\} \times \{1, \dots, n\}$ we have $\omega = \rho = \sigma$, further, as $k = \pi_2(\omega, k) = \pi_2(\sigma, l) = l$, we have $(\omega, k) = (\sigma, l)$.

surjectivity. If $k \in \{1, \dots, n\}$ then for $(\rho, k) \in I_\rho$ we have $\pi_2(\rho, k) = k$ proving surjectivity.

hence:

$$\forall \rho \in \{1, \dots, n\} \text{ we have } \pi_2: I_\rho \rightarrow \{1, \dots, n\} \text{ is a bijection}$$

so that

$$\begin{aligned}
 \sum_{k \in I_\rho} A_{k_1, k_2} &\stackrel{k \in I_\rho \Rightarrow k_1 = \rho}{=} \sum_{k \in I_\rho} A_{\rho, \pi_2(k)} \\
 &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{1, \dots, n\}} A_{\rho, i}
 \end{aligned}$$

which by substituting this in [eq: 11.147] gives

$$L(x_1, \dots, x_n) = \sum_{\rho \in \{1, \dots, n\}^m} \left(\sum_{k \in I_\rho} A_{k_1, k_2} \right) \quad (11.150)$$

If $\rho, \sigma \in \{1, \dots, n\}^m$ with $\rho \neq \sigma$ then if $k \in I_\rho \cap I_\sigma$ we have $(\rho, i) = k = (\sigma, j)$ so that $\rho = \sigma$ contradicting $\rho \neq \sigma$. Hence $I_\rho \cap I_\sigma = \emptyset$, so if we define

$$I = \bigcup_{\rho \in \{1, \dots, n\}^m} I_\rho$$

we have

$$\sum_{\rho \in I} A_{k_1, k_2} \stackrel{[\text{theorem: 11.44}]}{=} \sum_{\rho \in \{1, \dots, n\}^m} \left(\sum_{k \in I_\rho} A_{k_1, k_2} \right)$$

which combined with [eq: 11.150] gives

$$L(x_1, \dots, x_n) = \sum_{\rho \in I} A_{k_1, k_2} \quad (11.151)$$

Define now

$$\eta: \{1, \dots, n\}^{m+1} \rightarrow \bigcup_{\rho \in \{1, \dots, n\}^m} I_\rho = I \text{ by } \eta(k) = (k|_{\{1, \dots, m\}}, k(m+1)) = (k|_{\{1, \dots, m\}}, k_{m+1})$$

then we have:

injectivity. If $\eta(k) = \eta(l)$ then we have $(k|_{\{1, \dots, m\}}, k(m+1)) = (l|_{\{1, \dots, m\}}, l(m+1))$ then $k|_{\{1, \dots, m\}} = l|_{\{1, \dots, m\}} \Rightarrow \forall i \in \{1, \dots, m\} \ k(i) = l(i)$ and $k(m+1) = l(m+1)$ so that $k = l$.

surjectivity. If $z \in \bigcup_{\rho \in \{1, \dots, n\}^m} I_\rho$ then there exists a $\rho \in \{1, \dots, n\}^m$ so that $z \in I_\rho$. Hence there exists a $k \in \{1, \dots, n\}$ such that $z = (\rho, k)$. Define $l \in \{1, \dots, n\}^{m+1}$ by $l_i = \begin{cases} \rho_i & \text{if } i \in \{1, \dots, m\} \\ k & \text{if } i = m+1 \end{cases}$ then $\eta(l) = (l|_{\{1, \dots, m\}}, l(m+1)) = (\rho, k) = z$.

proving that

$$\eta: \{1, \dots, n\}^{m+1} \rightarrow I \text{ is a bijection}$$

Using [theorem: 11.36] we have

$$\sum_{k \in I} A_{k_1, k_2} = \sum_{k \in \{1, \dots, n\}^{m+1}} A_{\eta(k)_1, \eta(k)_2}$$

which combined with [eq: 11.151] proves

$$L(x_1, \dots, x_n) = \sum_{k \in \{1, \dots, n\}^{m+1}} A_{\eta(k)_1, \eta(k)_2} \quad (11.152)$$

Now for $k \in \{1, \dots, n\}^{m+1}$ we have by [eq: 11.148]

$$\begin{aligned} A_{\eta(k)_1, \eta(k)_2} &= \\ A_{k|_{\{1, \dots, m\}}, k_{m+1}} &= \\ \left(\left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{(k|_{\{1, \dots, m\}})_j} \right) \cdot \alpha_{m+1}^{k_{m+1}} \right) \cdot L(e_{(k|_{\{1, \dots, m\}})_1}, \dots, e_{(k|_{\{1, \dots, m\}})_m}, e_{k_{m+1}}, & \\ x_{m+2}, \dots, x_n) &= \\ \left(\left(\prod_{j \in \{1, \dots, m\}} \alpha_j^{k_j} \right) \cdot \alpha_{m+1}^{k_{m+1}} \right) \cdot L(e_{k_1}, \dots, e_{k_m}, e_{k_{m+1}}, x_{m+2}, \dots, x_n) &= \\ \left(\prod_{j \in \{1, \dots, m+1\}} \alpha_j^{k_j} \right) \cdot L(e_{k_1}, \dots, e_{k_{m+1}}, x_{m+2}, \dots, x_n) \end{aligned}$$

which combined with [eq: 11.152] gives

$$L(x_1, \dots, x_n) = \sum_{k \in \{1, \dots, n\}^{m+1}} \left(\prod_{j \in \{1, \dots, m+1\}} \alpha_j^{k_j} \right) \cdot L(e_{k_1}, \dots, e_{k_{m+1}}, x_{m+2}, \dots, x_n)$$

proving that $m+1 \in S$.

By mathematical induction we have that $S_n = \mathbb{N}$. So as $n \leq n$ we have

$$L(x_1, \dots, x_n) = \sum_{k \in \{1, \dots, n\}^n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{k_j} \right) \cdot L(e_{k_1}, \dots, x_n) \quad \square$$

11.7 Determinant Functions

We will make extensive use of the signature of a permutation in defining skew-symmetric functions. So it will be useful to have the concept of -1 in a field and examine its relation with $-1 \in \mathbb{Z}$.

Definition 11.273. If F is a field and 1 is the multiplicative unit then -1 is the additive inverse of 1 . Hence we have $1 + (-1) = 0 = (-1) + 1$

Proposition 11.274. *If F is a field then $(-1) \cdot (-1) = 1$*

Proof. $0 = (-1) \cdot 0 = (-1) \cdot ((-1) + 1) = (-1) \cdot (-1) + (-1) \cdot 1 = (-1) \cdot (-1) + (-1)$ so that

$$1 = 0 + 1 = ((-1) \cdot (-1) + (-1)) + 1 = (-1) \cdot (-1) \quad \square$$

Proposition 11.275. *If F is a field then $-1 \neq 0$ and $(-1)^{-1} = -1$.*

Proof. Assume that $-1 = 0$ then $1 = 1 + 0 = 1 + (-1) = 0$ contradicting that for a field $1 \neq 0$ [see definition: 4.51], so we must have that $(-1) \neq 0$ and thus that $(-1)^{-1}$ exist. Further, as $1 = (-1) \cdot (-1)$ we have that

$$(-1)^{-1} = (-1)^{-1} \cdot 1 = (-1)^{-1} \cdot ((-1) \cdot (-1)) = ((-1)^{-1} \cdot (-1)) \cdot (-1) = 1 \cdot (-1) = -1 \quad \square$$

Proposition 11.276. *If F is a field then $\forall f \in F$ we have $(-1) \cdot f = -f$*

Proof. As we have

$$f + (-1) \cdot f = (-1) \cdot f + f = (-1) \cdot f + 1 \cdot f = ((-1) + 1) \cdot f = 0 \cdot f = 0 \quad \square$$

Proposition 11.277. *If V is a vector space over a field F then $\forall x \in X$ we have $-x = (-1) \cdot x$*

Proof. Let $x \in X$ then

$$(-1) \cdot x + x = x + (-1) \cdot x = 1 \cdot x + (-1) \cdot x = (1 + (-1)) \cdot x = 0 \cdot x = 0$$

so that $-x = (-1) \cdot x \quad \square$

Next we define the power in a field by recursion.

Definition 11.278. *Let F be a field, $a \in F$ and $n \in \mathbb{N}_0$ then*

$$a^n = \begin{cases} 1 & \text{if } n = 0 \\ a \cdot a^{n-1} & \text{if } n \in \mathbb{N} \end{cases}$$

Proposition 11.279. *Let F be a field, $a \in F$ and $n, m \in \mathbb{N}_0$ then $a^n \cdot a^m = a^{n+m}$*

Proof. We prove this by induction on n . So let

$$S_m = \{n \in \mathbb{N} \mid a^n \cdot a^m = a^{n+m}\}$$

then we have:

$0 \in S_m$. Then $a^0 \cdot a^m = 1 \cdot a^m = a^m = a^{0+m}$ proving that $0 \in S_m$.

$n \in S_m \Rightarrow n+1 \in S_m$. We have

$$\begin{aligned} a^{n+1} \cdot a^m &= (a \cdot a^n) \cdot a^m \\ &= a \cdot (a^n \cdot a^m) \\ &\stackrel{n \in S}{=} a \cdot a^{n+m} \\ &= a^{(n+m)+1} \\ &= a^{(n+1)+m} \end{aligned}$$

proving that $n+1 \in S_m \quad \square$

Proposition 11.280. *Let F be a field then for $n \in \mathbb{N}_0$ we have*

$$\begin{aligned} 1^n &= 1 \\ (-1)^n &= -1 \text{ or } 1 \\ (-1)^n &= \begin{cases} -1 & \text{if } n \text{ is odd} \\ 1 & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

Further if $n \in \mathbb{N}$ then $0^n = 0$

Proof.

1. We use induction to prove this, so let

$$S = \{n \in \mathbb{N}_0 \mid 1^n = 1\}$$

then we have:

$0 \in S$. As $1^0 = 1$ by definition we have $0 \in S$

$n \in S \Rightarrow n+1 \in S$. As $1^{n+1} = 1 \cdot 1^n \underset{n \in S}{=} 1 \cdot 1 = 1$ proving that $n+1 \in S$

2. Again we use induction, so let

$$S = \{n \in \mathbb{N}_0 \mid (-1)^n \in \{-1, 1\}\}$$

then we have:

$0 \in S$. By definition $(-1)^0 = 1$ proving that $0 \in S$

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ we have either:

$(-1)^n = 1$. Then $(-1)^{n+1} = (-1) \cdot (-1)^n = (-1) \cdot 1 = -1$

$(-1)^n = -1$. Then $(-1)^{n+1} = (-1) \cdot (-1)^n = (-1) \cdot (-1) \underset{[\text{proposition: 11.274}]}{=} 1$

proving that $n+1 \in S$.

3. If n is even then there exists a $m \in \mathbb{N}_0$ such that $n = 2 \cdot m = m + m$. Now for $(-1)^m$ we have by (2) either:

$(-1)^m = 1$. Then $(-1)^n = (-1)^{m+m} \underset{[\text{theorem: 11.279}]}{=} (-1)^m \cdot (-1)^m = 1 \cdot 1 = 1$

$(-1)^m = -1$. Then

$(-1)^n = (-1)^{m+m} \underset{[\text{theorem: 11.279}]}{=} (-1)^m \cdot (-1)^m = (-1) \cdot (-1) \underset{[\text{theorem: 11.274}]}{=} 1$

hence

$$(-1)^n = 1$$

4. If n is odd then there exist a $m \in \mathbb{N}_0$ such that $n = 2 \cdot m + 1$ then we have

$$(-1)^n = (-1)^{2 \cdot m + 1} = (-1) \cdot (-1)^{2 \cdot m} \underset{(3)}{=} (-1) \cdot 1 = -1$$

5. If $n \in \mathbb{N}$ then $n-1 \in \mathbb{N}_0$ so that

$$0^n = 0^{(n-1)+1} = 0 \cdot 0^{n-1} = 0$$

□

Theorem 11.281. Let $\langle F, +, \cdot \rangle$ be a field and $n \in \mathbb{N}_0$ then we have that

1. $((-1)^n)^{-1} = (-1)^n$

2. $\forall m \in \mathbb{N}_0$ we have $(-1)^{n+2 \cdot m} = (-1)^n$

Proof.

1. $(-1)^n = 1 \cdot (-1)^n = (((-1)^n)^{-1} \cdot (-1)^n) \cdot (-1)^n = ((-1)^n)^{-1} ((-1)^n \cdot (-1)^n) = ((-1)^n)^{-1} 1 = (-1)^n$

2. $(-1)^{n+2 \cdot m} \underset{[\text{proposition: 11.279}]}{=} (-1)^n \cdot (-1)^{2 \cdot m} = (-1)^n$. □

All the above work is done so that we can define a mapping of $\{1, -1\} \subseteq \mathbb{Z}$ to F which we will use in the definition of skew symmetric multilinear mappings.

Definition 11.282. Let $\langle F, +, \cdot \rangle$ be a field then we define

$$\odot: \{1, -1\} \times F \rightarrow F \text{ by } z \odot f = \begin{cases} f & \text{if } z=1 \\ -f & \text{if } z=-1 \end{cases}$$

where $\{-1, 1\} \subseteq \mathbb{Z}$.

Theorem 11.283. *Let $n \in \mathbb{N}_0$ then*

$$\forall f \in F \text{ we have } \underbrace{(-1)^n}_{\in \{1, -1\} \subseteq \mathbb{Z}} \odot f = \underbrace{(-1)^n}_{\in \{1, -1\} \subseteq F} \cdot f$$

Proof. We use induction to prove this, so let $f \in F$ and define

$$S_f = \{n \in \mathbb{N}_0 \mid (-1)^n \odot f = (-1)^n \cdot f\}$$

then we have:

$$0 \in S. \text{ As } (-1)^0 \odot f = 1 \odot f \stackrel{\text{def}}{=} f = 1 \cdot f \stackrel{[\text{definition: 11.278}]}{=} (-1)^0 \cdot f \text{ proving that } 0 \in S$$

$n \in S \Rightarrow n+1 \in S$. Let $n \in S$ then we have either:

n is even. Then $n+1$ is odd

$$\begin{aligned} (-1)^{n+1} \odot f &= (-1) \odot f \\ &\stackrel{\text{def}}{=} -f \\ &\stackrel{[\text{theorem: 11.276}]}{=} (-1) \cdot f \\ &\stackrel{[\text{theorem: 11.280}]}{=} (-1)^{n+1} \cdot f \end{aligned}$$

proving that $n+1 \in S$ in this case.

n is odd. Then $n+1$ is even

$$\begin{aligned} (-1)^{n+1} \odot f &= 1 \odot f \\ &\stackrel{\text{def}}{=} f \\ &= 1 \cdot f \\ &\stackrel{[\text{theorem: 11.280}]}{=} (-1)^{n+1} \cdot f \end{aligned}$$

proving that $n+1 \in S$ □

From now on, to avoid excessive notation we use $\cdot$ instead of $\odot$ and rely on context to figure out which operator is used. More specific if $\sigma \in P_n$ then $\text{sign}(\sigma) \in \{-1, 1\} \subseteq \mathbb{Z}$ so that if $f \in F$ $\text{sign}(\sigma) \cdot f$ is actually $\text{sign}(\sigma) \odot f$.

Definition 11.284. *Let X, Y be vector spaces over a field F , $m \in \mathbb{N}$ then a n -linear mapping $L \in \text{Hom}^n(X; Y)$ [see definition: 11.262] is **symmetric** if $\forall \sigma \in P_n$ we have*

$$\sigma L = L$$

[see definition: 11.235].

More useful than symmetric n -linear mappings are skew-linear mappings.

Definition 11.285. *Let X, Y be vector spaces over a field F , $n \in \mathbb{N}$ then a n -linear mapping $L \in \text{Hom}^n(X; Y)$ [see definition: 11.262] is **skew-symmetric** if $\forall \sigma \in P_n$ we have*

$$\sigma L = \text{sign}(\sigma) \cdot L$$

[see definition: 11.235].

A trivial example of a mapping that is at the same time symmetric and skew-symmetric is the following.

Example 11.286. Let X, Y be vector space over a field F then if $L \in \text{Hom}^1(X; Y)$ L is symmetric and skew-symmetric.

Proof. If $\sigma \in P_1$ then $\sigma = \text{Id}_{\{1\}}$ so that $\text{sign}(\sigma) = 1$ and $\sigma L(x) = L(x \circ \sigma) = L$ proving that

$$\sigma L = L = 1 \cdot L = \text{sign}(\sigma) \cdot L$$

□

If $L \in \text{Hom}^n(X; Y)$ is symmetric then $\sigma L = L \in \text{Hom}^n(X; Y)$ and if L is skew-symmetric we have $\sigma L = \text{sign}(\sigma) \cdot L \in \text{Hom}^n(X; Y)$. Actually this is a general result valid for every n -linear mapping.

Theorem 11.287. *Let X, Y be vector spaces over a field F , $n \in \mathbb{N}$ and $L \in \text{Hom}^n(X; Y)$ a n -linear mapping then*

$$\forall \sigma \in P_n \text{ we have } \sigma L \in \text{Hom}^n(X; Y)$$

Proof. Let $i \in \{1, \dots, n\}$, $\alpha \in F$ and $u, v \in X_i$. Then if

$$z = \left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n \right) \in X^n$$

$$r = \left(x_1, \dots, \underbrace{u}_i, \dots, x_n \right) \in X^n$$

$$s = \left(x_1, \dots, \underbrace{v}_i, \dots, x_n \right) \in X^n$$

Let $k = \sigma^{-1}(i)$ then if $j \neq k$ we must have $\sigma(j) \neq i$ [otherwise if $\sigma(j) = i$ we have as σ is bijective that $j = \sigma^{-1}(i) = k$ a contradiction]. So $\forall j \in \{1, \dots, n\} \setminus \{k\}$ $\sigma(j) \neq i$ and we have:

$$(z \circ \sigma)_j = x(\sigma(j)) = (r \circ \sigma)_j \quad (11.153)$$

$$(z \circ \sigma)_j = x(\sigma(j)) = (s \circ \sigma)_j \quad (11.154)$$

Then

$$\begin{aligned} & (\sigma L) \left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n \right) = \\ & \quad (\sigma L)(z) = \\ & \quad L(z \circ \sigma) \quad [\text{theorem: 11.257}] \\ & \quad L \left((z \circ \sigma)_1, \dots, \underbrace{u + \alpha \cdot v}_{\sigma^{-1}(i)}, \dots, (z \circ \sigma)_n \right) = \\ & \quad L \left((z \circ \sigma)_1, \dots, \underbrace{u}_{\sigma^{-1}(i)}, \dots, (z \circ \sigma)_n \right) + \alpha \cdot L \left((z \circ \sigma)_1, \dots, \underbrace{v}_{\sigma^{-1}(i)}, \dots, (z \circ \sigma)_n \right) \quad [\text{eqs: 11.153, 11.154}] \\ & \quad L \left((r \circ \sigma)_1, \dots, \underbrace{u}_{\sigma^{-1}(i)}, \dots, (r \circ \sigma)_n \right) + \alpha \cdot L \left((s \circ \sigma)_1, \dots, \underbrace{v}_{\sigma^{-1}(i)}, \dots, (s \circ \sigma)_n \right) \quad [\text{theorem: 11.257}] \\ & \quad L(r \circ \sigma) + \alpha \cdot L(s \circ \sigma) = \\ & \quad (\sigma L)(r) + \alpha \cdot (\sigma L)(s) = \\ & \quad (\sigma L)(x_1, \dots, x_{i-1}, u, x_{i+1}, \dots, x_n) + \alpha \cdot (\sigma L)(x_1, \dots, x_{i-1}, v, x_{i+1}, \dots, x_n) \end{aligned}$$

proving that σL is multilinear. □

The next theorem shows we can construct a new skew-symmetric from a n -linear mapping. First we need a little lemma:

Lemma 11.288. *Let $n \in \mathbb{N}$ and $\rho \in P_n$ then*

$$T_\rho: P_n \rightarrow P_n \text{ defined by } T_\rho(\sigma) = \rho \circ \sigma$$

is a bijection.

Proof.

injectivity. If $T_\rho(\sigma) = T_\rho(\sigma')$ then

$$\begin{aligned} \rho \circ \sigma = \rho \circ \sigma' & \Rightarrow \rho^{-1} \circ (\rho \circ \sigma) = \rho^{-1} \circ (\rho \circ \sigma') \\ & \Rightarrow (\rho^{-1} \circ \rho) \circ \sigma = (\rho^{-1} \circ \rho) \circ \sigma' \\ & \Rightarrow \text{Id}_{\{1, \dots, n\}} \circ \sigma = \text{Id}_{\{1, \dots, n\}} \circ \sigma' \\ & \Rightarrow \sigma = \sigma' \end{aligned}$$

proving injectivity.

surjectivity. Let $\sigma \in P_n$ then $\rho^{-1} \circ \sigma \in P_n$ [see theorem: 11.224] and

$$T_\rho(\rho^{-1} \circ \sigma) = \rho \circ (\rho^{-1} \circ \sigma) = (\rho \circ \rho^{-1}) \circ \sigma = \text{Id}_{\{1, \dots, n\}} \circ \sigma = \sigma$$

proving surjectivity. □

Theorem 11.289. Let $n \in \mathbb{N}$, X, Y be vector spaces over a field F and a $L \in \text{Hom}^n(X; Y)$ a n -linear mapping then

$$\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L$$

is a skew-symmetric mapping:

Note 11.290. The sum is well defined as P_n is finite, $\forall \sigma \in P_n \sigma L \in \text{Hom}^n(X; Y) \Rightarrow \text{sign}(\sigma) \cdot \sigma L \in \text{Hom}^n(X; Y) \subseteq Y^{x^n}$ and Y^{X^n} is a vector space [see theorems: 11.223, 11.287 and 11.268].

Proof. Using [theorem: 11.287] we have $\forall \sigma \in P_n$ that $\text{sign}(\sigma) \cdot \sigma L \in \text{Hom}^n(X; Y)$ so that by [theorem: 11.83]

$$\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L \in \text{Hom}^n(X; Y) \text{ or } \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L \text{ is } n\text{-linear}$$

Next we have to prove that $\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L$ is skew-symmetric. So let $\rho \in P_n$ then we have for $x \in X^n$ that

$$\begin{aligned} \left(\rho \left(\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L \right) \right)(x) &= \left(\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L \right)(x \circ \rho) \\ &= \sum_{\sigma \in P_n} (\text{sign}(\sigma) \cdot \sigma L)(x \circ \rho) \\ &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot (\sigma L)(x \circ \rho) \\ &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot L((x \circ \rho) \circ \sigma) \\ &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot L(x \circ (\rho \circ \sigma)) \\ &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot ((\rho \circ \sigma)L)(x) \\ &\stackrel{\text{sign}(\rho) \cdot \text{sign}(\rho)=1}{=} \sum_{\sigma \in P_n} (\text{sign}(\rho) \cdot \text{sign}(\rho)) \cdot \text{sign}(\sigma) \cdot ((\rho \circ \sigma)L)(x) \\ &= \sum_{\sigma \in P_n} \text{sign}(\rho) \cdot ((\text{sign}(\rho) \cdot \text{sign}(\sigma)) \cdot ((\rho \circ \sigma)L)(x)) \\ &= \text{sign}(\rho) \cdot \sum_{\sigma \in P_n} (\text{sign}(\rho) \cdot \text{sign}(\sigma)) \cdot ((\rho \circ \sigma)L)(x) \\ &\stackrel{[\text{theorem: 11.251}]}{=} \text{sign}(\rho) \cdot \sum_{\sigma \in P_n} \text{sign}(\rho \circ \sigma) \cdot ((\rho \circ \sigma)L)(x) \\ &\stackrel{[\text{lemma: 11.288}]}{=} \text{sign}(\rho) \cdot \sum_{\sigma \in P_n} \text{sign}(T_\rho(\sigma)) \cdot (T_\rho(\sigma)L)(x) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \text{sign}(\rho) \cdot \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot (\sigma L)(x) \end{aligned}$$

proving that

$$\rho \left(\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L \right) = \text{sign}(\rho) \cdot \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma L$$

□

Theorem 11.291. *Let X, Y be vector spaces over a field F with characteristic zero [see definition: 10.47], $n \in \mathbb{N}$ and $L \in \text{Hom}^n(X; Y)$ then the following are equivalent:*

1. L is skew-symmetric.
2. For every $x \in X^n$ for which $\exists i, j \in \{1, \dots, n\}$ with $i \neq j$ such that $x_i = x_j$ we have $L(x) = 0$.
3. For every $x \in X^n$ such that $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ is linear dependent we have $L(x) = 0$.

Proof.

1 $\Rightarrow$ 2. Let $x \in X^n$ such that $\exists i, j \in \{1, \dots, n\}$ with $i \neq j$ and $x_i = x_j \Rightarrow x(i) = x(j)$. Then we have for $k \in \{1, \dots, n\}$ that

$$\begin{aligned} (x \circ (i \leftrightarrow j_n))(k) &= x((i \leftrightarrow j_n)(k)) \\ &= \begin{cases} x(i) & \text{if } k = j \\ x(j) & \text{if } k = i \\ x(k) & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases} \\ &\stackrel{x(i)=x(j)}{=} \begin{cases} x(k) & \text{if } k = j \\ x(k) & \text{if } k = i \\ x(k) & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases} \\ &= x(k) \end{aligned}$$

proving that

$$x \circ (i \leftrightarrow j_n) = x$$

Then

$$\begin{aligned} L(x) &= L(x \circ (i \leftrightarrow j_n)) \\ &\stackrel{L \text{ is skew-symmetric}}{=} \text{sign}((i \leftrightarrow j_n)) \cdot L(x) \\ &\stackrel{[\text{theorem: 11.251}]}{=} (-1) \cdot L(x) \\ &= -L(x) \end{aligned}$$

so that $L(x) + L(x) = 0$ or $(1 + 1) \cdot L(x) = 0$, as F has characteristic zero it follows that $L(x) = 0$.

2 $\Rightarrow$ 1. Assume that (2) hold then we prove the following

$$\forall x \in X^n \text{ we have } \forall i, j \in \{1, \dots, n\} \text{ with } i \neq j \text{ that } L(x \circ (i \leftrightarrow j_n)) = -L(x) \quad (11.155)$$

Proof. Let $x \in X^n$, $i, j \in \{1, \dots, n\}$ with $i \neq j$. Then

$$\begin{aligned} 0 &= L(x_1, \dots, \underbrace{x_i + x_j}_i, \dots, \underbrace{x_i + x_j}_j, \dots, x_n) \\ &= L(x_1, \dots, \underbrace{x_i + x_j}_i, \dots, x_i, \dots, x_n) + L(x_1, \dots, \underbrace{x_i + x_j}_i, \dots, x_j, \dots, x_n) \\ &= L(x_1, \dots, \underbrace{x_i}_i, \dots, \underbrace{x_i}_j, \dots, x_n) + L(x_1, \dots, \underbrace{x_j}_i, \dots, \underbrace{x_i}_j, \dots, x_n) + L(x_1, \dots, \underbrace{x_i}_i, \dots, \underbrace{x_j}_j, \dots, x_n) \\ &\quad + L(x_1, \dots, \underbrace{x_j}_i, \dots, \underbrace{x_j}_j, \dots, x_n) \\ &= 0 + L(x_1, \dots, \underbrace{x_j}_i, \dots, \underbrace{x_i}_j, \dots, x_n) + L(x_1, \dots, \underbrace{x_i}_i, \dots, \underbrace{x_j}_j, \dots, x_n) + 0 \\ &= L(x \circ (i \leftrightarrow j_n)) + L(x) \end{aligned}$$

proving that

$$L(x \circ (i \xleftrightarrow{n} j)) = -L(x) \quad \square$$

Next we prove that

$$\text{If } m \in \mathbb{N} \text{ \{if } (i_k \leftrightarrow j_k)\}_{k \in \{1, \dots, m\}} \text{ is a family of strict transformations then } ((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m))L = (-1)^m L \quad (11.156)$$

Proof. We proceed by induction, so let

$$S = \{m \in \mathbb{N} \mid \text{if } \{(i_k \leftrightarrow j_k)\}_{k \in \{1, \dots, m\}} \text{ is a family of \textbf{strict} transpositions then } ((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m))L = (-1)^m L\} \text{ then we have:}$$

$$1 \in S.$$

$$\begin{aligned} ((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_1 \xleftrightarrow{n} j_1))L &= (i_1 \xleftrightarrow{n} j_1)L \\ &\stackrel{[\text{eq: 11.155}]}{=} -L \\ &= (-1)^1 \cdot L \end{aligned}$$

proving that $1 \in S$.

$m \in S \Rightarrow m+1 \in S$. We have

$$\begin{aligned} &((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_{m+1} \xleftrightarrow{n} j_{m+1}))L &&= \\ &(((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m)) \circ (i_{m+1} \xleftrightarrow{n} j_{m+1}))L &&\stackrel{[\text{theorem: 11.238}]}{=} \\ &((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m))((i_{m+1} \xleftrightarrow{n} j_{m+1}))L &&\stackrel{[\text{eq: 11.155}]}{=} \\ &((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m))(-L) &&= \\ &((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m))((-1) \cdot L) &&\stackrel{[\text{proposition: 11.239}]}{=} \\ &(-1) \cdot ((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \xleftrightarrow{n} j_m))L &&\stackrel{m \in S}{=} \\ &(-1) \cdot ((-1)^m \cdot L) &&= \\ &((-1) \cdot (-1)^m) \cdot L &&= \\ &(-1)^{m+1} \cdot L && \end{aligned}$$

proving that $m+1 \in S$. $\square$

Let $\sigma \in P_n$ then by [theorem: 11.250] there exists a $\{(i_k \leftrightarrow j_k)\}_{k \in \{1, \dots, m\}}$ of **strict** transpositions such that

$$\sigma = (i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \leftrightarrow j_m) \text{ and } \text{sign}(\sigma) = (-1)^m$$

So

$$\sigma L = ((i_1 \xleftrightarrow{n} j_1) \circ \dots \circ (i_m \leftrightarrow j_m))L \stackrel{[\text{eq: 11.156}]}{=} (-1)^m \cdot L = \text{sign}(\sigma) \cdot L$$

2 $\Rightarrow$ 3. As $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ is linear dependent there exist by [theorem: 11.128] a $k \in \{1, \dots, n\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, n\} \setminus \{k\}} \subseteq F$ such that

$$x_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i \quad (11.157)$$

If $i \in \{1, \dots, n\} \setminus \{k\} \Rightarrow i \neq k$ we have

$$\left(x_1, \dots, \underbrace{x_i}_k, \dots, x_n\right)_k = x_i$$

so that by (2) $L\left(x_1, \dots, \underbrace{x_i}_{k}, \dots, x_n\right) = 0$. Hence

$$\forall i \in \{1, \dots, n\} \setminus \{k\} \text{ we have } L\left(x_1, \dots, \underbrace{x_i}_{k}, \dots, x_n\right) = 0 \quad (11.158)$$

Finally

$$\begin{aligned} L(x) &= L\left(x_1, \dots, \underbrace{x_k}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{eq: 11.157}]}{=} L\left(x_1, \dots, \underbrace{\sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{theorems: 11.266, 11.190}]}{=} \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot L\left(x_1, \dots, \underbrace{x_i}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{eq: 11.158}]}{=} 0 \end{aligned}$$

3 $\Rightarrow$ 2. If $x = (x_1, \dots, x_n) \in X^n$ is such that $\exists i, j \in \{1, \dots, n\}$ where $i \neq j$ and $x_i = x_j$ then by [theorem: 11.130] $\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent, hence by (3) $L(x) = 0$ $\square$

Theorem 11.292. Let X, Y be vector spaces over a field F with characteristic zero [see definition: 10.47], $n \in \mathbb{N}$ and $L \in \text{Hom}^n(X; Y)$ a skew-symmetric n -linear mapping then if $x = (x_1, \dots, x_n) \in X^n$ is such that $\{x_i | i \in \{1, \dots, n\}\}$ is a linear dependent set then $L(x) = 0$

Proof. Let $x = (x_1, \dots, x_n) \in X^n$ such that $W = \{x_i | i \in \{1, \dots, n\}\}$ is linear dependent, then as W is finite and linear dependent there exists by [theorem: 11.117] a $u \in W$ and a $\{\alpha_w\}_{w \in W \setminus \{u\}} \subseteq F$ such that

$$u = \sum_{w \in W \setminus \{u\}} \alpha_w \cdot w \quad (11.159)$$

As $u \in W = \{x_i | i \in \{1, \dots, n\}\}$ there exists a $k \in \{1, \dots, n\}$ such that $x_k = u$. Let $w \in W \setminus \{u\}$ then $\exists i \in \{1, \dots, n\}$ such that $w = x_i$ and $i \neq k$ [for if $i = k$ then $x_i = x_k = u$] hence by [theorem: 11.291] $L(x_1, \dots, x_{k-1}, w, x_{k+1}, \dots, x_n) = 0$. So

$$\exists k \in \{1, \dots, n\} \text{ so that } u = x_k \text{ and } \forall w \in W \setminus \{u\} \quad L\left(x_1, \dots, \underbrace{w}_{k}, \dots, x_n\right) = 0 \quad (11.160)$$

Finally

$$\begin{aligned} L(x) &= L\left(x_1, \dots, \underbrace{x_k}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{eq: 11.159}]}{=} L\left(x_1, \dots, \underbrace{u}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{eq: 11.160}]}{=} L\left(x_1, \dots, \underbrace{\sum_{w \in W \setminus \{u\}} \alpha_w \cdot w}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{theorems: 11.266, 11.190}]}{=} \sum_{w \in W \setminus \{u\}} \alpha_w \cdot L\left(x_1, \dots, \underbrace{w}_{k}, \dots, x_n\right) \\ &\stackrel{[\text{eq: 11.158}]}{=} 0 \end{aligned}$$

$\square$

Theorem 11.293. Let X, Y be vector spaces over a field F with characteristic zero, $n \in \mathbb{N}$ and $L: X^n \rightarrow Y$ a skew-symmetric n -linear mapping. If X is finite dimensional with $\dim(X) < n$ then

$$\forall x \in X \text{ we have } L(x) = 0$$

in another words

$$L = C_0$$

Proof. For $n \in \mathbb{N}$ we have either:

$n = 1$. Then, as $\dim(X) < n$, it follows that $\dim(X) = 0$ so that by [example: 11.153] $X = \{0\}$. Hence if $L \in \text{Hom}^1(X; Y)$ we have for $x \in X^1$ that $x = (0)$ so that $L(x) = L(0) \stackrel{[\text{theorem: 11.259}]}{=} 0$ proving the theorem.

$1 < n$. Let $x = (x_1, \dots, x_n) \in X^n$ defined by the function $x: \{1, \dots, n\} \rightarrow X^n$ then we have two cases to consider:

x is injective. $x: \{1, \dots, n\} \rightarrow \{x_i | i \in \{1, \dots, n\}\}$ is a bijection, so we have by [definition: 11.84] that

$$\{x_i | i \in \{1, \dots, n\}\} = \{x_1, \dots, x_n\} \quad (11.161)$$

As X is finite dimensional with $\dim(X) < n$ there exist a $B \subseteq X$ with $\text{card}(B) < n$ and $X = \text{span}(B)$. Assume that $\{x_i | i \in \{1, \dots, n\}\}$ is linear independent then using Steinitz lemma [see lemma: 11.125] together with [eq: 11.161] we have that $n \leq \text{card}(B)$ which, as $\text{card}(B) < n$, leads to the contradiction $n < n$. Hence we must have that $\{x_i | i \in \{1, \dots, n\}\}$ is linear dependent. So as L is skew-symmetric it follows from [theorem: 11.292] that

$$L(x) = 0$$

x is not injective. Then there exist $i, j \in \{1, \dots, n\}$ with $x_i = x_j$ such that $i \neq j$. So by [theorem: 11.292] we have

$$L(x) = 0$$

So in all cases we have $\forall x \in X$ that $L(x) = 0$. □

Theorem 11.294. Let $n \in \mathbb{N}$, X, Y vector spaces over the field F , $E = \{e_1, \dots, e_n\} \subseteq X$ a basis for X and $L \in \text{Hom}^n(X; Y)$ such that L is skew-symmetric then $\forall x \in X$ we have:

$$L(x_1, \dots, x_n) = \sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot (\sigma L)(e_1, \dots, e_n)$$

where $\forall i \in \{1, \dots, n\}$ $\{\alpha_j^i\}_{j \in \{1, \dots, n\}}$ is the **unique family** satisfying $x_i = \sum_{j \in \{1, \dots, n\}} \alpha_j^i \cdot e_j$ [which exists by [theorem: 11.143]]

Proof. As $P_n = \{\sigma \in \{1, \dots, n\}^{\{1, \dots, n\}} | \sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\} \text{ is a bijection}\}$ we have

$$P_n \subseteq \{1, \dots, n\}^{\{1, \dots, n\}} \stackrel{[\text{definition: 6.93}]}{=} \{1, \dots, n\}^n.$$

Take $\sigma \in \{1, \dots, n\}^n \setminus P_n$ then we have that $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is not a bijection. For $\sigma(\{1, \dots, n\})$ we have two cases to consider:

$\sigma(\{1, \dots, n\}) = \{1, \dots, n\}$. Then σ can not be injective [otherwise $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is a bijection].

$\sigma(\{1, \dots, n\}) \neq \{1, \dots, n\}$. Then as $\sigma(\{1, \dots, n\}) \subseteq \{1, \dots, n\}$ we have $\sigma(\{1, \dots, n\}) \subset \{1, \dots, n\}$ so that by [theorem: 10.91]

$$\text{card}(\sigma(\{1, \dots, n\})) < \text{card}(\{1, \dots, n\}) = n$$

Assume that σ is injective then $\sigma: \{1, \dots, n\} \rightarrow \sigma(\{1, \dots, n\})$ is a bijection so that $\{1, \dots, n\} \approx \sigma(\{1, \dots, n\})$ or $\text{card}(\sigma(\{1, \dots, n\})) = n$ contradicting $\text{card}(\sigma(\{1, \dots, n\})) < n$. So σ is not injective.

As σ is not injective, there exists $i, j \in \{1, \dots, n\}$ with $i \neq j$ such that $\sigma(i) = \sigma(j)$ or $e_{\sigma(i)} = e_{\sigma(j)}$. Hence by [theorem: 11.291 (2)] we have $L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = 0$. To summarize

$$\text{If } \sigma \in \{1, \dots, n\} \setminus P_n \text{ then } L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = 0 \quad (11.162)$$

Now if $x \in X$ we have

$$\begin{aligned} L(x_1, \dots, x_n) &\stackrel{[\text{theorem: 11.272}]}{=} \sum_{\sigma \in \{1, \dots, n\}^n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \\ &= \sum_{\sigma \in \{1, \dots, n\}^n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \\ &= \sum_{\sigma \in \{1, \dots, n\}^n \setminus P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) + \\ &\quad \sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \\ &= \sum_{\sigma \in \{1, \dots, n\}^n \setminus P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot 0 + \sum_{k \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot \\ &\quad L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \\ &= \sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \\ &= \sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot (\sigma L)(e_1, \dots, e_n) \end{aligned}$$

□

Corollary 11.295. Let $n \in \mathbb{N}$, X, Y vector spaces over the field F , $E = \{e_1, \dots, e_n\} \subseteq X$ a basis for X and $L \in \text{Hom}^n(X; Y)$ a L a skew-symmetric n -linear mapping then $\forall x \in X$ we have

$$L(x_1, \dots, x_n) = \left(\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_1, \dots, e_n)$$

where $\forall i \in \{1, \dots, n\}$ $\{\alpha_j^i\}_{j \in \{1, \dots, n\}}$ is the **unique family** satisfying $x_i = \sum_{j \in \{1, \dots, n\}} \alpha_j^i \cdot e_j$ [which exists by [theorem: 11.143]].

Proof. By [theorem: 11.294] we have

$$\begin{aligned} L(x_1, \dots, x_n) &= \sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot (\sigma L)(e_1, \dots, e_n) \\ &\stackrel{L \text{ is skew-symmetric}}{=} \sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot (\text{sign}(\sigma) \cdot L(e_1, \dots, e_n)) \\ &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot L(e_1, \dots, e_n) \\ &\stackrel{[\text{theorem: 11.80}]}{=} \left(\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \right) \cdot L(e_1, \dots, e_n) \end{aligned}$$

□

The above theorem proves that a skew-symmetric n -linear depends only on its values at

$(e_1, \dots, e_n)$ as is expressed in the following corollary.

Corollary 11.296. *Let $n \in \mathbb{N}$, X, Y vector spaces over the field F , $E = \{e_1, \dots, e_n\} \subseteq X$ a basis for X then we have:*

1. *If $L_1, L_2 \in \text{Hom}^n(X; Y)$ are skew-symmetric n -linear mappings such that*

$$L_1(e_1, \dots, e_n) = L_2(e_1, \dots, e_n)$$

then

$$L_1 = L_2$$

2. *If $L \in \text{Hom}^n(X; Y)$ is a skew-symmetric n -linear mapping such that $L(e_1, \dots, e_n) = 0$ then*

$$L = C_0$$

Proof.

1. Using [theorem: 11.295] we have for $x \in X$

$$\begin{aligned} L_1(x) &= L_1(x_1, \dots, x_n) \\ &= \left(\sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot \text{sign}(\sigma) \right) \cdot L_1(e_1, \dots, e_n) \\ &= \left(\sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot \text{sign}(\sigma) \right) \cdot L_2(e_1, \dots, e_n) \\ &= L_2(x_1, \dots, x_n) \\ &= L_2(x) \end{aligned}$$

where $\forall i \in \{1, \dots, n\}$ $\{\alpha_j^i\}_{j \in \{1, \dots, n\}}$ is the **unique family** satisfying $x_i = \sum_{j \in \{1, \dots, n\}} \alpha_j^i \cdot e_j$ [which exist by [theorem: 11.143]]. Proving that

$$L_1 = L_2$$

2. Using [theorem: 11.295] we have for $x \in X$ that

$$\begin{aligned} L(x) &= L(x_1, \dots, x_n) \\ &= \left(\sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot \text{sign}(\sigma) \right) \cdot L_1(e_1, \dots, e_n) \\ &= \left(\sum_{\sigma \in P_n} \left(\prod_{j \in \{1, \dots, n\}} \alpha_j^{\sigma(j)} \right) \cdot \text{sign}(\sigma) \right) \cdot 0 \\ &= 0 \end{aligned}$$

where $\forall i \in \{1, \dots, n\}$ $\{\alpha_j^i\}_{j \in \{1, \dots, n\}}$ is the **unique family** satisfying $x_i = \sum_{j \in \{1, \dots, n\}} \alpha_j^i \cdot e_j$ [which exists by [theorem: 11.143]]. So that

$$L = C_0$$

□

We are now ready to define determinant functions.

Definition 11.297. (determinant function) *Let $n \in \mathbb{N}$, X a n -dimensional vector space over a field with characteristic zero then a skew-symmetric mapping $\Delta \in \text{Hom}^n(X; F)$ is called a **determinant function** or **determinant mapping**.*

Example 11.298. Let $n \in \mathbb{N}$, X a n -dimensional vector space over a field with characteristic zero then $C_0: X^n \rightarrow F$ defined by $C_0(x) = 0$ is a determinant function. This determinant function is called the trivial determinant function.

Proof. If $i \in \{1, \dots, n\}$ $u, v \in X_i$ then

$$\begin{aligned} C_0\left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n\right) &= \\ &= 0 \\ &= 0 + 0 \\ C_0\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + C_0\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) &= \end{aligned}$$

and

$$\begin{aligned} C_0\left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n\right) &= 0 \\ &= \alpha \cdot 0 \\ &= \alpha \cdot C_0\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) \end{aligned}$$

So

$$C_0 \in \text{Hom}^n(X; Y)$$

Further if $\sigma \in P_n$ then for $x \in X$ we have

$$(\sigma C_0)(x) = C_0(x \circ \sigma) = 0 = \text{sign}(\sigma) \cdot 0 = \text{sign}(\sigma) \cdot C_0(x) \quad \square$$

The following theorem shows that there exist non trivial determinant function.

Theorem 11.299. *Let $n \in \mathbb{N}$, X a n -dimensional vector space over a field F of characteristic zero and $E = \{e_1, \dots, e_n\} \subseteq X$ a basis for X then there exists a determinant function $\Delta \in \text{Hom}(X^n; F)$ such that $\Delta(e_1, \dots, e_n) = 1$. This proves that there exist a non trivial determinant function.*

Proof. Let $i \in \{1, \dots, n\}$ and define $f_i: X \rightarrow F$ by $f_i(x) = \alpha_i$ where $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ is the **unique** family such that $x = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i$ [see theorem: 11.143]. Then we have that

$$x = \sum_{i \in \{1, \dots, n\}} f_i(x) \cdot e_i$$

Further if $x, y \in X$ and $\alpha \in F$ then we have

$$\begin{aligned} x + y &\stackrel{\text{def } f_i}{=} \sum_{i \in \{1, \dots, n\}} f_i(x) \cdot e_i + \sum_{i \in \{1, \dots, n\}} f_i(y) \cdot e_i \\ &\stackrel{[\text{theorem: 11.38}]}{=} \sum_{i \in \{1, \dots, n\}} (f_i(x) \cdot e_i + f_i(y) \cdot e_i) \\ &= \sum_{i \in \{1, \dots, n\}} (f_i(x) + f_i(y)) \cdot e_i \\ x + y &\stackrel{\text{def } f_i}{=} \sum_{i \in \{1, \dots, n\}} f_i(x + y) \cdot e_i \end{aligned}$$

so that by uniqueness we have

$$\forall i \in \{1, \dots, n\} \text{ that } f_i(x + y) = f_i(x) + f_i(y) \quad (11.163)$$

Likewise

$$\begin{aligned} \alpha \cdot x &\stackrel{\text{def } f_i}{=} \alpha \cdot \sum_{i \in \{1, \dots, n\}} f_i(x) \cdot e_i \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha \cdot f_i(x) \cdot e_i \\ &= \sum_{i \in \{1, \dots, n\}} (\alpha \cdot f_i(x)) \cdot e_i \\ \alpha \cdot x &\stackrel{\text{def } f_i}{=} \sum_{i \in \{1, \dots, n\}} f_i(\alpha \cdot x) \end{aligned}$$

so by uniqueness we have

$$\forall i \in \{1, \dots, n\} f_i(\alpha \cdot x) = \alpha \cdot f_i(x) \quad (11.164)$$

Let $j \in \{1, \dots, n\}$ then we have by [theorem: 11.165] that $e_j = \sum_{i \in \{1, \dots, n\}} \delta_{i,j} \cdot e_i$, which as $e_j = \sum_{i \in \{1, \dots, n\}} f_i(e_j) \cdot e_i$, proving by uniqueness

$$\forall i \in \{1, \dots, n\} f_i(e_j) = \delta_{i,j} \quad (11.165)$$

Define

$$\Psi: X^n \rightarrow F \text{ by } \Psi(x_1, \dots, x_n) = \prod_{i \in \{1, \dots, n\}} f_i(x_i)$$

then we have for $i \in \{1, \dots, n\}$, $u, v \in X_i$ and $\alpha \in F$ that for

$$\begin{aligned} y &= \left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n \right) \\ z &= \left(x_1, \dots, \underbrace{u}_i, \dots, x_n \right) \\ r &= \left(x_1, \dots, \underbrace{v}_i, \dots, x_n \right) \\ s &= \left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n \right) \\ \Psi\left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n\right) &= \prod_{j \in \{1, \dots, n\}} y_j \\ &= \left(\prod_{j \in \{i\}} y_j \right) \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} y_j \\ &= y_i \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} y_j \\ &= (u+v) \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} y_j \\ &= u \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} y_j + v \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} y_j \\ &= z_i \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} z_j + r_i \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} r_j \\ &= \prod_{j \in \{1, \dots, n\}} z_j + \prod_{j \in \{1, \dots, n\}} r_j \\ &= L\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + L\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) \end{aligned}$$

and

$$\begin{aligned} \Psi\left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n\right) &= \prod_{j \in \{1, \dots, n\}} s_j \\ &= \left(\prod_{j \in \{i\}} s_j \right) \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} s_j \\ &= s_i \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} s_j \\ &= (\alpha \cdot u) \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} s_j \\ &= \alpha \cdot \left(u \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} s_j \right) \end{aligned}$$

$$\begin{aligned}
&= \alpha \cdot \left(z_i \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} z_j \right) \\
&= \alpha \cdot \prod_{j \in \{1, \dots, n\}} z_j \\
&= \alpha \cdot L \left(x_1, \dots, \underbrace{u}_i, \dots, x_n \right)
\end{aligned}$$

So we have that

$$\Psi \in \text{Hom}^n(X; F) \text{ or } \Psi \text{ is } n\text{-linear}$$

Using [theorem: 11.289] we create the following skew symmetric n -linear mapping

$$\Delta: X^n \rightarrow F \text{ by } \Delta = \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \sigma\Psi \quad (11.166)$$

which is by definition a determinant function. Let $\sigma \in P_n$ then we have either:

$\sigma = \text{Id}_{\{1, \dots, n\}}$. Then

$$\sigma\Psi(e_1, \dots, e_n) = \Psi(e_1, \dots, e_n) = \prod_{i \in \{1, \dots, n\}} f_i(e_i) \stackrel{\forall i \in I f_i(i) = \delta_{i,i} = 1}{=} 1$$

$\sigma \in P_n \setminus \{\text{Id}_{\{1, \dots, n\}}\}$. Then there exists a $i \in \{1, \dots, n\}$ such that $i \neq \text{Id}_{\{1, \dots, n\}}(i) = \sigma(i)$ so that $f_i(e_{\sigma(i)}) \stackrel{[\text{eq: 11.165}]}{=} \delta_{i, \sigma(i)} = 0$ then

$$\begin{aligned}
\sigma\Psi(e_1, \dots, e_n) &= \Psi(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \\
&= \prod_{j \in \{1, \dots, n\}} f_j(e_{\sigma(j)}) \\
&= \left(\prod_{j \in \{i\}} f_j(e_{\sigma(j)}) \right) \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} f_j(e_{\sigma(j)}) \\
&= f_i(e_{\sigma(i)}) \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} f_j(e_{\sigma(j)}) \\
&= 0 \cdot \prod_{j \in \{1, \dots, n\} \setminus \{i\}} f_j(e_{\sigma(j)}) \\
&= 0
\end{aligned}$$

So that

$$\sigma\Psi(e_1, \dots, e_n) = \begin{cases} 1 & \text{if } \sigma = \text{Id}_{\{1, \dots, n\}} \\ 0 & \text{if } \sigma \in P_n \setminus \{\text{Id}_{\{1, \dots, n\}}\} \end{cases} \quad (11.167)$$

so that

$$\begin{aligned}
&\Delta(e_1, \dots, e_n) \stackrel{[\text{eq: 11.166}]}{=} \\
&\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot (\sigma\Psi)(e_1, \dots, e_n) \stackrel{[\text{theorem: 11.43}]}{=} \\
&\sum_{\sigma \in P_n \setminus \{\text{Id}_{\{1, \dots, n\}}\}} \text{sign}(\sigma) \cdot (\sigma\Psi)(e_1, \dots, e_n) + \sum_{\sigma \in \{\text{Id}_{\{1, \dots, n\}}\}} \text{sign}(\sigma) \cdot (\sigma\Psi)(e_1, \dots, e_n) \stackrel{[\text{eq: 11.167}]}{=} \\
&\sum_{\sigma \in P_n \setminus \{\text{Id}_{\{1, \dots, n\}}\}} \text{sign}(\sigma) \cdot 0 + \sum_{\sigma \in \{\text{Id}_{\{1, \dots, n\}}\}} \text{sign}(\sigma) \cdot (\sigma\Psi)(e_1, \dots, e_n) = \\
&\sum_{\sigma \in \{\text{Id}_{\{1, \dots, n\}}\}} \text{sign}(\sigma) \cdot (\sigma\Psi)(e_1, \dots, e_n) \stackrel{[\text{theorem: 11.34}]}{=} \\
&\text{sign}(\text{Id}_{\{1, \dots, n\}}) \cdot (\text{Id}_{\{1, \dots, n\}}\Psi)(e_1, \dots, e_n) \stackrel{[\text{eq: 11.167}]}{=} \\
&1 \cdot 1 \stackrel{[\text{eq: 11.167}]}{=} \\
&1
\end{aligned}$$

proving that

$$\Delta(e_1, \dots, e_n) = 1 \quad \square$$

We show now that every skew-symmetric n -linear function can be written as the scalar product of a determinant function and a vector in the target domain.

Theorem 11.300. *Let $n \in \mathbb{N}$, X a vector space over a field F of characteristic zero, Y a vector space over the same field F and Δ a non zero determinant function. Then for every skew-symmetric $L \in \text{Hom}^n(X; Y)$ there exist a **unique** $y \in Y$ such that*

$$\forall x \in X \text{ we have } L(x) = \Delta(x) \cdot y$$

Proof. Let $E = \{e_1, \dots, e_n\} \subseteq X$ be a basis of X . By [theorem: 11.295]

$$\Delta(x_1, \dots, x_n) = \left(\sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \prod_{j \in \{1, \dots, n\}} \alpha_j^{\alpha_j} \right) \cdot \Delta(e_1, \dots, e_n)$$

so as $\Delta \neq C_0$ we must have that

$$\Delta(e_1, \dots, e_n) \neq 0$$

Define

$$\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \setminus \{0\} \text{ by } \alpha_i = \begin{cases} \Delta(e_1, \dots, e_n)^{-1} & \text{if } i = 1 \\ 1 & \text{if } i \in \{2, \dots, n\} \end{cases}$$

then by [theorem: 11.144] we have that

$$h: \{1, \dots, n\} \rightarrow \{\alpha_i \cdot e_i \mid i \in \{1, \dots, n\}\} \text{ defined by } h(i) = \alpha_i \cdot e_i = \begin{cases} \Delta(e_1, \dots, e_n)^{-1} \cdot e_1 & \text{if } i = 1 \\ e_i & \text{if } i \in \{2, \dots, n\} \end{cases}$$

is a bijection defining the finite set

$$H = \{h_1, \dots, h_n\} = \{\Delta(e_1, \dots, e_n)^{-1} \cdot e_1, e_2, \dots, e_n\}$$

and that

$$H = \{\Delta(e_1, \dots, e_n)^{-1} \cdot e_1, e_2, \dots, e_n\} \text{ is a basis of } X \quad (11.168)$$

Now

$$\begin{aligned} \Delta(h_1, \dots, h_n) &= \Delta(\Delta(e_1, \dots, e_n)^{-1} \cdot e_1, e_2, \dots, e_n) \\ &\stackrel{\Delta \in \text{Hom}^n(X; Y)}{=} \Delta(e_1, \dots, e_n)^{-1} \cdot \Delta(e_1, \dots, e_n) \\ &= 1 \end{aligned} \quad (11.169)$$

Set

$$y = L(h_1, \dots, h_n) \quad (11.170)$$

and define

$$K: X^n \rightarrow Y \text{ by } K(x_1, \dots, x_n) = \Delta(x_1, \dots, x_n) \cdot y \quad (11.171)$$

Then for $i \in \{1, \dots, n\}$, $u, v \in X$ we have that

$$\begin{aligned} K\left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n\right) &= \\ \Delta\left(x_1, \dots, \underbrace{u+v}_i, \dots, x_n\right) \cdot y &= \\ \left(\Delta\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + \Delta\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right)\right) \cdot y &= \\ \Delta\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) \cdot y + \Delta\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) \cdot y &= \\ K\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) + K\left(x_1, \dots, \underbrace{v}_i, \dots, x_n\right) \end{aligned}$$

and

$$\begin{aligned}
 K\left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n\right) &= \Delta\left(x_1, \dots, \underbrace{\alpha \cdot u}_i, \dots, x_n\right) \cdot y \\
 &= \left(\alpha \cdot \Delta\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right)\right) \cdot y \\
 &= \alpha \cdot \left(\Delta\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right) \cdot y\right) \\
 &= \alpha \cdot K\left(x_1, \dots, \underbrace{u}_i, \dots, x_n\right)
 \end{aligned}$$

which proves that

$$K \in \text{Hom}^n(X; Y)$$

Further if $\sigma \in P_n$ then if $x \in X^n$ we have

$$(\sigma K)(x) = K(x \circ \sigma) = \Delta(x \circ \sigma) \cdot y = (\text{sign}(\sigma) \cdot \Delta(x)) \cdot y = \text{sign}(\sigma) \cdot (\Delta(x) \cdot y) = \text{sign}(\sigma) \cdot K(x)$$

proving that

K is a skew-symmetric linear function

As $K(h_1, \dots, h_n) = \Delta(h_1, \dots, h_n) \cdot y$ [eq: 11.169] $y = L(h_1, \dots, h_n)$ we have, as $\{h_1, \dots, h_n\}$ is a basis, by [theorem: 11.296] that $L = K$ or using the definition of K [see eq: 11.171] we have

$$\forall x \in X \text{ we have } L(x) = \Delta(x) \cdot y$$

Now for uniqueness, assume that there exists a $y' \in Y$ such that $\forall x \in X$ we have $L(x) = \Delta(x) \cdot y'$. Then $\Delta(f_1, \dots, f_n) \cdot y = \Delta(f_1, \dots, f_n) \cdot y'$ [eq: 11.169] $\Rightarrow 1 \cdot y = 1 \cdot y'$ proving that $y = y'$. $\square$

Corollary 11.301. *Let $n \in \mathbb{N}$, X a n -dimensional vector space over a field F of characteristic zero. Let Δ be a non zero determinant function then if Δ' is another determinant function there exist a $\alpha \in F$ such that*

$$\Delta' = \alpha \cdot \Delta$$

Proof. If Δ' is another determinant function then we have either:

$\Delta = C_0$. Take $\alpha = 0$ then

$$\Delta' = C_0 = 0 \cdot \Delta$$

$\Delta \neq C_0$. As Δ is a determinant function, hence a skew-symmetric n -linear mapping with range F , we have by the previous theorem [theorem: 11.300] that there exist a $\alpha \in F$ such that

$$\Delta' = \Delta \cdot \alpha = \alpha \cdot \Delta \quad \square$$

Definition 11.302. *Let $n \in \mathbb{N}$, X a set, $y \in X$, $i \in \{1, \dots, n\}$ and $x \in X^n$ then*

$$(y, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in X^n$$

is defined by

$$(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k = \begin{cases} y & \text{if } k = 1 \\ x_{k-1} & \text{if } k \in \{2, \dots, i\} \\ x_k & \text{if } k \in \{i+1, \dots, n\} \end{cases}$$

Lemma 11.303. *Let $n \in \mathbb{N}$, X a set, $i \in \{1, \dots, n\}$ and $x \in X^n$ then*

$$(x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = x \circ (i \rightsquigarrow_n 1)$$

Proof. For $i \in \{1, \dots, n\}$ we have either:

$i = 1$. Then by [definition: 11.252] we have that $(1 \rightsquigarrow_n i) = (1 \rightsquigarrow_n 1) = \text{Id}_{\{1, \dots, n\}}$. Then we have for $j \in \{1, \dots, n\}$

$$\begin{aligned}
 (x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_j &= \begin{cases} x_i & \text{if } j = 1 \\ x_{i-1} & \text{if } j \in \{2, \dots, i\} \\ x_k & \text{if } j \in \{i+1, \dots, n\} \end{cases} \\
 &\stackrel{i=1}{=} \begin{cases} x_1 & \text{if } j = 1 \\ x_{i-1} & \text{if } j \in \{2, \dots, 1\} \\ x_k & \text{if } j \in \{2, \dots, n\} \end{cases} \\
 &= \begin{cases} x_1 & \text{if } j = 1 \\ x_{i-1} & \text{if } j \in \{2, \dots, 1\} \\ x_k & \text{if } j \in \{2, \dots, n\} \end{cases} \\
 &\stackrel{\{2, \dots, 1\} = \emptyset}{=} \begin{cases} x_1 & \text{if } j = 1 \\ x_k & \text{if } j \in \{2, \dots, n\} \end{cases} \\
 &= x_j \\
 &= (x \circ \text{Id}_{\{1, \dots, n\}})_j \\
 &= (x \circ (i \rightsquigarrow_n 1))(j)
 \end{aligned}$$

proving that

$$(x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = x \circ (i \rightsquigarrow_n 1)$$

$i \in \{2, \dots, n\}$. Then $1 < i$ and for $j \in \{1, \dots, n\}$ we have either:

$j = 1$. Then

$$\begin{aligned}
 (x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_j &= (x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_1 \\
 &= x_i \\
 &= x(i) \\
 &\stackrel{j=1 \wedge [\text{definition: 11.252(3)}]}{=} x((i \rightsquigarrow_n 1)(j)) \\
 &= (x \circ (i \rightsquigarrow_n 1))(j)
 \end{aligned}$$

$j \in \{2, \dots, i\}$. Then

$$\begin{aligned}
 (x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_j &= x_{j-1} \\
 &= x(j-1) \\
 &\stackrel{1 < j \leq i \wedge [\text{definition: 11.252(3)}]}{=} x((i \rightsquigarrow_n 1)(j)) \\
 &= (x \circ (i \rightsquigarrow_n 1))(j)
 \end{aligned}$$

$j \in \{i+1, \dots, n\}$. Then

$$\begin{aligned}
 (x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_j &= x_j \\
 &= x(j) \\
 &\stackrel{i < i+1 \leq j \leq n \wedge [\text{definition: 11.252(3)}]}{=} x((i \rightsquigarrow_n 1)(j)) \\
 &= (x \circ (i \rightsquigarrow_n 1))(j)
 \end{aligned}$$

proving that

$$(x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = x \circ (i \rightsquigarrow_n 1)$$

□

Definition 11.304. Let $n \in \mathbb{N}$, X a n -dimensional vector space over a field F with characteristic zero and $\Delta \in \text{Hom}^n(X; F)$ a determinant function then we define:

$$\underline{\Delta}: X \times X^n \rightarrow X \text{ where } \underline{\Delta}(y, (x_1, \dots, x_n)) = \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i$$

where $(y, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ is defined by [see the previous definition: 11.302]

$$(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k = \begin{cases} y & \text{if } k = 1 \\ x_{k-1} & \text{if } k \in \{2, \dots, i\} \\ x_k & \text{if } k \in \{i+1, \dots, n\} \end{cases}$$

To calculate $\underline{\Delta}(y, (x_1, \dots, x_n))$ we need the following lemma.

Lemma 11.305. Let $n \in \mathbb{N}$, X a vector space over a field F , $i, j \in \{1, \dots, n\}$, $y \in X$ and $x = (x_1, \dots, x_n) \in X^n$ then for $z = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ we have:

1. If $j+1 \leq i$ then $\left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n\right) \circ (j+1 \rightsquigarrow_n i) = (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)$
2. If $i+1 \leq j$ then $\left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n\right) \circ (j \rightsquigarrow_n i+1) = (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)$

Proof. The prove is simple but rather elaborated because we have to check so many cases.

1. Let $k \in \{1, \dots, n\}$ and take $a_k = \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n\right) \circ ((j+1 \rightsquigarrow_n i)(k))$ then using the definition of $\rightsquigarrow_n$ [see definition: 11.252] we must look at the following cases for $j+1, i$

$j+1 = i$. Then by [definition: 11.252(1)] we have that $(j+1 \rightsquigarrow_n i) = \text{Id}_{\{1, \dots, n\}}$ so that

$$a_k = \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n\right)(k) \quad (11.172)$$

now for $k \in \{1, \dots, n\}$ we have to look at the following cases:

$k = 1$. Then

$$\begin{aligned} a_k & \stackrel{[\text{eq: 11.172}]}{=} \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n\right)_k \\ & \stackrel{k=1 < j+1}{=} z_k \\ & = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]\wedge k=1}{=} y \\ & \stackrel{[\text{definition: 11.302}]\wedge k=1}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

$1 < k < j$. Then

$$\begin{aligned} a_k & \stackrel{[\text{eq: 11.172}]}{=} \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n\right)_k \\ & \stackrel{k < j < j+1}{=} z_k \\ & = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]\wedge 1 < k < j < j+1=i}{=} x_{k-1} \\ & \stackrel{[\text{definition: 11.302}]\wedge 1 < k < j}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

$1 < k = j + 1$. Then

$$\begin{aligned}
 a_k & \stackrel{[\text{eq: 11.172}]}{=} \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_k \\
 & \stackrel{k=j+1}{=} x_i \\
 & \stackrel{k=j+1=i}{=} x_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge k=j+1}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$j + 1 < k \leq n$. Then

$$\begin{aligned}
 a_k & \stackrel{[\text{eq: 11.172}]}{=} \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_k \\
 & \stackrel{j+1 < k}{=} z_k \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge i=j+1 < k \Rightarrow i+1 \leq k}{=} x_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge j+1 < k}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$j + 1 < i$. Then [definition: 11.252(3)] applies and we have to look at the following cases:

$1 = k$. Then $1 = k < j + 1$ so that $(j + 1 \rightsquigarrow_n i)(k) \stackrel{[\text{definition: 11.252(2)}]}{=} k$ hence

$$\begin{aligned}
 a_k & = \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_k \\
 & \stackrel{k=1 \leq j < j+1}{=} z_k \\
 & \stackrel{k=1}{=} z_1 \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_1 \\
 & \stackrel{=}{=} y \\
 & \stackrel{[\text{definition: 11.302}]\wedge k=1}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$1 < k < j + 1$. Then $(j + 1 \rightsquigarrow_n i)(k) \stackrel{[\text{definition: 11.252(2)}]}{=} k$ hence

$$\begin{aligned}
 a_k & = \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_k \\
 & \stackrel{k=1 \leq j < j+1}{=} z_k \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k < j+1 \leq i}{=} x_{k-1} \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k < j+1 \Rightarrow 1 < k \leq j}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$j + 1 \leq k < i$. Then $(j + 1 \rightsquigarrow_n i)(k) \stackrel{[\text{definition: 11.252(2)}]}{=} k + 1$ hence

$$\begin{aligned}
 a_k & = \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_{k+1} \\
 & \stackrel{j+1 \leq k < k+1}{=} z_{k+1} \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{k+1} \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k+1 \leq i}{=} x_{(k+1)-1} \\
 & \stackrel{=}{=} x_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge j+1 \leq k}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$k = i$. Then $(j+1 \rightsquigarrow_n i)(k) \stackrel{[\text{definition: 11.252(2)}]}{=} j+1$ hence

$$\begin{aligned} a_k &= \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_{j+1} \\ &= x_i \\ &\stackrel{i=k}{=} x_k \\ &\stackrel{[\text{definition: 11.302}]}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

$i < k \leq n$. Then $(j+1 \rightsquigarrow_n i)(k) \stackrel{[\text{definition: 11.252(2)}]}{=} k$ hence

$$\begin{aligned} a_k &= \left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right)_k \\ &= z_k \\ &= (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\ &\stackrel{[\text{definition: 11.302}]}{=} x_k \\ &\stackrel{[\text{definition: 11.302}]}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

so in all cases we have $a_k = (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k$ proving that

$$\left(z_1, \dots, \underbrace{x_i}_{j+1}, \dots, z_n \right) \circ (j+1 \rightsquigarrow_n i) = (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)$$

2. Let $k \in \{1, \dots, n\}$ and take $b_k = \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right) ((j \rightsquigarrow_n i+1)(k))$ then using the definition of $\rightsquigarrow_n$ [see definition: 11.252] we must look at the following cases for $j, i+1$:

$i+1 = j$. Then by [definition: 11.252] we have that $(j \rightsquigarrow_n i+1) = \text{Id}_{\{1, \dots, n\}}$ so that

$$b_k = \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \quad (11.173)$$

Now for $k \in \{1, \dots, n\}$ we have to look at the following cases:

$k = 1$. Then

$$\begin{aligned} b_k &\stackrel{[\text{eq: 11.173}]}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\ &\stackrel{k=1 < i+1=j}{=} z_1 \\ &= (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_1 \\ &\stackrel{[\text{definition: 11.302}]}{=} y \\ &\stackrel{[\text{definition: 11.302}]}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

$1 < k < i$. Then

$$\begin{aligned} b_k &\stackrel{[\text{eq: 11.173}]}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\ &\stackrel{k < i < i+1=j}{=} z_k \\ &= (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\ &\stackrel{[\text{definition: 11.302}]}{=} x_{k-1} \\ &\stackrel{[\text{definition: 11.302}]}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

$1 < k = i$. Then

$$\begin{aligned}
 b_k & \stackrel{[\text{eq: 11.173}]}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\
 & \stackrel{k=i < i+1=j}{=} z_k \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k=i}{=} x_{k-1} \\
 & \stackrel{[\text{definition: 11.302}]\wedge k=i < i+1}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$k = i + 1$. Then

$$\begin{aligned}
 b_k & \stackrel{[\text{eq: 11.173}]}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\
 & \stackrel{k=i+1=j}{=} x_i \\
 & \stackrel{k=i+1}{=} x_{k-1} \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k=i+1=j}{=} (y, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k
 \end{aligned}$$

$i + 1 < k \leq n$. Then

$$\begin{aligned}
 b_k & \stackrel{[\text{eq: 11.173}]}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\
 & \stackrel{j=i+1 < k}{=} z_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge j=i+1 < k \Rightarrow j+1 \leq k}{=} (y, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k
 \end{aligned}$$

$i + 1 < j$. Then by [definition: 11.252(3)] we have look at the following cases:

$k = 1$. Then $(j \rightsquigarrow_n i + 1)(k) \stackrel{[\text{definition: 11.252(3)}]}{=} k$ hence

$$\begin{aligned}
 b_k & \stackrel{=}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\
 & \stackrel{k=1 < i+1 < j}{=} z_k \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge k=1}{=} y \\
 & \stackrel{[\text{definition: 11.302}]\wedge k=1}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$1 < k < i + 1$. Then $(j \rightsquigarrow_n i + 1)(k) \stackrel{[\text{definition: 11.252(3)}]}{=} k$ hence

$$\begin{aligned}
 b_k & \stackrel{=}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_k \\
 & \stackrel{k < i+1 < j}{=} z_k \\
 & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k < i+1 \Rightarrow 1 < k \leq i}{=} x_{k-1} \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k < i+1 < j}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$1 < k = i + 1$. Then $(j \rightsquigarrow_n i + 1)(k) \stackrel{[\text{definition: 11.252(3)}]}{=} j$ hence

$$\begin{aligned}
 b_k & \stackrel{11.252}{=} \left(z_1, \dots, \underbrace{x_i}_j, \dots, z_n \right)_j \\
 & \stackrel{=}{=} x_i \\
 & \stackrel{k=i+1}{=} x_{k-1} \\
 & \stackrel{[\text{definition: 11.302}]\wedge 1 < k=i+1 < j}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k
 \end{aligned}$$

$i + 1 < k \leq j$. Then $(j \rightsquigarrow_n i + 1)(k) \stackrel{[\text{definition: 11.252(3)}]}{=} k - 1$ hence

$$\begin{aligned} b_k & \stackrel{11.252}{=} (z_1, \dots, z_{j-1}, x_i, z_{j+1}, \dots, z_n)_{k-1} \\ & \stackrel{k \leq j \Rightarrow k-1 < j}{=} z_{k-1} \\ & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{k-1} \\ & \stackrel{[\text{definition: 11.302}]}{=} x_{k-1} \\ & \stackrel{[\text{definition: 11.302}]}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

$j < k \leq n$. Then $(j \rightsquigarrow_n i + 1)(k) \stackrel{[\text{definition: 11.252(3)}]}{=} k$ hence

$$\begin{aligned} b_k & \stackrel{11.252}{=} \left(z_1, \dots, \underbrace{x_i}_{j}, \dots, z_n \right)_k \\ & \stackrel{j < k}{=} z_k \\ & \stackrel{=}{=} (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]}{=} x_k \\ & \stackrel{[\text{definition: 11.302}]}{=} (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k \end{aligned}$$

So in all cases we have $b_k = (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n)_k$ proving that

$$\left(z_1, \dots, \underbrace{x_i}_{j}, \dots, z_n \right) \circ (j \rightsquigarrow_n i + 1) = (y, x_1, \dots, x_{j-1}, x_j, \dots, x_n) \quad \square$$

Theorem 11.306. Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field of characteristic zero with $\dim(X) = n$, $y \in X$, $(x_1, \dots, x_n) \in X^n$ and $\Delta \in \text{Hom}^n(X; F)$ a determinant function then

$$\underline{\Delta}(y, (x_1, \dots, x_n)) = \Delta(x_1, \dots, x_n) \cdot y$$

or using the definition of $\underline{\Delta}: X \times X^n \rightarrow X$ we have [see definition: 11.304]

$$\sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i = \Delta(x_1, \dots, x_n) \cdot y$$

Proof. Let $x = (x_1, \dots, x_n) \in X^n$ then we have for $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ the following cases to consider:

$\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent. Then by [theorem: 11.291] we have that

$$\Delta(x_1, \dots, x_n) = 0 \quad (11.174)$$

So if we prove that $\underline{\Delta}(y, (x_1, \dots, x_n)) = 0$ we are done for this case. Consider the following cases for $n \in \mathbb{N}$.

$n = 1$. As $\{x_i\}_{i \in \{1\}}$ is linear dependent there exists a $\{\alpha_i\}_{i \in \{1\}}$ not all zero [so $\alpha_1 \neq 0$] such that $0 = \sum_{i \in \{1\}} \alpha_i \cdot x_i = \alpha_1 \cdot x_1$. So, as $\alpha_1 \neq 0$, we have that $x_1 = 0$, hence

$$\Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_1 = \Delta(y) \cdot 0 = 0$$

So

$$\begin{aligned} \underline{\Delta}(y, (x_1, \dots, x_n)) &= \sum_{i \in \{1, \dots, 1\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\ &= (-1)^0 \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_1 \\ &= 0 \end{aligned}$$

$n = 2$. As $\{x_i\}_{i \in \{1,2\}}$ is linear dependent it follows by [theorem: 11.128] that there exists a $k \in \{1, 2\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, 2\} \setminus \{k\}}$ such that $x_k = \sum_{i \in \{1,2\} \setminus \{k\}} \alpha_i \cdot x_i$. For $k \in \{1, 2\}$ we have either:

$k = 1$. Then $x_1 = \sum_{i \in \{2\}} \alpha_i \cdot x_i = \alpha_2 \cdot x_2$ so that

$$\begin{aligned} \underline{\Delta}(y, (x_1, \dots, x_n)) &= \sum_{i \in \{1, \dots, 2\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\ &= (-1)^{1-1} \cdot \Delta(y, x_2) \cdot x_1 + (-1)^{2-1} \cdot \Delta(y, x_1) \cdot x_2 \\ &= \Delta(y, x_2) \cdot x_1 - \Delta(y, x_1) \cdot x_2 \\ &= \Delta(y, x_2) \cdot (\alpha_2 \cdot x_2) - \Delta(y, \alpha_2 \cdot x_2) \cdot x_2 \\ &= \alpha_2 \cdot (\Delta(y, x_2) \cdot x_2) - \alpha_2 \cdot \Delta(y, x_2) \cdot x_2 \\ &= 0 \end{aligned}$$

$k = 2$. Then $x_2 = \sum_{i \in \{1\}} \alpha_i \cdot x_i = \alpha_1 \cdot x_1$ so that

$$\begin{aligned} \underline{\Delta}(y, (x_1, \dots, x_n)) &= \sum_{i \in \{1, \dots, 2\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\ &= (-1)^{1-1} \cdot \Delta(y, x_2) \cdot x_1 + (-1)^{2-1} \cdot \Delta(y, x_1) \cdot x_2 \\ &= \Delta(y, x_2) \cdot x_1 - \Delta(y, x_1) \cdot x_2 \\ &= \Delta(y, \alpha_1 \cdot x_1) \cdot x_1 - \Delta(y, x_1) \cdot (\alpha_1 \cdot x_1) \\ &= \alpha_1 \cdot (\Delta(y, x_1) \cdot x_1) - \alpha_1 \cdot \Delta(y, x_1) \cdot x_1 \\ &= 0 \end{aligned}$$

$2 < n$. As $\{x_i\}_{i \in \{1, \dots, n\}}$ is linear dependent there exists by [theorem: 11.128] a $k \in \{1, \dots, n\}$ and a $\{\alpha_i\}_{i \in \{1, \dots, n\} \setminus \{k\}} \subseteq F$ such that

$$x_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i \quad (11.175)$$

So

$$\begin{aligned} \underline{\Delta}(y, (x_1, \dots, x_n)) &= \sum_{i \in \{1, \dots, n\}} ((-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)) \cdot x_i \\ &= A + B \end{aligned} \quad (11.176)$$

where

$$A = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \quad (11.177)$$

$$B = \sum_{i \in \{k\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \quad (11.178)$$

Now

$$\begin{aligned} B &= \sum_{i \in \{k\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\ &= (-1)^{k-1} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot x_k \\ &= (-1) \cdot ((-1)^k \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot x_k) \end{aligned} \quad (11.179)$$

For $i \in \{1, \dots, n\} \setminus \{k\}$ we have the following cases:

$k < i$. Let

$$z = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

then

$$\begin{aligned} z_{k+1} &= (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{k+1} \\ &\stackrel{[\text{definition: 11.302}]}{=} x_{(k+1)-1} \\ &= x_k \\ &\stackrel{[\text{eq: 11.175}]}{=} \sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot x_j \end{aligned}$$

So that

$$\begin{aligned}
 \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) &= \\
 &= \Delta(z) = \\
 \Delta\left(z_1, \dots, \underbrace{\sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot x_j}_{k+1}, \dots, z_n\right) &= \\
 \sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot \Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) &=
 \end{aligned} \tag{11.180}$$

Now for $j \in \{1, \dots, n\} \setminus \{k\}$ consider the following cases for i, j :

$j < i$. Then we have

$$\begin{aligned}
 \left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_{j+1} & \stackrel{j \neq k \Rightarrow j+1 \neq k+1}{=} \\
 & \stackrel{z_{j+1}}{=} \\
 (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{j+1} & \stackrel{[\text{definition: 11.302}]}{=} \wedge 1 < j+1 \leq i \\
 & \stackrel{x_j}{=} \\
 \left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_{k+1} &
 \end{aligned}$$

So we have $j \neq k \Rightarrow j+1 \neq k+1$ and

$$\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_{k+1} = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{j+1}$$

which by [theorem: 11.291] proves that

$$\Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) = 0 \tag{11.181}$$

$i < j$. Then we have

$$\begin{aligned}
 \left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_j & \stackrel{k < i \Rightarrow k+1 \leq i < j \Rightarrow k+1 \neq j}{=} \\
 & \stackrel{z_j}{=} \\
 (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_j & \stackrel{[\text{definition: 11.302}]}{=} \wedge i < j \Rightarrow i+1 \leq j \\
 & \stackrel{x_j}{=} \\
 \left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_{k+1} &
 \end{aligned}$$

So we have $k < i \Rightarrow k+1 \leq i < j \Rightarrow k+1 \neq j$ and

$$\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_j = \left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right)_{k+1}$$

which by [theorem: 11.291] proves that

$$\Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) = 0 \tag{11.182}$$

$i = j$. As $k < i \Rightarrow k+1 \leq i$ and $z = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ we have by [theorem: 11.305] that

$$\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right) \circ (k+1 \overset{n}{\rightsquigarrow} i) = (y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n)$$

Hence

$$\begin{aligned}
& \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) = \\
& \Delta\left(\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right) \circ (k+1 \rightsquigarrow_n i)\right) = \\
& ((k+1 \rightsquigarrow_n i) \Delta)\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right) \stackrel{[\text{theorem: 11.255}]}{=} \\
& (-1)^{|k+1-i|} \cdot \Delta\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right) \stackrel{k < i \Rightarrow k+1 \leq i}{=} \\
& (-1)^{i-k-1} \cdot \Delta\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right)
\end{aligned}$$

which by [ref: 11.281] gives

$$\Delta\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right) = (-1)^{i-k-1} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \quad (11.183)$$

Further, using the fact that $\{1, \dots, n\} \setminus \{k\}$ is the disjoint union of

$$\{1, \dots, i-1\} \setminus \{k\}, \{i\} \setminus \{k\} \text{ and } \{i+1, \dots, n\} \setminus \{k\}$$

we have

$$\sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot \Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) \stackrel{[\text{theorem: 11.43}]}{=} A_1 + A_2 + A_3 \quad (11.184)$$

where

$$\begin{aligned}
A_1 &= \sum_{j \in \{1, \dots, i-1\} \setminus \{k\}} \alpha_j \cdot \Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) \\
&\stackrel{[\text{eq: 11.181}]}{=} 0 \\
A_2 &= \sum_{j \in \{i\} \setminus \{k\}} \alpha_j \cdot \Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) \\
&\stackrel{i \neq k \Rightarrow \{i\} \setminus \{k\} = \{i\}}{=} \alpha_i \cdot \Delta\left(z_1, \dots, \underbrace{x_i}_{k+1}, \dots, z_n\right) \\
&\stackrel{[\text{eq: 11.183}]}{=} (\alpha_i \cdot (-1)^{i-k-1} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n)) \\
A_3 &= \sum_{j \in \{i+1, \dots, n\} \setminus \{k\}} \alpha_j \cdot \Delta\left(z_1, \dots, \underbrace{x_j}_{k+1}, \dots, z_n\right) \\
&\stackrel{[\text{eq: 11.182}]}{=} 0
\end{aligned}$$

proving by [eqs 11.180, 11.184] and the above that that

$$\begin{aligned}
& \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = \\
& \alpha_i \cdot (-1)^{i-k-1} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \quad (11.185)
\end{aligned}$$

$i < k$. Let

$$z = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

then

$$\begin{aligned}
z_k &= (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \\
&\stackrel{[\text{definition: 11.302}]}{=} x_k \wedge i < k \Rightarrow i+1 \leq k \\
&\stackrel{[\text{eq: 11.175}]}{=} \sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot x_j
\end{aligned}$$

So that

$$\begin{aligned}
 \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) &= \\
 &= \Delta(z) = \\
 \Delta\left(z_1, \dots, \underbrace{\sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot x_j}_{k}, \dots, z_n\right) &= \\
 \sum_{j \in \{1, \dots, n\} \setminus \{k\}} \alpha_j \cdot \Delta\left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right) &=
 \end{aligned} \tag{11.186}$$

Now for $j \in \{1, \dots, n\} \setminus \{k\}$ we have the following cases:

$j < i$. Then we have:

$$\begin{aligned}
 \left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_{j+1} & \stackrel{j < i \Rightarrow j+1 \leq i < k}{=} \\
 & \stackrel{z_{j+1}}{=} \\
 (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{j+1} & \stackrel{[\text{definition: 11.302}]}{=} \wedge j < i \Rightarrow 1 < j+1 \leq i \\
 & \stackrel{x_{(j+1)-1}}{=} \\
 & \stackrel{x_j}{=} \\
 & \stackrel{j \neq k}{=} \\
 \left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_k &
 \end{aligned}$$

So we have $j < i \Rightarrow j+1 \leq i < k \Rightarrow j+1 \neq k$ and

$$\left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_{j+1} = \left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_k$$

which by [theorem: 11.291] proves that

$$\Delta\left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right) = 0 \tag{11.187}$$

$i < j$. Then we have:

$$\begin{aligned}
 \left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_j & \stackrel{j \neq k}{=} \\
 & \stackrel{z_j}{=} \\
 (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) & \stackrel{[\text{definition: 11.302}]}{=} \wedge i < j \Rightarrow i+1 \leq j \\
 & \stackrel{x_j}{=} \\
 \left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_k &
 \end{aligned}$$

So we have $j \neq k$ and

$$\left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_j = \left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right)_k$$

which by [theorem: 11.291] proves that

$$\Delta\left(z_1, \dots, \underbrace{x_j}_{k}, \dots, z_n\right) = 0 \tag{11.188}$$

$i = j$. As $i < k \Rightarrow i+1 \leq k$ and $z = (y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ we have by [theorem: 11.305] that

$$\left(z_1, \dots, \underbrace{x_i}_{k}, \dots, z_n\right) \circ (k \overset{n}{\rightsquigarrow} i+1) = (y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n)$$

where

$$\begin{aligned}
B_1 &= \sum_{i \in \{1, \dots, k-1\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\
&\stackrel{[\text{eq: 11.191}]}{=} \sum_{i \in \{1, \dots, k-1\}} (-1)^{i-1} \cdot \alpha_i \cdot (-1)^{k-i-1} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot x_i \\
&= \sum_{i \in \{1, \dots, k-1\}} (-1)^{k-2} \cdot \alpha_i \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot x_i \\
&= (-1)^{k-2} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i \\
&= (-1)^2 \cdot (-1)^{k-2} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i \\
&= (-1)^k \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i \\
B_2 &= \sum_{i \in \{k+1, \dots, n\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\
&\stackrel{[\text{eq: 11.183}]}{=} \sum_{i \in \{k+1, \dots, n\}} (-1)^{i-1} \cdot \alpha_i \cdot (-1)^{i-k-1} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot x_i \\
&= \sum_{i \in \{k+1, \dots, n\}} (-1)^{2 \cdot i - k - 2} \cdot \alpha_i \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot x_i \\
&= (-1)^{2 \cdot i - k - 2} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot x_i \\
&= (-1)^{2 \cdot k} \cdot (-1)^{2 \cdot i - k - 2} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot x_i \\
&= (-1)^{2 \cdot i - 2 + k} \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot x_i \\
&= (-1)^k \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \cdot \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot x_i
\end{aligned}$$

So, as $x_k = \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \alpha_i \cdot x_i = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i + \sum_{i \in \{k+1, \dots, n\}} \alpha_i \cdot x_i$, we have that

$$\begin{aligned}
A &= (-1)^k \cdot \Delta(y, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \\
&\stackrel{[\text{eq: 11.179}]}{=} (-1) \cdot B
\end{aligned}$$

So that $A + B = 0$ proving by [eq: 11.176] that

$$\underline{\Delta}(y, (x_1, \dots, x_n)) = 0$$

So in all cases we have $\underline{\Delta}(y, (x_1, \dots, x_n)) = 0$ which by [eq: 11.174] proves that

$$\underline{\Delta}(y, (x_1, \dots, x_n)) = 0 \cdot y \stackrel{[\text{eq: 11.174}]}{=} \Delta(x_1, \dots, x_n) \cdot y$$

$\{x_i\}_{i \in \{1, \dots, n\}}$ is linear independent. Then by [corollary: 11.156] we have that

$$\{x_i | i \in \{1, \dots, n\}\} \text{ is a basis of } X$$

So if $y \in X$ there exists by [theorem: 11.143] a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$y = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i \tag{11.192}$$

By [definition: 11.304]

$$\underline{\Delta}(y, (x_1, \dots, x_n)) = \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \quad (11.193)$$

Let $i \in \{1, \dots, n\}$ then we have as Δ is n -linear that

$$\Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = \sum_{j \in \{1, \dots, n\}} \alpha_j \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \quad (11.194)$$

For i, j we have the following cases:

$i < j$. Then

$$\begin{aligned} (x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_j & \stackrel{[\text{theorem: 11.302}]}{=} \stackrel{\wedge i < j \Rightarrow i+1 \leq j}{=} x_j \\ & \stackrel{[\text{theorem: 11.302}]}{=} (x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_1 \end{aligned}$$

which as $1 \leq i < j \Rightarrow j \neq 1$ proves by [theorem: 11.291] that

$$\Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = 0 \quad (11.195)$$

$j < i$. Then

$$\begin{aligned} (x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_{j+1} & \stackrel{[\text{theorem: 11.302}]}{=} \stackrel{\wedge j < i \Rightarrow 1 < j+1 \leq i}{=} \\ & \stackrel{x_{(j+1)-1}}{=} \stackrel{x_j}{=} \stackrel{[\text{theorem: 11.302}]}{=} \\ & (x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_1 \end{aligned}$$

which as $1 < j+1$ proves by [theorem: 11.291] that

$$\Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = 0 \quad (11.196)$$

$i = j$. Let

$$A = (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (1 \rightsquigarrow_n j) \quad (11.197)$$

then we have the following cases to consider:

$j = 1$. Then $(1 \rightsquigarrow_n j) = (1 \rightsquigarrow_n 1) \stackrel{[\text{definition: 11.252}]}{=} \text{Id}_{\{1, \dots, n\}}$ so that $\forall k \in \{1, \dots, n\}$

$$\begin{aligned} A_k & = (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{theorem: 11.302}]}{=} \begin{cases} x_j & \text{if } k = 1 \\ x_{k-1} & \text{if } k \in \{2, \dots, j\} \\ x_k & \text{if } k \in \{j+1, \dots, n\} \end{cases} \\ & \stackrel{j=1 \wedge \{2, \dots, j\} = \emptyset}{=} \begin{cases} x_1 & \text{if } k = 1 \\ x_k & \text{if } k \in \{2, \dots, n\} \end{cases} \\ & = x_k \end{aligned}$$

so that

$$A = (x_1, \dots, x_n)$$

$j \in \{2, \dots, n\}$. Then $1 < j$ we have the following cases to consider (helped by [definition: 11.252]) for $k \in \{1, \dots, n\}$:

$1 \leq k < j$. Then

$$\begin{aligned} A_k & = (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(1 \rightsquigarrow_n j)(k)} \\ & \stackrel{[\text{definition: 11.252}]}{=} (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{k+1} \\ & \stackrel{[\text{theorem: 11.302}]}{=} \stackrel{\wedge 1 \leq k < j \Rightarrow 1 < k+1 \leq j}{=} x_{(k+1)-1} \\ & = x_k \end{aligned}$$

$k = j$. Then

$$\begin{aligned} A_k &= (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(1 \rightsquigarrow_n j)(k)} \\ &\stackrel{[\text{definition: 11.252}]}{=} (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_1 \\ &\stackrel{[\text{theorem: 11.302}]}{=} x_j \\ &\stackrel{=}{k=j} x_k \end{aligned}$$

$j < k \leq n$. Then

$$\begin{aligned} A_k &= (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(1 \rightsquigarrow_n j)(k)} \\ &\stackrel{[\text{definition: 11.252}]}{=} (x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ &\stackrel{[\text{theorem: 11.302}]}{=} x_k \quad \wedge j < k \Rightarrow j+1 \leq k \end{aligned}$$

so that

$$A = (x_1, \dots, x_n)$$

So in all cases $A = (x_1, \dots, x_n)$ proving that

$$(x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (1 \rightsquigarrow_n j) = (x_1, \dots, x_n) \quad (11.198)$$

So we have that

$$\begin{aligned} \Delta(x_1, \dots, x_n) &= \Delta((x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (1 \rightsquigarrow_n j)) \\ &= ((1 \rightsquigarrow_n j) \Delta)(x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \\ &= \text{sign}((1 \rightsquigarrow_n j)) \cdot \Delta(x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \\ &\stackrel{[\text{theorem: 11.255}]}{=} (-1)^{j-1} \cdot \Delta(x_j, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \\ &\stackrel{=}{i=j} (-1)^{i-1} \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \end{aligned}$$

proving that

$$\Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = (-1)^{i-1} \cdot \Delta(x_1, \dots, x_n) \quad (11.199)$$

Now as $\{1, \dots, n\}$ is the disjoint union of $\{1, \dots, i-1\}$, $\{x_i\}$ and $\{x_{i+1}, \dots, x_n\}$ we have

$$\sum_{j \in \{1, \dots, n\}} \alpha_j \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \stackrel{[\text{theorem: 11.43}]}{=} C_1 + C_2 + C_3$$

where

$$\begin{aligned} C_1 &= \sum_{j \in \{1, \dots, i-1\}} \alpha_j \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \\ &\stackrel{[\text{eq: 11.196}]}{=} 0 \\ C_2 &= \sum_{j \in \{i\}} \alpha_j \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \\ &= \alpha_i \cdot \Delta(x_i, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \\ &\stackrel{[\text{eq: 11.200}]}{=} (-1)^{i-1} \cdot \alpha_i \cdot \Delta(x_1, \dots, x_n) \\ C_3 &= \sum_{j \in \{i+1, \dots, n\}} \alpha_j \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \\ &\stackrel{[\text{eq: 11.195}]}{=} 0 \end{aligned}$$

So that

$$\sum_{j \in \{1, \dots, n\}} \alpha_j \cdot \Delta(x_j, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = (-1)^{i-1} \cdot \alpha_i \cdot \Delta(x_1, \dots, x_n)$$

Substituting the above in [eq: 11.194] proves

$$\Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i = \alpha_i \cdot (-1)^{i-1} \cdot \Delta(x_1, \dots, x_n) \quad (11.200)$$

So

$$\begin{aligned}
 \underline{\Delta}(y, (x_1, \dots, x_n)) &\stackrel{\text{def}}{=} \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \cdot x_i \\
 &\stackrel{[\text{eq: 11.200}]}{=} \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \alpha_i \cdot (-1)^{i-1} \cdot \Delta(x_1, \dots, x_n) \cdot x_i \\
 &= \Delta(x_1, \dots, x_n) \cdot \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i \\
 &\stackrel{[\text{eq: 11.192}]}{=} \Delta(x_1, \dots, x_n) \cdot y
 \end{aligned}$$

proving that

$$\underline{\Delta}(y, (x_1, \dots, x_n)) = \Delta(x_1, \dots, x_n) \cdot y$$

So in all the case we have proved that

$$\underline{\Delta}(y, (x_1, \dots, x_n)) = \Delta(x_1, \dots, x_n) \cdot y \quad \square$$

We show now how given a determinant function and a linear mapping we can create a new determinant function.

Definition 11.307. Let $n \in \mathbb{N}$, X a vector space over a field F such that $\dim(X) = n$, $\Delta \in \text{Hom}^n(X; F)$ a non trivial determinant function and $L \in \text{Hom}(X, X)$ [a linear transformation] then we define

$$\Delta_L: X^n \rightarrow F \text{ by } \Delta_L(x_1, \dots, x_n) = \Delta(L(x_1), \dots, L(x_n))$$

It turns out that Δ_L is also a determinant function.

Theorem 11.308. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$, $\Delta \in \text{Hom}^n(X; F)$ a non trivial determinant function and $L \in \text{Hom}(X, X)$ then

Δ_L is a determinant function

Proof. First we proof that Δ_L is n -linear. Let $i \in \{1, \dots, n\}$, $\alpha \in F$ and $u, v \in X$ then we have:

$$\begin{aligned}
 \Delta_L(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n) &= \\
 \Delta(L(x_1), \dots, \underbrace{L(u + \alpha \cdot v)}_i, \dots, L(x_n)) &= \\
 \Delta(L(x_1), \dots, \underbrace{L(u) + \alpha \cdot L(v)}_i, \dots, L(x_n)) &= \\
 \Delta(L(x_1), \dots, \underbrace{L(u)}_i, \dots, L(x_n)) + \alpha \cdot \Delta(L(x_1), \dots, \underbrace{L(v)}_i, \dots, L(x_n)) &= \\
 \Delta_L(x_1, \dots, \underbrace{u}_i, \dots, x_n) + \alpha \cdot \Delta_L(x_1, \dots, \underbrace{v}_i, \dots, x_n) &
 \end{aligned}$$

proving that

$$\Delta_L \in \text{Hom}^n(X; F) \text{ or } \Delta_L \text{ is } n\text{-linear}$$

As for skew-symmetry. Let $(x_1, \dots, x_n) \in X^n$ such that $\exists i, j \in \{1, \dots, n\}$ with $i \neq j$ and $x_i = x_j$ then $L(x_i) = L(x_j)$, hence

$$\begin{aligned}
 \Delta_L(x_1, \dots, x_n) &= \Delta(L(x_1), \dots, L(x_n)) \\
 &\stackrel{[\text{theorem: 11.291}]}{=} 0
 \end{aligned}$$

which by [theorem: 11.291] proves that

$$\Delta_L \text{ is skew-symmetric} \quad \square$$

Note 11.309. It is not true that Δ_L is a non trivial determinant function if Δ is a non trivial determinant function. For example, if $L = C_0: X \rightarrow X$ is defined by $C_0(x) = 0$ then for $(x_1, \dots, x_n) \in X^n$ we have $\Delta_L(x_1, \dots, x_n) = \Delta(L(x_1), \dots, L(x_n)) = \Delta(0, \dots, 0) = 0$, even if Δ is non trivial.

Theorem 11.310. Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with $\dim(X) = n$ and $L \in \text{Hom}(X, X)$ then there exist a $\alpha \in F$ such that for **every** non trivial determinant function $\Delta \in \text{Hom}^n(X; F)$ we have

$$\Delta_L = \alpha \cdot \Delta$$

Proof. Let Δ be the non trivial that exists by [theorem: 11.299], for Δ_L we have two possibilities:

$\Delta_L = C_0$. Then if we take $\alpha = 0$ we have that

$$\Delta_L = C_0 = 0 \cdot \Delta$$

$\Delta_L \neq C_0$. Then by [theorem: 11.308] Δ_L is also a determinant function, so by [theorem: 11.301] there exists a $\alpha \in F$ such that

$$\Delta_L = \alpha \cdot \Delta$$

So for L there exist a $\alpha \in F$ such that

$$\Delta_L = \alpha \cdot \Delta \quad (11.201)$$

proving existence. Let now Δ' be another non trivial determinant function then by [theorem: 11.301] there exist a $\lambda \in F$ such that

$$\Delta' = \lambda \cdot \Delta \quad (11.202)$$

Let $(x_1, \dots, x_n) \in X^n$ then

$$\begin{aligned} \Delta'_L(x_1, \dots, x_n) &= \Delta'(L(x_1), \dots, L(x_n)) \\ &\stackrel{[\text{eq: 11.202}]}{=} \lambda \cdot \Delta(L(x_1), \dots, L(x_n)) \\ &= \lambda \cdot \Delta_L(x_1, \dots, x_n) \end{aligned}$$

So

$$\begin{aligned} \Delta'_L &= \lambda \cdot \Delta_L \\ &\stackrel{[\text{eq: 11.201}]}{=} \lambda \cdot (\alpha \cdot \Delta) \\ &= \alpha \cdot (\lambda \cdot \Delta) \\ &\stackrel{[\text{eq: 11.202}]}{=} \alpha \cdot \Delta' \end{aligned}$$

proving that

$$\Delta'_L = \alpha \cdot \Delta_L \quad \square$$

The above theorem ensures that the following definition of the determinant of a linear function makes sense.

Definition 11.311. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$ and $L \in \text{Hom}(X, X)$ then we define the function

$$\det: \text{Hom}^n(X; F) \rightarrow F$$

where $\det(L) \in F$ is the scalar such that for every non trivial determinant function Δ we have

$$\Delta_L = \det(L) \cdot \Delta$$

Theorem 11.312. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$, $E = \{e_1, \dots, e_n\}$ basis for X and $L \in \text{Hom}(X, X)$ then

$$\det(L) = \Delta_L(e_1, \dots, e_n) = \Delta(L(e_1), \dots, L(e_n))$$

where $\Delta: X^n \rightarrow F$ is the determinant function such that $\Delta(e_1, \dots, e_n) = 1$ [see 11.299 for existence]

Proof. This follows from

$$\Delta(L(e_1), \dots, L(e_n)) = \Delta_L(e_1, \dots, e_n) \stackrel{\text{def}}{=} \det(L) \cdot \Delta(e_1, \dots, e_n) = \det(L) \cdot 1 = \det(L) \quad \square$$

Example 11.313. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$ and $\lambda \in F$ then we have for

$$\lambda \cdot \text{Id}_X: X \rightarrow X \text{ defined by } (\lambda \cdot \text{Id}_X)(x) = \lambda \cdot x$$

that

$$\det(\lambda \cdot \text{Id}_X) = \lambda^n$$

or in the case of $\lambda = 1$ that

$$\det(\text{Id}_X) = 1$$

Proof. Let Δ be a non trivial determinant function then

$$\begin{aligned} \Delta_{\lambda \cdot \text{Id}_X}(x_1, \dots, x_n) &= \Delta(\lambda \cdot \text{Id}_X(x_1), \dots, \lambda \cdot \text{Id}_X(x_n)) \\ &= \Delta(\lambda \cdot x_1, \dots, \lambda \cdot x_n) \\ &\stackrel{[\text{theorem: 11.270}]}{=} \lambda^n \cdot \Delta(x_1, \dots, x_n) \end{aligned}$$

proving that $\Delta_{\lambda \cdot \text{Id}_X} = \lambda^n \cdot \Delta$, hence

$$\det(\lambda \cdot \text{Id}_X) = \lambda^n \quad \square$$

Example 11.314. If X is a one dimensional vector space with a basis $\{e_1\}$ and $L \in \text{Hom}(X, X)$ then

$$\det(L) = \beta$$

where β is defined by

$$L(e_1) = \beta \cdot e_1$$

Proof. If $x = (x_1) \in X^1$ then as $x_1 \in X$ there exists a $\alpha \in F$ such that $x_1 = \alpha \cdot e_1$. Hence $L(x_1) = L(\alpha \cdot e_1) = \alpha \cdot L(e_1)$. As $L(e_1) \in X$ there exists a $\beta \in F$ such that $L(e_1) = \beta \cdot e_1$. So we have

$$\Delta_L(x_1) = \Delta(L(x_1)) = \Delta((\alpha \cdot \beta) \cdot e_1) = \beta \cdot \Delta(\alpha \cdot e_1) = \beta \cdot \Delta(x_1)$$

proving that

$$\det(L) = \beta \quad \square$$

Theorem 11.315. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$ then we have:

1. $\det(\text{Id}_X) = 1$
2. Let $L \in \text{Hom}(X, X)$ then L is a linear isomorphism $\Leftrightarrow \det(L) \neq 0$
3. If $L_1, L_2 \in \text{Hom}(X, X)$ then $\det(L_1 \circ L_2) = \det(L_1) \cdot \det(L_2)$

4. If $L \in \text{Hom}(X, X)$ is a linear isomorphism then $\det(L^{-1}) = (\det(L))^{-1}$

Proof.

1. Let $(x_1, \dots, x_n) \in X^n$ and Δ a non trivial determinant function then we have

$$\Delta_{\text{Id}_X}(x_1, \dots, x_n) = \Delta(\text{Id}_X(x_1), \dots, \text{Id}_X(x_n)) = \Delta(x_1, \dots, x_n)$$

proving that $\Delta_{\text{Id}_X} = \Delta = 1 \cdot \Delta$. So that $\det(\text{Id}) = 1$.

2. As $\dim(X) = n$ there exists a finite basis $E = \{e_1, \dots, e_n\}$ of X .

$\Rightarrow$. As $e: \{1, \dots, n\} \rightarrow E$ is a bijection and $L: X \rightarrow X$ is a bijection we have that $L \circ e$ is injective so that $L \circ e: \{1, \dots, n\} \rightarrow L(e(\{1, \dots, n\})) = L(E)$ is a bijection. Hence $\text{card}(L(E)) = n$ and $L(E) = \{(L \circ e)_1, \dots, (L \circ e)_n\} = \{L(e_1), \dots, L(e_n)\}$. As L is injective and E is linear independent it follows from [theorem: 11.199] that $L(E) = \{L(e_1), \dots, L(e_n)\}$ is linear independent. So applying [theorem: 11.155] it follows that

$$L(E) = \{L(e_1), \dots, L(e_n)\} \text{ is a basis for } X$$

Using [theorem: 11.299] there exist a non trivial determinant function Δ such that

$$\Delta(L(e_1), \dots, L(e_n)) = 1$$

So

$$1 = \Delta(L(e_1), \dots, L(e_n)) = \Delta_L(e_1, \dots, e_n) = \det(L) \cdot \Delta(e_1, \dots, e_n)$$

If now $\det(L) = 0$ then we have $1 = 0$, so we must have that

$$\det(L) \neq 0$$

$\Leftarrow$. Using [theorem: 11.199] there exist a non trivial determinant function Δ such that

$$\Delta(e_1, \dots, e_n) = 1$$

then

$$\begin{aligned} \Delta(L(e_1), \dots, L(e_n)) &= \Delta_L(e_1, \dots, e_n) \\ &= \det(L) \cdot \Delta(e_1, \dots, e_n) \\ &= \det(L) \end{aligned}$$

Assume that $\{L(e_i)\}_{i \in \{1, \dots, n\}} \subseteq X$ is linear dependent then it follows from [theorem: 11.291] that $\Delta(L(e_1), \dots, L(e_n)) = 0$, so that $\det(L) = 0$, contradicting $\det(L) \neq 0$. Hence we have that $\{L(e_i)\}_{i \in \{1, \dots, n\}}$ is linear independent. Let $x \in \ker(L)$ then $L(x) = 0$ and as $E = \{e_1, \dots, e_n\}$ is a basis we have that there exist a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $x = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i$ so that

$$\begin{aligned} 0 &= L(x) \\ &= L\left(\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i\right) \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot L(e_i) \end{aligned}$$

which as $\{L(e_i)\}_{i \in \{1, \dots, n\}}$ is linear independent proves that $\forall i \in \{1, \dots, n\} \alpha_i = 0$. So that $x = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i = 0$. Hence we have that $\ker(L) = \{0\}$ which by [theorem: 11.195] proves that

L is injective

As $\dim(X)$ is finite and $L: X \rightarrow X$ is injective it follows from [theorem: 11.201] that

$L: X \rightarrow X$ is a linear isomorphism

3. First we have if $(x_1, \dots, x_n) \in X^n$ and Δ a non trivial determinant function:

$$\begin{aligned}
 \Delta_{L_1 \circ L_2}(x_1, \dots, x_n) &= \Delta((L_1 \circ L_2)(x_1), \dots, (L_1 \circ L_2)(x_n)) \\
 &= \Delta(L_1(L_2(x_1)), \dots, L_1(L_2(x_n))) \\
 &= \Delta_{L_1}(L_2(x_1), \dots, L_2(x_n)) \\
 &= \det(L_1) \cdot \Delta(L_2(x_1), \dots, L_2(x_n)) \\
 &= \det(L_1) \cdot \Delta_{L_2}(x_1, \dots, x_n) \\
 &= \det(L_1) \cdot (\det(L_2) \cdot \Delta(x_1, \dots, x_n)) \\
 &= (\det(L_1) \cdot \det(L_2)) \cdot \Delta(x_1, \dots, x_n)
 \end{aligned}$$

so that $\Delta_{L_1 \circ L_2} = (\det(L_1) \cdot \det(L_2)) \cdot \Delta$ proving that

$$\det(\Delta_{L_1 \circ L_2}) = \det(L_1) \cdot \det(L_2)$$

4. As L is a linear isomorphism we have that

$$\begin{aligned}
 1 &\stackrel{(1)}{=} \det(\text{Id}_X) \\
 &= \det(L \circ L^{-1}) \\
 &\stackrel{(3)}{=} \det(L) \cdot \det(L^{-1})
 \end{aligned}$$

which as by (2) $\det(L) \neq 0$ proves that

$$\det(L^{-1}) = (\det(L))^{-1}$$

□

Corollary 11.316. *Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$, $m \in \mathbb{N}$ and $\{L_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Hom}(X, X)$ then*

$$\det(L_1 \circ \dots \circ L_m) \stackrel{\text{def}}{=} \det\left(\prod_{i=1}^m L_i\right) = \prod_{i \in \{1, \dots, m\}} \det(L_i)$$

Proof. We prove this by induction on m so let

$$S = \left\{ m \in \mathbb{N} \mid \text{If } \{L_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Hom}(X, X) \text{ then } \det\left(\prod_{i=1}^m L_i\right) = \prod_{i \in \{1, \dots, m\}} \det(L_i) \right\}$$

then we have:

1 $\in S$. If $\{L_i\}_{i \in \{1\}} \subseteq \text{Hom}(X, X)$ then we have

$$\det\left(\prod_{i=1}^1 L_i\right) = \det(L_1) = \prod_{i \in \{1\}} \det(L_i)$$

proving that **1** $\in S$.

m $\in S \Rightarrow m+1$ $\in S$. Let $\{L_i\}_{i \in \{1, \dots, m+1\}} \subseteq \text{Hom}(X, X)$ then we have

$$\begin{aligned}
 \det\left(\prod_{i=1}^{m+1} L_i\right) &= \det\left(\left(\prod_{i=1}^m L_i\right) \circ L_{m+1}\right) \\
 &\stackrel{[\text{theorem: 11.315}]}{=} \det\left(\prod_{i=1}^m L_i\right) \circ \det(L_{m+1}) \\
 &\stackrel{m \in S}{=} \left(\prod_{i \in \{1, \dots, m\}} \det(L_i)\right) \cdot \det(L_{m+1}) \\
 &= \prod_{i \in \{1, \dots, m+1\}} \det(L_i)
 \end{aligned}$$

proving that $m + 1 \in S$. □

Definition 11.317. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$, Δ a non trivial determinant function and $L \in \text{Hom}(X, X)$ then we define

$\overline{\Delta L}: X^n \rightarrow \text{Hom}(X, X)$ where $\overline{\Delta L}(x_1, \dots, x_n): X \rightarrow X$ is defined by

$$\overline{\Delta L}(x_1, \dots, x_n)(y) = \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(y, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i$$

Proof. Of course we must prove that $\overline{\Delta L}(x_1, \dots, x_n) \in \text{Hom}(X, X)$. So let $\alpha \in F$, $u, v \in X$ then we have

$$\begin{aligned} & \overline{\Delta L}(x_1, \dots, x_n)(u + v) = \\ & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(u + v, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i = \\ & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot (\Delta(u, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) + \Delta(v, L(x_1), \dots, \\ & L(x_{i-1}), L(x_{i+1}), \dots, L(x_n))) \quad \quad \quad \stackrel{[\text{theorem: 11.38}]}{=} \\ & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(u, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i + \sum_{i \in \{1, \dots, n\}} (-1)^i \cdot \\ & \Delta(v, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i = \\ & \overline{\Delta L}(x_1, \dots, x_n)(u) + \overline{\Delta L}(x_1, \dots, x_n)(v) \end{aligned}$$

and

$$\begin{aligned} & \overline{\Delta L}(x_1, \dots, x_n)(\alpha \cdot u) = \\ & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(\alpha \cdot u, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i = \\ & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \alpha \cdot \Delta(u, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i \quad \stackrel{[\text{theorem: 11.80}]}{=} \\ & \alpha \cdot \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(u, L(x_1), \dots, L(x_{i-1}), L(x_{i+1}), \dots, L(x_n)) \cdot x_i = \\ & \alpha \cdot \overline{\Delta L}(x_1, \dots, x_n)(u) \end{aligned}$$

□

Lemma 11.318. Let $n \in \mathbb{N}$, X a set, $x = (x_1, \dots, x_n) \in X^n$, $i, j \in \{1, \dots, n\}$ such that $i \neq j$ and $x_i = x_j$ and $t \in X$ then we have

1. If $j < i$ then

$$(t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (i \rightsquigarrow_n j + 1) = (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

2. If $i < j$ then

$$(t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (i + 1 \rightsquigarrow_n j) = (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

Proof.

1. Let $j < i$. Define

$$A = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (i \rightsquigarrow_n j + 1) \quad (11.203)$$

then we have as $j + 1 \leq i$ the following cases to consider for i :

$j + 1 = i$. Then by [definition: 11.252(1)] we have that $(i \rightsquigarrow_n j + 1) = \text{Id}_{\{1, \dots, n\}}$ so that

$$A = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \quad (11.204)$$

Now for $k \in \{1, \dots, n\}$ we have either (motivated by [definition: 11.302]) the following possible cases

$k = 1$. Then

$$\begin{aligned} A_k & \stackrel{[\text{eq: } \overline{11.204}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge k=1}{=} t \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge k=1}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$2 \leq k \leq j$. Then

$$\begin{aligned} A_k & \stackrel{[\text{eq: } \overline{11.204}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge 2 \leq k < j}{=} x_{k-1} \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge 2 \leq k \leq j = i-1 < i}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k = j + 1$. Then

$$\begin{aligned} A_k & \stackrel{[\text{eq: } \overline{11.204}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge i=j+1=k}{=} x_k \\ & \stackrel{k=j+1=i}{=} x_i \\ & \stackrel{x_i=x_j}{=} x_j \\ & \stackrel{j=k-1}{=} x_{k-1} \\ & \stackrel{[\text{definition: } \overline{11.302}] i=j+1=k}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$j + 1 < k$. Then

$$\begin{aligned} A_k & \stackrel{[\text{eq: } \overline{11.204}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: } \overline{11.302}] j+1 < k}{=} x_k \\ & \stackrel{[\text{definition: } \overline{11.302}] i=j+1 < k \Rightarrow i+1 \leq k}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$j + 1 < i$. Then we have motivated by [definition: 11.252(3)] the following possible cases for k :

$k = 1$. Then

$$\begin{aligned} A_k & = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i \rightsquigarrow_n j+1)(k)} \\ & \stackrel{[\text{definition: } \overline{11.252(3)}] \wedge k=1 < j+1}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge k=1}{=} t \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge k=1}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k \neq 1 \wedge k < j + 1$. Then

$$\begin{aligned} A_k & = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i \rightsquigarrow_n j+1)(k)} \\ & \stackrel{[\text{definition: } \overline{11.252(3)}] \wedge k < j+1}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge 1 < k < j+1 \Rightarrow 1 < k \leq j}{=} x_{k-1} \\ & \stackrel{[\text{definition: } \overline{11.302}] \wedge 1 < k < j+1=i}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k \neq 1 \wedge k = j + 1$. Then

$$\begin{aligned}
 A_k &= (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i \rightsquigarrow_n j+1)(k)} \\
 &\stackrel{[\text{definition: 11.252(3)}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_i \\
 &\stackrel{[\text{definition: 11.302}]}{=} x_i \quad \wedge j+1 < i \\
 &\stackrel{x_i = x_j}{=} x_j \\
 &\stackrel{j = k-1}{=} x_{k-1} \\
 &\stackrel{1 < k = j+1 < i}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k
 \end{aligned}$$

$k \neq 1 \wedge j + 1 < k \leq i$. Then

$$\begin{aligned}
 A_k &= (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i \rightsquigarrow_n j+1)(k)} \\
 &\stackrel{[\text{definition: 11.252(3)}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{k-1} \\
 &\stackrel{[\text{definition: 11.302}]}{=} x_{k-1} \quad \wedge j+1 < k \Rightarrow j+1 \leq k-1 \\
 &\stackrel{[\text{definition: 11.302}]}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \quad \wedge 1 < k \leq i
 \end{aligned}$$

$k \neq 1 \wedge i < k$. Then

$$\begin{aligned}
 A_k &= (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i \rightsquigarrow_n j+1)(k)} \\
 &\stackrel{[\text{definition: 11.252(3)}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \quad \wedge i < k \leq n \\
 &\stackrel{[\text{definition: 11.302}]}{=} x_k \quad \wedge j+1 < i < k \\
 &\stackrel{[\text{definition: 11.302}]}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \quad \wedge i < k \Rightarrow i+1 \leq k
 \end{aligned}$$

So in all possible cases we have

$$(t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i \rightsquigarrow_n j+1)(k)} = (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k$$

proving that we have

$$(t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (i \rightsquigarrow_n j+1) = (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

2. Let $i < j$. Define

$$B = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (i+1 \rightsquigarrow_n j) \quad (11.205)$$

then we have as $i+1 \leq j$ the following cases to consider for j :

$i+1 = j$. Then by [definition: 11.252(1)] we have that $(i \rightsquigarrow_n j+1) = \text{Id}_{\{1, \dots, n\}}$ so that

$$B = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \quad (11.206)$$

Now for $k \in \{1, \dots, n\}$ we have either (motivated by [definition: 11.302]) following cases:

$k = 1$. Then

$$\begin{aligned}
 B_k &\stackrel{[\text{eq: 11.206}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} t \quad \wedge k=1 \\
 &\stackrel{[\text{definition: 11.302}]}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \quad \wedge k=1
 \end{aligned}$$

$1 < k \leq i$. Then

$$\begin{aligned} B_k & \stackrel{[\text{eq: 11.206}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge 1 < k \leq i < i+1=j}{=} x_{k-1} \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge 1 < k \leq i}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k = i + 1$. Then

$$\begin{aligned} B_k & \stackrel{[\text{eq: 11.206}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge 1 < k = i+1=j}{=} x_{k-1} \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{k-1=i}{=} x_i \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{x_i=x_j}{=} x_j \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{j=i+1=k}{=} x_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge k=i+1}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$i + 1 < k$. Then

$$\begin{aligned} B_k & \stackrel{[\text{eq: 11.206}]}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge j=i+1 < k \Rightarrow j+1 \leq k}{=} x_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge i+1 < k}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$i + 1 < j$. Then we have motivated by [definition: 11.252(2)] the following possible cases for k :

$k = 1$. Then

$$\begin{aligned} B_k & = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i+1 \rightsquigarrow_n j)(k)} \\ & \stackrel{[\text{theorem: 11.252(2)}]}{=} \stackrel{\wedge 1=k < i+1}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge k=1}{=} t \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge k=1}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k \neq 1 \wedge k < i + 1$. Then

$$\begin{aligned} B_k & = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i+1 \rightsquigarrow_n j)(k)} \\ & \stackrel{[\text{definition: 11.252(2)}]}{=} \stackrel{\wedge 1 < k < i+1}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge 1 < k < i+1 < j}{=} x_{k-1} \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge 1 < k < i+1 \Rightarrow 1 < k \leq i}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k \neq 1 \wedge i + 1 \leq k < j$. Then

$$\begin{aligned} B_k & = (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i+1 \rightsquigarrow_n j)(k)} \\ & \stackrel{[\text{definition: 11.252(2)}]}{=} \stackrel{\wedge i+1 \leq k < j}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{k+1} \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge 1 < k < j \Rightarrow 1 < k+1 \leq j}{=} x_{(k+1)-1} \\ & = x_k \\ & \stackrel{[\text{definition: 11.302}]}{=} \stackrel{\wedge i+1 \leq k}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k \end{aligned}$$

$k \neq 1 \wedge k = j$. Then

$$\begin{aligned}
 B_k &= (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i+1 \rightsquigarrow_n j)(k)} \\
 &\stackrel{[\text{definition: 11.252(2)}] \wedge k=j}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{i+1} \\
 &\stackrel{[\text{definition: 11.302}] \wedge 1 < i+1 < j}{=} x_{(i+1)-1} \\
 &= x_i \\
 &\stackrel{x_i=x_j}{=} x_j \\
 &\stackrel{j=k}{=} x_k \\
 &\stackrel{[\text{definition: 11.302}] \wedge i < i+1 < j=k}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k
 \end{aligned}$$

$k \neq 1 \wedge j < k$. Then

$$\begin{aligned}
 B_k &= (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i+1 \rightsquigarrow_n j)(k)} \\
 &\stackrel{[\text{definition: 11.252}] \wedge j < k}{=} (t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_k \\
 &\stackrel{[\text{definition: 11.302}] \wedge j < k \rightarrow j+1 \leq k}{=} x_k \\
 &\stackrel{[\text{definition: 11.302}] \wedge i+1 < j < k}{=} (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k
 \end{aligned}$$

So in all possible cases we have

$$(t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)_{(i+1 \rightsquigarrow_n j)(k)} = (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)_k$$

proving that we have

$$(t, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \circ (i+1 \rightsquigarrow_n j) = (t, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \quad \square$$

The above lemma will be used to prove the following theorem that will be used to define the adjoint of a linear transformation.

Theorem 11.319. *Let $n \in \mathbb{N}$, X a vector space over a field of characteristic zero with $\dim(X) = n$, Δ a non trivial determinant function, $L \in \text{Hom}(X, X)$ then*

$$\overline{\Delta L} \in \text{Hom}^n(X; \text{Hom}(X, X)) \text{ and } \overline{\Delta L} \text{ is skew-symmetric}$$

Proof. Given $x \in X^n$, $t \in X$ then we have by definition

$$\begin{aligned}
 \overline{\Delta L}(x)(t) &= \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\
 &= \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot A_j(t, x)
 \end{aligned} \tag{11.207}$$

where

$$A_j(t, x) = \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \tag{11.208}$$

Let $x = (x_1, \dots, x_n) \in X^n$, $i \in \{1, \dots, n\}$, $\alpha \in F$ and $u, v, t \in X$. Define

$$\begin{aligned}
 y &= \left(x_1, \dots, \underbrace{u + \alpha \cdot v}_i, \dots, x_n \right) \\
 r &= \left(x_1, \dots, \underbrace{u}_i, \dots, x_n \right) \\
 s &= \left(x_1, \dots, \underbrace{v}_i, \dots, x_n \right)
 \end{aligned}$$

Let $j \in \{1, \dots, n\}$ and define

$$\begin{aligned} y^j &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n)) \\ r^j &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n)) \\ s^j &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n)) \end{aligned}$$

then

$$A_j(t, y) = \Delta(y^j) \cdot y_j \wedge A_j(t, r) = \Delta(r^j) \cdot r_j \wedge A_j(t, s) = \Delta(s^j) \cdot s_j \quad (11.209)$$

Consider the following possible cases for $j \in \{1, \dots, n\}$:

$j = i$. Then for $k \in \{1, \dots, n\}$ we have either:

$k = 1$. Then

$$\begin{aligned} (y^j)_k &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_1 \\ &\stackrel{[\text{definition: 11.302}] \wedge k=1}{=} t \\ (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_1 \\ &\stackrel{[\text{definition: 11.302}] \wedge k=1}{=} t \\ (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_1 \\ &\stackrel{[\text{definition: 11.302}] \wedge k=1}{=} t \end{aligned}$$

$2 \leq k \leq j$. Then

$$\begin{aligned} (y^j)_k &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \\ &\stackrel{[\text{definition: 11.302}] \wedge 2 \leq k \leq j}{=} L(y_{k-1}) \\ &\stackrel{k-1 < k \leq j=i \Rightarrow k-1 \neq i}{=} L(x_{k-1}) \\ (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\ &\stackrel{[\text{definition: 11.302}] \wedge 2 \leq k \leq j}{=} L(r_{k-1}) \\ &\stackrel{k-1 < k \leq j=i \Rightarrow k-1 \neq i}{=} L(x_{k-1}) \\ (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\ &\stackrel{[\text{definition: 11.302}] \wedge 2 \leq k \leq j}{=} L(s_{k-1}) \\ &\stackrel{k-1 < k \leq j=i \Rightarrow k-1 \neq i}{=} L(x_{k-1}) \end{aligned}$$

$j+1 \leq k$. Then

$$\begin{aligned} (y^j)_k &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \\ &\stackrel{[\text{definition: 11.302}] \wedge j+1 \leq k}{=} L(y_k) \\ &\stackrel{i=j < j+1 \leq k \Rightarrow i \neq k}{=} L(x_k) \\ (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\ &\stackrel{[\text{definition: 11.302}] \wedge j+1 \leq k}{=} L(r_k) \\ &\stackrel{i=j < j+1 \leq k \Rightarrow i \neq k}{=} L(x_k) \\ (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\ &\stackrel{[\text{definition: 11.302}] \wedge j+1 \leq k}{=} L(s_k) \\ &\stackrel{i=j < j+1 \leq k \Rightarrow i \neq k}{=} L(x_k) \end{aligned}$$

So $\forall k \in \{1, \dots, n\}$ we have $(y^j)_k = (r^j)_k = (s^j)_k$ or

$$y^j = r^j = s^j \quad (11.210)$$

Hence

$$\begin{aligned} A_j(t, y) &\stackrel{[\text{eq: 11.209}]}{=} \Delta(y^j) \cdot y_j \\ &\stackrel{=}{=} \Delta(y^j) \cdot y_i \\ &= \Delta(y^j) \cdot (u + \alpha \cdot v) \\ &= \Delta(y^j) \cdot u + \alpha \cdot \Delta(y^j) \cdot v \\ &\stackrel{[\text{eq: 11.210}]}{=} \Delta(r^j) \cdot u + \alpha \cdot \Delta(s^j) \cdot v \\ &= \Delta(r^j) \cdot r_i + \alpha \cdot \Delta(s^j) \cdot s_i \\ &\stackrel{=}{=} \Delta(r^j) \cdot r_j + \alpha \cdot \Delta(s^j) \cdot s_j \\ &= A_j(t, r) + A_j(t, s) \end{aligned}$$

proving that in this case

$$A_j(t, y) = A_j(t, r) + A_j(t, s) \quad (11.211)$$

$j < i$. Then we have

$$\begin{aligned} (y^j)_i &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_i \\ &\stackrel{[\text{definition: 11.302}]}{=} L(y_i) \wedge 1 < j+1 \leq i \\ &= L(u + \alpha \cdot v) \\ &= L(u) + \alpha \cdot L(v) \end{aligned}$$

hence $y^j = \left((y^j)_1, \dots, \underbrace{L(u) + \alpha \cdot L(v)}_i, \dots, (y^j)_n \right)$ so that

$$\begin{aligned} \Delta(y^j) &= \\ \Delta\left((y^j)_1, \dots, \underbrace{L(u) + \alpha \cdot L(v)}_i, \dots, (y^j)_n \right) &= \\ \Delta\left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^j)_n \right) + \alpha \cdot \Delta\left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^j)_n \right) &= \end{aligned} \quad (11.212)$$

Further we have for $k \in \{1, \dots, n\}$ either:

$k = 1$. Then

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^j)_n \right)_k &\stackrel{1 \leq j < i \Rightarrow i \neq 1}{=} \\ &\stackrel{=}{=} (y^j)_1 \\ (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_1 &\stackrel{[\text{definition: 11.302}]}{=} t \wedge k=1 \end{aligned}$$

and

$$\begin{aligned} (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n)) \\ &\stackrel{[\text{definition: 11.302}]}{=} t \wedge k=1 \end{aligned}$$

and

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(\mathbf{v})}_i, \dots, (y^i)_n \right)_k \quad 1 \leq j < \overline{i} \Rightarrow i \neq 1 \\
 & \quad (y^j)_1 = \\
 & (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_1 \quad [\text{definition: 11.302}] \wedge k=1 \\
 & \quad t
 \end{aligned}$$

and

$$\begin{aligned}
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n)) \\
 & \quad [\text{definition: 11.302}] \wedge k=1 \quad t
 \end{aligned}$$

Hence we have in this case that

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k = (r^j)_k \\
 & \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k = (s^j)_k
 \end{aligned}$$

$2 \leq k \leq j$. Then

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(\mathbf{u})}_i, \dots, (y^i)_n \right)_k \quad k \leq j < \overline{i} \Rightarrow k \neq i \\
 & \quad (y^j)_k = \\
 & (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \quad [\text{definition: 11.302}] \\
 & \quad L(y_{k-1}) \quad k-1 < k \leq j < \overline{i} \Rightarrow k \neq i \\
 & \quad L(x_{k-1})
 \end{aligned}$$

and

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(\mathbf{v})}_i, \dots, (y^i)_n \right)_k \quad k \leq j < \overline{i} \Rightarrow k \neq i \\
 & \quad (y^j)_k = \\
 & (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \quad [\text{definition: 11.302}] \\
 & \quad L(y_{k-1}) \quad k-1 < k \leq j < \overline{i} \Rightarrow k \neq i \\
 & \quad L(x_{k-1})
 \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 & \quad [\text{definition: 11.302}] \quad L(r_{k-1}) \\
 & \quad k-1 < k \leq j < \overline{i} \Rightarrow k \neq i \quad L(x_{k-1}) \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 & \quad [\text{definition: 11.302}] \quad L(s_{k-1}) \\
 & \quad k-1 < k \leq j < \overline{i} \Rightarrow k \neq i \quad L(x_{k-1})
 \end{aligned}$$

Hence we have that

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &= (r^j)_k \\ \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &= (s^j)_k \end{aligned}$$

$j+1 \leq k < i$. Then

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &\stackrel{k < i \Rightarrow k \neq i}{=} \\ (y^j)_k &= \\ (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k &\stackrel{[\text{definition: 11.302}]}{=} \\ L(y_k) &\stackrel{k < i \Rightarrow k \neq i}{=} \\ L(x_k) & \end{aligned}$$

and

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &\stackrel{k < i \Rightarrow k \neq i}{=} \\ (y^j)_k &= \\ (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k &\stackrel{[\text{definition: 11.302}]}{=} \\ L(y_k) &\stackrel{k < i \Rightarrow k \neq i}{=} \\ L(x_k) & \end{aligned}$$

and

$$\begin{aligned} (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\ &\stackrel{[\text{definition: 11.302}]}{=} L(r_k) \\ &\stackrel{k < i \Rightarrow k \neq i}{=} L(x_k) \\ (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\ &\stackrel{[\text{definition: 11.302}]}{=} L(s_k) \\ &\stackrel{k < i \Rightarrow k \neq i}{=} L(x_k) \end{aligned}$$

Hence we have that

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &= (r^j)_k \\ \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &= (s^j)_k \end{aligned}$$

$k = i$. Then

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &\stackrel{k=i}{=} L(u) \\ \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &\stackrel{k=i}{=} L(v) \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} \bigwedge_{j < i = k \Rightarrow j+1 \leq k} L(r_k) \\
 &\stackrel{i=k}{=} L(u) \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} \bigwedge_{j < i = k \Rightarrow j+1 \leq k} L(s_k) \\
 &= L(v)
 \end{aligned}$$

Hence we have that

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &= (r^j)_k \\
 \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &= (s^j)_k
 \end{aligned}$$

$i+1 \leq k$. Then

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &\stackrel{i < i+1 \leq k \Rightarrow k \neq i}{=} (y^j)_k \\
 &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \stackrel{[\text{definition: 11.302}]}{=} L(y_k) \\
 &\stackrel{i < i+1 \leq k \Rightarrow k \neq i}{=} L(x_k)
 \end{aligned}$$

and

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &\stackrel{i < i+1 \leq k \Rightarrow k \neq i}{=} (y^j)_k \\
 &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \stackrel{[\text{definition: 11.302}]}{=} L(y_k) \\
 &\stackrel{i < i+1 \leq k \Rightarrow k \neq i}{=} L(x_k)
 \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} L(r_k) \\
 &\stackrel{i < i+1 \leq k \Rightarrow k \neq i}{=} L(x_k) \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} L(s_k) \\
 &\stackrel{i < i+1 \leq k \Rightarrow k \neq i}{=} L(x_k)
 \end{aligned}$$

Hence we have that

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &= (r^j)_k \\
 \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &= (s^j)_k
 \end{aligned}$$

So we have proved that $\forall k \in \{1, \dots, n\}$

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right)_k &= (r^j)_k \\ \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right)_k &= (s^j)_k \end{aligned}$$

or

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_i, \dots, (y^i)_n \right) &= r^j \\ \left((y^j)_1, \dots, \underbrace{L(v)}_i, \dots, (y^i)_n \right) &= s^j \end{aligned}$$

which combined with [eq: 11.212] proves that

$$\Delta(y^j) = \Delta(r^j) + \alpha \cdot \Delta(s^j)$$

So that

$$\begin{aligned} A_j(t, y) &= \Delta(y^j) \cdot y_j \\ &= (\Delta(r^j) + \alpha \cdot \Delta(s^j)) \cdot y_j \\ &= \Delta(r^j) \cdot y_j + \alpha \cdot \Delta(s^j) \cdot y_j \\ &\stackrel{j < i \Rightarrow i \neq j}{=} \Delta(r^j) \cdot r_j + \alpha \cdot \Delta(s^j) \cdot s_j \\ &= A_j(t, r) + \alpha \cdot A_j(t, s) \end{aligned}$$

or summarized

$$A_j(t, y) = A_j(t, r) + \alpha \cdot A_j(t, s) \quad (11.213)$$

$i < j$. Then we have

$$\begin{aligned} (y^j)_{i+1} &= (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_{i+1} \\ &\stackrel{[\text{definition: 11.302}]}{=} L(y_{(i+1)-1}) \\ &= L(y_i) \\ &= L(u + \alpha \cdot v) \\ &= L(u) + \alpha \cdot L(v) \end{aligned}$$

hence $y^j = \left((y^j)_1, \dots, \underbrace{L(u) + \alpha \cdot L(v)}_{i+1}, \dots, (y^i)_n \right)$ so that

$$\begin{aligned} \Delta(y^j) &= \\ \Delta((y^j)_1, \dots, (y^j)_i, L(u) + \alpha \cdot L(v), (y^j)_{i+2}, \dots, (y^i)_n) &= \\ \Delta\left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right) + \alpha \cdot \Delta\left((y^j)_1, \dots, \underbrace{\alpha \cdot L(v)}_{i+1}, \dots, (y^i)_n \right) &= \end{aligned} \quad (11.214)$$

Further we have for $k \in \{1, \dots, n\}$ either:

$k = 1$. Then

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k &\stackrel{k=1 < i+1 \Rightarrow k \neq i+1}{=} \\ &= (y^j)_1 \\ (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_1 &\stackrel{[\text{definition: 11.302}]}{=} \\ &= t \end{aligned}$$

and

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k \quad k=1 < i+1 \Rightarrow k \neq i+1 \\
 & \quad (y^j)_1 = \\
 & (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_1 \quad [\text{definition: 11.302}] \\
 & \quad t
 \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 & \quad [\text{definition: 11.302}] \wedge k=1 \quad t \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 & \quad [\text{definition: 11.302}] \wedge k=1 \quad t
 \end{aligned}$$

Hence we have

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k = (r^j)_k \\
 & \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k = (s^j)_k
 \end{aligned}$$

$2 \leq k \leq i$. Then

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k \quad k \leq i < i+1 \Rightarrow k \neq i+1 \\
 & \quad (y^j)_k = \\
 & (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \quad [\text{definition: 11.302}] \\
 & \quad L(y_{k-1}) \quad k \leq i \Rightarrow k-1 \neq i \\
 & \quad L(x_{k-1})
 \end{aligned}$$

and

$$\begin{aligned}
 & \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k \quad k \leq i < i+1 \Rightarrow k \neq i+1 \\
 & \quad (y^j)_k = \\
 & (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k \quad [\text{definition: 11.302}] \\
 & \quad L(y_{k-1}) \quad k \leq i \Rightarrow k-1 \neq i \\
 & \quad L(x_{k-1})
 \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 & \quad [\text{definition: 11.302}] \quad L(r_{k-1}) \\
 & \quad k \leq i \Rightarrow k-1 \neq i \quad L(x_{k-1}) \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 & \quad [\text{definition: 11.302}] \quad L(s_{k-1}) \\
 & \quad k \leq i \Rightarrow k-1 \neq i \quad L(x_{k-1})
 \end{aligned}$$

Hence we have

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{\mathbf{L}(\mathbf{u})}_{i+1}, \dots, (y^i)_n \right)_k &= (r^j)_k \\ \left((y^j)_1, \dots, \underbrace{\mathbf{L}(\mathbf{v})}_{i+1}, \dots, (y^i)_n \right)_k &= (s^j)_k \end{aligned}$$

$k = i+1$. Then

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{\mathbf{L}(\mathbf{u})}_{i+1}, \dots, (y^i)_n \right)_k &\stackrel{=}{=}_{k=i+1} L(u) \\ \left((y^j)_1, \dots, \underbrace{\mathbf{L}(\mathbf{v})}_{i+1}, \dots, (y^i)_n \right)_k &\stackrel{=}{=}_{k=i+1} L(v) \end{aligned}$$

and

$$\begin{aligned} (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\ &\stackrel{=}{=}_{[\text{definition: 11.302}]} \wedge i < j \Rightarrow 2 \leq i+1 = k \leq j L(r_{k-1}) \\ &\stackrel{=}{=}_{k=i+1} L(r_i) \\ &= L(u) \\ (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\ &\stackrel{=}{=}_{[\text{definition: 11.302}]} \wedge i < j \Rightarrow 2 \leq i+1 = k \leq j L(s_{k-1}) \\ &\stackrel{=}{=}_{k=i+1} L(s_i) \\ &= L(v) \end{aligned}$$

Hence we have

$$\begin{aligned} ((y^j)_1, \dots, (y^j)_i, \mathbf{L}(\mathbf{u}), (y^j)_{i+2}, \dots, (y^i)_n)_k &= (r^j)_k \\ ((y^j)_1, \dots, (y^j)_i, \mathbf{L}(\mathbf{v}), (y^j)_{i+2}, \dots, (y^i)_n)_k &= (s^j)_k \end{aligned}$$

$i+1 < k \leq j$. Then

$$\begin{aligned} ((y^j)_1, \dots, (y^j)_i, \mathbf{L}(\mathbf{u}), (y^j)_{i+2}, \dots, (y^i)_n)_k &\stackrel{=}{=}_{i+1 < k \Rightarrow k \neq i+1} \\ &= (y^j)_k \\ (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k &\stackrel{=}{=}_{[\text{definition: 11.302}]} \\ &= L(y_{k-1}) \stackrel{=}{=}_{i+1 < k \Rightarrow i \neq k-1} \\ &= L(x_{k-1}) \end{aligned}$$

and

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{\mathbf{L}(\mathbf{v})}_{i+1}, \dots, (y^i)_n \right)_k &\stackrel{=}{=}_{i+1 < k \Rightarrow k \neq i+1} \\ &= (y^j)_k \\ (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k &\stackrel{=}{=}_{[\text{definition: 11.302}]} \\ &= L(y_{k-1}) \stackrel{=}{=}_{i+1 < k \Rightarrow i \neq k-1} \\ &= L(x_{k-1}) \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} L(r_{k-1}) \\
 &\stackrel{i+1 < k \Rightarrow i \neq k-1}{=} L(x_{k-1}) \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} L(s_{k-1}) \\
 &\stackrel{i+1 < k \Rightarrow i \neq k-1}{=} L(x_{k-1})
 \end{aligned}$$

Hence we have

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k &= (r^j)_k \\
 \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k &= (s^j)_k
 \end{aligned}$$

$j+1 \leq k$. Then

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k &\stackrel{i < j < j+1 \leq k \Rightarrow i+1 \neq k}{=} \\
 (y^j)_k &= \\
 (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k &\stackrel{[\text{definition: 11.302}]}{=} \\
 L(y_k) &\stackrel{i < j < j+1 \leq k \Rightarrow i \neq k}{=} \\
 L(x_k)
 \end{aligned}$$

and

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k &\stackrel{i < j < j+1 \leq k \Rightarrow i+1 \neq k}{=} \\
 (y^j)_k &= \\
 (t, L(y_1), \dots, L(y_{j-1}), L(y_{j+1}), \dots, L(y_n))_k &\stackrel{[\text{definition: 11.302}]}{=} \\
 L(y_k) &\stackrel{i < j < j+1 \leq k \Rightarrow i \neq k}{=} \\
 L(x_k)
 \end{aligned}$$

and

$$\begin{aligned}
 (r^j)_k &= (t, L(r_1), \dots, L(r_{j-1}), L(r_{j+1}), \dots, L(r_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} L(r_k) \\
 &\stackrel{i < j < j+1 \leq k \Rightarrow i \neq k}{=} L(x_k) \\
 (s^j)_k &= (t, L(s_1), \dots, L(s_{j-1}), L(s_{j+1}), \dots, L(s_n))_k \\
 &\stackrel{[\text{definition: 11.302}]}{=} L(s_k) \\
 &\stackrel{i < j < j+1 \leq k \Rightarrow i \neq k}{=} L(x_k)
 \end{aligned}$$

Hence we have

$$\begin{aligned}
 \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k &= (r^j)_k \\
 \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k &= (s^j)_k
 \end{aligned}$$

So we have proved that $\forall k \in \{1, \dots, n\}$

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right)_k &= (r^j)_k \\ \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right)_k &= (s^j)_k \end{aligned}$$

or

$$\begin{aligned} \left((y^j)_1, \dots, \underbrace{L(u)}_{i+1}, \dots, (y^i)_n \right) &= r^j \\ \left((y^j)_1, \dots, \underbrace{L(v)}_{i+1}, \dots, (y^i)_n \right) &= s^j \end{aligned}$$

which combined with [eq: 11.214] proves that

$$\Delta(y^j) = \Delta(r^j) + \alpha \cdot \Delta(s^j)$$

So that

$$\begin{aligned} A_j(t, y) &= \Delta(y^j) \cdot y_j \\ &= (\Delta(r^j) + \alpha \cdot \Delta(s^j)) \cdot y_j \\ &= \Delta(r^j) \cdot y_j + \alpha \cdot \Delta(s^j) \cdot y_j \\ &\stackrel{j < i \Rightarrow i \neq j}{=} \Delta(r^j) \cdot r_j + \alpha \cdot \Delta(s^j) \cdot s_j \\ &= A_j(t, r) + \alpha \cdot A_j(t, s) \end{aligned}$$

or summarized

$$A_j(t, y) = A_j(t, r) + \alpha \cdot A_j(t, s) \quad (11.215)$$

By [eqs: 11.213, 11.213 and 11.215] we have that

$$\forall j \in \{1, \dots, n\} \quad A_j(t, y) = A_j(t, r) + \alpha \cdot A_j(t, s)$$

Next

$$\begin{aligned} \overline{\Delta L}(y)(t) &\stackrel{[\text{theorem: 11.207}]}{=} \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot A_j(t, y) \\ &\stackrel{[\text{theorem: 11.215}]}{=} \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot (A_j(t, r) + \alpha \cdot A_j(t, s)) \\ &\stackrel{[\text{theorem: 11.38}]}{=} \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot A_j(t, r) + \alpha \cdot \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot A_j(t, s) \\ &= \overline{\Delta L}(r)(t) + \alpha \cdot \overline{\Delta L}(s)(t) \end{aligned}$$

proving, as $t \in X$ was chosen arbitrary, that $\overline{\Delta L}(y) = \overline{\Delta L}(r) + \alpha \cdot \overline{\Delta L}(s)$ or

$$\begin{aligned} \overline{\Delta L}(x_1, \dots, x_{i-1}, u + \alpha \cdot v, x_{i+1}, \dots, x_n) &= \\ \overline{\Delta L}(x_1, \dots, x_{i-1}, u, x_{i+1}, \dots, x_n) + \alpha \cdot \overline{\Delta L}(x_1, \dots, x_{i-1}, v, x_{i+1}, \dots, x_n) \end{aligned}$$

This finally proves that

$$\overline{\Delta L} \in \text{Hom}^n(X; \text{Hom}(X, X))$$

Next we have to prove that $\overline{\Delta L}$ is skew-symmetric. Let $t \in X$ and $x = (x_1, \dots, x_n) \in X^n$ such that $\exists (k, l) \in \{1, \dots, n\}$ satisfying $k \neq l$ and $x_k = x_l$. We may always assume that $k < l$ [otherwise exchange k and l]. Consider now the cases for $i, j \in \{1, \dots, n\}$

$j \neq k \wedge j \neq l$. Then we have that following sub cases to consider for j :

$j < k$. Then we have $j + 1 \leq k < l$ hence

$$\begin{aligned} (t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n))_k & \stackrel{[\text{definition: } 11.302]}{=} \stackrel{j+1 \leq k}{=} \\ & L(x_k) \\ & \stackrel{x_k = x_l}{=} \\ & L(x_l) \\ & \stackrel{[\text{definition: } 11.302]}{=} \stackrel{j+1 < l}{=} \\ (t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n))_l \end{aligned}$$

which as $k \neq l$ proves by [theorem: 11.291] that

$$\Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) = 0$$

$k < j < l$. Then we have $1 < k + 1 \leq j$ and $j + 1 \leq l$ so that

$$\begin{aligned} (t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n))_{k+1} & \stackrel{[\text{definition: } 11.302]}{=} \stackrel{2 \leq k+1 \leq j}{=} \\ & L(x_{(k+1)-1}) \\ & \stackrel{=}{=} \\ & L(x_k) \\ & \stackrel{x_k = x_l}{=} \\ & L(x_l) \\ & \stackrel{[\text{definition: } 11.302]}{=} \stackrel{j+1 \leq l}{=} \\ (t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n))_l \end{aligned}$$

which as $k < j < l \Rightarrow k + 1 \leq j < l \Rightarrow k + 1 \neq l$ proves by [theorem: 11.291] that

$$\Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) = 0$$

$l < j$. Then we have $1 < k + 1 \leq j$ and $1 < l + 1 \leq j$ so that

$$\begin{aligned} (t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n))_{k+1} & \stackrel{[\text{definition: } 11.302]}{=} \stackrel{k+1 \leq j}{=} \\ & L(x_{(k+1)-1}) \\ & \stackrel{=}{=} \\ & L(x_k) \\ & \stackrel{x_k = x_l}{=} \\ & L(x_l) \\ & \stackrel{=}{=} \\ & L(x_{(l+1)-1}) \\ & \stackrel{[\text{definition: } 11.302]}{=} \stackrel{2 \leq l+1 < j}{=} \\ (t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n))_{k+1} \end{aligned}$$

which as $k \neq l \Rightarrow k + 1 \neq l + 1$ proves by [theorem: 11.291] that

$$\Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) = 0$$

So in all cases we have

$$\Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) = 0 \quad (11.216)$$

$j = k \vee j = l$. For $k < l$ we have by [lemma: 11.318] and $L(x_k) = L(x_l)$ that

$$\begin{aligned} (t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n)) \circ (k \rightsquigarrow_n l + 1) & = \\ (t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) \end{aligned}$$

so that

$$\begin{aligned}
& \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) &= \\
& \Delta((t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n)) \circ (k \rightsquigarrow_n l+1)) &= \\
& ((k \rightsquigarrow_n l+1)\Delta)(t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n)) &= \\
& \text{sign}((k \rightsquigarrow_n l+1)) \cdot \Delta(t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n)) &\stackrel{[\text{theorem: 11.255}]}{=} \\
& (-1)^{l+1-k} \cdot \Delta(t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n))
\end{aligned}$$

So by [proposition: 11.281] we have that

$$\begin{aligned}
& \Delta(t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n)) = \\
& (-1)^{l+1-k} \cdot \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n))
\end{aligned} \tag{11.217}$$

Now

$$\begin{aligned}
& \sum_{j \in \{1, \dots, n\}} (-1)^j \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j &= \\
& \overline{\Delta L}(x_1, \dots, x_n)(t) &\stackrel{[\text{theorem: 11.43}]}{=} \\
& A + B + C
\end{aligned} \tag{11.218}$$

where

$$\begin{aligned}
A &= \sum_{j \in \{1, \dots, n\} \setminus \{k, l\}} (-1)^{j-1} \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\
&\stackrel{[\text{theorem: 11.216}]}{=} \sum_{j \in \{1, \dots, n\} \setminus \{k, l\}} (-1)^{j-1} \cdot 0 \cdot x_j \\
&= 0 \\
B &= \sum_{j \in \{k\}} (-1)^{j-1} \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\
&\quad (-1)^{k-1} \cdot \Delta(t, L(x_1), \dots, L(x_{k-1}), L(x_{k+1}), \dots, L(x_n)) \cdot x_k \\
&\stackrel{[\text{eq: 11.217}]}{=} (-1)^{k-1} \cdot (-1)^{l+1-k} \cdot \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) \cdot x_k \\
&= (-1)^l \cdot \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) \cdot x_k \\
&= (-1) \cdot (-1)^{l-1} \cdot \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) \cdot x_k \\
&\stackrel{[x_k = x_l]}{=} (-1) \cdot (-1)^{l-1} \cdot \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) \cdot x_l \\
C &= \sum_{j \in \{l\}} (-1)^{j-1} \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\
&= (-1)^{l-1} \cdot \Delta(t, L(x_1), \dots, L(x_{l-1}), L(x_{l+1}), \dots, L(x_n)) \cdot x_l \\
&= -B
\end{aligned}$$

so that using [eq: 11.218] $\overline{\Delta L}(x_1, \dots, x_n)(t) = 0$, which, as $t \in X$ was chosen arbitrary, that

$$\overline{\Delta L}(x_1, \dots, x_n) = C_0 \text{ [where } C_0 \text{ is the neutral element in } \text{Hom}(X, X)]$$

$$\overline{\Delta L} \text{ is skew-symmetric}$$

□

The above theorem allows us to define the adjoint of a linear map.

Definition 11.320. Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero such that $\dim(X) = n$, $L \in \text{Hom}(X, X)$ a linear transformation. then the function **adjoint** is defined by

$$\text{adjoint: } \text{Hom}(X, X) \rightarrow \text{Hom}(X, X)$$

where $\text{adjoint}(L) \in \text{Hom}(X, X)$ is such that for every non trivial determinant function Δ we have $\forall x \in X^n$ that

$$\overline{\Delta L}(x) = \Delta(x) \cdot \text{adjoint}(L)$$

Proof. Of course we must prove that given a $L \in \text{Hom}(X, X)$ there exist a $y \in \text{Hom}(X, X)$ such that $\overline{\Delta L}(x) = \Delta(x) \cdot \text{adjoint}(L)$ and that y is independent of the chosen Δ .

existence. Let $L \in \text{Hom}(X, X)$ then by the previous theorem [theorem: 11.319]

$$\overline{\Delta L} \in \text{Hom}^n(X; \text{Hom}(X, X))$$

is a skew-symmetric n -linear mapping. Using [theorem: 11.300] to get a $y \in \text{Hom}(X, X)$ such that

$$\forall x \in X^n \text{ we have } \overline{\Delta L}(x) = \Delta(x) \cdot y$$

uniqueness. Let Δ' another non trivial determinant function and $y' \in \text{Hom}(X, X)$ such that

$$\forall x \in X^n \text{ we have } \overline{\Delta' L}(x) = \Delta'(x) \cdot y'$$

Using [theorem: 11.301] there exists a $\lambda \in F$ such that

$$\Delta' = \lambda \cdot \Delta$$

Then we have $\forall x \in X^n, t \in X$ that

$$\begin{aligned} (\Delta'(x) \cdot y')(t) &= (\overline{\Delta' L}(x))(t) \\ &= \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot \Delta'(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\ &= \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot \lambda \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\ &= \lambda \cdot \sum_{j \in \{1, \dots, n\}} (-1)^{j-1} \cdot \Delta(t, L(x_1), \dots, L(x_{j-1}), L(x_{j+1}), \dots, L(x_n)) \cdot x_j \\ &= \lambda \cdot (\overline{\Delta L}(x))(t) \\ &= (\lambda \cdot (\Delta(x) \cdot y))(t) \\ &= ((\lambda \cdot \Delta)(x) \cdot y)(t) \\ &= (\Delta'(x) \cdot y)(t) \end{aligned}$$

which, as $t \in X$ is arbitrary chosen, proves that

$$\forall x \in X^n \text{ we have } \Delta'(x) \cdot y' = \Delta'(x) \cdot y$$

As Δ' is non trivial there exists a $x \in X$ such that $\Delta'(x) \neq 0$ which combined with the above proves that

$$y = y' \quad \square$$

Example 11.321. If X is a one dimensional space with basis $\{e_1\}$ then if $L \in \text{Hom}(X, X)$ we have

1. $\text{adjoint}(L) = \text{Id}_X$
2. $\text{adjoint}(L) \circ L = \det(L) \cdot \text{Id}_X$
3. $L \circ \text{adjoint}(L) = \det(L) \cdot \text{Id}_X$

Proof. If $(x_1) \in X^1$ then there exists a $\alpha \in F$ such that $x_1 = \alpha \cdot e_1$, further by [example: 11.314] we have that $L(e_1) = \det(L) \cdot e_1$ and if $t \in X$ then there exists a $\beta \in F$ such that $t = \beta \cdot e_1$. Let Δ be a determinant function then

$$\begin{aligned} \overline{\Delta L}(x_1)(t) &= \sum_{i \in \{1\}} (-1)^{i-1} \cdot \Delta(t) \cdot x_i \\ &= \Delta(t) \cdot x_1 \\ &= \Delta(\beta \cdot e_1) \cdot (\alpha \cdot e_1) \\ &= \alpha \cdot \beta \cdot \Delta(e_1) \cdot e_1 \\ &= \Delta(\alpha \cdot e_1) \cdot \beta \cdot e_1 \\ &= \Delta(x_1) \cdot t \\ &= \Delta(x_1) \cdot \text{Id}_X(t) \\ &= (\Delta(x_1) \cdot \text{Id}_X)(t) \end{aligned}$$

so that $\overline{\Delta L}(x_1) = \Delta(x_1) \cdot \text{Id}_X$, hence by definition

$$\text{adjoint}(l) = \text{Id}_X.$$

Next we have

1. $\text{adjoint}(L(x)) = \text{Id}_X(L(x)) = L(\alpha \cdot e_1) = \alpha \cdot L(e_1) = \alpha \cdot \det(L) \cdot e_1 = \det(L) \cdot (\alpha \cdot e_1) = \det(L) \cdot x = \det(L) \cdot \text{Id}_X(x)$ proving that

$$\text{adjoint}(L) \circ L = \det(L) \circ L$$

2. $L(\text{adjoint}(x)) = L(\text{Id}_X(x)) = L(x) = \alpha \cdot L(e_1) = \alpha \cdot \det(L) \cdot e_1 = \det(L) \cdot (\alpha \cdot e_1) = \det(L) \cdot x = \det(L) \cdot \text{Id}_X(x)$ proving that

$$L \circ \text{adjoint}(L) = \det(L) \cdot \text{Id}_X \quad \square$$

Actually the above example points to a more general result as is expressed in the following theorem.

Lemma 11.322. *Let $n \in \mathbb{N}$, X a vector space over a field F of characteristic zero such that $\dim(X) = n$ and $L \in \text{Hom}(X, X)$ then we have*

1. $\text{adjoint}(L) \circ L = \det(L) \cdot \text{Id}_X$
2. $L \circ \text{adjoint}(L) = \det(L) \cdot \text{Id}_X$

Proof. Let $\{e_1, \dots, e_n\}$ be a basis for X and let Δ be the non trivial determinant function such that $\Delta(e_1, \dots, e_n) = 1$ [see theorem: 11.299].

1. Let $x \in X$ then we have

$$\begin{aligned} &(\text{adjoint}(L) \circ L)(x) &&= \\ &\text{adjoint}(L)(L(x)) &&= \\ &(1 \cdot \text{adjoint}(L))(L(x)) &&= \\ &(\Delta(e_1, \dots, e_n) \cdot \text{adjoint}(L))(L(x)) &&\stackrel{[\text{definition: 11.320}]}{=} \\ &\overline{\Delta L}(e_1, \dots, e_n)(L(x)) &&\stackrel{[\text{definition: 11.317}]}{=} \\ &\sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(L(x), L(e_1), \dots, L(e_{i-1}), L(e_{i+1}), \dots, L(e_n)) \cdot e_i &&= \\ &\sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta_L(x, e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_n) \cdot e_i &&\stackrel{[\text{theor., def: 11.308, 11.304}]}{=} \\ &\underline{\Delta L}(x, (e_1, \dots, e_n)) &&\stackrel{[\text{theorem: 11.306}]}{=} \\ &\Delta_L(e_1, \dots, e_n) \cdot x &&\stackrel{[\text{theorem: 11.311}]}{=} \\ &(\det(L) \cdot \Delta(e_1, \dots, e_n)) \cdot x &&= \\ &\det(L) \cdot x &&= \\ &\det(L) \cdot \text{Id}_X(x) \end{aligned}$$

proving that

$$\text{adjoint}(L) \circ L = \det(L) \cdot \text{Id}_X$$

2. Let $x \in X$ then we have

$$\begin{aligned}
 & (L \circ \text{adjoint}(L))(x) &= \\
 & L(\text{adjoint}(L)(x)) &= \\
 & L(1 \cdot \text{adjoint}(L)(x)) &= \\
 & L(\Delta(e_1, \dots, e_n) \cdot \text{adjoint}(L)(x)) &= \text{[theorem: 11.320]} \\
 & L(\overline{\Delta L}(e_1, \dots, e_n)(x)) &= \text{[definition: 11.317]} \\
 & L\left(\sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(x, L(e_1), \dots, L(e_{i-1}), L(e_{i+1}), \dots, L(e_n)) \cdot e_i\right) &= \\
 & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(x, L(e_1), \dots, L(e_{i-1}), L(e_{i+1}), \dots, L(e_n)) \cdot L(e_i) &= \text{[def: 11.304]} \\
 & \Delta(x, (L(e_1), \dots, L(e_n))) &= \text{[theorem: 11.306]} \\
 & \Delta(L(e_1), \dots, L(e_n)) \cdot x &= \\
 & \Delta_L(e_1, \dots, e_n) \cdot x &= \text{[theorem: 11.311]} \\
 & (\det(L) \cdot \Delta(e_1, \dots, e_n)) \cdot x &= \\
 & \det(L) \cdot x &= \\
 & \det(L) \cdot \text{Id}_X(x) &=
 \end{aligned}$$

proving that

$$L \circ \text{adjoint}(L) = \det(L) \cdot \text{Id}_X \quad \square$$

The above lemma leads to a theorem that allows us to calculate the inverse of a linear mapping.

Theorem 11.323. *Let $n \in \mathbb{N}$, X a vector space over a field F of characteristic zero such that $\dim(X) = n$ and $L \in \text{Hom}(X, X)$ then we have:*

1. L is a isomorphism $\Leftrightarrow \det(L) \neq 0$
2. If L is a isomorphism then $L^{-1} = (\det(L))^{-1} \cdot \text{adjoint}(L)$

Proof.

1. This is already proved in [theorem: 11.315].
2. Using the previous lemma [lemma: 11.322] we have

$$\begin{aligned}
 L \circ \text{adjoint}(L) &= \det(L) \cdot \text{Id}_X &\Rightarrow & \text{adjoint}(L) = L^{-1} \circ (\det(L) \cdot \text{Id}_X) \\
 & &\Rightarrow & \text{adjoint}(L) = \det(L) \cdot L^{-1} \circ \text{Id}_X \\
 & &\Rightarrow & \text{adjoint}(L) = \det(L) \cdot L^{-1} \\
 & &\Rightarrow_{\det(L) \neq 0} & (\det(L))^{-1} \cdot \text{adjoint}(L) = L^{-1} \\
 & && \square
 \end{aligned}$$

11.8 Matrices

In finite dimensional vector spaces a linear mapping can be represented by a matrix. Composition of linear mappings becomes then multiplication of matrices. Further the adjoint and determinant of a linear mapping can be translated to the adjoint and determinant of matrices. One disadvantage of this approach is that the matrix representation of a linear mapping is dependent on the chosen basis and that this approach works only in finite dimensional vector spaces. However one big advantage is that all calculations can then be done using essential multiplication and addition in a field (in most cases the real or complex numbers). This is the main reason for the existence of this section.

11.8.1 Definition and properties

Definition 11.324. Let F be a field, $n, m \in \mathbb{N}$ then a $n \times m$ **matrix** in F is the graph of a mapping from $\{1, \dots, n\} \times \{1, \dots, m\}$ to F . The set of all $n \times m$ matrices is called $\mathcal{M}_{n,m}(F)$ or $\mathcal{M}_{n,m}$ if F is clear from the context. Hence

$$\mathcal{M}_{n,m}(F) = F^{\{1, \dots, n\} \times \{1, \dots, m\}} \stackrel{[\text{definition: 2.32}]}{=} \{M \mid M: \{1, \dots, n\} \times \{1, \dots, m\} \rightarrow F\}$$

As $1 \times n$ matrix is called a **row vector** and a $n \times 1$ matrix is called a **column vector**. Let $M \in \mathcal{M}_{n,m}(F)$ and $i, j \in \{1, \dots, n\} \times \{1, \dots, m\}$ then we use the following notation

$$M(i, j) \stackrel{\text{notation}}{=} M_{i,j}$$

Another notation for $M \in \mathcal{M}_{n,m}(F)$ is

$$M = \begin{pmatrix} M_{1,1} & \dots & M_{1,m} \\ \vdots & \ddots & \vdots \\ M_{n,1} & \dots & M_{n,m} \end{pmatrix}$$

For a row vector $R \in \mathcal{M}_{1,n}(F)$ we use the notation

$$R = (R_1 \quad \dots \quad R_n)$$

and for a column vector $C \in \mathcal{M}_{n,1}$ we use the notation

$$C = \begin{pmatrix} C_1 \\ \vdots \\ C_n \end{pmatrix}$$

In essence a $n \times m$ matrix can be said to be a collection of n $1 \times m$ row vectors or a collection of m $n \times 1$ column vectors. Elements of $\mathcal{M}_{n,n}$ [matrices where the number of column vectors is equal to the number of row vectors] are called **square matrices**.

Example 11.325. Let $n, m \in \mathbb{N}$, F be a field then $E \in \mathcal{M}_{n,m}(F)$ is defined by

$$\forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \text{ we have } E_{i,j} = \delta_{i,j} \text{ [see definition: 11.164]}$$

this matrix is called the **identity matrix**. So

$$E = \begin{pmatrix} 1 & 0 & \dots & 0 & 1 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & 0 & \ddots & 0 & 0 \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix}$$

Example 11.326. Let $n, m \in \mathbb{N}$, F a field with neutral element 0 then $0 \in \mathcal{M}_{n,m}(F)$ is defined by

$$\forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \text{ we have } 0_{i,j} = 0$$

this matrix is called the **null matrix**. SO

$$0 = \begin{pmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{pmatrix}$$

We define now the different arithmetic operations on matrices.

Definition 11.327. Let $n, m \in \mathbb{N}$ then we define

1. **(sum)** $+: \mathcal{M}_{n,m}(F) \times \mathcal{M}_{n,m}(F) \rightarrow \mathcal{M}_{n,m}(F)$ is defined by

$$A + B \in \mathcal{M}_{n,m}(F) \text{ where } \forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \quad (A + B)_{i,j} = A_{i,j} + B_{i,j}$$

2. **(scalar product)** $\cdot : F \times \mathcal{M}_{n,m}(F) \rightarrow \mathcal{M}_{n,m}(F)$ is defined by

$$\alpha \cdot A \in \mathcal{M}_{n,m}(F) \text{ where } \forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \quad (\alpha \cdot A)_{i,j} = \alpha \cdot A_{i,j}$$

3. **(inner product)** If $k \in \mathbb{N}$ then we define $\cdot : \mathcal{M}_{n,m}(F) \times \mathcal{M}_{m,k}(F) \rightarrow \mathcal{M}_{n,k}(F)$ by

$$A \cdot B \in \mathcal{M}_{n,k}(F) \text{ where } (A \cdot B)_{i,j} = \sum_{r \in \{1, \dots, m\}} A_{i,r} \cdot B_{r,j}$$

4. **(null matrix)** $0 \in \mathcal{M}_{n,m}(F)$ is defined by $\forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \quad 0_{i,j} = 0$

5. **(identity matrix)** $E \in \mathcal{M}_{n,m}(F)$ is defined by $\forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \quad E_{i,j} = \delta_{i,j}$

6. **(additive inverse)** If $M \in \mathcal{M}_{n,m}(F)$ then $-M$ is defined by $\forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\} \quad (-M)_{i,j} = -M_{i,j}$

Example 11.328. Then we have

$$\begin{aligned} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 4 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 2 & 6 \\ 4 & 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 4 & 9 \\ 8 & 5 & 3 \end{pmatrix} \\ 10 \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 4 & 2 \end{pmatrix} &= \begin{pmatrix} 10 & 20 & 30 \\ 40 & 40 & 20 \end{pmatrix} \\ \begin{pmatrix} 1 & 2 & 3 \\ 4 & 4 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 3 \\ 2 & 0 \end{pmatrix} &= \begin{pmatrix} 7 & 8 \\ 8 & 20 \end{pmatrix} \end{aligned}$$

Theorem 11.329. Let $n, m \in \mathbb{N}$ and F a field then, $\langle \mathcal{M}_{n,m}(F), +, \cdot \rangle$ is a vector space over the field F [using the operations in [definition: 11.327]]

Proof. First we prove the group axioms:

associativity. Let $A, B, C \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned} ((A + B) + C)_{i,j} &= (A + B)_{i,j} + C_{i,j} \\ &= (A_{i,j} + B_{i,j}) + C_{i,j} \\ &= A_{i,j} + (B_{i,j} + C_{i,j}) \\ &= A_{i,j} + (B + C)_{i,j} \\ &= (A + (B + C))_{i,j} \end{aligned}$$

proving that

$$(A + B) + C = A + (B + C)$$

commutativity. Let $A, B \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned} (A + B)_{i,j} &= A_{i,j} + B_{i,j} \\ &= B_{i,j} + A_{i,j} \\ &= (B + A)_{i,j} \end{aligned}$$

proving that

$$A + B = B + A$$

neutral element. Let $A \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$(A + 0)_{i,j} = A_{i,j} + 0_{i,j} = A_{i,j} + 0 = A_{i,j}$$

proving that

$$0 + A \underset{\text{commutativity}}{=} A + 0 = A$$

inverse element. Let $A \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$(A + (-A))_{i,j} = A_{i,j} + (-A)_{i,j} = A_{i,j} + (-A_{i,j}) = 0 = 0_{i,j}$$

proving that

$$(-A) + A \underset{\text{commutativity}}{=} A + (-A) = 0$$

Next we check the rest of the vector space axioms:

1. Let $A, B \in \mathcal{M}_{n,m}(F)$, $\alpha \in F$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned} (\alpha \cdot (A + B))_{i,j} &= \alpha \cdot (A + B)_{i,j} \\ &= \alpha \cdot (A_{i,j} + B_{i,j}) \\ &= \alpha \cdot A_{i,j} + \alpha \cdot B_{i,j} \\ &= (\alpha \cdot A)_{i,j} + (\alpha \cdot B)_{i,j} \\ &= (\alpha \cdot A + \alpha \cdot B)_{i,j} \end{aligned}$$

proving that

$$\alpha \cdot (A + B) = \alpha \cdot A + \alpha \cdot B$$

2. Let $A \in \mathcal{M}_{n,m}(F)$, $\alpha, \beta \in F$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned} ((\alpha + \beta) \cdot A)_{i,j} &= (\alpha + \beta) \cdot A_{i,j} \\ &= \alpha \cdot A_{i,j} + \beta \cdot A_{i,j} \\ &= (\alpha \cdot A)_{i,j} + (\beta \cdot A)_{i,j} \\ &= (\alpha \cdot A + \beta \cdot A)_{i,j} \end{aligned}$$

proving that

$$(\alpha + \beta) \cdot A = \alpha \cdot A + \beta \cdot A$$

3. Let $A \in \mathcal{M}_{n,m}(F)$, $\alpha, \beta \in F$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned} ((\alpha \cdot \beta) \cdot A)_{i,j} &= (\alpha \cdot \beta) \cdot A_{i,j} \\ &= \alpha \cdot (\beta \cdot A_{i,j}) \\ &= \alpha \cdot (\beta \cdot A)_{i,j} \\ &= (\alpha \cdot (\beta \cdot A))_{i,j} \end{aligned}$$

proving that

$$(\alpha \cdot \beta) \cdot A = \alpha \cdot (\beta \cdot A)$$

4. Let $A \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$(1 \cdot A)_{i,j} = 1 \cdot A_{i,j} = A_{i,j} = A_{i,j}$$

proving that

$$1 \cdot A = A$$

□

Theorem 11.330. Let $n \in \mathbb{N}$ and F a field then $\langle \mathcal{M}_{n,n}(F), \cdot \rangle$ is a semi-group with as neutral element the identity matrix E .

Proof.

associativity. Let $A, B, C \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned}
 ((A \cdot B) \cdot C)_{i,j} &= \sum_{k \in \{1, \dots, n\}} (A \cdot B)_{i,k} \cdot C_{k,j} \\
 &= \sum_{k \in \{1, \dots, n\}} \left(\sum_{l \in \{1, \dots, n\}} A_{i,l} \cdot B_{l,k} \right) \cdot C_{k,j} \\
 &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{k \in \{1, \dots, n\}} \left(\sum_{l \in \{1, \dots, n\}} (A_{i,l} \cdot B_{l,k}) \cdot C_{k,j} \right) \\
 &\stackrel{[\text{theorem: 11.41}]}{=} \sum_{l \in \{1, \dots, n\}} \left(\sum_{k \in \{1, \dots, n\}} (A_{i,l} \cdot B_{l,k}) \cdot C_{k,j} \right) \\
 &= \sum_{l \in \{1, \dots, n\}} \left(\sum_{k \in \{1, \dots, n\}} A_{i,l} \cdot (B_{l,k}) \cdot C_{k,j} \right) \\
 &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{l \in \{1, \dots, n\}} A_{i,l} \cdot \left(\sum_{k \in \{1, \dots, n\}} (B_{l,k}) \cdot C_{k,j} \right) \\
 &= \sum_{l \in \{1, \dots, n\}} A_{i,l} \cdot (B \cdot C)_{l,j} \\
 &= (A \cdot (B \cdot C))_{i,j}
 \end{aligned}$$

proving that

$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$

neutral element. Let $A \in \mathcal{M}_{n,m}(F)$ then for $(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ we have

$$\begin{aligned}
 (A \cdot E)_{i,j} &= \sum_{k \in \{1, \dots, n\}} A_{i,k} \cdot E_{k,j} \\
 &= \sum_{k \in \{1, \dots, n\}} A_{i,k} \cdot \delta_{k,j} \\
 &= A_{i,j} \\
 (E \cdot A)_{i,j} &= \sum_{k \in \{1, \dots, n\}} E_{i,k} \cdot A_{k,j} \\
 &= \sum_{k \in \{1, \dots, n\}} \delta_{i,k} \cdot A_{k,j} \\
 &= A_{i,j}
 \end{aligned}$$

proving that

$$A \cdot E = A = E \cdot A$$

□

Note 11.331. That the inner product is not commutative as the following example shows:

$$\begin{aligned}
 \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \cdot \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} &= \begin{pmatrix} 10 & 4 \\ 16 & 6 \end{pmatrix} \\
 \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} &= \begin{pmatrix} 6 & 16 \\ 4 & 10 \end{pmatrix}
 \end{aligned}$$

Also there exist non zero matrices that do not have a inverse, for example

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

and we have

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

so that $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has no inverse.

We define now the transpose of a matrix, which is essential the operation that interchange the column vectors and row vectors of a matrix.

Definition 11.332. Let $n, m \in \mathbb{N}$ and $M \in \mathcal{M}_{n,m}(F)$ then $M^T \in \mathcal{M}_{m,n}(F)$ is defined by

$$\forall (i, j) \in \{1, \dots, m\} \times \{1, \dots, n\} \quad (M^T)_{i,j} = M_{j,i}$$

The following definitions allows use to extract the rows and columns from a matrix and introduces the important concept of row (column) rank of a matrix that eventually leads to the important concept of the rank from a matrix.

Definition 11.333. Let F be a field, $n, m \in \mathbb{N}$ then we define:

1. $\text{row}: \mathcal{M}_{n,m}(F) \times \{1, \dots, n\} \rightarrow F^m$ where $\text{row}(M, i) \in F^m$ is defined by $(\text{row}(M, i))_j = M_{i,j}$ $\forall j \in \{1, \dots, m\}$. In other words this function extract the i -the row from M .
2. $\text{col}: \mathcal{M}_{n,m}(F) \times \{1, \dots, m\} \rightarrow F^n$ where $\text{col}(M, i) \in F^n$ is defined by $(\text{col}(M, i))_j = M_{j,i}$ $\forall j \in \{1, \dots, n\}$. In other words this function extracts the i -the column from M .
3. Let $M \in \mathcal{M}_{n,m}(F)$ then the set of rows of M is noted as $\text{rows}(M)$, hence

$$\text{rows}(M) = \{\text{row}(M, i) | i \in \{1, \dots, n\}\} \subseteq F^m$$

4. Let $M \in \mathcal{M}_{n,m}(F)$ then the set of columns of M is noted as $\text{cols}(M)$, hence:

$$\text{cols}(M) = \{\text{col}(M, i) | i \in \{1, \dots, m\}\} \subseteq F^n$$

As F^n, F^m are vector spaces over F with $\dim(F^n) = n$, $\dim(F^m) = m$ [see example: 11.167], and, given $M \in \mathcal{M}_{n,m}(F)$, $\text{span}(\text{rows}(M))$ and $\text{span}(\text{cols}(M))$ are sub spaces of F^m and F^n , we have that $\dim(\text{span}(\text{rows}(M)))$ and $\dim(\text{span}(\text{cols}(M)))$ are finite with $\dim(\text{span}(\text{rows}(M))) \leq m$ and $\dim(\text{span}(\text{cols}(M))) \leq n$ [see theorems: 11.110, 11.159]. This allows us to define:

1. If $M \in \mathcal{M}_{n,m}(F)$ then the row rank of M noted as $\text{rrank}(M)$ is defined by

$$\text{rrank}(M) = \dim(\text{span}(\text{rows}(M))) \leq m$$

2. If $M \in \mathcal{M}_{n,m}(F)$ then the column rank of M noted as $\text{crank}(M)$ is defined by

$$\text{crank}(M) = \dim(\text{span}(\text{cols}(M))) \leq n$$

It turns out that the column rank and row rank of a matrix are equal, which leads to the definition of the rank of a matrix.

Theorem 11.334. Let F a field, $n, m \in \mathbb{N}$ and $M \in \mathcal{M}_{n,m}(F)$ then

$$\text{rrank}(M) = \text{crank}(M)$$

Proof. Let

$$r = \text{rrank}(M) = \dim(\text{span}(\text{rows}(M))) \leq m \text{ and } c = \text{crank}(M) = \dim(\text{span}(\text{cols}(M))) \leq n$$

As $r = \dim(\text{span}(\text{rows}(M)))$ there exist a basis $E = \{e_1, \dots, e_r\} \subseteq \text{span}(\text{rows}(M)) \subseteq F^m$. Let $i \in \{1, \dots, n\}$ then there exist a $\{\lambda_{i,j}\}_{j \in \{1, \dots, r\}} \subseteq F$ such that

$$\text{row}(M, i) = \sum_{j \in \{1, \dots, r\}} \lambda_{i,j} \cdot e_j \tag{11.219}$$

now let $k \in \{1, \dots, m\}$ then

$$\begin{aligned}
 \text{col}(M, k)_i &= M_{i,k} \\
 &= \text{row}(M, i)_k \\
 &\stackrel{[\text{eq: 11.219}]}{=} \left(\sum_{j \in \{1, \dots, r\}} \lambda_{i,j} \cdot e_j \right)_k \\
 &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{j \in \{1, \dots, r\}} \lambda_{i,j} \cdot (e_j)_k \\
 &= \sum_{j \in \{1, \dots, r\}} (e_j)_k \cdot \lambda_{i,j}
 \end{aligned} \tag{11.220}$$

Define now $\{f_i\}_{i \in \{1, \dots, r\}} \subseteq F^n$ by $(f_i)_l = \lambda_{l,i} \forall l \in \{1, \dots, n\}$ then by [eq: 11.220] we have

$$\text{col}(M, k)_i = \sum_{j \in \{1, \dots, r\}} (e_j)_k \cdot (f_j)_i$$

so that

$$\text{col}(M, k) = \sum_{j \in \{1, \dots, r\}} (e_j)_k \cdot f_j$$

proving using [theorem: 11.107] that $\text{col}(M, k) \in \text{span}(\{f_i | i \in \{1, \dots, r\}\})$. Hence

$$\text{cols}(M) \subseteq \text{span}(\{f_i | i \in \{1, \dots, r\}\})$$

so that by [theorems: 11.109, 11.112]

$$\text{span}(\text{cols}(M)) \subseteq \text{span}(\{f_i | i \in \{1, \dots, r\}\})$$

Using [theorem: 11.159]

$$\dim(\text{span}(\text{cols}(M))) \leq \dim(\text{span}(\{f_i | i \in \{1, \dots, r\}\}))$$

As $\text{span}(\{f_i | i \in \{1, \dots, r\}\})$ is a vector space we have by [theorem: 11.154] that

$$\dim(\text{span}(\{f_i | i \in \{1, \dots, r\}\})) \leq r = \text{rrank}(M)$$

giving

$$\text{crank}(M) = \dim(\text{span}(\text{cols}(M))) \leq \text{rrank}(M) \tag{11.221}$$

Now we use the same reasoning to prove the opposite equation. As $c = \dim(\text{span}(\text{cols}(M)))$ there exist a basis $G = \{g_1, \dots, g_c\}$ of $\text{span}(\text{cols}(M))$. Let $i \in \{1, \dots, m\}$ then there exists a $\{\alpha_{i,j}\}_{j \in \{1, \dots, c\}} \subseteq F$ such that

$$\text{col}(M, i) = \sum_{j \in \{1, \dots, c\}} \alpha_{i,j} \cdot g_j \tag{11.222}$$

Now let $k \in \{1, \dots, n\}$ then we have

$$\begin{aligned}
 (\text{row}(M, k))_i &= M_{k,i} \\
 &= (\text{col}(M, i))_k \\
 &\stackrel{[\text{eq: 11.222}]}{=} \left(\sum_{j \in \{1, \dots, c\}} \alpha_{i,j} \cdot g_j \right)_k \\
 &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{j \in \{1, \dots, c\}} \alpha_{i,j} \cdot (g_j)_k \\
 &= \sum_{j \in \{1, \dots, c\}} (g_j)_k \cdot \alpha_{i,j}
 \end{aligned} \tag{11.223}$$

Define now $\{h_i\}_{i \in \{1, \dots, c\}} \subseteq F^m$ by $(h_i)_l = \alpha_{l,i} \forall l \in \{1, \dots, m\}$ then by [eq: 11.223] we have

$$(\text{row}(M, k))_i = \sum_{j \in \{1, \dots, c\}} (g_j)_k \cdot (h_j)_i$$

so that

$$\text{row}(M, k) = \sum_{j \in \{1, \dots, c\}} (g_j)_k \cdot h_j$$

proving, using [theorem: 11.107], that $\text{row}(M, k) \in \text{span}(\{h_i | i \in \{1, \dots, c\}\})$. Hence

$$\text{rows}(M) \subseteq \text{span}(\{h_i | i \in \{1, \dots, c\}\})$$

so that by that by [theorems: 11.109, 11.112]

$$\text{span}(\text{rows}(M)) \subseteq \text{span}(\{h_i | i \in \{1, \dots, c\}\})$$

Using [theorem: 11.159] we have:

$$\dim(\text{span}(\text{rows}(M))) \leq \dim(\text{span}(\{h_i | i \in \{1, \dots, c\}\}))$$

By [theorem: 11.154] it follows that

$$\dim(\text{span}(\{h_i | i \in \{1, \dots, c\}\})) \leq c.$$

So we have

$$\text{rrank}(M) = \dim(\text{span}(\text{rows}(M))) \leq c = \text{crank}(M)$$

combining the above with [eq: 11.221] proves

$$\text{rrank}(M) = \text{crank}(M)$$

□

The above theorem let us define the rank of a matrix.

Definition 11.335. Let $n, m \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,m}(F)$ then

$$\text{rank}(M) = \text{rrank}(M) \stackrel{[\text{theorem: 11.334}]}{=} \text{crank}(M)$$

Note: As by [definition: 11.333] $\text{rrank}(M) \leq m$ and $\text{crank}(M) \leq n$ it follows that

$$\text{rank}(M) \leq \min(n, m)$$

11.8.2 Matrices and linear mappings

First note that a linear mapping between finite dimensional spaces is uniquely determined by its values on the basis vectors of the domain.

Theorem 11.336. Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field F with bases $\{e_1, \dots, e_n\} \subseteq X, \{f_1, \dots, f_m\} \subseteq Y$ then if $L_1, L_2 \in \text{Hom}(X, Y)$ with $\forall i \in \{1, \dots, n\} L_1(e_i) = L_2(e_i)$ then $L_1 = L_2$.

Proof. Let $x \in X$ then there exists a unique family $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$x = \sum_{i \in \{1, \dots, n\}} x_i e_i$$

Then

$$\begin{aligned} L_1(x) &= L_1\left(\sum_{i \in \{1, \dots, n\}} x_i e_i\right) \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot L_1(e_i) \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot L_2(e_i) \\ &= L_2\left(\sum_{i \in \{1, \dots, n\}} x_i e_i\right) \\ &= L_2(x) \end{aligned}$$

proving that $L_1 = L_2$. □

Theorem 11.337. *Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field F with bases $\{e_1, \dots, e_n\} \subseteq X$ and $\{l_i\}_{i \in \{1, \dots, n\}} \subseteq Y$ then there exist a unique $L \in \text{Hom}(X, Y)$ such that $\forall i \in \{1, \dots, n\}$ $L(e_i) = l_i$.*

Proof. Define $L: X \rightarrow Y$ by $L(x) = \sum_{i \in \{1, \dots, n\}} x_i \cdot l_i$ where $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ is the unique family such that $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$. Then if $y, z \in X$ and $\alpha \in F$ there exist unique $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq F$ and $\{z_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $y = \sum_{i \in \{1, \dots, n\}} y_i \cdot e_i$ and $z = \sum_{i \in \{1, \dots, n\}} z_i \cdot e_i$. Then

$$y + \alpha \cdot z = \sum_{i \in \{1, \dots, n\}} y_i \cdot e_i + \alpha \cdot \sum_{i \in \{1, \dots, n\}} z_i \cdot e_i = \sum_{i \in \{1, \dots, n\}} (y_i + \alpha \cdot z_i) \cdot e_i$$

so that

$$L(y + \alpha \cdot z) = \sum_{i \in \{1, \dots, n\}} (y_i + \alpha \cdot z_i) \cdot l_i = \sum_{i \in \{1, \dots, n\}} y_i \cdot l_i + \alpha \cdot \sum_{i \in \{1, \dots, n\}} z_i \cdot l_i = L(y) + \alpha \cdot L(z)$$

Hence

$$L \in \text{Hom}(X, Y)$$

Further as $\forall i \in \{1, \dots, n\}$ we have that $e_i = \sum_{k \in \{1, \dots, n\}} \delta_{k,i} \cdot e_k$ it follows that

$$L(e_i) = \sum_{k \in \{1, \dots, n\}} \delta_{k,i} \cdot l_k = l_i$$

Finally if there exist a $K \in \text{Hom}(X, Y)$ such that $\forall i \in \{1, \dots, n\}$ $K(e_i) = l_i$ then $\forall i \in \{1, \dots, n\}$ $K(e_i) = L(e_i)$ so that by [theorem: 11.336] $L = K$. □

Theorem 11.338. *Let $n \in \mathbb{N}$, X a finite dimensional vector space with two bases $\{e_1, \dots, e_n\}$, $\{f_1, \dots, f_n\}$ then there exist a linear isomorphism $L \in \text{Hom}(X, X)$ such that $\forall i \in \{1, \dots, n\}$ we have that $L(e_i) = f_i$ [hence $L^{-1}(f_i) = L^{-1}(L(e_i)) = e_i$].*

Proof. Using the previous theorem [theorem: 11.337] there exist a linear mapping $L \in \text{Hom}(X, X)$ such that $\forall i \in \{1, \dots, n\}$ $L(e_i) = f_i$. Let $x \in X$ such that $L(x) = 0$ then as $\{e_1, \dots, e_n\}$ is a basis there exist a $\{x_i\}_{i \in \{1, \dots, n\}}$ such that $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$. Hence

$$0 = L(x) = L\left(\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i\right) = \sum_{i \in \{1, \dots, n\}} x_i \cdot L(e_i) = \sum_{i \in \{1, \dots, n\}} x_i \cdot f_i$$

which as $\{f_1, \dots, f_n\}$ is linear dependent proves that $\forall i \in \{1, \dots, n\}$ $x_i = 0$. Hence $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i = 0$. This together [theorem: 11.171] proves that $\ker(L) = \{0\}$ so that by [theorem: 11.201] L is a isomorphism. □

We show now how to associate a matrix with a linear mapping between finite dimensional spaces.

Definition 11.339. *Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field F with basis $\{e_1, \dots, e_n\} \subseteq X$, $\{f_1, \dots, f_m\} \subseteq Y$ and $L \in \text{Hom}(X, Y)$. Then we define $\mathcal{M}(L; E, F) \in \mathcal{M}_{m,n}(F)$ to be the **unique** matrix such that*

$$\forall i \in \{1, \dots, n\} \quad L(e_i) = \sum_{j \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{j,i} \cdot f_j$$

[theorem: 11.143 ensures uniqueness and existence]. $\mathcal{M}(L; E, F)$ is called the matrix of L for the basis E and F .

Note 11.340. The matrix $\mathcal{M}(L; E, F)$ depends clearly not only on L but also on the basis used. If it is clear from the context which basis are used then we use the notation $\mathcal{M}(L)$.

We show now how the matrix of a linear mapping can be used to calculate the result of applying the linear mapping.

Theorem 11.341. *Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field F with bases $\{e_1, \dots, e_n\} \subseteq X, \{f_1, \dots, f_m\} \subseteq Y$ and $L \in \text{Hom}(X, Y)$ then we have:*

1. *If $x \in X$ then by [theorem: 11.143] there exists **unique** $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ and $\{L(x)_i\}_{i \in \{1, \dots, m\}}$ such that $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$ and $L(x) = \sum_{i \in \{1, \dots, m\}} L(x)_i \cdot f_i$. Then we have*

$$\forall j \in \{1, \dots, m\} \quad L(x)_j = \sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; E, F)_{j,i} \cdot x_i$$

so if we define

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathcal{M}_{n,1}(F) \text{ the column vector uniquely defined by the components of } x$$

and

$$Y = (L(x)_1 \ \dots \ L(x)_m) \text{ the row vector uniquely defined by the components of } L(x)$$

then we have

$$Y = \mathcal{M}(L; E, F) \cdot X$$

reducing applying a linear map to multiplication of matrices, involving only operations in F .

2. $\text{rank}(L) = \text{rank}(\mathcal{M}(L; E, F))$

Proof.

1. We have

$$\begin{aligned} \sum_{j \in \{1, \dots, m\}} L(x)_j \cdot f_j &= L(x) \\ &= L\left(\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i\right) \\ &\stackrel{[\text{theorem: 11.190}]}{=} \sum_{i \in \{1, \dots, n\}} x_i \cdot L(e_i) \\ &\stackrel{\text{def}}{=} \sum_{i \in \{1, \dots, n\}} x_i \cdot \left(\sum_{j \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{j,i} \cdot f_j\right) \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{i \in \{1, \dots, n\}} \left(\sum_{j \in \{1, \dots, m\}} x_i \cdot (\mathcal{M}(L; E, F)_{j,i} \cdot f_j)\right) \\ &= \sum_{i \in \{1, \dots, n\}} \left(\sum_{j \in \{1, \dots, m\}} (x_i \cdot \mathcal{M}(L; E, F)_{j,i}) \cdot f_j\right) \\ &\stackrel{[\text{theorem: 11.41}]}{=} \sum_{j \in \{1, \dots, m\}} \left(\sum_{i \in \{1, \dots, n\}} (x_i \cdot \mathcal{M}(L; E, F)_{j,i}) \cdot f_j\right) \\ &\stackrel{[\text{theorem: 11.80}]}{=} \sum_{j \in \{1, \dots, m\}} \left(\sum_{i \in \{1, \dots, n\}} x_i \cdot \mathcal{M}(L; E, F)_{j,i}\right) f_j \\ &= \sum_{j \in \{1, \dots, m\}} \left(\sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; E, F)_{j,i} \cdot x_i\right) f_j \end{aligned}$$

which as $\{f_i | i \in \{1, \dots, m\}\}$ is a basis proves by the uniqueness of the expansion proves that

$$\forall j \in \{1, \dots, m\} \quad L(x)_j = \sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; E, F)_{j,i} \cdot x_i$$

2. Let $M = \mathcal{M}(L; E, F) \in \mathcal{M}_{m,n}(F)$. Let $y \in L(X)$ then there exists a $x \in X$ such that $y = L(x)$. As $\{e_1, \dots, e_n\}$ is a basis of X we have $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$ so that

$$y = L\left(\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i\right) \stackrel{[\text{theorem: 11.190}]}{=} \sum_{i \in \{1, \dots, n\}} x_i \cdot L(e_i)$$

proving by [theorem: 11.102] that $y \in \text{span}(\{L(e_i) | i \in \{1, \dots, n\}\})$, hence

$$L(X) \subseteq \text{span}(\{L(e_i) | i \in \{1, \dots, n\}\}) \quad (11.224)$$

If $y \in \text{span}(\{L(e_i) | i \in \{1, \dots, n\}\})$ then by [theorem: 11.102] there exists a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$y = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot L(e_i) \stackrel{[\text{theorem: 11.190}]}{=} L\left(\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i\right) \in L(X)$$

proving that $\text{span}(\{L(e_i) | i \in \{1, \dots, n\}\}) \subseteq L(X)$. Combining this with [eq: 11.224] gives

$$L(X) = \text{span}(\{L(e_i) | i \in \{1, \dots, n\}\}) \quad (11.225)$$

Using [theorem: 11.147] there exist a basis

$$B \subseteq \{L(e_i) | i \in \{1, \dots, n\}\} \text{ for } L(X)$$

Using the definition of the rank of a linear map we have that

$$\text{rank}(L) \stackrel{\text{def}}{=} \dim(L(X)) = \text{card}(B)$$

By [theorem: 11.95] there exist a bijection $\beta: \{1, \dots, \text{rank}(L)\} \rightarrow J \subseteq \{1, \dots, n\}$ such that

$$B = \{L(e_{\beta(1)}), \dots, L(e_{\beta(\text{rank}(L))})\} \text{ where } B \text{ is a basis of } L(X) \quad (11.226)$$

Define now

$$B_c = \{\text{col}(M, \beta(i)) | i \in \{1, \dots, \text{rank}(L)\}\} \subseteq \text{cols}(M)$$

then by [theorem: 11.109] we have that

$$\text{span}(B_c) \subseteq \text{span}(\text{cols}(M)) \quad (11.227)$$

For the opposite inclusion, let $c \in \text{span}(\text{cols}(M)) = \text{span}(\text{col}(M, i) | i \in \{1, \dots, n\})$ then by [theorem: 11.102] there exist a $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$c = \sum_{i \in \{1, \dots, n\}} \lambda_i \cdot \text{col}(M, i)$$

Let $k \in \{1, \dots, m\}$ then we have

$$\begin{aligned} c_k &= \left(\sum_{i \in \{1, \dots, n\}} \lambda_i \cdot \text{col}(M, i) \right)_k \\ &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{i \in \{1, \dots, n\}} \lambda_i \cdot (\text{col}(M, i))_k \\ &\stackrel{[\text{definition: 11.333}]}{=} \sum_{i \in \{1, \dots, n\}} \lambda_i \cdot M_{k,i} \end{aligned} \quad (11.228)$$

Consider now the sum $\sum_{i \in \{1, \dots, n\}} \lambda_i \cdot L(e_i)$, then we have by [theorem: 11.102] that

$$\sum_{i \in \{1, \dots, n\}} \lambda_i \cdot L(e_i) \in \text{span}(\{L(e_i) | i \in \{1, \dots, n\}\}) \stackrel{[\text{eq: 11.225}]}{=} L(X)$$

By the above, [eq: 11.226] and [theorem: 11.143] there exist a $\{\gamma_i\}_{i \in \{1, \dots, \text{rank}(L)\}} \subseteq F$ such that

$$\begin{aligned}
\sum_{i \in \{1, \dots, n\}} \lambda_i \cdot L(e_i) &= \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot L(e_{\beta(j)}) \\
&\stackrel{[\text{definition: 11.339}]}{=} \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot \left(\sum_{i \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{i, \beta(j)} \cdot f_i \right) \\
&= \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot \left(\sum_{i \in \{1, \dots, m\}} M_{i, \beta(j)} \cdot f_i \right) \\
&\stackrel{[\text{theorem: 11.80}]}{=} \sum_{j \in \{1, \dots, \text{rank}(L)\}} \left(\sum_{i \in \{1, \dots, m\}} \gamma_j \cdot (M_{i, \beta(j)} \cdot f_i) \right) \\
&\stackrel{[\text{theorem: 11.41}]}{=} \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot (M_{i, \beta(j)} \cdot f_i) \right) \\
&= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, \text{rank}(L)\}} (\gamma_j \cdot M_{i, \beta(j)}) \cdot f_i \right) \\
&\stackrel{[\text{theorem: 11.80}]}{=} \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot M_{i, \beta(j)} \right) \cdot f_i \\
\sum_{i \in \{1, \dots, n\}} \lambda_i \cdot L(e_i) &\stackrel{[\text{definition: 11.339}]}{=} \sum_{j \in \{1, \dots, n\}} \lambda_j \cdot \left(\sum_{i \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{i, j} \cdot f_i \right) \\
&= \sum_{j \in \{1, \dots, n\}} \lambda_j \cdot \left(\sum_{i \in \{1, \dots, m\}} M_{i, j} \cdot f_i \right) \\
&\stackrel{[\text{theorem: 11.80}]}{=} \sum_{j \in \{1, \dots, n\}} \left(\sum_{i \in \{1, \dots, m\}} \lambda_j \cdot (M_{i, j} \cdot f_i) \right) \\
&\stackrel{[\text{theorem: 11.41}]}{=} \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} \lambda_j \cdot (M_{i, j} \cdot f_i) \right) \\
&= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} (\lambda_j \cdot M_{i, j}) \cdot f_i \right) \\
&\stackrel{[\text{theorem: 11.80}]}{=} \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} \lambda_j \cdot M_{i, j} \right) \cdot f_i
\end{aligned}$$

Using the uniqueness of the expansion of a vector in a basis we have from the above that

$$\forall i \in \{1, \dots, n\} \text{ we have } \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot M_{i, \beta(j)} = \sum_{j \in \{1, \dots, n\}} \lambda_j \cdot M_{i, j}$$

and substituting this in [eq: 11.228] proves that $\forall k \in \{1, \dots, m\}$

$$\begin{aligned}
c_k &= \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot M_{k, \beta(j)} \\
&\stackrel{[\text{definition: 11.333}]}{=} \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot \text{col}(M, \beta(j))_k \\
&\stackrel{[\text{theorem: 11.50}]}{=} \left(\sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot \text{col}(M, \beta(j)) \right)_k
\end{aligned}$$

proving that

$$c = \sum_{j \in \{1, \dots, \text{rank}(L)\}} \gamma_j \cdot \text{col}(M, \beta(j)) \in \text{span}(B_c)$$

Hence $\text{span}(\text{cols}(M)) \subseteq \text{span}(B_c)$ which together with [eq: 11.227] gives $\text{span}(\text{cols}(M)) = \text{span}(B_c)$ or using the definition of B_c .

$$\text{span}\{\text{col}(M, \beta(i)) | i \in \{1, \dots, \text{rank}(L)\}\} = \text{span}(\text{cols}(M)) \quad (11.229)$$

Next we prove that $\{\text{col}(M, \beta(i))\}_{i \in \{1, \dots, \text{rank}(L)\}}$ is linear independent. Let $\{\alpha_i\}_{i \in \{1, \dots, \text{rank}(L)\}} \subseteq F$ such that

$$\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot \text{col}(M, \beta(i)) = 0$$

then $\forall k \in \{1, \dots, m\}$ we have

$$\begin{aligned} 0 &= \left(\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot \text{col}(M, \beta(i)) \right)_k \\ &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot \text{col}(M, \beta(i))_k \\ &\stackrel{[\text{definition: 11.333}]}{=} \sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot M_{k, \beta(i)} \end{aligned} \quad (11.230)$$

Now

$$\begin{aligned} &\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot L(e_{\beta(i)}) \stackrel{[\text{definition: 11.339}]}{=} \\ &\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot \left(\sum_{j \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{j, \beta(i)} \cdot f_j \right) = \\ &\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot \left(\sum_{j \in \{1, \dots, m\}} M_{j, \beta(i)} \cdot f_j \right) \stackrel{[\text{theorem: 11.80}]}{=} \\ &\sum_{i \in \{1, \dots, \text{rank}(L)\}} \left(\sum_{j \in \{1, \dots, m\}} \alpha_i \cdot (M_{j, \beta(i)} \cdot f_j) \right) \stackrel{[\text{theorem: 11.41}]}{=} \\ &\sum_{j \in \{1, \dots, m\}} \left(\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot (M_{j, \beta(i)} \cdot f_j) \right) = \\ &\sum_{j \in \{1, \dots, m\}} \left(\sum_{i \in \{1, \dots, \text{rank}(L)\}} (\alpha_i \cdot M_{j, \beta(i)}) \cdot f_j \right) \stackrel{[\text{theorem: 11.80}]}{=} \\ &\sum_{j \in \{1, \dots, m\}} \left(\sum_{i \in \{1, \dots, \text{rank}(L)\}} \alpha_i \cdot M_{j, \beta(i)} \right) \cdot f_j \stackrel{[\text{eq: 11.230}]}{=} \\ &\sum_{j \in \{1, \dots, m\}} 0 \cdot f_j = \\ &0 \end{aligned} \quad (11.231)$$

Now by [eq: 11.226] $\{L(e_{\beta(1)}), \dots, L(e_{\beta(\text{rank}(L))})\}$ is a basis, hence linear independent, so by [theorem: 11.120]

$$\forall i \in \{1, \dots, m\} \text{ we have } \alpha_i = 0$$

So $\{\text{col}(M, \beta(i))\}_{i \in \{1, \dots, \text{rank}(L)\}}$ is linear independent or by [theorem: 11.135]

$$\{\text{col}(M, \beta(i)) | i \in \{1, \dots, \text{rank}(L)\}\} \text{ is linear independent}$$

Combining this with [eq: 11.229] proves that

$$\{\text{col}(M, \beta(i)) | i \in \{1, \dots, \text{rank}(L)\}\} \text{ is a basis for } \text{span}(\text{cols}(M))$$

which by [theorem: 11.156] proves that

$$\dim(\text{span}(\text{cols}(M))) = \text{rank}(L)$$

So

$$\text{rank}(M) \stackrel{\text{def}}{=} \dim(\text{span}(\text{cols}(M))) = \text{rank}(L)$$

proving, as $M = \mathcal{M}(L; E, F)$, that

$$\text{rank}(\mathcal{M}(L; E, F)) = \text{rank}(L) \quad \square$$

Example 11.342. Let $n \in \mathbb{N}$, $\mathcal{F}$ a field and $E = \{e_1, \dots, e_n\}$ the canonical basis of $\mathcal{F}^n$ [see example: 11.167] and $L \in \text{Hom}(\mathcal{F}^n, \mathcal{F}^n)$ then we have that $\forall x = (x_1, \dots, x_n) \in \mathcal{F}^n$ we have that

$$\forall i \in \{1, \dots, n\} \text{ we have } L(x)_i = \sum_{j=1}^n \mathcal{M}(L; E, E)_{j,i} x_j$$

where $L(x) = (L(x)_1, \dots, L(x)_n)$.

Proof. By [example: 11.167] we have for $x = (x_1, \dots, x_n) \in \mathcal{F}^n$ and $L(x) = (L(x)_1, \dots, L(x)_n) \in \mathcal{F}^n$ that

$$x = \sum_{i=1}^n x_i \cdot e_i \text{ and } L(x) = \sum_{i=1}^n L(x)_i \cdot e_i$$

Hence by the previous theorem [theorem: 11.341] we have

$$L(x)_i = \sum_{j=1}^n \mathcal{M}(L; E, E)_{j,i} \cdot x_j \quad \square$$

Next we show that the mapping that associate a matrix with a linear mapping is linear.

Theorem 11.343. Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field $\mathcal{F}$ with bases $E = \{e_1, \dots, e_n\}, F = \{f_1, \dots, f_m\}$ for X, Y then

$$\mathcal{M}(E, F): \text{Hom}(X, Y) \rightarrow \mathcal{M}_{m,n}(\mathcal{F}) \text{ defined by } \mathcal{M}(E, F)(L) = \mathcal{M}(L; E, F)$$

satisfies:

1. $\forall \alpha \in \mathcal{F}$ and $L \in \text{Hom}(X, Y)$ we have $\mathcal{M}(E, F)(\alpha \cdot L) = \alpha \cdot \mathcal{M}(E, F)(L)$
2. $\forall K, L \in \text{Hom}(X, Y)$ we have $\mathcal{M}(E, F)(K + L) = \mathcal{M}(E, F)(K) + \mathcal{M}(E, F)(L)$
3. $\mathcal{M}(E, F)(C_0) = 0$
4. $\mathcal{M}(E, E)(\text{Id}_X) = E$ [be aware the last E is the identity matrix and the first two E 's specifies the basis in X]

in other words

$$\mathcal{M}(E, G) \in \text{Hom}(\text{Hom}(X, Y), \mathcal{M}_{m,n}(\mathcal{F}))$$

Proof.

1. Let $i \in \{1, \dots, n\}$ then

$$\begin{aligned} \sum_{j \in \{1, \dots, m\}} \mathcal{M}(\alpha \cdot L; E, F)_{j,i} \cdot f_j &= (\alpha \cdot L)(e_i) \\ &= \alpha \cdot L(e_i) \\ &= \alpha \cdot \sum_{j \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{j,i} \cdot f_j \\ &= \sum_{j \in \{1, \dots, m\}} (\alpha \cdot \mathcal{M}(L; E, F)_{j,i}) \cdot f_j \end{aligned}$$

proving by the uniqueness of the expansion that $\forall i, j \in \{1, \dots, m\}$

$$\mathcal{M}(\alpha \cdot L; E, F)_{j,i} = \alpha \cdot \mathcal{M}(L; E, F)_{j,i}$$

or

$$\mathcal{M}(E, F)(\alpha \cdot L) = \mathcal{M}(\alpha \cdot L; E, F) = \alpha \cdot \mathcal{M}(L; E, F) = \mathcal{M}(E, F)(L)$$

2. Let $i \in \{1, \dots, n\}$ then

$$\begin{aligned} \sum_{j \in \{1, \dots, m\}} \mathcal{M}(K + L; E, F)_{j, i} \cdot f_j &= \\ (K + L)(e_i) &= \\ K(e_i) + L(e_i) &= \\ \sum_{j \in \{1, \dots, m\}} \mathcal{M}(K; E, F)_{j, i} \cdot f_j + \sum_{j \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{j, i} \cdot f_j &= \\ \sum_{j \in \{1, \dots, m\}} (\mathcal{M}(K; E, F)_{j, i} \cdot f_j + \mathcal{M}(L; E, F)_{j, i} \cdot f_j) &= \\ \sum_{j \in \{1, \dots, m\}} (\mathcal{M}(K; E, F)_{j, i} + \mathcal{M}(L; E, F)_{j, i}) \cdot f_j & \end{aligned}$$

proving by the uniqueness of the expansion that $\forall j \in \{1, \dots, m\} \forall i, j \in \{1, \dots, m\}$

$$\mathcal{M}(K + L; E, F)_{j, i} = \mathcal{M}(K; E, F)_{j, i} + \mathcal{M}(L; E, F)_{j, i}$$

or

$$\mathcal{M}(E, F)(K + L) = \mathcal{M}(K + L; E, F) = \mathcal{M}(K; E, F) + \mathcal{M}(L; E, F) = \mathcal{M}(E, F)(K) + \mathcal{M}(E, F)(L)$$

3. Let $i \in \{1, \dots, n\}$ then

$$\begin{aligned} \sum_{j \in \{1, \dots, m\}} 0 \cdot f_j &= 0 \\ &= C_0(e_i) \\ &= \sum_{j \in \{1, \dots, m\}} \mathcal{M}(C_0; E, F)_{j, i} \cdot f_j \end{aligned}$$

proving by the uniqueness of the expansion that $\forall i, j \in \{1, \dots, m\}$

$$\mathcal{M}(C_0; E, F)_{j, i} = 0$$

or

$$\mathcal{M}(E, F)(C_0) = \mathcal{M}(C_0; E, F) = 0$$

4. Let $i \in \{1, \dots, n\}$ then

$$\begin{aligned} \sum_{j \in \{1, \dots, n\}} \delta_{j, i} \cdot e_j &= e_i \\ &= \text{Id}_X(e_i) \\ &= \sum_{j \in \{1, \dots, n\}} \mathcal{M}(\text{Id}_X; E, E)_{j, i} \cdot f_j \end{aligned}$$

proving by the uniqueness of the expansion that $\forall i, j \in \{1, \dots, n\}$

$$\mathcal{M}(\text{Id}_X; E, E)_{j, i} = \delta_{j, i} = E_{j, i}$$

or

$$\mathcal{M}(E, E)(\text{Id}_X) = \mathcal{M}(\text{Id}_X; E, E) = E \quad \square$$

Next we prove that $\mathcal{M}(E, F)$ is actually a bijection so that $\text{Hom}(X, Y)$ and $\mathcal{M}_{m, n}(F)$ are isomorphic.

Theorem 11.344. *Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field $\mathcal{F}$ with bases $E = \{e_1, \dots, e_n\}, F = \{f_1, \dots, f_m\}$ for X, Y then*

$$\mathcal{M}(E, F): \text{Hom}(X, Y) \rightarrow \mathcal{M}_{m, n}(F) \text{ where } \mathcal{M}(E, F)(L) = \mathcal{M}(L; E, F)$$

is a bijection. Further if $M \in \mathcal{M}_{m,n}(F)$ then $\mathcal{M}^{-1}(E, F)(M)$ is defined by

$$\mathcal{M}^{-1}(E, F)(M)(x) = \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot f_i$$

where $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ is the unique family such that $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$

Proof.

injectivity. Let $L_1, L_2 \in \text{Hom}(X, Y)$ such that $\mathcal{M}(E, F)(L_1) = \mathcal{M}(E, F)(L_2)$ then we have

$$\mathcal{M}(L_1; E, F) = \mathcal{M}(L_2; E, F) \quad (11.232)$$

Let $x \in X$ then there exists a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$, So

$$\begin{aligned} L_1(x) &= L_1 \left(\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right) \\ &\stackrel{[\text{theorem: 11.190}]}{=} \sum_{i \in \{1, \dots, n\}} x_i \cdot L_1(e_i) \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot \left(\sum_{j \in \{1, \dots, m\}} \mathcal{M}(L_1; E, F)_{j,i} \cdot f_j \right) \\ &\stackrel{[\text{eq: 11.232}]}{=} \sum_{i \in \{1, \dots, n\}} x_i \cdot \left(\sum_{j \in \{1, \dots, m\}} \mathcal{M}(L_2; E, F)_{j,i} \cdot f_j \right) \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot L_2(e_i) \\ &\stackrel{[\text{theorem: 11.190}]}{=} L_2 \left(\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right) \\ &= L_2(x) \end{aligned}$$

proving that $L_1 = L_2$.

surjectivity. Let $M \in \mathcal{M}_{m,n}(F)$ then define $L: X \rightarrow Y$ by

$$L(x) = \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot f_i \text{ where } x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$$

then we have for $x, y \in X$ and $\alpha \in F$ that

$$x + y = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i + \sum_{i \in \{1, \dots, n\}} y_i \cdot e_i = \sum_{i \in \{1, \dots, n\}} (x_i + y_i) \cdot e_i$$

and

$$\alpha \cdot x = \alpha \cdot \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i = \sum_{i \in \{1, \dots, n\}} (\alpha \cdot x_i) \cdot e_i$$

So that

$$\begin{aligned} L(x + y) &= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot (x_j + y_j) \right) \cdot f_i = \\ &= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j + \sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot y_j \right) \cdot f_i = \\ &= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot f_i + \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot y_j \right) \cdot f_i = \\ &= L(x) + L(y) \end{aligned}$$

and

$$\begin{aligned}
 L(\alpha \cdot x) &= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot (\alpha \cdot x_j) \right) \cdot f_i \\
 &= \sum_{i \in \{1, \dots, m\}} \left(\alpha \cdot \sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot f_i \\
 &= \alpha \cdot \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot f_i \\
 &= \alpha \cdot L(x)
 \end{aligned}$$

proving that

$$L \in \text{Hom}(X, Y)$$

Let $i \in \{1, \dots, n\}$ then

$$\begin{aligned}
 e_i &= \sum_{j \in \{1, \dots, m\}} \delta_{j,i} \cdot e_j \text{ so that } (e_i)_j = \delta_{j,i} \\
 \sum_{k \in \{1, \dots, m\}} \mathcal{M}(L; E, F)_{k,i} \cdot f_k &= L(e_i) \\
 &= \sum_{k \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{k,j} \cdot \delta_{j,i} \right) \cdot f_k \\
 &= \sum_{k \in \{1, \dots, m\}} M_{k,i} \cdot f_k
 \end{aligned}$$

So by uniqueness of the expansion in a basis we have that $\forall i \in \{1, \dots, n\}, \forall k \in \{1, \dots, m\}$

$$\mathcal{M}(L; E, F)_{k,i} = M_{k,i}$$

hence

$$\mathcal{M}(E, F)(L) = M$$

proving surjectivity and

$$\mathcal{M}(E, F)^{-1}(M) = L \quad \square$$

Corollary 11.345. Let $n, m \in \mathbb{N}$, X, Y finite dimensional vector spaces over a field $\mathcal{F}$ with bases $E = \{e_1, \dots, e_n\}, F = \{f_1, \dots, f_m\}$ for X, Y then

$$\mathcal{M}(E, F): \text{Hom}(X, Y) \rightarrow \mathcal{M}_{m,n}(\mathcal{F}) \text{ defined by } \mathcal{M}(E, F)(L) = M(L; E, F)$$

is a isomorphism. Hence $\text{Hom}(X, Y)$ is isomorphic with $\mathcal{M}_{m,n}(\mathcal{F})$

Proof. This follows from [theorem: 11.343] and [theorem: 11.344]. $\square$

Example 11.346. Let $\mathcal{F}$ be a field, $n, m \in \mathbb{N}$ and $E = \{e_1, \dots, e_n\} \subseteq \mathcal{F}^n$, $F = \{f_1, \dots, f_m\} \subseteq \mathcal{F}^m$ defined by $(e_i)_j = \delta_{i,j}^n$ and $\{f_i\}_j = \delta_{i,j}^m$ the canonical bases for the vector spaces $\mathcal{F}^n, \mathcal{F}^m$ over $\mathcal{F}$ then for $M \in \mathcal{M}_{m,n}(\mathcal{F})$ we have for $x \in \mathcal{F}^n$ that $\forall i \in \{1, \dots, m\}$

$$(\mathcal{M}(E, F)^{-1}(M)(x))_i = \sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j$$

or if we define $X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ then

$$\mathcal{M}(E, F)^{-1}(M)(x) = M \cdot X$$

Proof. Let $x \in \mathcal{F}^n$ then we have

$$x_i = \sum_{j \in \{1, \dots, n\}} x_j \cdot \delta_{j,i}^n = \sum_{j \in \{1, \dots, n\}} x_j \cdot (e_j)_i = \left(\sum_{j \in \{1, \dots, n\}} x_j \cdot e_j \right)_i$$

so that

$$x = \sum_{j \in \{1, \dots, n\}} x_j \cdot e_j$$

hence

$$\begin{aligned}
 (\mathcal{M}(E, F)^{-1}(M)(x))_k &\stackrel{[\text{theorem: 11.344}]}{=} \left(\sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot f_i \right)_k \\
 &\stackrel{[\text{theorem: 11.50}]}{=} \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot (f_i)_k \\
 &= \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in \{1, \dots, n\}} M_{i,j} \cdot x_j \right) \cdot \delta_{i,k}^m \\
 &= \sum_{j \in \{1, \dots, n\}} M_{k,j} \cdot x_j \\
 &\quad \square
 \end{aligned}$$

We have the following relation between the composition of linear mappings and the product of matrices.

Theorem 11.347. *Let $n, m, r \in \mathbb{N}$, X, Y, Z finite dimensional vector spaces over a field $\mathcal{F}$ with bases $E = \{e_1, \dots, e_n\}, F = \{f_1, \dots, f_m\}, G = \{g_1, \dots, g_r\}$ for X, Y, Z then we have for $L_1 \in \text{Hom}(X, Y)$ and $L_2 \in \text{Hom}(Y, Z)$ that*

$$\mathcal{M}(L_2 \circ L_1; E, G) = \mathcal{M}(L_2; F, G) \cdot \mathcal{M}(L_1; E, F)$$

or in other words

$$\mathcal{M}(E, G)(L_2 \circ L_1) = \mathcal{M}(F, G)(L_2) \cdot \mathcal{M}(E, F)(L_1)$$

Proof. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned}
 \sum_{k \in \{1, \dots, r\}} \mathcal{M}(L_2 \circ L_1; E, G)_{k,i} \cdot g_k &= \\
 (L_2 \circ L_1)(e_i) &= \\
 L_2(L_1(e_i)) &= \\
 L_2 \left(\sum_{j \in \{1, \dots, m\}} \mathcal{M}(L_1; E, F)_{j,i} \cdot f_j \right) &\stackrel{[\text{theorem: 11.190}]}{=} \\
 \sum_{j \in \{1, \dots, m\}} \mathcal{M}(L_1; F, G)_{j,i} \cdot L_2(f_j) &= \\
 \sum_{j \in \{1, \dots, m\}} \mathcal{M}(L_1; E, F)_{j,i} \cdot \left(\sum_{k \in \{1, \dots, r\}} \mathcal{M}(L_2; F, G)_{k,j} \cdot g_k \right) &\stackrel{[\text{theorem: 11.80}]}{=} \\
 \sum_{j \in \{1, \dots, m\}} \left(\sum_{k \in \{1, \dots, r\}} \mathcal{M}(L_1; E, F)_{j,i} \cdot (\mathcal{M}(L_2; F, G)_{k,j} \cdot g_k) \right) &\stackrel{[\text{theorem: 11.41}]}{=} \\
 \sum_{k \in \{1, \dots, r\}} \left(\sum_{j \in \{1, \dots, m\}} (\mathcal{M}(L_2; F, G)_{k,j} \cdot \mathcal{M}(L_1; E, F)_{j,i}) \cdot g_k \right) &\stackrel{[\text{theorem: 11.80}]}{=} \\
 \sum_{k \in \{1, \dots, r\}} (\mathcal{M}(L_2; F, G) \cdot \mathcal{M}(L_1; E, F))_{k,i} \cdot g_k &
 \end{aligned}$$

By the uniqueness of the expansion in a basis it follow then that $\forall i \in \{1, \dots, n\}$ and $\forall k \in \{1, \dots, r\}$ that

$$\mathcal{M}(L_2 \circ L_1; E, G)_{k,i} = (\mathcal{M}(L_2; F, G) \cdot \mathcal{M}(L_1; E, F))_{k,i}$$

proving

$$\mathcal{M}(L_2 \circ L_1; E, G) = (\mathcal{M}(L_2; F, G) \cdot \mathcal{M}(L_1; E, F))$$

□

11.8.3 Inverse, Determinant and Adjoint of matrices

For linear transformations we have the inverse of a linear transformation based on the composition operator $\circ$. Not every linear transformation has an inverse but we can use the determinant of a linear transformation to check if a linear transformation is invertible. Finally we can use the determinant and the adjoint of a linear transformation to calculate the inverse of an invertible linear transformation. Having just proved that the composition of linear mappings can be translated in the product of their associated matrices, it seems reasonable to define the inverse of a matrix in terms of the product of matrices and introduce the concept of the determinant and adjoint of matrices. It will also turn out that the determinant and adjoint of a linear transformation are related to the determinant and adjoint of a linear mapping.

Definition 11.348. Let $n \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,n}(F)$ then $N \in \mathcal{M}_{n,n}(F)$ is a **inverse** of M if

$$N \cdot M = E = M \cdot N$$

It turns out that a matrix can have only one inverse.

Theorem 11.349. Let $n \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,n}$ then if N_1 and N_2 are inverses of M we have that $N_1 = N_2$

Proof. Let N_1, N_2 be inverses of M then we have

$$N_1 = N_1 \cdot E = N_1 \cdot (M \cdot N_2) = (N_1 \cdot M) \cdot N_2 = E \cdot N_2 = N_2 \quad \square$$

The above leads to the following definition

Definition 11.350. Let $n \in \mathbb{N}$ and F a field then $M \in \mathcal{M}_{n,n}(F)$ is **invertible** if there exist a $N \in \mathcal{M}_{n,n}(F)$ that is the inverse of M . This **unique** inverse is noted as M^{-1} , so we have that

$$M^{-1} \cdot M = E = M \cdot M^{-1}$$

The inverse of an inverse is the invertible matrix itself.

Theorem 11.351. Let $n \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,n}(F)$ an invertible matrix then M^{-1} is also invertible and $(M^{-1})^{-1} = M$

Proof. This can be proved by a easy calculation:

$$M^{-1} \cdot M = E = M \cdot M^{-1}$$

hence M is the inverse of M^{-1} , so that $(M^{-1})^{-1} = M$. $\square$

Theorem 11.352. Let $n \in \mathbb{N}$, X be a vector space over a field, $F = \{f_1, \dots, f_n\}$ a basis for X and $L \in \text{Hom}(X, X)$ then we have

L has an inverse L^{-1} [or equivalently by [theorem: 2.73] L is an isomorphism]

$$\Updownarrow$$

$\mathcal{M}(L; F, F)$ is invertible

Further if L^{-1} exists then

$$\mathcal{M}(L^{-1}; F, F) = \mathcal{M}(L, F, F)^{-1}$$

Proof.

$\Rightarrow$. We have for L^{-1} that

$$\begin{aligned} E & \stackrel{[\text{theorem: 11.343 (4)}]}{=} \mathcal{M}[\text{Id}_X; F, F] \\ & = \mathcal{M}[L \circ L^{-1}; F, F] \\ & \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(L; F, F) \cdot \mathcal{M}(L^{-1}; F, F) \\ E & \stackrel{[\text{theorem: 11.343 (4)}]}{=} \mathcal{M}[\text{Id}_X; F, F] \\ & = \mathcal{M}[L^{-1} \circ L; F, F] \\ & \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(L^{-1}; F, F) \cdot \mathcal{M}(L; F, F) \end{aligned}$$

proving that $\mathcal{M}(L; F, F)$ has an inverse and that

$$\mathcal{M}(L; F, F)^{-1} = \mathcal{M}(L^{-1}; F, F)$$

$\Leftarrow$. Define $K \stackrel{=}{\text{[theorem: 11.344]}} \mathcal{M}(F, F)^{-1}(\mathcal{M}(L; F, F)^{-1})$ then we have

$$\begin{aligned} & K \circ L \stackrel{=}{\text{[theorem: 11.344]}} \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(K \circ L)) \stackrel{=}{\text{[theorem: 11.347]}} \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(K) \cdot \mathcal{M}(F, F)(L)) = \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(\mathcal{M}(F, F)^{-1}(\mathcal{M}(L; F, F)^{-1})) \cdot \mathcal{M}(F, F)(L)) = \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(L; F, F)^{-1} \cdot \mathcal{M}(F, F)(L)) = \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(L; F, F)^{-1} \cdot \mathcal{M}(L; F, F)) = \\ & \mathcal{M}(F, F)^{-1}(E) \stackrel{=}{\text{[theorem: 11.343]}} \\ & \text{Id}_X \end{aligned}$$

and

$$\begin{aligned} & L \circ K \stackrel{=}{\text{[theorem: 11.344]}} \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(L \circ K)) \stackrel{=}{\text{[theorem: 11.347]}} \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(L) \cdot \mathcal{M}(F, F)(K)) = \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(L) \cdot \mathcal{M}(F, F)(\mathcal{M}(F, F)^{-1}(\mathcal{M}(L; F, F)^{-1}))) = \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(L) \cdot \mathcal{M}(L; F, F)^{-1} \cdot) = \\ & \mathcal{M}(F, F)^{-1}(\mathcal{M}(L; F, F) \cdot \mathcal{M}(L; F, F)^{-1}) = \\ & \mathcal{M}(F, F)^{-1}(E) \stackrel{=}{\text{[theorem: 11.343]}} \\ & \text{Id}_X \end{aligned}$$

which proves that $K \circ L = \text{Id}_X = L \circ K$ so that

$$L \text{ is invertible with inverse } K = \mathcal{M}(F, F)^{-1}(\mathcal{M}(F, F)(L)) \quad \square$$

Corollary 11.353. Let $n \in \mathbb{N}$, X be a vector space over a field F and $E = \{e_1, \dots, e_n\}$, $F = \{f_1, \dots, f_n\}$ two bases of X then there exist a $M \in \mathcal{M}_{n,n}(F)$ such that M is invertible and $\forall i \in \{1, \dots, n\}$ $f_i = \sum_{j \in \{1, \dots, n\}} M_{j,i} \cdot e_j$ and $e_i = \sum_{j \in \{1, \dots, n\}} (M^{-1})_{j,i} \cdot f_j$.

Proof. Using [theorem: 11.338] there exist a linear isomorphism $L \in \text{Hom}(X, X)$ such that $\forall i \in \{1, \dots, n\}$ we have $f_i = L(e_i)$ and $e_i = L^{-1}(f_i)$. Take $M = \mathcal{M}(L; E, F) \in \mathcal{M}_{n,n}(F)$ then we have by [definition: 11.339] that

$$\forall i \in \{1, \dots, n\} \quad f_i = L(e_i) = \sum_{j \in \{1, \dots, n\}} M_{j,i} \cdot e_j$$

Further as L is a linear isomorphism it follows from [theoremL 11.352] that $M = \mathcal{M}(L; E, F)$ is invertible and $M^{-1} = \mathcal{M}(L; E, F)^{-1} = \mathcal{M}(L^{-1}; F, E)$ so that

$$\forall i \in \{1, \dots, n\} \quad e_i = L^{-1}(f_i) = \sum_{j \in \{1, \dots, n\}} \mathcal{M}(L^{-1}, F, E)_{j,i} \cdot f_j = \sum_{j \in \{1, \dots, n\}} (M^{-1})_{j,i} \cdot f_j \quad \square$$

We are now ready to define the determinant of a matrix.

Definition 11.354. (Determinant) Let $n \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,n}(F)$ a square matrix then

$$\det(M) = \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right)$$

Another notation for the determinant is $|M|$ so using this notation we have

$$|M| = \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right)$$

Example 11.355. If $n = 1$ then $M \in \mathcal{M}_{1,1}(F)$ is of the form $(M_{1,1})$ and $P_1 = \{\text{Id}_{\{1\}}\}$ so that

$$\det(M) = \sum_{\sigma \in \{\text{Id}_{\{1\}}\}} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1\}} M_{i, \sigma(i)} \right) = \text{sign}(\text{Id}_{\{1\}}) \left(\prod_{i \in \{1\}} M_{i, \text{Id}_{\{1\}}(i)} \right) = M_{11}$$

Definition 11.356. (diagonal matrix) Let $n, m \in \mathbb{N}$ and F a field then $M \in \mathcal{M}_{n,m}$ is a **diagonal matrix** if $\forall (i, j) \in \{1, \dots, n\} \cdot \{1, \dots, m\}$ $M_{i,j} = \delta_{i,j} \cdot M_{i,i}$

One nice property for diagonal square matrices is that the determinant is easy to calculate.

Theorem 11.357. Let $n \in \mathbb{N}$, F a field, $M \in \mathcal{M}_{n,n}(F)$ a diagonal matrix then

$$\det(M) = \prod_{i \in \{1, \dots, n\}} M_{i,i}$$

Proof. First if $\sigma \in P_n$ is such that $\sigma \neq \text{Id}_{\{1, \dots, n\}}$ then $\exists k \in \{1, \dots, n\}$ such that $k \neq \sigma(k)$. Hence

$$\begin{aligned} \prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} &= \left(\prod_{i \in \{1, \dots, n\} \setminus \{k\}} M_{i,j} \right) \cdot \left(\prod_{i \in \{k\}} M_{i, \sigma(i)} \right) \\ &= \left(\prod_{i \in \{1, \dots, n\} \setminus \{k\}} M_{i,j} \right) \cdot M_{k, \sigma(k)} \\ &= \left(\prod_{i \in \{1, \dots, n\} \setminus \{k\}} M_{i,j} \right) \cdot M_{k,k} \cdot \delta_{k, \sigma(k)} \\ &= \left(\prod_{i \in \{1, \dots, n\} \setminus \{k\}} M_{i,j} \right) \cdot M_{k,k} \cdot 0 \\ &= 0 \end{aligned}$$

So that

$$\begin{aligned} \det(M) &= \sum_{\sigma \in P_n} \left(\text{sign}(\sigma) \cdot \prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right) = \\ \sum_{\sigma \in P_n \setminus \{\text{Id}_{\{1, \dots, n\}}\}} \left(\text{sign}(\sigma) \cdot \prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right) &+ \sum_{\sigma \in \{\text{Id}_{\{1, \dots, n\}}\}} \left(\text{sign}(\sigma) \cdot \prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right) = \\ \sum_{\sigma \in P_n \setminus \{\text{Id}_{\{1, \dots, n\}}\}} \text{sign}(\sigma) \cdot 0 &+ \sum_{\sigma \in \{\text{Id}_{\{1, \dots, n\}}\}} \left(\text{sign}(\sigma) \cdot \prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right) = \\ \text{sign}(\text{Id}_{\{1, \dots, n\}}) \cdot \prod_{i \in \{1, \dots, n\}} M_{i, \text{Id}_{\{1, \dots, n\}}(i)} &= \\ \prod_{i \in \{1, \dots, n\}} M_{i,i} &= \end{aligned}$$

□

The determinant is invariant under the transpose operation on a matrix.

Theorem 11.358. *Let $n \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,n}(F)$ then*

$$\det(M) = \det(M^T)$$

Proof. First define $(\cdot)^{-1}: P_n \rightarrow P_n$ by $(\cdot)^{-1}(\sigma) = \sigma^{-1}$ then we have:

injectivity. If $(\cdot)^{-1}(\sigma) = (\cdot)^{-1}(\rho)$ then

$$\begin{aligned} \sigma^{-1} = \rho^{-1} &\Rightarrow \sigma \circ \sigma^{-1} = \sigma \circ \rho^{-1} \\ &\Rightarrow \text{Id}_{\{1, \dots, n\}} = \sigma \circ \rho^{-1} \\ &\Rightarrow \text{Id}_{\{1, \dots, n\}} \circ \rho = (\sigma \circ \rho^{-1}) \circ \rho \\ &\Rightarrow \rho = \sigma \end{aligned}$$

surjectivity. If $\sigma \in P_n$ then $(\cdot)^{-1}(\sigma^{-1}) = (\sigma^{-1})^{-1} = \sigma$

so that

$$(\cdot)^{-1}: P_n \rightarrow P_n \text{ is a bijection} \quad (11.233)$$

First note that $\langle F, \cdot \rangle$ is a Abelian semi group

$$\prod_{i \in \{1, \dots, n\}} M_{\sigma(i), i} = \prod_{i \in \{1, \dots, n\}} M_{\sigma(i), \sigma(\sigma^{-1}(i))} \stackrel{[\text{theorem: 11.36}]}{=} \prod_{i \in \{1, \dots, n\}} M_{i, \sigma^{-1}(i)} \quad (11.234)$$

So that

$$\begin{aligned} \det(M^T) &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} (M^T)_{i, \sigma(i)} \right) \\ &= \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{\sigma(i), i} \right) \\ &\stackrel{[\text{eq: 11.234}]}{=} \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{i, \sigma^{-1}(i)} \right) \\ &\stackrel{[\text{theorem: 11.251}]}{=} \sum_{\sigma \in P_n} \text{sign}(\sigma^{-1}) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{i, \sigma^{-1}(i)} \right) \\ &= \sum_{\sigma \in P_n} \text{sign}((\cdot)^{-1}(\sigma)) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{i, (\cdot)^{-1}(\sigma)(i)} \right) \\ &\stackrel{[\text{eq: 11.233 and theorem: 11.36}]}{=} \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{i, \sigma(i)} \right) \\ &= \det(M) \end{aligned}$$

□

Theorem 11.359. *Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero with basis $E = \{e_1, \dots, e_n\} \subseteq X$ and $L \in \text{Hom}(X, X)$ then we have:*

$$\det(L) = \det(\mathcal{M}(L; E, E)) = \det(\mathcal{M}(E, E)(L))$$

Proof. Using [theorem: 11.299] there exists a determinant function Δ in F^n such that $\Delta(e_1, \dots, e_n) = 1$. Then we have

$$\Delta_L(e, \dots, e_n) \stackrel{[\text{theorem: 11.311}]}{=} \det(L) \cdot \Delta(e_1, \dots, e_n) = \det(L) \quad (11.235)$$

Let $M = \mathcal{M}(L, E, E)$ then we have

$$\begin{aligned}
 \det(L) & \stackrel{[\text{eq: 11.235}]}{=} \Delta_L(e, \dots, e_n) \\
 & \stackrel{\text{def}}{=} \Delta(L(e_1), \dots, L(e_n)) \\
 & = \Delta\left(\sum_{i \in \{1, \dots, n\}} M_{i,1} \cdot e_i, \dots, \sum_{i \in \{1, \dots, n\}} M_{i,n} \cdot e_i\right) \\
 & \stackrel{[\text{theorem: 11.294}]}{=} \sum_{\sigma \in P_n} \left(\prod_{i \in \{1, \dots, n\}} M_{\sigma(i),i} \right) \cdot (\sigma \Delta)(e_1, \dots, e_n) \\
 & \stackrel{\Delta \text{ is skew-symmetry}}{=} \sum_{\sigma \in P_n} \left(\prod_{i \in \{1, \dots, n\}} M_{\sigma(i),i} \right) \cdot (\text{sign}(\sigma) \cdot \Delta(e_1, \dots, e_n)) \\
 & = \sum_{\sigma \in P_n} \text{sign}(\sigma) \cdot \left(\prod_{i \in \{1, \dots, n\}} M_{\sigma(i),i} \right) \\
 & = \det(M^T) \\
 & \stackrel{[\text{theorem: 11.358}]}{=} \det(M)
 \end{aligned}$$

□

Corollary 11.360. *Let $n \in \mathbb{N}$, X a vector space over a field F with characteristic zero with basis $E = \{e_1, \dots, e_n\} \subseteq X$ and $L \in \text{Hom}(X, X)$ then*

L is a linear isomorphism

$\Updownarrow$

$$\det(\mathcal{M}(L; E, E)) \neq 0$$

Proof. We have

$\Rightarrow$. If L is a linear isomorphism it follows from [theorem: 11.315] that $\det(L) \neq 0$. So that $\det(\mathcal{M}(L; E, E)) \stackrel{[\text{theorem: 11.359}]}{=} \det(L) \neq 0$.

$\Leftarrow$. As $\det(L) \stackrel{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}(L; E, E)) \neq 0$ it follows from [theorem: 11.315] that L is a linear isomorphism. □

We use now the above theorem to prove the following:

Theorem 11.361. *Let $n \in \mathbb{N}$ and $\mathcal{F}$ a field of characteristic zero then we have:*

1. *If $M_1, M_2 \in \mathcal{M}_{n,n}(\mathcal{F})$ then $\det(M_1 \cdot M_2) = \det(M_1) \cdot \det(M_2)$*
2. *$\det(E) = 1$*
3. *If M has a inverse M^{-1} then $\det(M) \neq 0$ and $\det(M^{-1}) = (\det(M))^{-1}$*

Proof. Let $\mathcal{F}^n$ be the vector space over $\mathcal{F}$ with the canonical basis $F = \{e_1, \dots, e_n\}$ defined by $\{e_i\}_k = \delta_{i,k}$ for $k \in \{1, \dots, n\}$ [see: theorem: 11.167].

1. Take then $L_1 = \mathcal{M}[F, F]^{-1}(M_1)$ and $L_2 = \mathcal{M}(F, F)^{-1}(M_2)$ then we have:

$$\mathcal{M}(F, F)(L_1) = M_1 \text{ and } \mathcal{M}(F, F)(L_2) = M_2 \tag{11.236}$$

$$\begin{aligned}
 \mathcal{M}(F, F)(L_1 \circ L_2) & \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(F, F)(L_1) \cdot \mathcal{M}(F, F)(L_2) \\
 & = \mathcal{M}(F, F)(\mathcal{M}[F, F]^{-1}(M_1)) \cdot \mathcal{M}(F, F)(\mathcal{M}[F, F]^{-1}(M_2)) \\
 & = M_1 \cdot M_2
 \end{aligned} \tag{11.237}$$

hence:

$$\begin{aligned}
 \det(M_1 \cdot M_2) &\stackrel{[\text{eq: 11.237}]}{=} \det(\mathcal{M}(F, F)(L_1 \circ L_2)) \\
 &\stackrel{[\text{theorem: 11.359}]}{=} \det(L_1 \circ L_2) \\
 &\stackrel{[\text{theorem: 11.315}]}{=} \det(L_1) \cdot \det(L_2) \\
 &\stackrel{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}(F, F)(L_1)) \cdot \det(\mathcal{M}(F, F)(L_2)) \\
 &\stackrel{[\text{eq: 11.236}]}{=} \det(M_1) \cdot \det(M_2)
 \end{aligned}$$

2. As $\mathcal{M}(F, F)(\text{Id}_{\mathcal{F}^n}) \stackrel{[\text{theorem: 11.343}]}{=} E$ we have

$$\det(E) = \det(\mathcal{M}(F, F)(\text{Id})) \stackrel{[\text{theorem: 11.359}]}{=} \det(\text{Id}_{\mathcal{F}^n}) \stackrel{[\text{theorem: 11.315}]}{=} 1$$

3. As $E = M \cdot M^{-1}$ we have $1 \stackrel{(2)}{=} \det(E) = \det(M \cdot M^{-1}) = \det(M) \cdot \det(M^{-1})$ hence $\det(M) \neq 0$ and $(\det(M))^{-1} = \det(M^{-1})$. $\square$

Theorem 11.362. Let $n \in \mathbb{N}$, F a field of characteristic zero, $M \in \mathcal{M}_{n,n}(F)$, $E = \{e_1, \dots, e_n\}$ the canonical basis for F^n [see theorem: 11.167] and Δ a determinant function in F^n such that $\Delta(e_1, \dots, e_n) = 1$ then

$$\det(M) = \Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) = \Delta(\text{row}(M, 1), \dots, \text{row}(M, n))$$

Proof. Let $M \in \mathcal{M}_{n,n}(F)$ and take $L = \mathcal{M}(E, E)^{-1}(M) \in \text{Hom}(F^n, F^n)$ then for $k \in \{1, \dots, n\}$ we have that

$$\begin{aligned}
 (L(e_i))_k &= (\mathcal{M}(E, E)^{-1}(M)(e_i))_k \\
 &\stackrel{[\text{example: 11.346}]}{=} \sum_{j \in \{1, \dots, n\}} M_{k,j} \cdot (e_i)_j \\
 &= \sum_{j \in \{1, \dots, n\}} M_{k,j} \cdot \delta_{j,i} \\
 &= M_{k,i} \\
 &= \text{col}(M, i)_k
 \end{aligned} \tag{11.238}$$

So

$$\begin{aligned}
 \Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) &\stackrel{[\text{eq: 11.238}]}{=} \Delta(L(e_1), \dots, L(e_n)) \\
 &= \Delta_L(e_1, \dots, e_n) \\
 &\stackrel{[\text{definition: 11.311}]}{=} \det(L) \cdot \Delta(e_1, \dots, e_n) \\
 &= \det(L) \cdot 1 \\
 &= \det(L) \\
 &\stackrel{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}(E, E)(L)) \\
 &= \det(M)
 \end{aligned}$$

proving that

$$\det(M) = \Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) \tag{11.239}$$

For the rows note that $\forall i, j \in \{1, \dots, n\}$ we have

$$\text{col}(M^T, i)_j = (M^T)_{j,i} = M_{i,j} = \text{row}(M, i)_j$$

so that

$$\text{col}(M^T, i) = \text{row}(M, i) \tag{11.240}$$

and

$$\begin{aligned} \det(M) &\stackrel{[\text{theorem: 11.358}]}{=} \det(M^T) \\ &\stackrel{[\text{eq: 11.239}]}{=} \Delta(\text{col}(M^T, 1), \dots, \text{col}(M^T, n)) \\ &\stackrel{[\text{eq: 11.240}]}{=} \Delta(\text{row}(M, 1), \dots, \text{row}(M, n)) \end{aligned}$$

giving finally

$$\Delta(\text{row}(M, 1), \dots, \text{row}(M, n)) = \det(M) \quad \square$$

The above can be used to determine what happens to the determinant of a matrix if we permute rows or columns of the matrix.

Definition 11.363. Let $n \in \mathbb{N}$, F a field, $M \in \mathcal{M}_{n,n}(F)$ and $\sigma \in P_n$ then:

1. $M_\sigma \in \mathcal{M}_{n,n}(F)$ is defined by $\forall i, j \in \{1, \dots, n\} \ (M_\sigma)_{i,j} = M_{\sigma(i),j}$ [essential we permute the rows of the matrix].
2. $M^\sigma \in \mathcal{M}_{n,n}(F)$ is defined by $\forall i, j \in \{1, \dots, n\} \ (M^\sigma)_{i,j} = M_{i,\sigma(j)}$ [essential we permute columns of the matrix].

Theorem 11.364. Let $n \in \mathbb{N}$, F a field, $M \in \mathcal{M}_{n,n}(F)$ and $\sigma \in P_n$ then:

1. $\det(M_\sigma) = \text{sign}(\sigma) \cdot \det(M)$
2. $\det(M^\sigma) = \text{sign}(\sigma) \cdot \det(M)$

Proof. Let $E = \{e_1, \dots, e_n\}$ be the canonical basis for F^n [see: theorem: 11.167] then by [theorem: 11.299] there exists a determinant function Δ in F^n such that

$$\Delta(e_1, \dots, e_n) = 1$$

1. $\forall i, j \in \{1, \dots, n\}$ we have $\text{row}(M_\sigma, i)_j = (M_\sigma)_{i,j} = M_{\sigma(i),j} = \text{row}(M, \sigma(i))_j$ proving that

$$\text{row}(M_\sigma, i) = \text{row}(M, \sigma(i)) \quad (11.241)$$

so that

$$\begin{aligned} \det(M_\sigma) &\stackrel{[\text{theorem: 11.362}]}{=} \Delta(\text{row}(M_\sigma, 1), \dots, \text{row}(M_\sigma, n)) \\ &\stackrel{[\text{eq: 11.241}]}{=} \Delta(\text{row}(M, \sigma(1)), \dots, \text{row}(M, \sigma(n))) \\ &\stackrel{\text{skew-symmetry}}{=} \text{sign}(\sigma) \cdot \Delta(\text{row}(M, 1), \dots, \text{row}(M, n)) \\ &\stackrel{[\text{theorem: 11.362}]}{=} \text{sign}(\sigma) \cdot \det(M) \end{aligned}$$

2. $\forall i, j \in \{1, \dots, n\}$ we have $\text{col}(M^\sigma, i)_j = (M^\sigma)_{j,i} = M_{j,\sigma(i)} = \text{col}(M, \sigma(i))_j$ proving that

$$\text{col}(M^\sigma, i) = \text{col}(M, \sigma(i)) \quad (11.242)$$

so that

$$\begin{aligned} \det(M^\sigma) &\stackrel{[\text{theorem: 11.362}]}{=} \Delta(\text{col}(M^\sigma, 1), \dots, \text{col}(M^\sigma, n)) \\ &\stackrel{[\text{eq: 11.241}]}{=} \Delta(\text{col}(M, \sigma(1)), \dots, \text{col}(M, \sigma(n))) \\ &\stackrel{\text{skew-symmetry}}{=} \text{sign}(\sigma) \cdot \Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) \\ &\stackrel{[\text{theorem: 11.362}]}{=} \text{sign}(\sigma) \cdot \det(M) \end{aligned}$$

□

The above allows us to construct a test to determine if the determinant of a matrix is zero.

Corollary 11.365. *Let $n \in \mathbb{N}$, F a field with characteristic zero and $M \in \mathcal{M}_{n,n}(F)$ then we have*

$$\begin{aligned} \det(M) &= 0 \\ \Updownarrow \\ \text{rank}(M) &< n \end{aligned}$$

Proof.

$\Rightarrow$. We prove this by contradiction, so assume that $n \leq \text{rank}(M)$ then, as by [definition: 11.335] $\text{rank}(M) \leq n$, we must have that $\text{rank}(M) = n \stackrel{[\text{definitions: 11.333, 11.335}]}{=} \dim(\text{span}(\text{cols}(M)))$.

Using [theorem: 11.158] it follows that $\text{cols}(M) = \{\text{col}(M, i) | i \in \{1, \dots, n\}\}$ is a basis for $\text{span}(\text{cols}(M)) \subseteq F^n$, hence $\{\text{col}(M, i) | i \in \{1, \dots, n\}\}$ is linear independent. Using [theorem: 11.155] and the fact that $\text{card}(\{\text{col}(M, i) | i \in \{1, \dots, n\}\}) = \dim(\text{span}(\text{cols}(M))) = n$ it follows that

$$\{\text{col}(M, i) | i \in \{1, \dots, n\}\} \text{ is a basis for } F^n$$

By [theorem: 11.299] there exists a determinant function Δ in F^n such that

$$\Delta(\text{cols}(M, 1), \dots, \text{cols}(M, n)) = 1$$

Using [theorem: 11.362] we have $\det(M) = \Delta(\text{cols}(M, 1), \dots, \text{cols}(M, n))$ so that $\det(M) = 1$ contradicting the hypothesis $\det(M) = 0$. Hence we must have that $\text{rank}(M) < n$.

$\Leftarrow$. Let $\{e_1, \dots, e_n\}$ be the canonical basis for F^n , then by [theorems: 11.299, 11.362] there exists a determinant function in F^n such that

$$\det(M) = \Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) \quad (11.243)$$

By the hypothesis we have $\dim(\text{span}(\text{cols}(M))) = \text{rank}(M) < n$. Define

$$\text{col}(M): \{1, \dots, n\} \rightarrow \text{cols}(M) \text{ by } \text{col}(M)(i) = \text{col}(M, i)$$

then we have two possibilities:

col(M) is injective. Then as

$$\text{col}(M) = \{\text{col}(M)(i) | i \in \{1, \dots, n\}\} = \{\text{col}(M, i) | i \in \{1, \dots, n\}\} = \text{cols}(M)$$

we have that $\text{col}(M): \{1, \dots, n\} \rightarrow \text{cols}(M)$ is a bijection so that $\text{card}(\text{cols}(M)) = n$. If $\text{cols}(M)$ is linear independent then it is a basis for $\text{span}(\text{cols}(M))$ so that $\dim(\text{cols}(M)) = n$, contradicting $\dim(\text{span}(\text{cols}(M))) = \text{rank}(M) < n$. Hence we must have that $\text{cols}(M) = \{\text{col}(M, i) | i \in \{1, \dots, n\}\}$ is linear dependent. So by [theorem: 11.131] $\{\text{col}(M, i)\}_{i \in \{1, \dots, n\}}$ is linear dependent, hence by [theorem: 11.291] we have that $\Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) = 0$, proving by [eq: 11.243] that

$$\det(M) = 0$$

col(M) is not injective. Then there exists $i, j \in \{1, \dots, n\}$ with $i \neq j$ and $\text{col}(M, i) = \text{col}(M, j)$. So that by [theorem: 11.291] we have that $\Delta(\text{col}(M, 1), \dots, \text{col}(M, n)) = 0$, proving by [eq: 11.243] that

$$\det(M) = 0 \quad \square$$

We will now develop the necessary tools to calculate the inverse of a matrix (if the matrix is invertible).

Definition 11.366. *Let $n \in \mathbb{N} \setminus \{1\}$, X a set and $a \in X$ then we define*

$$[+a]: X^{n-1} \rightarrow X^n \text{ by } [+a](x) \text{ where } \forall i \in \{1, \dots, n\} \quad ([+a](x))_i = \begin{cases} a & \text{if } i = 1 \\ x_{i-1} & \text{if } i \in \{2, \dots, n\} \end{cases}$$

In other words $[+a](x_1, \dots, x_{n-1}) = (a, x_1, \dots, x_{n-1})$

Lemma 11.367. *Let $n \in \mathbb{N} \setminus \{1\}$, F a field and $0 \in F$ the additive neutral element then*

$$[+0] \in \text{Hom}(F^{n-1}, F^n)$$

Proof. Let $\alpha \in F$, $x, y \in F^{n-1}$ then we have for $i \in \{1, \dots, n\}$

$$\begin{aligned} [+0](x+y) &= \begin{cases} 0 & \text{if } i=1 \\ (x+y)_{i-1} & \text{if } i \in \{2, \dots, n\} \end{cases} \\ &= \begin{cases} 0+0 & \text{if } i=1 \\ x_{i-1} + y_{i-1} & \text{if } i \in \{2, \dots, n\} \end{cases} \\ &= [+0](x) + [+0](y) \end{aligned}$$

and

$$[+0](\alpha \cdot x) = \begin{cases} 0 & \text{if } i=1 \\ (\alpha \cdot x)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases} = \begin{cases} \alpha \cdot 0 & \text{if } i=1 \\ \alpha \cdot x_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases} = \alpha \cdot ([+0](x)) \quad \square$$

We use the above now to generate a new determinant function from a existing determinant function.

Theorem 11.368. Let $n \in \mathbb{N}$, F a field with characteristic zero, $E = \{e_1, \dots, e_n\}$ the canonical basis for F^n [see: theorem: 11.167] then if

$$\Delta: (F^n)^n \rightarrow F$$

is a determinant function and we define

$$\Delta^-: (F^{n-1})^{n-1} \rightarrow F \text{ where } \Delta^-(x_1, \dots, x_{n-1}) = \Delta(e_1, [+0](x_1), \dots, [+0](x_{n-1}))$$

we have that

1. Δ^- is a determinant function.
2. If $\Delta(e_1, \dots, e_n) = 1$ then $\Delta^-(e_1, \dots, e_{n-1}) = 1$

Proof.

1. First we prove multilinearity. Let $x, y \in F^{n-1}$ and $\alpha \in F$ then for

$$(a_1, \dots, \underbrace{x + \alpha \cdot y}_i, \dots, a_{n-1}) \in (F^{n-1})^{n-1}$$

we have

$$\begin{aligned} \Delta^-(a_1, \dots, \underbrace{x + \alpha \cdot y}_i, \dots, a_{n-1}) &= \\ \Delta(e_1, [+0](a_1), \dots, \underbrace{[+0](x + \alpha \cdot y)}_i, \dots, [+0](a_n)) &= \\ \Delta(e_1, [+0](a_1), \dots, \underbrace{[+0](x) + \alpha \cdot [+0](y)}_i, \dots, [+0](a_n)) &= \\ \Delta(e_1, [+0](a_1), \dots, \underbrace{[+0](x)}_i, \dots, [+0](a_n)) + \alpha \cdot \Delta(e_1, [+0](a_1), \dots, \underbrace{[+0](y)}_i, \dots, [+0](a_n)) &= \\ \Delta^-(a_1, \dots, \underbrace{x}_i, \dots, a_{n-1}) + \alpha \cdot \Delta^-(a_1, \dots, \underbrace{y}_i, \dots, a_{n-1}) \end{aligned}$$

Next we prove skew-symmetry. Let $\{a_i\}_{i \in \{1, \dots, n-1\}} \subseteq F^{n-1}$ such that there exists a $i \neq j \in \{1, \dots, n-1\}$ with $a_i = a_j$ then $\forall k \in \{1, \dots, n\}$ we have

$$([+0](a_i))_k = \begin{cases} 0 & \text{if } k=1 \\ (a_i)_{k-1} & \text{if } k \in \{2, \dots, n\} \end{cases} = \begin{cases} 0 & \text{if } k=1 \\ (a_j)_{k-1} & \text{if } k \in \{2, \dots, n\} \end{cases} = ([+0](a_j))_k$$

proving that

$$[+0](a_i) = [+0](a_j)$$

Hence we have

$$0 = \Delta(e_1, [+0](a_1), \dots, [+0](a_n)) = \Delta^-(a_1, \dots, a_{i-1})$$

proving by [theorem: 11.291] that

Δ^- is skew-symmetric

2. Assume that $\Delta(e_1, \dots, e_n) = 1$. Let $i, j \in \{1, \dots, n-1\}$ then

$$\begin{aligned} ([+0](e_i))_j &= \begin{cases} 0 & \text{if } j = 1 \\ (e_i)_{j-1} & \text{if } i \in \{2, \dots, n\} \end{cases} \\ &= \begin{cases} 0 & \text{if } j = 1 \\ \delta_{i, j-1} & \text{if } i \in \{2, \dots, n\} \end{cases} \\ &\stackrel{i+1=j \Rightarrow i=j-1}{=} \delta_{i+1, j} \\ &= (e_{i+1})_j \end{aligned}$$

proving that $[+0](e_i) = e_{i+1}$ so that

$$\Delta^-(e_1, \dots, e_{n-1}) = \Delta(e_1, [+0](e_1), \dots, [+0](e_{n-1})) = \Delta(e_1, e_2, \dots, e_n) = 1 \quad \square$$

It will be useful to decrease the size of a matrix, for this we introduce the $[n \boxplus m]$ function that removes a row and a column from a matrix.

Definition 11.369. Let $n \in \mathbb{N} \setminus \{1\} = \{2, \dots, \infty\}$, F a field and $i, j \in \{1, \dots, n-1\}$ then we define:

$$[i \boxplus j]: \mathcal{M}_{n,n}(F) \rightarrow \mathcal{M}_{n-1,n-1}(F) \text{ where } ([i \boxplus j](M))_{k,l} = \begin{cases} M_{k,l} & \text{if } 1 \leq k < i \wedge 1 \leq l < j \\ M_{k+1,l} & \text{if } i \leq k \leq n-1 \wedge 1 \leq l < j \\ M_{k,l+1} & \text{if } 1 \leq k < i \wedge j \leq l \leq n-1 \\ M_{k+1,l+1} & \text{if } i \leq k \leq n-1 \wedge j \leq l \leq n-1 \end{cases}$$

In other words $[i \boxplus j](M)$ is the matrix that you get if you remove the i -the row and the j -the column of M .

Example 11.370.

$$[2 \boxplus 3] \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 10 & 20 & 30 & 40 \\ 50 & 60 & 70 & 80 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 10 & 20 & 40 \\ 50 & 60 & 80 \end{pmatrix}$$

We show not that removing the i -the row and j -the column of the transpose of a matrix is the same as transposing the result of removing the j -the row and i -the column from the matrix.

Theorem 11.371. Let $n \in \mathbb{N} \setminus \{1\}$, F a field, $i, j \in \{1, \dots, n\}$ and $M \in \mathcal{M}_{n,n}(F)$ then

$$[i \boxplus j](M^T) = ([j \boxplus i]M)^T$$

Proof. This is easily proved by considering all the possible cases, so if $k, l \in \{1, \dots, n-1\}$ then we have either:

$1 \leq k < i \wedge 1 \leq l < j$. Then

$$\begin{aligned} ([i \boxplus j](M^T))_{k,l} &= (M^T)_{k,l} \\ &= M_{l,k} \\ &= ([j \boxplus i]M)_{l,k} \\ &= (([j \boxplus i]M)^T)_{k,l} \end{aligned}$$

$i \leq k \leq n-1 \wedge 1 \leq l < j$. Then

$$\begin{aligned} ([i \boxplus j](M^T))_{k,l} &= (M^T)_{k+1,l} \\ &= M_{l,k+1} \\ &= ([j \boxplus i]M)_{l,k} \\ &= (([j \boxplus i]M)^T)_{k,l} \end{aligned}$$

$1 \leq k < i \wedge j \leq l \leq n-1$. Then

$$\begin{aligned} ([i \boxplus j](M^T))_{k,l} &= (M^T)_{k,l+1} \\ &= M_{l+1,k} \\ &= ([j \boxplus i]M)_{l,k} \\ &= (([j \boxplus i]M)^T)_{k,l} \end{aligned}$$

$i \leq k \leq n-1 \wedge j \leq l \leq n-1$. Then

$$\begin{aligned} ([i \boxplus j](M^T))_{k,l} &= (M^T)_{k+1,l+1} \\ &= M_{l+1,k+1} \\ &= ([j \boxplus i]M)_{l,k} \\ &= (([j \boxplus i]M)^T)_{k,l} \end{aligned}$$

so that

$$[i \boxplus j](M^T) = ([j \boxplus i]M)^T$$

□

Removing a row and column of a matrix reduces the number of rows and columns with one, allowing us later in this book to express the calculation of the determinant of a matrix by calculating the determinant of a sub matrix. We can also reduce the size of a matrix by removing all the rows or columns after a specified row or specified column. This is the purpose of the following definition.

Definition 11.372. Let $n \in \mathbb{N} \setminus \{1\}$, F a field and $M \in \mathcal{M}_{n,n}(F)$ then we define

1. If $m \in \{2, \dots, n\}$ then $[<m](M) \in \mathcal{M}_{m-1,m-1}$ by $\forall k, l \in \{1, \dots, m-1\}$

$$([<m](M))_{k,l} = M_{k,l}$$

2. If $m \in \{1, \dots, n-1\}$ then $[>m](M) \in \mathcal{M}_{n-m,n-m}(F)$ by $\forall k, l \in \{1, \dots, n-m\}$

$$([>m](M))_{k,l} = M_{k+m,l+m}$$

We have now the following properties for $[<m]$ and $[>m]$.

Theorem 11.373. Let $n \in \mathbb{N} \setminus \{1\}$, F a field then we have for $M \in \mathcal{M}_{n,n}(F)$

1. $[1 \boxplus 1](M) = [>1](M)$
2. $[n \boxplus n](M) = [<n](M)$
3. If $m_1 \in \{2, \dots, n\}$ and $m_2 \in \{2, \dots, m_1-1\}$ then $[<m_2]([<m_1](M)) = [<m_2](M)$
4. If $m_1 \in \{1, \dots, n-1\}$ and $m_2 \in \{1, \dots, n-m_1-1\}$ then $[>m_2]([>m_1](M)) = [>(m_1+m_2)](M)$
5. If $m \in \{2, \dots, n\}$ then $([<m](M))^T = [<m](M^T)$
6. If $m \in \{1, \dots, n-1\}$ then $([>m](M))^T = [>m](M^T)$

Proof.

1. $\forall i, j \in \{1, \dots, n-1\}$ we have

$$\begin{aligned} ([1 \boxplus 1](M))_{i,j} & \stackrel{1 \leq i, j \leq n-1 \wedge \text{definition: 11.369}}{=} M_{i+1,j+1} \\ & = ([>1]M)_{i,j} \end{aligned}$$

so we have $[1 \boxplus 1](M) = [>1]M$

2. $\forall i, j \in \{1, \dots, n-1\}$ we have

$$\begin{aligned} ([n \boxplus n](M))_{i,j} & \quad 1 \leq i, j \leq n-1 < n \wedge \overline{\overline{\text{definition: 11.369}}} & M_{i,j} \\ & \quad 1 \leq i, j \leq n-1 & ([<n](M))_{i,j} \end{aligned}$$

so we have $[n \boxplus n](M) = [<n](M)$

3. If $i, j \in \{1, \dots, m_2-1\}$ then we have

$$\begin{aligned} ([<m_2]([<m_1](M)))_{i,j} & \quad 1 \leq i, j \leq m_2-1 & ([<m_1](M))_{i,j} \\ & \quad 1 \leq i, j \leq m_2-1 < m_2 \leq m_1-1 & M_{i,j} \\ & \quad 1 \leq i, j \leq m_2-1 & ([<m_2]M)_{i,j} \end{aligned}$$

so we have $[<m_2]([<m_1](M)) = [<m_2](M)$

4. First note that

$$\begin{aligned} [>m_1](M) \in \mathcal{M}_{n-m_1, n-m_1}(F) & \Rightarrow [>m_2]([>m_1](M)) \in \mathcal{M}_{n-m_1-m_2, n-m_1-m_2}(F) \\ & \Rightarrow [>m_2]([>m_1](M)) \in \mathcal{M}_{n-(m_1+m_2), n-(m_1+m_2)}(F) \end{aligned}$$

and

$$[>(m_1+m_2)](M) \in \mathcal{M}_{n-(m_1+m_2), n-(m_1+m_2)}(F)$$

So that

$$[>(m_1+m_2)](M), [>m_2]([>m_1](M)) \in \mathcal{M}_{n-(m_1+m_2), n-(m_1+m_2)}(F)$$

Next $\forall i, j \in \{1, \dots, n-(m_1+m_2)\}$ we have

$$\begin{aligned} ([>m_2]([>m_1](M)))_{i,j} &= ([>m_1](M))_{i+m_2, j+m_2} \\ &= M_{i+m_2+m_1, j+m_2+m_1} \\ &= M_{i+(m_1+m_2), j+(m_1+m_2)} \\ &= ([>(m_1+m_2)](M))_{i,j} \end{aligned}$$

proving

$$[>m_2]([>m_1](M)) = [>(m_1+m_2)](M)$$

5. For $\forall i, j \in \{1, \dots, m-1\}$ we have

$$\begin{aligned} ([<m](M))^T_{i,j} &= ([<m](M))_{j,i} \\ &= M_{j,i} \\ &= (M^T)_{i,j} \\ &= ([<m](M^T))_{i,j} \end{aligned}$$

so

$$([<m](M))^T = [<m](M^T)$$

6. For $\forall i, j \in \{1, \dots, n-m\}$ we have

$$\begin{aligned} ([>m](M))^T_{i,j} &= ([>m](M))_{j,i} \\ &= M_{j+m, i+m} \\ &= (M^T)_{i+m, j+m} \\ &= ([>m](M^T))_{i,j} \end{aligned}$$

so

$$([>m](M))^T = [>m](M^T)$$

□

We want now to calculate the determinant of a $n \times n$ matrix in terms of a $(n-1) \times (n-1)$ submatrix. More specific we want to prove that

$$\begin{vmatrix} m_{1,1} & \dots & m_{1,i-1} & 0 & \dots & m_{1,i+1} & \dots & m_{1,n} \\ & & & \vdots & & & & \vdots \\ m_{j-1,1} & & & 0 & & & & \vdots \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 & 0 \\ m_{j+1,1} & & & 0 & & & & \vdots \\ \vdots & & & \vdots & & & & \vdots \\ m_{n,1} & \dots & \dots & 0 & \dots & \dots & & m_{n,n} \end{vmatrix}$$

is equal to

$$(-1)^{i+j} \cdot \begin{vmatrix} m_{1,1} & \dots & m_{1,i-1} & m_{1,i+1} & \dots & m_{1,n} \\ \vdots & & \vdots & \vdots & & \vdots \\ m_{j-1,1} & \dots & m_{j-1,i-1} & m_{j-1,i+1} & \dots & m_{j-1,n} \\ m_{j+1,1} & \dots & m_{j+1,i-1} & m_{j+1,i+1} & \dots & m_{j+1,n} \\ \vdots & & \vdots & \vdots & & \vdots \\ m_{n,1} & \dots & m_{n,i-1} & m_{n,i+1} & \dots & m_{n,n} \end{vmatrix}$$

First we prove that

$$\begin{vmatrix} 1 & 0 & \dots & 0 \\ 0 & m_{2,2} & \dots & m_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & m_{n,2} & \dots & m_{n,n} \end{vmatrix} = \begin{vmatrix} m_{2,2} & \dots & m_{2,n} \\ \vdots & \ddots & \vdots \\ m_{n,2} & \dots & m_{n,n} \end{vmatrix}$$

Lemma 11.374. Let $n \in \mathbb{N} \setminus \{1\}$ and $M \in M_{n,n}(F)$ such that $\forall i \in \{1, \dots, n\}$

$$\text{row}(M, 1)_i = \delta_{i,1} = \text{col}(M, 1)_i$$

then

$$\det(M) = \det([1 \boxplus 1](M))$$

Proof. Let $\{e_1, \dots, e_n\}$ be the canonical basis of F^n . Let $i \in \{1, \dots, n\}$ then we have either:

$i = 1$. Then we have $\forall j \in \{1, \dots, n\}$ that $\text{row}(M, 1)_j = \delta_{i,1} = (e_1)_j$ so that

$$\text{row}(M, 1) = e_1 \tag{11.244}$$

$i \in \{2, \dots, n\}$. Then for $j \in \{1, \dots, n\}$ we have either:

$j = 1$. Then

$$\begin{aligned} ([+0](\text{row}([1 \boxplus 1](M), i)))_j &= ([+0](\text{row}([1 \boxplus 1](M), i)))_1 \\ &\stackrel{[\text{definition: 11.366}]}{=} 0 \\ &\stackrel{i+1 \neq 1}{=} \delta_{i+1,1} \\ &= \text{col}(M, 1)_{i+1} \\ &= M_{i+1,1} \\ &= \text{row}(M, i+1)_1 \\ &= \text{row}(M, i+1)_j \end{aligned}$$

$j \in \{2, \dots, n\}$. Then $1 \leq j-1 \leq n \wedge 1 \leq i \leq n$

$$\begin{aligned}
 ([+0](\text{row}([1 \boxplus 1](M), i)))_j &\stackrel{[\text{definition: 11.366}] \wedge j \in \{2, \dots, n\}}{=} \text{row}([1 \boxplus 1](M), i)_{j-1} \\
 &\stackrel{[\text{definition: 11.369}]}{=} ([1 \boxplus 1](M))_{i, j-1} \\
 &= M_{i+1, j} \\
 &= \text{row}(M, i+1)_j
 \end{aligned}$$

So we have that

$$\text{row}(M, i+1) = [+0](\text{row}([1 \boxplus 1](M), i)) \quad (11.245)$$

Let Δ be a determinant function such that $\Delta(e_1, \dots, e_n) = 1$ [see theorem: 11.299] then we have

$$\begin{aligned}
 \det(M) &\stackrel{[\text{theorem: 11.362}]}{=} \Delta(\text{row}(M, 1), \text{row}(M, 2), \dots, \text{row}(M, n)) \\
 &\stackrel{[\text{eq: 11.244}]}{=} \Delta(e_1, \text{row}(M, 2), \dots, \text{row}(M, n)) \\
 &\stackrel{[\text{eq: 11.245}]}{=} \Delta(e_1, [+0](\text{row}([1 \boxplus 1](M), 1)), \dots, [+0](\text{row}([1 \boxplus 1](M), n-1))) \\
 &\stackrel{[\text{definition: 11.368}]}{=} \Delta^-(\text{row}([1 \boxplus 1](M), 1), \dots, \text{row}([1 \boxplus 1](M), n-1)) \quad (11.246)
 \end{aligned}$$

As Δ is a determinant function with $\Delta(e_1, \dots, e_n) = 1$ we have by [theorem: 11.368] that

$$\Delta^- \text{ is a determinant function and } \Delta^-(e_1, \dots, e_{n-1}) = 1$$

which as $\{e_1, \dots, e_{n-1}\}$ is the canonical basis of F^{n-1} proves by [theorem: 11.362] that

$$\Delta^-(\text{row}([1 \boxplus 1](M), 1), \dots, \text{row}([1 \boxplus 1](M), n-1)) = \det([1 \boxplus 1](M))$$

Substituting the above in [eq: 11.246] proves then

$$\det(M) = \det([1 \boxplus 1](M)) \quad \square$$

We extend now the above lemma to a more general case:

$$\begin{vmatrix} 0 & \dots & 0 & 1 & 0 & \dots & 0 \\ m_{2,1} & & m_{2,j-1} & 0 & m_{2,j+1} & \dots & m_{2,n} \\ \vdots & & & \vdots & & & \vdots \\ m_{n,1} & \dots & m_{n,j-1} & 0 & m_{n,j+1} & \dots & m_{n,n} \end{vmatrix}$$

is equal to

$$(-1)^{1+j} \cdot \begin{vmatrix} m_{2,1} & \dots & m_{2,n} \\ \vdots & \ddots & \vdots \\ m_{2,n} & \dots & m_{n,n} \end{vmatrix}$$

Theorem 11.375. Let $n \in \mathbb{N} \setminus \{1\}$, F a field with characteristic zero and $M \in \mathcal{M}_{n,n}(F)$ satisfying $\exists j \in \{1, \dots, n\}$ such that $\forall i \in \{1, \dots, n\}$ we have $\text{row}(M, 1)_i = \delta_{i,j}$ and $\text{col}(M, j)_i = \delta_{i,1}$ then

$$\det(M) = (-1)^{1+j} \cdot \det([1 \boxplus j](M))$$

Proof. For j we have the following cases:

$j = 1$. Then $\forall i \in \{1, \dots, n\}$ we have $\text{row}(M, 1)_i = \delta_{i,1} = \text{col}(M, 1)_i$ so that the conditions of the previous lemma [lemma: 11.374] are satisfied. Hence $\det(M) = \det([1 \boxplus 1](M))$ which, as $j = 1$ proves

$$\det(M) = (-1)^{1+j} \cdot \det([1 \boxplus j](M))$$

$j \in \{2, \dots, n\}$. Define M' by moving the j -th column before the first column [see definition: 11.252], in other words M' is defined by

$$\forall k, l \in \{1, \dots, n\} \quad M'_{k,l} = M_{k, (j \rightsquigarrow 1)(l)}$$

then using [definition: 11.363] we have that

$$M' = M^{(j \rightsquigarrow_n 1)} \quad (11.247)$$

Further $\forall k \in \{1, \dots, n\}$ we have

$$\begin{aligned} \text{row}(M', 1)_k &= M'_{1,k} \\ &= M_{1, (j \rightsquigarrow_n 1)(k)} \\ &\stackrel{1 < j \wedge [\text{definition: 11.252(3)}]}{=} \begin{cases} M_{1,k} & \text{if } 1 \leq k < 1 \\ M_{1,j} & \text{if } k = 1 \\ M_{1,k-1} & \text{if } 1 < k \leq j \\ M_{1,k} & \text{if } j < k \leq n \end{cases} \\ &= \begin{cases} M_{1,j} & \text{if } k = 1 \\ M_{1,k-1} & \text{if } 1 < k \leq j \\ M_{1,k} & \text{if } j < k \leq n \end{cases} \\ &= \begin{cases} \text{col}(M, j)_1 & \text{if } k = 1 \\ \text{row}(M, 1)_{k-1} & \text{if } 1 < k \leq j \\ \text{row}(M, 1)_k & \text{if } j < k \leq n \end{cases} \\ &= \begin{cases} \delta_{k,1} & \text{if } k = 1 \\ \delta_{k-1,j} & \text{if } 1 < k \leq j \\ \delta_{k,1} & \text{if } j < k \leq n \end{cases} \\ &= \begin{cases} 1 & \text{if } k = 1 \\ \delta_{k-1,j} & \text{if } 1 < k \leq j \\ \delta_{k,j} & \text{if } j < k \leq n \end{cases} \\ &\stackrel{k \leq j \Rightarrow k-1 < j \Rightarrow k-1 \neq j}{=} \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{if } 1 < k \leq j \\ \delta_{k,j} & \text{if } j < k \leq n \end{cases} \\ &\stackrel{j < k \Rightarrow j \neq k}{=} \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{if } 1 < k \leq j \\ 0 & \text{if } j < k \leq n \end{cases} \\ &= \delta_{k,1} \end{aligned}$$

and we have also

$$\begin{aligned} \text{col}(M', 1)_k &= M'_{k,1} \\ &= M_{k, (j \rightsquigarrow_n 1)(1)} \\ &\stackrel{1=1 \wedge [\text{definition: 11.252(1)}]}{=} M_{k,j} \\ &= \text{col}(M, j)_k \\ &= \delta_{k,1} \end{aligned}$$

Hence M' satisfies the conditions of the previous lemma [lemma: 11.374] giving us

$$\det(M') = \det([1 \boxplus 1](M')) \quad (11.248)$$

Now if $k, l \in \{1, \dots, n-1\}$ then $1 \leq k, l \leq n-1$ it follows from [definition: 11.369] that

$$([1 \boxplus 1](M'))_{k,l} = M'_{k+1,l+1} \quad (11.249)$$

Take $k \in \{1, \dots, n-1\}$ then for $l \in \{1, \dots, n-1\}$ we have the following possible cases:

$1 \leq l < j$. Then

$$\begin{aligned} M'_{k+1, l+1} &= M_{k+1, (j \rightsquigarrow_n 1)(l+1)} \\ &\stackrel{l < j \Rightarrow 1 < l+1 \leq j \wedge [\text{definition: 11.252(3)}]}{=} M_{k+1, l} \\ &\stackrel{1 \leq k \leq n-1, l < j \wedge [\text{definition: 11.369}]}{=} ([1 \boxplus j](M))_{k, l} \end{aligned}$$

$j \leq l \leq n-1$. Then

$$\begin{aligned} M'_{k+1, l+1} &= M_{k+1, (j \rightsquigarrow_n 1)(l+1)} \\ &\stackrel{j \leq l \Rightarrow j < l+1 \leq n \wedge [\text{definition: 11.252(3)}]}{=} M_{k+1, l+1} \\ &\stackrel{1 \leq k \leq n-1 \wedge j \leq l \leq n-1 \wedge [\text{definition: 11.369}]}{=} ([1 \boxplus j](M))_{k, l} \end{aligned}$$

So in all cases we gave $M'_{k+1, l+1} = ([1 \boxplus j](M))_{k, l}$, combining this with [eq: 11.249] results in $([1 \boxplus 1](M'))_{k, l} = ([1 \boxplus j](M))_{k, l}$. Hence we have $[1 \boxplus 1](M') = [1 \boxplus j](M)$. Substituting this result in [eq: 11.248] gives

$$\det(M') = \det([1 \boxplus j](M)) \quad (11.250)$$

Further

$$\begin{aligned} \det(M') &\stackrel{[\text{eq: 11.247}]}{=} \det(M^{(j \rightsquigarrow_n 1)}) \\ &\stackrel{[\text{theorem: 11.364}]}{=} \text{sign}((j \rightsquigarrow_n 1)) \cdot \det(M) \\ &\stackrel{[\text{theorem: 11.255}]}{=} (-1)^{j-1} \cdot \det(M) \end{aligned}$$

Hence by [theorem: 11.281]

$$\det(M) = (-1)^{j+1} \cdot \det(M') \stackrel{[\text{eq: 11.250}]}{=} (-1)^{1+j} \cdot \det([1 \boxplus j](M))$$

proving the lemma. $\square$

Finally we prove that

$$\begin{vmatrix} m_{1,1} & \dots & m_{1,i-1} & 0 & \dots & m_{1,i+1} & \dots & m_{1,n} \\ & & & \vdots & & & & \vdots \\ m_{j-1,1} & & & 0 & & & & \vdots \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 & 0 \\ m_{j+1,1} & & & 0 & & & & \vdots \\ \vdots & & & \vdots & & & & \vdots \\ m_{n,1} & \dots & \dots & 0 & \dots & \dots & & m_{n,n} \end{vmatrix}$$

is equal to

$$(-1)^{i+j} \cdot \begin{vmatrix} m_{1,1} & \dots & m_{1,i-1} & m_{1,i+1} & \dots & m_{1,n} \\ \vdots & & \vdots & \vdots & & \vdots \\ m_{j-1,1} & \dots & m_{j-1,i-1} & m_{j-1,i+1} & \dots & m_{j-1,n} \\ m_{j+1,1} & \dots & m_{j+1,i-1} & m_{j+1,i+1} & \dots & m_{j+1,n} \\ \vdots & & \vdots & \vdots & & \vdots \\ m_{n,1} & \dots & m_{n,i-1} & m_{n,i+1} & \dots & m_{n,n} \end{vmatrix}$$

Theorem 11.376. Let $n \in \mathbb{N} \setminus \{1\}$, F a field with characteristic zero and $M \in \mathcal{M}_{n,n}(F)$ satisfying $\exists i, j \in \{1, \dots, n\}$ such that $\forall k \in \{1, \dots, n\}$ we have $\text{row}(M, i)_k = \delta_{j,k}$ and $\text{col}(M, j)_k = \delta_{i,k}$ then

$$\det(M) = (-1)^{i+j} \cdot \det([i \boxplus j](M))$$

Proof. For $i \in \{1, \dots, n\}$ we have the following cases to consider:

$i = 1$. Then $\forall k \in \{1, \dots, n\}$ we have $\text{row}(M, 1)_k = \delta_{j,k}$ and $\text{col}(M, j)_k = \delta_{i,k}$ so that the conditions for the previous lemma [lemma: 11.375] are satisfied, hence we have that

$$\det(M) = (-1)^{1+j} \cdot \det([1 \boxplus j])$$

which as $i = 1$ proves that

$$\det(M) = (-1)^{i+j} \cdot \det([i \boxplus j](M))$$

$i \in \{2, \dots, n\}$. We move now the i -th row before the first row, so define M' by

$$M' = M_{(i \rightsquigarrow 1)} \quad (11.251)$$

or using [definition: 11.363] we have

$$\forall k, l \in \{1, \dots, n\} \text{ that } M'_{k,l} = M_{(i \rightsquigarrow 1)(k),l} \quad (11.252)$$

Let $k \in \{1, \dots, n\}$ then we have

$$\begin{aligned} \text{row}(M', 1)_k &= M'_{1,k} \\ &= M_{(i \rightsquigarrow 1_n)(1),k} \\ &\stackrel{1=1 < k \wedge [\text{definition: 11.252(3)}]}{=} M_{i,k} \\ &= \text{row}(M, i)_k \\ &= \delta_{j,k} \end{aligned}$$

and

$$\begin{aligned} \text{col}(M', j)_k &= M'_{k,j} \\ &= M_{(i \rightsquigarrow 1_n)(k),j} \\ &\stackrel{[\text{definition: 11.252(3)}]}{=} \begin{cases} M_{k,j} & \text{if } 1 \leq k < 1 \\ M_{i,j} & \text{if } k = 1 \\ M_{k-1,j} & \text{if } 1 \leq k \leq i \\ M_{k,j} & \text{if } i < k \leq n \end{cases} \\ &= \begin{cases} M_{i,j} & \text{if } k = 1 \\ M_{k-1,j} & \text{if } 1 \leq k \leq i \\ M_{k,j} & \text{if } i < k \leq n \end{cases} \\ &= \begin{cases} \text{row}(M, i)_j & \text{if } k = 1 \\ \text{col}(M, j)_{k-1} & \text{if } 1 \leq k \leq i \\ \text{col}(M, j)_k & \text{if } i < k \leq n \end{cases} \\ &= \begin{cases} \delta_{j,j} & \text{if } k = 1 \\ \delta_{i,k-1} & \text{if } 1 \leq k \leq i \\ \delta_{i,k} & \text{if } i < k \leq n \end{cases} \\ &= \begin{cases} 1 & \text{if } k = 1 \\ \delta_{i,k-1} & \text{if } 1 \leq k \leq i \\ \delta_{i,k} & \text{if } i < k \leq n \end{cases} \\ &\stackrel{k \leq i \Rightarrow k-1 < i \Rightarrow k-1 \neq i}{=} \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{if } 1 \leq k \leq i \\ \delta_{i,k} & \text{if } i < k \leq n \end{cases} \\ &\stackrel{i < k \Rightarrow i \neq k}{=} \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{if } 1 \leq k \leq i \\ 0 & \text{if } i < k \leq n \end{cases} \\ &= \delta_{1,k} \end{aligned}$$

So the conditions for the previous lemma [lemma: 11.375] are satisfied for M' , hence

$$\det(M') = (-1)^{1+j} \cdot \det([1 \boxplus j](M')) \quad (11.253)$$

Now if $k, l \in \{1, \dots, n-1\}$ we have the following possibilities for k, l :

$1 \leq k < i \wedge 1 \leq l < j$. Then

$$\begin{aligned} ([1 \boxplus j](M'))_{k,l} & \stackrel{1 \leq k \leq n-1 \wedge 1 \leq l < j \wedge [\text{definition: 11.369}]}{=} M'_{k+1,l} \\ & \stackrel{=}{=} M_{(i \rightsquigarrow_n 1)(k+1),l} \\ & \stackrel{1 \leq k < i \Rightarrow 1 < k+1 \leq i \wedge [\text{theorem: 11.252(3)}]}{=} M_{k,l} \\ & \stackrel{1 \leq k \leq i \wedge 1 \leq l < j \wedge [\text{definition: 11.369}]}{=} ([i \boxplus j](M))_{k,l} \end{aligned}$$

$i \leq k \leq n-1 \wedge 1 \leq l < j$. Then

$$\begin{aligned} ([1 \boxplus j](M'))_{k,l} & \stackrel{1 \leq k \leq n-1 \wedge 1 \leq l < j \wedge [\text{definition: 11.369}]}{=} M'_{k+1,l} \\ & \stackrel{=}{=} M_{(i \rightsquigarrow_n 1)(k+1),l} \\ & \stackrel{i \leq k < n-1 \Rightarrow i < k+1 \leq n \wedge [\text{theorem: 11.252(3)}]}{=} M_{k+1,l} \\ & \stackrel{i \leq k \leq n-1 \wedge 1 \leq l < j \wedge [\text{definition: 11.369}]}{=} ([i \boxplus j](M))_{k,l} \end{aligned}$$

$1 \leq k < i \wedge j \leq l < n-1$. Then

$$\begin{aligned} ([1 \boxplus j](M'))_{k,l} & \stackrel{1 \leq k \leq n-1 \wedge j \leq l < n-1 \wedge [\text{definition: 11.369}]}{=} M'_{k+1,l+1} \\ & \stackrel{=}{=} M_{(i \rightsquigarrow_n 1)(k+1),l+1} \\ & \stackrel{1 \leq k < i \Rightarrow 1 < k+1 \leq n \wedge [\text{theorem: 11.252(3)}]}{=} M_{k,l+1} \\ & \stackrel{1 \leq k < i \wedge j \leq l < n-1 \wedge [\text{definition: 11.369}]}{=} ([i \boxplus j](M))_{k,l} \end{aligned}$$

$i \leq k \leq n-1 \wedge j \leq l \leq n-1$. Then

$$\begin{aligned} ([1 \boxplus j](M'))_{k,l} & \stackrel{1 \leq k \leq n-1 \wedge j \leq l < n-1 \wedge [\text{definition: 11.369}]}{=} M'_{k+1,l+1} \\ & \stackrel{=}{=} M_{(i \rightsquigarrow_n 1)(k+1),l+1} \\ & \stackrel{i \leq k < n-1 \Rightarrow i < k+1 \leq n \wedge [\text{theorem: 11.252(3)}]}{=} M_{k+1,l+1} \\ & \stackrel{i \leq k \leq n-1 \wedge j \leq l < n-1 \wedge [\text{definition: 11.369}]}{=} ([i \boxplus j](M))_{k,l} \end{aligned}$$

So we have proved that $\forall k, l \in \{1, \dots, n-1\}$ $([1 \boxplus j](M'))_{k,l} = ([i \boxplus j](M))_{k,l}$ giving

$$[1 \boxplus j](M') = [i \boxplus j](M)$$

Substituting the above in [eq: 11.253] gives then

$$\det(M') = (-1)^{1+j} \cdot \det([i \boxplus j](M)) \quad (11.254)$$

Now

$$\begin{aligned} \det(M') & \stackrel{[\text{eq: 11.251}]}{=} \det(M_{(i \rightsquigarrow_n 1)}) \\ & \stackrel{[\text{theorem: 11.364}]}{=} \text{sign}((i \rightsquigarrow_n 1)) \cdot \det(M) \\ & \stackrel{[\text{theorem: 11.255}]}{=} (-1)^{i-1} \cdot \det(M) \end{aligned}$$

Substituting the above in [eq: 11.254] gives

$$(-1)^{i-1} \cdot \det(M) = (-1)^{1+j} \cdot \det([i \boxplus j](M)).$$

Multiplying both sides by $(-1)^{i-1}$ and using the fact that $(-1)^{i-1} \cdot (-1)^{i-1} = 1$ and $(-1)^{i-1} \cdot (-1)^{j+1} = (-1)^{i+j}$ we have finally that

$$\det(M) = (-1)^{i+j} \cdot \det([i \boxplus j](M)) \quad \square$$

For linear transformations we can calculate the inverse of a linear transformation using the determinant and adjoint of a linear transformation. We have introduced the determinant of a matrix and shows its relation with the determinant of the associated linear transformation. Now we do the same for the adjoint, so we will define the adjoint of a matrix and show its relation with the associated linear transformation.

Definition 11.377. Let $n, m \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,m}(F)$ then for $i, j \in \{1, \dots, n\}$ $[i \oplus j](M)$ is defined by

$$\forall k, l \in \{1, \dots, n\} \text{ we have } ([i \oplus j](M))_{k,l} = \begin{cases} M_{k,l} & \text{if } l \in \{1, \dots, n\} \setminus \{i\} \\ \delta_{k,j} & \text{if } l = i \end{cases}$$

In other words we replace the i -th column by the column $\begin{pmatrix} \delta_{1,j} \\ \vdots \\ \delta_{n,j} \end{pmatrix}$

Example 11.378.

$$[2 \oplus 3] \left(\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 10 & 20 & 30 & 40 \\ 11 & 12 & 13 & 14 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 & 3 & 4 \\ 5 & 0 & 7 & 8 \\ 10 & 1 & 30 & 40 \\ 11 & 0 & 13 & 14 \end{pmatrix}$$

Definition 11.379. (adjoint) Let $n \in \mathbb{N}$, F a field and $M \in \mathcal{M}_{n,n}(F)$ then we define

$$\text{adjoint: } \mathcal{M}_{n,n}(F) \rightarrow \mathcal{M}_{n,n}(F) \text{ where } \forall i, j \in \{1, \dots, n\} \text{ (adjoint}(M))_{i,j} = \det([i \oplus j](M))$$

The relation between the adjoint of a linear transformation and its matrix is expressed in the following theorem.

Theorem 11.380. Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field with characteristic zero, $E = \{e_1, \dots, e_n\}$ a basis of X and $L \in \text{Hom}(X, X)$ then

$$\mathcal{M}(E, E)(\text{adjoint}(L)) = \text{adjoint}(\mathcal{M}(E, E)(L))$$

or in other words

$$\mathcal{M}(\text{adjoint}(L); E, E) = \text{adjoint}(\mathcal{M}(L; E, E))$$

So the matrix of the adjoint of a linear transformation is the adjoint of the matrix of the linear transformation.

Proof. Using [theorem: 11.299] there exists a determinant function in X such that

$$\Delta(e_1, \dots, e_n) = 1$$

Now using the definition of the matrix of a linear map we have for $i \in \{1, \dots, n\}$

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} \mathcal{M}(\text{adjoint}(L); E, E)_{i,j} \cdot e_i & \stackrel{\text{def}}{=} \\ & \text{adjoint}(L)(e_j) \\ & 1 \cdot \text{adjoint}(L)(e_j) = \\ & \Delta(e_1, \dots, e_n) \cdot \text{adjoint}(L)(e_j) \stackrel{[\text{definition: 11.320}]}{=} \\ & \overline{\Delta L}(e_1, \dots, e_n)(e_j) \stackrel{[\text{definition: 11.317}]}{=} \\ & \sum_{i \in \{1, \dots, n\}} (-1)^{i-1} \cdot \Delta(e_j, L(e_1), \dots, L(e_{i-1}), L(e_{i+1}), \dots, L(e_n)) \cdot e_i \end{aligned}$$

From the uniqueness of expanding in a basis it follows that $\forall i, j \in \{1, \dots, n\}$ we have

$$\mathcal{M}(\text{adjoint}(L); E, E)_{i,j} = (-1)^{i-1} \cdot \Delta(e_j, L(e_1), \dots, L(e_{i-1}), L(e_{i+1}), \dots, L(e_n)) \quad (11.255)$$

Given $i, j \in \{1, \dots, n\}$ define

$$x^{(i,j)} \in X^n \text{ by } \forall k \in \{1, \dots, n\} \quad x_k^{(i,j)} = \begin{cases} e_j & \text{if } k = i \\ L(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \end{cases} \quad (11.256)$$

Then we have for $i, j \in \{1, \dots, n\}$ and $k \in \{1, \dots, n\}$ that

$$\begin{aligned} (e_j, L(e_1), \dots, L(e_{i-1}), L(e_{i+1}), \dots, L(e_n))_k &\stackrel{[\text{definition: 11.302}]}{=} \\ &\begin{cases} e_j & \text{if } k = 1 \\ L(e_{k-1}) & \text{if } k \in \{2, \dots, i\} \\ L(e_k) & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &\begin{cases} x_i^{(i,j)} & \text{if } k = 1 \\ x_{k-1}^{(i,j)} & \text{if } k \in \{2, \dots, i\} \\ x_k^{(i,j)} & \text{if } k \in \{i+1, \dots, n\} \end{cases} = \\ &(x_i^{(i,j)}, x_1^{(i,j)}, \dots, x_{i-1}^{(i,j)}, x_{i+1}^{(i,j)}, \dots, x_n^{(i,j)})_k \stackrel{[\text{theorem: 11.303}]}{=} \\ &\left(x_{(i \rightsquigarrow 1)(1)}^{(i,j)}, \dots, x_{(i \rightsquigarrow 1)(n)}^{(i,j)}\right) \end{aligned}$$

So substituting the above in [eq: 11.255] gives $\forall i, j \in \{1, \dots, n\}$ that

$$\begin{aligned} \mathcal{M}(\text{adjoint}(L); E, E)_{i,j} &= (-1)^{i-1} \cdot \Delta\left(x_{(i \rightsquigarrow 1)(1)}^{(i,j)}, \dots, x_{(i \rightsquigarrow 1)(n)}^{(i,j)}\right) \\ &= (-1)^{i-1} \cdot ((i \rightsquigarrow 1)\Delta)(x_1^{(i,j)}, \dots, x_n^{(i,j)}) \\ &= (-1)^{i-1} \cdot \text{sign}((i \rightsquigarrow 1)) \cdot \Delta(x_1^{(i,j)}, \dots, x_n^{(i,j)}) \\ &\stackrel{[\text{theorem: 11.255}]}{=} (-1)^{i-1} \cdot (-1)^{i-1} \cdot \Delta(x_1^{(i,j)}, \dots, x_n^{(i,j)}) \\ &= \Delta(x_1^{(i,j)}, \dots, x_n^{(i,j)}) \end{aligned} \quad (11.257)$$

Let $i, j \in \{1, \dots, n\}$ and define $M^{(i,j)} \in \mathcal{M}_{n,n}(F)$ by

$$M^{(i,j)} = [i \oplus j](\mathcal{M}(E, E)(L)) \quad (11.258)$$

so that by [definition: 11.377]

$$\forall r, s \in \{1, \dots, n\} \text{ we have } (M^{(i,j)})_{r,s} = \begin{cases} (\mathcal{M}(E, E)(L))_{r,s} & \text{if } s \in \{1, \dots, n\} \setminus \{i\} \\ \delta_{r,j} & \text{if } s = i \end{cases} \quad (11.259)$$

Let $L^{(i,j)}: X \rightarrow X$ be defined by $L^{(i,j)} = \mathcal{M}(E, E)^{-1}(M^{(i,j)})$ so that $\mathcal{M}(E, E)(L^{(i,j)}) = M^{(i,j)}$. Then we have

$$\forall r \in \{1, \dots, n\} \text{ we have } L^{(i,j)}(e_r) = \sum_{s \in \{1, \dots, n\}} (M^{(i,j)})_{s,r} \cdot e_s \quad (11.260)$$

For $r \in \{1, \dots, n\}$ we have either:

$r = i$. Then

$$\begin{aligned} L^{(i,j)}(e_r) &= L^{(i,j)}(e_i) \\ &\stackrel{[\text{eq: 11.260}]}{=} \sum_{s \in \{1, \dots, n\}} (M^{(i,j)})_{s,i} \cdot e_s \\ &\stackrel{[\text{eq: 11.259}]}{=} \sum_{s \in \{1, \dots, n\}} \delta_{s,j} \cdot e_s \\ &= e_j \\ &\stackrel{r=i \wedge [\text{eq: 11.256}]}{=} x_i^{(i,j)} \\ &= x_r^{(i,j)} \end{aligned}$$

$r \neq i$. Then

$$\begin{aligned}
 L^{(i,j)}(e_r) & \stackrel{[\text{eq: 11.260}]}{=} \sum_{s \in \{1, \dots, n\}} (M^{(i,j)})_{s,r} \cdot e_s \\
 & \stackrel{[\text{eq: 11.259}]}{=} \sum_{s \in \{1, \dots, n\}} (\mathcal{M}(E, E)(L))_{s,r} \cdot e_s \\
 & = \sum_{s \in \{1, \dots, n\}} (\mathcal{M}(L; E, E)(L))_{s,r} \cdot e_s \\
 & = L(e_r) \\
 & \stackrel{r \neq u \wedge [\text{eq: 11.256}]}{=} x_r^{i,j}
 \end{aligned}$$

So we have proved that

$$\forall r \in \{1, \dots, n\} \text{ we have } L^{(i,j)}(e_r) = x_r^{(i,j)} \quad (11.261)$$

Substituting the above in [eq: 11.257] results in $\forall i, \in \{1, \dots, n\}$

$$\begin{aligned}
 \mathcal{M}(\text{adjoint}(L); E, E)_{i,j} & = \Delta(L^{(i,j)}(e_1), \dots, L^{(i,j)}(e_n)) \\
 & = \Delta_{L^{(i,j)}}(e_1, \dots, e_n) \\
 & \stackrel{[\text{definition: 11.311}]}{=} \det(L^{(i,j)}) \cdot \Delta(e_1, \dots, e_n) \\
 & = \det(L^{(i,j)}) \\
 & \stackrel{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}[E, E](L^{(i,j)})) \\
 & = \det(\mathcal{M}[E, E](\mathcal{M}[E, E]^{-1}(M^{(i,j)}))) \\
 & = \det(M^{(i,j)}) \\
 & \stackrel{[\text{eq: 11.258}]}{=} \det([i \oplus j](\mathcal{M}[E, E](L))) \\
 & \stackrel{[\text{definition: 11.379}]}{=} (\text{adjoint}(\mathcal{M}[E, E])(L))_{i,j}
 \end{aligned}$$

proving finally that

$$\mathcal{M}(\text{adjoint}(L); E, E) = \text{adjoint}(\mathcal{M}[E, E])(L) \quad \square$$

The following theorem is a reformulation of [theorem: 11.322] in terms of matrices instead of linear transformations.

Theorem 11.381. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field with characteristic zero, $E = \{e_1, \dots, e_n\}$ a basis of X and $L \in \text{Hom}(X, X)$ then we have:*

1. $\mathcal{M}(E, E)(L) \cdot \text{adjoint}(\mathcal{M}(E, E)(L)) = \det(\mathcal{M}(E, E)(L)) \cdot E$
2. $\text{adjoint}(\mathcal{M}(E, E)(L)) \cdot \mathcal{M}(E, E)(L) = \det(\mathcal{M}(E, E)(L)) \cdot E$

or taking in account that $\mathcal{M}(E, E)(L) = \mathcal{M}(L; E, E)$ that:

1. $\mathcal{M}(L; E, E) \cdot \text{adjoint}(\mathcal{M}(L; E, E)) = \det(\mathcal{M}(L; E, E)) \cdot E$
2. $\text{adjoint}(\mathcal{M}(L; E, E)) \cdot \mathcal{M}(L; E, E) = \det(\mathcal{M}(L; E, E)) \cdot E$

or if we write out the matrix product:

1. $\forall i, j \in \{1, \dots, n\}$ we have

$$\sum_{k \in \{1, \dots, n\}} \mathcal{M}(L; E, E)_{i,k} \cdot \text{adjoint}(\mathcal{M}(L; E, E))_{k,j} = \det(\mathcal{M}(L; E, E)) \cdot \delta_{i,j}$$

2. $\forall i, j \in \{1, \dots, n\}$ we have

$$\sum_{k \in \{1, \dots, n\}} \text{adjoint}(\mathcal{M}(L; E, E))_{i,k} \cdot \mathcal{M}(L; E, E)_{k,j} = \det(\mathcal{M}(L; E, E)) \cdot \delta_{i,j}$$

Proof.

1. We have

$$\begin{aligned}
 \det(\mathcal{M}(E, E)(L)) \cdot E &\stackrel{[\text{theorem: 11.343}]}{=} \det(\mathcal{M}(E, E)(L)) \cdot \mathcal{M}(E, E)(\text{Id}_X) \\
 &\stackrel{[\text{theorem: 11.343}]}{=} \mathcal{M}(E, E)(\det(\mathcal{M}(E, E)(L)) \cdot \text{Id}_X) \\
 &\stackrel{[\text{theorem: 11.359}]}{=} \mathcal{M}(E, E)(\det(L) \cdot \text{Id}_X) \\
 &\stackrel{[\text{theorem: 11.322}]}{=} \mathcal{M}(E, E)(L \circ \text{adjoint}(L)) \\
 &\stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(E, E)(L) \cdot \mathcal{M}(E, E)(\text{adjoint}(L))
 \end{aligned}$$

2. We have

$$\begin{aligned}
 \det(\mathcal{M}(E, E)(L)) \cdot E &\stackrel{[\text{theorem: 11.343}]}{=} \det(\mathcal{M}(E, E)(L)) \cdot \mathcal{M}(E, E)(\text{Id}_X) \\
 &\stackrel{[\text{theorem: 11.343}]}{=} \mathcal{M}(E, E)(\det(\mathcal{M}(E, E)(L)) \cdot \text{Id}_X) \\
 &\stackrel{[\text{theorem: 11.359}]}{=} \mathcal{M}(E, E)(\det(L) \cdot \text{Id}_X) \\
 &\stackrel{[\text{theorem: 11.322}]}{=} \mathcal{M}(E, E)(\text{adjoint}(L) \circ L) \\
 &\stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(E, E)(\text{adjoint}(L)) \cdot \mathcal{M}(E, E)(L)
 \end{aligned}$$

□

Next we look at some properties of the determinant and adjoint of a matrix that helps us to calculate the determinant and adjoint of a matrix and in the end the inverse of invertible matrices.

Theorem 11.382. *Let $n \in \mathbb{N}$, F a field with characteristic zero and $M \in \mathcal{M}_{n,n}(F)$ then we have:*

1.

$$M \cdot \text{adjoint}(M) = \det(M) \cdot E,$$

hence $\forall i, j \in \{1, \dots, n\}$

$$\sum_{k \in \{1, \dots, n\}} M_{i,k} \cdot \text{adjoint}(M)_{k,j} = \det(M) \cdot \delta_{i,j}$$

2.

$$\text{adjoint}(M) \cdot M = \det(M) \cdot E$$

hence $\forall i, j \in \{1, \dots, n\}$

$$\sum_{k \in \{1, \dots, n\}} \text{adjoint}(M)_{i,k} \cdot M_{k,j} = \det(M) \cdot \delta_{i,j}$$

3. If $n > 1$ then $\forall i, j \in \{1, \dots, n\}$ we have

$$\text{adjoint}(M)_{i,j} = (-1)^{i+j} \cdot \det([j \boxplus i](M))$$

4. If $n > 1$ then we have $\forall j \in \{1, \dots, n\}$

$$\det(M) = \sum_{i \in \{1, \dots, n\}} (-1)^{i+j} \cdot M_{i,j} \cdot \det([i \boxplus j](M))$$

we call this the expansion of the determinant with respect to the j -the column.

5. If $n > 1$ then we have $\forall j \in \{1, \dots, n\}$

$$\det(M) = \sum_{i \in \{1, \dots, n\}} (-1)^{i+j} \cdot M_{j,i} \cdot \det([j \boxplus i](M))$$

we call this the expansion of the determinant with respect to the j -th row.

$$6. \det(E) = 1$$

$$7. \text{ If } M \in \mathcal{M}_{1,1}(F) \text{ then } \det(M) = M_{1,1} \text{ and } \text{adjoint}(M) = (1)$$

Proof.

1. Take $L = \mathcal{M}(E, E)^{-1}(M)$ so that $M = \mathcal{M}(E, E)(L)$ then we have that

$$\begin{aligned} M \cdot \text{adjoint}(M) &= \mathcal{M}(E, E)(L) \cdot \text{adjoint}(\mathcal{M}(E, E)(L)) \\ &\stackrel{[\text{theorem: 11.381}]}{=} \det(\mathcal{M}(E, E)(L)) \cdot E \\ &= \det(M) \cdot E \end{aligned}$$

2. Take $L = \mathcal{M}(E, E)^{-1}(M)$ so that $M = \mathcal{M}(E, E)(L)$ then we have that

$$\begin{aligned} \text{adjoint}(M) \cdot M &= \text{adjoint}(\mathcal{M}(E, E)(L)) \cdot \mathcal{M}(E, E)(L) \\ &\stackrel{[\text{theorem: 11.381}]}{=} \det(\mathcal{M}(E, E)(L)) \cdot E \\ &= \det(M) \cdot E \end{aligned}$$

3. First if $i, j \in \{1, \dots, n\}$ then we have

$$\begin{aligned} \det([i \oplus j](M)) &\stackrel{[\text{theorem: 11.358}]}{=} \det([i \oplus j](M))^T \\ &= \det([i \oplus j](M))^T \cdot \delta_{i,i} \\ &\stackrel{(1)}{=} \sum_{k \in \{1, \dots, n\}} ([i \oplus j](M))^T_{i,k} \cdot \text{adjoint}([i \oplus j](M))^T_{k,i} \\ &= \sum_{k \in \{1, \dots, n\}} ([i \oplus j](M))_{k,i} \cdot \text{adjoint}([i \oplus j](M))^T_{k,i} \\ &\stackrel{[\text{definition: 11.377}]}{=} \sum_{k \in \{1, \dots, n\}} \delta_{k,j} \cdot \text{adjoint}([i \oplus j](M))^T_{k,i} \\ &= \text{adjoint}([i \oplus j](M))^T_{j,i} \\ &\stackrel{[\text{definition: 11.379}]}{=} \det([j \oplus i]([i \oplus j](M))^T) \end{aligned}$$

giving

$$\det([i \oplus j](M)) = \det([j \oplus i]([i \oplus j](M))^T) \quad (11.262)$$

Then $\forall k \in \{1, \dots, n\}$ we have

$$\begin{aligned} \text{row}([j \oplus i]([i \oplus j](M))^T, i)_k &= ([j \oplus i]([i \oplus j](M))^T)_{i,k} \\ &\stackrel{[\text{definition: 11.377}]}{=} \begin{cases} ([i \oplus j](M))^T_{i,k} & \text{if } k \in \{1, \dots, n\} \setminus \{j\} \\ \delta_{i,i} & \text{if } k = j \end{cases} \\ &= \begin{cases} ([i \oplus j](M))_{k,i} & \text{if } k \in \{1, \dots, n\} \setminus \{j\} \\ 1 & \text{if } k = j \end{cases} \\ &\stackrel{[\text{definition: 11.377}]}{=} \begin{cases} \delta_{k,j} & \text{if } k \in \{1, \dots, n\} \setminus \{j\} \\ 1 & \text{if } k = j \end{cases} \\ &= \begin{cases} 0 & \text{if } k \in \{1, \dots, n\} \setminus \{j\} \\ 1 & \text{if } k = j \end{cases} \\ &= \delta_{k,j} \end{aligned}$$

giving

$$\text{row}([j \oplus i]([i \oplus j](M))^T, i)_k = \delta_{k,j} \quad (11.263)$$

Further

$$\begin{aligned} \text{col}([j \oplus i]([i \oplus j](M))^T, j)_k &= ([j \oplus i]([i \oplus j](M))^T)_{k,j} \\ &\stackrel{[\text{definition: 11.377}]}{=} \begin{cases} ([i \oplus j](M))^T_{k,j} & \text{if } j \in \{1, \dots, n\} \setminus \{j\} \\ \delta_{k,i} & \text{if } j = j \end{cases} \\ &= \delta_{k,i} \end{aligned}$$

giving

$$\text{col}([j \oplus i]([i \oplus j](M))^T, j)_k = \delta_{k,i} \quad (11.264)$$

From [eqs: 11.263, 11.264] it follows that the conditions of [theorem: 11.376] are satisfied, hence

$$\det([j \oplus i]([i \oplus j](M))^T) = (-1)^{i+j} \cdot \det((i \boxplus j)([j \oplus i]([i \oplus j](M))^T))$$

which combined with [eq: 11.262] results in

$$\det([i \oplus j](M)) = (-1)^{i+j} \cdot \det((i \boxplus j)([j \oplus i]([i \oplus j](M))^T)) \quad (11.265)$$

Let $k, l \in \{1, \dots, n-1\}$ then we have for $((i \boxplus j)([j \oplus i]([i \oplus j](M))^T))_{k,l}$ the following cases to consider for k, l :

$1 \leq k < i \wedge 1 \leq l < j$. Then

$$\begin{aligned} ((i \boxplus j)([j \oplus i]([i \oplus j](M))^T))_{k,l} &\stackrel{[\text{definition: 11.369}]}{=} \\ ([j \oplus i]([i \oplus j](M))^T)_{k,l} &\stackrel{l < j \Rightarrow l \neq j \wedge [\text{definition: 11.377}]}{=} \\ ([i \oplus j](M))^T_{k,l} &= \\ ([i \oplus j](M))_{l,k} &\stackrel{k < i \Rightarrow k \neq i \wedge [\text{definition: 11.377}]}{=} \\ M_{l,k} &= \\ (M^T)_{k,l} &\stackrel{[\text{definition: 11.369}]}{=} \\ ([i \boxplus j](M^T))_{k,l} & \end{aligned}$$

$i \leq k \leq n-1 \wedge 1 \leq l < j$. Then

$$\begin{aligned} ((i \boxplus j)([j \oplus i]([i \oplus j](M))^T))_{k,l} &\stackrel{[\text{definition: 11.369}]}{=} \\ ([j \oplus i]([i \oplus j](M))^T)_{k+1,l} &\stackrel{l < j \Rightarrow l \neq j \wedge [\text{definition: 11.377}]}{=} \\ ([i \oplus j](M))^T_{k+1,l} &= \\ ([i \oplus j](M))_{l,k+1} &\stackrel{i \leq k \Rightarrow i \neq k+1 \wedge [\text{definition: 11.377}]}{=} \\ M_{l,k+1} &= \\ (M^T)_{k+1,l} &\stackrel{[\text{definition: 11.369}]}{=} \\ ([i \boxplus j](M^T))_{k,l} & \end{aligned}$$

$1 \leq k < i \wedge j \leq l \leq n-1$. Then

$$\begin{aligned} ((i \boxplus j)([j \oplus i]([i \oplus j](M))^T))_{k,l} &\stackrel{[\text{definition: 11.369}]}{=} \\ ([j \oplus i]([i \oplus j](M))^T)_{k,l+1} &\stackrel{j \leq l \Rightarrow l+1 \neq j \wedge [\text{definition: 11.377}]}{=} \\ ([i \oplus j](M))^T_{k,l+1} &= \\ ([i \oplus j](M))_{l+1,k} &\stackrel{k < i \Rightarrow k \neq i \wedge [\text{definition: 11.377}]}{=} \\ M_{l+1,k} &= \\ (M^T)_{k,l+1} &\stackrel{[\text{definition: 11.369}]}{=} \\ ([i \boxplus j](M^T))_{k,l} & \end{aligned}$$

$i \leq k \leq n-1 \wedge j \leq l \leq n-1$. Then

$$\begin{aligned}
 & ((i \boxplus j)([j \oplus i]([i \oplus j](M))^T))_{k,l} && \stackrel{=}{=} && [\text{definition: 11.369}] \\
 & ([j \oplus i]([i \oplus j](M))^T)_{k+1,l+1} && j \leq l \Rightarrow j \neq l+1 \wedge && \stackrel{=}{=} [\text{definition: 11.377}] \\
 & (([i \oplus j](M))^T)_{k+1,l+1} && = && \\
 & ([i \oplus j](M))_{l+1,k+1} && i \leq k \Rightarrow i \neq k+1 \wedge && \stackrel{=}{=} [\text{definition: 11.377}] \\
 & M_{l+1,k+1} && = && \\
 & (M^T)_{k+1,l+1} && \stackrel{=}{=} && [\text{definition: 11.369}] \\
 & ([i \boxplus j](M^T))_{k,l} && &&
 \end{aligned}$$

The above proves that

$$(i \boxplus j)([j \oplus i]([i \oplus j](M))^T) = [i \boxplus j](M^T)$$

combining the above with [eq: 11.265] gives

$$\begin{aligned}
 \det([i \oplus j](M)) &= (-1)^{i+j} \cdot \det([i \boxplus j](M^T)) \\
 &\stackrel{=}{=}_{[\text{theorem: 11.371}]} (-1)^{i+j} \cdot \det([j \boxplus i](M))^T
 \end{aligned} \tag{11.266}$$

Finally

$$\begin{aligned}
 \text{adjoint}(M)_{i,j} &\stackrel{=}{=}_{[\text{definition: 11.379}]} \det([i \oplus j](M)) \\
 &\stackrel{=}{=}_{[\text{eq: 11.266}]} (-1)^{i+j} \cdot \det([j \boxplus i](M))^T \\
 &\stackrel{=}{=}_{[\text{theorem: 11.358}]} (-1)^{i+j} \cdot \det([j \boxplus i]M)
 \end{aligned}$$

4. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned}
 \det(M) &= \det(M) \cdot \delta_{i,i} \\
 &\stackrel{(2)}{=} \sum_{k \in \{1, \dots, n\}} \text{adjoint}(M)_{i,k} \cdot M_{k,i} \\
 &\stackrel{(3)}{=} \sum_{k \in \{1, \dots, n\}} (-1)^{i+k} \cdot \det([k \boxplus i](M)) \cdot M_{k,i} \\
 &= \sum_{k \in \{1, \dots, n\}} (-1)^{i+k} \cdot M_{k,i} \cdot \det([k \boxplus i](M))
 \end{aligned}$$

5. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned}
 \det(M) &= \det(M) \cdot \delta_{i,i} \\
 &\stackrel{(1)}{=} \sum_{k \in \{1, \dots, n\}} M_{i,k} \cdot \text{adjoint}(M)_{k,i} \\
 &\stackrel{(3)}{=} \sum_{k \in \{1, \dots, n\}} M_{i,k} \cdot (-1)^{i+k} \cdot \det([i \boxplus k](M)) \\
 &= \sum_{k \in \{1, \dots, n\}} (-1)^{i+k} \cdot M_{i,k} \cdot \det([i \boxplus k](M))
 \end{aligned}$$

6. This follows from [theorem: 11.382]

7. Using [theorem: 11.355] we have for $M \in \mathcal{M}_{1,1}(F)$ that

$$\det(M) = M_{1,1}$$

Further

$$\text{adjoint}(M) = \det([1 \oplus 1] \begin{pmatrix} M_{1,1} \end{pmatrix}) = \det((1)) = 1$$

□

We can use (4),(7) from the above to calculate the determinant of a matrix as show in the following example:

Example 11.383. Using [theorem: 11.382 (4)] we have

$$\begin{aligned}
 & \begin{vmatrix} -2 & 2 & 3 \\ -1 & 1 & 3 \\ 2 & 0 & 1 \end{vmatrix} = \\
 & (-1)^{1+1} \cdot M_{1,1} \cdot \det([1 \boxplus 1]M) + (-1)^{1+2} \cdot M_{2,1} \cdot \det([2 \boxplus 1] \cdot M) + (-1)^{1+3} \cdot M_{3,1} \cdot \det([3 \boxplus 1]M) = \\
 & -2 \cdot \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} + (-1) \cdot (-1) \cdot \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} + 2 \cdot \begin{vmatrix} 2 & 3 \\ 1 & 3 \end{vmatrix} = \\
 & -2 \cdot (1 \cdot | \ 1 \ | + (-1) \cdot 0 \cdot | \ 3 \ |) + (2 \cdot | \ 1 \ | + (-1) \cdot 0 \cdot | \ 3 \ |) + 2 \cdot (2 \cdot | \ 3 \ | + (-1) \cdot 1 \cdot | \ 3 \ |) = \\
 & -2 + 2 + 2 \cdot 3 = 6
 \end{aligned}$$

To simplify calculation of the adjoint of a matrix we introduce the co-factor matrix

Definition 11.384. Let $n \in \mathbb{N} \setminus \{1\}$, F a field with characteristic zero and $M \in \mathcal{M}_{n,n}(F)$ then the co-factor matrix noted by $\text{cofactor}(M)$ is defined by

$$\text{cofactor}(M) \in \mathcal{M}_{n,n}(M) \text{ where } \forall i, j \in \{1, \dots, n\} \text{cofactor}(M)_{i,j} = (-1)^{i+j} \cdot \det([i \boxplus j](M))$$

The use of the co-factor matrix is shown in the following theorem:

Theorem 11.385. Let $n \in \mathbb{N} \setminus \{1\}$, F a field with characteristic zero and $M \in \mathcal{M}_{n,n}(F)$ then

$$\text{adjoint}(M) = (\text{cofactor}(M))^T$$

Proof. Let $i, j \in \{1, \dots, n\}$ then

$$((\text{cofactor}(M))^T)_{i,j} = \text{cofactor}(M)_{j,i} = (-1)^{j+i} \cdot \det([j \boxplus i](M)) \stackrel{[\text{theorem: 11.382(3)}]}{=} \text{adjoint}(M)_{i,j}$$

proving that

$$\text{adjoint}(M) = (\text{cofactor}(M))^T \quad \square$$

Example 11.386.

$$\begin{aligned}
 \text{cofactor}\left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}\right) &= \begin{pmatrix} 1 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} & -1 \cdot \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} & 1 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ -1 \cdot \begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} & 1 \cdot \begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} & -1 \cdot \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} \\ 1 \cdot \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} & -1 \cdot \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} & 1 \cdot \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} \end{pmatrix} \\
 &= \begin{pmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{pmatrix}
 \end{aligned}$$

so that

$$\begin{aligned}
 \text{adjoint}\left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}\right) &= \begin{pmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{pmatrix}^T \\
 &= \begin{pmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{pmatrix}
 \end{aligned}$$

Sometimes there exists shortcuts for calculating the determinant of a matrix.

Corollary 11.387. *Let $n \in \mathbb{N} \setminus \{1\}$ then for $M \in \mathcal{M}_{n,n}(F)$ we have:*

1. *If $\exists m \in \mathbb{N}$ with $1 \leq m < n$ such that $\forall i \in \{1, \dots, m\}$ and $\forall j \in \{1, \dots, n\}$ we have*

$$\text{col}(M, i)_j = \delta_{i,j}$$

then

$$\det(M) = \det([>m](M))$$

2. *If $\exists m \in \mathbb{N}$ with $1 < m \leq n$ such that $\forall i \in \{m, \dots, n\}$ we have*

$$\text{col}(M, i)_j = \delta_{i,j}$$

then

$$\det(M) = \det([<m](M))$$

3. *If $\exists m \in \mathbb{N}$ with $1 \leq m < n$ such that $\forall i \in \{1, \dots, m\}$ and $\forall j \in \{1, \dots, n\}$ we have*

$$\text{row}(M, i)_j = \delta_{i,j}$$

then

$$\det(M) = \det([>m](M))$$

4. *If $\exists m \in \mathbb{N}$ with $1 < m \leq n$ such that $\forall i \in \{m, \dots, n\}$ and $\forall j \in \{1, \dots, n\}$ we have*

$$\text{row}(M, i)_j = \delta_{i,j}$$

then

$$\det(M) = \det([<m](M))$$

Proof.

1. *We prove this by induction, so let*

$S_n = \{m \in \mathbb{N} \mid \text{If } m < n \text{ and } M \in \mathcal{M}_{n,n}(F) \text{ satisfies } \forall i \in \{1, \dots, m\}, \forall j \in \{1, \dots, n\} \text{ col}(M, i)_j = \delta_{i,j} \text{ then } \det(M) = \det([>m](M))\}$

then we have:

1 $\in S_n$. *Let $M \in \mathcal{M}_{n,n}(F)$ be such that $\forall i \in \{1\}, \forall j \in \{1, \dots, n\} \text{ col}(M, i)_j = \delta_{j,i}$, hence $\text{col}(M, 1) = \delta_{1,i}$, then we have*

$$\begin{aligned} \det(M) &\stackrel{[\text{theorem: 11.382(4)}]}{=} \sum_{i \in \{1, \dots, n\}} (-1)^{i+1} \cdot M_{i,1} \cdot \det([i \boxplus 1](M)) \\ &= \sum_{i \in \{1, \dots, n\}} (-1)^{i+1} \cdot \delta_{i,1} \cdot \det([i \boxplus 1](M)) \\ &= (-1)^2 \cdot \det([1 \boxplus 1](M)) \\ &= \det([1 \boxplus 1](M)) \\ &\stackrel{[\text{theorem: 11.373}]}{=} \det([>1](M)) \end{aligned}$$

proving that $1 \in S_n$.

$m \in S_n \Rightarrow m+1 \in S_n$. *Let $m+1 < n$ then we have*

$$1 < n - m \Rightarrow 1 \leq n - m - 1 \Rightarrow 1 \in \{1, \dots, n - m - 1\} \quad (11.267)$$

Further let $M \in \mathcal{M}_{n,n}(F)$ be such that $\forall i \in \{1, \dots, m+1\}, \forall j \in \{1, \dots, n\}$ we have $\text{col}(M, i)_j = \delta_{i,j}$. As $m \in S_n$, $m < m+1 < n$ and $\forall i \in \{1, \dots, m\}, \forall j \in \{1, \dots, n\} \text{ col}(M, i)_j = \delta_{i,j}$ we have

$$\det(M) = \det([>m](M)) \quad (11.268)$$

If $j \in \{1, \dots, n-m\}$ then

$$\begin{aligned}
 \text{col}([>m](M), 1)_j &= ([>m](M))_{j,1} \\
 &\stackrel{[\text{definition: } 11.372]}{=} M_{j+m,1+m} \\
 &= \text{col}(1+m, j+m) \\
 &= \delta_{1+m, j+m} \\
 &= \delta_{j,1}
 \end{aligned} \tag{11.269}$$

Next

$$\begin{aligned}
 &\det(M) \stackrel{[\text{eq: } 11.268]}{=} \det([>m](M)) \stackrel{[\text{theorem: } 11.382(4)]}{=} \\
 &\sum_{i \in \{1, \dots, n\}} (-1)^{i+1} \cdot ([>m](M))_{i,1} \cdot \det([i \boxplus 1]([>m])(M)) = \\
 &\sum_{i \in \{1, \dots, n\}} (-1)^{i+1} \cdot \text{col}([>m](M), 1)_i \cdot \det([i \boxplus 1]([>m])(M)) \stackrel{[\text{eq: } 11.269]}{=} \\
 &\sum_{i \in \{1, \dots, n\}} (-1)^{i+1} \cdot \delta_{i,1} \cdot \det([i \boxplus 1]([>m])(M)) = \\
 &(-1)^2 \cdot \det([1 \boxplus 1]([>m])(M)) \stackrel{[\text{theorem: } 11.373]}{=} \\
 &\det([>1]([>m](M))) \stackrel{[\text{eq: } 11.267][\text{theorem: } 11.373]}{=} \\
 &\det([>m+1](M))
 \end{aligned}$$

proving that $m+1 \in S_n$.

Mathematical induction proves then that $S_n = \mathbb{N}$. So if $1 \leq m < n$ and $M \in \mathcal{M}_{n,n}(F)$ satisfies $\forall i \in \{1, \dots, m\}, \forall j \in \{1, \dots, n\} \text{ col}(M, i)_j = \delta_{i,j}$ then we have, as $m \in \mathbb{N} = S_n$, that $\det(M) = \det([>m](M))$.

2. We prove this by induction, so let

$S = \{n \in \{2, \dots, \infty\} \mid \text{If } M \in \mathcal{M}_{n,n}(F) \text{ satisfies } \exists m \in \{2, \dots, n\} \text{ such that } \forall i \in \{m, \dots, n\}, \forall j \in \{1, \dots, n\} \text{ we have } \text{col}(M, i)_j = \delta_{i,j} \text{ then } \det(M) = \det([<m](M))\}$

then we have:

2 ∈ S. Let $M \in \mathcal{M}_{2,2}(F)$ satisfying $\exists m \in \{2, \dots, 2\}$ such that $\forall i \in \{m, \dots, 2\}$ and $\forall j \in \{1, \dots, 2\}$ we have $\text{col}(M, i)_j = \delta_{i,j}$ then $\forall j \in \{1, \dots, 2\} \text{ col}(M, 2)_j = \delta_{2,j}$. So we have

$$\begin{aligned}
 \det(M) &\stackrel{[\text{theorem: } 11.382(4)]}{=} \sum_{i \in \{1, \dots, 2\}} (-1)^{i+2} \cdot M_{i,2} \cdot \det([i \boxplus 2](M)) \\
 &= \sum_{i \in \{1, \dots, 2\}} (-1)^{i+2} \cdot \text{col}(M, 2)_i \cdot \det([i \boxplus 2](M)) \\
 &= \sum_{i \in \{1, \dots, n\}} (-1)^{i+2} \cdot \delta_{2,i} \cdot \det([i \boxplus 2](M)) \\
 &= (-1)^{2+2} \cdot \det([2 \boxplus 2](M)) \\
 &= \det([2 \boxplus 2](M)) \\
 &\stackrel{[\text{theorem: } 11.373] \wedge n=2}{=} \det([<2](M))
 \end{aligned}$$

proving that $2 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $M \in \mathcal{M}_{n+1,n+1}(F)$ such that $\exists m \in \{2, \dots, n+1\}$ such that $\forall i \in \{m, \dots, n+1\}$ and $\forall j \in \{1, \dots, n+1\}$ we have $\text{col}(M, i)_j = \delta_{i,j}$. As

$$[n+1 \boxplus n+1](M) \stackrel{[\text{theorem: } 11.373]}{=} [<n+1](M) \tag{11.270}$$

it follows that $\forall i \in \{m, \dots, n\}$ and $\forall j \in \{1, \dots, n\}$ we have

$$\begin{aligned} \text{col}([n+1 \boxplus n+1](M), i)_j &= ([n+1 \boxplus n+1](M))_{j,i} \\ &\stackrel{i < n+1 \wedge j < n+1 \wedge \text{definition: 11.369}}{=} M_{j,i} \\ &= \text{col}(M, i)_j \\ &= \delta_{i,j} \end{aligned}$$

which as $[n+1 \boxplus n+1](M) \in \mathcal{M}_{n,n}(F)$ and $n \in S$ proves that

$$\det([n+1 \boxplus n+1](M)) = \det([<m]([n+1 \boxplus n+1](M)))$$

substituting [eq: 11.270] in the last term from the above gives

$$\det([n+1 \boxplus n+1](M)) = \det([<m](<n+1](M)))$$

As $m \in \{2, \dots, n+1\}$ it follows from [theorem: 11.373] that

$$[<m](<n+1](M)) = [<m](M)$$

so that

$$\det([n+1 \oplus n+1](M)) = \det([<m](M)) \quad (11.271)$$

Further we have

$$\begin{aligned} \det(M) &\stackrel{[\text{theorem: 11.382(4)}]}{=} \sum_{i \in \{1, \dots, n\}} (-1)^{i+(n+1)} \cdot M_{i,n+1} \cdot \det([i \boxplus n+1](M)) \\ &= \sum_{i \in \{1, \dots, n\}} (-1)^{i+(n+1)} \cdot \text{col}(M, n+1)_i \cdot \det([i \boxplus n+1](M)) \\ &= \sum_{i \in \{1, \dots, n\}} (-1)^{i+(n+1)} \cdot \delta_{i,n+1} \cdot \det([i \boxplus n+1](M)) \\ &= (-1)^{(n+1)+(n+1)} \cdot \det([n+1 \boxplus n+1](M)) \\ &= \det([n+1 \boxplus n+1](M)) \\ &\stackrel{[\text{eq: 11.271}]}{=} \det([<m](M)) \end{aligned}$$

proving that $n+1 \in S$.

3. As $\forall i \in \{1, \dots, m\}$ and $\forall j \in \{1, \dots, n\}$ we have $\delta_{i,j} = \text{row}(M, i)_j = M_{i,j} = (M^T)_{j,i} = \text{col}(M^T, i)_j$ we can use (1) to get

$$\det(M^T) = \det([>m](M^T)) \quad (11.272)$$

hence

$$\begin{aligned} \det(M) &\stackrel{[\text{theorem: 11.358}]}{=} \det(M^T) \\ &\stackrel{[\text{eq: 11.272}]}{=} \det([>m](M^T)) \\ &\stackrel{[\text{theorem: 11.373}]}{=} \det(([>m](M))^T) \\ &\stackrel{[\text{theorem: 11.358}]}{=} \det([<m](M)) \end{aligned}$$

4. As $\forall i \in \{m, \dots, n\}$ and $\forall j \in \{1, \dots, n\}$ we have $\delta_{i,j} = \text{row}(M, i)_j = M_{i,j} = (M^T)_{j,i} = \text{col}(M^T, i)_j$ we can use (2) to get

$$\det(M^T) = \det([<m](M^T)) \quad (11.273)$$

hence

$$\begin{aligned} \det(M) &\stackrel{[\text{theorem: 11.358}]}{=} \det(M^T) \\ &\stackrel{[\text{eq: 11.273}]}{=} \det([<m](M^T)) \\ &\stackrel{[\text{theorem: 11.373}]}{=} \det(([<m](M))^T) \\ &\stackrel{[\text{theorem: 11.358}]}{=} \det([>m](M)) \end{aligned}$$

□

The next theorem shows sufficient and necessary conditions for a matrix to be invertible and how to calculate the inverse of the matrix.

Theorem 11.388. *Let $n \in \mathbb{N}$, $M \in \mathcal{M}_{n,n}(F)$ then we have*

$$\begin{aligned} \text{rank}(M) = m & \\ \Updownarrow & \\ \det(M) \neq 0 & \\ \Updownarrow & \\ \exists M^{-1} \in \mathcal{M}_{n,n}(F) \text{ such that } M \cdot M^{-1} = E = M^{-1} \cdot M & \text{ [in other word } M \text{ is invertible]} \end{aligned}$$

Further if M is invertible then

$$M^{-1} = (\det(M))^{-1} \cdot \text{adjoint}(M)$$

Proof. We have from [theorem: 11.365] that

$$\text{rank}(M) = m \Leftrightarrow \det(M) \neq 0$$

If $\det(M) \neq 0$ then $(\det(M))^{-1}$ exists and we have

$$\begin{aligned} ((\det(M))^{-1} \cdot \text{adjoint}(M)) \cdot M &= (\det(M))^{-1} \cdot (\text{adjoint}(M) \cdot M) \\ &\stackrel{[\text{theorem: 11.382(2)}]}{=} (\det(M))^{-1} \cdot (\det(M) \cdot E) \\ &= E \\ M \cdot ((\det(M))^{-1} \cdot \text{adjoint}(M)) &= (\det(M))^{-1} \cdot (M \cdot \text{adjoint}(M)) \\ &\stackrel{[\text{theorem: 11.382(1)}]}{=} (\det(M))^{-1} \cdot (\det(M) \cdot E) \\ &= E \end{aligned}$$

proving

$$\det(M) \neq 0 \Rightarrow M^{-1} \cdot M = E = M \cdot M^{-1} \text{ where } M^{-1} = (\det(M))^{-1} \cdot \text{adjoint}(M)$$

Finally if there exists $M^{-1} \in \mathcal{M}_{n,n}(F)$ such that $M^{-1} \cdot M = E = M \cdot M^{-1}$ then

$$1 = \det(E) = \det(M \cdot M^{-1}) \stackrel{[\text{theorem: 11.361}]}{=} \det(M) \cdot \det(M^{-1})$$

hence we must have that $\det(M) \neq 0$. So we have

$$\text{If } \exists M^{-1} \in \mathcal{M}_{n,n}(F) \text{ such that } M^{-1} \cdot M = E = M \cdot M^{-1} \text{ then } \det(M) \neq 0 \quad \square$$

11.9 Nonsingular transformations

The purpose of this section is to prove that every invertible (nonsingular) transformation on a vector space can always be written as a composition of a **limited** set of linear transformations called the **elementary transformations**. So statements about a general nonsingular transformations can be proved by proving this statement for the elementary transformations (as long as these statements stay valid under composition of transformations).

Definition 11.389. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with $\dim(X) = n$ then a linear transformation $L \in \text{Hom}(X, X)$ is nonsingular if $\det(L) \neq 0$. Using [theorem: 11.315(2)] this is equivalent with saying that $L: X \rightarrow X$ is a linear isomorphism. The set of nonsingular transformations is noted as $\text{GL}(X)$ so*

$$\text{GL}(X) = \{L \in \text{Hom}(X, X) | \det(L) \neq 0\}$$

We show now that $\text{GL}(X)$ together with the composition forms a group

Theorem 11.390. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with $\dim(X) = n$ then*

$\langle \text{GL}(X), \circ \rangle$ is a sub semi-group of $\langle \text{Hom}(X, X), \circ \rangle$ [see theorem: 11.183]

Furthermore

$\langle \text{GL}(X), \circ \rangle$ is actually a group where L^{-1} is the inverse of $L \in \text{GL}(X)$

Proof. If $L_1, L_2 \in \text{GL}(X)$ then $\det(L_1) \neq 0 \wedge \det(L_2) \neq 0$ so that

$$\det(L_1 \circ L_2) \underset{[\text{theorem: 11.315}]}{=} \det(L_1 \circ L_2) = \det(L_1) \cdot \det(L_2) \neq 0$$

proving that

$$\forall L_1, L_2 \in \text{GL}(X) \text{ we have } L_1 \circ L_2 \in \text{GL}(X)$$

Further as $\det(\text{Id}_X) \underset{[\text{theorem: 11.315}]}{=} 1 \neq 0$ we have

$$\text{Id}_X \in \text{GL}(X) \text{ where } \text{Id}_X \text{ is the neutral element in } \langle \text{Hom}(X, X) \rangle$$

So by [definition: 4.12]

$$\text{GL}(X) \text{ is a sub-semi-group of } \langle \text{Hom}(X, X), \circ \rangle$$

Using [theorem: 4.14] that

$$\langle \text{GL}(X), \circ \rangle \text{ is a semi-group}$$

Finally if $L \in \text{GL}(X)$ then L is a linear isomorphism [see theorem: 11.315(2)], so that L^{-1} by [theorem: 11.178] L^{-1} is a linear isomorphism, hence $L^{-1} \in \text{GL}(X)$. Which as $L \circ L^{-1} = \text{Id}_X = L^{-1} \circ L$ proves that

$$\langle \text{GL}(X), \circ \rangle \text{ is a group} \quad \square$$

As $\langle \text{GL}(X), \circ \rangle$ is a group, hence a semi-group we can use [remark: 11.2] to define the finite product of elements in $\text{GL}(X)$ defined by the 'product' operator $\circ$.

We will now introduce a special class of nonsingular operators called elementary transformations.

Definition 11.391. *Let $n \in \mathbb{N}$, X a finite dimensional vector space with $\dim(X) = n$ over a field F and $E = \{e_1, \dots, e_n\}$ a basis for X then we define:*

1. *Let $i \in \{1, \dots, n\}$ and $\alpha \in F$ define*

$$M_{[i, \alpha]}^n \in \mathcal{M}_{n, n}(F) \text{ by } \forall k, l \in \{1, \dots, n\} \quad (M_{[i, \alpha]}^n)_{k, l} = \begin{cases} \delta_{k, l} & \text{if } l \in \{1, \dots, n\} \setminus \{i\} \\ \alpha \cdot \delta_{k, i} & \text{if } l = i \end{cases}$$

and

$$\sigma_{[i, \alpha]}^n \in \text{Hom}(X, X) \text{ by } \sigma_{[i, \alpha]}^n = \mathcal{M}(E, E)^{-1}(M_{[i, \alpha]}^n)$$

So that $\forall k \in \{1, \dots, n\} \setminus \{i\}$ we have

$$\begin{aligned} \sigma_{[i, \alpha]}^n(e_k) &= \sum_{j \in \{1, \dots, n\}} (M_{[i, \alpha]}^n)_{j, k} \cdot e_j \\ &= \sum_{j \in \{1, \dots, n\}} \delta_{j, k} \cdot e_j \\ &= e_k \end{aligned}$$

and

$$\begin{aligned}\sigma_{[i,\alpha]}^n(e_i) &= \sum_{j \in \{1, \dots, n\}} (M_{[i,\alpha]}^n)_{j,i} \cdot e_j \\ &= \sum_{j \in \{1, \dots, n\}} \alpha \cdot \delta_{j,i} \cdot e_j \\ &= \alpha \cdot e_i\end{aligned}$$

or more compact

$$\forall k \in \{1, \dots, n\} \text{ we have } \sigma_{[i,\alpha]}^n(e_k) = \begin{cases} e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ \alpha \cdot e_i & \text{if } k = i \end{cases}$$

2. Let $i, j \in \{1, \dots, n\}$ with $i \neq j$ and define

$$N_{[i,j]}^n \in \mathcal{M}_{n,n}(F) \text{ by } \forall k, l \in \{1, \dots, n\} \quad (N_{[i,j]}^n)_{k,l} = \begin{cases} \delta_{j,k} & \text{if } l = i \\ \delta_{i,k} & \text{if } l = j \\ \delta_{k,l} & \text{if } l \in \{1, \dots, n\} \setminus \{i, j\} \end{cases}$$

and

$$\tau_{[i,j]}^n \in \text{Hom}(X, X) \text{ by } \tau_{[i,j]}^n = \mathcal{M}(E, E)^{-1}(N_{[i,j]}^n)$$

So that $\forall k \in \{1, \dots, n\} \setminus \{i, j\}$ we have

$$\begin{aligned}\tau_{[i,j]}^n(e_k) &= \sum_{l \in \{1, \dots, n\}} (N_{[i,j]}^n)_{l,k} \cdot e_l \\ &= \sum_{l \in \{1, \dots, n\}} \delta_{l,k} \cdot e_l \\ &= e_k\end{aligned}$$

and

$$\begin{aligned}\tau_{[i,j]}^n(e_i) &= \sum_{l \in \{1, \dots, n\}} (N_{[i,j]}^n)_{l,i} \cdot e_l \\ &= \sum_{l \in \{1, \dots, n\}} \delta_{j,l} \cdot e_l \\ &= e_j\end{aligned}$$

and

$$\begin{aligned}\tau_{[i,j]}^n(e_j) &= \sum_{l \in \{1, \dots, n\}} (N_{[i,j]}^n)_{l,j} \cdot e_l \\ &= \sum_{l \in \{1, \dots, n\}} \delta_{i,l} \cdot e_l \\ &= e_i\end{aligned}$$

proving that

$$\forall k \in \{1, \dots, n\} \quad \tau_{[i,j]}^n(e_k) = \begin{cases} e_i & \text{if } k=j \\ e_j & \text{if } k=i \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases}$$

3. Let $i, j \in \{1, \dots, n\}$ with $i \neq j$, $\alpha \in F$ and define

$$O_{[i,j,\alpha]}^n \in \mathcal{M}_{n,n}(F) \text{ by } \forall k, l \in \{1, \dots, n\} \quad (O_{[i,j,\alpha]}^n)_{k,l} = \begin{cases} \delta_{k,i} + \alpha \cdot \delta_{j,k} & \text{if } l = i \\ \delta_{k,l} & \text{if } l \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

and

$$\beta_{[i,j,\alpha]}^n \in \text{Hom}(X, X) \text{ by } \beta_{[i,j,\alpha]}^n = \mathcal{M}[E, E]^{-1}(O_{[i,j,\alpha]}^n)$$

So that $\forall k \in \{1, \dots, n\} \setminus \{i\}$ we have

$$\begin{aligned}\beta_{[i,j,\alpha]}^n(e_k) &= \sum_{l \in \{1, \dots, n\}} (O_{[i,j,\alpha]}^n)_{l,k} \cdot e_l \\ &= \sum_{l \in \{1, \dots, n\}} \delta_{l,k} \cdot e_l \\ &= e_k\end{aligned}$$

and

$$\begin{aligned}\beta_{[i,j,\alpha]}^n(e_i) &= \sum_{l \in \{1, \dots, n\}} (O_{[i,j,\alpha]}^n)_{l,i} \cdot e_l \\ &= \sum_{l \in \{1, \dots, n\}} (\delta_{l,i} + \alpha \cdot \delta_{j,l}) \cdot e_l \\ &= \sum_{l \in \{1, \dots, n\}} \delta_{l,i} \cdot e_l + \sum_{l \in \{1, \dots, n\}} \alpha \cdot \delta_{j,l} \cdot e_l \\ &= e_i + \alpha \cdot e_j\end{aligned}$$

or in other words

$$\forall k \in \{1, \dots, n\} \text{ we have } \beta_{[i,j,\alpha]}^n(e_k) = \begin{cases} e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ e_i + \alpha \cdot e_j & \text{if } k=i \end{cases}$$

The above transformations have the following properties

Theorem 11.392. Let $n \in \mathbb{N}$, X a finite dimensional vector space with $\dim(X) = n$ over a field F with characteristic zero then we have:

1. $\forall i \in \{1, \dots, n\}$, $\forall \alpha \in F$ we have $\det(\sigma_{[i,\alpha]}^n) = \alpha$. Hence by [theorem: 11.315]

$$\sigma_{[i,\alpha]}^n \text{ is a linear isomorphism} \Leftrightarrow \alpha \neq 0$$

2. $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have

$$\det(\tau_{[i,j]}^n) = -1$$

so that by [theorem: 11.315] $\tau_{[i,j]}^n$ is a linear isomorphism.

3. $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ and $\forall \alpha \in F$ we have that $\det(\beta_{[i,j,\alpha]}^n) = 1$ so that by [theorem: 11.315] $\beta_{[i,j,\alpha]}^n$ is a linear isomorphism.

4. $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have $\tau_{[i,j]}^n \circ \tau_{[i,j]}^n = \text{Id}_{\{1, \dots, n\}}$ so that $(\tau_{[i,j]}^n)^{-1} = \tau_{[i,j]}^n$

Proof. Let $\{e_1, \dots, e_n\}$ be a basis for X and let $\Delta: X^n \rightarrow F$ be the determinant function such that $\Delta(e_1, \dots, e_n) = 1$ [see theorem: 11.299] then we have:

- 1.

$$\begin{aligned}\det(\sigma_{[i,\alpha]}^n) &\stackrel{[\text{theorem: 11.312}]}{=} \Delta(\sigma_{[i,\alpha]}^n(e_1), \dots, \sigma_{[i,\alpha]}^n(e_n)) \\ &= \Delta\left(\sigma_{[i,\alpha]}^n(e_1), \dots, \underbrace{\sigma_{[i,\alpha]}^n(e_i)}_i, \dots, \sigma_{[i,\alpha]}^n(e_n)\right) \\ &= \Delta\left(e_1, \dots, \underbrace{\alpha \cdot e_i}_i, \dots, e_n\right) \\ &= \alpha \cdot \Delta(e_1, \dots, e_n) \\ &= \alpha \cdot 1 \\ &= \alpha\end{aligned}$$

2. Let $k \in \{1, \dots, n\}$ then

$$\tau_{[i,j]}^n(e_k) \stackrel{[\text{definition: 11.391}]}{=} \begin{cases} e_i & \text{if } k = j \\ e_j & \text{if } k = i \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases} \stackrel{[\text{definition: 11.228}]}{=} e_{(i \leftrightarrow_n j)(k)}$$

so that

$$\begin{aligned} \det(\tau_{[i,j]}^n) &\stackrel{[\text{theorem: 11.312}]}{=} \Delta(\tau_{[i,j]}^n(e_1), \dots, \tau_{[i,j]}^n(e_n)) \\ &= \Delta(e_{(i \leftrightarrow_n j)(1)}, \dots, e_{(i \leftrightarrow_n j)(n)}) \\ &= ((i \leftrightarrow_n j)\Delta)(e_1, \dots, e_n) \\ &\stackrel{[\text{theorem: 11.285}]}{=} \text{sign}(i \leftrightarrow_n j) \cdot \Delta(e_1, \dots, e_n) \\ &\stackrel{i \neq j \wedge [\text{theorem: 11.251}]}{=} (-1) \cdot \Delta(e_1, \dots, e_n) \\ &= -1 \end{aligned}$$

3. We have

$$\begin{aligned} &\det(\beta_{[i,j,\alpha]}^n) \stackrel{[\text{theorem: 11.312}]}{=} \\ &\Delta(\beta_{[i,j,\alpha]}^n(e_1), \dots, \beta_{[i,j,\alpha]}^n(e_n)) = \\ &\Delta\left(\beta_{[i,j,\alpha]}^n(e_1) \dots \underbrace{\beta_{[i,j,\alpha]}^n(e_i)}_i \dots \beta_{[i,j,\alpha]}^n(e_n)\right) \stackrel{[\text{theorem: 11.312}]}{=} \\ &\Delta\left(e_1, \dots, \underbrace{e_i + \alpha \cdot e_j}_i, \dots, e_n\right) = \\ &\Delta\left(e_1, \dots, \underbrace{e_i}_i, \dots, e_n\right) + \alpha \cdot \Delta\left(e_1, \dots, \underbrace{e_j}_i, \dots, e_n\right) = \\ &\Delta(e_1, \dots, e_n) + \alpha \cdot \Delta\left(e_1, \dots, \underbrace{e_j}_i, \dots, e_n\right) \stackrel{i \neq j \wedge [\text{theorem: 11.291}]}{=} \\ &1 + \alpha \cdot 0 \\ &1 \end{aligned}$$

4. Let $k \in \{1, \dots, n\}$ then we have

$$\begin{aligned} \tau_{[i,j]}^n(\tau_{[i,j]}^n(e_k)) &= \begin{cases} \tau_{[i,j]}^n(e_j) & \text{if } k = i \\ \tau_{[i,j]}^n(e_i) & \text{if } k = j \\ \tau_{[i,j]}^n(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases} \\ &= \begin{cases} e_i & \text{if } k = i \\ e_j & \text{if } k = j \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \end{cases} \\ &= e_k \end{aligned}$$

Let $x \in X$ then there exists a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$ so that

$$(\tau_{[i,j]}^n \circ \tau_{[i,j]}^n)(x) = \sum_{i \in \{1, \dots, n\}} x_i \cdot (\tau_{[i,j]}^n \circ \tau_{[i,j]}^n)(e_i) = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i = x$$

proving that

$$\tau_{[i,j]}^n \circ \tau_{[i,j]}^n = \text{Id}_{\{1, \dots, n\}}$$

□

We are now ready to define elementary transformations:

Definition 11.393. Let $n \in \mathbb{N}$, X a finite dimensional vector space with $\dim(X) = n$ over a field F with characteristic zero then a nonsingular transformation $T \in \text{Hom}(X, X)$ is a **elementary transformation in X** if T is either of the form

$$\begin{aligned} T &= \text{Id}_X \\ T &= \sigma_{[i, \alpha]}^n \text{ where } i \in \{1, \dots, n\} \wedge \alpha \in F \\ T &= \tau_{[i, j]}^n \text{ where } i, j \in \{1, \dots, n\} \text{ with } i \neq j \\ T &= \beta_{[i, j, \alpha]}^n \text{ where } i, j \in \{1, \dots, n\} \text{ with } i \neq j \text{ and } \alpha \in F \end{aligned}$$

The set of all the elementary transformations is noted by $\text{Elem}(X)$, hence

$$\text{Elem}(X) = \{T \in \text{Hom}(X, X) | T \text{ is a elementary transformation}\}$$

We say that a linear transformation $L \in \text{Hom}(X, Y)$ is **composed of elementary transformations in X** if there exists a family $\{T_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Elem}(X)$, $m \in \mathbb{N}$ such that

$$L = \prod_{i=1}^n T_i$$

Lemma 11.394. Let X be a finite dimensional vector space over a field F with characteristic zero with $\dim(X) = n+1$, $n \in \mathbb{N}$, $E = \{e_1, \dots, e_{n+1}\}$ a basis for X , $m \in \{1, \dots, n\}$ and $\{\alpha_i\}_{i \in \{1, \dots, m\}} \subseteq F$ then we have:

1. $\forall k \in \{1, \dots, n\}$ we have

$$\left(\prod_{i=1}^m \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (e_k) = e_k$$

2.

$$\left(\prod_{i=1}^m \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (e_{n+1}) = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i + e_{n+1}$$

In other words $\forall k \in \{1, \dots, n+1\}$ we have

$$\left(\prod_{i=1}^m \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (e_k) = \begin{cases} e_k & \text{if } k \in \{1, \dots, n\} \\ \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot e_i + e_{n+1} & \text{if } k = n+1 \end{cases}$$

Proof.

1. We prove this by induction, so given $k \in \{1, \dots, n\}$ let

$$S_{n,k} = \left\{ m \in \mathbb{N} | \text{If } m \leq n \text{ then } \forall \{\alpha_i\}_{i \in \{1, \dots, m\}} \text{ we have } \left(\prod_{i=1}^m \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (e_k) = e_k \right\}$$

then:

$1 \in S_{n,k}$. Let $\{\alpha_i\}_{i \in \{1, \dots, 1\}} \subseteq F$ then

$$\left(\prod_{i=1}^1 \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (e_k) = \beta_{[n+1, 1, \alpha_1]}^{n+1} (e_k)_{k \neq n+1 \wedge [\text{definition: 11.391}]} = e_k$$

proving that $1 \in S_{n,k}$.

$m \in S_{n,k} \Rightarrow m+1 \in S_{n,k}$. Assume that $m+1 \leq n$ and let $\{\alpha_i\}_{i \in \{1, \dots, m+1\}} \subseteq F$ then we have:

$$\begin{aligned} \left(\prod_{i=1}^{m+1} \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (e_k) &= \left(\prod_{i=1}^m \beta_{[n+1, i, \alpha_i]}^{n+1} \circ \beta_{[n+1, m+1, \alpha_{m+1}]}^{n+1} \right) (e_k) \\ &= \left(\prod_{i=1}^m \beta_{[n+1, i, \alpha_i]}^{n+1} \right) (\beta_{[n+1, m+1, \alpha_{m+1}]}^{n+1} (e_k)) \end{aligned}$$

$$\begin{array}{c} \stackrel{=}{=} \\ k \neq n+1 [\text{theorem: 11.391}] \end{array} \left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_k) \\ \stackrel{=}{=} \\ m \in \overline{S_{n,k}} \quad e_k$$

proving that $m+1 \in S_{n,k}$.

2. We prove this by induction, so let

$$S_n = \left\{ m \in \mathbb{N} \mid \text{If } m \leq n \text{ then } \forall \{\alpha_i\}_{i \in \{1, \dots, m\}} \subseteq F \text{ we have } \left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{n+1}) = \sum_{i \in \{1, \dots, m\}} \alpha_i \cdot e_i + e_{n+1} \right\}$$

then we have:

$1 \in S_n$. If $\{\alpha_i\}_{i \in \{1\}}$ then

$$\begin{aligned} \left(\prod_{i=1}^1 \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{n+1}) &= \beta_{[n+1,1,\alpha_1]}^{n+1} (e_{n+1}) \\ &\stackrel{=}{=}_{k=n+1 [\text{definition: 11.391}]} e_{n+1} + \alpha_1 \cdot e_1 \\ &= \sum_{i \in \{1\}} \alpha_i \cdot e_i + e_{n+1} \end{aligned}$$

proving that $1 \in S_n$.

$m \in S_n \Rightarrow m+1 \in S_n$. Assume that $m+1 \leq n$ and let $\{\alpha_i\}_{i \in \{1, \dots, m+1\}} \subseteq F$ then we have

$$\begin{aligned} &\left(\prod_{i=1}^{m+1} \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{n+1}) = \\ &\left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \circ \beta_{[n+1,m+1,\alpha_{m+1}]}^{n+1} \right) (e_{n+1}) = \\ &\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} (\beta_{[n+1,m+1,\alpha_{m+1}]}^{n+1} (e_{n+1})) \stackrel{=}{=}_{k=n+1 \wedge [\text{definition: 11.391}]} \\ &\left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{n+1} + \alpha_{m+1} \cdot e_{m+1}) = \\ &\left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{n+1}) + \alpha_{m+1} \cdot \left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{m+1}) \stackrel{=}{=}_{m+1 \leq n < \overline{n+1} \text{ and (1)}} \\ &\left(\prod_{i=1}^m \beta_{[n+1,i,\alpha_i]}^{n+1} \right) (e_{n+1}) + \alpha_{m+1} \cdot e_{m+1} \stackrel{=}{=}_{m \in \overline{S_n}} \\ &\sum_{i=1}^m \alpha_i \cdot e_i + e_{n+1} + \alpha_{m+1} \cdot e_{m+1} = \\ &\sum_{i=1}^{m+1} \alpha_i \cdot e_i + e_{n+1} \end{aligned}$$

proving that $n+1 \in S_n$. □

Lemma 11.395. *Let X a finite dimensional vector space with $\dim(X) = n + 1$ over a field F with characteristic zero, $E = \{e_1, \dots, e_{n+1}\}$ a basis for X , $m \in \{1, \dots, n\}$ and $\{\alpha_i\}_{i \in \{1, \dots, m\}} \subseteq F$ then we have:*

1. $\forall k \in \{m + 1, \dots, n + 1\}$ we have

$$\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) = e_k$$

2. $\forall k \in \{1, \dots, m\}$ we have

$$\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) = e_k + \alpha_k \cdot e_{n+1}$$

In other words we have $\forall k \in \{1, \dots, n + 1\}$ that

$$\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) = \begin{cases} e_k + \alpha_k \cdot e_{n+1} & \text{if } k \in \{1, \dots, m\} \\ e_k & \text{if } k \in \{m + 1, \dots, n + 1\} \end{cases}$$

Proof.

1. We use induction on m for this proof, so let

$$S_n = \left\{ m \in \mathbb{N} \mid \text{If } m \leq n \text{ then } \forall \{\alpha_i\}_{i \in \{1, \dots, m\}} \text{ and } \forall k \in \{m + 1, \dots, n + 1\} \text{ we have} \right. \\ \left. \left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) = e_k \right\}$$

then we have:

$1 \in S_n$. Let $\{\alpha_i\}_{i \in \{1\}} \subseteq F$ then for $k \in \{1 + 1, \dots, n + 1\} \Rightarrow k \neq 1$ we have

$$\left(\prod_{i=1}^1 \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) = \beta_{[1, n+1, \alpha_1]}^{n+1} (e_k) \stackrel{k \neq 1 [\text{definition: 11.391}]}{=} e_k$$

proving that $1 \in S_n$.

$m \in S_n \Rightarrow m + 1 \in S_n$. Assume that $m + 1 \leq n$ and let $\{\alpha_i\}_{i \in \{1, \dots, m+1\}} \subseteq F$ then we have for $k \in \{(m + 1) + 1, \dots, n + 1\} \subseteq \{m + 1, \dots, n + 1\} \Rightarrow k \neq m + 1$ and $k \in \{m + 1, \dots, n + 1\}$. Further

$$\begin{aligned} & \left(\prod_{i=1}^{m+1} \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) = \\ & \left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \circ \beta_{[m+1, n+1, \alpha_{m+1}]}^{n+1} \right) (e_k) \\ & \left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (\beta_{[m+1, n+1, \alpha_{m+1}]}^{n+1} (e_k)) \stackrel{k \neq 1 \wedge [\text{definition: 11.391}]}{=} \\ & \left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) \stackrel{m \in S_n \wedge k \in \{m+1, \dots, n+1\}}{=} e_k \end{aligned}$$

proving that $m+1 \in S_n$.

2. This is proved by induction, so let

$$S_n = \left\{ m \in \mathbb{N} \mid \text{If } m \leq n \text{ then } \forall \{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F \text{ and } \forall k \in \{1, \dots, m\} \text{ we have} \right. \\ \left. \left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_k) = e_k + \alpha_k \cdot e_{n+1} \right\}$$

then we have:

$1 \in S$. Let $\{\alpha_i\}_{i \in \{1\}} \subseteq F$ then for $k \in \{1\} \Rightarrow k=1$ we have

$$\begin{aligned} \left(\prod_{i=1}^1 \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_1) &= (\beta_{[1, n+1, \alpha_1]}^{n+1})(e_1) \\ &\stackrel{k=1 \wedge [\text{definition: 11.391}]}{=} e_1 + \alpha_1 \cdot e_{n+1} \\ &\stackrel{k=1}{=} e_k + \alpha_k \cdot e_{n+1} \end{aligned}$$

proving that $1 \in S_n$.

$m \in S \Rightarrow m+1 \in S$. Assume that $m+1 \leq n$ and let $\{\alpha_i\}_{i \in \{1, \dots, m+1\}} \subseteq F$ then for $k \in \{1, \dots, m+1\}$ we have either:

$k = m+1$. Then

$$\begin{aligned} &\left(\prod_{i=1}^{m+1} \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_k) = \\ &\left(\prod_{i=1}^{m+1} \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_{m+1}) = \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \circ \beta_{[m+1, n+1, \alpha_{m+1}]}^{n+1} \right)(e_{m+1}) = \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(\beta_{[m+1, n+1, \alpha_{m+1}]}^{n+1}(e_{m+1})) \stackrel{[\text{definition: 11.391}]}{=} \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_{m+1} + \alpha_{m+1} \cdot e_{n+1}) = \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_{m+1}) + \alpha_{m+1} \cdot \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_{n+1}) \stackrel{(1) \wedge m+1, n+1 \in \{m+1, \dots, n+1\}}{=} \\ &\quad e_{m+1} + \alpha_{m+1} \cdot e_{n+1} \stackrel{m+1=k}{=} \\ &\quad e_k + \alpha_k \cdot e_{n+1} \end{aligned}$$

$k \in \{1, \dots, m\}$. Then

$$\begin{aligned} &\left(\prod_{i=1}^{m+1} \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(e_k) = \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \circ \beta_{[m+1, n+1, \alpha_{m+1}]}^{n+1} \right)(e_k) = \\ &\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right)(\beta_{[m+1, n+1, \alpha_{m+1}]}^{n+1}(e_k)) \stackrel{k \neq m+1 \wedge [\text{definition: 11.391}]}{=} \end{aligned}$$

$$\left(\prod_{i=1}^m \beta_{[i, n+1, \alpha_i]}^{n+1} \right) (e_k) \quad k \in \{1, \dots, m\} \wedge m \in S_n$$

$$e_k + \alpha_k \cdot e_{n+1}$$

So in all cases we have $(\prod_{i=1}^{m+1} \beta_{[i, n+1, \alpha_i]}^{n+1})(e_k) = e_k + \alpha_k \cdot e_{n+1}$ proving that

$$m+1 \in S_n.$$

□

We have the following trivial lemma about transformations that are compositions of elementary transformations.

Lemma 11.396. *Let $n \in \mathbb{N}$, X a finite dimensional vector space with $\dim(X) = n$ over a field F with characteristic zero then we have:*

1. *If $T \in \text{GL}(X)$ is a elementary transformation and $L \in \text{GL}(X)$ is composed of elementary transformations then $L \circ T$ and $T \circ L$ are composed of linear transformations*
2. *If $L_1, L_2 \in \text{GL}(X)$ are composed of elementary transformations then $L_1 \circ L_2$ is composed of elementary transformations.*

Proof.

1. As L is composed of elementary transformations there exists a $m \in \mathbb{N}$ and $\{E_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Elem}(X)$ such that $L = \prod_{i=1}^m E_i$. Define now

$$\{S_i\}_{i \in \{1, \dots, m+1\}} \subseteq \text{Elem}(X) \text{ by } S_i = \begin{cases} T & \text{if } i = 1 \\ E_{i-1} & \text{if } i \in \{2, \dots, m+1\} \end{cases}$$

and

$$\{T_i\}_{i \in \{1, \dots, m+1\}} \subseteq \text{Elem}(X) \text{ by } T_i = \begin{cases} E_i & \text{if } i \in \{1, \dots, m\} \\ T & \text{if } i = m+1 \end{cases}$$

Then we have

$$\begin{aligned} \prod_{i=1}^{m+1} S_i & \stackrel{[\text{definition: 11.12}]}{=} \prod_{i=0}^m S_{i+1} \\ & \stackrel{[\text{theorem: 11.22}]}{=} S_{0+1} \circ \prod_{i=1}^m S_{i+1} \\ & = S_1 \circ \prod_{i=1}^m S_{i+1} \\ & = T \circ \prod_{i=1}^m E_{(i-1)+1} \\ & = T \circ \prod_{i=1}^m E_i \\ & = T \circ L \end{aligned}$$

and

$$\begin{aligned} \prod_{i=1}^{m+1} T_i & \stackrel{[[\text{theorem: 11.15}]]}{=} \left(\prod_{i=1}^m T_i \right) \circ T_{m+1} \\ & = \left(\prod_{i=1}^m E_i \right) \circ T \\ & = L \circ T \end{aligned}$$

proving that $L \circ T$ and $T \circ L$ are composed of elementary transformations.

2. If L_1, L_2 are composed of elementary transformations then there exists $k, l \in \mathbb{N}$ and families of elementary transformations $\{T_i\}_{i \in \{1, \dots, k\}} \subseteq \text{Elem}(X)$, $\{S_i\}_{i \in \{1, \dots, l\}} \subseteq \text{Elem}(X)$ such that

$$L_1 = \prod_{i=1}^k T_i \text{ and } L_2 = \prod_{i=1}^l S_i$$

Define then the family $\{E_i\}_{i \in \{1, \dots, k+l\}} \subseteq \text{GL}(X)$ by

$$\forall i \in \{1, \dots, k+l\} \quad E_i = \begin{cases} T_i & \text{if } i \in \{1, \dots, k\} \\ S_{i-k} & \text{if } i \in \{k+1, \dots, k+l\} \end{cases}$$

then we have

$$\begin{aligned} L_1 \circ L_2 &= \prod_{i=1}^k T_i \circ \prod_{i=1}^l S_i \\ &= \prod_{i=1}^k T_i \circ \prod_{i=1}^l S_i \\ &\stackrel{[\text{theorem: 11.14}]}{=} \prod_{i=1}^k T_i \circ \prod_{i=k+1}^{k+l} S_{i-k} \\ &= \prod_{i=1}^k E_i \circ \prod_{i=k+1}^{k+l} E_i \\ &\stackrel{[\text{theorem: 11.23}]}{=} \prod_{i=1}^{k+l} E_i \end{aligned}$$

proving that $L_1 \circ L_2$ is composed of elementary transpositions. $\square$

To use mathematical induction on the dimension of a vector space we introduce the following definition.

Definition 11.397. Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field $\mathcal{F}$ with basis $E = \{e_1, \dots, e_{n+1}\}$, $Y = \text{span}(e_1, \dots, e_n)$ the vector space with basis $F = \{e_1, \dots, e_n\}$ and $L \in \text{Hom}(Y, Y)$ then we define $M_{[n, L]} \in \mathcal{M}_{n+1, n+1}(F)$ by

$$\forall k, l \in \{1, \dots, n+1\} \text{ we have } (M_{[n+1, L]})_{k, l} = \begin{cases} \mathcal{M}(L; F, F)_{k, l} & \text{if } (k, l) \in \{1, \dots, n\} \times \{1, \dots, n\} \\ \delta_{k, l} & \text{if } k = n+1 \vee l = n+1 \end{cases}$$

and define $L^{[n]} \in \text{Hom}(X, X)$ by

$$L^{[n+1]} = \mathcal{M}(E, E)^{-1}(M_{[n+1, L]})$$

then we have $\forall k \in \{1, \dots, n\}$ that

$$\begin{aligned} L^{[n+1]}(e_k) &= \sum_{i \in \{1, \dots, n+1\}} (M_{[n+1, L]})_{i, k} \cdot e_i \\ &= \sum_{i \in \{1, \dots, n\}} (M_{[n+1, L]})_{i, k} \cdot e_i + \sum_{i \in \{n+1\}} (M_{[n+1, L]})_{i, k} \cdot e_i \\ &= \sum_{i \in \{1, \dots, n\}} (M_{[n+1, L]})_{i, k} \cdot e_i + (M_{[n+1, L]})_{n+1, k} \cdot e_{n+1} \\ &= \sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i, k} \cdot e_i + \delta_{n+1, k} \cdot e_{n+1} \\ &= L(e_k) + \delta_{n+1, k} \cdot e_{n+1} \\ &\stackrel{k \neq n+1}{=} L(e_k) + 0 \\ &\stackrel{k \neq n}{=} L(e_k) \end{aligned}$$

and

$$\begin{aligned}
 L^{[n+1]}(e_{n+1}) &= \sum_{i \in \{1, \dots, n+1\}} (M_{[n+1, L]})_{i, n+1} \cdot e_i \\
 &= \sum_{i \in \{1, \dots, n+1\}} \delta_{i, n+1} \cdot e_i \\
 &= e_{n+1}
 \end{aligned}$$

proving that

$$\forall k \in \{1, \dots, n\} \text{ we have } L^{[n+1]}(e_k) = \begin{cases} L(e_k) & \text{if } k \in \{1, \dots, n\} \\ e_{n+1} & \text{if } k = n+1 \end{cases}$$

Lemma 11.398. Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field $\mathcal{F}$ with basis $E = \{e_1, \dots, e_{n+1}\}$, $Y = \text{span}(\{e_1, \dots, e_n\})$ a n -dimensional vector space with basis $F = \{e_1, \dots, e_n\}$ then we have:

1. $(\text{Id}_Y)^{[n+1]} = \text{Id}_X$
2. $\forall i \in \{1, \dots, n\}$ and $\alpha \in F$ we have $(\sigma_{[i, \alpha]}^n)^{[n+1]} = \sigma_{[i, \alpha]}^{n+1}$
3. $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have $(\tau_{[i, j]}^n)^{[n+1]} = \tau_{[i, j]}^{n+1}$
4. $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ and $\alpha \in F$ then $(\beta_{[i, j, \alpha]}^n)^{[n+1]} = \beta_{[i, j, \alpha]}^{n+1}$

in other words if $T \in \text{Elem}(X)$ is a elementary transformation in Y then $T^{[n+1]}$ is a elementary transformation in X .

Proof.

1. Let $k \in \{1, \dots, n+1\}$ then

$$\begin{aligned}
 (\text{Id}_X)^{[n+1]}(e_k) &= \begin{cases} \text{Id}_X(e_k) & \text{if } k \in \{1, \dots, n\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &= \begin{cases} e_k & \text{if } k \in \{1, \dots, n\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &= e_k \\
 &= \text{Id}_Y(e_k)
 \end{aligned}$$

so that by [theorem: 11.336] $(\text{Id}_X)^{[n+1]} = \text{Id}_Y$.

2. Let $k \in \{1, \dots, n+1\}$ then

$$\begin{aligned}
 (\sigma_{[i, \alpha]}^n)^{[n+1]}(e_k) &= \begin{cases} \sigma_{[i, \alpha]}^n(e_k) & \text{if } k \in \{1, \dots, n\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{[\text{definition: 11.391}]}{=} \begin{cases} \alpha \cdot e_i & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{\{1, \dots, n\} \setminus \{i\} \subseteq \{1, \dots, n+1\} \setminus \{i\}}{=} \begin{cases} \sigma_{[i, \alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \\ \sigma_{[i, \alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{i \neq n+1}{=} \begin{cases} \sigma_{[i, \alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \\ \sigma_{[i, \alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ \sigma_{[i, \alpha]}^{n+1}(e_k) & \text{if } k = n+1 \end{cases} \\
 &= \sigma_{[i, \alpha]}^{n+1}(e_k)
 \end{aligned}$$

so that by [theorem: 11.336] $\sigma_{[i,\alpha]}^n = \sigma_{[i,\alpha]}^{n+1}$.

3. Let $k \in \{1, \dots, n+1\}$ then

$$\begin{aligned}
 (\tau_{[i,j]}^n)^{[n+1]}(e_k) &= \begin{cases} \tau_{[i,j]}^n(e_k) & \text{if } k \in \{1, \dots, n\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{[\text{definition: 11.391}]}{=} \begin{cases} e_i & \text{if } k \in \{1, \dots, n\} \wedge k = j \\ e_j & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{\{1, \dots, n\} \setminus \{i, j\} \subseteq \overline{\{1, \dots, n+1\}} \setminus \{i, j\}}{=} \begin{cases} \tau_{[i,j]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = j \\ \tau_{[i,j]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ \tau_{[i,j]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{i, j \neq n+1}{=} \begin{cases} \tau_{[i,j]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = j \\ \tau_{[i,j]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ \tau_{[i,j]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i, j\} \\ \tau_{[i,j]}^{n+1}(e_k) & \text{if } k = n+1 \end{cases} \\
 &= \tau_{[i,j]}^{n+1}(e_k)
 \end{aligned}$$

proving by [theorem: 11.336] $\tau_{[i,j]}^n = \tau_{[i,j]}^{n+1}$.

4. Let $k \in \{1, \dots, n+1\}$ then

$$\begin{aligned}
 (\beta_{[i,j,\alpha]}^n)^{[n+1]}(e_k) &= \begin{cases} \beta_{[i,j,\alpha]}^n(x_k) & \text{if } k \in \{1, \dots, n\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{[\text{definition: 11.391}]}{=} \begin{cases} e_i + \alpha \cdot e_j & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &= \begin{cases} \beta_{[i,j,\alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ e_k & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{\{1, \dots, n\} \setminus \{i\} \subseteq \overline{\{1, \dots, n+1\}} \setminus \{i\}}{=} \begin{cases} \beta_{[i,j,\alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ \beta_{[i,j,\alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ e_{n+1} & \text{if } k = n+1 \end{cases} \\
 &\stackrel{i \neq n+1}{=} \begin{cases} \beta_{[i,j,\alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \wedge k = i \\ \beta_{[i,j,\alpha]}^{n+1}(e_k) & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \\ \beta_{[i,j,\alpha]}^{n+1}(e_k) & \text{if } k = n+1 \end{cases} \\
 &= \beta_{[i,j,\alpha]}^{n+1}(e_k)
 \end{aligned}$$

proving by [theorem: 11.336] $\beta_{[i,j,\alpha]}^n = \beta_{[i,j,\alpha]}^{n+1}$.

□

Extending a elementary transformation to a higher dimension also applies to nonsingular transformations that are composed of elementary transformation.

Lemma 11.399. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with basis $E = \{e_1, \dots, e_{n+1}\}$, $Y = \text{span}(\{e_1, \dots, e_n\})$ a n -dimensional vector space with basis $F = \{e_1, \dots, e_n\}$ then if $K, L \in \text{GL}(Y)$ we have that*

$$(K \circ L)^{[n+1]} = K^{[n+1]} \circ L^{[n+1]}$$

Proof. Let $k \in \{1, \dots, n+1\}$ then we have either:

$k \in \{1, \dots, n\}$. Then

$$\begin{aligned} (K \circ L)^{[n+1]}(e_k) &= (K \circ L)(e_k) \\ &= K(L(e_k)) \\ &= K\left(\sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i,k} \cdot e_i\right) \\ &= \sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i,k} \cdot K(e_i) \\ &= \sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i,k} \cdot K^{[n+1]}(e_i) \\ &= K^{[n+1]}\left(\sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i,k} \cdot e_i\right) \\ &= K^{[n+1]}\left(\sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i,k} \cdot e_i + 0 \cdot e_{n+1}\right) \\ &\stackrel{k \neq n+1}{=} K^{[n+1]}\left(\sum_{i \in \{1, \dots, n\}} \mathcal{M}(L; F, F)_{i,k} \cdot e_i + \delta_{n+1,k} \cdot e_{n+1}\right) \\ &\stackrel{[\text{see: 11.397}]}{=} K^{[n+1]}\left(\sum_{i \in \{1, \dots, n\}} (M_{[n+1,L]_{i,k}}) \cdot e_i + (M_{[n+1,L]_{n+1,k}}) \cdot e_{n+1}\right) \\ &= K^{[n+1]}\left(\sum_{i \in \{1, \dots, n+1\}} (M_{[n+1,L]_{i,k}}) \cdot e_i\right) \\ &= K^{[n+1]}(L^{[n+1]}(e_k)) \\ &= (K^{[n+1]} \circ L^{[n+1]})(e_k) \end{aligned}$$

$k = n+1$. Then

$$\begin{aligned} (K \circ L)^{[n+1]}(e_k) &= e_{n+1} \\ &= L^{[n+1]}(e_k) \\ &\stackrel{L^{[n+1]}(e_k) = e_{n+1}}{=} K^{[n+1]}(L^{[n+1]}(e_k)) \\ &= (K^{[n+1]} \circ L^{[n+1]})(e_k) \end{aligned}$$

So we have $\forall k \in \{1, \dots, n+1\}$ that $(K \circ L)^{[n+1]}(e_k) = (K^{[n+1]} \circ L^{[n+1]})(e_k)$ which by [theorem: 11.336] proves that

$$(K \circ L)^{[n+1]} = K^{[n+1]} \circ L^{[n+1]} \quad \square$$

Lemma 11.400. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with basis $E = \{e_1, \dots, e_{n+1}\}$, $Y = \text{span}(\{e_1, \dots, e_n\})$ a n -dimensional vector space with basis $F = \{e_1, \dots, e_n\}$ then if $m \in \mathbb{N}$, $\{L_i\}_{i \in \{1, \dots, m\}} \subseteq \text{GL}(Y)$ we have that*

$$\left(\prod_{i=1}^m L_i\right)^{[n+1]} = \prod_{i=1}^m (L_i)^{[n+1]}$$

Proof. We prove this by induction, so let

$$S = \left\{ m \in \mathbb{N} \mid \text{If } \{L_i\}_{i \in \{1, \dots, m\}} \subseteq \text{GL}(Y) \text{ then } \left(\prod_{i=1}^m L_i \right)^{[n+1]} = \prod_{i=1}^m (L_i)^{[n+1]} \right\}$$

then we have:

$$1 \in S. \text{ Let } \{L_i\}_{i \in \{1, \dots, 1\}} \subseteq \text{GL}(Y) \text{ then } \left(\prod_{i=1}^1 L_i \right)^{[n+1]} = (L_1)^{[n+1]} = \prod_{i=1}^1 (L_i)^{[n+1]}$$

$m \in S \Rightarrow m+1 \in S$. Let $\{L_i\}_{i \in \{1, \dots, m+1\}} \subseteq \text{GL}(X)$ then we have

$$\begin{aligned} \left(\prod_{i=1}^{m+1} L_i \right)^{[n+1]} &= \left(\left(\prod_{i=1}^m L_i \right) \circ L_{m+1} \right)^{[n+1]} \\ &\stackrel{[\text{lemma: 11.399}]}{=} \left(\prod_{i=1}^m L_i \right)^{[n+1]} \circ (L_{m+1})^{[n+1]} \\ &\stackrel{m \in S}{=} \left(\prod_{i=1}^m (L_i)^{[n+1]} \right) \circ (L_{m+1})^{[n+1]} \\ &= \prod_{i=1}^{m+1} (L_i)^{[n+1]} \end{aligned}$$

proving that $m+1 \in S$. □

Corollary 11.401. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with basis $E = \{e_1, \dots, e_{n+1}\}$, $Y = \text{span}(\{e_1, \dots, e_n\})$ a n -dimensional vector space with basis $F = \{e_1, \dots, e_n\}$ then if $L \in \text{GL}(Y)$ is composed of elementary transformations in Y it follows that $L^{[n+1]}$ is composed of elementary transformations in X .*

Proof. As $L \in \text{GL}(Y)$ is composed of linear transformations there exists a $m \in \mathbb{N}$ and a family $\{T_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Elem}(Y)$ of elementary transformations such that

$$L = \prod_{i=1}^m T_i$$

Using [lemma: 11.398] we have $\forall i \in \{1, \dots, m\}$ that $(L_i)^{[n+1]} \in \text{Elem}(X)$. So as

$$L^{[n+1]} \stackrel{[\text{theorem: 11.400}]}{=} \prod_{i=1}^m (T_i)^{[n+1]}$$

we have that $L^{[n+1]}$ is composed of elementary transformations in X . □

We are now ready to prove that every nonsingular transformation is composed of elementary transformations.

Lemma 11.402. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with characteristic zero, $L \in \text{GL}(X)$ a nonsingular transformation then L is composed of elementary transformations. In other words there exists a $m \in \mathbb{N}$ and $\{T_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Elem}(X)$ such that*

$$L = \prod_{i=1}^m T_i \stackrel{\text{def}}{=} T_1 \circ \dots \circ T_m$$

Proof. We prove this by induction on the dimension n of X , so let

$S = \{n \in \mathbb{N} \mid \text{If } L \in \text{GL}(X) \text{ where } X \text{ is a } n\text{-dimensional vector space over } F \text{ then } L \text{ is composed of elementary transformations in } X\}$

then we have:

$1 \in S$. As X is one dimensional it has a base $\{e_i\}_{i \in \{1, \dots, 1\}}$ then, if $L \in \text{GL}(X)$ we have $L(e_1) = \alpha \cdot e_1$, so $L = \sigma_{[1, \alpha]}^1 \in \text{Elem}(X)$. As $L = \sigma_{[1, \alpha]}^1 = \prod_{i=1}^1 \sigma_{[i, \alpha]}^1$ it follows that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in \text{GL}(X)$ where $\dim(X) = n+1$ is a $(n+1)$ -dimensional vector space over F and $E = \{e_1, \dots, e_{n+1}\}$ a basis for X . Let $M = \mathcal{M}(L; E, E) \in \mathcal{M}_{n+1, n+1}(F)$ be the matrix associated with the nonsingular linear transformation. Then

$$\forall i \in \{1, \dots, n\} \text{ we have } L(e_i) = \sum_{k \in \{1, \dots, n+1\}} M_{k,i} \cdot e_k \quad (11.274)$$

As $L \in \text{GL}(X)$ we have that $\det(L) \neq 0$ hence

$$\begin{aligned} 0 &\neq \det(L) \\ &\stackrel{[\text{theorem: 11.359}]}{=} \det(M) \\ &\stackrel{[\text{theorem: 11.382}]}{=} \sum_{i \in \{1, \dots, n+1\}} (-1)^{i+(n+1)} \cdot M_{i, n+1} \cdot \det([i \boxplus n+1](M)) \end{aligned}$$

If $\forall i \in \{1, \dots, n+1\} \det([i \boxplus n+1](M)) = 0$ then the above leads to the contradiction $0 \neq 0$, hence

$$\exists i_0 \in \{1, \dots, n+1\} \text{ such that } \det([i_0 \boxplus n+1](M)) \neq 0 \quad (11.275)$$

For i_0 we have now two possibilities to consider:

$i_0 = n+1$. Then by taking $L_1 = L$ and $T = \text{Id}_X$ [so that T is a elementary transformation] $L = T \circ L_1$ and $\det([n+1, n+1](\mathcal{M}(L_1; E, E))) \neq 0$ resulting in

$$\exists T \in \text{Elem}(X) \text{ such that } L = T \circ L_1 \text{ and } \det([n+1, n+1](\mathcal{M}(L_1; E, E))) \neq 0 \quad (11.276)$$

$i_0 \in \{1, \dots, n\}$. Take $k, l \in \{1, \dots, n\}$ and consider $([i_0 \boxplus n+1](M))_{k,l}$ then for k we have either:

$$1 \leq k < i_0. \text{ Then } ([i_0 \boxplus n+1](M))_{k,l} \stackrel{1 \leq l < n+1 \wedge [\text{definition: 11.369}]}{=} M_{k,l}$$

$$i_0 \leq k \leq n. \text{ Then } ([i_0 \boxplus n+1](M))_{k,l} \stackrel{1 \leq l < n+1 \wedge [\text{definition: 11.369}]}{=} M_{k+1,l}$$

proving that

$$\forall k, l \in \{1, \dots, n\} \text{ we have } ([i_0 \boxplus n+1](M))_{k,l} = \begin{cases} M_{k,l} & \text{if } 1 \leq k < i_0 \\ M_{k+1,l} & \text{if } i_0 \leq k \leq n \end{cases} \quad (11.277)$$

Use [definition: 11.363] to define $M' \in \mathcal{M}_{n,n}(F)$ by

$$M' = ([i_0 \boxplus n+1](M))_{(n \rightsquigarrow_{\text{ni}} i_0)} \text{ so that } \forall k, l \in \{1, \dots, n\} M'_{k,l} = ([i_0 \boxplus n+1](M))_{(n \rightsquigarrow_{\text{ni}} i_0)(k),l}$$

then we have for $i_0 \in \{1, \dots, n\}$ the following cases to consider:

$i_0 = n$. Then $\forall k, n \in \{1, \dots, n\}$ we have

$$\begin{aligned}
 M'_{k,l} &= ([i_0 \boxplus n + 1](M))_{(n \rightsquigarrow i_0)(k),l} \\
 &\stackrel{[\text{definition: 11.252(1)}]}{=} ([i_0 \boxplus n + 1](M))_{k,l} \\
 &\stackrel{[\text{eq: 11.277}]}{=} \begin{cases} M_{k,l} & \text{if } 1 \leq k < i_0 \\ M_{k+1,l} & \text{if } i_0 \leq k \leq n \end{cases} \\
 &\stackrel{i_0=n}{=} \begin{cases} M_{k,l} & \text{if } 1 \leq k < n \\ M_{k+1,l} & \text{if } k = i_0 \end{cases} \\
 &\stackrel{i_0=n}{=} \begin{cases} M_{k,l} & \text{if } k \in \{1, \dots, n\} \setminus \{i_0\} \\ M_{k+1,l} & \text{if } k = i_0 \end{cases}
 \end{aligned}$$

$i_0 \in \{1, \dots, n-1\}$. Then we have

$$\begin{aligned}
 M'_{k,l} &= ([i_0 \boxplus n + 1](M))_{(n \rightsquigarrow i_0)(k),l} \\
 &\stackrel{i_0 < n \wedge [\text{definition: 11.252(3)}]}{=} \begin{cases} ([i_0 \boxplus n + 1](M))_{k,l} & \text{if } 1 \leq k < i_0 \\ ([i_0 \boxplus n + 1](M))_{n,l} & \text{if } k = i_0 \\ ([i_0 \boxplus n + 1](M))_{k-1,l} & \text{if } i_0 < k \leq n \\ ([i_0 \boxplus n + 1](M))_{k,l} & \text{if } n < k \leq n \end{cases} \\
 &\stackrel{n < k \leq n \text{ is impossible}}{=} \begin{cases} ([i_0 \boxplus n + 1](M))_{k,l} & \text{if } 1 \leq k < i_0 \\ ([i_0 \boxplus n + 1](M))_{n,l} & \text{if } k = i_0 \\ ([i_0 \boxplus n + 1](M))_{k-1,l} & \text{if } i_0 < k \leq n \end{cases} \\
 &\stackrel{k < i_0 \wedge l \leq n < n+1 \wedge [\text{definition: 11.369}]}{=} \begin{cases} M_{k,l} & \text{if } 1 \leq k < i_0 \\ ([i_0 \boxplus n + 1](M))_{n,l} & \text{if } k = i_0 \\ ([i_0 \boxplus n + 1](M))_{k-1,l} & \text{if } i_0 < k \leq n \end{cases} \\
 &\stackrel{i_0 < n \wedge l \leq n < n+1 \wedge [\text{definition: 11.369}]}{=} \begin{cases} M_{k,l} & \text{if } 1 \leq k < i_0 \\ M_{n+1,l} & \text{if } k = i_0 \\ ([i_0 \boxplus n + 1](M))_{k-1,l} & \text{if } i_0 < k \leq n \end{cases} \\
 &\stackrel{l \leq n < n+1 \wedge [\text{definition: 11.369}]}{=} \begin{cases} M_{k,l} & \text{if } 1 \leq k < i_0 \\ M_{n+1,l} & \text{if } k = i_0 \\ M_{(k-1)+1,l} & \text{if } i_0 < k \leq n \end{cases} \\
 &= \begin{cases} M_{k,l} & \text{if } k \in \{1, \dots, n\} \setminus \{i_0\} \\ M_{k+1,l} & \text{if } k = i_0 \end{cases}
 \end{aligned}$$

So in all cases we have

$$\forall k, l \in \{1, \dots, n\} \text{ we have } M'_{k,l} = \begin{cases} M_{k,l} & \text{if } k \in \{1, \dots, n\} \setminus \{i_0\} \\ M_{k+1,l} & \text{if } k = i_0 \end{cases} \quad (11.278)$$

Further

$$\begin{aligned}
 \det(M') &= \det([i_0 \boxplus n + 1](M))_{(n \rightsquigarrow i_0)} \\
 &\stackrel{[\text{theorem: 11.364}]}{=} \text{sign}((n \rightsquigarrow_{\text{ni}} i_0)) \cdot \det([i_0 \boxplus n + 1](M)) \\
 &\stackrel{[\text{eq: 11.275}]}{\neq} 0
 \end{aligned}$$

proving that

$$\det(M') \neq 0 \quad (11.279)$$

Define

$$L_1 = \tau_{[n+1, i_0]}^{n+1} \circ L \quad (11.280)$$

then we have $\forall i \in \{1, \dots, \}$ that

$$\begin{aligned} \sum_{k \in \{1, \dots, n+1\}} \mathcal{M}(L_1, E, E)_{k,i} \cdot e_k &= \\ L_1(e_i) &= \\ \tau_{[n+1, i_0]}^{n+1}(L(e_i)) &= \\ \tau_{[n+1, i_0]}^{n+1} \left(\sum_{k \in \{1, \dots, n+1\}} M_{k,i} \cdot e_k \right) &= \\ \sum_{k \in \{1, \dots, n+1\}} M_{k,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_k) &= \\ \sum_{k \in \{1, \dots, n\} \setminus \{n+1, i_0\}} M_{k,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_k) + \sum_{k \in \{n+1\}} M_{k,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_k) + \sum_{k \in \{i_0\}} M_{k,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_k) &= \\ \sum_{k \in \{1, \dots, n\} \setminus \{n+1, i_0\}} M_{k,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_k) + M_{n+1,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_{n+1}) + M_{i_0,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_{i_0}) &= \\ \sum_{k \in \{1, \dots, n\} \setminus \{i_0\}} M_{k,i} \cdot \tau_{[n+1, i_0]}^{n+1}(e_k) + M_{n+1,i} \cdot e_{i_0} + M_{i_0,i} \cdot e_{n+1} &= \\ \sum_{k \in \{1, \dots, n\} \setminus \{i_0\}} M_{k,i} \cdot e_k + M_{n+1,i} \cdot e_{i_0} + M_{i_0,i} \cdot e_{n+1} \end{aligned}$$

which as the expansion in a basis is unique proves that

$$\forall k, l \in \{1, \dots, n+1\} \quad (\mathcal{M}(L_1; E, E))_{k,l} = \begin{cases} M_{k,l} & \text{if } k \in \{1, \dots, n\} \setminus \{i_0\} \\ M_{n+1,l} & \text{if } k = i_0 \\ M_{i_0,l} & \text{if } k = n+1 \end{cases} \quad (11.281)$$

Now $\forall k, l \in \{1, \dots, n\}$ we have

$$\begin{aligned} ([n+1 \boxplus n+1](\mathcal{M}(L_1; E, E)))_{k,l} &\stackrel{k, l \leq n+1}{=} \mathcal{M}(L_1; E, E)_{k,l} \\ &\stackrel{[\text{eq: 11.281}]}{=} \begin{cases} M_{k,l} & \text{if } k \in \{1, \dots, n\} \setminus \{i_0\} \\ M_{n+1,l} & \text{if } k = i_0 \\ M_{i_0,l} & \text{if } k = n+1 \end{cases} \\ &\stackrel{k, l < n+1}{=} \begin{cases} M_{k,l} & \text{if } k \in \{1, \dots, n\} \setminus \{i_0\} \\ M_{n+1,l} & \text{if } k = i_0 \end{cases} \\ &\stackrel{[\text{eq: 11.278}]}{=} M'_{k,l} \end{aligned}$$

so that $[n+1 \boxplus n+1](\mathcal{M}(L_1; E, E)) = M'$, hence by [eq: 11.279] that

$$\det([n+1 \boxplus n+1](\mathcal{M}(L_1; E, E))) \neq 0 \quad (11.282)$$

Next we have

$$\tau_{[n+1, i_0]}^{n+1} \circ L_1 \stackrel{[\text{eq: 11.280}]}{=} \tau_{[n+1, i_0]}^{n+1} \circ (\tau_{[n+1, i_0]}^{n+1} \circ L) \stackrel{[\text{theorem: 11.392}]}{=} \text{Id}_X \circ L = L$$

So if we take $T = \tau_{[n+1, i_0]}^{n+1} \in \text{Elem}(X)$ we have together with [eq: 11.282] that

$$\exists T \in \text{Elem}(X) \text{ such that } L = T \circ L_1 \text{ and } \det([n+1, n+1](\mathcal{M}(L_1; E, E))) \neq 0 \quad (11.283)$$

So in all cases we have by [eqs: 11.276, 11.283] that

$$\exists T \in \text{Elem}(X) \text{ such that } L = T \circ L_1 \text{ and } \det([n+1, n+1](\mathcal{M}(L_1; E, E))) \neq 0 \quad (11.284)$$

Let

$$N = \mathcal{M}(L_1; E, E) \quad (11.285)$$

Take now $Y = \text{span}(\{e_1, \dots, e_n\})$ so that $F = \{e_1, \dots, e_n\}$ is a basis for Y . Define now $L_2 \in \text{Hom}(Y, Y)$ by

$$L_2 = \mathcal{M}(F, F)^{-1}([n+1, n+1](N)) \quad (11.286)$$

so that $\det(L_2) \stackrel{[\text{theorem: 11.359}]}{=} \det([n+1, n+1](N)) = \det([n+1, n+1](\mathcal{M}(L_1; E, E))) \neq 0$ proving that $L_2 \in \text{GL}(Y)$. Hence as $n \in S$ we have that

$$L_2 \text{ is composed of elementary transformations in } Y \quad (11.287)$$

Using the above and [corollary: 11.401] it follows that

$$L_3 = L_2^{[n+1]} \text{ is composed of elementary transformations in } X \quad (11.288)$$

Define now

$$L_4 = \prod_{i=1}^n \beta_{[n+1, i, N_{i, n+1}]}^{n+1} \quad (11.289)$$

then as L_4 and L_3 are composed of linear transformations it follows from [lemma: 11.396] that

$$L_5 = L_4 \circ L_3 \text{ is composed of linear transformations} \quad (11.290)$$

If $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned} L_5(e_i) &= \left(\prod_{k=1}^n \beta_{[n+1, k, N_{k, n+1}]}^{n+1} \right) (L_3(e_i)) \\ &= \left(\prod_{k=1}^n \beta_{[n+1, k, N_{k, n+1}]}^{n+1} \right) (L_2^{[n+1]}(e_i)) \\ &\stackrel{[\text{definition: 11.397}]}{=} \left(\prod_{k=1}^n \beta_{[n+1, k, N_{k, n+1}]}^{n+1} \right) (L_2(e_i)) \\ &= \left(\prod_{k=1}^n \beta_{[n+1, k, N_{k, n+1}]}^{n+1} \right) \left(\sum_{j=\{1, \dots, n\}} ([n+1, n+1]N)_{j, i} \cdot e_j \right) \\ &\stackrel{[i, j \leq n+1 \wedge \text{definition: 11.369}]}{=} \left(\prod_{k=1}^n \beta_{[n+1, k, N_{k, n+1}]}^{n+1} \right) \left(\sum_{j=\{1, \dots, n\}} N_{j, i} \cdot e_j \right) \\ &= \sum_{j=\{1, \dots, n\}} N_{j, i} \cdot \left(\prod_{k=1}^n \beta_{[n+1, k, N_{k, n+1}]}^{n+1} \right) (e_j) \\ &\stackrel{[\text{theorem: 11.394}]}{=} \sum_{j=\{1, \dots, n\}} N_{j, i} \cdot e_j \end{aligned}$$

and

$$\begin{aligned}
 L_5(e_{n+1}) &= \left(\prod_{k=1}^n \beta_{[n+1,k,N_{k,n+1}]}^{n+1} \right) (L_3(e_i)) \\
 &= \left(\prod_{k=1}^n \beta_{[n+1,k,N_{k,n+1}]}^{n+1} \right) (L_2^{[n+1]}(e_{n+1})) \\
 &\stackrel{[\text{definition: 11.397}]}{=} \left(\prod_{k=1}^n \beta_{[n+1,k,N_{k,n+1}]}^{n+1} \right) (e_{n+1}) \\
 &\stackrel{[\text{theorem: 11.394}]}{=} \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + e_{n+1}
 \end{aligned}$$

proving that

$$\forall i \in \{1, \dots, n+1\} \quad L_5(e_i) = \begin{cases} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + e_{n+1} & \text{if } i = n+1 \end{cases} \quad (11.291)$$

Take $\alpha \in F$ [later we will choose a value for α] then as $\sigma_{[n+1,\alpha]}^{n+1} \in \text{Elem}(X)$ and L_5 is composed of elementary transformations [see eq: 11.290] we have by [lemma: 11.396] that

$$L_6 = \sigma_{[n+1,\alpha]}^{n+1} \circ L_5 \text{ is composed of elementary transformations in } X \quad (11.292)$$

For $i \in \{1, \dots, n\}$ we have

$$\begin{aligned}
 L_6(E_i) &= \sigma_{[n+1,\alpha]}^{n+1} (L_5(e_i)) \\
 &\stackrel{[\text{eq: 11.291}]}{=} \sigma_{[n+1,\alpha]}^{n+1} \left(\sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j \right) \\
 &= \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot \sigma_{[n+1,\alpha]}^{n+1} (e_j) \\
 &\stackrel{[\text{definition: 11.391}]}{=} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j
 \end{aligned}$$

and

$$\begin{aligned}
 L_6(e_{n+1}) &= \sigma_{[n+1,\alpha]}^{n+1} (L_5(e_{n+1})) \\
 &\stackrel{[\text{eq: 11.291}]}{=} \sigma_{[n+1,\alpha]}^{n+1} \left(\sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + e_{n+1} \right) \\
 &= \sigma_{[n+1,\alpha]}^{n+1} \left(\sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j \right) + \sigma_{[n+1,\alpha]}^{n+1} (e_{n+1}) \\
 &\stackrel{[\text{definition: 11.391}]}{=} \sigma_{[n+1,\alpha]}^{n+1} \left(\sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j \right) + \alpha \cdot e_{n+1} \\
 &= \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot \sigma_{[n+1,\alpha]}^{n+1} (e_j) + \alpha \cdot e_{n+1} \\
 &\stackrel{[\text{definition: 11.391}]}{=} \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + \alpha \cdot e_{n+1}
 \end{aligned}$$

proving that

$$\forall i \in \{1, \dots, n+1\} \text{ we have } L_6(e_i) = \begin{cases} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + \alpha \cdot e_{n+1} & \text{if } i = n+1 \end{cases} \quad (11.293)$$

Next given $B \in \mathcal{M}_{n+1,1}(F)$ [we will specify the content of B later] define

$$L_7 = \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) \circ L_6$$

Which as L_6 and $\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1}$ are composed of elementary transformations proves that

$$L_7 \text{ is composed of linear transformations.} \quad (11.294)$$

Further if $i \in \{1, \dots, n\}$ we have

$$\begin{aligned} L_7(e_i) &= \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) (L_6(e_i)) \\ &\stackrel{[\text{eq: } 11.293]}{=} \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) \left(\sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j \right) \\ &= \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) (e_j) \\ &\stackrel{[\text{theorem: } 11.395]}{=} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot (e_j + B_j \cdot e_{n+1}) \\ &= \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j + \sum_{j \in \{1, \dots, n\}} (N_{j,i} \cdot B_j) \cdot e_{n+1} \\ &= \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j + \left(\sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot B_j \right) \cdot e_{n+1} \end{aligned}$$

and

$$\begin{aligned} L_7(e_{n+1}) &\stackrel{[\text{eq: } 11.293]}{=} \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) (L_6(e_{n+1})) \\ &= \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) \left(\sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + \alpha \cdot e_{n+1} \right) \\ &= \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) \left(\sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j \right) + \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) (\alpha \cdot e_{n+1}) \\ &= \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) (e_j) + \alpha \cdot \left(\prod_{i \in \{1, \dots, n\}} \beta_{[i, n+1, B_i]}^{n+1} \right) (e_{n+1}) \\ &\stackrel{[\text{theorem: } 11.395]}{=} \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot (e_j + B_j \cdot e_{n+1}) + \alpha \cdot e_{n+1} \\ &= \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + \left(\sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot B_j \right) \cdot e_{n+1} + \alpha \cdot e_{n+1} \\ &= \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + \left(\alpha + \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot B_j \right) \cdot e_{n+1} \end{aligned}$$

proving that $\forall i \in \{1, \dots, n\}$

$$L_7(e_i) = \begin{cases} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j + (\sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot B_j) \cdot e_{n+1} & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + (\alpha + \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot B_j) \cdot e_{n+1} & \text{if } i = n+1 \end{cases} \quad (11.295)$$

As

$$\begin{aligned} \det([n+1 \boxplus n+1](N))^T & \stackrel{[\text{theorem: 11.358}]}{=} \det([n+1 \boxplus n+1](N)) \\ & \stackrel{[\text{eq: 11.285}]}{=} \det([n+1 \boxplus n+1](\mathcal{M}(L_1; E, E))) \\ & \stackrel{[\text{eq: 11.284}]}{\neq} 0 \end{aligned}$$

$([n+1 \boxplus n+1](N))^T$ has a inverse, so we can define

$$K = ([n+1 \boxplus n+1](N))^T{}^{-1}$$

Then we have $\forall i, j \in \{1, \dots, n\}$ we have

$$\begin{aligned} \delta_{i,j} &= ((([n+1 \boxplus n+1](N))^T \cdot K)_{i,j}) \\ &= \sum_{k \in \{1, \dots, n\}} ((([n+1 \boxplus n+1](N))^T)_{i,k} \cdot K_{k,j}) \\ &= \sum_{k \in \{1, \dots, n\}} ([n+1 \boxplus n+1](N))_{k,i} \cdot K_{k,j} \\ &\stackrel{[\text{theorem: 11.369}]}{=} \sum_{k \in \{1, \dots, n\}} N_{k,i} \cdot K_{k,j} \end{aligned} \quad (11.296)$$

Now we choose the value of $B \in \mathcal{M}_{n+1,1}(F)$ to be

$$\forall i \in \{1, \dots, n+1\} \quad B_i = \begin{cases} \sum_{k \in \{1, \dots, n\}} K_{i,k} \cdot N_{n+1,k} & \text{if } i \in \{1, \dots, n\} \\ 1 & \text{if } i = n+1 \end{cases}$$

then we have $\forall i \in \{1, \dots, n\}$ that

$$\begin{aligned} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot B_j &= \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot \left(\sum_{k \in \{1, \dots, n\}} K_{j,k} \cdot N_{n+1,k} \right) \\ &= \sum_{j \in \{1, \dots, n\}} \left(\sum_{k \in \{1, \dots, n\}} N_{j,i} \cdot K_{j,k} \cdot N_{n+1,k} \right) \\ &\stackrel{[\text{theorem: 11.45}]}{=} \sum_{k \in \{1, \dots, n\}} \left(\sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot K_{j,k} \cdot N_{n+1,k} \right) \\ &= \sum_{k \in \{1, \dots, n\}} \left(\sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot K_{j,k} \right) \cdot N_{n+1,k} \\ &\stackrel{[\text{eq: 11.296}]}{=} \sum_{k \in \{1, \dots, n\}} \delta_{i,k} \cdot N_{n+1,k} \\ &= N_{n+1,i} \end{aligned}$$

Substituting the above result in [eq: 11.295] results in $\forall i \in \{1, \dots, n+1\}$

$$L_7(e_i) = \begin{cases} \sum_{j \in \{1, \dots, n\}} N_{j,i} \cdot e_j + N_{n+1,i} \cdot e_{n+1} & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + (\alpha + \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot B_j) \cdot e_{n+1} & \text{if } i = n+1 \end{cases}$$

or

$$L_7(e_i) = \begin{cases} \sum_{j \in \{1, \dots, n+1\}} N_{j,i} \cdot e_j & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + (\alpha + \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot B_j) \cdot e_{n+1} & \text{if } i = n+1 \end{cases}$$

Next choose α to be $N_{n+1,n+1} - \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot B_j$ so that after substituting this in the above gives

$$\begin{aligned}
 L_7(e_i) &= \begin{cases} \sum_{j \in \{1, \dots, n+1\}} N_{j,i} \cdot e_j & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n\}} N_{j,n+1} \cdot e_j + N_{n+1,n+1} \cdot e_{n+1} & \text{if } i = n+1 \end{cases} \\
 &= \begin{cases} \sum_{j \in \{1, \dots, n+1\}} N_{j,i} \cdot e_j & \text{if } i \in \{1, \dots, n\} \\ \sum_{j \in \{1, \dots, n+1\}} N_{j,n+1} \cdot e_j & \text{if } i = n+1 \end{cases} \\
 &= \sum_{j \in \{1, \dots, n+1\}} N_{j,i} \cdot e_j \\
 &\stackrel{[\text{eq: 11.285}]}{=} \sum_{j \in \{1, \dots, n+1\}} \mathcal{M}(L_1; E, E)_{j,i} \cdot e_j \\
 &= L_1(e_i)
 \end{aligned}$$

So that $\forall i \in \{1, \dots, n+1\}$ we have $L_7(e_i) = L_1(e_i)$ proving by [theorem: 11.336] that $L_7 = L_1$, hence L_1 is composed of elementary transformations [see eq: 11.294]. As $L \stackrel{[\text{eq: 11.284}]}{=} T \circ L_1$ and $T \in \text{Elem}(X)$ we have by [lemma: 11.396] that L is composed of elementary transformations. Proving

$$n+1 \in S$$

Mathematical induction proves then that $\mathbb{N} = S = \{n \in \mathbb{N} \mid \text{If } L \in \text{GL}(X) \text{ where } X \text{ is a } n\text{-dimensional vector space over } F \text{ then } L \text{ is composed of elementary transformations in } X\}$ proving the theorem. $\square$

In the proof of the above theorem it is not guaranteed that all the elementary transformations in the composition are non singular, the following theorem prove that we may assume that they are all nonsingular.

Theorem 11.403. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over a field F with characteristic zero, $L \in \text{GL}(X)$ a nonsingular transformation then L is composed of non singular elementary transformations. In other words there exists a $m \in \mathbb{N}$ and $\{T_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Elem}(X)$ such that $\forall i \in \{1, \dots, m\} \det(T_i) \neq 0$ [or equivalent T_i is nonsingular]*

$$L = \prod_{i=1}^m T_i \stackrel{\text{def}}{=} T_1 \circ \dots \circ T_m$$

Proof. Let $L \in \text{GL}(X)$, using the previous lemma [lemma: 11.402] there exist a $\{T_i\}_{i \in \{1, \dots, m\}} \subseteq \text{Elem}(X)$ such that

$$L = \prod_{i=1}^m T_i$$

Assume that $\exists k \in \{1, \dots, m\}$ such that $\det(L_k) = 0$ then we have

$$\det(L) = \det\left(\prod_{i=1}^m T_i\right) \stackrel{[\text{theorem: 11.316}]}{=} \prod_{i \in \{1, \dots, m\}} \det(L_i) = \left(\prod_{i \in \{1, \dots, m\} \setminus \{k\}} \det(L_i)\right) \cdot \det(L_k) = 0$$

contradicting $L \in \text{GL}(X) \Rightarrow \det(L) \neq 0$. Hence $\forall i \in \{1, \dots, m\}$ we have $\det(L_i) \neq 0$. $\square$

Chapter 12

Internal Direct Sum

We will now extend the sum of two subsets of a vector space [see definition: 11.208] to a sum of a finite family of subsets of a vector space.

Definition 12.1. Let V be a vector space over a field F , I a finite set and $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ [a finite family of subsets of V] then the sum of $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ noted as $\sum_{i \in I} V_i$ is defined as

$$\sum_{i \in I} V_i = \left\{ v \in V \mid \exists \{v_i\}_{i \in I} \subseteq V \text{ such that } \forall i \in I, v_i \in V_i \text{ and } v = \sum_{i \in I} v_i \right\} \subseteq V$$

Example 12.2. Let V be a vector space over a field F , $\{V_i\}_{i \in \emptyset} \subseteq \mathcal{P}(V)$ a empty family of subspaces [see example: 2.113]

Proof. Use [example: 2.113] to get a $\{v_i\}_{i \in \emptyset} \subseteq V$ then we have $0 \stackrel{[\text{definition: 11.32}]}{=} \sum_{i \in \emptyset} v_i$ and vacuously $\forall i \in \emptyset, v_i \in V_i$, proving that $0 \in \sum_{i \in \emptyset} V_i$. Further if $v \in \sum_{i \in \emptyset} V_i$ then there exist a $\{v_i\}_{i \in \emptyset} \subseteq V$ such that $v = \sum_{i \in \emptyset} v_i \stackrel{[\text{definition: 11.32}]}{=} 0$ proving that $\sum_{i \in \emptyset} V_i \subseteq \{0\}$, Hence we have

$$\sum_{i \in \emptyset} V_i = \{0\}$$

□

Some of the properties of finite sums of elements of the vector space transfer to the finite sum of a family of non empty subsets.

Theorem 12.3. Let $\langle V, +, \cdot \rangle$ be a vector space over a field F then we have for a finite set I and a finite family $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ of sub-spaces of V that

1. If $I = \{k\}$ then we have $\sum_{i \in \{k\}} V_i = V_k$
2. If $\sigma: J \rightarrow I$ is a bijection then we have $\sum_{i \in J} V_{\sigma(i)} = \sum_{i \in I} V_i$
3. If $k \in I$ then $\sum_{i \in I} V_i = (\sum_{i \in I \setminus \{k\}} V_i) + V_k$
4. If $I = I_1 \cup I_2$ with $I_1 \cap I_2 = \emptyset$ then $\sum_{i \in I} V_i = \sum_{i \in \{1,2\}} (\sum_{j \in I_i} V_j)$
5. If $I = \bigcup_{i \in \{1, \dots, m\}} I_i$ with $\forall i, j \in I$ if $i \neq j$ then $I_i \cap I_j = \emptyset$ then

$$\sum_{i \in I} V_i = \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in I_i} V_j \right)$$

Proof.

1. Let $v \in \sum_{i \in \{k\}} V_i$ then $\exists \{v_i\}_{i \in \{k\}}$ such that $\forall i \in \{k\}$ we have $v_i \in V_i$ and

$$v = \sum_{i \in \{k\}} v_i \stackrel{[\text{theorem: 11.34}]}{=} v_k \in V_k$$

proving that

$$\sum_{i \in \{k\}} V_i \subseteq V_k \quad (12.1)$$

Let $v \in V_k$ and define $\{v_i\}_{i \in \{k\}} \subseteq V$ by $v_k = v$ then $\forall i \in \{k\}$ we have $v_i = v \in V_k$ and

$$v = v_k \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{k\}} v_i$$

proving that $v \in \sum_{i \in \{k\}} v_i$. Hence $V_k \subseteq \sum_{i \in \{k\}} V_i$ which combined with [eq: 12.1]

$$\sum_{i \in \{k\}} V_i = V_k$$

2. Let $v \in \sum_{i \in I} V_i$ then there exist a $\{v_i\}_{i \in I} \subseteq V$ with $\forall i \in I$ $v_i \in V_i$ and $v = \sum_{i \in I} v_i$. Define now

$$\{w_i\}_{i \in J} \subseteq V \text{ by } w_i = v_{\sigma(i)}$$

then $\forall i \in J$ we have $w_i = v_{\sigma(i)} \in V_{\sigma(i)}$ and

$$\sum_{i \in J} w_i = \sum_{i \in J} v_{\sigma(i)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in I} v_i = v$$

hence $v \in \sum_{i \in J} V_{\beta(i)}$ proving that

$$\sum_{i \in I} V_i \subseteq \sum_{i \in J} V_{\beta(i)} \quad (12.2)$$

If $v \in \sum_{i \in J} V_{\beta(i)}$ then there exist a $\{w_i\}_{i \in J} \subseteq V$ such that $\forall i \in J$ we have $w_i \in V_{\beta(i)}$ and $v = \sum_{i \in J} w_i$. Define now

$$\{v_i\}_{i \in I} \subseteq V \text{ by } v_i = w_{\sigma^{-1}(i)}$$

then $\forall i \in I$ we have $v_i = w_{\sigma^{-1}(i)} \in V_{\sigma^{-1}(i)} = V_i$ and

$$\sum_{i \in I} v_i = \sum_{i \in I} w_{\sigma^{-1}(i)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in J} w_i = v$$

proving that $v \in \sum_{i \in I} V_i$. Hence $\sum_{i \in J} V_{\beta(i)} \subseteq \sum_{i \in J} V_{\beta(i)}$ which combined with [eq: 12.2]

$$\sum_{i \in I} V_i = \sum_{i \in J} V_{\beta(i)}$$

3. If $v \in \sum_{i \in I} V_i$ then $\exists \{v_i\}_{i \in I}$ such that $\forall i \in I$ $v_i \in V_i$ and $v = \sum_{i \in I} v_i$. Let $w = \sum_{i \in I \setminus \{k\}} v_i$ then as trivially $\forall i \in I \setminus \{k\}$ $v_i \in V_i$ we have that

$$w \in \sum_{i \in I \setminus \{k\}} V_i$$

and

$$v = \sum_{i \in I} v_i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus \{k\}} v_i + \sum_{i \in \{k\}} v_i \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in I \setminus \{k\}} v_i + v_k = w + v_k$$

so, as $v_k \in V_k$, it follows that $v \in (\sum_{i \in I \setminus \{k\}} V_i) + V_k$. Hence

$$\sum_{i \in I} V_i \subseteq \left(\sum_{i \in I \setminus \{k\}} V_i \right) + V_k \quad (12.3)$$

On the other hand if $v \in (\sum_{i \in I \setminus \{k\}} V_i) + V_k$ then there exists a $u \in \sum_{i \in I \setminus \{k\}} V_i$ and $w \in V_k$ such that $v = u + w$. As $u \in \sum_{i \in I \setminus \{k\}} V_i$ there exist a $\{v_i\}_{i \in I \setminus \{k\}} \subseteq V$ such that $\forall i \in I \setminus \{k\}$ $v_i \in V_i$ and $u = \sum_{i \in I \setminus \{k\}} v_i$. Define now

$$\{w_i\}_{i \in I} \subseteq V \text{ by } w_i = \begin{cases} w & \text{if } i = k \\ v_i & \text{if } i \in I \setminus \{k\} \end{cases}$$

then we have

$$\sum_{i \in I} w_i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus \{k\}} w_i + \sum_{i \in k} w_i \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in I \setminus k} w_i + w_k = \sum_{i \in I \setminus k} v_i + w = u + w$$

which, as $\forall i \in I$ $w_i = \begin{cases} w \in V_k & \text{if } i = k \\ v_i \in V_i & \text{if } i \in I \setminus \{k\} \end{cases} \in V_i$, results in $v \in \sum_{i \in I} V_i$ proving that

$$\left(\sum_{i \in I \setminus \{k\}} V_i \right) + V_k \subseteq \sum_{i \in I} V_i$$

which combined with [eq: 12.3] proves that

$$\sum_{i \in I} V_i = \left(\sum_{i \in I \setminus \{k\}} V_i \right) + V_k$$

4. If $v \in \sum_{i \in I} V_i$ then there exist a $\{v_i\}_{i \in I} \subseteq V$ such that $\forall i \in I$ we have $v_i \in V_i$ and $v = \sum_{i \in I} v_i$. Define

$$\{w_i\}_{i \in \{1,2\}} \subseteq V \text{ by } w_i = \sum_{j \in I_i} v_j$$

If $i \in \{1,2\}$ then $\forall j \in I_i \subseteq I$ $v_j \in V_j$ so we have

$$\forall i \in \{1,2\} \text{ we have that } w_i \in \sum_{j \in I_i} V_j$$

further we have

$$\begin{aligned} \sum_{i \in \{1,2\}} w_i &= \sum_{i \in \{1,2\}} \left(\sum_{j \in I_i} v_j \right) \\ &\stackrel{[\text{theorem: 11.44}]}{=} \sum_{i \in I_1 \cup I_2} v_i \\ &= v \end{aligned}$$

proving that $v \in \sum_{i \in \{1,2\}} (\sum_{j \in I_i} v_j)$. Hence

$$\sum_{i \in I} V_i \subseteq \sum_{i \in \{1,2\}} \left(\sum_{j \in I_i} v_j \right) \quad (12.4)$$

If $v \in \sum_{i \in \{1,2\}} (\sum_{j \in I_i} V_j)$ then there exists a $\{v_i\}_{i \in \{1,2\}} \subseteq V$ with $\forall i \in \{1,2\}$ $v_i \in \sum_{j \in I_i} V_j$ such that $v = \sum_{i \in \{1,2\}} v_i$. Let $i \in \{1,2\}$ then there exist a $\{v_{i,j}\}_{j \in I_i} \subseteq V$ with $\forall j \in I_i$ $v_{i,j} \in V_j$ and $v_i = \sum_{j \in I_i} v_{i,j}$. We can then, as $I_1 \cap I_2 \neq \emptyset$, define

$$\{w_i\}_{i \in I} \subseteq V \text{ by } w_i = \begin{cases} v_{1,i} & \text{if } i \in I_1 \\ v_{2,i} & \text{if } i \in I_2 \end{cases}$$

then we have $\forall i \in I = I_1 \cup I_2$ that

$$w_i = \begin{cases} v_{1,i} \in V_i & \text{if } i \in I_1 \\ v_{2,i} \in V_i & \text{if } i \in I_2 \end{cases} \in V_i$$

and

$$\begin{aligned} \sum_{i \in I} w_i &\stackrel{[\text{theorem: 11.44}]}{=} \sum_{i \in I_1} w_i + \sum_{i \in I_2} w_i \\ &= \sum_{i \in I_1} v_{1,i} + \sum_{i \in I_2} v_{2,i} \\ &= v_1 + v_2 \\ &= \sum_{i \in \{1,2\}} v_i \\ &= v \end{aligned}$$

proving that $v \in \sum_{i \in I} V_i$. So $\sum_{i \in \{1,2\}} (\sum_{j \in I_i} v_j) \subseteq \sum_{i \in I} V_i$ which combined with [eq: 12.4] gives

$$\sum_{i \in I} V_i = \sum_{i \in \{1,2\}} \left(\sum_{j \in I_i} v_j \right)$$

5. This is proved by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } I = \bigcup_{i \in \{1, \dots, m\}} I_i \text{ with } \forall i, j \in I \text{ if } i \neq j \text{ then } I_i \cap I_j = \emptyset, \text{ then } \sum_{i \in I} V_i = \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in I_i} V_j \right) \right\}$$

then we have:

1 $\in S$. If $I = \bigcup_{i \in \{1\}} I_i$ [so we have automatically $\forall i, j \in I$ if $i \neq j$ then $I_i \cap I_j = \emptyset$] then

$$\sum_{i \in \{1\}} \left(\sum_{j \in I_i} V_j \right) \stackrel{(\overline{1})}{=} \sum_{j \in I_1} V_j \stackrel{I=I_1}{=} \sum_{j \in I} V_j$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $I = \bigcup_{i \in \{1, \dots, m+1\}} I_i$ with $\forall i, j \in I$ if $i \neq j$ then $I_i \cap I_j = \emptyset$. Let $J_1 = \bigcup_{i \in \{1, \dots, m\}} I_i$ and $J_2 = I_{m+1}$ then

$$J_1 \cap J_2 = \left(\bigcup_{i \in \{1, \dots, m\}} I_i \right) \cap J_2 = \bigcup_{i \in \{1, \dots, m\}} (I_i \cap I_{m+1}) = \emptyset$$

and $I = J_1 \cup J_2$. Hence we have

$$\begin{aligned} \sum_{i \in I} V_i &\stackrel{(\overline{4})}{=} \sum_{i \in \{1,2\}} \left(\sum_{j \in J_i} V_j \right) \\ &\stackrel{(\overline{3})}{=} \left(\sum_{i \in \{1\}} \left(\sum_{j \in J_i} V_j \right) \right) + \sum_{j \in J_2} V_j \\ &\stackrel{(\overline{1})}{=} \sum_{j \in J_1} V_j + \sum_{j \in J_2} V_j \\ &= \sum_{j \in J_1} V_j + \sum_{j \in I_{m+1}} V_j \\ &= \left(\sum_{j \in \bigcup_{i \in \{1, \dots, m\}} I_i} V_j \right) + \sum_{j \in I_{m+1}} V_j \\ &\stackrel{n \in S}{=} \sum_{i \in \{1, \dots, m\}} \left(\sum_{j \in I_i} V_j \right) + \sum_{j \in I_{m+1}} V_j \\ &\stackrel{(\overline{3})}{=} \sum_{i \in \{1, \dots, m+1\}} \left(\sum_{j \in I_i} V_j \right) \end{aligned}$$

proving that $m+1 \in S$. □

We consider now the case where we have a family of sub-spaces of a vector space.

Theorem 12.4. *If V is a vector space over a field F and $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ such that $\forall i \in I$ V_i is a sub space of V then*

1. $\sum_{i \in I} V_i$ is a sub space of V
2. If $J \subseteq I$ then $\sum_{i \in J} V_i \subseteq \sum_{i \in I} V_i$

3. If $k \in I$ then $V_k \subseteq \sum_{i \in I} V_i$

Proof.

1. As $\forall i \in I$ V_i is a sub space of V we have that $0 \in V_i$, so as $\sum_{i \in I} 0 = 0$ it follows that $0 \in \sum_{i \in I} V_i$ proving that

$$\emptyset \neq \sum_{i \in I} V_i$$

Let $\alpha \in F$ and $x, y \in \sum_{i \in I} V_i$ then there exists a $\{x_i\}_{i \in I} \subseteq V$ and $\{y_i\}_{i \in I} \subseteq V$ such that $\forall i \in I$ $x_i, y_i \in V_i$ and $x = \sum_{i \in I} x_i$, $y = \sum_{i \in I} y_i$. As $\forall i \in I$ V_i is a subspace we have that $\alpha \cdot x_i + y_i \in V_i$ and

$$\alpha \cdot x + y = \alpha \cdot \sum_{i \in I} x_i + \sum_{i \in I} y_i \stackrel{[\text{theorems: 11.80, 11.17}]}{=} \sum_{i \in I} (\alpha \cdot x_i + y_i)$$

proving that $\alpha \cdot x + y \in \sum_{i \in I} V_i$.

2. If $v \in \sum_{i \in J} V_i$ then there exist a $\{v_i\}_{i \in J} \subseteq V$ such that $\forall i \in J$ $v_i \in V_i$ and $v = \sum_{i \in J} v_i$. Define then

$$\{w_i\}_{i \in I} \subseteq V \text{ by } w_i = \begin{cases} 0 & \text{if } i \in I \setminus J \\ v_i & \text{if } i \in J \end{cases}$$

then $\forall i \in I$ we have $w_i \in V_i$ [as $0 \in V_i$] and

$$\begin{aligned} \sum_{i \in I} w_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus J} w_i + \sum_{i \in J} w_i \\ &= \sum_{i \in I \setminus J} 0 + \sum_{i \in J} v_i \\ &= v \end{aligned}$$

Hence $v \in \sum_{i \in I} V_i$ proving

$$\sum_{i \in J} V_i \subseteq \sum_{i \in I} V_i$$

3. Let $k \in I$ and $v \in V_k$, define

$$\{v_i\}_{i \in I} \text{ by } v_i = \begin{cases} v & \text{if } i = k \\ 0 & \text{if } i \in I \setminus \{k\} \end{cases}$$

then, as V_i 's are sub-spaces of V hence containing 0 we have

$$\forall i \in I \setminus \{k\} \text{ that } v_i = \begin{cases} v \in V_k & \text{if } i = k \\ 0 \in V_i & \text{if } i \in I \setminus \{k\} \end{cases} \in V_i$$

Further

$$\sum_{i \in I} v_i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus \{k\}} v_i + \sum_{i \in \{k\}} v_i \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in I \setminus \{k\}} v_i + v_k = \sum_{i \in I \setminus \{k\}} 0 + v = v$$

proving that $v \in \sum_{i \in I} V_i$. Hence

$$V_k \subseteq \sum_{i \in I} V_i$$

□

If $v \in \sum_{i \in I} V_i$ then we can write v as $\sum_{i \in I} v_i$ where $v_i \in V_i$, however this expansion is not unique, which is not very useful (compare this with the uniqueness of the expansion of a vector in a basis). To solve this we extend the concept of a direct sum of two sub-spaces [see definition: 11.209] to a direct sum of a finite family.

Definition 12.5. (Internal Direct Sum) If V is a vector space over a field then

$$V = \bigoplus_{i \in I} V_i$$

iff

1. I is a finite set
2. $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ satisfies $\forall i \in I$ V_i is a sub space of V
3. If $v \in V$ then there exist a **unique** family $\{v_i\}_{i \in I} \subseteq V$ such that $\forall i \in I$ $v_i \in V_i$ and $v = \sum_{i \in I} v_i$.

Example 12.6. Let V be a vector space over a field F , $\{V_i\}_{i \in \emptyset} \subseteq \mathcal{P}(V)$ a empty family of sub-spaces of V [see example: 2.113] then

$$\bigoplus_{i \in \emptyset} V_i = \{0\}$$

Proof. If $v \in \bigoplus_{i \in \emptyset} V_i$ then there exist a $\{v_i\}_{i \in \emptyset} \subseteq V$ such that $v = \sum_{i \in \emptyset} v_i$ [definition: 11.32] $= 0$ proving that

$$\bigoplus_{i \in \emptyset} V_i \subseteq \{0\},$$

Use [example: 2.113] to get a $\{v_i\}_{i \in \emptyset} \subseteq V$ then we have 0 [definition: 11.32] $= \sum_{i \in \emptyset} v_i$ and vacuously $\forall i \in \emptyset$ $v_i \in V_i$. If $\{w_i\}_{i \in \emptyset} \subseteq V$ is such that $0 = \sum_{i \in \emptyset} w_i$ then we have vacuously $v_i = w_i$ hence $\{v_i\}_{i \in \emptyset} = \{w_i\}_{i \in \emptyset}$ proving uniqueness. Hence $0 \in \bigoplus_{i \in \emptyset} V_i$ so that

$$\bigoplus_{i \in \emptyset} V_i = \{0\}$$

□

Example 12.7. If V is a vector space and W a sub-space then $W = \bigoplus_{i \in \{k\}} V_i$ where $V_k = W$.

Proof. $V_1 = W$ is a subspace of V and if $v \in V$ we have for $\{v_i\}_{i \in \{1\}} \subseteq V$ that $v = \sum_{i \in \{1\}} v_i$ and if $\{w_i\}_{i \in \{1\}}$ such that $v = \sum_{i \in \{1\}} w_i$ then $w_i = v = v_i \forall i \in \{1\}$. □

We can rephrase the above definition in another way as follows

Theorem 12.8. If V is a vector space over a field, I a finite set, $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ a finite family of sub-spaces of V then

$$\begin{aligned} V &= \bigoplus_{i \in I} V_i \\ &\Downarrow \\ V &= \sum_{i \in I} V_i \text{ and } \forall k \in I \text{ we have } V_k \cap \left(\sum_{i \in I \setminus \{k\}} V_i \right) = \{0\} \end{aligned}$$

Proof.

$\Rightarrow$. Let $v \in V$ then, as $V = \bigoplus_{i \in I} V_i$, there exist a **unique** $\{v_i\}_{i \in I} \subseteq V$ such that $\forall i \in I$ we have $v_i \in V_i$ and $v = \sum_{i \in I} v_i$, proving $V \subseteq \sum_{i \in I} V_i$. As by [definition: 12.1] $\sum_{i \in I} V_i \subseteq V$ we have

$$V = \sum_{i \in I} V_i$$

Let $k \in I$. If $v \in V_k \cap \left(\sum_{i \in I \setminus \{k\}} V_i \right)$ then $v \in V_k$ and $v \in \sum_{i \in I \setminus \{k\}} V_i$. So there exists a $\{v_i\}_{i \in I \setminus \{k\}} \subseteq V$ such that $\forall i \in I \setminus \{k\}$ $v_i \in V_i$ and $v = \sum_{i \in I \setminus \{k\}} v_i$. Define

$$\{w_i\}_{i \in I} \subseteq V \text{ by } w_i = \begin{cases} -v & \text{if } i = k \\ v_i & \text{if } i \in I \setminus \{k\} \end{cases} \in V_i$$

then

$$\forall i \in I \text{ we have } w_i \in \begin{cases} -v \in V_k & \text{if } i = k \\ v_i \in V_i & \text{if } i \in I \setminus \{k\} \end{cases} \in V_i$$

and

$$\begin{aligned}
 \sum_{i \in I} w_i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus \{k\}} w_i + \sum_{i \in \{k\}} w_i \\
 & \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in I \setminus \{k\}} w_i + w_k \\
 & = \sum_{i \in I \setminus \{k\}} v_i + (-v) \\
 & = v + (-v) \\
 & = 0
 \end{aligned}$$

Further for $\{u_i\}_{i \in I} \subseteq V$ defined by $u_i = 0$ we have $\forall i \in I \ u_i = 0 \in V_i$ and $0 \stackrel{[\text{theorem: 11.37}]}{=} \sum_{i \in I} u_i$, so, as $0 \in V$ and $V = \bigoplus_{i \in I} V_i$, we have by **uniqueness** that $\forall i \in I \ u_i = w_i$, hence $-v = w_k = u_k = 0$ proving that $v = 0$. As $v \in V_k \cap (\sum_{i \in I \setminus \{k\}} V_i)$ was chosen arbitrary we conclude that

$$V_k \cap \left(\sum_{i \in I \setminus \{k\}} V_i \right) \subseteq \{0\}$$

By [theorem: 12.4] $\sum_{i \in I \setminus \{k\}} V_i$ is a sub-space of V and by the hypothesis V_k is a subspace of V so we have $0 \in V_k \cap (\sum_{i \in I \setminus \{k\}} V_i)$ and it follows that

$$V_k \cap \left(\sum_{i \in I \setminus \{k\}} V_i \right) = \{0\}$$

$\Leftarrow$. Let $v \in V$ then as $V = \sum_{i \in I} V_i$ there exists a $\{v_i\}_{i \in I} \subseteq V$ with $\forall i \in I \ v_i \in V_i$ and $v = \sum_{i \in I} v_i$. Let $\{w_i\}_{i \in I} \subseteq V$ such that $\forall i \in I \ w_i \in V_i$ and $v = \sum_{i \in I} v_i$. Assume that $\{v_i\}_{i \in I} \neq \{w_i\}_{i \in I}$ then there exist a $k \in I$ such that $v_k \neq w_k$ and we have

$$\begin{aligned}
 0 & = v + (-v) \\
 & = \sum_{i \in I} v_i + \left(-\sum_{i \in I} w_i \right) \\
 & \stackrel{[\text{theorems: 11.38, 11.40}]}{=} \sum_{i \in I} (v_i - w_i) \\
 & = \sum_{i \in I \setminus \{k\}} (v_i - w_i) + \sum_{i \in \{k\}} (v_i - w_i) \\
 & = \sum_{i \in I \setminus \{k\}} (v_i - w_i) + (v_k - w_k)
 \end{aligned}$$

so that

$$w_k - v_k = \sum_{i \in I \setminus \{k\}} (v_i - w_i)$$

Now $w_k - v_k \in V_k$ and $\forall i \in I \setminus \{k\} \ v_i - w_i \in V_i$ so that $w_k - v_k \in V_k \cap \sum_{i \in I \setminus \{k\}} V_i$, hence by the hypothesis we have that $w_k - v_k = 0$ contradicting $v_k \neq w_k$. So we must have that $\{v_i\}_{i \in I} = \{w_i\}_{i \in I}$. Proving that

$$V = \bigoplus_{i \in I} V_i \quad \square$$

Theorem 12.9. Let V be a vector space over a field F , I_1, I_2 finite sets such that $I_1 \cap I_2 = \emptyset$, U, W sub-spaces of V , $\{V_i\}_{i \in I_1 \cup I_2} \subseteq \mathcal{P}(V)$ a finite family of sub-spaces of V such that

$$U = \bigoplus_{i \in I_1} V_i, \ W = \bigoplus_{i \in I_2} V_i \text{ and } V = U \oplus W$$

then

$$V = \bigoplus_{i \in I_1 \cup I_2} V_i$$

Proof. As $U = \bigoplus_{i \in I_1} V_i$, $W = \bigoplus_{i \in I_2} V_i$ we have by definition that

$$\forall i \in I_1 \ V_i \text{ is a sub-space of } U, V \text{ and } \forall i \in I_2 \ V_i \text{ is a sub-space of } W, W \quad (12.5)$$

Let $v \in V$ then there exists **unique** $u \in U$ and a $w \in W$ such that

$$v = u + w \quad (12.6)$$

As $U = \bigoplus_{i \in I_1} U_i$ and $W = \bigoplus_{i \in I_2} W_i$ there exists **unique** $\{u_i\}_{i \in I_1} \subseteq U \subseteq V$ and $\{w_i\}_{i \in I_2} \subseteq W \subseteq V$ such that

$$\forall i \in I_1 \ u_i \in V_i \wedge \forall i \in I_2 \ w_i \in V_i \wedge u = \sum_{i \in I_1} u_i \wedge w = \sum_{i \in I_2} w_i \quad (12.7)$$

As $I_1 \cap I_2 = \emptyset$ we can define

$$\{v_i\}_{i \in I_1 \cup I_2} \subseteq V \text{ by } v_i = \begin{cases} u_i & \text{if } i \in I_1 \\ w_i & \text{if } i \in I_2 \end{cases} \in V_i$$

then

$$\begin{aligned} \sum_{i \in I_1 \cup I_2} v_i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I_1} v_i + \sum_{i \in I_2} v_i \\ & = \sum_{i \in I_1} u_i + \sum_{i \in I_2} w_i \\ & = u + w \\ & \stackrel{[\text{eq: 12.6}]}{=} v \end{aligned}$$

proving that

$$v = \sum_{i \in I_1 \cup I_2} v_i \text{ and } \forall i \in I_1 \cup I_2 \ v_i = \begin{cases} u_i \in V_i & \text{if } i \in I_1 \\ w_i \in V_i & \text{if } i \in I_2 \end{cases} \in V_i$$

Let $\{z_i\}_{i \in I_1 \cup I_2} \subseteq V$ such that $\forall i \in I_1 \cup I_2 \ z_i \in V_i$ and $v = \sum_{i \in I_1 \cup I_2} z_i$. Define then u', w' by

$$u' = \sum_{i \in I_1} z_i \text{ and } w' = \sum_{i \in I_2} z_i \quad (12.8)$$

then we have that

$$u' + w' = \sum_{i \in I_1} z_i + \sum_{i \in I_2} z_i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I_1 \cup I_2} z_i = v$$

As $\forall i \in I_1 \ V_i$ is a sub-space of U , $\forall i \in I_2 \ V_i$ is a sub-space of W [see: 12.5] we have by [theorem: 12.4] that $u' \in U$ and $w' \in W$ so

$$u' + w' = v \stackrel{[\text{eq: 12.6}]}{=} u + w \wedge u, u' \in U \wedge w, w' \in W$$

Hence as $V = U \oplus W$ we have by the uniqueness of the decomposition that

$$u = u' \wedge w = w' \quad (12.9)$$

Combining [eqs: 12.7, 12.8, 12.9] we have

$$u = \sum_{i \in I_1} u_i = u' = \sum_{i \in I_1} z_i \wedge w = \sum_{i \in I_2} w_i = w' = \sum_{i \in I_2} z_i$$

Now as $U = \bigoplus_{i \in I_1} V_i$, $W = \bigoplus_{i \in I_2} V_i$ we have by the above and uniqueness that

$$\forall i \in I_1 \ z_i = u_i \text{ and } \forall i \in I_2 \ z_i = w_i$$

hence we have $\forall i \in I_1 \cup I_2$ that

$$v_i = \begin{cases} u_i & \text{if } i \in I_1 \\ w_i & \text{if } i \in I_2 \end{cases} = \begin{cases} z_i & \text{if } i \in I_1 \\ z_i & \text{if } i \in I_2 \end{cases} = z_i$$

proving that the decomposition of $v \in V$ is unique. Hence

$$V = \bigoplus_{i \in I_1 \cup I_2} V_i \quad \square$$

Corollary 12.10. *Let V be a vector space over a field F , I a finite set, $k \notin I$, U a sub-space of V $\{V_i\}_{i \in I \cup \{k\}} \subseteq \mathcal{P}(V)$ a finite family of sub-spaces of V such that*

$$\forall i \in I \ V_i \text{ is a sub-space of } U$$

and

$$U = \bigoplus_{i \in I} V_i \text{ and } V = U \bigoplus V_k$$

then

$$V = \bigoplus_{i \in I \cup \{k\}} V_i$$

Proof. Take $I_1 = I$ and $I_2 = \{k\}$ then we have as $k \notin I$ that $I_1 \cap I_2 = \emptyset$. Further $\forall i \in I_1 \ V_i$ is a sub-space of U and $\forall i \in I_2 \ V_i = V_k$ is a sub-space of V_k . Further as $V_k \stackrel{[\text{example: 12.7}]}{=} \bigoplus_{i \in \{k\}} W_i$ and by the hypothesis $U = \bigoplus_{i \in I} V_i$ and $V = U \bigoplus V_k$ it follows from [theorem: 12.9] that

$$V = \bigoplus_{i \in I \cup \{k\}} V_i \quad \square$$

Theorem 12.11. *Let V be a vector space over a field F , I a finite set and $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ a family of sub-spaces of V such that*

$$V = \bigoplus_{i \in I} V_i$$

then we have:

1. If $J \subseteq I$ we have that

$$\sum_{i \in J} V_i = \bigoplus_{i \in J} V_i$$

2. If $I = I_1 \cup I_2$ with $I_1 \cap I_2 = \emptyset$ then

$$\sum_{i \in I_1} V_i = \bigoplus_{i \in I_1} V_i \text{ and } \sum_{i \in I_2} V_i = \bigoplus_{i \in I_2} V_i$$

and

$$V = \left(\sum_{i \in I_1} V_i \right) \oplus \left(\sum_{i \in I_2} V_i \right)$$

3. If $k \in I$ then

$$\sum_{i \in I \setminus \{k\}} V_i = \bigoplus_{i \in I \setminus \{k\}} V_i$$

and

$$V = \left(\sum_{i \in I \setminus \{k\}} V_i \right) \oplus V_k$$

Note 12.12. By [theorem: 12.4] we have that $\sum_{i \in J} V_i$, $\sum_{i \in I_1} V_i$ and $\sum_{i \in I_2} V_i$ are sub-spaces of V as is required for (1) and (2).

Proof.

1. Let $j \in \sum_{i \in J} V_i$ then there exists a $\{v_i\}_{i \in J} \subseteq V$ such that $\forall i \in J \ v_i \in V_i$ and $v = \sum_{i \in J} v_i$. Assume that there exists also a $\{w_i\}_{i \in J} \subseteq V$ such that $\forall i \in J \ w_i \in V_i$ and $v = \sum_{i \in J} w_i$. Define then

$$\{v'_i\}_{i \in I} \subseteq V \text{ by } v'_i = \begin{cases} 0 & \text{if } i \in I \setminus J \\ v_i & \text{if } i \in J \end{cases} \text{ and } \{w'_i\}_{i \in I} \subseteq V \text{ by } w'_i = \begin{cases} 0 & \text{if } i \in I \setminus J \\ w_i & \text{if } i \in J \end{cases}$$

then we have $\forall i \in I$

$$v'_i = \begin{cases} 0 \in V_i & \text{if } i \in I \setminus J \\ u_i \in V_i & \text{if } i \in J \end{cases} \in V_i \text{ and } w'_i = \begin{cases} 0 \in V_i & \text{if } i \in I \setminus J \\ w_i \in V_i & \text{if } i \in J \end{cases} \in V_i \quad (12.10)$$

Further

$$\begin{aligned} \sum_{i \in I} v'_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus J} v'_i + \sum_{i \in J} v'_i \\ &= \sum_{i \in I \setminus J} 0 + \sum_{i \in J} v_i \\ &\stackrel{[\text{theorem: 11.37}]}{=} 0 + v \\ &= v \end{aligned} \quad (12.11)$$

$$\begin{aligned} \sum_{i \in I} w'_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I \setminus J} w'_i + \sum_{i \in J} w'_i \\ &= \sum_{i \in I \setminus J} 0 + \sum_{i \in J} w_i \\ &\stackrel{[\text{theorem: 11.37}]}{=} 0 + v \\ &= v \end{aligned} \quad (12.12)$$

As $V = \bigoplus_{i \in I} V_i$ we have by [eqs: 12.10, 12.11 and 12.12] that $\forall i \in I$ we have $v'_i = w'_i$. Hence

$$\forall j \in J \text{ we have } v_j = v'_j = w'_j = w_j$$

proving that

$$\sum_{i \in J} V_i = \bigoplus_{i \in J} V_i$$

2. As $I = I_1 \cup I_2$ we have I_1, I_2 so that by (1) we have

$$\sum_{i \in I_1} V_i = \bigoplus_{i \in I_1} V_i \text{ and } \sum_{i \in I_2} V_i = \bigoplus_{i \in I_2} V_i \quad (12.13)$$

As $V = \bigoplus_{i \in I} V_i$ we have by [theorem: 12.8] that $V = \sum_{i \in I} V_i$, further by [theorem: 12.3 (4)] $\sum_{i \in I} V_i = (\sum_{i \in I_1} V_i) + (\sum_{i \in I_2} V_i)$ so that

$$V = \left(\sum_{i \in I_1} V_i \right) + \left(\sum_{i \in I_2} V_i \right) \quad (12.14)$$

Let $v \in V$ then by the above there exist

$$v^1 \in \sum_{i \in I_1} V_i, v^2 \in \sum_{i \in I_2} V_i \text{ such that } v = v^1 + v^2 \quad (12.15)$$

hence there exists $\{v_i^1\}_{i \in I_1} \subseteq V$ and $\{v_i^2\}_{i \in I_2} \subseteq V$ such that

$$\forall i \in I_1 \ v_i^1 \in V_i, \forall i \in I_2 \ v_i^2 \in V_i, v^1 = \sum_{i \in I_1} v_i^1 \text{ and } v^2 = \sum_{i \in I_2} v_i^2 \quad (12.16)$$

Assume that there exist also

$$w^1 \in \sum_{i \in I_1} V_i, w^2 \in \sum_{i \in I_2} V_i \text{ such that } v = w^1 + w^2 \quad (12.17)$$

then there exists $\{w_i^1\}_{i \in I_1} \subseteq V$ and $\{w_i^2\}_{i \in I_2} \subseteq V$ such that

$$\forall i \in I_1 \ w_i^1 \in V_i, \forall i \in I_2 \ w_i^2 \in V_i, w^1 = \sum_{i \in I_1} w_i^1 \text{ and } w^2 = \sum_{i \in I_2} w_i^2 \quad (12.18)$$

As $I = I_1 \cup I_2$ and $I_1 \cap I_2 = \emptyset$ we can define

$$\{v_i\}_{i \in I} \subseteq V \text{ by } v_i = \begin{cases} v_i^1 & \text{by } i \in I_1 \\ v_i^2 & \text{by } i \in I_2 \end{cases} \text{ and } \{w_i\}_{i \in I} \subseteq V \text{ by } w_i = \begin{cases} w_i^1 & \text{by } i \in I_1 \\ w_i^2 & \text{by } i \in I_2 \end{cases}$$

then we have by [eqs: 12.16 and 12.18]

$$\forall i \in I \ v_i = \begin{cases} v_i^1 \in V_i & \text{if } i \in I_1 \\ v_i^1 \in V_i & \text{if } i \in I_2 \end{cases} \in V_i \text{ and } w_i = \begin{cases} w_i^1 \in V_i & \text{if } i \in I_1 \\ w_i^1 \in V_i & \text{if } i \in I_2 \end{cases} \in V_i \quad (12.19)$$

Further

$$\begin{aligned} \sum_{i \in I} v_i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I_1} v_i + \sum_{i \in I_2} v_i \\ & = \sum_{i \in I_1} v_i^1 + \sum_{i \in I_2} v_i^2 \\ & \stackrel{[\text{eq: 12.16}]}{=} v^1 + v^2 \\ & \stackrel{[\text{eq: 12.15}]}{=} v \end{aligned} \quad (12.20)$$

$$\begin{aligned} \sum_{i \in I} w_i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I_1} w_i + \sum_{i \in I_2} w_i \\ & = \sum_{i \in I_1} w_i^1 + \sum_{i \in I_2} w_i^2 \\ & \stackrel{[\text{eq: 12.18}]}{=} w^1 + w^2 \\ & \stackrel{[\text{eq: 12.17}]}{=} v \end{aligned} \quad (12.21)$$

As $V = \bigoplus_{i \in I} V_i$ we have by [eqs: 12.19, 12.20 and 12.21] that

$$\forall i \in I \ v_i = w_i \quad (12.22)$$

Next

$$\begin{aligned} v^1 & \stackrel{[\text{eq: 12.16}]}{=} \sum_{i \in I_1} v_i^1 \\ & = \sum_{i \in I_1} v_i \\ & \stackrel{[\text{eq: 12.22}]}{=} \sum_{i \in I_1} w_i \\ & = \sum_{i \in I_1} w_i^1 \\ & \stackrel{[\text{eq: 12.18}]}{=} w^1 \\ v^2 & \stackrel{[\text{eq: 12.16}]}{=} \sum_{i \in I_2} v_i^2 \\ & = \sum_{i \in I_2} v_i \\ & \stackrel{[\text{eq: 12.22}]}{=} \sum_{i \in I_2} w_i \\ & = \sum_{i \in I_2} w_i^2 \\ & \stackrel{[\text{eq: 12.18}]}{=} w^2 \end{aligned}$$

proving that

$$v = v^1 + v^2 = w^1 + w^2 \wedge v^1 = w^1 \wedge v^2 = w^2$$

So that we have, taking in account [eqs: 12.15, 12.18] and [definition: 11.209] that

$$V = \left(\sum_{i \in I_1} V_i \right) \oplus \left(\sum_{i \in I_2} V_i \right) \quad \square$$

Theorem 12.13. *If V is a vector space over a field, I finite, $\{V_i\}_{i \in I} \subseteq \mathcal{P}(V)$ a finite family of subspaces of V and $\sigma: J \rightarrow I$ a bijection such that $V = \bigoplus_{i \in J} V_{\sigma(i)}$ then $V = \bigoplus_{i \in I} V_i$.*

Proof. As $V = \bigoplus_{i \in J} V_{\sigma(i)}$ we have by [corollary: 12.8] that

$$V = \sum_{i \in J} V_{\sigma(i)} \quad (12.23)$$

and

$$\forall k \in I \text{ we have } V_{\sigma(k)} \bigcap \sum_{i \in I \setminus \{k\}} V_{\sigma(i)} = \{0\} \quad (12.24)$$

Now using [theorem: 12.3] we have that $\sum_{i \in J} V_{\sigma(i)} = \sum_{i \in I} V_i$ so that [eq: 12.23] becomes

$$V = \sum_{i \in I} V_i \quad (12.25)$$

Let $l \in I$ and take $k = \sigma^{-1}(l) \in J$ so that $l = \sigma(k)$. Let $i \in \sigma(J \setminus \{k\})$ then $\exists j \in J$ with $j \neq k$ such that $i = \sigma(j)$. Assume that $i = l$ then $\sigma(j) = i = l = \sigma(k)$ $\xRightarrow{\sigma \text{ is injective}}$ $j = k$ contradicting $j \neq k$, hence $i \neq l$ so that $i \in \sigma(J) \setminus \{l\}$, proving that

$$\sigma(J \setminus \{k\}) \subseteq \sigma(J) \setminus \{l\}$$

On the other hand if $i \in \sigma(J) \setminus \{l\}$ then $\exists j \in J$ such that $i = \sigma(j)$ and $i \neq l$, assume that $j = k$ then $i = \sigma(j) = \sigma(k) = l$ contradicting $i \neq l$, hence $j \neq k$ so that $i \in \sigma(J \setminus \{k\})$, proving, together with the above, that

$$\sigma(J) \setminus \{l\} = \sigma(J \setminus \{k\})$$

As $\sigma: J \rightarrow I$ is a bijection we have that $\sigma(J) = I$ so that $\sigma(J \setminus \{k\}) = I \setminus \{l\}$ hence

$$\sigma|_{J \setminus \{k\}}: J \setminus \{k\} \rightarrow \sigma(J \setminus \{k\}) = I \setminus \{l\} \text{ is a bijection}$$

Applying then [theorem: 12.3] we have that

$$\sum_{i \in J \setminus \{k\}} V_{\sigma(i)} = \sum_{i \in I \setminus \{l\}} V_i$$

so that

$$V_l \bigcap \sum_{i \in I \setminus \{l\}} V_i = V_{\sigma(k)} \bigcap \sum_{i \in J \setminus \{k\}} V_{\sigma(i)} \stackrel{[\text{eq: 12.24}]}{=} \{0\}$$

Combining this with [eq: 12.25] proves, using [theorem: 12.8], that

$$V = \bigoplus_{i \in I} V_i \quad \square$$

The following example shows that the direct integral sum defined in this section is indeed an extension of the internal direct sum in [definition: 11.209].

Example 12.14. Let $\{V_i\}_{i \in \{1, \dots, 2\}} \subseteq \mathcal{P}(V)$ be a family of two sub-spaces of a vector space over a field F then

$$V = \bigoplus_{i \in \{1, 2\}} V_i$$

then

$$V = V_1 \oplus V_2$$

Proof. If $V = \bigoplus_{i \in \{1, 2\}} V_i$ then we have

$$V \stackrel{[\text{theorem: 12.11}]}{=} \left(\sum_{i \in \{1\}} V_i \right) \oplus V_2 \stackrel{[\text{theorem: 12.3}]}{=} V_1 \oplus V_2$$

$\square$

Theorem 12.15. Let V be a vector space over a field F , $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of finite dimensional sub-spaces of V such that

$$V = \bigoplus_{i \in \{1, \dots, n\}} V_i$$

then V is finite dimensional and

$$\dim(V) = \sum_{i \in \{1, \dots, n\}} \dim(V_i)$$

Proof. We use mathematical induction for the proof, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } V = \bigoplus_{i \in \{1, \dots, n\}} V_i \text{ where } V_i \text{ are finite dimensional then } V \text{ is finite dimensional and} \right. \\ \left. \dim(V) = \sum_{i \in \{1, \dots, n\}} \dim(V_i) \right\}$$

then we have:

$1 \in S$. If $V = \bigoplus_{i \in \{1\}} V_i$ then by [example: 12.7] $V = V_1$, hence V is finite dimensional and

$$\dim(V) = \dim(V_1) = \sum_{i \in \{1\}} \dim(V_i)$$

proving that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Assume that $V = \bigoplus_{i \in \{1, \dots, n+1\}} V_i$ where $\forall i \in \{1, \dots, n+1\}$ V_i is a finite dimensional sub-space of F then we have by [theorem: 12.11] that

$$\sum_{i \in \{1, \dots, n\}} V_i = \bigoplus_{i \in \{1, \dots, n\}} V_i \text{ and } V = \left(\bigoplus_{i \in \{1, \dots, n\}} V_i \right) \oplus V_{n+1} \quad (12.26)$$

As $n \in S$ $\sum_{i \in \{1, \dots, n\}} V_i$ is finite dimensional and

$$\dim\left(\bigoplus_{i \in \{1, \dots, n\}} V_i\right) = \sum_{i \in \{1, \dots, n\}} \dim(V_i) \quad (12.27)$$

As V_{n+1} is also finite dimensional we can apply [theorem: 11.212] resulting in

$$V = \left(\bigoplus_{i \in \{1, \dots, n\}} V_i \right) \oplus V_{n+1} \text{ is finite dimensional}$$

and

$$\dim\left(\left(\bigoplus_{i \in \{1, \dots, n\}} V_i\right) \oplus V_{n+1}\right) = \dim\left(\bigoplus_{i \in \{1, \dots, n\}} V_i\right) + \dim(V_{n+1})$$

substituting the above in [eq: 12.27] gives

$$\dim(V) = \sum_{i \in \{1, \dots, n\}} \dim(V_i) + \dim(V_{n+1}) = \sum_{i \in \{1, \dots, n+1\}} \dim(V_i)$$

proving that $n+1 \in S$. □

Theorem 12.16. Let V, W be vector spaces over a field F such that $V = \bigoplus_{i \in I} V_i$ and $L: V \rightarrow W$ a linear isomorphism then

$$W = \bigoplus_{i \in I} L(V_i)$$

Proof. Let $w \in W$ then as L is bijective there exist a $v \in V$ such that $L(v) = w$. As $V = \bigoplus_{i \in I} V_i$ there exist a $\{v_i\}_{i \in I} \subseteq V$ such that $\forall i \in I$ $v_i \in V_i$ and $v = \sum_{i \in I} v_i$. Define then

$$\{w_i\}_{i \in I} \subseteq W \text{ by } w_i = L(v_i)$$

then $\forall i \in I$ $w_i = L(v_i) \in L(V_i)$ and $w = L(v) = L(\sum_{i \in I} v_i) = \sum_{i \in I} L(v_i) = \sum_{i \in I} w_i$. Let $\{w'_i\}_{i \in I} \subseteq W$ be such that $\forall i \in I$ $w'_i \in L(V_i)$ and $w = \sum_{i \in I} w'_i$. Define

$$\{v'_i\}_{i \in I} \subseteq V \text{ by } L^{-1}(w'_i)$$

then we have

$$\begin{aligned}
 v &= L^{-1}(w) \\
 &= L^{-1}\left(\sum_{i \in I} w'_i\right) \\
 &\stackrel{L^{-1} \text{ is linear}}{=} \sum_{i \in I} L^{-1}(w'_i) \\
 &= \sum_{i \in I} v'_i
 \end{aligned}$$

As $V = \bigoplus_{i \in I} V_i$ we have that $\forall i \in I$ $v'_i = v_i$ so that $w_i = L(v_i) = L(v'_i) = L(L^{-1}(w'_i)) = w'_i$. Hence we have that

$$W = \bigoplus_{i \in I} L(V_i) \quad \square$$

Chapter 13

Tensor product of vector spaces

Multilinear mappings are difficult to handle, the idea of the tensor product is to focus all attention a single pair of a vector space P and a fixed multilinear mapping between $\prod_{i \in \{1, \dots, n\}} V_i$ and P such that every multilinear mapping between $\prod_{i \in \{1, \dots, n\}} V_i$ and a vector space can be written as the composition of the fixed multilinear mapping and a linear mapping.

Definition 13.1. (Tensor Product) Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of vector spaces over a field F then a **tensor product of $V_1, \dots, V_n$** is a pair $\langle P, \nu \rangle$ where

1. P is a vector space over the field F
2. $\nu \in \text{Hom}(V_1, \dots, V_n; P)$ [see definition: 11.258]
3. $\text{span}(\nu(\prod_{i \in \{1, \dots, n\}} V_i)) = P$
4. **(Universal Factorization property)** If U is any vector space over F and $\varphi \in \text{Hom}(V_1, \dots, V_n; U)$ then there exist a $h \in \text{Hom}(P, U)$ such that $\varphi = h \circ \nu$

It turns out that the map h such that $\varphi = h \circ \nu$ is actually unique. Hence every multilinear mapping $\varphi \in \text{Hom}(V_1, \dots, V_n; U)$ can be represented uniquely by a linear mapping $h \in \text{Hom}(P, U)$ such that $\varphi = h \circ \nu$.

Theorem 13.2. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of vector spaces over a field F , U a vector space over the same field F and $\langle P, \nu \rangle$ a tensor product then $\forall \varphi \in \text{Hom}(V_1, \dots, V_n; U)$ there exist a **unique** linear mapping $h \in \text{Hom}(P, U)$ such that $\varphi = h \circ \nu$

Proof. As $\langle P, \nu \rangle$ is a tensor product P is a vector space, $\nu \in \text{Hom}(V_1, \dots, V_n; P)$, $P = \text{span}(\nu(\prod_{i \in \{1, \dots, n\}} V_i))$. Let $\varphi \in \text{Hom}(V_1, \dots, V_n; U)$ then by definition there exist a $h \in \text{Hom}(P, U)$ such that $\varphi = h \circ \nu$. Assume that there exist a $k \in \text{Hom}(P, U)$ such that $\varphi = k \circ \nu$. Let $x \in P$ then, as $P = \text{span}(\nu(\prod_{i \in \{1, \dots, n\}} V_i))$, there exist by [theorem: 11.107] a $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq \nu(\prod_{i \in \{1, \dots, n\}} V_i)$ and a $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that

$$x = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot y_i \quad (13.1)$$

If $i \in \{1, \dots, n\}$ then as $y_i \in \nu(\prod_{i \in \{1, \dots, n\}} V_i)$ there exist a $z \in \prod_{i \in \{1, \dots, n\}} V_i$ such that $y_i = \nu(z)$. Hence $h(y_i) = h(\nu(z)) = (h \circ \nu)(z) = \varphi(z) = (k \circ \nu)(z) = k(\nu(z)) = k(y_i)$ proving that

$$\forall i \in \{1, \dots, n\} \text{ we have } h(y_i) = k(y_i) \quad (13.2)$$

So we have

$$\begin{aligned} h(x) &\stackrel{[\text{eq: 13.1}]}{=} h\left(\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot y_i\right) \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot h(y_i) \\ &\stackrel{[\text{eq: 13.2}]}{=} \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot k(y_i) \\ &= k\left(\sum_{i \in \{1, \dots, n\}} \alpha_i \cdot y_i\right) \\ &= k(x) \end{aligned}$$

Which, as $x \in P$ was chosen arbitrary, proves that

$$h = k \quad \square$$

We show now that given a finite family of vector spaces $\{V_i\}_{i \in \{1, \dots, n\}}$ there exist a least one tensor product. First we introduce the concept of a free vector space which is a vector space created from a given set and field.

Definition 13.3. Let X be a non empty set and $\langle F, +, \cdot \rangle$ a field then $\mathcal{F}(X, F)$ is defined by

$$\mathcal{F}(X, F) = \{f \in F^X \mid f^{-1}(F \setminus \{0\}) \text{ is finite}\} \subseteq F^X$$

In other words $\mathcal{F}(X, F)$ is the set of graphs of functions from X to F such that the functions are non zero for a finite number of elements in F .

Next we define the operators needed for a vector space.

Definition 13.4. Let X be a non empty set and a field $\langle F, +, \cdot \rangle$ then we define:

1. If $f, g \in \mathcal{F}(X, F)$ then $f + g$ is defined by $(f + g)(x) = f(x) + g(x)$
2. If $f \in \mathcal{F}(X, F)$ and $\alpha \in F$ then $(\alpha \cdot f)(x) = \alpha \cdot f(x)$
3. $\forall a \in X$ we define $\delta_a: X \rightarrow F$ by $\delta_a(x) = \begin{cases} 1 & \text{if } x = a \\ 0 & \text{if } x \in X \setminus \{a\} \end{cases}$

Finally we define the free vector space.

Theorem 13.5. Let X be a set and $\langle F, +, \cdot \rangle$ then $\langle \mathcal{F}(X, F), +, \cdot \rangle$ [see the above definition] is a vector space over the field $\langle F, +, \cdot \rangle$ where the neutral element is C_0 . This vector space is called the **free vector space of X** .

Proof. Let $f, g \in \mathcal{F}(X, F)$. If $x \in (f + g)^{-1}(F \setminus \{0\})$ then we have $(f + g)(x) \in F \setminus \{0\}$ so that $f(x) + g(x) \neq 0$. So we must have $f(x) \neq 0 \vee g(x) \neq 0$ [otherwise $(f + g)(x) = f(x) + g(x) = 0$], hence $x \in f^{-1}(F \setminus \{0\}) \cup g^{-1}(F \setminus \{0\})$, proving that

$$(f + g)^{-1}(F \setminus \{0\}) \subseteq f^{-1}(F \setminus \{0\}) \cup g^{-1}(F \setminus \{0\})$$

As $f, g \in \mathcal{F}(X, F)$ we have that $f^{-1}(F \setminus \{0\}), g^{-1}(F \setminus \{0\})$ are finite hence using [theorems: 10.91, 6.34] we have that $(f + g)^{-1}(F \setminus \{0\})$ is a finite set so that $f + g \in \mathcal{F}(X, F)$ and

$$+: \mathcal{F}(X, F) \times \mathcal{F}(X, F) \rightarrow \mathcal{F}(X, F) \text{ is well defined}$$

Let $\alpha \in F$ and $f \in \mathcal{F}(X, F)$. If $x \in (\alpha \cdot f)^{-1}(F \setminus \{0\})$ then $\alpha \cdot f(x) \in F \setminus \{0\}$ so that $\alpha \cdot f(x) \neq 0$ hence we must have that $f(x) \neq 0$ or $x \in f^{-1}(F \setminus \{0\})$ proving $(\alpha \cdot f)^{-1}(F \setminus \{0\}) \subseteq f^{-1}(F \setminus \{0\})$ [a finite set], so by [theorem: 10.91] $f^{-1}(F \setminus \{0\})$ is finite or $\alpha \cdot f \in \mathcal{F}(X, F)$, proving that

$$\cdot: F \times \mathcal{F}(X, F) \rightarrow \mathcal{F}(X, F) \text{ is well defined}$$

Further for $C_0: X \rightarrow F$ defined by $C_0(x) = 0$ we have trivially that $(C_0)^{-1}(F \setminus \{0\}) = \emptyset$ [which is finite] hence

$$C_0 \in \mathcal{F}(X, F)$$

Next we check the axioms of a vector space:

1. $\langle \mathcal{F}(X, F), + \rangle$ is a Abelian group:

associativity. If $f, g, h \in \mathcal{F}(X, F)$ then we have $\forall x \in X$ that

$$\begin{aligned} (f + (g + h))(x) &= f(x) + (g + h)(x) \\ &= f(x) + (g(x) + h(x)) \\ &= (f(x) + g(x)) + h(x) \\ &= (f + g)(x) + h(x) \\ &= ((f + g) + h)(x) \end{aligned}$$

proving that $f + (g + h) = (f + g) + h$

commutativity. If $f, g \in \mathcal{F}(X, F)$ then we have $\forall x \in X$ that

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$$

proving that

$$f + g = g + f$$

neutral element. If $f \in \mathcal{F}(X, F)$ then we have $\forall x \in X$ that

$$(f + C_0)(x) = f(x) + C_0(x) = f(x) + 0 = f(x) = 0 + f(x) = C_0(x) + f(x) = (C_0 + f)(x)$$

proving that

$$f + C_0 = f = C_0 + f$$

inverse element. Let $f \in \mathcal{F}(X, F)$ then define $(-f): X \rightarrow F$ by $(-f)(x) = -f(x)$.

Let $x \in (-f)^{-1}(F \setminus \{0\})$ then $(-f)(x) \neq 0 \Rightarrow -f(x) \neq 0 \Rightarrow f(x) \neq 0$, proving that $x \in f^{-1}(F \setminus \{0\})$. Hence $(-f)^{-1}(F \setminus \{0\}) \subseteq f^{-1}(F \setminus \{0\})$ [a finite set], proving that $(-f)^{-1}(F \setminus \{0\})$ is finite so that $-f \in \mathcal{F}(X, F)$. Next $\forall x \in X$ we have

$$\begin{aligned} (f + (-f))(x) &= f(x) + (-f)(x) \\ &= f(x) + (-f(x)) \\ &= 0 \\ &= C_0(x) \end{aligned}$$

proving that $(-f) + f \underset{\text{commutativity}}{=} f + (-f) = C_0$

2. Additional vector space axioms:

a. If $\alpha \in F$ and $f, g \in \mathcal{F}(X, F)$ we have $\forall x \in X$ that

$$\begin{aligned} (\alpha \cdot (f + g))(x) &= \alpha \cdot (f + g)(x) \\ &= \alpha \cdot (f(x) + g(x)) \\ &= \alpha \cdot f(x) + \alpha \cdot g(x) \\ &= (\alpha \cdot f)(x) + (\alpha \cdot g)(x) \\ &= (\alpha \cdot f + \alpha \cdot g)(x) \end{aligned}$$

proving that $\alpha \cdot (f + g) = \alpha \cdot f + \alpha \cdot g$

b. If $\alpha, \beta \in F$ and $f \in \mathcal{F}(X, F)$ then we have $\forall x \in X$ that

$$\begin{aligned} ((\alpha + \beta) \cdot f)(x) &= (\alpha + \beta) \cdot f(x) \\ &= \alpha \cdot f(x) + \beta \cdot f(x) \\ &= (\alpha \cdot f)(x) + (\beta \cdot f)(x) \\ &= (\alpha \cdot f + \beta \cdot f)(x) \\ &= (\alpha \cdot f + \beta \cdot f)(x) \end{aligned}$$

proving that $(\alpha + \beta) \cdot f = \alpha \cdot f + \beta \cdot f$.

c. If $\alpha, \beta \in F$ and $f \in \mathcal{F}(X, F)$ then we have $\forall x \in X$ that

$$\begin{aligned} ((\alpha \cdot \beta) \cdot f)(x) &= (\alpha \cdot \beta) \cdot f(x) \\ &= \alpha \cdot (\beta \cdot f(x)) \\ &= \alpha \cdot (\beta \cdot f)(x) \\ &= (\alpha \cdot (\beta \cdot f))(x) \end{aligned}$$

proving that

$$(\alpha \cdot \beta) \cdot f = \alpha \cdot (\beta \cdot f)$$

d. If $f \in \mathcal{F}(X, Y)$ then $\forall x \in X$ we have

$$(1 \cdot f)(x) = 1 \cdot f(x) = f(x)$$

proving that $1 \cdot f = f$ □

We prove now that $\{\delta_a | a \in X\}$ is a basis for $\mathcal{F}(X, F)$ and that X can be embedded in $\mathcal{F}(X, F)$ using the injection $\delta: X \rightarrow \mathcal{F}(X, F)$.

Theorem 13.6. *Let X be a non empty set and $\langle F, +, \cdot \rangle$ a field then*

1. $\delta: X \rightarrow \mathcal{F}(X, F)$ defined by $\delta(a) = \delta_a$ is a injection, hence X is embedded in $\mathcal{F}(X, F)$ and

$$\delta: X \rightarrow \delta(X) \text{ is a bijection}$$

2. $\delta(X) = \{\delta(a) | a \in X\} = \{\delta_a | a \in X\}$ is a basis of $\mathcal{F}(X, F)$

Proof.

1. First we must ensure that $\forall a \in X \delta_a \in \mathcal{F}(X, F)$, so let $x \in (\delta_a)^{-1}(F \setminus \{0\})$ then we must have that $(\delta_a)(x) \neq 0$. As $(\delta_a)(x) = \begin{cases} 1 & \text{if } x = a \\ 0 & \text{if } x \in X \setminus \{a\} \end{cases}$, we must have that $x = a$, hence $(\delta_a)^{-1}(F \setminus \{0\}) \subseteq \{a\}$ a finite set. Hence

$$\forall a \in X \text{ we have } \delta_a \in \mathcal{F}(X, F)$$

and the following function is well defined

$$\delta: X \rightarrow \mathcal{F}(X, F) \text{ where } \delta(a) = \delta_a$$

If $\delta(a) = \delta(b)$ then $\delta_a = \delta_b$. Assume that $a \neq b$ then $\delta_a(b) \stackrel{b \in X \setminus \{a\}}{=} 0$ and $\delta_b(b) = 1$, as $1 \neq 0$ in a field, we have $\delta_a(b) \neq \delta_b(b)$ contradicting $\delta_a = \delta_b$, hence $a = b$. Hence

δ is injective

2. Define

$$B = \{\delta_a | a \in X\} \subseteq \mathcal{F}(X, F)$$

Let $I \subseteq B$, I finite and $\{\alpha_i\}_{i \in I} \subseteq F$ such that $\sum_{i \in I} \alpha_i \cdot i = 0$. Let $j \in I \subseteq B$ then $\exists a \in X$ such that $j = \delta(a)$. If $i \in I \setminus \{j\} \subseteq B$ then $\exists b \in X$ such that $i = \delta(b)$ and as $i \neq j$ we must have $a \neq b$ hence $i(a) = \delta(b)(a) = \delta_b(a) = 0$. So we have

$$\begin{aligned} 0 &= \left(\sum_{i \in I} \alpha_i \cdot i \right)(a) \\ &= \sum_{i \in I} \alpha_i \cdot i(a) \\ &= \sum_{i \in I \setminus \{j\}} \alpha_i \cdot i(a) + \sum_{i \in \{j\}} \alpha_i \cdot i(a) \\ &= \sum_{i \in I \setminus \{j\}} \alpha_i \cdot 0 + \alpha_j \cdot j(a) \\ &= \alpha_j \cdot j(a) \\ &= \alpha_j \cdot \delta(a)(a) \\ &= \alpha_j \cdot \delta_a(a) \\ &= \alpha_j \cdot 1 \\ &= \alpha_j \end{aligned}$$

So we have proved that

$$\forall j \in I \alpha_j = 0$$

from which it follows that

$$B \text{ is linear independent} \quad (13.3)$$

Next if $f \in \mathcal{F}(X, F)$ then $f^{-1}(F \setminus \{0\})$ is finite and we have two cases to consider:

$f^{-1}(F \setminus \{0\}) = \emptyset$. Then $f = C_0$ and, as X is non empty, there exist a $a \in X$. Then we can write f as $f = C_0 = 0 \cdot \delta_a$. Hence if we define $I = \{\delta_a\} \subseteq B$ and $\{\alpha_i\}_{i \in \{\delta_a\}} \subseteq F$ by $\alpha_{\delta_a} = 0$ we have that $f = 0 \cdot \delta_a = \sum_{i \in \{\delta_a\}} \alpha_i \cdot i$. Hence

$$f \in \text{span}(B)$$

$f^{-1}(F \setminus \{0\}) \neq \emptyset$. Then $I = \{\delta_a | a \in f^{-1}(F \setminus \{0\})\} \subseteq B$ is finite [see theorem: 6.45]. As δ is injective we have that

$$\delta: f^{-1}(F \setminus \{0\}) \rightarrow \delta(f^{-1}(F \setminus \{0\})) = \{\delta_a | a \in f^{-1}(F \setminus \{0\})\} \text{ is a bijection}$$

Define now

$$\{\alpha_i\}_{i \in I} \text{ by } \alpha_i = f(\delta^{-1}(i))$$

then we have for $x \in X$ either

$f(x) = 0$. Let $i \in I$ then $\exists a \in f^{-1}(F \setminus \{0\})$ such that $i = \delta_a$, as $f(a) \neq 0$ we can not have that $a = x$ [because then $f(a) = f(x) = 0$], hence $i(x) = \delta_a(x) = 0$. In other words

$$\forall i \in I \text{ we have } i(x) = 0$$

So

$$\begin{aligned} \left(\sum_{i \in I} \alpha_i \cdot i \right)(x) &= \sum_{i \in I} \alpha_i \cdot i(x) \\ &= \sum_{i \in I} \alpha_i \cdot 0 \\ &= 0 \\ &= f(x) \end{aligned}$$

$f(x) \neq 0$. As $f(x) \neq 0$ we have that $x \in f^{-1}(F \setminus \{0\})$ so that $\delta_x \in I$. Let $i \in I \setminus \{\delta_x\}$ then $\exists a \in f^{-1}(F \setminus \{0\})$ with $a \neq x$ such that $i = \delta_a$, hence $i(x) = \delta_a(x) \underset{x \neq a}{=} 0$. So

$$\begin{aligned} \left(\sum_{i \in I} \alpha_i \cdot i \right)(x) &= \sum_{i \in I} \alpha_i \cdot i(x) \\ &= \sum_{i \in I \setminus \{\delta_x\}} \alpha_i \cdot i(x) + \sum_{i \in \{\delta_x\}} \alpha_i \cdot i(x) \\ &= \sum_{i \in I \setminus \{\delta_x\}} \alpha_i \cdot i(x) + \alpha_{\delta(x)} \cdot \delta_x(x) \\ &= \sum_{i \in I \setminus \{\delta_{a_0}, \delta_x\}} \alpha_i \cdot 0 + f(\delta^{-1}(\delta_x)) \cdot \delta_x(x) \\ &= f(\delta^{-1}(\delta_x)) \cdot \delta_x(x) \\ &= f(x) \cdot 1 \\ &= f(x) \end{aligned}$$

So in all cases we have $(\sum_{i \in I} \alpha_i \cdot i)(x) = f(x)$ proving that $f = \sum_{i \in I} \alpha_i \cdot i$, so, as $I \subseteq B$, we have

$$f \in \text{span}(B)$$

So we have proved that $\forall f \in \mathcal{F}(X, F) \quad f \in \text{span}(B) \subseteq \mathcal{F}(X, F)$ proving that $\mathcal{F}(X, F) \subseteq \text{span}(B) \subseteq \mathcal{F}(X, F)$, hence $\mathcal{F}(X, F) = \text{span}(B)$ which as B is linear independent [see eq: 13.3] proves that

$$B = \delta(X) \text{ is a basis of } \mathcal{F}(X, F) \quad \square$$

Giving a family $\{V_i\}_{i \in \{1, \dots, n\}}$ of vector spaces over a field F , we can construct the free vector space $\langle \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i), +, \cdot \rangle$ with basis $B = \delta(\prod_{i \in \{1, \dots, n\}} V_i)$. We want now to create a new vector space and a surjective mapping from $\mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i)$ to this new vector space such that all elements of the form

$$\delta(v_1, \dots, v_{i-1}, x + \alpha \cdot y, v_{i+1}, \dots, v_n) - \delta(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_n) - \alpha \cdot \delta(v_1, \dots, v_{i-1}, y, v_{i+1}, \dots, v_n)$$

are mapped on the neutral element of this newer vector space. The composition of δ and this new mapping will be a multilinear mapping from $\prod_{i \in \{1, \dots, n\}} V_i$ to this new space. The standard way to do this is by creating a factor space based on a sub-space so that the projection mapping maps this sub-space on the neutral element in the factor space [see theorems: 11.78, 11.79].

Definition 13.7. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F and $\langle \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i), +, \cdot \rangle$ the free vector space of $\prod_{i \in \{1, \dots, n\}} V_i$ then

$$\mathcal{N}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right) \subseteq \mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i\right)$$

is defined by

$$\mathcal{N}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right) = \bigcup_{i \in \{1, \dots, n\}} \mathcal{N}_i$$

where

$$\mathcal{N}_i = \left\{ \delta[i \rightarrow x + \alpha \cdot y]v - \delta[i \rightarrow x]v - \alpha \cdot \delta[i \rightarrow y]v \mid v \in \prod_{i \in \{1, \dots, n\} \setminus \{i\}} V_i, x, y \in V_i, \alpha \in F \right\}$$

and for $i \in \{1, \dots, n\}$, $x \in \prod_{i \in \{1, \dots, n\} \setminus \{i\}} V_i$ and $y \in V_i$ we have

$$\forall k \in \{1, \dots, n\} \text{ we have } ([i \rightarrow y]x)_k = \begin{cases} y & \text{if } k = i \\ x_i & \text{if } k \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

in other words

$$[i \rightarrow y]x = \{x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n\}$$

Definition 13.8. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F and $\langle \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i), +, \cdot \rangle$ the free vector space of $\prod_{i \in \{1, \dots, n\}} V_i$ then $V_1 \odot \dots \odot V_n$ is the factor space [see theorem: 11.78]

$$\mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i\right) / \text{span}\left(\mathcal{N}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right)\right)$$

giving the vector space

$$\langle V_1 \odot \dots \odot V_n, +, \cdot \rangle = \left\langle \mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i\right) / \text{span}\left(\mathcal{N}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right)\right), +, \cdot \right\rangle$$

over the field F

Note 13.9. In the above definition we must use the span of $\mathcal{N}(\prod_{i \in \{1, \dots, n\}} V_i, F)$ because $\mathcal{N}(\prod_{i \in \{1, \dots, n\}} V_i, F)$ is not a sub-space of $\mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i)$.

Definition 13.10. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F then we define

$$\odot: \prod_{i \in \{1, \dots, n\}} V_i \rightarrow V_1 \odot \dots \odot V_n \text{ by } \odot = \pi \circ \delta$$

where:

1. $\delta: \prod_{i \in \{1, \dots, n\}} V_i \rightarrow \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i, F)$ is defined in [theorem: 13.6] by

$$\delta(x_1, \dots, x_n) = \delta_{(x_1, \dots, x_n)} \text{ and } \delta_{(x_1, \dots, x_n)}(r_1, \dots, r_n) = \begin{cases} 1 & \text{if } (x_1, \dots, x_n) = (r_1, \dots, r_n) \\ 0 & \text{if } (\prod_{i \in \{1, \dots, n\}} V_i) \setminus (x_1, \dots, x_n) \end{cases}$$

2. $\pi: \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i, F) \rightarrow \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i, F) / \text{span}(\mathcal{N}(\prod_{i \in \{1, \dots, n\}} V_i, F))$ is defined in [theorem: 11.79] by

$$\pi(x) = \left\{ y \mid x - y \in \text{span} \left(\mathcal{N} \left(\prod_{i \in \{1, \dots, n\}} V_i, F \right) \right) \right\}$$

If $(v_1, \dots, v_n) \in \prod_{i \in \{1, \dots, n\}} V_i$ then we note $\odot(v_1, \dots, v_n)$ as $v_1 \odot \dots \odot v_n$ and call this the tensor product of the vectors $v_1, \dots, v_n$.

Note 13.11. We use the word **tensor product** for two different things. Given a family $\{V_i\}_{i \in \{1, \dots, n\}}$ of vector spaces then we have a tensor product of $V_1, \dots, V_n$ [see definition: 13.1] or the tensor product of vectors in the above definition.

The following theorem will be needed to prove the universal factorization property of a tensor product.

Theorem 13.12. (Universal property) Let X, Z be vector spaces over a field F , Y a sub-space of X and $L \in \text{Hom}(X, Z)$ such that $Y \subseteq \ker(L)$ then there exist a **unique** $h \in \text{Hom}(X/Y, Z)$ such that $L = h \circ \pi_Y$. Here X/Y and π_Y are defined in [theorems: 11.78, 11.79].

Proof. Let $x, y \in X$ such that $\sim_Y[x] = \sim_Y[y]$ then $x - y \in Y$ hence, as $Y \subseteq \ker(L)$ we have $L(x - y) = 0$, so, as L is linear, $L(x) - L(y) = 0 \Rightarrow L(x) = L(y)$. This allows use to define the function

$$h: X/Y \rightarrow Z \text{ where } h(\sim[x]) = L(x)$$

Let $\sim_Y[x], \sim_Y[y] \in X/Y$ and $\alpha \in F$ then

$$h(\sim_Y[x] + \sim_Y[y]) = h(\sim_Y[x + y]) = L(x + y) = L(x) + L(y) = h(\sim_Y[x]) + h(\sim_Y[y])$$

and

$$h(\alpha \cdot \sim_Y[x]) = h(\sim_Y[\alpha \cdot x]) = L(\alpha \cdot x) = \alpha \cdot L(x) = \alpha \cdot h(\sim_Y[x])$$

proving that

$$h \in \text{Hom}(X/Y, Z)$$

Further let $x \in X$ then $(h \circ \pi_Y)(x) = h(\pi_Y(x)) = h(\sim_Y[x]) = L(x)$ so that

$$L = h \circ \pi_Y$$

Finally if $k \in \text{Hom}(X/Y, Z)$ is such that $L = k \circ \pi_Y$ then if $\sim_Y[x] \in X/Y$ we have

$$h(\sim_Y[x]) = L(x) = k(\pi_Y(x)) = k(\sim_Y[x])$$

so that

$$h = k$$

□

We are now ready to prove that for every family $\{V_i\}_{i \in \{1, \dots, n\}}$ of vector spaces there exists a tensor product of $V_1, \dots, V_n$.

Theorem 13.13. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F then

$$\langle V_1 \odot \dots \odot V_n, \odot \rangle$$

is a tensor product [see definition: 13.1] of $\{V_i\}_{i \in \{1, \dots, n\}}$. Using [definition: 13.1] we must prove that:

1. $V_1 \odot \dots \odot V_n$ is a vector space over F
2. $\odot \in \text{Hom}(V_1, \dots, V_n; V_1 \odot \dots \odot V_n)$
3. $\text{span}(\odot(\prod_{i \in \{1, \dots, n\}} V_i)) = V_1 \odot \dots \odot V_n$
4. **(Universal factorization property)** If U is a vector space over F and $\varphi \in \text{Hom}(V_1, \dots, V_n; U)$ then there exist a $h \in \text{Hom}(V_1 \odot \dots \odot V_n, U)$ such that $\varphi = h \circ \odot$.

Note 13.14. Using [theorem: 13.2] h is *unique*

Proof.

1. By [definition: 13.8] $\langle V_1 \odot \dots \odot V_n, +, \cdot \rangle$ is a vector space over the field F
2. Let $i \in \{1, \dots, n\}$, $\alpha \in F$, $x, y \in V_i$ and $v \in \prod_{i \in \{1, \dots, n\} \setminus \{i\}} V_i$ and take

$$(v_1, \dots, v_{i-1}, x + \alpha \cdot y, v_{i+1}, \dots, v_n) = [i \rightarrow x + \alpha \cdot y]v$$

$$(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_n) = [i \rightarrow x]v$$

$$(v_1, \dots, v_{i-1}, y, v_{i+1}, \dots, v_n) = [i \rightarrow y]v$$

then

$$\delta_{[i \rightarrow x + \alpha \cdot y]v} - \delta_{[i \rightarrow x]v} - \alpha \cdot \delta_{[i \rightarrow y]v} \in \mathcal{N} \left[\prod_{i \in \{1, \dots, n\}} V_i, F \right] \subseteq \text{span} \left(\mathcal{N} \left[\prod_{i \in \{1, \dots, n\}} V_i, F \right] \right)$$

so that by [theorem: 11.79]

$$\pi(\delta_{[i \rightarrow x + \alpha \cdot y]v} - \delta_{[i \rightarrow x]v} - \alpha \cdot \delta_{[i \rightarrow y]v}) = 0 \quad (13.4)$$

As by [theorem: 11.79] π is linear we have that

$$0 = \pi(\delta_{[i \rightarrow x + \alpha \cdot y]v} - \delta_{[i \rightarrow x]v} - \alpha \cdot \delta_{[i \rightarrow y]v}) = \pi(\delta_{[i \rightarrow x + \alpha \cdot y]v}) - \pi(\delta_{[i \rightarrow x]v}) - \alpha \cdot \pi(\delta_{[i \rightarrow y]v})$$

proving that

$$\pi(\delta_{[i \rightarrow x + \alpha \cdot y]v}) = \pi(\delta_{[i \rightarrow x]v}) + \alpha \cdot \pi(\delta_{[i \rightarrow y]v})$$

hence

$$\pi(\delta([i \rightarrow x + \alpha \cdot y]v)) = \pi(\delta([i \rightarrow x]v)) + \alpha \cdot \pi(\delta([i \rightarrow y]v))$$

which as $\odot = \pi \circ \delta$ gives

$$\odot([i \rightarrow x + \alpha \cdot y]v) = \odot([i \rightarrow x]v) + \alpha \cdot \odot([i \rightarrow y]v)$$

or

$$\begin{aligned} & \odot(v_1, \dots, v_{i-1}, x + \alpha \cdot y, v_{i+1}, \dots, v_n) = \\ & \odot(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_n) + \alpha \cdot \odot(v_1, \dots, v_{i-1}, y, v_{i+1}, \dots, v_n) \end{aligned}$$

proving by [theorem: 11.260] that

$$\odot \text{ is multilinear hence } \odot \in \text{Hom}(V_1, \dots, V_n; V_1 \odot \dots \odot V_n)$$

3. As $\odot: \prod_{i \in \{1, \dots, n\}} V_i \rightarrow V_1 \odot \dots \odot V_n$ we have already that $\odot(\prod_{i \in \{1, \dots, n\}} V_i) \subseteq V_1 \odot \dots \odot V_n$, so that by [theorems: 11.109, 11.111] that

$$\text{span} \left(\odot \left(\prod_{i \in \{1, \dots, n\}} V_i \right) \right) \subseteq V_1 \odot \dots \odot V_n \quad (13.5)$$

For the opposite inclusion, let $x \in V_1 \odot \cdots \odot V_n$ then as π is a surjection [see theorem: 11.79] there exists a $y \in \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i)$ such that $x = \pi(y)$. By [theorem: 13.6] the set $\{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\}$ is a basis of $\mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i)$, hence there exist a finite $I \subseteq \{1, \dots, n\}$ such that $y = \sum_{i \in I} \alpha_i \cdot i$. So

$$x = \pi(y) = \pi\left(\sum_{i \in I} \alpha_i \cdot i\right) \stackrel{\text{linearity of } \pi}{=} \sum_{i \in I} \alpha_i \cdot \pi(i) \quad (13.6)$$

Now if $i \in I \subseteq \{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\}$ then $\exists a \in \prod_{i \in \{1, \dots, n\}} V_i$ such that $i = \delta(a)$, hence $\pi(i) = \pi(\delta(a)) = \odot(a) \in \odot(\prod_{i \in \{1, \dots, n\}} V_i)$. Hence by [eq: 13.6] it follows that $x \in \text{span}(\odot(\prod_{i \in \{1, \dots, n\}} V_i))$. So we conclude that $V_1 \odot \cdots \odot V_n \subseteq \text{span}(\odot(\prod_{i \in \{1, \dots, n\}} V_i))$, which combined with [eq: 13.5] gives

$$\text{span}\left(\odot\left(\prod_{i \in \{1, \dots, n\}} V_i\right)\right) = V_1 \odot \cdots \odot V_n$$

4. Using [theorem: 13.6] we have that

$$\delta\left(\prod_{i \in \{1, \dots, n\}} V_i\right) = \left\{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\right\} \text{ is a basis for } \mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i\right)$$

and

$$\delta: \prod_{i \in \{1, \dots, n\}} V_i \rightarrow \delta\left(\prod_{i \in \{1, \dots, n\}} V_i\right) = \left\{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\right\} \text{ is a bijection}$$

Let $x \in \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i)$ then there exist a finite $I \subseteq \{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\}$ and a $\{\alpha_i\}_{i \in I} \subseteq F$ such that $x = \sum_{i \in I} \alpha_i \cdot i$. Assume that there exists another finite $J \subseteq \{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\}$ and $\{\beta_i\}_{i \in J} \subseteq F$ such that $x = \sum_{i \in J} \beta_i \cdot i$. Then by [theorem: 11.140] we have

$$\forall i \in I \setminus J \text{ that } \alpha_i = 0, \forall j \in J \setminus I \text{ that } \beta_j = 0 \text{ and } \forall i \in I \cap J \text{ that } \alpha_i = \beta_i \quad (13.7)$$

Hence

$$\begin{aligned} \sum_{i \in I} \alpha_i \cdot \varphi(\delta^{-1}(i)) &= \sum_{i \in I \setminus J} \alpha_i \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in I \cap J} \alpha_i \cdot \varphi(\delta^{-1}(i)) \\ &\stackrel{[\text{eq: 13.7}]}{=} \sum_{i \in I \setminus J} 0 \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in I \cap J} \beta_i \cdot \varphi(\delta^{-1}(i)) \\ &\stackrel{[\text{eq: 13.7}]}{=} \sum_{i \in I \setminus J} \beta_i \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in I \cap J} \beta_i \cdot \varphi(\delta^{-1}(i)) \\ &= \sum_{i \in I} \beta_i \cdot \varphi(\delta^{-1}(i)) \end{aligned}$$

The above ensures that the following function is well defined

$$\gamma: \mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right) \rightarrow U \text{ by } \gamma(x) = \sum_{i \in I} \alpha_i \cdot \varphi(\delta^{-1}(i)) \text{ where } x = \sum_{i \in I} \alpha_i \cdot i, I \text{ is finite}$$

If $x \in \prod_{i \in \{1, \dots, n\}} V_i$ then if we define $\{\alpha_i\}_{i \in \{\delta(x)\}} \subseteq F$ by $\alpha_{\delta(x)} = 1$ we have

$$\sum_{i \in \{\delta(x)\}} \alpha_i \cdot i = \alpha_{\delta(x)} \cdot \delta(x) = 1 \cdot \delta(x) = \delta(x)$$

so that

$$(\gamma \circ \delta)(x) = \gamma(\delta(x)) = \sum_{i \in \{\delta(x)\}} \alpha_i \cdot \varphi(\delta^{-1}(i)) = \alpha_{\delta(x)} \cdot \varphi(\delta^{-1}(\delta(x))) = 1 \cdot \varphi(x) = \varphi(x)$$

proving that

$$\gamma \circ \delta = \varphi \quad (13.8)$$

Let $x, y \in \mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i, F)$ and $\alpha \in F$ then there exists finite $I, J \subseteq \{\delta(a) | a \in \prod_{i \in \{1, \dots, n\}} V_i\}$, $\{\alpha_i\}_{i \in I} \subseteq F$ and $\{\beta_i\}_{i \in J} \subseteq F$ with $x = \sum_{i \in I} \alpha_i \cdot i$ and $y = \sum_{i \in J} \beta_i \cdot i$. Define then

$$\{\gamma_i\}_{i \in I \cup J} \subseteq F \text{ by } \gamma_i = \begin{cases} \alpha_i & \text{if } i \in I \setminus J \\ \alpha \cdot \beta_i & \text{if } i \in J \setminus I \\ \alpha_i + \alpha \cdot \beta_i & \text{if } i \in I \cap J \end{cases}$$

then we have

$$\begin{aligned} \sum_{i \in I \cup J} \gamma_i \cdot i &= \sum_{i \in I \setminus J} \gamma_i \cdot i + \sum_{i \in J \setminus I} \gamma_i \cdot i + \sum_{i \in I \cap J} \gamma_i \cdot i \\ &= \sum_{i \in I \setminus J} \alpha_i \cdot i + \sum_{i \in J \setminus I} \alpha \cdot \beta_i \cdot i + \sum_{i \in I \cap J} (\alpha_i + \alpha \cdot \beta_i) \cdot i \\ &= \sum_{i \in I \setminus J} \alpha_i \cdot i + \alpha \cdot \sum_{i \in J \setminus I} \beta_i \cdot i + \sum_{i \in I \cap J} \alpha_i \cdot i + \alpha \cdot \sum_{i \in I \cap J} \beta_i \cdot i \\ &= \left(\sum_{i \in I \setminus J} \alpha_i \cdot i + \sum_{i \in I \cap J} \alpha_i \cdot i \right) + \alpha \cdot \left(\sum_{i \in J \setminus I} \beta_i \cdot i + \sum_{i \in I \cap J} \beta_i \cdot i \right) \\ &= \sum_{i \in I \cup J} \alpha_i \cdot i + \sum_{i \in I \cup J} \beta_i \cdot i \\ &= x + \alpha \cdot y \end{aligned}$$

so that by definition of γ

$$\begin{aligned} \gamma(x + \alpha \cdot y) &= \sum_{i \in I \cup J} \gamma_i \cdot \varphi(\delta^{-1}(i)) \\ &= \sum_{i \in I \setminus J} \gamma_i \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in J \setminus I} \gamma_i \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in I \cap J} \gamma_i \cdot \varphi(\delta^{-1}(i)) = \\ &= \sum_{i \in I \setminus J} \alpha_i \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in J \setminus I} \alpha \cdot \beta_i \cdot \varphi(\delta^{-1}(i)) + \sum_{i \in I \cap J} (\alpha_i + \alpha \cdot \beta_i) \cdot \varphi(\delta^{-1}(i)) = \\ &= \underbrace{\sum_{i \in I \setminus J} \alpha_i \cdot \varphi(\delta^{-1}(i))}_{(1)} + \underbrace{\alpha \cdot \sum_{i \in J \setminus I} \beta_i \cdot \varphi(\delta^{-1}(i))}_{(2)} + \underbrace{\sum_{i \in I \cap J} \alpha_i \cdot \varphi(\delta^{-1}(i))}_{(1)} + \\ &= \underbrace{\alpha \cdot \sum_{i \in I \cap J} \beta_i \cdot \varphi(\delta^{-1}(i))}_{(2)} = \\ &= \underbrace{\sum_{i \in I} \alpha_i \cdot \varphi(\delta^{-1}(i))}_{(1)} + \underbrace{\alpha \cdot \sum_{i \in I} \beta_i \cdot \varphi(\delta^{-1}(i))}_{(2)} = \\ &= \gamma(x) + \alpha \cdot \gamma(y) \end{aligned}$$

proving that γ is linear, hence

$$\gamma \in \text{Hom}\left(\mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right), U\right) \quad (13.9)$$

Let $x \in \mathcal{N}(\prod_{i \in \{1, \dots, n\}} V_i, F) \subseteq \mathcal{F}(\prod_{i \in I} V_i, F)$ then by [definition: 13.7] there exist a $i \in \{1, \dots, n\}$, $v \in \prod_{i \in \{1, \dots, n\} \setminus \{i\}} V_i$, $x, y \in V_i$ and $\alpha \in F$ such that

$$x = \delta_{[i \mapsto x + \alpha \cdot y]v} - \delta_{[i \mapsto x]v} - \alpha \cdot \delta_{[i \mapsto y]v}$$

Then

$$\begin{aligned}
& \gamma(x) = \\
& \gamma(\delta_{[i \rightarrow x + \alpha \cdot y]v} - \delta_{[i \rightarrow x]v} - \alpha \cdot \delta_{[i \rightarrow y]v}) \stackrel{[\text{eq: 13.9}]}{=} \\
& \gamma(\delta_{[i \rightarrow x + \alpha \cdot y]v}) - \gamma(\delta_{[i \rightarrow x]v}) - \alpha \cdot \gamma(\delta_{[i \rightarrow y]v}) = \\
& \gamma(\delta([i \rightarrow x + \alpha \cdot y]v)) - \gamma(\delta([i \rightarrow x]v)) - \alpha \cdot \gamma(\delta([i \rightarrow y]v)) \stackrel{[\text{eq: 13.8}]}{=} \\
& \varphi([i \rightarrow x + \alpha \cdot y]v) - \varphi([i \rightarrow x]v) - \alpha \cdot \varphi([i \rightarrow y]v) = \\
& \varphi(v_1, \dots, v_{i-1}, x + \alpha \cdot y, v_{i+1}, \dots, v_n) - \varphi(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_n) - \\
& \alpha \cdot \varphi(v_1, \dots, v_{i-1}, y, v_{i+1}, \dots, v_n) \stackrel{\varphi \in \text{Hom}(V_1, \dots, V_n; U)}{=} \\
& 0 =
\end{aligned}$$

so that $\mathcal{N}(\prod_{i \in \{1, \dots, n\}} V_i, F) \subseteq \ker(\gamma)$. By [theorem: 11.193] $\ker(\gamma)$ is a sub-space, hence using [theorems: 11.109, 11.111] we have

$$\text{span}\left(\mathcal{N}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right)\right) \subseteq \ker(\gamma) \quad (13.10)$$

Finally [eqs: 13.9 and 13.10] allows us to apply [theorem: 13.12] to find a unique

$$h \in \text{Hom}\left(\mathcal{F}\left(\prod_{i \in \{1, \dots, n\}} V_i\right) / \text{span}\left(\mathcal{N}\left(\prod_{i \in \{1, \dots, n\}} V_i, F\right)\right), U\right)$$

such that

$$\gamma = h \circ \pi$$

As $\mathcal{F}(\prod_{i \in \{1, \dots, n\}} V_i) / \text{span}(\mathcal{N}(\prod_{i \in \{1, \dots, n\}} V_i, F)) \stackrel{\text{def}}{=} V_1 \odot \dots \odot V_n$ we have

$$\exists! h \in \text{Hom}(V_1 \odot \dots \odot V_n, U) \text{ with } \gamma = h \circ \pi$$

Using then [eq: 13.8] we have that $\varphi = \gamma \circ \delta = (h \circ \pi) \circ \delta = h \circ (\pi \circ \rho) \stackrel{[\text{definition: 13.10}]}{=} h \circ \odot$ giving:

$$\varphi = h \circ \odot \text{ where } h \in \text{Hom}(V_1 \odot \dots \odot V_n, U) \quad \square$$

So given a family $\{V_i\}_{i \in \{1, \dots, n\}}$ of vector spaces over a field we have proved that there exist a tensor product of $V_1, \dots, V_n$. Next we prove that this vector space is uniquely determined within a canonical linear isomorphism.

Theorem 13.15. *Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of vector space over a field F then if $\langle P, \nu \rangle$ and $\langle Q, \mu \rangle$ are tensor products of $V_1, \dots, V_n$ then there exists a **unique** linear isomorphism*

$$k: P \rightarrow Q$$

such that

$$k \circ \nu = \mu$$

Proof. As $\mu \in \text{Hom}(V_1, \dots, V_n; Q)$ we have, as $\langle P, \nu \rangle$ is a tensor product of $V_1, \dots, V_n$, that there exist a $k \in \text{Hom}(P, Q)$ such that

$$k \circ \nu = \mu \quad (13.11)$$

Likewise, as $\nu \in \text{Hom}(V_1, \dots, V_n; P)$ we have, as $\langle Q, \mu \rangle$ is a tensor product of $V_1, \dots, V_n$, that there exist a $h \in \text{Hom}(Q, P)$ such that

$$h \circ \mu = \nu \quad (13.12)$$

Next we have

$$\mu \stackrel{[\text{eq: 13.11}]}{=} k \circ \nu \stackrel{[\text{eq: 13.12}]}{=} k \circ (h \circ \mu) = (k \circ h) \circ \mu \text{ and } \nu \stackrel{[\text{eq: 13.12}]}{=} h \circ \mu \stackrel{[\text{eq: 13.11}]}{=} h \circ (k \circ \nu) = (h \circ k) \circ \nu \quad (13.13)$$

If $y \in P$ then as $P = \text{span}(\nu(\prod_{i \in \{1, \dots, n\}} V_i))$ there exist by [theorem: 11.101] a $\{\alpha_i\}_{i \in \{1, \dots, k\}} \subseteq F$ and $\{y_i\}_{i \in \{1, \dots, k\}} \subseteq \nu(\prod_{i \in \{1, \dots, n\}} V_i)$ such that $y = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot y_i$. As $\{y_i\}_{i \in \{1, \dots, k\}} \subseteq \nu(\prod_{i \in \{1, \dots, n\}} V_i)$ we have by a consequence of the Axiom of Choice [see theorem: 3.138] that there exist a $\{x_i\}_{i \in \{1, \dots, k\}} \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ such that $\forall i \in \{1, \dots, k\} \nu(x_i) = y_i$. Let

$$x = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot x_i \in \prod_{i \in \{1, \dots, n\}} V_i \text{ [a vector space by [theorem: 11.73]]}$$

then we have

$$\nu(x) = \nu\left(\sum_{i \in \{1, \dots, k\}} \alpha_i \cdot x_i\right) = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot \nu(x_i) = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot y_i = y$$

or

$$\forall y \in P \text{ there exist a } x \in \prod_{i \in \{1, \dots, n\}} V_i \text{ such that } \nu(x) = y \quad (13.14)$$

Now if $y \in P$ we have

$$\begin{aligned} (h \circ k)(y) &\stackrel{[\text{eq: 13.14}]}{=} (h \circ k)(\nu(x)) \\ &= ((h \circ k) \circ \nu)(x) \\ &\stackrel{[\text{eq: 13.13}]}{=} \nu(x) \\ &\stackrel{[\text{eq: 13.14}]}{=} y \\ &= \text{Id}_P(y) \end{aligned}$$

proving that $\forall y \in P$ we have $(h \circ k)(y) = \text{Id}_P(y)$ or

$$h \circ k = \text{Id}_P \quad (13.15)$$

If $y \in Q$ then as $Q = \text{span}(\mu(\prod_{i \in \{1, \dots, n\}} V_i))$ there exist by [theorem: 11.101] a $\{\alpha_i\}_{i \in \{1, \dots, k\}} \subseteq F$ and $\{y_i\}_{i \in \{1, \dots, k\}} \subseteq \mu(\prod_{i \in \{1, \dots, n\}} V_i)$ such that $y = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot y_i$. As $\{y_i\}_{i \in \{1, \dots, k\}} \subseteq \mu(\prod_{i \in \{1, \dots, n\}} V_i)$ we have by a consequence of the Axiom of Choice [see theorem: 3.138] that there exist a $\{x_i\}_{i \in \{1, \dots, k\}} \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ such that $\forall i \in \{1, \dots, k\} \mu(x_i) = y_i$. Let

$$x = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot x_i \in \prod_{i \in \{1, \dots, n\}} V_i \text{ [a vector space by [theorem: 11.73]]}$$

then we have

$$\mu(x) = \mu\left(\sum_{i \in \{1, \dots, k\}} \alpha_i \cdot x_i\right) = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot \mu(x_i) = \sum_{i \in \{1, \dots, k\}} \alpha_i \cdot y_i = y$$

or

$$\forall y \in Q \text{ there exist a } x \in \prod_{i \in \{1, \dots, n\}} V_i \text{ such that } \mu(x) = y \quad (13.16)$$

Now

$$\begin{aligned} (k \circ h)(y) &\stackrel{[\text{eq: 13.16}]}{=} (k \circ h)(\mu(x)) \\ &= ((k \circ h) \circ \mu)(x) \\ &\stackrel{[\text{eq: 13.13}]}{=} \mu(x) \\ &\stackrel{[\text{eq: 13.16}]}{=} y \\ &= \text{Id}_Q(y) \end{aligned}$$

proving that $\forall y \in P$ we have $(h \circ k)(y) = \text{Id}_P(y)$ or

$$k \circ h = \text{Id}_Q \quad (13.17)$$

From [eqs: 13.15, 13.17] it follows, using [theorem: 2.73], that

$$k: Q \rightarrow P \text{ is a bijection}$$

and as k is linear we have together with [eq: 13.11] that

$$k \text{ is a linear isomorphism and } k \circ \nu = \mu$$

Next we must prove that $k: Q \rightarrow P$ is unique. So assume that $k' \in \text{Hom}(P, Q)$ such that $k' \circ \nu = \mu$, then by [eq: 13.11]

$$k' \circ \nu = k \circ \nu \quad (13.18)$$

Let $y \in P$ then by [eq: 13.14] there exists a $x \in \prod_{i \in \{1, \dots, n\}} V_i$ such that $y = \nu(x)$ hence

$$k'(y) = k'(\nu(x)) = (k' \circ \nu)(x) \stackrel{[\text{eq: 13.18}]}{=} (k \circ \nu)(x) = k(\nu(x)) = k(y)$$

proving that

$$k = k' \quad \square$$

Corollary 13.16. *Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of vector space over a field F and $\langle \nu, Q \rangle$ a tensor product of $V_1, \dots, V_n$ then*

1. *There exist a **unique** linear isomorphism*

$$k: V_1 \odot \cdots \odot V_n \rightarrow Q$$

such that

$$k \circ \odot = \nu$$

2. *There exist a **unique** linear isomorphism*

$$h: Q \rightarrow V_1 \odot \cdots \odot V_n$$

such that

$$h \circ \nu = \odot$$

Proof. Using [theorem: 13.13] $\langle V_1 \odot \cdots \odot V_n, \odot \rangle$ is a tensor product of $V_1, \dots, V_n$.

1. By [theorem: 13.15] there exist a **unique** linear isomorphism

$$k: V_1 \odot \cdots \odot V_n \rightarrow Q$$

such that

$$k \circ \odot = \nu$$

2. By [theorem: 13.15] there exist a **unique** linear isomorphism

$$h: Q \rightarrow V_1 \odot \cdots \odot V_n$$

such that

$$h \circ \nu = \odot \quad \square$$

Theorem 13.17. *Let V be a vector space over a field F and $\{V_i\}_{i \in \{1\}}$ the family defined by $V_1 = V$ then if we define*

$$\mu: \prod_{i \in \{1\}} V_i \rightarrow V \text{ by } \mu((v)) = v$$

then we have:

1. $\langle V, \mu \rangle$ is a tensor product of $V_1, \dots, V_1$
2. If $\langle U, \nu \rangle$ is another tensor product of $V_1, \dots, V_1$ then there exist a linear isomorphism

$$k: V \rightarrow U$$

such that

$$\nu: \prod_{i \in \{1\}} V_i \rightarrow U \text{ is defined by } \nu((v)) = k(v)$$

3. If W is a sub-space of V and $\{W_i\}_{i \in \{1\}} \subseteq \mathcal{P}(V)$ is defined by $W_i = W$ then

$\langle k(W), \nu|_{\prod_{i \in \{1\}} W_i} \rangle$ is a tensor product and $k(W)$ is a sub-space of U

4. If $\{W_i\}_{i \in \{1, \dots, n\}}$ is a family of sub-spaces of V so that $V = \bigoplus_{i \in \{1, \dots, n\}} W_i$ then we have, if we define for $i \in \{1, \dots, n\}$ $\{W_j^i\}_{j \in \{1\}}$ by $W_1^i = W_i$ that

$$U = \bigoplus_{i \in \{1, \dots, n\}} k(W_i)$$

and $\forall i \in \{1, \dots, n\}$

$\langle k(W_i), \otimes|_{\prod_{j \in \{1\}} W_j^i} \rangle$ is a tensor product of $W_1^i, \dots, W_1^i$ and $k(W_i)$ is a sub-space of U

Proof.

1. If $u, v \in V_1, \alpha \in F$ then trivially $\mu((u + \alpha \cdot v)) = u + \alpha \cdot v = \mu((u)) + \alpha \cdot \mu((v))$, so we have

$$\mu \in \text{Hom}(V_1, \dots, V_1; V) \quad (13.19)$$

Let $v \in V$ then for $(v) \in \prod_{i \in \{1\}} V_i$ we have $\mu((v)) = v$ so that $V \subseteq \mu(\prod_{i \in \{1\}} V_i)$ which, as $\mu(\prod_{i \in \{1\}} V_i) \subseteq V$, proves that $V = \mu(\prod_{i \in \{1\}} V_i)$. Using [theorem: 11.111] we have then

$$V = \text{span} \left(\mu \left(\prod_{i \in \{1, \dots, 1\}} V_i \right) \right) \quad (13.20)$$

Let $\varphi \in \text{Hom}(V_1, \dots, V_1; X)$ and define

$$h: V \rightarrow X \text{ by } h(v) = \varphi((v))$$

then for $u, v \in V$ and $\alpha \in F$ we have

$$h(u + \alpha \cdot v) = \varphi((u + \alpha \cdot v)) = \varphi((u)) + \alpha \cdot \varphi((v)) = h(u) + \alpha \cdot h(v)$$

proving that

$$h \in \text{Hom}(V, X) \quad (13.21)$$

If $x = (v) \in \prod_{i \in \{1\}} V_i$ then

$$(h \circ \mu)(x) = h(\mu((v))) = h(v) = \varphi((v)) = \varphi(x)$$

resulting in

$$h \circ \mu = \varphi \quad (13.22)$$

By [eqs: 13.19, 13.20, 13.21 and 13.22] we have by the definition of the tensor product that

$$\langle V, \mu \rangle \text{ is a tensor product of } \{V_i\}_{i \in \{1\}} \quad (13.23)$$

2. If $\langle U, \nu \rangle$ is another tensor product of $\{V_i\}_{i \in \{1\}}$ then by [theorem: 13.15] there exist a **unique** linear isomorphism $k: V \rightarrow U$ such that $k \circ \mu = \nu$. Let $(v) \in \prod_{i \in \{1\}} V_i$ then

$$\nu((v)) = k(\mu((v))) = k(v)$$

so that

$$\nu: \prod_{i \in \{1\}} V_i \rightarrow U \text{ is defined by } \nu((v)) = k(v) \quad (13.24)$$

3. As W is a sub-space of V we have by [theorem: 11.193] and the fact that $k: V \rightarrow U$ is a isomorphism, hence a linear mapping, that

$$k(W) \text{ is a sub-space of } U \quad (13.25)$$

If $v \in \nu(\prod_{i \in \{1\}} W_i)$ then there exist a $(w) \in \prod_{i \in \{1\}} W_i \subseteq \prod_{i \in \{1\}} V_i$ such that

$$v = \nu((w)) \underset{\text{eq: 13.24}}{=} k(w) \in k(W)$$

proving that

$$\nu\left(\prod_{i \in \{1\}} W_i\right) \subseteq k(W), \quad (13.26)$$

Using [theorem: 11.109] on the above gives

$$\text{span}\left(\nu\left(\prod_{i \in \{1\}} W_i\right)\right) \subseteq \text{span}(k(W)) \stackrel{[\text{eq: 13.25}] \wedge [\text{theorem: 11.111}]}{=} k(W) \quad (13.27)$$

If $v \in k(W)$ then there exist a $w \in W \Rightarrow (w) \in \prod_{i \in \{1\}} W_i \subseteq \prod_{i \in \{1\}} V_i$ such that $v = k(w)$, hence by [eq: 13.24] $v = k(w) = \nu((w)) \in \nu\left(\prod_{i \in \{1\}} W_i\right)$. Hence

$$k(W) \subseteq \nu\left(\prod_{i \in \{1\}} W_i\right) \subseteq \text{span}\left(\nu\left(\prod_{i \in \{1\}} W_i\right)\right) = \text{span}\left(\nu_{|\prod_{i \in \{1\}} W_i}\left(\prod_{i \in \{1\}} W_i\right)\right) \quad (13.28)$$

which combined with [eq: 13.27] results in

$$k(W) = \text{span}\left(\nu_{|\prod_{i \in \{1\}} W_i}\left(\prod_{i \in \{1\}} W_i\right)\right) \quad (13.29)$$

If $u, v \in W_1 = W$ and $\alpha \in F$ then

$$\nu_{|\prod_{i \in \{1\}} W_i}((u + \alpha \cdot v)) = \nu((u + \alpha \cdot v)) = \nu((u)) + \alpha \cdot \nu((v)) = \nu_{|\prod_{i \in \{1\}} W_i}(u) + \alpha \cdot \nu_{|\prod_{i \in \{1\}} W_i}(v)$$

proving that [using [eq: 13.26]]

$$\nu_{|\prod_{i \in \{1\}} W_i} \in \text{Hom}(W_1, \dots, W_1; k(W)) \quad (13.30)$$

If $\varphi \in \text{Hom}(W_1, \dots, W_1; Y)$ then by [theorem: 11.271] there exist a $\psi \in \text{Hom}(V_1, \dots, V_1; Y)$ such that $\psi_{|\prod_{i \in \{1\}} W_i} = \varphi$. As $\langle U, \nu \rangle$ is a tensor product of $V_1, \dots, V_1$ there exist a $h: U \rightarrow Y$ such that $h \circ \nu = \psi$, define $f: k(W) \rightarrow Y$ by $f = h_{|k(W)}$. Let $x \in \prod_{i \in \{1\}} W_i$ then

$$\begin{aligned} (f \circ \nu_{|\prod_{i \in \{1\}} W_i})(x) &= f(\nu_{|\prod_{i \in \{1\}} W_i}(x)) \\ &= f(\nu(x)) \\ &= h_{|k(W)}(\nu(x)) \\ &\stackrel{[\text{eq: 13.26}]}{=} h(\nu(x)) \\ &= \psi(x) \\ &\stackrel{x \in \prod_{i \in \{1\}} W_i}{=} \psi_{|\prod_{i \in \{1\}} W_i}(x) \\ &= \varphi(x) \end{aligned}$$

proving that

$$\text{If } \varphi \in \text{Hom}(W_1, \dots, W_1) \text{ then } f \circ \nu_{|\prod_{i \in \{1\}} W_i} = \varphi \quad (13.31)$$

Finally [eqs: 13.25, 13.29, 13.30 and 13.31] proves that

$$\langle k(W), \nu_{|\prod_{i \in \{1\}} W_i} \rangle \text{ is a tensor space of } W_1, \dots, W_1 \text{ and } k(W) \text{ is a sub-space of } U$$

4. First using (3) we have $\forall i \in \{1, \dots, n\}$

$$\langle k(W_i), \otimes_{|\prod_{j \in \{1\}} W_j^i} \rangle \text{ is a tensor product of } W_1^i, \dots, W_1^i \text{ and } k(W_i) \text{ is a sub-space of } U$$

Further, as $V = \bigoplus_{i \in \{1, \dots, n\}} W_i$ and $k: V \rightarrow U$ is an isomorphism, we have by [theorem: 12.16] that

$$U = \bigoplus_{i \in \{1, \dots, n\}} k(W_i) \quad \square$$

If $\{V_i\}_{i \in \{1, \dots, n\}}$ is a family of vector spaces over a field and $\{W_i\}_{i \in \{1, \dots, n\}}$ is a family of sub-spaces then a tensor product of $W_1, \dots, W_n$ is in general not a sub-space of a tensor product of $V_1, \dots, V_n$. However given a tensor product of $V_1, \dots, V_n$ we can find a tensor product that is a sub-space of the given tensor product. This was proved in the previous theorem for the case $n = 1$ however it is also valid in the general case.

Theorem 13.18. *Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F , $\{W_i\}_{i \in \{1, \dots, n\}}$ a family such that $\forall i \in \{1, \dots, n\}$ W_i is a sub-space of V_i and $\langle V, \nu \rangle$ a tensor product of $V_1, \dots, V_n$. Define*

$$W = \text{span} \left(\nu \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \right)$$

then

$$W \text{ is a sub-space of } V \text{ [hence a vector space]}$$

and

$$\langle W, \nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \rangle \text{ is a tensor product of } W_1, \dots, W_n$$

Proof. By [theorem: 2.149] $\prod_{i \in \{1, \dots, n\}} W_i \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ so that

$$\nu \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \subseteq \nu \left(\prod_{i \in \{1, \dots, n\}} V_i \right) \subseteq \text{span} \left(\nu \left(\prod_{i \in \{1, \dots, n\}} V_i \right) \right) = V$$

so that by [theorems: 11.110, 11.61]

$$W \stackrel{\text{definition}}{=} \text{span} \left(\nu \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \right) \text{ is a vector space that is a sub-space of } V$$

If $i \in \{1, \dots, n\}$, $x \in \prod_{j \in \{1, \dots, n\} \setminus \{i\}} W_j$, $u, v \in X_i$ and $\alpha \in F$ then we have

$$\begin{aligned} \nu|_{\prod_{i \in \{1, \dots, n\}} W_i}(x_1, \dots, x_{i-1}, u + \alpha \cdot v, x_{i+1}, \dots, x_n) &= \\ \nu(x_1, \dots, x_{i-1}, u + \alpha \cdot v, x_{i+1}, \dots, x_n) &= \\ \nu(x_1, \dots, x_{i-1}, u, x_{i+1}, \dots, x_n) + \alpha \cdot \nu(x_1, \dots, x_{i-1}, v, x_{i+1}, \dots, x_n) &= \\ \nu|_{\prod_{i \in \{1, \dots, n\}} W_i}(x_1, \dots, x_{i-1}, u, x_{i+1}, \dots, x_n) + \alpha \cdot \nu|_{\prod_{i \in \{1, \dots, n\}} W_i}(x_1, \dots, x_{i-1}, v, x_{i+1}, \dots, x_n) \end{aligned}$$

proving that

$$\nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \in \text{Hom}(W_1, \dots, W_n; W) \quad (13.32)$$

Next

$$\text{span} \left(\nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \right) = \text{span} \left(\nu \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \right) \stackrel{\text{def}}{=} W \quad (13.33)$$

Let U be a vector space over the field F and $\varphi \in \text{Hom}(W_1, \dots, W_n; U)$ then by [theorem: 11.271] there exist a $\psi \in \text{Hom}(V_1, \dots, V_n; U)$ such that $\psi|_{\prod_{i \in \{1, \dots, n\}} W_i} = \varphi$. As $\langle V, \nu \rangle$ is a tensor product there exist a $k \in \text{Hom}(V, U)$ such that $\psi = k \circ \nu$. Define $h = k|_W$ then we have $\forall x \in \prod_{i \in \{1, \dots, n\}} W_i$ that

$$\begin{aligned} (h \circ \nu|_{\prod_{i \in \{1, \dots, n\}} W_i})(x) &= h(\nu(x)) \\ &= h|_{W}(\nu(x)) \\ &= k(\nu(x)) \\ &= \psi(x) \\ &= \psi|_{\prod_{i \in \{1, \dots, n\}} W_i}(x) \\ &= \varphi(x) \end{aligned}$$

proving that

$$\varphi = h \circ \nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \quad (13.34)$$

From [eqs: 13.32, 13.33 and 13.34] we conclude that

$$\langle W, \nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \rangle \text{ is a tensor product of } W_1, \dots, W_n$$

Finally

As $\prod_{i \in \{1, \dots, n\}} W_i \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ we have

$$\nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \left(\prod_{i \in \{1, \dots, n\}} W_i \right) = \nu \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \subseteq \nu \left(\prod_{i \in \{1, \dots, n\}} V_i \right)$$

so that

$$W = \text{span} \left(\nu|_{\prod_{i \in \{1, \dots, n\}} W_i} \left(\prod_{i \in \{1, \dots, n\}} W_i \right) \right) \subseteq \text{span} \left(\nu \left(\prod_{i \in \{1, \dots, n\}} V_i \right) \right) = V$$

□

To simplify notation we use the following notation convention.

Convention 13.19. If $\{V_i\}_{i \in \{1, \dots, n\}}$ is a family of vector spaces over a field then a tensor product is a pair $\langle V, \mu \rangle$ where V is a vector space and $\mu \in \text{Hom}(V_1, \dots, V_n; V)$. We have now two conventions:

Focus on the vector space of a vector product.

The notation

$$V_1 \otimes \cdots \otimes V_n$$

means that $V_1 \otimes \cdots \otimes V_n$ is a vector space and there exist a (unspecified) multilinear mapping $\mu \in \text{Hom}(V_1, \dots, V_n; V_1 \otimes \cdots \otimes V_n)$ such that $\langle V_1 \otimes \cdots \otimes V_n, \mu \rangle$ is a tensor product of $V_1, \dots, V_n$. More specific it means that:

1. $V_1 \otimes \cdots \otimes V_n$ is a vector space over the same field F as (1)
2. There exist a $\mu \in \text{Hom}(V_1, \dots, V_n; V_1 \otimes \cdots \otimes V_n)$ such that $\langle V_1 \otimes \cdots \otimes V_n, \mu \rangle$ is a tensor product of $V_1, \dots, V_n$ or more in detail:

- a. $V_1 \otimes \cdots \otimes V_n = \text{span}(\mu(\prod_{i \in \{1, \dots, n\}} V_i))$

- b. If U is a vector space and $\varphi \in \text{Hom}(V_1, \dots, V_n; U)$ then there exist a

$$h: \text{Hom}(V_1 \otimes \cdots \otimes V_n, U)$$

such that

$$\varphi = h \circ \mu$$

If $n = 2$ then we use the shorter notation

$$V_1 \otimes V_2 \text{ for } V_1 \otimes \cdots \otimes V_n$$

We slightly abuse our definition and says that $V_1 \otimes \cdots \otimes V_n$ is a tensor product of $V_1, \dots, V_n$ when we actually should say that there exist a $\mu \in \text{Hom}(V_1, \dots, V_n; V_1 \otimes \cdots \otimes V_n)$ such that $\langle V_1 \otimes \cdots \otimes V_n, \mu \rangle$ is a tensor product of $V_1, \dots, V_n$. In many cases we also use the notation $\otimes$ for the assumed multilinear mapping that makes $V_1 \otimes \cdots \otimes V_n$ part of a tensor product.

One vector space and more then one multilinear mapping.

Then we use $V_1 \otimes \cdots \otimes V_n$ for the vector space and different symbols like $\otimes, \otimes^*, \otimes_1, \otimes_i \dots$ for the multilinear mapping. Examples

$$\langle V_1 \otimes \cdots \otimes V_n, \otimes \rangle$$

$$\langle V_1 \otimes \cdots \otimes V_n, \otimes^* \rangle$$

$$\langle V_1 \otimes \cdots \otimes V_n, \otimes_i \rangle$$

...

More than one vector space and multilinear mapping.

Then we use different symbols $\otimes, \circledast, \odot, \otimes_i \dots$ as shown in the following examples

$$\langle V_1 \otimes \cdots \otimes V_n, \otimes \rangle$$

$$\langle V_1 \circledast \cdots \circledast V_n, \circledast \rangle$$

$$\langle V_1 \otimes_i \cdots \otimes_i V_n, \otimes_i \rangle$$

Using the above convention we can rephrase [theorems: 13.13, 13.15, 13.17 and 13.18].

Theorem 13.20. *Let $\{V_i\}_{i \in \{1, \dots, n\}}$ be a family of vector spaces over a field then there exists a tensor product*

$$V_1 \otimes \cdots \otimes V_n$$

Proof. See [theorem: 13.13] and [convention: 13.19]. □

Theorem 13.21. *Let $\{V_i\}_{i \in \{1, \dots, n\}}$ be a family of vector spaces over a field F then for*

$$\langle V_1 \otimes \cdots \otimes V_n, \otimes \rangle, \langle V_1 \otimes \cdots \otimes V_n, \circledast \rangle \text{ two tensor products based on } V_1, \dots, V_n$$

then there exists a unique linear isomorphism

$$k: V_1 \otimes \cdots \otimes V_n \rightarrow V_1 \circledast \cdots \circledast V_n$$

such that

$$k \circ \otimes = \circledast$$

Proof. See [theorem: 13.15] and [convention: 13.19]. □

Theorem 13.22. *Let V be a vector space over a field F [defining the family of vector spaces $\{V_i\}_{i \in \{1\}}$ where $V_1 = V$] and $\langle V_1 \otimes \cdots \otimes V_1, \otimes \rangle$ a tensor product of $V_1, \dots, V_1$. If we define*

$$\otimes: \prod_{i \in \{1\}} V_i \rightarrow V \text{ by } \otimes((v)) = v$$

then we have:

1. $\langle V, \otimes \rangle$ is a tensor product of $V_1, \dots, V_1$ in other words $V_1 \otimes \cdots \otimes V_1 = V$
2. If $\langle V_1 \otimes \cdots \otimes V_n, \otimes \rangle$ is another tensor product of $V_1, \dots, V_1$ then there exists a isomorphism

$$k: V \rightarrow V_1 \otimes \cdots \otimes V_1$$

such that

$$\otimes: \prod_{i \in \{1\}} V_i \rightarrow V_1 \otimes \cdots \otimes V_1 \text{ we have } \otimes((v)) = k(v)$$

3. If W is a sub-space of V then for $\{W_i\}_{i \in \{1\}} \subseteq \mathcal{P}(V)$ defined by $W_1 = W$ we have that there exist a tensor product $\langle W_1 \otimes \cdots \otimes W_1, \otimes_{\prod_{i \in \{1\}} W_i} \rangle$ of $W_1, \dots, W_1$ such that

$$a. W_1 \otimes \cdots \otimes W_1 = k(W)$$

$$b. W_1 \otimes \cdots \otimes W_1 \text{ is a sub-space of } V$$

4. If $V = \bigoplus W_i$ then, if we define $\forall i \in \{1, \dots, n\} \{W_j^i\}_{j \in \{1\}}$ by $W_1^i = W_i$, we have that

$$\forall i \in \{1, \dots, n\} \text{ there exist a tensor product } W_1^i \otimes \cdots \otimes W_1^i \text{ of } W_1^i, \dots, W_1^i$$

such that

$$W_1^i \otimes \cdots \otimes W_1^i \text{ is a subspace of } V_1 \otimes \cdots \otimes V_1$$

and

$$V_1 \otimes \cdots \otimes V_1 = \bigoplus_{i \in \{1, \dots, n\}} W_1^i \otimes \cdots \otimes W_1^i$$

Proof. See [theorem: 13.17] and [convention: 13.19] $\square$

Theorem 13.23. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a family of vector spaces over a field F , $\{W_i\}_{i \in \{1, \dots, n\}}$ a family such that $\forall i \in \{1, \dots, n\}$ W_i is a sub-space of V_i . If $\langle V_1 \otimes \dots \otimes V_n, \otimes \rangle$ is a tensor product of $V_1, \dots, V_n$ then if we define $W_1 \otimes \dots \otimes W_n = \text{span}(\otimes(\prod_{i \in \{1, \dots, n\}} V_i))$ we have that

$$\langle W_1 \otimes \dots \otimes W_n, \otimes|_{\prod_{i \in \{1, \dots, n\}} W_i} \rangle \text{ is a tensor product of } W_1, \dots, W_n$$

and

$$W_1 \otimes \dots \otimes W_n \subseteq V_1 \otimes \dots \otimes V_n$$

Proof. See [theorem: 13.18] and [convention: 13.19] $\square$

We prove now some useful theorems about tensor products

Theorem 13.24. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of vector space over a field F , a tensor product $V_1 \otimes \dots \otimes V_n$ and $(v_1, \dots, v_n) \in \prod_{i \in \{1, \dots, n\}} V_i$ then we have

$$v_1 \otimes \dots \otimes v_n = 0 \Leftrightarrow \exists i \in \{1, \dots, n\} \ v_i = 0$$

Proof.

$\Rightarrow$. Let $v_1 \otimes \dots \otimes v_n = 0$. Assume $\forall i \in \{1, \dots, n\}$ we have $v_i \neq 0$ then by [theorem: 11.187] there exist a $L_i \in \text{Hom}(V_i, F)$ such that $L_i(v_i) = 1$ then by [theorem: 11.261] $\prod_{i \in \{1, \dots, n\}} L_i \in \text{Hom}(V_1 \otimes \dots \otimes V_n, F)$. As $\langle V_1 \otimes \dots \otimes V_n, \otimes \rangle$ is a tensor product there exist a $h \in \text{Hom}(P, F)$ such that $\prod_{i \in \{1, \dots, n\}} L_i = h \circ \otimes$. Then

$$\begin{aligned} 0 & \underset{\mu_{v_1 \otimes \dots \otimes v_n = 0}}{=} h(v_1 \otimes \dots \otimes v_n) \\ & = h(\otimes(v_1, \dots, v_n)) \\ & = (h \circ \otimes)(v_1, \dots, v_n) \\ & = \left(\prod_{i \in \{1, \dots, n\}} L_i \right)(v_1, \dots, v_n) \\ & = \prod_{i \in \{1, \dots, n\}} L_i(v_i) \\ & \underset{[\text{theorem: 11.37}]}{=} 1 \end{aligned}$$

giving the contradiction $0 = 1$. Hence we must have that $\exists i \in \{1, \dots, n\}$ such that $v_i = 0$.

$\Leftarrow$. This follows from [theorem: 11.259] and the fact that $\otimes \in \text{Hom}(V_1, \dots, V_n; V_1 \otimes \dots \otimes V_n)$ $\square$

Lemma 13.25. Let $\{V_i\}_{i \in \{1, 2\}}$ family of two vector spaces over a field f with a tensor product $\langle V_1 \otimes V_2, \otimes \rangle$. Take $z \in V_1 \otimes V_2$ and define

$$K_z = \left\{ k \in \mathbb{N} \mid \exists \{u_i\}_{i \in \{1, \dots, k\}} \subseteq V_1, \{v_i\}_{i \in \{1, \dots, k\}} \subseteq V \text{ such that } z = \sum_{i \in \{1, \dots, k\}} u_i \otimes v_i \right\}$$

then

$$K_z \neq 0$$

so that by [theorem: 10.19]

$$k_z = \min(K_z) \text{ exist}$$

Then for every $\{u_i\}_{i \in \{1, \dots, k_z\}} \subseteq V_1, \{v_i\}_{i \in \{1, \dots, k_z\}} \subseteq V$ such that

$$z = \sum_{i \in \{1, \dots, k_z\}} u_i \otimes v_i$$

we have then that

$$\{u_i\}_{i \in \{1, \dots, k_z\}} \text{ and } \{v_i\}_{i \in \{1, \dots, k_z\}} \text{ are linear independent}$$

Proof. Let $z \in V_1 \otimes V_1$. As by definition $V_1 \otimes V_2 = \text{span}(\otimes(\prod_{i \in \{1, \dots, n\}} V_i))$ there exists by [theorem: 11.107] a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ and $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $z = \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i$. As $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ their exist by [theorem: 3.138] a $\{(u'_i, v_i)\}_{i \in \{1, \dots, n\}} \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ such that $\forall i \in \{1, \dots, n\}$ we have $x_i = \otimes(u'_i, v_i)$. Define $\{u_i\}_{i \in \{1, \dots, n\}} \subseteq V_1$ by $u_i = \alpha_i \cdot u'_i$ then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, n\}} u_i \otimes v_i &= \sum_{i \in \{1, \dots, n\}} (\alpha_i \cdot u'_i) \otimes v_i \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot (u'_i \otimes v_i) \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot \otimes(u'_i, v_i) \\ &= \sum_{i \in \{1, \dots, n\}} \alpha_i \cdot x_i \\ &= z \end{aligned}$$

proving that $n \in K_z$ or

$$K_z \neq 0$$

So by [theorem: 10.19] $k_z = \min(K_z)$ exist. Let $\{u_i\}_{i \in \{1, \dots, k_z\}} \subseteq V_1$ and $\{v_i\}_{i \in \{1, \dots, k_z\}} \subseteq V_2$ such that $z = \sum_{i \in \{1, \dots, k_z\}} u_i \otimes v_i$ then we have:

1. Assume that $\{u_i\}_{i \in \{1, \dots, k_z\}}$ is linear dependent then by [theorem: 11.128] there exists a $l \in \{1, \dots, k_z\}$ and $\{\alpha_i\}_{i \in \{1, \dots, k_z\} \setminus \{l\}} \subseteq F$ such that $u_l = \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} \alpha_i \cdot u_i$. Define now

$$\{w_i\}_{i \in \{1, \dots, k_z\} \setminus \{l\}} \subseteq V_2 \text{ by } w_i = v_i + \alpha_i \cdot v_l$$

then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes w_i &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes (v_i + \alpha_i \cdot v_l) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} (u_i \otimes v_i + \alpha_i \cdot (u_i \otimes v_l)) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} \alpha_i \cdot (u_i \otimes v_l) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} ((\alpha_i \cdot u_i) \otimes v_l) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + \left(\sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} \alpha_i \cdot u_i \right) \otimes v_l \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + u_l \otimes v_l \\ &= \sum_{i \in \{1, \dots, k_z\}} u_i \otimes v_i \\ &= z \end{aligned}$$

Using [corollary: 10.94] we have that $\text{card}(\{1, \dots, k_z\} \setminus \{l\}) = k_z - 1$ so that there exist a bijection $\beta: \{1, \dots, k_z - 1\} \rightarrow \{1, \dots, k_z\} \setminus \{l\}$ hence if we define

$$\{u'_i\}_{i \in \{1, \dots, k_z - 1\}} \subseteq V_1 \text{ by } u'_i = u_{\beta(i)} \text{ and } \{v'_i\}_{i \in \{1, \dots, k_z - 1\}} \subseteq V_2 \text{ by } v'_i = w_{\beta(i)}$$

then we have

$$\sum_{i \in \{1, \dots, k_z - 1\}} u'_i \otimes v'_i = \sum_{i \in \{1, \dots, k_z - 1\}} u_{\beta(i)} \otimes w_{\beta(i)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes w_i = z$$

so that $k_z - 1 \in K_z$ contradicting $k_z = \min(K_z)$. Hence $\{u_i\}_{i \in \{1, \dots, k_z\}}$ is not linear dependent, proving that $\{u_i\}_{i \in \{1, \dots, k_z\}}$ is linear dependent.

2. Assume that $\{v_i\}_{i \in \{1, \dots, k_z\}}$ is linear dependent then by [theorem: 11.128] there exists a $l \in \{1, \dots, k_z\}$ and $\{\alpha_i\}_{i \in \{1, \dots, k_z\} \setminus \{l\}} \subseteq F$ such that $u_l = \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} \alpha_i \cdot v_i$. Define now

$$\{w_i\}_{i \in \{1, \dots, k_z\} \setminus \{l\}} \subseteq V_1 \text{ by } w_i = u_i + \alpha_i \cdot u_l$$

then we have

$$\begin{aligned} \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} w_i \otimes v_i &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} (u_i + \alpha_i \cdot u_l) \otimes v_i \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} (u_i \otimes v_i + \alpha_i \cdot (u_l \otimes v_i)) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} \alpha_i \cdot (u_l \otimes v_i) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} (u_l \otimes (\alpha_i \cdot v_i)) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + u_l \otimes \left(\sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} \alpha_i \cdot v_i \right) \\ &= \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} u_i \otimes v_i + u_l \otimes v_l \\ &= \sum_{i \in \{1, \dots, k_z\}} u_i \otimes v_i \\ &= z \end{aligned}$$

Using [corollary: 10.94] we have that $\text{card}(\{1, \dots, k_z\} \setminus \{l\}) = k_z - 1$ so that there exist a bijection $\beta: \{1, \dots, k_z - 1\} \rightarrow \{1, \dots, k_z\} \setminus \{l\}$ hence if we define

$$\{u'_i\}_{i \in \{1, \dots, k_z - 1\}} \subseteq V_1 \text{ by } u'_i = w_{\beta(i)} \text{ and } \{v'_i\}_{i \in \{1, \dots, k_z - 1\}} \subseteq V_2 \text{ by } v'_i = v_{\beta(i)}$$

then we have

$$\sum_{i \in \{1, \dots, k_z - 1\}} u'_i \otimes v'_i = \sum_{i \in \{1, \dots, k_z - 1\}} w_{\beta(i)} \otimes v_{\beta(i)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{1, \dots, k_z\} \setminus \{l\}} w_i \otimes v_i = z$$

so that $k_z - 1 \in K_z$ contradicting $k_z = \min(K_z)$. Hence $\{u_i\}_{i \in \{1, \dots, k_z\}}$ is not linear dependent, proving that $\{u_i\}_{i \in \{1, \dots, k_z\}}$ is linear dependent. $\square$

The following theorems will be important for induction arguments.

Theorem 13.26. Let $n \in \mathbb{N}$ and $\{V_i\}_{i \in \{1, \dots, n+1\}}$ be a family of vector spaces over F ,

$$\langle V_1 \otimes_n \dots \otimes_n V_n, \otimes_n \rangle \text{ a tensor product of } V_1, \dots, V_n$$

and

$$\langle (V_1 \otimes_n \dots \otimes_n V_n) \otimes V_{n+1}, \otimes \rangle \text{ a tensor product of } V_1 \otimes_n \dots \otimes_n V_n, V_{n+1}$$

then there exist a tensor product

$$\langle V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}, \otimes_{n+1} \rangle \text{ of } V_1, \dots, V_{n+1}$$

such that

$$V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1} = (V_1 \otimes_n \dots \otimes_n V_n) \otimes V_{n+1}$$

Proof. Let $V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}$ be the vector space $(V_1 \otimes_n \dots \otimes_n V_n) \otimes V_{n+1}$, so we have by definition

$$V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1} = (V_1 \otimes_n \dots \otimes_n V_n) \otimes V_{n+1}$$

Define now

$$\otimes_{n+1}: \prod_{i \in \{1, \dots, n+1\}} V_i \rightarrow (V_1 \otimes_n \dots \otimes_n V_n) \otimes V_{n+1} \text{ where } \otimes_{n+1} \text{ is defined by} \quad (13.35)$$

$$\otimes_{n+1}(v_1, \dots, v_{n+1}) = \otimes(\otimes_n(v_1, \dots, v_n), v_{n+1})$$

Let $i \in \{1, \dots, n+1\}$ and take $\{v_j\}_{j \in \{1, \dots, n+1\} \setminus \{i\}}$ $\alpha \in F$, $u, v \in V_i$ then we have for

$$\otimes_{n+1}(v_1, \dots, v_i, u + \alpha \cdot v, v_{i+1}, \dots, v_n)$$

either:

$i \in \{1, \dots, n\}$. Then

$$\begin{aligned} \otimes_{n+1}(v_1, \dots, v_i, u + \alpha \cdot v, v_{i+1}, \dots, v_n) &= \\ \otimes(\otimes_n(v_1, \dots, v_i, u + \alpha \cdot v, v_{i+1}, \dots, v_n), v_{n+1}) &= \\ \otimes(\otimes_n(v_1, \dots, v_i, u, v_{i+1}, \dots, v_n) + \alpha \cdot (v_1, \dots, v_i, v, v_{i+1}, \dots, v_n), v_{n+1}) &= \\ \otimes(\otimes_n(v_1, \dots, v_i, u, v_{i+1}, \dots, v_n), v_{n+1}) + \alpha \cdot \otimes(\otimes_n(v_1, \dots, v_i, v, v_{i+1}, \dots, v_n), v_{n+1}) &= \\ \otimes_{n+1}(v_1, \dots, v_i, u, v_{i+1}, \dots, v_n) + \alpha \cdot \otimes_{n+1}(v_1, \dots, v_i, v, v_{i+1}, \dots, v_n) \end{aligned}$$

$i = n+1$. Then

$$\begin{aligned} \otimes_{n+1}(v_1, \dots, v_i, u + \alpha \cdot v, v_{i+1}, \dots, v_n) &= \\ \otimes(\otimes_n(v_1, \dots, v_n), u + \alpha \cdot v) &= \\ \otimes(\otimes_n(v_1, \dots, v_n), u) + \alpha \cdot \otimes(\otimes_n(v_1, \dots, v_n), v) &= \\ \otimes_{n+1}(v_1, \dots, v_i, u, v_{i+1}, \dots, v_n) + \alpha \cdot \otimes_{n+1}(v_1, \dots, v_i, v, v_{i+1}, \dots, v_n) \end{aligned}$$

which proves that

$$\otimes_{n+1} \in \text{Hom}(V_1, \dots, V_{n+1}; (V_1 \otimes_n \dots \otimes_n V_n) \otimes V_{n+1}) \quad (13.36)$$

If $v \in \otimes((\otimes(\prod_{i \in \{1, \dots, n\}} V_i)) \cdot V_{n+1})$ then there exists a $v_1 \in \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ and $v_2 \in V_{n+1}$ such that $v = \otimes(v_1, v_2)$. As $v_1 \in \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ there exist a $(v_{1,1}, \dots, v_{1,n}) \in \prod_{i \in \{1, \dots, n\}} V_i$ such that $v_1 = \otimes_n(v_{1,1}, \dots, v_{1,n})$. So

$$v = \otimes(v_1, v_2) = \otimes(\otimes_n(v_{1,1}, \dots, v_{1,n}), v_2) \stackrel{\text{def}}{=} \otimes_{n+1}(v_{1,1}, \dots, v_{1,n}, v_2) \in \otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right)$$

Hence we have that

$$\otimes\left(\left(\otimes_n\left(\prod_{i \in \{1, \dots, n\}} V_i\right)\right) \cdot V_{n+1}\right) \subseteq \otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right) \quad (13.37)$$

If $v \in \otimes((V_1 \otimes_n \dots \otimes_n V_n), V_{n+1})$ then there exists a $v_1 \in V_1 \otimes_n \dots \otimes_n V_n = \text{span}(\otimes_n(\prod_{i \in \{1, \dots, n\}} V_i))$ and $v_2 \in V_{n+1}$ such that $v = \otimes(v_1, v_2)$. As $v_1 \in \text{span}(\otimes_n(\prod_{i \in \{1, \dots, n\}} V_i))$ there exist a finite $I \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ and a $\{\alpha_u\}_{u \in I} \subseteq F$ such that $v_1 = \sum_{u \in I} \alpha_u \cdot u$. So

$$\begin{aligned} v &= \otimes(v_1, v_2) \\ &= \otimes\left(\sum_{u \in I} \alpha_u \cdot u, v_2\right) \\ &= \sum_{u \in I} \alpha_u \cdot \otimes(u, v_2) \\ &\subseteq \text{span}\left(\otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right)\right) \text{ [see eq: 13.37]} \end{aligned}$$

proving

$$\otimes((V_1 \otimes_n \dots \otimes_n V_n), V_{n+1}) \subseteq \text{span}\left(\otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right)\right)$$

Using [theorems: 11.109, 11.112] we have

$$(V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1} = \text{span}(\otimes((V_1 \otimes_n \cdots \otimes_n V_n) \cdot V_{n+1})) \subseteq \text{span}\left(\otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right)\right) \quad (13.38)$$

By the definition of $\otimes_{n+1}$ [see eq: 13.35]

$$\otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right) \subseteq (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}$$

hence by [theorems: 11.109]

$$\text{span}\left(\otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right)\right) \subseteq \text{span}((V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}) = (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}$$

Combining the above with [eq: 13.38] gives

$$\text{span}\left(\otimes_{n+1}\left(\prod_{i \in \{1, \dots, n+1\}} V_i\right)\right) = (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1} \quad (13.39)$$

Let $\varphi \in \text{Hom}(V_1, \dots, V_{n+1}; U)$. Given $v \in V_{n+1}$ define

$$\varphi_v: \prod_{i \in \{1, \dots, n\}} V_i \rightarrow U \text{ by } \varphi_v(v_1, \dots, v_n) = \varphi(v_1, \dots, v_n, v)$$

then φ_v is multilinear because φ is multilinear. As $\langle V_1 \otimes_n \cdots \otimes_n V_n, \otimes_n \rangle$ is a tensor product there exist a $k_v \in \text{Hom}(V_1 \otimes_n \cdots \otimes_n V_n, U)$ such that

$$k_v \circ \otimes_n = \varphi_v \quad (13.40)$$

Let $u, v \in V_{n+1}$, $\alpha \in F$ and $w \in \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ then there exist a $(v_1, \dots, v_n)$ such that $w = \otimes_n(v_1, \dots, v_n)$. Further

$$\begin{aligned} k_{u+\alpha \cdot v}(w) &= k_{u+\alpha \cdot v}(\otimes(v_1, \dots, v_n)) \\ &= \varphi_{u+\alpha \cdot v}(v_1, \dots, v_n) \\ &= \varphi(v_1, \dots, v_n, u + \alpha \cdot v) \\ &= \varphi(v_1, \dots, v_n, u) + \alpha \cdot \varphi(v_1, \dots, v_n, v) \\ &= \varphi_u(v_1, \dots, v_n) + \alpha \cdot \varphi_v(v_1, \dots, v_n) \\ &= k_u(\otimes_n(v_1, \dots, v_n)) + \alpha \cdot k_v(\otimes_n(v_1, \dots, v_n)) \\ &= k_u(w) + \alpha \cdot k_v(w) \end{aligned} \quad (13.41)$$

Let $w \in \text{span}(\otimes(\prod_{i \in \{1, \dots, n\}} V_i))$ then there exists a finite set $I \subseteq \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ and $\{\alpha_i\}_{i \in I} \subseteq F$ such that

$$w = \sum_{z \in I} \alpha_z \cdot z$$

Then

$$\begin{aligned} k_{u+\alpha \cdot v}(w) &= k_{u+\alpha \cdot v}\left(\sum_{z \in I} \alpha_z \cdot z\right) \\ &= \sum_{z \in I} \alpha_z \cdot k_{u+\alpha \cdot v}(z) \\ &= \sum_{z \in I} \alpha_z \cdot (k_u(z) + \alpha \cdot k_v(z)) \\ &= \sum_{z \in I} \alpha_z \cdot k_u(z) + \alpha \cdot \sum_{z \in I} \alpha_z \cdot k_v(z) \\ &= k_u(w) + \alpha \cdot k_v(w) \end{aligned}$$

proving that

$$\forall u, v \in V_{n+1}, \forall \alpha \in F \text{ we have } k_{u+\alpha \cdot v} = k_u + \alpha \cdot k_v \quad (13.42)$$

Define

$$\psi: (V_1 \otimes_n \cdots \otimes_n V_n) \cdot V_{n+1} \rightarrow U \text{ by } \psi(u, v) = k_v(u) \quad (13.43)$$

then we have for $x, y \in V_1 \otimes_n \cdots \otimes_n V_n$, $\alpha \in F$ and $v \in V_{n+1}$

$$\psi(x + \alpha \cdot y, v) = k_v(x + \alpha \cdot y) = k_v(x) + \alpha \cdot k_v(y) = \psi(x) + \alpha \cdot \psi(y)$$

and for $v \in V_1 \otimes_n \cdots \otimes_n V_n$, $x, y \in V_{n+1}$, $\alpha \in F$ that

$$\psi(v, x + \alpha \cdot y) = k_{x+\alpha \cdot y}(v) \stackrel{[\text{eq: 13.42}]}{=} k_x(v) + \alpha \cdot k_y(v) = \psi(v, x) + \alpha \cdot \psi(v, y)$$

From the above it follows that

$$\psi \in \text{Hom}(V_1 \otimes_n \cdots \otimes_n V_n, V_{n+1}; U) \quad (13.44)$$

As $\langle (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}, \otimes \rangle$ is a tensor product there exist a $h \in \text{Hom}((V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}, U)$ such that

$$h \circ \otimes = \psi \quad (13.45)$$

Let $(v_1, \dots, v_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} V_i$ then we have

$$\begin{aligned} (h \circ \otimes_{n+1})(v_1, \dots, v_{n+1}) &= h(\otimes_{n+1}(v_1, \dots, v_{n+1})) \\ &= h(\otimes(\otimes_n(v_1, \dots, v_n), v_{n+1})) \\ &\stackrel{[\text{eq: 13.45}]}{=} \psi(\otimes_n(v_1, \dots, v_n), v_{n+1}) \\ &\stackrel{[\text{eq: 13.43}]}{=} k_{v_{n+1}}(\otimes_n(v_1, \dots, v_n)) \\ &\stackrel{[\text{eq: 13.40}]}{=} \varphi_{v_{n+1}}(v_1, \dots, v_n) \\ &= \varphi(v_1, \dots, v_n, v_{n+1}) \end{aligned}$$

which proves that

$$h \circ \otimes_{n+1} = \varphi \quad (13.46)$$

Finally from [eqs: 13.39, 13.36 and 13.46] it follows that

$$\langle (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}, \otimes_{n+1} \rangle \text{ is a tensor product of } V_1, \dots, V_{n+1} \quad \square$$

Theorem 13.27. Let $n \in \mathbb{N}$ and $\{V_i\}_{i \in \{1, \dots, n+1\}}$ be a family of vector spaces over F . If

$$\langle V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}, \otimes_{n+1} \rangle \text{ and } \langle V_1 \otimes_n \cdots \otimes_n V_n, \otimes_n \rangle \text{ are tensor products}$$

then there exist a tensor product

$$\langle (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}, \otimes \rangle \text{ of } V_1 \otimes_n \cdots \otimes_n V_n \text{ and } V_{n+1}$$

such that

$$V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} = (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}$$

Proof. Take the family $\{W_i\}_{i \in \{1, 2\}}$ by $W_1 = V_1 \otimes_n \cdots \otimes_n V_n$ and $W_2 = V_{n+1}$ of vector spaces. Using [theorem: 13.13] there exist a tensor product $\langle P, \nu \rangle$ of $V_1 \otimes_n \cdots \otimes_n V_n$ and V_{n+1} , so

$$\nu \in \text{Hom}\left(\prod_{i \in \{1, 2\}} W_i; P\right) \quad (13.47)$$

$$P = \text{span}\left(\nu\left(\prod_{i \in \{1, 2\}} W_i\right)\right) \quad (13.48)$$

$$\forall \varphi \in \text{Hom}\left(\prod_{i \in \{1, 2\}} W_i; U\right) \text{ there exist a } h \in \text{Hom}(P, U) \text{ such that } \varphi = h \circ \nu \quad (13.49)$$

Let $w \in V_{n+1}$ and define

$$\beta_w: \prod_{i \in \{1, \dots, n\}} V_i \rightarrow V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1} \text{ by } \beta_w(v_1, \dots, v_n) = \otimes_{n+1}(v_1, \dots, v_n, w)$$

then for $i \in \{1, \dots, n\}$ and $\alpha \in F$, $u, v \in V_i$ we have

$$\begin{aligned} \beta_w(v_1, \dots, v_{i-1}, u + \alpha \cdot v, v_{i+1}, v_{i+1}, \dots, v_n) &= \\ \otimes_{n+1}(v_1, \dots, v_{i-1}, u + \alpha \cdot v, v_{i+1}, \dots, v_n, w) &= \\ \otimes_{n+1}(v_1, \dots, v_{i-1}, u, v_{i+1}, \dots, v_n, w) + \alpha \cdot \otimes_{n+1}(v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n, w) &= \\ \beta_w(v_1, \dots, v_{i-1}, u, v_{i+1}, v_{i+1}, \dots, v_n) + \alpha \cdot \beta_w(v_1, \dots, v_{i-1}, v, v_{i+1}, v_{i+1}, \dots, v_n) \end{aligned}$$

proving that

$$\forall w \in V_{n+1} \text{ we have } \beta_w \in \text{Hom}(V_1, \dots, V_n; V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}) \quad (13.50)$$

Further for $\alpha \in F$ and $u, v \in V_{n+1}$ we have

$$\begin{aligned} \beta_{u+\alpha \cdot v}(v_1, \dots, v_n) &= \otimes_{n+1}(v_1, \dots, v_n, u + \alpha \cdot v) \\ &= \otimes_{n+1}(v_1, \dots, v_n, u) + \alpha \cdot \otimes_{n+1}(v_1, \dots, v_n, v) \\ &= \beta_u(v_1, \dots, v_n) + \alpha \cdot \beta_v(v_1, \dots, v_n) \end{aligned}$$

proving that

$$\beta_{u+\alpha \cdot v} = \beta_u + \alpha \cdot \beta_v \quad (13.51)$$

As $V_1 \otimes_n \dots \otimes_n V_n$ is a tensor product of $\{V_i\}_{i \in \{1, \dots, n\}}$ we have from [eq: 13.50] that $\forall w \in V_{n+1}$ there exist a

$$g_w \in \text{Hom}(V_1 \otimes_n \dots \otimes_n V_n, V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}) \text{ such that } \beta_w = g_w \circ \otimes_n \quad (13.52)$$

Hence $\forall (v_1, \dots, v_n) \in \prod_{i \in \{1, \dots, n\}} V_i$, $\forall w \in V_{n+1}$ we have

$$g_w(\otimes_n(v_1, \dots, v_n)) = \beta_w(v_1, \dots, v_n) \stackrel{[\text{eq: 13.50}]}{=} \otimes_{n+1}(v_1, \dots, v_n, w) \quad (13.53)$$

Define now

$$\varphi: \prod_{i \in \{1, \dots, n+1\}} V_i \rightarrow P \text{ by } \varphi(v_1, \dots, v_{n+1}) = \nu(\otimes_n(v_1, \dots, v_n), v_{n+1}) \quad (13.54)$$

then we have for $i \in \{1, \dots, n+1\}$, $u, v \in V_i$ and $\alpha \in F$ either:

$$i \in \{1, \dots, n\}.$$

$$\begin{aligned} \varphi(v_1, \dots, v_{i-1}, u + \alpha \cdot v, v_{i+1}, \dots, v_n) &= \\ \nu(\otimes_n(v_1, \dots, v_{i-1}, u + \alpha \cdot v, v_{i+1}, \dots, v_n), v_{n+1}) &= \\ \nu(\otimes_n(v_1, \dots, v_{i-1}, u, v_{i+1}, \dots, v_n) + \alpha \cdot \otimes_n(v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n), v_{n+1}) &= \\ \nu(\otimes_n(v_1, \dots, v_{i-1}, u, v_{i+1}, \dots, v_n), v_{n+1}) + \alpha \cdot \nu(\otimes_n(v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n), v_{n+1}) &= \\ \varphi(v_1, \dots, v_{i-1}, u, v_{i+1}, \dots, v_n) + \alpha \cdot \varphi(v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n) \end{aligned}$$

$$i = n+1.$$

$$\begin{aligned} \varphi(v_1, \dots, v_{i-1}, u + \alpha \cdot v, v_{i+1}, \dots, v_n) &= \\ \nu(\otimes_n(v_1, \dots, v_n), u + \alpha \cdot v) &= \\ \nu(\otimes_n(v_1, \dots, v_n), u) + \alpha \cdot \nu(\otimes_n(v_1, \dots, v_n), v) &= \\ \varphi(v_1, \dots, v_{i-1}, u, v_{i+1}, \dots, v_n) + \alpha \cdot \varphi(v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n) \end{aligned}$$

proving that

$$\varphi \in \text{Hom}(V_1, \dots, V_{n+1}, P) \quad (13.55)$$

As $V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}$ is a tensor product of $V_1, \dots, V_{n+1}$ there exist a $h \in \text{Hom}(V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}, P)$ such that $\varphi = h \circ \otimes_{n+1}$ or $\forall (v_1, \dots, v_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} V_i$ we have

$$h(\otimes_{n+1}(v_1, \dots, v_{n+1})) = \nu(\otimes_n(v_1, \dots, v_n), v_{n+1}) \quad (13.56)$$

Define now

$$\psi: \prod_{i \in \{1,2\}} W_i \rightarrow V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} \text{ by } \psi(z, v) \stackrel{[\text{eq: 13.52}]}{=} g_v(z) \quad (13.57)$$

Then if $i \in \{1, 2\}$ we have for $\alpha \in F$ and $u, v \in W_i$ and either:

$i = 1$. Then, as $W_i = V_1 \otimes_n \cdots \otimes_n V_n$, we have $u, v \in V_1 \otimes_n \cdots \otimes_n V_n$, so that given $v_2 \in V_{n+1}$

$$\begin{aligned} \psi(u + \alpha \cdot v, v_2) &= g_{v_2}(u + \alpha \cdot v) \\ &\stackrel{[\text{eq: 13.52}]}{=} g_{v_2}(u) + \alpha \cdot g_{v_2}(v) \\ &= \psi(u, v_2) + \alpha \cdot \psi(v, v_2) \end{aligned}$$

$i = 2$. Then $W_2 = V_{n+1}$ and $u, v \in V_{n+1}$, let $v_1 \in V_1 \otimes_n \cdots \otimes_n V_n$. As $V_1 \otimes_n \cdots \otimes_n V_n = \text{span}(\otimes_n(\prod_{i \in \{1, \dots, n\}} V_i))$ there exists by [theorem: 11.107] a $\{u_i\}_{i \in \{1, \dots, l\}} \subseteq \otimes_n(\prod_{i \in \{1, \dots, n\}} V_i)$ and a $\{\alpha_i\}_{i \in \{1, \dots, l\}} \subseteq F$ such that $v_1 = \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot u_i$. If $i \in \{1, \dots, l\}$ then for u_i there exists a $x \in \prod_{i \in \{1, \dots, n\}} V_i$ such that $u_i = \otimes_n(x)$, hence

$$\begin{aligned} g_{u+\alpha \cdot v}(u_i) &= g_{u+\alpha \cdot v}(\otimes_n(x)) \\ &\stackrel{[\text{eq: 13.52}]}{=} \beta_{u+\alpha \cdot v}(x) \\ &\stackrel{[\text{eq: 13.51}]}{=} \beta_u(x) + \alpha \cdot \beta_v(x) \\ &\stackrel{[\text{eq: 13.52}]}{=} g_u(\otimes_n(x)) + \alpha \cdot g_v(\otimes_n(x)) \\ &= g_u(u_i) + \alpha \cdot g_v(u_i) \end{aligned}$$

So that

$$\begin{aligned} \psi(v_1, u + \alpha \cdot v) &= g_{u+\alpha \cdot v}(v_1) \\ &= g_{u+\alpha \cdot v}\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot u_i\right) \\ &\stackrel{[\text{eq: 13.52}]}{=} \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot g_{u+\alpha \cdot v}(u_i) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot (g_u(u_i) + \alpha \cdot g_v(u_i)) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot g_u(u_i) + \alpha \cdot \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot g_v(u_i) \\ &\stackrel{[\text{eq: 13.52}]}{=} g_u\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot u_i\right) + \alpha \cdot g_v\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot u_i\right) \\ &= g_u(v_1) + \alpha \cdot g_v(v_1) \\ &= \psi(v_1, u) + \alpha \cdot \psi(v_1, v) \end{aligned}$$

proving that ψ is multilinear or

$$\psi \in \text{Hom}(W_1, W_2; V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}) = \text{Hom}(V_1 \otimes_n \cdots \otimes_n V_n, V_{n+1}; V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}) \quad (13.58)$$

By [eq: 13.49] we have that there exist a

$$k \in \text{Hom}(P, V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}) \text{ such that } \psi = k \circ \nu \quad (13.59)$$

Let $(v_1, \dots, v_n) \in \prod_{i \in \{1, \dots, n\}} V_i$ and $v \in V_{n+1}$ then as $\otimes_n \in \text{Hom}(V_1, \dots, V_n; V_1 \otimes_n \cdots \otimes_n V_n)$ we have that $\otimes_n(v_1, \dots, v_n) \in V_1 \otimes_n \cdots \otimes_n V_n$ so that $\nu(\otimes_n(v_1, \dots, v_n), v) \in P$ is in the domain of k and we have

$$\begin{aligned} k(\nu(\otimes_n(v_1, \dots, v_n), v)) &\stackrel{[\text{eq: 13.59}]}{=} \psi(\otimes_n(v_1, \dots, v_n), v) \\ &\stackrel{[\text{eq: 13.57}]}{=} g_v(\otimes_n(v_1, \dots, v_n)) \\ &\stackrel{[\text{eq: 13.53}]}{=} \otimes_{n+1}(v_1, \dots, v_n, v) \end{aligned} \quad (13.60)$$

As $h \in \text{Hom}(V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}, P)$ and $k \in \text{Hom}(P, V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1})$ it follows that

$$k \circ h \in \text{Hom}(V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}, V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1})$$

If $z \in \otimes_{n+1}(\prod_{i \in \{1, \dots, n+1\}} V_i)$ then there exist a $(v_1, \dots, v_{n+1})$ such that $z = \otimes_{n+1}(v_1, \dots, v_{n+1})$ and

$$\begin{aligned} (k \circ h)(z) &= (k \circ h)(\otimes_{n+1}(v_1, \dots, v_{n+1})) \\ &= k(h(\otimes_{n+1}(v_1, \dots, v_{n+1}))) \\ &\stackrel{[\text{eq: 13.56}]}{=} k(\nu(\otimes_n(v_1, \dots, v_n), v_{n+1})) \\ &\stackrel{[\text{eq: 13.60}]}{=} \otimes_{n+1}(v_1, \dots, v_{n+1}) \\ &= z \end{aligned} \tag{13.61}$$

Next if $z \in V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1} = \text{span}(\otimes_{n+1}(\prod_{i \in \{1, \dots, n+1\}} V_i))$ there exist a finite $I \subseteq \otimes_{n+1}(\prod_{i \in \{1, \dots, n+1\}} V_i)$ and a $\{\alpha_i\}_{i \in I} \subseteq F$ such that $z = \sum_{i \in I} \alpha_i \cdot i$. Hence

$$(k \circ h)(z) = (k \circ h)\left(\sum_{i \in I} \alpha_i \cdot i\right) = \sum_{i \in I} \alpha_i \cdot (k \circ h)(i) \stackrel{[\text{eq: 13.61}]}{=} \sum_{i \in I} \alpha_i \cdot i = z$$

proving that

$$k \circ h = \text{Id}_{V_1 \otimes \dots \otimes V_{n+1}} \tag{13.62}$$

Again as $h \in \text{Hom}(V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1}, P)$ and $k \in \text{Hom}(P, V_1 \otimes_{n+1} \dots \otimes_{n+1} V_{n+1})$ it follows that

$$h \circ k \in \text{Hom}(P, P) \tag{13.63}$$

Let $v = (v_1, \dots, v_n) \in \prod_{i \in \{1, \dots, n\}} V_i$, $w \in V_{n+1}$ then we have

$$\begin{aligned} h(k(\nu(\otimes_n(v), w))) &= h(k(\nu(\otimes_n(v_1, \dots, v_n), w))) \\ &\stackrel{[\text{eq: 13.60}]}{=} h(\otimes_{n+1}(v_1, \dots, v_n, w)) \\ &\stackrel{[\text{eq: 13.56}]}{=} \nu(\otimes_n(v_1, \dots, v_n), w) \\ &= \nu(\otimes_n(v), w) \end{aligned} \tag{13.64}$$

Let $z \in V_1 \otimes_n \dots \otimes_n V_n = \text{span}(\otimes_n(\prod_{i \in \{1, \dots, n\}} V_i))$ then by [theorem: 11.107] there exist a $\{z_i\}_{i \in \{1, \dots, l\}} \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ and a $\{\alpha_i\}_{i \in \{1, \dots, l\}} \subseteq F$ such that $z = \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot z_i$. Using [theorem: 3.138] there exist a $\{y_i\}_{i \in \{1, \dots, l\}} \subseteq \prod_{i \in \{1, \dots, n\}} V_i$ such that $\forall i \in \{1, \dots, l\} \otimes_n(y_i) = z_i$. So given $w \in V_{n+1}$ we have

$$\begin{aligned} h(k(\nu(z, w))) &= h\left(k\left(\nu\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot z_i, w\right)\right)\right) \\ &\stackrel{\nu \text{ is multilinear}}{=} h\left(k\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot \nu(z_i, w)\right)\right) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot h(k(\nu(\otimes_n(y_i), w))) \\ &\stackrel{[\text{eq: 13.64}]}{=} \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot \nu(\otimes_n(y_i), w) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot \nu(z_i, w) \\ &\stackrel{\nu \text{ is multilinear}}{=} \nu\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot z_i, w\right) \\ &= \nu(z, w) \end{aligned}$$

proving that

$$\forall z \in V_1 \otimes_n \dots \otimes_n V_n \text{ and } w \in V_{n+1} \text{ that } h(k(\nu(z, w))) = \nu(z, w) \tag{13.65}$$

Let $z \in \prod_{i \in \{1,2\}} W_i$ then $z = (z_1, z_2)$ with $z_1 \in W_1 = V_1 \otimes_n \cdots \otimes_n V_n$ and $z_2 \in V_{n+1}$ then

$$h(k(\nu(z))) = h(k(\nu(z_1, z_2))) \stackrel{[\text{eq: 13.65}]}{=} \nu(z, w) = \nu(z) \quad (13.66)$$

If $z \in P = \text{span}(\nu(\prod_{i \in \{1,2\}} W_i))$ then by [theorem: 11.107] there exists a $\{z_i\}_{i \in \{1, \dots, l\}} \subseteq \nu(\prod_{i \in \{1,2\}} W)_i$ and a $\{\alpha_i\}_{i \in \{1, \dots, l\}} \subseteq F$ such that $z = \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot z_i$. Using [theorem: 3.138] there exist a $\{y_i\}_{i \in \{1, \dots, l\}} \subseteq \prod_{i \in \{1,2\}} W_i$ such that $\forall i \in \{1, \dots, l\}$ we have $z_i = \nu(y_i)$.

$$\begin{aligned} (h \circ k)(z) &= h(k(z)) \\ &= h\left(k\left(\sum_{i \in \{1, \dots, l\}} \alpha_i \cdot z_i\right)\right) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot h(k(z_i)) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot h(k(\nu(y_i))) \\ &\stackrel{[\text{eq: 13.66}]}{=} \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot \nu(y_i) \\ &= \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot z_i \\ &= z \end{aligned}$$

proving that

$$h \circ k = \text{Id}_P \quad (13.67)$$

Now we are ready to construct the tensor product, starting with the multilinear mapping. Define [see eq: 13.47, 13.59]

$$\otimes: \prod_{i \in \{1,2\}} W_i \rightarrow V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} \text{ by } \otimes(z, v) = k(\nu(z, v)) \quad (13.68)$$

If $i \in \{1, 2\}$ we have for $\alpha \in F$ and $u, v \in W_i$ either:

$i = 1$. Then as $W_1 = V_1 \otimes_n \cdots \otimes_n V_n$ we have $u, v \in V_1 \otimes_n \cdots \otimes_n V_n$ and, as ν is multilinear and k linear, we have for $v_2 \in V_{n+1}$

$$\begin{aligned} \otimes(u + \alpha \cdot v, v_2) &= k(\nu(u + \alpha \cdot v, v_2)) \\ &\stackrel{\nu \text{ is multilinear}}{=} k(\nu(u, v_2) + \alpha \cdot \nu(v, v_2)) \\ &\stackrel{k \text{ is linear}}{=} k(\nu(u, v_2)) + \alpha \cdot k(\nu(v, v_2)) \\ &= \otimes(u, v_2) + \alpha \cdot \otimes(v, v_2) \end{aligned}$$

$i = 2$. Then as $W_i = V_{n+1}$ we have $u, v \in V_{n+1}$ and, as ν is multilinear and k linear, that for $v_1 \in V_1 \otimes_n \cdots \otimes_n V_n$

$$\begin{aligned} \otimes(v_1, u + \alpha \cdot v) &= k(\nu(v_1, u + \alpha \cdot v)) \\ &\stackrel{\nu \text{ is multilinear}}{=} k(\nu(v_1, u) + \alpha \cdot \nu(v_1, v)) \\ &= k(\nu(v_1, u)) + \alpha \cdot k(\nu(v_1, v)) \\ &= \otimes(v_1, u) + \alpha \cdot \otimes(v_1, v) \end{aligned}$$

proving that

$$\otimes \in \text{Hom}(W_1, W_2; V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}) = \text{Hom}(V_1 \otimes_n \cdots \otimes_n V_n, V_{n+1}; V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}) \quad (13.69)$$

By the definition of $\otimes$ [see: 13.68] we have $\otimes(\prod_{i \in \{1,2\}} W_i) \subseteq V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}$ proving by [theorems: 11.109, 11.111] that

$$\text{span}\left(\otimes\left(\prod_{i \in \{1,2\}} W_i\right)\right) \subseteq V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} \quad (13.70)$$

If $v \in V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} = \text{span}(\otimes_{n+1}(\prod_{i \in \{1, \dots, n+1\}} V_i))$ then there exists $\{v_i\}_{i \in \{1, \dots, l\}} \subseteq \otimes_{n+1}(\prod_{i \in \{1, \dots, n+1\}} V_i)$ and $\{\alpha_i\}_{i \in \{1, \dots, l\}} \subseteq F$ such that

$$v = \sum_{i \in \{1, \dots, l\}} \alpha_i \cdot v_i$$

If $i \in \{1, \dots, l\}$ then as $v_i \in \otimes_{n+1}(\prod_{i \in \{1, \dots, n+1\}} V_i)$ there exists a $(u_1, \dots, u_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} V_i$ such that

$$\begin{aligned} v_i &= \otimes_{n+1}(u_1, \dots, u_{n+1}) \\ &\stackrel{[\text{eq: 13.62}]}{=} k(h(\otimes_{n+1}(u_1, \dots, u_{n+1}))) \\ &\stackrel{[\text{eq: 13.56}]}{=} k(\nu(\otimes_n(u_1, \dots, u_n), u_{n+1})) \\ &\stackrel{[\text{eq: 13.68}]}{=} \otimes(\otimes_n(u_1, \dots, u_n), u_{n+1}) \\ &\in \otimes\left(\prod_{i \in \{1, 2\}} W_i\right) \end{aligned}$$

which proves that $v \in \text{span}(\otimes(\prod_{i \in \{1, 2\}} W_i))$. Hence $V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} \subseteq \text{span}(\otimes(\prod_{i \in \{1, 2\}} W_i))$ and combining this with [eq: 13.70] gives

$$V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} = \text{span}\left(\otimes\left(\prod_{i \in \{1, 2\}} W_i\right)\right) \quad (13.71)$$

Let $\theta \in \text{Hom}(W_1, W_2; U) = \text{Hom}(V_1 \otimes_n \cdots \otimes_n V_n, V_{n+1}; U)$ then by [eq: 13.49] there exists a

$$f \in \text{Hom}(P, U) \text{ such that } \theta = f \circ \nu \quad (13.72)$$

as $f \in \text{Hom}(P, U)$ and $h \in \text{Hom}(V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}, P)$ we have that

$$g = f \circ h \in \text{Hom}(V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}, U) \quad (13.73)$$

Further if $z = (z_1, z_2) \in \prod_{i \in \{1, 2\}} W_i$ then we have

$$\begin{aligned} (g \circ \otimes)(z) &= g(\otimes(z_1, z_2)) \\ &\stackrel{[\text{eq: 13.68}]}{=} g(k(\nu(z_1, z_2))) \\ &\stackrel{[\text{eq: 13.73}]}{=} f(h(k(\nu(z_1, z_2)))) \\ &\stackrel{[\text{eq: 13.67}]}{=} f(\nu(z_1, z_2)) \\ &\stackrel{[\text{eq: 13.72}]}{=} \theta(z_1, z_2) \\ &= \theta(z) \end{aligned}$$

proving that

$$\forall \theta \in \text{Hom}(W_1, W_2; U) \text{ we have } \exists g \in \text{Hom}(V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}, U) \text{ such that } \theta = g \circ \otimes \quad (13.74)$$

To conclude using [eqs: 13.68, 13.69, 13.71, 13.74] it follows that

$$\langle V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1}, \otimes \rangle \text{ is a tensor product of } W_1, W_2$$

or using [convention: 13.19]

$$(V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1} \text{ is a tensor product and } V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} = (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1} \quad \square$$

Corollary 13.28. *Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n+1\}}$ be family of vector spaces over a field F and $\langle V_1 \otimes_{n+1}, \dots, \otimes_{n+1} V_{n+1}, \otimes_{n+1} \rangle$ a tensor product of $V_1, \dots, V_{n+1}$ then there exist a tensor product of $\langle V_1 \otimes_n \cdots \otimes_n V_n, \otimes_n \rangle$ of $V_1, \dots, V_n$ and a tensor product $\langle (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}, \otimes \rangle$ of $V_1 \otimes_n \cdots \otimes_n V_n, V_{n+1}$ such that*

$$V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} = (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}$$

Proof. By [theorem: 13.20] there exist a tensor product $V_1 \otimes \cdots \otimes V_n$ of $V_1, \dots, V_n$, hence using the previous theorem [theorem: 13.27] there exist also a tensor product $\langle (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1}, \otimes \rangle$ of $V_1 \otimes_n \cdots \otimes_n V_n, V_{n+1}$ such that

$$V_1 \otimes_{n+1} \cdots \otimes_{n+1} V_{n+1} = (V_1 \otimes_n \cdots \otimes_n V_n) \otimes V_{n+1} \quad \square$$

Lemma 13.29. *Let $n \in \mathbb{N}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ a finite family of finite sets then $\prod_{i \in \{1, \dots, n\}} X_i$ is finite and $\text{card}(\prod_{i \in \{1, \dots, n\}} X_i) = \prod_{i \in \{1, \dots, n\}} \text{card}(X_i)$*

Proof. We use induction to proof this, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{X_i\}_{i \in \{1, \dots, n\}} \text{ is a family of finite sets then } \prod_{i \in \{1, \dots, n\}} X_i \text{ is finite} \right.$$

$$\left. \text{and } \text{card}\left(\prod_{i \in \{1, \dots, n\}} X_i\right) = \prod_{i \in \{1, \dots, n\}} \text{card}(X_i) \right\}$$

then we have:

$1 \in S$. Let $\{X_i\}_{i \in \{1\}}$ is such that X_1 is finite. By [example: 2.147] we have that $\prod_{i \in \{1\}} X_i$ is bijective with X_1 , hence $\prod_{i \in \{1\}} X_i$ is finite and $\text{card}(\prod_{i \in \{1\}} X_i) = \text{card}(X_1) = \prod_{i \in \{1\}} \text{card}(X_i)$. so $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{X_i\}_{i \in \{1, \dots, n+1\}}$ be family of finite sets. As $n \in S$ we have that

$$\prod_{i \in \{1, \dots, n\}} X_i \text{ is finite and } \text{card}\left(\prod_{i \in \{1, \dots, n\}} X_i\right) = \prod_{i \in \{1, \dots, n\}} \text{card}(X_i) \quad (13.75)$$

Using [theorem: 10.92] and the fact that X_{n+1} is finite we have that

$$\begin{aligned} & \left(\prod_{i \in \{1, \dots, n\}} X_i\right) \times X_{n+1} \text{ is finite and } \text{card}\left(\left(\prod_{i \in \{1, \dots, n\}} X_i\right) \times X_{n+1}\right) = \\ & \text{card}\left(\prod_{i \in \{1, \dots, n\}} X_i\right) \cdot \text{card}(X_{n+1}) \stackrel{[\text{eq: 13.75}]}{=} \prod_{i \in \{1, \dots, n+1\}} \text{card}(X_i) \end{aligned}$$

By [theorem: 6.91] $\prod_{i \in \{1, \dots, n+1\}} X_i$ is bijective with $(\prod_{i \in \{1, \dots, n\}} X_i) \times X_{n+1}$ so that $(\prod_{i \in \{1, \dots, n\}} X_i) \times X_{n+1}$ is finite and $\text{card}((\prod_{i \in \{1, \dots, n\}} X_i) \times X_{n+1}) = \prod_{i \in \{1, \dots, n+1\}} \text{card}(X_i)$ proving that $n+1 \in S$. $\square$

We introduce now the concept of a multi-index

Definition 13.30. *Let $k \in \mathbb{N}$, $\{n_i\}_{i \in \{1, \dots, k\}} \subseteq \mathbb{N}$ be finite family of natural numbers then set of multi-indexes named $\Gamma(n_1, \dots, n_k)$ is defined by*

$$\Gamma(n_1, \dots, n_k) = \prod_{i \in \{1, \dots, k\}} \{1, \dots, n_i\}$$

Using the previous lemma [lemma: 13.29] we have that that

$$\text{card}(\Gamma(n_1, \dots, n_k)) = \prod_{i \in \{1, \dots, k\}} n_i$$

Example 13.31. $\Gamma(2, 3) = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$

Theorem 13.32. *Let $k \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, k\}}$ be a finite family of vector space over a field F , V a vector space over the same field, $\varphi \in \text{Hom}(V_1, \dots, V_k; V)$ a multilinear mapping, $\{n_i\}_{i \in \{1, \dots, k\}} \subseteq \mathbb{N}$ and $\{w_{i,j}\}_{j \in \{1, \dots, n_i\}} \subseteq V_i$, $i \in \{1, \dots, k\}$ then*

$$\varphi\left(\sum_{j \in \{1, \dots, n_1\}} w_{1,j}, \dots, \sum_{j \in \{1, \dots, n_k\}} w_{k,j}\right) = \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} \varphi(w_{1,\gamma(1)}, \dots, w_{k,\gamma(k)})$$

Proof. We prove this by induction on k , so let

$$S = \left\{ k \in \mathbb{N} \mid \text{If } \varphi \in \text{Hom}(V_1, \dots, V_k; V), \{n_i\}_{i \in \{1, \dots, k\}} \subseteq \mathbb{N}, i \in \{1, \dots, m\}, \{w_{i,j}\}_{j \in \{1, \dots, n_i\}} \text{ then} \right.$$

$$\left. \varphi \left(\sum_{j \in \{1, \dots, n_1\}} w_{1,j}, \dots, \sum_{j \in \{1, \dots, n_k\}} w_{k,j} \right) = \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} \varphi(w_{1,\gamma(1)}, \dots, w_{k,\gamma(k)}) \right\}$$

then we have:

1 $\in S$. Define $\tau: \{1, \dots, n_1\} \rightarrow \Gamma(n_1, \dots, n_1) = \prod_{i \in \{1\}} \{1, \dots, n_i\}$ where $\tau(i)$ is defined by $\tau(i) = (i) \Rightarrow \tau(i)_1 = \tau(i)(1) = i$ then we have:

injectivity. If $\tau(i) = \tau(j)$ then $i = \tau(i)(1) = \tau(j)(1) = j$

surjectivity. If $\gamma \in \Gamma(n_1, \dots, n_1)$ then $\gamma: \{1\} \rightarrow \{1, \dots, n_1\}$. Take $i = \gamma(1) \in \{1, \dots, n_1\}$ then $\forall j \in \{1\}$ we have $\tau(i)(j) = i = \gamma(1) = \gamma(j)$ proving that $\tau(i) = \gamma$.

proving that

$$\tau: \{1, \dots, n_1\} \rightarrow \Gamma(n_1, \dots, n_1) \text{ is a bijection}$$

Let $\{w_{1,j}\}_{j \in \{1, \dots, n_1\}} \subseteq V_1$ and $\varphi \in \text{Hom}(V_1, \dots, V_1; V)$ then

$$\begin{aligned} \varphi \left(\sum_{j \in \{1, \dots, n_1\}} w_{1,j} \right) & \stackrel{\varphi \text{ is multilinear}}{=} \sum_{j \in \{1, \dots, n_1\}} \varphi(w_{1,j}) \\ & = \sum_{j \in \{1, \dots, n_1\}} \varphi(w_{1,\tau(j)(1)}) \\ & \stackrel{[\text{theorem: 11.36}]}{=} \sum_{\gamma \in \Gamma(n_1, \dots, n_1)} \varphi(w_{1,\gamma(1)}) \end{aligned}$$

proving that $1 \in S$.

$k \in S \Rightarrow k+1 \in S$. Let $\varphi \in \text{Hom}(V_1, \dots, V_{k+1}; V)$, $\{n_i\}_{i \in \{1, \dots, k+1\}} \subseteq \mathbb{N}$, $\{w_{i,j}\}_{j \in \{1, \dots, n_i\}}$ where $i \in \{1, \dots, k+1\}$. Define

$$\psi: \prod_{i \in \{1, \dots, k\}} V_i \rightarrow V \text{ by } \psi \left(v_1, \dots, v_k, \sum_{j \in \{1, \dots, n_{k+1}\}} w_{k+1,j} \right)$$

then trivially $\psi \in \text{Hom}(V_1, \dots, V_k; V)$. Then we have

$$\begin{aligned} \varphi \left(\sum_{j \in \{1, \dots, n_1\}} w_{1,j}, \dots, \sum_{j \in \{1, \dots, n_{k+1}\}} w_{k+1,j} \right) & = \\ \psi \left(\sum_{j \in \{1, \dots, n_1\}} w_{1,j}, \dots, \sum_{j \in \{1, \dots, n_k\}} w_{k,j} \right) & \stackrel{k \in S}{=} \\ \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} \psi(w_{1,\gamma(1)}, \dots, w_{k,\gamma(k)}) & = \\ \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} \varphi \left(w_{1,\gamma(1)}, \dots, w_{k,\gamma(k)}, \sum_{j \in \{1, \dots, n_{k+1}\}} w_{k+1,j} \right) & = \\ \sum_{\gamma \in \Gamma(n_1, \dots, n_{k+1})} \left(\sum_{j \in \{1, \dots, n_{k+1}\}} \varphi(w_{1,\gamma(1)}, \dots, w_{k,\gamma(k)}, w_{k+1,j}) \right) & \stackrel{[\text{theorem: 11.45}]}{=} \\ \sum_{(\gamma,j) \in \Gamma(n_1, \dots, n_k) \times \{1, \dots, n_{k+1}\}} \varphi(w_{1,\gamma(1)}, \dots, w_{k,\gamma(k)}, w_{k+1,j}) & \quad (13.76) \end{aligned}$$

Define now

$$\chi: \Gamma(n_1, \dots, n_k) \times \{1, \dots, n_{k+1}\} \rightarrow \Gamma(n_1, \dots, n_{k+1})$$

where

$$\chi(\gamma, i) \text{ is defined by } \chi(\gamma, i)(j) = \begin{cases} i & \text{if } j = k+1 \\ \gamma(j) & \text{if } j \in \{1, \dots, k\} \end{cases}$$

then we have:

injectivity. If $\chi(\gamma, r) = \chi(\zeta, s)$ then $\forall j \in \{1, \dots, k\}$ we have

$$\gamma(j) = \chi(\gamma, r)(j) = \chi(\zeta, s)(j) = \zeta(j)$$

proving that $\gamma = \zeta$. Further $r = \chi(\gamma, r)(k+1) = \chi(\zeta, s)(k+1) = s$ so that we have that $(\gamma, r) = (\zeta, s)$.

surjectivity. If $\gamma \in \Gamma(n_1, \dots, n_{k+1})$ take then $\zeta = \gamma|_{\{1, \dots, k\}}$ and $r = \gamma(k+1)$ then we have $\chi(\zeta, r)(j) = \begin{cases} r & \text{if } j = k+1 \\ \zeta(j) & \text{if } j \in \{1, \dots, k\} \end{cases} = \begin{cases} \gamma(k+1) & \text{if } j = k+1 \\ \gamma(j) & \text{if } j \in \{1, \dots, k\} \end{cases} = \gamma(j)$. Hence $\chi(\zeta, r) = \gamma$ proving surjectivity.

Hence $\chi: \Gamma(n_1, \dots, n_k) \times \{1, \dots, n_{k+1}\} \rightarrow \Gamma(n_1, \dots, n_{k+1})$ is a bijection. So

$$\begin{aligned} & \sum_{(\gamma, j) \in \Gamma(n_1, \dots, n_k) \times \{1, \dots, n_{k+1}\}} \varphi(w_{1, \gamma(1)}, \dots, w_{k, \gamma(k)}, w_{k+1, j}) = \\ & \sum_{(\gamma, j) \in \Gamma(n_1, \dots, n_k) \times \{1, \dots, n_{k+1}\}} \varphi(w_{1, \chi(\gamma, j)(1)}, \dots, w_{k, \chi(\gamma, j)(k)}, w_{k+1, \chi(\gamma, j)(k+1)}) \stackrel{[\text{theorem: 11.36}]}{=} \\ & \sum_{\gamma \in \Gamma(n_1, \dots, n_{k+1})} \varphi(w_{1, \gamma(1)}, \dots, w_{k+1, \gamma(k+1)}) \end{aligned}$$

which combined with [eq: 13.76] gives

$$\varphi\left(\sum_{j \in \{1, \dots, n_1\}} w_{1, j}, \dots, \sum_{j \in \{1, \dots, n_{k+1}\}} w_{k+1, j}\right) = \sum_{\gamma \in \Gamma(n_1, \dots, n_{k+1})} \varphi(w_{1, \gamma(1)}, \dots, w_{k+1, \gamma(k+1)})$$

proving that $k+1 \in S$. $\square$

Lemma 13.33. Let $k \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, k\}}$ a finite family of vector spaces over a field F , $\langle V_1 \otimes \dots \otimes V_n, \otimes \rangle$ a tensor product of $V_1, \dots, V_n$, $\{n_i\}_{i \in \{1, \dots, k\}} \subseteq \mathbb{N}$ and $\forall i \in \{1, \dots, k\}$ a family $\{W_{i, j}\}_{j \in \{1, \dots, n_i\}}$ a satisfying

$$\forall j \in \{1, \dots, n_i\} \quad W_{i, j} \text{ is a sub-space of } V_i$$

and

$$V_i = \sum_{j \in \{1, \dots, n_i\}} W_{i, j} \text{ [see definition 12.1]}$$

then

$$V_1 \otimes \dots \otimes V_k = \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}$$

where $\forall \gamma \in \Gamma(n_1, \dots, n_k)$

$$W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)} = \text{span}\left(\otimes\left(\prod_{i \in \{1, \dots, k\}} W_{i, \gamma(i)}\right)\right)$$

and

$$\langle W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}, \otimes|_{\prod_{j \in \{1, \dots, k\}} W_{i, \gamma(i)}} \rangle \text{ is a tensor product of } W_{1, \gamma(1)}, \dots, W_{k, \gamma(k)}$$

with

$$W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)} \subseteq V_1 \otimes \dots \otimes V_n$$

[see theorem: 13.18].

Proof. Let $v = (v_1, \dots, v_k) \in \prod_{i \in \{1, \dots, k\}} V_i$ then by the hypothesis we have $\forall i \in \{1, \dots, k\}$ that there exist a $\{w_{i, j}\}_{j \in \{1, \dots, n_i\}} \subseteq W_i$ such that $v_i = \sum_{j \in \{1, \dots, n_i\}} w_{i, j}$. As $\otimes$ is multilinear we have

$$\begin{aligned} \otimes(v_1, \dots, v_k) &= \otimes\left(\sum_{j \in \{1, \dots, n_1\}} w_{1, j}, \dots, \sum_{j \in \{1, \dots, n_k\}} w_{k, j}\right) \\ &\stackrel{[\text{theorem: 13.32}]}{=} \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} \otimes(w_{1, \gamma(1)}, \dots, w_{k, \gamma(k)}) \end{aligned} \quad (13.77)$$

As $\forall i \in \{1, \dots, k\}$, $\forall j \in \{1, \dots, n_i\}$ we have $w_{i, \gamma(i)} \in W_{i, \gamma(i)}$ we have

$$\begin{aligned} \otimes(w_{1, \gamma(1)}, \dots, w_{k, \gamma(k)}) &\in \otimes\left(\prod_{i \in \{1, \dots, k\}} W_{i, \gamma(i)}\right) \\ &\subseteq \text{span}\left(\otimes\left(\prod_{i \in \{1, \dots, k\}} W_{i, \gamma(i)}\right)\right) \\ &= W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)} \end{aligned}$$

So using the above, [eq: 13.77], the definition of the sum of sets [see definition 12.1] it follows that

$$\otimes(v_1, \dots, v_k) \in \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}$$

or

$$\otimes\left(\prod_{i \in \{1, \dots, k\}} V_i\right) \subseteq \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}$$

which using [theorem: 11.109] gives

$$V_1 \otimes \dots \otimes V_k = \text{span}\left(\otimes\left(\prod_{i \in \{1, \dots, k\}} V_i\right)\right) \subseteq \text{span}\left(\sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}\right)$$

By [theorem: 13.18] $W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}$ is a sub-space of $V = V_1 \otimes \dots \otimes V_k$ so that by [theorem: 12.4] $\sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}$ is a sub-space of $V_1 \otimes \dots \otimes V_n$. Hence by [theorems: 11.109, 11.111]

$$\sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)} = \text{span}\left(\sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)}\right) \subseteq V_1 \otimes \dots \otimes V_k$$

Combining the last two results proves then

$$\sum_{\gamma \in \Gamma(n_1, \dots, n_k)} W_{1, \gamma(1)} \otimes \dots \otimes W_{k, \gamma(k)} = V_1 \otimes \dots \otimes V_k \quad \square$$

Lemma 13.34. Let $\{V_i\}_{i \in \{1, \dots, 2\}}$ be a family of two vector spaces over a field and $\langle V_1 \otimes V_2, \otimes \rangle$ a tensor product of V_1, V_2 then

1. If $V_1 = W_1 \oplus W_2$ [W_1, W_2 sub-spaces of V_1] then we have that

$$V_1 \otimes V_2 = (W_1 \otimes V_2) \oplus (W_2 \otimes V_2)$$

where we have used [theorem: 13.18] to define

$$W_1 \otimes V_2 = \text{span}(\otimes(W_1 \cdot V_2)) \subseteq V_1 \otimes V_2, \quad W_2 \otimes V_2 = \text{span}(\otimes(W_2 \cdot V_2)) \subseteq V_1 \otimes V_2$$

so that

$$\langle W_1 \otimes V_2, \otimes_{|W_1 \cdot V_2} \rangle \text{ and } \langle W_2 \otimes V_2, \otimes_{|W_2 \cdot V_2} \rangle \text{ are tensor products of } W_1, V_2 \text{ and } W_2, V_2$$

2. If $V_2 = W_1 \oplus W_2$ [W_1, W_2 sub-spaces of V_2] then we have that

$$V_1 \otimes V_2 = (V_1 \otimes W_1) \oplus (V_1 \otimes W_2)$$

where we have used [theorem: 13.18] to define

$$V_1 \otimes W_1 = \text{span}(\otimes(V_1 \cdot W_1)) \subseteq V_1 \otimes V_2, \quad V_1 \otimes W_2 = \text{span}(\otimes(V_1 \cdot W_2)) \subseteq V_1 \otimes V_2$$

so that

$$\langle V_1 \otimes W_1, \otimes_{|V_1 \cdot W_1} \rangle \text{ and } \langle V_1 \otimes W_2, \otimes_{|V_1 \cdot W_2} \rangle \text{ are tensor products of } V_1, W_1 \text{ and } V_1, W_2$$

Here we use [definition: 6.88] for $W_1 \cdot V_2, W_2 \cdot V_2, V_1 \cdot W_1$ and $V_1 \cdot W_2$

Proof.

1. If $v = (v_1, v_2) \in V_1 \times V_2$ then as $V_1 = W_1 \oplus W_2$ there exists **unique** $w_1 \in W_1$, $w_2 \in W_2$ such that $v_1 = w_1 + w_2$. Hence $\otimes(v) = \otimes(w_1 + w_2, v_2) = \otimes(w_1, v_2) + \otimes(w_2, v_2)$ where

$$\otimes(w_1, v_2) \in \otimes(W_1 \cdot V_2) \subseteq \text{span}(\otimes(W_1 \cdot V_2)) = W_1 \otimes V_2$$

and

$$\otimes(w_2, v_2) \in \otimes(W_2 \cdot V_2) \subseteq \text{span}(\otimes(W_2 \cdot V_2)) = W_2 \otimes V_2$$

hence $\otimes(v) \in W_1 \otimes V_2 + W_2 \otimes V_2$. So $\otimes(V_1 \cdot V_2) \subseteq W_1 \otimes V_2 + W_2 \otimes V_2$ which using [theorem: 11.109] gives

$$V_1 \otimes V_2 = \text{span}(\otimes(V_1 \cdot V_2)) \subseteq \text{span}(W_1 \otimes V_2 + W_2 \otimes V_2) \quad (13.78)$$

As $W_1 \otimes V_2$ and $W_2 \otimes V_2$ are sub spaces of $V_1 \otimes V_2$ [see theorem: 13.18] we have by [theorem: 12.4] that $W_1 \otimes V_2 + W_2 \otimes V_2$ is a sub-space of $V_1 \otimes V_2$, hence using [theorems: 11.109, 11.111] we have

$$W_1 \otimes V_2 + W_2 \otimes V_2 = \text{span}(W_1 \otimes V_2 + W_2 \otimes V_2) \subseteq V_1 \otimes V_2$$

combining this with [eq: 13.78] gives then

$$W_1 \otimes V_2 + W_2 \otimes V_2 = V_1 \otimes V_2 \quad (13.79)$$

Assume now that there exist a $x \in (W_1 \otimes V_2) \cap (W_2 \otimes V_2)$ with $x \neq 0$. Using [theorem: 13.25] there exists $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq W_1$, $\{v_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq V_2$, $\{w_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq W_2$, $\{v_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq V_2$ such that

$$x = \sum_{i \in \{1, \dots, k_x\}} w_{1,i} \otimes v_{1,i} \quad (13.80)$$

$$x = \sum_{i \in \{1, \dots, l_x\}} w_{2,i} \otimes v_{2,i} \quad (13.81)$$

and $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq W_1$, $\{v_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq V_2$, $\{w_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq W_2$, $\{v_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq V_2$ are linear independent. Using [theorem: 11.189] there exist a $L \in \text{Hom}(V_2, F)$ such that $L(v_{1,1}) = 1$ and $\forall i \in \{2, \dots, k_x\}$ $L(v_{1,i}) = 0$. Let now $K \in \text{Hom}(V_1, F)$ then

$$K \cdot L: V_1 \times V_2 \rightarrow F \text{ by } (K \cdot L)(v_1, v_2) = K(v_1) \cdot L(v_2)$$

then from the linearity of K, L it follows that $K \cdot L \in \text{Hom}(V_1 \otimes V_2, F)$. As $\langle V_1 \otimes V_2, \otimes \rangle$ is a tensor product of V_1, V_2 there exist a $h \in \text{Hom}(V_1 \otimes V_2, F)$ such that

$$K \cdot L = h \circ \otimes \quad (13.82)$$

$$\begin{aligned} h(x) & \stackrel{[\text{eq: 13.80}]}{=} h\left(\sum_{i \in \{1, \dots, k_x\}} w_{1,i} \otimes v_{1,i}\right) \\ & \stackrel{h \text{ is linear}}{=} \sum_{i \in \{1, \dots, k_x\}} h(w_{1,i} \otimes v_{1,i}) \\ & = \sum_{i \in \{1, \dots, k_x\}} h(\otimes(w_{1,i}, v_{1,i})) \\ & \stackrel{[\text{eq: 13.82}]}{=} \sum_{i \in \{1, \dots, k_x\}} (K \cdot L)(w_{1,i}, v_{1,i}) \\ & = \sum_{i \in \{1, \dots, k_x\}} K(w_{1,i}) \cdot L(v_{1,i}) \\ & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{2, \dots, k_x\}} K(w_{1,i}) \cdot L(v_{1,i}) + \sum_{i \in \{1\}} K(w_{1,i}) \cdot L(v_{1,i}) \\ & \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{2, \dots, k_x\}} K(w_{1,i}) \cdot L(v_{1,i}) + K(w_{1,1}) \cdot L(v_{1,1}) \\ & = \sum_{i \in \{2, \dots, k_x\}} 0 \cdot L(v_{1,i}) + K(w_{1,1}) \cdot 1 \\ & \stackrel{[\text{theorem: 11.37}]}{=} K(w_{1,1}) \end{aligned}$$

but we have also

$$\begin{aligned}
 h(x) &\stackrel{[\text{theorem: 13.81}]}{=} h\left(\sum_{i \in \{1, \dots, l_x\}} w_{2,i} \otimes v_{2,i}\right) \\
 &= \sum_{i \in \{1, \dots, l_x\}} h(w_{2,i} \otimes v_{2,i}) \\
 &= \sum_{i \in \{1, \dots, l_x\}} h(\otimes(w_{2,i}, v_{2,i})) \\
 &\stackrel{[\text{eq: 13.82}]}{=} \sum_{i \in \{1, \dots, l_x\}} (K \cdot L)((w_{2,i}, v_{2,i})) \\
 &= \sum_{i \in \{1, \dots, l_x\}} K(w_{2,i}) \cdot L(v_{2,i}) \\
 &= \sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot K(w_{2,i}) \\
 &\stackrel{K \text{ is linear}}{=} K\left(\sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i}\right)
 \end{aligned}$$

so we have

$$\forall K \in \text{Hom}(V_1, F) \text{ we have } K(w_{1,1}) = K\left(\sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i}\right) \quad (13.83)$$

So we can apply [theorem: 11.188] giving that $w_{1,1} = \sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i}$. As $w_{1,1} \in W_1$ and $\sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i} \in W_2$ [as W_2 is a sub-space of V_1] it follows that

$$w_{1,1} \in W_1 \cap W_2 \stackrel{W_1 = \overline{W_1 \oplus W_2}}{=} \{0\}$$

proving that $w_{1,1} = 0$. However this means by [theorem: 11.130] that $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}}$ is linear dependent contradicting the fact that $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}}$ is linear independent. Hence the assumption $x \neq 0$ is wrong and we must have that $x = 0$ or $(W_1 \otimes V_2) \cap (W_2 \otimes V_2) \subseteq \{0\}$. This together with fact that by $W_1 \otimes V_2$, $W_2 \otimes V_2$ are vector spaces so that $0 \in (W_1 \otimes V_2) \cap (W_2 \otimes V_2)$ proves that

$$(W_1 \otimes V_2) \cap (W_2 \otimes V_2) = \{0\}$$

which together with [eq: 13.78] proves that

$$V_1 \otimes V_2 = (W_1 \otimes V_2) \oplus (W_2 \otimes V_2)$$

2. The proof is similar to the proof of (1). If $v = (v_1, v_2) \in V_1 \cdot V_2$ then as $V_2 = W_1 \oplus W_2$ there exists **unique** $w_1 \in W_1$, $w_2 \in W_2$ such that $v_2 = w_1 + w_2$. Hence $\otimes(v) = \otimes(v_1, w_1 + w_2) = \otimes(v_1, w_1) + \otimes(v_1, w_2)$ where

$$\otimes(v_1, w_1) \in \otimes(V_1 \cdot W_1) \subseteq \text{span}(\otimes(V_1 \cdot W_1)) = V_1 \otimes W_1$$

and

$$\otimes(v_1, w_2) \in \otimes(V_1 \cdot W_2) \subseteq \text{span}(\otimes(V_1 \cdot W_2)) = V_1 \otimes W_2$$

hence $\otimes(v) \in V_1 \otimes W_1 + V_1 \otimes W_2$. So $\otimes(V_1 \cdot V_2) \subseteq V_1 \otimes W_1 + V_1 \otimes W_2$ which using [theorem: 11.109] gives

$$V_1 \otimes V_2 = \text{span}(\otimes(V_1 \cdot V_2)) \subseteq \text{span}(V_1 \otimes W_1 + V_1 \otimes W_2) \quad (13.84)$$

As $V_1 \otimes W_1$ and $V_1 \otimes W_2$ are sub spaces of $V_1 \otimes V_2$ [see theorem: 13.18] we have by [theorem: 12.4] that $V_1 \otimes W_1 + V_1 \otimes W_2$ is a sub-space of $V_1 \otimes V_2$, hence using [theorems: 11.109, 11.111] we have

$$V_1 \otimes W_1 + V_1 \otimes W_2 = \text{span}(V_1 \otimes W_1 + V_1 \otimes W_2) \subseteq V_1 \otimes V_2$$

combining this with [eq: 13.84] gives then

$$W_1 \otimes V_2 + W_2 \otimes V_2 = V_1 \otimes V_2 \quad (13.85)$$

Assume now that there exist a $x \in (V_1 \otimes W_1) \cap (V_1 \otimes W_2)$ with $x \neq 0$. Using [theorem: 13.25] there exists $\{v_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq V_1$, $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq W_1$, $\{v_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq V_1$, $\{w_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq W_2$ such that

$$x = \sum_{i \in \{1, \dots, k_x\}} v_{1,i} \otimes w_{1,i} \quad (13.86)$$

$$x = \sum_{i \in \{1, \dots, l_x\}} v_{2,i} \otimes w_{2,i} \quad (13.87)$$

and $\{v_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq V_1$, $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}} \subseteq W_1$, $\{v_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq V_1$, $\{w_{2,i}\}_{i \in \{1, \dots, l_x\}} \subseteq W_2$ are linear independent. Using [theorem: 11.189] there exist a $L \in \text{Hom}(V_1, F)$ such that $L(v_{1,1}) = 1$ and $\forall i \in \{2, \dots, k_x\}$ $L(v_{1,i}) = 0$. Let now $K \in \text{Hom}(V_2, F)$ then

$$L \cdot K: V_1 \cdot V_2 \rightarrow F \text{ by } (L \cdot K)(v_1, v_2) = L(v_1) \cdot K(v_2)$$

then from the linearity of L, K it follows that $L \cdot K \in \text{Hom}(V_1 \otimes V_2, F)$. As $\langle V_1 \otimes V_2, \otimes \rangle$ is a tensor product of V_1, V_2 there exist a $h \in \text{Hom}(V_1 \otimes V_2, F)$ such that

$$L \cdot K = h \circ \otimes \quad (13.88)$$

$$\begin{aligned} h(x) & \stackrel{[\text{eq: 13.86}]}{=} h\left(\sum_{i \in \{1, \dots, k_x\}} v_{1,i} \otimes w_{1,i}\right) \\ & \stackrel{h \text{ is linear}}{=} \sum_{i \in \{1, \dots, k_x\}} h(v_{1,i} \otimes w_{1,i}) \\ & = \sum_{i \in \{1, \dots, k_x\}} h(\otimes(v_{1,i}, w_{1,i})) \\ & \stackrel{[\text{eq: 13.88}]}{=} \sum_{i \in \{1, \dots, k_x\}} (L \cdot K)(v_{1,i}, w_{1,i}) \\ & = \sum_{i \in \{1, \dots, k_x\}} L(v_{1,i}) \cdot K(w_{1,i}) \\ & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{2, \dots, k_x\}} L(v_{1,i}) \cdot K(w_{1,i}) + \sum_{i \in \{1\}} L(v_{1,i}) \cdot K(w_{1,i}) \\ & \stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{2, \dots, k_x\}} L(v_{1,i}) \cdot K(w_{1,i}) + L(v_{1,1}) \cdot K(w_{1,1}) \\ & = \sum_{i \in \{2, \dots, k_x\}} 0 \cdot L(v_{1,i}) + K(w_{1,1}) \cdot 1 \\ & \stackrel{[\text{theorem: 11.37}]}{=} K(w_{1,1}) \end{aligned}$$

but we have also

$$\begin{aligned} h(x) & \stackrel{[\text{theorem: 13.87}]}{=} h\left(\sum_{i \in \{1, \dots, l_x\}} v_{2,i} \otimes w_{2,i}\right) \\ & = \sum_{i \in \{1, \dots, l_x\}} h(v_{2,i} \otimes w_{2,i}) \\ & = \sum_{i \in \{1, \dots, l_x\}} h(\otimes(v_{2,i}, w_{2,i})) \\ & \stackrel{[\text{eq: 13.88}]}{=} \sum_{i \in \{1, \dots, l_x\}} (L \cdot K)((v_{2,i}, w_{2,i})) \\ & = \sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot K(w_{2,i}) \\ & = \sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot K(w_{2,i}) \\ & \stackrel{K \text{ is linear}}{=} K\left(\sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i}\right) \end{aligned}$$

so we have

$$\forall K \in \text{Hom}(V_1, F) \text{ we have } K(w_{1,1}) = K\left(\sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i}\right) \quad (13.89)$$

So we can apply [theorem: 11.188] giving that $w_{1,1} = \sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i}$. As $w_{1,1} \in W_1$ and $\sum_{i \in \{1, \dots, l_x\}} L(v_{2,i}) \cdot w_{2,i} \in W_2$ [as W_2 is a sub-space of V_1] it follows that

$$w_{1,1} \in W_1 \cap W_2 \underset{V_1 = W_1 \oplus W_2}{=} \{0\}$$

proving that $w_{1,1} = 0$. However this means by [theorem: 11.130] that $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}}$ is linear dependent contradicting the fact that $\{w_{1,i}\}_{i \in \{1, \dots, k_x\}}$ is linear independent. Hence the assumption $x \neq 0$ is wrong and we must have that $x = 0$ or $(V_1 \otimes W_1) \cap (V_1 \otimes W_2) \subseteq \{0\}$. This together with fact that by $V_1 \otimes W_1, V_1 \otimes W_2$ are vector spaces so that $0 \in (V_1 \otimes W_1) \cap (V_1 \otimes W_2)$ proves that

$$(V_1 \otimes W_1) \cap (V_1 \otimes W_2) = \{0\}$$

which together with [eq: 13.78] proves that

$$V_1 \otimes V_2 = (V_1 \otimes W_1) \oplus (V_1 \otimes W_2) \quad \square$$

Lemma 13.35. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1,2\}}$ be vector spaces over a field F then

1. If $V_1 = \bigoplus_{i \in \{1, \dots, n\}} W_i$ then if $\langle V_1 \otimes V_2, \otimes \rangle$ is a tensor product of V_1, V_2 we have

$$V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} W_i \otimes V_2$$

where $\forall i \in \{1, \dots, n\}$ $W_i \otimes V_2 = \text{span}(\otimes(W_i \cdot V_2))$ so that $\langle W_i \otimes V_2, \otimes_{|W_i \cdot V_2} \rangle$ is a tensor product of W_i, V_2 [see theorem: 13.18]

2. If $V_2 = \bigoplus_{i \in \{1, \dots, n\}} W_i$ then if $\langle V_1 \otimes V_2, \otimes \rangle$ is a tensor product of V_1, V_2 we have

$$V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} V_1 \otimes W_i$$

where $\forall i \in \{1, \dots, n\}$ $V_1 \otimes W_i = \text{span}(\otimes(V_1 \cdot W_i))$ so that $\langle V_1 \otimes W_i, \otimes_{|V_1 \cdot W_i} \rangle$ is a tensor product of V, W_i [see theorem: 13.18]

Proof.

1. We proceed by mathematical induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } V_1, V_2 \text{ are vector spaces over a field } F \text{ such that } V_1 = \bigoplus_{i \in \{1, \dots, n\}} W_i \text{ then } V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} W_i \otimes V_2 \right\}$$

then we have that:

$1 \in S$. Then if $V_1 = \sum_{i \in \{1\}}^{\oplus} W_i$ we have by [example: 12.7] that $V_1 = W_1$ and

$$V_1 \otimes V_2 = W_1 \otimes V_2 \underset{[\text{example: 12.7}]}{=} \sum_{i \in \{1\}}^{\oplus} W_i \otimes V_2$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $V_1 = \bigoplus_{i \in \{1, \dots, n+1\}} W_i$ then we have by [theorem: 12.11] that

$$\sum_{i \in \{1, \dots, n\}} W_i = \bigoplus_{i \in \{1, \dots, n\}} W_i \text{ and } V_1 = \left(\sum_{i \in \{1, \dots, n\}} W_i \right) \oplus W_{n+1}$$

or, if we take $V_1' = \sum_{i \in \{1, \dots, n\}} W_i$, that

$$V_1' = \bigoplus_{i \in \{1, \dots, n\}} W_i \text{ and } V_1 = V_1' \oplus W_{n+1} \quad (13.90)$$

By [lemma: 13.34] we have then

$$V_1 \otimes V_2 = (V_1' \otimes V_2) \oplus (W_{n+1} \otimes V_2) \quad (13.91)$$

Further as $V_1' = \bigoplus_{i \in \{1, \dots, n\}} V_i$ and $n \in S$ we have that

$$V_1' \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} W_i \otimes V_2 \quad (13.92)$$

Using then [theorem: 12.9] on [eqs: 13.91 and 13.92] results in

$$V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n+1\}} W_i \otimes V_2$$

proving that $n+1 \in S$

2. We proceed by mathematical induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } V_1, V_2 \text{ are vector spaces over a field } F \text{ such that } V_2 = \bigoplus_{i \in \{1, \dots, n\}} W_i \text{ then } V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} V_1 \otimes W_i \right\}$$

then we have that:

$1 \in S$. Then if $V_2 = \sum_{i \in \{1\}}^{\oplus} W_i$ we have by [example: 12.7] that $V_2 = W_1$ and

$$V_1 \otimes V_2 = V_1 \otimes W_1 \stackrel{[\text{example: 12.7}]}{=} \bigoplus_{i \in \{1\}} V_1 \otimes W_i$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $V_2 = \bigoplus_{i \in \{1, \dots, n+1\}} W_i$ then we have by [theorem: 12.11] that

$$\sum_{i \in \{1, \dots, n\}} W_i = \bigoplus_{i \in \{1, \dots, n\}} W_i \text{ and } V_2 = \left(\sum_{i \in \{1, \dots, n\}} W_i \right) \oplus W_{n+1}$$

or, if we take $V_{12}' = \sum_{i \in \{1, \dots, n\}} W_i$, that

$$V_2' = \bigoplus_{i \in \{1, \dots, n\}} W_i \text{ and } V_2 = V_2' \oplus W_{n+1} \quad (13.93)$$

By [lemma: 13.34] we have then

$$V_1 \otimes V_2 = (V_1 \otimes V_2') \oplus (V_1 \otimes W_{n+1}) \quad (13.94)$$

Further as $V_2' = \bigoplus_{i \in \{1, \dots, n\}} V_i$ and $n \in S$ we have that

$$V_1 \otimes V_2' = \bigoplus_{i \in \{1, \dots, n\}} V_1 \otimes W_i \quad (13.95)$$

Using then [theorem: 12.9] on [eqs: 13.94 and 13.95] results in

$$V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n+1\}} V_1 \otimes W_i$$

proving that $n+1 \in S$ □

Corollary 13.36. *Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1,2\}}$ be vector spaces over a field F and I a finite set then*

1. *If $V_1 = \bigoplus_{i \in I} W_i$ then if $\langle V_1 \otimes V_2, \otimes \rangle$ is a tensor product of V_1, V_2 we have*

$$V_1 \otimes V_2 = \bigoplus_{i \in I} W_i \otimes V_2$$

where $\forall i \in I$ $W_i \otimes V_2 = \text{span}(\otimes(W_i \cdot V_2))$ so that $\langle W_i \otimes V_2, \otimes_{|W_i \cdot V_2} \rangle$ is a tensor product of W_i, V_2 [see theorem: 13.18]

2. *If $V_2 = \bigoplus_{i \in I} W_i$ then if $\langle V_1 \otimes V_2, \otimes \rangle$ is a tensor product of V_1, V_2 we have*

$$V_1 \otimes V_2 = \bigoplus_{i \in I} V_1 \otimes W_i$$

where $\forall i \in I$ $V_1 \otimes W_i = \text{span}(\otimes(V_1 \cdot W_i))$ so that $\langle V_1 \otimes W_i, \otimes_{|V_1 \cdot W_i} \rangle$ is a tensor product of V, W_i [see theorem: 13.18]

Proof. As I is finite there exist a $n \in \mathbb{N}$ and a bijection $\beta: I \rightarrow \{1, \dots, n\}$

1. Define $\{U_i\}_{i \in \{1, \dots, n\}}$ by $U_i = W_{\beta(i)}$ then we have

$$\bigoplus_{i \in \{1, \dots, n\}} U_i = \bigoplus_{i \in \{1, \dots, n\}} W_{\beta(i)} \stackrel{[\text{theorem: 12.13}]}{=} \bigoplus_{i \in I} W_i = V_1$$

Using the previous lemma [lemma: 13.34] we have then that

$$V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} U_i \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} W_{\beta(i)} \otimes V_2 \stackrel{[\text{theorem: 12.13}]}{=} \bigoplus_{i \in I} W_i \otimes V_2$$

and $\forall i \in \{1, \dots, n\}$ $U_i \otimes V_2 = \text{span}(\otimes(U_i \cdot V_2))$. Hence we have

$$V_1 \otimes V_2 = \bigoplus_{i \in I} W_i \otimes V_2$$

and if $i \in I$ we have, as $W_i = W_{\beta(\beta^{-1}(i))} = U_{\beta^{-1}(i)}$, that

$$W_i \otimes V_2 = U_{\beta^{-1}(i)} \otimes V_2 = \text{span}(\otimes(U_{\beta^{-1}(i)} \cdot V_2)) = \text{span}(\otimes(W_i \cdot V_2))$$

2. Define $\{U_i\}_{i \in \{1, \dots, n\}}$ by $U_i = W_{\beta(i)}$ then we have

$$\bigoplus_{i \in \{1, \dots, n\}} U_i = \bigoplus_{i \in \{1, \dots, n\}} W_{\beta(i)} \stackrel{[\text{theorem: 12.13}]}{=} \bigoplus_{i \in I} W_i = V_2$$

Using the previous lemma [lemma: 13.34] we have then that

$$V_1 \otimes V_2 = \bigoplus_{i \in \{1, \dots, n\}} V_1 \otimes U_i = \bigoplus_{i \in \{1, \dots, n\}} V_1 \otimes W_{\beta(i)} \stackrel{[\text{theorem: 12.13}]}{=} \bigoplus_{i \in I} V_1 \otimes W_i$$

and $\forall i \in \{1, \dots, n\}$ $V_1 \otimes U_i = \text{span}(\otimes(V_1 \cdot U_i))$. Hence we have

$$V_1 \otimes V_2 = \bigoplus_{i \in I} V_1 \otimes W_i$$

and if $i \in I$ we have, as $W_i = W_{\beta(\beta^{-1}(i))} = U_{\beta^{-1}(i)}$, that

$$V_1 \otimes W_i = U_{\beta^{-1}(i)} \otimes V_1 = \text{span}(\otimes(V_1 \cdot U_{\beta^{-1}(i)})) = \text{span}(\otimes(V_1 \cdot W_i))$$

□

Lemma 13.37. *Let V, W be two vector spaces over a field such that*

$$V = \bigoplus_{i \in \{1, \dots, k\}} V_i \text{ and } W = \bigoplus_{i \in \{1, \dots, l\}} W_i$$

then if $\langle V \otimes W, \otimes \rangle$ is a tensor product of V, W we have

$$V \otimes W = \bigoplus_{\gamma \in \Gamma(k, l)} V_{\gamma(1)} \otimes W_{\gamma(2)}$$

where as usual $\forall \gamma \in \Gamma(k, l)$ $V_{\gamma(1)} \otimes W_{\gamma(2)} = \text{span}(\otimes(V_{\gamma(1)} \cdot W_{\gamma(2)}))$ so that $\langle V_{\gamma(1)} \otimes W_{\gamma(2)}, \otimes_{V_{\gamma(1)} \cdot W_{\gamma(2)}} \rangle$ is a tensor product of $V_{\gamma(1)} \cdot W_{\gamma(2)}$ [see theorem: 13.18]

Proof. We use induction to prove this. Let V be a vector space over a field F such that $V = \bigoplus_{i \in \{1, \dots, k\}} V_i$. Define

$$S_V = \left\{ l \in \mathbb{N} \mid \text{If } W \text{ is a vector space over } F \text{ with } W = \bigoplus_{i \in \{1, \dots, l\}} W_i \text{ then } V \otimes W = \bigoplus_{\gamma \in \Gamma(k, l)} V_{\gamma(1)} \otimes W_{\gamma(2)} \right\}$$

then we have:

$1 \in S_V$. If $W = \bigoplus_{i \in \{1\}} W_i$ then we have by [example: 12.7] that $W = W_1$ so we can use [lemma: 13.35] to get

$$V \otimes W = \bigoplus_{i \in \{1, \dots, k\}} V_i \otimes W_1 \quad (13.96)$$

Define

$$\tau: \{1, \dots, k\} \rightarrow \Gamma(k, 1) \text{ where } \tau(i) \text{ is defined by } \tau(i) = (i, 1) \text{ so that } \tau(i)(1) = i \wedge \tau(i)(2) = 1$$

then we have:

injectivity. If $\tau(i) = \tau(j)$ then $i = \gamma(i)(1) = \gamma(j)(1) = j$ proving injectivity.

surjectivity. If $\gamma \in \Gamma(k, 1)$ then $\gamma(1) \in \{1, \dots, k\}$ and $\gamma(2) \in \{1\} \Rightarrow \gamma(2) = 1$, hence if we take $i = \gamma(1)$ we have $\tau(i)(1) = i = \gamma(1) \wedge \tau(i)(2) = 1 = \gamma(2)$ proving that $\tau(i) = \gamma$.

so we conclude that

$$\tau: \{1, \dots, k\} \rightarrow \Gamma(k, 1) \text{ is a bijection with } \tau(i)(1) = i \wedge \tau(i)(2) = 1 \quad (13.97)$$

Next

$$\begin{aligned} \bigoplus_{i \in \{1, \dots, k\}} V_i \otimes W_1 &= \bigoplus_{i \in \{1, \dots, k\}} V_{\tau(i)(1)} \otimes W_{\tau(i)(2)} \\ &\stackrel{[\text{theorem: 12.13}]}{=} \bigoplus_{\gamma \in \Gamma(k, 1)} V_{\gamma(1)} \otimes W_{\gamma(2)} \end{aligned}$$

which together with [eq: 13.96] proves that $1 \in S_V$.

$l \in S \Rightarrow l + 1 \in S_V$. If $W = \bigoplus_{i \in \{1, \dots, l+1\}} W_i$ then we have by [theorem: 12.11] that

$$\sum_{i \in \{1, \dots, l\}} W_i = \bigoplus_{i \in \{1, \dots, l\}} W_i \text{ and } W = \left(\sum_{i \in \{1, \dots, l\}} W_i \right) \oplus W_{l+1}$$

or

$$W' = \bigoplus_{i \in \{1, \dots, l\}} W_i \text{ and } W = W' \oplus W_{l+1} \text{ where } W' = \sum_{i \in \{1, \dots, l\}} W_i \quad (13.98)$$

Hence using [lemma: 13.34]

$$V \otimes W = (V \otimes W') \oplus (V \otimes W_{l+1}) \quad (13.99)$$

As $l \in S$ and $W' \stackrel{[\text{eq: 13.98}]}{=} \bigoplus_{i \in \{1, \dots, l\}} W_i$ we have that

$$V \otimes W' = \bigoplus_{\gamma \in \Gamma(k, l)} V_{\gamma(1)} \otimes W_{\gamma(2)} \quad (13.100)$$

Further, as $V = \bigoplus_{i \in \{1, \dots, k\}} V_i$, we have by [lemma: 13.35] that

$$V \otimes W_{l+1} = \bigoplus_{i \in \{1, \dots, k\}} V_i \otimes W_{l+1} \quad (13.101)$$

Define

$$\chi: \{1, \dots, k\} \rightarrow \{1, \dots, k\} \cdot \{l+1\} \text{ where } \chi(i) = (i, l+1) \Rightarrow \chi(i)(1) = i \wedge \chi(i)(2) = l+1$$

then we have:

injectivity. If $\chi(i) = \chi(j)$ then $i = \chi(i)(1) = \chi(j)(1) = j$ proving injectivity.

surjectivity. Let $\gamma \in \{1, \dots, k\} \cdot \{l+1\}$ and take $i = \gamma(1)$ then we have $\chi(i)(1) = i = \gamma(1)$ and $\chi(i)(2) = l+1 = \gamma(2)$ [as $\gamma(2) \in \{l+1\}$] so that $\chi(i) = \gamma$

So

$$\chi: \{1, \dots, k\} \rightarrow \{1, \dots, k\} \cdot \{l+1\} \text{ where } \chi(i)(1) = i \wedge \chi(i)(2) = l+1 \text{ is a bijection} \quad (13.102)$$

Using this bijection we have

$$\begin{aligned} V \otimes W_{l+1} & \stackrel{[\text{eq: 13.101}]}{=} \bigoplus_{i \in \{1, \dots, k\}} V_i \otimes W_{l+1} \\ & = \bigoplus_{i \in \{1, \dots, k\}} V_i \otimes W_{l+1} \\ & = \bigoplus_{i \in \{1, \dots, k\}} V_{\chi(i)(1)} \otimes W_{\chi(i)(2)} \\ & \stackrel{[\text{eq: 13.102}] \text{ and } [\text{theorem: 12.13}]}{=} \bigoplus_{\gamma \in \{1, \dots, k\} \cdot \{l+1\}} V_{\gamma(1)} \otimes W_{\gamma(2)} \end{aligned}$$

so that

$$V \otimes W_{l+1} = \bigoplus_{\gamma \in \{1, \dots, k\} \cdot \{l+1\}} V_{\gamma(1)} \otimes W_{\gamma(2)} \quad (13.103)$$

Now

$$\begin{aligned} \Gamma(k, l+1) & = \{1, \dots, k\} \cdot \{1, \dots, l+1\} \\ & = \{1, \dots, k\} \cdot (\{1, \dots, l\} \cup \{l+1\}) \\ & \stackrel{[\text{theorem: 6.89}]}{=} (\{1, \dots, k\} \cdot \{1, \dots, l\}) \cup (\{1, \dots, k\} \cdot \{l+1\}) \\ & = \Gamma(k, l) \cup (\{1, \dots, k\} \cdot \{l+1\}) \end{aligned}$$

and

$$\begin{aligned} \Gamma(k, l) \cap (\{1, \dots, k\} \cdot \{l+1\}) & = (\{1, \dots, k\} \cdot \{1, \dots, l\}) \cap (\{1, \dots, k\} \cdot \{l+1\}) \\ & \stackrel{[\text{theorem: 6.89}]}{=} (\{1, \dots, k\} \cap \{1, \dots, k\}) \cdot (\{1, \dots, l\} \cap \{l+1\}) \\ & = (\{1, \dots, k\} \cap \{1, \dots, k\}) \cdot \emptyset \\ & \stackrel{[\text{theorem: 2.146}]}{=} \emptyset \end{aligned}$$

So we have

$$\begin{aligned} \Gamma(k, l+1) & = \Gamma(k, l) \cup (\{1, \dots, k\} \cdot \{l+1\}) \wedge \Gamma(k, l) \cap (\{1, \dots, k\} \cdot \{l+1\}) = \emptyset \\ V \otimes W' & \stackrel{[\text{eq: 13.100}]}{=} \bigoplus_{\gamma \in \Gamma(k, l)} V_{\gamma(1)} \otimes W_{\gamma(2)} \\ V \otimes W_{l+1} & \stackrel{[\text{eq: 13.103}]}{=} \bigoplus_{\gamma \in \{1, \dots, k\} \cdot \{l+1\}} V_{\gamma(1)} \otimes W_{\gamma(2)} \\ V \otimes W & \stackrel{[\text{eq: 13.99}]}{=} (V \otimes W') \oplus (V \otimes W_{l+1}) \end{aligned}$$

which allows us to use [theorem: 12.9] to get

$$V \otimes W = \bigoplus_{\gamma \in \Gamma(k, l+1)} V_{\gamma(1)} \otimes W_{\gamma(2)}$$

which proves that

$$l+1 \in S \quad \square$$

Theorem 13.38. Let $n \in \mathbb{N}_0$, $\{V_i\}_{i \in \{1, \dots, n\}}$ be vector spaces over a field V such that $\forall i \in \{1, \dots, n\}$ $\{W_{i,j}\}_{j \in \{1, \dots, k_i\}} \subseteq \mathcal{P}(V_i)$ is a family of sub-spaces of V_i such that $V_i = \bigoplus_{j \in \{1, \dots, k_i\}} W_{i,j}$ then we have for a tensor product $\langle V_1 \otimes \dots \otimes V_n, \otimes \rangle$ that

$$V_1 \otimes \dots \otimes V_n = \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} (W_{1,\gamma(1)} \otimes \dots \otimes W_{n,\gamma(n)})$$

where

$$W_{1,\gamma(1)} \otimes \dots \otimes W_{n,\gamma(n)} \text{ is a tensor product of } W_{1,\gamma(1)}, \dots, W_{n,\gamma(n)}$$

Proof. We prove this by induction on n , so let

$$S = \left\{ n \in \mathbb{N} \mid \{V_i\}_{i \in \{1, \dots, n\}} \text{ are vector spaces, } \{W_{i,j}\}_{j \in \{1, \dots, k_i\}} \subseteq \mathcal{P}(V_i), V_i = \bigoplus_{j \in \{1, \dots, k_i\}} W_{i,j}, \right. \\ \left. i \in \{1, \dots, n\} \text{ and } \langle V_1 \otimes \dots \otimes V_n, \otimes \rangle \text{ a tensor product of } V_1, \dots, V_n \text{ then } V_1 \otimes \dots \otimes V_n = \right. \\ \left. \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} W_{1,\gamma(1)} \otimes \dots \otimes W_{n,\gamma(n)} \text{ where } W_{1,\gamma(1)} \otimes \dots \otimes W_{n,\gamma(n)} \text{ is a tensor product of } W_{1,\gamma(1)}, \dots, \right. \\ \left. W_{n,\gamma(n)} \right\}$$

then we have:

1 $\in S$. Let $\langle V_1 \otimes \dots \otimes V_1, \otimes \rangle$ be a tensor product of $V_1, \dots, V_1$ and $\{W_{1,j}\}_{j \in \{1, \dots, k_1\}}$ a family of sub-spaces of V_1 such that $V_1 = \bigoplus_{j \in \{1, \dots, k_1\}} W_{1,j}$. Using [theorem: 13.22] there exist for every $j \in \{1, \dots, k_1\}$ a tensor product $W_{1,j} \otimes \dots \otimes W_{1,j}$ of $W_{1,j}, \dots, W_{1,j}$ such that

$$V_1 \otimes \dots \otimes V_1 = \bigoplus_{j \in \{1, \dots, k_1\}} W_{1,j} \otimes \dots \otimes W_{1,j}$$

Define now

$$\kappa: \{1, \dots, k_1\} \rightarrow \Gamma(k_1) \text{ by } \kappa(i)(1) = i$$

then we have:

injectivity. If $i, j \in \{1, \dots, k_1\}$ satisfies $\kappa(i) = \kappa(j)$ then $i = \kappa(i)(1) = \kappa(j)(1) = j$ proving injectivity.

surjectivity. Let $\gamma \in \Gamma(k_1)$ then for $i = \gamma(1)$ we have $\kappa(i)(1) = i = \gamma(1)$ proving that $\kappa(i) = \gamma$ hence surjectivity.

proving that

$$\kappa: \{1, \dots, k\} \rightarrow \Gamma(k_1) \text{ where } \kappa(i)(1) = i$$

So

$$\begin{aligned} V_1 \otimes \dots \otimes V_1 &= \bigoplus_{j \in \{1, \dots, k_1\}} W_{1,j} \otimes \dots \otimes W_{1,j} \\ &= \bigoplus_{j \in \{1, \dots, k_1\}} W_{1,\kappa(j)(1)} \otimes \dots \otimes W_{1,\kappa(j)(1)} \\ &\stackrel{[\text{theorem: 12.13}]}{=} \bigoplus_{\gamma \in \Gamma(k_1)} W_{1,\gamma(1)} \otimes \dots \otimes W_{1,\gamma(1)} \end{aligned}$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\langle V_1 \otimes \cdots \otimes V_{n+1}, \otimes_{n+1} \rangle$ be a tensor product of $V_1, \dots, V_n$ and $\forall i \in \{1, \dots, n+1\}$ $\{W_{i,j}\}_{j \in \{1, \dots, k_{n+1}\}}$ a family of sub-spaces of V_i such that $V_i = \bigoplus_{j \in \{1, \dots, k_{n+1}\}} W_{i,j}$. Using [theorem: 13.28] we have that there exist a tensor product $V_1 \otimes \cdots \otimes V_n$ of $V_1, \dots, V_n$ and a tensor product $(V_1 \otimes \cdots \otimes V_n) \circ V_{n+1}$ of $V_1 \otimes \cdots \otimes V_n, V_{n+1}$ such that

$$V_1 \otimes \cdots \otimes V_{n+1} = (V_1 \otimes \cdots \otimes V_n) \otimes V_{n+1} \quad (13.104)$$

As $n \in S$ there exists $\forall \gamma \in \Gamma(k_1, \dots, k_n)$ a tensor product $W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)}$ of $W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)}$ such that

$$V_1 \otimes \cdots \otimes V_n = \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)} \quad (13.105)$$

So we have

$$\begin{aligned} & V_1 \otimes \cdots \otimes V_{n+1} \quad [\text{eq: 13.104}] \\ & (V_1 \otimes \cdots \otimes V_n) \otimes V_{n+1} \quad [\text{eq: 13.105}] \\ & \left(\bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)} \right) \otimes V_{n+1} = \\ & \left(\bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)} \right) \otimes \left(\bigoplus_{j \in \{1, \dots, k_{n+1}\}} W_{n+1,j} \right) \quad [\text{theorem: 13.35}] \\ & \bigoplus_{j \in \{1, \dots, k_{n+1}\}} \left(\left(\bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)} \right) \otimes W_{n+1,j} \right) \quad [\text{theorem: 13.36}] \\ & \bigoplus_{j \in \{1, \dots, k_{n+1}\}} \left(\bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} (W_{1,\gamma(1)} \otimes \cdots \otimes W_{n,\gamma(n)}) \otimes W_{n+1,j} \right) \end{aligned} \quad (13.106)$$

Given $j \in \{1, \dots, k_{n+1}\}$ define

$$\Gamma^j(k_1, \dots, k_{n+1}) = \{\gamma \in \Gamma(k_1, \dots, k_{n+1}) \mid \gamma(n+1) = j\} \subseteq \Gamma(k_1, \dots, k_{n+1})$$

If $\gamma \in \Gamma(k_1, \dots, k_{n+1})$ then clearly $\gamma(n+1) \in \{1, \dots, k_{n+1}\}$ so that $\gamma \in \Gamma^{\gamma(n+1)}(k_1, \dots, k_{n+1})$ proving that $\gamma \in \bigcup_{j \in \{1, \dots, k_{n+1}\}} \Gamma^j(k_1, \dots, k_{n+1})$. Hence

$$\Gamma(k_1, \dots, k_{n+1}) = \bigcup_{j \in \{1, \dots, k_{n+1}\}} \Gamma^j(k_1, \dots, k_{n+1})$$

Let $i, j \in \{1, \dots, k_{n+1}\}$ with $i \neq j$ then if $\gamma \in \Gamma^i(k_1, \dots, k_{n+1}) \cap \Gamma^j(k_1, \dots, k_{n+1})$ we would have the contradiction $\gamma(n+1) = i \neq j = \gamma(n+1)$ hence $\Gamma^i(k_1, \dots, k_{n+1}) \cap \Gamma^j(k_1, \dots, k_{n+1}) = \emptyset$. So

$$\Gamma(k_1, \dots, k_{n+1}) \text{ is the disjoint union of } \Gamma^j(k_1, \dots, k_{n+1}) \text{ } j \in \{1, \dots, k_{n+1}\} \quad (13.107)$$

Further if we define

$$\alpha_j: \Gamma^j(k_1, \dots, k_{n+1}) \rightarrow \Gamma(k_1, \dots, k_n) \text{ by } \alpha_j(\gamma) = \gamma|_{\{1, \dots, n\}} \text{ [hence } \alpha_j(\gamma)(i) = \gamma(i) \forall i \in \{1, \dots, n\}]$$

then we have:

injectivity. If $\alpha_j(\gamma) = \alpha_j(\delta)$ then $\forall i \in \{1, \dots, n\}$ $\gamma(i) = \alpha_j(\gamma)(i) = \alpha_j(\delta)(i) = \delta(i)$ and $\gamma(n+1) = j = \delta(n+1)$, proving that $\gamma = \delta$.

surjectivity. If $\gamma \in \Gamma(k_1, \dots, k_n)$ define then $\gamma' \in \Gamma^j(k_1, \dots, k_{n+1})$ by $\gamma'(i) = \begin{cases} j & \text{if } i = n+1 \\ \gamma(i) & \text{if } i \in \{1, \dots, n\} \end{cases}$ then $\alpha_j(\gamma') = \gamma$.

so we have that

$$\alpha_j: \Gamma^j(k_1, \dots, k_{n+1}) \rightarrow \Gamma(k_1, \dots, k_n) \text{ is a bijection with } \forall i \in \{1, \dots, n\} \alpha_j(\gamma)(i) = \gamma(i) \quad (13.108)$$

So we have by [theorem: 12.3] and [eq: 13.107] that

$$\begin{aligned}
 & \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_{n+1})} (W_{1, \gamma(1)} \otimes \dots \otimes W_{n, \gamma(n)}) \otimes W_{n+1, \gamma(n+1)} = \\
 & \bigotimes_{i \in \{1, \dots, k_{n+1}\}} \left(\bigoplus_{\gamma \in \Gamma^j(k_1, \dots, k_{n+1})} (W_{1, \gamma(1)} \otimes \dots \otimes W_{n, \gamma(n)}) \otimes W_{n+1, \gamma(n+1)} \right) \quad \gamma(n+1)=j \\
 & \bigotimes_{i \in \{1, \dots, k_{n+1}\}} \left(\bigoplus_{\gamma \in \Gamma^j(k_1, \dots, k_{n+1})} (W_{1, \gamma(1)} \otimes \dots \otimes W_{n, \gamma(n)}) \otimes W_{n+1, j} \right) \quad [\text{eq: 13.108}] \\
 & \bigotimes_{i \in \{1, \dots, k_{n+1}\}} \left(\bigoplus_{\gamma \in \Gamma^j(k_1, \dots, k_{n+1})} (W_{1, \alpha_j(\gamma)(1)} \otimes \dots \otimes W_{n, \alpha_j(\gamma)(n)}) \otimes W_{n+1, j} \right) \quad [\text{theorem: 12.13}] \\
 & \bigotimes_{i \in \{1, \dots, k_{n+1}\}} \left(\bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} (W_{1, \gamma(1)} \otimes \dots \otimes W_{n, \gamma(n)}) \otimes W_{n+1, j} \right) \quad [\text{eq: 13.106}] \\
 & V_1 \otimes \dots \otimes V_{n+1}
 \end{aligned}$$

or summarized

$$V_1 \otimes \dots \otimes V_{n+1} = \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_{n+1})} (W_{1, \gamma(1)} \otimes \dots \otimes W_{n, \gamma(n)}) \otimes W_{n+1, \gamma(n+1)} \quad (13.109)$$

Applying [theorem: 13.26] we have then that

$$(W_{1, \gamma(1)} \otimes \dots \otimes W_{n, \gamma(n)}) \otimes W_{n+1, \gamma(n+1)} = W_{1, \gamma(1)} \otimes \dots \otimes W_{n+1, \gamma(n+1)}$$

a tensor product of $W_{1, \gamma(1)}, \dots, W_{n+1, \gamma(n+1)}$. Hence we have

$$V_1 \otimes \dots \otimes V_{n+1} = \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_{n+1})} W_{1, \gamma(1)} \otimes \dots \otimes W_{n+1, \gamma(n+1)}$$

proving that $n+1 \in S$. $\square$

The above theorem can be used to construct a basis of a tensor product out of the bases of the factors. First we consider the one dimensional case.

Lemma 13.39. *Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, m\}}$ a family of one-dimensional vector spaces over a field F and $\forall i \in \{1, \dots, n\}$ $\{e_i\}$ a basis for V_i [so that $e_i \neq 0$ and $V_i = \text{span}(\{e_i\})$] then for every tensor product $\langle V_1 \otimes \dots \otimes V_n, \otimes \rangle$ of $V_1, \dots, V_n$ we have*

$$V_1 \otimes \dots \otimes V_n = \text{span}(\{\otimes(e_1, \dots, e_n)\}) \underset{\text{notation}}{=} \text{span}(\{e_1 \otimes \dots \otimes e_n\})$$

Proof. Define

$$P = \text{span}(\{\otimes(e_1, \dots, e_n)\})$$

and

$$\nu: \prod_{i \in \{1, \dots, n\}} V_i = \prod_{i \in \{1, \dots, n\}} \text{span}(\{e_i\}) \rightarrow P \text{ by } \nu(v_1, \dots, v_n) = \left(\prod_{i \in \{1, \dots, n\}} \lambda_i \right) \cdot \otimes(e_1, \dots, e_n)$$

where $\forall i \in \{1, \dots, n\}$ $v_i = \lambda_i \cdot e_i$ [the unique expansion of a vector in a given basis]. Then we proof that

$$\langle P, \nu \rangle \text{ is a tensor product of } V_1, \dots, V_n \quad (13.110)$$

Proof.

1. By [theorem: 11.110] $P = \text{span}(\{\otimes(e_1, \dots, e_n)\})$ is a vector space.

2. Let $i \in \{1, \dots, n\}$ $u, v \in V_i$, $\alpha \in F$ then for $(v_1, \dots, v_{i-1}, u + \alpha \cdot v_{i+1}, \dots, v_n)$ we have that there exists a unique $\{\lambda_j\}_{j \in \{1, \dots, n\}} \subseteq F$ such that $v_i = \lambda_i \cdot e_i$ and a $\alpha_u, \alpha_v \in F$ such that $u = \alpha_u \cdot e_i$ and $v = \alpha_v \cdot e_i$. Define then

$$\{\beta_j\}_{j \in \{1, \dots, n\}} \subseteq F \text{ by } \beta_j = \begin{cases} \alpha_u + \alpha \cdot \alpha_v & \text{if } j = i \\ \lambda_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

$$\{\zeta_j\}_{j \in \{1, \dots, n\}} \subseteq F \text{ by } \zeta_j = \begin{cases} \alpha_u & \text{if } j = i \\ \lambda_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

$$\{\gamma_j\}_{j \in \{1, \dots, n\}} \subseteq F \text{ by } \gamma_j = \begin{cases} \alpha_v & \text{if } j = i \\ \lambda_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

so that $\forall j \in \{1, \dots, n\}$ we have

$$\beta_j \cdot e_j = \begin{cases} \alpha_u \cdot e_i + \alpha \cdot \alpha_v \cdot e_i & \text{if } j = i \\ \lambda_j \cdot e_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases} = \begin{cases} u + \alpha \cdot v & \text{if } j = i \\ v_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

$$\zeta_j \cdot e_j = \begin{cases} \alpha_u \cdot e_i & \text{if } j = i \\ \lambda_j \cdot e_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases} = \begin{cases} u & \text{if } j = i \\ v_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

$$\gamma_j \cdot e_j = \begin{cases} \alpha_v \cdot e_i & \text{if } j = i \\ \lambda_j \cdot e_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases} = \begin{cases} v & \text{if } j = i \\ v_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

Hence

$$\begin{aligned} & \nu(v_1, \dots, v_{i-1}, u + \alpha \cdot v, v_{i+1}, \dots, v_n) = \\ & \left(\prod_{j \in \{1, \dots, n\}} \beta_j \right) \cdot \otimes(e_1, \dots, e_n) \stackrel{[\text{theorem: 11.43}]}{=} \\ & \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \beta_j \right) \cdot \left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \beta_j \right) \right) \cdot \otimes(e_1, \dots, e_n) \stackrel{[\text{theorem: 11.34}]}{=} \\ & \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \beta_j \right) \cdot \beta_i \right) \cdot \otimes(e_1, \dots, e_n) = \\ & \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \lambda_j \right) \cdot (\alpha_u + \alpha \cdot \alpha_v) \right) \cdot \otimes(e_1, \dots, e_n) = \\ & \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \lambda_j \right) \cdot \alpha_u \right) \cdot \otimes(e_1, \dots, e_n) + \alpha \cdot \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \lambda_j \right) \cdot \alpha_v \right) \cdot \\ & \otimes(e_1, \dots, e_n) = \\ & \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \zeta_j \right) \cdot \zeta_i \right) \cdot \otimes(e_1, \dots, e_n) + \alpha \cdot \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \gamma_j \right) \cdot \gamma_i \right) \cdot \\ & \otimes(e_1, \dots, e_n) \stackrel{[\text{theorem: 11.34}]}{=} \\ & \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \zeta_j \right) \cdot \left(\prod_{j \in \{i\}} \zeta_i \right) \right) \cdot \otimes(e_1, \dots, e_n) + \alpha \cdot \left(\left(\prod_{j \in [1, \dots, n] \setminus \{i\}} \gamma_j \right) \cdot \right. \\ & \left. \left(\prod_{j \in \{i\}} \gamma_j \right) \right) \cdot \otimes(e_1, \dots, e_n) \stackrel{[\text{theorem: 11.43}]}{=} \\ & \left(\prod_{j \in [1, \dots, n]} \zeta_j \right) \cdot \otimes(e_1, \dots, e_n) + \alpha \cdot \left(\prod_{j \in [1, \dots, n]} \gamma_j \right) \cdot \otimes(e_1, \dots, e_n) = \\ & \nu(v_1, \dots, v_{i-1}, u, v_{i+1}, \dots, v_n) + \alpha \cdot (v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n) \end{aligned}$$

proving that

$$\nu \in \text{Hom}(V_1, \dots, V_n; P)$$

3. As by definition of ν we have that $\nu(\prod_{i \in \{1, \dots, n\}} V_i) \subseteq P$ it follows by [theorems: 11.109, 11.111] that

$$\text{span}\left(\nu\left(\prod_{i \in \{1, \dots, n\}} V_i\right)\right) \subseteq P \quad (13.111)$$

Further if $v \in P$ then there exists a $\lambda \in F$ such that $v = \lambda \cdot \otimes(e_1, \dots, e_n)$, take then $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq F$ by $\lambda_i = \begin{cases} \lambda & \text{if } i = 1 \\ 1 & \text{if } i \in \{2, \dots, n\} \end{cases}$ then we have for $(\lambda_1 \cdot e_1, \dots, \lambda_n \cdot e_n)$ that

$$\begin{aligned} \nu(\lambda_1 \cdot e_1, \dots, \lambda_n \cdot e_n) &= \left(\prod_{i \in \{1, \dots, n\}} \lambda_i\right) \cdot \otimes(e_1, \dots, e_n) \\ &\stackrel{[\text{theorem: 11.43}]}{=} \left(\left(\prod_{i \in \{2, \dots, n\}} \lambda_i\right) \cdot \left(\prod_{i \in \{1\}} \lambda_i\right)\right) \cdot \otimes(e_1, \dots, e_n) \\ &\stackrel{[\text{theorem: 11.34}]}{=} \left(\left(\prod_{i \in \{2, \dots, n\}} 1\right) \cdot \lambda\right) \cdot \otimes(e_1, \dots, e_n) \\ &\stackrel{[\text{theorem: 11.37}]}{=} (1 \cdot \lambda) \cdot \otimes(e_1, \dots, e_n) \\ &= \lambda \cdot \otimes(e_1, \dots, e_n) \\ &= v \end{aligned}$$

proving that $P \subseteq \nu(P = \prod_{i \in \{1, \dots, n\}} V_i) \subseteq \text{span}(\nu(P = \prod_{i \in \{1, \dots, n\}} V_i))$. Combining this with [eq: 13.111] results in

$$P = \text{span}\left(\nu\left(\prod_{i \in \{1, \dots, n\}} V_i\right)\right)$$

4. Let $\varphi \in \text{Hom}(V_1, \dots, V_n; P) = \text{Hom}(\text{span}(\{e_1\}), \dots, \text{span}(\{e_n\}); U)$. Define

$$h: P = \text{span}(\{\otimes(e_1, \dots, e_n)\}) \rightarrow U \text{ by } h(v) = \lambda \cdot \varphi(e_1, \dots, e_n) \text{ where } v = \lambda \cdot \otimes(e_1, \dots, e_n)$$

If $u, v \in \text{span}(\{\otimes(e_1, \dots, e_n)\})$ and $\alpha \in F$ then there exists a $\lambda_u, \lambda_v \in F$ such that $u = \lambda_u \cdot \otimes(e_1, \dots, e_n)$, $v = \lambda_v \cdot \otimes(e_1, \dots, e_n)$ then we have

$$\begin{aligned} h(u + \alpha \cdot v) &= h(\lambda_u \cdot \otimes(e_1, \dots, e_n) + \alpha \cdot \lambda_v \cdot \otimes(e_1, \dots, e_n)) \\ &= h((\lambda_u + \alpha \cdot \lambda_v) \cdot \otimes(e_1, \dots, e_n)) \\ &= (\lambda_u + \alpha \cdot \lambda_v) \cdot \varphi(e_1, \dots, e_n) \\ &= \lambda_u \cdot \varphi(e_1, \dots, e_n) + \alpha \cdot \lambda_v \cdot \varphi(e_1, \dots, e_n) \\ &= h(u) + \alpha \cdot h(v) \end{aligned}$$

proving that

$$h \in \text{Hom}(P, U)$$

Let $(v_1, \dots, v_n) \in \prod_{i \in \{1, \dots, n\}} V_i = \prod_{i \in \{1, \dots, n\}} \text{span}(\{e_i\})$ then there exists a $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq F$ such that $(v_1, \dots, v_n) = (\lambda_1 \cdot e_1, \dots, \lambda_n \cdot e_n)$. So

$$\begin{aligned} (h \circ \nu)(v_1, \dots, v_n) &= h(\nu(v_1, \dots, v_n)) \\ &= h(\nu(\lambda_1 \cdot e_1, \dots, \lambda_n \cdot e_n)) \\ &= h\left(\left(\prod_{i \in \{1, \dots, n\}} \lambda_i\right) \cdot \otimes(e_1, \dots, e_n)\right) \\ &= \left(\prod_{i \in \{1, \dots, n\}} \lambda_i\right) \cdot \varphi(e_1, \dots, e_n) \\ &\stackrel{[\text{theorem: 11.270}]}{=} \varphi(\lambda_1 \cdot e_1, \dots, \lambda_n \cdot e_n) \\ &= \varphi(v_1, \dots, v_n) \end{aligned}$$

proving that

$$h \circ \nu = \varphi$$

Finally (1),(2),(3) and (4) completes the proof of [eq: 13.110]. $\square$

As $\langle V_1 \otimes \cdots \otimes V_n, \otimes \rangle$ is a tensor product we have that

$$V_1 \otimes \cdots \otimes V_n = \text{span} \left(\otimes \left(\prod_{i \in \{1, \dots, n\}} V_i \right) \right)$$

Hence as $\otimes(\{e_1, \dots, e_n\}) \subseteq \otimes(\prod_{i \in \{1, \dots, n\}} V_i)$ it follows from [theorem: 11.109] that

$$P = \text{span}(\{\otimes(e_1, \dots, e_n)\}) \subseteq \text{span} \left(\otimes \left(\prod_{i \in \{1, \dots, n\}} V_i \right) \right) = V_1 \otimes \cdots \otimes V_n \quad (13.112)$$

Further by [theorem: 13.24] and the fact that $\forall i \in \{1, \dots, n\} \ e_i \neq 0$ it follows that $\otimes(e_1, \dots, e_n) \neq 0$ hence

$$P = \text{span}(\{\otimes(e_1, \dots, e_n)\}) \text{ is one-dimensional}$$

As $\langle V_1 \otimes \cdots \otimes V_n, \otimes \rangle$ is also a tensor product of $V_1, \dots, V_n$ we have by [theorem: 13.15] that $V_1 \otimes \cdots \otimes V_n$ is linear isomorphic with P , hence by [theorem: 11.203] $V_1 \otimes \cdots \otimes V_n$ is also one dimensional so that there exists a $v_0 \in V_1 \otimes \cdots \otimes V_n$ such that

$$V_1 \otimes \cdots \otimes V_n = \text{span}(v_0)$$

As $\otimes(e_1, \dots, e_n) \in \otimes(\prod_{i \in \{1, \dots, n\}} V_i) \subseteq \text{span}(\otimes(\prod_{i \in \{1, \dots, n\}} V_i)) = V_1 \otimes \cdots \otimes V_n$ there exist a $\lambda \in F$ such that $\otimes(e_1, \dots, e_n) = \lambda \cdot v_0$. As $\otimes(e_1, \dots, e_n) \neq 0$ we must have that $\lambda \neq 0$ hence

$$v_0 = \lambda^{-1} \cdot \otimes(e_1, \dots, e_n)$$

Let $v \in V_1 \otimes \cdots \otimes V_n$ then there exists a $\gamma \in F$ such that $v = \gamma \cdot v_0 = (\gamma \cdot \lambda^{-1}) \cdot \otimes(e_1, \dots, e_n)$, proving that $v \in \text{span}(\{\otimes(e_1, \dots, e_n)\})$. Hence $V_1 \otimes \cdots \otimes V_n \subseteq \text{span}(\{\otimes(e_1, \dots, e_n)\})$ so combining this with [eq: 13.112] proves that

$$V_1 \otimes \cdots \otimes V_n = \text{span}(\{\otimes(v_1, \dots, v_n)\}) \quad \square$$

Theorem 13.40. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of finite vector spaces over a field F with $\forall i \in \{1, \dots, n\}$ bases $\{e_{i,1}, \dots, e_{i,k_i}\}$ of V_i [where $k_i = \dim(V_i)$] then

$$\{\otimes(e_{1,\gamma(1)}, \dots, e_{n,\gamma(n)}) \mid \gamma \in \Gamma(k_1, \dots, k_n)\} = \{e_{1,\gamma(1)} \otimes \cdots \otimes e_{n,\gamma(n)} \mid \gamma \in \Gamma(k_1, \dots, k_n)\}$$

is a basis of $V_1 \otimes \cdots \otimes V_n$. Hence by [theorem: 13.30] we have that

$$\dim(V_1 \otimes \cdots \otimes V_n) = \prod_{i \in \{1, \dots, n\}} k_i$$

Proof. Let $i \in \{1, \dots, n\}$ and $v \in V_i$ then by [theorem: 11.143] there exists a **unique** $\{\alpha_j\}_{j \in \{1, \dots, k_i\}} \subseteq F$ such that $v = \sum_{j \in \{1, \dots, k_i\}} \alpha_j \cdot e_{i,j}$. As $\forall j \in \{1, \dots, k_i\} \ \alpha_j \cdot e_j \in \text{span}(\{e_j\})$ it follows from the definition of a direct sum [see definition: 12.5] that

$$V_i = \bigoplus_{j \in \{1, \dots, k_i\}} \text{span}(\{e_{i,j}\})$$

So we have that

$$\begin{aligned} V_1 \otimes \cdots \otimes V_n &\stackrel{[\text{theorem: 13.38}]}{=} \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} (\text{span}(\{e_{1,\gamma(1)}\}) \otimes \cdots \otimes \text{span}(\{e_{n,\gamma(n)}\})) \\ &\stackrel{[\text{theorem: 13.39}]}{=} \bigoplus_{\gamma \in \Gamma(k_1, \dots, k_n)} \text{span}(\otimes(e_1, \dots, e_n)) \end{aligned}$$

Let $v \in V_1 \otimes \cdots \otimes V_n$ then by the definition of a direct sum [see definition: 12.5] there exist a **unique** $\{\alpha_i\}_{i \in \Gamma(k_1, \dots, k_n)}$ such that $v = \sum_{\Gamma(k_1, \dots, k_n)} \alpha_i \cdot \otimes(e_1, \dots, e_n)$. By [theorem: 11.142] it follows then that

$$\{\otimes(e_{1, \gamma(1)}, \dots, e_{n, \gamma(n)}) | \gamma \in \Gamma(k_1, \dots, k_n)\} \text{ is a basis for } V_1 \otimes \cdots \otimes V_n \quad \square$$

Once we have a basis of a vector space we can consider the expansion of its elements in the basis, this leads to the idea of the components of a tensor product.

Definition 13.41. Let $n \in \mathbb{N}$, $\{V_i\}_{i \in \{1, \dots, n\}}$ a finite family of finite vector spaces over a field F with $\forall i \in \{1, \dots, n\}$ bases $\{e_{i,1}, \dots, e_{i,k_i}\}$ of V_i [where $k_i = \dim(V_i)$] then by [theorem: 13.40] we have that basis

$$\{e_{1, \gamma(1)} \otimes \cdots \otimes e_{n, \gamma(n)} | \gamma \in \Gamma(k_1, \dots, k_n)\}$$

Let $T \in V_1 \otimes \cdots \otimes V_n$ then there exists **unique** $\{T_\gamma\}_{\gamma \in \Gamma(k_1, \dots, k_n)} \subseteq F$ such that

$$T = \sum_{\gamma \in \Gamma(k_1, \dots, k_n)} T_\gamma \cdot (e_{1, \gamma(1)} \otimes \cdots \otimes e_{n, \gamma(n)})$$

The $\prod_{i \in \{1, \dots, n\}} k_i$ T_γ are called the **components** of T . As $\Gamma(k_1, \dots, k_n) = \prod_{i \in \{1, \dots, n\}} \{1, \dots, k_i\}$ we can write a $\gamma \in \Gamma(k_1, \dots, k_n)$ as $\gamma = (\gamma_1, \dots, \gamma_n)$. Hence we also use the notation

$$\{T_{\gamma_1, \dots, \gamma_n}\}_{\gamma \in \Gamma(k_1, \dots, k_n)}$$

for the components so that

$$T = \sum_{\gamma \in \Gamma(k_1, \dots, k_n)} T_{\gamma_1, \dots, \gamma_n}$$

We introduce now the concept of p - q tensors.

Definition 13.42. Let V be a vector space over a field, $p, q \in \mathbb{N}_0$ then a p -contravariant, q -covariant tensor product noted as V_q^p is defined by

1. If $p \neq 0 \vee q \neq 0$ then

$$V_q^p = V_1 \otimes \cdots \otimes V_{p+q} \text{ where } \forall i \in \{1, \dots, p+q\} \quad V_i = \begin{cases} V & \text{if } i \in \{1, \dots, p\} \\ V^* & \text{if } i \in \{p+1, \dots, p+q\} \end{cases}$$

where $V^* = \text{Hom}(V, F)$ is the dual space of V [see definition: 11.186].

2. If $p = q = 0$ then we defined

$$V_q^p = F$$

Another notation used is

$$V_q^p = \underbrace{V \otimes \cdots \otimes V}_p \otimes \underbrace{V^* \otimes \cdots \otimes V^*}_q$$

elements of V_q^p are called p -contravariant, q -covariant tensors or in short p - q -tensors.

Chapter 14

Topology

14.1 Topological spaces

Definition 14.1. (Topological Space) Let X be a set then $\mathcal{T} \subseteq \mathcal{P}(X)$ is a **topology** on X if

1. $X \in \mathcal{T}$
2. $\emptyset \in \mathcal{T}$
3. $\forall U, V \in \mathcal{T}$ we have $U \cap V \in \mathcal{T}$
4. If $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ is a family of sets in $\mathcal{T}$ then $\bigcup_{i \in I} U_i \in \mathcal{T}$

Elements of $\mathcal{T}$ are called **open sets** and the pair $\langle X, \mathcal{T} \rangle$ is called a **topological space**.

The most trivial (and least usable) example of a topological space is $\langle \emptyset, \{\emptyset\} \rangle$.

Example 14.2. $\langle \emptyset, \{\emptyset\} \rangle$ is a topological space. Further the only topology on $\emptyset$ is $\{\emptyset\}$

Proof. As $\emptyset \subseteq \emptyset$ we have that $\emptyset \in \mathcal{P}(\emptyset)$ hence $\mathcal{T} = \{\emptyset\} \subseteq \mathcal{P}(\emptyset)$. Further

1. $X = \emptyset \in \{\emptyset\} = \mathcal{T}$
2. $\emptyset \in \{\emptyset\} = \mathcal{T}$
3. If $U, V \in \mathcal{T} = \{\emptyset\}$ then $U = \emptyset = V$ so that $U \cap V = \emptyset \in \{\emptyset\} = \mathcal{T}$
4. If $\{U_i\}_{i \in I} \subseteq \mathcal{T} = \{\emptyset\}$ then we have $\forall i \in I$ that $U_i = \emptyset$ hence $\bigcup_{i \in I} U_i = \emptyset \in \{\emptyset\} = \mathcal{T}$

Further if $\mathcal{T}$ is a topology on $\emptyset$ we have $\emptyset \in \mathcal{T}$ so that $\{\emptyset\} \subseteq \mathcal{T}$ and if $U \in \mathcal{T}$ then $U \subseteq \emptyset \Rightarrow U = \emptyset$ so that $\mathcal{T} \subseteq \{\emptyset\}$. Hence $\mathcal{T} = \{\emptyset\}$. $\square$

Example 14.3. If X is a set then $\langle X, \{X, \emptyset\} \rangle$ is a topological space

Proof. For $\mathcal{T} = \{X, \emptyset\}$ we have:

1. $X \in \{X, \emptyset\} = \mathcal{T}$
2. $\emptyset \in \{X, \emptyset\} = \mathcal{T}$
3. If $U, V \in \mathcal{T} = \{X, \emptyset\}$ then we have either:
 - $U = X \wedge V = X$. Then $U \cap V = X \in \{\emptyset, X\} = \mathcal{T}$
 - $U = \emptyset \wedge V = X$. Then $U \cap V = \emptyset \in \{\emptyset, X\} = \mathcal{T}$
 - $U = X \wedge V = \emptyset$. Then $U \cap V = \emptyset \in \{\emptyset, X\} = \mathcal{T}$
 - $U = \emptyset \wedge V = \emptyset$. Then $U \cap V = \emptyset \in \{\emptyset, X\} = \mathcal{T}$
4. If $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ then we have either:
 - $\forall i \in I$ we have $U_i = \emptyset$. Then $\bigcup_{i \in I} U_i = \emptyset \in \{\emptyset, X\} = \mathcal{T}$
 - $\exists i \in I$ with $U_i = X$. Then $\bigcup_{i \in I} U_i = X \in \{\emptyset, X\} = \mathcal{T}$

$\square$

Example 14.4. (Discrete Topology) If X is a set then $\langle X, \mathcal{P}(X) \rangle$ is a topological space. $\mathcal{P}(X)$ is called the **discrete topology** on X .

Proof. For $\mathcal{T} \in \mathcal{P}(X)$ we have:

1. As $X \subseteq X$ we have $X \in \mathcal{P}(X) = \mathcal{T}$
2. As $\emptyset \subseteq X$ we have $\emptyset \in \mathcal{P}(X) = \mathcal{T}$
3. If $U, V \subseteq \mathcal{P}(X)$ then $U, V \subseteq X$, hence by [theorem: 1.30] $U \cap V \subseteq X$ so that $U \cap V \in \mathcal{P}(X)$
4. If $\{U_i\}_{i \in I} \subseteq \mathcal{T} = \mathcal{P}(X)$ then $\forall i \in I U_i \in \mathcal{P}(X) \Rightarrow U_i \subseteq X$, hence by [theorem: 2.135] $\bigcup_{i \in I} U_i \subseteq X$ proving that $\bigcup_{i \in I} U_i \in \mathcal{P}(X) = \mathcal{T}$ $\square$

The last two examples shows that given a set X there could exist two different topologies topologies on X . In the examples both $\{\emptyset, X\}$ and $\mathcal{P}(X)$ are topologies on X . So if we say that a set in X is open we have to specify which topology is used.

Definition 14.5. Let X be a set and $\mathcal{T}_1, \mathcal{T}_2$ two topologies on X then we say that $\mathcal{T}_1$ is **finer** than $\mathcal{T}_2$ iff $\mathcal{T}_2 \subseteq \mathcal{T}_1$. In other words $\mathcal{T}_1$ is finer than $\mathcal{T}_2$ if every open set in $\mathcal{T}_2$ is also a open set in $\mathcal{T}_1$. Clearly if $\mathcal{T}_1$ is finer than $\mathcal{T}_2$ and $\mathcal{T}_2$ is finer than $\mathcal{T}_1$ then $\mathcal{T}_1 = \mathcal{T}_2$.

Using mathematical induction it follows that every **finite** intersection of open sets is open.

Theorem 14.6. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $I \neq \emptyset$ a finite set and $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ a finite family of open sets then $\bigcap_{i \in I} U_i \in \mathcal{T}$ [or in other words $\bigcap_{i \in I} U_i$ is open].

Proof. We use induction to prove this, so define

$$\mathcal{S} = \left\{ n \in \mathbb{N} \mid \text{If } I \text{ is finite with } \text{card}(I) = n \text{ and } \{U_i\}_{i \in I} \subseteq \mathcal{T} \text{ then } \bigcap_{i \in I} U_i \in \mathcal{T} \right\}$$

then we have:

$1 \in \mathcal{S}$. Then $I = \{i\}$ so that $\bigcap_{j \in \{i\}} U_j = U_i \in \mathcal{T}$ proving that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n + 1 \in \mathcal{S}$. Let I be a finite set with $\text{card}(I) = n + 1$ then there exists a bijection

$$\beta: \{1, \dots, n+1\} \rightarrow I$$

so that by [theorem: 2.99]

$$\beta|_{\{1, \dots, n\}}: \{1, \dots, n\} \rightarrow I \setminus \{\beta(n+1)\} \text{ is a bijection}$$

or $\text{card}(I \setminus \{n+1\}) = n$. As $n \in \mathcal{S}$ it follows that $\bigcap_{i \in I \setminus \{n+1\}} U_i \in \mathcal{T}$, hence, as $U_{n+1} \in \mathcal{T}$, we have

$$\bigcap_{i \in I} U_i = \left(\bigcap_{i \in I \setminus \{n+1\}} U_i \right) \cap U_{n+1} \in \mathcal{T}$$

proving that $n + 1 \in \mathcal{S}$. $\square$

Theorem 14.7. Let $\langle X, \mathcal{T} \rangle$ be a topological space then we have for $U \subseteq X$

$$U \in \mathcal{T} \Leftrightarrow \forall x \in U \text{ there exist a } V \in \mathcal{T} \text{ such that } x \in V \subseteq U$$

Proof.

$\Rightarrow$. If $U \in \mathcal{T}$ then we have either:

$U = \emptyset$. Then $\forall x \in U = \emptyset$ we have vacuously that there exist a $V \in \mathcal{T}$ such that $x \in V \subseteq U$.

$U \neq \emptyset$. Then $\forall x \in U$ we have $x \in U \subseteq U$ where $U \in \mathcal{T}$.

$\Leftarrow$. By the hypothesis we have that $\forall x \in X$ there exist a $V \in \mathcal{T}$ such that $x \in V \subseteq U$ this defines by [theorem: 3.108] a family $\{V_x\}_{x \in U} \subseteq \mathcal{T}$ such that $\forall x \in U x \in V_x \subseteq U$. By [theorem: 2.135] it follows that $\bigcup_{x \in U} V_x \subseteq U$. Further if $x \in U$ then $x \in V_x \subseteq \bigcup_{x \in U} V_x$, proving that $U \subseteq \bigcup_{x \in U} V_x$. Hence

$$U = \bigcup_{x \in U} V_x \in \mathcal{T} \quad \square$$

Given a topology on a set then we can induce a topology on a subset of this set.

Definition 14.8. (Subspace Topology) Let $\langle X, \mathcal{T} \rangle$ be a topological space and $Y \subseteq X$ then

$$\langle Y, \mathcal{T}_{|Y} \rangle \text{ where } \mathcal{T}_{|Y} = \{U \cap Y \mid U \in \mathcal{T}\}$$

is a topology called the **subspace topology on Y induced by $\mathcal{T}$** .

Note 14.9. $\mathcal{T}_{|X} = \{U \cap X \mid U \in \mathcal{T}\} \stackrel{U \in \mathcal{T} \Rightarrow U \subseteq X}{=} \{U \mid U \in \mathcal{T}\} = \mathcal{T}$. In other words the subspace topology on X induced by $\mathcal{T}$ is the topology of X itself.

Proof. Of course we must prove that $\mathcal{T}_{|Y}$ is indeed a topology on Y .

1. As $Y \subseteq X$ and $X \in \mathcal{T}$ it follows that $Y = X \cap Y \in \mathcal{T}_{|Y}$.
2. $\emptyset = \emptyset \cap Y \in \mathcal{T}_{|Y}$ because $\emptyset \in \mathcal{T}$.
3. If $V_1, V_2 \in \mathcal{T}_{|Y}$ then there exists $U_1, U_2 \in \mathcal{T}$ such that $V_1 = U_1 \cap Y$ and $V_2 = U_2 \cap Y$. Then we have

$$V_1 \cap V_2 = (U_1 \cap Y) \cap (U_2 \cap Y) = (U_1 \cap U_2) \cap Y \in \mathcal{T}_{|Y} \text{ because } U_1 \cap U_2 \in \mathcal{T}$$

4. Let $\{V_i\}_{i \in I} \subseteq \mathcal{T}_{|Y}$ then there exists a $\{U_i\}_{i \in I}$ such that $\forall i \in I \ V_i = U_i \cap Y$, hence

$$\bigcup_{i \in I} V_i = \bigcup_{i \in I} (U_i \cap Y) \stackrel{[\text{theorem: 2.138}]}{=} \left(\bigcup_{i \in I} U_i \right) \cap Y \in \mathcal{T}_{|Y} \text{ because } \bigcup_{i \in I} U_i \in \mathcal{T} \quad \square$$

The subspace topology of a subspace topology is again a subspace topology.

Theorem 14.10. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $Z \subseteq Y \subseteq X$ then $\mathcal{T}_{|Z} = (\mathcal{T}_{|Y})_{|Z}$

Proof. First we have

$$\begin{aligned} U \in (\mathcal{T}_{|Y})_{|Z} &\Rightarrow \exists V \in \mathcal{T}_{|Y} \text{ such that } U = V \cap Z \\ &\Rightarrow \exists W \in \mathcal{T} \text{ such that } V = W \cap Y \\ &\Rightarrow U = (W \cap Y) \cap Z = W \cap (Z \cap Y) \\ &\stackrel{Z \subseteq Y}{\Rightarrow} U = W \cap Z \\ &\Rightarrow U \in \mathcal{T}_{|Z} \end{aligned}$$

Second we have

$$\begin{aligned} U \in \mathcal{T}_{|Z} &\Rightarrow \exists W \in \mathcal{T} \text{ such that } U = W \cap Z \\ &\stackrel{Z \subseteq Y}{\Rightarrow} U = W \cap (Y \cap Z) = (W \cap Y) \cap Z \\ &\stackrel{W \cap Y \in \mathcal{T}_{|Y}}{\Rightarrow} U \in (\mathcal{T}_{|Y})_{|Z} \end{aligned} \quad \square$$

A subspace topology is in general not a subset of the topology that induced it, however if the subset defining the subspace topology is open this is the case.

Theorem 14.11. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $Y \subseteq X$ then

$$Y \text{ is open} \Leftrightarrow \mathcal{T}_{|Y} \subseteq \mathcal{T}$$

Proof.

$\Rightarrow$. Let $V \in \mathcal{T}_{|Y}$ then there exist a $U \in \mathcal{T}$ such that $V = U \cap Y$, as $Y, U \in \mathcal{T}$ we have that $V = U \cap Y \in \mathcal{T}$ proving that $\mathcal{T}_{|Y} \subseteq \mathcal{T}$.

$\Leftarrow$. As $\mathcal{T}|_Y$ is a topology on Y we have that $Y \in \mathcal{T}|_Y$ hence using $\mathcal{T}|_Y \subseteq \mathcal{T}$ we have $Y \in \mathcal{T}$ proving that Y is open. $\square$

Given a subset of a topological space we can find the largest open set that is contained in the set, this is the idea of the interior of a set.

Definition 14.12. (Interior) Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then the **interior** of A noted as A° is defined by

$$A^\circ = \bigcup_{U \in \{V \in \mathcal{T} | V \subseteq A\}} U$$

Note 14.13. If $U \in \{V \in \mathcal{T} | V \subseteq A\}$ then $U \in \mathcal{T}$ and $U \subseteq A$ so that $A^\circ = \bigcup_{U \in \{V \in \mathcal{T} | V \subseteq A\}} U \in \mathcal{T}$ and $A^\circ \subseteq A$.

Theorem 14.14. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq B \subseteq X$ then we have

$$A^\circ \subseteq B^\circ$$

Proof. If $U \in \{V \in \mathcal{T} | V \subseteq A\}$ then $U \subseteq A \subseteq B$ so that $U \in \{V \in \mathcal{T} | V \subseteq B\}$ which proves that

$$\{V \in \mathcal{T} | V \subseteq A\} \subseteq \{V \in \mathcal{T} | V \subseteq B\}$$

so that

$$A^\circ = \bigcup_{U \in \{V \in \mathcal{T} | V \subseteq A\}} U \subseteq \bigcup_{U \in \{V \in \mathcal{T} | V \subseteq B\}} U = B^\circ \quad \square$$

It turns out that the interior of a set is the biggest open subset of this set.

Theorem 14.15. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $A \subseteq X$ and U a open set such that $U \subseteq A$ then $U \subseteq A^\circ$.

Proof. If U is open and $U \subseteq A$ then $U \in \{V \in \mathcal{T} | V \subseteq A\}$ so that $U \subseteq \bigcup_{V \in \{V \in \mathcal{T} | V \subseteq A\}} V = A^\circ$ $\square$

Theorem 14.16. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $U \subseteq X$ then

$$U \text{ is open} \Leftrightarrow U = U^\circ$$

Proof.

$\Rightarrow$. If U is open then, as $U \subseteq U$, we have by the previous theorem [theorem: 14.15] that $U \subseteq U^\circ$ and by [definition: 14.12] $U^\circ \subseteq U$ Hence

$$U = U^\circ$$

$\Leftarrow$. As by [definition: 14.12] U° is open it follows from $U = U^\circ$ that U is open. $\square$

14.1.1 Closed Sets

Definition 14.17. Let $\langle X, \mathcal{T} \rangle$ be a topological space then the set of closed sets noted by $\mathcal{T}^c$ is defined by

$$\mathcal{T}^c = \{A \in \mathcal{P}(X) | X \setminus A \in \mathcal{T}\} \subseteq \mathcal{P}(X)$$

A element of $\mathcal{T}^c$ is called a **closed set**.

We have the following condition for closed sets in a subspace topology.

Theorem 14.18. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $Y \subseteq X$ then

$$(\mathcal{T}|_Y)^c = \{A \cap Y | A \in \mathcal{T}^c\}$$

In other words $C \subseteq Y$ is closed in $\mathcal{T}|_Y$ if there exist a closed set C' in $\mathcal{T}$ such that $C = C' \cap Y$

Proof. If $C \in (\mathcal{T}_Y)^c$ then $Y \setminus C \in \mathcal{T}_Y$ hence $\exists U \in \mathcal{T}$ such that $Y \setminus C = U \cap Y$. Hence

$$\begin{aligned} C &= Y \setminus (Y \setminus C) \\ &= Y \setminus (U \cap Y) \\ &\stackrel{[\text{theorem: 1.34}]}{=} (Y \setminus U) \cup (Y \setminus Y) \\ &= Y \setminus U \\ &\stackrel{Y \subseteq X}{=} (X \cap Y) \setminus U \\ &= Y \cap (X \setminus U) \end{aligned}$$

which as $X \setminus U \in \mathcal{T}^c$ proves that $C \in \{A \cap Y \mid A \in \mathcal{T}^c\}$. Hence

$$(\mathcal{T}_Y)^c \subseteq \{A \cap Y \mid A \in \mathcal{T}^c\} \quad (14.1)$$

If $C \in \{A \cap Y \mid A \in \mathcal{T}^c\}$ then there exist a $C' \in \mathcal{T}^c$ such that $C = C' \cap Y$ then we have

$$Y \setminus C = Y \setminus (C' \cap Y) \stackrel{[\text{theorem: 1.34}]}{=} (Y \setminus C') \cup (Y \setminus Y) = (Y \setminus C') = (X \cap Y) \setminus C' = (X \setminus C') \cap Y \in \mathcal{T}_Y$$

proving that $C \in (\mathcal{T}_Y)^c$. Hence $\{A \cap Y \mid A \in \mathcal{T}^c\} \subseteq (\mathcal{T}_Y)^c$ which combined with [eq: 14.1] results in

$$(\mathcal{T}_Y)^c = \{A \cap Y \mid A \in \mathcal{T}^c\} \quad \square$$

Example 14.19. Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\emptyset$ and X are open and closed sets

Proof. By definition of a topology $\emptyset, X$ are open sets. Further, as $X \setminus \emptyset = X \in \mathcal{T}$ and $X \setminus X = \emptyset \in \mathcal{T}$ we have that $\emptyset$ and X are also closed sets. $\square$

The above example shows that a set can be at the same time open and closed. You could also have open sets that are not closed and closed sets that are not open. The following example shows that you can also have sets that are neither open or closed.

Example 14.20. Take the topological space $\langle \{1, 2, 3\}, \{\emptyset, \{1, 2, 3\}\} \rangle$ [see example: 14.3] then $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$ are neither open or closed.

Proof. Using the topology $\mathcal{T} = \{\emptyset, \{1, 2, 3\}\}$ it is clear that $\mathcal{T}$ does not contain $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$ hence $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$ are not open. Further we have

$$\begin{aligned} \{1, 2, 3\} \setminus \{1\} &= \{2, 3\} \notin \mathcal{T} \\ \{1, 2, 3\} \setminus \{2\} &= \{1, 3\} \notin \mathcal{T} \\ \{1, 2, 3\} \setminus \{3\} &= \{1, 2\} \notin \mathcal{T} \\ \{1, 2, 3\} \setminus \{1, 2\} &= \{3\} \notin \mathcal{T} \\ \{1, 2, 3\} \setminus \{1, 3\} &= \{2\} \notin \mathcal{T} \\ \{1, 2, 3\} \setminus \{2, 3\} &= \{1\} \notin \mathcal{T} \end{aligned}$$

so that $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$ are not closed. $\square$

The above two example shows that we can have sets that are open and closed, sets that are only open, sets that are only closed and sets that are neither open or closed. So open and closed are not exclusive properties. However they are complementary as the following theorem shows.

Theorem 14.21. Let X be a set and $\mathcal{T} \subseteq \mathcal{P}(X)$ and

$$\mathcal{T}^c = \{A \in \mathcal{P}(X) \mid X \setminus A \in \mathcal{T}\}$$

then

$$\mathcal{T} = \{A \in \mathcal{P}(X) \mid X \setminus A \in \mathcal{T}^c\}$$

Proof. If $A \in \{A \in \mathcal{P}(X) \mid X \setminus A \in \mathcal{T}^c\}$ then $A \in \mathcal{P}(X) \Rightarrow A \subseteq X$ and $X \setminus A \in \mathcal{T}^c$ so that $A \stackrel{[\text{theorem: 1.36}]}{=} X \setminus (X \setminus A) \in \mathcal{T}$ proving

$$\{A \in \mathcal{P}(X) \mid X \setminus A \in \mathcal{T}^c\} \subseteq \mathcal{T} \quad (14.2)$$

If $A \in \mathcal{T}$ then as $\mathcal{T} \subseteq \mathcal{P}(X)$ we have $X \setminus (X \setminus A) \stackrel{[\text{theorem: 1.36}]}{=} A \in \mathcal{T}$ so that $X \setminus A \in \mathcal{T}^c$ proving that $A \in \{A \in \mathcal{P}(X) | X \setminus A \in \mathcal{T}^c\}$. Hence $\mathcal{T} \subseteq \{A \in \mathcal{P}(X) | X \setminus A \in \mathcal{T}^c\}$ which combined with [eq: 14.2] proves that

$$\mathcal{T} = \{A \in \mathcal{P}(X) | X \setminus A \in \mathcal{T}^c\} \quad \square$$

Theorem 14.22. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then the set of closed sets $\mathcal{T}^c$ satisfies the following:*

1. $\emptyset \in \mathcal{T}^c$
2. $X \in \mathcal{T}^c$
3. If $U \in \mathcal{T}$ then $X \setminus U \in \mathcal{T}^c$
4. If $A \in \mathcal{T}^c$ and $U \in \mathcal{T}$ then $A \setminus U \in \mathcal{T}^c$.
5. If $U \in \mathcal{T}$ and $A \in \mathcal{T}^c$ then $U \setminus A \in \mathcal{T}$
6. If $A, B \in \mathcal{T}^c$ then $A \cup B \in \mathcal{T}^c$
7. If I is finite and $\{A_i\}_{i \in I} \subseteq \mathcal{T}^c$ is a finite family of closed sets then $\bigcup_{i \in I} A_i \in \mathcal{T}^c$
8. If $\{A_i\}_{i \in I}$ is a family in $\mathcal{T}^c$ (a family of closed sets) then $\bigcap_{i \in I} A_i \in \mathcal{T}^c$

Furthermore when $\mathcal{C} \subseteq \mathcal{P}(X)$ is a set of subsets of X satisfying 1,2,3 and 5 of the above then $\mathcal{T} = \{X \setminus A | A \in \mathcal{C}\}$ is a topology on X with the set of closed sets $\mathcal{T}^c = \mathcal{C}$.

Proof.

1. $X \setminus \emptyset = X \in \mathcal{T} \Rightarrow \emptyset \in \mathcal{T}^c$
2. $X \setminus X = \emptyset \in \mathcal{T} \Rightarrow X \in \mathcal{T}^c$
3. This follows from [theorem: 14.21].
4. Let $U \in \mathcal{T}$ and $A \in \mathcal{T}^c$ then $X \setminus A \in \mathcal{T}$. Further

$$\begin{aligned} X \setminus (A \setminus U) &= X \cap (A \setminus U)^c \\ &= X \cap (A \cap U^c)^c \\ &= X \cap (A^c \cup (U^c)^c) \\ &= X \cap (A^c \cup U) \\ &= (X \cap A^c) \cup (X \cap U) \\ &\stackrel{U \subseteq X}{=} (X \cap A^c) \cup U \\ &= (X \setminus A) \cup U \\ &\in \mathcal{T} \end{aligned}$$

which proves that $A \setminus U \in \mathcal{T}^c$.

5. Let $U \in \mathcal{T}$ and $A \in \mathcal{T}^c$ then $X \setminus A \in \mathcal{T}$ and

$$U \setminus A = (U \cap X) \setminus A = (U \cap X) \cap A^c = U \cap (X \cap A^c) = U \cap (X \setminus A) \in \mathcal{T}$$

6. If $A, B \in \mathcal{T}^c$ then $X \setminus (A \cup B) \stackrel{[\text{theorem: 1.34}]}{=} (X \setminus A) \cap (X \setminus B) \in \mathcal{T}$ [as $X \setminus A, X \setminus B \in \mathcal{T}$] so that $A \cup B \in \mathcal{T}^c$.
7. If $\{A_i\}_{i \in I} \subseteq \mathcal{T}^c$ then $\forall i \in I$ we have that $X \setminus A_i$ is open. As I is finite we have by [theorem: 14.6] that $\bigcap_{i \in I} (X \setminus A_i)$ is open, hence $X \setminus (\bigcap_{i \in I} (X \setminus A_i))$ is closed. Finally

$$X \setminus \left(\bigcap_{i \in I} (X \setminus A_i) \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{i \in I} (X \setminus (X \setminus A_i)) \stackrel{[\text{theorem: 1.36}]}{=} \bigcup_{i \in I} A_i$$

8. If $\{A_i\}_{i \in I}$ is a family in $\mathcal{T}^c$ then $X \setminus (\bigcap_{i \in I} A_i) \stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{i \in I} (X \setminus A_i) \in \mathcal{T}$ [as $\forall i \in I$ we have $X \setminus A_i \in \mathcal{T}$].

Assume now that $\mathcal{C} \subseteq \mathcal{P}(X)$ fulfills 1,2,3 and 5 and define $\mathcal{T} = \{U \in \mathcal{P}(X) | X \setminus U \in \mathcal{C}\}$ then

1. $\emptyset = X \setminus X \Rightarrow \emptyset \in \mathcal{T}$
2. $X = X \setminus \emptyset \Rightarrow X \in \mathcal{T}$
3. If $U, V \in \mathcal{T}$ then $X \setminus U \in \mathcal{C} \wedge X \setminus V \in \mathcal{C}$ so that by (4) $(X \setminus U) \cup (X \setminus V) \in \mathcal{C}$. As by [theorem: 1.34] $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ it follows that $U \cap V \in \mathcal{T}$.
4. If $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ then $\forall i \in I$ we have $X \setminus U_i \in \mathcal{C}$ so that by (6) $\bigcap_{i \in I} (X \setminus U_i) \in \mathcal{C}$. As by [theorem: 2.140] $X \setminus (\bigcup_{i \in I} U_i) = \bigcap_{i \in I} (X \setminus U_i)$ it follows that $\bigcup_{i \in I} U_i \in \mathcal{T}$.

From the above it follows then that

$\mathcal{T}$ is a topology

If $A \in \mathcal{T}^c$ then $X \setminus A \in \mathcal{T}$ so $X \setminus (X \setminus A) \in \mathcal{C}$ which as $A \stackrel{[\text{theorem: 1.36}]}{=} X \setminus (X \setminus A)$ proves that $A \in \mathcal{C}$, hence

$$\mathcal{T}^c \subseteq \mathcal{C}$$

Further if $A \in \mathcal{C}$ then $X \setminus A \in \mathcal{T}$ so $A \in \mathcal{T}^c$ proving that $\mathcal{C} \subseteq \mathcal{T}^c$ which combined with the above proves that

$$\mathcal{T}^c = \mathcal{C} \quad \square$$

Just as we have defined the interior of a set as the largest open set containing in the set we can define the closure of a set as the smallest closed set containing the set.

Definition 14.23. (Closure of a Set) Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then the *closure* of A noted by $\bar{A}$ is defined by

$$\bar{A} = \bigcap_{C \in \{C \in \mathcal{T}^c | A \subseteq C\}} C$$

Note 14.24. If $C \in \{C \in \mathcal{T}^c | A \subseteq C\}$ then $A \subseteq C$ and $C \in \mathcal{T}^c$ hence by [theorem: 2.135] we have

$$A \subseteq \bar{A}$$

further by [theorem: 14.22] we have that

$$\bar{A} \in \mathcal{T}^c \text{ or } \bar{A} \text{ is closed}$$

Next we show that the closure of a set is the smallest closed set containing the set.

Theorem 14.25. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then if C is closed set such that $A \subseteq C$ it follows that $\bar{A} \subseteq C$.

Proof. If C is closed set such that $A \subseteq C$ then $C \in \{C \in \mathcal{T}^c | A \subseteq C\}$ so that $\bigcap_{B \in \{C \in \mathcal{T}^c | A \subseteq C\}} B \subseteq C$ $\square$

It turns out that a closed set is a set that is equal to its closure.

Theorem 14.26. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then

$$A \text{ is closed} \Leftrightarrow A = \bar{A}$$

Proof.

$\Rightarrow$. If A is closed then as $A \subseteq \bar{A}$ we have by the previous theorem [theorem: 14.25] that $\bar{A} \subseteq A$ and by [definition: 14.23] $A \subseteq \bar{A}$. Hence

$$A = \bar{A}$$

$\Leftarrow$. As by [definition: 14.23] $\bar{A}$ is closed it follows from $A = \bar{A}$ that A is closed. $\square$

Definition 14.27. If $\langle X, \mathcal{T} \rangle$ is a topological space and $A \subseteq X$ then $x \in X$ is a **limit point** or **accumulation point** of A if $\forall U \in \mathcal{T}$ with $x \in U$ we have $(A \setminus \{x\}) \cap U \neq \emptyset$. The set of all limit points of A is called the **derived set** of A and noted by A' hence

$$A' = \{x \in X \mid x \text{ is a limit point of } A\}$$

Theorem 14.28. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $A, B \subseteq X$ with $A \subseteq B$ then $A' \subseteq B'$

Proof. Let $x \in A'$ then $\forall U \in \mathcal{T}$ with $x \in U$ we have $(A \setminus \{x\}) \cap U \neq \emptyset$. As $A \subseteq B$ it follows that $(A \setminus \{x\}) \cap U \subseteq (B \setminus \{x\}) \cap U$ so that that $(B \setminus \{x\}) \cap U \neq \emptyset$ proving that $x \in B'$. $\square$

Theorem 14.29. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then

$$\bar{A} = A \cup A'$$

Proof. Let $x \in \bar{A}$ then we have either:

$x \in A$. Then trivially $x \in A \cup A'$

$x \notin A$. Assume that $x \notin A'$ then there exist a $U \in \mathcal{T}$ with $x \in U$ such that $(A \setminus \{x\}) \cap U = \emptyset$. As $x \notin A$ we have that $A \setminus \{x\} = A$ hence $A \cap U = \emptyset$. So $\forall a \in A$ we have that $a \notin U \Rightarrow a \in X \setminus U$ proving that $A \subseteq X \setminus U$ which, as $X \setminus U$ is closed, gives by [theorem: 14.25] that $\bar{A} \subseteq X \setminus U$. Hence if $x \in \bar{A}$ we have $x \notin U$ contradicting $x \in U$. So the assumption is wrong and we must have that $x \in A' \subseteq A \cup A'$.

As in all cases we have $x \in A \cup A'$ it follows that

$$\bar{A} \subseteq A \cup A' \tag{14.3}$$

Let $x \in A \cup A'$ then we have either:

$x \in A$. Then, as by [definition: 14.23] $A \subseteq \bar{A}$, we have $x \in \bar{A}$.

$x \notin A$. Then $A \setminus \{x\} = A$ and as $x \in A \cup A'$ we must have $x \in A'$. Assume that $x \notin \bar{A}$ then $x \in X \setminus \bar{A}$ a open set, so as $x \in A'$ we have by [definition: 14.27] that

$$\emptyset \neq (A \setminus \{x\}) \cap (X \setminus \bar{A}) \underset{A \setminus \{x\} = A}{=} A \cap (X \setminus \bar{A}) \subseteq \bar{A} \cap (X \setminus \bar{A}) = \emptyset$$

a contradiction. Hence we must have that $x \in \bar{A}$.

As in all cases $x \in \bar{A}$ it follows that $A \cup A' \subseteq \bar{A}$ which combined with [eq: 14.3] proves

$$\bar{A} = A \cup A' \tag{14.4}$$

Corollary 14.30. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then we have

$$\bar{A} = \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\}$$

Proof. Let $x \in \bar{A}$ then we have either:

$x \in A$. Then if $U \in \mathcal{T}$ with $x \in U$ we have $x \in A \cap U \Rightarrow A \cap U \neq \emptyset$. Hence

$$x \in \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\}$$

$x \notin A$. As $\bar{A} \underset{[\text{theorem: 14.29}]}{=} A \cup A'$ we must have that $x \in A'$, then $\forall U \in \mathcal{T}$ with $x \in U$ we have by [definition: 14.27] that $\emptyset \neq (A \setminus \{x\}) \cap U \subseteq A \cap U$. Hence

$$x \in \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\}$$

So we have that

$$\bar{A} \subseteq \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\} \tag{14.4}$$

Let $x \in \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\}$. Assume that $x \notin \bar{A}$ then $x \in X \setminus \bar{A} \in \mathcal{T}$ hence $\emptyset \neq A \cap (X \setminus \bar{A}) \subseteq \bar{A} \cap (X \setminus \bar{A}) = \emptyset$ a contradiction. So we must have that $x \in \bar{A}$ proving that $\{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\} \subseteq \bar{A}$, which combined with [eq: 14.4] proves

$$\bar{A} = \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\} \quad \square$$

Corollary 14.31. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then*

$$A \text{ is closed} \Leftrightarrow A = \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } A \cap U \neq \emptyset\}$$

Proof. This follows from [theorem: 14.26] and the previous theorem [theorem: 14.30]. $\square$

Corollary 14.32. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq B \subseteq X$ then*

$$\bar{A} \subseteq \bar{B}$$

Proof. Let $x \in \bar{A}$ then for every $U \in \mathcal{T}$ with $x \in U$ we have that $A \cap U \neq \emptyset$ so that as $A \cap U \subseteq B \cap U$ we have $B \cap U \neq \emptyset$ proving $x \in \bar{B}$. Hence $\bar{A} \subseteq \bar{B}$. $\square$

We have the following relation between the closure using a subspace topology and the original topology.

Theorem 14.33. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $U \in \mathcal{T}$ a open set in X and $A \subseteq U$ then we have that*

$$\text{cl}_U(A) = \text{cl}_X(A) \cap U$$

where $\text{cl}_X(A)$ is the closure of A using the topology $\mathcal{T}$ on X and $\text{cl}_U(A)$ is the closure of A using the subspace topology $\mathcal{T}_U$ on U .

Proof. Using the alternative definition of the closure of a set [see theorem: 14.30] we have that

$$\text{cl}_X(A) = \{x \in X \mid \forall V \in \mathcal{T} \text{ with } x \in V \text{ we have } A \cap V \neq \emptyset\} \quad (14.5)$$

$$\text{cl}_U(A) = \{x \in U \mid \forall V \in \mathcal{T}_U \text{ with } x \in V \text{ we have } A \cap V \neq \emptyset\} \quad (14.6)$$

Let $x \in \text{cl}_U(A)$ then by [eq: 14.6] $x \in U$. Let $V \in \mathcal{T}$ with $x \in V$ then we have $x \in V \cap U$ which, as $V \cap U \in \mathcal{T}_U$, proves by [eq: 14.6] that $\emptyset \neq (V \cap U) \cap A \subseteq V \cap A$. Hence $\emptyset \neq V \cap A$ which, as $V \in \mathcal{T}$ has been chosen arbitrary proves by [eq: 14.5] that $x \in \text{cl}_X(A)$. As $x \in U$ it follows that $x \in \text{cl}_X(A) \cap U$ so that

$$\text{cl}_U(A) \subseteq \text{cl}_X(A) \cap U \quad (14.7)$$

Let $x \in \text{cl}_X(A) \cap U$ then $x \in U \Rightarrow x \in X$ and $x \in \text{cl}_X(A)$. Let $V \in \mathcal{T}_U$ with $x \in V$ then, as by [theorem: 14.11] $\mathcal{T}_U \subseteq \mathcal{T}$ for $U \in \mathcal{T}$, we have that $V \in \mathcal{T}$ so that by [eq: 14.5] we have that $A \cap V \neq \emptyset$. Hence using [eq: 14.6] it follows that $x \in \text{cl}_U(A)$. Hence we have $\text{cl}_X(A) \cap U \subseteq \text{cl}_U(A)$ which combined with [eq: 14.7] proves finally

$$\text{cl}_U(A) = \text{cl}_X(A) \cap U \quad \square$$

14.1.2 Basis of a topological space

When dealing with a vector space we can prove many statements by limiting ourselves to a basis of a vector space because every vector can be written as a linear combination of the basis vectors. It turns out that we can do something similar with topological spaces.

Definition 14.34. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\mathcal{B} \subseteq \mathcal{P}(X)$ is a **basis for $\mathcal{T}$** if $\mathcal{B} \subseteq \mathcal{T}$ and $\forall U \in \mathcal{T}$ there exists a $\{B_i\}_{i \in I} \subseteq \mathcal{B}$ such that $U = \bigcup_{i \in I} B_i$. In other words $\mathcal{B}$ is a basis for $\mathcal{T}$ if $\mathcal{B}$ consist of open sets and every open set is the union of sets in $\mathcal{B}$.*

Every topology is a basis for itself as the next example shows.

Example 14.35. Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\mathcal{T}$ is a basis for $\mathcal{T}$.

Proof. Let $U \in \mathcal{T}$ then we can define $\{B_i\}_{i \in \{1\}} \subseteq \mathcal{T}$ by $B_1 = U$ so that $U = U_1 = \bigcup_{i \in \{1\}} U_i$. $\square$

Theorem 14.36. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $\mathcal{B}$ a basis for $\mathcal{T}$ then

$$U \in \mathcal{T} \Leftrightarrow \forall x \in U \text{ there exist a } B \in \mathcal{B} \text{ such that } x \in B \subseteq U$$

Proof.

$\Rightarrow$. As U is open and $\mathcal{B}$ is a basis there exist a $\{B_i\}_{i \in I} \subseteq \mathcal{B}$ such that $U = \bigcup_{i \in I} B_i$. Hence if $x \in U$ there exist a $i \in I$ such that $x \in B_i \subseteq \bigcup_{i \in I} B_i = U$ which as $B_i \in \mathcal{B}$ proves our assertion.

$\Leftarrow$. Assume that $\forall x \in U$ there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ then by a consequence of the Axiom of Choice [see theorem: 3.108] there exist a $\{B_x\}_{x \in U} \subseteq \mathcal{B}$ such that

$$\forall x \in U \text{ we have } x \in B_x \subseteq U$$

Hence using [theorem: 2.135] it follow that $\bigcup_{x \in U} B_x \subseteq U$, further if $x \in U$ then $x \in B_x$ so that $x \in \bigcup_{y \in U} B_y$, proving that $U \subseteq \bigcup_{y \in U} B_y$. Hence

$$U = \bigcup_{x \in U} B_x \text{ where } \{B_x\}_{x \in U} \subseteq \mathcal{B}$$

Finally as $\forall x \in U \ B_x \in \mathcal{B} \subseteq \mathcal{T} \Rightarrow B_x \in \mathcal{T}$ it follows that $U = \bigcup_{x \in U} B_x \in \mathcal{T}$, $\square$

Given a basis of a topology it is easy to construct a basis of a sub-topology.

Theorem 14.37. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\mathcal{B}$ a basis for $\mathcal{T}$ and $A \subseteq X$ then

$$\mathcal{B}|_A = \{B \cap A \mid B \in \mathcal{B}\} \text{ is a basis for } \mathcal{T}|_A = \{U \cap A \mid U \in \mathcal{T}\}$$

Proof. If $B \in \mathcal{B}_A$ then there exist a $B' \in \mathcal{B}$ such that $B = B' \cap A$, hence, as $\mathcal{B} \subseteq \mathcal{T} \Rightarrow B' \in \mathcal{T}$, we have that $B \in \mathcal{T}|_A$ proving that

$$\mathcal{B}|_A \subseteq \mathcal{T}|_A$$

If $U \in \mathcal{T}|_A$ then there exists a $U' \in \mathcal{T}$ such that $U = U' \cap A$. As $\mathcal{B}$ is a basis of $\mathcal{T}$ there exist a $\{B'_i\}_{i \in I} \subseteq \mathcal{B}$ such that $U' = \bigcup_{i \in I} B'_i$. Then

$$U = U' \cap A = \left(\bigcup_{i \in I} B'_i \right) \cap A \stackrel{[\text{theorem: 2.138}]}{=} \bigcup_{i \in I} (B'_i \cap A)$$

where $\{B'_i \cap A\}_{i \in I} \subseteq \mathcal{B}|_A$. $\square$

Theorem 14.38. Let $\langle X, \mathcal{T} \rangle$ be a topological space and let $\mathcal{B} \subseteq \mathcal{T}$ then

$$\mathcal{B} \text{ is a basis for } \mathcal{T} \Leftrightarrow \forall U \in \mathcal{T} \text{ we have } \forall x \in U \text{ that there exist a } B \in \mathcal{B} \text{ such that } x \in B \subseteq U$$

Proof.

$\Rightarrow$. Let $U \in \mathcal{T}$ then, as $\mathcal{B}$ is a basis for $\mathcal{T}$ we have by [theorem: 14.36] that $\forall x \in U$ there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

$\Leftarrow$. If $U \in \mathcal{T}$ then $\forall x \in U$ there exists a $B \in \mathcal{B} \subseteq \mathcal{T}$ such that $x \in B \subseteq U$, hence using [theorem: 3.108] there exist a $\{B_x\}_{x \in U} \subseteq \mathcal{B}$ such that $\forall x \in B_x \ x \in B_x \subseteq U$. As $\forall x \in U \ B_x \subseteq U$ we have $\bigcup_{x \in U} B_x \subseteq U$, further if $x \in U$ then $x \in B_x \in \mathcal{B}$ so that $x \in \bigcup_{y \in U} B_y$. Hence

$$U = \bigcup_{x \in U} B_x \text{ where } \{B_x\}_{x \in U} \subseteq \mathcal{B}$$

Proving that $\mathcal{B}$ is a basis for $\mathcal{T}$. □

Corollary 14.39. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\mathcal{B}$ a basis for $\mathcal{T}$ and $\mathcal{A}$ such that $\mathcal{B} \subseteq \mathcal{A} \subseteq \mathcal{T}$ then $\mathcal{A}$ is a basis of $\mathcal{T}$*

Proof. Let $U \in \mathcal{T}$ and $x \in U$ then as $\mathcal{B}$ is a basis of $\mathcal{T}$ there exist by [theorem: 14.38] a $B \in \mathcal{B}$ such that $x \in B \subseteq U$. As $\mathcal{B} \subseteq \mathcal{A}$ we have that $B \in \mathcal{A}$ proving by [theorem: 14.38] and the fact that $\mathcal{A} \subseteq \mathcal{T}$ that $\mathcal{A}$ is a basis for $\mathcal{T}$. □

Theorem 14.40. *Let X be a set and $\mathcal{T}_1, \mathcal{T}_2$ two topologies on X with basis $\mathcal{B}_1, \mathcal{B}_2$ then*

$$\mathcal{T}_2 \text{ is finer than } \mathcal{T}_1$$

$$\Updownarrow$$

$$\forall x \in X, \forall B \in \mathcal{B}_1 \text{ with } x \in B \text{ there exist a } B' \in \mathcal{B}_2 \text{ such that } x \in B' \subseteq B$$

Proof.

$\Rightarrow$. Let $x \in X$ and take $B \in \mathcal{B}_1$ such that $x \in B$. As $B \in \mathcal{B}_1 \subseteq \mathcal{T}_1 \subseteq \mathcal{T}_2 \Rightarrow B \in \mathcal{T}_2$ and $\mathcal{B}_2$ is a basis for $\mathcal{T}_2$ we have by [theorem: 14.36] that there exist a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B$.

$\Leftarrow$. Let $U \in \mathcal{T}_1$ and take $x \in U$ then by [theorem: 14.36] and the fact that $\mathcal{B}_1$ is a basis for $\mathcal{T}_1$ it follows that there exist a $B \in \mathcal{B}_1$ such that $x \in B \subseteq U$. Using the hypothesis there exist a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B \subseteq U$. Applying then [theorem: 14.36] again proves that $U \in \mathcal{T}_2$. Hence we have $\mathcal{T}_1 \subseteq \mathcal{T}_2$ or

$$\mathcal{T}_2 \text{ is finer than } \mathcal{T}_1$$

□

Corollary 14.41. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\mathcal{B}$ a basis for $\mathcal{T}$ and $A \subseteq X$ then*

$$A \text{ is closed} \Leftrightarrow \forall x \in X \setminus A \text{ there exists a } B \in \mathcal{B} \text{ such that } x \in B \subseteq X \setminus A$$

Proof.

$\Rightarrow$. As A is closed we have by definition $X \setminus A \in \mathcal{T}$. Hence if $x \in X \setminus A$ we have by the previous theorem [theorem: 14.36] a $B \in \mathcal{B}$ such that $x \in B \subseteq X \setminus A$.

$\Leftarrow$. If $x \in X \setminus A$ then $\exists B \in \mathcal{B}$ such that $x \in B \subseteq X \setminus A$ so by the previous theorem [theorem: 14.36] $X \setminus A$ is open, hence A is closed. □

Theorem 14.42. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $\mathcal{B}$ a basis for $\mathcal{T}$ then:*

1. $\forall x \in X$ there exist a $B \in \mathcal{B}$ such that $x \in B$.
2. $\forall B_1, B_2 \in \mathcal{B}$ we have $\forall x \in B_1 \cap B_2$ that $\exists B \in \mathcal{B}$ such that $x \in B \subseteq B_1 \cap B_2$

Proof.

1. As $X \in \mathcal{T}$ it follows from [theorem: 14.36] that for $x \in X$ there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq X$.
2. If $x \in B_1 \cap B_2$ then as $B_1, B_2 \in \mathcal{B} \subseteq \mathcal{T}$ we have that $B_1 \cap B_2 \in \mathcal{T}$, hence as $\mathcal{B}$ is a basis for $\mathcal{T}$ there exist by [theorem: 14.36] a $B \in \mathcal{B}$ such that $x \in B \subseteq B_1 \cap B_2$. □

The above show the necessary conditions that a basis must satisfies, the following shows that any set of subset fulfilling the above conditions can be the basis of a topology.

Theorem 14.43. *Let X be a set, $\mathcal{B} \subseteq \mathcal{P}(X)$ a set of subsets of X satisfying:*

1. $\forall x \in X$ there exist a $B \in \mathcal{B}$ such that $x \in B$
2. $\forall B_1, B_2 \in \mathcal{B}$ we have $\forall x \in B_1 \cap B_2$ that there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq B_1 \cap B_2$

then

$$\mathcal{T}[\mathcal{B}] = \{U \in \mathcal{P}(X) | \forall x \in U \text{ there exist a } B \in \mathcal{B} \text{ such that } x \in B \subseteq U\} \text{ is a topology}$$

and

$\mathcal{B}$ is a basis for $\mathcal{T}[\mathcal{B}]$

We call $\mathcal{B}$ the **generating basis** for $\mathcal{T}[\mathcal{B}]$ and $\mathcal{T}[\mathcal{B}]$ the **topology generated by $\mathcal{B}$** .

Proof.

- a) $\forall x \in \emptyset$ we have vacuously that there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq \emptyset$, hence $\emptyset \in \mathcal{T}[\mathcal{B}]$.
- b) Let $x \in X$ then by (1) there exists a $B \in \mathcal{B} \subseteq \mathcal{P}(X)$ such that $x \in B \subseteq X$ hence $X \in \mathcal{T}[\mathcal{B}]$.
- c) Let $U_1, U_2 \in \mathcal{T}[\mathcal{B}]$. If $x \in U_1 \cap U_2$ then $x \in U_1 \wedge x \in U_2$, so there exists $B_1, B_2 \in \mathcal{B}$ such that $x \in B_1 \subseteq U_1$ and $x \in B_2 \subseteq U_2$, so $x \in B_1 \cap B_2 \subseteq U_1 \cap U_2$. By (2) there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq B_1 \cap B_2 \subseteq U_1 \cap U_2$ proving that $U_1 \cap U_2 \in \mathcal{T}[\mathcal{B}]$.
- d) Let $\{U_i\}_{i \in I} \subseteq \mathcal{T}[\mathcal{B}]$, if $x \in \bigcup_{i \in I} U_i$ then there exist a $i \in I$ such that $x \in U_i$, hence there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq U_i$ which, as $U_i \subseteq \bigcup_{i \in I} U_i$ proves that $\bigcup_{i \in I} U_i \in \mathcal{T}[\mathcal{B}]$.

From (a),(b),(c) and (d) it follows that

$\mathcal{T}[\mathcal{B}]$ is a topology on X

If $B \in \mathcal{B}$ then $\forall x \in B$ we have $x \in B \subseteq B$ which proves that $B \in \mathcal{T}[\mathcal{B}]$, hence

$$\mathcal{B} \subseteq \mathcal{T}[\mathcal{B}]$$

Further if $U \in \mathcal{T}[\mathcal{B}]$ then if $x \in U$ there exist by definition of $\mathcal{T}[\mathcal{B}]$ a $B \in \mathcal{B}$ such that $x \in B \subseteq U$. By [theorem: 14.38] it follows that

$\mathcal{B}$ is a basis of $\mathcal{T}[\mathcal{B}]$ □

The above theorem shows how a subset of $\mathcal{P}(X)$ satisfying condition (1) and (2) can be used to generate a topology on X . What about a general subset of $\mathcal{P}(X)$ (without any conditions), can we construct a topology from it. It turns out that the answer is yes. Given a $\mathcal{S} \subseteq \mathcal{P}(X)$ we can create a $\mathcal{B} \subseteq \mathcal{P}(X)$ that satisfies (1) and (2) and then generate a topology from $\mathcal{B}$.

Theorem 14.44. Let X be a set and $\mathcal{S} \subseteq \mathcal{P}(X)$ then if we define $\mathcal{B}[\mathcal{S}]$ by

$$\mathcal{B}[\mathcal{S}] = \left\{ B \in \mathcal{P}(X) \mid \exists \{S_i\}_{i \in I} \subseteq \mathcal{S}, I \text{ finite and } I \neq \emptyset \text{ such that } B = \bigcap_{i \in I} S_i \right\} \cup \{X\}$$

we have that:

- 1. $\forall x \in X$ there exist a $B \in \mathcal{B}[\mathcal{S}]$ such that $x \in B$
- 2. $\forall B_1, B_2 \in \mathcal{B}[\mathcal{S}]$ we have $\forall x \in B_1 \cap B_2$ there exist a $B \in \mathcal{B}[\mathcal{S}]$ such that $x \in B \subseteq B_1 \cap B_2$
- 3. $\mathcal{S} \subseteq \mathcal{B}[\mathcal{S}]$

hence by [theorem: 14.43] we have that

$$\mathcal{T}[\mathcal{B}[\mathcal{S}]] = \{U \in \mathcal{P}(X) \mid \forall x \in U \text{ there exist a } B \in \mathcal{B}[\mathcal{S}] \text{ such that } x \in B \subseteq U\}$$

is a topology on X . This topology is called the **topology generated by the sub-basis $\mathcal{S}$** .

Proof.

- 1. As $\mathcal{B}[\mathcal{S}] = \{B \in \mathcal{P}(X) \mid \exists \{S_i\}_{i \in I} \subseteq \mathcal{S}, I \text{ finite and } I \neq \emptyset \text{ such that } B = \bigcap_{i \in I} S_i\} \cup \{X\}$ we have that $X \in \mathcal{B}[\mathcal{S}]$. Hence

$$\forall x \in X \text{ we have } x \in X \subseteq X \text{ where } X \in \mathcal{B}[\mathcal{S}]$$

- 2. Let $x \in B_1 \cap B_2$ where $B_1, B_2 \in \mathcal{B}[\mathcal{S}]$ then there exists finite non empty families $\{S_i\}_{i \in I_1} \subseteq \mathcal{S}$, $\{T_i\}_{i \in I_2} \subseteq \mathcal{S}$ such that $B_1 = \bigcap_{i \in I_1} S_i$ and $B_2 = \bigcap_{i \in I_2} T_i$. Define

$$\{R_i\}_{i \in (I_1 \times \{0\}) \cup (I_2 \times \{1\})} \subseteq \mathcal{S} \text{ by } R_{(i,j)} = \begin{cases} S_i & \text{if } j = 0 \\ T_i & \text{if } j = 1 \end{cases}$$

then we have

$$\begin{aligned}
 y \in \bigcap_{i \in (I_1 \times \{0\}) \cup (I_2 \times \{1\})} R_i &\Leftrightarrow \forall (i, j) \in (I_1 \times \{0\}) \cup (I_2 \times \{1\}) \text{ we have } y \in R_{(i,j)} \\
 &\Leftrightarrow [\forall (i, j) \in I_1 \times \{0\} \text{ we have } y \in R_{(i,j)}] \wedge [\forall (i, j) \in I_2 \times \{1\} \text{ we have } y \in R_{(i,j)}] \\
 &\Leftrightarrow [\forall i \in I_1 \text{ we have } y \in R_{(i,0)}] \wedge [\forall i \in I_2 \text{ we have } y \in R_{(i,1)}] \\
 &\Leftrightarrow [\forall i \in I_1 \text{ we have } y \in S_i] \wedge [\forall i \in I_2 \text{ we have } y \in T_i] \\
 &\Leftrightarrow y \in \bigcap_{i \in I_1} S_i \wedge y \in \bigcap_{i \in I_2} T_i \\
 &\Leftrightarrow y \in B_1 \bigcap B_2
 \end{aligned}$$

This proves that $B_1 \bigcap B_2 = \bigcap_{i \in (I_1 \times \{0\}) \cup (I_2 \times \{1\})} R_i \in \mathcal{B}$ and thus we have found a $B = B_1 \bigcap B_2 \in \mathcal{B}$ such that $x \in B \subseteq B_1 \bigcap B_2$.

3. If $B \in \mathcal{S}$ define $\{S_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ with $S_1 = B$ then $B = S_1 = \bigcap_{i \in \{1\}} S_i$ so that $B \in \mathcal{B}[\mathcal{S}]$. $\square$

Given a family of a topological spaces we can generate a topology for the products of the spaces. It turns out that we have two trivial choices: the box topology and the product topology. In the finite case these topologies are the same.

Definition 14.45. Let I be a non empty set, $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces then the **box topology** noted by $\mathcal{T}_{\text{box}}$ is the topology on $\prod_{i \in I} X_i$ generated by the basis

$$\mathcal{A} = \left\{ \prod_{i \in I} U_i \mid \{U_i\}_{i \in I} \text{ is a family such that } \forall i \in I \ U_i \in \mathcal{T}_i \right\}$$

or

$$\mathcal{T}_{\text{box}} = \mathcal{T}[\mathcal{A}]$$

Proof. Of course we must ensure that $\mathcal{A}$ satisfies the conditions specified in [theorem: 14.43]. First if $B \in \mathcal{A}$ then there exist a $\{U_i\}_{i \in I}$ satisfying $\forall i \in I$ we have $U_i \in \mathcal{T}_i \subseteq \mathcal{P}(X_i) \Rightarrow U_i \subseteq X_i$ such that $B = \prod_{i \in I} U_i$. Hence using [theorem: 2.149] we have that $B = \prod_{i \in I} U_i \subseteq \prod_{i \in I} X_i$, proving that $B \in \mathcal{P}(\prod_{i \in I} X_i)$ or

$$\mathcal{A} \subseteq \mathcal{P}\left(\prod_{i \in I} X_i\right)$$

Further we have:

1. As $\forall i \in I \ \prod_{i \in I} X_i \subseteq \bigcup_{i \in I} X_i \Rightarrow \prod_{i \in I} X_i \in \mathcal{P}(\prod_{i \in I} X_i)$ and $\forall i \in I \ X_i \in \mathcal{T}_i$ it follows that

$$\prod_{i \in I} X_i \in \mathcal{A}$$

Hence $\forall x \in \prod_{i \in I} X_i$ we have for $B = \prod_{i \in I} X_i \in \mathcal{A}$ that $x \in B \in \mathcal{A}$.

2. Let $B_1, B_2 \in \mathcal{A}$ then there exist $\{U_i\}_{i \in I}, \{V_i\}_{i \in I}$ satisfying $\forall i \in I \ U_i, V_i \in \mathcal{T}_i$ such that $B_1 = \prod_{i \in I} U_i$ and $B_2 = \prod_{i \in I} V_i$ then $\forall i \in I$ we have $U_i \cap V_i \in \mathcal{T}_i$ so that

$$B_1 \bigcap B_2 = \left(\prod_{i \in I} U_i \right) \bigcap \left(\prod_{i \in I} V_i \right) \stackrel{[\text{theorem: 2.150}]}{=} \prod_{i \in I} (U_i \cap V_i) \in \mathcal{A}$$

hence $\forall x \in B_1 \bigcap B_2$ we have $x \in B_1 \bigcap B_2 \subseteq B_1 \bigcap B_2$ where $B_1 \bigcap B_2 \in \mathcal{A}$.

By [theorem: 14.43] it follows that $\mathcal{T}[\mathcal{A}]$ is a topology on $\prod_{i \in I} X_i$. $\square$

We can construct the box topology based on the bases of $\mathcal{T}_i$.

Theorem 14.46. Let I be a non empty set, $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces with $\forall i \in I \ \mathcal{B}_i$ a basis for $\mathcal{T}_i$ then

$$\mathcal{B} = \left\{ \prod_{i \in I} B_i \mid \{B_i\}_{i \in I} \text{ is a family such that } \forall i \in I \ B_i \in \mathcal{B}_i \right\}$$

is a basis of $\mathcal{T}_{\text{box}}$ or

$$\mathcal{T}_{\text{box}} = \mathcal{T}[\mathcal{B}]$$

Proof. Using [theorem: 14.45] we have

$$\mathcal{T}_{\text{box}} = \mathcal{T}[\mathcal{A}] \text{ where } \mathcal{A} = \left\{ \prod_{i \in I} U_i \mid \{U_i\}_{i \in I} \text{ is a family such that } \forall i \in I U_i \in \mathcal{T}_i \right\}$$

So if $U \in \mathcal{T}_{\text{box}}$ and $x \in U$ then by [theorem: 14.43] there exist a $\{U_i\}_{i \in I}$ satisfying $\forall i \in I U_i \in \mathcal{T}_i$ such that

$$x \in \prod_{i \in I} U_i \subseteq U \text{ hence } \forall i \in I \text{ we have } x_i \in U_i$$

Let $i \in I$ then as $\mathcal{B}_i$ is a basis of $\mathcal{T}_i$ there exist by [theorem: 14.36] a $B \in \mathcal{B}_i$ such that $x_i \in B \subseteq U_i$. So if we define $\mathcal{A}_i = \{B \in \mathcal{B}_i \mid x_i \in B \subseteq U_i\} \subseteq \mathcal{B}_i$ then $\mathcal{A}_i \neq \emptyset$. Using the Axiom of Choice [see theorem: 3.135] there exist a function $B: I \rightarrow \bigcup_{i \in I} \mathcal{A}_i$ such that $\forall i \in I B_i \in \mathcal{A}_i \subseteq \mathcal{B}_i$. This defines a family $\{B_i\}_{i \in I}$ such that $\forall i \in I x_i \in B_i \subseteq U_i$ and thus by [theorem: 2.149] $x \in \prod_{i \in I} B_i \subseteq \prod_{i \in I} U_i = U$ where $\prod_{i \in I} B_i \in \mathcal{B}$. So by [theorem: 14.38] $\mathcal{B}$ is a basis of $\mathcal{T}_{\text{box}}$. $\square$

The second topology that we can define on a product of sets is the product topology.

Definition 14.47. Let I be a non empty set, $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces then the product topology $\mathcal{T}_{\text{product}}$ on $\prod_{i \in I} X_i$ is defined by the subbase

$$\mathcal{S} = \{\pi_i^{-1}(V) \mid i \in I \wedge V \in \mathcal{T}_i\}$$

where $\forall i \in I$

$$\pi_i: \prod_{j \in I} X_j \rightarrow X_i \text{ is the projection map defined by } \pi_i(x) = x_i$$

or in other terms

$$\mathcal{T}_{\text{product}} = \mathcal{T}[\mathcal{B}[\mathcal{S}]]$$

[see theorem: 14.44].

Note 14.48. As $\forall i \in I$ we have $\pi_i^{-1}(X_i) = X$ it follows that $X \in \mathcal{S}$ so the basis for $\mathcal{T}_{\text{product}}$ is by [theorem: 14.44]

$$\mathcal{B}[\mathcal{S}] = \left\{ B \in \mathcal{P}\left(\prod_{i \in I} X_i\right) \mid \exists \{S_i\}_{i \in J} \subseteq \mathcal{S}, J \text{ finite and } J \neq \emptyset \text{ such that } B = \bigcap_{i \in J} S_i \right\}$$

Theorem 14.49. Let I be a non empty set, $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces then if we define

$$\mathcal{B} = \left\{ \prod_{i \in I} U_i \mid \{U_i\}_{i \in I} \text{ is such that there exist a finite } J \subseteq I \text{ with } \forall i \in J U_i \in \mathcal{T}_i \text{ and } \forall i \in I \setminus J U_i = X_i \right\}$$

we have

$$\mathcal{B} = \mathcal{B}[\mathcal{S}]$$

where [see definition: 14.47]

$$\mathcal{B}[\mathcal{S}] = \left\{ B \in \mathcal{P}\left(\prod_{i \in I} X_i\right) \mid \exists \{S_i\}_{i \in J} \subseteq \mathcal{S}, J \text{ finite and } J \neq \emptyset \text{ such that } B = \bigcap_{i \in J} S_i \right\}$$

and

$$\mathcal{S} = \{\pi_i^{-1}(V) \mid i \in I \wedge V \in \mathcal{T}_i\}$$

So

$$\mathcal{T}_{\text{product}} \underset{\text{definition}}{=} \mathcal{T}[\mathcal{B}[\mathcal{S}]] = \mathcal{T}[\mathcal{B}]$$

Proof. Let $B \in \mathcal{B}[S]$. Then there must exist a $\{S_i\}_{i \in J} \subseteq \mathcal{S}$, $\emptyset \neq J$ finite such that $B = \bigcap_{i \in J} S_i$. Let $i \in J$ then as $S_i \in \mathcal{S}$ we have $\exists k_i \in I$ and a $U_{k_i} \in \mathcal{T}_{k_i}$ such that $S_i = \pi_{k_i}^{-1}(U_{k_i})$. Define

$$\forall i \in J \{B_{k_i, k}\}_{k \in I} \text{ by } B_{k_i, k} = \begin{cases} U_{k_i} \in \mathcal{T}_k & \text{if } k_i = k \\ X_k \in \mathcal{T}_k & \text{if } k \in I \setminus \{k_i\} \end{cases} \in \mathcal{T}_k$$

Let $x \in S_i = \pi_{k_i}^{-1}(U_{k_i})$ then $\forall k \in I$ we have either:

$k = k_i$. Then $\pi_k(x) = \pi_{k_i}(x) \in U_{k_i} = B_{k_i, k}$

$k \in I \setminus \{k_i\}$. Then $\pi_k(x) \in X_k = B_{k_i, k}$

proving that

$$S_i \subseteq \prod_{k \in I} B_{k_i, k} \quad (14.8)$$

Further if $x \in \prod_{k \in I} B_{k_i, k}$ then $\pi_{k_i}(x) \in B_{k_i, k_i} = U_{k_i}$ so that $x \in \pi_{k_i}^{-1}(U_{k_i}) = S_i$, hence $\prod_{k \in I} B_{k_i, k} \subseteq S_i$. Combining this with [eq: 14.8] gives

$$S_i = \prod_{k \in I} B_{k_i, k}$$

Then

$$\begin{aligned} B &= \bigcap_{i \in J} S_i \\ &= \bigcap_{i \in J} \left(\prod_{k \in I} B_{k_i, k} \right) \\ &\stackrel{[\text{theorem: 2.151}]}{=} \prod_{k \in I} \left(\bigcap_{i \in J} B_{k_i, k} \right) \\ &= \prod_{k \in I} C_k \end{aligned} \quad (14.9)$$

where $\{C_k\}_{k \in I}$ is defined by $C_k = \bigcap_{i \in J} B_{k_i, k}$. Let $k \in I$ then as $\emptyset \neq J$ is finite and $B_{k_i, k} \in \mathcal{T}_k$ it follows by [theorem: 14.6] that $C_k \in \mathcal{T}_k$, hence

$$\forall k \in I \text{ we have } C_k \in \mathcal{T}_k \quad (14.10)$$

Define $K = \{k_i | i \in J\} \subseteq I$ which is finite as J is finite [see theorem: 6.44], further as $J \neq \emptyset$ $K \neq \emptyset$. If $k \in I \setminus K$ then $\forall i \in J$ we have $k \neq k_i$ so that $B_{k_i, k} = X_k$ hence $C_k = \bigcap_{i \in J} X_k = X_k$. To conclude we have

$$B = \prod_{k \in I} C_k \text{ and } \forall k \in I \ C_k \in \mathcal{T}_k \text{ and } \forall k \in I \setminus K \ C_k = X_k \text{ where } K \text{ is finite and non empty}$$

So by definition of $\mathcal{B}$ we have $B \in \mathcal{B}$. Hence

$$\mathcal{B}(\mathcal{S}) \subseteq \mathcal{B} \quad (14.11)$$

Let $B \in \mathcal{B}$. Then $B = \prod_{i \in I} U_i$ where $\{U_i\}_{i \in I}$ is a family such that there exist a finite non empty $J \subseteq I$ such that

$$\forall i \in J \ U_i \in \mathcal{T}_i \text{ and } \forall i \in I \setminus J \text{ we have } U_i = X_i \quad (14.12)$$

Define

$$\{U_{i, j}\}_{(i, j) \in I \times J} \text{ by } U_{i, j} = \begin{cases} U_i & \text{if } i = j \\ X_i & \text{if } (i, j) \in I \times J \setminus \{(i, j) \in I \times J | i = j\} \end{cases} = \begin{cases} U_i & \text{if } i = j \\ X_i & \text{if } i \neq j \end{cases} \quad (14.13)$$

Let $i \in I$. If $x \in U_i$ then $\forall j \in J$ we have either $i = j$ then $x \in U_i = U_{i, j}$ or $i \neq j$ so that $x \in X_i = U_{i, j}$ proving that $x \in \bigcap_{j \in J} U_{i, j}$. Hence

$$U_i \subseteq \bigcap_{j \in J} U_{i, j} \quad (14.14)$$

If $x \in \bigcap_{j \in J} U_{i,j}$ then $\forall j \in J$ $x \in U_{i,j}$. For i we have either:

$i \in I \setminus J$. Then $\forall j \in J$ we have $i \neq j$ so that $x \in U_{i,j} \stackrel{[\text{eq: 14.13}]}{=} X_i \stackrel{[\text{eq: 14.12}]}{=} U_i$

$i \in J$. As $x \in \bigcap_{j \in J} U_{i,j}$ we have $x \in U_{i,i} \stackrel{[\text{eq: 14.13}]}{=} U_i$

Hence $\bigcap_{j \in J} U_{i,j} \subseteq U_i$ which by [eq: 14.14] proves that $U_i = \bigcap_{j \in J} U_{i,j}$. Summarized

$$\forall i \in I \text{ we have } U_i = \bigcap_{j \in J} U_{i,j} \quad (14.15)$$

Next

$$\begin{aligned} B &= \prod_{i \in I} U_i \\ &\stackrel{[\text{eq: 14.15}]}{=} \prod_{i \in I} \left(\bigcap_{j \in J} U_{i,j} \right) \\ &\stackrel{[\text{theorem: 2.151}]}{=} \bigcap_{j \in J} \left(\prod_{i \in I} U_{i,j} \right) \end{aligned} \quad (14.16)$$

Let $j \in J$. If $x \in \prod_{i \in I} U_{i,j}$ then $\pi_j(x) \in U_{j,j} \stackrel{[\text{eq: 14.13}]}{=} U_j$ so that $x \in \pi_j^{-1}(U_j)$ proving that

$$\prod_{i \in I} U_{i,j} \subseteq \pi_j^{-1}(U_j) \quad (14.17)$$

Let $x \in \pi_j^{-1}(U_j)$ then for $i \in I$ we have either:

$i = j$. Then $\pi_i(x) \stackrel{i=j}{=} \pi_j(x) \in U_j \stackrel{[\text{eq: 14.13}]}{=} U_{j,j} = U_{i,j}$

$i \neq j$. Then $\pi_i(x) \in X_i \stackrel{[\text{eq: 14.13}]}{=} U_{i,j}$

so that in all cases $\pi_i(x) \in U_{i,j}$ proving that $x \in \prod_{i \in I} U_{i,j}$. Hence $\pi_j^{-1}(U_j) \subseteq \prod_{i \in I} U_{i,j}$ which combined with [eq: 14.17] proves that

$$\prod_{i \in I} U_{i,j} = \pi_j^{-1}(U_j)$$

Substituting this in [eq: 14.16] gives

$$B = \bigcap_{j \in J} \pi_j^{-1}(U_j)$$

Further $\forall j \in J$ $U_j \in \mathcal{T}_j$ so that by definition of S $\pi_j^{-1}(U_j) \in S$ which together with the above and the fact that J is finite proves that $B \in \mathcal{B}[S]$. Hence $\mathcal{B} \subseteq \mathcal{B}[S]$ which combined with [eq: 14.11] gives

$$\mathcal{B} = \mathcal{B}[S]$$

proving the theorem. □

Corollary 14.50. Let I be a non empty set, $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces then

$$\mathcal{B} = \left\{ \bigcap_{i \in J} \pi_i^{-1}(U_i) \mid \{U_i\}_{i \in J} \text{ is a family such that } J \text{ is finite non empty and } \forall j \in J, U_j \in \mathcal{T}_j \right\}$$

is basis for the product topology $\mathcal{T}_{\text{product}}$ on $\prod_{i \in I} X_i$.

Proof. Using the definition of the product topology [definition: 14.47] we have

$$\mathcal{T}_{\text{product}} = \mathcal{T}[\mathcal{B}[S]] \text{ where } S = \{\pi_i^{-1}(V) \mid i \in I \wedge V \in \mathcal{T}_i\}$$

Let $B \in \mathcal{B}$ then there exist a finite non empty J and $\{U_i\}_{i \in J}$ satisfying $\forall j \in J$ $U_j \in \mathcal{T}_j$ such that $B = \bigcap_{i \in J} \pi_i^{-1}(U_i)$. If $i \in J$ then $\pi_i^{-1}(U_i) \in S \subseteq \mathcal{B}[S] \subseteq \mathcal{T}_{\text{product}}$ so that $\pi_i^{-1}(U_i) \in \mathcal{T}_{\text{product}}$. As J is finite non empty it follows from [theorem: 14.6] that $B = \bigcap_{i \in J} \pi_i^{-1}(U_i) \in \mathcal{T}_{\text{product}}$. Hence we have

$$\mathcal{B} \subseteq \mathcal{T}_{\text{product}} \quad (14.18)$$

Using [theorem: 14.49] we have that

$$\mathcal{A} = \left\{ \prod_{i \in I} U_i \mid \{U_i\}_{i \in I} \text{ such that there exist a finite non empty } J \subseteq I \text{ with } \forall i \in J U_i \in \mathcal{T}_i \text{ and } \forall i \in I \setminus J U_i = X_i \right\}$$

is a basis for the product topology $\mathcal{T}_{\text{product}}$ on $\prod_{i \in I} X_i$

If $B \in \mathcal{A}$ then $B = \prod_{i \in I} U_i$ where for $\{U_i\}_{i \in I}$ there exist a finite non empty $J \subseteq I$ such that $\forall i \in J U_i \in \mathcal{T}_i$ and $\forall i \in I \setminus J$ we have $U_i = X_i$. Then we have:

1. If $x \in \prod_{i \in I} U_i$ then $\forall i \in I$ we have that $\pi_i(x) \in U_i$ or $x \in \pi_i^{-1}(U_i)$ proving that $x \in \bigcap_{i \in I} \pi_i^{-1}(U_i)$ proving that

$$\prod_{i \in I} U_i \subseteq \bigcap_{i \in I} \pi_i^{-1}(U_i)$$

2. If $x \in \bigcap_{i \in I} \pi_i^{-1}(U_i)$ then $\forall i \in I$ we have either:

$$i \in J. \text{ Then } x \in \pi_i^{-1}(U_i) \text{ so that } \pi_i(x) \in U_i$$

$$i \in I \setminus J. \text{ Then } U_i = X_i \text{ and } \pi_i(x) \in X_i = U_i$$

so that $x \in \prod_{i \in I} U_i$. Hence we have

$$\bigcap_{i \in I} \pi_i^{-1}(U_i) \subseteq \prod_{i \in I} U_i$$

hence

$$B = \prod_{i \in I} U_i = \bigcap_{i \in I} \pi_i^{-1}(U_i)$$

proving that $B \in \mathcal{B}$. Hence we have that $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{T}_{\text{product}}$ which as $\mathcal{A}$ is a basis for $\mathcal{T}_{\text{product}}$ proves by [theorem: 14.39] that $\mathcal{B}$ is a basis for $\mathcal{T}_{\text{product}}$. $\square$

Using the above corollary it is easy to prove that in the finite case the box topology is equivalent with the product topology.

Theorem 14.51. *Let $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces then*

1. $\mathcal{T}_{\text{product}} \subseteq \mathcal{T}_{\text{box}}$ [in other words the box topology is finer then the product topology]
2. If I is finite then $\mathcal{T}_{\text{box}} = \mathcal{T}_{\text{product}}$

Proof. Consider the bases $\mathcal{B}_{\text{product}}, \mathcal{B}_{\text{box}}$ for the product and box topologies [see: 14.45 and 14.49]

$$\mathcal{B}_{\text{box}} = \left\{ \prod_{i \in I} U_i \mid \{U_i\}_{i \in I} \text{ is a family such that } \forall i \in I U_i \in \mathcal{T}_i \right\}$$

$$\mathcal{B}_{\text{product}} = \left\{ \prod_{i \in I} U_i \mid \{U_i\}_{i \in I} \text{ is such that there exist a finite } J \subseteq I \text{ with } \forall i \in J U_i \in \mathcal{T}_i \text{ and } \forall i \in I \setminus J U_i = X_i \right\}$$

1. Let $B \in \mathcal{B}_{\text{product}}$ then $B = \prod_{i \in I} U_i$ where $\{U_i\}_{i \in I}$ is such that there exist a finite $J \subseteq I$ with $\forall i \in J U_i \in \mathcal{T}_i$ and $\forall i \in I \setminus J U_i = X_i$. As $\forall i \in I \setminus J U_i = X_i \in \mathcal{T}_i$ we have $\forall i \in I$ that $U_i \in \mathcal{T}_i$ it follows that $B \in \mathcal{B}_{\text{box}}$. Hence

$$\mathcal{B}_{\text{product}} \subseteq \mathcal{B}_{\text{box}} \tag{14.19}$$

If $x \in X$ and $B \in \mathcal{B}_{\text{product}}$ satisfies $x \in B$ then as $\mathcal{B}_{\text{product}} \subseteq \mathcal{B}_{\text{box}}$ we have also $B \in \mathcal{B}_{\text{box}}$ hence $x \in B \subseteq B$ where $B \in \mathcal{B}_{\text{box}}$. So by [theorem: 14.40] we have that $\mathcal{T}_{\text{box}}$ is finer then $\mathcal{T}_{\text{product}}$ hence $\mathcal{T}_{\text{product}} \subseteq \mathcal{T}_{\text{box}}$.

2. Assume that I is finite. Take $B \in \mathcal{B}_{\text{box}}$ then $B = \prod_{i \in I} U_i \subseteq \prod_{i \in I} X_i$ is such that $\forall i \in I U_i \in \mathcal{T}_i$ which as I is finite and $I \setminus I = \emptyset$ means that $B \in \mathcal{B}_{\text{product}}$. Hence $\mathcal{B}_{\text{box}} \subseteq \mathcal{B}_{\text{product}}$ so combining this with [eq: 14.19] gives

$$\mathcal{B}_{\text{product}} = \mathcal{B}_{\text{box}}$$

hence

$$\mathcal{T}_{\text{box}} = \mathcal{B}[\mathcal{B}_{\text{box}}] = \mathcal{B}[\mathcal{B}_{\text{product}}] = \mathcal{T}_{\text{product}} \quad \square$$

Theorem 14.52. Let $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces and let $\forall i \in I A_i \subseteq X_i$ be equipped with the subspace topology $(\mathcal{T}_i)_{|A_i}$ then the subspace topology $(\mathcal{T}_{\text{product}})_{|\prod_{i \in I} A_i}$ for $\prod_{i \in I} A_i$ is the product topology of $\{\langle A_i, (\mathcal{T}_i)_{|A_i} \rangle\}_{i \in I}$.

Proof. Let $\mathcal{B}_{\text{product}}$ be the basis of $\mathcal{T}_{\text{product}}$ then by [theorem: 14.37]

$$(\mathcal{B}_{\text{product}})_{|\prod_{i \in I} A_i} = \left\{ B \cap \prod_{i \in I} A_i \mid B \in \mathcal{B}_{\text{product}} \right\} \text{ is a basis for } (\mathcal{T}_{\text{product}})_{|\prod_{i \in I} A_i}$$

further let $\mathcal{T}^{\text{product}}$ be the product topology of $\{\langle A_i, (\mathcal{T}_i)_{|A_i} \rangle\}_{i \in I}$ and $\mathcal{B}^{\text{product}}$ the basis van $\mathcal{T}^{\text{product}}$.

1. If $B \in (\mathcal{B}_{\text{product}})_{|\prod_{i \in I} A_i}$ then by [definition: 14.49] $B = (\prod_{i \in I} U_i) \cap (\prod_{i \in I} A_i)$ where $U_i = X_i$ except for a finite subset J of I where $U_i \in \mathcal{T}$. Hence $B = \prod_{i \in I} (U_i \cap A_i)$ where for $i \in I \setminus J$ $U_i \cap A_i = X_i \cap A_i = A_i$ and for $i \in J$ $U_i \cap A_i \in (\mathcal{T}_i)_{|A_i}$. So by [definition: 14.49] it follows that $B \in \mathcal{B}^{\text{product}}$ the basis of $\mathcal{T}^{\text{product}}$ [the product topology of $\{\langle A_i, (\mathcal{T}_i)_{|A_i} \rangle\}_{i \in I}$]. Hence we have

$$(\mathcal{B}_{\text{product}})_{|\prod_{i \in I} A_i} \subseteq \mathcal{B}^{\text{product}}$$

2. If $B \in \mathcal{B}^{\text{product}}$ then by [definition: 14.49] $B = \prod_{i \in I} U_i$ where $U_i = A_i = X_i \cap A_i$ except for a finite $J \subseteq I$ where $U_i \in (\mathcal{T}_i)_{|A_i} \Rightarrow U_i = W_i \cap A_i$ where $W_i \in \mathcal{T}_i$. Then if we define $\{V_i\}_{i \in I}$ by $V_i = \begin{cases} X_i & \text{if } i \in I \setminus J \\ W_i & \text{if } i \in J \end{cases}$ then $B = \prod_{i \in I} (V_i \cap A_i) \stackrel{[\text{theorem: 2.150}]}{=} (\prod_{i \in I} V_i) \cap (\prod_{i \in I} A_i)$. By [definition: 14.49] $\prod_{i \in I} V_i \in \mathcal{B}_{\text{product}}$ so that $B \in (\mathcal{B}_{\text{product}})_{|\prod_{i \in I} A_i}$. Hence we have

$$\mathcal{B}^{\text{product}} \subseteq (\mathcal{B}_{\text{product}})_{|\prod_{i \in I} A_i}$$

Using (1) and (2) it follows that $(\mathcal{B}_{\text{product}})_{|\prod_{i \in I} A_i} = \mathcal{B}^{\text{product}}$ proving that

$$(\mathcal{T}_{\text{product}})_{|\prod_{i \in I} A_i} = \mathcal{T}^{\text{product}} \text{ the product topology of } \{\langle A_i, \mathcal{T}_i \rangle\}_{i \in I} \quad \square$$

14.1.3 Dense sets

Definition 14.53. (Dense set) Let $\langle X, \mathcal{T} \rangle$ be a topological space then $A \subseteq X$ is a **dense subset** of X if $\bar{A} = X$ [see definition: 14.23 for $\bar{A}$].

Theorem 14.54. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then

$$A \text{ is a dense subset of } X$$

$$\Updownarrow$$

$$\forall x \in X \text{ we have } \forall U \in \mathcal{T} \text{ with } x \in U \text{ that there } \exists a \in A \text{ such that } a \in U.$$

Proof.

$\Rightarrow$. If A is dense then $X = \bar{A} \stackrel{[\text{theorem: 14.30}]}{=} \{x \in X \mid \forall U \in \mathcal{T} \text{ with } x \in U \text{ we have } U \cap A \neq \emptyset\}$. So if $x \in X$ then $\forall U \in \mathcal{T}$ with $x \in U$ we have $U \cap A \neq \emptyset \Rightarrow \exists a \in A$ such that $a \in U$.

$\Leftarrow$. If $x \in X$ then if $U \in \mathcal{T}$ with $x \in U$ there exist by the hypothesis a $a \in A$ such that $a \in U$, hence $a \in U \cap A$ proving that $U \cap A \neq \emptyset$. So by [theorem: 14.30] $x \in \bar{A}$. Hence $X \subseteq \bar{A}$ which as $\bar{A} \subseteq X$ proves that $X = \bar{A}$ or that A is dense in X . $\square$

We can rephrase the previous theorem in terms of a basis of a topology.

Theorem 14.55. *Let $\langle X, \mathcal{T} \rangle$ be a topological space with basis $\mathcal{B}$ and $A \subseteq X$ then*

A is dense in X

$\Updownarrow$

$\forall x \in X$ we have $\forall B \in \mathcal{B}$ with $x \in B$ there exist a $a \in A$ such that $a \in B$

Proof.

$\Rightarrow$. Assume that A is dense in X . Let $B \in \mathcal{B}$ be such that $x \in B$ then as $\mathcal{B} \subseteq \mathcal{T}$ we have by the previous theorem [theorem: 14.54] that there exists a $a \in A$ such that $a \in B$.

$\Leftarrow$. Let $x \in X$ then if $U \in \mathcal{T}$ satisfies $x \in U$ we have, as $\mathcal{B}$ is a basis for $\mathcal{T}$, by [theorem: 14.36] that $x \in B \subseteq U$. Using the hypothesis there exist a $a \in A$ such that $a \in B \subseteq U$. Hence using the previous theorem [theorem: 14.54] it follows that A is dense in X . $\square$

Theorem 14.56. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then*

$A^\circ = \emptyset \Leftrightarrow X \setminus A$ is dense in X

Proof.

$\Rightarrow$. Assume that $A^\circ = \emptyset$. Let $x \in X$ and $U \in \mathcal{T}$ such that $x \in U$. Assume that $U \subseteq A$ then by [definition: 14.12] $x \in A^\circ$ contradicting $A^\circ = \emptyset$, hence we must have that $A \not\subseteq U$. So there exist a $y \in U$ such that $y \notin A \Rightarrow y \in X \setminus A$ proving that $y \in U \cap (X \setminus A)$ or $U \cap (X \setminus A) \neq \emptyset$. By [theorem: 14.30] it follows that $x \in \overline{X \setminus A}$, hence $X \subseteq \overline{X \setminus A}$ which as $\overline{X \setminus A} \subseteq X$ proves that $X = \overline{X \setminus A}$. Hence $X \setminus A$ is dense in X .

$\Leftarrow$. Assume that $x \in A^\circ$ then as A° is open and by the hypothesis $X \setminus A$ dense there exist by [theorem: 14.54] a $a \in X \setminus A$ such that $a \in A^\circ$, hence $\emptyset \neq A^\circ \cap (X \setminus A) \subseteq A \cap (X \setminus A) = \emptyset$, leading to the contradiction $\emptyset \neq \emptyset$, Hence the assumption is wrong and we must have that

$A^\circ = \emptyset$

$\square$

Definition 14.57. *A topological space $\langle X, \mathcal{T} \rangle$ is a Baire space if for every $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ satisfying $\forall i \in \mathbb{N}$ A_i is closed and $(A_i)^\circ = \emptyset$ we have that $(\bigcup_{i \in \mathbb{N}} A_i)^\circ = \emptyset$. In other words the union of any family of closed sets with empty interior has also a empty interior.*

The most trivial and least usefully example of a Baire set is $\langle \emptyset, \{\emptyset\} \rangle$

Example 14.58. Let $X = \emptyset$ then by [theorem: 14.2] the only topology on X is $\{\emptyset\}$. The topological $\langle \emptyset, \{\emptyset\} \rangle$ is a Baire space.

Proof. If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X) = \{\emptyset\}$ then $\forall i \in \mathbb{N}$ $A_i = \emptyset$ so that so that $(\bigcup_{i \in \mathbb{N}} A_i)^\circ \subseteq \bigcup_{i \in \mathbb{N}} A_i = \emptyset$. $\square$

Theorem 14.59. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then*

$\langle X, \mathcal{T} \rangle$ is a Baire space

$\Updownarrow$

For every family $\{U_i\}_{i \in \mathbb{N}} \subseteq \mathcal{T}$ such that $\forall i \in \mathbb{N}$ U_i is open en dense in X we have that $\bigcap_{i \in \mathbb{N}} U_i$ is dense in X

Proof.

$\Rightarrow$. Let $\{U_i\}_{i \in \mathbb{N}}$ be a family of open sets that are dense in X . Define $\{A_i\}_{i \in \mathbb{N}}$ by $A_i = X \setminus U_i$ then $\forall i \in \mathbb{N}$ A_i is closed and as $X \setminus A_i = X \setminus (X \setminus U_i) = U_i$ is dense in X we have by [theorem: 14.56] that $(A_i)^\circ = \emptyset$. Hence by the definition of a Baire space we have

$$\left(\bigcup_{i \in \mathbb{N}} A_i \right)^\circ = \emptyset$$

Using [theorem: 14.56] again we have that

$$X \setminus \left(\bigcup_{i \in \mathbb{N}} A_i \right) \text{ is dense in } X$$

As $X \setminus \left(\bigcup_{i \in \mathbb{N}} A_i \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{i \in \mathbb{N}} (X \setminus A_i) = \bigcap_{i \in \mathbb{N}} (X \setminus (X \setminus U_i)) = \bigcap_{i \in \mathbb{N}} U_i$ proving by the above that

$$\bigcap_{i \in \mathbb{N}} U_i \text{ is dense in } X$$

$\Leftarrow$. Let $\{A_i\}_{i \in \mathbb{N}}$ be a family of closed sets with $\forall i \in \mathbb{N} (A_i)^\circ = \emptyset$. Define $\{U_i\}_{i \in \mathbb{N}}$ by $U_i = X \setminus A_i$ then $\forall i \in \mathbb{N}$ we have that U_i is open and as $(X \setminus U_i)^\circ = (X \setminus (X \setminus A_i))^\circ = (A_i)^\circ = \emptyset$ we have by [theorem: 14.56] that $U_i = X \setminus (X \setminus U_i)$ is dense in X . By the hypothesis we have then that $\bigcap_{i \in \mathbb{N}} U_i$ is dense in X . As $\bigcap_{i \in \mathbb{N}} U_i = \bigcap_{i \in \mathbb{N}} (X \setminus A_i) \stackrel{[\text{theorem: 2.140}]}{=} X \setminus \left(\bigcup_{i \in \mathbb{N}} A_i \right)$ it follows that

$$X \setminus \left(\bigcup_{i \in \mathbb{N}} A_i \right) \text{ is dense in } X$$

Using [theorem: 14.56] again it follows that

$$\left(\bigcup_{i \in \mathbb{N}} A_i \right)^\circ = \emptyset$$

Hence by definition $\langle X, d \rangle$ is a Baire space. □

14.2 Metric spaces

A metric space is a set with a concept of a distance that allows us to define a topology.

Definition 14.60. (Pseudo Metric Space) A pseudo metric space $\langle X, d \rangle$ is a **non empty** set X together with a function

$$d: X \times X \rightarrow \mathbb{R}$$

such that

1. $\forall x \in X \ d(x, x) = 0$
2. $\forall x, y \in X$ we have $d(x, y) = d(y, x)$
3. $\forall x, y, z \in X$ we have $d(x, z) \leq d(x, y) + d(y, z)$

the function $d: X \times X \rightarrow \mathbb{R}$ is called a **pseudo metric**.

Definition 14.61. (Metric Space) A pseudo metric space $\langle X, d \rangle$ is called a **metric space** if we have $\forall x, y \in X$ we have $d(x, y) = 0 \Leftrightarrow x = y$. In other words

$$d: X \times X \rightarrow \mathbb{R}$$

satisfies:

1. $\forall x, y \in X$ we have $d(x, y) = d(y, x)$
2. $\forall x, y, z \in X$ we have $d(x, z) \leq d(x, y) + d(y, z)$
3. $\forall x, y \in X$ we have $d(x, y) = 0 \Leftrightarrow x = y$ [hence $\forall x \in X \ d(x, x) = 0$]

the function $d: X \times X \rightarrow \mathbb{R}$ is called a **metric**.

Theorem 14.62. If $\langle X, d \rangle$ is a pseudo metric space then $\forall x, y \in X$ we have $0 \leq d(x, y)$

Proof. If $x, y \in X$ then $0 = d(x, x) \leq d(x, y) + d(y, x) = d(x, y) + d(x, y) = 2 \cdot d(x, y)$ hence

$$0 \leq d(x, y)$$

□

To define a topology we need the concept of balls.

Definition 14.63. (Open Ball) Let $\langle X, d \rangle$ be a metric space, $\varepsilon \in \mathbb{R}^+$ and $x \in X$ then a **open ball centered around x** noted by $B_d(x, \varepsilon)$ is defined by

$$B_d(x, \varepsilon) = \{y \in X \mid d(x, y) < \varepsilon\} \subseteq X$$

Definition 14.64. (Closed Ball) Let $\langle X, d \rangle$ be a metric space, $\varepsilon \in \mathbb{R}^+$ and $x \in X$ then a **closed ball centered around x** noted by $\overline{B_d(x, \varepsilon)}$ is defined by

$$\overline{B_d(x, \varepsilon)} = \{y \in X \mid d(x, y) \leq \varepsilon\} \subseteq X$$

We proceed now to define topology based on a pseudo metric.

Lemma 14.65. Let $\langle X, d \rangle$ be a pseudo metric space then if $x_1, x_2 \in X$ and $\varepsilon_1, \varepsilon_2 \in \mathbb{R}^+$ we have that

$$\forall x \in B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2) \text{ there } \exists \varepsilon \in \mathbb{R}^+ \text{ such that } x \in B_d(x, \varepsilon) \subseteq B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2)$$

Proof. Let $x \in B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2)$ then $d(x, x_1) < \varepsilon_1$ and $d(x, x_2) < \varepsilon_2$. Define

$$\varepsilon = \min(\varepsilon_1 - d(x_1, x), \varepsilon_2 - d(x_2, x))$$

then as $0 < \varepsilon_1 - d(x_1, x) \wedge 0 < \varepsilon_2 - d(x_2, x)$ we have that

$$\varepsilon \in \mathbb{R}^+$$

Let $y \in B_d(x, \varepsilon)$ then $d(x, y) < \varepsilon$ so that

$$d(x_1, y) \leq d(x_1, x) + d(x, y) < d(x_1, x) + \varepsilon \leq d(x_1, x) + \varepsilon_1 - d(x_1, x) = \varepsilon_1$$

and

$$d(x_2, y) \leq d(x_2, x) + d(x, y) < d(x_2, x) + \varepsilon < d(x_2, x) + \varepsilon_2 - d(x_2, x) = \varepsilon_2$$

so that $y \in B_d(x_1, \varepsilon_1) \wedge y \in B_d(x_2, \varepsilon_2)$ or $y \in B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2)$, Hence

$$B_d(x, \varepsilon) \subseteq B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2)$$

Finally as $d(x, x) = 0 < \varepsilon$ we have that

$$x \in B_d(x, \varepsilon)$$

□

Corollary 14.66. Let $\langle X, d \rangle$ be a pseudo metric space, $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then

$$\forall y \in B_d(x, \varepsilon) \text{ there exist a } \delta \in \mathbb{R}^+ \text{ such that } y \in B_d(y, \delta) \subseteq B_d(x, \varepsilon)$$

Proof. As $y \in B_d(x, \varepsilon) = B_d(x, \varepsilon) \cap B_d(x, \varepsilon)$ we can use the previous lemma [lemma: 14.65] to find a $\delta \in \mathbb{R}^+$ such that $B_d(y, \delta) \subseteq B_d(x, \varepsilon) \cap B_d(x, \varepsilon) = B_d(x, \varepsilon)$. □

Next we show that open balls can be used to define a topology on a metric space.

Theorem 14.67. Let $\langle X, d \rangle$ be a pseudo metric space then if we define $\mathcal{B}_d$ by

$$\mathcal{B}_d = \{B_d(x, \varepsilon) \mid x \in X \wedge \varepsilon \in \mathbb{R}^+\}$$

we have:

1. $\forall x \in X$ there exists a $B \in \mathcal{B}_d$ such that $x \in B$.
2. $\forall B_1, B_2 \in \mathcal{B}_d$ we have $\forall x \in B_1 \cap B_2$ that there exist a $B \in \mathcal{B}_d$ such that $x \in B \subseteq B_1 \cap B_2$

so that by [theorem: 14.43]

$$\mathcal{T}_d = \mathcal{T}[\mathcal{B}_d] \stackrel{\text{def}}{=} \{U \in \mathcal{P}(X) \mid \forall x \in U \text{ there exist a } B \in \mathcal{B}_d \text{ such that } x \in B \subseteq U\}$$

is a topology with $\mathcal{B}_d$ as a basis.

Proof.

1. $\forall x \in X$ we have as $d(x, x) = 0 < 1$ that $x \in B_d(x, 1) \in \mathcal{B}_d$.
2. If $B_1, B_2 \in \mathcal{B}_d$ then there exist $x_1, x_2 \in X$ and $\varepsilon_1, \varepsilon_2 \in \mathbb{R}^+$ such that $B_1 = B_d(x_1, \varepsilon_1)$ and $B_2 = B_d(x_2, \varepsilon_2)$. So if $x \in B_1 \cap B_2$ we have $x \in B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2)$ hence by [lemma: 14.65] there exist a $\varepsilon \in \mathbb{R}^+$ such that for $B = B_d(x, \varepsilon) \in \mathcal{B}_d$ we have

$$x \in B = B_d(x, \varepsilon) \subseteq B_d(x_1, \varepsilon_1) \cap B_d(x_2, \varepsilon_2) = B_1 \cap B_2 \quad \square$$

Corollary 14.68. Let $\langle X, d \rangle$ be a pseudo metric space that

$$U \in \mathcal{T}_d \Leftrightarrow \forall x \in U \text{ there exist a } \varepsilon \in \mathbb{R}^+ \text{ such that } x \in B_d(x, \varepsilon) \subseteq U$$

Proof.

- $\Rightarrow$. If $x \in U$ open in the metric topology then by the definition of a basis there exists a $y \in X$ and a $\varepsilon > 0$ such that $x \in B_d(y, \varepsilon) \subseteq U$. Using [corollary: 14.66] there exists a $\delta > 0$ such that $x \in B_d(x, \delta) \subseteq B_d(y, \varepsilon) \subseteq U$.
- $\Leftarrow$. If for every $x \in U$ there exists a $\delta > 0$ such that $x \in B_d(x, \delta) \subseteq U$ we have as $B_d(x, \delta) \in \mathcal{B}_d$ that $U \in \mathcal{T}_d$ [using theorem: 14.36]. $\square$

The reason why $\overline{B_d(x, \varepsilon)}$ is called a closed ball because it is closed using the metric topology.

Theorem 14.69. Let $\langle X, d \rangle$ be a pseudo metric space with the metric topology $\mathcal{T}_d$, $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then $\overline{B_d(x, \varepsilon)}$ is closed.

Proof. Let $y \in X \setminus \overline{B_d(x, \varepsilon)}$ then we must have that $\varepsilon < d(x, y)$. Take $\delta = d(x, y) - \varepsilon > 0$ then if $z \in B_d(y, \delta)$ we have $d(y, z) < \delta = d(x, y) - \varepsilon$. Assume that $d(x, z) \leq \varepsilon$ then we have

$$d(x, y) \leq d(x, z) + d(y, z) < \varepsilon + d(x, y) - \varepsilon < d(x, y)$$

leading to the contradiction $d(x, y) < d(x, y)$. Hence the assumption is wrong and we must have $\varepsilon < d(x, z)$ so that $z \in X \setminus \overline{B_d(x, \varepsilon)}$ proving that $y \in B_d(y, \delta) \subseteq X \setminus \overline{B_d(x, \varepsilon)}$. So $X \setminus \overline{B_d(x, \varepsilon)}$ is open from which it follows that $\overline{B_d(x, \varepsilon)}$ is closed. $\square$

Actually we can say even more, the closed ball is the closure of the open ball which justifies the notation $\overline{B_d(x, \varepsilon)}$.

If we have a metric space $\langle X, d \rangle$ with the metric topology $\mathcal{T}_d$ then the subspace topology has a very simple form as is show in the next theorem:

Theorem 14.70. Let $\langle X, d \rangle$ be a pseudo metric space with the metric topology $\mathcal{T}_d$ and $A \subseteq X$ then for the restricted function $d|_{A \times A}: A \times A \rightarrow \mathbb{R}$ we have

1. $\langle A, d|_{A \times A} \rangle$ is a pseudo metric space.
2. If $\langle X, d \rangle$ is a metric space then $\langle A, d|_{A \times A} \rangle$ is a metric space.
3. $\mathcal{T}_{d|_{A \times A}} = (\mathcal{T}_d)|_A$ [the induced subspace topology on A is the metric topology of the restricted metric topology].

Proof.

1. We have:

$$a. \forall x \in A \quad d|_{A \times A}(x, x) = d(x, x) = 0$$

$$b. \forall x, y \in A \quad d|_{A \times A}(x, y) = d(x, y) = d(y, x) = d|_{A \times A}(y, x)$$

- c. $\forall x, y, z \in A \quad d_{|A \times A}(x, z) = d(x, z) \leq d(x, y) + d(y, z) = d_{|A \times A}(x, y) + d_{|A \times A}(y, z)$
2. Let $x, y \in A$ such that $d_{|A \times A}(x, y) = 0$ then $d(x, y) = 0$ hence $x = y$.
3. Let $x \in A$ and $\varepsilon \in \mathbb{R}^+$ then we have

$$\begin{aligned}
 y \in B_{d_{|A \times A}}(x, \varepsilon) &\Leftrightarrow y \in A \wedge d_{|A \times A}(x, y) < \varepsilon \\
 &\Leftrightarrow y \in A \wedge d(x, y) < \varepsilon \\
 &\stackrel{A \subseteq X \wedge x \in A}{\Leftrightarrow} y \in A \wedge x \in X \wedge d(x, y) < \varepsilon \\
 &\Leftrightarrow y \in A \wedge x \in B_d(x, \varepsilon) \\
 &\Leftrightarrow y \in B_d(x, \varepsilon) \bigcap A
 \end{aligned}$$

proving that

$$\forall x \in A \text{ and } \forall \varepsilon \in \mathbb{R}^+ \quad B_{d_{|A \times A}}(x, \varepsilon) = B_d(x, \varepsilon) \bigcap A \quad (14.20)$$

Let $U \in (\mathcal{T}_d)_{|A}$ then there exist a $V \in \mathcal{T}_d$ such that $U = V \bigcap A$. Let $x \in U$ then $x \in V$ so that that by [corollary: 14.68] there exist a $\varepsilon \in \mathbb{R}^+$ such that $x \in B_d(x, \varepsilon) \subseteq V$, hence $x \in B_d(x, \varepsilon) \bigcap A \subseteq V \bigcap A = U$ or using [eq: 14.20] $x \in B_{d_{|A \times A}}(x, \varepsilon) \subseteq U$. So using [corollary: 14.68] again it follows that $U \in \mathcal{T}_{d_{|A \times A}}$ giving

$$(\mathcal{T}_d)_{|A} \subseteq \mathcal{T}_{d_{|A \times A}} \quad (14.21)$$

Let $U \in \mathcal{T}_{d_{|A \times A}}$ and $x \in U$ then as $\mathcal{T}_{d_{|A \times A}}$ is a topology on A we have $U \subseteq A$, hence $x \in A$. By [corollary: 14.68] there exist a $\varepsilon \in \mathbb{R}^+$ such that $x \in B_{d_{|A \times A}}(x, \varepsilon) \subseteq U \stackrel{[\text{eq: 14.20}]}{\Rightarrow} x \in B_d(x, \varepsilon) \bigcap A \subseteq U$. As $B_d(x, \varepsilon) \bigcap A \in (\mathcal{B}_d)_{|A}$ a basis for $(\mathcal{T}_d)_{|A}$ [see theorem: 14.37] it follows from [theorem: 14.36] that $U \in (\mathcal{T}_d)_{|A}$. Hence $\mathcal{T}_{d_{|A \times A}} \subseteq (\mathcal{T}_d)_{|A}$ which combined with [eq: 14.21] result in

$$(\mathcal{T}_d)_{|A} = \mathcal{T}_{d_{|A \times A}} \quad \square$$

Definition 14.71. Two pseudo metrics d_1, d_2 on a set X are equivalent if $\mathcal{T}_{d_1} = \mathcal{T}_{d_2}$. In other words two pseudo metrics are equivalent iff there generated topologies are the same.

Theorem 14.72. Let d_1, d_2 be two pseudo metrics on a set X then

$$\mathcal{T}_{d_2} \text{ is a finer then } \mathcal{T}_{d_1}$$

$$\Updownarrow$$

$$\forall x \in X, \forall \varepsilon \in \mathbb{R}^+ \text{ there exist } \delta \in \mathbb{R}^+ \text{ such that } x \in B_{d_2}(x, \delta) \subseteq B_{d_1}(x, \varepsilon)$$

Proof.

$\Rightarrow$. If $x \in X$ and $\varepsilon \in \mathbb{R}^+$ we have for $B_{d_1}(x, \varepsilon) \in \mathcal{B}_{d_1}$ using [theorem: 14.40] that there exist a $y \in X$ and a $\delta' \in \mathbb{R}^+$ such that $x \in B_{d_2}(y, \delta') \subseteq B_{d_1}(x, \varepsilon)$. Hence using [theorem: 14.66] there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_{d_2}(x, \delta) \subseteq B_{d_2}(y, \delta') \subseteq B_{d_1}(x, \varepsilon)$.

$\Leftarrow$. Let $B \in \mathcal{B}_{d_1}$ then $\exists x \in X, \varepsilon \in \mathbb{R}^+$ such that $B = B_{d_1}(x, \varepsilon)$, using the hypothesis there exists a $\delta \in \mathbb{R}^+$ such that $x \in B_{d_2}(x, \delta) \subseteq B_{d_1}(x, \varepsilon)$. As $B_{d_2}(x, \varepsilon) \in \mathcal{B}_{d_2}$ it follows from [theorem: 14.40] that $\mathcal{T}_{d_2}$ is finer then $\mathcal{T}_{d_1}$. $\square$

Definition 14.73. Let $\langle X, d_X \rangle$ and $\langle Y, d_Y \rangle$ be two pseudo metric spaces then a function

$$\varphi: X \rightarrow Y$$

is a **isometry** if $\forall x, y \in X$ we have

$$d_Y(\varphi(x), \varphi(y)) = d_X(x, y)$$

If in addition $\varphi: X \rightarrow Y$ is **bijective** then we say that $\varphi: X \rightarrow Y$ is a **isometric isomorphism**.

If between two pseudo metric spaces there exist a **isometric isomorphism** then we say that these pseudo metric spaces are **isometric**. We note this relation as

$$X \approx Y$$

In essence two metric spaces are **isometric** if there exist a bijection between these spaces that conserves distances.

Note that a isometry between a metric space [not just pseudo metric space] and a pseudo metric space is automatically injective.

Theorem 14.74. Let $\langle X, d_X \rangle$ be a metric space, $\langle Y, d_Y \rangle$ a pseudo metric space and

$$\varphi: X \rightarrow Y \text{ a isometry}$$

then

$$\varphi \text{ is injective}$$

Hence if in addition $\varphi: X \rightarrow Y$ is surjective it follows that $\varphi: X \rightarrow Y$ is a bijection so making $\varphi: X \rightarrow Y$ a isometric isomorphism.

Proof. If for $x, y \in X$ we have that $\varphi(x) = \varphi(y)$ then $d_X(x, y) = d_Y(\varphi(x), \varphi(y)) = 0$. As d_X is a metric it follows that $x = y$ proving injectivity. $\square$

Theorem 14.75. Let $\langle X, d_X \rangle, \langle Y, d_Y \rangle$ be two pseudo metric spaces and $\varphi: X \rightarrow Y$ a **isometric isomorphism** then

$$\varphi^{-1}: Y \rightarrow X$$

is a isometry.

Proof. By [theorem: 2.74] $\varphi^{-1}: Y \rightarrow X$ is a function. Further if $x, y \in Y$ then $d_Y(x, y) = d_Y(\varphi(\varphi^{-1}(x)), \varphi(\varphi^{-1}(y))) \stackrel{\varphi \text{ is a isometry}}{=} d_X(\varphi^{-1}(x), \varphi^{-1}(y))$. $\square$

Theorem 14.76. Let $\langle X, d_X \rangle, \langle Y, d_Y \rangle, \langle Z, d_Z \rangle$ be pseudo metric spaces and $\varphi: X \rightarrow Y, \psi: Y \rightarrow Z$ isometries then

$$\psi \circ \varphi: X \rightarrow Z \text{ is a isometry}$$

If in addition φ and ψ are isometric isomorphisms then

$$\psi \circ \varphi: X \rightarrow Z \text{ is a isometric isomorphism}$$

Proof. First we have

$$d_Z((\psi \circ \varphi)(x), (\psi \circ \varphi)(y)) = d_Z(\psi(\varphi(x)), \psi(\varphi(y))) = d_Y(\varphi(x), \varphi(y)) = d_X(x, y)$$

proving that $\psi \circ \varphi: X \rightarrow Z$ is a isometry. Further if φ, ψ isometric isomorphisms, hence bijective, then by [theorem: 2.78] $\psi \circ \varphi$ is a bijection, so $\psi \circ \varphi$ is a isometric isomorphism. $\square$

Theorem 14.77. Let $\langle X, d_X \rangle, \langle Y, d_Y \rangle$ be two pseudo metric spaces and $\varphi: X \rightarrow Y$ a isometric isomorphism then we have:

1. $\forall x \in X, \varepsilon \in \mathbb{R}^+$ we have $\varphi(B_{d_X}(x, \varepsilon)) = B_{d_Y}(\varphi(x), \varepsilon)$
2. $\forall x \in Y, \varepsilon \in \mathbb{R}^+$ we have $\varphi^{-1}(B_{d_Y}(y, \varepsilon)) = B_{d_X}(\varphi^{-1}(y), \varepsilon)$

Proof.

1. If $y \in \varphi(B_{d_X}(x, \varepsilon))$ then there exist a $z \in B_{d_X}(x, \varepsilon)$ such that $y = \varphi(z)$, hence $d_X(x, z) < \varepsilon$ or as φ is a isometry

$$d_Y(\varphi(x), y) = d_Y(\varphi(x), \varphi(z)) = d_X(x, z) < \varepsilon$$

proving that $y \in B_{d_Y}(\varphi(x), \varepsilon)$. On the other hand if $y \in B_{d_Y}(\varphi(x), \varepsilon)$ then $d_Y(\varphi(x), y) < \varepsilon$, hence

$$d_X(x, \varphi^{-1}(y)) = d_Y(\varphi(x), \varphi(\varphi^{-1}(y))) = d_Y(\varphi(x), y) < \varepsilon$$

so that $\varphi^{-1}(y) \in B_{d_X}(x, \varepsilon)$.

2. As $\psi = \varphi^{-1}: Y \rightarrow X$ is also a isometry we have

$$B_{d_X}(\varphi^{-1}(y), \varepsilon) = B_{d_X}(\psi(y), \varepsilon) \stackrel{(1)}{=} \psi(B_{d_Y}(y, \varepsilon)) = \varphi^{-1}(B_{d_Y}(y, \varepsilon)) \quad \square$$

Theorem 14.78. *Let $\langle X, d_X \rangle$, $\langle Y, d_Y \rangle$ be two pseudo metric spaces and $\varphi: X \rightarrow Y$ a isometric isomorphism. Then*

1. $\mathcal{T}_{d_X} = \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\} = \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\}$
2. $\mathcal{T}_{d_Y} = \{\varphi(U) | U \in \mathcal{T}_{d_X}\} = \{V \subseteq Y | \varphi^{-1}(V) \in \mathcal{T}_{d_X}\}$

Proof.

1. Let $U \in \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\}$ then there exist a $V \in \mathcal{T}_{d_Y}$ such that $U = \varphi^{-1}(V)$, hence $\varphi(U) = V \in \mathcal{T}_{d_Y}$ proving that

$$\{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\} \subseteq \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\} \quad (14.22)$$

If $U \in \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\}$ then $\varphi(U) \in \mathcal{T}_{d_Y}$ so, as $U = \varphi^{-1}(\varphi(U))$, we have $U \in \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\}$, proving that $\{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\} \subseteq \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\}$. Combining this with [eq: 14.22] proves that

$$\{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\} = \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\} \quad (14.23)$$

Let $U \in \mathcal{T}_{d_X}$ and take $y \in \varphi(U)$ then $\exists x \in U$ such that $y = \varphi(x)$. Using [theorem: 14.68] there exist a $\varepsilon \in \mathbb{R}^+$ such that $x \in B_{d_X}(x, \varepsilon) \subseteq U$ hence

$$y = \varphi(x) \in \varphi(B_{d_X}(x, \varepsilon)) \stackrel{[\text{theorem: 14.77}]}{=} B_{d_Y}(\varphi(x), \varepsilon) \subseteq \varphi(U)$$

Hence, using [theorem: 14.68] it follows that $\varphi(U) \in \mathcal{T}_{d_Y}$ giving $U \in \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\}$. So we conclude that

$$\mathcal{T}_{d_X} \subseteq \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\} \quad (14.24)$$

Let $U \in \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\}$ and $x \in U$ then there exist a $V \in \mathcal{T}_{d_Y}$ such that $x \in U = \varphi^{-1}(V) \Rightarrow \varphi(x) \in V$. As V is open there exist by [theorem: 14.68] a $\varepsilon \in \mathbb{R}^+$ such that

$$\varphi(x) \in B_{d_Y}(\varphi(x), \varepsilon) \subseteq V$$

Hence

$$x = \varphi^{-1}(\varphi(x)) \in \varphi^{-1}(B_{d_Y}(\varphi(x), \varepsilon)) \subseteq \varphi^{-1}(V) = U$$

As $\varphi^{-1}(B_{d_Y}(\varphi(x), \varepsilon)) \stackrel{[\text{theorem: 14.77}]}{=} B_{d_X}(\varphi^{-1}(\varphi(x)), \varepsilon) = B_{d_X}(x, \varepsilon)$ it follows from the above that

$$x \in B_{d_X}(x, \varepsilon) \subseteq U$$

So by [theorem: 14.68] we have $U \in \mathcal{T}_{d_X}$. Hence $\{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\} \subseteq \mathcal{T}_{d_X}$ and combining this with [eq: 14.24] gives

$$\mathcal{T}_{d_X} \subseteq \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\} \stackrel{[\text{eq: 14.23}]}{=} \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\} \subseteq \mathcal{T}_{d_X}$$

or

$$\mathcal{T}_{d_X} = \{U \subseteq X | \varphi(U) \in \mathcal{T}_{d_Y}\} = \{\varphi^{-1}(V) | V \in \mathcal{T}_{d_Y}\}$$

2. As $\psi: Y \rightarrow X$ where $\psi = \varphi^{-1}$ is a isometry it follows from (1) that

$$\begin{aligned} \mathcal{T}_{d_Y} &= \{U \subseteq Y | \psi(U) \in \mathcal{T}_{d_X}\} \\ &= \{U \subseteq Y | \varphi^{-1}(U) \in \mathcal{T}_{d_X}\} \\ \mathcal{T}_{d_Y} &= \{\psi^{-1}(V) | V \in \mathcal{T}_{d_X}\} \\ &= \{\varphi(V) | V \in \mathcal{T}_{d_X}\} \end{aligned}$$

hence

$$\mathcal{T}_{d_Y} = \{U \subseteq Y \mid \varphi^{-1}(U) \in \mathcal{T}_{d_X}\} = \{\varphi(V) \mid V \in \mathcal{T}_{d_X}\} \quad \square$$

Definition 14.79. Let $\langle X, d \rangle$ be a pseudo metric space and $A \subseteq X$ then A is **bounded** if

$$\exists M \in \mathbb{R}^+ \text{ such that } \forall x, y \in A \text{ we have } d(x, y) \leq M$$

If A is bounded and **non empty** then using the fact that $\langle \mathbb{R}, \leq \rangle$ is conditional complete [see theorem: 10.18] it follows that

$$\sup(\{d(x, y) \mid x, y \in A\}) \text{ exist for } A$$

The **diameter of A** noted as $\text{diam}(A)$ is then defined by

$$\text{diam}(A) = \sup(\{d(x, y) \mid x, y \in A\})$$

Example 14.80. Let $\langle X, d \rangle$ be a metric space then $\emptyset$ is bounded.

Proof. $\forall x, y \in \emptyset$ we have vacuously that $d(x, y) \leq 1$ $\square$

Theorem 14.81. Let $\langle X, d \rangle$ a pseudo metric space and $U \subseteq X$ a bounded set then every subset of U is also bounded.

Proof. As U is bounded there exist a $M \in \mathbb{R}^+$ such that $\forall x, y \in U$ we have $d(x, y) \leq M$. Let $V \subseteq U$ then $\forall x, y \in V$ we have as $V \subseteq U$ that $x, y \in U$ hence $d(x, y) \leq M$, proving that V is bounded. $\square$

Theorem 14.82. Let $\langle X, d \rangle$ be a pseudo metric space then $\forall x \in X, \forall \varepsilon \in \mathbb{R}^+$ we have that

$$B_d(x, \varepsilon) \text{ and } \overline{B_d(x, \varepsilon)} \text{ are bounded}$$

and

$$\text{diam}(B_d(x, \varepsilon)) \leq 2 \cdot \varepsilon \wedge \text{diam}(\overline{B_d(x, \varepsilon)}) \leq 2 \cdot \varepsilon$$

Proof.

1. Let $y, z \in B_d(x, \varepsilon)$ then $d(y, z) \leq d(y, x) + d(x, z) < \varepsilon + \varepsilon = 2 \cdot \varepsilon$ proving that $B_d(x, \varepsilon)$ is bounded and $\text{diam}(B_d(x, \varepsilon)) = \sup(\{d(x, y) \mid x, y \in B_d(x, \varepsilon)\}) \leq 2 \cdot \varepsilon$.
2. Let $y, z \in \overline{B_d(x, \varepsilon)}$ then $d(y, z) \leq d(y, x) + d(x, z) \leq \varepsilon + \varepsilon = 2 \cdot \varepsilon$ proving that $\overline{B_d(x, \varepsilon)}$ is bounded and $\text{diam}(\overline{B_d(x, \varepsilon)}) = \sup(\{d(x, y) \mid x, y \in \overline{B_d(x, \varepsilon)}\}) \leq 2 \cdot \varepsilon$. $\square$

Theorem 14.83. Let $\{\langle X_i, d_i \rangle\}_{i \in I}$ be a **finite** family of pseudo metric spaces then for

$$d_{\max}: \left(\prod_{i \in I} X_i \right) \times \left(\prod_{i \in I} X_i \right) \rightarrow \mathbb{R} \text{ defined by } d_{\max}(x, y) = \max(\{d_i(\pi_i(x), \pi_i(y)) \mid i \in I\})$$

where $\forall i \in I$ π_i is the projection map defined by

$$\pi_i: \prod_{j \in I} X_j \rightarrow X_i \text{ is defined by } \pi_i(x) = x(i) \underset{\text{note}}{=} x_i$$

we have:

1. $\langle \prod_{i \in I} X_i, d_{\max} \rangle$ is a pseudo metric space.
2. If $\forall i \in I$ $\langle X_i, d_i \rangle$ is a metric space then $\langle \prod_{i \in I} X_i, d_{\max} \rangle$ is a metric space
3. $\mathcal{T}_{d_{\max}} = \mathcal{T}_{\text{box}} = \mathcal{T}_{\text{product}}$ [the metric topology is equal to the box and product topology of $\prod_{i \in I} X_i$.

Note 14.84. As I is finite $\max(\{d_i(\pi_i(x), \pi_i(y)) \mid i \in I\})$ exist by [theorem: 6.49].

Proof.

1. Let $x, y, z \in \prod_{i \in I} X_i$ then

- a. $d_{\max}(x, x) = \max(\{d_i(\pi_i(x), \pi_i(x)) | i \in I\}) = \max(\{0\}) = 0$
- b. $d_{\max}(x, y) = \max(\{d_i(\pi_i(x), \pi_i(y)) | i \in I\}) = \max(\{d_i(\pi_i(y), \pi_i(x)) | i \in I\}) = d(y, x)$
- c. We have

$$\begin{aligned}
 d_{\max}(x, z) &= \max(\{d_i(\pi_i(x), \pi_i(z)) | i \in I\}) \\
 &\leq_{[\text{theorem: 3.69}]} \max(\{d_i(\pi_i(x), \pi_i(y)) + d(\pi_i(y), \pi_i(z)) | i \in I\}) \\
 &\leq_{[\text{theorem: 3.69}]} \max(\{\max(\{d_i(\pi_i(x), \pi_i(y)) | i \in I\}) + d(\pi_i(y), \pi_i(z)) | i \in I\}) \\
 &\stackrel{[\text{theorem: 10.26}]}{=} \max(\{d_i(\pi_i(x), \pi_i(y)) | i \in I\}) + \max(\{d(\pi_i(y), \pi_i(z)) | i \in I\}) \\
 &= d_{\max}(x, y) + d_{\max}(y, z)
 \end{aligned}$$

2. Let $d(x, y) = 0$ then as $\forall i \in I$ we have

$$0 \leq d_i(\pi_i(x), \pi_i(y)) \leq \max(\{d_i(\pi_i(x), \pi_i(y)) | i \in I\}) = d(x, y) = 0$$

so that $d_i(\pi_i(x), \pi_i(y)) = 0$. As d_i is a metric it follows that $\pi_i(x) = \pi_i(y) \Rightarrow x(i) = y(i)$ proving finally that $x = y$.

3. As I is finite we have by [theorem: 14.51] that

$$\mathcal{T}_{\text{box}} = \mathcal{T}_{\text{product}} \quad (14.25)$$

The bases of $\mathcal{T}_d$ and $\mathcal{T}_{\text{box}}$ are [see definitions: 14.67, 14.46]

$$\mathcal{B}_{d_{\max}} = \{B_{d_{\max}}(x, \varepsilon) | x \in X \wedge \varepsilon \in \mathbb{R}^+\}$$

and

$$\mathcal{B}_{\text{box}} = \left\{ \prod_{i \in I} B_i | \{B_i\}_{i \in I} \text{ such that } \forall i \in I B_i \in \mathcal{B}_{d_i} \right\}$$

Let $x \in \prod_{i \in I} X_i$ and $B \in \mathcal{B}_{d_{\max}}$ such that $x \in B$. Then there exists a $y \in X$, $\varepsilon \in \mathbb{R}^+$ such that $x \in B_{d_{\max}}(y, \varepsilon) \subseteq B$ or using [theorem: 14.66] there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{d_{\max}}(x, \delta) \subseteq B_{d_{\max}}(y, \varepsilon) \subseteq B$$

Take $\prod_{i \in I} B_{d_i}(\pi_i(x), \delta) \in \mathcal{B}_{\text{box}}$ then, as $\forall i \in I$ we have $d_i(\pi_i(x), \pi_i(x)) = 0 < \delta$, it follows that $\pi_i(x) \in B_{d_i}(\pi_i(x), \delta)$, proving

$$x \in \prod_{i \in I} B_{d_i}(\pi_i(x), \delta)$$

Let $z \in \prod_{i \in I} B_{d_i}(\pi_i(x), \delta)$ then $\forall i \in I$ we have $\pi_i(z) \in B_{d_i}(\pi_i(x), \delta)$ so that $d_i(\pi_i(x), \pi_i(z)) < \delta$. Hence $d_{\max}(x, z) = \max(\{d_i(\pi_i(x), \pi_i(z)) | i \in I\}) < \delta$ which proves that $z \in B_{d_{\max}}(x, \delta)$, giving

$$\prod_{i \in I} B_{d_i}(\pi_i(x), \delta) \subseteq B_{d_{\max}}(x, \delta) \subseteq B$$

So if we take $B' = \prod_{i \in I} B_{d_i}(\pi_i(x), \delta) \in \mathcal{B}_{\text{box}}$ then we have

$$\forall x \in \prod_{i \in I} X_i \text{ and } \forall B \in \mathcal{B}_{d_{\max}} \text{ there exist a } B' \in \mathcal{B}_{\text{box}} \text{ such that } x \in B' \subseteq B \quad (14.26)$$

Let $x \in \prod_{i \in I} X_i$ and $B \in \mathcal{B}_{\text{box}}$ such that $x \in B$. So $B = \prod_{i \in I} B_i$ where $\forall i \in I B_i \in \mathcal{B}_{d_i} \subseteq \mathcal{T}_{d_i}$ and $\pi_i(x) \in B_i$. Using [theorem: 14.68] there exist a $\delta_i \in \mathbb{R}^+$ such that $\pi_i(x) \in B_{d_i}(\pi_i(x), \delta_i) \subseteq B_i$. Take $\delta = \min(\{\delta_i | i \in I\}) \in \mathbb{R}^+$ then for $z \in B_{d_{\max}}(x, \delta)$ we have

$$\forall i \in I d_i(\pi_i(x), \pi_i(z)) \leq \max(\{d_i(\pi_i(x), \pi_i(x)) | i \in I\}) = d(x, z) < \delta \leq \delta_i$$

so that $\forall i \in I \pi_i(z) \in B_{d_i}(\pi_i(x), \pi_i(z)) \subseteq B_i$. So we have that $z \in \prod_{i \in I} B_i$ proving that $B_{d_{\max}}(x, \delta) \subseteq \prod_{i \in I} B_i = B$. So if we take $B' = B_{d_{\max}}(x, \delta)$ then we have proved that

$$\forall x \in \prod_{i \in I} X_i \text{ and } \forall B \in \mathcal{B}_{\text{box}} \text{ there exist a } B' \in \mathcal{B}_{d_{\max}} \text{ such that } x \in B' \subseteq B \quad (14.27)$$

Combining now [eqs: 14.26, 14.27] with [theorem: 14.40] proves

$$\mathcal{T}_{d_{\max}} = \mathcal{T}_{\text{box}} \quad \square$$

For metric spaces we have a simple test to determine if a set is dense in a topological space.

Theorem 14.85. *Let $\langle X, d \rangle$ a pseudo metric space and $A \subseteq X$ then we have*

A is dense in X [using the metric topology $\mathcal{T}_d$]

$$\Updownarrow$$

$\forall x \in X$ we have $\forall \varepsilon \in \mathbb{R}^+$ that there exist a $a \in A$ such that $d(x, a) < \varepsilon$

Proof.

$\Rightarrow$. Let $x \in X$ then $\forall \varepsilon \in \mathbb{R}^+ x \in B_d(x, \varepsilon)$ so that by [theorem: 14.55] there exist a $a \in A$ such that $a \in B_d(x, \varepsilon)$, hence $d(x, a) < \varepsilon$.

$\Leftarrow$. Let $x \in X$ and let $B \in \mathcal{B}_d$ such that $x \in B$. As $B \in \mathcal{B}_d$ there exists a $y \in X$ and a $\varepsilon \in \mathbb{R}^+$ such that $x \in B = B_d(y, \varepsilon)$. By [theorem: 14.66] there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_d(x, \delta) \subseteq B_d(y, \varepsilon) = B.$$

By the hypotheses there exist a $a \in A$ such that $d(x, a) < \delta$ or $a \in B_d(x, \delta) \subseteq B$. So

$$\forall x \in X \text{ we have } \forall B \in \mathcal{B}_d \text{ with } x \in B \text{ there exist a } a \in A \text{ such that } a \in B$$

which by [theorem: 14.55] implies that A is dense in X . $\square$

14.3 Normed space

The idea of a (pseudo) normed space is that of a real or complex vector space where the concept of a length of a vector is defined.

Notation 14.86. *From now on $\mathbb{K}$ represent either the field $\mathbb{R}$ or the field $\mathbb{C}$*

Definition 14.87. (Pseudo Norm) *A **pseudo normed** space noted as $\langle X, ||| \rangle$ is a real [or complex] vector space X together with a function $|||: X \rightarrow \mathbb{R}$ called the **norm** such that*

1. $\forall x \in X$ we have $0 \leq ||x||$
2. $\forall x \in X, \forall \alpha \in \mathbb{K}$ we have $||\alpha \cdot x|| = |\alpha| \cdot ||x||$
3. $\forall x, y \in X$ we have $||x + y|| \leq ||x|| + ||y||$

Theorem 14.88. *Let $\langle X, ||| \rangle$ be a pseudo normed space then $||0|| = 0$*

Proof. We have $||0|| = ||0 \cdot 0|| = |0| \cdot ||0|| = 0 \cdot ||0|| = 0$ $\square$

Definition 14.89. (Norm) *A pseudo normed space $\langle X, ||| \rangle$ is a normed space if we have also*

$$\forall x \in X \text{ with } ||x|| = 0 \text{ we have } x = 0$$

Theorem 14.90. *Let $\langle X, ||| \rangle$ be a pseudo normed space and $\{x_i\}_{i \in I} \subseteq X$, I finite then*

$$\left\| \sum_{i \in I} x_i \right\| \leq \sum_{i \in I} ||x_i||$$

Proof. We first prove this for $I = \{1, \dots, n\}$, so let

$$\mathcal{S} = \left\{ n \in \mathbb{N}_0 \mid \forall \{x_i\}_{i \in \{1, \dots, n\}} \subseteq X \text{ we have } \left\| \sum_{i \in \{1, \dots, n\}} x_i \right\| \leq \sum_{i \in \{1, \dots, n\}} \|x_i\| \right\}$$

then we have:

$0 \in \mathcal{S}$. As $\{1, \dots, 0\} = \emptyset$ we have

$$\left\| \sum_{i \in \{1, \dots, 0\}} x_i \right\| = \left\| \sum_{i \in \emptyset} x_i \right\| = \|0\| = 0 = \sum_{i \in \emptyset} \|x_i\| = \sum_{i \in \{1, \dots, 0\}} \|x_i\|$$

hence $0 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{x_i\}_{i \in \{1, \dots, n+1\}} \subseteq X$ then we have

$$\begin{aligned} \left\| \sum_{i \in \{1, \dots, n+1\}} x_i \right\| &\stackrel{[\text{theorem: 11.43}]}{=} \left\| \sum_{i \in \{1, \dots, n\}} x_i + \sum_{i \in \{n+1\}} x_i \right\| \\ &\stackrel{[\text{theorem: 11.34}]}{=} \left\| \sum_{i \in \{1, \dots, n\}} x_i + x_{n+1} \right\| \\ &\leq \left\| \sum_{i \in \{1, \dots, n\}} x_i \right\| + \|x_{n+1}\| \\ &\leq_{n \in \mathcal{S}} \sum_{i \in \{1, \dots, n\}} \|x_i\| + \|x_{n+1}\| \\ &\stackrel{[\text{theorem: 11.34}]}{=} \sum_{i \in \{1, \dots, n\}} \|x_i\| + \sum_{i \in \{n+1\}} \|x_{n+1}\| \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{1, \dots, n+1\}} \|x_i\| \end{aligned}$$

proving that $n+1 \in \mathcal{S}$.

Using mathematical induction proves that

$$\forall n \in \mathbb{N}_0 \text{ we have if } \{x_i\}_{i \in \{1, \dots, n\}} \subseteq X \text{ that } \left\| \sum_{i \in \{1, \dots, n\}} x_i \right\| \leq \sum_{i \in \{1, \dots, n\}} \|x_i\| \quad (14.28)$$

Let now I be a finite set and $\{x_i\}_{i \in I} \subseteq X$ then there exist a bijection $\beta: \{1, \dots, n\} \rightarrow I$ so that we have

$$\begin{aligned} \left\| \sum_{i \in I} x_i \right\| &\stackrel{[\text{theorem: 11.36}]}{=} \left\| \sum_{i \in \{1, \dots, n\}} x_{\beta(i)} \right\| \\ &\leq_{[\text{eq: 14.28}]} \sum_{i \in \{1, \dots, n\}} \|x_{\beta(i)}\| \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in I} \|x_i\| \end{aligned}$$

□

Theorem 14.91. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space then $\forall x, y \in X$ we have

$$\|x\| - \|y\| \leq \|x + y\| \quad \text{and} \quad \|x\| - \|y\| \leq \|x - y\|$$

Proof. Let $x, y \in X$ then then we we have

$$\|x\| = \|x + y - y\| \leq \|x + y\| + \|-y\| = \|x + y\| + \|y\| \quad (14.29)$$

$$\|x\| = \|x - y + y\| \leq \|x - y\| + \|y\| \quad (14.30)$$

$$\|y\| = \|x + y - x\| \leq \|x + y\| + \|-x\| = \|x + y\| + \|x\| \quad (14.31)$$

$$\|y\| = \|x - y + x\| \leq \|x - y\| + \|x\| \quad (14.32)$$

For $\|x\| - \|y\|$ we have either

$\|x\| - \|y\| < 0$. Then

$$|\|x\| - \|y\|| = \|y\| - \|x\| \leq \|x + y\| \quad [\text{eq: 14.31}]$$

$$|\|x\| - \|y\|| = \|y\| - \|x\| \leq \|x - y\| \quad [\text{eq: 14.32}]$$

$0 \leq \|x\| - \|y\|$. Then

$$|\|x\| - \|y\|| = \|x\| - \|y\| \leq \|x + y\| \quad [\text{eq: 14.29}]$$

$$|\|x\| - \|y\|| = \|x\| - \|y\| \leq \|x - y\| \quad [\text{eq: 14.30}]$$

□

Given a normed space we can define a metric and thus also a topology.

Theorem 14.92. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space and define

$$d_{\|\cdot\|}: X \times X \rightarrow \mathbb{R} \text{ by } d_{\|\cdot\|}(x, y) = \|x - y\|$$

then

$$\langle X, d_{\|\cdot\|} \rangle \text{ is a pseudo metric space}$$

Further if $\langle X, \|\cdot\| \rangle$ is a normed space then $\langle X, d_{\|\cdot\|} \rangle$ is a metric space.

Proof. If $\langle X, \|\cdot\| \rangle$ is a pseudo normed space then we have for $x, y, z \in X$

1. $d_{\|\cdot\|}(x, y) = \|x - y\| \geq 0$
2. $d_{\|\cdot\|}(x, y) = \|x - y\| = \|(-1) \cdot (y - x)\| = |-1| \cdot \|y - x\| = d_{\|\cdot\|}(y, x)$
3. $d_{\|\cdot\|}(x, z) = \|x - z\| = \|x - y + y - z\| \leq \|x - y\| + \|y - z\| = d_{\|\cdot\|}(x, y) + d_{\|\cdot\|}(y, z)$

If in addition if $\langle X, \|\cdot\| \rangle$ is a normed space then if $d_{\|\cdot\|}(x, y) = 0$ we have $\|x - y\| = 0$ or as $\|\cdot\|$ is a norm that $x - y = 0 \Rightarrow x = y$. Hence $d_{\|\cdot\|}$ is a metric. □

As a pseudo metric defines a topology and a pseudo norm defines a pseudo metric a pseudo norm can be used to define a topology.

Definition 14.93. Let $\langle X, \|\cdot\| \rangle$ be pseudo normed space, $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then the open ball $B_{\|\cdot\|}(x, \varepsilon)$ in the pseudo normed space is defined by

$$B_{\|\cdot\|}(x, \varepsilon) = \{y \in X \mid \|x - y\| < \varepsilon\}$$

the set of all open balls in the pseudo normed space is noted by $\mathcal{B}_{\|\cdot\|}$ hence

$$\mathcal{B}_{\|\cdot\|} = \{B_{\|\cdot\|}(x, \varepsilon) \mid x \in X \wedge \varepsilon \in \mathbb{R}^+\}$$

Note 14.94. As $d_{\|\cdot\|}(x, y) = \|x - y\|$ we have that

$$B_{d_{\|\cdot\|}}(x, \varepsilon) = \{y \in X \mid d_{\|\cdot\|}(x, y) < \varepsilon\} = \{y \in X \mid \|x - y\| < \varepsilon\} = B_{\|\cdot\|}(x, \varepsilon)$$

and further

$$\mathcal{B}_{d_{\|\cdot\|}} = \{B_{d_{\|\cdot\|}}(x, \varepsilon) \mid x \in X \wedge \varepsilon \in \mathbb{R}^+\} = \{B_{\|\cdot\|}(x, \varepsilon) \mid x \in X \wedge \varepsilon \in \mathbb{R}^+\} = \mathcal{B}_{\|\cdot\|}$$

Theorem 14.95. Let $\langle X, \|\cdot\|_X \rangle$ be a pseudo normed space then we have:

1. If $\varepsilon \in \mathbb{R}^+$ then $B_{\|\cdot\|}(0, \varepsilon) = \varepsilon \cdot B_{\|\cdot\|}(0, 1)$
2. If $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then $B_{\|\cdot\|}(x, \varepsilon) = x + B_{\|\cdot\|}(0, \varepsilon)$

Proof.

1. If $x \in B_{\|\cdot\|}(0, \varepsilon)$ then $\|x\| < \varepsilon$, define $x' = \frac{1}{\varepsilon} \cdot x$ so that $x = \varepsilon \cdot x'$ and

$$\|x'\| = \left\| \frac{1}{\varepsilon} \cdot x \right\| = \frac{1}{\varepsilon} \|x\| < \frac{1}{\varepsilon} \cdot \varepsilon = 1$$

proving that $x \in \varepsilon \cdot B_{\|\cdot\|}(0, 1)$. Further if $x \in \varepsilon \cdot B_{\|\cdot\|}(0, 1)$ then $x = \varepsilon \cdot x'$ where $x' \in B_{\|\cdot\|}(0, 1)$ so that $\|x'\| < 1$ or $\|x\| = \varepsilon \cdot \|x'\| < \varepsilon \cdot 1 = \varepsilon$ proving that $x \in B_{\|\cdot\|}(0, \varepsilon)$. Hence we conclude that

$$B_{\|\cdot\|}(0, \varepsilon) = \varepsilon \cdot B_{\|\cdot\|}(0, 1)$$

2. $y \in B_{\|\cdot\|}(x, \varepsilon) \Leftrightarrow \|y - x\| < \varepsilon \Leftrightarrow y - x \in B_{\|\cdot\|}(0, \varepsilon) \Leftrightarrow y = x + B_{\|\cdot\|}(0, \varepsilon)$ □

Definition 14.96. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space then the norm topology $\mathcal{T}_{\|\cdot\|}$ on X is defined by

$$\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{d_{\|\cdot\|}}$$

where

$$d_{\|\cdot\|}: X \times X \rightarrow \mathbb{R} \text{ is defined by } d_{\|\cdot\|}(x, y) = \|x - y\|$$

So

$$\mathcal{T}_{\|\cdot\|} = \mathcal{T}[\mathcal{B}_{d_{\|\cdot\|}}]$$

where

$$\mathcal{B}_{d_{\|\cdot\|}} = \{B_{d_{\|\cdot\|}}(x, \varepsilon) | x \in X \wedge \varepsilon \in \mathbb{R}^+\} = \{B_{\|\cdot\|}(x, \varepsilon) | x \in X \wedge \varepsilon \in \mathbb{R}^+\} \stackrel{\text{def}}{=} \mathcal{B}_{\|\cdot\|}$$

Using the above definition we have the equivalent version of [theorem: 14.68].

Theorem 14.97. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space then we have

$$U \in \mathcal{T}_{\|\cdot\|} \Leftrightarrow \forall x \in U \exists \varepsilon \in \mathbb{R}^+ \text{ such that } x \in B_{\|\cdot\|}(x, \varepsilon) \subseteq U$$

Proof. This follows from [theorem: 14.68] and the fact that $\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{d_{\|\cdot\|}}$ and $B_{\|\cdot\|}(x, \varepsilon) = B_{d_{\|\cdot\|}}(x, \varepsilon)$. □

Giving a finite family of normed spaces we can create a norm for the product of these spaces.

Theorem 14.98. Let $\langle X_i, \|\cdot\|_i \rangle_{i \in I}$, I finite be a finite family of normed spaces then

$$\|\cdot\|_{\max}: \prod_{i \in I} X_i \rightarrow \mathbb{R} \text{ defined by } \|x\|_{\max} = \max(\{\|\pi_i(x)\|_i | i \in I\})$$

is a norm called the **maximum norm**. Further the generated topology $\mathcal{T}_{\|\cdot\|_{\max}}$ is the same as the product topology or

$$\mathcal{T}_{\|\cdot\|_{\max}} = \mathcal{T}_{\text{product}}$$

Proof. First we prove that $\|\cdot\|_{\max}$ is a norm.

1. Let $x \in \prod_{i \in I} X_i$ then as $\forall i \in I \ 0 \leq \|x_i\|_i$ we have $0 \leq \max(\{\|x_i\|_i | i \in I\}) = \|x\|_{\max}$.
2. Let $x \in \prod_{i \in I} X_i$ and $\alpha \in \mathbb{K}$ then

$$\begin{aligned} \|\alpha \cdot x\|_{\max} &= \max(\{\|\pi_i(\alpha \cdot x)\|_i | i \in I\}) \\ &\stackrel{[\text{example: 11.175}]}{=} \max(\{\|\alpha \cdot \pi_i(x)\|_i | i \in I\}) \\ &= \max(\{|\alpha| \cdot \|\pi_i(x)\|_i | i \in I\}) \\ &\stackrel{[\text{theorem: 10.27}]}{=} |\alpha| \cdot \max(\{\|\pi_i(x)\|_i | i \in I\}) \\ &= |\alpha| \cdot \|x\|_{\max} \end{aligned}$$

3. Let $x, y \in \prod_{i \in I} X_i$ then

$$\begin{aligned}
 \|x + y\|_{\max} &= \max(\{\|\pi_i(x + y)\|_i | i \in I\}) \\
 &\stackrel{[\text{example: 11.175}]}{=} \max(\{\|\pi_i(x) + \pi_i(y)\|_i | i \in I\}) \\
 &\leq [\text{theorem: 3.69}] \wedge \|\cdot\|_i \text{ is a norm } \max(\{\|\pi_i(x)\|_i + \|\pi_i(y)\|_i | i \in I\}) \\
 &\stackrel{[\text{theorem: 10.25}]}{=} \max(\{\|\pi_i(x)\|_i | i \in I\}) + \max(\{\|\pi_i(y)\|_i | i \in I\}) \\
 &= \|x\|_{\max} + \|y\|_{\max}
 \end{aligned}$$

4. If $\|x\|_{\max} = 0$ then $\forall i \in I$ we have $0 \leq \|\pi_i(x)\|_i \leq \|x\|_{\max} = 0$, hence $\forall i \in I$ $\pi_i(x) = 0$ so that $x = 0$.

Finally $\forall i \in I$ and $x, y \in X$ we have that

$$\begin{aligned}
 d_{\|\cdot\|_{\max}}(x, y) &= \|x - y\|_{\max} \\
 &= \max(\{\|\pi_i(x - y)\|_i | i \in I\}) \\
 &\stackrel{[\text{example: 11.175}]}{=} \max(\{\|\pi_i(x) - \pi_i(y)\|_i | i \in I\}) \\
 &= \max(\{d_{\|\cdot\|_i}(x, y) | i \in I\}) \\
 &\stackrel{[\text{theorem: 14.83}]}{=} d_{\max}(x, y)
 \end{aligned}$$

so that we have by [theorem: 14.83] that

$$\mathcal{T}_{d_{\|\cdot\|_{\max}}} = \mathcal{T}_{d_{\max}} = \mathcal{T}_{\text{box}} = \mathcal{T}_{\text{product}}$$

□

Example 14.99. $\langle \mathbb{R}, \|\cdot\| \rangle$ is a real normed space where $\|\cdot\|$ is defined by

$$\|\cdot\|: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \text{ by } |x| = \begin{cases} x & \text{if } 0 \leq x \\ -x & \text{if } x < 0 \end{cases}$$

Note 14.100. If $0 \leq x$ then $x = |x| \Rightarrow x < |x|$ and if $x < 0$ then $x < 0 \leq -|x|$ so $\forall x \in X$ we have $x < |x|$.

Proof. We have:

1. If $x \in \mathbb{R}$ then we have either:

$0 \leq x$. Then $0 \leq x = |x|$

$x < 0$. Then $0 < -x = |x|$

so in all cases we have $0 \leq |x|$

2. Let $\alpha \in \mathbb{R}$ and $x \in \mathbb{R}$ then we have either:

$0 \leq \alpha \wedge 0 \leq x$. Then $0 \leq \alpha \cdot x$ so that $|\alpha \cdot x| = \alpha \cdot x = |\alpha| \cdot |x|$.

$\alpha < 0 \wedge 0 \leq x$. Then $\alpha \cdot x \leq 0$ so that $|\alpha \cdot x| = -(\alpha \cdot x) = (-\alpha) \cdot x = |\alpha| \cdot |x|$.

$0 \leq \alpha \wedge x < 0$. Then $\alpha \cdot x \leq 0$ so that $|\alpha \cdot x| = -(\alpha \cdot x) = \alpha \cdot (-x) = |\alpha| \cdot |x|$.

$\alpha < 0 \wedge x < 0$. Then $0 < \alpha \cdot x$ so that $|\alpha \cdot x| = (-\alpha) \cdot (-x) = |\alpha| \cdot |x|$.

so in all cases we have

$$|\alpha \cdot x| = |\alpha| \cdot |x|$$

3. Let $x, y \in \mathbb{R}$ then we have for $x + y$ either:

$0 \leq x + y$. Then $|x + y| = x + y \leq |x| + |y|$

$x + y < 0$. Then $|x + y| = -(x + y) = (-x) + (-y) \leq |-x| + |-y| \stackrel{(2)}{=} |-1| \cdot |x| + |-1| \cdot |y| = |x| + |y|$,

so in all cases we have

$$|x + y| \leq |x| + |y|$$

4. If $|x| = 0$ then we have for $x \in \mathbb{R}$ either:

$0 \leq x$. Then $x = |x| = 0$

$x < 0$. Then $-x = |x| = 0 \Rightarrow x = 0$ □

Example 14.101. Let $n \in \mathbb{N}$ then $\langle \mathbb{R}^n, |||_n \rangle$ where $|||_n = \max(\{|x_i| | i \in \{1, \dots, n\}\})$ is a normed space with $\mathcal{T}_{|||_n} = \mathcal{T}_{\text{product}}$. This topology is called the **standard** or **canonical** topology on $\mathbb{R}^n$.

Proof. Take $\langle X_i, |||_i \rangle_{i \in \{1, \dots, n\}}$ by $X_i = \mathbb{R}$ and $|||_i = | \cdot |$ then $\prod_{i \in \{1, \dots, n\}} X_i = \mathbb{R}^n$ and $|||_{\max} = |||_n$ so that by [theorem: 14.98] $\langle \mathbb{R}^n, |||_n \rangle$ is a normed space with $\mathcal{T}_{|||_n} = \mathcal{T}_{\text{product}}$ □

Example 14.102. $\langle \mathbb{C}, ||| \rangle$ is a complex normed space where $|||$ is the complex norm [see definition: 10.84]

Proof. This was proved in [theorem: 10.85] □

Example 14.103. $\langle \mathbb{C}, |||_{\mathbb{C}} \rangle$ is a normed space where

$$|||_{\mathbb{C}}: \mathbb{C} \rightarrow \mathbb{R} \text{ is defined by } |||_{\mathbb{C}} = \max(|\text{Re}(z)|, |\text{Im}(z)|)$$

Proof. We have:

1. If $z \in \mathbb{C}$ then $0 \leq |\text{Re}(z)|$ and $0 \leq |\text{Im}(z)|$ so that $0 \leq \max(|\text{Re}(z)|, |\text{Im}(z)|) = |||_{\mathbb{C}}$
2. Let $\alpha \in \mathbb{R}$ and $z \in \mathbb{C}$ then

$$|\text{Im}(\alpha \cdot z)| \stackrel{[\text{theorem: 10.80}]}{=} |\alpha \cdot \text{Im}(z)| \stackrel{[\text{example: 14.99}]}{=} |\alpha| \cdot |\text{Im}(z)|$$

$$|\text{Re}(\alpha \cdot z)| \stackrel{[\text{theorem: 10.80}]}{=} |\alpha \cdot \text{Re}(z)| \stackrel{[\text{example: 14.99}]}{=} |\alpha| \cdot |\text{Re}(z)|$$

so that

$$|||\alpha \cdot z|||_{\mathbb{C}} = \max(|\text{Re}(\alpha \cdot z)|, |\text{Im}(\alpha \cdot z)|) = |\alpha| \cdot \max(|\text{Re}(z)|, |\text{Im}(z)|) = |\alpha| \cdot |||_{\mathbb{C}}$$

3. Let $z, w \in \mathbb{C}$ then we have

$$\begin{aligned} |\text{Re}(z+w)| &\stackrel{[\text{theorem: 10.80}]}{=} |\text{Re}(z) + \text{Re}(w)| \\ &\leq |\text{Re}(z)| + |\text{Re}(w)| \\ &\leq \max(|\text{Re}(z)|, |\text{Im}(z)|) + \max(|\text{Re}(w)|, |\text{Im}(w)|) \\ &= |||_{\mathbb{C}} + |||_{\mathbb{C}} \\ |\text{Im}(z+w)| &\stackrel{[\text{theorem: 10.80}]}{=} |\text{Im}(z) + \text{Im}(w)| \\ &\leq |\text{Im}(z)| + |\text{Im}(w)| \\ &\leq \max(|\text{Re}(z)|, |\text{Im}(z)|) + \max(|\text{Re}(w)|, |\text{Im}(w)|) \\ &= |||_{\mathbb{C}} + |||_{\mathbb{C}} \end{aligned}$$

so that

$$\begin{aligned} |||z+w|||_{\mathbb{C}} &= \max(|\text{Re}(z+w)|, |\text{Im}(z+w)|) \\ &\leq |||_{\mathbb{C}} + |||_{\mathbb{C}} \end{aligned}$$

4. If $|z|_2 = 0$ then

$$\begin{aligned} 0 &\leq |\text{Re}(z)| \leq \max(|\text{Re}(z)|, |\text{Im}(z)|) = |||_{\mathbb{C}} = 0 \\ 0 &\leq |\text{Im}(z)| \leq \max(|\text{Re}(z)|, |\text{Im}(z)|) = |||_{\mathbb{C}} = 0 \end{aligned}$$

so that $\text{Re}(z) = \text{Im}(z) = 0$. Hence $z = \text{Re}(z) + i \cdot \text{Im}(z) = 0$. □

For $\langle \mathbb{R}, ||| \rangle$ we have the following equivalences for the basis of the normed topology based on the absolute value norm.

Theorem 14.104. *Given the normed space $\langle \mathbb{R}, || \rangle$ we have for the basis $\mathcal{B}_{||}$ of the normed topology $\mathcal{T}_{||}$ that*

$$\begin{aligned}\mathcal{B}_{||} &\stackrel{\text{def}}{=} \{B_{||}(x, \varepsilon) | x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\} \\ &= \{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\} \\ &= \{]a, b[| a, b \in \mathbb{R} \wedge a < b\}\end{aligned}$$

Proof. First we have for $x \in \mathbb{R}$ and $\varepsilon \in \mathbb{R}^+$ that

$$\begin{aligned}y \in B_{||}(x, \varepsilon) &\Leftrightarrow |x - y| < \varepsilon \\ &\Leftrightarrow x - y < \varepsilon \wedge y - x < \varepsilon \\ &\Leftrightarrow x - \varepsilon < y \wedge y < x + \varepsilon \\ &\Leftrightarrow y \in]x - \varepsilon, x + \varepsilon[\end{aligned}$$

proving that

$$B_{||}(x, \varepsilon) =]x - \varepsilon, x + \varepsilon[$$

From this it follows that

$$\{B_{||}(x, \varepsilon) | x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\} = \{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\}$$

If $B \in \{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\}$ then there exist a $x \in \mathbb{R}$ and a $\varepsilon \in \mathbb{R}^+$ such that $B =]x - \varepsilon, x + \varepsilon[$, which, as $x - \varepsilon, x + \varepsilon \in \mathbb{R} \wedge x - \varepsilon < x + \varepsilon$, proves that $B \in \{]a, b[| a, b \in \mathbb{R} \wedge a < b\}$. Hence

$$\{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\} \subseteq \{]a, b[| a, b \in \mathbb{R} \wedge a < b\} \quad (14.33)$$

If $B \in \{]a, b[| a, b \in \mathbb{R} \wedge a < b\}$ then there exist a $a, b \in \mathbb{R}$ with $a < b$ such that $B =]a, b[$. Take then $\varepsilon = (b - a)/2$ and $x = a + \varepsilon$ then we have

$$\begin{aligned}y \in]x - \varepsilon, x + \varepsilon[&\Leftrightarrow x - \varepsilon < y \wedge y < x + \varepsilon \\ &\Leftrightarrow (a + \varepsilon) - \varepsilon < y \wedge y < (a + \varepsilon) + \varepsilon \\ &\Leftrightarrow a < y \wedge y < a + 2 \cdot \varepsilon \\ &\Leftrightarrow a < y \wedge y < a + b - a \\ &\Leftrightarrow a < y \wedge y < b \\ &\Leftrightarrow y \in]a, b[\end{aligned}$$

proving that $B =]x - \varepsilon, x + \varepsilon[\in \{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\}$. Hence $\{]a, b[| a, b \in \mathbb{R} \wedge a < b\} \subseteq \{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\}$, combining this with [eq: 14.33] results in

$$\{]x - \varepsilon, x + \varepsilon[| x \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+\} = \{]a, b[| a, b \in \mathbb{R} \wedge a < b\} \quad \square$$

Theorem 14.105. *Let $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ be the topological space generated by the norm $||$, $a, b \in \mathbb{R}$ then the following intervals are open in $\mathcal{T}_{||}$:*

$$\begin{aligned}&]a, b[\\ &]-\infty, a[\\ &]a, \infty[\\ &]-\infty, \infty[\end{aligned}$$

the following intervals are closed:

$$\begin{aligned}&[a, b] \\ &[a, \infty[\\ &]-\infty, a] \\ &]-\infty, \infty[\end{aligned}$$

and the following intervals are neither open or closed:

$$\begin{aligned}&[a, b[\quad \text{where } a < b \\ &]a, b] \quad \text{where } a < b\end{aligned}$$

Proof. Let $a, b \in \mathbb{R}$ then we have:

$]a, b[\in \mathcal{T}_{||}$. For a, b we have either:

$a \leq b$. Then by [theorem: 14.104] $]a, b[\in \mathcal{B}_{||} \subseteq \mathcal{T}_{||}$

$b \leq a$. Then $]a, b[= \emptyset \in \mathcal{T}_{||}$

$] -\infty, a[\in \mathcal{T}_{||}$. If $x \in] -\infty, a[$ then for $]x - 1, a[\in \mathcal{B}_{||}$ we have $x \in]x - 1, a[\subseteq] -\infty, a[$ proving by [theorem: 14.36] that $] -\infty, a[\in \mathcal{T}_{||}$.

$]a, \infty[$. If $x \in]a, \infty[$ then for $]a, x + 1[\in \mathcal{B}_{||}$ we have $x \in]a, x + 1[\subseteq]a, \infty[$ proving by [theorem: 14.36] that $]a, \infty[\in \mathcal{T}_{||}$.

$] -\infty, \infty[\in \mathcal{T}_{||}$. $] -\infty, \infty[$ is open and closed.

$[a, b]$ is closed. We have

$$\begin{aligned} x \in \mathbb{R} \setminus [a, b] &\Leftrightarrow x \in \mathbb{R} \wedge \neg(a \leq x \wedge x \leq b) \\ &\Leftrightarrow x \in \mathbb{R} \wedge (x < a \vee b < x) \\ &\Leftrightarrow (x \in \mathbb{R} \wedge x < a) \vee (x \in \mathbb{R} \wedge b < x) \\ &\Leftrightarrow x \in] -\infty, a[\cup]b, \infty[\end{aligned}$$

proving that $\mathbb{R} \setminus [a, b] =] -\infty, a[\cup]b, \infty[\in \mathcal{T}_{||}$ hence $[a, b]$ is closed.

$[a, \infty[$ is closed. We have

$$\begin{aligned} x \in \mathbb{R} \setminus [a, \infty[&\Leftrightarrow x \in \mathbb{R} \wedge \neg(a \leq x) \\ &\Leftrightarrow x \in \mathbb{R} \wedge x < a \\ &\Leftrightarrow x \in] -\infty, a[\end{aligned}$$

proving that $\mathbb{R} \setminus [a, \infty[=] -\infty, a[\in \mathcal{T}_{||}$ so that $[a, \infty[$ is closed.

$] -\infty, a]$ is closed. We have

$$\begin{aligned} x \in \mathbb{R} \setminus] -\infty, a] &\Leftrightarrow x \in \mathbb{R} \wedge \neg(x \leq a) \\ &\Leftrightarrow x \in \mathbb{R} \wedge a < x \\ &\Leftrightarrow x \in]a, \infty[\end{aligned}$$

proving that $\mathbb{R} \setminus] -\infty, a] =]a, \infty[\in \mathcal{T}_{||}$ so that $] -\infty, a]$ is closed.

Assume that $a < b$ then we have

$[a, b[$ is neither open/closed. Assume that $[a, b[$ is open then as $a \leq a < b$ we have $a \in [a, b[$, so by [theorem: 14.36] there exist a $]x, y[\in \mathcal{B}_{||} \stackrel{=}{=}_{[\text{theorem: 14.104}]} \{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $a \in]x, y[\subseteq [a, b[$. Hence $x < a < y$, using [theorem: 10.32] there exist a $z \in \mathbb{R}$ such that $x < z < a < y \Rightarrow z \in]x, y[\subseteq [a, b[$ hence $a \leq z$ contradicting $z < a$, hence

$[a, b[$ is not open

Assume that $\mathbb{R} \setminus [a, b[$ is open then as $b \in \mathbb{R} \setminus [a, b[$ we have by [theorem: 14.36] that there exist a $]x, y[\in \mathcal{B}_{||} \stackrel{=}{=}_{[\text{theorem: 14.104}]} \{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $b \in]x, y[\subseteq \mathbb{R} \setminus [a, b[$ so that $x < b < y$. As also $a < b$ we have that $x \leq \max(a, x) < b < y$, using [theorem: 10.32] there exists $z \in \mathbb{R}$ such that

$$x \leq \max(a, x) < z < b < y \tag{14.34}$$

so that $z \in]x, y[\subseteq \mathbb{R} \setminus [a, b[$. Hence we have $z < a \vee b \leq z$, as $a \leq \max(a, x) < z$ we can't have $z < a$ so we must have that $b \leq z$ contradicting $z < b$ [see eq: 14.34] hence the assumption is wrong and $\mathbb{R} \setminus [a, b[$ is not open proving that

$[a, b[$ is not closed

$]a, b]$ is neither open/closed. Assume that $]a, b]$ is open then as $a < b \leq b$ we have $b \in]a, b[$, so by [theorem: 14.36] there exist a $]x, y[\in \mathcal{B}_{||} \stackrel{=}{=}_{[\text{theorem: 14.104}]} \{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $b \in]x, y[\subseteq]a, b]$. Hence $x < b < y$, using [theorem: 10.32] there exist a $z \in \mathbb{R}$ such that $x < b < z < y \Rightarrow z \in]x, y[\subseteq]a, b]$ giving $z \leq b$ contradicting $b < z$, hence

$]a, b[$ is not open

Assume that $\mathbb{R} \setminus]a, b]$ is open then as $a \in \mathbb{R} \setminus]a, b]$ we have by [theorem: 14.36] that there exist a $]x, y[\in \mathcal{B}_{||} \stackrel{=}{=}_{[\text{theorem: 14.104}]} \{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $a \in]x, y[\subseteq \mathbb{R} \setminus]a, b]$ so that $x < a < y$. As also $a < b$ we have that $x < a < \min(b, y) \leq y$, using [theorem: 10.32] there exists a $z \in \mathbb{R}$ such that

$$x < a < z < \min(b, y) \leq y \quad (14.35)$$

so that $z \in]x, y[\subseteq \mathbb{R} \setminus]a, b]$ hence we have $z \leq a \vee b < z$, as $z < \min(b, y) < b$ we can't have $b < z$ so we must have $z \leq a$ contradicting $a < z$ [see eq: 14.35]. So the assumption is wrong and $\mathbb{R} \setminus]a, b]$ is not open proving that

$]a, b]$ is not closed □

Corollary 14.106. Let $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ be the topological space generated by the norm $||$ and I a **non empty open generalized interval** then I is one of the following sets

$$\begin{aligned}]a, b[& \quad a, b \in \mathbb{R} \text{ where } a < b \text{ [otherwise } I = \emptyset] \\]-\infty, a[& \quad a \in \mathbb{R} \\]a, \infty[& \quad a \in \mathbb{R} \\]-\infty, \infty[& \end{aligned}$$

Proof. Let I be a **non empty** generalized interval then by [theorem: 3.143] I is of the following forms, where $a, b \in \mathbb{R}$

$]a, b[\wedge a < b$. Then by the previous theorem [theorem: 14.105]

$I =]a, b[$ is open

$]a, b] \wedge a \leq b$. Then by the previous theorem [theorem: 14.105]

$I =]a, b[$ is not open

$[a, b[\wedge a \leq b$. Then by the previous theorem [theorem: 14.105]

$I = [a, b[$ is not open

$[a, b] \wedge a \leq b$. As $[a, b] = I \neq \emptyset$ we must have $a \leq b$. Assume that $[a, b]$ is open then as $a \leq a \leq b$ we have $a \in [a, b]$, so by [theorem: 14.36] there exist a $]x, y[\in \mathcal{B}_{||} \stackrel{=}{=}_{[\text{theorem: 14.104}]} \{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $a \in]x, y[\subseteq [a, b]$, hence $x < a < y$ and using [theorem: 10.32] there exists a $z \in \mathbb{R}$ such that $x < z < a < y$. So $z \in]x, y[\subseteq [a, b]$ resulting in $a \leq z$ contradicting $z < a$. Hence the assumption is wrong and we have that

$I = [a, b]$ is not open

$[a, \infty[$. Assume that $[a, \infty[$ is open then as $a \in [a, \infty[$ there exist by [theorem: 14.36] a $]x, y[\in \mathcal{B}_{||} \stackrel{=}{=}_{[\text{theorem: 14.104}]} \{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $a \in]x, y[\subseteq [a, \infty[$, hence $x < a < y$. Using [theorem: 10.32] there exist a $z \in \mathbb{R}$ such that $x < z < a < y$ hence $z \in]x, y[\subseteq [a, \infty[$ giving $a \leq z$ which contradict $z < a$, hence the assumption is wrong and

$I = [a, \infty[$ is not open

$]a, \infty[$. Then by the previous theorem [theorem: 14.105]

$I =]a, \infty[$ is open

$] -\infty, a]$. Assume that $] -\infty, a]$ is open then as $a \in] -\infty, a]$ there exist by [theorem: 14.36] a $]x, y[\in \mathcal{B}_{||} \stackrel{[\text{theorem: 14.104}]}{=} \{x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $a \in]x, y[\subseteq] -\infty, a]$, hence $x < a < y$. Using [theorem: 10.32] there exist a $z \in \mathbb{R}$ such that $x < a < z < y$ hence $z \in]x, y[\subseteq] -\infty, a]$ giving $z \leq a$ which contradict $a < z$, hence the assumption is wrong and

$$I =] -\infty, a] \text{ is not open}$$

$] -\infty, a[$. Then by the previous theorem [theorem: 14.105]

$$I =] -\infty, a[\text{ is open}$$

$] -\infty, \infty[$. Then by the previous theorem [theorem: 14.105]

$$I =] -\infty, \infty[\text{ is open} \quad \square$$

Corollary 14.107. Let $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ be the topological space generated by the norm $||$ and I a **non empty closed generalized interval** then I is one of the following sets.

$$\begin{aligned} &[a, b] \text{ where } a, b \in \mathbb{R} \text{ with } a \leq b \text{ [otherwise } I = \emptyset] \\ &[a, \infty[\text{ where } a \in \mathbb{R} \\ &]-\infty, a] \text{ where } a \in \mathbb{R} \end{aligned}$$

Proof. Let I be a **non empty** generalized interval then by [theorem: 3.143] I is of the following forms, where $a, b \in \mathbb{R}$

$]a, b[\wedge a < b$. Assume that $]a, b[$ is closed then $\mathbb{R} \setminus]a, b[$ is open. As $a \notin]a, b[$ hence $a \in \mathbb{R} \setminus]a, b[$, we have by [theorems: 14.36, 14.104] that there exists $r, s \in \mathbb{R}$ such that $a \in]r, s[\subseteq \mathbb{R} \setminus]a, b[$ so that $r < a < s$ and thus $a < \min(s, b)$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $r < a < x < \min(s, b)$ so that $r < x < s \wedge a < x < b$ hence $x \in]r, s[\cap]a, b[$ contradicting $]r, s[\subseteq \mathbb{R} \setminus]a, b[$. Hence the assumption is wrong and $]a, b[$ is not closed.

$]a, b] \wedge a \leq b$. Then by the previous theorem [theorem: 14.105]

$$I =]a, b[\text{ is not closed}$$

$[a, b[\wedge a \leq b$. Then by the previous theorem [theorem: 14.105]

$$I = [a, b[\text{ is not closed}$$

$[a, b] \wedge a \leq b$. Then by the previous theorem [theorem: 14.105] $I = [a, b]$ is closed.

$[a, \infty[$. Then by the previous theorem [theorem: 14.105] $I = [a, \infty[$ is closed.

$]a, \infty[$. Assume that $]a, \infty[$ is closed then $\mathbb{R} \setminus]a, \infty[$ is open. As $a \notin]a, \infty[$ we have $a \in \mathbb{R} \setminus]a, \infty[$ so that by [theorems: 14.36, 14.104] there exists $r, s \in \mathbb{R}$ with $a \in]r, s[\subseteq \mathbb{R} \setminus]a, \infty[$. Hence $r < a < s$ so that by [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $r < a < x < s$, hence $x \in]r, s[\cap]a, \infty[$ contradicting $]r, s[\subseteq \mathbb{R} \setminus]a, \infty[$. So the assumption is wrong and $]a, \infty[$ is not closed.

$] -\infty, a]$. Then by the previous theorem [theorem: 14.105] $I =] -\infty, a]$ is closed.

$] -\infty, a[$. Assume that $] -\infty, a[$ is closed then $\mathbb{R} \setminus] -\infty, a[$ is open. As $a \notin] -\infty, a[$ we have $a \in \mathbb{R} \setminus] -\infty, a[$ so that by [theorems: 14.36, 14.104] there exists $r, s \in \mathbb{R}$ with $a \in]r, s[\subseteq \mathbb{R} \setminus] -\infty, a[$. Hence $r < a < s$ so that by [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $r < x < a < s$, hence $x \in]r, s[\cap] -\infty, a[$ contradicting $]r, s[\subseteq \mathbb{R} \setminus] -\infty, a[$. So the assumption is wrong and $] -\infty, a[$ is not closed.

$] -\infty, \infty[$. Then by the previous theorem [theorem: 14.105]

$$I =] -\infty, \infty[\text{ is closed} \quad \square$$

Note 14.108. The condition that $I \neq \emptyset$ in the above theorem is essential because if $I = \emptyset$ then I is open and closed and either of the form:

$$\begin{aligned}]a, b[& \text{ where } b \leq a \\ [a, b] & \text{ where } b < a \\ [a, b[& \text{ where } b < a \\]a, b] & \text{ where } b < a \end{aligned}$$

so that for the empty interval there are more forms then stated in the previous two corollaries. We prove now that every open set in $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ is a countable union of open intervals.

Theorem 14.109. *Given $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ the set of real numbers endowed with the topology generated by the absolute value norm and $U \in \mathcal{T}_{||}$ a open set then there exists a $\mathcal{U} \subseteq \mathcal{T}_{||}$ where $\mathcal{U}$ is countable and $\forall I \in \mathcal{U}$ we have that I is a non empty open generalized interval such that*

$$U = \bigcup_{I \in \mathcal{U}} I$$

and

$$\forall I, J \in \mathcal{U} \text{ with } I \neq J \text{ we have } I \cap J = \emptyset$$

Proof. Define the relation $\sim \in U \times U$ by $x \sim y$ iff $] \min(x, y), \max(x, y)[\subseteq U$ then we have

reflexivity. $\forall x \in U] \min(x, x), \max(x, x)[=]x, x[= \emptyset \subseteq U$ so that $x \sim x$

symmetry. If $x \sim y$ then $] \min(x, y), \max(x, y)[\subseteq U$ so that

$$] \min(y, x), \max(y, x)[=] \min(x, y), \max(x, y)[\subseteq U$$

giving $y \sim x$.

transitivity. If $x \sim y$ and $y \sim z$ then we have

$$] \min(x, y), \max(x, y)[\subseteq U \text{ and }] \min(y, z), \max(y, z)[\subseteq U \quad (14.36)$$

now for $x, y, z \in U$ we have either:

$x \leq y \leq z$. Then by [eq: 14.36] we have

$$]x, y[=] \min(x, y), \max(x, y)[\subseteq U \wedge]y, z[=] \min(y, z), \max(y, z)[\subseteq U$$

and as $y \in U$

$$] \min(x, z), \max(x, z)[=]x, z[=]x, y[\bigcup \{y\} \bigcup]y, z[\subseteq U$$

proving that $x \sim z$.

$x \leq z \leq y$. Then by [eq: 14.36] we have

$$]x, y[=] \min(x, y), \max(x, y)[\subseteq U \wedge]z, y[=] \min(y, z), \max(y, z)[\subseteq U$$

$$] \min(x, z), \max(x, z)[=]x, z[\subseteq]x, y[\subseteq U$$

proving that $x \sim z$.

$y \leq x \leq z$. Then by [eq: 14.36] we have

$$]y, x[=] \min(x, y), \max(x, y)[\subseteq U \wedge]y, z[=] \min(y, z), \max(y, z)[\subseteq U$$

$$] \min(x, z), \max(x, z)[=]x, z[\subseteq]y, z[\subseteq U$$

proving that $x \sim z$.

$y \leq z \leq x$. Then by [eq: 14.36] we have

$$]y, x[=] \min(x, y), \max(x, y)[\subseteq U \wedge]y, z[=] \min(y, z), \max(y, z)[\subseteq U$$

$$] \min(x, z), \max(x, z)[=]z, x[\subseteq]y, x[\subseteq U$$

proving that $x \sim z$.

$z \leq x \leq y$. Then by [eq: 14.36] we have

$$\begin{aligned}]x, y[&=]\min(x, y), \max(x, y)[\subseteq U \wedge]z, y[=]\min(y, z), \max(y, z)[\subseteq U \\ &] \min(x, z), \max(x, z)[=]z, x[\subseteq]z, y[\subseteq U \end{aligned}$$

proving that $x \sim z$.

$z \leq y \leq x$. Then by [eq: 14.36] we have

$$]y, x[=]\min(x, y), \max(x, y)[\subseteq U \wedge]z, y[=]\min(y, z), \max(y, z)[\subseteq U$$

and as $y \in U$ we have that

$$] \min(x, z), \max(x, z)[=]z, x[=]z, y[\cup \{y\} \cup]y, x[\subseteq U$$

proving that $x \sim z$.

So in all cases we have $x \sim z$.

Hence we have that

$\sim$ is a equivalence relation

Define then $\mathcal{U} = \{\sim[x] \mid x \in U\} = I / \sim$ the set of equivalence classes defined by $\sim$ then we have using [theorem: 3.18] that

$$U = \bigcup_{I \in \mathcal{U}} I \text{ and } \forall I, J \in \mathcal{U} \text{ with } I \neq J \quad I \cap J = \emptyset \quad (14.37)$$

Let $I \in \mathcal{U}$ then there exist a $x \in U$ such that $I = \sim[x]$. Let $y \in \sim[x] \subseteq U$ then by [theorem: 14.36] there exist a $]a, b[\in \mathcal{B}_{||}$ [theorem: 14.104] $\{]x, y[\mid x, y \in \mathbb{R} \wedge x < y\}$ such that $y \in]a, b[\subseteq U$ then $\forall z \in]a, b[$ we have either:

$y \leq z$. Then $a < y \leq z < b$ so that $] \min(y, z), \max(y, z)[=]y, z[\subseteq]a, b[\subseteq U$

$z < y$. Then $a < z < y < b$ so that $] \min(y, z), \max(y, z)[=]z, y[\subseteq]a, b[\subseteq U$

proving that $y \sim z$. From $y \in \sim[x]$ we have $y \sim x$ so by transitivity and symmetry that $z \sim x$ hence $z \in \sim[x] = I$ proving that $]a, b[\subseteq I$ or as $y \in]a, b[$ that $y \in]a, b[\subseteq I$. To summarize

$$\forall y \in I = \sim[x] \text{ there exist a }]a, b[\in \mathcal{B}_{||} \text{ such that } y \in]a, b[\subseteq I$$

proving by [definition: 14.34] that

$$I = \sim[x] \text{ is open and as } x \in \sim[x] = I \quad I \neq \emptyset \quad (14.38)$$

Let $r, s \in I = \sim[x]$ with $r < s$ then $r \sim x \wedge s \sim x$ so that by symmetry and transitivity we have $s \sim r$ so that $]r, s[=] \min(r, s), \max(r, s)[\subseteq U$. If $t \in]r, s[\Rightarrow r < t < s$ then

$$] \min(t, r), \max(t, r)[=]r, t[\subseteq]r, s[\subseteq U$$

proving that $t \sim r$, as $r \sim x$ it follows that $t \sim x$ so that $t \in \sim[x] = I$. Hence

$$\forall r, s \in I \text{ with } r < s \text{ we have that }]r, s[\subseteq I$$

which by [definition: 3.141] proves that

$$I \text{ is a generalized interval} \quad (14.39)$$

As by [eq: 14.38] I is open it follows from [corollary: 14.106] that

$$\forall I \in \mathcal{U} \text{ we have } I =]a, b[\text{ or } I =]a, \infty[\text{ or } I =]-\infty, a[\text{ or } I =]-\infty, \infty[\text{ where } a, b \in \mathbb{R}$$

Now if $I \in \mathcal{U}$ then $\exists x \in U$ so that $I = \sim[x] \neq \emptyset$, by the above we have the following cases to consider:

$I =]a, b[$. Then as $I \neq \emptyset$ we have $a < b$ so using the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] there exist a $q \in \mathbb{Q}$ such that $a < q < b$, hence $q \in]a, b[= I$ so that

$$\mathbb{Q} \cap I \neq \emptyset$$

$I =]-\infty, a[$. Then as $a - 1 < a$ we have by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] that there exist a $q \in \mathbb{Q}$ such that $a - 1 < q < a$ hence $q \in]-\infty, a[= I$ proving

$$\mathbb{Q} \cap I \neq \emptyset$$

$I =]a, \infty[$. Then as $a < a + 1$ we have by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] that there exist a $q \in \mathbb{Q}$ such that $a < q < a + 1$ hence $q \in]a, \infty[= I$ proving

$$\mathbb{Q} \cap I \neq \emptyset$$

$I =]-\infty, \infty[$. Then as $0 \in \mathbb{Q} \subseteq \mathbb{R} =]-\infty, \infty[$ we have

$$\mathbb{Q} \cap I \neq \emptyset$$

So we have

$\forall I \in \mathcal{U}$ we have that $\mathbb{Q} \cap I \neq \emptyset$ so that $\{\mathbb{Q} \cap I\}_{I \in \mathcal{U}} \subseteq \mathcal{P}(\mathbb{R})$ is a family of non empty subsets of $\mathbb{R}$

Using a consequence of the Axiom of Choice [see theorem 3.106] there exist a function

$$f: \mathcal{U} \rightarrow \bigcup_{I \in \mathcal{U}} (\mathbb{Q} \cap I) \text{ such that } f(I) \in \mathbb{Q} \cap I$$

Let $I, J \in \mathcal{U}$ such that $f(I) = f(J)$. As $f(I) \in \mathbb{Q} \cap I$ and $f(J) \in \mathbb{Q} \cap J$ we have as $f(I) = f(J)$ that $f(I) \in (\mathbb{Q} \cap I) \cap (\mathbb{Q} \cap J) = \mathbb{Q} \cap (I \cap J)$ so that $I \cap J \neq \emptyset$. Then by [eq: 14.37] we must have $I = J$ which proves as $\bigcup_{I \in \mathcal{U}} (\mathbb{Q} \cap I) \subseteq \mathbb{Q}$ that

$$f: \mathcal{U} \rightarrow \mathbb{Q} \text{ is injective}$$

Now as by [theorem: 10.5] $\mathbb{Q}$ is denumerable we have by [theorem: 6.75] that

$\mathcal{U}$ is countable and by [eqs: 14.37, 14.38, 14.39] $U = \bigcup_{I \in \mathcal{U}} I$ and $\forall I \in \mathcal{U}$ I is a open generalized interval that is not empty. $\square$

Corollary 14.110. *Given $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ the set of reals endowed with the topology generated by the absolute value norm and $U \in \mathcal{T}_{||}$ a open set then there exist a countable set $\mathcal{I}$ and a family*

$$\{]a_i, b_i[\}_{i \in \mathcal{I}} \subseteq \mathcal{T}_{||} \text{ such that } U = \bigcup_{i \in \mathcal{I}}]a_i, b_i[$$

such that $\forall i \in \mathcal{I}$ $]a_i, b_i[\neq \emptyset$

Proof. Using the previous theorem [theorem: 14.109] there exists a countable set $\mathcal{U} \subseteq \mathcal{T}_{||}$ such that $\forall I \in \mathcal{U}$ I is a open generalized interval and non empty and

$$U = \bigcup_{I \in \mathcal{U}} I \tag{14.40}$$

So let $I \in \mathcal{U}$ then by [corollary: 14.106] I is either

$I =]a, b[$. Define then $\mathcal{I}_I = \{1\}$ and $\{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathcal{I}_I}$ by $a_1 = a \wedge b_1 = b$ so that

$$\mathcal{I}_I \text{ is countable [because it is finite] and } I = \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$$

Further as $]a_1, b_1[= I \neq \emptyset$ we have $\forall i \in \mathcal{I}_I$ that $]a_{(I,i)}, b_{(I,i)}[\neq \emptyset$.

$I =]-\infty, a[$. Take $\mathcal{I}_I = \mathbb{N}$ and $\{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathcal{I}_I} = \{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathbb{N}_0}$ by $a_{(I,i)} = -i + a \wedge b_{(I,i)} = a$. Then $\forall i \in \mathcal{I}_I$ we have $\forall x \in]a_{(I,i)}, b_{(I,i)}[=]-i + a, a[\Rightarrow -i + a < x < a \Rightarrow x \in]-\infty, a[= I$ proving that

$$\bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[\subseteq I \tag{14.41}$$

Further if $x \in I$ then $x < a$ and by a consequence of the Archimedean Property of the real numbers [see theorem: 10.30] there exist a $n \in \mathbb{N} = \mathcal{I}_I$ such that $-n < x - a$ hence $-n + a < x$ so that $x \in]-n + a, a[=]a_{(I,n)}, b_{(I,n)}[\subseteq \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$. Proving that $I \subseteq \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$ which combined with 14.41] proves that

$$\mathcal{I}_I \text{ is countable and } I = \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$$

Further $\forall i \in \mathcal{I}_I = \mathbb{N}$ we have as $-i + a < a$ that $\emptyset \neq]-i + a, a[=]a_{(I,i)}, b_{(I,i)}[$.

$I =]a, \infty[$. Take $\mathcal{I}_I = \mathbb{N}$ and $\{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathcal{I}_I} = \{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathbb{N}_0}$ by $a_{(I,i)} = a \wedge b_{(I,i)} = a + i$. Then $\forall i \in \mathcal{I}_I$ we have $\forall x \in]a_{(I,i)}, b_{(I,i)}[=]a, a + i[\Rightarrow a < x < a + i \Rightarrow x \in]a, \infty[$ proving that

$$\bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[\subseteq I \quad (14.42)$$

Further if $x \in I$ then $a < x$ and by a consequence of the Archimedean Property of the real numbers [see theorem: 10.30] there exist a $n \in \mathbb{N} = \mathcal{I}_I$ such that $x - a < n$ hence $x < n + a$ or $x \in]a, a + n[=]a_{(I,n)}, b_{(I,n)}[\subseteq \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$. Proving that $I \subseteq \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$ which combined with 14.42] proves that

$$\mathcal{I}_I \text{ is countable and } I = \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$$

Further $\forall i \in \mathcal{I}_I = \mathbb{N}$ we have as $a < a + i$ that $\emptyset \neq]a, a + i[=]a_{(I,i)}, b_{(I,i)}[$

$I =]-\infty, \infty[$. Then $I = \mathbb{R}$. Take $\mathcal{I}_I = \mathbb{N}$ and $\{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathcal{I}_I} = \{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathbb{N}_0}$ by $a_{(I,i)} = -i \wedge b_{(I,i)} = i$. Then $\forall i \in \mathcal{I}_I$ we have $]a_{(I,i)}, b_{(I,i)}[\subseteq \mathbb{R} = I$ so that

$$\bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[\subseteq I \quad (14.43)$$

Further if $x \in \mathbb{R}$ then by [theorem: 10.30] there exist a $n \in \mathbb{N}_0 = \mathcal{I}_I$ such that $|x| < n$ hence $x < |x| < n$ and $-x < |x| < n \Rightarrow -n < x$ proving that $-n < x < n \Rightarrow x \in]-n, n[=]a_n, b_n[\subseteq \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$. Hence $I \subseteq \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$ which combined with [eq: 14.43] proves that

$$\mathcal{I}_I \text{ is countable and } I = \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[$$

Further $\forall i \in \mathcal{I}_I$ we have $\emptyset \neq]-i, i[=]a_{(I,i)}, b_{(I,i)}[$

So we have that $\forall I \in \mathcal{U}$ there exist a countable $\mathcal{I}_I$ and $\{]a_{(I,i)}, b_{(I,i)}[\}_{i \in \mathcal{I}_I} \subseteq \mathcal{T}_I$ so that

$$I = \bigcup_{i \in \mathcal{I}_I}]a_{(I,i)}, b_{(I,i)}[\text{ and } \forall i \in I \text{ we have that }]a_{(I,i)}, b_{(I,i)}[\neq \emptyset \quad (14.44)$$

Now we have

$$\begin{aligned} x \in \bigcup_{r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I}]a_r, b_r[& \Rightarrow x \in \bigcup_{r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I}]a_{(r_1, r_2)}, b_{(r_1, r_2)}[\\ & \Rightarrow \exists r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I \text{ such that } x \in]a_{(r_1, r_2)}, b_{(r_1, r_2)}[\\ & \Rightarrow \exists I \in \mathcal{U} \text{ such that } r \in \{I\} \times \mathcal{I}_I \text{ such that } x \in]a_{(r_1, r_2)}, b_{(r_1, r_2)}[\\ & \stackrel{[\text{eq: 14.44}]}{\Rightarrow} x \in I \\ & \Rightarrow x \in \bigcup_{I \in \mathcal{U}} I \end{aligned}$$

proving that

$$\bigcup_{r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I}]a_r, b_r[\subseteq \bigcup_{I \in \mathcal{U}} I \quad (14.45)$$

and

$$\begin{aligned}
x \in \bigcup_{I \in \mathcal{U}} I &\Rightarrow \exists I \in \mathcal{U} \text{ such that } x \in I \\
&\stackrel{[\text{eq: 14.44}]}{\Rightarrow} \exists i \in \mathcal{I}_I \text{ such that } x \in]a_{(I,i)}, b_{(I,i)}[\\
&\Rightarrow x \in]a_{(r_1, r_2)}, b_{(r_1, r_2)}[\text{ where } r = (I, i) \in \{I\} \times \mathcal{I}_I \\
&\Rightarrow x \in]a_r, b_r[\text{ where } r = (I, i) \in \{I\} \times \mathcal{I}_I \subseteq \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I \\
&\Rightarrow x \in \bigcup_{r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I}]a_r, b_r[
\end{aligned}$$

proving that $\bigcup_{I \in \mathcal{U}} I \subseteq \bigcup_{r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I}]a_r, b_r[$. Combining this with [eqs: 14.40, 14.45] results in

$$U = \bigcup_{I \in \mathcal{U}} I = \bigcup_{r \in \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I}]a_r, b_r[\quad (14.46)$$

Using [theorem: 6.81] together with [eq: 14.44] we have $\forall I \in \mathcal{U}$ that $\{I\} \times \mathcal{I}_I$ is countable. As $\mathcal{U}$ is countable it follows from [theorem: 6.77] that $\bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I$ is countable. Hence if we take $\mathcal{I} = \bigcup_{I \in \mathcal{U}} \{I\} \times \mathcal{I}_I$ we have that

$$\mathcal{I} \text{ is countable, } U \stackrel{[\text{eq: 14.46}]}{=} \bigcup_{r \in \mathcal{I}}]a_r, b_r[\text{ and } \forall i \in I \text{ we have }]a_i, b_i[\neq \emptyset \quad \square$$

Theorem 14.111. *Given $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ the set of real numbers endowed with the topology generated by the absolute value norm and $S \subseteq \mathbb{R}$ a non empty closed set then we have:*

1. *If $\sup(S)$ exist then $\sup(S) \in S$*
2. *If $\inf(S)$ exist then $\inf(S) \in S$*

Proof.

1. Assume that $\sup(S) \notin S$ then $\sup(S) \in X \setminus S \in \mathcal{T}_{||}$, hence by [theorems: 14.97, 14.104] there exist a $\varepsilon \in \mathbb{R}^+$ such that $\sup(S) \in]\sup(S) - \varepsilon, \sup(S) + \varepsilon[\subseteq X \setminus S$. As $\sup(S) - \varepsilon < \sup(S)$ there exist a $s \in S$ such that $\sup(S) - \varepsilon < s \leq \sup(S) < \sup(S) + \varepsilon$ so that $s \in X \setminus S$ contradicting $s \in S$. Hence we must have $\sup(S) \in S$.
2. Assume that $\inf(S) \notin S$ then $\inf(S) \in X \setminus S \in \mathcal{T}_{||}$, hence by [theorems: 14.97, 14.104] then there exist a $\varepsilon \in \mathbb{R}^+$ such that $\inf(S) \in]\inf(S) - \varepsilon, \inf(S) + \varepsilon[\subseteq X \setminus S$. As $\inf(S) < \inf(S) + \varepsilon$ there exist a $s \in S$ such that $\inf(S) - \varepsilon < \inf(S) \leq s < \inf(S) + \varepsilon$ so that $s \in X \setminus S$ contradicting $s \in S$. Hence we must have $\inf(S) \in S$. $\square$

The concept of a bounded set [see definition: 14.79] in a metric space becomes simpler if the metric space happens to be a normed space.

Theorem 14.112. *Let $\langle X, |||| \rangle$ be a pseudo normed space, $A \subseteq X$ then*

$$A \text{ is bounded} \Leftrightarrow \exists M \in \mathbb{R}^+ \text{ such that } \forall x \in X \text{ we have } ||x|| \leq M$$

Proof.

$\Rightarrow$. If A is a bounded set then we have either:

$A = \emptyset$. Then $\forall x \in A$ we have vacuously that $||x|| \leq 1$.

$A \neq \emptyset$. Then $\exists x_0 \in A$. As A is bounded there exist a $N \in \mathbb{R}$ such that $\forall x, y \in A$ we have $||x - y|| = d_{|||}(x, y) \leq N$. Take $M = N + ||x_0||$ then if $z \in A$ we have $||z|| = ||z - x_0 + x_0|| \leq ||z - x_0|| + ||x_0|| \leq N + ||x_0|| = M$

$\Leftarrow$. Let $N \in \mathbb{R}$ such that $\forall x \in A$ we have $||x|| \leq N$ and set $M = 2 \cdot N$. Then $\forall x, y \in A$ we have $d_{|||}(x, y) = ||x - y|| \leq ||x|| + ||y|| = N + N = 2 \cdot N = M$ $\square$

Example 14.113. Let $\langle \mathbb{R}, || \rangle$ be the normed space of the real numbers then we have:

1. $\forall a, b \in \mathbb{R}$ with $a \leq b$ we have that
 - a. $[a, b]$ is bounded
 - b. $[a, b[$ is bounded
 - c. $]a, b]$ is bounded
 - d. $]a, b[$ is bounded
2. $\forall a \in \mathbb{R}$ we have that
 - a. $[a, \infty[$ is not bounded
 - b. $]a, \infty[$ is not bounded
 - c. $] -\infty, a]$ is not bounded
 - d. $] -\infty, a[$ is not bounded

Proof.

1. Let $M = \max(|a|, |b|)$ then
 - a. Then for $x \in [a, b]$ we have either:
 - $0 \leq x$. Then $|x| = x \leq b \leq |b| \leq \max(|a|, |b|) = M$
 - $x < 0$. Then $|x| = -x \leq -a \leq |a| \leq \max(|a|, |b|) = M$
 - b. Then for $x \in [a, b[$ we have either:
 - $0 \leq x$. Then $|x| = x < b \leq |b| \leq \max(|a|, |b|) = M$
 - $x < 0$. Then $|x| = -x \leq -a \leq |a| \leq \max(|a|, |b|) = M$
 - c. Then for $x \in]a, b]$ we have either:
 - $0 \leq x$. Then $|x| = x \leq b \leq |b| \leq \max(|a|, |b|) = M$
 - $x < 0$. Then $|x| = -x < -a \leq |a| \leq \max(|a|, |b|) = M$
 - d. Then for $x \in]a, b[$ we have either:
 - $0 \leq x$. Then $|x| = x < b \leq |b| \leq \max(|a|, |b|) = M$
 - $x < 0$. Then $|x| = -x < -a \leq |a| \leq \max(|a|, |b|) = M$
2.
 - a. Assume that $[a, \infty[$ is bounded then there exist a M such that $\forall x \geq a$ we have $|x| \leq M$. As $a \leq |a|$ it follows that $|a| \leq M$ hence $a < |a| \leq M < M + 1$ so that $|M + 1| = M + 1 \leq M$, a contradiction. Hence $[a, \infty[$ is not bounded.
 - b. Assume that $]a, \infty[$ is bounded then there exist a M such that $\forall x > a$ we have $|x| \leq M$. As $a \leq |a| < |a| + 1$ it follows that $|a| + 1 \leq M$ hence $a < |a| + 1 \leq M < M + 1$ so that $|M + 1| = M + 1 \leq M$, a contradiction. Hence $]a, \infty[$ is not bounded.
 - c. Assume that $] -\infty, a]$ is bounded then there exist a M such that $\forall x \leq a$ we have that $|x| \leq M$. As $-|a| \leq a$ it follows that $|a| = -|a| \leq M < M + 1$ so that $-(M + 1) < -|a| \leq a$, hence $M + 1 = |-(M + 1)| \leq M$, a contradiction. So $] -\infty, a]$ is not bounded.
 - d. Assume that $] -\infty, a[$ is bounded then there exist a M such that $\forall x < a$ we have that $|x| \leq M$. As $-|a| - 1 < -|a| \leq a$ it follows that $|a| + 1 = -|a| - 1 \leq M < M + 1$ so that $-(M + 1) < -|a| - 1 < -|a| \leq a$, hence $M + 1 = |-(M + 1)| \leq M$, a contradiction. So $] -\infty, a[$ is not bounded. $\square$

Theorem 14.114. Let $\langle X, ||| \rangle$ be a pseudo normed space with norm topology $\mathcal{T}_{|||}$ and $A \subseteq X$ where A is a sub-space of X then for $||| |_A$ defined by

$$||| |_A: A \rightarrow \mathbb{R} \text{ defined by } \|x\|_A = \|x\|$$

we have

1. $\langle A, |||_{|A} \rangle$ is a pseudo normed space.
2. If $\langle A, ||| \rangle$ is a normed space $\langle A, |||_{|A} \rangle$ is also a normed space
3. $\mathcal{T}_{|||_{|A}} = (\mathcal{T}_{|||})_{|A}$

Proof.

1. We have:

- a. $\forall x \in A$ we have $0 \leq \|x\| = \|x\|_{|A}$
- b. $\forall x \in A$ we have $\forall \alpha \in \mathbb{R}$ [or $\alpha \in \mathbb{C}$] that $\|\alpha \cdot x\|_{|A} = |\alpha| \|x\| = |\alpha| \cdot \|x\|_{|A}$
- c. $\forall x, y \in A$ we have $\|x + y\|_{|A} = \|x + y\| \leq \|x\| + \|y\| \leq \|x\|_{|A} + \|y\|_{|A}$

2. Let $x \in A$ such that $\|x\|_{|A} = 0$ then $\|x\| = 0$ hence as $|||$ is a norm we have $x = 0$.

3. Let $x, y \in A$ then $(d_{|||})_{|A \times A}(x, y) = d_{|||}(x, y) = \|x - y\| = \|x - y\|_{|A} = d_{|||_{|A}}(x, y)$ proving that

$$(d_{|||})_{|A} = d_{|||_{|A}} \quad (14.47)$$

So we have

$$\mathcal{T}_{|||_{|A}} \stackrel{\text{def}}{=} \mathcal{T}_{d_{|||_{|A}}} \stackrel{\text{eq: 14.47}}{=} \mathcal{T}_{(d_{|||})_{|A \times A}} \stackrel{\text{theorem: 14.70}}{=} (\mathcal{T}_{d_{|||}})_{|A} \stackrel{\text{def}}{=} (\mathcal{T}_{|||})_{|A} \quad \square$$

In the above theorem we have the requirement that A is a sub-space of X needed to ensure that A is a vector space otherwise we can not say that $\langle A, |||_{|A} \rangle$ is a pseudo normed or normed space. However the topology of a every pseudo normed space is defined by the metric $d_{|||}(x, y) = \|x - y\|$ [see definition: 14.96] and by [theorem: 14.70] the sub-space topology on a arbitrary subset is a metric space based on the restriction of the metric to the subset. This is made formal in the following theorem.

Theorem 14.115. Let $\langle X, ||| \rangle$ be a pseudo normed space with the topology $\mathcal{T}_{|||}$, $A \subseteq X$ then we have for the subspace topology $(\mathcal{T}_{|||})_{|A}$ that $(\mathcal{T}_{|||})_{|A} = \mathcal{T}_{(d_{|||})_{|A}}$ where

$$\forall x, y \in A \text{ we have } (d_{|||})_{|A}(x, y) = \|x - y\|$$

Further $\forall x \in X \wedge \varepsilon \in \mathbb{R}^+$ we have

$$A \cap B_{|||}(x, \varepsilon) = B_{d_{|||}}(x, \varepsilon) = \{y \in A \mid \|x - y\| < \varepsilon\}$$

Proof. By [definition: 14.96] $\mathcal{T}_{|||} = \mathcal{T}_{d_{|||}}$ where

$$d_{|||}: X \times X \rightarrow \mathbb{R} \text{ is defined by } d_{|||}(x, y) = \|x - y\|$$

Next using [theorem: 14.70] we have that

$$(\mathcal{T}_{d_{|||}})_{|A} = \mathcal{T}_{(d_{|||})_{|A}}$$

where $(d_{|||})_{|A}$ is the metric defined by

$$(d_{|||})_{|A}: A \rightarrow \mathbb{R} \quad (d_{|||})_{|A}(x, y) = d_{|||}(x, y) = \|x - y\|$$

Finally we have

$$\begin{aligned} y \in B_{|||}(x, \varepsilon) \cap A &\Leftrightarrow y \in A \wedge y \in B_{|||}(x, \varepsilon) \\ &\Leftrightarrow y \in A \wedge \|y - x\| < \varepsilon \\ &\Leftrightarrow \{y \in A \mid \|y - x\| < \varepsilon\} \\ y \in B_{d_{|||}}(x, \varepsilon) \cap A &\Leftrightarrow y \in A \wedge y \in B_{d_{|||}}(x, \varepsilon) \\ &\Leftrightarrow y \in A \wedge d_{|||}(x, y) < \varepsilon \\ &\Leftrightarrow y \in A \wedge \|x - y\| < \varepsilon \\ &\Leftrightarrow \{y \in A \mid \|y - x\| < \varepsilon\} \end{aligned}$$

which proves that

$$A \cap B_{|||}(x, \varepsilon) = B_{d_{|||}}(x, \varepsilon) = \{y \in A \mid \|x - y\| < \varepsilon\}$$

□

Theorem 14.116. Let $\langle X, ||| \rangle$ be a pseudo normed space, $M \subseteq X$ then $\forall \alpha \in \mathbb{K}$ with $\alpha \neq 0$ and $\forall x \in X$ we have:

1. $\overline{\alpha \cdot M} = \alpha \cdot \overline{M}$
2. If M is closed in $\mathcal{T}_{|||}$ then $\alpha \cdot M$ is closed in $\mathcal{T}_{|||}$
3. If M is open in $\mathcal{T}_{|||}$ then $\alpha \cdot M$ is open in $\mathcal{T}_{|||}$
4. If M is open in $\mathcal{T}_{|||}$ then $x + M$ is open in $\mathcal{T}_{|||}$
5. If M is open in $\mathcal{T}_{|||}$ and $A \subseteq X$ then $A + M$ is open in $\mathcal{T}_{|M}$
6. If $x \in X$ then $x + \overline{M} \subseteq \overline{x + M}$
7. $\overline{M} - \overline{M} \subseteq \overline{M - M}$

See [definition: 11.74] for the definition of $\alpha \cdot M$, $x + M$, $A + M$ and $M - M$

Proof.

1. Let $x \in \overline{\alpha \cdot M}$ then as $\alpha \neq 0$ we can define $z = 1/\alpha \cdot x$ so that $x = \alpha \cdot z$. Assume that U is a open set such that $z \in U$ then by [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that

$$z \in B_{|||}(z, \delta) \subseteq U \quad (14.48)$$

Further we have trivially $x \in B_{|||}(x, |\alpha| \cdot \delta)$ which is open so by [theorem: 14.30] and $x \in \overline{\alpha \cdot M}$ we have that $B_{|||}(x, |\alpha| \cdot \delta) \cap (\alpha \cdot M) \neq \emptyset$, hence there exist a $m \in M$ such that $\alpha \cdot m \in B_{|||}(x, |\alpha| \cdot \delta)$, from which it follows that $\|x - \alpha \cdot m\| < |\alpha| \cdot \delta$ or as $x = \alpha \cdot z$ that $|\alpha| \|z - m\| = \|\alpha \cdot z - \alpha \cdot m\| = \|x - \alpha \cdot m\| < |\alpha| \cdot \delta$ from which we conclude, as $|\alpha| \neq 0$, that $\|z - m\| < \delta$ or that $m \in B_{|||}(z, \delta) \subseteq_{\text{eq: 14.48}} U$. Hence $U \cap M \neq \emptyset$ which, as $z \in U$, means by [theorem: 14.30], that $z \in \overline{M}$ so that $x = \alpha \cdot z \in \alpha \cdot \overline{M}$. So we have proved that

$$\overline{\alpha \cdot M} \subseteq \alpha \cdot \overline{M} \quad (14.49)$$

For the opposite inclusion, assume that $x \in \alpha \cdot \overline{M}$. If U is a open set such that $x \in U$ then by [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{|||}(x, \delta) \subseteq U \quad (14.50)$$

As $x \in \alpha \cdot \overline{M}$ there exist a $z \in \overline{M}$ such that $x = \alpha \cdot z$, further we have trivially $z \in B_{|||}(z, \delta/|\alpha|)$ which is a open set so that by [theorem: 14.30] we have that $M \cap B_{|||}(z, \delta/|\alpha|) \neq \emptyset$, hence there exist a $m \in M$ such that $m \in B_{|||}(z, \delta/|\alpha|) \Rightarrow \|z - m\| < \delta/|\alpha|$. So $\|x - \alpha \cdot m\| = \|\alpha \cdot z - \alpha \cdot m\| = |\alpha| \cdot \|z - m\| < |\alpha| \cdot (\delta/|\alpha|) = \delta$ proving that $\alpha \cdot m \in B_{|||}(x, \delta) \subseteq U$ hence, as $\alpha \cdot m \in \alpha \cdot M$ we have that $U \cap (\alpha \cdot M) \neq \emptyset$, proving by [theorem: 14.30] that $x \in \overline{\alpha \cdot M}$. So $\alpha \cdot \overline{M} \subseteq \overline{\alpha \cdot M}$ which combined with [eq: 14.49] gives

$$\overline{\alpha \cdot M} = \alpha \cdot \overline{M}$$

2. If A is closed then by [theorem: 14.26] $A = \overline{A}$, hence $\alpha \cdot A = \alpha \cdot \overline{A} \stackrel{(1)}{=} \overline{\alpha \cdot A}$ proving by [theorem: 14.26] that $\alpha \cdot A$ is closed.
3. If $x \in \alpha \cdot M$ then $x = \alpha \cdot m$ where $m \in M$. As M is open there exists by [theorem: 14.68] a $\delta \in \mathbb{R}^+$ such that $m \in B_{|||}(m, \delta) \subseteq M$. If $z \in B_{|||}(x, |\alpha| \cdot \delta)$ then $\|z - x\| < |\alpha| \cdot \delta$ so that

$$\|m - (1/\alpha) \cdot z\| = \|(1/\alpha) \cdot x - (1/\alpha) \cdot z\| = (1/|\alpha|) \cdot \|x - z\| < (1/|\alpha|) \cdot |\alpha| \cdot \delta = \delta$$

proving that $(1/\alpha) \cdot z \in B_{|||}(m, \delta) \subseteq M \Rightarrow z \in \alpha \cdot M$. Hence $x \in B_{|||}(x, |\alpha| \cdot \delta) \subseteq \alpha \cdot M$ proving by [theorem: 14.68] that $\alpha \cdot M$ is open.

4. Let $z \in x + M$ then $z - x \in M$ a open set, hence there exist by [theorem: 14.68] a $\delta \in \mathbb{R}^+$ such that

$$z - x \in B_{|||}(z - x, \delta) \subseteq M$$

Take $y \in B_{|||}(z, \delta)$ then $\|z - y\| < \delta \Rightarrow \|(y - x) - (z - x)\| = \|y - z\| = \|z - y\| < \delta$ proving that $y - x \in B_{|||}(z - x, \delta) \subseteq M \Rightarrow y \in x + M$. Hence $z \in B_{|||}(z, \delta) \subseteq x + M$ proving by [theorem: 14.36] that $x + M$ is open.

5. This follows from (4) as $A + M = \bigcup_{x \in A} (x + M)$ that is a union of open sets, which is open by definition.
6. Let $z \in x + \overline{M}$ then $z = x + y$ where $y \in \overline{M}$. If U is a open set such that $z \in U$ then by [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that

$$z \in B_{|||}(z, \delta) \subseteq U$$

As $y \in B_{|||}(y, \delta) \in \mathcal{T}_{|||}$ and $y \in \overline{M}$ it follows from [theorem: 14.30] that $B_{|||}(y, \delta) \cap M \neq \emptyset$, hence there exist a $y_1 \in B_{|||}(y, \delta) \cap M$. Take $z_1 = x + y_1$ then $z_1 \in x + M$ and $\|z - z_1\| = \|x + y - x - y_1\| = \|y - y_1\| < \delta$ so that $z_1 \in B_{|||}(z, \delta) \subseteq U$. Hence $z_1 \in U \cap (x + M)$ so that $U \cap (x + M) \neq \emptyset$ proving by [theorem: 14.30] that $z \in \overline{x + M}$. Hence

$$x + \overline{M} \subseteq \overline{x + M}$$

7. If $x \in \overline{M} - \overline{M}$ then $x = y_1 - y_2$ where $y_1, y_2 \in \overline{M}$. Let U be a open set such that $x \in U$ then by [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{|||}(x, \delta) \subseteq U \quad (14.51)$$

As $y_1, y_2 \in \overline{M}$ we have, as $y_1 \in B_{|||}(y_1, \delta/2) \in \mathcal{T}_{|||}$, $y_2 \in B_{|||}(y_2, \delta/2) \in \mathcal{T}_{|||}$, by [theorem: 14.30] that there exists a $z_1 \in B_{|||}(y_1, \delta/2) \cap M$ and a $z_2 \in B_{|||}(y_2, \delta/2) \cap M$. Take $z = z_1 - z_2 \in M - M$ then

$$\|x - z\| = \|(y_1 - y_2) - (z_1 - z_2)\| = \|(y_1 - z_1) + (z_2 - y_2)\| \leq \|y_1 - z_1\| + \|y_2 - z_2\| < \delta$$

proving that $z \in B_{|||}(x, \delta)$. As $z \in M - M$ it follows that $\emptyset \neq B_{|||}(x, \delta) \cap (M - M) \subseteq U \cap (M - M)$. So applying [theorem: 14.30] proves then that $\overline{M} - \overline{M} \subseteq \overline{M} - \overline{M}$. $\square$

Theorem 14.117. Let $\langle X, d \rangle$ a pseudo normed space and $A \subseteq X$ then we have

$$A \text{ is dense in } X \text{ [using the norm topology } \mathcal{T}_{|||}]$$

$$\Updownarrow$$

$$\forall x \in X \text{ we have } \forall \varepsilon \in \mathbb{R}^+ \text{ there exist a } a \in A \text{ such that } \|x - a\| < \varepsilon$$

Proof.

$\Rightarrow$. Let $x \in X$ then $\forall \varepsilon \in \mathbb{R}^+$ such that $x \in B_{d_{|||}}(x, \varepsilon)$ we have by [theorem: 14.55] that $\exists a \in A$ such that $a \in B_{d_{|||}}(x, \varepsilon)$, hence $\|x - a\| < \varepsilon$.

$\Leftarrow$. Let $x \in X$ and let $B \in \mathcal{B}_{d_{|||}}$ such that $x \in B$. As $B \in \mathcal{B}_{d_{|||}}$ there exists a $y \in X$ and a $\varepsilon \in \mathbb{R}^+$ such that $x \in B = B_{d_{|||}}(y, \varepsilon)$. By [theorem: 14.66] there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_{d_{|||}}(x, \delta) \subseteq B_{d_{|||}}(y, \varepsilon) = B$. By the hypothesis there exist a $a \in A$ such that $\|x - a\| < \delta$ or $a \in B_{d_{|||}}(x, \delta) \subseteq B_{d_{|||}}(y, \varepsilon) = B$. So

$$\forall x \in X \text{ we have } \forall B \in \mathcal{B}_{d_{|||}} \text{ with } x \in B \text{ that there exist a } a \in A \text{ such that } a \in B$$

Applying [theorem: 14.55] on the above proves that A is dense in X . $\square$

Theorem 14.118. Let $\|\cdot\|_1, \|\cdot\|_2$ two norms on a vector space X and $\mathcal{T}_{\|\cdot\|_1}, \mathcal{T}_{\|\cdot\|_2}$ the associated topologies then $\mathcal{T}_{\|\cdot\|_2}$ is finer than $\mathcal{T}_{\|\cdot\|_1}$

$$\mathcal{T}_{\|\cdot\|_1} \text{ is finer than } \mathcal{T}_{\|\cdot\|_2} \Leftrightarrow \exists M \in \mathbb{R}^+ \text{ such that } \forall x \in X \text{ we have } \|x\|_1 \leq M \cdot \|x\|_2$$

Proof.

$\Rightarrow$. As $\mathcal{T}_{\|\cdot\|_1}$ is finer than $\mathcal{T}_{\|\cdot\|_2}$ we have by [theorem: 14.72] there exist a $\delta \in \mathbb{R}^+$ such that

$$0 \in B_{\|\cdot\|_2}(0, \delta) \subseteq B_{\|\cdot\|_1}(0, 1) \quad (14.52)$$

Take $M = 2/\delta$. Let $x \in X$ then we have either:

$x = 0$. Then $\|x\|_1 = 0 \leq 0 = M \cdot 0 = M \cdot \|x\|_2$

$x \neq 0$. Take $y = (1/(M \cdot \|x\|_2)) \cdot x$ [as $\|\cdot\|_2$ is a norm $\|x\|_2 \neq 0$] then we have

$$\|y\|_2 = |1/(M \cdot \|x\|_2)| \cdot \|x\|_2 = 1/M = \delta/2 < \delta$$

so that by [eq: 14.52] $y \in B_{\|\cdot\|_1}(0, 1)$, hence $\|y\|_1 < 1$ or as

$$1 > \|y\|_1 = \|(1/(M \cdot \|x\|_2)) \cdot x\|_1 = |1/(M \cdot \|x\|_2)| \cdot \|x\|_1 = \|x\|_1 / (M \cdot \|x\|_2)$$

so that $\|x\|_1 / (M \cdot \|x\|_2) < 1$ proving that

$$\|x\|_1 \leq M \cdot \|x\|_2$$

hence

$$\forall x \in X \text{ we have } \|x\|_1 \leq M \cdot \|x\|_2$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$ and $x \in B_{\|\cdot\|_1}(x, \varepsilon)$. As $M \in \mathbb{R}^+$ we can take $\delta = \varepsilon/M \in \mathbb{R}^+$ then if $y \in B_{\|\cdot\|_2}(x, \delta)$ we have $\|x - y\|_2 < \delta$ and by the hypothesis

$$\|x - y\|_1 \leq M \cdot \|x - y\|_2 < M \cdot \delta = M \cdot (\varepsilon/M) = \varepsilon$$

so that $y \in B_{\|\cdot\|_1}(x, \varepsilon)$. Hence $B_{\|\cdot\|_2}(x, \delta) \subseteq B_{\|\cdot\|_1}(x, \varepsilon)$ which by [theorem: 14.72] proves that

$$\mathcal{T}_{\|\cdot\|_2} \text{ is finer than } \mathcal{T}_{\|\cdot\|_1} \quad \square$$

Definition 14.119. Let X be a real or complex vector space then two pseudo norms are equivalent if

$$\mathcal{T}_{\|\cdot\|_1} = \mathcal{T}_{\|\cdot\|_2}$$

Theorem 14.120. Let X be a real or complex vector space and

$$\mathcal{N}(X) = \{\|\cdot\| \in \mathbb{R}^{X \times X} \mid \|\cdot\| \text{ is a pseudo norm}\}$$

then if we define $\approx \subseteq \mathcal{N}(X) \times \mathcal{N}(X)$ by $\|\cdot\|_1 \approx \|\cdot\|_2$ if $\|\cdot\|_1$ is equivalent with $\|\cdot\|_3$ we have that

$$\approx \text{ is a equivalence relation}$$

Proof. The proof is trivial because set equality is a equivalence relation

reflexivity. If $\|\cdot\|$ is a pseudo norm on X then clearly $\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{\|\cdot\|}$ so that $\|\cdot\| \approx \|\cdot\|$

symmetry. If $\|\cdot\|_1 \approx \|\cdot\|_2$ then $\mathcal{T}_{\|\cdot\|_1} = \mathcal{T}_{\|\cdot\|_2} \Rightarrow \mathcal{T}_{\|\cdot\|_2} = \mathcal{T}_{\|\cdot\|_1}$ so that $\|\cdot\|_2 \approx \|\cdot\|_1$

transitivity. If $\|\cdot\|_1 \approx \|\cdot\|_2$ and $\|\cdot\|_2 \approx \|\cdot\|_3$ then $\mathcal{T}_{\|\cdot\|_1} = \mathcal{T}_{\|\cdot\|_2} \wedge \mathcal{T}_{\|\cdot\|_2} \approx \mathcal{T}_{\|\cdot\|_3}$, hence $\mathcal{T}_{\|\cdot\|_1} = \mathcal{T}_{\|\cdot\|_3}$ proving that $\|\cdot\|_1 \approx \|\cdot\|_3$ $\square$

Theorem 14.121. Let X be a real or complex vector space and $\|\cdot\|_1, \|\cdot\|_2$ pseudo norms on X then

$$\|\cdot\|_1 \text{ is equivalent with } \|\cdot\|_2 \Leftrightarrow \exists M_1, M_2 \in \mathbb{R}^+ \text{ such that } \forall x \in X \ M_1 \cdot \|x\|_1 \leq \|x\|_2 \leq M_2 \cdot \|x\|_1$$

Proof. Let $\mathcal{T}_{\|\cdot\|_1}, \mathcal{T}_{\|\cdot\|_2}$ be the topologies generated by $\|\cdot\|_1, \|\cdot\|_2$ then we have

$\Rightarrow$. Then as $\mathcal{T}_{\|\cdot\|_1} = \mathcal{T}_{\|\cdot\|_2}$ we have that $\mathcal{T}_{\|\cdot\|_2}$ is finer than $\mathcal{T}_{\|\cdot\|_1}$ and $\mathcal{T}_{\|\cdot\|_1}$ is finer than $\mathcal{T}_{\|\cdot\|_2}$ so by [theorem: 14.118] there exists $M'_1, M_2 \in \mathbb{R}^+$ such that $\forall x \in X \ \|x\|_1 \leq M'_1 \cdot \|x\|_2$ and $\|x\|_2 \leq M_2 \cdot \|x\|_1$ hence if we take $M_1 = 1/M'_1$ then we have

$$M_1 \cdot \|x\|_1 \leq \|x\|_2 \leq M_2 \cdot \|x\|_1$$

$\Leftarrow$. Let $x \in X$ then as $M_1 \cdot \|x\|_1 \leq \|x\|_2 \leq M_2 \cdot \|x\|_1$ we have $\|x\|_1 \leq (1/M_1) \cdot \|x\|_2$ and $\|x\|_2 \leq M_2 \cdot \|x\|_1$ so that by [theorem: 14.118] we have that $\mathcal{T}_{\|\cdot\|_1}$ is finer than $\mathcal{T}_{\|\cdot\|_2}$ and $\mathcal{T}_{\|\cdot\|_2}$ is finer than $\mathcal{T}_{\|\cdot\|_1}$. So $\mathcal{T}_{\|\cdot\|_1} = \mathcal{T}_{\|\cdot\|_2}$ or $\|\cdot\|_1$ is equivalent with $\|\cdot\|_2$. $\square$

Example 14.122. In the real vector space $\mathbb{R}$ all norms are equivalent.

Proof. Note that by [example: 14.99]

$$\|\cdot\| \text{ is a norm on } \mathbb{R}$$

Let $\|\cdot\|$ be a norm on $\mathbb{R}$ then given $x \in X$ we have $\|x\| = \|1 \cdot x\| = |x| \cdot \|1\|$, so if we take $N = M = \|1\| \in \mathbb{R}^+$ [because $1 \neq 0$ and $\|\cdot\|$ is a norm] we have

$$N \cdot |x| \leq N \cdot |x| = \|1\| \cdot |x| = \|x\| = \|1\| \cdot |x| = M \cdot |x| \leq M \cdot |x|$$

so that $\|\cdot\|$ is equivalent with $|\cdot|$. Hence every norm on $\mathbb{R}$ is equivalent with $|\cdot|$. As norm equivalence is a equivalence relation [see theorem: 14.120] it follows that all norms on $\mathbb{R}$ are equivalent. $\square$

Example 14.123. In the complex vector space $\mathbb{C}$ all norms are equivalent. Hence using [examples: 14.102, 14.103] we have that $\|\cdot\|$ and $\|\cdot\|_{\mathbb{C}}$ are equivalent norms so that $\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{\|\cdot\|_{\mathbb{C}}}$.

Proof. Note that by [example: 14.99]

$$\|\cdot\| \text{ is a norm on } \mathbb{C}$$

Let $\|\cdot\|$ be a norm on $\mathbb{C}$ then given $x \in X$ we have $\|x\| = \|1 \cdot x\| = |x| \cdot \|1\|$, so if we take $N = M = \|1\| \in \mathbb{R}^+$ [because $1 \neq 0$ and $\|\cdot\|$ is a norm] we have

$$N \cdot |x| \leq N \cdot |x| = \|1\| \cdot |x| = \|x\| = \|1\| \cdot |x| = M \cdot |x| \leq M \cdot |x|$$

so that $\|\cdot\|$ is equivalent with $|\cdot|$. Hence every norm on $\mathbb{C}$ is equivalent with $|\cdot|$. As norm equivalence is a equivalence relation [see theorem: 14.120] it follows that all norms on $\mathbb{C}$ are equivalent. $\square$

Definition 14.124. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces then $\varphi: X \rightarrow Y$ is a **linear isometry** if

1. $\varphi: X \rightarrow Y$ is a linear mapping
2. $\forall x \in X$ we have $\|x\|_X = \|\varphi(x)\|_Y$

If in addition $\varphi: X \rightarrow Y$ is a **bijection** then

$$\varphi: X \rightarrow Y \text{ is a linear isometric isomorphism}$$

If between $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ there exist a linear isometric isomorphism then we says that $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ **are isometric**. We note this as

$$X \approx Y$$

Theorem 14.125. Let $\langle X, \|\cdot\|_X \rangle$ a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a pseudo normed spaces and $\varphi: X \rightarrow Y$ a **linear isometry** then φ is **injective**. Hence if $\varphi: X \rightarrow Y$ is also surjective then $\varphi: X \rightarrow Y$ is a bijection and thus a **linear isometric isomorphism**.

Proof. Let $x, y \in X$ such that $\varphi(x) = \varphi(y)$ then

$$\|x - y\|_X = \|\varphi(x - y)\|_Y = \|\varphi(x) - \varphi(y)\|_Y = \|0\|_Y = 0$$

which as $\|\cdot\|_X$ is a norm proves that $x - y = 0$ hence $x = y$. $\square$

Theorem 14.126. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces and $\varphi: X \rightarrow Y$ a linear isometric isomorphism then $\varphi^{-1}: Y \rightarrow X$ is a linear isometric isomorphism.

Proof. Because of [theorem: 11.178] we have that $\varphi^{-1}: Y \rightarrow X$ is a linear isomorphism. Further if $y \in Y$ then $\|y\|_Y = \|\varphi(\varphi^{-1}(y))\|_Y = \|\varphi^{-1}(y)\|_X$. Hence $\varphi^{-1}: Y \rightarrow X$ is a linear isometry. $\square$

Theorem 14.127. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ and $\langle Z, \|\cdot\|_Z \rangle$ be pseudo metric spaces and $\varphi: X \rightarrow Y, \psi: Y \rightarrow Z$ linear isometries then

$$\psi \circ \varphi: X \rightarrow Z \text{ is a linear isometry}$$

If in addition φ, ψ are linear isometric isomorphisms then

$$\psi \circ \varphi: X \rightarrow Z \text{ is a linear isometric isomorphism}$$

Proof. Using [theorem: 11.180] $\psi \circ \varphi: X \rightarrow Z$ is a linear mapping. Further given $x \in X$ we have

$$\|(\psi \circ \varphi)(x)\|_Z = \|\psi(\varphi(x))\|_Z = \|\varphi(x)\|_Y = \|x\|_X$$

hence

$$\psi \circ \varphi: X \rightarrow Z \text{ is a linear isometry}$$

Further if φ, ψ are linear isometric isomorphisms, then they are bijection, hence by [theorem: 2.78] $\psi \circ \varphi$ is a bijection, so that $\psi \circ \varphi$ is a linear isometric isomorphism. $\square$

We have the following relation between a linear isometry in a normed space and a linear isometry in its associated metric space.

Theorem 14.128. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces with associated metric spaces and $\varphi: X \rightarrow Y$ a linear isometry then*

$$\forall x, y \in X \text{ we have } d_{\|\cdot\|_X}(x, y) = d_{\|\cdot\|_Y}(\varphi(x), \varphi(y)) \Leftrightarrow \forall x \in X \text{ we have } \|x\|_X = \|\varphi(x)\|_Y$$

In other words if φ is a linear mapping then

$$\varphi \text{ is a isometry} \Leftrightarrow \varphi \text{ is a linear isometry}$$

and

$$\varphi \text{ is a isometric isomorphism} \Leftrightarrow \varphi \text{ is a linear isometric isomorphism}$$

Proof.

$\Rightarrow$. Let $x \in X$ then we have

$$\|x\|_X = \|x - 0\|_X = d_{\|\cdot\|_X}(x, 0) = d_{\|\cdot\|_Y}(\varphi(x), \varphi(0)) = \|\varphi(x) - \varphi(0)\|_Y = \|\varphi(x - 0)\|_Y = \|\varphi(x)\|_Y$$

$\Leftarrow$. Let $x, y \in X$ then we have

$$d_{\|\cdot\|_X}(x, y) = \|x - y\|_X = \|\varphi(x - y)\|_Y = \|\varphi(x) - \varphi(y)\|_Y = d_{\|\cdot\|_Y}(\varphi(x), \varphi(y)) \quad \square$$

Theorem 14.129. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $\varphi: X \rightarrow Y$ a linear isometric isomorphism then*

$$\mathcal{T}_{\|\cdot\|_X} = \{\varphi^{-1}(V) | V \in \mathcal{T}_{\|\cdot\|_Y}\} = \{U \subseteq X | \varphi(U) \in \mathcal{T}_{\|\cdot\|_Y}\}$$

and

$$\mathcal{T}_{\|\cdot\|_Y} = \{V \subseteq Y | \varphi^{-1}(V) \in \mathcal{T}_{\|\cdot\|_X}\} = \{\varphi(U) | U \in \mathcal{T}_{\|\cdot\|_X}\}$$

Proof. This follows from [theorem: 14.78] and [theorem: 14.128] $\square$

14.4 Continuous functions

14.4.1 Continuous and open functions

Definition 14.130. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $x \in X$ then a function*

*$f: X \rightarrow Y$ is **continuous at x** if $\forall V \in \mathcal{T}_Y$ with $f(x) \in V$ there exist a $U \in \mathcal{T}_X$ such that $x \in U \wedge f(U) \subseteq V$*

Note 14.131. Be aware that the notation of continuity does depends on the function but also on the chosen topologies.

Definition 14.132. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $A \subseteq X$ then a function $f: X \rightarrow Y$ is continuous on A iff $\forall x \in A$ f is continuous at x*

Theorem 14.133. Let $\langle X, \mathcal{T}_X \rangle$, $\langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $\mathcal{B}_X$ a topological basis for $\mathcal{T}_X$, $\mathcal{B}_Y$ a topological basis for $\mathcal{T}_Y$, $x \in X$ and $f: X \rightarrow Y$ a function then we have the following equivalences:

1. f is continuous at x
2. $\forall V \in \mathcal{B}_Y$ with $f(x) \in V$ there exist a $U \in \mathcal{T}_X$ such that $x \in U$ and $f(U) \subseteq V$
3. $\forall V \in \mathcal{B}_Y$ with $f(x) \in V$ there exist a $U \in \mathcal{B}_X$ such that $x \in U$ and $f(U) \subseteq V$

Proof.

1 $\Rightarrow$ 2. This is trivial as $\mathcal{B}_Y \in \mathcal{T}_Y$.

2 $\Rightarrow$ 3. Let $V \in \mathcal{B}_Y$ with $f(x) \in V$ then by (2) there exist a $U' \in \mathcal{T}_X$ such that $x \in U' \wedge f(U') \subseteq V$, by [theorem: 14.36] there exist a $U \in \mathcal{B}_X$ with $x \in U \subseteq U'$ so that $f(U) \subseteq f(U') \subseteq V$.

3 $\Rightarrow$ 1. Let $V \in \mathcal{T}_Y$ with $f(x) \in V$ then by [theorem: 14.36] there exist a $W \in \mathcal{B}_Y$ such that $f(x) \in W \subseteq V$. Hence by (3) there exist a $U \in \mathcal{B}_X$ such that $x \in U$ and $f(U) \subseteq W \subseteq V$ proving that f is continuous at x . $\square$

If we work with metric or normed spaces then we can use the classical ε - δ definition of functions continuous at a point.

Theorem 14.134. Let $\langle X, d_X \rangle$, $\langle Y, d_Y \rangle$ be metric spaces, $x \in X$ and $f: X \rightarrow Y$ then we have

$f: X \rightarrow Y$ is continuous at x using the topologies $\mathcal{T}_{d_X}$, $\mathcal{T}_{d_Y}$

$\Updownarrow$

$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+$ such that $\forall z \in X$ satisfying $d_X(x, z) < \delta$ we have $d_Y(f(x), f(z)) < \varepsilon$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ then $f(x) \in B_{d_Y}(f(x), \varepsilon) \in \mathcal{B}_{d_Y}$, hence by [theorem: 14.133] there exists a $B \in \mathcal{B}_{d_X}$ such that $f(x) \in B$ and $f(B) \subseteq B_{d_Y}(f(x), \varepsilon)$, hence there exist a $y \in X$ and a $\delta' \in \mathbb{R}^+$ such that $x \in B_{d_X}(y, \delta') \in \mathcal{B}_{d_X}$ and $f(B_{d_X}(y, \delta')) \subseteq B_{d_Y}(f(x), \varepsilon)$. Using [theorem: 14.66] there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_{d_X}(x, \delta) \subseteq B_{d_X}(y, \delta')$. So we have

$$f(B_{d_X}(x, \delta)) \subseteq f(B_{d_X}(y, \delta')) \subseteq B_{d_Y}(f(x), \varepsilon)$$

So if $z \in X$ satisfies $d_X(x, z) < \delta$ then $z \in B_{d_X}(x, \delta)$ so that $f(z) \in B_{d_Y}(f(x), \varepsilon)$ or $d_Y(f(x), f(z)) < \varepsilon$.

$\Leftarrow$. Let $V \in \mathcal{T}_Y$ such that $f(x) \in V$ then by [theorem: 14.68] there exist a $\varepsilon \in \mathbb{R}^+$ such that $f(x) \in B_{d_Y}(f(x), \varepsilon) \subseteq V$. By the hypothesis there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall z \in X \text{ with } d_X(x, z) < \delta \text{ we have } d_Y(f(x), f(z)) < \varepsilon$$

If $y \in f(B_{d_X}(x, \delta))$ then there exist a $z \in X$ such that $d_X(x, z) < \delta$ and $y = f(z)$ hence using the above we have that $d_Y(f(x), f(z)) < \varepsilon$ or $y = f(z) \in B_{d_Y}(f(x), \varepsilon) \subseteq V$. As $y \in f(B_{d_X}(x, \delta))$ was chosen arbitrary it follows that

$$f(B_{d_X}(x, \delta)) \subseteq V \text{ where } x \in B_{d_X}(x, \delta) \in \mathcal{T}_X$$

So we have that $\forall V \in \mathcal{T}_Y$ with $f(x) \in V$ we found a $B_{d_X}(x, \delta) \in \mathcal{T}_X$ such that $f(B_{d_X}(x, \delta)) \subseteq V$ proving that f is continuous at x . $\square$

Theorem 14.135. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces, $x \in X$ and $f: X \rightarrow Y$ then we have

$f: X \rightarrow Y$ is continuous at x using the topologies $\mathcal{T}_{\|\cdot\|_X}$, $\mathcal{T}_{\|\cdot\|_Y}$

$\Updownarrow$

$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+$ such that $\forall y \in X$ satisfying $\|x - y\|_X < \delta$ we have $\|f(x) - f(y)\|_Y < \varepsilon$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 14.134] there exist a $\delta \in \mathbb{R}^+$ such that $\forall y \in X$ with $d_X(x, y) < \delta$ we have $d_Y(f(x), f(y)) < \varepsilon$. Hence if $\|x - y\|_X < \delta$ then $d_X(x, y) < \delta$ so that $\|f(x) - f(y)\|_Y = d_Y(f(x), f(y)) < \varepsilon$.

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$ then by the hypothesis there exist a $\delta \in \mathbb{R}^+$ such that for all $y \in X$ with $\|x - y\|_X < \delta$ we have $\|f(x) - f(y)\|_Y < \varepsilon$. So if $y \in X$ satisfies $d_{\|\cdot\|_X}(x, y) < \delta$ then $\|x - y\|_X < \delta$ and we have $d_{\|\cdot\|_Y}(f(x), f(y)) = \|f(x) - f(y)\|_Y < \varepsilon$. Hence by [theorem: 14.134] it follows that f is continuous at x . $\square$

The above works also for the subspace topology.

Corollary 14.136. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ then using the subspace topology $(\mathcal{T}_{\|\cdot\|_X})|_A$ on A and $\mathcal{T}_{\|\cdot\|_Y}$ on Y we have:

$$f: A \rightarrow Y \text{ is continuous at } x$$

$$\Updownarrow$$

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall y \in A \text{ satisfying } \|x - y\|_X < \delta \text{ we have } \|f(x) - f(y)\|_Y < \varepsilon$$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ and consider the open set $B_{\|\cdot\|_Y}(f(x), \varepsilon)$. As $f: A \rightarrow Y$ is continuous we have that there exist a $U \in (\mathcal{T}_{\|\cdot\|_X})|_A$ with $x \in U$ such that

$$f(U) \subseteq B_{\|\cdot\|_Y}(f(x), \varepsilon) \quad (14.53)$$

As $U \in (\mathcal{T}_{\|\cdot\|_X})|_A$ there exist a $U' \in \mathcal{T}_{\|\cdot\|_X}$ such that $U = U' \cap A$. As $x \in U$ we must have that $x \in U'$ so that, using [theorem: 14.97], there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_{\|\cdot\|_X}(x, \delta) \subseteq U'$. As $x \in A$ we have $x \in B_{\|\cdot\|_X}(x, \delta) \cap A \subseteq U' \cap A = U$ so that

$$f(x) \in f(B_{\|\cdot\|_X}(x, \delta) \cap A) \subseteq f(U) \underset{[\text{eq: 14.53}]}{\subseteq} B_{\|\cdot\|_Y}(f(x), \varepsilon) \quad (14.54)$$

So if $y \in A$ such that $\|x - y\|_X < \delta$ then

$$y \in B_{\|\cdot\|_X}(x, \delta) \cap A \Rightarrow f(y) \in f(B_{\|\cdot\|_X}(x, \delta) \cap A) \underset{[\text{eq: 14.54}]}{\subseteq} B_{\|\cdot\|_Y}(f(x), \varepsilon)$$

from which it follows that $\|f(x) - f(y)\|_Y < \varepsilon$.

$\Leftarrow$. Let $V \in \mathcal{T}_{\|\cdot\|_Y}$ with $f(x) \in V$. Using [theorem: 14.97] there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$f(x) \in B_{\|\cdot\|_Y}(f(x), \varepsilon) \subseteq V \quad (14.55)$$

By the hypothesis there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall y \in A \text{ with } \|x - y\|_X < \delta \text{ we have } \|f(x) - f(y)\|_Y < \varepsilon$$

As $B_{\|\cdot\|_X}(x, \delta) \in \mathcal{T}_{\|\cdot\|_X}$ we have that $x \in B_{\|\cdot\|_X}(x, \delta) \cap A \in (\mathcal{T}_{\|\cdot\|_X})|_A$. Let $z \in f(B_{\|\cdot\|_X}(x, \delta) \cap A)$ then there exist a $y \in B_{\|\cdot\|_X}(x, \delta) \cap A$ such that $z = f(y)$. As $y \in B_{\|\cdot\|_X}(x, \delta) \cap A$ we have $y \in A$ and $\|x - y\|_X < \delta$ so that $\|f(x) - z\|_Y = \|f(x) - f(y)\|_Y < \varepsilon$ from which it follows that $z \in B_{\|\cdot\|_Y}(f(x), \varepsilon) \subseteq V$. Hence

$$f(B_{\|\cdot\|_X}(x, \delta) \cap A) \subseteq V$$

In other words we found a $B_{\|\cdot\|_X}(x, \delta) \cap A \in (\mathcal{T}_{\|\cdot\|_X})|_A$ such that $x \in B_{\|\cdot\|_X}(x, \delta) \cap A$ and $f(B_{\|\cdot\|_X}(x, \delta) \cap A) \subseteq V$ which proves that f is continuous using the topologies $(\mathcal{T}_{\|\cdot\|_X})|_A$ and $\mathcal{T}_{\|\cdot\|_Y}$. $\square$

Example 14.137. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $A \subseteq X$, $\alpha \in \mathbb{R}$ and $z \in X$ then

$$f: A \rightarrow X \text{ defined by } f(x) = \alpha \cdot x + z$$

is continuous using the subspace topology $(\mathcal{T}_{|||})|_A$ on A and $\mathcal{T}_{|||}$ on X

Note 14.138. If $A = X$ then $(\mathcal{T}_{|||})|_A = \mathcal{T}_{|||}$ so that in that case we have that

$$f: X \rightarrow X \text{ defined by } f(x) = \alpha \cdot x + z$$

is a continuous function.

Proof. Let $x \in X$ then given $\varepsilon \in \mathbb{R}^+$ take $\delta = \frac{\varepsilon}{1+|\alpha|}$ then we have for $y \in A$ such that $\|x - y\| < \delta$ that

$$\|f(x) - f(y)\| = \|\alpha \cdot x + z - (\alpha \cdot y + z)\| = \|\alpha \cdot (x - y)\| = |\alpha| \cdot \|x - y\| < |\alpha| \cdot \delta = |\alpha| \cdot \frac{\varepsilon}{1+|\alpha|} < \varepsilon$$

which proves by [corollary: 14.136] that f is continuous at x . As $x \in A$ is chosen arbitrary it follows that f is a continuous function. $\square$

Example 14.139. Let $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ be the normed space of the real numbers and $\langle \mathbb{C}, \mathcal{T}_{||} \rangle$ be the topological space of the complex numbers then

$$\text{Re}: \mathbb{C} \rightarrow \mathbb{R} \text{ and } \text{Im}: \mathbb{C} \rightarrow \mathbb{R}$$

are continuous functions.

Proof. Using [examples: 14.99, 14.123] we have that $\mathcal{T}_{||}$ is generated by the absolute value norm and $\mathcal{T}_{\mathbb{C}}$ by the norm $|||_{\mathbb{C}}$ defined by $\|z\|_{\mathbb{C}} = \max(|\text{Re}(z)|, |\text{Im}(z)|)$. Let $x \in \mathbb{C}$ then for $\varepsilon \in \mathbb{R}^+$ we have that for $y \in \mathbb{C}$ such that $\|x - y\|_{\mathbb{C}} < \varepsilon$ that $\max(|\text{Re}(x - y)|, |\text{Im}(x - y)|) < \varepsilon$ so that

$$|\text{Re}(x) - \text{Re}(y)| \Big|_{\text{theorem: 10.80}} = |\text{Re}(x - y)| \leq \max(|\text{Re}(x - y)|, |\text{Im}(x - y)|) < \varepsilon$$

and

$$|\text{Im}(x) - \text{Im}(y)| \Big|_{\text{theorem: 10.80}} = |\text{Im}(x - y)| \leq \max(|\text{Re}(x - y)|, |\text{Im}(x - y)|) < \varepsilon$$

which proves that Re, Im are continuous at x . As $x \in \mathbb{C}$ was chosen arbitrary it follows that Re and Im are continuous functions. $\square$

Definition 14.140. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces then a function $f: X \rightarrow Y$ is **continuous** if $\forall x \in X$ f is continuous at x [in other words f is continuous on $\text{dom}(f)$]. The set of all the graphs of continuous functions from X to Y is noted as $\mathcal{C}(X, Y)$. So

$$\mathcal{C}(X, Y) = \{f \in Y^X \mid f: X \rightarrow Y \text{ is continuous}\}$$

Theorem 14.141. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces and $f: X \rightarrow Y$ a function then

$$\forall x \in X \text{ } f \text{ is continuous at } x \Leftrightarrow \forall V \in \mathcal{T}_Y \text{ we have } f^{-1}(V) \in \mathcal{T}_X$$

or using the above definition:

$$f \text{ is continuous} \Leftrightarrow \forall V \in \mathcal{T}_Y \text{ we have } f^{-1}(V) \in \mathcal{T}_X$$

Proof.

$\Rightarrow$. Let $V \in \mathcal{T}_Y$ then $\forall x \in f^{-1}(V) \subseteq X$ we have $f(x) \in V$ so that by continuity at x there exist a $U \in \mathcal{T}_X$ such that $x \in U \wedge f(U) \subseteq V$. Hence by a consequence of the axiom of choice [see: 3.108] there exist a $\{U_x\}_{x \in f^{-1}(V)} \subseteq \mathcal{T}_X$ such that $\forall x \in f^{-1}(V)$ $x \in U_x$ and $f(U_x) \subseteq V$. So $U_x \subseteq f^{-1}(f(U_x)) \subseteq f^{-1}(V)$ so that

$$\bigcup_{x \in f^{-1}(V)} U_x \subseteq f^{-1}(V)$$

Further if $x \in f^{-1}(V)$ then $x \in U_x \subseteq \bigcup_{x \in f^{-1}(V)} U_x$ so that $f^{-1}(V) \subseteq \bigcup_{x \in f^{-1}(V)} U_x$ or combining this with the above that

$$f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x \text{ where } \{U_x\}_{x \in f^{-1}(V)} \subseteq \mathcal{T}_X$$

so that by definition

$$f^{-1}(V) \in \mathcal{T}_X$$

$\Leftarrow$. Let $x \in X$ and $V \in \mathcal{T}_Y$ such that $f(x) \in V$ then $x \in f^{-1}(V)$. By the hypothesis we have $f^{-1}(V) \in \mathcal{T}_X$, so if we take $U = f^{-1}(V)$ then $U \in \mathcal{T}_X$ and $x \in U$. Further by [theorem: 2.26] we have $f(f^{-1}(V)) \subseteq V$ so that $f(U) = f(f^{-1}(V)) \subseteq V$ proving that f is continuous at x . $\square$

Example 14.142. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $y \in Y$ then $C_y: X \rightarrow Y$ defined by $C_y(x) = y$ is continuous.

Proof. If $V \in \mathcal{T}_X$ then we have either:

$y \in V$. Then $f^{-1}(V) = X \in \mathcal{T}_X$

$y \notin V$. Then $f^{-1}(V) = \emptyset \in \mathcal{T}_X$

So using [theorem: 14.141] it follows that C_y is continuous. $\square$

Example 14.143. Let $\langle X, \mathcal{T}_X \rangle$ be a topological space then $\text{Id}_X: X \rightarrow X$ is continuous.

Proof. Let $V \in \mathcal{T}_X$ then $(\text{Id}_X)^{-1}(V) = V \in \mathcal{T}_X$ proving by [theorem: 14.141] that Id_X is continuous. $\square$

Example 14.144. Let $\langle X, ||| \rangle$ be a normed space, $x \in X$ then the **translation function** defined by

$$\tau_x: X \rightarrow X \text{ where } \tau_x(y) = x + y$$

is continuous.

Proof. Let U be a open set in X . If $y \in \tau_x^{-1}(U)$ then $x + y = \tau_x(y) \in U$, $y = -x + (x + y) \in -x + U$ [see [definition: 11.74] for the definition of $-x + U$]. On the other hand if $y \in -x + U$ then there exist a $u \in U$ such that $y = -x + u \Rightarrow \tau_x(y) = x + y = x + (-x + u) = u \in U$ hence $y \in \tau_x^{-1}(U)$. So we have that

$$\tau_x^{-1}(U) = -x + U \quad \square$$

Theorem 14.145. Let $\langle X, \mathcal{T}_X \rangle$ and $\langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $B \subseteq Y$ and $f: X \rightarrow Y$ a function such that $f(X) \subseteq B$ then we have

$$\begin{aligned} f: X \rightarrow Y \text{ is continuous using the topologies } \mathcal{T}_X \text{ on } X \text{ and } \mathcal{T}_Y \text{ on } Y \\ \Updownarrow \\ f: X \rightarrow B \text{ is continuous using the topologies } \mathcal{T}_X \text{ on } X \text{ and } (\mathcal{T}_Y)_{|B} \text{ on } B \end{aligned}$$

Proof.

$\Rightarrow$. Let $V \in (\mathcal{T}_Y)_{|B}$ then there exist a open set $V' \in \mathcal{T}_Y$ such that $V = V' \cap B$. As $f: X \rightarrow Y$ is continuous we have by [theorem: 14.8] that

$$f^{-1}(V') \in \mathcal{T}_X$$

As $f(X) \subseteq B$ we have that $f^{-1}(B) = X$ so that

$$f^{-1}(V' \cap B) \stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(V') \cap f^{-1}(B) = f^{-1}(V') \cap X = f^{-1}(V') \in \mathcal{T}_X$$

So using [theorem: 14.8] again it follows that $f: X \rightarrow B$ is continuous using the topology $\mathcal{T}_X$ on X and the subspace topology $(\mathcal{T}_Y)_{|B}$ on B .

$\Leftarrow$. Assume that $f: X \rightarrow B$ is continuous [using the topology $\mathcal{T}_X$ on X and $(\mathcal{T}_Y)_{|B}$ on B]. Let $V \in \mathcal{T}_Y$ then $V \cap Y \in \mathcal{T}_Y$ so that $f^{-1}(V \cap Y) \in \mathcal{T}_X$. As

$$f^{-1}(V \cap Y) \stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(V) \cap f^{-1}(Y) = f^{-1}(V) \cap X = f^{-1}(V)$$

so that $f^{-1}(V) \in \mathcal{T}_X$ which proves that $f: X \rightarrow Y$ is continuous. $\square$

Theorem 14.146. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $A \subseteq X$, $x \in A$ and $f: X \rightarrow Y$ a function that is continuous at x then

$$f|_A: A \rightarrow Y$$

is continuous at x using the subspace topology $(\mathcal{T}_X)|_A = \{A \cap U \mid U \in \mathcal{T}_X\}$ on A and the topology $\mathcal{T}_Y$ on Y .

Proof. Let $V \in \mathcal{T}_Y$ such that $f|_A(x) \in V$. Then $f(x) \in V$ and by continuity of f at x there exist a $U \in \mathcal{T}_X$ with $x \in U$ and $f(U) \subseteq V$. As $x \in A$ we have that $x \in A \cap U \in (\mathcal{T}_X)|_A$ and $f|_A(A \cap U) \stackrel{[theorem: 2.92]}{=} f(A \cap U) \subseteq f(U) \subseteq V$ proving that $f|_A$ is continuous at x . $\square$

Corollary 14.147. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $A, B \subseteq X$ with $B \subseteq A$, $x \in B$ and $f: A \rightarrow Y$ a function that is continuous at x [using the topologies $(\mathcal{T}_X)|_A$ on A and $\mathcal{T}_Y$ on Y] then $f|_B: B \rightarrow Y$ is continuous at x [using the topologies $(\mathcal{T}_X)|_B$ on B and $\mathcal{T}_Y$ on Y]

Proof. By the previous theorem [theorem: 14.146] $f|_B$ is continuous at x using the topologies $((\mathcal{T}_X)|_A)|_B$ on B and $\mathcal{T}_Y$ on Y . By [theorem: 14.10] $((\mathcal{T}_X)|_A)|_B = (\mathcal{T}_X)|_B$ so that $f|_B$ is continuous at x using the topologies $(\mathcal{T}_X)|_B$ on B and $\mathcal{T}_Y$ on Y . $\square$

Theorem 14.148. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $A \subseteq X$ and $f: X \rightarrow Y$ a function that is continuous on A then $f|_A: A \rightarrow Y$ is continuous using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$.

Proof. Let $x \in A$ then, as f is continuous on A , f is continuous at x . Using [theorem: 14.146] it follows then that $f|_A: A \rightarrow Y$ is continuous at x using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$. As this is true for every $x \in A$ it follows that $f|_A$ is continuous using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$. $\square$

The opposite of [theorem: 14.146] is true if A is open showing that continuity at a point is a local property.

Theorem 14.149. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $U \in \mathcal{T}_X$, $x \in U$ and $f: X \rightarrow Y$ a function such that $f|_U: U \rightarrow Y$ is continuous at x then f is continuous at x .

Proof. Let $V \in \mathcal{T}_Y$ such that $f(x) \in V$ then as $f|_U$ is continuous at x there exist a $W' \in (\mathcal{T}_X)|_U$ such that $x \in W'$ and $f|_U(W') \subseteq V \Rightarrow f(W') \subseteq V$. As $W' \in (\mathcal{T}_X)|_U$ there exist a $W \in \mathcal{T}_X$ such that $W' = W \cap U$, as $U \in \mathcal{T}_X$ it follows that $W' \in \mathcal{T}_X$. So we found a $W' \in \mathcal{T}_X$ such that $x \in W'$ and $f(W') \subseteq V$ proving that f is continuous at x . $\square$

Corollary 14.150. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $A \subseteq X$, $U \in \mathcal{T}_X$ with $U \subseteq A$, $x \in U$ and $f: A \rightarrow Y$ a function such that $f|_U: U \rightarrow Y$ is continuous at x using the topologies $(\mathcal{T}_X)|_U$ on U and $\mathcal{T}_Y$ on Y then $f: A \rightarrow Y$ is continuous at x using the topologies $(\mathcal{T}_X)|_A$ on A and $\mathcal{T}_Y$ on Y .

Proof. Let $V \in \mathcal{T}_Y$ such that $f|_U(x) \in V$ then by continuity there exist a open a $W' \in (\mathcal{T}_X)|_U$ such that $x \in W'$ and $f|_U(W') \subseteq V \Rightarrow f(W') \subseteq V$. As $W' \in (\mathcal{T}_X)|_U$ there exist a $W \in \mathcal{T}_X$ such that $W' = W \cap U$. As $W, U \in \mathcal{T}_X$ we have that $W' \in \mathcal{T}_X$. So if we take $O = W' \cap A$ then $O \in (\mathcal{T}_X)|_A$, $x \in O$ and $f(O) \subseteq f(W') \subseteq V$. Hence f is continuous at x using the topologies $(\mathcal{T}_X)|_A$ on A and $\mathcal{T}_Y$ on Y . $\square$

Theorem 14.151. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $U \in \mathcal{T}_X$ and $f: X \rightarrow Y$ a function then we have

$$\begin{aligned} f \text{ is continuous on } U \\ \Updownarrow \\ f|_U \text{ is continuous using the topologies } (\mathcal{T}_X)|_U \text{ and } \mathcal{T}_Y \end{aligned}$$

Proof.

$\Rightarrow$. This follows from [theorem: 14.148].

$\Leftarrow$. Let $x \in U$ then by the continuity of $f|_U$ it follows from [theorem: 14.149] that f is continuous at x . As $x \in U$ has been chosen arbitrary we must conclude that f is continuous on U . $\square$

Theorem 14.152. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $A \subseteq X$ and $f: X \rightarrow Y$ a continuous function then

$$f|_A: A \rightarrow Y$$

is a continuous function using the subspace topology $(\mathcal{T}_X)|_A = \{A \cap U \mid U \in \mathcal{T}_X\}$ on A and the topology $\mathcal{T}_Y$ on Y .

Proof. Let $V \in \mathcal{T}_Y$ then we have $(f|_A)^{-1}(V) \stackrel{[\text{theorem: 2.92}]}{=} A \cap f^{-1}(V)$. As f is continuous $f^{-1}(V) \in \mathcal{T}_X$ so that $A \cap f^{-1}(V) \in (\mathcal{T}_X)|_A$ proving that $(f|_A)^{-1}(V) \in (\mathcal{T}_X)|_A$, hence by [theorem: 14.141] $f|_A$ is continuous. $\square$

Corollary 14.153. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $B \subseteq A \subseteq X$ and $f: X \rightarrow Y$ a function such that $f|_A: A \rightarrow Y$ is continuous using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$ then $f|_B: B \rightarrow Y$ is continuous using the topologies $(\mathcal{T}_X)|_B$ and $\mathcal{T}_Y$.

Proof. As $f|_A: A \rightarrow Y$ is continuous using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$ then by [theorem: 14.152] $(f|_A)|_B: B \rightarrow Y$ is continuous using the topologies $((\mathcal{T}_X)|_A)|_B$ and $\mathcal{T}_Y$. By [theorem: 2.92(3)] $(f|_A)|_B = f|_B$ and by [theorem: 14.10] $((\mathcal{T}_X)|_A)|_B = (\mathcal{T}_X)|_B$. Hence $f|_B: B \rightarrow Y$ is continuous using the topologies $(\mathcal{T}_X)|_B$ and $\mathcal{T}_Y$. $\square$

Theorem 14.154. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces then the following are equivalent:

1. f is continuous
2. $\forall A \subseteq X$ we have $f(\overline{A}) \subseteq \overline{f(A)}$
3. $\forall F \subseteq Y$ F closed we have that $f^{-1}(F)$ is closed

Proof.

1 $\Rightarrow$ 2. Let $y \in f(\overline{A})$ then there exist a $x \in \overline{A}$ such that $y = f(x)$. Let $V \in \mathcal{T}_Y$ such that $y \in V$ then, as f is continuous, there exist a $U \in \mathcal{T}_X$ such that $x \in U$ and $f(U) \subseteq V$. Given that $x \in \overline{A}$ we have by [theorem: 14.31] that $A \cap U \neq \emptyset$ so that $\emptyset \neq f(A \cap U)$. By [theorem: 2.103] $f(A \cap U) \subseteq f(A) \cap f(U)$ so that $\emptyset \neq f(A) \cap f(U) \subseteq f(A) \cap V$ proving that $y \in \overline{f(A)}$. As $y \in f(\overline{A})$ was chosen arbitrary it follows that

$$f(\overline{A}) \subseteq \overline{f(A)}$$

2 $\Rightarrow$ 3. Let $A \subseteq Y$ be closed then by (2)

$$f(\overline{f^{-1}(A)}) \subseteq \overline{f(f^{-1}(A))} \tag{14.56}$$

Further by [theorem: 2.26] we have that $f(f^{-1}(A)) \subseteq A$ so that by [corollary: 14.32] we have $\overline{f(f^{-1}(A))} \subseteq \overline{A} \stackrel{[\text{theorem: 14.26}]}{=} A$ or combining this with [eq: 14.56] gives

$$f(\overline{f^{-1}(A)}) \subseteq A$$

Hence by [theorem: 2.102] we have that $f^{-1}(f(\overline{f^{-1}(A)})) \subseteq f^{-1}(A)$ and by [theorem: 2.26] $\overline{f^{-1}(A)} \subseteq f^{-1}(f(\overline{f^{-1}(A)}))$ so that $\overline{f^{-1}(A)} \subseteq f^{-1}(A)$. As by [theorem: 14.29] $f^{-1}(A) \subseteq \overline{f^{-1}(A)}$ it follows that $f^{-1}(A) = \overline{f^{-1}(A)}$. Hence by [theorem: 14.26] that

$$f^{-1}(A) \text{ is closed}$$

3 $\Rightarrow$ 1. If $V \in \mathcal{T}_Y$ then by [theorem: 14.21] $Y \setminus V$ is closed so by (3) we have that $f^{-1}(Y \setminus V)$ is closed so that $X \setminus f^{-1}(Y \setminus V) \in \mathcal{T}_X$. Finally by [theorem: 2.102] we have that

$$f^{-1}(V) = f^{-1}(Y \setminus (Y \setminus V)) = X \setminus f^{-1}(Y \setminus V) \in \mathcal{T}_X$$

proving that for every $V \in \mathcal{T}_Y$ we have $f^{-1}(V) \in \mathcal{T}_X$. Hence by [theorem: 14.141] $f: X \rightarrow Y$ is continuous. $\square$

The following lemma shows how we can stitch together two continuous functions to a continuous function.

Lemma 14.155. (Pasting Lemma) *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be two topological spaces, A, B two closed sets in X such that $f: A \rightarrow Y$ and $g: B \rightarrow Y$ two functions such that:*

1. *f is continuous using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$.*
2. *g is continuous using the topologies $(\mathcal{T}_X)|_B$ and $\mathcal{T}_Y$.*
3. *$\forall x \in A \cap B$ we have that $f(x) = g(x)$*

then we have that [see theorem: 2.90]

$$h: A \cup B \rightarrow Y \text{ defined by } h = f \cup g$$

is a well defined continuous function [using the topologies $(\mathcal{T}_X)|_{A \cup B}, \mathcal{T}_Y$

Note 14.156. By the note in [theorem: 2.90] we can define h also by $h(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B \end{cases}$ which is a less abstract definition of h .

Proof. First using [theorem: 2.90(1)] it follows that

$$h: A \cup B \rightarrow Y$$

is indeed a function. Let C be a closed set in Y then, as f and g are continuous, we have that $f^{-1}(C)$ is a closed set in $(\mathcal{T}_X)|_A$ and $g^{-1}(C)$ is a closed set in $(\mathcal{T}_X)|_B$. Hence, using [theorem: 14.18] there exist closed sets C_f, C_g in $\mathcal{T}_X$ such that $f^{-1}(C) = C_f \cap A$ and $g^{-1}(C) = C_g \cap B$. As A, B are closed sets in $\mathcal{T}_X$ it follows that $C_f \cap A$ and C_g are closed sets in $\mathcal{T}_X$. Hence we have that $f^{-1}(C), g^{-1}(C)$ are closed sets in $\mathcal{T}_X$ so that

$$f^{-1}(C) \cup g^{-1}(C) \text{ is closed in } \mathcal{T}_X$$

Using then [theorem: 14.18] it follows that

$$(f^{-1}(C) \cup g^{-1}(C)) \cap (A \cup B) \text{ is closed in } (\mathcal{T}_X)|_{A \cup B}$$

Now, as $f^{-1}(C) \subseteq A, g^{-1}(C) \subseteq B$, it follows that $f^{-1}(C) \cup g^{-1}(C) \subseteq A \cup B$ so that

$$(f^{-1}(C) \cup g^{-1}(C)) \cap (A \cup B) = f^{-1}(C) \cup g^{-1}(C)$$

which proves that

$$f^{-1}(C) \cup g^{-1}(C) \text{ is closed in } (\mathcal{T}_X)|_{A \cup B}$$

Now using [theorem: 2.90(2)] it follows that $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$ so that

$$h^{-1}(C) \text{ is closed in } (\mathcal{T}_X)|_{A \cup B}$$

To summarize we have proved that

$$\forall C \text{ closed in } \mathcal{T}_Y \text{ we have that } h^{-1}(C) \text{ is closed in } (\mathcal{T}_X)|_{A \cup B}$$

Applying then [theorem: 14.154] that

$$h \text{ is continuous using the topologies } (\mathcal{T}_X)|_{A \cup B} \text{ and } \mathcal{T}_Y$$

$\square$

If a function is continuous then the preimage of a open set is open, we can consider also functions so that the image of a open set is open. This is the idea of a open function.

Definition 14.157. (Open Function) *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces then a function*

$$f: X \rightarrow Y \text{ is open}$$

if $\forall U \in \mathcal{T}_X$ we have that $f(U) \in \mathcal{T}_Y$

The composition of two continuous or open functions is again a continuous or open function.

Theorem 14.158. *Let $\langle X, \mathcal{T}_X \rangle$, $\langle Y, \mathcal{T}_Y \rangle$ and $\langle Z, \mathcal{T}_Z \rangle$ be topological spaces then we have:*

1. *If $f: X \rightarrow Y$ is a function that is continuous at x and $g: Y \rightarrow Z$ a function that is continuous at $f(x)$ then $g \circ f$ is continuous at x .*
2. *If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are continuous functions then $g \circ f: X \rightarrow Z$ is continuous.*
3. *If $A \subseteq Y$ and $f: X \rightarrow Y$ is a continuous function such that $f(X) \subseteq A$ and $g: A \rightarrow Z$ a continuous function [using the topologies $(\mathcal{T}_Y)|_A$ on A and $\mathcal{T}_Z$ on Z] then $g \circ f: X \rightarrow Z$ is continuous.*
4. *If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are open functions then $g \circ f: X \rightarrow Z$ is open*

Proof.

1. Let $W \in \mathcal{T}_Z$ such that $(g \circ f)(x) \in W$ then $g(f(x)) \in W$. As g is continuous at $f(x)$ there exist a open set $V \in \mathcal{T}_Y$ such that $f(x) \in V$ and $g(V) \subseteq W$. Using the fact that f is continuous at x there exist a open set $U \in \mathcal{T}_X$ such that $x \in U$ and $f(U) \subseteq V$. Further we have

$$(g \circ f)(U) \stackrel{[\text{theorem: 2.23}]}{=} g(f(U)) \stackrel{[\text{theorem: 2.17}]}{\subseteq} g(V) \subseteq W$$

which proves that $g \circ f$ is continuous at x .

2. Let $V \in \mathcal{T}_Z$ then, as g is continuous we have by [theorem: 14.141] that $g^{-1}(V) \in \mathcal{T}_Y$, applying [theorem: 14.141] again results in $f^{-1}(g^{-1}(V)) \in \mathcal{T}_X$. As

$$(g \circ f)^{-1}(V) \stackrel{[\text{theorem: 2.23}]}{=} f^{-1}(g^{-1}(V)) \in \mathcal{T}_X$$

we have by [theorem: 14.141] that $g \circ f$ is continuous.

3. As $f: X \rightarrow Y$ is continuous and $f(X) \subseteq A$ we have by [theorem: 14.145] that $f: X \rightarrow A$ is continuous using the topologies $\mathcal{T}_X$ on X and $(\mathcal{T}_Y)|_A$. Hence we have by (2) that $g \circ f: X \rightarrow Z$ is continuous.
4. Let $U \in \mathcal{T}_Z$ then, as f is open, $f(U) \in \mathcal{T}_Y$, hence as g is open we have that

$$(g \circ f)(U) \stackrel{[\text{theorem: 2.23}]}{=} g(f(U)) \in \mathcal{T}_Z$$

proving that $g \circ f$ is open. □

We have a more relaxed version of the open theorem

Theorem 14.159. *Let $\langle X, \mathcal{T}_X \rangle$, $\langle Y, \mathcal{T}_Y \rangle$ and $\langle Z, \mathcal{T}_Z \rangle$ be topological spaces, $V \in \mathcal{T}_Y$ and $f: X \rightarrow Y$, $g: V \rightarrow Z$ functions with $f(X) \subseteq V$ then we have:*

1. *If $x \in X$ and f is continuous at x [using the topologies $\mathcal{T}_X, \mathcal{T}_Y$] and g is continuous at $f(x)$ [using the the topologies $(\mathcal{T}_Y)|_V, \mathcal{T}_Z$] then $g \circ f$ is continuous at x [using the topologies $\mathcal{T}_X, \mathcal{T}_Z$.*
2. *If f is continuous using the topologies $\mathcal{T}_X, \mathcal{T}_Y$ and g is continuous using the topologies $(\mathcal{T}_Y)|_V, \mathcal{T}_Z$ then $g \circ f$ is continuous using the topologies $(\mathcal{T}_X)|_U, \mathcal{T}_Z$.*

Proof.

1. Let $W \in \mathcal{T}_Z$ such that $g(f(x)) \in W$ then as g is continuous at $f(x)$ it follows that there exist a $V' \in (\mathcal{T}_Y)|_V$ such that $f(x) \in V'$ and $g(V') \subseteq W$. As $V \in \mathcal{T}_Y$ it follows from [theorem: 14.11] that $(\mathcal{T}_Y)|_V \subseteq \mathcal{T}_Y$ so that $V' \in \mathcal{T}_Y$. Hence as f is continuous at x there exist a $U \in \mathcal{T}_X$ such that $x \in U$ and $f(U) \subseteq V'$. Further we have

$$(g \circ f)(U) \stackrel{[\text{theorem: 2.23}]}{=} g(f(U)) \stackrel{[\text{theorem: 2.17}]}{\subseteq} g(V') \subseteq W$$

which by definition means that $g \circ f$ is continuous at x .

2. Let W be a open set in $\mathcal{T}_Z$ then as g is continuous it follows from [theorem: 14.141] that $g^{-1}(W) \in (\mathcal{T}_Y)_|V$. As $V \in \mathcal{T}_Y$ it follows from [theorem: 14.11] that $g^{-1}(W) \in \mathcal{T}_Y$. Hence as f is continuous it follows from [theorem: 14.141] that $f^{-1}(g^{-1}(W)) \in \mathcal{T}_X$. So

$$(g \circ f)^{-1}(W) \stackrel{[\text{theorem: 2.23}]}{=} f^{-1}(g^{-1}(W)) \in \mathcal{T}_X$$

which by [theorem: 14.141] proves that that $g \circ f$ is continuous using the topologies $\mathcal{T}_X, \mathcal{T}_Z$. $\square$

One of the reasons to use the product topology on a product of topological spaces is that the projection map is open and continuous.

Theorem 14.160. *Let $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces and $\langle \prod_{i \in I} X_i, \mathcal{T}_{\text{product}} \rangle$ be the product topological space [see definition: 14.47] then $\forall i \in I$ we have that the projection map*

$$\pi_i: \prod_{j \in I} X_j \rightarrow X_i \text{ defined by } \pi_i(x) = x_i$$

is open and continuous.

Proof. Let $S = \{\pi_i^{-1}(U) | i \in I \wedge U \in \mathcal{T}_i\}$ then by [theorem: 14.44] we have

$$S \subseteq \mathcal{B}[S] \subseteq \mathcal{T}[\mathcal{B}[S]] \stackrel{[\text{definition: 14.47}]}{=} \mathcal{T}_{\text{product}}$$

So if $i \in I$ then for $U \in \mathcal{T}_i$ we have $\pi_i^{-1}(U) \in S \subseteq \mathcal{T}_{\text{product}}$ so that

$$\forall i \in I \text{ we have } \pi_i \text{ is continuous} \quad (14.57)$$

Let $U \in \mathcal{T}_{\text{product}}$ and let $t \in \pi_i(U)$ then there exist a $x \in U$ so that $t = \pi_i(x) = x_i$. By [theorem: 14.49] we have that

$$\mathcal{B}[S] = \left\{ \prod_{i \in I} U_i | \{U_i\}_{i \in I} \subseteq \mathcal{P}\left(\prod_{i \in I} X_i\right) \text{ is such that there exist a finite } J \subseteq I \text{ with } \forall i \in J U_i \in \mathcal{T}_i \text{ and } \forall i \in I \setminus J U_i = X_i \right\}$$

Hence there exist J finite and a $\{U_i\}_{i \in I}$ satisfying $U_i \in \mathcal{T}_i$ if $i \in J$ and $U_i = X_i \in \mathcal{T}_i$ if $i \in I \setminus J$ such that

$$x \in \prod_{i \in I} U_i \subseteq U$$

so $t = \pi_i(x) = x_i \in U_i \subseteq \pi_i(\prod_{i \in I} U_i) \subseteq \pi_i(U)$. To summarize we have

$$\forall t \in \pi_i(U) \text{ there exist a } U_i \in \mathcal{T}_i \text{ such that } t \in U_i \subseteq \pi_i(U)$$

which by [theorem: 14.7] proves that $\pi_i(U) \in \mathcal{T}_i$. Hence

$$\forall i \in I \text{ we have that } \pi_i \text{ is open} \quad \square$$

Corollary 14.161. *Let $\langle X, \mathcal{T}_X \rangle$ be a topological space, $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces, $\langle \prod_{i \in I} X_i, \mathcal{T}_{\text{product}} \rangle$ be the product topological space [see definition: 14.47] $x \in U$ and*

$$f: X \rightarrow \prod_{i \in I} X_i \text{ a function}$$

then

$$f \text{ is continuous at } x \Leftrightarrow \forall i \in I f_i = \pi_i \circ f \text{ is continuous at } x$$

Proof. Let $x \in X$

$\Rightarrow$. Let $i \in I$ then as f is continuous at x and by [theorem: 14.160] π_i is continuous hence continuous at $f(x)$ we have by [theorem: 14.158] that $\pi_i \circ f$ is continuous at x .

$\Leftarrow$. Using [definitions: 14.47] we have that the basis for $\mathcal{T}_{\text{product}}$ is

$$\mathcal{B}_{\text{product}} = \left\{ B \in \mathcal{P}(X) | \exists \{S_i\}_{i \in J} \subseteq \mathcal{S}, J \text{ finite and } J \neq \emptyset \text{ such that } B = \bigcap_{i \in J} S_i \right\}$$

where

$$S = \{\pi_i^{-1}(U) | i \in I \wedge U \in \mathcal{T}_i\}$$

Let $x \in X$ and $V \in \mathcal{T}_{\text{product}}$ such that $f(x) \in V$ then by [theorem: 14.36] there exists a finite J , $\{S_i\}_{i \in J} \subseteq S$ such that

$$f(x) \in \bigcap_{i \in J} S_i \subseteq V \text{ and } \bigcap_{i \in J} S_i \in \mathcal{B}_{\text{product}} \subseteq \mathcal{T}_{\text{product}} \quad (14.58)$$

Let $i \in J$ then $\exists i_j \in I$ and a $U_j \in \mathcal{T}_{i_j}$ such that $S_i = \pi_{i_j}^{-1}(U_j)$ and as $f(x) \in \bigcap_{i \in J} S_i \subseteq V$ we have also $f(x) \in S_i$. Hence we have

$$x \in f^{-1}(S_i) = f^{-1}(\pi_{i_j}^{-1}(U_j)) = (\pi_{i_j} \circ f)^{-1}(U_j) = (f_{i_j})^{-1}(U_j)$$

As f_{i_j} is continuous at x and $U_j \in \mathcal{T}_j$ we have that $f_{i_j}^{-1}(U_j) \in \mathcal{T}_X$ so, as J is finite, it follows that, if we take $U = \bigcap_{i \in J} f^{-1}(S_i)$, that

$$U = \bigcap_{i \in J} f^{-1}(S_i) = \bigcap_{i \in J} (f_{i_j})^{-1}(U_j) \in \mathcal{T}_X$$

and as $x \in f^{-1}(S_i) \forall i \in J$ that

$$x \in \bigcap_{i \in J} f^{-1}(S_i) = U$$

Further

$$\begin{aligned} f(U) &= f\left(\bigcap_{i \in J} f^{-1}(S_i)\right) \\ &\subseteq_{[\text{theorem: 2.144}]} \bigcap_{i \in J} f(f^{-1}(S_i)) \\ &\subseteq_{[\text{theorem: 2.26}]} \bigcap_{i \in J} S_i \\ &\subseteq_{[\text{eq: 14.58}]} V \end{aligned}$$

Hence we have proved that $\forall V \in \mathcal{T}_{\text{product}}$ with $x \in V$ there exist a $U \in \mathcal{T}_X$ such that $x \in U$ and $f(U) \subseteq V$ which proves that f is continuous at x . $\square$

Theorem 14.162. Let $\langle X_1, \mathcal{T}_1 \rangle, \langle X_2, \mathcal{T}_2 \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $\langle \prod_{i \in \{1,2\}} X_i, \mathcal{T}_{\text{product}} \rangle$ the topological space based on the product of X_1, X_2 and

$$f: \prod_{i \in \{1,2\}} X_i \rightarrow Y \text{ a continuous function}$$

then we have

1. $\forall x \in X_1$ $f_1(x): X_2 \rightarrow Y$ defined by $f_1(x)(y) = f(x, y)$ is continuous.
2. $\forall x \in X_2$ $f_2(x): X_1 \rightarrow Y$ defined by $f_2(x)(y) = f(y, x)$ is continuous.

Proof. First as $\{1, 2\}$ is finite we have that $\mathcal{T}_{\text{product}} \stackrel{[\text{theorem: 14.51}]}{=} \mathcal{T}_{\text{box}}$ so that the basis for $\mathcal{T}_{\text{product}}$ is

$$\mathcal{B} \stackrel{[\text{definition: 14.45}]}{=} \left\{ \prod_{i \in I} U_i | \{U_i\}_{i \in I} \subseteq \mathcal{P}\left(\prod_{i \in I} X_i\right) \text{ is a family such that } \forall i \in I U_i \in \mathcal{T}_i \right\} \quad (14.59)$$

1. Let $x \in X_1$ and let $V \in \mathcal{T}_Y$ then by continuity of f we have that $f^{-1}(V) \in \mathcal{T}_{\text{product}}$. Take now $y \in f_1(x)^{-1}(V)$ then $f(x, y) = f_1(x)(y) \in V$ so that $(x, y) \in f^{-1}(V)$ and by [eq: 14.59] there exists a $U_1 \in \mathcal{T}_1, U_2 \in \mathcal{T}_2$ such that $(x, y) \in \prod_{i \in \{1,2\}} U_i \subseteq f^{-1}(V)$. Hence $x \in U_1$ and $y \in U_2$. If $t \in U_2$ then $(x, t) \in \prod_{i \in \{1,2\}} U_i \subseteq f^{-1}(V)$ hence $f_1(x)(t) = f(x, t) \in V$ or $t \in f_1(x)^{-1}(V)$ proving that $y \in U_2 \subseteq f_1(x)^{-1}(V)$ which by [theorem: 14.7] proves that $f_1(x)^{-1}(V)$ is open. Hence by [theorem: 14.141]

$f_1(x)$ is continuous

2. Let $x \in X_2$ and let $V \in \mathcal{T}_Y$ then by continuity of f we have that $f^{-1}(V) \in \mathcal{T}_{\text{product}}$. Take now $y \in f_2(x)^{-1}(V)$ then $f(y, x) = f_2(x)(y) \in V$ so that $(y, x) \in f^{-1}(V)$ and by [eq: 14.59] there exists a $U_1 \in \mathcal{T}_1$, $U_2 \in \mathcal{T}_2$ such that $(y, x) \in \prod_{i \in \{1,2\}} U_i \subseteq f^{-1}(V)$. Hence $x \in U_2$ and $y \in U_1$. If $t \in U_1$ then $(t, x) \in \prod_{i \in \{1,2\}} U_i \subseteq f^{-1}(V)$ hence $f_2(x)(t) = f(t, x) \in V$ or $t \in f_2(x)^{-1}(V)$ proving that $y \in U_1 \subseteq f_2(x)^{-1}(V)$ which by [theorem: 14.7] proves that $f_2(x)^{-1}(V)$ is open. Hence by [theorem: 14.141]

$f_2(x)$ is continuous

□

Theorem 14.163. Let $\langle X, ||| \rangle$ be a pseudo normed space then

$$\cdot: \mathbb{K} \times X \rightarrow X \text{ defined by } \cdot(\alpha, x) = \alpha \cdot x$$

is continuous [using the product topology based on $\mathcal{T}_{||}$, $\mathcal{T}_{|||}$ and $\mathcal{T}_{|||}$]

Proof. Using [theorem: 14.98] the product topology on $\mathcal{T}_{\text{product}} = \mathcal{T}_{||| \max}$ where for $(\alpha, x) \in \mathbb{K} \times X$ $||(\alpha, x)||_{\max} = \max(\{|\alpha|, \|x\|\})$. Let $(\alpha, x) \in \mathbb{K} \times X$ and take $\varepsilon \in \mathbb{R}^+$. If $(\beta, y) \in \mathbb{K} \times X$ is such that $\|x - y\| \leq \delta$ and $|\alpha - \beta| < \delta$ then we have

$$\begin{aligned} \|\cdot(\alpha, x) - \cdot(\beta, y)\| &= \|\alpha \cdot x - \beta \cdot y\| \\ &= \|\alpha \cdot x - \beta \cdot x + \beta \cdot x - \beta \cdot y\| \\ &= \|(\alpha - \beta) \cdot x + \beta \cdot (x - y)\| \\ &\leq |\alpha - \beta| \cdot \|x\| + |\beta| \cdot \|x - y\| \\ &= |\alpha - \beta| \cdot \|x\| + |\beta - \alpha + \alpha| \cdot \|x - y\| \\ &\leq |\alpha - \beta| \cdot \|x\| + (|\beta - \alpha| + |\alpha|) \cdot \|x - y\| \\ &= |\alpha - \beta| \cdot \|x\| + |\beta - \alpha| \cdot \|x - y\| + |\alpha| \cdot \|x - y\| \\ &< \delta \cdot \|x\| + \delta \cdot \delta + |\alpha| \cdot \delta \\ &= \delta \cdot (\|x\| + \delta + |\alpha|) \end{aligned}$$

Take now $\delta = \min(1, \varepsilon / (\|x\| + |\alpha| + 1)) > 0$ then we have that

$$\begin{aligned} \delta \cdot (\|x\| + \delta + |\alpha|) &<_{\delta \leq 1} \delta \cdot (\|x\| + 1 + |\alpha|) \\ &< (\varepsilon / (\|x\| + |\alpha| + 1)) \cdot (\|x\| + 1 + |\alpha|) \\ &= \varepsilon \end{aligned}$$

So we have proved that for $\delta = \min(1, \varepsilon / (\|x\| + |\alpha| + 1))$ and $(\beta, y) \in \mathbb{K} \times X$ such that $\|x - y\| \leq \delta$ and $|\alpha - \beta| < \delta$ that

$$\|\cdot(\alpha, x) - \cdot(\beta, y)\| < \varepsilon$$

So if we take $(\beta, y) \in \mathbb{K} \times X$ such that $||(\alpha, x) - (\beta, y)||_{\max} < \delta$ then $|\alpha - \beta| < \delta$ and $\|x - y\| < \delta$ resulting in $\|\cdot(\alpha, x) - \cdot(\beta, y)\| < \varepsilon$. Applying then [theorem: 14.135] proves that $\cdot$ is continuous at $(\alpha, x) \in \mathbb{K} \times X$ which as (α, x) was chosen arbitrary proves continuity. □

Corollary 14.164. Let $\langle X, ||| \rangle$ be a pseudo normed space over the field $\mathbb{K}$ then we have:

1. $\forall \alpha \in \mathbb{K}$ the function $\mu_\alpha: X \rightarrow X$ defined by $\mu_\alpha(x) = \alpha \cdot x$ is continuous
2. $\forall x \in X$ the function $\nu_x: \mathbb{K} \rightarrow X$ defined by $\nu_x(\alpha) = \alpha \cdot x$ is continuous

Proof. This follows from [theorem: 14.162] and [theorem: 14.163]. □

Theorem 14.165. Let $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ be pseudo normed spaces then we have:

1. If $f: X \rightarrow Y$, $g: X \rightarrow Y$ are continuous functions at $x \in X$ then

$$f + g: X \rightarrow Y \text{ where } (f + g)(x) = f(x) + g(x)$$

is continuous at x . So if f, g are continuous then $f + g$ is continuous.

2. If $\alpha \in \mathbb{K}$ and $f: X \rightarrow Y$ is a continuous function at x then

$$\alpha \cdot f: X \rightarrow Y \text{ where } (\alpha \cdot f)(x) = \alpha \cdot f(x)$$

is continuous at x . So if f are continuous then $\alpha \cdot f$ is continuous.

Proof. We use the ε - δ -definition of continuity [see theorem: 14.135] to proof this theorem. So take $x \in X$ and $\varepsilon \in \mathbb{R}^+$.

1. As f, g are continuous at x there exists $\delta_1, \delta_2 \in \mathbb{R}^+$ such that $\forall x' \in X$ such that $\|x - x'\|_X < \delta_1$ we have $\|f(x) - f(x')\|_Y < \frac{\varepsilon}{2}$ and $\forall x' \in X$ such that $\|x - x'\|_X < \delta_2$ we have $\|g(x) - g(x')\|_Y < \frac{\varepsilon}{2}$. So if $\|x - x'\|_X < \delta$ then

$$\begin{aligned} \|(f+g)(x) - (f+g)(x')\|_Y &= \|f(x) + g(x) - f(x') - g(x')\|_Y \\ &\leq \|f(x) - f(x')\|_Y + \|g(x) - g(x')\|_Y \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

proving that $f + g$ is continuous at x .

2. As f is continuous at x there exist a $\delta \in \mathbb{R}^+$ such that if $\|x - x'\|_X < \delta$ then

$$\|f(x) - f(x')\|_Y \leq \varepsilon / (|\alpha| + 1)$$

then we have

$$\|(\alpha \cdot f)(x) - (\alpha \cdot f)(x')\|_Y = \|\alpha \cdot (f(x) - f(x'))\|_Y = |\alpha| \cdot \|f(x) - f(x')\|_Y < \alpha \cdot (\varepsilon / (|\alpha| + 1)) < \varepsilon$$

proving that $\alpha \cdot f$ is continuous at x . $\square$

Using induction we can extend the above to a finite family of functions.

Theorem 14.166. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{C}(X, Y)$ a family of continuous functions then

$$\sum_{i=1}^n f_i: X \rightarrow Y \text{ defined by } \left(\sum_{i=1}^n f_i \right)(x) = \sum_{i=1}^n f_i(x)$$

is continuous.

Proof. Let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{C}(X, Y) \text{ then } \sum_{i=1}^n f_i \in \mathcal{C}(X, Y) \right\}$$

then we have:

- $1 \in S$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{C}(X, Y)$ then $\forall x \in X$ we have $(\sum_{i=1}^1 f_i)(x) = \sum_{i=1}^1 f_i(x) = f_1(x)$, proving that $\sum_{i=1}^1 f_i = f_1 \in \mathcal{C}(X, Y)$. Hence

$$1 \in S$$

$n \in S \Rightarrow n + 1 \in S$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{C}(X, Y)$ then we have $\forall x \in X$ that

$$\begin{aligned} \left(\sum_{i=1}^{n+1} f_i \right)(x) &= \sum_{i=1}^{n+1} f_i(x) \\ &= \left(\sum_{i=1}^n f_i(x) \right) + f_{n+1}(x) \\ &= \left(\sum_{i=1}^n f_i \right)(x) + f_{n+1}(x) \\ &= \left(\left(\sum_{i=1}^n f_i \right) + f_{n+1} \right)(x) \end{aligned}$$

proving that $\sum_{i=1}^{n+1} f_i = (\sum_{i=1}^n f_i) + f_{n+1}$. As $n \in S$ $\sum_{i=1}^n f_i$ is continuous, so by the continuity of f_{n+1} we have by [theorem: 14.165] that $\sum_{i=1}^{n+1} f_i$ is continuous. Hence $n+1 \in S$. $\square$

Theorem 14.167. Let $\langle X, \|\cdot\|_X \rangle$ be a pseudo normed space, $\langle Y, \|\cdot\|_Y \rangle$ a pseudo normed space over $\mathbb{K}$ and $f: X \rightarrow \mathbb{K}$, $g: X \rightarrow Y$ functions that are continuous at x then

$$f \cdot g: X \rightarrow Y \text{ defined by } (f \cdot g)(x) = f(x) \cdot g(x)$$

is continuous at x . Hence if f, g are continuous functions then $f \cdot g$ is a continuous function.

Note 14.168. The product used in this theorem is the scalar product $\cdot: \mathbb{K} \times Y \rightarrow Y$, if $Y = \mathbb{K}$ then this product reduces to the product in the field $\mathbb{K}$.

Proof. We use the ε - δ -definition of continuity [see theorem: 14.135] to proof this theorem. So take $x \in X$ and $\varepsilon \in \mathbb{R}^+$. Then as f is continuous at x there exist a $\delta_1 \in \mathbb{R}^+$ such that $\forall x' \in X$ with $\|x - x'\|_X < \delta_1$ we have $|f(x) - f(x')| < 1$. Then

$$|f(x')| \leq |f(x') - f(x) + f(x)| \leq |f(x)| + |f(x) - f(x')| < |f(x)| + 1 \quad (14.60)$$

Using continuity again there exist a $\delta_2 \in \mathbb{R}^+$ such that $\forall x \in X$ with $\|x - x'\| < \delta_2$ we have

$$|f(x) - f(x')| < \frac{\varepsilon}{2 \cdot (1 + \|g(x)\|_Y)} \quad (14.61)$$

As g is also continuous at x there exists a $\delta_3 \in \mathbb{R}^+$ such that $\forall x' \in X$ with $\|x - x'\| < \delta_3$

$$\|g(x) - g(x')\|_Y < \frac{\varepsilon}{2 \cdot (1 + |f(x)|)} \quad (14.62)$$

Take $\delta = \min(\delta_1, \delta_2, \delta_3) \in \mathbb{R}^+$ then

$$\begin{aligned} & \| (f \cdot g)(x) - f(x') \cdot g(x') \|_Y &= \\ & \| f(x) \cdot g(x) - f(x') \cdot g(x') \|_Y &= \\ & \| f(x) \cdot g(x) - f(x') \cdot g(x) + f(x') \cdot g(x) - f(x') \cdot g(x') \|_Y &\leq \\ & \| f(x) \cdot g(x) - f(x') \cdot g(x) \|_Y + \| f(x') \cdot g(x) - f(x') \cdot g(x') \|_Y &= \\ & \| (f(x) - f(x')) \cdot g(x) \|_Y + \| f(x') \cdot (g(x) - g(x')) \|_Y &= \\ & |f(x) - f(x')| \cdot \|g(x)\|_Y + |f(x')| \cdot \|g(x) - g(x')\|_Y &\leq \\ & & \text{[eq: 14.60]} \\ & |f(x) - f(x')| \cdot \|g(x)\|_Y + (|f(x)| + 1) \cdot \|g(x) - g(x')\|_Y &\leq \\ & & \text{[eqs: 14.61, 14.62]} \\ & \frac{\varepsilon}{2 \cdot (1 + \|g(x)\|_Y)} \cdot \|g(x)\|_Y + (|f(x)| + 1) \cdot \frac{\varepsilon}{2 \cdot (1 + |f(x)|)} &< \\ & & \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ & & \varepsilon \end{aligned}$$

proving that $f \cdot g$ is continuous at x . $\square$

Using mathematical induction we can then prove the following corollary

Corollary 14.169. Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ a pseudo normed space and $f: X \rightarrow \mathbb{K}$ a function continuous at x then

$$f^n: X \rightarrow \mathbb{K} \text{ defined by } f^n(x) = (f(x))^n$$

is continuous at x . So if f is continuous then f^n is also continuous.

Proof. Take

$$S = \{n \in \mathbb{N}_0 \mid f^n \text{ is continuous at } x\}$$

then we have:

$0 \in S$. As $\forall x \in X$ $(f^0)(x) = (f(x))^0 = 1$ we have that $f^0 = C_1$ which is continuous by [example: 14.142]. Hence $0 \in S$

$n \in S \Rightarrow n+1 \in S$. As $\forall x \in X$ $f^{n+1}(x) = (f(x))^{n+1} = (f(x))^n \cdot f(x) = (f^n \cdot f)(x)$ we have that $f^{n+1} = f^n \cdot f$. From $n \in S$ we have that f^n is continuous at x so that by the continuity of f at x and [theorem: 14.167] $f^{n+1} = f^n \cdot f$ is continuous at x . Hence $n+1 \in S$ $\square$

Using the identity function Id_X in the above corollary proves then that the power function itself is continuous.

Corollary 14.170. *Let $n \in \mathbb{N}_0$ then for $\langle \mathbb{K}, || \rangle$ we have that the function*

$$\cdot^n: \mathbb{K} \rightarrow \mathbb{K} \text{ defined by } (\cdot^n)(x) = x^n$$

is continuous.

Proof. $\forall x \in X$ we have $(\cdot^n)(x) = x^n = (\text{Id}_X(x))^n = (\text{Id}_X)^n(x)$ which proves that $\cdot^n = (\text{Id}_X)^n$. So we can apply [example: 14.143] and the previous corollary [corollary: 14.169]. $\square$

Theorem 14.171. *Let $\langle X, ||| \rangle$ be a pseudo normed space and $f: X \rightarrow \mathbb{K}$ a continuous function such that $\forall x \in X$ $f(x) \neq 0$ then $\frac{1}{f}: X \rightarrow \mathbb{K}$ defined by $\left(\frac{1}{f}\right)(x) = \frac{1}{f(x)}$ is continuous.*

Proof. Let $x \in X$ then as f is continuous there exist a $\delta_1 \in \mathbb{R}^+$ such that $\forall y \in X$ with $|x - y| < \delta_1$ we have $|f(x) - f(y)| < \frac{|f(x)|}{2}$ so that

$$|f(x)| = |f(x) - f(y) + f(y)| \leq |f(x) - f(y)| + |f(y)| < \frac{|f(x)|}{2} + |f(y)|$$

so $\frac{|f(x)|}{2} = |f(x)| - \frac{|f(x)|}{2} < |f(y)|$ or $\frac{1}{|f(y)|} < \frac{2}{|f(x)|}$. Summarized we have proved that

$$\forall y \in X \text{ with } |x - y| < \delta_1 \text{ we have } \frac{1}{|f(y)|} < \frac{2}{|f(x)|} \quad (14.63)$$

Let $\varepsilon \in \mathbb{R}^+$ then as f is continuous there exist a $\delta_2 \in \mathbb{R}^+$ such that

$$\forall y \in X \text{ with } |x - y| < \delta_2 \text{ we have } |f(x) - f(y)| < \varepsilon \cdot \frac{|f(x)|^2}{2} \quad (14.64)$$

So if we take $\delta = \min(\delta_1, \delta_2) \in \mathbb{R}^+$ then we have

$$\begin{aligned} \left| \frac{1}{f(x)} - \frac{1}{f(y)} \right| &= \left| \frac{f(y) - f(x)}{|f(x) \cdot f(y)|} \right| \\ &= \frac{|f(y) - f(x)|}{|f(x)| \cdot |f(y)|} \\ &<_{[\text{eq: 14.63}]} \frac{2}{|f(x)|^2} \cdot |f(y) - f(x)| \\ &<_{[\text{eq: 14.64}]} \frac{2}{|f(x)|^2} \cdot \varepsilon \cdot \frac{|f(x)|^2}{2} \\ &= \varepsilon \end{aligned}$$

proving that $\frac{1}{f(x)}$ is continuous. $\square$

Example 14.172. Let $\langle X, ||| \rangle$ be a pseudo normed space and $f: X \rightarrow \mathbb{K}$ a continuous function such that $\forall x \in X$ $f(x) \neq 0$ then $\frac{1}{f^2}: X \rightarrow \mathbb{K}$ defined by $\left(\frac{1}{f^2}\right)(x) = \frac{1}{f(x)^2}$ is continuous.

Proof. As f is continuous it follows from [theorem: 14.169] it follows that f^2 is continuous, so using [theorem: 14.171] it follows that $\frac{1}{f^2}$ is continuous. $\square$

Theorem 14.173. *Let $n \in \mathbb{N}$ and $\langle \mathbb{R}^n, |||_n \rangle$ the normed space with the maximum norm $|||_n$ defined by $\|x\|_n = \max(\{|\pi_i(x)| \mid i \in \{1, \dots, n\}\})$ and $|||$ another norm on $\mathbb{R}^n$ then*

$$|||: \mathbb{R}^n \rightarrow \mathbb{R} \text{ defined by } |||(x) = \|x\|$$

is continuous [using the topological spaces $\langle \mathbb{R}^n, |||_n \rangle$ and $\langle \mathbb{R}, || \rangle$]

Proof. Let $E = \{e_1, \dots, e_n\}$ be the canonical basis of $\mathbb{R}^n$ defined by $(e_i)_j = \delta_{i,j}$ [see: 11.167] then $\forall x = (x_1, \dots, x_n) \in \mathbb{R}^n$ we have $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$. Hence

$$\begin{aligned}
 \|x\| &= \left\| \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\| \\
 &\leq_{[\text{theorem: 14.90}]} \sum_{i \in \{1, \dots, n\}} \|x_i \cdot e_i\| \\
 &= \sum_{i \in \{1, \dots, n\}} |x_i| \cdot \|e_i\| \\
 &\leq \sum_{i \in \{1, \dots, n\}} \max \{|x_i| | i \in \{1, \dots, n\}\} \cdot \|e_i\| \\
 &= \sum_{i \in \{1, \dots, n\}} \max \{|\pi_i(x)| | i \in \{1, \dots, n\}\} \cdot \|e_i\| \\
 &= \sum_{i \in \{1, \dots, n\}} \|x\|_n \cdot \|e_i\| \\
 &= \|x\|_n \cdot \sum_{i \in \{1, \dots, n\}} \|e_i\| \\
 &= \|x\|_n \cdot A \text{ where } A = \sum_{i \in \{1, \dots, n\}} \|e_i\|
 \end{aligned}$$

hence we have that

$$\forall x \in \mathbb{R}^n \text{ we have } \|x\| \leq \|x\| \cdot A$$

Let $x \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}^+$ then if we take $\delta = \frac{\varepsilon}{A+1}$ then if $\|x - y\|_n \leq \delta$ we have

$$\|x - y\| \leq A \cdot \|x - y\|_n < A \cdot \delta = A \cdot \frac{\varepsilon}{1+A} < \varepsilon$$

proving that $\|\cdot\|$ is continuous [see theorem: 14.135]. $\square$

14.4.2 Uniform and Lipschitz continuity

We have a stronger form of continuity in the case of normed spaces called uniform continuity.

Definition 14.174. Let $\langle X, d_X \rangle, \langle Y, d_Y \rangle$ be topological spaces, then a function $f: X \rightarrow Y$ is **uniform continuous** if $\forall \varepsilon \in \mathbb{R}^+$ there exists a $\delta \in \mathbb{R}^+$ such that $\forall x, y \in X$ with $d_X(x, y) < \delta$ we have $d_Y(f(x), f(y)) < \varepsilon$

Theorem 14.175. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be topological spaces, then a function $f: X \rightarrow Y$ is **uniform continuous** if $\forall \varepsilon \in \mathbb{R}^+$ there exists a $\delta \in \mathbb{R}^+$ such that $\forall x, y \in X$ with $\|x - y\|_X < \delta$ we have $\|f(x) - f(y)\|_Y < \varepsilon$

Proof. This follows from the fact that $d_X(x, y) = \|x - y\|_X$ $\square$

The difference with normal continuity at x is that there exist a δ **which can depend** on x such that $\forall y \in Y$ with $\|x - y\|_X$ we have $\|f(x) - f(y)\|_Y < \varepsilon$. However uniform continuity implies continuity as is mentioned in the following theorem.

Theorem 14.176. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces and $f: X \rightarrow Y$ a uniform continuous function then f is continuous.

Proof. Let $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then there exist a $\delta \in \mathbb{R}^+$ such that for $x, y \in X$ with $\|x - y\|_X < \delta$ we have $\|f(x) - f(y)\|_Y < \varepsilon$. Hence if $x \in X$ then for every $y \in X$ such that $\|x - y\|_X < \delta$ we have $\|f(x) - f(y)\|_Y < \varepsilon$ proving that f is continuous at x . As $x \in X$ was chosen arbitrary it follows that f is continuous. $\square$

Example 14.177. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space then

$$\|\cdot\|: X \rightarrow \mathbb{R} \text{ where } \|\cdot\|(x) = \|x\|$$

is **uniform continuous** [hence by [theorem: 14.174] continuous] using the topological spaces $\langle X, \mathcal{T}_{|||} \rangle$ and $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$

Proof. Let $x \in X$ then if $\varepsilon \in \mathbb{R}^+$ we have for $\delta = \varepsilon \in \mathbb{R}^+$ that $\forall x, y \in X$ such that $\|x - y\| < \delta = \varepsilon$

$$|||(x) - |||(y)| = |||x - y||| \leq_{[\text{theorem: 14.91}]} \|x - y\| < \delta = \varepsilon$$

proving by [theorem: 14.135] that $|||$ is uniform continuous at x . $\square$

Example 14.178. Given the topological spaces $\langle \mathbb{C}, \mathcal{T}_{||} \rangle$ and $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ [see examples: 14.99 and 14.102] then

1. $\text{Re}: \mathbb{C} \rightarrow \mathbb{R}$ is uniform continuous [hence by [theorem: 14.176] continuous]
2. $\text{Im}: \mathbb{C} \rightarrow \mathbb{R}$ is uniform continuous [hence by [theorem: 14.176] continuous]

Proof. Let $\varepsilon \in \mathbb{R}^+$, take $\delta = \varepsilon \in \mathbb{R}^+$ then $\forall x, y \in \mathbb{C}$ such that $|x - y| < \delta = \varepsilon$ we have

1. $|\text{Re}(x) - \text{Re}(y)| \stackrel{[\text{theorem: 10.80}]}{=} |\text{Re}(x - y)| \leq_{[\text{theorem: 10.85}]} |x - y| < \delta = \varepsilon$ proving that Re is uniform continuous at x .
2. $|\text{Im}(x) - \text{Im}(y)| \stackrel{[\text{theorem: 10.80}]}{=} |\text{Im}(x - y)| \leq_{[\text{theorem: 10.85}]} |x - y| < \delta = \varepsilon$ proving that Im is uniform continuous at x . $\square$

Theorem 14.179. Let $\langle X, ||| \rangle$ be a pseudo normed space, $\mathcal{T}_{|||}$ the norm topology on X and $\mathcal{T}_{\text{product}}$ the product topology on $X \times X$ then $+: X \times X \rightarrow X$ is uniform continuous [hence by [theorem: 14.176] continuous] where $+(x, y) = x + y$.

Proof. Using [theorem: 14.98] the product topology on $\mathcal{T}_{\text{product}} = \mathcal{T}_{|||_{\max}}$ where for $x = (x_1, x_2) \in X \times X$ $\|x\|_{\max} = \|(x_1, x_2)\|_{\max} = \max(\{\|x_i\| \mid i \in \{1, 2\}\})$. Let $\varepsilon \in \mathbb{R}^+$ and take $\delta = \frac{\varepsilon}{2}$ then $\forall x = (x_1, x_2), y = (y_1, y_2) \in X \times X$ with $\|x - y\|_{\max} < \delta$ we have $\max(\{\|x_1 - y_1\|, \|x_2 - y_2\|\}) < \delta$ so that

$$\|x_1 - y_1\| < \delta \wedge \|x_2 - y_2\| < \delta$$

Next

$$\begin{aligned} \|(x + y) - (x + y)\| &= \|(x_1 + x_2) - (y_1 + y_2)\| \\ &= \|(x_1 - y_1) + (x_2 - y_2)\| \\ &\leq \|x_1 - y_1\| + \|x_2 - y_2\| \\ &< \delta + \delta \\ &= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

which proves that $+: X \times X \rightarrow X$ is uniform continuous. $\square$

Definition 14.180. Let $\langle X, |||_X \rangle, \langle Y, |||_Y \rangle$ be pseudo normed spaces then a function is **Lipschitz continuous** if $\forall x, y \in X$ we have $\|f(x) - f(y)\|_Y \leq \|x - y\|_X$

It turns out that Lipschitz continuous is a stronger condition than uniform continuity.

Theorem 14.181. Let $\langle X, |||_X \rangle, \langle Y, |||_Y \rangle$ be pseudo normed spaces and $f: X \rightarrow Y$ a **Lipschitz continuous function** then f is **uniform continuous** and by [theorem: 14.176] f is continuous.

Proof. Let $\varepsilon \in \mathbb{R}^+$ and take $\delta = \varepsilon \in \mathbb{R}^+$ then if $x, y \in X$ satisfies $\|x - y\|_X < \delta$ we have

$$\|f(x) - f(y)\|_Y \leq \|x - y\|_X < \delta = \varepsilon$$

proving uniform continuity. $\square$

Theorem 14.182. Let $\langle X, ||| \rangle$ be a pseudo normed space then $\forall x \in X$ we have that

$$\delta_x: X \rightarrow X \text{ defined by } \delta_x(y) = x + y$$

is Lipschitz continuous [hence by [theorem: 14.181] continuous and uniform continuous]

Proof. Let $x \in X$ and $y, z \in X$ then

$$\|\delta_x(y) - \delta_x(z)\| = \|x + y - (x + z)\| = \|y - z\| \quad \square$$

An example of a Lipschitz continuous function is the set distance function to a non empty set.

Definition 14.183. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space and $\emptyset \neq S \subseteq X$ a non empty set then the **set distance function** noted by δ_S is defined by

$$\delta_S: X \rightarrow \mathbb{R} \text{ where } \delta_S(x) = \inf(\{\|x - y\| \mid y \in S\})$$

Note 14.184. $\forall d \in \{\|x - y\| \mid y \in S\}$ we have $0 \leq d$, hence 0 is a lower bound of $\{\|x - y\| \mid y \in S\}$, further $S \neq \emptyset \Rightarrow \{\|x - y\| \mid y \in S\} \neq \emptyset$. So using the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] it follows that $\inf(\{\|x - y\| \mid y \in S\})$ exists.

Further if $x \in S$ then $0 \leq \inf(\{\|x - y\| \mid y \in S\}) \leq 0$ proving that

$$\forall x \in S \text{ we have } \delta_S(x) = 0$$

Theorem 14.185. Let $\langle X, \|\cdot\| \rangle$ be a pseudo normed space $\emptyset \neq S \subseteq X$ then $\delta_S: X \rightarrow \mathbb{R}$ is Lipschitz continuous [hence uniform continuous and thus continuous [see theorems: 14.176, 14.181]]

Proof. Let $x, y \in X$ and $\varepsilon \in \mathbb{R}^+$ then as $\delta_S(x) = \inf(\{\|x - z\| \mid z \in S\}) < \inf(\{\|x - z\| \mid z \in S\}) + \varepsilon$ there exist a $z_\varepsilon \in S$ such that $\delta_S(x) \leq \|x - z_\varepsilon\| < \delta_S(x) + \varepsilon$ hence

$$-\delta_S(x) < \varepsilon - \|x - z_\varepsilon\|.$$

Further, as $z_\varepsilon \in S$, we have $\delta_S(y) = \inf(\{\|y - z\| \mid z \in S\}) \leq \|y - z_\varepsilon\|$ hence we have that

$$\begin{aligned} \delta_S(y) - \delta_S(x) &< \|y - z_\varepsilon\| - \|x - z_\varepsilon\| + \varepsilon \\ &= \|y - x + x - z_\varepsilon\| - \|x - z_\varepsilon\| + \varepsilon \\ &\leq \|y - x\| + \|x - z_\varepsilon\| - \|x - z_\varepsilon\| + \varepsilon \\ &= \|y - x\| + \varepsilon \end{aligned}$$

So we have

$$\forall \varepsilon \in \mathbb{R}^+ \text{ we have } \delta_S(y) - \delta_S(x) < \|y - x\| + \varepsilon$$

which by [theorem: 10.31] proves that $\delta_S(y) - \delta_S(x) \leq \|y - x\|$. Interchanging x and y gives $-(\delta_S(y) - \delta_S(x)) = \delta_S(x) - \delta_S(y) \leq \|x - y\| = \|y - x\|$, hence we have

$$|\delta_S(y) - \delta_S(x)| \leq \|y - x\|$$

proving Lipschitz continuity. $\square$

14.4.3 Homeomorphism

Ring-, group, order- and linear- isomorphisms allows use to identify two rings, groups, partial ordered sets and vector spaces. We can do the same thing for topological spaces.

Definition 14.186. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be two topological spaces then a **homeomorphism** between $\langle X, \mathcal{T}_X \rangle$ and $\langle Y, \mathcal{T}_Y \rangle$ is a **bijection** $f: X \rightarrow Y$ such that $f: X \rightarrow Y$ and $f^{-1}: Y \rightarrow X$ are **continuous**. If there exist a homeomorphism between two topological space then these spaces are called **homeomorphic** spaces.

We have trivially the following equivalent definition of a homeomorphism

Theorem 14.187. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be two topological spaces and $f: X \rightarrow Y$ a bijection then

$$f \text{ is a homeomorphism} \Leftrightarrow f \text{ is continuous and open}$$

Proof.

$\Rightarrow$. As f is a homeomorphism we have that $f: X \rightarrow Y$ is continuous and $f^{-1}: Y \rightarrow X$ is continuous. So if $U \in \mathcal{T}_X$ we have $(f^{-1})^{-1}(U) \in \mathcal{T}_Y$ which, as $(f^{-1})^{-1}(U) \stackrel{[\text{theorem: 2.70}]}{=} f(U)$, proves that $f(U) \in \mathcal{T}_Y$. Hence $f: X \rightarrow Y$ is continuous and open.

$\Leftarrow$. Let $U \in \mathcal{T}_X$ then as $f: X \rightarrow Y$ is open we have $f(U) \in \mathcal{T}_Y$, which as $(f^{-1})^{-1}(U) \stackrel{[\text{theorem: 2.70}]}{=} f(U)$ proves that $(f^{-1})^{-1}(U) \in \mathcal{T}_Y$. So $f^{-1}: Y \rightarrow X$ is continuous, as by the hypothesis $f: X \rightarrow Y$ is also continuous, it follows that $f: X \rightarrow Y$ is a homeomorphism. $\square$

A trivial example of a homeomorphism is the following:

Example 14.188. Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\text{Id}_X: X \rightarrow X$ defined by $\text{Id}_x(x) = x$ is a homeomorphism. In other words $\langle X, \mathcal{T} \rangle$ is homeomorphic to itself.

Proof. Clearly Id_X is a bijection [see example: 2.66] further if U is a open set we have that $\text{Id}_X^{-1}(U) = U$ and $\text{Id}_X(U) = U$ so that Id_X is continuous and open. $\square$

Another rather trivial example of a homeomorphism is the following.

Example 14.189. Let $\langle \mathbb{R}, \|\cdot\| \rangle$ and $\langle \mathbb{R}^1, \|\cdot\|_1 \rangle$ be the normed spaces of the real numbers and $\mathbb{R}^1$ [see examples: 14.99, 14.101] then

$\varphi: \mathbb{R} \rightarrow \mathbb{R}^1$ defined by $\varphi(x) = (x)$ [here (x) is the tuple with only one element such that $(x)_1 = x$] is a homeomorphism.

Proof. Define

$$\psi: \mathbb{R}^1 \rightarrow \mathbb{R} \text{ by } \psi(x) = x_1$$

then we have $\forall x \in \mathbb{R}$ that $(\psi \circ \varphi)(x) = \psi(\varphi(x)) = (x)_1 = x = \text{Id}_{\mathbb{R}}(x)$ so that

$$\psi \circ \varphi = \text{Id}_{\mathbb{R}}$$

Further $\forall x \in \mathbb{R}^1$ we have $(\varphi \circ \psi)(x) = \varphi(\psi(x)) = (x_1) = x = \text{Id}_{\mathbb{R}^1}(x)$ so that

$$\varphi \circ \psi = \text{Id}_{\mathbb{R}^1}$$

which proves by [theorems: 2.73, 2.75] that

$$\varphi \text{ is a bijection with } \varphi^{-1} = \psi \tag{14.65}$$

Let $x \in \mathbb{R}$ then $\forall \varepsilon \in \mathbb{R}^+$ we have for $y \in \mathbb{R}$ with $|x - y| < \varepsilon$ that

$$\begin{aligned} \|\varphi(x) - \varphi(y)\|_1 &= \max(\{ |(\varphi(x) - \varphi(y))_i| \mid i \in \{1\} \}) \\ &= \max(\{ |(\varphi(x) - \varphi(y))_1| \}) \\ &= |(\varphi(x) - \varphi(y))_1| \\ &= |(x) - (y))_1| \\ &= |(x - y)_1| \\ &= |x - y| \\ &< \varepsilon \end{aligned}$$

which proves, as $x \in \mathbb{R}$ is choosen arbitrary, that

$$\varphi \text{ is continuous} \tag{14.66}$$

Let $x \in \mathbb{R}^1$ then $\forall \varepsilon \in \mathbb{R}^+$ we have for $y \in \mathbb{R}^1$ with $\|x - y\|_1 < \varepsilon$ that

$$\begin{aligned} |\psi(x) - \psi(y)| &= |x_1 - y_1| \\ &= |(x - y)_1| \\ &= \max(\{ |(x - y)_i| \mid i \in \{1\} \}) \\ &= \|x - y\|_1 \\ &< \varepsilon \end{aligned}$$

which proves, as $x \in \mathbb{R}^1$ is chosen arbitrary, that

$$\varphi^{-1} = \psi \text{ is continuous} \quad (14.67)$$

From [eqs: 14.65, 14.66 and 14.67] it follows that

$$\varphi \text{ is a homeomorphism} \quad \square$$

We have trivially that the the inverse of a homeomorphism is a homeomorphism.

Theorem 14.190. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be two topological spaces and $f: X \rightarrow Y$ a homeomorphism then $f^{-1}: Y \rightarrow X$ is a homeomorphism.*

Proof. This follows from the definition and [theorems: 2.74 and 2.75]. $\square$

Theorem 14.191. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces and $f: X \rightarrow Y$ a homeomorphism then we have $\forall A \in \mathcal{T}_Y$ that $\overline{f^{-1}(A)} = f^{-1}(\overline{A})$*

Proof. Let $A \subseteq Y$. As $\overline{A}$ is closed it follows from [theorem: 14.154] that $f^{-1}(A)$ is closed, further as $A \subseteq \overline{A}$ we have that $f^{-1}(A) \subseteq f^{-1}(\overline{A})$ so that by [14.25]

$$\overline{f^{-1}(A)} \subseteq f^{-1}(\overline{A}) \quad (14.68)$$

As f is a homeomorphism f^{-1} is a homeomorphism hence continuous. Hence by [theorem: 14.154] we have that $f^{-1}(\overline{A}) \stackrel{[\text{theorem: 2.70}]}{=} (f^{-1})(\overline{A}) \subseteq \overline{(f^{-1})(A)} \stackrel{[\text{theorem: 2.70}]}{=} \overline{f^{-1}(A)}$ which combined with [eq: 14.68] proves

$$\overline{f^{-1}(A)} = f^{-1}(\overline{A}) \quad \square$$

Theorem 14.192. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ and $\langle Z, \mathcal{T}_Z \rangle$ be topological spaces and $f: X \rightarrow Y, g: Y \rightarrow Z$ be homeomorphisms then $g \circ f: X \rightarrow Z$ is a homeomorphism.*

Proof. This follows from the definition and [theorems: 2.78, 14.158]. $\square$

Theorem 14.193. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be two topological spaces, $A \subseteq X$ a subset of X , $f: X \rightarrow Y$ a homeomorphism then $f|_A: A \rightarrow f(A) = f|_A(A)$ is a homeomorphism [using the subspace topologies $(\mathcal{T}_X)|_A$ and $(\mathcal{T}_Y)|_{f(A)}$]*

Proof. First, by [theorem: 2.98]

$$f|_A: A \rightarrow f(A) = f|_A(A) \text{ is a bijection} \quad (14.69)$$

Further if $V \in (\mathcal{T}_Y)|_{f(A)}$ then there exist a $V' \in \mathcal{T}_Y$ such that $V = V' \cap f(A)$. So

$$\begin{aligned} (f|_A)^{-1}(V) &= (f|_A)^{-1}(V' \cap f(A)) \\ &\stackrel{[\text{theorem: 2.92}]}{=} A \cap f^{-1}(V' \cap f(A)) \\ &\stackrel{[\text{theorem: 2.103}]}{=} A \cap (f^{-1}(V') \cap f^{-1}(f(A))) \\ &\stackrel{[\text{theorem: 2.57}]}{=} A \cap (f^{-1}(V') \cap A) \\ &= f^{-1}(V') \cap A \end{aligned}$$

As f is continuous and V' open we have that $f^{-1}(V') \in \mathcal{T}_X$ hence $f^{-1}(V') \cap A \in (\mathcal{T}_X)|_A$ proving that $(f|_A)^{-1}(V) \in (\mathcal{T}_X)|_A$. Hence

$$f|_A: A \rightarrow f(A) \text{ is continuous} \quad (14.70)$$

Let $U \in (\mathcal{T}_X)|_A$ then there exist a $U' \in \mathcal{T}_X$ such that $U = U' \cap A$ then

$$f|_A(U) = f|_A(U' \cap A) \stackrel{U' \cap A \subseteq U}{=} f(U' \cap A) \stackrel{[\text{theorem: 2.103}] \wedge f|_A \text{ is bijective}}{=} f(U') \cap f(A)$$

By [theorem: 14.187] f is open so that $f(U') \in \mathcal{T}_Y$ hence $f(U') \cap f(A) \in (\mathcal{T}_Y)_{|f(A)}$ proving that $f|_A(U) \in (\mathcal{T}_Y)_{|f(A)}$. Hence we have that

$$f|_A: A \rightarrow f(A) \text{ is open} \quad (14.71)$$

Finally using [theorem: 14.187] on [eqs: 14.69, 14.70 and 14.71] proves that

$$f|_A: A \rightarrow f(A) \text{ is a homeomorphism using the topologies } (\mathcal{T}_X)_{|A} \text{ and } (\mathcal{T}_Y)_{|f(A)} \quad \square$$

Every linear isometric isomorphism [see definition: 14.128] is automatically a homeomorphism.

Theorem 14.194. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be pseudo normed spaces over $\mathbb{K}$ and $f: X \rightarrow Y$ a linear isometric isomorphism then $f: X \rightarrow Y$ is a homeomorphism using the topologies $\mathcal{T}_{\|\cdot\|_X}, \mathcal{T}_{\|\cdot\|_Y}$. In other words isometric spaces are homeomorphic.*

Proof. As $f: X \rightarrow Y$ is a linear isometric isomorphism it follows that

$$f: X \rightarrow Y \text{ is a bijection}$$

Further as $\forall x, y \in X$ we have $\|f(x) - f(y)\|_Y = \|f(x - y)\|_Y = \|x - y\|_X$ it follows that $f: X \rightarrow Y$ is Lipschitz continuous, hence continuous. Further by [theorem: 14.126] $f^{-1}: Y \rightarrow X$ is also a linear isometry hence continuous. So $f: X \rightarrow Y$ and $f^{-1}: Y \rightarrow X$ are continuous which proves that $f: X \rightarrow Y$ is homeomorphism. $\square$

Theorem 14.195. *Let $\langle X, \mathcal{T} \rangle$ be a topological space. Y a set and $f: X \rightarrow Y$ a bijection. Then we have:*

1. $\mathcal{T}_{f, \mathcal{T}} = \{f(U) | U \in \mathcal{T}\}$ forms a topology on Y
2. $f: X \rightarrow Y$ is a homeomorphism using the topologies $\mathcal{T}$ and $\mathcal{T}_{f, \mathcal{T}}$
3. If $g: X \rightarrow Y$ satisfies $f = g \circ h$ where $h: X \rightarrow X$ is a homeomorphism using the topologies $\mathcal{T}$ and $\mathcal{T}$ then g is a homeomorphism and $\mathcal{T}_{f, \mathcal{T}} = \mathcal{T}_{g, \mathcal{T}}$

Proof.

1.

- a. As $f(\emptyset) = \emptyset$ it follows that

$$\emptyset \in \mathcal{T}_{f, \mathcal{T}}$$

- b. As $f: X \rightarrow Y$ is a bijection we have $f(X) = Y$ hence as $X \in \mathcal{T}$

$$Y \in \mathcal{T}_{f, \mathcal{T}}$$

- c. If $\{V_i\}_{i \in I} \subseteq \mathcal{T}_{f, \mathcal{T}}$ then $\forall i \in I$ there exist a $U \in \mathcal{T}$ such that $f(U) = V_i$, hence by a consequence of the Axiom of Choice [see theorem: 3.108] there exist a $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ such that $\forall i \in I$ $f(U_i) = V_i$. Hence

$$\bigcup_{i \in I} V_i = \bigcup_{i \in I} f(U_i) \stackrel{[\text{theorem: 2.144}]}{=} f\left(\bigcup_{i \in I} U_i\right)$$

which, as $\bigcup_{i \in I} U_i \in \mathcal{T}$, proves that

$$\bigcup_{i \in I} V_i \in \mathcal{T}_{f, \mathcal{T}}$$

- d. Let $V_1, V_2 \in \mathcal{T}_{f, \mathcal{T}}$ then $\exists U_1, U_2 \in \mathcal{T}$ such that $V_1 = f(U_1), V_2 = f(U_2)$ then

$$V_1 \cap V_2 = f(U_1) \cap f(U_2) \stackrel{[\text{theorem: 2.103}]}{=} f(U_1 \cap U_2)$$

which, as $U_1 \cap U_2 \in \mathcal{T}$, proves that

$$V_1 \cap V_2 \in \mathcal{T}_{f, \mathcal{T}}$$

2. If $U \in \mathcal{T}$ then by definition $f(U) \in \mathcal{T}_{f, \mathcal{T}}$ proving that

f is open

Further if $V \in \mathcal{T}_{f, \mathcal{T}}$ then there exist a $U \in \mathcal{T}$ so that $V = f(U)$, so

$$f^{-1}(V) = f^{-1}(f(U)) \stackrel{[\text{theorem: 2.57}]}{=} U \in \mathcal{T}$$

proving that

f is continuous

Hence as $f: X \rightarrow Y$ is a bijection we have that

$f: X \rightarrow Y$ is a homeomorphism

3. First as $h: X \rightarrow X$ is a homeomorphism we have by [theorem: 14.190] that $h^{-1}: X \rightarrow X$ is a homeomorphism. Further

$$g = g \circ \text{Id}_X = g \circ (h \circ h^{-1}) = (g \circ h) \circ h^{-1} = f \circ h^{-1} \quad (14.72)$$

so, as f is a homeomorphism by (2) and h^{-1} is a homeomorphism, we have by [theorem: 14.192] that

$g: X \rightarrow Y$ is a homeomorphism

Next if $V \in \mathcal{T}_{g, \mathcal{T}}$ then there exist a $U \in \mathcal{T}$ such that $V = g(U)$, as

$$V = g(U) \stackrel{[\text{eq: 14.72}]}{=} (f \circ h^{-1})(U) = f((h^{-1})(U)) \in \mathcal{T}_{f, \mathcal{T}} \text{ [as } h^{-1} \text{ is open } (h^{-1})(U) \in \mathcal{T}] \quad (14.73)$$

So

$$\mathcal{T}_{g, \mathcal{T}} \subseteq \mathcal{T}_{f, \mathcal{T}} \quad (14.74)$$

If $V \in \mathcal{T}_{f, \mathcal{T}}$ then there exist a $U \in \mathcal{T}$ such that $V = f(U)$, as

$$f(U) = (g \circ h)(U) = g(h(U)) \in \mathcal{T}_{g, \mathcal{T}} \text{ [as } h \text{ is open } h(U) \in \mathcal{T}]$$

Proving that $\mathcal{T}_{f, \mathcal{T}} \subseteq \mathcal{T}_{g, \mathcal{T}}$, combining this with [eq: 14.74] gives finally

$$\mathcal{T}_{f, \mathcal{T}} = \mathcal{T}_{g, \mathcal{T}} \quad \square$$

Theorem 14.196. Let X be a set, $\langle Y, \mathcal{T} \rangle$ a topological space and $f: X \rightarrow Y$ a function then

$$\mathcal{T}_{f, \mathcal{T}}^{-1} = \{f^{-1}(U) | U \in \mathcal{T}\} \text{ is a topology on } X$$

This topology is called the **inverse induced topology of $\mathcal{T}$ by f** . Further $f: X \rightarrow Y$ is continuous using the topologies $\mathcal{T}_{f, \mathcal{T}}^{-1}$ and $\mathcal{T}$.

Proof. First we have

1. As $\emptyset = f^{-1}(\emptyset)$ we have

$$\emptyset \in \mathcal{T}_{f, \mathcal{T}}^{-1}$$

2. As $X = f^{-1}(Y)$ we have

$$X \in \mathcal{T}_{f, \mathcal{T}}^{-1}$$

3. Let $\{V_i\}_{i \in I} \subseteq \mathcal{T}_{f, \mathcal{T}}^{-1}$ then $\forall i \in I$ there exist a $U \in \mathcal{T}$ such that $f^{-1}(U) = V_i$, hence by a consequence of the Axiom of Choice [see theorem: 3.108] there exist a $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ such that $\forall i \in I$ $f^{-1}(U_i) = V_i$. Hence

$$\bigcup_{i \in I} V_i = \bigcup_{i \in I} f^{-1}(U_i) \stackrel{[\text{theorem: 2.144}]}{=} f^{-1}\left(\bigcup_{i \in I} U_i\right) \in \mathcal{T}_{f, \mathcal{T}}^{-1} \text{ [as } \bigcup_{i \in I} U_i \in \mathcal{T}]$$

4. If $V_1, V_2 \in \mathcal{T}_{f, \mathcal{T}}^{-1}$ then $\exists U_1, U_2 \in \mathcal{T}$ such that $f^{-1}(U_1) = V_1$, $f^{-1}(U_2) = V_2$. So

$$V_1 \cap V_2 = f^{-1}(U_1) \cap f^{-1}(U_2) \stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(U_1 \cap U_2) \text{ [as } U_1 \cap U_2 \in \mathcal{T}] \quad \square$$

Theorem 14.197. Let $\langle X, \|\cdot\|_X \rangle$ be a finite dimensional pseudo normed vector space with $\dim(X) = n$ then we have given a basis $E = \{e_1, \dots, e_n\}$ for E that

1. $\|\cdot\|_{\langle E \rangle}: \mathbb{K}^n \rightarrow \mathbb{R}$ defined by $\|(x_1, \dots, x_n)\|_{\langle E \rangle} = \left\| \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\|_X$ is a pseudo norm on $\mathbb{K}^n$. Further if $\|\cdot\|_X$ is a norm then $\|\cdot\|_{\langle E \rangle}$ is a norm.
2. $\varphi: \mathbb{K}^n \rightarrow X$ defined by $\varphi(x_1, \dots, x_n) = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$ is a linear isometric isomorphism using the norms $\|\cdot\|_{\langle E \rangle}$ and $\|\cdot\|_X$

Hence by [theorem: 14.194] φ is a homeomorphism so that $\mathbb{K}^n$ is homeomorphic with X .

Proof.

1. To prove that $\|\cdot\|_{\langle E \rangle}$ is a norm

- a. $\forall x = (x_1, \dots, x_n) \in \mathbb{K}^n$ we have $\|x\|_{\langle E \rangle} = \left\| \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\|_X \geq 0$
- b. If $\alpha \in \mathbb{K}$, $x = (x_1, \dots, x_n) \in \mathbb{K}^n$ we have

$$\begin{aligned}
 \|\alpha \cdot x\|_{\langle E \rangle} &= \|(\alpha \cdot x_1, \dots, \alpha \cdot x_n)\|_{\langle E \rangle} \\
 &= \left\| \sum_{i \in \{1, \dots, n\}} (\alpha \cdot x_i) \cdot e_i \right\|_X \\
 &= \left\| \sum_{i \in \{1, \dots, n\}} \alpha \cdot (x_i \cdot e_i) \right\|_X \\
 &= \left\| \alpha \cdot \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\|_X \\
 &= |\alpha| \cdot \left\| \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\|_X \\
 &= |\alpha| \cdot \|x\|_{\langle E \rangle}
 \end{aligned}$$

- c. Let $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n) \in \mathbb{K}^n$ then we have

$$\begin{aligned}
 \|x + y\|_{\langle E \rangle} &= \|(x_1 + y_1, \dots, x_n + y_n)\|_{\langle E \rangle} \\
 &= \left\| \sum_{i \in \{1, \dots, n\}} (x_i + y_i) \right\|_X \\
 &= \left\| \sum_{i \in \{1, \dots, n\}} x_i + \sum_{i \in \{1, \dots, n\}} y_i \right\|_X \\
 &\leq \left\| \sum_{i \in \{1, \dots, n\}} x_i \right\|_X + \left\| \sum_{i \in \{1, \dots, n\}} y_i \right\|_X \\
 &= \|x\|_{\langle E \rangle} + \|y\|_{\langle E \rangle}
 \end{aligned}$$

Assume that $\|\cdot\|_X$ is a norm. If $x = (x_1, \dots, x_n) \in \mathbb{K}^n$ satisfies $\|x\|_{\langle E \rangle} = 0$ then

$$\left\| \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\|_X = \|x\|_{\langle E \rangle} = 0.$$

So as, $\|\cdot\|_X$ is a norm, $0 = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i$. As $\{e_1, \dots, e_n\}$ is a basis we have by linear independency that $\forall i \in \{1, \dots, n\}$ $x_i = 0$ or $x = 0$.

2. For $\varphi: \mathbb{K}^n \rightarrow X$ we have:

injectivity. Let $\varphi(x) = \varphi(y)$ then $\sum_{i \in \{1, \dots, n\}} x_i \cdot e_i = \sum_{i \in \{1, \dots, n\}} y_i \cdot e_i$, hence, as $\{e_1, \dots, e_n\}$ is a basis we have by [theorem: 11.143] that $\forall i \in \{1, \dots, n\}$ $x_i = y_i$. Hence $x = y$.

surjectivity. If $y \in X$ then by [theorem: 11.143] there exists a $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ such that $y = \sum_{i \in \{1, \dots, n\}} y_i \cdot e_i$. So if we take $x = (y_1, \dots, y_n)$ then $\varphi(x) = y$.

From the above it follows that

$$\varphi: \mathbb{K}^n \rightarrow X \text{ is a bijection}$$

Let $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n) \in \mathbb{K}^n$ and $\alpha \in \mathbb{K}$ then we have

$$\begin{aligned} \varphi(x + \alpha \cdot y) &= \sum_{i \in \{1, \dots, n\}} (x_i + \alpha \cdot y_i) \cdot e_i \\ &= \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i + \alpha \cdot \sum_{i \in \{1, \dots, n\}} y_i \cdot e_i \\ &= \varphi(x) + \alpha \cdot \varphi(y) \end{aligned}$$

proving that

$$\varphi \text{ is linear}$$

Finally if $x = (x_1, \dots, x_n) \in \mathbb{K}^n$ then we have

$$\|\varphi(x)\|_X = \left\| \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i \right\|_X = \|x\|_{\langle E \rangle}$$

proving that

$$\varphi: \mathbb{K}^n \rightarrow X \text{ is a linear isometric isomorphism}$$

□

14.5 Linear mappings and continuity

Definition 14.198. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces then the set of linear continuous function graphs is noted as $L(X, Y)$. So

$$L(X, Y) = \{L \in \text{Hom}(X, Y) \mid L: X \rightarrow Y \text{ is continuous using the topologies } \mathcal{T}_{\|\cdot\|_X} \text{ and } \mathcal{T}_{\|\cdot\|_Y}\}$$

Example 14.199. Let $\langle X, \|\cdot\| \rangle$ be a normed space then $\text{Id}_X \in L(X, X)$

Proof. This follows from [theorem: 11.183] and [example: 14.143].

□

Theorem 14.200. Let $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces over $\mathbb{K}$ then $L(X, Y)$ is a subspace of the vector space $\text{Hom}(X, Y)$ [see theorem: 11.185] with neutral element C_0 . Hence by [theorem: 11.61] $L(X, Y)$ is a vector space with neutral element C_0 .

Proof. Let $L_1, L_2 \in L(X, Y)$ and $\alpha \in \mathbb{K}$ then by [theorem: 14.165] $L_1 + \alpha \cdot L_2 \in L(X, Y)$. Further as C_0 is continuous [see theorem: 14.142] we have that $C_0 \in L(X, Y)$ hence $L(X, Y) \neq \emptyset$. □

We examine now the conditions for a linear mapping to be continuous in the normed topologies.

Theorem 14.201. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $L \in \text{Hom}(X, Y)$ then we have the following equivalences:

1. L is continuous [in other words $L \in L(X, Y)$]
2. L is continuous at $0 \in X$
3. $\exists M \in \mathbb{R}_0^+$ such that $\forall x \in X$ with $\|x\|_X = 1$ we have $\|L(x)\|_Y \leq M$
4. $\exists M \in \mathbb{R}_0^+$ such that $\forall x \in X$ we have $\|L(x)\|_Y \leq M \cdot \|x\|_X$

Note 14.202. Because of (3) elements of $L(X, Y)$ are sometimes called bounded linear functions instead of linear continuous functions.

Note 14.203. If $Y = \mathbb{K}$ and $\|\cdot\|_Y = \|\cdot\|$ [the absolute value norm on $\mathbb{K}$] then elements of $L(X, \mathbb{K})$ are called continuous linear functionals or bounded linear functionals.

Proof.

1 $\Rightarrow$ 2. This follows from the definition of continuity.

2 $\Rightarrow$ 3. As L is continuous at 0 we have by [theorem: 14.135] that there exist a $\delta \in \mathbb{R}^+$ such that $\forall x \in X$ satisfying $\|x\|_X = \|x - 0\|_X < \delta$ we have $\|L(x)\|_Y = \|L(x) - L(0)\|_Y < 1$. As $0 < \delta$ there exist by [theorem: 10.30] a $\delta' \in \mathbb{R}$ such that $0 < \delta' < \delta$. If $x \in X$ such that $\|x\|_X = 1$ then $\|\delta' \cdot x\|_X = |\delta'| \cdot \|x\|_X = \delta' \cdot 1 = \delta' < \delta$ so that $\delta' \cdot \|L(x)\|_Y = \|\delta' \cdot L(x)\|_Y = \|L(\delta' \cdot x)\|_Y < 1$, hence $\|L(x)\|_Y < \frac{1}{\delta'}$. So if we take $M = \frac{1}{\delta'} \in \mathbb{R}^+ \subseteq \mathbb{R}_0^+$ then $\forall x \in X$ with $\|x\|_X = 1$ we have $\|L(x)\|_Y < M \Rightarrow \|L(x)\|_Y \leq M$.

3 $\Rightarrow$ 4. By (3) there exist a $M \in \mathbb{R}_0^+$ such that $\forall x \in X$ with $\|x\|_X = 1$ we have $\|L(x)\|_Y \leq M$. Let $x \in X$ then we have either:

$\|x\|_X = 0$. Then as $\|\cdot\|_X$ is a norm $x = 0$ hence $\|L(x)\|_Y = \|L(0)\|_Y = \|0\|_Y = 0 = M \cdot \|x\|_X$

$\|x\|_X \neq 0$. Then we have $\left\| \frac{1}{\|x\|_X} \cdot x \right\|_X = \left| \frac{1}{\|x\|_X} \right| \cdot \|x\|_X = \frac{\|x\|_X}{\|x\|_X} = 1$ so that

$$\frac{1}{\|x\|_X} \cdot \|L(x)\|_Y = \left\| \frac{1}{\|x\|_X} \cdot L(x) \right\|_Y = \left\| L\left(\frac{1}{\|x\|_X} \cdot x \right) \right\|_Y \leq M$$

Hence

$$\|L(x)\|_Y \leq M \cdot \|x\|_X$$

4 $\Rightarrow$ 1. Let $M \in \mathbb{R}_0^+$ such that $\forall x \in X$ we have $\|L(x)\|_Y \leq M \cdot \|x\|_X$. Let $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then, if we take $\delta = \frac{\varepsilon}{M+1}$, we have for $y \in X$ with $\|x - y\|_X < \delta$ that

$$\|L(x) - L(y)\|_Y = \|L(x - y)\|_Y \leq M \cdot \|x - y\|_X \leq M \cdot \frac{\varepsilon}{M+1} < \varepsilon$$

So L is continuous at x and as $x \in X$ was chosen arbitrary it follows that f is continuous. $\square$

A consequence of the above are the following corollaries.

Corollary 14.204. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed spaces equipped with the absolute value norm then every linear mapping $L: \mathbb{K} \rightarrow \mathbb{K}$ is continuous.

Proof. As $L: \mathbb{K} \rightarrow \mathbb{K}$ is a linear mapping it follows from [example: 11.179] that $L = L(1) \cdot \text{Id}_{\mathbb{K}}$. Hence $\forall x \in \mathbb{K}$ we have $|L(x)| = |L(1) \cdot x| = |L(1)| \cdot |x|$ so that by [theorem: 14.201] L is continuous. $\square$

Corollary 14.205. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed spaces equipped with the absolute value norm then every linear isomorphism $L: \mathbb{K} \rightarrow \mathbb{K}$ is a homeomorphism.

Proof. As L is a linear isomorphism it follows from [theorem: 11.178] that also L^{-1} is a linear isomorphism. Hence using [corollary: 14.204] it follows that L and L^{-1} are continuous which proves that L is a homeomorphism. $\square$

Corollary 14.206. Let $n, m \in \mathbb{N}$ and $\langle \mathbb{K}^n, \|\cdot\|_n \rangle, \langle \mathbb{K}^m, \|\cdot\|_m \rangle$ where

$$\|x\|_n = \max(\{|\pi_i(x)| \mid i \in \{1, \dots, n\}\}), \quad \|x\|_m = \max(\{|\pi_i(x)| \mid i \in \{1, \dots, m\}\})$$

are the maximum norms then every linear function $L: \mathbb{K}^n \rightarrow \mathbb{K}^m$ is continuous.

Proof. Let $\{e_i\}_{i \in \{1, \dots, n\}}$ be the canonical basis of $\mathbb{K}^n$ [see 11.167] then $\forall x = (x_1, \dots, x_n) \in \mathbb{K}^n$ we have $x = \sum_{i \in \{1, \dots, n\}} x_i \cdot e_i = \sum_{i \in \{1, \dots, n\}} \pi_i(x) \cdot e_i$ where $\pi_i(x) = x_i$. So that

$$\|L(x)\|_m = \left\| \sum_{i \in \{1, \dots, n\}} \pi_i(x) \cdot L(e_i) \right\|_m \leq \sum_{i \in \{1, \dots, n\}} |\pi_i(x)| \cdot \|L(e_i)\|_m \leq M' \cdot \sum_{i \in \{1, \dots, n\}} |\pi_i(x)|$$

where $M' = \max(\{\|L(e_i)\|_m \mid i \in \{1, \dots, n\}\})$. As $\forall i \in \{1, \dots, n\}$ we have

$$|\pi_i(x)| \leq \max(\{|\pi_i(x)| \mid i \in \{1, \dots, n\}\}) = \|x\|_n$$

it follows that $\sum_{i \in \{1, \dots, n\}} |\pi_i(x)| \leq \sum_{i \in \{1, \dots, n\}} \|x\|_m = n \cdot \|x\|_m$. So if $M = n \cdot M'$ then

$$\|L(x)\|_m \leq M \cdot \|x\|_m$$

By the previous theorem [theorem: 14.201] it follows then that

$$L: \mathbb{K}^n \rightarrow \mathbb{K}^m \text{ is continuous}$$

□

Corollary 14.207. *Let $n, m \in \mathbb{N}$ and $\langle \mathbb{R}^n, \|\cdot\|_n \rangle$ where $\|\cdot\|_n$ is the maximum norm then every linear isomorphism $L: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a homeomorphism*

Proof. As L is a linear isomorphism it follows from [theorem: 11.178] that also L^{-1} is a linear isomorphism. Hence using [corollary: 14.206] it follows that L and L^{-1} are continuous which proves that L is a homeomorphism. □

Theorem 14.208. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $L \in L(X, Y)$ then for $A_{L,s}$ and $A_{L,r}$ defined by*

$$A_{L,s} = \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ with } \|x\|_X = 1 \text{ we have } \|L(x)\|_Y \leq M\}$$

and

$$A_{L,r} = \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ we have } \|L(x)\|_Y \leq M \cdot \|x\|_X\}$$

we have

1. $A_{L,s} = A_{L,r}$
2. $\inf(A_{L,r})$ exist and $0 \leq \inf(A_{L,r})$ [or by (1) $\inf(A_{L,r})$ exist and $0 \leq \inf(A_{L,r})$]

Proof.

1. Let $M \in A_{L,s}$ then $\forall x \in X$ we have either:

$\|x\|_X = 0$. Then as $\|\cdot\|_X$ is a norm $x = 0$ and thus

$$\|L(x)\|_Y = \|L(0)\|_Y = \|0\|_Y = 0 \leq M \cdot 0 = M \cdot \|x\|_X$$

$\|x\|_X \neq 0$. Then $\left\| \frac{1}{\|x\|_X} \cdot x \right\|_X = \frac{1}{\|x\|_X} \cdot \|x\|_X = 1$ so that as $M \in A_{L,s}$, we have

$$\frac{1}{\|x\|_X} \cdot \|L(x)\|_Y = \left\| \frac{1}{\|x\|_X} \cdot L(x) \right\|_Y = \left\| L\left(\frac{1}{\|x\|_X} \cdot x\right) \right\|_Y \leq M$$

hence

$$\|L(x)\|_Y \leq M \cdot \|x\|_X$$

So as in all cases $\|L(x)\|_Y \leq M \cdot \|x\|_X$ for $x \in X$, hence it follows that $M \in A_{L,r}$ proving that

$$A_{L,s} \subseteq A_{L,r} \tag{14.75}$$

If $M \in A_{L,r}$ then if $x \in X$ with $\|x\|_X = 1$ we have $\|L(x)\|_Y \leq M \cdot \|x\|_X = M \cdot 1 = M$ so that $M \in A_{L,s}$, hence $A_{L,r} \subseteq A_{L,s}$. Combining this with [eq: 14.75] proves that

$$A_{L,s} = A_{L,r}$$

2. As $L \in L(X, Y)$ we have by [theorem: 14.201] that there exist a $M \in \mathbb{R}_0^+$ such that $\forall x \in X$ with $\|x\|_X = 1$ $\|L(x)\|_Y \leq M$ proving that $M \in A_{L,r}$. Hence we have

$$A_{L,r} \neq \emptyset$$

Further, as $A_{L,r} \subseteq \mathbb{R}_0^+$, it follows that $A_{L,r}$ is bounded below by 0, so as $\mathbb{R}$ is conditional complete [see theorem: 10.18] we have that

$$\inf(A_{L,r}) \text{ exist and } 0 \leq \inf(A_{L,r})$$

□

The above theorem allows the following definition.

Definition 14.209. (Operator Norm) Let $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $L \in L(X, Y)$ then the **operator norm**, noted as $\|L\|_{L(X, Y)}$ is defined by

$$\|L\|_{L(X, Y)} = \inf(A_{L, s}) \stackrel{[\text{theorem: 14.208}]}{=} \inf(A_{L, r})$$

where

$$A_{L, s} = \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ with } \|x\|_X = 1 \text{ we have } \|L(x)\|_Y \leq M\}$$

and

$$A_{L, r} = \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ we have } \|L(x)\|_Y \leq M \cdot \|x\|_X\}$$

By [theorem: 14.200] $L(X, Y)$ is a vector space, we prove now that $\|L\|_{L(X, Y)}$ is a norm on $L(X, Y)$.

Theorem 14.210. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces over a field $\mathbb{K}$ then we have that:

1. $\forall x \in X$ we have $\|L(x)\|_Y \leq \|L\|_{L(X, Y)} \cdot \|x\|_X$
2. $\|C_0\|_{L(X, Y)} = 0$ where $C_0: X \rightarrow Y$ is defined by $C_0(x) = 0$ [the neutral element in $L(X, Y)$]
3. $\|L\|_{L(X, Y)}$ is a norm on $L(X, Y)$ making $\langle L(X, Y), \|L\|_{L(X, Y)} \rangle$ a normed space.

Proof.

1. Let $x \in X$ then we have either:

$\|x\|_X = 0$. Then

$$\|L(x)\|_Y = \|L(0)\|_Y = \|0\|_Y = 0 = \|L\|_{L(X, Y)} \cdot 0 = \|L\|_{L(X, Y)} \cdot \|x\|_X$$

$\|x\|_X \neq 0$. Assume that $\|L\|_{L(X, Y)} < \frac{1}{\|x\|_X} \cdot \|L(x)\|_Y$ then, as $\|L\|_{L(X, Y)} = \inf(A_{L, r})$, there exist a

$$M \in A_{L, r} = \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ we have } \|L(x)\|_Y \leq M \cdot \|x\|_X\}$$

such that $\|L\|_{L(X, Y)} \leq M < \frac{1}{\|x\|_X} \cdot \|L(x)\|_Y$ so that $M \cdot \|x\|_X < \|L(x)\|_Y$ contradicting the fact that $M \in A_{L, r} \Rightarrow \|L(x)\|_Y \leq M \cdot \|x\|_X$. So we must have that $\frac{1}{\|x\|_X} \cdot \|L(x)\|_Y \leq \|L\|_{L(X, Y)}$ or

$$\|L(x)\|_Y \leq \|L\|_{L(X, Y)} \cdot \|x\|_X$$

So in all cases we have

$$\|L(x)\|_Y \leq \|L\|_{L(X, Y)} \cdot \|x\|_X$$

2. $\forall x \in X$ we have $\|C_0(x)\|_Y = \|0\|_Y = 0 \leq 0 \cdot \|x\|_X$ proving that $0 \in A_{C_0, r}$, hence

$$0 \leq \inf(A_{C_0, r}) \leq 0$$

resulting in

$$\|C_0\|_{L(X, Y)} = \inf(A_{C_0, r}) = 0$$

3. Let $L \in L(X, Y)$ then by [theorem: 14.208] we have

$$\forall L \in L(X, Y) \text{ that } 0 \leq \|L\|_{L(X, Y)} \tag{14.76}$$

Further if $L_1, L_2 \in L(X, Y)$ then for $x \in X$ we have by (1) that

$$\|L_1(x)\|_Y \leq \|L\|_{L(X, Y)} \cdot \|x\|_X \text{ and } \|L_2(x)\|_Y \leq \|L_2\|_{L(X, Y)} \cdot \|x\|_X \tag{14.77}$$

Further

$$\begin{aligned} \|(L_1 + L_2)(x)\|_Y &= \|L_1(x) + L_2(x)\|_Y \\ &\leq \|L_1(x)\|_Y + \|L_2(x)\|_Y \\ &\stackrel{[\text{eq: 14.77}]}{\leq} \|L\|_{L(X, Y)} \cdot \|x\|_X + \|L_2\|_{L(X, Y)} \cdot \|x\|_X \\ &= (\|L\|_{L(X, Y)} + \|L_2\|_{L(X, Y)}) \cdot \|x\|_X \end{aligned}$$

proving that $\|L\|_{L(X,Y)} + \|L\|_{L(X,Y)} \in A_{(L_1+L_2),r}$ so that

$$\|L_1 + L_2\|_{L(X,Y)} = \inf(A_{(L_1+L_2),r}) \leq \|L\|_{L(X,Y)} + \|L\|_{L(X,Y)} \quad (14.78)$$

Let $\alpha \in \mathbb{K}$ and $L \in L(X,Y)$. By (1) we have $\forall x \in X$ that $\|L(x)\|_Y \leq \|L\|_{L(X,Y)} \cdot \|x\|_X$ then

$$\|(\alpha \cdot L)(x)\|_Y = \|\alpha \cdot L(x)\|_Y = |\alpha| \cdot \|L(x)\|_Y \leq |\alpha| \cdot \|L\|_{L(X,Y)} \cdot \|x\|_X$$

proving that $|\alpha| \cdot \|L\|_{L(X,Y)} \in A_{\alpha \cdot L,r}$. Hence

$$\|\alpha \cdot L\|_{L(X,Y)} = \inf(A_{\alpha \cdot L,r}) \leq |\alpha| \cdot \|L\|_{L(X,Y)} \quad (14.79)$$

For the opposite inequality we have two cases for $|\alpha|$ to consider:

$|\alpha| = 0$. Then, as $\|\cdot\|$ is a norm in $\mathbb{K}$, we have $\alpha = 0$ so that $\alpha \cdot L = C_0$, hence

$$\|\alpha \cdot L\|_{L(X,Y)} = \|C_0\|_{L(X,Y)} \stackrel{(2)}{=} 0 = |\alpha| \cdot \|L\|_{L(X,Y)}$$

or

$$|\alpha| \cdot \|L\|_{L(X,Y)} = 0 = \|\alpha \cdot L\|_{L(X,Y)} \Rightarrow |\alpha| \cdot \|L\|_{L(X,Y)} \leq \|\alpha \cdot L\|_{L(X,Y)}$$

$|\alpha| \neq 0$. As $\forall x \in X$ we have $\|(\alpha \cdot L)(x)\|_Y = \|\alpha \cdot L(x)\|_Y = |\alpha| \cdot \|L(x)\|_Y \leq |\alpha| \cdot \|L\|_{L(X,Y)} \cdot \|x\|_X$ so that

$$\|L(x)\|_Y = \frac{\|(\alpha \cdot L)(x)\|_Y}{|\alpha|} \leq \frac{\|\alpha \cdot L\|_{L(X,Y)} \cdot \|x\|_X}{|\alpha|}$$

proving that $\frac{\|\alpha \cdot L\|_{L(X,Y)}}{|\alpha|} \in A_{L,r}$, hence $\|L\|_{L(X,Y)} = \inf(A_{L,r}) \leq \frac{\|\alpha \cdot L\|_{L(X,Y)}}{|\alpha|}$. So

$$|\alpha| \cdot \|L\|_{L(X,Y)} \leq \|\alpha \cdot L\|_{L(X,Y)}$$

So in all cases we have $|\alpha| \cdot \|L\|_{L(X,Y)} \leq \|\alpha \cdot L\|_{L(X,Y)}$ which combined with [eq: 14.79] results in

$$\|\alpha \cdot L\|_{L(X,Y)} = |\alpha| \cdot \|L\|_{L(X,Y)} \quad (14.80)$$

Using [eqs: 14.76, 14.78 and 14.80] it follows that

$$\|\cdot\|_{L(X,Y)} \text{ is a pseudo norm}$$

To prove that $\|\cdot\|_{L(X,Y)}$ is a norm let $L \in L(X,Y)$ such that $\|L\|_{L(X,Y)} = 0$. Assume that $L \neq C_0$ [the neutral element in $L(X,Y)$] then there exist a $x \in X$ such that $L(x) \neq C_0(x) = 0$. As $\|\cdot\|_Y$ is a norm we have $\|L(x)\|_Y \neq 0$, but then as $0 \neq \|L(x)\|_Y \leq_{(1)} \|L\|_{L(X,Y)} \cdot \|x\|_X = 0$ resulting in a contradiction. So the assumption is wrong and we must have that $L = C_0$ which proves that

$$\|\cdot\|_{L(X,Y)} \text{ is a norm on } L(X,Y) \quad \square$$

Up to now the operator norm $\|\cdot\|_{L(X,Y)}$ is defined using an infimum, the following theorem shows that we can also define the operator norm as a supremum.

Theorem 14.211. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces where X is not trivial [there exists a $x \in X$ with $x \neq 0$], $L \in L(X,Y)$ and define

$$A_{L,t} = \{\|L(x)\|_Y \mid x \in X \text{ with } \|x\|_X = 1\}$$

and

$$A_{L,u} = \{\|L(x)\|_Y \mid x \in X \text{ with } \|x\|_X \leq 1\}$$

then

$$\|L\|_{L(X,Y)} = \sup(A_{L,t}) = \sup(A_{L,u})$$

Proof. If $M \in A_{L,t}$ then $\exists x \in X$ with $\|x\|_X = 1$ such that $M = \|L(x)\|_Y$ which as $\|x\|_X = 1 \leq 1$ proves that $M \in A_{L,u}$. Hence

$$A_{L,t} \subseteq A_{L,u} \quad (14.81)$$

As $\exists x_0 \in X$ with $x_0 \neq 0$ we have as $\|\cdot\|_X$ is a norm that $\left\| \frac{1}{\|x_0\|} \cdot x_0 \right\|_X = \frac{1}{\|x_0\|_X} \cdot \|x_0\|_X = 1$ so that $\|L(x_0)\|_Y \in A_{L,t}, A_{L,u}$ proving that

$$\emptyset \neq A_{L,t} \wedge \emptyset = A_{L,u} \quad (14.82)$$

Now if $M \in A_{L,u}$ then $\exists x \in X$ with $\|x\|_X \leq 1$ such that $M = \|L(x)\|_Y$, using [theorem: 14.210] we have that $\|L(x)\|_Y \leq \|L\|_{L(X,Y)} \cdot \|x\|_X \leq \|L\|_{L(X,Y)} \cdot 1 = \|L\|_{L(X,Y)}$, proving that $\|L\|_{L(X,Y)}$ is an upper bound for $A_{L,u}$ and by [eq: 14.81] $A_{L,u}$. Hence by [theorem: 10.18] $\sup(A_{L,t}), \sup(A_{L,u})$ exists and $\sup(A_{L,t}), \sup(A_{L,u}) \leq \|L\|_{L(X,Y)}$. Further by [eq: 14.81] and [theorem: 3.77] we have that $\sup(A_{L,t}) \leq \sup(A_{L,u})$. Summarized we have that

$$\sup(A_{L,t}) \leq \sup(A_{L,u}) \leq \|L\|_{L(X,Y)}$$

Now $\forall x \in X$ with $\|x\|_X = 1$ we have that $\|L(x)\|_Y \in A_{L,t}$ so that $\|L(x)\|_Y \leq \sup(A_{L,t})$ or using the definition of $A_{L,s}$ [see: 14.209] that $\sup(A_{L,t}) \in A_{L,s}$. Hence

$$\|L\|_{L(X,Y)} = \inf(A_{L,s}) \leq \sup(A_{L,t}) \leq \sup(A_{L,u}) \leq \|L\|_{L(X,Y)}$$

proving that

$$\sup(A_{L,t}) = \sup(A_{L,u}) = \|L\|_{L(X,Y)} \quad \square$$

To summarize the the above theorems about $L(X, Y)$ we have

Theorem 14.212. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces then

$$\langle L(X, Y), \|\cdot\|_{L(X,Y)} \rangle$$

is a normed space where

$$\forall L \in L(X, Y) \text{ we have } \|L\|_{L(X,Y)} = \inf(A_{L,s}) = \inf(A_{L,r}) = \sup(A_{L,t}) = \sup(A_{L,u})$$

and

$$\begin{aligned} A_{L,s} &= \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ with } \|x\|_X = 1 \text{ we have } \|L(x)\|_Y \leq M\} \\ A_{L,r} &= \{M \in \mathbb{R}_0^+ \mid \forall x \in X \text{ we have } \|L(x)\|_Y \leq M \cdot \|x\|_X\} \\ A_{L,t} &= \{\|L(x)\|_Y \mid x \in X \text{ with } \|x\|_X = 1\} \\ A_{L,u} &= \{\|L(x)\|_Y \mid x \in X \text{ with } \|x\|_X \leq 1\} \end{aligned}$$

Note 14.213. If there exist a $M \in \mathbb{R}_0^+$ such that $\forall x \in X$ with $\|x\|_X = 1$ we have $\|L(x)\|_Y \leq M$ then $M \in A_{L,s}$ so that $\|L\|_{L(X,Y)} = \inf(A_{L,s}) \leq M$.

Note 14.214. If there exist a $M \in \mathbb{R}_0^+$ such that $\forall x \in X$ we have $\|L(x)\|_Y \leq M \cdot \|x\|_X$ then $M \in A_{L,r}$ so that $\|L\|_{L(X,Y)} = \inf(A_{L,r}) \leq M$.

Proof. This follows from [theorems: 14.200, 14.210, 14.211]. $\square$

Example 14.215. Let $\langle X, \|\cdot\| \rangle$ be a normed space then $\text{Id}_X \in L(X, X)$ and $\|\text{Id}_X\|_{L(X,X)} \leq 1$.

Proof. From [example: 14.215] $\text{Id}_X \in L(X, X)$. Further for $x \in X$ we have $\|\text{Id}_X(x)\| = \|x\|$ which, using [theorem: 14.212], proves that $\|\text{Id}_X\|_{L(X,X)} \leq 1$. $\square$

Example 14.216. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $\langle \mathbb{K}, \|\cdot\| \rangle$ be normed space of real or complex numbers and $x \in X$ then if we define

$$(\cdot x): \mathbb{K} \rightarrow X \text{ by } (\cdot x)(r) = r \cdot x$$

we have that

$$(\cdot x) \in L(\mathbb{K}, X)$$

with

$$\|(\cdot x)\|_{L(\mathbb{K}, X)} = \|x\|$$

Proof. Let $r, s \in \mathbb{K}$ and $\alpha \in \mathbb{K}$ then we have

$$(\cdot x)(r + \alpha \cdot s) = (r + \alpha \cdot s) \cdot x = r \cdot x + (\alpha \cdot s) \cdot x = r \cdot x + \alpha \cdot (s \cdot x) = (\cdot x)(r) + \alpha \cdot (\cdot x)(s)$$

proving that

$$(\cdot x) \in \text{Hom}(\mathbb{K}, X)$$

Let $r \in \mathbb{K}$ be such that $|r| = 1$ then $\|(\cdot x)(r)\| = \|r \cdot x\| = |r| \cdot \|x\| = \|x\|$ so that by [theorem: 14.201]

$$(\cdot x) \in L(\mathbb{K}, X)$$

Further

$$A_{(\cdot x), t} = \{\|(\cdot x)(r)\| \mid |r| = 1\} = \{\|x\|\}$$

so that

$$\|(\cdot x)\|_{L(\mathbb{K}, X)} \underset{[\text{theorem: 14.212}]}{=} \sup(A_{(\cdot x), t}) = \sup(\{\|x\|\}) = \|x\| \quad \square$$

Example 14.217. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a family of normed spaces and $\langle \prod_{i \in \{1, \dots, n\}} X_i, \|\cdot\|_{\max} \rangle$ the product space with the maximum norm [see theorem: 14.98] then

$$\forall i \in \{1, \dots, n\} \quad \pi_i \in L\left(\prod_{j \in \{1, \dots, n\}} X_j, X_i\right) \text{ and } \|\pi_i\|_{L(\prod_{j \in \{1, \dots, n\}} X_j, X_i)} \leq 1$$

Proof. By [theorem: 11.175] we have

$$\forall i \in \{1, \dots, n\} \quad \pi_i: \prod_{j \in \{1, \dots, n\}} X_j \rightarrow X_i \text{ is a linear mapping}$$

Further if $x \in \prod_{j \in \{1, \dots, n\}} X_j$ then

$$\|\pi_i(x)\|_i \leq \max(\{\|\pi_i(x)\|_i \mid i \in \{1, \dots, n\}\}) = \|x\|_{\max} \leq 1 \cdot \|x\|_{\max}$$

so that by [theorem: 14.201] π_i is continuous. Further $1 \in A_{L, r}$ so that $\|\pi_i\|_{L(\prod_{j \in \{1, \dots, n\}} X_j, X_i)} \leq 1$. Hence we have

$$\forall i \in \{1, \dots, n\} \quad \pi_i \in L\left(\prod_{j \in \{1, \dots, n\}} X_j, X_i\right) \text{ and } \|\pi_i\|_{L(\prod_{j \in \{1, \dots, n\}} X_j, X_i)} \leq 1 \quad \square$$

Theorem 14.218. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ and $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces over $\mathbb{K}$, $L_1 \in L(X, Y)$, $L_2 \in L(Y, Z)$ then $L_2 \circ L_1 \in L(X, Z)$ and $\|L_2 \circ L_1\|_{L(X, Z)} \leq \|L_1\|_{L(X, Y)} \cdot \|L_2\|_{L(Y, Z)}$

Proof. By [theorems: 11.180 and 14.158] it follows that

$$L_2 \circ L_1 \in L(X, Z)$$

Further $\forall x \in X$ we have

$$\|(L_2 \circ L_1)(x)\|_Z = \|L_2(L_1(x))\|_Z \leq \|L_2\|_{L(Y, Z)} \cdot \|L_1(x)\|_Y \leq \|L_1\|_{L(X, Y)} \cdot \|L_2\|_{L(Y, Z)} \cdot \|x\|_X$$

Hence $\|L_1\|_{L(X, Y)} \cdot \|L_2\|_{L(Y, Z)} \in A_{L_2 \circ L_1, r}$ so that

$$\|L_2 \circ L_1\|_{L(X, Z)} = \inf(A_{L_2 \circ L_1, r}) \leq \|L_1\|_{L(X, Y)} \cdot \|L_2\|_{L(Y, Z)} \quad \square$$

Theorem 14.219. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces over K , $\langle W, \|\cdot\|_W \rangle$ the normed space where W is a sub-space of X and $\|\cdot\|_W: W \rightarrow \mathbb{R}$ is defined by $\|z\|_W = \|z\|_X$ then we have

$$\forall L \in L(X, Y) \text{ that } L|_W \in L(W, Y) \text{ and } \|L|_W\|_{L(W, Y)} \leq \|L\|_X$$

Proof. First by [theorem: 11.169] it follows that

$$L|_W \in \text{Hom}(W, Y)$$

Let $x \in W$ then we have

$$\|L|_W(x)\|_Y = \|L(x)\|_Y \leq \|L\|_{L(X,Y)} \cdot \|x\|_X = \|L\|_{L(X,Y)} \cdot \|x\|_{|W}$$

which proves by [theorem: 14.201] that

$$L|_W \in L(W, Y)$$

Further using [theorem: 14.212] we have that

$$\|L|_W\|_{L(W,Y)} \leq \|L\|_{L(X,Y)} \quad \square$$

For open linear mappings we have:

Theorem 14.220. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces over $\mathbb{K}$ then for a linear mapping*

$$L: X \rightarrow Y$$

we have

$$L \text{ is open} \Leftrightarrow \exists \delta \in \mathbb{R}^+ \text{ such that } B_{\|\cdot\|_Y}(0, \delta) \subseteq L(B_{\|\cdot\|_X}(0, 1))$$

Proof.

$\Rightarrow$. As $B_{\|\cdot\|_X}(0, 1) \in \mathcal{T}_{\|\cdot\|_X}$ and L is open it follows that $L(B_{\|\cdot\|_X}(0, 1)) \in \mathcal{T}_{\|\cdot\|_Y}$, further $0 \in B_{\|\cdot\|_X}(0, 1) \Rightarrow 0 = L(0) \in L(B_{\|\cdot\|_X}(0, 1))$. So by [theorem: 14.97] there exist a $\delta \in \mathbb{R}^+$ such that

$$0 \in B_{\|\cdot\|_Y}(0, \delta) \subseteq L(B_{\|\cdot\|_X}(0, 1)).$$

$\Leftarrow$. Let $U \in \mathcal{T}_{\|\cdot\|_X}$ and let $y \in L(U)$ then there exist a $x \in U$ such that $y = L(x)$, by [theorem: 14.97] there exists a $\delta' \in \mathbb{R}^+$ such that

$$x \in B_{\|\cdot\|_X}(x, \delta') \subseteq U \Rightarrow L(x) \in L(B_{\|\cdot\|_X}(x, \delta')) \subseteq L(U).$$

By the hypothesis there exist a $\delta'' \in \mathbb{R}^+$ such that $0 \in B_{\|\cdot\|_Y}(0, \delta'') \subseteq L(B_{\|\cdot\|_X}(0, 1))$. Then we have for $\delta = \delta'' \cdot \delta'$ that

$$\begin{aligned} z \in B_{\|\cdot\|_Y}(y, \delta) &\Rightarrow z \in B_{\|\cdot\|_Y}(y, \delta'' \cdot \delta') \\ &\Rightarrow z \in B_{\|\cdot\|_Y}(L(x), \delta'' \cdot \delta') \\ &\Rightarrow \|z - L(x)\|_Y < \delta'' \cdot \delta' \\ &\Rightarrow \frac{1}{\delta'} \cdot \|z - L(x)\|_Y < \delta'' \\ &\Rightarrow \left\| \frac{1}{\delta'} \cdot (z - L(x)) \right\|_Y < \delta'' \\ &\Rightarrow \frac{1}{\delta'} \cdot (z - L(x)) \in B_{\|\cdot\|_Y}(0, \delta'') \subseteq L(B_{\|\cdot\|_X}(0, 1)) \\ &\Rightarrow \exists x_0 \in X \text{ such that } \|x_0\|_X < 1 \text{ and } L(x_0) = \frac{1}{\delta'} \cdot (z - L(x)) \\ &\Rightarrow z = \delta' \cdot L(x_0) + L(x) \wedge \|x_0\|_X < 1 \\ &\Rightarrow z = L(\delta' \cdot x_0 + x) \wedge \|x_0\|_X < 1 \\ &\Rightarrow z = L(\delta' \cdot x_0 + x) \wedge \|\delta' \cdot x_0\|_X < \delta' \\ &\Rightarrow z = L(\delta' \cdot x_0 + x) \wedge \|(\delta' \cdot x_0 + x) - x\|_X < \delta' \\ &\Rightarrow z = L(\delta' \cdot x_0 + x) \wedge \delta' \cdot x_0 + x \in B_{\|\cdot\|_X}(x, \delta') \\ &\Rightarrow z \in L(B_{\|\cdot\|_X}(x, \delta')) \subseteq L(U) \end{aligned}$$

proving that $B_{\|\cdot\|_Y}(y, \delta) \subseteq L(U)$. In other words

$$\forall y \in L(U) \text{ there exist a } \delta \in \mathbb{R}^+ \text{ such that } y \in B_{\|\cdot\|_Y}(y, \delta) \subseteq L(U)$$

So by [theorem: 14.97] it follows that $L(U) \in \mathcal{T}_{\|\cdot\|_Y}$ proving that L is open. $\square$

14.5.1 The Hahn-Banach theorem

We prove now a important result in functional analysis, the Hahn-Banach theorem. This theorem states that every continuous linear functional on a sub-space of a real normed space can be extended to a continuous linear functional on the normed space. The proof is rather elaborated and uses Zorn's lemma, so we split the proof in different lemma's. We start with proving Hahn-Banach for a one dimensional extension.

Lemma 14.221. *Let $\langle X, \|\cdot\| \rangle$ be a real normed linear space, $Y \subseteq X$ a sub space of X and $x \in X \setminus Y$. Consider*

$\langle Y, \|\cdot\|_Y \rangle$ the normed space based on the sub-space norm $\|\cdot\|_Y$ [see theorem: 14.114]

$Y_x = \text{span}(Y \cup \{x\})$ [which by [theorem: 11.110] is a sub-space of X]

$\langle Y_x, \|\cdot\|_{Y_x} \rangle$ the normed space based on the sub-space norm $\|\cdot\|_{Y_x}$ [see theorem: 14.114]

then for every $L \in L(Y, \mathbb{R})$ there exist a $L_x \in L(Y_x, \mathbb{R})$ such that

$$(L_x)|_Y = L \text{ and } \|L_x\|_{L(Y_x, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})}$$

*In other words if Y is a **proper** subset of X then every continuous linear functional on Y can be extended to a sub-space, one dimensional larger, perserving the operator norm.*

Proof. For $\|L\|_{L(Y, \mathbb{R})}$ we have either:

$\|L\|_{L(Y, \mathbb{R})} = 0$. Then L must be the neutral element of $L(Y, \mathbb{R})$, take then L_x the neutral element in $L(Y_x, \mathbb{R})$, then

$$L_x \in L(Y_x, \mathbb{R})$$

and we have by the properties of norms that

$$\|L\|_{L(Y, \mathbb{R})} = 0 = \|L_x\|_{L(Y_x, \mathbb{R})}$$

Further $\forall y \in Y$ we have $L_x(y) \stackrel{[\text{theorem: 14.200}]}{=} 0 \stackrel{[\text{theorem: 14.200}]}{=} L(y)$ which proves that

$$(L_x)|_Y = L$$

$\|L\|_{L(Y, \mathbb{R})} \neq 0$. Define then $K \in L(Y, \mathbb{R})$ [note that $L(X, \mathbb{R})$ is a vector space] by

$$K = \frac{1}{\|L\|_{L(Y, \mathbb{R})}} \cdot L \tag{14.83}$$

so that we have

$$\|K\|_{L(Y, \mathbb{R})} = \left\| \frac{1}{\|L\|_{L(Y, \mathbb{R})}} \cdot L \right\|_{L(Y, \mathbb{R})} = \frac{1}{\|L\|_{L(Y, \mathbb{R})}} \cdot \|L\|_{L(Y, \mathbb{R})} = 1 \tag{14.84}$$

Let $u, v \in Y$ then we have

$$\begin{aligned} K(u) + K(v) &\leq |K(u) + K(v)| \\ &= |K(u + v)| \\ &\leq \|K\|_{L(Y, \mathbb{R})} \cdot \|u + v\|_Y \\ &\stackrel{[\text{eq: 14.84}]}{=} \|u + v\|_Y \\ &= \|u + v\| \\ &= \|u - x + x + v\| \\ &\leq \|u - x\| + \|x + v\| \end{aligned}$$

so that

$$\forall u, v \in Y \text{ we have } K(u) - \|u - x\| \leq \|v + x\| - K(v)$$

Define $S = \{K(u) - \|u - x\| \mid u \in Y\}$. Then by the above it follows that S_v is bounded above by $\|v + x\| - K(v) \forall v \in Y$, further $0 \in Y \Rightarrow Y \neq \emptyset$ from which it follows that $S \neq \emptyset$. Hence using the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] we have that

$$M = \sup(S) = \sup(\{K(u) - \|u - x\| \mid u \in Y\}) \text{ exist and } M \leq \|v + x\| - K(v) \forall v \in Y \quad (14.85)$$

Now $\forall y \in Y_x = \text{span}(Y \cup \{x\})$ we have by [theorem: 11.215] that there exist **unique** $z_y \in Y$ and $\lambda_y \in \mathbb{R}$ such that $y = z_y + \lambda_y \cdot x$ which allows use to define the function

$$K_x: Y_x \rightarrow \mathbb{R} \text{ by } K_x(y) = \lambda_y \cdot M + K(z_y) \text{ where } \lambda_y \in \mathbb{R}, z_y \in Y \text{ are such that } y = z_y + \lambda_y \cdot x$$

Let $u, v \in Y_x$ and $\alpha \in \mathbb{R}$ then $u = z_u + \lambda_u \cdot x$, $v = z_v + \lambda_v \cdot x$ and

$$u + \alpha \cdot v = (z_u + \alpha \cdot z_v) + (\lambda_u + \alpha \cdot \lambda_v) \cdot x$$

so that

$$\begin{aligned} K_x(u + \alpha \cdot v) &= (\lambda_u + \alpha \cdot \lambda_v) \cdot M + K(z_u + \alpha \cdot z_v) \\ &= \lambda_u \cdot M + K(z_u) + \alpha \cdot \lambda_v \cdot M + \alpha \cdot K(z_v) \\ &= K_x(u) + \alpha \cdot K_x(v) \end{aligned}$$

which prove that

$$K_x \in \text{Hom}(Y_x, \mathbb{R}) \quad (14.86)$$

Let $u \in Y$ then $K(u) - \|u - x\| \underset{\text{[eq: 14.85]}}{\leq} M$ and $M \underset{\text{[eq: 14.85]}}{\leq} \|u + x\| - K(u)$ so that

$$\forall u \in Y \text{ we have } K(u) - M \leq \|u - x\| \wedge M + K(u) \leq \|u + x\| \quad (14.87)$$

If $\lambda \in \mathbb{R}^+$ then $\forall u \in Y$ we have $\frac{1}{\lambda} \cdot u \in Y$ so that by [eq: 14.87]

$$K\left(\frac{1}{\lambda} \cdot u\right) - M \leq \left\| \frac{1}{\lambda} \cdot u - x \right\| \wedge M + K\left(\frac{1}{\lambda} \cdot u\right) \leq \left\| \frac{1}{\lambda} \cdot u + x \right\|$$

which after multiplying both sides of the above by λ , taking in account that $0 < \lambda$, we have

$$K(u) - \lambda \cdot M \leq \|u - \lambda \cdot x\| \wedge \lambda \cdot M + K(u) \leq \|u + \lambda \cdot x\|$$

which proves that

$$\forall \lambda \in \mathbb{R}^+ \forall u \in Y \text{ we have } K(u) - \lambda \cdot M \leq \|u - \lambda \cdot x\| \wedge \lambda \cdot M + K(u) \leq \|u + \lambda \cdot x\| \quad (14.88)$$

As $\forall u \in Y$ we have $-u \in Y$ so it follows from [eq: 14.88] that

$$\forall \lambda \in \mathbb{R}^+ \forall u \in Y \text{ we have } -(K(u) + \lambda \cdot M) \leq \|-u - \lambda \cdot x\| = \|u + \lambda \cdot x\| \quad (14.89)$$

Let $\lambda \in \mathbb{R}^-$ then $-\lambda \in \mathbb{R}^+$ so using [eqs: 14.88] we have

$$\forall \lambda \in \mathbb{R}^- \forall u \in Y \text{ we have } K(u) + \lambda \cdot M \leq \|u + \lambda \cdot x\| \wedge -\lambda \cdot M + K(u) \leq \|u - \lambda \cdot x\| \quad (14.90)$$

As $\forall u \in Y$ we have $-u \in Y$ so it follows from [eq: 14.90] that

$$\forall \lambda \in \mathbb{R}^- \forall u \in Y \text{ we have } -(\lambda \cdot M + K(u)) \leq \|-u - \lambda \cdot x\| = \|u + \lambda \cdot x\| \quad (14.91)$$

We use the above now in calculating the value of $|K_x(y)|$. Let $y \in Y_x$ then there exist unique $z_y \in Y$ and $\lambda_y \in \mathbb{R}$ such that $y = z_y + \lambda_y \cdot x$ so that

$$K_x(y) = \lambda_y \cdot M + K(z_y)$$

For λ_y we have either:

$\lambda_y = 0$. Then $y = z_y$ and

$$|K_x(y)| = |K(z_y) + 0 \cdot M| = |K(z_y)| \leq \|K\|_{L(Y, \mathbb{R})} \cdot \|z_y\|_{|Y|} \underset{\text{[eq: 14.84]}}{=} \|z_y\|_{|Y|} = \|z_y\| = \|y\|$$

$0 < \lambda_y$. Then

$$K_x(y) = \lambda_y \cdot M + K(z_y) \underset{\text{[eq: 14.88]}}{\leq} \|z_y + \lambda_y \cdot x\| = \|y\|$$

and

$$-K_x(y) = -(\lambda_y \cdot M - K(z_y)) \underset{[\text{eq: 14.89}]}{\leq} \|z_y + \lambda_y \cdot x\| = \|y\|$$

which proves that

$$|K_x(y)| \leq \|y\|$$

$\lambda_y < 0$. Then

$$K_x(y) = \lambda_y \cdot M + K(z_y) \underset{[\text{eq: 14.90}]}{\leq} \|z_y + \lambda_y \cdot x\| = \|y\|$$

and

$$-K_x(y) = -(\lambda_y \cdot M - K(z_y)) \underset{[\text{eq: 14.91}]}{\leq} \|z_y + \lambda_y \cdot x\| = \|y\|$$

which proves that

$$|K_x(y)| \leq \|y\|$$

So we have proved that $\forall y \in Y_x \quad |K_x(y)| \leq \|y\| = \|y\|_{|Y_x}$ which proves by [theorems: 14.201, 14.212] that

$$K_x \in L(Y_x, \mathbb{R}) \text{ and } \|K_x\|_{L(Y_x, \mathbb{R})} \leq 1 \quad (14.92)$$

Let $y \in Y$ then $y = y + 0 = y + 0 \cdot x$ so that $K_x(y) = 0 \cdot M + K(y) = K(y)$ which proves that

$$(K_x)_{|Y} = K \quad (14.93)$$

Assume now that $\|K_x\|_{L(X, \mathbb{R})} < 1$. As $\forall y \in Y$ we have $|K(y)| = |K_x(y)| \leq \|K_x\|_{L(Y_x, \mathbb{R})} \cdot \|y\|_{Y_x}$ it follows from [theorem: 14.212] that $\|K\|_{L(Y, \mathbb{R})} \leq \|K_x\|_{L(Y_x, \mathbb{R})} < 1$ which contradicts $\|K\|_{L(X, Y)} = 1$. Hence we must have that

$$\|K_x\|_{L(Y_x, \mathbb{R})} = 1 \quad (14.94)$$

Let $L_x = \|L\|_{L(Y, \mathbb{R})} \cdot K_x$ then as $L(Y_x, \mathbb{R})$ is a vector space it follows from [eq: 14.92] that

$$L_x \in L(Y_x, \mathbb{R})$$

Further we have

$$\|L_x\|_{L(Y_x, \mathbb{R})} = \| \|L\|_{L(Y, \mathbb{R})} \cdot K_x \|_{L(Y_x, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})} \cdot \|K_x\|_{L(Y_x, \mathbb{R})} \underset{[\text{eq: 14.94}]}{=} \|L\|_{L(Y, \mathbb{R})}$$

Finally $\forall y \in Y$ we have

$$L_x(y) = \|L\|_{L(Y, \mathbb{R})} \cdot K_x(y) \underset{[\text{eq: 14.93}]}{=} \|L\|_{L(Y, \mathbb{R})} \cdot K(y) \underset{[\text{eq: 14.83}]}{=} L(y)$$

from which it follows that

$$(L_x)_{|Y} = L \quad \square$$

We use now Zorn's lemma to extend the above lemma to the Hahn-Banach theorem. We start with creating a partial ordered set.

Definition 14.222. Let $\langle X, \|\cdot\| \rangle$ be a normed space, Y a sub-space of X and $L \in L(Y, \mathbb{R})$ then we define

$$\mathcal{E}_{X, Y, L} = \left\{ (Z, K) \mid \begin{array}{l} Z \text{ is a sub-space of } X, K \in L(Z, \mathbb{R}) \text{ with } \\ Y \subseteq Z \\ K_{|Y} = L \\ \|K\|_{L(Z, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})} \end{array} \right\}$$

Note 14.223. As $Y \subseteq Y \subseteq X$, $L \in (Y, \mathbb{R})$, $L_{|Y} = L$ and $\|L\|_{L(Y, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})}$ it follows that

$$(Y, L) \in \mathcal{E}_{X, Y, L}$$

We define now a partial order on $\mathcal{E}_{X, Y, L}$.

Lemma 14.224. Let $\langle X, ||| \rangle$ be a real normed space, Y a sub-space of X and $L \in L(Y, \mathbb{R})$ then if we define $\preceq \in \mathcal{E}_{X,Y,L} \times \mathcal{E}_{X,Y,L}$ by

$$(Z, K) \preceq (Z', K') \Leftrightarrow Z \subseteq Z' \text{ and } K|_Z = K'$$

then $\preceq$ is a partial order on $\mathcal{E}_{X,Y,L}$ [see definitions: 3.27, 3.28] making $\langle \mathcal{E}_{X,Y,L}, \preceq \rangle$ a partial ordered set.

Proof. We have

reflexivity. Let $(Z, K) \in \mathcal{E}_{X,Y,L}$ then we have trivially $Z \subseteq Z$ and $K|_Z = K$ which proves that

$$(Z, K) \preceq (Z, K)$$

anti-symmetry. If $(Z, K) \preceq (Z', K')$ and $(Z', K') \preceq (Z, K)$ then we have $Z \subseteq Z' \wedge Z' \subseteq Z$ so that $Z = Z'$. Further $\forall y \in Z = Z'$ we have $K(y) = (K')|_Z(y) = K'(y)$ so that $K = K'$. Hence we have that $(Z, K) = (Z', K')$.

transitivity. If $(Z, K) \preceq (Z', K') \wedge (Z', K') \preceq (Z'', K'')$ then we have $Z \subseteq Z' \subseteq Z''$ proving that $Z \subseteq Z''$. Further we have that $K = (K')|_Z$ and $K' = (K'')|_{Z'}$ so that $\forall y \in Z$ we have

$$K(y) = (K')|_Z(y) = K'(y) = (K'')|_{Z'}(y) = K''(y)$$

which proves that $K = (K'')|_Z$. Hence we have $(Z, K) \preceq (Z'', K'')$. $\square$

Lemma 14.225. Let $\langle X, ||| \rangle$ be a real normed space, Y a sub-space of X and $L \in L(Y, \mathbb{R})$ then for the partial ordered set $\langle \mathcal{E}_{X,Y,L}, \preceq \rangle$ we have that every **chain** $\mathcal{F} \subseteq \mathcal{E}_{X,Y,L}$ has a upper bound in $\mathcal{E}_{X,Y,L}$.

Proof. Let $\mathcal{F}$ be a chain in $\mathcal{E}_{X,Y,L}$ we have either:

$\mathcal{F} = \emptyset$. Then we have $\forall (Z, K) \in \mathcal{F}$ that $(Z, K) \preceq (Y, L)$ vacuously which as $(Y, L) \in \mathcal{E}_{X,Y,L}$ proves that $\mathcal{F} = \emptyset$ has a upper bound in $\mathcal{E}_{X,Y,L}$.

$\mathcal{F} \neq \emptyset$. Define

$$R = \bigcup_{(Z,K) \in \mathcal{F}} Z \subseteq X \quad (14.95)$$

As $\forall (Z, K) \in \mathcal{F}$ Z is a vector space we have $0 \in Z$ so that $0 \in R$ which proves that

$$R \neq \emptyset \quad (14.96)$$

Let $x, y \in R$ and $\alpha \in \mathbb{R}$ then there exists $(Z_x, K_x), (Z_y, K_y) \in \mathcal{F}$ such that $x \in Z_x \wedge y \in Z_y$. As $\mathcal{F}$ is a chain we have either:

$(Z_x, K_x) \preceq (Z_y, K_y)$. Then $Z_x \subseteq Z_y$ so that $x, y \in Z_y \Rightarrow x + \alpha \cdot y \in Z_y \subseteq R$.

$(Z_y, K_y) \preceq (Z_x, K_x)$. Then $Z_y \subseteq Z_x$ so that $x, y \in Z_x \Rightarrow x + \alpha \cdot y \in Z_x \subseteq R$.

so in all cases we have $x + \alpha \cdot y \in R$. This together with [eq: 14.96] proves [see theorem: 11.59] that

$$R \text{ is a sub-space of } X \quad (14.97)$$

Further we have $\forall (Z, K) \in \mathcal{F} \subseteq \mathcal{E}_{X,Y,L}$ that $Y \subseteq Z$ so that we have

$$Y \subseteq R \quad (14.98)$$

Define now $G = \bigcup_{(Z,K) \in \mathcal{F}} K \subseteq R \times \mathbb{R}$ [note that as K is a function from Z to $\mathbb{R}$ so that $K \subseteq Z \times \mathbb{R} \subseteq R \times \mathbb{R}$]. If $(x, y), (x, y') \in G$ then there exist $(Z_1, K_1), (Z_2, K_2) \in \mathcal{F}$ such that $(x, y) \in K_1 \wedge (x, y') \in K_2$ or using the definition of functions that $y = K_1(x) \wedge y' = K_2(x)$, further we have $x \in Z_1 \cap Z_2$. As $\mathcal{F}$ is a chain we have either:

$(Z_1, K_1) \preceq (Z_2, K_2)$. Then $K_1 = (K_2)|_{Z_1}$ so that $y = K_1(x) = (K_2)|_{Z_1}(x) = K_2(x) = y'$

$(Z_2, K_2) \preceq (Z_1, K_1)$. Then $K_2 = (X_1)_{|Z_2}$ so that $y = K_1(x) = (K_1)_{|Z_2}(x) = K_2(x) = y'$ so in all cases we have $y = y'$ which proves that

$G: R \rightarrow \mathbb{R}$ is a partial function

If $x \in R$ then there exist a $(Z, K) \in \mathcal{F}$ such that $x \in Z = \text{dom}(K)$ so there exist a $y \in \mathbb{R}$ such that $(x, y) \in K \subseteq \bigcup_{(Z, K) \in \mathcal{F}} K = G$ which proves that $x \in \text{dom}(G)$. Hence $R \subseteq \text{dom}(G)$ and using [theorem: 2.27] it follows that

$$G: R \rightarrow \mathbb{R} \text{ is a function} \quad (14.99)$$

If $x \in R$ then

$$\begin{aligned} y = G(x) &\Leftrightarrow (x, y) \in G \\ &\Leftrightarrow \exists (Z, K) \in \mathcal{F} \text{ with } (x, y) \in K \\ &\Leftrightarrow \exists (Z, K) \in \mathcal{F} \text{ with } x \in Z \wedge y = K(x) \end{aligned}$$

so we have

$$G: R \rightarrow \mathbb{R} \text{ is defined by } G(x) = K(z) \text{ where } (Z, K) \in \mathcal{F} \wedge x \in Z \quad (14.100)$$

Let $x, y \in R$ and $\alpha \in \mathbb{R}$ then there exists $(Z_1, K_1), (Z_2, K_2) \in \mathcal{F}$ such that $x \in Z_1 \wedge y \in Z_2$. As $\mathcal{F}$ is a chain we have either:

$(Z_1, K_1) \preceq (Z_2, K_2)$. Then $x, y \in Z_2 \Rightarrow x + \alpha \cdot y \in Z_2$ so that

$$G(x + \alpha \cdot y) = K_2(x + \alpha \cdot y) = K_2(x) + \alpha \cdot K_2(y) = G(x) + \alpha \cdot G(y)$$

$(Z_2, K_2) \preceq (Z_1, K_1)$. Then $x, y \in Z_1 \Rightarrow x + \alpha \cdot y \in Z_1$ so that

$$G(x + \alpha \cdot y) = K_1(x + \alpha \cdot y) = K_1(x) + \alpha \cdot K_1(y) = G(x) + \alpha \cdot G(y)$$

which proves that G is a linear function. Hence

$$G \in \text{Hom}(R, \mathbb{R}) \quad (14.101)$$

As $\mathcal{F} \neq \emptyset$ there exist a $(Z_0, K_0) \in \mathcal{F} \subseteq \mathcal{E}_{X, Y, L}$ so that $(K_0)_{|Y} = L$. Hence $\forall x \in Y \subseteq Z_0$ we have $L(x) = K_0(x)$ so that by [eq: 14.100] $L(x) = G(x)$ which proves that

$$G_{|Y} = L \quad (14.102)$$

Let $x \in R$ then $G(x) = K(x)$ where $(Z, K) \in \mathcal{F}$ and $x \in Z$ so that

$$|G(x)| = |K(x)| \leq \|K\|_{L(Z, \mathbb{R})} \cdot \|x\|_{|Z_{(Z, K) \in \mathcal{E}_{X, Y, Z}}} = \|L\|_{L(Y, Z)} \cdot \|x\|_{|Z} = \|L\|_{L(Y, Z)} \cdot \|x\|_{|R}$$

In other words we have proved that $\forall x \in R \ |G(x)| \leq \|L\|_{L(Y, Z)} \cdot \|x\|_{|R}$ which by [theorems: 14.201 and 14.212] proves that

$$G \in L(R, \mathbb{R}) \text{ and } \|G\|_{L(R, \mathbb{R})} \leq \|L\|_{L(R, \mathbb{R})} \quad (14.103)$$

Assume that $\|G\|_{L(R, \mathbb{R})} < \|L\|_{L(R, \mathbb{R})}$ then we have $\forall x \in Y$ that

$$|L(x)|_{\text{eq: 14.102}} |G(x)| \leq \|G\|_{L(R, \mathbb{R})}$$

so that by [theorem: 14.212] $\|L\|_{L(Y, \mathbb{R})} \leq \|G\|_{L(R, \mathbb{R})} < \|L\|_{L(R, \mathbb{R})}$ which is a contradiction. Hence the assumption is false and we must have $\|L\|_{L(R, \mathbb{R})} \leq \|G\|_{L(R, \mathbb{R})}$ which combined with [eq: 14.103] proves that

$$G \in L(R, \mathbb{R}) \text{ and } \|G\|_{L(R, \mathbb{R})} = \|L\|_{L(R, \mathbb{R})} \quad (14.104)$$

Using the definition of $\mathcal{E}_{X, Y, L}$ together with [eqs: 14.97, 14.98, 14.104, 14.102] we conclude that

$$(R, G) \in \mathcal{E}_{X, Y, L} \quad (14.105)$$

Let now $(Z, K) \in \mathcal{F}$ then $Z \subseteq \bigcup_{(Z, K) \in \mathcal{F}} Z = R$ and $\forall x \in Z$ we have by [eq: 14.100] that $G(x) = K(x)$ from which it follows that $G|_Z = K$, hence we have that $(Z, K) \preccurlyeq (G, R)$. Combining this with [eq: 14.105] proves finally that

$$(R, G) \text{ is an upper bound of } \mathcal{F} \quad \square$$

We are now ready to prove the Hahn-Banach theorem.

Theorem 14.226. (Hahn-Banach Theorem) *Let $\langle X, \|\cdot\| \rangle$ be a real normed space, $Y \subseteq X$ a sub-space of X , $L \in L(Y, \mathbb{R})$ then there exist a $K \in L(X, \mathbb{R})$ such that $K|_Y = L$ and $\|K\|_{L(X, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})}$*

Proof. From [lemma: 14.225] it follows that every chain in the partial ordered set $\langle \mathcal{E}_{X, Y, L}, \preccurlyeq \rangle$ has an upper bound so that by Zorn's lemma [see lemma: 3.128] there exist a maximal element $\langle Z_m, K_m \rangle \in \mathcal{E}_{X, Y, L}$. As $\langle Z_m, K_m \rangle \in \mathcal{E}_{X, Y, L}$ we have that

$$Z_m \text{ is a sub space of } X \wedge Y \subseteq Z_m \wedge K_m \in L(Z_m, \mathbb{R}) \wedge (K_m)|_Y = L \wedge \|K_m\|_{L(Z_m, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})} \quad (14.106)$$

Assume now that $Z_m \neq X$ then there exist a $x \in X \setminus Z_m$ and we can use [lemma: 14.221] from which it follows that for $Z = \text{span}(Z_m \cup \{x\})$, a sub space of X , there exist a $K \in L(Z, \mathbb{R})$ such that

$$K|_{Z_m} = K_m \text{ and } \|K\|_{L(Z, \mathbb{R})} = \|K_m\|_{L(Z_m, \mathbb{R})} \quad (14.107)$$

Now we have $Y \subseteq Z_m \subseteq Z_m \cup \{x\} \xrightarrow{[\text{theorem: 11.108}]} \text{span}(Z_m \cup \{x\})$ so that we have

$$Y \xrightarrow{[\text{theorem: 11.111}]} \text{span}(Y) \xrightarrow{[\text{theorem: 11.109}]} \text{span}(\text{span}(Z_m \cup \{x\})) \xrightarrow{[\text{theorem: 11.112}]} \text{span}(Z_m \cup \{x\}) = Z$$

which proves that

$$Y \subseteq Z \text{ where } Z \text{ is a sub-space of } X$$

Further $\forall x \in Y$ we have $L(x) \xrightarrow{(K_m)|_Y = L} K_m(x) \xrightarrow{[\text{eq: 14.107}]} K(x)$ so that

$$K|_Y = L$$

and

$$\|L\|_{L(Y, \mathbb{R})} \xrightarrow{[\text{eq: 14.106}]} \|K_m\|_{L(Z_m, \mathbb{R})} \xrightarrow{[\text{eq: 14.107}]} \|K\|_{L(Z, \mathbb{R})}$$

So we have proved that $\langle Z, K \rangle \in \mathcal{E}_{X, Y, L}$ which, as $\langle Z_m, K_m \rangle$ is the maximal element of $\mathcal{E}_{X, Y, L}$, means that $\langle Z, K \rangle \preccurlyeq \langle Z_m, K_m \rangle$. By the definition of $\preccurlyeq$ [see lemma: 14.224] we must then have that $Z \subseteq Z_m$. Now $x \in \text{span}(Z_m, \{x\}) = Z \subseteq Z_m$ which contradicts with the fact that $x \in X \setminus Z_m$, hence our assumption is wrong and we must have that $Z_m = X$. So we have by [eq: 14.106] that

$$K_m \in L(Z, \mathbb{R}) \wedge (K_m)|_Y = L \wedge \|K_m\|_{L(X, \mathbb{R})} = \|L\|_{L(Y, \mathbb{R})}$$

which completes the proof of the theorem. $\square$

Theorem 14.227. *Let $\langle X, \|\cdot\| \rangle$ be a Banach space, $x \in X$ with $x \neq 0$ then there exist a $L \in L(X, \mathbb{R})$ such that $\|L\|_{L(X, \mathbb{R})} = 1$ and $L(x) = \|x\|$*

Proof. Define $Y = \{t \cdot x | t \in \mathbb{R}\} \subseteq X$ then by [example: 11.60]

$$Y \text{ is a sub-space of } X$$

If $y \in Y$ then there exist a $t \in \mathbb{R}$ such that $y = t \cdot x$, let $t' \in \mathbb{R}$ such that $y = t' \cdot x$ hence $0 = (t - t') \cdot x$. Assume that $t \neq t'$ then $t - t' \neq 0$ so that $0 = \frac{1}{t - t'} \cdot (t - t') \cdot x = x$ contradicting the fact that $x \neq 0$, hence we must have that $t = t'$. So we can define the following function.

$$L': Y \rightarrow \mathbb{R} \text{ by } L(y) = t \cdot \|x\| \text{ where } y = t \cdot x$$

Let $y, z \in Y, \alpha \in \mathbb{R}$ then there exist $t_y, t_z \in \mathbb{R}$ such that $y = t_y \cdot x \wedge z = t_z \cdot x$ so that

$$\begin{aligned} L'(y + \alpha \cdot z) &= L'(t_y \cdot x + \alpha \cdot (t_z \cdot x)) \\ &= L'((t_y + \alpha \cdot t_z) \cdot x) \\ &= (t_y + \alpha \cdot t_z) \cdot x \\ &= t_y \cdot x + \alpha \cdot (t_z \cdot x) \\ &= L'(y) + \alpha \cdot L'(z) \end{aligned}$$

which proves that

$$L' \in \text{Hom}(Y, \mathbb{R})$$

Further for $y = t_y \cdot x \in Y$ with $\|y\| = 1$ we have

$$|L'(y)| = |L'(t_y \cdot x)| = |t_y \cdot \|x\|| = |t_y| \cdot \|x\| = \|t_y \cdot x\| = \|y\| = 1$$

so that by [theorems: 14.201]

$$L' \in L(Y, \mathbb{R})$$

and

$$\|L'\|_{L(X, \mathbb{R})} = \sup(\{\|L'(y)\| \mid y \in Y \wedge \|y\| = 1\}) = \sup(\{1\}) = 1$$

Using the Hahn-Banach theorem [theorem: 14.226] there exist a $L \in L(X, \mathbb{R})$ such that

$$\|L\|_{L(X, \mathbb{R})} = \|L'\|_{L(Y, \mathbb{R})} = 1 \quad \text{and} \quad L|_Y = L'$$

Further as $x = 1 \cdot x$ we have $x \in Y$ so that

$$L(x) = L|_Y(x) = L'(x) = 1 \cdot \|x\| = \|x\|$$

□

14.6 Multilinear mappings and continuity

We prove now that the set of continuous mappings is a sub space of the set of multilinear mappings.

Theorem 14.228. *Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_{i \in \{1, \dots, n\}} \rangle\}$ a family of normed vector spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed vector space over $\mathbb{K}$ and $L \in \text{Hom}(X_1, \dots, X_n; Y)$ then we have the following equivalences:*

1. *L is continuous [using the product topology on $\prod_{i \in \{1, \dots, n\}} X_i$ [see theorem: 14.98] and the topology $\mathcal{T}_{\|\cdot\|_Y}$ on Y].*
2. *L is continuous at 0.*
3. *$\exists M \in \mathbb{R}_0^+$ such that $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ satisfying $\forall i \in \{1, \dots, n\} \ \|x_i\|_i = 1$ we have that $\|L(x)\|_Y \leq M$.*
4. *$\exists M \in \mathbb{R}_0^+$ such that $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ we have $\|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$.*

Proof. First by [theorem: 14.98] we have $\mathcal{T}_{\text{product}} = \mathcal{T}_{\|\cdot\|_{\max}}$ where

$$\|x\|_{\max} = \max(\{\|\pi_i(x)\|_i \mid i \in \{1, \dots, n\}\}) = \max(\{\|x_i\|_i \mid i \in \{1, \dots, n\}\})$$

Secondly we have:

1 $\Rightarrow$ 2. This is trivial.

2 $\Rightarrow$ 3. Using [theorem: 14.135] there exist a $\delta \in \mathbb{R}^+$ such that $\forall y = (y_1, \dots, y_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ with $\|y\|_{\max} < \delta$ we have $\|L(y)\|_Y = \|L(y) - L(0)\|_Y < 1$. Let $x \in \prod_{i \in \{1, \dots, n\}} X_i$ such that $\forall i \in \{1, \dots, n\} \ \|x_i\|_i = 1$ then $\|x\|_{\max} = \max(\{\|x_i\|_i \mid i \in \{1, \dots, n\}\}) = 1$ so

$$\left\| \frac{\delta}{2} \cdot x \right\|_{\max} = \frac{\delta}{2} \cdot \|x\|_{\max} = \frac{\delta}{2} < \delta$$

and we must have that $\left\|L\left(\frac{\delta}{2} \cdot x\right)\right\|_Y < 1$. So if we take $M = \left(\frac{2}{\delta}\right)^n$ then

$$\begin{aligned}
 \frac{1}{M} \cdot \|L(x)\|_Y &= \left(\frac{\delta}{2}\right)^n \cdot \|L(x)\|_Y \\
 &= \left\|\left(\frac{\delta}{2}\right)^n \cdot L(x)\right\|_Y \\
 &= \left\|\left(\frac{\delta}{2}\right)^n \cdot L(x_1, \dots, x_n)\right\|_Y \\
 &\stackrel{[\text{theorem: 11.270}]}{=} \left\|L\left(\frac{\delta}{2} \cdot x_1, \dots, \frac{\delta}{2} \cdot x_n\right)\right\|_Y \\
 &= \left\|L\left(\frac{\delta}{2} \cdot (x_1, \dots, x_n)\right)\right\|_Y \\
 &= \left\|L\left(\frac{\delta}{2} \cdot x\right)\right\|_Y \\
 &< 1
 \end{aligned}$$

proving that

$$\|L(x)\|_Y < M$$

3 $\Rightarrow$ 4. Let $M \in \mathbb{R}_0^+$ be such that $\forall y \in \prod_{i \in \{1, \dots, n\}} X_i$ with $\forall i \in \{1, \dots, n\} \|y_i\|_i = 1$ we have $\|L(y)\|_Y \leq M$. Then given $x \in \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$ we have two cases possible:

$\exists i \in \{1, \dots, n\}$ with $x_i = 0$. Then by [theorem: 11.259] $L(x) = 0$ so that

$$\|L(x)\|_Y = \|0\|_Y = 0 \leq 0 = M \cdot 0 = M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

$\forall i \in \{1, \dots, n\}$ we have $x_i \neq 0$. Define then $y \in \prod_{i \in \{1, \dots, n\}} X_i$ by $y_i = \frac{1}{\|x_i\|_i} \cdot x_i$ then

$$\|y_i\|_i = \left\|\frac{1}{\|x_i\|_i} \cdot x_i\right\|_i = \frac{1}{\|x_i\|_i} \cdot \|x_i\|_i = 1.$$

Further

$$\begin{aligned}
 \frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot \|L(x_1, \dots, x_n)\|_Y &= \\
 \left\|\frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot L(x_1, \dots, x_n)\right\|_Y &\stackrel{[\text{theorem: 11.270}]}{=} \\
 \left\|L\left(\frac{1}{\|x\|_1} \cdot x_1, \dots, \frac{1}{\|x_n\|_n} \cdot x_n\right)\right\|_Y &= \\
 \|L(y_1, \dots, y_n)\|_Y &\leq M
 \end{aligned}$$

so that

$$\|L(x_1, \dots, x_n)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

hence in all cases we have

$$\|L(x_1, \dots, x_n)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

4 $\Rightarrow$ 1. By the hypothesis there exist a $M \in \mathbb{R}_0^+$ such that

$$\forall x \in \prod_{i \in \{1, \dots, n\}} X_i \text{ we have } \|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

Let $x \in \prod_{i \in \{1, \dots, n\}} X_i$ and $\varepsilon \in \mathbb{R}^+$ take then $\delta = \min\left(1, \frac{\varepsilon}{M+1}\right)$ then we have if $\|x - y\|_{\max} < \delta$ that $\forall i \in \{1, \dots, n\} \|x_i - y_i\|_i < \delta$ so that

$$\begin{aligned} \|L(x) - L(y)\|_Y &= \|L(x - y)\|_Y \\ &\leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i - y_i\|_i \\ &= M \cdot \left(\prod_{i \in \{1, \dots, n-1\}} \|x_i - y_i\|_i \right) \cdot \|x_n - y_n\|_n \\ &< M \cdot 1 \cdot \|x_n - y_n\|_n \\ &< M \cdot \frac{\varepsilon}{M+1} \\ &< \varepsilon \end{aligned}$$

proving that L is continuous at x . As $x \in \prod_{i \in \{1, \dots, n\}} X_i$ was chosen arbitrary it follows that L is continuous. $\square$

Example 14.229. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the complex [real] vector space over $\mathbb{K}$ then we have:

1. $\forall x = (x_1, \dots, x_n) \in \mathbb{K}^n$ we have $|\prod_{i \in \{1, \dots, n\}} x_i| = \prod_{i \in \{1, \dots, n\}} |x_i|$
2. $g: \mathbb{K}^n \rightarrow \mathbb{K}$ defined by $g(x_1, \dots, x_n) = \prod_{i \in \{1, \dots, n\}} x_i$ is multilinear and continuous [using the norms $\|_n = \max(\{|x_i| | i \in \{1, \dots, n\}\})$ and $\|\cdot\|$].
3. Let $\langle X, \|\cdot\|_X \rangle$ be a normed space and $z \in X$ then

$$h_z: \mathbb{K}^n \rightarrow X \text{ defined by } h_z(x) = z \cdot g(x) = \left(\prod_{i \in \{1, \dots, n\}} x_i \right) \cdot z$$

is multilinear and continuous [using the norms $\|_n = \max(\{|x_i| | i \in \{1, \dots, n\}\})$ and $\|\cdot\|_X$].

Proof.

1. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } (x_1, \dots, x_n) \in \mathbb{K}^n \text{ then } \left| \prod_{i \in \{1, \dots, n\}} x_i \right| = \prod_{i \in \{1, \dots, n\}} |x_i| \right\}$$

then we have:

1 $\in S$. If $(x_1, \dots, x_1) \in \mathbb{K}^1$ then $|\prod_{i \in \{1\}} x_i| = |x_1| = \prod_{i \in \{1\}} |x_i|$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $(x_1, \dots, x_{n+1}) \in \mathbb{K}^{n+1}$ then $(x_1, \dots, x_n) \in \mathbb{K}^n$ so that, as $n \in S$, we have $|\prod_{i \in \{1, \dots, n\}} x_i| = \prod_{i \in \{1, \dots, n\}} |x_i|$. So

$$\begin{aligned} \left| \prod_{i \in \{1, \dots, n+1\}} x_i \right| &= \left| \left(\prod_{i \in \{1, \dots, n\}} x_i \right) \cdot x_{n+1} \right| \\ &= \left| \prod_{i \in \{1, \dots, n\}} x_i \right| \cdot |x_{n+1}| \\ &= \left(\prod_{i \in \{1, \dots, n\}} |x_i| \right) \cdot |x_{n+1}| \\ &= \prod_{i \in \{1, \dots, n+1\}} |x_i| \end{aligned}$$

proving that $n+1 \in S$.

2. First using [theorem: 11.263] it follows that

$g: \mathbb{K}^n \rightarrow \mathbb{K}$ is multi linear

Secondly let $(x_1, \dots, x_n) \in \mathbb{K}^n$ then we have

$$|g(x_1, \dots, x_n)| = \left| \prod_{i \in \{1, \dots, n\}} x_i \right| \stackrel{(1)}{=} \prod_{i \in \{1, \dots, n\}} |x_i| = 1 \cdot \prod_{i \in \{1, \dots, n\}} |x_i|$$

which by [theorem: 14.228] proves that

$$g: \mathbb{K}^n \rightarrow \mathbb{K} \text{ is continuous}$$

3. Let $\alpha, x, y \in \mathbb{K}$, $i \in \{1, \dots, n\}$ then for $(x_1, \dots, \underbrace{x + \beta \cdot y}_i, \dots, x_n) \in \mathbb{K}^n$ we have:

$$\begin{aligned} h_z(x_1, \dots, \underbrace{x + \beta \cdot y}_i, \dots, x_n) &= \\ g(x_1, \dots, \underbrace{x + \beta \cdot y}_i, \dots, x_n) \cdot z &\stackrel{(2)}{=} \\ \left(g(x_1, \dots, \underbrace{x}_i, \dots, x_n) + \beta \cdot g(x_1, \dots, \underbrace{y}_i, \dots, x_n) \right) \cdot z &= \\ h_z(x_1, \dots, \underbrace{x}_i, \dots, x_n) + \beta \cdot h_z(x_1, \dots, \underbrace{y}_i, \dots, x_n) \end{aligned}$$

proving that

$$h_z: \mathbb{K}^n \rightarrow X \text{ is multilinear}$$

Further we have

$$\begin{aligned} \|h_z(x_1, \dots, x_n)\|_X &= \|g(x_1, \dots, x_n) \cdot z\| \\ &= |g(x_1, \dots, x_n)| \cdot \|z\|_X \\ &\leq_{(2)} \|z\|_X \cdot \prod_{i \in \{1, \dots, n\}} |x_i| \end{aligned}$$

proving that

$$h_z: \mathbb{K}^n \rightarrow X \text{ is continuous} \quad \square$$

Definition 14.230. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a family of normed spaces and $\langle Y, \|\cdot\|_Y \rangle$ a normed space then the set of the graphs of multilinear continuous mappings between $\prod_{i \in \{1, \dots, n\}} X_i$ and Y is noted as $L(X_1 \dots X_n; Y)$ hence

$$L(X_1 \dots X_n; Y) = \{L \in \text{Hom}(X_1 \dots X_n; Y) | L \text{ is continuous}\}$$

Theorem 14.231. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a family of normed spaces and $\langle Y, \|\cdot\|_Y \rangle$ a normed space then $L(X_1, \dots, X_n; Y)$ is a subspace of $\text{Hom}(X_1 \dots X_n; Y)$ [and thus by [theorem: 11.61] a vector space].

Proof. Let $C_0: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Y$ be defined by $C_0(x) = 0$. By [theorem: 11.268]

$$C_0 \in \text{Hom}(X_1 \dots X_n; Y) \quad (14.108)$$

Further as $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ we have that $\|C_0(x)\|_Y = \|0\|_Y = 0 \leq 1 \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$ it follows by [theorem: 14.228] that

$$C_0 \text{ is continuous} \quad (14.109)$$

Hence by [eqs: 14.108, 14.109] $C_0 \in L(X_1, \dots, X_n; Y)$ proving that

$$L(X_1 \dots X_n; Y) \neq \emptyset$$

Further if $\alpha \in \mathbb{K}$ and $L_1, L_2 \in L(X_1 \dots X_n; Y)$ then by [theorem: 11.268]

$$L_1 + \alpha \cdot L_2 \in \text{Hom}(X_1 \dots X_n; Y) \quad (14.110)$$

Also using [theorem: 14.228] we have that there exists $M_1, M_2 \in \mathbb{R}^+$ such that $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ $\|L_1(x)\|_Y \leq M_1 \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$ and $\|L_2(x)\|_Y \leq M_2 \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$. Hence

$$\begin{aligned} \|(L_1 + \alpha \cdot L_2)(x)\|_Y &= \|L_1(x) + \alpha \cdot L_2(x)\|_Y \\ &\leq \|L_1(x)\|_Y + \|\alpha \cdot L_2(x)\|_Y \\ &= \|L_1(x)\|_Y + |\alpha| \cdot \|L_2(x)\|_Y \\ &\leq M_1 \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i + |\alpha| \cdot M_2 \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \\ &= (M_1 + |\alpha| \cdot M_2) \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \end{aligned}$$

proving by [theorem: 14.228] that

$$L_1 + \alpha \cdot L_2 \text{ is continuous} \quad (14.111)$$

Combining [eqs: 14.110, 14.111] proves

$$L_1 + \alpha \cdot L_2 \in L(X_1, \dots, X_n; Y) \quad \square$$

Next we proceed to create a norm for the vector space $L(X_1, \dots, X_n; Y)$.

Theorem 14.232. *Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of normed vector spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed spaces and $L \in L(X_1 \dots X_n)$ and define $M_{L,s}$ and $M_{L,r}$ by*

$$M_{L,s} = \left\{ M \in \mathbb{R}_0^+ \mid \forall x \in \prod_{i \in \{1, \dots, n\}} X_i \vdash \forall i \in \{1, \dots, n\} \text{ we have } \|x_i\|_i = 1 \text{ then we have } \|L(x)\|_Y \leq M \right\}$$

and

$$M_{L,r} = \left\{ M \in \mathbb{R}_0^+ \mid \forall x \in \prod_{i \in \{1, \dots, n\}} X_i \text{ we have } \|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \right\}$$

then we have:

1. $M_{L,s} = M_{L,r}$
2. $\inf(M_{L,s})$ exists [hence by (1) $\inf(M_{L,r})$ exist and $\inf(M_{L,s}) = \inf(M_{L,r})$]

Proof.

1. First we prove that $M_{L,s} = M_{L,r}$. Let $M \in M_{L,s}$ then if $x = (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have to look at the following possibilities for x :

$\exists i \in \{1, \dots, n\}$ with $\|x_i\|_i = 0$. Then as $\|\cdot\|_i$ is a norm it follows that $x_i = 0$. So by [theorem: 11.259] we have $L(x) = 0$, hence

$$\|L(x)\|_Y = \|0\|_Y = 0 = M \cdot 0 \stackrel{[\text{theorem: 11.54}]}{=} M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

$\forall i \in \{1, \dots, n\}$ we have $\|x_i\|_i \neq 0$. Then we can define $\{y_i\}_{i \in I}$ by $y_i = \frac{1}{\|x_i\|_i} \cdot x_i$ so that $\|y_i\|_i = 1$. Hence we have as $M \in M_{L,s}$

$$\begin{aligned} \frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot \|L(x)\|_Y &= \frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot \|L(x_1, \dots, x_n)\|_Y \\ &= \left\| \frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot L(x_1, \dots, x_n) \right\|_Y \\ &\stackrel{[\text{theorem: 11.270}]}{=} \left\| L\left(\frac{1}{\|x_1\|_1} \cdot x_1, \dots, \frac{1}{\|x_n\|_n} \cdot x_n\right) \right\|_Y \\ &= \|L(y_1, \dots, y_n)\|_Y \\ &\leq M \end{aligned}$$

hence we have

$$\|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

So in all cases we have

$$\|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

proving that $M \in M_{L,r}$ or that

$$M_{L,s} \subseteq M_{L,r} \quad (14.112)$$

For the opposite inclusion let $M \in M_{L,r}$ then for $x \in \prod_{i \in \{1, \dots, n\}} X_i$ satisfying $\forall i \in \{1, \dots, n\} \|x_i\|_i = 1$ we have

$$\|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i = M \cdot 1 = M$$

which proves that $M \in M_{L,s}$, hence $M_{L,r} \subseteq M_{L,s}$ which combined with [eq: 14.112] results in

$$M_{L,s} = M_{L,r}$$

2. Using [theorem: 14.228] there exists a $M \in \mathbb{R}^+$ such that $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ we have $\|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$, so $M \in M_{L,r}$ proving that $M_{L,r} \neq 0$. As $\forall M \in M_{L,r} 0 \leq M$ and the real numbers are conditional complete [see theorem: 10.18] it follows that

$$\inf(M_{L,s}) \text{ exists and } 0 \leq \inf(M_{L,s}) \quad \square$$

The above theorem ensures that the following definition makes sense.

Definition 14.233. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of normed vector spaces over the field $\mathbb{K}$, $\langle Y, \|\cdot\|_Y \rangle$ is a normed spaces over $\mathbb{K}$ and $L \in L(X_1 \dots X_n; Y)$ then the operator norm of L noted as $\|L\|_{L(X_1 \dots X_n; Y)}$ is defined as

$$\|L\|_{L(X_1 \dots X_n; Y)} = \inf(M_{L,r}) \stackrel{[theorem: 14.232]}{=} \inf(M_{L,s})$$

It turns out that $\|L\|_{L(X_1, \dots, X_n; Y)}$ is a norm on $L(X_1, \dots, X_n; Y)$.

Theorem 14.234. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of normed vector spaces over the field $\mathbb{K}$ and $\langle Y, \|\cdot\|_Y \rangle$ a normed spaces over $\mathbb{K}$ then we have

1. If $L \in L(X_1 \dots X_n; Y)$ then $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$\|L(x)\|_Y \leq \|L\|_{L(X_1 \dots X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

2. If $L \in L(X_1 \dots X_n; Y)$ then $\forall M \in \mathbb{R}^+$ such that $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$

$$\|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

we have

$$\|L\|_{L(X_1 \dots X_n; Y)} \leq M$$

3. If $L \in L(X_1 \dots X_n; Y)$ then $\forall M \in \mathbb{R}^+$ such that $\|L(x)\|_Y \leq M \forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ with $\forall i \in \{1, \dots, n\} \|x_i\|_i = 1$ we have

$$\|L\|_{L(X_1 \dots X_n; Y)} \leq M$$

4. $\|\cdot\|_{L(X_1, \dots, X_n; Y)}: L(X_1 \dots X_n; Y) \rightarrow \mathbb{R}$ defined by $\|\cdot\|_{L(X_1 \dots X_n; Y)}(L) = \|L\|_{L(X_1 \dots X_n; Y)}$ is a norm on $L(X_1 \dots X_n; Y)$. In other words $\langle L(X_1 \dots X_n; Y), \|\cdot\|_{L(X_1 \dots X_n; Y)} \rangle$ is a normed space.

Proof.

1. Let $x \in \prod_{i \in \{1, \dots, n\}} X_i$ then we have the following possibilities for x

$\exists i \in \{1, \dots, n\}$ with $\|x_i\|_i = 0$. Then by [theorem: 11.259] $L(x) = 0$ so that

$$\begin{aligned} \|L(x)\|_Y &= \|0\|_Y \\ &= 0 \\ &= \|L\|_{L(X_1 \dots X_n; Y)} \cdot 0 \\ &\stackrel{[\text{theorem: 11.54}]}{=} \|L\|_{L(X_1 \dots X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \end{aligned}$$

$\forall i \in \{1, \dots, n\}$ we have $\|x_i\|_i \neq 0$. Then by [theorem: 11.53] $0 < \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$.

Assume that $\|L\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i < \|L(x)\|_Y$ then

$$\|L\|_{L(X_1, \dots, X_n; Y)} < \frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot \|L(x)\|_Y$$

As $\|L\|_{L(X_1, \dots, X_n; Y)} = \inf(M_{L,r})$ there exist a $M \in M_{L,r}$ such that

$$\|L\|_{L(X_1 \dots X_n; Y)} \leq M < \frac{1}{\prod_{i \in \{1, \dots, n\}} \|x_i\|_i} \cdot \|L(x)\|_Y$$

hence $M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i < \|L(x)\|_Y$ contradicting $M \in M_{L,r}$. Hence we must have that

$$\|L(x)\|_Y \leq \|L\|_{L(X_1 \dots X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

So in all cases we have

$$\|L(x)\|_Y \leq \|L\|_{L(X_1 \dots X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

2. This follows from

$$\begin{aligned} &\|L\|_{L(X_1 \dots X_n; Y)} = \\ &\inf(M_{L,r}) = \\ &\inf\left(\left\{M \in \mathbb{R}_+ \mid \forall x \in \prod_{i \in \{1, \dots, n\}} X_i \text{ we have } \|L(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i\right\}\right) = \end{aligned}$$

3. This follows from

$$\begin{aligned} &\|L\|_{L(X_1 \dots X_n; Y)} = \\ &\inf(M_{L,s}) = \\ &\inf\left(M \in \mathbb{R}_+ \mid \forall x \in \prod_{i \in \{1, \dots, n\}} X_i \vdash \forall i \in \{1, \dots, n\} \|x_i\|_i = 1 \text{ we have } \|L(x)\|_Y \leq M\right) \end{aligned}$$

4. Let $L_1, L_2 \in L(X_1 \dots X_n; Y)$ then by [theorem: 14.231] we have $L_1 + L_2 \in L(X_1 \dots X_n; Y)$. Further we have for $x \in \prod_{i \in \{1, \dots, n\}} X_i$ with $\forall i \in \{1, \dots, n\} \|x_i\|_i = 1$ that

$$\begin{aligned} &\|(L_1 + L_2)(x)\|_Y = \\ &\|L_1(x) + L_2(x)\|_Y \leq \\ &\|L_1(x)\|_Y + \|L_2(x)\|_Y \leq_{(1)} \\ &\|L_1\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i + \|L_2\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i = \\ &(\|L_1\|_{L(X_1, \dots, X_n; Y)} + \|L_2\|_{L(X_1, \dots, X_n; Y)}) \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \end{aligned}$$

which by (2) proves that

$$\|L_1 + L_2\|_{L(X_1 \dots X_n; Y)} \leq \|L_1\|_{L(X_1 \dots X_n; Y)} + \|L_2\|_{L(X_1 \dots X_n; Y)} \quad (14.113)$$

For C_0 we have by [theorem: 14.231] that $C_0 \in L(X_1 \dots X_n; Y)$. Further for $x \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$\|C_0(x)\|_Y = \|0\|_Y = 0 = 0 \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

so that by (1)

$$\|C_0\|_{L(X_1 \dots X_n; Y)} = 0 \quad (14.114)$$

Assume that for $L \in L(X_1 \dots X_n; Y)$ we have that $\|L\|_{L(X_1 \dots X_n; Y)} = 0$ then by (1) we have $\forall x \in \prod_{i \in \{1, \dots, n\}} X_i$ that $0 \leq \|L(x)\|_Y \leq \|L\|_{L(X_1 \dots X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i = 0$. Hence $\|L\|_Y = 0$ which as $\|\cdot\|_Y$ is a norm proves that $L = C_0$ [the neutral element of $L(X_1 \dots X_n; Y)$]. So we have

$$L = C_0 \Leftrightarrow \|L\|_{L(X_1 \dots X_n; Y)} = 0 \quad (14.115)$$

If $L \in L(X_1 \dots X_n; Y)$ and $\alpha \in \mathbb{K}$ then by [theorem: 14.231] $\alpha \cdot L \in L(X_1 \dots X_n; Y)$. Let $x \in \prod_{i \in \{1, \dots, n\}} X_i$ then we have

$$\|(\alpha \cdot L)(x)\|_Y = \|\alpha \cdot L(x)\|_Y = |\alpha| \cdot \|L(x)\|_Y \leq |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)} \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

which proves by (1) that

$$\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} \leq |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)} \quad (14.116)$$

Now for the opposite inequality for $\alpha \in \mathbb{K}$ we have either:

$\alpha = 0$. Then

$$\begin{aligned} |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)} &= 0 \\ &\stackrel{[\text{eq: 14.114}]}{=} \|C_0\|_{L(X_1 \dots X_n; Y)} \\ &= \|(0 \cdot L)\|_{L(X_1 \dots X_n; Y)} \\ &= \|(\alpha \cdot L)\|_{L(X_1 \dots X_n; Y)} \end{aligned}$$

$\alpha \neq 0$. Assume that $\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} < |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)}$. Then as we have $\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} = \inf(M_{\alpha \cdot L, r})$ it follows that there exist a $M \in M_{\alpha \cdot L, r}$ such that

$$\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} \leq M < |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)} \quad (14.117)$$

If $x \in \prod_{i \in \{1, \dots, n\}} X_i$ we have, as $M \in M_{\alpha \cdot L, r}$ that

$$|\alpha| \cdot \|L(x)\|_Y = \|\alpha \cdot L(x)\|_Y = \|(\alpha \cdot L)(x)\|_Y \leq M \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

so that

$$\|L(x)\|_Y \leq \frac{M}{|\alpha|} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i$$

proving that $\frac{M}{|\alpha|} \in M_{L, r}$ hence

$$\|L\|_{L(X_1 \dots X_n; Y)} = \inf(M_{L, r}) \leq \frac{M}{|\alpha|} \stackrel{[\text{eq: 14.117}]}{<} \frac{|\alpha|}{|\alpha|} \cdot \|L\|_{L(X_1 \dots X_n; Y)}$$

leading to the contradiction $\|L\|_{L(X_1 \dots X_n; Y)} > \|L\|_{L(X_1 \dots X_n; Y)}$. Hence we must have that $\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} \leq |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)}$.

So in both cases we have $\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} \leq |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)}$ which combined with [eq: 14.116] proves that

$$\|\alpha \cdot L\|_{L(X_1 \dots X_n; Y)} = |\alpha| \cdot \|L\|_{L(X_1 \dots X_n; Y)} \quad (14.118)$$

Finally using [eqs: 14.113, 14.115 and 14.118] it follows that

$$\|\cdot\|_{L(X_1 \dots X_n; Y)} \text{ is a norm in } L(X_1 \dots X_n; Y) \quad \square$$

If all $\langle X_i, \|\cdot\|_i \rangle$ are the same normed space then we use the following definition.

Definition 14.235. Let $n \in \mathbb{N}$ and $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces over $\mathbb{K}$ then we define

$$L^n(X; Y) = \begin{cases} L(X, Y) & \text{if } n = 1 \\ L\left(\underbrace{X \dots X}_n; Y\right) & \text{if } 1 < n \end{cases}$$

and for $L \in L^n(X; Y)$

$$\|L\|_{L^n(X; Y)} = \begin{cases} \|L\|_{L(X, Y)} & \text{if } n = 1 \\ \|L\|_{L\left(\underbrace{X \dots X}_n; Y\right)} & \text{if } 1 < n \end{cases}$$

Theorem 14.236. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$ a normed space and $L \in L^n(\mathbb{K}^n; X)$ then

$$\|L\|_{L^n(\mathbb{K}^n; X)} \leq \left\| L\left(\underbrace{1, \dots, 1}_n\right) \right\|$$

Proof. If $x \in \mathbb{K}^n$ then

$$\begin{aligned} \|L(x_1, \dots, x_n)\| &= \|L(x_1 \cdot 1, \dots, x_n \cdot 1)\| \\ &\stackrel{[\text{theorem: 11.270}]}{=} \left\| \left(\prod_{i=1}^n x_i \right) \cdot L\left(\underbrace{1, \dots, 1}_n\right) \right\| \\ &= \left\| L\left(\underbrace{1, \dots, 1}_n\right) \right\| \cdot \left| \prod_{i=1}^n x_i \right| \\ &= \left\| L\left(\underbrace{1, \dots, 1}_n\right) \right\| \cdot \prod_{i=1}^n |x_i| \end{aligned}$$

so that $\left\| L\left(\underbrace{1, \dots, 1}_n\right) \right\| \in M_{L, r}$. Hence we have

$$\|L\|_{L^n(\mathbb{K}^n; X)} = \inf(M_{L, r}) \leq \left\| L\left(\underbrace{1, \dots, 1}_n\right) \right\|$$

□

Theorem 14.237. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of normed vector spaces, $\langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle$ normed spaces, $L \in L(X_1 \dots X_n; Y)$ and $K \in L(Y, Z)$ then

1. $K \circ L \in L(X_1 \dots X_n; Z)$
2. $\|K \circ L\|_{L(X_1 \dots X_n; Z)} \leq \|L\|_{L(X_1 \dots X_n; Y)} \cdot \|K\|_{L(Y, Z)}$

Proof.

1. From [theorem: 11.269] it follows that $K \circ L \in \text{Hom}(X_1 \dots X_n; Z)$, Further as the composition of continuous mappings is continuous [see theorem: 14.158] we have that $K \circ L$ is continuous. Hence

$$K \circ L \in L(X_1 \dots X_n; Z)$$

2. Let $x \in \prod_{i \in \{1, \dots, n\}} X_i$ then we have

$$\begin{aligned} \|(K \circ L)(x)\|_Z &= \|K(L(x))\|_Z \\ &\leq_{[\text{theorem: 14.210}]} \|K\|_{L(Y, Z)} \cdot \|L(x)\|_Y \\ &\leq_{[\text{theorem: 14.234}]} \|K\|_{L(Y, Z)} \cdot \|L\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \end{aligned}$$

proving by [theorem: 14.234] that

$$\|K \circ L\|_{L(X_1 \dots X_n; Z)} \leq \|L\|_{L(X_1 \dots X_n; Y)} \cdot \|K\|_{L(Y, Z)}$$

□

We show now that the composition of linear mappings is a multilinear mapping itself.

Theorem 14.238. *Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ and $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces over $\mathbb{K}$ then if we define*

$$(\circ): L(Y, Z) \cdot L(X, Y) \rightarrow L(X, Z) \text{ defined by } (\circ)(L_2, L_1) = L_2 \circ L_1$$

then we have that

1. $(\circ) \in L(L(Y, Z), L(X, Y); L(X, Z))$ [in other words $(\circ)$ is multilinear and continuous]
2. $\|(\circ)\|_{L(L(Y, Z), L(X, Y); L(X, Z))} \leq 1$

Proof. Let $\alpha \in \mathbb{K}$. Then we have for $L_1, L_2 \in L(Y, Z)$ and $L \in L(X, Y)$ that $\forall x \in X$ we have

$$\begin{aligned} (\circ)(L_1 + \alpha \cdot L_2, L)(x) &= ((L_1 + \alpha \cdot L_2) \circ L)(x) \\ &= (L_1 + \alpha \cdot L_2)(L(x)) \\ &= L_1(L(x)) + \alpha \cdot L_2(L(x)) \\ &= (L_1 \circ L)(x) + \alpha \cdot (L_2 \circ L)(x) \\ &= ((L_1 \circ L) + \alpha \cdot (L_2 \circ L))(x) \\ &= (\circ(L_1, L) + \alpha \cdot \circ(L_2, L))(x) \end{aligned}$$

proving that

$$\circ(L_1 + \alpha \cdot L_2, L) = \circ(L_1, L) + \alpha \cdot \circ(L_2, L)$$

Further if $L \in L(Y, Z)$ and $L_1, L_2 \in L(X, Y)$ then $\forall x \in X$ we have

$$\begin{aligned} \circ(L, L_1 + \alpha \cdot L_2)(x) &= (L \circ (L_1 + \alpha \cdot L_2))(x) \\ &= L((L_1 + \alpha \cdot L_2)(x)) \\ &= L(L_1(x) + \alpha \cdot L_2(x)) \\ &= L(L_1(x)) + \alpha \cdot L(L_2(x)) \\ &= (L \circ L_1)(x) + \alpha \cdot (L \circ L_2)(x) \\ &= ((L \circ L_1) + \alpha \cdot (L \circ L_2))(x) \\ &= ((\circ)(L, L_1) + \alpha \cdot (\circ)(L, L_2))(x) \end{aligned}$$

proving that

$$(\circ)(L, L_1 + \alpha \cdot L_2) = (\circ)(L, L_1) + \alpha \cdot (\circ)(L, L_2)$$

Hence $\circ$ is multilinear or

$$(\circ) \in \text{Hom}(L(Y, Z), L(X, Y); L(X, Z))$$

For continuity note that for $L_1 \in L(Y, Z)$, $L_2 \in L(X, Y)$ and $x \in X$ we have

$$\begin{aligned} \|(\circ)(L_1, L_2)(x)\|_Z &= \|(L_1 \circ L_2)(x)\|_Z \\ &= \|L_1(L_2(x))\|_Z \\ &\leq_{[\text{theorem: 14.210}]} \|L_1\|_{L(Y, Z)} \cdot \|L_2(x)\|_Y \\ &\leq_{[\text{theorem: 14.210}]} \|L_1\|_{L(Y, Z)} \cdot \|L_2\|_{L(X, Y)} \cdot \|x\|_X \end{aligned}$$

which proves by [theorem: 14.234] that

$$\|(\circ)(L_1, L_2)\|_{L(X, Z)} \leq \|L_1\|_{L(Y, Z)} \cdot \|L_2\|_{L(X, Y)} = 1 \cdot \|L_1\|_{L(Y, Z)} \cdot \|L_2\|_{L(X, Y)}$$

so that by [theorem: 14.228] $(\circ)$ is continuous. Hence

$$(\circ) \in L(L(Y, Z), L(X, Y); L(X, Z))$$

and by [theorem 14.234] that

$$\|(\circ)\|_{L(L(Y, Z), L(X, Y); L(X, Z))} \leq 1.$$

□

Theorem 14.239. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ and $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces over $\mathbb{K}$ then we can construct a linear isometric isomorphism

$$\varphi: L(X, Y; Z) \rightarrow L(X, L(Y, Z))$$

Likewise if we take $\psi = \varphi^{-1}$ then

$$\psi: L(X, L(Y, Z)) \rightarrow L(X, Y; Z) \text{ is a linear isometric isomorphism}$$

In other words $L(X, L(Y, Z))$ and $L(X, Y; Z)$ are isometric [see definition: 14.124] or

$$L(X, L(Y, Z)) \approx L(X, Y; Z)$$

Further we have $\forall (x, y) \in X \cdot Y$ that

$$\forall L \in L(X, Y; Z) \quad \varphi(L)(x)(y) = L(x, y)$$

and

$$\forall L \in L(X, (L(Y, Z))) \text{ that } \psi(L)(x, y) = L(x)(y)$$

Proof. Let $L \in L(X, Y; Z)$ and given $x \in X$ define

$$L_x: Y \rightarrow Z \text{ by } L_x(y) = L(x, y) \quad (14.119)$$

If $r, s \in Y$ and $\alpha \in \mathbb{K}$ then

$$L_x(r + \alpha \cdot s) = L(x, r + \alpha \cdot s) \underset{\text{multilinearity}}{=} L(x, r) + \alpha \cdot L(x, s) = L_x(r) + \alpha \cdot L_x(s)$$

proving that

$$L_x \in \text{Hom}(Y, Z) \quad (14.120)$$

Further for $r \in Y$ we have

$$\|L_x(r)\|_Z = \|L(x, r)\|_Z \leq_{[\text{theorem: 14.234}]} \|L\|_{L(X, Y; Z)} \cdot \|x\|_X \cdot \|r\|_Y$$

proving by [theorem: 14.201] and [definition: 14.209] that

$$\forall x \in X \quad L_x \in L(Y, Z) \text{ and } \|L_x\|_{L(Y, Z)} \leq \|L\|_{L(X, Y; Z)} \cdot \|x\|_X \quad (14.121)$$

The above allows us, given $L \in L(X, Y; Z)$, define

$$\varphi_L: X \rightarrow L(Y, Z) \text{ by } \varphi_L(x) = L_x \quad (14.122)$$

Then if $x, y \in X$ and $\alpha \in \mathbb{K}$ we have for $r \in Y$ that

$$\begin{aligned} (\varphi_L(x + \alpha \cdot y))(r) &= L_{x + \alpha \cdot y}(r) \\ &\underset{[\text{eq: 14.119}]}{=} L(x + \alpha \cdot y, r) \\ &\underset{\text{multilinearity}}{=} L(x, r) + \alpha \cdot L(y, r) \\ &\underset{[\text{eq: 14.119}]}{=} L_x(r) + \alpha \cdot L_y(r) \\ &= (\varphi_L(x))(r) + \alpha \cdot (\varphi_L(y))(r) \\ &= (\varphi_L(x) + \alpha \cdot \varphi_L(y))(r) \end{aligned}$$

proving that $\varphi_L(x + \alpha \cdot y) = \varphi_L(x) + \alpha \cdot \varphi_L(y)$. Hence

$$\varphi_L \in \text{Hom}(X, L(Y, Z)) \quad (14.123)$$

For $x \in X$ we have

$$\|\varphi_L(x)\|_{L(Y, Z)} \underset{[\text{eq: 14.122}]}{=} \|L_x\|_{L(Y, Z)} \leq_{[\text{eq: 14.121}]} \|L\|_{L(X, Y; Z)} \cdot \|x\|_X$$

proving by [theorem: 14.201] and [definition: 14.209] that

$$\varphi_L \in L(X, L(Y, Z)) \text{ and } \|\varphi_L\|_{L(X, L(Y, Z))} \leq \|L\|_{L(X, Y; Z)} \quad (14.124)$$

The above let us define

$$\varphi: L(X, Y; Z) \rightarrow L(X, L(Y, Z)) \text{ by } \varphi(L) = \varphi_L \quad (14.125)$$

Let $L_1, L_2 \in L(X, Y; Z)$ and $\alpha \in \mathbb{K}$ then for $x \in X$ we have $\forall r \in Y$ that

$$\begin{aligned} (L_1 + \alpha \cdot L_2)_x(r) &\stackrel{[\text{eq: 14.119}]}{=} (L_1 + \alpha \cdot L_2)(x, r) \\ &= L_1(x, r) + \alpha \cdot L_2(x, r) \\ &= (L_1)_x(r) + \alpha \cdot (L_2)_x(r) \\ &= ((L_1)_x + \alpha \cdot (L_2)_x)(r) \end{aligned}$$

proving that $(L_1 + \alpha \cdot L_2)_x = (L_1)_x + \alpha \cdot (L_2)_x$. Hence

$$\varphi_{L_1 + \alpha \cdot L_2}(x) = (L_1 + \alpha \cdot L_2)_x = (L_1)_x + \alpha \cdot (L_2)_x = \varphi_{L_1}(x) + \alpha \cdot \varphi_{L_2}(x) = (\varphi_{L_1} + \alpha \cdot \varphi_{L_2})(x)$$

proving that $\varphi_{L_1 + \alpha \cdot L_2} = \varphi_{L_1} + \alpha \cdot \varphi_{L_2}$. So

$$\varphi(L_1 + \alpha \cdot L_2) = \varphi_{L_1 + \alpha \cdot L_2} = \varphi_{L_1} + \alpha \cdot \varphi_{L_2} = \varphi(L_1) + \alpha \cdot \varphi(L_2)$$

proving that

$$\varphi \in \text{Hom}(L(X, Y; Z), L(X, L(Y, Z))) \quad (14.126)$$

Further giving $L \in L(X, Y; Z)$ we have

$$\|\varphi(L)\|_{L(X, L(Y, Z))} = \|\varphi L\|_{L(X, L(Y, Z))} \leq_{[\text{eq: 14.124}]} \|L\|_{L(X, Y; Z)}$$

giving by [theorem: 14.201]

$$\varphi \in L(L(X, Y; Z), L(X, L(Y, Z))) \text{ and } \|\varphi(L)\|_{L(X, L(Y, Z))} \leq \|L\|_{L(X, Y; Z)} \quad (14.127)$$

Next we will define the inverse mapping of φ . Let $L \in L(X, L(Y, Z))$ and define

$$\psi_L: X \cdot Y \rightarrow Z \text{ by } \psi_L(x, y) = (L(x))(y) \quad (14.128)$$

Let $x, y \in X$, $r \in Y$ and $\alpha \in \mathbb{K}$ then we have

$$\begin{aligned} \psi_L(x + \alpha \cdot y, r) &\stackrel{[\text{eq: 14.128}]}{=} (L(x + \alpha \cdot y))(r) \\ &\stackrel{L \in L(X, L(Y, Z))}{=} (L(x) + \alpha \cdot L(y))(r) \\ &= (L(x))(r) + \alpha \cdot (L(y))(r) \\ &\stackrel{[\text{eq: 14.128}]}{=} \psi_L(x, r) + \alpha \cdot \psi_L(y, r) \end{aligned}$$

and for $x \in X$, $r, s \in Y$ and $\alpha \in \mathbb{K}$ we have

$$\begin{aligned} \psi_L(x, r + \alpha \cdot s) &\stackrel{[\text{eq: 14.128}]}{=} (L(x))(r + \alpha \cdot s) \\ &\stackrel{L(x) \in L(Y, Z)}{=} (L(x))(r) + \alpha \cdot (L(x))(s) \\ &\stackrel{[\text{eq: 14.128}]}{=} \psi_L(x, r) + \alpha \cdot \psi_L(x, s) \end{aligned}$$

proving that

$$\psi_L \in \text{Hom}(X, Y; Z)$$

If $(x, y) \in X \cdot Y$ then

$$\|\psi_L(x, y)\|_Z = \|(L(x))(y)\|_Z \leq_{L(x) \in L(Y, Z)} \|L(x)\|_{L(Y, Z)} \cdot \|y\|_Y \leq \|L\|_{L(X, L(Y, Z))} \cdot \|x\|_X \cdot \|y\|_Y$$

proving by [theorems: 14.228, 14.234] that

$$\psi_L \in L(X, Y; Z) \text{ and } \|\psi_L\|_{L(X, Y; Z)} \leq \|L\|_{L(X, L(Y, Z))} \quad (14.129)$$

The above allows use to define

$$\psi: L(X, L(Y, Z)) \rightarrow L(X, Y; Z) \text{ by } \psi(L) = \psi_L \quad (14.130)$$

where by [eq: 14.129]

$$\forall L \in L(X, L(Y, Z)) \quad \|\psi(L)\|_{L(X, Y; Z)} = \|\psi_L\|_{L(X, Y; Z)} \leq \|L\|_{L(X, L(Y, Z))} \quad (14.131)$$

Let $L \in L(X, L(Y, Z))$ and $x \in X, y \in Y$ then we have

$$\begin{aligned}
 (((\varphi \circ \psi)(L))(x))(y) &= ((\varphi(\psi(L)))(x))(y) \\
 &\stackrel{[\text{eq: 14.125}]}{=} ((\varphi_{\psi(L)})(x))(y) \\
 &\stackrel{[\text{eq: 14.122}]}{=} (\psi(L))_x(y) \\
 &\stackrel{[\text{eq: 14.119}]}{=} \psi(L)(x, y) \\
 &\stackrel{[\text{eq: 14.130}]}{=} \psi_L(x, y) \\
 &\stackrel{[\text{eq: 14.128}]}{=} (L(x))(y)
 \end{aligned}$$

so that $((\varphi \circ \psi)(L))(x) = L(x)$ from which it follows that $(\varphi \circ \psi)(L) = L$. Hence we have that

$$\varphi \circ \psi = \text{Id}_{L(X, L(Y, Z))} \quad (14.132)$$

On the other hand if $L \in L(X, Y; Z)$ and $(x, y) \in X \cdot Y$ then we have

$$\begin{aligned}
 ((\psi \circ \varphi)(L))(x, y) &= (\psi(\varphi(L)))(x, y) \\
 &\stackrel{[\text{eq: 14.130}]}{=} \psi_{\varphi(L)}(x, y) \\
 &\stackrel{[\text{eq: 14.128}]}{=} ((\varphi(L))(x))(y) \\
 &\stackrel{[\text{eq: 14.125}]}{=} ((\varphi_L)(x))(y) \\
 &\stackrel{[\text{eq: 14.122}]}{=} L_x(y) \\
 &\stackrel{[\text{eq: 14.119}]}{=} L(x, y)
 \end{aligned}$$

proving that $(\psi \circ \varphi)(L) = L$, hence

$$\psi \circ \varphi = \text{Id}_{L(X, Y; Z)} \quad (14.133)$$

From [eqs: 14.132, 14.133] it follows by [theorem: 2.73] that $\varphi: L(X, Y; Z) \rightarrow L(X, L(Y, Z))$ is a bijection. Combining this with [eq: 14.126] proves that

$$\varphi \text{ is a linear isomorphism with } \psi = \varphi^{-1} \quad (14.134)$$

Finally we have for $L \in L(X, Y; Z)$ that

$$\begin{aligned}
 \|L\|_{L(X, Y; Z)} &= \|\text{Id}_{L(X, Y; Z)}(L)\|_{L(X, Y; Z)} \\
 &\stackrel{[\text{eq: 14.133}]}{=} \|(\psi \circ \varphi)(L)\|_{L(X, Y; Z)} \\
 &= \|\psi(\varphi(L))\|_{L(X, Y; Z)} \\
 &\leq_{[\text{eq: 14.131}]} \|\varphi(L)\|_{L(X, L(Y, Z))}
 \end{aligned}$$

which combined with [eq: 14.127] proves that $\|\varphi(L)\|_{L(X, Y; Z)} = \|L\|_{L(X, Y; Z)}$. This together with [eq: 14.134] proves that

$$\varphi: L(X, Y; Z) \rightarrow L(X, L(Y, Z)) \text{ is a linear isometric isomorphism}$$

As $\psi = \varphi^{-1}$ we have by [theorem: 14.126] that

$$\psi: L(X, L(Y, Z)) \rightarrow L(X, Y; Z) \text{ is a linear isometric isomorphism}$$

Finally if $x \in X, y \in Y$ then we have for $L \in L(X, Y; Z)$ that

$$\varphi(L)(x)(y) \stackrel{[\text{eq: 14.125}]}{=} \varphi_L(x)(y) \stackrel{[\text{eq: 14.122}]}{=} L_x(y) \stackrel{[\text{eq: 14.119}]}{=} L(x, y)$$

so that

$$\forall L \in L(X, L(Y, Z)) \text{ we have } \varphi(L)(x)(y) = L(x, y) \quad (14.135)$$

Further if $L \in L(X, L(Y, Z))$ then

$$\begin{aligned}
 L(x)(y) &= ((\varphi \circ \psi)(L))(x)(y) \\
 &= (\varphi(\psi(L)))(x)(y) \\
 &\stackrel{[\text{eq: 14.135}]}{=} \psi(L)(x, y)
 \end{aligned}$$

so that

$$\forall L \in L(X, L(Y, Z)) \text{ we have } \psi(L)(x, y) = (L(x))(y)$$

□

If $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ are the same in the above then we have

$$L(X, L(X, Z)) \approx L(X, X; Y) \underset{[\text{definition: 14.235}]}{=} L^2(X; Z)$$

actually we can extend this to arbitrary $n \in \mathbb{N}$ as is show in the next theorem. This result will be very important if we define higher order derivatives.

Theorem 14.240. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces over $\mathbb{K}$ then we can construct a **unique** linear isometric isomorphism*

$$\varphi_{[X, Y, n]}: L^{n+1}(X; Y) \rightarrow L(X, L^n(X; Y))$$

and if we take $\psi_{[X, Y, n]} = \varphi_{[X, Y, n]}^{-1}$ we have the linear isometric isomorphism

$$\psi_{[X, Y, n]}: L(X, L^n(X; Y)) \rightarrow L^{n+1}(X; Y)$$

such that for every $(x_1, \dots, x_{n+1}) \in X^n$

$$\forall L \in L^{n+1}(X; Y) \quad \varphi_{[X, Y, n]}(L)(x_1)(x_2, \dots, x_{n+1}) = L(x_1, \dots, x_{n+1})$$

and

$$\forall L \in L(X, L^n(X; Y)) \quad \psi_{[X, Y, n]}(L)(x_1, \dots, x_{n+1}) = L(x_1)(x_2, \dots, x_{n+1})$$

Further if there exist a $\varphi': L^{n+1}(X; Y) \rightarrow L(X, L^n(X; Y))$ such that $\forall (x_1, \dots, x_{n+1}) \in X^n$ $\varphi(L)(x_1)(x_2, \dots, x_{n+1}) = L(x_1, \dots, x_{n+1})$ then

$$\varphi_{[X, Y, n]} = \varphi'$$

and if there exist a $\psi': L(X, L^n(X; Y)) \rightarrow L^{n+1}(X; Y)$ such that $\forall (x_1, \dots, x_{n+1}) \in X^n$ $\psi(L)(x_1, \dots, x_{n+1}) = L(x_1)(x_2, \dots, x_{n+1})$ then

$$\psi_{[X, Y, n]} = \psi'$$

In other words using [definition: 14.235] it follows that

$$L(X, L^n(X; Y)) \approx L^{n+1}(X; Y)$$

so we can identify $L(X, L^n(X; Y))$ with $L^{n+1}(X; Y)$ using a **unique linear isometric isomorphism**.

Proof. For $n \in \mathbb{N}$ we have either:

$n = 1$. Then by [theorem: 14.239] we have that there exist linear isometric isomorphisms

$$\varphi L(X, X; Y) \rightarrow L(X, L(X, Y))$$

$$\psi: L(X, L(X, Y)) \rightarrow L(X, X; Y)$$

and for $(x_1, x_2) \in X \cdot X = X^2$ that

$$\forall L \in L(X \cdot X; Y) \text{ we have } \varphi(L)(x_1)(x_2) = L(x_1, x_2)$$

and

$$\forall L \in L(X, L(X, Y)) \text{ we have } \psi(L)(x_1, x_2) = L(x_1)(x_2)$$

Which, as by [definition: 14.235] $L(X, Y) = L^1(X, Y)$ and $L(X, X; Y) = L^2(X, Y)$, proves the first part of the theorem for the case $n = 1$.

$1 < n$. To prove the theorem for this case we use a slightly modified version of the proof of the previous theorem. Let $L \in L^{n+1}(X; Z)$ and $x \in X$ define

$$L_x: X^n \rightarrow Y \text{ by } L_x(x_1, \dots, x_n) = L(x, x_1, \dots, x_n) \quad (14.136)$$

Given $r, s \in X$ and $\alpha \in \mathbb{K}$ then we have for $i \in \{1, \dots, n\}$ that

$$\begin{aligned} L_x \left(x_1, \dots, \underbrace{r + \alpha \cdot s}_{i+1}, \dots, x_n \right) &= \\ L \left(x, x_1, \dots, \underbrace{r + \alpha \cdot s}_{i+1}, \dots, x_n \right) &\stackrel{\text{multilinearity}}{=} \\ L \left(x, x_1, \dots, \underbrace{r}_{i+1}, \dots, x_n \right) + \alpha \cdot L \left(x, x_1, \dots, \underbrace{s}_{i+1}, \dots, x_n \right) &= \\ L_x \left(x_1, \dots, \underbrace{r}_i, \dots, x_n \right) + \alpha \cdot L_x \left(x_1, \dots, \underbrace{s}_i, \dots, x_n \right) \end{aligned}$$

proving that

$$L_x \in \text{Hom}^n(X; Y) \stackrel{\text{def}}{=} \text{Hom} \left(\underbrace{X, \dots, X}_n; Y \right)$$

Further for $r \in X^n$ we have if we define $(y_1, \dots, y_{n+1}) \in X^{n+1}$ by $y_i = \begin{cases} x & \text{if } i=1 \\ r_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$

$$\begin{aligned} \|L_x(r_1, \dots, r_n)\|_Y &= \|L(x, r_1, \dots, r_n)\|_Y \\ &= \|L(y_1, \dots, y_{n+1})\|_Y \\ &\stackrel{[\text{theorem: 14.234}]}{\leq} \|L\|_{L \left(\underbrace{X, \dots, X}_{n+1}; Y \right)} \cdot \prod_{i \in \{1, \dots, n+1\}} \|y_i\|_X \\ &\stackrel{\text{def}}{=} \|L\|_{L^{n+1}(X; Y)} \cdot \prod_{i \in \{1, \dots, n+1\}} \|y_i\|_X \\ &= \|L\|_{L^{n+1}(X; Y)} \cdot \|y_1\|_X \cdot \prod_{i \in \{2, \dots, n+1\}} \|y_i\|_X \\ &= \|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X \cdot \prod_{i \in \{2, \dots, n+1\}} \|r_{i-1}\|_X \\ &= (\|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X) \cdot \prod_{i \in \{1, \dots, n\}} \|r_i\|_X \end{aligned}$$

proving by [theorem: 14.228] and [definition: 14.209] that

$$\forall x \in X \quad L_x \in L^n(X; Y) \quad \text{and} \quad \|L_x\|_{L^n(X; Y)} \leq \|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X \quad (14.137)$$

The above allows us, given $L \in L^{n+1}(X; Y)$, to define

$$\varphi_L: X \rightarrow L^n(X; Y) \quad \text{by} \quad \varphi_L(x) = L_x \quad (14.138)$$

Then if $x, y \in X$ and $\alpha \in \mathbb{K}$ we have for $r \in X^n$ that

$$\begin{aligned} (\varphi_L(x + \alpha \cdot y))(r) &\stackrel{[\text{eq: 14.138}]}{=} L_{x + \alpha \cdot y}(r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.136}]}{=} L(x + \alpha \cdot y, r_1, \dots, r_n) \\ &= L(x, r_1, \dots, r_n) + \alpha \cdot L(y, r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.136}]}{=} L_x(r_1, \dots, r_n) + \alpha \cdot L_y(r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.138}]}{=} \varphi_L(x)(r) + \alpha \cdot \varphi_L(y)(r) \\ &= (\varphi_L(x) + \alpha \cdot \varphi_L(y))(r) \end{aligned}$$

proving that $\varphi_L(x + \alpha \cdot y) = \varphi_L(x) + \alpha \cdot \varphi_L(y)$. Hence

$$\varphi_L \in \text{Hom}(X, L^n(X; Y)) \quad (14.139)$$

For $x \in X$ we have

$$\|\varphi_L(x)\|_{L^n(X; Y)} \stackrel{[\text{eq: 14.138}]}{=} \|L_x\|_{L^n(X; Y)} \stackrel{[\text{eq: 14.137}]}{\leq} \|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X$$

proving by [theorem: 14.201] and [definition: 14.209] that

$$\varphi_L \in L(X, L^n(X; Y)) \quad \text{and} \quad \|\varphi_L\|_{L(X, L^n(X; Y))} \leq \|L\|_{L^{n+1}(X; Y)} \quad (14.140)$$

The above let us define

$$\varphi: L^{n+1}(X; Y) \rightarrow L(X, L^n(X; Y)) \text{ by } \varphi(L) = \varphi_L \quad (14.141)$$

Let $L_1, L_2 \in L^{n+1}(X; Y)$ and $\alpha \in \mathbb{K}$ then for $x \in X$ and $\forall r \in X^n$ we have

$$\begin{aligned} (L_1 + \alpha \cdot L_2)_x(r) &= (L_1 + \alpha \cdot L_2)(x, r_1, \dots, r_n) \\ &= L_1(x, r_1, \dots, r_n) + \alpha \cdot L_2(x, r_1, \dots, r_n) \\ &= (L_1)_x(r_1, \dots, r_n) + \alpha \cdot (L_2)_x(r_1, \dots, r_n) \\ &= ((L_1)_x + \alpha \cdot (L_2)_x)(r_1, \dots, r_n) \\ &= ((L_1)_x + \alpha \cdot (L_2)_x)(r) \end{aligned}$$

proving that $(L_1 + \alpha \cdot L_2)_x = (L_1)_x + \alpha \cdot (L_2)_x$. Hence

$$\varphi_{L_1 + \alpha \cdot L_2}(x) = (L_1 + \alpha \cdot L_2)_x = (L_1)_x + \alpha \cdot (L_2)_x = \varphi_{L_1}(x) + \alpha \cdot \varphi_{L_2}(x)$$

so

$$\varphi_{L_1 + \alpha \cdot L_2} = \varphi_{L_1} + \alpha \cdot \varphi_{L_2}$$

proving by [eq: 14.141]

$$\varphi(L_1 + \alpha \cdot L_2) \underset{[\text{eq: 14.141}]}{=} \varphi(L_1) + \alpha \cdot \varphi(L_2)$$

so that

$$\varphi \in \text{Hom}(L^{n+1}(X; Y), L(X, L^n(X; Y))) \quad (14.142)$$

Further given $L \in L^{n+1}(X; Y)$ we have

$$\|\varphi(L)\|_{L(X, L^n(X; Y))} \underset{[\text{eq: 14.141}]}{=} \|\varphi_L\|_{L(X, L^n(X; Y))} \leqslant_{[\text{eq: 14.140}]} \|L\|_{L^{n+1}(X; Y)}$$

giving by [theorem: 14.201]

$$\varphi \in L(L^{n+1}(X; Y), L(X, L^n(X; Y))) \text{ and } \|\varphi(L)\|_{L(X, L^n(X; Y))} \leqslant \|L\|_{L^{n+1}(X; Y)} \quad (14.143)$$

Next we must define the inverse mapping of φ . Let $L \in L(X, L^n(X; Y))$ and define

$$\psi_L: X^{n+1} \rightarrow Y \text{ by } \psi_L(x_1, \dots, x_{n+1}) = L(x_1)(x_2, \dots, x_{n+1}) \quad (14.144)$$

Let $i \in \{1, \dots, n+1\}$, $y, z \in X$, $x \in \prod_{j \in \{1, \dots, n+1\} \setminus \{i\}} X$ and $\alpha \in \mathbb{K}$ then for i we have either:

$i = 1$. Then

$$\begin{aligned} &\psi_L\left(x_1, \dots, \underbrace{y + \alpha \cdot z}_i, \dots, x_{n+1}\right) \underset{i=1}{=} \\ &\psi_L(y + \alpha \cdot z, x_2, \dots, x_{n+1}) \underset{[\text{eq: 14.144}]}{=} \\ &L(y + \alpha \cdot z)(x_2, \dots, x_{n+1}) \underset{L \in L(X, \overline{L}^n(X; Y))}{=} \\ &(L(y) + \alpha \cdot L(z))(x_2, \dots, x_{n+1}) = \\ &(L(y))(x_2, \dots, x_{n+1}) + \alpha \cdot (L(z))(x_2, \dots, x_{n+1}) \underset{[\text{eq: 14.144}]}{=} \\ &\psi_L(y, x_2, \dots, x_{n+1}) + \alpha \cdot \psi_L(z, x_2, \dots, x_{n+1}) \underset{i=1}{=} \\ &\psi_L\left(x_1, \dots, \underbrace{y}_i, \dots, x_{n+1}\right) + \alpha \cdot \psi_L\left(x_1, \dots, \underbrace{z}_i, \dots, x_{n+1}\right) \end{aligned}$$

$i \in \{2, \dots, n+1\}$. Then

$$\begin{aligned} &\psi_L\left(x_1, \dots, \underbrace{y + \alpha \cdot z}_i, \dots, x_{n+1}\right) \underset{[\text{eq: 14.144}] \wedge i \neq 1}{=} \\ &(L(x_1))\left(x_2, \dots, \underbrace{y + \alpha \cdot z}_{i-1}, \dots, x_{n+1}\right) \underset{L(x_1) \in \overline{L}^n(X; Y)}{=} \\ &(L(x_1))\left(x_2, \dots, \underbrace{y}_{i-1}, \dots, x_{n+1}\right) + \alpha \cdot (L(x_1))\left(x_2, \dots, \underbrace{z}_{i-1}, \dots, x_{n+1}\right) \underset{[\text{eq: 14.144}]}{=} \\ &\psi_L\left(x_1, \dots, \underbrace{y}_i, \dots, x_{n+1}\right) + \alpha \cdot \psi_L\left(x_1, \dots, \underbrace{z}_i, \dots, x_{n+1}\right) \end{aligned}$$

proving that

$$\psi_L \in \text{Hom}^{n+1}(X; Y) \quad (14.145)$$

Further if $x \in X^{n+1}$ then

$$\begin{aligned} \|\psi_L(x_1, \dots, x_{n+1})\|_Y &\stackrel{[\text{eq: 14.144}]}{=} \|L(x_1)(x_2, \dots, x_{n+1})\|_Y \\ &\leq_{L(x_1) \in L^n(X; Y)} \|L(x_1)\|_{L^n(X; Y)} \cdot \prod_{i \in \{2, \dots, n+1\}} \|x_i\|_X \\ &\leq_{L \in} \|L\|_{L(X, L^n(X; Y))} \cdot \|x_1\|_X \cdot \prod_{i \in \{2, \dots, n+1\}} \|x_i\|_X \\ &= \|L\|_{L(X, L^n(X; Y))} \cdot \prod_{i \in \{1, \dots, n+1\}} \|x_i\|_X \end{aligned}$$

proving by [theorems: 14.228, 14.234] that

$$\psi_L \in L^{n+1}(X; Y) \text{ and } \|\psi_L\|_{L^{n+1}(X; Y)} \leq \|L\|_{L(X, L^n(X; Y))} \quad (14.146)$$

The above allows us to define

$$\psi: L(X, L^n(X; Y)) \rightarrow L^{n+1}(X; Y) \text{ by } \psi(L) = \psi_L \quad (14.147)$$

where by [eq: 14.146]

$$\forall L \in L(X, L^n(X; Y)) \text{ we have } \|\psi(L)\|_{L^{n+1}(X; Y)} \leq \|L\|_{L(X, L^n(X; Y))} \quad (14.148)$$

Let $L \in L(X, L^n(X; Y))$ then for $x \in X$ and $r \in X^n$ we have

$$\begin{aligned} (((\varphi \circ \psi)(L))(x))(r) &= (((\varphi \circ \psi)(L))(x))(r_1, \dots, r_n) \\ &= (((\varphi(\psi(L)))(x)))(r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.141}]}{=} (\varphi_{\psi(L)}(x))(r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.138}]}{=} (\psi(L))_x(r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.136}]}{=} (\psi(L))(x, r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.147}]}{=} \psi_L(x, r_1, \dots, r_n) \\ &\stackrel{[\text{eq: 14.144}]}{=} (L(x))(r_1, \dots, r_n) \\ &= (L(x))(r) \end{aligned}$$

from which it follows that $((\varphi \circ \psi)(L))(x) = L(x)$, hence $(\varphi \circ \psi)(L) = L$ proving that

$$\varphi \circ \psi = \text{Id}_{L(X, L^n(X; Y))} \quad (14.149)$$

On the other hand if $L \in L^{n+1}(X; Y)$ and $x \in X^{n+1}$ then we have

$$\begin{aligned} ((\psi \circ \varphi)(L))(x) &= ((\psi \circ \varphi)(L))(x_1, \dots, x_{n+1}) \\ &= (\psi(\varphi(L)))(x_1, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.147}]}{=} \psi_{\varphi(L)}(x_1, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.144}]}{=} ((\varphi(L))(x_1))(x_2, \dots, x_n) \\ &\stackrel{[\text{eq: 14.141}]}{=} (\varphi_L(x_1))(x_2, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.138}]}{=} L_{x_1}(x_2, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.136}]}{=} L(x_1, x_2, \dots, x_{n+1}) \\ &= L(x) \end{aligned}$$

proving that $(\psi \circ \varphi)(L) = L$ or

$$\psi \circ \varphi = \text{Id}_{L^{n+1}(X; Y)} \quad (14.150)$$

From [eqs: 14.149, 14.150] it follows by [theorem: 2.73] that

$$\varphi: L^{n+1}(X; Y) \rightarrow L(X, L^n(X; Y))$$

is a bijection. Combining this with [eq: 14.142] proves that

$$\varphi \text{ is a linear isomorphism} \quad (14.151)$$

Finally we have for $L \in L^{n+1}(X; Y)$ that

$$\begin{aligned} \|L\|_{L^{n+1}(X; Y)} &= \|\text{Id}_{L^{n+1}(X; Y)}(L)\|_{L^{n+1}(X; Y)} \\ &\stackrel{[\text{eq: 14.150}]}{=} \|(\psi \circ \varphi)(L)\|_{L^{n+1}(X; Y)} \\ &= \|\psi(\varphi(L))\|_{L^{n+1}(X; Y)} \\ &\leq_{[\text{eq: 14.148}]} \|\varphi(L)\|_{L(X, L^n(X; Y))} \end{aligned}$$

which combined with [eq: 14.143] proves that $\|\varphi(L)\|_{L(X, L^n(X; Y))} = \|L\|_{L^{n+1}(X; Y)}$. This together with [eq: 14.151] proves that

$$\varphi: L^{n+1}(X; Y) \rightarrow L(X, L^n(X; Y)) \text{ is a linear isometric isomorphism}$$

and as $\psi = \varphi^{-1}$ we have by [theorem: 14.126] that

$$\psi: L(X, L^n(X; Y)) \rightarrow L^{n+1}(X; Y) \text{ is a linear isometric isomorphism}$$

Let $(x_1, \dots, x_{n+1}) \in X^{n+1}$. Take $L \in L^{n+1}(X; Y)$ then we have

$$\begin{aligned} \varphi(L)(x_1)(x_2, \dots, x_{n+1}) &\stackrel{[\text{eq: 14.141}]}{=} \varphi_L(x_2, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.138}]}{=} L_{x_1}(x_2, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.136}]}{=} L(x_1, \dots, x_{n+1}) \end{aligned}$$

proving that

$$\forall L \in L^{n+1}(X; Y) \text{ we have } \varphi(L)(x_1)(x_2, \dots, x_{n+1}) = L(x_1, \dots, x_{n+1}) \quad (14.152)$$

further if $L \in L(X, L^n(X; Y))$ then

$$\begin{aligned} L(x_1)(x_2, \dots, x_{n+1}) &= ((\varphi \circ \psi)(L))(x_1)(x_2, \dots, x_{n+1}) \\ &= (\varphi(\psi(L)))(x_1)(x_2, \dots, x_{n+1}) \\ &\stackrel{[\text{eq: 14.152}]}{=} \psi(L)(x_1, x_2, \dots, x_{n+1}) \end{aligned}$$

proving that

$$\forall L \in L(X, L^n(X; Y)) \text{ we have } \psi(L)(x_1, x_2, \dots, x_{n+1}) = L(x_1)(x_2, \dots, x_{n+1})$$

proving the first part of the theorem for the case $1 < n$.

As for uniqueness if $\varphi': L^{n+1}(X; Y) \rightarrow L(X, L^n(X; Y))$ satisfies

$$\varphi(L)(x_1)(x_2, \dots, x_{n+1}) = L(x_1, \dots, x_{n+1})$$

then we have $\forall L \in L^{n+1}(X; Y) \forall x \in X$ that $\forall y \in X^n$

$$\varphi(L)(x)(y) = L(x, y_1, \dots, y_n) = \varphi(L)(x)(y)$$

so that $\varphi'(L)(x) = \varphi(L)(x)$ or $\varphi'(L) = \varphi(L)$ proving that

$$\varphi' = \varphi$$

Likewise if $\psi': L(X, L^n(X; Y)) \rightarrow L^{n+1}(X; Y)$ satisfies

$$\psi'(L)(x_1, x_2, \dots, x_{n+1}) = L(x_1)(x_2, \dots, x_{n+1})$$

then we have $\forall L \in L(X, L^n(X; Y)) \forall x \in X^{n+1}$ that $\forall y \in X^n$

$$\psi'(L)(x_1, \dots, x_{n+1}) = L(x_1)(x_2, \dots, x_{n+1}) = \psi(L)(x_1, \dots, x_{n+1})$$

so that $\psi'(L) = \psi(L)$ proving that

$$\psi' = \psi$$

Taking $\varphi_{[X,Y,n]} = \varphi$ and $\psi_{[X,Y,n]} = \psi$ completes then the proof if the theorem. $\square$

14.7 Separation

Definition 14.241. A topological space $\langle X, \mathcal{T} \rangle$ is **Hausdorff** if $\forall x, y \in X$ with $x \neq y$ there exists $U, V \in \mathcal{T}$ such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$. In other words a topological space is **Hausdorff** if two different points can be separated by disjoint open sets.

It turns out that all metric and normed spaces are Hausdorff

Theorem 14.242. If $\langle X, d \rangle$ is a metric space then $\langle X, \mathcal{T}_d \rangle$ is Hausdorff [using the metric topology $\mathcal{T}_d$]

Note 14.243. If $\langle X, \|\cdot\| \rangle$ is a normed space then $\langle X, \mathcal{T}_{\|\cdot\|} \rangle \stackrel{\text{definition: 14.96}}{=} \langle X, \mathcal{T}_{d_{\|\cdot\|}} \rangle$ so that every normed space is also Hausdorff.

Proof. Let $x, y \in X$ such that $x \neq y$ then as d is a metric we have that $0 < d(x, y)$ [for if $d(x, y) = 0$ then $x = y$]. Take $\varepsilon = d(x, y) > 0$ then $x \in B_d(x, \frac{\varepsilon}{2})$ and $y \in B_d(y, \frac{\varepsilon}{2})$. Assume that

$$z \in B_d\left(x, \frac{\varepsilon}{2}\right) \cap B_d\left(y, \frac{\varepsilon}{2}\right)$$

then $\varepsilon = d(x, y) \leq d(x, z) + d(z, y) = d(x, z) + d(y, z) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ leading to the contradiction $\varepsilon < \varepsilon$. Hence the assumption is wrong so that

$$B_d\left(x, \frac{\varepsilon}{2}\right) \cap B_d\left(y, \frac{\varepsilon}{2}\right) = \emptyset \text{ and } x \in B_d\left(x, \frac{\varepsilon}{2}\right) \in \mathcal{T}_d, y \in B_d\left(y, \frac{\varepsilon}{2}\right) \in \mathcal{T}_d \quad \square$$

One benefit of a Hausdorff spaces is that every finite set is closed in a Hausdorff space.

Theorem 14.244. If $\langle X, \mathcal{T} \rangle$ is a Hausdorff topological space then every finite set is closed.

Proof. Let $x \in X$ then if $y \in X \setminus \{x\}$ we have $y \neq x$ so by the Hausdorff condition there exists $U, V \in \mathcal{T}$ with $x \in U$, $y \in V$ and $U \cap V = \emptyset$. Hence $x \notin V$ so that $y \in V \subseteq X \setminus \{x\}$ proving by [theorem: 14.7] that $X \setminus \{x\}$ is open so that $\{x\}$ is closed. Hence we have that

$$\forall x \in X \text{ we have that } \{x\} \text{ is closed} \quad (14.153)$$

Now if $A \subseteq X$ is finite then we have either:

$A = \emptyset$. Then A is trivially closed.

$A \neq \emptyset$. Then by [theorem: 10.87] there exist a bijection $\beta: \{1, \dots, n\} \rightarrow A$ so that

$$A = \beta(\{1, \dots, n\}) = \beta\left(\bigcup_{i \in \{1, \dots, n\}} \{i\}\right) = \bigcup_{i \in \{1, \dots, n\}} \beta(\{i\}) = \bigcup_{i \in \{1, \dots, n\}} \{\beta(i)\}$$

proving by [eq: 14.153] and [theorem: 14.22] that A is closed. $\square$

Definition 14.245. A topological space $\langle X, \mathcal{T} \rangle$ is **regular** if for every closed set A and $x \in X \setminus A$ there exists $U, V \in \mathcal{T}$ such that $x \in U$, $A \subseteq V$ and $U \cap V = \emptyset$. In other words a topological space is **regular** if every closed set and a point outside the closed set can be separated by disjoint open sets.

All metric and normed spaces are not only Hausdorff but also regular.

Theorem 14.246. If $\langle X, d \rangle$ is a metric space then $\langle X, \mathcal{T}_d \rangle$ is regular [using the metric topology $\mathcal{T}_d$]

Note 14.247. If $\langle X, \|\cdot\| \rangle$ is a normed space then $\langle X, \mathcal{T}_{\|\cdot\|} \rangle \stackrel{\text{definition: 14.96}}{=} \langle X, \mathcal{T}_{d_{\|\cdot\|}} \rangle$ so that every normed space is also regular.

Proof. Let A be a closed set and $x \in X \setminus A$. As A is closed $X \setminus A$ is open, hence by [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_d(x, \delta) \subseteq X \setminus A \text{ hence } B_d(x, \delta) \cap A = \emptyset \quad (14.154)$$

Given $a \in A$ assume that $z \in B_d\left(x, \frac{\delta}{2}\right) \cap B_d(a, \delta_2)$ then $d(z, x) < \frac{\delta}{2} \wedge d(z, a) < \frac{\delta}{2}$ then

$$d(x, a) \leq d(x, z) + d(z, a) < \frac{\delta}{2} + \frac{\delta}{2} = \delta$$

so that $a \in B_d(x, \delta)$, hence $a \in A \cap B_d(x, \delta) = \emptyset$ contradicting [eq: 14.154]. Hence it follows that

$$B_d\left(x, \frac{\delta}{2}\right) \cap B_d\left(a, \frac{\delta}{2}\right) = \emptyset \quad (14.155)$$

Take $V = \bigcup_{a \in A} B_d\left(a, \frac{\delta}{2}\right) \in \mathcal{T}_d$ and $U = B_d\left(x, \frac{\delta}{2}\right) \in \mathcal{T}_d$ then clearly

$$A \subseteq V \text{ and } x \in U$$

Further

$$V \cap U = \left(\bigcup_{a \in A} B_d\left(a, \frac{\delta}{2}\right) \right) \cap B_d\left(x, \frac{\delta}{2}\right) = \bigcup_{a \in A} \left(B_d\left(a, \frac{\delta}{2}\right) \cap B_d\left(x, \frac{\delta}{2}\right) \right) \stackrel{[\text{eq: 14.155}]}{=} \emptyset \quad \square$$

Theorem 14.248. Let $\langle X, \mathcal{T} \rangle$ be a **regular** topological space, U a open set, $x \in U$ then there exist a $V \in \mathcal{T}$ such that $x \in V$ and $\bar{V} \subseteq U$.

Proof. Take $x \in U$ then as $X \setminus U$ is a closed set and $x \in U \stackrel{[\text{theorem: 1.36}]}{=} X \setminus (X \setminus U)$ there exist a $V, W \in \mathcal{T}$ such that $x \in V$, $X \setminus U \subseteq W$ and $V \cap W = \emptyset$. As $V \cap W = \emptyset$ we have that $V \subseteq X \setminus W$, as $X \setminus W$ is closed it follows from [theorem: 14.25] that $\bar{V} \subseteq X \setminus W$. As $X \setminus U \subseteq W$ we have $X \setminus W \subseteq X \setminus (X \setminus U) = U$ so that

$$\bar{V} \subseteq U \quad \square$$

Corollary 14.249. Let $\langle X, \mathcal{T} \rangle$ be a regular topological space, A a closed set, $x \in X \setminus A$ and U a open set with $x \in U$ then there exist a open set $V \in \mathcal{T}$ such that

$$x \in V, \bar{V} \cap A = \emptyset \text{ and } \bar{V} \subseteq U$$

Proof. As A is a closed set and $x \notin A$ there exists by regularity two open sets $V_1, V_2 \in \mathcal{T}$ such that

$$x \in V_1, A \subseteq V_2 \text{ and } V_1 \cap V_2 = \emptyset$$

As $x \in V_1 \cap U \in \mathcal{T}$ there exist by [theorem: 14.248] a open set $V \in \mathcal{T}$ such that

$$x \in V \text{ and } \bar{V} \subseteq V_1 \cap U \subseteq U$$

so that

$$\bar{V} \cap A \subseteq (V_1 \cap U) \cap A \subseteq V_1 \cap A \subseteq V_1 \cap V_2 = \emptyset$$

Hence we have

$$V \in \mathcal{T}, x \in V, \bar{V} \subseteq U \text{ and } \bar{V} \cap A = \emptyset \quad \square$$

Definition 14.250. A topological space $\langle X, \mathcal{T} \rangle$ is **normal** if for every pair of distinct closed sets A, B there exist $U, V \in \mathcal{T}$ such that $A \subseteq U$, $B \subseteq V$ and $U \cap V = \emptyset$. In other words a topological space is **normed** if every disjoint pair of closed sets can be separated by a disjoint pair of open sets.

Theorem 14.251. Let $\langle X, \mathcal{T} \rangle$ be a topological space then we have:

1. If $\langle X, \mathcal{T} \rangle$ is a normal space such that $\forall x \in X \{x\}$ is closed then $\langle X, \mathcal{T} \rangle$ is regular

2. If $\langle X, \mathcal{T} \rangle$ is a regular space such that $\forall x \in X$ $\{x\}$ is closed then $\langle X, \mathcal{T} \rangle$ is Hausdorff.

Proof.

1. Let A be a closed set and $x \in X \setminus A$ then by the hypothesis $\{x\}$ is closed, so by normality there exists $U, V \in \mathcal{T}$ such that $A \subseteq U$, $x \in \{x\} \subseteq V$ and $U \cap V = \emptyset$. Hence $\langle X, \mathcal{T} \rangle$ is regular.
2. Let $x, y \in X$ such that $x \neq y$ then $\{x\}$ and $\{y\}$ are closed and $\{x\} \cap \{y\} = \emptyset$ so that by normality there exists $U, V \in \mathcal{T}$ such that $x \in \{x\} \subseteq U$, $y \in \{y\} \subseteq V$ and $U \cap V = \emptyset$. Hence $\langle X, \mathcal{T} \rangle$ is Hausdorff. $\square$

Definition 14.252. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $x \in X$ then $A \subseteq X$ is a **neighborhood of x** iff there exist a open set $U \in \mathcal{T}$ such that $x \in U \subseteq A$. If A itself is open then A is called a **open neighborhood of x** .

Definition 14.253. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $x \in X$ then a **fundamental system of neighborhoods of x** is a set $\mathcal{N}$ of **neighborhoods of x** such that for every neighborhood A of x there exist a $N \in \mathcal{N}$ such that $x \in N \subseteq A$.

Definition 14.254. A topological space $\langle X, \mathcal{T} \rangle$ is **first countable** if every element of X has a **countable fundamental system of neighborhoods**.

Theorem 14.255. If $\langle X, d \rangle$ is a metric space then $\langle X, \mathcal{T}_d \rangle$ is **first countable** [using the metric topology $\mathcal{T}_d$].

Note 14.256. If $\langle X, \|\cdot\| \rangle$ is a normed space then $\langle X, \mathcal{T}_{\|\cdot\|} \rangle \stackrel{\text{definition: 14.96}}{=} \langle X, \mathcal{T}_{d_{\|\cdot\|}} \rangle$ so that every normed space is also first countable.

Proof. Given $x \in X$ define $\mathcal{N}_x = \{B_d(x, \frac{1}{n}) | n \in \mathbb{N}\}$ which is countable [see theorem: 6.75]. If A is a neighborhood of x then there exist a $U \in \mathcal{T}$ such that $x \in U \subseteq A$. Using [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_d(x, \delta) \subseteq U$. As $0 < \delta$ there exist a $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \delta$ [see theorem: 10.30] hence we have that $x \in B_d(x, \frac{1}{n}) \subseteq B_d(x, \delta) \subseteq U \subseteq A$ which as $B_d(x, \frac{1}{n}) \in \mathcal{N}_x$ proves first countability. $\square$

Definition 14.257. A topological space $\langle X, \mathcal{T} \rangle$ is **second countable** if $\mathcal{T}$ has a **countable basis**.

TODO check this

Example 14.258. $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ is second countable

Proof. Let $I = \{(q, r) \in \mathbb{Q} \times \mathbb{Q} | 0 < r\}$ then by [theorems: 10.5, 6.81, 6.71] I is countable. So that by [corollary: 6.74]

$$\mathcal{B} = \{B_{||}(q, \rho) | (q, r) \in I\} \subseteq \mathcal{T}_{||} \text{ is countable}$$

Let U be a open set in $\mathbb{R}$ and $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{||}(x, \delta) \subseteq U$$

As $0 < \delta$ there exist by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] a $r \in \mathbb{Q}$ such that $0 < r < \delta$ hence we have

$$x \in B_{||}(x, r) \subseteq B_{||}(x, \delta) \subseteq U$$

As $x - \frac{r}{3} < x + \frac{r}{3}$ there exist by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] a $q \in \mathbb{Q}$ such that $x - \frac{r}{3} < q < x + \frac{r}{3}$ so that $|q - x| < \frac{r}{3}$. Let $y \in B_{||}(q, \frac{r}{3})$ then we have that $|y - q| < \frac{r}{3}$ so that

$$|x - y| < |x - q + q - y| < |x - q| + |q - y| < \frac{r}{3} + \frac{r}{3} < r$$

which proves that $y \in B_{||}(q, r)$. Hence we have that

$$B_{||}\left(q, \frac{r}{3}\right) \subseteq B_{||}(x, r) \subseteq U$$

Further as $|q - x| < \frac{r}{3}$ it follows that

$$x \in B_{||}\left(q, \frac{r}{3}\right) \subseteq B_{||}(x, r) \subseteq U$$

Finally as $q \in \mathbb{Q}$ and $r \in \mathbb{Q} \Rightarrow \frac{r}{3} \in \mathbb{Q}$ we have found for $x \in U$ a $B_{||}\left(q, \frac{r}{3}\right) \in \mathcal{B}$ such that $x \in B_{||}\left(q, \frac{r}{3}\right) \subseteq U$. Hence by [theorem: 14.38] $\mathcal{B}$ is a countable basis for $\mathcal{T}_{||}$. $\square$

Example 14.259. Let $n \in \mathbb{N}$ then $\langle \mathbb{R}^n, \mathcal{T}_{|||n} \rangle$ is second countable

Proof. Let $I = \{(q, r) \in \mathbb{Q}^n \times \mathbb{Q} | 0 < r\} \subseteq \mathbb{Q}^n \times \mathbb{Q}$ then by [theorems: 10.5, 6.82, 6.81, 6.71] I is countable. So that by [corollary: 6.74]

$$\mathcal{B} = \{B_{|||n}(q, \rho) | (q, r) \in I\} \subseteq \mathcal{T}_{|||n} \text{ is countable}$$

Let U be a open set in $\mathbb{R}^n$ and $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{|||n}(x, \delta) \subseteq U$$

As $0 < \delta$ there exist by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] a $r \in \mathbb{Q}$ such that $0 < r < \delta$ hence we have

$$x \in B_{|||n}(x, r) \subseteq B_{|||n}(x, \delta) \subseteq U$$

Let $i \in \{1, \dots, n\}$ then as $x_i - \frac{r}{3} < x_i + \frac{r}{3}$ there exist by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] a $q_i \in \mathbb{Q}$ such that $x_i - \frac{r}{3} < q_i < x_i + \frac{r}{3}$ so that $|q_i - x_i| < \frac{r}{3}$. Let $q = (q_1, \dots, q_n)$ and take $y \in B_{|||n}\left(q, \frac{r}{3}\right)$ then we have $\forall i \in \{1, \dots, n\}$ that

$$|y_i - q_i| \leq \max(\{|y_i - q_i| | i \in \{1, \dots, n\}\}) = \|y_i - q_i\|_n < \frac{r}{3}$$

Hence we have

$$|x_i - y_i| < |x_i - q_i + q_i - y_i| < |x_i - q_i| + |q_i - y_i| < \frac{r}{3} + \frac{r}{3} < r$$

so that $\|x - y\|_n = \max(\{|x_i - y_i| | i \in \{1, \dots, n\}\}) < r$ which proves that $y \in B_{|||n}(x, r)$. Hence

$$B_{|||n}\left(q, \frac{r}{3}\right) \subseteq B_{|||n}(x, r) \subseteq U$$

Further as $\|q - x\|_n = \max(\{|q_i - x_i| | i \in \{1, \dots, n\}\}) < \frac{r}{3}$ we have

$$x \in B_{|||n}\left(q, \frac{r}{3}\right) \subseteq U$$

Finally as $q \in \mathbb{Q}^n$ and $r \in \mathbb{Q} \Rightarrow \frac{r}{3} \in \mathbb{Q}$ we have found for $x \in U$ a $B_{|||n}\left(q, \frac{r}{3}\right) \in \mathcal{B}$ such that $x \in B_{|||n}\left(q, \frac{r}{3}\right) \subseteq U$. Hence by [theorem: 14.38] $\mathcal{B}$ is a countable basis for $\mathcal{T}_{|||n}$. $\square$

Theorem 14.260. Let $\langle X, \mathcal{T} \rangle$ be a **second countable** topological space then $\langle X, \mathcal{T} \rangle$ is **first countable**.

Proof. Let $\langle X, \mathcal{T} \rangle$ be a second countable topological space and let $\mathcal{B} \subseteq \mathcal{T}$ be the countable basis of $\mathcal{T}$ then if $x \in X$ and A a neighborhood of x then there exists a $U \in \mathcal{T}$ such that $x \in U \subseteq A$. As $\mathcal{B}$ is a basis there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U \subseteq A$ [see theorem: 14.38]. So $\mathcal{B}$ is a fundamentally system of neighborhoods of every element of x . $\square$

Theorem 14.261. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then we have for the subspace topology $\langle A, \mathcal{T}_{|A} \rangle$ where $\mathcal{T}_{|A} = \{V \cap A | V \in \mathcal{T}\}$ that

1. If $\langle X, \mathcal{T} \rangle$ is Hausdorff then $\langle A, \mathcal{T}_{|A} \rangle$ is Hausdorff.
2. If $\langle X, \mathcal{T} \rangle$ is second countable then $\langle A, \mathcal{T}_{|A} \rangle$ is second countable.

Proof.

1. Let $x, y \in A$ with $x \neq y$ then as $\langle X, \mathcal{T} \rangle$ is Hausdorff there exist $V, W \in \mathcal{T}$ such that $x \in V$, $y \in W$ and $V \cap W = \emptyset$. From this it follows that $x \in A \cap V \in \mathcal{T}_{|A}$, $y \in A \cap W \in \mathcal{T}_{|A}$ and $(A \cap V) \cap (A \cap W) = A \cap (V \cap W) = A \cap \emptyset = \emptyset$ which proves that $\langle A, \mathcal{T}_{|A} \rangle$ is Hausdorff.

2. As $\langle X, \mathcal{T} \rangle$ is second countable there exist a countable basis $\mathcal{B} \subseteq \mathcal{T}$ for $\mathcal{T}$. Define now

$$\mathcal{B}|_U = \{B \cap A \mid B \in \mathcal{B}\}$$

Let $W \in \mathcal{T}|_U$ then there exist a open set $O \in \mathcal{T}$ such that

$$W = O \cap A$$

Let $x \in W$ then $x \in O \in \mathcal{T}$, $x \in A$ and, as $\mathcal{B}$ is a basis for $\mathcal{T}$, we have by [theorem: 14.38] a $B \in \mathcal{B}$ such that $x \in B \subseteq O$. Hence $x \in B \cap A \subseteq O \cap A = W$ which, as $B \cap A \in \mathcal{B}|_U$ proves by [theorem: 14.38] that

$$\mathcal{B}|_U \text{ is a basis for } \mathcal{T}|_A$$

Finally as $\mathcal{B}$ is countable it follows from [theorem: 6.74] that

$$\mathcal{B}|_U = \{B \cap A \mid B \in \mathcal{B}\} \text{ is countable}$$

□

Theorem 14.262. *Let $\langle X, \mathcal{T} \rangle$ be a second countable topological space and $\mathcal{U} \subseteq \mathcal{T}$ a set of open sets such that $X = \bigcup_{U \in \mathcal{U}} U$ then there exist a countable subset $\mathcal{C} \subseteq \mathcal{U}$ such that $X = \bigcup_{U \in \mathcal{C}} U$. In other words every covering of X with open sets contains a countable subset covering X .*

Proof. As $\langle X, \mathcal{T} \rangle$ is second countable there exist a countable basis $\mathcal{B} \subseteq \mathcal{T}$ of $\mathcal{T}$. Define now

$$\mathcal{B}' = \{B \in \mathcal{B} \mid \exists U \in \mathcal{U} \text{ such that } B \subseteq U\} \subseteq \mathcal{B}$$

Define now

$$\{\mathcal{U}_B\}_{B \in \mathcal{B}'}, \text{ where } \mathcal{U}_B = \{U \in \mathcal{U} \mid B \subseteq U\}$$

then for $B \in \mathcal{B}'$ we have that $\exists U \in \mathcal{U}$ such that $B \subseteq U$ hence $U \in \mathcal{U}_B \Rightarrow \mathcal{U}_B \neq \emptyset$. In other words

$$\forall B \in \mathcal{B}' \text{ we have } \mathcal{U}_B \neq \emptyset$$

Using now the Axiom of Choice [see theorem: 3.106] there exist a $\{U_B\}_{B \in \mathcal{B}'} \subseteq \bigcup_{B \in \mathcal{B}'} \mathcal{U}_B$ such that $\forall B \in \mathcal{B}' U_B \in \mathcal{U}_B \Rightarrow B \subseteq U_B$. As $\mathcal{B}' \subseteq \mathcal{B}$ and $\mathcal{B}$ is countable it follow from [theorem: 6.71] that $\mathcal{B}'$ is countable hence using [corollary: 6.74] it follows that

$$\{U_B \mid B \in \mathcal{B}'\} \text{ is countable}$$

Let $x \in X$ then as $X = \bigcup_{U \in \mathcal{U}} U$ there exist a $U \in \mathcal{U}$ such that $x \in U$, as $\mathcal{B}$ is a basis of $\mathcal{T}$ there exist by [theorem: 14.38] a $B \in \mathcal{B}$ such that $x \in B \subseteq U$, as $B \subseteq U$ $B \in \mathcal{B}'$ so that $x \in B \subseteq U_B \subseteq \bigcup_{B \in \mathcal{B}'} U_B$. Hence we must conclude that $X \subseteq \bigcup_{B \in \{U_B \mid B \in \mathcal{B}'\}} B$ which as $\forall B \in \mathcal{B}' B \subseteq X$ proves that

$$X = \bigcup_{B \in \{U_B \mid B \in \mathcal{B}'\}} B$$

□

14.8 Compact Spaces

Definition 14.263. (Compact Space) *A topological space $\langle X, \mathcal{T} \rangle$ is **compact** if for every $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ satisfying $X = \bigcup_{i \in I} U_i$ there exist a finite $J \subseteq I$ such that $X = \bigcup_{i \in J} U_i$. In other words a topological space is compact if every covering of the space by a family of open sets contains a finite subfamily of sets that covers the space.*

In addition to compact topological spaces we have also compact subsets of a topological space.

Definition 14.264. (Compact subset) *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $C \subseteq X$ then C is a **compact subset** if $\langle X, \mathcal{T}|_C \rangle$ is a **compact topological space** where $\mathcal{T}|_C = \{U \cap C \mid U \in \mathcal{T}\}$ is the subspace topology on C [see definition: 14.8]*

Definition 14.265. (Precompact Set) *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then A is **precompact** if its closure $\bar{A}$ is a compact set.*

We show now that a compact topological spaces is a topological space that is its own compact subspace.

Theorem 14.266. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then we have*

$$\langle X, \mathcal{T} \rangle \text{ is compact} \Leftrightarrow X \text{ is a compact subset of } \langle X, \mathcal{T} \rangle$$

Proof. First note that $\mathcal{T}|_X \stackrel{[\text{definition: 14.8}]}{=} X$ so that $\langle X, \mathcal{T} \rangle = \langle X, \mathcal{T}|_X \rangle$

$\Rightarrow$. If $\langle X, \mathcal{T} \rangle$ is compact then $\langle X, \mathcal{T}|_X \rangle$ is compact so that X is a compact subset of $\langle X, \mathcal{T} \rangle$.

$\Leftarrow$. If X is a compact subset of $\langle X, \mathcal{T} \rangle$ then $\langle X, \mathcal{T}|_X \rangle$ is compact so that $\langle X, \mathcal{T} \rangle$ is compact. $\square$

A alternative definition of a compact subset is given in the following theorem.

Theorem 14.267. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $C \subseteq X$ then*

C is compact

$\Updownarrow$

For every $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ satisfying $C \subseteq \bigcup_{i \in I} U_i$ there exist a finite $J \subseteq I$ such that $C \subseteq \bigcup_{i \in J} U_i$

Proof.

$\Rightarrow$. If $\{U_i\}_{i \in I}$ is a collection of open sets such that $C \subseteq \bigcup_{i \in I} U_i$ then $C = \bigcup_{i \in I} (C \cap U_i)$ where $\forall i \in I$ we have that $U_i \cap C \in \mathcal{T}|_C$ so that by the fact that $\langle C, \mathcal{T}|_C \rangle$ is compact there exists a finite $J \subseteq I$ such that $C = \bigcup_{i \in J} (U_i \cap C) \subseteq \bigcup_{i \in J} U_i$.

$\Leftarrow$. Assume that $\{V_i\}_{i \in I}$ is a collection of open sets in $\mathcal{T}|_C$ such that $C = \bigcup_{i \in I} V_i$, then by the definition of the subspace topology we have $\forall i \in I$ there exists a U_i such that $V_i = U_i \cap C$, so that $C = \bigcup_{i \in I} (U_i \cap C) \subseteq \bigcup_{i \in I} U_i$. Hence by the hypothesis there exists a finite $J \subseteq I$ such that $C \subseteq \bigcup_{i \in J} U_i \Rightarrow C = C \cap (\bigcup_{i \in J} U_i) = \bigcup_{i \in J} (U_i \cap C) = \bigcup_{i \in J} V_i$ proving that $\langle C, \mathcal{T}|_C \rangle$ is a compact topological space. Hence by definition C is a topological subset of $\langle X, \mathcal{T} \rangle$. $\square$

We can also express compactness using a basis for the topology.

Theorem 14.268. *Let $\langle X, \mathcal{T} \rangle$ be a topological space with a basis $\mathcal{B}$ and $C \subseteq X$ then*

C is a compact subset $\Leftrightarrow \forall \{B_i\}_{i \in I} \subseteq \mathcal{B}$ with $C \subseteq \bigcup_{i \in I} B_i$ there exist a finite J we have $C \subseteq \bigcup_{i \in J} B_i$

Proof.

$\Rightarrow$. This follows from $\mathcal{B} \subseteq \mathcal{T}$ and [theorem: 14.267].

$\Leftarrow$. Let $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ be a family of open sets such that $C \subseteq \bigcup_{i \in I} U_i$. Then $\forall c \in C$ there exist a $i_c \in I$ such that $c \in U_{i_c}$ as $\mathcal{B}$ is a basis it follows from [theorem: 14.38] that there exists a B_c such that $c \in B_c \subseteq U_{i_c}$. So $C \subseteq \bigcup_{c \in C} B_c$ and by the hypothesis there exists a finite set $F \subseteq C$ such that $C \subseteq \bigcup_{c \in F} B_c$. Define then the finite set $J = \{i_c | c \in F\} \subseteq I$ then if $c \in C$ there exist a $c \in F$ such that $c \in B_c \subseteq U_{i_c}$ where $i_c \in J$ proving that

$$C \subseteq \bigcup_{i \in J} U_i \quad \square$$

Example 14.269. If $\langle X, \mathcal{T} \rangle$ is a topological space then $\emptyset$ is a compact set.

Proof. If $\{U_i\}_{i \in I}$ is a non empty family of open sets covering $\emptyset$ then we have for I either

$I = \emptyset$. then $J = \emptyset = I$ is finite and $\emptyset = \bigcup_{i \in \emptyset} U_i = \bigcup_{i \in J} U_i$ so $\{U_i\}_{i \in J}$ is a finite covering of $\emptyset$

$I \neq \emptyset$. then $\exists i \in I$ so if we take $J = \{i\} \subseteq I$ then J is finite and $\emptyset \subseteq E_i = \bigcup_{j \in \{i\}} U_i$ $\square$

Example 14.270. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $x \in X$ then $\{x\}$ is a compact subset of X .

Proof. Let $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ be such that $\{x\} \subseteq \bigcup_{i \in I} U_i$ then $x \in \bigcup_{i \in I} U_i$ so there exist a $i \in I$ such that $x \in U_i$. Hence $\{x\} \subseteq U_i = \bigcup_{j \in \{i\}} U_j$ which as $\{i\}$ is finite and $\{i\} \subseteq I$ proves that $\{x\}$ is compact. $\square$

Theorem 14.271. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $n \in \mathbb{N}$ and $\{C_i\}_{i \in \{1, \dots, n\}}$ a finite family of compact subsets of X then $\bigcup_{i \in \{1, \dots, n\}} C_i$ is a compact subset.

Proof. Let $\bigcup_{i \in I} U_i$ be a open cover of $\bigcup_{i \in \{1, \dots, n\}} C_i$ then $\forall i \in \{1, \dots, n\}$ we have

$$C_i \subseteq \bigcup_{i \in \{1, \dots, n\}} C_i \subseteq \bigcup_{j \in I} U_j \xRightarrow{\text{[theorem: 14.267]}} \exists \text{finite } I_i \subseteq I \text{ such that } C_i \subseteq \bigcup_{j \in I_i} U_j$$

so that

$$\bigcup_{i \in \{1, \dots, n\}} C_i \subseteq \bigcup_{i \in \{1, \dots, n\}} \left(\bigcup_{j \in I_i} U_j \right)$$

If now $x \in \bigcup_{i \in \{1, \dots, n\}} \left(\bigcup_{j \in I_i} U_j \right)$ then there exist a $i \in \{1, \dots, n\}$ such that $x \in \bigcup_{j \in I_i} U_j$, hence there exists a $j \in I_i \subseteq \bigcup_{k \in \{1, \dots, n\}} I_k$ with $x \in U_j$, proving that $x \in \bigcup_{j \in \bigcup_{k \in \{1, \dots, n\}} I_k} U_j$. So we have

$$\bigcup_{i \in \{1, \dots, n\}} \left(\bigcup_{j \in I_i} U_j \right) \subseteq \bigcup_{j \in \bigcup_{k \in \{1, \dots, n\}} I_k} U_j$$

hence

$$\bigcup_{i \in \{1, \dots, n\}} C_i \subseteq \bigcup_{j \in \bigcup_{k \in \{1, \dots, n\}} I_k} U_j$$

where $\bigcup_{k \in \{1, \dots, n\}} I_k \subseteq I$ and by [theorem: 6.36] $\bigcup_{k \in \{1, \dots, n\}} I_k$ is finite. $\square$

Theorem 14.272. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $A \subseteq X$ and $C \subseteq A$ then

$$C \text{ is compact in } \langle A, \mathcal{T}|_A \rangle \Leftrightarrow C \text{ is compact in } \langle X, \mathcal{T} \rangle$$

Proof.

$\Rightarrow$. If $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ satisfies $C \subseteq \bigcup_{i \in I} U_i$ then $C \subseteq_{C \subseteq A} C \cap A \subseteq \bigcup_{i \in I} (U_i \cap A)$ where $\forall i \in I$ $U_i \cap A \in \mathcal{T}|_A$. As C is compact in $\langle A, \mathcal{T}|_A \rangle$, it follows from [theorem: 14.267] that there exist a finite $J \subseteq I$ so that $C \subseteq \bigcup_{i \in J} (U_i \cap A) \subseteq \bigcup_{i \in J} U_i$ proving by [theorem: 14.267] that C is compact in $\langle X, \mathcal{T} \rangle$.

$\Leftarrow$. If $\{V_i\}_{i \in I} \subseteq \mathcal{T}|_A$ satisfies $C \subseteq \bigcup_{i \in I} V_i$ then $\forall i \in I$ we have that $\exists U_i \in \mathcal{T}$ such that $V_i = U_i \cap A \subseteq U_i$. Hence $C \subseteq \bigcup_{i \in I} U_i$ which as C is compact in $\langle X, \mathcal{T} \rangle$ proves by [theorem: 14.267] that there exist a $J \subseteq I$ such that $C \subseteq \bigcup_{i \in J} U_i$. So

$$C = C \cap A \subseteq \left(\bigcup_{i \in J} U_i \right) \cap A = \bigcup_{i \in J} (U_i \cap A) = \bigcup_{i \in J} V_i$$

proving by [theorem: 14.267] that C is compact in $\langle A, \mathcal{T}|_A \rangle$ $\square$

In pseudo metric [or normed spaces] every compact set must be bounded.

Theorem 14.273. Let $\langle X, d \rangle$ be a pseudo metric space and $C \subseteq X$ a compact subset then C is bounded.

Proof. Let C be a compact subset of X then for $\{B_d(c, 1)\}_{c \in C} \subseteq \mathcal{T}_d$ we have $C \subseteq \bigcup_{c \in C} B_d(c, 1)$. As C is compact in $\langle X, \mathcal{T}_d \rangle$ we have by [theorem: 14.267] that there exist a finite set $J \subseteq C$ such that

$$C \subseteq \bigcup_{c \in J} B_d(c, 1) \quad (14.156)$$

We have now two cases for J :

$J = \emptyset$. Then $C \subseteq \bigcup_{c \in J} B_d(c, 1) = \emptyset$ hence by [example: 14.80] C is bounded

$J \neq \emptyset$. Then $\{d(c, t) | (c, t) \in J \times J\} \neq \emptyset$ and by [theorems: 6.41, 6.46] $\{d(c, t) | (c, t) \in J \times J\}$ is finite, so that by [theorem: 6.49] $N = \max(\{d(c, t) | (c, t) \in J \times J\})$ exists. Take then

$$M = N + 2$$

Let $x, y \in C$ then by [eq: 14.156] there exists $c_x, c_y \in J$ such that $x \in B_d(c_x, 1)$, $y \in B_d(c_y, 1)$, hence

$$\begin{aligned} d(x, y) &\leq d(x, c_x) + d(c_x, y) \\ &\leq d(x, c_x) + d(c_x, c_y) + d(c_y, y) \\ &< 1 + N + 1 \\ &= M \end{aligned}$$

proving that C is bounded. $\square$

Corollary 14.274. *Let $\langle X, ||| \rangle$ be a pseudo normed space and $C \subseteq X$ a compact subset then there exist a $M \in \mathbb{R}^+$ such that $\forall c \in C$ we have $||c|| \leq M$*

Proof. This follows from the previous theorem [theorem: 14.273] and [theorem: 14.112] $\square$

Continuous maps preserves compactness.

Theorem 14.275. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, $f: X \rightarrow Y$ a continuous function and C a compact set in $\langle X, \mathcal{T}_X \rangle$ then $f(C)$ is compact in $\langle Y, \mathcal{T}_Y \rangle$.*

Proof. Let $\{V_i\}_{i \in I}$ be a family of open sets in Y so that $f(C) \subseteq \bigcup_{i \in I} V_i$. Let $x \in C$ then we have $f(x) \in f(C) \subseteq \bigcup_{i \in I} V_i$ and thus $\exists i \in I$ such that $f(x) \in V_i$. By continuity of f there exists a U_i open in X with $x \in U_i$ and $f(U_i) \subseteq V_i$, hence $x \in \bigcup_{i \in I} U_i$. As $x \in C$ was chosen arbitrary it follows that

$$C \subseteq \bigcup_{i \in I} U_i$$

By compactness of C there exist a finite $J \subseteq I$ with

$$C \subseteq \bigcup_{i \in J} U_i \Rightarrow f(C) \subseteq f\left(\bigcup_{i \in J} U_i\right) = \bigcup_{i \in J} f(U_i) \subseteq \bigcup_{i \in J} V_i$$

proving that $f(C)$ is compact. $\square$

The following two theorems shows the relation between compactness and closedness.

Theorem 14.276. *Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space then every compact subset is closed.*

Proof. Let C be a compact subset of X . Take $y \in X \setminus C$ then $\forall c \in C$ we have by the Hausdorff condition that there exists $U_c, V_c \in \mathcal{T}$ such that $c \in U_c$, $y \in V_c$ and $\emptyset \neq U_c \cap V_c$. So $C \subseteq \bigcup_{c \in C} U_c$, hence, as C is compact, there exist a finite $F \subseteq C$ satisfying $C \subseteq \bigcup_{c \in F} U_c$. As $\forall c \in F \subseteq C$ we have $y \in V_c$ it follows that $y \in \bigcap_{c \in F} V_c$ which is open because F is finite. Further

$$\begin{aligned} \left(\bigcap_{c \in F} V_c\right) \cap C &\subseteq \left(\bigcap_{c \in F} V_c\right) \cap \left(\bigcup_{x \in F} U_x\right) \\ &= \bigcup_{x \in F} \left(\left(\bigcap_{c \in F} V_c\right) \cap U_x\right) \\ &\subseteq \bigcup_{x \in F} (V_x \cap U_x) \\ &= \bigcup_{x \in F} \emptyset \\ &= \emptyset \end{aligned}$$

So for every $y \in X \setminus C$ we have found a open set $\bigcap_{c \in F} V_c$ such that $y \in \bigcap_{c \in F} V_c \subseteq X \setminus C$ proving by [theorem: 14.7] that $X \setminus C$ is open or that C is closed. $\square$

Theorem 14.277. Let $\langle X, \mathcal{T} \rangle$ be a topological space, C a compact subset of X , $F \subseteq C$ a closed subset of C then C is compact.

Proof. Let $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ be a family of open sets such that $F \subseteq \bigcup_{i \in I} U_i$. As F is closed we have that $X \setminus F$ is open, using [theorem: 1.74] there exist a k such that $k \notin I$. Define then

$$\{V_i\}_{i \in I \cup \{k\}} \subseteq \mathcal{T} \text{ by } V_i = \begin{cases} U_i & \text{if } i \in I \\ X \setminus F & \text{if } i = k \end{cases}$$

Then for $x \in C$ we have either:

$x \in F$. Then $\exists i \in I \subseteq I \cup \{k\}$ such that $x \in U_i = V_i$ hence $x \in \bigcup_{i \in I \cup \{k\}} V_i$

$x \notin F$. Then $x \in X \setminus F = V_k$ such that $x \in \bigcup_{i \in I \cup \{k\}} V_i$

this proves that $C \subseteq \bigcup_{i \in I \cup \{k\}} V_i$. As C is compact there exist a finite set $J \subseteq I$ such that $C \subseteq \bigcup_{i \in J} V_i$ which as $F \subseteq C$ gives

$$F \subseteq \bigcup_{i \in J} V_i$$

Let $x \in F$ then there exist a $i \in J$ such that $x \in V_i$. Assume that $i = k$ then $x \in V_k = X \setminus F$ contradicting $x \in F$, hence $i \in J \setminus \{k\}$ and $x \in V_i = U_i$. So $F \subseteq \bigcup_{i \in J \setminus \{k\}} U_i$ which as $J \setminus \{k\}$ is finite proves that F is compact. $\square$

Theorem 14.278. Let $\langle X, \mathcal{T} \rangle$ be a **Hausdorff** topological space, $U \in \mathcal{T}$ and $A \subseteq U$ a precompact set [using the subspace topology $\mathcal{T}|_U$ [see definition: 14.8] on U then A is precompact using the topology $\mathcal{T}$ on X .

Proof. Let $\text{cl}_U(A)$ be the closure of A using the topology $\mathcal{T}|_U$ and $\text{cl}_X(A)$ the closure of A using the topology $\mathcal{T}$ on X . As A is precompact using $\mathcal{T}|_U$ we have that $\text{cl}_X(A)$ is compact using $\mathcal{T}|_U$, hence applying [theorem: 14.272] it follows that

$$\text{cl}_U(A) \text{ is compact using } \mathcal{T} \quad (14.157)$$

As $\mathcal{T}$ is Hausdorff it follows from [theorem: 14.276] and the above that $\text{cl}_U(A)$ is closed using the topology $\mathcal{T}$. Hence using [theorem: 14.26] we have that

$$\text{cl}_X(\text{cl}_U(A)) = \text{cl}_U(A) \quad (14.158)$$

As $A \subseteq \text{cl}_U(A)$ we have by [theorem: 14.32] that $\text{cl}_X(A) \subseteq \text{cl}_X(\text{cl}_U(A)) \stackrel{[\text{eq: 14.158}]}{=} \text{cl}_U(A)$, further using [theorem: 14.33] we have that $\text{cl}_U(A) = \text{cl}_X(A) \cap U \subseteq \text{cl}_X(A)$ which proves that

$$\text{cl}_X(A) = \text{cl}_U(A)$$

which combined with [eq: 14.157] proves that

$$\text{cl}_X(A) \text{ is compact using } \mathcal{T}$$

or in other words

$$A \text{ is precompact using } \mathcal{T} \quad \square$$

We have seen that for a Hausdorff space a compact set is closed and bounded. We show now that the opposite is true for the real numbers.

Theorem 14.279. (Heine Borel) Let $a, b \in \mathbb{R}$ then $[a, b]$ is a compact subset of $\mathbb{R}$ [using the norm topology $\mathcal{T}_||$ generated by the absolute value $||\cdot||$].

Proof. First if $b < a$ then $[a, b] = \emptyset$ which is compact by [example: 14.269] hence we have only to prove the remaining case $a \leq b$. Let $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ be family of open sets such that $[a, b] \subseteq \bigcup_{i \in I} U_i$. Define

$$\mathcal{G} = \left\{ x \in \mathbb{R} \mid a \leq x \text{ and there exist a finite } J \subseteq I \text{ such that } [a, x] \subseteq \bigcup_{i \in J} U_i \right\}$$

If $x \in \mathcal{G}$ then there exist a finite $J \subseteq I$ such that $[a, x] \subseteq \bigcup_{i \in J} U_i$, let $y \in [a, x]$ then

$$a \leq y \text{ and } [a, y] \subseteq [a, x] \subseteq [a, x] \subseteq \bigcup_{i \in J} U_i$$

proving that $y \in \mathcal{G}$. Hence we have proved that

$$\forall x \in \mathcal{G} \quad [a, x] \subseteq \mathcal{G} \quad (14.159)$$

As $a \in [a, b] \subseteq \bigcup_{i \in I} U_i$ there exist a $k \in I$ such that $a \in U_k$, by [theorems: 14.38, 14.104] there exist a $\delta \in \mathbb{R}^+$ such that $]a - \delta, a + \delta[\subseteq U_k = \bigcup_{i \in \{k\}} U_i$. Hence $\forall y \in [a, a + \delta[$ we have $y \in U_k = \bigcup_{i \in \{k\}} U_i$ which as $\{k\}$ is finite, $\{k\} \subseteq I$ and $a \leq y$ proves that $y \in \mathcal{G}$. So we have

$$\exists \delta \in \mathbb{R}^+ \text{ such that } [a, a + \delta[\subseteq \mathcal{G} \text{ and as } a \in [a, a + \delta[\quad \mathcal{G} \neq \emptyset \quad (14.160)$$

For $\mathcal{G}$ we have now two possible cases:

$\forall u \in \mathbb{R} \exists x \in \mathcal{G}$ such that $u \leq x$. In particular there exist a $x \in \mathcal{G}$ such that $b \leq x$ so that by [eq: 14.159] $[a, x] \subseteq \mathcal{G}$ which as $b \in [a, x]$ [because $a \leq b \leq x$] proves that

$$b \in \mathcal{G}$$

$\exists u \in \mathbb{R} \forall x \in \mathcal{G}$ we have $x \leq u$. Then $\mathcal{G}$ is bounded above by u hence, as by [eq: 14.160] $\mathcal{G} \neq \emptyset$, we have by the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] that

$$s = \sup(\mathcal{G}) \text{ exist}$$

Assume that $s \leq b$. Then as $a + \frac{\delta}{2} \in [a, a + \delta[\subseteq_{\text{eq: 14.160}} \mathcal{G}$ it follows that $a < a + \frac{\delta}{2} \leq s$ proving that

$$a < s \leq b$$

hence $s \in [a, b] \subseteq \bigcup_{i \in I} U_i$. So there exist a $l \in I$ such that $s \in U_l$, hence by [theorems: 14.38, 14.104] there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$s \in]s - \varepsilon, s + \varepsilon[\subseteq U_l \Rightarrow s \in \left[s - \frac{\varepsilon}{2}, s + \frac{\varepsilon}{2} \right] \subseteq]s - \varepsilon, s + \varepsilon[\subseteq U_l$$

As $s - \frac{\varepsilon}{2} < s = \sup(\mathcal{G})$ there exist a $c \in \mathcal{G}$ with

$$s - \frac{\varepsilon}{2} < c \leq s. \quad (14.161)$$

As $c \in \mathcal{G}$ there exist a finite $J \subseteq I$ such that $[a, c] \subseteq \bigcup_{i \in J} U_i$ and $a \leq c$. Hence

$$[a, c] \cup \left[s - \frac{\varepsilon}{2}, s + \frac{\varepsilon}{2} \right] \subseteq \bigcup_{i \in J \cup \{l\}} U_i \quad (14.162)$$

If $y \in [a, s + \frac{\varepsilon}{2}]$ then $a \leq y \leq s + \frac{\varepsilon}{2}$ and for y we have either:

$$y \leq c. \text{ Then } y \in [a, c] \subseteq [a, c] \cup \left[s - \frac{\varepsilon}{2}, s + \frac{\varepsilon}{2} \right]$$

$$c < y. \text{ Then by [eq: 14.161] } s - \frac{\varepsilon}{2} < c < y \leq s + \frac{\varepsilon}{2} \text{ proving that } y \in \left[s - \frac{\varepsilon}{2}, s + \frac{\varepsilon}{2} \right] \subseteq [a, c] \cup \left[s - \frac{\varepsilon}{2}, s + \frac{\varepsilon}{2} \right]$$

proving using [eq: 14.162] that

$$\left[a, s + \frac{\varepsilon}{2} \right] \subseteq \bigcup_{i \in J \cup \{l\}} U_i$$

which as $J \cup \{l\}$ is finite, $J \cup \{l\} \subseteq I$ and $c \in \mathcal{G} \Rightarrow a \leq c \leq s < s + \frac{\varepsilon}{2}$ proves that $s + \frac{\varepsilon}{2} \in \mathcal{G}$. Hence as $s = \sup(\mathcal{G})$ we must have that $s + \frac{\varepsilon}{2} \leq s$ a contradiction. Hence the assumption is wrong and we must have that

$$b < s$$

As $s = \sup(\mathcal{G})$ there exist a $d \in \mathcal{G}$ such that $b < d \leq s$, using [eq: 14.159] we have then that $[a, d] \subseteq \mathcal{G}$. From $a \leq b < d$ it follows that $b \in [a, d]$ proving that

$$b \in \mathcal{G}$$

So in all cases we have $b \in \mathcal{G}$ which means by the definition of $\mathcal{G}$ that there exist a finite $J \subseteq I$ such that $[a, b] \subseteq \bigcup_{i \in J} U_i$. Hence $[a, b]$ is compact. $\square$

Corollary 14.280. *If $C \subseteq \mathbb{R}$ then using the topology $\mathcal{T}_{||}$*

$$C \text{ is compact} \Leftrightarrow C \text{ is closed and bounded}$$

Proof.

$\Rightarrow$. Using [theorem: 14.273] C is bounded, further by [theorem: 14.242] $\mathcal{T}_{||_{\max}}$ is Hausdorff, so using [theorem: 14.276] C is closed.

$\Leftarrow$. As C is bounded there exist a $M \in \mathbb{R}^+$ such that $\forall x \in C \ |x| \leq M$ so we have $x \leq M$ and $-x \leq M \Rightarrow -M \leq x$ proving that $C \subseteq [-M, M]$. By [theorem: 14.279] $[-M, M]$ is compact which as C is closed proves by [theorem: 14.277] that C is compact. $\square$

Theorem 14.281. (Extreme Value Theorem) *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $f: X \rightarrow \mathbb{R}$ a continuous function [using the topology $\mathcal{T}_{||}$ on $\mathbb{R}$] and $C \subseteq X$ a compact subset of X then there exist a $M \in \mathbb{R}^+$ such that*

$$\forall x \in C \text{ we have } |f(x)| \leq M$$

Proof. As C is compact and f continuous we have by [theorem: 14.275] that $f(C)$ is a compact subset of $\mathbb{R}$, hence using [corollary: 14.280] $f(C)$ is bounded. So by [theorem: 14.274] there exist a $M \in \mathbb{R}^+$ such that $\forall y \in f(C)$ we have $|f(y)| \leq M$, hence if $x \in C \Rightarrow f(x) \in f(C)$ we have that $|f(x)| \leq M$. $\square$

Theorem 14.282. (Extreme Value Theorem) *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\emptyset \neq C \subseteq X$ a non empty compact subset of X and $f: X \rightarrow \mathbb{R}$ a continuous function [using the topology $\mathcal{T}_{||}$ on $\mathbb{R}$] then there exists $m, M \in C$ such that*

$$\forall x \in C \text{ we have } f(m) \leq f(x) \leq f(M)$$

Proof. As C is compact and f continuous we have by [theorem: 14.275] that

$$f(C) \text{ is compact}$$

As by [theorem: 14.242] $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ is Hausdorff and $f(C)$ is compact it follows from [theorem: 14.276] that

$$f(C) \text{ is closed}$$

Using [theorem: 14.274] there exist a $N \in \mathbb{R}^+$ such that $\forall x \in f(C)$ we have $|x| \leq N \Rightarrow -N \leq x \leq N$ proving that $f(C)$ is bounded above and below. As further $f(C) \neq \emptyset$ we have using the fact that $\mathbb{R}$ is conditional complete [see theorem: 10.18] that $\sup(f(C))$ and $\inf(f(C))$ exist. As $f(C)$ is closed and non empty we have by [theorem: 14.111] that $\sup(f(C)) \in f(C)$ and $\inf(f(C)) \in f(C)$. Hence there exists $m, M \in C$ such that $\sup(f(C)) = f(M)$ and $\inf(f(C)) = f(m)$. So we have $\forall x \in C$ that

$$f(m) = \inf(f(C)) \leq f(x) \leq \sup(f(C)) = f(M) \quad \square$$

Theorem 14.283. *Let $\langle X, \mathcal{T} \rangle$ be a compact Hausdorff space then it is regular and normal*

Proof. Let A be a closed set and $x \in X \setminus A$. As A is closed we have we have by [theorem: 14.277] that

$$A \text{ is compact}$$

then by the Hausdorff condition we have $\forall a \in A$ that there exists $U_a, V_a \in \mathcal{T}$ such that $a \in U_a$, $x \in V_a$ and $U_a \cap V_a = \emptyset$. So $A \subseteq \bigcup_{a \in A} U_a$ which as A is compact means that there exist a finite $I \subseteq A$ such that $A \subseteq \bigcup_{a \in I} U_a$. Take now $U = \bigcup_{a \in I} U_a \in \mathcal{T}$ and $V = \bigcap_{a \in I} V_a \in \mathcal{T}$ [as I is finite] then we have $A \subseteq U$, $x \in V$ and

$$U \cap V = \left(\bigcup_{a \in I} U_a \right) \cap \left(\bigcap_{b \in I} V_b \right) = \bigcup_{a \in I} \left(U_a \cap \left(\bigcap_{b \in I} V_b \right) \right) \stackrel{U_a \cap (\bigcap_{b \in I} V_b) \subseteq U_a \cap V_a = \emptyset}{=} \emptyset$$

proving that

$$\langle X, \mathcal{T} \rangle \text{ is regular} \quad (14.163)$$

Let A, B be closed sets such that $A \cap B = \emptyset$. Then $\forall a \in A$ we have as $a \in X \setminus B$ by [eq: 14.163] that there exists $U_a, V_a \in \mathcal{T}$ with $a \in U_a$, $B \subseteq V_a$ and $U_a \cap V_a = \emptyset$. As by [theorem: 14.277] A is compact and $A \subseteq \bigcup_{a \in A} U_a$ there exist a finite $I \subseteq A$ such that $A \subseteq \bigcup_{a \in I} U_a$. So if we take $U = \bigcup_{a \in I} U_a \in \mathcal{T}$ and $V = \bigcap_{a \in I} V_a \in \mathcal{T}$ [as I is finite] then we have $A \subseteq U$ and $B \subseteq V$ and

$$U \cap V = \left(\bigcup_{a \in I} U_a \right) \cap \left(\bigcap_{b \in I} V_b \right) = \bigcup_{a \in I} \left(U_a \cap \left(\bigcap_{b \in I} V_b \right) \right) \stackrel{U_a \cap (\bigcap_{b \in I} V_b) \subseteq U_a \cap V_a = \emptyset}{=} \emptyset$$

proving that

$$\langle X, \mathcal{T} \rangle \text{ is normal} \quad \square$$

A weaker version of compactness is sequential compactness.

Definition 14.284. (Limit Point Compact Set) Let $\langle X, \mathcal{T} \rangle$ be a topological space then a subset $C \subseteq X$ is **limit point compact** if every **infinite** $A \subseteq C$ has a limit point [see definition: 14.27]

Theorem 14.285. Every compact subspace of a topological space $\langle X, \mathcal{T} \rangle$ is limit point compact.

Proof. We proceed by contradiction. Let C be a compact subset and let $A \subseteq C$ be an infinite subset of C that does not have a limit point. Then by the definition of a limit point [see definition: 14.27] we have $\forall x \in X$ that there exist a $U_x \in \mathcal{T}$ with $x \in U_x$ so that $U_x \cap (A \setminus \{x\}) = \emptyset$. Let $y \in U_x \cap A$ then if $y \neq x$ we have $y \in (U_x \cap A) \setminus \{x\} = U_x \cap (A \setminus \{x\}) = \emptyset$ a contradiction, hence we have that $y = x \in \{x\}$, proving that

$$U_x \cap A \subseteq \{x\} \quad (14.164)$$

As $C \subseteq \bigcup_{x \in C} U_x$ we have, as C is compact, that there exist a finite $F \subseteq C$ such that $C \subseteq \bigcup_{x \in F} U_x$. Further from $A \subseteq C$ we have

$$A = C \cap A \subseteq \left(\bigcup_{x \in F} U_x \right) \cap A = \bigcup_{x \in F} (U_x \cap A) \subseteq_{[\text{eq: 14.164}]} \bigcup_{x \in F} \{x\} = F$$

proving that A is finite contradicting the fact that we have taken A to be infinite. So we must have that every infinite subset of C has a limit point. $\square$

Definition 14.286. (Compact Class) Let X be a set then $C \subseteq \mathcal{P}(X)$ a set of subsets of X is a **compact class** if for any sequence $\{K_i\}_{i \in \mathbb{N}} \subseteq C$ with $\bigcap_{i \in \mathbb{N}} K_i = \emptyset$ there exist a $N \in \mathbb{N}$ such that $\bigcap_{i \in \{1, \dots, N\}} K_i = \emptyset$.

Theorem 14.287. Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space and C a set of compact subsets of X then C is a compact class.

Proof. We prove this by contradiction. Let $\{K_i\}_{i \in \mathbb{N}} \subseteq C$ be a sequence of subsets of C such that $\bigcap_{i \in \mathbb{N}} K_i = \emptyset$. Assume that $\forall n \in \mathbb{N}$ we have $\bigcap_{i \in \{1, \dots, n\}} K_i \neq \emptyset$. Define

$$\{E_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X) \text{ by } E_i = \bigcap_{j \in \{1, \dots, i\}} K_j \quad (14.165)$$

so that

$$E_n \subseteq E_m \text{ for } n, m \in \mathbb{N} \text{ with } m \leq n \quad (14.166)$$

Let $j \in \mathbb{N}$ then K_j is compact hence, as $\mathcal{T}$ is Hausdorff it follows from [theorem: 14.276] that K_j is closed, hence for $i \in \mathbb{N}$ we have by [theorem: 14.22] that $E_i = \bigcap_{j \in \{1, \dots, i\}} K_j$ is closed, hence $X \setminus E_i$ is open proving that

$$\{X \setminus E_i\}_{i \in \mathbb{N}} \text{ is a sequence of open sets in other words } \{X \setminus E_i\}_{i \in \mathbb{N}} \subseteq \mathcal{T}$$

Further as $\forall i \in \mathbb{N}$ we have that $E_i = \bigcap_{j \in \{1, \dots, i\}} K_j \subseteq K_i$ it follows from [theorem: 1.36] that $X \setminus K_i \subseteq X \setminus E_i$. Let $x \in K_1$ then as $\bigcap_{i \in \mathbb{N}} K_i = \emptyset$ there exist a $n \in \mathbb{N}$ such that $x \notin K_n$ hence $x \in X \setminus K_n \subseteq X \setminus E_n$ proving that $K_1 \subseteq \bigcup_{n \in \mathbb{N}} (X \setminus E_n)$. As K_1 is compact there exist a finite $I \subseteq \mathbb{N}$ such that

$$K_1 \subseteq \bigcup_{n \in I} (X \setminus E_n) \quad (14.167)$$

Take $m = \max(I)$ then as $\forall i \in I$ we have $i \leq m$ so that by [eq: 14.166] $E_m \subseteq E_i \Rightarrow X \setminus E_i \subseteq X \setminus E_m$ which using [eq: 14.167] proves that

$$K_1 \subseteq X \setminus E_m \quad (14.168)$$

Hence

$$E_m \stackrel{[\text{eq: 14.165}]}{=} \bigcap_{i \in \{1, \dots, m\}} K_i \subseteq K_1 \stackrel{[\text{eq: 14.168}]}{\subseteq} X \setminus E_m$$

from which it follows that $E_m = \emptyset$ [for if $y \in E_m$ then $y \in X \setminus E_m$ so that $y \notin E_m$]. As $E_m = \bigcap_{i \in \{1, \dots, m\}} K_i$ we have $\emptyset = \bigcap_{i \in \{1, \dots, m\}} K_i$ which contradicts the assumption that $\forall n \in \mathbb{N}$ we have $\bigcap_{i \in \{1, \dots, n\}} K_i \neq \emptyset$. So the assumption is wrong and we must have that there exist a $N \in \mathbb{N}$ such that $\bigcap_{i \in \{1, \dots, N\}} K_i = \emptyset$ from which it follows that C is a compact class. $\square$

Theorem 14.288. Let $\langle X, d_X \rangle$, $\langle Y, d_Y \rangle$, K a non empty compact subset of X and $f: K \rightarrow Y$ a continuous function then f is uniform continuous [see definition: 14.174].

Proof. Let $\varepsilon \in \mathbb{R}^+$. Take $x \in K$ then by the continuity of f there exist by [theorem: 14.134] a $\delta_x \in \mathbb{R}^+$ [depending on x] such that for every $y \in K$ with $(d_X)_{|K \times K}(x, y) < \delta_x$ we have $d_Y(f(x), f(y)) < \frac{\varepsilon}{2}$. As $d_X(x, y) = (d_X)_{|K \times K}(x, y) < \delta_x$ we have $x \in B_{d_X}(x, \delta_x)$ so that

$$K \subseteq \bigcup_{x \in K} B_{d_X}\left(x, \frac{\delta_x}{2}\right), \quad (14.169)$$

hence, as K is compact, there exist a finite $I \subseteq K$ such that $K \subseteq \bigcup_{x \in I} B_{d_X}\left(x, \frac{\delta_x}{2}\right)$. As $K \neq \emptyset$ and $\left\{\frac{\delta_x}{2} | x \in I\right\}$ is finite [see theorem: 6.46] it follows that

$$\delta = \min\left(\left\{\frac{\delta_x}{2} | x \in I\right\}\right) \text{ exist} \quad (14.170)$$

Let now $x, y \in K$ be such that $(d_X)_{|K \times K}(x, y) < \delta \Rightarrow d_X(x, y) < \delta$ then, as $x \in K$, there exist by [eq: 14.169] a $z \in I$ such that

$$x \in B_{d_X}\left(z, \frac{\delta_z}{2}\right) \text{ or } (d_X)_{|K \times K}(x, z) = d_X(x, z) < \frac{\delta_z}{2} \quad (14.171)$$

Then

$$(d_X)_{|K \times K}(z, y) = d_X(z, y) \leq d_X(z, x) + d_X(x, y) < \frac{\delta_z}{2} + \delta \leq_{[\text{eq: 14.170}]} \frac{\delta_z}{2} + \frac{\delta_z}{2} = \delta_z$$

so that

$$d_Y(f(z), f(y)) < \frac{\varepsilon}{2} \quad (14.172)$$

As $(d_X)_{|K \times K}(x, z) <_{[\text{eq: 14.171}]} \frac{\delta_z}{2}$ we have

$$d_Y(f(x), f(z)) < \frac{\varepsilon}{2} \quad (14.173)$$

Finally we have

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(z)) + d_Y(f(z), f(y)) <_{[\text{eqs: 14.172, 14.173}]} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving uniform continuity. $\square$

14.8.1 Locally Compact and Paracompact spaces

Definition 14.289. A topological space $\langle X, \mathcal{T} \rangle$ is **locally compact** iff $\forall x \in X$ there exist a $U \in \mathcal{T}$ and a compact subset C such that $x \in U \subseteq C$.

Theorem 14.290. *Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space then we have the following equivalences:*

1. $\langle X, \mathcal{T} \rangle$ is locally compact.
2. $\forall x \in X$ there exist a precompact set $U \in \mathcal{T}$ such that $x \in U$.
3. $\langle X, \mathcal{T} \rangle$ has a basis $\mathcal{B} \subseteq \mathcal{T}$ such that $\forall B \in \mathcal{B}$ we have that B is precompact.

Proof.

1 $\Rightarrow$ 2. Let $x \in X$ then by the local compactness there exist a open set $U \in \mathcal{T}$ and a compact set C such that $x \in U \subseteq C$. As $\langle X, \mathcal{T} \rangle$ is Hausdorff it follows from [theorem: 14.276] that C is closed. Hence we have that $x \in U \subseteq \bar{U} \subseteq \bar{C} \stackrel{[\text{theorem: 14.32}]}{=} C$ is closed and $\stackrel{[\text{theorem: 14.26}]}{=} C$. As $\bar{U}$ is closed and $\bar{U} \subseteq C$ it follows from [theorem: 14.277] that $\bar{U}$ is compact. Hence U is a precompact set [see definition: 14.265] with $x \in U \in \mathcal{T}$.

2 $\Rightarrow$ 3. Define $\mathcal{B} = \{V \in \mathcal{T} \mid V \text{ is precompact}\} \subseteq \mathcal{T}$. Let $U \in \mathcal{T}$ and take $x \in U$ then by (2) there exist a **precompact** $V \in \mathcal{T}$ such that $x \in V$. Take $W = U \cap V$ then $x \in W$, $W \in \mathcal{T}$ and $W \subseteq V$, $W \subseteq U$. Hence by [theorem: 14.32] $\bar{W} \subseteq \bar{V}$ which, as $\bar{V}$ is compact [because it is precompact], proves by [theorem: 14.277] that $\bar{W}$ is compact. Hence W is precompact so that $W \in \mathcal{B}$. To summarize we have $\forall U \in \mathcal{T}$, $\forall x \in U$ that there exist a $B \in \mathcal{B}$ such that $x \in B \subseteq U$. Using [theorem: 14.38] it follows then that $\mathcal{B}$ is a basis for $\mathcal{T}$.

3 $\Rightarrow$ 1. Let $\mathcal{B}$ be a basis of precompact sets of $\mathcal{T}$ [so that $\mathcal{B} \subseteq \mathcal{T}$]. Take $x \in X$ then as $X \in \mathcal{T}$ there exist a precompact set $U \in \mathcal{B} \subseteq \mathcal{T}$ such that $x \in U$. By the definition of precompact sets $\bar{U}$ is compact. Hence $\forall x \in X$ we found a $U \in \mathcal{T}$ such that $x \in U \subseteq \bar{U}$ where $\bar{U}$ is compact which proves that $\langle X, \mathcal{T} \rangle$ is locally compact. $\square$

Definition 14.291. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ is a **exhausting of X by compact sets** if $\forall i \in \mathbb{N}$ C_i is a compact subset of X , $C_i \subseteq (C_{i+1})^\circ$ and $X = \bigcup_{i \in \mathbb{N}} C_i$.*

Theorem 14.292. *Let $\langle X, \mathcal{T} \rangle$ be a second countable locally compact Hausdorff space then there exist a exhausting of X by compact sets $\{C_i\}_{i \in \mathbb{N}}$.*

Proof. Because $\langle X, \mathcal{T} \rangle$ is locally compact and Hausdorff it follows from [theorem: 14.290] there exist a basis $\mathcal{B}$ such that

$$\forall U \in \mathcal{B} \text{ } U \text{ is precompact and open [because a basis is a subset of } \mathcal{T}] \quad (14.174)$$

Hence we have as X is a open set by [definition: 14.34] that there exist a $\mathcal{W} \subseteq \mathcal{B}$ so that

$$X = \bigcup_{U \in \mathcal{W}} U$$

As $\langle X, \mathcal{T} \rangle$ is second countable it follows from [theorem: 14.262] and the above that there exist a countable $\mathcal{C} \subseteq \mathcal{W}$ such that

$$X = \bigcup_{U \in \mathcal{C}} U \quad (14.175)$$

As $\mathcal{C}$ is countable there exist a bijection $\beta: \mathbb{N} \rightarrow \mathcal{C}$, allowing us to define $\{U_i\}_{i \in \mathbb{N}}$ by $U_i = \beta(i)$. Then if $x \in X$ there exist a $U \in \mathcal{C}$ such that $x \in U$, so that there exist a $i \in \mathbb{N}$ such that $U = \beta(i) = U_i$ so that $x \in U_i \subseteq \bigcup_{i \in \mathbb{N}} U_i$. Hence $X \subseteq \bigcup_{i \in \mathbb{N}} U_i$ which as trivially $\bigcup_{i \in \mathbb{N}} U_i \subseteq X$ proves that

$$X = \bigcup_{i \in \mathbb{N}} U_i \quad (14.176)$$

Further using [eq: 14.174] we have that

$$\forall i \in \mathbb{N} \text{ } U_i \in \mathcal{T} \text{ and } U_i \text{ is precompact} \Rightarrow \bar{U}_i \text{ is compact} \quad (14.177)$$

We define now $\{C_n\}_{n \in \mathbb{N}}$ such that $\forall n \in \mathbb{N}$ we have $U_n \subseteq C_n \wedge C_n$ is compact and $C_n \subseteq (C_{n+1})^\circ$ using recursion:

1 = n. Let $C_1 = \bar{U}_1$ then we have $U_1 \subseteq C_1$ and by [eq: 14.177] C_1 is compact.

$1 < n$. Then given $\{C_i\}_{i \in \{1, \dots, n\}}$ such that $\forall i \in \{1, \dots, k\} U_i \subseteq C_i \wedge C_i$ is compact and $\forall i \in \{1, \dots, n-1\} C_i \subseteq (C_i)^\circ$ we construct a C_{n+1} such that C_{n+1} is compact, $U_{n+1} \subseteq C_{n+1}$ and $C_n \subseteq (C_n)^\circ$ as follows. As C_n is compact and by [eqs: 14.176, 14.177] $C_n \subseteq \bigcup_{i \in \mathbb{N}} U_i$ where $U_i \in \mathcal{T}$ it follows that there exist a finite set $I \subseteq \mathbb{N}$ such that $C_n \subseteq \bigcup_{i \in I} U_i$. As $I \cup \{1\}$ is finite and non empty it follows from [theorem: 6.49] that $\max(I \cup \{1\})$ exist, tale then

$$m_n = \max(n+1, \max(I \cup \{1\}))$$

then $n+1 \in \{1, \dots, m_n\}$ and $\forall i \in I i \leq m_n$ so that

$$C_n \subseteq \bigcup_{i \in I} U_i \subseteq \bigcup_{i \in \{1, \dots, m_n\}} U_i \text{ and } U_{n+1} \subseteq \bigcup_{i \in \{1, \dots, m_n\}} U_i \quad (14.178)$$

Define now C_{n+1} by

$$C_{n+1} = \bigcup_{i \in \{1, \dots, m_n\}} \overline{U_i}$$

then by [eq: 14.178] we have that

$$U_{n+1} \subseteq C_{n+1} \quad (14.179)$$

Further, as $\forall i \in \{1, \dots, m_n\} \overline{U_i}$ is compact [see eq: 14.177], it follows from [theorem: 14.271] that

$$C_{n+1} \text{ is a compact set} \quad (14.180)$$

As $\forall i \in \mathbb{N} U_i \in \mathcal{T}$ [see eq: 14.177], it follows that $\bigcup_{i \in \{1, \dots, m_n\}} U_i \in \mathcal{T}$ so that by [theorem: 14.16]

$$\bigcup_{i \in \{1, \dots, m_n\}} U_i = \left(\bigcup_{i \in \{1, \dots, m_n\}} U_i \right)^\circ$$

Further as $\bigcup_{i \in \{1, \dots, m_n\}} U_i \subseteq \bigcup_{i \in \{1, \dots, m_n\}} \overline{U_i} \subseteq C_{n+1}$ it follows from [theorem: 14.14] that

$$\left(\bigcup_{i \in \{1, \dots, m_n\}} U_i \right)^\circ \subseteq (C_{n+1})^\circ$$

so that $\bigcup_{i \in \{1, \dots, m_n\}} U_i \subseteq (C_{n+1})^\circ$ which combined with [eq: 14.178] proves that

$$C_n \subseteq (C_{n+1})^\circ \quad (14.181)$$

Summarizing [eqs: 14.179, 14.180, 14.181] we have constructed a C_{n+1} such that

$$U_{n+1} \subseteq C_{n+1}, C_{n+1} \text{ is compact and } C_n \subseteq (C_{n+1})^\circ$$

Let $x \in X$ then as $X = \bigcup_{i \in \mathbb{N}} U_i$ [see 14.175] there exist a $i \in \mathbb{N}$ such that $x \in U_i$, as by construction $U_i \subseteq C_i$ it follows that $x \in C_i$ hence $X \subseteq \bigcup_{i \in \mathbb{N}} U_i$. As further $U_i \subseteq X$ [because $U_i \in \mathcal{T}$] $\forall i \in \mathbb{N}$ it follows that

$$C = \bigcup_{i \in \mathbb{N}} U_i$$

Hence the constructed sequence $\{C_n\}_{n \in \mathbb{N}}$ is then the exhausting of X by compact sets. $\square$

14.8.2 Product of Compact sets

The purpose of this section is to prove that the product of compact sets is compact, this is know as Tychonoff's theorem. The best way to do this is to develop the framework of filter bases.

14.8.2.1 Filter bases

Definition 14.293. (Filter Base) Let X be a set then a non empty family $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ of a subsets of X is a **filter base in X** if

1. $\forall \alpha \in I$ we have $A_\alpha \neq \emptyset$

2. $\forall \alpha, \beta \in I$ there exist a $\gamma \in I$ such that $A_\gamma \subseteq A_\alpha \cap B_\beta$

Using mathematical induction we can extend (2) of the above definition.

Theorem 14.294. Let X be a set and $\mathfrak{A} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base then for a finite non empty $J \subseteq I$ there exist a $\gamma \in I$ such that $A_\gamma \subseteq \bigcap_{\alpha \in J} A_\alpha$

Proof. Define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } J \subseteq I \text{ with } \text{card}(J) = n \text{ then there exist a } \gamma \in I \text{ such that } A_\gamma \subseteq \bigcap_{\alpha \in J} A_\alpha \right\}$$

then we have:

$1 \in S$. If $J \subseteq I$ with $\text{card}(J) = 1$ then there exist a $\gamma \in I$ such that $J = \{\gamma\}$. As $A_\gamma \subseteq A_\gamma = \bigcup_{\alpha \in J} A_\alpha$ it follows that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $J \subseteq I$ with $\text{card}(J) = n+1$ then there exist a bijection

$$\beta: \{1, \dots, n+1\} \rightarrow J$$

By [theorem: 2.99] $\beta|_{\{1, \dots, n\}}: \{1, \dots, n\} \rightarrow J \setminus \{\beta(n+1)\}$ is a bijection so that $\text{card}(J \setminus \{\beta(n+1)\}) = n$. As $n \in S$ there exist a $\zeta \in J \setminus \{\beta(n+1)\}$ such that

$$A_\zeta \subseteq \bigcap_{\alpha \in J \setminus \{\beta(n+1)\}} A_\alpha$$

By definition of a filter base there exist a $\gamma \in I$ such that

$$A_\gamma \subseteq A_\zeta \cap A_{\beta(n+1)} \subseteq \left(\bigcap_{\alpha \in J \setminus \{\beta(n+1)\}} A_\alpha \right) \cap A_{\beta(n+1)} = \bigcap_{\alpha \in J} A_\alpha$$

proving that $n+1 \in S$.

By mathematical induction it follows that $S = \mathbb{N}$. Hence if $J \subseteq I$ is finite we have as $\text{card}(J) \in \mathbb{N}$ that there exist a $\gamma \in I$ such that $A_\gamma \subseteq \bigcap_{\alpha \in J} A_\alpha$. $\square$

Example 14.295. Let X be a set, $\emptyset \neq A \subseteq X$ then $\mathfrak{A} = \{A_\alpha\}_{\alpha \in \{A\}} \subseteq \mathcal{P}(X)$ defined by $A_\alpha = A$ is a filter base in X

Proof. First as $\{A\} \neq \emptyset$ $\{A_\alpha\}_{\alpha \in \{A\}}$ is a non empty family of subsets in X . Further

1. $\forall \alpha \in \{A\}$ we have $A_\alpha = A = A \neq \emptyset$.

2. If $\alpha, \beta \in \{A\}$ then $\alpha = \beta = A$ so for $\gamma = A$ we have $A_\gamma = A \subseteq A \cap A = A_\alpha \cap A_\beta$ $\square$

Definition 14.296. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $x \in X$ then $\mathfrak{U}(x) = \{U\}_{U \in \{U \in \mathcal{T} \mid x \in U\}} \subseteq \mathcal{P}(X)$ [a family of open sets containing x] is called a **neighborhood filter base of x** .

The name used in the above definition suggest that $\mathfrak{U}(x)$ is a filter base in X which is proved in the following theorem.

Theorem 14.297. Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\mathfrak{U}(x)$ is a filter base in X

Proof. Let $x \in X$ then as $x \in \mathcal{T}$ we have that $x \in \{U \in \mathcal{T} \mid x \in U\}$ so that $\mathfrak{U}(x)$ is a non empty family. Further we have

1. If $U \in \{U \in \mathcal{T} \mid x \in U\}$ then $x \in U$ so that $U \neq \emptyset$,

2. If $U, V \in \{U \in \mathcal{T} \mid x \in U\}$ then $x \in U \cap V \in \mathcal{T}$ so that for $W = U \cap V \in \{U \in \mathcal{T} \mid x \in U\}$ we have

$$W \subseteq U \cap V$$

proving by [definition: 14.293] that $\mathfrak{U}(x)$ is a filter base in X . $\square$

Theorem 14.298. Let X be a set and let $\mathfrak{A} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ and $\mathfrak{B} = \{B_\alpha\}_{\alpha \in J} \subseteq \mathcal{P}(X)$ filter bases then we have:

1. $\mathfrak{A} \cup \mathfrak{B}$ is a filter base in X where $\mathfrak{A} \cup \mathfrak{B} = \{A_\alpha \cup B_\beta\}_{(\alpha, \beta) \in I \times J}$
2. If $\forall (\alpha, \beta) \in I \times J$ $A_\alpha \cap B_\beta \neq \emptyset$ then $\mathfrak{A} \cap \mathfrak{B}$ is a filter base where $\mathfrak{A} \cap \mathfrak{B} = \{A_\alpha \cap B_\beta\}_{(\alpha, \beta) \in I \times J}$

Proof. First as $I \neq \emptyset$ and $J \neq \emptyset$ we have that $I \times J \neq \emptyset$, further we have:

$\mathfrak{A} \cup \mathfrak{B}$ is a filter base. Because

1. $\forall (\alpha, \beta) \in I \times J$ we have $A_\alpha \neq \emptyset \wedge B_\beta \neq \emptyset$ so that $A_\alpha \cup B_\beta \neq \emptyset$
2. $\forall (\alpha_1, \beta_1), (\alpha_2, \beta_2) \in I \times J$ there exists a $\gamma_1 \in I$ and a $\gamma_2 \in J$ such that $A_{\gamma_1} \subseteq A_{\alpha_1} \cap A_{\alpha_2}$ and $B_{\gamma_2} \subseteq B_{\beta_1} \cap B_{\beta_2}$, hence for $(\gamma_1, \gamma_2) \in I \times J$ we have

$$A_{\gamma_1} \cup B_{\gamma_2} \subseteq (A_{\alpha_1} \cap A_{\alpha_2}) \cup (B_{\beta_1} \cap B_{\beta_2}) \quad (14.182)$$

Further

$$\begin{aligned} A_{\alpha_1} \cap A_{\alpha_2} &\subseteq A_{\alpha_1} \subseteq A_{\alpha_1} \cup B_{\beta_1} \\ A_{\alpha_1} \cap A_{\alpha_2} &\subseteq A_{\alpha_2} \subseteq A_{\alpha_2} \cup B_{\beta_2} \end{aligned}$$

so that

$$A_{\alpha_1} \cap A_{\alpha_2} \subseteq (A_{\alpha_1} \cup B_{\beta_1}) \cap (A_{\alpha_2} \cup B_{\beta_2}) \quad (14.183)$$

and

$$\begin{aligned} B_{\beta_1} \cap B_{\beta_2} &\subseteq B_{\beta_1} \subseteq A_{\alpha_1} \cup B_{\beta_1} \\ B_{\beta_1} \cap B_{\beta_2} &\subseteq B_{\beta_2} \subseteq A_{\alpha_2} \cup B_{\beta_2} \end{aligned}$$

so that

$$B_{\beta_1} \cap B_{\beta_2} \subseteq (A_{\alpha_1} \cup B_{\beta_1}) \cap (A_{\alpha_2} \cup B_{\beta_2}) \quad (14.184)$$

Combining [eqs: 14.182, 14.183 and 14.184] gives finally

$$A_{\gamma_1} \cup B_{\gamma_2} \subseteq (A_{\alpha_1} \cup B_{\beta_1}) \cap (A_{\alpha_2} \cup B_{\beta_2})$$

$\mathfrak{A} \cap \mathfrak{B}$ is a filter base. Because

1. $\forall (\alpha, \beta) \in I \times J$ we have $A_\alpha \cap B_\beta \neq \emptyset$ by the definition of $\mathfrak{A} \cap \mathfrak{B}$
2. $\forall (\alpha_1, \beta_1), (\alpha_2, \beta_2) \in I \times J$ there exists a $\gamma_1 \in I$ and a $\gamma_2 \in J$ such that $A_{\gamma_1} \subseteq A_{\alpha_1} \cap A_{\alpha_2}$ and $B_{\gamma_2} \subseteq B_{\beta_1} \cap B_{\beta_2}$, hence for $(\gamma_1, \gamma_2) \in I \times J$ we have

$$A_{\gamma_1} \cap B_{\gamma_2} \subseteq (A_{\alpha_1} \cap A_{\alpha_2}) \cap (B_{\beta_1} \cap B_{\beta_2}) = (A_{\alpha_1} \cap B_{\beta_1}) \cap (A_{\alpha_2} \cap B_{\beta_2}) \quad \square$$

Theorem 14.299. Let X be a set and let $\mathfrak{A} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base in X then it has the **finite intersection property**, meaning that every **finite** $J \subseteq I$ we have $\bigcap_{\alpha \in J} A_\alpha \neq \emptyset$

Proof. Using [theorem: 14.294] there exist a $\gamma \in I$ such that $A_\gamma \subseteq \bigcap_{\alpha \in J} A_\alpha$, as by the definition of a filter base $A_\gamma \neq \emptyset$ it follows that $\bigcap_{\alpha \in J} A_\alpha \neq \emptyset$. $\square$

Definition 14.300. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $x \in X$ and $\mathfrak{A} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base in X then we say that:

1. $\mathfrak{A}$ **converges to** x noted as $\mathfrak{A} \rightarrow x$ if $\forall U \in \mathcal{T}$ with $x \in U$ there exist a $\alpha \in I$ such that $A_\alpha \subseteq U$
2. $\mathfrak{A}$ **accumulates at** x noted as $\mathfrak{A} \succ x$ if $\forall U \in \mathcal{T}$ with $x \in U$ we have that $\forall \alpha \in I$ $A_\alpha \cap U \neq \emptyset$

Theorem 14.301. Let $\langle X, \mathcal{T} \rangle$ be a topological space then

$\mathcal{T}$ is Hausdorff $\Leftrightarrow$ each filter base that converges converges to exactly one point

Proof.

$\Rightarrow$. Assume that $\mathcal{T}$ is Hausdorff and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ be such that for $x, y \in X$ $\mathfrak{U} \rightarrow x$ and $\mathfrak{U} \rightarrow y$. If $x \neq y$ then as $\mathcal{T}$ is Hausdorff there exist $U, V \in \mathcal{T}$ such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$. From $\mathfrak{U} \rightarrow x$ and $\mathfrak{U} \rightarrow y$ it follows that there exists $\alpha, \beta \in I$ such that $A_\alpha \subseteq U$ and $A_\beta \subseteq V$. Then there exist a $\gamma \in I$ such that $\emptyset \neq A_\gamma \subseteq A_\alpha \cap A_\beta \subseteq U \cap V$, hence $\emptyset \neq U \cap V$ contradicting $U \cap V = \emptyset$. So we must have that $x = y$ proving uniqueness.

$\Leftarrow$. Assume that $\mathcal{T}$ is not Hausdorff then $\exists x, y \in X$ with $x \neq y$ such that $\forall U, V \in \mathcal{T}$ with $x \in U \wedge y \in V$ we have $U \cap V \neq \emptyset$. We construct now a filter base that converges to x and y . By [theorem: 14.298] and [definition: 14.296] we have that $\mathfrak{W} = \mathfrak{U}(x) \cap \mathfrak{U}(y)$ is a filter base in X where

$$\mathfrak{U}(x) = \{U\}_{U \in \{U \in \mathcal{T} | x \in U\}} \text{ and } \mathfrak{U}(y) = \{U\}_{U \in \{U \in \mathcal{T} | y \in U\}}$$

If $U \in \mathcal{T}$ with $x \in U$ then as $y \in X \in \mathcal{T}$ we have that $(U, X) \in \{U \in \mathcal{T} | x \in U\} \times \{V \in \mathcal{T} | y \in V\}$ so that $U \cap X = U \subseteq U$ proving that

$$\mathfrak{W} \rightarrow x$$

Further if $V \in \mathcal{T}$ with $y \in V$ then as $x \in X \in \mathcal{T}$ we have that $(X, V) \in \{U \in \mathcal{T} | x \in U\} \times \{V \in \mathcal{T} | y \in V\}$ so that $X \cap V = V \subseteq V$ proving that

$$\mathfrak{W} \rightarrow y$$

By the hypothesis it follows then that $x = y$ contradicting $x \neq y$. So the assumption that $\mathcal{T}$ is not Hausdorff is wrong proving that $\mathcal{T}$ is Hausdorff. $\square$

Theorem 14.302. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $x \in X$ and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base in X then

$$\mathfrak{U} \succ x \Leftrightarrow x \in \bigcap_{\alpha \in I} \overline{A_\alpha}$$

Proof.

$\Rightarrow$. If $\mathfrak{U} \succ x$ then $\forall U \in \mathcal{T}$ with $x \in U$ we have $\forall \alpha \in I$ $A_\alpha \cap U \neq \emptyset$, so that by [theorem: 14.30] $\forall \alpha \in I$ $x \in \overline{A_\alpha}$ hence $x \in \bigcap_{\alpha \in I} \overline{A_\alpha}$.

$\Leftarrow$. Let $U \in \mathcal{T}$ with $x \in U$ then as $x \in \bigcap_{\alpha \in I} \overline{A_\alpha}$ we have that $\forall \alpha \in I$ $x \in \overline{A_\alpha}$ so that by [theorem: 14.30] $A_\alpha \cap U \neq \emptyset$. Hence $\mathfrak{U} \succ x$. $\square$

Example 14.303. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $x \in X$ then $\mathfrak{U}(x) \rightarrow x$

Proof. By [definition: 14.296] $\mathfrak{U}(x) = \{U\}_{U \in \{U \in \mathcal{T} | x \in U\}}$. So if $U \in \mathcal{T}$ with $x \in U$ then $U \in \{U \in \mathcal{T} | x \in U\}$ and $U \subseteq U$ proving that $\mathfrak{U}(x) \rightarrow x$. $\square$

Definition 14.304. Let X be a set and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$, $\mathfrak{W} = \{B_\beta\}_{\beta \in J} \subseteq \mathcal{P}(X)$ two filter bases in X then $\mathfrak{W}$ is **subordinate** to $\mathfrak{U}$ noted as $\mathfrak{W} \gg \mathfrak{U}$ or $\mathfrak{U} \ll \mathfrak{W}$ iff

$$\forall \alpha \in I \text{ we have that } \exists \beta \in J \text{ such that } B_\beta \subseteq A_\alpha$$

Definition 14.305. Let X be a set and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$, $\mathfrak{W} = \{B_\beta\}_{\beta \in J} \subseteq \mathcal{P}(X)$ two filter bases in X then $\mathfrak{U} \sqsubseteq \mathfrak{W}$ iff

$$\forall \alpha \in I \text{ there exist a } \beta \in J \text{ such that } A_\alpha = B_\beta$$

Theorem 14.306. Let X be a set then we have:

1. If $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$, $\mathfrak{W} = \{B_\beta\}_{\beta \in J} \subseteq \mathcal{P}(X)$ are two filter bases in X such that $\mathfrak{U} \sqsubseteq \mathfrak{W}$ then $\mathfrak{W} \gg \mathfrak{U}$
2. If $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$, $\mathfrak{W} = \{B_\beta\}_{\beta \in J} \subseteq \mathcal{P}(X)$ are two filter bases in X such that $\mathfrak{W} \gg \mathfrak{U}$ then $\forall \beta \in J$ and $\forall \alpha \in I$ we have that $A_\alpha \cap B_\beta \neq \emptyset$

3. If $\langle X, \mathcal{T} \rangle$ is a topological space, $x \in X$ and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base in X

$$\mathfrak{U} \rightarrow x \Leftrightarrow \mathfrak{U} \geq \mathfrak{U}(x)$$

4. If $\mathfrak{U}_1 = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$, $\mathfrak{U}_2 = \{B_\alpha\}_{\alpha \in J} \subseteq \mathcal{P}(X)$ and $\mathfrak{U}_3 = \{C_\alpha\}_{\alpha \in K} \subseteq \mathcal{P}(X)$ are filter bases in X such that $\mathfrak{U}_1 \gg \mathfrak{U}_2$ and $\mathfrak{U}_2 \gg \mathfrak{U}_3$ then $\mathfrak{U}_1 \gg \mathfrak{U}_3$

5. If $\mathfrak{U}$ is a filter base in X then $\mathfrak{U} \gg \mathfrak{U}$

Proof.

1. Let $\alpha \in I$ then as $\mathfrak{U} \subseteq \mathfrak{W}$ there exist a $\beta \in J$ such that $A_\alpha = B_\beta$ so that $B_\beta \subseteq A_\alpha$ proving that $\mathfrak{W} \gg \mathfrak{U}$.

2. We proceed by contradiction. So assume that $\exists \beta \in J$ and $\exists \alpha \in I$ such that $A_\alpha \cap B_\beta = \emptyset$. As $\mathfrak{W} \gg \mathfrak{U}$ there exist a $\gamma \in J$ such that $B_\gamma \subseteq A_\alpha$ and by the definition of a filter base there exist a $\delta \in J$ such that $\emptyset \neq B_\delta \subseteq B_\gamma \cap B_\beta \subseteq A_\alpha \cap B_\beta$ proving that $A_\alpha \cap B_\beta \neq \emptyset$ a contradiction. Hence the assumption is wrong and we have $\forall \beta \in J$ and $\forall \alpha \in I$ $A_\alpha \cap B_\beta \neq \emptyset$.

3. By definition $\mathfrak{U}(x) = \{U\}_{U \in \{U \in \mathcal{T} | x \in U\}} \subseteq \mathcal{P}(X)$ and

$$\begin{aligned} \mathfrak{U} \rightarrow x &\Leftrightarrow \forall U \in \mathcal{T} \text{ with } x \in U \exists \alpha \in I \text{ such that } A_\alpha \subseteq U \\ &\Leftrightarrow \forall U \in \{U \in \mathcal{T} | x \in U\} \exists \alpha \in I \text{ such that } A_\alpha \subseteq U \\ &\Leftrightarrow \mathfrak{U} \gg \mathfrak{U}(x) \end{aligned}$$

4. Let $\gamma \in K$ then as $\mathfrak{U}_2 \gg \mathfrak{U}_3$ there exist a $\beta \in J$ such that $B_\beta \subseteq C_\gamma$, as $\mathfrak{U}_1 \gg \mathfrak{U}_2$ there exist a $\alpha \in I$ such that $A_\alpha \subseteq B_\beta$. Hence $A_\alpha \subseteq C_\gamma$ proving that $\mathfrak{U}_1 \gg \mathfrak{U}_3$.

5. If $\mathfrak{U} = \{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ is a filter base in X then $\forall i \in I$ we have $A_i = A_i \Rightarrow A_i \subseteq A_i$, hence $\mathfrak{U} \gg \mathfrak{U}$. $\square$

Theorem 14.307. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$, $\mathfrak{W} = \{B_\alpha\}_{\alpha \in J} \subseteq \mathcal{P}(X)$ be filter bases in X then we have for $x, y \in X$ that

1. If $\mathfrak{U} \rightarrow x$ then $\mathfrak{U} \succ x$
2. If X is Hausdorff, $\mathfrak{U} \rightarrow x$ and $\mathfrak{U} \succ y$ then $x = y$
3. If $\mathfrak{W} \gg \mathfrak{U}$ then we have:

- a. If $\mathfrak{U} \rightarrow x$ then $\mathfrak{W} \rightarrow x$
- b. If $\mathfrak{W} \succ x$ then $\mathfrak{U} \succ x$

Proof.

1. Let $U \in \mathcal{T}$ with $x \in U$ then as $\mathfrak{U} \rightarrow x$ there exist a $\alpha \in I$ such that $A_\alpha \subseteq U$. Given $\beta \in I$ then as there exist a $\gamma \in I$ with $\emptyset \neq A_\gamma \subseteq A_\alpha \cap A_\beta \subseteq U \cap A_\beta \Rightarrow U \cap A_\beta \neq \emptyset$ proving that $\mathfrak{U} \succ x$.

2. Assume that $x \neq y$ then as X is Hausdorff there exists $U, V \in \mathcal{T}$ with $x \in U \wedge y \in V$ such that $U \cap V = \emptyset$. As $\mathfrak{U} \rightarrow x$ there exist a $\alpha \in I$ such that $A_\alpha \subseteq U$ so that

$$A_\alpha \cap V \subseteq U \cap V = \emptyset$$

As $\mathfrak{U} \rightarrow y$ we have that $A_\alpha \cap V \neq \emptyset$ contradicting the above hence we must have $x = y$.

3. Let $\mathfrak{W} \gg \mathfrak{U}$ then we have:

- a. Let $U \in \mathcal{T}$ with $x \in U$ then as $\mathfrak{U} \rightarrow x$ there exist a $\alpha \in I$ such that $A_\alpha \subseteq U$. As $\mathfrak{W} \gg \mathfrak{U}$ there exist a $\beta \in J$ such that $B_\beta \subseteq A_\alpha \subseteq U$ proving that $\mathfrak{W} \rightarrow x$
- b. Let $U \in \mathcal{T}$ with $x \in U$ then as $\mathfrak{W} \succ x$ we have $\forall \beta \in J$ $U \cap B_\beta \neq \emptyset$. Let $\alpha \in I$ then as $\mathfrak{W} \gg \mathfrak{U}$ there exist a $\gamma \in J$ such that $B_\gamma \subseteq A_\alpha$ hence $\emptyset \neq U \cap B_\gamma \subseteq U \cap A_\alpha$ proving that $\mathfrak{U} \succ x$. $\square$

Definition 14.308. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $A \subseteq X$ and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base then we say that $\mathfrak{U}$ is **on** A iff $\forall \alpha \in I$ we have $A_\alpha \subseteq A$.

Theorem 14.309. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $A \subseteq X$ then

$$x \in \bar{A} \Leftrightarrow \text{there exist a filter base } \mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X) \text{ on } A \text{ with } \mathfrak{U} \rightarrow x$$

Proof.

$\Rightarrow$. Let $x \in \bar{A}$ then by [theorem: 14.30] $\forall U \in \mathcal{T}$ with $x \in U$ we have $A \cap U \neq \emptyset$. As $\emptyset \neq A \cap U \subseteq A \Rightarrow A \neq \emptyset$ we have using [example: 14.295] that $\mathfrak{U} = \{A_\alpha\}_{\alpha \in \{A\}}$ where $A_\alpha = A$ is a filter base in X , further by [theorem: 14.297] $\mathfrak{U}(x) = \{U\}_{U \in \{U \in \mathcal{T} | x \in U\}}$ is also a filter base in X . Define $\mathfrak{W} = \{A_\alpha \cap U\}_{(\alpha, U) \in \{A\} \times \{U \in \mathcal{T} | x \in U\}}$. As $\forall (\alpha, U) \in \{A\} \times \{U \in \mathcal{T} | x \in U\}$ we have that $\emptyset \neq A \cap U \subseteq U$ it follows from [theorem: 14.298] that

$$\mathfrak{W} \text{ is a filter base on } A$$

Further as $\forall U \in \mathcal{T}$ with $x \in U$ we have that for $(A, U) \in \{A\} \times \{U \in \mathcal{T} | x \in U\}$ $A \cap U \subseteq U$ it follows that

$$\mathfrak{W} \rightarrow x$$

$\Leftarrow$. By the hypothesis there exist a filter base $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ on A with $\mathfrak{U} \rightarrow x$. Then if $U \in \mathcal{T}$ with $x \in U$ there exist a $\alpha \in I$ such that $A_\alpha \subseteq U$ [because $\mathfrak{U} \rightarrow x$], further, as $\mathfrak{U}$ is a filter base on A , $A_\alpha \subseteq A$. So it follows that $\emptyset \neq A_\alpha = A_\alpha \cap A \subseteq U \cap A$ which by [theorem: 14.30] proves that $x \in \bar{A}$. $\square$

Theorem 14.310. Let $f: X \rightarrow Y$ be a function between sets X and Y . If $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ is a filter base in X then if we define $f[\mathfrak{U}]$ by

$$f[\mathfrak{U}] = \{f(A_\alpha)\}_{\alpha \in I} \subseteq \mathcal{P}(Y)$$

we have that

$$f[\mathfrak{U}] \text{ is a filter base in } Y$$

Proof. As $\mathfrak{U}$ is a filter base we have $I \neq \emptyset$ and

1. $\forall \alpha \in I$ $A_\alpha \neq \emptyset$ so that

$$f(A_\alpha) \neq \emptyset$$

2. If $\alpha, \beta \in I$ then there exist a $\gamma \in I$ such that $A_\gamma \subseteq A_\alpha \cap A_\beta$ so that using [theorem: 2.103]

$$f(A_\gamma) \subseteq f(A_\alpha \cap A_\beta) \subseteq f(A_\alpha) \cap f(A_\beta)$$

proving that $f[\mathfrak{U}] = \{f(A_\alpha)\}_{\alpha \in I} \subseteq \mathcal{P}(Y)$ is a filter base of Y . $\square$

Filter bases allows us to have a equivalent definition of a continuous function.

Theorem 14.311. Let $\langle X, \mathcal{T} \rangle, \langle Y, \mathcal{S} \rangle$ be topological spaces and $f: X \rightarrow Y$ is a function then

$$f \text{ is continuous at } x \Leftrightarrow f[\mathfrak{U}(x)] \rightarrow f(x)$$

where

$$f[\mathfrak{U}(x)] = \{f(U)\}_{U \in \{U \in \mathcal{T} | x \in U\}} \text{ is a filter base by [theorem: 14.310]}$$

Proof. First note that $\mathfrak{U}(x) = \{U\}_{U \in \{U \in \mathcal{T} | x \in U\}} \subseteq \mathcal{P}(X)$ so that

$$f[\mathfrak{U}(x)] = \{f(U)\}_{U \in \{U \in \mathcal{T} | x \in U\}}$$

$\Rightarrow$. Let $V \in \mathcal{S}$ such that $f(x) \in V$ then as f is continuous at x there exist a $U \in \mathcal{T}$ with $x \in U$ such that $f(U) \subseteq V$. In other words $\exists U \in \{U \in \mathcal{T} | x \in U\}$ such that $f(U) \subseteq V$ which by [definition: 14.300] proves that

$$f[\mathfrak{U}(x)] \rightarrow f(x)$$

$\Leftarrow$. Let $V \in \mathcal{S}$ with $f(x) \in V$ then as $f[\mathfrak{U}(x)] \rightarrow f(x)$ there exist a $U \in \{U \in \mathcal{T} | x \in U\}$ such that

$$f(U) \subseteq V$$

proving that f is continuous at x . $\square$

Theorem 14.312. Let $\langle X, \mathcal{T} \rangle, \langle Y, \mathcal{S} \rangle$ be topological spaces and $f: X \rightarrow Y$ a function then

f is continuous on $X \Leftrightarrow \forall x \in X$ and for every filter base $\mathfrak{U}$ in X with $\mathfrak{U} \rightarrow x$ we have $f[\mathfrak{U}] \rightarrow f(x)$

Proof.

$\Rightarrow$. Let $x \in X$ and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ such that $\mathfrak{U} \rightarrow x$ then by [theorem: 14.306] $\mathfrak{U} \gg \mathfrak{U}(x)$. So if $U \in \{U \in \mathcal{T} | x \in U\}$ by [definition: 14.304] there exist a $\alpha \in I$ such that $A_\alpha \subseteq U$, hence $f(A_\alpha) \subseteq f(U)$. So by [definition: 14.304]

$$f[\mathfrak{U}] \gg f[\mathfrak{U}(x)] \quad (14.185)$$

As f is continuous, hence continuous at x , we have by the previous theorem [theorem: 14.311] that

$$f[\mathfrak{U}(x)] \rightarrow f(x) \quad (14.186)$$

Using [theorem: 14.307] together with [eqs: 14.185, 14.186] proves that

$$f[\mathfrak{U}] \rightarrow f(x) \quad (14.187)$$

$\Leftarrow$. Take $A \subseteq X$ and let $y \in f(\bar{A})$ then $\exists x \in \bar{A}$ such that $y = f(x)$. Using [theorem: 14.309] there exist a filter base $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ on A such that $\mathfrak{U} \rightarrow x$. As $\mathfrak{U}$ is a filter base on A we have that $\forall \alpha \in I$ $A_\alpha \subseteq A$ so that $f(A_\alpha) \subseteq f(A)$ proving that $f[\mathfrak{U}]$ is on $f(A)$. This together with $f[\mathfrak{U}] \rightarrow f(x)$ [by the hypothesis] gives by [theorem: 14.309] that $y = f(x) \in \overline{f(A)}$ proving that

$$f(\bar{A}) \subseteq \overline{f(A)}$$

Finally using [theorem: 14.154] on the above proves that f is continuous. $\square$

Theorem 14.313. Let $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces, $\langle X, \mathcal{T}_{\text{product}} \rangle$ the topological space defined by $X = \prod_{i \in I} X_i$ together with the product topology $\mathcal{T}_{\text{product}}$ [see theorem: 14.47], $x \in X$ and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in J} \subseteq \prod_{i \in I} X_i$ a filter base in X and $x \in X$ then

$\mathfrak{U} \rightarrow x \Leftrightarrow \forall i \in I$ we have for $\pi_i: X \rightarrow X_i$ the projection function that $\pi_i[\mathfrak{U}] \rightarrow \pi_i(x) = x_i$

Proof.

$\Rightarrow$. Let $i \in I$ as $\mathfrak{U} \rightarrow x$ we have, as $\pi_i: \prod_{i \in I} X_i \rightarrow X_i$ is continuous [see theorem: 14.160], by the previous theorem [theorem: 14.312] that $\pi_i[\mathfrak{U}] \rightarrow \pi_i(x) = x_i$.

$\Leftarrow$. Let $x \in \prod_{i \in I} X_i$ and $U \in \mathcal{T}_{\text{product}}$ such that $x \in U$. By the definition of the product topology [see theorem: 14.47] there exist a finite J and $\{S_i\}_{i \in J} \subseteq \{\pi_i^{-1}(V) | i \in I \wedge V \in \mathcal{T}_i\}$ such that

$$x \in \bigcap_{j \in J} S_j \subseteq U \quad (14.188)$$

Let $j \in J$ then $\exists i_j \in I$ and a $U_j \in \mathcal{T}_{i_j}$ such that $S_j = \pi_{i_j}^{-1}(U_j)$, so $\pi_{i_j}(S_j) \subseteq U_j$, hence $\pi_{i_j}(x) \in \pi_{i_j}(S_j) \subseteq U_j \in \mathcal{T}_{i_j}$. By the hypothesis $\pi_{i_j}[\mathfrak{U}] \rightarrow \pi_{i_j}(x)$ so there exist a $\alpha_j \in J$ such that $\pi_{i_j}(A_{\alpha_j}) \subseteq U_j$ hence $A_{\alpha_j} \subseteq \pi_{i_j}^{-1}(U_j) = S_j$. So $\bigcap_{j \in J} A_{\alpha_j} \subseteq \bigcap_{j \in J} S_j \subseteq U$. As J is finite it follows from [theorem: 14.294] that $\exists \gamma \in I$ such that

$$A_\gamma \subseteq \bigcap_{j \in J} A_{\alpha_j} \subseteq U$$

proving that

$$\mathfrak{U} \rightarrow x \quad \square$$

Definition 14.314. Let X a set then a filter base $\mathfrak{U}$ in X is called **maximal** or a **ultra filter** if for every filter base $\mathfrak{W}$ in X with $\mathfrak{W} \gg \mathfrak{U}$ we have that $\mathfrak{U} \gg \mathfrak{W}$.

Theorem 14.315. Let X a set then and $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base in X then

$\mathfrak{U}$ is maximal

$\Updownarrow$

$\forall A \subseteq X$ we have $(\exists \beta \in I \text{ with } A_\beta \subseteq A \wedge \forall \alpha \in I A_\alpha \not\subseteq X \setminus A) \vee (\forall \alpha \in I A_\alpha \not\subseteq A \wedge \exists \beta \in I \text{ with } A_\beta \subseteq X \setminus A)$

In other words

$\mathfrak{U}$ is maximal $\Leftrightarrow \forall A \subseteq X$ only one of the sets $A, X \setminus A$ contains a member of $\mathfrak{U}$

Proof.

$\Rightarrow$. Let $A \subseteq X$ and assume that there exist $\alpha, \beta \in I$ such that $A_\alpha \subseteq A$ and $A_\beta \subseteq X \setminus A$ then $A_\alpha \cap A_\beta \subseteq A \cap (X \setminus A) = \emptyset$ proving that $A_\alpha \cap A_\beta = \emptyset$. By definition of a filter base there exist a $\gamma \in I$ such that $\emptyset \neq A_\gamma \subseteq A_\alpha \cap A_\beta = \emptyset$ leading to the contradiction $\emptyset \neq \emptyset$. Hence

$$\neg(\exists \alpha, \beta \in I \text{ such that } A_\alpha \subseteq A \wedge A_\beta \subseteq X \setminus A) \quad (14.189)$$

We have now two possibilities to consider for $A \subseteq X$:

$\forall \alpha \in I A_\alpha \not\subseteq A$. Then $\forall \alpha \in I$ there exist a $x \in A_\alpha$ so that $x \notin A$, hence $(X \setminus A) \cap A_\alpha \neq \emptyset$. Further if $\alpha, \beta \in I$ then as $\mathfrak{U}$ is a filter base there exist a $\gamma \in I$ such that $A_\gamma \subseteq A_\alpha \cap A_\beta$ so that $(X \setminus A) \cap A_\gamma \subseteq (X \setminus A) \cap (A_\alpha \cap A_\beta) = ((X \setminus A) \cap A_\alpha) \cap ((X \setminus A) \cap A_\beta)$. Hence we have that

$$\mathfrak{W} = \{(X \setminus A) \cap A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X) \text{ is a filter base in } X$$

Further as $\forall \alpha \in I (X \setminus A) \cap A_\alpha \subseteq A_\alpha$ we have by [definition: 14.304] that $\mathfrak{W} \gg \mathfrak{U}$, as $\mathfrak{U}$ is maximal it follows that $\mathfrak{U} \gg \mathfrak{W}$. Hence $\forall \alpha \in I$ there exist a $\beta \in I$ such that $A_\beta \subseteq (X \setminus A) \cap A_\alpha \subseteq X \setminus A$. So in this case we have proved that

$$\forall \alpha \in I A_\alpha \not\subseteq A \wedge \exists \beta \in I \text{ with } A_\beta \subseteq X \setminus A$$

$\exists \beta \in I$ with $A_\beta \subseteq A$. Then by [eq: 14.189] we must have that $\forall \alpha \in I$ that $A_\alpha \not\subseteq X \setminus A$ proving that in this case

$$\exists \beta \in I \text{ with } A_\beta \subseteq A \wedge \forall \alpha \in I A_\alpha \not\subseteq X \setminus A$$

$\Leftarrow$. Let $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ be a filter bases such that

$$\forall A \subseteq X \text{ one and only one of the two sets } A, X \setminus A \text{ contains a member of } \mathfrak{U} \quad (14.190)$$

Let then $\mathfrak{W} = \{B_\beta\}_{\beta \in J}$ such that $\mathfrak{W} \gg \mathfrak{U}$. If $\beta \in J$ then for B_β we have by [eq: 14.190] that there exist a $\alpha \in I$ such that either $A_\alpha \subseteq B_\beta$ or $A_\alpha \subseteq X \setminus B_\beta$. The last case would mean that $A_\alpha \cap B_\beta = \emptyset$ contradicting [theorem: 14.306 (2)], so we must have that $A_\alpha \subseteq B_\beta$ proving that $\mathfrak{U} \gg \mathfrak{W}$ and thus maximality of $\mathfrak{U}$. $\square$

Theorem 14.316. Let X be a set, $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ a filter base in X then there exist a maximal filter base $\mathfrak{W}$ in X such that $\mathfrak{W} \gg \mathfrak{U}$

Proof. We use a extension of Zorn's lemma [see theorem: 3.136] in this proof. To use this we must first define a pre-order [see definition: 3.25]. Let

$$\mathcal{A} = \{\mathfrak{W} | \mathfrak{W} \text{ is a filter base in } X \text{ with } \mathfrak{W} \gg \mathfrak{U}\}$$

Using [theorem: 14.306] we have that $\mathfrak{U} \gg \mathfrak{U}$ so that

$$\mathfrak{U} \in \mathcal{A} \text{ hence } \mathcal{A} \neq \emptyset$$

Define now the following relation $\leq$ on $\mathcal{A}$

$$\leq \subseteq \mathcal{A} \times \mathcal{A} \text{ by } \mathfrak{V} \leq \mathfrak{W} \text{ iff } \mathfrak{W} \gg \mathfrak{V}$$

then we have:

reflexivity. Let $\mathfrak{W} \in \mathcal{A}$ then by then by [theorem: 14.306] we have $\mathfrak{W} \leq \mathfrak{W}$.

transitivity. Assume that $\mathfrak{W}_1 \leq \mathfrak{W}_2 \wedge \mathfrak{W}_2 \leq \mathfrak{W}_3$ then $\mathfrak{W}_2 \gg \mathfrak{W}_1 \wedge \mathfrak{W}_3 \gg \mathfrak{W}_2$ hence by [theorem: 14.306] we have $\mathfrak{W}_3 \gg \mathfrak{W}_1$ so that $\mathfrak{W}_1 \leq \mathfrak{W}_3$.

Hence we have that

$$\langle \mathcal{A}, \leq \rangle \text{ is a pre-ordered set}$$

Let $\mathcal{C} \subseteq \mathcal{A}$ be any **chain** in $\langle \mathcal{A}, \leq \rangle$ then we must consider two cases for $\mathcal{C}$

$\mathcal{C} = \emptyset$. Then we have $\forall \mathfrak{C} \in \mathcal{C}$ that $\mathfrak{C} \leq \mathfrak{U}$ is satisfied vacuously proving that $\mathfrak{U}$ is an upper bound of $\mathcal{C}$.

$\mathcal{C} \neq \emptyset$. Given $\mathfrak{C} = \{C_\alpha\}_{\alpha \in J} \in \mathcal{C}$ define $S_{\mathfrak{C}} = \{C_\alpha | \alpha \in J\} \neq \emptyset$ [as $J \neq \emptyset$] and define

$$\mathbb{B} = \bigcup_{\mathfrak{C} \in \mathcal{C}} S_{\mathfrak{C}} \neq \emptyset \text{ [as } \mathcal{C} \neq \emptyset \text{]} \text{ and } \mathfrak{B} = \{B\}_{B \in \mathbb{B}} \subseteq \mathcal{P}(X) \quad (14.191)$$

then we have

$$\forall \mathfrak{C} = \{C_\alpha\}_{\alpha \in J} \in \mathcal{C} \text{ we have } \forall \alpha \in J \text{ that } C_\alpha \in \mathbb{B} \quad (14.192)$$

Next we have

1. $\forall B \in \mathbb{B}$ there exist a $\mathfrak{C} = \{C_\alpha\}_{\alpha \in J} \in \mathcal{C}$ such that $B \in S_{\mathfrak{C}}$, hence $\exists \alpha \in J$ such that $B = C_\alpha \neq \emptyset$ [as $\mathfrak{C}$ is a filter base]. Hence we have

$$\forall B \in \mathbb{B} \text{ that } B \neq \emptyset \quad (14.193)$$

2. Let $B_1, B_2 \in \mathbb{B}$ then there exists $\mathfrak{C}_1 = \{C_\alpha^1\}_{\alpha \in J_1} \in \mathcal{C}$, $\mathfrak{C}_2 = \{C_\alpha^2\}_{\alpha \in J_2} \in \mathcal{C}$ such that $\exists \gamma_1 \in J_1$, $\exists \gamma_2 \in J_2$ with $B_1 = C_{\gamma_1}^1$, $B_2 = C_{\gamma_2}^2$. Now as $\mathcal{C}$ is a chain we have the following possible cases:

$\mathfrak{C}_1 \leq \mathfrak{C}_2$. Then $\mathfrak{C}_2 \gg \mathfrak{C}_1$, hence there exist a $\gamma \in J_2$ such that $C_\gamma^2 \subseteq C_{\gamma_1}^1 = B_1$. As $\mathfrak{C}_2$ is a filter base there exist a $\beta \in J_2$ such that $C_\beta^2 \subseteq C_\gamma^2 \cap C_{\gamma_2}^2 \subseteq B_1 \cap B_2$. Take $B_3 = C_\beta^2 \in \mathbb{B}$ then $B_3 \subseteq B_1 \cap B_2$.

$\mathfrak{C}_2 \leq \mathfrak{C}_1$. Then $\mathfrak{C}_1 \gg \mathfrak{C}_2$, hence there exist a $\gamma \in J_1$ such that $C_\gamma^1 \subseteq C_{\gamma_2}^2 = B_2$. As $\mathfrak{C}_1$ is a filter base there exist a $\beta \in J_1$ such that $C_\beta^1 \subseteq C_\gamma^1 \cap C_{\gamma_1}^1 \subseteq B_1 \cap B_2$. Take $B_3 = C_\beta^1 \in \mathbb{B}$ then $B_3 \subseteq B_1 \cap B_2$.

So in all cases we have

$$\forall B_1, B_2 \in \mathbb{B} \text{ there exist a } B_3 \in \mathbb{B} \text{ such that } B_3 \subseteq B_1 \cap B_2 \quad (14.194)$$

Combining [eqs: 14.191, 14.193 and 14.194] proves that

$\mathfrak{B}$ is a filter base

As $\mathcal{C} \neq \emptyset$ there exist a $\mathfrak{C} = \{C_\alpha\}_{\alpha \in J} \in \mathcal{C} \subseteq \mathcal{A}$ so that $\mathfrak{C} \gg \mathfrak{U}$, hence if $\alpha \in I$ then there exist a $\gamma \in J$ such that $C_\gamma \subseteq A_\alpha$. By [eq: 14.192] we have that $C_\gamma \in \mathbb{B}$ proving that $\mathfrak{B} \gg \mathfrak{U}$, hence

$$\mathfrak{B} \in \mathcal{A}$$

Further if $\mathfrak{D} = \{D_\alpha\}_{\alpha \in K} \in \mathcal{C}$ then $\forall \alpha \in K$ $D_\alpha \subseteq D_\alpha$ hence, as by [eq: 14.192] $D_\alpha \in \mathbb{B}$, we have that $\mathfrak{B} \gg \mathfrak{D}$ proving that $\mathfrak{D} \leq \mathfrak{B}$. Hence

$\mathfrak{B}$ is an upper bound of $\mathcal{C}$

So in all cases we have that $\mathcal{C}$ has an upper bound. In other words we have proved that every chain in $\mathcal{A}$ has an upper bound. So using a variant of Zorn's lemma [theorem: 3.136] there exist a $\mathfrak{M} \in \mathcal{A}$ such that $\forall \mathfrak{N} \in \mathcal{A}$ with $\mathfrak{M} \leq \mathfrak{N}$ we have $\mathfrak{N} \leq \mathfrak{M}$. Hence

$$\forall \mathfrak{N} \in \mathcal{A} \text{ with } \mathfrak{N} \gg \mathfrak{M} \text{ we have } \mathfrak{M} \gg \mathfrak{N} \text{ and as } \mathfrak{M} \in \mathcal{A} \text{ we have also } \mathfrak{M} \gg \mathfrak{U} \quad (14.195)$$

Now if $\mathfrak{D}$ is a filter base in X with $\mathfrak{D} \gg \mathfrak{W}$ we have as $\mathfrak{W} \geq \mathfrak{U}$ by [theorem: 14.306] that $\mathfrak{D} \gg \mathfrak{U}$ so that $\mathfrak{D} \in \mathcal{A}$, hence by [eq: 14.195] it follows that $\mathfrak{M} \gg \mathfrak{D}$. Hence

$\mathfrak{M}$ is a ultra filter or maximal filter base with $\mathfrak{M} \gg \mathfrak{U}$ $\square$

Theorem 14.317. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $\mathfrak{U} = \{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ a maximal filter then $\forall x \in X$*

$$\mathfrak{U} \succ x \Leftrightarrow \mathfrak{U} \rightarrow x$$

Proof.

$\Rightarrow$. Let $U \in \mathcal{T}$ with $x \in U$. Using [theorem: 14.315] there exist a $\alpha \in I$ such that either $A_\alpha \subseteq U$ or $A_\alpha \subseteq X \setminus U$. Now as $\mathfrak{U} \succ x$ we must have that $A_\alpha \cap U \neq \emptyset$ contradicting $A_\alpha \subseteq X \setminus U$ hence we must have that $A_\alpha \subseteq U$, so $\mathfrak{U} \rightarrow x$.

$\Leftarrow$. This follows from [theorem: 14.307]. $\square$

Theorem 14.318. *Let X, Y be sets and $f: X \rightarrow Y$ a function then if $\mathfrak{U} = \{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ is a maximal filter base in X then $f[\mathfrak{U}]$ is a maximal filter base in Y .*

Proof. Using [theorem: 14.310] $f[\mathfrak{U}] = \{f(A_i)\}_{i \in I}$ is a filter base in Y . To prove maximality we use [theorem: 14.315]. So let $A \subseteq Y$ and consider $f^{-1}(A)$ then by the maximality of $\mathfrak{U}$ we have either:

$\exists \beta \in I$ with $A_\beta \subseteq f^{-1}(A) \wedge \forall \alpha \in I$ $A_\alpha \not\subseteq X \setminus f^{-1}(A)$. Then $f(A_\beta) \subseteq A$ and $\forall \alpha \in I$ there exist a $x \in A_\alpha$ such that $x \notin X \setminus f^{-1}(A) \Rightarrow x \in f^{-1}(A) \Rightarrow f(x) \in A \Rightarrow f(x) \notin Y \setminus A$ so that $f(A_\alpha) \not\subseteq Y \setminus A$.

$\forall \alpha \in I$ $A_\alpha \not\subseteq f^{-1}(A) \wedge \exists \beta \in I$ with $A_\beta \subseteq X \setminus f^{-1}(A)$. Then $\forall \alpha$ we have that $\exists x \in A_\alpha$ such that $x \notin f^{-1}(A) \Rightarrow f(x) \notin A$ proving that $f(A_\alpha) \not\subseteq A$. Further $\forall x \in A_\beta$ we have $x \in X \setminus f^{-1}(A) \Rightarrow x \notin f^{-1}(A) \Rightarrow f(x) \notin A$ proving that $f(A_\beta) \subseteq Y \setminus A$.

Using [theorem: 14.315] it follows then that $f[\mathfrak{U}]$ is a maximal filter base in Y . $\square$

14.8.2.2 Tychonoff's theorem

All this hard work is to prove Tychonoff's theorem on the product of compact spaces. The following theorem creates the relation between compactness and filter bases.

Theorem 14.319. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then the following are equivalent*

1. X is compact
2. If $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}^c$ is a non empty family of closed sets in X satisfying $\bigcap_{\alpha \in I} A_\alpha = \emptyset$ then there exist a non empty finite $J \subseteq I$ such that $\bigcap_{\alpha \in J} A_\alpha = \emptyset$.
3. If $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}^c$ is a non empty family of closed sets in X such that $\forall J \subseteq I$ J finite and non empty we have $\bigcap_{\alpha \in J} A_\alpha \neq \emptyset$ then $\bigcap_{\alpha \in I} A_\alpha \neq \emptyset$.
4. If $\mathfrak{U}$ is a filter base in X then there exist a $x \in X$ such that $\mathfrak{U} \succ x$
5. If $\mathfrak{U}$ is a maximal filter base in X then there exist a $x \in X$ such that $\mathfrak{U} \rightarrow x$

Proof.

$1 \Rightarrow 2$. Let $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}^c$ be a family of closed sets in X such that $\bigcap_{\alpha \in I} A_\alpha = \emptyset$ then

$$X = X \setminus \emptyset = X \setminus \left(\bigcap_{\alpha \in I} A_\alpha \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{\alpha \in I} (X \setminus A_\alpha)$$

As $\forall \alpha \in I$ $X \setminus A_\alpha \in \mathcal{T}$ and X is compact there exist a finite $J \subseteq I$ such that

$$X = \bigcup_{\alpha \in J} (X \setminus A_\alpha)$$

hence

$$\emptyset = X \setminus X = X \setminus \left(\bigcup_{\alpha \in J} (X \setminus A_\alpha) \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{\alpha \in J} (X \setminus (X \setminus A_\alpha)) = \bigcap_{\alpha \in J} A_\alpha$$

2 $\Rightarrow$ 1. Let $\{U_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}$ be a family of open sets such that $X = \bigcup_{\alpha \in I} U_\alpha$ then

$$\emptyset = X \setminus X = X \setminus \left(\bigcup_{\alpha \in I} U_\alpha \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{\alpha \in I} (X \setminus U_\alpha)$$

As $\forall \alpha \in I$ we have that $X \setminus U_\alpha$ is closed we have by the hypothesis that there exist a non empty finite $J \subseteq I$ such that $\bigcap_{\alpha \in J} (X \setminus U_\alpha) = \emptyset$. Hence we have

$$X = X \setminus \emptyset = X \setminus \left(\bigcap_{\alpha \in J} (X \setminus U_\alpha) \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{\alpha \in J} (X \setminus (X \setminus U_\alpha)) = \bigcup_{\alpha \in J} U_\alpha$$

proving that X is compact.

2 $\Rightarrow$ 3. Let $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}^c$ be a family of closed sets in X such that $\forall J \subseteq I$ J finite and non empty we have $\bigcap_{\alpha \in J} A_\alpha \neq \emptyset$. Assume that $\bigcap_{\alpha \in I} A_\alpha = \emptyset$ then by (2) there exist a finite non empty $J \subseteq I$ such that $\bigcap_{\alpha \in J} A_\alpha = \emptyset$ contradiction the fact that $\bigcap_{\alpha \in J} A_\alpha \neq \emptyset$. Hence we must have that

$$\bigcap_{\alpha \in I} A_\alpha \neq \emptyset$$

3 $\Rightarrow$ 2. Let $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}^c$ be a family of closed sets in X such that $\bigcap_{\alpha \in I} A_\alpha = \emptyset$. Assume that $\forall J \subseteq I$ finite non empty we have $\bigcap_{\alpha \in J} A_\alpha \neq \emptyset$ then by (3) $\bigcap_{\alpha \in I} A_\alpha \neq \emptyset$ contradicting $\bigcap_{\alpha \in I} A_\alpha = \emptyset$. Hence there must exist a finite non empty $J \subseteq I$ such that $\bigcap_{\alpha \in J} A_\alpha = \emptyset$, proving (3).

3 $\Rightarrow$ 4. Let $\mathfrak{U} = \{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$ be a filter base, then for every finite $J \subseteq I$ we have by [theorem: 14.294] that there exist a $\gamma \in I$ such that $\emptyset \neq A_\gamma \subseteq \bigcap_{\alpha \in J} A_\alpha$ proving that $\emptyset \neq \bigcap_{\alpha \in J} A_\alpha$ hence, as $\forall \alpha \in J$ we have $A_\alpha \subseteq \overline{A_\alpha}$, $\emptyset \neq \bigcap_{\alpha \in J} \overline{A_\alpha}$. Using (3) it follows then that $\emptyset \neq \bigcap_{\alpha \in I} \overline{A_\alpha}$, hence there exist a $x \in X$ such that $x \in \bigcap_{\alpha \in I} \overline{A_\alpha}$. Applying [theorem: 14.302] proves that

$$\mathfrak{U} \succ x$$

4 $\Rightarrow$ 5. Let $\mathfrak{U}$ be a maximal filter base then by (4) there exist a $x \in X$ such that $\mathfrak{U} \succ x$, by [theorem: 14.317] we have then that

$$\mathfrak{U} \rightarrow x$$

5 $\Rightarrow$ 3. Let $\{F_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}^c$ a family of closed sets in X such that for every finite non empty $J \subseteq I$ we have $\bigcap_{\alpha \in J} F_\alpha \neq \emptyset$. Take $K = \{\bigcap_{\alpha \in J} F_\alpha \mid J \subseteq I, J \text{ finite non empty}\}$ and define the family

$$\mathfrak{U} = \{A\}_{A \in K} \subseteq \mathcal{P}(X)$$

We prove now that $\mathfrak{U}$ is a filter base. First, as $I \neq \emptyset$ there exist a $i \in I$, as $\{i\}$ is finite and $\{i\} \subseteq I$ it follows that $\bigcap_{\alpha \in \{i\}} F_\alpha \in K$ so that

$$K \neq \emptyset \tag{14.196}$$

Second, if $A \in K$ then there exist a finite $J \subseteq I$ such that $A = \bigcap_{\alpha \in J} F_\alpha$, as $\bigcap_{\alpha \in J} F_\alpha \neq \emptyset$ it follows that $A \neq \emptyset$. So we have proves that

$$\forall A \in K \text{ we have } A \neq \emptyset \tag{14.197}$$

Third, let $A_1, A_2 \in K$ then there exist finite non empty $J_1, J_2 \subseteq I$ such that $A_1 = \bigcap_{\alpha \in J_1} F_\alpha$, $A_2 = \bigcap_{\beta \in J_2} F_\beta$. If $x \in A_1 \cap A_2 = (\bigcap_{\alpha \in J_1} F_\alpha) \cap (\bigcap_{\beta \in J_2} F_\beta)$ then if $\gamma \in J_1 \cup J_2$ we have either or both $\gamma \in J_1 \Rightarrow x \in F_\gamma$ or $\gamma \in J_2 \Rightarrow x \in F_\gamma$ so that $x \in \bigcap_{\gamma \in J_1 \cup J_2} F_\gamma$, hence

$$A_1 \cap A_2 \subseteq \bigcap_{\gamma \in J_1 \cup J_2} F_\gamma \tag{14.198}$$

If $x \in \bigcap_{\gamma \in J_1 \cup J_2} F_\gamma$ then $\forall \alpha \in J_1$ we have $\alpha \in J_1 \cup J_2$ so that $x \in F_\alpha$, hence $x \in \bigcap_{\alpha \in J_1} F_\alpha = A_1$ and $\forall \beta \in J_2$ we have $\beta \in J_1 \cup J_2$ so that $x \in F_\beta$, hence $x \in \bigcap_{\alpha \in J_2} F_\beta = A_2$. So $x \in A_1 \cap A_2$ proving that $\bigcap_{\gamma \in J_1 \cup J_2} F_\gamma \subseteq A_1 \cap A_2$, combining this with [eq: 14.198] proves that

$$A_1 \cap A_2 = \bigcap_{\gamma \in J_1 \cup J_2} F_\gamma$$

As $J_1 \cup J_2$ is finite non empty it follows that $A_3 = \bigcap_{\gamma \in J_1 \cup J_2} F_\gamma \in K$. So we have proved that

$$\forall A_1, A_2 \in K \text{ there exist a } A_3 \in K \text{ such that } A_3 = A_1 \cap A_2 \in K \quad (14.199)$$

From [eqs: 14.196, 14.197, 14.199] it follows that

$$\mathfrak{U} = \{A\}_{A \in K} \subseteq \mathcal{P}(X) \text{ is a filter base in } X$$

Using [theorem: 14.316] there exist a maximal filter base $\mathfrak{M}$ in X such that $\mathfrak{M} \gg \mathfrak{U}$. By (5) there exist a $x \in X$ such that $\mathfrak{M} \rightarrow x$, using [theorem: 14.307] it follows that $\mathfrak{M} \succ x$, finally as $\mathfrak{M} \gg \mathfrak{U}$ it follows from [theorem: 14.307] that

$$\mathfrak{U} \succ x$$

By [theorem: 14.302] we have then $x \in \bigcap_{A \in K} \bar{A}$ hence $\forall A \in K$ we have $x \in \bar{A}$. For every $\alpha \in I$ $\{\alpha\} \subseteq I$ and $\{\alpha\}$ is finite so that $F_\alpha = \bigcap_{a \in \{\alpha\}} F_a \in K$, hence $x \in \bar{F_\alpha} \stackrel{F_\alpha \text{ is closed}}{=} F_\alpha$. So we have proved that

$$x \in \bigcap_{\alpha \in I} F_\alpha \Rightarrow \bigcap_{\alpha \in I} F_\alpha \neq \emptyset \quad \square$$

We are now ready for Tychonoff's theorem.

Theorem 14.320. *Let $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of non empty topological spaces then for the topological space $\langle \prod_{i \in I} X_i, \mathcal{T}_{\text{product}} \rangle$ based on the product topology we have*

$$\prod_{i \in I} X_i \text{ is compact} \Leftrightarrow \forall i \in I \ X_i \text{ is compact}$$

Proof.

$\Rightarrow$. Let $i \in I$ then by [theorem: 14.160] $\pi_i: \prod_{i \in I} X_i \rightarrow X_i$ is continuous, hence using [theorem: 14.275] it follows that $\pi_i(\prod_{i \in I} X_i)$ is compact. Further as $\forall i \in I \ X_i \neq \emptyset$ we have by [theorem: 3.109] $\pi_i: \prod_{i \in I} X_i \rightarrow X_i$ is surjective so that $\pi_i(\prod_{i \in I} X_i) = X_i$. Hence we have that X_i is compact.

$\Leftarrow$. Let $\mathfrak{U}$ be a maximal filter base in $\prod_{i \in I} X_i$. Given $i \in I$ we have by [theorem: 14.318] that $\pi_i[\mathfrak{U}]$ is a maximal filter base in X_i , as X_i is compact it follows from [theorem: 14.319 (5)] that $\exists x_i \in X_i$ such that $\pi_i[\mathfrak{U}] \rightarrow x_i$. This defines a $x \in \prod_{i \in I} X_i$ so that $\pi_i[\mathfrak{U}] \rightarrow x_i = \pi_i(x)$. Using [theorem: 14.313] it follows that $\mathfrak{U} \rightarrow x$ so by [theorem: 14.319 (5)] again we have that $\prod_{i \in I} X_i$ is compact. $\square$

Corollary 14.321. *Let $\{\langle X_i, \mathcal{T}_i \rangle\}_{i \in I}$ be a family of topological spaces then if $\forall i \in I$ we have $\emptyset \neq C_i \subseteq X_i$ we have that*

$$\prod_{i \in I} C_i \text{ is compact} \Leftrightarrow \forall i \in I \ C_i \text{ is compact}$$

Proof. Note that the subspace topology of $\prod_{i \in I} C_i$ is equal to the product topology of the subspace topologies [see theorem 14.52]. As by definition a subset is compact if it is compact in the subspace topology we can use Tychonoff's [theorem: 14.320] to prove this theorem. $\square$

Note 14.322. If in the above corollary there exist a $i \in I$ such that $C_i = \emptyset$ then $\prod_{i \in I} C_i = \emptyset$ hence by [example: 14.269] compact even if the rest of the C_i 's are not compact. Hence

14.8.2.3 Consequences of Tychonoff's Theorem

Theorem 14.323. Let $\langle \mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n} \rangle$ be the topological space of tuples of real numbers where the norm $\|\cdot\|_n$ is defined by $\|x\|_n = \max(\{|\pi_i(x)| | i \in \{1, \dots, n\}\})$ then for $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ and $\{b_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ with $\forall i \in \{1, \dots, n\} \ a_i \leq b_i$ we have that

$$\prod_{i \in \{1, \dots, n\}} [a_i, b_i] \text{ is compact}$$

Proof. First using [theorem: 14.98] we have that $\mathcal{T}_{\|\cdot\|_n}$ is the product topology based on the product of the normed spaces $\langle \mathbb{R}, \|\cdot\| \rangle$. Now by Heine Borel for the real numbers [see theorem: 14.279] we have that $\forall i \in \{1, \dots, n\} \ [a_i, b_i]$ is compact, so using Tychonoff's theorem [theorem: 14.321] it follows that $\prod_{i \in \{1, \dots, n\}} [a_i, b_i]$ is compact. $\square$

Theorem 14.324. (Heine-Borel) Let $\langle \mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n} \rangle$ be the topological space of the tuples of real numbers based on the maximum norm and $C \subseteq \mathbb{R}^n$ then we have

$$C \text{ is compact} \Leftrightarrow C \text{ is bounded and closed}$$

Proof.

$\Rightarrow$. Using [theorem: 14.273] C is bounded, further by [theorem: 14.242] $\mathcal{T}_{\|\cdot\|_n}$ is Hausdorff, so using [theorem: 14.276] C is closed.

$\Leftarrow$. If C is empty then C is compact by [example: 14.269] so we may assume that $C \neq \emptyset$. Hence there exist a $c \in C$. As C is bounded there exist a $M \in \mathbb{R}^+$ such that

$$\forall x \in C \text{ we have } \max(\{|\pi_i(x - c)| | i \in \{1, \dots, n\}\}) = \|x - c\|_n \leq M$$

So given $x \in C$ and $i \in \{1, \dots, n\}$ we have that $|x_i - c_i| = |\pi_i(x) - \pi_i(c)| = |\pi_i(x - c)| \leq M$ or $x_i \in [c_i - M, c_i + M]$, hence we have that $x \in \prod_{i \in \{1, \dots, n\}} [c_i - M, c_i + M]$ proving that

$$C \subseteq \prod_{i \in \{1, \dots, n\}} [c_i - M, c_i + M]$$

Using the previous theorem [theorem: 14.323] $\prod_{i \in \{1, \dots, n\}} [c_i - M, c_i + M]$ is compact and as C is closed we have by [theorem: 14.277] that C is compact. $\square$

A important consequence of the Heine-Borel theorem is that all norms on $\mathbb{R}^n$ are equivalent [see definition: 14.119 and theorem: 14.121].

Theorem 14.325. let $n \in \mathbb{N}$ then all norms on $\mathbb{R}^n$ are equivalent.

Proof. Let $\|\cdot\|_n$ be the maximum norm defined as $\|x\|_n = \max(\{|\pi_i(x)| | i \in \{1, \dots, n\}\})$ and let $\|\cdot\|$ be another norm defined on $\mathbb{R}^n$. Using [theorem: 14.173] it follows that

$$\|\cdot\|: \mathbb{R}^n \rightarrow \mathbb{R} \text{ is continuous using the norms } \|\cdot\|_n \text{ and } \|\cdot\| \quad (14.200)$$

Further using [example: 14.177] it follows that

$$\|\cdot\|_n: \mathbb{R}^n \rightarrow \mathbb{R} \text{ is continuous using the norms } \|\cdot\|_n \text{ and } \|\cdot\| \quad (14.201)$$

Define now

$$S = \{x \in \mathbb{R}^n | \|x\|_n = 1\}$$

so that

$$S = \|\cdot\|_n^{-1}(\{1\})$$

As $\langle \mathbb{R}, \|\cdot\| \rangle$ is Hausdorff [see theorem: 14.242] we have by [theorem: 14.244] that $\{1\}$ is closed. Using the continuity of $\|\cdot\|_n$ [see eq: 14.201] it follows from [theorem: 14.154] that $\|\cdot\|_n^{-1}(\{1\})$ is closed, hence

$$S \text{ is closed} \quad (14.202)$$

Further if $x, y \in S$ then $\|x - y\|_n \leq \|x\|_n + \|y\|_n \leq 1 + 1 = 2$ which proves that

$$S \text{ is bounded} \quad (14.203)$$

So [eqs: 14.202, 14.203] allows us to use Heine-Borel [theorem: 14.324] to prove that

$$S \text{ is compact}$$

Using the above and the fact that $\|\cdot\|$ is continuous [eq: 14.200] allows us to use the extreme value theorem [theorem: 14.282], hence there exists $x_1, x_2 \in S$ such that $\forall x \in S$ we have $\|x_1\| \leq \|x\| \leq \|x_2\|$. Define $M_1 = \|x_1\|$ and $M_2 = \|x_2\|$ then as $\|x_1\|_n = 1 \Rightarrow x_1 \neq 0$ and $\|x_2\|_n = 1 \Rightarrow x_2 \neq 0$ we have

$$\exists \alpha, \beta \in \mathbb{R}^+ \text{ such that } \forall x \in \mathbb{R}^n \text{ with } \|x\|_n = 1 \text{ we have } M_1 \leq \|x\| \leq M_2 \text{ where } M_1, M_2 \in \mathbb{R}^+ \quad (14.204)$$

Let $x \in X$ then we have either

$x = \mathbf{0}$. Then $\|x\|_n = 0 = \|x\|$ so that $M_1 \cdot \|x\|_n = 0 = \|x\| = 0 = M_2 \cdot \|x\|_n$ hence

$$M_1 \cdot \|x\|_n \leq \|x\| \leq M_2 \cdot \|x\|_n$$

$x \neq \mathbf{0}$. Then $\|x\|_n \neq 0$ so that $\left\| \frac{1}{\|x\|_n} \cdot x \right\|_n = \frac{1}{\|x\|_n} \cdot \|x\|_n = 1$ so that by [eq: 14.204]

$$M_1 \leq \left\| \frac{1}{\|x\|_n} \cdot x \right\| \leq M_2 \Rightarrow M_1 \leq \frac{1}{\|x\|_n} \cdot \|x\| \leq M_2$$

giving

$$M_1 \cdot \|x\|_n \leq \|x\| \leq M_2 \cdot \|x\|_n$$

so in all cases we have $M_1 \cdot \|x\|_n \leq \|x\| \leq M_2 \cdot \|x\|_n$. Applying then [theorem: 14.121] proves that

$$\|\cdot\|_n \text{ and } \|\cdot\| \text{ are equivalent norms}$$

As $\|\cdot\|$ was chosen arbitrary we have proved that every norm in $\mathbb{R}^n$ is equivalent with $\|\cdot\|_n$. Hence by [theorem: 14.120] all the norms in $\mathbb{R}^n$ are equivalent. $\square$

Corollary 14.326. Let $n \in \mathbb{N}$ and $\|\cdot\|$ a norm on $\mathbb{R}^n$ then $\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{\|\cdot\|_n}$ where $\|\cdot\|_n$ is the maximum norm defined by $\|x\|_n = \max(\{\pi_i(x) | i \in \{1, \dots, n\}\})$

Proof. Using the above theorem [theorem: 14.325] we have that $\|\cdot\|$ and $\|\cdot\|_n$ are equivalent norms. Hence using [definition: 14.119] it follows that $\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{\|\cdot\|_n}$ $\square$

Corollary 14.327. Let $n \in \mathbb{N}$, $\|\cdot\|$ a norm on $\mathbb{R}^n$ then we have $\forall x \in \mathbb{R}^n$ and $\delta \in \mathbb{R}^+$ that

1. $\overline{B_{\|\cdot\|}(x, \delta)}$ is compact [using the canonical topology $\mathcal{T}_{\|\cdot\|_n}$]
2. If $A \subseteq B_{\|\cdot\|}(x, \delta)$ then A is precompact [using the canonical topology $\mathcal{T}_{\|\cdot\|_n}$]

Note 14.328. As $B_{\|\cdot\|}(x, \delta) \subseteq B_{\|\cdot\|_n}(x, \delta)$ it follows that $B_{\|\cdot\|}(x, \delta)$ is precompact.

Proof.

1. As by [theorem: 14.325] $\|\cdot\|$ and $\|\cdot\|_n$ are equivalent there exist $M_1, M_2 \in \mathbb{R}^+$ such that

$$\forall x \in \mathbb{R}^n \quad M_1 \cdot \|x\|_n \leq \|x\| \leq M_2 \cdot \|x\|_n$$

Further we have by [corollary: 14.326] that

$$\mathcal{T}_{\|\cdot\|} = \mathcal{T}_n$$

Let $x \in \mathbb{R}^n$ and $\delta \in \mathbb{R}^+$ then for $y, z \in \overline{B_{\|\cdot\|}(x, \delta)}$ we have

$$\|y - z\| = \|y - x + x - z\| \leq \|y - x\| + \|x - z\| \leq \delta + \delta = 2 \cdot \delta$$

As $M_1 \cdot \|y - x\|_n \leq \|y - x\| \Rightarrow \|y - x\|_n \leq \frac{1}{M_1} \cdot \|y - x\|$ so that by the above we have that

$$\|y - z\| \leq \frac{2 \cdot \delta}{M}$$

which proves that

$$\overline{B_{|||}(x, \delta)} \text{ is bounded using the norm } |||_n$$

By [theorem: 14.69] we have also that

$$\overline{B_{|||}(x, \delta)} = \overline{B_{d_{|||}}(x, \delta)} \text{ is closed in } \mathcal{T}_{|||} = \mathcal{T}_{|||_n}$$

Hence using Heine Borel [see theorem: 14.324] it follows that

$$\overline{B_{|||}(x, \delta)} \text{ is compact}$$

2. Let $A \subseteq B_{|||}(x, \delta) \subseteq \overline{B_{|||}(x, \delta)}$ we have

$$\overline{A} \underset{[\text{theorem: 14.32}]}{\subseteq} \overline{\overline{B_{|||}(x, \delta)}} \underset{[\text{theorem: 14.26}]}{=} \overline{B_{|||}(x, \delta)}$$

which, as $\overline{B_{|||}(x, \delta)}$ is bounded, proves by [theorem: 14.81] that $\overline{A}$ is bounded. This, together with the fact that $\overline{A}$ is closed, allowed us to use Heine Borel [see theorem: 14.324] to prove that

$$\overline{A} \text{ is compact in } \mathcal{T}_{|||} = \mathcal{T}_{|||_n}$$

Hence using the definition of precompact sets [see definition: 14.265] it follows that

$$A \text{ is precompact} \quad \square$$

Corollary 14.329. *Let X be a finite dimensional vector space over $\mathbb{R}$ then all norms on X are equivalent*

Proof. Let $|||_1, |||_2$ two norms on X then by [theorem: 14.197] there exist two norms $|||_{(1)}$ and $|||_{(2)}$ on $\mathbb{R}^n$ and a **isometric** isomorphism (depending only on the basis chosen in X)

$$\varphi: \mathbb{R}^n \rightarrow X \text{ so that } \forall x \in \mathbb{R}^n \text{ we have } \|\varphi(x)\|_1 = \|x\|_{(1)} \wedge \|\varphi(x)\|_2 = \|x\|_{(2)}$$

By the previous theorem [theorem: 14.325] $|||_{(1)}$ and $|||_{(2)}$ are equivalent, hence by [theorem: 14.121] there exists $M_1, M_2 \in \mathbb{R}^+$ such that

$$\forall x \in \mathbb{R}^n \text{ we have } M_1 \cdot \|x\|_{(1)} \leq \|x\|_{(2)} \leq M_2 \cdot \|x\|_{(1)}$$

Let $x \in X$ then $x = \varphi(\varphi^{-1}(x))$ so that

$$\|x\|_1 = \|\varphi(\varphi^{-1}(x))\|_1 = \|\varphi^{-1}(x)\|_{(1)} \text{ and } \|x\|_2 = \|\varphi(\varphi^{-1}(x))\|_2 = \|\varphi^{-1}(x)\|_{(2)}$$

hence we have

$$M_1 \cdot \|x\|_1 = M_1 \cdot \|\varphi^{-1}(x)\|_{(1)} \leq \|\varphi^{-1}(x)\|_{(2)} = \|x\|_2$$

and

$$\|x\|_2 = \|\varphi^{-1}(x)\|_{(2)} \leq M_2 \cdot \|\varphi^{-1}(x)\|_{(1)} = M_2 \cdot \|x\|_1$$

giving

$$M_1 \cdot \|x\|_1 \leq \|x\|_2 \leq M_2 \cdot \|x\|_1$$

proving by [theorem: 14.121] that $|||_1$ and $|||_2$ are equivalent. $\square$

Theorem 14.330. *Let $\langle X, ||| \rangle$ be a normed space over $\mathbb{R}$, $n \in \mathbb{N}$, $L: \mathbb{R}^n \rightarrow X$ a linear isomorphism and $C \subseteq X$ then*

$$C \text{ is compact} \Leftrightarrow C \text{ is closed and bounded}$$

Proof. Define

$$|||*: \mathbb{R}^n \rightarrow \mathbb{R} \text{ by } \|x\|^* = \|L(x)\|$$

then we have

$$1. \text{ If } x \in \mathbb{R}^n \text{ then } 0 \leq \|L(x)\| = \|x\|^*$$

2. If $\alpha \in \mathbb{R}$, $x \in \mathbb{R}^n$ then

$$\|\alpha \cdot x\|^* = \|L(\alpha \cdot x)\| = \|\alpha \cdot L(x)\| = |\alpha| \cdot \|L(x)\| = |\alpha| \cdot \|x\|^*$$

3. If $x, y \in \mathbb{R}^n$ then

$$\|x + y\|^* = \|L(x + y)\| = \|L(x) + L(y)\| \leq \|L(x)\| + \|L(y)\| = \|x\|^* + \|y\|^*$$

4. If $\|x\|^* = 0$ then $\|L(x)\| = 0$ which as $\|\cdot\|$ is a norm means that $L(x) = 0$, hence as L is a isomorphism it follows that $x = 0$.

So we have proved that

$$\|\cdot\|^* \text{ is a norm on } \mathbb{R}^n \text{ and } L \text{ is a linear isometric isomorphism between } \langle \mathbb{R}^n, \|\cdot\|^* \rangle \text{ and } \langle X, \|\cdot\| \rangle \quad (14.205)$$

Using [theorem: 14.194] it follows that

$$L: \mathbb{R}^n \rightarrow X \text{ is a homeomorphism using the topologies } \mathcal{T}_{\|\cdot\|^*} \text{ on } \mathbb{R}^n \text{ and } \mathcal{T}_{\|\cdot\|} \text{ on } X$$

As by [theorem: 14.325] all norms are equivalent we have that $\mathcal{T}_{\|\cdot\|_n} = \mathcal{T}_{\|\cdot\|^*}$ where $\|\cdot\|_n$ is the maximum norm on $\mathbb{R}^n$. Hence we have that

$$L: \mathbb{R}^n \rightarrow X \text{ is a homeomorphism using the topologies } \mathcal{T}_{\|\cdot\|_n} \text{ on } \mathbb{R}^n \text{ and } \mathcal{T}_{\|\cdot\|} \text{ on } X \quad (14.206)$$

Further as $\|\cdot\|^*$ and $\|\cdot\|_n$ are equivalent [see theorem: 14.325] there exist by [theorem: 14.121] a $N_1, N_2 \in \mathbb{R}^+$ such that

$$\forall x \in \mathbb{R}^n \quad \|x\|^* \leq N_1 \cdot \|x\|_n \text{ and } \|x\|_n \leq N_2 \cdot \|x\|^* \quad (14.207)$$

Assume now that $C \subseteq X$ is compact. Then, as $K = L^{-1}: X \rightarrow \mathbb{R}^n$ is a continuous mapping between $\langle X, \mathcal{T}_{\|\cdot\|} \rangle$ and $\langle \mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n} \rangle$, we have using [theorem: 14.275] that $K(C)$ is compact in $\langle \mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n} \rangle$, hence using Heine-Borel [see theorem: 14.324] we have that

$$K(C) \text{ is closed and bounded in } \langle \mathbb{R}^n, \|\cdot\|_n \rangle$$

As K is continuous it follows from [theorem: 14.154] we have that $K^{-1}(K(C))$ is closed, which as $K = L^{-1}$ is a bijection, proves by [theorem: 2.57] that $K^{-1}(K(C)) = C$, hence we have that

$$C \text{ is closed in } \langle X, \|\cdot\| \rangle$$

Further as $K(C)$ is bounded there exist a $M \in \mathbb{R}^+$ such that $\forall x, y \in K(C)$ we have $\|x - y\|_n \leq M$. Let $x, y \in C$ then

$$\|x - y\| = \|L(L^{-1}(x - y))\| = \|L(K(x - y))\| = \|K(x - y)\|^* \leq_{[\text{eq: 14.207}]} N_1 \cdot \|K(x - y)\|_n$$

as $K(x), K(y) \in K(C)$ we have $\|K(x - y)\|_n = \|K(x) - K(y)\|_n \leq M$, hence $\|x - y\| \leq N_1 \cdot M$. So we have proved that

$$C \text{ is bounded in } \langle X, \|\cdot\| \rangle$$

Summarized we have proved that

$$C \text{ is compact} \Rightarrow C \text{ is closed and bounded} \quad (14.208)$$

For the opposite implication assume that C is closed and bounded. As by [eq: 14.206] L is a continuous map between $\langle \mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n} \rangle$ and $\langle X, \mathcal{T}_{\|\cdot\|} \rangle$ it follows from [theorem: 14.154] that

$$L^{-1}(C) \text{ is closed in } \mathcal{T}_{\|\cdot\|_n} \quad (14.209)$$

Further as C is bounded there exist a $M \in \mathbb{R}^+$ such that $\forall x, y \in C$ we have $\|x - y\| \leq M$. Now if $x, y \in L^{-1}(C)$ then as $L(x), L(y) \in C$ we have

$$\|x - y\|_n \leq_{[\text{eq: 14.207}]} N_2 \cdot \|x - y\| = N_2 \cdot \|L(x - y)\| = N_2 \cdot \|L(x) - L(y)\| \leq N_2 \cdot M$$

proving that

$$L^{-1}(C) \text{ is bounded in } \langle \mathbb{R}^n, \|\cdot\|_n \rangle \quad (14.210)$$

Using Heine-Borel [see theorem: 14.324] on [eqs: 14.209, 14.210] proves that $L^{-1}(C)$ is compact in $\mathcal{T}_{|||n}$. As by [eq: 14.206] $L: \mathbb{R}^n \rightarrow X$ is continuous [using $\mathcal{T}_{|||n}$ and $\mathcal{T}_{|||}$ on $\mathbb{R}^n$ and X] it follows by [theorem: 14.275] that $C \stackrel{[\text{theorem: 2.57}]}{=} L(L^{-1}(C))$ is compact. Hence we have proved that

$$C \text{ is closed and bounded} \Rightarrow C \text{ is compact} \quad (14.211)$$

Combining [eqs: 14.208, 14.211] proves that

$$C \text{ is compact} \Leftrightarrow C \text{ is closed and bounded} \quad \square$$

A application of the above theorem is the following

Corollary 14.331. *Let $\langle \mathbb{C}, || \rangle$ be the complex space equipped with the absolute value norm $||$ and $C \subseteq \mathbb{C}$ then*

$$C \text{ is compact} \Leftrightarrow C \text{ is closed and bounded}$$

Proof. Using [example: 11.177] there exist a linear isomorphism between $\mathbb{R}^2$ and $\mathbb{C}$, hence applying the previous theorem [see 14.330] proves the theorem. $\square$

Corollary 14.332. *Let $\langle \mathbb{C}, ||| \rangle$ be the complex space equipped with a norm $|||$ then*

$$C \text{ is compact using } ||| \Leftrightarrow C \text{ is closed and bounded using } |||$$

Proof. By [theorem: 14.123] we have that $|||$ and $||$ are equivalent norms so that there exist $M_1, M_2 \in \mathbb{R}^+$ such that

$$\forall z \in \mathbb{C} \quad M_1 \cdot ||z|| \leq |z| \leq M_2 \cdot ||z|| \quad (14.212)$$

If C is bounded using $||$ then there exist a M such that $\forall z \in C$ we have $|z| \leq M$ hence $M_1 \cdot ||z|| < M$ or $||z|| \leq \frac{M}{M_1}$ proving that C is bounded using $|||$. If C is bounded using $|||$ then there exist a M such that $\forall z \in C$ we have $||z|| \leq M$ hence $|z| \leq M_2 \cdot ||z|| \leq M_2 \cdot M$ proving that C is bounded using $||$. So we have proved that

$$C \text{ is bounded using } || \Leftrightarrow C \text{ is bounded using } |||$$

Further using [theorem: 14.119] it follows that

$$C \text{ is closed using } || \Leftrightarrow C \text{ is closed using } |||$$

and

$$C \text{ is compact using } || \Leftrightarrow C \text{ is compact using } |||$$

Hence using [theorem: 14.332] it follows that

$$C \text{ is compact using } ||| \Leftrightarrow C \text{ is closed and bounded using } ||| \quad \square$$

Theorem 14.333. *Let $\langle X, ||| \rangle$ be a finite dimensional normed space over $\mathbb{R}$ and $C \subseteq X$ then*

$$C \text{ is compact} \Leftrightarrow C \text{ is closed and bounded}$$

Proof. Using [theorem: 14.197] there exist a isometric isomorphism between X and $\mathbb{R}^n$ where $n = \dim(X)$, hence applying the previous theorem [see 14.330] proves the theorem. $\square$

Theorem 14.334. *Let $\langle X, |||_X \rangle, \langle Y, |||_Y \rangle$ be finite dimensional spaces over $\mathbb{R}$ then*

$$\text{Hom}(X, Y) = L(X, Y)$$

in other words every linear mapping is continuous.

Proof. Using [theorem: 14.197] there exist linear isomorphisms $\varphi_1: \mathbb{R}^n \rightarrow X$, $\varphi_2: \mathbb{R}^m \rightarrow Y$ and norms $|||_{\langle n \rangle}, |||_{\langle m \rangle}$ on $\mathbb{R}^n, \mathbb{R}^m$ such that

$$\varphi_1: \mathbb{R}^n \rightarrow X \text{ is a homeomorphism using the norms } |||_{\langle n \rangle}, |||_X \text{ on } \mathbb{R}^n, X \quad (14.213)$$

$$\varphi_2: \mathbb{R}^m \rightarrow Y \text{ is a homeomorphism using the norms } |||_{\langle m \rangle}, |||_Y \text{ on } \mathbb{R}^m, Y \quad (14.214)$$

Let $L \in \text{Hom}(X, Y)$ and define

$$L': \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ by } L' = \varphi_2^{-1} \circ L \circ \varphi_1$$

Then as φ_1 , φ_2^{-1} and L are linear mappings it follows from [theorem: 11.180] that

$$L': \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ is a linear mapping}$$

Using [theorem: 14.206]

$$L': \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ is linear continuous using the maximum norms } |||_{\langle n \rangle}, |||_{\langle m \rangle}$$

As by [theorem: 14.325] it follows that $T_{|||_{\langle n \rangle}} = T_{|||_n}$ and $T_{|||_{\langle m \rangle}} = T_m$ proving that

$$L': \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ is linear continuous using the norms } |||_{\langle n \rangle}, |||_{\langle m \rangle} \quad (14.215)$$

Using [theorem: 14.218] on [eqs: 14.213, 14.214 and 14.215] that

$$\varphi_2 \circ L' \circ \varphi_1^{-1} \in L(X, Y)$$

Finally, as $\varphi_2 \circ L' \circ \varphi_1^{-1} = \varphi_2 \circ \varphi_2^{-1} \circ L \circ \varphi_1 \circ \varphi_1^{-1} = \text{Id}_Y \circ L \circ \text{Id}_X = L$, it follows from the above that

$$L \in L(X, Y) \quad \square$$

14.9 Convergence

Convention 14.335. For simplicity we use the following convention if $N \in \mathbb{N}_0$

$$\begin{aligned} \forall n \geq N & \text{ is a shorthand for } \forall n \in \mathbb{N}_0 \text{ with } n \geq N \\ \forall n > N & \text{ is a shorthand for } \forall n \in \mathbb{N}_0 \text{ with } n > N \end{aligned}$$

14.9.1 Sequences and limits

Definition 14.336. Let X be a set, $k \in \mathbb{N}_0$ then a family $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ is called a **sequence in X** .

A special kind of sequence in a partial ordered set are increasing or decreasing sequences.

Definition 14.337. Let $\langle X, \leq \rangle$ be a partial ordered set, $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence in X then we say:

1. $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **increasing** if $\forall i \in \{k, \dots, \infty\} \ x_i \leq x_{i+1}$.
2. $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **strictly increasing** if $\forall i \in \{k, \dots, \infty\} \ x_i < x_{i+1}$.
3. $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **decreasing** if $\forall i \in \{k, \dots, \infty\} \ x_{i+1} \leq x_i$.
4. $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **strictly decreasing** if $\forall i \in \{k, \dots, \infty\} \ x_{i+1} < x_i$.

Using mathematical induction on the above definition gives the following theorem.

Theorem 14.338. Let $\langle X, \leq \rangle$ be a partial ordered set, $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence in X then

1. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **increasing** then $\forall n, m \in \{k, \dots, \infty\}$ with $n \leq m$ we have $x_n \leq x_m$.
2. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **strictly increasing** then $\forall n, m \in \{k, \dots, \infty\}$ with $n < m$ we have $x_n < x_m$.
3. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **decreasing** then $\forall n, m \in \{k, \dots, \infty\}$ with $n \leq m$ we have $x_m \leq x_n$.
4. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **strictly decreasing** then $\forall n, m \in \{k, \dots, \infty\}$ with $n < m$ we have $x_m < x_n$.

Proof.

1. Let $n \in \{k, \dots, n\}$ and take

$$S_n = \{m \in \mathbb{N}_0 \mid x_n \leq x_{n+m}\}$$

then we have:

$0 \in S_n$. As $x_n = x_{n+0}$ it follows that $0 \in S_n$

$m \in S_n \Rightarrow m+1 \in S_n$. As $m \in S_n$ we have $x_n \leq x_{n+m}$ which as $x_{n+m} \leq x_{(n+m)+1}$ proves that $x_n \leq x_{(n+m)+1}$, hence $m+1 \in S_n$.

Using mathematical induction $S_n = \mathbb{N}_0$ so if $n \in \{k, \dots, \infty\}$ and $m \in \{k, \dots, \infty\}$ with $n \leq m$ then $m-n \in \mathbb{N}_0 = S_n$ so that $x_n \leq x_{n+(m-n)} = x_m$.

2. Let $n \in \{k, \dots, n\}$ and take

$$S_n = \{m \in \mathbb{N} \mid x_n < x_{n+m}\}$$

then we have:

$1 \in S_n$. As $x_n < x_{n+1}$ it follows that $1 \in S_n$

$m \in S_n \Rightarrow m+1 \in S_n$. As $m \in S_n$ we have $x_n < x_{n+m}$ which as $x_{n+m} < x_{(n+m)+1}$ proves that $x_n < x_{(n+m)+1}$, hence $m+1 \in S_n$.

Using mathematical induction $S_n = \mathbb{N}$ so if $n \in \{k, \dots, \infty\}$ and $m \in \{k, \dots, \infty\}$ with $n < m$ then $m-n \in \mathbb{N} = S_n$ so that $x_n < x_{n+(m-n)} = x_m$.

3. Let $n \in \{k, \dots, n\}$ and take

$$S_n = \{m \in \mathbb{N}_0 \mid x_{n+m} \leq x_n\}$$

then we have:

$0 \in S_n$. As $x_n = x_{n+0}$ it follows that $0 \in S_n$

$m \in S_n \Rightarrow m+1 \in S_n$. As $m \in S_n$ we have $x_{n+m} \leq x_n$ which as $x_{(n+m)+1} \leq x_m$ proves that $x_{(n+m)+1} \leq x_n$, hence $m+1 \in S_n$.

Using mathematical induction $S_n = \mathbb{N}_0$ so if $n \in \{k, \dots, \infty\}$ and $m \in \{k, \dots, \infty\}$ with $n \leq m$ then $m-n \in \mathbb{N}_0 = S_n$ so that $x_m = x_{n+(m-n)} \leq x_n$.

4. Let $n \in \{k, \dots, n\}$ and take

$$S_n = \{m \in \mathbb{N} \mid x_{n+m} < x_n\}$$

then we have:

$1 \in S_n$. As $x_{n+1} < x_n$ it follows that $1 \in S_n$

$m \in S_n \Rightarrow m+1 \in S_n$. As $m \in S_n$ we have $x_{n+m} < x_n$ which as $x_{(n+m)+1} < x_m$ proves that $x_{(n+m)+1} < x_n$, hence $m+1 \in S_n$.

Using mathematical induction $S_n = \mathbb{N}$ so if $n \in \{k, \dots, \infty\}$ and $m \in \{k, \dots, \infty\}$ with $n < m$ then $m-n \in \mathbb{N} = S_n$ so that $x_m = x_{n+(m-n)} < x_n$. $\square$

Corollary 14.339. Let $k \in \mathbb{N}_0$, $\{A_i\}_{i \in \{k, \dots, \infty\}}$ be a sequence of sets then we have

1. If $\forall i \in \{k, \dots, \infty\} A_i \subseteq A_{i+1}$ then $\forall n, m \in \{k, \dots, \infty\}$ with $n \leq m$ we have $A_n \subseteq A_m$
2. If $\forall i \in \{k, \dots, \infty\} A_{i+1} \subseteq A_i$ then $\forall n, m \in \{k, \dots, \infty\}$ with $n \leq m$ we have $A_m \subseteq A_n$

Proof. This follows from [theorem: 3.33] and [theorem: 14.338]. $\square$

We introduce now the concept of convergence of a sequence.

Definition 14.340. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $k \in \mathbb{N}_0$, $x \in X$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence then we say that $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is **convergent to x** if $\forall U \in \mathcal{T}$ with $x \in U$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in U$. We call $\{x_i\}_{i \in \{k, \dots, \infty\}}$ a **convergent sequence**.

We show now that convergence is independent of the start point of the domain of the family.

Theorem 14.341. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $k \in \mathbb{N}_0$, $x \in X$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence then we have:*

1. $\forall m \in \{k, \dots, \infty\}$ we have $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges to $x \Leftrightarrow \{x_i\}_{i \in \{m, \dots, \infty\}}$ converges to x
2. $\forall m \in \mathbb{N}_0$ if $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges to x then $\{x_{i+m}\}_{i \in \{k, \dots, \infty\}}$ converges to x

Proof.

1. We have

$\Rightarrow$. Let $U \in \mathcal{T}$ such that $x \in U$ then, as $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges to x , there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in U$. Hence if we take $M = \max(N, m)$ then $M \in \{m, \dots, \infty\}$ and we have $\forall n \geq M$ that $x_n \in U$. So $\{x_i\}_{i \in \{m, \dots, \infty\}}$ converges to x .

$\Leftarrow$. Let $U \in \mathcal{T}$ such that $x \in U$ then as $\{x_i\}_{i \in \{m, \dots, \infty\}}$ converges to x there exist a $N \in \{m, \dots, \infty\}$ such that $\forall n \geq M$ we have $x_n \in U$. As $m \in \{k, \dots, \infty\}$ we have that $N \in \{k, \dots, \infty\}$ hence $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges to x .

2. Let $U \in \mathcal{T}$ such that $x \in U$ then as $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges to x there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in U$. As $N \leq n \leq n+m$ we have that $x_{n+m} \in U$ proving that $\{x_{n+m}\}_{n \in \{k, \dots, \infty\}}$ converges to x . $\square$

Theorem 14.342. *Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space, $k \in \mathbb{N}_0$, $x, y \in X$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence that is **convergent to x and y** then $x = y$.*

Proof. Assume that $x \neq y$ then as $\mathcal{T}$ is Hausdorff there exist a $U, V \in \mathcal{T}$ such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$. By the definition of convergence there exist a $N_x, N_y \in \{k, \dots, \infty\}$ such that $\forall n \geq N_x$ $x_n \in U$ and $\forall n \geq N_y$ we have $x_n \in V$. So for $N = \max(N_x, N_y)$ we have $x_N \in U \cap V$ contradicting $U \cap V = \emptyset$. Hence we must have

$$x = y \quad \square$$

Definition 14.343. *Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space, $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence that is convergent to the **unique** $x \in X$ then we note this x as*

$$\lim_{n \rightarrow \infty} x_n$$

and say that x is the **limit** of $\{x_i\}_{i \in \{k, \dots, \infty\}}$. So if we say that

$$\lim_{n \rightarrow \infty} x_n = x$$

then we mean that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is convergent to $x \in X$.

We have the following equivalent definition for the limit in a metric space.

Theorem 14.344. *Let $\langle X, d \rangle$ be a metric space [which by theorem: 14.242 is Hausdorff], $k \in \mathbb{N}_0$, $x \in X$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ then*

$$\lim_{n \rightarrow \infty} x_n = x \Leftrightarrow \forall \varepsilon \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\} \text{ such that } \forall n \in \mathbb{N} \text{ with } N \leq n \ d(x, x_n) < \varepsilon$$

Further if $\langle X, ||| \rangle$ is a normed space we have as $d(x, x_n) = \|x_n - x\|$ that

$$\lim_{n \rightarrow \infty} x_n = x \Leftrightarrow \forall \varepsilon \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\} \text{ such that } \forall n \in \mathbb{N} \text{ with } N \leq n \ \|x_n - x\| < \varepsilon$$

Proof. If $\langle X, d \rangle$ is a metric space then we have

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ then as $x \in B_d(x, \varepsilon) \in \mathcal{T}$ we have from $\lim_{n \rightarrow \infty} x_n = x$ that there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in B_d(x, \varepsilon)$, hence $d(x, x_n) < \varepsilon$.

$\Leftarrow$. Let $U \in \mathcal{T}$ be such that $x \in U$ then by [theorem: 14.68] there exist a $\varepsilon \in \mathbb{R}^+$ such that $x \in B_d(x, \varepsilon) \subseteq U$. By the hypothesis there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $d(x, x_n) < \varepsilon$, hence $x \in B_d(x, \varepsilon) \subseteq U$, proving that $\lim_{n \rightarrow \infty} x_n = x$.

If $\langle X, ||| \rangle$ is a normed space then we have

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ then as $x \in B_{|||}(x, \varepsilon) \in \mathcal{T}$ we have from $\lim_{n \rightarrow \infty} x_n = x$ that there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in B_{|||}(x, \varepsilon)$, hence $\|x_n - x\| < \varepsilon$.

$\Leftarrow$. Let $U \in \mathcal{T}$ be such that $x \in U$ then by [theorem: 14.97] there exist a $\varepsilon \in \mathbb{R}^+$ such that $x \in B_{|||}(x, \varepsilon) \subseteq U$. By the hypothesis there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|x_n - x\| < \varepsilon$, hence $x \in B_{|||}(x, \varepsilon) \subseteq U$, proving that $\lim_{n \rightarrow \infty} x_n = x$. $\square$

If norms in a normed space are equivalent then the limit is independent of the chosen norm.

Theorem 14.345. *If X is a set, $k \in \mathbb{N}_0$ and $|||_1, |||_2$ are equivalent norms and $\{x_n\}_{n \in \{k, \dots, \infty\}}$ then we have*

$$x = \lim_{n \rightarrow \infty} x_n \text{ using } |||_1 \Leftrightarrow x = \lim_{n \rightarrow \infty} x_n \text{ using } |||_2$$

Proof. As $|||_1$ and $|||_2$ are equivalent it follows from [theorem: 14.121] that $\exists M_1, M_2 \in \mathbb{R}^+$ such that

$$\forall x \in X \text{ we have } M_1 \cdot \|x\|_1 \leq \|x\|_2 \leq M_2 \cdot \|x\|_1 \quad (14.216)$$

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $x = \lim_{n \rightarrow \infty} x_n$ using $|||_1$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\}$ we have $\|x_n - x\|_1 < \frac{\varepsilon}{M_2}$, hence $\|x - x_n\|_2 \underset{[\text{eq: 14.216}]}{\leq} M_2 \cdot \|x - x_n\|_1 < M_2 \cdot \frac{\varepsilon}{M_2} = \varepsilon$, which proves that $x = \lim_{n \rightarrow \infty} x_n$ using $|||_2$.

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $x = \lim_{n \rightarrow \infty} x_n$ using $|||_2$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\}$ we have $\|x_n - x\|_2 < M_1 \cdot \varepsilon$, hence $\|x - x_n\|_1 \underset{[\text{eq: 14.216}]}{\leq} \frac{1}{M_1} \cdot \|x - x_n\|_2 < \frac{1}{M_1} \cdot M_1 \cdot \varepsilon = \varepsilon$, which proves that $x = \lim_{n \rightarrow \infty} x_n$ using $|||_1$. $\square$

We can reformulate every limit to a element to a limit to 0 in $\mathbb{R}$.

Theorem 14.346. *If $\langle X, d \rangle$ is a metric space, $k \in \mathbb{N}_0$, $x \in X$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a sequence then we have*

$$\lim_{n \rightarrow \infty} x_n = x \text{ [defined in } \langle X, d \rangle] \Leftrightarrow \lim_{n \rightarrow \infty} d(x, x_n) = 0 \text{ [defined in } \langle \mathbb{R}, || \rangle]$$

If $\langle X, ||| \rangle$ is a normed space, $k \in \mathbb{N}_0$, $x \in X$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a sequence then we have

$$\lim_{n \rightarrow \infty} x_n = x \text{ [defined in } \langle X, ||| \rangle] \Leftrightarrow \lim_{n \rightarrow \infty} \|x_n - x\| = 0 \text{ [defined in } \langle \mathbb{R}, || \rangle]$$

Proof. If $\langle X, d \rangle$ is a metric space then we have

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ then, as $\lim_{n \rightarrow \infty} x_n = x$, there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $d(x, x_n) < \varepsilon$. Hence $|d(x, x_n) - 0| = |d(x, x_n)| = d(x, x_n) < \varepsilon$ proving that $\{d(x, x_n)\}_{n \in \{k, \dots, \infty\}}$ converges to 0 or that

$$\lim_{n \rightarrow \infty} d(x, x_n) = 0$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$ then there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have

$$d(x, x_n) = |d(x, x_n)| = |d(x, x_n) - 0| < \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} x_n = x$$

If $\langle X, ||| \rangle$ is a normed space then we have

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$ then, as $\lim_{n \rightarrow \infty} x_n = x$, there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ $\|x_n - x\| < \varepsilon$. Hence $|||x_n - x|| - 0| = |||x_n - x|| = \|x_n - x\| < \varepsilon$ proving that $\{|||x_n - x||\}_{n \in \{k, \dots, \infty\}}$ converges to 0 or that

$$\lim_{n \rightarrow \infty} |||x_n - x|| = 0$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$ then there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have

$$|||x_n - x|| = |||x_n - x|| = |||x_n - x|| - 0| < \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} x_n = x \quad \square$$

Example 14.347. If $\langle X, ||| \rangle$ is a normed space, $k \in \mathbb{N}_0$, $x \in X$ then we have for $\{x_i\}_{i \in \{k\}} \subseteq X$ defined by $x_i = x$ that

$$\lim_{n \rightarrow \infty} x_n = x$$

Proof. Let $\varepsilon \in \mathbb{R}^+$ and take $N = k$ then we have $\forall n \geq N$ that

$$|||x_n - x|| = |||x - x|| = 0 < \varepsilon$$

proving that $\lim_{n \rightarrow \infty} x_n = x$. $\square$

Example 14.348. If $b \in \mathbb{R}$ with $0 \leq b < 1$ then $\{b^n\}_{n \in \mathbb{N}}$ converges to 0.

Proof. We have to consider two cases for b

$b = 0$. Then $\forall n \in \mathbb{N}$ we have $b^n = 0$ so that by [example: 14.347] $\lim_{n \rightarrow \infty} b^n = 0$.

$0 < b$. Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.64] there exist a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have $0 < b^n < \varepsilon$ hence $|b^n - 0| = |b^n| = b^n < \varepsilon$, proving that $\lim_{n \rightarrow \infty} b^n = 0$. $\square$

Example 14.349. If $a \in \mathbb{R}$, $k \in \mathbb{N}_0$ with $0 < a + k$ so that $\left\{\frac{1}{a+n}\right\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ is well defined then

$$\lim_{n \rightarrow \infty} \frac{1}{a+n} = 0$$

Proof. Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.30] there exist a $N \in \mathbb{N}$ such that $\max\left(\frac{1}{\varepsilon} - a, -a\right) < N$ so if $N \leq n$ then we have $\frac{1}{\varepsilon} - a < N \leq n \Rightarrow \frac{1}{\varepsilon} < n + a \Rightarrow \frac{1}{a+n} < \varepsilon$ and $-a < n \Rightarrow 0 < a + n$. Hence $0 < \frac{1}{a+n} < \varepsilon$, so that

$$\left| \frac{1}{a+n} - 0 \right| = \left| \frac{1}{a+n} \right| = \frac{1}{a+n} < \varepsilon$$

proving

$$\lim_{n \rightarrow \infty} \frac{1}{a+n} = 0 \quad \square$$

Example 14.350. Let $k \in \mathbb{N}_0$, $a, b \in \mathbb{R}$ with $0 < b + k$ so that $\left\{\frac{a+n}{b+n}\right\}_{n \in \{k, \dots, \infty\}}$ is well defined then

$$\lim_{n \rightarrow \infty} \frac{a+n}{b+n} = 1$$

Proof. Let $\varepsilon \in \mathbb{R}^+$ then by the Archimedean property of the real numbers [see theorem: 10.30] there exist a $M \in \{k, \dots, \infty\}$ such that $\frac{|a-b|}{\varepsilon} - b < M$. Take $N = \max(k, M)$ then for $\forall n \geq N$ we have

$$\frac{|a-b|}{\varepsilon} - b < n \Rightarrow \frac{|a-b|}{\varepsilon} < n + b \xrightarrow{0 < b+k < b+n} \frac{|a-b|}{b+n} < \varepsilon \Rightarrow \left| \frac{a-b}{b+n} \right| < \varepsilon$$

Next

$$\left| \frac{a+n}{b+n} - 1 \right| = \left| \frac{a+n-(b+n)}{b+n} \right| = \left| \frac{a-b}{b+n} \right| < \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} \frac{a+n}{b+n} = 1 \quad \square$$

There exist a nice relation between the closure of a set in a metric [or normed space] and convergent sequences.

Theorem 14.351. *Let $\langle X, d \rangle$ be a metric space $A \subseteq X$ then we have*

$$x \in \bar{A} \Leftrightarrow x \in X \text{ and } \exists \{x_n\}_{n \in \mathbb{N}} \subseteq A \text{ such that } \lim_{n \rightarrow \infty} x_n = x$$

Note 14.352. *As a normed space $\langle X, ||| \rangle$ is a metric space equipped with the metric $d_{|||}$ defined by $d_{|||}(x, y) = \|x - y\|$ the theorem is also valid for normed spaces*

Proof.

$\Rightarrow$. Let $x \in \bar{A}$. Given $n \in \mathbb{N}$ we have by [theorem: 14.30] that $A \cap B_d(x, \frac{1}{n}) \neq \emptyset$ hence we have, if we use the Axiom of choice [see theorem: 3.106], that there exist a $\{x_n\}_{n \in \mathbb{N}} \subseteq \bigcup_{n \in \mathbb{N}} (A \cap B_d(x, \frac{1}{n})) \subseteq A$ such that that $\forall n \in \mathbb{N} \ x_n \in A \cap B_d(x, \frac{1}{n})$. Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.30] there exist a $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$. Hence if $n \in \{N, \dots, \infty\}$ we have

$$d(x, x_n) \underset{x_n \in B_d(x, \frac{1}{n})}{<} \frac{1}{n} \leq \frac{1}{N} < \varepsilon$$

which proves that

$$\lim_{n \rightarrow \infty} x_n = x$$

$\Leftarrow$. Suppose there exist a sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq A$ such that $\lim_{n \rightarrow \infty} x_n = x$. Let U be a open set such that $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_d(x, \delta) \subseteq U$. As $\lim_{n \rightarrow \infty} x_n = x$ there exist a $N \in \mathbb{N}$ such that $d(x, x_N) < \delta \Rightarrow x_N \in B_d(x, \delta) \subseteq U$ from which it folloes that $A \cap U \neq \emptyset$. Applying then [theorem: 14.30] it follows that $x \in \bar{A}$. $\square$

Theorem 14.353. *Let $\langle X, d \rangle$ be a metric space, Y a sub-space of the vector space X then*

$$\bar{Y} \text{ is a sub-space of } X$$

Note 14.354. *As a normed space $\langle X, ||| \rangle$ is a metric space equipped with the metric $d_{|||}$ defined by $d_{|||}(x, y) = \|x - y\|$ the theorem is also valid for normed spaces.*

Proof. As $0 \in Y \subseteq \bar{Y}$ we have that

$$\bar{Y} \neq \emptyset \quad (14.217)$$

Further let $x, y \in \bar{Y}$ and $\alpha \in \mathbb{K}$. Then by the previous theorem [theorem: 14.351] there exist sequences $\{x_n\}_{n \in \mathbb{N}} \subseteq Y$, $\{y_n\}_{n \in \mathbb{N}} \subseteq Y$ such that $\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} y_n = y$. As Y is a sub-space we have that $\{x_n + \alpha \cdot y_n\}_{n \in \mathbb{N}} \subseteq Y$. Using [theorems: 14.358, 14.357] we have that

$$\lim_{n \rightarrow \infty} (x_n + \alpha \cdot y_n) = x + \alpha \cdot y$$

so that by [theorem: 14.351] we have $x + \alpha \cdot y \in \bar{Y}$. Hence we have

$$\forall x, y \in \bar{Y}, \forall \alpha \in \mathbb{K} \text{ that } x + \alpha \cdot y \in \bar{Y} \quad (14.218)$$

From [eqs: 14.217, 14.218] it follows that

$$\bar{Y} \text{ is a sub-space of } X \quad \square$$

14.9.2 Properties of the limit

Theorem 14.355. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a normed space, $\{s_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ such that $\lim_{i \rightarrow \infty} s_i = 0$ exist and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ satisfies that $\forall i \in \{k, \dots, \infty\} \|x_i\| < s_i$ then $\lim_{i \rightarrow \infty} x_i = 0$ exists.

Proof. Let $\varepsilon \in \mathbb{R}^+$ then there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N |s_n| = |s_n - 0| < \varepsilon$. Hence if $n \geq N$ we have $\|x_n - 0\| = \|x_n\| \leq s_n < \varepsilon$ proving that

$$\lim_{i \rightarrow \infty} x_i = 0 \text{ exists} \quad \square$$

Theorem 14.356. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $k \in \mathbb{N}_0$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a sequence such that

$$\lim_{n \rightarrow \infty} x_n = x$$

then for $a \in X$ we have for $\{x_n + a\}_{n \in \{k, \dots, \infty\}} \subseteq X$ that

$$\lim_{n \rightarrow \infty} (x_n + a) = x + a$$

In other words

$$\lim_{n \rightarrow \infty} (x_n + a) = \lim_{n \rightarrow \infty} x_n + a$$

Proof. As $\lim_{n \rightarrow \infty} x_n = x$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|x_n - x\| < \varepsilon$. Hence we have

$$\|(x_n + a) - (x + a)\| = \|x_n - x\| < \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} (x_n + a) = x + a \quad \square$$

Theorem 14.357. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $k, l \in \mathbb{N}_0$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$, $\{y_n\}_{n \in \{k, \dots, \infty\}} \subseteq Y$ sequences such that

$$\lim_{n \rightarrow \infty} x_n = x \text{ and } \lim_{n \rightarrow \infty} y_n = y$$

then for $\{x_n + y_n\}_{n \in \{\max(k, l), \dots, \infty\}} \subseteq X$ we have

$$\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$$

In other words

$$\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$$

Proof. As $\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} y_n = y$ there exists $N_x \in \{k, \dots, \infty\}$, $N_y \in \{l, \dots, \infty\}$ such that $\forall n \geq N_1$ we have $\|x_n - x\| < \frac{\varepsilon}{2}$ and $\forall n \in N_2$ we have $\|y_n - y\| < \frac{\varepsilon}{2}$. Hence if we take $N = \max(N_1, N_2)$ then $\forall n \geq N$ we have

$$\|(x_n + y_n) - (x + y)\| \leq \|x_n - x\| + \|y_n - y\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} (x_n + y_n) = x + y \quad \square$$

Theorem 14.358. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $k \in \mathbb{N}_0$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a sequence such that

$$\lim_{n \rightarrow \infty} x_n = x$$

then given $\alpha \in \mathbb{K}$ we have for the sequence $\{\alpha \cdot x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ that

$$\lim_{n \rightarrow \infty} (\alpha \cdot x_n) = \alpha \cdot x$$

In other words

$$\lim_{n \rightarrow \infty} (\alpha \cdot x_n) = \alpha \cdot \lim_{n \rightarrow \infty} x_n$$

Proof. For $\alpha \in \mathbb{K}$ we have either:

$\alpha = 0$. Then $\forall n \in \{k, \dots, \infty\}$ we have $\alpha \cdot x_n = 0$ so that by [example: 14.347]

$$\lim_{n \rightarrow \infty} (\alpha \cdot x_n) = 0 = 0 \cdot x = \alpha \cdot x$$

$\alpha \neq 0$. As $\lim_{n \rightarrow \infty} x_n = x$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|x_n - x\| < \frac{\varepsilon}{|\alpha|}$. Hence we have

$$\|(\alpha \cdot x_n) - (\alpha \cdot x)\| = \|\alpha(x_n - x)\| = |\alpha| \cdot \|x_n - x\| < |\alpha| \cdot \frac{\varepsilon}{|\alpha|} = \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} (\alpha \cdot x_n) = \alpha \cdot x \quad \square$$

Theorem 14.359. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $k \in \mathbb{N}_0$, $\{\lambda_n\}_{n \in \mathbb{N}} \subseteq \mathbb{K}$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ such that $\lim_{n \rightarrow \infty} \lambda_n$ and $\lim_{n \rightarrow \infty} x_n$ exists then $\lim_{n \rightarrow \infty} (\lambda_n \cdot x_n)$ exist and $\lim_{n \rightarrow \infty} (\lambda_n \cdot x_n) = \left(\lim_{n \rightarrow \infty} \lambda_n\right) \cdot \left(\lim_{n \rightarrow \infty} x_n\right)$.

Proof. Let $\lambda = \lim_{n \rightarrow \infty} \lambda_n$ and $x = \lim_{n \rightarrow \infty} x_n$ and take $\varepsilon \in \mathbb{R}^+$. As $\lambda = \lim_{n \rightarrow \infty} \lambda_n$ there exists a $N_1 \in \{k, \dots, \infty\}$ such that

$$\forall n \geq N_1 \text{ we have } |\lambda_n - \lambda| < 1 \Rightarrow |\lambda_n| = |\lambda_n - \lambda + \lambda| \leq |\lambda_n - \lambda| + |\lambda| < 1 + |\lambda| \quad (14.219)$$

Further as $1 + \|\lambda\| \in \mathbb{R}^+$ there exist a $N_2 \in \{k, \dots, \infty\}$ such that

$$\forall n \geq N_2 \text{ we have } |\lambda_n - \lambda| < \frac{\varepsilon}{2 \cdot (1 + \|\lambda\|)} \quad (14.220)$$

As $x = \lim_{n \rightarrow \infty} x_n$ and $1 + |\lambda| \in \mathbb{R}^+$ there exist a $N_3 \in \{k, \dots, \infty\}$ such that

$$\forall n \geq N_3 \text{ we have } \|x_n - x\| < \frac{\varepsilon}{2 \cdot (1 + |\lambda|)} \quad (14.221)$$

Let $N = \max(N_1, N_2, N_3)$ then we have $\forall n \geq N$ we have

$$\begin{aligned} \|\lambda_n \cdot x_n - \lambda \cdot x\| &= \|\lambda_n \cdot x_n - \lambda_n \cdot x + \lambda_n \cdot x - \lambda \cdot x\| \\ &= \|\lambda_n \cdot (x_n - x) + (\lambda_n - \lambda) \cdot x\| \\ &\leq \|\lambda_n \cdot (x_n - x)\| + \|(\lambda_n - \lambda) \cdot x\| \\ &= |\lambda_n| \cdot \|x_n - x\| + |\lambda_n - \lambda| \cdot \|x\| \\ &\stackrel{[\text{eqs: 14.220, 14.221}]}{<} |\lambda_n| \cdot \frac{\varepsilon}{2 \cdot (1 + |\lambda|)} + \|x\| \cdot \frac{\varepsilon}{2 \cdot (1 + \|\lambda\|)} \\ &< |\lambda_n| \cdot \frac{\varepsilon}{2 \cdot (1 + |\lambda|)} + \frac{\varepsilon}{2} \\ &\stackrel{[\text{eq: 14.219}]}{<} (1 + |\lambda|) \cdot \frac{\varepsilon}{2 \cdot (1 + |\lambda|)} + \frac{\varepsilon}{2} \\ &= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

which proves that

$$\lim_{n \rightarrow \infty} (x_n \cdot y_n) \text{ exist and } \lim_{n \rightarrow \infty} (x_n \cdot y_n) = x \cdot y = \left(\lim_{n \rightarrow \infty} x_n\right) \cdot \left(\lim_{n \rightarrow \infty} y_n\right) \quad \square$$

Corollary 14.360. Let $k \in \mathbb{N}_0$, $n \in \mathbb{N}$ and $\{x_{i,j}\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{K}$ where $j \in \{1, \dots, n\}$ such that $\forall j \in \{1, \dots, n\}$ we have that $\lim_{i \rightarrow \infty} x_{i,j}$ exist then $\lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n\}} x_{i,j}\right)$ exist and

$$\lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n\}} x_{i,j}\right) = \prod_{j \in \{1, \dots, n\}} \left(\lim_{i \rightarrow \infty} x_{i,j}\right)$$

Proof. We use induction to prove this so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{x_{i,j}\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{K} \text{ satisfies that } \forall j \in \{1, \dots, n\} \lim_{i \rightarrow \infty} x_{i,j} \text{ exist then} \right.$$

$$\left. \lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n\}} x_{i,j} \right) \text{ exist and } \lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n\}} x_{i,j} \right) = \prod_{j \in \{1, \dots, n\}} \left(\lim_{i \rightarrow \infty} x_{i,j} \right) \right\}$$

then we have:

$1 \in S$. If $\{x_{i,j}\}_{i \in \{k, \dots, \infty\}}$ satisfies that for $j \in \{1\}$ $\lim_{i \rightarrow \infty} x_{i,j}$ exist then we have as $\prod_{j \in \{1\}} x_{i,j} = x_{i,1}$ that $\lim_{i \rightarrow \infty} (\prod_{j \in \{1\}} x_{i,j})$ exist and

$$\lim_{i \rightarrow \infty} \left(\prod_{j \in \{1\}} x_{i,j} \right) = \lim_{i \rightarrow \infty} x_{i,1} = \prod_{j \in \{1\}} \left(\lim_{i \rightarrow \infty} x_{i,j} \right)$$

which proves that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{x_{i,j}\}_{i \in \{k, \dots, \infty\}}$ be such that for $j \in \{1, \dots, n+1\}$ $\lim_{i \rightarrow \infty} x_{i,j}$ exist. Then as $n \in S$ we have that $\lim_{i \rightarrow \infty} (\prod_{j \in \{1, \dots, n\}} x_{i,j})$ exist and $\lim_{i \rightarrow \infty} (\prod_{j \in \{1, \dots, n\}} x_{i,j}) = \prod_{j \in \{1, \dots, n\}} (\lim_{i \rightarrow \infty} x_{i,j})$. Hence using [theorem: 14.359] we have that

$$\lim_{i \rightarrow \infty} \left(\left(\prod_{j \in \{1, \dots, n\}} \left(\lim_{i \rightarrow \infty} x_{i,j} \right) \right) \cdot x_{i,n+1} \right)$$

exist and

$$\lim_{i \rightarrow \infty} \left(\left(\prod_{j \in \{1, \dots, n\}} \left(\lim_{i \rightarrow \infty} x_{i,j} \right) \right) \cdot x_{i,n+1} \right) = \left(\lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n\}} x_{i,j} \right) \right) \cdot \lim_{i \rightarrow \infty} x_{i,n+1}$$

So that

$$\lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n+1\}} x_{i,j} \right) \text{ exist and } \lim_{i \rightarrow \infty} \left(\prod_{j \in \{1, \dots, n+1\}} x_{i,j} \right) = \prod_{j \in \{1, \dots, n+1\}} \left(\lim_{i \rightarrow \infty} x_{i,j} \right)$$

which proves that $n+1 \in S$ □

Theorem 14.361. Let $k \in \mathbb{N}_0$ and $\{z_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{C}$ a sequence of complex numbers then if $\|$ is the absolute value norm on $\mathbb{R}$ and $\|_c$ is the canonical norm on $\mathbb{C}$ [see definition: 10.84] then we have that

$$\{z_n\}_{n \in \{k, \dots, \infty\}} \text{ converges in } \langle \mathbb{C}, \|_c \rangle$$

$$\Updownarrow$$

$$\{\text{Re}(z_n)\}_{n \in \{k, \dots, \infty\}} \text{ and } \{\text{Im}(z_n)\}_{n \in \{k, \dots, \infty\}} \text{ converges in } \langle \mathbb{R}, \| \rangle$$

Further if $\{\text{Re}(z_n)\}_{n \in \{k, \dots, \infty\}}, \{\text{Im}(z_n)\}_{n \in \{k, \dots, \infty\}}$ converges [or equivalently $\{z_n\}_{n \in \{k, \dots, \infty\}}$ converges] then we have that

$$\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} \text{Re}(z_n) + i \cdot \lim_{n \rightarrow \infty} \text{Im}(z_n)$$

Proof. We have

$\Rightarrow$. As $\{z_n\}_{n \in \{k, \dots, \infty\}}$ converges there exist a $z \in \mathbb{C}$ such that $\forall \varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, n\}$ such that

$$\forall n \geq N \text{ we have } |z_n - z|_c < \varepsilon$$

Now using [theorem: 10.85] we have that

$$|\text{Re}(z_n) - \text{Re}(z)| = |\text{Re}(z_n - z)| \leq |z_n - z|_c < \varepsilon$$

$$|\text{Im}(z_n) - \text{Im}(z)| = |\text{Im}(z_n - z)| < |z_n - z|_c < \varepsilon$$

proving that $\{\operatorname{Re}(z_n)\}_{n \in \{k, \dots, \infty\}}$ converges to $\operatorname{Re}(x)$ and $\{\operatorname{Im}(z_n)\}_{n \in \{k, \dots, \infty\}}$ converges to $\operatorname{Im}(z)$.

$\Leftarrow$. As $\{\operatorname{Re}(z_n)\}_{n \in \{k, \dots, \infty\}}$ and $\{\operatorname{Im}(z_n)\}_{n \in \{k, \dots, \infty\}}$ converges there exists $x, y \in \mathbb{R}$ such that $\forall \varepsilon \in \mathbb{R}^+$ there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that $\forall n \geq N_1$ $|\operatorname{Re}(z_n) - x| < \frac{\varepsilon}{2}$ and $\forall m \geq N_2$ $|\operatorname{Im}(z_n) - y| < \frac{\varepsilon}{2}$. Hence if we take $z = x + i \cdot y$ and $m \geq \max(N_1, N_2)$ then we have that

$$|z_n - z|_c \leq_{[\text{theorem: 10.85}]} |\operatorname{Re}(z_n - z)| + |\operatorname{Im}(z_n - z)| = |\operatorname{Re}(z_n) - x| + |\operatorname{Im}(z_n) - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving that $\{z_n\}_{n \in \{k, \dots, \infty\}}$ converges to z . Further

$$\lim_{n \rightarrow \infty} z_n = z = x + i \cdot y = \lim_{n \rightarrow \infty} \operatorname{Re}(z_n) + i \cdot \lim_{n \rightarrow \infty} \operatorname{Im}(z_n) \quad \square$$

Theorem 14.362. Let $k \in \mathbb{N}_0$, $\{x_n\}_{i \in \{n, \dots, \infty\}} \subseteq \mathbb{C}$ be a convergent sequence of complex numbers then $\{\overline{x_n}\}_{n \in \{k, \dots, \infty\}}$ is convergent and $\lim_{n \rightarrow \infty} \overline{x_n} = \overline{\lim_{n \rightarrow \infty} x_n}$

Proof. Let $x = \lim_{n \rightarrow \infty} x_n$ then for $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $|x_n - x| < \varepsilon$. Now $|\overline{x_n} - \overline{x}| \stackrel{[\text{theorem: 10.83}]}{=} |\overline{x_n - x}| \stackrel{[\text{theorem: 10.85}]}{=} |x_n - x| < \varepsilon$ proving that

$$\lim_{n \rightarrow \infty} \overline{x_n} = \overline{x} = \overline{\lim_{n \rightarrow \infty} x_n} \quad \square$$

Theorem 14.363. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a normed space and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a convergent sequence such that $\lim_{n \rightarrow \infty} x_n = x$ then we have for $A \in \mathbb{R}$:

1. If $\exists M \in \{k, \dots, \infty\}$ such that $\forall n \geq M$ $\|x_n\| \leq A$ then $\|x\| \leq A$.
2. If $\exists M \in \{k, \dots, \infty\}$ such that $\forall n \geq M$ $A \leq \|x_n\|$ then $A \leq \|x\|$.

Proof.

1. We proceed by contradiction. So assume that $A < \|x\|$ then $\varepsilon = \|x\| - A \in \mathbb{R}^+$ hence, as $\lim_{n \rightarrow \infty} x_n = x$, there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ $\|x_n - x\| < \varepsilon$ or as $N \leq \max(N, M)$

$$\|x\| \leq \|x - x_{\max(N, M)}\| + \|x_{\max(N, M)}\| < \varepsilon + \|x_{\max(N, M)}\| \leq \varepsilon + A = \|x\| - A + A = \|x\|$$

leading to the contradiction $\|x\| < \|x\|$. Hence we must have $\|x\| \leq A$.

2. We proceed by contradiction. So assume that $\|x\| < A$ then $\varepsilon = A - \|x\| \in \mathbb{R}^+$ hence, as $\lim_{n \rightarrow \infty} x_n = x$, there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ $\|x_n - x\| < \varepsilon$ or as $N \leq \max(N, M)$

$$A \leq \|x_{\max(M, N)}\| \leq \|x_{\max(M, N)} - x\| + \|x\| < \varepsilon + \|x\| = A - \|x\| + \|x\| = A$$

leading to the contradiction $A < A$. Hence we must have $A \leq \|x\|$. $\square$

Theorem 14.364. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $k \in \mathbb{N}_0$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a convergent sequence then there exist a $K \in \mathbb{R}^+$ such that $\forall n \in \{k, \dots, \infty\}$ we have $\|x_n\| \leq K$.

Proof. Let $x = \lim_{n \rightarrow \infty} x_i$ then as $1 \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|x_n\| = \|x_n - x + x\| \leq \|x_n - x\| + \|x\| < 1 + \|x\|$. Hence if we define

$$K = \max(1 + \|x\|, \max(\{x_i | i \in \{k, \dots, N\}\}))$$

we have

$$\forall n \in \{k, \dots, \infty\} \text{ that } \|x_n\| \leq K \quad \square$$

Theorem 14.365. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ a normed space, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a **convergent** sequence in X and $\{s_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ a **convergent** sequence such that $\forall n \in \{k, \dots, \infty\}$ we have

$$\|x_n\| \leq s_n \text{ then } \left\| \lim_{n \rightarrow \infty} x_n \right\| \leq \lim_{n \rightarrow \infty} s_n$$

Proof. Let $x = \lim_{n \rightarrow \infty} x_n$ and $s = \lim_{n \rightarrow \infty} s_n = s$. As $\forall n \in \{k, \dots, \infty\}$ we have $0 \leq \|x_n\| < s_n$ it follows from [theorem: 14.363] that

$$0 \leq s$$

Assume that $s < \|x\|$ then $\varepsilon = \|x\| - s \in \mathbb{R}^+$. Using the definition of a limit there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that $\forall n \geq N_1$ $\|x_n - x\| < \frac{\varepsilon}{2}$ and $\forall m \geq N_2$ $|s_n - s| < \frac{\varepsilon}{2}$. Take $N = \max(N_1, N_2)$ then we have

$$\|x\| \leq \|x - x_N\| + \|x_N\| < \frac{\varepsilon}{2} + \|x_N\| \leq \frac{\varepsilon}{2} + s_N \leq \frac{\varepsilon}{2} + |s_N| \leq \frac{\varepsilon}{2} + |s_N - s| + |s| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} + |s| = \varepsilon + s = \|x\|$$

leading to the contradiction $\|x\| < \|x\|$. Hence we must have that $\|x\| \leq s$. $\square$

Example 14.366. Let $k \in \mathbb{N}$ then $\left\{\frac{(-1)^n}{n}\right\}_{n \in \{k, \dots, \infty\}}$ has a limit where $\lim_{n \rightarrow \infty} \left(\frac{(-1)^n}{n}\right) = 0$

Proof. As $\forall n \in \{k, \dots, \infty\}$ $\left|\frac{(-1)^n}{n}\right| = \frac{1}{n}$ and by [example: 14.349] $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ exists it follows from [theorem: 14.365] that $\lim_{n \rightarrow \infty} \left(\frac{(-1)^n}{n}\right)$ exists and $0 \leq \left|\lim_{n \rightarrow \infty} \left(\frac{(-1)^n}{n}\right)\right| \leq \lim_{n \rightarrow \infty} \frac{1}{n} = 0$ so that

$$\lim_{n \rightarrow \infty} \left(\frac{(-1)^n}{n}\right) = 0 \quad \square$$

Theorem 14.367. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed space, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$, $\{y_n\}_{n \in \{k, \dots, \infty\}} \subseteq Y$ convergent sequences such that $\forall n \in \{k, \dots, \infty\}$ $\|x_n\|_X \leq \|y_n\|_Y$ then

$$\left\|\lim_{n \rightarrow \infty} x_n\right\|_X \leq \left\|\lim_{n \rightarrow \infty} y_n\right\|_Y$$

Proof. Let $x = \lim_{n \rightarrow \infty} x_n$, $y = \lim_{n \rightarrow \infty} y_n$ and assume that $\|y\|_Y < \|x\|_X$ so that $\varepsilon = \|x\|_X - \|y\|_Y \in \mathbb{R}^+$. Find $N_1, N_2 \in \{k, \dots, \infty\}$ such that $\forall n \geq N_1$ $\|x_n - x\|_X < \frac{\varepsilon}{2}$ and $\forall m \geq N_2$ $\|y_m - y\|_Y < \frac{\varepsilon}{2}$. Take $N = \max(N_1, N_2)$ then we have:

$$\begin{aligned} \|x\|_X &\leq \|x_N - x\|_X + \|x_N\|_X \\ &\leq \|x_N - x\|_X + \|y_N\|_Y \\ &\leq \|x_N - x\|_X + \|y_N - y\|_Y + \|y\|_Y \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} + \|y\|_Y \\ &= \varepsilon + \|y\|_Y \\ &= \|x\|_X \end{aligned}$$

giving the contradiction $\|x\|_X < \|x\|_X$. So the assumption is wrong and we must have that $\|x\|_X \leq \|y\|_Y$. $\square$

Theorem 14.368. Let $k \in \mathbb{N}_0$, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ a convergent sequence of non negative numbers then $0 \leq \lim_{n \rightarrow \infty} x_n$.

Proof. Assume that $x = \lim_{i \rightarrow \infty} x_i < 0$ then $\varepsilon = -x \in \mathbb{R}^+$. By the definition of a limit there exists a $N \in \{k, \dots, \infty\}$ such that $|x_N - x| < \frac{\varepsilon}{2}$. As $0 \leq x_N \wedge x < 0 \Rightarrow 0 < -x$ we have that $0 < x_N - x$ so that $x_N - x = |x_N - x| < \frac{\varepsilon}{2} \Rightarrow x_N < \frac{\varepsilon}{2} + x = \frac{\varepsilon}{2} - \varepsilon = -\frac{\varepsilon}{2} < 0$ contradicting $0 \leq x_N$ so we must have that $0 \leq \lim_{i \rightarrow \infty} x_i$ $\square$

Corollary 14.369. Let $\{x_n\}_{n \in \{k, \dots, \infty\}}, \{y_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ be convergent sequences with $\forall n \in \{k, \dots, \infty\}$ $x_n \leq y_n$ then $\lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} y_n$

Proof. Using the assumption we have that $\{y_i - x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ and using [theorems: 14.357, 14.358] it follows that $\lim_{n \rightarrow \infty} (y_n - x_n) = \lim_{n \rightarrow \infty} y_n - \lim_{n \rightarrow \infty} x_n$. From [theorem: 14.368] we have that $0 \leq \lim_{n \rightarrow \infty} (y_n - x_n)$ so that $\lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} y_n$. $\square$

Theorem 14.370. Let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R} \setminus \{0\}$ be a sequence of non zero numbers such that

$$\lim_{n \rightarrow \infty} \left| \frac{x_{n+1}}{x_n} \right| \text{ converges with } \lim_{n \rightarrow \infty} \left| \frac{x_{n+1}}{x_n} \right| < 1$$

then $\lim_{n \rightarrow \infty} x_n$ converges with $\lim_{n \rightarrow \infty} x_n = 0$

Proof. Let $L = \lim_{n \rightarrow \infty} \left| \frac{x_{n+1}}{x_n} \right|$ then by [theorem: 14.363] $0 \leq L$ and as $L < 1$ we have by [theorem: 10.32] that there exist a $r \in \mathbb{R}$ such that $L < r < 1$. Take $\zeta = r - L \in \mathbb{R}^+$ then there exist a $N_1 \in \{k, \dots, \infty\}$ such that

$$\forall n \in \{N_1, \dots, \infty\} \left| \left| \frac{x_{n+1}}{x_n} \right| - L \right| < \left| \left| \frac{x_{n+1}}{x_n} \right| - L \right| < r - L$$

so that

$$\forall n \in \{N_1, \dots, \infty\} \text{ we have } \left| \frac{x_{n+1}}{x_n} \right| < r \text{ or } |x_{n+1}| < r \cdot |x_n| \quad (14.222)$$

we prove now by induction that

$$\forall l \in \mathbb{N} \quad |x_{N_1+l}| < r^l \cdot |x_{N_1}| \quad (14.223)$$

Proof. Let

$$S = \{l \in \mathbb{N} \mid |x_{N_1+l}| < r^l \cdot |x_{N_1}|\}$$

then we have

$1 \in S$. Using [eq: 14.222] we have that $|x_{N_1+1}| < r \cdot |x_{N_1}| = r^1 \cdot |x_{N_1}|$ which proves that $1 \in S$.

$l \in S \Rightarrow l+1 \in S$. As $l \in S$ we have that $|x_{N_1+l}| < r^l \cdot |x_{N_1}|$, by [eq: 14.222] we have that $|x_{N_1+(l+1)}| < r \cdot |x_{N_1+l}|$. Hence we have, taking in account that $0 \leq L < r \Rightarrow 0 < r$,

$$|x_{N_1+(l+1)}| < r \cdot (r^l) \cdot |x_{N_1}| = r^{l+1} \cdot |x_{N_1}|$$

which proves that

$$l+1 \in S \quad \square$$

Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.64] there exist a N_2 such that

$$\forall n \in \{N_2, \dots, \infty\} \text{ we have } r^n < \frac{\varepsilon}{|x_{N_1}|} \xRightarrow{\text{[eq: 14.223]}} |x_{N_1+n}| < \varepsilon$$

Take now $N = N_1 + N_2$ then as $n \in \{N, \dots, \infty\}$ we have that $n \geq N_1 + N_2 \Rightarrow n - N_1 \geq N_2$ so that

$$|x_n - 0| = |x_{N_1+(n-N_1)}| < \varepsilon$$

which proves that

$$\lim_{n \rightarrow \infty} x_n = 0 \quad \square$$

There is a relation between convergence and continuity of a function.

Theorem 14.371. If $\langle X, \mathcal{T}_X \rangle$, $\langle Y, \mathcal{T}_Y \rangle$ and $f: X \rightarrow Y$ a continuous function at x then

$$\forall \{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X \text{ with } \lim_{n \rightarrow \infty} x_n = x \text{ we have that } \lim_{n \rightarrow \infty} f_n(x) = f(x)$$

Proof. As f is continuous at x we have for $V \in \mathcal{T}_Y$ with $f(x) \in V$ that there exist a $U \in \mathcal{T}_X$ such that $x \in U$ and $f(x) \in f(U) \subseteq V$. If $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ satisfies $\lim_{n \rightarrow \infty} x_n = x$ then there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in U$, hence as $f(U) \subseteq V$ we have $f(x_n) \in V$. So we have

$$\lim_{n \rightarrow \infty} f(x_n) = f(x) \quad \square$$

For metric and normed spaces we have also the opposite of the above implication.

Theorem 14.372. If $\langle X, d_x \rangle$ and $\langle Y, d_Y \rangle$ are metric spaces, $f: X \rightarrow Y$ a function then we have

$$f \text{ is continuous at } x \Leftrightarrow \forall \{x_n\}_{n \in \mathbb{N}} \subseteq X \text{ with } \lim_{n \rightarrow \infty} x_n = x \text{ we have } \lim_{n \rightarrow \infty} f(x_n) = f(x)$$

Note 14.373. As a normed space is a metric space the theorem is also applicable to normed spaces.

Proof.

$\Rightarrow$. This follows from the previous theorem [theorem: 14.371].

$\Leftarrow$. Assume that f is not continuous at x then by [theorem: 14.134] there exist a $\varepsilon > 0$ such that $\forall \delta \in \mathbb{R}^+$ there exist a $y \in X$ with $d_X(x, y) < \delta$ such that $\varepsilon \leq d_Y(f(x), f(y))$. Hence $\forall n \in \mathbb{N}_0$ there exist a $x_n \in X$ with $d_X(x, x_n) < \frac{1}{n}$ such that $\varepsilon \leq d_Y(f(x), f(x_n))$. Take $\zeta \in \mathbb{R}^+$ then by the Archimedean property [see theorem: 10.29] there exist a $N \in \mathbb{N}$ such that $\frac{1}{N} < \zeta$, hence $\forall n \geq N$ we have

$$d_X(x, x_n) < \frac{1}{n} \leq \frac{1}{N} < \zeta$$

proving that

$$\lim_{n \rightarrow \infty} x_n = x$$

Hence by the hypothesis we have that

$$\lim_{n \rightarrow \infty} f(x_n) = f(x)$$

So there exist a $M \in \mathbb{N}$ such that $d_Y(f(x), f(x_M)) < \varepsilon$ contradicting the fact that $\forall n \in \mathbb{N}$ $\varepsilon \leq d_Y(f(x), f(x_n))$. So the assumption is wrong and we must have that f is continuous at x . $\square$

Corollary 14.374. Let $k \in \mathbb{N}_0$, $\langle X, ||| \rangle$ a normed space and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence such that $\lim_{n \rightarrow \infty} x_n$ exist then $\lim_{n \rightarrow \infty} |||x_n|||$ exist and $\lim_{n \rightarrow \infty} |||x_n||| = |||\lim_{n \rightarrow \infty} x_n|||$

Proof. Using [example: 14.177] $|||: X \rightarrow \mathbb{R}$ defined by $|||(x) = \|x\|$ is continuous. So as $\lim_{n \rightarrow \infty} x_n$ converges to x we have by the previous theorem [theorem: 14.372] that $\lim_{n \rightarrow \infty} |||(x_n)$ converges to $|||(x) = \|x\|$. In other words we have $\lim_{n \rightarrow \infty} |||x_n||| = |||\lim_{n \rightarrow \infty} x_n|||$, $\square$

14.9.3 Sequences of real numbers

In this section we always assume that we work with the normed space $\langle \mathbb{R}, ||| \rangle$. The basic property that sets $\mathbb{R}$ apart from other normed spaces is the fact that $\langle \mathbb{R}, \leq \rangle$ is conditional complete.

For the real numbers there is a nice relation between the supremum and infimum and increasing or decreasing sequences.

Theorem 14.375. Let $k \in \mathbb{N}_0$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ then we have

1. If $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is increasing then we have

$$\sup(\{x_n | n \in \{k, \dots, \infty\}\}) \text{ exists} \Leftrightarrow \lim_{n \rightarrow \infty} x_n \text{ exists}$$

further if the supremum or the limit exists we have that

$$\lim_{n \rightarrow \infty} x_n = \sup(\{x_n | n \in \{k, \dots, \infty\}\})$$

2. If $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is decreasing then we have

$$\inf(\{x_n | n \in \{k, \dots, \infty\}\}) \text{ exists} \Leftrightarrow \lim_{n \rightarrow \infty} x_n \text{ exists}$$

further if the supremum or the limit exists we have that

$$\lim_{n \rightarrow \infty} x_n = \inf(\{x_n | n \in \{k, \dots, \infty\}\})$$

Proof.

1. We have

$\Rightarrow$. Let $x = \sup(\{x_n | n \in \{k, \dots, \infty\}\})$ and take $\varepsilon \in \mathbb{R}^+$ then $x - \varepsilon < x$ so that there exist a $N \in \{k, \dots, \infty\}$ such that $x - \varepsilon < x_N$. By [theorem: 14.338] we have $\forall n \leq N$ that $x - \varepsilon < x_N \leq x_n \leq x < x + \varepsilon$, hence $x - x_n < \varepsilon \wedge x_n - x < \varepsilon \Rightarrow |x_n - x| < \varepsilon$, proving that

$$\lim_{n \rightarrow \infty} x_n \text{ exists and } \lim_{n \rightarrow \infty} x_n = x = \sup(\{x_n | n \in \{k, \dots, \infty\}\})$$

$\Leftarrow$. Let $x = \lim_{n \rightarrow \infty} x_n$ then as $1 \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $|x_n - x| < 1 \Rightarrow x_n - x < 1 \Rightarrow x_n < x + 1$. So $\forall n \in \{k, \dots, \infty\}$ we have that $x_n \leq \max(\{x_i | i \in \{1, \dots, N\} \cup \{x + 1\}\})$ or in other words $\{x_n | n \in \{k, \dots, \infty\}\}$ is bounded above. Using the conditional completeness of the real numbers [see theorem: 10.18] it follows that

$$\sup(\{x_n | n \in \{k, \dots, \infty\}\}) \text{ exist}$$

2. We have

$\Rightarrow$. Let $x = \inf(\{x_n | n \in \{k, \dots, \infty\}\})$ and take $\varepsilon \in \mathbb{R}^+$ then $x < x + \varepsilon$ so that there exist a $N \in \{k, \dots, \infty\}$ such that $x_N < x + \varepsilon$. By [theorem: 14.338] we have $\forall n \leq N$ that $x - \varepsilon < x \leq x_n \leq x_N < x + \varepsilon$, hence $x - x_n < \varepsilon \wedge x_n - x < \varepsilon \Rightarrow |x_n - x| < \varepsilon$, proving that

$$\lim_{n \rightarrow \infty} x_n \text{ exists and } \lim_{n \rightarrow \infty} x_n = x = \inf(\{x_n | n \in \{k, \dots, \infty\}\})$$

$\Leftarrow$. Let $x = \lim_{n \rightarrow \infty} x_n$ then as $1 \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $|x_n - x| < 1 \Rightarrow x - x_n < 1 \Rightarrow x - 1 < x_n$. So $\forall n \in \{k, \dots, \infty\}$ we have that $\min(\{x_i | i \in \{1, \dots, N\} \cup \{x - 1\}\}) \leq x_n$ or in other words $\{x_n | n \in \{k, \dots, \infty\}\}$ is bounded below. Using the conditional completeness of the real numbers [see theorem: 10.18] it follows that

$$\inf(\{x_n | n \in \{k, \dots, \infty\}\}) \text{ exist}$$

□

Corollary 14.376. Let $k \in \mathbb{N}_0$ and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ then we have

1. If $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is increasing and there exist a $M \in \mathbb{R}$ and a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ $x_n \leq M$ then $\lim_{n \rightarrow \infty} x_n$ exist and $\lim_{n \rightarrow \infty} x_n \leq M$.
2. If $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is decreasing and there exist a $M \in \mathbb{R}$ and a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ $M \leq x_n$ then $\lim_{n \rightarrow \infty} x_n$ exist and $M \leq \lim_{n \rightarrow \infty} x_n$.

Proof.

1. As $\forall n \geq N$ $x_n \leq M$ we have that $\emptyset \neq \{x_n | n \in \{N, \dots, \infty\}\} \leq M$ it follows from the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] that $\sup(\{x_n | n \in \{N, \dots, \infty\}\})$ exist and

$$\sup(\{x_n | n \in \{N, \dots, \infty\}\}) \leq M.$$

Hence by [theorem: 14.375] $\{x_n\}_{n \in \{N, \dots, \infty\}}$ converges to $\sup(\{x_n | n \in \{N, \dots, \infty\}\})$. Which using [theorem: 14.341] proves that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ converges to $\sup(\{x_n | n \in \{N, \dots, \infty\}\})$. In other words

$$\lim_{n \rightarrow \infty} x_n \text{ exist and } \lim_{n \rightarrow \infty} x_n \leq M$$

2. As $\forall n \geq N$ $M \leq x_n$ we have that $\emptyset \neq \{x_n | n \in \{N, \dots, \infty\}\} \geq M$ it follows from the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] that $\inf(\{x_n | n \in \{N, \dots, \infty\}\})$ exist and

$$M \leq \inf(\{x_n | n \in \{N, \dots, \infty\}\}).$$

Hence by [theorem: 14.375] $\{x_n\}_{n \in \{N, \dots, \infty\}}$ converges to $\inf(\{x_n \in \{N, \dots, \infty\}\})$. Which using [theorem: 14.341] proves that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ converges to $\inf(\{x_n \in \{N, \dots, \infty\}\})$. In other words

$$\lim_{n \rightarrow \infty} x_n \text{ exist and } M \leq \lim_{n \rightarrow \infty} x_n \quad \square$$

We introduce now a extension of the concept of a limit.

Definition 14.377. Let $k \in \mathbb{N}_0$, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ a sequence of real numbers then the limit superior and the limit inferior noted as $\limsup_{n \rightarrow \infty} x_i$ and $\liminf_{n \rightarrow \infty} x_i$ are defined as follows:

1. $\limsup_{n \rightarrow \infty} x_n$ exist if

- a. $\forall n \in \{k, \dots, \infty\} \sup(\{x_i | i \in \{n, \dots, \infty\}\})$ exist
- b. $\lim_{n \rightarrow \infty} \sup(\{x_i | i \in \{n, \dots, \infty\}\})$ exist

If $\limsup_{n \rightarrow \infty} x_n$ exist then $\limsup_{n \rightarrow \infty} x_n = \lim_{m \rightarrow \infty} \sup(\{x_i | i \in \{n, \dots, \infty\}\})$

2. $\liminf_{n \rightarrow \infty} x_n$ exist if

- a. $\forall n \in \{k, \dots, \infty\} \inf(\{x_i | i \in \{n, \dots, \infty\}\})$ exist
- b. $\lim_{n \rightarrow \infty} \inf(\{x_i | i \in \{n, \dots, \infty\}\})$ exist

If $\liminf_{n \rightarrow \infty} x_n$ exist then $\liminf_{n \rightarrow \infty} x_n = \lim_{m \rightarrow \infty} \inf(\{x_i | i \in \{n, \dots, \infty\}\})$

Theorem 14.378. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ such that $\liminf_{n \rightarrow \infty} x_i$ and $\limsup_{n \rightarrow \infty} x_i$ exists then

$$\liminf_{n \rightarrow \infty} x_i \leq \limsup_{n \rightarrow \infty} x_i$$

Proof. Let $n \in \{k, \dots, \infty\}$ then we have

$$\inf(\{x_i | i \in \{n, \dots, \infty\}\}) \leq x_n \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

So that by [theorem: 14.369]

$$\liminf_{n \rightarrow \infty} x_i \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \leq \lim_{n \rightarrow \infty} \sup(\{x_i | i \in \{n, \dots, \infty\}\}) \stackrel{\text{def}}{=} \limsup_{n \rightarrow \infty} x_i \quad \square$$

We have the following equivalent definition of the $\limsup$ and $\liminf$ of a sequence.

Theorem 14.379. Let $k \in \mathbb{N}$, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ a sequence of real numbers then

1.

$$\limsup_{n \rightarrow \infty} x_n \text{ exist}$$

$$\Updownarrow$$

$$\forall n \in \{k, \dots, \infty\} \sup(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exist}$$

and

$$\inf(\{\sup(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) \text{ exist}$$

Further if $\limsup_{n \rightarrow \infty} x_n$ exist then

$$\limsup_{n \rightarrow \infty} x_n = \inf(\{\sup(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\})$$

2.

$$\liminf_{n \rightarrow \infty} x_n \text{ exist}$$

$$\Updownarrow$$

$$\forall n \in \{k, \dots, \infty\} \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exist}$$

and

$$\sup(\{\inf(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) \text{ exist}$$

Further if $\liminf_{n \rightarrow \infty} x_n$ exist then

$$\liminf_{n \rightarrow \infty} x_n = \sup(\{\inf(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\})$$

Proof.

1. Assume that $\forall n \in \{k, \dots, \infty\} \sup(\{x_i | i \in \{n, \dots, \infty\}\})$ exist. Then $\forall n \in \{k, \dots, \infty\}$ we have $\{x_i | i \in \{n+1, \dots, \infty\}\} \subseteq \{x_i | i \in \{n, \dots, \infty\}\}$ so that by [theorem: 3.77]

$$\sup(\{x_i | i \in \{n+1, \dots, \infty\}\}) \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

proving that $\{\sup(\{x_i | i \in \{n, \dots, \infty\}\})\}_{n \in \{k, \dots, \infty\}}$ is a decreasing sequence. The rest of the proof follows then from [definition: 14.377] and [theorem: 14.375].

2. Assume that $\forall n \in \{k, \dots, \infty\} \inf(\{x_i | i \in \{n, \dots, \infty\}\})$ exist. Then $\forall n \in \{k, \dots, \infty\}$ we have $\{x_i | i \in \{n+1, \dots, \infty\}\} \subseteq \{x_i | i \in \{n, \dots, \infty\}\}$ so that by [theorem: 3.77]

$$\inf(\{x_i | i \in \{n, \dots, \infty\}\}) \leq \inf(\{x_i | i \in \{n+1, \dots, \infty\}\})$$

proving that $\{\inf(\{x_i | i \in \{n, \dots, \infty\}\})\}_{n \in \{k, \dots, \infty\}}$ is an increasing sequence. The rest of the proof follows then from [definition: 14.377] and [theorem: 14.375]. $\square$

We have the following relation between the limit of a sequence and the limit inferior and the limit superior of a sequence.

Theorem 14.380. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ then we have that

$$\lim_{n \rightarrow \infty} x_n \text{ exist} \Leftrightarrow \liminf_{n \rightarrow \infty} x_n \text{ and } \limsup_{n \rightarrow \infty} x_n \text{ exists and } \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n$$

Further if $\lim_{n \rightarrow \infty} x_i$ exist then

$$\lim_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_i = \limsup_{n \rightarrow \infty} x_n$$

Proof.

$\Rightarrow$. As $\lim_{n \rightarrow \infty} x_i$ exist we have by [theorem: 14.364] that there exist a $K \in \mathbb{R}^+$ such that $\forall i \in \{k, \dots, \infty\} |x_i| \leq K \Rightarrow -x_i, x_i \leq K \Rightarrow -K \leq x_i \leq K$. Hence $\forall n \in \{k, \dots, \infty\} \{x_i | i \in \{n, \dots, \infty\}\}$ is bounded below by $-K$ and above by K . So by the fact that $\mathbb{R}$ is conditional complete [see theorem: 10.18] it follows that

$$\sup(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ and } \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exists}$$

and

$$\inf(\{x_i | i \in \{n, \dots, \infty\}\}) \leq x_n \leq K \text{ and } -K \leq x_n \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

Using the above we have that $\{\inf(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}$ is bounded above by K and $\{\sup(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}$ is bounded below by $-K$. Using [theorem: 10.18] again it follows that $\sup(\{\inf(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\})$ and $\inf(\{\sup(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\})$ exists. Using then [theorem: 14.379] it follows that

$$\liminf_{n \rightarrow \infty} x_i \text{ and } \limsup_{n \rightarrow \infty} x_i \text{ exist} \tag{14.224}$$

and

$$\liminf_{n \rightarrow \infty} x_n = \sup (\{\inf (\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) \quad (14.225)$$

$$\limsup_{n \rightarrow \infty} x_n = \inf (\{\sup (\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) \quad (14.226)$$

Let $x = \lim_{n \rightarrow \infty} x_n$. Then giving $\varepsilon \in \mathbb{R}^+$ there exist a $n \in \{k, \dots, \infty\}$ such that $\forall i \in \{n, \dots, \infty\}$ we have $|x_i - x| < \varepsilon \Rightarrow x - \varepsilon < x_i < x + \varepsilon$. Hence $x - \varepsilon \leq \inf (\{x_i | i \in \{n, \dots, \infty\}\})$ or $x \leq \inf (\{x_i | i \in \{n, \dots, \infty\}\}) + \varepsilon$ and $\sup (\{x_i | i \in \{n, \dots, \infty\}\}) \leq x + \varepsilon$. Hence using [theorem: 10.31] we have that

$$x \leq \inf (\{x_i | i \in \{n, \dots, \infty\}\}) \text{ and } \sup (\{x_i | i \in \{n, \dots, \infty\}\}) \leq x$$

hence

$$x \leq \sup (\{\inf (\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) \stackrel{[\text{eq: 14.225}]}{=} \liminf_{n \rightarrow \infty} x_n$$

and

$$\limsup_{n \rightarrow \infty} x_n \stackrel{[\text{eq: 14.226}]}{=} \inf (\{\sup (\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) \leq x$$

which as by [theorem: 14.378] $\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n$ proves that

$$x \leq \liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n \leq x$$

so that

$$\lim_{n \rightarrow \infty} x_n = x = \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n$$

$\Leftarrow$. By the hypothesis $\liminf_{n \rightarrow \infty} x_n$ and $\limsup_{n \rightarrow \infty} x_n$ exist and $x = \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n$. Let $\varepsilon \in \mathbb{R}^+$.

Then $x - \varepsilon < x = \liminf_{n \rightarrow \infty} x_n \stackrel{[\text{theorem: 14.379}]}{=} \sup (\{\inf (\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\})$ so there exist a $N \in \{k, \dots, \infty\}$ such that $x - \varepsilon < \inf (\{x_i | i \in \{N, \dots, \infty\}\})$. In other words

$$\forall i \geq N \text{ we have } x - \varepsilon < x_i \quad (14.227)$$

Further $\inf (\{\sup (\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) = \limsup_{n \rightarrow \infty} x_n = x < x + \varepsilon$ so there exist a $M \in \{k, \dots, \infty\}$ such that $\sup (\{x_i | i \in \{M, \dots, \infty\}\}) < x + \varepsilon$. In other words

$$\forall i \geq M \text{ we have } x_i < x + \varepsilon \quad (14.228)$$

Let $i \geq \max(M, N)$ then by [eqs: 14.227 and 14.228] we have $x - \varepsilon < x_i < x + \varepsilon \Rightarrow |x_i - x| < \varepsilon$ which as $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary proves that

$$\lim_{n \rightarrow \infty} x_n \text{ exists and } \lim_{n \rightarrow \infty} x_n = x \quad \square$$

Theorem 14.381. Let $k \in \mathbb{N}$, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ a sequence of real numbers then we have

1. If $\liminf_{n \rightarrow \infty} x_n$ exists then $\limsup_{n \rightarrow \infty} (-x_n)$ exists and $\liminf_{n \rightarrow \infty} x_n = -\limsup_{n \rightarrow \infty} (-x_n)$
2. If $\limsup_{n \rightarrow \infty} x_n$ exists then $\liminf_{n \rightarrow \infty} (-x_n)$ exists and $\limsup_{n \rightarrow \infty} x_n = -\liminf_{n \rightarrow \infty} (-x_n)$
3. If $\limsup_{n \rightarrow \infty} x_n$ exists and $\alpha \in [0, \infty[$ then $\limsup_{n \rightarrow \infty} (\alpha \cdot x_n)$ exists and

$$\limsup_{n \rightarrow \infty} (\alpha \cdot x_n) = \alpha \cdot \limsup_{n \rightarrow \infty} x_n$$

4. If $\liminf_{n \rightarrow \infty} x_n$ exists and $\alpha \in [0, \infty[$ then $\liminf_{n \rightarrow \infty} (\alpha \cdot x_n)$ exists and $\liminf_{n \rightarrow \infty} (\alpha \cdot x_n) = \alpha \cdot \liminf_{n \rightarrow \infty} x_n$
5. If $x \in \mathbb{R}$ and $\liminf_{n \rightarrow \infty} x_n$ exists then for $\{x_n + x\}_{n \in \{k, \dots, \infty\}}$ $\liminf_{n \rightarrow \infty} (x_n + x)$ exists and

$$\liminf_{n \rightarrow \infty} (x_n + x) = \liminf_{n \rightarrow \infty} x_n + x$$

6. If $x \in \mathbb{R}$ and $\limsup_{n \rightarrow \infty} x_n$ exists then for $\{x_n + x\}_{n \in \{k, \dots, \infty\}}$ $\limsup_{n \rightarrow \infty} (x_n + x)$ exists and

$$\limsup_{n \rightarrow \infty} (x_n + x) = \limsup_{n \rightarrow \infty} x_n + x$$

Proof.

1. As $\liminf_{n \rightarrow \infty} x_i$ exists we have

$$\forall n \in \{k, \dots, \infty\} \text{ that } \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exists} \quad (14.229)$$

Using [theorem: 10.21] we have then that $\sup(\{-x_i | i \in \{n, \dots, \infty\}\})$ exists and

$$\forall n \in \mathbb{N}_0 \in \{k, \dots, \infty\} \sup(\{-x_i | i \in \{n, \dots, \infty\}\}) = -\inf(\{x_i | i \in \{n, \dots, \infty\}\}) \quad (14.230)$$

Again using the existence of $\liminf_{n \rightarrow \infty} x_i$ we have the existence of $\liminf_{n \rightarrow \infty} (\{x_i | i \in \{n, \dots, \infty\}\})$

and further we have

$$\begin{aligned} \liminf_{n \rightarrow \infty} x_n &= \liminf_{n \rightarrow \infty} (\{x_i | i \in \{n, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 14.230}]}{=} \lim_{n \rightarrow \infty} (-\sup(\{-x_i | i \in \{n, \dots, \infty\}\})) \\ &\stackrel{[\text{theorem: 14.358}]}{=} -\lim_{n \rightarrow \infty} (\sup(\{-x_i | i \in \{n, \dots, \infty\}\})) \\ &\stackrel{\text{def}}{=} -\limsup_{n \rightarrow \infty} (-x_n) \end{aligned}$$

2. As $\limsup_{n \rightarrow \infty} x_i$ exists we have

$$\forall n \in \{k, \dots, \infty\} \sup(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exists} \quad (14.231)$$

Using [theorem: 10.21] we have then that $\inf(\{-x_i | i \in \{n, \dots, \infty\}\})$ exists and

$$\forall n \in \mathbb{N}_0 \in \{k, \dots, \infty\} \inf(\{-x_i | i \in \{n, \dots, \infty\}\}) = -\sup(\{x_i | i \in \{n, \dots, \infty\}\}) \quad (14.232)$$

Again using the existence of $\limsup_{n \rightarrow \infty} x_i$ we have the existence of $\limsup_{n \rightarrow \infty} (\{x_i | i \in \{n, \dots, \infty\}\})$,

further we have

$$\begin{aligned} \limsup_{n \rightarrow \infty} x_n &= \limsup_{n \rightarrow \infty} (\{x_i | i \in \{n, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 14.232}]}{=} \lim_{n \rightarrow \infty} (-\inf(\{-x_i | i \in \{n, \dots, \infty\}\})) \\ &\stackrel{[\text{theorem: 14.358}]}{=} -\lim_{n \rightarrow \infty} (\inf(\{-x_i | i \in \{n, \dots, \infty\}\})) \\ &\stackrel{\text{def}}{=} -\liminf_{n \rightarrow \infty} (-x_n) \end{aligned}$$

3. As $\limsup_{n \rightarrow \infty} x_i$ exists we have

$$\forall n \in \{k, \dots, \infty\} \sup(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exists}$$

Using [theorem: 10.22] we have that $\sup(\{\alpha \cdot x_i | i \in \{n, \dots, \infty\}\})$ exists and is equal to $\alpha \cdot \sup(\{x_i | i \in \{n, \dots, \infty\}\})$. So as $\limsup_{n \rightarrow \infty} (\{x_i | i \in \{n, \dots, \infty\}\})$ exists we have by [theorem:

14.358] that $\lim_{n \rightarrow \infty} (\alpha \cdot \sup(\{x_i | i \in \{n, \dots, \infty\}\}))$ exists and

$$\begin{aligned} \alpha \cdot \limsup_{n \rightarrow \infty} x_n &= \alpha \cdot \limsup_{n \rightarrow \infty} (\{\alpha \cdot x_i | i \in \{n, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 14.358}]}{=} \lim_{n \rightarrow \infty} (\alpha \cdot \sup(\{x_i | i \in \{n, \dots, \infty\}\})) \\ &= \lim_{n \rightarrow \infty} (\sup(\{\alpha \cdot x_i | i \in \{n, \dots, \infty\}\})) \\ &= \limsup_{n \rightarrow \infty} (\alpha \cdot x_n) \end{aligned}$$

4. As $\liminf_{n \rightarrow \infty} x_i$ exists we have

$$\forall n \in \{k, \dots, \infty\} \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exists}$$

Using [theorem: 10.22] we have that $\inf(\{\alpha \cdot x_i | i \in \{n, \dots, \infty\}\})$ exists and is equal to $\alpha \cdot \inf(\{x_i | i \in \{n, \dots, \infty\}\})$. So as $\liminf_{n \rightarrow \infty}(\{x_i | i \in \{n, \dots, \infty\}\})$ exists we have by [theorem: 14.358] that $\lim_{n \rightarrow \infty}(\alpha \cdot \inf(\{x_i | i \in \{n, \dots, \infty\}\}))$ exists and

$$\begin{aligned} \alpha \cdot \liminf_{n \rightarrow \infty} x_n &= \alpha \cdot \lim_{n \rightarrow \infty} \inf(\{\alpha \cdot x_i | i \in \{n, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 14.358}]}{=} \lim_{n \rightarrow \infty} (\alpha \cdot \inf(\{x_i | i \in \{n, \dots, \infty\}\})) \\ &= \lim_{n \rightarrow \infty} (\inf(\{\alpha \cdot x_i | i \in \{n, \dots, \infty\}\})) \\ &= \liminf_{n \rightarrow \infty} (\alpha \cdot x_n) \end{aligned}$$

5. As $\liminf_{n \rightarrow \infty} x_i$ exists we have

$$\forall n \in \{k, \dots, \infty\} \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exist}$$

Using [theorem: 10.24] we have that $\inf(\{x_i + x | i \in \{n, \dots, \infty\}\})$ exists and

$$\inf(\{x_i + x | i \in \{n, \dots, \infty\}\}) = \inf(\{x_i | i \in \{n, \dots, \infty\}\}) + x$$

Hence

$$\begin{aligned} \liminf_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} (\inf(\{x_i + x | i \in \{n, \dots, \infty\}\})) \\ &= \lim_{n \rightarrow \infty} (\inf(\{x_i | i \in \{n, \dots, \infty\}\}) + x) \\ &\stackrel{[\text{theorem: 14.356}]}{=} \lim_{n \rightarrow \infty} \inf(\{x_i | i \in \{n, \dots, \infty\}\}) + x \\ &= \liminf_{n \rightarrow \infty} x_i + x \end{aligned}$$

6. As $\limsup_{n \rightarrow \infty} x_i$ exists we have

$$\forall n \in \{k, \dots, \infty\} \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

Using [theorem: 10.24] we have that $\sup(\{x_i + x | i \in \{n, \dots, \infty\}\})$ exists and

$$\sup(\{x_i + x | i \in \{n, \dots, \infty\}\}) = \sup(\{x_i | i \in \{n, \dots, \infty\}\}) + x.$$

Hence

$$\begin{aligned} \limsup_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} (\sup(\{x_i + x | i \in \{n, \dots, \infty\}\})) \\ &= \lim_{n \rightarrow \infty} (\sup(\{x_i | i \in \{n, \dots, \infty\}\}) + x) \\ &\stackrel{[\text{theorem: 14.356}]}{=} \lim_{n \rightarrow \infty} \sup(\{x_i | i \in \{n, \dots, \infty\}\}) + x \\ &= \limsup_{n \rightarrow \infty} x_i + x \end{aligned}$$

□

14.10 Complete spaces

Next we define a necessary condition for a sequence to be convergent in a metric or normed space.

Definition 14.382. (Cauchy Condition) Let $\langle X, d \rangle$ be a metric space, $k \in \mathbb{N}_0$ then a sequence $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ is **Cauchy** if

$\forall \varepsilon \in \mathbb{R}^+$ there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \in \mathbb{N}$ with $N \leq n, m$ we have $d(x_n, x_m) < \varepsilon$

For a normed space the above definition becomes

Definition 14.383. (Cauchy condition) Let $\langle X, ||| \rangle$ be a metric space, $k \in \mathbb{N}_0$ then a sequence $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ is **Cauchy** if

$\forall \varepsilon \in \mathbb{R}^+$ there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \in \mathbb{N}_0$ with $N \leq n, m$ we have $\|x_n - x_m\| < \varepsilon$

Convergence and Cauchy are topological constructs as is show in the following theorem.

Theorem 14.384. Let X be a set with two equivalent norms $|||_1$ and $|||_2$ then for a sequence $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ we have

1. $\{x_n\}_{n \in \{k, \dots, \infty\}}$ converges to x using $|||_1 \Leftrightarrow \{x_n\}_{n \in \{k, \dots, \infty\}}$ converges to x using $|||_2$
2. $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy using $|||_1 \Leftrightarrow \{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy using $|||_2$

Proof. First note that we only have to prove $\Rightarrow$ (as $\Leftarrow$ follows from applying the theorem in the opposite direction. Next as $|||_1$ and $|||_2$ are equivalent we have by [theorem: 14.121] there exists $M_1, M_2 \in \mathbb{R}^+$ such that $\forall x \in X$ we have $M_1 \cdot \|x\|_1 \leq \|x\|_2 \leq M_2 \cdot \|x\|_1$. We have then:

1. Let $\varepsilon > 0$ then by convergence in $|||_1$ there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|x - x_n\|_1 < \frac{\varepsilon}{M_2}$ so that $\|x - x_n\|_2 \leq M_2 \cdot \|x - x_n\|_1 < M_2 \cdot \frac{\varepsilon}{M_2} = \varepsilon$
2. Let $\varepsilon > 0$ then by the Cauchy property in $|||_1$ there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have $\|x_n - x_m\|_1 < \frac{\varepsilon}{M_2}$ and thus $\|x_n - x_m\|_2 \leq M_2 \cdot \|x_n - x_m\|_1 < M_2 \cdot \frac{\varepsilon}{M_2} = \varepsilon \quad \square$

A Cauchy sequence in a normed space is bounded.

Theorem 14.385. Let $\langle X, ||| \rangle$ be a normed space, $k \in \mathbb{N}$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a Cauchy sequence then there exists a $K \in \mathbb{R}^+$ such that $\forall i \in \{k, \dots, \infty\}$ we have $\|x_i\| \leq K$

Proof. As $1 \in \mathbb{R}^+$ then exist a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \in \mathbb{N}_0$ with $N \leq n, m$ we have $\|x_n - x_m\| < 1$. Let $M = \max(\max(\{\|x_i\| \mid i \in \{k, \dots, N\}\}), \|x_N\| + 1)$ then for $i \in \{k, \dots, \infty\}$ we have either:

$i \in \{k, \dots, N\}$. Then $\|x_i\| \leq \max(\{\|x_i\| \mid i \in \{k, \dots, N\}\}) \leq M$

$N < i$. Then $\|x_i\| = \|x_i - x_N + x_N\| \leq \|x_i - x_N\| + \|x_N\| < 1 + \|x_N\| \leq M$

proving that $\forall i \in \{k, \dots, \infty\}$ we have $\|x_i\| \leq M$. $\square$

We prove now that the Cauchy condition is indeed necessary for convergence.

Theorem 14.386. If $\langle X, d \rangle$ is a metric space then every convergent sequence is Cauchy. Likewise if $\langle X, ||| \rangle$ is a normed space then every convergent sequence is Cauchy.

Proof. Let $\langle X, d \rangle$ be a metric space and let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ be a sequence that converges to x then for $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $d(x, x_n) < \frac{\varepsilon}{2}$. So if $n, m \in \mathbb{N}$ with $n, m \geq N$ then we have

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy. Likewise let $\langle X, ||| \rangle$ be a normed space and let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ be a sequence that converges to x then for $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|x_n - x\| < \frac{\varepsilon}{2}$. So if $n, m \in \mathbb{N}$ with $n, m \geq N$ then we have

$$\|x_m - x_n\| = \|x_m - x + x - x_n\| \leq \|x_m - x\| + \|x - x_n\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy. $\square$

Example 14.387. Take $]0, \infty[$ and define $d:]0, \infty[\times]0, \infty[\rightarrow \mathbb{R}$ by $d(x, y) = |x - y|$ then

$\langle]0, \infty[\times]0, \infty[, d \rangle$ is a metric space

and further $\{\frac{1}{n}\}_{n \in \mathbb{N}}$ is Cauchy but does not converges

Proof. As $\langle \mathbb{R}, d_{||} \rangle$ is a metric space and $d = (d_{||})_{||0, \infty[\times]0, \infty[}$ it follows from [theorem: 14.70] that $\langle]0, \infty[, d \rangle$ is a metric space. As by [example: 14.349] $\{\frac{1}{n}\}_{n \in \mathbb{N}}$ converges to 0 in $\langle \mathbb{R}, d_{||} \rangle$ it is Cauchy in $\langle \mathbb{R}, d_{||} \rangle$. Hence $\forall \varepsilon \in \mathbb{R}^+$ there exist a $N \in \mathbb{R}^+$ such that $\forall n, m \in \{N, \dots, \infty\}$ we have $d(x_n, x_m) = d_{||}(x_n, x_m) < \varepsilon$ proving that $\{\frac{1}{n}\}_{n \in \mathbb{N}}$ is Cauchy in $\langle]0, \infty[, d \rangle$. However as $\{\frac{1}{n}\}_{n \in \mathbb{N}}$ converges to 0 in $\langle \mathbb{R}, d_{||} \rangle$ and $0 \notin]0, \infty[$ it follows that $\{\frac{1}{n}\}_{n \in \mathbb{N}}$ does not converges in $\langle]0, \infty[, d \rangle$. $\square$

The above example proves that although every convergent sequence is Cauchy the opposite is not always true. So if we want that Cauchy is a sufficient condition for converging we must state this as a extra property for the metric or the norm.

Definition 14.388. A metric space $\langle X, d \rangle$ or a normed space $\langle X, ||| \rangle$ is **complete** if every Cauchy sequence is convergent. A complete normed space is called a **Banach** space.

Theorem 14.389. Let $n \in \mathbb{N}$, $\{\langle X_i, |||_i \rangle\}_{i \in \{1, \dots, n\}}$ a family of Banach spaces then if $|||_{\max}$ is the maximum norm [see theorem: 14.98] then

$$\left\langle \prod_{i \in \{1, \dots, n\}} X_i, |||_{\max} \right\rangle \text{ is a Banach space}$$

Proof. Using [theorem: 14.98] we have that $\langle \prod_{i \in \{1, \dots, n\}} X_i, |||_{\max} \rangle$ is a normed space. Let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \prod_{i \in \{1, \dots, n\}} X_i$ be a Cauchy sequence. Take $\varepsilon \in \mathbb{R}^+$ then there exists a $N \in \{k, \dots, \infty\}$ such that $\forall k, l \geq N$ we have $\max(\{\|\pi_i(x_k - x_l)\|_i | i \in \{1, \dots, n\}\}) = \|x_k - x_l\|_{\max} < \varepsilon$. Hence $\forall i \in \{1, \dots, n\}$ we have $\|\pi_i(x_k) - \pi_i(x_l)\|_i = \|\pi_i(x_k - x_l)\|_i < \varepsilon$ which as $\langle X_i, |||_i \rangle$ is Banach proves that there exist a $y_i \in X_i$ such that $\lim_{k \rightarrow \infty} \pi_i(x_k) = y_i$. In other words given $\varepsilon \in \mathbb{R}^+$ there exist a $N_i \in \{k, \dots, \infty\}$ such that $\forall k \geq N_i$ we have $\|\pi_i(x_k) - y_i\|_i < \varepsilon$. Take $y = (y_1, \dots, y_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ then we have that for $k \geq \max(\{N_i | i \in \{1, \dots, n\}\})$ that

$$\|x_k - y\|_{\max} = \max(\{\|\pi_i(x_k - y)\|_i | i \in \{1, \dots, n\}\}) = \max(\{\|\pi_i(x_k) - y_i\|_i | i \in \{1, \dots, n\}\}) < \varepsilon$$

proving that $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges. $\square$

Next we prove that in every metric space the limit point of $\{x_n | n \in \{k, \dots, \infty\}\}$ where $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is a Cauchy sequence. To prove this we first need a little lemma.

Lemma 14.390. Let $k \in \mathbb{N}_0$, $\langle X, d \rangle$ be a metric space, $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a sequence and x a limit point of $\{x_n | n \in \{k, \dots, \infty\}\}$ then $\forall \varepsilon \in \mathbb{R}^+$, $\forall N \in \{k, \dots, \infty\}$ there exist a $m > N$ such that $d(x, x_m) < \varepsilon$

Proof. Let $\varepsilon \in \mathbb{R}^+$ and define

$$A_\varepsilon = \{n \in \{k, \dots, \infty\} | 0 < d(x, x_i) < \varepsilon\} \subseteq \{k, \dots, \infty\}$$

Assume that A_ε is finite. As x is a limit point of $\{x_n | n \in \{k, \dots, \infty\}\}$ and $x \in B_d(x, \varepsilon)$ a open set we have that $B_d(x, \varepsilon) \cap (\{x_n | n \in \{k, \dots, \infty\}\} \setminus \{x\}) \neq \emptyset$, hence there exists a $i \in \{k, \dots, \infty\}$ such that

$$x_i \neq x \Rightarrow 0 < d(x_i, x) \text{ and } d(x, x_i) < \varepsilon$$

proving that

$$A_\varepsilon \neq \emptyset$$

Define $B_\varepsilon = \{d(x, x_n) | n \in A_\varepsilon\} \xrightarrow{A_\varepsilon \neq \emptyset} B_\varepsilon \neq \emptyset$ and $f: A_\varepsilon \rightarrow B_\varepsilon$ by $f(n) = d(x, x_n)$ is a surjection, as A_ε is assumed to be finite it follows from [theorem: 6.44] that

$$\emptyset \neq B_\varepsilon = \{d(x, x_n) | n \in A_\varepsilon\} \text{ is finite}$$

Hence using [theorem: 6.49]

$$\delta = \min(B_\varepsilon) \text{ exists}$$

As $\delta \in B_\varepsilon$ there exist a $n \in A_\varepsilon$ such that $\delta = d(x, x_n)$ hence

$$0 < \delta = d(x, x_n) < \varepsilon$$

Again using the fact that x is a limit point of $\{x_n | n \in \{k, \dots, \infty\}\}$ and $x \in B_d(x, \delta)$ there exist a $j \in \{k, \dots, \infty\}$ such that

$$x_j = x \Rightarrow 0 < d(x, x_j) \text{ and } d(x, x_j) < \delta < \varepsilon$$

hence $j \in A_\varepsilon$ so that $d(x, x_j) \in B_\varepsilon$, as $\delta = \min(B_\varepsilon)$ we have $\delta \leq d(x, x_j) < \delta$ leading to the contradiction $\delta < \delta$. So the assumption that A_ε is finite is false proving that

A_ε is infinite

Assume now that there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n > N$ we have $\varepsilon \leq d(x, x_n)$ then $n \notin A_\varepsilon$ hence $A_\varepsilon \subseteq \{k, \dots, N\} \Rightarrow A_\varepsilon$ is finite which contradicts the fact that A_ε is infinite. So the assumption is wrong and we have that

$$\forall N \in \{k, \dots, \infty\} \text{ there exist a } n > N \text{ such that } d(x, x_n) < \varepsilon$$

proving the theorem. □

Theorem 14.391. Let $k \in \mathbb{N}_0, \langle X, d \rangle$ a metric space and $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ a Cauchy sequence then if x and y are limit points of $\{x_n | n \in \{k, \dots, \infty\}\}$ we have that $x = y$.

Proof. Let x, y be limit points of $\{x_n | n \in \{k, \dots, \infty\}\}$ and assume that $x \neq y$. As $x \neq y$ we have that $\varepsilon = d(x, y) \in \mathbb{R}^+$. By the Cauchy condition there exist a $N \in \{k, \dots, \infty\}$ such that $\forall i, j \geq N$ we have $d(x_i, x_j) < \frac{\varepsilon}{3}$. Using the previous lemma [lemma: 14.390] there exists $n, m > N$ such that $d(x, x_n) < \frac{\varepsilon}{3}$ and $d(y, x_m) < \frac{\varepsilon}{3}$. Hence

$$\varepsilon = d(x, y) \leq d(x, x_n) + d(x_n, y) \leq d(x, x_n) + d(x_n, x_m) + d(x_m, y) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \quad \square$$

14.10.1 Examples of complete spaces

First we prove that every compact metric space is complete, to do this we first need a lemma about limit points and sequences.

Theorem 14.392. Every compact metric space $\langle X, d \rangle$ is complete.

Proof. Let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ be a Cauchy sequence in X and consider $S = \{x_n | n \in \{k, \dots, \infty\}\}$ then for S we must consider the following cases:

S is finite. Assume that $\forall N \in \{k, \dots, \infty\}$ there exists $i_0, j_0 \geq N$ with $x_{i_0} \neq x_{j_0}$. Define

$$A = \{d(x, y) | (x, y) \in S \times S \wedge x \neq y\} \subseteq \{d(x, y) | (x, y) \in S\}$$

Now $S \times S$ is finite by [theorem: 6.41] and $f: S \times S \rightarrow \{d(x, y) | (x, y) \in S\}$ defined by $f(x, y) = d(x, y)$ is a surjection so that by [theorem: 6.44] $\{d(x, y) | (x, y) \in S\}$ is finite and thus that A is finite. Further by the assumption there exist $i, j \geq k$ such that $x_i \neq x_j$ proving that $d(x_i, x_j) \in A$ and thus that $A \neq \emptyset$. So $\varepsilon = \min(A)$ exist and $0 < \varepsilon$ [for if $d \in A \Rightarrow \exists x, y \in S$ with $x \neq y \Rightarrow d = d(x, y) > 0$]. As $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy there exist a $N \in \{k, \dots, \infty\}$ such that $\forall i, j \geq N$ we have $d(x_i, x_j) < \varepsilon$. By the assumption there exist a $i_0, j_0 \geq N$ such that $x_{i_0} \neq x_{j_0} \Rightarrow d(x_{i_0}, x_{j_0}) \in A$, hence $\varepsilon = \min(S) \leq d(x_{i_0}, y_{i_0}) < \varepsilon$ a contradiction. So the assumption must be false meaning that

$$\exists N \in \{k, \dots, \infty\} \text{ such that } \forall i, j \geq N \ x_i = x_j \Rightarrow d(x_i, x_j) = 0$$

or taking $j = N$ and $x = x_N$ we have that

$$\forall i \geq N \ d(x_i, x) = 0$$

proving as $\forall \varepsilon \in \mathbb{R}^+ 0 < \varepsilon$ that

$$\lim_{n \rightarrow \infty} x_n = x$$

S is infinite. Then as $S \subseteq X$ a compact space we have using [theorem: 14.285] and [definition: 14.284] that there exist a limit point for S which because of [theorem: 14.391] is unique. Let's call this unique limit point of S x . Take $\varepsilon > 0$ then by the Cauchy property there exist a $N \in \{k, \dots, \infty\}$ such that $\forall i, j \geq N$ we have $d(x_i, x_j) < \frac{\varepsilon}{2}$. By [lemma: 14.390] there exist a $n > N$ such that $d(x, x_n) < \frac{\varepsilon}{2}$. Hence if $m \geq N$ we have that

$$d(x, x_m) \leq d(x, x_n) + d(x_n, x_m) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \quad \square$$

Theorem 14.393. Let $\langle X, d \rangle$ be a complete space and $A \subseteq X$ where A is closed then $\langle A, d|_A \rangle$ is complete.

Proof. Let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq A$ be a Cauchy sequence in $\langle A, d|_A \rangle$ then given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have that $d(x_n, x_m) = d|_A(x_n, x_m) < \varepsilon$. So $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy in X a complete space, hence $\{x_n\}_{n \in \{k, \dots, \infty\}}$ converges to $x \in X$ or

$$\forall \varepsilon \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\} \text{ such that } d(x_n, x) < \varepsilon \quad (14.233)$$

Let $U \in \mathcal{T}_d$ such that $x \in A$ then there exist a $\delta \in \mathbb{R}^+$ such that $x \in B_d(x, \delta) \subseteq U$. By [eq: 14.233] there exist a $N \in \{k, \dots, \infty\}$ such that $d(x_N, x) < \delta$ hence $x_N \in B_d(x, \varepsilon) \subseteq U$ proving as $x_N \in A$ that $A \cap U \neq \emptyset$. So using [theorem: 14.31] it follows that

$$x \in A$$

Hence if $\zeta \in \mathbb{R}^+$ we have by [eq: 14.233] that there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ $d(x, x_n) < \varepsilon \Rightarrow_{x_n, x \in A} d|_A(x, x_n) = d(x_n, x) < \zeta$ proving that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ converges in A . $\square$

A very important example of a complete normed space is the set of real numbers with the absolute value norm.

Theorem 14.394. The normed space $\langle \mathbb{R}, || \rangle$ is a Banach space.

Proof. Let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ be a Cauchy sequence in $\mathbb{R}$ then by [theorem: 14.385] there exist a $K \in \mathbb{R}^+$ such that $\forall n \in \{k, \dots, \infty\} |x_n| \leq K$ so that

$$\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq [-K, K] \quad (14.234)$$

By [theorem: 14.279] $[-K, K]$ is a compact subset of $\mathbb{R}$, hence by definition $[-K, K]$ is a compact space using the sub-space topology on $[-K, K]$. By the previous theorem [theorem: 14.392] it follows that

$$\langle [-K, K], (d|_{[-K, K]}) \rangle \text{ is complete}$$

Using [theorem: 14.115] the sub-space topology is defined by the metric

$$(d|_{[-K, K]}): [-K, K] \times [-K, K] \rightarrow \mathbb{R} \text{ where } (d|_{[-K, K]})(x, y) = |x - y| \quad (14.235)$$

Given $\varepsilon \in \mathbb{R}^+$ we have as $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy a $N \in \{k, \dots, \infty\}$ such that $\forall i, j \geq N$ we have

$$|x_i - x_j| < \varepsilon \Rightarrow_{[\text{eqs: 14.234, 14.235}]} (d|_{[-K, K]})(x_i - x_j) < \varepsilon$$

proving that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy in $\langle [-K, K], (d|_{[-K, K]}) \rangle$. Hence there exist a $x \in [-K, K]$ such that $\lim_{n \rightarrow \infty} x_n = x$ [using $(d|_{[-K, K]})$]. Hence given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$

$$|x_n - x| \underset{[\text{eqs: 14.234, 14.235}]}{=} (d|_{[-K, K]})(x_n - x) < \varepsilon$$

proving that

$$\{x_n\}_{n \in \{k, \dots, \infty\}} \text{ converges to } x \text{ using } \langle \mathbb{R}, || \rangle \quad \square$$

Corollary 14.395. *Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$ then $\langle \mathbb{R}^n, \|\cdot\| \rangle$ is a Banach space.*

Proof. Let $\|\cdot\|_n$ be the maximum norm defined by $\|x\|_n = \max(\{|\pi_i(x)| | i \in \{1, \dots, n\}\})$ then as $\langle \mathbb{R}, \|\cdot\| \rangle$ is a Banach space by the previous theorem [theorem: 14.394] we have by [theorem: 14.389] that

$$\langle \mathbb{R}^n, \|\cdot\|_n \rangle \text{ is a Banach space} \quad (14.236)$$

Now using [theorem: 14.325]

$$\|\cdot\|_n \text{ is equivalent with } \|\cdot\| \quad (14.237)$$

Let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}^n$ be a Cauchy sequence using $\|\cdot\|$ then by [theorem: 14.384] and [eq: 14.237] it follows that $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is a Cauchy sequence, hence by [eq: 14.236] $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges. Finally using [theorem: 14.384] and [eq: 14.237] it follows that

$$\{x_i\}_{i \in \{k, \dots, \infty\}} \text{ converges in } \langle \mathbb{R}^n, \|\cdot\| \rangle$$

proving that $\langle \mathbb{R}^n, \|\cdot\| \rangle$ is a Banach space. $\square$

Corollary 14.396. *Every finite dimensional normed vector space over $\mathbb{R}$ is a Banach space*

Proof. Let $\langle X, \|\cdot\|_X \rangle$ be a n -dimensional normed vector space then using [theorem: 14.197] there exist a norm $\|\cdot\|$ on $\mathbb{R}^n$ and a linear isometric isomorphism $\varphi: X \rightarrow \mathbb{R}^n$ using the norms $\|\cdot\|_X$ and $\|\cdot\|$. Let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ be a Cauchy sequence in $\langle X, \|\cdot\|_X \rangle$. Then given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall i, j \geq N$ $\|x_i - x_j\|_X < \varepsilon$. Consider now the sequence $\{\varphi(x_i)\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}^n$ then $\|\varphi(x_i) - \varphi(x_j)\| = \|\varphi(x_i - x_j)\| = \|x_i - x_j\|_X < \varepsilon$ proving that $\{\varphi(x_i)\}_{i \in \{k, \dots, \infty\}}$ is a Cauchy sequence in $\langle \mathbb{R}^n, \|\cdot\| \rangle$. As $\langle \mathbb{R}^n, \|\cdot\| \rangle$ is a Banach space by [theorem: 14.395] so that $\{\varphi(x_i)\}_{i \in \{k, \dots, \infty\}}$ converges to a $y \in \mathbb{R}^n$. Hence given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m \geq N$ we have $\|\varphi(x_m) - y\| < \varepsilon$. Take $x = \varphi^{-1}(y)$ then we have $\|x_m - x\|_X = \|\varphi(x_m - x)\| = \|\varphi(x_m) - \varphi(x)\| = \|\varphi(x_m) - y\| < \varepsilon$ proving that $\{x_i\}_{i \in \{k, \dots, \infty\}}$ converges to x . Hence $\langle X, \|\cdot\|_X \rangle$ is a Banach space. $\square$

Theorem 14.397. *The normed space $\langle \mathbb{C}, \|\cdot\| \rangle$ is a Banach space [where $\langle \mathbb{C}, \|\cdot\| \rangle$ is defined in [example: 14.102]]*

Proof. To avoid any confusion between the norm on $\mathbb{R}$ and $\mathbb{C}$ we use $|\cdot|$ for the absolute value norm on $\mathbb{R}$ and $\|\cdot\|_c$ for the canonical norm on $\mathbb{C}$. Let $\{z_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathbb{C}$ be a Cauchy sequence in $\langle \mathbb{C}, \|\cdot\|_c \rangle$ then $\forall n \in \{k, \dots, \infty\}$ we have that $z_n = x_n + i \cdot y_n$ where $x_n, y_n \in \mathbb{R}$. Now $\forall n, m \in \{k, \dots, \infty\}$ we have by [theorem: 10.85 (3)] that

$$|x_m - x_n| = |\operatorname{Re}(z_m - z_n)| < |z_m - z_n|_c \text{ and } |y_m - y_n|_c = |\operatorname{Im}(z_m - z_n)| < |z_m - z_n|_c \quad (14.238)$$

By the Cauchy condition there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have that $|z_n - z_m|_c < \varepsilon$, combining this with [eqs: 14.238] gives $|x_m - x_n| < \varepsilon \wedge |y_m - y_n| < \varepsilon$. Hence

$$\{x_n\}_{n \in \{k, \dots, \infty\}}, \{y_n\}_{n \in \{k, \dots, \infty\}} \text{ are Cauchy in } \langle \mathbb{R}, \|\cdot\| \rangle$$

As by [theorem: 14.394] $\langle \mathbb{R}, \|\cdot\| \rangle$ is complete there exists $x, y \in \mathbb{R}$ such that

$$\lim_{n \rightarrow \infty} x_n = x \wedge \lim_{n \rightarrow \infty} y_n = y$$

Hence using [theorem: 14.361] it follows that $\{z_n\}_{n \in \{k, \dots, \infty\}}$ converges to z . $\square$

Theorem 14.398. *Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space then the normed space $\langle L(X, Y), \|\cdot\|_{L(X, Y)} \rangle$ [see theorem: 14.210] is a Banach space.*

Proof. Let $\{L_n\}_{n \in \{k, \dots, \infty\}} \subseteq L(X, Y)$ be a Cauchy sequence. Let $x \in X$ then given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have $\|L_n - L_m\|_{L(X, Y)} < \frac{\varepsilon}{\|x\| + 1}$. Hence

$$\|L_n(x) - L_m(x)\|_Y = \|(L_n - L_m)(x)\|_Y \leq_{\text{[theorem: 14.210]}} \|L_n - L_m\|_{L(X, Y)} \cdot \|x\|_X < \frac{\varepsilon}{\|x\| + 1} \cdot \|x\|_X < \varepsilon$$

proving that

$$\{L_n(x)\}_{n \in \{k, \dots, \infty\}} \text{ is a Cauchy sequence in } \langle Y, \|\cdot\|_Y \rangle \quad (14.239)$$

As $\langle Y, \|\cdot\|_Y \rangle$ is a Banach space $\lim_{n \rightarrow \infty} L_n(x)$ exist, allowing us to define

$$L: X \rightarrow Y \text{ by } L(x) = \lim_{n \rightarrow \infty} L_n(x) \quad (14.240)$$

Let $x, y \in X$ and $\alpha \in \mathbb{K}$ then we have $\{L_n(x + \alpha \cdot y)\}_{n \in \{k, \dots, \infty\}} = \{L_n(x) + \alpha \cdot L_n(y)\}_{n \in \{k, \dots, \infty\}}$. Using [theorems: 14.358, 14.357] $\{L_n(x + \alpha \cdot y)\}_{n \in \{k, \dots, \infty\}}$ converges and

$$\lim_{n \rightarrow \infty} L_n(x + \alpha \cdot y) = \lim_{n \rightarrow \infty} L_n(x) + \alpha \cdot \lim_{n \rightarrow \infty} L_n(y)$$

proving that

$$L(x + \alpha \cdot y) = L(x) + \alpha \cdot L(y)$$

or that

$$L \in \text{Hom}(X, Y) \quad (14.241)$$

As $\{L_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy there exist by [theorem: 14.385] a $K \in \mathbb{R}^+$ such that $\forall n \in \{k, \dots, \infty\}$ we have $\|L_n\|_{L(X, Y)} \leq K$. Hence $\forall n \in \{k, \dots, \infty\}$ we have for $x \in X$

$$\|L_n(x)\|_Y \leq_{[\text{theorem: 14.210}]} \|L_n\|_{L(X, Y)} \cdot \|x\|_X \leq K \cdot \|x\|_X$$

which by [theorem: 14.363] proves

$$\|L(x)\|_Y = \left\| \lim_{n \rightarrow \infty} L_n(x) \right\|_Y \leq K \cdot \|x\|_X \quad (14.242)$$

Hence using [eq: 14.241] and [theorem: 14.201] it follows that

$$L \in L(X, Y) \quad (14.243)$$

Let $\varepsilon \in \mathbb{R}^+$ then for $x \in X$ with $\|x\|_X = 1$ we have, as $L(x) = \lim_{n \rightarrow \infty} L_n(x)$, that there exist a $N_x \in \{k, \dots, \infty\}$ such that

$$\forall n \geq N_x \text{ we have } \|L_n(x) - L(x)\| < \frac{\varepsilon}{4}$$

As $\{L_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have $\|L_n - L_m\|_{L(X, Y)} < \frac{\varepsilon}{4}$ hence

$$\|L_n(x) - L_m(x)\|_Y = \|(L_n - L_m)(x)\| \leq \|L_n - L_m\|_{L(X, Y)} \cdot \|x\|_X = \|L_n - L_m\|_{L(X, Y)} < \frac{\varepsilon}{4}$$

Let $n \geq N$ then we have

$$\begin{aligned} \|(L_n - L)(x)\|_Y &= \|L_n(x) - L(x)\|_Y \\ &\leq \|L_n(x) - L_{\max(N, N_x)} + L_{\max(N, N_x)} - L(x)\|_Y \\ &\leq \|L_n(x) - L_{\max(N, N_x)}\|_Y + \|L_{\max(N, N_x)} - L(x)\|_Y \\ &< \frac{\varepsilon}{4} + \frac{\varepsilon}{4} \\ &= \frac{\varepsilon}{2} \end{aligned}$$

Hence using [theorem: 14.209] it follows that $\|L_n - L\|_{L(X, Y)} < \frac{\varepsilon}{2} < \varepsilon$ proving that

$$\{L_n\}_{n \in \{k, \dots, \infty\}} \text{ converges to } L \text{ in } \langle L(X, Y), \|\cdot\|_{L(X, Y)} \rangle$$

which implies that $\langle L(X, Y), \|\cdot\|_{L(X, Y)} \rangle$ is complete and thus a Banach space. $\square$

We have a similar theorem for multilinear mappings.

Theorem 14.399. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a sequence of normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space then the normed space $\langle L(X_1, \dots, X_n; Y), \|\cdot\|_{L(X_1, \dots, X_n; Y)} \rangle$ [see theorem: 14.234] is a Banach space.

Proof. Let $\{L_i\}_{i \in \{k, \dots, \infty\}} \subseteq L(X_1, \dots, X_n; Y)$ be a Cauchy sequence. Let $x = (x_1, \dots, x_n) \in \prod_{l \in \{1, \dots, n\}} X_l$ then given $\varepsilon \in \mathbb{R}^+$ we have that $\exists N \in \{k, \dots, \infty\}$ such that $\forall i, j \geq N$ we have

$$\|L_i - L_j\|_{L(X_1, \dots, X_n; Y)} < \frac{\varepsilon}{1 + \prod_{l \in \{1, \dots, n\}} \|x_l\|_l}$$

So that

$$\begin{aligned} \|L_i(x) - L_j(y)\|_Y &= \|(L_i - L_j)(x)\|_Y \\ &\leq \|L_i - L_j\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{l \in \{1, \dots, n\}} \|x_l\|_l \\ &< \frac{\varepsilon}{1 + \prod_{l \in \{1, \dots, n\}} \|x_l\|_l} \cdot \prod_{l \in \{1, \dots, n\}} \|x_l\|_l \\ &< \varepsilon \end{aligned}$$

which proves that

$$\{L_i(x)\}_{i \in \{1, \dots, n\}} \text{ is a Cauchy sequence in } \langle Y, \|\cdot\|_Y \rangle$$

As $\langle Y, \|\cdot\|_Y \rangle$ is a Banach space it follows that $\lim_{i \rightarrow \infty} L_i(x)$ exist, this allows to define the function

$$L: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Y \text{ where } L(x) = \lim_{i \rightarrow \infty} L_i(x) \quad (14.244)$$

Let $\alpha \in \mathbb{K}, i \in \{1, \dots, n\}, x, y \in X_i$ and $(x_1, \dots, \underbrace{x + \alpha \cdot y}_{i}, \dots, x_n) \in \prod_{j \in \{1, \dots, n\} \setminus \{i\}} X_j$ then we have that

$$\begin{aligned} &\left\{ L_l \left(x_1, \dots, \underbrace{x + \alpha \cdot y}_{i}, \dots, x_n \right) \right\}_{l \in \{k, \dots, \infty\}} = \\ &\left\{ L_l \left(x_1, \dots, \underbrace{x}_{i}, \dots, x_n \right) + \alpha \cdot L_l \left(x_1, \dots, \underbrace{y}_{i}, \dots, x_n \right) \right\}_{l \in \{k, \dots, \infty\}} \end{aligned}$$

Using [theorems: 14.358, 14.357] $\left\{ L_l \left(x_1, \dots, \underbrace{x + \alpha \cdot y}_{i}, \dots, x_n \right) \right\}_{l \in \{k, \dots, \infty\}}$ converges and

$$\begin{aligned} &\lim_{l \rightarrow \infty} L_l \left(x_1, \dots, \underbrace{x + \alpha \cdot y}_{i}, \dots, x_n \right) = \\ &\lim_{l \rightarrow \infty} L_l \left(x_1, \dots, \underbrace{x}_{i}, \dots, x_n \right) + \alpha \cdot \lim_{l \rightarrow \infty} L_l \left(x_1, \dots, \underbrace{y}_{i}, \dots, x_n \right) \end{aligned}$$

proving that

$$L \in \text{Hom}(X_1, \dots, X_n; Y) \quad (14.245)$$

As $\{L_i\}_{i \in \{k, \dots, \infty\}}$ is Cauchy there exist by [theorem: 14.385] a $K \in \mathbb{R}^+$ such that $\forall i \in \{k, \dots, \infty\}$ we have $\|L_i\|_{L(X, Y)} \leq K$. Hence $\forall i \in \{k, \dots, \infty\}$ we have for $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ that

$$\|L_i(x)\|_Y \leq \|L_i\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{j \in \{1, \dots, n\}} \|x_j\|_j \leq K \cdot \prod_{j \in \{1, \dots, n\}} \|x_j\|_j$$

which by [theorem: 14.363] proves

$$\|L(x)\|_Y = \left\| \lim_{i \rightarrow \infty} L_i(x) \right\|_Y \leq K \cdot \prod_{j \in \{1, \dots, n\}} \|x_j\|_j$$

Hence using [eq: 14.245] and [theorem: 14.228] it follows that

$$L \in L(X_1, \dots, X_n; Y) \quad (14.246)$$

Let $\varepsilon \in \mathbb{R}^+$ then for $x = (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ with $\forall i \in \{1, \dots, n\} \|x_i\|_i = 1$ we have by [eq: 14.244] that there exist a $N_x \in \{k, \dots, \infty\}$ such that

$$\forall n \geq N_x \text{ we have } \|L_n(x) - L(x)\|_Y < \frac{\varepsilon}{4}$$

Further as $\{L_i\}_{i \in \{k, \dots, \infty\}}$ is Cauchy there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m, l \geq N$ we have $\|L_m - L_l\|_{L(X_1, \dots, X_n; Y)} < \frac{\varepsilon}{4}$ so that

$$\begin{aligned} \|L_m(x) - L_l(x)\|_Y &= \|(L_m - L_l)(x)\|_Y \\ &\leq \|L_m - L_l\|_{L(X_1, \dots, X_n; Y)} \cdot \prod_{i \in \{1, \dots, n\}} \|x_i\|_i \\ &= \|L_m - L_l\|_{L(X_1, \dots, X_n; Y)} \\ &< \frac{\varepsilon}{4} \end{aligned}$$

Hence if $m \geq N$ then we have

$$\begin{aligned} \|L_m(x) - L(x)\|_Y &= \|L_m(x) - L_{\max(N, N_x)}(x) + L_{\max(N, N_x)}(x) - L(x)\|_Y \\ &\leq \|L_m(x) - L_{\max(N, N_x)}(x)\|_Y + \|L_{\max(N, N_x)}(x) - L(x)\|_Y \\ &< \frac{\varepsilon}{4} + \frac{\varepsilon}{4} \\ &= \frac{\varepsilon}{2} \end{aligned}$$

which by [theorem: 14.234] proves that $\|L_m - L\|_{L(X_1, \dots, X_n; Y)} \leq \frac{\varepsilon}{2} < \varepsilon$. So $\{L_i\}_{i \in \{k, \dots, \infty\}}$ converges to L in $\langle L(X_1, \dots, X_n; Y), \|\cdot\|_{L(X_1, \dots, X_n; Y)} \rangle$ proving that $\langle L(X_1, \dots, X_n; Y), \|\cdot\|_{L(X_1, \dots, X_n; Y)} \rangle$ is a Banach space. $\square$

Corollary 14.400. Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, $n \in \mathbb{N}$ then $\langle L^n(X; Y), \|\cdot\|_{L^n(X; Y)} \rangle$ is a Banach space.

Proof. For $n \in \mathbb{N}$ we have either

$n = 1$. By [definition: 14.235] $\langle L^1(X; Y), \|\cdot\|_{L^1(X; Y)} \rangle = \langle L(X, Y), \|\cdot\|_{L(X, Y)} \rangle$ which is Banach by the [theorem: 14.398].

$1 < n$. Then by [definition: 14.235]

$$\langle L^n(X; Y), \|\cdot\|_{L^n(X; Y)} \rangle = \left\langle L^n\left(\underbrace{X, \dots, X}_n; Y\right), \|\cdot\|_{L^n\left(\underbrace{X, \dots, X}_n; Y\right)} \right\rangle$$

which is Banach by [theorem: 14.399]. $\square$

14.10.2 Uniform convergence and Bounded functions

Definition 14.401. Let $\emptyset \neq X$ be a set, $\langle Y, \|\cdot\| \rangle$ a normed space then a function

$$f: X \rightarrow Y \text{ is bounded if } f(X) \text{ is bounded}$$

In other words, using [theorem: 14.112], $f: X \rightarrow Y$ is bounded if there exist a $M \in \mathbb{R}^+$ such that $\forall x \in X$ we have $\|f(x)\| \leq M$. The set of all bounded functions between X and Y is noted by $\mathcal{B}(X, Y)$ hence

$$\mathcal{B}(X, Y) = \{f \in Y^X \mid f: X \rightarrow Y \text{ is bounded}\} \subseteq Y^X$$

Theorem 14.402. Let $\emptyset \neq X$ be a set, $\langle Y, \|\cdot\| \rangle$ a normed space over the field F then $\langle \mathcal{B}(X, Y), +, \cdot \rangle$ is a sub-space of $\langle Y^X, +, \cdot \rangle$ [see theorem: 11.69]. Hence by [theorem: 11.61] $\langle \mathcal{B}(X, Y), +, \cdot \rangle$ is a vector space.

Proof. For the neutral element C_0 in $\langle Y^X, +, \cdot \rangle$ we have $\forall x \in X$ that $\|C_0(x)\| = \|0\| = 0 < 1$ so that

$$C_0 \in \mathcal{B}(X, Y) \Rightarrow \emptyset \neq \mathcal{B}(X, Y)$$

Further if $f, g \in \mathcal{B}(X, Y)$ and $\alpha \in F$ then there exists $M_1, M_2 \in \mathbb{R}^+$ such that $\forall x \in X$ we have $\|f(x)\| \leq M_1 \wedge \|g(x)\| \leq M_2$. Take $M = \max(M_1, M_2, |\alpha| \cdot M_2 + 1)$ then we have

$$\|(f + \alpha \cdot g)(x)\| = \|f(x) + \alpha \cdot g(x)\| \leq \|f(x)\| + \|\alpha \cdot g(x)\| = \|f(x)\| + |\alpha| \cdot \|g(x)\| \leq M_1 + |\alpha| \cdot M_2 \leq M$$

proving that $f + \alpha \cdot g \in \mathcal{B}(X, Y)$. □

We define now a norm on $\mathcal{B}(X, Y)$

Theorem 14.403. *Let $\emptyset \neq X$ be a set, $\langle Y, ||| \rangle$ a normed space over F then $|||_\infty$ defined by*

$$|||_\infty: \mathcal{B}(X, Y) \rightarrow \mathbb{R} \text{ by } |||_\infty = \sup(\{||f(y)|||y \in X\})$$

is well defined and $|||_\infty$ is a norm on $\mathcal{B}(X, Y)$ making $\langle \mathcal{B}(X, Y), |||_\infty \rangle$ a normed space. $|||_\infty$ is called the supremum norm.

Proof. As $X \neq \emptyset$ there exist a $x \in X$ so that $\emptyset \neq \{||f(y)|||y \in X\}$. If $f \in \mathcal{B}(X, Y)$ then there exist a $M \in \mathbb{R}^+$ such that $\forall x \in X$ we have $||f(x)|| \leq M$, hence $\{||f(y)|||y \in X\}$ is bounded above. As $\mathbb{R}$ is conditional complete [see theorem: 10.18] $\sup(\{||f(y)|||y \in X\})$ exist. Now we have

1. $\forall f \in \mathcal{B}(X, Y)$ we have as $0 \leq ||f(x)|| \forall x \in X$ that $0 \leq \sup(\{||f(y)|||y \in X\})$.
2. $\forall f \in \mathcal{B}(X, Y)$ and $\forall \alpha \in F$ then

$$\{||(\alpha \cdot f)(y)|||y \in X\} = \{|\alpha| \cdot ||f(y)|||y \in X\} = |\alpha| \cdot \{||f(y)|||y \in X\}$$

so that

$$||\alpha \cdot f||_\infty = \sup(|\alpha| \cdot \{||f(y)|||y \in X\}) \stackrel{[\text{theorem: 10.22}]}{=} |\alpha| \cdot \sup(\{||f(y)|||y \in X\}) = |\alpha| \cdot ||f||_\infty$$

3. If $f, g \in \mathcal{B}(X, Y)$ then

$$\begin{aligned} ||f + g||_\infty &= \sup(\{||(f + g)(y)|||y \in X\}) \\ &= \sup(\{||f(y) + g(y)|||y \in X\}) \\ &\leq_{[\text{theorem: 3.78}]} \sup(\{||f(y)|| + ||g(y)|||y \in X\}) \\ &\leq_{[\text{theorem: 3.78}]} \sup(\{||f(y)|||y \in X\} + \{||g(y)|||y \in X\}) \\ &\stackrel{[\text{theorem: 10.23}]}{=} \sup(\{||f(x)|||x \in X\}) + \sup(\{||g(x)|||x \in X\}) \\ &\leq ||f||_\infty + ||g||_\infty \end{aligned}$$

4. If $||f||_\infty = 0$ then $\sup(\{||f(x)|||x \in X\}) = 0$ so that $\forall x \in X$ $0 \leq ||f(x)|| \leq 0 \Rightarrow ||f(x)|| = 0$ proving, as $|||$ is a norm, that $f(x) = 0 = C_0(x)$. Hence $f = C_0$. □

We introduce now a stronger version of convergence for a sequence of functions to a normed space.

Definition 14.404. *Let $k \in \mathbb{N}_0, \emptyset \neq X$ be a set, $\langle Y, ||| \rangle$ a normed space and $\{f_n\}_{n \in \{k, \dots, \infty\}} \subseteq Y^X$ then $\{f_n\}_{n \in \{k, \dots, \infty\}}$ **converges uniformly** to a $f \in Y^X$ if $\forall \varepsilon \in \mathbb{R}$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\forall x \in X$ that*

$$||f_n(x) - f(x)|| < \varepsilon$$

Note 14.405. *The N in the definition is not dependent on the x which would be the case in pointwise convergence. Hence uniform convergence is stronger than pointwise convergence.*

Theorem 14.406. *Let $\langle X, |||_X \rangle$ be a normed space, $\langle Y, |||_Y \rangle$ a normed space then if $\{f_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathcal{B}(X, Y)$ is a sequence of continuous functions from X to Y uniformly converging to a function $f: X \rightarrow Y$ then f is continuous.*

Proof. Let $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then by uniform convergence that there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have

$$\forall x \in X \quad ||f_n(x) - f(x)||_Y < \frac{\varepsilon}{3}$$

As f_N is continuous we have that there exist a $\delta_N \in \mathbb{R}^+$ such that if $||x - y||_X < \delta_N$ then

$$||f_N(x) - f_N(y)||_Y < \frac{\varepsilon}{3}$$

Let $\|x - y\|_X < \delta_N$ then we have

$$\begin{aligned}\|f(x) - f(y)\|_Y &= \|f(x) - f_N(x) + f_N(x) - f_N(y) + f_N(y) - f(y)\|_Y \\ &\leq \|f(x) - f_N(x)\|_Y + \|f_N(x) - f_N(y)\|_Y + \|f_N(y) - f(y)\|_Y \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon\end{aligned}$$

proving by [theorem: 14.135] that f is continuous. $\square$

It turns out that convergence in the supremum norm $\|\cdot\|_\infty$ is equal to uniform convergence.

Theorem 14.407. *Let $X \neq \emptyset$ and $\langle Y, \|\cdot\| \rangle$ a normed space and $\{f_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathcal{B}(X, Y)$ then*

$$\lim_{n \rightarrow \infty} f_n = f \text{ using } \|\cdot\|_\infty \Leftrightarrow \{f_n\}_{n \in \{k, \dots, \infty\}} \text{ converges uniformly to } f$$

Proof.

$\Rightarrow$. Given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$

$$\sup(\{\|(f_n - f)(x)\| \mid x \in X\}) = \|f_n - f\|_\infty < \varepsilon$$

hence $\forall x \in X$ we have

$$\|f_n(x) - f(x)\| \leq \sup(\{\|(f_n - f)(x)\| \mid x \in X\}) < \varepsilon$$

proving uniform convergence.

$\Leftarrow$. Given $\varepsilon > 0$ there exists a $N \in \mathbb{N}$ such that if $n \geq N$ we have $\forall x \in X$ that $\|f_n(x) - f(x)\| < \varepsilon$ proving that ε is an upper bound of $\{\|f_n(x) - f(x)\| \mid x \in X\}$ so that

$$\|f_n - f\|_\infty = \sup(\{\|f_n(x) - f(x)\| \mid x \in X\}) < \varepsilon$$

proving that $\lim_{n \rightarrow \infty} f_n = f$ $\square$

Theorem 14.408. *Let $X \neq \emptyset$ and $\langle Y, \|\cdot\| \rangle$ a Banach space then $\langle \mathcal{B}(X, Y), \|\cdot\|_\infty \rangle$ [see theorem: 14.403] is a Banach space.*

Proof. Let $\{f_n\}_{n \in \{k, \dots, \infty\}} \subseteq \mathcal{B}(X, Y)$ be a Cauchy sequence in $\mathcal{B}(X, Y)$. Let $\varepsilon \in \mathbb{R}^+$ then there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have

$$\sup(\|f_n(x) - f_m(x)\| \mid x \in X) = \|f_n - f_m\|_\infty < \varepsilon$$

Hence given $x \in X$ we have $\|f_n(x) - f_m(x)\| \leq \sup(\|f_n(x) - f_m(x)\| \mid x \in X) < \varepsilon$ proving that $\{f_n(x)\}_{n \in \{k, \dots, \infty\}}$ is a Cauchy sequence, hence as Y is complete there exist a unique $\lim_{n \rightarrow \infty} f_n(x)$. This allows us to define a function

$$f: X \rightarrow Y \text{ by } f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad (14.247)$$

Next we prove that f is bounded so that $f \in \mathcal{B}(X, Y)$. Let $x \in X$ then as $1 \in \mathbb{R}^+$ we have by the definition of the limit that there exist a $N_x \in \{k, \dots, \infty\}$ [depending on x] such that $\forall n \geq N_x$ we have $\|f_n(x) - f(x)\| < 1$ so that

$$\|f(x)\| \leq \|f_n(x) - f(x)\| + \|f_n(x)\| < 1 + \|f_n(x)\| \leq 1 + \sup(\{\|f_n(x)\| \mid x \in X\}) = 1 + \|f_n\|_\infty$$

hence

$$\forall x \in X \exists N_x \in \mathbb{R}^+ \text{ such that } \forall n \geq N_x \|f(x)\| < 1 + \|f_n\|_\infty \quad (14.248)$$

Further as $\{f_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy in $\mathcal{B}(X, Y)$ and $1 \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m \geq N \Rightarrow m, N \geq N$ we have $\|f_m - f_N\|_\infty < 1$, hence $\|f_m\|_\infty \leq \|f_N\|_\infty + \|f_m - f_N\|_\infty < \|f_N\|_\infty + 1$. Summarized

$$\exists N \in \{k, \dots, \infty\} \text{ such that } \forall m \geq N \text{ we have } \|f_m\|_\infty < \|f_N\|_\infty + 1 \quad (14.249)$$

Let $x \in X$ and take $M_x = \max(N, N_x)$ then we have

$$\begin{aligned} \|f(x)\| &\leq_{[\text{eq: 14.248}]} 1 + \|f_{M_x}\|_\infty \\ &\leq_{[\text{eq: 14.249}]} 1 + 1 + \|f_N\|_\infty \\ &\leq M \end{aligned}$$

where $M = 2 + \|f_N\|_\infty$ independent of x , proving that $\|f\|_\infty = \sup(\{\|f(x)\| \mid x \in X\}) \leq M$. Hence we conclude that f is bounded or

$$f \in \mathcal{B}(X, Y) \quad (14.250)$$

Next we must prove that $\lim_{n \rightarrow \infty} f_n = f$ using the norm $\|\cdot\|_\infty$. Let $\varepsilon \in \mathbb{R}^+$ then for $x \in X$ we have using [eq: 14.247] that there exists a $N_x \in \{k, \dots, \infty\}$ such that $\forall n \geq N_x$ we have $\|f(x) - f_n(x)\| < \frac{\varepsilon}{2}$. Further as $\{f_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy [using the norm $\|\cdot\|_\infty$] there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have $\|f_n - f_m\| < \frac{\varepsilon}{2}$. Let $x \in X$ then for $n \geq N$ we have $\max(n, N_x) \geq N, N_x$ so that

$$\begin{aligned} \|(f_n - f)(x)\| &= \|f_n(x) - f(x)\| \\ &= \|f_n(x) - f_{\max(N, N_x)}(x) + f_{\max(N, N_x)}(x) - f(x)\| \\ &\leq \|f_n(x) - f_{\max(N, N_x)}(x)\| + \|f_{\max(N, N_x)}(x) - f(x)\| \\ &\leq \|(f_n - f_{\max(N, N_x)})(x)\| + \|f_{\max(N, N_x)}(x) - f(x)\| \\ &\leq \sup(\{\|(f_n - f_{\max(N, N_x)})(x)\| \mid x \in X\}) + \|f_{\max(N, N_x)}(x) - f(x)\| \\ &= \|f_n - f_{\max(N, N_x)}\|_\infty + \|f_{\max(N, N_x)}(x) - f(x)\| \\ &<_{N, N_x \leq \max(N, N_x)} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

which as $x \in X$ was chosen arbitrary proves that $\|f_n - f\|_\infty = \sup(\{\|(f_n - f)(x)\| \mid x \in X\}) < \varepsilon$. Hence it follows that

$$\lim_{n \rightarrow \infty} f_n \rightarrow f \text{ using the norm } \|\cdot\|_\infty \quad \square$$

14.10.3 Series

Series are an extension of finite sums to infinite sums. Note that in contrast with finite sums infinite sums do not have in general the properties of commutativity and associativity.

Definition 14.409. (Series) Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a normed space and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence such that $\lim_{n \rightarrow \infty} (\sum_{i=k}^n x_i)$ exist then we say that the **series** $\sum_{i=k}^\infty x_i$ **converges** and note the limit as $\sum_{i=k}^\infty x_i$, hence

$$\sum_{i=k}^\infty x_i = \lim_{n \rightarrow \infty} \left(\sum_{i=k}^n x_i \right)$$

Remark 14.410. To avoid excessive notation instead of saying ' $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is a sequence such that the series $\sum_{i=k}^\infty x_i$ converges' we just say 'the series $\sum_{i=k}^\infty x_i$ converges' where the sequence $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is implicit assumed.

An example of a series in $\mathbb{R}$ is the power series, to prove this we first need a little lemma.

Lemma 14.411. Let $\lambda \in \mathbb{R} \setminus \{1\}$ then if $k \in \mathbb{N}$ and $n \in \mathbb{N}_0$ we have that

$$\sum_{i=1}^k \lambda^{n+(i-1)} = \lambda^n \cdot \frac{1 - \lambda^k}{1 - \lambda}$$

In particular we have

$$1. \text{ If } n=0 \text{ then } \sum_{i=1}^k \lambda^{(i-1)} = \frac{1 - \lambda^k}{1 - \lambda}$$

2. If $n=1$ then $\sum_{i=1}^k \lambda^i = \lambda \cdot \frac{1-\lambda^k}{1-\lambda}$

Proof. We prove this by induction so let

$$S = \left\{ k \in \mathbb{N} \mid \sum_{i=1}^k \lambda^{n+(i-1)} = \lambda^n \cdot \frac{1-\lambda^k}{1-\lambda} \right\}$$

then we have:

$1 \in S$. If $k=1$ then $\sum_{i=1}^1 \lambda^{n+(i-1)} = \lambda^n = \lambda^n \cdot \frac{1-\lambda}{1-\lambda} = \lambda^n \cdot \frac{1-\lambda^1}{1-\lambda}$ so that $1 \in S$

$k \in S \Rightarrow k+1 \in S$. Then we have

$$\begin{aligned} \sum_{i=1}^{k+1} \lambda^{n+(i-1)} &= \left(\sum_{i=1}^k \lambda^{n+(i-1)} \right) + \lambda^{n+((k+1)-1)} \\ &= \left(\sum_{i=1}^k \lambda^{n+(i-1)} \right) + \lambda^{n+k} \\ &\stackrel{k \in S}{=} \lambda^n \cdot \frac{1-\lambda^k}{1-\lambda} + \lambda^{n+k} \\ &= \frac{\lambda^n \cdot (1-\lambda^k) + (1-\lambda) \cdot \lambda^{n+k}}{1-\lambda} \\ &= \frac{\lambda^n \cdot (1-\lambda^k + (1-\lambda) \cdot \lambda^k)}{1-\lambda} \\ &= \frac{\lambda^n \cdot (1-\lambda^k + \lambda^k - \lambda^{k+1})}{1-\lambda} \\ &= \lambda^n \cdot \frac{1-\lambda^{k+1}}{1-\lambda} \end{aligned}$$

proving that $k+1 \in S$ □

Theorem 14.412. Let $\lambda \in \mathbb{R}$ with $0 \leq \lambda < 1$ then $\sum_{i=1}^{\infty} \lambda^i$ converges and $\sum_{i=1}^{\infty} \lambda^i = \frac{\lambda}{1-\lambda}$

Proof. Let $k \in \mathbb{N}$ then we have that

$$\begin{aligned} \left| \sum_{i=1}^k \lambda^i - \frac{\lambda}{1-\lambda} \right| &= \left| \sum_{i=1}^k \lambda \cdot \lambda^{i-1} - \frac{\lambda}{1-\lambda} \right| \\ &= \left| \lambda \cdot \sum_{i=1}^k \lambda^{i-1} - \frac{\lambda}{1-\lambda} \right| \\ &\stackrel{[\text{theorem: 14.411}]}{=} \left| \lambda \cdot \frac{1-\lambda^k}{1-\lambda} - \frac{\lambda}{1-\lambda} \right| \\ &= \left| \frac{\lambda - \lambda^{k+1} - \lambda}{1-\lambda} \right| \\ &= \left| \frac{-\lambda^{k+1}}{1-\lambda} \right| \\ &= \frac{\lambda^{k+1}}{1-\lambda} \\ &= \frac{\lambda}{1-\lambda} \cdot \lambda^k \end{aligned}$$

Using [example: 14.348] we have that $\lim_{k \rightarrow \infty} \lambda^k = 0$, hence given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{1, \dots, \infty\}$ such that $\forall k \geq N$ $\lambda^k = |\lambda^k - 0| < \varepsilon \cdot \frac{1-\lambda}{1+\lambda}$ so that $\frac{\lambda}{1-\lambda} \cdot \lambda^k < \frac{\lambda}{1-\lambda} \cdot \frac{1-\lambda}{1+\lambda} \cdot \varepsilon = \frac{\lambda}{1+\lambda} \cdot \varepsilon < \varepsilon$, hence

$$\left| \sum_{i=1}^k \lambda^i - \frac{\lambda}{1-\lambda} \right| < \varepsilon$$

proving that

$$\lim_{k \rightarrow \infty} \sum_{i=1}^k \lambda^i = \frac{\lambda}{1-\lambda} \quad \square$$

If a series is convergent then the terms must tend to zero as is show in the following theorem.

Theorem 14.413. *Let $k \in \mathbb{N}_0$, $\langle X, ||| \rangle$ be a normed space then if $\sum_{i=k}^{\infty} x_i$ converges then*

1. $\lim_{i \rightarrow \infty} x_i = 0$
2. $\exists K \in \mathbb{R}^+$ such that $\forall i \in \{k, \dots, \infty\}$ we have $\|x_i\| \leq K$

Proof.

1. Let $\varepsilon \in \mathbb{R}^+$. As $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ exist we have that $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}}$ is Cauchy so there exist a $N \in \{k, \dots, \infty\}$ such that $n, m \geq N$ we have that $\|\sum_{i=k}^n x_i - \sum_{i=k}^m x_i\| < \varepsilon$. In particular we have $\forall n \geq N \Rightarrow n+1 \geq N$ that $\|x_n - 0\| = \|x_n\| = \|\sum_{i=k}^{n+1} x_i - \sum_{i=k}^n x_i\| < \varepsilon$. Hence

$$\lim_{n \rightarrow \infty} x_n = 0$$

2. As $\lim_{n \rightarrow \infty} x_n$ exist it follows from [theorem: 14.364] that there exist a $K \in \mathbb{R}^+$ such that

$$\forall i \in \{k, \dots, \infty\} \text{ we have } \|x_i\| \leq K \quad \square$$

Proof. Let $\varepsilon \in \mathbb{R}^+$. As $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ exist we have that $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}}$ is Cauchy so there exist a $N \in \{k, \dots, \infty\}$ such that $n, m \geq N$ we have that $\|\sum_{i=k}^n x_i - \sum_{i=k}^m x_i\| < \varepsilon$. In particular we have $\forall n \geq N \Rightarrow n+1 \geq N$ that $\|x_n - 0\| = \|x_n\| = \|\sum_{i=k}^{n+1} x_i - \sum_{i=k}^n x_i\| < \varepsilon$. Hence

$$\lim_{n \rightarrow \infty} x_n = 0 \quad \square$$

In a Banach space we have the following sufficient and necessary condition for convergence of a series.

Theorem 14.414. *Let $k \in \mathbb{N}_0$, $\langle X, ||| \rangle$ be a Banach space then*

$$\sum_{i=k}^{\infty} x_i \text{ converges} \Leftrightarrow \forall \varepsilon \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\} \text{ such that } \forall n, m \geq N \text{ with } n \leq m \text{ we have } \left\| \sum_{i=n}^m x_i \right\| < \varepsilon$$

Proof. Take $\{s_i\}_{i \in \{k, \dots, \infty\}}$ defined by $s_i = \sum_{j=k}^i x_j$ then

$\Rightarrow$. If $\sum_{i=k}^{\infty} x_i$ converges we have that $\lim_{i \rightarrow \infty} s_i$ exists, take $\varepsilon > 0$, then using [theorem: 14.386], there exists a $N' \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N'$ we have $\|s_m - s_n\| < \varepsilon$. Take $n, m \geq N = N' + 1$ with $n \leq m$ then $N' \leq n - 1, m$ and $\|\sum_{i=n}^m x_i\| \stackrel{11.23}{=} \|\sum_{i=k}^m x_i - \sum_{i=k}^{n-1} x_i\| = \|s_m - s_{n-1}\| < \varepsilon$.

$\Leftarrow$. Let $\varepsilon > 0$ then there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ with $n \leq m$ we have $\|\sum_{i=n}^m x_i\| < \varepsilon$, as $\sum_{i=n+1}^m x_i \stackrel{11.23}{=} \sum_{i=k}^m x_i - \sum_{i=k}^n x_i = s_m - s_n$ we conclude that $\|s_m - s_n\| = \|\sum_{i=n+1}^m x_i\| < \varepsilon$. From this it follows by definition that $\{s_i\}_{i \in \{k, \dots, \infty\}}$ is Cauchy and as X is a Banach space and thus complete $\lim_{i \rightarrow \infty} s_i$ exists proving that $\sum_{i=k}^{\infty} x_i$ converges. $\square$

Theorem 14.415. *Let $k, l \in \mathbb{N}_0$, $\langle X, ||| \rangle$ a normed space and $\{x_i\}_{i \in \{k+l, \dots, \infty\}} \subseteq X$ a sequence in X then we have*

$$\sum_{i=k}^{\infty} x_{i+l} \text{ converges} \Leftrightarrow \sum_{i=k+l}^{\infty} x_i \text{ converges}$$

Further if $\sum_{i=k}^{\infty} x_{i+l}$ or $\sum_{i=k+l}^{\infty} x_i$ converges then

$$\sum_{i=k}^{\infty} x_{i+l} = \sum_{i=k+l}^{\infty} x_i$$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $\sum_{i=k}^{\infty} x_{i+l}$ converges there exist a $x \in X$ such that there exist a $N' \in \{k, \dots, \infty\}$ such that $\forall n \geq N'$ we have that $\|\sum_{i=k}^n x_{i+l} - x\| < \varepsilon$. Hence if we take $N = N' + l \in \{k+l, \dots, \infty\}$ then if $n \geq N$ we have that $n \geq N' + l \Rightarrow n-l \geq N'$ so that

$$\left\| \sum_{i=k+l}^n x_i - x \right\| = \left\| \sum_{i=k}^{n-l} x_{i+l} - x \right\| < \varepsilon.$$

This proves that $\sum_{i=k+l}^{\infty} x_i$ converges to x or

$$\sum_{i=k+l}^{\infty} x_i = x = \sum_{i=k}^{\infty} x_{i+l}$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $\sum_{i=k+l}^{\infty} x_i$ converges there exist a $x \in X$ and a $N' \in \{k+l, \dots, \infty\}$ such that $\forall n \geq N'$ we have $\|\sum_{i=k+l}^n x_i - x\| < \varepsilon$. Take $N = N' - l \in \{k, \dots, \infty\}$ then $\forall n \geq N$ we have $n+l \geq N'$ so that

$$\left\| \sum_{i=k}^n x_{i+l} - x \right\| = \left\| \sum_{i=k+l}^{n+l} x_i - x \right\| < \varepsilon$$

proving that $\sum_{i=k}^{\infty} x_{i+l}$ converges to x or

$$\sum_{i=k}^{\infty} x_{i+l} = x = \sum_{i=k+l}^{\infty} x_i \quad \square$$

Theorem 14.416. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ a normed space, $m \in \{k+1, \dots, \infty\}$ and $\{x_i\}_{i \in \{k, \dots, n\}} \subseteq X$ then

$$\sum_{i=k}^{\infty} x_i \text{ converges} \Leftrightarrow \sum_{i=m}^{\infty} x_i \text{ converges}$$

Further if $\sum_{i=k}^{\infty} x_i$ or $\sum_{i=m}^{\infty} x_i$ converges then

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{m-1} x_i + \sum_{i=m}^{\infty} x_i$$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $\sum_{i=k}^{\infty} x_i$ converges there exist a $x \in X$ and a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|\sum_{i=k}^n x_i - x\| < \varepsilon$, hence we have for $n \geq \max(m, N)$ that

$$\left\| \sum_{i=m}^n x_i - \left(x - \sum_{i=k}^{m-1} x_i \right) \right\| = \left\| \sum_{i=m}^n x_i + \sum_{i=k}^{m-1} x_i - x \right\| \stackrel{[\text{theorem: 11.23}]}{=} \left\| \sum_{i=k}^n x_i - x \right\| < \varepsilon$$

proving that $\sum_{i=m}^{\infty} x_i$ converges and that $\sum_{i=m}^{\infty} x_i = x - \sum_{i=k}^{m-1} x_i = \sum_{i=k}^{\infty} x_i - \sum_{i=k}^{m-1} x_i$. Hence

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{m-1} x_i + \sum_{i=m}^{\infty} x_i$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $\sum_{i=m}^{\infty} x_i$ converges there exist a $x \in X$ and a $N \in \{m, \dots, \infty\}$ such that $\forall n \geq N$ we have $\|\sum_{i=m}^n x_i - x\| < \varepsilon$. Then we have

$$\left\| \sum_{i=k}^n x_i - \left(\sum_{i=k}^{m-1} x_i + x \right) \right\| = \left\| \sum_{i=k}^n x_i - \sum_{i=k}^{m-1} x_i - x \right\| = \left\| \sum_{i=m}^n x_i - x \right\| < \varepsilon$$

proving that $\sum_{i=k}^{\infty} x_i$ converges and that

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{m-1} x_i + x = \sum_{i=k}^{m-1} x_i + \sum_{i=m}^{\infty} x_i \quad \square$$

Theorem 14.417. Let $\langle X, ||| \rangle$ be a normed space and $\sum_{i=k}^{\infty} x_i$ a convergent space [so that by [theorem: 14.416] $\forall m \in \{k, \dots, \infty\}$ $\sum_{i=m}^{\infty} x_i$ converges and thus exist] then

$$\lim_{m \rightarrow \infty} \left(\sum_{i=m}^{\infty} x_i \right) = 0$$

Proof. Let $\varepsilon \in \mathbb{R}^+$. As $\sum_{i=k}^{\infty} x_i$ converges there exist a $N \in \{k, \dots, \infty\}$ so that $\forall n \geq N$ we have $\|\sum_{i=1}^n x_i - x\| < \varepsilon$ where $x = \sum_{i=k}^{\infty} x_i$. Let $m \geq N + 1 \Rightarrow m - 1 \geq N$ then

$$\left\| \sum_{i=m}^{\infty} x_i - 0 \right\| = \left\| \sum_{i=m}^{\infty} x_i \right\| \stackrel{[\text{theorem: 14.416}]}{=} \left\| \sum_{i=k}^{\infty} x_i - \sum_{i=k}^{m-1} x_i \right\| = \left\| \sum_{i=k}^{m-1} x_i - x \right\| < \varepsilon$$

proving that $\{\sum_{i=m}^{\infty} x_i\}_{m \in \{k, \dots, \infty\}}$ converges to 0, Hence

$$\lim_{m \rightarrow \infty} \left(\sum_{i=m}^{\infty} x_i \right) = 0 \quad \square$$

The sum and scalar product of series is a series.

Theorem 14.418. Let $\langle X, ||| \rangle$ be a normed space then we have:

1. If $\sum_{i=k}^{\infty} x_i$ and $\sum_{i=k}^{\infty} y_i$ are convergent series then $\sum_{i=k}^{\infty} (x_i + y_i)$ is convergent and

$$\sum_{i=k}^{\infty} (x_i + y_i) = \sum_{i=k}^{\infty} x_i + \sum_{i=k}^{\infty} y_i$$

2. If $\sum_{i=k}^{\infty} x_i$ and $\sum_{i=k}^{\infty} y_i$ are convergent series with $\forall i \in \{k, \dots, \infty\} x_i \leq y_i$ then

$$\sum_{i=k}^{\infty} x_i \leq \sum_{i=k}^{\infty} y_i$$

3. If $\sum_{i=k}^{\infty} x_i$ is convergent and $\alpha \in \mathbb{K}$ then $\sum_{i=k}^{\infty} (\alpha \cdot x_i)$ is convergent and

$$\sum_{i=k}^{\infty} (\alpha \cdot x_i) = \alpha \cdot \sum_{i=k}^{\infty} x_i$$

Proof.

1. As $\sum_{i=k}^{\infty} x_i$ and $\sum_{i=k}^{\infty} y_i$ are convergent

$$\lim_{n \rightarrow \infty} \sum_{i=k}^{\infty} x_i \text{ and } \lim_{n \rightarrow \infty} \sum_{i=k}^n y_i \text{ exists}$$

Hence using [theorem: 14.357] $\lim_{n \rightarrow \infty} \sum_{i=k}^{\infty} (x_i + y_i) = \lim_{n \rightarrow \infty} (\sum_{i=k}^n x_i + \sum_{i=k}^n y_i)$ exists and is equal to $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i + \lim_{n \rightarrow \infty} \sum_{i=k}^n y_i$. Hence by definition $\sum_{i=k}^{\infty} (x_i + y_i)$ is convergent and

$$\sum_{i=k}^{\infty} (x_i + y_i) = \sum_{i=k}^{\infty} x_i + \sum_{i=k}^{\infty} y_i$$

2. As $\forall n \in \{k, \dots, \infty\}$ we have by [theorem: 11.55] $\sum_{i=k}^n x_i \leq \sum_{i=k}^n y_i$ that

$$\sum_{i=k}^{\infty} x_i = \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \stackrel{[\text{theorem: 14.369}]}{\leq} \lim_{n \rightarrow \infty} \sum_{i=k}^n y_i = \sum_{i=k}^{\infty} y_i$$

3. As $\sum_{i=k}^{\infty} x_i$ is convergent $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ exist, so that by [theorem: 14.358] $\lim_{n \rightarrow \infty} (\sum_{i=k}^n \alpha \cdot x_i) = \lim_{n \rightarrow \infty} \alpha \cdot (\sum_{i=k}^n x_i)$ exist and is equal to $\alpha \cdot \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$. Hence by definition $\sum_{i=k}^{\infty} (\alpha \cdot x_i)$ is convergent and

$$\sum_{i=k}^{\infty} (\alpha \cdot x_i) = \alpha \cdot \sum_{i=k}^{\infty} x_i$$

□

Corollary 14.419. Let $\langle \mathbb{C}, || \rangle$ be the normed space of complex numbers with the norm $||$ and $\{z_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{C}$ then

$$\sum_{i=k}^{\infty} x_i \text{ converges} \Leftrightarrow \sum_{i=k}^{\infty} \text{Im}(z_i) \text{ and } \sum_{i=k}^{\infty} \text{Re}(z_i) \text{ converges}$$

Further if $\sum_{i=k}^{\infty} z_i$ converges then

$$\text{Im}\left(\sum_{i=k}^{\infty} z_i\right) = \sum_{i=k}^{\infty} \text{Im}(z_i) \text{ and } \text{Re}\left(\sum_{i=k}^{\infty} z_i\right) = \sum_{i=k}^{\infty} \text{Re}(z_i)$$

so that

$$\sum_{i=k}^{\infty} z_i = \sum_{i=k}^{\infty} \text{Re}(z_i) + i \cdot \sum_{i=k}^{\infty} \text{Im}(z_i)$$

Proof.

$\Rightarrow$. Assume that $\sum_{i=k}^{\infty} z_i$ converges to z then given $\varepsilon > 0$ there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\} |\sum_{i=k}^n z_i - z| < \varepsilon$. Now

$$\begin{aligned} \left| \left(\sum_{i=k}^n \text{Re}(z_i) \right) - \text{Re}(z) \right| &= \left| \text{Re} \left(\sum_{i=k}^n z_i - z \right) \right| \\ &\leq_{[\text{theorem: 10.85}]} \left| \sum_{i=k}^n z_i - z \right| \\ &< \varepsilon \end{aligned}$$

proving that

$$\sum_{i=k}^{\infty} \text{Re}(z_i) \text{ converges to } \text{Re}(z) = \text{Re} \left(\sum_{i=k}^{\infty} z_i \right)$$

Similar

$$\begin{aligned} \left| \left(\sum_{i=k}^n \text{Im}(z_i) \right) - \text{Im}(z) \right| &= \left| \text{Im} \left(\sum_{i=k}^n z_i - z \right) \right| \\ &\leq_{[\text{theorem: 10.85}]} \left| \sum_{i=k}^n z_i - z \right| \\ &< \varepsilon \end{aligned}$$

proving that

$$\sum_{i=k}^{\infty} \text{Im}(z_i) \text{ converges to } \text{Im}(z) = \text{Im} \left(\sum_{i=k}^{\infty} z_i \right)$$

$\Leftarrow$. Assume that $\sum_{i=k}^{\infty} \text{Re}(z_i)$ and $\sum_{i=k}^{\infty} \text{Im}(z_i)$ converges. As $\forall i \in \{k, \dots, \infty\} z_i = \text{Re}(z_i) + i \cdot \text{Im}(z_i)$ it follows from the previous theorem [see theorem: 14.418] that $\sum_{i=k}^{\infty} z_i$ converges. □

14.10.3.1 Series of non negative numbers

Series are in general not commutative, however series of non negative real numbers (and later absolute convergent series) are commutative. First we look at some equivalences of convergence of series of non negative real numbers

Theorem 14.420. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ be a sequence of non negative real numbers then we have the following equivalences:

1. $\sum_{i=k}^{\infty} x_i = x$ [in other words $\sum_{i=k}^{\infty} x_i$ converges to x]
2. $\sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$ exists and $x = \sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$
3. $\sup(\{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\})$ exists and

$$x = \sup\left(\left\{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\right\}\right)$$

4. $\forall \varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m \in \mathbb{N}$ and $n \geq N$ we have $\sum_{i=n+1}^{n+m} x_i < \varepsilon$
5. $\forall \varepsilon \in \mathbb{R}^+$ there exist a finite $K \subseteq \{k, \dots, \infty\}$ such that $\forall H \subseteq \{k, \dots, \infty\}$ with H finite and $K \cap H = \emptyset$ we have $\sum_{i \in H} x_i < \varepsilon$

Proof. First if $s \in \{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\}$ then $\exists n \in \{k, \dots, \infty\}$ so that

$$s = \sum_{i=k}^n x_i \stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{k, \dots, n\}} x_i \in \left\{ \sum_{i \in K} x_i | K \subseteq \{k, \dots, \infty\} \wedge K \text{ is finite} \right\}$$

proving that

$$\left\{ \sum_{i=k}^n x_i | n \in \{k, \dots, \infty\} \right\} \subseteq \left\{ \sum_{i \in K} x_i | K \subseteq \{k, \dots, \infty\} \wedge K \text{ is finite} \right\} \quad (14.251)$$

Next

- 1 $\Leftrightarrow$ 2. Let $n \in \{k, \dots, \infty\}$ then $\sum_{i=k}^n x_i \leq \sum_{i=k}^n x_i + x_{n+1} = \sum_{i=k}^{n+1} x_i$, hence $\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\}$ is a increasing sequence. So

$$\begin{aligned} \sum_{i=k}^{\infty} x_i \text{ exists and } \sum_{i=k}^{\infty} x_i = x & \stackrel{\Leftrightarrow}{\text{definition}} \\ \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \text{ exists } \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i = x & \stackrel{[\text{theorem: 14.375}]}{\Leftrightarrow} \end{aligned}$$

$$\sup\left(\left\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\right\}\right) \text{ exist and } \sup\left(\left\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\right\}\right) = x$$

- 2 $\Rightarrow$ 3. Let $s \in \{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$ then there exists a finite $K \subseteq \{k, \dots, \infty\}$ so that $s = \sum_{i \in K} x_i$. Take $n = \max(K)$ then $K \subseteq \{k, \dots, m\}$ so that

$$\sum_{i \in K} x_i \leq \sum_{i \in K} x_i + \sum_{i \in \{k, \dots, m\} \setminus K} x_i = \sum_{i \in \{k, \dots, m\}} x_i = \sum_{i=k}^m x_i \leq \sup\left(\left\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\right\}\right)$$

hence $\{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$ is bounded above by $\sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$ so that by [theorem: 10.18]

$$\sup\left(\left\{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\right\}\right) \text{ exists} \quad (14.252)$$

and

$$\sup\left\{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\right\} \leq \sup\left(\left\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\right\}\right) \quad (14.253)$$

Combining [eq: 14.251] with [theorem: 3.77] gives

$$\sup\left(\left\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\right\}\right) \leq \sup\left\{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\right\}$$

which combined with [eq: 14.253] proves that

$$\sup \left(\left\{ \sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\} \right\} \right) = \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \right)$$

3 $\Rightarrow$ 2. Let $s \in \{\sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\}\}$ then from [eq: 14.251] $s \in \{\sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$ so that $s \leq \sup(\{\sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\})$ hence by [theorem: 10.18]

$$\sup \left(\left\{ \sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\} \right\} \right) \text{ exists}$$

further as $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}}$ is increasing we have by [theorem: 14.375] that

$$\sup \left(\left\{ \sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\} \right\} \right) = \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i = \sum_{i=k}^{\infty} x_i$$

1 $\Rightarrow$ 4. As $\sum_{i=k}^{\infty} x_i$ converges $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ exists so that by [theorem: 14.386] $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}}$ is Cauchy. Hence given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have $|\sum_{i=k}^n x_i - \sum_{i=k}^m x_i| < \varepsilon$. So if $m \in \mathbb{N}$ and $n \geq N$ then we have $N + m, n \geq N$ hence

$$\sum_{i=n+1}^{n+m} x_i = \left| \sum_{i=n+1}^{n+m} x_i \right| = \left| \sum_{i=k}^{n+m} x_i - \sum_{i=k}^n x_i \right| < \varepsilon$$

4 $\Rightarrow$ 1. Let $\varepsilon \in \mathbb{R}^+$ then by the hypothesis there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m \in \mathbb{N}$ and $\forall n \geq m$ that $\sum_{i=n+1}^{n+m} x_i < \varepsilon$. Take $r, s \geq N$ then without losing generality we may assume that $s \leq r$. If $r = s$ then $|\sum_{i=k}^r x_i - \sum_{i=k}^s x_i| = 0 < \varepsilon$ and if $s < r$ then

$$\left| \sum_{i=k}^r x_i - \sum_{i=k}^s x_i \right| = \left| \sum_{i=s+1}^r x_i \right| = \sum_{i=s+1}^r x_i = \sum_{i=s+1}^{(r-s)+s} x_i < 0 < r-s \varepsilon$$

proving that $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}}$ is a Cauchy sequence. As by [theorem: 14.394] $(\mathbb{R}, \|\cdot\|)$ is a Banach space hence complete it follows that $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ exists or that

$$\sum_{i=1}^{\infty} x_i \text{ converges}$$

4 $\Rightarrow$ 5. Let $\varepsilon \in \mathbb{R}^+$ then by the hypothesis there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m \in \mathbb{N}$ and $\forall n \geq m$ that $\sum_{i=n+1}^{n+m} x_i < \varepsilon$. Take $K = \{k, \dots, N\}$ then if $H \subseteq \{k, \dots, \infty\}$ is finite and $K \cap H = \emptyset$ we must have that $H \subseteq \{N+1, \dots, \infty\}$. We must now consider two cases for H :

$H = \emptyset$. Then $\sum_{i \in H} x_i = 0 < \varepsilon$

$H \neq \emptyset$. Then $N < \min(H) \leq \max(H)$ so that $m = \max(H) - N > 0$ and

$$H \subseteq \{N+1, \dots, \max(H)\} = \{N+1, \dots, N+m\}$$

so that

$$\sum_{i \in H} x_i \leq \sum_{i \in H} x_i + \sum_{i \in \{N+1, \dots, N+m\} \setminus H} x_i = \sum_{i \in \{N+1, \dots, N+m\}} x_i < \varepsilon$$

5 $\Rightarrow$ 4. Let $\varepsilon > 0$ then there exists a finite $K \subseteq \{k, \dots, \infty\}$ such that $\forall H \subseteq \{k, \dots, \infty\}$ with H finite and $K \cap H = \emptyset$ we have $\sum_{i \in H} x_i < \varepsilon$. Take $K' = K \cup \{k\}$ then $K' \neq \emptyset$ so that $N = \max(K')$ exists. Then $\forall n \geq N$ and $m \in \mathbb{N}$ we have

$$K \cap \{n+1, \dots, n+m\} \subseteq K' \cap \{n+1, \dots, n+m\} \underset{\max(K') \leq N \leq n < n+1}{=} \emptyset$$

hence for $H = \{n+1, \dots, n+m\}$ we have $K \cap H = \emptyset$ so that

$$\sum_{i=n+1}^{n+m} x_i = \sum_{i \in \{n+1, \dots, n+m\}} x_i = \sum_{i \in H} x_i < \varepsilon \quad \square$$

Corollary 14.421. *Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ be a sequence of non negative real numbers such that $\sum_{i=k}^{\infty} x_i$ converges then $0 \leq \sum_{i=k}^{\infty} x_i$.*

Proof. As $\sum_{i=k}^{\infty} x_i$ converges it follows from the previous theorem [theorem: 14.420] it follows that $\sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$ exists and $\sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\}) = \sum_{i=k}^{\infty} x_i$. As $\{x_i | i \in \{k, \dots, \infty\}\} \subseteq [0, \infty[$ it follows that $0 \leq \sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$ so that $0 \leq \sum_{i=k}^{\infty} x_i$. $\square$

Actually we can also have a strict inequality if one of the terms is strictly positive.

Corollary 14.422. *Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ be a sequence of non negative real numbers such that $\exists l \in \{k, \dots, \infty\}$ with $0 < x_l$ then if $\sum_{i=k}^{\infty} x_i$ converges we have $0 < \sum_{i=k}^{\infty} x_i$.*

Proof. By the previous theorem [theorem: 14.420] it follows that $\sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$ exists and $\sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\}) = \sum_{i=k}^{\infty} x_i$. As $0 < x_l \leq \sup(\{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\})$ it follows that $0 < \sum_{i=k}^{\infty} x_i$. $\square$

We are now ready to prove commutativity of series of positive numbers.

Theorem 14.423. *Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ be a sequence of non negative real numbers and $\beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ a bijection then we have*

$$\sum_{i=k}^{\infty} x_i \text{ converges} \Leftrightarrow \sum_{i=k}^{\infty} x_{\beta(i)} \text{ converges}$$

Further if $\sum_{i=k}^{\infty} x_i$ converges then $\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{\infty} x_{\beta(i)}$

Proof. Let $s \in \{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$ then there exist a finite $K \subseteq \{k, \dots, \infty\}$ such that $s = \sum_{i \in K} x_i$. Take $L = \beta^{-1}(K)$ then as $\beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a bijection

$$\beta|_L: L \rightarrow \beta(L) = \beta(\beta^{-1}(K)) = K \text{ is a bijection}$$

Hence

$$s = \sum_{i \in K} x_i \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in L} x_{\beta(i)} = \sum_{i \in L} x_{\beta(i)}$$

As K is finite L must be finite [otherwise as β is a bijection $K = \beta(\beta^{-1}(K)) = \beta(L)$ is infinite], hence $s = \sum_{i \in L} x_{\beta(i)} \in \{\sum_{i \in K} x_{\beta(i)} | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$ proving that

$$\left\{ \sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \subseteq \left\{ \sum_{i \in K} x_{\beta(i)} | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \quad (14.254)$$

Let $s \in \{\sum_{i \in K} x_{\beta(i)} | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$ then there exist a finite $K \subseteq \{k, \dots, \infty\}$ such that $s = \sum_{i \in K} x_{\beta(i)}$. As $\beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ we have that $\beta|_K: K \rightarrow \beta(K)$ is a bijection. Hence we have

$$s = \sum_{i \in K} x_{\beta(i)} = \sum_{i \in K} x_{\beta|_K(i)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \beta(K)} x_i$$

Which as $\beta(K)$ is finite [for K is finite and β is a bijection] means that $s = \sum_{i \in \beta(K)} x_i \in \{\sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\}$. So that

$$\left\{ \sum_{i \in K} x_{\beta(i)} | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \subseteq \left\{ \sum_{i \in K} x_i | K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\}$$

which combined with [eq: 14.254] prove that

$$\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} = \left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \quad (14.255)$$

Now we have:

$\Rightarrow$. As $\sum_{i=k}^{\infty} x_i$ converges it follows from [theorem: 14.420] that

$$\sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \right) \text{ exists}$$

so that by [eq: 14.255] $\sup(\{\sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\})$ exists, hence using [theorem: 14.420] again it follows that

$$\sum_{i=k}^{\infty} x_{\beta(i)} \text{ converges}$$

Further we have

$$\begin{aligned} \sum_{i=k}^{\infty} x_i & \stackrel{[\text{theorem: 14.420}]}{=} \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \right) \\ & \stackrel{[\text{eq: 14.255}]}{=} \sup \left(\left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \right) \\ & \stackrel{[\text{theorem: 14.420}]}{=} \sum_{i=k}^{\infty} x_{\beta(i)} \end{aligned}$$

$\Leftarrow$. As $\sum_{i=k}^{\infty} x_{\beta(i)}$ converges it follows from [theorem: 14.420] that

$$\sup \left(\left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite} \right\} \right) \text{ exists}$$

so that by [eq: 14.255] $\sup(\{\sum_{i \in K} x_i \mid K \in \mathcal{P}(\{k, \dots, \infty\}) \wedge K \text{ is finite}\})$ exists, hence using [theorem: 14.420] again it follows that

$$\sum_{i=k}^{\infty} x_i \text{ converges} \quad \square$$

Just as commutativity in finite sums allows us to define a more general form of a sum over elements of a finite set [see definition: 11.32] we can use the above theorem to define a infinite sum over a countable set of non negative numbers.

Theorem 14.424. *Let I be a denumerable set [see definition: 6.24], $\{x_i\}_{i \in I} \subseteq [0, \infty[$ a denumerable family of non negative numbers. If there exist a $\beta: \mathbb{N}_0 \rightarrow I$ such that $\sum_{i=0}^{\infty} x_{\beta(i)} = x$ exist then for every bijection $\alpha: \mathbb{N}_0 \rightarrow I$ we have that $\sum_{i=0}^{\infty} x_{\alpha(i)} = x$.*

Proof. As $\beta: \mathbb{N}_0 \rightarrow I$ is a bijection we have that $\beta^{-1}: I \rightarrow \mathbb{N}_0$ is a bijection so that $\beta^{-1} \circ \alpha: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is a bijection. Hence, as $\sum_{i=0}^{\infty} x_{\beta(i)} = x$, it follows from [theorem: 14.423] that $\sum_{i=0}^{\infty} x_{\beta((\beta^{-1} \circ \alpha)(i))} = x$. As $\forall i \in \mathbb{N}_0$ we have $x_{\beta((\beta^{-1} \circ \alpha)(i))} = x_{\beta(\beta^{-1}(\alpha(i)))} = x_{\alpha(i)}$ it follows that $\sum_{i=0}^{\infty} x_{\alpha(i)} = x$. $\square$

The above theorem ensures that the following definition makes set

Definition 14.425. *Let I be a denumerable set [see definition: 6.24], $\{x_i\}_{i \in I} \subseteq [0, \infty[$ a denumerable family of non negative numbers then we say that*

$$\sum_{i \in I} x_i \text{ converges to } x \text{ or } \sum_{i \in I} x_i = x$$

if there exists a bijection $\beta: \mathbb{N}_0 \rightarrow [0, \infty[$ and a $x \in \mathbb{R}$ such that $\sum_{i=0}^{\infty} x_{\beta(i)}$ converges to x .

As a countable family is either a finite family or a denumerable family of non negative real number we can combine definitions [definition: 11.32] and [definition: 14.424] to define the sum of a countable family of non negative real numbers.

Definition 14.426. Let I be a countable set [so by [definition: 6.25] I is finite or denumerable] and $\{x_i\}_{i \in I} \subseteq [0, \infty[$ then we say that $\sum_{i \in I} x_i$ converges if either

I is finite and then $\sum_{i \in I} x_i = \sum_{i=0}^n x_{\beta(i)}$ where $\beta: \{0, \dots, n\} \rightarrow I$ is a bijection [see definition: 11.32]

or

I is denumerable and there exist a $x \in \mathbb{R}$ such that $\sum_{i \in I} x_i = x$ [see definition: 14.424]

Note 14.427. In the finite case $\sum_{i \in I} x_i$ is guaranteed to be defined, however in the infinite case convergence is needed to ensure that $\sum_{i \in I} x_i$ exist.

Example 14.428. If $\{x_i\}_{i \in \emptyset} \subseteq [0, \infty[$ then $\sum_{i \in \emptyset} x_i$ converges to 0

Proof. As $\emptyset$ is finite this follows from [definition: 11.32]. □

Theorem 14.429. Let I, J be countable sets, $I \subseteq J$ and $\{x_i\}_{i \in J} \subseteq [0, \infty[$ such that $\sum_{i \in J} x_i$ converges then $\sum_{i \in I} x_i$ converges and $\sum_{i \in I} x_i \leq \sum_{i \in J} x_i$.

Proof. For J we have either:

J is finite. Then as $I \subseteq J$ I is also finite so that $\sum_{i \in I} x_i, \sum_{i \in J} x_i$ are defined by [theorem: 11.32] hence

$$\sum_{i \in I} x_i \leq \sum_{i \in I} x_i + \sum_{i \in J \setminus I} x_i \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in J} x_i$$

J is denumerable. By definition there exist a bijection $\beta: \mathbb{N}_0 \rightarrow J$ such that $\sum_{i=0}^{\infty} x_{\beta(i)}$ converges to a real number x and $\sum_{i \in J} x_i = x = \sum_{i=0}^{\infty} x_{\beta(i)}$. For I we have now either:

I is finite. As $\beta: \mathbb{N}_0 \rightarrow J$ is a bijection and I is finite we have that $\beta^{-1}(I) \stackrel{[\text{theorem: 2.70}]}{=} (\beta^{-1})(I)$ is finite [see theorem: 6.45] so that $\beta|_{\beta^{-1}(I)}: \beta^{-1}(I) \rightarrow \beta(\beta^{-1}(I)) = I$ is a bijection. Hence we have

$$\sum_{i \in I} x_i \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \beta^{-1}(I)} x_{\beta|_{\beta^{-1}(I)}(i)} = \sum_{i \in \beta^{-1}(I)} x_{\beta(i)}$$

As $\sum_{i \in I} x_i = \sum_{i \in \beta^{-1}(I)} x_{\beta(i)} \in \sup(\{\sum_{i \in K} x_{\beta(i)} | K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\})$ we have that

$$\sum_{i \in I} x_i \leq \sup\left(\left\{\sum_{i \in K} x_{\beta(i)} | K \subseteq \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\right\}\right) \stackrel{[\text{theorem: 14.420}]}{=} \sum_{i=0}^{\infty} x_{\beta(i)} = \sum_{i \in J} x_i$$

hence

$$\sum_{i \in I} x_i \leq \sum_{i \in J} x_i$$

I is denumerable. Then there exist a bijection $\alpha: \mathbb{N}_0 \rightarrow I \subseteq J$. Take a finite $K \subseteq \mathbb{N}_0$ then $\alpha(K)$ is finite and $\alpha(K) \subseteq J$ so that $\beta^{-1}(\alpha(K)) \stackrel{[\text{theorem: 2.70}]}{=} (\beta^{-1})(\alpha(K))$ is finite [see theorem: 6.45]. Define then the following bijections:

$$\alpha|_K: K \rightarrow \alpha(K) \text{ and } \beta|_{\beta^{-1}(\alpha(K))}: \beta^{-1}(\alpha(K)) \rightarrow \beta(\beta^{-1}(\alpha(K))) = \alpha(K)$$

Then we have

$$\begin{aligned}
\sum_{i \in K} x_{\alpha(i)} &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \alpha(K)} x_i \\
&\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \beta^{-1}(\alpha(K))} x_{\beta(i)} \\
&\leq \sup \left(\left\{ \sum_{i \in L} x_{\beta(i)} \mid L \in \mathcal{P}(\mathbb{N}_0) \wedge L \text{ is finite} \right\} \right) \\
&\stackrel{[\text{theorem: 14.420}]}{=} \sum_{i=0}^{\infty} x_{\beta(i)} \\
&= \sum_{i \in J} x_i
\end{aligned}$$

Hence $\{\sum_{i \in K} x_{\alpha(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\}$ is bounded above by $\sum_{i \in J} x_i$ and trivially non empty. So by [theorem: 10.18] $\sup(\{\sum_{i \in K} x_{\alpha(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\})$ exist and $\sup(\{\sum_{i \in K} x_{\alpha(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\}) \leq \sum_{i \in J} x_i$. So using [theorem: 14.420] it follows that $\sum_{i=0}^{\infty} x_{\alpha(i)}$ exist and

$$\sum_{i=0}^{\infty} x_{\alpha(i)} = \sup \left(\left\{ \sum_{i \in K} x_{\alpha(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite} \right\} \right) \leq \sum_{i \in J} x_i$$

Finally using [definition: 14.424] we have that $\sum_{i \in I} x_i = \sum_{i=0}^{\infty} x_{\alpha(i)}$ so that

$$\sum_{i \in I} x_i \leq \sum_{i \in J} x_i \quad \square$$

For countable families we have also the following equivalent definition of convergence.

Theorem 14.430. *Let I be a countable set and $\{x_i\}_{i \in I} \subseteq [0, \infty[$ a countable family of non negative real numbers then*

$$\sum_{i \in I} x_i \text{ converges} \Leftrightarrow \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right) \text{ exist}$$

Further if $\sum_{i \in I} x_i$ converges then

$$\sum_{i \in I} x_i = \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right)$$

Proof. As I is countable we have two possibilities to consider:

I is finite. Then for a finite $K \subseteq I$ we have by [theorem: 14.429] that $\sum_{i \in K} x_i \leq \sum_{i \in I} x_i$ so that $\{\sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite}\}$ is bounded above by $\sum_{i \in I} x_i$. Hence by [theorem: 10.18] $\sup(\{\sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite}\})$ exist and

$$\sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right) \leq \sum_{i \in I} x_i \quad (14.256)$$

As $\sum_{i \in I} x_i$ converges by definition if I is finite we have

$$\sum_{i \in I} x_i \text{ converges} \Leftrightarrow \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right) \text{ exist}$$

As I is finite we have clearly $\sum_{i \in I} x_i \in \{\sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite}\}$ so that

$$\sum_{i \in I} x_i \leq \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right)$$

which combined with [eq: 14.256] gives

$$\sum_{i \in I} x_i = \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right)$$

I is denumerable. Then there exist a bijection $\beta: \mathbb{N}_0 \rightarrow I$. Let $s \in \{\sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\}$ then there exist a finite $K \subseteq \mathbb{N}_0$ such that $s = \sum_{i \in K} x_{\beta(i)}$. As $\beta|_K: K \rightarrow \beta(K)$ is a bijection we have

$$s = \sum_{i \in K} x_{\beta(i)} = \sum_{i \in K} x_{\beta|_K(i)} \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \beta(K)} x_i$$

which as $\beta(K)$ is finite [see theorem: 6.45] and $\beta(K) \subseteq I$ proves that $s \in \{\sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite}\}$ hence

$$\left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite} \right\} \subseteq \left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \quad (14.257)$$

Let $s \in \{\sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite}\}$ then there exist a finite $K \subseteq I$ such that $s = \sum_{i \in K} x_i$. As $\beta|_{\beta^{-1}(K)}: \beta^{-1}(K) \rightarrow \beta(\beta^{-1}(K)) = K$ is a bijection we have

$$s = \sum_{i \in K} x_i \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \beta^{-1}(K)} x_{\beta|_{\beta^{-1}(K)}(i)} = \sum_{i \in \beta^{-1}(K)} x_{\beta(i)}$$

which as $\beta^{-1}(K) \stackrel{[\text{theorem: 2.70}]}{=} (\beta^{-1})(K)$ is finite by [theorem: 6.45] and $\beta^{-1}(K) \subseteq \mathbb{N}_0$ proves that $s \in \{\sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite}\}$. Hence

$$\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \subseteq \left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite} \right\}$$

which combined with [eq: 14.257] proves that

$$\left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite} \right\} = \left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \quad (14.258)$$

Finally we have

$$\begin{aligned} & \sum_{i \in I} x_i \text{ converges to } x \stackrel{[\text{definition: 14.424}]}{\Leftrightarrow} \\ & \sum_{i=0}^{\infty} x_{\beta(i)} \text{ converges to } x \stackrel{[\text{theorem: 14.420}]}{\Leftrightarrow} \\ & \sup \left(\left\{ \sum_{i \in K} x_{\beta(i)} \mid K \in \mathcal{P}(\mathbb{N}_0) \wedge K \text{ is finite} \right\} \right) \text{ exists and is equal to } x \stackrel{[\text{eq: 14.258}]}{\Leftrightarrow} \\ & \sup \left(\left\{ \sum_{i \in K} x_i \mid K \in \mathcal{P}(I) \wedge K \text{ is finite} \right\} \right) \text{ exists and is equal to } x \end{aligned}$$

□

A trivial example of the use of the above theorem is the following

Example 14.431. Let I be a countable set and $\{x_i\}_{i \in I} \subseteq [0, \infty[$ such that $\forall i \in I$ $x_i = 0$ then $\sum_{i \in I} x_i$ converges to 0.

Proof. If $K \subseteq I$ finite then we have as $\forall i \in K$ $x_i = 0$ that $\sum_{i \in K} x_i = 0$ so that $\sup(\{\sum_{i \in K} x_i \mid K \subseteq I \wedge K \text{ is finite}\}) = 0$ which by the previous theorem proves the example. □

Convergent series of non negative numbers are associative, to prove this we need first a little lemma.

Lemma 14.432. Let $n \in \mathbb{N}_0$ and $\{A_i\}_{i \in \{0, \dots, n\}} \subseteq \mathcal{P}(\mathbb{R}) \setminus \{\emptyset\}$ a family of non empty sets of real numbers then if there exist a $s \in \mathbb{R}$ such that $\forall (x_0, \dots, x_n) \in \prod_{i \in \{0, \dots, n\}} A_i$ $\sum_{i=0}^n x_i \leq s$ it follows that $\forall i \in \{0, \dots, n\}$ $\sup(A_i)$ exist and $\sum_{i=0}^n \sup(A_i) \leq s$

Proof. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{A_i\}_{i \in \{0, \dots, n\}} \subseteq \mathcal{P}(\mathbb{R}) \setminus \{\emptyset\} \text{ satisfies that for a } s \in \mathbb{R} \text{ with } \forall x \in \prod_{i \in \{0, \dots, n\}} A_i \text{ we have } \sum_{i=0}^n x_i \leq s \text{ then we have that } \forall i \in \{0, \dots, n\} \sup(A_i) \text{ exist and } \sum_{i=0}^n \sup(A_i) \leq s \right\}$$

then we have:

$0 \in S$. Let $y \in A_0$ then $y = (x) \in \prod_{i \in \{0\}} A_i$ so that $x = \sum_{i=0}^0 y_0 \leq s$ proving that A_0 is bounded above by s , hence, as also $A_0 \neq \emptyset$, it follows from the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] that $\sup(A_0)$ exists and $\sup(A_0) \leq s \Rightarrow \sum_{i=0}^0 \sup(A_i) = \sup(A_0) \leq s$. Hence $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{A_i\}_{i \in \{0, \dots, n+1\}} \subseteq \mathcal{P}(\mathbb{R}) \setminus \{\emptyset\}$ and let $s \in \mathbb{R}$ be such that $\forall x \in \prod_{i \in \{0, \dots, n+1\}} A_i$ we have $\sum_{i=0}^{n+1} x_i \leq s$. Let $u \in A_{n+1}$ then we have $\forall x \in \prod_{i \in \{0, \dots, n\}} A_i$ that $y = (x_0, \dots, x_n, u) \in \prod_{i \in \{0, \dots, n+1\}} A_i$ so that $u + \sum_{i=0}^n x_i = \sum_{i=0}^{n+1} y_i \leq s$, hence

$$\forall x \in \prod_{i \in \{0, \dots, n\}} A_i \text{ we have } \sum_{i=0}^n x_i \leq s - u$$

So as $n \in S$ we have that

$$\forall i \in \{0, \dots, n\} \sup(A_i) \text{ exist and } \sum_{i=0}^n \sup(A_i) \leq s - u \quad (14.259)$$

From the above it follows that for $u \in A_{n+1}$ $u \leq s - \sum_{i=0}^n \sup(A_i)$ proving that A_{n+1} is bounded above by $s - \sum_{i=0}^n \sup(A_i)$, as further $\emptyset \neq A_{n+1}$ it follows from [theorem: 10.18] that

$$\sup(A_{n+1}) \text{ exist and } \sup(A_{n+1}) \leq s - \sum_{i=0}^n \sup(A_i) \quad (14.260)$$

So by [eqs: 14.259, 14.260] we have

$$\forall i \in \{0, \dots, n+1\} \sup(A_i) \text{ exist and } \sum_{i=0}^{n+1} \sup(A_i) = \sup(A_{n+1}) + \sum_{i=0}^n \sup(A_i) \leq s$$

proving that $n+1 \in S$. □

Lemma 14.433. Let I be a countable set, $\{x_i\}_{i \in I} \subseteq [0, \infty[$ such that $\sum_{i \in I} x_i$ converges, $n \in \mathbb{N}_0$ and $\{N_i\}_{i \in \{0, \dots, n\}} \subseteq \mathcal{P}(I)$ such that $\forall i, j \in \{0, \dots, n\}$ with $i \neq j$ we have $N_i \cap N_j = \emptyset$ and $I = \bigcup_{i \in \{0, \dots, n\}} N_i$ then $\forall i \in \{0, \dots, n\}$ we have that

$$\sum_{j \in N_i} x_j \text{ converges}$$

and

$$\sum_{i \in I} x_i = \sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right)$$

Proof. As $\forall i \in \{0, \dots, n\}$ $N_i \subseteq I$ and $\sum_{i \in I} x_i$ converges it follows from [theorem: 14.429] that

$$\forall i \in \{0, \dots, n\} \sum_{j \in N_i} x_j \text{ converges} \quad (14.261)$$

As $\sum_{i \in I} x_i$ converges it follows from [theorem: 14.430] that $\sup(\{\sum_{i \in K} x_i | K \subseteq I \wedge K \text{ finite}\})$ exist and

$$\sum_{i \in I} x_i = \sup \left(\left\{ \sum_{i \in K} x_i | K \subseteq I \wedge K \text{ finite} \right\} \right) \quad (14.262)$$

Let $\varepsilon \in \mathbb{R}^+$ then, as $\sum_{i \in I} x_i - \varepsilon < \sup(\{\sum_{i \in K} x_i | K \subseteq I \wedge K \text{ finite}\})$, there exist a finite $K \subseteq I$ such that $\sum_{i \in I} x_i - \varepsilon < \sum_{i \in K} x_i$ or

$$\sum_{i \in I} x_i < \sum_{i \in K} x_i + \varepsilon \quad (14.263)$$

Now given $i \in \{0, \dots, n\}$ we have that $N_i \cap K$ is finite and as $N_i \cap K \subseteq N_i$ it follows from [theorem: 14.429] that

$$\sum_{j \in N_i \cap K} x_j \leq \sum_{j \in N_i} x_j \quad (14.264)$$

Further as $K \subseteq I$

$$K = I \cap K = \left(\bigcup_{i \in \{0, \dots, n\}} N_i \right) \cap K = \bigcup_{i \in \{0, \dots, n\}} (N_i \cap K)$$

and $\forall i, j \in \{0, \dots, n\}$ with $i \neq j$ we have

$$(N_i \cap K) \cap (N_j \cap K) = (N_i \cap N_j) \cap K = \emptyset \cap K = \emptyset$$

so using [theorem: 11.43] we have

$$\begin{aligned} \sum_{i \in K} x_i & \stackrel{[\text{theorem: 11.43}]}{=} \sum_{i=0}^n \left(\sum_{j \in N_i \cap K} x_j \right) \\ & \leq_{[\text{eq: 14.264}]} \sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right) \end{aligned}$$

Combining the above with [eq: 14.263] results in $\sum_{i \in I} x_i < \sum_{i=0}^n (\sum_{j \in N_i} x_j) + \varepsilon$ which as $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary allows us to use [theorem: 10.31] giving

$$\sum_{i \in I} x_i \leq \sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right) \quad (14.265)$$

Take now $s \in \prod_{i \in \{0, \dots, n\}} \{\sum_{j \in K} x_j | K \subseteq N_i \wedge K \text{ finite}\}$ then there exist a family of finite sets $\{K_i\}_{i \in \{0, \dots, n\}}$ with $\forall i \in \{0, \dots, n\} K_i \subseteq N_i$ [so the sets are finite and pairwise disjoint] and $s_i = \sum_{j \in K_i} x_j$. Hence

$$\sum_{i=0}^n s_i = \sum_{i=0}^n \left(\sum_{j \in K_i} x_j \right) \stackrel{[\text{theorem: 11.43}]}{=} \sum_{j \in \bigcup_{i \in \{0, \dots, n\}} K_i} x_i \quad (14.266)$$

As $\bigcup_{i \in \{0, \dots, n\}} K_i \subseteq \bigcup_{i \in \{0, \dots, n\}} N_i = I$ we have by [theorem: 14.429] as $\sum_{i \in I} x_i$ is convergent that $\sum_{j \in \bigcup_{i \in \{0, \dots, n\}} K_i} x_i \leq \sum_{i \in I} x_i$, hence by [eq: 14.266]

$$\sum_{i=0}^n s_i \leq \sum_{i \in I} x_i$$

So we can use the previous lemma [lemma: 14.432] proving that

$$\forall i \in \{0, \dots, n\} \sup \left(\sum_{j \in K} x_j | K \subseteq N_i \wedge K \text{ finite} \right) \text{ exist and } \sum_{i=0}^n \sup \left(\sum_{j \in K} x_j | K \subseteq N_i \wedge K \text{ finite} \right) \leq \sum_{i \in I} x_i$$

Using [theorem: 14.430] we have $\forall i \in \{0, \dots, n\} \sum_{j \in N_i} x_j = \sup(\sum_{j \in K} x_j | K \subseteq N_i \wedge K \text{ finite})$ which combined with the above proves that

$$\sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right) \leq \sum_{i \in I} x_i$$

which combined with [eq: 14.265] proves

$$\sum_{i \in I} x_i = \sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right) \quad \square$$

Lemma 14.434. *Let I be a countable set, $\{x_i\}_{i \in I} \subseteq [0, \infty[$ such that $\sum_{i \in I} x_i$ converges, $\{N_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{P}(I)$ be such that $\forall i, j \in \mathbb{N}_0$ with $i \neq j$ $N_i \cap N_j = \emptyset$ and $I = \bigcup_{i \in \mathbb{N}_0} N_i$ then we have that*

$$\forall i \in \mathbb{N}_0 \sum_{j \in N_i} x_j \text{ converges}$$

and

$$\sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right) \text{ converges and } \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right) = \sum_{i \in I} x_i$$

Proof. First as $\forall i \in \mathbb{N}_0$ $N_i \subseteq I$ we have by [theorem: 14.429] that

$$\forall i \in \mathbb{N}_0 \sum_{j \in N_i} x_j \text{ converges} \quad (14.267)$$

Let $n \in \mathbb{N}_0$ and define $K_n = \bigcup_{i \in \{0, \dots, n\}} N_i \subseteq I$ then by [theorem: 14.429] again we have that

$$\forall n \in \mathbb{N}_0 \sum_{i \in K_n} x_i \text{ converges and } \sum_{i \in K_n} x_i \leq \sum_{i \in I} x_i \quad (14.268)$$

Further as I is countable it follows by [theorem: 6.71] that $\forall n \in \mathbb{N}_0$ K_n is countable and as $\{N_i\}_{i \in \{0, \dots, n\}}$ is pairwise disjoint it follows from the previous lemma [lemma: 14.433] that

$$\sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right) \stackrel{[\text{lemma: 14.433}]}{=} \sum_{i \in K_n} x_i \leq_{[\text{eq: 14.268}]} \sum_{i \in I} x_i$$

so that $\{\sum_{i=0}^n (\sum_{j \in N_i} x_j) | n \in \mathbb{N}_0\}$ is bounded above by $\sum_{i \in I} x_i$, hence we have by [theorem 10.18] that $\sup(\{\sum_{i=0}^n (\sum_{j \in N_i} x_j) | n \in \mathbb{N}_0\})$ exist and

$$\sup \left(\left\{ \sum_{i=0}^n \left(\sum_{j \in N_i} x_j \right) | n \in \mathbb{N}_0 \right\} \right) \leq \sum_{i \in I} x_i$$

Using then [theorem: 14.420] we have that

$$\sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right) \text{ converges and } \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right) \leq \sum_{i \in I} x_i \quad (14.269)$$

For the opposite equality, let $\varepsilon \in \mathbb{R}^+$ then as $\sum_{i \in I} x_i - \varepsilon < \sum_{i \in I} x_i$ and

$$\sum_{i \in I} x_i \stackrel{[\text{theorem: 14.430}]}{=} \sup \left(\left\{ \sum_{i \in K} x_i | K \subseteq I \wedge K \text{ is finite} \right\} \right)$$

there exist a finite $K \subseteq I$ such that $\sum_{i \in I} x_i - \varepsilon < \sum_{i \in K} x_i$ or

$$\sum_{i \in I} x_i < \sum_{i \in K} x_i + \varepsilon \quad (14.270)$$

Take now $J = \{i \in \mathbb{N}_0 | K \cap N_i \neq \emptyset\} \subseteq \mathbb{N}_0$ so that by [theorem: 6.71] J is countable. If $x \in K$ then as $K = K \cap I = K \cap (\bigcup_{i \in \mathbb{N}_0} N_i) = \bigcup_{i \in \mathbb{N}_0} (K \cap N_i)$ there exist a $i \in \mathbb{N}_0$ such that $x \in K \cap N_i \Rightarrow K \cap N_i \neq \emptyset$ hence $i \in J$ proving that $x \in \bigcup_{i \in J} (K \cap N_i)$. Hence $K \subseteq \bigcup_{i \in J} (K \cap N_i)$ which as trivially $\bigcup_{i \in J} (K \cap N_i) \subseteq K$ proves that

$$K = \bigcup_{i \in J} (K \cap N_i) \quad (14.271)$$

We have now two cases to consider for J :

$J = \emptyset$. Then by [eq: 14.271] $K = \emptyset$ hence $\sum_{i \in K} x_i = 0$ so that by [eq: 14.270] $\sum_{i \in I} x_i < \varepsilon$ which by [theorem: 10.31] proves that

$$\sum_{i \in I} x_i \leq 0 \leq \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right)$$

$J \neq \emptyset$. Assume that J is not finite then as $\{K \cap N_i\}_{i \in J}$ is a family of pairwise disjoint non empty sets it follows from [theorem: 6.48] that $K = \bigcup_{i \in J} (K \cap N_i)$ is infinite, contradicting the fact that K was chosen to be finite. Hence the assumption is wrong and J is finite and non empty so that by [theorem: 6.49]

$$m = \max(J) \text{ exist}$$

Let $x \in \bigcup_{i \in \{0, \dots, m\}} (K \cap N_i)$ then $\exists i \in \{0, \dots, m\}$ such that $x \in K \cap N_i \Rightarrow K \cap N_i \neq \emptyset$ so that $i \in J$ proving that $x \in \bigcup_{i \in J} (K \cap N_i) \stackrel{[\text{eq: 14.134}]}{=} K$ proving that

$$\bigcup_{i \in \{0, \dots, m\}} (K \cap N_i) \subseteq K \quad (14.272)$$

On the other hand if $x \in K$ then by [eq: 14.271] there exist a $i \in J$ such that $x \in K \cap N_i$ which as $J \subseteq \{0, \dots, \max(J)\} = \{0, \dots, m\}$ proves that $x \in \bigcup_{i \in \{0, \dots, m\}} (K \cap N_i)$ or $K \subseteq \bigcup_{i \in \{0, \dots, m\}} (K \cap N_i)$, combining this with [eq: 14.272] gives

$$K = \bigcup_{i \in \{0, \dots, m\}} (K \cap N_i)$$

As $\{K \cap N_i\}_{i \in J}$ is pairwise disjoint we have by the above and [theorem: 11.43] that

$$\begin{aligned} \sum_{i \in K} x_i &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i=0}^m \left(\sum_{j \in K \cap N_i} x_j \right) \\ &\leq [\text{theorem: 14.429}] \sum_{i=0}^m \left(\sum_{j \in N_i} x_j \right) \\ &\leq \sup \left(\left\{ \sum_{i=0}^m \left(\sum_{j \in N_i} x_j \right) \mid m \in \mathbb{N}_o \right\} \right) \\ &\stackrel{[\text{theorem: 14.420}]}{=} \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right) \end{aligned}$$

So we have

$$\sum_{i \in I} x_i \stackrel{[\text{eq: 14.270}]}{<} \sum_{i \in K} x_i + \varepsilon \leq \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right) + \varepsilon$$

so that by [theorem: 10.31] we have

$$\sum_{i \in I} x_i \leq \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right)$$

So in all cases we have $\sum_{i \in I} x_i \leq \sum_{i=0}^{\infty} (\sum_{j \in N_i} x_j)$ which combined with [eq: 14.269] it follows that

$$\sum_{i \in I} x_i = \sum_{i=0}^{\infty} \left(\sum_{j \in N_i} x_j \right)$$

□

Finally we are ready to prove the general case of countable families.

Theorem 14.435. *Let I, K be countable sets, $I \neq \emptyset$, $\{K_i\}_{i \in I}$ a countable family of countable sets such that $K = \bigcup_{i \in I} K_i$ and $\forall i, j \in I$ with $i \neq j$ $K_i \cap K_j = \emptyset$ and $\{x_i\}_{i \in K} \subseteq [0, \infty[$ a countable family of non negative numbers such that $\sum_{i \in K} x_i$ converges then*

$$\forall i \in I \sum_{j \in K_i} x_j \text{ converges and } \sum_{i \in K} x_i = \sum_{i \in I} \left(\sum_{j \in K_i} x_j \right)$$

Proof. As I is countable we have two possible cases to consider:

I is finite. Then there exists a $n \in \mathbb{N}_0$ and a bijection $\beta: \{0, \dots, n\} \rightarrow I$. Define then

$$\{L_i\}_{i \in \{0, \dots, n\}} \text{ by } L_i = K_{\beta(i)}$$

then we have

$$K = \bigcup_{i \in I} K_i \stackrel{[\text{theorem: 2.127}]}{=} \bigcup_{i \in \{0, \dots, n\}} K_{\beta(i)} = \bigcup_{i \in \{0, \dots, n\}} L_i$$

Further if $i, j \in \{0, \dots, n\}$ with $i \neq j$ then as β is a bijection $\beta(i) \neq \beta(j)$ so that $L_i \cap L_j = K_{\beta(i)} \cap K_{\beta(j)} = \emptyset$. So we can use [lemma: 14.433] giving

$$\forall i \in \{0, \dots, n\} \sum_{j \in L_i} x_j \text{ converges and } \sum_{i=0}^n \left(\sum_{j \in L_i} x_j \right) = \sum_{i \in K} x_i \quad (14.273)$$

Finally $\forall i \in I$ we have that $K_i = K_{\beta(\beta^{-1}(i))} = L_{\beta^{-1}(i)}$ so that

$$\sum_{j \in K_i} x_j = \sum_{j \in L_{\beta^{-1}(i)}} x_j \text{ converges}$$

and

$$\begin{aligned} \sum_{i \in K} x_i &\stackrel{[\text{eq: 14.273}]}{=} \sum_{i=0}^n \left(\sum_{j \in L_i} x_j \right) \\ &= \sum_{i=0}^n \left(\sum_{j \in K_{\beta(i)}} x_j \right) \\ &\stackrel{[\text{definition: 11.32}]}{=} \sum_{i \in I} \left(\sum_{j \in K_i} x_j \right) \end{aligned}$$

I is denumerable. Then there exist a bijection $\beta: \mathbb{N}_0 \rightarrow I$. Define then

$$\{L_i\}_{i \in \mathbb{N}_0} \text{ by } L_i = K_{\beta(i)}$$

then we have

$$K = \bigcup_{i \in I} K_i \stackrel{[\text{theorem: 2.127}]}{=} \bigcup_{i \in \mathbb{N}_0} K_{\beta(i)} = \bigcup_{i \in \mathbb{N}_0} L_i$$

Further if $i, j \in \mathbb{N}_0$ with $i \neq j$ then as β is a bijection $\beta(i) \neq \beta(j)$ so that $L_i \cap L_j = K_{\beta(i)} \cap K_{\beta(j)} = \emptyset$. So we can use [lemma: 14.434] giving

$$\forall i \in \mathbb{N}_0 \sum_{j \in L_i} x_j \text{ converges and } \sum_{i=0}^{\infty} \left(\sum_{j \in L_i} x_j \right) = \sum_{i \in K} x_i \quad (14.274)$$

Finally $\forall i \in I$ we have that $K_i = K_{\beta(\beta^{-1}(i))} = L_{\beta^{-1}(i)}$ so that

$$\sum_{j \in K_i} x_j = \sum_{j \in L_{\beta^{-1}(i)}} x_j \text{ converges}$$

and

$$\begin{aligned}
 \sum_{i \in K} x_i & \stackrel{[\text{eq: 14.274}]}{=} \sum_{i=0}^{\infty} \left(\sum_{j \in L_j} x_j \right) \\
 & = \sum_{i=0}^{\infty} \left(\sum_{j \in K_{\beta(i)}} x_j \right) \\
 & \stackrel{[\text{definition: 14.425}]}{=} \sum_{i \in I} \left(\sum_{j \in K_i} x_j \right)
 \end{aligned}$$

□

14.10.3.2 Absolute Convergent Series

Definition 14.436. (Absolute Convergence) Let $\langle X, ||| \rangle$ be a normed space and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence then $\sum_{i=k}^{\infty} x_i$ is **absolute convergent** if $\sum_{i=k}^{\infty} \|x_i\|$ is convergent.

The following theorem shows that in a Banach space every series where the norm of its terms is dominated by the terms of a convergent series is convergent.

Theorem 14.437. (Dominant Convergence) Let $k \in \mathbb{N}_0$, $\langle X, ||| \rangle$ be a Banach space, $\{s_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ such that $\sum_{i=k}^{\infty} s_i$ is convergent and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ satisfies $\forall i \in \{k, \dots, \infty\} \ \|x_i\| \leq s_i$ then

$$\sum_{i=k}^{\infty} x_i \text{ converges}$$

and

$$\left\| \sum_{i=k}^{\infty} x_i \right\| \leq \sum_{i=k}^{\infty} s_i$$

Proof. Let $\varepsilon \in \mathbb{R}^+$ then as $\sum_{i=k}^{\infty} s_i$ converges we have that $\{\sum_{i=k}^n s_i\}_{n \in \{k, \dots, \infty\}}$ convergence, hence by [theorem: 14.386] there exist $N \in \{k, \dots, \infty\}$ such that $\forall n, m \geq N$ we have

$$\left| \sum_{i=k}^n s_i - \sum_{i=k}^m s_i \right| < \varepsilon$$

For n, m we have the following possible cases to consider:

$n = m$. Then $\|\sum_{i=k}^n x_i - \sum_{i=k}^m x_i\| = \|\sum_{i=k}^n x_i - \sum_{i=k}^n x_i\| = \|0\| = 0 < \varepsilon$

$n < m$. Then

$$\begin{aligned}
 \left\| \sum_{i=k}^n x_i - \sum_{i=k}^m x_i \right\| & = \left\| \sum_{i=k}^m x_i - \sum_{i=k}^n x_i \right\| \\
 & = \left\| \sum_{i=n+1}^m x_i \right\| \\
 & \stackrel{[\text{theorem: 14.90}]}{\leq} \sum_{i=n+1}^m \|x_i\| \\
 & \leq \sum_{i=n+1}^m s_i \\
 & = \left| \sum_{i=n+1}^m s_i \right| \\
 & = \left| \sum_{i=k}^m s_i - \sum_{i=k}^n s_i \right| \\
 & < \varepsilon
 \end{aligned}$$

$m < n$.

$$\begin{aligned}
 \left\| \sum_{i=k}^n x_i - \sum_{i=k}^m x_i \right\| &= \left\| \sum_{i=n+1}^m x_i \right\| \\
 &\leq_{[\text{theorem: 14.90}]} \sum_{i=n+1}^m \|x_i\| \\
 &\leq \sum_{i=n+1}^m s_i \\
 &= \left| \sum_{i=n+1}^m s_i \right| \\
 &= \left| \sum_{i=k}^m s_i - \sum_{i=k}^n s_i \right| \\
 &< \varepsilon
 \end{aligned}$$

Hence in all cases $\|\sum_{i=k}^n x_i - \sum_{i=k}^m x_i\| < \varepsilon$ proving that $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}}$ is Cauchy. As $\langle X, \|\cdot\| \rangle$ is a Banach space $\lim_{n \rightarrow \infty} x_n$ exist so that by definition

$$\sum_{i=k}^{\infty} x_i \text{ converges}$$

Further if $n \in \{k, \dots, \infty\}$ then we have

$$\left\| \sum_{i=k}^n x_i \right\| \leq \sum_{i=k}^n \|x_i\| \leq \sum_{i=k}^n s_i$$

Using [theorem: 14.367] it follows that

$$\lim_{n \rightarrow \infty} \left\| \sum_{i=k}^n x_i \right\| \leq \lim_{n \rightarrow \infty} \sum_{i=k}^n s_i = \sum_{i=k}^{\infty} s_i$$

As by [theorem: 14.374] $\lim_{n \rightarrow \infty} \|\sum_{i=k}^n x_i\| = \left\| \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \right\| \stackrel{\text{def}}{=} \|\sum_{i=k}^{\infty} x_i\|$ it follows that

$$\left\| \sum_{i=k}^{\infty} x_i \right\| \leq \sum_{i=k}^{\infty} s_i$$

□

We have a variant of the dominant convergence theorem where the inequality is strict, this variant will be used in the proof of the open mapping theorem.

Corollary 14.438. *Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a Banach space, $\{s_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty[$ such that $\sum_{i=k}^{\infty} s_i$ is convergent and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence in X such that:*

1. $\forall i \in \{k, \dots, \infty\} \quad \|x_i\| \leq s_i$
2. $\exists n_0 \in \{k, \dots, \infty\}$ such that $\|x_{n_0}\| < s_{n_0}$

then we have

$$\sum_{i=k}^{\infty} x_i \text{ converges}$$

and

$$\left\| \sum_{i=k}^{\infty} x_i \right\| < \sum_{i=k}^{\infty} s_i$$

Proof. Using [theorem: 14.416] it follows that

$$\sum_{i=n_0+1}^{\infty} s_i \text{ converges and } \sum_{i=k}^{\infty} s_i = \sum_{i=k}^{n_0} s_i + \sum_{i=n_0+1}^{\infty} s_i \quad (14.275)$$

As $\forall i \in \{n_0 + 1, \dots, \infty\}$ $\|x_i\| \leq s_i$ we can use dominant convergence [theorem: 14.437] that

$$\sum_{i=n_0+1}^{\infty} x_i \text{ converges and } \left\| \sum_{i=n_0+1}^{\infty} x_i \right\| \leq \sum_{i=n_0+1}^{\infty} s_i \quad (14.276)$$

So using [theorem: 14.416] we have that

$$\sum_{i=k}^{\infty} x_i \text{ converges and } \sum_{i=k}^{\infty} x_i = \sum_{i=k}^{n_0} x_i + \sum_{i=n_0+1}^{\infty} x_i \quad (14.277)$$

We have also that

$$\begin{aligned} \left\| \sum_{i=k}^{n_0} x_i \right\| &\leq \sum_{i=k}^{n_0} \|x_i\| \\ &= \sum_{i=k}^{n_0-1} \|x_i\| + \|x_{n_0}\| \\ &\leq \sum_{i=k}^{n_0-1} s_i + \|x_{n_0}\| \\ &< \sum_{i=k}^{n_0-1} s_i + s_{n_0} \\ &= \sum_{i=k}^{n_0} s_i \end{aligned}$$

proving that

$$\left\| \sum_{i=k}^{n_0} x_i \right\| < \sum_{i=k}^{n_0} s_i \quad (14.278)$$

So we have that $\sum_{i=k}^{\infty} x_i$ converges and

$$\begin{aligned} \left\| \sum_{i=k}^{\infty} x_i \right\| &\stackrel{[\text{eq: 14.277}]}{=} \left\| \sum_{i=k}^{n_0} x_i + \sum_{i=n_0+1}^{\infty} x_i \right\| \\ &\leq \left\| \sum_{i=k}^{n_0} x_i \right\| + \left\| \sum_{i=n_0+1}^{\infty} x_i \right\| \\ &\stackrel{[\text{eq: 14.276}]}{\leq} \sum_{i=k}^{n_0} s_i + \sum_{i=n_0+1}^{\infty} s_i \\ &<_{[\text{eq: 14.278}]} \sum_{i=k}^{n_0} s_i + \sum_{i=n_0+1}^{\infty} s_i \\ &\stackrel{[\text{eq: 14.275}]}{=} \sum_{i=k}^{\infty} s_i \end{aligned}$$

proving

$$\left\| \sum_{i=k}^{\infty} x_i \right\| < \sum_{i=k}^{\infty} s_i$$

□

A consequence of the dominant convergence theorem is that every absolute convergent series is convergent.

Corollary 14.439. *Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a Banach space and $\sum_{i=k}^{\infty} x_i$ a **absolute convergent series** then $\sum_{i=k}^{\infty} x_i$ converges and $\left\| \sum_{i=k}^{\infty} x_i \right\| \leq \sum_{i=k}^{\infty} \|x_i\|$*

Proof. As $\sum_{i=k}^{\infty} x_i$ is absolute convergent we have by definition that $\sum_{i=k}^{\infty} \|x_i\|$ converges and as trivially $\forall i \in \{k, \dots, \infty\} \|x_i\| \leq \|x_i\|$ it follows from the previous theorem [theorem: 14.437] that $\sum_{i=k}^{\infty} x_i$ converges and that $\|\sum_{i=k}^{\infty} x_i\| \leq \sum_{i=k}^{\infty} \|x_i\|$. $\square$

Theorem 14.440. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a normed space and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence such that there exist a convergent series $\sum_{i=k}^{\infty} s_i$ with $\forall i \in \{k, \dots, \infty\} \|x_i\| \leq s_i$ then $\sum_{i=k}^{\infty} \|x_i\|$ converges [using the normed space $\langle \mathbb{R}, \|\cdot\| \rangle$].

Proof. As $\{\|x_i\|\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ is a sequence in the Banach space $\langle \mathbb{R}, \|\cdot\| \rangle$ [see theorem: 14.394] it follows from [theorem: 14.437] that $\sum_{i=k}^{\infty} \|x_i\|$ converges. $\square$

The following theorem show that only in a Banach space convergence and absolute convergence is the same.

Theorem 14.441. Let $\langle X, \|\cdot\| \rangle$ be a normed space then

$$\langle X, \|\cdot\| \rangle \text{ is Banach} \Leftrightarrow \text{every absolute convergent series converges}$$

Proof.

$\Rightarrow$. This follows from [corollary: 14.439]

$\Leftarrow$. Let $\{x_n\}_{n \in \{k, \dots, \infty\}} \subseteq X$ be a Cauchy sequence. Then $\forall n \in \mathbb{N}$ there exist a $N'_n \in \{k, \dots, \infty\}$ such that $\forall r, s \geq N'_n$ we have $\|x_r - x_s\| < \frac{1}{2^{n+1}}$ define $N_n = \max(N'_n, n)$ then $N_n + 1, N_n \geq N'_n$ so that

$$\|x_{N_n+1} - x_{N_n}\| < \frac{1}{2^{n+1}}$$

Hence we have found a sequence $\{x_{N_n}\}_{n \in \mathbb{N}} \subseteq X$ such that

$$\forall n \in \mathbb{N} \|x_{N_n+1} - x_{N_n}\| < \frac{1}{2^{n+1}} \text{ and } N_n \geq n \quad (14.279)$$

Define

$$\{y_i\}_{i \in \mathbb{N}} \text{ by } y_i = \begin{cases} x_{N_1} & \text{if } i = 1 \\ x_{N_i} - x_{N_{i-1}} & \text{if } i \in \{2, \dots, \infty\} \end{cases}$$

then we have

$$\forall i \in \{2, \dots, \infty\} \|y_i\| = \|x_{N_i} - x_{N_{i-1}}\| < \frac{1}{2^{(i-1)+1}} = \frac{1}{2^i} \quad (14.280)$$

Using [theorem: 14.412] $\sum_{i=1}^{\infty} \frac{1}{2^i}$ is convergent so that by [theorem: 14.416] $\sum_{i=2}^{\infty} \frac{1}{2^i}$ converges. Using then [theorem: 14.440] on [eq: 14.280] proves that $\sum_{i=2}^{\infty} \|y_i\|$ converges, further by [theorem: 14.416] we have that $\sum_{i=1}^{\infty} \|y_i\|$ converges. Hence $\{y_i\}_{i \in \mathbb{N}}$ is absolute convergent. By the hypothesis every absolute convergent series converges hence we have that

$$\sum_{i=1}^{\infty} y_i \text{ converges or } \lim_{n \rightarrow \infty} \sum_{i=1}^n y_i \text{ exist} \quad (14.281)$$

Let $n \in \mathbb{N}$ then we have that

$$\begin{aligned} \sum_{i=1}^n y_i &= \begin{cases} y_1 & \text{if } n = 1 \\ y_1 + \sum_{i=2}^n y_i & \text{if } 1 < n \end{cases} \\ &= \begin{cases} x_{N_1} & \text{if } n = 1 \\ x_{N_1} + \sum_{i=2}^n (x_{N_i} - x_{N_{i-1}}) & \text{if } 1 < n \end{cases} \\ &\stackrel{[\text{theorem: 11.21}]}{=} \begin{cases} x_{N_1} & \text{if } n = 1 \\ x_{N_1} + x_{N_n} - x_{N_1} & \text{if } 1 < n \end{cases} \\ &= x_{N_n} \end{aligned}$$

Combining the above with [eq: 14.281] proves that

$$\{x_{N_i}\}_{i \in \mathbb{N}} \text{ converges to a limit } x \quad (14.282)$$

Take $\varepsilon \in \mathbb{R}^+$ then as $\{x_n\}_{n \in \{k, \dots, \infty\}}$ is Cauchy there exist a $M_1 \in \{k, \dots, \infty\}$ such that $\forall n, m \geq M_1$ we have

$$\|x_n - x_m\| < \frac{\varepsilon}{2}$$

Further as $\lim_{n \rightarrow \infty} x_{N_n} = x$ there exist a $M_2 \in \mathbb{N}$ such that $\forall n \geq M_2$ we have

$$\|x_{N_n} - x\| < \frac{\varepsilon}{2}$$

Let $N = \max(M_1, M_2)$ then if $n \geq N$ we have $n \geq M_1$ [eq: 14.279] $N_n \geq n \geq M_1$ and $n \geq M_2$ so that

$$\|x_n - x\| = \|x_n - x_{N_n} + x_{N_n} - x\| \leq \|x_n - x_{N_n}\| + \|x_{N_n} - x\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving that $\{x_n\}_{n \in \{k, \dots, \infty\}}$ converges to x . As $\{x_n\}_{n \in \{k, \dots, \infty\}}$ was a arbitrary Cauchy sequence it follows that $\langle X, \|\cdot\| \rangle$ is a Banach space. $\square$

We have the following equivalent definitions for absolute convergent series.

Theorem 14.442. *Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ a normed space and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq X$ a sequence then the following are equivalent:*

1. $\sum_{i=k}^{\infty} x_i$ is absolute convergent
2. $\forall \varepsilon \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\forall m \in \mathbb{N}$ and $\forall n \geq N$ we have $\sum_{i=n+1}^{n+m} \|x_i\| < \varepsilon$
3. $\forall \varepsilon \in \mathbb{R}^+$ there exist a finite $K \subseteq \{k, \dots, \infty\}$ such that $\forall H \subseteq \{k, \dots, \infty\}$ with H finite and $K \cap H = \emptyset$ we have $\sum_{i \in H} \|x_i\| < \varepsilon$

Proof. As $\sum_{i=k}^{\infty} \|x_i\|$ is a series of non negative numbers and absolute convergences of $\sum_{i=k}^{\infty} x_i$ is the same as convergence of $\sum_{i=k}^{\infty} \|x_i\|$ the proof follows from [theorem: 14.420]. $\square$

Lemma 14.443. *Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ be a Banach space and $\sum_{i=k}^{\infty} x_i$ a absolute convergent series then $\forall \varepsilon \in \mathbb{R}^+$ there exist a finite $K \subseteq \{k, \dots, \infty\}$ such that*

1. $\forall L \subseteq \{k, \dots, \infty\}$ with L finite and $K \cap L = \emptyset$ we have $\sum_{i \in L} \|x_i\| < \varepsilon$
2. $\forall H \subseteq \{k, \dots, \infty\}$ with H finite and $K \subseteq H$ we have $\|\sum_{i=k}^{\infty} x_i - \sum_{i \in H} x_i\| < \varepsilon$

Proof. Let $\varepsilon \in \mathbb{R}^+$. As by [theorem: 14.441] absolute convergence implies convergence we have that $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ converges to $\sum_{i=k}^{\infty} x_i$, hence there exist a $N \in \{k, \dots, \infty\}$ so that

$$\forall n \geq N \text{ we have } \left\| \sum_{i=k}^n x_i - \sum_{i=k}^{\infty} x_i \right\| < \frac{\varepsilon}{2} \quad (14.283)$$

Using absolute convergence and the previous theorem [see theorem: 14.442] there exist a finite $K' \subseteq \{k, \dots, \infty\}$ such that for every finite $L \subseteq \{k, \dots, \infty\}$ with $K' \cap L = \emptyset$ we have that $\sum_{i \in L} \|x_i\| < \frac{\varepsilon}{2}$. Take $K = \{k, \dots, N\} \cup K'$ then we have $\{k, \dots, N\} \subseteq K \subseteq \{k, \dots, \infty\}$ and for every finite $L \subseteq \{k, \dots, \infty\}$ with $K \cap L = \emptyset$ we have, as $K' \subseteq L$, that $K' \cap L \neq \emptyset$ so that $\sum_{i \in L} \|x_i\| < \frac{\varepsilon}{2}$. So we have the existence of a $K \subseteq \{k, \dots, \infty\}$ such that

$$\{k, \dots, N\} \subseteq K \wedge \forall L \subseteq \{k, \dots, \infty\} \text{ with } L \text{ finite and } K \cap L = \emptyset \text{ we have } \sum_{i \in L} \|x_i\| < \frac{\varepsilon}{2} \quad (14.284)$$

proving (1).

Take $H \subseteq \{k, \dots, \infty\}$ with $K \subseteq H \wedge H$ finite. Let $m \stackrel{\text{is finite}}{=} \max(H)$ so that $H \subseteq \{k, \dots, m\}$ then

$$\begin{aligned}
 \left\| \sum_{i=k}^{\infty} x_i - \sum_{i \in H} x_i \right\| &= \left\| \sum_{i \in H} x_i - \sum_{i=k}^{\infty} x_i \right\| \\
 &= \left\| \sum_{i \in \{k, \dots, m\}} x_i - \sum_{i \in \{k, \dots, m\} \setminus H} x_i - \sum_{i=k}^{\infty} x_i \right\| \\
 &= \left\| \sum_{i=k}^m x_i - \sum_{i \in \{k, \dots, m\} \setminus H} x_i - \sum_{i=k}^{\infty} x_i \right\| \\
 &\leq \left\| \sum_{i=k}^m x_i - \sum_{i=k}^{\infty} x_i \right\| + \left\| \sum_{i \in \{k, \dots, m\} \setminus H} x_i \right\|
 \end{aligned} \tag{14.285}$$

As $\{k, \dots, m\} \setminus H \subseteq \{k, \dots, \infty\}$ and

$$K \cap (\{k, \dots, m\} \setminus H) = K \cap (\{k, \dots, m\} \cap H^c) = (K \cap H^c) \cap \{k, \dots, m\} \stackrel{K \subseteq H}{=} \emptyset \cap \{k, \dots, m\} = \emptyset$$

so that by [eq: 14.284] we have

$$\left\| \sum_{i \in \{k, \dots, m\} \setminus H} x_i \right\| < \frac{\varepsilon}{2} \tag{14.286}$$

As $\{k, \dots, N\} \subseteq K \subseteq H$ we have that $N \leq \max(H) = m$ so that by [eq: 14.283] we have

$$\left\| \sum_{i=k}^m x_i - \sum_{i=k}^{\infty} x_i \right\| < \frac{\varepsilon}{2} \tag{14.287}$$

combining [eqs: 14.285, 14.286 and 14.287] proves that

$$\left\| \sum_{i=k}^{\infty} x_i - \sum_{i \in H} x_i \right\| < \varepsilon$$

and (2). □

Theorem 14.444. Let $k \in \mathbb{N}_0$, $\langle X, \|\cdot\| \rangle$ a Banach space and $\sum_{i=k}^{\infty} x_i$ a absolute convergent series then we have:

1. If $\beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a bijection then $\sum_{i=k}^{\infty} x_{\beta(i)}$ is absolute convergent [and by [theorem: 14.441] it follows that $\sum_{i=k}^{\infty} x_{\beta(i)}$ is also convergent]
2. If $\alpha, \beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ are bijections then $\sum_{i=k}^{\infty} x_{\alpha(i)} = \sum_{i=k}^{\infty} x_{\beta(i)}$
3. If $\beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a bijection then $\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{\infty} x_{\beta(i)}$

Proof.

1. As $\sum_{i=k}^{\infty} x_i$ is absolute convergent $\sum_{i=k}^{\infty} \|x_i\|$ is a convergent series of non negative real numbers, hence using [theorem: 14.423] $\sum_{i=k}^{\infty} \|x_{\beta(i)}\|$ is also convergent proving that $\sum_{i=k}^{\infty} x_{\beta(i)}$ is absolute convergent.
2. Let $s = \sum_{i=k}^{\infty} x_{\alpha(i)}$ and $t = \sum_{i=k}^{\infty} x_{\beta(i)}$. Assume that $s \neq t$ then $\|s - t\| \in \mathbb{R}^+$. Using [lemma: 14.443] there exists finite $K_{\alpha}, K_{\beta} \subseteq \{k, \dots, \infty\}$ such that for every finite H_{α}, H_{β} with $K_{\alpha} \subseteq H_{\alpha}, K_{\beta} \subseteq H_{\beta}$ we have

$$\left\| \sum_{i \in H_{\alpha}} x_{\alpha(i)} - s \right\| < \frac{\|s - t\|}{4} \wedge \left\| \sum_{i \in H_{\beta}} x_{\beta(i)} - t \right\| < \frac{\|s - t\|}{4} \tag{14.288}$$

Define now $P_{\alpha\beta} = K_\alpha \cup \alpha^{-1}(\beta(K_\beta))$ and $P_{\beta\alpha} = K_\beta \cup \beta^{-1}(\alpha(K_\alpha))$. As α and β are bijections and K_α and K_β are finite it follows from [theorems: 2.70, 6.45 and 6.34] that $P_{\alpha\beta}$ and $P_{\beta\alpha}$ are finite and $K_\alpha \subseteq P_{\alpha\beta}$ and $K_\beta \subseteq P_{\beta\alpha}$ so that by [eq: 14.288]

$$\left\| \sum_{i \in P_{\alpha\beta}} x_{\alpha(i)} - s \right\| < \frac{\|s - t\|}{4} \wedge \left\| \sum_{i \in P_{\beta\alpha}} x_{\beta(i)} - t \right\| < \frac{\|s - t\|}{4} \quad (14.289)$$

Further as $\alpha: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ and $\beta: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ are bijections we have that

$$\alpha|_{P_{\alpha\beta}}: P_{\alpha\beta} \rightarrow \alpha(P_{\alpha\beta}) = \alpha(K_\alpha) \cup \beta(K_\beta) \text{ and } \beta|_{P_{\beta\alpha}}: P_{\beta\alpha} \rightarrow \beta(P_{\beta\alpha}) = \beta(K_\beta) \cup \alpha(K_\alpha)$$

are bijections so that we can apply [theorem: 11.36]

$$\begin{aligned} \sum_{i \in P_{\alpha\beta}} x_{\alpha(i)} &= \sum_{i \in P_{\alpha\beta}} x_{\alpha|_{P_{\alpha\beta}}(i)} \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \alpha(P_{\alpha\beta})} x_i \\ &= \sum_{i \in \alpha(K_\alpha) \cup \beta(K_\beta)} x_i \\ &= \sum_{i \in \beta(K_\beta) \cup \alpha(K_\alpha)} x_i \\ &= \sum_{i \in \beta(P_{\beta\alpha})} x_i \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in P_{\beta\alpha}} x_{\beta|_{P_{\beta\alpha}}(i)} \\ &= \sum_{i \in P_{\beta\alpha}} x_{\beta(i)} \end{aligned}$$

proving that

$$\sum_{i \in P_{\alpha\beta}} x_{\alpha(i)} = \sum_{i \in P_{\beta\alpha}} x_{\beta(i)} \quad (14.290)$$

Now

$$\begin{aligned} \|s - t\| &\stackrel{[\text{eq: 14.290}]}{=} \left\| s - \sum_{i \in P_{\alpha\beta}} x_{\alpha(i)} + \sum_{i \in P_{\beta\alpha}} x_{\beta(i)} - t \right\| \\ &\leq \left\| s - \sum_{i \in P_{\alpha\beta}} x_{\alpha(i)} \right\| + \left\| \sum_{i \in P_{\beta\alpha}} x_{\beta(i)} - t \right\| \\ &<_{[\text{eq: 14.289}]} \frac{\|s - t\|}{4} + \frac{\|s - t\|}{4} \\ &< \frac{\|s - t\|}{2} \\ &< \|s - t\| \end{aligned}$$

a contradiction. So the assumption is incorrect and we must have $s = t$ or

$$\sum_{i=k}^{\infty} x_{\alpha(i)} = \sum_{i=k}^{\infty} x_{\beta(i)}$$

3. As $\text{Id}_{\{k, \dots, \infty\}}: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a bijection we have

$$\sum_{i=k}^{\infty} x_{\alpha(i)} \stackrel{(2)}{=} \sum_{i=k}^{\infty} x_{\text{Id}_{\{k, \dots, \infty\}}(i)} = \sum_{i=k}^{\infty} x_i \quad \square$$

14.10.4 Properties of complete spaces.

Definition 14.445. Let $\langle X, d \rangle$ be a metric space then a function $f: X \rightarrow X$ is a **contraction** if there exist a $\lambda \in [0, 1[$ such that $\forall x, y \in X$ we have $d(f(x), f(y)) \leq \lambda \cdot d(x, y)$

Theorem 14.446. *Let $\langle X, d \rangle$ be a metric space and $f: X \rightarrow X$ a contraction then f is continuous.*

Proof. As $f: X \rightarrow X$ is a contraction there exist a $\lambda \in [0, 1[$ such that

$$\forall x, y \in X \quad d(f(x), f(y)) \leq \lambda \cdot d(x, y)$$

Let $x \in X$ and $\varepsilon \in \mathbb{R}^+$. If $y \in X$ is such that $d(x, y) < \varepsilon$ then we have

$$d(f(x), f(y)) \leq \lambda \cdot d(x, y) < d(x, y) < \varepsilon$$

which by [theorem: 14.134] proves that f is continuous at x . Hence as x was chosen arbitrary $f: X \rightarrow X$ is continuous. $\square$

Contractions are special because they guarantee the existence of a fixed point, that is a point that is mapped on itself by the contraction, this is proved in the Banach Fixed Point theorem.

Theorem 14.447. (Banach Fixed Point Theorem) *Let $\langle X, d \rangle$ be a **complete** metric space and $f: X \rightarrow X$ a contraction then there exists a **unique** $x_0 \in X$ such that $f(x_0) = x_0$. Further if we define for $n \in \mathbb{N}$ $f^{(n)}$ recursively by*

$$\begin{aligned} f^{(1)} &= f \\ f^{(n+1)} &= f \circ f^{(n)} \end{aligned}$$

then $\forall x \in X$ we have

$$\lim_{n \rightarrow \infty} f^{(n)}(x) = x_0$$

Proof. As $f: X \rightarrow X$ is a contraction there exist a $\lambda \in [0, 1[$ such that

$$\forall x, y \in X \quad d(f(x), f(y)) \leq \lambda \cdot d(x, y)$$

As a metric space is non empty there exist a $x \in X$. We prove now by induction that

$$\forall n \in \mathbb{N} \text{ we have } d(f^{(n)}(x), f^{(n+1)}(x)) \leq \lambda^n \cdot d(x, f(x)) \quad (14.291)$$

Proof. Define

$$S = \{n \in \mathbb{N} \mid d(f^{(n)}(x), f^{(n+1)}(x)) \leq \lambda^n \cdot d(x, f(x))\}$$

then we have:

$1 \in S$. As

$$d(f^{(1)}(x), f^{(1+1)}(x)) = d(f(x), (f \circ f)(x)) = d(f(x), f(f(x))) \leq \lambda \cdot d(x, f(x)) = \lambda^1 \cdot d(x, f(x))$$

it follows that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. We have

$$\begin{aligned} d(f^{(n+1)}(x), f^{((n+1)+1)}(x)) &= d((f \circ f^{(n)})(x), (f \circ f^{(n+1)})(x)) \\ &= d(f(f^{(n)}(x)), f(f^{(n+1)}(x))) \\ &\leq \lambda \cdot d(f^{(n)}(x), f^{(n+1)}(x)) \\ &\leq_{n \in S} \lambda \cdot \lambda^n \cdot d(x, f(x)) \\ &= \lambda^{n+1} \cdot d(x, f(x)) \end{aligned}$$

proving that $n+1 \in S$.

Mathematical induction proves [eq: 14.291]. $\square$

Next we prove by induction that

$$\forall n, k \in \mathbb{N} \text{ we have } d(f^{(n)}(x), f^{(n+k)}(x)) \leq \left(\sum_{i=1}^k \lambda^{n+(i-1)} \right) \cdot d(x, f(x)) \quad (14.292)$$

Proof. Given $n \in \mathbb{N}$ define

$$T_n = \left\{ k \in \mathbb{N} \mid d(f^{(n)}(x), f^{(n+k)}(x)) \leq \left(\sum_{i=1}^k \lambda^{n+(i-1)} \right) \cdot d(x, f(x)) \right\}$$

then we have:

$1 \in T_n$. We have by [eq: 14.291] that

$$\begin{aligned} d(f^{(n)}(x), f^{(n+1)}(x)) &\leq \lambda^n \cdot d(x, f(x)) \\ &= \lambda^{n+(1-1)} \cdot d(x, f(x)) \\ &= \left(\sum_{i=1}^1 \lambda^{n+(i-1)} \right) \cdot d(x, f(x)) \end{aligned}$$

proving that $1 \in T_n$.

$k \in T_n \Rightarrow k+1 \in T_n$. Then we have:

$$\begin{aligned} d(f^{(n)}(x), f^{(n+(k+1))}(x)) &\leq \\ d(f^{(n)}(x), f^{(n+k)}(x)) + d(f^{(n+k)}(x), f^{(n+(k+1))}(x)) &\leq_{k \in T_n} \\ \left(\sum_{i=1}^k \lambda^{n+(i-1)} \right) \cdot d(x, f(x)) + d(f^{(n+k)}(x), f^{(n+(k+1))}(x)) &\leq_{[\text{eq: 14.291}]} \\ \left(\sum_{i=1}^k \lambda^{n+(i-1)} \right) \cdot d(x, f(x)) + \lambda^{n+k} \cdot d(x, f(x)) &= \\ \left(\sum_{i=1}^k \lambda^{n+(i-1)} + \lambda^{n+k} \right) \cdot d(x, f(x)) &= \\ \left(\sum_{i=1}^k \lambda^{n+(i-1)} + \lambda^{n+((k+1)-1)} \right) \cdot d(x, f(x)) &= \\ \left(\sum_{i=1}^{k+1} \lambda^{n+(i-1)} \right) \cdot d(x, f(x)) \end{aligned}$$

proving that $k+1 \in T_n$.

Mathematical induction proves [eq: 14.292] □

As $\lambda \in [0, 1[$ we have that $\lambda \neq 1$ so that for $k \in \mathbb{N}$

$$\sum_{i=1}^k \lambda^{n+(i-1)} \stackrel{[\text{lemma: 14.411}]}{=} \lambda^n \cdot \frac{1 - \lambda^k}{1 - \lambda}$$

which combined with [eq: 14.292] proves that

$$\forall k, n \in \mathbb{N} \text{ that } d(f^{(n)}(x), f^{(n+k)}(x)) \leq \lambda^n \cdot \frac{1 - \lambda^k}{1 - \lambda} \cdot d(x, f(x)) < \frac{\lambda^n}{1 - \lambda} \cdot d(x, f(x)) \quad (14.293)$$

Let $\varepsilon \in \mathbb{R}^+$ then as $0 \leq \lambda < 1$ we have by [theorem: 14.348] that there exist a $N \in \mathbb{N}$ such that $\forall n \geq N \lambda^n = |\lambda^n - 0| < \frac{\varepsilon}{\frac{1}{1-\lambda} \cdot d(x, f(x))}$. If $n, m \geq N$ then we may always assume that $m \geq n$ so that

$$\begin{aligned} d(f^{(n)}(x), f^{(m)}(x)) &= d(f^{(n)}(x), f^{(n+(m-n))}(x)) \\ &\stackrel{[\text{eq: 14.293}]}{<} \frac{\lambda^n}{1 - \lambda} \cdot d(x, f(x)) \\ &< \frac{\varepsilon}{\frac{1}{1-\lambda} \cdot d(x, f(x))} \cdot \frac{1}{1 - \lambda} \cdot d(x, f(x)) \\ &= \varepsilon \end{aligned}$$

This proves that $\{f^{(n)}(x)\}_{n \in \{1, \dots, \infty\}}$ is Cauchy, as $\langle X, d \rangle$ is complete it follows that

$$\lim_{n \rightarrow \infty} f^{(n)}(x) = x_0 \text{ exist} \quad (14.294)$$

As by [theorem: 14.446] $f: X \rightarrow X$ is continuous it follows from the above and [theorem: 14.372] that

$$\lim_{n \rightarrow \infty} f^{(n+1)}(x) = \lim_{n \rightarrow \infty} f(f^{(n)}(x)) = f\left(\lim_{n \rightarrow \infty} f^{(n)}(x)\right) = f(x_0) \quad (14.295)$$

So given $\varepsilon \in \mathbb{R}^+$ there exist by [eq: 14.294] a $N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$ we have

$$d(f^{(n)}(x), x_0) < \frac{\varepsilon}{2}$$

and by [eq: 14.295] there exist a $N_2 \in \mathbb{N}$ such that $\forall n \geq N_2$ we have

$$d(f^{(n+1)}(x), f(x_0)) < \frac{\varepsilon}{2}$$

Hence if $n \geq \max(N_1, N_2 + 1)$ then we have

$$\begin{aligned} d(f(x_0), x_0) &\leq d(f(x_0), f^{(n)}(x)) + d(f^{(n)}(x), x_0) \\ &= d(f^{(n)}(x), f(x_0)) + d(f^{(n)}(x), x_0) \\ &<_{[\text{eq: 14.294}]} d(f^{(n)}(x), f(x_0)) + \frac{\varepsilon}{2} \\ &= d(f^{((n-1)+1)}(x), f(x_0)) + \frac{\varepsilon}{2} \\ &<_{n > N_2 + 1 \Rightarrow n-1 \geq N_2} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

As $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary we can use [theorem: 10.31] on the above giving $d(f(x_0), x_0) = 0$, hence

$$f(x_0) = x_0 \text{ where } \lim_{n \rightarrow \infty} f^{(n)}(x) = x_0$$

Finally to prove uniqueness, assume that there exist a $x_1 \in X$ with $x_0 \neq x_1$ such that $f(x_1) = x_1$. As $\lambda < 1$ and $0 < d(x_0, x_1)$ it follows that $\lambda \cdot d(x_1, x_0) < 1 \cdot d(x_1, x_0) = d(x_1, x_0)$, further we have

$$0 < d(x_1, x_0) = d(f(x_1), f(x_0)) \leq \lambda \cdot d(x_1, x_0) < d(x_1, x_0)$$

giving the contradiction $d(x_1, x_0) < d(x_1, x_0)$. So our assumption must be wrong proving that x_0 is unique. $\square$

Lemma 14.448. *Let $\langle X, d \rangle$ be a complete metric space and let $\{A_i\}_{i \in \mathbb{N}}$ be a sequence of non empty bounded closed sets such that $\forall i \in \mathbb{N}$ we have $A_{i+1} \subseteq A_i$ and $\lim_{n \rightarrow \infty} \text{diam}(A_i) = 0$ then we have that*

$$\bigcap_{i \in \mathbb{N}} A_i \neq \emptyset$$

Proof. As $\{A_i\}_{i \in \mathbb{N}}$ is a family of non empty sets it follows from the Axiom of Choice [see theorem: 3.106] there exist a function $x: \mathbb{N} \rightarrow \bigcup_{i \in \mathbb{N}} A_i$ such that $\forall i \in \mathbb{N}$ $x(i) \in A_i$. This defines a family

$$\{x_i\}_{i \in \mathbb{N}} \text{ such that } \forall i \in \mathbb{N} \ x_i \in A_i$$

Let $\varepsilon \in \mathbb{R}^+$ then as $\lim_{n \rightarrow \infty} \text{diam}(A_i) = 0$ there exist a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have

$$\text{diam}(A_n) = |\text{diam}(A_n) - 0| < \varepsilon$$

If $n, m \geq N$ then as $\{A_n\}_{n \in \mathbb{N}}$ is decreasing it follows from [theorem: 14.339] that $A_n \subseteq A_N \wedge A_m \subseteq A_N$ so that $x_n \in A_N \wedge x_m \in A_N$, hence

$$d(x_n, x_m) \leq \sup(\{d(x, y) | x, y \in A_N\}) = \text{diam}(A_N) < \varepsilon$$

which proves that

$$\{x_i\}_{i \in \mathbb{N}} \text{ is a Cauchy sequence}$$

As $\langle X, d \rangle$ is complete it follows that

$$y = \lim_{i \rightarrow \infty} x_i \text{ exists}$$

Let $n \in \mathbb{N}$, $\varepsilon \in \mathbb{R}^+$. Give $U \in \mathcal{T}$ a open set such that $y \in U$ then by [theorem: 14.68] there exist a $\delta \in \mathbb{R}^+$ such that

$$y \in B_d(y, \delta) \subseteq U$$

As $y = \lim_{i \rightarrow \infty} x_i$ there exist a $M \in \mathbb{N}$ such that $\forall m \geq M$ we have

$$d(x_m, y) < \delta \text{ or } x_m \in B_d(y, \delta) \subseteq U$$

If $K = \max(n, M)$ then $n \leq K$ so that $x_K \in A_K \subseteq A_n$, $M \leq K \Rightarrow x_K \in U$ proving that $x_K \in A_n \cap U$ or $A_n \cap U \neq \emptyset$. Using [theorem: 14.30] we conclude then that $y \in \overline{A_n}$. A_n is closed $\wedge$ [theorem: 14.26] $\overline{A_n} = A_n$. So we have proved that $\forall n \in \mathbb{N}$ $y \in A_n$ or $y \in \bigcap_{n \in \mathbb{N}} A_n$, hence

$$\bigcap_{n \in \mathbb{N}} A_n \neq \emptyset \quad \square$$

Theorem 14.449. (Baire Category Theorem)

1. If $\langle X, \mathcal{T} \rangle$ is a compact Hausdorff topological space then X is a Baire space.
2. If $\langle X, d \rangle$ is a complete metric space then X is a Baire space

See [definition: 14.57] for the definition of a Baire space.

Proof. Let $\{A_n\}_{n \in \mathbb{N}}$ be a countable family of sets in X such that $\forall n \in \mathbb{N}$ A_n is closed and $(A_n)^\circ = \emptyset$ then we must prove that $(\bigcup_{n \in \mathbb{N}} A_n)^\circ = \emptyset$

1. As $\langle X, \mathcal{T} \rangle$ is compact and Hausdorff it follows from [theorem: 14.283] that

$$\langle X, \mathcal{T} \rangle \text{ is regular} \quad (14.296)$$

Let $U_0 \in \mathcal{T}$ be a open set such that $U_0 \neq \emptyset$. Then we use recursion to construct a family of non empty open sets $\{U_n\}_{n \in \mathbb{N}} \subseteq \mathcal{T}$ such that

$$\forall n \in \mathbb{N} \text{ we have } \emptyset \neq U_n, \overline{U_n} \subseteq U_{n-1} \text{ and } \overline{U_n} \bigcap A_n = \emptyset \quad (14.297)$$

Proof. As $(A_1)^0 = \emptyset$ we have that $U_0 \not\subseteq A_1$ [for if $U_0 \subseteq A_1$ then $\emptyset \neq U_0 \subseteq (A_1)^\circ = \emptyset$ a contradiction] so there exist a $y \in U_0$ such that $y \notin A_1$. Using the fact that A_1 is closed and X is regular we can use [theorem: 14.249] to find a $U_1 \in \mathcal{T}$ such that

$$y \in U_1 \Rightarrow U_1 \neq \emptyset, \overline{U_1} \subseteq U_0 \text{ and } \overline{U_1} \bigcap A_1 = \emptyset$$

For the recursion step. Let U_{n-1} a non empty open set, as $(A_n)^\circ = \emptyset$, we have that $U_{n-1} \not\subseteq A_n$ [for if $U_{n-1} \subseteq A_n$ then $\emptyset \neq U_{n-1} \subseteq (A_n)^\circ = \emptyset$ a contradiction], hence there exist a $y \in U_{n-1}$ such that $y \notin A_n$. Using [theorem: 14.249] there exist a $U_n \in \mathcal{T}$ such that

$$y \in U_n \Rightarrow U_n \neq \emptyset, \overline{U_n} \subseteq U_{n-1} \text{ and } \overline{U_n} \bigcap A_n = \emptyset \quad \square$$

By construction we have $\forall n \in \mathbb{N}$ that $\overline{U_{n+1}} \subseteq U_n \subseteq \overline{U_n}$. Let I be a finite non empty subset of $\mathbb{N}_0$ then $m = \max(I)$ exist, so $\forall n \in I$ $n \leq m$ and we have by [theorem: 14.339] $\overline{U_m} \subseteq \overline{U_n}$ proving that $\overline{U_m} \subseteq \bigcap_{n \in I} \overline{U_n}$. As $\emptyset \neq U_m \subseteq \overline{U_m}$ it follows that $\bigcap_{n \in I} \overline{U_n} \neq \emptyset$ proving that $\{\overline{U_n}\}_{n \in \mathbb{N}}$ has the finite intersection property. As X is compact it follows from [theorem: 14.319]

$$\bigcap_{n \in \mathbb{N}} \overline{U_n} \neq \emptyset \text{ or there exist a } x \in \bigcap_{n \in \mathbb{N}} \overline{U_n}$$

As $1 \in \mathbb{N}$ we have $x \in \overline{U_1} \subseteq U_0$, further, as $\forall n \in \mathbb{N}$ $x \in \overline{U_n} \Rightarrow \overline{U_n} \bigcap A_n = \emptyset \Rightarrow x \notin A_n$, it follows that $x \notin \bigcup_{n \in \mathbb{N}} A_n$. Hence we have proved that for every non empty open set U_0 $U_0 \not\subseteq \bigcup_{i \in \mathbb{N}} A_i$ which, as $(\bigcup_{n \in \mathbb{N}} A_n)^\circ \subseteq \bigcup_{n \in \mathbb{N}} A_n$, proves that $(\bigcup_{n \in \mathbb{N}} A_n)^\circ = \emptyset$. Hence $\langle X, \mathcal{T} \rangle$ is a Baire space.

2. As $\langle X, d \rangle$ is a metric space it follows from [theorem: 14.246] that

$$\langle X, d \rangle \text{ is regular} \quad (14.298)$$

Let $U_0 \in \mathcal{T}$ be a open set such that $U_0 \neq \emptyset$. Then we use recursion to construct a family of non empty open sets $\{U_n\}_{n \in \mathbb{N}} \subseteq \mathcal{T}$ such that

$$\forall n \in \mathbb{N} \text{ we have } \emptyset \neq U_n, \bar{U}_n \subseteq U_{n-1}, \bar{U}_n \cap A_n = \emptyset, U_n \text{ bounded and } \text{diam}(U_n) \leq \frac{1}{n} \quad (14.299)$$

Proof. As $(A_1)^0 = \emptyset$ we have that $U_0 \not\subseteq A_1$ [for if $U_0 \subseteq A_1$ then $\emptyset \neq U_0 \subseteq (A_1)^\circ = \emptyset$ a contradiction] so there exist a $y \in U_0$ such that $y \notin A_1$. Using the fact that A_1 is closed and X is regular we can use [theorem: 14.249] to find a $V_1 \in \mathcal{T}$ such that

$$y \in V_1 \Rightarrow V_1 \neq \emptyset, \bar{V}_1 \subseteq U_0 \text{ and } \bar{V}_1 \cap A_1 = \emptyset$$

Take $U_1 = V_1 \cap B_d(y, \frac{1}{4}) \ni y$ then

$$U_1 \subseteq V_1 \Rightarrow \bar{U}_1 \cap A_1 \subseteq \bar{V}_1 \cap A_1 = \emptyset \text{ and } \bar{U}_1 \subseteq \bar{V}_1 \subseteq U_0$$

$$U_1 \subseteq B_d\left(y, \frac{1}{2}\right) \Rightarrow \bar{U}_1 \subseteq \overline{B_d\left(y, \frac{1}{2}\right)}.$$

Using [example: 14.82] we have that $\overline{B_d\left(y, \frac{1}{2}\right)}$ is bounded with $\text{diam}(\overline{B_d\left(y, \frac{1}{2}\right)}) \leq 2 \cdot \frac{1}{2}$, so, as $\bar{U}_1 \subseteq \overline{B_d\left(y, \frac{1}{2}\right)}$, it follows that $\bar{U}_1$ is bounded with $\text{diam}(\bar{U}_1) \leq 1$. Summarized

$$\emptyset \neq U_1, \bar{U}_1 \subseteq U_0, \bar{U}_1 \cap A_1 = \emptyset, \bar{U}_1 \text{ is bounded and } \text{diam}(\bar{U}_1) \leq 1$$

For the recursion step, for U_{n-1} a non empty open set, as $(A_n)^\circ = \emptyset$, we have that $U_{n-1} \not\subseteq A_n$ [for if $U_{n-1} \subseteq A_n$ then $\emptyset \neq U_{n-1} \subseteq (A_n)^\circ = \emptyset$ a contradiction], hence there exist a $y \in U_{n-1}$ such that $y \notin A_n$. Using [theorem: 14.249] there exist a $V_n \in \mathcal{T}$ such that

$$y \in V_n \Rightarrow V_n \neq \emptyset, \bar{V}_n \subseteq U_{n-1} \text{ and } \bar{V}_n \cap A_n = \emptyset$$

Take $U_n = V_n \cap B_d(y, \frac{1}{2 \cdot n}) \ni y$ then

$$U_n \subseteq V_n \Rightarrow \bar{U}_n \cap A_n \subseteq \bar{V}_n \cap A_n = \emptyset \text{ and } \bar{U}_n \subseteq \bar{V}_n \subseteq U_{n-1}$$

$$U_n \subseteq B_d\left(y, \frac{1}{2 \cdot n}\right) \Rightarrow \bar{U}_n \subseteq \overline{B_d\left(y, \frac{1}{2 \cdot n}\right)}$$

By [example: 14.82] we have that $\overline{B_d\left(y, \frac{1}{2 \cdot n}\right)}$ is bounded with $\text{diam}(\overline{B_d\left(y, \frac{1}{2 \cdot n}\right)}) \leq \frac{2}{2 \cdot n} = \frac{1}{n}$, so, as $\bar{U}_n \subseteq \overline{B_d\left(y, \frac{1}{2 \cdot n}\right)}$, it follows that $\bar{U}_n$ is bounded and $\text{diam}(\bar{U}_n) < \frac{1}{n}$. Summarized

$$\emptyset \neq U_n, \bar{U}_n \subseteq U_{n-1}, \bar{U}_n \cap A_n = \emptyset, \bar{U}_n \text{ is bounded and } \text{diam}(\bar{U}_n) < \frac{1}{n} \quad \square$$

Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.30] there exist a $N \in \mathbb{N}$ such that $0 < \frac{1}{N} < \varepsilon$ so if $n \geq N$ we have

$$|\text{diam}(\bar{U}_n) - 0| = \text{diam}(\bar{U}_n) \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon$$

proving that

$$\lim_{n \rightarrow \infty} \text{diam}(\bar{U}_n) = 0$$

Hence as $\forall n \in \mathbb{N} \bar{U}_{n+1} \subseteq \bar{U}_n$ and $\emptyset \neq U_n \subseteq \bar{U}_n$ it follows, using [lemma: 14.448], that

$$\bigcap_{n \in \mathbb{N}} \bar{U}_n \neq \emptyset \text{ or there exist a } x \in \bigcap_{n \in \mathbb{N}} \bar{U}_n$$

As $1 \in \mathbb{N}$ we have $x \in \bar{U}_1 \subseteq U_0$, further, as $\forall n \in \mathbb{N} x \in \bar{U}_n \xrightarrow{\bar{U}_n \cap A_n = \emptyset} x \notin A_n$, it follows that $x \notin \bigcup_{n \in \mathbb{N}} A_n$. Hence we have proved that for every non empty open set U_0 $U_0 \not\subseteq \bigcup_{i \in \mathbb{N}} A_i$ which, as $(\bigcup_{n \in \mathbb{N}} A_n)^\circ \subseteq \bigcup_{n \in \mathbb{N}} A_n$, proves that $(\bigcup_{n \in \mathbb{N}} A_n)^\circ = \emptyset$. Hence $\langle X, \mathcal{T} \rangle$ is a Baire space. $\square$

Lemma 14.450. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be Banach spaces and $L: X \rightarrow Y$ a surjective linear function then we have:*

$$L \text{ is open} \Leftrightarrow \exists \varepsilon \in \mathbb{R}^+ \text{ such that } B_{\|\cdot\|_Y}(0, \varepsilon) \subseteq L(B_{\|\cdot\|_X}(0, 1))$$

Proof.

$\Rightarrow$. As L is open and $B_{\|\cdot\|_X}(0, 1)$ is a open set we have that $L(B_{\|\cdot\|_X}(0, 1))$ is a open set, further as $L(0) = 0 \in L(B_{\|\cdot\|_X}(0, 1))$. Hence by [theorem: 14.97] there exist a $\varepsilon \in \mathbb{R}^+$ such that $B_{\|\cdot\|_Y}(0, \varepsilon) \subseteq L(B_{\|\cdot\|_X}(0, 1))$.

$\Leftarrow$. By the hypothesis there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$B_{\|\cdot\|_Y}(0, \varepsilon) \subseteq L(B_{\|\cdot\|_X}(0, 1)) \quad (14.300)$$

Let $r \in \mathbb{R}^+$. We use now r to scale [eq: 14.300]. Let $y \in B_{\|\cdot\|_Y}(0, r \cdot \varepsilon)$ then $\|y\|_Y < r \cdot \varepsilon$ so that $\|\frac{1}{r} \cdot y\|_Y = \frac{1}{r} \cdot \|y\|_Y < \frac{1}{r} \cdot r \cdot \varepsilon = \varepsilon$ so that $\frac{1}{r} \cdot y \in B_{\|\cdot\|_Y}(0, \varepsilon)$. Hence by [eq: 14.300] there exist a $x \in B_{\|\cdot\|_X}(0, 1)$ such that $\frac{1}{r} \cdot y = L(x)$ or $y = r \cdot L(x) = L(r \cdot x)$. As $\|r \cdot x\|_X = r \cdot \|x\|_X < r \cdot 1 = r$ we have $r \cdot x \in B_{\|\cdot\|_X}(0, r)$ so that $y = L(r \cdot x) \in L(B_{\|\cdot\|_X}(0, r))$. Hence we have that

$$B_{\|\cdot\|_Y}(0, r \cdot \varepsilon) \subseteq L(B_{\|\cdot\|_X}(0, r)) \quad (14.301)$$

Let $b \in Y$. As L is a surjection there exist a $a \in X$ such that $b = L(a)$. Let $y \in B_{\|\cdot\|_Y}(b, r \cdot \varepsilon)$ then $\|y - L(a)\|_Y = \|y - b\|_Y < r \cdot \varepsilon$, hence $y - L(a) \in B_{\|\cdot\|_Y}(0, r \cdot \varepsilon)$ so that by [eq: 14.301] we have that $y - L(a) \in L(B_{\|\cdot\|_X}(0, r))$. Hence there exist a $x \in B_{\|\cdot\|_X}(0, r)$ such that $y - L(a) = L(x)$ or $y = L(x) + L(a) = L(x + a)$. Further $\|(x + a) - a\|_X = \|x\|_X < r$ so that $x + a \in B_{\|\cdot\|_X}(a, r)$ proving that $y \in L(B_{\|\cdot\|_X}(a, r))$ or

$$B_{\|\cdot\|_Y}(b, r \cdot \varepsilon) \subseteq L(B_{\|\cdot\|_X}(a, r))$$

Summarized we have

$$\forall b \in Y, \forall r \in \mathbb{R}^+ \text{ we have } B_{\|\cdot\|_Y}(b, r \cdot \varepsilon) \subseteq L(B_{\|\cdot\|_X}(a, r)) \text{ where } b = L(a) \quad (14.302)$$

Let U be a open set in X . Given $y \in L(U)$ there exist a $x \in U$ such that $y = L(x)$. As U is open there exist by [theorem: 14.97] a $\delta \in \mathbb{R}^+$ such that $x \in B_{\|\cdot\|_X}(x, \delta) \subseteq U$. Take then in [eq: 14.302] $r = \delta$ and $a = x \Rightarrow b = L(a) = L(x) = y$ then we have

$$y \in B_{\|\cdot\|_Y}(y, \delta \cdot \varepsilon) \subseteq L(B_{\|\cdot\|_X}(x, \delta)) \subseteq L(U)$$

proving by [theorem: 14.97] that $L(U)$ is open. Hence $L: X \rightarrow Y$ is a open mapping $\square$

Lemma 14.451. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be Banach spaces and $L: X \rightarrow Y$ a surjective continuous linear function then $\exists \lambda \in \mathbb{R}^+$ such that $B_{\|\cdot\|_Y}(0, \lambda) \subseteq L(B_{\|\cdot\|_X}(0, 1))$.*

Proof. First by the Baire Category theorem [see theorem: 14.449] we have that

$$\langle Y, \|\cdot\|_Y \rangle \text{ is a Baire Space}$$

Let $x \in X$ then by the Archimedean property of $\mathbb{R}$ [see theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $\|x\| < n$ hence $x \in B_{\|\cdot\|_X}(0, n) \subseteq \bigcup_{n \in \mathbb{N}} B_{\|\cdot\|_X}(0, n)$. Hence $X \subseteq \bigcup_{n \in \mathbb{N}} B_{\|\cdot\|_X}(0, n)$ so that

$$Y \stackrel{L \text{ is surjective}}{=} L(X) \subseteq L\left(\bigcup_{n \in \mathbb{N}} B_{\|\cdot\|_X}(0, n)\right) \stackrel{[\text{theorem: 2.144}]}{=} \bigcup_{n \in \mathbb{N}} L(B_{\|\cdot\|_X}(0, n)) \subseteq \bigcup_{n \in \mathbb{N}} \overline{L(B_{\|\cdot\|_X}(0, n))} \subseteq Y$$

proving that

$$Y = \bigcup_{n \in \mathbb{N}} \overline{L(B_{\|\cdot\|_X}(0, n))} \quad (14.303)$$

If now $\forall n \in \mathbb{N} \ (\overline{L(B_{\|\cdot\|_X}(0, n))})^\circ = \emptyset$ then, as Y is a Baire Space, we have that $(\bigcup_{n \in \mathbb{N}} \overline{L(B_{\|\cdot\|_X}(0, n))})^\circ = \emptyset$ hence $Y^\circ = \emptyset$, as Y is open we have by [theorem: 14.16] that $Y = Y^\circ$, so that $Y = \emptyset$ contradicting $0 \in Y$. Hence there exist a $n_0 \in \mathbb{N}$ such that

$$(\overline{L(B_{\|\cdot\|_X}(0, n_0))})^\circ \neq \emptyset$$

Now

$$\begin{aligned} \overline{L(B_{\parallel x}(0, n_0))} &\stackrel{[\text{theorem: 14.95}]}{=} \overline{L(n_0 \cdot B_{\parallel x}(0, 1))} \\ &\stackrel{[\text{theorem: 11.170}]}{=} \overline{n_0 \cdot L(B_{\parallel x}(0, 1))} \\ &\stackrel{[\text{theorem: 14.116}]}{=} \overline{n_0 \cdot L(B_{\parallel x}(0, 1))} \end{aligned}$$

so that we have

$$\emptyset \neq (n_0 \cdot \overline{L(B_{\parallel x}(0, 1))})^\circ$$

Hence there exist a $v'_0 \in Y$ such that $v'_0 \in (n_0 \cdot \overline{L(B_{\parallel x}(0, 1))})^\circ$, using [theorem: 14.97] there exist a $r' \in \mathbb{R}^+$ such that $v'_0 \in B_{\parallel Y}(v'_0, r') \subseteq (n_0 \cdot \overline{L(B_{\parallel x}(0, 1))})^\circ \subseteq n_0 \cdot \overline{L(B_{\parallel x}(0, 1))}$. Take $r = \frac{r'}{n_0}$ and $v_0 = \frac{1}{n_0} \cdot v'_0$ then if $y \in B_{\parallel Y}(v_0, r)$ we have

$$\|v_0 - y\|_Y < r = \frac{r'}{n_0}$$

As

$$\frac{1}{n_0} \cdot \|v'_0 - n_0 \cdot y\|_Y = \left\| \frac{1}{n_0} \cdot (v'_0 - n_0 \cdot y) \right\|_Y = \|v_0 - y\|_Y$$

it follows that $\frac{1}{n_0} \cdot \|v'_0 - n_0 \cdot y\|_Y < \frac{r'}{n_0}$ so that $\|v'_0 - n_0 \cdot y\|_Y < r'$ hence

$$n_0 \cdot y \in B_{\parallel Y}(v'_0, r') \subseteq n_0 \cdot \overline{L(B_{\parallel x}(0, 1))} \Rightarrow y \in \overline{L(B_{\parallel x}(0, 1))}$$

This proves that $B_{\parallel Y}(v_0, r) \subseteq \overline{L(B_{\parallel x}(0, 1))}$ or if we take $\delta = \frac{r}{4}$ this results in

$$\exists v_0 \in Y, \delta \in \mathbb{R}^+ \text{ such that } v_0 \in B_{\parallel Y}(v_0, 4 \cdot \delta) \subseteq \overline{L(B_{\parallel x}(0, 1))} \quad (14.304)$$

As $v_0 \in \overline{L(B_{\parallel x}(0, 1))}$ we have by [theorem: 14.30] that $B_{\parallel Y}(v_0, 2 \cdot \delta) \cap L(B_{\parallel x}(0, 1)) \neq \emptyset$. So there exist a $v_1 \in B_{\parallel Y}(v_0, 2 \cdot \delta) \cap L(B_{\parallel x}(0, 1))$ or

$$\|v_1 - v_0\|_Y < 2 \cdot \delta \text{ and there exist a } u_1 \in B_{\parallel x}(0, 1) \Rightarrow \|u_1\|_X < 1 \text{ such that } v_1 = L(u_1) \quad (14.305)$$

If $x \in B_{\parallel Y}(v_1, 2 \cdot \delta)$ then we have

$$\|x - v_0\|_Y \leq \|x - v_1\|_Y + \|v_1 - v_0\|_Y < 2 \cdot \delta + 2 \cdot \delta = 4 \cdot \delta$$

so that $x \in B_{\parallel Y}(v_0, 4 \cdot \delta)$. Hence

$$B_{\parallel Y}(v_1, 2 \cdot \delta) \subseteq B_{\parallel Y}(v_0, 4 \cdot \delta) \subseteq \overline{L(B_{\parallel x}(0, 1))} \quad (14.306)$$

Let $v \in B_{\parallel Y}(0, \delta)$ then $\|(2 \cdot v + v_1) - v_1\|_Y = \|2 \cdot v\|_Y = 2 \cdot \|v\|_Y < 2 \cdot \delta$ so that $2 \cdot v + v_1 \in B_{\parallel Y}(v_1, 2 \cdot \delta)$ or

$$v + \frac{1}{2} \cdot v_1 = \frac{1}{2} \cdot (2 \cdot v + v_1) \in \frac{1}{2} \cdot B_{\parallel Y}(v_1, 2 \cdot \delta) \subseteq_{[\text{eq: 14.306}]} \frac{1}{2} \cdot \overline{L(B_{\parallel x}(0, 1))}$$

so that

$$v \in -\frac{1}{2} \cdot v_1 + \frac{1}{2} \cdot \overline{L(B_{\parallel x}(0, 1))}$$

As

$$\begin{aligned} \frac{1}{2} \cdot \overline{L(B_{\parallel x}(0, 1))} &\stackrel{[\text{theorem: 14.116}]}{=} \overline{\frac{1}{2} \cdot L(B_{\parallel x}(0, 1))} \\ &\stackrel{[\text{theorem: 11.170}]}{=} \overline{L\left(\frac{1}{2} \cdot B_{\parallel x}(0, 1)\right)} \\ &\stackrel{[\text{theorem: 14.95}]}{=} \overline{L\left(B_{\parallel x}\left(0, \frac{1}{2}\right)\right)} \end{aligned}$$

it follows, taking account that $v_1 \stackrel{[\text{eq: 14.305}]}{=} L(u_1)$, that

$$v \in -\frac{1}{2} \cdot L(u_1) + \overline{L\left(B_{\parallel x}\left(0, \frac{1}{2}\right)\right)}$$

Now

$$\begin{aligned} -\frac{1}{2} \cdot L(u_1) + \overline{L\left(B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)} &\subseteq_{[\text{theorem: 14.116}]} \overline{-\frac{1}{2} \cdot L(u_1) + L\left(B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)} \\ &\stackrel{[\text{theorem: 11.170}]}{=} \overline{L\left(-\frac{1}{2} \cdot u_1\right) + L\left(B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)} \\ &\stackrel{[\text{theorem: 11.170}]}{=} \overline{L\left(-\frac{1}{2} \cdot u_1 + B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)} \end{aligned}$$

so that

$$v \in \overline{L\left(-\frac{1}{2} \cdot u_1 + B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)}$$

If $x \in -\frac{1}{2} \cdot u_1 - B_{\|\cdot\|_X}(0, \frac{1}{2})$ then there exist a $y \in B_{\|\cdot\|_X}(0, \frac{1}{2})$ such that $x = -\frac{1}{2} \cdot u_1 + y$, hence

$$\|x\|_X = \left\| -\frac{1}{2} \cdot u_1 \right\|_X + \|y\|_X < \left\| \frac{1}{2} \cdot u_1 \right\|_X + \frac{1}{2} <_{[\text{eq: 14.305}]} \frac{1}{2} + \frac{1}{2} = 1$$

proving that $-\frac{1}{2} \cdot u_1 - B_{\|\cdot\|_X}(0, \frac{1}{2}) \subseteq B_{\|\cdot\|_X}(0, 1)$ so that we have

$$v \in \overline{L(B_{\|\cdot\|_X}(0, 1))}$$

Therefore as $v \in B_{\|\cdot\|_Y}(0, \delta)$ was chosen arbitrary it follows that

$$B_{\|\cdot\|_Y}(0, \delta) \subseteq \overline{L(B_{\|\cdot\|_X}(0, 1))} \quad (14.307)$$

Let $n \in \mathbb{N}$ then we have that

$$\begin{aligned} B_{\|\cdot\|_Y}\left(0, \left(\frac{1}{2}\right)^n \cdot \delta\right) &\stackrel{[\text{theorem: 14.95}]}{=} \left(\frac{1}{2}\right)^n \cdot B_{\|\cdot\|_Y}(0, \delta) \\ &\subseteq_{[\text{eq: 14.307}]} \left(\frac{1}{2}\right)^n \cdot \overline{L(B_{\|\cdot\|_X}(0, 1))} \\ &\stackrel{[\text{theorem: 14.116}]}{=} \overline{\left(\frac{1}{2}\right)^n \cdot L(B_{\|\cdot\|_X}(0, 1))} \\ &\stackrel{[\text{theorem: 14.95}]}{=} \overline{L\left(B_{\|\cdot\|_X}\left(0, \left(\frac{1}{2}\right)^n\right)\right)} \end{aligned}$$

proving that

$$B_{\|\cdot\|_Y}\left(0, \left(\frac{1}{2}\right)^n \cdot \delta\right) \subseteq \overline{L\left(B_{\|\cdot\|_X}\left(0, \left(\frac{1}{2}\right)^n\right)\right)} \quad (14.308)$$

Let $y \in B_{\|\cdot\|_Y}(0, \frac{\delta}{2})$. Then using recursion we construct a sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ such that

$$\forall n \in \mathbb{N} \text{ we have } \left\| y - \sum_{i=1}^n L(x_i) \right\|_Y < \left(\frac{1}{2}\right)^{n+1} \cdot \delta \text{ and } \|x_n\|_X < \left(\frac{1}{2}\right)^n \quad (14.309)$$

Proof.

Case 1. As $y \in B_{\|\cdot\|_Y}(0, \frac{\delta}{2}) = B_{\|\cdot\|_Y}(0, (\frac{\delta}{2})^1)$ we have by [eq: 14.308] that

$$y \in \overline{L\left(B_{\|\cdot\|_X}\left(0, \left(\frac{1}{2}\right)^1\right)\right)} = \overline{L\left(B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)}.$$

Hence as $y \in B_{\|\cdot\|_Y}(y, \frac{\delta}{4}) \in \mathcal{T}_{\|\cdot\|_Y}$ we have by [theorem: 14.30] that

$$B_{\|\cdot\|_Y}\left(y, \frac{\delta}{4}\right) \cap \overline{L\left(B_{\|\cdot\|_X}\left(0, \frac{1}{2}\right)\right)} \neq \emptyset$$

hence there exist a $x_1 \in B_{\|\cdot\|_X}(0, \frac{1}{2})$ such that $\|y - L(x_1)\|_Y < \frac{\delta}{4}$ or

$$\left\| y - \sum_{i=1}^1 L(x_i) \right\|_Y < \left(\frac{1}{2} \right)^{1+1} \cdot \delta \text{ and } \|x_1\|_X < \left(\frac{1}{2} \right)^1$$

Case n+1. As $\|y - \sum_{i=1}^n L(x_i)\|_Y < \left(\frac{1}{2} \right)^{n+1} \cdot \delta$ we have $y - \sum_{i=1}^n L(x_i) \in B_{\|\cdot\|_Y}(0, \left(\frac{1}{2} \right)^{n+1} \cdot \delta)$ hence using [eq: 14.308] we have that $y - \sum_{i=1}^n L(x_i) \in L(B_{\|\cdot\|_X}(0, \left(\frac{1}{2} \right)^{n+1}))$. As

$$y - \sum_{i=1}^n L(x_i) \in B_{\|\cdot\|_Y} \left(y - \sum_{i=1}^n L(x_i), \left(\frac{1}{2} \right)^{(n+1)+1} \cdot \delta \right) \in \mathcal{T}_{\|\cdot\|_Y}$$

it follows from [theorem: 14.30] that

$$B_{\|\cdot\|_Y} \left(y - \sum_{i=1}^n L(x_i), \left(\frac{1}{2} \right)^{(n+1)+1} \cdot \delta \right) \cap L \left(\overline{B_{\|\cdot\|_X} \left(0, \left(\frac{1}{2} \right)^{n+1} \right)} \right) \neq \emptyset$$

Hence there exist a $x_{n+1} \in B_{\|\cdot\|_X}(0, \left(\frac{1}{2} \right)^{n+1})$ such that

$$\left\| y - \sum_{i=1}^n L(x_i) - L(x_{n+1}) \right\|_Y < \left(\frac{1}{2} \right)^{(n+1)+1} \cdot \delta$$

or

$$\left\| y - \sum_{i=1}^{n+1} L(x_i) \right\|_Y < \left(\frac{1}{2} \right)^{(n+1)+1} \cdot \delta \text{ and } \|x_{n+1}\|_X < \left(\frac{1}{2} \right)^{n+1} \quad \square$$

As $\forall n \in \mathbb{N} \|x_n\|_X < \left(\frac{1}{2} \right)^n$ [see eq: 14.309], $\sum_{i=1}^{\infty} \left(\frac{1}{2} \right)^n \underset{[\text{theorem: 14.412}]}{=} \frac{1/2}{1-1/2} = 1$ [see theorem: 14.412], $\langle X, \|\cdot\|_X \rangle$ is a Banach space it follows from the dominant convergence [theorem: 14.438] that

$$x = \sum_{i=1}^{\infty} x_i \text{ exist and } \|x\| < 1 \text{ so that } x \in B_{\|\cdot\|_X}(0, 1) \quad (14.310)$$

Further if $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.64] there exist a $N \in \mathbb{N}$ such that $\left(\frac{1}{2} \right)^N < \frac{\varepsilon}{\delta}$. Let $n \geq N$ then we have

$$\left\| \sum_{i=1}^n L(x_i) - y \right\|_Y < \left(\frac{1}{2} \right)^{n+1} \cdot \delta <_{[\text{theorem: 10.62}]} \left(\frac{1}{2} \right)^N \cdot \delta < \frac{\varepsilon}{\delta} \cdot \delta = \varepsilon$$

proving that

$$\sum_{i=1}^{\infty} L(x_i) = y \quad (14.311)$$

Now as L is continuous we have using [theorem: 14.371] that

$$\begin{aligned} L(x) & \underset{[\text{eq: 14.310}]}{=} L \left(\sum_{i=1}^{\infty} x_i \right) \\ & = L \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i \right) \\ & \underset{[\text{theorem: 14.371}]}{=} \lim_{n \rightarrow \infty} L \left(\sum_{i=1}^n x_i \right) \\ & = \lim_{n \rightarrow \infty} \sum_{i=1}^n L(x_i) \\ & = \sum_{i=1}^{\infty} L(x_i) \\ & \underset{[\text{eq: 14.311}]}{=} y \end{aligned}$$

Which, as $x \in B_{\|\cdot\|_X}(0, 1)$ [see eq: 14.310], proves that $y \in L(B_{\|\cdot\|_X}(0, 1))$, hence as $y \in B_{\|\cdot\|_Y}(0, \frac{\delta}{2})$ was chosen arbitrary this proves that $B_{\|\cdot\|_Y}(0, \frac{\delta}{2}) \subseteq L(B_{\|\cdot\|_X}(0, 1))$ or if we take $\lambda = \frac{\delta}{2}$ that

$$B_{\|\cdot\|_Y}(0, \lambda) \subseteq L(B_{\|\cdot\|_X}(0, 1)) \quad \square$$

We are now finally ready to proof the Open Mapping Theorem stating that every continuous linear surjective mapping between two Banach spaces is open.

Theorem 14.452. (Open Mapping Theorem) *Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be Banach spaces and $L: X \rightarrow Y$ a surjective continuous linear function then $L: X \rightarrow Y$ is a **open mapping**.*

Proof. Using [lemma: 14.451] there exist a $\lambda \in \mathbb{R}^+$ such that

$$B_{\|\cdot\|_Y}(0, \lambda) \subseteq L(B_{\|\cdot\|_X}(0, 1))$$

Applying then [lemma: 14.450] proves that L is open. $\square$

Corollary 14.453. *Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be Banach spaces and $L: X \rightarrow Y$ a continuous linear isomorphism then $L: X \rightarrow Y$ is a homeomorphism.*

Proof. As $L: X \rightarrow Y$ is a isomorphism it is surjective. Hence as L is continuous and linear it follows from [theorem: 14.452] that $L: X \rightarrow Y$ is a open mapping. Hence if U is a open set in X we have for $L^{-1}: Y \rightarrow X$ that $(L^{-1})^{-1}(U) \stackrel{[\text{theorem: 2.70}]}{=} L(U)$ is open. So that $L^{-1}: Y \rightarrow X$ is continuous which by the continuity of $L: X \rightarrow Y$ proves that $L: X \rightarrow Y$ is a homeomorphism. $\square$

14.11 Connected Sets

14.11.1 Connected Sets and Connected Components

Definition 14.454. *A topological space $\langle X, \mathcal{T} \rangle$ is **connected** if $\forall U_1, U_2 \in \mathcal{T}$ with $U_1 \cap U_2 = \emptyset$ and $X = U_1 \cup U_2$ we have that either $U_1 = \emptyset$ or $U_2 = \emptyset$. In other words a topological space is connected if it is not the union of two disjoint non empty sets.*

Definition 14.455. *A topological space is **disconnected** if it is not connected or in other words there exists $U_1, U_2 \in \mathcal{T}$ with $U_1 \neq \emptyset$, $U_2 \neq \emptyset$, $U_1 \cap U_2 = \emptyset$ and $X = U_1 \cup U_2$.*

Definition 14.456. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then A is a connected subset if $\langle A, \mathcal{T}_A \rangle$ is connected where $\mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}\}$ is the subspace topology on A .*

We have the following equivalent definition of connected subsets

Theorem 14.457. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then we have that*

A is a connected subset

$\Updownarrow$

$\forall U_1, U_2 \in \mathcal{T}$ with $(U_1 \cap U_2) \cap A = \emptyset$ and $A \subseteq U_1 \cup U_2$ we have $U_1 \cap A = \emptyset$ or $U_2 \cap A = \emptyset$

Proof.

$\Rightarrow$. Let $U_1, U_2 \in \mathcal{T}$ with $(U_1 \cap U_2) \cap A = \emptyset$ and $A \subseteq U_1 \cup U_2$ then we have

$$U_1 \cap A \in \mathcal{T}_A \text{ and } U_2 \cap A \in \mathcal{T}_A$$

$$(U_1 \cap A) \cap (U_2 \cap A) = (U_1 \cap U_2) \cap A = \emptyset$$

$$A \stackrel{A \subseteq U_1 \cup U_2}{=} A \cap (U_1 \cup U_2) = (U_1 \cap A) \cup (U_2 \cap A)$$

As A is a connected subset $\langle A, \mathcal{T}|_A \rangle$ is connected and it follows from the above that $U_1 \cap A = \emptyset$ or $U_2 \cap A = \emptyset$.

$\Leftarrow$. Let $U_1, U_2 \in \mathcal{T}|_A$ such that $U_1 \cap U_2 = \emptyset$ and $A = U_1 \cup U_2$. Then there exists $U'_1, U'_2 \in \mathcal{T}$ such that $U_1 = U'_1 \cap A$ and $U_2 = U'_2 \cap A$. Hence we have that

$$\begin{aligned} (U'_1 \cap U'_2) \cap A &= (U'_1 \cap A) \cap (U'_2 \cap A) = U_1 \cap U_2 = \emptyset \\ A &= U_1 \cup U_2 = (U'_1 \cap A) \cup (U'_2 \cap A) \subseteq U'_1 \cup U'_2 \end{aligned}$$

so that by the hypothesis $U_1 = U'_1 \cap A = \emptyset$ or $U_2 = U'_2 \cap A = \emptyset$. Hence we have by definition that $\langle A, \mathcal{T}|_A \rangle$ is connected proving that A is a connected subset. $\square$

Example 14.458. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $x \in X$ then $\{x\}$ is a connected subset

Proof. Let $U_1, U_2 \in \mathcal{T}$ such that $(U_1 \cap U_2) \cap \{x\} = \emptyset$ and $\{x\} \subseteq U_1 \cup U_2$. As $\{x\} \subseteq U_1 \cup U_2$ we have $x \in U_1 \cup U_2$ so that we have either:

$x \in U_1$. Assume that $U_2 \cap \{x\} \neq \emptyset$ then there exist a $y \in U_2 \cap \{x\}$ so that $y \in U_2 \wedge y \in \{x\}$ hence $x = y \in U_2$. Hence $x \in (U_1 \cap U_2) \cap \{x\} = \emptyset$ leading to the contradiction $x \in \emptyset$. Hence the assumption is false and we must have that $U_2 \cap \{x\} = \emptyset$.

$x \in U_2$. Assume that $U_1 \cap \{x\} \neq \emptyset$ then there exist a $y \in U_1 \cap \{x\}$ so that $y \in U_1 \wedge y \in \{x\}$ hence $x = y \in U_1$. Hence $x \in (U_1 \cap U_2) \cap \{x\} = \emptyset$ leading to the contradiction $x \in \emptyset$. Hence the assumption is false and we must have that $U_1 \cap \{x\} = \emptyset$.

so we have $U_1 \cap \{x\} = \emptyset$ or $U_2 \cap \{x\} = \emptyset$ which proves that $\{x\}$ is a connected subset. $\square$

The following set relations will be useful in the next theorem.

Lemma 14.459. Let A, B, C be classes such that $A \cap B = \emptyset$ and $C = A \cup B$ then we have:

1. $A = C \setminus B$
2. $B = C \setminus A$
3. $A \setminus B = A$

Proof.

1. If $x \in A \subseteq A \cup B = C \Rightarrow x \in C$ and if $x \in B$ we would have $x \in A \cap B = \emptyset$ a contradiction so $x \notin B \Rightarrow x \in C \setminus B \Rightarrow A \subseteq C \setminus B$. If $x \in C \setminus B$ then $x \in C = A \cup B$ so $x \in A \vee x \in B$, as $x \notin B$ we must have $x \in A \Rightarrow C \setminus B \subseteq A$.
2. If $x \in B \subseteq A \cup B = C \Rightarrow x \in C$ and if $x \in A$ we would have $x \in A \cap B = \emptyset$ a contradiction so $x \notin A \Rightarrow x \in C \setminus A \Rightarrow B \subseteq C \setminus A$. If $x \in C \setminus A$ then $x \in C = A \cup B$ so $x \in A \vee x \in B$, as $x \notin A$ we must have $x \in B \Rightarrow C \setminus A \subseteq B$.
3. If $x \in A$ then, as $A \cap B = \emptyset$, we must have $x \notin B$ so that $x \in A \setminus B$ hence $A \subseteq A \setminus B \subseteq A$ so that $A \setminus B = A$. $\square$

Theorem 14.460. Let $\langle X, \mathcal{T} \rangle$ be a topological space then the following are equivalent:

1. $\emptyset$ and X are the only subsets of X that are open and closed.
2. X is connected
3. For every pair of closed sets A_1, A_2 with $A_1 \cap A_2 = \emptyset$ and $X = A_1 \cup A_2$ we have either $A_1 = \emptyset$ or $A_2 = \emptyset$.

Proof.

- 1 $\Rightarrow$ 2. Assume that U_1, U_2 are **disjoint open** sets with $X = U_1 \cup U_2$ then by [lemma: 14.459] $U_2 \stackrel{[lemma: 14.459]}{=} X \setminus U_1$ so that U_2 is open and closed. According to (1) we must then have that either:

$$U_2 = \emptyset. \text{ Then } U_1 = \emptyset \vee U_2 = \emptyset$$

$U_2 = X$. Then $U_1 \stackrel{[\text{lemma: 14.459}]}{=} X \setminus U_2 = X \setminus X = \emptyset$ so that $U_1 = \emptyset \vee U_2 = \emptyset$

2 $\Rightarrow$ 3. Assume that we have two **disjoint closed** sets A_1, A_2 with $X = A_1 \cup A_2$ then $A_1 \stackrel{[\text{lemma: 14.459}]}{=} X \setminus A_2$ is open and $A_2 \stackrel{[\text{lemma: 14.459}]}{=} X \setminus A_1$ is open, so, as X is connected, we must have either $A_1 = \emptyset$ or $A_2 = \emptyset$. A is open and closed then $X \setminus A$ is closed, $A \cap (X \setminus A) = \emptyset$ and $X = A \cup (X \setminus A)$. By (3) we must have either $A = \emptyset$ or $X \setminus A = \emptyset \Rightarrow A = X$. $\square$

Theorem 14.461. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $U, V \in \mathcal{T}$ non empty open sets such that $U \cap V = \emptyset$ and $X = U \cup V$ and $Y \subseteq X$ is a connected subset of X then we have $Y \subseteq U$ or $Y \subseteq V$.

Proof. As $U, V \in \mathcal{T}$ we have that $U \cap Y, V \cap Y \in \mathcal{T}_Y$, further $(U \cap Y) \cap (V \cap Y) \subseteq U \cap V = \emptyset$. As Y is a connected subset we have by definition that $\langle Y, \mathcal{T}_Y \rangle$ is a connected space so that we have by [definition: 14.454] either:

$U \cap Y = \emptyset$. Then $Y \stackrel{Y \subseteq X}{=} X \cap Y = (U \cup V) \cap Y = (U \cap Y) \cup (V \cap Y) = \emptyset \cup V \cap Y = V \cap Y \subseteq V$
so that

$$Y \subseteq V$$

$V \cap Y = \emptyset$. Then $Y \stackrel{Y \subseteq X}{=} X \cap Y = (U \cup V) \cap Y = (U \cap Y) \cup (V \cap Y) = (U \cap Y) \cup \emptyset = U \cap Y \subseteq U$
so that

$$Y \subseteq U$$

hence we conclude that either $Y \subseteq U$ or $Y \subseteq V$. $\square$

We use the above theorem to prove that the union of connected subsets that has a common element are connected.

Theorem 14.462. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\{Y_i\}_{i \in I}$ be a family of connected subsets of X such that $\bigcap_{i \in I} Y_i \neq \emptyset$ then $\bigcup_{i \in I} Y_i$ is a connected subset of X .

Proof. As $\bigcap_{i \in I} Y_i \neq \emptyset$ there exist a $p \in \bigcap_{i \in I} Y_i$ hence

$$\forall i \in I \text{ we have } p \in Y_i \quad (14.312)$$

Assume that $\bigcup_{i \in I} Y_i$ is not a connected subset then $\langle \bigcup_{i \in I} Y_i, \mathcal{T}_{|\bigcup_{i \in I} Y_i} \rangle$ is not connected. Hence there exist $U, V \in \mathcal{T}_{|\bigcup_{i \in I} Y_i}$ such that

$$U \cap V = \emptyset \wedge \bigcup_{i \in I} Y_i = U \cup V \wedge U \neq \emptyset \neq V$$

Let $i \in I$ then, as Y_i is a connected subset of X $\langle Y_i, \mathcal{T}_{|Y_i} \rangle$ is a connected space, as $Y_i \subseteq \bigcup_{i \in I} Y_i \subseteq X$ it follows from [theorem: 14.10] that $\mathcal{T}_{|Y_i} = (\mathcal{T}_{|\bigcup_{i \in I} Y_i})_{|Y_i}$ so that $\langle Y_i, (\mathcal{T}_{|\bigcup_{i \in I} Y_i})_{|Y_i} \rangle$ is a connected space hence a connected subset of $\bigcup_{i \in I} Y_i$. Hence using [theorem: 14.461] we have either $Y_i \subseteq U$ or $Y_i \subseteq V$. To summarize we have

$$\forall i \in I \text{ we have } Y_i \subseteq U \text{ or } Y_i \subseteq V \quad (14.313)$$

As $p \in \bigcap_{i \in I} Y_i \subseteq \bigcup_{i \in I} Y_i = U \cup V$ we have either:

$p \in U$. Assume that there exist a $i \in I$ such that $Y_i \subseteq V$ then by [eq: 14.312] $p \in Y_i \subseteq V$ so that $p \in U \cap V = \emptyset$ a contradiction. Hence $\forall i \in I$ we have $Y_i \not\subseteq V$ so that by [eq: 14.313] $Y_i \subseteq U$. Hence we have

$$\bigcup_{i \in I} Y_i \subseteq U$$

So we have

$$V = (U \cup V) \cap V = \left(\bigcup_{i \in I} Y_i \right) \cap V \subseteq U \cap V = \emptyset$$

which contradicts $V \neq \emptyset$.

$p \in V$. Assume that there exist a $i \in I$ such that $Y_i \subseteq U$ then by [eq: 14.312] $p \in Y_i \subseteq U$ so that $p \in \bigcup \bigcap V = \emptyset$ a contradiction. Hence $\forall i \in I$ we have $Y_i \not\subseteq U$ so that by [eq: 14.313] $Y_i \subseteq V$. Hence we have

$$\bigcup_{i \in I} Y_i \subseteq U$$

So we have

$$U = (U \cup V) \cap U = \left(\bigcup_{i \in I} Y_i \right) \cap U \subseteq V \cap U = \emptyset$$

which contradicts $V \neq \emptyset$.

As we have a contradiction in all possible case the assumption is false and we must have that

$$\bigcup_{i \in I} Y_i \text{ is a connected subset} \quad \square$$

Theorem 14.463. Let $\langle X, \mathcal{T}_X \rangle$ be connected topological space, $\langle Y, \mathcal{T}_Y \rangle$ a topological space and $f: X \rightarrow Y$ a continuous function

then

$f(X)$ is a connected subset of Y

Proof. We prove this by contradiction. So assume that $f(X)$ is not connected then there exist open sets $V_1, V_2 \in \mathcal{T}_Y$ with $V_1 \cap f(X) \neq \emptyset$, $V_2 \cap f(X) \neq \emptyset$, $(V_1 \cap f(X)) \cap (V_2 \cap f(X)) = \emptyset$ and $(V_1 \cap f(X)) \cup (V_2 \cap f(X)) = f(X)$. As f is continuous we have by [theorem: 14.141] that

$$U_1 = f^{-1}(V_1) \in \mathcal{T}_X \text{ and } U_2 = f^{-1}(V_2) \in \mathcal{T}_X \quad (14.314)$$

As $V_1 \cap f(X), V_2 \cap f(X) \neq \emptyset$ there exists a $y_1 \in V_1 \cap f(X)$ and a $y_2 \in V_2 \cap f(X)$. Hence there exist $x_1, x_2 \in X$ such that $f(x_1) = y_1 \in V_1$ and $f(x_2) = y_2 \in V_2$ so that $x_1 \in f^{-1}(V_1) = U_1$ and $x_2 \in f^{-1}(V_2) = U_2$, hence

$$U_1 \neq \emptyset \text{ and } U_2 \neq \emptyset \quad (14.315)$$

Let $x \in X$ then $f(x) \in f(X) = (V_1 \cap f(X)) \cup (V_2 \cap f(X)) = V_1 \cup V_2$ so that

$$x \in f^{-1}(V_1 \cup V_2) \stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(V_1) \cup f^{-1}(V_2) = U_1 \cup U_2 \subseteq X$$

proving

$$X = U_1 \cup U_2 \quad (14.316)$$

Further if $x \in U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2)$ then $f(x) \in V_1 \cap V_2$ which as $f(x) \in f(X)$ results in $f(x) \in (V_1 \cap f(X)) \cap (V_2 \cap f(X)) = \emptyset$ a contradiction hence we must have that

$$U_1 \cap U_2 = \emptyset \quad (14.317)$$

Combining [eqs: 14.314, 14.315 and 14.317] we conclude that X is not connected contradicting the fact that X was assumed to be connected. Hence the assumption that $f(X)$ is not connected is wrong and we must have that $f(X)$ is connected. $\square$

A very important example of a connected subset of $\mathbb{R}$ is the closed interval (later we will show that every generalized interval is connected).

Lemma 14.464. Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of real numbers then $\forall a, b \in \mathbb{R}$ with $a \leq b$ we have that $[a, b]$ is connected [using the topology $(\mathcal{T}_{\|\cdot\|})_{|[a, b]}$]

Proof. Note that $[a, b] = \left[\frac{a+b}{2} - \frac{b-a}{2}, \frac{a+b}{2} + \frac{b-a}{2} \right] = \overline{B\left(\frac{a+b}{2}, \frac{b-a}{2}\right)}$ which is a closed set in $\mathcal{T}_{\|\cdot\|}$ by [theorem: 14.69]. Let $A \subseteq [a, b]$ be a closed set in the subspace topology $(\mathcal{T}_{\|\cdot\|})_{|[a, b]}$ then there exist by [theorem: 14.18] a closed set A' in $\mathcal{T}_{\|\cdot\|}$ such that $A = A' \cap [a, b]$ which is then closed in $\mathcal{T}_{\|\cdot\|}$. Hence it follows that

$$\text{every closed set in } (\mathcal{T}_{\|\cdot\|})_{|[a, b]} \text{ is closed in } \mathcal{T}_{\|\cdot\|} \quad (14.318)$$

We proceed now by contradiction. So assume that $[a, b]$ is not connected. Then by [theorem: 14.460] there exists sets $A_1, A_2 \subseteq [a, b]$ such that A_1, A_2 are closed in $(\mathcal{T}_{||})_{|[a, b]}$, hence also closed in $\mathcal{T}_{||}$, $A_1 \neq \emptyset$, $A_2 \neq \emptyset$, $A_1 \cap A_2 = \emptyset$ and $[a, b] = A_1 \cup A_2$. We can always assume that $b \in A_2$ [otherwise switch A_1 and A_2]. As $A_1 \subseteq A_1 \cup A_2 = [a, b]$ it follows that A_1 is bounded above by b which together with $A_1 \neq \emptyset$ allows us to use the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] proving that

$$c = \sup(A_1) \text{ exist}$$

Let $U \in \mathcal{T}_{||}$ such that $c \in U$ then there exist a $\varepsilon \in \mathbb{R}^+$ with $c \in]c - \varepsilon, c + \varepsilon[= B_{||}(c, \varepsilon) \subseteq U$. As $c - \varepsilon < c = \sup(A_1)$ there exist a $c' \in A_1$ such that $c - \varepsilon < c' \leq c < c + \varepsilon$ so that

$$c' \in]c - \varepsilon, c + \varepsilon[\cap A_1 = B_{||}(c, \varepsilon) \cap A_1 \subseteq U \cap A_1 \Rightarrow U \cap A_1 \neq \emptyset \quad (14.319)$$

proving by [theorem: 14.31] and the fact that A_1 is closed that

$$c \in A_1 \quad (14.320)$$

As $c \in A_1 \subseteq [a, b]$ we have $a \leq c < b$ or

$$]c, b] \subseteq [a, b] \quad (14.321)$$

Assume that $x \in]c, b] \cap A_1$ then $c < x$ and $x \leq \sup(A_1) = c$ giving the contradiction $x < c$, hence $]c, b] \cap A_1 = \emptyset$. Combining this with [eq: 14.321] results in

$$]c, b] \subseteq [a, b] \setminus A_1 = (A_1 \cup A_2) \setminus A_1 \stackrel{[\text{lemma: 14.459}]}{=} A_2$$

As $c \in A_1$, $b \in A_2$, $A_1 \cap A_2 = \emptyset$ we have that $c \neq b$ which as b is a upper bound of A_1 proves

$$c < b \Rightarrow 0 < b - c$$

Let $U \in \mathcal{T}_{|||}$ such that $c \in U$ then there exist a $\varepsilon \in \mathbb{R}^+$ such that $c \in]c - \varepsilon, c + \varepsilon[\subseteq U$. Take $\delta = \min(b - c, \varepsilon) > 0$ then we have $c < c + \frac{\delta}{2} < c + \delta \leq c + b - c = b$ so that $c + \frac{\delta}{2} \in]c, b] \subseteq A_2$ and as $c - \varepsilon < c < c + \frac{\delta}{2} \leq c + \frac{\varepsilon}{2} < c + \varepsilon$ we have that $c + \frac{\delta}{2} \in]c - \varepsilon, c + \varepsilon[\cap A_2 \subseteq U \cap A_2$ proving that $U \cap A_2 \neq \emptyset$. Using [theorem: 14.31] it follows that $c \in A_2$ which combined with [eq: 14.320] gives $c \in A_1 \cap A_2$ contradicting $A_1 \cap A_2 \stackrel{[\text{eq: 14.319}]}{=} \emptyset$. Hence the assumption that $[a, b]$ is not connected is wrong so that we must have that $[a, b]$ is connected. $\square$

Actually it turns out that the only connected sets in $[a, b]$ are generalized intervals [see definition: 3.141 and theorem: 3.144].

Theorem 14.465. *Let $\langle \mathbb{R}, || \rangle$ be the normed spaces of real numbers and $I \subseteq \mathbb{R}$ then we have*

$$I \text{ is connected} \Leftrightarrow I \text{ is a generalized interval}$$

Proof.

$\Rightarrow$. We prove this by contradiction. So assume that I is a connected set but not a generalized interval. As $\langle \mathbb{R}, \leq \rangle$ is conditional complete we have by [theorem: 3.142] that there exists $x, y \in I$ such that $]x, y[\not\subseteq I$. Hence there exist a $z \in]x, y[$ such that $z \in \mathbb{R} \setminus I$. Now as $I \subseteq \mathbb{R} =]-\infty, z[\cup \{z\} \cup]z, \infty[$ and $z \notin I$ it follows that $I \subseteq]-\infty, z[\cup]z, \infty[$ so that

$$I \subseteq (]-\infty, z[\cap I) \cup (]z, \infty[\cap I) \quad (14.322)$$

We have also

$$(]-\infty, z[\cap I) \cap (]z, \infty[\cap I) = I \cap (]-\infty, z[\cap]z, \infty[) = I \cap \emptyset = \emptyset \quad (14.323)$$

As $x, y \in I$ and $z \in]x, y[\Rightarrow x < z < y \Rightarrow x \in]-\infty, z[\cap I \wedge y \in]z, \infty[\cap I$ it follows that

$$]-\infty, z[\cap I \neq \emptyset \text{ and }]z, \infty[\cap I \neq \emptyset \quad (14.324)$$

Finally using [theorem: 14.105] we have that $]-\infty, z[\in \mathcal{T}_{||}$ and $]z, \infty[\in \mathcal{T}_{||}$ so that

$$]-\infty, z[\cap I \in (\mathcal{T}_{||})_{|I} \text{ and }]z, \infty[\cap I \in (\mathcal{T}_{||})_{|I} \quad (14.325)$$

Hence using [eqs: 14.322, 14.323, 14.324 and 14.325] it follows that I is not connected contradicting the hypothesis that I is connected. So our assumption that I is not a generalized interval is wrong, proving that I is indeed a generalized interval.

⇐. We prove this also by contradiction. So assume that I is a generalized interval that is not connected. Then there exists open sets $U_1, U_2 \in \mathcal{T}_{||}$ such that $(U_1 \cap I) \neq \emptyset = (U_2 \cap I)$, $I = (U_1 \cap I) \cup (U_2 \cap I)$ and $(U_1 \cap I) \cap (U_2 \cap I) = \emptyset$. Then there exists $x \in U_1 \cap I$ and a $y \in U_2 \cap I$. As $(U_1 \cap I) \cap (U_2 \cap I) = \emptyset$ it follows that $x \neq y$ and we can always assume that $x < y$ [if $y < x$ interchange x and y]. As I is a generalized interval we have by [theorem: 3.141] that $[x, y] \subseteq I$. So that

$$\begin{aligned} [x, y] &= [x, y] \cap I \\ &= [x, y] \cap ((U_1 \cap I) \cup (U_2 \cap I)) \\ &= ([x, y] \cap U_1 \cap I) \cup ([x, y] \cap U_2 \cap I) \\ &= ([x, y] \cap U_1) \cup ([x, y] \cap U_2) \end{aligned}$$

or

$$[x, y] = ([x, y] \cap U_1) \cup ([x, y] \cap U_2) \quad (14.326)$$

Further as $[x, y] \subseteq I$ we have $([x, y] \cap U_1) \cap ([x, y] \cap U_2) \subseteq (I \cap U_1) \cap (I \cap U_2) = \emptyset$ proving that

$$([x, y] \cap U_1) \cap ([x, y] \cap U_2) = \emptyset \quad (14.327)$$

Further as $x \in U_1 \cap I \Rightarrow x \in U_1 \xrightarrow{x \in [x, y]} x \in U_1 \cap [x, y]$ and $y \in U_2 \cap I \Rightarrow y \in U_2 \xrightarrow{y \in [x, y]} y \in U_2 \cap [x, y]$ we have

$$U_1 \cap [x, y] \neq \emptyset \text{ and } U_2 \cap [x, y] \neq \emptyset \quad (14.328)$$

Finally by definition we have that

$$U_1 \cap [x, y] \in (\mathcal{T}_{||})|_{[x, y]} \text{ and } U_2 \cap [x, y] \in (\mathcal{T}_{||})|_{[x, y]} \quad (14.329)$$

From [eqs: 14.326, 14.327, 14.328 and [eq: 14.329]] it follows that $[x, y]$ is not connected contradicting the fact that by [lemma: 14.464] $[x, y]$ is connected. Hence the assumption that I is not connected is wrong so that I must be connected. $\square$

A important consequence of the fact that generalized intervals in $\mathbb{R}$ are connected is the Intermediate Value Theorem. First we need a little lemma proving that the image of a closed interval of a continuous real function on the interval is again a closed interval.

Lemma 14.466. *Let $\langle \mathbb{R}, || \rangle$ be the normed spaces of real numbers, $a, b \in \mathbb{R}$ with $a \leq b$ and*

$$f: [a, b] \rightarrow \mathbb{R} \text{ a continuous function}$$

then there exist $c, d \in \mathbb{R}$ such that

$$f([a, b]) = [c, d]$$

So there exists $x, y \in [a, b]$ such that $c = f(x)$ and $d = f(y)$ and $\forall z \in [a, b]$ $f(x) \leq f(z) \leq f(y)$

Proof. Using Heine Borell [theorem: 14.279] it follows that $[a, b]$ is a compact set. As $f: [a, b] \rightarrow \mathbb{R}$ is continuous it follows from [theorem: 14.275] that

$$f([a, b]) \text{ is compact}$$

As compact sets in $\mathbb{R}$ are closed and bounded [see theorem: 14.280] it follows that

$$f([a, b]) \text{ is closed and bounded}$$

As $[a, b]$ is connected [see lemma: 14.464] we have, using [theorem: 14.463] by the continuity of f that

$$f([a, b]) \text{ is connected}$$

Hence using the previous theorem [theorem: 14.465] it follows that

$$f([a, b]) \text{ is a generalized interval}$$

As $f([a, b])$ is closed and not empty it follows from [theorem: 14.107] that

$$f([a, b]) \text{ is either } [c, d],]-\infty, c] \text{ or } [c, \infty[$$

By [example: 14.113] $]-\infty, c]$ and $[c, \infty[$ are unbounded while $[c, d]$ is bounded, hence we have that

$$f([a, b]) = [c, d] \quad \square$$

Theorem 14.467. (Intermediate Value Theorem) Let $\langle \mathbb{R}, || \rangle$ be the normed space of real numbers, $a, b \in \mathbb{R}$ with $a \leq b$ and $f: [a, b] \rightarrow \mathbb{R}$ a continuous function then we have:

1. If $y \in \mathbb{R}$ satisfies $f(a) < y < f(b)$ then there exist a $x \in [a, b]$ such that $y = f(x)$.
2. If $y \in \mathbb{R}$ satisfies $f(b) < y < f(a)$ then there exist a $x \in [a, b]$ such that $y = f(x)$.

Proof. Using [lemma: 14.466] we have that $f([a, b]) = [c, d]$, hence $f(a), f(b) \in [c, d]$. Let $y \in \mathbb{R}$ then we have:

$f(a) < y < f(b)$. Then $y \in [f(a), f(b)]$ and $f(a) < f(b)$ where $f(a), f(b) \in [c, d]$. As $[c, d]$ is a generalized interval it follows from [theorem: 3.141] that $[f(a), f(b)] \subseteq [c, d] = f([a, b])$. Hence $y \in f([a, b])$ so there exist a $x \in [a, b]$ such that $y = f(x)$.

$f(b) < y < f(a)$. Then $y \in [f(b), f(a)]$ and $f(b) < f(a)$ where $f(a), f(b) \in [c, d]$. As $[c, d]$ is a generalized interval it follows from [theorem: 3.141] that $[f(b), f(a)] \subseteq [c, d] = f([a, b])$. Hence $y \in f([a, b])$ so there exist a $x \in [a, b]$ such that $y = f(x)$. $\square$

Next we show that a topological space can be divided in maximal connected subsets called components.

Definition 14.468. (Component) Let $\langle X, \mathcal{T} \rangle$ be a topological space then the relation $\sim \in X \times X$ defined by $x \sim y$ iff there exist a connected subset A such that $x \in A \wedge y \in A$ is a equivalence relation. Given $x \in X$ the equivalence class [see definition: 3.9] $\sim[x] = \{y \in X | y \sim x\}$ is called a **component**. Using [definition: 3.16] and [theorem: 3.14] it follows that the set of all classes $X/\sim$ partitions X is a disjoint union of partition classes hence we have that

$$X = \bigsqcup_{C \in X/\sim} C$$

Proof. Of course we have to prove that $\sim$ is indeed a equivalence relation.

reflexivity. Let $x \in X$ then as by [example: 14.458] $\{x\}$ is a connected subset and we have $x, x \in \{x\}$ it follows that $x \sim x$.

symetricity. Let $x \sim y$ then there exist a connected subset A such that $x, y \in A \Rightarrow y, x \in A$ proving that $y \sim x$.

transitivity. Let $x \sim y$ and $y \sim z$ then there exists connected subsets A, B such that $x, y \in A$ and $y, z \in B$. Hence $y \in A \cap B$ so that by [theorem: 14.462] $A \cup B$ is a connected. As also $x, z \in A \cup B$ it follows that $x \sim z$. $\square$

We show now that a component of a topological space is a maximal connected subset of X .

Theorem 14.469. Let $\langle X, \mathcal{T} \rangle$ be a topological space and C a component of X then we have

1. If A is a connected subset such that $C \subseteq A$ then $C = A$.
2. If A is a connected subset such that $C \cap A \neq \emptyset$ then $A \subseteq C$
3. C is a connected subset

Proof. If C is a component then there exist a $x \in X$ such that $C = \sim[x] = \{y \in X | y \sim x\}$.

1. Let A be a connected set such that $C \subseteq A$. Let $y \in A$ then as $x \in C \subseteq A \Rightarrow x \in A$ we have $x, y \in A$ so that by the definition of $\sim$ that $y \sim x$ so that $y \in C$. Hence $A \subseteq C \subseteq A$ which proves that

$$C = A$$

2. As $C \cap A \neq \emptyset$ there exist a $z \in A \cap C$. As $z \in C$ we have $z \sim x$. Let $y \in A$ then, as $z \in A$ we have that $y \sim z$ hence by transitivity of $\sim$ we have $y \sim x$ so that $y \in \sim[x]$ which proves that

$$A \subseteq C$$

3. Let $y \in C$ then $y \sim x$ so that there exist a connected subset A_y such that $y, x \in A_y$ hence by (2) we have that $A_y \subseteq C$. Take $\{A_y\}_{y \in C}$ then we have $\forall y \in C$ that $x, y \in A_y \subseteq C$ so that $C = \bigcup_{y \in C} A_y$ and $x \in \bigcap_{y \in C} A_y$. Using [theorem: 14.462] it follows that $\bigcup_{y \in C} A_y$ is a connected subset proving that

C is a connected subset □

14.11.2 Path Connected Space

Definition 14.470. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $x, y \in X$ then x is path connected to y if there exist a continuous [using the topologies $\mathcal{T}_{|[0,1]}$ and $\mathcal{T}$] $f: [0, 1] \rightarrow X$ such that $f(0) = x$ and $f(1) = y$.

Definition 14.471. (Path Connected Space) Let $\langle X, \mathcal{T} \rangle$ be a topological space and $A \subseteq X$ then A is path connected if $\forall x, y \in A$ x is path connected to y . If X is path connected then $\langle X, \mathcal{T} \rangle$ is a path connected space.

Every path connected space is a connected space so path connectedness is a stronger condition than connectedness.

Theorem 14.472. Let $\langle X, \mathcal{T} \rangle$ be a path connected space then $\langle X, \mathcal{T} \rangle$ is a connected space.

Proof. Let $\langle X, \mathcal{T} \rangle$ be a path connected space. Assume that $\langle X, \mathcal{T} \rangle$ is not connected then there exist open sets U, V with $U \cap V = \emptyset$ and $X = U \cup V$ such that $U \neq \emptyset \neq V$. So there exists a $x \in U$ and a $y \in V$ then there exist a continuous function $f: [0, 1] \rightarrow X$ such that $f(0) = x$ and $f(1) = y$. As $[0, 1]$ is connected [see theorem: 14.464] it follows from [theorem: 14.463] $f([0, 1])$ is a connected subset of X . Using [theorem: 14.461] it follows that

$$f([0, 1]) \subseteq U \text{ or } f([0, 1]) \subseteq V$$

If $f([0, 1]) \subseteq U$ then $x = f(0) \in U$ and $y = f(1) \in U$ which as $y \in V$ means that $y \in U \cap V = \emptyset$ which is a contradiction. Likewise if $f([0, 1]) \subseteq V$ then $x = f(0) \in V$ and $y = f(1) \in V$ which as $x \in U$ means that $x \in U \cap V = \emptyset$ again a contradiction. As in all possible cases we reach a contradiction the assumption is false and we must have that $\langle X, \mathcal{T} \rangle$ is a connected space. □

A import example of path connected sets are the convex sets.

Definition 14.473. Let $\langle X, \|\cdot\| \rangle$ be a normed space then $C \subseteq X$ is **convex** if $\forall x, y \in C$ we have $\forall t \in [0, 1]$ that $x + t \cdot (y - x) = (1 - t) \cdot x + t \cdot y \in C$

Example 14.474. Let $\langle X, \|\cdot\| \rangle$ be a normed space and C a convex subset of X then C is a connected set [hence by 14.472]

Proof. Let $x, y \in C$ and define

$$f: [0, 1] \rightarrow X \text{ where } f(t) = x + t \cdot (y - x)$$

Let $t \in [0, 1]$ then for $\varepsilon \in \mathbb{R}^+$ take $\delta = \frac{\varepsilon}{1 + \|y - x\|}$ then for $s \in [0, 1]$ such that $|t - s| < \delta$ then we have

$$\begin{aligned} \|f(t) - f(s)\| &= \|x + t \cdot (y - x) - x - s \cdot (y - x)\| \\ &= \|(t - s) \cdot (y - x)\| \\ &= |t - s| \cdot \|y - x\| \\ &\leq \frac{\varepsilon}{1 + \|y - x\|} \cdot \|y - x\| \\ &< \varepsilon \end{aligned}$$

which by [theorem: 14.136] proves that

$$f \text{ is continuous using } (\mathcal{T}_{||})_{[0,1]}, \mathcal{T}_{|||}$$

Further we have that

$$f(0) = x + 0 \cdot (y - x) = x \text{ and } f(1) = x + 1 \cdot (y - x) = y$$

Hence we have that x is path connected to y . As $x, y \in C$ was choosen arbitrary it follows that C is path connected. $\square$

Just as the image of a connected set by a continuous function is connected [see theorem: 14.463] the image of a path connected space by a continuous function is path connected.

Theorem 14.475. *Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces, A a path connected set in X and $f: A \rightarrow Y$ a continuous function [using the topologies $(\mathcal{T}_X)|_A$ and $\mathcal{T}_Y$] then $f(A)$ is path connected.*

Proof. Let $x, y \in f(A)$ then there exists $x', y' \in A$ such that

$$f(x') = x \wedge f(y') = y$$

As A is path connected there exist a continuous [using the topologies $(\mathcal{T}_{||})_{[0,1]}, \mathcal{T}_X$] function

$$g: A \rightarrow X$$

such that

$$g(0) = x' \wedge g(1) = y'$$

Using [theorem: 14.159] it follows that $f \circ g: [0, 1] \rightarrow Y$ is a continuous [using the topologies $(\mathcal{T}_{||})_{[0,1]}, \mathcal{T}_Y$] function. Further we have

$$(f \circ g)(0) = f(g(0)) = f(x') = x \wedge (f \circ g)(1) = f(g(1)) = f(y') = y$$

which proves that x is path connected to y . As $x, y \in f(A)$ has been choosen arbitrary it follows that $f(A)$ is path connected. $\square$

To introduce pathwise components using a equivalence relation we need the following lemmas.

Lemma 14.476. *Let $a, b, c, d \in \mathbb{R}$ with $a < b \wedge c < d$ then there exists a homeomorphism*

$$h: [a, b] \rightarrow [c, d]$$

[using the topologies $(\mathcal{T}_{||})_{[a,b]}$ and $(\mathcal{T}_{||})_{[c,d]}$ such that $h(a) = c$ and $h(b) = d$]

Proof. Define

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ by } f(x) = \frac{d-c}{b-a} \cdot x + \frac{b \cdot c - a \cdot d}{b-a}$$

then we have

$$\begin{aligned} f(a) &= \frac{d-c}{b-a} \cdot a + \frac{b \cdot c - a \cdot d}{b-a} \\ &= \frac{a \cdot d - a \cdot c + b \cdot c - a \cdot d}{b-a} \\ &= \frac{b \cdot c - a \cdot c}{b-a} \\ &= c \\ f(b) &= \frac{d-c}{b-a} \cdot b + \frac{b \cdot c - a \cdot d}{b-a} \\ &= \frac{b \cdot d - b \cdot c + b \cdot c - a \cdot d}{b-a} \\ &= \frac{b \cdot d - a \cdot d}{b-a} \\ &= d \end{aligned}$$

Hence we have

$$f(a) = c \wedge f(b) = d \tag{14.330}$$

Let $x \in [a, b]$ then $a \leq x \leq b$, as $0 < \frac{d-c}{b-a}$ it follows that

$$\frac{d-c}{b-a} \cdot a \leq \frac{d-c}{b-a} \cdot x \leq \frac{d-c}{b-a} \cdot b$$

hence

$$\frac{d-c}{b-a} \cdot a + \frac{b \cdot c - a \cdot d}{b-a} \leq \frac{d-c}{b-a} \cdot x + \frac{b \cdot c - a \cdot d}{b-a} \leq \frac{d-c}{b-a} \cdot b + \frac{b \cdot c - a \cdot d}{b-a}$$

or using the definition of f

$$c_{[\text{eq: 14.330}]}^{\equiv} f(a) \leq f(x) \leq f(b)_{[\text{eq: 14.330}]}^{\equiv} d$$

which proves that

$$f([a, b]) \subseteq [c, d]. \quad (14.331)$$

Define

$$g: \mathbb{R} \rightarrow \mathbb{R} \text{ by } g(x) = \frac{b-a}{d-c} \cdot x + \frac{d \cdot a - c \cdot b}{d-c}$$

then we have

$$\begin{aligned} g(c) &= \frac{b-a}{d-c} \cdot c + \frac{d \cdot a - c \cdot b}{d-c} \\ &= \frac{b \cdot c - a \cdot c + d \cdot a - c \cdot b}{d-c} \\ &= \frac{d \cdot a - a \cdot c}{d-c} \\ &= a \\ g(d) &= \frac{b-a}{d-c} \cdot d + \frac{d \cdot a - c \cdot b}{d-c} \\ &= \frac{b \cdot d - a \cdot d + d \cdot a - c \cdot b}{d-c} \\ &= \frac{b \cdot d - c \cdot b}{d-c} \\ &= b \end{aligned}$$

in other words

$$g(c) = a \wedge g(d) = b \quad (14.332)$$

Let $x \in [c, d]$ then $c \leq x \leq d$, as $0 < \frac{b-a}{d-c}$ it follows that

$$\frac{b-a}{d-c} \cdot c \leq \frac{b-a}{d-c} \cdot x \leq \frac{b-a}{d-c} \cdot d$$

so that

$$\frac{b-a}{d-c} \cdot c + \frac{d \cdot a - c \cdot b}{d-c} \leq \frac{b-a}{d-c} \cdot x + \frac{d \cdot a - c \cdot b}{d-c} \leq \frac{b-a}{d-c} \cdot d + \frac{d \cdot a - c \cdot b}{d-c}$$

or using the definition of g we have that

$$a_{[\text{eq: 14.331}]}^{\equiv} g(c) \leq g(x) \leq g(d)_{[\text{eq: 14.331}]}^{\equiv} b$$

which proves that

$$g([c, d]) \subseteq [a, b] \quad (14.333)$$

Let $x \in \mathbb{R}$ then

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) \\ &= g\left(\frac{d-c}{b-a} \cdot x + \frac{b \cdot c - a \cdot d}{b-a}\right) \\ &= \left(\frac{d-c}{b-a} \cdot x + \frac{b \cdot c - a \cdot d}{b-a}\right) \cdot \frac{b-a}{d-c} + \frac{d \cdot a - c \cdot b}{d-c} \\ &= x + \frac{b \cdot c - a \cdot d}{d-c} + \frac{d \cdot a - c \cdot b}{d-c} \\ &= x \\ &= \text{Id}_{\mathbb{R}}(x) \end{aligned}$$

which proves that

$$g \circ f = \text{Id}_{\mathbb{R}}$$

Let $x \in [c, d]$ then

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f\left(\frac{b-a}{d-c} \cdot x + \frac{d \cdot a - c \cdot b}{d-c}\right) \\ &= \left(\frac{b-a}{d-c} \cdot x + \frac{d \cdot a - c \cdot b}{d-c}\right) \cdot \frac{d-c}{b-a} + \frac{b \cdot c - a \cdot d}{b-a} \\ &= x + \frac{d \cdot a - c \cdot b}{b-a} + \frac{b \cdot c - a \cdot d}{b-a} \\ &= x \\ &= \text{Id}_{\mathbb{R}}(x) \end{aligned}$$

which proves that

$$f \circ g = \text{Id}_{\mathbb{R}}$$

Hence using [theorem: 2.73] it follows that

$$f \text{ is a bijection with } (f)^{-1} = g \quad (14.334)$$

By [example: 14.137] it follows that f, g are continuous function which together with the above proves that

f is a homeomorphism

Now $[c, d] = (f \circ g)([c, d]) = f(g([c, d])) \subseteq f([a, b])$ which combined with [eq: 14.331] proves that

$$f([a, b]) = [c, d] \quad (14.335)$$

Hence using [theorem: 14.193] it follows that

$$h = f|_{[a, b]}: [a, b] \rightarrow [c, d]$$

is a homeomorphism between $[a, b] \rightarrow [c, d]$. $\square$

Lemma 14.477. *The function $f: [0, 1] \rightarrow \mathbb{R}$ defined by $\sigma(x) = -x + 1$ has the following properties:*

1. σ is continuous using the topologies $(\mathcal{T}_{||})|_{[0, 1]}, \mathcal{T}_{||}$
2. $\sigma(0) = 1 \wedge \sigma(1) = 0$
3. $\sigma([0, 1]) \subseteq [0, 1]$

Proof.

1. This follows from [example: 14.137].
2. $\sigma(0) = -0 + 1 = 1$ and $\sigma(1) = -1 + 1 = 0$
3. Let $x \in [0, 1]$ then $0 \leq x \leq 1$ so that $-1 \leq -x \leq 0$ hence we have $0 = -1 + 1 \leq -x + 1 \leq 0 + 1 = 1$ so that $0 \leq \sigma(x) \leq 1$ from which it follows that $\sigma(x) \in [0, 1]$. Hence we have that

$$\sigma([0, 1]) \subseteq [0, 1] \quad \square$$

We are now ready to define path connected components.

Definition 14.478. *Let $\langle X, \mathcal{T} \rangle$ be a topological space then the relation $\sim_p \subseteq X \times X$ defined by $x \sim_p y$ iff x is path connected to y is a equivalence relation. Given $x \in X$ the equivalence class [see definition: 3.9] $\sim_p[x] = \{y \in X | y \sim_p x\}$ is called a **path connected component**. Using [definition: 3.16] and [theorem: 3.14] it follows that the set of all classes $X / \sim_p$ partitions X is a disjoint union of partition classes hence we have that*

$$X = \bigsqcup_{C \in X / \sim_p} C$$

Proof. Of course we must prove that $\sim_p$ is a equivalence relation.

reflexivity. Let $x \in X$ then the constant function $C_x: [0, 1] \rightarrow X$ defined by $C_x(t) = x$ is continuous by [theorem: 14.142], further $C_x(0) = x = C_x(1)$. Hence $x \sim_p x$.

symetricity. If $x \sim_p y$ then there exist a continuous [using $\mathcal{T}_{|[0,1]}$ and $\mathcal{T}$] $f: [0, 1] \rightarrow X$ such that $f(0) = x \wedge f(1) = y$. Take $\sigma: [0, 1] \rightarrow [0, 1]$ the function defined in [lemma: 14.477] then

$$\sigma \text{ is continuous [using } (\mathcal{T}_{|[0,1]}, \mathcal{T}) \wedge \sigma([0, 1]) \subseteq [0, 1] \wedge f(0) = 1 \wedge f(1) = 0$$

Hence using [theorem: 14.159] we have

$$f \circ \sigma: [0, 1] \rightarrow X \text{ is continuous [using } (\mathcal{T}_{|[0,1]}, \mathcal{T})$$

Further $(f \circ \sigma)(0) = f(\sigma(0)) = f(1) = y$ and $(f \circ \sigma)(1) = f(\sigma(1)) = f(0) = x$. Hence we have that

$$y \sim_p x$$

transitive. If $x \sim_p y$ and $y \sim_p z$ then there exist two continuous [using $(\mathcal{T}_{|[0,1]}, \mathcal{T})$] $f: [0, 1] \rightarrow X$ and $g: [0, 1] \rightarrow X$ such that $f(0) = x \wedge f(1) = y = g(0) \wedge g(1) = z$. Using [lemma: 14.476] there exists two homeomorphic functions $\varphi: [0, \frac{1}{2}] \rightarrow [0, 1]$ and $\psi: [\frac{1}{2}, 1] \rightarrow [0, 1]$ such that

$$\varphi(0) = 0 \wedge \varphi\left(\frac{1}{2}\right) = 1 \wedge \psi\left(\frac{1}{2}\right) = 0 \wedge \psi(1) = 1$$

Using then [theorem: 14.159, 14.145] we have that

$$f \circ \varphi: \left[0, \frac{1}{2}\right] \rightarrow X \text{ is continuous [using } (\mathcal{T}_{|[0, \frac{1}{2}]}, \mathcal{T})$$

$$g \circ \psi: \left[\frac{1}{2}, 1\right] \rightarrow X \text{ is continuous [using } (\mathcal{T}_{|[\frac{1}{2}, 1]}), \mathcal{T})$$

As $[0, \frac{1}{2}] \cap [\frac{1}{2}, 1] = \{\frac{1}{2}\}$ and

$$(f \circ \varphi)\left(\frac{1}{2}\right) = f\left(\varphi\left(\frac{1}{2}\right)\right) = f(1) = y = g(0) = g\left(\psi\left(\frac{1}{2}\right)\right) = (g \circ \psi)\left(\frac{1}{2}\right)$$

we can use the pasting lemma [see lemma: 14.155] so that the function

$$h = (f \circ \varphi) \bigcup (g \circ \psi): [0, 1] \rightarrow X \text{ defined by } h(x) = \begin{cases} (f \circ \varphi)(x) & \text{if } x \in [0, \frac{1}{2}] \\ (g \circ \psi)(x) & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

is continuous using the topologies $(\mathcal{T}_{|[0,1]}, \mathcal{T})$. Further we have that

$$h(0) \underset{0 \in [0, \frac{1}{2}]}{=} (f \circ \varphi)(1) = f(\varphi(0)) = f(0) = x$$

$$h(1) \underset{1 \in [\frac{1}{2}, 1]}{=} (g \circ \psi)(1) = g(\psi(1)) = g(1) = z$$

which proves that

$$x \sim_p z$$

□

We prove now that a path connected component is a maximal component subset of X .

Theorem 14.479. Let $\langle X, \mathcal{T} \rangle$ be a topological space and C a path connected component then we have:

1. C is path connected.
2. If A is a path connected subset such that $C \subseteq A$ then $C = A$
3. If A is a path connected subset such that $C \cap A \neq \emptyset$ then $A \subseteq C$

Proof. If C is a path connected component of X then there exist a $x \in X$ such that

$$C = \sim_p[x] = \{y \in X \mid y \sim_p x\}$$

1. Let $y, z \in C$ then $y \sim_p x \wedge z \sim_p x$ so that by transitivity and symmetry $y \sim_p z$ which proves that C is path connected.
2. Let A be a path connected set such that $C \subseteq A$. As $x \sim_p x$ $x \in C \Rightarrow x \in A$. Let $y \in A$ then as A is path connected we have that $y \sim_p x$ which proves that $y \in C$. Hence we have that $A \subseteq C \subseteq A$ so that

$$C = A$$

3. Let A be a path connected set such that $A \cap C \neq \emptyset$ then there exist a z such that $z \in A \wedge z \in C$ so that $z \sim_p x$. Let $y \in A$ then as A is path connected $y \sim_p z$ so that by transitivity $y \sim_p x$ which proves that $y \in C$. Hence we have that

$$A \subseteq C$$

□

Next we introduce the concept of local connectivity.

Definition 14.480. Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\langle X, \mathcal{T} \rangle$ is locally connected if there exist a basis $\mathcal{B} \subseteq \mathcal{T}$ for $\mathcal{T}$ such that $\forall B \in \mathcal{B}$ B is connected. In other words a locally connected space is a space whose topology has a basis of connected sets.

Likewise we have the concept of a locally path connected space.

Definition 14.481. Let $\langle X, \mathcal{T} \rangle$ be a topological space then X is locally path connected if there exist a basis $\mathcal{B} \subseteq \mathcal{T}$ for $\mathcal{T}$ such that $\forall B \in \mathcal{B}$ B is path connected. In other words a locally path connected space is a space whose topology has a basis of path connected sets.

Theorem 14.482. Let $\langle X, \mathcal{T} \rangle$ be locally path connected space then we have:

1. Every path connected component of X is open in X . In other words $\forall x \in X$ $\sim_p[x] \in \mathcal{T}$.
2. $\forall x \in X$ $\sim_p[x] = \sim[x]$. In loose terms the path connected components and components are equal.
3. Every component of X is open in X . In other words $\forall x \in X$ $\forall x \in X$ $\sim[x] \in \mathcal{T}$.
4. If $X \neq \emptyset$ then we have that

$$X \text{ is connected} \Leftrightarrow X \text{ is path connected}$$

5. $\forall U \in \mathcal{T}$ we have that U is locally path connected.

Proof. As X is locally path connected there exist a basis $\mathcal{B} \subseteq \mathcal{T}$ such that $\forall B \in \mathcal{B}$ B is path connected.

1. Let C be a path connected component then there exist a $x \in X$ such that $C = \sim_p[x] = \{y \in X \mid y \sim_p x\}$. Let $y \in C$ then as X is open, $y \in X$ and $\mathcal{B}$ is a basis for $\mathcal{T}$ there exist a open path connected set $U \in \mathcal{B}$ such that $y \in U$. Hence $y \in U \cap C \Rightarrow U \cap C \neq \emptyset$, using then [theorem: 14.479] it follows that $U \subseteq C$. Hence $\forall y \in C$ we found a open set U with $x \in U \subseteq C$ which proves by [theorem: 14.7] that U is open in X .
2. Let $x \in X$ then $\sim_p[x]$ is path connected by [theorem: 14.479], further by [theorem: 14.472] $\sim_p[x]$ is connected. By [theorem: 14.469] $\sim[x]$ is also connected. As $x \in \sim[x] \cap \sim_p[x]$ it follows that $\sim[x] \cap \sim_p[x] \neq \emptyset$ so that by [theorem: 14.469]

$$\sim_p[x] \subseteq \sim[x] \tag{14.336}$$

Now for the opposite in inclusion. Consider the subspace topology $\mathcal{T}_{|\sim[x]} = \{U \cap \sim[x] \mid U \in \mathcal{T}\}$. As by (1) $\sim_p[x] \in \mathcal{T}$ and $\sim_p[x]_{|eq: 14.336} \sim_p[x] \cap \sim[x] \in \mathcal{T}_{|\sim[x]}$ it follows that

$$\sim_p[x] \in \mathcal{T}_{|\sim[x]} \tag{14.337}$$

Let $y \in \sim[x] \setminus \sim_p[x]$ then as $y \in X$ and X is locally path connected there exist a path connected open set $W \in \mathcal{B} \subseteq \mathcal{T}$ such that $y \in W$. If $W \cap \sim_p[x] \neq \emptyset$ then by [theorem: 14.479] we have that $W \subseteq \sim_p[x]$ from which it follows that $y \in \sim_p[x]$ contradicting $y \in \sim[x] \setminus \sim_p[x]$. Hence we must have that $W \cap \sim_p[x] = \emptyset$ hence $(W \cap \sim[x]) \cap \sim_p[x] = \emptyset$ which proves that

$$y \in (W \cap \sim[x]) \subseteq \sim[x] \setminus \sim_p[x] \text{ where } (W \cap \sim[x])_{W \in \mathcal{T}} \in \mathcal{T}_{|\sim[x]}$$

As $y \in \sim[x] \setminus \sim_p[x]$ has been chosen arbitray it follows from [theorem: 14.7] that $\sim[x] \setminus \sim_p[x]$ is open from which it follows that $\sim_p[x]$ is closed. Combining this with [eq: 14.337] gives

$$\sim_p[x] \text{ is open and closed in } \mathcal{T}_{|\sim[x]}$$

As $\sim[x]$ is connected it follows from [theorem: 14.460] that $\sim_p[x] = \emptyset$ or $\sim_p[x] = \sim[x]$. As $x \in \sim_p[x]$ it follows that we must have that

$$\sim_p[x] = \sim[x]$$

3. If C is a component then there exist a $x \in X$ so that $C = \sim[x]$, using (2) we have that $\sim[x] = \sim_p[x]$ and by (1) that $\sim_p[x]$ is open in X . Hence C is open in X .

4.

$\Rightarrow$. As $X \neq \emptyset$ there exist a $x_0 \in X$.

$$U = \sim_p[x_0] \stackrel{\text{def}}{=} \{z \in X \mid z \sim_p x_0\} \quad (14.338)$$

and

$$V = \{z \in X \mid \neg(z \sim_p x_0)\} \quad (14.339)$$

Then we have that

$$X = U \cup V \text{ and } U \cap V = \emptyset \quad (14.340)$$

Further by (1) we have that

$$U = \sim_p[x_0] \in \mathcal{T} \quad (14.341)$$

Let $z \in V$ then as $z \in X$ is a open set and X is locally path connected there exist a path connected set $W \in \mathcal{T}$ such that $z \in W$. Assume that $U \cap W \neq \emptyset$ then there exist a u such that $u \in U$ and $u \in W$. As $U \stackrel{\text{eq: 14.338}}{=} \sim_p[x_0]$ we have $u \sim_p x_0$ and. as W is path connected, we have $z \sim_p u$, hence using transitivity it follows that $z \sim_p x_0$ so that $z \in U$ or $z \in V \cap U \stackrel{\text{eq: 14.340}}{=} \emptyset$ a contradiction. Hence the assumption is false and we must have that $U \cap W = \emptyset$. As

$$W = W \cap X = W \cap (U \cup V) = (W \cap U) \cup (W \cap V) = \emptyset \cup (W \cap V) \subseteq V$$

we have that $\forall z \in V$ there exist a $W \in \mathcal{T}$ such that $z \in W \subseteq V$. Using [theorem: 14.7] it follows that

$$V \in \mathcal{T} \quad (14.342)$$

As X is connected, $U, V \in \mathcal{T}$ [see eqs: 14.341, 14.342] and $X = U \cup V \wedge U \cap V = \emptyset$ [see eq: 14.340] it follows from [definition: 14.454] that $U = \emptyset \vee V = \emptyset$. Given that $x_0 \in \sim_p[x_0] = U$ it follows that $V = \emptyset$. So that $X = U$ which, as $U = \sim_p[x_0]$ is path connected by [theorem: 14.479], proves that

X is path connected

$\Leftarrow$. This follows from [theorem: 14.472].

5. Let $U \in \mathcal{T}$ and define $\mathcal{C} = \{B \in \mathcal{B} \mid B \subseteq U\} \subseteq \mathcal{B} \subseteq \mathcal{T}$ then for $B \in \mathcal{C}$ we have that $B \stackrel{\text{eq: 14.337}}{=} B \cap U \in \mathcal{T}_{|U}$ so that

$$\mathcal{C} \subseteq \mathcal{T}_{|U} \quad (14.343)$$

Further as every set in $\mathcal{B}$ is path connected we have that

$$\forall C \in \mathcal{C} \ C \text{ is path connected} \quad (14.344)$$

Let $V \in \mathcal{T}_{|U}$ then there exist a $V' \in \mathcal{T}$ such that

$$V = V' \bigcap U \text{ and as } V', U \in \mathcal{T} \text{ we have also } V \in \mathcal{T}$$

Let $x \in V$ then as $V \in \mathcal{T}$ and $\mathcal{B}$ is a basis for $\mathcal{T}$ there exist a $W \in \mathcal{B}$ such that $x \in W \subseteq V$. As $V \subseteq U$ we have also $W \subseteq U$ so that $W \in \mathcal{C}$. Hence

$$\mathcal{C} \text{ is a basis for } \mathcal{T}_{|U} \quad (14.345)$$

From leqs: 14.343, 14.344 and 14.345] it follows that

$$\langle U, \mathcal{T}_{|U} \rangle \text{ is locally path connected} \quad \square$$

Chapter 15

Integration in Banach spaces

There are many ways to define integrals of functions, Riemann integrals, Lebesgue Integrals, Gauge Integrals, etc ... In this chapter we define an extension of the classic Riemann Darboux integral to functions whose range is a Banach space. Lebesgue and Gauge Integrals (like Henstock integrals) are more general, however to prove later the Fundamental Theorem of Calculus later we must define integrals before we introduce the concept of differentiation. For this reason we define Riemann integrals here and more general forms of integration in later chapters.

First we introduce the concept of partitioning a closed interval in sub intervals with no gaps. The way to do this is by specifying an ordered family of boundary points for the sub intervals. For example the interval $[1, 12]$ can be partitioned in the following way:

$$[1, 12] = [1, 4] \cup [4, 8] \cup [8, 12]$$

by the family $\{t_i\}_{i \in \{1, \dots, 4\}}$ where

$$\begin{aligned} t_1 &= 1 \\ t_2 &= 4 \\ t_3 &= 8 \\ t_4 &= 12 \end{aligned}$$

We use underscores to show the index of the boundary points of the sub intervals. Hence we use the following notation.

$$[1, 12] = [1_1, 4_2] \cup [4_2, 8_3] \cup [8_3, 12_4]$$

This is the idea of the following definition.

Definition 15.1. Let $a, b \in \mathbb{R}$ with $a < b$ then a family $\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ is a **partition of $[a, b]$** if $n \in \mathbb{N} \setminus \{1\}$, $t_1 = a$, $t_n = b$ and $\forall i \in \{1, \dots, n-1\}$ we have $t_i < t_{i+1}$.

Note 15.2. The strict inequality $a < b$ is needed for if $a = b$ we would have the contradiction $a = t_1 < t_2 = b$.

Lemma 15.3. If $n \in \mathbb{N} \setminus \{1\}$ and $\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $\forall i \in \{1, \dots, n-1\}$ $t_i < t_{i+1}$ then we have

1. $\forall i, j \in \{1, \dots, n\}$ with $i < j$ that $t_i < t_j$
2. $\min(\{t_i | i \in \{1, \dots, n\}\}) = t_1$
3. $\max(\{t_i | i \in \{1, \dots, n\}\}) = t_n$

Proof.

1. We use induction to prove this, so given $i \in \{1, \dots, n-1\}$ define

$$S_{n,i} = \{k \in \mathbb{N} | \text{If } i+k \leq n \text{ then } t_i < t_{i+k}\}$$

then we have:

$1 \in S_{n,i}$. If $i+1 \leq n$ then $i \in \{1, \dots, n-1\}$ and by definition $t_i < t_{i+1}$ so that

$k \in S_{n,i} \Rightarrow k+1 \in S_{n,i}$. If $i+(k+1) \leq n$ then $i+k \leq n-1$ so that $i+k \in \{1, \dots, n-1\}$, hence $t_{i+k} < t_{i+(k+1)}$ by definition. As $k \in S_{n,i}$ and $i+k \leq i+(k+1) \leq n$ we have $t_i < t_{i+k}$ so that $t_i < t_{i+(k+1)}$. Proving that $k+1 \in S_{n,i}$.

Using induction we have that $S_{n,i} = \mathbb{N}$. So if $i, j \in \{1, \dots, n\}$ with $i < j$ then $k = j - i \in \mathbb{N} \in S_{n,i}$ and $i+k = j \leq n$ so that $t_i < t_{i+k} = t_j$.

2. If $x \in \{t_i | i \in \{1, \dots, n\}\}$ then there exist a $i \in \{1, \dots, n\}$ such that $x = t_i$. As $1 \leq i$ it follows then from (1) that $t_1 \leq x$ which, as $t_1 \in \{t_i | i \in \{1, \dots, n\}\}$, proves that $t_1 = \min(\{t_i | i \in \{1, \dots, n\}\})$.
3. If $x \in \{t_i | i \in \{1, \dots, n\}\}$ then there exist a $i \in \{1, \dots, n\}$ such that $x = t_i$. As $i \leq n$ it follows then from (1) that $x \leq t_n$ which, as $t_n \in \{t_i | i \in \{1, \dots, n\}\}$, proves that $t_n = \max(\{t_i | i \in \{1, \dots, n\}\})$. $\square$

Theorem 15.4. Let $a, b \in \mathbb{R}$ with $a < b$ and $\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ a partition of $[a, b]$ then we have:

1. $\forall i, j \in \{1, \dots, n\}$ with $i < j$ we have $t_i < t_j$
2. $\forall i \in \{1, \dots, n\}$ we have $t_i \in [a, b]$
3. If $t_i = t_j$ then $i = j$
4. If $i \in \{1, \dots, n-1\}$ and $k \in \{1, \dots, n\}$ such that $t_k \in [t_i, t_{i+1}]$ then $k = i$ or $k = i+1$
5. $\forall i \in \{1, \dots, n-1\}$ we have $\forall k \in \{1, \dots, n\}$ we have $t_k \notin]t_i, t_{i+1}[$

Proof.

1. This follows from [lemma: 15.3].
2. If $i \in \{1, \dots, n\}$ then we have either:
 - $i = 1$. Then we have $t_i = a \in [a, b]$
 - $i = n$. Then we have $t_i = b \in [a, b]$
 - $i \in \{2, \dots, n-1\}$. Then we have by (1) $a = t_1 < t_i < t_n = b$ so that $t_i \in [a, b]$
3. If $t_i = t_j$ then if $i \neq j$ we have either $i < j$ or $j < i$ we have by (1) that $t_i < t_j$ or $t_j < t_i$ contradicting $t_i = t_k$. So we must have $i = j$.
4. If $t_k \in [t_i, t_{i+1}]$ then $t_i \leq t_k \leq t_{i+1}$. If now $k < i$ then using (1) we have $t_k < t_i$ contradicting $t_i \leq t_k$, if $i+1 < k$ we have using (1) $t_{i+1} < t_k$ contradicting $t_k \leq t_{i+1}$. So we must have $i \leq k \leq i+1$ which means $i = k$ or $k = i+1$.
5. Assume that $t_k \in]t_i, t_{i+1}[$ then $t_i < t_k < t_{i+1}$. For k we have either:
 - $k \leq i$. Then by (1) we have $t_k \leq t_i$ contradicting $t_i < t_k$.
 - $i < k$. Then $i+1 \leq k$ so that by (1) we have $t_{i+1} \leq t_k$ contradicting $t_k < t_{i+1}$. $\square$

The reason for the name partition of a interval is that is can be used to divide the interval in a union of sub intervals.

Theorem 15.5. Let $a, b \in \mathbb{R}$ with $a < b$ and $\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ a partition of $[a, b]$ then we have that

1. $[a, b[= \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}[$ and $\forall i, j \in \{1, \dots, n-1\}$ with $i \neq j$ we have $[t_i, t_{i+1}[\cap [t_j, t_{j+1}[= \emptyset$
2. $[a, b] = \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}]$

Proof.

1. Let $i \in \{1, \dots, n-1\}$ then by [theorem: 15.4] we have $t_i \in [a, b]$ so that $a \leq t_i < t_{i+1} \leq t_n = b$. If $x \in [t_i, t_{i+1}[$ then $t_i \leq x < t_{i+1}$ so that $a \leq x < b$ or $x \in [a, b[$. Hence $[t_i, t_{i+1}[\subseteq [a, b[$ so that

$$\bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}[\subseteq [a, b[\quad (15.1)$$

For the opposite inequality, let $x \in [a, b[$ then $a \leq x < b$. Define $M_x = \{i \in \{1, \dots, n-1\} | t_i \leq x\}$ then, as $M_x \subseteq \{1, \dots, n-1\}$ M_x is finite, further as $t_1 = a \leq x$ we have $1 \in M_x$ so that $M_x \neq \emptyset$. Hence $m = \max(M_x)$ exist. As $m \in M_x$ we have that $t_m \leq x$. Now for m we have either:

$m = n - 1$. Then $m + 1 = n$ and as $x < b$ we have $t_m \leq x < b = t_{m+1}$ so that

$$x \in [t_m, t_{m+1}[\subseteq \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}[$$

$m < n - 1$. If $t_{m+1} \leq x$ then, as $m + 1 \in \{1, \dots, n-1\}$ we have $m + 1 \in M_x$ so that $m \leq \max(M_x) = m$ leading to the contradiction $m < m$. Hence we must have $x < t_{m+1}$ so that $t_m \leq x < t_{m+1}$ or

$$x \in [t_m, t_{m+1}[\subseteq \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}[$$

So we have that $[a, b[\subseteq \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}[$ which combined with [eq: 15.1] proves

$$[a, b] = \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}[$$

Let $i, j \in \{1, \dots, n-1\}$ with $i \neq j$ then we may assume that without loosing generality that $i < j$ [if not exchange i, j]. Let $x \in [t_i, t_{i+1}[\cap [t_j, t_{j+1}[$ then $t_i \leq x < t_{i+1}$ and $t_j \leq x < t_{j+1}$. As $i < j \Rightarrow t + 1 \leq j$ so that $t_{i+1} \leq t_j$, hence $x < t_{i+1} \leq t_j \leq x$ giving the contradiction $x < x$. So we must have that

$$[t_i, t_{i+1}[\cap [t_j, t_{j+1}[= \emptyset$$

2. Let $i \in \{1, \dots, n-1\}$ then by [theorem: 15.4] we have $t_i \in [a, b]$ so that $a \leq t_i < t_{i+1} \leq t_n = b$. If $x \in [t_i, t_{i+1}]$ then $t_i \leq x \leq t_{i+1}$ so that $a \leq x \leq b$ or $x \in [a, b]$. Hence $[t_i, t_{i+1}] \subseteq [a, b]$ so that

$$\bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}] \subseteq [a, b] \quad (15.2)$$

For the opposite inequality, let $x \in [a, b]$ then $a \leq x \leq b$. Define $M_x = \{i \in \{1, \dots, n-1\} | t_i \leq x\}$ then, as $M_x \subseteq \{1, \dots, n-1\}$ M_x is finite, further as $t_1 = a \leq x$ we have $1 \in M_x$ so that $M_x \neq \emptyset$. Hence $m = \max(M_x)$ exist. As $m \in M_x$ we have that $t_m \leq x$. Now for m we have either:

$m = n - 1$. Then $m + 1 = n$ and as $x \leq b$ we have $t_m \leq x \leq b = t_{m+1}$ so that

$$x \in [t_m, t_{m+1}] \subseteq \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}]$$

$m < n - 1$. If $t_{m+1} \leq x$ then, as $m + 1 \in \{1, \dots, n-1\}$ we have $m + 1 \in M_x$ so that $m \leq \max(M_x) = m$ leading to the contradiction $m < m$. Hence we must have $x < t_{m+1}$ so that $t_m \leq x < t_{m+1}$ or

$$x \in [t_m, t_{m+1}[\subseteq [t_m, t_{m+1}] \subseteq \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}]$$

So we have that $[a, b] \subseteq \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}]$ which combined with [eq: 15.2] proves

$$[a, b] = \bigcup_{i \in \{1, \dots, n-1\}} [t_i, t_{i+1}] \quad \square$$

If we have two partitions of a interval then we can always find a partition that is 'composed' of the two partitions and is finer than the two partitions. With finer we mean that the boundary points of this partition is based on the boundary points of the two given partitions.

$\mathcal{P}_1 = [1, 4_2] \cup [4_2, 8_3] \cup [8_3, 12_4]$, boundary points $\{1, 4_2, 8_3, 12_4\}$

$\mathcal{P}_2 = [1, 2_2] \cup [2_2, 5_3] \cup [5_3, 6_4] \cup [6_4, 12_5]$, boundary points $\{1, 2_2, 5_3, 6_4, 12_5\}$

then the following partition can be based on the boundary points of $\mathcal{P}_1$ and $\mathcal{P}_2$

$\mathcal{P} = [1_1, 2_2] \cup [2_2, 4_3] \cup [4_3, 5_4] \cup [5_4, 6_5] \cup [6_5, 8_6] \cup [8_6, 12_7]$, boundary points

$$\{1_1, 2_2, 4_3, 5_4, 6_5, 8_6, 12_7\} = \{1_1, 4_2, 8_3, 12_4\} \cup \{1_1, 2_2, 5_3, 6_4, 12_5\}$$

It will turn out that there only exist a such partition that satisfies this condition about the boundary points. Although intuitively such a partition exist and is unique, proving this is rather elaborate. This is done in the next two lemmas. This partition $\mathcal{P}$ based on $\mathcal{P}_1$ and $\mathcal{P}_2$ is noted as $\mathcal{P}_1 \square \mathcal{P}_2$

Lemma 15.6. *Let $a, b \in \mathbb{R}$ with $a < b$ and $\mathcal{P}_1 = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}$, $\mathcal{P}_2 = \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}$ two partitions of $[a, b]$ then there exist a partition $\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that*

$$\{t_i | i \in \{1, \dots, n\}\} = \{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}$$

Proof. Define $P = \{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}$ then as $\{t_i^{(1)} | i \in \{1, \dots, n_1\}\}$, $\{t_i^{(2)} | i \in \{1, \dots, n_2\}\}$ are finite [using [theorem: 6.46] we have by [theorem: 6.34] that P is finite. Hence there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, n\} \rightarrow P$ which defines a family $\{\beta(i)\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$. Using [theorem: 6.54] there exist a bijection $\alpha: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that $\forall i \in \{1, \dots, n-1\}$ we have $\beta(\alpha(i)) \leq \beta(\alpha(i+1))$. If $\beta(\alpha(i)) = \beta(\alpha(i+1)) \xRightarrow{\beta \text{ is injective}} \alpha(i) = \alpha(i+1) \xRightarrow{\alpha \text{ is injective}} i = i+1$ a contradiction, so we must have that $\beta(\alpha(i)) < \beta(\alpha(i+1))$. Hence if we define

$$\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R} \text{ by } t_i = (\beta \circ \alpha)(i) = \beta(\alpha(i))$$

Then

$$\forall i \in \{1, \dots, n-1\} \quad t_i < t_{i+1} \quad (15.3)$$

If $t \in \{t_i | i \in \{1, \dots, n\}\}$ then there exist a $i \in \{1, \dots, n\}$ such that $t = t_i = \beta(\alpha(i)) \in P$. Further if $x \in P$ then as $\beta \circ \alpha: \{1, \dots, n\} \rightarrow P$ is a bijection there exist a $i \in \{1, \dots, n\}$ such that $x = (\beta \circ \alpha)(i) = t_i \in \{t_i | i \in \{1, \dots, n\}\}$. Hence we have

$$\{t_i | i \in \{1, \dots, n\}\} = P = \{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\} \quad (15.4)$$

Using the above we have

$$\begin{aligned} & \min(\{t_i | i \in \{1, \dots, n\}\}) && \stackrel{[lemma: 15.3]}{=} t_1 \\ & \min(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}) && \stackrel{[eq: 15.4]}{=} \min(\{t_i | i \in \{1, \dots, n\}\}) \\ & \min(\min(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\}), \min(\{t_i^{(2)} | i \in \{1, \dots, n_2\}\})) && \stackrel{[theorem: 3.66]}{=} \min(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}) \\ & \min(\min(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\}), \min(\{t_i^{(2)} | i \in \{1, \dots, n_2\}\})) && \stackrel{[lemma: 15.3]}{=} \min(\{t_i | i \in \{1, \dots, n\}\}) \\ & \min(\{t_1^{(1)}, t_1^{(2)}\}) && = \\ & \min(\{a\}) && = \\ & a && \\ & \max(\{t_i | i \in \{1, \dots, n\}\}) && \stackrel{[lemma: 15.3]}{=} t_n \\ & \max(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}) && \stackrel{[eq: 15.4]}{=} \max(\{t_i | i \in \{1, \dots, n\}\}) \\ & \max(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}) && \stackrel{[theorem: 3.66]}{=} \max(\min(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\}), \min(\{t_i^{(2)} | i \in \{1, \dots, n_2\}\})) \\ & \max(\max(\{t_i^{(1)} | i \in \{1, \dots, n_1\}\}), \max(\{t_i^{(2)} | i \in \{1, \dots, n_2\}\})) && \stackrel{[lemma: 15.3]}{=} \max(\{t_i | i \in \{1, \dots, n\}\}) \\ & \max(\{t_{n_1}^{(1)}, t_{n_2}^{(2)}\}) && = \\ & \max(\{b\}) && = \end{aligned}$$

so that $a = t_1$ and $b = t_n$ which combined with [eqs: 15.3] proves that

$$\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R} \text{ is a partition of } [a, b] \quad \square$$

Lemma 15.7. *Let $a, b \in \mathbb{R}$ with $a < b$, $\mathcal{P}_1 = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}$, $\mathcal{P}_2 = \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}$ two partitions of $[a, b]$ then if $\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ and $Q = \{s_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$ are two partitions of $[a, b]$ such that*

$$\{t_i | i \in \{1, \dots, n\}\} = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\} = \{s_i | i \in \{1, \dots, m\}\}$$

then $\mathcal{P} = Q$.

Proof. Define the functions

$$s: \{1, \dots, n\} \rightarrow \{s_i | i \in \{1, \dots, n\}\} \text{ by } s(i) = s_i \text{ and } t: \{1, \dots, m\} \rightarrow \{t_i | i \in \{1, \dots, m\}\} \text{ by } t(i) = t_i$$

then these functions are bijective [injectivity follows from [see theorem: 15.4 (3)] and surjectivity from the definition of the functions]. As $\{s_i | i \in \{1, \dots, n\}\} = \{t_i | i \in \{1, \dots, m\}\}$ it follows that we have the bijection

$$t^{-1} \circ s: \{1, \dots, n\} \rightarrow \{1, \dots, m\}$$

from which it follows that

$$n = m$$

Next we use induction to prove that the above functions are equal. So define

$$S = \{k \in \mathbb{N} | \text{If } k \in \{1, \dots, n\} \text{ then } \forall i \in \{1, \dots, k\} \text{ we have } s_i = t_i\}$$

then we have:

$1 \in S$. This follows from the fact that $s_1 = a = t_1$.

$k \in S \Rightarrow k + 1 \in S$. Assume that $s_{k+1} \neq t_{k+1}$ then we have either:

$s_{k+1} < t_{k+1}$. As $\{s_i | i \in \{1, \dots, n\}\} = \{t_i | i \in \{1, \dots, n\}\}$ there exist a $l \in \{1, \dots, n\}$ such that

$$s_{k+1} = t_l \quad (15.5)$$

So that $t_l = s_{k+1} < t_{k+1}$ from which it follows that $l < k + 1$ [otherwise $t_{k+1} \leq t_l$], hence we have that $l \leq k$. As $k \in S$ we have that

$$s_l = t_l$$

Hence using [eq: 15.5] on the above it follows that $s_{k+1} = s_l$ from which it follows by [theorem: 15.4 (3)] that $k + 1 = l$ so that we have by [eq: 15.5] that $s_{k+1} = t_{k+1}$ contradicting $s_{k+1} < t_{k+1}$.

$t_{k+1} < s_{k+1}$. As $\{s_i | i \in \{1, \dots, n\}\} = \{t_i | i \in \{1, \dots, n\}\}$ there exist a $l \in \{1, \dots, n\}$ such that

$$t_{k+1} = s_l \quad (15.6)$$

So that $s_l < t_{k+1} < s_{k+1}$ from which it follows that $l < k + 1$ [otherwise $s_{k+1} \leq s_l$], hence we have that $l \leq k$. As $k \in S$ we have that

$$s_l = t_l$$

Hence using [eq: 15.6] it follows that $t_{k+1} = t_k$, using [theorem: 15.4 (3)] it follows that $k + 1 = l$ and using [eq: 15.6] this results in $s_{k+1} = t_{k+1}$ which contradicts $t_{k+1} < s_{k+1}$.

As we have a contradiction in both cases the assumption must be wrong, hence we have that $s_{k+1} = t_{k+1}$, proving that $k + 1 \in S$. $\square$

Combing the two above lemmas gives the following theorem.

Theorem 15.8. *Let $a, b \in \mathbb{R}$ with $a < b$ and $\mathcal{P}_1 = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}$, $\mathcal{P}_2 = \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}$ two partitions of $[a, b]$ then there exist a **unique** partition $\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that*

$$\{t_i | i \in \{1, \dots, n\}\} = \{t_i^{(1)} | i \in \{1, \dots, n_1\}\} \cup \{t_i^{(2)} | i \in \{1, \dots, n_2\}\}$$

This **unique** partition is noted as $\mathcal{P}_1 \square \mathcal{P}_2$

Proof. This follows from [lemma: 15.6] and [lemma: 15.7]. □

Consider the following two partitions of $[1, 12]$

$$\begin{aligned}\mathcal{P}_1 &= [1_1, 4_2] \cup [4_2, 8_3] \cup [8_3, 12_4], \text{ boundary points } \{1_1, 4_2, 8_3, 12_4\} \\ \mathcal{P}_2 &= [1_1, 2_2] \cup [2_2, 5_3] \cup [5_3, 6_4] \cup [6_4, 12_5], \text{ boundary points } \{1_1, 2_2, 5_3, 6_4, 12_5\}\end{aligned}$$

and the 'combined' partition

$$\begin{aligned}\mathcal{P}_1 \square \mathcal{P}_2 &= [1_1, 2_2] \cup [2_2, 4_3] \cup [4_3, 5_4] \cup [5_4, 6_5] \cup [6_5, 8_6] \cup [8_6, 12_7], \text{ boundary points} \\ &\quad \{1_1, 2_2, 4_3, 5_4, 6_5, 8_6, 12_7\} = \{1_1, 4_2, 8_3, 12_4\} \cup \{1_1, 2_2, 5_3, 6_4, 12_5\}\end{aligned}$$

Note that for every sub interval I in $\mathcal{P}_1 \square \mathcal{P}_2$ we can find a sub interval in $\mathcal{P}_1$ that contains I and a sub interval in $\mathcal{P}_2$ that contains I . Actually we can find a function $I_{\mathcal{P}_1}$ that maps the index of the first boundary point of I to the index of the first boundary point of the containing sub interval in $\mathcal{P}_1$ and a function $I_{\mathcal{P}_2}$ that map the index of the first boundary point of I to index of the first boundary point of the containing sub interval in $\mathcal{P}_2$.

For $\mathcal{P}_1 \square \mathcal{P}_2$ and $\mathcal{P}_1$ we have:

$$\begin{aligned}[1_1, 2_2], [2_2, 4_3] &\subseteq [1_1, 4_2] \Rightarrow I_{\mathcal{P}_1}(1) = 1 \\ &\quad I_{\mathcal{P}_1}(2) = 1 \\ [4_3, 5_4], [5_4, 6_5], [6_5, 8_6] &\subseteq [4_2, 8_3] \Rightarrow I_{\mathcal{P}_1}(3) = 2 \\ &\quad I_{\mathcal{P}_1}(4) = 2 \\ &\quad I_{\mathcal{P}_1}(5) = 2 \\ [8_6, 12_7] &\subseteq [8_3, 12_4] \Rightarrow I_{\mathcal{P}_1}(6) = 3\end{aligned}$$

For $\mathcal{P}_1 \square \mathcal{P}_2$ and $\mathcal{P}_2$ we have:

$$\begin{aligned}[1_1, 2_2] &\subseteq [1_1, 2_2] \Rightarrow I_{\mathcal{P}_2}(1) = 1 \\ [2_2, 4_3], [4_3, 5_4] &\subseteq [2_2, 5_3] \Rightarrow I_{\mathcal{P}_2}(2) = 2 \\ &\quad I_{\mathcal{P}_2}(3) = 2 \\ [5_4, 6_5] &\subseteq [5_3, 6_4] \Rightarrow I_{\mathcal{P}_2}(4) = 3 \\ [6_5, 8_6], [8_6, 12_7] &\subseteq [6_4, 12_5] \Rightarrow I_{\mathcal{P}_2}(5) = 4 \\ &\quad I_{\mathcal{P}_2}(6) = 4\end{aligned}$$

Hence we have that

$$\begin{aligned}I_{\mathcal{P}_1}: \{1, \dots, 7-1\} &\rightarrow \{1, \dots, 4-1\} \text{ is surjective} \\ I_{\mathcal{P}_2}: \{1, \dots, 7-1\} &\rightarrow \{1, \dots, 5-1\} \text{ is surjective}\end{aligned}$$

Further we have

$$\begin{aligned}I_{\mathcal{P}_1}^{-1}(\{1\}) &= \{1\} = \{1, \dots, 1\} \\ I_{\mathcal{P}_1}^{-1}(\{2\}) &= \{3, 4, 5\} = \{3, \dots, 5\} \\ I_{\mathcal{P}_1}^{-1}(\{3\}) &= \{6\} = \{6, \dots, 6\} \\ I_{\mathcal{P}_2}^{-1}(\{1\}) &= \{1\} = \{1, \dots, 1\} \\ I_{\mathcal{P}_2}^{-1}(\{2\}) &= \{2, 3\} = \{2, \dots, 3\} \\ I_{\mathcal{P}_2}^{-1}(\{3\}) &= \{4\} = \{4, \dots, 4\} \\ I_{\mathcal{P}_2}^{-1}(\{4\}) &= \{5, 6\} = \{5, \dots, 6\}\end{aligned}$$

so that

$$\begin{aligned} I_{\mathcal{P}_1}^{-1}(\{1\}), I_{\mathcal{P}_1}^{-1}(\{2\}), I_{\mathcal{P}_1}^{-1}(\{3\}) & \text{ are pairwise disjoint} \\ I_{\mathcal{P}_2}^{-1}(\{1\}), I_{\mathcal{P}_2}^{-1}(\{2\}), I_{\mathcal{P}_2}^{-1}(\{3\}), I_{\mathcal{P}_2}^{-1}(\{4\}), & \text{ are pairwise disjoint} \end{aligned}$$

The exact proof of the above in general is done in the following two lemmas.

Lemma 15.9. *Let $a, b \in \mathbb{R}$ with $a < b$ and $\mathcal{P}_1 = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}$, $\mathcal{P}_2 = \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}$ two partitions of $[a, b]$, let $\mathcal{P}_1 \sqcup \mathcal{P}_2 = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ then we have that*

*$\forall i \in \{1, \dots, n-1\}$ there exist a **unique** $i_{\mathcal{P}_1} \in \{1, \dots, n_1-1\}$ and a **unique** $i_{\mathcal{P}_2} \in \{1, \dots, n_2-1\}$ such that*

$$[t_i, t_{i+1}] \subseteq [t_{i_{\mathcal{P}_1}}^{(1)}, t_{i_{\mathcal{P}_1}+1}^{(1)}] \text{ and } [t_i, t_{i+1}] \subseteq [t_{i_{\mathcal{P}_2}}^{(2)}, t_{i_{\mathcal{P}_2}+1}^{(2)}]$$

This allows us to define two functions

$$\begin{aligned} I_{\mathcal{P}_1}: \{1, \dots, n-1\} & \rightarrow \{1, \dots, n_1\} \text{ by } I_{\mathcal{P}_1}(i) = i_{\mathcal{P}_1} \\ I_{\mathcal{P}_2}: \{1, \dots, n-1\} & \rightarrow \{1, \dots, n_2\} \text{ by } I_{\mathcal{P}_2}(i) = i_{\mathcal{P}_2} \end{aligned}$$

such that

$$[t_i, t_{i+1}] \subseteq [t_{I_{\mathcal{P}_1}(i)}^{(1)}, t_{I_{\mathcal{P}_1}(i)+1}^{(1)}] \text{ and } [t_i, t_{i+1}] \subseteq [t_{I_{\mathcal{P}_2}(i)}^{(2)}, t_{I_{\mathcal{P}_2}(i)+1}^{(2)}]$$

Proof. We prove this for $i_{\mathcal{P}_1}$ [the proof for $i_{\mathcal{P}_2}$ is exact the same as this proof by exchanging 1 with 2]. Let $i \in \{1, \dots, n-1\}$ then we have

$$t_i \neq t_n = b = t_n^{(1)} = t_n^{(2)} \text{ so that } t_i \neq t_n^{(1)} \text{ and } t_i \neq t_n^{(2)}$$

Define $B = \{j \in \{1, \dots, n_1\} | t_j^{(1)} \leq t_i\} \subseteq \{1, \dots, n_1\}$ then as $t_1^{(1)} = a \leq t_i$ we have $1 \in B$ so that $B \neq \emptyset$. As B is finite $m = \max(B) \in \{1, \dots, n_1\}$ exist. If $m = n_1$ then $b = t_{n_1}^{(1)} = t_m^{(1)} \leq t_i$ which as $t_i \neq b$ is a contradiction. Hence we must have that $m \neq n_1$ so that

$$m \in \{1, \dots, n_1-1\} \quad (15.7)$$

As $m = \max(B)$ we have $m+1 \notin B$ so that $t_i < t_{m+1}^{(1)}$ and, as $m \in B$, we have $t_m^{(1)} \leq t_i$ or

$$t_m^{(1)} \leq t_i < t_{m+1}^{(1)} \quad (15.8)$$

By [theorem: 15.8] there exist a $l \in \{1, \dots, n\}$ such that

$$t_{m+1}^{(1)} = t_l \quad (15.9)$$

Assume that $t_{m+1}^{(1)} < t_{i+1}$, then by the above and [eq: 15.8] we have $t_i < t_l < t_{i+1}$ so that $t_l \in]t_i, t_{i+1}[$ which is impossible by [theorem: 15.4 (5)]. Hence the assumption is wrong and we must have $t_{i+1} \leq t_{m+1}^{(1)}$. Combining this with [eq: 15.8] gives $t_m^{(1)} \leq t_i < t_{i+1} \leq t_{m+1}^{(1)}$ or $[t_i, t_{i+1}] \subseteq [t_m^{(1)}, t_{m+1}^{(1)}]$. So if we take $i_{\mathcal{P}_1} = m$ then we have

$$i_{\mathcal{P}_1} \in \{1, \dots, n_1-1\} \text{ and } [t_i, t_{i+1}] \subseteq [t_{i_{\mathcal{P}_1}}^{(1)}, t_{i_{\mathcal{P}_1}+1}^{(1)}]$$

This proves existence, next we prove uniqueness. So assume that there exist a $k \in \{1, \dots, n_1-1\}$ such that $[t_i, t_{i+1}] \subseteq [t_k^{(1)}, t_{k+1}^{(1)}]$. Assume that $k \neq i_{\mathcal{P}_1}$ then we have two cases to consider:

$k < i_{\mathcal{P}_1}$. Then $t_k^{(1)} < t_{i_{\mathcal{P}_1}}^{(1)} \leq t_i < t_{i+1} \leq t_{k+1}^{(1)}$ so that $t_{i_{\mathcal{P}_1}}^{(1)} \in]t_k^{(1)}, t_{k+1}^{(1)}[$ which is impossible by [theorem: 15.4 (5)].

$i_{\mathcal{P}_1} < k$. Then $t_{i_{\mathcal{P}_1}}^{(1)} < t_k^{(1)} \leq t_i < t_{i+1} \leq t_{i_{\mathcal{P}_1}+1}^{(1)}$ so that $t_k^{(1)} \in]t_{i_{\mathcal{P}_1}}^{(1)}, t_{i_{\mathcal{P}_1}+1}^{(1)}[$ which is impossible by [theorem: 15.4 (5)].

So as in all cases we have a contradiction the assumption must be wrong. Hence $k = i_{\mathcal{P}_1}$ which proves uniqueness. $\square$

Lemma 15.10. *Let $a, b \in \mathbb{R}$ with $a < b$ and $\mathcal{P}_1 = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}$, $\mathcal{P}_2 = \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}$ two partitions of $[a, b]$, let $\mathcal{P}_1 \sqcup \mathcal{P}_2 = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ then for the functions*

$$I_{\mathcal{P}_1}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_1-1\}$$

$$I_{\mathcal{P}_2}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_2-1\}$$

functions such that

$$\forall i \in \{1, \dots, n-1\} \text{ we have } [t_i, t_{i+1}] \subseteq [t_{I_{\mathcal{P}_1}(i)}^{(1)}, t_{I_{\mathcal{P}_1}(i)+1}^{(1)}] \text{ and } [t_{I_{\mathcal{P}_2}(i)}^{(2)}, t_{I_{\mathcal{P}_2}(i)+1}^{(2)}]$$

[who exist by [lemma: 15.9]] we have that:

1.

a. $I_{\mathcal{P}_1}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_1-1\}$ is surjective

b. $I_{\mathcal{P}_2}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_2-1\}$ is surjective

and $I_{\mathcal{P}_2}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_2-1\}$ are surjective.

2.

a. $\forall i \in \{1, \dots, n_1-1\}$ $I_{\mathcal{P}_1}^{-1}(\{i\}) = \{m_i^{(1)}, \dots, M_i^{(1)}\}$ where $t_{m_i^{(1)}} = t_i^{(1)}$ and $t_{M_i^{(1)}+1} = t_{i+1}^{(1)}$

b. $\forall i \in \{1, \dots, n_2-1\}$ $I_{\mathcal{P}_2}^{-1}(\{i\}) = \{m_i^{(2)}, \dots, M_i^{(2)}\}$ where $t_{m_i^{(2)}} = t_i^{(2)}$ and $t_{M_i^{(2)}+1} = t_{i+1}^{(2)}$

3.

a. $\forall i, j \in \{1, \dots, n_1-1\}$ with $i \neq j$ we have $I_{\mathcal{P}_1}^{-1}(\{i\}) \cap I_{\mathcal{P}_1}^{-1}(\{j\}) = \emptyset$

b. $\forall i, j \in \{1, \dots, n_2-1\}$ with $i \neq j$ we have $I_{\mathcal{P}_2}^{-1}(\{i\}) \cap I_{\mathcal{P}_2}^{-1}(\{j\}) = \emptyset$

Proof.

1.

a. Let $k \in \{1, \dots, n_1-1\}$ then for $t_k^{(1)}$ there exist by [theorem: 15.8] there exist a $i \in \{1, \dots, n\}$ such that $t_k^{(1)} = t_i$. As $k < n_1$ we have that $t_k^{(1)} < t_n^{(1)} = b$ so that $t_i \neq b$ proving that $i \neq n$ or

$$i \in \{1, \dots, n-1\} \text{ and } t_i = t_k^{(1)} \quad (15.10)$$

Assume that $t_{k+1}^{(1)} < t_{i+1}$. By [theorem: 15.8] there exist a $j \in \{1, \dots, n\}$ such that $t_j = t_{k+1}^{(1)}$. Then we have $t_i = t_k^{(1)} < t_{k+1}^{(1)} = t_j = t_{k+1}^{(1)} < t_{i+1}$ or $t_j \in]t_i, t_{i+1}[$, which is impossible by [theorem: 15.4 (5)], so that assumption must be wrong and we must have $t_{i+1} \leq t_{k+1}^{(1)}$, combining this with [eq: 15.10] results in $t_k^{(1)} = t_i < t_{i+1} \leq t_{k+1}^{(1)}$ or $[t_i, t_{i+1}] \subseteq [t_k^{(1)}, t_{k+1}^{(1)}]$. This proves that $k = I_{\mathcal{P}_1}(i)$ and thus surjectivity.

b. This is similar as the proof of (1.a) by replacing 1 by 2 in the proof.

2.

a. Let $i \in \{1, \dots, n_1-1\}$ then, as $I_{\mathcal{P}_1}$ is surjective we have that $\emptyset = I_{\mathcal{P}_1}^{-1}(\{i\}) \subseteq \{1, \dots, n-1\}$. Hence then minimum and maximum exist, take

$$m_i^{(1)} = \min(I_{\mathcal{P}_1}^{-1}(\{i\})) \text{ and } M_i^{(1)} = \max(I_{\mathcal{P}_1}^{-1}(\{i\})) \quad (15.11)$$

By the definition of min, max we have

$$I_{\mathcal{P}_1}^{-1}(\{i\}) \subseteq \{m_i^{(1)}, \dots, M_i^{(1)}\} \text{ and } m_i^{(1)}, M_i^{(1)} \in I_{\mathcal{P}_1}^{-1}(\{i\}) \quad (15.12)$$

If $k \in \{m_i^{(1)}, \dots, M_i^{(1)}\}$ then $m_i^{(1)} \leq k \leq M_i^{(1)} \Rightarrow k+1 \leq M_i^{(1)}$ so that

$$t_{m_i^{(1)}} \leq t_k \text{ and } t_{k+1} \leq t_{M_i^{(1)}+1} \quad (15.13)$$

As we have $m_i^{(1)}, M_i^{(1)} \in I_{\mathcal{P}_1}^{-1}(\{i\})$ it follows that $I_{\mathcal{P}_1}(m_i^{(1)}) = i = I_{\mathcal{P}_1}(M_i^{(1)})$ so that $[t_{m_i^{(1)}}, t_{m_i^{(1)}+1}] \subseteq [t_i^{(1)}, t_{i+1}^{(1)}]$ and $[t_{M_i^{(1)}}, t_{M_i^{(1)}+1}] \subseteq [t_i^{(1)}, t_{i+1}^{(1)}]$ so that

$$t_i^{(1)} \leq t_{m_i^{(1)}} \leq t_k < t_{k+1} \leq t_{M_i^{(1)}+1} \leq t_{i+1}^{(1)} \text{ and } t_i^{(1)} \leq t_{M_i^{(1)}} < t_{M_i^{(1)}+1} \leq t_{i+1}^{(1)} \quad (15.14)$$

or $[t_k, t_{k+1}] \subseteq [t_i^{(1)}, t_{i+1}^{(1)}]$, hence $I_{\mathcal{P}_1}(k) = i$ or $k \in I_{\mathcal{P}_1}^{-1}(\{i\})$. So we have proved that $\{m_i^{(1)}, \dots, M_i^{(1)}\} \subseteq I_{\mathcal{P}_1}^{-1}(\{i\})$, combining this with [eq: 15.12] gives

$$I_{\mathcal{P}_1}^{-1}(\{i\}) = \{m_i^{(1)}, \dots, M_i^{(1)}\} \quad (15.15)$$

Assume that $t_i^{(1)} < t_{m_i^{(1)}}$. By [theorem: 15.8] there exist a $l \in \{1, \dots, n\}$ such that

$$t_l = t_i^{(1)}$$

giving $t_l < t_{m_i^{(1)}}$, resulting in

$$l < m_i^{(1)}, \quad (15.16)$$

hence $l+1 < m_i^{(1)}+1$. From this it follows that

$$t_i^{(1)} = t_l < t_{l+1} < t_{m_i^{(1)}+1} \leq_{[\text{eq: 15.14}]} t_{i+1}^{(1)}$$

So that $[t_l, t_{l+1}] \subseteq [t_i^{(1)}, t_{i+1}^{(1)}]$ proving that $I_{\mathcal{P}_1}(l) = i$ or $l \in I_{\mathcal{P}_1}^{-1}(\{i\}) = \{m_i^{(1)}, \dots, M_i^{(1)}\}$, hence

$$m_i^{(1)} \leq l <_{[\text{eq: 15.16}]} m_i^{(1)}$$

a contradiction. From this we conclude that the assumption is wrong, so $t_{m_i^{(1)}} \leq t_i^{(1)}$, combining this with [eq: 15.14] proves that

$$t_{m_i^{(1)}} = t_i^{(1)} \quad (15.17)$$

Assume that $t_{M_i^{(1)}+1} < t_{i+1}^{(1)}$. By [theorem: 15.8] there exist a $k \in \{1, \dots, n\}$ such that

$$t_k = t_{i+1}^{(1)}$$

giving $t_{M_k^{(1)}+1} < t_k$ so that resulting in $M_k^{(1)}+1 < k$ or

$$M_k^{(1)} < k-1 \quad (15.18)$$

so that

$$t_i^{(1)} \leq_{[\text{eq: 15.14}]} t_{M_k^{(1)}} < t_{k-1} < t_k = t_{i+1}^{(1)}$$

So that $[t_{k-1}, t_k] \subseteq [t_i^{(1)}, t_{i+1}^{(1)}]$ proving that $I_{\mathcal{P}_1}(k-1) = i$ or $k-1 \in I_{\mathcal{P}_1}^{-1}(\{i\}) = \{m_i^{(1)}, \dots, M_i^{(1)}\}$ hence $k-1 \leq M_i^{(1)} <_{[\text{eq: 15.18}]} k-1$ a contradiction. Hence the assumption must be wrong and we have $t_{i+1}^{(1)} \leq t_{M_i^{(1)}+1}$, combining this with [eq: 15.14] gives

$$t_{M_i^{(1)}+1} = t_{i+1}^{(1)} \quad (15.19)$$

Finally (2.a) is proved by [eqs: 15.15, 15.17 and 15.19].

b. This is similar as the proof of (2.a) by replacing 1 by 2 in the proof.

3.

a. Let $i, j \in \{1, \dots, n_1-1\}$ with $i \neq j$. Assume that $k \in I_{\mathcal{P}_1}^{-1}(\{i\}) \cap I_{\mathcal{P}_1}^{-1}(\{j\})$ then $I_{\mathcal{P}_1}(\{k\}) = i$ and $I_{\mathcal{P}_1}(\{k\}) = j$ so that by the definition of a function we have $i = j$ a contradiction. Hence $I_{\mathcal{P}_1}^{-1}(\{i\}) \cap I_{\mathcal{P}_1}^{-1}(\{j\}) = \emptyset$.

b. This is similar as the proof of (3.a) by replacing 1 by 2 in the proof. $\square$

To be able to define a integral of a function on a closed interval we need a limiting process where the length of the intervals in the partitions become smaller and smaller so that more details of the function are used in the calculation of the integral. To quantify this we introduce a norm of a partition. Beware this norm is unrelated to norms on normed spaces.

Definition 15.11. Let $a, b \in \mathbb{R}$ with $a < b$ and $\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ a partition of $[a, b]$ then the **norm** of the partition noted as $\mu(\mathcal{P})$ is defined by

$$\mu(\mathcal{P}) = \max(\{|t_{i+1} - t_i| | i \in \{1, \dots, n-1\}\}) = \max_{t_i < t_{i+1}}(\{t_{i+1} - t_i | i \in \{1, \dots, n-1\}\})$$

Definition 15.12. Let $a, b \in \mathbb{R}$ with $a < b$ and $\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ a partition of $[a, b]$ then a **tag** on $\mathcal{P}$ is a family $\{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R}$ such that $\forall i \in \{1, \dots, n-1\}$ we have $s_i \in [t_i, t_{i+1}]$. A **tagged partition** on $[a, b]$ is a pair of a partition and a tag on the partition. So a tagged partition is a pair

$$\langle \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}, \{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R} \rangle$$

such that:

1. $n \in \mathbb{N} \setminus \{1\}$
2. $\forall i \in \{1, \dots, n-1\}$ we have $t_i < t_{i+1}$ and $s_i \in [t_i, t_{i+1}]$
3. $t_1 = a$ and $t_n = b$

The norm of a tagged partition is the norm of the partition of the tagged partition in other words if

$$\langle \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}, \{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R} \rangle$$

is a tagged partition then

$$\mu(\langle \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}, \{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R} \rangle) = \mu(\{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R})$$

Theorem 15.13. Let $a, b \in \mathbb{R}^+$ with $a < b$, $\varepsilon \in \mathbb{R}^+$ then there exist a tagged partition $\mathbb{P}$ on $[a, b]$ such that $\mu(\mathbb{P}) < \varepsilon$

Proof. Using [theorem: 10.30] there exist a $N \in \mathbb{N}$ such that $\frac{1}{N} < \frac{\varepsilon}{b-a}$ and take $\delta = \frac{b-a}{N}$. Then we have $\frac{b-a}{N} = (b-a) \cdot \frac{1}{N} < (b-a) \cdot \frac{\varepsilon}{b-a} = \varepsilon$ so that

$$\delta < \varepsilon \tag{15.20}$$

Define now

$$\{t_i\}_{i \in \{1, \dots, N+1\}} \text{ by } t_i = a + (i-1) \cdot \delta \text{ and } \{s_i\}_{i \in \{1, \dots, N+1\}} \text{ by } s_i = t_i$$

then $\forall i \in \{1, \dots, (N+1)-1\}$ we have

$$s_i \in [t_i, t_{i+1}]$$

and

$$t_i = a + (i-1) \cdot \delta < a + (i-1) \cdot \delta + \delta = a + ((i+1)-1) \cdot \delta = t_{i+1}$$

and

$$t_{i+1} - t_i = a + ((i+1)-1) \cdot \delta - (a + (i-1) \cdot \delta) = \delta < \varepsilon$$

so that

$$\mu(\{t_i\}_{i \in \{1, \dots, N+1\}}) = \max(\{t_{i+1} - t_i | i \in \{1, \dots, (N+1)-1\}\}) = \max(\{\delta\}) = \delta < \varepsilon$$

Further

$$t_1 = a + (1-1) \cdot \delta = a \text{ and } t_{N+1} = a + ((N+1)-1) \cdot \delta = a + N \cdot \delta = a + \frac{b-a}{N} \cdot N = b$$

Hence

$$\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, N+1\}}, \{s_i\}_{i \in \{1, \dots, N+1\}} \rangle$$

is a tagged partition on $[a, b]$ with $\mu(\mathbb{P}) < \varepsilon$. $\square$

We are now ready to define a Riemann sum that will be used to approximate the integral of a function.

Definition 15.14. Let $a, b \in \mathbb{R}$ with $a < b$, $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}, \{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R} \rangle$ a tagged partition on $[a, b]$, $\langle X, \|\cdot\| \rangle$ a normed space and $f: [a, b] \rightarrow X$ a function then the **Riemann sum of f using the tagged partition $\mathbb{P}$** noted as $\mathcal{S}(f, \mathbb{P})$ is defined by

$$\mathcal{S}(f, \mathbb{P}) = \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_i)$$

Theorem 15.15. Let $a, b \in \mathbb{R}$ with $a < b$, $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}, \{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R} \rangle$ a tagged partition on $[a, b]$, $\alpha \in \mathbb{K}$ and $f: [a, b] \rightarrow X$, $g: [a, b] \rightarrow X$ functions then

$$\mathcal{S}(\alpha \cdot f + g, \mathbb{P}) = \alpha \cdot \mathcal{S}(f, \mathbb{P}) + \mathcal{S}(g, \mathbb{P})$$

Proof. We have

$$\begin{aligned} \mathcal{S}(\alpha \cdot f + g, \mathbb{P}) &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (\alpha \cdot f + g)(s_i) \\ &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (\alpha \cdot f(s_i) + g(s_i)) \\ &= \sum_{i=1}^{n-1} (\alpha \cdot (t_{i+1} - t_i) \cdot f(s_i) + (t_{i+1} - t_i) \cdot g(s_i)) \\ &= \alpha \cdot \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_i) + \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot g(s_i) \\ &= \alpha \cdot \mathcal{S}(f, \mathbb{P}) + \mathcal{S}(g, \mathbb{P}) \end{aligned}$$

$\square$

Theorem 15.16. Let $a, b \in \mathbb{R}$ with $a < b$, $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}, \{s_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathbb{R} \rangle$ a tagged partition on $[a, b]$, $\langle \mathbb{R}, \|\cdot\| \rangle$ the normed space of real numbers and $f: [a, b] \rightarrow \mathbb{R}$, $g: [a, b] \rightarrow \mathbb{R}$ functions such that $\forall x \in [a, b]$ we have $f(x) \leq g(x)$ then $\mathcal{S}(f, \mathbb{P}) \leq \mathcal{S}(g, \mathbb{P})$

Proof. As $\forall i \in \{1, \dots, n\}$ we have $0 \leq t_{i+1} - t_i$, $f(s_i) \leq g(s_i)$ it follows that $f(s_i) \cdot (t_{i+1} - t_i) \leq g(s_i) \cdot (t_{i+1} - t_i)$ so that $\mathcal{S}(f, \mathbb{P}) = \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_i) = \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot g(s_i) = \mathcal{S}(g, \mathbb{P})$. $\square$

Lemma 15.17. Let $a, b \in \mathbb{R}$ with $a < b$,

$$\mathbb{P}_1 = \langle \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}, \{s_i^{(1)}\}_{i \in \{1, \dots, n_1-1\}} \rangle$$

$$\mathbb{P}_2 = \langle \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}, \{s_i^{(2)}\}_{i \in \{1, \dots, n_2-1\}} \rangle$$

tagged partitions on $[a, b]$, $\langle X, \|\cdot\| \rangle$ a normed space and

$$f: [a, b] \rightarrow X \text{ a function}$$

then for

$$\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \text{ defined by } \mathcal{P} = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \sqcup \{t_i^{(1)}\}_{i \in \{1, \dots, n_2\}}$$

we have:

1. $S(f, \mathbb{P}_1) = \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_{I_{\mathcal{P}_1}(i)}^{(1)})$
2. $S(f, \mathbb{P}_2) = \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_{I_{\mathcal{P}_2}(i)}^{(2)})$
3. $S(f, \mathbb{P}_1) - S(f, \mathbb{P}_2) = \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (f(s_{I_{\mathcal{P}_1}(i)}^{(1)}) - f(s_{I_{\mathcal{P}_2}(i)}^{(2)}))$

where

$$I_{\mathcal{P}_1}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_1-1\} \text{ and } I_{\mathcal{P}_2}: \{1, \dots, n-1\} \rightarrow \{1, \dots, n_2-1\}$$

are the functions defined in lemmas [lemma: 15.9] and [lemma: 15.10].

Proof.

1. As $\{1, \dots, n_1-1\} = \bigcup_{k \in \{1, \dots, n_1-1\}} \{k\}$ we have

$$\begin{aligned} \{1, \dots, n-1\} &= I_{\mathcal{P}_1}^{-1}(\{1, \dots, n_1-1\}) \\ &= I_{\mathcal{P}_1}^{-1}\left(\bigcup_{k \in \{1, \dots, n_1-1\}} \{k\}\right) \\ &\stackrel{[\text{theorem: 2.144}]}{=} \bigcup_{k \in \{1, \dots, n_1-1\}} I_{\mathcal{P}_1}^{-1}(\{k\}) \end{aligned}$$

and by [lemma: 15.10] we have

$$\forall i, j \in \{1, \dots, n_1-1\} \text{ with } i \neq j \text{ we have } I_{\mathcal{P}_1}^{-1}(\{i\}) \cap I_{\mathcal{P}_1}^{-1}(\{j\}) = \emptyset$$

allowing us to apply [theorem: 11.43] in the following.

$$\begin{aligned} \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_{I_{\mathcal{P}_1}(i)}^{(1)}) &= \sum_{i \in \{1, \dots, n-1\}} (t_{i+1} - t_i) \cdot f(s_{I_{\mathcal{P}_1}(i)}^{(1)}) \\ &= \sum_{i \in \bigcup_{k \in \{1, \dots, n_1-1\}} I_{\mathcal{P}_1}^{-1}(\{k\})} (t_{i+1} - t_i) \cdot f(s_{I_{\mathcal{P}_1}(i)}^{(1)}) \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{k=1}^{n_1-1} \left(\sum_{i \in I_{\mathcal{P}_1}^{-1}(\{k\})} (t_{i+1} - t_i) \cdot f(s_{I_{\mathcal{P}_1}(i)}^{(1)}) \right) \\ &= \sum_{k=1}^{n_1-1} \left(\sum_{i \in I_{\mathcal{P}_1}^{-1}(\{k\})} (t_{i+1} - t_i) \cdot f(s_k^{(1)}) \right) \\ &= \sum_{k=1}^{n_1-1} \left(\sum_{i \in I_{\mathcal{P}_1}^{-1}(\{k\})} (t_{i+1} - t_i) \right) \cdot f(s_k^{(1)}) \\ &\stackrel{[\text{lemma: 15.10}]}{=} \sum_{k=1}^{n_1-1} \left(\sum_{i \in \{m_k^{(1)}, \dots, M_k^{(1)}\}} (t_{i+1} - t_i) \right) \cdot f(s_k^{(1)}) \\ &= \sum_{k=1}^{n_1-1} \left(\sum_{i \in m_k^{(1)}}^{M_k^{(1)}} (t_{i+1} - t_i) \right) \cdot f(s_k^{(1)}) \\ &\stackrel{[\text{theorem: 11.21}]}{=} \sum_{k=1}^{n_1-1} (t_{M_k^{(1)}+1} - t_{m_k^{(1)}}) \cdot f(s_k^{(1)}) \\ &\stackrel{[\text{lemma: 15.10}]}{=} \sum_{k=1}^{n_1-1} (t_{k+1}^{(1)} - t_k^{(1)}) \cdot f(s_k^{(1)}) \\ &= S(f, \mathbb{P}_1) \end{aligned}$$

2. The proof is similar to the proof of (1) just replace 1 with 2.

3. We have that

$$\begin{aligned} S(f, \mathbb{P}_1) - S(f, \mathbb{P}_2) &\stackrel{(1,2)}{=} \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_{I_{\mathbb{P}_1}(i)}^{(1)}) - \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_{I_{\mathbb{P}_2}(i)}^{(2)}) \\ &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (f(s_{I_{\mathbb{P}_1}(i)}^{(1)}) - f(s_{I_{\mathbb{P}_2}(i)}^{(2)})) \end{aligned}$$

□

Lemma 15.18. *Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ a normed space and*

$$f: [a, b] \rightarrow X \text{ a continuous function}$$

then for every $\varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that for every two tagged partitions $\mathbb{P}_1, \mathbb{P}_2$ on $[a, b]$ with $\mu(\mathbb{P}_1) < \delta$ and $\mu(\mathbb{P}_2) < \delta$ we have $\|S(f, \mathbb{P}_1) - S(f, \mathbb{P}_2)\| < \varepsilon$

Proof. As $[a, b]$ is compact [see theorem: 14.279] and f is continuous we can use [theorem: 14.288] that f is uniform continuous on $[a, b]$. So there exist a $\delta' \in \mathbb{R}^+$ such that

$$\forall x, x' \in [a, b] \text{ with } |x - x'| < \delta' \text{ we have } \|f(x) - f(x')\| < \frac{\varepsilon}{b - a} \quad (15.21)$$

Take $\delta = \frac{\delta'}{2}$ and let

$$\begin{aligned} \mathbb{P}_1 &= \langle \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \subseteq \mathbb{R}, \{s_i^{(1)}\}_{i \in \{1, \dots, n_1-1\}} \rangle \\ \mathbb{P}_2 &= \langle \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}} \subseteq \mathbb{R}, \{s_i^{(2)}\}_{i \in \{1, \dots, n_2-1\}} \rangle \end{aligned}$$

be two tagged partitions on $[a, b]$ such that $\mu(\mathbb{P}_1) < \delta$ and $\mu(\mathbb{P}_2) < \delta$. Consider

$$\mathcal{P} = \{t_i\}_{i \in \{1, \dots, n\}} \text{ defined by } \mathcal{P} = \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}} \sqcup \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}}$$

then we have by [lemma: 15.17] that

$$\|S(f, \mathbb{P}_1) - S(f, \mathbb{P}_2)\| = \left\| \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (f(s_{I_{\mathbb{P}_1}(i)}^{(1)}) - f(s_{I_{\mathbb{P}_2}(i)}^{(2)})) \right\| \quad (15.22)$$

Let $i \in \{1, \dots, n-1\}$ then by [lemma: 15.9] $t_i \in [t_i, t_{i+1}] \subseteq [t_{I_{\mathbb{P}_1}(i)}^{(1)}, t_{I_{\mathbb{P}_1}(i)+1}^{(1)}] \cap [t_{I_{\mathbb{P}_2}(i)}^{(2)}, t_{I_{\mathbb{P}_2}(i)+1}^{(2)}]$ so that

$$t_{I_{\mathbb{P}_1}(i)}^{(1)} \leq t_i \leq t_{I_{\mathbb{P}_1}(i)+1}^{(1)} \text{ and } t_{I_{\mathbb{P}_2}(i)}^{(2)} \leq t_i \leq t_{I_{\mathbb{P}_2}(i)+1}^{(2)}$$

Further we have $s_{I_{\mathbb{P}_1}(i)}^{(1)} \in [t_{I_{\mathbb{P}_1}(i)}^{(1)}, t_{I_{\mathbb{P}_1}(i)+1}^{(1)}]$ and $s_{I_{\mathbb{P}_2}(i)}^{(2)} \in [t_{I_{\mathbb{P}_2}(i)}^{(2)}, t_{I_{\mathbb{P}_2}(i)+1}^{(2)}]$ so that

$$t_{I_{\mathbb{P}_1}(i)}^{(1)} \leq s_{I_{\mathbb{P}_1}(i)}^{(1)} \leq t_{I_{\mathbb{P}_1}(i)+1}^{(1)} \text{ and } t_{I_{\mathbb{P}_2}(i)}^{(2)} \leq s_{I_{\mathbb{P}_2}(i)}^{(2)} \leq t_{I_{\mathbb{P}_2}(i)+1}^{(2)}$$

So $s_{I_{\mathbb{P}_1}(i)}^{(1)} - t_i \leq t_{I_{\mathbb{P}_1}(i)+1}^{(1)} - t_{I_{\mathbb{P}_1}(i)}^{(1)} \leq \mu(\mathbb{P}_1) < \delta$ and $t_i - s_{I_{\mathbb{P}_2}(i)}^{(2)} \leq t_{I_{\mathbb{P}_1}(i)+1}^{(1)} - t_{I_{\mathbb{P}_1}(i)}^{(1)} \leq \mu(\mathbb{P}_1) < \delta$ proving that

$$|s_{I_{\mathbb{P}_1}(i)}^{(1)} - t_i| < \delta$$

Also we have $s_{I_{\mathbb{P}_2}(i)}^{(2)} - t_i \leq t_{I_{\mathbb{P}_2}(i)+1}^{(2)} - t_{I_{\mathbb{P}_2}(i)}^{(2)} \leq \mu(\mathbb{P}_2) < \delta$ and $t_i - s_{I_{\mathbb{P}_2}(i)}^{(2)} \leq t_{I_{\mathbb{P}_2}(i)+1}^{(2)} - t_{I_{\mathbb{P}_2}(i)}^{(2)} \leq \mu(\mathbb{P}_2) < \delta$ proving

$$|s_{I_{\mathbb{P}_2}(i)}^{(2)} - t_i| < \delta$$

So that

$$|s_{I_{\mathbb{P}_1}(i)}^{(1)} - s_{I_{\mathbb{P}_2}(i)}^{(2)}| \leq |s_{I_{\mathbb{P}_1}(i)}^{(1)} - t_i| + |t_i - s_{I_{\mathbb{P}_2}(i)}^{(2)}| < \delta + \delta = 2 \cdot \delta = \delta'$$

Hence by [eq: 15.21] it follows that

$$\|f(s_{I_{\mathbb{P}_1}(i)}^{(1)}) - f(s_{I_{\mathbb{P}_2}(i)}^{(2)})\| < \frac{\varepsilon}{b - a} \quad (15.23)$$

Now

$$\begin{aligned}
& \left\| \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (f(s_{I_{P_1}(i)}^{(1)}) - f(s_{I_{P_2}(i)}^{(2)})) \right\| && \leq \\
& \sum_{i=1}^{n-1} \left\| (t_{i+1} - t_i) \cdot (f(s_{I_{P_1}(i)}^{(1)}) - f(s_{I_{P_2}(i)}^{(2)})) \right\| && = \\
& \sum_{i=1}^{n-1} (|t_{i+1} - t_i| \cdot \| (f(s_{I_{P_1}(i)}^{(1)}) - f(s_{I_{P_2}(i)}^{(2)})) \|) && <_{[\text{eq: 15.23}]} \\
& \sum_{i=1}^{n-1} |t_{i+1} - t_i| \cdot \frac{\varepsilon}{b-a} && = \\
& \frac{\varepsilon}{b-a} \cdot \sum_{i=1}^{n-1} |t_{i+1} - t_i| && \stackrel{t_i < t_{i+1}}{=} \\
& \frac{\varepsilon}{b-a} \cdot \sum_{i=1}^{n-1} (t_{i+1} - t_i) && \stackrel{[\text{theorem: 11.21}]}{=} \\
& \frac{\varepsilon}{b-a} \cdot (t_n - t_1) && = \\
& \frac{\varepsilon}{b-a} \cdot (b-a) && = \\
& \varepsilon && =
\end{aligned}$$

□

Lemma 15.19. *Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ a normed space and $f: [a, b] \rightarrow X$ a function and $I_1, I_2 \in X$ satisfying:*

1. $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that for every tagged partition $\mathbb{P}$ on $[a, b]$ with $\mu(\mathbb{P}) < \delta$ we have $\|I_1 - \mathcal{S}(f, \mathbb{P})\| < \varepsilon$
2. $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that for every tagged partition $\mathbb{P}$ on $[a, b]$ with $\mu(\mathbb{P}) < \delta$ we have $\|I_2 - \mathcal{S}(f, \mathbb{P})\| < \varepsilon$

then we have that $I_1 = I_2$.

Proof. Assume that $I_1 \neq I_2$ then we have $\varepsilon = \|I_1 - I_2\| \in \mathbb{R}^+$. By (1),(2) there exist $\delta_1, \delta_2 \in \mathbb{R}^+$ such that for every tagged partition $\mathbb{P}_1$ with $\mu(\mathbb{P}_1) < \delta_1$ and for every tagged partition $\mathbb{P}_2$ with $\mu(\mathbb{P}_2) < \delta_2$ we have $\|I_1 - \mathcal{S}(f, \mathbb{P}_1)\| < \frac{\varepsilon}{2}$ and $\|I_2 - \mathcal{S}(f, \mathbb{P}_2)\| < \frac{\varepsilon}{2}$. Using [theorem: 15.13] there exist a partition $\mathbb{P}$ such that $\mu(\mathbb{P}) < \min(\delta_1, \delta_2)$, so that we have

$$\|I_1 - \mathcal{S}(f, \mathbb{P})\| < \frac{\varepsilon}{2} \text{ and } \|I_2 - \mathcal{S}(f, \mathbb{P})\| < \frac{\varepsilon}{2}$$

so that

$$\varepsilon = \|I_1 - I_2\| \leq \|I_1 - \mathcal{S}(f, \mathbb{P})\| + \|\mathcal{S}(f, \mathbb{P}) - I_2\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

which is a contradiction. Hence the assumption that $I_1 \neq I_2$ is wrong and we must have $I_1 = I_2$. □

We are now ready for the definition of a Riemann Integrable function and the Riemann Integral of this function.

Definition 15.20. *Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ then a function $f: [a, b] \rightarrow X$ is **Riemann Integrable** if there exist a $I \in X$ such that $\forall \varepsilon \in \mathbb{R}$ there exist a $\delta \in \mathbb{R}^+$ such that for every tagged partition $\mathbb{P}$ on $[a, b]$ with $\mu(\mathbb{P}) < \varepsilon$ we have $\|I - \mathcal{S}(f, \mathbb{P})\| < \varepsilon$. If f is Riemann Integrable then by [lemma: 15.19] I is unique. We call this I [if it exist] the **Riemann integral** of f and not this as $\int_a^b f$. The set of graphs of Riemann Integrable function is noted by $\mathcal{R}[[a, b], X]$ in other words:*

$$\mathcal{R}([a, b], X) = \{f \in X^{[a, b]} \mid f: [a, b] \rightarrow X \text{ is Riemann Integrable}\}$$

The integral $\int_a^b$ is then a function from $\mathcal{R}[[a, b], X]$ to X defined by

$$\int_a^b : \mathcal{R}[[a, b], X] \rightarrow X \text{ where } \left(\int_a^b \right)(f) = \int_a^b f$$

Note 15.21. Another notation that you find in the literature is $\int_a^b f(x)dx$ instead of $\int_a^b f$. One benefit of this notation is if there is an expression to calculate the value of a function based on an expression. For example instead of using

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ is defined by } f(x) = \frac{\cos(x) + 1}{x + 1}$$

and then referring to the integral using

$$\int_a^b f$$

we can simply use

$$\int_a^b \frac{\cos(x) + 1}{x + 1} dx$$

Also if we have a function $f: [a, b] \times Y \rightarrow \mathbb{R}$ by $f(t, x) = \cos(t) \cdot x$ and then defining for a $x \in Y$

$$g_x: [a, b] \rightarrow \mathbb{R} \text{ by } g_x(t) = f(t, x) = \cos(t) \cdot x$$

so that we can talk about $\int_a^b g_x$ we can simplify this by

$$\int_a^b \cos(t) \cdot x dt$$

Example 15.22. Let $a, b \in \mathbb{R}$ such that $a < b$, $\langle X, \|\cdot\| \rangle$ a normed space, $x \in X$ then

$$C_x: [a, b] \rightarrow X \text{ defined by } C_x(i) = x$$

is Riemann Integrable and

$$\int_a^b C_x = (b - a) \cdot x$$

Proof. Let $\varepsilon \in \mathbb{R}^+$ and let $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}}, \{s_i\}_{i \in \{1, \dots, n-1\}} \rangle$ be any tagged partition on $[a, b]$ with $\mu(\mathbb{P}) < 1$ then we have:

$$\begin{aligned} \mathcal{S}(f, \mathbb{P}) &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot C_x(s_i) \\ &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot x \\ &= \left(\sum_{i=1}^{n-1} (t_{i+1} - t_i) \right) \cdot x \\ &\stackrel{[\text{theorem: 11.10}]}{=} (t_n - t_1) \cdot x \\ &= (b - a) \cdot x \end{aligned}$$

so that

$$\|S(f, \mathbb{P}) - (b - a) \cdot x\| = 0 < \varepsilon$$

Hence C_x is Riemann Integrable and $\int_a^b C_x = (b - a) \cdot x$

□

Lemma 15.23. Let $a, b \in \mathbb{R}$ with $a < b$ then there exist a sequence $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ of tagged partitions on $[a, b]$ such that

$$\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$$

Proof. Let $k \in \mathbb{N}$ and define

$$\{t^{(k)}\}_{i \in \{1, \dots, k+1\}} \text{ by } t_i^{(k)} = a + (i-1) \cdot \frac{b-a}{k} \text{ and } \{s_i^{(k)}\}_{i \in \{1, \dots, k\}} \text{ by } s_i^{(k)} = t_i^{(k)}$$

allowing us to define

$$\mathbb{P} = \langle \{t^{(k)}\}_{i \in \{1, \dots, k+1\}}, \{s_i^{(k)}\}_{i \in \{1, \dots, k\}} \rangle$$

Let $i \in \{1, \dots, (k+1) - 1\}$ then we have

$$t_i^{(k)} = a + (i-1) \cdot \frac{b-a}{k} < a + (i-1) \cdot \frac{b-a}{k} + \frac{b-a}{k} = a + ((i+1) - i) \cdot \frac{b-a}{k} = t_{i+1}^{(k)}$$

and

$$t_{i+1}^{(k)} - t_i^{(k)} = a + ((i+1) - i) \cdot \frac{b-a}{k} - \left(a + (i-1) \cdot \frac{b-a}{k} \right) = \frac{b-a}{k}$$

and

$$s_i^{(k)} \subseteq [t_{i+1}^{(k)} - t_i^{(k)}]$$

So that

$$\mu(\{t^{(k)}\}_{i \in \{1, \dots, k+1\}}) = \max(\{t_{i+1}^{(k)} - t_i^{(k)} | i \in \{1, \dots, (k+1) - 1\}\}) = \max\left(\left\{\frac{b-a}{k}\right\}\right) = \frac{b-a}{k}$$

Hence $\forall k \in \mathbb{N}$ we have that $\mathbb{P}_k$ is a tagged partition on $[a, b]$ and

$$\lim_{k \rightarrow \infty} \mu(\mathbb{P}_k) = \lim_{k \rightarrow \infty} \frac{b-a}{k} \stackrel{[\text{theorem: 14.358}]}{=} (b-a) \cdot \lim_{k \rightarrow \infty} \frac{1}{k} \stackrel{[\text{example: 14.349}]}{=} 0 \quad \square$$

Theorem 15.24. Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ a normed space and $f: [a, b] \rightarrow X$ a **continuous** function then the following are equivalent:

1. f is Riemann Integrable with integral $\int_a^b f$
2. There exist a $I \in X$ such that for every sequence of tagged partitions $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ with

$$\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$$

we have

$$\lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) = I$$

Furthermore we have $I = \int_a^b f$

Proof.

- 1 $\Rightarrow$ 2.** Let $\varepsilon \in \mathbb{R}^+$ then as f is Riemann Integrable with integral $\int_a^b f$ there exist a $\delta \in \mathbb{R}^+$ such that for any tagged partition on $[a, b]$ with $\mu(\mathbb{P}) < \delta$ we have $\|\mathcal{S}(f, \mathbb{P}) - \int_a^b f\| < \varepsilon$. Let $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ be a family of tagged partitions on $[a, b]$ such that $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$. Then there exist a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have $\mu(\mathbb{P}_n) = |\mu(\mathbb{P}_n) - 0| < \delta$ so that $\|\mathcal{S}(f, \mathbb{P}_n) - \int_a^b f\| < \varepsilon$ which proves that

$$\lim_{n \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_n) = \int_a^b f$$

- 2 $\rightarrow$ 1.** Using [lemma: 15.231] there exists a sequence $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ of tagged partitions on $[a, b]$ such that $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$. Then by the hypothesis we have that there exist a $I \in X$ such that

$$\lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) = I$$

Let $\varepsilon \in \mathbb{R}^+$ then there exist a $N_1 \in \mathbb{N}$ such that

$$\forall n \geq N_1 \text{ we have } \|\mathcal{S}(f, \mathbb{P}_n) - I\| < \frac{\varepsilon}{2}$$

Using [lemma: 15.18] there exist a $\delta \in \mathbb{R}^+$ such that for every two partitions $\mathbb{P}', \mathbb{P}''$ on $[a, b]$ with $\mu(\mathbb{P}') < \delta$ and $\mu(\mathbb{P}'') < \delta$ we have

$$\|\mathcal{S}(f, \mathbb{P}') - \mathcal{S}(f, \mathbb{P}'')\| < \frac{\varepsilon}{2} \quad (15.24)$$

As $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$ there exist a $N_2 \in \mathbb{N}$ such that $\forall n \geq N_2$ we have that

$$\mu(\mathbb{P}_n) = |\mu(\mathbb{P}_n) - 0| < \delta \quad (15.25)$$

Take $N = \max(N_1, N_2)$. Let $\mathbb{P}$ be a partition on $[a, b]$ with $\mu(\mathbb{P}) < \delta$ then

$$\begin{aligned} \|\mathcal{S}(f, \mathbb{P}) - I\| &= \|\mathcal{S}(f, \mathbb{P}) - \mathcal{S}(f, \mathbb{P}_N) + \mathcal{S}(f, \mathbb{P}_N) - I\| \\ &\leq \|\mathcal{S}(f, \mathbb{P}) - \mathcal{S}(f, \mathbb{P}_N)\| + \|\mathcal{S}(f, \mathbb{P}_N) - I\| \\ &<_{N > N_2 \wedge [\text{eqs: 15.24, 15.25}]} \frac{\varepsilon}{2} + \|\mathcal{S}(f, \mathbb{P}_N) - I\| \\ &<_{N > N_1} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

proving that f is Riemann Integrable and by uniqueness [see lemma: 15.19] that $\int_a^b f = I$. $\square$

We show now that every continuous function from $[a, b]$ to a Banach space is Riemann Integrable.

Theorem 15.25. *Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ a Banach space and $f: [a, b] \rightarrow X$ a continuous function then f is Riemann Integrable. In other words we have $\mathcal{C}([a, b], X) \subseteq \mathcal{R}([a, b], X)$.*

Proof. Let $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ be a sequence of tagged partitions on $[a, b]$ with $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$. Let $\varepsilon \in \mathbb{R}^+$ then by [lemma: 15.18] there exist a $\delta \in \mathbb{R}^+$ such that for any two tagged partitions $\mathbb{P}, \mathbb{P}'$ with $\mu(\mathbb{P}) < \delta$ and $\mu(\mathbb{P}') < \delta$ we have

$$\|\mathcal{S}(f, \mathbb{P}) - \mathcal{S}(f, \mathbb{P}')\| < \varepsilon \quad (15.26)$$

As $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$ there exist a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have $\mu(\mathbb{P}_n) = |\mu(\mathbb{P}_n) - 0| < \delta$. So if $n, m \geq N$ we have $\mu(\mathbb{P}_n) < \delta$ and $\mu(\mathbb{P}_m) < \delta$, so that by [eq: 15.26] $\|\mathcal{S}(f, \mathbb{P}_n) - \mathcal{S}(f, \mathbb{P}_m)\| < \varepsilon$. This proves that $\{\mathcal{S}(f, \mathbb{P}_i)\}_{i \in \mathbb{N}}$ is a Cauchy sequence in X . As X is a Banach space

$$\lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) \text{ exist}$$

Hence by [theorem: 15.24] it follows that f is Riemann Integrable. $\square$

Theorem 15.26. *Let $a, b \in \mathbb{R}$ with $a < b$, $\langle \mathbb{R}, \|\cdot\| \rangle$ the normed space of the real numbers and $f, g \in \mathcal{C}([a, b], \mathbb{R})$ [so that by [theorem: 15.25] f, g are Riemann Integrable] such that $\forall x \in [a, b]$ we have $f(x) \leq g(x)$*

then we have

$$\int_a^b f \leq \int_a^b g$$

Proof. Using [lemma: 15.231] there exists $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ such that $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$. Using [theorem: 15.16] we have

$$\forall i \in \mathbb{N} \quad \mathcal{S}(f, \mathbb{P}_i) \leq \mathcal{S}(g, \mathbb{P}_i) \quad (15.27)$$

Further we have using [theorem: 15.24] that

$$\begin{aligned} \int_a^b f &\stackrel{[\text{theorem: 15.24}]}{=} \lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) \\ &\leq_{[\text{eq: 15.27} \text{ and } [\text{theorem: 14.369}]]} \lim_{i \rightarrow \infty} \mathcal{S}(g, \mathbb{P}_i) \\ &\stackrel{[\text{theorem: 15.24}]}{=} \int_a^b g \end{aligned}$$

$\square$

Up to now we have always assumed that $a < b$ in the definition of the Riemann integral, we remove now this restriction by extending the definition of the Riemann integral.

Definition 15.27. Let $a, b \in \mathbb{R}$ with $a \leq b$, $\langle X, ||| \rangle$ a normed space and $f: [a, b] \rightarrow X$ is Riemann Integrable with integral $\int_a^b f$ if

1. If $a < b$ then we use [definition: 15.20]
2. If $a = b$ then f is Riemann Integrable and $\int_a^b f = 0$

The set of Riemann Integrable function graphs between is noted again by $\mathcal{R}([a, b], X)$ so that

$$\mathcal{R}([a, b], X) = \begin{cases} X^{[a, b]} & \text{if } a = b \\ \{f \in X^{[a, b]} \mid f: [a, b] \rightarrow X \text{ is Riemann Integrable as defined in [definition: 15.20]}\} & \text{if } a < b \end{cases}$$

We show now that $\int_a^b$ is a linear mapping.

Theorem 15.28. Let $a, b \in \mathbb{R}$ with $a \leq b$, $\langle X, ||| \rangle$ a normed space then

$$\int_a^b : \mathcal{R}([a, b], X) \rightarrow X \text{ is a linear mapping}$$

In other words If $\alpha \in \mathbb{K}$ and $f, g \in \mathcal{R}([a, b], X)$ then we have that $\alpha \cdot f + g \in \mathcal{R}([a, b], X)$ and

$$\int_a^b (\alpha \cdot f + g) = \alpha \cdot \int_a^b f + \int_a^b g$$

Proof. We have to consider the following cases for a, b .

$a = b$. Then by definition $\alpha \cdot f + g \in \mathcal{R}([a, b], X)$ and

$$\int_a^b (\alpha \cdot f + g) = 0 = 0 + 0 = \alpha \cdot \int_a^b f + \int_a^b g$$

$a < b$. Let $\varepsilon \in \mathbb{R}^+$ then as f, g are integrable there exists $\delta_1, \delta_2 \in \mathbb{R}^+$ such that for every tagged partition $\mathbb{P}'$ on $[a, b]$ with $\mu(\mathbb{P}') < \delta_1$ we have

$$\left\| S(f, \mathbb{P}') - \int_a^b f \right\| < \frac{\varepsilon}{2 \cdot (1 + |\alpha|)} \quad (15.28)$$

and for every tagged partition $\mathbb{P}''$ on $[a, b]$ with $\mu(\mathbb{P}'') < \delta_2$ we have

$$\left\| S(g, \mathbb{P}'') - \int_a^b g \right\| < \frac{\varepsilon}{2} \quad (15.29)$$

Let now $\mathbb{P}$ be a tagged partition on $[a, b]$ such that $\mu(\mathbb{P}) < \min(\delta_1, \delta_2)$ then we have

$$\begin{aligned} & \left\| S(\alpha \cdot f + g, \mathbb{P}) - \alpha \cdot \int_a^b f - \int_a^b g \right\| \stackrel{[\text{theorem: 15.15}]}{=} \\ & \left\| \alpha \cdot S(f, \mathbb{P}) + S(g, \mathbb{P}) - \alpha \cdot \int_a^b f - \int_a^b g \right\| \leq \\ & \left\| \alpha \cdot S(f, \mathbb{P}) - \alpha \cdot \int_a^b f \right\| + \left\| S(g, \mathbb{P}) - \int_a^b g \right\| = \\ & |\alpha| \cdot \left\| S(f, \mathbb{P}) - \int_a^b f \right\| + \left\| S(g, \mathbb{P}) - \int_a^b g \right\| <_{[\text{eqs: 15.28, 15.29}]} \\ & |\alpha| \cdot \frac{\varepsilon}{2 \cdot (1 + |\alpha|)} + \frac{\varepsilon}{2} < \\ & \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

Which proves that $\alpha \cdot f + g$ is Riemann integrable and that

$$\int_a^b (\alpha \cdot f + g) = \alpha \cdot \int_a^b f + \int_a^b g \quad \square$$

Theorem 15.29. *Let $a, b \in \mathbb{R}$ with $a \leq b$, $\langle X, \|\cdot\| \rangle$ a Banach space and $f: [a, b] \rightarrow X$ a function then we have:*

1. *If $f \in \mathcal{R}([a, b], X)$ then $\forall L \in L(X, \mathbb{R})$ we have that $L \circ f$ is integrable and $\int_a^b (L \circ f) = L(\int_a^b f)$*
2. *If f is continuous [so that by [theorem: 15.25] f is Riemann integrable] then $\|f\|: [a, b] \rightarrow \mathbb{R}$ defined by $\|f\|(x) = \|f(x)\|$ is Riemann integrable and $\|\int_a^b f\| \leq \int_a^b \|f\|$*
3. *If $x \in X$ then $C_x: [a, b] \rightarrow X$ by $C_x(i) = x$ is Riemann integrable and $\int_a^b C_x = (b - a) \cdot x$*
4. *If f is continuous [hence by [theorem: 15.25] Riemann integrable] and $m \in \mathbb{R}$ such that $\forall x \in [a, b]$ we have $\|f(x)\| \leq m$ then*

$$\left\| \int_a^b f \right\| \leq m \cdot (b - a)$$

Proof. We have to look at the following cases for a, b :

$a = b$. Then we have:

1. By definition $L \circ f$ is Riemann integrable and

$$\int_a^b (L \circ f) = 0 = L(0) = L\left(\int_a^b f\right)$$

2. By definition $\|f\|$ is Riemann integrable and

$$\int_a^b \|f\| = 0 = \|0\| = \left\| \int_a^b f \right\|$$

3. By definition C_x is Riemann integrable and

$$\int_a^b C_x = 0 = 0 \cdot x = (b - a) \cdot x$$

4. By definition f is Riemann integrable and

$$\left\| \int_a^b f \right\| = \|0\| = m \cdot 0 = m \cdot (b - a)$$

$a < b$.

1. Let $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}}, \{s_i\}_{i \in \{1, \dots, n-1\}} \rangle$ be any tagged partition on $[a, b]$ then we have:

$$\begin{aligned} \mathcal{S}(L \circ f, \mathbb{P}) &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot (L \circ f)(s_i) \\ &= \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot L(f(s_i)) \\ &= L\left(\sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_i)\right) \\ &= L(\mathcal{S}(f, \mathbb{P})) \end{aligned} \quad (15.30)$$

Let $\varepsilon \in \mathbb{R}^+$. As $L \in L(X, \mathbb{R})$ L is continuous, hence there exist a $\delta' \in \mathbb{R}^+$ such that if $\|x\| = \|x - 0\| < \delta'$ then

$$|L(x)| < \varepsilon \quad (15.31)$$

As f is Riemann integrable there exist a $\delta \in \mathbb{R}^+$ such that for every tagged partition $\mathbb{P}$ on $[a, b]$ with $\mu(\mathbb{P}) < \delta$ we have $\|\mathcal{S}(f, \mathbb{P}) - \int_a^b f\| < \delta'$. So

$$\begin{aligned} \left| \mathcal{S}(L \circ f, \mathbb{P}) - L\left(\int_a^b f\right) \right| &\stackrel{[\text{eq: 15.30}]}{=} \left| L(\mathcal{S}(f, \mathbb{P})) - L\left(\int_a^b f\right) \right| \\ &= \left| L\left(\mathcal{S}(f, \mathbb{P}) - \int_a^b f\right) \right| \\ &\stackrel{[\text{eq: 15.31}]}{<} \varepsilon \end{aligned}$$

proving that $L \circ f$ is Riemann integrable and that

$$\int_a^b (L \circ f) = L\left(\int_a^b f\right)$$

2. By [example: 14.173] $\|\cdot\|: X \rightarrow \mathbb{R}$ is continuous, hence as $\|f\| = \|\cdot\| \circ f$, it follows from [theorem: 14.158] it follows that

$$\|f\|: [a, b] \rightarrow \mathbb{R} \text{ is continue}$$

Hence using [theorem: 15.25] it follows that

$$\|f\| \text{ is Riemann integrable}$$

Let $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}}, \{s_i\}_{i \in \{1, \dots, n-1\}} \rangle$ be any tagged partition on $[a, b]$ then we have that

$$\begin{aligned} \|\mathcal{S}(f, \mathbb{P})\| &= \left\| \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_i) \right\| \\ &\leq \sum_{i=1}^{n-1} \|(t_{i+1} - t_i) \cdot f(s_i)\| \\ &= \sum_{i=1}^{n-1} |t_{i+1} - t_i| \cdot \|f(s_i)\| \\ &\stackrel{t_i < t_{i+1}}{=} \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot \|f(s_i)\| \\ &= \mathcal{S}(\|f\|, \mathbb{P}) \end{aligned} \tag{15.32}$$

Let $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ be a sequence of tagged partitions on $[a, b]$ such that $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$. As $\|f\|$ is continuous and Riemann integrable it follows from [theorem: 15.24] that

$$\lim_{i \rightarrow \infty} \mathcal{S}(\|f\|, \mathbb{P}_i) \text{ exist and } \lim_{i \rightarrow \infty} \mathcal{S}(\|f\|, \mathbb{P}_i) = \int_a^b \|f\| \tag{15.33}$$

As also f is continuous and Riemann integrable it follows from [theorem: 15.24] that

$$\lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) \text{ exist and } \lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) = \int_a^b f \tag{15.34}$$

By [eq: 15.32] we have that $\forall i \in \mathbb{N}$ that $\|\mathcal{S}(f, \mathbb{P}_i)\| \leq \mathcal{S}(\|f\|, \mathbb{P}_i)$ so that we can apply [theorem: 14.365] together with [eqs: 15.33, 15.34] gives

$$\left\| \int_a^b f \right\| = \left\| \lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) \right\| \leq \lim_{i \rightarrow \infty} \mathcal{S}(\|f\|, \mathbb{P}_i) = \int_a^b \|f\|$$

3. This follows from [example: 15.22].

4. Let $\mathbb{P} = \langle \{t_i\}_{i \in \{1, \dots, n\}}, \{s_i\}_{i \in \{1, \dots, n-1\}} \rangle$ be any tagged partition on $[a, b]$. then we have:

$$\begin{aligned}
 \|\mathcal{S}(f, \mathbb{P})\| &= \left\| \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot f(s_i) \right\| \\
 &\leq \sum_{i=1}^{n-1} \|(t_{i+1} - t_i) \cdot f(s_i)\| \\
 &= \sum_{i=1}^{n-1} |t_{i+1} - t_i| \cdot \|f(s_i)\| \\
 &\stackrel{t_i < t_{i+1}}{=} \sum_{i=1}^{n-1} (t_{i+1} - t_i) \cdot \|f(s_i)\| \\
 &\leq \sum_{i=1}^{n-1} ((t_{i+1} - t_i) \cdot m) \\
 &= \left(\sum_{i=1}^{n-1} (t_{i+1} - t_i) \right) \cdot m \\
 &\stackrel{[\text{theorem: 11.10}]}{=} (t_n - t_1) \cdot m \\
 &= (b - a) \cdot m
 \end{aligned}$$

proving that

$$\|\mathcal{S}(f, \mathbb{P})\| \leq (b - a) \cdot m \quad (15.35)$$

Let $\{\mathbb{P}_i\}_{i \in \mathbb{N}}$ be a sequence of tagged partitions on $[a, b]$ with $\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i) = 0$ then as f is continuous we have that

$$\lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) = \int_a^b f$$

exist. Using the above together with [eq: 15.35] allows us to apply [theorem: 14.363] giving

$$\left\| \int_a^b f \right\| = \left\| \lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i) \right\| \leq (b - a) \cdot m \quad \square$$

Lemma 15.30. Let $a, b \in \mathbb{R}$ with $a \leq b$, $c \in]a, b[$, $\mathbb{P}_1 = \langle \{t_i^{(1)}\}_{i \in \{1, \dots, n_1\}}, \{s_i^{(1)}\}_{i \in \{1, \dots, n_1-1\}} \rangle$ a tagged partition on $[a, c]$ and $\mathbb{P}_2 = \langle \{t_i^{(2)}\}_{i \in \{1, \dots, n_2\}}, \{s_i^{(2)}\}_{i \in \{1, \dots, n_2-1\}} \rangle$ a tagged partition on $[c, b]$ then for $\mathbb{P}_1 \sqcup \mathbb{P}_2$ defined by

$$\mathbb{P}_1 \sqcup \mathbb{P}_2 = \langle \{t_i\}_{i \in \{1, \dots, n_1+n_2-1\}}, \{s_i\}_{i \in \{1, \dots, n_1+n_2-2\}} \rangle$$

where

$$\forall i \in \{1, \dots, n_1+n_2-1\} \text{ we have } t_i = \begin{cases} t_i^{(1)} & \text{if } i \in \{1, \dots, n_1-1\} \\ t_{i-n_1+1}^{(2)} & \text{if } i \in \{n_1, \dots, n_1+n_2-1\} \end{cases}$$

and

$$\forall i \in \{1, \dots, n_1+n_2\} \text{ we have } s_i = \begin{cases} s_i^{(1)} & \text{if } i \in \{1, \dots, n_1-1\} \\ s_{i-n_1+1}^{(2)} & \text{if } i \in \{n_1, \dots, n_1+n_2-2\} \end{cases}$$

then we have:

1. $\mathbb{P}_1 \sqcup \mathbb{P}_2$ is a tagged partition on $[a, b]$
2. $\mu(\mathbb{P}_1 \sqcup \mathbb{P}_2) = \max(\mu(\mathbb{P}_1), \mu(\mathbb{P}_2))$
3. If $\langle X, \|\cdot\| \rangle$ is a normed space and $f: [a, b] \rightarrow X$ a function then

$$\mathcal{S}(f, \mathbb{P}_1 \sqcup \mathbb{P}_2) = \mathcal{S}[f|_{[a, c]}, \mathbb{P}_1] + \mathcal{S}[f|_{[c, b]}, \mathbb{P}_2]$$

Proof.

1. First we have that

$$t_1 = t_1^{(1)} = a \quad (15.36)$$

Second, using [definition: 15.12] $1 < n_2 \Rightarrow 0 \leq n_2 - 1$ we have $n_1 = n_1 \leq n_1 + n_2 - 1$, so that

$$t_{n_1+n_2-1} = t_{(n_1+n_2-1)-n_1+1}^{(2)} = t_{n_2}^{(2)} = b \quad (15.37)$$

Third for $i \in \{1, \dots, n_1 + n_2 - 2\}$ we have either:

$i \in \{1, \dots, n_1 - 2\}$. Then

$$t_i = t_i^{(1)} < t_{i+1}^{(1)} \underset{i < n_1-2 \leq n_1-1}{=} t_{i+1}$$

and

$$s_i = s_i^{(1)} \in [t_i^{(1)}, t_{i+1}^{(1)}] \underset{i < n_1-2 \leq n_1-1}{=} [t_i, t_{i+1}]$$

$i = n_1 - 1$. Then

$$t_i = t_{n_1-1} = t_{n_1-1}^{(1)} < t_{n_1}^{(1)} = c = t_1^{(2)} = t_{n_1-n_1+1}^{(2)} = t_{n_1} = t_{i+1}$$

and as

$$s_{n_1-1} = s_{n_1-1}^{(1)} \in [t_{n_1-1}^{(1)}, t_{n_1}^{(1)}] = [t_{n_1-1}, t_{n_1}^{(1)}] = [t_{n_1-1}, c] = [t_{n_1-1}, t_1^{(2)}] = [t_{n_1-1}, t_{n_1}]$$

we have

$$s_i \in [t_i, t_{i+1}]$$

$i \in \{n_1, \dots, n_1 + n_2 - 2\}$. Then

$$t_i = t_{i-n_1+1}^{(2)} < t_{(i+1)-n_1+1}^{(2)} = t_{i+1}$$

and

$$s_i = s_{i-n_1+1}^{(2)} \in [t_{i-n_1+1}^{(2)}, t_{(i+1)-n_1+1}^{(2)}] = [t_i, t_{i+1}]$$

Hence in all cases we have

$$t_i < t_{i+1} \text{ and } s_i \in [t_i, t_{i+1}] \quad (15.38)$$

So using [eqs: 15.36, 15.37 and 15.38] it follows that

$$\mathbb{P}_1 \bigsqcup \mathbb{P}_2 \text{ is a tagged partition on } [a, b]$$

2. Note that

$$\begin{aligned} & \{t_{i+1} - t_i | i \in \{1, \dots, n_1 + n_2 - 2\}\} = \\ & \{t_{i+1} - t_i | i \in \{1, \dots, n_1 - 2\}\} \cup \{t_{i+1} - t_i | i \in \{n_1 - 1\}\} \cup \{t_{i+1} - t_i | i \in \{n_1, \dots, n_1 + n_2 - 2\}\} = \\ & \{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 2\}\} \cup \{t_1^{(2)} - t_{n_1-1}^{(1)}\} \cup \{t_{(i+1)-n_1+1}^{(2)} - t_{i-n_1+1}^{(2)} | i \in \{n_1, \dots, \\ & n_1 + n_2 - 2\}\} = \\ & \{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 2\}\} \cup \{c - t_{n_1-1}^{(1)}\} \cup \{t_{(i+1)-n_1+1}^{(2)} - t_{i-n_1+1}^{(2)} | i \in \{n_1, \dots, \\ & n_1 + n_2 - 2\}\} = \\ & \{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 2\}\} \cup \{t_{n_1}^{(1)} - t_{n_1-1}^{(1)}\} \cup \{t_{(i+1)-n_1+1}^{(2)} - t_{i-n_1+1}^{(2)} | i \in \{n_1, \dots, \\ & n_1 + n_2 - 2\}\} = \\ & \{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 1\}\} \cup \{t_{(i+1)-n_1+1}^{(2)} - t_{i-n_1+1}^{(2)} | i \in \{n_1, \dots, n_1 + n_2 - 2\}\} = \\ & \{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 1\}\} \cup \{t_{i+1}^{(2)} - t_i^{(2)} | i \in \{1, \dots, n_2 - 1\}\} \end{aligned}$$

so that

$$\begin{aligned} & \mu(\mathbb{P}_1 \bigsqcup \mathbb{P}_2) = \\ & \max(\{t_{i+1} - t_i | i \in \{1, \dots, n_1 + n_2 - 2\}\}) = \\ & \max(\{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 1\}\} \cup \{t_{i+1}^{(2)} - t_i^{(2)} | i \in \{1, \dots, n_2 - 1\}\}) = \\ & \max(\max(\{t_{i+1}^{(1)} - t_i^{(1)} | i \in \{1, \dots, n_1 - 1\}\}), \max(\{t_{i+1}^{(2)} - t_i^{(2)} | i \in \{1, \dots, n_2 - 1\}\})) \underset{[\text{theor:3.66}]}{=} \\ & \max(\mu(\mathbb{P}_1), \mu(\mathbb{P}_2)) \end{aligned}$$

3. We have

$$\begin{aligned}
 \mathcal{S}(f, \mathbb{P}) &= \sum_{i=1}^{n_1+n_2-2} (t_{i+1} - t_i) \cdot f(s_i) = \\
 &= \sum_{i=1}^{n_1-2} (t_{i+1} - t_i) \cdot f(s_i) + \sum_{i=n_1-1}^{n_1-1} (t_{i+1} - t_i) \cdot f(s_i) + \sum_{i=n_1}^{n_1+n_2-2} (t_{i+1} - t_i) \cdot f(s_i) = \\
 &= \sum_{i=1}^{n_1-2} (t_{i+1}^{(1)} - t_i^{(1)}) \cdot f(s_i^{(1)}) + (t_{n_1} - t_{n_1-1}) \cdot f(s_{n_1-1}) + \sum_{i=n_1}^{n_1+n_2-2} (t_{i+1}^{(2)} - t_{i-n_1+1}^{(2)}) \cdot \\
 &= f(s_{i-n_1+1}^{(2)}) \\
 &= \sum_{i=1}^{n_1-2} (t_{i+1}^{(1)} - t_i^{(1)}) \cdot f(s_i^{(1)}) + (t_1^{(2)} - t_{n_1-1}^{(1)}) \cdot f(s_{n_1-1}^{(1)}) + \sum_{i=1}^{n_2-1} (t_{i+1}^{(2)} - t_i^{(2)}) \cdot f(s_i^{(2)}) = \\
 &= \sum_{i=1}^{n_1-2} (t_{i+1}^{(1)} - t_i^{(1)}) \cdot f(s_i^{(1)}) + (c - t_{n_1-1}^{(1)}) \cdot f(s_{n_1-1}^{(1)}) + \sum_{i=1}^{n_2-1} (t_{i+1}^{(2)} - t_i^{(2)}) \cdot f(s_i^{(2)}) = \\
 &= \sum_{i=1}^{n_1-2} (t_{i+1}^{(1)} - t_i^{(1)}) \cdot f(s_i^{(1)}) + (t_{n_1}^{(1)} - t_{n_1-1}^{(1)}) \cdot f(s_{n_1-1}^{(1)}) + \sum_{i=1}^{n_2-1} (t_{i+1}^{(2)} - t_i^{(2)}) \cdot f(s_i^{(2)}) = \\
 &= \sum_{i=1}^{n_1-1} (t_{i+1}^{(1)} - t_i^{(1)}) \cdot f(s_i^{(1)}) + \sum_{i=1}^{n_2-1} (t_{i+1}^{(2)} - t_i^{(2)}) \cdot f(s_i^{(2)}) = \\
 &= \sum_{i=1}^{n_1-1} (t_{i+1}^{(1)} - t_i^{(1)}) \cdot f_{|[a,c]}(s_i^{(1)}) + \sum_{i=1}^{n_2-1} (t_{i+1}^{(2)} - t_i^{(2)}) \cdot f_{|[c,b]}(s_i^{(2)}) = \\
 &= \mathcal{S}(f_{|[a,c]}, \mathbb{P}_1) + \mathcal{S}(f_{|[c,b]}, \mathbb{P}_2)
 \end{aligned}$$

□

We prove now a theorem that will be critical to prove later the Fundamental Theorem of Calculus.

Theorem 15.31. *Let $a, b \in \mathbb{R}$ with $a \leq b$, $c \in [a, b]$, $\langle X, \|\cdot\| \rangle$ a Banach space and $f \in \mathcal{C}([a, b], X)$ [so that by theorem: 15.25 is Riemann integrable] then*

$$f|_{[a,c]}: [a, c] \rightarrow X, f|_{[c,b]}: [c, b] \rightarrow X \text{ are Riemann integrable}$$

and

$$\int_a^b f = \int_a^c f + \int_c^b f$$

Note 15.32. *To simplify notation we use the convention that if $f \in \mathcal{C}([a, b], X)$ and $x \in [a, b]$ that $\int_a^x f$ is the same as $\int_a^x f|_{[a,x]}$ and $\int_x^b f$ is the same as $\int_x^b f|_{[x,b]}$. So that we can write*

$$\int_a^b f = \int_a^c f + \int_c^b f$$

where the theorem ensures that $\int_a^c f$, $\int_c^b f$ exist and are well defined.

Proof. We have the following cases to consider for a, b, c :

$a = b$. Then by definition we have that f , $f|_{[a,c]}$, $f|_{[c,b]}$ are Riemann integrable and

$$\int_a^b f = 0 = 0 + 0 = \int_a^c f + \int_c^b f$$

$a < b \wedge a = c$. Then $f|_{[a,c]}$ is Riemann integrable by definition, $[c, b] = [a, b]$ and

$$\int_a^b f = 0 + \int_a^b f = 0 + \int_c^b f = \int_a^c f + \int_c^b f$$

$a < b \wedge b = c$. Then $f|_{[b,c]}$ is Riemann integrable by definition, $[a, c] = [a, b]$ and

$$\int_a^b f = \int_a^c f|_{[a,c]} = \int_a^c f|_{[a,c]} + 0 = \int_a^c f|_{[a,c]} + \int_c^b f|_{[c,b]}$$

$a < c < b$. As $f: [a, b] \rightarrow X$ is continuous it follows from [theorem: 14.146] that $f|_{[a,c]}: [a, c] \rightarrow X$ and $f|_{[c,b]}: [c, b] \rightarrow X$ are continuous. Using [theorem: 15.25] we conclude that $f|_{[a,c]}$ and $f|_{[c,b]}$ are Riemann integrable. By [lemma: 15.231] there exists a sequence of tagged partitions $\{\mathbb{P}_i^{(1)}\}_{i \in \mathbb{N}}$ on $[a, c]$ and a sequence of tagged partitions $\{\mathbb{P}_i^{(2)}\}_{i \in \mathbb{N}}$ on $[c, b]$ such that

$$\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i^{(1)}) = 0 \text{ and } \lim_{i \rightarrow \infty} \mu(\mathbb{P}_i^{(2)}) = 0$$

Let $\varepsilon \in \mathbb{R}^+$ then there exists $N_1, N_2 \in \mathbb{N}$ such that for $n \geq N_1$ we have

$$\mu(\mathbb{P}_n^{(1)}) = |\mu(\mathbb{P}_n^{(1)}) - 0| < \varepsilon \text{ and } \mu(\mathbb{P}_n^{(2)}) = |\mu(\mathbb{P}_n^{(2)}) - 0| < \varepsilon$$

So if $n \geq \max(N_1, N_2)$

$$|\mu(\mathbb{P}_n^{(1)} \sqcup \mathbb{P}_n^{(2)}) - 0| = \mu(\mathbb{P}_n^{(1)} \sqcup \mathbb{P}_n^{(2)}) = \max(\mu(\mathbb{P}_n^{(1)}), \mu(\mathbb{P}_n^{(2)})) < \varepsilon$$

which proves that

$$\lim_{i \rightarrow \infty} \mu(\mathbb{P}_i^{(1)} \sqcup \mathbb{P}_i^{(2)}) = 0$$

Now using [theorem: 15.24] and the fact that $f, f|_{[a,c]}$ and $f|_{[c,b]}$ are Riemann integrable and continuous, it follows that

$$\lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i^{(1)} \sqcup \mathbb{P}_i^{(2)}), \lim_{i \rightarrow \infty} \mathcal{S}(f|_{[a,c]}, \mathbb{P}_i^{(1)}) \text{ and } \lim_{i \rightarrow \infty} \mathcal{S}(f|_{[c,b]}, \mathbb{P}_i^{(2)}) \text{ exists} \quad (15.39)$$

and

$$\int_a^b f = \lim_{i \rightarrow \infty} \mathcal{S}(f, \mathbb{P}_i^{(1)} \sqcup \mathbb{P}_i^{(2)}), \int_a^c f|_{[a,c]} = \lim_{i \rightarrow \infty} \mathcal{S}(f|_{[a,c]}, \mathbb{P}_i^{(1)}), \int_c^b f|_{[c,b]} = \lim_{i \rightarrow \infty} \mathcal{S}(f|_{[c,b]}, \mathbb{P}_i^{(2)}) \quad (15.40)$$

Further we have $\forall i \in \mathbb{N}$ that

$$\mathcal{S}(f, \mathbb{P}_i^{(1)} \sqcup \mathbb{P}_i^{(2)}) \stackrel{[\text{lemma: 15.30}]}{=} \mathcal{S}(f|_{[a,b]}, \mathbb{P}_i^{(1)}) + \mathcal{S}(f|_{[c,b]}, \mathbb{P}_i^{(2)}) \quad (15.41)$$

Finally using [theorem: 14.357] together with [eqs: 15.39, 15.40 and 15.41] proves that

$$\int_a^b f = \int_a^c f|_{[a,c]} + \int_c^b f|_{[c,b]} \quad \square$$

Chapter 16

Differentiation in Normed Spaces

Note 16.1. We assume from now on that all the normed spaces are non trivial, in other words we assume that every normed space contains at least one non zero element.

Note 16.2. Unless stated otherwise all the normed spaces in this chapter are assumed to be vector space over the same field $\mathbb{K}$. For example if we say that $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ are normed spaces then we assume that either X and Y are both vector spaces over the field $\mathbb{R}$ or they are both vector spaces over the field $\mathbb{C}$.

16.1 Fréchet differentiation

16.1.1 The classical derivative

First look at the definition of the derivative that you find in most basic calculus books. Let $A \subseteq \mathbb{R}$, $x \in A$ then a function $f: A \rightarrow \mathbb{R}$ has a derivative f'_x at x if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exist and the derivative is then define by

$$f'_x = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The above definition that you find in many basic calculus books is very imprecise, for example in taking the limit, how can we ensure that $x+h$ is in the domain, how can we avoid dividing by 0. For this we have to do some extra work in defining the limit operation $\lim_{h \rightarrow 0}$. First we introduce the concept of the convergence of a function to a value when it approaches a point its domain.

We introduce now convergence of a function to a point.

Definition 16.3. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in X$ and $f: A \rightarrow Y$ a function then we say that **f converges to y at x** if $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that $\forall z \in A$ with $0 < \|x - z\|_X < \delta$ we have $\|f(z) - y\|_Y < \varepsilon$.

Note 16.4. As $0 < \|x - z\|_X$ we have that $x \neq z$, so in the above definition we never evaluate f at x , hence **x must not be in the domain of f** .

There is a problem with the above definition because it is possible that f converges to two different values at x as the following example shows.

Example 16.5. Consider the normed space $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ and take the function

$$f: \{1\} \rightarrow \mathbb{R} \text{ defined by } f(x) = |x|$$

Let $y \in \mathbb{R}$ and take $\varepsilon \in \mathbb{R}^+$ then for $\delta = 1$ we have $\forall z \in \{1\}$ such that $0 < |1 - z| < 1$ that, as $z \in \{1\} \Rightarrow z = 1 \Rightarrow 0 < |0| < 1 \Rightarrow 0 < 0$, we reach a contradiction hence we have vacuously that $|f(z) - f(1)| < \varepsilon$. Hence we have $\forall y \in \mathbb{R}$ that f converges to y at x by $\{1\}$.

To avoid this problem we require an extra condition on the point x namely that x is a limit point of the domain of A . The following lemma shows then that in that case we have uniqueness.

Lemma 16.6. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A \cap A'$ where A' is the set of limit points of A [see definition: 14.27] and $f: A \rightarrow Y$ a function such that*

$$f \text{ converges to } y_1 \text{ at } x \text{ by } A \text{ and } f \text{ converges to } y_2 \text{ at } x \text{ by } A$$

then we have that

$$y_1 = y_2$$

Proof. Assume that $y_1 \neq y_2$ then $\|y_1 - y_2\|_Y \in \mathbb{R}^+$. As f converges to y_1 at x and f converges to y_2 at x there exists $\delta_1, \delta_2 \in \mathbb{R}^+$ such that

$$\forall z \in A \text{ with } 0 < \|x - z\|_X < \delta_1 \text{ we have } \|f(z) - y_1\|_Y < \frac{\|y_1 - y_2\|_Y}{2} \quad (16.1)$$

$$\forall z \in A \text{ with } 0 < \|x - z\|_X < \delta_2 \text{ we have } \|f(z) - y_2\|_Y < \frac{\|y_1 - y_2\|_Y}{2} \quad (16.2)$$

Take $\delta = \min(\delta_1, \delta_2) \in \mathbb{R}^+$ then, as $x \in B_{\|\cdot\|_X}(x, \delta)$ a open set and x is a limit point of A , there exist by [definition: 14.27] a $z \in (A \setminus \{x\}) \cap B_{\|\cdot\|_X}(x, \delta)$. Hence $z \in A$ and $\|x - z\|_X < \delta \leq \delta_1, \delta_2$ so that by [eqs: 16.1, 16.2]

$$\|f(z) - y_1\|_Y + \|f(z) - y_2\|_Y < \frac{\|y_1 - y_2\|_Y}{2} + \frac{\|y_1 - y_2\|_Y}{2} = \|y_1 - y_2\|_Y \quad (16.3)$$

Now

$$\|y_1 - y_2\|_Y = \|y_1 - f(z) + f(z) - y_2\|_Y \leq \|f(z) - y_1\|_Y + \|f(z) - y_2\|_Y \underset{\text{eq: 16.3}}{<} \|y_1 - y_2\|_Y$$

leading to the contradiction $\|y_1 - y_2\|_Y < \|y_1 - y_2\|_Y$. Hence the assumption is wrong and we must have that $y_1 = y_2$. $\square$

From the above it follows that it is useful to find cases where x is a limit point of a set.

Theorem 16.7. *Let $\langle X, \|\cdot\| \rangle$ be a normed space and $U \in \mathcal{T}_{\|\cdot\|}$ a open set in X then every element of U is a limit point of U . In other words $U \subseteq U'$.*

Proof. Let $x \in U$ then by [theorem: 14.97] there exist a $\delta_1 \in \mathbb{R}^+$ such that $x \in B_{\|\cdot\|}(x, \delta_1) \subseteq U$. Let V be a open set such that $x \in V$ then by [theorem: 14.97] again there exist a $\delta_2 \in \mathbb{R}^+$ such that $x \in B_{\|\cdot\|}(x, \delta_2) \subseteq V$. As X is non trivial [see note 16.1] there exist a $x_0 \in X$ with $x_0 \neq 0$. Take $\delta = \min\left(\frac{\delta_1}{2}, \frac{\delta_2}{2}\right) \in \mathbb{R}^+$ and $y = x + \frac{\delta}{\|x_0\|} \cdot x_0$ then $y \neq x$

$$\|y - x\| = \left\| \frac{\delta}{\|x_0\|} \cdot x_0 \right\| = \left| \frac{\delta}{\|x_0\|} \right| \cdot \|x_0\| = \frac{\delta}{\|x_0\|} \cdot \|x_0\| = \delta \quad (16.4)$$

As $\delta \neq 0$ we have by the above that $x \neq y$ and $\|y - x\| = \delta \leq \frac{\delta_1}{2}, \frac{\delta_2}{2} \leq \delta_1, \delta_2$ so that $y \in U \wedge y \in V$. Hence $y \in (U \setminus \{x\}) \cap V \Rightarrow (U \setminus \{x\}) \cap V \neq \emptyset$ which by [definition: 14.27] proves that x is a limit point of U . $\square$

Corollary 16.8. *Let $\langle X, \|\cdot\| \rangle$ be a normed space $A \subseteq X$. If $x \in A^\circ$ [see definition: 14.12] then x is a limit point of A . In other words every interior point of a set is a limit point or $A^\circ \subseteq A'$.*

Proof. As $x \in A^\circ$ a open set, it follows from the previous theorem [theorem: 16.7] that x is a limit point of A° . As $A^\circ \subseteq A$ it follows from [theorem: 14.28] that x is a limit point of A . $\square$

Corollary 16.9. *Let $\langle X, \|\cdot\| \rangle$ be a normed space and $\delta \in \mathbb{R}^+$ then $\forall x \in X$ we have*

1. *Every element of $B_{\|\cdot\|}(x, \delta)$ is a limit point of $B_{\|\cdot\|}(x, \delta)$*

2. Every element of $\overline{B_{|||}(x, \delta)}$ is a limit point of $\overline{B_{|||}(x, \delta)}$

Proof.

1. As $B_{|||}(x, \delta)$ is open this follows trivially from [theorem: 16.7].

2. Let $y \in \overline{B_{|||}(x, \delta)}$ then we have $\|y - x\| \leq \delta$, hence we have two cases to consider:

$\|y - x\| < \delta$. Then $y \in B_{|||}(x, \delta)$ so that by (1) y is a limit point of $B_{|||}(x, \delta)$. As $B_{|||}(x, \delta) \subseteq \overline{B_{|||}(x, \delta)}$ it follows from [theorem: 14.28] that x is a limit point of $\overline{B_{|||}(x, \delta)}$.

$\|y - x\| = \delta$. Let U be an open set such that $y \in U$ then by [theorem: 14.97] there exist a $\delta_1 \in \mathbb{R}^+$ such that $y \in B_{|||}(y, \delta_1) \subseteq U$. As $\|y - x\| = \delta > 0$ we can define $\lambda = \min\left(\frac{1}{2}, \frac{\delta_1}{2 \cdot \|y - x\|}\right)$ and $z = y + \lambda \cdot (x - y)$. Then we have

$$\begin{aligned} \|z - x\| &= \|y + \lambda \cdot (x - y) - x\| \\ &= \|y - x - \lambda \cdot (y - x)\| \\ &= \|(1 - \lambda) \cdot (y - x)\| \\ &= |1 - \lambda| \cdot \|y - x\| \\ &= (1 - \lambda) \cdot \|y - x\| \\ &= (1 - \lambda) \cdot \delta \end{aligned}$$

As $\lambda \leq \frac{1}{2}$ we have $(1 - \lambda) \in \mathbb{R}^+$ so that $\|z - x\| \neq 0$ hence $z \neq x$. Further we have $\|z - x\| = (1 - \lambda) \cdot \delta \leq \left(1 - \frac{1}{2}\right) \cdot \delta = \frac{\delta}{2} < \delta$ proving that $z \in \overline{B_{|||}(x, \delta)}$. To summarize we have

$$z \in \overline{B_{|||}(x, \delta)} \setminus \{x\} \quad (16.5)$$

Further we have

$$\begin{aligned} \|z - y\| &= \|y + \lambda \cdot (x - y) - y\| \\ &= \|\lambda \cdot (x - y)\| \\ &= |\lambda| \cdot \|x - y\| \\ &= \lambda \cdot \|x - y\| \\ &\leq \frac{\delta_1}{2 \cdot \|y - x\|} \cdot \|x - y\| \\ &= \frac{\delta_1}{2} \\ &< \delta_1 \end{aligned}$$

so that

$$z \in B_{|||}(y, \delta_1) \subseteq U$$

which combined with [eq: 16.5] proves that

$$(\overline{B_{|||}(x, \delta)} \setminus \{x\}) \cap U \neq \emptyset$$

which proves that x is a limit point of $\overline{B_{|||}(x, \delta)}$ □

Corollary 16.10. Let $\langle \mathbb{R}, || \rangle$ be the normed space of the real numbers with the absolute value norm then we have

1. $\forall x \in [a, \infty[$ where $a \in \mathbb{R}$ x is a limit point of $[a, \infty[$ in other words $[a, \infty[\subseteq ([a, \infty[)'$.
2. $\forall x \in]a, \infty[$ where $a \in \mathbb{R}$ x is a limit point of $]a, \infty[$ in other words $]a, \infty[\subseteq (]a, \infty[)'$.
3. $\forall x \in]-\infty, a]$ where $a \in \mathbb{R}$ x is a limit point of $] -\infty, a]$ in other words $] -\infty, a] \subseteq (]-\infty, a])'$.
4. $\forall x \in]-\infty, a[$ where $a \in \mathbb{R}$ x is a limit point of $] -\infty, a[$ in other words $] -\infty, a[\subseteq (]-\infty, a[)'$.

5. $\forall x \in [a, b]$ where $a, b \in \mathbb{R}$ where $a < b$ x is a limit point of $[a, b]$ in other words $[a, b] \subseteq ([a, b])'$.
6. $\forall x \in [a, b[$ where $a, b \in \mathbb{R}$ where $a < b$ x is a limit point of $[a, b[$ in other words $[a, b[\subseteq ([a, b])'$.
7. $\forall x \in]a, b]$ where $a, b \in \mathbb{R}$ where $a < b$ x is a limit point of $]a, b]$ in other words $]a, b] \subseteq ([a, b])'$.
8. $\forall x \in]a, b[$ where $a, b \in \mathbb{R}$ where $a < b$ x is a limit point of $]a, b[$ in other words $]a, b[\subseteq ([a, b])'$.

Proof.

1. Let $x \in [a, \infty[$ then $a \leq x$. If U is a open set with $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $x \in]x - \delta, x + \delta[\subseteq U$, then $x - \delta < x < x + \frac{\delta}{2} < x + \delta$ and $a \leq x < x + \frac{\delta}{2}$ from which it follows, as $x + \frac{\delta}{2} \neq x$, that $x + \frac{\delta}{2} \in ([a, \infty[\setminus \{x\}) \cap]x - \delta, x + \delta[\subseteq ([a, \infty[\setminus \{x\}) \cap U$. Hence $([a, \infty[\setminus \{x\}) \cap U \neq \emptyset$ which proves that x is a limit point of $[a, \infty[$.
2. Let $x \in]a, \infty[$, using [theorem: 14.105] $]a, \infty[$ is a open set so that by [theorem: 16.7] x is a limit point of $]a, \infty[$.
3. Let $x \in]-\infty, a]$ then $x \leq a$. If U is a open set with $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $x \in]x - \delta, x + \delta[\subseteq U$, then $x - \delta < x - \frac{\delta}{2} < x < x + \delta$ and $x - \frac{\delta}{2} < x \leq a$ from which it follows, as $x - \frac{\delta}{2} \neq x$, that $x - \frac{\delta}{2} \in (]-\infty, a] \setminus \{x\}) \cap]x - \delta, x + \delta[\subseteq (]-\infty, a] \setminus \{x\}) \cap U$. Hence $(]-\infty, a] \setminus \{x\}) \cap U \neq \emptyset$ which proves that x is a limit point of $]-\infty, a]$.
4. Let $x \in]-\infty, a[$, using [theorem: 14.105] $]-\infty, a[$ is a open set so that by [theorem: 16.7] x is a limit point of $]-\infty, a[$.
5. Let $x \in [a, b] = \left[\left(a + \frac{b-a}{2} \right) - \frac{b-a}{2}, \left(a + \frac{b-a}{2} \right) + \frac{b-a}{2} \right] = B_{||} \left(a + \frac{b-a}{2}, \frac{b-a}{2} \right)$ so that by [theorem: 16.9] x is a limit point of $[a, b]$.
6. Let $x \in [a, b]$ then $a \leq x < b$. If U is open with $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $x \in]x - \delta, x + \delta[\subseteq U$. Take $\varepsilon = \min \left(\frac{(b-x)}{2}, \frac{\delta}{2} \right) \in \mathbb{R}^+$ then we have $x \neq x + \varepsilon$, $a \leq x < x + \varepsilon \leq x + \frac{b-x}{2} < x + (b-x) = b$ and $x - \delta < x < x + \varepsilon < x + \frac{\delta}{2} < x + \delta$ so that $x + \varepsilon \in ([a, b] \setminus \{x\}) \cap]x - \delta, x + \delta[\subseteq ([a, b] \setminus \{x\}) \cap U$. Hence $([a, b] \setminus \{x\}) \cap U \neq \emptyset$ which proves that x is a limit point of $[a, b]$.
7. Let $x \in]a, b]$ then $a < x \leq b$. If U is open with $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $x \in]x - \delta, x + \delta[\subseteq U$. Take $\varepsilon = \min \left(\frac{(x-a)}{2}, \frac{\delta}{2} \right) \in \mathbb{R}^+$ then we have $x \neq x - \varepsilon$, $a = x - (x-a) < x - \frac{x-a}{2} \leq x - \varepsilon < x \leq b$ and $x - \delta < x - \frac{\delta}{2} < x < x + \delta$ so that $x - \varepsilon \in ([a, b] \setminus \{x\}) \cap]x - \delta, x + \delta[\subseteq ([a, b] \setminus \{x\}) \cap U$. Hence $([a, b] \setminus \{x\}) \cap U \neq \emptyset$ which proves that x is a limit point of $]a, b]$.
8. Let $x \in]a, b[$, using [theorem: 14.105] $]a, b[$ is a open set so that by [theorem: 16.7] x is a limit point of $]a, b[$. $\square$

We can now define the limit of a function at a point.

Definition 16.11. Let $\langle X, |||_X \rangle, \langle Y, |||_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A \cap A'$ where A' is the set of limit points of A [see definition: 14.27] and $f: A \rightarrow Y$ a function that converges to y at x by A then this **unique** y [by lemma: 16.6] is noted as

$$\lim_{\substack{z \rightarrow x \\ A}} f(z)$$

If we say that

$$y = \lim_{\substack{z \rightarrow x \\ A}} f(z) \text{ exist}$$

then this is equivalent with saying that f converges to y at x by A or equivalently

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall z \in A \text{ with } 0 < ||x - z||_X < \delta \text{ we have } ||f(z) - y||_Y < \varepsilon$$

Note 16.12. Using [theorem: 16.7], [corollary: 16.8], [corollary: 16.9] and [corollary: 16.10] it follows that this definition is valid in the following cases:

1. A is a open set in X or $A = B_{\|\cdot\|_X}(z, \delta)$ or $A = \overline{B_{\|\cdot\|_X}(z, \delta)}$
2. $\langle X, \|\cdot\|_X \rangle = \langle \mathbb{R}, \|\cdot\| \rangle$ and A is of the form
 - a. $[a, \infty[$
 - b. $]a, \infty[$
 - c. $] -\infty, a]$
 - d. $] -\infty, a[$
 - e. $[a, b]$
 - f. $[a, b[$
 - g. $]a, b]$
 - h. $]a, b[$
3. $x \in A^\circ$

We are now almost ready to define the derivative of a function. As we want to define f'_x as $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, we have to look at the domain D of the function defined by $h \rightarrow \frac{f(x+h) - f(x)}{h}$. To ensure that $\frac{f(x+h) - f(x)}{h}$ is well defined we have to assume that $0 \notin D$ and $\forall h \in D$ that $x+h \in \text{dom}(f)$. This is done in the following.

Definition 16.13. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $A \subseteq X$ and $x \in A$ then

$$A_x = \{h \in X \mid x+h \in A\}$$

Note 16.14. Note as $x+0 = x \in A$ we have

$$0 \in A_x$$

The following is a equivalent definition for A_x .

Theorem 16.15. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $A \subseteq X$ and $x \in A$ then

$$A_x = -x + A = A - x$$

where

$$-x + A = \{-x + a \mid a \in A\} \text{ and } A - x = \{a - x \mid a \in A\}$$

Proof. We have

$$\begin{aligned} h \in A_x &\Leftrightarrow x+h \in A \\ &\Leftrightarrow \exists u \in A \text{ with } x+h = u \\ &\Leftrightarrow \exists u \in A \text{ with } h = (-x) + u \\ &\Leftrightarrow h \in (-x) + A \\ h \in (-x) + A &\Leftrightarrow \exists u \in A \text{ with } h = (-x) + u \\ &\Leftrightarrow \exists u \in A \text{ with } h = u + (-x) \\ &\Leftrightarrow h \in A + (-x) \end{aligned}$$

□

Corollary 16.16. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $x \in X$ and U a open set in X then U_x is open in X

Proof. By [theorem: 16.15] $U_x = -x + U$ hence using [theorem: 14.116] U_x is open. □

Theorem 16.17. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $x \in X$ and $\delta \in \mathbb{R}^+$ then $B_{\|\cdot\|}(0, \delta) = B_{\|\cdot\|}(x, \delta)_x$.

Proof.

$$\begin{aligned}
 x \in B_{|||}(x, \delta)_x &\Leftrightarrow x + h \in B_{|||}(x, \delta) \\
 &\Leftrightarrow x \in X \wedge \|(x + h) - x\| < \delta \\
 &\Leftrightarrow x \in X \wedge \|h\| < \delta \\
 &\Leftrightarrow h \in B_{|||}(0, \delta) \\
 &\quad \square
 \end{aligned}$$

Lemma 16.18. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $A \subseteq X$ and $x \in A'$ a limit point of A then x is a limit point of $A \setminus \{x\}$. In other words we have $A' \subseteq (A \setminus \{x\})'$.

Proof. Let $U \in \mathcal{T}$ with $x \in U$ then by [definition: 14.27] we have that $(A \setminus \{x\}) \cap U \neq \emptyset$. As $(A \setminus \{x\}) \setminus \{x\} \stackrel{[\text{theorem: 1.34}]}{=} A \setminus (\{x\} \cup \{x\}) = A \setminus \{x\}$ we have that $((A \setminus \{x\}) \setminus \{x\}) \cap U \neq \emptyset$. As $U \in \mathcal{T}$ has been chosen arbitrary it follows by [definition: 14.27] that x is a limit point of $A \setminus \{x\}$. $\square$

Lemma 16.19. Let $\langle X, ||| \rangle$ be a normed space and $A \subseteq X$ then if $x \in A$ is a limit point of A then $0 \in A_x$ is a limit point of A_x so that by the previous lemma [lemma: 16.18] 0 is a limit point of $A_x \setminus \{0\}$.

Proof. Let U be a open set such that $0 \in U$. Then $x = x + 0 \in x + U$ which by [theorem: 14.116] is a open set in X . As x is a limit point of A we have that $(A \setminus \{x\}) \cap (x + U) \neq \emptyset$ so that there exist a $z \in (A \setminus \{x\}) \cap (x + U)$. As $(z - x) + x = z \underset{z \in A \setminus \{x\}}{\in} A$ we have $(z - x) \in A_x$, further $z - x \neq 0$ [as $x \in A \setminus \{x\}$] and $z \in x + U \Rightarrow z - x \in U$ it follows that $z - x \in (A_x \setminus \{0\}) \cap U$ so that $(A_x \setminus \{0\}) \cap U \neq \emptyset$. Hence 0 is a limit point of A_x . $\square$

Definition 16.20. Let $\langle \mathbb{K}, || \rangle$ be the normed space of the real [or complex] numbers with the canonical norm $||$, $\langle Y, ||| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow Y$ a function then **f has a derivative at x noted as f'_x** if we have

1. $x \in A'$ where A' is the set of limit points of A [see definition: 14.27].
2. we have for the function

$$\Delta_x f: A_x \setminus \{0\} \rightarrow Y \text{ defined by } \Delta_x f(h) = \frac{f(x+h) - f(x)}{h}$$

that

$$f'_x = \lim_{\substack{h \rightarrow 0 \\ A_x \setminus \{0\}}} \Delta_x f(h) \text{ exist}$$

in most books this is summarized as

$$f'_x = \lim_{\substack{h \rightarrow 0 \\ A_x \setminus \{0\}}} \frac{f(x+h) - f(x)}{h} \text{ exist}$$

Note 16.21. Using [theorem: 16.7], [corollary: 16.8], [corollary: 16.9] and [corollary: 16.10] it follows that (1) is satisfied automatically in the following cases

1. A is a open set in $\langle \mathbb{K}, || \rangle$ or $A = B_{||}(z, \delta) =]z - \delta, z + \delta[$ or $A = \overline{B_{|||x}(z, \delta)} = [z - \delta, z + \delta]$
2. $\langle \mathbb{K}, || \rangle = \langle \mathbb{R}, || \rangle$ and A is of the form
 - a. $[a, \infty[$
 - b. $]a, \infty[$
 - c. $] - \infty, a]$
 - d. $] - \infty, a[$
 - e. $[a, b]$
 - f. $[a, b[$
 - g. $]a, b]$

$$h.]a, b[$$

$$3. x \in A^\circ$$

We can reformulate the above definition using the definition of the $\lim_{\substack{h \rightarrow 0 \\ A_x \setminus \{0\}}}$.

Theorem 16.22. *Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real or complex numbers, $\langle Y, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$, $\langle Y, \|\cdot\| \rangle$ a normed space and $f: A \rightarrow Y$ a function. Then we have*

$$f \text{ has a derivative } f'_x \text{ at } x$$

$$\Updownarrow$$

$$x \in A'$$

$$\text{and}$$

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ so that } \forall h \in A_x \text{ with } 0 < |h| < \delta \quad \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| < \varepsilon$$

Proof. Define

$$\Delta_x f: A_x \setminus \{0\} \rightarrow Y \text{ where } \Delta_x f(h) = \frac{f(x+h) - f(x)}{h}$$

As f has a derivative f'_x at x we have that

$$x \in A' \text{ and } f'_x = \lim_{\substack{h \rightarrow 0 \\ A_x \setminus \{0\}}} \Delta_x f(h) \text{ exist}$$

which by [definition: 16.11] means that

$$\forall \varepsilon \in \mathbb{R}^+ \forall \delta \in \mathbb{R}^+ \text{ we have } \forall h \in A_x \setminus \{0\} \text{ with } 0 < |h| < \delta \quad \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\|_Y < \varepsilon$$

which, as $h \in A_x \setminus \{0\}$ with $0 < |h| < \delta$ is equivalent with $h \in A_x$ with $0 < |h| < \delta$, is equivalent with

$$\forall \varepsilon \in \mathbb{R}^+ \forall \delta \in \mathbb{R}^+ \text{ we have } \forall h \in A_x \setminus \{0\} \text{ with } 0 < |h| < \delta \quad \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\|_Y < \varepsilon \quad \square$$

Another equivalent definition of derivating is the following.

Theorem 16.23. *Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real or complex numbers, $\langle Y, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow Y$ a function. Then we have*

$$f \text{ has a derivative } f'_x \text{ at } x$$

$$\Updownarrow$$

$$x \in A'$$

$$\text{and}$$

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall h \in A_x \text{ with } |h| < \delta \text{ we have } \|f(x+h) - f(x) - h \cdot f'_x\| \leq \varepsilon \cdot |h|$$

Proof.

$\Rightarrow$. As f'_x exist we have that

$$x \in A'$$

Let $\varepsilon \in \mathbb{R}^+$ then as f'_x exists it follows from the previous theorem [theorem: 16.22] that $\exists \delta \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } 0 < |h| < \delta \text{ we have } \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| < \varepsilon \quad (16.6)$$

Let now $h \in A_x$ with $|h| < \delta$ then we have either:

$h = 0$. Then

$$\|f(x+h) - f(x) - h \cdot f'_x\| = \|f(x) - f(x) - 0 \cdot f'_x\| = \|0\| = 0 = \varepsilon \cdot 0 = \varepsilon \cdot |h|$$

$h \neq 0$. Then $0 < |h| < \delta$

$$\begin{aligned}
 \|f(x+h) - f(x) - h \cdot f'_x\| &= \frac{|h|}{|h|} \cdot \|f(x+h) - f(x) - h \cdot f'_x\| \\
 &\stackrel{[\text{eq: 10.85(10)}] \text{ if } \mathbb{K}=\mathbb{C}}{=} |h| \cdot \left\| \frac{1}{h} \cdot \|f(x+h) - f(x) - h \cdot f'_x\| \right\| \\
 &= |h| \cdot \left\| \frac{f(x+h) - f(x) - h \cdot f'_x}{h} \right\| \\
 &= |h| \cdot \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| \\
 &\stackrel{[\text{eq: 16.6}]}{<} |h| \cdot \varepsilon
 \end{aligned}$$

hence in all cases we have that

$$\|f(x+h) - f(x) - h \cdot f'_x\| \leq \varepsilon \cdot |h|$$

$\Leftarrow$. By the hypothesis we have that

$$x \in A'$$

Let $\varepsilon \in \mathbb{R}^+$ then by the hypothesis there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } |h| < \delta \quad \|f(x+h) - f(x) - h \cdot f'_x\| \leq \frac{\varepsilon}{2} \cdot |h| \quad (16.7)$$

Let now $h \in A_x$ with $0 < |h| < \delta$ then $h \neq 0$ and we can divide h giving

$$\begin{aligned}
 \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| &= \left\| \frac{f(x+h) - f(x) - h \cdot f'_x}{h} \right\| \\
 &= \frac{1}{|h|} \cdot \|f(x+h) - f(x) - h \cdot f'_x\| \\
 &\stackrel{[\text{eq: 16.7}]}{\leq} \frac{1}{|h|} \cdot \frac{\varepsilon}{2} \cdot |h| \\
 &= \frac{\varepsilon}{2} \\
 &< \varepsilon
 \end{aligned}$$

which prove by [theorem: 16.22] that f'_x exist.

□

Up to now we have looked at the derivative f'_x at a single limit point in the domain of a function which is a element of the co-domain of f . Now we define the derivative function f' .

Definition 16.24. (Derivative) Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers], $A \subseteq \mathbb{K}$, $\langle X, \|\cdot\| \rangle$ a normed space and $f: A \rightarrow X$ a function. The set of all points of A for which f has a derivative at x is noted as $\mathcal{D}_f$ in other words

$$\mathcal{D}_f = \{x \in A \mid f \text{ has a derivative at } x\} \subseteq A$$

The derivative f' of f is then defined by

$$f': \mathcal{D}_f \rightarrow Y \text{ where } f'(x) = f'_x$$

Note 16.25. As the existence of a derivative at x implies that $x \in A'$ [see definition: 16.22] it follows that

$$\mathcal{D}_f \subseteq A \cap A' \subseteq A, A'$$

Note 16.26. If $\mathcal{D}_f = \emptyset$ then $f' \subseteq \mathcal{D}_f \times Y = \emptyset$ and $f': \mathcal{D}_f \rightarrow Y$ is the empty function $\emptyset: \emptyset \rightarrow Y$ [see example: 2.46].

Note 16.27. The benefit of this definition is that

1. $f': \mathcal{D}_f \rightarrow X$ is always defined even if f has never a derivative.

2. $\mathcal{D}_f$ is the set of all elements of the domain of f for which the derivative exists so its the most natural definition of domain of f'
3. If we refer to f' where $f: A \rightarrow Y$ is a function then we can automatically assume that f' is the graph of the function

$$f': \mathcal{D}_f \rightarrow Y$$

f' will be used later to define recursively higher order derivatives.

Definition 16.28. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers], $A \subseteq \mathbb{K}$, $B \subseteq A$ $\langle X, \|\cdot\| \rangle$ a normed space and $f: A \rightarrow X$ a function then we say that

f has derivatives on B

iff

$$\forall x \in B \text{ } f \text{ has a derivative at } x$$

Example 16.29. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers] and $\langle X, \|\cdot\| \rangle$ a normed space then for $f: \emptyset \rightarrow Y$ we have that $\mathcal{D}_f = \emptyset$ so that $f': \mathcal{D}_f \rightarrow Y$ is again the empty function.

Example 16.30. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ a normed space, $y \in X$ and $A \subset X$ either a open set or of the form $[a, \infty[$, $]a, \infty[$, $]-\infty, a]$, $]-\infty, a[$, $[a, b]$, $[a, b[$, $]a, b]$, $]a, b[$. Let

$$C_y: A \rightarrow X \text{ defined by } C_y(x) = y$$

then

$$\mathcal{D}_{C_y} = A$$

and for

$$(C_y)': A \rightarrow X \text{ defined by } (C_y)'(x) = (C_y)'_x$$

we have that

$$(C_y)' = C_0$$

Proof. Using [theorems: 16.7, 16.10] it follows that $A \subseteq A'$. Let $x \in A$ then $x \in A'$, further if $\varepsilon \in \mathbb{R}^+$ then for $h \in A_x$ with $0 < |h| < 1$ we have

$$\left\| \frac{C_y(x+h) - C_y(x)}{h} - 0 \right\| = \left\| \frac{y - y}{h} - 0 \right\| = \|0\| = 0 < \varepsilon$$

so that by [theorem: 16.22] C_y has a derivative at x with $(C_y)'_x = 0$. Hence $x \in \mathcal{D}_{C_y}$ which proves that $A \subseteq \mathcal{D}_{C_y} \subseteq A$ or

$$\mathcal{D}_{C_y} = A$$

Further we have $\forall x \in A$ that $(C_y)'(x) = (C_y)'_x = 0 = C_0(x)$ which proves that

$$(C_y)' = C_0 \quad \square$$

Example 16.31. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ the normed space of real [or complex] numbers and $A \subset X$ either a open set or of the form $[a, \infty[$, $]a, \infty[$, $]-\infty, a]$, $]-\infty, a[$, $[a, b]$, $[a, b[$, $]a, b]$, $]a, b[$. Let

$$\text{Id}_A: A \rightarrow \mathbb{K} \text{ defined by } \text{Id}_A(x) = x$$

then we have

$$\mathcal{D}_{\text{Id}_A} = A$$

and for

$$(\text{Id}_A)': \mathcal{D}_{\text{Id}_A} \rightarrow Y \text{ defined by } (\text{Id}_A)'(x) = (\text{Id}_A)'_x$$

we have

$$(\text{Id}_A)' = C_1$$

Proof. Using [theorems: 16.7, 16.10] it follows that $A \subseteq A'$. Let $x \in A$ then $x \in A'$, further if $\varepsilon \in \mathbb{R}^+$ then for $h \in A_x$ with $0 < |h| < 1$ we have

$$\left| \frac{\text{Id}_A(x+h) - \text{Id}_A(x)}{h} - 1 \right| = \left| \frac{x+h-x}{h} - 1 \right| = |0| = 0 < \varepsilon$$

So that by [theorem: 16.22] Id_A has a derivative $(\text{Id}_A)'_x$ at x with $(\text{Id}_A)'_x = 1$. Hence $x \in \mathcal{D}_{C_y}$ which proves that $A \subseteq \mathcal{D}_{C_y} \subseteq A$ or

$$\mathcal{D}_{C_y} = A$$

Further we have $\forall x \in A$ that $(C_y)'(x) = (C_y)'_x = 1 = C_1(x)$ which proves that

$$(C_y)' = C_1 \quad \square$$

Example 16.32. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ be a normed space and $A \subseteq \mathbb{R}$ either a open set or of the form $[a, \infty[,]a, \infty[,]-\infty, a],]-\infty, a[, [a, b], [a, b[,]a, b],]a, b[$ if $\mathbb{K} = \mathbb{R}$. Let $x, y \in X$ and define

$$f: A \rightarrow X \text{ defined by } f(t) = t \cdot x + y$$

then we have

$$\mathcal{D}_f = A$$

and for

$$f': \mathcal{D}_{\text{Id}_A} \rightarrow Y \text{ defined by } f'(x) = f'_x$$

we have

$$f' = C_x$$

Proof. Using [theorems: 16.7, 16.10] it follows that $A \subseteq A'$. Let $t \in A$ then $t \in A'$, further if $\varepsilon \in \mathbb{R}^+$ then for $h \in A_x$ with $0 < |h| < 1$ we have

$$\left| \frac{f(t+h) - f(t)}{h} - x \right| = \left| \frac{(t+h) \cdot x + y - t \cdot x - y}{h} - x \right| = \left| \frac{h \cdot x}{h} - x \right| = 0 < \varepsilon$$

So that by [theorem: 16.22] f has a derivative f'_t at t with $f'_t = x$. Hence $t \in \mathcal{D}_f$ which proves that $A \subseteq \mathcal{D}_f \subseteq A$ or

$$\mathcal{D}_f = A$$

Further we have $\forall x \in A$ that $f'(t) = f'_t = x = C_x(t)$ which proves that

$$f' = C_x \quad \square$$

16.1.2 Properties of the derivative

The existence of a derivative implies continuity

Theorem 16.33. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers], $A \subseteq \mathbb{K}$, $\langle X, \|\cdot\| \rangle$ a normed space, $x \in A$ and $f: A \rightarrow Y$ a function that has a derivative f'_x at x then f is continuous at x .

Proof. Let $\varepsilon \in \mathbb{R}^+$. As f has a derivative f'_x at x it follows from [theorem: 16.23] that there exist a $\delta_1 \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $|h| < \delta_1$ we have

$$\|f(x+h) - f(x) - h \cdot f'_x\| \leq \frac{\varepsilon}{2} \cdot |h|$$

Let $\delta = \min \left(\delta_1, 1, \frac{\varepsilon}{2 \cdot (1 + \|f'_x\|)} \right) \in \mathbb{R}^+$ then we have for $y \in A$ with $|y - x| < \delta$ that $y - x \in A_x$ [as $(y - x) + x = y \in A$] and $|y - x| < \delta_1, 1, \frac{\varepsilon}{1 + \|f'_x\|}$ then we have

$$\|f(y) - f(x) - (y - x) \cdot f'_x\| = \|f(x + (y - x)) - f(x) - (y - x) \cdot f'_x\| \leq \frac{\varepsilon}{2} \cdot |y - x| \leq \frac{\varepsilon}{2}$$

and

$$\|(y-x) \cdot f'_x\| = |y-x| \cdot \|f'_x\| \leq \frac{\varepsilon}{2 \cdot (1 + \|f'_x\|)} \cdot \|f'_x\| < \frac{\varepsilon}{2}$$

So that

$$\begin{aligned} \|f(y) - f(x)\| &= \|f(y) - f(x) - (y-x) \cdot f'_x + (y-x) \cdot f'_x\| \\ &\leq \|f(y) - f(x) - (y-x) \cdot f'_x\| + \|(y-x) \cdot f'_x\| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &< \varepsilon \end{aligned}$$

So by [theorem: 14.136] it follows that f is continuous at x . $\square$

We look now at the dependency of the derivative of a function on the domain of the function.

Theorem 16.34. *Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ a normed space, $A, B \subseteq \mathbb{K}$ with $B \subseteq A$, $x \in B$ such that x is a limit point of B and $f: A \rightarrow X$ a function such that f has a derivative f'_x at x then $f|_B$ has a derivative $(f|_B)'_x$ at x with $(f|_B)'_x = f'_x$.*

Note 16.35. *The requirement that x is a limit point is needed, the existence of f'_x ensures that x is a limit point of A but in general this does not ensure that x is a limit point of B .*

Proof. Let $\varepsilon \in \mathbb{R}^+$ then as f'_x exist we have by [theorem: 16.22] that there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $0 < |h| < \delta$ we have that

$$\left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| < \varepsilon$$

Let $h \in B_x$ with $0 < |h| < \delta$ then $x+h \in B \subseteq A \Rightarrow h \in A_x$ so that

$$\left\| \frac{f|_B(x+h) - f|_B(x)}{h} - f'_x \right\| = \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| < \varepsilon$$

which proves, as $x \in B'$, that $f|_B$ has a derivative at x and $(f|_B)'_x = f'_x$. $\square$

If we restrict ourselves to interior points in B then the opposite of the above theorem is also true so that the existence of a derivative at a inner point depends only on the values of the functions in a neighborhood of this point. This is the so called **locally** principle.

Theorem 16.36. *Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq X$, $B \subseteq A$, $x \in B^\circ$ and $f: A \rightarrow X$ a function then we have that*

$$\begin{aligned} &f \text{ has a derivative } f'_x \text{ at } x \\ &\quad \Updownarrow \\ &f|_B \text{ has a derivative } (f|_B)'_x \text{ at } x \end{aligned}$$

Further if either f'_x exist or $(f|_B)'_x$ exist then $f'_x = (f|_B)'_x$.

Note 16.37. As $x \in B^\circ \subseteq_{B \subseteq A} A^\circ$ it follows using [theorem: 16.8] that $x \in B' \wedge x \in A'$ which is needed for the definition of the derivatives.

Proof.

$\Rightarrow$. This follows from the previous theorem [theorem: 16.34].

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $f|_B$ has a derivative at x there exist a $\delta_1 \in \mathbb{R}^+$ such that $\forall h \in B_x$ with $0 < |h| < \delta_1$ we have

$$\left\| \frac{f|_B(x+h) - f|_B(x)}{h} - (f|_B)'_x \right\| < \varepsilon$$

As $x \in B^\circ$ there exist by [definition: 14.12] a $U \in \mathcal{T}_||$ such that $x \in U \subseteq B$. Hence there exist a $\delta_2 \in \mathbb{R}^+$ such that $x \in B_{||}(x, \delta_2) \subseteq U \subseteq B \subseteq A$. Take then $\delta = \min(\delta_1, \delta_2)$ then if we have $h \in A_x$ with $0 < |h| < \delta$ we have that $|x+h-x| = |h| < \delta_2$ so that $x+h \in B_{||}(x, \delta_2) \subseteq B \Rightarrow h \in B_x$. Hence we have, as also $0 < |h| < \delta \leq \delta_1$, that

$$\left\| \frac{f(x+h) - f(x)}{h} - (f|_B)'_x \right\|_{x \in B, x+h \in B} = \left\| \frac{f|_B(x+h) - f|_B(x)}{h} - (f|_B)'_x \right\| < \varepsilon$$

which proves that f has a derivative at x and $f'_x = (f|_B)'_x$. $\square$

The following example shows that the requirement of inner points is essential.

Example 16.38. Consider the normed space $\langle \mathbb{R}, || \rangle$ and define

$$B = \{0\} \cup \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R} \text{ and } A = B \cup \left\{ \frac{1}{n} + \frac{1}{n^2} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$$

and

$$f: A \rightarrow \mathbb{R} \text{ defined by } f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 0 & \text{if } x \in B \\ 1 & \text{if } x \in A \setminus B \end{cases}$$

then

1. $0 \in B \subseteq A$
2. 0 is a limit point of B
3. $(f|_B)'_0$ exist with $(f|_B)'_0 = 0$
4. f has no derivatives at 0

Proof.

1. This follows trivially from the definition of A, B .
2. If U is open set with 0 then there exist a $\delta \in \mathbb{R}^+$ such that $0 \in B_{||}(0, \delta) \subseteq U$. As $0 < \delta$ there exist a $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \delta$ [see theorem: 10.30] hence $\frac{1}{n} \in B_{||}(0, \delta) \subseteq U$ as $0 \neq \frac{1}{n} \in B$ it follows that $\frac{1}{n} \in (B \setminus \{0\}) \cup U \Rightarrow (B \setminus \{0\}) \cap U \neq \emptyset$.
3. Let $\varepsilon \in \mathbb{R}^+$ and take $\delta = 1$ then $\forall h \in B_0$ with $0 < |h| < 1$ we have that $h = 0 + h \in B \Rightarrow h \in B$ so that

$$\left| \frac{f(0+h) - f(0)}{h} - 0 \cdot h \right| = \left| \frac{f(h) - f(0)}{h} \right| = \left| \frac{0 - 0}{h} \right| = 0$$

which proves that $(f|_B)'_0$ exist with $(f|_B)'_0 = 0$.

4. First we prove that $\forall n \in \mathbb{N}$ we have that $\frac{1}{n} + \frac{1}{n^2} \notin \left\{ \frac{1}{m} \mid m \in \mathbb{N} \right\}$. Assume that there exist $n, m \in \mathbb{N}$ such that

$$\frac{1}{n} + \frac{1}{n^2} = \frac{1}{m}$$

then $\frac{1}{m} = \frac{n+1}{n^2}$ so that

$$m = \frac{n^2}{n+1}$$

Now as $(n+1) \cdot (n-1) + 1 = n^2 - 1 + 1 = n^2$ so that

$$\frac{n^2}{n+1} = \frac{(n+1) \cdot (n-1) + 1}{n+1} = (n-1) + \frac{1}{n+1}$$

hence $\frac{n^2}{n+1} \notin \mathbb{N}$ and the assumption must be false. Hence we have that

$$\forall n \in \mathbb{N} \text{ that } \frac{1}{n} + \frac{1}{n^2} \notin \left\{ \frac{1}{m} \mid m \in \mathbb{N} \right\}$$

Take $\delta \in \mathbb{R}^+$ then as $0 < \delta$ there exist by [theorem: 10.30] a $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \frac{\delta}{2} < \delta$ and as $\frac{1}{n} < 1 \Rightarrow \frac{1}{n^2} < \frac{1}{n} < \frac{\delta}{2}$ we have that $0 < \frac{1}{n} + \frac{1}{n^2} < \delta$ and $0 < \frac{1}{n} < \delta$. Hence $\forall \delta \in \mathbb{R}^+$ there exist a $n \in \mathbb{N}$ such that $\frac{1}{n} \in B \subseteq A = A_0 \wedge 0 < \left| \frac{1}{n} \right| < \delta$ and a $\frac{1}{n} + \frac{1}{n^2} \in A \setminus B \wedge 0 < \left| \frac{1}{n} + \frac{1}{n^2} \right| < \delta$. As

$$\frac{f\left(0 + \frac{1}{n}\right) - f(0)}{\frac{1}{n}} = \frac{0 - 0}{\frac{1}{n}} = 0$$

and

$$\frac{f\left(0 + \frac{1}{n} + \frac{1}{n^2}\right) - f(0)}{\frac{1}{n} + \frac{1}{n^2}} = \frac{1 - 0}{\frac{1}{n} + \frac{1}{n^2}} > 0$$

Hence we can never find a single $v \in \mathbb{R}$ such that given a $\varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_0 = A$ with $0 < |h| < \delta$ we have

$$\left| \frac{f(x+h) - f(x) - h \cdot v}{h} \right| < \varepsilon \cdot |h|$$

So f has no derivative at 0. □

Using the locally principle we have another equivalence for the existence of derivatives on a open sets.

Corollary 16.39. *Let $\langle \mathbb{K}, || \rangle$ be the normed space of real [or complex numbers], $A \subseteq \mathbb{K}$, $U \in \mathcal{T}_{||}$ with $U \subseteq A$, $\langle X, ||| \rangle$ a normed space and $f: A \rightarrow X$ a function then we have the following equivalences:*

$$\begin{aligned} & f \text{ has derivatives on } U \\ & \Updownarrow \\ & \forall x \in U \text{ there exist a } V \in \mathcal{T}_{||} \text{ with } x \in V \subseteq U \text{ such that } f|_V \text{ has a derivative at } x \end{aligned}$$

Proof.

$\Rightarrow$. Let $x \in U$ then as f has derivatives on U it follows that f has a derivative at x , further by [theorem: 16.7] $x \in U'$. Hence by [theorem: 16.34] we must conclude that $f|_U$ has a derivative at x . So by taking $V = U$ we found a $V \in \mathcal{T}_{||}$ with $x \in V$ such that $f|_V$ has a derivative at x .

$\Leftarrow$. Let $x \in U$ then by the hypothesis there exist a $V \in \mathcal{T}_{||}$ with $x \in V \subseteq U$ such that $f|_V$ has a derivative at x . As V is open it follows from [theorem: 14.16] that $V = V^\circ$ so that $x \in V^\circ$. Using [theorem: 16.36] it follows then that f has a derivative at x . As $x \in U$ has been chosen arbitrary it follows that

$$f \text{ has derivatives on } U$$

□

Differentiating a function is a linear operation as is show in the following theorem.

Theorem 16.40. *Let $\langle \mathbb{K}, || \rangle$ be the normed space of real or complex numbers, $\langle X, ||| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ then we have:*

1. *If $f: A \rightarrow X$ and $g: A \rightarrow X$ are functions that have derivatives at x then $f + g$ has a derivative at x and $(f + g)'_x = f'_x + g'_x$.*
2. *If $k \in \mathbb{N}$, $\{f_i: A \rightarrow X\}_{i \in \{1, \dots, k\}}$ be a family of functions that has a derivative at x then $\sum_{i=1}^n f_i$ has a derivative at x with $(\sum_{i=1}^k f_i)'_x = \sum_{i=1}^k (f_i)'_x$*
3. *If $f: A \rightarrow X$ has a derivative f'_x at x and $\alpha \in \mathbb{K}$ then $\alpha \cdot f$ has a derivative at x and $(\alpha \cdot f)'_x = \alpha \cdot f'_x$.*

Proof.

1. As f has a derivative at x it follows that

$$x \in A'$$

Let $\varepsilon \in \mathbb{R}^+$. As f and g has a derivative at x it follows from [theorem: 16.22] that there exists $\delta_f, \delta_g \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } 0 < |h| < \delta_f \text{ we have } \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| < \frac{\varepsilon}{2}$$

$$\forall h \in A_x \text{ with } 0 < |h| < \delta_g \text{ we have } \left\| \frac{g(x+h) - g(x)}{h} - g'_x \right\| < \frac{\varepsilon}{2}$$

Let $\delta = \min(\delta_1, \delta_2)$ then we have $\forall h \in A_x$ with $0 < |h| < \delta$ that

$$\begin{aligned} \left\| \frac{(f+g)(x+h) - (f+g)(x)}{h} - (f'_x + g'_x) \right\| &= \\ \left\| \frac{f(x+h) - f(x)}{h} - f'_x + \frac{g(x+h) - g(x)}{h} - g'_x \right\| &\leq \\ \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| + \left\| \frac{g(x+h) - g(x)}{h} - g'_x \right\| &< \\ \frac{\varepsilon}{2} + \frac{\varepsilon}{2} &= \\ \varepsilon & \end{aligned}$$

which proves by [theorem: 16.22] that $f+g$ has a derivative $(f+g)'_x$ with

$$(f+g)'_x = f'_x + g'_x$$

2. We prove this by induction so define

$$S = \left\{ k \in \mathbb{N} \mid \text{If } \{f_i: A \rightarrow X\}_{i \in \{1, \dots, k\}} \text{ satisfies that } \forall i \in \{1, \dots, k\} \text{ } f_i \text{ has a derivative } (f_i)'_x \text{ at } x \text{ then } \sum_{i=1}^k f_i \text{ has a derivative at } x \text{ with } \left(\sum_{i=1}^k f_i \right)'_x = \sum_{i=1}^k (f_i)'_x \right\}$$

then we have

1 $\in S$. If $\{f_i|A \rightarrow Y\}_{i \in \{1\}}$ is a family of functions with a n -th derivative at x then as $\sum_{i=1}^1 f_i = f_1$ and it follows that $\sum_{i=1}^1 f_i$ has a n -th derivative at x , further

$$\left(\sum_{i=1}^1 f_i \right)'_x = (f_1)'_x = \sum_{i=1}^1 (f_i)'_x$$

So it follows that $1 \in S$.

$k \in S \Rightarrow k+1 \in S$. Let $\{f_i: A \rightarrow Y\}_{i \in \{1, \dots, k+1\}}$ be a family of functions that have a derivative at x then we have as $k \in S$ that $\sum_{i=1}^k f_i$ has a derivative at x with $(\sum_{i=1}^k f_i)'_x = \sum_{i=1}^k (f_i)'_x$. As also f_{k+1} has a n -th derivative at x it follows from (1) that $\sum_{i=1}^{k+1} f_i = (\sum_{i=1}^k f_i) + f_{k+1}$ has a n -th derivative at x with

$$\begin{aligned} \left(\sum_{i=1}^{k+1} f_i \right)'_x &= \left(\sum_{i=1}^k f_i + f_{k+1} \right)'_x \\ &\stackrel{(1)}{=} \left(\sum_{i=1}^k f_i \right)'_x + (f_{k+1})'_x \\ &\stackrel{n \in S}{=} \sum_{i=1}^k (f_i)'_x + (f_{k+1})'_x \\ &= \sum_{i=1}^{k+1} (f_i)'_x \end{aligned}$$

proving that

$$n+1 \in S$$

3. As f has a derivative at x it follows that

$$x \in A'$$

Let $\varepsilon \in \mathbb{R}^+$. As f has a derivative f'_x at x there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $0 < |h| < \delta$ we have

$$\left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| < \frac{\varepsilon}{|\alpha| + 1}$$

so that

$$\begin{aligned} \left\| \frac{(\alpha \cdot f)(x+h) - (\alpha \cdot f)(x)}{h} - \alpha \cdot f'_x \right\| &= \left\| \alpha \cdot \left(\frac{f(x+h) - f(x)}{h} - f'_x \right) \right\| \\ &= |\alpha| \cdot \left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\| \\ &< |\alpha| \cdot \frac{\varepsilon}{|\alpha| + 1} \\ &< \varepsilon \end{aligned}$$

which proves by [theorem: 16.22] that $\alpha \cdot f$ has a derivative $(\alpha \cdot f)'_x$ with

$$(\alpha \cdot f)'_x = \alpha \cdot f'_x$$

□

Theorem 16.41. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real numbers, $\langle X, \|\cdot\| \rangle$ a normed space over $\mathbb{K}$, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow \mathbb{K}$ and $g: A \rightarrow X$ functions that have a derivative at x then $f \cdot g$ has a derivative at x and

$$(f \cdot g)'_x = f(x) \cdot g'_x + f'_x \cdot g(x)$$

Note 16.42. The multiplication operator is the scalar product $\cdot: \mathbb{K} \times Y \rightarrow Y$. If $Y = \mathbb{K}$ then this reduce to the product operator in the field $\mathbb{K}$.

Proof. As f has a derivative at x it follows that

$$x \in A'$$

Let $\varepsilon \in \mathbb{R}^+$ then we have by [theorem: 16.23] that there exists $\delta_1, \delta_2 \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } |h| < \delta_1 \text{ we have } |f(x+h) - f(x) - h \cdot f'_x| \leq \frac{\varepsilon}{3 \cdot (1 + \|g'_x\|)} \cdot |h| \quad (16.8)$$

$$\forall h \in A_x \text{ with } |h| < \delta_2 \text{ we have } \|g(x+h) - g(x) - h \cdot g'_x\| \leq \frac{\varepsilon}{3 \cdot (1 + |f(x)|)} \cdot |h| \quad (16.9)$$

Further we have by [theorem: 16.33] that

$$f \text{ is continuous at } x \quad (16.10)$$

By continuity of f at x there exist a $\delta_3 \in \mathbb{R}^+$ such that $\forall y \in A$ with $|x - y| < \delta_3$ we have that $|f(y) - f(x)| < \min\left(1, \frac{1}{1 + \|g'_x\|}\right) \in \mathbb{R}^+$. If $h \in A_x$ with $|h| < \delta$ then $x+h \in A$ and $|(x+h) - x| < \delta_3$ so that we have

$$\forall h \in A_x \text{ with } |h| < \delta_3 \text{ we have } |f(x+h) - f(x)| < \frac{\varepsilon}{3 \cdot (1 + \|g'_x\|)} \quad (16.11)$$

and

$$\forall h \in A_x \text{ with } |h| < \delta_3 \text{ we have } |f(x+h)| \leq |f(x+h) - f(x)| + |f(x)| < 1 + |f(x)| \quad (16.12)$$

Let now $h \in A_x$ with $\delta = \min(\delta_1, \delta_2, \delta_3)$ that

$$\begin{aligned}
 & \| (f \cdot g)(x+h) - (f \cdot g)(x) - h \cdot (f'_x \cdot g(x) + f(x) \cdot g'_x) \| \\
 &= \left\| \underbrace{f(x+h) \cdot g(x+h)}_1 - \underbrace{f(x) \cdot g(x)}_2 - \underbrace{(h \cdot f'_x) \cdot g(x)}_3 - \underbrace{(h \cdot f(x)) \cdot g'_x}_4 \right\| \\
 &= \left\| \underbrace{f(x+h) \cdot g(x+h)}_1 - \underbrace{f(x+h) \cdot g(x)}_5 - \underbrace{(h \cdot f(x+h)) \cdot g'_x}_6 + \underbrace{f(x+h) \cdot g(x)}_5 + \right. \\
 &\quad \left. \underbrace{(h \cdot f(x+h)) \cdot g'_x}_6 - \underbrace{f(x) \cdot g(x)}_2 - \underbrace{(h \cdot f'_x) \cdot g(x)}_3 - \underbrace{(h \cdot f(x)) \cdot g'_x}_4 \right\| \\
 &= \left\| \underbrace{f(x+h) \cdot (g(x+h) - g(x) - h \cdot g'_x)}_{1,5,6} + \underbrace{(f(x+h) - f(x) - h \cdot f'_x) \cdot g(x)}_{5,2,3} + \right. \\
 &\quad \left. \underbrace{h \cdot (f(x+h) - f(x)) \cdot g'_x}_{6,4} \right\| \\
 &\leq |f(x+h)| \cdot \|g(x+h) - g(x) - h \cdot g'_x\| + |f(x+h) - f(x) - h \cdot f'_x| \cdot \|g(x)\| + \\
 &\quad |h| \cdot |f(x+h) - f(x)| \cdot \|g'_x\| \quad [\text{eq: 16.12}] \\
 &\leq (1 + |f(x)|) \cdot \|g(x+h) - g(x) - h \cdot g'_x\| + |f(x+h) - f(x) - h \cdot f'_x| \cdot \|g(x)\| + \\
 &\quad |h| \cdot |f(x+h) - f(x)| \cdot \|g'_x\| \quad [\text{eq: 16.9}] \\
 &\leq (1 + |f(x)|) \cdot \frac{\varepsilon}{3 \cdot (1 + |f(x)|)} \cdot |h| + |f(x+h) - f(x) - h \cdot f'_x| \cdot \|g(x)\| + \\
 &\quad |h| \cdot |f(x+h) - f(x)| \cdot \|g'_x\| \\
 &\leq \frac{\varepsilon}{3} \cdot |h| + |f(x+h) - f(x) - h \cdot f'_x| \cdot \|g(x)\| + |h| \cdot |f(x+h) - f(x)| \cdot \|g'_x\| \quad [\text{eq: 16.8}] \\
 &\leq \frac{\varepsilon}{3} \cdot |h| + \frac{\varepsilon}{3 \cdot (1 + \|g(x)\|)} \cdot \|g(x)\| \cdot |h| + |h| \cdot |f(x+h) - f(x)| \cdot \|g'_x\| \\
 &\leq \frac{\varepsilon}{3} \cdot |h| + \frac{\varepsilon}{3} \cdot |h| + |h| \cdot |f(x+h) - f(x)| \cdot \|g'_x\| \quad [\text{eq: 16.11}] \\
 &\leq \frac{\varepsilon}{3} \cdot |h| + \frac{\varepsilon}{3} \cdot |h| + |h| \cdot \frac{1}{3 \cdot (1 + \|g'_x\|)} \cdot \|g'_x\| \\
 &= \frac{\varepsilon}{3} \cdot |h| + \frac{\varepsilon}{3} \cdot |h| + \frac{\varepsilon}{3} \cdot |h| = \varepsilon \cdot |h|
 \end{aligned}$$

Hence we have that

$$f \cdot g \text{ has a derivative at } x \text{ with } (f \cdot g)'_x = f'_x \cdot g(x) + f(x) \cdot g'_x$$

□

16.1.3 Fréchet differential of a function

We want now to extend the definition of a derivative of a function $f: A \rightarrow X$ where $A \subseteq K$ to the general case where $A \subseteq X$ a normed space. As you can not divide by a $h \in X$ we can not define the extended derivative based on a limit of $\left\| \frac{f(x+h) - f(x)}{h} - f'_x \right\|_Y$. Hence we base our extension on the equivalent definition [theorem: 16.23] of the derivative where we do not use division by h .

Definition 16.43. (Fréchet approximation) Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $a \in A$ and $f: A \rightarrow Y$ a function then L is a **Fréchet approximation of f at x** if $L \in L(X, Y)$ and $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } \|h\|_X < \delta \text{ we have } \|f(x+h) - f(x) - L(h)\|_Y \leq \varepsilon \cdot \|h\|_X$$

Next we have to ensure that the **Fréchet approximation** of a function is unique. Unfortunately this is not as simple as requiring that x is a limit point as the following example shows.

Example 16.44. Consider $\langle \mathbb{R}^2, \|\cdot\|_2 \rangle, \langle \mathbb{R}, \|\cdot\| \rangle$, $A = \{(x, y) \in \mathbb{R}^2 | y = 0\}$ and take

$$f: A \rightarrow \mathbb{R} \text{ defined by } f((x, y)) = 0$$

then

$$(0, 0) \text{ is a limit point of } A$$

Further for

$$L_1: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ defined by } L_1((x, y)) = y$$

$$L_2: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ defined by } L_2((x, y)) = 2 \cdot y$$

we have that

$$L_1, L_2 \text{ are Fréchet approximations of } f \text{ at } (0, 0)$$

but

$$L_1 \neq L_2$$

Note that

$$\forall x \in A \text{ we have that } L_1(x) = 0 = L_2(x)$$

Proof. Let U be a open set such that $(0, 0) \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $(0, 0) \in B_{\|\cdot\|_2}((0, 0), \delta) \subseteq U$. Take $y = \left(\frac{\delta}{2}, 0\right)$ then $y \in A$ and $y \neq (0, 0)$. Further

$$\|y - (0, 0)\|_2 = \|(\delta/2, 0 - (0, 0))\|_2 = \|(f/2, 0)\|_2 = \delta/2 < \delta$$

so that $y \in (A \setminus \{(0, 0)\}) \cap B_{\|\cdot\|_2}((0, 0), \delta) \subseteq (A \setminus \{(0, 0)\}) \cap U$ which proves that

$$(0, 0) \text{ is a limit point of } U$$

If $(x, y), (x', y') \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$ then

$$\begin{aligned} L_1((x, y) + \alpha \cdot (x', y')) &= L_1((x + \alpha \cdot x', y + \alpha \cdot y')) \\ &= y + \alpha \cdot y' \\ &= L_1((x, y)) + \alpha \cdot L_1((x', y')) \\ L_2((x, y) + \alpha \cdot (x', y')) &= L_2((x + \alpha \cdot x', y + \alpha \cdot y')) \\ &= 2 \cdot (y + \alpha \cdot y') \\ &= 2 \cdot y + \alpha \cdot 2 \cdot y' \\ &= L_2((x, y)) + \alpha \cdot L_2((x', y')) \end{aligned}$$

so that

$$L_1, L_2 \in \text{Hom}(\mathbb{R}^2, \mathbb{R})$$

Further

$$\begin{aligned} |L_1(x, y)| &= |y| \\ &\leq \max(|x|, |y|) \\ &= \|(x, y)\|_2 \\ |L_2(x, y)| &= |2 \cdot y| \\ &\leq 2 \cdot \max(|x|, |y|) \\ &= 2 \cdot \|(x, y)\|_2 \end{aligned}$$

which proves that

$$L_1, L_2 \in L(\mathbb{R}^2, \mathbb{R})$$

Let $\varepsilon \in \mathbb{R}^+$ then for $(h_1, h_2) \in A_{(0,0)}$ with $\|(h_1, h_2)\|_2 < 1$ we have $(h_1, h_2) = (h_1, h_2) + (0, 0) \in A$ from which it follows that $h_2 = 0$. Hence we have

$$|f((0, 0) + (h_1, h_2)) - f((0, 0)) - L_1((h_1, h_2))| = |0 - 0 - h_2|_{h_2=0} = 0 \leq \varepsilon \cdot \|(h_1, h_2)\|_2$$

$$|f((0, 0) + (h_1, h_2)) - f((0, 0)) - L_2((h_1, h_2))| = |0 - 0 - 2 \cdot h_2|_{h_2=0} = 0 \leq \varepsilon \cdot \|(h_1, h_2)\|_2$$

so

L_1, L_2 are Fréchet approximation of f at $(0,0)$

Further as $L_1((1,1)) = 1 \neq 2 = L_2((1,1))$ it follows that

$$L_1 \neq L_2 \quad \square$$

In books about optimization this problem is solved by the concept of a tangent cone. The basic idea is to find given the domain of the function and the point to differentiate at and find the set of points in X where the Fréchet approximations are the same and define then the differential to be the restriction of a Fréchet approximation to this set. In the above example we have that $(L_1)|_A = (L_2)|_A$. The problem with this approach is that it is extremely complex and is not easy to extend to higher order differentiation. So the approach used in this book is to limit differentiability to interior points [see definition: 14.12] of the domain of a function.

Lemma 16.45. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A^\circ$ and $f: A \rightarrow Y$ a function then if $L_1, L_2 \in \mathcal{L}[X; Y]$ two Fréchet approximations of f at x then $L_1 = L_2$.*

Proof. Assume that $L_1 \neq L_2$ then there exist a $h_0 \in X$ such that $L_1(h_0) \neq L_2(h_0)$ hence we have that

$$\varepsilon = \|L_1(h_0) - L_2(h_0)\|_Y \in \mathbb{R}^+$$

Now using the hypothesis there exists $\delta_1, \delta_2 \in \mathbb{R}^+$ such that

$$\forall k \in U_x \text{ with } \|k\|_X < \delta_1 \text{ we have } \|f(x+k) - f(x) - L_1(k)\|_Y \leq \frac{\varepsilon}{2 \cdot (\|h_0\|_X + 1)} \cdot \|k\| \quad (16.13)$$

and

$$\forall k \in U_x \text{ with } \|k\|_X < \delta_2 \text{ we have } \|f(x+k) - f(x) - L_2(k)\|_Y \leq \frac{\varepsilon}{2 \cdot (\|h_0\|_X + 1)} \cdot \|k\| \quad (16.14)$$

As $x \in A^\circ$ there exist by [definition: 14.12] there exist a open set U such that $x \in U \subseteq A$ hence there exist a $\delta_3 \in \mathbb{R}^+$ so that $x \in B_{\|\cdot\|_X}(x, \delta_3) \subseteq U \subseteq A$. Let $k \in B_{\|\cdot\|_X}(0, \delta_3)$ then $\|(x+k) - x\|_X = \|k\|_X < \delta_3$ so that $x+k \in B_{\|\cdot\|_X}(x, \delta_3) \subseteq A$ which proves that

$$0 \in B_{\|\cdot\|_X}(0, \delta_3) \subseteq A_x$$

Take $\delta = \min(\delta_1, \delta_2, \delta_3)$ and define $h_1 = \frac{\delta}{(\|h_0\|_X + 1)} \cdot h_0$ then

$$\|h_1\|_X = \left\| \frac{\delta}{(\|h_0\|_X + 1)} \cdot h_0 \right\|_X = \delta \cdot \frac{\|h_0\|_X}{\|h_0\|_X + 1} < \delta \leq \delta_1, \delta_2, \delta_3 \quad (16.15)$$

so that $h_1 \in A_x$ and

$$\begin{aligned} & \frac{\delta}{\|h_0\|_X + 1} \cdot \|L_1(h_0) - L_2(h_0)\|_Y &= \\ & \left\| \frac{\delta}{\|h_0\|_X + 1} \cdot L_1(h_0) - \frac{\delta}{\|h_0\|_X + 1} \cdot L_2(h_0) \right\|_Y &= \\ & \left\| L_1\left(\frac{\delta}{\|h_0\|_X + 1} \cdot h_0\right) \right\|_Y + \left\| L_2\left(\frac{\delta}{\|h_0\|_X + 1} \cdot h_0\right) \right\|_Y &= \\ & \|L_1(h_1) - L_2(h_1)\|_Y &= \\ & \|f(x+h_1) - f(x) - L_2(h_1) - (f(x+h_1) - f(x) - L_1(h_1))\|_Y &\leq \\ & \|f(x+h_1) - f(x) - L_2(h_1)\|_Y + \|f(x+h_1) - f(x) - L_1(h_1)\|_Y &\leq \\ & \frac{\varepsilon}{2 \cdot (\|h_0\|_X + 1)} \cdot \|h_1\|_X + \frac{\varepsilon}{2 \cdot (\|h_0\|_X + 1)} \cdot \|h_1\|_X &= \\ & \frac{\varepsilon}{\|h_0\|_X + 1} \cdot \|h_1\|_X &= \\ & \frac{\varepsilon}{\|h_0\|_X + 1} \cdot \left\| \frac{\delta}{(\|h_0\|_X + 1)} \cdot h_0 \right\|_X &= \\ & \frac{\varepsilon}{\|h_0\|_X + 1} \cdot \frac{\delta}{(\|h_0\|_X + 1)} \cdot \|h_0\|_X &= \end{aligned}$$

[eqs: 16.13, 16.14, 16.15]

so that after dividing both sides by $\frac{\delta}{(\|h_0\|_X + 1)} > 0$

$$\|L_1(h_0) - L_2(h_0)\|_Y \leq \frac{\varepsilon}{\|h_0\|_X + 1} \cdot \|h_0\|_X < \varepsilon = \|L_1(h) - L_2(h)\|_Y$$

a contradiction. So we must have

$$L_1 = L_2$$

□

We can now define the Fréchet differential of a function at a point.

Definition 16.46. (Fréchet differentiability) Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $x \in A$ then a function

$$f: A \rightarrow Y$$

is Fréchet differentiable at x if $x \in A^\circ$ and there exist a $L \in L(X, Y)$ such that L is a Fréchet approximation of f at x . Using [definition: 16.43] this is equivalent with

$$x \in A^\circ$$

and

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall h \in A_x \text{ with } \|h\|_X < \delta \text{ we have } \|f(x+h) - f(x) - L(x)\|_Y \leq \varepsilon \cdot \|h\|_X$$

The unique L [see lemma: 16.45] is called the **Fréchet differential of f at x** and noted as $D_x f$.

Note 16.47. The requirement $x \in A^\circ$ is needed to ensure that $D_x f$ is well defined.

Note that the Fréchet differential $D_x f$ at $x \in A^\circ$ is a element of $L(X, Y)$ hence $D_x f$ is a continuous linear function. However we can also look at a different function who maps x to $D_x f$ which is the differential of f .

Definition 16.48. (Differential) Let $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $f: A \rightarrow Y$ a function. Then we define

$$\mathfrak{D}_f = \{x \in A \mid f \text{ is Fréchet differentiable at } x\} \subseteq A$$

and

$$Df: \mathfrak{D}_f \rightarrow L(X, Y) \text{ where } Df(x) = D_x f$$

Df is called the differential of f .

Note 16.49. As the differentiability of f at x implies by definition that $x \in A^\circ$ it follows that

$$\mathfrak{D}_f \subseteq A^\circ \subseteq A$$

Note 16.50. If $\mathfrak{D}_f = \emptyset$ then Df is the empty function [see example: 2.46].

Note 16.51. The benefit of this definition is that

1. $Df: \mathfrak{D}_f \rightarrow Y$ is always defined even if f is never differentiable at a point in A .
2. $\mathfrak{D}_f$ is the set of all elements of the domain of f for which f is differentiable at hence it is the most natural definition of the domain of Df .
3. If we refer to Df where $f: A \rightarrow Y$ is a function then we can automatically assume that Df is the graph of the function

$$Df: \mathfrak{D}_f \rightarrow Y$$

Definition 16.52. Let $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{\|X\|}$ with $U \subseteq A$ and $f: A \rightarrow Y$ a function then we say

f is Fréchet differentiable on U

iff

$\forall x \in U$ we have that f is Fréchet differentiable at x

In the special case where f is Fréchet differentiable on its domain we say that

f is Fréchet differentiable

Example 16.53. Let $\langle X, \|\cdot\| \rangle$ be a normed space, U a open set in X and

$$\text{Id}_U: U \rightarrow X \text{ defined by } \text{Id}_U(x) = x$$

then

$$\mathfrak{D}_{\text{Id}_U} = U \text{ in other words } \text{Id}_U \text{ is Fréchet differentiable on } U$$

and for

$$D\text{Id}_U: \mathfrak{D}_{\text{Id}_U} \rightarrow L(X, Y) \text{ where } D\text{Id}_U(x) = D_x\text{Id}_U$$

we have

$$D\text{Id}_U = C_{\text{Id}_X}$$

Proof. Using [example: 14.215] it follows that

$$\text{Id}_X \in L(X, Y)$$

As U is open we have by [theorem: 14.16] that $U = U^\circ$. Let $x \in U$ and take $\varepsilon \in \mathbb{R}^+$ then for $h \in U_x$ with $\|h\|_X < 1$ we have

$$\|\text{Id}_U(x+h) - \text{Id}_U(x) - \text{Id}_X(h)\|_Y = \|x+h - x - h\|_Y = 0 \leq \varepsilon \cdot \|h\|_x$$

so that $D_x\text{Id}_U = \text{Id}_X$. Hence

$$\mathfrak{D}_{\text{Id}_U} = U$$

and $\forall x \in U$ we have $D(\text{Id}_U)(x) = D_x\text{Id}_U = \text{Id}_X = C_{\text{Id}_X}(x)$ which proves that

$$D(\text{Id}_U) = C_{\text{Id}_X} \quad \square$$

Example 16.54. Let $\langle X, \|\cdot\| \rangle, \langle Y, \|\cdot\|_Y \rangle$ be a normed space, U a open set in X , $y \in Y$

$$C_y: U \rightarrow Y \text{ defined by } C_y(x) = y$$

then

$$\mathfrak{D}_{C_y} = U \text{ in other words } C_y \text{ is Fréchet differentiable}$$

and for

$$DC_y: \mathfrak{D}_{C_y} \rightarrow L(X, Y) \text{ where } D(C_y)(x) = D_xC_y$$

we have

$$DC_y = C_0$$

Proof. Using [example: 14.200] it follows that

$$C_0 \in L(X, Y)$$

As U is open we have by [theorem: 14.16] that $U = U^\circ$. Let $x \in U$ and take $\varepsilon \in \mathbb{R}^+$ then for $h \in U_x$ with $\|h\|_X < 1$ we have

$$\|C_y(x+h) - C_y(x) - C_0(h)\|_Y = \|y - y - 0\|_Y = 0 \leq \varepsilon \cdot \|h\|_x$$

so that $D_xC_y = C_0$. Hence

$$\mathfrak{D}_{C_y} = U$$

and $\forall x \in U$ we have $D(C_y)(x) = D_xC_y = C_0 = C_{C_0}(x)$ which proves that

$$D(C_y) = C_{C_0} \quad \square$$

Example 16.55. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle, L \in L(X, Y)$ then we have

$$\mathfrak{D}_L = X \text{ in other words } L \text{ is Fréchet differentiable}$$

and for

$$DL: X \rightarrow X \text{ defined by } DL(x) = D_x L$$

that

$$DL = C_L$$

Proof. As X is open we have by [theorem: 14.16] that $X = X^\circ$. Let $x \in X$ and take $\varepsilon \in \mathbb{R}^+$ then for $h \in X_x$ with $\|h\|_X < 1$ we have

$$\|L(x+h) - L(x) - L(h)\|_Y = \|L(x) + L(h) - L(x) - L(h)\|_Y = 0 \leq \varepsilon \cdot \|h\|_x$$

so that $D_x L = C_0$. Hence

$$\mathfrak{D}_{C_y} = X$$

and $\forall x \in X$ we have $DL(x) = D_x L = L = C_L(x)$ which proves that

$$DL = C_L \quad \square$$

Although Fréchet differentiability and the differential is defined using norms, the definition is actually dependent on the topology generated by the norms. So if two norms generate the same topology Fréchet differentiability is independent of the chosen norm.

Theorem 16.56. Let X, Y be vector spaces over $\mathbb{K}$ with equivalent norms $\|\cdot\|_{X_1}, \|\cdot\|_{X_2}$ on X and $\|\cdot\|_{Y_1}, \|\cdot\|_{Y_2}$ on Y , $A \subseteq X$, $x \in A$ [using the topology $\mathcal{T}_{\|\cdot\|_{X_1}}$ norms are equivalent $\mathcal{T}_{\|\cdot\|_{X_2}}$] and $f: A \rightarrow X$ a function then we have that if

f is Fréchet differentiable function at x with differential $D_x f$ at x using $\|\cdot\|_{X_1}$ and $\|\cdot\|_{Y_1}$

then

f is Fréchet differentiable function at x with differential $D_x f$ at x using $\|\cdot\|_{X_2}$ and $\|\cdot\|_{Y_2}$

Proof. As the norms are equivalent we have by [theorem: 14.121] there exists $\alpha_X, \alpha_Y, \beta_X, \beta_Y \in \mathbb{R}^+$ such that $\forall x \in X$ and $\forall y \in Y$ we have

$$\alpha_X \cdot \|x\|_{X_2} \leq \|x\|_{X_1} \leq \beta_X \cdot \|x\|_{X_2} \text{ and } \alpha_Y \cdot \|y\|_{Y_1} \leq \|y\|_{Y_2} \leq \beta_Y \cdot \|y\|_{Y_1}$$

As f is Fréchet differentiable at x using $\|\cdot\|_{X_1}, \|\cdot\|_{Y_1}$ it follows that $x \in A^\circ$ [using $\mathcal{T}_{\|\cdot\|_{X_1}}$], as the norms are equivalent we have that $\mathcal{T}_{\|\cdot\|_{X_1}} = \mathcal{T}_{\|\cdot\|_{X_2}}$ so that

$$x \in A^\circ \text{ [using } \mathcal{T}_{\|\cdot\|_{X_1}}]$$

Let $\varepsilon \in \mathbb{R}^+$ then there exist a $\delta' \in \mathbb{R}^+$ such that $\forall h \in U_x$ with $\|h\|_{X_1} < \delta'$ we have

$$\|f(x+h) - f(x) - D_x f\|_{Y_1} \leq \frac{\varepsilon}{\beta_Y \cdot \beta_X} \cdot \|h\|_{X_1}$$

Take $\delta = \frac{\delta'}{\beta_X}$ then $\forall h \in U_x$ with $\|h\|_{X_2} < \delta$ we have $\|h\|_{X_1} \leq \beta_X \cdot \|h\|_{X_2} < \delta \cdot \beta_X = \delta'$ so that from the above we have

$$\begin{aligned} \|f(x+h) - f(x) - D_x f\|_{Y_1} &\leq \frac{\varepsilon}{\beta_Y \cdot \beta_X} \cdot \|h\|_{X_1} \\ &\leq \frac{\varepsilon}{\beta_Y \cdot \beta_X} \cdot \beta_X \cdot \|h\|_{X_2} \\ &= \frac{\varepsilon}{\beta_Y} \cdot \|h\|_{X_2} \end{aligned}$$

so that

$$\|f(x+h) - f(x) - D_x f(h)\|_{Y_2} \leq \beta_Y \cdot \|f(x+h) - f(x) - D_x f(h)\|_{Y_1} \leq \beta_Y \cdot \frac{\varepsilon}{\beta_Y} \cdot \|h\|_{X_2} = \varepsilon \cdot \|h\|_{X_2}$$

This together with the fact $D_x f$ is also a continuous linear function between X and Y using the norms $\|\cdot\|_{X_2}, \|\cdot\|_{Y_2}$ [because the norms are equivalent] proves that f is Fréchet differentiable at x with differential $D_x f$ using the norms $\|\cdot\|_{X_2}, \|\cdot\|_{Y_2}$. $\square$

Differentiability of a function at a point is a local property it only depends on the function values for a arbitrary small open neighborhood of this point not on the domain of the function.

Theorem 16.57. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $B \subseteq A$, $x \in B$ and $f: A \rightarrow Y$ is a function then we have*

$$x \in B^\circ \text{ and } f \text{ is Fréchet differentiable at } x$$

$$\Updownarrow$$

$$f|_B \text{ is Fréchet differentiable at } x$$

Further for either of the above cases we have $D_x f = D_x f|_B$.

Note 16.58. *If B is open in X then $B \stackrel{[\text{theorem: 14.16}]}{=} B^\circ$ so that $x \in B^\circ$ in the above is automatically satisfied and can be removed.*

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$. Then, as f is Fréchet differentiable at x with differential $D_x f \in L(X, Y)$, there exist a $\delta \in \mathbb{R}^+$ such that $\forall y \in A_x$ with $\|h\|_X < \delta$ we have

$$\|f(x+h) - f(x) - D_x f(h)\|_Y \leq \varepsilon \cdot \|h\|_X$$

Let $h \in B_x \Rightarrow x+h \in B \subseteq A \Rightarrow h \in B_x$ so if $\|h\|_X < \delta$ we have

$$\|f|_B(x+h) - f|_B(x) - D_x f(h)\|_Y \stackrel{=}{=}_{x, x+h \in B} \|f(x+h) - f(x) - D_x f(h)\| \leq \varepsilon \cdot \|h\|_X$$

which proves, as $x \in B^\circ$, that $f|_B$ is Fréchet differentiable at x with $D_x f|_B = D_x f$.

$\Leftarrow$. As $f|_B$ is Fréchet differentiable at x we have by [definition: 16.46] that

$$x \in B^\circ$$

As $x \in B^\circ$ there exist a U open such that $x \in U \subseteq B$ hence there exist a $\delta_1 \in \mathbb{R}^+$ such that

$$x \in B_{\|\cdot\|_X}(x, \delta_1) \subseteq B \subseteq A$$

Let $\varepsilon \in \mathbb{R}^+$. As $f|_B$ is Fréchet differentiable at x with differential $D_x f|_B$ there exist a $\delta_2 \in \mathbb{R}^+$ such that $\forall h \in B_x$ with $\|h\|_X < \delta$ we have

$$\|f|_B(x+h) - f|_B(x) - D_x f|_B(h)\|_Y \leq \varepsilon \cdot \|h\|_X$$

Take $\delta = \min(\delta_1, \delta_2) \in \mathbb{R}^+$ then we have $\forall h \in U_x$ with $\|h\|_X \leq \delta$ that

$$\|(x+h) - x\|_X = \|h\|_X < \delta \leq \delta_1 \Rightarrow x+h \in B_{\|\cdot\|_X}(x, \delta_1) \subseteq B \Rightarrow h \in B_x$$

so as also $\|h\|_X < \delta \leq \delta_2$ we have that

$$\|f(x+h) - f(x) - D_x f_V(x)\|_Y \stackrel{=}{=}_{x, x+h \in B} \|f|_B(x+h) - f|_B(x) - D_x f|_B(h)\|_Y \leq \varepsilon \cdot \|h\|_X$$

which proves, as $x \in B^\circ \subseteq A^\circ$, that f is Fréchet differentiable at x with $D_x f = D_x f|_B$. $\square$

Using the locally principle we have another way to specify that a function is differentiable on a set.

Theorem 16.59. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U \in \mathcal{T}_X$ with $U \subseteq A$ and $f: A \rightarrow Y$ a function then we have the following equivalences*

$$f \text{ is Fréchet differentiable on } U$$

$$\Updownarrow$$

$$\forall x \in U \text{ there exist a } V \in \mathcal{T}_X \text{ with } x \in V \subseteq U \text{ such that } f|_V \text{ is Fréchet differentiable at } x$$

Proof.

$\Rightarrow$. Let $x \in U$ then as f is Fréchet differentiable on U it follows that f is Fréchet differentiable at x . As U is open we have by [theorem: 14.16] that $U = U^\circ$ so that $x \in U^\circ \subseteq A$. Hence using [theorem: 16.57] it follows that $f|_U$ is Fréchet differentiable at x . Taking $V = U$ we have that $x \in V \subseteq U$ such that $f|_V$ is Fréchet differentiable at x .

$\Leftarrow$. Let $x \in U$ then by the hypothesis there exist a $V \in \mathcal{T}_X$ with $x \in V \subseteq U$ such that $f|_V$ is Fréchet differentiable at x . As V is open have by [theorem: 14.16] that $V = V^\circ$ so that $x \in V^\circ \subseteq A$. Hence using [theorem: 16.57] it follows that f is Fréchet differentiable at x which, as $x \in U$ has been chosen arbitrary, proves that

f is Fréchet differentiable on U

□

We look now at alternative definitions of Fréchet differentiability at a point, first we introduce the concept of a ε -mapping.

Definition 16.60. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $0 \in A$ then a ε -mapping is a function $\varepsilon: A \rightarrow Y$ which is continuous at $0 \in A$ [using the subspace topology on A] and satisfies $\varepsilon(0) = 0$.

Theorem 16.61. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X, x \in A$ and $f: A \rightarrow Y$ a function then the following are equivalent for $L \in L(X, Y)$

1. L is a Fréchet approximation of f at x [see definition: 16.43]

2. There exist a ε -mapping $\varepsilon: A_x \rightarrow Y$ such that $\forall h \in A_x$ we have

$$f(x+h) - f(x) - L(h) = \|h\|_X \cdot \varepsilon(h)$$

3. $\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $0 < \|h\|_X < \delta$ we have

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} < \varepsilon$$

4. $\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+$ such that $\forall y \in A \setminus \{x\}$ with $\|y - x\|_X < \delta$ we have

$$\frac{\|f(y) - f(x) - L(y-x)\|_Y}{\|y-x\|_X} < \varepsilon$$

5. $\forall \{h_i\}_{i \in \mathbb{N}} \subseteq A_x$ with $\forall i \in \mathbb{N} 0 < \|h_i\|_X$ and $\lim_{i \rightarrow \infty} h_i = 0$ we have

$$\lim_{n \rightarrow \infty} \frac{\|f(x+h_i) - f(x) - L(h_i)\|_Y}{\|h_i\|_X} = 0$$

Hence

f is Fréchet differentiable at x

$\Updownarrow$

$x \in A^\circ$ and either 1, 2, 3, 4 or 5 is true with $L = D_x f$

Proof.

1 $\Rightarrow$ 2. Define

$$\varepsilon: A_x \rightarrow Y \text{ by } \varepsilon(h) = \begin{cases} 0 & \text{if } h = 0 \\ \frac{f(x+h) - f(x) - L(h)}{\|h\|_X} & \text{if } h \in A_x \setminus \{0\} \end{cases}$$

then we have $\forall h \in A_x$ that

$$\begin{aligned} \|h\|_X \cdot \varepsilon(h) &= \begin{cases} 0 & \text{if } h = 0 \\ f(x+h) - f(x) - L(h) & \text{if } h \in A_x \setminus \{0\} \end{cases} \\ &= \begin{cases} f(x+0) - f(x) - L(0) & \text{if } h = 0 \\ f(x+h) - f(x) - L(h) & \text{if } h \in A_x \setminus \{0\} \end{cases} \\ &= f(x+h) - f(x) - L(h) \end{aligned}$$

proving that

$$\forall h \in U_x \text{ we have } f(x+h) - f(x) - L(h) = \|h\|_X \cdot \varepsilon(h)$$

Rest us to prove that ε is continuous at 0. Let $\zeta \in \mathbb{R}^+$. As L is a Fréchet differentiable approximation of f at x there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $\|h\|_X < \delta$ we have

$$\|f(x+h) - f(x) - L(h)\|_Y < \zeta \cdot \|h\|_X$$

Hence we have

$$\begin{aligned} \|\varepsilon(h) - \varepsilon(0)\|_Y &= \|\varepsilon(h)\|_Y \\ &= \left\| \begin{cases} \|0\|_Y < \zeta \text{ if } h=0 \\ \frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} < \zeta \text{ if } h \in A_x \setminus \{0\} \end{cases} \right\|_Y \\ &< \zeta \end{aligned}$$

which by [theorem: 14.136] proves that ε is continuous at 0 in the subspace topology of A_x .

2 $\Rightarrow$ 3. Let $\varepsilon: A_x \rightarrow Y$ be a ε -mapping such that $\forall h \in A_x$

$$f(x+h) - f(x) - L(h) = \|h\|_X \cdot \varepsilon(h)$$

Take $\zeta \in \mathbb{R}^+$. As $\varepsilon: A_x \rightarrow Y$ is continuous we can use [theorem: 14.136] to find a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $\|h\|_X = \|h - 0\|_X < \delta$ we have $\|\varepsilon(h)\|_Y = \|\varepsilon(h) - \varepsilon(0)\|_Y < \zeta$. Hence if $0 < \|h\|_X < \delta$ then

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} = \frac{\|h\|_X \cdot \varepsilon(h)}{\|h\|_X} = \varepsilon(h) < \zeta$$

3 $\Rightarrow$ 1. Assume that $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ so that $\forall h \in A_x$ with $0 < \|h\|_X < \delta$ we have

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} < \varepsilon$$

Now $\forall h \in A_x$ with $\|h\|_X < \delta$ we have two cases to consider:

$\|h\|_X = 0$. Then $h = 0$ and

$$\|f(x+h) - f(x) - L(h)\|_Y = \|f(x+0) - f(x) - L(0)\|_Y = \|0\|_Y = \varepsilon \cdot 0 = \varepsilon \cdot \|h\|_X$$

$0 < \|h\|_X$. Then

$$\|f(x+h) - f(x) - L(h)\|_Y = \|h\|_X \cdot \frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} \leq \|h\|_X \cdot \varepsilon = \varepsilon \cdot \|h\|_X$$

so that in all cases we have

$$\|f(x+h) - f(x) - L(h)\|_Y \leq \varepsilon \cdot \|h\|_X$$

Hence L is a Fréchet approximation of f at x .

3 $\Rightarrow$ 4. Assume that $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ so that $\forall h \in A_x$ with $0 < \|h\|_X < \delta$ we have

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} < \varepsilon \tag{16.16}$$

Let $y \in A \setminus \{x\} \Rightarrow y - x \in A_x$ [as $(y-x) + x = y \in A$] such that $0 < \|y-x\| < \delta$ then we have

$$\frac{\|f(y) - f(x) - L(y-x)\|_Y}{\|y-x\|_X} = \frac{\|f(x+(y-x)) - f(x) - L(y-x)\|_Y}{\|(y-x)\|_X} \underset{[\text{eq 16.16}]}{<} \varepsilon$$

4 $\Rightarrow$ 3. Assume that $\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+$ such that $\forall y \in A \setminus \{x\}$ with $\|y-x\|_X < \delta$ we have

$$\frac{\|f(y) - f(x) - L(y-x)\|_Y}{\|y-x\|_X} < \varepsilon$$

Hence if $h \in A_x$ with $0 < \|h\|_X < \delta$ we have, if we take $y = x + h$, that $0 < \|y - x\|_X < \delta$ and $y \in A \setminus \{x\}$ so that

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} = \frac{\|f(y) - f(x) - L(y-x)\|_Y}{\|y-x\|_X} < \varepsilon$$

3 $\Rightarrow$ 5. Let $\{h_i\}_{i \in \mathbb{N}} \subseteq A_x$ be a sequence with $\forall i \in \mathbb{N} \ 0 < \|h_i\|_X$ such that $\lim_{i \rightarrow \infty} h_i = 0$. Take $\varepsilon \in \mathbb{R}^+$ then by (3) there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in U_x$ with $0 < \|h\|_X < \delta$ we have

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} < \varepsilon$$

As $\lim_{i \rightarrow \infty} h_i = 0$ there exist a $N \in \mathbb{N}$ such that $\forall i \geq N$ we have that $0 < \|h_i\|_X = \|h_i - 0\|_X < \delta$, so that

$$\left| \frac{\|f(x+h_i) - f(x) - L(h_i)\|_Y}{\|h_i\|_X} - 0 \right| = \frac{\|f(x+h_i) - f(x) - L(h_i)\|_Y}{\|h_i\|_X} < \varepsilon$$

This proves that

$$\lim_{i \rightarrow \infty} \frac{\|f(x+h_i) - f(x) - L(h_i)\|_Y}{\|h_i\|_X} = 0$$

5 $\Rightarrow$ 3. Assume that (3) is false. Then there exist a $\varepsilon_0 \in \mathbb{R}^+$ such that $\forall \delta \in \mathbb{R}^+$ there exist a $h \in A_x$ with $0 < \|h\|_X < \delta$ such that

$$\frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} > \varepsilon_0$$

Using [theorem: 3.108] it follows that $\forall n \in \mathbb{N}$ we have, as $\frac{1}{n} \in \mathbb{R}^+$, that there exist a $h_n \in A_x$ with $0 < \|h_n\|_X < \frac{1}{n}$ such that

$$\frac{\|f(x+h_n) - f(x) - L(h_n)\|_Y}{\|h_n\|_X} > \varepsilon_0 \quad (16.17)$$

Now using [example: 14.349] together with [theorem: 14.355] it follows that

$$\lim_{n \rightarrow \infty} h_n = 0$$

Hence by (5) we have

$$\lim_{n \rightarrow \infty} \frac{\|f(x+h_n) - f(x) - L(h_n)\|_Y}{\|h_n\|_X} = 0$$

So there exist a $N \in \mathbb{N}$ such that

$$\frac{\|f(x+h_N) - f(x) - L(h_N)\|_Y}{\|h_N\|_X} < \varepsilon_0$$

contradicting [eq: 16.17]. Hence the assumption is wrong and (3) follows.

To summarize we have

$$1 \Leftrightarrow 2 \Leftrightarrow 3 \Leftrightarrow 4 \Leftrightarrow 5 \quad (16.18)$$

Finally we have:

$\Rightarrow$. If f is Fréchet differentiable at x then $x \in A^\circ$ and (1) is true hence by [eq: 16.18] $x \in A^\circ$ and either 1,2,3,4 or 5 is true with $L = D_x f$.

$\Leftarrow$. If $x \in A^\circ$ and either 1,2,3,4,5 is true then by [eq: 16.18] $x \in A^\circ$ and 1 is true $\square$

The following corollary will be used later in the proof of the inverse function theorem.

Corollary 16.62. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $U \subseteq X$ a open set, $x \in U$ and $f: U \rightarrow Y$ a function then the following are equivalent:

1. f is Fréchet differentiable at x

2. There exist a $L \in L(X, Y)$, a $\delta \in \mathbb{R}^+$ such that $B_{\|\cdot\|_X}(x, \delta) \subseteq U$ and a ε -mapping $\varepsilon: B_{\|\cdot\|_X}(0, \delta) \rightarrow Y$ such that

$$\forall h \in B_{\|\cdot\|_X}(0, \delta) \quad f(x+h) - f(x) - L(h) = \|h\|_X \cdot \varepsilon(h)$$

Further if (1) or (2) is true then $D_x f = L$

Proof.

1 $\Rightarrow$ 2. As $x \in U$ and U is open we have that there exists a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{\|\cdot\|_X}(x, \delta) \subseteq U$$

As $B_{\|\cdot\|_X}(x, \delta)$ is open it follows from [theorem: 14.16] that

$$x \in (B_{\|\cdot\|_X}(x, \delta))^\circ \subseteq U$$

so that by [theorem: 16.57] and the fact that f is Fréchet differentiability at x we have

$$f|_{B_{\|\cdot\|_X}(x, \delta)} \text{ is Fréchet differentiable at } x \text{ with } D_x f|_{B_{\|\cdot\|_X}(x, \delta)} = D_x f$$

Using the previous theorem [theorem: 16.61(2)] there exist a ε -mapping

$$\varepsilon: (B_{\|\cdot\|_X}(x, \delta))_x \stackrel{[\text{theorem: 16.17}]}{=} B_{\|\cdot\|_X}(0, \delta) \rightarrow Y$$

such that $\forall h \in (B_{\|\cdot\|_X}(x, \delta))_x \stackrel{[\text{theorem: 16.17}]}{=} B_{\|\cdot\|_X}(0, \delta)$ we have

$$f|_{B_{\|\cdot\|_X}(x, \delta)}(x+h) - f|_{B_{\|\cdot\|_X}(x, \delta)}(x) - D_x f|_{B_{\|\cdot\|_X}(x, \delta)}(h) = \|h\|_X \cdot \varepsilon(h)$$

As $x, x+h \in B_{\|\cdot\|_X}(x, \delta)$ and $D_x f|_{B_{\|\cdot\|_X}(x, \delta)} = D_x f$ we have

$$f(x+h) - f(x) - D_x f(h) = \|h\|_X \cdot \varepsilon(h)$$

Hence $L = D_x f$ is the required $L \in L(X, Y)$.

2 $\Rightarrow$ 1. By (2) there exist a $L \in L(X, Y)$ and a ε -mapping

$$\varepsilon: B_{\|\cdot\|_X}(0, \delta) \stackrel{[\text{theorem: 16.17}]}{=} (B_{\|\cdot\|_X}(x, \delta))_x \rightarrow Y$$

such that

$$\forall h \in B_{\|\cdot\|_X}(0, \delta) \stackrel{[\text{theorem: 16.17}]}{=} (B_{\|\cdot\|_X}(x, \delta))_x$$

we have

$$f|_{B_{\|\cdot\|_X}(x, \delta)}(x+h) - f|_{B_{\|\cdot\|_X}(x, \delta)}(x) - L(h) = f(x+h) - f(x) - L(h) = \|h\|_X \cdot \varepsilon(h)$$

so that by [theorem: 16.61(2)] $f|_{B_{\|\cdot\|_X}(x, \delta)}$ is Fréchet differentiable with

$$L = D_x f|_{B_{\|\cdot\|_X}(x, \delta)}$$

As $B_{\|\cdot\|_X}(x, \delta)$ is open it follows from [theorem: 14.16] that

$$x \in (B_{\|\cdot\|_X}(x, \delta))^\circ \subseteq U$$

Using then [theorem: 16.57] it follows that f is differentiable at x with

$$L = D_x f|_{B_{\|\cdot\|_X}(x, \delta)} = D_x f$$

□

Differentiability implies continuity as the following theorem proves.

Theorem 16.63. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ a function that is Fréchet differentiable at x then f is continuous at x .

Proof. Let $\varepsilon \in \mathbb{R}^+$. As f is Fréchet differentiable at x there exist a $L \in L(X, Y)$ and a $\delta_1 \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $\|h\|_X < \delta_1$ we have

$$\|f(x+h) - f(x) - L(h)\|_Y \leq \frac{\varepsilon}{2} \cdot \|h\|_X \quad (16.19)$$

As $L \in L(X, Y)$ we have [see theorem: 14.210] that

$$\forall h \in X \quad \|L(h)\|_Y \leq \|L\|_{L(X, Y)} \cdot \|h\|_X \quad (16.20)$$

Define now $\delta = \min\left(\delta_1, 1, \frac{\varepsilon}{2 \cdot (\|L\|_{L(X, Y)} + 1)}\right) \in \mathbb{R}^+$ then for every $y \in A$ with $\|y - x\|_X < \delta$ we have $y - x \in A_x$ [as $(y - x) + x = y \in A$] and $\|y - x\|_X \leq \delta_1, 1, \frac{\varepsilon}{2 \cdot (\|L\|_{L(X, Y)} + 1)}$. Using [eq: 16.19] we have

$$\begin{aligned} \|f(y) - f(x) - L(y - x)\|_Y &= \|f(x + (y - x)) - f(x) - L(y - x)\|_Y \\ &\leq \frac{\varepsilon}{2} \cdot \|y - x\|_X \\ &\leq \frac{\varepsilon}{2} \end{aligned}$$

and by [eq: 16.20] it follows that

$$\|L(y - x)\|_Y \leq \|L\|_{L(X, Y)} \cdot \|y - x\|_X \leq \|L\|_{L(X, Y)} \cdot \frac{\varepsilon}{2 \cdot (\|L\|_{L(X, Y)} + 1)} < \frac{\varepsilon}{2}$$

Hence we have $\forall y \in A$ with $\|y - x\|_X < \delta$ that

$$\begin{aligned} \|f(y) - f(x)\|_Y &= \|f(y) - f(x) - L(y - x) + L(y - x)\|_Y \\ &\leq \|f(y) - f(x) - L(y - x)\|_Y + \|L(y - x)\|_Y \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

So by [theorem: 14.136] it follows that f is continuous at x . □

We look now at the relation between the Fréchet differential of a function and the derivative of a function.

Theorem 16.64. *Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of the real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow X$ a function then we have*

$$x \in A^\circ \text{ and } f \text{ has a derivative } f'_x \text{ at } x$$

$$\Updownarrow$$

f is Fréchet differentiable at x with Fréchet differential $D_x f$

Further in either case we have

$$f'_x = D_x f(1)$$

and

$$\forall h \in \mathbb{K} \text{ we have } D_x f(h) = h \cdot f'_x$$

Note 16.65. *If f has a derivative at x we have by definition that $x \in A'$ but this does not mean that $x \in A^\circ$ which is a requirement for the differentiability of f at x . Hence the condition $x \in A^\circ$ is required.*

Proof.

$\Rightarrow$. Define

$$L: \mathbb{K} \rightarrow X \text{ by } L(h) = h \cdot f'_x$$

then we have by [example: 14.216] that

$$L \in L(\mathbb{K}, X)$$

Let $\varepsilon \in \mathbb{R}^+$. As f has a derivative f'_x at x it follows from [theorem: 16.23] that there exists a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $|h| < \delta$ we have

$$\|f(x + h) - f(x) - h \cdot f'_x\| \leq \varepsilon \cdot |h|$$

Hence

$$\|f(x + h) - f(x) - L(h)\| = \|f(x + h) - f(x) - h \cdot f'_x\| \leq \varepsilon \cdot |h|$$

which proves, as $x \in A^\circ$, that f is Fréchet differentiable at x with $D_x f = L$. Hence

$$\forall h \in \mathbb{K} \text{ we have } D_x f = h \cdot f'_x \text{ and } D_x f(1) = f'_x$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$. As f is Fréchet differentiable at x with differential $D_x f$

$$x \in A^\circ$$

and there exist a $\delta \in \mathbb{R}^+$ such that $\forall \delta \in A_x$ with $|h| < \delta$ we have that

$$\|f(x+h) - f(x) - D_x f(h)\| \leq \varepsilon \cdot |h|$$

Hence we have that

$$\|f(x+h) - f(x) - h \cdot D_x f(1)\| = \|f(x+h) - f(z) - D_x f(h)\| \leq \varepsilon \cdot |h|$$

Further as $x \in A^\circ$ we have by [theorem: 16.8] that

$$x \in A'$$

So by [theorem: 16.23] that f has a derivative f'_x at x with $f'_x = D_x f(1)$. $\square$

16.1.4 Properties of the Fréchet differential.

Theorem 16.66. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A$ then we have:

1. If $f: A \rightarrow Y$ is Fréchet differentiable at x and $g: A \rightarrow Y$ is Fréchet differentiable at x then

$$f + g: A \rightarrow Y \text{ is Fréchet differentiable with } D_x(f + g) = D_x f + D_x g$$

2. If $n \in \mathbb{N}$, $\{f_i: A \rightarrow Y\}$ be a family that are Fréchet differentiable at x then $\sum_{i=1}^n f_i$ is Fréchet differentiable at x with

$$D_x \left(\sum_{i=1}^n f_i \right) = \sum_{i=1}^n D_x f_i$$

3. If $f: A \rightarrow Y$ is Fréchet differentiable at x and $\alpha \in \mathbb{K}$ then

$$\alpha \cdot f: A \rightarrow Y \text{ is Fréchet differentiable at } x \text{ with Fréchet differential } D_x(\alpha \cdot f) = \alpha \cdot D_x f$$

Proof.

1. Let $\varepsilon \in \mathbb{R}^+$. As f, g are Fréchet differentiable at x

$$x \in A^\circ$$

and there exist $\delta_1, \delta_2 \in \mathbb{R}^+$ such that

$$\forall h \in U_x \text{ with } \|h\|_x < \delta_1 \text{ we have } \|f(x+h) - f(x) - D_x f(h)\|_Y \leq \frac{\varepsilon}{2} \cdot \|h\|_x$$

$$\forall h \in U_x \text{ with } \|h\|_x < \delta_2 \text{ we have } \|g(x+h) - g(x) - D_x g(h)\|_Y \leq \frac{\varepsilon}{2} \cdot \|h\|_x$$

As $D_x f, D_x g \in L(X, Y)$ we have by [theorem: 14.200] that $D_x f + D_x g \in L(X, Y)$. Further if $h \in U_x$ with $\|h\|_x < \min(\delta_1, \delta_2)$ then we have

$$\begin{aligned} \|(f+g)(x+h) - (f+g)(x) - (D_x f + D_x g)(h)\|_Y &= \\ \|f(x+h) + g(x+h) - f(x) - g(x) - (D_x f(h) + D_x g(h))\|_Y &\leq \\ \|f(x+h) - f(x) - D_x f(h)\|_Y + \|g(x+h) - g(x) - D_x g(h)\|_Y &\leq \\ \frac{\varepsilon}{2} \cdot \|h\|_x + \frac{\varepsilon}{2} \cdot \|h\|_x &= \\ \varepsilon \cdot \|h\|_x & \end{aligned}$$

proving that $f + g$ is Fréchet differentiable at x with Fréchet differential

$$D_x(f + g) = D_x f + D_x g$$

2. This is easily proved by induction, let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i: A \rightarrow Y\}_{i \in \{1, \dots, n\}} \text{ is a family of functions that are Fréchet differentiable at } x \text{ then } \sum_{i=1}^n f_i \text{ is Fréchet differentiable at } x \text{ with } D_x \left(\sum_{i=1}^n f_i \right) = \sum_{i=1}^n D_x f_i \right\}$$

then we have:

1 $\in S$. If $\{f_i: A \rightarrow Y\}_{i \in \{1\}}$ is a family of Fréchet differentiable at x then we have that $\sum_{i=1}^1 f_i = f_1$ is Fréchet differentiable at x with $D_x(\sum_{i=1}^1 f_i) = D_x f_1 = \sum_{i=1}^1 D_x f_i$. Hence $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $\{f_i: A \rightarrow Y\}_{i \in \{1, \dots, n+1\}}$ is a family of Fréchet differentiable at x then, as $n \in S$, $\sum_{i=1}^n f_i$ is Fréchet differentiable at x with $D_x(\sum_{i=1}^n f_i) = \sum_{i=1}^n D_x f_i$. As also f_{n+1} is Fréchet differentiable at x we have by (1) that

$$\sum_{i=1}^{n+1} f_i = f_{n+1} + \sum_{i=1}^n f_i$$

that

$$D_x \left(\sum_{i=1}^{n+1} f_i \right) = D_x f_{n+1} + D_x \left(\sum_{i=1}^n f_i \right) = D_x f_{n+1} + \sum_{i=1}^n D_x f_i = \sum_{i=1}^{n+1} D_x f_i$$

which proves that

$$n+1 \in S$$

3. Let $\varepsilon \in \mathbb{R}^+$. As f is Fréchet differentiable at x

$$x \in A^\circ$$

and there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall h \in U_x \text{ with } \|h\|_X < \delta_1 \text{ we have } \|f(x+h) - f(x) - D_x f(h)\|_Y \leq \frac{\varepsilon}{|\alpha|+1} \cdot \|h\|_X$$

As $D_x f \in L(X, Y)$ it follows from [theorem: 14.200] that $\alpha \cdot D_x f \in L(X, Y)$. Further if $h \in U_x$ with $\|h\|_X < \delta$ then we have

$$\begin{aligned} \|(\alpha \cdot f)(x+h) - (\alpha \cdot f)(x) - (\alpha \cdot D_x f)(h)\|_Y &= \\ \|\alpha \cdot f(x+h) - \alpha \cdot f(x) - \alpha \cdot D_x f(h)\|_Y &= \\ \|\alpha \cdot (f(x+h) - f(x) - D_x f(h))\|_Y &= \\ |\alpha| \cdot \|f(x+h) - f(x) - D_x f(h)\|_Y &\leq \\ |\alpha| \cdot \frac{\varepsilon}{|\alpha|+1} \cdot \|h\|_X &< \\ \varepsilon \cdot \|h\|_X & \end{aligned}$$

proving that $\alpha \cdot f$ is Fréchet differentiable at x with Fréchet differential $\alpha \cdot D_x f$.

□

We introduce now the most important rule for Fréchet differentiable the Chain Rule.

Theorem 16.67. (Chain Rule) Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $A \subseteq X$, $x \in A$, $B \subseteq Y$, $f: A \rightarrow Y$, $g: B \rightarrow Z$ functions such that $f(A) \subseteq B$ and $f(x) \in B$, f is Fréchet differentiable at x and g is Fréchet differentiable at $f(x)$ then

$$g \circ f: A \rightarrow Z \text{ is Fréchet differentiable at } x$$

and

$$D_x(g \circ f) = D_{f(x)} g \circ D_x f$$

Proof. As f is Fréchet differentiable at x we have by [theorem: 16.61] that

$$x \in A^\circ \tag{16.21}$$

and there exist a ε -mapping $\varepsilon_f: A_x \rightarrow Y$ such that

$$\forall h \in A_x \text{ we have } f(x+h) - f(x) - D_x f(h) = \|h\|_X \cdot \varepsilon_f(h) \quad (16.22)$$

Further as g is Fréchet differentiable at $f(x)$ we can use [theorem: 16.61] again, giving

$$f(x) \in B^\circ$$

and the existence of a ε -mapping $\varepsilon_g: B_{f(x)} \rightarrow Z$ satisfying

$$\forall k \in B_{f(x)} \text{ we have } g(f(x)+k) - g(f(x)) - D_{f(x)}g(k) = \|k\|_Y \cdot \varepsilon_g(k) \quad (16.23)$$

Let $h \in A_x$. As $(f(x+h) - f(x)) + f(x) = f(x+h) \in f(A) \subseteq B$ we have that $f(x+h) - f(x) \in B_{f(x)}$, combining this with [eq: 16.23] gives

$$\begin{aligned} & \|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) \stackrel{[\text{eq: 16.23}]}{=} \\ & g(f(x) + (f(x+h) - f(x))) - g(f(x)) - D_{f(x)}g(f(x+h) - f(x)) = \\ & g(f(x+h)) - g(f(x)) - D_{f(x)}g(f(x+h) - f(x)) \stackrel{[\text{eq: 16.22}]}{=} \\ & g(f(x+h)) - g(f(x)) - D_{f(x)}g(\|h\|_X \cdot \varepsilon_f(h) + D_x f(h)) = \\ & g(f(x+h)) - g(f(x)) - D_{f(x)}g(\|h\|_X \cdot \varepsilon_f(h)) - D_{f(x)}g(D_x f(h)) = \\ & (g \circ f)(x+h) - (g \circ f)(x) - D_{f(x)}g(\|h\|_X \cdot \varepsilon_f(h)) - (D_{f(x)}g \circ D_x f)(h) \end{aligned}$$

proving that

$$\begin{aligned} & (g \circ f)(x+h) - (g \circ f)(x) - (D_{f(x)}g \circ D_x f)(h) = \\ & \|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) + D_{f(x)}g(\|h\|_X \cdot \varepsilon_f(h)) = \\ & \|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) + \|h\|_X \cdot D_{f(x)}g(\varepsilon_f(h)) \end{aligned} \quad (16.24)$$

The next logical step is to find a ε -mapping $\zeta: A_x \rightarrow Y$ such that

$$\zeta(h) \cdot \|h\|_X = \|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) + \|h\|_X \cdot D_{f(x)}g(\varepsilon_f(h))$$

because we can then use [theorem: 16.61] to finish the proof. So define

$$\zeta: A_x \rightarrow Y \text{ by } \begin{cases} \frac{\|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x))}{\|h\|_X} + D_{f(x)}g(\varepsilon_f(h)) & \text{if } h \in A_x \setminus \{0\} \\ 0 & \text{if } h = 0 \end{cases}$$

Then for $h \in A_x$ we have either:

$h = 0$. Then

$$\begin{aligned} & \|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) + \|h\|_X \cdot D_{f(x)}g(\varepsilon_f(h)) = \\ & \|f(x+0) - f(x)\|_Y \cdot \varepsilon_g(f(x+0) - f(x)) + \|0\|_X \cdot D_{f(x)}g(\varepsilon_f(0)) = \\ & 0 = \\ & 0 \cdot \zeta(0) = \\ & \|h\|_X \cdot \zeta(h) \end{aligned}$$

$h \neq 0$. Then

$$\begin{aligned} & \|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) + \|h\|_X \cdot D_{f(x)}g(\varepsilon_f(h)) = \\ & \|h\|_X \cdot \frac{\|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x)) + \|h\|_X \cdot D_{f(x)}g(\varepsilon_f(h))}{\|h\|_X} = \\ & \|h\|_X \cdot \left(\frac{\|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x))}{\|h\|_X} + D_{f(x)}g(\varepsilon_f(h)) \right) = \\ & \|h\|_X \cdot \zeta(h) \end{aligned}$$

Combining the above with [eq: 16.24] gives us the desired

$$(g \circ f)(x+h) - (g \circ f)(x) - (D_{f(x)}g \circ D_x f)(h) = \|h\|_X \cdot \zeta(h) \quad (16.25)$$

Next we must prove that ζ is continuous at 0. Let $\varepsilon \in \mathbb{R}^+$. As ε_f is a ε -mapping, it is continuous at 0. So by [theorem: 14.136] there exist a $\delta_f \in \mathbb{R}^+$ such that if $h \in A_x$ with $\|h\|_X < \delta_f$ then

$$\|\varepsilon_f(h)\|_Y = \|\varepsilon_f(h) - 0\|_Y < 1 \quad (16.26)$$

So if additional $0 < \|h\|_X$ we have

$$\begin{aligned} \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} &\stackrel{[\text{eq: 16.22}]}{=} \frac{\|h\|_X \cdot \varepsilon_f(h) + D_x f(h)\|_Y}{\|h\|_X} \\ &\leq \frac{\|h\|_X \cdot \|\varepsilon_f(h)\|_Y + \|D_x f(h)\|_Y}{\|h\|_X} \\ &= \frac{\|h\|_X \cdot \|\varepsilon_f(h)\|_Y + \|D_x f(h)\|_Y}{\|h\|_X} \\ &\leq \frac{\|h\|_X \cdot \|\varepsilon_f(h)\|_Y + \|D_x f\|_{L(X,Y)} \cdot \|h\|_X}{\|h\|_X} \\ &= \|\varepsilon_f(h)\|_Y + \|D_x f\|_{L(X,Y)} \\ &\stackrel{[\text{eq: 16.26}]}{<} 1 + \|D_x f\|_{L(X,Y)} \end{aligned}$$

hence

$$\forall h \in A_x \text{ with } 0 < \|h\|_X < \delta_f \text{ we have that } \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} < 1 + \|D_x f\|_{L(X,Y)} \quad (16.27)$$

As ε_g is a ε -mapping it is continuous at 0. So by [theorem: 14.136] there exist a $\delta_g \in \mathbb{R}^+$ such that

$$\forall k \in B_{f(x)} \text{ with } \|k\|_Y < \delta_g \text{ we have } \|\varepsilon_g(k)\|_Z < \frac{\varepsilon}{2 \cdot (1 + \|D_x f\|_{L(X,Y)})} \quad (16.28)$$

As f is Fréchet differentiable at x we have by [theorem: 16.63] that f is continuous at x , hence there exist by [theorem: 14.136] a $\delta_1 \in \mathbb{R}^+$ such that $\forall y \in A$ with $\|y - x\|_X < \delta_1$ we have $\|f(y) - f(x)\|_Y < \delta_g$. So if $h \in A_x$ with $\|h\|_X < \delta_1$ we have $x+h \in A$ and $\|(x+h) - x\|_X = \|h\|_X < \delta_1$ so that $\|f(x+h) - f(x)\|_Y < \delta_g$ which by [eq: 16.28] results in $\|\varepsilon_g(f(x+h) - f(x))\|_Z < \frac{\varepsilon}{2 \cdot (1 + \|D_x f\|_{L(X,Y)})}$. Summarized

$$\forall h \in A_x \text{ with } \|h\|_X < \delta_1 \text{ we have } \|\varepsilon_g(f(x+h) - f(x))\|_Z < \frac{\varepsilon}{2 \cdot (1 + \|D_x f\|_{L(X,Y)})}$$

Combining this with [eq: 16.27] we have $\forall h \in A_x$ with $0 < \|h\|_X < \min(\delta_1, \delta_f)$ that

$$\begin{aligned} \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} \cdot \|\varepsilon_g(f(x+h) - f(x))\|_Z &\leq \\ (1 + \|D_x f\|_{L(X,Y)}) \cdot \|\varepsilon_g(f(x+h) - f(x))\|_Z &< \\ (1 + \|D_x f\|_{L(X,Y)}) \cdot \frac{\varepsilon}{2 \cdot (1 + \|D_x f\|_{L(X,Y)})} &= \\ \frac{\varepsilon}{2} \end{aligned}$$

or

$$\forall h \in A_x \text{ with } 0 < \|h\|_X < \min(\delta_1, \delta_f) \|\varepsilon_g(f(x+h) - f(x))\|_Z \cdot \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} < \frac{\varepsilon}{2} \quad (16.29)$$

As $\varepsilon_f: A_x \rightarrow Y$ is a ε -mapping it is continuous at 0, so by [theorem: 14.136] there exist a $\delta_2 \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $\|h\|_X < \delta_2$ we have $\|\varepsilon_f(h)\|_Y < \frac{\varepsilon}{2 \cdot (1 + \|D_{f(x)} g\|_{L(Y,Z)})}$. Hence

$$\begin{aligned} \|D_{f(x)} g(\varepsilon_f(h))\|_Z &\leq \|D_{f(x)} g\|_{L(Y,Z)} \cdot \|\varepsilon_f(h)\|_Y \\ &\leq \|D_{f(x)} g\|_{L(Y,Z)} \cdot \frac{\varepsilon}{2 \cdot (1 + \|D_{f(x)} g\|_{L(Y,Z)})} \\ &< \frac{\varepsilon}{2} \end{aligned}$$

In other words

$$\forall h \in A_x \text{ with } \|h\|_X < \delta_2 \text{ we have } \|D_{f(x)} g(\varepsilon_f(h))\|_Z < \frac{\varepsilon}{2} \quad (16.30)$$

Take now $\delta = \min(\delta_f, \delta_1, \delta_2)$ and $h \in A_x$ with $\|h\|_X = \|h - 0\|_X < \delta$ then we have either:

$h = 0$. Then $\|\zeta(h) - 0\|_Z = \|\zeta(0)\|_Z = \|0\|_Z < \varepsilon$

$h \neq 0$. Then $0 < \|h\|_X < \delta_f, \delta_1, \delta_2$ so that

$$\begin{aligned}
 \|\zeta(h) - 0\|_Z &= \left\| \frac{\|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x))}{\|h\|_X} + D_{f(x)}g(\varepsilon_f(h)) \right\|_Z \\
 &\leq \left\| \frac{\|f(x+h) - f(x)\|_Y \cdot \varepsilon_g(f(x+h) - f(x))}{\|h\|_X} \right\|_Z + \|D_{f(x)}g(\varepsilon_f(h))\|_Z \\
 &= \frac{\|f(x+h) - f(x)\|_Y}{\|h\|_X} \cdot \|\varepsilon_g(f(x+h) - f(x))\|_Z + \|D_{f(x)}g(\varepsilon_f(h))\|_Z \\
 &\stackrel{[\text{eq: 16.29}]}{<} \frac{\varepsilon}{2} + \|D_{f(x)}g(\varepsilon_f(h))\|_Z \\
 &\stackrel{[\text{eq: 16.30}]}{<} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
 &= \varepsilon
 \end{aligned}$$

So we have in all cases that $\|\zeta(h) - 0\|_Z < \varepsilon$ which as $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary proves that ζ is continuous at 0. As also $\zeta(0) = 0$ it follows by definition that $\zeta: A_x \rightarrow Z$ is a ε -mapping. Hence as, by [eq: 16.25] $(g \circ f)(x+h) - (g \circ f)(x) - (D_{f(x)}g \circ D_x f)(h) = \|h\|_X \cdot \zeta(h)$ and by [eq: 16.21] $x \in A^\circ$, it follows from [theorem: 16.61] that

$$g \circ f \text{ is Fréchet differentiable with differential } D_x(g \circ f) = D_{f(x)}g \circ D_x f \quad \square$$

Corollary 16.68. *Let $\langle X, \|\cdot\| \rangle$ be a normed space, $\langle \mathbb{K}, \|\cdot\| \rangle$ the normed space of real [or complex] numbers, $A \subseteq \mathbb{K}$, $x \in A^\circ$, $B \subseteq \mathbb{K}$ and $f: A \rightarrow \mathbb{K}$, $g: B \rightarrow X$ functions such that $f(A) \subseteq B$ and $f(x) \in B^\circ$, f has a derivative f'_x at x and g has a derivative $g'_{f(x)}$ at $f(x)$. Then $g \circ f: A \rightarrow X$ has a derivative at x and*

$$(g \circ f)'_x = f'_x \cdot g'_{f(x)}$$

Proof. Using [theorem: 16.64] it follows that f is Fréchet differentiable at x and g is Fréchet differentiable at $f(x)$. Using the Chain Rule [theorem: 16.67] we have that $g \circ f$ is Fréchet differentiable at x . Hence, using [theorem: 16.64], we have that $g \circ f$ has a derivative at x . Further we have

$$\begin{aligned}
 (g \circ f)'_x &\stackrel{[\text{theorem: 16.64}]}{=} D_x(g \circ f)(1) \\
 &\stackrel{[\text{theorem: 16.67}]}{=} (D_{f(x)}g \circ D_x f)(1) \\
 &= D_{f(x)}g(D_x f(1)) \\
 &\stackrel{[\text{theorem: 16.64}]}{=} D_{f(x)}g(f'_x) \\
 &\stackrel{[\text{theorem: 16.64}]}{=} f'_x \cdot g'_{f(x)}
 \end{aligned}$$

□

Corollary 16.69. *Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$, $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $A \subseteq X$, $x \in A$, $f: A \rightarrow Y$ a function that is Fréchet differentiable at x and $L \in L(Y, Z)$ then*

$$L \circ f: U \rightarrow Z \text{ is Fréchet differentiable at } x$$

and

$$D_x(L \circ f) = L \circ D_x f$$

Proof. Using [example: 16.55] L is Fréchet differentiable at $f(x)$ and $D_{f(x)}L = L$. Hence using the chain rule [see theorem: 16.67] $L \circ f$ is Fréchet differentiable at x and

$$D_x(L \circ f) = D_{f(x)}L \circ D_x f = L \circ D_x f \quad \square$$

16.1.5 Partial differentials

The partial differential of a function of several variable is the differential with respect to one of these variables while keeping the other variables constant. In essence given a function $f(x_1, \dots, x_n)$ we create a new function defined by $t \mapsto f(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_n)$ and calculate the differential of this new function at x_i . To do this we write this new function as the composition of f and a differentiable function and apply the chain rule.

Definition 16.70. Let $n \in \mathbb{N} \setminus \{1\}$, $\{X_i\}_{i \in \{1, \dots, n\}}$ a finite family of sets, $x \in \prod_{i \in \{1, \dots, n\}} X_i$, $i \in \{1, \dots, n\}$, $t \in X_i$ then $I^{[i, x]}: X_i \rightarrow \prod_{i \in \{1, \dots, n\}} X_i$ is defined by

$$I^{[i, x]}(t) = \left(x_1, \dots, \underbrace{t}_i, \dots, x_n \right)$$

or more formally

$$(I^{[i, x]}(t))_j = \begin{cases} t & \text{if } j = i \\ x_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

The following lemma show now the important properties of this function.

Lemma 16.71. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of normed spaces, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98] then we have $\forall i \in \{1, \dots, n\}$ that:

1. $I^{[i, 0]} \in L(X_i, \prod_{j \in \{1, \dots, n\}} X_j)$ and $\forall t \in X_i$ we have $\|t\|_i = \|I^{[i, 0]}(t)\|_{\max}$.
2. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then $I^{[i, x]}(x_i) = x$.
3. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then $x = \sum_{i=1}^n I^{[i, 0]}(x_i)$.
4. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then $\forall t \in X_i$ $I^{[i, x]}$ is Fréchet differentiable at x with $D_t I^{[i, x]} = I^{[i, 0]}$

Note 16.72. As X_j is open we have by [theorem: 14.16] that $t \in (X_j)^\circ$ which is required for the differentiability.

5. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then $I^{[i, x]}$ is continuous.
6. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$, $A \subseteq X_i$ and $t \in A^\circ$ then $(I^{[i, x]})|_A$ is Fréchet differentiable at t with $D_t(I^{[i, x]})|_A = I^{[i, 0]}$.
7. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$, $A \subseteq X_j$ then $(I^{[i, x]})|_A$ is continuous.
8. If $x = (x_1, \dots, x_n) \in \prod_{j \in \{1, \dots, n\}} X_j$ then for every open set U in $\prod_{j \in \{1, \dots, n\}} X_j$ we have that $(I^{[i, x]})^{-1}(U)$ is open in X_i .
9. $\forall i, k \in \{1, \dots, n\}$, $\forall t \in X_i$ we have $(I^{[i, 0]}(t))_k = \delta_{i, k} \cdot t$

Proof.

1. Let $r, s \in X_i$, $\alpha \in \mathbb{K}$ then

$$\begin{aligned} I^{[i, 0]}(r + \alpha \cdot s) &= \left(0 \dots \underbrace{r + \alpha \cdot s}_i \dots 0 \right) \\ &= \left(0 \dots \underbrace{r}_i \dots 0 \right) + \alpha \cdot \left(0 \dots \underbrace{s}_i \dots 0 \right) \\ &= I^{[i, 0]}(r) + \alpha \cdot I^{[i, 0]}(s) \\ &= L(r) + \alpha \cdot L(s) \end{aligned}$$

proving that

$$L \in \text{Hom} \left(X_i, \prod_{j \in \{1, \dots, n\}} X_j \right)$$

Further if $t \in X_i$ then for $j \in \{1, \dots, n\}$ we have

$$\pi_j(I^{[i,0]}(t)) = \pi_j\left(\left(0, \dots, \underbrace{t}_j, \dots, 0\right)\right) = \begin{cases} t & \text{if } j = i \\ 0_j & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

so that

$$\|\pi_j(I^{[i,0]}(t))\|_j = \begin{cases} \|t\|_i & \text{if } j = i \\ 0 & \text{if } j \in \{1, \dots, n\} \setminus \{i\} \end{cases}$$

so that

$$\begin{aligned} \|I^{[i,0]}(t)\|_{\max} &= \max(\{\|\pi_j(I^{[i,0]}(t))\|_j | j \in \{1, \dots, n\}\}) \\ &= \max(\{\|\pi_j(I^{[i,0]}(t))\|_j | j \in \{1, \dots, n\} \setminus \{i\}\} \cup \{\|\pi_j(I^{[i,0]}(t))\|_j | j \in \{i\}\}) \\ &= \max(\{0\} \cup \{\|t\|_i\}) \\ &= \max(\{0, \|t\|_i\}) \\ &= \|t\|_i \end{aligned}$$

proving that

$$\|I^{[i,0]}(t)\|_{\max} = \|t\|_i = 1 \cdot \|t\|_i$$

Hence by [theorem: 14.201] it follows that

$$I^{[i,0]} \in L\left(X_i, \prod_{j \in \{1, \dots, n\}} X_j\right)$$

2. This follows from $I^{[x,i]}(x_i) = \left(x_1, \dots, \underbrace{x_i}_i, \dots, x_n\right) = x$

3. Let $x \in \prod_{j \in \{1, \dots, n\}} X_j$ and $j \in \{1, \dots, n\}$ then

$$\begin{aligned} \left(\sum_{i=1}^n I^{[i,0]}(x_i)\right)_j &= \left(\sum_{i \in \{1, \dots, n\}} I^{[i,0]}(x_i)\right)_j \\ &= \pi_j\left(\sum_{i \in \{1, \dots, n\}} I^{[i,0]}(x_i)\right) \\ &\stackrel{[\text{theorem: 11.175}]}{=} \sum_{i \in \{1, \dots, n\}} \pi_j(I^{[i,0]}(x_i)) \\ &= \sum_{i \in \{1, \dots, n\} \setminus \{j\}} \pi_j(I^{[i,0]}(x_i)) + \sum_{i \in \{j\}} \pi_j(I^{[i,0]}(x_i)) \\ &\stackrel{(2)}{=} 0 + x_j \\ &= x_j \end{aligned}$$

proving that

$$x = \sum_{i=1}^n I^{[i,0]}(x_i)$$

4. Let $\varepsilon \in \mathbb{R}^+$ then we have $\forall h \in A_t$ that

$$\begin{aligned} &\|I^{[i,x]}(t+h) - I^{[i,x]}(t) - I^{[i,0]}(h)\|_{\max} = \\ &\left\| \left(x_1, \dots, \underbrace{t+h}_i, \dots, x_n\right) - \left(x_1, \dots, \underbrace{t}_i, \dots, x_n\right) - I^{[i,0]}(h) \right\|_{\max} = \\ &\left\| \left(0_1, \dots, \underbrace{h}_i, \dots, 0_n\right) - I^{[i,0]}(h) \right\|_{\max} = \\ &\left\| \left(0_1, \dots, \underbrace{h}_i, \dots, 0_n\right) - \left(0_1, \dots, \underbrace{h}_i, \dots, 0_n\right) \right\|_{\max} = \\ &\|0_1, \dots, 0_n\|_{\max} = \\ &0 < \\ &\varepsilon \cdot \|h\|_i \stackrel{(1)}{=} \end{aligned}$$

which, as by (1) $I^{[i,0]} \in L(X_j, \prod_{j \in \{1, \dots, n\}} X_j)$ and $t \in X_j|_{[\text{theorem: 14.16}]} = (X_j)^\circ$ proves that

$$I^{[i,k]} \text{ is Fréchet differentiable at } t \text{ with } D_t I^{[i,x]} = I^{[i,0]}$$

5. Let $t \in X_i$ then by (4) $I^{[i,k]}$ is Fréchet differentiable at t which by [theorem: 16.63] proves that $I^{[i,x]}$ is continuous at t . As t has been chosen arbitrary it follows that $I^{[i,x]}$ is a continuous function.
6. This follows from [theorem: 16.57] and (4).
7. This follows from [theorem: 14.146].
8. Let U be a open set in $\prod_{j \in \{1, \dots, n\}} X_j$ and by (5) $I^{[i,x]}: X_i \rightarrow \prod_{j \in \{1, \dots, n\}} X_j$ is continuous it follows that $(I^{[i,x]})^{-1}(U)$ is open.
9. Let $i \in \{1, \dots, n\}$ and $t \in X_i$ then we have for $k \in \{1, \dots, n\}$

$$(I^{[i,0]}(t))_k = \left(0, \dots, \underbrace{t}_i, \dots, 0\right)_k = \delta_{i,k} \cdot t \quad \square$$

Lemma 16.73. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of normed spaces, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98] and $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$ and $x \in A^\circ$ then $\forall i \in \{1, \dots, n\}$ we have $x_i \in ((I^{[i,x]})^{-1}(A))^\circ$.

Proof. Let As $x \in A^\circ$ there exist a open set $U \in \mathcal{T}_{\|\cdot\|_{\max}}$ with $x \in U \subseteq A$. As by [lemma: 16.71] $I^{[i,x]}$ is continuous it follows that $(I^{[i,x]})^{-1}(U) \in \mathcal{T}_i$. Let $i \in \{1, \dots, n\}$ then as $I^{[i,x]}(x_i)|_{[\text{lemma: 16.71}]} x \in U \subseteq A$ hence

$$x_i \in (I^{[i,x]})^{-1}(U) \subseteq (I^{[i,x]})^{-1}(A) \text{ where } (I^{[i,x]})^{-1}(U) \in \mathcal{T}_i$$

which proves that

$$x_i \in ((I^{[i,x]})^{-1}(A))^\circ \quad \square$$

Definition 16.74. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$, $x \in A$, $i \in \{1, \dots, n\}$ and

$$f: A \rightarrow Y \text{ a function}$$

then

$$f \text{ is } i\text{-partial differentiable at } x \text{ with } i\text{-partial differential } D_{x,i}f$$

if the function

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(A) \rightarrow Y$$

is Fréchet differentiable at x_i and $D_{x,i}f$ is then defined to be

$$D_{x,i}f = D_{x_i}(f \circ I^{[i,x]})$$

Note 16.75. As $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$ and $I^{[i,x]}: X_i \rightarrow \prod_{j \in \{1, \dots, n\}} X_j$ it follows from [corollary: 2.83] that

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(A) \rightarrow Y$$

is indeed a function.

Note 16.76. As $f \circ I^{[i,x]}$ is Fréchet differentiability at x_i it follows by [definition: 16.46] that $x_i \in ((I^{[i,x]})^{-1}(A))^\circ$.

Remark 16.77. Instead of saying

$$f \text{ is } i\text{-partial differentiable at } x \text{ with } i\text{-partial differential } D_{x,i}f$$

we use sometimes the shorter form

$$D_{x,i}f \text{ exists}$$

Just as we have introduced the derivative of a function and the differential of a function we can define the partial differential of a function.

Definition 16.78. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$, $x \in A$, $i \in \{1, \dots, n\}$ and $f: A \rightarrow Y$ a function. Then we define

$$\mathfrak{D}_{i,f} = \{x \in A \mid f \text{ is } i\text{-partial differentiable at } x\}$$

and

$$D_{i,f}: \mathfrak{D}_{i,f} \rightarrow L(X, Y) \text{ where } D_{i,f}(x) = D_{x,i}f$$

$D_{i,f}$ is called the i -partial differential of f .

Note 16.79. If $\mathfrak{D}_{i,f} = \emptyset$ then $D_{i,f}$ is the empty function [see example: 2.46].

Note 16.80. The benefit of this definition is that

1. $D_{i,f}: \mathfrak{D}_{i,f} \rightarrow Y$ is always defined even if f is never i -partial differentiable at a point in A .
2. $\mathfrak{D}_{i,f}$ is the set of all elements of the domain of f for which f is i -partial differentiable at hence it is the most natural definition of the domain of $D_{i,f}$.
3. If we refer to $D_{i,f}$ where $f: A \rightarrow Y$ is a function then we can automatically assume that $D_{i,f}$ is the graph of the function

$$D_{i,f}: \mathfrak{D}_{i,f} \rightarrow Y$$

Definition 16.81. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$, $U \in \mathcal{T}_{\|\cdot\|_{\max}}$ with $U \subseteq A$, $i \in \{1, \dots, n\}$ and $f: A \rightarrow Y$ a function. Then we say that **f is i -partial differentiable on U** iff

$$\forall x \in U \ f \text{ is } i\text{-partial differentiable at } x$$

Just as differentiability is a local property [see theorem: 16.57] partial is a local property.

Theorem 16.82. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$, $B \subseteq A$, $x = (x_1, \dots, x_n) \in A$, $i \in \{1, \dots, n\}$ and

$$f: A \rightarrow Y \text{ a function}$$

then we have

$$x_i \in ((I^{[i,x]})^{-1}(A))^{\circ} \text{ and } f \text{ is } i\text{-partial differentiable at } x$$

$$\Updownarrow$$

$$f|_B \text{ is } i\text{-partial differentiable at } x$$

Further if f or $f|_B$ is i -partial differentiable at x then $D_{x,i}f = D_{x,i}f|_B$.

Proof. Let $t \in (I^{[i,x]})^{-1}(B) \Rightarrow I^{[i,x]}(t) \in B$ it follows that

$$\begin{aligned} (f|_B \circ I^{[i,x]})(t) &= f|_B(I^{[i,x]}(t)) \\ &\stackrel{I^{[i,x]}(t) \in B}{=} f(I^{[i,x]}(t)) \\ &= (f \circ I^{[i,x]})(t) \\ &\stackrel{t \in (I^{[i,x]})^{-1}(B)}{=} (f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)} \end{aligned}$$

which proves that

$$f|_B \circ I^{[i,x]} = (f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)} \quad (16.31)$$

Hence we have

$\Rightarrow$. If f is i -partial differentiable at x with i -partial differential $D_{x,i}f$ then by [definition: 16.74] $f \circ I^{[i,x]}$ is Fréchet differentiable at x_i with

$$D_{x,i}f = D_{x_i}(f \circ I^{[i,x]}) \quad (16.32)$$

As by the hypothesis $x_i \in ((I^{[i,x]})^{-1}(B))^\circ$ it follows from [theorem: 16.57] that $(f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)}$ is Fréchet differentiable at x_i with

$$D_{x_i}(f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)} = D_{x_i}(f \circ I^{[i,x]})$$

Using [eq: 16.31] on the above we have that $f|_B \circ I^{[i,x]}$ is Fréchet differentiable at x_i and

$$D_{x_i}(f|_B \circ I^{[i,x]}) = D_{x_i}(f \circ I^{[i,x]})|_{[eq: 16.32]} = D_{x,i}f$$

so that by applying [definition: 16.74] we have

$$f|_B \text{ is } i\text{-partial differentiable at } x \text{ with } D_{x,i}f|_B = D_{x,i}f$$

$\Leftarrow$. If $f|_B$ is i -partial differentiable at x with i -partial differential $D_{x,i}f|_B$ then by [definition: 16.74] $f|_B \circ I^{[i,x]}$ is Fréchet differentiable at x_i with

$$D_{x,i}f|_B = D_{x_i}(f|_B \circ I^{[i,x]}) \quad (16.33)$$

By [eq: 16.31] it follows that $(f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)}$ is Fréchet differentiable at x_i with

$$D_{x_i}(f|_B \circ I^{[i,x]}) = D_{x_i}(f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)} \quad (16.34)$$

It follows then from [theorem: 16.57] that $f \circ I^{[i,x]}$ is Fréchet differentiable at x_i with

$$D_{x_i}(f \circ I^{[i,x]}) = D_{x_i}(f \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(B)}|_{[eqs: 16.33, 16.34]} = D_{x,i}f|_B$$

and that

$$x_i \in ((I^{[i,x]})^{-1}(A))^\circ$$

Hence using [definition: 16.74] we have that

$$f \text{ is } i\text{-partial differentiable at } x \text{ with } D_{x,i}f = D_{x,i}f|_B \quad \square$$

We show now that Fréchet differentiability of a function with many variables ensures that all the partial differentials exist.

Theorem 16.83. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$, $x \in A$ and $f: A \rightarrow Y$ a function that is Fréchet differentiable at x then $\forall i \in \{1, \dots, n\}$ f is i -partial differentiable at x with i -partial differential

$$D_{x,i}f = D_x f \circ I^{[i,0]}$$

Further we have

$$D_x f = \sum_{i=1}^n (D_{x,i}f \circ \pi_i)$$

In other words $\forall h \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$D_x f(h) = \sum_{i=1}^n D_{x,i}f(h_i)$$

Proof. Let $i \in \{1, \dots, n\}$. Using [lemma: 16.71(4)] it follows that

$$I^{[i,x]}: X_i \rightarrow \prod_{j \in \{1, \dots, n\}} X_j \text{ is Fréchet differentiable at } x_i \text{ with } D_x(I^{[i,x]}) = I^{[i,0]}$$

As f is Fréchet differentiable at x it follows from [definition: 16.46] that $x \in A^\circ$ so that by [lemma: 16.73] $x_i \in ((I^{[i,x]})^{-1}(A))^\circ \subseteq X_i$ hence we have using locality [see theorem: 16.57] on the above that

$$(I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}: (I^{[i,x]})^{-1}(A) \rightarrow \prod_{j \in \{1, \dots, n\}} X_j \text{ is Fréchet differentiable at } x_i$$

with

$$D_{x_i}((I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}) = D_{x_i}(I^{[i,x]}) = I^{[i,0]} \quad (16.35)$$

As $f: A \rightarrow Y$ is Fréchet differentiable at x , $x \in A \Rightarrow x_i \in (I^{[i,x]})^{-1}(A)$ so that $(I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}(x_i) = I^{[i,x]}(x_i) = x$ and $(I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}((I^{[i,x]})^{-1}(A)) \subseteq A$ it follows from the Chain Rule [theorem: 16.67] that

$$f \circ (I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)} \text{ is Fréchet differentiable at } x_i$$

with

$$\begin{aligned} D_{x_i}(f \circ (I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}) &= D_{(I^{[i,x]}(x_i))_{|(I^{[i,x]})^{-1}(A)}} f \circ D_{x_i}((I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}) \\ &= D_{I^{[i,x]}(x_i)} f \circ D_{x_i}((I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}) \\ &= D_x f \circ D_{x_i}((I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}) \\ &\stackrel{[\text{eq: 16.35}]}{=} D_x f \circ I^{[i,0]} \end{aligned}$$

As by [corollary: 2.83] $f \circ I^{[i,x]} = f \circ (I^{[i,x]})_{|(I^{[i,x]})^{-1}(A)}$ it follows that

$$f \circ I^{[i,x]} \text{ is Fréchet differentiable at } x_i \text{ with } D_{x_i}(f \circ I^{[i,x]}) = D_x f \circ I^{[i,0]}$$

Using then [definition: 16.74] we have that

$$f \text{ is } i\text{-partial differentiable at } x \text{ with } D_{x,i}f = D_x f \circ I^{[i,0]} \quad (16.36)$$

Further if $h \in \prod_{j \in \{1, \dots, n\}} X_j$ then we have

$$\begin{aligned} D_x f(h) &\stackrel{[\text{lemma: 16.71(3)}]}{=} D_x f\left(\sum_{i=1}^n I^{[i,0]}(h_i)\right) \\ &= \sum_{i=1}^n D_x f(I^{[i,0]}(h_i)) \\ &= \sum_{i=1}^n (D_x f \circ I^{[i,0]})(h_i) \\ &\stackrel{[\text{eq: 16.36}]}{=} \sum_{i=1}^n D_{x,i}f(h_i) \\ &= \sum_{i=1}^n D_{x,i}f(\pi_i(h)) \\ &= \sum_{i=1}^n (D_{x,i}f \circ \pi_i)(h) \\ &= \left(\sum_{i=1}^n (D_{x,i}f \circ \pi_i)\right)(h) \end{aligned}$$

so that we have

$$D_x f = \sum_{j=1}^n (D_{x,j}f \circ \pi_j)$$

□

Example 16.84. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq \prod_{j \in \{1, \dots, n\}} X_j$, $x \in A^0$ and $y \in Y$ then $\forall i \in \{1, \dots, n\}$ we have that the constant function

$$C_y: A \rightarrow Y \text{ defined by } C_y(x) = y$$

has a i -partial differential at x with

$$D_{y,i}C_i = C_0$$

where

$$C_0: X_i \rightarrow Y \text{ is defined by } C_0(h) = 0$$

Proof. By [example: 16.54] C_y is Fréchet differentiable at x with $D_x C_y = C_0$ where

$$C_0: X \rightarrow Y \text{ is defined by } C_0(h) = 0$$

Hence using [theorem: 16.83] we have $\forall i \in \{1, \dots, n\}$ that C_y has a i -partial differential with

$$D_{x,i}C_y = C_0 \circ I^{[i,0]} = C_0$$

[because $\forall h \in X_i$ $(C_0 \circ I^{[i,0]})(h) = C_0(I^{[i,0]}(h)) = 0$]. □

Although Fréchet differentiability of a multi parameter function implies that all the partial differentials exist the opposite is not true as the following example proves.

Example 16.85. Define the function

$$f: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ by } f(x, y) = \begin{cases} \frac{x \cdot y}{x^2 + y^2} & \text{if } (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\} \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

then f is not Fréchet differentiable at $(0, 0)$ however f is 1-partial and 2-partial differentiable at $(0, 0)$

Proof. Let $\varepsilon = \frac{1}{4}$, $\delta \in \mathbb{R}^+$ and consider $z = \left(\frac{\delta}{2}, \frac{\delta}{2}\right)$ then we have

$$\|z\|_{\max} = \max\left(\left|\frac{\delta}{2}\right|, \left|\frac{\delta}{2}\right|\right) = \frac{\delta}{2} < \delta$$

However

$$\begin{aligned} |f(z) - f(0, 0)| & \underset{(\delta, \delta) \neq (0, 0)}{=} \left| \frac{\frac{\delta}{2} \cdot \frac{\delta}{2}}{\left(\frac{\delta}{2}\right)^2 + \left(\frac{\delta}{2}\right)^2} - 0 \right| \\ & = \left| \frac{1}{2} \right| \\ & > \varepsilon \end{aligned}$$

which proves that f is not continuous at $(0, 0)$. So f can not be Fréchet differentiable at $(0, 0)$ because Fréchet differentiable functions are continuous [see theorem: 16.63]. However we have that

$$\begin{aligned} (f \circ I^{[1,0]})(t) &= f(t, 0) = \begin{cases} \frac{t \cdot 0}{t^2 + 0^2} & \text{if } (t, 0) \neq (0, 0) \\ 0 & \text{if } (t, 0) = 0 \end{cases} = 0 = C_0(t) \\ (f \circ I^{[2,0]})(t) &= f(0, t) = \begin{cases} \frac{0 \cdot t}{0^2 + t^2} & \text{if } (0, t) \neq (0, 0) \\ 0 & \text{if } (0, t) = 0 \end{cases} = 0 = C_0(t) \end{aligned}$$

proving that

$$f^{[1, [(0, 0)]]} = C_0 = f^{[2, [(0, 0)]]}$$

By [example: 16.54] $f \circ I^{[1,0]}$, $f \circ I^{[2,0]}$ are Fréchet differentiable at 0 with

$$D_0(f \circ I^{[1,0]}) = C_0 = D_0 f \circ I^{[2,0]}$$

Hence f is 1-partial and 2-partial differentiable at $(0,0)$ with

$$D_{(0,0),1}f = C_0 = D_{(0,0),2}f$$

□

As the partial differential is based on the Fréchet differential of $f \circ I^{[i,x]}$ many properties of the Fréchet differential also applies for the partial differential.

Theorem 16.86. *Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle \prod_{j \in \{1, \dots, n\}} X_j, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $\langle Y, \|\cdot\| \rangle$ a normed space, $A \subseteq \prod_{i \in \{1, \dots, n\}} X_i$, $x \in A$ and $i \in \{1, \dots, n\}$ then we have:*

1. *If $f: A \rightarrow Y$ and $g: A \rightarrow Y$ are functions that are i -partial differentiable at x then*

$$f + g: A \rightarrow Y$$

is i -partial differentiable at x with

$$D_{x,i}(f + g) = D_{x,i}f + D_{x,i}g$$

2. *If $\alpha \in \mathbb{K}$ and $f: A \rightarrow Y$ is a function that is i -partial differentiable at x then*

$$\alpha \cdot f: A \rightarrow Y$$

is i -partial differentiable at x with

$$D_{x,i}(\alpha \cdot f) = \alpha \cdot D_{x,i}f$$

3. *If $f: X \rightarrow Y$ is such that $f \circ I^{[i,x]} \in L(X_i, Y)$ then f is i -partial differentiable and*

$$D_{x,i}f = f \circ I^{[i,x]}$$

Proof.

1. As f, g are i -partial differentiable at x we have by definition that $f \circ I^{[i,x]}$ and $g \circ I^{[i,x]}$ are Fréchet differentiable at x_i with

$$D_{x,i}f = D_{x_i}(f \circ I^{[i,x]}) \text{ and } D_{x,i}g = D_{x_i}(g \circ I^{[i,x]}) \quad (16.37)$$

Using [theorem: 16.66] we have that

$$f \circ I^{[i,x]} + g \circ I^{[i,x]} \text{ is Fréchet differentiable at } x_i$$

with

$$D_{x_i}(f \circ I^{[i,x]} + g \circ I^{[i,x]}) = D_{x_i}(f \circ I^{[i,x]}) + D_{x_i}(g \circ I^{[i,x]})$$

As $(f + g) \circ I^{[i,x]} = f \circ I^{[i,x]} + g \circ I^{[i,x]}$ it follows that $f + g$ is Fréchet differentiable at x_i with

$$D_{x_i}((f + g) \circ I^{[i,x]}) = D_{x_i}(f \circ I^{[i,x]}) + D_{x_i}(g \circ I^{[i,x]}) \stackrel{[\text{eq: 16.37}]}{=} D_{x,i}f + D_{x,i}g$$

Hence by definition

$$f + g \text{ is } i\text{-partial differentiable with } D_{x,i}(f + g) = D_{x,i}f + D_{x,i}g$$

2. As f is i -partial differentiable we have by definition that $f \circ I^{[i,x]}$ is Fréchet differentiable at x_i with

$$D_{x,i}f = D_{x_i}(f \circ I^{[i,x]}) \quad (16.38)$$

Using [theorem: 16.66] we have that

$$\alpha \cdot (f \circ I^{[i,x]}) \text{ is Fréchet differentiable at } x_i$$

with

$$D_{x_i}(\alpha \cdot (f \circ I^{[i,x]})) = \alpha \cdot D_{x_i}(f \circ I^{[i,x]})$$

As $(\alpha \cdot f) \circ I^{[i,x]} = \alpha \cdot (f \circ I^{[i,x]})$ it follows that $\alpha \cdot f$ is Fréchet differentiable at x_i with

$$D_{x_i}((\alpha f) \circ I^{[i,x]}) = \alpha \cdot D_{x_i}(f \circ I^{[i,x]}) \stackrel{[\text{eq: 16.38}]}{=} \alpha \cdot D_{x,i}f$$

Hence by definition

$$\alpha \cdot f \text{ is } i\text{-partial differentiable with } D_{x,i}(\alpha \cdot f) = \alpha \cdot D_{x,i}f$$

3. As $f \circ I^{[i,x]} \in L(X_i, Y)$ we have by [example: 16.55] that $f \circ I^{[i,x]}$ is Fréchet differentiable at x_i with $D_{x_i}(f \circ I^{[i,x]}) = f \circ I^{[i,x]}$. Hence by definition we have that

$$f \text{ is } i\text{-partial differentiable at } x \text{ with } D_{x,i}f(x) = f \circ I^{[i,x]} \quad \square$$

We have a special version of the Chain rule for partial differentiation.

Theorem 16.87. *Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of vector spaces, $\langle \prod_{i \in \{1, \dots, n\}} X_i, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98], $\langle Y, \|\cdot\|_Y \rangle$, $\langle Z, \|\cdot\|_Z \rangle$ normed spaces, $A \subseteq \prod_{i \in \{1, \dots, n\}} X_i$, $B \subseteq Y$, $x \in A$, $i \in \{1, \dots, n\}$,*

$f: A \rightarrow Y$ a function that is i -partial differentiable at x such that $f(A) \subseteq B$ and $f(x) \in B$

and

$g: B \rightarrow Z$ a function that is Fréchet differentiable at $f(x)$

then

$$g \circ f: A \rightarrow Z \text{ is } i\text{-partial differentiable at } x \text{ and } D_{x,i}(g \circ f) = D_{f(x)}g \circ D_{x,i}f$$

Proof. As f is i -partial differentiable at x we have by [definition: 16.74] that

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(A) \rightarrow Y \text{ is Fréchet differentiable at } x_i$$

and

$$D_{x,i}f = D_{x_i}(f \circ I^{[i,x]}) \quad (16.39)$$

As $(f \circ I^{[i,x]})((I^{[i,x]})^{-1}(A)) \stackrel{[\text{theorem: 2.23}]}{=} f(I^{[i,x]}((I^{[i,x]})^{-1}(A))) \stackrel{[\text{theorem: 2.26}]}{\subseteq} f(A) \subseteq B$ and $g: B \rightarrow Z$ is Fréchet differentiable at $f(x) = (f \circ I^{[i,x]})(x_i)$ it follows from the chain rule [theorem: 16.67] that

$$g \circ (f \circ I^{[i,x]}) \text{ is Fréchet differentiable at } x_i \quad (16.40)$$

with

$$D_{x_i}(g \circ (f \circ I^{[i,x]})) = D_{g((f \circ I^{[i,x]})(x_i))}g \circ D_{x_i}(f \circ I^{[i,x]}) \stackrel{[\text{eq: 16.39}]}{=} D_{g((f \circ I^{[i,x]})(x_i))}g \circ D_{x,i}f \quad (16.41)$$

As

$$(g \circ f) \circ I^{[i,x]} \stackrel{[\text{theorem: 2.21}]}{=} g \circ (f \circ I^{[i,x]})$$

and

$$g((f \circ I^{[i,x]})(x_i)) = g(f(I^{[i,x]}(x_i))) \stackrel{[\text{lemma: 16.71}]}{=} g(f(x))$$

[eqs: 16.40, 16.41] becomes

$$(g \circ f) \circ I^{[i,x]} \text{ is Fréchet differentiable at } x_i \text{ and } D_{x_i}((g \circ f) \circ I^{[i,x]}) = D_{g(f(x))}g \circ D_{x,i}f$$

Hence we have by [definition: 16.74] that

$$g \circ f \text{ is partial differentiable at } x \text{ and } D_{x,i}(g \circ f) = D_{g(f(x))}g \circ D_{x,i}f \quad \square$$

16.1.6 Partial Derivatives

Definition 16.88. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ where $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$ equipped with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $x \in A$, $i \in \{1, \dots, n\}$ and

$$f: A \rightarrow X \text{ a function}$$

then f has a ***i*-partial derivative at x** noted by $\partial_{x,i}f(x) \in X$ iff

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(A) \rightarrow X$$

has a derivative at x_i , the ***i*-partial derivative at x** is defined by

$$\partial_{x,i}f = (f \circ I^{[i,x]})'_{x_i} \in X$$

Note 16.89. The existence of the derivative of $f \circ I^{[i,x]}$ at x_i implies by [definition: 16.22] that

$$x \in ((I^{[i,x]})^{-1}(A))'$$

Note 16.90. By [corollary: 2.83] $f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(A) \rightarrow X$ is indeed a function.

Using the above definition we can define the partial derivative of a function.

Definition 16.91. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ equipped with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}$ and $f: A \rightarrow X$ then we define:

$$\mathcal{D}_{i,f} = \{x \in A \mid f \text{ has a } i\text{-partial derivative at } x\}$$

and

$$\partial_i f: \mathcal{D}_{i,f} \rightarrow X \text{ where } \partial_i f(x) = \partial_{x,i}f$$

Note 16.92. If $\mathcal{D}_{i,f} = \emptyset$ then $\partial_i f$ is the empty function [see example: 2.46].

Note 16.93. The benefit of this definition is that

1. $\partial_i f: \mathcal{D}_{i,f} \rightarrow X$ is always defined even if f has never a *i*-partial derivative at a point in A .
2. $\mathcal{D}_{i,f}$ is the set of all elements of the domain of f for which f has a *i*-partial derivative at, hence it is the most natural definition of the domain of $\partial_i f$.
3. If we refer to $\partial_i f$ where $f: A \rightarrow Y$ is a function then we can automatically assume that $\partial_i f$ is the graph of the function

$$\partial_i f: \mathcal{D}_{i,f} \rightarrow Y$$

Definition 16.94. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ equipped with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}$, $U \in \mathcal{T}_{\|\cdot\|_n}$ with $U \subseteq A$ and $f: A \rightarrow X$ then we say that

$$f \text{ has a } i\text{-partial derivative on } U$$

iff

$$\forall x \in U \text{ } f \text{ has a } i\text{-partial derivative at } x$$

We show now that partial derivatives are actual partial differentials in the special case where $\prod_{i \in \{1, \dots, n\}} X_i = \mathbb{K}^n$.

Theorem 16.95. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle \mathbb{K}, \|\cdot\|_n \rangle$ the vector space $\mathbb{K}^n$ equipped with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $x \in A$, $i \in \{1, \dots, n\}$ and $f: A \rightarrow X$ a function then we have:

1. If f has a *i*-partial differential at x then

$$f \text{ has a } i\text{-partial derivative with } \partial_{x,i}f = D_{x,i}f(1)$$

and

$$\forall h \in \mathbb{K} \text{ we have } D_{x,i}f(h) = h \cdot \partial_{x,i}f$$

2. If $x \in A^\circ$ and f has a i -partial derivative at x then

$$f \text{ has a } i\text{-partial differential at } x \text{ with } \partial_{x,i}f = D_{x,i}f(1)$$

and

$$\forall h \in \mathbb{K} \text{ we have } D_{x,i}f(h) = h \cdot \partial_{x,i}f$$

Proof.

1. As f has a i -partial differential at x , it follows from [definition: 16.74] that

$$f \circ I^{[i,x]} \text{ is Fréchet differentiable at } x_i \text{ with } D_{x,i}f = D_{x_i}(f \circ I^{[i,x]}) \quad (16.42)$$

Using [theorem: 16.64] on the above it follows that

$$f \circ I^{[i,x]} \text{ has a derivative at } x_i \text{ with } (f \circ I^{[i,x]})'_{x_i} = D_{x_i}(f \circ I^{[i,x]})(1) \quad (16.43)$$

and

$$\forall h \in \mathbb{K} \text{ we have } D_x(f \circ I^{[i,x]})(h) = h \cdot (f \circ I^{[i,x]})'_{x_i} \quad (16.44)$$

So by [definition: 16.88] it follows that

$$f \text{ has a } i\text{-partial derivative at } x$$

with

$$\partial_{x,i}f = (f \circ I^{[i,x]})'_{x_i} \quad (16.45)$$

Hence $\partial_{x,i}f = (f \circ I^{[i,x]})'_{x_i} \underset{[\text{eq: 16.43}]}{=} D_{x_i}(f \circ I^{[i,x]})(1) \underset{[\text{eq: 16.42}]}{=} D_{x,i}f(1)$ so that

$$\partial_{x,i}f = D_{x,i}f(1)$$

Further if $h \in \mathbb{K}$ then we have

$$D_{x,i}f(h) \underset{[\text{eq: 16.42}]}{=} D_{x_i}(f \circ I^{[i,x]})(h) \underset{[\text{eq: 16.44}]}{=} h \cdot (f \circ I^{[i,x]})'_{x_i} \underset{[\text{eq: 16.45}]}{=} h \cdot \partial_{x,i}f$$

so that

$$\forall h \in \mathbb{K} \text{ we have } D_{x,i}f(h) = h \cdot \partial_{x,i}f$$

2. As f has a i -partial derivative at x it follows from [definition: 16.88] that

$$f \circ I^{[i,x]} \text{ has a derivative at } x_i$$

with

$$\partial_{x,i}f = (f \circ I^{[i,x]})'_{x_i} \quad (16.46)$$

Using them [theorem: 16.64] it follows that

$$f \circ I^{[i,x]} \text{ is Fréchet differentiable at } x_i \quad (16.47)$$

$$(f \circ I^{[i,x]})'_{x_i} = D_{x_i}(f \circ I^{[i,x]})(1) \quad (16.48)$$

$$\forall h \in \mathbb{R} \ D_{x_i}(f \circ I^{[i,x]})(h) = h \cdot (f \circ I^{[i,x]})'_{x_i} \quad (16.49)$$

Using [definition: 16.74] on [eq: 16.47] we have then that

$$f \text{ is } i\text{-partial differentiable at } x$$

with

$$D_{x,i}f = D_{x_i}(f \circ I^{[i,x]}) \quad (16.50)$$

Further $\partial_{x,i}f \underset{[\text{eq: 16.46}]}{=} (f \circ I^{[i,x]})'_{x_i} \underset{[\text{eq: 16.48}]}{=} D_{x_i}(f \circ I^{[i,x]})(1) \underset{[\text{eq: 16.50}]}{=} D_{x,i}f(1)$ so that

$$\partial_{x,i}f = D_{x,i}f(1)$$

Finally $\forall h \in \mathbb{K}$ we have

$$D_{x,if}(h) \stackrel{[\text{eq: 16.50}]}{=} D_{x_i}(f \circ I^{[i,x]})(h) \stackrel{[\text{eq: 16.49}]}{=} h \cdot (f \circ I^{[i,x]})'_{x_i} \stackrel{[\text{eq: 16.46}]}{=} h \cdot \partial_{x,if}$$

so that

$$\forall h \in \mathbb{R} \text{ we have } D_{x,if}(h) = h \cdot \partial_{x,if}$$

□

We can rephrase [theorem: 16.83] as follows for partial derivatives.

Theorem 16.96. *Let $n \in \mathbb{N} \setminus \{1\}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ the vector space $\mathbb{K}^n$ equipped with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $x \in A$ and*

$f: A \rightarrow X$ a function that is Fréchet differentiable at x

then we have:

1. $\forall i \in \{1, \dots, n\}$ f has a i -partial derivative $\partial_{x,if}$ at x and a i -partial differential $D_{x,if} \in L(\mathbb{R}, X)$ which is defined by

$$D_{x,if}(h) = h \cdot \partial_{x,if}$$

so that

$$\partial_{x,if} = D_{x,if}(1)$$

2. Let $\{e_1, \dots, e_n\}$ be the canonical basis on $\mathbb{R}^n$ defined by $(e_i)_j = \delta_{i,j}$ [see theorem: 11.167] then

$$\forall i \in \{1, \dots, n\} \text{ we have } \partial_{x,if} = D_x f(e_i)$$

and

$$\forall h \in \mathbb{R}^n \text{ we have } D_x f(h) = \sum_{i=1}^n h_i \cdot \partial_{x,if}$$

If we use matrix notation where

$$h = \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix} \in \mathcal{M}_{n,1}(\mathbb{R}) \text{ is a column vector}$$

and

$$\partial_x f(x) = (\partial_{x,1f} \ \dots \ \partial_{x,nf}) \text{ is a row vector}$$

then it follows that we can write this using matrix notation as

$$D_x f(h) = \partial_x f \cdot h$$

Proof.

1. As f is Fréchet differentiable at x it follows from [definition: 16.46] that

$$x \in A^\circ$$

and by [theorem: 16.83] that $\forall i \in \{1, \dots, n\}$ f is i -partial differentiable at x . Using [theorem: 16.95] we conclude that f has a i -partial derivative with

$$D_{x,if}(h) = h \cdot \partial_{x,if} \tag{16.51}$$

and

$$\partial_{x,if} = D_{x,if}(1) \tag{16.52}$$

2. We have for $i, j \in \{1, \dots, n\}$ that

$$(I^{[i,0]}(1))_j = \left(0, \dots, 0, \underbrace{1}_i, 0, \dots, 0 \right)_j = \delta_{i,j} = (e_i)_j$$

or

$$I^{[i,0]}(1) = e_i \quad (16.53)$$

so that

$$\begin{aligned} \partial_{x,i} f &\stackrel{[\text{eq: 16.52}]}{=} D_{x,i} f(1) \\ &\stackrel{[\text{theorem: 16.83}]}{=} (D_x f \circ I^{[i,0]})(1) \\ &= D_x f(I^{[i,0]}(1)) \\ &\stackrel{[\text{eq: 16.53}]}{=} D_x f(e_i) \end{aligned}$$

proving that

$$\forall i \in \{1, \dots, n\} \quad \partial_{x,i} f = D_x f(e_i)$$

Further give $h \in \mathbb{R}^n$ we have

$$\begin{aligned} D_x f(h) &\stackrel{[\text{theorem: 16.83}]}{=} \sum_{i=1}^n D_{x,i} f(h_i) \\ &\stackrel{[\text{eq: 16.51}]}{=} \sum_{i=1}^n h_i \cdot \partial_{x,i} f \end{aligned}$$

□

16.1.7 Differentiation of vector valued functions

After examine the differentiation of a function of many variables we look now at the differentiability of vector valued function [functions whose values live in a finite product of normed spaces].

Definition 16.97. Let $n \in \mathbb{N} \setminus \{1\}$, X a set, $\{X_i\}_{i \in \{1, \dots, n\}}$ a finite family of sets and $\{f_i\}_{i \in \{1, \dots, n\}}$ a family such that $f_i \in X_i^X$ then we define the following function

$$(f_1, \dots, f_n)_*: X \rightarrow \prod_{i \in \{1, \dots, n\}} X_i \text{ by } (f_1, \dots, f_n)_*(x) = (f_1(x), \dots, f_n(x))$$

Note 16.98. We use the notation $(f_1, \dots, f_n)_*$ so that we don't confuse the above function with the tuple $(f_1, \dots, f_n)$ which is a element of $\prod_{i \in \{1, \dots, n\}} X_i^X$ and not a function from X to $\prod_{i \in \{1, \dots, n\}} X_i$.

Theorem 16.99. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle X, \|\cdot\| \rangle$ a normed space, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle \prod_{i \in \{1, \dots, n\}} X_i, \|\cdot\|_{\max} \rangle$ the product space equipped with the maximum norm [see theorem: 14.98] and $\{L_i\}_{i \in \{1, \dots, n\}}$ a family such that $\forall i \in \{1, \dots, n\} \quad L_i \in L(X, X_i)$ then

$$(L_1, \dots, L_n)_* \in L\left(X, \prod_{i \in \{1, \dots, n\}} X_i\right)$$

and

$$\|(L_1, \dots, L_n)_*\|_{L(X, \prod_{i \in \{1, \dots, n\}} X_i)} \leq \max(\{\|L_i\|_{L(X, X_i)} \mid i \in \{1, \dots, n\}\})$$

Proof. Let $x, y \in X$ and $\alpha \in \mathbb{K}$ then $\forall i \in \{1, \dots, n\}$ we have

$$\begin{aligned} ((L_1, \dots, L_n)_*(x + \alpha \cdot y))_i &= (L_1(x + \alpha \cdot y), \dots, L_i(x + \alpha \cdot y))_i \\ &= L_i(x + \alpha \cdot y) \\ &\stackrel{L_i \in L(X, X_i)}{=} L_i(x) + \alpha \cdot L_i(y) \\ &= (L_1(x), \dots, L_n(x))_i + \alpha \cdot (L_1(y), \dots, L_n(y))_i \\ &= ((L_1, \dots, L_n)_*(x))_i + (\alpha \cdot (L_1, \dots, L_n)_*(y))_i \\ &= ((L_1, \dots, L_n)_*(x) + \alpha \cdot (L_1, \dots, L_n)_*(y))_i \end{aligned}$$

proving that $(L_1, \dots, L_n)_*(x + \alpha \cdot y) = (L_1, \dots, L_n)_*(x) + \alpha \cdot (L_1, \dots, L_n)_*(y)$. Hence

$$(L_1, \dots, L_n)_* \in \text{Hom}\left(X, \prod_{i \in \{1, \dots, n\}} X_i\right)$$

Further if $x \in X$ then we have

$$\begin{aligned} \|(L_1, \dots, L_n)_*(x)\|_{\max} &= \|(L_1(x), \dots, L_n(x))\|_{\max} \\ &= \max(\{\|L_i(x)\|_i | i \in \{1, \dots, n\}\}) \\ &\leq \max(\{\|L_i\|_{L(X, X_i)} \cdot \|x\| | i \in \{1, \dots, n\}\}) \\ &\stackrel{[\text{theorem: 3.69}]}{\leq} \max(\{\|L_i\|_{L(X, X_i)} | i \in \{1, \dots, n\}\}) \cdot \|x\| \\ &\stackrel{[\text{theorem: 10.27}]}{\leq} \end{aligned}$$

proving by [theorem: 14.201] that

$$(L_1, \dots, L_n)_* \in L\left(X, \prod_{i \in \{1, \dots, n\}} X_i\right)$$

and by [theorem: 14.212]

$$\|(L_1, \dots, L_n)_*\|_{L(X, \prod_{i \in \{1, \dots, n\}} X_i)} \leq \max(\{\|L_i\|_{L(X, X_i)} | i \in \{1, \dots, n\}\}) \quad \square$$

Theorem 16.100. Let $n \in \mathbb{N} \setminus \{1\}$, X a set and $\{X_i\}_{i \in \{1, \dots, n\}}$ a family of sets and

$$f: X \rightarrow \prod_{i \in \{1, \dots, n\}} X_i \text{ a function}$$

then

$$f = (\pi_1 \circ f, \dots, \pi_n \circ f)_*$$

Proof. Let $x \in X$ then we have $\forall i \in \{1, \dots, n\}$ that

$$(f(x))_i = \pi_i(f(x)) = (\pi_i \circ f)(x)$$

so that

$$f(x) = ((\pi_1 \circ f)(x), \dots, (\pi_n \circ f)(x)) = (\pi_1 \circ f, \dots, \pi_n \circ f)_*(x)$$

proving that

$$f = (\pi_1 \circ f, \dots, \pi_n \circ f)_* \quad \square$$

Theorem 16.101. Let $n \in \mathbb{N}_0 \setminus \{1\}$, $\langle X, \|\cdot\| \rangle$ be a normed space, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a family of normed spaces, $\langle \prod_{i \in \{1, \dots, n\}} X_i, \|\cdot\|_{\max} \rangle$ the normed product space equipped with the maximum norm [see theorem: 14.98], $A \subseteq X$, $x \in A$ and

$$f: A \rightarrow \prod_{i \in \{1, \dots, n\}} X_i \text{ a function}$$

then we have

$$f \text{ is Fréchet differentiable at } x \Leftrightarrow \forall i \in \{1, \dots, n\} \pi_i \circ f \text{ is Fréchet differentiable at } x$$

Further if f is Fréchet differentiable at x then

$$\forall i \in \{1, \dots, n\} \pi_i \circ D_x f = D_x(\pi_i \circ f)$$

or in other words

$$D_x f = (D_x(\pi_1 \circ f), \dots, D_x(\pi_n \circ f))_*$$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}^+$. As f is Fréchet differentiable at x we have

$$x \in A^\circ$$

and for $D_x f \in L(X, \prod_{i \in \{1, \dots, n\}} X_i)$ that

$$\exists \delta \in \mathbb{R}^+ \text{ such that } \forall h \in A_x \text{ with } \|h\| < \delta \quad \|f(x+h) - f(x) - D_x f(h)\|_{\max} \leq \varepsilon \cdot \|h\| \quad (16.54)$$

Let $i \in \{1, \dots, n\}$ then we have $\forall h \in A_x$ with $\|h\| < \delta$ that

$$\begin{aligned} \|(\pi_i \circ f)(x+h) - (\pi_i \circ f)(x) - (\pi_i \circ D_x f)(h)\|_i &= \\ \|(\pi_i(f(x+h)) - \pi_i(f(x)) - \pi_i(D_x f(h)))\|_i &= \\ \|\pi_i(f(x+h) - f(x) - D_x f(h))\|_i &\leq \\ \max(\{\|\pi_i(f(x+h) - f(x) - D_x f(h))\|_i | i \in \{1, \dots, n\}\}) &\stackrel{\text{def}}{=} \\ \|f(x+h) - f(x) - D_x f(h)\|_{\max} &\stackrel{[eq: 16.54]}{\leq} \\ \varepsilon \cdot \|h\| & \end{aligned}$$

Further, using [example: 14.217] and [theorem: 14.218], it follows that

$$\pi_i \circ D_x f \in L(X, X_i)$$

Hence we have by [definition: 16.46] that $\pi_i \circ f$ is Fréchet differentiable at x with

$$D_x(\pi_i \circ f) = \pi_i \circ D_x f$$

$\Leftarrow$. Let $i \in \{1, \dots, n\}$ then as $\pi_i \circ f$ is Fréchet differentiable at x we have that

$$x \in A^\circ$$

and there exist a $D_x(\pi_i \circ f) \in L(X, X_i)$ such that $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta_{i,\varepsilon}$ such that

$$\forall h \in A_x \text{ with } \|h\| < \delta_{i,\varepsilon} \quad \|(\pi_i \circ f)(x+h) - (\pi_i \circ f)(x) - D_x(\pi_i \circ f)(h)\|_i \leq \varepsilon \cdot \|h\| \quad (16.55)$$

Using [theorem: 16.99]

$$L = (D_x(\pi_1 \circ f), \dots, D_x(\pi_n \circ f))_* \in L\left(X, \prod_{i \in \{1, \dots, n\}} X_i\right) \quad (16.56)$$

Let $\varepsilon \in \mathbb{R}^+$ and take $\delta = \min(\{\delta_{i,\varepsilon} | i \in \{1, \dots, n\}\})$ then we have

$$\begin{aligned} \|f(x+h) - f(x) - L(h)\|_{\max} &= \\ \max(\{\|\pi_i(f(x+h) - f(x) - L(h))\|_i | i \in \{1, \dots, n\}\}) &= \\ \max(\{\|\pi_i(f(x+h)) - \pi_i(f(x)) - \pi_i(L(h))\|_i | i \in \{1, \dots, n\}\}) &= \\ \max(\{\|(\pi_i \circ f)(x+h) - (\pi_i \circ f)(x) - D_x(\pi_i \circ f)(h)\|_i | i \in \{1, \dots, n\}\}) &\stackrel{[eq: 16.55]}{\leq} \\ \varepsilon \cdot \|h\| & \end{aligned}$$

hence

f is Fréchet differentiable at x

with

$$D_x f = L = (D_x(\pi_1 \circ f), \dots, D_x(\pi_n \circ f))_*$$

□

Using the above we have the following chain rule for the differential of vector valued functions.

Theorem 16.102. Let $n \in \mathbb{N}_0 \setminus \{1\}$, $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\{\langle Y_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a family of normed spaces, $\langle \prod_{i \in \{1, \dots, n\}} Y_i, \|\cdot\|_{\max} \rangle$ the normed spaces equipped with the maximum norm [see theorem: 14.98], $\langle Z, \|\cdot\|_Z \rangle$ a normed space, $A \subseteq X$, $x \in A$, $B \subseteq \prod_{i \in \{1, \dots, n\}} Y_i$ and

$$f: A \rightarrow \prod_{i \in \{1, \dots, n\}} Y_i \text{ and } g: B \rightarrow Z$$

functions such that

$$f(A) \subseteq B, f \text{ is Fréchet differentiable at } x \text{ and } g \text{ Fréchet differentiable at } f(x)$$

then

$$g \circ f \text{ is Fréchet differentiable at } x$$

and

$$D_x(g \circ f) = \sum_{i=1}^n (D_{f(x),i} g \circ D_x(\pi_i \circ f))$$

Proof. First using the Chain Rule [see theorem: 16.67] it follows that

$$g \circ f \text{ is Fréchet differentiable at } x \text{ with } D_x(g \circ f) = D_{f(x)} g \circ D_x f$$

Let $h \in X$ then

$$\begin{aligned} D_x(g \circ f)(h) &= (D_{f(x)} g \circ D_x f)(h) \\ &= D_{f(x)} g(D_x f(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} \sum_{i=1}^n D_{f(x),i} g((D_x f(h))_i) \\ &= \sum_{i=1}^n D_{f(x),i} g(\pi_i(D_x f(h))) \\ &= \sum_{i=1}^n D_{f(x),i} g((\pi_i \circ D_x f)(h)) \\ &\stackrel{[\text{theorem: 16.101}]}{=} \sum_{i=1}^n D_{f(x),i} g((D_x(\pi_i \circ f))(h)) \\ &= \sum_{i=1}^n (D_{f(x),i} g \circ D_x(\pi_i \circ f))(h) \\ &= \left(\sum_{i=1}^n (D_{f(x),i} g \circ D_x(\pi_i \circ f)) \right)(h) \end{aligned}$$

proving that

$$D_x(g \circ f) = \sum_{i=1}^n (D_{f(x),i} g \circ D_x(\pi_i \circ f)) \quad \square$$

We can now look at the differentiability of functions between $\mathbb{K}^n$ and $\mathbb{K}^m$.

Theorem 16.103. (Jacobian matrix) Let $n, m \in \mathbb{N} \setminus \{1\}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$, $\langle \mathbb{K}^m, \|\cdot\|_m \rangle$ two normed spaces [based on the maximum norms] over the field $\mathbb{K}$, $A \subseteq \mathbb{K}^n$, $x \in A$ and

$f: A \rightarrow \mathbb{K}^m$ a function that is Fréchet differentiable at x

then $\forall i \in \{1, \dots, n\}$, $\forall j \in \{1, \dots, m\}$ we have that $\pi_j \circ f: A \rightarrow \mathbb{K}$ has a i -partial derivative and

$$\forall h \in \mathbb{K}^n \text{ we have } (D_x f(h))_j \stackrel{\text{def}}{=} \pi_j(D_x f(h)) = \sum_{i=1}^n \partial_{x,i}(\pi_j \circ f) \cdot h_i$$

If we define the following matrices

$$h = \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix} \in \mathcal{M}_{n,1}(\mathbb{R})$$

and

$$D_x f(h) = \begin{pmatrix} (D_x f(h))_1 \\ \vdots \\ (D_x f(h))_m \end{pmatrix} \in \mathcal{M}_{m,1}(\mathbb{R})$$

and

$$\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x = \begin{pmatrix} \partial_{x,1}(\pi_1 \circ f) & \dots & \partial_{x,n}(\pi_1 \circ f) \\ \vdots & \ddots & \vdots \\ \partial_{x,1}(\pi_m \circ f) & \dots & \partial_{x,n}(\pi_m \circ f) \end{pmatrix} \in \mathcal{M}_{m,n}(\mathbb{R})$$

or

$$\forall i \in \{1, \dots, m\}, \forall j \in \{1, \dots, n\} \text{ we have } \left(\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \right)_{i,j} = \partial_{x,j}(\pi_i \circ f)$$

then we can express the Fréchet differential of f at x as the following matrix product

$$D_x f(h) = \left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \cdot h$$

The matrix $\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x$ is called the **Jacobian matrix of f at x** . Further if $E = \{e_1, \dots, e_n\}$ defined by $(e_i)_j = \delta_{i,j}$ is the canonical basis of $\mathbb{R}^n$ and $E' = \{e'_1, \dots, e'_m\}$ defined by $(e'_i)_j = \delta_{i,j}$ is the canonical basis of $\mathbb{R}^m$ then

$$\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x = \mathcal{M}(D_x f; E, E') \text{ [see definition: 11.339]}$$

Proof. Let $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$. As f is Fréchet differentiable at x it follows from [theorem: 16.101] that $\pi_j \circ f: A \rightarrow \mathbb{R}$ is Fréchet differentiable at x with

$$D_x(\pi_j \circ f) = \pi_j \circ D_x f \quad (16.57)$$

Using [theorem: 16.96] it follows that $\pi_j \circ f$ has a i -partial derivative at x with

$$D_x(\pi_j \circ f)(h) = \sum_{i=1}^n h_i \cdot \partial_{x,i}(\pi_j \circ f) \quad (16.58)$$

Further for $h \in \mathbb{R}^n$ we have

$$\begin{aligned} (D_x f(h))_j &= \pi_j(D_x f(h)) \\ &= (\pi_j \circ D_x f)(h) \\ &\stackrel{[\text{eq: 16.57}]}{=} D_x(\pi_j \circ f)(h) \\ &\stackrel{[\text{eq: 16.58}]}{=} \sum_{i=1}^n h_i \cdot \partial_{x,i}(\pi_j \circ f) \\ &\stackrel{\partial_{x,i}(\pi_j \circ f) \in \mathbb{K}}{=} \sum_{i=1}^n \partial_{x,i}(\pi_j \circ f) \cdot h_i \end{aligned}$$

proving that

$$(D_x f(h))_j = \sum_{i=1}^n \partial_{x,i}(\pi_j \circ f) \cdot h_i \quad (16.59)$$

Next for the matrix notation we have for $j \in \{1, \dots, m\}$ that

$$\begin{aligned} \left(\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \cdot h \right)_j &= \begin{pmatrix} \partial_{x,1}(\pi_1 \circ f) & \dots & \partial_{x,n}(\pi_1 \circ f) \\ \vdots & \ddots & \vdots \\ \partial_{x,1}(\pi_m \circ f) & \dots & \partial_{x,n}(\pi_m \circ f) \end{pmatrix} \cdot \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix} \\ &= \sum_{i=1}^n \left(\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \right)_{j,i} \cdot h_i \\ &= \sum_{i=1}^n \partial_{x,i}(\pi_j \circ f) \cdot h_i \\ &\stackrel{[\text{eq: 16.59}]}{=} (D_x f(h))_j \\ &= \begin{pmatrix} (D_x f(h))_1 \\ \vdots \\ (D_x f(h))_m \end{pmatrix}_j \end{aligned}$$

so that

$$\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \cdot h = D_x f(h)$$

Finally if $i \in \{1, \dots, n\}$, $j \in \{1, \dots, m\}$ that we have by [definition: 11.339] that

$$\begin{aligned} (D_x f(e_i))_j &= \left(\sum_{k=1}^m \mathcal{M}(D_x f; E, E')_{k,i} \cdot e'_k \right)_j \\ &= \sum_{k=1}^m \mathcal{M}(D_x f; E, E')_{k,i} \cdot (e'_k)_j \\ &= \sum_{k=1}^m \mathcal{M}(D_x f; E, E')_{k,i} \cdot \delta_{k,j} \\ &= \mathcal{M}(D_x f; E, F)_{j,i} \end{aligned}$$

Further we have

$$\begin{aligned} (D_x f(e_i))_j &\stackrel{[\text{eq: 16.59}]}{=} \sum_{k=1}^n \partial_{x,k}(\pi_j \circ f) \cdot (e_i)_k \\ &= \sum_{k=1}^n \partial_{x,k}(\pi_j \circ f) \cdot \delta_{i,k} \\ &= \partial_{x,i}(\pi_j \circ f) \\ &= \left(\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \right)_{j,i} \end{aligned}$$

so that

$$\mathcal{M}(D_x f; E, E')_{j,i} = \left(\left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \right)_{j,i}$$

or

$$\mathcal{M}(D_x f; E, E') = \left[\frac{\partial(f_1, \dots, f_m)}{\partial(1, \dots, n)} \right]_x \quad \square$$

We turn our attention now to the differentiability of the scalar product of differentiable functions. For this we look at the differentiability of bi-linear functions.

Theorem 16.104. *Let $\langle X_1, \|\cdot\|_1 \rangle$, $\langle X_2, \|\cdot\|_2 \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $\langle X_1 \cdot X_2, \|\cdot\|_{\max} \rangle$ the normed space of the product space $X_1 \cdot X_2$ with the maximum norm $\|\cdot\|_{\max}$, $L \in L(X_1 \cdot X_2; Y)$ [a bi-linear continuous mapping], $x = (x_1, x_2) \in X_1 \cdot X_2$ then L is differentiable at x and*

$$D_x L = D_{(x_1, x_2)} L = L(x_1, *) + L(*, x_2) \in L(X_1 \cdot X_2, Y)$$

where

$$L(x_1, *): X_1 \cdot X_2 \rightarrow Y \text{ is defined by } L(x_1, \star)(h) = L(x_1, *) (h_1, h_2) = L(x_1, h_2)$$

and

$$L(*, x_2): X_1 \cdot X_2 \rightarrow Y \text{ is defined by } L(\star, x_2)(h) = L(*, x_2) (h_1, h_2) = L(h_1, x_2)$$

Proof. First we prove that $L(x_1, *) + L(*, x_2)$ is linear. So let $(h_1, h_2), (g_1, g_2) \in X_1 \cdot X_2$ and $\alpha \in \mathbb{K}$ then we have

$$\begin{aligned} (L(x_1, *) + L(*, x_2))((h_1, h_2) + \alpha \cdot (g_1, g_2)) &= \\ (L(x_1, *) + L(*, x_2))(h_1 + \alpha \cdot g_1, h_2 + \alpha \cdot g_2) &= \\ L(x_1, h_2 + \alpha \cdot g_2) + L(h_1 + \alpha \cdot g_1, x_2) &= \\ L(x_1, h_2) + \alpha \cdot L(x_1, g_2) + L(h_1, x_2) + \alpha \cdot L(g_1, x_2) &= \\ L(x_1, h_2) + L(h_1, x_2) + \alpha \cdot (L(x_1, g_2) + L(g_1, x_2)) &= \\ (L(x_1, *) + L(*, x_2))(h_1, h_2) + \alpha \cdot ((L(x_1, *) + L(*, x_2))(g_1, g_2)) & \end{aligned}$$

Hence we have that

$$L(x_1, *) + L(*, x_2) \in \text{Hom}(X_1 \cdot X_2, Y) \quad (16.60)$$

Next we prove that $L(x_1, *) + L(*, x_2)$ is continuous. So let $(h_1, h_2) \in X_1 \cdot X_2$ with

$$\|h_1\|_1 = 1 = \|h_2\|_2 \quad (16.61)$$

then we have

$$\begin{aligned}
 & \| (L(x_1, *) + L(*, x_2))(h_1, h_2) \|_Y = \\
 & \| L(x_1, h_2) + L(h_1, x_2) \|_Y \leq \\
 & \| L(x_1, h_2) \|_Y + \| L(h_1, x_2) \|_Y \stackrel{[\text{theorem: 14.234}]}{\leq} \\
 & \| L \|_{L(X_1, X_2; Y)} \cdot \| x_1 \|_1 \cdot \| h_2 \|_2 + \| L \|_{L(X_1, X_2; Y)} \cdot \| x_2 \|_2 \cdot \| h_1 \|_1 \stackrel{[\text{eq: 16.61}]}{=} \\
 & \| L \|_{L(X_1, X_2; Y)} \cdot \| x_1 \|_1 + \| L \|_{L(X_1, X_2; Y)} \cdot \| x_2 \|_2
 \end{aligned}$$

proving by [theorem: 14.228] that $L(x_1, *) + L(*, x_2)$ is continuous, combining this with [eq: 16.60] results in

$$L(x_1, *) + L(*, x_2) \in L(X_1 \cdot X_2, Y) \quad (16.62)$$

Finally Let $\varepsilon \in \mathbb{R}^+$ and take $\delta = \frac{\varepsilon}{\|L\|_{L(X_1, X_2; Y)} + 1} \in \mathbb{R}^+$ then, if $(h_1, h_2) \in X_1 \cdot X_2 = (X_1 \cdot X_2)_x$ and $\|(h_1, h_2)\|_{\max} < \delta$, we have

$$\|h_1\|_1, \|h_2\|_2 \leq \max(\{\|h_i\|_i | i \in \{1, 2\}\}) = \|(h_1, h_2)\|_{\max} < \delta \quad (16.63)$$

and

$$\begin{aligned}
 & \| L((x_1, x_2) + (h_1, h_2)) - L(x_1, x_2) - (L(x_1, *) + L(*, x_2))(h_1, h_2) \|_Y = \\
 & \| L(x_1 + h_1, x_2 + h_2) - L(x_1, x_2) - L(x_1, h_2) - L(h_1, x_2) \|_Y = \\
 & \| L(x_1, x_2 + h_2) + L(h_1, x_2 + h_2) - L(x_1, x_2) - L(x_1, h_2) - L(h_1, x_2) \|_Y = \\
 & \left\| \underbrace{L(x_1, x_2)}_1 + \underbrace{L(x_1, h_2)}_2 + \underbrace{L(h_1, x_2)}_3 + L(h_1, h_2) - \underbrace{L(x_1, x_2)}_1 - \underbrace{L(x_1, h_2)}_2 - \underbrace{L(h_1, x_2)}_3 \right\|_Y = \\
 & \| L(h_1, h_2) \|_Y \stackrel{[\text{theorem: 14.234}]}{\leq} \\
 & \| L \|_{L(X_1, X_2; Y)} \cdot \| h_1 \|_1 \cdot \| h_2 \|_2 \stackrel{[\text{eq: 16.63}]}{<} \\
 & \| L \|_{L(X_1, X_2; Y)} \cdot \frac{\varepsilon}{\|L\|_{L(X_1, X_2; Y)} + 1} \cdot \| h_2 \|_2 < \\
 & \varepsilon \cdot \| h_2 \|_2 \stackrel{[\text{eq: 16.63}]}{\leq} \\
 & \varepsilon \cdot \|(h_1, h_2)\|_{\max}
 \end{aligned}$$

which proves that $L(x_1, *) + L(*, x_2)$ is a Fréchet approximation of L at x , further $x \in X_1 \cdot X_2 \stackrel{[\text{theorem: 14.16}]}{=} (X_1 \cdot X_2)^\circ$. Hence we have by [definition: 16.46] that

$$L \text{ is Fréchet differentiable at } x \text{ with } D_x L = L(x_1, *) + L(*, x_2) \quad \square$$

We return now to the differentiability of the scalar product of differentiable real or complex valued functions which is an extension of [theorem: 16.41]

Theorem 16.105. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ a normed space over $\mathbb{K}$, $\langle \mathbb{K}, \|\cdot\| \rangle$, $A \subseteq X$ open with $x \in A$ and

$f: A \rightarrow \mathbb{K}$ a function that is Fréchet differentiable at x

$g: A \rightarrow Y$ a function that is Fréchet differentiable at x

then

$$f \cdot g: A \rightarrow X \text{ defined by } (f \cdot g)(x) = f(x) \cdot g(x)$$

is Fréchet differentiable at x with

$$D_x(f \cdot g) = f(x) \cdot D_x g + D_x f \cdot g(x)$$

where $f(x) \cdot D_x g + D_x f \cdot g(x): X \rightarrow Y$ is defined by

$$(f(x) \cdot D_x g + D_x f \cdot g(x))(t) = f(x) \cdot D_x g(t) + D_x f(t) \cdot g(x)$$

Note 16.106. The multiplication operator is the scalar product $\cdot: \mathbb{K} \cdot Y \rightarrow Y$. If $Y = \mathbb{K}$ then this reduce to the product operator in the field $\mathbb{K}$.

Proof. Define the function

$$L: \mathbb{K} \cdot Y \rightarrow Y \text{ by } L(x, y) = x \cdot y$$

Let $y, z \in Y, \alpha, x \in \mathbb{K}$ then we have

$$L(x, y + \alpha \cdot z) = x \cdot (y + \alpha \cdot z) = x \cdot y + \alpha \cdot (x \cdot z) = L(x, y) + \alpha \cdot L(y, z)$$

and for $\alpha, x, y \in \mathbb{K}$ and $z \in Y$ we have

$$L(x + \alpha \cdot y, z) = (x + \alpha \cdot y) \cdot z = (x \cdot z) + \alpha \cdot (y \cdot z) = L(x, z) + \alpha \cdot L(y, z)$$

proving that L is bi-linear or

$$L \in \text{Hom}(\mathbb{K}, Y; Y) \quad (16.64)$$

Further if $h = (h_1, h_2) \in \mathbb{K} \cdot Y$ with $|h_1| = \|h_2\|_Y = 1$ then we have

$$\|L(h)\|_Y = \|L(h_1, h_2)\|_Y = \|h_1 \cdot h_2\|_Y = |h_1| \cdot \|h_2\| = 1$$

proving by [theorem: 14.228] that L is continuous. Combining this with [eq: 16.64] proves that $L \in L(\mathbb{K}, Y; Y)$. Using the previous theorem [see theorem: 16.104] it follows that

$$\forall y = (y_1, y_2) \in \mathbb{K} \cdot Y \text{ } L \text{ is differentiable at } y \text{ with } D_{(y_1, y_2)}L = L(y_1, *) + L(*, y_2) \quad (16.65)$$

Define

$$h: A \rightarrow \mathbb{K} \cdot Y \text{ by } h(x) = (f(x), g(x))$$

then $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$. As f, g are Fréchet differentiable at x it follows that $\pi_1 \circ h, \pi_2 \circ h$ are Fréchet differentiable at x with $D_x(\pi_1 \circ h) = D_x f$ and $D_x(\pi_2 \circ h) = D_x g$. So we can use [theorem: 16.101] giving

$$h \text{ is Fréchet differentiable at } x \text{ with } \pi_1 \circ D_x h = D_x f \text{ and } \pi_2 \circ D_x h = D_x g \quad (16.66)$$

Now $\forall y \in A$ we have

$$(L \circ h)(y) = (L(h(y))) = L(f(y), g(y)) = f(y) \cdot g(y) = (f \cdot g)(y)$$

proving that

$$L \circ h = f \cdot g$$

Using the Chain Rule [see theorem: 16.67] with [eqs: 16.65 and 16.66] we have that $L \circ h$ is Fréchet differentiable at x so that

$$f \cdot g \text{ is Fréchet differentiable at } x$$

and $\forall y \in X$ we have

$$\begin{aligned} D_x(g \cdot f)(y) &= D_x(L \circ h)(y) \\ &= (D_{h(x)}L \circ D_x h)(y) \\ &= D_{h(x)}L(D_x h(y)) \\ &= D_{(f(x), g(x))}L(D_x h(y)) \\ &\stackrel{[\text{theorem: 16.101}]}{=} D_{(f(x), g(x))}L((D_x(\pi_1 \circ h), D_x(\pi_2 \circ h))_*(y)) \\ &\stackrel{[\text{eq: 16.66}]}{=} D_{(f(x), g(x))}L((D_x f, D_x g)_*(y)) \\ &= D_{(f(x), g(x))}L(D_x f(y), D_x g(y)) \\ &\stackrel{[\text{eq: 16.65}]}{=} (L(f(x), *) + L(*, g(x)))(D_x f(y), D_x g(y)) \\ &= L(f(x), D_x g(y)) + L(D_x f(y), g(x)) \\ &= f(x) \cdot D_x g(y) + (D_x f(y)) \cdot g(x) \end{aligned} \quad (16.67)$$

So if we define $f(x) \cdot D_x g + D_x f \cdot g(x): X \rightarrow Y$ is by

$$(f(x) \cdot D_x g + D_x f \cdot g(x))(y) = f(x) \cdot D_x g(y) + D_x f(y) \cdot g(x)$$

then we have by [eq: 16.67]

$$D_x(g \cdot f) = f(x) \cdot D_x g + D_x f \cdot g(x)$$

□

16.1.8 Usefully derivatives

Theorem 16.107. Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of real numbers, $x \in \mathbb{R} \setminus \{0\}$ then we have that

$$\left(\frac{1}{*}\right): \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \text{ defined by } \left(\frac{1}{*}\right)(x) = \frac{1}{x}$$

has a derivative at x with

$$\left(\frac{1}{*}\right)'_x = -\frac{1}{x^2}$$

Proof. As $\{0\}$ is a closed set [see theorems: 14.242, 14.244] it follows that $\mathbb{R} \setminus \{0\}$ is an open set, hence, as $x \in \mathbb{R} \setminus \{0\}$, we have from [theorem: 16.7] that

$$x \text{ is a limit point of } \mathbb{R} \setminus \{0\} \quad (16.68)$$

Let $\varepsilon \in \mathbb{R}^+$. Then we have for $h \in (\mathbb{R} \setminus \{0\})_x$ with

$$\begin{aligned} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} - \left(-\frac{1}{x^2}\right) &= \frac{\frac{1}{x+h} - \frac{1}{x}}{h} + \frac{1}{x^2} \\ &= \frac{\frac{x-x-h}{(x+h) \cdot x}}{h} + \frac{1}{x^2} \\ &= \frac{\frac{-h}{(x+h) \cdot x}}{h} + \frac{1}{x^2} \\ &= \frac{-1}{(x+h) \cdot x} + \frac{1}{x^2} \\ &= \frac{-x + (x+h)}{(x+h) \cdot x^2} \\ &= \frac{h}{x^2 \cdot (x+h)} \end{aligned} \quad (16.69)$$

Take now $\delta = \min\left(\frac{\varepsilon}{2} \cdot |x|^3, \frac{|x|}{2}\right) \in \mathbb{R}^+$ [as $x \neq 0$] then we have for $h \in (\mathbb{R} \setminus \{0\})_x$ with $0 < |h| < \delta$ that

$$|h| < \frac{|x|}{2} < |x| \Rightarrow 0 < |x| - |h| \text{ and } \frac{|x|}{2} = |x| - \frac{|x|}{2} < |x| - |h| \Rightarrow \frac{1}{|x| - |h|} < \frac{2}{|x|} \quad (16.70)$$

so that

$$|x| = |x+h-h| \leq |x+h| + |h| \Rightarrow |x| - |h| < |x+h| \stackrel{[\text{eq: 16.70}]}{\Rightarrow} \frac{1}{|x+h|} < \frac{1}{|x| - |h|} < \frac{2}{|x|} \quad (16.71)$$

and

$$|h| < \frac{\varepsilon}{2} \cdot |x|^3 \quad (16.72)$$

Hence we have

$$\begin{aligned} \left| \frac{\frac{1}{x+h} - \frac{1}{x}}{h} - \left(-\frac{1}{x^2}\right) \right| &\stackrel{[\text{eq: 16.69}]}{=} \left| \frac{h}{x^2 \cdot (x+h)} \right| \\ &= \frac{|h|}{|x|^2 \cdot |x+h|} \\ &<_{[\text{eq: 16.71}]} \frac{|h| \cdot 2}{|x|^2 \cdot |x|} \\ &= \frac{|h| \cdot 2}{|x|^3} \\ &<_{[\text{eq: 16.72}]} \frac{\varepsilon \cdot |x|^3}{2 \cdot |x|^3} \\ &= \varepsilon \end{aligned}$$

which proves that

$$\left(\frac{1}{*}\right) \text{ has a derivative at } x \text{ and } \left(\frac{1}{*}\right)'_x = -\frac{1}{x^2}$$

□

Theorem 16.108. *We have*

1. *If $n \in \mathbb{N}$ then $\forall x \in \mathbb{R}$*

$$(\cdot)^n: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } (\cdot)^n(x) = x^n$$

has a derivative at x and

$$((\cdot)^n)'_x = n \cdot (\cdot)^{n-1}(x)$$

or in a shorter notation

$$(x^n)'_x = n \cdot x^{n-1}$$

2. *If $z \in \{-n \mid n \in \mathbb{N}\}$ then $\forall x \in \mathbb{R} \setminus \{0\}$ [a open subset of $\mathbb{R}$] then*

$$(\cdot): \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \text{ defined by } (\cdot)^z(x) = \frac{1}{x^{(-z)}}$$

has a derivative at x and

$$((\cdot)^z)'_x = z \cdot (\cdot)^{z-1}$$

or in a shorter notation

$$(x^{-n})'_x = -n \cdot x^{-(n+1)} = -n \cdot x^{-n-1}$$

Proof.

1. First note that

$$x \in \mathbb{R}_{[\text{theorem: 14.16}]} = \mathbb{R}^\circ$$

We use induction to prove that $(\cdot)^n$ has a derivative at x

$$S = \{n \in \mathbb{N} \mid (\cdot)^n \text{ has a derivative at } x \text{ and } ((\cdot)^n)'_x = n \cdot (\cdot)^{n-1}(x)\}$$

then we have:

1 $\in S$. If $n = 1$ then $(\cdot)^1 = (\cdot)^1 = \text{Id}_{\mathbb{R}}$ so that by [example: 16.31] and the fact that $\mathbb{R}$ is open

$$(\cdot)^1 \text{ has a derivative at } x \text{ and}$$

and

$$((\cdot)^1)'_x = (\text{Id}_{\mathbb{R}})'_x = 1 = 1 \cdot (\cdot)^0(x) = 1 \cdot (\cdot)^{1-1}(x)$$

proving that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Given $x \in X$ we have

$$(\cdot)^{n+1}(x) = x^{n+1} = x \cdot x^n = \text{Id}_X(x) \cdot (\cdot)^n(x)$$

so that

$$(\cdot)^{n+1} = \text{Id}_{\mathbb{R}} \cdot (\cdot)^n$$

As $n \in S$ we have that $(\cdot)^n$ has a derivative at x with

$$((\cdot)^n)'_x = n \cdot (\cdot)^{n-1}(x)$$

So $(\cdot)^{n+1}$ is the product of two functions that have a derivative at x , hence by [theorem: 16.41] we have that

$$(\cdot)^{n+1} \text{ has a derivative at } x$$

and

$$\begin{aligned} ((\cdot)^{n+1})'_x &= (\text{Id}_{\mathbb{R}} \cdot (\cdot)^n)'_x \\ &= \text{Id}_{\mathbb{R}}(x) \cdot ((\cdot)^n)'_x + (\cdot)^n(x) \cdot (\text{Id}_{\mathbb{R}})'_x \\ &= x \cdot n \cdot x^{n-1} + x^n \cdot 1 \\ &= n \cdot x^n + 1 \cdot x^n \\ &= (n+1) \cdot x^n \\ &= (n+1) \cdot x^{(n+1)-1} \\ &= (n+1) \cdot (\cdot)^{(n+1)-1}(x) \end{aligned}$$

proving that $n+1 \in S$.

2. Let $z = -n \in \{-n | n \in \mathbb{N}\}$ then for every $y \in \mathbb{R} \setminus \{0\}$ we have

$$\left(\left(\frac{1}{*} \right) \circ (\cdot)^n \right) (y) = \left(\frac{1}{*} \right) ((\cdot)^n(y)) = \left(\frac{1}{*} \right) (y^n) = \frac{1}{y^n} = \frac{1}{y^{-z}} = (\cdot)^z(y) \quad (16.73)$$

So $(\cdot)^z$ is the composition of $\left(\frac{1}{*} \right)$ and $(\cdot)^n$ where these functions have a derivative at x and

$$((\cdot)^n)'_x = n \cdot x^{n-1} \text{ and } \left(\frac{1}{*} \right)'_{x \text{ [eq: 16.107]}} = -\frac{1}{x^2} \quad (16.74)$$

As $\{0\}$ is a closed set [see theorems: 14.242, 14.244] it follows that $\mathbb{R} \setminus \{0\}$ is a open set, hence by [theorem: 14.16] we have

$$x \in (\mathbb{R} \setminus \{0\})^\circ \quad (16.75)$$

Hence using [theorem: 16.68] we have that $(\cdot)^z$ has a derivative at x and

$$\begin{aligned} ((\cdot)^z)'_x &= \left(\left(\frac{1}{*} \right) \circ (\cdot)^n \right)'_x \\ &\stackrel{\text{[theorem: 16.68]}}{=} ((\cdot)^n)'_x \cdot \left(\frac{1}{*} \right)'_{(\cdot)^n(x)} \\ &\stackrel{\text{[eq: 16.74]}}{=} n \cdot x^{n-1} \cdot \frac{-1}{((\cdot)^n(x))^2} \\ &= n \cdot x^{n-1} \cdot \frac{-1}{x^{2 \cdot n}} \\ &= -n \cdot x^{n-1} \cdot \frac{1}{x^{n-1} \cdot x^{2 \cdot n - (n-1)}} \\ &= -n \cdot \frac{1}{x^{n+1}} \\ &= z \cdot \frac{1}{x^{n+1}} \\ &= z \cdot (\cdot)^{-(n+1)}(x) \\ &= z \cdot (\cdot)^{z-1}(x) \\ &\quad \square \end{aligned}$$

Theorem 16.109. Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of real (complex) numbers, $A \subseteq \mathbb{R}$, $x \in A^\circ$ and $f: A \rightarrow \mathbb{K}$ and $g: A \rightarrow \mathbb{K}$ are functions such that $g(A) \subseteq \mathbb{R} \setminus \{0\}$, f has a derivative at x and g has a derivative at x then the function

$$\frac{f}{g}: A \rightarrow \mathbb{R} \text{ defined by } \frac{f}{g}(x) = \frac{f(x)}{g(x)}$$

has a derivative at x and

$$\left(\frac{f}{g} \right)' = \frac{f'_x \cdot g(x) - g'_x \cdot f(x)}{(g(x))^2}$$

Proof. Let $x \in A^\circ$. As $g: A \rightarrow \mathbb{K}$ has a derivative g'_x at x , $g(A) \subseteq \mathbb{R} \setminus \{0\}$ and $\left(\frac{1}{*} \right): \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ has a derivative at $g(x)$ [see theorem: 16.107] it follows from the chain rule [see theorem: 16.68] that

$$\left(\left(\frac{1}{*} \right) \circ g \right): \mathbb{R} \rightarrow \mathbb{R}$$

has a derivative at x with

$$\left(\left(\frac{1}{*} \right) \circ g \right)'_x = \left(\frac{1}{*} \right)'_{g(x)} \cdot g'_x \stackrel{\text{[theorem: 16.107]}}{=} -\frac{1}{(g(x))^2} \cdot g'_x \quad (16.76)$$

Using the above and the fact that f has also a derivative at x it follow from the product rule [theorem: 16.41] that $f \cdot \left(\left(\frac{1}{*}\right) \circ g\right)$ has a derivative at x and

$$\begin{aligned}
 \left(f \cdot \left(\left(\frac{1}{*}\right) \circ g\right)\right)'_x &= f'_x \cdot \left(\left(\frac{1}{*}\right) \circ g\right)(x) + f(x) \cdot \left(\left(\frac{1}{*}\right) \circ g\right)'_x \\
 &= f'_x \cdot \frac{1}{g(x)} + f(x) \cdot \left(\left(\frac{1}{*}\right) \circ g\right)'_x \\
 &\stackrel{[\text{eq: 16.76}]}{=} f'_x \cdot \frac{1}{g(x)} + f(x) \cdot \left(-\frac{1}{(g(x))^2}\right) \cdot g'_x \\
 &= \frac{f'_x}{g(x)} - f(x) \cdot \frac{g'_x}{(g(x))^2} \\
 &= \frac{f'_x \cdot g(x) - f(x) \cdot g'_x}{(g(x))^2}
 \end{aligned}$$

Finally we have $\forall t \in \mathbb{R}$ that $(f \cdot \left(\left(\frac{1}{*}\right) \circ g\right))(t) = f(t) \cdot \left(\frac{1}{*}\right)(g(t)) = f(t) \cdot \frac{1}{g(t)} = \left(\frac{f}{g}\right)(t)$ proving that $\frac{f}{g} = f \cdot \left(\left(\frac{1}{*}\right) \circ g\right)$, hence

$$\left(\frac{f}{g}\right)'_x = \frac{f'_x \cdot g(x) - f(x) \cdot g'_x}{(g(x))^2}$$

□

16.2 Higher order differentials

16.2.1 Linear mappings and multi-linear mappings

The idea of higher order differentiation looks simple, just differentiate the function that maps a point to it's differential, but it actually rather complex. If a function $f: A \rightarrow Y$ is Fréchet differentiable on A then $\forall x \in A$ $D_x f \in L(X, Y)$ exist so that we can define the function

$$Df: A \rightarrow L(X, Y) \text{ by } (Df)(x) = D_x f$$

and ask if that function is Fréchet differentiable at $x \in U$. If so then $D_x(Df) \in L(X, L(X, Y))$ is defined to be the second order differential of f noted as $D_x^2 f$. Proceeding in this way we can define the third order differential $D_x^3 f$ at x and so on. There is however a problem, if we look at the types of the differentials we have

$$\begin{aligned}
 D_x f &\in L(X, Y) \\
 D_x^2 f &\in L(X, L(X, Y)) \\
 D_x^3 f &\in L(X, L(X, L(X, Y))) \\
 &\dots
 \end{aligned}$$

So we have to learn to work with sets like $L(X, L(X, Y))$, $L(X, L(X, L(X, Y))) \dots$. Evaluating the higher order differentials is rather elaborated because we have to use

$$\begin{aligned}
 (Df(x))(y) &\in Y \\
 ((D^2 f(x))(y_1))(y_2) &\in Y \\
 (((D^3 f(x))(y_1))(y_2))(y_3) &\in Y \\
 &\dots
 \end{aligned}$$

It would be nicer if the higher order differentials are multi-linear functions so that we have

$$\begin{aligned}
 D_x f(y) &\in Y \\
 D_x^2 f(y_1, y_2) &\in Y \\
 D_x^3 f(y_1, y_2, y_3) &\in Y \\
 &\dots \\
 D_x^n f(y_1, \dots, y_n) &\in Y
 \end{aligned}$$

To be able to do this we must do the following identifications

$$\begin{aligned} L(X, L(X, Y)) & \text{ is identified with } L^2(X; Y) \\ L(X, L(X, L(X, Y))) & \text{ is identified with } L^3(X; Y) \\ & \dots \end{aligned}$$

where the identification maintains linearity, continuity and the norm. This is the focus of this section. First we need some recursive definitions.

Definition 16.110. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces then we define $L_n(X; Y)$ recursively as follows.

$$L_n(X; Y) = \begin{cases} L(X, Y) & \text{if } n = 1 \\ L(X, L_{n-1}(X; Y)) & \text{if } 1 < n \end{cases}$$

Example 16.111. Using the above definition we have

$$\begin{aligned} L_1(X; Y) &= L(X, Y) \\ L_2(X; Y) &= L(X, L_1(X; Y)) \\ &= L(X, L(X, Y)) \\ L_3(X; Y) &= L(X, L_2(X; Y)) \\ &= L(X, L(X, L(X, Y))) \\ &\dots \dots \dots \\ L_n(X; Y) &= L(X, L(X, \dots, (L(X, Y)))) \end{aligned}$$

Lemma 16.112. Let $n, m \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces then

$$L_{n+m}(X; Y) = L_m(X; L_n(X; Y))$$

Proof. We prove this by induction, so define

$$S_n = \{m \in \mathbb{N} \mid \text{If } \langle X, \|\cdot\| \rangle \text{ is a normed space then } L_{n+m}(X; Y) = L_m(X; L_n(X; Y))\}$$

then we have:

$$1 \in S_n. \text{ As } L_{n+1}(X; Y) \stackrel{\text{def}}{=} L(X, L_n(X; Y)) \stackrel{\text{def}}{=} L_1(X; L_n(X; Y)) \text{ it follows that } 1 \in S_n.$$

$$m \in S_n \Rightarrow m+1 \in S_n. \text{ We have}$$

$$\begin{aligned} L_{n+(m+1)}(X; Y) &= L_{(n+m)+1}(X; Y) \\ &= L(X, L_{n+m}(X; Y)) \\ &\stackrel{m \in S_n}{=} L(X, L_m(X; L_n(X; Y))) \\ &\stackrel{\text{def}}{=} L_{m+1}(X; L_n(X; Y)) \end{aligned}$$

proving that $m+1 \in S$. □

Corollary 16.113. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces then

$$L_n(X; Y) = L_{n-1}(X; L_1(X; Y)) \stackrel{L_1(X; Y) = L(X, Y)}{=} L_{n-1}(X; L(X, Y))$$

Proof. We have

$$\begin{aligned} L_n(X; Y) &= L_{(n-1)+1}(X; Y) \\ &\stackrel{n-1 \in \mathbb{N} \wedge [\text{lemma: 16.112}]}{=} L_{n-1}(X; L_1(X; Y)) \end{aligned}$$

□

We want now to show that $L_n(X; Y)$ is isometric to $L^n(X; Y)$, first we show how elements of $L_n(X; Y)$ can act on tuples of elements of X .

Definition 16.114. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $L \in L_n(X; Y)$, $m \in \{1, \dots, n\}$ and $x = (x_1, \dots, x_m) \in X^m$ then we define $L(x_1: \dots: x_m)$ recursively as

$$L(x_1: \dots: x_m) = \begin{cases} L(x_1) & \text{if } m = 1 \\ (L(x_1))(x_2: \dots: x_m) & \text{if } 1 < m \leq n \end{cases}$$

Note 16.115. This is well defined as for $1 < m \leq n$ we have that $1 < n$ so that

$$L \in L_n(X; Y) = L(X, L_{n-1}(X; Y)) \text{ hence } L(x_1) \in L_{n-1}(X; Y)$$

Example 16.116. Let $n = 4$ then for $L \in L_4(X; Y) = L(X, L_3(X; Y))$ we have

$$\begin{aligned} L(x_1: \dots: x_4) &= (L(x_1))(x_2: \dots: x_4) \text{ where } L(x_1) \in L_3(X; Y) = L(X, L_2(X; Y)) \\ &= ((L(x_1))(x_2))(x_3: \dots: x_4) \text{ where } (L(x_1))(x_2) \in L_2(X; Y) = L(X, L_1(X; Y)) \\ &= (((L(x_1))(x_2))(x_3))(x_4: \dots: x_4) \text{ where } (((L(x_1))(x_2))(x_3)) \in L_1(X; Y) = L(X, Y) \\ &= (((L(x_1))(x_2))(x_3))(x_4) \in Y \\ L(x_1: \dots: x_3) &= (L(x_1))(x_2: \dots: x_3) \text{ where } L(x_1) \in L_3(X; Y) = L(X, L_2(X; Y)) \\ &= ((L(x_1))(x_2))(x_3: \dots: x_3) \text{ where } (L(x_1))(x_2) \in L_2(X; Y) = L(X, L_1(X; Y)) \\ &= ((L(x_1))(x_2))(x_3) \in L_1(X; Y) \end{aligned}$$

The above example suggests the following theorem that also ensures that [definition: 16.114] is valid.

Lemma 16.117. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $L \in L_n(X; Y)$, $m \in \{1, \dots, n\}$ and $x = (x_1, \dots, x_m) \in X^m$ then we have

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n \\ L_{n-m}(X; Y) & \text{if } m < n \end{cases}$$

Proof. We use induction to prove this so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } m \in \{1, \dots, n\} \text{ then } \forall L \in L_n(X; Y) \text{ we have } \forall (x_1, \dots, x_m) \in X^m \text{ that} \right.$$

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n \\ L_{n-m}(X; Y) & \text{if } m < n \end{cases} \left. \right\}$$

then we have:

$1 \in S$. If $m \in \{1, \dots, 1\} = \{1\}$ then $m = 1 = n$. Let $L \in L_1(X; Y) = L(X, Y)$ then we have for every $(x_1) \in X^1$ that $L(x_1: \dots: x_1) = L(x_1) \in Y$. Hence as $m = n$ we have

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n \\ L_{n-m}(X; Y) & \text{if } m < n \end{cases}$$

proving that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $L \in L_{n+1}(X; Y)$ then for $m \in \{1, \dots, n + 1\}$ we have for m either:

$m = 1$. As $L_{n+1}(X; Y) = L(X, L_n(X; Y))$ and $L(x_1: \dots: x_m) = L(x_1: \dots: x_1) = L(x_1)$ it follows that

$$L(x_1: \dots: x_m) \in L_n(X; Y) = L_{(n+1)-m}(X; Y)$$

proving as $m = 1 < n + 1$ that

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n + 1 \\ L_{(n+1)-m}(X; Y) & \text{if } m < n + 1 \end{cases}$$

$m = n + 1$. Given $(x_1, \dots, x_{n+1}) \in X^m = X^{n+1}$ we have, as $L_{n+1}(X; Y) = L(X, L_n(X; Y))$, that $L(x_1) \in L_n(X; Y)$. As $n \in S$, $(x_2, \dots, x_{n+1}) \in X^n$ it follows that

$$(L(x_1))(x_2: \dots: x_{n+1}) \in Y$$

hence

$$L(x_1: \dots: x_m) \underset{m=n+1}{=} L(x_1: \dots: x_{n+1}) = (L(x_1))(x_2: \dots: x_{n+1}) \in Y$$

proving as $m \not\leq n+1$ that

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n+1 \\ L_{(n+1)-m}(X; Y) & \text{if } m < n+1 \end{cases}$$

$m \in \{2, \dots, n\}$. Given $(x_1, \dots, x_m) \in X^m$ we have, as $L_{n+1}(X; Y) = L(X, L_n(X; Y))$, that $L(x_1) \in L_n(X; Y)$. As $n \in S$, $(x_2, \dots, x_m) \in X^{m-1}$ and $1 \leq m-1 < n$ it follows that

$$(L(x_1))(x_2: \dots: x_m) \in L_{n-(m-1)}(X; Y) = L_{(n+1)-m}(X; Y)$$

hence

$$L(x_1: \dots: x_m) = (L(x_1))(x_2: \dots: x_m) \in L_{(n+1)-m}(X; Y)$$

proving as $1 < m < n+1$

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n+1 \\ L_{(n+1)-m}(X; Y) & \text{if } m < n+1 \end{cases}$$

So in all cases we have

$$\forall (x_1, \dots, x_m) \text{ that } L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n+1 \\ L_{(n+1)-m}(X; Y) & \text{if } m < n+1 \end{cases}$$

proving that $n+1 \in S$. □

In case $X = \mathbb{K}$ we have a simpler expression for $L(x_1: \dots: x_n)$.

Theorem 16.118. Let $n \in \mathbb{N}$, $\langle X, ||| \rangle$ a normed space, $L \in L_n(\mathbb{K}; X)$ and $x = (x_1, \dots, x_n) \in \mathbb{K}^n$ then we have

$$L(x_1: \dots: x_n) = \left(\prod_{i=1}^n x_i \right) \cdot L\left(\underbrace{1: \dots: 1}_n\right)$$

Proof. We prove this by induction, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(\mathbb{K}; X) \text{ and } (x_1, \dots, x_n) \in \mathbb{K}^n \text{ then } L(x_1: \dots: x_n) = \left(\prod_{i=1}^n x_i \right) \cdot L\left(\underbrace{1: \dots: 1}_n\right) \right\}$$

then we have:

$1 \in S$. If $L \in L_1(\mathbb{K}; X) = L(\mathbb{K}, X)$ then for $(x_1) \in \mathbb{K}^1$ we have

$$L(x_1: \dots: x_1) = L(x_1) = L(x_1 \cdot 1) \underset{L \in L(\mathbb{K}, X)}{=} x_1 \cdot L(1) = \left(\prod_{i=1}^1 x_i \right) \cdot L\left(\underbrace{1: \dots: 1}_1\right)$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $L \in L_{n+1}(\mathbb{K}; X) = L(\mathbb{K}, L_n(\mathbb{K}; X))$ then we have for $(x_1, \dots, x_{n+1}) \in \mathbb{K}^{n+1}$ that $L(x_1) \in L_n(\mathbb{K}; X)$. So as $n \in S$ and $(x_2, \dots, x_{n+1}) \in \mathbb{K}^n$ we have

$$\begin{aligned} L(x_1)(x_2: \dots: x_{n+1}) &= \left(\prod_{i=2}^{n+1} x_i \right) \cdot (L(x_1))\left(\underbrace{1: \dots: 1}_n\right) \\ &\underset{L \in L(X, L_n(\mathbb{K}; X))}{=} \left(\prod_{i=2}^{n+1} x_i \right) \cdot x_1 \cdot (L(1))\left(\underbrace{1: \dots: 1}_n\right) \\ &= \left(\prod_{i=1}^{n+1} x_i \right) \cdot (L(1))\left(\underbrace{1: \dots: 1}_n\right) \\ &= \left(\prod_{i=1}^{n+1} x_i \right) \cdot L\left(\underbrace{1: \dots: 1}_{n+1}\right) \end{aligned}$$

proving that

$$L(x_1: \dots: x_{n+1}) = L(x_1)(x_2: \dots: x_{n+1}) = L\left(\underbrace{1: \dots: 1}_{n+1}\right) \cdot \prod_{i=1}^n x_i$$

hence

$$n+1 \in S$$

□

Lemma 16.119. *Let $n \in \mathbb{N} \setminus \{1\}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces and $L \in L_n(X; Y)$ then for $(x_1, \dots, x_n) \in X^n$ we have*

$$L(x_1: \dots: x_n) = (L(x_1: \dots: x_{n-1}))(x_n)$$

Proof. We use induction to prove this, so define

$$S = \{n \in \{2, \dots, \infty\} \mid \text{If } L \in L_n(X; Y) \text{ then for } (x_1, \dots, x_n) \in X^n \text{ we have } L(x_1: \dots: x_n) = (L(x_1: \dots: x_{n-1}))(x_n)\}$$

then we have:

2 $\in S$. If $L \in L_2(X; Y) = L(X, L_1(X; Y)) = L(X, L(X, Y))$ then for $(x_1, x_2) \in X^2$ we have

$$L(x_1: \dots: x_2) = (L(x_1))(x_2: \dots: x_2) = (L(x_1))(x_2) = (L(x_1: \dots: x_1))(x_2) = (L(x_1: \dots: x_{2-1}))(x_2)$$

proving that $2 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X; Y)$ then for $(x_1, \dots, x_{n+1}) \in X^{n+1}$ we have, as $L_{n+1}(X; Y) = L(X, L_n(X; Y))$, that $L(x_1) \in L_n(X; Y)$. As $n \in S$ and $(x_2, \dots, x_{n+1}) \in X^n$ we have that

$$(L(x_1))(x_2: \dots: x_{n+1}) \stackrel{n \in S}{=} ((L(x_1))(x_2: \dots: x_n))(x_{n+1}) = (L(x_1: \dots: x_n))(x_{n+1})$$

so that

$$L(x_1: \dots: x_{n+1}) = (L(x_1))(x_2: \dots: x_{n+1}) = (L(x_1: \dots: x_n))(x_{n+1})$$

proving that

$$n+1 \in S$$

□

Lemma 16.120. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_n(X; Y)$ then $\forall (x_1, \dots, x_n) \in X^n$ we have*

$$(L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) = L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n)$$

Proof. The proof is by induction, so let

$$S = \{n \in \mathbb{N} \mid \forall \alpha \in \mathbb{K}, \forall L_1, L_2 \in L_n(X; Y) \text{ we have } \forall (x_1, \dots, x_n) \in X^n \text{ that } (L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) = L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n)\}$$

then we have:

1 $\in S$. Take $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_1(X; Y) = L(X, Y)$ then we have for $(x_1) \in X^1$ that

$$\begin{aligned} (L_1 + \alpha \cdot L_2)(x_1: \dots: x_1) &= (L_1 + \alpha \cdot L_2)(x_1) \\ &\stackrel{\text{pointwise definition}}{=} L_1(x_1) + \alpha \cdot L_2(x_1) \\ &= L_1(x_1: \dots: x_1) + \alpha \cdot L_2(x_1: \dots: x_1) \end{aligned}$$

proving that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Take $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_{n+1}(X; Y) = L(X, L_n(X; Y))$ then we have

$$(L_1 + \alpha \cdot L_2)(x_1) \stackrel{\text{pointwise definition}}{=} L_1(x_1) + \alpha \cdot L_2(x_1)$$

hence

$$\begin{aligned} (L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) &= ((L_1 + \alpha \cdot L_2)(x_1))(x_2: \dots: x_n) \\ &= (L_1(x_1) + \alpha \cdot L_2(x_1))(x_2: \dots: x_n) \end{aligned} \tag{16.77}$$

As $L_1(x_1), L_2(x_1) \in L_n(X; Y)$ and $n \in S$ it follows that

$$\begin{aligned} (L_1(x_1) + \alpha \cdot L_2(x_1))(x_2: \dots: x_n) &\stackrel{n \in S}{=} (L_1(x_1))(x_2: \dots: x_n) + \alpha \cdot (L_2(x_1))(x_2: \dots: x_n) \\ &= L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n) \end{aligned}$$

which combined with [eq: 16.77] proves that

$$(L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) = L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n)$$

Hence we have that

$$n+1 \in S$$

□

Lemma 16.121. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces and $L \in L_n(X; Y)$, $i \in \{1, \dots, n\}$ then for $i \in \{1, \dots, n\}$ we have for $\left(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n\right) \in X^n$ that*

$$L\left(x_1: \dots, \underbrace{r + \alpha \cdot t}_i, \dots: x_n\right) = L\left(x_1: \dots, \underbrace{r}_i, \dots: x_n\right) + \alpha \cdot L\left(x_1: \dots, \underbrace{t}_i, \dots: x_n\right)$$

Proof. We use mathematical induction for the proof, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(X; Y) \text{ then } \forall i \in \{1, \dots, n\} \text{ we have for } \left(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n\right) \in X^n \text{ that} \right. \\ \left. L\left(x_1: \dots, \underbrace{r + \alpha \cdot t}_i, \dots: x_n\right) = L\left(x_1: \dots, \underbrace{r}_i, \dots: x_n\right) + \alpha \cdot L\left(x_1: \dots, \underbrace{t}_i, \dots: x_n\right) \right\}$$

then we have:

$1 \in S$. Let $L \in L_1(X; Y) = L(X, Y)$ then if $i \in \{1, \dots, 1\} = \{1\}$ we have $i = 1$ and given

$$(r + \alpha \cdot t) = \left(x_1, \dots, \underbrace{r + \alpha \cdot t}_1, \dots, x_1\right) \in X^1$$

that

$$\begin{aligned} L\left(x_1: \dots, \underbrace{r + \alpha \cdot t}_i, \dots: x_n\right) &= \\ &\stackrel{L \in L(X, Y)}{=} L(r + \alpha \cdot t) \\ &\stackrel{L(r) + \alpha \cdot L(t)}{=} L(r) + \alpha \cdot L(t) \\ L\left(x_1: \dots, \underbrace{r}_1, \dots: x_n\right) + \alpha \cdot L\left(x_1: \dots, \underbrace{t}_1, \dots: x_n\right) &= \end{aligned}$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X; Y) = L(X, L_n(X; Y))$ and take $i \in \{1, \dots, n+1\}$ then we have either:

$i = 1$. Then for $\left(x_1: \dots, \underbrace{r + \alpha \cdot t}_1, \dots: x_{n+1}\right) \in X^{n+1}$ we have

$$\begin{aligned} L\left(x_1: \dots, \underbrace{r + \alpha \cdot t}_1, \dots: x_{n+1}\right) &= \\ &\stackrel{L \in L(X, L_n(X; Y))}{=} L(r + \alpha \cdot t; x_2: \dots: x_{n+1}) \\ &\stackrel{[lemma: 16.120]}{=} (L(r) + \alpha \cdot L(t))(x_2: \dots: x_{n+1}) \\ &\stackrel{L(r)(x_2: \dots: x_{n+1}) + \alpha \cdot L(t)(x_2: \dots: x_{n+1})}{=} \\ &\stackrel{L(r; x_2: \dots: x_{n+1}) + \alpha \cdot L(t; x_2: \dots: x_{n+1})}{=} \\ L\left(x_1: \dots, \underbrace{r}_i, \dots: x_{n+1}\right) + \alpha \cdot L\left(x_1: \dots, \underbrace{t}_i, \dots: x_{n+1}\right) &= \end{aligned}$$

proving that in this case that

$$\begin{aligned}
 n+1 &\in S \\
 i \in \{2, \dots, n+1\}. \text{ Then for } &\left(x_1 \dots \underbrace{r + \alpha \cdot t}_{i-1} \dots x_{n+1}\right) \in X^{n+1} \text{ we have} \\
 \left(x_2 \dots \underbrace{r + \alpha \cdot t}_{i-1} \dots x_{n+1}\right) &\in X^n \text{ which as } L(x_1) \in L_n(X; Y) \text{ and } n \in S \text{ results in} \\
 L\left(x_1: \dots \underbrace{r + \alpha \cdot t}_{i-1} \dots x_{n+1}\right) &= \\
 (L(x_1))\left(x_2: \dots \underbrace{r + \alpha \cdot t}_{i-1} \dots x_n\right) & \\
 (L(x_1))\left(x_2: \dots \underbrace{r}_{i-1} \dots x_n\right) + \alpha \cdot (L(x_1))\left(x_2: \dots \underbrace{s}_{i-1} \dots x_n\right) &\stackrel{n \in S}{=} \\
 L\left(x_1: \dots \underbrace{r}_{i-1} \dots x_n\right) + \alpha \cdot L\left(x_1: \dots \underbrace{t}_{i-1} \dots x_n\right) &=
 \end{aligned}$$

so that in this case we have also

$$n+1 \in S$$

So in all cases we have

$$n+1 \in S$$

□

Lemma 16.122. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $L \in L_n(X; Y)$ and $(x_1, \dots, x_n) \in X^n$ [so that by lemma: 16.117] $L(x_1: \dots: x_n) \in Y$ then

$$\|L(x_1: \dots: x_n)\|_Y \leq \|L\|_{L_n(X; Y)} \cdot \prod_{i=1}^n \|x_i\|_X$$

Proof. We prove this by induction, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(X; Y) \text{ then } \forall (x_1, \dots, x_n) \in X^n \text{ we have } \|L(x_1: \dots: x_n)\|_Y \leq \|L\|_{L_n(X; Y)} \cdot \prod_{i=1}^n \|x_i\|_X \right\}$$

then we have:

$1 \in S$. If $L \in L_1(X; Y) = L(X, Y)$ then $\forall (x_1) \in X^1$ we have

$$\|L(x_1: \dots: x_1)\|_Y = \|L(x_1)\|_Y \leq \|L\|_{L(X; Y)} \cdot \|x_1\|_X = \|L\|_{L_1(X; Y)} \cdot \|x_1\|_X = \|L\|_{L_1(X; Y)} \cdot \prod_{i=1}^1 \|x_i\|_X$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X; Y) = L(X, L_n(X; Y))$ and $(x_1, \dots, x_{n+1}) \in X^{n+1}$ then, as $L(x_1) \in L_n(X; Y)$, $(x_2, \dots, x_{n+1}) \in X^n$ and $n \in S$, it follows that

$$\|L(x_1)(x_2: \dots: x_{n+1})\|_Y \leq \|L(x_1)\|_{L_n(X; Y)} \cdot \prod_{i=1}^n \|x_{i+1}\|_X = \|L(x_1)\|_{L_n(X; Y)} \cdot \prod_{i=2}^{n+1} \|x_i\|_X \quad (16.78)$$

Next we have

$$\begin{aligned}
 \|L(x_1: \dots: x_{n+1})\|_Y &= \|(L(x_1))(x_2: \dots: x_{n+1})\|_Y \\
 &\stackrel{[\text{eq: 16.78}]}{\leq} \|L(x_1)\|_{L_n(X; Y)} \cdot \prod_{i=2}^{n+1} \|x_i\|_X \\
 &\stackrel{L \in L_{n+1}(X; Y) = L(X, L_n(X; Y))}{\leq} \|L\|_{L_{n+1}(X; Y)} \cdot \|x_1\|_X \cdot \prod_{i=2}^{n+1} \|x_i\|_X \\
 &\leq \|L\|_{L_{n+1}(X; Y)} \cdot \prod_{i=1}^{n+1} \|x_i\|_X
 \end{aligned}$$

proving that

$$n+1 \in S$$

□

Lemma 16.123. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $L_1, L_2 \in L_n(X; Y)$ such that $\forall (x_1, \dots, x_n) \in X^n$ we have $L_1(x_1: \dots: x_n) = L_2(x_1: \dots: x_n)$ then $L_1 = L_2$.*

Proof. We use induction to prove this, so define

$$S = \{n \in \mathbb{N} \mid \text{If } L_1, L_2 \in L_n(X; Y) \text{ satisfies } \forall (x_1, \dots, x_n) \ L_1(x_1: \dots: x_n) = L_2(x_1: \dots: x_n) \text{ then } L_1 = L_2\}$$

then we have:

1 $\in S$. Let $L_1, L_2 \in L_1(X; Y) = L(X, Y)$ be such that $\forall (x_1) \in X^1$ we have $L_1(x_1: \dots: x_1) = L_2(x_1: \dots: x_1)$. Then $\forall x \in X$ we have for $y = (x) \in X^1$ [so that $y_1 = x$] that

$$L_1(x) = L_1(y_1) = L_1(y_1: \dots: y_1) = L_2(y_1: \dots: y_1) = L_2(y_1) = L_2(x)$$

proving that $L_1 = L_2$, hence $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L_1, L_2 \in L_{n+1}(X; Y) = L(X, L_n(X; Y))$ such that

$$\forall x = (x_1, \dots, x_{n+1}) \in X^{n+1} \ L_1(x_1: \dots: x_{n+1}) = L_2(x_1: \dots: x_{n+1})$$

Let $y \in X$ then $L_1(y), L_2(y) \in L_n(X; Y)$ and $\forall (x_1, \dots, x_n) \in X^n$ we have

$$L_1(y)(x_1: \dots: x_n) = L_1(y: x_1: \dots: x_n) = L_2(y: x_1: \dots: x_n) = L_2(y)(x_1: \dots: x_n)$$

which as $n \in S$ proves that $L_1(y) = L_2(y)$. As y is chosen arbitrary we conclude that $L_1 = L_2$ proving

$$n+1 \in S$$

□

Lemma 16.124. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces and $L \in L_n(X; Y)$. If $\forall (x_1, \dots, x_n) \in X^n$ we have that $\|L(x_1: \dots: x_n)\|_Y \leq M \cdot \prod_{i=1}^n \|x_i\|_X$ then it follows that*

$$\|L\|_{L_n(X; Y)} \leq M$$

Proof. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(X; Y) \text{ satisfies } \forall (x_1, \dots, x_n) \in X^n \ \|L(x_1: \dots: x_n)\|_Y \leq M \cdot \prod_{i=1}^n \|x_i\|_X \text{ then } \|L\|_{L_n(X; Y)} \leq M \right\}$$

then we have:

1 $\in S$. Let $L \in L_1(X; Y) = L(X, Y)$ such that $\forall (x_1) \in X^1$ we have $\|L(x_1: \dots: x_1)\|_Y \leq M \cdot \prod_{i=1}^1 \|x_i\|_X$. Then given $y \in X$ we have for $x = (y) \in X^1$ that

$$\|L(y)\|_Y = \|L(x_1)\|_Y = \|L(x_1: \dots: x_1)\|_Y \leq M \cdot \prod_{i=1}^1 \|x_i\|_X = M \cdot \|y\|_X$$

so that by [definition: 14.209] $\|L\|_{L_1(X; Y)} = \|L\|_{L(X, Y)} \leq M$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X; Y) = L(X, L_n(X; Y))$ such that $\forall (x_1, \dots, x_{n+1}) \in X^{n+1}$ we have

$$\|L(x_1: \dots: x_{n+1})\|_Y \leq M \cdot \prod_{i=1}^{n+1} \|x_i\|_X$$

Fix $x \in X$ then $\forall (y_1, \dots, y_n) \in X^n$ we have

$$\|(L(x))(y_1: \dots: y_n)\|_Y = \|L(x: y_1: \dots: y_n)\|_Y \leq M \cdot \|x\|_X \cdot \prod_{i=1}^n \|y_i\|_X = (M \cdot \|x\|_X) \cdot \prod_{i=1}^n \|y_i\|_X$$

As $n \in S$ and $L(x) \in L_n(X; Y)$ it follows that $\|L(x)\|_{L_n(X; Y)} \leq M \cdot \|x\|_X$. Hence by [definition: 14.209] it follows that $\|L\|_{L_{n+1}(X; Y)} \leq M$ proving

$$n+1 \in S$$

□

Lemma 16.125. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces and $L \in L^n(X; Y)$ [see definition: 14.235] then there exist a $K \in L_n(X; Y)$ such that $\forall (x_1, \dots, x_n) \in X^n$ we have $K(x_1, \dots, x_n) = L(x_1, \dots, x_n)$*

Proof. We use induction to prove this so let

$S = \{n \in \mathbb{N} \mid \text{If } L \in L^n(X; Y) \text{ then there exist a } K \in L_n(X; Y) \text{ such that } \forall (x_1, \dots, x_n) \in X^n \text{ we have } K(x_1, \dots, x_n) = L(x_1, \dots, x_n)\}$

then we have:

$1 \in S$. If $L \in L^1(X; Y)$ then as $L^1(X; Y) \stackrel{[\text{definition: 14.235}]}{=} L(X, Y) = L_1(X; Y)$ we have that $L \in L_1(X; Y)$ so if we take $K = L$ then we have $K \in L_1(X; Y)$ and for $(x_1) \in X^1$ we have $K(x_1) = L(x_1) = L(x_1, \dots, x_1)$. So we have that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L^{n+1}(X; Y)$. Take $x \in X$ and define

$$L_x: X^n \rightarrow Y \text{ by } L_x(x_1, \dots, x_n) = L(x, x_1, \dots, x_n)$$

If $i \in \{1, \dots, n\}$ and $(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n) \in X^n$ then

$$\begin{aligned} & L_x(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n) = \\ & L(x, x_1, \dots, \underbrace{r + \alpha \cdot t}_{i+1}, \dots, x_n) \stackrel{[\text{lemma: 16.121}]}{=} \\ & L(x, x_1, \dots, \underbrace{r}_{i+1}, \dots, x_n) + \alpha \cdot L(x, x_1, \dots, \underbrace{t}_{i+1}, \dots, x_n) = \\ & L_x(x_1, \dots, \underbrace{r}_i, \dots, x_n) + \alpha \cdot L_x(x_1, \dots, \underbrace{s}_i, \dots, x_n) \end{aligned}$$

which proves that

$$L_x \in \text{Hom}^n(X; Y) \tag{16.79}$$

Let $(x_1, \dots, x_n) \in X^n$ then for $(y_1, \dots, y_{n+1})$ defined by $y_i = \begin{cases} x & \text{if } i=1 \\ x_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$ we have

$$\begin{aligned} \|L_x(x_1, \dots, x_n)\|_Y &= L(x, x_1, \dots, x_n) \\ &= L(y_1, \dots, y_{n+1}) \\ &\leq_{L \in L^{n+1}(X; Y)} \|L\|_{L^{n+1}(X; Y)} \cdot \prod_{i=1}^{n+1} \|y_i\|_X \\ &= \|L\|_{L^{n+1}(X; Y)} \cdot \|y_1\|_X \cdot \prod_{i=2}^{n+1} \|y_i\|_X \\ &= \|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X \cdot \prod_{i=2}^{n+1} \|x_{i-1}\|_X \\ &= (\|x\|_X \cdot \|L\|_{L^{n+1}(X; Y)}) \cdot \prod_{i=1}^n \|x_i\|_X \end{aligned}$$

proving by [theorem: 14.228] and [eq: 16.79] that L_x is continuous. Hence

$$L_x \in L^n(X; Y) \tag{16.80}$$

As $n \in S$ there exist a $K_x \in L_n(X; Y)$ such that $\forall (x_1, \dots, x_n) \in X^n$ we have

$$K_x(x_1: \dots: x_n) = L_x(x_1, \dots, x_n) = L(x, x_1, \dots, x_n) \quad (16.81)$$

This allows us to define

$$K: X \rightarrow L_n(X; Y) \text{ where } K(x) = K_x$$

then we have for $(x_1, \dots, x_{n+1}) \in X^{n+1}$

$$K(x_1)(x_2: \dots: x_{n+1}) = K_{x_1}(x_2: \dots: x_{n+1}) \stackrel{[\text{eq: 16.81}]}{=} L(x_1, \dots, x_{n+1}) \quad (16.82)$$

If $x, y \in X$ and $\alpha \in \mathbb{K}$ then we have $\forall (x_1, \dots, x_n) \in X^n$ that

$$\begin{aligned} K(x + \alpha \cdot y)(x_1: \dots: x_n) &\stackrel{[\text{eq: 16.82}]}{=} L(x + \alpha \cdot y, x_1, \dots, x_n) \\ &\stackrel{L \in L^{n+1}(X; Y)}{=} L(x, x_1, \dots, x_n) + \alpha \cdot L(y, x_1, \dots, x_n) \\ &\stackrel{[\text{eq: 16.82}]}{=} K(x)(x_1: \dots: x_n) + \alpha \cdot K(y)(x_1: \dots: x_n) \\ &\stackrel{[\text{lemma: 16.120}]}{=} (K(x) + \alpha \cdot K(y))(x_1: \dots: x_n) \end{aligned}$$

which by [lemma: 16.123] proves that

$$K(x + \alpha \cdot y) = K(x) + \alpha \cdot K(y)$$

Hence we have that

$$K \in \text{Hom}(X, L_n(X; Y)) \quad (16.83)$$

Let $x \in X$ then we have for $(y_1, \dots, y_n) \in X^n$ that

$$\begin{aligned} \|K(x)(y_1: \dots: y_n)\|_Y &= \|K_x(y_1: \dots: y_n)\|_Y \\ &\stackrel{[\text{eq: 16.82}]}{=} \|L(x, y_1, \dots, y_n)\|_Y \\ &\leq_{L \in L^{n+1}(X; Y)} \|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X \cdot \prod_{i=1}^n \|y_i\|_X \\ &= (\|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X) \cdot \prod_{i=1}^n \|y_i\|_X \end{aligned}$$

which as $K(x) \in L_n(X; Y)$ proves using [lemma: 16.124] that

$$\|K(x)\|_{L_n(X; Y)} \leq \|L\|_{L^{n+1}(X; Y)} \cdot \|x\|_X$$

Hence using [eq: 16.83] together with [theorem: 14.201] proves that K is continuous or

$$K \in L(X, L_n(X; Y)) = L_{n+1}(X; Y)$$

further

$$K(x_1: \dots: x_{n+1}) = K(x_1)(x_2: \dots: x_{n+1}) \stackrel{[\text{eq: 16.82}]}{=} L(x_1, \dots, x_{n+1})$$

The above proves that

$$n+1 \in S$$

□

We are now ready to define the identification mapping between $L_n(X; Y)$ and $L^n(X; Y)$.

Theorem 16.126. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, then*

$\mathcal{I}_{n, X, Y}: L_n(X; Y) \rightarrow L^n(X; Y)$ where $\mathcal{I}_{n, X, Y}(L): X^n \rightarrow Y$ is defined by

$$(\mathcal{I}_{n, X, Y}(L))(x_1, \dots, x_n) = L(x_1: \dots: x_n)$$

is a linear isometric isomorphism. In other words

1. $\forall L \in L_n(X; Y) \quad \mathcal{I}_{n, X, Y}(L) \in L^n(X; Y)$ so that $\mathcal{I}_{n, X, Y}: L_n(X; Y) \rightarrow L^n(X; Y)$ is indeed a function

2. $\mathcal{I}_{n,X,Y}: L_n(X;Y) \rightarrow L^n(X;Y)$ is a bijection.
3. $\forall L_1, L_2 \in L_n(X;Y)$ and $\alpha \in \mathbb{K}$ we have $\mathcal{I}_{n,X,Y}(L_1 + \alpha \cdot L_2) = \mathcal{I}_{n,X,Y}(L_1) + \alpha \cdot \mathcal{I}_{n,X,Y}(L_2)$
4. $\forall L \in L_n(X;Y)$ we have $\|L\|_{L_n(X;Y)} = \|\mathcal{I}_{n,X,Y}(L)\|_{L^n(X;Y)}$

Note 16.127. If $n=1$ then as $\forall(x_1) \in X^1$ we have $\mathcal{I}_{1,X,Y}(L)(x_1) = L(x_1: \dots: x_1) = L(x_1)$ and $L^1(X;Y) = L(X,Y) = L_1(X;Y)$ it follows that

$$\mathcal{I}_{1,X,Y} = \text{Id}_{L(X,Y)}$$

Note 16.128. If $K \in L^n(X;Y)$ then we have, as $\mathcal{I}_{n,X,Y}$ is a bijection, that $K = \mathcal{I}_{n,X,Y}((\mathcal{I}_{n,X,Y})^{-1}(K))$ so that for $x \in X^n$ we have

$$K(x_1, \dots, x_n) = \mathcal{I}_{n,X,Y}((\mathcal{I}_{n,X,Y})^{-1}(K))(x_1, \dots, x_n) = (\mathcal{I}_{n,X,Y})^{-1}(K)(x_1: \dots: x_n)$$

proving that

$$\forall x \in X^n \quad (\mathcal{I}_{n,X,Y})^{-1}(K)(x_1: \dots: x_n) = K(x_1, \dots, x_n)$$

Proof. We have

1. Let $L \in L_n(X;Y)$ then for $i \in \{1, \dots, n\}$ and $(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n) \in X^n$ we have

$$\begin{aligned} \mathcal{I}_{n,X,Y}(L)\left(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n\right) &= \\ L\left(x_1: \dots: \underbrace{r + \alpha \cdot t}_i: \dots: x_n\right) &\stackrel{[\text{lemma: 16.121}]}{=} \\ L\left(x_1: \dots: \underbrace{r}_i: \dots: x_n\right) + \alpha \cdot L\left(x_1: \dots: \underbrace{t}_i: \dots: x_n\right) &= \\ \mathcal{I}_{n,X,Y}(L)\left(x_1, \dots, \underbrace{r}_i, \dots, x_n\right) + \alpha \cdot \mathcal{I}_{n,X,Y}(L)\left(x_1, \dots, \underbrace{t}_i, \dots, x_n\right) \end{aligned}$$

proving that

$$\mathcal{I}_{n,X,Y}(L) \in \text{Hom}^n(X;Y) \quad [\text{see definition: 11.262}] \quad (16.84)$$

Further if $(x_1, \dots, x_n) \in X^n$ then we have:

$$\begin{aligned} \|\mathcal{I}_{n,X,Y}(L)(x_1, \dots, x_n)\|_Y &= \|L(x_1: \dots: x_n)\|_Y \\ &\leq_{[\text{lemma: 16.122}]} \|L\|_{L_n(X;Y)} \cdot \prod_{i=1}^n \|x_i\| \end{aligned}$$

Using then [theorems: 14.228 and 14.234] together with [eq: 16.84] it follows that

$$\mathcal{I}_{n,X,Y}(L) \in L^n(X;Y) \text{ and } \|\mathcal{I}_{n,X,Y}(L)\|_{L^n(X;Y)} \leq \|L\|_{L_n(X;Y)} \quad (16.85)$$

so that

$$\mathcal{I}_{n,X,Y}: L_n(X;Y) \rightarrow L^n(X;Y) \text{ is a well defined function}$$

2. To prove bijectivity we have to prove:

injectivity. Let $L_1, L_2 \in L_n(X;Y)$ be such that $\mathcal{I}_{n,X,Y}(L_1) = \mathcal{I}_{n,X,Y}(L_2)$. Then $\forall(x_1, \dots, x_n) \in X^n$ we have

$$L_1(x_1: \dots: x_n) = \mathcal{I}_{n,X,Y}(L_1)(x_1, \dots, x_n) = \mathcal{I}_{n,X,Y}(L_2)(x_1, \dots, x_n) = L_2(x_1: \dots: x_n)$$

Using [lemma: 16.123] it follows then that $L_1 = L_2$ which proves injectivity.

surjectivity. Let $K \in L^n(X;Y)$ then using [lemma: 16.125] there exist a $L \in L_n(X;Y)$ such that $\forall(x_1, \dots, x_n) \in X^n$ we have $K(x_1: \dots: x_n) = L(x_1, \dots, x_n)$. Hence

$$\mathcal{I}_{n,X,Y}(L)(x_1, \dots, x_n) = K(x_1: \dots: x_n) = L(x_1, \dots, x_n)$$

from which it follows that that $\mathcal{I}_{n,X,Y}(K) = L$ and thus surjectivity.

3. Let $L_1, L_2 \in L_n(X; Y)$ and $\alpha \in \mathbb{K}$ then $\forall (x_1, \dots, x_n) \in X^n$ we have

$$\begin{aligned} \mathcal{I}_{n,X,Y}(L_1 + \alpha \cdot L_2)(x_1, \dots, x_n) &= \\ (L_1 + \alpha \cdot L_2)(x_1, \dots, x_n) &\stackrel{[\text{lemma: 16.120}]}{=} \\ L_1(x_1, \dots, x_n) + \alpha \cdot L_2(x_1, \dots, x_n) &= \\ \mathcal{I}_{n,X,Y}(L_1)(x_1, \dots, x_n) + \alpha \cdot \mathcal{I}_{n,X,Y}(L_2)(x_1, \dots, x_n) &= \\ (\mathcal{I}_{n,X,Y}(L_1) + \alpha \cdot \mathcal{I}_{n,X,Y}(L_2))(x_1, \dots, x_n) & \end{aligned}$$

proving that

$$\mathcal{I}_{n,X,Y}(L_1 + \alpha \cdot L_2) = \mathcal{I}_{n,X,Y}(L_1) + \alpha \cdot \mathcal{I}_{n,X,Y}(L_2)$$

4. Let $L \in L_n(X; Y)$ then by [eq: 16.85] we have that

$$\|\mathcal{I}_{n,X,Y}(L)\|_{L^n(X;Y)} \leq \|L\|_{L_n(X;Y)} \quad (16.86)$$

For the opposite inequality, let $(x_1, \dots, x_n) \in X^n$ then

$$\|L(x_1, \dots, x_n)\|_Y = \|\mathcal{I}_{n,X,Y}(L)(x_1, \dots, x_n)\|_Y \stackrel{\substack{\leq \\ \mathcal{I}_{n,X,Y}(L) \in L^n(X,Y)}}{\leq} \|\mathcal{I}_{n,X,Y}(L)\|_{L^n(X;Y)} \cdot \prod_{i=1}^n \|x_i\|_X$$

so that by [lemma: 16.124] it follows that

$$\|L\|_{L_n(X;Y)} \leq \|\mathcal{I}_{n,X,Y}(L)\|_{L^n(X;Y)}$$

which combined with [eq: 16.86] proves that

$$\|L\|_{L_n(X;Y)} = \|\mathcal{I}_{n,X,Y}(L)\|_{L^n(X;Y)} \quad \square$$

16.2.2 Higher order Fréchet differentiation.

Definition 16.129. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $f: A \rightarrow Y$ a function then the n -times differential $D^{[n]}f: \mathfrak{D}_f^n \rightarrow L_n(X; Y)$ is recursively defined by

$n=1$. Then

$$\mathfrak{D}_f^1 = \mathfrak{D}_f$$

and

$$D^{[1]}f: \mathfrak{D}_f^1 \rightarrow L(X; Y) \stackrel{[\text{definition: 16.110}]}{=} L_1(X; Y)$$

is defined by

$$D^{[1]}f = Df$$

where $\mathfrak{D}_f$ and Df is defined by [definition: 16.48]. In other words

$$\mathfrak{D}_f^1 = \{x \in A \mid f \text{ is Fréchet differentiable at } x\}$$

and $D^{[1]}f$ is defined by

$$D^{[1]}f: \mathfrak{D}_f^1 \rightarrow L_1(X; Y) \text{ where } D^{[1]}f(x) = D_x f$$

$1 < n$. Then

$$\mathfrak{D}_f^n = \mathfrak{D}_{D^{[n-1]}f} \stackrel{[\text{definition: 16.48}]}{=} \{x \in \mathfrak{D}_f^{n-1} \mid D^{[n-1]}f \text{ is Fréchet differentiable at } x\}$$

and

$$D^{[n]}f: \mathfrak{D}_f^n \rightarrow L(X, L_n(X; Y)) \stackrel{[\text{definition: 16.110}]}{=} L_n(X; Y)$$

is defined by

$$D^{[n]}f = D(D^{[n-1]}f)$$

Note 16.130. It is always possible that $\mathfrak{D}_f^n = \emptyset$ so that $D^n f$ is the empty function [see example: 2.46].

Note 16.131. If $\mathfrak{D}_f^n = \emptyset$ then

$$\mathfrak{D}_f^{n+1} = \mathfrak{D}_{D^{[n]}f} = \{x \in \mathfrak{D}^n = \emptyset \mid D^{[n]}f \text{ is Fréchet differentiable at } x\} = \emptyset$$

Convention 16.132. If $A \subseteq X$ and $f: A \rightarrow Y$ is a function then $D^{[n]}f$ is always the graph of the function $D^{[n]}f: \mathfrak{D}_f^n \rightarrow L(Y)$. In other words the domain of $D^{[n]}f$ is always $\mathfrak{D}_f^n$ and the codomain is always $L_n(X; Y)$.

The problem with $D_x^{[n]}f$ is that it is an element of $L_n(X; Y)$ so that for example

$$D_x^{[3]}f \in L(X, L(X, L(X, Y)))$$

and we have to deal with expressions like $((D_x^{[3]}f(h_1))(h_2))(h_3)$ which we note as $D_x^{[3]}f(h_1: \dots: h_3)$. Lucky we can use [theorem: 16.126] to turn $D^{[n]}f(x)$ into a multilinear mapping.

Definition 16.133. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$ and $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ a function that is n -times differentiable at x then

$$D_x^n f = \mathcal{I}_{n, X, Y}(D_x^{[n]}f) \in L^n(X; Y)$$

Note 16.134. Using [theorem: 16.126] we have that $D_x^1 f = D_x^{[1]}f \stackrel{[\text{theorem: 16.140}]}{=} D_x f$ and for $n \in \mathbb{N} \setminus \{1\}$

$$D_x^n f(h_1, \dots, h_n) = D_x^{[n]}f[h_1: \dots: h_n].$$

We use in general $D_x^{[n]}f$ because the proofs are simpler, however if we have to calculate the n -th differential we can use the multi-linear version $D_x^n f$.

The domains $\mathfrak{D}_f^n$ of the differential $D^{[n]}f$ is decreasing with n .

Theorem 16.135. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ a function then we have $\forall n, m \in \mathbb{N}$ with $n \leq m$ that

$$\mathfrak{D}_f^n \subseteq \mathfrak{D}_f^m \subseteq \mathfrak{D}_f \subseteq A^\circ \subseteq A$$

Further we have $\forall n, m \in \mathbb{N}$ with $n < m$ that

$$\mathfrak{D}_f^n \subseteq (\mathfrak{D}_f^m)^\circ$$

Proof. Let $n \in \mathbb{N}$ then by [definition: 16.129] we have that

$$\mathfrak{D}_f^{n+1} = \mathfrak{D}_{D^{[n]}f} \stackrel{[\text{definition: 16.48}]}{=} \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is Fréchet differentiable at } x\} \subseteq \mathfrak{D}_f^n$$

As $m \leq n$ implies $m \in \{1, \dots, n\}$ it follows from the above and [theorem: 10.36] that

$$\mathfrak{D}_f^n \subseteq \mathfrak{D}_f^m \subseteq \mathfrak{D}_f^1 \stackrel{[\text{definition: 16.129}]}{=} \mathfrak{D}_f \stackrel{[\text{theorem: 16.48}]}{\subseteq} A^\circ \subseteq A$$

Assume further that $n, m \in \mathbb{N}$ with $m < n$. If $x \in \mathfrak{D}_f^{m+1}$ then $x \in \mathfrak{D}_f^m$ and $D_f^{[m]}$ is Fréchet differentiable at x , hence it follows from [definition: 16.46] that $x \in (\text{dom}(D^{[m]}f))^\circ = (\mathfrak{D}_f^m)^\circ$. Hence we have that

$$\mathfrak{D}_f^{m+1} \subseteq (\mathfrak{D}_f^m)^\circ$$

So that as $m < n \Rightarrow m+1 \leq n$

$$\mathfrak{D}_f^n \subseteq \mathfrak{D}_f^{m+1} \subseteq (\mathfrak{D}_f^m)^\circ$$

□

Once the n -times differential is defined we can define n -times differentiability at a point and the n -times differential at a point.

Definition 16.136. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ a function then we say that

$$f \text{ is } n\text{-times differentiable at } x \text{ iff } x \in \mathfrak{D}_f^n$$

The n -times differential of f at x noted as $D_x^{[n]}f$ is then defined to be

$$D_x^{[n]}f = D^{[n]}f(x) \in L_n(X; Y)$$

Convention 16.137. If we say that

$$D_x^{[n]}f \text{ exist}$$

then we mean that

$$f \text{ is } n\text{-times differentiable at } x \text{ with differential } D_x^{[n]}f = D^{[n]}f(x) \text{ at } x$$

Theorem 16.138. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ a function that is n -times differentiable at x then f is Fréchet differentiable at x and $\forall m \in \{1, \dots, n\}$ we have that f is m -times differentiable at x .

Proof. Using [theorem: 16.135] we have that

$$\mathfrak{D}_f^n \subseteq \mathfrak{D}_f^m \subseteq \mathfrak{D}_f$$

If f is n -times differentiable at x we have by [definition: 16.136] that $x \in \mathfrak{D}_f^n$ which, as $\mathfrak{D}_f^n \subseteq \mathfrak{D}_f^m$, proves that $x \in \mathfrak{D}_f^m$. Hence using [definition: 16.136] f is m -times differentiable at x . Further, as $x \in \mathfrak{D}_f^1 \subseteq \mathfrak{D}_f = \{x \in A \mid f \text{ is Fréchet differentiable at } x\}$, f is Fréchet differentiable at x . $\square$

We consider now ∞ -times differentiability.

Definition 16.139. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, a differential domain set $A \subseteq X$ and $f: A \rightarrow Y$ a function then

$$f \text{ is } \infty\text{-times differentiable at } x \in A \text{ iff } \forall n \in \mathbb{N} \text{ } f \text{ is } n\text{-times differentiable at } x$$

Further if $U \in \mathcal{T}_{\|\cdot\|_X}$ is a open set in X then

$$f \text{ is } \infty\text{-times differentiable on } U \text{ iff } \forall n \in \mathbb{N} \text{ } f \text{ is } n\text{-times differentiable on } U$$

We have now the following equivalences for differentiability at a point and differentiability on a open set.

Theorem 16.140. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $f: A \rightarrow Y$ a function then we have the following equivalences:

1. f is n -times differentiable at x
2. We have for n either:

$$n = 1. \text{ } f \text{ is Fréchet differentiable at } x$$

$$1 < n. \text{ } D^{[n-1]}f \text{ is Fréchet differentiable at } x$$

Further we have that if (1) is true [which implies that (2) is true] that

$$D_x^{[n]}f = \begin{cases} D_x f & \text{if } n = 1 \\ D_x^{[n]}f = D_x(D^{[n-1]}f) & \end{cases}$$

Proof.

$1 \Rightarrow 2$. As f is n -times differentiable at x we have by [definition: 16.136]

$$x \in \mathfrak{D}_f^n \text{ and } D_x^{[n]}f = D^{[n]}f(x) \in L_n(X; Y) \quad (16.87)$$

for $n \in \mathbb{N}$ we have either:

$n = 1$. Then by [definition: 16.129] $\mathfrak{D}_f^1 = \mathfrak{D}_f = \{x \in A \mid f \text{ is Fréchet differentiable at } x\}$
Hence as by [eq: 16.87] $x \in \mathfrak{D}_f^1$

f is Fréchet differentiable at x

and

$$D_x^{[1]}f \underset{[\text{eq: 16.87}]}{=} D^{[1]}f(x) = Df(x) \underset{[\text{definition: 16.48}]}{=} D_x f$$

$1 < n$. Then by [definition: 16.129]

$$\mathfrak{D}_f^{[n]} = D_{D^{[n-1]}f}(\mathfrak{D}_f^{[n-1]}) = \{x \in \mathfrak{D}^{[n-1]}f \mid D^{[n-1]}f \text{ is Fréchet differentiable at } x\}$$

and

$$D^{[n]}f = D(D^{[n-1]}f) \quad (16.88)$$

Hence as by [eq: 16.87] $x \in \mathfrak{D}^n f$ we have that

$$D^{[n-1]}f \text{ is Fréchet differentiable at } x \quad (16.89)$$

and

$$D_x^{[n]}f \underset{[\text{eq: 16.87}]}{=} D^{[n]}f(x) \underset{[\text{eq: 16.88}]}{=} (D(D^{[n-1]}f))(x) = D_x(D^{[n-1]}f) \quad (16.90)$$

$2 \Rightarrow 1$. For $n \in \mathbb{N}$ we have either:

$n = 1$. As f is Fréchet differentiable at x it follows by [definition: 16.48] that

$$x \in \mathfrak{D}_f \underset{[\text{definition: 16.129}]}{=} \mathfrak{D}_f^1$$

which by [definition: 16.136] means that

f is 1-times differentiable at x

$1 < n$. By the hypothesis

$D^{[n-1]}f$ is Fréchet differentiable at x

So that $x \in \mathfrak{D}_{D^{[n-1]}f} \underset{[\text{definition: 16.129}]}{=} \mathfrak{D}_f^n$ which by [definition: 16.136] proves that

f is n -times differentiable at x □

Definition 16.141. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $f: A \rightarrow Y$ a function and a open set $U \in \mathcal{T}_{\|\cdot\|_X}$ then we say that

f is n -times differentiable on U

iff

$\forall x \in U$ we have that f is n -times differentiable at x

As f is n -times differentiable at x iff $x \in \mathfrak{D}_f^n$ it follows that

$$f \text{ is } n\text{-times differentiable on } U \Leftrightarrow U \subseteq \mathfrak{D}_f^n$$

Note 16.142. If $\text{dom}(f) = A = U$ then we say that f is n -times differentiable.

Note 16.143. You can ask yourself why we limit our self to open sets and not use more general sets. Assume that B is a set such that $\forall x \in B$ f is n -times differentiable at x . Let $x \in B$ then by [theorem: 16.138] f is 1-times differentiable at x and by [theorem: 16.140] f is Fréchet differentiable at x . Hence by the definition of Fréchet differentiability [see definition: 16.46] $x \in B^\circ$. So we have that $B \subseteq B^\circ \subseteq B$ so that by [theorem: 14.16] B is a open set. In other words we gain nothing by removing the condition that B must be open.

Using the above definition with [theorem: 16.140] gives another criteria for a function to be n -times differentiable at a point.

Theorem 16.144. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, x and $f: A \rightarrow Y$ a function then we have

$$\begin{aligned} & f \text{ is } (n+1)\text{-times differentiable at } x \\ & \quad \Updownarrow \\ & \exists U \in \mathcal{T}_{\|\cdot\|_X} \text{ with } x \in U \subseteq A \text{ such that } f \text{ is } n\text{-times differentiable on } U \\ & \quad \text{and} \\ & (D^{[n]}f)|_U \text{ is Fréchet differentiable at } x \end{aligned}$$

Further we have in both cases that $D_x^{[n+1]}f = D_x((D^{[n]}f)|_U)$

Proof.

$\Rightarrow$. As f is $(n+1)$ -times differentiable at x it follows from [theorem: 16.140] that

$$D^{[n]}f \text{ is Fréchet differentiable at } x \text{ with } D_x^{[n+1]}f = D_x(D^{[n]}f) \quad (16.91)$$

Using the definition of Fréchet differentiability [see definition: 16.46] it follows that $x \in (\mathfrak{D}_f^n)^\circ$, hence there exist a open set U in the topology $\mathcal{T}_{\|\cdot\|_X}$ such that

$$x \in U \subseteq \mathfrak{D}_f^n$$

From the above it follows by [definition: 16.141] that

$$f \text{ is } n\text{-times differentiable on } U$$

Further as $D^{[n]}f$ is Fréchet differentiable at x [see eq: 16.91] and $U \subseteq \mathfrak{D}_f^n = \text{dom}(D^{[n]}f)$ we have by [theorem: 16.57] that

$$(D^{[n]}f)|_U \text{ is Fréchet differentiable at } x$$

with $D_x(D^{[n]}f) = D_x((D^{[n]}f)|_U)$. Combining this with [eq: 16.91] proves that

$$D_x^{[n+1]}f = D_x((D^{[n]}f)|_U)$$

$\Leftarrow$. As f is n -times differentiable on U we have by [definition: 16.141] that $U \subseteq \mathfrak{D}_f^n = \text{dom}(D^{[n]}f)$ which together with the fact that $(D^{[n]}f)|_U$ is Fréchet differentiable at x proves by [theorem: 16.57] that

$$D^{[n]}f \text{ is Fréchet differentiable at } x$$

Hence using [theorem: 16.140] it follows that

$$f \text{ is } (n+1)\text{-times differentiable at } x \quad \square$$

Next we show that n -times differentiability at a point is a local property and only depends on the value of a function in a neighborhood of this point.

Theorem 16.145. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $B \subseteq A$, $x \in B$ and $f: A \rightarrow Y$ a function then we have

$$\begin{aligned} & x \in B^\circ \text{ and } f \text{ is } n\text{-times differentiable at } x \\ & \quad \Updownarrow \\ & f|_B \text{ is } n\text{-times differentiable at } x \end{aligned}$$

Further we have in either cases that

$$D_x^{[n]}f = D_x^{[n]}f|_B$$

Note 16.146. If B is a open set in the topology $\mathcal{T}_{\|\cdot\|_X}$ of X then we have that $B \stackrel{[\text{theorem: 14.16}]}{=} B^\circ$ so that in this case we can replace $x \in B^\circ$ by $x \in B$.

Proof.

$\Rightarrow$. We use induction to prove this, so let

$S = \{n \in \mathbb{N} \mid \text{If } f \text{ is } n\text{-times differentiable at } x \in A \text{ then we have that } f|_B \text{ is } n\text{-times differentiable at } x \text{ and } D_x^n f = D_x^n f|_B\}$

then we have

1 $\in S$. Then using [theorem: 16.140] we have that f is Fréchet differentiable at $x \in B^\circ$. Applying then [theorem: 16.57] it follows that $f|_B$ is Fréchet differentiable at x with $D_x f = D_x f|_B$. Finally using [theorem: 16.140] again we have that $f|_B$ is 1-times differentiable with $D_x^{[1]} f = D_x^{[1]} f|_B$ which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. As f is $(n+1)$ -times differentiable at x it follows from [theorem: 16.140] that

$$D^{[n]} f \text{ is Fréchet differentiable at } x \text{ with } D_x^{[n+1]} f = D_x(D^{[n]} f)$$

Let $y \in \mathfrak{D}_f^n$ then f is n -times differentiable at y [see definition: 16.136], hence as $n \in S$ it follows that $f|_B$ is n -times differentiable at y or using [definition: 16.136] $y \in \mathfrak{D}_{f|_B}^n$ with $D_y^{[n]} f = D_y^{[n]} f|_B$ $\xRightarrow{\text{definition: 16.136}}$ $D^{[n]} f(y) = D^{[n]} f|_B(y)$. Hence we have that

$$\mathfrak{D}_f^n \subseteq \mathfrak{D}_{f|_B}^n \text{ with } D^{[n]} f = (D^{[n]} f|_B)|_{\mathfrak{D}_f^n} \quad (16.92)$$

As $D^{[n]} f$ is Fréchet differentiable at x it follows from the above that

$$(D^{[n]} f|_B)|_{\mathfrak{D}_f^n} \text{ is Fréchet differentiable at } x$$

Using [theorem: 16.57] it follows then that

$$D^{[n]} f|_B \text{ is Fréchet differentiable at } x$$

with

$$D_x(D^{[n]} f|_B) = D_x((D^{[n]} f|_B)|_{\mathfrak{D}_f^n}) \stackrel{\text{eq: 16.92}}{=} D_x(D^{[n]} f)$$

Hence using [theorem: 16.140] we have that

$$f|_B \text{ is } (n+1)\text{-times differentiable with } D_x^{[n+1]} f = D_x^{[n+1]} f|_B$$

$$n+1 \in S$$

$\Leftarrow$. First as $f|_B$ is n -times differentiable at x we have by [theorem: 16.138] that $f|_B$ is 1-times differentiable at x or, using [theorem: 16.140] $f|_B$ is Fréchet differentiable at x . Using then the definition of Fréchet differentiation [see definition: 16.46] it follows that

$$x \in B^\circ$$

We use induction for the rest of the proof. so let

$S = \{n \in \mathbb{N} \mid \text{If } f|_B \text{ is } n\text{-times differentiable at } x \text{ then } f \text{ is } n\text{-times differentiable at } x \text{ and } D_x^n f = D_x^n f|_B\}$

then we have

1 $\in S$. Then using [theorem: 16.140] we have that $f|_B$ is Fréchet differentiable at x . Applying then [theorem: 16.57] it follows that f is Fréchet differentiable at x with $D_x f = D_x f|_B$. Finally using [theorem: 16.140] again we have that f is 1-times differentiable with $D_x^{[1]} f = D_x^{[1]} f|_B$ which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $f|_B$ is $(n+1)$ -times differentiable at x it follows from [theorem: 16.140] that

$$D^{[n]} f|_B \text{ is Fréchet differentiable at } x \text{ with } D_x^{[n+1]} f|_B = D_x(D^{[n]} f|_B)$$

Let $y \in \mathfrak{D}_{f|_B}^n$ then by [definition: 16.136] $f|_B$ is n -times differentiable at x . Hence as $n \in S$ it follows that f is n -times differentiable at x or using [definition: 16.136] that $x \in \mathfrak{D}_f^n$ and $D_x^{[n]}f = D_x^{[n]}f|_B \xRightarrow{[\text{definition: 16.136}]} D^{[n]}f(x) = D^{[n]}f|_B(x)$. Hence we have that

$$\mathfrak{D}_{f|_B}^n \subseteq \mathfrak{D}_f^n \text{ and } D^{[n]}f|_B = (D^{[n]}f)|_{\mathfrak{D}_{f|_B}^n} \quad (16.93)$$

As $D^{[n]}f|_B$ is Fréchet differentiable at x it follows from the above that

$$(D^{[n]}f)|_{\mathfrak{D}_{f|_B}^n} \text{ is Fréchet differentiable at } x$$

Using [theorem: 16.57] it follows then that

$$D^{[n]}f \text{ is Fréchet differentiable at } x$$

and

$$D_x(D^{[n]}f) = D_x\left((D^{[n]}f)|_{\mathfrak{D}_{f|_B}^n}\right) \stackrel{[\text{eq. 16.93}]}{=} D_x(D^{[n]}f|_B)$$

Hence using [theorem: 16.140] we have that

$$f \text{ is } (n+1)\text{-times differentiable at } z \text{ and } D_x^{[n+1]}f = D_x^{[n+1]}f|_B$$

which proves that

$$n+1 \in S \quad \square$$

We can apply the above theorem to n -times differentiability on a open set.

Corollary 16.147. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{\|\cdot\|_X}$ with $U \subseteq A$ and $f: A \rightarrow Y$ a function then we have*

$$\begin{aligned} f \text{ is } n\text{-times differentiable on } U \\ \Updownarrow \\ f|_U \text{ is } n\text{-times differentiable on } U \end{aligned}$$

Further if f or $f|_U$ is n -times differentiable on U then we have

$$\forall x \in U \text{ we have that } D_x^{[n]}f = D_x^{[n]}f|_U$$

Proof. We have that

$$\begin{aligned} f \text{ is } n\text{-times differentiable on } U & \Leftrightarrow \forall x \in U \text{ } f \text{ is } n\text{-times differentiable at } x \\ & \stackrel{[\text{theorem: 16.147}]}{\Leftrightarrow} \forall x \in U \text{ } f|_U \text{ is } n\text{-times differentiable at } x \\ & \Leftrightarrow f|_U \text{ is } n\text{-times differentiable on } U \end{aligned}$$

Further if $x \in U$ then by [theorem: 16.147] we have that

$$D_x^{[n]}f = D_x^{[n]}f|_U \quad \square$$

Corollary 16.148. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ normed spaces, $U \in \mathcal{T}_{\|\cdot\|_X}$ and $f: U \rightarrow Y$ a function then we have*

$$\begin{aligned} f \text{ is differentiable on } U \\ \Updownarrow \\ \forall x \in U \exists V \in \mathcal{T}_{\|\cdot\|_X} \text{ with } x \in V \subseteq U \text{ such that } f|_V \text{ is differentiable on } V \end{aligned}$$

Proof.

$\Rightarrow$. Let $x \in U$ then if we take $V = U$ we have that $x \in V \subseteq U$ and $f|_V = f$ so that $f|_V$ is n -times differentiable on V .

$\Leftarrow$. Let $x \in U$ then by the hypothesis there exist a $V_x \in \mathcal{T}_{|||_x}$ with $x \in V_x \subseteq U$ such that $f|_{V_x}$ is n -times differentiable on V_x . Using [corollary: 16.147] it follows that f is n -times differentiable on V_x . Using the [definition: 16.141] it follows that $V_x \subseteq \mathfrak{D}_f^n$ and by [theorem: 16.135] $\mathfrak{D}_f^n$. As this is true for every $x \in U$ we have that $U \subseteq \bigcup_{x \in U} V_x \subseteq \mathfrak{D}_f^n \subseteq U$ which proves that

$$U = \mathfrak{D}_f^n$$

so that

f is differentiable on U

□

We define now the concept of a function of class C^n on a open set that combines differentiability with continuity.

Definition 16.149. Let $n \in \mathbb{N}_0$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ normed spaces, $A \subseteq X$. $U \in \mathcal{T}_{|||_X}$ a open set in X with $U \subseteq X$ and $f: A \rightarrow Y$ a function then we say that f is of class C^n on U iff for the following cases of n we have:

$n = 0$. Then $f|_U$ is continuous

$n \in \mathbb{N}$. Then f is n -times differentiable on U and $(D^{[n]}f)|_U$ is continuous

We can extend the above definition to functions of class C^∞ on a open set.

Definition 16.150. Let $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{|||_X}$ a open set in X and $f: A \rightarrow Y$ a function then f is of class C^∞ on U iff $\forall n \in \mathbb{N}_0$ f is of class C^n

In the special case where the open set is the domain of the function we say that the function is of class C^n as is show in the following definition.

Definition 16.151. Let $n \in \mathbb{N}_0 \cup \{\infty\}$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ normed spaces, $U \in \mathcal{T}_{|||_X}$ a open set in X and $f: U \rightarrow Y$ a function then f is of class C^n iff f is of class C^n on U .

If the domain of f is a open set and f is of class C^n we don't have to work with restricted functions so we have the following equivalent definition of a function of class C^n .

Theorem 16.152. Let $n \in \mathbb{N}_0$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ normed spaces, $U \in \mathcal{T}_{|||_X}$ a open set in X and $f: U \rightarrow Y$ a function then we have

$$\begin{aligned} & f \text{ is of class } C^n \\ & \Updownarrow \\ & n = 0 \text{ and } f \text{ is continuous} \\ & \text{or} \\ & n \in \mathbb{N} \text{ and } U = \mathfrak{D}_f^n \text{ and } D^{[n]}f \text{ is continuous} \end{aligned}$$

Proof.

$\Rightarrow$. For n we have either

$n = 0$. Then, as f is of class C^0 on U , $f|_U$ is continuous. As $U = \text{dom}(f)$ we have that $f = f|_U$ so that f is continuous.

$n \in \mathbb{N}$. Then, as f is of class C^n on U , f is n -times differentiable on U and $(D^{[n]}f)|_U$ is continuous. Hence we have that $U \subseteq \mathfrak{D}_f^{[n]} \subseteq U^0 \subseteq U$ which proves that [theorem: 16.135]

$$U = \mathfrak{D}_f^n = \text{dom}(D^{[n]}f)$$

Hence $(D^{[n]}f)|_U = D^{[n]}f$ which proves that

$$D^{[n]}f \text{ is continuous}$$

$\Leftarrow$. For n we have either

$n = 0$. Then, as $U = \text{dom}(f)$ we have that $f = f|_U$ so that, as f is continuous, $f|_U$ is continuous. Hence

f is of class C^0 on U

$n \in \mathbb{N}$. As $U = \mathfrak{D}_f^n$ it follows that

f is n -times differentiable on U

and

$$(D^{[n]}f)|_U = D^{[n]}f$$

which, as $D^{[n]}f$ is continuous proves that

$$(D^{[n]}f)|_U \text{ is continuous}$$

Hence

f is of class C^n

□

For the cases where $n=1$ we have the following simpler definition for functions of class C^1 on U .

Theorem 16.153. *Let $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ normed spaces, $A \subseteq X$. $U \in \mathcal{T}_{|||_X}$ a open set in X with $U \subseteq A$ and $f: A \rightarrow Y$ a function then*

f is of class C^1 on U

$\Updownarrow$

f is Fréchet differentiable on U and $(Df)|_U: U \rightarrow L(X, Y)$ is continuous

Proof.

$\Rightarrow$. As f is of class C^1 on U then we have by definition that f is 1-times differentiable on U and $(D^{[1]}f)|_U$ is continuous. Hence we have by [definition: 16.141] that $\forall x \in U$ f is 1-times differentiable at x so that by [theorem: 16.140] f is Fréchet differentiable at x . Hence

f is Fréchet differentiable on U

Further as $(D^{[1]}f)|_U$ is continuous it follows from [definition: 16.129] that

$(Df)|_U$ is continuous

$\Leftarrow$. As f is Fréchet differentiable on U we have by definition that $\forall x \in U$ f is Fréchet differentiable at x so that by [theorem: 16.140] f is 1-times differentiable at x . Hence we have that

f is 1-times differentiable on U

Further as $(Df)|_U$ is continuous it follows from [definition: 16.129] that

$(D^{[1]}f)|_U$ is continuous

□

Theorem 16.154. *Let $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ normed spaces, $U \in \mathcal{T}_{|||_X}$ a open set in X and $f: U \rightarrow Y$ a function then*

f is of class C^1

$\Updownarrow$

f is Fréchet differentiable on U and $Df: U \rightarrow L(X, Y)$ is continuous

Proof.

$\Rightarrow$. As f is of class C^1 then we have by definition that f is 1-times differentiable on U and $D^{[1]}f$ is continuous. Hence we have by [definition: 16.141] that $\forall x \in U$ f is 1-times differentiable at x so that by [theorem: 16.140] f is Fréchet differentiable at x . Hence

f is Fréchet differentiable on U

Further as $D^{[1]}f$ is continuous it follows from [definition: 16.129] that

$$Df \text{ is continuous}$$

$\Leftarrow$. As f is Fréchet differentiable on U we have by definition that $\forall x \in U$ f is Fréchet differentiable at x so that by [theorem: 16.140] f is 1-times differentiable at x . Hence we have that

$$f \text{ is 1-times differentiable on } U$$

Further as Df is continuous it follows from [definition: 16.129] that

$$D^{[1]}f \text{ is continuous} \quad \square$$

Theorem 16.155. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{\|\cdot\|_X}$ with $U \subseteq A$ and $f: A \rightarrow Y$ a function then we have*

$$\begin{aligned} f \text{ is of class } C^n \text{ on } U \\ \Updownarrow \\ f|_U \text{ is of class } C^n \text{ on } U \end{aligned}$$

Proof.

$\Rightarrow$. By definition f is n -times differentiable on U and $(D^{[n]}f)|_U$ is continuous. Using [corollary: 16.147] it follows that

$$f|_U \text{ is } n\text{-times differentiable on } U \quad (16.94)$$

and

$$\forall x \in U \text{ we have } D_x^{[n]}f = D_x^{[n]}f|_U \quad (16.95)$$

Hence we have $\forall x \in U$ that

$$D^{[n]}f(x) \underset{\text{[definition: 16.136]}}{=} D_x^{[n]}f \underset{\text{[eq: 16.95]}}{=} D_x^{[n]}f|_U \underset{\text{[definition: 16.136]}}{=} D^{[n]}f|_U(x)$$

which proves that

$$(D^{[n]}f)|_U = (D^{[n]}f|_U)|_U$$

As $(D^{[n]}f)|_U$ is continuous it follows from the above that $(D^{[n]}f|_U)|_U$ is continuous, combining this with [eq: 16.94] proves that

$$f|_U \text{ is of class } C^n \text{ on } U$$

$\Leftarrow$. By definition $f|_U$ is n -times differentiable on U and $(D^{[n]}f|_U)|_U$ is continuous. Using [corollary: 16.147] it follows that

$$f \text{ is } n\text{-times differentiable on } U \quad (16.96)$$

and

$$\forall x \in U \text{ we have } D_x^{[n]}f = D_x^{[n]}f|_U \quad (16.97)$$

Hence we have $\forall x \in U$ that

$$D^{[n]}f(x) \underset{\text{[definition: 16.136]}}{=} D_x^{[n]}f \underset{\text{[eq: 16.97]}}{=} D_x^{[n]}f|_U \underset{\text{[definition: 16.136]}}{=} D^{[n]}f|_U(x)$$

which proves that

$$(D^{[n]}f)|_U = (D^{[n]}f|_U)|_U$$

As $(D^{[n]}f|_U)|_U$ is continuous it follows from the above that $(D^{[n]}f)|_U$ is continuous, combining this with [eq: 16.96] proves that

$$f \text{ is of class } C^n \text{ on } U \quad \square$$

Corollary 16.156. *Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U, V \in \mathcal{T}_{\|\cdot\|_X}$ with $V \subseteq U \subseteq A$ and $f: A \rightarrow Y$ a function that is of class C^n on U then $f|_V$ is of class C^n on V .*

Proof. For $n \in \mathbb{N}_0$ we have either:

$n = 0$. Then by [definition: 16.149] $f|_U$ is continuous, so that by [corollary: 14.153] $f|_V$ is continuous. Hence $f|_V$ is of class C^0 on V .

$n \in \mathbb{N}$. Then by [definition: 16.149] f is n -times differentiable on U and $(D^{[n]}f)|_U$ is continuous. As f is n -times differentiable on U we have by [definition: 16.141] that $U \subseteq \mathfrak{D}_f^n$, hence as $V \subseteq U$, $V \subseteq \mathfrak{D}_f^n$ from which it follows that

f is n -times differentiable on V

Further as $(D^{[n]}f)|_U$ is continuous and $V \subseteq U$ it follows from [corollary: 14.153] that

$(D^{[n]}f)|_V$ is continuous

Hence we have by [definition: 16.149] that

f is of class C^n on V

Finally using the previous theorem [see theorem: 16.155]

$f|_V$ is of class C^n on V □

Every ∞ -times differentiable function on a open set is C^∞ on this set as is proved in the following theorem.

Theorem 16.157. Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{\|\cdot\|_X}$ a open set in X and $f: A \rightarrow Y$ a function then we have

f is ∞ -times differentiable on U

$\Updownarrow$

f is of class C^∞ on U

Proof.

$\Rightarrow$. Let $n \in \mathbb{N}_0$ then we have either:

$n = 0$. Let $x \in U$ then, as f is ∞ -times differentiable on U , f is 1-times differentiable on U , hence 1-times differentiable at x . Using [theorem: 16.140] it follows that f is Fréchet differentiable at x . Using [theorem: 16.63] it follows that f is continuous at x . By [theorem: 14.146] it follows that $f|_U$ is continuous at x . As this is valued for every $x \in U$ it follows that

$f|_U$ is a continuous function

or that

f is of class C^0 on U

$n \in \mathbb{N}$. As f is ∞ -times differentiable on U it follows that

f is n -times differentiable on U or equivalently $U \subseteq \mathfrak{D}_f^n$ (16.98)

Further as f is ∞ -times differentiable on U it follows that

f is $(n+1)$ -times differentiable on U

Let $x \in U$ then by the above f is $(n+1)$ -times differentiable at x . Hence using [theorem: 16.140] it follows that $D^{[n]}f$ is differentiable at x so that by [theorem: 16.63] $D^{[n]}f$ is continuous at x . Hence using [theorem: 14.146] it follows that $(D^{[n]}f)|_U$ is continuous at x . As this is valued for every $x \in U$ it follows that

$(D^{[n]}f)|_U$ is a continuous function (16.99)

From [eqs: 16.98, 16.99] it follows that

$$f \text{ is of class } C^n \text{ on } U$$

As $\forall n \in \mathbb{N}_0$ f is of class C^n on U we have that

$$f \text{ is of class } C^\infty \text{ on } U$$

$\Leftarrow$. Let $n \in \mathbb{N}$ then as f is of class C^n on U we have by definition that f is n -times differentiable on U . $\square$

Theorem 16.158. Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be two normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{\|\cdot\|_X}$ with $U \subseteq A$ and $f: A \rightarrow Y$ a function that is of class C^n on U then $\forall m \in \{0, \dots, n\}$ f is of class C^m on U .

Proof. For $m \in \mathbb{N}_0$ we have either:

$m = 0$. Then for n we have either:

$n = 0$. Then $m = n$ so that f is of class C^n on U .

$0 < n$. Let $x \in U$. As f is of class C^n f is n -times differentiable at x , so that by [theorem: 16.138] f is 1-times differentiable at x . Using [theorem: 16.140] proves that f is Fréchet differentiable at x so that by [theorem: 16.63] f is continuous at x . By [theorem: 14.146] it follows that $f|_U$ is continuous at x . As this is true for every $x \in U$ we have that $f|_U$ is continuous which proves that f is of class C^0 .

$1 < m$. Then for n we have either:

$n = m$.

Then f is of class C^m on U .

$m < n$. Then $m+1 \leq n$. As f is of class C^n on U we have that f is n -times differentiable on U so that $U \subseteq \mathfrak{D}_f^n$, which, as by [theorem: 16.135] $\mathfrak{D}^n \subseteq \mathfrak{D}^{m+1} \subseteq \mathfrak{D}_f^m$, proves that

$$f \text{ is } m\text{-times differentiable on } U \quad (16.100)$$

and

$$f \text{ is } (m+1)\text{-times differentiable on } U$$

Let $x \in U$ then, by the above, f is $(m+1)$ -times differentiable at x . Using [theorem: 16.140] it follows that $D^{[m]}f$ is Fréchet differentiable at x . Hence using [theorem: 16.63] it follows that $D^{[m]}f$ is continuous at x . By [theorem: 14.146] it follows that $(D^{[m]}f)|_U$ is continuous at x . As this is true for every $x \in U$ we have that $(D^{[m]}f)|_U$ is continuous. Combining this result with [eq: 16.100] proves that

$$f \text{ is of class } C^m \quad \square$$

Theorem 16.159. Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be two normed spaces, $U \in \mathcal{T}_{\|\cdot\|_X}$ and $f: U \rightarrow Y$ a function then we have

$$f \text{ is of class } C^n$$

$$\Updownarrow$$

$$\forall x \in U \exists V \in \mathcal{T}_{\|\cdot\|_X} \text{ with } x \in V \subseteq U \text{ such that } f|_V \text{ is of class } C^n \text{ on } V$$

Proof.

$\Rightarrow$. As f is of class C^n on U we have $\forall x \in U$ that for $V = U$ $x \in V \subseteq U$ and $f|_V = f$ is of class C^n on V .

$\Leftarrow$. For n we have two possible case:

$n = 0$. Let $x \in U$ then by the hypothesis there exist a $V \in \mathcal{T}_{\|\cdot\|_X}$ with $x \in V \subseteq U$ such that $f|_V$ is continuous [see definition: 16.149]. Hence $f|_V$ is continuous at x which as V is open proves by [theorem: 14.150] that f is continuous at x . As this is true $\forall x \in U$ it follows that f is continuous. Hence f is of class C^n [see theorem: 16.152].

$n \in \mathbb{N}$. Let $x \in U$ then by the hypothesis $\exists V_x \in \mathcal{T}_{|||_X}$ with $x \in V_x \subseteq U$ such that $f|_{V_x}$ is of class C^n on V . Using [theorem: 16.155] it follows that f is of class C^n on V_x . Hence by [definitions: 16.149, 16.141] it follows that

$$x \in V_x \subseteq \mathfrak{D}_f^n \text{ and } (D^{[n]}f)|_{V_x} \text{ is continuous} \quad (16.101)$$

As $\forall x \in U \ x \in V_x \subseteq \bigcup_{x \in U} V_x \subseteq U$ it follows that $U = \bigcup_{x \in U} V_x$ which combined with [eq: 16.101] proves that

$$U = \mathfrak{D}_f^n \text{ or } f \text{ is } n\text{-times differentiable on } U$$

Further as by [eq: 16.101] $\forall x \in U \ (D^{[n]}f)|_{V_x}$ is continuous and $x \in V_x$ it follows that $(D^{[n]}f)|_{V_x}$ is continuous at x , using then [theorem: 14.150] it follows that $D^{[n]}f$ is continuous at x . Hence we have that

$$D^{[n]}f \text{ is continuous}$$

so that

$$f \text{ is of class } C^n \quad \square$$

16.2.3 Higher order derivatives

There is a interesting relation between functions of class C^1 on a open set and the derivative of a function. First we need a little lemma.

Lemma 16.160. *Let $\langle \mathbb{K}, || \rangle$ be the normed space of the real [or complex] numbers, $\langle X, ||| \rangle$ a normed space over $\mathbb{K}$, $A \subseteq \mathbb{K}$, $x \in A$ and*

$$f: A \rightarrow L(\mathbb{K}, X) \text{ a function}$$

and define

$$f(*) (1): A \rightarrow X \text{ by } (f(*) (1))(y) = f(y)(1)$$

then we have

$$f \text{ is continuous at } x \Leftrightarrow f(*) (1) \text{ is continuous at } x$$

Proof.

$\Rightarrow$. Let $x \in A$ and $\varepsilon \in \mathbb{R}^+$. As f is continuous at x there exist by [corollary: 14.136] a $\delta \in \mathbb{R}^+$ such that if $y \in A$ with $|y - x| < \delta$ then we have $\|f(y) - f(x)\|_{L(\mathbb{K}, X)} < \varepsilon$. Hence

$$\begin{aligned} \|(f(*) (1))(y) - (f(*) (1))(x)\| &= \|f(y)(1) - f(x)(1)\| \\ &= \|(f(y) - f(x))(1)\| \\ &\leq \|f(y) - f(x)\|_{L(\mathbb{K}, X)} \cdot |1| \\ &= \|f(y) - f(x)\|_{L(\mathbb{K}, X)} \\ &< \varepsilon \end{aligned}$$

Applying then [corollary: 14.136] proves that $f(*) (1)$ is continuous at x .

$\Leftarrow$. Let $x \in A$ and $\varepsilon \in \mathbb{R}^+$ then, as $f(*) (1)$ is continuous at x , we have by [corollary: 14.136] that there exist a $\delta \in \mathbb{R}^+$ such that for $y \in A$ with $|y - x| < \delta$

$$\|(f(*) (1))(y) - (f(*) (1))(x)\| < \frac{\varepsilon}{2} \quad (16.102)$$

Hence we have $\forall t \in \mathbb{K}$ that

$$\begin{aligned} \|(f(y) - f(x))(t)\| &= \|(f(y) - f(x))(t \cdot 1)\| \\ &= \|t \cdot (f(y) - f(x))(1)\| \\ &= |t| \cdot \|(f(y) - f(x))(1)\| \\ &= |t| \cdot \|f(y)(1) - f(x)(1)\| \\ &= |t| \cdot \|(f(*) (1))(y) - (f(*) (1))(x)\| \\ &\leq \frac{\varepsilon}{2} \cdot |t| \\ &\quad \text{[eq: 16.102]} \end{aligned}$$

so that by [theorem: 14.212] we have $\|f(y) - f(x)\|_{L(\mathbb{K}, X)} \leq \frac{\varepsilon}{2} < \varepsilon$, hence f is continuous at x [see corollary: 14.136]. $\square$

Theorem 16.161. *Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of the real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ a normed space, U a open set in $\mathbb{K}$, $f: U \rightarrow X$ a function then*

$$\begin{aligned} & f \text{ is of class } C^1 \\ & \Updownarrow \\ & \forall x \in U \text{ } f \text{ has a derivative at } x \\ & f' \text{ is continuous} \end{aligned}$$

Further if f is C^1 on U then

$$f' = D^{[1]}f(*) (1)$$

where

$$D^{[1]}f(*) (1): U \rightarrow X \text{ is defined by } (D^{[1]}f(*) (1))(x) = D^{[1]}f(x)(1)$$

Proof.

$\Rightarrow$. As f is of class C^1 we have by [theorem: 16.152] that

$$U = \mathfrak{D}_f^{[1]} \text{ and } D^{[1]}f \text{ is continuous} \quad (16.103)$$

Let $x \in U$ then by the above and [definition: 16.136] f is 1-times differentiable at x or using [theorem: 16.140] f is Fréchet differentiable at x with $D_x^{[1]}f = D_x f$. Using [theorem: 16.64] it follows then that f has a derivative f'_x at x with $f'_x = D_x f(1) = D_x^{[1]}f(1)$. Summarized we have

$$\forall x \in U \text{ } f \text{ has a derivative at } x \text{ [so that } U = \mathfrak{D}_f] \text{ with } f'_x = D_x^{[1]}f(1) \quad (16.104)$$

Define now [see lemma: 16.160]

$$D^{[1]}f(*) (1): U \rightarrow X \text{ by } (D^{[1]}f(*) (1))(x) = D^{[1]}f(x)(1) \quad (16.105)$$

then we have $\forall x \in U$ that

$$\begin{aligned} f'(x) & \stackrel{[\text{eq: 16.104}]}{=} D_x^{[1]}f(1) \\ & \stackrel{[\text{definition: 16.136}]}{=} D^{[1]}f(x)(1) \\ & \stackrel{[\text{eq: 16.105}]}{=} (D^{[1]}f(*) (1))(x) \end{aligned}$$

from which it follows that

$$f' = D^{[1]}f(*) (1) \quad (16.106)$$

As $D^{[1]}f$ is continuous [see eq: 16.103] it follows from [lemma: 16.160] that $D^{[1]}f(*) (1)$ is continuous. Combining this with [eq: 16.106] proves that

$$f' \text{ is continuous}$$

$\Leftarrow$. Let $x \in U$ then by the hypothesis f has a derivative at x and the function f' is continuous. As $x \in U \stackrel{[\text{theorem: 14.16}]}{=} U$ it follows from [theorem: 16.64] that f is Fréchet differentiable at x with $f'_x = D_x f(1)$. Using then [theorem: 16.140] we have that f is 1-times differentiable at x and $f'_x = D_x^{[1]}f \stackrel{[\text{definition: 16.136}]}{=} D^{[1]}f(x)(1)$. To summarize

$$\forall x \in U \text{ } f \text{ is 1-times differentiable at } x \text{ [hence } U = \mathfrak{D}_f^{[1]}] \text{ with } f'_x = D^{[1]}f(x)(1) \quad (16.107)$$

Define now

$$D^{[1]}f(*) (1): U \rightarrow X \text{ by } (D^{[1]}f(*) (1))(x) = D^{[1]}f(x)(1) \quad (16.108)$$

then we have for $x \in U$ that

$$\begin{aligned} (D^{[1]}f)(*)(1))(x) & \stackrel{[\text{eq: 16.108}]}{=} D^{[1]}f(x)(1) \\ & \stackrel{[\text{eq: 16.107}]}{=} f'_x \\ & \stackrel{[\text{definition: 16.24}]}{=} f'(x) \end{aligned}$$

so that

$$f' = D^{[1]}f)(*)(1)$$

As f' is continuous it follows from the above that $D^{[1]}f)(*)(1)$ is continuous. Using then [lemma: 16.160] it follows that

$$D^{[1]}f \text{ is continuous}$$

Combing this result with [eq: 16.107] proves, using [theorem: 16.152], that

$$f \text{ is of class } C^1 \quad \square$$

We use now recursion to define higher order derivatives.

Definition 16.162. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers based on the norm $\|\cdot\|$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$ and $f: A \rightarrow X$ a function then we define the n -times derivative $f^{(n)}: \mathcal{D}_f^n \rightarrow X$ of f recursively as follows

$n = 1$. Then

$$\mathcal{D}_f^1 = \mathcal{D}_f = \{x \in A \mid f \text{ has a derivative at } x\}$$

and

$$f^{(1)} = f': \mathcal{D}_f \rightarrow X$$

See [definition: 16.24] for the definition of f' and $\mathcal{D}_f$

$1 < n$. Then

$$\mathcal{D}_f^n = \mathcal{D}_{f^{(n-1)}} \stackrel{[\text{definition: 16.24}]}{=} \{x \in \mathcal{D}_f^{n-1} \mid f^{(n-1)} \text{ has a derivative at } x\}$$

and

$$f^{(n)} = (f^{(n-1)})': \mathcal{D}_f^n \rightarrow X$$

Convention 16.163. If $A \subseteq \mathbb{K}$ and $f: A \rightarrow X$ is a function then $f^{(n)}$ is always the graph of the function $f^{(n)}: \mathcal{D}_f^n \rightarrow X$. In other words the domain of $f^{(n)}$ is always $\mathcal{D}_f^n$ and the codomain is always X .

The domain of $\mathcal{D}_f^n$ of $f^{(n)}$ shrinks as n increases as the following theorem shows

Theorem 16.164. Let $n, m \in \mathbb{N}$ with $m \leq n$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers based on the norm $\|\cdot\|$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$ and $f: A \rightarrow X$ a function then we have that

$$\mathcal{D}_f^n \subseteq \mathcal{D}_f^m \subseteq \mathcal{D}_f \subseteq A \bigcap A'$$

Proof. Let $n \in \mathbb{N}$ then we have that

$$\mathcal{D}_f^{n+1} = \mathcal{D}_f = \{x \in \mathcal{D}_f^n \mid f^{(n-1)} \text{ has a derivative at } x\} \subseteq \mathcal{D}_f^n$$

As $m \leq n$ implies $m \in \{1, \dots, n\}$ it follows from the above and [theorem: 10.36] that

$$\mathcal{D}_f^n \subseteq \mathcal{D}_f^m \subseteq \mathcal{D}_f^1 \stackrel{[\text{definition: 16.162}]}{=} \mathcal{D}_f \stackrel{[\text{theorem: 16.24}]}{\subseteq} A \bigcap A' \quad \square$$

Once we have defined we can define n -times derivatives at a point.

Definition 16.165. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers based on the norm $\|\cdot\|$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow X$ a function then we say that

f has a n -times derivative at x

iff

$$x \in \mathcal{D}_f^n$$

and the derivative $f_x^{(n)}$ of f at x is defined as

$$f_x^{(n)} = f^{(n)}(x)$$

Definition 16.166. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers based on the norm $\|\cdot\|$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $B \subseteq A$ then $f: A \rightarrow X$ has n -times derivatives on B iff

$$B \subseteq \mathcal{D}_f^{[n]}$$

or equivalently using [definition: 16.165]

$\forall x \in B$ we have that f has a n -times derivative at x

We have now the following equivalent definition of the n -times derivative at a point.

Theorem 16.167. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers based on the norm $\|\cdot\|$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow X$ a function then we have the following equivalences:

1. f has a n -times derivative at x .
2. We have for n either

$n = 1$. We have f has a derivative at x and $f_x^{(1)} = f'_x$.

$1 < n$. $f^{(n-1)}$ has a derivative at x and $f_x^{(n)} = (f^{(n-1)})'_x$

Proof.

$1 \Rightarrow 2$. Then by [definition: 16.165]

$$x \in \mathcal{D}_f^n \text{ and } f_x^{(n)} = f^{(n)}(x) \quad (16.109)$$

For n we have either

$n = 1$. Using [definition: 16.162] we have that $\mathcal{D}_f^1 = \mathcal{D}_f = \{x \in A \mid f \text{ has a derivative at } x\}$ and $f^{(1)} = f'$. Hence we have by [eq: 16.109]

$$f \text{ has a derivative at } x \text{ and } f_x^{(1)} \stackrel{[\text{eq: 16.109}]}{=} f^{(1)}(x) = f'(x) \stackrel{[\text{definition: 16.24}]}{=} f'_x$$

$1 < n$. Using [definition: 16.162] we have that

$$\mathcal{D}_f^n = \mathcal{D}_{f^{(n-1)}}(x \in \mathcal{D}_f^{n-1} \mid f^{(n-1)} \text{ has a derivative at } x) \text{ and } f^{(n)} = (f^{(n-1)})' \quad (16.110)$$

Hence we have by [eq: 16.109] that

$$f^{(n-1)} \text{ has a derivative at } x$$

and

$$f_x^{(n)} \stackrel{[\text{eq: 16.109}]}{=} f^{(n)}(x) \stackrel{[\text{eq: 16.110}]}{=} (f^{(n-1)})'(x) \stackrel{[\text{definition: 16.24}]}{=} (f^{(n-1)})'_x$$

$2 \Rightarrow 1$. For n we have either:

$n = 1$. As f has a derivative at x it follows that

$$x \in \{x \in A \mid f \text{ has a derivative at } x\} \stackrel{[\text{definition: 16.24}]}{=} \mathcal{D}_f \stackrel{[\text{definition: 16.162}]}{=} \mathcal{D}_f^1$$

so that by [definition: 16.165]

f has a 1-times derivative at x

$1 < n$. As $f^{(n-1)}$ has a derivative at x we have that

$$x \in \{x \in \mathcal{D}_f^{n-1} \mid f^{(n-1)} \text{ has a derivative at } x\} \stackrel{[\text{definition: 16.162}]}{=} \mathcal{D}_f^n$$

so that by [definition: 16.165]

f has a n -times derivative at x

□

The following theorem shows the effect that restricting a function has on the existence of derivatives.

Theorem 16.168. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$ a normed space, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers], $A \subseteq \mathbb{K}$, $B \subseteq A$ such that $B \subseteq B'$, $x \in B$ and $f: A \rightarrow X$ a function such that $B \subseteq \mathcal{D}_f^n$ [hence by [definition: 16.166] f has n -times derivatives on B] then $B = \mathcal{D}_{f|_B}^n$ [hence by [definition: 16.166] $f|_B$ has n -times derivatives on B] with $(f|_B)^{(n)} = (f^{(n)})|_B$.*

Proof. We use induction to prove this, so define

$$S = \{n \in \mathbb{N} \mid \text{If } B \subseteq \mathcal{D}_f^n \text{ then } B = \mathcal{D}_{f|_B}^n \text{ and } (f|_B)^{(n)} = (f^{(n)})|_B\}$$

then we have

$1 \in S$. Assume that $B \subseteq \mathcal{D}_f^1$. Then we have by [theorem: 16.164] that

$$B \subseteq \mathcal{D}_f$$

Let $y \in B$ then by the above $y \in \mathcal{D}_f$ so that by [definition: 16.24] f has a derivative at y . As $y \in B \subseteq B'$ it follows from [theorem: 16.34] that

$$f|_B \text{ has a derivative at } y \text{ with } (f|_B)'_y = f'_y$$

Using [theorem: 16.167] it follows that

$$f|_B \text{ has a 1-times derivative at } y \text{ with } (f|_B)^{(1)}_y = f^{(1)}_y$$

By [definitions: 16.165, 16.24] the above becomes

$$y \in \mathcal{D}_{f|_B}^1 \text{ and } (f|_B)^{(1)}(y) = f^{(1)}(y)$$

As $y \in B$ is chosen arbitrary it follows from the above that

$$B \subseteq \mathcal{D}_{f|_B}^1 \text{ and } ((f|_B)^{(1)})|_B = (f^{(1)})|_B$$

Using [theorem: 16.164] it follows that $\mathcal{D}_{f|_B}^1 \subseteq B$ so that

$$B = \mathcal{D}_{f|_B}^1 \text{ and } (f|_B)^{(1)} = (f^{(1)})|_B$$

which proves that

$$1 \in S$$

$n \in S \Rightarrow n+1 \in S$. Assume that $B \subseteq \mathcal{D}_f^{n+1}$. Then we have by [theorem: 16.164] that

$$B \subseteq \mathcal{D}_f^{n+1} \subseteq \mathcal{D}_f^n$$

which as $n \in S$ proves that

$$B = \mathcal{D}_{f|_B}^n \text{ and } (f|_B)^{(n)} = (f^{(n)})|_B \quad (16.111)$$

Let $y \in B$ then $y \in \mathcal{D}_f^{n+1}$ so that by [definition: 16.165] f has a $(n+1)$ -times derivative at y . Using [theorem: 16.167] it follows that

$$f^{(n)} \text{ has a derivative at } y \text{ with } f^{(n+1)}_y = (f^{(n)})'_y \quad (16.112)$$

As $y \in B \subseteq B'$ y is a limit point of B , hence using [theorem: 16.34] it follows that

$$(f^{(n)})|_B \text{ has a derivative at } y \text{ with } ((f^{(n)})|_B)'_y = (f^{(n)})'_y$$

substituting [eq: 16.111] in the above gives

$$(f|_B)^n \text{ has a derivative at } y \text{ with } ((f|_B)^{(n)})'_y = (f^{(n)})'_y \stackrel{\text{eq: 16.112}}{=} f_y^{(n+1)}$$

Using [theorem: 16.167] on the above it follows that

$$f|_B \text{ has a } (n+1)\text{-times derivative at } y \text{ with } (f|_B)^{(n+1)}_y = f_y^{(n+1)}$$

Hence using [definitions: 16.165, 16.24] we have that

$$y \in \mathcal{D}_{f|_B}^{n+1} \text{ and } (f|_B)^{(n+1)}(y) = f^{(n+1)}(y)$$

As $y \in B$ has been chosen arbitrary it follows that

$$B \subseteq \mathcal{D}_{f|_B}^{n+1} \text{ and } ((f|_B)^{(n+1)})|_B = (f^{(n+1)})|_B$$

Further, as by [theorem: 16.164] $\mathcal{D}_{f|_B}^{n+1} \subseteq B$, it follows from the above that

$$B = \mathcal{D}_{f|_B}^{n+1} \text{ and } (f|_B)^{(n+1)} = (f^{n+1})|_B$$

which proves that

$$n+1 \in S$$

□

Theorem 16.169. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, || \rangle$ be the normed space of real [or complex] numbers based on the norm $||$, $\langle X, ||| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow X$ a function such that f has a n -times derivative at x then $\forall m \in \{1, \dots, n\}$ f has a m -times derivative at x .

Proof. If f has a n -times derivative at x we have by [definition: 16.165] that $x \in \mathcal{D}_f^n$, which, as by [theorem: 16.164] $\mathcal{D}_f^n \subseteq \mathcal{D}_f^n$, proves that $x \in \mathcal{D}_f^n$. Hence it follows using [definition: 16.165] that f has a m -times derivative at x . □

Let's examine now the relation between n -times differentiability and the existence of the n -th derivative. First we need a little lemma.

Lemma 16.170. Let $n \in \mathbb{N}$, $\langle X, ||| \rangle$ be a normed space, $\langle \mathbb{K}, || \rangle$ the normed space of real (complex) numbers based on the canonical norm $||$ and $L \in L_n(\mathbb{K}; X)$ then

$$\|L\|_{L_n(\mathbb{K}; X)} \leq \left\| L \left(\underbrace{1, \dots, 1}_n \right) \right\|$$

Proof. As usually we prove this by induction on n , so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(\mathbb{K}; X) \text{ then } \|L\|_{L_n(\mathbb{K}; X)} \leq \left\| L \left(\underbrace{1, \dots, 1}_n \right) \right\| \right\}$$

then we have:

$1 \in S$. If $L \in L_1(\mathbb{K}; X) = L(\mathbb{K}, X)$ then if $x \in X^1$ satisfies $|x| = 1$ we have

$$\|L(x)\| = \|L(x \cdot 1)\| = \|x \cdot L(1)\| = |x| \cdot \|L(1)\| = \left\| L \left(\underbrace{1}_1 \right) \right\|$$

so that by [theorem: 14.212] $\|L\|_{L_1(\mathbb{K}; X)} \leq \left\| L \left(\underbrace{1}_1 \right) \right\|$, hence $1 \in S$

$n \in S \Rightarrow n+1 \in S$. If $L \in L_{n+1}(\mathbb{K}; X) = L(\mathbb{K}, L_n(\mathbb{K}; X))$ then $L(1) \in L_n(\mathbb{K}; X)$. As $n \in S$ we have that

$$\|L(1)\|_{L_n(\mathbb{K}; X)} \leq \left\| L(1) \left(\underbrace{1, \dots, 1}_n \right) \right\| = \left\| L \left(\underbrace{1, \dots, 1}_{n+1} \right) \right\| \quad (16.113)$$

If $x \in \mathbb{K}$ such that $|x| = 1$ then as $L \in L(\mathbb{K}, L_n(\mathbb{K}; X))$ we have $L(x) \in L_n(\mathbb{K}; X)$ so that

$$\begin{aligned}
 \|L(x)\|_{L_n(\mathbb{K}; X)} &= \|L(x \cdot 1)\|_{L_n(\mathbb{K}; X)} \\
 &= \|x \cdot L(1)\|_{L_n(\mathbb{K}; X)} \\
 &= |x| \cdot \|L(1)\|_{L_n(\mathbb{K}; X)} \\
 &= \|L(1)\|_{L_n(\mathbb{K}; X)} \\
 &\stackrel{[\text{eq: 16.113}]}{\leq} \left\| L\left(\underbrace{1, \dots, 1}_n\right) \right\|
 \end{aligned} \tag{16.114}$$

Hence using [theorem: 14.212] $\|L\|_{L_{n+1}(\mathbb{K}; X)} \leq \left\| L\left(\underbrace{1, \dots, 1}_{n+1}\right) \right\|$ which proves that $n+1 \in S$. $\square$

Theorem 16.171. Let $n \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers based on the norm $\|\cdot\|$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$, $x \in A$ and $f: A \rightarrow X$ a function that is n -times differentiable at x then f has a n -times derivative at x with

$$f_x^{(n)} = D_x^{[n]} f\left(\underbrace{1, \dots, 1}_n\right) \stackrel{[\text{definition: 16.140}]}{=} D_x^n f\left(\underbrace{1, \dots, 1}_n\right)$$

Note 16.172. Note the opposite is not always true. For $1 < n$ the n -times differentiability of f at x requires that $x \in (\mathfrak{D}_f^{n-1})^\circ$ [see theorem: 16.135] which does not follow from the existence of a derivative of f at x [which requires that x is a limit point of $\mathfrak{D}_f^{n-1}$]. One solution would be to redefine derivatives so that the existence of a derivative at a point requires that the point is an inner point of the domain of the function for which the derivative is calculated [just as is done in differentiation of a function]. The problem with this approach is that then, for functions defined on closed intervals $[a, b]$, we can not have derivatives at a and b . So we keep our definitions and accept the fact that the existence of derivatives does not guarantee the existence of differentials.

Proof. We prove this by induction, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } f \text{ is } n\text{-times differentiable at } x \in A \text{ then we have that } f \text{ has a } n\text{-times derivative at } x \text{ with } f_x^{(n)} = D_x^{[n]} f\left(\underbrace{1, \dots, 1}_n\right) \right\}$$

then we have:

$1 \in S$. If f is 1-times differentiable at x it follows from [theorem: 16.140] that f is Fréchet differentiable at x with $D_x^{[1]} f = D_x f$. By [theorem: 16.64] it follows that f has a derivative at x with $f'_x = D_x f(1)$. Using [theorem: 16.167] it follows that

f has a 1-times derivative at x

with $f_x^{(1)} = f'_x$. Hence we have

$$f_x^{(1)} = D_x f(1) = D_x^{[1]} f(1) = D_x^{[1]} f\left(\underbrace{1}_1\right)$$

which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Assume that f is $(n+1)$ -times differentiable at x then by [theorem: 16.140] we have that

$$D^{[n]} f \text{ is Fréchet differentiable at } x \text{ with } D_x^{[n+1]} f = D_x(D^{[n]} f) \tag{16.115}$$

Let $y \in \mathfrak{D}_f^n$ then by [definition: 16.136] f is n -times differentiable at y so that, as $n \in S$, we have that

$$f \text{ has a } n\text{-times derivative at } y \text{ with } f_y^{(n)} = D_y^{[n]} f\left(\underbrace{1, \dots, 1}_n\right) \tag{16.116}$$

As $D^{[n]}f$ is Fréchet differentiable at x it follows from [definition: 16.46] that $x \in (\mathfrak{D}_f^{[n]})^\circ \subseteq \mathfrak{D}_f^{[n]}$, applying then [theorem: 16.8] it follows that x is a limit point of $\mathfrak{D}_f^{[n]}$. Hence we have

$$x \in (\mathfrak{D}_f^{[n]})' \text{ and } x \in (\mathfrak{D}_f^{[n]})^\circ \subseteq \mathfrak{D}_f^{[n]} \quad (16.117)$$

Let $\varepsilon \in \mathbb{R}^+$ then by the Fréchet differentiability of f at x [see eq: 16.115] there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in (\mathfrak{D}_f^{[n]})_x$ with $|h| < \delta$ we have

$$\|D^{[n]}f(x+h) - D^{[n]}f(x) - D_x(D^{[n]}f)(h)\|_{L_n(\mathbb{K};X)} \leq \varepsilon \cdot |h| \quad (16.118)$$

Now

$$\begin{aligned} & \left\| (f^{(n)})_{|\mathfrak{D}_f^{[n]}|}(x+h) - (f^{(n)})_{|\mathfrak{D}_f^{[n]}|}(x) - h \cdot D_x^{[n+1]}f\left(\underbrace{1;\dots;1}_{n+1}\right) \right\| = \\ & \left\| f^{(n)}(x+h) - f^{(n)}(x) - h \cdot D_x^{[n+1]}f\left(\underbrace{1;\dots;1}_{n+1}\right) \right\| \\ & \left\| f^{(n)}(x+h) - f^{(n)}(x) - h \cdot D_x^{[n+1]}f(1)\left(\underbrace{1;\dots;1}_n\right) \right\| = \\ & \left\| f^{(n)}(x+h) - f^{(n)}(x) - D_x^{[n+1]}f(h \cdot 1)\left(\underbrace{1;\dots;1}_n\right) \right\| = \\ & \left\| f^{(n)}(x+h) - f^{(n)}(x) - D_x^{[n+1]}f(h)\left(\underbrace{1;\dots;1}_n\right) \right\| \stackrel{[\text{theorem: 16.165}]}{=} \\ & \left\| f_{x+h}^{(n)} - f_x^{(n)} - D_x^{[n+1]}f(h)\left(\underbrace{1;\dots;1}_n\right) \right\| \stackrel{[\text{eq: 16.116}]}{=} \\ & \left\| D_{x+h}^{[n]}f\left(\underbrace{1;\dots;1}_n\right) - D_x^{[n]}f\left(\underbrace{1;\dots;1}_n\right) - D_x^{[n+1]}f(h)\left(\underbrace{1;\dots;1}_n\right) \right\| \stackrel{[\text{lemma: 16.120}]}{=} \\ & \left\| (D_{x+h}^{[n]}f - D_x^{[n]}f - D_x^{[n+1]}f(h))\left(\underbrace{1;\dots;1}_n\right) \right\| \stackrel{[\text{lemma: 16.122}]}{\leq} \\ & \|D_{x+h}^{[n]}f - D_x^{[n]}f - D_x^{[n+1]}f(h)\|_{L_n(\mathbb{K};X)} \cdot \prod_{i=1}^n |h| = \\ & \|D_{x+h}^{[n]}f - D_x^{[n]}f - D_x^{[n+1]}f(h)\|_{L_n(\mathbb{K};X)} \stackrel{[\text{eq: 16.115}]}{=} \\ & \|D_{x+h}^{[n]}f - D_x^{[n]}f - D_x(D^{[n]}f)(h)\|_{L_n(\mathbb{K};X)} \stackrel{[\text{theorem: 16.136}]}{=} \\ & \|D^{[n]}f(x+h) - D^{[n]}f(x) - D_x(D^{[n]}f)(h)\|_{L_n(\mathbb{K};X)} \stackrel{[\text{eq: 16.118}]}{\leq} \\ & \varepsilon \cdot |h| \end{aligned}$$

To summarize, let $\varepsilon \in \mathbb{R}^+$ then there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in (\mathfrak{D}_f^{[n]})_x$ with $|h| < \delta$ we have

$$\left\| (f^{(n)})_{|\mathfrak{D}_f^{[n]}|}(x+h) - (f^{(n)})_{|\mathfrak{D}_f^{[n]}|}(x) - h \cdot D_x^{[n+1]}f\left(\underbrace{1;\dots;1}_{n+1}\right) \right\| \leq \varepsilon \cdot |h|$$

Using now [eq: 16.117] together with the above allows us to use [theorem: 16.23] proving that

$$(f^{(n)})_{|\mathfrak{D}_f^{[n]}|} \text{ has a derivative at } x \text{ with } ((f^{(n)})_{|\mathfrak{D}_f^{[n]}|})'_x = D_x^{[n+1]}f\left(\underbrace{1;\dots;1}_{n+1}\right)$$

Using now [theorem: 16.36] together with [eq: 16.117] it follows from the above that

$$f^{(n)} \text{ has a derivative at } x \text{ with } (f^{(n)})'_x = D_x^{[n+1]}f\left(\underbrace{1;\dots;1}_{n+1}\right)$$

Finally using [theorem: 16.167] we have that

$$f \text{ has a } (n+1)\text{-times derivative at } z \text{ with } f_x^{(n+1)} = D_x^{[n+1]} f \left(\underbrace{1: \dots : 1}_{n+1} \right)$$

which proves that

$$n+1 \in S$$

□

16.2.4 Properties of higher order differentials and derivatives.

Theorem 16.173. *Let $n, m \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $f: A \rightarrow Y$ a function then we have that*

$$\mathfrak{D}_f^{n+m} = \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is } m\text{-times differentiable at } x\} \text{ and } D^{[n+m]}f = D^{[m]}(D^{[n]}f)$$

Proof. We use induction to prove this, so given $n \in \mathbb{N}$ define

$$S_n = \{m \in \mathbb{N} \mid \mathfrak{D}_f^{n+m} = \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is } m\text{-times differentiable at } x\} \text{ and } D^{[n+m]}f = D^{[m]}(D^{[n]}f)\}$$

then we have:

$1 \in S_n$. Using [definition: 16.129] we have that

$$\begin{aligned} \mathfrak{D}_f^{n+1} &= \mathfrak{D}_{D^{[n]}f} \\ &= \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is Fréchet differentiable at } x\} \\ &\stackrel{[\text{theorem: 16.140}]}{=} \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is 1-times differentiable at } x\} \end{aligned}$$

and

$$D^{[n+1]}f \stackrel{[\text{definition: 16.129}]}{=} D(D^{[n]}f) \stackrel{[\text{definition: 16.129}]}{=} D^{[1]}(D^{[n]}f)$$

which proves that

$$1 \in S_n$$

$m \in S_n \Rightarrow m+1 \in S_n$. We have for $\mathfrak{D}_f^{n+(m+1)}$ and $D^{n+(m+1)}f$ that

$$\begin{aligned} \mathfrak{D}_f^{n+(m+1)} &= \mathfrak{D}_f^{(n+m)+1} \\ &\stackrel{[\text{definition: 16.129}]}{=} \{x \in \mathfrak{D}_f^{n+m} \mid D^{[n+m]}f \text{ is Fréchet differentiable at } x\} \end{aligned}$$

and $D^{[n+(m+1)]}f = D^{[(n+m)+1]}f \stackrel{[\text{definition: 16.129}]}{=} D(D^{[n+m]}f)$. Summarized we have

$$\mathfrak{D}_f^{n+(m+1)} = \{x \in \mathfrak{D}_f^{n+m} \mid D^{[n+m]}f \text{ is Fréchet differentiable at } x\} \quad (16.119)$$

$$D^{[n+(m+1)]}f = D(D^{[n+m]}f) \quad (16.120)$$

As $m \in S_n$ we have that

$$\mathfrak{D}_f^{n+m} = \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is } m\text{-times differentiable at } x\} \quad (16.121)$$

$$D^{[n+m]}f = D^{[m]}(D^{[n]}f) \quad (16.122)$$

Let $x \in \mathfrak{D}_f^{n+(m+1)}$ then by [eq: 16.119] we have that

$$x \in \mathfrak{D}_f^{n+m} \text{ and } D^{[n+m]}f \text{ is Fréchet differentiable at } x$$

Using [eqs: 16.121, 16.122] on the above we have that

$$x \in \mathfrak{D}_f^n \text{ and } D^{[m]}(D^{[n]}f) \text{ is Fréchet differentiable at } x$$

Using [theorem: 16.140] on the above it follows that $D^{[n]}f$ is $(m+1)$ -times differentiable at x so that we have that $x \in \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is } (m+1)\text{-times differentiable at } x\}$. Hence we have

$$\mathfrak{D}_f^{n+(m+1)} \subseteq \{x \in \mathfrak{D}_f^n \mid D^{[n]}f \text{ is } (m+1)\text{-times differentiable at } x\} \quad (16.123)$$

For the opposite inclusion. Let $y \in \{x \in \mathfrak{D}_f^{[n]} \mid D^{[n]}f \text{ is } (m+1)\text{-times differentiable at } x\}$ then we have

$$y \in \mathfrak{D}_f^{[n]} \text{ and } D^{[n]}f \text{ is } (m+1)\text{-times differentiable at } y$$

Using [theorem: 16.138] it follows that $D^{[n]}f$ is m -times differentiable at y , hence

$$y \in \{x \in \mathfrak{D}_f^{[n]} \mid D^{[n]}f \text{ is } m\text{-times differentiable at } x\} \stackrel{[\text{eq: 16.121}]}{=} \mathfrak{D}_f^{n+m} \quad (16.124)$$

Further as $D^{[n]}f$ is $(m+1)$ -times differentiable at y it follows from [theorem: 16.140] that $D^{[m]}(D^{[n]}f)$ is Fréchet differentiable at y . Hence we have that

$$y \stackrel{[\text{eq: 16.124}]}{\in} \{x \in \mathfrak{D}_f^{[n+m]} \mid D^{[m]}(D^{[n]}f) \text{ is Fréchet differentiable at } x\} \stackrel{[\text{eq: 16.119}]}{=} \mathfrak{D}_f^{n+(m+1)}$$

which prove that

$$\{x \in \mathfrak{D}_f^{[n]} \mid D^{[n]}f \text{ is } (m+1)\text{-times differentiable at } x\} \subseteq \mathfrak{D}_f^{n+(m+1)}$$

Combining the above with [eq: 16.123] results in

$$\mathfrak{D}_f^{n+(m+1)} = \{x \in \mathfrak{D}_f^{[n]} \mid D^{[n]}f \text{ is } (m+1)\text{-times differentiable at } x\} \quad (16.125)$$

Further we have

$$D^{[n+(m+1)]}f \stackrel{[\text{eq: 16.120}]}{=} D[D^{[n+m]}f] \stackrel{[\text{eq: 16.122}]}{=} D(D^{[m]}(D^{[n]}f)) \stackrel{[\text{definition: 16.129}]}{=} D^{[m+1]}(D^{[n]}f)$$

so that

$$D^{[n+(m+1)]}f = D^{[m+1]}(D^{[n]}f)$$

which together with [eq: 16.125] proves that

$$m+1 \in S$$

□

Corollary 16.174. *Let $n, m \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $f: A \rightarrow Y$ a function then we have:*

1. *If $x \in A$ then we have*

$$\begin{aligned} &f \text{ is } (n+m)\text{-times differentiable at } x \\ &\quad \Updownarrow \\ &D^{[n]}f \text{ is } m\text{-times differentiable at } x \end{aligned}$$

Further in either case we have

$$D_x^{[n+m]}f = D_x^{[n]}(D^{[n]}f)$$

2. *If $U \in \mathcal{T}_{\|\cdot\|_X}$ with $U \subseteq A$ then we have*

$$\begin{aligned} &f \text{ is } (n+m)\text{-times differentiable on } U \\ &\quad \Updownarrow \\ &D^{[n]}f \text{ is } m\text{-times differentiable on } U \end{aligned}$$

Proof.

1. We have

$$\begin{aligned} &f \text{ is } (n+m)\text{-times differentiable at } x \stackrel{[\text{definition: 16.136}]}{\Leftrightarrow} \\ &\quad x \in \mathfrak{D}_f^{n+m} \stackrel{[\text{theorem: 16.173}]}{\Leftrightarrow} \\ &x \in \{x \in \mathfrak{D}_f^{[n]} \mid D^{[n]}f \text{ is } m\text{-times differentiable at } x\} \Leftrightarrow \\ &\quad D^{[n]}f \text{ is } m\text{-times differentiable at } x \end{aligned}$$

Further if f is $(n+m)$ -times differentiable at x then

$$D_x^{[n+m]}f \underset{[\text{definition: 16.136}]}{=} D^{[n+m]}f(x) \underset{[\text{theorem: 16.173}]}{=} D^{[m]}(D^{[n]}f)(x) \underset{[\text{definition: 16.136}]}{=} D_x^{[m]}(D^{[n]}f)$$

or

$$D_x^{[n+m]}f = D_x^{[m]}(D^{[n]}f)$$

2. Then

$$\begin{aligned} f \text{ is } (n+m)\text{-times differentiable on } U &\Leftrightarrow \\ \forall x \in U \ f \text{ is } (n+m)\text{-times differentiable at } x &\underset{(1)}{\Leftrightarrow} \\ \forall x \in U \ D^{[n]}f \text{ is } m\text{-times differentiable at } x &\Leftrightarrow \\ D^{[n]}f \text{ is } m\text{-times differentiable on } U & \\ \square \end{aligned}$$

Theorem 16.175. Let $n, m \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$, $U \in \mathcal{T}_{\|\cdot\|_X}$ with $U \subseteq A$ and $f: A \rightarrow Y$ a function then we have

$$\begin{aligned} f \text{ is of class } C^{n+m} \text{ on } U \\ \Updownarrow \\ D^{[n]}f \text{ is of class } C^m \text{ on } U \end{aligned}$$

Proof.

$\Rightarrow$. As f is of class C^{n+m} we have by definition that f is $(n+m)$ -times differentiable on U and $(D^{[n+m]}f)|_U$ is continuous. Using [theorem: 16.174(2)] we have that

$$D^{[n]}f \text{ is } m\text{-times differentiable on } U \quad (16.126)$$

Further using [theorem: 16.173] we have that $D^{[n+m]}f = D^{[m]}(D^{[n]}f)$ so that

$$(D^{[n+m]}f)|_U = (D^{[m]}(D^{[n]}f))|_U$$

which as $(D^{[n+m]}f)|_U$ is continuous proves that

$$(D^{[m]}(D^{[n]}f)) \text{ is continuous} \quad (16.127)$$

Combining [eqs: 16.126, 16.127] it follows that

$$D^{[n]}f \text{ is of class } C^m \text{ on } U$$

$\Leftarrow$. As $D^{[n]}f$ is of class C^m we have by definition that $D^{[n]}f$ is m -times differentiable on U and $(D^{[m]}(D^{[n]}f))|_U$ is continuous. Using [theorem: 16.174(2)] we have that

$$f \text{ is } (n+m)\text{-times differentiable on } U \quad (16.128)$$

Further using [theorem: 16.173] we have that $D^{[n+m]}f = D^{[m]}(D^{[n]}f)$ so that

$$(D^{[n+m]}f)|_U = (D^{[m]}(D^{[n]}f))|_U$$

which as $(D^{[m]}(D^{[n]}f))|_U$ is continuous proves that

$$(D^{[n+m]}f)|_U \text{ is continuous} \quad (16.129)$$

Combining [eqs: 16.128, 16.129] proves that

$$f \text{ is of class } C^{n+m} \text{ on } U \quad \square$$

We show now that we have similar theorems for higher order derivatives.

Theorem 16.176. Let $n, m \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers], $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$ and $f: A \rightarrow X$ a function then we have that

$$\mathcal{D}_f^{n+m} = \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } m\text{-times derivative at } x\}$$

and

$$f^{(n+m)} = (f^{(n)})^{(m)}$$

Proof. We use induction to prove this, so given $n \in \mathbb{N}$ define

$$S_n = \{m \in \mathbb{N} \mid \mathcal{D}_f^{n+m} = \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } m\text{-times derivative at } x\} \text{ and } f^{(n+m)} = (f^{(n)})^{(m)}\}$$

then we have:

$1 \in S_n$. Using [definition: 16.162] we have that

$$\begin{aligned} \mathcal{D}_f^{n+1} &= \mathcal{D}_{f^{(n)}} \\ &= \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a derivative at } x\} \\ &\stackrel{[\text{theorem: 16.167}]}{=} \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a 1-times derivative at } x\} \end{aligned}$$

and

$$f^{(n+1)} \stackrel{[\text{definition: 16.162}]}{=} (f^n)' \stackrel{[\text{definition: 16.162}]}{=} (f^{(n)})^{(1)}$$

which proves that

$$1 \in S_n$$

$m \in S_n \Rightarrow m+1 \in S_n$. We have for $\mathcal{D}^{n+(m+1)}$ and $f^{(n+(m+1))}$ that

$$\begin{aligned} \mathcal{D}_f^{n+(m+1)} &= \mathcal{D}_f^{(n+m)+1} \\ &\stackrel{[\text{definition: 16.162}]}{=} \{x \in \mathcal{D}_f^{n+m} \mid f^{(n+m)} \text{ has a derivative at } x\} \end{aligned}$$

and $(f)^{n+(m+1)} = (f)^{(n+m)+1} \stackrel{[\text{definition: 16.162}]}{=} (f^{(n+m)})'$. Summarized we have

$$\mathcal{D}_f^{n+(m+1)} = \{x \in \mathcal{D}_f^{n+m} \mid f^{(n+m)} \text{ has a derivative at } x\} \quad (16.130)$$

$$(f)^{(n+(m+1))} = (f^{(n+m)})' \quad (16.131)$$

As $m \in S_n$ we have that

$$\mathcal{D}_f^{n+m} = \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } m\text{-times derivative at } x\} \quad (16.132)$$

$$f^{(n+m)} = (f^{(n)})^{(m)} \quad (16.133)$$

Let $x \in \mathcal{D}^{n+(m+1)}$ then by [eq: 16.130] we have that

$$x \in \mathcal{D}_f^{n+m} \text{ and } f^{(n+m)} \text{ has a derivative at } x$$

Using [eqs: 16.132, 16.133] on the above we have that

$$x \in \mathcal{D}_f^n \text{ and } (f^{(n)})^{(m)} \text{ has a derivative at } x$$

Using [theorem: 16.167] on the above it follows that $f^{(m)}$ has a $(m+1)$ -times differential at x so that we have $x \in \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } (m+1)\text{-times derivative at } x\}$. Hence we have that

$$\mathcal{D}^{n+(m+1)} \subseteq \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } (m+1)\text{-times derivative at } x\} \quad (16.134)$$

For the opposite inclusion. Let $y \in \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } (m+1)\text{-times derivative at } x\}$ then we have

$$y \in \mathcal{D}_f^n \text{ and } f^{(n)} \text{ has a } (m+1)\text{-times derivative at } y$$

Using [theorem: 16.169] it follows that $f^{(n)}$ has a m -times derivative at y , hence

$$y \in \{x \in \mathcal{D}_f^n \mid f^{(n)} \text{ has a } m\text{-times derivative at } x\} \stackrel{[\text{eq: 16.132}]}{=} \mathcal{D}_f^{n+m} \quad (16.135)$$

Further as $f^{(n)}$ has a $(m+1)$ -times derivative at y it follows from [theorem: 16.167] that $(f^{(n)})^{(m)}$ has a derivative at x . Hence we have that

$$y \stackrel{[\text{eq: 16.135}]}{\in} \{y \in \mathcal{D}_f^{n+m} \mid (f^{(n)})^{(m)} \text{ has a derivative at } x\} \stackrel{[\text{eq: 16.130}]}{=} \mathcal{D}_f^{n+(m+1)}$$

which proves that

$$\{x \in \mathcal{D}_f^n | f^{(n)} \text{ has a } (m+1)\text{-times derivative at } x\} \subseteq \mathcal{D}_f^{n+(m+1)}$$

Combining the above with [eq: 16.134] results in

$$\mathcal{D}_f^{n+(m+1)} = \{x \in \mathcal{D}_f^n | f^{(n)} \text{ has a } (m+1)\text{-times derivative at } x\} \quad (16.136)$$

Further we have

$$f^{(n+(m+1))} \underset{[\text{eq: 16.131}]}{=} (f^{(n+m)})' \underset{[\text{eq: 16.133}]}{=} ((f^{(n)})^{(m)})' \underset{[\text{definition: 16.162}]}{=} (f^{(n)})^{(m+1)}$$

so that

$$f^{(n+(m+1))} = (f^{(n)})^{(m+1)}$$

which together with [eq: 16.136] proves that

$$m+1 \in S \quad \square$$

Corollary 16.177. *Let $n, m \in \mathbb{N}$, $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex numbers], $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}$ and $f: A \rightarrow X$ a function then we have that*

1. *If $x \in A$ then*

$$\begin{aligned} &f \text{ has a } (n+m)\text{-times derivative at } x \\ &\quad \Downarrow \\ &f^{(n)} \text{ has a } m\text{-times derivative at } x \end{aligned}$$

Further in all cases we have

$$f_x^{(n+m)} = (f^{(n)})_x^{(m)}$$

2. *If $U \in \mathcal{T}_\parallel$ with $U \subseteq A$ then we have*

$$\begin{aligned} &f \text{ has } (n+m)\text{-derivatives on } U \\ &\quad \Downarrow \\ &f^{(n)} \text{ has } m\text{-derivatives on } U \end{aligned}$$

Proof.

1. We have

$$\begin{aligned} &f \text{ has a } (n+m)\text{-times differentiable at } x \quad \Leftrightarrow \quad [\text{definition: 16.165}] \\ &\quad \quad \quad x \in \mathcal{D}_f^{n+m} \quad \Leftrightarrow \quad [\text{theorem: 16.176}] \\ &x \in \{x \in \mathcal{D}_f^{[n]} | f^{(n)} \text{ has a } m\text{-times derivative at } x\} \quad \Leftrightarrow \\ &\quad \quad \quad f^{(n)} \text{ has a } m\text{-times derivative at } x \end{aligned}$$

Further if f has a $(n+m)$ -times derivative at x then

$$f_x^{(n+m)} \underset{[\text{definition: 16.165}]}{=} f^{(n+m)}(x) \underset{[\text{theorem: 16.176}]}{=} (f^{(n)})^{(m)}(x) \underset{[\text{definition: 16.165}]}{=} (f^{(n)})_x^{(m)}$$

or

$$f_x^{(n+m)} = (f^{(n)})_x^{(m)}$$

2. Then

$$\begin{aligned} &f \text{ has } (n+m)\text{-times derivatives on } U \quad \Leftrightarrow \\ &\forall x \in U \quad f \text{ has a } (n+m)\text{-times derivative at } x \quad \Leftrightarrow \quad (I) \\ &\forall x \in U \quad f^{(n)} \text{ has a } m\text{-times derivative at } x \quad \Leftrightarrow \\ &\quad \quad \quad f^{(n)} \text{ has } m\text{-times derivatives on } U \end{aligned}$$

$\square$

Just as the Fréchet differential and derivatives are linear [see theorems: 16.40, 16.66], we have the same for higher order differentials and derivatives. To prove this linearity we need the following theorem.

Theorem 16.178. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $A \subseteq X$ then we have:*

1. *If $f: A \rightarrow Y$, $g: A \rightarrow Y$ are functions such that f, g are n -times differentiable at $x \in A$ then $f + g$ is n -times differentiable at x and $D_x^{[n]}(f + g) = D_x^{[n]}f + D_x^{[n]}g$.*
2. *If $k \in \mathbb{N}$ and $\{f_i: A \rightarrow Y\}_{i \in \{1, \dots, k\}}$ is a family of functions such that $\forall i \in \{1, \dots, k\}$ f_i is n -times differentiable at $x \in A$ then $\sum_{i=1}^k f_i$ is n -times differentiable at x and*

$$D_x^{[n]}\left(\sum_{i=1}^k f_i\right) = \sum_{i=1}^k D_x^{[n]}f_i$$

3. *If $\alpha \in \mathbb{K}$, $f: A \rightarrow Y$ is a function that is n -times differentiable at $x \in A$ then $\alpha \cdot f$ is n -times differentiable at x and $D_x^{[n]}(\alpha \cdot f) = \alpha \cdot D_x^{[n]}f$.*
4. *If U is a open set in the topology $\mathcal{T}_{\|\cdot\|_X}$ with $U \subseteq A$ and $f: A \rightarrow Y$ and $g: A \rightarrow Y$ functions that are of class C^n on U then $f + g$ is of class C^n .*
5. *If U is a open set in the topology $\mathcal{T}_{\|\cdot\|_X}$ and $\{f_i: A \rightarrow Y\}_{i \in \{1, \dots, k\}}$ is a family of functions such that $\forall i \in \{1, \dots, k\}$ f_i is of class C^n on U then $\sum_{i=1}^k f_i$ is of class C^n on U .*
6. *If U is a open set in the topology $\mathcal{T}_{\|\cdot\|_X}$, $\alpha \in \mathbb{K}$ and $f: A \rightarrow Y$ is of class C^n on U then $\alpha \cdot f$ is of class C^n of U .*

Proof.

1. We use induction on n to prove this, so let

$S = \{n \in \mathbb{N} \mid \text{If } A \subseteq X \text{ and } f, g: A \rightarrow Y \text{ are } n\text{-times differentiable at } x \in A \text{ then } f + g \text{ is } n\text{-times differentiable at } x \text{ and } D_x^{[n]}(f + g) = D_x^{[n]}f + D_x^{[n]}g\}$

then we have:

1 $\in S$. If $f, g: A \rightarrow Y$ are 1-times differentiable at x then by [theorem: 16.140] f, g are Fréchet differentiable at x and $D_x^{[1]}f = D_x f$, $D_x^{[1]}g = D_x g$. Using [theorem: 16.66] it follows that $f + g$ is Fréchet differentiable at x [hence by [theorem: 16.140] 1-times differentiable at x] and

$$D_x^{[1]}(f + g) \underset{[\text{theorem: 16.140}]}{=} D_x(f + g) = D_x f + D_x g = D_x^{[1]}f + D_x^{[1]}g$$

which proves that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. As f and g are $(n + 1)$ -times differentiable at x and $2 \in \{1, \dots, n + 1\}$ we have by [theorem: 16.138] that f, g are 2-times differentiable at x . Using [theorem: 16.144] that there exist open sets U, V with $x \in U \subseteq A, x \in V \subseteq A$ such that

f is 1-times differentiable on U and g is 1-times differentiable on V

Take the open set $W = U \cap V$ then $x \in W \subseteq A$ and

f, g are 1-times differentiable on W

or by [definition: 16.136]

$$W \subseteq \mathfrak{D}_f^1 \text{ and } W \subseteq \mathfrak{D}_g^1 \tag{16.137}$$

Let $y \in W$ then f, g are 1-times differentiable at y hence by [theorem: 16.140] Fréchet differentiable at y so that by [theorem: 16.66]

$f + g$ is Fréchet differentiable at y with $D_y(f + g) = D_y f + D_y g$

or using [theorem: 16.140] again we have

$f + g$ is 1-times differentiable at y with $D_y^{[1]}(f + g) = D_y^{[1]}f + D_y^{[1]}g$

Using [definition: 16.136] we have that

$$y \in \mathfrak{D}_{f+g}^1 \text{ and } D^{[1]}(f+g)(y) = D^{[1]}f(y) + D^{[1]}g(y)$$

Hence we have using the above and [eq: 16.137] that

$$(D^{[1]}(f+g))|_W = (D^{[1]}f)|_W + (D^{[1]}g)|_W = (D^{[1]}f + D^{[1]}g)|_W \quad (16.138)$$

As f and g are $(n+1)$ -times differentiable at x we have by [theorem: 16.174] that

$$D^{[1]}f \text{ is } n\text{-times differentiable at } x \text{ and } D_x^{[n+1]}f = D_x^{[n]}(D^{[1]}f) \quad (16.139)$$

$$D^{[1]}g \text{ is } n\text{-times differentiable at } x \text{ and } D_x^{[n+1]}g = D_x^{[n]}(D^{[1]}g) \quad (16.140)$$

As $n \in S$, it follows from the above that

$$D^{[1]}f + D^{[1]}g \text{ is } n\text{-times differentiable at } x$$

and

$$D_x^{[n]}(D^{[1]}f + D^{[1]}g) = D_x^{[n]}(D^{[1]}f) + D_x^{[n]}(D^{[1]}g) \quad (16.141)$$

Using then [theorem: 16.145] on the above it follows that

$$(D^{[1]}f + D^{[1]}g)|_W \text{ is } n\text{-times differentiable at } x \quad (16.142)$$

and

$$\begin{aligned} D_x^{[n]}((D^{[1]}f + D^{[1]}g)|_W) &\stackrel{[\text{theorem: 16.145}]}{=} D_x^{[n]}(D^{[1]}f + D^{[1]}g) \\ &\stackrel{[\text{eq: 16.141}]}{=} D_x^{[n]}(D^{[1]}f) + D_x^{[n]}(D^{[1]}g) \\ &\stackrel{[\text{eqs: 16.139, 16.140}]}{=} D_x^{[n+1]}f + D_x^{[n+1]}g \end{aligned}$$

so that

$$D_x^{[n]}((D^{[1]}f + D^{[1]}g)|_W) = D_x^{[n+1]}f + D_x^{[n+1]}g \quad (16.143)$$

Hence using [eq: 16.138] on [eqs: 16.142, 16.143] we have that

$$(D^{[1]}(f+g))|_W \text{ is } n\text{-times differentiable at } x$$

$$D_x^{[n]}((D^{[1]}(f+g))|_W) = D_x^{[n+1]}f + D_x^{[n+1]}g$$

which by [theorem: 16.145] results in

$$D^{[1]}(f+g) \text{ is } n\text{-times differentiable at } x$$

$$D_x^{[n]}(D^{[1]}(f+g)) = D_x^{[n+1]}f + D_x^{[n+1]}g$$

Using [theorem: 16.174] again on the above proves that

$$f+g \text{ is } n\text{-times differentiable at } x \text{ with } D_x^{[n+1]}(f+g) = D_x^{[n+1]}f + D_x^{[n+1]}g$$

so that

$$n+1 \in S$$

2. We prove this by induction on k , so define

$$S = \left\{ k \in \mathbb{N} \mid \text{If } \{f_i \mid A \rightarrow Y\}_{i \in \{1, \dots, k\}} \text{ is a family of } n\text{-times differentiable functions at } x \in A \right.$$

$$\left. \text{then } \sum_{i=1}^k f_i \text{ is } n\text{-times differentiable at } x \text{ with } D_x^{[n]} \left(\sum_{i=1}^k f_i \right) = \sum_{i=1}^k D_x^{[n]} f_i \right\}$$

then we have:

1 ∈ S. Let $\{f_i|A \rightarrow Y\}_{i \in \{1\}}$ be a family of functions that are n -times differentiable at x . As $\sum_{i=1}^1 f_i = f_1$, it follows that $\sum_{i=1}^1 f_i$ is n -times differentiable at x , further

$$D_x^{[n]}\left(\sum_{i=1}^1 f_i\right) = D_x^{[n]}f_1 = \sum_{i=1}^1 D_x^{[n]}f_i$$

So it follows that $1 \in S$.

$k \in S \Rightarrow k+1 \in S$. Let $\{f_i: A \rightarrow Y\}_{i \in \{1, \dots, k+1\}}$ be a family of n -times differentiable functions at x then we have as $k \in S$ that $\sum_{i=1}^k f_i$ is n -times differentiable at x with $D_x^{[n]}(\sum_{i=1}^k f_i) = \sum_{i=1}^k D_x^{[n]}f_i$. As also f_{k+1} is n -times differentiable at x it follows from (1) that $\sum_{i=1}^{k+1} f_i = (\sum_{i=1}^k f_i) + f_{k+1}$ is n -times differentiable at x with

$$\begin{aligned} D_x^{[n]}\left(\sum_{i=1}^{k+1} f_i\right) &= D_x^{[n]}\left(\sum_{i=1}^k f_i + f_{k+1}\right) \\ &\stackrel{(1)}{=} D_x^{[n]}\left(\sum_{i=1}^k f_i\right) + D_x^{[n]}f_{k+1} \\ &\stackrel{n \in S}{=} \sum_{i=1}^k D_x^{[n]}f_i + D_x^{[n]}f_{k+1} \\ &= \sum_{i=1}^{k+1} D_x^{[n]}f_i \end{aligned}$$

proving that

$$n+1 \in S$$

3. We use induction on n , so let

$S = \{n \in \mathbb{N} | \text{If } f: A \rightarrow Y \text{ is } n\text{-times differentiable at } x \in A \text{ then } \alpha \cdot f \text{ is } n\text{-times differentiable at } x \text{ and } D_x^{[n]}(\alpha \cdot f) = \alpha \cdot D_x^{[n]}f\}$

then we have:

1 ∈ S. 1 ∈ S. If $f: A \rightarrow Y$ is 1-times differentiable at x then by [theorem: 16.140] f is Fréchet differentiable at x and $D_x^{[1]}f = D_x f$. Using [theorem: 16.66] it follows that $\alpha \cdot f$ is Fréchet differentiable at x and $D_x(\alpha \cdot f) = \alpha \cdot D_x f$. Hence using [theorem: 16.140] again $\alpha \cdot f$ is 1-times differentiable and

$$D_x^{[1]}(\alpha \cdot f) = D_x(\alpha \cdot f) = \alpha \cdot D_x f = \alpha \cdot D_x^{[1]}f$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $y \in \mathfrak{D}_f^n$ then by [definition: 16.136] f is n -times differentiable at x . As $n \in S$ it follows that

$$\alpha \cdot f \text{ is } n\text{-times differentiable at } y \text{ with } D_y^{[n]}(\alpha \cdot f) = \alpha \cdot D_y^{[n]}f$$

Hence by [definition: 16.136]

$$y \in \mathfrak{D}_{\alpha \cdot f}^n \text{ and } D^{[n]}(\alpha \cdot f)(y) = \alpha \cdot D^{[n]}f(y) = (\alpha \cdot D_f^{[n]})(y)$$

As this is true for all $y \in \mathfrak{D}_f^n$ we have that

$$\mathfrak{D}_f^n \subseteq \mathfrak{D}_{\alpha \cdot f}^n \text{ and } \alpha \cdot D^{[n]}f = (D^{[n]}(\alpha \cdot f))|_{\mathfrak{D}_f^n} \quad (16.144)$$

As f is $(n+1)$ -times differentiable at x it follows from [theorem: 16.140] that $D^{[n]}f$ is Fréchet differentiable at x with $D_x^{[n+1]}f = D_x(D^{[n]}f)$. Using [theorem: 16.66] it follows that

$$\alpha \cdot D^{[n]}f \text{ is Fréchet differentiable at } x$$

with

$$D_x(\alpha \cdot D^{[n]}f) = \alpha \cdot D_x(D^{[n]}f) = \alpha \cdot D_x^{[n+1]}f$$

Using [eq: 16.144] on the above it follows that

$$(D^{[n]}(\alpha \cdot f))|_{\mathfrak{D}_f^n} \text{ is Fréchet differentiable at } x$$

with

$$D_x((D^{[n]}(\alpha \cdot f))|_{\mathfrak{D}_f^n}) = \alpha \cdot D_x^{[n+1]}f$$

Using the locality of Fréchet differentiability [see theorem: 16.57] it follows that

$$D^{[n]}(\alpha \cdot f) \text{ is Fréchet differentiable at } x$$

with

$$D_x(D^{[n]}(\alpha \cdot f)) = D_x((D^{[n]}(\alpha \cdot f))|_{\mathfrak{D}_f^n}) = \alpha \cdot D_x^{[n+1]}f$$

Hence using [theorem: 16.140] it follows that

$$\alpha \cdot f \text{ is } (n+1)\text{-times differentiable at } x \text{ with } D_x^{[n+1]}(\alpha \cdot f) = \alpha \cdot D_x^{[n+1]}f$$

which proves that

$$n+1 \in S$$

4. If $f: A \rightarrow Y$ and $g: A \rightarrow Y$ are of class C^n on U then we have two cases to consider for n :

$n = 0$. Then f, g are continuous so that by [theorem: 14.165] $f + g$ is continuous. Hence $f + g$ is C^0 at x .

$1 \leq n$. As f and g are of class C^n on U we have by [definition: 16.149] that f and g are n -times differentiable on U [or equivalently $U \subseteq \mathfrak{D}_f^n \wedge U \subseteq \mathfrak{D}_g^n$] and that $(D^{[n]}f)|_U$ and $(D^{[n]}g)|_U$ are continuous. Hence $\forall x \in U$ we have that f, g are differentiable at x , so that by (1) $f + g$ is n -times differentiable at x with $D_x^{[n]}(f + g) = D_x^{[n]}f + D_x^{[n]}g$. In other words using [definition: 16.141]

$$f + g \text{ is } n\text{-times differentiable on } U \text{ or equivalent } U \subseteq \mathfrak{D}_{f+g}^n$$

and using [definition: 16.136]

$$\forall x \in U \text{ we have } D^{[n]}(f + g)(x) = D^{[n]}f(x) + D^{[n]}g(x)$$

So that

$$(D^{[n]}(f + g))|_U = (D^{[n]}f)|_U + (D^{[n]}g)|_U$$

As $(D^{[n]}f)|_U$ and $(D^{[n]}g)|_U$ are continuous it follows from [theorem: 14.165] that

$$(D^{[n]}(f + g))|_U \text{ is continuous}$$

Hence we have that

$$f + g \text{ is of class } C^n \text{ on } U$$

5. We use induction to prove this. So let

$$S = \left\{ k \in \mathbb{N} \mid \{f_i: A \rightarrow Y\}_{i \in \{1, \dots, k\}} \text{ are function that are of class } C^n \text{ on } U \text{ then } \sum_{i=1}^k f_i \text{ is of class } C^n \text{ on } U \right\}$$

then we have:

$1 \in S$. As $\sum_{i=1}^k f_i = f_1$ and f_1 is of class C^n on U it follows that $\sum_{i=1}^1 f_i$ is of class C^n on U .

$k \in S \Rightarrow k+1 \in S$. Let $\{f_i: A \rightarrow Y\}_{i \in \{1, \dots, k\}}$ be a family of functions that are of class C^n on U . As $k \in S$ $\sum_{i=1}^k f_i$ is of class C^n on U and clearly f_{k+1} is of class C^n on U , we have by (4) that $f_{k+1} + \sum_{i=1}^k f_i$ is of class C^n on U . 'Finally as $\sum_{i=1}^{k+1} f_i = f_{k+1} + \sum_{i=1}^k f_i$ it follows that $\sum_{i=1}^{k+1} f_i$ is of class C^n on U which prove that $k+1 \in S$.

6. As f is of class C^n on U we have by [definition: 16.149] two cases to consider:

$n=0$. Then f is continuous so that by [theorem: 14.165] $\alpha \cdot f$ is continuous which proves that $\alpha \cdot f$ is of class C^0 on U .

$1 < n$. Then f is n -times differentiable on U [or equivalently $U \subseteq \mathfrak{D}_f^n$] and $(D^{[n]}f)|_U$ is continuous. Hence $\forall x \in U$ we have that f is n -times differentiable at x which by (3) proves that $\alpha \cdot f$ is n -times differentiable at x with $D_x^{[n]}(\alpha \cdot f) = \alpha \cdot D_x^{[n]}f$. In other words using [definition: 16.141]

$$\alpha \cdot f \text{ is } n\text{-times differentiable on } U \text{ or equivalently } U \subseteq \mathfrak{D}_{\alpha \cdot f}^n$$

and using [definition: 16.136]

$$\forall x \in U \text{ we have } D^{[n]}(\alpha \cdot f)(x) = \alpha \cdot D^{[n]}f(x)$$

so that

$$(D^{[n]}(\alpha \cdot f))|_U = \alpha \cdot (D^{[n]}f)|_U$$

As $(D^{[n]}f)|_U$ is continuous it follows from [theorem: 14.165] that $\alpha \cdot (D^{[n]}f)|_U$ is continuous which by the above proves that $(D^{[n]}(\alpha \cdot f))|_U$ is continuous. Hence we have that

$$\alpha \cdot f \text{ is of class } C^n \text{ on } U$$

□

16.2.5 Examples of ∞ -times differentiable functions

We show now some example of ∞ -times differentiable functions or equivalent functions that are of class C^∞ or equivalently ∞ -times differentiable [see theorem: 16.157].

Example 16.179. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $U \in \mathcal{T}_{\|\cdot\|_X}$ and $y \in Y$ then the constant function $C_y: U \rightarrow Y$ defined by $C_y(x) = y$ is ∞ -times differentiable on U [hence by [theorem: 16.157] C_y is of class C^∞ on U]. Furthermore $\forall n \in \mathbb{N}, \forall x \in U$ we have that $D_x^{[n]}C_y = 0_n$ where $0_n \in L_n(X; Y)$ is the neutral element in $L_n(X; Y)$.

Proof. We prove this by induction on n , so define

$$S = \{n \in \mathbb{N} | C_y: U \rightarrow Y \text{ is } n\text{-times differentiable on } U \text{ with } \forall x \in U D_x^{[n]}C_y = 0_n\}$$

then we have:

$1 \in S$. Using [example: 16.54] C_y is Fréchet differentiable for every $x \in U$ with $D_x C_y = 0_1$, hence by [theorem: 16.140] C_y is 1-times differentiable at x with $D_x^{[1]}C_y = 0_1$ which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ it follows that

$$\forall x \in U C_y \text{ is } n\text{-times differentiable at } x \text{ with } D_x^{[n]}C_y = 0_n \in L_n(X; Y)$$

Hence $U \subseteq \mathfrak{D}_{C_y}^n \underset{\text{[theorem: 16.135]}}{\subseteq} U$ so that $U = \mathfrak{D}_{C_y}^n$ and

$$\forall x \in U \text{ we have } D^{[n]}C_y(x) \underset{\text{[definition: 16.136]}}{=} D_x^{[n]}C_y = 0_n$$

which proves that

$$D^{[n]}f = C_{0_n} \tag{16.145}$$

Take $x \in U$. Let $\varepsilon \in \mathbb{R}^+$ then for $h \in A_x$ with $|h| < 1$ we have

$$\begin{aligned} \|(D^{[n]}C_y)(x+h) - (D^{[n]}C_n)(x) - 0_{n+1}(h)\|_{L_n(X;Y)} &\stackrel{[\text{eq: 16.145}]}{=} \|0_n - 0_n - 0_{n+1}(h)\|_{L_n(X;Y)} \\ &= \|0_n - 0_n - 0_n\|_{L_n(X;Y)} \\ &= \|0_n\|_{L_n(X;Y)} \\ &= 0 \\ &\leq \varepsilon \cdot \|h\|_x \end{aligned}$$

proving that $D^{[n]}C_y$ is Fréchet differentiable at x with

$$D_x(D^{[n]}C_y) = 0_{n+1}$$

Hence by [theorem: 16.140] f is $(n+1)$ -times differentiable at x with $D_x^{[n+1]}C_y = 0_{n+1}$. So

$$f \text{ is } (n+1)\text{-times differentiable on } U \text{ with } \forall x \in U \ D_x(D^{[n]}C_y) = 0_{n+1}$$

$$n+1 \in S$$

□

Example 16.180. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $L \in L(X, Y)$ then we have:

1. L is ∞ -times differentiable on X [hence by [theorem: 16.157] L is of class C^∞]
2. $\forall n \in \mathbb{N}$ we have $\forall x \in X$ that

$$D_x^{[n]}L = \begin{cases} L & \text{if } n = 1 \\ 0_n & \text{if } 1 < n \end{cases}$$

Proof. For $n \in \mathbb{N}$ we have either:

$n = 1$. Then by [example: 16.55] we have that

$$\forall x \in X \ L \text{ is Fréchet differentiable at } x \text{ with } D_x L = L$$

or using [theorem: 16.140]

$$\forall x \in X \ L \text{ is 1-times differentiable at } x \text{ with } D_x^{[1]}L = L$$

Hence $X \subseteq \mathfrak{D}_L^1 \subseteq X$ so that $X = \mathfrak{D}_L^1$ and $\forall x \in X$ we have

$$\forall x \in X \text{ we have } D^{[1]}L(x) \stackrel{[\text{definition: 16.136}]}{=} D_x^{[1]}L = L = C_L(x)$$

which proves that

$$D^{[1]}L = C_L \tag{16.146}$$

$1 < n$. As $D^{[1]}L \stackrel{[\text{eq: 16.146}]}{=} C_L \in L_1(X; Y)$ it follows from [example: 16.179] that

$$D^{[1]}L \text{ is } (n-1)\text{-times differentiable on } X \text{ with } \forall x \in X \ D_x^{[n-1]}(D^{[1]}L) = 0_n$$

Using then [theorem: 16.174] it follows that

$$L \text{ is } n\text{-times differentiable on } X \text{ with } \forall x \in X \ D_x^{[n]}L = 0_n$$

□

Example 16.181. Let $\langle X, \|\cdot\| \rangle$ be a normed space, U a open set in X then $\text{Id}_U: U \rightarrow X$ is ∞ -times differentiable on U [hence by [theorem: 16.157] Id_U is of class C^∞]

Proof. Let $n \in \mathbb{N}$. Using [theorem: 14.215] $\text{Id}_X \in L(X, X)$ so that by [example: 16.180] Id_X is n -times differentiable on X . Let $x \in U$ then by [theorem: 16.145] we have that $(\text{Id}_X)|_U$ is n -times differentiable at x . As clearly $(\text{Id}_X)|_U = \text{Id}_U$ it follows that Id_U is n -times differentiable at x . Summarized $\forall n \in \mathbb{N}$ we have that Id_U is n -times differentiable on U . □

Example 16.182. Let $\langle \mathbb{K}, \|\cdot\| \rangle$ be the normed space of real [or complex] numbers, $\langle X, \|\cdot\| \rangle$ a normed space over $\mathbb{K}$, U a open set in $\mathbb{K}$, $x, y \in X$ then for

$$\varphi: U \rightarrow X \text{ where } \varphi(t) = t \cdot x + y$$

we have that φ is ∞ -times differentiable on U [hence by [theorem: 16.157] φ is of class C^∞] and $\forall t \in U$ we have $D_t^{[1]}\varphi = D_t\varphi = L$ where $L \in L(\mathbb{K}, X)$ is defined by $L(r) = r \cdot x$. So by [theorem: 16.171] we have that φ has a derivative for every $t \in A$ with $\varphi'_t = D_t^{[1]}(1) = L(1) = x$.

Proof. Define $L: \mathbb{K} \rightarrow X$ by $L(r) = r \cdot x$ then by [example: 14.216] it follows that $L \in L(\mathbb{K}, X)$. Let $t \in U$ then given $\varepsilon \in \mathbb{R}^+$ we have for $h \in U_x$ with $|h| < 1$ that

$$\|\varphi(t+h) - \varphi(t) - L(h)\| = \|(t+h) \cdot x + y - (t \cdot x + y) - h \cdot x\| = \|h \cdot x - h \cdot x\| = \|0\| \leq \varepsilon \cdot |h|$$

which proves that

$$\forall t \in U \quad \varphi \text{ is Fréchet differentiable at } t \text{ with } D_t\varphi = L$$

Using [theorem: 16.140] it follows that

$$\forall t \in U \text{ is 1-times differentiable at } t \text{ with } D_t^{[1]}\varphi = L$$

Hence we have that $U \subseteq \mathfrak{D}_\varphi^1 \subseteq U \Rightarrow U = \mathfrak{D}_U^1$ so that

$$\varphi \text{ is 1-times differentiable on } U \text{ and } D^{[1]}\varphi = C_L$$

Let $n \in \{2, \dots, \infty\}$ then by [example: 16.179] it follows that $D^{[1]}\varphi$ is $(n-1)$ -times differentiable on U . Hence using [theorem: 16.174] it follows that

$$\varphi \text{ is } n\text{-times differentiable on } U \quad \square$$

Example 16.183. Let $k \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, k\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\| \rangle$ a normed space, $i \in \{1, \dots, k\}$, $x \in \prod_{j \in \{1, \dots, k\}} X_j$ then

$$I^{[i,x]}: X_i \rightarrow \prod_{j \in \{1, \dots, k\}} X_j \text{ defined by } I^{[i,x]}(t) = \left(x_1, \dots, \underbrace{t}_i, \dots, x_n \right) \text{ [see definition: 16.70]}$$

is ∞ -times differentiable on X_i [hence by [theorem: 16.157] C^∞ on X_i]. Further $\forall n \in \mathbb{N}$

$$D_x^{[n]} I^{[i,x]}(x) = \begin{cases} I^{[i,0]} & \text{if } n=1 \\ 0_n & \text{if } 1 < n \end{cases}$$

Proof. Let $x \in \prod_{j \in \{1, \dots, k\}} X_j$ and $i \in \{1, \dots, k\}$. If $t \in X_i$ then

$$\begin{aligned} \left(I^{[i,0]} + C_{\left(x_1, \dots, \underbrace{0}_i, \dots, x_n \right)} \right)(t) &= I^{[i,0]}(t) + C_{(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_n)}(t) \\ &= \left(0, \dots, \underbrace{t}_i, \dots, 0_i \right) + (x_1, \dots, \underbrace{0}_i, \dots, x_n) \\ &= \left(x_1, \dots, \underbrace{t}_i, \dots, x_n \right) \\ &= I^{[i,x]}(t) \end{aligned}$$

so that

$$I^{[i,x]} = I^{[i,0]} + C_{\left(x_1, \dots, \underbrace{0}_i, \dots, x_n \right)}$$

By [lemma: 16.71] $I^{[i,0]} \in L(X_i, \prod_{j \in \{1, \dots, k\}} X_j)$ so that by [examples: 16.180 and 16.179] $I^{[i,0]}$ and $C_{\left(x_1, \dots, \underbrace{0}_i, \dots, x_n \right)}$ are ∞ -times differentiable at $t \in X_i$. Hence using [theorem: 16.178] it follows that

$$I^{[i,x]} \text{ is } \infty\text{-times differentiable at } t$$

and $\forall n \in \mathbb{N}$ we have

$$D_t^{[n]} I^{[i,x]} = D_t^{[n]} I^{[i,0]} + D_t^{[n]} C_{\left(x_1, \dots, \underbrace{0}_i, \dots, x_n \right)} \stackrel{\text{[example: 16.179]}}{=} D_t^{[n]} I^{[i,0]} \quad (16.147)$$

If $n \in \mathbb{N}$ then we have either:

$n = 1$. Then by [example: 16.180] and $I^{[i,0]} \in L(X_i, \prod_{j \in \{1, \dots, k\}} X_j)$ $D^{[1]}I^{[i,0]}(t) = I^{[i,0]}$

$1 < n$. Then by [example: 16.180] and $I^{[i,0]} \in L(X_i, \prod_{j \in \{1, \dots, k\}} X_j)$ $D^{[1]}I^{[i,0]}(t) = 0_n$ $\square$

Example 16.184. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $x \in X$ then for the translation function τ_x defined by

$$\tau_x: X \rightarrow X \text{ where } \tau_x(y) = x + y$$

we have that:

1. τ_x is ∞ -times differentiable on X [hence by [theorem: 16.157] C^∞ on X]

2. $\forall n \in \mathbb{N}, \forall y \in X$ we have

$$D_y^{[n]}\tau_x = \begin{cases} \text{Id}_X & \text{if } n = 1 \\ 0_n & \text{if } 1 < n \end{cases}$$

Proof. Let $x \in X$ then $\forall y \in X$ we have $(\text{Id}_X + C_x)(y) = \text{Id}_X(y) + C_x(y) = y + x = \tau_x(y)$ proving that

$$\tau_x = \text{Id}_X + C_x$$

As $\text{Id}_X \in L(X, X)$ [see example: 14.199] we have by [example: 16.180] that Id_X is ∞ -times differentiable, further by [example: 16.179] we have that C_x is ∞ -times differentiable. Using [theorem: 16.178] it follows that τ_x is ∞ -times differentiable and

$$D_y^{[n]}\tau_x = D_y^{[n]}\text{Id}_X + D_y^{[n]}C_x \underset{\text{[example: 16.179]}}{=} D_y^{[n]}\text{Id}_X + 0_n = D_y^{[n]}\text{Id}_X$$

Hence using [example: 16.180] it follows that

$$D_y^{[n]}\tau_x = \begin{cases} \text{Id}_X & \text{if } n = 1 \\ 0_n & \text{if } 1 < n \end{cases} \quad \square$$

Example 16.185. Let $k \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, k\}}$ be a finite family of normed spaces, $i \in \{1, \dots, k\}$ then the projection map $\pi_i: \prod_{j \in \{1, \dots, k\}} X_j \rightarrow X_i$ is ∞ -times differentiable on $\prod_{j \in \{1, \dots, k\}} X_j$ [hence by [theorem: 16.157] of class C^∞ and $\forall n \in \mathbb{N}$ we have

$$D_t^{[n]}\pi_i = \begin{cases} \pi_i & \text{if } n = 1 \\ 0_n & \text{if } 1 < n \end{cases}$$

Proof. As by [theorem: 14.217] $\pi_i \in L(\prod_{j \in \{1, \dots, k\}} X_j, X_i)$ the proof follows from [example: 16.180]. $\square$

Theorem 16.186. Let $\langle X_1, \|\cdot\|_1 \rangle$, $\langle X_2, \|\cdot\|_2 \rangle$ and $\langle Y, \|\cdot\| \rangle$ be normed spaces, $\langle X_1 \cdot X_2, \|\cdot\|_{\max} \rangle$ the normed space based on $X_1 \cdot X_2$ using the maximum norm and $L \in L(X_1, X_2; Y)$ be a bi-linear mapping then we have

1. L is ∞ -times differentiable on $X_1 \cdot X_2$ [hence L is of class C^∞]

2. $\forall (x_1, x_2) = x \in X_1 \cdot X_2$ $fD_x^{[1]}L \underset{x=(x_1, x_2)}{=} D_{(x_1, x_2)}^{[1]}L = L(x_1, *) + L(*, x_2)$ where

$$L(x_1, *) + L(*, x_2): X_1 \cdot X_2 \rightarrow Y \text{ is defined by } (L(x_1, *) + L(*, x_2))(r, s) = L_1(x_1, s) + L(r, x_2)$$

3. $\forall (x_1, x_2) = x \in X_1 \cdot X_2$ $D_x^{[2]}L \underset{x=(x_1, x_2)}{=} D_{(x_1, x_2)}^{[2]}L = D^{[1]}L$ where

$$D^{[1]}L: X_1 \cdot X_2 \rightarrow L_2(X_1 \cdot X_2, L(X_1 \cdot X_2; Y)) \text{ is defined by } (D^{[1]}L)(x_1, x_2) = D_{(x_1, x_2)}^{[1]}L$$

4. If $n \in \{3, \dots, \infty\}$ then $\forall (x_1, x_2) = x \in X_1 \cdot X_2$ $D^{[n]}L(x_1, x_2) = 0_n$

Proof. $\forall (x_1, x_2) \in X_1 \cdot X_2$ we have by [theorem: 16.104] L is Fréchet differentiable at (x_1, x_2) with

$$D_{(x_1, x_2)}L = L(x_1, *) + L(*, x_2)$$

hence L is 1-times differentiable at (x_1, x_2) and $D_{(x_1, x_2)}^{[1]}L = L(x_1, *) + L(*, x_2)$. Hence by [definition: 16.129]

$$\mathfrak{D}_L^1 = X_1 \cdot X_2$$

and $D^{[1]}L$ is defined by

$$D^{[1]}L: X_1 \cdot X_2 \rightarrow L(X_1 \cdot X_2, Y) \text{ by } D^{[1]}L(x_1, x_2) = L(x_1, *) + L(*, x_2) \quad (16.148)$$

so that by [definition: 16.141]

$$L \text{ is 1-times differentiable on } X_1 \cdot X_2 \text{ and } \forall (x_1, x_2) \ D_{(x_1, x_2)}^{[1]}L = L(x_1, *) + L(*, x_2) \quad (16.149)$$

Next we prove that $D^{[1]}L$ is a linear continuous mapping. If $(x_1, x_2), (y_1, y_2) \in X_1 \cdot X_2$, $\alpha \in \mathbb{K}$ then for every $(r, s) \in X_1 \cdot X_2$ we have

$$\begin{aligned} D^{[1]}L((x_1, x_2) + \beta \cdot (y_1, y_2))(r, s) &= D^{[1]}L(x_1 + \beta \cdot y_1, x_2 + \beta \cdot y_2)(r, s) \\ &= (L(x_1 + \beta \cdot y_1, *) + L(*, x_2 + \beta \cdot y_2))(r, s) \\ &= L(x_1 + \beta \cdot y_1, s) + L(r, x_2 + \beta \cdot y_2) \\ &= L(x_1, s) + \beta \cdot L(y_1, s) + L(r, x_2) + \beta \cdot L(r, y_2) \\ &= (L(x_1, s) + L(r, x_2)) + \beta \cdot (L(y_1, s) + L(r, y_2)) \\ &= D^{[1]}L(x_1, x_2)(r, s) + \beta \cdot D^{[1]}L(y_1, y_2)(r, s) \\ &= (D^{[1]}L(x_1, x_2) + D^{[1]}L(y_1, y_2))(r, s) \end{aligned}$$

proving that

$$D^{[1]}L((x_1, x_2) + \beta \cdot (y_1, y_2)) = D^{[1]}L(x_1, x_2) + D^{[1]}L(y_1, y_2)$$

Hence we have that

$$D^{[1]}L \text{ is linear} \quad (16.150)$$

Further we have that $\forall (x_1, x_2), (r, s) \in X_1 \cdot X_2$

$$\begin{aligned} \|D^{[1]}L(x_1, x_2)(r, s)\| &= \\ \|L(x_1, s) + L(r, x_2)\| &\leq \\ \|L(x_1, s)\| + \|L(r, x_2)\| &\leq \quad [\text{theorem: 14.228}] \\ \|L\|_{L(X_1, X_2; Y)} \cdot \|x\|_1 \cdot \|s\|_2 + \|L\|_{L(X_1, X_2; Y)} \cdot \|r\|_1 \cdot \|x_2\|_2 &\leq \\ \|L\|_{L(X_1, X_2; Y)} \cdot (\|x\|_1 \cdot \|s\|_2 + \|r\|_1 \cdot \|x_2\|_2) &\leq \\ \|L\|_{L(X_1, X_2; Y)} \cdot (\|x\|_1 \cdot \max(\|r\|_1, \|s\|_2) + \|x_2\|_2 \cdot \max(\|r\|_1, \|s\|_2)) &= \\ \|L\|_{L(X_1, X_2; Y)} \cdot (\|x_1\|_1 + \|x_2\|_2) \cdot \max(\|r\|_1, \|s\|_2) &= \\ \|L\|_{L(X_1, X_2; Y)} \cdot (\|x_1\|_1 + \|x_2\|_2) \cdot \|(r, s)\|_{\max} &\leq \\ \|L\|_{L(X_1, X_2; Y)} \cdot (\max(\|x_1\|_1, \|x_2\|_2) + \max(\|x_1\|_1, \|x_2\|_2)) \cdot \|(r, s)\|_{\max} &= \\ \|L\|_{L(X_1, X_2; Y)} \cdot 2 \cdot \max(\|x_1\|_1, \|x_2\|_2) \cdot \|(r, s)\|_{\max} &= \\ (2 \cdot \|L\|_{L(X_1, X_2; Y)} \cdot \|(x_1, x_2)\|_{\max}) \cdot \|(r, s)\|_{\max} & \end{aligned}$$

proving, as $D^{[1]}L(x_1, x_2) \in L(X_1 \cdot X_2, Y)$, by [definition: 14.209] that

$$\|D^{[1]}L(x_1, x_2)\|_{L(X_1 \cdot X_2, Y)} \leq 2 \cdot \|L\|_{L(X_1, X_2; Y)} \cdot \|(x_1, x_2)\|_{\max}$$

so that by [theorem: 14.228] it follows that

$$D^{[1]}L \in L(X_1 \cdot X_2, L(X_1 \cdot X_2, Y)) = L_2(X_1 \cdot X_2; Y) \quad (16.151)$$

Let $n \in \{2, \dots, \infty\}$ then using [example: 16.180] it follows that

$$D^{[1]}L \text{ is } (n-1)\text{-times differentiable on } X_1 \cdot X_2$$

and

$$\forall (x_1, x_3) \text{ we have } D_{(x_1, x_2)}^{[n-1]}(D^{[1]}L) = \begin{cases} D^{[1]}L & \text{if } n-1=1 \Rightarrow n=2 \\ 0_n & \text{if } 1 < n-1 \Rightarrow n \in \{3, \dots, \infty\} \end{cases}$$

Using then [theorem: 16.174] on the above it follows that

$$L \text{ is } n\text{-times differentiable on } X_1 \cdot X_2$$

and

$$\forall (x_1, x_2) \in X_1 \cdot X_2 \text{ we have } D_{(x_1, x_2)}^{[n]} = \begin{cases} D^{[1]}L & \text{if } n = 2 \\ 0_n & \text{if } n \in \{3, \dots, \infty\} \end{cases} \quad \square$$

16.2.6 The chain rule for higher order differentials and derivatives

The purpose of this section is to prove that the composition of n -times differentiable functions is again n -times differentiable. As a preparation for this goal we need to prove and state some supporting lemma's.

Lemma 16.187. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $L \in L(Y, Z)$ then if we define*

$$(L \circ *) : L(X, Y) \rightarrow L(X, Z) \text{ by } (L \circ *)(T) = L \circ T$$

we have that

$$(L \circ *) \in L(L(X, Y), L(X, Z))$$

Proof. First note that if $L \in L(Y, Z)$ and $T \in L(X, Y)$ then by [theorem: 14.218] $L \circ T \in L(X, Z)$ so that $(*) \circ L$ is indeed a function between $L(X, Y)$ and $L(X, Z)$. Next we need to prove linearity. Let $T_1, T_2 \in L(X, Y)$, $\alpha \in \mathbb{K}$ then for $z \in X$ we have

$$\begin{aligned} ((L \circ *)(T_1 + \alpha \cdot T_2))(z) &= (L \circ (T_1 + \alpha \cdot T_2))(z) \\ &= L((T_1 + \alpha \cdot T_2)(z)) \\ &= L(T_1(z) + \alpha \cdot T_2(z)) \\ &= L(T_1(z)) + \alpha \cdot L(T_2(z)) \\ &= (L \circ T_1)(z) + \alpha \cdot (L \circ T_2)(z) \\ &= ((L \circ T_1) + \alpha \cdot (L \circ T_2))(z) \\ &= ((L \circ *)(T_1) + \alpha \cdot (L \circ *)(T_2))(z) \end{aligned}$$

proving that $(L \circ *)(T_1 + \alpha \cdot T_2) = (L \circ *)(T_1) + \alpha \cdot (L \circ *)(T_2)$ or

$$(L \circ *) \in \text{Hom}(L(X, Y), L(X, Z))$$

As for continuity. Let $T \in L(X, Y)$ then as $L \in L(Y, Z)$ we have by [theorem: 14.218] that

$$\|(L \circ *)(T)\|_{L(X, Z)} = \|L \circ T\|_{L(X, Z)} \leq \|L\|_{L(Y, Z)} \cdot \|T\|_{L(X, Y)}$$

which proves by [theorem: 14.201] that

$$(L \circ *) \in L(L(X, Y), L(X, Z)) \quad \square$$

We prove now a limited version of the chain rule for higher order differentiation.

Lemma 16.188. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ is a function that is n -times differentiable at x then $L \circ f$ is n -times differentiable at x .*

Proof. We use induction to prove this, so define

$S_X = \{n \in \mathbb{N} \mid \text{If } \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle \text{ are normed spaces, } f: A \rightarrow Y \text{ is } n\text{-times differentiable at } x \text{ and } L \in L(Y, Z) \text{ then } L \circ f \text{ is } n\text{-times differentiable at } x\}$

then we have:

$1 \in S_X$. If f is n -times differentiable at x then by [theorems: 16.138] f is 1-times differentiable at x , so that by [theorem: 16.140] f is Fréchet differentiable at x . Hence if $L \in L(Y, Z)$ we have by [corollary: 16.69] that $L \circ f$ is Fréchet differentiable at x , so that by [theorem: 16.140] $L \circ f$ is 1-times differentiable at x , which proves that $1 \in S_X$.

$n \in \mathcal{S}_X \Rightarrow n+1 \in \mathcal{S}_X$. Let $f: A \rightarrow Y$ be $(n+1)$ -times differentiable at x and let $L \in L(Y, Z)$. Using [lemma: 16.187] we have for the function

$$(L \circ *): L(X, Y) \rightarrow L(X, Z) \text{ defined by } (L \circ *)(T) = L \circ T \quad (16.152)$$

that

$$(L \circ *) \in L(L(X, Y), L(X, Z)) \quad (16.153)$$

Let $y \in \mathfrak{D}_f^1$ then by [definition: 16.129] f is Fréchet differentiable at y . Using [corollary: 16.69] $L \circ f$ is Fréchet differentiable at y , so that by [definition: 16.129] $y \in \mathfrak{D}_{L \circ f}^1$, and $D_y(L \circ f) = L \circ D_y f \stackrel{[\text{eq: 16.152}]}{=} (L \circ *)(D_y f)$. Hence we have

$$\mathfrak{D}_f^1 \subseteq \mathfrak{D}_{L \circ f}^1 \text{ and } \forall y \in D_y(L \circ f) = (L \circ *)(D_y f) \quad (16.154)$$

Let $y \in \mathfrak{D}_f^1$ then we have

$$\begin{aligned} D^{[1]}(L \circ f)(y) &\stackrel{[\text{definition: 16.136}]}{=} D_y^{[1]}(L \circ f) \\ &\stackrel{[\text{theorem: 16.140}]}{=} D_y(L \circ f) \\ &\stackrel{[\text{eq: 16.154}]}{=} (L \circ *)(D_y f) \\ &\stackrel{[\text{definition: 16.136}]}{=} (L \circ *)(Df(y)) \\ &= ((L \circ *) \circ Df)(y) \\ &\stackrel{[\text{definition: 16.129}]}{=} ((L \circ *) \circ D^{[1]} f)(y) \end{aligned}$$

so that, taking in account $\mathfrak{D}_f^1 \subseteq \mathfrak{D}_{L \circ f}^1$, we have

$$(D^{[1]}(L \circ f))|_{\mathfrak{D}_f^1} = ((L \circ *) \circ D^{[1]} f) \quad (16.155)$$

As f is $(n+1)$ -times differentiable at x it follows from [theorem: 16.174] that $D^{[1]}f$ is n -times differentiable at x . As $n \in \mathcal{S}_X$ and $(L \circ *) \in L(L(X, Y), L(X, Z))$ it follows that $((L \circ *) \circ D^{[1]} f)$ is n -times differentiable at x . Which by [eq: 16.155] proves that $D^{[1]}(L \circ f)|_{\mathfrak{D}_f^1}$ is n -times differentiable. Using now locality [see theorem: 16.145] it follows that $D^{[1]}(L \circ f)$ is n -times differentiable at x . Finally using [theorem: 16.174] again it follows that

$$L \circ f \text{ is } (n+1)\text{-times differentiable at } x$$

which proves that

$$n+1 \in \mathcal{S}_X \quad \square$$

Lemma 16.189. Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$, $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $L \in L(Y, Z)$, U a open set in X and $f: U \rightarrow Y$ is function that is of class C^n then $L \circ f$ is of class C^n .

Note 16.190. Remember that a function is of class C^n if its domain is open and the function is of class C^n on its domain.

Proof. For $n \in \mathbb{N}_0$ we have either:

$n = 0$. As f is C^0 we have that f is continuous, further $L \in L(Y, Z) \Rightarrow L$ is continuous so that by [theorem: 14.158] $L \circ f$ is continuous hence $L \circ f$ is of class C^0 .

$n \in \mathbb{N}$. We use induction to prove this, so define

$\mathcal{S}_{X,U} = \{n \in \mathbb{N} \mid \text{If } \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle \text{ are normed spaces, } f: U \rightarrow Y \text{ a function that is of class } C^n \text{ and } L \in L(Y, Z) \text{ then } L \circ f \text{ is of class } C^n\}$

then we have:

$1 \in \mathcal{S}_{X,U}$. As f is of class C^1 we have that f is 1-times differentiable on U and $D^{[1]}f$ is continuous. Hence $U \subseteq \mathfrak{D}_f^1$ which, as by [theorem: 16.135] $\mathfrak{D}_f^1 \subseteq U$, proves that

$$\mathfrak{D}_f^1 = U \quad (16.156)$$

Let $x \in U$ then f is 1-times differentiable at x , hence Fréchet differentiable at x [see theorem: 16.140]. Further using [corollary: 16.69] it follows that $L \circ f$ is Fréchet differentiable at x with $D_x(L \circ f) = L \circ D_x f$. Hence using [theorem: 16.140] again it follows that

$$L \circ f \text{ is 1-times differentiable at } x \text{ with } D_x^{[1]}(L \circ f) = L \circ D_x^{[1]} f \quad (16.157)$$

As this is true for every $x \in U$ we have that $U \subseteq \mathfrak{D}_{L \circ f}^1$ which, as by [theorem: 16.135] $\mathfrak{D}_{L \circ f}^1 \subseteq U$, proves that $U = \mathfrak{D}_{L \circ f}^1$. Summarized we have proved that

$$\mathfrak{D}_{L \circ f}^1 = U \stackrel{[\text{eq: 16.156}]}{=} \mathfrak{D}_f^1 \text{ so that } L \circ f \text{ is 1-times differentiable on } U \quad (16.158)$$

Define [see lemma: 16.187]

$$(L \circ *): L(X, Y) \rightarrow L(X, Z) \text{ where } (L \circ *)(T) = L \circ T \quad (16.159)$$

then by [lemma: 16.187]

$$(L \circ *) \in L(L(X, Y), L(X, Z)) \quad (16.160)$$

Now

$$\begin{aligned} D^{[1]}(L \circ f)(x) &\stackrel{[\text{definition: 16.136}]}{=} D_x^{[1]}(L \circ f) \\ &\stackrel{[\text{eq: 16.157}]}{=} L \circ D_x^{[1]} f \\ &\stackrel{[\text{eq: 16.159}]}{=} (L \circ *) (D_x^{[1]} f) \\ &\stackrel{[\text{definition: 16.136}]}{=} (L \circ *) (D^{[1]} f(x)) \\ &= ((L \circ *) \circ D^{[1]} f)(x) \end{aligned}$$

which, as this is true for every $x \in U = \mathfrak{D}_{L \circ f}^1 = \mathfrak{D}_f^1$ [see eq: 16.158], proves that

$$D^{[1]}(L \circ f) = (L \circ *) \circ D^{[1]} f$$

As $D^{[1]} f$ is continuous and $(L \circ *)$ is continuous [see eq: 16.160] it follows that $D^{[1]}(L \circ f)$ is continuous. Combining this with [eq: 16.158] proves that

$$L \circ f \text{ is of class } C^1$$

which proves that

$$1 \in S_{X, U}$$

$n \in S_{X, U} \Rightarrow n + 1 \in S_{X, U}$. Let $f: A \rightarrow Y$ be of class C^{n+1} and let $L \in L(Y, Z)$. Using [lemma: 16.187] we have for the function

$$(L \circ *): L(X, Y) \rightarrow L(X, Z) \text{ by } (L \circ *)(T) = L \circ T \quad (16.161)$$

that

$$(L \circ *) \in L(L(X, Y), L(X, Z)) \quad (16.162)$$

Let $y \in U$ then f is $(n + 1)$ -times differentiable at y , hence 1-times differentiable at y by [theorem: 16.138]. Using then [theorem: 16.140] f is Fréchet differentiable at y so that by [corollary: 16.69]

$$L \circ f \text{ is Fréchet differentiable at } y \text{ and } D_y(L \circ f) = L \circ D_y f \stackrel{[\text{eq: 16.161}]}{=} (L \circ *) (D_y f)$$

Using [theorem: 16.140] on the above produces

$$L \circ f \text{ is 1-times differentiable at } y \text{ and } D_y^{[1]}(L \circ f) = (L \circ *) (D_y^{[1]} f)$$

As this is true for every $y \in U$ it follows that $U \subseteq \mathfrak{D}_{L \circ f}^1$ which, as by [theorem: 16.135] $\mathfrak{D}_{L \circ f}^1 \subseteq U$, proves that

$$U = \mathfrak{D}_{L \circ f}^1 \text{ and } \forall y \in D_y^{[1]}(L \circ f) = (L \circ *) (D_y^{[1]} f) \quad (16.163)$$

Note also that, as $\forall y \in U$ f is 1-times differentiable at y , we have by [definition: 16.136] that $U \subseteq \mathfrak{D}_f^1$. Further by [theorem: 16.135] we have that $\mathfrak{D}_{L \circ f}^1 \subseteq U$ so that we must conclude that

$$U = \mathfrak{D}_f^1 \quad (16.164)$$

Let $y \in U$ then we have

$$\begin{aligned} D^{[1]}(L \circ f) &\stackrel{[\text{definition: 16.136}]}{=} D_y^{[1]}(L \circ f) \\ &\stackrel{[\text{eq: 16.163}]}{=} (L \circ *) (D_y^{[1]} f) \\ &\stackrel{[\text{definition: 16.136}]}{=} (L \circ *) (D^{[1]} f(y)) \\ &= ((L \circ *) \circ D^{[1]} f)(y) \end{aligned}$$

so that, taking in account $U = \mathfrak{D}_{L \circ f}^1$ and $U = \mathfrak{D}_f^1$, we have

$$D^{[1]}(L \circ f) = ((L \circ *) \circ D^{[1]} f) \quad (16.165)$$

As f is of class C^{n+1} it follows from [theorem: 16.174] that $D^{[1]}f$ is of class C^n . As $n \in S_{X,U}$ and $(L \circ *) \in L((L(X, Y), L(X, Z)))$ [see eq: 16.162] it follows that $(L \circ *) \circ D^{[1]}f$ is of class C^n on U . Which by [eq: 16.165] proves that

$$D^{[1]}(L \circ f) \text{ is of class } C^n \text{ on } U$$

or using [theorem: 16.174] that

$$L \circ f \text{ is of class } C^{n+1} \text{ on } U$$

which proves that

$$n+1 \in \mathcal{S}_{X,U} \quad \square$$

Lemma 16.191. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$ a normed space, $\{\langle Y_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces then if we define [see lemma: 16.99]

$$\mathcal{K}: \prod_{i \in \{1, \dots, n\}} L(X, Y_i) \rightarrow L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right) \text{ by } \mathcal{K}(L_1, \dots, L_n) = (L_1, \dots, L_n)_*$$

we have

$$\mathcal{K} \in L\left(\prod_{i \in \{1, \dots, n\}} L(X, Y_i), L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right)\right)$$

where $\langle \prod_{i \in \{1, \dots, n\}} L(X, Y_i), \|\cdot\|_{\max} \rangle$ is the normed space with the maximum norm $\|\cdot\|_{\max}$ defined to be $\|(L_1, \dots, L_n)\|_{\max} = \max(\{\|L_i\|_{L(X, Y_i)} | i \in \{1, \dots, n\}\})$

Proof. Let $\alpha \in \mathbb{K}$, $(L_1, \dots, L_n), (K_1, \dots, K_n) \in \prod_{i \in \{1, \dots, n\}} L(X, Y_i)$ then we have for $x \in X$ that

$$\begin{aligned} \mathcal{K}((L_1, \dots, L_n) + \alpha \cdot (K_1, \dots, K_n))(x) &= \mathcal{K}((L_1 + \alpha \cdot K_1, \dots, L_n + \alpha \cdot K_n))(x) \\ &= (L_1 + \alpha \cdot K_1, \dots, L_n + \alpha \cdot K_n)_*(x) \\ &= ((L_1 + \alpha \cdot K_1)(x), \dots, (L_n + \alpha \cdot K_n)(x)) \\ &= (L_1(x) + \alpha \cdot K_1(x), \dots, L_n(x) + \alpha \cdot K_n(x)) \\ &= (L_1(x), \dots, L_n(x)) + \alpha \cdot (K_1(x), \dots, K_n(x)) \\ &= (L_1, \dots, L_n)_*(x) + \alpha \cdot (K_1, \dots, K_n)_*(x) \\ &= \mathcal{K}(L_1, \dots, L_n)(x) + \alpha \cdot \mathcal{K}(K_1, \dots, K_n)(x) \\ &= (\mathcal{K}(L_1, \dots, L_n) + \alpha \cdot \mathcal{K}(K_1, \dots, K_n))(x) \end{aligned}$$

proving that $\mathcal{K}((L_1, \dots, L_n) + \alpha \cdot (K_1, \dots, K_n)) = \mathcal{K}(L_1, \dots, L_n) + \alpha \cdot \mathcal{K}(K_1, \dots, K_n)$. So we have

$$\mathcal{K} \in \text{Hom}\left(\prod_{i \in \{1, \dots, n\}} L(X, Y_i), L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right)\right)$$

Let $(L_1, \dots, L_n) \in \prod_{i \in \{1, \dots, n\}} L(X, Y_i)$ then we have that

$$\begin{aligned} \|\mathcal{K}((L_1, \dots, L_n))\|_{L(X, \prod_{i \in \{1, \dots, n\}} Y_i)} &= \|(L_1, \dots, L_n)_*\|_{L(X, \prod_{i \in \{1, \dots, n\}} Y_i)} \\ &\leq \max(\{\|L_i\|_{L(X, Y_i)} \mid i \in \{1, \dots, n\}\}) \\ &\stackrel{[\text{lemma: 16.99}]}{=} \|(L_1, \dots, L_n)\|_{\max} \end{aligned}$$

proving that

$$\mathcal{K} \in L\left(\prod_{i \in \{1, \dots, n\}} L(X, Y_i), L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right)\right)$$

□

Theorem 16.192. Let $n \in \mathbb{N}$, $k \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$ a normed space, $\{\langle Y_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, k\}}$ a finite family of normed spaces, $A \subseteq X$, $x \in A$ and $f: A \rightarrow \prod_{i \in \{1, \dots, k\}} Y_i$ then we have

f is n -times differentiable at x

$\Updownarrow$

$\forall i \in \{1, \dots, k\}$ we have $\pi_i \circ f$ is n -times differentiable at x

Proof.

$\Rightarrow$. Using [example: 14.217] it follows that $\forall i \in \{1, \dots, k\}$ $\pi_i \in L(\prod_{j \in \{1, \dots, k\}} Y_j, Y_i)$. So, as f is n -times differentiable at x , it follows from [lemma: 16.188] that $\pi_i \circ f$ is n -times differentiable at x .

$\Leftarrow$. We use induction to prove this, so let

$$S_X = \left\{ n \in \mathbb{N} \mid \text{Let } \{\langle Y_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, k\}} \text{ be normed spaces, } x \in A \text{ and } f: A \rightarrow \prod_{j \in \{1, \dots, k\}} Y_j \right.$$

a function such that $\forall i \in \{1, \dots, k\}$ $\pi_i \circ f$ is n -times differentiable at x then f is n -times differentiable at x }

then we have:

$1 \in S_X$. As $\forall i \in \{1, \dots, k\}$ $\pi_i \circ f$ is 1-times differentiable at x it follows that $\pi_i \circ f$ is Fréchet differentiable at x . Hence using [theorem: 16.101] f is Fréchet differentiable at x proving by [theorem: 16.140] that f is 1-times differentiable at x . So we conclude that $1 \in S$.

$n \in S_X \Rightarrow n+1 \in S_X$. First we have

$$\begin{aligned} y \in \mathfrak{D}_f^1 &\stackrel{[\text{definition: 16.129}]}{\Leftrightarrow} f \text{ is Fréchet differentiable at } y \\ &\stackrel{[\text{theorem: 16.101}]}{\Leftrightarrow} \forall i \in \{1, \dots, k\} \pi_i \circ f \text{ is Fréchet differentiable at } y \\ &\stackrel{[\text{definition: 16.129}]}{\Leftrightarrow} \forall i \in \{1, \dots, k\} y \in \mathfrak{D}_{\pi_i \circ f}^1 \\ &\Leftrightarrow y \in \bigcap_{i \in \{1, \dots, k\}} \mathfrak{D}_{\pi_i \circ f}^1 \end{aligned}$$

which proves that

$$\mathfrak{D}_f^1 = \bigcap_{i \in \{1, \dots, k\}} \mathfrak{D}_{\pi_i \circ f}^1 \text{ and } \forall i \in \{1, \dots, k\} \mathfrak{D}_f^1 \subseteq \mathfrak{D}_{\pi_i \circ f}^1 \quad (16.166)$$

Let $y \in \mathfrak{D}_f^1 \stackrel{[\text{eq: 16.166}]}{=} \bigcap_{i \in \{1, \dots, k\}} \mathfrak{D}_{\pi_i \circ f}^1$ then $\forall i \in \{1, \dots, k\}$ $\pi_i \circ f$ is 1-times differentiable at y , so that by [definition: 16.129], $\pi_i \circ f$ is Fréchet differentiable at y . Using [theorem: 16.101] it follows that

$$D_y f = (D_y(\pi_1 \circ f), \dots, D_y(\pi_k \circ f))_*$$

Using [theorem: 16.140] on the above we have that

$$\forall y \in \mathfrak{D}_f^1 \text{ we have } D_y^{[1]}f = (D_y^{[1]}(\pi_1 \circ f), \dots, D_y^{[1]}(\pi_k \circ f))_* \quad (16.167)$$

Using [lemma: 16.191] we have for

$$\mathcal{K}: \prod_{i \in \{1, \dots, n\}} L(X, Y_i) \rightarrow L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right) \text{ by } \mathcal{K}(L_1, \dots, L_n) = (L_1, \dots, L_n)_*$$

that

$$\mathcal{K} \in L\left(\prod_{i \in \{1, \dots, n\}} L(X, Y_i), L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right)\right) \quad (16.168)$$

Define

$$g: \mathfrak{D}_f^1 \rightarrow \prod_{i \in \{1, \dots, k\}} L(X, Y_i) \text{ by } g(y) = (D^{[1]}(\pi_1 \circ f)(y), \dots, D^{[1]}(\pi_k \circ f)(y)) \quad (16.169)$$

them we have $\forall y \in \mathfrak{D}_f^1$ that

$$\begin{aligned} (\mathcal{K} \circ g)(y) &= \\ \mathcal{K}(g(y)) &\stackrel{[\text{eq: 16.169}]}{=} \\ \mathcal{K}(D^{[1]}(\pi_1 \circ f)(y), \dots, D^{[1]}(\pi_k \circ f)(y)) &\stackrel{[\text{definition: 16.136}]}{=} \\ \mathcal{K}(D_y^{[1]}(\pi_1 \circ f), \dots, D_y^{[1]}(\pi_k \circ f)) &= \\ (D_y^{[1]}(\pi_1 \circ f), \dots, D_y^{[1]}(\pi_k \circ f))_* &\stackrel{[\text{eq: 16.167}]}{=} \\ D_y^{[1]}f &\stackrel{[\text{definition: 16.136}]}{=} \\ D^{[1]}f(y) \end{aligned}$$

which proves that

$$D^{[1]}f = \mathcal{K} \circ g \quad (16.170)$$

Let $i \in \{1, \dots, k\}$ then as $\pi_i \circ f$ is $(n+1)$ -times differentiable at x and $2 \in \{1, \dots, n+1\}$ it follows from [theorem: 16.169] that $\pi_i \circ f$ is 2-times differentiable at x . Hence using [theorem: 16.140] $D^1(\pi_i \circ f)$ is Fréchet differentiable at x which by [definition: 16.46] proves that $x \in (\mathfrak{D}_{\pi_i \circ f}^1)^\circ$. Hence $x \in \bigcap_{i \in \{1, \dots, k\}} (\mathfrak{D}_{\pi_i \circ f}^1)^\circ \subseteq \bigcap_{i \in \{1, \dots, k\}} \mathfrak{D}_{\pi_i \circ f}^1 = \mathfrak{D}_f^1$. As $\bigcap_{i \in \{1, \dots, k\}} (\mathfrak{D}_{\pi_i \circ f}^1)^\circ$ is a open set [because it is a **finite** union of open sets] it follows that

$$x \in (\mathfrak{D}^1 f)^\circ \quad (16.171)$$

Let $i \in \{1, \dots, k\}$ then as $\pi_i \circ f$ is $(n+1)$ -times differentiable at x it follows from [theorem: 16.173] that $D^{[1]}(\pi_i \circ f)$ is n -times differentiable at x . Using now the fact that $x \in (\mathfrak{D}^1 f)^\circ \subseteq \mathfrak{D}_{\pi_i \circ f}^1$ [see eqs: 16.166, 16.171] it follows from [theorem: 16.145] that

$$D^{[1]}(\pi_i \circ f)|_{\mathfrak{D}_f^1} \text{ is } n\text{-times differentiable at } x \quad (16.172)$$

Let $i \in \{1, \dots, k\}$ then we have for $y \in \mathfrak{D}_f^1$ that

$$(\pi_i \circ g)(y) \stackrel{[\text{eq: 16.169}]}{=} \pi_i \circ ((D^{[1]}(\pi_1 \circ f)(y), \dots, D^{[1]}(\pi_k \circ f)(y))) = D^{[1]}(\pi_i \circ f)(y)$$

which, as $\mathfrak{D}_f^1 \stackrel{[\text{eq: 16.166}]}{\subseteq} \mathfrak{D}_{\pi_i \circ f}^1$, proves that

$$\pi_i \circ g = (D^{[1]}(\pi_i \circ f))|_{\mathfrak{D}_f^1}$$

Hence using [eq: 16.172] we have

$$\forall i \in \{1, \dots, k\} \text{ } \pi_i \circ g \text{ is } n\text{-times differentiable at } x$$

As $n \in S_X$ it follows then from the above that

$$g \text{ is } n\text{-times differentiable at } x$$

Using now [lemma: 16.189] together with [eq: 16.168] on the above it follows that

$$\mathcal{K} \circ g \text{ is } n\text{-times differentiable at } x$$

Combining this with [eq: 16.170] proves that $D^{[1]}f$ is n -times differentiable at x . Finally using [theorem: 16.174]

$$f \text{ is } (n+1)\text{-times differentiable at } x$$

which proves that

$$n+1 \in S_X$$

□

Theorem 16.193. *Let $n \in \mathbb{N}_0$, $k \in \mathbb{N}$, $\langle X, \|\cdot\| \rangle$ a normed space, $\{\langle Y_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, k\}}$ a finite family of normed spaces, $U \in \mathcal{T}_{\|\cdot\|_X}$ and $f: U \rightarrow \prod_{i \in \{1, \dots, n\}} Y_i$ a function then we have*

$$f \text{ is of class } C^n$$

$$\Updownarrow$$

$$\forall i \in \{1, \dots, k\} \text{ we have } \pi_i \circ f \text{ is of class } C^n$$

Note 16.194. *Remember that a function is of class C^n if its domain is open and the function is of class C^n on its domain.*

Proof.

$\Rightarrow$. Using [example: 14.217] it follows that $\forall i \in \{1, \dots, k\} \pi_i \in L(\prod_{j \in \{1, \dots, n\}} Y_j, Y_i)$. So, as f is of class C^n , it follows from [lemma: 16.189] that $\pi_1 \circ f$ is of class C^n .

$\Leftarrow$. For $n \in \mathbb{N}_0$ we have either:

$n = 0$. Then $\forall i \in \{1, \dots, k\} \pi_i \circ f$ is of class C^0 so that $\pi_i \circ f$ is continuous. Using [theorem: 14.161] it follows that f is continuous hence by definition of class C^0 .

$n \in \mathbb{N}$. We prove this by induction, so let

$$S_X = \left\{ n \in \mathbb{N} \mid \text{Let } \{\langle Y_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, k\}} \text{ be a family of normed spaces, } f: U \rightarrow \prod_{i \in \{1, \dots, k\}} Y_i \right. \\ \left. \text{a function such that } \forall i \in \{1, \dots, k\} \pi_i \circ f \text{ is of class } C^n \text{ then } f \text{ is of class } C^n \right\}$$

then we have:

$1 \in S_X$. As $\forall i \in \{1, \dots, k\} \pi_i \circ f$ is of class C^1 $\pi_i \circ f$ is 1-times differentiable on U and $D^{[1]}(\pi_i \circ f)$ is continuous. Using [definition: 16.141] and [theorem: 16.135] it follows that

$$\forall i \in \{1, \dots, k\} \text{ we have } U = \mathfrak{D}_{\pi_i \circ f}^1$$

Let $x \in U$ then $\forall i \in \{1, \dots, k\} \pi_i \circ f$ is 1-times differentiable at x hence $\pi_i \circ f$ is Fréchet differentiable at x . Using [theorem 16.101] it follows then that

$$f \text{ is Fréchet differentiable at } x \text{ with } D_x f = (D_x(\pi_1 \circ f), \dots, D_x(\pi_k \circ f))_*$$

Using [theorem: 16.140] on the above we have that f is 1-times differentiable at x with $D_x^{[1]}f = (D_x^{[1]}(\pi_1 \circ f), \dots, D_x^{[1]}(\pi_k \circ f))_*$. As this is true $\forall x \in U$ it follows that $U \subseteq \mathfrak{D}_f^1 \underset{[\text{theorem: 16.135}]}{\subseteq} U$, hence we have that

$$U = \mathfrak{D}_f^1 \text{ and } \forall x \in U \ D_x^{[1]}f = (D_x^{[1]}(\pi_1 \circ f), \dots, D_x^{[1]}(\pi_k \circ f))_* \quad (16.173)$$

From $U = \mathfrak{D}_f^1$ it follows by [definition: 16.141]

$$f \text{ is 1-times differentiable on } U$$

So the only thing left to prove is the continuity of $D^{[1]}f$. Using [lemma: 16.191] we have for

$$\mathcal{K}: \prod_{i \in \{1, \dots, n\}} L(X, Y_i) \rightarrow L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right) \text{ by } \mathcal{K}(L_1, \dots, L_n) = (L_1, \dots, L_n)_*$$

that

$$\mathcal{K} \in L\left(\prod_{i \in \{1, \dots, n\}} L(X, Y_i), L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right)\right) \quad (16.174)$$

Define

$$g: U \rightarrow \prod_{i \in \{1, \dots, k\}} L(X, Y_i) \quad g(x) = (D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x)) \quad (16.175)$$

then we have $\forall x \in U$ that

$$\begin{aligned} (\mathcal{K} \circ g)(x) &= \mathcal{K}(g(x)) \\ &\stackrel{[\text{eq: 16.175}]}{=} \mathcal{K}(D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x)) \\ &= (D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x))_* \\ &\stackrel{[\text{definition: 16.136}]}{=} (D_x^{[1]}(\pi_1 \circ f), \dots, D_x^{[1]}(\pi_k \circ f))_* \\ &\stackrel{[\text{eq: 16.173}]}{=} D_x^{[1]}f \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}f(x) \end{aligned}$$

which, taking in account that $U \stackrel{[\text{eq: 16.173}]}{=} \mathfrak{D}_f^1$, proves that

$$D^{[1]}f = \mathcal{K} \circ g \quad (16.176)$$

Let $i \in \{1, \dots, k\}$ then we have $\forall x \in U$ that

$$\begin{aligned} (\pi_i \circ g)(x) &= \pi_i(g(x)) \\ &= \pi_i(D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x)) \\ &= D^{[1]}(\pi_i \circ f)(x) \end{aligned}$$

hence, taking in account that $U = \mathfrak{D}_{\pi_i \circ f}^1$, we have that

$$\pi_i \circ g = D^{[1]}(\pi_i \circ f)$$

which, as $D^{[1]}(\pi_i \circ f)$ is continuous [because $\pi_i \circ f$ is of class C^1], proves that

$$\pi_i \circ g \text{ is continuous } \forall i \in \{1, \dots, l\}$$

Using [theorem: 14.161] it follows that g is continuous. Hence, as $\mathcal{K}$ is continuous by [eq: 16.174] we have that $\mathcal{K} \circ g$ is continuous. Combining this with [eq: 16.176] proves that

$$D^{[1]}f \text{ is continuous}$$

from which it follows that

$$f \text{ is of class } C^1$$

or

$$1 \in \mathcal{S}_X$$

$n \in \mathcal{S}_X \Rightarrow n+1 \in \mathcal{S}_X$. Let $x \in U$ then $\forall i \in \{1, \dots, k\}$ we have, as $\pi_i \circ f$ is of class C^{n+1} that by [theorem: 16.158] $\pi_i \circ f$ is of class C^1 from which it follows that $\pi_1 \circ f$ is 1-times differentiable on U . Hence we have by [definition: 16.141] that $U \subseteq \mathfrak{D}_{\pi_i \circ f}^1$ and by [theorem: 16.135] $\mathfrak{D}_{\pi_i \circ f}^1 \subseteq U$ so that

$$\forall i \in \{1, \dots, k\} \text{ we have } U = \mathfrak{D}_{\pi_i \circ f}^1$$

So we can define $g: U \rightarrow \prod_{i \in \{1, \dots, k\}} L(X, Y_i)$ by

$$g(x) = (D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x)) \quad (16.177)$$

Hence $\forall i \in \{1, \dots, k\}$ we have that

$$\begin{aligned} (\pi_i \circ g)(x) &= \pi_i(g(x)) \\ &\stackrel{[\text{eq: 16.177}]}{=} \pi_i(D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x)) \\ &= D^{[1]}(\pi_i \circ f)(x) \end{aligned}$$

which proves that

$$\pi_i \circ g = D^{[1]}(\pi_i \circ f) \quad (16.178)$$

As $\pi_i \circ f$ is of class C^{n+1} on U it follows from [theorem: 16.175] that $D^{[1]}(\pi_i \circ f)$ is of class C^n which by [eq: 16.178] proves that $\pi_i \circ g$ is of class C^n . Summarized

$$\forall i \in \{1, \dots, n\} \text{ we have that } \pi_i \circ g \text{ is of class } C^n$$

As $n \in S_X$ it follows from the above that

$$g \text{ is of class } C^n \quad (16.179)$$

Let $x \in U$ then, as $\forall i \in \{1, \dots, k\}$ $\pi_i \circ f$ is of class C^n , it follows from [theorem: 16.158] that $\pi_i \circ f$ is of class C^1 . Hence we have that $\pi_i \circ f$ is 1-times differentiable at x so that by [theorem: 16.140] $\pi_i \circ f$ is Fréchet differentiable at x . Using [theorem: 16.101] it follows then that

$$f \text{ is Fréchet differentiable at } x \text{ and } D_x f = (D_x(\pi_1 \circ f), \dots, D_x(\pi_2 \circ f))_*$$

Using [theorem: 16.140] it follows that

$$f \text{ is 1-times differentiable at } x \text{ and } D_x^{[1]} f = (D_x^{[1]}(\pi_1 \circ f), \dots, D_x^{[1]}(\pi_2 \circ f))_*$$

which, as this is true for every $x \in U$, proves that $U \subseteq \mathfrak{D}_f^1$. Further by [theorem: 16.135] $\mathfrak{D}_f^1 \subseteq U$. Hence we have

$$U = \mathfrak{D}_f^1 \text{ so that } f \text{ is 1-times differentiable on } U \quad (16.180)$$

$$\forall x \in U \ D_x^{[1]} f = (D_x^{[1]}(\pi_1 \circ f), \dots, D_x^{[1]}(\pi_2 \circ f))_* \quad (16.181)$$

Using [lemma: 16.191] we have for

$$\mathcal{K}: \prod_{i \in \{1, \dots, n\}} L(X, Y_i) \rightarrow L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right) \text{ by } \mathcal{K}(L_1, \dots, L_n) = (L_1, \dots, L_n)_*$$

that

$$\mathcal{K} \in L\left(\prod_{i \in \{1, \dots, n\}} L(X, Y_i), L\left(X, \prod_{i \in \{1, \dots, n\}} Y_i\right)\right) \quad (16.182)$$

Now $\forall x \in U$ we have

$$\begin{aligned} (\mathcal{K} \circ g)(x) &= \mathcal{K}(g(x)) \\ &\stackrel{[\text{eq: 16.177}]}{=} \mathcal{K}(D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x)) \\ &= (D^{[1]}(\pi_1 \circ f)(x), \dots, D^{[1]}(\pi_k \circ f)(x))_* \\ &\stackrel{[\text{definition: 16.136}]}{=} (D_x^{[1]}(\pi_1 \circ f), \dots, D_x^{[1]}(\pi_k \circ f))_* \\ &\stackrel{[\text{eq: 16.181}]}{=} D_x^{[1]} f \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]} f(x) \end{aligned}$$

which, as $U = \mathfrak{D}_f^1$ by [eq: 16.180], proves that

$$D^{[1]}f = \mathcal{K} \circ g \quad (16.183)$$

As by [eq: 16.179] g is of class C^n and $\mathcal{K}$ is linear and continuous [see eq: 16.182] it follows from [lemma: 16.189] that $\mathcal{K} \circ g$ is of class C^n so that by [eq: 16.183] $D^{[1]}f$ is of class C^n on U . Finally applying [theorem: 16.175] we have that

$$f \text{ is of class } C^{n+1}$$

which proves that

$$n+1 \in S_X \quad \square$$

Lemma 16.195. *Let $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$, $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces then*

$$(\circ): L(Y, Z) \cdot L(X, Y) \rightarrow L(X, Z) \text{ defined by } (\circ)(L, K) = L \circ K$$

is ∞ -times differentiable on $L(Y, Z) \cdot L(X, Y)$. Further using [theorem: 16.157] $(\circ)$ is of class C^∞ .

Proof. Using [theorem: 14.238] we have that $(\circ) \in L((Y, Z), L(X, Y); L(X, Z))$ so that by [lemma: 16.186] $(\circ)$ is ∞ -times differentiable at every element of $L(Y, Z) \cdot L(X, Y)$. Hence $(\circ)$ is ∞ -times differentiable on $L(Y, Z) \cdot L(X, Y)$ $\square$

We are now ready to prove the Chain Rules for higher order differentiation.

Theorem 16.196. (General Chain Rule (1)) *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$, $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $A \subseteq X$, $x \in A$, $B \subseteq Y$ and $f: A \rightarrow Y$, $g: B \rightarrow Z$ functions such that $f(A) \subseteq B$, f is n -times differentiable at x and g is n -times differentiable on $f(x)$ then*

$$g \circ f: A \rightarrow Z \text{ is } n\text{-times differentiable at } x$$

Proof. We prove this by induction on n , so let

$S_{X,A} = \{n \in \mathbb{N} \mid \text{Let } \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle \text{ be normed spaces, } B \subseteq Y, f: A \rightarrow Y \text{ a function that is } n\text{-times differentiable at } x \in A \text{ with } f(A) \subseteq B \text{ and } g: B \rightarrow Z \text{ a function that is } n\text{-times differentiable at } f(x) \text{ then } g \circ f \text{ is } n\text{-times differentiable at } x\}$

then we have:

$1 \in S_{X,A}$. Using [theorem: 16.140] we have that f is Fréchet differentiable at x and g is Fréchet differentiable at $f(x)$. Hence using the chain rule [see theorem: 16.67] it follows that $g \circ f$ is Fréchet differentiable at x , so that by [theorem: 16.140] $g \circ f$ is 1-times differentiable at x which proves that $1 \in S_{X,A}$.

$n \in S_{X,A} \Rightarrow n+1 \in S_{X,A}$. Define

$$\mathfrak{D} = \mathfrak{D}_f^1 \bigcap f^{-1}(\mathfrak{D}_g^1) \text{ so that } f(\mathfrak{D}) \subseteq \mathfrak{D}_g^1 \quad (16.184)$$

As f is $(n+1)$ -differentiable at x we have by [theorem: 16.169] that f is 2-times differentiable at x . Using [theorem: 16.140] it follows that

$$D^{[1]}f \text{ is Fréchet differentiable at } x \quad (16.185)$$

Hence using the definition of Fréchet differentiability [see definition: 16.46] it follows that

$$x \in (\mathfrak{D}_f^1)^\circ \quad (16.186)$$

As $D^{[1]}f$ is Fréchet differentiable at x [see eq: 16.185] it follows from [theorem: 16.63] that f is continuous at x . Hence there exist a open set U with $x \in U$ such that $f(U) \subseteq (\mathfrak{D}_g^1)^\circ$ so that $x \in U \subseteq f^{-1}((\mathfrak{D}_g^1)^\circ) \subseteq f^{-1}(\mathfrak{D}_g^1)$ which proves that

$$x \in (f^{-1}(\mathfrak{D}_g^1))^\circ$$

combining this with [eq: 16.186] proves that

$$x \in (\mathfrak{D}_f^1)^\circ \cap (f^{-1}(\mathfrak{D}_g^1))^\circ \subseteq \mathfrak{D}_f^1 \cap f^{-1}(\mathfrak{D}_g^1) \stackrel{[\text{eq: 16.184}]}{=} \mathfrak{D}$$

which as $(\mathfrak{D}_f^1)^\circ \cap (f^{-1}(\mathfrak{D}_g^1))^\circ$ is open proves that

$$x \in \mathfrak{D}^\circ = (\mathfrak{D}_f^1 \cap f^{-1}(\mathfrak{D}_g^1))^\circ \quad (16.187)$$

Let $y \in \mathfrak{D} = \mathfrak{D}_f^1 \cap f^{-1}(\mathfrak{D}_g^1)$ then f is 1-times differentiable at y , hence Fréchet differentiable at y and $f(y) \in \mathfrak{D}_g^1 \Rightarrow g$ is 1-times differentiable at $f(y)$ hence Fréchet differentiable at $f(y)$. So using the chain rule we have that

$$g \circ f \text{ is Fréchet differentiable at } y \text{ and } D_y(g \circ f) = D_{f(y)}g \circ D_y f$$

Using now [theorem: 16.140] on the above we have that

$$g \circ f \text{ is 1-times differentiable at } y \text{ and } D_y^{[1]}(g \circ f) = D_{f(y)}^{[1]}g \circ D_y^{[1]}f \quad (16.188)$$

Define now

$$h: \mathfrak{D} \rightarrow L(Y, Z) \cdot L(X, Y) \text{ by } h(y) = (D_{f(y)}^{[1]}g, D_y^{[1]}f) \quad (16.189)$$

then we have $\forall y \in \mathfrak{D}$ that

$$\begin{aligned} (\pi_1 \circ h)(y) &= \pi_1(h(y)) \\ &\stackrel{[\text{eq: 16.189}]}{=} \pi_1(D_{f(y)}^{[1]}g, D_y^{[1]}f) \\ &= D_{f(y)}^{[1]}g \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}g(f(y)) \\ &\stackrel{y \in \mathfrak{D}}{=} D^{[1]}g(f|_{\mathfrak{D}}(y)) \\ &= (D^{[1]}g \circ f|_{\mathfrak{D}})(y) \\ (\pi_2 \circ h)(y) &= \pi_2(h(y)) \\ &\stackrel{[\text{eq: 16.189}]}{=} \pi_2(D_{f(y)}^{[1]}g, D_y^{[1]}f) \\ &= D_y^{[1]}f \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}f(y) \end{aligned}$$

which proves that

$$\pi_1 \circ h = D^{[1]}g \circ f|_{\mathfrak{D}} \text{ and } \pi_2 \circ h = (D^{[1]}f)|_{\mathfrak{D}} \quad (16.190)$$

As f is $(n+1)$ -times differentiable at x it follows from [theorem: 16.138] that f is n -times differentiable at x . As $x \in \mathfrak{D}^\circ$ [see eq: 16.187] it follows from [theorem: 16.145] that

$$f|_{\mathfrak{D}} \text{ is } n\text{-times differentiable at } x$$

Further as g is $(n+1)$ -times differentiable at $f(x)$ it follows from [theorem: 16.174]

$$D^{[1]}g \text{ is } n\text{-times differentiable at } f(x)$$

As further $f(\mathfrak{D}) \subseteq \mathfrak{D}_g^1$ [see 16.184] and $n \in S_{X,A}$ it follows from the above that $D^{[1]}g \circ f|_{\mathfrak{D}}$ is n -times differentiable at x . Combining this with [eq: 16.190] proves that

$$\pi_1 \circ h \text{ is } n\text{-times differentiable at } x \quad (16.191)$$

As f is $(n+1)$ -times differentiable at x it follows from [theorem: 16.174] that $D^{[1]}f$ is n -times differentiable at x , as $x \in \mathfrak{D}^\circ$ [see eq 16.187], it follows from locality [see theorem: 16.145] that $(D^{[1]}f)|_{\mathfrak{D}}$ is n -times differentiable at x . Hence using [eq: 16.190] we have that

$$\pi_2 \circ h \text{ is } n\text{-times differentiable at } x \quad (16.192)$$

Using now [eqs: 16.191, 16.192] allows us to apply [theorem: 16.192] to prove that

$$h \text{ is } n\text{-times differentiable at } x \quad (16.193)$$

Let $y \in \mathfrak{D}$ then we have

$$\begin{aligned} ((\circ) \circ h)(y) &= (\circ)(h(y)) \\ &= (\circ)(D_{f(y)}^{[1]}g, D_y^{[1]}f) \\ &= D_{f(y)}^{[1]}g \circ D_y^{[1]}f \\ &\stackrel{[\text{eq: 16.188}]}{=} D_y^{[1]}(g \circ f) \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}(g \circ f)(y) \end{aligned}$$

which proves that

$$(D^{[1]}(g \circ f))|_{\mathfrak{D}} = ((\circ) \circ h) \quad (16.194)$$

As $h(\mathfrak{D}) \subseteq L(Y, Z) \cdot L(X, Y)$, $(\circ)$ is n -times differentiable at $h(x)$ [because by [lemma: 16.195] $(\circ)$ is ∞ times differentiable on $L(Y, Z) \cdot L(X, Y)$], h is n -times differentiable at x and $n \in S_{X,A}$ it follows that $((\circ) \circ h)$ is n -times differentiable at x . Hence, using [eq: 16.194], $(D^{[1]}(g \circ f))|_{\mathfrak{D}}$ is n -times differentiable at x . Using locality [see theorem: 16.147] it follow that

$$D^{[1]}(g \circ f) \text{ is } n\text{-times differentiable at } x$$

Finally using [theorem: 16.174] it follows that

$$g \circ f \text{ is } (n+1)\text{-times differentiable at } x$$

which proves that

$$n+1 \in S_{X,A} \quad \square$$

Theorem 16.197. (General Chain Rule (2)) Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$, $\langle Z, \|\cdot\|_Z \rangle$ be normed spaces, $U \subseteq X$ a open set in the topology $\mathcal{T}_{\|\cdot\|_X}$, $V \subseteq Y$ a open set in the topology $\mathcal{T}_{\|\cdot\|_Y}$, $f: U \rightarrow Y$ a function of class C^n such that $f(U) \subseteq V$ and $g: V \rightarrow Z$ a function of class C^n then

$$g \circ f \text{ is of class } C^n$$

Note 16.198. Remember that a function is of class C^n if its domain is open and the function is of class C^n on its domain.

Proof. For $n \in \mathbb{N}_0$ we have either:

$n = 0$. As f is of class C^0 and g is of class C^0 it follows that f is continuous and g is continuous. As $f(U) \subseteq V$ it follows, using [theorem: 14.159], that $g \circ f$ is continuous which proves that $g \circ f$ is of class C^0 .

$n \in \mathbb{N}$. We use induction to prove this case, so define

$S_{X,U} = \{n \in \mathbb{N} | \text{Let } \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle \text{ be normed spaces, } V \in \mathcal{T}_{\|\cdot\|_Y}, f: U \rightarrow Y \text{ a function of class } C^n \text{ such that } f(U) \subseteq V \text{ and } g: V \rightarrow Z \text{ a function of class } C^n \text{ then } g \circ f \text{ is of class } C^n\}$
then we have:

1 $\in S_{X,U}$. Let $x \in U$ then, as f is of class C^1 , it follows that f is 1-times differentiable at x so that by [theorem: 16.140] f is Fréchet differentiable at x . Likewise as $f(x) \in f(U) \subseteq V$ and g is of class C^1 it follows that g is 1-times differentiable at $f(x)$ so that by [theorem: 16.140] g is Fréchet differentiable at $f(x)$. Hence as $f(U) \subseteq V$ it follows from the chain rule [see theorem: 16.67] that

$$g \circ f \text{ is Fréchet differentiable at } x \text{ with } D_x(g \circ f) = D_{f(x)}g \circ D_xf$$

or using [theorem: 16.140]

$$g \circ f \text{ is 1-times differentiable at } x \text{ and } D_x^{[1]}(g \circ f) = D_{f(x)}^{[1]}g \circ D_x^{[1]}f$$

As this is true for every $x \in U$ it follows that $U \subseteq \mathfrak{D}_{g \circ f}^1$ which, as by [theorem: 16.135] $\mathfrak{D}_{g \circ f}^1 \subseteq U$, proves that

$$\mathfrak{D}_{g \circ f}^1 = U \text{ so that } g \circ f \text{ is differentiable on } U \quad (16.195)$$

$$\forall x \in U = \mathfrak{D}_{g \circ f}^1 \text{ we have } D_x^{[1]}(g \circ f) = D_{f(x)}^{[1]}g \circ D_x^{[1]}f \quad (16.196)$$

So the only thing left to prove is the continuity of $D^{[1]}(g \circ f)$. As f is of class C^1 hence 1-times differentiable on U , $f(U) \subseteq V$ and g is of class C^1 hence 1-times differentiable on V we can define the following function

$$h: U \rightarrow L(Y, Z) \cdot L(X, Y) \text{ by } h(x) = (D_{f(x)}^{[1]}g, D_x^{[1]}f) \quad (16.197)$$

Then we have $\forall x \in U$ that

$$\begin{aligned} (\pi_1 \circ h)(x) &= \pi_1(h(x)) \\ &\stackrel{[\text{eq: 16.197}]}{=} \pi_1(D_{f(x)}^{[1]}g, D_x^{[1]}f) \\ &= D_{f(x)}^{[1]}g \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}g(f(x)) \\ &= (D^{[1]}g \circ f)(x) \\ (\pi_2 \circ h)(x) &= \pi_2(h(x)) \\ &\stackrel{[\text{eq: 16.197}]}{=} \pi_2(D_{f(x)}^{[1]}g, D_x^{[1]}f) \\ &= D_x^{[1]}f \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}f(x) \end{aligned}$$

which proves that

$$\pi_1 \circ h = D^{[1]}g \circ f \text{ and } \pi_2 \circ h = D^{[1]}f$$

As f is of class C^1 it follows from [theorem: 16.158] that f is of class C^0 proving that f is continuous, further as g is of class C^1 D^1g is continuous. Hence using [theorem: 14.159], $D^{[1]}g \circ f$ is continuous which proves that $\pi_1 \circ h$ is continuous. Further as f is of class C^1 $D^{[1]}f$ is continuous so that $\pi_2 \circ h$ is continuous. Finally using [theorem: 14.161] it follows that

$$h \text{ is continuous} \quad (16.198)$$

Let $x \in U$ then we have

$$\begin{aligned} ((\circ) \circ h)(x) &= (\circ)(h(x)) \\ &\stackrel{[\text{eq: 16.197}]}{=} (\circ)(D_{f(x)}^{[1]}g, D_x^{[1]}f) \\ &= D_{f(x)}^{[1]}g \circ D_x^{[1]}f \\ &\stackrel{[\text{eq: 16.196}]}{=} D_x^{[1]}(g \circ f) \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}(g \circ f)(x) \end{aligned}$$

which proves that

$$D^{[1]}(g \circ f) = (\circ) \circ h$$

By [lemma: 16.195] $(\circ)$ is of class C^∞ hence by definition of class C^0 from which it follows that $(\circ)$ is continuous. Hence, as h is continuous [see eq: 16.198], we have by [theorem: 14.159] that $(\circ) \circ h$ is continuous. This proves by the above that

$$D^{[1]}(g \circ f) \text{ is continuous}$$

so that

$$g \circ f \text{ is of class } C^1$$

which proves that

$$1 \in S_{X,U}$$

$n \in S_{X,U} \Rightarrow n+1 \in S_{X,U}$. Let $x \in U$ then, as f is of class C^{n+1} it follows from [theorem: 16.158] that f is of class C^1 so that f is 1-times differentiable at x . Hence by [theorem: 16.140] we have that f is Fréchet differentiable at x . As g is of class C^{n+1} it follows from [theorem: 16.158] that g is of class C^1 hence 1-times differentiable on V which, as $f(x) \in f(U) \subseteq V$, means that g is 1-times differentiable at $f(x)$. Using [theorem: 16.140] it follows that g is Fréchet differentiable at $f(x)$. So we can use the chain rule [see theorem: 16.67] to get

$$g \circ f \text{ is Fréchet differentiable at } x \text{ with } D_x(g \circ f) = D_{f(x)}g \circ D_x f$$

or using [theorem: 16.140]

$$g \circ f \text{ is 1-times differentiable at } x \text{ and } D_x^{[1]}(g \circ f) = D_{f(x)}^{[1]}g \circ D_x^{[1]}f$$

As this is true for every $x \in U$ it follows that $U \subseteq \mathfrak{D}_{g \circ f}^1$ which, as by [theorem: 16.135] $\mathfrak{D}_{g \circ f}^1 \subseteq U$, proves that

$$\mathfrak{D}_{g \circ f}^1 = U \text{ so that } g \circ f \text{ is differentiable on } U \quad (16.199)$$

$$\forall x \in U = \mathfrak{D}_{g \circ f}^1 \text{ we have } D_x^{[1]}(g \circ f) = D_{f(x)}^{[1]}g \circ D_x^{[1]}f \quad (16.200)$$

As f and g are of class C^{n+1} it follows from [theorem: 16.158] that f and g are of class C^1 hence f is 1-times differentiable on U and g is 1-times differentiable on V . So, as $f(U) \subseteq V$, we can define the following function

$$h: U \rightarrow L(Y, Z) \cdot L(X, Y) \text{ by } h(x) = (D_{f(x)}^{[1]}g, D_x^{[1]}f) \quad (16.201)$$

then we have $\forall x \in U$

$$\begin{aligned} (\pi_1 \circ h)(x) &= \pi_1(h(x)) \\ &\stackrel{[\text{eq: 16.201}]}{=} \pi_1(D_{f(x)}^{[1]}g, D_x^{[1]}f) \\ &= D_{f(x)}^{[1]}g \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}g(f(x)) \\ &= (D^{[1]}g \circ f)(x) \\ (\pi_2 \circ h)(x) &= \pi_2(h(x)) \\ &\stackrel{[\text{eq: 16.201}]}{=} \pi_2(D_{f(x)}^{[1]}g, D_x^{[1]}f) \\ &= D_x^{[1]}f \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}f(x) \end{aligned}$$

which proves that

$$\pi_1 \circ h = D^{[1]}g \circ f \text{ and } \pi_2 \circ h = D^{[1]}f \quad (16.202)$$

As f is of class C^{n+1} we have by [theorem: 16.158] that f is of class C^n , further as g is of class C^{n+1} it follows from [theorem: 16.175] it follows that $D^{[1]}g$ is of class C^n . Hence as $n \in S_{X,U}$ it follows that $D^{[1]}g \circ f$ is of class C^n so that by [eq: 16.202] $\pi_1 \circ h$ is of class C^n . Further, as f is of class C^{n+1} , it follows from [theorem: 16.175] that $D^{[1]}f$ is of class C^n so that by [eq: 16.202] $\pi_2 \circ h$ is of class C^n . Using now [lemma: 16.193] it follows that

$$h \text{ is of class } C^n \quad (16.203)$$

Now $\forall x \in U$ we have

$$\begin{aligned}
 ((\circ) \circ h)(x) &= (\circ)(h(x)) \\
 &= (\circ)(D_{f(x)}^{[1]}g, D_x^{[1]}f) \\
 &= D_{f(x)}^{[1]}g \circ D_x^{[1]}f \\
 &\stackrel{[\text{eq: 16.200}]}{=} D_x^{[1]}(g \circ f) \\
 &\stackrel{[\text{definition: 16.136}]}{=} D^{[1]}(g \circ f)(x)
 \end{aligned}$$

proving that

$$D^{[1]}(g \circ f) = (\circ) \circ h \quad (16.204)$$

As $(\circ)$ is of class C^∞ it follows by definition that $(\circ)$ is of class C^n , further h is of class C^n [see eq: 16.203]. Hence as $n \in S_{X,U}$ it follows that $D^{[1]}(g \circ f)$ is of class C^n , Finally using [theorem: 16.175] we prove that

$$g \circ f \text{ is of class } C^{n+1}$$

from which it follows that

$$n+1 \in S_{X,U} \quad \square$$

As a example of the general chain rule we prove that the product of n -times differentiable functions or functions of class C^n is n -times n -times differentiable or of class C^n .

Theorem 16.199. *Let $n \in \mathbb{N}$, $\langle X, ||| \rangle$ be a normed space, $\langle \mathbb{K}, || \rangle$ be the normed space of real (complex) numbers, $A \subseteq X$, $x \in A$ and $f: A \rightarrow \mathbb{K}$, $g: A \rightarrow \mathbb{K}$ functions that are n -times differentiable at x then*

$$f \cdot g: A \rightarrow \mathbb{K}$$

is n -times differentiable at x .

Proof. Define

$$\varphi: A \rightarrow \mathbb{K} \cdot \mathbb{K} \text{ by } \varphi(x) = (f(x), g(x))$$

then we have for $x \in A$ that

$$\begin{aligned}
 (\pi_1 \circ \varphi)(x) &= \pi_1(\varphi(x)) \\
 &= \pi_1((f(x), g(x))) \\
 &= f(x) \\
 (\pi_2 \circ \varphi)(x) &= \pi_2(\varphi(x)) \\
 &= \pi_2((f(x), g(x))) \\
 &= g(x)
 \end{aligned}$$

proving that

$$\pi_1 \circ \varphi = f \text{ and } \pi_2 \circ \varphi = g$$

Using [theorem: 16.192] and the fact f, g are n -times differentiable at x it follows that

$$\varphi \text{ is } n\text{-times differentiable at } x$$

Consider the function

$$(\cdot): \mathbb{K} \cdot \mathbb{K} \rightarrow \mathbb{K} \text{ by } (\cdot)(r, s) = r \cdot s$$

then we have for $r, s, t, \alpha \in \mathbb{K}$ that

$$(\cdot)(r + \alpha \cdot s, t) = (r + \alpha \cdot s) \cdot t = r \cdot t + \alpha \cdot (s \cdot t) = (\cdot)(r, t) + \alpha \cdot (\cdot)(s, t)$$

and

$$(\cdot)(r, s + \alpha \cdot t) = r \cdot (s + \alpha \cdot t) = r \cdot s + \alpha \cdot (r \cdot t) = (\cdot)(r, s) + \alpha \cdot (\cdot)(r, t)$$

proving that

$$(\cdot) \in \text{Hom}(\mathbb{K}, \mathbb{K}; \mathbb{K})$$

Further we have for $r, s \in \mathbb{K}$ that

$$|(\cdot)(r, s)| = |r \cdot s| = |r| \cdot |s|$$

proving by [theorem: 14.228] that

$$(\cdot) \in L(\mathbb{K}, \mathbb{K}; \mathbb{K})$$

Hence by [theorem: 16.186]

$(\cdot)$ is ∞ -times differentiable on $\mathbb{K}$

Let $x \in A$ then

$$\begin{aligned} ((\cdot) \circ \varphi)(x) &= (\cdot)(\varphi(x)) \\ &= (\cdot)(f(x), g(x)) \\ &= f(x) \cdot g(x) \\ &= (f \cdot g)(x) \end{aligned}$$

proving that $(\cdot) \circ \varphi = f \cdot g$. So using the generalized chain rule [theorem: 16.196] it follows that

$f \cdot g$ is n -times differentiable at x □

Theorem 16.200. Let $n \in \mathbb{N}$, $\langle X, ||| \rangle$ be a normed space, $\langle \mathbb{K}, || \rangle$ be the normed space of real (complex) numbers, U a open set in X , $f: U \rightarrow \mathbb{K}$, $g: U \rightarrow \mathbb{K}$ functions that are of class C^n then

$f \cdot g: A \rightarrow \mathbb{K}$ is of class C^n

Proof. For $n \in \mathbb{N}_0$ we have two cases to consider:

$n = 0$. Then f, g are continuous so that by [theorem: 14.167] $f \cdot g$ is continuous, hence of class C^0 .

$n \in \mathbb{N}$. Define

$$\varphi: A \rightarrow \mathbb{K} \cdot \mathbb{K} \text{ by } \varphi(x) = (f(x), g(x))$$

then we have for $x \in A$ that

$$\begin{aligned} (\pi_1 \circ \varphi)(x) &= \pi_1(\varphi(x)) \\ &= \pi_1((f(x), g(x))) \\ &= f(x) \\ (\pi_2 \circ \varphi)(x) &= \pi_2(\varphi(x)) \\ &= \pi_2((f(x), g(x))) \\ &= g(x) \end{aligned}$$

proving that

$$\pi_1 \circ \varphi = f \text{ and } \pi_2 \circ \varphi = g$$

Using [lemma: 16.193] and the fact f, g are of class C^n it follows that

φ is of class C^n

Consider the function

$$(\cdot): \mathbb{K} \cdot \mathbb{K} \rightarrow \mathbb{K} \text{ by } (\cdot)(r, s) = r \cdot s$$

then we have for $r, s, t, \alpha \in \mathbb{K}$ that

$$(\cdot)(r + \alpha \cdot s, t) = (r + \alpha \cdot s) \cdot t = r \cdot t + \alpha \cdot (s \cdot t) = (\cdot)(r, t) + \alpha \cdot (\cdot)(s, t)$$

and

$$(\cdot)(r, s + \alpha \cdot t) = r \cdot (s + \alpha \cdot t) = r \cdot s + \alpha \cdot (r \cdot t) = (\cdot)(r, s) + \alpha \cdot (\cdot)(r, t)$$

proving that

$$(\cdot) \in \text{Hom}(\mathbb{K}, \mathbb{K}; \mathbb{K})$$

Further we have for $r, s \in \mathbb{K}$ that

$$|(\cdot)(r, s)| = |r \cdot s| = |r| \cdot |s|$$

proving by [theorem: 14.228] that

$$(\cdot) \in L(\mathbb{K}, \mathbb{K}; \mathbb{K})$$

Hence by [theorem: 16.186]

$$(\cdot) \text{ is of class } C^n$$

Let $x \in U$ then

$$\begin{aligned} ((\cdot) \circ \varphi)(x) &= (\cdot)(\varphi(x)) \\ &= (\cdot)(f(x), g(x)) \\ &= f(x) \cdot g(x) \\ &= (f \cdot g)(x) \end{aligned}$$

proving that $(\cdot) \circ \varphi = f \cdot g$. So using the chain rule [see theorem: 16.197] it follows that

$$f \cdot g \text{ is of class } C^n \quad \square$$

Theorem 16.201. Let $n \in \mathbb{N}_0$, $m \in \mathbb{N}$, U a open set in $\mathbb{K}$ and $f: U \rightarrow \mathbb{K}$ a function of class C^m then

$$f^m: U \rightarrow \mathbb{K} \text{ defined by } f^m(x) = (f(x))^m \text{ is of class } C^m$$

Proof. We prove this by induction so let

$$S_f = \{m \in \mathbb{N} \mid f^m: U \rightarrow \mathbb{K} \text{ defined by } f^m(x) = (f(x))^m \text{ is of class } C^n\}$$

then we have:

$1 \in S_f$. As $\forall x \in U$ $f^1(x) = (f(x))^1 = f(x)$ it follows that $f^1 = f$ so that f^1 is of class C^n proving that $1 \in S_f$.

$m \in S_f \Rightarrow m+1 \in S_f$. As $m \in S_f$ it follows that $f^m: U \rightarrow \mathbb{K}$ defined by $f^m(x) = (f(x))^m$ is of class C^n . Now $\forall x \in U$ we have that

$$(f \cdot f^m)(x) = f(x) \cdot f^m(x) = f(x) \cdot (f(x))^m = (f(x))^{m+1} = f^{(m+1)}(x)$$

from which it follows that $f^{m+1} = f \cdot f^m$. Applying then [theorem: 16.199] proves that $f \cdot f^m$ is of class C^n so that f^{m+1} is of class C^n . Hence

$$m+1 \in S_f \quad \square$$

16.3 Intermediate value and main value theorems.

16.3.1 Derivatives, extremums, concavity, and convexity

Definition 16.202. Let $\langle X, ||| \rangle$ be a normed space, $\langle \mathbb{R}, || \rangle$ be the normed space of real numbers, $A \subseteq X$ and $f: A \rightarrow \mathbb{R}$ then

1. $x \in A$ is a **local weak minimum of f** if there exist a $\delta \in \mathbb{R}^+$ such that $B_{|||}(x, \delta) \subseteq A$ and $\forall y \in B_{|||}(x, \delta)$ $f(x) \leq f(y)$.
2. $x \in A$ is a **local weak maximum of f** if there exist a $\delta \in \mathbb{R}^+$ such that $B_{|||}(x, \delta) \subseteq A$ and $\forall y \in B_{|||}(x, \delta)$ $f(y) \leq f(x)$.

A **local weak minimum** or **local weak maximum** of f is called a **local extremum**.

Note 16.203. As $B_{|||}(x, \delta)$ is a open set and $x \in B_{|||}(x, \delta) \subseteq A$ it follows from [theorem: 14.15] that $x \in A^\circ$ hence a local weak extremum is automatically a interior point of A .

Theorem 16.204. Let $\langle \mathbb{R}, || \rangle$ be the normed space of real numbers, $A \subseteq \mathbb{R}$ and

$$f: A \rightarrow \mathbb{R}$$

a function with a local extremum x then if f has a derivative at x it follows that $f'(x) = 0$.

Proof. If x is a local extremum of f then we have either:

x is a local weak minimum. Then there exist a $\delta \in \mathbb{R}^+$ such that $]x - \delta, x + \delta[\subseteq A$ and $\forall y \in]x - \delta, x + \delta[$ we have $f(x) \leq f(y)$. Assume that $f'(x) \neq 0$ then we have either:

$0 < f'(x)$. As $f'(x) \in \mathbb{R}^+$ there exist a $\delta_{f'(x)} \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } 0 < |h| < \delta_{f'(x)} \text{ we have } \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < f'(x) \quad (16.205)$$

Let $k = \frac{\min(\delta, \delta_{f'(x)})}{2} \in \mathbb{R}^+$ then $0 < |-k| = k < \delta, \delta_{f'(x)}$ so that $x - \delta < x - k < x + \delta$ or $x - k \in]x - \delta, x + \delta[$ so that

$$f(x) \leq f(x - k) \Rightarrow 0 \leq f(x - k) - f(x) \Rightarrow 0 \leq -\left(\frac{f(x - k) - f(x)}{-k} \right) \quad (16.206)$$

As $x - k \in]x - \delta, x + \delta[\subseteq A$ we have that $-k \in A_x$ so that by [eq: 16.205]

$$\left| \frac{f(x - k) - f(x)}{-k} - f'(x) \right| < f'(x)$$

combining this with [eq: 16.206]

$$f'(x) \leq f'(x) - \frac{f(x - k) - f(x)}{-k} \leq \left| \frac{f(x - k) - f(x)}{-k} - f'(x) \right| < f'(x)$$

leading to the contradiction $f'(x) < f'(x)$.

$f'(x) < 0$. As $-f'(x) \in \mathbb{R}^+$ there exist $\delta_{-f'(x)} \in \mathbb{R}^+$ such that we have

$$\forall h \in A_x \text{ with } 0 < |h| < \delta_{-f'(x)} \text{ we have } \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < -f'(x) \quad (16.207)$$

Let $k = \frac{\min(\delta, \delta_{-f'(x)})}{2} \in \mathbb{R}^+$ then $0 < |k| = k < \delta, \delta_{-f'(x)}$ so that $x - \delta < x < x + k < x + \delta$ or $x + k \in]x - \delta, x + \delta[$ so that

$$f(x) \leq f(x + k) \Rightarrow 0 \leq f(x + k) - f(x) \Rightarrow 0 \leq \frac{f(x + k) - f(x)}{k} \quad (16.208)$$

As $x + k \in]x - \delta, x + \delta[\subseteq A$ we have that $k \in A_x$ so that by [eq: 16.207]

$$\left| \frac{f(x + k) - f(x)}{k} - f'(x) \right| < -f'(x)$$

Hence using [eq: 16.208]

$$-f'(x) \leq -f'(x) + \frac{f(x + k) - f(x)}{k} \leq \left| \frac{f(x + k) - f(x)}{k} - f'(x) \right| < -f'(x)$$

leading to the contradiction $-f'(x) < -f'(x)$.

As in all case we have a contradiction the assumption is wrong, hence we must have that

$$f'(x) = 0.$$

x is a local weak maximum. Then there exist a $\delta \in \mathbb{R}^+$ such that $]x - \delta, x + \delta[\subseteq A$ and $\forall y \in]x - \delta, x + \delta[$ we have $f(y) \leq f(x)$. Assume that $f'(x) \neq 0$ then we have either:

$0 < f'(x)$. As $f'(x) \in \mathbb{R}^+$ there exist by [theorem: 16.22] a $\delta_{f'(x)} \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } 0 < |h| < \delta_{f'(x)} \text{ we have } \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < f'(x) \quad (16.209)$$

Let $k = \frac{\min(\delta, \delta_{f'(x)})}{2} \in \mathbb{R}^+$ then $0 < |k| = k < \delta, \delta_{f'(x)}$ so that $x - \delta < x < x + k < x + \delta$ or $x + k \in]x - \delta, x + \delta[$ so that

$$f(x + k) \leq f(x) \Rightarrow 0 \leq -(f(x + k) - f(x)) \Rightarrow 0 \leq -\frac{f(x + k) - f(x)}{k} \quad (16.210)$$

As $x + k \in]x - \delta, x + \delta[\subseteq A$ we have that $k \in A_x$ so that by [eq: 16.209]

$$\left| \frac{f(x+k) - f(x)}{k} - f'(x) \right| < f'(x)$$

Hence using [eq: 16.210]

$$f'(x) \leq f'(x) - \frac{f(x+k) - f(x)}{k} \leq \left| \frac{f(x+k) - f(x)}{k} - f'(x) \right| < f'(x)$$

leading to the contradiction $f'(x) < f'(x)$.

$f'(x) < 0$. As $-f'(x) \in \mathbb{R}^+$ there exist a $\delta_{-f'(x)} \in \mathbb{R}^+$ such that

$$\forall h \in A_x \text{ with } 0 < |h| < \delta_{-f'(x)} \text{ so that } \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < -f'(x) \quad (16.211)$$

Let $k = \frac{\min(\delta, \delta_{f'(x)})}{2}$ then $0 < |-k| = k < \delta, \delta_{-f'(x)}$ so that $x - \delta < x - k < x < x + \delta$ or $x - k \in]x - \delta, x + \delta[$ so that

$$f(x-k) \leq f(x) \Rightarrow 0 \leq -(f(x-k) - f(x)) \Rightarrow 0 \leq \frac{f(x-k) - f(x)}{-k} \quad (16.212)$$

As $x - k \in]x - \delta, x + \delta[\subseteq A$ we have that $-k \in A_x$ so that by [eq: 16.211]

$$\left| \frac{f(x-k) - f(x)}{-k} - f'(x) \right| < -f'(x)$$

Hence using [eq: 16.212]

$$-f'(x) \leq -f'(x) + \frac{f(x-k) - f(x)}{-k} \leq \left| \frac{f(x-k) - f(x)}{-k} - f'(x) \right| < -f'(x)$$

leading to the contradiction that $-f'(x) < -f'(x)$

As in all case we have a contradiction the assumption is wrong, hence we must have that

$$f'(x) = 0.$$

□

Theorem 16.205. (Rolle's theorem) Let $(\mathbb{R}, \|\cdot\|)$ be the normed space of real numbers, $a, b \in \mathbb{R}$ with $a < b$ and $f \in \mathcal{C}([a, b], \mathbb{R})$, the set of continuous functions between $[a, b]$ and $\mathbb{R}$ [see definition: 14.140], such that $f(a) = f(b)$ and f has a derivative at every $x \in]a, b[$ then there exist a $\zeta \in]a, b[$ such that $f'_\zeta = 0$.

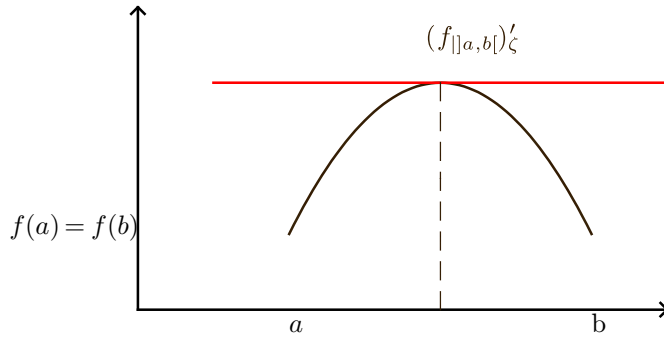


Figure 16.1.

Note 16.206. As $]a, b[$ is a open set we have by [theorem: 14.16] that $]a, b[=]a, b[^\circ$ so that by [theorem: 16.64]

$$\forall x \in]a, b[\text{ } f \text{ has a derivative at } x$$

$$\Updownarrow$$

$$f \text{ is Fréchet differentiable on }]a, b[$$

Proof. Take $k = f(a) = f(b)$. As f is continuous on $[a, b]$ we can use [theorem: 14.466] to find $c, d \in \mathbb{R}$ such that

$$f([a, b]) = [c, d]$$

Hence as $a \in [a, b]$ it follows that $k = f(a) \in [c, d]$ so that $c \leq k \leq d$ and we have the following cases to consider:

$c = d = k$. Then $\forall x \in [a, b]$ we have $f(x) \in f([a, b]) = [c, d] = \{k\}$ so that $f = C_k$. As $a < b$ there exist a $\zeta \in]a, b[$ and we have for ζ by [example: 16.30] that

$$f'_\zeta = f'(\zeta) = C_0(\zeta) = 0$$

$c < k$. As $f([a, b]) = [c, d]$ there exist a $\zeta \in [a, b]$ such that $f(\zeta) = c < k = f(a) = f(b)$ so that $\zeta \neq a, b$, from which it follows that $\zeta \in]a, b[$. As $]a, b[$ is open there exist a $\delta \in \mathbb{R}^+$ such that $\zeta \in B_{||}(\zeta, \delta) \subseteq]a, b[$. Further $\forall x \in B_{||}(\zeta, \delta)$ we have $f(x) \in f([a, b]) = [c, d]$ so that $f(\zeta) = c \leq f(x)$ which proves that ζ is a local weak minimum of f . Hence using [theorem: 16.204] it follows that $f'_\zeta = 0$.

$k < d$. As $f([a, b]) = [c, d]$ there exist a $\zeta \in [a, b]$ such that $f(\zeta) = d > k = f(a) = f(b)$ so that $\zeta \neq a, b$ from which it follows that $\zeta \in]a, b[$. As $]a, b[$ is open there exist a $\delta \in \mathbb{R}^+$ such that $\zeta \in B_{||}(\zeta, \delta) \subseteq]a, b[$. Further $\forall x \in B_{||}(\zeta, \delta)$ we have $f(x) \in f([a, b]) = [c, d]$ so that $f(x) \leq d = f(\zeta)$ which proves that ζ is a local weak maximum of $f|_{]a, b[}$. Hence using [theorem: 16.204] it follows that $f'_\zeta = 0$.

□

Theorem 16.207. (Lagrange's Mean Value theorem) Let $\langle \mathbb{R}, || \rangle$ be the normed space of real numbers, $a, b \in \mathbb{R}$ with $a < b$ and $f \in \mathcal{C}([a, b], \mathbb{R})$ such that f has a derivative at every $x \in]a, b[$ then there exist a $\xi \in]a, b[$ such that

$$f(b) - f(a) = f'_\xi \cdot (b - a)$$

Note 16.208. As $]a, b[$ is a open set we have by [theorem: 14.16] that $]a, b[=]a, b[^\circ$ so that by [theorem: 16.64]

$$\begin{aligned} \forall x \in]a, b[\quad f \text{ has a derivative at } x \\ \Updownarrow \\ f \text{ is Fréchet differentiable on }]a, b[\end{aligned}$$

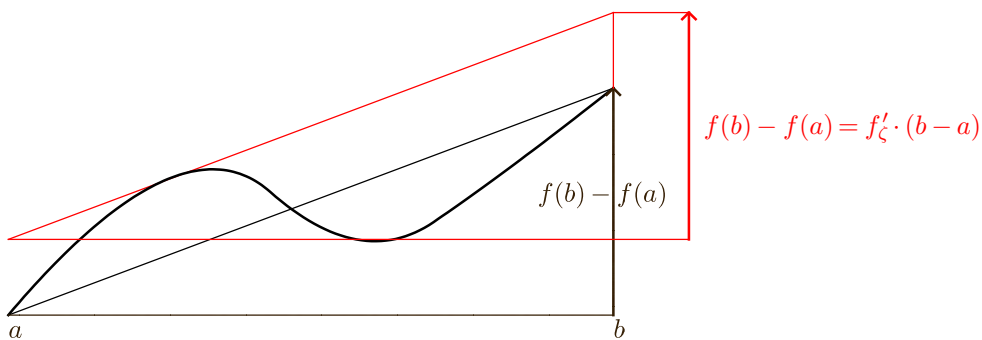


Figure 16.2.

Proof. As $a < b$ we can define

$$g: [a, b] \rightarrow \mathbb{R} \text{ by } g(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} \cdot (x - a) \right)$$

then by [examples: 14.142, 14.143] and [theorems: 14.165, 14.146] it follows that

$$g \in \mathcal{C}([a, b], \mathbb{R})$$

Further using [examples: 16.30, 16.31] and [theorem: 16.40] it follows that

$$\forall x \in]a, b[\text{ } g \text{ is Fréchet differentiable at } x \text{ with } g'_x = f'_x - \frac{f(b) - f(a)}{b - a}$$

Now

$$\begin{aligned} g(a) &= f(a) - \left(f(a) + \frac{f(b) - f(a)}{b - a} \cdot (a - a) \right) = f(a) - f(a) = 0 \\ g(b) &= f(b) - \left(f(a) + \frac{f(b) - f(a)}{b - a} \cdot (b - a) \right) = f(b) - f(a) + f(b) - f(a) = 0 \end{aligned}$$

so we can apply Rolle's theorem [theorem: 16.205] to find a $\zeta \in]a, b[$ such that

$$g'_\zeta = 0$$

As

$$g'_\zeta = f'_\zeta - \frac{f(b) - f(a)}{b - a}$$

it follows that

$$f(b) - f(a) = f'_\zeta \cdot (b - a) \quad \square$$

Corollary 16.209. *Let $\langle \mathbb{R}, || \rangle$ be the normed space of real numbers, $A \subseteq \mathbb{R}$, $a, b \in \mathbb{R}$ with $a < b$ such that $[a, b] \subseteq A$ and $f: A \rightarrow \mathbb{R}$ a function that has a derivative at every $x \in [a, b]$ then there exist a $\xi \in]a, b[$ such that*

$$f(b) - f(a) = f'_\xi \cdot (b - a)$$

Note 16.210. *Let $x \in [a, b]$ then by [theorem: 16.10] x is a limit point of $[a, b]$ which is needed for the derivative at x to even exist.*

Proof. Let $x \in [a, b]$ then, as f has a derivative at x so that by [theorem: 16.33] f is continuous at x . Hence by [theorem: 14.146] we have that $f|_{[a, b]}$ is continuous at x . So we have proved that

$$f|_{[a, b]} \in \mathcal{C}([a, b], \mathbb{R}) \quad (16.213)$$

Further if $x \in]a, b[$ then, as $]a, b[\subseteq [a, b]$, f has a derivative at x . As $]a, b[$ is open it follows from [theorem: 16.36] that $f|_{[a, b]}$ has a derivative at x with $(f|_{[a, b]})'_x = f'_x$. Summarized we have

$$\forall x \in]a, b[\text{ } f|_{[a, b]} \text{ has a derivative at } x \text{ with } (f|_{[a, b]})'_x = f'_x \quad (16.214)$$

The conditions for Lagrange's theorem [theorem: 16.207] are satisfied by [eqs: 16.213, 16.214] so that there exist a $\zeta \in]a, b[$ such that

$$f|_{[a, b]}(b) - f|_{[a, b]}(a) = (f|_{[a, b]})'_\zeta \cdot (b - a) \quad (16.215)$$

So we have

$$\begin{aligned} f(b) - f(a) &\stackrel{a, b \in [a, b]}{=} f|_{[a, b]}(b) - f|_{[a, b]}(a) \\ &\stackrel{[\text{eq: 16.215}]}{=} (f|_{[a, b]})'_\zeta \cdot (b - a) \\ &\stackrel{[\text{eq: 16.215}]}{=} f'_\zeta \cdot (b - a) \end{aligned} \quad \square$$

Corollary 16.211. *Let $\langle \mathbb{R}, || \rangle$ be the normed space of the real numbers, $A \subseteq \mathbb{R}$, $I \subseteq A$*

I of the forms $[a, b], [a, b[,]a, b],]a, b[, [a, \infty[,]a, \infty[,]-\infty, a],]-\infty, a[$ where $a, b \in \mathbb{R}$ and $a < b$ and

$$f: A \rightarrow \mathbb{R}$$

a function that has a derivative f'_x at every point $x \in I$ then we have:

1. If $\forall t \in I \ 0 < f'_t$ then f is strictly increasing on I
2. If $\forall t \in I \ 0 \leq f'_t$ then f is increasing on I .
3. If $\forall t \in I \ f'_t < 0$ then f is strictly decreasing on I .
4. If $\forall t \in I \ f'_t \leq 0$ then f is decreasing on I .

Proof. Let $x, y \in I$ with $x < y$ then as I is a generalized interval [see theorem: 3.144] it follows from [definition: 3.141] that $[x, y] \subseteq I \subseteq A$, hence by the previous corollary [corollary: 16.209] there exist a $\zeta \in]x, y[$ such that $f(y) - f(x) = f'_\zeta \cdot (y - x)$. Summarized we have

$$\forall x, y \in I \text{ with } x < y \text{ there exist a } \zeta \in]x, y[\text{ such that } f(y) - f(x) = f'_\zeta \cdot (y - x) \quad (16.216)$$

So we have that

1. If $\forall t \in I \ 0 < f'_t$ it follows that $0 < f'_\zeta$ so that, as $x < y$, $0 < f'_\zeta \cdot (y - x) \stackrel{[\text{eq: 16.216}]}{=} f(y) - f(x)$ so that $f(x) < f(y)$. Hence f is strictly increasing.
2. If $\forall t \in I \ 0 \leq f'_t$ it follows that $0 \leq f'_\zeta$ so that, as $x < y$, $0 \leq f'_\zeta \cdot (y - x) \stackrel{[\text{eq: 16.216}]}{=} f(y) - f(x)$ so that $f(x) \leq f(y)$. Hence f is increasing.
3. If $\forall t \in I \ f'_t < 0$ it follows that $f'_\zeta < 0$ so that, as $x < y$, $0 > f'_\zeta \cdot (y - x) \stackrel{[\text{eq: 16.216}]}{=} f(y) - f(x)$ so that $f(x) > f(y)$. Hence f is strictly decreasing.
4. If $\forall t \in I \ f'_t \leq 0$ it follows that $f'_\zeta \leq 0$ so that, as $x < y$, $0 \geq f'_\zeta \cdot (y - x) \stackrel{[\text{eq: 16.216}]}{=} f(y) - f(x)$ so that $f(x) \geq f(y)$. Hence f is decreasing. $\square$

Definition 16.212. Let $(\mathbb{R}, \|\cdot\|)$ be the normed space of the real numbers, $A \subseteq \mathbb{R}$, $a, b \in \mathbb{R}$ with $a < b$ and $[a, b] \subseteq A$ and

$$f: A \rightarrow \mathbb{R}$$

a function then we say:

1. f is convex on $[a, b]$ if $\forall x \in [a, b]$ we have

$$f(x) \leq \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} = f(a) + \frac{f(b) - f(a)}{b - a} \cdot (x - a)$$

2. f is concave on $[a, b]$ if $\forall x \in [a, b]$ we have

$$f(a) + \frac{f(b) - f(a)}{b - a} \cdot (x - a) = \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} \leq f(x)$$

In other words a function is concave on a interval if the function curve lies below the line connecting $(a, f(a))$ and $(b, f(b))$ and convex on a interval if the function curve lies above the line connecting $(a, f(a))$ and $(b, f(b))$.

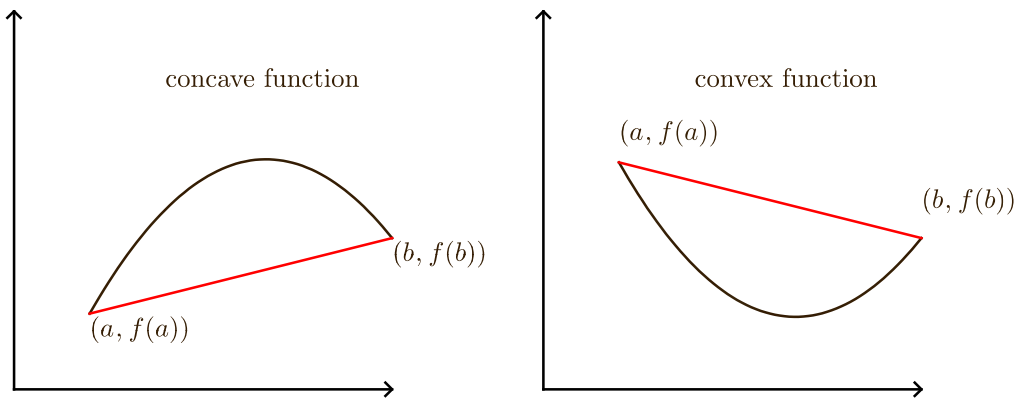


Figure 16.3.

Theorem 16.213. Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of the real numbers, $A \subseteq \mathbb{R}$, $a, b \in \mathbb{R}$ with $a < b$ and $[a, b] \subseteq A$ and $f: A \rightarrow \mathbb{R}$ a function then we have:

1.

$$\begin{aligned} & f \text{ is convex on } [a, b] \\ & \quad \Downarrow \\ & \forall \alpha, \beta \in [0, \infty[\text{ with } \alpha + \beta = 1 \text{ we have } f(\alpha \cdot a + \beta \cdot b) \leq \alpha \cdot f(a) + \beta \cdot f(b) \end{aligned}$$

2.

$$\begin{aligned} & f \text{ is concave on } [a, b] \\ & \quad \Downarrow \\ & \forall \alpha, \beta \in \mathbb{R}^+ \text{ with } \alpha + \beta = 1 \text{ we have } \alpha \cdot f(a) + \beta \cdot f(b) \leq f(\alpha \cdot a + \beta \cdot b) \end{aligned}$$

Proof.

1.

$\Rightarrow$. Let $x = \alpha \cdot a + \beta \cdot b$ then we have

$$\begin{aligned} x &= \alpha \cdot a + \beta \cdot b \\ &= \alpha \cdot a + (1 - \alpha) \cdot b \\ &= \alpha \cdot (a - b) + b \\ &\leq_{(a-b) < 0 \wedge 0 \leq \alpha \Rightarrow \alpha \cdot (a-b) \leq 0} 0 + b \\ &= b \\ \\ x &= \alpha \cdot a + \beta \cdot b \\ &= (1 - \beta) \cdot a + \beta \cdot b \\ &= a + \beta \cdot (b - a) \\ &\geq_{0 < b-a \wedge 0 \leq \beta \Rightarrow \beta \cdot (b-a) \geq 0} a + 0 \\ &= a \end{aligned}$$

So that

$$x \in [a, b]$$

As f is convex on $[a, b]$ we have that

$$\begin{aligned} f(\alpha \cdot a + \beta \cdot b) &= f(x) \\ &\leq \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} \\ &= \frac{f(a) \cdot (b - \alpha \cdot a - \beta \cdot b) + f(b) \cdot (\alpha \cdot a + \beta \cdot b - a)}{b - a} \\ &= \frac{f(a) \cdot ((1 - \beta) \cdot b - \alpha \cdot a) + f(b) \cdot ((\alpha - 1) \cdot a + \beta \cdot b)}{b - a} \\ &= \frac{f(a) \cdot (\alpha \cdot b - \alpha \cdot a) + f(b) \cdot (-\beta \cdot a + \beta \cdot b)}{b - a} \\ &= \frac{f(a) \cdot \alpha \cdot (b - a) + f(b) \cdot \beta \cdot (b - a)}{b - a} \\ &= \alpha \cdot f(a) + \beta \cdot f(b) \end{aligned}$$

$\Leftarrow$. Let $x \in [a, b]$ then $\alpha = \frac{b-x}{b-a} \in [0, \dots, \infty[$ and $\beta = \frac{x-a}{b-a} \in [0, \dots, \infty[$ and

$$\alpha + \beta = \frac{b-x}{b-a} + \frac{x-a}{b-a} = \frac{b-a}{b-a} = 1$$

so that

$$f(\alpha \cdot a + \beta \cdot b) \leq \alpha \cdot f(a) + \beta \cdot f(b) \quad (16.217)$$

Further

$$\begin{aligned} \alpha \cdot a + \beta \cdot b &= \frac{b-x}{b-a} \cdot a + \frac{x-a}{b-a} \cdot b \\ &= \frac{(b-x) \cdot a + (x-a) \cdot b}{b-a} \\ &= \frac{b \cdot a - x \cdot a + x \cdot b - a \cdot b}{b-a} \\ &= \frac{x \cdot (b-a)}{b-a} = x \\ \alpha \cdot f(a) + \beta \cdot f(b) &= \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} \end{aligned}$$

So substituting the above in [eq: 16.217] proves that $f(x) \leq \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a}$ so that f is convex.

2.

$\Rightarrow$. Let $x = \alpha \cdot a + \beta \cdot b$ then we have

$$\begin{aligned} x &= \alpha \cdot a + \beta \cdot b \\ &= \alpha \cdot a + (1-\alpha) \cdot b \\ &= \alpha \cdot (a-b) + b \\ &\leq_{(a-b)<0 \wedge 0 \leq \alpha \Rightarrow \alpha \cdot (a-b) \leq 0} 0 + b \\ &= b \\ \\ x &= \alpha \cdot a + \beta \cdot b \\ &= (1-\beta) \cdot a + \beta \cdot b \\ &= a + \beta \cdot (b-a) \\ &\geq_{0 < b-a \wedge 0 \leq \beta \Rightarrow \beta \cdot (b-a) \geq 0} a + 0 \\ &= a \end{aligned}$$

So that

$$x \in [a, b]$$

As f is concave on $[a, b]$ we have that

$$\begin{aligned} f(\alpha \cdot a + \beta \cdot b) &= f(x) \\ &\geq \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} \\ &= \frac{f(a) \cdot (b-\alpha \cdot a - \beta \cdot b) + f(b) \cdot (\alpha \cdot a + \beta \cdot b - a)}{b-a} \\ &= \frac{f(a) \cdot ((1-\beta) \cdot b - \alpha \cdot a) + f(b) \cdot ((\alpha-1) \cdot a + \beta \cdot b)}{b-a} \\ &= \frac{f(a) \cdot (\alpha \cdot b - \alpha \cdot a) + f(b) \cdot (-\beta \cdot a + \beta \cdot b)}{b-a} \\ &= \frac{f(a) \cdot \alpha \cdot (b-a) + f(b) \cdot \beta \cdot (b-a)}{b-a} \\ &= \alpha \cdot f(a) + \beta \cdot f(b) \end{aligned}$$

$\Leftarrow$. Let $x \in [a, b]$ then $\alpha = \frac{b-x}{b-a} \in [0, \dots, \infty[$ and $\beta = \frac{x-a}{b-a} \in [0, \dots, \infty[$ and

$$\alpha + \beta = \frac{b-x}{b-a} + \frac{x-a}{b-a} = \frac{b-a}{b-a} = 1$$

so that

$$f(\alpha \cdot a + \beta \cdot b) \geq \alpha \cdot f(a) + \beta \cdot f(b) \quad (16.218)$$

Further

$$\begin{aligned}
 \alpha \cdot a + \beta \cdot b &= \frac{b-x}{b-a} \cdot a + \frac{x-a}{b-a} \cdot b \\
 &= \frac{(b-x) \cdot a + (x-a) \cdot b}{b-a} \\
 &= \frac{b \cdot a - x \cdot a + x \cdot b - a \cdot b}{b-a} \\
 &= \frac{x \cdot (b-a)}{b-a} = x \\
 \alpha \cdot f(a) + \beta \cdot f(b) &= \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a}
 \end{aligned}$$

So substituting the above in [eq: 16.218] proves that $f(x) \geq \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a}$ so that f is concave. $\square$

The next theorem shows the relation between the second derivative and convexity and concavity.

Theorem 16.214. Let $\langle \mathbb{R}, || \rangle$ be the normed space of real numbers, $A \subseteq \mathbb{R}$, $a, b \in \mathbb{R}$ with $a < b$ and $[a, b] \subseteq A$, $f: A \rightarrow \mathbb{R}$ a function such that $f|_{[a,b]}$ has a 2-times derivative on $[a, b]$ then

1. If $\forall x \in]a, b[\ 0 \leq (f|_{[a,b]})_x^{(2)}$ then f is convex on $[a, b]$.
2. If $\forall x \in]a, b[\ (f|_{[a,b]})_x^{(2)} \leq 0$ then f is concave on $[a, b]$.

Proof. Let $x \in [a, b]$ then we have either

$x = a$. Then $\frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} = f(a) = f(x)$. As this result is independent of the sign of $(f|_{[a,b]})_x^{(2)}$ we have

$$\begin{cases} \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} \geq f(x) & \text{if } \forall x \in]a, b[\ (f|_{[a,b]})_x^{(2)} \geq 0 \\ \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} \leq f(x) & \text{if } \forall x \in]a, b[\ (f|_{[a,b]})_x^{(2)} \leq 0 \end{cases}$$

$x = b$. Then $\frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} = f(b) = f(x)$. As this result is independent of the sign of $(f|_{[a,b]})_x^{(2)}$ we have

$$\begin{cases} \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} \geq f(x) & \text{if } f_x^{(2)} \geq 0 \\ \frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} \leq f(x) & \text{if } f_x^{(2)} \leq 0 \end{cases}$$

$x \in]a, b[$. Then

$$\frac{x-a}{b-a} + \frac{b-x}{b-a} = \frac{x-a+b-x}{b-a} = 1$$

so by multiplying both sides by $f(x)$ we have $f(x) = \frac{x-a}{b-a} \cdot f(x) + \frac{b-x}{b-a} \cdot f(x)$ so that

$$\begin{aligned}
 &\frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} - f(x) = \\
 &\frac{f(a) \cdot (b-x) + f(b) \cdot (x-a)}{b-a} - \frac{x-a}{b-a} \cdot f(x) - \frac{b-x}{b-a} \cdot f(x) = \\
 &\frac{b-x}{b-a} \cdot f(a) + \frac{x-a}{b-a} \cdot f(b) - \frac{x-a}{b-a} \cdot f(x) - \frac{b-x}{b-a} \cdot f(x) = \\
 &\frac{(x-a)}{b-a} \cdot (f(b) - f(x)) - \frac{b-x}{b-a} \cdot (f(x) - f(a)) = \\
 &\frac{(x-a) \cdot (b-x)}{b-a} \cdot \frac{f(b) - f(x)}{b-x} - \frac{(b-x) \cdot (x-a)}{b-a} \cdot \frac{f(x) - f(a)}{x-a} = \\
 &\frac{(x-a) \cdot (b-x)}{b-a} \cdot \left(\frac{f(b) - f(x)}{b-x} - \frac{f(x) - f(a)}{x-a} \right) \tag{16.219}
 \end{aligned}$$

As $f|_{[a,b]}$ has a 2-times derivative at x we have using the definition of higher order derivatives that $\forall y \in [a, b]$ $f|_{[a,b]}$ has a derivative at y and the function

$$(f|_{[a,b]})': [a, b] \rightarrow \mathbb{R} \text{ defined by } (f|_{[a,b]})'(y) = (f|_{[a,b]})'_y \quad (16.220)$$

has a derivative at x with

$$((f|_{[a,b]})')'_x = (f|_{[a,b]})^{(2)}_x \quad (16.221)$$

As $x \in [a, b]$ was chosen arbitrary we can define

$$(f|_{[a,b]})^{(2)}: [a, b] \rightarrow \mathbb{R} \text{ where } (f|_{[a,b]})^{(2)}(x) = (f|_{[a,b]})^{(2)}_x \quad (16.222)$$

and have

$$\forall x \in [a, b] \text{ that } (f|_{[a,b]})^{(2)}(x) = (f|_{[a,b]})^{(2)}_x \stackrel{[\text{eq: 16.221}]}{=} ((f|_{[a,b]})')'_x \quad (16.223)$$

Let $x \in [a, b]$. As $[a, x] \subseteq [a, b]$, $[x, b] \subseteq [a, b]$ we have by Lagrange's theorem [theorem: 16.209] that there exists $y_1 \in]a, x[$, $y_2 \in]x, b[$ such that

$$f|_{[a,b]}(x) - f|_{[a,b]}(a) = (f|_{[a,b]})'_{y_1} \cdot (x - a) \text{ and } f|_{[a,b]}(b) - f|_{[a,b]}(x) = (f|_{[a,b]})'_{y_2} \cdot (b - x)$$

which at $f|_{[a,b]}(x) \stackrel{[a,b]}{=} f(x)$, $f|_{[a,b]}(a) = a$ and $f|_{[a,b]}(b) = f(b)$ gives

$$f(x) - f(a) = (f|_{[a,b]})'_{y_1} \cdot (x - a) \text{ and } f(b) - f(x) = (f|_{[a,b]})'_{y_2} \cdot (b - x) \quad (16.224)$$

Substituting [eq: 16.224] in [eq: 16.219] gives

$$\begin{aligned} & \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} - f(x) = \\ & \frac{(x - a) \cdot (b - x)}{b - a} \cdot \left(\frac{(f|_{[a,b]})'_{y_2} \cdot (b - x)}{b - x} - \frac{(f|_{[a,b]})'_{y_1} \cdot (x - a)}{x - a} \right) = \\ & \frac{(x - a) \cdot (b - x)}{b - a} \cdot ((f|_{[a,b]})'_{y_2} - (f|_{[a,b]})'_{y_1}) \stackrel{[\text{eq: 16.220}]}{=} \\ & \frac{(x - a) \cdot (b - x)}{b - a} \cdot ((f|_{[a,b]})'(y_2) - (f|_{[a,b]})'(y_1)) \end{aligned} \quad (16.225)$$

As $y_1 \in]a, x[$, $y_2 \in]x, b[$ we have $a < y_1 < x$ and $x < y_2 < b$ so that $y_1 < y_2$ and $[y_1, y_2] \subseteq [a, b]$ hence we can use Lagrange's theorem [theorem: 16.209] using $(f|_{[a,b]})'$ to find a $z \in [y_1, y_2]$

$$(f|_{[a,b]})'(y_2) - (f|_{[a,b]})'(y_1) = ((f|_{[a,b]})'_z)' \cdot (y_2 - y_1) \stackrel{[\text{eq: 16.223}]}{=} (f|_{[a,b]})^{(2)}_z$$

Substituting the above in [eq: 16.225] gives

$$\frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} - f(x) = \frac{(x - a) \cdot (b - x)}{b - a} \cdot (f|_{[a,b]})^{(2)}_z$$

As $x \in [a, b]$ we have that $0 \leq \frac{(x - a) \cdot (b - x)}{b - a}$ so that

$$\begin{cases} \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} \geq f(x) & \text{if } \forall x \in]a, b[(f|_{[a,b]})^{(2)}(x) \geq 0 \\ \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} \leq f(x) & \text{if } \forall x \in]a, b[(f|_{[a,b]})^{(2)}(x) \leq 0 \end{cases}$$

So in all cases we have $\begin{cases} \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} \geq f(x) & \text{if } \forall x \in]a, b[f^{(2)}(x) \geq 0 \\ \frac{f(a) \cdot (b - x) + f(b) \cdot (x - a)}{b - a} \leq f(x) & \text{if } \forall x \in]a, b[f^{(2)}(x) \leq 0 \end{cases}$ proving that

1. If $\forall x \in]a, b[0 \leq (f|_{[a,b]})^{(2)}(x)$ then f is convex on $[a, b]$

2. If $\forall x \in]a, b[(f|_{[a,b]})^{(2)}(x) \leq 0$ then f is concave on $[a, b]$

□

16.3.2 Fundamental theorem of Calculus for the Riemann Integral

Theorem 16.215. (Fundamental Theorem of Calculus (I)) Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of the real numbers, $\langle X, \|\cdot\| \rangle$ be a **Banach space**, $a, b \in \mathbb{R}$ with $a < b$ and $f: [a, b] \rightarrow X$ a continuous function [so that by [theorem: 15.31] $\forall x \in [a, b]$ $\int_a^x f|_{[a, x]}$ exist] then for the function

$$F: [a, b] \rightarrow X \text{ defined by } F(x) = \int_a^x f|_{[a, x]}$$

we have that F has a derivative on $[a, b]$ and $F' = f$ where

$$F': [a, b] \rightarrow X \text{ is defined by } F'(x) = F'_x$$

Proof. Let $x \in [a, b]$ then by [theorem: 16.10] x is a limit point of $[a, b]$, hence $x \in [a, b]'$. As $[a, b] \subseteq A$ we have by [theorem: 14.28] that $[a, b]' \subseteq A'$ so that

$$\forall x \in [a, b] \text{ we have that } x \text{ is a limit point of } A \text{ or } x \in A' \quad (16.226)$$

Let $\varepsilon \in \mathbb{R}^+$. As f is continuous at x there exist a $\delta \in \mathbb{R}^+$ such that for $y \in [a, b]$ with $|y - x| < \delta$ it follows that $\|f(y) - f(x)\| < \frac{\varepsilon}{2}$. Let $h \in [a, b]_x \Rightarrow x + h \in [a, b]$ with $0 < |h| < \delta$ then we have for h either:

0 < h. If $y \in [x, x + h]$ then $|y - x| = y - x \leq x + h - x = h = |h| < \delta$ so that $\|f(y) - f(x)\| < \frac{\varepsilon}{2}$. Hence, as $f(y) - f(x) = f(y) - C_{f(x)}(y) = (f - C_{f(x)})(y)$ it follows that

$$\forall y \in [x, x + h] \text{ we have } \|(f - C_{f(x)})|_{[x, x + h]}(y)\| = \|(f - C_{f(x)})(y)\| < \frac{\varepsilon}{2} \quad (16.227)$$

Next

$$\begin{aligned} & \left\| \frac{F(x + h) - F(x)}{h} - f(x) \right\| && = \\ & \left\| \frac{\left(\int_a^{x+h} f|_{[a, x+h]} \right) - \left(\int_a^x f|_{[a, x]} \right)}{h} - f(x) \right\| && \stackrel{[\text{theorem: 15.31}]}{=} \\ & \left\| \frac{\left(\int_a^x f|_{[a, x]} \right) + \left(\int_x^{x+h} f|_{[x, x+h]} \right) - \left(\int_a^x f|_{[a, x]} \right)}{h} - f(x) \right\| && = \\ & \left\| \frac{\int_x^{x+h} f|_{[x, x+h]} - h \cdot f(x)}{h} \right\| && = \\ & \left\| \frac{\left(\int_x^{x+h} f|_{[x, x+h]} \right) - \int_x^{x+h} (C_{f(x)})|_{[x, x+h]}}{h} \right\| && \stackrel{[\text{example: 15.22}]}{=} \\ & \left\| \frac{\left(\int_x^{x+h} f|_{[x, x+h]} \right) - \int_x^{x+h} (C_{f(x)})|_{[x, x+h]}}{h} \right\| && \stackrel{[\text{theorem: 15.28}]}{=} \\ & \left\| \frac{\int_x^{x+h} (f|_{[x, x+h]} - (C_{f(x)})|_{[x, x+h]})}{h} \right\| && = \\ & \left\| \frac{\int_x^{x+h} (f - C_{f(x)})|_{[x, x+h]}}{h} \right\| && \stackrel{0 < h}{=} \\ & \left\| \frac{\int_x^{x+h} (f - C_{f(x)})|_{[x, x+h]}}{h} \right\| && \stackrel{[\text{eq: 16.227}] \wedge [\text{theorem: 15.29}]}{\leq} \\ & \frac{\varepsilon}{2} \cdot \frac{x + h - x}{h} && = \\ & \frac{\varepsilon}{2} && < \\ & \varepsilon && \end{aligned}$$

$h < 0$. Then $h = -|h|$. If $y \in [x+h, x] = [x-|h|, x]$ then $|y-x| = x-y \leq x-(x-|h|) = |h| < \delta$ and thus $\|f(y) - f(x)\| < \frac{\varepsilon}{2}$. Hence as $f(y) - f(x) = f(y) - C_{f(x)}(y) = (f - C_{f(x)})(h)$ it follows that

$$\forall y \in [x+h, x] \text{ we have } \|(f - C_{f(x)})|_{[x, x+h]}(y)\| = \|(f - C_{f(x)})(y)\| < \frac{\varepsilon}{2} \quad (16.228)$$

Next

$$\begin{aligned} & \left\| \frac{F(x+h) - F(x)}{h} - f(x) \right\| = \\ & \left\| \frac{F(x-|h|) - F(x)}{-|h|} - f(x) \right\| = \\ & \left\| \frac{\left(\int_a^{x-|h|} f|_{[a, x-|h|]} \right) - \left(\int_a^x f|_{[a, x]} \right)}{-|h|} - f(x) \right\| \stackrel{[\text{theorem: 15.31}]}{=} \\ & \left\| \frac{\left(\int_a^{x-|h|} f|_{[a, x-|h|]} \right) - \left(\int_a^{x-|h|} f|_{[a, x-|h|]} \right) - \left(\int_{x-|h|}^x f|_{[x-|h|, x]} \right)}{-|h|} - f(x) \right\| = \\ & \left\| \frac{-\left(\int_{x-|h|}^x f|_{[x-|h|, x]} \right)}{-|h|} - f(x) \right\| = \\ & \left\| \frac{\left(\int_{x-|h|}^x f|_{[x-|h|, x]} \right) - |h| \cdot f(x)}{|h|} \right\| \stackrel{[\text{example: 15.22}]}{=} \\ & \left\| \frac{\left(\int_{x-|h|}^x f|_{[x-|h|, x]} \right) - \left(\int_{x-|h|}^x (C_{f(x)})|_{[x-|h|, x]} \right)}{|h|} \right\| \stackrel{[\text{theorem: 15.28}]}{=} \\ & \left\| \frac{\int_{x-|h|}^x (f|_{[x-|h|, x]} - (C_{f(x)})|_{[x-|h|, x]})}{|h|} \right\| = \\ & \left\| \frac{\int_{x-|h|}^x (f - C_{f(x)})|_{[x-|h|, x]}}{|h|} \right\| = \\ & \left\| \frac{\int_{x-|h|}^x (f - C_{f(x)})|_{[x-|h|, x]}}{|h|} \right\| \stackrel{[\text{eq: 16.228}]}{\leq} \\ & \frac{\varepsilon}{2} \cdot \frac{(x - (x-|h|))}{|h|} \stackrel{[\text{theorem: 15.29}]}{=} \\ & \frac{\varepsilon}{2} < \varepsilon \end{aligned}$$

Hence we have in all cases that $\left\| \frac{F(x+h) - F(x)}{h} - f(x) \right\| < \varepsilon$ which together with [eq: 16.226] proves by [theorem: 16.22] that F has a derivative at x with

$$F'_x = f(x)$$

As this is true for every x in $[a, b]$ we have if we define $F': [a, b] \rightarrow X$ by $F'(x) = F'_x$ that

$$F' = f \quad \square$$

We use now the above theorem to prove the Fundamental Theorem of Calculus.

Theorem 16.216.

(Fundamental Theorem of Calculus (II)) Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of real numbers, $\langle X, \|\cdot\| \rangle$ be a **Banach space**, $a, b \in \mathbb{R}$ with $a < b$ and $f: [a, b] \rightarrow X$ a function such that f has derivatives on $[a, b]$ and **f' is continuous** then

$$f(b) - f(a) = \int_a^b f'$$

Proof. Given that $\langle X, \|\cdot\| \rangle$ is a Banach space, f has a derivative on $[a, b]$ and $f': [a, b] \rightarrow X$ is continuous, it follows from [theorem: 15.31] that $\forall x \in [a, b]$ $\int_a^x f'$ is well defined. This allows us to define

$$g: [a, b] \rightarrow X \text{ by } g(x) = f(a) + \int_a^x f'$$

So if we define

$$F: [a, b] \rightarrow X \text{ by } F(x) = \int_a^x f'$$

then we have that

$$g = C_{f(a)} + F$$

Let $x \in [a, b]$. By [example: 16.30] $C_{f(a)}$ has derivatives on $[a, b]$ with $(C_{f(a)})' = C_0$ and using [theorem: 16.215] F has derivatives on $[a, b]$ with $F' = f'$. Using then the linearity of derivation [see theorem: 16.40] it follows that g has derivatives on $[a, b]$ with

$$g' = (C_{f(a)})' + F' = C_0 + f' = f' \quad (16.229)$$

Note also that $g(a) = C_{f(a)}(a) + F(a) = f(a) + \int_a^a f' \stackrel{\text{[definition: 15.27]}}{=} f(a) + 0 = f(a)$ or

$$f(a) = g(a) \quad (16.230)$$

Define now the function φ by

$$\varphi: [a, b] \rightarrow X \text{ where } \varphi(x) = \|f(x) - g(x)\|$$

Let $x \in [a, b]$ then as f and g has a derivative at x we have that $f - g$ has a derivative at x [see theorem: 16.40] so that by [theorem: 16.33] $f - g$ is continuous at x . Hence $f - g: [a, b] \rightarrow X$ is a continuous function. Further by [theorem: 14.177] $\|\cdot\|: X \rightarrow \mathbb{R}$ defined by $\|\cdot\|(x) = \|\cdot\|$ is a continuous function. Hence, using 14.159]

$$\|\cdot\| \circ (f - g): [a, b] \rightarrow \mathbb{R} \text{ is a continuous function}$$

As $\forall x \in [a, b]$ $(\|\cdot\| \circ (f - g))(x) = \|\cdot\|((f - g)(x)) = \|f(x) - g(x)\| = \varphi(x)$ it follows that $\varphi = \|\cdot\| \circ (f - g)$, which proves that

$$\varphi \in \mathcal{C}([a, b], \mathbb{R})$$

Let $x \in]a, b[$ then as $]a, b[$ is open there exist a $\delta \in \mathbb{R}^+$ such that $B_{\|\cdot\|}(x, \delta) \subseteq]a, b[$. If $0 < |h| < \delta$ then $|x + h - x| = |h| < \delta$ so that $x + h \in B_{\|\cdot\|}(x, \delta) \subseteq]a, b[$. Then we have

$$\begin{aligned} \left| \frac{\varphi(x+h) - \varphi(x)}{h} \right| &= \left| \frac{\|f(x+h) - g(x+h)\| - \|f(x) - g(x)\|}{h} \right| \\ &= \frac{|\|f(x+h) - g(x+h)\| - \|f(x) - g(x)\||}{|h|} \\ &\stackrel{[\text{theorem: 14.91}]}{\leq} \frac{\|f(x+h) - g(x+h) - (f(x) - g(x))\|}{|h|} \\ &= \frac{\|f(x+h) - g(x+h) + (g(x) - f(x))\|}{|h|} \\ &\stackrel{[\text{eq: 16.229}] \Rightarrow f' = g'}{=} \left\| \frac{f(x+h) - f(x)}{h} - f'(x) - \left(\frac{g(x+h) - g(x)}{h} - g'(x) \right) \right\| \\ &\leq \left\| \frac{f(x+h) - f(x)}{h} - f'(x) \right\| + \left\| \frac{g(x+h) - g(x)}{h} - g'(x) \right\| \end{aligned}$$

Summarized, if $0 < |h| < \delta$ then

$$\left| \frac{\varphi(x+h) - \varphi(x)}{h} \right| \leq \left\| \frac{f(x+h) - f(x)}{h} - f'(x) \right\| + \left\| \frac{g(x+h) - g(x)}{h} - g'(x) \right\| \quad (16.231)$$

Let $\varepsilon \in \mathbb{R}^+$. As f has derivatives on $[a, b]$ f has a derivative at x . Hence there exist a $\delta_1 \in \mathbb{R}^+$ such that if $0 < |h| < \delta_1$ and $h \in]a, b[_x \Rightarrow x, x+h \in]a, b[_x$ that

$$\left\| \frac{f(x+h) - f(x)}{h} - f'(x) \right\| < \frac{\varepsilon}{2}$$

Likewise as g has derivatives on $[a, b]$ we have by definition that g has a derivative at x . Hence there exist a $\delta_2 \in \mathbb{R}^+$ such that if $0 < |h| < \delta_2$ and $h \in]a, b[_x \Rightarrow x, x+h \in]a, b[_x$ that

$$\left\| \frac{g(x+h) - g(x)}{h} - g'(x) \right\| < \frac{\varepsilon}{2}$$

So if $0 < |h| < \min(\delta, \delta_1, \delta_2)$ and $h \in]a, b[_x$ we have using the above and [eq: 16.231] that

$$\left| \frac{\varphi(x+h) - \varphi(x)}{h} - 0 \right| = \left| \frac{\varphi(x+h) - \varphi(x)}{h} \right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Hence

$$\forall x \in]a, b[_x \quad \varphi \text{ has a derivative at } x \text{ and } \varphi'_x = 0$$

Let $x \in [a, b]$ then $[a, x] \subseteq [a, b]$ so that we can use Lagrange's theorem [see theorem: 16.209] to find a $\zeta \in]a, b[_x$ with

$$\varphi(x) - \varphi(a) = \varphi'_\zeta \cdot (x - a) = 0 \cdot (x - a) = 0$$

This prove that $\forall x \in [a, b] \quad \varphi(x) = \varphi(a) = \|f(a) - g(a)\| \stackrel{[\text{eq: 16.230}]}{=} 0$. Hence $\forall x \in [a, b]$ that

$$\|f(x) - g(x)\| = \varphi(x) = 0$$

proving that $f(x) = g(x)$ so that $f(b) = g(b) = f(a) + \int_a^b f'$. Hence

$$f(b) - f(a) = \int_a^b f' \quad \square$$

Theorem 16.217. (Integration by Parts) Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of real numbers, $\langle X, \|\cdot\| \rangle$ be a Banach space, $a, b \in \mathbb{R}$ with $a < b$ and $f: [a, b] \rightarrow \mathbb{R}$, $g: [a, b] \rightarrow X$ functions such that:

1. f has derivatives on $[a, b]$.
2. $f': [a, b] \rightarrow \mathbb{R}$ defined by $f'(x) = f'_x$ is continuous.
3. g has derivatives on $[a, b]$.
4. $g': [a, b] \rightarrow X$ defined by $f'(x) = f'_x$ is continuous.

then we have that $f', g', f \cdot g'$ and $f' \cdot g$ are Riemann integrable and

$$\begin{aligned} \int_a^b (f' \cdot g) &= f(b) \cdot g(b) - f(a) \cdot g(a) - \int_a^b (f \cdot g') \\ \int_a^b (f \cdot g') &= f(b) \cdot g(b) - f(a) \cdot g(a) - \int_a^b (f' \cdot g) \end{aligned}$$

Proof. As f has derivatives on $[a, b]$ it follows from [theorem: 16.33] it follows that f is continuous, likewise g is continuous. Hence using [theorem: 14.167] it follows that

$$f \cdot g', f' \cdot g \text{ are continuous} \quad (16.232)$$

Using then [theorem: 15.25] it follows that

$$f \cdot g', f' \cdot g \text{ are Riemann integrable}$$

Using [theorem: 16.41] we have that $f \cdot g$ has derivatives on $[a, b]$ and

$$\forall x \in [a, b] \text{ we have } (f \cdot g)'_x = f(x) \cdot g'_x + f'_x \cdot g(x) = f(x) \cdot g'(x) + f'(x) \cdot g(x)$$

or if we define

$$(f \cdot g)': [a, b] \rightarrow X \text{ by } (f \cdot g)'(x) = (f \cdot g)'(x) = (f \cdot g)'_x \quad (16.233)$$

it follows that

$$(f \cdot g)' = f \cdot g' + f' \cdot g \quad (16.234)$$

Hence using [theorem: 16.40] and [eq: 16.232] it follows that $(f \cdot g)'$ is continuous so that by [theorem: 15.25]

$$(f \cdot g)' \text{ is Riemann integrable}$$

Finally we have

$$\begin{aligned} f(b) \cdot g(b) - f(a) \cdot g(a) &\stackrel{[\text{theorem: 16.216}]}{=} \int_a^b (f \cdot g)' \\ &\stackrel{[\text{eq: 16.234}]}{=} \int_a^b (f \cdot g' + f' \cdot g) \\ &\stackrel{[\text{theorem: 15.28}]}{=} \int_a^b (f \cdot g') + \int_a^b (f' \cdot g) \end{aligned}$$

so that

$$\int_a^b (f' \cdot g) = f(b) \cdot g(b) - f(a) \cdot g(a) - \int_a^b (f \cdot g')$$

and

$$\int_a^b (f \cdot g') = f(b) \cdot g(b) - f(a) \cdot g(a) - \int_a^b (f' \cdot g) \quad \square$$

16.3.3 Mean Value Theorems

The mean value theorems shows how function values varies based on the variation of the function arguments. In essence they specify for a function $f: X \rightarrow Y$ a relation between $f(x) - f(y)$ and $x - y$. A example that is already proved is the Lagrange's Mean Value Theorem [see theorem: 16.207] which states that $f(b) - f(a) = f'_\zeta \cdot (b - a)$ where $\zeta \in [a, b]$. For these new Mean Value Theorems we first need a little lemma that says that translating and scaling is essential continuous differentiable.

Lemma 16.218. *Let $\langle X, ||| \rangle$ be a normed vector space, $\langle \mathbb{R}, || \rangle$ the normed spaces of real numbers, $x, y \in X$ and a, b with $a < b$ then we have that for the function*

$$\sigma: [a, b] \rightarrow X \text{ defined by } \sigma(t) = x + t \cdot (y - x)$$

that

1. σ is of class C^1 on $]a, b[$ where $\forall t \in]a, b[$ we have $D_t \sigma$ is defined by $D_t \sigma(s) = s \cdot (y - x)$
2. σ has a derivative on $[a, b]$ and $\sigma' = C_{y-x}$ so that by [theorem: 14.142] is continuous

Proof.

1. This follows from [example: 16.182]
2. This follows from [example: 16.32]. □

Theorem 16.219. (Mean Value Theorem (I)) *Let $\langle X, ||| \rangle$ be a normed vector space, U a open set in X , $\langle \mathbb{R}, || \rangle$ the normed spaces of real numbers, $x, y \in U$ such that $\{(1-t) \cdot x + t \cdot y | t \in [0, 1]\} \subseteq U$ and $f: U \rightarrow \mathbb{R}$ a function such that $\forall t \in [0, 1]$ f is Fréchet differentiable at $(1-t) \cdot x + t \cdot y$ then there exists a $\zeta \in]0, 1[$ such that*

$$f(y) - f(x) = D_z f(y - x) \text{ where } z = x + \zeta \cdot (y - x)$$

Proof. We have to be Riemann here, derivating is defined on $[a, b]$ but differentiating is only defined on the open set $]0, 1[$, Define

$$\sigma: [0, 1] \rightarrow X \text{ by } \sigma(t) = x + t \cdot (y - x) = (1 - t) \cdot x + t \cdot y$$

then we have by the hypothesis that

$$\sigma([0, 1]) = \{x + t \cdot (y - x) | t \in [0, 1]\} \subseteq U$$

and we can define

$$f \circ \sigma: [0, 1] \rightarrow \mathbb{R}$$

For $t \in [0, 1]$ we have by the hypothesis that f is Fréchet differentiable at $\sigma(t)$ which by [theorem: 16.63] proves that

$$\forall t \in [0, 1] \text{ } f \text{ is continuous at } \sigma(t)$$

For $t \in [0, 1]$ we have by [lemma: 16.218] that σ has a derivative at t which by [theorem: 16.33] proves that

$$\forall t \in [0, 1] \text{ } \sigma \text{ is continuous at } t$$

Hence using [theorem: 14.158] it follows that

$$\forall t \in [0, 1] \text{ } f \circ \sigma \text{ is continuous at } t \text{ or } f \circ \sigma \in \mathcal{C}([0, 1], \mathbb{R}) \quad (16.235)$$

Let $t \in]0, 1[$ then by [lemma: 16.218] σ is Fréchet differentiable at t which, as by the hypothesis f is Fréchet differentiable at $\sigma(t)$, proves by the chain rule [see theorem: 16.67] that

$$f \circ \sigma \text{ is Fréchet differentiable at } t \text{ with } D_t(f \circ \sigma) = D_{\sigma(t)}f \circ D_t\sigma \quad (16.236)$$

As $t \in]0, 1[=]0, 1[^\circ$ it follows from [theorem: 16.64] that $f \circ \sigma$ has a derivative at t with

$$\begin{aligned} (f \circ \sigma)'_t &= D_t(f \circ \sigma)(1) \\ &\stackrel{[\text{eq: 16.236}]}{=} (D_{\sigma(t)}f \circ D_t\sigma)(1) \\ &= D_{\sigma(t)}f(D_t\sigma(1)) \\ &\stackrel{[\text{lemma: 16.218}]}{=} D_{\sigma(t)}f(1 \cdot (y - x)) \\ &= D_{\sigma(t)}f(y - x) \end{aligned}$$

Summarized we have

$$\forall t \in]0, 1[\text{ } f \circ \sigma \text{ has a derivative at } t \text{ with } (f \circ \sigma)'_t = D_{\sigma(t)}f(y - x) \quad (16.237)$$

[eqs: 16.236, 16.237] allows us to use Lagrange's theorem [theorem: 16.207] giving

$$\exists \zeta \in]0, 1[\text{ such that } (f \circ \sigma)(1) - (f \circ \sigma)(0) = (f \circ \sigma)'_{\zeta} \cdot (1 - 0) \stackrel{[\text{eq: 16.237}]}{=} D_{\sigma(\zeta)}f(y - x) \quad (16.238)$$

Finally

$$\begin{aligned} f(y) - f(x) &= f(x + 1 \cdot (y - x)) - f(x + 0 \cdot (y - x)) \\ &= f(\sigma(1)) - f(\sigma(0)) \\ &= (f \circ \sigma)(1) - (f \circ \sigma)(0) \\ &\stackrel{[\text{eq: 16.238}]}{=} D_{\sigma(\zeta)}f(y - x) \end{aligned}$$

which, if we take $z = \sigma(\zeta) = x + \zeta \cdot (y - x)$ proves that

$$f(y) - f(x) = D_z f(y - x) \text{ where } z = \sigma(\zeta) = x + \zeta \cdot (y - x) \quad \square$$

We generalize now the the above Mean Value Theorem to functions where the codomain of the function is a arbitrary normed space. To do this we need the concept of convex sets.

Definition 16.220. (Convex Sets) Let X be a vector space over $\mathbb{K}$ then $C \subseteq X$ is **convex** if $\forall x, y \in C$ we have $\forall t \in [0, 1]$ that $x + t \cdot (y - x) = (1 - t) \cdot x + t \cdot y \in C$.

Best examples of complex sets are open and closed balls in a normed space as is proved in the next theorems.

Theorem 16.221. *Let $\langle X, \|\cdot\| \rangle$ be a normed space, $x \in X$ and $\delta \in \mathbb{R}^+$ then $B_{\|\cdot\|}(x, \delta)$ and $\overline{B_{\|\cdot\|}(x, \delta)}$ are convex.*

Proof. Let $x \in X$, $\delta \in \mathbb{R}^+$, $y, z \in X$ and $t \in [0, 1]$ then we have

$$\begin{aligned}
 \|(1-t) \cdot y + t \cdot z - x\| &= \|(1-t) \cdot y + t \cdot z - x + t \cdot x - t \cdot x\| \\
 &= \|(1-t) \cdot y - x \cdot (1-t) + t \cdot z - t \cdot x\| \\
 &= \|(1-t) \cdot (y-x) + t \cdot (z-x)\| \\
 &\leq \|(1-t) \cdot (y-x)\| + \|t \cdot (z-x)\| \\
 &= |1-t| \cdot \|y-x\| + |t| \cdot \|z-x\| \\
 &\stackrel{0 \leq 1-t, t}{=} (1-t) \cdot \|y-x\| + t \cdot \|z-x\|
 \end{aligned} \tag{16.239}$$

Hence if $y, z \in B_{\|\cdot\|}(x, \delta)$ then

$$\begin{aligned}
 \|(1-t) \cdot y + t \cdot z - x\| &\stackrel{[\text{eq: 16.239}]}{\leq} (1-t) \cdot \|y-x\| + t \cdot \|z-x\| \\
 &< (1-t) \cdot \delta + t \cdot \delta \\
 &= \delta
 \end{aligned}$$

proving that

$$(1-t) \cdot y + t \cdot z \in B_{\|\cdot\|}(x, \delta)$$

Further if $y, z \in \overline{B_{\|\cdot\|}(x, \delta)}$ then

$$\begin{aligned}
 \|(1-t) \cdot y + t \cdot z - x\| &\stackrel{[\text{eq: 16.239}]}{\leq} (1-t) \cdot \|y-x\| + t \cdot \|z-x\| \\
 &\leq (1-t) \cdot \delta + t \cdot \delta \\
 &= \delta
 \end{aligned}$$

so that

$$(1-t) \cdot y + t \cdot z \in \overline{B_{\|\cdot\|}(x, \delta)} \quad \square$$

Theorem 16.222. (Mean Value Theorem (II)) *Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X and $f: U \rightarrow Y$ a function of class C^1 . Assume that there is a convex subset $C \subseteq U$ and a $k \in \mathbb{R}$ such that $\forall x \in C \quad \|D_x f\|_{L(X, Y)} \leq k$ then*

$$\forall x, y \in C \text{ we have } \|f(y) - f(x)\|_Y \leq k \cdot \|y - x\|_X$$

Note 16.223. *As f is of class C^1 f is 1-times differentiable on U , hence Fréchet differentiable on U , so that $\forall x \in C \subseteq U \quad \|D_x f\|_{L(X, Y)}$ is well defined.*

Proof. Let $x, y \in C$. Define the function

$$\sigma: \mathbb{R} \rightarrow X \text{ by } \sigma(t) = x + t \cdot (y - x)$$

then if $t \in [0, 1]$ we have $\sigma(t) = x + t \cdot (y - x) = x \cdot (1-t) + t \cdot y \in C$ [as C is convex] proving that

$$\sigma([0, 1]) \subseteq C \subseteq U \tag{16.240}$$

By [exercise: 16.182] σ is of class C^∞ , hence by definition also of class C^0 , so that σ is continuous from which it follows that

$$W = \sigma^{-1}(U) \text{ is a open set in } \mathbb{R}$$

Take now $\sigma|_W$ then $\sigma|_W$ is defined by

$$\sigma|_W: W \rightarrow X \text{ where } \sigma|_W(t) = \sigma(t) = x + t \cdot (y - x)$$

so that by [exercise: 16.182]

$$\sigma|_W \text{ is of class } C^\infty \text{ with } \forall t \in W \quad D_t \sigma|_W(1) = y - x \tag{16.241}$$

Hence, as f is of class C^1 , it follows from the chain rule [see theorem: 16.197] that

$$h = f \circ \sigma|_W: W \rightarrow Y \text{ is of class } C^1$$

Using then [theorem: 16.161] it follows that

$$\forall t \in W \text{ } h \text{ has a derivative at } t \text{ with } h'_t = D_t h(1) \text{ and } h' \text{ is continuous} \quad (16.242)$$

As f is of class C^1 f is 1-times differentiable on U , hence Fréchet differentiable on U . Likewise, as $\sigma|_W$ is of class C^1 , $\sigma|_W$ is 1-times differentiable on W hence Fréchet differentiable on W . Using then the chain rule [see theorem: 16.67] it follows that

$$\forall t \in W D_t h \text{ exist with } D_t h = D_t(f \circ \sigma|_W) = D_{\sigma|_W(t)} f \circ D_t \sigma|_W \quad (16.243)$$

As by [eq: 16.240] $\sigma([0, 1]) \subseteq U$ we have

$$[0, 1] \subseteq \sigma^{-1}(U) = W$$

Take

$$g = h|_{[0, 1]}: [0, 1] \rightarrow Y$$

Let $t \in [0, 1]$ then, as h has a derivative h'_t at t [see eq: 16.242] and by [theorem: 16.10] t is a limit point of $[0, 1]$, it follows from [theorem: 16.34] that $h|_{[0, 1]}$ has a derivative at t with $(h|_{[0, 1]})'_t = h'_t$. Summarized

$$\forall t \in [0, 1] \text{ we have that } g \text{ has a derivative at } t \text{ with } g'_t = h'_t \quad (16.244)$$

From the above it follows that $g' = (h')|_{[0, 1]}$ so that, as h' is continuous [see eq: 16.242], we have by [theorem: 14.146] that

$$g' \text{ is continuous} \quad (16.245)$$

[eqs: 16.244, 16.245] allows us to apply the fundamental theorem of calculus [theorem: 16.216] giving

$$g(1) - g(0) = \int_0^1 g' \quad (16.246)$$

Now

$$\begin{aligned} g(1) - g(0) &= h|_{[0, 1]}(1) - h|_{[0, 1]}(0) \\ &= h(1) - h(0) \\ &= (f \circ \sigma|_W)(1) - (f \circ \sigma|_W)(0) \\ &= f(\sigma|_W(1)) - f(\sigma|_W(0)) \\ &= f(\sigma(1)) - f(\sigma(0)) \\ &= f(x + 1 \cdot (y - x)) = f(x + 0 \cdot (y - x)) \\ &= f(y) - f(x) \end{aligned}$$

after substituting the above in [eq: 16.246] we get

$$f(y) - f(x) = \int_0^1 g' \quad (16.247)$$

Let $t \in [0, 1]$ then we have

$$\begin{aligned} \|g'(t)\|_Y &= \|g'_t\|_Y \\ &\stackrel{[\text{eq: 16.244}]}{=} \|h'_t\|_Y \\ &\stackrel{[\text{eq: 16.242}]}{=} \|D_t h(1)\|_Y \\ &\stackrel{[\text{eq: 16.243}]}{=} \|(D_{\sigma|_W(t)} f \circ D_t(\sigma|_W))(1)\|_Y \\ &= \|D_{\sigma(t)} f(D_t(\sigma|_W)(1))\|_Y \\ &\stackrel{[\text{eq: 16.241}]}{=} \|D_{\sigma(t)} f(y - x)\|_Y \\ &\leq \|D_{\sigma(t)} f\|_{L(X, Y)} \cdot \|y - x\|_X \\ &\stackrel{\sigma(t) \in C \text{ by [eq: 16.240]}}{\leq} k \cdot \|y - x\|_X \end{aligned}$$

hence we have

$$\forall t \in [0, 1] \quad \|g'(t)\| \leq k \cdot \|y - x\|_X \quad (16.248)$$

Finally we have

$$\begin{aligned} \|f(y) - f(x)\|_Y & \stackrel{[\text{eq: 16.247}]}{=} \left\| \int_0^1 g' \right\|_Y \\ & \stackrel{[\text{eq: 16.248}] \text{ and } [\text{theorem: 15.29(4)}]}{\leq} (k \cdot \|y - x\|_X) \cdot (1 - 0) \\ & = k \cdot \|y - x\| \end{aligned}$$

which completes the proof. $\square$

For the next Mean Value theorems we introduce the concept of line segments.

Definition 16.224. Let X be a vector space, $a, b \in X$ then the **line segment connecting a to b** noted as $L_{a,b}$ is defined to be the set

$$L_{a,b} = \{(1-t) \cdot a + t \cdot b \mid t \in [0, 1]\} = \{a + t \cdot (b - a) \mid t \in [0, 1]\} \subseteq X$$

Note 16.225. By taking $t = 0, 1$ it follows that $a, b \in L_{a,b}$

Example 16.226. $[0, 1] = L_{0,1}$ in $\mathbb{R}$

Proof. If $x \in L_{0,1}$ then there exist a $t \in [0, 1]$ such that $x = (1-t) \cdot 0 + t \cdot 1 = t \in [0, 1]$ so that

$$L_{0,1} \subseteq [0, 1]$$

On the other hand if $x \in [0, 1]$ then if we take $t = x$ we have $x = t = (1-t) \cdot 0 + t \cdot 1 \in L_{0,1}$ which proves that $[0, 1] \subseteq L_{0,1}$. Hence we have

$$L_{0,1} = [0, 1] \quad \square$$

Lemma 16.227. Let $\langle X, \|\cdot\| \rangle$ be a normed space and $a, b \in X$ then $L_{a,b}$ is convex, compact and bounded.

Proof. Let $x, y \in L_{a,b}$ then there exists $t_x, t_y \in [0, 1]$ such that

$$x = (1-t_x) \cdot a + t_x \cdot b \text{ and } y = (1-t_y) \cdot a + t_y \cdot b$$

Let $t \in [0, 1]$ then we have

$$\begin{aligned} (1-t) \cdot x + t \cdot y &= (1-t) \cdot ((1-t_x) \cdot a + t_x \cdot b) + t \cdot ((1-t_y) \cdot a + t_y \cdot b) \\ &= (1-t) \cdot (1-t_x) \cdot a + (1-t) \cdot t_x \cdot b + t \cdot (1-t_y) \cdot a + t \cdot t_y \cdot b \\ &= ((1-t) \cdot (1-t_x) + t \cdot (1-t_y)) \cdot a + ((1-t) \cdot t_x + t \cdot t_y) \cdot b \\ &= (1-t_x - t + t \cdot t_x + t - t \cdot t_y) \cdot a + (t_x - t \cdot t_x + t \cdot t_y) \cdot b \\ &= (1 - (t_x - t \cdot t_x + t \cdot t_y)) \cdot a + (t_x - t \cdot t_x + t \cdot t_y) \cdot b \\ &= (1-s) \cdot a + s \cdot b \text{ where } s = t_x - t \cdot t_x + t \cdot t_y \end{aligned}$$

where $s = t_x - t \cdot t_x + t \cdot t_y$. In other words

$$(1-t) \cdot a + t \cdot b \text{ where } s = t_x - t \cdot t_x + t \cdot t_y = (1-t) \cdot t_x + t \cdot t_y$$

Now as $0 \leq t, t_x, t_y \leq 1$ $0 \leq 1 - t \stackrel{t_x \leq 1}{\Rightarrow} 0 \leq (1-t) \cdot t_x \leq (1-t)$ and $t \cdot t_x \leq t$ so that $0 \leq (1-t) \cdot t_x + t \cdot t_y \leq (1-t) + t = 1$ hence we have

$$(1-t) \cdot a + t \cdot b \text{ where } s \in [0, 1]$$

which prove that $(1-t) \cdot x + t \cdot y \in L_{a,b}$. Hence we have that

$$L_{a,b} \text{ is convex}$$

Define

$$\sigma: \mathbb{R} \rightarrow X \text{ by } \sigma(t) = (1-t) \cdot a + t \cdot b$$

then trivially

$$\sigma([0, 1]) = L_{a,b}$$

Further given $s \in \mathbb{R}$ let $\varepsilon \in \mathbb{R}^+$ then if we take $\delta = \frac{\varepsilon}{\|a-b\|_X + 1}$ then we have for $t \in \mathbb{R}$ with $|t-s| < \delta$ we have

$$\begin{aligned} \|\sigma(s) - \sigma(t)\|_X &= \|(1-s) \cdot a + s \cdot b - ((1-t) \cdot a + t \cdot b)\|_X \\ &= \|(1-s) \cdot a - (1-t) \cdot a + s \cdot b - t \cdot b\|_X \\ &= \|(1-s-1+t) \cdot a + (s-t) \cdot b\|_X \\ &= \|(s-t) \cdot (b-a)\|_X \\ &= |s-t| \cdot \|b-a\|_X \\ &\leq \frac{\varepsilon}{\|a-b\|_X + 1} \cdot \|b-a\|_X \\ &< \varepsilon \end{aligned}$$

proving that σ is continuous at s . As $s \in \mathbb{R}$ was chosen arbitrary it follows that σ is continuous. As $[0, 1]$ is compact [see theorem: 14.279] and σ is continuous it follows from [theorem: 14.275] that

$$L_{a,b} = \sigma([0, 1]) \text{ is compact}$$

Applying then [theorem: 14.273] proves that

$$L_{a,b} \text{ is bounded} \quad \square$$

Using the above we have the following corollary to the Mean Value Theorem (II) [see theorem: 16.222].

Corollary 16.228. *Let $\langle X, \|\cdot\|_X \rangle$ a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X , $f: U \rightarrow Y$ a function that is of class C^1 and $x, y \in X$ such that*

$$1. L_{x,y} \subseteq U$$

$$2. \exists k \in \mathbb{R} \text{ such that } \forall z \in L_{x,y} \text{ we have } \|D_z f\|_{L(X,Y)} \leq k \text{ [in other words } Df \text{ is bounded on } L_{x,y}]$$

then

$$\forall r, s \in L_{x,y} \text{ we have } \|f(s) - f(r)\|_Y \leq k \cdot \|s - r\|_X$$

Proof. Using [lemma: 16.227] $L_{x,y}$ is convex so it follows from the Mean Value Theorem (II) [see theorem: 16.222] that

$$\forall r, s \in L_{x,y} \text{ we have } \|f(s) - f(r)\|_Y \leq k \cdot \|s - r\|_X \quad \square$$

Theorem 16.229. (Mean Value Theorem (III)) *Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X , $x, y \in X$ such that $L_{x,y} \subseteq U$ and $f: U \rightarrow Y$ a function that is C^1 on U then*

$$\sup (\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\})$$

exists and

$$\|f(y) - f(x)\|_Y \leq \sup (\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\}) \cdot \|y - x\|_X$$

Proof. As f is of class C^1 we have by [definition: 16.152] that f is 1-times differentiable on U and that $D^{[1]}f$ is continuous. As by [definition: 16.129] $D^1 f = Df$ it follows that Df is continuous. Further by the previous lemma [lemma: 16.227] $L_{x,y}$ is compact. So we can use [theorem: 14.275] to prove that $Df(L_{x,y})$ is compact, applying then [theorem: 14.273] it follows that $Df(L_{x,y})$ is bounded. So there exist a $M \in \mathbb{R}^+$ such that $\forall L_1, L_2 \in Df(L_{x,y})$ we have that $\|L_1 - L_2\|_{L(X,Y)} \leq M$. Let $\zeta \in L_{x,y} \subseteq U$ then we have

$$\begin{aligned} \|Df(\zeta)\|_{L(X,Y)} &= \|Df(\zeta) - Df(x) + Df(x)\|_{L(X,Y)} \\ &\leq \|Df(\zeta) - Df(x)\|_{L(X,Y)} + \|Df(x)\|_{L(X,Y)} \\ &\leq M + \|Df(x)\|_{L(X,Y)} \end{aligned}$$

proving that $\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\} = \{\|Df(\zeta)\| | \zeta \in L_{x,y}\}$ is bounded above by $M + \|Df(x)\|_{L(X,Y)}$. As $\mathbb{R}$ is conditionally complete [see theorem: 10.18] and $\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\} \neq \emptyset$ [because $x \in L_{x,y} \Rightarrow L_{x,y} \neq \emptyset$] it follows that

$$\sup(\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\}) \text{ exist}$$

As $L_{x,y}$ is convex [see lemma: 16.227], $L_{x,y} \subseteq U$ and $\forall \zeta \in L_{x,y}$

$$\|D_\zeta f\|_{L(X,Y)} \leq \sup(\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\})$$

it follows from [theorem: 16.222] that

$$\|f(y) - f(x)\|_{L(X,Y)} \leq \sup(\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{x,y}\}) \cdot \|y - x\|_X \quad \square$$

Corollary 16.230. Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X , $x \in U$, $\delta \in \mathbb{R}^+$ such that $B_{\|\cdot\|_X}(x, \delta) \subseteq U$, $f: U \rightarrow Y$ a function that is of class C^1 then $\forall y, z \in B_{\|\cdot\|_X}(x, \delta)$ we have that

$$\sup(\{\|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} | t \in [0, 1]\})$$

exist and

$$\|f(z) - f(y)\|_Y \leq \sup(\{\|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} | t \in [0, 1]\}) \cdot \|z - y\|_X$$

Proof. Let $y, z \in B_{\|\cdot\|_X}(x, \delta)$ then as open balls are convex [see theorem: 16.221] we have that $L_{z,y} = \{y + t \cdot (z - y) | t \in [0, 1]\} \subseteq B_{\|\cdot\|_X}(x, \delta) \subseteq U$. So by the previous theorem [theorem: 16.229]

$$\sup(\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{z,y}\})$$

exists and

$$\|f(z) - f(y)\|_Y \leq \sup(\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{z,y}\}) \cdot \|z - y\|_X$$

Now

$$\begin{aligned} r \in \{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{z,y}\} &\Leftrightarrow \exists \zeta \in L_{z,y} \text{ with } r = \|D_\zeta f\|_{L(X,Y)} \\ &\Leftrightarrow \exists t \in [0, 1] \text{ with } \zeta = y + t \cdot (z - y) \wedge r = \|D_\zeta f\|_{L(X,Y)} \\ &\Leftrightarrow \exists t \in [0, 1] \text{ with } r = \|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} \\ &\Leftrightarrow r \in \{\|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} | t \in [0, 1]\} \end{aligned}$$

So $\{\|D_\zeta f\|_{L(X,Y)} | \zeta \in L_{z,y}\} = \{\|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} | t \in [0, 1]\}$ proving that

$$\sup(\{\|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} | t \in [0, 1]\})$$

exist and

$$\|f(z) - f(y)\|_Y \leq \sup(\{\|D_{y+t \cdot (z-y)} f\|_{L(X,Y)} | t \in [0, 1]\}) \cdot \|z - y\|_X \quad \square$$

Theorem 16.231. (Mean Value Theorem (IV)) Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X , $a, b \in X$ such that $L_{a,b} \subseteq U$ and $f: U \rightarrow Y$ a function that is of class C^1 then $\forall x \in U$ we have

$$\|f(a) - f(b) - D_x f(a - b)\|_Y \leq \sup(\{\|D_\zeta f - D_x f\|_{L(X,Y)} | \zeta \in L_{a,b}\}) \cdot \|a - b\|_X$$

Proof. Let $x \in U$ and define

$$g: U \rightarrow Y \text{ by } g = f - (D_x f)|_U \quad (16.249)$$

As $D_x f$ is of class C^∞ [see example: 16.180] it follows by definition that $D_x f$ is of class C^1 . Hence $D_x f$ is 1-times differentiable on X and $D_x f$ is continuous. Using [theorem: 16.145] it follows that $(D_x f)|_U$ is 1-times differentiable on U and using [theorem: 14.146] $(D_x f)|_U$ is continuous, hence $(D_x f)|_U$ is of class C^1 . As f is also of class C^1 it follows from [theorem: 16.178] that g is of class C^1 . Hence we have that g is differentiable at $\zeta \in U$ with

$$\begin{aligned} D_\zeta g &\stackrel{[\text{theorem: 16.178}]}{=} D_\zeta f - D_\zeta (D_x f)|_U \\ &\stackrel{[\text{theorem: 16.147}]}{=} D_\zeta f - D_\zeta (D_x f) \\ &\stackrel{[\text{example: 16.55}]}{=} D_\zeta f - D_x f \end{aligned}$$

so that

$$D_{\zeta}g = D_{\zeta}f - D_{\zeta}(D_x f) \stackrel{[\text{example: 16.55}]}{=} D_{\zeta}f - D_x f \quad (16.250)$$

By the Mean Value Theorem (III) [theorem: 16.229] it follows that

$$\|g(a) - g(b)\|_Y \leq \sup(\{\|D_{\zeta}g\|_{L(X,Y)} | \zeta \in L_{a,b}\}) \cdot \|a - b\|$$

Hence

$$\begin{aligned} \|f(a) - f(b) - D_x f(a - b)\|_Y &= \|f(a) - f(b) - (D_x f(a) - D_x f(b))\|_Y \\ &\stackrel{[\text{eq: 16.249}]}{=} \|g(a) - g(b)\|_Y \\ &\leq \sup(\{\|D_{\zeta}g\|_{L(X,Y)} | \zeta \in L_{a,b}\}) \cdot \|a - b\|_X \\ &\stackrel{[\text{eq: 16.250}]}{=} \sup(\{\|D_{\zeta}f - D_x f\|_{L(X,Y)} | \zeta \in L_{a,b}\}) \cdot \|a - b\|_X \end{aligned}$$

□

Lemma 16.232. *Let $a, b \in \mathbb{R}$ with $a < b$, $c \in]a, b]$, $\beta \in \mathbb{R}$ and $f: [a, b] \rightarrow \mathbb{R}$ a continuous function such that $\forall x \in [a, c[$ we have $f(x) \leq \beta$ then $f(c) \leq \beta$.*

Proof. We prove this by contradiction. Assume that $\beta < f(c)$. Define $\varepsilon = f(c) - \beta \in \mathbb{R}^+$ then as f is continuous at $c \in [a, b]$ there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall x \in [a, b] \text{ with } |c - x| < \delta \text{ we have } |f(c) - f(x)| < \varepsilon = f(c) - \beta \quad (16.251)$$

As $c \in]a, b]$ we have $a < c$ from which it follows that $\min(c - a, \frac{\delta}{2}) \in \mathbb{R}^+$. Hence we have

$$c > c - \min\left(c - a, \frac{\delta}{2}\right) \geq c - (c - a) = a \Rightarrow c - \min\left(c - a, \frac{\delta}{2}\right) \in [a, c[\quad (16.252)$$

and

$$\left|c - \left(c - \min\left(c - a, \frac{\delta}{2}\right)\right)\right| = c - \left(c - \min\left(c - a, \frac{\delta}{2}\right)\right) = \min\left(c - a, \frac{\delta}{2}\right) \leq \frac{\delta}{2} < \delta$$

so

$$f(c) - f\left(c - \min\left(c - a, \frac{\delta}{2}\right)\right) \leq \left|f(c) - f\left(c - \min\left(c - a, \frac{\delta}{2}\right)\right)\right| \stackrel{[\text{eq: 16.251}]}{<} f(c) - \beta$$

from which it follows that $-f\left(c - \min\left(c - a, \frac{\delta}{2}\right)\right) < -\beta$ or

$$\beta < f\left(c - \min\left(c - a, \frac{\delta}{2}\right)\right)$$

As by [eq: 16.252] $c - \min\left(c - a, \frac{\delta}{2}\right) \in [a, c[$ it follows that $f\left(c - \min\left(c - a, \frac{\delta}{2}\right)\right) \leq \beta$ which contradicts the above. Hence the assumption is wrong and we must have that

$$f(c) \leq \beta \quad \square$$

Lemma 16.233. *Let $a, b \in \mathbb{R}$ with $a < b$ and $h: [a, b]$ a continuous function such that $0 < h(a)$ then there exist a $\rho \in \mathbb{R}^+$ so that $a + \rho < b$ and $\forall x \in [a, a + \rho[$ $0 < h(x)$*

Proof. Take $\varepsilon = h(a) \in \mathbb{R}^+$ then as h is continuous at a there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall x \in [a, b] \text{ with } |x - a| < \delta \text{ we have } |h(x) - h(a)| < \varepsilon = h(a) \quad (16.253)$$

Let $\rho = \min\left(\delta, \frac{b-a}{2}\right) \in \mathbb{R}^+$ then $a + \rho \leq a + \frac{b-a}{2} = \frac{2 \cdot a + b - a}{2} = \frac{a+b}{2} < \frac{b+b}{2} = b$ so that

$$a + \rho < b$$

Let $x \in [a, a + \rho[$ then $0 \leq x - a \wedge x < a + \rho$ so that $|x - a| = x - a < \rho < \delta$ so that by [eq: 16.253]

$$h(a) - h(x) \leq |h(x) - h(a)| < h(a)$$

from which it follows that $-h(x) < 0$ so that

$$0 < h(x) \quad \square$$

Theorem 16.234. (Mean Value Theorem (V)) Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ a normed space, $f: [a, b] \rightarrow X$, $\varphi: [a, b] \rightarrow \mathbb{R}$ functions such that

$$f, \varphi \text{ are continuous}$$

and

$$\forall x \in]a, b[\quad f \text{ and } \varphi \text{ have a derivative at } x \text{ with } \|f'_x\| \leq \varphi'_x$$

then

$$\|f(b) - f(a)\| \leq \varphi(b) - \varphi(a)$$

Proof. We first prove that given $\varepsilon \in \mathbb{R}^+$

$$\|f(b) - f(a)\| \leq \varphi(b) - \varphi(a) + \varepsilon \cdot (b - a) \quad (16.254)$$

Proof. Define

$$A = \{x \in [a, b] \mid \forall y \in [a, x[\quad \|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)\} \subseteq [a, b] \quad (16.255)$$

For $a \in [a, b]$ we have $[a, a[= \emptyset$ so we have vacuously that $\forall y \in [a, a[= \emptyset$

$$\|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)$$

which proves that

$$a \in A \quad (16.256)$$

Further if $x \in A$ and $y \in [a, x[$ then we have either:

y = a. Then by [eq: 16.256] $y \in A$

y ∈]a, x[. Then if $z \in [a, y[$ we have as $[a, y[\subseteq [a, x[$ and $x \in A$ that

$$\|f(z) - f(a)\| \leq \varphi(z) - \varphi(a) + \varepsilon \cdot (z - a)$$

which proves that $y \in A$.

From the above we conclude that $[a, x[\subseteq A$ which as $x \in A$ gives

$$\forall x \in A \text{ we have } [a, x] \subseteq A \quad (16.257)$$

As $\emptyset \neq A$ and A is bounded above by b [because $A \subseteq [a, b]$] it follows from [theorem: 10.18] that

$$\gamma = \sup(A) \text{ exist}$$

Further we have, as $a \in A$ and b is a upper bound of A , that

$$\gamma \in [a, b] \quad (16.258)$$

If $y \in [a, \gamma[$ then $y < \gamma = \sup(A)$ so that $\exists z \in A$ with $y < z \Rightarrow y \in [a, z[$ so that by definition of A

$$\|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)$$

which as $y \in [a, \gamma[$ was chosen arbitrary proves that

$$\gamma \in A$$

As $\forall x \in A \subseteq [a, b]$ we have $a \leq x \leq \sup(A) = \gamma$ it follows that $A \subseteq [a, \gamma]$, combining this with $\gamma \in A$ and [eq: 16.257] proves that

$$A = [a, \gamma] \quad (16.259)$$

Now for γ we can have either:

γ = a. Then

$$\begin{aligned} \|f(\gamma) - f(a)\| &= \|f(a) - f(a)\| \\ &= 0 \\ &= \varphi(a) - \varphi(a) + \varepsilon \cdot (a - a) \\ &= \varphi(\gamma) - \varphi(a) + \varepsilon \cdot (\gamma - a) \end{aligned}$$

a < γ. Then $\gamma \in]a, b]$. Define

$$g: [a, b] \rightarrow \mathbb{R} \text{ where } g(y) = \|f(y) - f(a)\| - (\varphi(y) - \varphi(a))$$

then as f, φ are continuous, $\|\cdot\|$ is continuous [see theorem: 14.173] it follows from [theorems: 14.159, 14.165] that

g is continuous

Further $\forall y \in [a, \gamma[$ we have by the definition of A and $\gamma \in A$ that

$$\|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon(y - a)$$

so that

$$\begin{aligned} g(y) &= \|f(y) - f(a)\| - (\varphi(y) - \varphi(a)) \\ &\leq \varepsilon \cdot (y - a) \\ &\underset{y < \gamma}{<} \varepsilon \cdot (\gamma - a) \end{aligned}$$

so that by [lemma: 16.232] $g(\gamma) \leq \varepsilon \cdot (\gamma - a)$ which proves that

$$\|f(\gamma) - f(a)\| \leq \varphi(\gamma) - \varphi(a) + \varepsilon(\gamma - a)$$

so in all cases we have that

$$\|f(\gamma) - f(a)\| \leq \varphi(\gamma) - \varphi(a) + \varepsilon(\gamma - a) \quad (16.260)$$

We prove now that $a < \gamma$. Let $\mu \in \mathbb{R}^+$

$$h: [a, b] \rightarrow \mathbb{R} \text{ by } h(y) = \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a) + \mu - \|f(y) - f(a)\|$$

then as f, φ are continuous, $\|\cdot\|$ is continuous [see theorem: 14.173] it follows from [theorems: 14.159, 14.165] that

h is continuous

Further we have that $h(a) = \varphi(a) - \varphi(a) + \varepsilon \cdot (a - a) + \mu - \|f(a) - f(a)\| = \mu$ so that

$$h(a) = \mu > 0$$

Using now [lemma: 16.233] there exist a $\rho \in \mathbb{R}^+$ such that $a + \rho < b$ and $\forall y \in [a, a + \rho[$ we have $0 < h(y)$. So using the definition of h we have $\forall y \in [a, a + \rho[$ that

$$\|f(y) - f(a)\| < \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a) + \mu$$

as this is true for every $\mu \in \mathbb{R}^+$ it follows from [theorem: 10.31] that

$$\|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)$$

which by definition of A proves that $a + \rho \in A$. Hence $a < a + \rho \leq \gamma$ which proves that

$$a < \gamma$$

We prove now by contradiction that $\gamma = b$. Assume that $\gamma < b$. As $\gamma \in]a, b[$ f'_γ and φ'_γ exist, so that by [theorem: 16.23] there exists $\delta_1, \delta_2 \in \mathbb{R}^+$ such that

$$\forall h \in [a, b]_\gamma \text{ with } |h| < \delta_1 \text{ we have } \|f(\gamma + h) - f(\gamma) - h \cdot f'_\gamma\| \leq \frac{\varepsilon}{2} \cdot |h| \quad (16.261)$$

$$\forall h \in [a, b]_\gamma \text{ with } |h| < \delta_1 \text{ we have } |\varphi(\gamma + h) - \varphi(\gamma) - h \cdot \varphi'_\gamma| \leq \frac{\varepsilon}{2} \cdot |h| \quad (16.262)$$

Let $\delta = \min(\delta_1, \delta_2, b - \gamma) \underset{\gamma < b \Rightarrow 0 < b - \gamma}{\in} \mathbb{R}^+$ and define

$$\zeta = \gamma + \delta \text{ so that } \gamma < \zeta \quad (16.263)$$

Then for $y \in [\gamma, \zeta[\Rightarrow \gamma \leq y < \zeta$ we have for $h = y - \gamma \geq 0$ that

$$|h| = |y - \gamma| = y - \gamma < \zeta - \gamma = \delta \leq \delta_1, \delta_2$$

and $a \underset{[\text{eq: 16.258}]}{\leq} \gamma \leq y = (\gamma + (y - \gamma)) = \gamma + h = y < \zeta = \gamma + \delta \leq \gamma + (b - \gamma) = b$ so that $\gamma + h \in [a, b]$ or

$$h \in [a, b]_\gamma$$

Hence using [eqs: 16.261, 16.262] we have that

$$\|f(\gamma + h) - f(\gamma) - h \cdot f'_\gamma\| \leq \frac{\varepsilon}{2} \cdot |h| \text{ and } |\varphi(\gamma + h) - \varphi(\gamma) - h \cdot \varphi'_\gamma| \leq \frac{\varepsilon}{2} \cdot |h|$$

so that, as $h = y - \gamma \geq 0$

$$\|f(y) - f(\gamma) - (y - \gamma) \cdot f'_\gamma\| \leq \frac{\varepsilon}{2} \cdot (y - \gamma) \text{ and } |\varphi(y) - \varphi(\gamma) - (y - \gamma) \cdot \varphi'_\gamma| \leq \frac{\varepsilon}{2} \cdot (y - \gamma) \quad (16.264)$$

Now

$$\begin{aligned} (y - \gamma) \cdot \varphi'_\gamma - (\varphi(y) - \varphi(\gamma)) &\leq |(y - \gamma) \cdot \varphi'_\gamma - (\varphi(y) - \varphi(\gamma))| \\ &= |\varphi(y) - \varphi(\gamma) - (y - \gamma) \cdot \varphi'_\gamma| \\ &\leq \frac{\varepsilon}{2} \cdot (y - \gamma) \end{aligned}$$

[eq: 16.264]

so that

$$(y - \gamma) \cdot \varphi'_\gamma \leq \varphi(y) - \varphi(\gamma) + \frac{\varepsilon}{2} \cdot (y - \gamma) \quad (16.265)$$

Further

$$\begin{aligned} \|f(y) - f(\gamma)\| &= \|f(y) - f(\gamma) - (y - \gamma) \cdot f'_\gamma + (y - \gamma) \cdot f'_\gamma\| \\ &\leq \|f(y) - f(\gamma) - (y - \gamma) \cdot f'_\gamma\| + \|(y - \gamma) \cdot f'_\gamma\| \\ &\leq \frac{\varepsilon}{2} \cdot (y - \gamma) + \|(y - \gamma) \cdot f'_\gamma\| \\ &\stackrel{0 \leq y - \gamma}{=} \frac{\varepsilon}{2} \cdot (y - \gamma) + (y - \gamma) \cdot \|f'_\gamma\| \\ &\stackrel{\text{hypothesis}}{\leq} \frac{\varepsilon}{2} \cdot (y - \gamma) + (y - \gamma) \cdot \varphi'_\gamma \\ &\stackrel{[eq: 16.265]}{\leq} \frac{\varepsilon}{2} \cdot (y - \gamma) + \varphi(y) - \varphi(\gamma) + \frac{\varepsilon}{2} \cdot (y - \gamma) \\ &= \varphi(y) - \varphi(\gamma) + \varepsilon \cdot (y - \gamma) \end{aligned}$$

proving that

$$\|f(y) - f(\gamma)\| \leq \varphi(y) - \varphi(\gamma) + \varepsilon \cdot (y - \gamma) \quad (16.266)$$

Further

$$\begin{aligned} \|f(y) - f(a)\| &= \|f(y) - f(\gamma) + f(\gamma) - f(a)\| \\ &\leq \|f(y) - f(\gamma)\| + \|f(\gamma) - f(a)\| \\ &\stackrel{[eq: 16.260]}{\leq} \|f(y) - f(\gamma)\| + \varphi(\gamma) - \varphi(a) + \varepsilon \cdot (\gamma - a) \\ &\stackrel{[eq: 16.266]}{\leq} \varphi(y) - \varphi(\gamma) + \varepsilon \cdot (y - \gamma) + \varphi(\gamma) - \varphi(a) + \varepsilon(\gamma - a) \\ &= \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a) \end{aligned}$$

to summarize we have

$$\forall y \in [\gamma, \zeta[\quad \|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)$$

As $\gamma \stackrel{[eq: 16.259]}{\in} A$ we have by the definition of A that

$$\forall y \in [a, \gamma[\quad \text{we have } \|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)$$

so that

$$\forall y \in [a, \zeta[\quad \text{we have } \|f(y) - f(a)\| \leq \varphi(y) - \varphi(a) + \varepsilon \cdot (y - a)$$

which by definition of A proves that $\zeta \in A$. Hence $\zeta \leq \sup(A) = \gamma$ which, as by [eq: 16.263] $\gamma < \zeta$, leads to the contradiction $\zeta < \zeta$. Hence the assumption $\gamma < b$ is wrong and we must have that

$$\gamma = b$$

Substituting this in [eq: 16.260] proves that

$$\|f(b) - f(a)\| \leq \varphi(b) - \varphi(a) + \varepsilon \cdot (b - a)$$

which finishes the proof of [eq: 16.254]. □

As $\|f(b) - f(a)\| \leq \varphi(b) - \varphi(a) + \varepsilon \cdot (b - a)$ and $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary it follows from [theorem: 10.31] that

$$\|f(b) - f(a)\| \leq \varphi(b) - \varphi(a) \quad \square$$

Theorem 16.235. (Mean Value Theorem (VI)) Let $a, b \in \mathbb{R}$ with $a < b$, $\langle X, \|\cdot\| \rangle$ a normed space, $f: [a, b] \rightarrow X$ a **continuous** function such that $\forall x \in]a, b[$ f'_x exist and there exist a $K \in \mathbb{R}$ such that $\forall x \in]a, b[$ $\|f'_x\| \leq K$ then

$$\|f(b) - f(a)\| \leq K \cdot (b - a)$$

Proof. Define

$$\varphi: [a, b] \rightarrow \mathbb{R} \text{ by } \varphi(x) = K \cdot x$$

then φ is continuous [see: 14.164] and by [example: 16.31], [theorem: 16.40] we have that $\forall x \in]a, b[$ φ has a derivative at x with $\varphi'_x = K$. Hence $\forall x \in]a, b[$ we have that $\|f'_x\| \leq K = \varphi'_x$, so using the previous theorem [theorem: 16.234] it follows that

$$\|f(b) - f(a)\| \leq \varphi(b) - \varphi(a) = K \cdot b - K \cdot a = K \cdot (b - a) \quad \square$$

Theorem 16.236. (Mean Value Theorem (VII)) Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X, C \subseteq A$ a convex subset of A and $f: A \rightarrow Y$ a function that is Fréchet differentiable on A such that there exist a $K \in \mathbb{R}$ such that $\forall x \in C$ $\|D_x f\|_{L(X, Y)} \leq K$ then we have

$$\forall x, y \in C \quad \|f(y) - f(x)\|_Y \leq K \cdot \|y - x\|_X$$

Proof. As f is Fréchet differentiable on C it follows from [theorem: 16.63] that

$$f \text{ is continuous}$$

Let $x, y \in C$ and define the function

$$\sigma: [0, 1] \rightarrow X \text{ by } \sigma(t) = x + t \cdot (y - x)$$

Let $z \in \sigma([0, 1])$ then there exist a $t \in [0, 1]$ such that $z = \sigma(t) = x + t \cdot (y - x)$, as $x, y \in C$ and C is convex it follows that $z \in C$. Hence we have

$$\sigma([0, 1]) \subseteq C \subseteq A$$

As f is continuous and σ is continuous [see examples: 14.143, 14.164 and theorem: 14.165] it follows from [theorem: 14.159] that

$$f \circ \sigma \text{ is continuous}$$

Further by [lemma: 16.218] σ is Fréchet differentiable on $[0, 1]$ with

$$\forall t \in [0, 1] \quad D_t \varphi \text{ is defined by } D_t \varphi: \mathbb{R} \rightarrow X \text{ where } D_t \varphi(h) = h \cdot (y - x)$$

Let $t \in [0, 1]$ then using the chain rule [theorem: 16.67] $f \circ \sigma: [0, 1] \rightarrow Y$ is Fréchet differentiable at t with

$$D_t(f \circ \sigma) = D_{\sigma(t)} f \circ D_t \varphi = D_{\sigma(t)} f \circ D_t \varphi$$

Using [theorem: 16.64] it follows then that $(f \circ \sigma)'_t$ exist and $(f \circ \sigma)'_t = D_t(f \circ \sigma)(1)$, hence

$$\begin{aligned} \|(f \circ \sigma)'_t\|_Y &= \|(D_{\sigma(t)} f \circ D_t \varphi)(1)\|_Y \\ &= \|D_{\sigma(t)}(D_t \varphi(1))\|_Y \\ &= \|D_{\sigma(t)} f(y - x)\|_Y \\ &\leq \|D_{\sigma(t)} f\|_{L(X, Y)} \cdot \|y - x\|_X \\ &\leq K \cdot \|y - x\|_X \end{aligned}$$

To summarize $f \circ \sigma$ is continuous and $\forall t \in [0, 1]$ $(f \circ \sigma)'_t$ exist with $\|(f \circ \sigma)'_t\|_Y \leq K \cdot \|y - x\|_X$. Using then the previous theorem [theorem: 16.235] it follows that

$$(f \circ \sigma)(1) - (f \circ \sigma)(0) \leq (K \cdot \|y - x\|_X) \cdot (1 - 0) = K \cdot \|y - x\|_X$$

As

$$(f \circ \sigma)(1) = f(\sigma(1)) = f(x + 1 \cdot (y - x)) = f(y)$$

and

$$(f \circ \sigma)(0) = f(\sigma(0)) = f(x + 0 \cdot (y - x)) = f(x)$$

we have finally

$$f(y) - f(x) \leq K \cdot \|y - x\|_X \quad \square$$

16.4 Symmetry of Higher Order Differentials

In this section we will prove that the n -times differential of a function is symmetric in its arguments. More specific if $\sigma \in P_n$ is a permutation of $(1, \dots, n)$ then we have that

$$D_x^{[n]} f(h_1, \dots, h_n) = D_x^{[n]} f(h_{\sigma(1)}, \dots, h_{\sigma(n)})$$

or using the multilinear representation $D_x^n f$ of the higher order differential [see definition: 16.133] that

$$D_x^n f(h_1, \dots, h_n) = D_x^n f(h_{\sigma(1)}, \dots, h_{\sigma(n)})$$

The proof is rather elaborated, so to simplify it we divide the proof in different lemma's. First we show that we have to prove this only for transpositions. Then we prove it for 2-times differentials and finally extend this to n -times differentials.

Lemma 16.237. *Let $n \in \mathbb{N} \setminus \{1\}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $L \in L_n(X; Y)$ and $(x_1, \dots, x_n) \in X^n$ then if $\forall i \in \{1, \dots, n-1\}$ $L(x_1, \dots, x_n) = L(x_{(i \leftrightarrow i+1)(1)}, \dots, x_{(i \leftrightarrow i+1)(n)})$ it follows that $\forall \sigma \in P_n$ $L(x_1, \dots, x_n) = L(x_{\sigma(1)}, \dots, x_{\sigma(n)})$.*

Proof. Using [theorem: 16.126] there exist a linear isometric isomorphism

$$\mathcal{I}_{n,X,Y}: L_n(X; Y) \rightarrow L^n(X; Y)$$

such that $\forall (x_1, \dots, x_n) \in X^n$ $(\mathcal{I}_{n,X,Y}(L))(x_1, \dots, x_n) = L(x_1, \dots, x_n)$. Take $K = \mathcal{I}_{n,X,Y}(L)$ then $\forall i \in \{1, \dots, n-1\}$ we have

$$\begin{aligned} \left(\left(i \leftrightarrow i+1 \right)_n K \right) (x_1, \dots, x_n) &= \left(K(x_{(i \leftrightarrow i+1)(1)}, \dots, x_{(i \leftrightarrow i+1)(n)}) \right) \\ &= L(x_{(i \leftrightarrow i+1)(1)}, \dots, x_{(i \leftrightarrow i+1)(n)}) \\ &= L(x_1, \dots, x_n) \\ &= K(x_1, \dots, x_n) \end{aligned}$$

proving that $\left(i \leftrightarrow i+1 \right)_n K = K$. Hence if $\sigma \in P_n$ we have by [corollary: 11.242] that $\sigma K = K$. Hence

$$\begin{aligned} L(x_{\sigma(1)}, \dots, x_{\sigma(n)}) &= K(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \\ &= (\sigma K)(x_1, \dots, x_n) \\ &= K(x_1, \dots, x_n) \\ &= L(x_1, \dots, x_n) \end{aligned}$$

□

Lemma 16.238. *Let $\langle X, \|\cdot\|_X \rangle$ be a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X and $f: U \rightarrow Y$ a function that is of class C^2 then $\forall x \in X$ we have $\forall u, v \in X$ that*

$$(D_x^{[2]} f)(v)(w) = (D_x^{[2]} f)(w)(v)$$

Proof. Let $x \in U$ then as U is open there exist a $r \in \mathbb{R}^+$ such that

$$x \in B_{\|\cdot\|_X}(x, r) \subseteq U$$

Let $v, w \in X$ such that $\|v\|_X < \frac{r}{4}$, $\|w\|_X < \frac{r}{4}$ and define

$$\varphi_{v,w}:]-2, 2[\rightarrow X \text{ by } \varphi_{v,w}(t) = x + t \cdot v \text{ and } \psi_{v,w}:]-2, 2[\rightarrow X \text{ by } \psi_{v,w}(t) = x + w + t \cdot v \quad (16.267)$$

If $t \in]-2, 2[$ then we have

$$\|\varphi_{v,w}(t) - x\|_X = \|t \cdot v\|_X = |t| \cdot \|v\|_X \leq 2 \cdot \frac{r}{4} < r$$

and

$$\|\psi_{v,w}(t) - x\|_X = \|w + t \cdot v\|_X \leq \|w\|_X + |t| \cdot \|v\|_X \leq \frac{r}{4} + 2 \cdot \frac{r}{4} < r$$

proving that

$$\varphi_{v,w}(-2, 2] \subseteq U \wedge \psi_{v,w}(-2, 2] \subseteq U \quad (16.268)$$

The above allows us to define

$$g_{v,w}:]-2, 2[\rightarrow Y \text{ by } g_{v,w} = f \circ \psi_{v,w} - f \circ \varphi_{v,w} \quad (16.269)$$

As by [example: 16.182] $\psi_{v,w}, \varphi_{v,w}$ are C^∞ on $]-2, 2[$ and f is C^2 on U , it follows from [theorems: 16.197, 16.178] that $g_{v,w}$ is C^2 on $]-2, 2[$. Hence by [theorem: 16.158] $g_{v,w}$ is C^1 on U . So, as $L_{0,1} \stackrel{[example: 16.226]}{=} [0, 1] \subseteq]-2, 2[$ and $0 \in]-2, 2[$, we can use the second mean value theorem (IV) [see theorem: 16.231] to get

$$\|g_{v,w}(0) - g_{v,w}(1) - D_0 g_{v,w}(0-1)\|_Y \leq \sup(\{\|D_\zeta g_{v,w} - D_0 g_{v,w}\|_{L(\mathbb{R}, Y)} | \zeta \in L_{0,1}\}) \cdot |0-1|$$

As $L_{0,1} = [0, 1]$ and $0-1 = -1$ we have

$$\|g_{v,w}(0) - g_{v,w}(1) - D_0 g_{v,w}(-1)\|_Y \leq \sup(\{\|D_\zeta g_{v,w} - D_0 g_{v,w}\|_{L(\mathbb{K}, Y)} | \zeta \in [0, 1]\}) \quad (16.270)$$

As $D_0 g_{v,w}(1) \stackrel{[theorem: 16.64]}{=} (g_{v,w})'_0$ we have that

$$\begin{aligned} \|g_{v,w}(0) - g_{v,w}(1) - D_0 g_{v,w}(-1)\|_Y &= \|g_{v,w}(0) - g_{v,w}(1) + D_0 g_{v,w}(1)\|_Y \\ &= \|g_{v,w}(1) - g_{v,w}(0) - D_0 g_{v,w}(1)\|_Y \\ &= \|g_{v,w}(1) - g_{v,w}(0) - (g_{v,w})'_0\|_Y \end{aligned}$$

so that by [eq: 16.270]

$$\|g_{v,w}(1) - g_{v,w}(0) - (g_{v,w})'_0\|_Y \leq \sup(\{\|D_\zeta g_{v,w} - D_0 g_{v,w}\|_{L(\mathbb{K}, Y)} | \zeta \in [0, 1]\}) \quad (16.271)$$

Now given $h \in \mathbb{R}$ we have

$$\begin{aligned} \|(D_\zeta g_{v,w} - D_0 g_{v,w})(h)\|_Y &= \|D_\zeta g_{v,w}(h) - D_0 g_{v,w}(h)\|_Y \\ &\stackrel{[theorem: 16.64]}{=} \|h \cdot (g_{v,w})'_\zeta - h \cdot (g_{v,w})'_0\|_Y \\ &= \|(g_{v,w})'_\zeta - (g_{v,w})'_0\|_Y \cdot |h| \end{aligned}$$

proving by the definition of the norm of a linear continuous mapping [see theorem: 14.212] that $\|D_\zeta g_{v,w} - D_0 g_{v,w}\|_{L(\mathbb{K}, Y)} \leq \|(g_{v,w})'_\zeta - (g_{v,w})'_0\|_Y$, hence by [eq: 16.271] it follows that

$$\|g_{v,w}(1) - g_{v,w}(0) - (g_{v,w})'_0\|_Y \leq \sup(\{\|(g_{v,w})'_\zeta - (g_{v,w})'_0\|_Y | \zeta \in [0, 1]\}) \quad (16.272)$$

Let $\zeta \in [0, 1]$ then we have using the chain rule that

$$\begin{aligned} (g_{v,w})'_\zeta &\stackrel{[theorem: 16.64]}{=} D_\zeta g_{v,w}(1) \\ &= D_\zeta(f \circ \psi_{v,w} - f \circ \varphi_{v,w})(1) \\ &\stackrel{[theorem: 16.66]}{=} D_\zeta(f \circ \psi_{v,w})(1) - D_\zeta(f \circ \varphi_{v,w})(1) \\ &\stackrel{[theorem: 16.67]}{=} (D_{\psi_{v,w}(\zeta)} f \circ D_\zeta \psi_{v,w})(1) - (D_{\varphi_{v,w}(\zeta)} f \circ D_\zeta \varphi_{v,w})(1) \\ &= D_{\psi_{v,w}(\zeta)} f(D_\zeta \psi_{v,w}(1)) - D_{\varphi_{v,w}(\zeta)} f(D_\zeta \varphi_{v,w}(1)) \\ &\stackrel{[example: 16.182]}{=} D_{\psi_{v,w}(\zeta)} f(v) - D_{\varphi_{v,w}(\zeta)} f(v) \\ &= D_{x+\zeta \cdot v+w} f(v) - D_{x+\zeta \cdot v} f(v) \end{aligned}$$

so that

$$\forall \zeta \in [0, 1] \quad (g_{v,w})'_\zeta = D_{x+\zeta \cdot v+w} f(v) - D_{x+\zeta \cdot v} f(v) \quad (16.273)$$

As f is C^2 on U we have that f is 2-times differentiable on U , so that by [theorem: 16.140] $D^{[1]}f$ is Fréchet differentiable on U with $\forall x \in U \quad D_x^{[2]}f = D_x(D^{[1]}f)$. As $D^1f = Df$ [see definition: 16.129] we have that

$$Df \text{ is Fréchet differentiable on } U \text{ and } \forall x \in U \quad D_x^{[2]}f = D_x(Df)$$

Then given $\varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that if $h \in U_x$ and $\|h\|_X < \delta$ then

$$\|Df(x+h) - Df(x) - D_x^{[2]}f(h)\|_{L(X,Y)} \leq \varepsilon \cdot \|h\|_X \quad (16.274)$$

Let $v, w \in X$ with $\|v\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)$, $\|w\|_X \leq \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)$ then $\forall \zeta \in [0, 1]$ we have

$$\|\zeta \cdot v + w\|_X \leq |\zeta| \cdot \|v\|_X + \|w\|_X \leq 1 \cdot \frac{\delta}{4} + \frac{\delta}{4} = \frac{\delta}{2} < \delta \text{ and } \|\zeta \cdot v\|_X = |\zeta| \cdot \|v\|_X \leq \frac{\delta}{4} < \delta$$

As $\|v\|_X, \|w\|_X < \frac{r}{4}$ and $|\zeta| \leq 1$ we have that

$$\|x + \zeta \cdot v + w - x\|_X \leq |\zeta| \cdot \|v\|_X + \|w\|_X \leq \frac{r}{4} + \frac{r}{4} < r \text{ and } \|x + \zeta \cdot v - x\|_X \leq |\zeta| \cdot \|v\|_X < r$$

so that $x + \zeta \cdot v + w, x + \zeta \cdot v \in B_{\|x\|_X}(x, r) \subseteq U$ or

$$\zeta \cdot v + w, \zeta \cdot v \in U_x$$

Using the above on [eq: 16.274] we have that

$$\|Df(x + \zeta \cdot v + w) - Df(x) - D_x^{[2]}f(\zeta \cdot v + w)\|_{L(X,Y)} \leq \varepsilon \cdot \|\zeta \cdot v + w\|_X \quad (16.275)$$

$$\|Df(x + \zeta \cdot v) - Df(x) - D_x^{[2]}f(\zeta \cdot v)\|_{L(X,Y)} \leq \varepsilon \cdot \|\zeta \cdot v\|_X \quad (16.276)$$

Also

$$D_x^{[2]}f(\zeta \cdot v + w)(v) - D_x^{[2]}f(\zeta \cdot v)(v) \stackrel{D_x^2 f \in L(\overline{\overline{X}}, L(X, Y))}{=} D_x^{[2]}(w)(v) \quad (16.277)$$

Now for $\zeta \in [0, 1]$

$$\begin{aligned} & \| (g_{v,w})'_\zeta - D_x^{[2]}f(w)(v) \|_Y \stackrel{[\text{eq: 16.273}]}{=} \\ & \| D_{x+\zeta \cdot v+w}f(v) - D_{x+\zeta \cdot v}f(v) - D_x^{[2]}f(w)(v) \|_Y \stackrel{[\text{definition: 16.48}]}{=} \\ & \| Df(x + \zeta \cdot v + w)(v) - Df(x + \zeta \cdot v)(v) - D_x^{[2]}f(w)(v) \|_Y \stackrel{[\text{eq: 16.277}]}{=} \\ & \left\| \underbrace{Df(x + \zeta \cdot v + w)(v)}_1 - \underbrace{Df(x + \zeta \cdot v)(v)}_2 - \left(\underbrace{D_x^{[2]}f(\zeta \cdot v + w)(v)}_3 - \underbrace{D_x^{[2]}f(\zeta \cdot v)(v)}_4 \right) \right\|_Y = \\ & \left\| \underbrace{Df(x + \zeta \cdot v + w)(v) - Df(x)(v) - D_x^{[2]}f(\zeta \cdot v + w)(v)}_1 - \left(\underbrace{Df(x + \zeta \cdot v)(v) - Df(x)(v) - D_x^{[2]}f(\zeta \cdot v)(v)}_2 \right) \right\|_Y \leq \\ & \| Df(x + \zeta \cdot v + w)(v) - Df(x)(v) - D_x^{[2]}f(\zeta \cdot v + w)(v) \|_Y + \| Df(x + \zeta \cdot v)(v) - Df(x)(v) - D_x^{[2]}f(\zeta \cdot v)(v) \|_Y \leq \\ & \| Df(x + \zeta \cdot v + w) - Df(x) - D_x^{[2]}f(\zeta \cdot v + w) \|_{L(X,Y)} \cdot \|v\|_X + \| Df(x + \zeta \cdot v) - Df(x) - D_x^{[2]}f(\zeta \cdot v) \|_{L(X,Y)} \cdot \|v\|_X \stackrel{[\text{eqs: 16.275, 16.276}]}{\leq} \\ & \varepsilon \cdot (\|\zeta \cdot v + w\|_X + \|\zeta \cdot v\|_X) \cdot \|v\|_X \leq \\ & \varepsilon \cdot \|v\|_X \cdot (|\zeta| \cdot \|v\|_X + \|w\|_X + |\zeta| \cdot \|v\|_X) \stackrel{\zeta \in [0,1]}{\leq} \\ & \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X + \|v\|_X) \leq \\ & \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X + \|w\|_X + \|v\|_X) = \\ & 2 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) \end{aligned}$$

proving that

$$\forall \zeta \in [0, 1] \text{ we have } \|(g_{v,w})'_\zeta - D_x^{[2]}f(w)(v)\|_Y \leq 2 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) \quad (16.278)$$

Further for $t \in [0, 1]$ we have:

$$\begin{aligned}
& \|(g_{v,w})'_t - (g_{v,w})'_0\|_Y = \\
& \|((g_{v,w})'_t - D_x^{[2]}f(w)(v) - ((g_{v,w})'_0 - D_x^{[2]}f(w)(v)))\|_Y \leq \\
& \|(g_{v,w})'_t - D_x^{[2]}f(w)(v)\|_Y + \|(g_{v,w})'_0 - D_x^{[2]}f(w)(v)\|_Y \stackrel{t, 0 \in [0, 1] \wedge \text{eq: 16.278}}{\leq} \\
& 2 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) + 2 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) = \\
& 4 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X)
\end{aligned}$$

proving

$$\|(g_{v,w})'_t - (g_{v,w})'_0\|_Y \leq 4 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) \quad (16.279)$$

As $g_{v,w}$ is C^1 on $] -2, 2[$, it follows from [theorem: 16.161] that the function

$$g'_{v,w}:] -2, 2[\rightarrow Y$$

is continuous. Define

$$h:] -2, 2[\rightarrow \mathbb{R} \text{ by } h(\zeta) = \|g'_{v,w}(\zeta) - g'_{v,w}(0)\|_Y = (\|\cdot\| \circ (g'_{v,w} - C_{g'_{v,w}(0)}))(\zeta)$$

then as $g'_{v,w}$ is continuous we have by [example: 14.142] and [theorems: 14.173, 14.158] that h is continuous. Then as $[0, 1]$ is compact and $[0, 1] \subseteq] -2, 2[$ it follows from the extreme value theorem [theorem: 14.282] that there exist a $\xi_0 \in [0, 1]$ such that $\forall t \in [0, 1] \ h(t) \leq h(\xi_0)$. Hence we have $h(\xi_0) \leq \sup(h[0, 1]) \leq h(\xi_0)$ or

$$\|g'_{v,w}(\xi_0) - g'_{v,w}(0)\|_Y = h(\xi_0) = \sup(h([0, 1])) \quad (16.280)$$

As $h([0, 1]) = \{h(\zeta) | \zeta \in [0, 1]\} = \{\|g'_{v,w}(\zeta) - g'_{v,w}(0)\|_Y | \zeta \in [0, 1]\}$ it follows that

$$h(\xi_0) = \sup(\{\|g'_{v,w}(\zeta) - g'_{v,w}(0)\|_Y | \zeta \in [0, 1]\}) = \sup(\{\|(g_{v,w})'_\zeta - (g_{v,w})'_0\|_Y | \zeta \in [0, 1]\}) \quad (16.281)$$

Next

$$\begin{aligned}
& \|g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(w)(v)\|_Y = \\
& \|g_{v,w}(1) - g_{v,w}(0) - (g_{v,w})'_0 + (g_{v,w})'_0 - (g_{v,w})'_{\xi_0} + (g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y \leq \\
& \|g_{v,w}(1) - g_{v,w}(0) - (g_{v,w})'_0\|_Y + \|(g_{v,w})'_0 - (g_{v,w})'_{\xi_0}\|_Y + \|(g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y \stackrel{\text{eq: 16.272}}{\leq} \\
& \sup(\{\|(g_{v,w})'_\zeta - (g_{v,w})'_0\|_Y | \zeta \in [0, 1]\}) + \|(g_{v,w})'_0 - (g_{v,w})'_{\xi_0}\|_Y + \|(g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y \stackrel{\text{eq: 16.281, 16.280}}{=} \\
& \|g'_{v,w}(\xi_0) - g'_{v,w}(0)\|_Y + \|(g_{v,w})'_0 - (g_{v,w})'_{\xi_0}\|_Y + \|(g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y = \\
& \|(g_{v,w})'_{\xi_0} - (g_{v,w})'_0\|_Y + \|(g_{v,w})'_0 - (g_{v,w})'_{\xi_0}\|_Y + \|(g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y = \\
& 2 \cdot \|(g_{v,w})'_{\xi_0} - (g_{v,w})'_0\|_Y + \|(g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y \stackrel{\text{eq: 16.279}}{\leq} \\
& 8 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) + \|(g_{v,w})'_{\xi_0} - D_x^{[2]}f(w)(v)\|_Y \stackrel{\text{eq: 16.278}}{\leq} \\
& 8 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) + 2 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) = \\
& 10 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X)
\end{aligned}$$

proving that $\forall v, w \in X$ with $\|v\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}\right)$, $\|w\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}\right)$ we have

$$\|g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(w)(v)\|_Y \leq 10 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) \quad (16.282)$$

As for $v' = w$ and $w' = v$ we have also $\|v'\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}\right)$, $\|w'\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}\right)$ it follows that

$$\|g_{v',w'}(1) - g_{v',w'}(0) - D_x^{[2]}f(w')(v')\|_Y \leq 10 \cdot \varepsilon \cdot \|v'\|_X \cdot (\|v'\|_X + \|w'\|_X)$$

or

$$\|g_{w,v}(1) - g_{w,v}(0) - D_x^{[2]}f(v)(w)\|_Y \leq 10 \cdot \varepsilon \cdot \|w\|_X \cdot (\|v\|_X + \|w\|_X) \quad (16.283)$$

Now

$$\begin{aligned}
g_{v,w}(1) - g_{v,w}(0) &= (f \circ \psi_{v,w})(1) - (f \circ \varphi_{v,w})(1) - ((f \circ \psi_{v,w})(0) - (f \circ \varphi_{v,w})(0)) \\
&= f(\psi_{v,w}(1)) - f(\varphi_{v,w}(1)) - f(\psi_{v,w}(0)) + f(\varphi_{v,w}(0)) \\
&= f(x + 1 \cdot v + w) - f(x + 1 \cdot v) - f(x + 0 \cdot v + w) + f(x + 0 \cdot v) \\
&= f(x + v + w) - f(x + v) - f(x + w) + f(x) \\
&= f(x + w + v) - f(x + w) - f(x + v) + f(x) \\
&= f(x + 1 \cdot w + v) - f(x + 1 \cdot w) - f(x + 0 \cdot w + v) + f(x + 0 \cdot w) \\
&= f(\psi_{w,v}(1)) - f(\varphi_{w,v}(1)) - f(\psi_{w,v}(0)) + f(\varphi_{w,v}(0)) \\
&= (f \circ \psi_{w,v})(1) - (f \circ \varphi_{w,v})(1) - ((f \circ \psi_{w,v})(0) - (f \circ \varphi_{w,v})(0)) \\
&= g_{w,v}(1) - g_{w,v}(0)
\end{aligned}$$

substituting this in [eq: 16.283] gives

$$\|g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(v)(w)\|_Y \leq 10 \cdot \varepsilon \cdot \|w\|_X \cdot (\|v\|_X + \|w\|_X) \quad (16.284)$$

Next taking in account that $\|v\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right) \Rightarrow \|v\|_X \leq 1$, $\|w\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right) \Rightarrow \|w\|_X \leq 1$ we have

$$\begin{aligned}
&\|D_x^{[2]}f(v)(w) - D_x^{[2]}f(w)(v)\|_Y = \\
&\|g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(w)(v) - (g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(v)(w))\|_Y \leq \\
&\|g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(w)(v)\|_Y + \|g_{v,w}(1) - g_{v,w}(0) - D_x^{[2]}f(v)(w)\|_Y \stackrel{[\text{eq: 16.282, 16.284}]}{\leq} \\
&10 \cdot \varepsilon \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) + 10 \cdot \varepsilon \cdot \|w\|_X \cdot (\|v\|_X + \|w\|_X) = \\
&\varepsilon \cdot [10 \cdot \|v\|_X \cdot (\|v\|_X + \|w\|_X) + 10 \cdot \|w\|_X \cdot (\|v\|_X + \|w\|_X)] \stackrel{\|v\|_X \leq 1 \wedge \|w\|_X \leq 1}{\leq} \\
&\varepsilon \cdot [10 \cdot 1 \cdot (1 + 1) + 10 \cdot 1 \cdot (1 + 1)] = \\
&\varepsilon \cdot 40
\end{aligned}$$

As $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary it follows from [theorem: 10.31] that $\|D_x^{[2]}f(v)(w) - D_x^{[2]}f(w)(v)\|_Y \leq 0$ so that $\|D_x^{[2]}f(v)(w) - D_x^{[2]}f(w)(v)\|_Y = 0$ proving that $D_x^{[2]}f(v)(w) = D_x^{[2]}f(w)(v)$. Hence

$$\forall v, w \in X \text{ with } \|v\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right), \|w\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right) \text{ we have } D_x^{[2]}f(v)(w) = D_x^{[2]}f(w)(v)$$

Let $u, v \in X$ then for $v' = \frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|v\|_X + 1} \cdot v$, $w' = \frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|w\|_X + 1} \cdot w$ we have $\|v'\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)$, $\|w'\|_X < \min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)$ so that

$$\begin{aligned}
\frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|v\|_X + 1} \cdot \frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|w\|_X + 1} \cdot D_x^{[2]}f(v)(w) &= D_x^{[2]}f\left(\frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|v\|_X + 1} \cdot v\right)\left(\frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|w\|_X + 1} \cdot w\right) \\
&= D_x^{[2]}f(v')(w') \\
&= D_x^{[2]}f(w')(v') \\
&= D_x^{[2]}f\left(\frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|w\|_X + 1} \cdot w\right)\left(\frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|v\|_X + 1} \cdot v\right) \\
&= \frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|v\|_X + 1} \cdot \frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|w\|_X + 1} \cdot D_x^{[2]}f(w)(v)
\end{aligned}$$

proving, as $\frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|v\|_X + 1} \cdot \frac{\min\left(\frac{\delta}{4}, \frac{r}{4}, 1\right)}{\|w\|_X + 1} \neq 0$, that

$$D_x^{[2]}f(v)(w) = D_x^{[2]}f(w)(v) \text{ for any } v, w \in X.$$

□

Next we define the evaluation operator and prove that is linear and continuous hence C^∞ on its domain.

Definition 16.239. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $(x_1, \dots, x_n) \in X^n$ then the evaluation operator $\text{ev}_{x_1, \dots, x_n}$ is defined by

$$\text{ev}_{x_1, \dots, x_n}: L_n(X; Y) \rightarrow Y \text{ where } \text{ev}_{x_1, \dots, x_n}(L) = L(x_1: \dots: x_n)$$

Lemma 16.240. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $(x_1, \dots, x_n) \in X^n$ then

$$\text{ev}_{x_1, \dots, x_n} \in L(L_n(X; Y), Y)$$

Hence by [example: 16.180] $\text{ev}_{x_1, \dots, x_n}$ is C^∞ on $L_n(X; Y)$ and

$$\forall L \in L_n(X; Y) \text{ we have } D_L^{[n]} \text{ev}_{x_1, \dots, x_n} = \begin{cases} \text{ev}_{x_1, \dots, x_n} & \text{if } n = 1 \\ 0_n & \text{if } 1 < n \end{cases}$$

Proof. Let $\beta \in \mathbb{K}$, $L_1, L_2 \in L_n(X; Y)$ then

$$\begin{aligned} \text{ev}_{x_1, \dots, x_n}(L_1 + \beta \cdot L_2) &= (L_1 + \beta \cdot L_2)(x_1: \dots: x_n) \\ &\stackrel{[\text{theorem: 16.120}]}{=} L_1(x_1: \dots: x_n) + \beta \cdot L_2(x_1: \dots: x_n) \\ &= \text{ev}_{x_1, \dots, x_n}(L_1) + \beta \cdot \text{ev}_{x_1, \dots, x_n}(L_2) \end{aligned}$$

proving that

$$\text{ev}_{x_1, \dots, x_n} \in \text{Hom}(L_n(X; Y), Y)$$

Further if $L \in L_n(X; Y)$ then

$$\begin{aligned} \|\text{ev}_{x_1, \dots, x_n}(L)\|_Y &= \|L(x_1: \dots: x_n)\|_Y \\ &\stackrel{[\text{lemma: 16.122}]}{\leq} \|L\|_{L_n(X; Y)} \cdot \prod_{i=1}^n \|x_i\|_X \\ &= \left(\prod_{i=1}^n \|x_i\|_X \right) \cdot \|L\|_{L_n(X; Y)} \end{aligned}$$

proving by [theorem: 14.201] that $\text{ev}_{x_1, \dots, x_n}$ is continuous hence

$$\text{ev}_{x_1, \dots, x_n} \in L(L_n(X; Y), Y) \quad \square$$

Lemma 16.241. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , $(x_1, \dots, x_n) \in X^n$ and $f: U \rightarrow Y$ a function that is of class C^{n+1} then for the function

$$g: U \rightarrow Y \text{ defined by } g(y) = D_y^{[n]} f(x_1: \dots: x_n)$$

we have that g is of class C^1 and $\forall y \in U$ we have $\forall h \in X$ that $D_y g(h) = D_y^{[n+1]} f(h: x_1: \dots: x_n)$

Note 16.242. As f is of class C^{n+1} it follows from [theorem: 16.158] that f is of class C^n so that f is n -times differentiable on U . This proves that $\forall x \in U$ $D_x^{[n]} f$ exist so that g is well defined.

Proof. As f is of class C^{n+1} it follows that f is $(n+1)$ -times differentiable on U and the function

$$D^{[n+1]} f: U \rightarrow L_{n+1}(X; Y) \text{ is continuous}$$

Further using [theorem: 16.175] it follows that

$$D^{[n]} f \text{ is of class } C^1$$

Let $y \in U$ then we have

$$\begin{aligned} g(y) &= D_y^{[n]} f(x_1: \dots: x_n) \\ &= \text{ev}_{x_1, \dots, x_n}(D_y^{[n]} f) \\ &\stackrel{[\text{definition: 16.136}]}{=} \text{ev}_{x_1, \dots, x_n}(D^{[n]} f(y)) \\ &= (\text{ev}_{x_1, \dots, x_n} \circ D^{[n]} f)(y) \end{aligned}$$

proving that

$$g = (\text{ev}_{x_1, \dots, x_n} \circ D^{[n]}f)$$

By [lemma: 16.240] $\text{ev}_{x_1, \dots, x_n}$ is of class C^∞ and as $D^{[n]}f$ is of class C^1 , so it follows from the chain rule [see theorem: 16.197] that

$$g = \text{ev}_{x_1, \dots, x_n} \circ D^{[n]}f \text{ is of class } C^1 \quad (16.285)$$

and $\forall y \in U$ we have

$$\begin{aligned} D_y g &= D_y (\text{ev}_{x_1, \dots, x_n} \circ D^{[n]}f) \\ &\stackrel{[\text{theorem: 16.67}]}{=} D_{D^{[n]}f(y)} \text{ev}_{x_1, \dots, x_n} \circ D_y (D^{[n]}f) \\ &\stackrel{[\text{lemma: 16.240}]}{=} \text{ev}_{x_1, \dots, x_n} \circ D_y (D^{[n]}f) \\ &\stackrel{[\text{theorem: 16.140}]}{=} \text{ev}_{x_1, \dots, x_n} \circ D_y^{[n+1]}f \end{aligned}$$

proving that

$$D_y g = \text{ev}_{x_1, \dots, x_n} \circ D_y^{[n+1]}f$$

Hence if $h \in X$ we have

$$\begin{aligned} D_y g(h) &= (\text{ev}_{x_1, \dots, x_n} \circ D_y^{[n+1]}f)(h) \\ &= \text{ev}_{x_1, \dots, x_n}(D_y^{[n+1]}f(h)) \\ &= D_y^{[n+1]}f(h)(x_1 : \dots : x_n) \\ &= D_y^{[n+1]}f(h; x_1 : \dots : x_n) \end{aligned}$$

The lemma is then proved by the above and [eq: 16.285]. □

Lemma 16.243. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces then*

$$\text{ev}: L(X, Y) \cdot X \rightarrow Y \text{ defined by } \text{ev}(L, x) = L(x)$$

is of class C^∞ . Further $\forall L \in L(X, Y)$ and $x \in X$ we have

$$D_{(L, x)} \text{ev} \stackrel{[\text{definition: 16.129}]}{=} D_{(L, x)}^{[1]} \text{ev} = \text{ev}(L, *) + \text{ev}(*, x)$$

[see also theorem: 16.186].

Proof. Using [theorem: 16.186] it is enough to prove that ev is a bi-linear continuous function or in other words $\text{ev} \in L(L(X, Y), X; Y)$. Let $L_1, L_2 \in L(X, Y)$, $x \in X$ and $\alpha \in \mathbb{K}$ then

$$\begin{aligned} \text{ev}(L_1 + \alpha \cdot L_2, x) &= (L_1 + \alpha \cdot L_2)(x) \\ &= L_1(x) + \alpha \cdot L_2(x) \\ &= \text{ev}(L_1, x) + \alpha \cdot \text{ev}(L_2, x) \end{aligned}$$

and for $L \in L(X, Y)$, $x, y \in X$ and $\alpha \in \mathbb{K}$

$$\begin{aligned} \text{ev}(L, x + \alpha \cdot y) &= L(x + \alpha \cdot y) \\ &= L(x) + \alpha \cdot L(y) \\ &= \text{ev}(L, x) + \alpha \cdot \text{ev}(L, y) \end{aligned}$$

proving that

$$\text{ev} \in \text{Hom}(L(X, Y), X; Y)$$

Let $(L, x) \in L(X, Y) \cdot X$ then

$$\begin{aligned} \|\text{ev}(L, x)\|_Y &= \|L(x)\|_Y \\ &\leq \|L\|_{L(X, Y)} \cdot \|x\|_X \end{aligned}$$

proving by [theorem: 14.228] that

$$\text{ev} \in L(L(X, Y), X; Y) \quad \square$$

Lemma 16.244. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , $n \in \mathbb{N} \setminus \{1\}$ and

$$f: U \rightarrow Y \text{ a function that is of class } C^n$$

then for $h \in X$ the function

$$Df(*) (h): U \rightarrow Y \text{ defined by } (Df(*) (h))(y) = D_y f(h)$$

is of class C^{n-1} [hence, as $1 < n \Rightarrow 1 \in \{0, \dots, n-1\}$, it follows by [theorem: 16.158] that $Df(*) (h)$ is of class C^1]. Further $\forall x \in U \forall k, h \in X$ we have

$$(D_x(Df(*) (h)))(k) = ((D_x^{[2]} f)(k))(h)$$

Proof. As f is of class C^n and $1 < n$ we have by [theorem: 16.158] that f is of class C^1 hence 1-times differentiable on U . So given $h \in X$ we can define

$$\varphi_h: U \rightarrow L(X, Y) \cdot X \text{ by } \varphi_h(y) = (D_y^{[1]} f, h) \underset{\text{definition: 16.136}}{=} (D^{[1]} f(y), h) \quad (16.286)$$

Then we have for $y \in X$

$$\begin{aligned} (\pi_1 \circ \varphi_h)(y) &= \pi_1(\varphi_h(y)) \\ &= \pi_1((D^{[1]} f(y), h)) \\ &= D^{[1]} f(y) \\ (\pi_2 \circ \varphi_h)(y) &= \pi_2(\varphi_h(y)) \\ &= \pi_2((D^{[1]} f(y), h)) \\ &= h \\ &= C_h(y) \end{aligned}$$

proving that

$$\pi_1 \circ \varphi_h = D^{[1]} f \text{ and } \pi_2 \circ \varphi_h = C_h \quad (16.287)$$

As f is of class C^{n+1} it follows from [theorem: 16.175] that $D^{[1]} f$ is of class C^{n-1} . Hence it follows from [theorem: 16.197] that $\pi_1 \circ \varphi_h$ is of class C^{n-1} . Further as by [example: 16.179] C^∞ it follows that $\pi_2 \circ \varphi_h$ is also of class C^{n-1} . From [theorem: 16.193] it follows then that

$$\varphi_h \text{ is of class } C^{n-1} \quad (16.288)$$

As $1 < n$ it follows that $1 \in \{1, \dots, n-1\}$ so that by [theorem: 16.158] φ_h is of class C^1 hence Fréchet differentiable on U . Let $x \in U$ then φ_h is differentiable at x so that by [theorem: 16.101] $\pi_1 \circ \varphi_h, \pi_2 \circ \varphi_h$ are Fréchet differentiable at x and

$$\begin{aligned} \pi_1 \circ D_x \varphi_h &\underset{\text{theorem: 16.101}}{=} D_x(\pi_1 \circ \varphi_h) \underset{\text{eq: 16.287}}{=} D_x(D^{[1]} f) = D_x^{[2]} f \\ \pi_2 \circ D_x \varphi_h &\underset{\text{theorem: 16.101}}{=} D_x(\pi_2 \circ \varphi_h) \underset{\text{eq: 16.287}}{=} D_x C_h \underset{\text{example: 16.179}}{=} C_0 \end{aligned}$$

So that by [theorem: 16.101]

$$\forall x \in U \ D_x \varphi_h = (D_x^{[2]} f, C_0)_* \quad (16.289)$$

Let $y \in U$ then by [theorem: 16.138] f is 1-times differentiable at y [hence Fréchet differentiable at y], so we can define the function

$$Df(*) (h): U \rightarrow Y \text{ by } (Df(*) (h))(y) = D_y f(h)$$

Then we have

$$\begin{aligned} (Df(*) (h))(y) &= D_y f(h) \\ &= \text{ev}(D_y f, h) \\ &= \text{ev}(\varphi_h(y)) \\ &= (\text{ev} \circ \varphi_h)(y) \end{aligned}$$

proving that

$$Df(*) (h) = \text{ev} \circ \varphi_h \quad (16.290)$$

As ev is of class C^∞ on $L(X, Y) \cdot X$ by [lemma: 16.243] and φ_h is of class C^{n-1} on U by [eq: 16.288] it follows from the chain rule [see theorem: 16.197] that $\text{ev} \circ \varphi_h$ is of class C^{n-1} on U hence we have that

$$Df(*) (h) \text{ is } C^{n-1} \text{ on } U$$

Let $x \in U$ then by the above $Df(*) (h)$ is $(n-1)$ -times differentiable at x . Given that $1 < n$ it follows that $1 \in \{1, \dots, n-1\}$ so that by [theorem: 16.138] $Df(*) (h)$ is 1-times differentiable at x . Then for $k \in X$ we have

$$\begin{aligned} (D_x(Df(*) (h)))(k) &\stackrel{[\text{eq: 16.290}]}{=} (D_x(\text{ev} \circ \varphi_h))(k) \\ &\stackrel{[\text{theorem: 16.67}]}{=} (D_{\varphi_h(x)} \text{ev} \circ D_x \varphi_h)(k) \\ &\stackrel{[\text{eq: 16.286}]}{=} (D_{(D^{[1]} f(x), h)} \text{ev} \circ D_x \varphi_h)(k) \\ &\stackrel{[\text{lemma: 16.243}]}{=} ((\text{ev}(D^{[1]} f(x), *) + \text{ev}(*, h)) \circ D\varphi_h(x))(k) \\ &\stackrel{[\text{eq: 16.289}]}{=} ((\text{ev}(D^{[1]} f(x), *) + \text{ev}(*, h)) \circ (D_x^{[2]} f, C_0)_*)(k) \\ &= (\text{ev}(D^{[1]} f(x), *) + \text{ev}(*, h))((D_x^{[2]} f, C_0)_*(k)) \\ &= (\text{ev}(D^{[1]} f(x), *) + \text{ev}(*, h))(D_x^{[2]} f(k), C_0(k)) \\ &= (\text{ev}(D^{[1]} f(x), *) + \text{ev}(*, h))(D_x^{[2]} f(k), 0) \\ &= \text{ev}(D^{[1]} f(x), 0) + \text{ev}(D_x^{[2]} f(k), h) \\ &= D^{[1]} f(x)(0) + ((D_x^{[2]} f)(k))(h) \\ &= (D_x^{[2]} f(k))(h) \end{aligned}$$

proving that

$$(D_x(Df(*) (h)))(k) = (D_x^{[2]} f(k))(h) \quad \square$$

Lemma 16.245. Let $n \in \mathbb{N}$ with $2 < n$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , $(x_1, \dots, x_{n-2}) \in X^{n-2}$ and $f: U \rightarrow Y$ a function that is of class C^n then the function

$$g: U \rightarrow Y \text{ defined by } g(y) = D_y^{[n-2]} f(x_1: \dots: x_{n-2})$$

is of class C^2 and for $u, v \in X$ we have $D_x^{[2]} g(v)(u) = D_x^{[n]} f(v: u: x_1: \dots: x_{n-2})$

Proof. As f is of class C^n and $n-1, n-2 \in \{1, \dots, n\}$ it follows from [theorem: 16.158] that f is of class C^{n-2} and of class C^{n-1} so that

$$D^{[n-1]} f: U \rightarrow L_{n-1}(X; Y) \text{ is continuous and } D^{[n-2]} f: U \rightarrow L_{n-2}(X; Y) \text{ is continuous}$$

Let $y \in U$ then we have

$$\begin{aligned} g(y) &= D_y^{[n-2]} f(x_1: \dots: x_{n-2}) \\ &\stackrel{[\text{definition: 16.136}]}{=} D^{[n-2]} f(y)(x_1: \dots: x_{n-2}) \\ &\stackrel{[\text{definition: 16.239}]}{=} \text{ev}_{x_1, \dots, x_{n-2}}(D^{[n-2]} f(y)) \\ &= (\text{ev}_{x_1, \dots, x_{n-2}} \circ D^{[n-2]} f)(y) \end{aligned}$$

proving that

$$g = \text{ev}_{x_1, \dots, x_{n-2}} \circ D^{[n-2]} f \quad (16.291)$$

As $n = (n - 2) + 2$ it follows from [theorem: 16.173] that $D^{[n-2]}f$ is of class C^2 , further by [lemma: 16.240] $\text{ev}_{x_1, \dots, x_{n-2}}$ is of class C^∞ . Hence using the chain rule [theorem: 16.197] $\text{ev}_{x_1, \dots, x_{n-2}} \circ D^{[n-2]}f$ is of class C^2 which together with [eq: 16.291] proves that

g is of class C^2

From the above it follows by [theorem: 16.158] it follows that g is of class C^1 on U hence g is 1-times differentiable [or Fréchet differentiable] on U so that the following make sense
Let $y \in U$ then we have:

$$\begin{aligned}
 Dg(y) &= D_y g \\
 &\stackrel{[\text{eq: 16.291}]}{=} D_y (\text{ev}_{x_1, \dots, x_{n-2}} \circ D^{[n-2]}f) \\
 &\stackrel{[\text{theorem: 16.67}]}{=} D_{D^{[n-2]}f(y)} \text{ev}_{x_1, \dots, x_{n-2}} \circ D_y (D^{[n-2]}f) \\
 &\stackrel{[\text{lemma: 16.240}]}{=} \text{ev}_{x_1, \dots, x_{n-2}} \circ D_y (D^{[n-2]}f) \\
 &= \text{ev}_{x_1, \dots, x_{n-2}} \circ D_y^{[n-1]}f
 \end{aligned} \tag{16.292}$$

Let $u \in X$ then we have

$$\begin{aligned}
 Dg(y)(u) &\stackrel{[\text{eq: 16.292}]}{=} (\text{ev}_{x_1, \dots, x_{n-2}} \circ D_y^{[n-1]}f)(u) \\
 &= \text{ev}_{x_1, \dots, x_{n-2}} (D_y^{[n-1]}f(u)) \\
 &= D_y^{[n-1]}f(u)(x_1: \dots: x_{n-2}) \\
 &= D_y^{[n-1]}f(u: x_1: \dots: x_{n-2}) \\
 &= \text{ev}_{u, x_1, \dots, x_{n-2}} (D_y^{[n-1]}f) \\
 &= \text{ev}_{u, x_1, \dots, x_{n-2}} (D^{[n-1]}f(y)) \\
 &= (\text{ev}_{u, x_1, \dots, x_{n-2}} \circ D^{[n-1]}f)(y)
 \end{aligned} \tag{16.293}$$

Define now

$$Dg(*) (u): U \rightarrow Y \text{ by } (Dg(*) (u))(y) = Dg(y)(u)$$

then we have $\forall y \in U$ $(Dg(*) (u))(y) = Dg(y)(u) \stackrel{[\text{eq: 16.293}]}{=} (\text{ev}_{u, x_1, \dots, x_{n-2}} \circ D^{[n-1]}f)(y)$ so that

$$Dg(*) (u) = \text{ev}_{u, x_1, \dots, x_{n-2}} \circ D^{[n-1]}f \tag{16.294}$$

As $D^{[n-1]}f$ is of class C^1 [see theorem: 16.175] and $\text{ev}_{x_1, \dots, x_{n-2}}$ is of class C^∞ [see lemma: 16.240], it follows from the chain rule [theorem: 16.197] and the above that

$$Dg(*) (u) \text{ is of class } C^1 \text{ hence differentiable on } U$$

and for $y \in U$ that

$$\begin{aligned}
 D_y (Dg(*) (u)) &\stackrel{[\text{eq: 16.294}]}{=} D_y (\text{ev}_{u, x_1, \dots, x_{n-2}} \circ D^{[n-1]}f) \\
 &\stackrel{[\text{theorem: 16.67}]}{=} D_{D^{[n-1]}f(y)} \text{ev}_{u, x_1, \dots, x_{n-2}} \circ D_y (D^{[n-1]}f) \\
 &\stackrel{[\text{lemma: 16.240}]}{=} \text{ev}_{u, x_1, \dots, x_{n-2}} \circ D_y (D^{[n-1]}f) \\
 &= \text{ev}_{u, x_1, \dots, x_{n-2}} \circ D_y^{[n]}f
 \end{aligned} \tag{16.295}$$

Let additional $v \in X$ then

$$\begin{aligned}
 D_y^{[2]}g(v)(u) &\stackrel{[\text{lemma: 16.244}]}{=} (D_y (Dg(*) (u)))(v) \\
 &\stackrel{[\text{eq: 16.295}]}{=} (\text{ev}_{u, x_1, \dots, x_{n-2}} \circ D_y^{[n]}f)(v) \\
 &= \text{ev}_{u, x_1, \dots, x_{n-2}} (D_y^{[n]}f(v)) \\
 &= D_y^{[n]}f(v)(u: x_1: \dots: x_{n-2}) \\
 &= D_y^{[n]}f(v: u: x_1: \dots: x_{n-2})
 \end{aligned}$$

proving the last part of the lemma. $\square$

Lemma 16.246. Let $n \in \mathbb{N} \setminus \{1\}$, $\langle X, \|\cdot\|_X \rangle$ a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set in X , $f: U \rightarrow Y$ a function that is of class C^n then $\forall x \in U$ and $\forall i \in \{1, \dots, n-1\}$ we have for $(u_1, \dots, u_n) \in X^n$ that

$$D_x^n f(u_1: \dots: u_n) = D_x^n f(u_{(i \leftrightarrow i+1)(1)}: \dots: u_{(i \leftrightarrow i+1)(n)})$$

Proof. We prove this by induction on n , so let

$S = \{n \in \{2, \dots, \infty\} \mid \text{If } f: U \rightarrow Y \text{ is of class } C^n \text{ then } \forall x \in U \wedge \forall i \in \{1, \dots, n-1\} \text{ we have that for } (u_1, \dots, u_n) \in X^n \ D_x^n f(u_1: \dots: u_n) = D_x^n f(u_{(i \leftrightarrow i+1)(1)}: \dots: u_{(i \leftrightarrow i+1)(n)})\}$

then we have:

$2 \in S$. Let $f: U \rightarrow Y$ be a function that is of class C^2 , $x \in U$ then for $i \in \{1, \dots, 2-1\} = \{1\} \Rightarrow i = 1$ and $(u_1, u_2) \in X^2$ we have

$$\begin{aligned} D_x^{[2]} f(u_1: \dots: u_2) &= D_x^{[2]} f(u_1)(u_2) \\ &\stackrel{[\text{theorem: 16.238}]}{=} D_x^{[2]} f(u_2)(u_1) \\ &= D_x^{[2]} f(u_2: \dots: u_1) \\ &= D_x^{[2]} f(u_{(1 \leftrightarrow 2)(1)}: \dots: u_{(1 \leftrightarrow 2)(2)}) \end{aligned}$$

proving that $2 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $f: U \rightarrow Y$ be a function that is of class C^{n+1} , $x \in U$, $(u_1, \dots, u_{n+1}) \in X^{n+1}$ then for $i \in \{1, \dots, (n+1)-1\} = \{1, \dots, n\}$ we have the following cases to consider:

$i = 1$. Then $(i \leftrightarrow i+1)_{n+1} = (1 \leftrightarrow 2)_{n+1} \in P_{n+1}$, so that $(i \leftrightarrow i+1)_{n+1}(1) = 2$, $(i \leftrightarrow i+1)_{n+1}(2) = 1$ and $\forall k \in \{3, \dots, n+1\} \ (i \leftrightarrow i+1)_{n+1}(k) = k$. Define

$$g: U \rightarrow Y \text{ by } g(y) = D_y^{[(n+1)-2]} f(u_3: \dots: u_{n+1})$$

then by the previous lemma [lemma: 16.245] g is of class C^2 and

$$\begin{aligned} D_x^{[n+1]} f(u_1: u_2: \dots: u_{n+1}) &\stackrel{[\text{lemma: 16.245}]}{=} D_x^{[2]} g(u_1)(u_2) \\ &\stackrel{[\text{theorem: 16.238}]}{=} D_x^{[2]} g(u_2)(u_1) \\ &\stackrel{[\text{lemma: 16.245}]}{=} D_x^{[n+1]} f(u_2: u_1: \dots: u_{n+1}) \\ &= D_x^{[n+1]} f(u_{(i \leftrightarrow i+1)(1)}: u_{(i \leftrightarrow i+1)(2)}: \dots: u_{(i \leftrightarrow i+1)(n+1)}) \end{aligned}$$

proving that in this case $n+1 \in S$.

$i \in \{2, \dots, n\}$. Then $\forall k \in \{1, \dots, n-1\}$ we have

$$\begin{aligned} (i-1 \leftrightarrow i)_n(k) + 1 &= 1 + \begin{cases} i-1 & \text{if } k=i \\ i & \text{if } k=i-1 \\ k & \text{if } k \neq i-1, i \end{cases} \\ &= \begin{cases} i & \text{if } k=i \\ i+1 & \text{if } k=i-1 \\ k+1 & \text{if } k \neq i-1, i \end{cases} \end{aligned} \tag{16.296}$$

further

$$\begin{aligned}
 \left(i \overset{\leftrightarrow}{n+1} i+1\right)(k+1) &= \begin{cases} i & \text{if } k+1 = i+1 \\ i+1 & \text{if } k+1 = i \\ k+1 & \text{if } k+1 \neq i+1, i \end{cases} \\
 &= \begin{cases} i & \text{if } k = i \\ i+1 & \text{if } k = i-1 \\ k+1 & \text{if } k \neq i-1, i \end{cases} \\
 &\stackrel{[\text{eq: 16.296}]}{=} (i-1 \overset{\leftrightarrow}{n} i)(k) + 1
 \end{aligned}$$

so that

$$\forall k \in \{1, \dots, n-1\} \text{ we have } (i-1 \overset{\leftrightarrow}{n} i)(k) + 1 = \left(i \overset{\leftrightarrow}{n+1} i+1\right)(k+1) \quad (16.297)$$

Let $y \in U$ then, as f is of class C^{n+1} , we have also that f is of class C^n [see theorem: 16.158], so we have, if we define $(y_1, \dots, y_n)$ by $y_i = u_{i+1}$, that

$$\begin{aligned}
 D_y^{[n]} f(u_2 : \dots : u_{n+1}) &= \\
 D_y^{[n]} f(y_1 : \dots : y_n) &\quad n \in S, i-1 \in \overline{\{1, \dots, n-1\}} \\
 D_y^{[n]} f(y_{(i-1 \overset{\leftrightarrow}{n} i)(1)} : \dots : y_{(i-1 \overset{\leftrightarrow}{n} i)(n)}) &= \\
 D_y^{[n]} f(u_{(i-1 \overset{\leftrightarrow}{n} i)(1)+1} : \dots : u_{(i-1 \overset{\leftrightarrow}{n} i)(n)+1}) &\quad [\text{eq: 16.297}] \\
 D_y^{[n]} f(u_{(i \overset{\leftrightarrow}{n+1} i+1)(2)} : \dots : u_{(i \overset{\leftrightarrow}{n+1} i+1)(n+1)}) &\quad (16.298)
 \end{aligned}$$

Define then g, h [as in [lemma: 16.241]]

$$g: U \rightarrow Y \text{ by } g(y) = D_y^{[n]} f(u_2 : \dots : u_{n+1}) \quad (16.299)$$

$$h: U \rightarrow Y \text{ by } h(y) = D_y^{[n]} f(u_{(i \overset{\leftrightarrow}{n+1} i+1)(2)} : \dots : u_{(i \overset{\leftrightarrow}{n+1} i+1)(n+1)})$$

then we have by [eq: 16.298] that

$$g = h \quad (16.300)$$

As $1 < 2 \leq i < i+1$ we have that $\left(i \overset{\leftrightarrow}{n+1} i+1\right)(1) = 1$ so that

$$\begin{aligned}
 D_x^{[n+1]} f(u_1 : \dots : u_{n+1}) &= D_x^{[n+1]} f(u_{(i \overset{\leftrightarrow}{n+1} i+1)(1)} : u_2 : \dots : u_{n+1}) \\
 &\stackrel{[\text{lemma: 16.241}]}{=} D_x g(u_{(i \overset{\leftrightarrow}{n+1} i+1)(1)}) \\
 &\stackrel{[\text{eq: 16.300}]}{=} D_y h(u_{(i \overset{\leftrightarrow}{n+1} i+1)(1)}) \\
 &\stackrel{[\text{lemma: 16.241}]}{=} D_x^{[n+1]} f(u_{(i \overset{\leftrightarrow}{n+1} i+1)(1)} : \dots : u_{(i \overset{\leftrightarrow}{n+1} i+1)(n+1)})
 \end{aligned}$$

proving that in this case we have also $n+1 \in S$.

Mathematical induction proves then the theorem. $\square$

Having done all the heavy lifting with the previous lemma's we can now prove that higher order differentials are symmetric in there arguments.

Theorem 16.247. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$ a normed space, $\langle Y, \|\cdot\|_Y \rangle$ a Banach space, U a open set and $f: U \rightarrow Y$ a function that is of class C^n then $\forall x \in U, \forall \sigma \in P_n$ and $\forall (x_1, \dots, x_n) \in X^n$ we have

$$D_x^{[n]} f(x_1 : \dots : x_n) = D_x^{[n]} f(x_{\sigma(1)} : \dots : x_{\sigma(n)})$$

Note 16.248. As $D_x^n f(y_1, \dots, y_n) = D_x^{[n]} f(y_1 : \dots : y_n)$ we have also

$$D_x^n f(x_1, \dots, x_n) = D_x^n f(x_{\sigma(1)}, \dots, x_{\sigma(n)})$$

Proof. For $n \in \mathbb{N}$ we have two possibilities to consider:

$n = 1$. Then $P_n = \{\text{Id}_{\{1\}}\}$ so that we have trivially

$$D_x^{[1]} f\left(\underbrace{x_1 \dots x_1}_1\right) = D_x^{[1]} f\left(\underbrace{x_{\sigma(1)} \dots x_{\sigma(1)}}_1\right)$$

$n \in \mathbb{N} \setminus \{1\}$. Then $\forall i \in \{1, \dots, n-1\}$ we have by [lemma: 16.246] that

$$D_x^{[n]} f(x_1 \dots x_n) = D_x^{[n]} f(x_{(i \leftrightarrow i+1)(1)} \dots x_{(i \leftrightarrow i+1)(n)})$$

Let $\sigma \in P_n$ then by the above and [lemma: 16.237] it follows that

$$D_x^{[n]} f(x_1 \dots x_n) = D_x^{[n]} f(x_{(\sigma 1)} \dots x_{(\sigma n)}) \quad \square$$

16.5 Higher Order Partial Differentiation

16.5.1 Linear and multilinear mappings.

In order to define higher order partial differentiability we work with functions $f: A \rightarrow Y$ where $A \subseteq \prod_{i \in \{1, \dots, n\}} X_i$. For this we need to extend the definition of $L_n(X; Y)$ [see definition: 16.110].

Definition 16.249. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space then we define $L_n(X_1 \dots X_n; Y)$ recursively as follows

$$L_n(X_1 \dots X_n; Y) = \begin{cases} L(X_1, Y) & \text{if } n = 1 \\ L(X_1, L_{n-1}(X_2 \dots X_n)) & \text{if } 1 < n \end{cases}$$

Note 16.250. If $\forall i \in \{1, \dots, n\} X_i = X$ then $L_n(X_1, \dots, X_n; Y)$ reduces to $L_n(X; Y)$

Example 16.251.

$$\begin{aligned} L_4(X_1 \dots X_4; Y) &= L(X_1, L_3(X_2 \dots X_4; Y)) \\ &= L(X_1, L(X_2, L_2(X_3 \dots X_4; Y))) \\ &= L(X_1, L(X_2, L(X_3, L_1(X_4 \dots X_4; Y)))) \\ &= L(X_1, L(X_2, L(X_3, L(X_4, Y)))) \end{aligned}$$

Further we have

$$\begin{aligned} L_2(X_1 \dots X_2; L_2(X_3 \dots X_4; Y)) &= L(X_1, L_1(X_2 \dots X_2; L_2(X_3 \dots X_4; Y))) \\ &= L(X_1, L(X_2, L_2(X_3 \dots X_4; Y))) \\ &= L(X_1, L(X_2, L(X_3, L_1(X_4 \dots X_4; Y)))) \\ &= L(X_1, L(X_2, L(X_3, L(X_4, Y)))) \\ &= L_4(X_1 \dots X_4; Y) \end{aligned}$$

The above example suggest the following two lemmas.

Lemma 16.252. Let $n \in \mathbb{N}$ and $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in n+1}$ be a finite family of normed spaces then we have

$$L_{n+1}(X_1 \dots X_{n+1}; Y) = L_n(X_1 \dots X_n; L(X_{n+1}, Y))$$

Proof. We prove this by induction so let

$S = \{n \in \mathbb{N} \mid \text{If } \{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n+1\}} \text{ is a family of normed spaces and } \langle Y, \|\cdot\|_Y \rangle \text{ a normed space then } L_{n+1}(X_1 \dots X_{n+1}; Y) = L_n(X_1 \dots X_n; (L(X_{n+1}, Y)))\}$

then we have:

$1 \in \mathbf{S}$. If $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, 2\}}$ is a family of normed spaces and $\langle Y, \|\cdot\|_Y \rangle$ a normed space then

$$L_2(X_1 \dots X_2; Y) = L(X_1, L_1(X_2 \dots X_2; Y)) = L_1\left(\underbrace{X_1 \dots X_1}_1; L(X_2, Y)\right)$$

proving that $1 \in S$.

$n \in \mathbf{S} \Rightarrow n+1 \in \mathbf{S}$. If $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, (n+1)+1\}}$ is a family of normed spaces and $\langle Y, \|\cdot\|_Y \rangle$ a normed space. Define $\{Y_i\}_{i \in \{1, \dots, n+1\}}$ by $Y_i = X_{i+1}$ then we have

$$\begin{aligned} L_{(n+1)+1}(X_1 \dots X_{(n+1)+1}; Y) &\stackrel{\text{def}}{=} L(X_1, L_{n+1}(X_2 \dots X_{(n+1)+1}; Y)) \\ &= L(X_1, L_{n+1}(Y_1 \dots Y_{n+1}; Y)) \\ &\stackrel{n \in S}{=} L(X_1, L_n(Y_1, \dots, Y_n; L(Y_{n+1}, Y))) \\ &= L(X_1, L_n(X_2 \dots X_{n+1}; L(X_{(n+1)+1}, Y))) \\ &\stackrel{\text{def}}{=} L_{n+1}(X_1 \dots X_{n+1}; L(X_{(n+1)+1}, Y)) \end{aligned}$$

proving that

$$n+1 \in S$$

□

Lemma 16.253. Let $n, m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n+m\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space then

$$L_{n+m}(X_1 \dots X_{n+m}) = L_n(X_1 \dots X_n; L_m(X_{n+1} \dots X_{n+m}; Y))$$

Proof. For $n \in \mathbb{N}$ we have two cases to consider:

$n = 1$. Then

$$\begin{aligned} L_{1+m}(X_1, \dots, X_{1+m}; Y) &= L(X_1, L_m(X_2 \dots X_{1+m}; Y)) \\ &= L_1(X_1 \dots X_1; L_m(X_2 \dots X_{1+m}; Y)) \end{aligned}$$

proving the lemma for this case.

$1 < n$. We prove this case by induction on m , so let

$S_n = \{m \in \mathbb{N} \mid \text{If } \{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n+m\}} \text{ is a family of normed spaces then } L_{n+m}(X_1 \dots X_{n+m}; Y) = L_n(X_1 \dots X_n; L_m(X_{n+1} \dots X_{n+m}; Y))\}$

then we have:

$1 \in S_n$. If $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n+1\}}$ is a family of normed spaces then

$$L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1; L_n(X_2 \dots X_{n+1}; Y)) = L_1(X_1 \dots X_1; L_n(X_2 \dots X_{n+1}; Y))$$

proving that $1 \in S_n$.

$m \in S_n \Rightarrow m+1 \in S_n$. Let $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n+(m+1)\}}$ be a family of normed spaces. Define $\{Y_i\}_{i \in \{1, \dots, n+m\}}$ by $Y_i = X_{i+1}$ then we have

$$\begin{aligned} L_{n+(m+1)}(X_1 \dots X_{n+(m+1)}; Y) &= \\ L(X_1, L_{n+m}(X_2 \dots X_{n+m+1}; Y)) &= \\ L(X_1, L_{n+m}(Y_1 \dots Y_{n+m}; Y)) &\stackrel{m \in S_n}{=} \\ L(X_1, L_n(Y_1 \dots Y_n; L_m(Y_{n+1} \dots Y_{n+m}; Y))) &\stackrel{1 < n \wedge \text{lemma: 16.252}}{=} \\ L(X_1, L_{n-1}(Y_1 \dots Y_{n-1}; L(Y_n, L_m(Y_{n+1} \dots Y_{n+m}; Y)))) &\stackrel{\text{def}}{=} \\ L(X_1, L_{n-1}(Y_1 \dots Y_{n-1}; L_{m+1}(Y_n \dots Y_{n+m}; Y))) &= \\ L(X_1, L_{n-1}(X_2, \dots, X_n; L_{m+1}(X_{n+1} \dots X_{n+m+1}; Y))) &\stackrel{\text{def}}{=} \\ L_n(X_1 \dots X_n; L_{m+1}(X_{n+1} \dots X_{n+(m+1)}; Y)) & \end{aligned}$$

proves that $n + 1$

Mathematical induction proves the lemma for the case $1 < n$. $\square$

We extend now the definition of $L(x_1 : \dots : x_n)$ [see definition: 16.114] to the more general case.

Definition 16.254. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n+m\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $m \in \mathbb{N}$ with $m \leq n$ then if $L \in L_n(X_1 \dots X_n; Y)$ and $(x_1, \dots, x_m) \in \prod_{i \in \{1, \dots, m\}} X_i$ then we define

$$L(x_1 : \dots : x_m) = \begin{cases} L(x_1) & \text{if } m = 1 \\ (L(x_1))(x_2 : \dots : x_m) & \text{if } 1 < m \end{cases}$$

Lemma 16.255. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $m \in \mathbb{N}$ with $m \leq n$ then if $L \in L_n(X_1 \dots X_n; Y)$ and $(x_1, \dots, x_m) \in \prod_{i \in \{1, \dots, m\}} X_i$ then

$$L(x_1 : \dots : x_m) \in \begin{cases} Y & \text{if } m = n \\ L_{n-m}(X_{m+1} \dots X_n; Y) & \text{if } m < n \end{cases}$$

Proof. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}} \text{ is a family of normed space and } L \in L_n(X_1 \dots X_n; Y) \right.$$

then for every $m \in \{1, \dots, n\}$ we have for every $(x_1, \dots, x_m) \in \prod_{i \in \{1, \dots, m\}} X_i$ that

$$L(x_1 : \dots : x_m) \in \begin{cases} Y & \text{if } m = n \\ L_{n-m}(X_{m+1} \dots X_n; Y) & \text{if } m < n \end{cases}$$

then we have:

$1 \in S$. If $m \in \{1, \dots, 1\} = \{1\}$ then for $L \in L_1(X_1 \dots X_1; Y) = L(X_1, Y)$ we have for $(x_1) \in \prod_{i \in \{1\}} X_i$ that $L(x_1 : \dots : x_1) = L(x_1) \in Y$ which as $m = 1 = n$ proves that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $L \in L_{n+1}(X_1 \dots X_{n+1}; Y)$ and $m \in \{1, \dots, n + 1\}$ then we must consider the following cases:

$m = 1$. Let $(x_1) \subseteq X^1$ then, as $L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$, we have

$$L\left(\underbrace{x_1 : \dots : x_1}_1\right) = L(x_1) \in L_n(X_2 \dots X_{n+1}; Y) \stackrel{m=1}{=} L_{(n+1)-m}(X_{m+1} \dots X_{n+1}; Y)$$

$m = n + 1$. Let $(x_1, \dots, x_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} X_i$ then, as

$$L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$$

we have

$$L(x_1) \in L_n(X_2 \dots X_{n+1}; Y)$$

Define $\{\langle Y_i, \|\cdot\|_{Y,i} \rangle\}_{i \in \{1, \dots, n\}}$ by $\langle Y_i, \|\cdot\|_{Y,i} \rangle = \langle X_{i+1}, \|\cdot\|_{i+1} \rangle$ and $(y_1, \dots, y_n) \in \prod_{i \in \{1, \dots, n\}} Y_i$ by $y_i = x_{i+1}$ then $L(x_1) \in L_n(Y_1 \dots Y_n; Y)$. As $n \in S$ we have that $L(x_1)(y_1 : \dots : y_n) \in Y$, so that $L(x_1)(x_2 : \dots : x_{n+1}) \in Y$. Hence

$$L(x_1 : \dots : x_m) = L(x_1 : \dots : x_{n+1}) = L(x_1)(x_2 : \dots : x_{n+1}) \in Y$$

$m \in \{2, \dots, n\}$. Let $(x_1, \dots, x_m) \in \prod_{i \in \{1, \dots, m\}} X_i$ then as

$$L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$$

we have that

$$L(x_1) \in L_n(X_2 \dots X_{n+1}; Y)$$

Define $\{\langle Y_i, \|\cdot\|_{Y,i} \rangle\}_{i \in \{1, \dots, n\}}$ by $\langle Y_i, \|\cdot\|_{Y,i} \rangle = \langle X_{i+1}, \|\cdot\|_{i+1} \rangle$ and $(y_1, \dots, y_{m-1}) \in \prod_{i \in \{1, \dots, n\}} Y_i$ by $y_i = x_{i+1}$ then

$$L(x_1) \in L(Y_1 \dots Y_n; Y)$$

so, as $1 \leq m-1 < n$ and $n \in S$ it follows that

$$L(x_1)(y_1: \dots: y_{m-1}) \in L_{n-(m-1)}(Y_{(m-1)+1} \dots Y_n; Y)$$

hence

$$L(x_1)(x_2: \dots: x_m) \in L_{(n+1)-m}(X_{m+1} \dots X_{n+1}; Y)$$

so that

$$L(x_1: \dots: x_m) = L(x_1)(x_2: \dots: x_m) \in L_{(n+1)-m}(X_{m+1} \dots X_{n+1}; Y)$$

So we have proved that

$$L(x_1: \dots: x_m) \in \begin{cases} Y & \text{if } m = n+1 \\ L_{(n+1)-m}(X_{m+1} \dots X_{n+1}; Y) & \text{if } m < n \end{cases}$$

from which it follows that

$$n+1 \in S$$

□

Lemma 16.256. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_n(X_1 \dots X_n; Y)$ then $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$(L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) = L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n)$$

Proof. The proof is by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \forall \alpha \in \mathbb{K} \text{ and } \forall L_1, L_2 \in L_n(X; Y) \text{ we have } \forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i \text{ that } (L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) = L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n) \right\}$$

then we have:

$1 \in S$. Take $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_1(X_1 \dots X_1; Y) = L(X, Y)$ then we have for $(x_1) \in \prod_{i \in \{1\}} X_i$ that

$$\begin{aligned} (L_1 + \alpha \cdot L_2)(x_1: \dots: x_1) &= (L_1 + \alpha \cdot L_2)(x_1) \\ &\stackrel{\text{pointwise definition}}{=} L_1(x_1) + \alpha \cdot L_2(x_1) \\ &= L_1(x_1: \dots: x_1) + \alpha \cdot L_2(x_1: \dots: x_1) \end{aligned}$$

proving that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Take $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$ and $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ then

$$(L_1 + \alpha \cdot L_2)(x_1) \stackrel{\text{pointwise definition}}{=} L_1(x_1) + \alpha \cdot L_2(x_1)$$

hence

$$\begin{aligned} (L_1 + \alpha \cdot L_2)(x_1: \dots: x_n) &= ((L_1 + \alpha \cdot L_2)(x_1))(x_2: \dots: x_n) \\ &= (L_1(x_1) + \alpha \cdot L_2(x_1))(x_2: \dots: x_n) \end{aligned} \tag{16.301}$$

As $L_1(x_1), L_2(x_1) \in L_n(X_1 \dots X_n; Y)$ and $n \in S$ it follows that

$$\begin{aligned} (L_1(x_1) + \alpha \cdot L_2(x_1))(x_2: \dots: x_n) &= (L_1(x_1))(x_2: \dots: x_n) + \alpha \cdot (L_2(x_1))(x_2: \dots: x_n) \\ &= L_1(x_1: \dots: x_n) + \alpha \cdot L_2(x_1: \dots: x_n) \end{aligned}$$

which combined with [eq: 16.301] proves that

$$(L_1 + \alpha \cdot L_2)(x_1 : \dots : x_n) = L_1(x_1 : \dots : x_n) + \alpha \cdot L_2(x_1 : \dots : x_n)$$

Hence we have that

$$n+1 \in S$$

□

Lemma 16.257. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $L \in L_n(X_1 \dots X_n; Y)$ and $i \in \{1, \dots, n\}$ then we have for

$$\begin{aligned} & \left(x_1 \dots \underbrace{r + \alpha \cdot t}_{i} \dots, x_n \right) \in \prod_{i \in \{1, \dots, n\}} X_i \\ \text{that} \quad & L\left(x_1 : \dots \underbrace{r + \alpha \cdot t}_{i} : x_n\right) = L\left(x_1 : \dots \underbrace{r}_{i} : x_n\right) + \alpha \cdot L\left(x_1 : \dots \underbrace{t}_{i} : x_n\right) \end{aligned}$$

Proof. We use mathematical induction on n for the proof, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(X_1 \dots X_n; Y) \text{ then } \forall i \in \{1, \dots, n\} \text{ we have for } \left(x_1, \dots, \underbrace{r + \alpha \cdot t}_{i}, \dots, x_n \right) \in \prod_{i \in \{1, \dots, n\}} X_i \text{ that } L\left(x_1 : \dots \underbrace{r + \alpha \cdot t}_{i} : x_n\right) = L\left(x_1 : \dots \underbrace{r}_{i} : x_n\right) + \alpha \cdot L\left(x_1 : \dots \underbrace{t}_{i} : x_n\right) \right\}$$

then we have:

$1 \in S$. Let $L \in L_1(X_1 \dots X_1; Y) = L(X_1, Y)$ then if $i \in \{1, \dots, 1\} = \{1\}$ we have $i = 1$. So given

$$(r + \alpha \cdot t) = \left(x_1, \dots, \underbrace{r + \alpha \cdot t}_{i}, \dots, x_n \right) \in X^1$$

we have

$$\begin{aligned} L\left(x_1 : \dots \underbrace{r + \alpha \cdot t}_{i} : x_n\right) &= \\ & \quad L(r + \alpha \cdot t) \quad L \in L(\overline{X_1}, Y) \\ & \quad L(r) + \alpha \cdot L(t) \\ L\left(x_1 : \dots \underbrace{r}_{i} : x_n\right) + \alpha \cdot L\left(x_1 : \dots \underbrace{t}_{i} : x_n\right) &= \end{aligned}$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$ and take $i \in \{1, \dots, n+1\}$ then we have either:

$i = 1$. Then for $\left(x_1, \dots, \underbrace{r + \alpha \cdot t}_{i}, \dots, x_{n+1} \right) = (r + \alpha \cdot t, x_2, \dots, x_{n+1})$ we have

$$\begin{aligned} L\left(x_1 : \dots \underbrace{r + \alpha \cdot t}_{i} : x_{n+1}\right) &= \\ & \quad L(r + \alpha \cdot t; x_2 : \dots : x_{n+1}) \\ & \quad (L(r + \alpha \cdot t))(x_2 : \dots : x_{n+1}) \\ & \quad (L(r) + \alpha \cdot L(t))(x_2 : \dots : x_{n+1}) \quad [\text{lemma: 16.256}] \\ & \quad (L(r))(x_2 : \dots : x_{n+1}) + \alpha \cdot L(t)(x_2 : \dots : x_{n+1}) \\ & \quad L(r; x_2 : \dots : x_{n+1}) + \alpha \cdot L(t; x_2 : \dots : x_{n+1}) \\ L\left(x_1 : \dots \underbrace{r}_{i} : x_{n+1}\right) + \alpha \cdot L\left(x_1 : \dots \underbrace{t}_{i} : x_{n+1}\right) &= \end{aligned}$$

proving

$$n+1 \in S$$

$i \in \{2, \dots, n+1\}$. Then for $\left(x_1, \dots, \underbrace{r + \alpha \cdot t}_{i}, \dots, x_{n+1}\right) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have
 $\left(x_2, \dots, \underbrace{r + \alpha \cdot t}_{i-1}, \dots, x_{n+1}\right) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ which as $L(x_1) \in L_n(X_2 \dots X_{n+1}; Y)$
 and $n \in S$ results in

$$\begin{aligned} L\left(x_1: \dots \underbrace{r + \alpha \cdot t}_{i} \dots x_{n+1}\right) &= \\ (L(x_1))\left(x_2: \dots \underbrace{r + \alpha \cdot t}_{i-1} \dots x_n\right) &\stackrel{n \in S}{=} \\ (L(x_1))\left(x_2: \dots \underbrace{r}_{i-1} \dots x_n\right) + \alpha \cdot (L(x_1))\left(x_2: \dots \underbrace{t}_{i-1} \dots x_n\right) &= \\ L\left(x_1: \dots \underbrace{r}_{i} \dots x_n\right) + \alpha \cdot L\left(x_1: \dots \underbrace{t}_{i} \dots x_n\right) &= \end{aligned}$$

so that in this case we have also

$$n+1 \in S$$

So in all cases we have

$$n+1 \in S$$

□

Lemma 16.258. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $L \in L_n(X_1 \dots X_n; Y)$ and $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ then we have

$$\|L(x_1: \dots: x_n)\|_Y \leq \|L\|_{L_n(X_1 \dots X_n; Y)} \cdot \prod_{i=1}^n \|x_i\|_i$$

Proof. We prove this by induction, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(X_1 \dots X_n; Y) \text{ then } \forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i \text{ we have } \|L(x_1: \dots: x_n)\|_Y \leq \|L\|_{L_n(X; Y)} \cdot \prod_{i=1}^n \|x_i\|_i \right\}$$

then we have:

$1 \in S$. If $L \in L_1(X_1 \dots X_1; Y) = L(X_1, Y)$ then $\forall (x_1) \in \prod_{i \in \{1\}} X_i$ we have

$$\begin{aligned} \|L(x_1: \dots: x_1)\|_Y &= \|L(x_1)\|_Y \\ &\leq \|L\|_{L(X_1, Y)} \cdot \|x_1\|_1 \\ &= \|L\|_{L_1(X_1 \dots X_1; Y)} \cdot \|x_1\|_1 \\ &= \|L\|_{L_1(X \dots X_1; Y)} \cdot \prod_{i=1}^1 \|x_i\|_i \end{aligned}$$

proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$
 and $(x_1, \dots, x_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} X_i$ then, as $L(x_1) \in L_n(X_2 \dots X_{n+1}; Y)$,
 $(x_2, \dots, x_{n+1}) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ and $n \in S$, we have

$$\begin{aligned} \|L(x_1)(x_2: \dots: x_{n+1})\|_Y &\leq \|L(x_1)\|_{L_n(X_2 \dots X_{n+1}; Y)} \cdot \prod_{i=1}^n \|x_{i+1}\|_{i+1} \\ &= \|L(x_1)\|_{L_n(X_2 \dots X_{n+1}; Y)} \cdot \prod_{i=2}^{n+1} \|x_i\|_i \end{aligned} \quad (16.302)$$

Next we have

$$\begin{aligned}
\|L(x_1: \dots: x_{n+1})\|_Y &= \|(L(x_1))(x_2: \dots: x_{n+1})\|_Y \\
&\leq_{[\text{eq: 16.302}]} \|L(x_1)\|_{L_n(X_2 \dots X_{n+1}; Y)} \cdot \prod_{i=2}^{n+1} \|x_i\|_X \\
&\leq_{L \in L(X_1, L_n(X_2 \dots X_{n+1}; Y))} \|L\|_{L_{n+1}(X_1 \dots X_{n+1}; Y)} \cdot \|x_1\|_X \cdot \prod_{i=2}^{n+1} \|x_i\|_X \\
&\leq \|L\|_{L_{n+1}(X_1 \dots X_{n+1}; Y)} \cdot \prod_{i=1}^{n+1} \|x_i\|_X
\end{aligned}$$

proving that

$$n+1 \in S$$

□

Lemma 16.259. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $L_1, L_2 \in L_n(X_1 \dots X_n; Y)$ such that $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$L_1(x_1: \dots: x_n) = L_2(x_1: \dots: x_n)$$

then

$$L_1 = L_2$$

Proof. We use induction to prove this, so define

$$S = \{n \in \mathbb{N} \mid \text{If } L_1, L_2 \in L_n(X_1 \dots X_n; Y) \text{ satisfies } \forall (x_1, \dots, x_n) \ L_1(x_1: \dots: x_n) = L_2(x_1: \dots: x_n) \text{ then } L_1 = L_2\}$$

then we have:

$1 \in S$. Let $L_1, L_2 \in L_1(X_1; Y) = L(X_1, Y)$ be such that $\forall (x_1) \in X^1$ we have $L_1(x_1: \dots: x_1) = L_2(x_1: \dots: x_1)$. Then $\forall x \in X$ we have for $y = (x) \in X^1$ [so that $y_1 = x$] that

$$L_1(x) = L_1(y_1) = L_1(\underbrace{y_1: \dots: y_1}_1) = L_2(\underbrace{y_1: \dots: y_1}_1) = L_2(y_1) = L_2(x)$$

proving that $L_1 = L_2$. Hence $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L_1, L_2 \in L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$ such that

$$\forall x = (x_1, \dots, x_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} X_i \text{ we have } L_1(x_1: \dots: x_{n+1}) = L_2(x_1: \dots: x_{n+1})$$

Let $y \in X$ then $L_1(y), L_2(y) \in L_n(X; Y)$ and $\forall (x_1, \dots, x_n) \in X^n$ we have

$$L_1(y)(x_1: \dots: x_n) = L_1(y: x_1: \dots: x_n) = L_2(y: x_1: \dots: x_n) = L_2(y)(x_1: \dots: x_n)$$

which, as $n \in S$, proves that $L_1(y) = L_2(y)$. As y is chosen arbitrary we conclude that $L_1 = L_2$. Hence we have proved that

$$n+1 \in S$$

□

Lemma 16.260. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space. If $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$\|L(x_1: \dots: x_n)\|_Y \leq M \cdot \prod_{i=1}^n \|x_i\|_i$$

then it follows that

$$\|L\|_{L_n(X_1 \dots X_n; Y)} \leq M$$

Proof. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If for } L \in L_n(X_1 \dots X_n; Y) \text{ we have } \forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i \text{ that } \|L(x_1: \dots: x_n)\|_Y \leq M \cdot \prod_{i=1}^n \|x_i\|_i \text{ then } \|L\|_{L_n(X_1 \dots X_n; Y)} \leq M \right\}$$

then we have:

$1 \in S$. Let $L \in L_1\left(\underbrace{X_1 \dots X_1}_1; Y\right) = L(X_1, Y)$ such that $\forall (x_1) \in \prod_{i \in \{1\}} X_i$ we have

$\|L(x_1: \dots: x_1)\|_Y \leq M \cdot \prod_{i=1}^1 \|x_i\|_i$. Then given $y \in X_1$ we have for $x = (y) \in \prod_{i \in \{1\}} X_i$ that

$$\|L(y)\|_Y = \|L(x_1)\|_Y = \left\| L\left(\underbrace{x_1: \dots: x_1}_1\right) \right\|_Y \leq M \cdot \prod_{i=1}^1 \|x_i\|_i = M \cdot \|y\|_X$$

so that by [definition: 14.209] $\|L\|_{L_1(X_1 \dots X_1; Y)} = \|L\|_{L(X_1, Y)} \leq M$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L_{n+1}(X_1 \dots X_{n+1}; Y) = L(X_1, L_n(X_2 \dots X_{n+1}; Y))$ such that $\forall (x_1, \dots, x_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} X_i$ we have:

$$\|L(x_1: \dots: x_{n+1})\|_Y \leq M \cdot \prod_{i=1}^{n+1} \|x_i\|_i$$

Fix $x \in X$ then $\forall (x_2, \dots, x_{n+1}) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ we have

$$\|(L(x))(x_2: \dots: x_{n+1})\|_Y = \|L(x: x_2: \dots: x_{n+1})\|_Y \leq M \cdot \|x\|_1 \cdot \prod_{i=2}^{n+1} \|x_i\|_i = (M \cdot \|x\|_1) \cdot \prod_{i=2}^{n+1} \|x_i\|_i$$

As $n \in S$ and $L(x_1) \in L_n(X_2 \dots X_{n+1}; Y)$ it follows that $\|L(x)\|_{L_n(X_2 \dots X_{n+1}; Y)} \leq M \cdot \|x\|_1$. Hence by [definition: 14.209] it follows that $\|L\|_{L_{n+1}(X_1 \dots X_{n+1}; Y)} \leq M$ proving

$$n+1 \in S$$

□

Lemma 16.261. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $L \in L(X_1 \dots X_n; Y)$ [see definition: 14.230] then there exist a $K \in L_n(X_1 \dots X_n; L)$ such that $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$

$$K(x_1: \dots: x_n) = L(x_1, \dots, x_n)$$

Proof. We use induction to prove this, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L(X_1 \dots X_n; Y) \text{ then there exist a } K \in L_n(X_1 \dots X_n; Y) \text{ such that } \forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i \text{ we have } K(x_1: \dots: x_n) = L(x_1, \dots, x_n) \right\}$$

then we have:

$1 \in S$. If $L \in L\left(\underbrace{X_1 \dots X_1}_1; Y\right)$ then, as $L_1\left(\underbrace{X_1 \dots X_1}_1; Y\right) = L(X_1, Y) = L\left(\underbrace{X_1 \dots X_1}_1; Y\right)$, we have that $L \in L_1(X_1 \dots X_1, Y)$ so if we take $K = L$ then we have $K \in L_1(X_1 \dots X_1, Y)$ and for $(x_1) \in \prod_{i \in \{1\}} X_i$ we have $K\left(\underbrace{x_1: \dots: x_1}_1\right) = K(x_1) = L(x_1) = L\left(\underbrace{x_1, \dots, x_1}_1\right)$. So we have that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $L \in L(X_1 \dots X_{n+1}; Y)$. Take $x \in X_1$ and define

$$L_x: \prod_{i \in \{2, \dots, n+1\}} X_i \rightarrow Y \text{ by } L_x(x_1, \dots, x_n) = L(x, x_1, \dots, x_n)$$

If $i \in \{1, \dots, n\}$ and $\left(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n\right) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ then

$$\begin{aligned} L_x \left(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n \right) &= \\ L \left(x, x_1, \dots, \underbrace{r + \alpha \cdot t}_{i+1}, \dots, x_n \right) &= \\ L \left(x, x_1, \dots, \underbrace{r}_{i+1}, \dots, x_n \right) + \alpha \cdot L \left(x, x_1, \dots, \underbrace{t}_{i+1}, \dots, x_n \right) &= \\ L_x \left(x_1, \dots, \underbrace{r}_i, \dots, x_n \right) + \alpha \cdot L_x \left(x_1, \dots, \underbrace{t}_i, \dots, x_n \right) \end{aligned}$$

proving that

$$L_x \in \text{Hom}(X_2 \dots X_{n+1}; Y) \quad (16.303)$$

Let $(x_1, \dots, x_n) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ then for $(y_1, \dots, y_{n+1})$ where $y_i = \begin{cases} x & \text{if } i = 1 \\ x_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$ we have

$$\begin{aligned} \|L_x(x_1, \dots, x_n)\|_Y &= \|L(x, x_1, \dots, x_n)\|_Y \\ &= \|L(y_1, \dots, y_{n+1})\|_Y \\ &\leq_{L \in L(X_1 \dots X_{n+1}; Y)} \|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \prod_{i=1}^{n+1} \|y_i\|_i \\ &= \|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \|y_1\|_1 \cdot \prod_{i=2}^{n+1} \|y_i\|_i \\ &= \|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \|x\|_1 \cdot \prod_{i=2}^{n+1} \|x_{i-1}\|_i \\ &= (\|x\|_X \cdot \|L\|_{L^{n+1}(X; Y)}) \cdot \prod_{i=1}^n \|x_i\|_{i+1} \end{aligned}$$

proving by [theorem: 14.228] and [eq: 16.303] that L_x is continuous. Hence

$$L_x \in L(X_2 \dots X_{n+1}; Y) \quad (16.304)$$

As $n \in S$ there exist a $K_x \in L_n(X_2 \dots X_{n+1}; Y)$ such that $\forall (x_1, \dots, x_n) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ we have

$$K_x(x_1; \dots; x_n) = L_x(x_1, \dots, x_n) = L(x, x_1, \dots, x_n) \quad (16.305)$$

This allows us to define

$$K: X_1 \rightarrow L_n(X_2 \dots X_{n+1}; Y) \text{ where } K(x) = K_x$$

then by [eq: 16.305] we have that for $(x_1, \dots, x_{n+1}) \in \prod_{i \in \{1, \dots, n+1\}} X_i$

$$K(x_1)(x_2; \dots; x_{n+1}) = K_{x_1}(x_2; \dots; x_{n+1}) = L(x_1, \dots, x_{n+1}) \quad (16.306)$$

If $x, y \in X_1$ and $\alpha \in \mathbb{K}$ then we have $\forall (x_1, \dots, x_n) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ that

$$\begin{aligned} K(x + \alpha \cdot y)(x_1; \dots; x_n) &\stackrel{[\text{eq: 16.306}]}{=} L(x + \alpha \cdot y, x_1, \dots, x_n) \\ &\stackrel{L \in L(X_1 \dots X_{n+1}; Y)}{=} L(x, x_1, \dots, x_n) + \alpha \cdot L(y, x_1, \dots, x_n) \\ &\stackrel{[\text{eq: 16.306}]}{=} K(x)(x_1; \dots; x_n) + \alpha \cdot K(y)(x_1; \dots; x_n) \\ &\stackrel{[\text{lemma: 16.120}]}{=} (K(x) + \alpha \cdot K(y))(x_1; \dots; x_n) \end{aligned}$$

which by [lemma: 16.123] proves that

$$K(x + \alpha \cdot y) = K(x) + \alpha \cdot K(y)$$

Hence we have that

$$K \in \text{Hom}(X, L_n(X_1 \dots X_{n+1}; Y)) \quad (16.307)$$

Let $x_1 \in X_1$ then we have for $(x_2, \dots, x_{n+1}) \in \prod_{i \in \{2, \dots, n+1\}} X_i$ that

$$\begin{aligned} \|K(x_1)(x_2; \dots; x_{n+1})\|_Y &= \|K_{x_1}(x_2; \dots; x_{n+1})\|_Y \\ &\stackrel{[\text{eq: 16.306}]}{=} \|L(x_1, x_2, \dots, x_{n+1})\|_Y \\ &\leq_{L \in L(X_1 \dots X_{n+1}; Y)} \|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \prod_{i=1}^{n+1} \|x_i\|_i \\ &= \|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \|x_1\|_1 \cdot \prod_{i=2}^{n+1} \|x_i\|_i \\ &= (\|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \|x_1\|_1) \cdot \prod_{i=2}^{n+1} \|x_i\|_i \end{aligned}$$

which as $K(x_1) \in L_n(X_2 \dots X_{n+1}; Y)$ proves using [lemma: 16.260] that

$$\|K(x_1)\|_{L_n(X_2 \dots X_{n+1}; Y)} \leq \|L\|_{L(X_1 \dots X_{n+1}; Y)} \cdot \|x_1\|_1$$

Hence using [eq: 16.307] together with [theorem: 14.201] proves that K is continuous or

$$K \in L(X_1, L_n(X_2 \dots X_{n+1}; Y)) = L_{n+1}(X_1 \dots X_{n+1}; Y)$$

further

$$K(x_1; \dots; x_{n+1}) = K(x_1)(x_2; \dots; x_{n+1}) \stackrel{[\text{eq: 16.306}]}{=} L(x_1, \dots, x_{n+1})$$

The above proves that

$$n+1 \in S$$

□

Theorem 16.262. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space then

$\mathcal{I}_{X_1 \dots X_n, Y}: L_n(X_1 \dots X_n; Y) \rightarrow L(X_1 \dots X_n; Y)$ where $\mathcal{I}_{X_1 \dots X_n, Y}(L): \prod_{i \in \{1, \dots, n\}} X_i \rightarrow Y$ is defined by

$$(\mathcal{I}_{X_1 \dots X_n, Y}(L))(x_1, \dots, x_n) = L(x_1; \dots; x_n)$$

is a linear isometric isomorphism. In other words

1. $\forall L \in L_n(X_1 \dots X_n; Y)$ $\mathcal{I}_{X_1 \dots X_n, Y}(L) \in L(X_1 \dots X_n; Y)$ so that

$$\mathcal{I}_{X_1 \dots X_n, Y}: L_n(X_1 \dots X_n; Y) \rightarrow L(X_1 \dots X_n; Y)$$

is indeed a function

2. $\mathcal{I}_{X_1 \dots X_n, Y}: L_n(X_1 \dots X_n; Y) \rightarrow L(X_1 \dots X_n; Y)$ is a bijection.
3. For $L \in L(X_1 \dots X_n; Y)$ we have $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ that

$$(\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L)(x_1; \dots; x_n) = L(x_1, \dots, x_n)$$

4. $\forall L_1, L_2 \in L_n(X_1 \dots X_n; Y)$ and $\alpha \in \mathbb{K}$ we have

$$\mathcal{I}_{X_1 \dots X_n, Y}(L_1 + \alpha \cdot L_2) = \mathcal{I}_{X_1 \dots X_n, Y}(L_1) + \alpha \cdot \mathcal{I}_{X_1 \dots X_n, Y}(L_2)$$

5. $\forall L \in L_n(X_1 \dots X_n; Y)$ we have $\|L\|_{L_n(X_1 \dots X_n; Y)} = \|\mathcal{I}_{X_1 \dots X_n, Y}(L)\|_{L(X_1 \dots X_n; Y)}$

Note 16.263. Using [theorem: 14.126] it follows that

$$(\mathcal{I}_{X_1 \dots X_n, Y})^{-1}: L(X_1 \dots X_n; Y) \rightarrow L_n(X_1 \dots X_n; Y)$$

is a linear isometric isomorphism. In other words:

1. $(\mathcal{I}_{X_1 \dots X_n, Y})^{-1}$ is a bijection.
2. $\forall L_1, L_2 \in L(X_1 \dots X_n; Y)$ and $\alpha \in \mathbb{K}$ we have

$$(\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L_1 + \alpha \cdot L_2) = (\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L_1) + \alpha \cdot (\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L_2)$$

3. $\forall L \in L(X_1, \dots, X_n)$ we have $\|L\|_{L(X_1 \dots X_n; Y)} = \|(\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L)\|_{L_n(X_1 \dots X_n; Y)}$

Proof.

1. Let $L \in L_n(X_1 \dots X_n; Y)$ then for $i \in \{1, \dots, n\}$ and $(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$\begin{aligned} \mathcal{I}_{X_1 \dots X_n, Y}(L) \left(x_1, \dots, \underbrace{r + \alpha \cdot t}_i, \dots, x_n \right) &= \\ L \left(x_1 : \dots : \underbrace{r + \alpha \cdot t}_i : \dots : x_n \right) &\stackrel{[\text{lemma: 16.257}]}{=} \\ L \left(x_1 : \dots : \underbrace{r}_i : \dots : x_n \right) + \alpha \cdot L \left(x_1 : \dots : \underbrace{t}_i : \dots : x_n \right) &= \\ \mathcal{I}_{X_1 \dots X_n, Y}(L) \left(x_1, \dots, \underbrace{r}_i, \dots, x_n \right) + \alpha \cdot \mathcal{I}_{X_1 \dots X_n, Y}(L) \left(x_1, \dots, \underbrace{s}_i, \dots, x_n \right) \end{aligned}$$

proving that

$$\mathcal{I}_{X_1 \dots X_n, Y}(L) \in \text{Hom}(X_1, \dots, X_n; Y) \quad (16.308)$$

Further if $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ then we have:

$$\begin{aligned} \|\mathcal{I}_{X_1 \dots X_n, Y}(L)(x_1, \dots, x_n)\|_Y &= \|L(x_1 : \dots : x_n)\|_Y \\ &\stackrel{[\text{lemma: 16.258}]}{\leq} \|L\|_{L_n(X_1 \dots X_n; Y)} \cdot \prod_{i=1}^n \|x_i\|_i \end{aligned}$$

Using [theorems: 14.228 and 14.234] it follows then that

$$\mathcal{I}_{X_1 \dots X_n, Y}(L) \text{ is continuous and } \|\mathcal{I}_{X_1 \dots X_n, Y}(L)\|_{L(X_1, \dots, X_n; Y)} \leq \|L\|_{L_n(X_1 \dots X_n; Y)} \quad (16.309)$$

Combining the above with [eq: 16.308] we have

$$\mathcal{I}_{X_1 \dots X_n, Y}(L) \in L(X_1, \dots, X_n; Y)$$

2. For bijectivity we have to prove:

injectivity. If $L_1, L_2 \in L_n(X_1 \dots X_n; Y)$ is such that $\mathcal{I}_{X_1 \dots X_n, Y}(L_1) = \mathcal{I}_{X_1 \dots X_n, Y}(L_2)$ then we have $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ that

$$L_1(x_1 : \dots : x_n) = \mathcal{I}_{X_1 \dots X_n, Y}(L_1)(x_1, \dots, x_n) = \mathcal{I}_{X_1 \dots X_n, Y}(L_2)(x_1, \dots, x_n) = L_2(x_1 : \dots : x_n)$$

Hence by [lemma: 16.259] it follows that $L_1 = L_2$.

surjectivity. Let $L \in L(X_1 \dots X_n; Y)$ then using [lemma: 16.261] there exist a $K \in L_n(X_1 \dots X_n; Y)$ such that $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have

$$K(x_1 : \dots : x_n) = L(x_1, \dots, x_n)$$

Hence

$$\mathcal{I}_{X_1 \dots X_n, Y}(K)(x_1, \dots, x_n) = K(x_1 : \dots : x_n) = L(x_1, \dots, x_n)$$

which proves that

$$\mathcal{I}_{X_1 \dots X_n, Y}(K) = L$$

so that $\mathcal{I}_{X_1 \dots X_n, Y}$ is surjective.

3. Let $L \in L(X_1 \dots X_n; Y)$ and take $K = (\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L) \in L_n(X_1 \dots X_n; Y)$ then we have that $L = \mathcal{I}_{X_1 \dots X_n, Y}(K)$ so that

$$(\mathcal{I}_{X_1 \dots X_n, Y})^{-1}(L)(x_1, \dots, x_n) = K(x_1, \dots, x_n) = \mathcal{I}_{X_1 \dots X_n, Y}(K)(x_1, \dots, x_n) = L(x_1, \dots, x_n).$$

4. Let $L_1, L_2 \in L_n(X_1 \dots X_n; Y)$ and $\alpha \in \mathbb{K}$ then $\forall (x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ we have:

$$\begin{aligned} \mathcal{I}_{X_1 \dots X_n, Y}(L_1 + \alpha \cdot L_2)(x_1, \dots, x_n) &= \\ (L_1 + \alpha \cdot L_2)(x_1, \dots, x_n) &\stackrel{[\text{lemma: 16.256}]}{=} \\ L_1(x_1, \dots, x_n) + \alpha \cdot L_2(x_1, \dots, x_n) &= \\ \mathcal{I}_{X_1 \dots X_n, Y}(L_1)(x_1, \dots, x_n) + \alpha \cdot \mathcal{I}_{X_1 \dots X_n, Y}(L_2)(x_1, \dots, x_n) &= \\ (\mathcal{I}_{X_1 \dots X_n, Y}(L_1) + \alpha \cdot \mathcal{I}_{X_1 \dots X_n, Y}(L_2))(x_1, \dots, x_n) &= \end{aligned}$$

proving that

$$\mathcal{I}_{X_1 \dots X_n, Y}(L_1 + \alpha \cdot L_2) = \mathcal{I}_{X_1 \dots X_n, Y}(L_1) + \alpha \cdot \mathcal{I}_{X_1 \dots X_n, Y}(L_2)$$

5. Let $L \in L_n(X_1 \dots X_n; Y)$ then by [eq: 16.309] we have

$$\|\mathcal{I}_{X_1 \dots X_n, Y}(L)\|_{L(X_1 \dots X_n; Y)} \leq \|L\|_{L_n(X_1 \dots X_n; Y)} \quad (16.310)$$

For the opposite inequality, let $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ then

$$\begin{aligned} \|L(x_1, \dots, x_n)\|_Y &= \|\mathcal{I}_{X_1 \dots X_n, Y}(L)(x_1, \dots, x_n)\|_Y \\ &\leq \|\mathcal{I}_{X_1 \dots X_n, Y}(L)\|_{L(X_1 \dots X_n; Y)} \cdot \prod_{i=1}^n \|x_i\|_{X_i} \end{aligned}$$

so that by [lemma: 16.260]

$$\|L\|_{L_n(X_1 \dots X_n; Y)} \leq \|\mathcal{I}_{X_1 \dots X_n, Y}(L)\|_{L(X_1 \dots X_n; Y)}$$

which combined with [eq: 16.310] proves that

$$\|L\|_{L_n(X_1 \dots X_n; Y)} = \|\mathcal{I}_{X_1 \dots X_n, Y}(L)\|_{L(X_1 \dots X_n; Y)} \quad \square$$

16.5.2 Higher Order Partial Differentiation

Definition 16.264. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow Y$ a function then we define for $k \in \{1, \dots, m\}$ $\mathfrak{D}_{[i_k \dots i_m], f}$ and $D_{[i_k \dots i_m]}f$ recursively as follows for k

$k = m$. Then

$$\mathfrak{D}_{[i_m \dots i_m], f} = \mathfrak{D}_{i_m, f}$$

and

$$D_{[i_m \dots i_m]}f = D_{i_m}f$$

Using [definition: 16.78] for the definition of $\mathfrak{D}_{i_m, f}$ and $D_{i_m}f$ we have that

$$\mathfrak{D}_{[i_m \dots i_m], f} = \mathfrak{D}_{i_m, f} = \{x \in A \mid f \text{ is } i_m\text{-partial differentiable at } x\} \subseteq A$$

and

$$D_{[i_m \dots i_m]}f: \mathfrak{D}_{[i_m \dots i_m], f} \rightarrow L(X_{i_m}, Y) \text{ is defined by } D_{[i_m \dots i_m]}f(x) = D_{x, i_m}f$$

$k < m$. Then

$$\mathfrak{D}_{[i_k \dots i_m], f} = \{x \in \mathfrak{D}_{[i_{k+1} \dots i_m], f} \mid D_{[i_{k+1} \dots i_m]}f \text{ has a } i_k\text{-partial differential at } x\}$$

and

$$D_{[i_k \dots i_m]} f: \mathfrak{D}_{[i_{k+1} \dots i_m], f} \rightarrow L(X_{i_k}, L_{m-k}(X_{i_{k+1}}, \dots, X_{i_m}; Y)) = L_{m-k+1}(X_k, \dots, X_m; Y)$$

is defined by

$$D_{i_1, \dots, i_m}^{[m]} f = D_{i_1}(D_{i_2 \dots i_m}^{[m-1]} f)$$

Convention 16.265. If $A \subseteq X = \prod_{j \in \{1, \dots, n\}} X_j$ and $f: A \rightarrow Y$ is a function then $D_{[i_1 \dots i_m]}^{[m]} f$ is always the graph of the function $D_{[i_1 \dots i_m]}^{[m]} f: \mathfrak{D}_{[i_1 \dots i_m], f}^m \rightarrow L_m(X_{i_1}, \dots, X_{i_m}; Y)$. In other words the domain of $D_{[i_1 \dots i_m]}^{[m]} f$ is always $\mathfrak{D}_{[i_1 \dots i_m], f}^m$ and the codomain is always $L_m(X_{i_1}, \dots, X_{i_m}; Y)$.

The domains of higher order partial is decreasing with increasing order.

Theorem 16.266. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow Y$ a function then we have that $\forall k, l \in \{1, \dots, m\}$ with $k \leq l$ that

$$\mathfrak{D}_{[i_l \dots i_m], f} \subseteq \mathfrak{D}_{[i_k \dots i_m], f} \subseteq \mathfrak{D}_{[i_m \dots i_m], f} = \mathfrak{D}_{i_m, f} \subseteq A$$

Proof. Let $l \in \{1, \dots, m\}$ then we have that $\forall r \in \{1, \dots, l-1\}$

$$\mathfrak{D}_{[i_r \dots i_m], f} = \{x \in \mathfrak{D}_{[i_{r+1} \dots i_m], f} \mid D_{[i_{r+1} \dots i_m]} f \text{ has a } i_l\text{-partial differential at } x\} \subseteq \mathfrak{D}_{[i_{r+1} \dots i_m], f}$$

so that by [theorem: 10.36] we have that $\forall k \in \{1, \dots, l\}$

$$\mathfrak{D}_{[i_l \dots i_m], f} \subseteq \mathfrak{D}_{[i_k \dots i_m], f}$$

Further we have $\forall k \in \{1, \dots, m\}$ either

$$\mathbf{k = m.} \text{ Then } \mathfrak{D}_{[i_k \dots i_m], f} = \mathfrak{D}_{[i_m \dots i_m], f} = \mathfrak{D}_{i_m, f} \subseteq A$$

$$\mathbf{k < m.} \text{ Then } \mathfrak{D}_{[i_k \dots i_m], f} \subseteq \mathfrak{D}_{[i_m \dots i_m], f} = \mathfrak{D}_{i_m, f} \subseteq A$$

so in all cases we have

$$\mathfrak{D}_{[i_k \dots i_m], f} \subseteq \mathfrak{D}_{[i_m \dots i_m], f} = \mathfrak{D}_{i_m, f} \subseteq A \quad \square$$

We can define now the concept of functions that have partial differential at a point.

Definition 16.267. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $x \in A$, $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow Y$ a function then we say that

$$f \text{ has a partial differential } D_{x, [i_1 \dots i_m]} f \text{ at } x \text{ iff } x \in \mathfrak{D}_{[i_1 \dots i_m], f}$$

the partial differential $D_{x, [i_1 \dots i_m]} f$ at x is then defined by

$$D_{x, [i_1 \dots i_m]} f = D_{[i_1 \dots i_m]} f(x)$$

Remark 16.268. Instead of saying

$$f \text{ has a partial differential } D_{x, [i_1 \dots i_m]} f \text{ at } x$$

we sometimes use the term

$$D_{x, [i_1 \dots i_m]} f \text{ exist}$$

Definition 16.269. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $U \in \mathcal{T}_{\|\cdot\|_{\max}}$ a open set in X with $U \subseteq A$, $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow Y$ a function then we say that

f has $D_{[i_1 \dots i+m]}f$ partial differentials on U

iff

$\forall x \in U$ f has a partial differential $D_{x, [i_1 \dots i_m]}f$ at x

Using [definition: 16.267]

f has $D_{[i_1 \dots i+m]}f$ partial differentials on U

is equivalent with

$$U \subseteq \mathfrak{D}_{f, [i_1 \dots i_m]}$$

In the cases where $U = \text{dom}(f)$ we have the following definition

Definition 16.270. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $U \in \mathcal{T}_{\|\cdot\|_{\max}}$ a open set in X , $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $f: U \rightarrow Y$ a function then we say that

f has $D_{[i_1 \dots i+m]}f$ partial differentials

iff

f has $D_{[i_1 \dots i+m]}f$ partial differentials on U

Note 16.271. If f has $D_{[i_1 \dots i_m]}f$ partial differentials then by [definition: 16.269] $U \subseteq \mathfrak{D}_{[i_1 \dots i_m], f}$ which, as $\mathfrak{D}_{[i_1 \dots i_m], f} \subseteq U$ by [theorem: 16.266], proves that in this case we have

$$\mathfrak{D}_{[i_1 \dots i_m], f} = U$$

Theorem 16.272. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ then we have for $k \in \{1, \dots, m\}$ that

1. If $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ a function that has a $D_{x, [i_1 \dots i_m]}f$ partial differential at x then f has a $D_{x, [i_k \dots i_m]}f$ partial differential at x .
2. If $A \subseteq X$, U a open set and $f: A \rightarrow Y$ a function that has $D_{[i_1 \dots i_m]}f$ partial differentials on U then f has $D_{[i_k \dots i_m]}f$ partial differentials on U .
3. If U is a open set in X and $f: U \rightarrow Y$ has $D_{[i_1 \dots i_m]}f$ partial differentials then f has $D_{[i_k \dots i_m]}f$ partial differentials.

Proof.

1. Using [definition: 16.267] we have that $x \in \mathfrak{D}_{[i_1, \dots, i+m], f}$ which, as by [theorem: 16.266] $\mathfrak{D}_{[i_1 \dots i_m], f} \subseteq \mathfrak{D}_{[i_k \dots i+m], f}$, proves that $x \in \mathfrak{D}_{[i_k \dots i_m], f}$. Hence using [definition: 16.267] again it follows that f has a partial differential $D_{x, [i_k \dots i_m]}f$ at x .
2. Using [definition: 16.269] we have that $U \subseteq \mathfrak{D}_{[i_1, \dots, i+m], f}$ which, as by [theorem: 16.266] $\mathfrak{D}_{[i_1 \dots i_m], f} \subseteq \mathfrak{D}_{[i_k \dots i+m], f}$, proves that $U \subseteq \mathfrak{D}_{[i_k \dots i_m], f}$. Hence using [definition: 16.269] again it follows that f has a partial differential $D_{x, [i_k \dots i_m]}f$ on U .
3. This follows from (2) using [definition: 16.270]. □

We have now a theorem similar to [theorem: 16.140]

Theorem 16.273. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $x \in A$, $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow Y$ a function then we have that the following are equivalent:

1. f has a $D_{x, [i_1 \dots i_m]}f$ partial differential at x
2. We have for m either:
 - $m = 1$. Then f is i_1 -partial differentiable at x
 - $1 < m$. Then

$D_{[i_2 \dots i_m]}f$ is i_1 -partial differentiable at x

Further we have if (1) is true that

$$D_{x, [i_1 \dots i_m]}f = \begin{cases} D_{x, i_1}f & \text{if } m = 1 \\ D_{x, i_1}(D_{[i_2 \dots i_m]}f) & \text{if } 1 < m \end{cases}$$

Proof.

$1 \Rightarrow 2$. For $m \in \mathbb{N}$ we have either:

$m = 1$. By [definition: 16.264] we have that

$$\mathfrak{D}_{f, [i_1 \dots i_1]} = \{x \in A \mid f \text{ is } i_1\text{-partial differentiable at } x\} \text{ and } D_{[i_1 \dots i_1]}f = D_{i_1}f$$

As f has a partial differential $D_{x, [i_1 \dots i_1]}f$ at x we have by [definition: 16.267] that

$$x \in \mathfrak{D}_{f, [i_1 \dots i_1]} \text{ and } D_{x, [i_1 \dots i_1]}f = D_{[i_1 \dots i_1]}f(x)$$

so that

$$f \text{ is } i_1\text{-partial differentiable at } x$$

with

$$D_{x, [i_1 \dots i_1]}f = D_{i_1}f(x) \stackrel{[\text{definition: 16.78}]}{=} D_{x, i_1}f$$

$1 < m$. By [definition: 16.264] we have that

$$\mathfrak{D}_{f, [i_1 \dots i_m]} = \{x \in \mathfrak{D}_{f, [i_2 \dots i_m]} \mid D_{[i_2 \dots i_m]}f \text{ is } i_1\text{-partial differentiable at } x\}$$

and

$$D_{[i_1 \dots i_m]}f = D_{i_1}(D_{[i_2 \dots i_m]}f)$$

As f has a partial differential $D_{x, [i_1 \dots i_m]}f$ at x it follows from [definition: 16.267] that

$$x \in \mathfrak{D}_{f, [i_1 \dots i_m]} \text{ and } D_{x, [i_1 \dots i_m]}f = D_{[i_1 \dots i_m]}f(x)$$

so that

$$D_{[i_2 \dots i_m]}f \text{ is } i_1\text{-partial differentiable at } x$$

with

$$D_{x, [i_1 \dots i_m]}f = D_{i_1}(D_{[i_2 \dots i_m]}f)(x) \stackrel{[\text{definition: 16.78}]}{=} D_{x, i_1}(D_{[i_2 \dots i_m]}f)$$

$2 \Rightarrow 1$. For $m \in \mathbb{N}$ we have either:

$m = 1$. As f is i_1 -partial differentiable at x it follows that

$$x \in \{x \in A \mid f \text{ is } i_1\text{-partial differentiable at } x\}$$

which by [definition: 16.264] proves that

$$x \in \mathfrak{D}_{f, [i_1 \dots i_1]}$$

so that by [definition: 16.267]

$$f \text{ has a partial differential } D_{x, [i_1 \dots i_1]}f \text{ at } x$$

$1 < m$. As $D_{[i_2 \dots i_m]}f$ is i_1 -partial differentiable at x it follows that

$$x \in \{x \in \mathfrak{D}_{f, [i_2 \dots i_m]} \mid D_{[i_2 \dots i_m]}f \text{ is } i_1\text{-partial differentiable at } x\}$$

which by [theorem: 16.264] proves that

$$x \in \mathfrak{D}_{f, [i_1 \dots i_m]}$$

so that by [definition: 16.267]

$$f \text{ has a partial differential } D_{x, [i_1 \dots i_m]}f \text{ at } x \quad \square$$

We prove now that higher order differentiability implies higher order partial differentiability. First we look at 1-times differentiability.

Lemma 16.274. *Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $x \in A$ and $f: A \rightarrow Y$ that is 1-times differentiable at x then $\forall i \in \{1, \dots, n\}$ we have that*

$$f \text{ has a partial derivative } D_{x, [i \dots i]}f \text{ at } x$$

and

$$D_{x, [i \dots i]}f = D_x^{[1]}f \circ I^{[i, 0]}$$

further

$$D_x^{[1]}f = \sum_{i=1}^n (D_{x, [i \dots i]}f \circ \pi_i)$$

Proof. As f is 1-times differentiable at x it follows from [theorem: 16.140] that f is Fréchet differentiable at x . Hence using [theorem: 16.83] it follows that $\forall i \in \{1, \dots, n\}$

$$f \text{ is } i\text{-partial differentiable at } x$$

with

$$D_{x, i}f = D_x f \circ I^{[i, x]} \text{ and } D_x f = \sum_{i=1}^n (D_{x, i}f \circ \pi_i)$$

Using then [theorems: 16.136, 16.273] that

$$f \text{ has a partial differential } D_{x, [i \dots i]}f \text{ at } x$$

with

$$D_{x, [i \dots i]}f = D_x^{[1]}f \text{ and } D_x^{[1]}f = \sum_{i=1}^n (D_{x, [i \dots i]}f \circ \pi_i) \quad \square$$

To extend the above to n -times differentiability we need a new lemma.

Lemma 16.275. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98] and $i \in \{1, \dots, n\}^m$ then there exist a*

$$\Phi \in L(L_m(X; Y), L_m(X_{i_1} \dots X_{i_m}; Y))$$

such that for every $L \in L_m(X; Y)$ and $\forall (h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ we have

$$\Phi(L)(h_1 : \dots : h_m) = L(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m))$$

Proof. Let $L \in L_m(X; Y)$ and define

$$\Psi_L: \prod_{j \in \{1, \dots, m\}} X_{i_j} \rightarrow Y \text{ by } \Psi_L(h_1, \dots, h_m) = L(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \in Y \quad (16.311)$$

Let $j \in \{1, \dots, m\}$ and

$$\left(h_1, \dots, \underbrace{k + \alpha \cdot l}_j, \dots, h_m \right) \in \prod_{k \in \{1, \dots, m\}} X_{i_k}$$

where $k, l \in X_{i_{m-j+1}}$ and $k \in \mathbb{K}$ then we have

$$\begin{aligned} & \Psi_L \left(h_1, \dots, \underbrace{k + \alpha \cdot l}_j, \dots, h_m \right) = \\ & L \left(I^{[i_1, 0]}(h_1) : \dots : \underbrace{I^{[i_j, 0]}(k + \alpha \cdot l)}_j : \dots : I^{[i_m, 0]}(h_m) \right) \stackrel{[\text{lemma: 16.71}]}{=} \\ & L \left(I^{[i_1, 0]}(h_1) : \dots : \underbrace{I^{[i_j, 0]}(k) + \alpha \cdot I^{[i_j, 0]}(l)}_j : \dots : I^{[i_m, 0]}(h_m) \right) \stackrel{[\text{lemma: 16.257}]}{=} \\ & L \left(I^{[i_1, 0]}(h_1) : \dots : \underbrace{I^{[i_j, 0]}(k)}_j : \dots : I^{[i_m, 0]}(h_m) \right) + \alpha \cdot L \left(I^{[i_1, 0]}(h_1) : \dots : \underbrace{I^{[i_j, 0]}(l)}_j : \dots : I^{[i_m, 0]}(h_m) \right) = \\ & \Psi_L \left(h_1, \dots, \underbrace{k}_j, \dots, h_m \right) + \alpha \cdot \Psi_L \left(h_1, \dots, \underbrace{l}_j, \dots, h_m \right) \end{aligned}$$

proving that

$$\Psi_L \in \text{Hom}(X_{i_1} \dots X_{i_m}; Y) \quad (16.312)$$

Further if $(h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ then we have

$$\begin{aligned} \|\Psi_L(h_1, \dots, h_m)\|_Y &= \|L(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m))\|_Y \\ &\stackrel{[\text{lemma: 16.258}]}{\leq} \|L\|_{L^m(X; Y)} \cdot \prod_{j=1}^m \|I^{[i_j, 0]}(h_j)\|_{i_j} \\ &\stackrel{[\text{lemma: 16.71}]}{=} \|L\|_{L^m(X; Y)} \cdot \prod_{j=1}^m \|h_j\|_{i_j} \end{aligned}$$

proving by [theorems: 14.228, 14.212] and [eq: 16.312] that

$$\Psi_L \in L(X_{i_1} \dots X_{i_m}; Y) \text{ and } \|\Psi_L\|_{L(X_{i_1} \dots X_{i_m}; Y)} \leq \|L\|_{L^m(X; Y)} \quad (16.313)$$

Define now the function

$$\Psi: L_m(X; Y) \rightarrow L(X_{i_1} \dots X_{i_m}; Y) \text{ by } \Psi(L) = \Psi_L \quad (16.314)$$

Let $\alpha \in \mathbb{K}$, $L_1, L_2 \in L_m(X; Y)$ and $(h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ then we have

$$\begin{aligned} & (\Psi(L_1 + \alpha \cdot L_2))(h_1 : \dots : h_m) \stackrel{[\text{eq: 16.314}]}{=} \\ & \Psi_{L_1 + \alpha \cdot L_2}(h_1 : \dots : h_m) \stackrel{[\text{eq: 16.311}]}{=} \\ & (L_1 + \alpha \cdot L_2)(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \stackrel{[\text{lemma: 16.120}]}{=} \\ & L_1(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) + \alpha \cdot L_2(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \stackrel{[\text{eq: 16.311}]}{=} \\ & \Psi_{L_1}(h_1 : \dots : h_m) + \alpha \cdot \Psi_{L_2}(h_1 : \dots : h_m) = \\ & (\Psi_{L_1} + \alpha \cdot \Psi_{L_2})(h_1 : \dots : h_m) \stackrel{[\text{eq: 16.314}]}{=} \\ & (\Psi(L_1) + \alpha \cdot \Psi(L_2))(h_1 : \dots : h_m) \end{aligned}$$

proving by [lemma: 16.259] that

$$\Psi(L_1 + \alpha \cdot L_2) = \Psi(L_1) + \alpha \cdot \Psi(L_2) \quad (16.315)$$

Using [theorem: 16.262] we can define the function

$$\Phi: L_m(X; Y) \rightarrow L_m(X_{i_1} \dots X_{i_m}; Y) \text{ by } \Phi = (\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \circ \Psi \quad (16.316)$$

then we have

$$\begin{aligned}
 \Phi(L_1 + \alpha \cdot L_2) & \stackrel{[\text{eq: 16.316}]}{=} ((\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \circ \Psi)(L_1 + \alpha \cdot L_2) \\
 & = (\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1}(\Psi(L_1 + \alpha \cdot L_2)) \\
 & \stackrel{[\text{eq: 16.315}]}{=} (\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1}(\Psi(L_1) + \alpha \cdot \Psi(L_2)) \\
 & \stackrel{[\text{theorem: 16.262}]}{=} (\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1}(\Psi(L_1)) + \alpha \cdot (\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1}(\Psi(L_2)) \\
 & = ((\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \circ \Psi)(L_1) + \alpha \cdot ((\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \circ \Psi)(L_2) \\
 & = \Phi(L_1) + \alpha \cdot \Phi(L_2)
 \end{aligned}$$

proving that

$$\Phi \in \text{Hom}(L_m(X; Y), L_m(X_{i_1} \dots X_{i_m}; Y)) \quad (16.317)$$

If $L \in L_m(X; Y)$ then

$$\begin{aligned}
 \|\Phi(L)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} & \stackrel{[\text{eq: 16.316}]}{=} \|((\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \circ \Psi)(L)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \\
 & = \|(\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1}(\Psi(L))\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \\
 & \stackrel{[\text{theorem: 16.262}]}{=} \|\Psi(L)\|_{L(X_{i_1} \dots X_{i_m}; Y)} \\
 & \stackrel{[\text{eq: 16.314}]}{=} \|\Psi_L\|_{L(X_{i_1} \dots X_{i_m}; Y)} \\
 & \leq \|L\|_{L^m(X; Y)} \\
 & \stackrel{[\text{eq: 16.313}]}{=} \|L\|_{L^m(X; Y)}
 \end{aligned}$$

proving by [theorem: 14.201] together with [eq: 16.317] that

$$\Phi \in L(L_m(X; Y), L_m(X_{i_1} \dots X_{i_m}; Y)) \quad (16.318)$$

Finally if $L \in L_m(X; Y)$ and $(h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ then we have

$$\begin{aligned}
 \Phi(L)(h_1: \dots: h_m) & \stackrel{[\text{eq: 16.316}]}{=} (((\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \circ \Psi)(L))(h_1: \dots: h_m) \\
 & = ((\mathcal{I}_{X_{i_1} \dots X_{i_m}, Y})^{-1} \Psi(L))(h_1: \dots: h_m) \\
 & \stackrel{[\text{theorem: 16.262}]}{=} \Psi(L)(h_1, \dots, h_m) \\
 & \stackrel{[\text{eq: 16.314}]}{=} \Psi_L(h_1, \dots, h_m) \\
 & \stackrel{[\text{eq: 16.311}]}{=} L(I^{[i_1, 0]}(h_1): \dots: I^{[i_m, 0]}(h_m))
 \end{aligned}$$

proving

$$\Phi(L)(h_1: \dots: h_m) = L(I^{[i_1, 0]}(h_1): \dots: I^{[i_m, 0]}(h_m)) \quad \square$$

We are now ready to prove m -times partial differentiability of f at x implies that f has partial differentials $D_{x, [i_1 \dots i_m]} f$ for every $i \in \{1, \dots, n\}^m$.

Theorem 16.276. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], $A \subseteq X$, $x \in X$ and $f: A \rightarrow Y$ a function that is m -times differentiable at x then $\forall i \in \{1, \dots, n\}^m$ f has a partial differential $D_{x, [i_1 \dots i_m]} f$. Further $\forall (h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ we have*

$$D_{x, [i_1 \dots i_m]} f(h_1: \dots: h_m) = D_x^{[m]} f(I^{[i_1, 0]}(h_1): \dots: I^{[i_m, 0]}(h_m))$$

Proof. We prove this by induction, so let

$$\begin{aligned}
 S = & \left\{ m \in \mathbb{N} \mid \text{If } A \subseteq X \text{ and } f: A \rightarrow Y \text{ is a function that is } m\text{-times differentiable at } x \in A \text{ then} \right. \\
 & \forall i \in \{1, \dots, n\}^m \text{ } f \text{ has a partial differential } D_{x, [i_1 \dots i_m]} \text{ at } x \text{ and } \forall (h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j} \text{ we} \\
 & \left. \text{have } D_{x, [i_1 \dots i_m]} f(h_1: \dots: h_m) = D_x^{[m]} f(I^{[i_1, 0]}(h_1): \dots: I^{[i_m, 0]}(h_m)) \right\}
 \end{aligned}$$

then we have:

$1 \in \mathbf{S}$. If f is 1-times differentiable at x then by [lemma: 16.274] we have $\forall i \in \{1, \dots, n\}^1$ that

$$D_{x, [i_1 \dots i_1]} f \text{ exist}$$

and

$$D_{x, [i_1 \dots i_1]} f = D_x^{[1]} \circ I^{[i_1, 0]}$$

So $\forall h \in \prod_{j \in \{1\}} X_{i_j}$ we have

$$\begin{aligned} D_{x, [i_1 \dots i_1]}^{[1]} f(h_1 : \dots : h_1) &= D_{x, [i_1 \dots i_1]}^{[1]} f(h_1) \\ &= (D_{x, [i_1 \dots i_1]}^{[1]} \circ I^{[i_1, 0]})(h_1) \\ &= D_{x, [i_1 \dots i_1]}^{[1]}(I^{[i_1, 0]}(h_1)) \\ &= D_{x, [i_1 \dots i_1]}^{[1]}(I^{[i_1, 0]}(h_1) : \dots : I^{[i_1, 0]}(h_1)) \end{aligned}$$

which proves that

$$1 \in \mathbf{S}$$

$m \in \mathbf{S} \Rightarrow m + 1 \in \mathbf{S}$. As f is $(m + 1)$ -times differentiable at x it follows from [theorem: 16.140] that

$$D^{[m]} f \text{ is Fréchet differentiable at } x \quad (16.319)$$

Let $k \in \{1, \dots, n\}^{m+1}$. Define then

$$i \in \{1, \dots, n\}^m \text{ by } i_j = k_{j+1} \quad (16.320)$$

Let $y \in \mathfrak{D}_f^m$ then by [definition: 16.136] f is m -times differentiable at y , as $m \in \mathbf{S}$ it follows that

$$D_{y, [i_1 \dots i_m]} f \text{ exist} \quad (16.321)$$

and $\forall (h_1 \dots h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ we have

$$D_{y, [i_1 \dots i_m]} f(h_1 : \dots : h_m) = D_y^{[m]} f(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \quad (16.322)$$

As $\forall y \in \mathfrak{D}_f^m$ $D_{y, [i_1 \dots i_m]} f$ exist, hence by [definition: 16.136] $y \in \mathfrak{D}_{[i_1 \dots i_m], f}$ it follows that

$$\mathfrak{D}_f^m \subseteq \mathfrak{D}_{[i_1, \dots, i_m], f} \quad (16.323)$$

Using [lemma: 16.275] there exist a $\Phi \in L(L_m(X; Y), L_m(X_{i_1} \dots X_{i_m}; Y))$ such that $\forall L \in L_m(X; Y)$ and $(h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ we have

$$\Phi(L)(h_1 : \dots : h_m) = L(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \quad (16.324)$$

Let $(h_1 \dots h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ then as $D^{[m]} f(y) \stackrel{[\text{definition: 16.136}]}{=} D_y^{[m]} f \in L_m(X; Y)$ we have that

$$\begin{aligned} ((\Phi \circ D^{[m]} f)(y))(h_1 : \dots : h_m) &= \Phi(D^{[m]} f(y))(h_1 : \dots : h_m) \\ &= (D^{[m]} f(y))(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \\ &= (D_y^{[m]} f)(I^{[i_1, 0]}(h_1) : \dots : I^{[i_m, 0]}(h_m)) \\ &\stackrel{[\text{eq: 16.322}]}{=} (D_{y, [i_1 \dots i_m]} f)(h_1 : \dots : h_m) \\ &= (D_{[i_1 \dots i_m]} f(y))(h_1 : \dots : h_m) \end{aligned}$$

which proves by [lemma: 16.259] that

$$(\Phi \circ D^{[m]} f)(y) = D_{[i_1, \dots, i_m]} f(y).$$

As this is true $\forall y \in \mathfrak{D}_f^m \stackrel{[\text{eq: 16.323}]}{\subseteq} \mathfrak{D}_{[i_1 \dots i_m], f}$ it follows that

$$\Phi \circ D^{[m]} f = (D_{[i_1 \dots i_m]} f)|_{\mathfrak{D}_f^m} \quad (16.325)$$

As by [eq: 16.319] $D^{[m]}f$ is Fréchet differentiable at x and $\Phi \in L(L_m(X; Y), L_m(X_{i_1} \dots X_{i_m}; Y))$ so that by [example: 16.180] Φ is ∞ -differentiable on $L_m(X; Y)$ it follows from the chain rule [theorem: 16.67] that

$$\Phi \circ D^{[m]}f \text{ is Fréchet differentiable at } x \quad (16.326)$$

with

$$\begin{aligned} D_x(\Phi \circ D^{[m]}f) &\stackrel{[\text{theorem: 16.67}]}{=} D_{D^{[m]}f(x)}\Phi \circ D_x(D^{[m]}f) \\ &\stackrel{[\text{theorem: 16.140}]}{=} D_{D^{[m]}f(x)}\Phi \circ D_x^{[m+1]}f \\ &\stackrel{[\text{example: 16.180}]}{=} \Phi \circ D_x^{[m+1]}f \end{aligned}$$

so that

$$D_x(\Phi \circ D^{[m]}f) = \Phi \circ D_x^{[m+1]}f \quad (16.327)$$

Substituting [eq: 16.325] in the above proves that

$$(D_{[i_1 \dots i_m]}f)|_{\mathfrak{D}_c^m} \text{ is Fréchet differentiable at } x \text{ with } D_x(D_{[i_1 \dots i_m]}f)|_{\mathfrak{D}_c^m} = \Phi \circ D_x^{[m+1]}f$$

Using then the locality of Fréchet differentiability [see theorem: 16.57] on the above gives

$$D_{[i_1 \dots i_m]}f \text{ is Fréchet differentiable at } x \text{ with } D_x(D_{[i_1 \dots i_m]}f) = \Phi \circ D_x^{[m+1]}f$$

which after using [eq: 16.320] becomes

$$D_{[k_2 \dots k_{m+1}]}f \text{ is Fréchet differentiable at } x \text{ with } D_x(D_{[k_2 \dots k_{m+1}]}f) = \Phi \circ D_x^{[m+1]}f \quad (16.328)$$

Using now [theorem: 16.83] on the above proves that

$$D_{[k_2 \dots k_{m+1}]}f \text{ is } k_1\text{-partial differentiable at } x$$

with $D_{x, k_1}(D_{[k_2 \dots k_{m+1}]}f) = D_x(D_{[k_2 \dots k_{m+1}]}f) \circ I^{[k_1, 0]}$ or, using [eq: 16.328]

$$D_{x, k_1}(D_{[k_2 \dots k_{m+1}]}f) = (\Phi \circ D_x^{[m+1]}f) \circ I^{[k_1, 0]}$$

Using now [theorem: 16.273] on the it follows that

$$f \text{ has a partial } D_{x, [k_1 \dots k_{m+1}]}f \text{ at } x \quad (16.329)$$

with

$$D_{x, [k_1 \dots k_{m+1}]}f = (\Phi \circ D_x^{[m+1]}f) \circ I^{[k_1, 0]} \quad (16.330)$$

Now, as $D_x^{[m+1]}f \in L_{m+1}(X; Y) = L(X, L_m(X; Y))$, it follows that for $h_1 \in X_{k_1}$ we have

$$D_x^{[m+1]}f(I^{[k_1, 0]}(h_1)) \in L_m(X; Y) \quad (16.331)$$

So that for $(h_1, \dots, h_{m+1}) \in \prod_{j \in \{1, \dots, m+1\}} X_{k_j}$ we have

$$\begin{aligned} D_{x, [k_1 \dots k_{m+1}]}f(h_1: \dots: h_{m+1}) &\stackrel{[\text{eq: 16.330}]}{=} \\ &((\Phi \circ D_x^{[m+1]}f) \circ I^{[k_1, 0]})(h_1: \dots: h_{m+1}) \\ &(((\Phi \circ D_x^{[m+1]}f) \circ I^{[k_1, 0]})(h_1))(h_2: \dots: h_{m+1}) \\ &(\Phi(D_x^{[m+1]}f(I^{[k_1, 0]}(h_1))))(h_2: \dots: h_{m+1}) \\ &\stackrel{[\text{eqs: 16.324, 16.331}]}{=} \\ &(D_x^{[m+1]}f(I^{[k_1, 0]}(h_1)))(I^{[k_2, 0]}(h_2) \dots I^{[k_{m+1}, 0]}(h_{m+1})) \\ &D_x^{[m+1]}f(I^{[k_1, 0]}(h_1): \dots: I^{[k_{m+1}, 0]}(h_{m+1})) \end{aligned}$$

which proves that

$$D_{x, [k_1 \dots k_{m+1}]}f(h_1: \dots: h_{m+1}) = D_x^{[m+1]}f(I^{[k_1, 0]}(h_1): \dots: I^{[k_{m+1}, 0]}(h_{m+1}))$$

Finally combining the above with [eq: 16.329] proves that

$$m+1 \in S$$

□

The following theorem shows the relation between functions that are of class C^m and higher order partial derivatives.

Theorem 16.277. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], U a open set in X and*

$$f: U \rightarrow Y \text{ a function that is of class } C^m$$

then $\forall i \in \{1, \dots, n\}^m$ we have that

$$U = \mathfrak{D}_{f, [i_1 \dots i_m]}$$

and

$$D_{[i_1 \dots i_m]} f: \mathfrak{D}_{f, [i_1 \dots i_m]} = U \rightarrow L_m(X_{i_1}) \text{ is continuous}$$

Proof. As f is of class C^m we have by [theorem: 16.152] that

$$U = \mathfrak{D}_f^m \quad (16.332)$$

so that by [definition: 16.136]

$$\forall x \in U \text{ } f \text{ is } m\text{-times differentiable at } x \text{ with } D_x^{[m]} f = D^{[m]} f(x) \quad (16.333)$$

Let $x \in U$ then by the above, f is m -times differentiable at x , so that by the previous theorem [theorem: 16.276] we have that $\forall i \in \{1, \dots, n\}^m$ f has a partial differential $D_{x, [i_1 \dots i_m]} f$ at x . Hence by [definition: 16.267] $x \in \mathfrak{D}_{[i_1 \dots i_m], f}$ which as this is true for every $x \in U$ proves that $U \subseteq \mathfrak{D}_{[i_1 \dots i_m], f}$, as further by [theorem: 16.266] $\mathfrak{D}_{[i_1 \dots i_m], f} \subseteq U$ it follows that $\mathfrak{D}_{[i_1 \dots i_m], f} = U$. So we have that

$$\forall i \in \{1, \dots, n\}^m \quad U = \mathfrak{D}_{[i_1 \dots i_m], f} \quad (16.334)$$

Using lemma: 16.275] there exist a $\Phi \in L(L_m(X; Y), L_m(X_{i_m} \dots X_{i_1}; Y))$ such that for $L \in L_m(X; Y)$

$$\Phi(L)(h_1: \dots: h_m) = L(I^{[i_m, 0]}(h_1): \dots: I^{[i_1, 0]}(h_m)) \quad (16.335)$$

Let $x \in U$, $i \in \{1, \dots, n\}^m$ and take $(h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ then we have as $D^{[m]} f(x) \in L_m(X; Y)$ that

$$\begin{aligned} (\Phi \circ D^{[m]} f)(x)(h_1 \dots h_m) &= \Phi(D^{[m]} f(x))(h_1: \dots: h_m) \\ &= D^{[m]} f(x)(I^{[i_1, 0]}(h_1) \dots I^{[i_m, 0]}(h_m)) \\ &\stackrel{[\text{theorem: 16.276}]}{=} D_{x, [i_1 \dots i_m]} f(h_1: \dots: h_m) \\ &\stackrel{[\text{definition: 16.267}]}{=} D_{[i_1 \dots i_m]} f(x)(h_1: \dots: h_m) \end{aligned}$$

which proves by [lemma: 16.259] that $(\Phi \circ D^{[m]} f)(x) = D_{[i_1 \dots i_m]} f(x)$. As this is true $\forall x \in U$ it follows that

$$\Phi \circ D^{[m]} f = D_{[i_1 \dots i_m]} f$$

As $\Phi \in L(L_m(X; Y), L_m(X_{i_m} \dots X_{i_1}; Y))$, hence continuous and $D^{[m]} f$ is continuous [because f is of class C^m] it follows from [theorem: 14.158] that $\Phi \circ D^{[m]} f$ is continuous which proves that

$$D_{[i_1 \dots i_m]} f \text{ is continuous} \quad \square$$

Next we show that the order in which you do the partial differentiation does not matter.

Theorem 16.278. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a **Banach** space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], U a open set in X , $x \in U$ and $f: U \rightarrow Y$ a function that is of class C^m , $i \subseteq \{1, \dots, n\}^m$ and $(h_1, \dots, h_m) \in \prod_{j \in \{1, \dots, m\}} X_{i_j}$ then $\forall \sigma \in P_n$ we have*

$$D_{x, [i_1 \dots i_m]} f(h_1: \dots: h_m) = D_{[i_{\sigma(1)} \dots i_{\sigma(m)}]} f(h_{\sigma(1)}: \dots: h_{\sigma(m)})$$

Note 16.279. Existence of the partial differentials of order m at x is guaranteed by [theorem: 16.276].

Note 16.280. If $\forall i \in \{1, \dots, m\} \ h_i = 1$ then $\forall i \in \{1, \dots, n\} \ h_{\sigma(i)} = 1 = h_i$ so that in this case we have

$$D_{x, i_1, \dots, i_m} f(1: \dots : 1) = D_{i_{\sigma(1)} \dots i_{\sigma(m)}} f(1: \dots : 1)$$

Proof. We have

$$\begin{aligned} D_{x, [i_{\sigma(1)} \dots i_{\sigma(m)}]} f(h_{\sigma(1)}: \dots : h_{\sigma(m)}) &\stackrel{[\text{theorem: 16.276}]}{=} \\ D_x^{[m]} f(I^{[i_{\sigma(1)}, 0]}(h_{\sigma(1)}): \dots : I^{[i_{\sigma(m)}, 0]}(h_{\sigma(m)})) &\stackrel{[\text{theorem: 16.247}]}{=} \\ D_x^{[m]} (I^{[i_1, 0]}(h_1): \dots : I^{[i_m, 0]}(h_m)) &\stackrel{[\text{theorem: 16.276}]}{=} \\ D_{x, [i_1 \dots i_m]} f(h_1: \dots : h_m) & \square \end{aligned}$$

It was show in [example: 16.85] that the existence of partial differentials at a point does not mean that the function is differentiable at the point. We examine now under which extra conditions the existence of partial differentials ensure the existence of the Fréchet differential. First we need some extra definitions and lemma's.

Definition 16.281. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98] then given $h = (h_1, \dots, h_n) \in X$ and $i \in \{0, \dots, n\}$ we define

$$h^{\{i\}} = \begin{cases} \sum_{j=1}^i I^{[j, 0]}(h_j) & \text{if } i \in \{1, \dots, n\} \\ 0 & \text{if } i = 0 \end{cases}$$

Example 16.282. Let $(h_1, h_2, h_3) \in \prod_{i \in \{1, \dots, n\}} X$ then

$$\begin{aligned} h^{\{0\}} &= (0, 0, 0) \\ h^{\{1\}} &= \sum_{j=1}^1 I^{[j, 0]}(h_j) = I^{[1, 0]}(h_1) = (h_1, 0, 0) \\ h^{\{2\}} &= \sum_{j=1}^2 I^{[j, 0]}(h_j) = I^{[1, 0]}(h_1) + I^{[2, 0]}(h_2) = (h_1, 0, 0) + (0, h_2, 0) = (h_1, h_2, 0) \\ h^{\{3\}} &= \sum_{j=1}^3 I^{[j, 0]}(h_j) = I^{[1, 0]}(h_1) + I^{[2, 0]}(h_2) + I^{[3, 0]}(h_3) = (h_1, 0, 0) + (0, h_2, 0) + (0, 0, h_3) = (h_1, h_2, h_3) \end{aligned}$$

The above example suggest the following lemma.

Lemma 16.283. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98] then given $h \in X$ we have

1. $h^{\{0\}} = 0$
2. $h^{\{n\}} = h$
3. $\forall i \in \{1, \dots, n\}$ we have $\forall k \in \{1, \dots, n\}$ that $(h^{\{i\}})_k = \begin{cases} 0 & \text{if } i < k \\ h_k & \text{if } k \leq i \end{cases}$. In other words we have

$$h^{\{i\}} = \begin{cases} \left(\underbrace{h_1, \dots, h_i}_i, \underbrace{0, \dots, 0}_{n-i} \right) & \text{if } i \in \{1, \dots, n\} \\ \left(\underbrace{0, \dots, 0}_n \right) & \text{if } i = 0 \end{cases}$$

4. Let $x, h \in \prod_{i \in \{1, \dots, n\}} X_i$ then $\forall i \in \{1, \dots, n\}$ we have

$$a. I^{[i, x+h^{\{i\}}]}(x_i + h_i) = x + h^{\{i\}}$$

$$b. I^{[i, x+h^{\{i\}}]}(x_i) = x + h^{\{i-1\}}$$

5. $\forall i \in \{0, \dots, n\}$ we have $\|h^{\{i\}}\|_{\max} \leq \|h\|_{\max}$

Proof.

1. This follows from the definition.

$$2. h^{\{n\}}_{n \in \mathbb{N} \Rightarrow 0 \neq n} = \sum_{j=1}^n I^{[j, 0]} h_j \stackrel{[\text{lemma: 16.71}]}{=} h$$

3. Let $i \in \{1, \dots, n\}$ then for $k \in \{1, \dots, n\}$ we have either:

$k \leq i$. Then

$$\begin{aligned} (h^{\{i\}})_k &= \left(\sum_{j=1}^i I^{[j, 0]}(h_j) \right)_k \\ &= \sum_{j=1}^i (I^{[j, 0]}(h_j))_k \\ &\stackrel{[\text{lemma: 16.71}]}{=} \sum_{j=1}^i \delta_{j, k} \cdot h_j \\ &\stackrel{k \in \{1, \dots, i\}}{=} h_k \end{aligned}$$

$i < k$. Then

$$\begin{aligned} (h^{\{i\}})_k &= \left(\sum_{j=1}^i I^{[j, 0]}(h_j) \right)_k \\ &= \sum_{j=1}^i (I^{[j, 0]}(h_j))_k \\ &\stackrel{[\text{lemma: 16.71}]}{=} \sum_{j=1}^i \delta_{j, k} \cdot h_j \\ &\stackrel{\forall j \in \{1, \dots, i\} \quad k \neq j}{=} 0 \end{aligned}$$

so that

$$(h^{\{i\}})_k = \begin{cases} 0 & \text{if } i < k \\ h_k & \text{if } k \leq i \end{cases}$$

4. Let $x, h \in \prod_{i \in \{1, \dots, n\}} X_i$ and $i \in \{1, \dots, n\}$ then we have:

a. Then we have for $k \in \{1, \dots, n\}$ either:

$k = i$. Then

$$\begin{aligned} (I^{[i, x+h^{\{i\}}]}(x_i + h_i))_k &\stackrel{k=i}{=} (I^{[i, x+h^{\{i\}}]}(x_i + h_i))_i \\ &\stackrel{[\text{definition: 16.70}]}{=} x_i + h_i \\ &\stackrel{i=k}{=} x_k + h_k \\ &\stackrel{(3) \wedge k \leq i}{=} x_k + (h^{\{i\}})_k \\ &= (x + h^{\{i\}})_k \end{aligned}$$

$k \neq i$. Then

$$(I^{[i, x+h^{\{i\}}]}(x_i + h_i))_k \stackrel{[\text{definition: 16.70}]}{=} (x + h^{\{i\}})_k$$

proving that in all cases

$$I^{[i, x+h^{\{i\}}]}(x_i + h_i) = x + h^{\{i\}}$$

b. Then we have for $k \in \{1, \dots, n\}$ either:

$k = i$.

$$\begin{aligned} (I^{[i, x+h^{\{i\}}]}(x_i))_k & \stackrel{=}{=}_{k=i} (I^{[i, x+h^{\{i\}}]}(x_i))_i \\ & \stackrel{=}{=}_{[\text{definition: 16.70}]} x_i \\ & = x_i + 0 \\ & \stackrel{=}{=}_{i=k} x_k + 0 \\ & \stackrel{=}{=}_{i-1 < i=k} x_k + (h^{\{i-1\}})_k \\ & = (x + h^{\{i-1\}})_k \end{aligned}$$

$k < i$.

$$\begin{aligned} (I^{[i, x+h^{\{i\}}]}(x_i))_k & \stackrel{=}{=}_{[\text{definition: 16.70}]} (x + h^{\{i\}})_k \\ & = x_k + (h^{\{i\}})_k \\ & \stackrel{=}{=}_{(3) \wedge k < i} x_k + h_k \\ & \stackrel{=}{=}_{(3) \wedge k \leq i-1} x_k + (h^{\{i-1\}})_k \\ & = (x + h^{\{i-1\}})_k \end{aligned}$$

$i < k$.

$$\begin{aligned} (I^{[i, x+h^{\{i\}}]}(x_i))_k & \stackrel{=}{=}_{[\text{definition: 16.70}]} (x + h^{\{i\}})_k \\ & = x_k + (h^{\{i\}})_k \\ & \stackrel{=}{=}_{(3) \wedge i < k} x_k + 0 \\ & \stackrel{=}{=}_{(3) \wedge i-1 < i < k} x_k + (h^{\{i-1\}})_k \\ & = (x + h^{\{i-1\}})_k \end{aligned}$$

proving that

$$I^{[i, x+h^{\{i\}}]}(x_i) = x + h^{\{i-1\}}$$

5. Given $i \in \{0, \dots, n\}$ then we have either:

$i = 0$. Then

$$\|h^{\{i\}}\|_{\max} = \|h^{\{0\}}\|_{\max} \stackrel{=}{=}_{(1)} \|0\|_{\max} = 0 \leq \|h\|_{\max}$$

$i \in \{1, \dots, n\}$. Then for $k \in \{1, \dots, n\}$ we have

$k \in \{1, \dots, i\}$. Then

$$\begin{aligned} (h^{\{i\}})_k & = \left(\sum_{j=1}^i I^{[j, 0]}(h_j) \right)_k \\ & = \sum_{j=1}^i (I^{[j, 0]}(h_j))_k \\ & \stackrel{=}{=}_{[\text{lemma: 16.71}]} \sum_{j=1}^i \delta_{j,k} \cdot h_j \\ & = h_k \end{aligned}$$

so that $\|(h^{\{i\}})_k\|_k = \|h_k\|_k$

$k \in \{i+1, \dots, n\}$. Then

$$\begin{aligned} (h^{(i)})_k &= \left(\sum_{j=1}^i I^{[j,0]}(h_j) \right)_k \\ &= \sum_{j=1}^i (I^{[j,0]}(h_j))_k \\ &\stackrel{[\text{lemma: 16.71}]}{=} \sum_{j=1}^i \delta_{j,k} \cdot h_j \\ &\stackrel{\forall j \in \{1, \dots, n\} \quad k \neq j}{=} 0 \end{aligned}$$

so that $\|(h^{(i)})_k\|_k = \|0\|_k \leq \|h_k\|_k$

proving that $\forall k \in \{1, \dots, n\} \quad \|(h^{(i)})_k\|_k \leq \|h_k\|_k$. Hence

$$\|h^{(i)}\|_{\max} = \max(\{\|h^{(i)}\|_k \mid k \in \{1, \dots, n\}\}) \leq \max(\{\|h_k\|_k \mid k \in \{1, \dots, n\}\}) = \|h\|_{\max}$$

So we have that

$$\forall i \in \{0, \dots, n\} \quad \|h^{(i)}\|_{\max} \leq \|h\|_{\max} \text{ for every } h \in X \quad \square$$

Lemma 16.284. Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a **Banach** space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], U a open set in X , $x \in U$, $\delta \in \mathbb{R}^+$ with $B_{\|\cdot\|_{\max}}(x, \delta) \subseteq U$ and a function

$$f: U \rightarrow Y$$

such that $\forall i \in \{1, \dots, n\}$

$$U = \mathfrak{D}_{i,f}$$

and

$$D_i f: \mathfrak{D}_{i,f} \rightarrow L(X_i, Y) \text{ is continuous}$$

Then $\forall u, v \in B_{\|\cdot\|_i}(x_i, \delta)$ we have that

$$\sup(\{\|D_i f(I^{[i,x]}(u + t \cdot (v - u)))\|_{L(X_i, Y)} \mid t \in [0, 1]\}) \text{ exists}$$

and

$$\|f(I^{[i,x]}(u)) - f(I^{[i,x]}(v))\|_Y \leq \|u - v\|_i \cdot \sup(\{\|D_i f(I^{[i,x]}(u + t \cdot (v - u)))\|_{L(X_i, Y)} \mid t \in [0, 1]\})$$

Note 16.285. Here $\mathfrak{D}_{i,f}$ and $D_i f$ are defined in [definition: 16.75] to be

$$\mathfrak{D}_{i,f} = \{y \in U \mid f \text{ has a } i\text{-partial differential at } y\}$$

and

$$D_i f: \mathfrak{D}_{i,f} \rightarrow L(X_i, Y) \text{ where } D_i f(x) = D_{x,i} f$$

Proof. Let $i \in \{1, \dots, n\}$. As $x \in U$ we have by the hypothesis that f has a i -partial differential at x so that by the definition of partial differentials [see definition: 16.74]

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(U) \rightarrow Y \text{ is Fréchet differentiable at } x_i \quad (16.336)$$

and

$$D_{i,x} f = D_{x_i}(f \circ I^{[i,x]}) \quad (16.337)$$

Let $t \in (I^{[i,x]})^{-1}(U)$ and take $y_t = I^{[i,x]}(t) \in U$ then it follows from the hypothesis that f has a partial differential at y_t , hence using [definition: 16.74]

$$f \circ I^{[i,y_t]}: (I^{[i,y_t]})^{-1}(U) \rightarrow Y \text{ is Fréchet differentiable at } (y_t)_i \quad (16.338)$$

and

$$D_{i,y_t}f = D_{(y_t)_i}(f \circ I^{[i,y_t]}) \quad (16.339)$$

Now $\forall s \in X_i$ we have

$$I^{[i,y_t]}(s) = \left(y_1, \dots, \underbrace{s}_i, \dots, y_n \right)_{y_t = I^{[i,x]}(t)} \left(x_1, \dots, \underbrace{s}_i, \dots, x_n \right) = I^{[i,x]}(s)$$

which proves that

$$I^{[i,y_t]} = I^{[i,x]}$$

Substituting the above in [eqs: 16.338, 16.339] gives

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(U) \rightarrow Y \text{ is Fréchet differentiable at } (y_t)_i \quad (16.340)$$

$$D_{i,y_t}f = D_{(y_t)_i}(f \circ I^{[i,x]}) \quad (16.341)$$

As $y_t = I^{[i,x]}(t)$ we have that $(y_t)_i = (I^{[i,x]}(t))_i = t$ so substituting this in [eq: 16.340, 16.341] gives

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(U) \rightarrow Y \text{ is Fréchet differentiable at } t \quad (16.342)$$

$$D_{i,I^{[i,x]}(t)}f = D_t(f \circ I^{[i,x]}) \quad (16.343)$$

Further as U is open we have by [lemma: 16.71(8)] that

$$(I^{[i,x]})^{-1}(U) \text{ is open} \quad (16.344)$$

As $t \in (I^{[i,x]})^{-1}(U)$ was chosen arbitrary we have that

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(U) \rightarrow Y \text{ is Fréchet differentiable on } (I^{[i,x]})^{-1}(U) \quad (16.345)$$

$$\forall t \in (I^{[i,x]})^{-1}(U) \text{ we have } D_{i,I^{[i,x]}(t)}f = D_t(f \circ I^{[i,x]}) \in L(X_i, Y) \quad (16.346)$$

Let $t \in (I^{[i,x]})^{-1}(U)$ then

$$\begin{aligned} D(f \circ I^{[i,x]})(t) &\stackrel{[\text{definition: 16.78}]}{=} D_t(f \circ I^{[i,x]}) \\ &\stackrel{[\text{eq: 16.346}]}{=} D_{i,I^{[i,x]}(t)}f \\ &\stackrel{[\text{definition: 16.78}]}{=} D_if(I^{[i,x]}(t)) \\ &= (D_tf \circ I^{[i,x]})(t) \end{aligned}$$

which proves that

$$D(f \circ I^{[i,x]}) = ((D_if) \circ I^{[i,x]})|_{(I^{[i,x]})^{-1}(U)} \stackrel{[\text{theorem: 2.96}]}{=} D_if \circ (I^{[i,x]})|_{(I^{[i,x]})^{-1}(U)} \quad (16.347)$$

Using [lemma: 16.71(7)] it follows that $(I^{[i,x]})|_{(I^{[i,x]})^{-1}(U)}$ is continuous, further by the hypothesis D_if is continuous. Hence $D_if \circ (I^{[i,x]})|_{(I^{[i,x]})^{-1}(U)}$ is continuous which proves that $D(f \circ I^{[i,x]})$ is continuous. Combining this with [eq: 16.345] proves that

$$f \circ I^{[i,x]}: (I^{[i,x]})^{-1}(U) \rightarrow Y \text{ is of class } C^1 \quad (16.348)$$

Now we have

$$\begin{aligned} t \in (I^{[i,x]})^{-1}(B_{\|\cdot\|_{\max}}(x, \delta)) &\Leftrightarrow I^{[i,x]}(t) \in B_{\|\cdot\|_{\max}}(x, \delta) \\ &\Leftrightarrow \|I^{[i,x]}(t) - x\|_{\max} < \delta \\ &\Leftrightarrow \left\| \left(x_1, \dots, \underbrace{t}_i, \dots, x_n \right) - \left(x_1, \dots, \underbrace{x_i}_i, \dots, x_n \right) \right\|_{\max} < \delta \\ &= \left\| \left(0, \dots, \underbrace{t - x_i}_i, \dots, 0 \right) \right\|_{\max} < \delta \\ &\Leftrightarrow \|t - x_i\|_i < \delta \\ &\Leftrightarrow t \in B_{\|\cdot\|_i}(x_i, \delta) \end{aligned}$$

so that

$$B_{\|\cdot\|_i}(x_i, \delta) = (I^{[i,x]})^{-1}(B_{\|\cdot\|_{\max}}(x, \delta)) \quad (16.349)$$

Using [eqs: 16.344, 16.348] we can apply the Mean Value Theorem [see corollary: 16.230] which proves that $\forall u, v \in B_{\|\cdot\|_i}(x_i, \delta)$ we have that

$$\sup \{ \|D_{u+t \cdot (v-u)}(f \circ I^{[i,x]})\|_{L(X_i, Y)} | t \in [0, 1] \}$$

exist and

$$\|(f \circ I^{[i,x]})(u) - (f \circ I^{[i,x]})(v)\|_Y \leq \|u - v\|_i \cdot \sup \{ \|D_{u+t \cdot (v-u)}(f \circ I^{[i,x]})\|_{L(X_i, Y)} | t \in [0, 1] \}$$

As

$$\begin{aligned} D_{u+t \cdot (v-u)}(f \circ I^{[i,x]}) &\stackrel{[\text{definition: 16.78}]}{=} D(f \circ I^{[i,x]})(u + t \cdot (v - u)) \\ &\stackrel{[\text{eq: 16.347}]}{=} (D_i f \circ (I^{[i,x]})_{|(I^{[i,x]})^{-1}(U)})(u + t \cdot (v - u)) \\ &= D_i f(I^{[i,x]}(u + t \cdot (v - u))) \end{aligned}$$

it follows that

$$\sup \{ \|D_i(f(I^{[i,x]}(u + t \cdot (v - u))))\|_{L(X_i, Y)} | t \in [0, 1] \} \text{ exists}$$

and

$$\|f(I^{[i,x]}(u)) - f(I^{[i,x]}(v))\|_Y \leq \|u - v\|_i \cdot \sup \{ \|D_i(f(I^{[i,x]}(u + t \cdot (v - u))))\|_{L(X_i, Y)} | t \in [0, 1] \} \quad \square$$

We are now ready to prove under which conditions the existence of partial differentials implies differentiability.

Theorem 16.286. *Let $n \in \mathbb{N} \setminus \{1\}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], U a open set in X and*

$$f: U \rightarrow Y$$

a function such that $\forall i \in \{1, \dots, n\}$

$$U = \mathfrak{D}_{i,f}$$

and

$$D_i f: \mathfrak{D}_{i,f} = U \rightarrow L(X_i, Y) \text{ is continuous}$$

then

$$f \text{ is of class } C^1$$

and $\forall x \in U = \mathfrak{D}_{i,f}$

$$D_x f = \sum_{i=1}^n (D_{x,i} f \circ \pi_i)$$

or using [definitions: 16.48, 16.78]

$$Df(x) = \sum_{i=1}^n (D_i f(x) \circ \pi_i)$$

Proof. By the hypothesis we have that

$$\forall x \in U \quad \forall i \in \{1, \dots, n\} \quad D_{x,i} f \text{ exist} \quad (16.350)$$

and

$$\forall i \in \{1, \dots, n\} \quad D_i f: U \rightarrow L(X_i, Y) \text{ defined by } D_i f(x) = D_{x,i} f \text{ is continuous} \quad (16.351)$$

Let $x \in U$ then as U is open there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{\|\cdot\|_{\max}}(x, \delta) \subseteq U$$

Define

$$K: X \rightarrow Y \text{ by } K(y) = \sum_{j=1}^n D_j f(x)(y_j) \quad (16.352)$$

We proceed now to prove that K is the differential of f at x , first we prove that $K \in L(X, Y)$. Let $y, z \in X$ and $\alpha \in \mathbb{K}$ then we have

$$\begin{aligned} K(y + \alpha \cdot z) &= \sum_{j=1}^n D_j f(x)(y + \alpha \cdot z)_j \\ &= \sum_{j=1}^n D_j f(x)(y_j + \alpha \cdot z_j) \\ &= \sum_{j=1}^n (D_j f(x)(y_j) + \alpha \cdot D_j f(x)(z_j)) \\ &= \sum_{j=1}^n D_j f(x)(y_j) + \alpha \cdot \sum_{j=1}^n D_j f(x)(z_j) \\ &= K(y) + \alpha \cdot K(z) \end{aligned}$$

proving that

$$K \in \text{Hom}(X, Y) \quad (16.353)$$

Further for every $y \in X$ we have

$$\begin{aligned} \|K(y)\|_Y &= \left\| \sum_{j=1}^n D_j f(x)(y_j) \right\|_Y \\ &\leq \sum_{j=1}^n \|D_j f(x)(y_j)\|_Y \\ &\leq \sum_{j=1}^n \|D_j f(x)\|_{L(X_j, Y)} \cdot \|y_j\|_j \\ &\leq \sum_{j=1}^n \|D_j f(x)\|_{L(X_j, Y)} \cdot \|y\|_{\max} \\ &\leq \left(\sum_{j=1}^n \|D_j f(x)\|_{L(X_j, Y)} \right) \cdot \|y\|_{\max} \end{aligned}$$

So using [theorem: 14.201] and [eq: 16.353] it follows that

$$K \in L(X, Y) \quad (16.354)$$

Let $i \in \{1, \dots, n\}$. From the above it follows using [example: 16.180] that K is ∞ -times differentiable on X and $\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \subseteq X$ $D_y K = K$. Applying [theorem: 16.83] on K gives then

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \text{ } D_{y,i} K \text{ exists and } D_{y,i} K = K \circ I^{[i,0]} \quad (16.355)$$

Now for $k \in X_i$ we have

$$\begin{aligned} (K \circ I^{[i,0]})(k) &= K(I^{[i,0]}(k)) \\ &= \sum_{j=1}^n D_j f(x)(I^{[i,0]}(k))_j \\ &\stackrel{[\text{lemma: 16.71}]}{=} \sum_{j=1}^n D_j f(x)(\delta_{i,j} \cdot k) \\ &= \sum_{j=1}^n \delta_{i,j} \cdot D_j f(x)(k) \\ &= D_i f(x)(k) \end{aligned}$$

proving that $K \circ I^{[i,0]} = D_i f(x)$. Substituting this result in [eq: 16.355] gives

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \text{ we have that } D_{y,i}K \text{ exists and } D_{y,i}K = D_i f(x) \quad (16.356)$$

Further applying [theorem: 16.82] on [eq: 16.350] taking account that $B_{\|\cdot\|_{\max}}(x, \delta)$ is open we have that

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \ D_{y,i}(f|_{B_{\|\cdot\|_{\max}}(x, \delta)}) \text{ exist and } D_{y,i}(f|_{B_{\|\cdot\|_{\max}}(x, \delta)}) = D_{y,i}f \stackrel{[\text{eq: 16.78}]}{=} D_i f(y) \quad (16.357)$$

Also, using [example: 16.84], it follows that

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \ D_{y,i}C_{f(x)} \text{ exist and } D_{y,i}C_{f(x)} = C_0 \quad (16.358)$$

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \ D_{y,i}C_{\sum_{j=1}^n D_j f(x) \cdot x_j} \text{ exist and } D_{y,i}C_{\sum_{j=1}^n D_j f(x) \cdot x_j} = C_0 \quad (16.359)$$

Define

$$L: B_{\|\cdot\|_{\max}}(x, \delta) \rightarrow Y \text{ by } L(y) = \sum_{j=1}^n D_j f(x)(y_j - x_j) = \left(\sum_{j=1}^n D_j f(x)(y_j) \right) - \sum_{j=1}^n D_j f(x)(x_j)$$

then as, $K(y) \stackrel{[\text{eq: 16.352}]}{=} \sum_{j=1}^n D_j f(x)(y_j)$, we have

$$L = K - C_{\sum_{j=1}^n D_j f(x) \cdot x_j} \quad (16.360)$$

Using [theorem: 16.86] together with [eqs: 16.356, 16.359] it follows that

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \ D_{y,x}L \text{ exists and } D_{y,x}L = D_i f(x) \quad (16.361)$$

Define

$$g: B_{\|\cdot\|_{\max}}(x, \delta) \rightarrow Y \text{ by } g(y) = f(y) - f(x) - \sum_{j=1}^n D_j f(x)(y_j - x_j) \quad (16.362)$$

then

$$g = f|_{B_{\|\cdot\|_{\max}}(x, \delta)} - C_{f(x)} - L$$

Hence by [eqs: 16.357, 16.358, 16.361] and [theorem: 16.86] it follows that

$$\forall y \in B_{\|\cdot\|_{\max}}(x, \delta) \ D_{y,i}g \text{ exist and } D_{y,i}g = D_{y,i}f - D_i f(x) \stackrel{[\text{definition: 16.78}]}{=} D_i f(y) - D_i f(x) \quad (16.363)$$

As $\text{dom}(g) = B_{\|\cdot\|_{\max}}(x, \delta)$ it follows from [definition: 16.78] that $\mathfrak{D}_{i,g} = B_{\|\cdot\|_{\max}}(x, \delta)$ and $D_i g$ is defined by

$$D_i g: \mathfrak{D}_{i,g} = B_{\|\cdot\|_{\max}}(x, \delta) \rightarrow L(X_i, Y) \text{ by } D_i g(y) = D_{y,i}g$$

Further $\forall y \in B_{\|\cdot\|_{\max}}(x, \delta)$

$$\begin{aligned} D_i g(y) &\stackrel{[\text{definition: 16.78}]}{=} D_{y,i}g \\ &\stackrel{[\text{eq: 16.363}]}{=} D_i f(y) - D_i f(x) \\ &= D_i f(y) - C_{D_i f(x)}(y) \\ &= (D_i f)|_{B_{\|\cdot\|_{\max}}(x, \delta)}(y) - C_{D_i f(x)}(y) \\ &= ((D_i f)|_{B_{\|\cdot\|_{\max}}(x, \delta)} - C_{D_i f(x)})(y) \end{aligned}$$

which proves that

$$D_i g = (D_i f)|_{B_{\|\cdot\|_{\max}}(x, \delta)} - C_{D_i f(x)}.$$

As $D_i f$ is continuous it follows from [theorem: 14.152] that $(D_i f)|_{B_{\|\cdot\|_{\max}}(x, \delta)}$ is continuous which together with the continuity of $-C_{D_i f(x)}$ proves that

$$D_i g \text{ is continuous} \quad (16.364)$$

Further we have

$$g(x) \stackrel{[\text{eq: 16.362}]}{=} f(x) - f(x) - \sum_{j=1}^n D_j f(x_j - x_j) = 0 \quad (16.365)$$

Take $\varepsilon \in \mathbb{R}^+$. As $D_i f$ is continuous at x there exist a $\delta_{i,2} \in \mathbb{R}^+$ such that $\forall y \in U$ with $\|y - x\|_{\max} < \delta_{i,2}$ we have $\|D_i f(y) - D_i f(x)\|_{L(X_i, Y)} < \frac{\varepsilon}{n}$. Define $\delta_2 = \min(\delta, \min(\{\delta_{j,2} | j \in \{1, \dots, n\}\})) \in \mathbb{R}^+$ then we have [using eq: 16.363] that

$$\forall i \in \{1, \dots, n\} \quad \forall y \in B_{\|\cdot\|_{\max}}(x, \delta_2) \text{ we have } \|D_i g(y)\|_{L(X_i, Y)} < \frac{\varepsilon}{n} \quad (16.366)$$

Let $h \in U_x$ with $\|h\|_{\max} < \delta_2$. Then $h \in B_{\|\cdot\|_{\max}}(0, \delta_2)$ and $\forall i \in \{0, \dots, n\}$ we have that

$$\|(x + h^{\{i\}}) - x\|_{\max} = \|h^{\{i\}}\|_{\max} \underset{[\text{lemma: 16.283}]}{\leq} \|h\|_{\max} < \delta_2$$

so that

$$\forall i \in \{0, \dots, n\} \quad x + h^{\{i\}} \in B_{\|\cdot\|_{\max}}(x, \delta_2) \text{ and by [eq: 16.366]} \quad \|D_i g(x + h^{\{i\}})\|_{L(X_i, Y)} < \frac{\varepsilon}{n} \quad (16.367)$$

Next

$$\begin{aligned} g(x + h) &\underset{[\text{lemma: 16.283}]}{=} g(x + h^{\{n\}}) \\ &\underset{[\text{eq: 16.365}]}{=} g(x + h^{\{n\}}) - g(x) \\ &\underset{[\text{lemma: 16.283}]}{=} g(x + h^{\{n\}}) - g(x + h^{\{0\}}) \\ &\underset{[\text{theorem: 11.21}]}{=} \sum_{i=1}^n (g(x + h^{\{i\}}) - g(x + h^{\{i-1\}})) \end{aligned}$$

so that

$$\|g(x + h)\|_Y \leq \sum_{i=1}^n \|g(x + h^{\{i\}}) - g(x + h^{\{i-1\}})\|_Y \quad (16.368)$$

Let $i \in \{1, \dots, n\}$ then, as $\|x_i + h_i - x_i\|_i = \|h_i\|_i \leq \|h\|_{\max} < \delta_2$, it follows that $x_i + h_i, x_i \in B_{\|\cdot\|_i}(x_i, \delta)$, hence, as $D_i g$ is continuous [see eq: 16.364] and $x + h^{\{i\}}, x_i \in B_{\|\cdot\|_{\max}}(x, \delta_2) \subseteq U$, we can use [lemma: 16.284] to get

$$\begin{aligned} &\|g(I^{[i, x+h^{\{i\}}]}(x_i + h_i)) - g(I^{[i, x+h^{\{i\}}]}(x_i))\|_Y \leq \\ &\|x_i + h_i - x_i\|_i \cdot \sup(\{\|D_i g(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))\|_{L(X_i, Y)} | t \in [0, 1]\}) = \\ &\|h_i\|_i \cdot \sup(\{\|D_i g(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))\|_{L(X_i, Y)} | t \in [0, 1]\}) \end{aligned}$$

As $I^{[i, x+h^{\{i\}}]}(x + h_i) \underset{[\text{lemma: 16.283}]}{=} x + h^{\{i\}}$ and $I^{[i, x+h^{\{i\}}]}(x_i) \underset{[\text{lemma: 16.283}]}{=} x + h^{\{i-1\}}$ it follows that

$$\begin{aligned} &\|g(x + h^{\{i\}}) - g(x + h^{\{i-1\}})\|_Y \leq \\ &\|h_i\|_i \cdot \sup(\{\|D_i g(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))\|_{L(X_i, Y)} | t \in [0, 1]\}) \end{aligned} \quad (16.369)$$

To simplify the above inequality we look at $I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i)$. For $k \in \{1, \dots, n\}$ and $t \in [0, 1]$ we have either:

$k = i$. Then

$$\begin{aligned} \|(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i) - x)_k\|_k &= \|(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i) - x)_i\|_i \\ &= \|(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))_i - x_i\|_i \\ &= \|x_i + h_i - t \cdot h_i - x_i\|_i \\ &= \|h_i \cdot (1 - t)\|_i \\ &\underset{0 \leq t \leq 1}{=} (1 - t) \cdot \|h\|_i \\ &\leq \|h_i\|_i \\ &\leq \|h\|_{\max} \\ &< \delta_2 \end{aligned}$$

$k \neq i$. Then

$$\begin{aligned}
 \|(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i) - x)_k\|_k &= \|(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))_k - x_k\|_k \\
 &= \|(x + h^{\{i\}})_k - x_k\|_k \\
 &= \|x_k + (h^{\{i\}})_k - x_k\|_k \\
 &= \|(h^{\{i\}})_k\|_k \\
 &\leq \|h^{\{i\}}\|_{\max} \\
 &\stackrel{[\text{lemma: 16.283}]}{=} \|h\|_{\max} \\
 &\stackrel{[\text{lemma: 16.283}]}{\leq} \|h\|_{\max} \\
 &< \delta_2
 \end{aligned}$$

so that

$$\begin{aligned}
 \|I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i) - x\|_{\max} &= \max(\{\|(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i) - x)_k\|_k | k \in \{1, \dots, n\}\}) \\
 &< \delta_2
 \end{aligned}$$

Hence $I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i) \in B_{\|\cdot\|_{\max}}(x, \delta_2)$ so that by [eq: 16.366] we have

$$\|D_i g(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))\|_{L(X_i, Y)} < \frac{\varepsilon}{n}$$

so that

$$\sup(\{\|D_i g(I^{[i, x+h^{\{i\}}]}(x_i + h_i - t \cdot h_i))\|_{L(X_i, Y)} | t \in [0, 1]\}) < \frac{\varepsilon}{n}$$

which by [eq: 16.369] proves that

$$\|g(x + h^{\{i\}}) - g(x + h^{\{i-1\}})\|_Y < \|h_i\|_i \cdot \frac{\varepsilon}{n} \leq \|h\|_{\max} \cdot \frac{\varepsilon}{n}$$

Substituting the above in [eq: 16.368] results then in

$$\|g(x + h)\|_Y \leq \sum_{i=1}^n \|g(x + h^{\{i\}}) - g(x + h^{\{i-1\}})\| < \sum_{i=1}^n \frac{\varepsilon}{n} \cdot \|h\|_{\max} = \varepsilon \cdot \|h\|_{\max} \quad (16.370)$$

Further

$$\begin{aligned}
 g(x + h) &\stackrel{[\text{eq: 16.362}]}{=} f(x + h) - f(x) - \sum_{i=1}^n D_i f(x)((x_i + h_i) - x_i) \\
 &= f(x + h) - f(x) - \sum_{i=1}^n D_i f(x)(h_i)
 \end{aligned}$$

so that

$$\left\| f(x + h) - f(x) - \sum_{i=1}^n D_i f(x)(h_i) \right\|_Y \leq \varepsilon \cdot \|h\|_{\max}$$

or as $K(h) \stackrel{[\text{eq: 16.352}]}{=} \sum_{i=1}^n D_i f(x)(h_i)$ we have that $\forall h \in (B_{\|\cdot\|_{\max}}(x, \delta))_x$ with $\|h\|_{\max} < \delta_2$ we have

$$\|f(x + h) - f(x) - K(h)\|_Y \leq \varepsilon \cdot \|h\|_{\max}$$

As U is open so that $x \in U^\circ$, $K \stackrel{[\text{eq: 16.354}]}{\in} L(X, Y)$ it follows from [definition: 16.46]

$$f \text{ is Fréchet differentiable at } x \text{ with } D_x f = K$$

so that

$$\forall h \in X \text{ we have } D_x f(h) = K(h) = \sum_{i=1}^n D_i f(x)(h_i) = \left(\sum_{i=1}^n D_i f(x) \circ \pi_i \right)(h)$$

As $x \in U$ was chosen arbitrary, f is Fréchet differentiable on U and we can define the function

$$Df: U \rightarrow L(X, Y) \text{ by } Df(x) = D_x f$$

where

$$Df(x) = D_x f = \sum_{i=1}^n D_i f(x) \circ \pi_i$$

Finally we have to prove that Df is continuous. Let $x \in U$ and $\varepsilon \in \mathbb{R}^+$ then as $D_i f$ is continuous there exist a $\delta_i \in \mathbb{R}^+$ such that for $y \in U$ with $\|x - y\|_{\max} < \delta_i$ $\|D_i f(x) - D_i f(y)\|_{L(X_i, Y)} < \frac{\varepsilon}{2 \cdot n}$. Then we have if we take $\delta = \min(\{\delta_i | i \in \{1, \dots, n\}\})$ for $y \in U$ with $\|x - y\|_{\max} < \delta$ that

$$\begin{aligned} \|(Df(x) - Df(y))(h)\|_Y &= \|Df(x)(h) - Df(y)(h)\|_Y \\ &= \left\| \sum_{i=1}^n D_i f(x)(h_i) - \sum_{i=1}^n D_i f(y)(h_i) \right\|_Y \\ &= \left\| \sum_{i=1}^n (D_i f(x)(h_i) - D_i f(y)(h_i)) \right\|_Y \\ &= \left\| \sum_{i=1}^n (D_i f(x) - D_i f(y))(h_i) \right\|_Y \\ &\leq \sum_{i=1}^n \|(D_i f(x) - D_i f(y))(h_i)\|_Y \\ &\leq \sum_{i=1}^n \|D_i f(x) - D_i f(y)\|_{L(X_i, Y)} \cdot \|h\|_i \\ &\leq \sum_{i=1}^n (\|D_i f(x) - D_i f(y)\|_{L(X_i, Y)} \cdot \|h\|_{\max}) \\ &= \left(\sum_{i=1}^n \|D_i f(x) - D_i f(y)\|_{L(X_i, Y)} \right) \cdot \|h\|_{\max} \\ &\leq \left(\sum_{i=1}^n \frac{\varepsilon}{2 \cdot n} \right) \cdot \|h\|_{\max} \\ &= \frac{\varepsilon}{2} \cdot \|h\|_{\max} \end{aligned}$$

so that by [theorem: 14.212] $\|Df(x) - Df(y)\|_{L(X, Y)} \leq \frac{\varepsilon}{2} < \varepsilon$. Hence Df is continuous at x and as x was chosen arbitrary we have that Df is continuous which by definition proves that

$$f \text{ is } C^1 \text{ on } U$$

□

We want now to use induction to extend the above theorem to the case of m -times partial differentials. If $\forall x < U$ we can express $D^{[m]}f$ as a finite sum of partial differentials $D_{[i_1 \dots i_m]}f$ then we can use the existence and continuity of $D_{[i_1 \dots i_{m+1}]}f$ to prove that $D^{[m]}f$ has continuous partial differentials proving by induction that $D^{[m+1]}f$ exist and is continuous. There is a problem however $D^{[m]}f(x) \in L_m(X; Y)$ but $D_{[i_1 \dots i_m]}f(x) \in L_m(X_{i_1} \dots X_{i_m}; Y)$ so we can not write $D^{[m]}f$ as a finite sum of $D_{[i_1 \dots i_m]}f(x)$. To solve this we examine how we can transform elements of $L_m(X_{i_1} \dots X_{i_m}; Y)$ to $L_m(X; Y)$.

Lemma 16.287. *Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_n \rangle$ be the normed space where $X = \prod_{i=1}^n X_i$ and $\|\cdot\|_{\max}$ is the maximum norm on X , $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ and $L \in L_m(X_{i_1}, \dots, X_{i_m}; Y)$ then there exist a **unique** function*

$$\pi^{[i_1 \dots i_m]}(L) \in L_m(X; Y)$$

such that

$$\forall x \in X^m \text{ we have } \pi^{[i_1 \dots i_m]}(L)(x_1 : \dots : x_m) = L(\pi_{i_1}(x_1) : \dots : \pi_{i_m}(x_m))$$

and

$$\|\pi^{[i_1, \dots, i_m]}(L)\|_{L_m(X; Y)} \leq \|L\|_{L_m(X_{i_1}, \dots, X_{i_m}; Y)}$$

Thanks to the existence and uniqueness of $\pi^{[i_1, \dots, i_m]}(L)$ for every $L \in L_m(X_{i_1} \dots X_{i_m}; Y)$ we can define a function

$$\pi^{[i_1, \dots, i_m]}: L_m(X_{i_1} \dots X_{i_m}; Y) \rightarrow L_m(X; Y)$$

where

$$\forall x \in X^m \text{ we have } \pi^{[i_1, \dots, i_m]}(L)(x_1: \dots: x_m) = L(\pi_{i_1}(x_1): \dots: \pi_{i_m}(x_m))$$

$$\|\pi^{[i_1, \dots, i_m]}(L)\|_{L_m(X; Y)} \leq \|L\|_{L_m(X_{i_1}, \dots, X_{i_m}; Y)}$$

Proof. Using [theorem: 16.262] we have for $L \in L(X_{i_1}, \dots, X_{i_m}; Y)$

$$K_L = \mathcal{I}_{X_{i_1} \dots X_{i_m}, Y}(L): \prod_{j \in \{1, \dots, m\}} X_{i_j} \rightarrow Y \text{ defined by } K_L(x_1, \dots, x_m) = L(x_1: \dots: x_m)$$

that

$$K_L \in L(X_{i_1} \dots X_{i_m}; Y)$$

Define now

$$M_L: X^m \rightarrow Y \text{ by } M_L(x_1, \dots, x_m) = K_L(\pi_{i_1}(x_1), \dots, \pi_{i_m}(x_m))$$

then we have for $k \in \{1, \dots, m\}$, $(x_1, \dots, \underbrace{y + \alpha \cdot z}_{k} \dots, x_m) \in X^m$ and $\alpha \in \mathbb{R}$ then

$$\begin{aligned} & M_L(x_1, \dots, \underbrace{y + \alpha \cdot z}_{k} \dots, x_m) = \\ & K_L(\pi_{i_1}(x_1), \dots, \underbrace{\pi_{i_k}(y + \alpha \cdot z)}_{k}, \dots, \pi_{i_m}(x_m)) = \\ & K_L(\pi_{i_1}(x_1), \dots, \underbrace{\pi_{i_k}(y) + \alpha \cdot \pi_{i_k}(z)}_{k}, \dots, \pi_{i_m}(x_m)) \stackrel{K_L \in L(X_{i_1} \dots X_{i_m}; Y)}{=} \\ & K_L(\pi_{i_1}(x_1), \dots, \underbrace{\pi_{i_k}(y)}_{k}, \dots, \pi_{i_m}(x_m)) + \alpha \cdot K_L(\pi_{i_1}(x_1), \dots, \underbrace{\pi_{i_k}(z)}_{k}, \dots, \pi_{i_m}(x_m)) = \\ & M_L(x_1, \dots, \underbrace{y}_{k}, \dots, x_m) + \alpha \cdot M_L(x_1, \dots, \underbrace{z}_{k}, \dots, x_m) \end{aligned}$$

which proves that

$$M_L \in \text{Hom}\left(\underbrace{X \dots X}_m; Y\right) \quad (16.371)$$

Further we have

$$\begin{aligned} \|M_L(x_1, \dots, x_m)\|_Y &= \|K_L(\pi_{i_1}(x_1), \dots, \pi_{i_m}(x_m))\|_Y \\ &\leq \|K_L\|_{L(X_{i_1} \dots X_{i_m}; Y)} \cdot \prod_{j=1}^m \|\pi_{i_j}(x_j)\|_{i_j} \\ &\stackrel{[\text{example: 14.217}]}{\leq} \|K_L\|_{L(X_{i_1} \dots X_{i_m}; Y)} \cdot \prod_{j=1}^m \|x_j\|_X \end{aligned}$$

which by [theorem: 14.201] and [eq: 16.371] proves that

$$M_L \in L\left(\underbrace{X \dots X}_m; Y\right)$$

Using then [theorem: 16.262] we have for $\pi^{[i_1, \dots, i_m]}(L)$ defined by

$$\pi^{[i_1, \dots, i_m]}(L) = \left(\mathcal{I}_{\underbrace{X, \dots, X}_m, Y}\right)^{-1}(M_L)$$

that

$$\pi^{[i_1, \dots, i_m]}(L) \in L_m\left(\underbrace{X \dots X}_m; Y\right) = L_m(X; Y)$$

and $\forall x \in X^m$

$$\begin{aligned} \pi^{[i_1, \dots, i_m]}(L)(x_1: \dots: x_m) &= \left(\mathcal{I}_{\underbrace{X, \dots, X}_m, Y}\right)^{-1}(M_L)(x_1: \dots: x_m) \\ &\stackrel{[\text{theorem: 16.262}]}{=} M_L(x_1, \dots, x_m) \\ &= K_L(\pi_{i_1}(x_1), \dots, \pi_{i_m}(x_m)) \\ &= L(\pi_{i_1}(x_1): \dots: \pi_{i_m}(x_m)) \end{aligned}$$

Hence we have

$$\forall x \in X^m \text{ that } \pi^{[i_1, \dots, i_m]}(L)(x_1: \dots: x_m) = L(\pi_{i_1}(x_1): \dots: \pi_{i_m}(x_m)) \quad (16.372)$$

Let $H \in L_m\left(\underbrace{X \dots X}_m; Y\right) = L_m(X; Y)$ be such that $\forall x \in X^m$ $H(x_1: \dots: x_m) = L(\pi_{i_1}(x_1): \dots: \pi_{i_m}(x_m))$ then we have that $\forall x \in X^m$ $\pi^{[i_1, \dots, i_m]}(L)(x_1: \dots: x_m) = H(x_1: \dots: x_m)$ so that by [lemma: 16.259] $\pi^{[i_1, \dots, i_m]}(L) = H$. So we conclude that

$$\pi^{[i_1, \dots, i_m]}(L) \text{ is the unique function in } L_m\left(\underbrace{X \dots X}_m; Y\right) \text{ so that [eq: 16.372] is true}$$

Finally let $x \in X^m$ then

$$\begin{aligned} \|\pi^{[i_1, \dots, i_m]}(L)(x_1: \dots: x_m)\|_Y &= \|L(\pi_{i_1}(x_1): \dots: \pi_{i_m}(x_m))\|_Y \\ &\leq \|L\|_{L_m(X_1 \dots X_m; Y)} \cdot \prod_{j=1}^m \|\pi_{i_j}(x_j)\|_{X_{i_j}} \\ &\stackrel{[\text{example: 14.217}]}{\leq} \|L\|_{L_m(X_1 \dots X_m; Y)} \cdot \prod_{j=1}^m \|x_j\|_X \end{aligned}$$

Hence using [lemma: 16.260] it follows that

$$\|\pi^{[i_1, \dots, i_m]}(L)\|_{L_m(X; Y)} \leq \|L\|_{L_m(X_1, \dots, X_m; Y)} \quad \square$$

Lemma 16.288. Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_n \rangle$ be the normed space where $X = \prod_{i=1}^n X_i$ and $\|\cdot\|_{\max}$ is the maximum norm on X , $m \in \mathbb{N}$, $i \in \{1, \dots, n\}^m$ then we have for

$$\pi^{[i_1, \dots, i_m]}: L_m(X_{i_1}: \dots: X_{i_m}; Y) \rightarrow L_m(X; Y_m) \text{ [see lemma: 16.287]}$$

that

$$\pi^{[i_1 \dots i_m]} \in L(L_m(X_{i_1}: \dots: X_{i_m}; Y), L_m(X; Y_m))$$

with

$$\|\pi^{[i_1 \dots i_m]}\|_{L(L_m(X_{i_1}: \dots: X_{i_m}; Y), L_m(X; Y_m))} \leq 1$$

Proof. Let $L, K \in L_m(X_{i_1}: \dots: X_{i_m}; Y)$, $\alpha \in \mathbb{K}$ then we have for every $y \in X^m$ that

$$\begin{aligned} (\pi^{[i_1 \dots i_m]}(L + \alpha \cdot K))(y_1: \dots: y_m) &= \\ (L + \alpha \cdot K)(\pi_{i_1}(y_1): \dots: \pi_{i_m}(y_m)) &\stackrel{[\text{theorem: 16.256}]}{=} \\ L(\pi_{i_1}(y_1): \dots: \pi_{i_m}(y_m)) + \alpha \cdot K(\pi_{i_1}(y_1): \dots: \pi_{i_m}(y_m)) &= \\ (\pi^{[i_1 \dots i_m]}(L))(y_1, \dots, y_m) + \alpha \cdot (\pi^{[i_1 \dots i_m]}(K))(y_1, \dots, y_m) &= \\ (\pi^{[i_1 \dots i_m]}(L) + \alpha \cdot \pi^{[i_1 \dots i_m]}(K))(y_1 \dots y_m) & \end{aligned}$$

which proves by [lemma: 16.259] that

$$\pi^{[i_1 \dots i_m]}(L + \alpha \cdot K) = \pi^{[i_1 \dots i_m]}(L) + \alpha \cdot \pi^{[i_1 \dots i_m]}(K)$$

which proves that

$$\pi^{[i_1 \dots i_m]} \in \text{Hom}(L_m(X_{i_1} : \dots : X_{i_m}; Y), L_m(X; Y))$$

Further we have for $L \in L_m(X_{i_1} : \dots : X_{i_m}; Y)$ that

$$\|\pi^{[i_1, \dots, i_m]}(L)\|_{L_m(X; Y)} \underset{[\text{lemma: 16.287}]}{\leq} \|L\|_{L_m(X_1, \dots, X_m; Y)}$$

which by [theorem: 14.201] proves that

$$\pi^{[i_1, \dots, i_m]} \in L(L_m(X_{i_1} : \dots : X_{i_m}; Y), L_m(X_m; Y_m))$$

and by [theorem: 14.212] that

$$\|\pi^{[i_1, \dots, i_m]}\|_{L(L_m(X_{i_1} \dots X_{i_m}; Y))} \leq 1 \quad \square$$

Lemma 16.289. *Let $n \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_n \rangle$ be the normed space where $X = \prod_{i=1}^n X_i$ and $\|\cdot\|_{\max}$ is the maximum norm on X then we have*

1. *If $i \in \{1, \dots, n\}^1$ and $L \in L_1(X_{i_1} \dots X_{i_1}; Y)$ then $\pi^{[i_1 \dots i_1]}(L) = L \circ \pi_{i_1}$*
2. *If $k \in \{1, \dots, n\}$, $i \in \{1, \dots, n\}^m$ and $L \in L_{m+1}(X_k, X_{i_1} \dots X_{i_m}; Y)$ then*

$$\forall x \in X \text{ we have } (\pi^{[k, i_1 \dots i_m]}(L))(x) = \pi^{[i_1 \dots i_m]}(L(\pi_k(x)))$$

Proof.

1. Let $x \in X$ then $\pi^{[i_1 \dots i_1]}(L)(x) = L(\pi_{i_1}(x)) = (L \circ \pi_i)(x)$ so that $\pi^{[i_1 \dots i_1]}(L) = L \circ \pi_{i_1}$.
2. Let $x \in X$ then we have, as $L \in L_{m+1}(X_k, X_{i_1} \dots, X_{i_m}) = L(X_k, L_m(X_{i_1}, \dots, X_{i_m}; Y))$, that

$$L(\pi_k(x)) \in L_m(X_{i_1}, \dots, X_{i_m}; Y) \quad (16.373)$$

Hence for every $y \in X^m$ we have

$$\begin{aligned} (((\pi^{[k, i_1 \dots i_m]}(L)))(x))(y_1 : \dots : y_m) & \underset{\text{def}}{=} (\pi^{[k, i_1 \dots i_m]}(L))(x : y_1 \dots y_m) \\ & = L(\pi_k(x) : \pi_{i_1}(y_1) : \dots : \pi_{i_m}(y_m)) \\ & = L(\pi_k(x))(\pi_{i_1}(y_1) : \dots : \pi_{i_m}(y_m)) \\ & \underset{[\text{eq: 16.373}]}{=} (\pi^{[i_1 \dots i_m]}(L(\pi_k(x))))(y_1 : \dots : y_m) \end{aligned}$$

so using [lemma: 16.259] we have

$$((\pi^{[k, i_1 \dots i_m]}(L)))(x) = (\pi^{[i_1 \dots i_m]}(L(\pi_k(x)))) \quad \square$$

Lemma 16.290. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$, U a open set in X , $x \in U$, $i \in \{1, \dots, n\}^m$, $k \in \{1, \dots, n\}$ and*

$$f : U \rightarrow L_m(X_{i_1} \dots X_{i_m}; Y)$$

a function such that is k -partial differentiable at x . Then we have for the following function

$$g : U \rightarrow L_m\left(\underbrace{X \dots X}_m; Y\right) = L_m(X; Y) \text{ defined by } g(x) = \pi^{[i_1, \dots, i_m]}(f(x))$$

that g is k -partial differentiable at x where $D_{x,k}f$ is defined by

$$D_{x,k}g : X_k \rightarrow L_m(X; Y) \text{ where } D_{x,k}g(h) = \pi^{[i_1, \dots, i_m]}(D_{x,k}f(h))$$

Further if instead of the condition that f is k -partial differentiable at x we have the stronger condition that

$$U = \mathfrak{D}_{k,f} = \{y \in U \mid f \text{ is } k\text{-partial differentiable at } y\}$$

and

$$D_k f: \mathfrak{D}_{k,f} = U \rightarrow L(X_k, L_m(X_{i_1} \dots X_{i_m}; Y)) \stackrel{\text{def}}{=} L_{m+1}(X_k, X_{i_1} \dots X_{i_m}; Y) \text{ where } D_k f(y) = D_{x,k} f$$

is **continuous** then

$$\mathfrak{D}_{k,g} \stackrel{\text{def}}{=} \{y \in U \mid g \text{ is } k\text{-partial differentiable at } y\} = U$$

and

$$D_k g: \mathfrak{D}_{k,g} = U \rightarrow L(X_k, L_m(X; Y)) \text{ defined by } D_k g(x) = D_{x,gg} \text{ is } \mathbf{continuous}$$

Note 16.291. By definition we have for $x \in U$

$$D_{x,k} f \in L(X_k, L_m(X_{i_1} \dots X_{i_m}; Y)) = L_{m+1}(X_k, X_{i_1} \dots X_{i_m}; Y)$$

so that for $h \in X_k$

$$D_{x,k} f(h) \in L_m(X_{i_1} \dots X_{i_m}; Y)$$

from which it follows that

$$\pi^{[i_1, \dots, i_m]}(D_{x,k} f(h)) \in L_m(X; Y)$$

Further $D_{x,k} g \in L(X_k, L_m(X; Y))$ so that

$$D_{x,k} g(h) \in L_m(X; Y)$$

which prove that the types check out for the equation

$$D_{x,k} g(h) = \pi^{[i_1, \dots, i_m]}(D_{x,k} f(h))$$

Proof. Let $x \in \mathfrak{D}_{k,f}$ or equivalently f is k -partial differentiable at x . First we prove that the function defined by

$$L: X_k \rightarrow L_m(X; Y) \text{ where } D_{x,g}(h) = \pi^{[i_1, \dots, i_m]}(D_{x,k} f(h))$$

is a element of $L(X_k, L_m(X; Y))$. Let $h, l \in X_k$ and $\alpha \in \mathbb{K}$ then we have that

$$\begin{aligned} L(h + \alpha \cdot l) &= \pi^{[i_1 \dots i_k]}(D_{x,k} f(h + \alpha \cdot l)) \\ &= \pi^{[i_1 \dots i_k]}(D_{x,k} f(h) + \alpha \cdot D_{x,k} f(l)) \\ &\stackrel{[\text{lemma: 16.288}]}{=} \pi^{[i_1 \dots i_k]}(D_{x,k} f(h)) + \alpha \cdot \pi^{[i_1 \dots i_k]}(D_{x,k} f(l)) \\ &= L(h) + \alpha \cdot L(l) \end{aligned}$$

so that

$$L \in \text{Hom}(X_k, L_m(X; Y)) \quad (16.374)$$

Let $h \in X_k$ then we have

$$\begin{aligned} \|L(h)\|_{L_m(X; Y)} &= \|\pi^{[i_1 \dots i_m]}(D_{x,k} f(h))\|_{L_m(X; Y)} \\ &\leq \|\pi^{[i_1 \dots i_m]}\|_{L(L_m(X_{i_1} \dots X_{i_m}; Y), L_m(X; Y))} \cdot \|D_{x,k} f(h)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \\ &\stackrel{[\text{lemma: 16.288}]}{\leq} \|D_{x,k} f(h)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \\ &\stackrel{[\text{lemma: 16.288}]}{\leq} \|D_{x,k} f\|_{L(X_k, L(X_{i_1} \dots X_{i_m}; Y))} \cdot \|h\|_k \end{aligned}$$

which by [eq: 16.374] and [theorem: 14.201] proves that

$$L \in L(X_k, L_m(X; Y)) \quad (16.375)$$

Next we prove differentiability. As f is k -partial differentiable at x with k -partial differential $D_{x,k} f$ at x we have by [definition: 16.74] that

$$f \circ I^{[k,x]}: (I^{[k,x]})^{-1}(U) \rightarrow Y \quad (16.376)$$

is Fréchet differentiable at x_i with differential $D_{x_k}(f \circ I^{[k,x]})$ at x_k where $x_k = \pi_k(x)$. Hence using [definition: 16.46] we have

$$x_i \in (I^{[k,x]})^{-1}(U)^\circ \quad (16.377)$$

and given $\varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in ((I^{[k,x]})^{-1}(U))_{x_k}$ with $\|h\|_k < \delta$ we have

$$\|f(I^{[k,x]}(x_k + h)) - f(I^{[k,x]}(x_k)) - D_{x,k}f(h)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \leq \varepsilon \cdot \|h\|_k \quad (16.378)$$

Hence we have that

$$\begin{aligned} & \|g(I^{[k,x]}(x_k + h)) - g(I^{[k,x]}(x_k)) - L(h)\|_{L_m(X; Y)} = \\ & \|\pi^{[i_1 \dots i_m]}(f(I^{[k,x]}(x_k + h))) - \pi^{[i_1 \dots i_m]}(f(I^{[k,x]}(x_k))) - \pi^{[i_1 \dots i_m]}(D_{x,k}f(h))\|_{L_m(X; Y)} \stackrel{[\text{lemma: 16.288}]}{=} \\ & \|\pi^{[i_1 \dots i_m]}(f(I^{[k,x]}(x_k + h)) - f(I^{[k,x]}(x_k)) - D_{x,k}f(h))\|_{L_m(X; Y)} \stackrel{[\text{lemma: 16.287}]}{\leq} \\ & \|f(I^{[k,x]}(x_k + h)) - f(I^{[k,x]}(x_k)) - D_{x,k}f(h)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \stackrel{[\text{eq: 16.378}]}{\leq} \\ & \varepsilon \cdot \|h\|_k \end{aligned}$$

The above together with [eqs: 16.375, 16.377] proves that $g \circ I^{[k,x]}$ is Fréchet differentiable at x_k with differential $D_{x_k}(g \circ I^{[k,x]}) = L$. Hence if $x \in \mathfrak{D}_{k,f}$

$$g \text{ is } k\text{-partial differentiable at } x \text{ with } D_{k,x}g = D_{x_k}(g \circ I^{[k,x]}) = L \quad (16.379)$$

so that $D_{k,x}g$ is defined by

$$D_{x,k}g: X_k \rightarrow L_m(X; Y) \text{ where } D_{x,k}g(h) = \pi^{[i_1, \dots, i_m]}(D_{x,k}f(h)) \quad (16.380)$$

Now for the final part. Assume now that that

$$U = \mathfrak{D}_{k,f} = \{y \in U \mid f \text{ is } k\text{-partial differentiable at } y\}$$

If $x \in U$ then g is k -partial differentiable at x [see eq: 16.379] so that

$$U \subseteq \mathfrak{D}_{k,g} = \{y \in U \mid g \text{ is } k\text{-partial differentiable at } y\} \subseteq U$$

hence

$$U = \mathfrak{D}_{k,g} = \{y \in U \mid g \text{ is } k\text{-partial differentiable at } y\}$$

and $D_k g$ is defined by

$$D_k g: \mathfrak{D}_{k,g} \rightarrow L(X_k, L_m(X; Y)) \text{ where } D_k g(x) = D_{x,k}g$$

Let $x \in U$ and $\varepsilon \in \mathbb{R}^+$ then as $D_k f$ is continuous there exist a $\delta \in \mathbb{R}^+$ such that

$$\forall y \in U \text{ with } \|x - y\|_k < \delta \text{ we have } \|D_k f(x) - D_k f(y)\|_{L(X_k, X_{i_1} \dots X_{i_m}; Y)} < \frac{\varepsilon}{2} \quad (16.381)$$

Let $h \in X_k$ then we have

$$\begin{aligned} \|(D_k g(x) - D_k g(y))(h)\|_{L_m(X; Y)} &= \|D_k g(x)(h) - D_k g(y)(h)\|_{L_m(X; Y)} \\ &= \|D_{x,k}g(h) - D_{y,k}g(h)\|_{L_m(X; Y)} \\ &\stackrel{[\text{eq: 16.380}]}{=} \|\pi^{[i_1 \dots i_m]}(D_{x,k}f(h)) - \pi^{[i_1 \dots i_m]}(D_{y,k}f(h))\|_{L_m(X; Y)} \\ &\stackrel{[\text{lemma: 16.288}]}{=} \|\pi^{[i_1 \dots i_m]}(D_{x,k}f(h) - D_{y,k}f(h))\|_{L_m(X; Y)} \\ &\stackrel{[\text{lemma: 16.287}]}{\leq} \|D_{x,k}f(h) - D_{y,k}f(h)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \\ &= \|(D_{x,k}f - D_{y,k}f)(h)\|_{L_m(X_{i_1} \dots X_{i_m}; Y)} \\ &\leq \|D_k f(x) - D_k f(y)\|_{L(X_k, X_{i_1} \dots X_{i_m}; Y)} \cdot \|h\|_k \\ &\leq \frac{\varepsilon}{2} \cdot \|h\|_k \end{aligned}$$

so that by [theorem: 14.212] we have

$$\|D_k g(x) - D_k g(y)\|_{L(X_k, L_m(X; Y))} \leq \frac{\varepsilon}{2} < \varepsilon$$

which proves that $D_k g$ is continuous at x . As $x \in U$ was chosen arbitrary it follows that

$$D_x g \text{ is continuous} \quad \square$$

We can now write an explicit formula for the m -times differential of a function in terms of the m -times partial differentials of this function.

Lemma 16.292. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$, U a open set in X and $f: U \rightarrow Y$ a function such that*

$$U = \mathfrak{D}_f^m$$

then we have

$$\forall x \in U \text{ that } D_x^{[m]} f = \sum_{k \in \{1, \dots, n\}^m} \pi^{[k_1, \dots, k_m]}(D_{x, [k_1 \dots k_m]} f)$$

Note 16.293. *As $D_x^{[m]} f$ exist it follows from [theorem: 16.276] that $\forall k \in \{1, \dots, n\}^m$ $D_{x, [k_1 \dots k_m]} f$ exist so that the formula for $D_x^{[m]} f$ is well defined.*

Proof. We use induction to prove this, so define

$$S = \left\{ m \in \mathbb{N} \mid \text{If } U = \mathfrak{D}_f^m \text{ exist then } \forall x \in U \ D_x^{[m]} f = \sum_{k \in \{1, \dots, n\}^m} \pi^{[k_1, \dots, k_m]}(D_{x, [k_1 \dots k_m]} f) \right\}$$

then we have:

1 $\in S$. Let $x \in U = \mathfrak{D}_f^1$ then $D_x^{[1]} f$ exist and we have by definition that f is Fréchet differentiable at x so that we have by [theorem: 16.83] that $\forall i \in \{1, \dots, n\}$ $D_{x, i} f$ exist and

$$D_x f = \sum_{i=1}^n (D_{x, i} f \circ \pi_i)$$

or using [theorem: 16.273] we have

$$D_x f = \sum_{i=1}^n (D_{x, [i \dots i]} f \circ \pi_i) = \sum_{k \in \{1, \dots, n\}^1} (D_{x, [k_1 \dots k_1]} f \circ \pi_i)$$

Using now [lemma: 16.289(1)] it follows that

$$D_x f = \sum_{k \in \{1, \dots, n\}^1} \pi^{[k_1 \dots k_1]}(D_{x, k_1, \dots, k_1}^{[1]} f)$$

which proves that $1 \in S$.

$m \in S \Rightarrow m+1 \in S$. Assume that $U = \mathfrak{D}_f^{m+1}$. Let $l \in \{1, \dots, n\}^m$ and take $x \in U$ then as f is $(m+1)$ -differentiable at x it follows from [theorem: 16.138] that f is m -times differentiable at x . Using [theorem: 16.276] it follows that f has a partial differential $D_{x, [l_1 \dots l_m]} f$ at x so that $x \in \mathfrak{D}_{[l_1 \dots l_m], f}$. Hence $U \subseteq \mathfrak{D}_{[l_1 \dots l_m], f} \subseteq U$ which proves that [theorem: 16.266]

$$U = \mathfrak{D}_{[l_1 \dots l_m], f}$$

so that $D_{[l_1 \dots l_m]} f$ is of the form

$$D_{[l_1 \dots l_m]} f: U = \mathfrak{D}_{[l_1 \dots l_m]} \rightarrow L_m(X_{l_1} \dots X_{l_m}) \quad (16.382)$$

Let $k \in \{1, \dots, n\}$ so that $(k, l_1, \dots, l_m) \in \{1, \dots, n\}^{m+1}$. Take $x \in U$ then, as f is $(m+1)$ -differentiable at x , it follows from [theorem: 16.276] that f has a partial differential $D_{x, [k, l_1 \dots l_m]} f$ at x . Using then [theorem: 16.273] we have that $D_{[l_1 \dots l_m]} f$ is k -partial differentiable at x with $D_{x, [k, l_1 \dots l_m]} f = D_{x, k}(D_{[l_1 \dots l_m]} f)$. Summarized we have

$$\forall x \in U \ D_{x, [k, l_1 \dots l_m]} f \text{ exist with } D_{x, [k, l_1 \dots l_m]} f = D_{x, k}(D_{[l_1 \dots l_m]} f) \quad (16.383)$$

Using now [lemma: 16.290] on [eqs: 16.382, 16.383] it follows that for

$$g_{l_1 \dots l_m}: U \rightarrow L_m(X; Y) \text{ defined by } g_{l_1 \dots l_m}(y) = \pi^{[l_1 \dots l_m]}(D_{[l_1 \dots l_m]} f(y)) \quad (16.384)$$

we have that

$$g_{l_1 \dots l_m} \text{ is } k\text{-partial differentiable at } x$$

where $D_{x, k} g_{l_1 \dots l_m}$ is defined by $\forall h \in X_k$

$$\begin{aligned} D_{x, k} g_{l_1 \dots l_m}(h) &= \pi^{[l_1 \dots l_m]}((D_{x, k}(D_{[l_1 \dots l_m]} f))(h)) \\ &\stackrel{[\text{eq: 16.383}]}{=} \pi^{[l_1 \dots l_m]}((D_{x, [k, l_1 \dots l_m]} f)(h)) \end{aligned} \quad (16.385)$$

Let $y \in X$ then $\pi_k(y) \in X_k$ so that

$$\begin{aligned} (D_{x, k} g_{l_1 \dots l_m} \circ \pi_k)(y) &= D_{x, k} g_{l_1 \dots l_m}(\pi_k(y)) \\ &\stackrel{[\text{eq: 16.385}]}{=} \pi^{[l_1 \dots l_m]}((D_{x, [k, l_1 \dots l_m]} f)(\pi_k(y))) \\ &\stackrel{[\text{lemma: 16.289(2)}]}{=} (\pi^{[k, l_1 \dots l_m]}(D_{x, [k, l_1 \dots l_m]} f))(y) \end{aligned}$$

which proves that

$$D_{x, k} g_{l_1 \dots l_m} \circ \pi_k = \pi^{[k, l_1 \dots l_m]}(D_{x, [k, l_1 \dots l_m]} f) \quad (16.386)$$

As $\forall x \in U \ D_x^{[m+1]} f$ exist we have by [theorem: 16.138] that $\forall x \in U \ D_x^{[n]} f$ exist which proves that

$$U = \mathfrak{D}_f^m = \{x \in U \mid f \text{ is } m\text{-times differentiable at } x\}$$

so that $D^{[m]} f$ is of the form

$$D^{[m]} f: \mathfrak{D}_f^m = U \rightarrow L_m(X, Y)$$

Again as $D_x^{[m+1]} f$ exist it follows from [theorem: 16.140] that

$$D^{[m]} f \text{ is Fréchet differentiable at } x \text{ with } D_x^{[m+1]} = D_x(D^{[m]} f)$$

Now

$$\begin{aligned} D^{[m]} f(x) &\stackrel{[\text{theorem: 16.136}]}{=} D_x^{[m]} f \\ &\stackrel{m \in S}{=} \sum_{l \in \{1, \dots, n\}^m} \pi^{[l_1 \dots l_m]}(D_{x, [l_1 \dots l_m]} f) \\ &\stackrel{[\text{definition: 16.267}]}{=} \sum_{l \in \{1, \dots, n\}^m} \pi^{[l_1 \dots l_m]}(D_{[l_1 \dots l_m]} f(x)) \\ &\stackrel{[\text{eq: 16.384}]}{=} \sum_{l \in \{1, \dots, n\}^m} g_{l_1 \dots l_m}(x) \\ &= \left(\sum_{l \in \{1, \dots, n\}^m} g_{l_1 \dots l_m} \right)(x) \end{aligned}$$

which as this is true $\forall x \in U$ that

$$D^{[m]} f = \sum_{l \in \{1, \dots, n\}^m} g_{l_1 \dots l_m} \quad (16.387)$$

As $D^{[m]}f$ is Fréchet differentiable at x with $D_x^{[m+1]}f = D_x(D^{[m]}f)$ we have by [theorem: 16.83] that

$$\begin{aligned}
 D_x^{[m+1]}f &= \sum_{k=1}^n ((D_{x,k}(D^{[m]}f)) \circ \pi_k) \\
 &\stackrel{[\text{eq: 16.387}]}{=} \sum_{k=1}^n \left(\left(D_{x,k} \left(\sum_{l \in \{1, \dots, m\}} g_{l_1 \dots l_m} \right) \right) \circ \pi_k \right) \\
 &= \sum_{k=1}^n \left(\sum_{l \in \{1, \dots, n\}^m} D_{x,k} g_{l_1 \dots l_m} \circ \pi_k \right) \\
 &\stackrel{[\text{eq: 16.386}]}{=} \sum_{k=1}^n \left(\sum_{l \in \{1, \dots, n\}^m} \pi^{[k, l_1 \dots l_m]}(D_{x, [k, l_1 \dots l_m]}f) \right) \\
 &= \sum_{k \in \{1, \dots, n\}^{m+1}} \pi^{[k_1 \dots k_{m+1}]}(D_{x, [k_1 \dots k_{m+1}]}f)
 \end{aligned}$$

which proves that

$$m+1 \in S$$

□

Finally we can now extend [theorem: 16.286].

Theorem 16.294. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{ \langle X_i, \|\cdot\|_i \rangle \}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], U a open set in X , $f: U \rightarrow Y$ a function such that $\forall i \in \{1, \dots, n\}^m$ we have that

$$U = \mathfrak{D}_{[l_1 \dots l_m], f}$$

and

$$D_{[i_1 \dots i_k]}f: U = \mathfrak{D}_{[i_1 \dots i_k], f} \rightarrow L_k(X_{i_1} \dots X_{i_k}; Y) \text{ is continuous}$$

then

$$f \text{ is of class } C^m$$

Proof. We use induction to prove this, so let

$S = \{m \in \mathbb{N} \mid \text{If } \forall i \in \{1, \dots, n\}^k U = \mathfrak{D}_{[i_1 \dots i_m], f} \text{ } D_{[i_1 \dots i_m]}f: U \rightarrow L_m(X_{i_1} \dots X_{i_m}; Y) \text{ is continuous then } f \text{ is of class } C^m\}$

then we have:

$1 \in S$. Assume that $\forall i \in \{1, \dots, n\}^1$ we have that

$$U = \mathfrak{D}_{[i_1 \dots i_1], f} \text{ and } D_{[i_1 \dots i_1]}f \text{ is continuous}$$

it follows from [definition: 16.264] that

$$U = \mathfrak{D}_{i_1, f} \text{ and } D_{i_1}f \text{ is continuous}$$

Hence by [theorem: 16.286] it follows that f is of class C^1 .

$m \in S \Rightarrow m+1 \in S$. Assume that we have

$$\forall i \in \{1, \dots, n\}^{m+1} U = \mathfrak{D}_{[i_1 \dots i_{m+1}], f} \text{ and } D_{[i_1 \dots i_{m+1}]}f \text{ is continuous} \quad (16.388)$$

Let $l \in \{1, \dots, n\}$ and $i \in \{1, \dots, n\}^m$ then $(l, i_1, \dots, i_m) \in \{1, \dots, n\}^{m+1}$ so that

$$U \stackrel{[\text{eq: 16.388}]}{=} \mathfrak{D}_{[l, i_1 \dots i_m], f} \stackrel{[\text{theorem: 16.266}]}{\subseteq} \mathfrak{D}_{[i_1 \dots i_m], f} \subseteq U$$

which proves that

$$\forall i \in \{1, \dots, n\}^m \text{ that } U = \mathfrak{D}_{[l_1 \dots l_m], f} \quad (16.389)$$

So that the function $D_{[i_1, \dots, i_m]}f$ is of the form

$$D_{[i_1, \dots, i_m]}f: \mathfrak{D}_{[i_1, \dots, i_m], f} = U \rightarrow L_m(X_{i_1}, \dots, X_{i_m}; Y) \quad (16.390)$$

and using [theorem: 16.264]

$$D_{[l, i_1, \dots, i_m]}f = D_l(D_{[i_1, \dots, i_m]}f) \quad (16.391)$$

so that by [eq: 16.388]

$$U = \mathfrak{D}_{l, D_{[i_1, \dots, i_m]}f} \text{ and } D_l(D_{[i_1, \dots, i_m]}f) \text{ is continuous} \quad (16.392)$$

Use [eq: 16.390] to define

$$g_{i_1, \dots, i_m}: U \rightarrow L_m(X; Y) \text{ by } g_{i_1, \dots, i_m}(x) = \pi^{[i_1, \dots, i_m]}(D_{[i_1, \dots, i_m]}f(x)) \quad (16.393)$$

We can use now [lemma: 16.290] together with [eq: 16.392] to get

$$U = \mathfrak{D}_{k, g_{i_1, \dots, i_m}} = \{x \in U \mid g_{i_1, \dots, i_m} \text{ is } l\text{-partial differentiable at } x\}$$

and

$$D_l g_{i_1, \dots, i_m}: U \rightarrow L_m(X; Y) \text{ defined by } D_l g_{i_1, \dots, i_m}(x) = D_{x, l} g_{i_1, \dots, i_m} \text{ is continuous}$$

Hence we have, using [theorems: 16.86, 14.166], for the **finite** sum

$$g = \sum_{i \in \{1, \dots, n\}^m} g_{i_1, \dots, i_m}$$

that $\forall l \in \{1, \dots, n\} \forall x \in U$ the l -partial derivative $D_{x, l}g$ exist and that

$$D_l g: U \rightarrow L(X_l, L_m(X, Y)) \text{ defined by } D_l g(x) = D_{x, l}g \text{ is continuous}$$

Applying then [theorem: 16.286] proves that

$$g \text{ is of class } C^1 \quad (16.394)$$

As $m \in \mathcal{S}_n$ we have, using [eq: 16.389] that f is of class C^m so that $\forall x \in U D_x^{[m]}f$ exist, hence we can define

$$D^{[m]}f: U \rightarrow L_m(X; Y) \text{ by } D^{[m]}f(x) = D_x^{[m]}f$$

$m \in \mathcal{S} \Rightarrow m+1 \in \mathcal{S}$. Assume that $\forall k \in \{1, \dots, m+1\}, \forall i \in \{1, \dots, n\}^k$ we have $\forall x \in U$ that $D_{x, i_1, \dots, i_k}^{[k]}f$ exist and

$$D_{i_1, \dots, i_k}^{[k]}f: U \rightarrow L_k(X_{i_1}, \dots, X_{i_k}; Y) \text{ defined by } D_{i_1, \dots, i_k}^{[k]}f(x) = D_{x, i_1, \dots, i_k}^{[k]}f \text{ is continuous} \quad (16.395)$$

Let $l \in \{1, \dots, n\}$ then, as $D_{x, l, i_1, \dots, i_m}^{[m+1]}f$ exist, the function

$$D_{x, i_1, \dots, i_m}^{[m]}f: U \rightarrow L_m(X_{i_1}, \dots, X_{i_m}; Y) \text{ defined by } D_{i_1, \dots, i_m}^{[m]}f(x) = D_{x, i_1, \dots, i_m}^{[m]}f$$

has a l -partial differential $D_{x, l}(D_{x, i_1, \dots, i_m}^{[m]}f)$ at x with

$$D_{x, l, i_1, \dots, i_m}^{[m+1]}f = D_{x, l}(D_{x, i_1, \dots, i_m}^{[m]}f)$$

Further as $m+1 \in \{1, \dots, m+1\}$ we have that

$$D_{l, i_1, \dots, i_{m+1}}^{[m+1]}f: U \rightarrow L_{m+1}(X_l, X_{i_1}, \dots, X_{i_m}; Y) \text{ defined by } D_{l, i_1, \dots, i_m}^{[m+1]}f(x) = D_{x, l, i_1, \dots, i_m}^{[m+1]}f$$

is continuous. Define now

$$g_{i_1, \dots, i_m}: U \rightarrow L_m(X; Y) \text{ by } g_{[i_1, \dots, i_m]}(x) = \pi^{[i_1, \dots, i_m]}(D_{i_1, \dots, i_m}^{[m]}f(x)) \quad (16.396)$$

then it follows from [lemma: 16.290] and [eq: 16.395] that

$$\forall x \in U \text{ we have that } D_{x, l}g_{i_1, \dots, i_m} \text{ exist} \quad (16.397)$$

and that

$$D_{x, l}g_{i_1, \dots, i_m}: U \rightarrow L_m(X; Y) \text{ defined by } D_l g_{i_1, \dots, i_m}(x) = D_{x, l, i_1, \dots, i_m}f \text{ is continuous} \quad (16.398)$$

Hence we have, using [theorems: 16.86, 14.166], for the **finite** sum

$$g = \sum_{i \in \{1, \dots, n\}^m} g_{i_1 \dots i_m} \quad (16.399)$$

that $\forall l \in \{1, \dots, n\} \forall x \in U$ the l -partial derivative $D_{x,l}g$ exist and that

$$D_l g: U \rightarrow L(X_l, L_m(X, Y)) \text{ define by } D_l g(x) = D_{x,l}g \text{ is continuous}$$

Applying then [theorem: 16.286] proves that

$$g \text{ is of class } C^1 \quad (16.400)$$

As $m \in \mathcal{S}_n$ we have, using [eq: 16.389] that f is of class C^m so that f is m -times differentiable on U . Using [definition: 16.141] it follows that $U \subseteq \mathfrak{D}_f^m$ and by [theorem: 16.135] $\mathfrak{D}_f^m$ so that

$$U = \mathfrak{D}_F^m$$

Let $x \in U$ then we have using [lemma: 16.292] and the above that

$$\begin{aligned} D^{[m]}f(x) &= D_x^{[m]}f \\ &\stackrel{[\text{lemma: 16.292}]}{=} \sum_{i \in \{1, \dots, n\}^m} \pi^{[i_1 \dots i_m]}(D_{x, i_1 \dots i_m}^{[m]}f) \\ &= \sum_{i \in \{1, \dots, n\}^m} \pi^{[i_1 \dots i_m]}(D_{i_1 \dots i_m}^{[m]}f(x)) \\ &\stackrel{[\text{eq: 16.396}]}{=} \sum_{i \in \{1, \dots, n\}^m} g_{i_1 \dots i_m}(x) \\ &\stackrel{[\text{eq: 16.399}]}{=} g(x) \end{aligned}$$

which proves that $D^{[m]}f = g$ or using [eq: 16.400] that $D^{[m]}f$ is of class C^1 . Applying then [theorem: 16.175] prove that f is of class C^{m+1} . Hence we have proved that

$$m+1 \in S \quad \square$$

Combining the above and [theorem: 16.277] we have the following fundamental theorem that links together higher order differentials and higher order continuous partial differentials.

Theorem 16.295. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\{\langle X_i, \|\cdot\|_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of normed spaces, $\langle Y, \|\cdot\|_Y \rangle$ a normed space, $\langle X, \|\cdot\|_{\max} \rangle$ the normed space based on the product $X = \prod_{i \in \{1, \dots, n\}} X_i$ with the maximum norm $\|\cdot\|_{\max}$ [see definition: 14.98], U a open set in X , $f: U \rightarrow Y$ a function then the following are equivalent:*

1. f is of class C^m
2. $\forall i \in \{1, \dots, n\}^m$ we have that $U = \mathfrak{D}_{[i_1 \dots i_m], f}$ [or equivalently $\forall x \in U$ $D_{x, [i_1 \dots i_m]}f$ exist] and

$$D_{[i_1 \dots i_m]}f: \mathfrak{D}_{[i_1 \dots i_m], f} = U \rightarrow L_k(X_{i_1} \dots X_{i_k}; Y) \text{ is continuous}$$

Proof.

$1 \Rightarrow 2$. This follows from [theorem: 16.277].

$2 \Rightarrow 1$. This is proved in the previous theorem [theorem: 16.294]. $\square$

16.5.3 Higher Order Partial Derivatives

Definition 16.296. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ the normed space with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow X$ a function then for $k \in \{1, \dots, m\}$ we define $\mathcal{D}_{[i_1 \dots i_m]f}$ and $D_{[i_k \dots i_m]}f$ recursively as follows for k*

$k = m$. Then

$$\mathcal{D}_{[i_m \dots i_m], f} = D_{i_m, f}$$

and

$$\partial_{[i_m \dots i_m]} f = \partial_{i_m} f$$

Using [definition: 16.91] we have that

$$\mathcal{D}_{i_m, f} = \{x \in A \mid f \text{ has a } i_m\text{-partial derivative at } x\} \subseteq A$$

and

$$\partial_{i_m} f: \mathcal{D}_{i_m, f} \rightarrow X \text{ is defined by } \partial_{i_m} f(x) = \partial_{x, i_m} f$$

$k < m$. Then

$$\partial_{[i_k \dots i_m], f} = \{x \in \mathcal{D}_{[i_{k+1} \dots i_m], f} \mid \partial_{[i_{k+1} \dots i_m]} f \text{ has a } i_k\text{-partial derivative at } x\}$$

and

$$\partial_{[i_k \dots i_m]} f: \mathcal{D}_{[i_k \dots i_m]} \rightarrow X$$

is defined by

$$\partial_{[i_k, \dots, i_m]} f = \partial_{i_k} (\partial_{[i_{k+1} \dots i_m]} f)$$

Using [definition: 16.91] it follows that

$$\partial_{[i_k \dots i_m]} f: \mathcal{D}_{[i_k \dots i_m]} \rightarrow Y \text{ is defined by } \partial_{[i_k \dots i_m]} f(x) = \partial_{x, i_k} (\partial_{[i_{k+1} \dots i_m]} f)$$

Convention 16.297. If $A \subseteq \mathbb{K}^n$ and $f: A \rightarrow X$ is a function then $\partial_{[i_1 \dots i_m]} f$ is always the graph of the function $\partial_{[i_1 \dots i_m]} f: \mathcal{D}_{[i_1 \dots i_m], f} \rightarrow X$. In other words the domain of $\partial_{[i_1 \dots i_m]} f$ is always $\mathcal{D}_{[i_1 \dots i_m], f}$ and the codomain is always X .

We have the following relation between the domains $\mathcal{D}_{[i_k \dots i_m], f}$ of $\partial_{[i_k \dots i_m]} f$

Theorem 16.298. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, |||_n \rangle$ the normed space with the maximum norm $|||_n$, $\langle X, ||| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow X$ a function then for $k, l \in \{1, \dots, m\}$ with $k \leq l$ we have that

$$\mathcal{D}_{[i_k \dots i_m], f} \subseteq \mathcal{D}_{[i_l \dots i_m], f} \subseteq \mathcal{D}_{[i_m \dots i_m], f} = \mathcal{D}_{i_m, f} \subseteq A$$

Proof. Let $l \in \{1, \dots, m\}$ then we have that $\forall r \in \{1, \dots, l-1\}$

$$\mathcal{D}_{[i_r \dots i_m], f} = \{x \in \mathcal{D}_{i_{r+1} \dots i_m} \mid \partial_{[i_{r+1} \dots i_m]} f \text{ has a } i_r\text{-partial derivative at } x\} \subseteq \mathcal{D}_{[i_{r+1} \dots i_m], f}$$

so that by [theorem: 10.36] we have that $\forall k \in \{1, \dots, l\}$

$$\mathcal{D}_{[i_k \dots i_m], f} \subseteq \mathcal{D}_{[i_l \dots i_m], f}$$

Further we have $\forall k \in \{1, \dots, m\}$ either

$$k = m. \text{ Then } \mathcal{D}_{[i_k \dots i_m], f} = \mathcal{D}_{[i_m \dots i_m], f} \subseteq A$$

$$k < m. \text{ Then } \mathcal{D}_{[i_k \dots i_m], f} \subseteq \mathcal{D}_{[i_m \dots i_m], f} \subseteq A$$

so in all cases we have

$$\mathcal{D}_{[i_k \dots i_m], f} \subseteq \mathcal{D}_{[i_m \dots i_m], f} = \mathcal{D}_{i_m, f} \subseteq A \quad \square$$

We can now define the concept of higher order partial derivatives at a point.

Definition 16.299. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, |||_n \rangle$ the normed space with the maximum norm $|||_n$, $\langle X, ||| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $x \in A$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow X$ a function then we say that

$$f \text{ has a partial derivative } \partial_{x, [i_1 \dots i_m]} f \text{ at } x \text{ [or equivalently } \partial_{x, [i_1 \dots i_m]} f \text{ exist]}$$

iff

$$x \in \mathcal{D}_{[i_1 \dots i_m], f}$$

The partial derivative $\partial_{x,[i_1 \dots i_m]}f$ at x is then defined by

$$\partial_{x,[i_1 \dots i_m]}f = \partial_{[i_1 \dots i_m]}f(x)$$

Definition 16.300. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ the normed space with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, U a open set in $\mathbb{K}^n$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow X$ a function then we say that

$$f \text{ has partial derivatives } \partial_{[i_1 \dots i_m]}f \text{ on } U$$

iff

$$U \subseteq \mathcal{D}_{[i_1 \dots i_m], f}$$

Using [definition: 16.299] this equivalent with

$$\forall x \in U \text{ } f \text{ has a partial derivative } \partial_{x,[i_1 \dots i_m]}f \text{ at } x$$

or

$$\forall x \in U \text{ } \partial_{x,[i_1 \dots i_m]}f \text{ exists}$$

Theorem 16.301. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ the normed space with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $x \in A$ $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow Y$ a function then we have that the following are equivalent

1. f has a partial derivative $\partial_{x,[i_1 \dots i_m]}f$ at x
2. For m we have either:
 - $m = 1$. Then f has a i_m -partial derivative at x .
 - $1 < m$. Then $\partial_{[i_2, \dots, i_m]}f$ has a i_1 -partial derivative at x

Further if (1) is valid then

$$\partial_{x,[i_1 \dots i_m]}f = \begin{cases} \partial_{x,i_1}f & \text{if } m = 1 \\ \partial_{x,i_1}(\partial_{[i_2 \dots i_m]}f) & \text{if } 1 < m \end{cases}$$

Proof.

$1 \Rightarrow 2$. For m we have either:

$m = 1$. Then by [definition: 16.299] we have that

$$x \in \mathcal{D}_{[i_1 \dots i_1], f} \stackrel{[\text{definition: 16.296}]}{=} \{x \in A \mid f \text{ has a } i_1\text{-partial derivative at } x\}$$

and

$$\partial_{x,[i_1 \dots i_1]}f = \partial_{[i_1 \dots i_1]}f(x) \stackrel{[\text{definition: 16.296}]}{=} \partial_{x,i_1}f$$

so that

$$f \text{ has a } i_1\text{-partial derivative at } x \text{ and } \partial_{x,[i_1 \dots i_m]}f = \partial_{x,i_1}f$$

$1 < m$. Then by [definition: 16.299] we have that

$$x \in \mathcal{D}_{[i_1 \dots i_m], f} \stackrel{[\text{def: 16.296}]}{=} \{x \in \mathcal{D}_{[i_2 \dots i_m], f} \mid \partial_{[i_2 \dots i_m]}f \text{ has a } i_1\text{-partial derivative at } x\}$$

so that

$$\partial_{[i_2 \dots i_m]}f \text{ has a } i_1\text{-partial derivative at } x$$

and

$$\partial_{x,[i_1 \dots i_m]}f = \partial_{x,i_1}(\partial_{[i_2 \dots i_m]}f)$$

$2 \Rightarrow 1$. For m we have either:

$m = 1$. Then as f has a i_1 -partial derivative at x it follows that

$$x \in \{x \in A \mid f \text{ has a } i_1\text{-partial derivative at } x\} \stackrel{[\text{definition: 16.296}]}{=} \mathcal{D}_{[i_1 \dots i_1], f}$$

which proves by [definition: 16.299] that

$$f \text{ has a } \partial_{x, [i_1 \dots i_1]} f \text{ derivative at } x$$

1 < m. Then as $\partial_{[i_2 \dots i_m]} f$ has a i_1 -partial derivative at x it follows that

$$x \in \{x \in \mathcal{D}_{[i_2 \dots i_m], f} \mid \partial_{[i_2 \dots i_m]} f \text{ has a } i_1\text{-partial derivative at } x\} \stackrel{[\text{definition: 16.296}]}{=} \mathcal{D}_{[i_1 \dots i_m], f}$$

which proves by [definition: 16.299] that

$$f \text{ has a } \partial_{x, [i_1 \dots i_m]} f \text{ derivative at } x$$

□

Just as there is a relation between partial derivatives and partial differentials [see theorem: 16.95] the same applies to partial derivatives and partial differentials of higher order.

Theorem 16.302. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, \|\cdot\|_n \rangle$ the normed space with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, $A \subseteq \mathbb{K}^n$, $x \in A$, $i \in \{1, \dots, n\}^m$ and $f: A \rightarrow X$ a function that has a partial differential $D_{x, [i_1 \dots i_m]} f$ at x then f has a partial derivative $\partial_{x, [i_1 \dots i_m]} f$ at x*

$$\partial_{x, [i_1 \dots i_m]} f = D_{x, [i_1 \dots i_m]} f \left(\underbrace{1: \dots : 1}_m \right)$$

Proof. We prove this by induction, so let

$$S = \left\{ m \in \mathbb{N} \mid \text{If } i \in \{1, \dots, n\}^m \text{ } y \in A \text{ and } f \text{ has a partial differential } D_{x, [i_1 \dots i_m]} f \text{ at } x \text{ then } f \text{ has a partial derivative } \partial_{x, [i_1 \dots i_m]} f \text{ at } x \text{ with } \partial_{x, [i_1 \dots i_m]} f = D_{x, [i_1 \dots i_m]} f \left(\underbrace{1: \dots : 1}_m \right) \right\}$$

then we have:

1 ∈ S. Let $i \in \{1, \dots, n\}^1$ and $x \in A$ such that f has a partial differential $D_{x, [i_1]} f$ at x then by [theorem: 16.273]

$$f \text{ has a } i_1\text{-partial differential at } x \text{ with } D_{x, [i_1]} f = D_{x, i_1} f$$

Using [theorem: 16.95] it follows that

$$f \text{ has a } i_1\text{-partial derivative at } x \text{ with } \partial_{x, i_1} f = D_{x, i_1} f(1) = D_{x, [i_1]} f(1)$$

Hence using [theorem: 16.301] it follows that

$$f \text{ has a partial derivative } \partial_{x, [i_1]} f \text{ at } x \text{ with } \partial_{x, [i_1]} f = \partial_{x, i_1} f = D_{x, [i_1]} f(1)$$

which proves that $1 \in S$.

m ∈ S ⇒ m + 1 ∈ S. Let $i \in \{1, \dots, n\}^{m+1}$, $x \in A$ and assume that f has a partial differential $D_{x, [i_1 \dots i_{m+1}]} f$ at x . Define

$$k \in \{1, \dots, n\}^m \text{ by } k_j = i_{j+1}$$

Consider now the mapping defined in [definition: 16.264]

$$D_{x, [k_1 \dots k_m]} f: \mathfrak{D}_{[k_1 \dots k_m], f} \rightarrow L_m(\mathbb{K}; Y)$$

Let $y \in \mathfrak{D}_{[k_1 \dots k_m], f}$ then we have by [definition: 16.267] that f has a partial differential $D_{x, [k_1 \dots k_m]} f$ at y . As $m \in S$ it follows that

$$f \text{ has a partial derivative } \partial_{y, [k_1 \dots k_m]} f \text{ at } y \tag{16.401}$$

with

$$\partial_{y, [k_1 \dots k_m]} f = D_{y, [k_1 \dots k_m]} f \left(\underbrace{1: \dots : 1}_n \right) \tag{16.402}$$

Using [definition: 16.403] on [eq: 16.401] it follows that $y \in \mathcal{D}_{[k_1 \dots k_m], f}$ which proves that

$$\mathfrak{D}_{[k_1 \dots k_m], f} \subseteq \mathcal{D}_{[k_1 \dots k_m], f} \quad (16.403)$$

Consider now the mapping defined in [definition: 16.296]

$$\partial_{[k_1 \dots k_m]}: \mathcal{D}_{[k_1 \dots k_m]} \rightarrow Y$$

and the evaluation operator from [definition: 16.239]

$$\text{ev}_{\underbrace{1 \dots 1}_m}: L_m(\mathbb{K}; X) \rightarrow X \text{ where } \text{ev}_{\underbrace{1 \dots 1}_m}(L) = L\left(\underbrace{1 \dots 1}_m\right)$$

then we have $\forall y \in \mathfrak{D}_{[k_1 \dots k_m], f}$ that

$$\begin{aligned} D_{y, [k_1 \dots k_m]} f\left(\underbrace{1 \dots 1}_m\right) &= \text{ev}_{\underbrace{1 \dots 1}_m}(D_{y, [k_1 \dots k_m]} f) \\ &\stackrel{[\text{eq: 16.267}]}{=} \text{ev}_{\underbrace{1 \dots 1}_m}(D_{[k_1 \dots k_m]} f(y)) \\ &= \left(\text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[k_1 \dots k_m]} f\right)(y) \end{aligned} \quad (16.404)$$

Hence

$$\begin{aligned} \partial_{k_1 \dots k_m} f(y) &\stackrel{[\text{definition: 16.299}]}{=} \partial_{y, [k_1 \dots k_m]} f \\ &\stackrel{[\text{eq: 16.402}]}{=} D_{y, [k_1 \dots k_m]} f\left(\underbrace{1 \dots 1}_m\right) \\ &\stackrel{[\text{eq: 16.404}]}{=} \left(\text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[k_1 \dots k_m]} f\right)(y) \end{aligned}$$

which combined with [eq: 16.403] proves that

$$(\partial_{[k_1 \dots k_m]} f)|_{\mathfrak{D}_{[k_1 \dots k_m], f}} = \text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[k_1 \dots k_m]} f \quad (16.405)$$

As f has a partial differential $D_{x, [i_1 \dots i_{m+1}]} f$ at x it follows from [theorem: 16.273] that $D_{[i_2 \dots i_{m+1}]} f$ is i_1 -partial differentiable at x with $D_{x, [i_1 \dots i_{m+1}]} f = D_{x, i_1}(D_{[i_2 \dots i_{m+1}]} f)$ or using the definition of $k \in \{1, \dots, n\}^m$ that

$$D_{[k_1 \dots k_m]} f \text{ is } i_1\text{-partial differentiable at } x \text{ with } D_{x, [i_1 \dots i_{m+1}]} f = D_{x, i_1}(D_{[k_1 \dots k_m]} f) \quad (16.406)$$

As by [lemma: 16.240] $\text{ev}_{\underbrace{1 \dots 1}_m}$ is Fréchet differentiable on $L_m(\mathbb{K}; X)$ it follows from the chain rule for partial differentiation [theorem: 16.87] that $\text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[i_2 \dots i_{m+1}]}^{[m]} f$ has a i_{m+1} -partial derivative at x and

$$\begin{aligned} D_{x, i_1}\left(\text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[k_1 \dots k_m]} f\right) &= D_{D_{[k_1 \dots k_m]} f(x)} \text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{x, i_1}(D_{[k_1 \dots k_m]} f) \\ &\stackrel{[\text{theorem: 16.140}]}{=} D_{D_{[k_1 \dots k_m]} f(x)}^{[1]} \text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{x, i_1}(D_{[k_1 \dots k_m]} f) \\ &\stackrel{[\text{lemma: 16.240}]}{=} \text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{x, i_1}(D_{[k_1 \dots k_m]} f) \\ &\stackrel{[\text{eq: 16.406}]}{=} \text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{x, [i_1 \dots i_{m+1}]} f \\ &= D_{x, [i_1 \dots i_{m+1}]} f\left(\underbrace{1 \dots 1}_m\right) \end{aligned}$$

Combining this with [eq: 16.405] proves that

$$(\partial_{[k_1 \dots k_m]} f)|_{\mathfrak{D}_{[k_1 \dots k_m], f}} \text{ has a } i_1\text{-partial differential at } x$$

with

$$D_{x, i_1}((\partial_{[k_1 \dots k_m]} f)|_{\mathfrak{D}_{[k_1 \dots k_m], f}}) = D_{x, [i_1 \dots i_{m+1}]} f\left(\underbrace{1 \dots 1}_m\right)$$

Using the locality of partial differentiation [see theorem: 16.82] it follows that

$$\partial_{[k_1 \dots k_m]} f \text{ has a } i_1\text{-partial differential at } x$$

with

$$D_{x, i_1}(\partial_{[k_1 \dots k_m]} f) = D_{x, [i_1 \dots i_{m+1}]} f \left(\underbrace{1 \dots 1}_m \right)$$

or using the definition of k that

$$\partial_{[i_2 \dots i_{m+1}]} f \text{ has a } i_1\text{-partial differential at } x$$

with

$$D_{x, i_1}(\partial_{[i_2 \dots i_{m+1}]} f) = D_{x, [i_1 \dots i_{m+1}]} f \left(\underbrace{1 \dots 1}_m \right) \quad (16.407)$$

Using [theorem: 16.95] it follows that

$$\partial_{[i_2 \dots i_{m+1}]} f \text{ has a } i_1\text{-partial derivative at } x$$

with

$$\begin{aligned} \partial_{x, i_1}(\partial_{[i_2 \dots i_m]} f) &= D_{x, i_1}(\partial_{[i_2 \dots i_m]} f)(1) \\ &\stackrel{[\text{eq: 16.407}]}{=} D_{x, [i_1 \dots i_{m+1}]} f \left(\underbrace{1 \dots 1}_m \right) (1) \\ &\stackrel{[\text{lemma: 16.119}]}{=} D_{x, [i_1 \dots i_{m+1}]} f \left(\underbrace{1 \dots 1}_{m+1} \right) \end{aligned}$$

Finally by [theorem: 16.301] it follows that

$$f \text{ has a partial derivative } \partial_{x, [i_1 \dots i_{m+1}]} f \text{ at } x$$

with

$$\partial_{x, [i_1 \dots i_{m+1}]} f = D_{x, [i_1 \dots i_{m+1}]} f \left(\underbrace{1 \dots 1}_{m+1} \right)$$

proving that

$$m+1 \in S \quad \square$$

The opposite of the above is not always true, the existence of a partial derivative at a point does not always implies that the corresponding partial differential exists. For example if $\partial_{x, [i_1 \dots i_1]} f$ exist then by [theorem: 16.301] f has a i -partial derivative at x . Unfortunately we can not use [theorem: 16.95] to conclude that f is a i -partial differentiable at x , unless $x \in A^\circ$. As by [theorem: 16.273] the existence of the i -partial differential at x is required for the existence of the partial differential $D_{x, [i_1 \dots i_1]} f$ it follows that we have a problem here [unless $x \in A^\circ$]. The root cause of the problem is that Fréchet differentiability at a point requires that this point is a inner point of the domain of the function while for derivating the point must be a limit point of the domain. Luckily if the domain of f is a open set and f has derivatives on U the reverse is possible. To prove this we need some definitions and lemmas.

Definition 16.303. Let $m \in \mathbb{N}$, $\langle \mathbb{K}^m, \|\cdot\|_m \rangle$ the normed space with the maximum norm $\|\cdot\|_m$, $\langle X, \|\cdot\| \rangle$ a normed space and $a \in X$ then we define

$$h_{m,a}: \mathbb{K}^m \rightarrow X \text{ by } h_{m,a}(x) = \left(\prod_{i=1}^m x_i \right) \cdot a$$

Using [example: 14.229] it follows that $h_{m,a} \in L^m(\mathbb{K}, X)$ so that we can use

$$\mathcal{I}_{m, \mathbb{K}, X}: L_m(\mathbb{K}; X) \rightarrow L^m(\mathbb{K}; X) \text{ defined in [theorem: 16.126]}$$

to finally define $\mathcal{P}[m, a] \in L_m(\mathbb{K}; X)$ by

$$\mathcal{P}[m, a] = (\mathcal{I}_{m, \mathbb{K}, X})^{-1}(h_{m,a}) \in L_m(\mathbb{K}; X)$$

Note 16.304. If $x = (x_1, \dots, x_m) \in \mathbb{K}^m$ then we have

$$\mathcal{P}[m, a](x_1 : \dots : x_m) = (\mathcal{I}_{m, \mathbb{K}, X})^{-1}(h_{m, a})(x_1 : \dots : x_m) \stackrel{[\text{theorem: 16.126}]}{=} h_{m, a}(x_1, \dots, x_m) = \left(\prod_{i=1}^m x_i \right) \cdot a$$

so that

$$\mathcal{P}[m, a]\left(\underbrace{1 : \dots : 1}_m\right) = a$$

Lemma 16.305. Let $m \in \mathbb{N}$, $\langle \mathbb{K}^m, \|\cdot\|_m \rangle$ the normed space with the maximum norm $\|\cdot\|_m$, $\langle X, \|\cdot\| \rangle$ a normed space then if we define

$$\mathcal{P}[m]: X \rightarrow L_m(\mathbb{K}; X) \text{ by } \mathcal{P}[m](y) = \mathcal{P}[m, y]$$

then

$$\mathcal{P}[m] \in L(X, L_m(\mathbb{K}; X))$$

so that by [example: 16.180] $\mathcal{P}[m]$ is ∞ -times differentiable on X and $\forall y \in X$

$$D_y \mathcal{P}[m] = D_y^{[1]} \mathcal{P}[m] = \mathcal{P}[m]$$

Proof. First we prove linearity, so let $\alpha \in \mathbb{K}$ and $x, y \in X$ then we have for $h \in \mathbb{K}^m$ that

$$\begin{aligned} (\mathcal{P}[m](x + \alpha \cdot y))(h_1 : \dots : h_m) &= \mathcal{P}[m, x + \alpha \cdot y](h_1 : \dots : h_m) \\ &\stackrel{[\text{note: 16.304}]}{=} \left(\prod_{i=1}^m h_i \right) \cdot (x + \alpha \cdot y) \\ &= \left(\prod_{i=1}^m h_i \right) \cdot x + \alpha \cdot \left(\left(\prod_{i=1}^m h_i \right) \cdot y \right) \\ &\stackrel{[\text{note: 16.304}]}{=} (\mathcal{P}[m](x))(h_1 : \dots : h_m) + (\alpha \cdot \mathcal{P}[m](y))(h_1 : \dots : h_m) \\ &\stackrel{[\text{lemma: 16.120}]}{=} (\mathcal{P}[m](x) + \alpha \cdot \mathcal{P}[m](y))(h_1 : \dots : h_m) \end{aligned}$$

proving by [lemma: 16.123] that $\mathcal{P}[m](x + \alpha \cdot y) = \mathcal{P}[m](x) + \alpha \cdot \mathcal{P}[m](y)$. Hence

$$\mathcal{P}[m] \in \text{Hom}(X; L_m(\mathbb{K}; X))$$

Further we have

$$\begin{aligned} \|(\mathcal{P}[m](x))(h_1 : \dots : h_m)\|_X &= \|\mathcal{P}[m, x](h_1 \dots h_m)\| \\ &\stackrel{[\text{note: 16.304}]}{=} \left\| \left(\prod_{i=1}^m h_i \right) \cdot x \right\| \\ &= \left| \prod_{i=1}^m h_i \right| \cdot \|x\| \\ &= \|x\| \cdot \prod_{i=1}^m |h_i| \end{aligned}$$

so that by [lemma: 16.124] we have $\|\mathcal{P}[m](x)\|_{L_m(\mathbb{K}; X)} \leq \|x\|$. Hence by [theorem: 14.201] $\mathcal{P}[m]$ is continuous proving that

$$\mathcal{P}[m] \in L(X, L_m(\mathbb{K}; X)) \quad \square$$

Theorem 16.306. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{R}^n, \|\cdot\|_n \rangle$ the normed space with the maximum norm $\|\cdot\|_n$, $\langle X, \|\cdot\| \rangle$ a normed space, a open set $U \subseteq \mathbb{R}^n$, $i \in \{1, \dots, n\}^m$ and $f: U \rightarrow X$ a function that has partial derivatives $\partial_{[i_1 \dots i_m]} f$ on U then f has partial differentials $D_{[i_1 \dots i_m]} f$ on U with

$$\forall x \in U \quad D_{x, [i_1 \dots i_m]} f = \mathcal{P}[m](\partial_{x, [i_1 \dots i_m]} f) \stackrel{[\text{lemma: 16.305}]}{=} \mathcal{P}[m, \partial_{x, [i_1 \dots i_m]} f]$$

Proof. We prove this by induction so let

$S = \{m \in \mathbb{N} \mid \text{If } i \in \{1, \dots, n\}^m \text{ and } f \text{ has } \partial_{[i_1 \dots i_m]} f \text{ partial derivatives on } U \text{ then } f \text{ has } D_{[i_1 \dots i_m]} f \text{ partial differentials on } U \text{ and } \forall x \in U \text{ we have } D_{x, [i_1 \dots i_m]} f = \mathcal{P}[m](\partial_{x, [i_1 \dots i_m]} f)\}$

then we have:

$1 \in S$. Let $i \in \{1, \dots, n\}^1$ and assume that f has $\partial_{[i_1 \dots i_1]} f$ derivatives on U . Let $x \in U$ then $\partial_{x, [i_1 \dots i_1]} f$ exist so that by [theorem: 16.301] f has a i_1 -partial derivative at x . As $x \in U = U^\circ$ we can use [theorem: 16.95] from which it follows that f has a i_1 -partial differential at x where

$$D_{x, i_1} f: \mathbb{K} \rightarrow X \text{ is defined by } D_{x, i_1} f(h) = h \cdot \partial_{x, i_1} f \quad (16.408)$$

Using then [theorem: 16.273] it follows that

$$f \text{ has a } D_{x, [i_1 \dots i_1]} f \text{ partial differential at } x \text{ with } D_{x, [i_1 \dots i_1]} f = D_{x, i_1} f \quad (16.409)$$

Let $h_1 \in \mathbb{K}$ then

$$\begin{aligned} \mathcal{P}[1](\partial_{x, [i_1 \dots i_1]} f)(h_1) &\stackrel{[\text{lemma: 16.305}]}{=} \mathcal{P}[1, \partial_{x, [i_1 \dots i_1]} f](h_1) \\ &= \mathcal{P}[1, \partial_{x, [i_1 \dots i_1]} f]\left(\underbrace{h_1 \dots h_1}_1\right) \\ &\stackrel{[\text{note: 16.304}]}{=} \left(\prod_{i=1}^1 h_i\right) \cdot \partial_{x, [i_1 \dots i_1]} f \\ &= h_1 \cdot \partial_{x, [i_1 \dots i_1]} f \\ &\stackrel{[\text{theorem: 16.301}]}{=} h_1 \cdot \partial_{x, i_1} f \\ &\stackrel{[\text{eq: 16.408}]}{=} D_{x, i_1} f(h_1) \\ &\stackrel{[\text{eq: 16.409}]}{=} D_{x, [i_1 \dots i_1]} f(h_1) \end{aligned}$$

which proves that

$$\mathcal{P}[1](\partial_{x, [i_1 \dots i_1]} f) = D_{x, [i_1 \dots i_1]} f$$

As this is true $\forall x \in U$ we have that

$$f \text{ has } D_{x, [i_1 \dots i_1]} f \text{ partial differentials on } U$$

with

$$\forall x \in U \mathcal{P}[1](\partial_{x, [i_1 \dots i_1]} f) = D_{x, [i_1 \dots i_1]} f$$

which proves that

$$1 \in S$$

$m \in S \Rightarrow m+1 \in S$. Let $i \in \{1, \dots, n\}^{m+1}$ and assume that f has a $D_{[i_1 \dots i_{m+1}]} f$ derivative on U then we have by [definition: 16.300] that $U \subseteq \mathcal{D}_{[i_1 \dots i_{m+1}], f}$ and by [theorem: 16.298] that $\mathcal{D}_{[i_1 \dots i_{m+1}], f} \subseteq \mathcal{D}_{[i_2 \dots i_{m+1}], f} \subseteq U$ so that

$$U = \mathcal{D}_{[i_1 \dots i_{m+1}], f} = \mathcal{D}_{[i_2 \dots i_{m+1}], f} \quad (16.410)$$

Define now

$$k \in \{1, \dots, n\}^m \text{ by } k_j = i_{j+1}$$

then by [eq: 16.410]

$$U = \mathcal{D}_{[k_1 \dots k_m], f} \quad (16.411)$$

As $U = \mathcal{D}_{[k_1 \dots k_m], f}$ we have by [definition: 16.300] that f has $\partial_{[k_1 \dots k_m]} f$ partial derivatives on U , which, as $m \in S$, proves that

$$f \text{ has } D_{[k_1 \dots k_m]} f \text{ partial differentials on } U \quad (16.412)$$

and

$$\forall x \in U \text{ we have } D_{x, [k_1 \dots k_m]} f = \mathcal{P}[m](\partial_{x, [k_1 \dots k_m]} f) \quad (16.413)$$

Using [definition: 16.269] on [eq: 16.412] proves that $U \subseteq \mathfrak{D}_{[k_1 \dots k_m], f}$ which as by [theorem: 16.266] $\mathfrak{D}_{[k_1 \dots k_m], f} \subseteq U$ proves that

$$U = \mathfrak{D}_{[k_1 \dots k_m], f} \stackrel{[\text{eq: 16.411}]}{=} \mathcal{D}_{[k_1 \dots k_m], f} \quad (16.414)$$

Let $x \in \mathfrak{D}_{[k_1 \dots k_m], f} = U$ then

$$\begin{aligned} D_{[k_1 \dots k_m]} f(x) &\stackrel{[\text{definition: 16.267}]}{=} D_{x, [k_1 \dots k_m]} f \\ &\stackrel{[\text{eq: 16.413}]}{=} \mathcal{P}[m, \partial_{x, [k_1 \dots k_m]} f] \\ &\stackrel{[\text{definition: 16.299}]}{=} \mathcal{P}[m, \partial_{[k_1 \dots k_m]} f(x)] \\ &\stackrel{[\text{lemma: 16.305}]}{=} \mathcal{P}[m](\partial_{[k_1 \dots k_m]} f(x)) \\ &= (\mathcal{P}[m] \circ \partial_{[k_1 \dots k_m]} f)(x) \end{aligned}$$

which together with [eq: 16.414] proves that

$$D_{[k_1 \dots k_m]} f = \mathcal{P}[m] \circ \partial_{[k_1 \dots k_m]} f \quad (16.415)$$

Let $x \in U$ then by [eq: 16.410] $x \in \mathcal{D}_{[i_1 \dots i_{m+1}], f}$ so that by [definition: 16.299] f has a partial derivative $\partial_{x, [i_1, \dots, i_{m+1}]} f$ at x , hence using [theorem: 16.301] it follows that

$$\partial_{[k_1 \dots k_m]} f = \partial_{[i_2 \dots i_{m+1}]} f \text{ has a } i_1\text{-partial derivative at } x$$

with

$$\partial_{x, [i_1 \dots i_{m+1}]} f = \partial_{x, i_1}(\partial_{[i_2 \dots i_{m+1}]} f) = \partial_{x, i_1}(\partial_{[k_1 \dots k_m]} f) \quad (16.416)$$

As $x \in U \stackrel{[\text{theorem: 14.16}]}{=} \stackrel{[\text{eq: 16.411}]}{=} U^\circ \stackrel{[\text{eq: 16.411}]}{=} \left(\mathcal{D}_{[k_2 \dots k_m], f} \right)^\circ$ it follows from [theorem: 16.95] that

$$\partial_{[k_1 \dots k_m]} f \text{ is } i_1\text{-partial differentiable at } x \quad (16.417)$$

where

$$D_{x, i_1}(\partial_{[k_1 \dots k_m]} f): \mathbb{K} \rightarrow X \text{ is defined by } D_{x, i_1}(\partial_{[k_1 \dots k_m]} f)(h) = h \cdot \partial_{x, i_1}(\partial_{[k_1 \dots k_m]} f) \quad (16.418)$$

As $\mathcal{P}[m]$ is Fréchet differentiable on X [see lemma: 16.305] and $\partial_{[k_1 \dots k_m]} f$ has a i_1 -partial differential at x we can apply the chain rule for partial differentials [see theorem: 16.87] to prove that

$$\mathcal{P}[m] \circ \partial_{[k_1 \dots k_m]} f \text{ is } i_1\text{-partial differentiable at } x$$

with

$$\begin{aligned} D_{x, i_1}(\mathcal{P}[m] \circ \partial_{[k_1 \dots k_m]} f) &\stackrel{[\text{theorem: 16.87}]}{=} D_{\partial_{[k_1 \dots k_m]} f(x)} \mathcal{P}[m] \circ D_{x, i_1} \partial_{[k_1 \dots k_m]} f \\ &\stackrel{[\text{lemma: 16.305}]}{=} \mathcal{P}[m] \circ D_{x, i_1} \partial_{[k_1 \dots k_m]} f \end{aligned}$$

Substituting [eq: 16.415] in the above it follows that

$$D_{[k_1 \dots k_m]} f \text{ is } i_1\text{-partial differentiable at } x$$

with

$$D_{x, i_1} D_{[k_1 \dots k_m]} f = \mathcal{P}[m] \circ D_{x, i_1} \partial_{[k_1 \dots k_m]} f$$

Using the definition of k on the above we have

$$D_{[i_2 \dots i_{m+1}]} f \text{ is } i_1\text{-partial differentiable at } x$$

with

$$D_{x, i_1}(D_{[i_2 \dots i_{m+1}]} f) = \mathcal{P}[m] \circ D_{x, i_1} \partial_{[i_2 \dots i_{m+1}]} f$$

By [theorem: 16.273] the above becomes

$$f \text{ has a partial differential } D_{x, [i_1 \dots i_{m+1}]} f \text{ at } x \quad (16.419)$$

with

$$D_{x,[i_1 \dots i_{m+1}]}f = \mathcal{P}[m] \circ D_{x,i_1} \partial_{[i_2 \dots i_{m+1}]}f \quad (16.420)$$

Let $h \in \mathbb{K}^{m+1}$ then we have

$$\begin{aligned} & D_{x,[i_1 \dots i_{m+1}]}f(h_1: \dots: h_{m+1}) && \stackrel{[eq: 16.420]}{=} \\ & (\mathcal{P}[m] \circ D_{x,i_1} \partial_{[i_2 \dots i_{m+1}]}f)(h_1: \dots: h_{m+1}) && = \\ & ((\mathcal{P}[m] \circ D_{x,i_1} \partial_{[i_2 \dots i_{m+1}]}f)(h_1))(h_2: \dots: h_{m+1}) && = \\ & (\mathcal{P}[m]((D_{x,i_1} \partial_{[i_2 \dots i_{m+1}]}f)(h_1)))(h_2: \dots: h_{m+1}) && \stackrel{[eq: 16.418]}{=} \\ & (\mathcal{P}[m](h_1 \cdot \partial_{x,i_1}(\partial_{[i_2 \dots i_{m+1}]}f)))(h_2: \dots: h_{m+1}) && \stackrel{[eq: 16.416]}{=} \\ & (\mathcal{P}[m](h_1 \cdot \partial_{x,[i_1 \dots i_{m+1}]}f))(h_2: \dots: h_{m+1}) && \stackrel{\mathcal{P}[m] \in L(X; L_m(\mathbb{R}; X))}{=} \\ & h_1 \cdot \mathcal{P}[m](\partial_{x,[i_1 \dots i_{m+1}]}f)(h_2: \dots: h_{m+1}) && \stackrel{[note: 16.304]}{=} \\ & h_1 \cdot \left(\left(\prod_{j=2}^{m+1} h_j \right) \cdot \partial_{x,[i_1 \dots i_{m+1}]}f \right) && = \\ & \left(\prod_{j=1}^{m+1} h_j \right) \cdot \partial_{x,[i_1 \dots i_{m+1}]}f && \stackrel{[note: 16.304]}{=} \\ & (\mathcal{P}[m+1](\partial_{x,[i_1 \dots i_{m+1}]}f))(h_1: \dots: h_{m+1}) \end{aligned}$$

hence, using [lemma: 16.123] we have

$$D_{x,[i_1 \dots i_{m+1}]}f = \mathcal{P}[m+1](\partial_{x,[i_1 \dots i_{m+1}]}f)$$

As this is true $\forall x \in U$ it follows that

$$f \text{ has a partial differential } D_{x,[i_1 \dots i_{m+1}]}f \text{ on } U$$

and

$$\forall x \in U \text{ we have } D_{x,[i_1 \dots i_{m+1}]}f = \mathcal{P}[m+1](\partial_{x,[i_1 \dots i_{m+1}]}f)$$

which proves that

$$m+1 \in S$$

□

Combing [theorems: 16.302, 16.306] we have the following theorem.

Theorem 16.307. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, ||_n \rangle$ the normed space with the maximum norm $|||_n$, $\langle X, ||| \rangle$ a normed space, a open set $U \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}^m$ and $f: U \rightarrow X$ a function then we have*

$$f \text{ has partial differentials } D_{[i_1 \dots i_m]}f \text{ on } U$$

$$\Updownarrow$$

$$f \text{ has partial derivatives } \partial_{[i_1 \dots i_m]}f \text{ on } U$$

Further we have that if f has partial differentials $D_{[i_1 \dots i_m]}f$ on U [or equivalently has partial derivatives $\partial_{[i_1 \dots i_m]}f$ on U] then we have that $\forall x \in U$ that

$$\partial_{x,[i_1 \dots i_m]}f = D_{x,[i_1 \dots i_m]}f \left(\underbrace{1: \dots: 1}_m \right)$$

and

$$D_{x,[i_1 \dots i_m]}f \in L_m(\mathbb{K}; X) \text{ is defined by } D_{x,[i_1 \dots i_m]}f = \mathcal{P}[m, \partial_{x,[i_1 \dots i_m]}f]$$

so that for $h \in \mathbb{R}^m$ we have

$$D_{x,[i_1 \dots i_m]}f(h_1: \dots: h_m) = \left(\prod_{i=1}^m h_i \right) \cdot \partial_{x,[i_1 \dots i_m]}f$$

Proof.

$\Rightarrow$. Let $x \in U$ then, as f has partial differentials $D_{[i_1 \dots i_m]}f$ on U , f has a partial differential $D_{x, [i_1 \dots i_m]}f$ at x , hence it follows from [theorem: 16.302] that f has a partial derivative $\partial_{x, [i_1 \dots i_m]}f$ at x with

$$\partial_{x, [i_1 \dots i_m]}f = D_{x, [i_1 \dots i_m]}f \left(\underbrace{1: \dots : 1}_m \right)$$

As this is true $\forall x \in U$ we have

f has partial derivatives $\partial_{[i_1 \dots i_m]}f$ on U

with $\forall x \in U$

$$\partial_{x, [i_1 \dots i_m]}f = D_{x, [i_1 \dots i_m]}f \left(\underbrace{1: \dots : 1}_m \right)$$

$\Leftarrow$. Using [theorem: 16.306] it follows directly that

f has partial derivatives

with $\forall x \in U$

$$D_{x, [i_1 \dots i_m]}f = \mathcal{P}[m, \partial_{x, [i_1 \dots i_m]}]$$

so that for $h \in \mathbb{K}^m$ we have

$$D_{x, [i_1 \dots i_m]}f(h_1: \dots : h_m) = \left(\prod_{i=1}^m h_i \right) \cdot \partial_{x, [i_1 \dots i_m]}f \quad \square$$

Now in the context of [theorem: 16.295] we look at continuity of $D_{[i_1 \dots i_m]}f$ and $\partial_{[i_1 \dots i_m]}f$.

Theorem 16.308. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, ||_n \rangle$ the normed space with the maximum norm $|||_n$, $\langle X, ||| \rangle$ a normed space, a open set $U \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}^m$ and $f: U \rightarrow X$ a function then we have*

f has partial differentials $D_{[i_1 \dots i_m]}f$ on U and $D_{[i_1 \dots i_m]}f$ is continuous

$\Updownarrow$

f has partial derivatives $\partial_{[i_1 \dots i_m]}f$ on U and $\partial_{[i_1 \dots i_m]}f$ is continuous

Proof.

$\Rightarrow$. If f has partial differentials $D_{[i_1 \dots i_m]}f$ on U then by [theorem: 16.307]

f has partial derivatives $\partial_{[i_1 \dots i_m]}f$ on U

and

$$\forall x \in U \text{ we have } \partial_{x, [i_1 \dots i_m]}f = D_{x, [i_1 \dots i_m]}f \left(\underbrace{1: \dots : 1}_m \right)$$

Let $x \in U$ then we have that

$$\begin{aligned} \partial_{[i_1 \dots i_m]}f(x) &\stackrel{[\text{definition: 16.299}]}{=} \partial_{x, [i_1 \dots i_m]}f \\ &= D_{x, [i_1 \dots i_m]}f \left(\underbrace{1: \dots : 1}_m \right) \\ &\stackrel{[\text{definition: 16.239}]}{=} \text{ev}_{\underbrace{1 \dots 1}_m}(D_{x, [i_1 \dots i_m]}f) \\ &\stackrel{[\text{theorem: 16.299}]}{=} \text{ev}_{\underbrace{1 \dots 1}_m}(D_{[i_1 \dots i_m]}f(x)) \\ &= \left(\text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[i_1 \dots i_m]}f \right)(x) \end{aligned}$$

which proves that

$$\partial_{[i_1 \dots i_m]}f = \text{ev}_{\underbrace{1 \dots 1}_m} \circ D_{[i_1 \dots i_m]}f$$

As $\text{ev}_{\underbrace{1, \dots, 1}_m}$ is continuous [see lemma: 16.240] and $D_{[i_1 \dots i_m]}f$ is assumed to be continuous it follows that $\text{ev}_{\underbrace{1, \dots, 1}_m} \circ D_{[i_1 \dots i_m]}f$ is continuous, hence we have that

$$\partial_{[i_1 \dots i_m]}f \text{ is continuous}$$

$\Leftarrow$. If f has partial derivatives $\partial_{[i_1 \dots i_m]}f$ on U then by [theorem: 16.307]

$$f \text{ has partial differentials } D_{[i_1 \dots i_m]}f \text{ on } U$$

and

$$\forall x \in U \text{ we have } D_{x, [i_1 \dots i_m]}f = \mathcal{P}[m, \partial_{x, [i_1 \dots i_m]}f]$$

Let $x \in U$ then we have that

$$\begin{aligned} D_{[i_1 \dots i_m]}f(x) &\stackrel{[\text{theorem: 16.267}]}{=} D_{x, [i_1 \dots i_m]}f \\ &= \mathcal{P}[m, \partial_{x, [i_1 \dots i_m]}f] \\ &= \mathcal{P}[m](\partial_{x, [i_1 \dots i_m]}f) \\ &\stackrel{[\text{theorem: 16.267}]}{=} \mathcal{P}[m](\partial_{[i_1 \dots i_m]}f(x)) \\ &= (\mathcal{P}[m] \circ \partial_{[i_1 \dots i_m]}f)(x) \end{aligned}$$

which proves that

$$D_{[i_1 \dots i_m]}f = \mathcal{P}[m] \circ \partial_{[i_1 \dots i_m]}f$$

As $\mathcal{P}[m]$ is continuous [see lemma: 16.305] and $\partial_{[i_1 \dots i_m]}f$ is assumed to be continuous it follows that $\mathcal{P}[m] \circ \partial_{[i_1 \dots i_m]}f$ is continuous or

$$D_{[i_1 \dots i_m]}f \text{ is continuous} \quad \square$$

Next we show that f is of class C^m if and only if $\forall i \in \{1, \dots, n\}^m$ f has continuous partial derivatives $\partial_{[i_1 \dots i_m]}f$ on U .

Theorem 16.309. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, |||_n \rangle$ the normed space with the maximum norm $|||_n$, $\langle X, ||| \rangle$ a normed space, a open set $U \subseteq \mathbb{K}^n$, $i \in \{1, \dots, n\}^m$ and $f: U \rightarrow X$ a function then we have*

$$\begin{aligned} &f \text{ is of class } C^m \\ &\quad \Updownarrow \\ &\forall i \in \{1, \dots, n\}^m \text{ we have that } f \text{ has partial differentials } \partial_{[i_1 \dots i_m]}f \text{ on } U \\ &\quad \text{and} \\ &\quad \partial_{[i_1 \dots i_m]}f \text{ is continuous} \end{aligned}$$

Proof. This follows directly from [theorem: 16.295] and [theorem: 16.308]. $\square$

We use the above theorem to show that higher order partial derivatives are symmetric.
TODO

Theorem 16.310. *Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{K}^n, ||| \rangle$ be the vector space $\mathbb{K}^n$ with the maximum norm, $\langle X, ||| \rangle$ a Banach space, a open set $U \subseteq \mathbb{K}^n$, $f: U \rightarrow X$ a function that is of class C^m then*

$$\forall i \in \{1, \dots, n\}^m \text{ } f \text{ has partial derivatives } \partial_{[i_1 \dots i_m]}f \text{ on } U$$

and $\forall \sigma \in \mathcal{P}_n$ we have $\forall x \in U$ we have

$$\partial_{x, i_1 \dots i_m}f = \partial_{x, [i_{\sigma(1)} \dots i_{\sigma(m)}]}f$$

Proof. Let $i \in \{1, \dots, n\}^m$ then by [theorem: 16.295] it follows that

$$f \text{ has partial differentials } D_{[i_1, \dots, i_m]}f \text{ on } U$$

and by [theorem: 16.278] that

$$\forall x \in U, \forall \sigma \in \mathcal{P}_n \text{ and } \forall h \in \mathbb{K}^n \quad D_{x, [i_1 \dots i_m]} f(h_1 : \dots : h_n) = D_{x, [i_{\sigma(1)} \dots i_{\sigma(m)}]} f(h_{\sigma(1)} : \dots : h_{\sigma(m)}) \quad (16.421)$$

Using [theorem: 16.307] it follows that

$$\forall x \in U \quad \partial_{x, [i_1 \dots i_m]} f = D_{x, [i_1 \dots i_m]} f \left(\underbrace{1 : \dots : 1}_m \right) \quad (16.422)$$

Let $x \in U$, $\sigma \in \mathcal{P}_n$ then

$$\begin{aligned} \partial_{x, [i_1 \dots i_m]} f &\stackrel{[\text{eq: 16.422}]}{=} D_{x, [i_1 \dots i_m]} f \left(\underbrace{1 : \dots : 1}_m \right) \\ &\stackrel{[\text{eq: 16.421}]}{=} D_{x, [i_{\sigma(1)} \dots i_{\sigma(m)}]} f(1 : \dots : 1) \\ &\stackrel{[\text{eq: 16.422}]}{=} \partial_{x, [i_{\sigma(1)} \dots i_{\sigma(m)}]} f \\ &\quad \square \end{aligned}$$

As m -times differentiability implies m -times partial differentiability [see theorem: 16.276] and m -times partial differentiability implies m -times partial derivatives [see theorem: 16.302] we can express the m -times differential in terms of the m -times partial differential and the m -times partial derivatives. To prove this we first need the following lemma.

Lemma 16.311. *Let $n \in \mathbb{N}$, $m \in \mathbb{N}$, $\langle \mathbb{R}^n, |||_N \rangle$ be the vector space of $\mathbb{R}^n$ with the maximum norm, $\langle X, ||| \rangle$ a normed space and $L \in L_m(\mathbb{R}^n; X)$ then for $\{h_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}^n$ we have*

$$L(h_1 : \dots : h_m) = \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=1}^m (h_j)_{i_j} \right) \cdot L(e_{i_1} : \dots : e_{i_m})$$

where $\{e_i\}_{i \in \{1, \dots, n\}}$ is the canonical basis on $\mathbb{R}^n$ [see theorem: 11.167] defined by $(e_i)_j = \delta_{i,j}$.

Proof. We proof this by induction on m so define

$$\begin{aligned} S = & \left\{ m \in \mathbb{N} \mid \text{If } \{h_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}^n \text{ and } L \in L_m(\mathbb{R}^n; X) \text{ then } L(h_1 : \dots : h_m) = \right. \\ & \left. \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=1}^m (h_j)_{i_j} \right) \cdot L(e_{i_1} : \dots : e_{i_m}) \right\} \end{aligned}$$

then we have:

$1 \in S$. If $\{h_i\}_{i \in \{1\}} \subseteq \mathbb{R}^n$ and $L \in L_1(\mathbb{R}^n; X) = L(\mathbb{R}^n, X)$ then we have

$$\begin{aligned} L\left(\underbrace{h_1 : \dots : h_1}_1\right) &= L(h_1) \\ &\stackrel{[\text{theorem: 11.167}]}{=} L\left(\sum_{i=1}^n (h_1)_i \cdot e_i\right) \\ &= \sum_{i=1}^n (h_1)_i \cdot L(e_i) \\ &= \sum_{i \in \{1, \dots, n\}^1} (h_1)_{i_1} \cdot L(e_{i_1}) \\ &= \sum_{i \in \{1, \dots, n\}^1} \left(\prod_{j=1}^1 (h_j)_{i_j} \right) \cdot L\left(\underbrace{e_{i_1} : \dots : e_{i_1}}_1\right) \end{aligned}$$

which proves that $1 \in S$.

$\mathbf{m} \in S \Rightarrow \mathbf{m} + \mathbf{1} \in S$. If $\{h_i\}_{i \in \{1, \dots, m+1\}} \subseteq \mathbb{R}^n$ and $L \in L_{m+1}(\mathbb{R}^n; X)$ then for $i \in \{1, \dots, n\}$ $L(e_i) \in L_m(\mathbb{R}^n; X)$ so, as $m \in S$, we have

$$L(e_i)(h_2: \dots: h_{m+1}) = \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=2}^{m+1} (h_j)_{i_j} \right) \cdot L(e_i)(e_{i_1}: \dots: e_{i_{m+1}}) \quad (16.423)$$

Next we have

$$\begin{aligned} & L(h_1: \dots: h_{m+1}) = \\ & L(h_1)(h_2: \dots: h_{m+1}) \quad [\text{theorem: 11.167}] \\ & L\left(\sum_{k=1}^n (h_1)_k \cdot e_k\right)(h_2: \dots: h_{m+1}) = \\ & \sum_{k \in \{1, \dots, n\}} (h_1)_k \cdot L(e_k)(h_2: \dots: h_{m+1}) \quad [\text{eq: 16.423}] \\ & \sum_{k \in \{1, \dots, n\}} \left((h_1)_k \cdot \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=2}^{m+1} (h_j)_{i_j} \right) \cdot L(e_k)(e_{i_1}: \dots: e_{i_{m+1}}) \right) = \\ & \sum_{k \in \{1, \dots, n\}} \left(\sum_{i \in \{1, \dots, n\}^m} (h_1)_k \cdot \left(\prod_{j=2}^{m+1} (h_j)_{i_j} \right) \cdot L(e_k)(e_{i_1}: \dots: e_{i_{m+1}}) \right) \quad [\text{theorem: 11.46}] \\ & \sum_{i \in \{1, \dots, n\}^{m+1}} \left((h_1)_{i_1} \cdot \prod_{j=2}^{m+1} (h_j)_{i_j} \right) \cdot L(e_{i_1})(e_{i_2}: \dots: e_{i_{m+1}}) = \\ & \sum_{i \in \{1, \dots, n\}^{m+1}} \left(\prod_{j=1}^{m+1} (h_j)_{i_j} \right) \cdot L(e_{i_1})(e_{i_2}: \dots: e_{i_{m+1}}) \end{aligned}$$

proving that $m+1 \in S$ □

Theorem 16.312. Let $n \in \mathbb{N} \setminus \{1\}$, $m \in \mathbb{N}$, $\langle \mathbb{R}^n, ||| \rangle$ be the vector space $\mathbb{R}^n$ with the maximum norm, $\langle X, ||| \rangle$ a normed space, a open set $U \subseteq \mathbb{R}^n$, $x \in U$, $f: U \rightarrow X$ a function that is m -times differentiable at x then for $\forall i \in \{1, \dots, n\}^m$ $D_{x, [i_1 \dots i_m]} f$ and $\partial_{x, [i_1 \dots i_m]} f$ exists and we have

$$\partial_{x, [i_1 \dots i_m]} f = D_{x, [i_1 \dots i_m]} f \left(\underbrace{1: \dots: 1}_m \right) = D_x f(e_{i_1}: \dots: e_{i_m})$$

and for $\{h_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}^n$ that

$$D_x^{(m)} f(h_1, \dots, h_m) = D_x^{[m]} f(h_1: \dots: h_m) = \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=1}^m (h_j)_{i_j} \right) \cdot \partial_{x, [i_1 \dots i_m]} f$$

where $\{e_i\}_{i \in \{1, \dots, n\}}$ is the canonical basis on $\mathbb{R}^n$ defined by $(e_i)_j = \delta_{i,j}$.

Proof. Let $i \in \{1, \dots, n\}^m$. Using [theorem: 16.276] it follows that $D_{x, i_m \dots i_1}^{[m]} f$ exist and for $h \in \mathbb{R}^m$

$$D_{x, [i_1 \dots i_m]} f(h_1: \dots: h_m) = D_x^{[m]} f(I^{[i_1, 0]}(h_1): \dots: I^{[i_m, 0]}(h_m)) \quad (16.424)$$

Next by [theorem: 16.302] $\partial_{x, [i_1 \dots i_m]} f$ exist and

$$\partial_{x, [i_1 \dots i_m]} f = D_{x, [i_1 \dots i_m]} f \left(\underbrace{1: \dots: 1}_m \right) \quad (16.425)$$

Let $k \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$ then $(I^{[i_j, 0]}(1))_k \stackrel{=}{=}_{[\text{lemma: 16.71}]} \delta_{i_j, k} \cdot 1 = \delta_{i_j, k} = (e_{i_j})_k$ so that

$$I^{[i_j, 0]}(1) = e_{i_j} \quad (16.426)$$

$$\begin{aligned} \partial_{x.[i_1 \dots i_m]} f &\stackrel{[\text{eqs: 16.424, 16.425}]}{=} D_x^{[m]} f \left(\underbrace{I^{[i_1,0]}(1) : \dots : I^{[i_m,0]}(1)}_m \right) \\ &\stackrel{[\text{eq: 16.426}]}{=} D_x^{[m]} f(e_{i_1} : \dots : e_{i_m}) \end{aligned} \quad (16.427)$$

Finally we have

$$\begin{aligned} D_x^{[m]} f(h_1 : \dots : h_m) &\stackrel{[\text{lemma: 16.311}]}{=} \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=1}^m (h_j)_{i_j} \right) \cdot D_x^{[m]} f(e_{i_1} : \dots : e_{i_m}) \\ &\stackrel{[\text{eq: 16.427}]}{=} \sum_{i \in \{1, \dots, n\}^m} \left(\prod_{j=1}^m (h_j)_{i_j} \right) \cdot \partial_{x.[i_1 \dots i_m]} f \end{aligned}$$

□

16.6 Inverse Function Theorem

16.6.1 Prerequisites

The purpose of this section is to prove the inverse function theorem that gives sufficient conditions for a function between Banach spaces to be invertible. First we need to introduce some concepts needed to prove the Inverse Function Theorem. We start with the concept of topological vector spaces which are essential vector spaces where the sum and scalar products are continuous.

Definition 16.313. A topological vector space is a vector space $\langle X, +, \cdot \rangle$ over $\mathbb{K}$ together with a topology $\mathcal{T}$ on X such that

1. $+: X \times X \rightarrow X$ defined by $(x, y) \rightarrow +(x, y) = x + y$ is continuous
2. $\cdot: \mathbb{K} \times X \rightarrow X$ defined by $(\alpha, x) \rightarrow \cdot(\alpha, x) = \alpha \cdot x$ is continuous

Note 16.314. Here we use the product topology on $X \times X$ based on the topology $\mathcal{T}$ of X and the product topology on $\mathbb{K} \times X$ based on the absolute norm topology $\mathcal{T}_{||}$ of $\mathbb{K}$ and the topology $\mathcal{T}$ of X .

Example 16.315. If $\langle X, |||| \rangle$ is a normed space over $\mathbb{K}$ then $\langle X, \mathcal{T}_{|||} \rangle$ is a topological vector space.

Proof. This follows from [theorems: 14.163 and 14.179] □

Definition 16.316. Let X, Y be topological vector spaces then a **toplinear isomorphism** is a function $L: X \rightarrow Y$ satisfying

1. L is a linear isomorphism, meaning that
 - a. L is a bijection
 - b. $L(x + y) = L(x) + L(y)$
 - c. $L(\alpha \cdot x) = \alpha \cdot L(x)$
2. L and L^{-1} are continuous

In other words a toplinear isomorphism is a linear homeomorphism [see definition: 14.186].

Note 16.317. Using [theorems: 11.178, 2.74] L^{-1} is also a continuous bijective linear isomorphism. Hence $L^{-1} \in L(Y, X)$ if L is a toplinear isomorphism.

We have the following characterization of toplinear isomorphisms.

Theorem 16.318. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces then $\forall L \in L(X, Y)$ we have that L is a toplinear isomorphism if and only if L is a bijection and L^{-1} is continuous.*

Proof. This follows trivial from the definition of $L(X, Y)$ and a toplinear isomorphism. $\square$

We can use a consequence of the open mapping theorem [see corollary: 14.453] to formulate the following theorem.

Theorem 16.319. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be Banach spaces, $L: X \rightarrow Y$ a continuous isomorphism [in other words $L \in L(X, Y)$ and L is a bijection] then L is a toplinear isomorphism.*

Proof. By [corollary: 14.453] L is a homeomorphism so that L^{-1} is continuous, hence by [theorem: 16.318] L is a toplinear isomorphism. $\square$

We define now the set of toplinear isomorphisms on a Banach space

Definition 16.320. *If $\langle X, \|\cdot\| \rangle$ is a normed space then $\mathcal{GL}(X)$ is defined by*

$$\mathcal{GL}(X) = \{L \in L(X, X) | L \text{ is a toplinear isomorphism}\} \subseteq L(X, X)$$

The set of toplinear isomorphisms on a Banach space forms a group under composition as stated in the next theorem.

Theorem 16.321. *If $\langle X, \|\cdot\| \rangle$ is a normed space then $\mathcal{GL}(X)$ is a group with the composition $\circ$ as the operator and the identity function Id_X [see definition: 2.49] as neutral element and for every $L \in \mathcal{GL}(X)$ L^{-1} as inverse of L .*

Proof. If $L_1, L_2 \in \mathcal{GL}(X)$ then by definition $L_1, L_2 \in L(X, X)$ and L_1, L_2 are toplinear isomorphism. Hence by [theorem: 16.318] L_1, L_2 are bijections and L_1^{-1}, L_2^{-1} are continuous. As the composition of bijections is a bijection [see theorem: 2.78] $L_1 \circ L_2$ is bijective, further by [theorem: 14.218] we have $L_1 \circ L_2 \in L(X, X)$. As $(L_1 \circ L_2)^{-1} \stackrel{[\text{theorem: 2.78}]}{=} L_2^{-1} \circ L_1^{-1}$ we have by [theorem: 14.158] that $(L_1 \circ L_2)^{-1}$ is continuous. Hence $L_1 \circ L_2 \in \mathcal{GL}(X)$ so that

$$\circ: \mathcal{GL}(X) \times \mathcal{GL}(X) \rightarrow \mathcal{GL}(X)$$

is well defined. Next we have:

neutral element. As Id_X is a bijection [see theorem: 2.66], $\text{Id}_X \in L(X, X)$ [see theorem: 14.199] and $(\text{Id}_X)^{-1} = \text{Id}_X \Rightarrow (\text{Id}_X)^{-1}$ is continuous it follows that $\text{Id}_X \in \mathcal{GL}(X)$. Finally we have trivially that $\forall L \in L(X, X)$ $\text{Id}_X \circ L = L = L \circ \text{Id}_X$.

inverse element. If $L \in \mathcal{GL}(X)$ then L is a linear isomorphism and L^{-1} is continuous. As by [theorem: 11.178] L^{-1} is a linear isomorphism it follows that $L^{-1} \in L(X, X)$, further $(L^{-1})^{-1} = L$ so that $(L^{-1})^{-1}$ is continuous. proving that $L^{-1} \in \mathcal{GL}(X)$. Finally $L \circ L^{-1} = \text{Id}_X = L^{-1} \circ L$.

associativity. This is trivial as composition of function is associative. $\square$

We will need the following theorem about limits in $L(X, X)$ and composition.

Theorem 16.322. *Let $\langle X, \|\cdot\| \rangle$ be a normed space, $L \in L(X, X)$ and $\{K_n\}_{n \in \mathbb{N}} \subseteq L(X, X)$ a countable family such that $\lim_{n \rightarrow \infty} K_n$ exist. Then we have*

$$\lim_{n \rightarrow \infty} (L \circ K_n) \text{ exist and } \lim_{n \rightarrow \infty} (L \circ K_n) = L \circ \left(\lim_{n \rightarrow \infty} K_n \right)$$

$$\lim_{n \rightarrow \infty} (K_n \circ L) \text{ exist and } \lim_{n \rightarrow \infty} (K_n \circ L) = \left(\lim_{n \rightarrow \infty} K_n \right) \circ L$$

Note 16.323. [theorem: 14.218] ensures that $L \circ K_n, K_n \circ L \in L(X, X)$ for every $n \in \mathbb{N}$.

Proof. As $\lim_{n \rightarrow \infty} K_n$ exist we have for $K = \lim_{n \rightarrow \infty} K_n$ that $\forall \varepsilon \in \mathbb{R}^+$ there exist a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have

$$\|K_n - K\|_{L(X, X)} < \frac{\varepsilon}{1 + \|L\|_{L(X, X)}}$$

Further we have

$$L \circ K_n - L \circ K \stackrel{[\text{theorem: 14.238}]}{=} L \circ (K_n - K) \text{ and } K_n \circ L - K \circ L \stackrel{[\text{theorem: 14.238}]}{=} (K_n - K) \circ L \quad (16.428)$$

Next if $n \geq N$ then we have

$$\begin{aligned} \|L \circ K_n - L \circ K\|_{L(X, Y)} &\stackrel{[\text{eq: 16.428}]}{=} \|L \circ (K_n - K)\|_{L(X, Y)} \\ &\stackrel{[\text{theorem: 14.218}]}{\leq} \|L\|_{L(X, X)} \cdot \|K_n - K\|_{L(X, X)} \\ &\leq \|L\|_{L(X, X)} \cdot \frac{\varepsilon}{1 + \|L\|_{L(X, X)}} \\ &< \varepsilon \\ \|K_n \circ L - K \circ L\|_{L(X, X)} &\stackrel{[\text{eq: 16.428}]}{=} \|(K_n - K) \circ L\|_{L(X, X)} \\ &\stackrel{[\text{theorem: 14.218}]}{\leq} \|K_n - K\|_{L(X, X)} \cdot \|L\|_{L(X, X)} \\ &\leq \frac{\varepsilon}{1 + \|L\|_{L(X, X)}} \cdot \|L\|_{L(X, X)} \\ &< \varepsilon \end{aligned}$$

which proves that

$$\lim_{n \rightarrow \infty} (L \circ K_n) = L \circ K = L \circ \left(\lim_{n \rightarrow \infty} K_n \right)$$

and

$$\lim_{n \rightarrow \infty} (K_n \circ L) = K \circ L = \left(\lim_{n \rightarrow \infty} K_n \right) \circ L \quad \square$$

Definition 16.324. Let $\langle X, \|\cdot\| \rangle$ be a normed space and $H \in L(X, X)$, $n \in \mathbb{N}$ then $H^n: X \rightarrow X$ is defined by

$$H^n = \begin{cases} H & \text{if } n = 1 \\ H \circ H^{n-1} & \text{if } 1 < n \end{cases}$$

in other words

$$H^n = \underbrace{H \circ \dots \circ H}_n$$

Theorem 16.325. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $n \in \mathbb{N}$ and $H \in L(X, X)$ then we have:

1. $H^n \circ H = H^{n+1}$
2. $\forall \alpha \in \mathbb{K}$ we have $(\alpha \cdot H)^n = (\alpha)^n \cdot H^n$
3. $H^n \in L(X, X)$ and $\|H^n\|_{L(X, X)} \leq \|H\|_{L(X, X)}^n$
4. $\sum_{i=1}^n H^i \in L(X, X)$ and $\|\sum_{i=1}^n H^i\|_{L(X, X)} \leq \sum_{i=1}^n \|H\|_{L(X, X)}^i$

Proof.

1. We prove this by induction so let

$$S = \{n \in \mathbb{N} | H^n \circ H = H^{n+1}\}$$

then we have:

1 $\in S$. As $H^1 \circ H = H \circ H = H \circ H^1 = H^{1+1}$ it follows that $1 \in S$

$n \in S \Rightarrow n + 1 \in S$. We have

$$\begin{aligned}
 H^{n+1} \circ H &= (H \circ H^n) \circ H \\
 &= (H^1 \circ H^n) \circ H \\
 &\stackrel{\circ \text{ is associative}}{=} H^1 \circ (H^n \circ H) \\
 &\stackrel{n \in S}{=} H^1 \circ H^{n+1} \\
 &= H \circ H^{n+1} \\
 &= H^{(n+1)+1}
 \end{aligned}$$

proving that $n + 1 \in S$.

2. This is easily proved by induction, so let

$$S = \{n \in \mathbb{N} | (\alpha \cdot H)^n = \alpha^n \cdot H^n\}$$

then we have:

1 $\in S$. Let $x \in X$ then $(\alpha \cdot H)^1(x) = (\alpha \cdot H)(x) = \alpha \cdot H(x) = \alpha^1 \cdot H^1(x) = (\alpha \cdot H^1)(x)$ proving that $(\alpha \cdot H)^1 = \alpha^1 \cdot H^1$. Hence $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $x \in X$ then we have as $1 < n + 1$

$$\begin{aligned}
 (\alpha \cdot H)^{n+1}(x) &= ((\alpha \cdot H) \circ (\alpha \cdot H)^n)(x) \\
 &= (\alpha \cdot H)((\alpha \cdot H)^n(x)) \\
 &\stackrel{n \in S}{=} (\alpha \cdot H)(\alpha^n \cdot H^n(x)) \\
 &= (\alpha \cdot H)(\alpha^n \cdot H^n(x)) \\
 &= \alpha \cdot H(\alpha^n \cdot H^n(x)) \\
 &= \alpha \cdot \alpha^n \cdot H(H^n(x)) \\
 &= \alpha^{n+1} \cdot ((H \circ H^n)(x)) \\
 &= \alpha^{n+1} \cdot H^{n+1}(x) \\
 &= (\alpha^{n+1} \cdot H^{n+1})(x)
 \end{aligned}$$

proving that $(\alpha \cdot H)^{n+1} = \alpha^{n+1} \cdot H^{n+1}$. Hence

$$n + 1 \in S$$

3. We prove this by induction, so let

$$S = \{n \in \mathbb{N} | H^n \in L(X, X) \text{ and } \|H^n\|_{L(X, X)} \leq \|H\|_{L(X, X)}^n\}$$

then we have:

1 $\in S$. As $H^1 = H \in L(X, X)$ and $\|H^1\|_{L(X, X)} = \|H\|_{L(X, X)} = \|H\|_{L(X, X)}^1$ it follows that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. As $n \in S$ it follows that $H^n \in L(X, X)$ and $\|H^n\|_{L(X, X)} \leq \|H\|_{L(X, X)}^n$. So as $H \in L(X, X)$ it follows from [theorem: 14.218] that

$$H^{n+1} = H \circ H^n \in L(X, X)$$

and

$$\begin{aligned}
 \|H^{n+1}\| &= \|H \circ H^n\|_{L(X, X)} \\
 &\leq \|H\|_{L(X, X)} \cdot \|H^n\|_{L(X, X)} \\
 &\leq \|H\|_{L(X, X)} \cdot \|H\|_{L(X, X)}^n \\
 &\stackrel{n \in S}{=} \|H\|_{L(X, X)}^{n+1}
 \end{aligned}$$

which proves that

$$n + 1 \in S$$

4. Let $\beta \in \mathbb{K}$ and $x, y \in X$

$$\begin{aligned} \left(\sum_{i=1}^n H^i \right) (x + \beta \cdot y) &\stackrel{\text{def}}{=} \sum_{i=1}^n H^i(x + \beta \cdot y) \\ &\stackrel{(3)}{=} \sum_{i=1}^n (H^i(x) + \beta \cdot H^i(y)) \\ &= \sum_{i=1}^n H^i(x) + \beta \cdot \sum_{i=1}^n H^i(y) \\ &= \left(\sum_{i=1}^n H^i \right) (x) + \beta \cdot \left(\sum_{i=1}^n H^i \right) (y) \end{aligned}$$

proving that

$$\sum_{i=1}^n H^i \in \text{Hom}(X, X)$$

Further

$$\begin{aligned} \left\| \left(\sum_{i=1}^n H^i \right) (x) \right\| &= \left\| \sum_{i=1}^n H^i(x) \right\| \\ &\leq \sum_{i=1}^n \|H^i(x)\| \\ &\leq \sum_{i=1}^n (\|H^i\| \cdot \|x\|) \\ &\stackrel{(3)}{\leq} \sum_{i=1}^n (\|H\|_{L(X, X)}^i \cdot \|x\|) \\ &= \left(\sum_{i=1}^n \|H\|_{L(X, Y)}^i \right) \cdot \|x\| \end{aligned}$$

so that by [theorem: 14.201]

$$\sum_{i=1}^n H^i \in L(X, X)$$

and by [theorem: 14.212] that

$$\left\| \sum_{i=1}^n H^i \right\|_{L(X, X)} \leq \sum_{i=1}^n \|H\|_{L(X, Y)}^i$$

□

Lemma 16.326. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $k \in \mathbb{N}$ and $H \in L(X, X)$ such that $\sum_{i=k}^{\infty} H^i$ converges [using the norm $\|\cdot\|_{L(X, X)}$] then $\forall x \in X$ we have that $\sum_{i=k}^{\infty} H^i(x)$ converges and $(\sum_{i=k}^{\infty} H^i)(x) = \sum_{i=k}^{\infty} H^i(x)$.

Proof. Let $x \in X$ and $\varepsilon \in \mathbb{R}^+$ then there exist a $N \in \mathbb{N}$ such that for $n \geq N$ we have that

$$\left\| \sum_{i=k}^n H^i - \sum_{i=k}^{\infty} H^i \right\|_{L(X, X)} < \frac{\varepsilon}{1 + \|x\|}$$

then we have

$$\begin{aligned} \left\| \sum_{i=k}^n H^i(x) - \left(\sum_{i=k}^{\infty} H^i \right) (x) \right\| &= \left\| \left(\sum_{i=k}^n H^i - \sum_{i=k}^{\infty} H^i \right) (x) \right\| \\ &\leq \left\| \sum_{i=k}^n H^i - \sum_{i=k}^{\infty} H^i \right\|_{L(X, X)} \cdot \|x\| \\ &< \frac{\varepsilon}{1 + \|x\|} \cdot \|x\| \\ &< \varepsilon \end{aligned}$$

proving that $\lim_{n \rightarrow \infty} \sum_{i=k}^n H^i(x)$ converges and

$$\lim_{n \rightarrow \infty} \sum_{i=k}^n H^i(x) = \left(\sum_{i=k}^{\infty} H^i \right)(x)$$

so that

$$\sum_{i=k}^{\infty} H^i(x) = \left(\sum_{i=k}^{\infty} H^i \right)(x) \quad \square$$

Lemma 16.327. Let $\langle X, \|\cdot\| \rangle$ be a Banach space and $H \in B_{\|\cdot\|_{L(X,X)}}(0, 1)$ then

$$\sum_{i=1}^{\infty} H^i \text{ converges}$$

$$\text{Id}_X - H \in \mathcal{GL}(X)$$

$$(\text{Id}_X - H)^{-1} = \text{Id}_X + \sum_{i=1}^{\infty} H^i$$

Note 16.328. In this lemma and the rest of this subsection the neutral element of $L(X, X)$ is noted as 0. Context can be used to determine if 0 is the neutral element of $\mathbb{K}$, X or $L(X, X)$.

Proof. As $H \in B_{\|\cdot\|_{L(X,X)}}(0, 1)$ it follows that $\|H\|_{L(X,X)} < 1$ so that by [example: 14.412]

$$\sum_{i=1}^{\infty} \|H\|_{L(X,X)}^i \text{ exist and } \sum_{i=1}^{\infty} \|H\|_{L(X,X)}^i = \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}}$$

As X is a Banach space it follows from [theorem: 14.398] that $L(X, X)$ is a Banach space. Further for $i \in \mathbb{N}$ we have $\|H^i\|_{L(X,X)} \leq_{[\text{theorem: 16.325}]} \|H\|_{L(X,X)}^i$. So we can use dominant convergence [see 14.437] to prove that

$$\sum_{i=1}^{\infty} H^i \text{ exist, } \sum_{i=1}^{\infty} H^i \in L(X, X) \text{ and } \left\| \sum_{i=1}^{\infty} H^i \right\|_{L(X,X)} \leq \sum_{i=1}^{\infty} \|H\|_{L(X,X)}^i = \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} \quad (16.429)$$

From the above and the fact that $\text{Id}_X \in L(X, X)$ [see example: 14.215] it follows that

$$\text{Id}_X + \sum_{i=1}^{\infty} H^i \in L(X, X) \quad (16.430)$$

Further as $\sum_{i=1}^{\infty} H^i$ converges it follows from [theorem: 14.413] that

$$\lim_{n \rightarrow \infty} H^n = 0 \quad (16.431)$$

Next

$$\begin{aligned} \left\| \text{Id}_X + \sum_{i=1}^{\infty} H^i \right\|_{L(X,X)} &\leq \left\| \text{Id}_X \right\|_{L(X,X)} + \left\| \sum_{i=1}^{\infty} H^i \right\|_{L(X,X)} \\ &\stackrel{[\text{example: 14.215}]}{\leq} 1 + \left\| \sum_{i=1}^{\infty} H^i \right\|_{L(X,X)} \\ &\stackrel{[\text{eq: 16.429}]}{\leq} 1 + \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} \\ &= \frac{1 - \|H\|_{L(X,X)} + \|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} \\ &= \frac{1}{1 - \|H\|_{L(X,X)}} \end{aligned} \quad (16.432)$$

Next we prove by induction on n that

$$\forall n \in \mathbb{N} \text{ we have } (\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) = \text{Id}_X - H^{n+1} \quad (16.433)$$

Proof. Let

$$S = \left\{ n \in \mathbb{N} \mid (\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) = \text{Id}_X - H^{n+1} \right\}$$

then we have:

$1 \in S$. Let $x \in X$ then we have

$$\begin{aligned} \left((\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^1 H^i \right) \right)(x) &= (\text{Id}_X - H) \left(\left(\text{Id}_X + \sum_{i=1}^1 H^i \right)(x) \right) \\ &= (\text{Id}_X - H) \left(\text{Id}_X(x) + \left(\sum_{i=1}^1 H^i \right)(x) \right) \\ &= (\text{Id}_X - H)(x + H(x)) \\ &= \text{Id}_X(x + H(x)) - H(x + H(x)) \\ &= x + H(x) - H(x) - H(H(x)) \\ &= x - (H \circ H)(x) \\ &= x - H^2(x) \\ &= \text{Id}_X(x) - H^2(x) \\ &= (\text{Id}_X - H^2)(x) \end{aligned}$$

proving that

$$(\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^1 H^i \right) = \text{Id}_X - H^{(1+1)}$$

from which it follows that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $x \in X$ then

$$\begin{aligned} \left((\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^{n+1} H^i \right) \right)(x) &= \\ (\text{Id}_X - H) \left(\left(\text{Id}_X + \sum_{i=1}^{n+1} H^i \right)(x) \right) &= \\ (\text{Id}_X - H) \left(\text{Id}_X(x) + \left(\sum_{i=1}^{n+1} H^i \right)(x) \right) &= \\ (\text{Id}_X - H) \left(\text{Id}_X(x) + \sum_{i=1}^{n+1} H^i(x) \right) &= \\ (\text{Id}_X - H) \left(\text{Id}_X(x) + \sum_{i=1}^n H^i(x) + H^{n+1}(x) \right) &= \\ (\text{Id}_X - H) \left(\text{Id}_X(x) + \sum_{i=1}^n H^i(x) \right) + (\text{Id}_X - H)(H^{n+1}(x)) &= \\ (\text{Id}_X - H) \left(\left(\text{Id}_X + \sum_{i=1}^n H^i \right)(x) \right) + (\text{Id}_X - H)(H^{n+1}(x)) &= \\ \left((\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \right)(x) + \text{Id}_X(H^{n+1}(x)) - H(H^{n+1}(x)) &\stackrel{n \in S}{=} \\ (\text{Id}_X - H^{n+1})(x) + H^{n+1}(x) - (H \circ H^{n+1})(x) &= \\ \text{Id}_X(x) - H^{(n+1)+1}(x) &= \\ (\text{Id}_X - H^{(n+1)+1})(x) & \end{aligned}$$

proving that $(\text{Id}_X - H) \circ (\text{Id}_X + \sum_{i=1}^{n+1} H^i) = (\text{Id}_X - H^{(n+1)+1})$. Hence

$$n+1 \in S$$

Mathematical induction completes the proof of [eq: 16.433]. $\square$

Next using induction we prove that

$$\forall n \in \mathbb{N} \text{ we have } \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) = \text{Id}_X - H^{n+1} \quad (16.434)$$

Proof. Define

$$S = \left\{ n \in \mathbb{N} \mid \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) = \text{Id}_X - H^{n+1} \right\}$$

then we have:

$1 \in S$. Let $x \in X$ then

$$\begin{aligned} \left(\left(\text{Id}_X + \sum_{i=1}^1 H^i \right) \circ (\text{Id}_X - H) \right)(x) &= ((\text{Id}_X + H) \circ (\text{Id}_X - H))(x) \\ &= (\text{Id}_X + H)((\text{Id}_X - H)(x)) \\ &= (\text{Id}_X + H)(\text{Id}_X(x) - H(x)) \\ &= \text{Id}_X(\text{Id}_X(x) - H(x)) + H(\text{Id}_X(x) - H(x)) \\ &= \text{Id}_X(x) - H(x) + H(\text{Id}_X(x)) - H(H(x)) \\ &= \text{Id}_X(x) - H(x) + H(x) - (H \circ H)(x) \\ &= \text{Id}_X(x) - H^2(x) \\ &= (\text{Id}_X - H^2)(x) \end{aligned}$$

proving that $(\text{Id}_X + \sum_{i=1}^1 H^i) \circ (\text{Id}_X - H) = \text{Id}_X - H^{1+1}$. Hence $1 \in S$.

$n \in S \Rightarrow n+1$. Let $x \in X$ then

$$\begin{aligned} \left(\left(\text{Id}_X + \sum_{i=1}^{n+1} H^i \right) \circ (\text{Id}_X - H) \right)(x) &= \\ \left(\text{Id}_X + \sum_{i=1}^{n+1} H^i \right)((\text{Id}_X - H)(x)) &= \\ \left(\text{Id}_X + \sum_{i=1}^n H^i + H^{n+1} \right)((\text{Id}_X - H)(x)) &= \\ \left(\text{Id}_X + \sum_{i=1}^n H^i \right)((\text{Id}_X - H)(x)) + H^{n+1}((\text{Id}_X - H)(x)) &= \\ \left(\left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) \right)(x) + H^{n+1}(x - H(x)) &\stackrel{n \in S}{=} \\ (\text{Id}_X - H^{n+1})(x) + H^{n+1}(x - H(x)) &= \\ \text{Id}_X(x) - (H^{n+1} \circ H)(x) &\stackrel{[\text{theorem: 16.325}]}{=} \\ \text{Id}_X(x) - H^{(n+1)+1}(x) &= \\ (\text{Id}_X - H^{(n+1)+1})(x) & \end{aligned}$$

proving that $(\text{Id}_X + \sum_{i=1}^{n+1} H^i) \circ (\text{Id}_X - H) = \text{Id}_X - H^{(n+1)+1}$. Hence

$$n+1 \in S$$

Mathematical induction proves then [eq: 16.434]. $\square$

To summarize [eqs: 16.433, 16.434] we have $\forall n \in \mathbb{N}$

$$(\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) = \text{Id}_X - H^{n+1} = \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) \quad (16.435)$$

Let $\varepsilon \in \mathbb{R}^+$ then as $\lim_{n \rightarrow \infty} H^i = 0$ [see eq: 16.431] there exist a $N \in \mathbb{N}$ such that for $n > N$

$$\|H^n\|_{L(X, X)} < \varepsilon$$

then

$$\begin{aligned} \left\| (\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) - \text{Id}_X \right\|_{L(X, X)} &\stackrel{[\text{eq: 16.435}]}{=} \|\text{Id}_X - H^{n+1} - \text{Id}_X\|_{L(X, Y)} \\ &= \|H^{n+1}\|_{L(X, Y)} \\ &\stackrel{[n+1 > n \geq N]}{<} \varepsilon \end{aligned}$$

and

$$\begin{aligned} \left\| \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) - \text{Id}_X \right\|_{L(X, X)} &\stackrel{[\text{eq: 16.435}]}{=} \|\text{Id}_X - H^{n+1} - \text{Id}_X\|_{L(X, Y)} \\ &= \|H^{n+1}\|_{L(X, Y)} \\ &\stackrel{[n+1 > n \geq N]}{<} \varepsilon \end{aligned}$$

So that

$$\lim_{n \rightarrow \infty} \left(\left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) \right) = \text{Id}_X = \lim_{n \rightarrow \infty} \left((\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \right) \quad (16.436)$$

Next we have, as $\sum_{i=1}^{\infty} H^i$ exist [see eq: 16.429], by [theorem: 14.356] that $\lim_{n \rightarrow \infty} (\text{Id}_X + \sum_{i=1}^n H^i)$ exist. Hence we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\left(\text{Id}_X + \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) \right) &\stackrel{[\text{theorem: 16.322}]}{=} \left(\lim_{n \rightarrow \infty} \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \right) \circ (\text{Id}_X - H) \\ &\stackrel{[\text{theorem: 14.356}]}{=} \left(\text{Id}_X + \lim_{n \rightarrow \infty} \sum_{i=1}^n H^i \right) \circ (\text{Id}_X - H) \\ &= \left(\text{Id}_X + \sum_{i=1}^{\infty} H^i \right) \circ (\text{Id}_X - H) \end{aligned}$$

and

$$\begin{aligned} \lim_{n \rightarrow \infty} \left((\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \right) &\stackrel{[\text{theorem: 16.322}]}{=} (\text{Id}_X - H) \circ \left(\lim_{n \rightarrow \infty} \left(\text{Id}_X + \sum_{i=1}^n H^i \right) \right) \\ &\stackrel{[\text{theorem: 14.356}]}{=} (\text{Id}_X - H) \circ \left(\text{Id}_X + \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n H^i \right) \right) \\ &= (\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^{\infty} H^i \right) \end{aligned}$$

which combined with [eq: 16.436] proves that

$$(\text{Id}_X - H) \circ \left(\text{Id}_X + \sum_{i=1}^{\infty} H^i \right) = \text{Id}_X = \left(\text{Id}_X + \sum_{i=1}^{\infty} H^i \right) \circ (\text{Id}_X - H)$$

hence

$$\text{Id}_X - H \text{ is a bijection and } (\text{Id}_X - H)^{-1} = \text{Id}_X + \sum_{i=1}^{\infty} H^i \in L(X, X)$$

As $\text{Id}_X \in L(X, X)$, $H \in B_{\|L(X, X)\|}(0, 1) \subseteq L(X, X)$ so that $\text{Id}_X - H \in L(X, X)$ and X is a Banach space it follows from [theorem: 16.319] that $\text{Id}_X - H$ is a toplinear isomorphism, so that

$$\text{Id}_X - H \in \mathcal{GL}(X) \quad \square$$

Corollary 16.329. *Let $\langle X, \|\cdot\| \rangle$ be a Banach space, $H \in B_{\|L(X, X)\|}(0, 1)$ then $\sum_{i=1}^{\infty} (-1)^i \cdot H^i$ converges, $\text{Id}_X + H \in \mathcal{GL}(X)$ and $(\text{Id}_X + H)^{-1} = \text{Id}_X + \sum_{i=1}^{\infty} (-1)^i \cdot H^i$.*

Proof. If $H \in B_{\|L(X, X)\|}(0, 1)$ then $\| -H \|_{L(X, X)} = \| (-1) \cdot H \|_{L(X, X)} = \| H \|_{L(X, X)} < 1$ so that $-H \in B_{\|L(X, X)\|}(0, 1)$. Hence by the previous theorem [theorem: 16.327] we have that

$$\text{Id}_X + H = \text{Id}_X - (-H) \in \mathcal{GL}(X)$$

and

$$\sum_{i=1}^{\infty} (-1)^i \cdot H^i \stackrel{[\text{theorem: 16.325}]}{=} \sum_{i=1}^{\infty} (-H)^i \text{ exist}$$

and

$$(\text{Id}_X + H)^{-1} = (\text{Id}_X - (-H))^{-1} = \text{Id}_X + \sum_{i=1}^{\infty} (-H)^i \stackrel{[\text{theorem: 16.325}]}{=} \text{Id}_X + \sum_{i=1}^{\infty} (-1)^i \cdot H^i \quad \square$$

Theorem 16.330. *Let $\langle X, \|\cdot\| \rangle$ be a Banach space then we have:*

1. $\mathcal{GL}(X)$ is a open set in $L(X, X)$
2. The function

$$\tau: \mathcal{GL}(X) \rightarrow L(X, X) \text{ defined by } \tau(L) = L^{-1}$$

is Fréchet differentiable on $\mathcal{GL}(X)$ and $\forall L \in \mathcal{GL}(X)$ we have that

$$D_L \tau \in L(L(X, X), L(X, X)) \text{ is defined by } D_L \tau(H) = -L^{-1} \circ H \circ L$$

3. The function τ is of class C^∞ .

Proof.

1. Let $L \in \mathcal{GL}(X) \subseteq L(X, X)$ then for any $H \in L(X, X)$ we have

$$\begin{aligned} H &= L + (H - L) \\ &= L \circ \text{Id}_X + \text{Id}_X \circ (H - L) \\ &= L \circ \text{Id}_X + (L \circ L^{-1}) \circ (H - L) \\ &= L \circ \text{Id}_X + L \circ (L^{-1} \circ (H - L)) \\ &\stackrel{[\text{theorem: 14.238}]}{=} L \circ (\text{Id}_X + L^{-1} \circ (H - L)) \end{aligned} \quad (16.437)$$

Take $\delta_L = \frac{1}{\|L^{-1}\|_{L(X, X)} + 1} \in \mathbb{R}^+$ then if $\|H - L\|_{L(X, X)} < \delta_L$ we have that

$$\begin{aligned} \|L^{-1} \circ (H - L)\|_{L(X, X)} &\stackrel{[\text{theorem: 14.218}]}{\leq} \|L^{-1}\|_{L(X, X)} \cdot \|H - L\|_{L(X, X)} \\ &< \|L^{-1}\|_{L(X, X)} \cdot \delta_L \\ &= \frac{\|L^{-1}\|_{L(X, X)}}{\|L^{-1}\|_{L(X, X)} + 1} \\ &< 1 \end{aligned}$$

Hence $\|L^{-1} \circ (H - L)\|_{L(X, X)} \in B_{\|L(X, X)\|}(0, 1)$ so that by [corollary: 16.329]

$$\text{Id}_X + L^{-1} \circ (H - L) \in \mathcal{GL}(X)$$

and as $L \in \mathcal{GL}(X)$ we have by [theorem: 16.321] that $L \circ (\text{Id}_X + L^{-1} \circ (H - L)) \in \mathcal{GL}(X)$, so by [eq: 16.437] $H \in \mathcal{GL}(X)$. In other words if $H \in B_{\|L(X, X)\|}(L, \delta_L)$ then $H \in \mathcal{GL}(X)$ so that

$$\forall L \in \mathcal{GL}(X) \text{ we have } L \in B_{\|L(X, X)\|}(L, \delta_L) \subseteq \mathcal{GL}(X) \quad (16.438)$$

which proves that

$$\mathcal{GL}(X) \text{ is a open set}$$

2. If $H \in B_{\|L(X, X)\|}(\text{Id}_X, 1)$ then $\|H - \text{Id}_X\|_{L(X, X)} < 1$ hence $H - \text{Id}_X \in B_{\|L(X, X)\|}(0, 1)$ so that by [corollary: 16.329]

$$H = \text{Id}_X + (H - \text{Id}_X) \in \mathcal{GL}(X)$$

so that

$$B_{\|L(X, X)\|}(\text{Id}_X, 1) \subseteq \mathcal{GL}(X) \quad (16.439)$$

To prove differentiability of τ we will rely on [corollary: 16.62], hence the need for a ε -mapping [see definition: 16.60]. Let $H \in B_{\|L(X, X)\|}(0, 1)$ then by [corollary: 16.329] $\sum_{i=1}^{\infty} (-1)^i \cdot H^i$ converges, so applying [theorem: 14.416] we have that

$$\sum_{i=2}^{\infty} (-1)^i \cdot H^i \in L(X, X) \text{ converges and } \sum_{i=1}^{\infty} (-1)^i H^i = -H + \sum_{i=2}^{\infty} (-1)^i \cdot H^i \quad (16.440)$$

As 0, $\sum_{i=2}^{\infty} (-1)^i \cdot H^i \in L(X, X)$ it follows that the following function is well defined

$$\varepsilon: B_{\|L(X, X)\|}(0, 1) \rightarrow L(X, X) \text{ by } \varepsilon(H) = \begin{cases} 0 & \text{if } H = 0 \\ \frac{1}{\|H\|_{L(X, X)}} \cdot \sum_{i=2}^{\infty} (-1)^i \cdot H^i & \text{if } H \neq 0 \end{cases} \quad (16.441)$$

Let $x \in X$ then we have

$$\begin{aligned} \|(\varepsilon(H))(x)\| &= \begin{cases} \|0(x)\| & \text{if } H = 0 \\ \left\| \frac{1}{\|H\|_{L(X, X)}} \cdot \left(\sum_{i=2}^{\infty} (-1)^i \cdot H^i \right)(x) \right\| & \text{if } H \neq 0 \end{cases} \\ &= \begin{cases} 0 & \text{if } H = 0 \\ \frac{1}{\|H\|_{L(X, X)}} \cdot \left\| \left(\sum_{i=2}^{\infty} (-1)^i \cdot H^i \right)(x) \right\| & \text{if } H \neq 0 \end{cases} \\ &\leq \begin{cases} 0 \cdot \|x\| & \text{if } H = 0 \\ \frac{1}{\|H\|_{L(X, X)}} \cdot \left\| \sum_{i=2}^{\infty} (-1)^i \cdot H^i \right\|_{L(X, X)} \cdot \|x\| & \text{if } H \neq 0 \end{cases} \\ &= \|x\| \cdot \begin{cases} 0 & \text{if } H = 0 \\ \frac{1}{\|H\|_{L(X, X)}} \cdot \left\| \sum_{i=2}^{\infty} (-1)^i \cdot H^i \right\|_{L(X, X)} & \text{if } H \neq 0 \end{cases} \end{aligned}$$

proving by [theorem: 14.212] that

$$\|\varepsilon(H)\|_{L(X, X)} \leq \begin{cases} 0 & \text{if } H = 0 \\ \frac{1}{\|H\|_{L(X, X)}} \cdot \left\| \sum_{i=2}^{\infty} (-1)^i \cdot H^i \right\|_{L(X, X)} & \text{if } H \neq 0 \end{cases} \quad (16.442)$$

For $H \in B_{\|_{L(X,X)}}(0, 1)$ we have $\|H\|_{L(X,X)} < 1$ so that by [example: 14.412] $\sum_{i=1}^{\infty} \|H\|_{L(X,X)}^i = \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}}$. Hence we have that

$$\begin{aligned}
 \sum_{i=2}^{\infty} \|H\|_{L(X,X)}^i & \stackrel{[\text{theorem: 14.416}]}{=} \left(\sum_{i=1}^{\infty} \|H\|_{L(X,X)}^i \right) - \|H\|_{L(X,X)} \\
 & = \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} - \|H\|_{L(X,X)} \\
 & = \frac{\|H\|_{L(X,X)} - \|H\|_{L(X,X)} \cdot (1 - \|H\|_{L(X,X)})}{1 - \|H\|_{L(X,X)}} \\
 & = \frac{\|H\|_{L(X,X)}^2}{1 - \|H\|_{L(X,X)}} \tag{16.443}
 \end{aligned}$$

As by [theorem: 16.325] we have $\|(-1)^i H^i\|_{L(X,X)} = \|H^i\|_{L(X,X)} \leq \|H\|_{L(X,X)}^i$ we can apply the dominant convergence theorem [theorem: 14.437] proving that $\sum_{i=2}^{\infty} (-1)^i \cdot H^i$ converges and $\|\sum_{i=2}^{\infty} (-1)^i \cdot H^i\|_{L(X,X)} \leq \frac{\|H\|_{L(X,X)}^2}{1 - \|H\|_{L(X,X)}}$. Substituting this result in [eq: 16.442] gives

$$\begin{aligned}
 \|\varepsilon(H)\|_{L(X,X)} & \leq \begin{cases} 0 & \text{if } H = 0 \\ \frac{1}{\|H\|_{L(X,X)}} \cdot \frac{\|H\|_{L(X,X)}^2}{1 - \|H\|_{L(X,X)}} & \text{if } H \neq 0 \end{cases} \\
 & = \begin{cases} 0 & \text{if } H = 0 \\ \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} & \text{if } H \neq 0 \end{cases}
 \end{aligned}$$

so that

$$\forall H \in B_{\|_{L(X,X)}}(0, 1) \text{ we have } \|\varepsilon(H)\|_{L(X,X)} \leq \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} \tag{16.444}$$

Let $\zeta \in \mathbb{R}^+$ then if $\|H\|_{L(X,X)} < \min\left(\frac{1}{2}, \frac{\zeta}{2}\right) < 1$ it follows that

$$\begin{aligned}
 \|\varepsilon(H) - \varepsilon(0)\|_{L(X,X)} & = \|\varepsilon(H)\|_{L(X,X)} \\
 & \stackrel{[\text{eq: 16.444}]}{\leq} \frac{\|H\|_{L(X,X)}}{1 - \|H\|_{L(X,X)}} \\
 & \leq \frac{\|H\|_{L(X,X)}}{1 - \frac{1}{2}} \\
 & < \frac{\zeta}{2} \cdot \frac{2}{1} \\
 & = \zeta
 \end{aligned}$$

which proves that

$$\varepsilon \text{ is continuous at } 0 \tag{16.445}$$

or using [definition: 16.60]

$$\varepsilon \text{ is a } \varepsilon\text{-mapping} \tag{16.446}$$

If $H \in B_{\|_{L(X,X)}}(0, 1)$ then

$$\begin{aligned}
 \tau(\text{Id}_X + H) - \tau(\text{Id}_X) & = (\text{Id}_X + H)^{-1} - (\text{Id}_X)^{-1} \\
 & = (\text{Id}_X + H)^{-1} - \text{Id}_X \\
 & \stackrel{[\text{corollary: 16.329}]}{=} \text{Id}_X + \sum_{i=1}^{\infty} (-1)^i \cdot H^i - \text{Id}_X \\
 & = \sum_{i=1}^{\infty} (-1)^i \cdot H^i \\
 & \stackrel{[\text{theorem: 14.416}]}{=} -H + \sum_{i=2}^{\infty} (-1)^i \cdot H^i
 \end{aligned}$$

As $\tau(\text{Id}_X + 0) - \tau(\text{Id}_X) - 0 = 0$ we have using the above that

$$\begin{aligned} \tau(\text{Id}_X + H) - \tau(\text{Id}_X) - (-H) &= \\ \begin{cases} 0 & \text{if } H = 0 \\ \sum_{i=2}^{\infty} (-1)^i \cdot H^i & \text{if } H \neq 0 \end{cases} &= \\ \begin{cases} \|H\|_{L(X,X)} \cdot 0 & \text{if } H = 0 \\ \|H\|_{L(X,X)} \cdot \left(\frac{1}{\|H\|_{L(X,X)}} \cdot \sum_{i=2}^{\infty} (-1)^i \cdot H^i \right) & \text{if } H \neq 0 \end{cases} & \text{[eq: 16.441]} \\ &= \|H\|_{L(X,X)} \cdot \varepsilon(H) \end{aligned}$$

so that

$$\forall H \in B_{\|_{L(X,X)}}(0, 1) \text{ we have } \tau(\text{Id}_X + H) - \tau(\text{Id}_X) - (-H) = \|H\|_{L(X,X)} \cdot \varepsilon(H) \quad (16.447)$$

Consider $L \in \mathcal{GL}(X)$ then we have the following cases to consider:

$L = \text{Id}_X$.

As $-\text{Id}_X \circ H \circ \text{Id}_X = -H$ we have that

$$\tau(\text{Id}_X + H) - \tau(\text{Id}_X) - (-\text{Id}_X \circ H \circ \text{Id}_X) \stackrel{\text{[eq: 16.447]}}{=} \|H\|_{L(X,X)} \cdot \varepsilon(H) \quad (16.448)$$

Note that $-\text{Id}_{L(L(X,X), L(X,X))} \in L(L(X, X), L(X, X))$ and

$$-\text{Id}_{L(L(X,X), L(X,X))}(H) = -H = -\text{Id}_X \circ H \circ \text{Id}_X$$

So that $\forall H \in B_{\|_{L(X,X)}}(0, 1)$ we have

$$\tau(\text{Id}_X + H) - \tau(\text{Id}_X) - (-\text{Id}_{L(L(X,X), L(X,X))})(H) = \|H\|_{L(X,X)} \cdot \varepsilon(H) \quad (16.449)$$

Using the above, the fact that ε is a ε -mapping [see eq: 16.446] and $B_{\|_{L(X,X)}}(\text{Id}_X, 1) \subseteq \mathcal{GL}(X)$ [see eq: 16.439] we can apply [corollary: 16.62] resulting in

$$D_{\text{Id}_X} \tau \text{ exist and } D_{\text{Id}_X} \tau = -\text{Id}_{L(L(X,X), L(X,X))}$$

where $D_{\text{Id}_X} \tau \in L(L(X, X), L(X, X))$ is defined by

$$D_{\text{Id}_X} \tau(H) = -\text{Id}_{L(L(X,X), L(X,X))}(H) = -H = -\text{Id}_X \circ H \circ \text{Id}_X$$

$L \neq \text{Id}_X$. Take $\delta = \min\left(\frac{1}{1 + \|L^{-1}\|_{L(X,X)}}, 1\right)$ then if $H \in B_{\|_{L(X,X)}}(0, \delta)$ we have

$$\begin{aligned} \|H \circ L^{-1}\|_{L(X,X)} &\stackrel{\text{[theorem: 14.218]}}{\leq} \|H\|_{L(X,X)} \cdot \|L^{-1}\|_{L(X,X)} \\ &\leq \delta \cdot \|L^{-1}\|_{L(X,X)} \\ &\leq \frac{1}{1 + \|L^{-1}\|_{L(X,X)}} \cdot \|L^{-1}\|_{L(X,X)} \\ &< 1 \end{aligned}$$

Hence using [corollary: 16.329] we conclude that

$$\forall H \in B_{\|_{L(X,X)}}(0, \delta) \quad \text{Id}_X + H \circ L^{-1} \in \mathcal{GL}(X) \quad (16.450)$$

As $L \in \mathcal{GL}(X)$ we have that $(\text{Id}_X + H \circ L^{-1}) \circ L \in \mathcal{GL}(X)$ [see theorem: 16.321], further we have that

$$(\text{Id}_X + H \circ L^{-1}) \circ L \stackrel{\text{[theorem: 14.238]}}{=} \text{Id}_X \circ L + (H \circ L^{-1}) \circ L = \text{Id}_X \circ L + H = L + H$$

hence it follows that

$$\forall H \in B_{\|_{L(X,X)}}(0, \delta) \quad L + H = (\text{Id}_X + H \circ L^{-1}) \circ L \in \mathcal{GL}(X) \quad (16.451)$$

If $K \in B_{\|L(X,X)\|}(L, \delta)$ then $\|K - L\|_{L(X,X)} < \delta$ so that $L - K \in B_{\|L(X,X)\|}(0, \delta)$ and by [eq: 16.451] that $L = (L - K) + K \in \mathcal{GL}(X)$, proving that

$$B_{\|L(X,X)\|}(L, \delta) \subseteq \mathcal{GL}(X) \quad (16.452)$$

If $H \in B_{\|L(X,X)\|}(0, \delta)$ then we have

$$\begin{aligned} \|H \circ L^{-1}\|_{L(X,X)} &\stackrel{[\text{theorem: 14.218}]}{\leq} \|H\|_{L(X,X)} \cdot \|L^{-1}\|_{L(X,X)} \\ &\leq \delta \cdot \|L^{-1}\|_{L(X,X)} \\ &\leq \frac{1}{1 + \|L^{-1}\|_{L(X,X)}} \cdot \|L^{-1}\|_{L(X,X)} \\ &< 1 \end{aligned}$$

so that

$$H \circ L^{-1} \in B_{\|L(X,X)\|}(0, 1)$$

proving, as $\text{dom}(\varepsilon) \stackrel{[\text{eq: 16.441}]}{=} B_{\|L(X,X)\|}(0, 1)$, that

$$\forall H \in B_{\|L(X,X)\|}(0, \delta) \text{ we have that } \varepsilon(H \circ L^{-1}) \text{ is defined}$$

and by inserting $H \circ L^{-1}$ for H in [eq: 16.447]

$$\tau(\text{Id}_X + H \circ L^{-1}) - \tau(\text{Id}_X) = -H \circ L^{-1} + \|H \circ L^{-1}\|_{L(X,X)} \cdot \varepsilon(H \circ L^{-1}) \quad (16.453)$$

Hence

$$\begin{aligned} \tau(L + H) - \tau(L) &= \\ (L + H)^{-1} - L^{-1} &\stackrel{[\text{eq: 16.451}]}{=} \\ ((\text{Id}_X + H \circ L^{-1}) \circ L)^{-1} - L^{-1} &= \\ L^{-1} \circ (\text{Id}_X + H \circ L^{-1})^{-1} - L^{-1} &= \\ L^{-1} \circ (\text{Id}_X + H \circ L^{-1})^{-1} - L^{-1} \circ \text{Id}_X &\stackrel{[\text{theorem: 14.238}]}{=} \\ L^{-1} \circ ((\text{Id}_X + H \circ L^{-1})^{-1} - \text{Id}_X) &= \\ L^{-1} \circ (\tau(\text{Id}_X + H \circ L^{-1}) - \text{Id}_X) &= \\ L^{-1} \circ (\tau(\text{Id}_X + H \circ L^{-1}) - \text{Id}_X^{-1}) &= \\ L^{-1} \circ (\tau(\text{Id}_X + H \circ L^{-1}) - \tau(\text{Id}_X)) &\stackrel{[\text{eq: 16.453}]}{=} \\ L^{-1} \circ (-H \circ L^{-1} + \|H \circ L^{-1}\|_{L(X,X)} \cdot \varepsilon(H \circ L^{-1})) &\stackrel{[\text{theorem: 14.238}]}{=} \\ -L \circ H \circ L^{-1} + \|H \circ L^{-1}\|_{L(X,X)} \cdot L^{-1} \circ \varepsilon(H \circ L^{-1}) \end{aligned}$$

proving that

$$\tau(L + H) - \tau(H) = -L \circ H \circ L^{-1} + \|H \circ L^{-1}\|_{L(X,X)} \cdot L^{-1} \circ \varepsilon(H \circ L^{-1}) \quad (16.454)$$

As $0 \in L(X, X)$ and if $H \neq 0$ by [theorem: 14.218]

$$\frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot L^{-1} \circ \varepsilon(H \circ L^{-1}) \in L(X, X)$$

we can define the following function

$$\varepsilon': B_{\|L(X,X)\|}(0, \delta) \rightarrow L(X, X) \text{ by } \varepsilon'(H) = \begin{cases} 0 & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot L^{-1} \circ \varepsilon(H \circ L^{-1}) & \text{if } H \neq 0 \end{cases}$$

then $\forall H \in B_{\|L(X,X)\|}(0, \delta)$

$$\|H\|_{L(X,X)} \cdot \varepsilon'(H) \stackrel{=}{\varepsilon(0 \circ L^{-1}) = \varepsilon(0) = \varepsilon'(0)} \|H \circ L^{-1}\|_{L(X,X)} \cdot L^{-1} \circ \varepsilon(H \circ L^{-1}) \quad (16.455)$$

Further for $H \in B_{\|L(X,X)}(0, \delta)$ and $x \in X$ we have

$$\begin{aligned}
& \|\varepsilon'(H)(x)\| = \\
& \left\| \begin{cases} 0(x) & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot (L^{-1} \circ \varepsilon(H \circ L^{-1}))(x) & \text{if } H \neq 0 \end{cases} \right\| = \\
& \left\| \begin{cases} 0 & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot (L^{-1} \circ \varepsilon(H \circ L^{-1}))(x) & \text{if } H \neq 0 \end{cases} \right\| = \\
& \begin{cases} \|x\| \cdot 0 & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot \|(L^{-1} \circ \varepsilon(H \circ L^{-1}))(x)\| & \text{if } H \neq 0 \end{cases} \leq \\
& \begin{cases} \|x\| \cdot 0 & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot \|(L^{-1} \circ \varepsilon(H \circ L^{-1}))\|_{L(X,X)} \cdot \|x\| & \text{if } H \neq 0 \end{cases} = \\
& \|x\| \cdot \begin{cases} 0 & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot \|(L^{-1} \circ \varepsilon(H \circ L^{-1}))\|_{L(X,X)} & \text{if } H \neq 0 \end{cases} \stackrel{[\text{theorem 14.218}]}{\leq} \\
& \|x\| \cdot \begin{cases} 0 & \text{if } H = 0 \\ \frac{\|H \circ L^{-1}\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot \|L^{-1}\|_{L(X,X)} \cdot \|\varepsilon(H \circ L^{-1})\|_{L(X,X)} & \text{if } H \neq 0 \end{cases} \stackrel{[\text{theorem 14.218}]}{\leq} \\
& \|x\| \cdot \begin{cases} 0 & \text{if } H = 0 \\ \frac{\|H\|_{L(X,X)}}{\|H\|_{L(X,X)}} \cdot \|L^{-1}\|_{L(X,X)}^2 \cdot \|\varepsilon(H \circ L^{-1})\|_{L(X,X)} & \text{if } H \neq 0 \end{cases} = \\
& \|x\| \cdot \begin{cases} 0 & \text{if } H = 0 \\ \|L^{-1}\|_{L(X,X)}^2 \cdot \|\varepsilon(H \circ L^{-1})\|_{L(X,X)} & \text{if } H \neq 0 \end{cases} \leq \\
& \|x\| \cdot \|L^{-1}\|_{L(X,X)}^2 \cdot \|\varepsilon(H \circ L^{-1})\|_{L(X,X)}
\end{aligned}$$

proving by [theorem: 14.212] that

$$\|\varepsilon'(H)\|_{L(X,X)} \leq \|L^{-1}\|_{L(X,X)}^2 \cdot \|\varepsilon(H \circ L^{-1})\|_{L(X,X)} \quad (16.456)$$

Let $\zeta \in \mathbb{R}^+$ then as ε is continuous at 0 [see 16.445] there exist a $\delta_1 \in \mathbb{R}^+$ such that if $\|H - 0\|_{L(X,X)} = \|H\|_{L(X,X)} < \delta_1$ then

$$\|\varepsilon(H)\|_{L(X,X)} \stackrel{\varepsilon(0)=0}{\leq} \|\varepsilon(H) - \varepsilon(0)\|_{L(X,X)} < \frac{\zeta}{\|L^{-1}\|_{L(X,X)}^2 + 1}$$

Take $\delta_2 = \frac{\delta_1}{\|L^{-1}\|_{L(X,X)} + 1}$ then if $\|H\|_{L(X,X)} < \delta_2$ we have

$$\begin{aligned}
\|H \circ L^{-1}\|_{L(X,X)} & \stackrel{[\text{theorem: 14.218}]}{\leq} \|H\|_{L(X,X)} \cdot \|L^{-1}\|_{L(X,X)} \\
& \leq \frac{\delta_1}{\|L^{-1}\|_{L(X,X)} + 1} \cdot \|L^{-1}\|_{L(X,X)} \\
& < \delta_1
\end{aligned}$$

so that

$$\|\varepsilon(H \circ L^{-1})\|_{L(X,X)} < \frac{\zeta}{\|L^{-1}\|_{L(X,X)}^2 + 1}$$

Substituting the above in [eq: 16.456] we have for $\|H\|_{L(X,Y)} < \min(\delta, \delta_2)$ that

$$\|\varepsilon'(H) - \varepsilon'(0)\| = \|\varepsilon'(H)\|_{L(X,X)} \leq \|L^{-1}\|_{L(X,X)}^2 \cdot \frac{\zeta}{\|L^{-1}\|_{L(X,X)}^2 + 1} < \zeta$$

which proves that ε' is continuous at 0. Hence

$$\varepsilon' \text{ is a } \varepsilon\text{-mapping at } 0 \quad (16.457)$$

Define now

$$\psi: L(X, X) \rightarrow L(X, X) \text{ by } \psi(H) = -L^{-1} \circ H \circ L$$

Then for $H_1, H_2 \in L(X, X)$ and $\alpha \in \mathbb{K}$ we have

$$\begin{aligned} \psi(H_1 + \alpha \cdot H_2) &= -L^{-1} \circ (H_1 + \alpha \cdot H_2) \circ L \\ &\stackrel{\text{associativity}}{=} -L^{-1} \circ ((H_1 + \alpha \cdot H_2) \circ L) \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L^{-1} \circ (H_1 \circ L + \alpha \cdot H_2 \circ L) \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L^{-1} \circ H_1 \circ L + \alpha \cdot (-L^{-1} \circ H_2 \circ L) \\ &= \psi(H_1) + \alpha \cdot \psi(H_2) \end{aligned}$$

proving that

$$\psi \in \text{Hom}(L(X, X), L(X, X))$$

Further for $H \in L(X, X)$ we have

$$\begin{aligned} \|\psi(H)\|_{L(X,X)} &= \|-L \circ H \circ L\|_{L(X,X)} \\ &\stackrel{[\text{theorem: 14.218}]}{\leq} \|-L \circ H\|_{L(X,X)} \cdot \|L\|_{L(X,X)} \\ &\leq \|L\|_{L(X,X)} \cdot \|H\|_{L(X,X)} \cdot \|L\|_{L(X,X)} \\ &= \|L\|_{L(X,X)}^2 \cdot \|H\|_{L(X,X)} \end{aligned}$$

proving by [theorem: 14.212] that

$$\psi \in L(L(X, X), L(X, X)) \text{ and } \|\psi\|_{L(L(X,X), L(X,X))} \leq \|L\|_{L(X,X)}^2 \quad (16.458)$$

Next

$$\begin{aligned} \tau(L + H) - \tau(L) - \psi(H) &= \tau(L + H) - \tau(L) - (-L \circ H \circ L) \\ &\stackrel{[\text{eq: 16.454}]}{=} \|H \circ L^{-1}\|_{L(X,X)} \cdot L^{-1} \circ \varepsilon(H \circ L^{-1}) \\ &\stackrel{[\text{eq: 16.455}]}{=} \|H\|_{L(X,X)} \cdot \varepsilon'(H) \end{aligned}$$

Finally using [eqs: 16.452, 16.457 and the above allows us to] apply [corollary: 16.62] proving that τ is Fréchet differentiable at L and

$$D_L \tau = \psi$$

so that $D_L \tau \in L(L(X, X), L(X, X))$ is defined by $D_L \tau(H) = \psi(H) = -L \circ H \circ L$.

So in all cases we have that $\forall L \in \mathcal{GL}(X)$ τ is Fréchet differentiable at L and $D_L \tau \in L(L(X, X), L(X, X))$ is defined by $D_L \tau(H) = -L \circ H \circ L$.

3. Using (2) we have that $\forall L \in \mathcal{GL}(X)$ τ has a Fréchet differential $D_L \tau$ at L where $D_L \tau \in L(L(X, X), L(X, X))$ is defined by $D_L \tau(H) = -L \circ H \circ L$. Hence we have $\mathfrak{D}_\tau = \mathcal{GL}(X)$ and for

$$D^{[1]} \tau: \mathfrak{D}_\tau = \mathcal{GL}(X) \rightarrow L(L(X, X), L(X, X)) \text{ defined by } D^{[1]} \tau(L) = D_L \tau$$

that

$$D_L \tau \text{ is defined by } D_L(\tau)(H) = -L \circ H \circ L \quad (16.459)$$

We proceed now to show that $D^{[1]} \tau$ is the composition of C^∞ mappings so that we can apply the chain rule to prove that $D^{[1]} \tau$ is C^∞ on $\mathcal{GL}(X)$. Define

$$\chi: \mathcal{GL}(X) \rightarrow \mathcal{GL}(X) \cdot \mathcal{GL}(X) \text{ by } \chi(L) = (L, L)$$

then $\forall L \in \mathcal{GL}(X)$ we have

$$(\pi_1 \circ \chi)(L) = \pi_1(\chi(L)) = \pi_1((L, L)) = L = \text{Id}_{\mathcal{GL}(X)}(L)$$

and

$$(\pi_2 \circ \chi)(L) = \pi_2(\chi(L)) = \pi_2((L, L)) = L = \text{Id}_{\mathcal{GL}(X)}(L)$$

so that $\pi_1 \circ \chi = \text{Id}_{\mathcal{GL}(X)}$, $\pi_2 \circ \chi = \text{Id}_{\mathcal{GL}(X)}$. As $\text{Id}_{\mathcal{GL}(X)}$ is C^∞ on $\mathcal{GL}(X)$ [see theorem: 16.179], it follows from [theorem: 16.101] that

$$\mathcal{X} \text{ is } C^\infty \text{ on } \mathcal{GL}(X) \quad (16.460)$$

Given $L, K \in L(X, X)$ define

$$\varphi_{L,K}: L(X, X) \rightarrow L(X, X) \text{ by } \varphi_{L,K}(H) = -L \circ H \circ K \quad (16.461)$$

Let $H_1, H_2 \in L(X, X)$, $\alpha \in \mathbb{K}$ then we have

$$\begin{aligned} \varphi_{L,K}(H_1 + \alpha \cdot H_2) &= -L \circ (H_1 + \alpha \cdot H_2) \circ K \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L \circ (H_1 \circ K + \alpha \cdot H_2 \circ K) \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L \circ H_1 \circ K + \alpha \cdot (-L \circ H_2 \circ K) \\ &= \varphi_{L,K}(H_1) + \alpha \cdot \varphi_{L,K}(H_2) \end{aligned}$$

proving that

$$\varphi_{L,K} \in \text{Hom}(L(X, X), L(X, X))$$

Further if $H \in L(X, X)$ then

$$\begin{aligned} \|\varphi_{L,K}(H)\|_{L(X,X)} &= \|-L \circ H \circ K\|_{L(X,X)} \\ &= \|-L \circ (H \circ K)\|_{L(X,X)} \\ &\leq \|L\|_{L(X \times X)} \cdot \|H \circ K\|_{L(X,X)} \\ &\stackrel{[\text{theorem: 14.218}]}{\leq} \|L\|_{L(X \times X)} \cdot \|H\|_{L(X,X)} \cdot \|K\|_{L(X,X)} \\ &\stackrel{[\text{theorem: 14.218}]}{=} (\|L\|_{L(X,X)} \cdot \|K\|_{L(X,X)}) \cdot \|H\|_{L(X,X)} \end{aligned}$$

hence, using [theorems: 14.201 and 14.212] it follows that

$$\varphi_{L,K} \in L(L(X, X), L(X, X)) \wedge \|\varphi_{L,K}\|_{L(L(X,X), L(X,X))} \leq \|L\|_{L(X,X)} \cdot \|K\|_{L(X,X)} \quad (16.462)$$

So we can define

$$\varphi: L(X, X) \cdot L(X, X) \rightarrow L(L(X, X), L(X, X)) \text{ by } \varphi(L, K) = \varphi_{L,K} \quad (16.463)$$

Then we have for $L, L_1, L_2 \in L(X, X)$ and $\alpha \in \mathbb{K}$ that

$$\begin{aligned} \varphi(L, L_1 + \alpha \cdot L_2)(H) &\stackrel{[\text{eq: 16.463}]}{=} \varphi_{L, L_1 + \alpha \cdot L_2}(H) \\ &\stackrel{[\text{eq: 16.461}]}{=} -L \circ H \circ (L_1 + \alpha \cdot L_2) \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L \circ (H \circ L_1 + \alpha \cdot H \circ L_2) \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L \circ H \circ L_1 + \alpha \cdot (-L \circ H \circ L_2) \\ &= \varphi(L, L_1)(H) + \alpha \cdot \varphi(L, L_2)(H) \\ &= (\varphi(L, L_1) + \alpha \cdot \varphi(L, L_2))(H) \\ \varphi(L_1 + \alpha \cdot L_2, L)(H) &= \varphi_{L_1 + \alpha \cdot L_2, L}(H) \\ &\stackrel{[\text{eq: 16.461}]}{=} -(L_1 + \alpha \cdot L_2) \circ H \circ L \\ &= -(L_1 + \alpha \cdot L_2) \circ H \circ L \\ &\stackrel{[\text{theorem: 14.238}]}{=} (-L_1 \circ H + \alpha \cdot (-L_2 \circ H)) \circ L \\ &\stackrel{[\text{theorem: 14.238}]}{=} -L_1 \circ H \circ L + \alpha \cdot (-L_2 \circ H \circ L) \\ &= \varphi(L_1, L)(H) + \alpha \cdot \varphi(L_2, L)(H) \\ &= (\varphi(L_1, L) + \alpha \cdot \varphi(L_2, L))(H) \end{aligned}$$

which proves that

$$\varphi(L, L_1 + \alpha \cdot L_2) = \varphi(L_1, L) + \alpha \cdot \varphi(L_2, L) \text{ and } \varphi(L_1 + \alpha \cdot L_2, L) = \varphi(L_1, L) + \alpha \cdot \varphi(L_2, L)$$

Hence

$$\varphi \in \text{Hom}(L(X, X), L(X, X); L(L(X, X), L(X, X)))$$

Further if $(L, K) \in L(X, X) \cdot L(X, X)$ then we have

$$\begin{aligned} \|\varphi(L, K)\|_{L(L(X, X), L(X, X))} &\stackrel{[\text{eq: 16.463}]}{=} \|\varphi_{L, K}\|_{L(L(X, X), L(X, X))} \\ &\stackrel{[\text{eq: 16.462}]}{\leq} \|L\|_{L(X, X)} \cdot \|K\|_{L(X, X)} \end{aligned}$$

which by [theorem: 14.228] proves that

$$\varphi \in L(L(X, X), L(X, X); L(L(X, X), L(X, X)))$$

Hence by [theorem: 16.186] we have that

$$\varphi \text{ is } C^\infty \text{ on } L(X, X) \cdot L(X, X) \quad (16.464)$$

Take $L \in \mathcal{GL}(X)$ then for $H \in L(X, X)$ we have

$$\begin{aligned} ((\varphi \circ \chi)(L))(H) &= (\varphi(\chi(L)))(H) \\ &= (\varphi(L, L))(H) \\ &\stackrel{[\text{eq: 16.463}]}{=} \varphi_{L, L}(H) \\ &\stackrel{[\text{eq: 16.461}]}{=} -L \circ H \circ L \\ &\stackrel{[\text{eq: 16.459}]}{=} D_L \tau(H) \end{aligned}$$

proving that $\forall L \in \mathcal{GL}(X) (\varphi \circ \chi)(L) = D_L \tau = D^{[1]} \tau(L)$. Hence we have

$$D^{[1]} \tau = \varphi \circ \chi \quad (16.465)$$

As φ and χ are C^∞ [see eqs: 16.460, 16.464] it follows from [theorem: 16.196] that $\varphi \circ \chi$ is C^∞ on $\mathcal{GL}(X)$ so that

$$D^{[1]} \tau \text{ is } C^\infty \text{ on } \mathcal{GL}(X)$$

Finally using [theorem: 16.175] it follows that

$$\tau \text{ is } C^\infty \text{ on } \mathcal{GL}(X) \quad \square$$

We introduce now the concept of differentiable functions who have a differentiable inverse.

Definition 16.331. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , V a open set in Y and $f: U \rightarrow V$ a function then we say that

1. f is a **diffeomorphism** iff
 - a. $f: U \rightarrow V$ is a bijection
 - b. $f: U \rightarrow Y$ is differentiable on U
 - c. $f^{-1}: V \rightarrow X$ is differentiable on V
2. f is a **diffeomorphism of class C^n** where $n \in \mathbb{N}_0 \cup \{\infty\}$ iff
 - a. $f: U \rightarrow V$ is a bijection
 - b. $f: U \rightarrow Y$ is C^n on U
 - c. $f^{-1}: V \rightarrow X$ is C^n on V

Theorem 16.332. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , V a open set in Y and $f: U \rightarrow V$ a function then we have:

1.

f is a diffeomorphism $\Leftrightarrow f^{-1}$ is a diffeomorphism

2. Let $n \in \mathbb{N}_0 \cup \{\infty\}$

f is a diffeomorphism of class $C^n \Leftrightarrow f^{-1}$ is a diffeomorphism of class C^n

Proof.

1.

$\Rightarrow$. As f is a diffeomorphism we have that:

- a. $f: U \rightarrow V$ is a bijection so that by [theorem: 2.74] $f^{-1}: V \rightarrow U$ is a bijection.
- b. $f^{-1}: V \rightarrow X$ is differentiable on V
- c. $(f^{-1})^{-1} \underset{[\text{theorem: 2.75}]}{=} f: U \rightarrow Y$ is differentiable on V

so that by definition f^{-1} is diffeomorphism.

$\Leftarrow$. As f^{-1} is a diffeomorphism we have that:

- a. $f^{-1}: V \rightarrow U$ is a bijection so that by [theorem: 2.74]

$$f \underset{[\text{theorem: 2.75}]}{=} (f^{-1})^{-1}: U \rightarrow V$$

is a bijection.

- b. Then $f \underset{[\text{theorem: 2.75}]}{=} (f^{-1})^{-1}: U \rightarrow Y$ is differentiable on U .
- c. Then $f^{-1}: V \rightarrow X$ is differentiable on V .

so that definition f is diffeomorphism.

2.

$\Rightarrow$. As f is a diffeomorphism of class C^n we have that:

- a. $f: U \rightarrow V$ is a bijection so that by [theorem: 2.74] $f^{-1}: V \rightarrow U$ is a bijection.
- b. $f^{-1}: V \rightarrow X$ is diffeomorphism of class C^n .
- c. $(f^{-1})^{-1} \underset{[\text{theorem: 2.75}]}{=} f: U \rightarrow Y$ is diffeomorphism of class C^n .

so that by definition f^{-1} diffeomorphism of class C^n .

$\Leftarrow$. As f^{-1} is a diffeomorphism of class C^n we have that:

- a. $f^{-1}: V \rightarrow U$ is a bijection so that by [theorem: 2.74]

$$f \underset{[\text{theorem: 2.75}]}{=} (f^{-1})^{-1}: U \rightarrow V$$

is a bijection.

- b. Then $f \underset{[\text{theorem: 2.75}]}{=} (f^{-1})^{-1}: U \rightarrow Y$ is diffeomorphism of class C^n .
- c. Then $f^{-1}: V \rightarrow X$ is a diffeomorphism of class C^n .

so that definition f is a diffeomorphism of class C^n . □

Example 16.333. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces then a homeomorphism $f: X \rightarrow Y$ is a diffeomorphism of class C^0

Proof. This follows trivially from the definitions of a homeomorphism [definition: 14.186] and diffeomorphism of class C^0 [definition: 16.331]. □

Example 16.334. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be Banach spaces and $L: X \rightarrow Y$ a continuous linear isomorphism [in other words $L \in L(X, Y)$ and L is a bijection] then L is a diffeomorphism of class C^∞ .

Proof. First using [theorem: 14.453] L is a homeomorphism so that L, L^{-1} are continuous. Taking in account [theorem: 11.178] we have also that L^{-1} is linear. So $L \in L(X, Y)$ and $L^{-1} \in L(Y, X)$. Finally by [example: 16.180] it follows that L, L^{-1} are of class C^∞ proving that L is a diffeomorphism of class C^∞ . $\square$

Example 16.335. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X then the identity function $\text{Id}_U: U \rightarrow U$ is a diffeomorphism of class C^∞ .

Proof. Using [theorems: 2.66, 2.76] it follows that $\text{Id}_U: U \rightarrow U$ is a bijection with $(\text{Id}_U)^{-1} = \text{Id}_U$. Hence using [theorem: 16.181] it follows that $\text{Id}_U: U \rightarrow U$ and $(\text{Id}_U)^{-1}: U \rightarrow U$ are of class C^∞ . Hence we have by definition that $\text{Id}_U: U \rightarrow U$ is a diffeomorphism of class C . $\square$

The restriction of a diffeomorphism of class C^n to a open set is again a diffeomorphism of class C^n on this set as the following theorem shows.

Theorem 16.336. Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , V a open set in Y and $f: U \rightarrow V$ a diffeomorphism of class C^n then for every open set $W \subseteq U$ we have that

$$f|_W: W \rightarrow f(W)$$

is a diffeomorphism of class C^n .

Proof. Using [theorem: 2.98] it follows that

$$f|_W: W \rightarrow f(W) \text{ is a bijection}$$

As f is a diffeomorphism of class C^n $f: U \rightarrow V$ is of class C^n on U and $f^{-1}: V \rightarrow X$ is of class C^n on V . Using [theorem: 16.156] it follows that

$$f|_W: W \rightarrow Y \text{ is of class } C^n \text{ on } W$$

Further as $f^{-1}: V \rightarrow X$ is of class C^n on V it is of class C^0 on V [see theorem: 16.158] hence continuous. So $f(W) \stackrel{[\text{theorem: 2.70}]}{=} (f^{-1})^{-1}(W)$ is open in the subspace topology of V hence open in the topology of Y [see theorem: 14.11]. Applying further [theorem: 16.156] it follows that $(f^{-1})|_{f(W)}: f(W) \rightarrow X$ is of class C^n on $f(W)$. Finally by [theorem: 2.100] $(f^{-1})|_{f(W)} = (f|_W)^{-1}$ so that

$$(f|_W)^{-1}: f(W) \rightarrow X \text{ is of class } C^n \text{ on } W$$

Hence

$$f|_W: W \rightarrow f(W) \text{ is a diffeomorphism of class } C^n$$

$\square$

Theorem 16.337. Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , V a open set in Y and $f: U \rightarrow V$ a bijection then

$$f: U \rightarrow V \text{ is a diffeomorphism of class } C^n$$

$$\Updownarrow$$

$\forall x \in U \exists W$ open in X so that $x \in W$, $f(W)$ is open in Y and $f|_W: W \rightarrow f(W)$ is a diffeomorphism of class C^n

Proof.

$\Rightarrow$. Let $x \in U$ then if we take $W = U$ we have that $x \in W$, $f(W) = V$ open in Y and the function $f|_W: W \rightarrow f(W)$ is equal to the function $f: U \rightarrow V$ from which it follows that $f|_W: W \rightarrow f(W)$ is a diffeomorphism of class C^n .

$\Leftarrow$. Let $x \in U$ then there exist a W open in X such that $x \in W$, $f(W)$ is open in Y and $f|_W: W \rightarrow f(W)$ is a diffeomorphism of class C^m . Hence using [definition: 16.331] it follows that $f|_W: W \rightarrow Y$ is of class C^n . As this is true for all $x \in U$ it follows from [theorem: 16.159] that

$$f: U \rightarrow Y \text{ is of class } C^n \quad (16.466)$$

Let $y \in V$ then as f is a bijection there exist a $x \in U$ such that $y = f(x)$. Hence there exist a W open in X such that

$$x \in W, f(W) \text{ is open in } Y \text{ and } f|_W: W \rightarrow f(W) \text{ is a diffeomorphism of class } C^n$$

Using then [definition: 16.331] it follows that

$$(f^{-1})|_{f(W)} \underset{[\text{theorem: 2.95}]}{=} (f|_W)^{-1}: f(W) \rightarrow X \text{ is of class } C^n$$

and as $x \in W \Rightarrow y = f(x) \in f(W)$ and $W \subseteq U \Rightarrow f(W) \subseteq f(U) = V$ that

$$y \in f(W) \subseteq V$$

To summarize we have $\forall y \in V$ found a $f(W)$ open in Y such that $y \in f(W) \subseteq V$ and $(f^{-1})|_{f(W)}: f(W) \rightarrow X$ is of class C^n . Using then [theorem: 16.159] it follows that

$$f^{-1}: V \rightarrow X \text{ is of class } C^n \quad (16.467)$$

As $f: U \rightarrow V$ is a bijection we have, using [eqs: 16.466, 16.467], by [definition: 16.331] that

$$f: U \Rightarrow V \text{ is a diffeomorphism of class } C^n \quad \square$$

We examine now what happens when diffeomorphism's are composed.

Theorem 16.338. *Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , V a open set in Y and $f: U \rightarrow V$ a diffeomorphism then $\forall x \in U$ we have that*

$$D_{f(x)}(f^{-1}) \circ D_x f = \text{Id}_X \text{ and } D_x f \circ D_{f(x)}(f^{-1}) = \text{Id}_Y$$

so that

$$D_{f(x)}(f^{-1}) = (D_x f)^{-1}$$

Further $D_{f(x)}(f^{-1})$ and $D_x f$ are toplinear isomorphisms [see definition: 16.316]

Proof. Using [example: 16.53] Id_U, Id_V are Fréchet differentiable on U, V and $\forall x \in U$ we have $D_x \text{Id}_U = \text{Id}_X$ and $D_{f(x)} \text{Id}_V = \text{Id}_Y$. Further using the chain rule [see theorem: 16.67] we have

$$\text{Id}_X = D_x \text{Id}_U = D_x(f^{-1} \circ f) = D_{f(x)}(f^{-1}) \circ D_x f$$

and

$$\text{Id}_Y = D_{f(x)} \text{Id}_V = D_{f(x)}(f \circ f^{-1}) = D_{f^{-1}(f(x))} f \circ D_{f(x)}(f^{-1}) = D_x f \circ D_{f(x)}(f^{-1})$$

Further as by the above $D_{f(x)}(f^{-1}) \circ D_x f = \text{Id}_X$ and $D_x f \circ D_{f(x)}(f^{-1}) = \text{Id}_Y$ it follows from [theorem: 2.73, 2.75] that $D_{f(x)}(f^{-1}), D_x f$ are bijections and $(D_x f)^{-1} = D_{f(x)}(f^{-1}) \in L(Y, X)$, $(D_{f(x)}(f^{-1}))^{-1} = D_x f \in L(X, Y)$ proving that $(D_x f)^{-1}, (D_{f(x)} g)^{-1}$ are continuous. Applying then [theorem: 16.318] proves that $D_{f(x)} g$ and $D_x f$ are toplinear isomorphisms. $\square$

Composition of diffeomorphism's is again a diffeomorphism as is show in the next theorem.

Theorem 16.339. *Let $n \in \mathbb{N}_0$, $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle, \langle Z, \|\cdot\|_Z \rangle$ be normed spaces, U a open set in X , V a open set in Y , W a open set in Z then*

1. *If $f: U \rightarrow V$ and $g: V \rightarrow W$ are diffeomorphisms with then $g \circ f: U \rightarrow W$ is a diffeomorphism*
2. *If $n \in \mathbb{N}_0$ and $f: U \rightarrow V$ and $g: V \rightarrow W$ are diffeomorphisms of class C^n then $g \circ f: U \rightarrow W$ is a diffeomorphism of class C^n .*

Proof.

1. Using [theorem: 2.78] it follows that $g \circ f$ is a bijection and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. If f, g are diffeomorphism's then f is differentiable on U , f^{-1}, g are differentiable on V and g^{-1} is differentiable on W . Using the chain rule [theorem: 16.67] $g \circ f$ is differentiable on U and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ is differentiable on W , hence $g \circ f$ is a diffeomorphism.
2. Likewise if f, g are diffeomorphism's of class C^n then f, g, f^{-1}, g^{-1} are C^n on their domains, hence, using the chain rule [theorem: 16.197], $g \circ f$ and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ are C^n on their domains. From this it follows that $g \circ f$ are differentiable of class C^n . $\square$

We introduce now a weaker version of a diffeomorphism the concept of a local diffeomorphism.

Definition 16.340. Let $n \in \mathbb{N}_0$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ be normed spaces, U a open set in X , $x \in U$ and $f: U \rightarrow Y$ a function then we define:

1. **f is a local diffeomorphism at x** if there exist a open set U_x in X with $x \in U_x \subseteq U$ and a open set $V_{f(x)}$ in Y with $f(x) \in V_{f(x)} \subseteq V$ such that

$$f|_{U_x}: U_x \rightarrow V_{f(x)} \text{ is a diffeomorphism}$$

2. **f is a local diffeomorphism at x of class C^n** if there exist a open set U_x in X with $x \in U_x \subseteq U$ and a open set $V_{f(x)}$ in Y with $f(x) \in V_{f(x)} \subseteq V$ such that

$$f|_{U_x}: U_x \rightarrow V_{f(x)} \text{ is a diffeomorphism of class } C^n$$

We are now ready to prove the Inverse Function Theorem.

Theorem 16.341. (Inverse Function Theorem) Let $n \in \mathbb{N}$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ be Banach spaces, U a open set in X , $x_0 \in U$ and $f: U \rightarrow Y$ a function such that

1. f is of class C^n
2. $D_{x_0}f \in L(X, Y)$ is a bijection

then f is a **local diffeomorphism of class C^n at x_0** . In other words using the definition of a local diffeomorphism of class C^n there exist a open set U_{x_0} in X and a open set $V_{f(x_0)}$ in Y with $x \in U_{x_0} \subseteq U$ and $f(x_0) \in V_{f(x_0)}$ such that

$$f|_{U_{x_0}}: U_{x_0} \rightarrow V_{f(x_0)} \text{ is a diffeomorphism of class } C^n$$

Note 16.342. As f is of class C^n and $n \in \mathbb{N}$ it follows from [theorem: 16.158] that f is of class C^1 , hence f is 1-times differentiable at x or using [theorem: 16.140] f is Fréchet differentiable at x_0 so that $D_{x_0}f$ in (2) exist.

Proof. The proof is very very long and will span several pages, so we split it up in different propositions and conjectures. First note that by the definition of the Fréchet differential [see definition: 16.46] that $D_{x_0}f \in L(X, Y)$. As by the hypothesis $D_{x_0}f$ is a bijection and X, Y are Banach spaces, it follows from [theorem: 16.319] that

$$D_{x_0}f \text{ is a toplinear isomorphism} \quad (16.468)$$

Next we prove a limited version of the Inverse Function Theorem where $Y = X$ and $x_0 = 0$ and extend it later to the more general case where Y could be different from X .

Proposition 16.343. Let $n \in \mathbb{N}$, $\langle X, |||_X \rangle$ be a Banach space and U a open set in X with $0 \in U$ and $f: U \rightarrow X$ a function such that f is of class C^n , $D_0f = \text{Id}_X$ and $f(0) = 0$ then f is a local diffeomorphism of class C^n at 0.

Proof. Again the proof of this proposition is very long so we split the proof in different conjectures. Define

$$T = \text{Id}_U - f: U \rightarrow X \quad (16.469)$$

Then

$$T(0) = \text{Id}_U(0) - f(0) \underset{f(0)=0}{=} 0$$

By [example: 16.181] Id_U is of class C^∞ hence of class C^n , so using [theorem: 16.178] it follows that

$$T \text{ is of class } C^n \text{ [hence by [theorem: 16.158] of class } C^1] \quad (16.470)$$

and using [theorem: 16.158] it follows that T is of class C^1 so that by [theorem: 16.154] T is Fréchet differentiable on U and

$$DT: U \rightarrow L(X, X) \text{ defined by } DT(x) = D_x T \text{ is continuous} \quad (16.471)$$

Further $\forall x \in U$ we have

$$DT(x) \underset{\text{eq: 16.471}}{=} D_x T \underset{\text{theorem: 16.178}}{=} D_x \text{Id}_U - D_x f = \text{Id}_X - D_x f \quad (16.472)$$

so that

$$DT(0) = \text{Id}_X - D_0 f \underset{D_0 f = \text{Id}_X}{=} \text{Id}_X - \text{Id}_X = 0 \in L(X, X) \quad (16.473)$$

As DT is continuous at 0 [by eq: 16.471] there exist a $\rho_1 \in \mathbb{R}^+$ such that $\forall x \in U$ with $\|x\|_X < \rho_1$ we have $\|DT(x)\|_{L(X, X)} = \|DT(x) - 0\|_{L(X, X)} \underset{\text{eq: 16.473}}{=} \|DT(x) - DT(0)\|_{L(X, X)} < \frac{1}{2}$. Further as $0 \in U$ there exist a $\rho_2 \in \mathbb{R}^+$ such that $0 \in B_{\|\cdot\|_X}(0, \delta_2) \subseteq U$. Take $\rho_3 = \min(\rho_1, \rho_2) \in \mathbb{R}^+$ then if $x \in B_{\|\cdot\|_X}(0, \rho_3)$ we have $\|x\|_X < \delta_2 \Rightarrow x \in U$ and $\|x\| \leq \delta_1 \Rightarrow \|DT(x)\|_X < \frac{1}{2}$. Take $\rho = \frac{\rho_3}{2}$ then we have

$$\overline{B_{\|\cdot\|_X}(0, \rho)} \subseteq B_{\|\cdot\|_X}(0, \rho_3) \subseteq U \text{ and } \forall x \in \overline{B_{\|\cdot\|_X}(0, \rho)} \text{ we have } \|DT(x)\|_{L(X, X)} < \frac{1}{2} < 1 \quad (16.474)$$

If $x \in B_{\|\cdot\|_X}(0, \rho)$ then by the above $DT(x) \in B_{\|\cdot\|_{L(X, X)}}(0, 1)$ so that by [lemma: 16.327] $\text{Id}_X - DT(x) \in \mathcal{GL}(X)$ which using [eq: 16.472] proves that $D_x f \in \mathcal{GL}(X)$. Summarized we have

$$\forall x \in B_{\|\cdot\|_X}(0, \rho) \quad D_x f \in \mathcal{GL}(X) \text{ [hence by [theorem: 16.321] } (D_x f)^{-1} \in \mathcal{GL}(X) \subseteq L(X, X)] \quad (16.475)$$

As $\overline{B_{\|\cdot\|_X}(0, \rho)}$ is convex [see theorem: 16.221], $\overline{B_{\|\cdot\|_X}(0, \rho)} \subseteq U$ we can use [eq: 16.474] and the fact that T is of class C^1 to apply the mean value theorem [theorem: 16.222] giving

$$\forall x \in \overline{B_{\|\cdot\|_X}(0, \rho)} \text{ we have } \|T(x)\| = \|T(x) - T(0)\| \leq \frac{1}{2} \cdot \|x - 0\| = \frac{1}{2} \cdot \|x\| \quad (16.476)$$

To finish this proposition we need to prove the following four conjectures.

Conjecture 16.344. *If $0 < \rho' \leq \rho$ then $\forall y \in \overline{B_{\|\cdot\|_X}(0, \frac{\rho'}{2})}$ there exist a **uniquely determined** $x \in \overline{B_{\|\cdot\|_X}(0, \rho')}$ such that $y = f(x)$.*

Proof. Let $y \in \overline{B_{\|\cdot\|_X}(0, \frac{\rho'}{2})}$ and define

$$T_y: U \rightarrow X \text{ by } T_y(x) = y + T(x) \underset{\text{eq: 16.469}}{=} y + \text{Id}_U(x) - f(x) = y + x - f(x) \quad (16.477)$$

in other words

$$T_y = C_y + T$$

Hence, as C_y is C^∞ on X [see example: 16.179] and T is of class C^n [hence of class C^1] [see eq: 16.470], it follows from [theorem: 16.178] that

$$T_y \text{ is of class } C^n \text{ [hence of class } C^1] \quad (16.478)$$

Let $x \in \overline{B_{\|\cdot\|_X}(0, \rho')} \subseteq \overline{B_{\|\cdot\|_X}(0, \rho)} \underset{\text{eq: 16.474}}{\subseteq} U$ then we have

$$\|T_y(x)\|_X = \|y + T(x)\|_X \leq \|y\|_X + \|T(x)\|_X \underset{\text{eq: 16.476}}{\leq} \|y\|_X + \frac{1}{2} \cdot \|x\|_X \leq \frac{\rho'}{2} + \frac{1}{2} \cdot \|x\| \leq \frac{\rho'}{2} + \frac{\rho'}{2} = \rho'$$

proving that

$$\forall x \in \overline{B_{\| \| X}(0, \rho')} \text{ we have } \|T_y(x)\|_X \leq \rho' \text{ so that } T_y(x) \in \overline{B_{\| \| X}(0, \rho')} \quad (16.479)$$

As $D_x T_y = D_x(C_y + T) = D_x C_y + D_x T = 0 + D_x T = D_x T$ we have using [eq: 16.474] that

$$\forall x \in \overline{B_{\| \| X}(0, \rho')} \subseteq \overline{B_{\| \| X}(0, \rho)} \quad \|D_x T_y\|_{L(X, X)} = \|D_x T\|_{L(X, X)} < \frac{1}{2} \quad (16.480)$$

As X is a Banach space, T_y is of class C^1 and $\overline{B_{\| \| X}(0, \rho')}$ is convex we can apply the mean value theorem [theorem: 16.222] on the above to get

$$\forall x_1, x_2 \in \overline{B_{\| \| X}(0, \rho')} \text{ we have } \|T_y(x_1) - T_y(x_2)\|_X \leq \frac{1}{2} \cdot \|x_1 - x_2\| \quad (16.481)$$

Using [eq: 16.479] it follows that the following function is well defined

$$S_y = (T_y)_{|\overline{B_{\| \| X}(0, \rho')}}: \overline{B_{\| \| X}(0, \rho')} \rightarrow \overline{B_{\| \| X}(0, \rho')}$$

and further by [eq: 16.481] that this function is a contraction [see definition: 14.445]. As $\overline{B_{\| \| X}(0, \rho')}$ is closed and X is a Banach space it follows from [theorem: 14.393] that $\overline{B_{\| \| X}(0, \rho')}$ is a complete metric space [note that a normed space is a metric space]. So we can apply the Banach Fixed Point theorem [see theorem: 14.447] giving

$$\text{There exist a } \mathbf{unique} \ x \in \overline{B_{\| \| X}(0, \rho')} \text{ such that } T_y(x)_{S_y = (T_y)_{|\overline{B_{\| \| X}(0, \rho')}}} = S_y(x) = x \quad (16.482)$$

From the definition of T_y [see eq: 16.477] we have that $f(x) = y + x - T_y(x)_{[\text{eq: 16.482}]}$ $y + x - x = y$ proving that

$$y = f(x)$$

As for uniqueness assume that there exist a $x' \in \overline{B_{\| \| X}(0, \rho')}$ such that $y = f(x')$ then we have that $T_y(x') = y + x' - f(x') = y + x' - y = x'$ so that by the uniqueness in [eq: 16.482] $x = x'$. This concludes the proof of [conjecture: 16.344]. $\square$

Conjecture 16.345. *For the open set*

$$U_\rho = B_{\| \| X}\left(0, \frac{\rho}{2}\right) \cap f^{-1}\left(B_{\| \| X}\left(0, \frac{\rho}{4}\right)\right)$$

[as f is C^n on U it is continuous hence $f^{-1}(B_{\| \| X}(0, \frac{\rho}{4}))$ is open] we have that

$$0 \in U_\rho \quad (16.483)$$

and

$$f|_{U_\rho}: U_\rho \rightarrow B_{\| \| X}\left(0, \frac{\rho}{4}\right)$$

is a bijection.

Proof. First as $f(0) = 0 \in B_{\| \| X}(0, \frac{\rho}{4})$ it follows that

$$0 \in B_{\| \| X}\left(0, \frac{\rho}{2}\right) \cap f^{-1}\left(B_{\| \| X}\left(0, \frac{\rho}{4}\right)\right) = U_\rho$$

Second for the function

$$f|_{U_\rho}: U_\rho \rightarrow B_{\| \| X}\left(0, \frac{\rho}{4}\right)$$

we have:

surjectivity. Let $y \in B_{\| \| X}(0, \frac{\rho}{4}) \subseteq \overline{B_{\| \| X}(0, \frac{\rho}{4})}$ then by taking $\rho' = \frac{\rho}{2}$ in the previous conjecture [conjecture: 16.344] there exist a unique $x \in \overline{B_{\| \| X}(0, \frac{\rho}{2})}$ such that

$$f(x) = y \quad (16.484)$$

We prove now by contradiction that

$$\|x\|_X \neq \frac{\rho}{2} \quad (16.485)$$

So assume that $\|x\|_X = \frac{\rho}{2}$. As $y \in B_{\|x\|_X}(0, \frac{\rho}{4})$ we have that $\|f(x)\|_X = \|y\|_X < \frac{\rho}{4}$ so that $\varepsilon = \frac{\rho}{4} - \|y\|_X \in \mathbb{R}^+$. Given that f is continuous [because f is C^n on U hence C^0 on U by [theorem: 16.158]] there exist a $\delta' \in \mathbb{R}^+$ such that if $z \in U$ with $\|x - z\|_X < \delta'$ then $\|f(x) - f(z)\|_X < \varepsilon$. Define $\delta = \min(\delta', \frac{\rho}{2}) \in \mathbb{R}^+$ and take $x_1 = \left(1 + \frac{\delta}{2 \cdot \rho}\right) \cdot x$ then we have, as $1 < 1 + \frac{\delta}{2 \cdot \rho}$, and $\|x\|_X = \frac{\rho}{2} \neq 0$ that

$$\|x\|_X < \left(1 + \frac{\delta}{2 \cdot \rho}\right) \cdot \|x\|_X = \|x_1\|_X$$

Further

$$\|x_1\|_X = \left(1 + \frac{\delta}{2 \cdot \rho}\right) \cdot \|x\|_X = \left(1 + \frac{\delta}{2 \cdot \rho}\right) \cdot \frac{\rho}{2} = \frac{\rho}{2} + \frac{\delta}{2 \cdot \rho} \cdot \frac{\rho}{2} = \frac{\rho}{2} + \frac{\delta}{4} \leq \frac{\rho}{2} + \frac{\rho}{4 \cdot 2} < \frac{\rho}{2} + \frac{\rho}{2} = \rho$$

giving

$$\frac{\rho}{2} = \|x\|_X < \|x_1\|_X < \rho \text{ so that by [eq: 16.474] } x_1 \in U \quad (16.486)$$

Further we have

$$\|x - x_1\|_X = \left\|x - \left(1 + \frac{\delta}{2 \cdot \rho}\right) \cdot x\right\|_X = \left\|\frac{\delta}{2 \cdot \rho} \cdot x\right\|_X = \frac{\delta}{2 \cdot \rho} \cdot \|x\|_X = \frac{\delta}{2 \cdot \rho} \cdot \frac{\rho}{2} = \frac{\delta}{4} < \delta \leq \delta'$$

so that

$$\|f(x) - f(x_1)\|_X < \varepsilon = \frac{\rho}{4} - \|y\|_X$$

Hence

$$\begin{aligned} \|f(x_1)\|_X &= \|f(x_1) - f(x) + f(x)\|_X \\ &\leq \|f(x_1) - f(x)\|_X + \|f(x)\|_X \\ &< \frac{\rho}{4} - \|y\|_X + \|f(x)\|_X \\ &\stackrel{[\text{eq: 16.484}]}{=} \frac{\rho}{4} - \|y\|_X + \|y\|_X \\ &= \frac{\rho}{4} \end{aligned}$$

proving that $f(x_1) \in B_{\|x\|_X}(0, \frac{\rho}{4}) \subseteq \overline{B_{\|x\|_X}(0, \frac{\rho}{4})}$. Using the previous conjecture [conjecture: 16.344] there exist a $x_2 \in \overline{B_{\|x\|_X}(0, \frac{\rho}{2})}$ such that $f(x_1) = f(x_2)$. As

$$f(x_2) = f(x_1) \in B_{\|x\|_X}(0, \frac{\rho}{4}) \subseteq \overline{B_{\|x\|_X}(0, \frac{\rho}{2})} \text{ and } \|x_1\|_X \stackrel{[\text{eq: 16.486}]}{<} \rho \wedge \|x_2\|_X \leq \frac{\rho}{2} < \rho$$

it follows from the uniqueness in [conjecture: 16.344] that $x_1 = x_2$. So we have

$$\frac{\rho}{2} \stackrel{[\text{eq: 16.486}]}{<} \|x_1\|_X = \|x_2\|_X \leq \frac{\rho}{2}$$

leading to the contradiction $\frac{\rho}{2} < \frac{\rho}{2}$. Hence our assumption is wrong and we must have that

$$\|x\|_X \neq \frac{\rho}{2}$$

So as $x \in \overline{B_{\|x\|_X}(0, \frac{\rho}{2})}$ we have $x \in B_{\|x\|_X}(0, \frac{\rho}{2})$, further as $f(x) = y \in B_{\|x\|_X}(0, \frac{\rho}{4})$ it follows that $x \in f^{-1}(B_{\|x\|_X}(0, \frac{\rho}{4}))$, proving that $x \in B_{\|x\|_X}(0, \frac{\rho}{2}) \cap f^{-1}(B_{\|x\|_X}(0, \frac{\rho}{4})) = U_\rho$. Hence we have found a $x \in U_\rho$ such that $f(x) = y$.

injectivity. Assume that there exist $x_1, x_2 \in U_\rho \subseteq B_{\|x\|_X}(0, \frac{\rho}{4})$ such that $f(x_1) = f(x_2) \in B_{\|x\|_X}(0, \frac{\rho}{4})$ then by the uniqueness of the previous conjecture [conjecture: 16.344] $x_1 = x_2$.

So $f|_{U_\rho}: U_\rho \rightarrow B_{\|x\|_X}(0, \frac{\rho}{4})$ is a bijection which together with [eq: 16.483] proves [conjecture: 16.345]. $\square$

Proceeding with the proof of [proposition: 16.343] we have as $f|_{U_\rho}: U_\rho \rightarrow B_{|||x}(0, \frac{\rho}{4})$ is a bijection by [conjecture: 16.345] that $f|_{U_\rho}$ has an inverse which we call g

$$g = (f|_{U_\rho})^{-1}: B_{|||x}(0, \frac{\rho}{4}) \rightarrow U_\rho \text{ where } U_\rho = B_{|||x}(0, \frac{\rho}{2}) \cap f^{-1}(B_{|||x}(0, \frac{\rho}{4})) \quad (16.487)$$

We prove now that g is continuous.

Conjecture 16.346. g is continuous.

Proof. Let $x \in U_\rho$ then $x = x - f(x) + f(x) \stackrel{[\text{eq: 16.469}]}{=} T(x) + f(x)$, so if $x_1, x_2 \in U_\rho$ then

$$\begin{aligned} \|x_1 - x_2\|_X &= \|T(x_1) + f(x_1) - T(x_2) - f(x_2)\|_X \\ &\leq \|T(x_1) - T(x_2)\|_X + \|f(x_1) - f(x_2)\|_X \end{aligned}$$

As by [eq: 16.487] $x_1, x_2 \in U_\rho \subseteq B_{|||x}(0, \frac{\rho}{2}) \subseteq \overline{B_{|||x}(0, \frac{\rho}{2})}$ which is convex [see theorem: 16.221], T is of class C^1 [see eq: 16.478] and $\forall x \in \overline{B_{|||x}(0, \frac{\rho}{2})} \subseteq B_{|||x}(0, \rho)$ we have $\|D_x T\|_{L(X, X)} < \frac{1}{2}$ [see eq: 16.474] we can apply the main value theorem [theorem: 16.222] giving $\|T(x_1) - T(x_2)\|_X \leq \frac{1}{2} \cdot \|x_1 - x_2\|_X$ so that

$$\|x_1 - x_2\|_X \leq \frac{1}{2} \cdot \|x_1 - x_2\|_X + \|f(x_1) - f(x_2)\|_X$$

hence

$$\|x_1 - x_2\|_X \leq 2 \cdot \|f(x_1) - f(x_2)\|_X$$

To summarize we have proved that

$$\forall x_1, x_2 \in U_\rho \quad \|x_1 - x_2\|_X \leq 2 \cdot \|f(x_1) - f(x_2)\|_X \quad (16.488)$$

Let $y_1, y_2 \in B_{|||x}(0, \frac{\rho}{4})$ then by [eq: 16.487] $g(y_1), g(y_2) \in U_\rho$ so that by [eq: 16.488] we have

$$\|g(y_1) - g(y_2)\|_X \leq 2 \cdot \|f(g(y_1)) - f(g(y_2))\|_X \stackrel{[\text{eq: 16.487}]}{=} 2 \cdot \|y_1 - y_2\|_X \quad (16.489)$$

Let $\varepsilon \in \mathbb{R}^+$ and $y \in B_{|||x}(0, \frac{\rho}{4})$ then for all $y' \in B_{|||x}(0, \frac{\rho}{4})$ with $\|y - y'\|_X < \frac{\varepsilon}{2}$ we have

$$\|g(y) - g(y')\|_X \leq 2 \cdot \|y - y'\|_X = 2 \cdot \frac{\varepsilon}{2} = \varepsilon$$

proving that g is continuous at y . As $y \in B_{|||x}(0, \frac{\rho}{4})$ was chosen arbitrary we must conclude that g is continuous. $\square$

Next we prove that g is differentiable on $B_{|||x}(0, \frac{\rho}{4})$.

Conjecture 16.347. g is differentiable on $B_{|||x}(0, \frac{\rho}{4})$ and $\forall y \in B_{|||x}(0, \frac{\rho}{4})$ we have $D_y g = (D_{g(y)} f)^{-1}$.

Proof. Let $y \in B_{|||x}(0, \frac{\rho}{4})$ then by [eq: 16.487] $g(y) \in U_\rho \subseteq B_{|||x}(0, \frac{\rho}{2}) \subseteq B_{|||x}(0, \rho) \stackrel{[\text{eq: 16.474}]}{\subseteq} U$. Further using [eq: 16.475] we have that $D_{g(y)} f$ is a toplinear isomorphism so that

$$(D_{g(y)} f)^{-1} \text{ exists and } (D_{g(y)} f)^{-1} \in L(X, X) \quad (16.490)$$

Take $h \in (B_{|||x}(0, \frac{\rho}{4}))_y$ so that $y + h \in B_{|||x}(0, \frac{\rho}{4})$ then for k defined by

$$k = g(y + h) - g(y) \quad (16.491)$$

we have

$$k + g(y) = g(y + h) \in g(B_{|||x}(0, \frac{\rho}{4})) \stackrel{[\text{eq: 16.487}]}{=} U_\rho \subseteq B_{|||x}(0, \frac{\rho}{2}) \subseteq B_{|||x}(0, \rho) \stackrel{[\text{eq: 16.474}]}{\subseteq} U$$

So that

$$k \in U_{g(y)} \stackrel{\text{def}}{=} \{h \in X \mid g(y) + h \in U\}$$

As f is Fréchet differentiable at $g(y)$ [for $g(y) \in U_\rho \subseteq B_{\|x\|}(0, \frac{\rho}{2}) \subseteq U$] there exist by [theorem: 16.61] a ε -mapping $\zeta: U_{g(y)} \rightarrow X$ such that

$$f(g(y) + k) - f(g(y)) - D_{g(y)}f(k) = \|k\|_X \cdot \zeta(k) \quad (16.492)$$

As $f(g(y) + k) \stackrel{[eq: 16.491]}{=} f(g(y) + g(y+h) - g(y)) = f(g(y+h)) \stackrel{[eq: 16.487]}{=} y+h$ and $f(g(y)) \stackrel{[eq: 16.487]}{=} y$ it follows that $y+h - y - D_{g(y)}f(k) = \|k\|_X \cdot \zeta(k)$. Hence we have

$$h = D_{g(y)}f(k) + \|k\|_X \cdot \zeta(k) \quad (16.493)$$

Applying $(D_{g(y)}f)^{-1}$ to both sides of [eq: 16.493] and using linearity we have

$$\begin{aligned} (D_{g(y)}f)^{-1}(h) &= (D_{g(y)}f)^{-1}(D_{g(y)}f(k) + \|k\|_X \cdot \zeta(k)) \\ &= (D_{g(y)}f)^{-1}(D_{g(y)}f(k)) + \|k\|_X \cdot (D_{g(y)}f)^{-1}(\zeta(k)) \\ &= k + \|k\|_X \cdot (D_{g(y)}f)^{-1}(\zeta(k)) \\ &\stackrel{[eq: 16.491]}{=} g(y+h) - g(y) + \|g(y+h) - g(y)\|_X \cdot (D_{g(y)}f)^{-1}(\zeta(g(y+h) - g(y))) \end{aligned}$$

so that

$$g(y+h) - g(y) - (D_{g(y)}f)^{-1}(h) = -\|g(y+h) - g(y)\|_X \cdot (D_{g(y)}f)^{-1}(\zeta(g(y+h) - g(y))) \quad (16.494)$$

The above suggest that we use [theorem: 16.61] again, so define

$$\varepsilon: \left(B_{\|x\|}\left(0, \frac{\rho}{4}\right) \right)_y \rightarrow X \text{ by } \begin{cases} 0 & \text{if } h = 0 \\ -\frac{\|g(y+h) - g(y)\|_X}{\|h\|_X} \cdot (D_{g(y)}f)^{-1}(\zeta(g(y+h) - g(y))) & \text{if } h \neq 0 \end{cases}$$

Using the fact that $g(y+0) - g(0) - (D_{g(y)}f)^{-1}(0) = 0$ we have from the above definition and [eq: 16.494] that

$$g(y+h) - g(y) - (D_{g(y)}f)^{-1}(h) = \|h\|_X \cdot \varepsilon(h) \quad (16.495)$$

As $y, y+h \in B_{\|x\|}(0, \frac{\rho}{4})$ it follows from [eq: 16.489] that

$$\|g(y+h) - g(y)\|_X \leq 2 \cdot \|y+h - y\|_X = 2 \cdot \|h\|_X \quad (16.496)$$

Let $\chi \in \mathbb{R}^+$ then by continuity of the ε -mapping ζ [see definition: 16.60] there exist a $\delta'_\chi \in \mathbb{R}^+$ such that for

$$\forall h' \in U_{g(y)} \text{ with } \|h'\|_X = \|h' - 0\|_X < \delta'_\chi \text{ we have } \|\zeta(h)\|_X = \|\zeta(h-0)\|_X < \chi \quad (16.497)$$

By the continuity of g [see conjecture: 16.346] there exist a $\delta_\chi \in \mathbb{R}^+$ such that $\forall h'' \in B_{\|x\|}(0, \frac{\rho}{4})$ with $\|h''\|_X < \delta_\chi$ we have that

$$\forall h'' \in B_{\|x\|}\left(0, \frac{\rho}{4}\right) \text{ with } \|y - h''\|_X < \delta_\chi \text{ we have } \|g(y) - g(h'')\|_X < \delta'_\chi \quad (16.498)$$

Let $h \in \left(B_{\|x\|}\left(0, \frac{\rho}{4}\right) \right)_y$ with $\|h\|_X < \delta_\chi$ then we have $y+h \in B_{\|x\|}(0, \frac{\rho}{4})$ and $\|y - (y+h)\|_X < \delta_\chi$ so that by [eq: 16.498]

$$\|g(y+h) - g(y)\|_X < \delta'_\chi$$

Further we have $(g(y+h) - g(y)) + g(y) = g(y+h) \in U_\rho \subseteq B_{\|x\|}(0, \frac{\rho}{2}) \subseteq B_{\|x\|}(0, \rho) \stackrel{[eq: 16.474]}{\subseteq} U$ so that

$$g(y+h) - g(y) \in U_{g(y)}$$

Combining these two above results with [eq: 16.497] proves that

$$\|\zeta(g(y+h) - g(y))\|_X < \chi$$

To summarize we have proved that

$$\forall \chi \in \mathbb{R}^+ \exists \delta_\chi \in \mathbb{R}^+ \text{ such that } \forall h \in \left(B_{\|x\|}\left(0, \frac{\rho}{4}\right) \right)_y \text{ with } \|h\|_X < \delta_\chi \quad \|\zeta(g(y+h) - g(y))\|_X < \chi \quad (16.499)$$

Take $\mu \in \mathbb{R}^+$ then given $\chi = \frac{\mu}{2 \cdot (\|(D_{g(y)}f)^{-1}\|_{L(X,Y)} + 1)} \in \mathbb{R}^+$ we have for $h \in B_{\|\cdot\|_X}(0, \frac{\rho}{4})_y$ with $\|h\|_X < \delta_\chi$ that

$$\begin{aligned}
& \|\varepsilon(h) - \varepsilon(0)\|_X = \\
& \|\varepsilon(h)\|_X = \\
& \begin{cases} \|0\|_X & \text{if } h = 0 \\ \left\| -\frac{\|g(y+h) - g(y)\|_X}{\|h\|_X} \cdot (D_{g(y)}f)^{-1}(\zeta(g(y+h) - g(y))) \right\|_X & \text{if } h \neq 0 \end{cases} = \\
& \begin{cases} 0 & \text{if } h = 0 \\ \frac{\|g(y+h) - g(y)\|_X}{\|h\|_X} \cdot \|(D_{g(y)}f)^{-1}(\zeta(g(y+h) - g(y)))\|_X & \text{if } h \neq 0 \end{cases} \leq \\
& \begin{cases} 0 & \text{if } h = 0 \\ \frac{\|g(y+h) - g(y)\|_X}{\|h\|_X} \cdot \|(D_{g(y)}f)^{-1}\|_{L(X,X)} \|\zeta(g(y+h) - g(y))\|_X & \text{if } h \neq 0 \end{cases} \stackrel{[\text{eq: 16.496}]}{\leq} \\
& \begin{cases} 0 & \text{if } h = 0 \\ \frac{2 \cdot \|h\|_X}{\|h\|_X} \cdot \|(D_{g(y)}f)^{-1}\|_{L(X,X)} \cdot \|\zeta(g(y+h) - g(y))\|_X & \text{if } h \neq 0 \end{cases} \stackrel{[\text{eq: 16.499}]}{\leq} \\
& \begin{cases} 0 & \text{if } h = 0 \\ \frac{2 \cdot \|h\|_X}{\|h\|_X} \cdot \|(D_{g(y)}f)^{-1}\|_{L(X,X)} \cdot \chi & \text{if } h \neq 0 \end{cases} = \\
& \begin{cases} 0 & \text{if } h = 0 \\ 2 \cdot \|(D_{g(y)}f)^{-1}\|_{L(X,X)} \cdot \chi & \text{if } h \neq 0 \end{cases} = \\
& \begin{cases} 0 & \text{if } h = 0 \\ 2 \cdot \|(D_{g(y)}f)^{-1}\|_{L(X,X)} \cdot \frac{\mu}{2 \cdot (\|(D_{g(y)}f)^{-1}\|_{L(X,Y)} + 1)} & \text{if } h \neq 0 \end{cases} < \\
& \begin{cases} 0 & \text{if } h = 0 \\ \mu & \text{if } h \neq 0 \end{cases} < \\
& \mu
\end{aligned}$$

proving that ε is continuous at 0. Hence ε is a ε -mapping, as further by [eq: 16.475] $(D_{g(y)}f)^{-1} \in L(X, X)$ it follows from [eq: 16.495] and [corollary: 16.62] that g is Fréchet differentiable at y with differential $D_y g = (D_{g(y)}f)^{-1}$. This concludes the proof of [conjecture: 16.347]. $\square$

To complete the proof of [proposition: 16.343] we need to proof that g is C^n on $B_{\|\cdot\|_X}(0, \frac{\rho}{4})$. Consider the inversion mapping:

$$\tau: \text{GL}(X) \rightarrow L(X, X) \text{ defined by } \tau(L) = L^{-1} \text{ [see theorem: 16.330]}$$

As $f: U \rightarrow X$ is of class C^n and $B_{\|\cdot\|_X}(0, \rho) \subseteq U$ it follows from [theorem: 16.156] that [eq: 16.474]

$$f|_{B_{\|\cdot\|_X}(0, \rho)}: B_{\|\cdot\|_X}(0, \rho) \rightarrow X \text{ is of class } C^n$$

Further we have by [theorem: 16.158] that f is of class C^1 so that by [theorem: 16.156] again $f|_{B_{\|\cdot\|_X}(0, \rho)}$ is of class C^1 and using [theorem: 16.153]

$$\forall y \in B_{\|\cdot\|_X}(0, \rho) \subseteq U \text{ we have } D_y f|_{B_{\|\cdot\|_X}(0, \rho)} \stackrel{[\text{theorem: 16.57}]}{=} D_y f \quad (16.500)$$

As by [eq: 16.475] $D_y f \in \mathcal{GL}(X) \forall y \in B_{\|\cdot\|_X}(0, \rho)$ we have for

$$Df|_{B_{\|\cdot\|_X}(0, \rho)}: B_{\|\cdot\|_X}(0, \rho) \rightarrow L(X, X) \text{ by } Df|_{B_{\|\cdot\|_X}(0, \rho)}(x) = D_x f|_{B_{\|\cdot\|_X}(0, \rho)}$$

that

$$Df|_{B_{\|\cdot\|_X}(0, \rho)}(B_{\|\cdot\|_X}(0, \rho)) \subseteq \mathcal{GL}(X)$$

Further by the previous conjecture [conjecture: 16.347] we have

$$Dg: B_{|||X}\left(0, \frac{\rho}{4}\right) \rightarrow L(X, X) \text{ by } Dg(y) = D_y g$$

As $g(B_{|||X}(0, \frac{\rho}{4})) \stackrel{[\text{eq: 16.487}]}{=} U_\rho \subseteq B_{|||X}(0, \frac{\rho}{2}) \subseteq B_{|||X}(0, \rho)$ the following function is well defined

$$\tau \circ Df|_{B_{|||X}(0, \rho)} \circ g: B_{|||X}\left(0, \frac{\rho}{4}\right) \rightarrow L(X, X)$$

is well defined. If $y \in B_{|||X}(0, \frac{\rho}{4})$ then we have

$$\begin{aligned} (\tau \circ Df|_{B_{|||X}(0, \rho)} \circ g)(y) &= \tau(Df|_{B_{|||X}(0, \rho)}(g(y))) \\ &= \tau(D_{g(y)}f|_{B_{|||X}(0, \rho)}) \\ &\stackrel{[\text{eq: 16.500}]}{=} \tau(D_{g(y)}f) \\ &= (D_{g(y)}f)^{-1} \\ &\stackrel{[\text{conjecture: 16.347}]}{=} D_y g \\ &= Dg(y) \end{aligned}$$

proving that

$$\tau \circ Df|_{B_{|||X}(0, \rho)} \circ g = Dg \quad (16.501)$$

Let $m \in \{1, \dots, n\}$ then as f is of class C^n it follows from [theorem: 16.158] that f is of class C^m , further as the open set $B_{|||X}(0, \rho) \subseteq U$ it follows from [theorem: 16.156] that $f|_{B_{|||X}(0, \rho)}$ is of class C^m . To summarize

$$\forall m \in \{1, \dots, n\} \text{ we have } f|_{B_{|||X}(0, \rho)} \text{ is of class } C^m \quad (16.502)$$

Next we use mathematical induction to prove that g is of class C^n . So define

$$S = \left\{ m \in \mathbb{N} \mid \text{If } m \leq n \text{ then } g \text{ is of class } C^m \text{ on } B_{|||X}\left(0, \frac{\rho}{4}\right) \right\}$$

then we have:

1 $\in S$. By [theorem: 16.330] τ is C^∞ on $GL(X)$ hence continuous, further by [conjecture: 16.346] g is continuous and by [eq: 16.502] $f|_{B_{|||X}(0, \rho)}$ is C^1 on $B_{|||X}(0, \rho)$ so that $Df|_{B_{|||X}(0, \rho)}$ is continuous. As further $B_{|||X}(0, \rho)$ and $GL(X)$ are open it follows from [theorem: 14.159] that $\tau \circ Df|_{B_{|||X}(0, \rho)} \circ g$ is continuous proving by [eq: 16.501] that Dg is continuous, hence g is of class C^1 . So $1 \in S$.

$m \in S \Rightarrow m+1 \in S$. If $m+1 \leq n$ then by [eq: 16.502] $f|_{B_{|||X}(0, \rho)}$ is of class C^{m+1} on $B_{|||X}(0, \rho)$ so that by [theorem: 16.175] and [definition: 16.129] $Df|_{B_{|||X}(0, \rho)}$ is of class C^m . Further as τ is of class C^∞ and g is of class C^m [because $m \in S$] it follows from the chain rule [theorem: 16.197] that $\tau \circ Df|_{B_{|||X}(0, \rho)} \circ g$ is of class C^m . Hence using [eq: 16.501] we conclude that $D^{[1]}g = Dg$ is of class C^m , which by [theorem: 16.175] proves that g is of class C^{m+1} . Hence it follows that $m+1 \in S$.

Using induction it follows that $S = \mathbb{N}$ so that, as $n \in \mathbb{N}$ and $n \leq n$, g is of class C^n . As g is the inverse of $f|_{U_\rho}: U_\rho \rightarrow B_{|||X}(0, \frac{\rho}{4})$, $0 \in U_\rho$ [see conjecture: 16.345], it follows that f is a local diffeomorphism of class C^n at 0. This concludes the proof of [proposition: 16.343]. $\square$

Next we must extend [proposition: 16.343] to the more general case where $X \neq Y$.

Proposition 16.348. *Let $n \in \mathbb{N}$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ be Banach spaces, U a open set in X , $0 \in U$ and $f: U \rightarrow Y$ a function such that f is of class C^n , $f(0) = 0$ and $D_0 f \in L(X, Y)$ is a bijection then f is a local diffeomorphism of class C^n at 0.*

Proof. As $D_0 f \in L(X, Y)$ is a bijection it follows from [theorem: 16.319] that

$$D_0 f \text{ is a toplinear isomorphism} \quad (16.503)$$

so that

$$(D_0f)^{-1}: Y \rightarrow X$$

is well defined and is linear and continuous, hence by [theorem: 16.180] $(D_0f)^{-1}$ is C^∞ on Y . As f is C^n on U it follows from the chain rule [theorem: 16.197] that

$$h = (D_0f)^{-1} \circ f: U \rightarrow X \text{ is of class } C^n \text{ on } U$$

Further

$$\begin{aligned} h(0) &= ((D_0f)^{-1} \circ f)(0) \\ &= (D_0f)^{-1}(f(0)) \\ &\stackrel{f(0)=0}{=} (D_0f)^{-1}(0) \\ &\stackrel{\text{linearity}}{=} 0 \end{aligned} \tag{16.504}$$

Also as h is of class C^n it is of class C^1 hence by [theorem: 16.153] differentiable on U and by the chain rule we have

$$\begin{aligned} D_0h &= D_0((D_0f)^{-1} \circ f) \\ &\stackrel{[\text{theorem: 16.67}]}{=} D_{f(0)}(D_0f)^{-1} \circ D_0f \\ &\stackrel{[\text{theorem: 16.180}]}{=} (D_{f(0)}f)^{-1} \circ D_0f \\ &\stackrel{f(0)=0}{=} (D_0f)^{-1} \circ D_0f \\ &= \text{Id}_X \end{aligned} \tag{16.505}$$

So the conditions for [proposition: 16.343] are satisfied for h and it follows that h is a local diffeomorphism of class C^n at 0. Hence there exist open sets U_0, V'_0 in X with $0 \in U_0 \subseteq U$ and $0 = h(0) \in V'_0$ such that

$$h|_{U_0}: U_0 \rightarrow V'_0 \text{ is a diffeomorphism of class } C^n \tag{16.506}$$

As D_0f is a toplinear isomorphism [see eq: 16.503] we have that

$$V_0 = (D_0f)(V'_0) \stackrel{[\text{theorem: 2.70}]}{=} ((D_0f)^{-1})^{-1}(V'_0) \text{ is open in } Y \tag{16.507}$$

Further as $D_0f \in L(X, Y)$ and is bijective it follows from [example: 16.334] that D_0f is a diffeomorphism of class C^∞ . Applying then [theorem: 16.336] proves that

$$(D_0f)|_{V'_0}: V'_0 \rightarrow D_0f(V'_0) = V_0 \text{ is a diffeomorphism of class } C^n \tag{16.508}$$

Next as $0 \in V'_0$ and D_0f is linear we have that $0 = (D_0f)|_{V'_0}(0) = D_0f(0) \in D_0f(V'_0) = V_0$ so that

$$0 \in V_0$$

Take

$$g: (D_0f)|_{V'_0} \circ h|_{U_0}: U_0 \rightarrow V_0$$

then as $h|_{U_0}, (D_0f)|_{V'_0}$ are diffeomorphism of class C^n it follows from [theorem: 16.339] that

$$g \text{ is a diffeomorphism of class } C^n \tag{16.509}$$

Let now $x \in U_0$ then we have

$$\begin{aligned} g(x) &= ((D_0f)|_{V'_0} \circ h|_{U_0})(x) \\ &= (D_0f)|_{V'_0}(h|_{U_0}(x)) \\ &= (D_0f)(h|_{U_0}(x)) \\ &= (D_0f)(h(x)) \\ &= (D_0f)((D_0f)^{-1} \circ f)(x) \\ &= (D_0f)(D_0f)^{-1}(f(x)) \\ &= f(x) \\ &\stackrel{x \in U_0}{=} f|_{U_0}(x) \end{aligned}$$

so that

$$g = f|_{U_0}$$

Hence $f|_{U_0}: U_0 \rightarrow V_0$ is a diffeomorphism of class C^n which proves that f is a local diffeomorphism of class C^n at 0 concluding the proof of [proposition: 16.348] $\square$

Finally we extend the previous proposition to prove the inverse function theorem. To increase readability we repeat here the inverse function theorem as a conjecture, so as this conjecture is proved we have completed the proof of the inverse function theorem.

Conjecture 16.349. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be Banach spaces, U a open set in X , $x_0 \in U$ and $f: U \rightarrow Y$ a function such that*

1. f is C^n on U
2. $D_{x_0}f \in L_n(X; Y)$ is a bijection

*then f is a **local diffeomorphism of class C^n at x_0** . In other words using the definition of a local diffeomorphism of class C^n there exist a open set U_{x_0} in X with $x_0 \in U_{x_0} \subseteq U$ and a open set $V_{f(x_0)}$ in Y with $f(x_0) \in V_{f(x_0)}$ such that*

$$f|_{U_{x_0}}: U_{x_0} \rightarrow V_{f(x_0)} \text{ is a diffeomorphism of class } C^n$$

Proof. Define $\mathcal{T}_1: X \rightarrow X$ by $\mathcal{T}_1(x) = x + x_0$ then by [example: 16.184] $\mathcal{T}_1$ is of class C^∞ with $D_0\mathcal{T}_1 = \text{Id}_X$. Further

$$\mathcal{T}_1(0) = x_0 \in U \text{ and we have } 0 \in W \text{ where } W = \mathcal{T}_1^{-1}(U) \quad (16.510)$$

As $\mathcal{T}_1$ is of class C^∞ hence continuous, W is a open set, so applying [theorem: 16.156] we have

$$(\mathcal{T}_1)|_W: W \rightarrow X \text{ is of class } C^\infty \text{ with } D_0(\mathcal{T}_1)|_W \stackrel{[\text{eq: 16.57}]}{=} D_0\mathcal{T}_1 = \text{Id}_X \quad (16.511)$$

Further

$$(\mathcal{T}_1)|_W(W) \stackrel{[\text{theorem: 2.92}]}{=} \mathcal{T}_1(W) = \mathcal{T}_1(\mathcal{T}_1^{-1}(U)) \stackrel{[\text{theorem: 2.26}]}{\subseteq} U \quad (16.512)$$

Further define $\mathcal{T}_2: Y \rightarrow Y$ by $\mathcal{T}_2(x) = x - f(x_0)$ then by [example: 16.184] $\mathcal{T}_2$ is of class C^∞ with $D_0\mathcal{T}_2 = \text{Id}_Y$. So we can construct the function

$$h = \mathcal{T}_2 \circ f \circ (\mathcal{T}_1)|_W: W \rightarrow Y \quad (16.513)$$

Using the chain rule [theorems: 16.197] it follows that h is of class C^n hence differentiable at $0 \in W$ and

$$\begin{aligned} D_0h &= D_0(\mathcal{T}_2 \circ f \circ (\mathcal{T}_1)|_W) \\ &= D_0((\mathcal{T}_2 \circ f) \circ (\mathcal{T}_1)|_W) \\ &\stackrel{[\text{theorem: 16.67}]}{=} D_{(\mathcal{T}_1)|_W(0)}(\mathcal{T}_2 \circ f) \circ D_0(\mathcal{T}_1)|_W \\ &\stackrel{[\text{eq: 16.511}]}{=} D_{(\mathcal{T}_1)|_W(0)}(\mathcal{T}_2 \circ f) \circ \text{Id}_X \\ &= D_{(\mathcal{T}_1)|_W(0)}(\mathcal{T}_2 \circ f) \\ &\stackrel{[\text{eq: 16.510}]}{=} D_{x_0}(\mathcal{T}_2 \circ f) \\ &\stackrel{[\text{theorem: 16.67}]}{=} D_{f(x_0)}\mathcal{T}_2 \circ D_{x_0}f \\ &= \text{Id}_Y \circ D_{x_0}f \\ &= D_{x_0}f \end{aligned}$$

As $D_{x_0}f$ is a bijection it follows from the above that

$$D_0h \text{ is a bijection}$$

Further we have

$$\begin{aligned}
 h(0) &= (\mathcal{T}_2 \circ f \circ (\mathcal{T}_1|_W))(0) \\
 &= \mathcal{T}_2(f((\mathcal{T}_1|_W)(0))) \\
 &= \mathcal{T}_2(f(x_0)) \\
 &= \mathcal{T}_2(f(x_0)) \\
 &= f(x_0) - f(x_0) \\
 &= 0
 \end{aligned}$$

So the conditions for the previous proposition [proposition: 16.348] are satisfied for h and we have that h is a local diffeomorphism of class C^n at 0. Hence there exist by [definition: 16.340] open sets U_0, V_0 such that

$$0 \in U_0 \subseteq W, 0 = h(0) \in V_0 \text{ and } h|_{U_0}: U_0 \rightarrow V_0 \text{ is a diffeomorphism of class } C^n \quad (16.514)$$

Define now

$$\mathcal{T}_3: X \rightarrow X \text{ by } \mathcal{T}_3(x) = x - x_0$$

then we have that for $x \in X$

$$\begin{aligned}
 (\mathcal{T}_3 \circ \mathcal{T}_1)(x) &= \mathcal{T}_3(\mathcal{T}_1(x)) = \mathcal{T}_3(x + x_0) = x + x_0 - x_0 = x = \text{Id}_X(x) \\
 (\mathcal{T}_1 \circ \mathcal{T}_3)(x) &= \mathcal{T}_1(\mathcal{T}_3(x)) = \mathcal{T}_1(x - x_0) = x - x_0 + x_0 = x = \text{Id}_X(x)
 \end{aligned}$$

proving that $\mathcal{T}_3 \circ \mathcal{T}_1 = \text{Id}_X = \mathcal{T}_3 \circ \mathcal{T}_1$ so that

$$\mathcal{T}_3, \mathcal{T}_1 \text{ are bijections and } (\mathcal{T}_3)^{-1} = \mathcal{T}_1 \wedge (\mathcal{T}_1)^{-1} = \mathcal{T}_3 \quad (16.515)$$

Let $U_{x_0} = (\mathcal{T}_3)^{-1}(U_0)$ then as $\mathcal{T}_3$ is a bijection we have that $\mathcal{T}_3(U_{x_0}) \stackrel{[\text{theorem: 2.57}]}{=} U_0$ hence

$$(\mathcal{T}_3)|_{U_{x_0}}: U_{x_0} \rightarrow U_0 \text{ is a bijection} \quad (16.516)$$

Further we have that

$$((\mathcal{T}_3)|_{U_{x_0}})^{-1} \stackrel{[\text{theorem: 2.100}]}{=} ((\mathcal{T}_3)^{-1})|_{\mathcal{T}_3(U_{x_0})} = ((\mathcal{T}_3)^{-1})|_{U_0} \stackrel{[\text{eq: 16.515}]}{=} (\mathcal{T}_1)|_{U_0}: U_0 \rightarrow U_{x_0} \quad (16.517)$$

We have also

$$U_{x_0} = (\mathcal{T}_3)^{-1}(U_0) \stackrel{[\text{eq: 16.514}]}{\subseteq} (\mathcal{T}_3)^{-1}(W) = ((\mathcal{T}_1)^{-1})^{-1}(W) \stackrel{[\text{theorem: 2.70}]}{=} \mathcal{T}_1(W) \stackrel{[\text{eq: 16.512}]}{\subseteq} U$$

or

$$U_{x_0} \subseteq U \quad (16.518)$$

By [eq: 16.514] $0 \in U_0$ so that $\mathcal{T}_3(x_0) = x_0 - x_0 = 0 \in U_0$ proving that

$$x_0 \in (\mathcal{T}_3)^{-1}(U_0) = U_{x_0} \quad (16.519)$$

Using [example: 16.184] it follows that $\mathcal{T}_3, \mathcal{T}_1 = (\mathcal{T}_3)^{-1}$ are of class C^∞ . So applying [theorem: 16.156] we have that

$$(\mathcal{T}_3)|_{U_{x_0}}: U_{x_0} \rightarrow X, ((\mathcal{T}_3)^{-1})|_{U_0}: U_0 \rightarrow X \text{ are of class } C^\infty \quad (16.520)$$

As $\mathcal{T}_3$ is of class C^∞ $\mathcal{T}_3$ is continuous so that

$$U_{x_0} = (\mathcal{T}_3)^{-1}(U_0) \text{ is a open set} \quad (16.521)$$

Hence by definition of a diffeomorphism [see definition: 16.331] it follows from [eqs: 16.516, 16.520, 16.521] that

$$(\mathcal{T}_3)|_{U_{x_0}}: U_{x_0} \rightarrow U_0 \text{ is a diffeomorphism of class } C^n \quad (16.522)$$

Next define

$$\mathcal{T}_4: Y \rightarrow Y \text{ by } \mathcal{T}_4(y) = y + f(x_0) \text{ and } \mathcal{T}_5: Y \rightarrow Y \text{ by } \mathcal{T}_5(y) = y - f(x_0)$$

then we have that for $y \in Y$

$$\begin{aligned}
 (\mathcal{T}_4 \circ \mathcal{T}_5)(y) &= \mathcal{T}_4(\mathcal{T}_5(y)) = \mathcal{T}_4(y - f(x_0)) = y - f(x_0) + f(x_0) = y = \text{Id}_Y(y) \\
 (\mathcal{T}_5 \circ \mathcal{T}_4)(y) &= \mathcal{T}_5(\mathcal{T}_4(y)) = \mathcal{T}_5(y + f(x_0)) = y + f(x_0) - f(x_0) = y = \text{Id}_Y(y)
 \end{aligned}$$

proving that $\mathcal{T}_4 \circ \mathcal{T}_5 = \text{Id}_Y = \mathcal{T}_5 \circ \mathcal{T}_4$ so that

$$\mathcal{T}_4, \mathcal{T}_5 \text{ are bijections and } (\mathcal{T}_4)^{-1} = \mathcal{T}_5 \wedge (\mathcal{T}_5)^{-1} = \mathcal{T}_4$$

Let $V_{f(x_0)} = \mathcal{T}_4(V_0)$ then we have

$$(\mathcal{T}_4)|_{V_0}: V_0 \rightarrow V_{f(x_0)} \text{ is a bijection} \quad (16.523)$$

further we have

$$V_{f(x_0)} = \mathcal{T}_4(V_0) = ((\mathcal{T}_5)^{-1})(V_0) \stackrel{[\text{theorem: 2.70}]}{=} (\mathcal{T}_5)^{-1}(V_0) \quad (16.524)$$

and

$$((\mathcal{T}_4)|_{V_0})^{-1} \stackrel{[\text{theorem: 2.100}]}{=} ((\mathcal{T}_4)^{-1})|_{\mathcal{T}_4(V_0)} = ((\mathcal{T}_4)^{-1})|_{V_{f(x_0)}}: V_{f(x_0)} \rightarrow V_0 \quad (16.525)$$

Using [example: 16.184] it follows that $\mathcal{T}_4, \mathcal{T}_5$ are of class C^∞ . So applying [theorem: 16.156] we have that

$$(\mathcal{T}_4)|_{V_0}: V_0 \rightarrow Y, ((\mathcal{T}_4)|_{V_0})^{-1}: V_{f(x_0)} \rightarrow Y \text{ are of class } C^\infty \quad (16.526)$$

As $\mathcal{T}_5$ is C^∞ of class, hence continuous, it follows that

$$V_{f(x_0)} \stackrel{[\text{eq: 16.524}]}{=} (\mathcal{T}_5)^{-1}(V_0) \text{ is a open set} \quad (16.527)$$

Hence by definition of a diffeomorphism [see definition: 16.331] it follows from [eq: 16.523, 16.526, 16.527] that

$$(\mathcal{T}_4)|_{V_0}: V_0 \rightarrow V_{f(x_0)} \text{ is a diffeomorphism of class } C^n \quad (16.528)$$

Define now

$$g = (\mathcal{T}_4)|_{V_0} \circ h|_{U_0} \circ (\mathcal{T}_3)|_{U_{x_0}}: U_{x_0} \rightarrow V_{f(x_0)}$$

then using [eqs 16.522, 16.514, 16.528] and [theorem: 16.339] it follows that

$$g: U_{x_0} \rightarrow V_{f(x_0)} \text{ is a diffeomorphism of class } C^n \quad (16.529)$$

Further we have for $x \in U_{x_0}$ that

$$\begin{aligned} g(x) &= ((\mathcal{T}_4)|_{V_0} \circ h|_{U_0} \circ (\mathcal{T}_3)|_{U_{x_0}})(x) \\ &= (\mathcal{T}_4)|_{V_0}(h|_{U_0}((\mathcal{T}_3)|_{U_{x_0}}(x))) \\ &= \mathcal{T}_4(h(\mathcal{T}_3(x))) \\ &= \mathcal{T}_4(h(x - x_0)) \\ &\stackrel{[\text{eq: 16.513}]}{=} \mathcal{T}_4((\mathcal{T}_2 \circ f \circ (\mathcal{T}_1)|_W)(x - x_0)) \\ &= \mathcal{T}_4((\mathcal{T}_2 \circ f \circ \mathcal{T}_1)(x - x_0)) \\ &= \mathcal{T}_4(\mathcal{T}_2(f(\mathcal{T}_1(x - x_0)))) \\ &= \mathcal{T}_4(\mathcal{T}_2(f(x - x_0 + x_0))) \\ &= \mathcal{T}_4(\mathcal{T}_2(f(x))) \\ &= \mathcal{T}_4(f(x) - f(x_0)) \\ &= f(x) - f(x_0) + f(x_0) \\ &= f(x) \\ &= f|_{U_{x_0}}(x) \end{aligned} \quad (16.530)$$

which proves that $g = f|_{U_{x_0}}$. Hence by [eq: 16.529] $f|_{U_{x_0}}: U_{x_0} \rightarrow V_{x_0}$ is a diffeomorphism of class C^n . As $x_0 \in U_{x_0}$ [see eq: 16.519], $f(x_0) \stackrel{x_0 \in U_{x_0}}{=} g(x_0) \in V_{f(x_0)}$, $U_{x_0} \subseteq U$ [see eq: 16.518] it follows from the definition of a local diffeomorphism [see definition: 16.340] that f is a local diffeomorphism of class C^n at x_0 . $\square$

Finally as the above conjecture [conjecture: 16.349] is the same as the Inverse Function Theorem [theorem: 16.341] we have proved the Inverse Function Theorem. $\square$

Applying the Inverse Function Theorem on the case where $X = Y = \mathbb{R}^n$ results in the following corollary.

Corollary 16.350. Let $n \in \mathbb{N}$, $\langle \mathbb{R}^n, \|\cdot\|_n \rangle$ be the normed space equipped with the maximum norm, U a open set in $\mathbb{R}^n$, $x_0 \in U$ and $f: U \rightarrow \mathbb{R}^n$ a function such that f is of class C^n and $\det\left(\left[\frac{\partial(f_1, \dots, f_n)}{\partial(1, \dots, n)}\right]_{x_0}\right) \neq 0$ [see theorem: 16.103] then f is a local diffeomorphism of class C^n at x_0 . In other words there exist a open set U_{x_0} in $\mathbb{R}^n$ with $x_0 \in U_{x_0} \subseteq U$ and a open set $V_{f(x_0)}$ in $\mathbb{R}^n$ with $f(x_0) \in V_{f(x_0)}$ such that

$$f|_{U_{x_0}}: U_{x_0} \rightarrow V_{f(x_0)} \text{ is a diffeomorphism of class } C^n$$

Proof. Using the canonical basis $E = \{e_1, \dots, e_n\}$ defined by $(e_i)_j = \delta_{i,j}$ we have by [theorem: 16.103] that

$$\mathcal{M}(D_{x_0}f; E, E) = \left[\frac{\partial(f_1, \dots, f_n)}{\partial(1, \dots, n)} \right]_{x_0}$$

As $\det\left(\left[\frac{\partial(f_1, \dots, f_n)}{\partial(1, \dots, n)}\right]_{x_0}\right) \neq 0$ it follows that $\det(\mathcal{M}(D_{x_0}f; E, E)) \neq 0$. So by [theorem: 11.360] $D_{x_0}f$ is a bijection. The rest of the proof follows by applying the inverse function theorem [theorem: 16.341]. $\square$

If f is a function between $\mathbb{R}$ and $\mathbb{R}$ then a consequence of the Inverse Function Theorem is the Inverse Function Theorem found in most calculus books.

Corollary 16.351. Let $\langle \mathbb{R}, \|\cdot\| \rangle$ be the normed space of the real numbers, $U \subseteq \mathbb{R}$ a open set, $x_0 \in U$ and $f: U \rightarrow \mathbb{R}$ is a function such that $\forall x \in U$ f has a derivative f'_x at x and the function

$$f': U \rightarrow \mathbb{R} \text{ defined by } f'(x) = f'_x$$

is continuous with $f'_{x_0} = f'(x_0) \neq 0$ then there exists open sets $U_{x_0}, V_{f(x_0)} \subseteq \mathbb{R}$ with $x_0 \in U_{x_0} \subseteq U$ and $f(x_0) \in V_{f(x_0)}$ such that:

1. $f|_{U_{x_0}}: U_{x_0} \rightarrow V_{f(x_0)}$ is a bijection
2. $\forall y \in V_{f(x_0)}$ we have that $(f|_{U_{x_0}})^{-1}$ has a derivative $((f|_{U_{x_0}})^{-1})'_y$ at y such that
$$((f|_{U_{x_0}})^{-1})': V_{f(x_0)} \rightarrow \mathbb{R} \text{ defined by } ((f|_{U_{x_0}})^{-1})'(y) = ((f|_{U_{x_0}})^{-1})'_y$$
is continuous.
3. $\forall x \in U_{x_0}$ we have that

$$f'_x \neq 0 \text{ and } ((f|_{U_{x_0}})^{-1})'_{f(x)} = \frac{1}{f'_x}$$

Proof. Using [theorem: 16.161] it follows that

$$f \text{ is } C^1 \text{ on } U$$

As $f'_{x_0} \neq 0$ we can define the function $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(t) = \frac{t}{f'_{x_0}}$ then we have $\forall t \in \mathbb{R}$

$$\begin{aligned} (D_{x_0}f \circ g)(t) &= D_{x_0}f(g(t)) \\ &= D_{x_0}f\left(\frac{t}{f'_{x_0}}\right) \\ &= \frac{1}{f'_{x_0}} \cdot D_{x_0}f(t) \\ &\stackrel{[\text{theorem: 16.64}]}{=} \frac{t}{f'_{x_0}} \cdot f'_{x_0} \\ &= t \\ &= \text{Id}_{\mathbb{R}}(t) \\ (g \circ D_{x_0}f)(t) &= g(D_{x_0}f(t)) \\ &\stackrel{[\text{theorem: 16.64}]}{=} g(t \cdot f'_{x_0}) \\ &= \frac{t \cdot f'_{x_0}}{f'_{x_0}} \\ &= t \\ &= \text{Id}_{\mathbb{R}}(t) \end{aligned}$$

proving that $D_{x_0}f \circ g = \text{Id}_{\mathbb{K}} = g \circ D_{x_0}f$. Hence

$$D_{x_0}f \text{ is a bijection and } (D_{x_0}f)^{-1} = g \quad (16.531)$$

So the conditions for the inverse function theorem [theorem: 16.341] are fulfilled and there exists open sets $U_{x_0}, V_{f(x_0)}$ with $x_0 \in U_{x_0} \subseteq U$, $f(x_0) \in V_{f(x_0)}$ such that

$$f|_{U_{x_0}}: U_{x_0} \rightarrow V_{f(x_0)} \text{ is a diffeomorphism of class } C^1$$

Hence we have that

$$(f|_{U_{x_0}})^{-1}: V_{f(x_0)} \rightarrow \mathbb{R} \text{ is } C^1 \text{ on } V_{f(x_0)}$$

so that by [theorem: 16.161] $\forall y \in V_{f(x_0)}$ we have that $(f|_{U_{x_0}})^{-1}$ has a derivative $((f|_{U_{x_0}})^{-1})'_y$ at y and the function

$$((f|_{U_{x_0}})^{-1})': V_{f(x_0)} \rightarrow \mathbb{R} \text{ with } ((f|_{U_{x_0}})^{-1})'(x) = ((f|_{U_{x_0}})^{-1})'_x$$

is continuous. If $x \in U_{x_0}$ then $f(x) = f|_{U_{x_0}}(x) \in V_{f(x_0)}$ and, as $f|_{U_{x_0}} \circ (f|_{U_{x_0}})^{-1} = \text{Id}_{V_{f(x_0)}}$, we have by the chain rule that

$$\begin{aligned} \text{Id}_{V_{f(x_0)}} & \stackrel{[\text{theorem: 16.53}]}{=} D_{f(x)}(\text{Id}_{V_{f(x_0)}}) \\ & = D_{f(x)}(f|_{U_{x_0}} \circ (f|_{U_{x_0}})^{-1}) \\ & \stackrel{[\text{theorem: 16.67}]}{=} D_{(f|_{U_{x_0}})^{-1}(f(x))} f|_{U_{x_0}} \circ D_{f(x)}((f|_{U_{x_0}})^{-1}) \\ & = D_x f|_{U_{x_0}} \circ D_{f(x)}((f|_{U_{x_0}})^{-1}) \end{aligned}$$

Hence

$$\begin{aligned} 1 & = \text{Id}_{V_{f(x_0)}}(1) \\ & = (D_x f|_{U_{x_0}} \circ D_{f(x)}((f|_{U_{x_0}})^{-1}))(1) \\ & = D_x f|_{U_{x_0}}(D_{f(x)}((f|_{U_{x_0}})^{-1})(1)) \\ & \stackrel{[\text{theorem: 16.64}]}{=} D_x f|_{U_{x_0}}(1 \cdot ((f|_{U_{x_0}})^{-1})'_{f(x)}) \\ & = ((f|_{U_{x_0}})^{-1})'_{f(x)} \cdot D_x f|_{U_{x_0}}(1) \\ & \stackrel{[\text{theorem: 16.64}]}{=} ((f|_{U_{x_0}})^{-1})'_{f(x)} \cdot (f|_{U_{x_0}})'_x \\ & \stackrel{[\text{theorem: 16.36}]}{=} ((f|_{U_{x_0}})^{-1})'_{f(x)} \cdot f'_x \end{aligned}$$

so that $f'_x \neq 0$ [otherwise we have $1 = 0$] and

$$((f|_{U_{x_0}})^{-1})'_{f(x)} = \frac{1}{f'_x} \quad \square$$

16.6.2 Implicit Function Theorem

The Inverse Function Theorem allows us to define a function implicit based on a relation that should hold. To illustrate this consider the function

$$f: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ defined by } f(x, y) = x^2 + y^2$$

then we can consider the circle $C = \{(x, y) | f(x, y) = 1\} = \{(x, y) | x^2 + y^2 = 1\}$ and a point $(x_0, y_0) \in C$ [hence $1 = f(x_0, y_0) = x_0^2 + y_0^2$]. Can we find a open set U containing x_0 and a function $g: U \rightarrow \mathbb{R}$ such that it's graph $\{(x, g(x)) | x \in U\}$ is equal to a arc $A \subseteq \{(x, y) | f(x, y) = 1\}$ [see figure: 16.4]. In other words does g satisfies $\forall x \in U$ that $f(x, g(x)) = 1$. If $2 \cdot y_0 = \partial_{(x_0, y_0), 2} f \neq 0$ [or equivalently $D_{(x_0, y_0), 2} f$ is a bijection] like in [figure: 16.4] the answer is clearly yes. However if $2 \cdot y_0 = \partial_{(x_0, y_0), 2} f = 0$ [or equivalently $D_{(x_0, y_0), 2} f$ is not a bijection] like in [figure: 16.5] the answer is no.

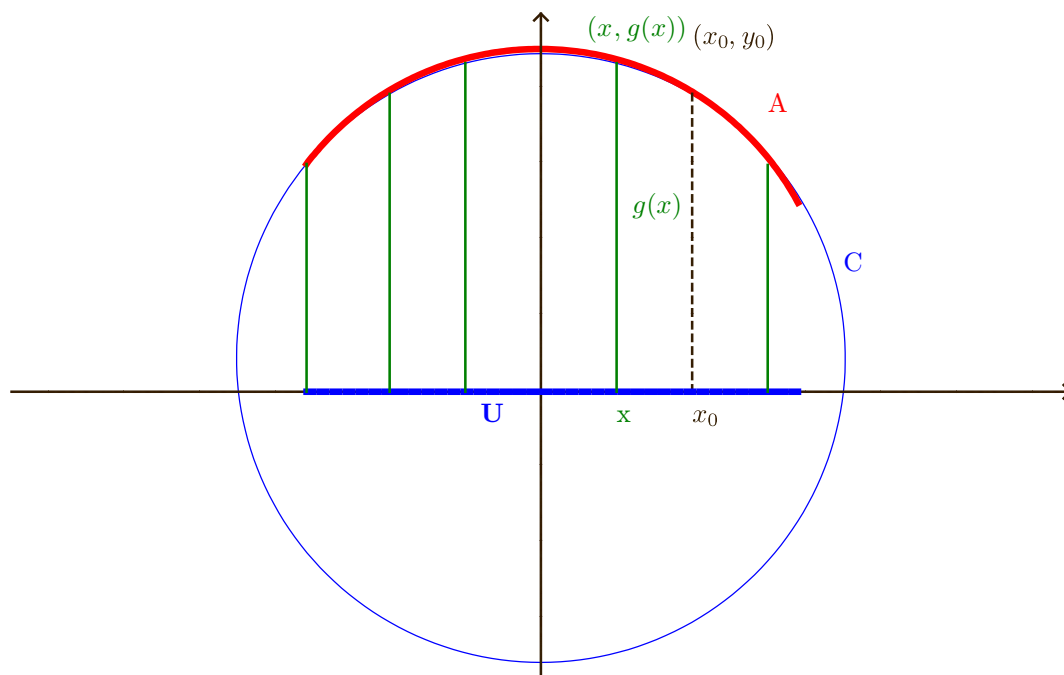


Figure 16.4.

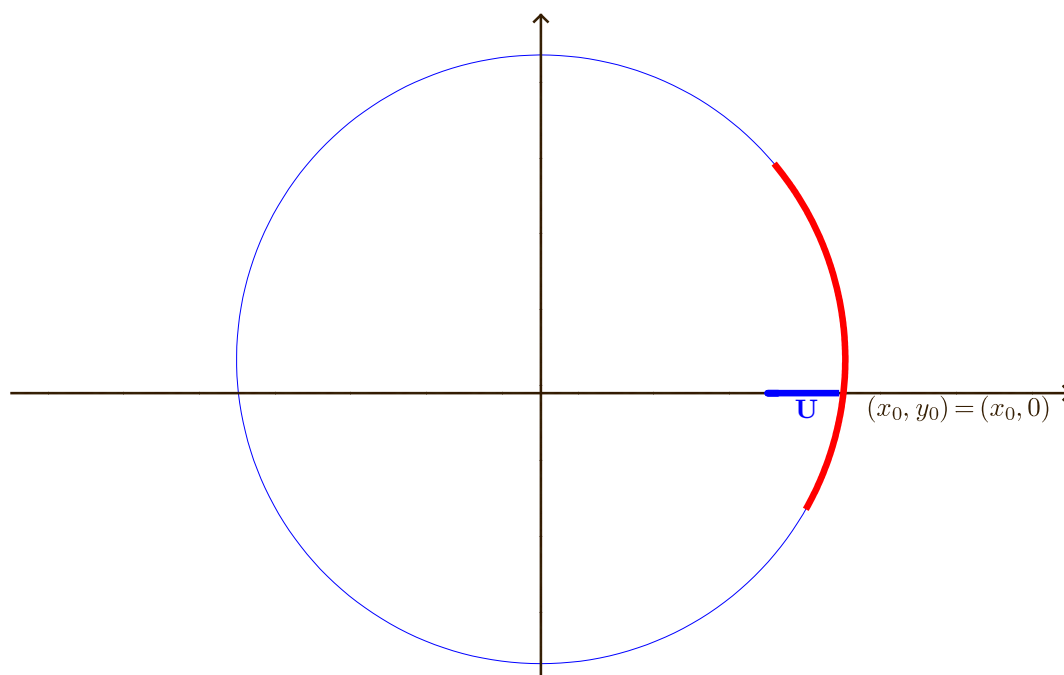


Figure 16.5.

The above discussion is about the implicit function theorem [theorem: 16.354] that is covered in most textbooks, there is however a more general version [see theorem: 16.353] that we will prove first and then derives the more specific version from the general version. First we need a little lemma.

Lemma 16.352. *Let X, Y, Z be vector spaces over the field $\mathbb{K}$. $L \in \text{Hom}(X, Z)$ and $H \in \text{Hom}(Y, Z)$ a linear isomorphism then*

$$T: X \cdot Y \rightarrow X \cdot Z \text{ defined by } T(x, y) = (x, L(x) + H(y))$$

is a linear isomorphism with inverse $S = T^{-1}$ defined by

$$S: X \cdot Z \rightarrow X \cdot Y \text{ by } S(x, z) = (x, -H^{-1}(L(x)) + H^{-1}(z)) = (x, (-H^{-1} \circ L)(x) + H^{-1}(z))$$

Proof. Let $(x_1, y_1), (x_2, y_2) \in X \cdot Y$ and $\alpha \in \mathbb{K}$ then

$$\begin{aligned} T((x_1, y_1) + \alpha \cdot (x_2, y_2)) &= T(x_1 + \alpha \cdot x_2, y_1 + \alpha \cdot y_2) \\ &= (x_1 + \alpha \cdot x_2, L(x_1 + \alpha \cdot x_2) + H(y_1 + \alpha \cdot y_2)) \\ &= (x_1 + \alpha \cdot x_2, L(x_1) + \alpha \cdot L(x_2) + H(y_1) + \alpha \cdot H(y_2)) \\ &= (x_1, L(x_1) + H(y_1)) + (\alpha \cdot x_2, \alpha \cdot L(x_2) + \alpha \cdot H(y_2)) \\ &= (x_1, L(x_1) + H(y_1)) + \alpha \cdot (x_2, L(x_2) + H(y_2)) \\ &= T(x_1, y_1) + \alpha \cdot T(x_2, y_2) \end{aligned}$$

proving that

T is a linear mapping

Let $x \in X$, $y \in Y$ and $z \in Z$ then

$$\begin{aligned} (T \circ S)(x, z) &= T(S(x, z)) \\ &= T(x, -H^{-1}(L(x)) + H^{-1}(z)) \\ &= (x, L(x) + H(-H^{-1}(L(x)) + H^{-1}(z))) \\ &= (x, L(x) - H(H^{-1}(L(x))) + H(H^{-1}(z))) \\ &= (x, L(x) - L(x) + z) \\ &= (x, z) \\ &= \text{Id}_{X \cdot Z}(x, z) \\ (S \circ T)(x, z) &= S(T(x, z)) \\ &= S(x, L(x) + H(z)) \\ &= (x, -H^{-1}(L(x)) + H^{-1}(L(x) + H(z))) \\ &= (x, -H^{-1}(L(x)) + H^{-1}(L(x)) + H^{-1}(H(z))) \\ &\stackrel{[\text{theorem: 11.178}]}{=} (x, z) \\ &= \text{Id}_{X \cdot Y}(x, z) \end{aligned}$$

proving that

$$S, T \text{ are bijections and } S = T^{-1}$$

□

There are actually two versions of the Implicit Function Theorems, we start with the more general one and use this to prove a simpler version.

Theorem 16.353. (Implicit Function Theorem (1)) Let $n \in \mathbb{N}$, $\langle X, |||_X \rangle$, $\langle Y, |||_Y \rangle$ and $\langle Z, |||_Z \rangle$ be Banach spaces, U a open set in X , V a open set in Y and $f: U \cdot V \rightarrow Z$ be a function that is C^n on the open set $U \cdot V$. So that using [theorems: 16.158, 16.277] and [definition: 16.264]

$$D_1 f: U \cdot V \rightarrow L(X, Z) \text{ defined by } D_1 f(x, y) = D_{(x, y), 1} f$$

$$D_2 f: U \cdot V \rightarrow L(Y, Z) \text{ defined by } D_2 f(x, y) = D_{(x, y), 2} f$$

are well defined and continuous. Then if there exist a $(x_0, y_0) \in U \cdot V$ such that $D_2 f(x_0, y_0)$ is a bijection then we have:

1. There exist a open set $W_0 \subseteq U \cdot Z$ and a open set $V_0 \subseteq V$ with $(x_0, f(x_0, y_0)) \in W_0$ such that there exist a map $g: W_0 \rightarrow V_0$ for which

$$a. \ g \text{ is } C^n \text{ on } W_0$$

$$b. \ g(x_0, f(x_0, y_0)) = y_0$$

$$c. \ \forall (x, z) \in W_0 \text{ we have } f(x, g(x, z)) = z$$

2. $\forall (x, z) \in W_0$ we have $D_2 f(x, g(x, z))$ is a bijection [so that $(D_2 f(x, g(x, z)))^{-1}$ exist].

3. As g is C^n on W_0 we have by [theorem: 16.158, 16.277] and [definition: 16.264] that

$D_1g: W_0 \rightarrow L(X, Y)$ defined by $D_1g(x) = D_{1,x}g$ is well defined and continuous

$D_2g: W_0 \rightarrow L(Z, Y)$ defined by $D_2g(x) = D_{2,x}g$ is well defined and continuous

Then we have for the above functions that $\forall (x, z) \in W_0$

$$a. D_1g(x, z) = -(D_2f(x, g(x, z)))^{-1} \circ D_1f(x, g(x, z))$$

$$b. D_2g(x, z) = (D_2f(x, g(x, z)))^{-1}$$

Proof. First, as we have to deal with different set products hence different projections, we introduce the following notations for the different projections:

$$\pi_1^{X \cdot Y}: X \cdot Y \rightarrow X \text{ defined by } \pi_1^{X \cdot Y}(x, y) = x$$

$$\pi_2^{X \cdot Y}: X \cdot Y \rightarrow Y \text{ defined by } \pi_2^{X \cdot Y}(x, y) = y$$

$$\pi_1^{X \cdot Z}: X \cdot Z \rightarrow X \text{ defined by } \pi_1^{X \cdot Z}(x, y) = x$$

$$\pi_2^{X \cdot Z}: X \cdot Z \rightarrow Z \text{ defined by } \pi_2^{X \cdot Z}(x, y) = y$$

Note that by [theorems: 16.185, 16.156] we have:

$$\pi_1^{X \cdot Y}, \pi_2^{X \cdot Y}, \pi_1^{X \cdot Z}, \pi_2^{X \cdot Z} \text{ and their restrictions to open sets are } C^\infty \text{ on their domains} \quad (16.532)$$

Define now the function

$$\varphi: U \cdot V \rightarrow X \cdot Z \text{ by } \varphi(x, y) \rightarrow (x, f(x, y)) \quad (16.533)$$

then $\forall (x, y) \in U \cdot V$ we have

$$\begin{aligned} (\pi_1^{X \cdot Z} \circ \varphi)(x, y) &= \pi_1^{X \cdot Z}(\varphi(x, y)) \\ &= \pi_1^{X \cdot Z}(x, f(x, y)) \\ &= x \\ &= \pi_1^{X \cdot Z}(x, y) \end{aligned}$$

proving, taking in account [eq: 16.532], that

$$\pi_1^{X \cdot Z} \circ \varphi = (\pi_1^{X \cdot Z})|_{U \cdot V} \text{ and } \pi_1^{X \cdot Z} \circ \varphi \text{ is } C^\infty \text{ on } U \cdot V \quad (16.534)$$

Further we have

$$\begin{aligned} (\pi_2^{X \cdot Z} \circ \varphi)(x, y) &= \pi_2^{X \cdot Z}(\varphi(x, y)) \\ &= \pi_2^{X \cdot Z}(x, f(x, y)) \\ &= f(x, y) \end{aligned}$$

proving that

$$\pi_2^{X \cdot Z} \circ \varphi = f \text{ so that } \pi_2^{X \cdot Z} \circ \varphi \text{ is } C^n \text{ on } U \cdot V \quad (16.535)$$

Using [theorem: 16.193] together with [eq: 16.534] and [eq: 16.535] proves that

$$\varphi \text{ is } C^n \text{ on } U \cdot V \quad (16.536)$$

So for $(x, y) \in U \cdot V$ we have that φ is Fréchet differentiable at (x, y) and we have

$$\begin{aligned} D_{(x,y)}\varphi &\stackrel{[\text{theorem: 16.101}]}{=} (D_{(x,y)}(\pi_1^{X \cdot Z} \circ \varphi), D_{(x,y)}(\pi_2^{X \cdot Z} \circ \varphi))_* \\ &\stackrel{[\text{eq: 16.534, 16.535}]}{=} (D_{(x,y)}(\pi_1^{X \cdot Z})|_{U \cdot V}, D_{(x,y)}f)_* \\ &\stackrel{[\text{theorem: 16.57}]}{=} (D_{(x,y)}\pi_1^{X \cdot Z}, D_{(x,y)}f)_* \\ &\stackrel{[\text{theorem: 16.185}]}{=} (\pi_1^{X \cdot Z}, D_{(x,y)}f)_* \\ &\stackrel{[\text{theorem: 16.83}]}{=} (\pi_1^{X \cdot Z}, (D_{(x,y)}1)f) \circ \pi_1^{X \cdot Z} + (D_{(x,y)}2f) \circ \pi_2^{X \cdot Z})_* \\ &= (\pi_1^{X \cdot Z}, D_1f(x, y) \circ \pi_1^{X \cdot Z} + D_2f(x, y) \circ \pi_2^{X \cdot Z})_* \end{aligned} \quad (16.537)$$

So if $h = (r, s) \in X \cdot Y$ we have that

$$\begin{aligned}
 D_{(x,y)}\varphi(r, s) &\stackrel{[\text{eq: 16.537}]}{=} (\pi_1^{X \cdot Z}, D_1f(x, y) \circ \pi_1^{X \cdot Z} + D_2f(x, y) \circ \pi_2^{X \cdot Z})_*(r, s) \\
 &= (\pi_1^{X \cdot Z}(r, s), (D_1f(x, y) \circ \pi_1^{X \cdot Z} + D_2f(x, y) \circ \pi_2^{X \cdot Z})(r, s)) \\
 &= (r, (D_1f(x, y) \circ \pi_1^{X \cdot Z})(r, s) + (D_2f(x, y) \circ \pi_2^{X \cdot Z})(r, s)) \\
 &= (r, D_1f(x, y)(\pi_1^{X \cdot Z}(r, s)) + D_2f(x, y)(\pi_2^{X \cdot Z}(r, s))) \\
 &= (r, D_1f(x, y)(r) + D_2f(x, y)(s))
 \end{aligned}$$

Hence we have:

$$\forall (x, y) \in U \cdot V, \forall (r, s) \in X \cdot Y \text{ we have } D_{(x,y)}\varphi(r, s) = (r, D_1f(x, y)(r) + D_2f(x, y)(s)) \quad (16.538)$$

Take $L = D_1f(x_0, y_0) \in L(X, Z) \subseteq \text{Hom}(X, Y)$ and $H = D_2f(x_0, y_0) \in L(Y, Z) \subseteq \text{Hom}(Y, Z)$ then by [eq: 16.538] we have

$$D_{(x_0, y_0)}\varphi(r, s) = (r, L(r) + H(s))$$

Further as $H = D_2f(x_0, y_0)$ is supposed to be a bijection it follows from [lemma: 16.352] that

$$D_{(x_0, y_0)}\varphi \text{ is a bijection} \quad (16.539)$$

with $\forall (r, s) \in X \cdot Y$

$$(D_{(x_0, y_0)}\varphi)^{-1}(r, s) = (r, -H^{-1}(L(r)) + H^{-1}(s))$$

so that

$$(D_{(x_0, y_0)}\varphi)^{-1}: X \cdot Z \rightarrow X \cdot Y$$

is defined by

$$(D_{(x_0, y_0)}\varphi)^{-1}(r, s) = (r, -((D_2f(x_0, y_0))^{-1} \circ D_1f(x_0, y_0))(r) + (D_2f(x_0, y_0))^{-1}(s)) \quad (16.540)$$

Using the Inverse Function Theorem [theorem: 16.341] on [eqs: 16.536, 16.539] it follows that there exist an open set W_1 in $X \cdot Y$ with $(x_0, y_0) \in W_1 \subseteq U \cdot V$ and an open set W_2 in $X \cdot Z$ with $\varphi(x_0, y_0) \in W_2$ such that

$$\varphi|_{W_1}: W_1 \rightarrow W_2 \text{ is a diffeomorphism of class } C^n \quad (16.541)$$

Using the definition of the product topology on $X \cdot Y$ [see: definition: 14.45] and [theorems: 14.51, 14.98] there exists an open set U_0 in X and an open set in V_0 in Y such that

$$(x_0, y_0) \in U_0 \cdot V_0 \subseteq W_1. \quad (16.542)$$

Using [theorem: 16.336] we have, if we take $W_0 = \varphi(U_0 \cdot V_0)$, that

$$\varphi|_{U_0 \cdot V_0} \stackrel{[\text{theorem: 2.92}]}{=} (\varphi|_{W_1})|_{U_0 \cdot V_0}: U_0 \cdot V_0 \rightarrow W_0 \text{ is a diffeomorphism of class } C^n \quad (16.543)$$

So

$$\forall (x, y) \in W_0 \text{ we have } \pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, y)) \in \pi_1^{X \cdot Y}(U_0 \cdot V_0) = U_0 \quad (16.544)$$

$$\pi_2^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, y)) \in \pi_2^{X \cdot Y}(U_0 \cdot V_0) = V_0 \quad (16.545)$$

Further if $(x, y) \in U_0 \cdot V_0$ then by [eq: 16.542] $(x, y) \in U_0 \cdot V_0 \subseteq W_1 \subseteq U \cdot V$ so that $x \in U$, as further $\varphi(x, y) = (x, f(x, y)) \in X \cdot Z$ it follows that $\varphi(x, y) \in U \cdot Z$ so that

$$W_0 = \varphi(U_0 \cdot V_0) \subseteq U \cdot Z \quad (16.546)$$

Further we have that

$$(x_0, f(x_0, y_0)) = \varphi(x_0, y_0) \in W_0 \quad (16.547)$$

If $y \in V_0$ then as $x_0 \in U_0$ we have $(x_0, y) \in U_0 \cdot V_0 \subseteq W_1 \subseteq U \cdot V$ so that $y \in V$ hence we have

$$V_0 \subseteq V \quad (16.548)$$

Using [eq: 16.543]

$$(\varphi|_{U_0 \cdot V_0})^{-1}: W_0 \rightarrow U_0 \cdot V_0 \text{ exists and is of class } C^n \quad (16.549)$$

If $(x, z) \in W_0$ then

$$\begin{aligned} (x, z) &= \varphi|_{U_0 \cdot V_0}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)) \\ &= \varphi((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)) \\ &= \varphi(\pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)), \pi_2^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z))) \\ &= (\pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)), f(\pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)), \pi_2^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)))) \end{aligned}$$

so that

$$x = \pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)) \in U_0 \text{ [see eq: 16.544]} \quad (16.550)$$

$$z = f(\pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)), \pi_2^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z))) \quad (16.551)$$

Define

$$g: W_0 \rightarrow V_0 = \pi_2^{X \cdot Y}(U_0 \cdot V_0) \text{ by } g = \pi_2^{X \cdot Y} \circ (\varphi|_{U_0 \cdot V_0})^{-1} \quad (16.552)$$

Then we have given $(x, z) \in W_0$ that

$$\begin{aligned} z &\stackrel{\text{[eq: 16.551]}}{=} f(\pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)), \pi_2^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z))) \\ &= f(\pi_1^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x, z)), g(x, z)) \\ &\stackrel{\text{[eq: 16.550]}}{=} f(x, g(x, z)) \end{aligned}$$

proving

$$\forall (x, z) \in W_0 \text{ that } f(x, g(x, z)) = z \quad (16.553)$$

As $(x_0, y_0) \in U_0 \cdot V_0$ $\varphi|_{U_0 \cdot V_0}(x_0, y_0) = \varphi(x_0, y_0) = (x_0, f(x_0, y_0))$ it follows that

$$(x_0, y_0) = (\varphi|_{U_0 \cdot V_0})^{-1}(x_0, f(x_0, y_0))$$

so that $y_0 = \pi_2^{X \cdot Y}((\varphi|_{U_0 \cdot V_0})^{-1}(x_0, f(x_0, y_0))) \stackrel{\text{[eq: 16.552]}}{=} g(x_0, f(x_0, y_0))$ proving that

$$g(x_0, f(x_0, y_0)) = y_0 \quad (16.554)$$

As $\pi_2^{X \cdot Y}$ is C^∞ on $X \cdot Y$ [see eq: 16.532] and by [eq: 16.549] $(\varphi|_{U_0 \cdot V_0})^{-1}$ is of class C^n it follows from the chain rule [theorem: 16.197] that

$$g \text{ is of class } C^n \quad (16.555)$$

As $\varphi|_{U_0 \cdot V_0}: U_0 \cdot V_0 \rightarrow W_0$ is a diffeomorphism of class C^n [see eq: 16.543] it follows from [theorem: 16.338] that

$$\forall (x, y) \in U_0 \cdot V_0 \text{ we have that } D_{(x, y)}\varphi \stackrel{\text{[theorem: 16.57]}}{=} D_{(x, y)}\varphi|_{U_0 \cdot V_0} \text{ is bijective} \quad (16.556)$$

So **part (1)** of the theorem is proved by [eqs: 16.552, 16.553, 16.554, 16.547, 16.546, 16.548 and 16.555].

Now for the next part. Take $r, s \in Y$ such that $D_2f(x, y)(r) = D_2f(x, y)(s)$ then we have

$$\begin{aligned} D_{(x, y)}\varphi(0, r) &\stackrel{\text{[eq: 16.538]}}{=} (0, D_1f(x, y)(0) + D_2f(x, y)(r)) \\ &= (0, D_1f(x, y)(0) + D_2f(x, y)(s)) \\ &\stackrel{\text{[eq: 16.538]}}{=} D\varphi(x, y)(0, s) \end{aligned}$$

so that by [eq: 16.556] $(0, r) = (0, s)$ from which it follows that $r = s$ proving that

$$D_2f(x, y) \text{ is injective} \quad (16.557)$$

If $z \in Z$ then as $D_{(x,y)}\varphi$ is a bijection [see eq: 16.556] there exist a $(r, s) \in X \cdot Y$ such that $D_{(x,y)}\varphi(r, s) = (0, z)$, hence by [eq: 16.538] we have $(r, D_1f(x, y)(r) + D_2f(x, y)(s)) = (0, z)$ so that $r = 0 \wedge D_1f(x, y)(r) + D_2f(x, y)(s) = z$. So we have $z = \underset{r=0}{D_1f(x, y)(0)} + D_2f(x, y)(s) = D_2f(x, y)(s)$ which proves that

$$D_2f(x, y) \text{ is surjective}$$

Combining this with [eq: 16.557] proves that

$$\forall (x, y) \in U_0 \cdot V_0 \text{ we have that } D_2f(x, y) \text{ is bijective}$$

Take now $(x, z) \in W_0$ then by [eq: 16.550] $x \in U_0$ and by [eq: 16.552] $g(x, z) \in V_0$ so that by the above we have

$$\forall (x, z) \in W_0 \text{ we have } D_2f(x, g(x, z)) \text{ is bijective so that } (D_2f(x, g(x, z)))^{-1} \text{ exist} \quad (16.558)$$

Which proves **part (2)** of the theorem.

For the next part. Define

$$\psi: W_0 \rightarrow X \cdot Y \text{ by } \psi(x, z) = (x, g(x, z))$$

Then we have for $(x, z) \in W_0$ that

$$\begin{aligned} (\pi_1^{X \cdot Y} \circ \psi)(x, z) &= \pi_1^{X \cdot Y}(\psi(x, z)) = \pi_1^{X \cdot Y}(x, g(x, z)) = x = \pi_1^{X \cdot Y}(x, z) \\ (\pi_2^{X \cdot Y} \circ \psi)(x, z) &= \pi_2^{X \cdot Y}(\psi(x, z)) = \pi_2^{X \cdot Y}(x, g(x, z)) = g(x, z) \end{aligned}$$

proving that

$$\pi_1^{X \cdot Y} \circ \psi = \pi_1^{X \cdot Y} \text{ and } \pi_2^{X \cdot Y} \circ \psi = g$$

As $\pi_1^{X \cdot Y}$ is of class C^∞ and by [eq: 16.555] g is of class C^n , we have using the above and [theorem: 16.193] that

$$\psi \text{ is } C^n \text{ on } W_0 \text{ hence by [theorem: 16.153] Fréchet differentiable on } W_0 \quad (16.559)$$

Consider

$$f \circ \psi: W_0 \rightarrow Z$$

then we have for $(x, z) \in W_0$ that

$$\begin{aligned} (f \circ \psi)(x, z) &= f(\psi(x, z)) \\ &= f(x, g(x, z)) \\ &\stackrel{[\text{eq: 16.553}]}{=} z \\ &= \pi_2^{X \cdot Z}(x, z) \end{aligned}$$

proving that

$$f \circ \psi = \pi_2^{X \cdot Z} \quad (16.560)$$

Further for $t \in X$ we have

$$((f \circ \psi) \circ I^{[1, (x, z)]})(t) = (f \circ \psi)(I^{[1, (x, z)]}(t)) \stackrel{[\text{definition: 16.70}]}{=} (f \circ \psi)(t, z) \stackrel{[\text{eq: 16.560}]}{=} z = C_z(t)$$

proving that

$$(f \circ \psi) \circ I^{[1, (x, z)]} = C_z$$

So that by [example: 16.54] $(f \circ \psi) \circ I^{[1, (x, z)]}$ is Fréchet differentiable at t which by [definition: 16.74] proves that $f \circ \psi$ has a 1-partial differential at t with

$$D_{(x, z), 1}(f \circ \psi) \stackrel{[\text{definition: 16.74}]}{=} D_x((f \circ \psi) \circ I^{[1, (x, z)]}) = D_x C_z \stackrel{[\text{example: 16.54}]}{=} 0 \in L(X, Z) \quad (16.561)$$

Also

$$\begin{aligned}
 ((\pi_1^{X \cdot Z} \circ \psi) \circ I^{[1, (x, z)]})(t) &= \pi_1^{X \cdot Z}(\psi(I^{[1, (x, z)]}(t))) \\
 &= \pi_1^{X \cdot Z}(\psi(t, z)) \\
 &= \pi_1^{X \cdot Z}(t, g(t, z)) \\
 &= t \\
 &= \text{Id}_X(t) \\
 ((\pi_2^{X \cdot Z} \circ \psi) \circ I^{[1, (x, z)]})(t) &= \pi_2^{X \cdot Z}(\psi(I^{[1, (x, z)]}(t))) \\
 &= \pi_2^{X \cdot Z}(\psi(t, z)) \\
 &= \pi_2^{X \cdot Z}(t, g(t, z)) \\
 &= g(t, z) \\
 &= g(I^{[1, (x, z)]}(t)) \\
 &= (g \circ I^{[1, (x, z)]})(t)
 \end{aligned}$$

so that

$$(\pi_1^{X \cdot Z} \circ \psi) \circ I^{[1, (x, z)]} = \text{Id}_X \text{ and } (\pi_2^{X \cdot Z} \circ \psi) \circ I^{[1, (x, z)]} = g \circ I^{[1, (x, z)]} \quad (16.562)$$

As g is of class C^n [see eq: 16.555] g is Fréchet differentiable on W_0 hence by [theorems: 16.158, 16.277] and [definition: 16.264] that

$D_1g: W_0 \rightarrow L(X, Y)$ defined by $D_1g(x) = D_{1,x}g$ is well defined and continuous

$D_2g: W_0 \rightarrow L(Z, Y)$ defined by $D_2g(x) = D_{2,x}g$ is well defined and continuous

Given $h \in X$ we have

$$\begin{aligned}
 D_{(x, z), 1}(f \circ \psi)(h) &\stackrel{[\text{eq: 16.559}]}{=} \\
 &\stackrel{[\text{theorem: 16.87}]}{=} \\
 (D_{\psi(x, z)}f \circ D_{(x, z), 1}\psi)(h) &= \\
 (D_{\psi(x, z)}f \circ (D_{(x, z)}(\psi \circ I^{[1, (x, z)]}))(h) &\stackrel{[\text{theorem: 16.101}]}{=} \\
 (D_{\psi(x, z)}f \circ (D_{(x, z)}(\pi_1^{X \cdot Z} \circ (\psi \circ I^{[1, (x, z)]})), D_{(x, z)}(\pi_2^{X \cdot Z} \circ (\psi \circ I^{[1, (x, z)]})))_{\star}(h) &\stackrel{[\text{eq: 16.562}]}{=} \\
 (D_{\psi(x, z)}f \circ (D_{(x, z)}\text{Id}_X, D_{(x, z)}(g \circ I^{[1, (x, z)]}))_{\star}(h) &\stackrel{[\text{example: 16.53}]}{=} \\
 (D_{\psi(x, z)}f \circ (\text{Id}_X, D_{(x, z)}(g \circ I^{[1, (x, z)]}))_{\star}(h) &= \\
 (D_{\psi(x, z)}f \circ (\text{Id}_X, D_{(x, z), 1}g))_{\star}(h) &= \\
 D_{\psi(x, z)}f((\text{Id}_X, D_{(x, z), 1}g)_{\star}(h)) &= \\
 D_{\psi(x, z)}f(h, D_{(x, z), 1}g(h)) &\stackrel{[\text{theorem: 16.83}]}{=} \\
 D_{\psi(x, z), 1}f(h) + D_{\psi(x, z), 2}f(D_{(x, z), 1}g(h)) &= \\
 D_{\psi(x, z), 1}f(h) + (D_{\psi(x, z), 2}f \circ D_{(x, z), 1}g)(h) &= \\
 (D_{\psi(x, z), 1}f + D_{\psi(x, z), 2}f \circ D_{(x, z), 1}g)(h) &= \\
 (D_1f(\psi(x, z)) + D_2f(\psi(x, z)) \circ D_1g(x, z))(h) &= \\
 (D_1f(x, g(x, z)) + D_2f(x, g(x, z)) \circ D_1g(x, z))(h)
 \end{aligned}$$

proving that

$$D_1f(x, g(x, z)) + D_2f(x, g(x, z)) \circ D_1g(x, z) = D_{(x, z), 1}(f \circ \psi) \stackrel{[\text{eq: 16.561}]}{=} 0$$

so that

$$D_2f(x, g(x, z)) \circ D_1g(x, z) = -D_1f(x, g(x, z))$$

as $(D_2f(x, g(x, z)))^{-1}$ exist [see eq: 16.558] and is linear by [theorem: 11.178] we have using the above

$$\forall (x, z) \in W_0 \quad D_1g(x, z) = -(D_2f(x, g(x, z)))^{-1} \circ D_1f(x, g(x, z)) \quad (16.563)$$

which proves **part (3.a)** of the theorem.

For the remaining part (3.b) of the theorem note that for $(x, z) \in W_0$ we have for $t \in Z$ that

$$(\pi_2^{X \cdot Z} \circ I^{[2, (x, z)]})(t) = \pi_2^{X \cdot Z}(I^{[2, (x, z)]}(t)) = \pi_2^{X \cdot Z}(x, t) = t = \text{Id}_Z(t)$$

so that

$$\pi_2^{X \cdot Z} \circ I^{[2, (x, z)]} = \text{Id}_Z \quad (16.564)$$

Further

$$\begin{aligned} D_{(x, z), 2}(f \circ \psi) &\stackrel{[\text{eq: 16.560}]}{=} D_{(x, z), 2}(\pi_2^{X \cdot Z}) \\ &= D_{(x, z)}(\pi_2^{X \cdot Z} \circ I^{[2, (x, z)]}) \\ &\stackrel{[\text{eq: 16.564}]}{=} D_{(x, z)}\text{Id}_Z \\ &\stackrel{[\text{eq: 16.53}]}{=} \text{Id}_Z \end{aligned} \quad (16.565)$$

Further we have

$$\begin{aligned} (\pi_1^{X \cdot Z} \circ \psi \circ I^{[2, (x, z)]})(t) &= \pi_1^{X \cdot Z}(\psi(I^{[2, (x, z)]}(t))) \\ &= \pi_1^{X \cdot Z}(\psi(x, t)) \\ &= \pi_1^{X \cdot Z}(x, g(x, t)) \\ &= x \\ &= C_x(t) \\ (\pi_2^{X \cdot Z} \circ \psi \circ I^{[2, (x, z)]})(t) &= \pi_2^{X \cdot Z}(\psi(I^{[2, (x, z)]}(t))) \\ &= \pi_2^{X \cdot Z}(\psi(x, t)) \\ &= \pi_2^{X \cdot Z}(x, g(x, t)) \\ &= g(x, t) \\ &= g(I^{[2, (x, z)]}(t)) \\ &= (g \circ I^{[2, (x, z)]})(t) \end{aligned}$$

proving that

$$\pi_1^{X \cdot Z} \circ \psi \circ I^{[2, (x, z)]} = C_x \wedge \pi_2^{X \cdot Z} \circ \psi \circ I^{[2, (x, z)]} = g \circ I^{[2, (x, z)]} \quad (16.566)$$

We use the above to calculate $D_{(x, z), 2}(f \circ \psi)$ in another way, let $h \in Z$ then

$$\begin{aligned} D_{(x, z), 2}(f \circ \psi)(h) &\stackrel{[\text{eq: 16.559}]}{=} \\ &\stackrel{[\text{theorem: 16.87}]}{=} (D_{\psi(x, z)}f \circ D_{(x, z), 2}\psi)(h) \\ &\stackrel{[\text{theorem: 16.101}]}{=} (D_{\psi(x, z)}f \circ D_{(x, z)}(\psi \circ I^{[2, (x, z)]}))(h) \\ &\stackrel{[\text{eq: 16.566}]}{=} (D_{\psi(x, z)}f \circ (D_{(x, z)}(\pi_1^{X \cdot Z} \circ (\psi \circ I^{[2, (x, z)]})), D_{(x, z)}(\pi_2^{X \cdot Z} \circ (\psi \circ I^{[2, (x, z)]})))_*(h) \\ &\stackrel{[\text{example: 16.54}]}{=} (D_{\psi(x, z)}f \circ (D_{(x, z)}C_x, D_{(x, z)}(g \circ I^{[2, (x, z)]})))_*(h) \\ &\stackrel{[\text{theorem: 16.83}]}{=} (D_{\psi(x, z)}f \circ (0, D_{(x, z)}(g \circ I^{[2, (x, z)]})))_*(h) \\ &\stackrel{[\text{theorem: 16.83}]}{=} (D_{\psi(x, z)}f \circ (0, D_{(x, z), 2}g)_*)(h) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_{\psi(x, z)}f((0, D_{(x, z), 2}g)_*(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_{\psi(x, z)}f(0(h), D_{(x, z), 2}g(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_{\psi(x, z)}f(0, D_{(x, z), 2}g(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_{\psi(x, z), 1}f(0) + D_{\psi(x, z), 2}f(D_{(x, z), 2}g(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_{\psi(x, z), 2}f(D_{(x, z), 2}g(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_2f(\psi(x, z)(D_2g(x, z)(h))) \\ &\stackrel{[\text{theorem: 16.83}]}{=} D_2f(x, g(x, z))(D_2g(x, z)(h)) \\ &\stackrel{[\text{theorem: 16.83}]}{=} (D_2f(x, g(x, z)) \circ D_2g(x, z))(h) \end{aligned}$$

proving that

$$D_2f(x, g(x, z)) \circ D_2g(x, z) = D_{(x,z),2}(f \circ \psi) \stackrel{[\text{eq: 16.565}]}{=} \text{Id}_Z$$

Hence we have, as $(D_2f(x, g(x, z)))^{-1}$ exist [see eq: 16.558], that

$$\forall (x, z) \in W_0 \text{ we have } D_2g(x, z) = (D_2f(x, g(x, z)))^{-1} \quad (16.567)$$

which proves **part (3.b)** of the theorem. $\square$

Next we have a simpler version of the Implicit Function Theorem that you find in most books.

Theorem 16.354. *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ and $\langle Z, \|\cdot\|_Z \rangle$ be Banach spaces, U a open set in X , V a open set in Y and $f: U \cdot V \rightarrow Z$ a function that is of class C^n . Using [theorems: 16.158, 16.277] and [definition: 16.264] we have that*

$$D_1f: U \cdot V \rightarrow L(X, Z) \text{ defined by } D_1f(x, y) = D_{(x,y),1}f$$

$$D_2f: U \cdot V \rightarrow L(Y, Z) \text{ defined by } D_2f(x, y) = D_{(x,y),2}f$$

are well defined and continuous. If $\exists (x_0, y_0) \in U \cdot V$ such that $D_2f(x_0, y_0)$ is bijective then there exist a open set W with $x_0 \in W \subseteq U$ and a function $g: W \rightarrow Y$ that is C^n on W such that

$$g(x_0) = y_0 \text{ and } \forall x \in W \text{ we have } f(x, g(x)) = f(x_0, y_0)$$

and

$$\forall x \in W \ (D_2f(x, g(x)))^{-1} \text{ exist and } Dg(x) = -(D_2f(x, g(x)))^{-1} \circ D_1f(x, g(x))$$

where $Dg: W \rightarrow L(X, Y)$ is defined by $Dg(x) = D_xg$.

Proof. Using the implicit function theorem [theorem: 16.353] there exist a open set $W_0 \subseteq U \cdot Z$ and a open set $V_0 \subseteq V$ with $(x_0, y_0) \in W_0$ such that there exist a map $h: W_0 \rightarrow V_0$ that is C^n on W_0 and satisfies

$$(x_0, f(x_0, y_0)) \in W_0 \quad (16.568)$$

$$h(x_0, f(x_0, y_0)) = y_0 \quad (16.569)$$

and $\forall (x, z) \in W_0$

$$f(x, h(x, z)) = z \quad (16.570)$$

$$D_1h(x, z) = -(D_2f(x, h(x, z)))^{-1} \circ D_1f(x, h(x, z)) \quad (16.571)$$

$$D_2h(x, z) = (D_2f(x, h(x, z)))^{-1} \quad (16.572)$$

where

$$D_1h: W_0 \rightarrow L(X, Y) \text{ is defined by } D_1h(x, z) = D_{(x,z),1}h$$

$$D_2h: W_0 \rightarrow L(Z, Y) \text{ is defined by } D_2h(x, z) = D_{(x,z),2}h$$

Using the definition of the product topology on $X \cdot Z$ [see: definition: 14.45] and [theorems: 14.51, 14.98] there exists a open set W in X and a open set O in Z such that

$$(x_0, f(x_0, y_0)) \in W \cdot O \subseteq W_0 \subseteq U \cdot Z$$

From the above it follows that $f(x_0, y_0) \in O$ so if $x \in W$ then by the above $(x, f(x_0, y_0)) \in W \cdot O \subseteq W_0$ which is the domain of h so the following function is well defined

$$g: W \rightarrow V \text{ where } g(x) = h(x, f(x_0, y_0)) \quad (16.573)$$

then we have

$$g(x_0) = h(x_0, f(x_0, y_0)) \stackrel{[\text{eq: 16.569}]}{=} y_0 \quad (16.574)$$

Further for $x \in W$ we have $(x, f(x_0, y_0)) \in W \cdot O \subseteq W_0$ so that

$$f(x, g(x)) = f(x, h(x, f(x_0, y_0))) \stackrel{[\text{eq: 16.570}]}{=} f(x_0, y_0)$$

hence

$$\forall x \in W \text{ we have } f(x, g(x)) = f(x_0, y_0) \quad (16.575)$$

Further we have for $t \in W$ that

$$\begin{aligned} (h \circ I^{[1, (x, f(x_0, y_0))]})(t) &= h(I^{[1, (x, f(x_0, y_0))]}(t)) \\ &= h(t, f(x_0, y_0)) \\ &\stackrel{[\text{eq: 16.573}]}{=} g(t) \end{aligned}$$

proving that

$$g = h \circ I^{[1, (x, f(x_0, y_0))]}$$

Using the chain rule [see theorem: 16.197], the fact that h is of class C^n and $I^{[1, (x, f(x_0, y_0))]}$ is of class C^∞ [see example: 16.183] it follows that

$$g \text{ is of class } C^n \quad (16.576)$$

so that

$$Dg: W \rightarrow L(X, Y) \text{ defined by } Dg(x) = D_x g$$

is well defined. Further we have for $x \in W$ that

$$\begin{aligned} Dg(x) &= D_x g \\ &= D_x (h \circ I^{[1, (x, f(x_0, y_0))]})(x) \\ &\stackrel{[\text{definition: 16.74}]}{=} D_{(x, f(x_0, y_0))} h \\ &= D_1 h(x, f(x_0, y_0)) \\ &\stackrel{[\text{eq: 16.571}]}{=} -(D_2 f(x, h(x, f(x_0, y_0))))^{-1} \circ D_1 f(x, h(x, f(x_0, y_0))) \\ &\stackrel{[\text{eq: 16.573}]}{=} -(D_2 f(x, g(x)))^{-1} \circ D_1 f(x, g(x)) \end{aligned} \quad (16.577)$$

The theorem is then proved by [eqs: 16.576, 16.574, 16.575 and 16.577]. $\square$

16.7 Taylor Expansion

If $A \subseteq \mathbb{R}$, $\langle X, \|\cdot\| \rangle$ a normed space and $f: X \rightarrow Y$, $x \in A$ and $f: A \rightarrow X$ a function that has a derivative f'_x at x then we have for $\varepsilon \in \mathbb{R}^+$ that there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in A_x$ with $|h| < \delta$ we have that

$$\|f(x+h) - f(x) - h \cdot f'_x\| \leq \varepsilon \cdot |h|$$

Hence if we take $\rho = \min(\delta, 1)$ then we have that $\forall h \in A_x$ with $|h| < \delta$ that

$$\|f(x+h) - f(x) - h \cdot f'_x\| < \varepsilon$$

Take $y \in U \cap B_{\|\cdot\|}(x, \rho)$ then $x + (y - x) \in U \Rightarrow (y - x) \in U_x$ and $|y - x| < \rho$ so that

$$\|f(y) - (f(x) + (y - x) \cdot f'_x)\| = \|f(x + (y - x)) - f(x) - (y - x) \cdot f'_x\| < \varepsilon$$

from which it follows that the polynomial of first order $f(x) + (y - x) \cdot f'_x$ is a approximation of f in a neighborhood of x . This suggest that if f has n -times derivatives at x that we can find a better approximation of f using a polynomial of higher order. This is the idea of a Taylor expansion.

Theorem 16.355. (Taylor's Theorem (I)) *Let $\langle X, \|\cdot\| \rangle$ be a normed space, $a, b \in \mathbb{R}$ with $a < b$, $x_0 \in]a, b[$ and $f: [a, b] \rightarrow X$ a function that is n -times differentiable at x_0 then we have*

$$\begin{aligned} \forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall h \in [a, b]_{x_0} \text{ with } |h| < \delta \text{ we have} \\ \left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \leq \varepsilon \cdot |h|^n \end{aligned}$$

Note 16.356. As f is n -times differentiable at x we have by [theorem: 16.171] that f has n -times derivatives at x so that by [theorem: 16.169] $f_{x_0}^{(k)}$ exist for $k \in \{1, \dots, n\}$.

Proof. We prove this by induction on n , so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } a, b \in \mathbb{R} \text{ with } a < b, x_0 \in]a, b[\text{ and } f: [a, b] \rightarrow X \text{ a function that is } n\text{-times differentiable at } x_0 \text{ then } \forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall h \in [a, b]_{x_0} \text{ with } |h| < \delta \text{ we have } \left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \leq \varepsilon \cdot |h|^n \right\}$$

then we have:

$1 \in S$. As f is 1-times differentiable at x_0 it follows from [theorem: 16.140] that f is Fréchet differentiable at x_0 . Using then [theorem: 16.64] it follows that f has a derivative f'_{x_0} at x_0 . Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 16.23] there exists a $\delta \in \mathbb{R}^+$ such that $\forall h \in [a, b]_{x_0}$ with $|h| < \delta$ we have

$$\|f(x_0 + h) - f(x_0) - h \cdot f'_{x_0}\| \leq \varepsilon \cdot |h|$$

hence

$$\begin{aligned} \left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^1 \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| &= \left\| f(x_0 + h) - f(x_0) - \frac{h^1}{1!} \cdot f_{x_0}^{(1)} \right\| \\ &= \|f(x_0 + h) - f(x_0) - h \cdot f'_{x_0}\| \\ &\leq \varepsilon \cdot |h| \\ &= \varepsilon \cdot |h|^1 \end{aligned}$$

which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. As f is $(n+1)$ -times differentiable at x_0 and $2 \in \{1, \dots, n+1\}$ it follows from [theorem: 16.138] that f is 2-times differentiable at x_0 so that by [theorem: 16.140] $D^{[1]}f$ is Fréchet differentiable at x_0 . Hence using the definition of Fréchet differentiability [see definition: 16.46] $x_0 \in (\mathfrak{D}_f^1)^\circ$ the domain of $D^{[1]}f$ so there exist a $\delta_1 \in \mathbb{R}^+$ such that $x_0 \in B_{||}(x_0, \delta_1) \subseteq \mathfrak{D}_f^1$. Let $y \in B_{||}(x_0, \delta_1) \subseteq \mathfrak{D}_f^1$ then by [definitions: 16.136] and [theorem: 16.140] f is Fréchet differentiable at y so that by [theorem: 16.64] f has derivative at y . To summarize we have that

$$\forall y \in B_{||}(x_0, \delta_1) \text{ we have that } f'_y \text{ exist} \quad (16.578)$$

From the fact that f is $(n+1)$ -times differentiable at x_0 it follows from [theorem: 16.171] that f has $(n+1)$ -times derivatives at x_0 . Then $\forall m \in \{2, \dots, n+1\}$ we have by [theorem: 16.177] that $f^{(1)}$ has a $(m-1)$ -derivative at x_0 with $f_{x_0}^{(m)} = (f^{(1)})_{x_0}^{(m-1)}$. Hence using [definition: 16.162] we have that

$$f' \text{ has a } (m-1)\text{-times derivative at } x_0 \text{ with } f_{x_0}^{(m)} = (f')_{x_0}^{(m-1)} \quad (16.579)$$

In particular as $2 \leq n+1$ it follows that

$$f' \text{ has a } n\text{-time derivative } (f')_{x_0}^{(n)} \text{ at } x_0 \text{ with } (f')_{x_0}^n = f_{x_0}^{(n+1)}$$

Let $\varepsilon \in \mathbb{R}^+$ then as $n \in S$ it follows from the above that there exist a $\delta_2 \in \mathbb{R}^+$ such that $\forall h \in [a, b]_{x_0}$ with $|h| < \delta_2$ we have

$$\left\| f'(x_0 + h) - f'(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot (f')_{x_0}^{(k)} \right\| \leq (n+1) \cdot \varepsilon \cdot |h|^n \quad (16.580)$$

Take $\delta = \min\left(\frac{\delta_1}{2}, \delta_2\right)$ then we have for $h \in [a, b]_{x_0}$ with $|h| < \delta$ the following possibilities to consider:

$0 \leq h$. Then $|h|^n = h^n$ and [eq: 16.580] becomes

$$\left\| f'(x_0 + h) - f'(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot (f')_{x_0}^{(k)} \right\| \leq (n+1) \cdot \varepsilon \cdot h^n \quad (16.581)$$

Further if $y \in [0, h]$ then $|x_0 + y - x_0| \leq h \leq \delta \leq \frac{\delta_1}{2} < \delta_1$ it follows from [eq: 16.578] that

$$\forall h \in [0, h] \quad f \text{ has a derivative at } x_0 + y$$

So if we define

$$g: [0, h] \rightarrow X \text{ by } g(y) = f(x_0 + y) - f(x_0) - f'(x_0) \cdot \frac{y}{1!} - \sum_{k=2}^{n+1} \frac{y^k}{k!} \cdot f_{x_0}^{(k)}$$

then we have by [example: 16.30] and [theorems: 16.40, 16.108, 16.68, 16.34] that

$$\forall y \in [0, h] \quad g \text{ has a derivative at } y$$

with

$$\begin{aligned} g'_y &= 1 \cdot f'_{x_0+y} - 0 - f'(x_0) \cdot \frac{1}{1!} - \sum_{k=2}^{n+1} \frac{k \cdot y^{k-1}}{k!} \cdot f_{x_0}^{(k)} \\ &= f'(x_0 + y) - f'(x_0) - \sum_{k=2}^{n+1} \frac{y^{k-1}}{(k-1)!} \cdot f_{x_0}^{(k)} \\ &\stackrel{[\text{eq: 16.579}]}{=} f'(x_0 + y) - f'(x_0) - \sum_{k=2}^{n+1} \frac{y^{k-1}}{(k-1)!} \cdot (f')_{x_0}^{(k-1)} \\ &= f'(x_0 + y) - f'(x_0) - \sum_{k=1}^n \frac{y^k}{k!} \cdot (f')_{x_0}^{(k)} \end{aligned}$$

so that

$$\forall y \in [0, h] \text{ we have } g'_y = f'(x_0 + y) - f'(x_0) - \sum_{k=1}^n \frac{y^k}{k!} \cdot (f')_{x_0}^{(k)} \quad (16.582)$$

Define further

$$\varphi: [0, \delta] \rightarrow \mathbb{R} \text{ by } \varphi(y) = \varepsilon \cdot y^{n+1}$$

then we have by [theorems: 16.108, 16.34] together with the fact that $\forall y \in [0, y]$ y is a limit point of $[0, h]$ [see theorem: 16.10] that

$$\forall y \in [0, h] \quad \varphi \text{ has a derivative at } y \text{ with } \varphi'_y = (n+1) \cdot \varepsilon \cdot y^n \quad (16.583)$$

Hence we have $\forall y \in [0, h]$ that

$$\begin{aligned} \|g'_y\| &\stackrel{[\text{eq: 16.582}]}{=} \left\| f'(x_0 + y) - f'(x_0) - \sum_{k=1}^n \frac{y^k}{k!} \cdot (f')_{x_0}^{(k)} \right\| \\ &\stackrel{[\text{eq: 16.581}]}{\leq} (n+1) \cdot \varepsilon \cdot y^n \\ &\stackrel{[\text{eq: 16.583}]}{=} \varphi'_y \end{aligned}$$

so that

$$\forall y \in [0, h] \text{ we have } \|g'_y\| \leq \varphi'_y$$

Using the Mean Value Theorem [theorem: 16.234] it follows then that

$$\|g(h) - g(0)\| \leq \varphi(h) - \varphi(0) \quad (16.584)$$

so that

$$\begin{aligned} & \left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^{n+1} \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| = \\ & \left\| f(x_0 + h) - f(x_0) - f'(x_0) \cdot \frac{h}{1!} - \sum_{k=2}^{n+1} \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| = \\ & \|g(h)\| \\ & \|g(h) - 0\| = \\ & \left\| g(h) - \left(f(x_0 + 0) - f(x_0) - f'(x_0) \cdot \frac{0}{1!} - \sum_{k=2}^{n+1} \frac{0^k}{k!} \cdot f_{x_0}^{(k)} \right) \right\| = \\ & \|g(h) - g(0)\| \stackrel{[\text{eq: 16.584}]}{\leq} \\ & \varphi(h) - \varphi(0) = \\ & \varepsilon \cdot h^{n+1} - \varepsilon \cdot 0^{n+1} = \\ & \varepsilon \cdot h^{n+1} \stackrel{0 \leq h}{=} \\ & \varepsilon \cdot |h|^{n+1} \end{aligned}$$

which prove that

$$\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^{n+1} \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \leq \varepsilon \cdot |h|^{n+1}$$

$h \leq 0$. Then $|h|^n = (-h)^n = (-1)^n \cdot h^n$ so that [eq: 16.580] becomes

$$\left\| f'(x_0 + h) - f'(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot (f')_{x_0}^{(k)} \right\| \leq (n+1) \cdot \varepsilon \cdot (-1)^n \cdot h^n \quad (16.585)$$

Further if $y \in [h, 0]$ $|x_0 + y - x_0| \leq h \leq \delta \leq \frac{\delta_1}{2} < \delta_1$ it follows from [eq: 16.578] that

$$\forall h \in [h, 0] \quad f \text{ has a derivative at } x_0 + y$$

so if we define

$$g: [h, 0] \rightarrow X \text{ by } g(y) = f(x_0 + y) - f(x_0) - f'(x_0) \cdot \frac{y}{1!} - \sum_{k=2}^{n+1} \frac{y^k}{k!} \cdot f_{x_0}^{(k)}$$

then by [example: 16.30] and [theorems: 16.40, 16.108, 16.68, 16.34] it follows that

$$\forall y \in [h, 0] \quad g \text{ has a derivative at } y$$

with

$$\begin{aligned} g'_y &= 1 \cdot f'_{x_0+y} - 0 - f'(x_0) \cdot \frac{1}{1!} - \sum_{k=2}^{n+1} \frac{k \cdot y^{k-1}}{k!} \cdot f_{x_0}^{(k)} \\ &= f'(x_0 + y) - f'(x_0) - \sum_{k=2}^{n+1} \frac{y^{k-1}}{(k-1)!} \cdot f_{x_0}^{(k)} \\ &\stackrel{[\text{eq: 16.579}]}{=} f'(x_0 + y) - f'(x_0) - \sum_{k=2}^{n+1} \frac{y^{k-1}}{(k-1)!} \cdot (f')_{x_0}^{(k-1)} \\ &= f'(x_0 + y) - f'(x_0) - \sum_{k=1}^n \frac{y^k}{k!} \cdot (f')_{x_0}^{(k)} \end{aligned}$$

so that

$$\forall y \in [h, 0] \text{ we have } g'_y = f'(x_0 + y) - f'(x_0) - \sum_{k=1}^n \frac{y^k}{k!} \cdot (f')_{x_0}^{(k)} \quad (16.586)$$

Define now

$$\varphi: [h, 0] \rightarrow \mathbb{R} \text{ by } \varphi(y) = \varepsilon \cdot (-1)^n \cdot y^{n+1}$$

then we have by [theorems: 16.108, 16.34] that

$$\forall y \in [h, 0] \varphi \text{ has a derivative at } y \text{ with } \varphi'_y = (n+1) \cdot \varepsilon \cdot (-1)^n \cdot y^n \quad (16.587)$$

Hence we have $\forall h \in [y, 0]$ that

$$\begin{aligned} \|g'_y\| &\stackrel{[\text{eq: 16.586}]}{=} \left\| f'(x_0 + y) - f'(x_0) - \sum_{k=1}^n \frac{y^k}{k!} \cdot (f')_{x_0}^{(k)} \right\| \\ &\stackrel{[\text{eq: 16.585}]}{\leq} (n+1) \cdot \varepsilon \cdot (-1)^n \cdot y^n \\ &\stackrel{[\text{eq: 16.587}]}{=} \varphi'_y \end{aligned}$$

so that

$$\forall y \in [h, 0] \text{ we have } \|g'_y\| \leq \varphi'_y$$

Using the Mean Value Theorem [theorem: 16.234] it follows then that

$$\|g(0) - g(h)\| \leq \varphi(0) - \varphi(h) \quad (16.588)$$

so that

$$\begin{aligned} &\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^{n+1} \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \\ &\left\| f(x_0 + h) - f(x_0) - f'(x_0) \cdot \frac{h}{1!} - \sum_{k=2}^{n+1} \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| = \\ &\|g(h)\| = \\ &\|g(h) - 0\| = \\ &\left\| g(h) - \left(f(x_0 + 0) - f(x_0) - f'(x_0) \cdot \frac{0}{1!} - \sum_{k=2}^{n+1} \frac{0^k}{k!} \cdot f_{x_0}^{(k)} \right) \right\| = \\ &\|g(h) - g(0)\| = \\ &\|g(0) - g(h)\| \stackrel{[\text{eq: 16.588}]}{\leq} \\ &\varphi(0) - \varphi(h) = \\ &\varepsilon \cdot (-1)^n \cdot 0^{n+1} - \varepsilon \cdot (-1)^n \cdot h^{n+1} = \\ &- \varepsilon \cdot (-1)^n \cdot h^{n+1} \leq \\ &\varepsilon \cdot |h|^{n+1} \end{aligned}$$

which prove that

$$\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^{n+1} \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \leq \varepsilon \cdot |h|^{n+1}$$

□

We have the following variants on the first form of Taylor's theorem.

Corollary 16.357. (Taylor Theorem(I.1)) Let $\langle X, ||| \rangle$ be a normed space, $a, b \in \mathbb{R}$ with $a < b$, $x_0 \in]a, b[$ and $f: [a, b] \rightarrow X$ a function that is n -times differentiable at x_0 then

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall x \in [a, b]_{x_0} \text{ with } |x - x_0| < \delta \text{ we have}$$

$$\left\| f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} \right\| \leq \varepsilon \cdot |x - x_0|^n$$

Note 16.358. As f is n -times differentiable at x we have by [theorem: 16.171] that f has n -times derivatives at x so that by [theorem: 16.169] $f_{x_0}^{(k)}$ exist for $k \in \{1 \dots n\}$.

Proof. Let $\varepsilon \in \mathbb{R}^+$. Using [theorem: 16.355] there exist $\delta \in \mathbb{R}^+$ such that $\forall h \in [a, b]_x$ with $|h| < \delta$ we have

$$\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \leq \varepsilon \cdot |h|^n$$

Let $x \in [a, b]$ such that $|x - x_0| < \delta$ then for $h = x - x_0$ we have $|h| < \delta$ and $x_0 + h = x \in [a, b] \Rightarrow h \in [a, b]_{x_0}$ so that by the above we have

$$\begin{aligned} \left\| f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} \right\| &= \left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{h^k}{k!} \cdot f_{x_0}^{(k)} \right\| \\ &\leq \varepsilon \cdot |x - x_0|^n \\ &\square \end{aligned}$$

The above theorem can be formulated in the language of $\text{ord}(|(x - x_0)^n|)$
 $\xrightarrow[A]{x \rightarrow x_0}$

Definition 16.359. Let $\langle X, |||_X \rangle, \langle Y, |||_Y \rangle$ be two normed spaces, $A \subseteq X$, $x_0 \in A$ and $f: A \rightarrow Y$ and $g: A \rightarrow [0, \infty]$ be two functions then we define

1. $f(x) = \text{ord}_{\xrightarrow[A]{x \rightarrow x_0}}(g(x))$ iff $\forall \varepsilon \in \mathbb{R}^+$ there exist at $\delta \in \mathbb{R}^+$ such that $\forall x \in A$ with $0 < \|x - x_0\|_X < \delta$ we have

$$\|f(x)\|_Y \leq \varepsilon \cdot g(x)$$

2. If $h: A \rightarrow Y$ is another function then

$$f(x) = h(x) + \text{ord}_{\xrightarrow[A]{x \rightarrow x_0}}(g(x))$$

iff

$$f(x) - h(x) = \text{ord}_{\xrightarrow[A]{x \rightarrow x_0}}(g(x))$$

Using this definition we can rephrase [theorem: 16.357] as follows.

Theorem 16.360. (Taylor's Theorem (I.2)) Let $\langle X, ||| \rangle$ be a normed space, $a, b \in \mathbb{R}$ with $a < b$, $x_0 \in]a, b[$ and $f: [a, b] \rightarrow X$ a function that is a n -times differentiable at x_0 then

$$f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f^{(n)}(x_0) = \text{ord}_{\xrightarrow[A]{x \rightarrow x_0}}(|(x - x_0)^n|)$$

or in other words

$$f(x) = f(x_0) + \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f^{(n)}(x_0) + \text{ord}_{\xrightarrow[A]{x \rightarrow x_0}}(|(x - x_0)^n|)$$

Proof. This follows from [definition: 16.359] and Taylor's Theorem (I.1) [theorem: 16.357] $\square$

There is a link between $f(x) = o(|x - x_0|^n)$ and a limit of the quotient $\frac{f(x)}{(x - x_0)^n}$.

Theorem 16.361. Let $\langle X, \|\cdot\| \rangle$ be a normed space, $n \in \mathbb{N}$, $a, b \in \mathbb{R}$ with $a < b$, $x_0 \in [a, b]$ and $f: [a, b] \rightarrow X$ a function then

$$f = \operatorname{ord}_{x \xrightarrow{[a,b]} x_0} (|x - x_0|^n) \Leftrightarrow \lim_{x \xrightarrow{[a,b]} x_0} \frac{f(x)}{(x - x_0)^n} = 0$$

Note 16.362. By [theorem: 16.10] x_0 is a limit point of $[a, b]$ so that $\lim_{x \xrightarrow{A} x_0} \frac{f(x)}{(x - x_0)^n}$ is well defined.

Note 16.363. In essence the theorem says that in the limit $\|f(x)\|$ decreases faster than $|x - x_0|^n$

Proof.

$\Rightarrow$. Let $\varepsilon \in \mathbb{R}$. As $f = \operatorname{ord}_{x \xrightarrow{A} x_0} (|x - x_0|^n)$ there exist a $\delta \in \mathbb{R}^+$ such that $\forall x \in A$ with $0 < |x - x_0| < \delta$ we have

$$\|f(x)\| \leq \frac{\varepsilon}{2} \cdot |x - x_0|^n \quad (16.589)$$

As $0 < |x - x_0|$ we have that $|x - x_0| \neq 0$ so that we can divide by $|x - x_0|^n$, $(x - x_0)^n$ so that

$$\begin{aligned} \left\| \frac{f(x)}{(x - x_0)^n} - 0 \right\| &= \left\| \frac{f(x)}{(x - x_0)^n} \right\| \\ &= \left| \frac{1}{(x - x_0)^n} \right| \cdot \|f(x)\| \\ &\stackrel{[\text{eq: 16.589}]}{\leq} \frac{1}{|x - x_0|^n} \cdot \frac{\varepsilon}{2} \cdot |x - x_0|^n \\ &= \frac{\varepsilon}{2} \\ &< \varepsilon \end{aligned}$$

which proves that $\lim_{x \xrightarrow{A} x_0} \frac{f(x)}{(x - x_0)^n} = 0$.

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$. As $\lim_{x \xrightarrow{A} x_0} \frac{f(x)}{(x - x_0)^n} = 0$ there exist a $\delta \in \mathbb{R}^+$ such that $\forall x \in A$ with $0 < |x - x_0| < \delta$ we have

$$\left\| \frac{f(x)}{(x - x_0)^n} \right\| = \left\| \frac{f(x)}{(x - x_0)^n} - 0 \right\| < \varepsilon$$

so that

$$\|f(x)\| = \left\| \frac{(x - x_0)^n}{(x - x_0)^n} \cdot f(x) \right\| = |x - x_0|^n \cdot \left\| \frac{f(x)}{(x - x_0)^n} \right\| < \varepsilon \cdot |x - x_0|^n$$

which proves that $f = \operatorname{ord}_{x \xrightarrow{A} x_0} (|x - x_0|^n)$ □

As a consequence of the above theorem we have the following:

Theorem 16.364. (Taylor's Theorem (I.3)) Let $\langle X, \|\cdot\| \rangle$ be a normed space, $a, b \in \mathbb{R}$ with $a < b$, $x_0 \in [a, b]$ and $f: [a, b] \rightarrow X$ a function that is a n -times differentiable at x_0 then we have

$$\lim_{x \xrightarrow{[a,b]} x_0} \frac{f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)}}{(x - x_0)} = 0$$

Proof. Using [theorem: 16.360] we have that

$$f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f^{(n)}(x_0) = \operatorname{ord}_{x \xrightarrow{A} x_0} (|x - x_0|^n)$$

so that by [theorem: 16.361]

$$\lim_{x \xrightarrow{[a,b]} x_0} \frac{f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)}}{(x - x_0)} = 0 \quad \square$$

Next we want to extend Taylor's theorem to functions between two Banach spaces. The main problem lies in the terms of the form $\frac{h^k}{k!} \cdot f_{x_0}^{[k]}$ in the Taylor's expansion because the power h^n is not defined in a general normed space. Further derivatives are not defined for functions between general normed spaces. So we have to use terms of the form $\frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right)$ and $\frac{1}{k!} \cdot D_{x_0}^{[k]} \left(\underbrace{x+h, \dots, x+h}_k \right)$. To get started we first need a definition and some lemma's.

Definition 16.365. Let X be a set, $n \in \mathbb{N}$ then for $x \in X$ and $h \in X$ we define for $p, q \in \mathbb{N}_0$ with $p+q=n$

$$x^p h^q \in X^n \text{ where } (x^p h^q)_i = \begin{cases} x & \text{if } i \in \{1, \dots, p\} \\ h & \text{if } i \in \{p+1, \dots, q\} \end{cases}$$

Example 16.366. Let $n=3$ then we have

$$\begin{aligned} x^0 h^3 &= (h, h, h) \\ x^1 h^2 &= (x, h, h) \\ x^2 h^1 &= (x, x, h) \\ x^3 h^0 &= (x, x, x) \end{aligned}$$

Lemma 16.367. Let X be a set, $n \in \mathbb{N}$ and $x, h \in X$ then we have for p, q with $p+q=n$ that

1. We have for

$$y = (y_1, \dots, y_{n+1}) \text{ defined by } y_i = \begin{cases} x & \text{if } i = 1 \\ (x^p h^q)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

that

$$y = x^{p+1} h^q$$

2. There **exist** a $\sigma \in \mathcal{P}_{n+1}$ such that for

$$y = (y_1, \dots, y_{n+1}) \text{ defined by } y_i = \begin{cases} h & \text{if } i = 1 \\ (x^p h^q)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

we have

$$(y_{\sigma(1)}, \dots, y_{\sigma(n+1)}) = x^p h^{q+1}$$

3. $\forall i \in \{1, \dots, n\} (x^n h^0)_i = x$

Proof.

1. Let $p, q \in \mathbb{N}_0$ with $p+q=n$ then $(p+1)+q=n+1$ and for $i \in \{1, \dots, n+1\}$ we have either:

$i \in \{1, \dots, p+1\}$. Then for i we have either:

$i = 1$. Then $y_i = x$ and as $i \in \{1, \dots, p+1\}$ $(x^{p+1} h^q)_i = x$ it follows that

$$y_i = (x^{p+1} h^q)_i$$

$i \in \{2, \dots, p+1\}$. Then $i-1 \in \{1, \dots, p\}$ so that $y_i = (x^p h^q)_{i-1} = x$, further as $i \in \{2, \dots, p+1\} \subseteq \{1, \dots, p+1\}$ we have that $(x^{p+1} h^q)_i = x$. Hence

$$y_i = (x^{p+1} h^q)_i$$

$i \in \{p+2, \dots, n+1\}$. Then $i-1 \in \{p+1, \dots, n\}$ so that $y_i = (x^p h^q)_{i-1} = h$, further as $i \in \{(p+1)+1, \dots, n+1\}$ $(x^{p+1} h^q)_i = h$. Hence

$$y_i = (x^{p+1} h^q)_i$$

As in all possible cases we have $y_i = (x^{p+1} h^q)_i$ it follows that

$$y = (x^{p+1} h^q)$$

2. Let $p, q \in \mathbb{N}_0$ with $p + q = n$ then $p + (q + 1) = n + 1$ and for p we have either:

$p = 0$. Then for $i \in \{1, \dots, n + 1\}$ we have either:

$i = 1$. Then $y_i = h$, further as $1 \in \{1, \dots, n + 1\} = \{p + 1, \dots, n + 1\}$ that $(x^p h^{q+1})_i = h$.
Hence

$$y_i = (x^p h^{q+1})_i$$

$i \in \{2, \dots, n + 1\}$. Then $i - 1 \in \{1, \dots, n\} = \{p + 1, \dots, n\}$ so that $y_i = (x^p h^q)_{i-1} = h$. Further as $i \in \{2, \dots, n + 1\} = \{(p + 1) + 1, \dots, n + 1\}$ $(x^p h^{q+1})_i = h$. Hence

$$y_i = (x^p h^{q+1})_i$$

So we have that $y = x^p h^{q+1}$, take now $\sigma = \text{Id}_{\{1, \dots, n+1\}} \in P_{n+1}$ then we have

$$(y_{\sigma(1)}, \dots, y_{\sigma(n)}) = (y_1, \dots, y_n) = x^p h^{q+1}$$

$p \neq 0$. Define $\sigma = \left(1 \leftrightarrow_{n+1} p + 1\right) \in P_{n+1}$ then we have for $i \in \{1, \dots, n + 1\}$ either:

$i = 1$. Then $y_{\sigma(i)} = y_{\sigma(1)} = y_{p+1} =_{p+1 \in \{p+1, \dots, n+1\}} (x^p h^{q+1})_i$, hence

$$y_{\sigma(i)} = (x^p h^{q+1})_i$$

$i \in \{2, \dots, p - 1\}$. Then $y_{\sigma(i)} = y_i = (x^p h^q)_{i-1}$ $i-1 \in \{1, \dots, p-1\} \subseteq \{1, \dots, p\}$ x . As $(x^p h^q)_{i-1} =_{i-1 \in \{1, \dots, p\}} x$ it follows that

$$y_{\sigma(i)} = (x^p h^{q+1})_i$$

$i = p + 1$. Then $y_{\sigma(i)} = y_{\sigma(p+1)} = y_1 = h$. As $i = p + 1 \in \{p + 1, \dots, n + 1\}$ $(x^p h^{q+1})_i = h$ so that

$$y_{\sigma(i)} = (x^p h^{q+1})_i$$

$i \in \{p + 2, \dots, n + 1\}$. Then $y_{\sigma(i)} = y_i = (x^p h^q)_{i-1}$ $i-1 \in \{p+1, \dots, n\}$ h , further as $\{p + 2, \dots, n + 1\} \subseteq \{p + 1, \dots, n + 1\}$ we have $(x^p h^{q+1})_i = h$ which proves that

$$y_{\sigma(i)} = (x^p h^{q+1})_i$$

Hence we have

$$(y_{\sigma(1)}, \dots, y_{\sigma(n+1)}) = x^p h^{q+1}$$

3. Let $i \in \{1, \dots, n\}$ then if $q = 0$ we have, as $p + q = n$, that $p = n$ so that $i \in \{1, \dots, n\} = \{1, \dots, p\}$ from which it follows that $(x^p h^q)_i = x$. $\square$

Lemma 16.368. Let $n \in \mathbb{N}$, $L \in L_{n+1}(X; Y)$ such that $\forall \sigma \in P_{n+1}$ and $\forall (x_1, \dots, x_{n+1}) \in X^n$ $L(x_1 : \dots : x_{n+1}) = L(x_{\sigma(1)} : \dots : x_{\sigma(n+1)})$ then we have $\forall x \in X$, $\forall \tau \in P_n$, $\forall (y_1, \dots, y_n) \in X^n$ that $L(x)(y_1 : \dots : y_n) = L(x)(y_{\tau(1)} : \dots : y_{\tau(n)})$.

Proof. Let $x \in X$, $\tau \in P_n$ and $(y_1, \dots, y_n) \in X^n$. Define now

$$(x_1, \dots, x_{n+1}) \in X^{n+1} \text{ by } x_i = \begin{cases} x & \text{if } i = 1 \\ y_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

$$\sigma: \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\} \text{ by } \sigma(i) = \begin{cases} 1 & \text{if } i = 1 \\ \tau(i-1) + 1 & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

then we have for σ :

injectivity. Let $i, j \in \{1, \dots, n+1\}$ such that $\sigma(i) = \sigma(j)$ then for i, j we have either:

$i = 1$. Assume that $j \in \{2, \dots, n+1\}$ then

$$\sigma(j) = \tau(j-1) + 1 \in_{\tau: \{1, \dots, n\} \rightarrow \{1, \dots, n\}} \{2, \dots, n+1\}$$

so that $\sigma(j) \neq 1 = \sigma(1) = \sigma(i)$ and we reach a contradiction. So we must have $i = 1 = j$.
 $j = 1$. Assume that $i \in \{2, \dots, n+1\}$ then

$$\sigma(i) = \tau(i-1) + 1 \underset{\tau: \{1, \dots, n\} \rightarrow \{1, \dots, n\}}{\in} \{2, \dots, n+1\}$$

so that $\sigma(i) \neq 1 = \sigma(1) = \sigma(j)$ and we reach a contradiction. So we must have $i = 1 = j$.
 $i, j \in \{2, \dots, n+1\}$. Then $\tau(i-1) + 1 = \sigma(i) = \sigma(j) = \tau(j-1) + 1$ so that $\tau(i-1) = \tau(j-1)$. As τ is injective it follows that $i-1 = j-1$ hence $i = j$.

surjectivity. Let $k \in \{1, \dots, n+1\}$ then we have either:

$k = 1$. Then $k = \sigma(1)$

$k \in \{2, \dots, n+1\}$. Then $k-1 \in \{1, \dots, n\}$ and as τ is surjection there exist a $j \in \{1, \dots, n\}$ such that $\tau(j) = k-1$. Hence for $i = j+1 \in \{2, \dots, n+1\}$ we have

$$\sigma(i) = \tau(i-1) + 1 = \tau(j) + 1 = k$$

So it follows that

$$\sigma \in \mathcal{P}_{n+1}$$

Further

$$\begin{aligned} L(x)(y_1: \dots: y_n) &= L(x_1)(x_2: \dots: x_{n+1}) \\ &= L(x_1: \dots: x_{n+1}) \\ &= L(x_{\sigma(1)}: \dots: x_{\sigma(n+1)}) \\ &= L(x_{\sigma(1)})(x_{\sigma(2)}: \dots: x_{\sigma(n+1)}) \\ &= L(x_1)(x_{\sigma(2)}: \dots: x_{\sigma(n+1)}) \\ &= L(x)(x_{\tau(1)+1}: \dots: x_{\tau(n)+1}) \\ &= L(x)(y_{\tau(1)}: \dots: y_{\tau(n)}) \end{aligned}$$

proving that

$$L(x)(y_1: \dots: y_n) = L(x)(y_{\tau(1)}: \dots: y_{\tau(n)}) \quad \square$$

Lemma 16.369. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $L \in L_n(X; Y)$ such that $\forall \sigma \in P_n$ and $\forall (x_1, \dots, x_n) \in X^n$ $L(x_1: \dots: x_n) = L(x_{\sigma(1)}: \dots: x_{\sigma(n)})$ then we have for $x, h \in X$ that there exist a $k \in \mathbb{N}_0$ and a $\{(p_i, q_i)\}_{i \in \{1, \dots, k\}}$ with $\forall i \in \{1, \dots, k\}$ $2 \leq q_i$, $n = p_i + q_i$ such that

$$L\left(\underbrace{x+h: \dots: x+h}_n\right) = T_1 + T_2 + T_3$$

where

$$\begin{aligned} T_1 &= L((x^n h^0)_1: \dots: (x^n h^0)_n) \\ T_2 &= n \cdot L((x^{n-1} h^1)_1: \dots: (x^{n-1} h^1)_n) \\ T_3 &= \sum_{i \in \{1, \dots, k\}} L((x^{p_i} h^{q_i})_1: \dots: (x^{p_i} h^{q_i})_n) \end{aligned}$$

Proof. We prove this by induction so let

$S = \left\{ n \in \mathbb{N} \mid \text{If } L \in L_n(X; Y) \text{ satisfies } \forall \sigma \in P_n, \forall y \in X^n \ L(y_{\sigma(1)}: \dots: y_{\sigma(n)}) = L(y_1: \dots: y_n) \text{ then} \right.$

there exist a $\{(p_i, q_i)\}_{i \in \{1, \dots, k\}}$ with $k \in \mathbb{N}_0$ with $\forall i \in \{1, \dots, k\}$ $2 \leq q_i \wedge n = p_i + q_i$ such that

$L\left(\underbrace{x+h: \dots: x+h}_n\right) = T_1 + T_2 + T_3$ where $T_1 = L((x^n h^0)_1: \dots: (x^n h^0)_n)$, $T_2 = n \cdot L((x^{n-1} h^1)_1: \dots: (x^{n-1} h^1)_n)$ and $T_3 = \sum_{i \in \{1, \dots, k\}} L((x^{p_i} h^{q_i})_1: \dots: (x^{p_i} h^{q_i})_n) \left. \right\}$

then we have:

$1 \in \mathbf{S}$. Let $L \in L_1(X; Y) = L(X, Y)$. Take $k=0$ then $\{1, \dots, k\} = \emptyset$ so that for $\{x^{p_i}h^{q_i}\}_{i \in \emptyset}$ we have vacuously that $\forall i \in \emptyset$ $2 \leq q_i$ $n = p_i + q_i$. Further

$$\sum_{i \in \{1, \dots, k\}} L((x^{p_i}h^{q_i})_1: \dots (x^{p_i}h^{q_i})_n) = \sum_{i \in \emptyset} L((x^{p_i}h^{q_i})_1: \dots (x^{p_i}h^{q_i})_n) = 0$$

For $x, h \in X$ we have

$$\begin{aligned} L\left(\underbrace{x+h: \dots : x+h}_1\right) &= L(x+h) \\ &= L(x) + L(h) \\ &= L(x) + L(h) + 0 \\ &= T_1 + T_2 + T_3 \end{aligned}$$

where

$$\begin{aligned} T_1 &= L(x) = L((x^1h^0)_1) = L\left(\underbrace{(x^1h^0)_1: \dots : (x^1h^0)_1}_1\right) \\ T_2 &= 1 \cdot L(h) = 1 \cdot L((x^0h^1)_1) = 1 \cdot L((x^0h^1)_1: \dots : L(x^0h^1)_1) \\ T_3 &= \sum_{(q,p) \in \emptyset} L((x^p h^q)_1: \dots (x^p h^q)_n) = 0 \end{aligned}$$

which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $x, h \in X$ and $L \in L_{n+1}(X; Y) = L(X, L_n(X; Y))$ such that $\forall \sigma \in P_{n+1}$, and $\forall (x_1, \dots, x_n) \in X^n$ we have $L(x_1: \dots : x_{n+1}) = L(x_{\sigma(1)}: \dots : x_{\sigma(n+1)})$. As $L \in L(X, L_n(X; Y))$ it follows from [lemma: 16.368] that $\forall \sigma \in P_n$, $\forall (y_1, \dots, y_n) \in X^n$

$$L(x+h)(y_{\sigma(1)}, \dots, y_{\sigma(n)}) = L(x+h)(y_1, \dots, y_n)$$

Hence as $n \in S$ it follows that there exist a

$$\{(p_i, q_i)\}_{i \in \{1, \dots, k\}} \text{ with } \forall i \in \{1, \dots, k\} \ 2 \leq q_i \wedge n = p_i + q_i$$

such that

$$L(x+h)\left(\underbrace{x+h: \dots : x+h}_n\right) = T_1 + T_2 + T_3 \quad (16.590)$$

where

$$\begin{aligned} T_1 &= L(x+h)((x^n h^0)_1: \dots : (x^n h^0)_n) \\ T_2 &= n \cdot L(x+h)((x^{n-1} h^1)_1: \dots : (x^{n-1} h^1)_n) \\ T_3 &= \sum_{i \in \{1, \dots, k\}} L(x+h)((x^{p_i} h^{q_i})_1: \dots : (x^{p_i} h^{q_i})_n) \end{aligned}$$

Define

$$u = (u_1, \dots, u_{n+1}) \text{ by } u_i = \begin{cases} x & \text{if } i = 1 \\ (x^n h^0)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

then we have

$$\begin{aligned} L(x)((x^n h^0)_2: \dots : (x^n h^0)_n) &= L(u_1)(u_2, \dots, u_{n+1}) \\ &= L(u_1: \dots : u_{n+1}) \\ &\stackrel{[\text{lemma: 16.367(1)}]}{=} L((x^{p+1} h^0)_1: \dots : (x^{p+1} h^0)_{n+1}) \end{aligned} \quad (16.591)$$

Define

$$v = (v_1, \dots, v_{n+1}) \text{ by } v_i = \begin{cases} h & \text{if } i = 1 \\ (x^n h^0)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

then by [lemma: 16.367(2)] there exist a $\sigma \in P_{n+1}$ so that

$$(v_{\sigma(1)}, \dots, v_{\sigma(n+1)}) = x^n h^1 = x^{(n+1)-1} h^1$$

hence

$$\begin{aligned}
 L(h)((x^n h^0)_2 : \dots : (x^n h^0)_n) &= L(v_1)(v_2 : \dots : v_{n+1}) \\
 &= L(v_1 : \dots : v_{n+1}) \\
 &= L(v_{\sigma(1)} : \dots : v_{\sigma(n+1)}) \\
 &= L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1})
 \end{aligned} \tag{16.592}$$

Define

$$w = (w_1, \dots, w_n) \text{ by } w_i = \begin{cases} x & \text{if } i = 1 \\ (x^{n-1} h^1)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

then we have

$$\begin{aligned}
 L(x)((x^{n-1} h^1)_2 : \dots : (x^{n-1} h^1)_n) &= \\
 L(w_1)(w_2 : \dots : w_{n+1}) &= \\
 L(w_1 : \dots : w_{n+1}) &\stackrel{[\text{lemma: 16.367(1)}]}{=} \\
 L((x^n h^1)_1 : \dots : (x^n h^1)_{n+1}) &= \\
 L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) &
 \end{aligned} \tag{16.593}$$

Define

$$y = (y_1, \dots, y_{n+1}) \text{ by } y_i = \begin{cases} h & \text{if } i = 1 \\ (x^{n-1} h^1)_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

then by [lemma: 16.367(2)] there exist a $\tau \in P_{n+1}$ such that

$$(y_{\tau(1)}, \dots, y_{\tau(n+1)}) = x^{n-1} h^2$$

Then

$$\begin{aligned}
 L(h)((x^{n-1} h^1)_2 : \dots : (x^{n-1} h^1)_n) &= L(y_1)(y_2 : \dots : y_{n+1}) \\
 &= L(y_1 : \dots : y_{n+1}) \\
 &= L(y_{\tau(1)} : \dots : y_{\tau(n+1)}) \\
 &= L((x^{n-1} h^2)_1 : \dots : (x^{n-1} h^2)_{n+1})
 \end{aligned} \tag{16.594}$$

Let $i \in \{1, \dots, k\}$ and define

$$r^i = (r_1^i, \dots, r_{n+1}^i) \text{ by } r_j^i = \begin{cases} x & \text{if } j = 1 \\ (x^{p_i} h^{q_i})_{j-1} & \text{if } j \in \{2, \dots, n+1\} \end{cases}$$

then we have

$$\begin{aligned}
 L(x)((x^{p_i} y^{q_i})_2 : \dots : (x^{p_i} y^{q_i})_n) &= L(r_1^i)(r_2^i : \dots : r_{n+1}^i) \\
 &= L(r_1^i : \dots : r_{n+1}^i) \\
 &\stackrel{[\text{lemma: 16.367(1)}]}{=} L((x^{p_i+1} h^{q_i})_1 : \dots : (x^{p_i+1} h^{q_i})_{n+1})
 \end{aligned} \tag{16.595}$$

Define

$$z^i = (z_1^i, \dots, z_{n+1}^i) \text{ by } z_j^i = \begin{cases} h & \text{if } i = 1 \\ (x^{p_i} h^{q_i})_{i-1} & \text{if } i \in \{2, \dots, n+1\} \end{cases}$$

By [lemma: 16.367(2)] there exist a $\rho \in P_{n+1}$ such that

$$(z_{\rho(1)}^i, \dots, z_{\rho(n+1)}^i) = ((x^{p_i} h^{q_i+1})_1, \dots, (x^{p_i} h^{q_i+1})_{n+1})$$

hence

$$\begin{aligned}
 L(h)((x^{p_i} h^{q_i})_2 : \dots : (x^{p_i} h^{q_i})_{n+1}) &= L(z_1^i)(z_2^i : \dots : z_{n+1}^i) \\
 &= L(z_1^i : \dots : z_{n+1}^i) \\
 &= L(z_{\rho(1)}^i : \dots : z_{\rho(n+1)}^i) \\
 &= L((x^{p_i} h^{q_i+1})_1 : \dots : (x^{p_i} h^{q_i+1})_{n+1})
 \end{aligned} \tag{16.596}$$

Now

$$\begin{aligned} T_1 &= L(x+h)((x^n h^0)_1: \dots: (x^n h^0)_n) \\ &= L(x)((x^n h^0)_2: \dots: (x^n h^0)_n) + L(h)((x^n h^0)_2: \dots: (x^n h^0)_n) \\ &= T_{1,1} + T_{1,2} \end{aligned}$$

where by [eqs: 16.591, 16.592]

$$\begin{aligned} T_{1,1} &= L((x^{n+1} h^0)_1: \dots: (x^{n+1} h^0)_{n+1}) \\ T_{1,2} &= L((x^{(n+1)-1} h^1)_1: \dots: (x^{(n+1)-1} h^1)_{n+1}) \end{aligned}$$

Next

$$\begin{aligned} T_2 &= n \cdot L(x+h)((x^{n-1} h^1)_2: \dots: (x^{n-1} h^1)_n) \\ &= n \cdot L(x)((x^{n-1} h^1)_2: \dots: (x^{n-1} h^1)_n) + n \cdot L(h)((x^{n-1} h^1)_2: \dots: (x^{n-1} h^1)_n) \\ &= T_{2,1} + T_{2,2} \end{aligned}$$

where by [eqs: 16.593, 16.594] we have

$$\begin{aligned} T_{2,1} &= n \cdot L((x^{(n+1)-1} h^1)_1: \dots: (x^{(n+1)-1} h^1)_{n+1}) \\ T_{2,2} &= n \cdot L((x^{n-1} h^2)_1: \dots: (x^{n-1} h^2)_{n+1}) \end{aligned}$$

Further

$$\begin{aligned} T_3 &= \sum_{i \in \{1, \dots, k\}} L(x+h)((x^{p_i} y^{q_i})_2: \dots: (x^{p_i} y^{q_i})_n) \\ &= \sum_{i \in \{1, \dots, k\}} L(x)((x^{p_i} y^{q_i})_2: \dots: (x^{p_i} y^{q_i})_n) + \sum_{i \in \{1, \dots, k\}} L(h)((x^{p_i} y^{q_i})_2: \dots: (x^{p_i} y^{q_i})_n) \\ &= T_{3,1} + T_{3,2} \end{aligned}$$

where by [eqs: 16.595, 16.596]

$$\begin{aligned} T_{3,1} &= \sum_{i \in \{1, \dots, k\}} L((x^{p_i+1} h^{q_i})_1: \dots: (x^{p_i+1} h^{q_i})_{n+1}) \\ T_{3,2} &= \sum_{i \in \{1, \dots, k\}} L((x^{p_i} h^{q_i+1})_1: \dots: (x^{p_i} h^{q_i+1})_{n+1}) \end{aligned}$$

Define now $\{(P_i, Q_i)\}_{i \in \{1, \dots, 2 \cdot k + n\}}$ by

$$(P_i, Q_i) = \begin{cases} (p_i + 1, q_i) & \text{if } i \in \{1, \dots, k\} \\ (p_{i-k}, q_{i-k} + 1) & \text{if } i \in \{k+1, \dots, 2 \cdot k\} \\ (n-1, 2) & \text{if } i \in \{2 \cdot k+1, \dots, 2 \cdot k + n\} \end{cases}$$

then we have $\forall i \in \{1, \dots, 2 \cdot k + 1\}$ either:

$i \in \{1, \dots, k\}$. Then $(P_i, Q_i) = (p_1 + 1, q_i)$ so that $2 \leq q_i = Q_i$ and $P_i + Q_i = p_i + 1 + q_i = n + 1$.

$i \in \{k+1, \dots, 2 \cdot k\}$. Then $(P_i, Q_i) = (p_{i-k}, q_{i-k} + 1)$ so that $2 \leq q_{i-k} \leq q_{i-k} + 1 = Q_i$ and $P_i + Q_i = p_{i-k} + q_{i-k} + 1 = n + 1$.

$i \in \{2 \cdot k+1, \dots, 2 \cdot k + n\}$. Then $(P_i, Q_i) = (n-1, 2)$ so that $2 \leq Q_i$ and $P_i + Q_i = n-1 + 2 = n + 1$.

so that

$$\forall i \in \{1, \dots, 2 \cdot k + 1\} \text{ we have } 2 \leq Q_i \wedge P_i + Q_i = n + 1 \quad (16.597)$$

Further

$$\sum_{i \in \{1, \dots, 2 \cdot k + n\}} L((x^{Q_i} h^{P_i})_1: \dots: (x^{Q_i} h^{P_i})_{n+1}) = S_1 + S_2 + S_3$$

where

$$\begin{aligned}
S_1 &= \sum_{i \in \{1, \dots, k\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1}) \\
&= \sum_{i \in \{1, \dots, k\}} L((x^{q_i} h^{p_i+1})_1 : \dots : (x^{q_i} h^{p_i+1})_{n+1}) \\
&= T_{3,1} \\
S_2 &= \sum_{i \in \{k+1, \dots, 2 \cdot k\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1}) \\
&= \sum_{i \in \{k+1, \dots, 2 \cdot k\}} L((x^{p_i-k} h^{q_i-k+1})_1 : \dots : (x^{p_i-k} h^{q_i-k+1})_{n+1}) \\
&= \sum_{i \in \{1, \dots, k\}} L((x^{p_i} h^{q_i+1})_1 : \dots : (x^{p_i} h^{q_i+1})_{n+1}) \\
&= T_{3,2} \\
S_3 &= \sum_{i \in \{2 \cdot k+1, \dots, 2 \cdot k+n\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1}) \\
&= \sum_{i \in \{2 \cdot k+1, \dots, 2 \cdot k+n\}} L((x^{n-1} h^2)_1 : \dots : (x^{n-1} h^2)_{n+1}) \\
&= n \cdot L((x^{n-1} h^2)_1 : \dots : (x^{n-1} h^2)_{n+1}) \\
&= T_{2,2}
\end{aligned}$$

Hence

$$T_{2,2} + T_{3,1} + T_{3,2} = \sum_{i \in \{1, \dots, 2 \cdot k+n\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1}) \quad (16.598)$$

So

$$\begin{aligned}
L\left(\underbrace{x+h : \dots : x+h}_{n+1}\right) &= L(x+h) \left(\underbrace{x+h : \dots : x+h}_n\right) \\
&= T_1 + T_2 + T_3 \\
&= T_{1,1} + T_{1,2} + T_{2,1} + T_{2,2} + T_{3,1} + T_{3,2} \\
&= L((x^{n+1} h^0)_1 : \dots : (x^{n+1} h^0)_{n+1}) + \\
&\quad L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) + \\
&\quad n \cdot L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) + \\
&\quad T_{2,2} + T_{3,1} + T_{3,2} \\
&\stackrel{[\text{eq: 16.598}]}{=} L((x^{n+1} h^0)_1 : \dots : (x^{n+1} h^0)_{n+1}) + \\
&\quad L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) + \\
&\quad n \cdot L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) + \\
&\quad \sum_{i \in \{1, \dots, 2 \cdot k+n\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1}) \\
&= L((x^{n+1} h^0)_1 : \dots : (x^{n+1} h^0)_{n+1}) + \\
&\quad (n+1) \cdot L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) + \\
&\quad \sum_{i \in \{1, \dots, 2 \cdot k+n\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1})
\end{aligned}$$

Hence

$$L\left(\underbrace{x+h : \dots : x+h}_{n+1}\right) = K_1 + K_2 + K_3$$

where

$$\begin{aligned}
K_1 &= L((x^{n+1} h^0)_1 : \dots : (x^{n+1} h^0)_{n+1}) \\
K_2 &= (n+1) \cdot L((x^{(n+1)-1} h^1)_1 : \dots : (x^{(n+1)-1} h^1)_{n+1}) + \\
K_3 &= \sum_{i \in \{1, \dots, 2 \cdot k+n\}} L((x^{Q_i} h^{P_i})_1 : \dots : (x^{Q_i} h^{P_i})_{n+1})
\end{aligned}$$

This together with [eq: 16.597] proves that

$$n+1 \in S$$

□

Lemma 16.370. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $L \in L_n(X; Y)$ such that $\forall \sigma \in P_n$, $\forall y = (y_1, \dots, y_n)$ we have $L(y_{\sigma(1)} : \dots : y_{\sigma(n)}) = L(y_1, \dots, y_n)$ then we have that

$$\Phi: X \rightarrow Y \text{ defined by } \Phi(x) = L\left(\underbrace{x : \dots : x}_n\right)$$

is Fréchet differentiable at every $x \in X$ where

$$D_x \Phi: X \rightarrow Y$$

is defined by

$$D_x \Phi(h) = n \cdot L((x^{n-1}h^1)_1 : \dots : (x^{n-1}h^1)_n) = n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : h\right)$$

Proof. Let $x \in X$ then by the previous lemma [lemma: 16.369] there exist a $\{q_i, p_i\}_{i \in \{1, \dots, k\}}$ with $\forall i \in \{1, \dots, k\}$ $2 \leq q_i$, $n = p_i + q_i$ such that $\forall h \in X$

$$L(x+h : \dots : x+h) = T_1 + T_2 + T_3$$

where

$$\begin{aligned} T_1 &= L((x^n h^0)_1 : \dots : (x^n h^0)_n) \\ T_2 &= n \cdot L((x^{n-1} h^1)_1 : \dots : (x^{n-1} h^1)_n) \\ T_3 &= \sum_{i \in \{1, \dots, k\}} L((x^{p_i} h^{q_i})_1 : \dots : (x^{p_i} h^{q_i})_n) \end{aligned}$$

Now

$$\begin{aligned} \Phi(x+h) &= L(x+h : \dots : x+h) \\ \Phi(x) &= L\left(\underbrace{x : \dots : x}_n\right) \\ &\stackrel{[\text{lemma: 16.367(3)}]}{=} L((x^n h^0)_1, \dots, (x^n h^0)_n) \\ &= T_1 \end{aligned}$$

hence we have

$$\Phi(x+h) - \Phi(x) - n \cdot L((x^{n-1}h^1)_1 : \dots : (x^{n-1}h^1)_n) = T_3 \quad (16.599)$$

Now

$$\begin{aligned} \|T_3\|_Y &= \left\| \sum_{i \in \{1, \dots, k\}} L((x^{p_i} h^{q_i})_1 : \dots : (x^{p_i} h^{q_i})_n) \right\|_Y \\ &\leq \sum_{i \in \{1, \dots, k\}} \|L((x^{p_i} h^{q_i})_1 : \dots : (x^{p_i} h^{q_i})_n)\| \\ &\leq \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \Pi_{j \in \{1, \dots, n\}} \|(x^{p_i} h^{q_i})_j\|_X) \\ &= \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot (\Pi_{j \in \{1, \dots, p_i\}} \|(x^{p_i} h^{q_i})_j\|_X) \cdot (\Pi_{j \in \{p_i+1, \dots, n\}} \|(x^{p_i} h^{q_i})_j\|_X)) \\ &= \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot (\Pi_{j \in \{1, \dots, p_i\}} \|x\|_X) \cdot (\Pi_{j \in \{p_i+1, \dots, n\}} \|h\|_X)) \\ &= \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i} \cdot \|h\|_X^{q_i}) \\ &= \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i} \cdot \|h\|_X^{q_i-2} \cdot \|h\|_X^2) \\ &= \left(\sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i} \cdot \|h\|_X^{q_i-2}) \right) \cdot \|h\|_X^2 \end{aligned}$$

which proves that

$$\|T_3\|_Y \leq \left(\sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i} \cdot \|h\|_X^{q_i-2}) \right) \cdot \|h\|_X^2 \quad (16.600)$$

Assume that $\|h\| < 1$ then as $\forall i \in \{1, \dots, k\}$ $2 \leq q_i$ we have $\|h\|_X^{q_i-2} \leq 1$ so that

$$\forall h \in X \text{ with } \|h\|_X < 1 \text{ we have } \|T_3\|_Y \leq \left(\sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i}) \right) \cdot \|h\|_X^2 \quad (16.601)$$

Let $\varepsilon \in \mathbb{R}^+$. Take $\delta_\varepsilon = \min \left(1, \frac{\varepsilon}{1 + \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i})} \right)$ then for $h \in X = X_x$ with $\|h\|_X < \delta_\varepsilon$ we have

$$\begin{aligned} \|\Phi(x+h) - \Phi(x) - n \cdot L((x^{n-1}h^1)_1 : \dots : (x^{n-1}h^1)_n)\|_Y &\stackrel{[\text{eq: 16.599}]}{=} \|T_3\|_Y \\ &\stackrel{[\text{eq: 16.601}]}{\leq} \left(\sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i}) \right) \cdot \|h\|_X^2 = \\ &\left(\left(\sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i}) \right) \cdot \|h\|_X \right) \cdot \|h\|_X \leq \\ &\left(\left(\sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i}) \right) \cdot \frac{\varepsilon}{1 + \sum_{i \in \{1, \dots, k\}} (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{p_i})} \right) \cdot \|h\|_X \leq \\ &\varepsilon \cdot \|h\|_x \end{aligned}$$

Hence $\forall \varepsilon \in \mathbb{R}^+$ we have $\forall h \in X = X_x$ with $\|h\|_X < \delta_\varepsilon$ that

$$\|\Phi(x+h) - \Phi(x) - n \cdot L((x^{n-1}h^1)_1 : \dots : (x^{n-1}h^1)_n)\|_Y \leq \varepsilon \cdot \|h\|_X \quad (16.602)$$

Define

$$K: X \rightarrow Y \text{ by } K(h) = n \cdot L((x^{n-1}h^1)_1 : \dots : (x^{n-1}h^1)_n)$$

As $\forall i \in \{1, \dots, n-1\}$ we have $(x^{n-1}h^1)_i = x$ and $(x^{n-1}h^1)_n = h$ it follows that

$$\forall h \in X \quad K(h) = n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : h\right)$$

Hence for $h, k \in X$, $\alpha \in \mathbb{K}$ we have

$$K(h + \alpha \cdot k) = n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : h + \alpha \cdot k\right) = n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : h\right) + \alpha \cdot n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : k\right) = K(h) + \alpha \cdot K(k)$$

which proves that

$$K \in \text{Hom}(X, Y)$$

Further $\forall h \in X$ we have $\|K(x)\|_Y = \left\| n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : h\right) \right\|_Y \leq n \cdot (\|L\|_{L_n(X; Y)} \cdot \|x\|_X^{n-1}) \cdot \|h\|_X$ which proves that K is continuous, hence

$$K \in L(X; Y)$$

As further X is open it follows from the definition of Fréchet differentiability [see definition: 16.46] that Φ is Fréchet differentiable at x with $D_x\Phi = K$. Further $D_x\Phi$ is defined by

$$\forall h \in X \quad D_x\Phi(h) = K(h) = n \cdot L((x^{n-1}h^1)_1 : \dots : (x^{n-1}h^1)_n) = n \cdot L\left(\underbrace{x : \dots : x}_{n-1} : h\right) \quad \square$$

We can now prove a version of Taylor's theorem for functions between normed spaces.

Theorem 16.371. (Taylor's Theorem (I.4)) *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , $x \in U$ and $f: U \rightarrow Y$ a function that is n -times differentiable at x_0 then $\forall \varepsilon \in \mathbb{R}^+$ there exist a $\delta \in \mathbb{R}^+$ such that $\forall h \in U_{x_0}$ with $\|h\|_X < \delta$ we have*

$$\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f\left(\underbrace{h : \dots : h}_k\right) \right\|_Y \leq \varepsilon \cdot \|h\|_X^n$$

Note 16.372. *Let $x \in U$ then as f is n -times differentiable at x_0 it follows from [theorem: 16.138] that $\forall m \in \{1, \dots, n\}$ f is m -times differentiable at x_0 so that $\sum_{k \in \{1, \dots, n-1\}} \frac{1}{k!} \cdot D_{x_0}^{[k]} f\left(\underbrace{h : \dots : h}_k\right)$ is well defined.*

Note 16.373. *Using [definition: 16.359] this equivalent with saying that*

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f\left(\underbrace{h : \dots : h}_k\right) = \text{ord}_{h \xrightarrow{U_x} 0}(\|h\|_X^n)$$

Proof. We prove this by induction so let

$$S_X = \left\{ n \in \mathbb{N} \mid \text{If } \langle Y, \|\cdot\|_Y \rangle \text{ is a normed space and } f: U \rightarrow Y \text{ is } n\text{-times differentiable at } x_0 \text{ then } \forall \varepsilon \in \mathbb{R}^+ \right. \\ \left. \exists \delta \in \mathbb{R}^+ \text{ such that } \forall h \in U_{x_0} \text{ with } \|h\|_X < \delta \text{ we have } \left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f\left(\underbrace{h : \dots : h}_k\right) \right\|_Y \leq \varepsilon \cdot \|h\|_X^n \right\}$$

then we have:

$1 \in S$. If f is 1-times differentiable at x_0 then by [theorem: 16.140] f is Fréchet differentiable at x_0 with $D_{x_0}^{[1]}f = D_{x_0}f$. Let $\varepsilon \in \mathbb{R}^+$ then there exists a $\delta \in \mathbb{R}^+$ such that $\forall h \in U_{x_0}$ with $\|h\|_X < \delta$ we have

$$\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^1 \frac{1}{k!} \cdot D_{x_0}^{[k]} f\left(\underbrace{h : \dots : h}_1\right) \right\|_Y = \|f(x_0 + h) - f(x_0) - D_{x_0}f(h)\|_Y \leq \varepsilon \cdot \|h\|_X$$

which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. As f is $(n+1)$ -times differentiable at x_0 and $2 \in \{1, \dots, n+1\}$ we have by [theorem: 16.138] that f is 2-times differentiable at x_0 . By [theorem: 16.140] it follows that $D^{[1]}f$ is Fréchet differentiable at x_0 , so by the definition of Fréchet differentiability [see definition: 16.46] it follows that $x_0 \in (\mathfrak{D}_f^1)^\circ$ where $\mathfrak{D}_f^1$ is the domain of $D^{[1]}f$. Hence there exist a $\delta_1 \in \mathbb{R}^+$ such that

$$B_{\|\cdot\|_X}(x_0, \delta_1) \subseteq \mathfrak{D}_f^1 \underset{[\text{theorem: 16.135}]}{\subseteq} U$$

Hence using [definition: 16.141] and [theorem: 16.140] we have that $\forall x \in B_{\|\cdot\|_X}(x_0, \delta)$ f is Fréchet differentiable at x or equivalently

$$\forall x \in B_{\|\cdot\|_X}(0, \delta_1) \quad f \text{ is Fréchet differentiable at } x_0 + x \quad (16.603)$$

Further as f is $(n+1)$ -times differentiable at x_0 it follows that $\forall k \in \{1, \dots, n+1\}$ $D_{x_0}^{[k]}f$ exist. Hence we can define the following function.

$$\varphi: B_{\|\cdot\|_X}(0, \delta_1) \rightarrow Y \text{ where } \varphi(x) = f(x_0 + x) - f(x_0) - \sum_{k=1}^{n+1} \frac{1}{k!} \cdot D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_k\right) \quad (16.604)$$

Let $k \in \{1, \dots, n+1\}$ and define

$$\Phi_k: B_{\|\cdot\|_X}(0, \delta_1) \rightarrow Y \text{ by } \Phi_k(x) = D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_k\right)$$

then it follows from [theorem: 16.57], and [lemma: 16.370] that $\forall x \in B_{\|\cdot\|_X}(0, \delta_1)$ Φ_k is Fréchet differentiable at x where $D_x\Phi_k$ is defined by

$$D_x\Phi_k: X \rightarrow Y \text{ where } D_x\Phi_k(h) = k \cdot D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_{k-1} : h\right)$$

Define

$$\varphi_2: B_{\|\cdot\|_X}(0, \delta_1) \rightarrow Y \text{ by } \varphi_2(x) = -\sum_{k=1}^{n+1} \frac{1}{k!} \cdot D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_k\right) \quad (16.605)$$

then $\varphi_2 = -\sum_{k=1}^{n+1} \frac{1}{k!} \cdot \Phi_k$ so that by [theorem: 16.66] φ_2 is Fréchet differentiable at $x \in B_{\|\cdot\|_X}(0, \delta_1)$ where $D_x\varphi_2 = -\sum_{k=1}^{n+1} \frac{1}{k!} \cdot D_x\Phi_k$ so that $D_x\varphi_2$ is defined by $\forall h \in X$ we have

$$D_x\varphi_2(h) = -\sum_{k=1}^{n+1} \frac{1}{k!} \cdot k \cdot D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_{k-1} : h\right) = -\sum_{k=1}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_{k-1} : h\right) \quad (16.606)$$

Define

$$\varphi_1: B_{\|\cdot\|_X}(0, \delta_1) \rightarrow Y \text{ by } \varphi_1(x) = f(x_0 + x) - f(x_0) = (f \circ (\text{Id}_{B_{\|\cdot\|_X}} + C_{x_0}) + C_{-f(x_0)})(x) \quad (16.607)$$

By [examples: 16.54, 16.53], [theorems: 16.67, 16.66] and [eq: 16.603] it follows that φ_1 is Fréchet differentiable for every $x \in B_{\|\cdot\|_X}(0, \delta_1)$ with

$$D_x\varphi_1 = D_{x_0+x}f \quad (16.608)$$

Combining [eqs: 16.604, 16.605, 16.607] results in

$$\varphi = \varphi_1 + \varphi_2$$

So that by [theorem: 16.66] and [eqs: 16.606, 16.608] we have

$$\forall x \in B_{\|\cdot\|_X}(0, \delta_1) \text{ is Fréchet differentiable at } x \quad (16.609)$$

where

$$D_x\varphi \in L(X, Y)$$

is defined by

$$D_x\varphi: X \rightarrow Y \text{ where } D_x\varphi(h) = D_{x_0+x}f(h) - \sum_{k=1}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{x: \dots: x}_{k-1} : h\right) \quad (16.610)$$

As f is $(n+1)$ differentiable at x_0 it follows from [theorem: 16.174] that $D^{[1]}f$ is n -times differentiable at x_0 with

$$D_{x_0}^{[n+1]}f = D_{x_0}^{[n]}(D^{[1]}f)$$

Let $\varepsilon \in \mathbb{R}^+$. As $n \in \mathbb{N}$ there exist a $\delta_2 \in \mathbb{R}^+$ such that $\forall h \in U_{x_0}$ with $\|h\|_X < \delta_2$ we have

$$\left\| D^{[1]}f(x_0 + h) - D^{[1]}f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]}(D^{[1]}f)\left(\underbrace{h: \dots: h}_k\right) \right\|_{L(X, Y)} \leq \varepsilon \cdot \|h\|_X^n \quad (16.611)$$

Let $\delta = \min(\delta_1, \delta_2)$ and take $h \in U_{x_0}$ with $\|h\|_X < \delta$. Given $y \in X$ we have

$$\begin{aligned}
 & \left(D^{[1]}f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]}(D^{[1]}f)\left(\underbrace{h: \dots: h}_k\right) \right)(y) \stackrel{[\text{theorem: 16.174}]}{=} \\
 & \left(D^{[1]}f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k+1]}f\left(\underbrace{h: \dots: h}_k\right) \right)(y) = \\
 & \left(D_{x_0}^{[1]}f + \sum_{k=2}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{h: \dots: h}_{k-1}\right) \right)(y) = \\
 & D_{x_0}^{[1]}f(y) + \sum_{k=2}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{h: \dots: h}_{k-1}\right)(y) \stackrel{[\text{lemma: 16.119}]}{=} \\
 & D_{x_0}^{[1]}f(y) + \sum_{k=2}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{h: \dots: h: y}_{k-1}\right) = \\
 & \frac{1}{(1-1)!} \cdot D_{x_0}^{[1]}f\left(\underbrace{h: \dots: h: y}_{1-1}\right) + \sum_{k=2}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{h: \dots: h: y}_{k-1}\right) = \\
 & \sum_{k=1}^{n+1} \frac{1}{(k-1)!} \cdot D_{x_0}^{[k]}f\left(\underbrace{h: \dots: h: y}_{k-1}\right) \stackrel{[\text{eq: 16.610}]}{=} \\
 & D_{x_0+h}f(y) - D_h\varphi(y) = \\
 & D_{x_0+h}^{[1]}f(y) - D_h\varphi(y) \tag{16.612}
 \end{aligned}$$

hence we have $\forall y \in X$ that

$$\begin{aligned}
 & \left(D^{[1]}f(x_0+h) - D^{[1]}f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]}(D^{[1]}f)\left(\underbrace{h: \dots: h}_k\right) \right)(y) \stackrel{[\text{eq: 16.612}]}{=} \\
 & D^{[1]}f(x_0+h)(y) - D_{x_0+h}^{[1]}f(y) + D_h\varphi(y) = \\
 & D_{x_0+h}^{[1]}f(h) - D_{x_0+h}^{[1]}f(h) + D_h\varphi(y) = \\
 & D_h\varphi(y)
 \end{aligned}$$

so that $\forall y \in X$ we have

$$\begin{aligned}
 & \left\| \left(D^{[1]}f(x_0+h) - D^{[1]}f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]}(D^{[1]}f)\left(\underbrace{h: \dots: h}_k\right) \right)(y) \right\|_Y \stackrel{\|D_h\varphi(y)\|_Y}{=} \\
 & \left\| D^{[1]}f(x_0+h) - D^{[1]}f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]}(D^{[1]}f)\left(\underbrace{h: \dots: h}_k\right) \right\|_{L(X,Y)} \cdot \|y\|_X \stackrel{[\text{eq: 16.611}]}{\leq} \\
 & \varepsilon \cdot \|h\|_X^n \cdot \|y\|_X
 \end{aligned}$$

Hence using [theorem: 14.212] it follows that

$$\|D_{x_0}\varphi\|_{L(X,Y)} \leq \varepsilon \cdot \|h\|_X^n$$

As $B_{\|\cdot\|_X}(0, \delta) \subseteq B_{\|\cdot\|_X}(0, \delta_1)$ φ is Fréchet differentiable on $B_{\|\cdot\|_X}(0, \delta)$ [see eq: 16.609] and $B_{\|\cdot\|_X}(0, \delta)$ is convex [see theorem: 16.221] we can use the Mean Value Theorem [theorem: 16.236] to get for $0, h \in B_{\|\cdot\|_X}(0, \delta)$ that

$$\|\varphi(h) - \varphi(0)\|_Y \leq \varepsilon \cdot \|h\|_X^n \cdot \|h - 0\|_X = \varepsilon \cdot \|h\|_X^{n+1}$$

As

$$\begin{aligned}
 \varphi(h) - \varphi(0) &= \\
 \varphi(h) - \left(f(x_0) - f(x_0) - \sum_{k=1}^{n+1} \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{0: \dots : 0}_k \right) \right) &= \\
 \varphi(h) - 0 &= \\
 \varphi(h) &= \\
 f(x_0 + h) - f(x_0) - \sum_{k=1}^{n+1} \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h: \dots : h}_k \right)
 \end{aligned}$$

it follows that

$$\left\| f(x_0 + h) - f(x_0) - \sum_{k=1}^{n+1} \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h: \dots : h}_k \right) \right\|_Y \leq \varepsilon \cdot \|h\|_X^{n+1}$$

which finally proves that $n+1 \in S$. $\square$

The previous versions of Taylor's theorem are asymptotic forms they do not give a formula or a estimation of the rest term

$$R_{x_0, x, n} = f(x) - f(x_0) - \sum_{k=1}^n \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)}$$

The next versions of Taylor's theorem solves give a formula or estimation of the rest terms.

Theorem 16.374. (Taylor's Theorem (II.1)) Let $n \in \mathbb{N} \setminus \{1\}$, $a, b \in \mathbb{R}$ with $a < b$, $x, x_0 \in \mathbb{R}$ with $x \neq x_0$ and $[m, M] \subseteq]a, b[$ where $m = \min(x_0, x)$, $M = \max(x_0, x)$ and $f: [a, b] \rightarrow \mathbb{R}$ a function such that

1. $\forall y \in [m, M]$ f has a $(n-1)$ -times derivative at y .
2. $f^{(n-1)}: [m, M] \rightarrow \mathbb{R}$ defined by $f^{(n-1)}(y) = f_y^{(n-1)}$ is continuous.
3. $\forall y \in]m, M[$ f has a n -times derivative at y .

then $\exists \theta \in]0, 1[$ such that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} + \frac{(x - x_0)^n}{n!} \cdot f_{(m+\theta \cdot (M-m))}^{(n)}$$

Note 16.375. As $x \neq x_0$ we have that $\min(x, x_0) \neq \max(x, x_0)$ so that $m < M$.

Note 16.376. As $0 < \theta < 1$ we have that $m < m + \theta \cdot (M - m) < m + 1 \cdot (M - m) = M$ so that $m + \theta \cdot (M - m) \in]m, M[$ which ensures that in (3) $f_{(m+\theta \cdot (M-m))}^{(n)}$ exist.

Note 16.377. $\forall k \in \{1, \dots, n-1\}$ we have by (1) and [theorem: 16.169] that $f_{x_0}^{(k)}$ exist so that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} + \frac{(x - x_0)^n}{n!} \cdot f_{(x_0 + \theta \cdot (x - x_0))}^{(n)}$$

is well defined.

Proof. Let $k \in \{1, \dots, n-1\}$ then as $\forall y \in [m, M]$ $f_y^{(n-1)}$ exist it follows from [theorem: 16.169] that $f_y^{(k)}$ exist so that we can define

$$\forall k \in \{1, \dots, n-1\} \quad f^{(k)}: [m, M] \rightarrow \mathbb{R} \text{ by } f^{(k)}(y) = f_y^{(k)} \quad (16.613)$$

Using $k = 1$ in the above together with [theorem: 16.33] proves that

$$\forall y \in [m, M] \ f'_y \text{ exist and } f \text{ is continuous at } y \quad (16.614)$$

Let $k \in \{1, \dots, n-1\}$ then we have either:

$k \in \{1, \dots, n-2\}$. Then $k+1 \in \{1, \dots, n-1\}$ so that $\forall y \in [m, M]$ we have that $f_y^{(k+1)}$ exist. Hence by definition $(f^{(k)})'_y$ exist and $f^{(k+1)}(y) = f_y^{(k+1)} = (f^{(k)})'_y$ and by [theorem: 16.33] $f^{(k)}$ is continuous.

$k = n-1$. As $\forall y \in]m, M[\ f_y^{(n)}$ exist it follows that $(f^{(n-1)})'_y$ exist $f_y^{(n)} = (f^{(n-1)})'_y$, further by (2) $f^{(n-1)}$ is continuous.

To summarize we have that

$$\forall k \in \{1, \dots, n-1\} \ f^{(k)} \text{ is continuous and } \forall y \in]m, M[\ (f^{(k)})'_y \text{ exist and } f_y^{(k+1)} = (f^{(k)})'_x \quad (16.615)$$

Let $\lambda \in \mathbb{R}$ [later we choose a specific value for λ] then by the above we can define the following function

$$g: [m, M] \rightarrow \mathbb{R} \text{ where } g(t) = f(t) + \sum_{k=1}^{n-1} \frac{(x-t)^k}{k!} \cdot f^{(k)}(t) + \lambda \cdot \frac{(x-t)^n}{n!} \quad (16.616)$$

then by [eqs: 16.614, 16.615], [example: 14.142] and [theorems: 14.165, 14.166, 14.167, 14.169, 14.146 and 14.166] it follows that

g is continuous

Further using [examples: 16.30, 16.31], [theorems: 16.40, 16.108, 16.68, 16.36] and [eqs: 16.614, 16.615] it follows that $\forall t \in]m, M[$ has a derivative g'_t at t with

$$\begin{aligned} g'_t &= f'_t + \sum_{k=1}^{n-1} \frac{1}{k!} (-k \cdot (x-t)^{k-1} \cdot f^{(k)}(t) + (x-t)^k \cdot f^{(k+1)}(t)) - \lambda \cdot \frac{n \cdot (x-t)^{n-1}}{n!} \\ &= f'_t + \sum_{k=1}^{n-1} \frac{(x-t)^k}{k!} f_t^{(k+1)} - \sum_{k=1}^{n-1} \frac{(x-t)^{k-1}}{(k-1)!} \cdot f_t^{(k)} - \lambda \cdot \frac{(x-t)^{n-1}}{(n-1)!} \\ &= f'_t + \sum_{k=1}^{n-1} \frac{(x-t)^k}{k!} f_t^{(k+1)} - \frac{(x-t)^0}{0!} \cdot f_t^{(1)} - \sum_{k=2}^{n-1} \frac{(x-t)^{k-1}}{(k-1)!} \cdot f_t^{(k)} - \lambda \cdot \frac{(x-t)^{n-1}}{(n-1)!} \\ &= f'_t + \sum_{k=1}^{n-1} \frac{(x-t)^k}{k!} f_t^{(k+1)} - f'_t - \sum_{k=2}^{n-1} \frac{(x-t)^{k-1}}{(k-1)!} \cdot f_t^{(k)} - \lambda \cdot \frac{(x-t)^{n-1}}{(n-1)!} \\ &= \sum_{k=1}^{n-1} \frac{(x-t)^k}{k!} f_t^{(k+1)} - \sum_{k=2}^{n-1} \frac{(x-t)^{k-1}}{(k-1)!} \cdot f_t^{(k)} - \lambda \cdot \frac{(x-t)^{n-1}}{(n-1)!} \\ &= \sum_{k=1}^{n-1} \frac{(x-t)^k}{k!} f_t^{(k+1)} - \sum_{k=1}^{n-2} \frac{(x-t)^k}{k!} \cdot f_t^{(k+1)} - \lambda \cdot \frac{(x-t)^{n-1}}{(n-1)!} \\ &= \frac{(x-t)^{(n-1)}}{(n-1)!} f_t^n - \lambda \cdot \frac{(x-t)^{(n-1)}}{(n-1)!} \end{aligned}$$

so that

$$\forall t \in]m, M[\text{ we have } g'_t = \frac{(x-t)^{(n-1)}}{(n-1)!} f_t^n - \lambda \cdot \frac{(x-t)^{(n-1)}}{(n-1)!} \quad (16.617)$$

As $x_0 \neq x$ we can choose λ to be

$$\lambda = \frac{n!}{(x-x_0)^n} \cdot \left(- \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + f(x) - f(x_0) \right)$$

then we have

$$\begin{aligned}
 & g(x_0) \quad [\text{eq: 16.616}] \\
 & f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + \lambda \cdot \frac{(x-x_0)^n}{n!} = \\
 & f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + \frac{n!}{(x-x_0)^n} \cdot \left(- \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + f(x) - f(x_0) \right) \cdot \\
 & \frac{(x-x_0)^n}{n!} = \\
 & f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) - \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + f(x) - f(x_0) = \\
 & f(x)
 \end{aligned}$$

so with this choice of λ we have

$$g(x_0) = f(x)$$

Further we have

$$\begin{aligned}
 g(x) & \quad [\text{eq: 16.616}] \\
 & f(x) + \sum_{k=1}^{n-1} \frac{(x-x)^k}{k!} \cdot f^{(k)}(x) + \lambda \cdot \frac{(x-x)^n}{n!} \\
 & = f(x)
 \end{aligned}$$

To summarize we have

$$g(x) = g(x_0) = f(x) \quad (16.618)$$

The above together with the fact that we have proved that g is continuous and have a derivative g'_t at $t \in]m, M[$ allows use to apply Rolle's theorem [theorem: 16.205] from which it follows that there exist a $\zeta \in]m, M[$ such that $g'(\zeta) = 0$. Combing this with [eq: 16.617] gives

$$\frac{(x-\zeta)^{(n-1)}}{(n-1)!} f_\zeta^n - \lambda \cdot \frac{(x-\zeta)^{(n-1)}}{(n-1)!} = 0$$

As $\zeta \in]m, M[\Rightarrow \zeta \neq \min(x, x_0), \max(x, x_0)$ we have that $(x-\zeta)^{(n-1)} \neq 0$ so we can divide the above by $\frac{(x-\zeta)^{(n-1)}}{(n-1)!}$ which proves that

$$f_\zeta^{(n)} = \lambda \quad (16.619)$$

Hence we have that

$$\begin{aligned}
 & f(x) \quad [\text{eq: 16.618}] \quad g(x_0) \\
 & \quad [\text{eq: 16.616}] \quad f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + \lambda \cdot \frac{(x-x_0)^n}{n!} \\
 & \quad [\text{eq: 16.619}] \quad f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + \frac{(x-x_0)^n}{n!} \cdot f_\zeta^{(n)} \quad (16.620)
 \end{aligned}$$

As $\zeta \in]m, M[\Rightarrow m < \zeta < M$ we have that $m < M$ so we can then define $\theta = \frac{\zeta-m}{M-m}$ then we have as $0 < (\zeta-m), 0 < (M-m)$ that $0 < \theta = \frac{\zeta-m}{M-m} < \frac{M-m}{M-m} = 1$ so that $\theta \in]0, 1[$. Further $m + \theta \cdot (M-m) = m + (\zeta-m) = \zeta$. Substituting this result in [eq: 16.620] proves finally that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f^{(k)}(x_0) + \frac{(x-x_0)^n}{n!} \cdot f_{(m+\theta \cdot (M-m))}^{(n)} \quad \text{where } \theta \in]0, 1[\quad \square$$

The above version of Taylor's Theorem is based on Rolle's theorem hence it work only for functions between a open subset of real numbers and the real numbers. In the next version of Taylor's theorem we deal with functions between a open subset of a Banach space and a Banach space. First we need to prove some lemma's.

Lemma 16.378. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, a open set $U \subseteq X$, $x_0 \in U$, $f: U \rightarrow Y$ a function that is n -times differentiable on U and $(n+1)$ -times differentiable at x_0 , $\{h_i\}_{i \in \{1, \dots, n\}} \subseteq X$ then the function

$$\Theta: U \rightarrow Y \text{ defined by } \Theta(x) = D_x^{[n]} f(h_1: \dots: h_n)$$

is Fréchet differentiable at x_0 where

$$D_{x_0} \Theta: X \rightarrow Y \text{ is defined by } D_{x_0} \Theta(k) = D_{x_0}^{[k+1]} f(k)(h_1: \dots: h_n) = D_{x_0}^{[k+1]} f(k: h_1: \dots: h_n)$$

Proof. Define

$$\psi: L_n(X; Y) \rightarrow Y \text{ by } \psi(L) = L(h_1: \dots: h_n)$$

then we have for $L_1, L_2 \in L_n(X; Y)$ and $\alpha \in \mathbb{K}$ that

$$\begin{aligned} \psi(L_1 + \alpha \cdot L_2) &= (L_1 + \alpha \cdot L_2)(h_1: \dots: h_n) \\ &\stackrel{[\text{theorem: 16.120}]}{=} L_1(h_1: \dots: h_n) + \alpha \cdot L_2(h_1: \dots: h_n) \\ &= \psi(L_1) + \alpha \cdot \psi(L_2) \end{aligned}$$

which proves that

$$\psi \in \text{Hom}(L_n(X; Y), Y)$$

Further we have

$$\|\psi(L)\|_Y = \|L(h_1: \dots: h_n)\|_Y \stackrel{[\text{lemma: 16.122}]}{\leq} \left(\prod_{i=1}^n \|h_i\|_X \right) \cdot \|L\|_{L_n(X; Y)}$$

which proves by [theorem: 14.201] that

$$\psi \in L(L_n(X; Y), Y)$$

Hence using [example: 16.55] that ψ is differentiable on $L_n(X; Y)$ with

$$\forall L \in L_n(X; Y) \quad D_L \psi = \psi \tag{16.621}$$

As f is n -times differentiable on U we have by [definition: 16.141] and [theorem: 16.135] that

$$\text{dom}(D^{[n]} f) = \mathfrak{D}_n^n = U$$

Further as f is $(n+1)$ -times differentiable at x_0 we have by [theorem: 16.140] that

$$D^{[n]} f: U \rightarrow L_n(X; Y) \text{ is Fréchet differentiable at } x_0 \text{ with } D_{x_0}^{[n+1]} f = D_{x_0}(D^{[n]} f) \tag{16.622}$$

Now for $x \in U$ we have that

$$(\psi \circ D^{[n]} f)(x) = \psi(D^{[n]} f(x)) = \psi(D_x^{[n]} f) = D_x^{[n]} f(h_1: \dots: h_n) = \Theta(x)$$

which proves that

$$\Theta = \psi \circ D^{[n]} f$$

Using now the chain rule [see theorem: 16.67] it follows that Θ is Fréchet differentiable at x_0 with

$$\begin{aligned} D_{x_0} \Theta &= D_{x_0}(\psi \circ D^{[n]} f) \\ &= D_{D^{[n]} f(x_0)} \psi \circ D_{x_0}(D^{[n]} f) \\ &\stackrel{[\text{eq: 16.621}]}{=} \psi \circ D_{x_0}(D^{[n]} f) \\ &\stackrel{[\text{eq: 16.622}]}{=} \psi \circ D_{x_0}^{[n+1]} f \end{aligned}$$

Hence we have $\forall k \in X$ that

$$D_{x_0} \Theta(k) = (\psi \circ D_{x_0}^{[n+1]} f)(k) = \psi(D_{x_0}^{[n+1]} f(k)) = D_{x_0}^{[n+1]} f(k)(h_1: \dots: h_n) \quad \square$$

Lemma 16.379. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, $A \subseteq X$ and $f: A \rightarrow L_n(X; Y)$ a continuous function then for $h \in X$ we have that

$$\Theta: A \rightarrow Y \text{ defined by } \Theta(x) = f(x) \left(\underbrace{h: \dots: h}_n \right)$$

is continuous.

Proof. Let $x \in A$ and take $\varepsilon \in \mathbb{R}^+$ then as f is continuous [hence continuous at x] there exist a $\delta \in \mathbb{R}^+$ such that $\forall y \in A$ with $\|x - y\|_X < \delta$ we have

$$\|f(x) - f(x_n)\|_{L_n(X; Y)} < \frac{\varepsilon}{\|h\|_X^n + 1}$$

Now

$$\begin{aligned} \|\Theta(x) - \Theta(y)\|_Y &= \left\| f(x) \left(\underbrace{h: \dots: h}_n \right) - f(y) \left(\underbrace{h: \dots: h}_n \right) \right\|_Y \\ &\stackrel{[\text{lemma: 16.120}]}{=} \left\| (f(x) - f(y)) \left(\underbrace{h: \dots: h}_n \right) \right\|_Y \\ &\stackrel{[\text{lemma: 16.122}]}{\leq} \|f(x) - f(y)\|_{L_n(X; Y)} \cdot \|h\|_X^n \\ &\leq \frac{\varepsilon}{\|h\|_X^n + 1} \cdot \|h\|_X^n \\ &< \varepsilon \end{aligned}$$

which proves that Θ is continuous at x . Hence as $x \in A$ was chosen arbitrary it follows that

Θ is continuous □

Lemma 16.380. Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , $x_0 \in U$, $h \in X$ such that the line segment $L_{x_0, x_0+h} \subseteq U$ and $f: U \rightarrow Y$ a function that is $(n+1)$ -differentiable on U then the function

$$\varphi: [0, 1] \rightarrow Y \text{ defined by } \varphi(t) = f(x_0 + t \cdot h) + \sum_{k=1}^n \frac{(1-t)^k}{k!} \cdot D_{x_0+t \cdot h}^{[k]} f \left(\underbrace{h: \dots: h}_k \right)$$

is continuous on $[0, 1]$ and has derivatives on $[0, 1]$ with

$$\forall t \in [0, 1] \quad \varphi'_t = \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots: h}_{n+1} \right)$$

Note 16.381. By [theorem: 16.138] we have that $\forall k \in \{1, \dots, n+1\}$ f is k -times differentiable on U , further we have by [definition: 16.224]

$$U \supseteq L_{x_0, x_0+h} = \{x_0 + t \cdot ((x_0 + h) - x_0) | t \in [0, 1]\} = \{x_0 + t \cdot h | t \in [0, 1]\}$$

which proves that φ is well defined.

Proof. Let $x_0 \in U$, $h \in X$. Define

$$\varphi: [0, 1] \rightarrow Y \text{ by } \varphi(t) = f(x_0 + t \cdot h) + \sum_{k=1}^n \frac{(1-t)^k}{k!} \cdot D_{x_0+t \cdot h}^{[k]} f \left(\underbrace{h: \dots: h}_k \right)$$

We decompose now f different sub-functions that are continuous. So define

$$\tau: [0, 1] \rightarrow X \text{ by } \tau(t) = x_0 + t \cdot h \tag{16.623}$$

then we have

$$\tau([0, 1]) = \{x_0 + t \cdot h | t \in [0, 1]\} = L_{x_0, x_0+h} \subseteq U \tag{16.624}$$

Using [examples: 14.164, 14.142] and [theorems: 14.152, 14.165] it follows that

$$\tau \text{ is continuous} \tag{16.625}$$

As f is $(n+1)$ -times differentiable on U it follows from [theorem: 16.138] that f is 1-times differentiable on U . Hence by [theorem: 16.140] f is Fréchet differentiable on U so that by [theorem: 16.63] f is continuous. Using now [theorem: 14.159] proves that

$$f \circ \tau \text{ is continuous} \quad (16.626)$$

Define now for $k \in \{1, \dots, n\}$

$$\gamma_k: [0, 1] \rightarrow \mathbb{R} \text{ by } \gamma_k(t) = (1-t)^k \quad (16.627)$$

then we have by [examples: 14.143, 14.142], [theorem: 14.152] and [corollary: 14.169] that

$$\gamma_k \text{ is continuous} \quad (16.628)$$

Let $k \in \{1, \dots, n\}$ then as f is $(n+1)$ -times differentiable on U it follows from [theorem: 16.136] that $D^{[k]}f$ is Fréchet differentiable on U so that by [theorem: 16.63]

$$D^{[k]}f \text{ is continuous} \quad (16.629)$$

Define now

$$\Theta_k: U \rightarrow Y \text{ by } \Theta_k(x) = D^{[k]}f(x) \left(\underbrace{h: \dots: h}_k \right) \quad (16.630)$$

then by [lemma: 16.379] it follows that

$$\Theta_k \text{ is continuous}$$

Using [theorem: 14.159] on the above together with [eq: 16.625] it follows that

$$\Theta_k \circ \tau \text{ is continuous} \quad (16.631)$$

Let $t \in [0, 1]$ then

$$\begin{aligned} & \left((f \circ \tau) + \sum_{k=1}^n \frac{\gamma_k}{k!} \cdot (\Theta_k \circ \tau) \right)(t) = \\ & (f \circ \tau)(t) + \sum_{k=1}^n \frac{\gamma_k(t)}{k!} \cdot (\Theta_k \circ \tau)(t) = \\ & f(\tau(t)) + \sum_{k=1}^n \frac{\gamma_k(t)}{k!} \cdot \Theta_k(\tau(t)) = \\ & f(x_0 + t \cdot h) + \sum_{k=1}^n \frac{(1-t)^k}{k!} \cdot D^{[k]}f(\tau(t)) \left(\underbrace{h: \dots: h}_k \right) = \\ & f(x_0 + t \cdot h) + \sum_{k=1}^n \frac{(1-t)^k}{k!} \cdot D^{[k]}f(x_0 + t \cdot h) \left(\underbrace{h: \dots: h}_k \right) \quad [\text{eq: 16.136}] \\ & f(x_0 + t \cdot h) + \sum_{k=1}^n \frac{(1-t)^k}{k!} \cdot D_{x_0+t \cdot h}^{[k]}f \left(\underbrace{h: \dots: h}_k \right) = \\ & \varphi(t) \end{aligned}$$

which proves that

$$\varphi = (f \circ \tau) + \sum_{k=1}^n \frac{\gamma_k}{k!} \cdot (\Theta_k \circ \tau) \quad (16.632)$$

Using then [theorems: 14.165, 14.166, 14.167] together with [eqs: 16.626, 16.628 and 16.631] on the above proves that

$$\varphi \text{ is continuous on } [0, 1]$$

which proves the first part of the lemma. Using [example: 16.32] it follows that

$$\forall t \in]0, 1[\quad \tau \text{ has a derivative at } t \text{ with } \tau'_t = h \quad (16.633)$$

Further as f is $(n+1)$ -times differentiable on U it follows from [theorems: 16.138, 16.140] that f is Fréchet differentiable on U . Hence using [theorem: 16.64] it follows that

$$\forall t \in U \quad f \text{ has a derivative at } t \text{ with } f'_t = D_t f(1) = D_t^{[1]}f(1)$$

As $\tau([0, 1]) \subseteq U$ [see eq: 16.624] we can use the chain rule [see theorem: 16.68] that

$$\forall t \in]0, 1[\ (f \circ \tau)'_t \text{ exist with } (f \circ \tau)'_t = \tau'_t \cdot f'_{\tau(t)} = h \cdot D_{x_0+t \cdot h}^{[1]} f(1) = D_{x_0+t \cdot h}^{[1]} f(h) \quad (16.634)$$

Let $k \in \{1, \dots, n\}$ then by [examples: 16.30, 16.31] it follows that

$$\forall t \in]0, 1[\ (\gamma_k)'_t \text{ with } (\gamma_k)'_t = -k \cdot (1-t)^{k-1} \quad (16.635)$$

Let $k \in \{1, \dots, n\} \Rightarrow k+1 \in \{2, \dots, k+1\}$ then as f is $(n+1)$ -times differentiable on U it follows from [theorem: 16.136] that f is $(k+1)$ -times differentiable on U . Using [lemma: 16.378] it follows that

$$\forall t \in U \ \Theta_k \text{ is Fréchet differentiable at } t \text{ with } \forall y \in X \ D_t \Theta_k(y) = D_t^{[k+1]} f(y) \left(\underbrace{h: \dots: h}_k \right) \quad (16.636)$$

Let $t \in]0, 1[$ then as τ has a derivative at t [see eq: 16.633] it follows from [theorem: 16.64] that

$$\tau \text{ is Fréchet differentiable at } t \text{ where } D_t \tau \text{ is defined by } \forall y \in \mathbb{R} \ D_t \tau(y) = y \cdot t'_{t[\text{eq: 16.633}]} = y \cdot h \quad (16.637)$$

Using the chain rule with the above and [eq: 16.636] proves that

$$\Theta_k \circ \tau \text{ is Fréchet differentiable at } t \text{ with } D_t(\Theta_k \circ \tau) = D_{\tau(t)} \Theta_k \circ D_t \tau \quad (16.638)$$

Using [theorem: 16.64] it follows from the above that

$$\Theta_k \circ \tau \text{ has a derivative at } t \text{ with } (\Theta_k \circ \tau)'_t = (D_{\tau(t)} \Theta_k \circ D_t \tau)(1)$$

Now

$$\begin{aligned} (\Theta_k \circ \tau)'_t &= (D_{\tau(t)} \Theta_k \circ D_t \tau)(1) \\ &= (D_{x_0+t \cdot h} \Theta_k \circ D_t \tau)(1) \\ &= D_{x_0+t \cdot h} \Theta_k(D_t \tau(1)) \\ &\stackrel{[\text{eq: 16.637}]}{=} D_{x_0+t \cdot h} \Theta_k(1 \cdot h) \\ &= D_{x_0+t \cdot h} \Theta_k(h) \\ &\stackrel{[\text{eq: 16.636}]}{=} D_{x_0+t \cdot h}^{[k+1]} f(h) \left(\underbrace{h: \dots: h}_k \right) \\ &= D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) \end{aligned}$$

To summarize we have that

$$\forall t \in]0, 1[\ (\Theta_k \circ \tau)'_t \text{ exists with } (\Theta_k \circ \tau)'_t = D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) \quad (16.639)$$

Using now [theorem: 16.41] together with [eq: 16.635] and [eq: 16.639] proves that $\forall t \in]0, 1[$

$$(\gamma_k \cdot \Theta_k \circ \tau)'_t \text{ exist} \quad (16.640)$$

with

$$\begin{aligned} (\gamma_k \cdot \Theta_k \circ \tau)' &= \gamma_k(t) \cdot (\Theta_k \circ \tau)'_t + (\gamma_k)'_t \cdot (\Theta_k \circ \tau)(t) \\ &\stackrel{[\text{eq: 16.635}]}{=} \gamma_k(t) \cdot (\Theta_k \circ \tau)'_t + (-k \cdot (1-t)^{k-1}) \cdot (\Theta_k \circ \tau)(t) \\ &= \gamma_k(t) \cdot (\Theta_k \circ \tau)'_t + (-k \cdot (1-t)^{k-1}) \cdot \Theta_k(\tau(t)) \\ &= \gamma_k(t) \cdot (\Theta_k \circ \tau)'_t + (-k \cdot (1-t)^{k-1}) \cdot \Theta_k(x_0+t \cdot h) \\ &\stackrel{[\text{eq: 16.630}]}{=} \gamma_k(t) \cdot (\Theta_k \circ \tau)'_t + (-k \cdot (1-t)^{k-1}) \cdot D^{[k]} f(x_0+t \cdot h) \left(\underbrace{h: \dots: h}_k \right) \\ &= \gamma_k(t) \cdot (\Theta_k \circ \tau)'_t + (-k \cdot (1-t)^{k-1}) \cdot D^{[k]} f_{x_0+t \cdot h} \left(\underbrace{h: \dots: h}_k \right) \\ &\stackrel{[\text{eq: 16.639}]}{=} \gamma_k(t) \cdot D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) + (-k \cdot (1-t)^{k-1}) \cdot D^{[k]} f_{x_0+t \cdot h} \left(\underbrace{h: \dots: h}_k \right) \\ &= (1-t)^k \cdot D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) - k \cdot (1-t)^{k-1} \cdot D^{[k]} f_{x_0+t \cdot h} \left(\underbrace{h: \dots: h}_k \right) \end{aligned}$$

To summarize

$$\forall t \in]0, 1[\quad (1-t)^k \cdot D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) - k \cdot (1-t)^{k-1} \cdot D_{x_0+t \cdot h}^{[k]} f \left(\underbrace{h: \dots: h}_k \right) \quad (16.641)$$

Using now [theorem: 16.40] on [eqs: 16.632, 16.634 and 16.640] it follows that $\forall t \in]0, 1[$

φ has a derivative at t

with

$$\begin{aligned} \varphi'_t &= \left((f \circ \tau) + \sum_{k=1}^n \frac{\gamma_k}{k!} \cdot (\Theta_k \circ \tau) \right)'_t = \\ &= (f \circ \tau)'_t + \sum_{k=1}^n \left(\frac{\gamma_k}{k!} \cdot (\Theta_k \circ \tau) \right)'_t = \\ &= (f \circ \tau)'_t + \sum_{k=1}^n \frac{1}{k!} \cdot (\gamma_k \cdot (\Theta \circ \tau))'_t = \\ &= D_{x_0+t \cdot h}^{[1]} f(h) + \sum_{k=1}^n \frac{1}{k!} \cdot (\gamma_k \cdot (\Theta \circ \tau))'_t = \\ &= D_{x_0+t \cdot h}^{[1]} f(h) + \sum_{k=1}^n \frac{1}{k!} \cdot \left((1-t)^k \cdot D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) - k \cdot (1-t)^{k-1} \cdot D_{x_0+t \cdot h}^{[k]} f \left(\underbrace{h: \dots: h}_k \right) \right) = \\ &= D_{x_0+t \cdot h}^{[1]} f(h) + \sum_{k=1}^n \left(\frac{(1-t)^k}{k!} \cdot D_{x_0+t \cdot h}^{[k+1]} f \left(\underbrace{h: \dots: h}_{k+1} \right) - \frac{(1-t)^{k-1}}{(k-1)!} \cdot D_{x_0+t \cdot h}^{[k]} f \left(\underbrace{h: \dots: h}_k \right) \right) = \\ &= D_{x_0+t \cdot h}^{[1]} f(h) + \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots: h}_{n+1} \right) - \frac{(1-t)^{1-1}}{(1-1)!} \cdot D_{x_0+t \cdot h}^{[1]} f \left(\underbrace{h: \dots: h}_1 \right) = \\ &= D_{x_0+t \cdot h}^{[1]} f(h) + \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots: h}_{n+1} \right) - D_{x_0+t \cdot h}^{[1]} f(h) = \\ &= \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots: h}_{n+1} \right) \end{aligned}$$

Summarized we have that

$$\forall t \in]0, 1[\quad \varphi \text{ has a derivative at } t \text{ with } \varphi'_t = \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots: h}_{n+1} \right) \quad \square$$

Theorem 16.382. (Taylor's Theorem II.2 (Lagrange's remainder)) Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U a open set in X , $x_0 \in U$, $h \in X$ such that the line segment $L_{x_0, x_0+h} \subseteq U$ and $f: U \rightarrow Y$ a function that is $(n+1)$ -times differentiable on U and for which

$$\sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right) \text{ exist}$$

then

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h: \dots: h}_k \right) + R_{x_0, h, n}$$

where

$$\|R_{x_0, h, n}\|_Y \leq \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right)$$

Proof. Define

$$\varphi: [0, 1] \rightarrow Y \text{ by } \varphi(t) = f(x_0 + t \cdot h) + \sum_{k=1}^n \frac{(1-t)^k}{k!} \cdot D_{x_0+t \cdot h}^{[k]} f \left(\underbrace{h: \dots: h}_k \right)$$

then by the previous [lemma 16.380] we have

φ is continuous on $[0, 1]$

and

$$\forall t \in]0, 1[\text{ that } \varphi'_t \text{ exist with } \varphi'_t = \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots : h}_{n+1} \right)$$

Hence $\forall]0, 1[$ we have that

$$\begin{aligned} \|\varphi'_t\|_Y &= \left\| \frac{(1-t)^n}{n!} \cdot D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots : h}_{n+1} \right) \right\|_Y \\ &= \left| \frac{(1-t)^n}{n!} \right| \cdot \left\| D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots : h}_{n+1} \right) \right\|_Y \\ &\stackrel{t \in]0, 1[}{=} \frac{(1-t)^n}{n!} \cdot \left\| D_{x_0+t \cdot h}^{[n+1]} f \left(\underbrace{h: \dots : h}_{n+1} \right) \right\|_Y \\ &\stackrel{[\text{lemma: 16.122}]}{\leq} \frac{(1-t)^n}{n!} \cdot \|D_{x_0+t \cdot h}^{[n+1]}\|_{L_{n+1}(X;Y)} \cdot \|h\|_X^{n+1} \\ &\leq \frac{(1-t)^n}{n!} \cdot \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right) \cdot \|h\|_X^{n+1} \end{aligned} \quad (16.642)$$

Define now

$$\psi: [0, 1] \rightarrow \mathbb{R} \text{ by } \psi(t) = -\frac{(1-t)^{n+1}}{(n+1)!} \cdot \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right) \cdot \|h\|_X^{n+1}$$

then by [examples: 14.143, 14.142], [theorems: 14.165, 14.152] and [corollary: 14.169] it follows that

ψ is continuous

and by [example: 16.32] it follows that $\forall t \in]0, 1[$ ψ'_t exists with

$$\psi'_t = \frac{(1-t)^n}{n!} \cdot \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right) \cdot \|h\|_X^{n+1}$$

Substituting the above in [eq: 16.642] we have that

$$\forall t \in]0, 1[\|\varphi'_t\|_Y \leq \psi'_t$$

So we can use the mean value theorem [theorem: 16.234] to get

$$\|\varphi(1) - \varphi(0)\|_Y \leq \psi(1) - \psi(0) \quad (16.643)$$

Define now

$$R_{x_0, h, n} = f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h: \dots : h}_k \right) \quad (16.644)$$

so that

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h: \dots : h}_k \right) + R_{x_0, h, n}$$

then we have

$$\begin{aligned} \varphi(1) - \varphi(0) &= f(x_0 + 1 \cdot h) + \sum_{k=1}^n \frac{(1-0)^k}{k!} \cdot D_{x_0+1 \cdot h}^{[k]} f \left(\underbrace{h: \dots : h}_k \right) - \varphi(0) \\ &= f(x_0 + 1 \cdot h) - \varphi(0) \\ &= f(x_0 + h) - f(x_0 + 0 \cdot h) - \sum_{k=1}^n \frac{(1-0)^k}{k!} \cdot D_{x_0+0 \cdot h}^{[k]} f \left(\underbrace{h: \dots : h}_k \right) \\ &= f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h: \dots : h}_k \right) \\ &\stackrel{[\text{eq: 16.644}]}{=} R_{x_0, h, n} \end{aligned} \quad (16.645)$$

Further

$$\begin{aligned}
 \psi(1) - \psi(0) &= -\frac{(1-1)^{n+1}}{(n+1)!} \cdot \sup(\{\|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} | t \in [0, 1]\}) \cdot \|h\|_X^{n+1} - \psi(0) \\
 &= -\psi(0) \\
 &= \frac{(1-0)^{n+1}}{(n+1)!} \cdot \sup(\{\|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} | t \in [0, 1]\}) \cdot \|h\|_X^{n+1} \\
 &= \frac{1}{(n+1)!} \cdot \sup(\{\|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} | t \in [0, 1]\}) \cdot \|h\|_X^{n+1} \quad (16.646)
 \end{aligned}$$

Finally combining [eqs: 16.643, 16.645, 16.646] gives

$$\|R_{x_0, h, n}\|_Y \leq \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot \sup(\{\|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} | t \in [0, 1]\}) \quad \square$$

Theorem 16.383. (Taylor's Theorem (II.3) (Lagrange's remainder) *Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, U an open set in X , $x_0 \in U$, $h \in X$ such that the line segment $L_{x_0, x_0+h} \subseteq U$ and $f: U \rightarrow Y$ a function that is of class C^{n+1} then*

$$\sup(\{\|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} | t \in [0, 1]\}) \text{ exist}$$

and

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f(\underbrace{h, \dots, h}_k) + R_{x_0, h, n}$$

where

$$\|R_{x_0, h, n}\|_Y \leq \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot \sup(\{\|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} | t \in [0, 1]\})$$

Proof. As f is of class C^{n+1} it follows that f is $(n+1)$ -differentiable on U and

$$D^{[n+1]} f: U \rightarrow L_{n+1}(X; Y) \text{ defined by } D^{[n+1]} f(x) = D_x^{[n+1]} f \text{ is continuous}$$

Define now

$$\tau: [0, 1] \rightarrow X \text{ by } \tau(t) = x_0 + t \cdot h$$

then

$$\tau([0, 1]) = L_{x_0, x_0+h} \subseteq U$$

Further using [examples: 14.164, 14.142] and [theorems: 14.152, 14.165] it follows that τ is continuous. Hence using [theorem: 14.159]

$$D^{[n+1]} f \circ \tau: [0, 1] \rightarrow L_{n+1}(X; Y)$$

is continuous. By Heine Borell [theorem: 14.279] $[0, 1]$ is a compact set so that by [theorem: 14.275] $(D^{[n+1]} f \circ \tau)([0, 1])$ is compact. Applying then [theorem: 14.273] proves that $(D^{[n+1]} f \circ \tau)([0, 1])$ is bounded. Hence there exist a $M \in \mathbb{R}^+$ such that $\forall L_1, L_2 \in (D^{[n+1]} f \circ \tau)([0, 1])$ we have $\|L_1 - L_2\|_{L_{n+1}(X;Y)} \leq M$. Let $t \in [0, 1]$ then we have

$$\begin{aligned}
 \|(D^{[n+1]} f \circ \tau)(t)\|_{L_n(X;Y)} &= \\
 \|(D^{[n+1]} f \circ \tau)(t) - (D^{[n+1]} f \circ \tau)(0) + (D^{[n+1]} f \circ \tau)(0)\|_{L_{n+1}(X;Y)} &\leq \\
 \|(D^{[n+1]} f \circ \tau)(t) - (D^{[n+1]} f \circ \tau)(0)\|_{L_{n+1}(X;Y)} + \|(D^{[n+1]} f \circ \tau)(0)\|_{L_{n+1}(X;Y)} &\leq \\
 M + \|(D^{[n+1]} f \circ \tau)(0)\|_{L_{n+1}(X;Y)} &
 \end{aligned}$$

which proves that

$$\{\|(D^{[n+1]} f \circ \tau)(t)\|_{L_{n+1}} | t \in [0, 1]\} \text{ is bounded above}$$

Further we have

$$\begin{aligned}
 s \in \{ \| (D^{[n+1]} f \circ \tau)(t) \|_{L_{n+1}(X;Y)} | t \in [0, 1] \} &\Leftrightarrow \exists t \in [0, 1] \text{ s.t. } \| (D^{[n+1]} f \circ \tau)(t) \|_{L_{n+1}(X;Y)} \\
 &\Leftrightarrow \exists t \in [0, 1] \text{ s.t. } \| D^{[n+1]} f(\tau(t)) \|_{L_{n+1}(X;Y)} \\
 &\Leftrightarrow \exists t \in [0, 1] \text{ s.t. } \| D^{[n+1]} f(x_0 + t \cdot h) \|_{L_{n+1}(X;Y)} \\
 &\Leftrightarrow \exists t \in [0, 1] \text{ s.t. } \| D_{x_0+t \cdot h}^{[n+1]} f \|_{L_{n+1}(X;Y)} \\
 &\Leftrightarrow s \in \{ \| D_{x_0+t \cdot h}^{[n+1]} f \|_{L_{n+1}(X;Y)} | t \in [0, 1] \}
 \end{aligned}$$

which proves that $\{ \| (D^{[n+1]} f \circ \tau)(t) \|_{L_{n+1}(X;Y)} | t \in [0, 1] \} = \{ \| D_{x_0+t \cdot h}^{[n+1]} f \|_{L_{n+1}(X;Y)} | t \in [0, 1] \}$. Hence $\{ \| D_{x_0+t \cdot h}^{[n+1]} f \|_{L_{n+1}(X;Y)} | t \in [0, 1] \}$ is bounded above. As $\mathbb{R}$ is conditionally complete [see theorem: 10.18] and $\{ \| D_{x_0+t \cdot h}^{[n+1]} f \|_{L_{n+1}(X;Y)} | t \in [0, 1] \} \neq \emptyset$ it follows that

$$\sup \{ \| D_{x_0+t \cdot h}^{[n+1]} f \|_{L_{n+1}(X;Y)} | t \in [0, 1] \} \text{ exist}$$

Applying now [theorem: 16.382] finishes the proof. $\square$

We introduce now a version of the Taylor's theorem where the remainder is a Riemann integral.

Lemma 16.384. *Let $\langle X, \| \cdot \| \rangle$ be a **Banach** space, $n \in \mathbb{N} \setminus \{1\}$, $a, b \in \mathbb{R}$ with $a < b$ and $f: [a, b] \rightarrow Y$ a function such that*

1. $[a, b] = \mathcal{D}_f^n$ [or equivalently by [definition: 16.166] $\forall t \in [a, b] \ f_t^{(n)}$ exist]
2. $f^{(n)}: \mathcal{D}_f^n \rightarrow X$ is continuous.

then for the function

$$R_{n,b,f}: [a, b] \rightarrow X \text{ defined by } R_{n,b,f}(t) = (b-t)^{n-1} \cdot f^{(n)}(t)$$

we have that $R_{n,b,f}$ is Riemann integrable and

$$f(b) = f(a) + \sum_{k=1}^{n-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{(n-1)!} \cdot \int_a^b R_{n,b,f}$$

Note 16.385. Using [theorem: 16.169] we have that $\forall m \in \{1, \dots, m-1\}$ $f_a^{(m)}$ exist which ensures that the sum is well defined.

Proof. We use induction to prove this so let

$$S = \left\{ n \in \{2, \dots, \infty\} \mid \text{If } [a, b] = \mathcal{D}_f^n \text{ and } f^{(n)} \text{ is continuous then } R_{n,b,f} \text{ is Riemann integrable} \right.$$

$$\text{and } f(b) = f(a) + \sum_{k=1}^{n-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{(n-1)!} \cdot \int_a^b R_{n,b,f} \text{ where } R_{n,b,f}: [a, b] \rightarrow X \text{ is defined by}$$

$$R_{n,b,f}(t) = (b-t)^{n-1} \cdot f^{(n)}(t) \left. \right\}$$

then we have:

2 $\in S$. Assume that $[a, b] = \mathcal{D}_f^2$ and $f^{(2)}$ is continuous. Using [theorem: 16.164] it follows that $[a, b] \subseteq \mathcal{D}_f^2 \subseteq \mathcal{D}_f^1 \subseteq [a, b]$ so that we have

$$\mathcal{D}_{f|_{\text{definition: 16.162}}} = \mathcal{D}_f^1 = \mathcal{D}_f^2 = [a, b]$$

Further using [definition: 16.162] twice we have that

$$f^{(2)} = (f^{(1)})' = (f')'$$

and

$$\mathcal{D}_f^2 = \{ x \in \mathcal{D}_f^1 = [a, b] \mid f' \text{ has a derivative at } x \}$$

so that by [theorem: 16.33]

$$f' \text{ is continuous}$$

Further as $f^{(2)}$ is assumed to be continuous and $(f')' = f^{(2)}$ it follows that

$$(f')' \text{ is continuous}$$

Then by [theorem: 16.167] we have that f has derivatives on $[a, b]$ and

$$f': [a, b] \rightarrow X \text{ defined by } f'(t) = f_t$$

Further if we define

$$h: [a, b] \rightarrow \mathbb{R} \text{ by } h(t) = b - t$$

then we have by [examples: 16.30, 16.31] and [theorem: 16.40] that $\forall t \in [a, b]$ h has a derivative h'_t at t with $h'_t = -1$. So we have if we define

$$h': [a, b] \rightarrow \mathbb{R} \text{ by } h'(t) = h'_t$$

that $\forall t \in [a, b]$ $h'(t) = -1$ which proves

$$h' = C_{-1}$$

and by [example: 14.142] that

$$h' \text{ is continuous}$$

Define

$$R_{2,b,f}: [a, b] \rightarrow X \text{ by } R_{2,b,f}(t) = (b-t)^{(2-1)} \cdot f^{(2)}(t) = (b-t) \cdot f^{(2)}(t)$$

then $\forall t \in [a, b]$ we have $R_{2,b,f}(t) = (b-t) \cdot f^{(2)}(t) = h(t) \cdot f^{(2)}(t) = h(t) \cdot (f')'$ so that

$$R_{2,b,f} = h \cdot f^{(2)} = h \cdot (f')'$$

Let $t \in [a, b]$ then $-f'(t) = C_{-1}(t) \cdot f'(t) = h'(t) \cdot f'(t)$ which proves that

$$-f' = h' \cdot f'$$

Using integration by parts [see theorem: 16.217] it follows that $h \cdot (f')'$ is Riemann integrable which proves that

$$R_{2,b,f} = h' \cdot (f')' \text{ is Riemann integrable}$$

Further

$$\begin{aligned} f(a) - f(b) &= -(f(b) - f(a)) \\ &\stackrel{[\text{theorem: 16.216}]}{=} -\int_a^b f' \\ &\stackrel{[\text{theorem: 15.28}]}{=} \int_a^b -f' \\ &= \int_a^b h' \cdot f' \\ &\stackrel{[\text{theorem: 16.217}]}{=} h(b) \cdot f'(b) - h(a) \cdot f'(a) - \int_a^b h \cdot (f')' \\ &= (b-b) \cdot f'(b) - (b-a) \cdot f'(a) - \int_a^b R_{2,b,f} \\ &= -(b-a) f'(a) - \int_a^b R_{2,b,f} \end{aligned}$$

so that

$$\begin{aligned} f(b) &= f(a) + (b-a) \cdot f'(a) + \int_a^b R_{2,b,f} \\ &= f(a) + \frac{(b-a)^1}{1!} \cdot f^{(1)}(a) + \frac{1}{(2-1)!} \int_a^b R_{2,b,f} \\ &= f(a) + \sum_{k=1}^1 \frac{(b-a)^k}{k!} + \frac{1}{(2-1)!} \int_a^b R_{2,b,f} \end{aligned}$$

which proves that $2 \in S$.

$n \in S \Rightarrow n+1 \in S$. Assume that $\mathcal{D}_f^{n+1} = [a, b]$ and that $f^{(n+1)}$ is continuous. Using [theorem: 16.164] it follows that $[a, b] \subseteq \mathcal{D}_f^{n+1} \subseteq \mathcal{D}_f^n \subseteq [a, b]$ so that

$$\mathcal{D}_f^{n+1} = \mathcal{D}_f^n = [a, b]$$

Using [definition: 16.162] it follows that

$$f^{(n+1)} = (f^{(n)})'$$

and that $\mathcal{D}_f^{n+1} = \{x \in \mathcal{D}_f^n = [a, b] \mid f^{(n)} \text{ is Fréchet differentiable at } x\}$ so that by [theorem: 16.33]

$$f^{(n)} \text{ is continuous}$$

Further as $f^{(n+1)}$ is continuous and $f^{(n+1)} = (f^{(n)})'$ it follows that

$$(f^{(n)})' \text{ is continuous}$$

Define now

$$h: [a, b] \rightarrow \mathbb{R} \text{ by } h(t) = (b-t)^n$$

then using [examples: 14.142, 14.143] and [theorems: 14.165, 14.169 and 14.152] it follows that

$$h \text{ is continuous}$$

Further we have by [examples: 16.30, 16.31] and [theorems: 16.40, 16.108 and 16.68] that $\forall t \in [a, b]$ h'_t exist with $h'_t = -n \cdot (b-t)^{n-1}$. So if we define

$$h': [a, b] \rightarrow \mathbb{R} \text{ by } h'(t) = h'_t$$

then

$$h' = -n \cdot g \text{ where } g: [a, b] \rightarrow \mathbb{R} \text{ is defined by } g(t) = (b-t)^{n-1}$$

Further by [examples: 14.142, 14.143] and [theorems: 14.165, 14.170] it follows that

$$g, h' \text{ are continuous}$$

and by [theorem: 14.167] that

$$g \cdot f^{(n)} \text{ is continuous}$$

Now using integration by parts [see theorem: 16.217] it follows that

$$h \cdot (f^{(n)})' = h \cdot f^{(n+1)} \text{ is Riemann integrable} \quad (16.647)$$

Now

$$\begin{aligned} -n \int_a^b g \cdot f^{(n)} &\stackrel{[\text{theorem: 15.28}]}{=} \int_a^b (-n \cdot g) \cdot f^{(n)} \\ &= \int_a^b h' \cdot f^{(n)} \\ &\stackrel{[\text{theorem: 16.217}]}{=} h(b) \cdot f^{(n)}(b) - h(a) \cdot f^{(n)}(a) - \int_a^b h \cdot (f^{(n)})' \\ &= (b-b)^n \cdot f^{(n)}(b) - (b-a)^n \cdot f^{(n)}(a) - \int_a^b h \cdot f^{(n+1)} \\ &= -(b-a)^n \cdot f^{(n)}(a) - \int_a^b h \cdot f^{(n+1)} \end{aligned}$$

so that

$$\int_a^b g \cdot f^{(n)} = \frac{1}{n} \cdot \left((b-a)^n \cdot f^{(n)}_a + \int_a^b h \cdot f^{(n+1)} \right) \quad (16.648)$$

As $n \in S$ we have that

$$f(b) = f(a) + \sum_{k=1}^{n-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{(n-1)!} \cdot \int_a^b R_{n,b,f}$$

where

$$R_{n,b,f}: [a, b] \rightarrow \mathbb{R} \text{ is defined by } R_{n,b,f}(t) = (b-t)^{n-1} \cdot f^{(n)}(t) = (g \cdot f^{(n)})(t)$$

is Riemann integrable. So

$$\begin{aligned} f(b) &= f(a) + \sum_{k=1}^{n-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{(n-1)!} \int_a^b g \cdot f^{(n)} \\ &\stackrel{[\text{eq: 16.648}]}{=} f(a) + \sum_{k=1}^{n-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{(n-1)!} \cdot \frac{1}{n} \cdot \left((b-a)^n \cdot f_a^{(n)} + \int_a^b h \cdot f^{(n+1)} \right) \\ &= f(a) + \sum_{k=1}^{n-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{n!} \cdot (b-a)^n \cdot f_a^{(n)} + \frac{1}{n!} \cdot \int_a^b h \cdot f^{(n+1)} \\ &= f(a) + \sum_{k=1}^{(n+1)-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{((n+1)-1)!} \int_a^b h \cdot f^{(n+1)} \end{aligned}$$

Define now

$$R_{n+1,b,f}: [a, b] \rightarrow X \text{ by } R_{n+1,b,f}(t) = (b-t)^n \cdot f^{(n+1)}(t) = (h \cdot f^{(n+1)})(t)$$

so that by [eq: 16.647]

$R_{n+1,b,f}$ is Riemann integrable

and

$$f(b) = f(a) + \sum_{k=1}^{(n+1)-1} \frac{(b-a)^k}{k!} \cdot f_a^{(k)} + \frac{1}{((n+1)-1)!} \int_a^b R_{n+1,b,f}$$

which proves that $n+1 \in S$. □

Theorem 16.386. (Taylor's Theorem III) Let $\langle X, \|\cdot\| \rangle$ be a normed space, $n \in \mathbb{N} \setminus \{1\}$, $A \subseteq \mathbb{R}$ with $x_0, x \in \mathbb{R}$ with $x_0 < x$, $[x_0, x] \subseteq A$ and $f: A \rightarrow Y$ a function such that

1. $[x_0, x] \subseteq \mathcal{D}_f^n$ [or equivalently by [definition: 16.166] $\forall t \in [a, b]$ $f_t^{(n)}$ exist].
2. $(\mathcal{D}^n f)|_{[x_0, x]}$ is continuous.

then for the function

$$R_{n,x,f}: [x_0, x] \rightarrow X \text{ defined by } R_{n,x,f}(t) = (x-t)^{n-1} \cdot f^{(n)}(t)$$

we have

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n,x,f}$$

Note 16.387. Using [theorem: 16.169] we have that $\forall m \in \{1, \dots, m-1\}$ $f_x^{(m)}$ exist which ensures that the sum is well defined.

Note 16.388. Using [notation: 15.21] we have

$$\int_{x_0}^x R_{n,x,f} = \int_{x_0}^x (x-t)^{n-1} \cdot f^{(n)}(t) dt$$

so that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x (x-t)^{n-1} \cdot f^{(n)}(t) dt$$

Proof. As $[x_0, x] \subseteq \mathcal{D}_f^n$ and by [theorem: 16.10] $[x_0, x] \subseteq [x_0, x]'$ it follows from [theorem: 16.168] that

$$[x_0, x] = \mathcal{D}_{f|_{[x_0, x]}}^n \text{ and } (f|_{[x_0, x]})^{(n)} = (f^{(n)})|_{[x_0, x]} \quad (16.649)$$

Hence we have $\forall t \in [x_0, x]$

$$f_{|[x_0, x]}(t) = f(t) \text{ and } (f_{|[x_0, x]})_t^{(n)} = (f_{|[x_0, x]})^{(n)} = (f^{(n)})_{|[x_0, x]}(t) = f^{(n)}(t) = f_t^{(n)}$$

so that

$$\forall t \in [x_0, x] \quad f_{|[x_0, x]}(t) = f(t) \text{ and } (f_{|[x_0, x]})_t^{(n)} = f_t^{(n)} \quad (16.650)$$

As $(f^{(n)})_{|[x_0, x]}$ is continuous it follows from [theorem: 14.146] that $(f^{(n)})_{|[x_0, x]}$ is continuous or using [eq: 16.649] that

$$(f_{|[x_0, x]})^{(n)} \text{ is continuous} \quad (16.651)$$

Using [lemma: 16.384] together with [eqs: 16.649, 16.651] gives for

$$R_{n, x, f_{|[x_0, x]}} : [x_0, x] \rightarrow X \text{ is defined by } R_{n, x, f_{|[x_0, x]}}(t) = (x - t)^{n-1} \cdot (f_{|[x_0, x]})^n(t) \quad (16.652)$$

that

$$f_{|[x_0, x]}(x) = f_{|[x_0, x]}(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot (f_{|[x_0, x]})_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n, x, f_{|[x_0, x]}} \quad (16.653)$$

Let $t \in [x_0, x]$ then

$$\begin{aligned} R_{n, x, f_{|[x_0, x]}}(t) &= (x - t)^{n-1} \cdot (f_{|[x_0, x]})^n(t) \\ &\stackrel{[\text{eq: 16.649}]}{=} (x - t)^{n-1} \cdot (f^{(n)})_{|[x_0, x]}(t) \\ &= (x - t)^{n-1} \cdot f^{(n)}(t) \\ &= R_{n, x, f}(t) \end{aligned}$$

so that

$$R_{n, x, f_{|[x_0, x]}} = R_{n, x, f} \quad (16.654)$$

Further

$$\begin{aligned} f(x) &\stackrel{[\text{eq: 16.650}]}{=} f_{|[x_0, x]}(x) \\ &\stackrel{[\text{eq: 16.653}]}{=} f_{|[x_0, x]}(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot (f_{|[x_0, x]})_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n, x, f_{|[x_0, x]}} \\ &\stackrel{[\text{eq: 16.654}]}{=} f_{|[x_0, x]}(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot (f_{|[x_0, x]})_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n, x, f} \\ &\stackrel{[\text{eq: 16.650}]}{=} f(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n, x, f} \end{aligned}$$

so that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} + \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n, x, f} \quad \square$$

The above form of Taylor's theorem allows us to introduce Taylor series where a function can be written as a power series.

Theorem 16.389. *Let $\langle X, ||| \rangle$ be a normed space, $A \subseteq \mathbb{R}$, $x_0, x \in \mathbb{R}$ with $x_0 < x$, $[x_0, x] \subseteq A$ and $f: A \rightarrow Y$ a function such that $\forall n \in \mathbb{N}$ $[x_0, x] \subseteq \mathcal{D}_f^n$ for equivalently [see definition: 16.166] f has n -times derivatives on $[x_0, x]$ then we have that*

$$1. \quad \forall n \in \mathbb{N} \setminus \{1\} \quad f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} \cdot f_{x_0}^{(k)} + T_n(x) \text{ where}$$

$$T_n(x) = \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n, x, f}$$

$$R_{n, x, f} : [x_0, x] \rightarrow X \text{ by } R_{n, x, f}(t) = (x - t)^{n-1} \cdot f^{(n)}(t)$$

2.

$$\begin{aligned} \lim_{n \rightarrow \infty} T_n(x) \text{ exist and } \lim_{n \rightarrow \infty} T_n(x) = 0 \\ \Updownarrow \\ \sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} \text{ exist and } f(x) = f(x_0) + \sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} \end{aligned}$$

Proof.

1. Let $n \in \mathbb{N} \setminus \{1\}$ then by the hypothesis we have

$$[x_0, x] \subseteq \mathcal{D}_f^n \text{ and also } [x_0, x] \subseteq \mathcal{D}_f^{n+1}$$

Hence if $y \in [x_0, x]$ then $y \in \mathcal{D}_f^{n+1}$ so that by [definition: 16.165] f has a $(n+1)$ -times derivative at y . Using then [theorem: 16.167] it follows that $f^{(n)}$ has a derivative at y so that by [theorem: 16.33] it is continuous at y . As this is valid $\forall y \in [x_0, x]$ it follows that

$f^{(n)}$ is continuous

So all the conditions for Taylor's Theorem III [see theorem: 16.386] are satisfied from which it follows that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} + T_n(x)$$

where

$$T_n(x) = \frac{1}{(n-1)!} \cdot \int_{x_0}^x R_{n,x,f}$$

$$R_{n,x,f}: [x_0, x] \rightarrow X \text{ by } R_{n,x,f}(t) = (x-t)^n \cdot f^{(n)}(t)$$

2. Using (1) we have that

$$T_n(x) = f(x) - f(x_0) - \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)}$$

Define then $\{u_n\}_{n \in \mathbb{N}}$ by $u_1 = f(x) - f(x_0)$ and for $n \in \mathbb{N} \setminus \{1\}$ $u_n = -\frac{(x-x_0)^{n-1}}{(n-1)!} \cdot f_{x_0}^{(n-1)}$ then for $n \in \mathbb{N} \setminus \{1\}$ we have

$$\begin{aligned} \sum_{k=1}^n u_k &= u_1 + \sum_{k=2}^n u_k \\ &= f(x) - f(x_0) - \sum_{k=2}^n \frac{(x-x_0)^{k-1}}{(k-1)!} \cdot f_{x_0}^{(k-1)} \\ &= f(x) - f(x_0) - \sum_{k=1}^{n-1} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} \\ &= T_n(x) \end{aligned}$$

so that

$$\forall n \in \mathbb{N} \setminus \{1\} \quad T_n(x) = \sum_{k=1}^n u_k \tag{16.655}$$

$\Rightarrow$. If $\lim_{n \rightarrow \infty} T_n(x) = 0$ exist then by [eq: 16.655]

$$\sum_{k=1}^{\infty} u_k = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n u_k \right) = \lim_{n \rightarrow \infty} T_n(x) = 0 \text{ exist}$$

Hence

$$\begin{aligned}
 0 &= \sum_{k=1}^{\infty} u_k \\
 &\stackrel{[\text{theorem: 14.416}]}{=} u_1 + \sum_{k=2}^{\infty} u_k \\
 &\stackrel{[\text{theorem: 14.415}]}{=} u_1 + \sum_{k=1}^{\infty} u_{k+1} \\
 &= f(x) - f(x_0) + \sum_{k=1}^{\infty} \left(-\frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} \right) \\
 &\stackrel{[\text{theorem: 14.418}]}{=} f(x) - f(x_0) - \sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)}
 \end{aligned}$$

so that $\sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)}$ exist and

$$f(x) = f(x_0) + \sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)}$$

$\Leftarrow$. If $\sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)}$ exist and $f(x) = f(x_0) + \sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)}$ then we have

$$\begin{aligned}
 0 &= \\
 f(x) - f(x_0) - \sum_{k=1}^{\infty} \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} &= \\
 f(x) - f(x_0) - \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} &\stackrel{[\text{theorem: 14.356}]}{=} \\
 \lim_{n \rightarrow \infty} \left(f(x) - f(x_0) - \sum_{k=1}^n \frac{(x-x_0)^k}{k!} \cdot f_{x_0}^{(k)} \right) &= \\
 \lim_{n \rightarrow \infty} T_n(x) &= \\
 \square
 \end{aligned}$$

We have also a variant of the Taylor Series for a C^∞ function based on the Taylor's Theorem with Lagrange's remainder [see theorem: 16.383]. First we need a little lemma that will be used to prove that the Lagrange remainder $\|R_{x_0, h, n}\|_Y$ in [theorem: 16.383] converges to 0.

Lemma 16.390. *Let $\langle X, \|\cdot\| \rangle$ be a normed space, $h \in X$ with $0 < \|h\|$ then $\lim_{n \rightarrow \infty} \frac{\|h\|^{n+1}}{(n+1)!}$ converges with $\lim_{n \rightarrow \infty} \frac{\|h\|^{n+1}}{(n+1)!} = 0$.*

Proof. Define $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^+$ by $x_n = \frac{\|h\|^{n+1}}{(n+1)!}$ then we have for $n \in \mathbb{N}$ that

$$\left| \frac{x_{n+1}}{x_n} \right| = \frac{x_{n+1}}{x_n} = \frac{\|h\|^{n+2}}{(n+2)!} \cdot \frac{(n+1)!}{\|h\|^{n+1}} = \|h\| \cdot \frac{1}{n+1}$$

then we have by [example: 14.349] and [theorem: 14.358] that

$$\lim_{n \rightarrow \infty} \left| \frac{x_{n+1}}{x_n} \right| = 0 < 1$$

so that by [theorem: 14.370] $\lim_{n \rightarrow \infty} x_n$ converges with $\lim_{n \rightarrow \infty} x_n = 0$. Hence using the definition of $\{x_n\}_{n \in \mathbb{N}}$ it follows that

$$\lim_{n \rightarrow \infty} \frac{\|h\|^{n+1}}{(n+1)!} \text{ converges with } \lim_{n \rightarrow \infty} \frac{\|h\|^{n+1}}{(n+1)!} = 0$$

□

Theorem 16.391. (Taylor's Theorem (IV)) Let $n \in \mathbb{N}$, $\langle X, \|\cdot\|_X \rangle$, $\langle Y, \|\cdot\|_Y \rangle$ be normed spaces, a open set U in X , $x_0 \in U$, $h \in X$ such that the line segment $L_{x_0, x_0+h} \subseteq U$ and $f: U \rightarrow Y$ a function that is of class C^∞ such that there exist a $K \in [0, \infty[$

$$\forall n \in \mathbb{N} \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right) \leq K$$

then

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^{\infty} \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right)$$

Note 16.392. The existence of $\sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right)$ is guaranteed by [theorem: 16.383]

Proof. Let $n \in \mathbb{N}$ then we have that f is of class C^{n+1} so that by [theorem: 16.383]

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right) + R_{x_0, h, n}$$

where

$$\|R_{x_0, h, n}\|_Y \leq \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right)$$

Using the hypothesis we have that $\forall n \in \mathbb{N} \sup \left(\left\{ \|D_{x_0+t \cdot h}^{[n+1]} f\|_{L_{n+1}(X;Y)} \mid t \in [0, 1] \right\} \right) \leq K$ so that we have

$$\forall n \in \mathbb{N} \quad \|R_{x_0, h, n}\|_Y \leq \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot K \quad (16.656)$$

Using [lemma: 16.390] it follows that $\lim_{n \rightarrow \infty} \frac{\|h\|_X^{n+1}}{(n+1)!} = 0$ so that by [theorem: 14.358] we have that

$$\lim_{n \rightarrow \infty} \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot K = 0 \cdot K = 0$$

Let $\varepsilon \in \mathbb{R}^+$ then there exist a $N \in \mathbb{N}$ such that $\forall n \in \{N, \dots, \infty\}$ we have that

$$\frac{\|h\|_X^{n+1}}{(n+1)!} \cdot K = \left\| \frac{\|h\|_X^{n+1}}{(n+1)!} \cdot K - 0 \right\| < \varepsilon$$

so that by [eq: 16.656]

$$\|R_{x_0, h, n} - 0\|_Y = \|R_{x_0, h, n}\|_Y < \varepsilon$$

which prove that

$$\lim_{n \rightarrow \infty} R_{x_0, h, n} = 0$$

or

$$\lim_{n \rightarrow \infty} \left(f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right) \right) = 0$$

Now

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right) \right) \stackrel{0}{=} \\ & \stackrel{[theorem: 14.356]}{=} \lim_{n \rightarrow \infty} \left(f(x_0 + h) - f(x_0) - \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right) \right) \\ & = f(x_0 + h) - f(x_0) - \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right) \\ & = f(x_0 + h) - f(x_0) - \sum_{k=1}^{\infty} \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right) \end{aligned}$$

so that

$$f(x_0 + h) = f(x_0) + \sum_{k=1}^{\infty} \frac{1}{k!} \cdot D_{x_0}^{[k]} f \left(\underbrace{h, \dots, h}_k \right)$$

□

Chapter 17

Fundamental theorem of algebra

The purpose of this chapter is to prove the fundamental theorem of algebra. This states that every higher order equation of the form

$$a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \cdots + a_0 = 0$$

has a solution in $\mathbb{C}$. In other words there exist a $z_0 \in \mathbb{C}$ such that

$$a_n \cdot z_0^n + a_{n-1} \cdot z_0^{n-1} + \cdots + a_0 = 0$$

In many books this is done by integration in the complex plane. However in this book we will use algebraic and analytic operations to prove the theorem. First to define the concept of a higher order equation we introduce the concept of polynomials.

17.1 Prerequisites

17.1.1 Polynomials

Definition 17.1. If $n \in \mathbb{N}_0$ then a function $p: \mathbb{C} \rightarrow \mathbb{C}$ is a **polynomial of order n** if there exist a family $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$, called the **coefficients of p** , with $a_n \neq 0$ so that p is defined by

$$p: \mathbb{C} \rightarrow \mathbb{C} \text{ where } p(z) = \sum_{i=0}^n a_i \cdot z^i$$

The collection of complex polynomials of order n is noted as $\mathcal{P}[n]$ so that

$$\mathcal{P}[n] = \{p \in \mathbb{C}^{\mathbb{C}} | p \text{ is a polynomial of order } n\}$$

The collection of all polynomials is noted as $\mathcal{P}$ so that

$$\mathcal{P} = \bigcup_{n \in \mathbb{N}_0} \mathcal{P}[n]$$

Note that in principle different coefficients of different order define the same polynomial function. We will prove however that this is not the case and that every polynomial can be represented by a unique set of coefficients.

Lemma 17.2. Let $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ such that $\forall x \in \mathbb{R} \subseteq \mathbb{C}$ we have $\sum_{i=0}^n a_i \cdot x^i = 0$ then $\forall i \in \{0, \dots, n\}$ we have $a_i = 0$

Proof. Assume that $\exists k \in \{0, \dots, n\}$ such that $a_k \neq 0$. Define $\mathcal{N} = \{i \in \{0, \dots, n\} | a_i \neq 0\}$ then $k \in \mathcal{N} \neq \emptyset$ hence $m = \max(\mathcal{N})$ exist. So $\forall i \in \{m+1, \dots, n\}$ we have $a_i = 0$. So that

$$\forall x \in \mathbb{R} \text{ we we have } 0 = p(x) = \sum_{i=0}^n a_i \cdot x^i = \sum_{i \in \{0, \dots, m\}} a_i \cdot x^i + \sum_{i \in \{m+1, \dots, n\}} a_i \cdot x^i = \sum_{i=0}^m a_i \cdot x^i \quad (17.1)$$

$m = 0$. Then we have $0 = p(1) = \sum_{i=0}^0 a_0 \cdot 1^0 = a_0 \neq 0$ a contradiction.

$0 < m$. Then as $a_m \neq 0$ we can take $x = \frac{\sum_{i=0}^{m-1} |a_i|}{|a_m|} + 1$ then $1 \leq x$ and we have by [theorem: 10.63] that $\forall j \in \{0, \dots, m-1\}$ we have $x^j \leq x^{m-1}$. Hence

$$\begin{aligned}
 \left| \sum_{i=0}^{m-1} a_i \cdot x^i \right| &\leq \sum_{i=0}^{m-1} |a_i \cdot x^i| \\
 &\stackrel{[\text{theorem: 10.85}]}{=} \sum_{i=0}^{m-1} |a_i| \cdot |x|^i \\
 &\leq \sum_{i=0}^{m-1} |a_i| \cdot |x|^{m-1} \\
 &= |x|^{m-1} \cdot \sum_{i=0}^{m-1} |a_i| \\
 &\stackrel{x = \frac{\sum_{i=0}^{m-1} |a_i|}{|a_m|} + 1}{=} |x|^{m-1} \cdot |a_m| \cdot (x-1) \\
 &<_{x-1 < x} |x|^{m-1} \cdot |a_m| \cdot x \\
 &\stackrel{0 < 1 \leq x}{=} |x|^m \cdot |a_m| \\
 &\stackrel{[\text{theorem: 10.85}]}{=} |-a_m \cdot x^m|
 \end{aligned}$$

Hence $\sum_{i=0}^{m-1} a_i \cdot x^i \neq -a_m x^m$ so that $\sum_{i=0}^m a_i \cdot x^i = \sum_{i=0}^{m-1} a_i \cdot x^i + a_m \cdot x^m \neq 0$ contradicting $0 \stackrel{[\text{eq: 17.1}]}{=} p(x) = \sum_{i=0}^m a_i \cdot x^i$.

As in all possible cases we have a contradiction the assumption is wrong and thus $\forall i \in \{0, \dots, n\}$ we have $a_i = 0$. $\square$

Note 17.3. The function $C_0: \mathbb{C} \rightarrow \mathbb{C}$ defined by $C_0(z) = 0$ is not a polynomial.

Proof. Assume that C_0 is a polynomial then there exist a $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $a_n \neq 0$ such that $\forall z \in \mathbb{C} \sum_{i=0}^n a_i \cdot z^i = C_0 = 0$. Then by [lemma: 17.2] we must have $a_n = 0$ contradicting $a_n \neq 0$. $\square$

Corollary 17.4. Let $n \in \mathbb{N}_0$ and $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$, $\{b_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ be finite families such that $\forall z \in \mathbb{C}$ we have

$$\sum_{i=0}^n a_i \cdot z^i = \sum_{i=0}^n b_i \cdot z^i$$

then we have $\forall i \in \{0, \dots, n\}$ $a_i = b_i$.

Proof. As $\forall z \in \mathbb{C}$ we have that $\sum_{i=0}^n a_i \cdot z^i = \sum_{i=0}^n b_i \cdot z^i$ it follows that

$$0 = \sum_{i=0}^n a_i \cdot z^i - \sum_{i=0}^n b_i \cdot z^i = \sum_{i=0}^n (a_i \cdot z^i - b_i \cdot z^i) = \sum_{i=0}^n (a_i - b_i) \cdot z^i$$

hence by [lemma: 17.2] it follows that $\forall i \in \{0, \dots, n\}$ $a_i = b_i$. $\square$

Lemma 17.5. Let $n, m \in \mathbb{N}_0$, $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ a family such that $a_n \neq 0$ and $\{b_i\}_{i \in \{0, \dots, m\}} \subseteq \mathbb{C}$ a family such that $b_m \neq 0$. Then if $\forall z \in \mathbb{C} \sum_{i=1}^n a_i \cdot z^i = \sum_{i=1}^m b_i \cdot z^i$ it follows that $n = m$ and $\forall i \in \{0, \dots, n\}$ we have $a_i = b_i$ [in other words $\{a_i\}_{i \in \{0, \dots, n\}} = \{b_i\}_{i \in \{0, \dots, m\}}$].

Proof. For $n, m \in \mathbb{N}_0$ we have the following different cases:

$n = m$. Then by [corollary: 17.4] we have $\forall i \in \{0, \dots, n\}$ $a_i = b_i$ and trivially $n = m$.

$n < m$. Define then $\{a'_i\}_{i \in \{0, \dots, m\}} \subseteq \mathbb{C}$ by $a'_i = \begin{cases} 0 & \text{if } n < i \\ a_i & \text{if } i \in \{0, \dots, n\} \end{cases}$ then we have $\forall z \in \mathbb{C}$

$$\begin{aligned} \sum_{i=0}^m a'_i \cdot z^i &= \sum_{i=0}^n a'_i \cdot z^i + \sum_{i=n+1}^m a'_i \cdot z^i \\ &= \sum_{i=0}^n a_i \cdot z^i + \sum_{i=n+1}^m 0 \cdot z^i \\ &= \sum_{i=0}^n a_i \cdot z^i \\ &= \sum_{i=1}^m b_i \cdot z^i \end{aligned}$$

so that by [corollary: 17.4] we have that $0 = a'_m = b_m$. As $b_m \neq 0$ we reach a contradiction. Hence this case will never occur.

$m < n$. Define then $\{b'_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ by $b'_i = \begin{cases} 0 & \text{if } m < i \\ b_i & \text{if } i \in \{0, \dots, m\} \end{cases}$ then we have $\forall z \in \mathbb{C}$

$$\begin{aligned} \sum_{i=0}^n b'_i \cdot z^i &= \sum_{i=0}^m b'_i \cdot z^i + \sum_{i=m+1}^n b'_i \cdot z^i \\ &= \sum_{i=0}^m b_i \cdot z^i + \sum_{i=m+1}^n 0 \cdot z^i \\ &= \sum_{i=0}^m b_i \cdot z^i \\ &= \sum_{i=1}^n a_i \cdot z^i \end{aligned}$$

so that by [corollary: 17.4] we have that $0 = b'_n = a_n$. As $a_n \neq 0$ we reach a contradiction. Hence this case will never occur.

So the only valid cases is $n = m$ proving the lemma. $\square$

The above ensures that the following functions are valid.

Definition 17.6. Let $\mathcal{C} = \{\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C} \mid a_n \neq 0\}$ then we define

$$\text{ord}_{\text{coef}}: \mathcal{C} \rightarrow \mathbb{N}_0 \text{ by } \text{ord}_{\text{coef}}\left(\sum_{i=0}^n a_i \cdot z^i\right) = n$$

$\text{coef}: \mathcal{P} \rightarrow \mathcal{C}$ where $\text{coef}(p) = \{a_i\}_{i \in \{1, \dots, n\}} \in \mathcal{C}$ is the unique family [see lemma: 17.5] such that

$$\forall z \in \mathbb{C} \text{ we have } p(z) = \sum_{i=0}^n a_i \cdot z^i$$

and

$$\text{ord}: \mathcal{P} \rightarrow \mathbb{N}_0 \text{ where } \text{ord}(p) = \text{ord}(\text{coef}(p))$$

Actually it turns out that the relation between a polynomial and its coefficients is a bijection

Theorem 17.7. The function

$$\text{coef}: \mathcal{P} \rightarrow \mathcal{C} \text{ where } \text{coef}(p) = \{a_i\}_{i \in \{1, \dots, n\}} \in \mathcal{C} \text{ such that } \forall z \in \mathbb{C} \text{ we have } p(z) = \sum_{i=0}^n a_i \cdot z^i$$

is a bijection. In a sense we can identify a polynomial with its coefficients.

Proof.

injectivity. If $\text{coef}(p) = \text{coef}(q)$ then there exist a $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ such that $\forall z \in \mathbb{C}$

$$p(z) = \sum_{i=0}^n a_i \cdot z^i = q(z)$$

so that $p = q$.

surjectivity. If $\{a_i\}_{i \in \{1, \dots, n\}} \in \mathcal{C}$ then $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ and $a_n \neq 0$. So if we define $p: \mathbb{C} \rightarrow \mathbb{C}$ by $p(z) = \sum_{i=0}^n a_i \cdot z^i$ then $\text{coef}(p) = \{a_i\}_{i \in \{1, \dots, n\}}$. $\square$

Note 17.8. To simplify notation we use the following convention. Instead of saying that the polynomial p is defined by the coefficients $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $a_n \neq 0$ such that $\forall z \in \mathbb{C} \ p(z) = \sum_{i=0}^n a_i \cdot z^i$ we simply say that $p(z) = \sum_{i=0}^n a_i \cdot z^i$.

Definition 17.9. A polynomial $p \in \mathcal{P}$ is **non constant** if $0 < \text{ord}(p)$, a polynomial p is **constant** if $\text{ord}(p) = 0$.

The following is a very trivial lemma that shows how to construct new polynomials from another polynomial.

Lemma 17.10. Let $p \in \mathcal{P}$ a polynomial and $c \in \mathbb{C} \setminus \{0\}$ then $c \cdot p: \mathbb{C} \rightarrow \mathbb{C}$ defined by $(c \cdot p)(x) = c \cdot p(x)$ is a polynomial with $\text{ord}(c \cdot p) = \text{ord}(p)$.

Proof. As p is a polynomial there exist a family $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $a_n \neq 0$ where $n = \text{ord}(p)$. Define $\{b_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ by $b_i = c \cdot a_i$ then as $c \neq 0 \wedge a_n \neq 0$ we have that $b_n \neq 0$ so that

$$q: \mathbb{C} \rightarrow \mathbb{C} \text{ defined by } q(z) = \sum_{i=0}^n b_i \cdot z^i$$

is a polynomial with $\text{ord}(q) = n$. As $\forall z \in \mathbb{C}$ we have

$$q(z) = \sum_{i=0}^n b_i \cdot z^i = \sum_{i=0}^n c \cdot a_i \cdot z^i = c \cdot \sum_{i=0}^n a_i \cdot z^i = c \cdot p(z) = (c \cdot p)(z)$$

it follows that $q = c \cdot p$ so that $c \cdot p$ is a polynomial with $\text{ord}(c \cdot p) = n = \text{ord}(p)$. $\square$

We will show now that the product of polynomials is again a polynomial. First we need a little lemma.

Lemma 17.11. Let $m, n \in \mathbb{N}_0$ then

$$\{0, \dots, n\} \times \{0, \dots, m\} = \bigcup_{i \in \{0, \dots, n+m\}} \{(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\}$$

and $\forall i, j \in \{0, \dots, n+m\}$ with $i \neq j$ we have that

$$\{(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\} \cap \{(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=j\} = \emptyset$$

Proof. Given $i \in \{0, \dots, n+m\}$ define

$$I_i = \{(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\} \subseteq \{0, \dots, n\} \times \{0, \dots, m\}$$

then we have trivially that

$$\bigcup_{i \in \{0, \dots, n+m\}} I_i \subseteq \{0, \dots, n\} \times \{0, \dots, m\} \quad (17.2)$$

If $(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\}$ then $0 \leq r \leq n \wedge 0 \leq s \leq m$ so that $0 \leq r+s \leq n+m$ hence

$$(r, s) \in I_{r+s} \subseteq \bigcup_{i \in \{0, \dots, n+m\}} I_i$$

Combining with [eq: 17.2] proves that

$$\{0, \dots, n\} \times \{0, \dots, m\} = \bigcup_{i \in \{0, \dots, n+m\}} I_i$$

Let $i, j \in \{0, \dots, n+m\}$ with $i \neq j$ then if $(r, s) \in I_i \cap I_j$ we have that $i = r + s = j$ contradicting $i \neq j$ so that

$$I_i \cap I_j = \emptyset \quad \square$$

Theorem 17.12. *Let $p, q \in \mathcal{P}$ be two polynomials then*

$$p \cdot q: \mathbb{C} \rightarrow \mathbb{C} \text{ defined by } (p \cdot q)(z) = p(z) \cdot q(z)$$

is also a polynomial. If $p(z) = \sum_{i=0}^n a_i \cdot z^i$ and $q(z) = \sum_{i=0}^m b_i \cdot z^i$ then $p \cdot q$ is defined by

$$(p \cdot q)(z) = \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=i\}} a_k \cdot b_l \right) \cdot z^i$$

from which it follows that

$$\text{ord}(p \cdot q) = \text{ord}(p) + \text{ord}(q)$$

Proof. Let $z \in \mathbb{C}$ then

$$\begin{aligned} (p \cdot q)(z) &= p(z) \cdot q(z) \\ &= \left(\sum_{k=0}^n a_k \cdot z^k \right) \cdot \left(\sum_{l=0}^m b_l \cdot z^l \right) \\ &= \left(\sum_{k \in \{0, \dots, n\}} a_k \cdot z^k \right) \cdot \left(\sum_{l \in \{0, \dots, m\}} b_l \cdot z^l \right) \\ &\stackrel{[\text{theorem: 11.48}]}{=} \sum_{(k,l) \in \{0, \dots, n\} \times \{0, \dots, m\}} a_k \cdot b_l \cdot z^k \cdot z^l \\ &= \sum_{(k,l) \in \{0, \dots, n\} \times \{0, \dots, m\}} a_k \cdot b_l \cdot z^{k+l} \\ &\stackrel{[\text{lemma: 17.11, theorem: 11.43}]}{=} \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=i\}} a_k \cdot b_l \cdot z^{k+l} \right) \\ &= \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=i\}} a_k \cdot b_l \cdot z^i \right) \\ &= \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=i\}} a_k \cdot b_l \right) \cdot z^i \end{aligned}$$

To finish the proof we must prove that $\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=n+m\}} a_k \cdot b_l \neq 0$. Now if $(r, s) \in \{(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=n+m\}$ then $r+s=n+m$. If $r \neq n$ then $r < n$ and as $s \leq m$ we have then $r+s < n+m$ contradicting $r+s=n+m$ so we must have that $r=n$ which as $r+s=n+m$ means that $s=m$. Hence $(k, l) \in \{(r, s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=n+m\} = \{(n, m)\}$ so that $\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=n+m\}} a_k \cdot b_l = a_n \cdot b_m$, as $a_n \neq 0 \neq b_m$ it follows that

$$\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} | r+s=n+m\}} a_k \cdot b_l \neq 0 \quad \square$$

We can now use mathematical induction to prove that a finite product of polynomials is again a polynomial.

Theorem 17.13. Let $n \in \mathbb{N}$ and $\{p_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}$ a finite family of polynomials then if we define

$$\prod_{i=1}^n p_i: \mathbb{C} \rightarrow \mathbb{C} \text{ by } \left(\prod_{i=1}^n p_i \right)(z) = \prod_{i=1}^n p_i(z)$$

then $\prod_{i=1}^n p_i$ is a polynomial with

$$\text{ord} \left(\prod_{i=1}^n p_i \right) = \sum_{i=1}^n \text{ord}(p_i)$$

Proof. We use mathematical induction to prove this, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{p_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P} \text{ then } \prod_{i=1}^n p_i \in \mathcal{P} \text{ and } \text{ord} \left(\prod_{i=1}^n p_i \right) = \sum_{i=1}^n \text{ord}(p_i) \right\}$$

then we have:

$1 \in S$. Then $\forall z \in \mathbb{C}$ we have $(\prod_{i=1}^1 p_i)(z) = p_1(z)$ proving that $\prod_{i=1}^1 p_i = p_1$ [which is a polynomial] and $\text{ord}(\prod_{i=1}^1 p_i) = \text{ord}(p_1) = \sum_{i=1}^1 \text{ord}(p_i)$ hence $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Then $\forall z \in \mathbb{C}$ we have that

$$\left(\prod_{i=1}^{n+1} p_i \right)(z) = \prod_{i=1}^{n+1} p_i(z) = p_{n+1}(z) \cdot \left(\prod_{i=1}^n p_i \right)(z)$$

proving that $\prod_{i=1}^{n+1} p_i = p_{n+1} \cdot \prod_{i=1}^n p_i$ which as $n \in S$ is a product of polynomials and thus by [lemma: 17.12] is a polynomial, further

$$\begin{aligned} \text{ord} \left(\prod_{i=1}^{n+1} p_i \right) &= \text{ord} \left(p_{n+1} \cdot \prod_{i=1}^n p_i \right) \\ &\stackrel{[\text{lemma: 17.13}]}{=} \text{ord}(p_{n+1}) + \text{ord} \left(\prod_{i=1}^n p_i \right) \\ &\stackrel{n \in S}{=} \text{ord}(p_{n+1}) + \sum_{i=1}^n \text{ord}(p_i) \\ &= \sum_{i=1}^{n+1} \text{ord}(p_i) \end{aligned}$$

□

17.1.2 Divergent limits

In the proof of the fundamental theorem of algebra we need the concept not of convergent limits but the concept of divergent limits.

Definition 17.14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function then

1. $\lim_{x \rightarrow \infty} f(x) = \infty$ if $\forall M \in \mathbb{R}$ there exist a $N \in \mathbb{R}$ such that $\forall x \geq N$ we have $f(x) \geq M$
2. $\lim_{x \rightarrow \infty} f(x) = -\infty$ if $\forall M \in \mathbb{R}$ there exists a $N \in \mathbb{R}$ such that $\forall x \geq N$ we have $f(x) \leq M$

Theorem 17.15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function then for

$$-f: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } (-f)(x) = -f(x)$$

we have:

1. If $\lim_{x \rightarrow \infty} f(x) = \infty$ then $\lim_{x \rightarrow \infty} (-f)(x) = -\infty$
2. If $\lim_{x \rightarrow \infty} f(x) = -\infty$ then $\lim_{x \rightarrow \infty} (-f)(x) = \infty$

Proof.

1. Take $M \in \mathbb{R}$ then there exists a N such that $\forall x \geq N$ we have $-M \leq f(x) \Rightarrow (-f)(x) \leq M$ proving that $\lim_{x \rightarrow \infty} (-f)(x) = -\infty$
2. Take $M \in \mathbb{R}$ then there exists a N such that $\forall x \geq N$ we have $f(x) \leq -M \Rightarrow M \leq (-f)(x)$ proving that $\lim_{x \rightarrow \infty} (-f)(x) = \infty$ $\square$

Lemma 17.16. Let $n \in \mathbb{N}$, $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{R}$ then

1. If $a_n > 0$ we have $\lim_{x \rightarrow \infty} \left(\sum_{i=0}^n a_i \cdot x^i \right) = \infty$
2. If $a_n < 0$ we have $\lim_{x \rightarrow \infty} \left(\sum_{i=0}^n a_i \cdot x^i \right) = -\infty$

Proof.

1. we prove this by induction so let

$$\mathcal{S} = \left\{ n \in \mathbb{N} \mid \forall \{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{R} \text{ with } a_n > 0 \text{ we have } \lim_{x \rightarrow \infty} \left(\sum_{i=0}^n a_i \cdot x^i \right) = \infty \right\}$$

then we have

1 $\in \mathcal{S}$. Let $M \in \mathbb{R}$ and take $N = \frac{M - a_0}{a_1}$ then if $x \geq N$ we have

$$M - a_0 = a_1 \cdot N \leq a_1 \cdot x = a_1 \cdot x^1$$

hence $M \leq a_1 \cdot x^1 + a_0 = \sum_{i=0}^1 a_i \cdot x^i$ proving that $\lim_{x \rightarrow \infty} \sum_{i=0}^1 a_i \cdot x^i = \infty$. Hence $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{a_i\}_{i \in \{0, \dots, n+1\}} \subseteq \mathbb{R}$ with $a_{n+1} > 0$ then

$$\sum_{i=0}^{n+1} a_i \cdot x^i = \left(\sum_{i=1}^{n+1} a_i \cdot x^i \right) + a_0 = x \cdot \left(\sum_{i=1}^{n+1} a_i \cdot x^{i-1} \right) + a_0 = x \cdot \left(\sum_{i=0}^n a_{i+1} \cdot x^i \right) + a_0,$$

hence if we define $\{b_i\}_{i \in \{0, \dots, n\}}$ by $b_i = a_{i+1}$, then we have

$$\sum_{i=1}^{n+1} a_i \cdot x^i = x \cdot \left(\sum_{i=0}^n b_i \cdot x^i \right) + a_0 \wedge b_n = a_{n+1} > 0 \quad (17.3)$$

Take $M \in \mathbb{R}$ then as $n \in \mathcal{S}$ there exists a $N' \in \mathbb{R}$ such that if $N' \leq x$ we have that

$$0 < \max(1, M - a_0) \leq \sum_{i=0}^n b_i \cdot x^i$$

Take then $N = \max(N', 1)$ then we have if $N \leq x$ that $1 \leq x$ so that

$$\max(1, M - a_0) \leq x \cdot \max(1, M - a_0)$$

and

$$x \cdot \max(1, M - a_0) \leq x \cdot \sum_{i=0}^n b_i \cdot x^i$$

so that

$$M - a_0 \leq \max(1, M - a_0) \leq x \cdot \sum_{i=0}^n b_i \cdot x^i$$

or

$$M \leq a_0 + x \cdot \sum_{i=0}^n b_i \cdot x^i \stackrel{[\text{eq: 17.3}]}{=} \sum_{i=1}^{n+1} a_i \cdot x^i$$

hence $\lim_{x \rightarrow \infty} \sum_{i=1}^{n+1} a_i \cdot x^i$ proving that $n+1 \in \mathcal{S}$.

2. If $a_n < 0$ then $0 < (-a_n)$ and we have by (1) that $\lim_{x \rightarrow \infty} -\sum_{i=0}^n a_i \cdot x^i = \lim_{z \rightarrow \infty} \sum_{i=0}^n (-a_i) \cdot x^i = \infty$ and thus by [theorem: 17.15] that $\lim_{x \rightarrow \infty} \left(\sum_{i=0}^n a_i \cdot x^i \right) = \lim_{x \rightarrow \infty} \left(-\sum_{i=0}^n a_i \cdot x^i \right) = -\infty$. $\square$

17.1.3 Properties of $\mathbb{C}$ needed for the fundamental theorem

First we define a norm on $\mathbb{C}$ that does not need the square root.

Theorem 17.17. *The function $\|\cdot\|: \mathbb{C} \rightarrow \mathbb{R}$ defined by $\|z\| = |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$ is a norm on the vector space $\mathbb{C}$ over $\mathbb{R}$.*

Proof. We have:

1. If $z \in \mathbb{C}$ then $\|z\| = |\operatorname{Re}(z)| + |\operatorname{Im}(z)| \geq 0$
2. If $\alpha \in \mathbb{R}$ then for z we have

$$\begin{aligned} \|\alpha \cdot z\| &= |\operatorname{Re}(\alpha \cdot z)| + |\operatorname{Im}(\alpha \cdot z)| \\ &\stackrel{[\text{theorem: 10.80}]}{=} |\alpha \cdot \operatorname{Re}(z)| + |\alpha \cdot \operatorname{Im}(z)| \\ &= |\alpha| \cdot |\operatorname{Re}(z)| + |\alpha| \cdot |\operatorname{Im}(z)| \\ &= |\alpha| \cdot (|\operatorname{Re}(z)| + |\operatorname{Im}(z)|) \\ &= |\alpha| \cdot \|z\| \end{aligned}$$

3. If $z_1, z_2 \in \mathbb{C}$ then we have

$$\begin{aligned} \|z_1 + z_2\| &= |\operatorname{Re}(z_1 + z_2)| + |\operatorname{Im}(z_1 + z_2)| \\ &\stackrel{[\text{theorem: 10.80}]}{=} |\operatorname{Re}(z_1) + \operatorname{Re}(z_2)| + |\operatorname{Im}(z_1) + \operatorname{Im}(z_2)| \\ &\leq |\operatorname{Re}(z_1)| + |\operatorname{Re}(z_2)| + |\operatorname{Im}(z_1)| + |\operatorname{Im}(z_2)| \\ &= (|\operatorname{Re}(z_1)| + |\operatorname{Im}(z_1)|) + (|\operatorname{Re}(z_2)| + |\operatorname{Im}(z_2)|) \\ &= \|z_1\| + \|z_2\| \end{aligned}$$

4. If $z \in \mathbb{C}$ satisfies $\|z\| = 0$ then $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 0$ so that $|\operatorname{Re}(z)| = 0 = |\operatorname{Im}(z)|$ hence $\operatorname{Re}(z) = \operatorname{Im}(z) = 0$ so that $z = 0$. $\square$

The above norm has the following properties.

Lemma 17.18. *The norm $\|\cdot\|$ has the following properties:*

1. If $z \in \mathbb{R}$ then $\|z\| = |z|$
2. $\forall z \in \mathbb{C}$ we have that $\|\bar{z}\| = \|z\|$
3. $\forall z, w \in \mathbb{C}$ we have that $\frac{\|z\| \cdot \|w\|}{2} \leq \|z \cdot w\| \leq \|z\| \cdot \|w\|$
4. $\forall z, w \in \mathbb{C}$ we have $\|\operatorname{Re}(z \cdot w)\| \leq \|z\| \cdot \|w\|$
5. $\forall n \in \mathbb{N}, z \in \mathbb{C}$ then $\|z^n\| \leq \|z\|^n$
6. $\forall n \in \mathbb{N}, z \in \mathbb{C}$ then $\|\operatorname{Re}(z^n)\| \leq \|z\|^n$
7. $\forall n \in \mathbb{N}, z \in \mathbb{C}$ then $\frac{\|z\|^n}{2^n} \leq \|z^n\|$

Proof.

1. If $z \in \mathbb{R}$ then by [theorem: 10.80] $\operatorname{Im}(z) = 0$ so that $\|z\| = |\operatorname{Re}(z)| + |\operatorname{Im}(z)| = |\operatorname{Re}(z)|$
2. Let $z = \operatorname{Re}(z) + i\operatorname{Im}(z) \in \mathbb{C}$ then

$$\|z\| = |\operatorname{Re}(z)| + |\operatorname{Im}(z)| = |\operatorname{Re}(z)| + |-\operatorname{Im}(z)| = \|\operatorname{Re}(z) - i \cdot \operatorname{Im}(z)\| = \|\bar{z}\|$$

3. If $a, b, c, d \in \mathbb{R}$ then we have $0 \leq (|a| - |b|)^2 = |a|^2 + |b|^2 - 2 \cdot |a| \cdot |b|$ so that

$$2 \cdot |a| \cdot |b| \leq |a|^2 + |b|^2 \quad (17.4)$$

So $(|a| + |b|)^2 = |a|^2 + |b|^2 + 2 \cdot |a| \cdot |b| \leq_{[\text{eq: 17.4}]} |a|^2 + |b|^2 + |a|^2 + |b|^2 = 2 \cdot (|a|^2 + |b|^2)$ proving that

$$(|a| + |b|)^2 \leq 2 \cdot (|a|^2 + |b|^2) \quad (17.5)$$

Similar we have $0 \leq (|c| - |d|)^2 = |c|^2 + |d|^2 - 2 \cdot |c| \cdot |d|$ so that

$$2 \cdot |c| \cdot |d| < |c|^2 + |d|^2 \quad (17.6)$$

So $(|c| + |d|)^2 = |c|^2 + |d|^2 + 2 \cdot |c| \cdot |d| \leq_{[\text{eq: 17.6}]} |c|^2 + |d|^2 + |c|^2 + |d|^2 = 2 \cdot (|c|^2 + |d|^2)$ proving that

$$(|c| + |d|)^2 \leq 2 \cdot (|c|^2 + |d|^2) \quad (17.7)$$

Hence we have by [eqs: 17.5, 17.7]

$$\begin{aligned} (|a| + |b|)^2 \cdot (|c| + |d|)^2 &\leq_{[\text{eq: 17.5}]} 2 \cdot (|a|^2 + |b|^2) \cdot (|c| + |d|)^2 \\ &\leq_{[\text{eq: 17.7}]} 2 \cdot (|a|^2 + |b|^2) \cdot 2 \cdot (|c|^2 + |d|^2) \\ &= 4 \cdot (|a|^2 + |b|^2) \cdot (|c|^2 + |d|^2) \end{aligned}$$

proving that

$$(|a| + |b|)^2 \cdot (|c| + |d|)^2 \leq 4 \cdot (|a|^2 + |b|^2) \cdot (|c|^2 + |d|^2) \quad (17.8)$$

Now

$$\begin{aligned} (a \cdot c - b \cdot d)^2 + (a \cdot d + b \cdot c)^2 &= a^2 \cdot c^2 + b^2 \cdot d^2 - 2 \cdot a \cdot c \cdot b \cdot d + a^2 \cdot d^2 + b^2 \cdot c^2 + a \cdot d \cdot b \cdot c \\ &= a^2 \cdot c^2 + b^2 \cdot d^2 + a^2 \cdot d^2 + b^2 \cdot c^2 \\ &= a^2 \cdot (c^2 + d^2) + b^2 \cdot (d^2 + c^2) \\ &= (a^2 + b^2) \cdot (c^2 + d^2) \\ &\stackrel{a, b, c, d \in \mathbb{R}}{=} (|a|^2 + |b|^2) \cdot (|c|^2 + |d|^2) \end{aligned}$$

substituting the above in [eq: 17.8] we have that

$$\begin{aligned} (|a| + |b|)^2 \cdot (|c| + |d|)^2 &\leq 4 \cdot ((a \cdot c - b \cdot d)^2 + (a \cdot d + b \cdot c)^2) \\ &= 4 \cdot (|a \cdot c - b \cdot d|^2 + |a \cdot d + b \cdot c|^2) \\ &= 4 \cdot |a \cdot c - b \cdot d|^2 + 4 \cdot |a \cdot d + b \cdot c|^2 \\ &\leq 4 \cdot |a \cdot c - b \cdot d|^2 + 4 \cdot |a \cdot d + b \cdot c|^2 + 8 \cdot |a \cdot c - b \cdot d| \cdot |a \cdot d + b \cdot c| \\ &= 4 \cdot (|a \cdot c - b \cdot d| + |a \cdot d + b \cdot c|)^2 \end{aligned}$$

Using the fact that the square root is increasing on $\mathbb{R}^+$ [see theorem: 10.71] it follows that

$$(|a| + |b|) \cdot (|c| + |d|) \leq 2 \cdot (|a \cdot c - b \cdot d| + |a \cdot d + b \cdot c|) \quad (17.9)$$

Let $z = a + i \cdot b \in \mathbb{C}$ and $w = c + i \cdot d \in \mathbb{C}$ then $z \cdot w = (a \cdot c - b \cdot d) + i \cdot (a \cdot d + b \cdot c)$ so that

$$\begin{aligned} \|z \cdot w\| &= |a \cdot c - b \cdot d| + |a \cdot d + b \cdot c| \\ &\geq_{[\text{eq: 17.9}]} \frac{1}{2} \cdot (|a| + |b|) \cdot (|c| + |d|) \\ &= \frac{1}{2} \cdot \|z\| \cdot \|w\| \end{aligned}$$

proving that

$$\frac{1}{2} \cdot \|z\| \cdot \|w\| \leq \|z \cdot w\| \quad (17.10)$$

Further we have

$$\begin{aligned} \|z \cdot w\| &= |a \cdot c - b \cdot d| + |a \cdot d + b \cdot c| \\ &\leq |a| \cdot |c| + |b| \cdot |d| + |a| \cdot |d| + |b| \cdot |c| \\ &= |a| \cdot (|c| + |d|) + |b| \cdot (|d| + |c|) \\ &= (|a| + |b|) \cdot (|c| + |d|) \\ &= \|z\| \cdot \|w\| \end{aligned}$$

proving that

$$\|z \cdot w\| \leq \|z\| \cdot \|w\|$$

(3) is proved by the above and [eq: 17.10].

4. If $z = a + i \cdot b$ and $w = c + i \cdot d$ so that $z \cdot w = (a \cdot c - b \cdot d) + i \cdot (a \cdot d + b \cdot c)$ then we have

$$\begin{aligned} \|\operatorname{Re}(z \cdot w)\| &= |\operatorname{Re}(z \cdot w)| \\ &= |a \cdot c - b \cdot d| \\ &\leq |a \cdot c - b \cdot d| + |a \cdot d + b \cdot c| \\ &= \|z \cdot w\| \end{aligned}$$

5. We prove this by induction, so define

$$S = \{n \in \mathbb{N} \mid \|z^n\| \leq \|z\|^n\}$$

then we have:

$1 \in S$. As $\|z^1\| = \|z\| = \|z\|^1$ it follows that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. As $\|z^{n+1}\| = \|z \cdot z^n\| \leq_{(3)} \|z\| \cdot \|z^n\| \leq_{n \in S} \|z\| \cdot \|z\|^n = \|z\|^{n+1}$ proving that $n + 1 \in S$.

6. For $z = x + i \cdot y$ we for $n \in \mathbb{N}$ two cases:

$n = 1$. Then $\|\operatorname{Re}(z^1)\| = \|\operatorname{Re}(z)\| = |x| < |x| + |y| = \|z\| = \|z\|^1$ so that $\|\operatorname{Re}(z^1)\| \leq \|z\|^1$

$1 < n$. Then $1 \leq n - 1$ and we have

$$\|\operatorname{Re}(z^n)\| = \|\operatorname{Re}(z \cdot z^{n-1})\| \leq_{(4)} \|z\| \cdot \|z^{n-1}\| \leq_{(5)} \|z\| \cdot \|z\|^{n-1} = \|z\|^n$$

so that

$$\|\operatorname{Re}(z^n)\| \leq \|z\|^n$$

7. This is proved by induction, so define

$$S = \left\{ n \in \mathbb{N} \mid \frac{\|z\|^n}{2^n} \leq \|z^n\| \right\}$$

then we have:

$1 \in S$. As $\frac{1}{2} < 1$ we have $\frac{\|z\|^1}{2} \leq \|z^1\| = \|z\| = \|z\|^1$ so that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Then we have

$$\frac{\|z\| \cdot \|z^n\|}{2} \leq_{(3)} \|z \cdot z^n\| = \|z^{n+1}\| \quad (17.11)$$

As $n \in S$ we have $\frac{\|z\|^n}{2^n} \leq \|z^n\|$. So that

$$\frac{\|z\|^{n+1}}{2^{n+1}} = \frac{\|z\| \cdot \|z^n\|}{2 \cdot 2^n} \leq_{n \in S} \frac{\|z\|}{2} \cdot \|z\|^n \leq_{[\text{eq: 17.11}]} \|z^{n+1}\|$$

giving $n + 1 \in S$. □

In the proof of the fundamental theorem of algebra we use the extreme value theorem [theorem: 14.282], so we need to look at continuous functions using the above norm. This is the subject of the following theorem.

Theorem 17.19. *The following functions are continuous in the normed space $\langle \mathbb{C}, \|\cdot\| \rangle$*

1. Given $\alpha \in \mathbb{C}$ then function $(\alpha \cdot): \mathbb{C} \rightarrow \mathbb{C}$ defined by $(\alpha \cdot)(x) = \alpha \cdot x$ is continuous.
2. Given $n \in \mathbb{N}_0$ we have that $(\cdot)^n: \mathbb{C} \rightarrow \mathbb{C}$ defined by $(\cdot)^n(z) = z^n$ is continuous.
3. Given $n \in \mathbb{N}$, $\{\alpha_i\}_{i \in \{0, \dots, n\}}$ then the function $p: \mathbb{C} \rightarrow \mathbb{C}$ defined by $p(z) = \sum_{i=0}^n \alpha_i \cdot z^i$ is continuous.
4. The function $\langle z \rangle: \mathbb{C} \rightarrow \mathbb{R}$ defined by $\langle z \rangle = \langle \rangle(z) = z \cdot \bar{z} = |z|^2$ is continuous

5. Given $n \in \mathbb{N}_0$, $\{a_i\}_{i \in \{0, \dots, n\}}$ the function

$$p \cdot \bar{p}: \mathbb{C} \rightarrow \mathbb{R} \text{ defined by } (p \cdot \bar{p})(z) = \left(\sum_{i=0}^n \alpha_i \cdot z^i \right) \cdot \overline{\left(\sum_{i=0}^n \alpha_i \cdot z^i \right)}$$

is continuous.

6. The function $(\bar{\cdot}): \mathbb{C} \rightarrow \mathbb{C}$ defined by $(\bar{\cdot})(z) = \bar{z}$ is continuous.

7. Given $z \in \mathbb{C}$ then the function $(\cdot z): \mathbb{R} \rightarrow \mathbb{C}$ defined by $(\cdot z)(\alpha) = \alpha \cdot z$ is continuous.

8. If $p: \mathbb{C} \rightarrow \mathbb{C}$ is a polynomial and $z \in \mathbb{C}$ then $p_z: \mathbb{R} \rightarrow \mathbb{C}$ defined by $p_z(\alpha) = p(\alpha \cdot z)$ is continuous.

9. The function $\text{Re}: \mathbb{C} \rightarrow \mathbb{R}$ is continuous

10. If $f: \mathbb{C} \rightarrow \mathbb{C}$ and $g: \mathbb{C} \rightarrow \mathbb{C}$ are continuous functions in $\langle \mathbb{C}, ||| \rangle$ then

$$f \cdot g: \mathbb{C} \rightarrow \mathbb{C} \text{ define by } (f \cdot g) \text{ is continuous in } \langle \mathbb{C}, ||| \rangle$$

Proof.

1. As $\forall z \in \mathbb{C}$ we have $(\alpha \cdot)(z) = \alpha \cdot z = \alpha \cdot \text{Id}_{\mathbb{C}}(z)$ we have $(\alpha \cdot) = \alpha \cdot \text{Id}_{\mathbb{C}}$. By [example: 14.143] $\text{Id}_{\mathbb{C}}$ is continuous so that by [theorem: 14.165] $(\alpha \cdot)$ is continuous.

2. As $\forall z \in \mathbb{C}$ we have that $(\cdot)^n(z) = z^n = (\text{Id}_{\mathbb{C}}(z))^n$ we have, as $\text{Id}_{\mathbb{C}}$ is continuous, by [theorem: 14.169] that $(\cdot)^n$ is continuous.

3. Let $i \in \{1, \dots, n\}$ then by (1) and (2) $(\alpha_i \cdot)$ and $(\cdot)^i$ are continuous. So, as the composition of continuous functions is continuous [see theorem: 14.158], it follows that $(\alpha_i \cdot) \circ (\cdot)^i$ is continuous. Using now [theorem: 14.166] it follows that

$$\sum_{i=0}^n ((\alpha_i \cdot) \circ (\cdot)^i)$$

is continuous. Let $z \in \mathbb{C}$ then

$$\left(\sum_{i=0}^n ((\alpha_i \cdot) \circ (\cdot)^i) \right)(z) = \sum_{i=0}^n ((\alpha_i \cdot) \circ (\cdot)^i)(z) = \sum_{i=0}^n (\alpha_i \cdot)((\cdot)^i(z)) = \sum_{i=0}^n \alpha_i \cdot z^i = p(z)$$

it follows that $p = \sum_{i=0}^n ((\alpha_i \cdot) \circ (\cdot)^i)$. Hence p is continuous.

4. Let $y \in \mathbb{C}$ and $\varepsilon \in \mathbb{R}^+$. Take $\delta = \min\left(1, \frac{\varepsilon}{2 \cdot \|y\| + 1}\right) \in \mathbb{R}^+$ then if $z \in \mathbb{C}$ with $\|z - y\| < \delta$ then if we define $h = z - y$ we have $\|h\| < \delta$ and

$$\begin{aligned} |\langle z \rangle - \langle y \rangle| &\stackrel{[\text{lemma: 17.18}]}{=} \|\langle z \rangle - \langle y \rangle\| \\ &= \|\langle y + h \rangle - \langle y \rangle\| \\ &\stackrel{[\text{theorem: 10.85}]}{=} \|(y + h) \cdot (\overline{y + h}) - y \cdot \bar{y}\| \\ &= \|y \cdot \bar{y} + y \cdot \bar{h} + h \cdot \bar{y} + h \cdot \bar{h} - y \cdot \bar{y}\| \\ &= \|y \cdot \bar{h} + h \cdot \bar{y} + h \cdot \bar{h}\| \\ &\leq \|y \cdot \bar{h}\| + \|h \cdot \bar{y}\| + \|h \cdot \bar{h}\| \\ &\stackrel{[\text{lemma: 17.18(3)}]}{\leq} \|y\| \cdot \|\bar{h}\| + \|h\| \cdot \|\bar{y}\| + \|h\| \cdot \|\bar{h}\| \\ &\stackrel{[\text{lemma: 17.18(2)}]}{=} \|y\| \cdot \|h\| + \|h\| \cdot \|y\| + \|h\| \cdot \|h\| \\ &= \|h\| \cdot (2 \cdot \|y\| + \|h\|) \\ &\stackrel{\leq \|h\| < \delta \leq 1}{\leq} \|h\| \cdot (2 \cdot \|y\| + 1) \\ &< \frac{\varepsilon}{2 \cdot \|y\| + 1} \cdot (2 \cdot \|y\| + 1) \\ &= \varepsilon \end{aligned}$$

proving continuity of $\langle \cdot \rangle$.

5. Let $z \in \mathbb{C}$ then

$$\begin{aligned} (\langle \cdot \rangle \circ p)(z) &= \langle \cdot \rangle(p(z)) \\ &= \langle \cdot \rangle \left(\sum_{i=0}^n \alpha_i \cdot z^i \right) \\ &= \left(\sum_{i=0}^n \alpha_i \cdot z^i \right) \cdot \overline{\left(\sum_{i=0}^n \alpha_i \cdot z^i \right)} \\ &= (p \cdot \bar{p})(z) \end{aligned}$$

so that

$$(\langle \cdot \rangle \circ p) = p \cdot \bar{p}$$

By (3),(4) p and $\langle \cdot \rangle$ are continuous, hence using [theorem: 14.158] $p \cdot \bar{p}$ is continuous.

6. Let $\varepsilon \in \mathbb{R}$, $z \in \mathbb{C}$ then if $z' \in \mathbb{C}$ with $\|z - z'\| < \varepsilon$ we have

$$\|\bar{z} - \bar{z}'\| \stackrel{[\text{theorem: 10.83}]}{=} \|\overline{z - z'}\| \stackrel{[\text{lemma: 17.18}]}{=} \|z - z'\| < \varepsilon$$

7. Let $z \in \mathbb{C}$, $\alpha \in \mathbb{R}$, given $\varepsilon > 0$ define $\delta = \frac{\varepsilon}{\|z\| + 1} \in \mathbb{R}^+$ then we have $\forall \alpha' \in \mathbb{R}$ with $|\alpha - \alpha'| < \delta$ that

$$\|(\cdot z)(\alpha) - (\cdot z)(\alpha')\| = \|\alpha \cdot z - \alpha' \cdot z\| = \|(\alpha - \alpha') \cdot z\| = |\alpha - \alpha'| \cdot \|z\| < \frac{\varepsilon}{\|z\| + 1} \cdot \|z\| < \varepsilon$$

proving that $(\cdot z)$ is continuous.

8. Given $\alpha \in \mathbb{R}$ we have $(p \cdot z)(\alpha) = p(\alpha \cdot z) = p((\cdot z)(\alpha)) = (p \circ (\cdot z))(\alpha)$ so that $p \cdot z = p \circ (\cdot z)$. As $p \cdot z = p \circ (\cdot z)$ is the composition of two continuous functions [see (3) and (7)] we have that $p \cdot z$ is continuous.

9. Let $z \in \mathbb{C}$ then given $\varepsilon \in \mathbb{R}^+$ we have for $\delta = \varepsilon$ that $\forall z' \in \mathbb{C}$ with $\|z - z'\| < \delta$ so that

$$|\operatorname{Re}(z) - \operatorname{Re}(z')| = |\operatorname{Re}(z - z')| \leq |\operatorname{Re}(z - z')| + |\operatorname{Im}(z - z')| = \|z - z'\| < \delta = \varepsilon$$

proving continuity.

10. Let $z \in \mathbb{C}$ and $\varepsilon \in \mathbb{R}^+$ then as f, g are continuous there exist a $\delta_1, \delta_2 \in \mathbb{R}^+$ such that

$$\forall z' \in \mathbb{C} \text{ with } \|z - z'\| < \delta_1 \text{ we have } \|f(z) - f(z')\| < \min \left(1, \frac{\varepsilon}{2 \cdot (1 + \|g(z)\|)} \right) \quad (17.12)$$

$$\forall z' \in \mathbb{C} \text{ with } \|z - z'\| < \delta_2 \text{ we have } \|g(z) - g(z')\| < \frac{\varepsilon}{2 \cdot (1 + \|f(z)\|)} \quad (17.13)$$

Define $\delta = \min(\delta_1, \delta_2) \in \mathbb{R}^+$ then if $z' \in \mathbb{C}$ with $\|z'\| < \delta$ we have

$$\|f(z')\| \leq \|f(z) - (f(z) - f(z'))\| \leq \|f(z) - f(z')\| + \|f(z)\| < 1 + \|f(z)\| \quad (17.14)$$

Now

$$\begin{aligned} \|(f \cdot g)(z) - (f \cdot g)(z')\| &= \\ \|f(z) \cdot g(z) - f(z') \cdot g(z')\| &= \\ \|f(z) \cdot g(z) - f(z') \cdot g(z) + f(z') \cdot g(z) - f(z') \cdot g(z')\| &\leq \\ \|f(z) \cdot g(z) - f(z') \cdot g(z)\| + \|f(z') \cdot g(z) - f(z') \cdot g(z')\| &= \\ \| (f(z) - f(z')) \cdot g(z) \| + \| f(z') \cdot (g(z) - g(z')) \| &\leq [\text{lemma: 17.18(4)}] \\ \|f(z) - f(z')\| \cdot \|g(z)\| + \|f(z')\| \cdot \|g(z) - g(z')\| &< [\text{eq: 17.14}] \\ \|f(z) - f(z')\| \cdot \|g(z)\| + (1 + \|f(z)\|) \cdot \|g(z) - g(z')\| &< [\text{eqs: 17.12, 17.13}] \\ \frac{\varepsilon}{2 \cdot (1 + \|g(z)\|)} \cdot \|g(z)\| + (1 + \|f(z)\|) \cdot \frac{\varepsilon}{2 \cdot (1 + \|f(z)\|)} &< \\ \frac{\varepsilon}{2} + \frac{\varepsilon}{2} &= \\ \varepsilon & \end{aligned}$$

proving that $f \cdot g$ is continuous at z . As $z \in \mathbb{C}$ was chosen arbitrary $f \cdot g$ is continuous. $\square$

The following very specific lemma will be used in the proof of the fundamental theorem of algebra.

Lemma 17.20. *Let $n \in \mathbb{N}_0$ then*

1. $(1+i)^{4 \cdot n+2} = 2 \cdot (-4)^n \cdot i$
2. $(1-i)^{4 \cdot n+2} = -2 \cdot (-4)^n \cdot i$

Proof. This is easily proved by induction.

1. Take $S = \{n \in \mathbb{N}_0 \mid (1+i)^{4 \cdot n+2} = 2 \cdot (-4)^n \cdot i\}$ then we have:

$0 \in S$. $(1+i)^{4 \cdot 0+2} = (1+i)^2 = 1^2 + 2 \cdot i + i^2 = 1 + 2 \cdot i - 1 = 2 \cdot i = 2 \cdot (-4)^0 \cdot i$ so that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. We have

$$\begin{aligned}
 (1+i)^{4 \cdot (n+1)+2} &= (1+i)^{(4 \cdot n+2)+4} \\
 &= (1+i)^{(4 \cdot n+2)} \cdot (1+i)^4 \\
 &\stackrel{n \in S}{=} 2 \cdot (-4)^n \cdot i \cdot (1+i)^4 \\
 &= 2 \cdot (-4)^n \cdot i \cdot (1+i)^2 \cdot (1+i)^2 \\
 &= 2 \cdot (-4)^n \cdot i \cdot (1+2 \cdot i + i^2) \cdot (1+2 \cdot i + i^2) \\
 &= 2 \cdot (-4)^n \cdot i \cdot (2 \cdot i) \cdot (2 \cdot i) \\
 &= 2 \cdot (-4)^n \cdot i \cdot (4 \cdot i^2) \\
 &= 2 \cdot (-4)^n \cdot i \cdot (-4) \\
 &= 2 \cdot (-4)^{n+1} \cdot i
 \end{aligned}$$

proving that $n+1 \in S$.

2. Take $S = \{n \in \mathbb{N}_0 \mid (1-i)^{4 \cdot n+2} = -2 \cdot (-4)^n \cdot i\}$ then we have:

$0 \in S$. $(1-i)^{4 \cdot 0+2} = (1-i)^2 = 1^2 - 2 \cdot i + i^2 = 1 - 2 \cdot i - 1 = -2 \cdot i = -2 \cdot (-4)^0 \cdot i$ so that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. We have

$$\begin{aligned}
 (1-i)^{4 \cdot (n+1)+2} &= (1-i)^{(4 \cdot n+2)+4} \\
 &= (1-i)^{(4 \cdot n+2)} \cdot (1-i)^4 \\
 &\stackrel{n \in S}{=} -2 \cdot (-4)^n \cdot i \cdot (1-i)^4 \\
 &= -2 \cdot (-4)^n \cdot i \cdot (1-i)^2 \cdot (1-i)^2 \\
 &= -2 \cdot (-4)^n \cdot i \cdot (1-2 \cdot i + i^2) \cdot (1-2 \cdot i + i^2) \\
 &= -2 \cdot (-4)^n \cdot i \cdot (-2 \cdot i) \cdot (-2 \cdot i) \\
 &= -2 \cdot (-4)^n \cdot i \cdot (4 \cdot i^2) \\
 &= -2 \cdot (-4)^n \cdot (-4) \cdot i \\
 &= -2 \cdot (-4)^{n+1} \cdot i
 \end{aligned}$$

proving that $n+1 \in S$. $\square$

We also use the binomial expansion of $(a+b)^n$ so we need the following definitions and theorems.

Definition 17.21. *Given $n \in \mathbb{N}_0$ then we define $n! \in \mathbb{N}$ by*

$$n! = \begin{cases} 1 & \text{if } n=0 \\ n \cdot (n-1) & \text{if } 0 < n \end{cases} > 0$$

Definition 17.22. Given $n \in \mathbb{N}_0$ and $k \in \{0, \dots, n\}$ then

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

Theorem 17.23. Let $n \in \mathbb{N}_0$ then we have

1. $\binom{n}{0} = 1$
2. $\binom{n}{n} = 1$
3. If $n \in \mathbb{N}$ and $0 < k \leq n$ then $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$

Proof.

1. $\binom{n}{0} = \frac{n!}{0! \cdot (n-0)!} = \frac{n!}{1 \cdot n!} = 1$
2. $\binom{n}{n} = \frac{n!}{n! \cdot (n-n)!} = \frac{n!}{n! \cdot 0!} = \frac{n!}{n! \cdot 1} = 1$
3. We have

$$\begin{aligned} \binom{n}{k} + \binom{n}{k-1} &= \frac{n!}{k! \cdot (n-k)!} + \frac{n!}{(k-1)! \cdot (n-(k-1))!} \\ &= \frac{n! \cdot ((n+1)-k)}{k! \cdot (n-k)! \cdot ((n+1)-k)} + \frac{n! \cdot k}{k \cdot (k-1)! \cdot (n+1-k)} \\ &= \frac{n! \cdot (n+1-k)}{k! \cdot ((n+1)-k)!} + \frac{n! \cdot k}{k! \cdot ((n+1)-k)!} \\ &= \frac{n! \cdot (n+1-k) + n! \cdot k}{k! \cdot ((n+1)-k)!} \\ &= \frac{n! \cdot (n+1)}{k! \cdot ((n+1)-k)!} \\ &= \frac{(n+1)!}{k! \cdot ((n+1)-k)!} \\ &= \binom{n+1}{k} \end{aligned}$$

□

We prove now the binomial formula.

Theorem 17.24. (Binomial Formula) Let F be a field, $n \in \mathbb{N}$, $a, b \in F$ then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k}$$

Proof. We prove this by induction, so defined

$$S = \left\{ n \in \mathbb{N}_0 \mid (a+b)^n = \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k} \right\}$$

then we have:

$$0 \in S. \quad \text{As } \sum_{k=0}^0 \binom{0}{k} \cdot a^k \cdot b^{0-k} = \binom{0}{0} \cdot a^0 \cdot b^{0-0} = \underset{[theorem: 17.23]}{=} 1 \cdot 1 \cdot 1 = 1 = (a+b)^0$$

$n \in S \Rightarrow n+1 \in S$. We have

$$\begin{aligned}
 (a+b)^{n+1} &= \\
 (a+b)^n \cdot (a+b) &\stackrel{n \in S}{=} \\
 \left(\sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k} \right) \cdot (a+b) &= \\
 \sum_{k=0}^n \binom{n}{k} \cdot a \cdot a^k \cdot b^{n-k} + \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b \cdot b^{n-k} &= \\
 \sum_{k=0}^n \binom{n}{k} \cdot a^{k+1} \cdot b^{n-k} + \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} &= \\
 \sum_{k=1}^{n+1} \binom{n}{k-1} \cdot a^k \cdot b^{n-(k-1)} + \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} &= \\
 \sum_{k=1}^{n+1} \binom{n}{k-1} \cdot a^k \cdot b^{(n+1)-k} + \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} &= \\
 \binom{n}{n+1-1} \cdot a^{n+1} \cdot b^{(n+1)-(n+1)} + \sum_{k=1}^n \binom{n}{k-1} \cdot a^k \cdot b^{(n+1)-k} + \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} &= \\
 a^{n+1} + \sum_{k=1}^n \binom{n}{k-1} \cdot a^k \cdot b^{(n+1)-k} + \sum_{k=0}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} &= \\
 a^{n+1} + \sum_{k=1}^n \binom{n}{k-1} \cdot a^k \cdot b^{(n+1)-k} + \sum_{k=1}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} + \binom{n}{0} \cdot a^0 \cdot b^{n-0+1} &= \\
 a^{n+1} + \sum_{k=1}^n \binom{n}{k-1} \cdot a^k \cdot b^{(n+1)-k} + \sum_{k=1}^n \binom{n}{k} \cdot a^k \cdot b^{n-k+1} + b^{n+1} &= \\
 a^{n+1} + b^{n+1} + \sum_{k=1}^n \left(\binom{n}{k-1} + \binom{n}{k} \right) \cdot a^k \cdot b^{(n+1)-k} &= \\
 a^{n+1} + b^{n+1} + \sum_{k=1}^n \binom{n+1}{k} \cdot a^k \cdot b^{(n+1)-k} &= \\
 \binom{n+1}{n+1} \cdot a^{(n+1)} \cdot b^{(n+1)-(n+1)} + \sum_{k=1}^n \binom{n+1}{k} \cdot a^k \cdot b^{(n+1)-k} &= \\
 \sum_{k=0}^{n+1} \binom{n+1}{k} \cdot a^k \cdot b^{(n+1)-k} &
 \end{aligned}$$

proving that $n+1 \in S$. □

Lemma 17.25. $p \in \mathcal{P}$ be a polynomial [see definition: 17.9] and $z_0 \in \mathbb{C}$ then

$$p_{(+z_0)}: \mathbb{C} \rightarrow \mathbb{C} \text{ defined by } p_{(+z_0)}(z) = p(z_0 + z)$$

is a polynomial with $\text{ord}(p_{(+z_0)}) = \text{ord}(p)$.

Proof. We prove this by induction so let

$$S = \{n \in \mathbb{N}_0 \mid \text{If } p \in \mathcal{P} \text{ with } \text{ord}(p) = n \text{ then } p_{(+z_0)} \in \mathcal{P} \text{ with } \text{ord}(p) = \text{ord}(p_{(+z_0)})\}$$

then we have:

$0 \in S$. If $p \in \mathcal{P}$ with $\text{ord}(p) = 0$ then $p = \sum_{i=0}^0 a_i \cdot z^i = a_0 \cdot z^0 = a_0$. If $z \in \mathbb{C}$ then $p_{(+z_0)}(z) = p(z+z_0) = a_0 = p(z)$ so that $p_{(+z_0)} = p$. Hence $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $p \in \mathcal{P}$ with $\text{ord}(p) = n+1$ then there exist a $\{a_i\}_{i \in \{0, \dots, n+1\}} \subseteq \mathbb{C}$ such that $a_{n+1} \neq 0$ such that $\forall z \in \mathbb{C} \ p(z) = \sum_{i=0}^{n+1} a_i \cdot z^i$. Define

$$\{b_i\}_{i \in \{0, \dots, n\}} \text{ by } b_i = a_{i+1}$$

then $b_n = a_{n+1} \neq 0$ so that $q = \sum_{i=0}^n b_i \cdot z^i$ is a polynomial with $\text{ord}(q) = n$. Further we have for $z \in \mathbb{C}$ that:

$$\begin{aligned}
 p(z) &= \sum_{i=0}^{n+1} a_i \cdot z^i \\
 &= a_0 + \sum_{i=1}^{n+1} a_i \cdot z^i \\
 &= a_0 + \sum_{i=1}^{n+1} a_i \cdot z \cdot z^{i-1} \\
 &= a_0 + z \cdot \sum_{i=1}^{n+1} a_i \cdot z^{i-1} \\
 &= a_0 + z \cdot \sum_{i=0}^n a_{i+1} \cdot z^i \\
 &= a_0 + z \cdot \sum_{i=0}^n b_i \cdot z^i \\
 &= a_0 + z \cdot q(z)
 \end{aligned} \tag{17.15}$$

As $n \in S$ we have that $q_{(+z_0)}$ is a polynomial of order n , hence there exist a $\{c_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $c_n \neq 0$ such that

$$\forall z \in \mathbb{C} \text{ we have } q(z_0 + z) = q_{(+z_0)}(z) = \sum_{i=0}^n c_i \cdot z^i \tag{17.16}$$

Further we have

$$\begin{aligned}
 p_{(+z_0)}(z) &= p(z_0 + z) \\
 &\stackrel{[\text{eq: 17.15}]}{=} a_0 + (z_0 + z) \cdot q(z_0 + z) \\
 &\stackrel{[\text{eq: 17.16}]}{=} a_0 + (z_0 + z) \cdot \sum_{i=0}^n c_i \cdot z^i \\
 &= a_0 + z_0 \cdot \sum_{i=0}^n c_i \cdot z^i + z \cdot \sum_{i=0}^n c_i \cdot z^i \\
 &= a_0 + \sum_{i=0}^n c_i \cdot z_0 \cdot z^i + \sum_{i=0}^n c_i \cdot z^{i+1} \\
 &= a_0 + \sum_{i=0}^n c_i \cdot z_0 \cdot z^i + \sum_{i=1}^{n+1} c_{i-1} \cdot z^i \\
 &= a_0 + c_0 \cdot z_0 + \sum_{i=1}^n c_i \cdot z_0 \cdot z^i + \sum_{i=1}^{n+1} c_{i-1} \cdot z^i \\
 &= a_0 + c_0 \cdot z_0 + \sum_{i=1}^n c_i \cdot z_0 \cdot z^i + \sum_{i=1}^n c_{i-1} \cdot z^i + c_n \cdot z^{n+1} \\
 &= (a_0 + c_0 \cdot z_0) + \sum_{i=1}^n (c_i \cdot z_0 + c_{i-1}) \cdot z^i + c_n \cdot z^{n+1} \\
 &= \sum_{i=1}^{n+1} d_i \cdot z^i
 \end{aligned}$$

where

$$\{d_i\}_{i \in \{0, \dots, n+1\}} \text{ is defined by } d_i = \begin{cases} a_0 + c_0 \cdot z_0 & \text{if } i = 0 \\ c_i \cdot z_0 + c_{i-1} & \text{if } i \in \{1, \dots, n\} \\ c_n & \text{if } i = n+1 \end{cases}$$

As $d_{n+1} = c_n \neq 0$ it follows that $p_{(+z_0)}(z)$ is a polynomial with $\text{ord}(p_{(+z_0)}) = n + 1 = \text{ord}(p)$ which proves that $n + 1 \in S$. $\square$

Lemma 17.26. *Let $p: \mathbb{C} \rightarrow \mathbb{C}$ be a polynomial of order $n \in \mathbb{N}$ then there exists a $1 \leq m \leq n$ and a polynomial q of order $n - m$ such that $p(z) = p(0) + z^m \cdot q(z)$ and $q(0) \neq 0$.*

Proof. As p is a polynomial of order n there exist a $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $a_n \neq 0$ so that $\forall z \in \mathbb{C}$

$$p(z) = \sum_{i=0}^n a_i \cdot z^i$$

Then we have that $p(0) = \sum_{i=0}^n a_i \cdot 0^i = a_0 + \sum_{i=1}^n a_i \cdot 0^i = a_0 + \sum_{i=1}^n a_i \cdot 0 = a_0$ proving that

$$a_0 = p(0) \quad (17.17)$$

Define $K = \{i \in \{1, \dots, n\} | a_i \neq 0\}$ then as $1 \leq n$ and $a_n \neq 0$ $K \neq \emptyset$ so that $m = \min(K)$ exist and $\forall i \in \{0, \dots, m-1\}$ we have $a_i = 0$ and $a_m \neq 0$. Define now

$$\{c_i\}_{i \in \{0, \dots, n-m\}} \text{ by } c_i = a_{i+m} \text{ then } c_{n-m} = a_n \neq 0$$

Define the polynomial q of order $n - m$ by

$$q: \mathbb{C} \rightarrow \mathbb{C} \text{ where } q(z) = \sum_{i=0}^{n-m} c_i \cdot z^i \quad (17.18)$$

then we have for $z \in \mathbb{C}$

$$\begin{aligned} p(z) &= \sum_{i=0}^n a_i \cdot z^i \\ &= \sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \\ &= a_0 + \sum_{i \in \{1, \dots, n\}} a_i \cdot z^i \\ &= a_0 + \sum_{i \in \{1, \dots, m-1\}} a_i \cdot z^i + \sum_{i \in \{m, \dots, n\}} a_i \cdot z^i \\ &= a_0 + \sum_{i \in \{1, \dots, m-1\}} 0 \cdot z^i + \sum_{i \in \{m, \dots, n\}} a_i \cdot z^i \\ &= a_0 + \sum_{i=m}^n a_i \cdot z^i \\ &= a_0 + \sum_{i=m}^n a_i \cdot z^m \cdot z^{i-m} \\ &= a_0 + z^m \cdot \sum_{i=m}^n a_i \cdot z^{i-m} \\ &= a_0 + z^m \cdot \sum_{i=0}^{n-m} a_{i+m} \cdot z^i \\ &= a_0 + z^m \cdot \sum_{i=0}^{n-m} c_i \cdot z^i \\ &\stackrel{[\text{eqs: 17.17, 17.18}]}{=} p(0) + z^m \cdot q(z) \end{aligned} \quad (17.19)$$

Finally we have $q(0) = \sum_{i=0}^{n-m} c_i \cdot 0^i = c_0 + \sum_{i=1}^{n-m} c_i \cdot 0^i = c_0 = a_m \neq 0$ $\square$

The following lemma shows how we can split up the sets $\{0, \dots, 2 \cdot n\}$ and $\{0, \dots, 2 \cdot n - 1\}$ in disjoint sets of even and odd numbers.

Lemma 17.27. *Let $n \in \mathbb{N}$ then we have the following **disjoint** unions*

1. $\{0, \dots, 2 \cdot n\} = \{2 \cdot k | k \in \{0, \dots, n\}\} \sqcup \{2 \cdot k + 1 | k \in \{0, \dots, n-1\}\}$
2. $\{0, \dots, 2 \cdot n - 1\} = \{2 \cdot k | k \in \{0, \dots, n-1\}\} \sqcup \{2 \cdot k + 1 | k \in \{0, \dots, n-1\}\}$

Proof.

1. We use induction in the proof, so let

$$S = \{n \in \mathbb{N} \mid \{0, \dots, 2 \cdot n\} = \{2 \cdot k \mid k \in \{0, \dots, n\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}\}$$

then we have:

$1 \in S$. As $\{0, \dots, 2 \cdot 1\} = \{0, 1, 2\} = \{0, 2\} \cup \{1\} = \{2 \cdot k \mid k \in \{0, 1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0\}\}$
we have that $1 \in S$

$n \in S \Rightarrow n + 1 \in S$. We have

$$\begin{aligned} \{0, \dots, 2 \cdot (n+1)\} &= \\ \{0, \dots, 2 \cdot n\} \cup \{2 \cdot n + 1, 2 \cdot n + 2\} &= \\ \{0, \dots, 2 \cdot n\} \cup \{2 \cdot n + 1, 2 \cdot (n+1)\} &\stackrel{n \in S}{=} \\ \{2 \cdot k \mid k \in \{0, \dots, n\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\} \cup \{2 \cdot n + 1, 2 \cdot (n+1)\} &= \\ \{2 \cdot k \mid k \in \{0, \dots, n+1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n\}\} &= \\ \{2 \cdot k \mid k \in \{0, \dots, n+1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, (n+1)-1\}\} & \end{aligned}$$

proving that $n + 1 \in S$.

Mathematical induction proves then that $\forall n \in \mathbb{N}$ we have

$$\{0, \dots, 2 \cdot n\} = \{2 \cdot k \mid k \in \{0, \dots, n\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}$$

Further if $m \in \{2 \cdot k \mid k \in \{0, \dots, n\}\} \cap \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}$ then there exist a $k \in \{0, \dots, n\}$ and a $l \in \{0, \dots, n-1\}$ such that $2 \cdot k = m = 2 \cdot l + 1$ so that by [theorem: 7.52] m is even and odd [= not even by [definition: 7.50]] leading to a contradiction. Hence we must have that

$$\{2 \cdot k \mid k \in \{0, \dots, n\}\} \cap \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\} = \emptyset$$

so that

$$\{0, \dots, 2 \cdot n\} = \{2 \cdot k \mid k \in \{0, \dots, n\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}$$

2. We use induction in the proof, so let

$$S = \{n \in \mathbb{N} \mid \{0, \dots, 2 \cdot n - 1\} = \{2 \cdot k \mid k \in \{0, \dots, n-1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}\}$$

then we have:

$1 \in S$. As $\{0, \dots, 2 \cdot 1 - 1\} = \{0, 1\} = \{0\} \cup \{1\} = \{2 \cdot k \mid k \in \{0\}\} \cup \{2 \cdot k + 1 \mid k \in \{0\}\}$ we have that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. We have

$$\begin{aligned} \{0, \dots, 2 \cdot (n+1) - 1\} &= \\ \{0, \dots, 2 \cdot n + 1\} &= \\ \{0, \dots, 2 \cdot n - 1\} \cup \{2 \cdot n, 2 \cdot n + 1\} &\stackrel{n \in S}{=} \\ \{2 \cdot k \mid k \in \{0, \dots, n-1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\} \cup \{2 \cdot n, 2 \cdot n + 1\} &= \\ \{2 \cdot k \mid k \in \{0, \dots, n\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n\}\} &= \\ \{2 \cdot k \mid k \in \{0, \dots, (n+1)-1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, (n+1)-1\}\} & \end{aligned}$$

so that $n + 1 \in S$.

Mathematical induction proves then that $\forall n \in \mathbb{N}$ we have

$$\{0, \dots, 2 \cdot n - 1\} = \{2 \cdot k \mid k \in \{0, \dots, n-1\}\} \cup \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}$$

Further if $m \in \{2 \cdot k \mid k \in \{0, \dots, n-1\}\} \cap \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\}$ then there exist a $k \in \{0, \dots, n-1\}$ and a $l \in \{0, \dots, n-1\}$ such that $2 \cdot k = m = 2 \cdot l + 1$ so that by [theorem: 7.52] m is even and odd [= not even by [definition: 7.50]] leading to a contradiction. Hence we must have that

$$\{2 \cdot k \mid k \in \{0, \dots, n-1\}\} \cap \{2 \cdot k + 1 \mid k \in \{0, \dots, n-1\}\} = \emptyset$$

so that

$$\{0, \dots, 2 \cdot n - 1\} = \{2 \cdot k | k \in \{0, \dots, n - 1\}\} \sqcup \{2 \cdot k + 1 | k \in \{0, \dots, n - 1\}\} \quad \square$$

Lemma 17.28. *Let $n \in \mathbb{N}$ then if we define*

$$A = \{(k, l) \in \{0, \dots, n\}^2 | k < l\} \text{ and for } m \in \{1, \dots, 2 \cdot n - 1\} B_m = \{(k, l) \in A | k + l = m\} \subseteq A$$

we have that

$$A = \bigsqcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m$$

Proof. As $\forall m \in \{1, \dots, 2 \cdot n - 1\} B_m \subseteq A$ it follows that

$$\bigcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m \subseteq A \quad (17.20)$$

Further if $(k, l) \in A$ then $0 \leq k < l \leq n$ so that $k + l < l + l \leq 2 \cdot n$ hence $k + l \leq 2 \cdot n - 1$. Further as $0 < l$ we have $0 < k + l$ or $1 \leq k + l$. Hence if we take $m = k + l$ we have $m \in \{1, \dots, 2 \cdot n - 1\}$ proving that $(k, l) \in B_m \subseteq \bigcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m$ or that $A \subseteq \bigcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m$. Combining this with [eq: 17.20] gives us

$$A = \bigcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m$$

Let $m, m' \in \{1, \dots, 2 \cdot n - 1\}$ with $m \neq m'$ and assume that $(k, l) \in B_m \cap B_{m'}$ then $m = k + l = m'$ contradicting $m \neq m'$, so we must have that $B_m \cap B_{m'} = \emptyset$. Hence

$$A = \bigsqcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m \quad \square$$

Lemma 17.29. *Let $n \in \mathbb{N}$ then for $k = 2 \cdot n$ and $\zeta = \left(1 + \frac{i}{k}\right)^2$ we have that $\operatorname{Re}(\zeta^k) < 0 < \operatorname{Im}(\zeta^k)$.*

Proof. We have

$$\begin{aligned} \zeta^k &= \\ \left(\left(1 + \frac{i}{k} \right)^2 \right)^k &\stackrel{[\text{theorem: 10.53}]}{=} \\ \left(1 + \frac{i}{k} \right)^{2 \cdot k} &\stackrel{k=2 \cdot n}{=} \\ \left(1 + \frac{i}{2 \cdot n} \right)^{4 \cdot n} &\stackrel{[\text{theorem: 17.24}]}{=} \\ \sum_{l=0}^{4 \cdot n} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n} \right)^l \cdot 1^{4 \cdot n - l} &= \\ \sum_{l=0}^{4 \cdot n} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n} \right)^l &= \\ \sum_{l \in \{0, \dots, 2 \cdot (2 \cdot n)\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n} \right)^l &\stackrel{[\text{lemma: 17.27}]}{=} \\ \sum_{l \in \{2 \cdot j | j \in \{0, \dots, 2 \cdot n\}\} \sqcup \{2 \cdot j + 1 | j \in \{0, \dots, 2 \cdot n - 1\}\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n} \right)^l &= \\ \sum_{l \in \{2 \cdot j | j \in \{0, \dots, 2 \cdot n\}\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n} \right)^l + \sum_{l \in \{2 \cdot j + 1 | j \in \{0, \dots, 2 \cdot n - 1\}\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n} \right)^l \end{aligned}$$

so that we have

$$\zeta^k = A + B \quad (17.21)$$

where

$$A = \sum_{l \in \{2 \cdot j | j \in \{0, \dots, 2 \cdot n\}\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n}\right)^l \quad (17.22)$$

$$B = \sum_{l \in \{2 \cdot j + 1 | j \in \{0, \dots, 2 \cdot n - 1\}\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n}\right)^l \quad (17.23)$$

For A we have

$$\begin{aligned} & \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} + \sum_{l \in \{2 \cdot j + 1 | j \in \{0, \dots, n - 1\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \\ &= \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n\}\} \sqcup \{2 \cdot j + 1 | j \in \{0, \dots, n - 1\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \\ &= \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} + \sum_{l \in \{2 \cdot j + 1 | j \in \{0, \dots, n - 1\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \end{aligned}$$

So that

$$A = C + D \quad (17.24)$$

where

$$C = \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \quad (17.25)$$

$$D = \sum_{l \in \{2 \cdot j + 1 | j \in \{0, \dots, n - 1\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \quad (17.26)$$

For C we have

$$\begin{aligned} & \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \\ &= \sum_{l \in \{0, \dots, n\}} \binom{4 \cdot n}{2 \cdot (2 \cdot l)} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot (2 \cdot l)} \\ &= \sum_{l \in \{0, \dots, n\}} \binom{4 \cdot n}{4 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{4 \cdot l} \\ &= \sum_{l \in \{0, \dots, n\}} \binom{4 \cdot n}{4 \cdot l} \cdot \frac{i^{4 \cdot l}}{(2 \cdot n)^{4 \cdot l}} \quad [\text{theorem: 10.81}] \\ &= \sum_{l \in \{0, \dots, n\}} \binom{4 \cdot n}{4 \cdot l} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l}} \\ &= \binom{4 \cdot n}{4 \cdot 0} \cdot \frac{1}{(2 \cdot n)^{4 \cdot 0}} + \binom{4 \cdot n}{4 \cdot 1} \cdot \frac{1}{(2 \cdot n)^{4 \cdot 1}} + \sum_{l \in \{2, \dots, n\}} \binom{4 \cdot n}{4 \cdot l} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l}} \\ &= 1 + \binom{4 \cdot n}{4} \cdot \frac{1}{(2 \cdot n)^4} + \sum_{l \in \{2, \dots, n\}} \binom{4 \cdot n}{4 \cdot l} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l}} \\ &= 1 + \binom{4 \cdot n}{4} \cdot \frac{1}{(2 \cdot n)^4} + \sum_{l \in \{1, \dots, n - 1\}} \binom{4 \cdot n}{4 \cdot (l + 1)} \cdot \frac{1}{(2 \cdot n)^{4 \cdot (l + 1)}} \quad \stackrel{2 \cdot n = k}{=} \\ &= 1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} + \sum_{l \in \{1, \dots, n - 1\}} \binom{4 \cdot n}{4 \cdot (l + 1)} \cdot \frac{1}{(2 \cdot n)^{4 \cdot (l + 1)}} \end{aligned}$$

so that

$$C = 1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} + \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot (l+1)} \cdot \frac{1}{(2 \cdot n)^{4 \cdot (l+1)}} \quad (17.27)$$

For D we have

$$\begin{aligned} D & \stackrel{[\text{eq: 17.26}]}{=} \sum_{l \in \{2 \cdot j+1 | j \in \{0, \dots, n-1\}\}} \binom{4 \cdot n}{2 \cdot l} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l} \stackrel{[\text{theorem: 11.36}]}{=} \\ & \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{2 \cdot (2 \cdot l+1)} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot (2 \cdot l+1)} = \\ & \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \left(\frac{i}{2 \cdot n}\right)^{4 \cdot l+2} = \\ & \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{i^{4 \cdot l+2}}{(2 \cdot n)^{4 \cdot l+2}} \stackrel{[\text{theorem: 10.81}]}{=} \\ & - \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+2}} = \\ & - \binom{4 \cdot n}{4 \cdot 0+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot 0+2}} - \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+2}} = \\ & - \binom{4 \cdot n}{2} \cdot \frac{1}{(2 \cdot n)^2} - \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+2}} \stackrel{k=2 \cdot n}{=} \\ & - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} - \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+2}} \end{aligned}$$

So that

$$D = - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} - \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+2}} \quad (17.28)$$

So we have from [eqs: 17.27, 17.28] that

$$A = C + D = 1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} + E \quad (17.29)$$

where

$$E = \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot (l+1)} \cdot \frac{1}{(2 \cdot n)^{4 \cdot (l+1)}} - \sum_{l \in \{1, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+2} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+2}} \quad (17.30)$$

First we look the term $1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2}$ and calculate $\binom{2 \cdot k}{4}$ in terms of $\binom{2 \cdot k}{2}$. First notice that $2 \leq 2 \cdot n = k$ so that $2 < 3 < 4 \leq 2 \cdot k$ and thus $0 < 2 \cdot k - 3 \wedge 0 < 2 \cdot k - 2$ which we use in the following calculation:

$$\begin{aligned} \binom{2 \cdot k}{4} & = \frac{(2 \cdot k)!}{4! \cdot (2 \cdot k - 4)!} \\ & = \frac{(2 \cdot k)!}{4 \cdot 3 \cdot 2! \cdot (2 \cdot k - 4)!} \\ & \stackrel{0 < 2 \cdot k - 3, 2 \cdot k - 2}{=} \frac{(2 \cdot k)! \cdot (2 \cdot k - 2) \cdot (2 \cdot k - 3)}{4 \cdot 3 \cdot 2! \cdot (2 \cdot k - 4)! \cdot (2 \cdot k - 2) \cdot (2 \cdot k - 3)} \\ & = \frac{(2 \cdot k)! \cdot (2 \cdot k - 2) \cdot (2 \cdot k - 3)}{4 \cdot 3 \cdot 2! \cdot (2 \cdot k - 2) \cdot (2 \cdot k - 3) \cdot (2 \cdot k - 4)!} \\ & = \frac{(2 \cdot k)! \cdot (2 \cdot k - 2) \cdot (2 \cdot k - 3)}{4 \cdot 3 \cdot 2! \cdot (2 \cdot k - 2)!} \end{aligned}$$

$$\begin{aligned}
&= \frac{(2 \cdot k)! \cdot (2 \cdot k - 2) \cdot (2 \cdot k - 3)}{2! \cdot (2 \cdot k - 2)! \cdot 4 \cdot 3} \\
&= \binom{2 \cdot k}{2} \cdot \frac{(2 \cdot k - 2) \cdot (2 \cdot k - 3)}{4 \cdot 3} \\
&= \binom{2 \cdot k}{2} \cdot \frac{4 \cdot k^2 - 6 \cdot k - 4 \cdot k + 6}{12} \\
&= \binom{2 \cdot k}{2} \cdot \frac{4 \cdot k^2 - 10 \cdot k + 6}{12} \\
&= \binom{2 \cdot k}{2} \cdot \frac{2 \cdot k^2 - 5 \cdot k + 3}{6}
\end{aligned}$$

so that

$$\begin{aligned}
1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} &= 1 + \binom{2 \cdot k}{2} \cdot \frac{2 \cdot k^2 - 5 \cdot k + 3}{6} \cdot \frac{1}{k^4} - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} \\
&= 1 + \binom{2 \cdot k}{2} \cdot \left(\frac{2 \cdot k^2 - 5 \cdot k + 3}{6} \cdot \frac{1}{k^4} - \frac{1}{k^2} \right) \\
&= 1 + \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} \cdot \left(\frac{2 \cdot k^2 - 5 \cdot k + 3}{6 \cdot k^2} - 1 \right) \\
&= 1 + \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} \cdot \left(\frac{2 \cdot k^2 - 5 \cdot k + 3 - 6 \cdot k^2}{6 \cdot k^2} \right) \\
&= 1 + \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} \cdot \left(\frac{-4 \cdot k^2 - 5 \cdot k + 3}{6 \cdot k^2} \right) \\
&= 1 + \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} \cdot \left(-\frac{2}{3} + \frac{-5 \cdot k + 3}{6 \cdot k^2} \right) \\
&= 1 + \frac{(2 \cdot k)!}{2! \cdot (2 \cdot k - 2)!} \cdot \frac{1}{k^2} \cdot \left(-\frac{2}{3} + \frac{-5 \cdot k + 3}{6 \cdot k^2} \right) \\
&= 1 + \frac{2 \cdot k \cdot (2 \cdot k - 1) \cdot (2 \cdot k - 2)!}{2 \cdot (2 \cdot k - 2)!} \cdot \frac{1}{k^2} \cdot \left(-\frac{2}{3} + \frac{-5 \cdot k + 3}{6 \cdot k^2} \right) \\
&= 1 + k \cdot (2 \cdot k - 1) \cdot \frac{1}{k^2} \cdot \left(-\frac{2}{3} + \frac{-5 \cdot k + 3}{6 \cdot k^2} \right) \\
&= 1 + \left(2 - \frac{1}{k} \right) \cdot \left(-\frac{2}{3} + \frac{-5 \cdot k + 3}{6 \cdot k^2} \right) \\
&= 1 - \frac{4}{3} + \frac{-5 \cdot k + 3}{3 \cdot k^2} + \frac{2}{3 \cdot k} - \frac{5 \cdot k + 3}{6 \cdot k^3} \\
&= -\frac{1}{3} + \frac{-10 \cdot k^2 + 3 \cdot k + 4 \cdot k^2 - 5 \cdot k + 3}{6 \cdot k^3} \\
&= -\frac{1}{3} + \frac{-6 \cdot k^2 - 2 \cdot k + 3}{6 \cdot k^3} \\
&= -\frac{1}{3} + \frac{-6 \cdot k^2}{6 \cdot k^3} + \frac{-2 \cdot k + 3}{6 \cdot k^3} \\
&= -\frac{1}{3} - \frac{1}{k} + \frac{-2 \cdot k + 3}{6 \cdot k^3}
\end{aligned}$$

So that

$$1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} = -\frac{1}{3} - \frac{1}{k} + \frac{-2 \cdot k + 3}{6 \cdot k^3} \quad (17.31)$$

As $2 \leq 2 \cdot n = k$ we have $4 \leq 2 \cdot k \Rightarrow 3 < 2 \cdot k$ or $3 - 2 \cdot k < 0$, hence using the above we have that

$$1 + \binom{2 \cdot k}{4} \cdot \frac{1}{k^4} - \binom{2 \cdot k}{2} \cdot \frac{1}{k^2} < 0 \quad (17.32)$$

Define

$$F = \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l + 2} \cdot (4 \cdot l + 2)! \cdot (4 \cdot n - 4 \cdot l - 4)!} \cdot \frac{1}{(4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot (4 \cdot n - 4 \cdot l - 2) \cdot (4 \cdot n - 4 \cdot l - 3) \cdot (2 \cdot n)^2}$$

then, as $l \leq n-1 \Rightarrow 4 \cdot l \leq 4 \cdot n - 4 \Rightarrow 0 \leq 4 \cdot n - 4 \cdot l - 4 < 4 \cdot n - 4 \cdot l - 3 < 4 \cdot n - 4 \cdot l - 2$, we have that

$$0 < F \text{ and } E_l = F \cdot (4 \cdot n - 4 \cdot l - 2) \cdot (4 \cdot n - 4 \cdot l - 3) - (4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot (2 \cdot n)^2 \quad (17.34)$$

Now as $l \in \{0, \dots, n-1\}$ we have that $0 < l < n$, further we have $1 \leq n \Rightarrow 1 \leq n^2 \Rightarrow -16 \cdot n^2 \leq -16$

$$l^2 < n^2, \quad l < n \text{ and } -16 \cdot n^2 \leq -16$$

which will be use in the following calculation

$$\begin{aligned} 16 \cdot n^2 - \underbrace{16 \cdot n \cdot l}_1 - \underbrace{12 \cdot n}_2 - \underbrace{16 \cdot n \cdot l}_1 + 16 \cdot l^2 + \underbrace{12 \cdot l}_3 - \underbrace{8 \cdot n}_2 + \underbrace{8 \cdot l}_3 + 6 - (4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot (2 \cdot n)^2 &= \\ 16 \cdot n^2 - \underbrace{32 \cdot n \cdot l}_1 - \underbrace{20 \cdot n}_2 + 16 \cdot l^2 + \underbrace{20 \cdot l}_3 + 6 - (4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot (2 \cdot n)^2 &= \\ 16 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - (4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot (2 \cdot n)^2 &= \\ 16 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - (4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot 4 \cdot n^2 &= \\ 16 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 4 \cdot n^2 \cdot (4 \cdot l + 4) \cdot (4 \cdot l + 3) &= \\ 16 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 4 \cdot n^2 \cdot (16 \cdot l^2 + 12 \cdot l + 16 \cdot l + 12) &= \\ 16 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 4 \cdot n^2 \cdot (16 \cdot l^2 + 28 \cdot l + 12) &= \\ 16 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l - 48 \cdot n^2 &= \\ \underbrace{16 \cdot n^2}_1 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l - \underbrace{48 \cdot n^2}_1 &= \\ -\underbrace{32 \cdot n^2}_1 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &= \\ -32 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot l^2 + 20 \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &<_{l^2 < n^2} \\ -32 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot n^2 + 20 \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &<_{l < n} \\ -32 \cdot n^2 - 32 \cdot n \cdot l - 20 \cdot n + 16 \cdot n^2 + 20 \cdot n + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &= \\ -\underbrace{32 \cdot n^2}_1 - 32 \cdot n \cdot l - \underbrace{20 \cdot n}_2 + \underbrace{16 \cdot n^2}_1 + \underbrace{20 \cdot n}_2 + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &= \\ \underbrace{-16 \cdot n^2}_1 - 32 \cdot n \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &<_{-16n^2 \leftarrow 16} \\ -16 - 32 \cdot n \cdot l + 6 - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &= \\ -10 - 32 \cdot n \cdot l - 64 \cdot n^2 \cdot l^2 - 112 \cdot n^2 \cdot l &< \\ 0 & \end{aligned}$$

Which proves that $(4 \cdot n - 4 \cdot l - 2) \cdot (4 \cdot n - 4 \cdot l - 3) - (4 \cdot l + 4) \cdot (4 \cdot l + 3) \cdot (2 \cdot n)^2 < 0$ and after combining this with [eq: 17.34] that $\forall l \in \{1, \dots, n-1\} \quad E_l < 0$ so that by [eq: 17.33]

$$E < 0 \quad (17.35)$$

Combining then [eqs: 17.35, 17.32 and 17.29] proves that

$$A \in \mathbb{R} \text{ and } A < 0 \quad (17.36)$$

We calculate now B

$$\begin{aligned}
& B \quad [\text{eq: } \overline{17.23}] \\
& \sum_{l \in \{2 \cdot j+1 | j \in \{0, \dots, 2 \cdot n-1\}\}} \binom{4 \cdot n}{l} \cdot \left(\frac{i}{2 \cdot n}\right)^l \quad [\text{theorem: } \overline{11.36}] \\
& \sum_{l \in \{0, \dots, 2 \cdot n-1\}} \binom{4 \cdot n}{2 \cdot l+1} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l+1} \quad [\text{lemma: } \overline{17.27}] \\
& \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n-1\}\} \sqcup \{2 \cdot j+1 | j \in \{0, \dots, n-1\}\}} \binom{4 \cdot n}{2 \cdot l+1} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l+1} = \\
& \sum_{l \in \{2 \cdot j | j \in \{0, \dots, n-1\}\}} \binom{4 \cdot n}{2 \cdot l+1} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l+1} + \sum_{l \in \{2 \cdot j+1 | j \in \{0, \dots, n-1\}\}} \binom{4 \cdot n}{2 \cdot l+1} \cdot \\
& \left(\frac{i}{2 \cdot n}\right)^{2 \cdot l+1} \quad [\text{theorem: } \overline{11.36}] \\
& \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{2 \cdot (2 \cdot l)+1} \cdot \left(\frac{i}{2 \cdot n}\right)^{2 \cdot (2 \cdot l)+1} + \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{2 \cdot (2 \cdot l+1)+1} \cdot \\
& \left(\frac{i}{2 \cdot n}\right)^{2 \cdot (2 \cdot l+1)+1} = \\
& \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+1} \cdot \left(\frac{i}{2 \cdot n}\right)^{4 \cdot l+1} + \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+3} \cdot \left(\frac{i}{2 \cdot n}\right)^{4 \cdot l+3} = \\
& \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{i^{4 \cdot l+1}}{(2 \cdot n)^{4 \cdot l+1}} + \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{i^{4 \cdot l+3}}{(2 \cdot n)^{4 \cdot l+3}} \quad [\text{theorem: } \overline{10.81}] \\
& \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{i}{(2 \cdot n)^{4 \cdot l+1}} + \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{-i}{(2 \cdot n)^{4 \cdot l+3}} = \\
& i \cdot \left(\sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+1}} - \sum_{l \in \{0, \dots, n-1\}} \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+3}} \right) = \\
& i \cdot \sum_{l \in \{0, \dots, n-1\}} \left(\binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+1}} - \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+3}} \right)
\end{aligned}$$

So that

$$B = i \cdot \sum_{l \in \{0, \dots, n-1\}} B_l \text{ where } B_l = \binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+1}} - \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+3}} \quad (17.37)$$

Now for $l \in \{0, \dots, n-1\}$ we have

$$\begin{aligned}
& B_l \quad [\text{eq: } \overline{17.37}] \\
& \binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+1}} - \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+3}} = \\
& \binom{4 \cdot n}{4 \cdot l+1} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+1}} - \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{1}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2} = \\
& \frac{1}{(2 \cdot n)^{4 \cdot l+1}} \cdot \left(\binom{4 \cdot n}{4 \cdot l+1} - \binom{4 \cdot n}{4 \cdot l+3} \cdot \frac{1}{(2 \cdot n)^2} \right) = \\
& \frac{1}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2} \cdot \left(\binom{4 \cdot n}{4 \cdot l+1} \cdot (2 \cdot n)^2 - \binom{4 \cdot n}{4 \cdot l+3} \right) = \\
& \frac{1}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2} \cdot \left(\frac{(4 \cdot n)! \cdot (2 \cdot n)^2}{(4 \cdot l+1)! \cdot (4 \cdot n - (4 \cdot l+1))!} - \frac{(4 \cdot n)!}{(4 \cdot l+3)! \cdot (4 \cdot n - (4 \cdot l+3))!} \right) = \\
& \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2} \cdot \left(\frac{(2 \cdot n)^2}{(4 \cdot l+1)! \cdot (4 \cdot n - 4 \cdot l - 1)!} - \frac{1}{(4 \cdot l+3)! \cdot (4 \cdot n - 4 \cdot l - 3)!} \right) =
\end{aligned}$$

$$\begin{aligned}
& \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2} \cdot \left(\frac{(2 \cdot n)^2}{(4 \cdot l+1)! \cdot (4 \cdot n-4 \cdot l-1)!} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2) \cdot (4 \cdot l+1)! \cdot (4 \cdot n-4 \cdot l-3)!} \right) = \\
& \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2 \cdot (4 \cdot l+1)!} \cdot \left(\frac{(2 \cdot n)^2}{(4 \cdot n-4 \cdot l-1)!} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2) \cdot (4 \cdot n-4 \cdot l-3)!} \right) = \\
& \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2 \cdot (4 \cdot l+1)!} \cdot \left(\frac{(2 \cdot n)^2}{(4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2) \cdot (4 \cdot n-4 \cdot l-3)!} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2) \cdot (4 \cdot n-4 \cdot l-3)!} \right) = \\
& \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2 \cdot (4 \cdot l+1)! \cdot (4 \cdot n-4 \cdot l-3)!} \cdot \left(\frac{(2 \cdot n)^2}{(4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2)} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2)} \right) =
\end{aligned}$$

Define $G_l = \frac{(4 \cdot n)!}{(2 \cdot n)^{4 \cdot l+1} \cdot (2 \cdot n)^2 \cdot (4 \cdot l+1)! \cdot (4 \cdot n-4 \cdot l-3)!}$ then

$$0 < G \text{ and } B_l = G \cdot \left(\frac{(2 \cdot n)^2}{(4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2)} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2)} \right) \quad (17.38)$$

Now for $l \in \{1, \dots, n-1\}$ we have

$$l \leq n-1 \Rightarrow 0 \leq n-l-1 \Rightarrow 0 \leq 4 \cdot n-4 \cdot l-4 < 4 \cdot n-4 \cdot l-2 < 4 \cdot n-4 \cdot l-1 \quad (17.39)$$

Further

$$\begin{aligned}
& \frac{(2 \cdot n)^2}{(4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2)} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2)} = \\
& \frac{(2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - (4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2)}{(4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2) \cdot (4 \cdot l+3) \cdot (4 \cdot l+2)}
\end{aligned}$$

Define $H_l = (4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2) \cdot (4 \cdot l+3) \cdot (4 \cdot l+2)$ then using [eq: 17.39]

$$\begin{aligned}
0 < H_l \quad \text{and} \quad \frac{(2 \cdot n)^2}{(4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2)} - \frac{1}{(4 \cdot l+3) \cdot (4 \cdot l+2)} = \\
\frac{(2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - (4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2)}{H_l} \quad (17.40)
\end{aligned}$$

As $l \in \{1, \dots, n-1\}$ we have $l < n \Rightarrow -l > -n$ and from $1 \leq n \Rightarrow 1 \leq n^2 \Rightarrow -n^2 \leq -1$ so that

$$-l > -n, -2 \geq -2 \cdot n^2 \quad (17.41)$$

which we will use in the following:

$$\begin{aligned}
& (2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - (4 \cdot n-4 \cdot l-1) \cdot (4 \cdot n-4 \cdot l-2) = \\
& (2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - 16 \cdot n^2 + 16 \cdot n \cdot l + 8 \cdot n + 16 \cdot l \cdot n - 16 \cdot l^2 - 8 \cdot l + 4 \cdot n - 4 \cdot l - 2 = \\
& (2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - 16 \cdot n^2 + \underbrace{16 \cdot n \cdot l}_1 + \underbrace{8 \cdot n}_2 + \underbrace{16 \cdot l \cdot n}_1 - 16 \cdot l^2 - \underbrace{8 \cdot l}_3 + \underbrace{4 \cdot n}_2 - \underbrace{4 \cdot l - 2}_3 = \\
& (2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - 16 \cdot n^2 + \underbrace{32 \cdot n \cdot l}_1 + \underbrace{12 \cdot n}_2 - 16 \cdot l^2 - \underbrace{12 \cdot l}_2 - 2 = \\
& (2 \cdot n)^2 \cdot (4 \cdot l+3) \cdot (4 \cdot l+2) - 16 \cdot n^2 + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 = \\
& (2 \cdot n)^2 \cdot (16 \cdot l^2 + 8 \cdot l + 12 \cdot l + 6) - 16 \cdot n^2 + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 = \\
& 4 \cdot n^2 \cdot (16 \cdot l^2 + 20 \cdot l + 6) - 16 \cdot n^2 + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 = \\
& 64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 24 \cdot n^2 - 16 \cdot n^2 + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 =
\end{aligned}$$

$$\begin{aligned}
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + \underbrace{24 \cdot n^2}_{1} - \underbrace{16 \cdot n^2}_{1} + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + \underbrace{8 \cdot n^2}_{1} + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 8 \cdot n^2 + 32 \cdot n \cdot l + 12 \cdot n - 16 \cdot l^2 - 12 \cdot l - 2 &>_{\text{eq: 17.41}} \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 8 \cdot n^2 + 32 \cdot n \cdot l + \underbrace{12 \cdot n}_{1} - 16 \cdot l^2 - \underbrace{12 \cdot n}_{1} - 2 &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 8 \cdot n^2 + 32 \cdot n \cdot l - 16 \cdot l^2 - 2 &\geq_{\text{eq: 17.41}} \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 8 \cdot n^2 + 32 \cdot n \cdot l - 16 \cdot l^2 - 2 \cdot n^2 &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + \underbrace{8 \cdot n^2}_{1} + 32 \cdot n \cdot l - 16 \cdot l^2 - \underbrace{2 \cdot n^2}_{1} &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + \underbrace{6 \cdot n^2}_{1} + 32 \cdot n \cdot l - 16 \cdot l^2 &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 6 \cdot n^2 + 32 \cdot n \cdot l - 16 \cdot l^2 &>_{n>l} \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 6 \cdot n_1^2 + 32 \cdot l^2 - 16 \cdot l^2 &= \\
64 \cdot n^2 \cdot l^2 + 80 \cdot n^2 \cdot l + 6 \cdot n_1^2 + 16 \cdot l^2 &> \\
0 &
\end{aligned}$$

Using the above in [eq: 17.40] proves that $0 < \frac{(2 \cdot n)^2}{(4 \cdot n - 4 \cdot l - 1) \cdot (4 \cdot n - 4 \cdot l - 2)} - \frac{1}{(4 \cdot l + 3) \cdot (4 \cdot l + 2)}$ which by [eq: 17.38] proves that $0 < B_l$ so that by [eq: 17.37]

$$B = i \cdot \sum_{l \in \{1, \dots, n-1\}} B_l \text{ where } 0 < \sum_{l \in \{1, \dots, n-1\}} B_l \in \mathbb{R} \quad (17.42)$$

Combining the above with [eqs: 17.21, 17.36 and 17.42] proves that

$$\zeta^k = A + i \cdot \sum_{l \in \{1, \dots, n-1\}} B$$

where $A, \sum_{l \in \{1, \dots, n-1\}} B \in \mathbb{R}$, $A < 0 < \sum_{l \in \{1, \dots, n-1\}} B$ so that

$$\text{Re}(\zeta^k) < 0 < \text{Im}(\zeta^k) \quad \square$$

17.1.4 Proof of the fundamental theorem of algebra

After all this work we are now ready to prove the fundamental theorem of algebra.

Theorem 17.30. *Let $n \in \mathbb{N}$, $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $a_n \neq 0$ and $p: \mathbb{C} \rightarrow \mathbb{C}$ the polynomial defined by $p(z) = \sum_{i=0}^n a_i \cdot z^i$ then there exist a $z_0 \in \mathbb{C}$ such that $p(z_0) = 0$. In other words every non constant polynomial has a root.*

Proof. First define

$$p \cdot \bar{p}: \mathbb{C} \rightarrow \mathbb{C} \text{ by } (p \cdot \bar{p})(z) = p(z) \cdot \overline{p(z)}$$

then for $z \in \mathbb{C}$ we have

$$\begin{aligned}
(p \cdot \bar{p})(z) &= \\
p(z) \cdot \overline{p(z)} &= \\
\left(\sum_{i=0}^n a_i \cdot z^i \right) \cdot \overline{\left(\sum_{i=0}^n a_i \cdot z^i \right)} &= \\
\left(\sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \right) \cdot \overline{\left(\sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \right)} & \stackrel{[\text{theorem: 10.83}]}{=} \\
\left(\sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \right) \cdot \left(\sum_{i \in \{0, \dots, n\}} \overline{a_i \cdot z^i} \right) & \stackrel{[\text{theorem: 11.48}]}{=}
\end{aligned}$$

$$\begin{aligned}
& \sum_{(i,j) \in \{0, \dots, n\}^2} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} = \\
& \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k=l\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + \\
& \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | l < k\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} = \\
& \sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \cdot \overline{a_i \cdot z^i} + \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + \\
& \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | l < k\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} = \\
& \sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \cdot \overline{a_i \cdot z^i} + \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + \\
& \sum_{(j,i) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} a_i \cdot \overline{a_j} \cdot z^i \cdot (\overline{z})^j = \\
& \sum_{i \in \{0, \dots, n\}} a_i \cdot z^i \cdot \overline{a_i \cdot z^i} + \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + \\
& \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} a_j \cdot z^j \cdot \overline{a_i \cdot z^i} = \\
& \sum_{i \in \{0, \dots, n\}} a_i \cdot \overline{a_i} \cdot z^i \cdot (\overline{z})^i + \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} (a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + a_j \cdot z^j \cdot \overline{a_i \cdot z^i}) \quad [\text{theorem: 10.83}] \\
& \sum_{i \in \{0, \dots, n\}} a_i \cdot \overline{a_i} \cdot z^i \cdot (\overline{z})^i + \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} (a_i \cdot z^i \cdot \overline{a_j \cdot z^j} + \overline{a_i \cdot z^i \cdot a_j \cdot z^j}) \quad [\text{theorem: 10.83}] \\
& \sum_{i \in \{0, \dots, n\}} a_i \cdot \overline{a_i} \cdot z^i \cdot (\overline{z})^i + 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) = \\
& \sum_{i \in \{0, \dots, n\}} |a_i|^2 \cdot |z|^{2 \cdot i} + 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j})
\end{aligned}$$

so we have that

$$(p \cdot \overline{p})(z) = \sum_{i \in \{0, \dots, n\}} |a_i|^2 \cdot |z|^{2 \cdot i} + 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) \quad (17.43)$$

Further we have

$$\begin{aligned}
& \frac{(p \cdot \overline{p})(z)}{p(z) \cdot \overline{p(z)}} = \\
& \frac{|p(z) \cdot \overline{p(z)}|}{\|p(z) \cdot \overline{p(z)}\|} \quad \begin{array}{l} 0 \leq p(z) \cdot \overline{p(z)} \in \mathbb{R} \\ [\text{lemma: 17.18}] \\ [\text{eq: 17.43}] \end{array} \\
& \left\| \sum_{i \in \{0, \dots, n\}} |a_i|^2 \cdot |z|^{2 \cdot i} + 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) \right\| \geq \quad [\text{theorem: 14.91}] \\
& \left\| \sum_{i \in \{0, \dots, n\}} |a_i|^2 \cdot |z|^{2 \cdot i} \right\| - \left\| 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) \right\| \quad [\text{lemma: 17.18}] \\
& \left| \sum_{i \in \{0, \dots, n\}} |a_i|^2 \cdot |z|^{2 \cdot i} \right| - \left\| 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) \right\| = \\
& \sum_{i \in \{0, \dots, n\}} |a_i|^2 \cdot |z|^{2 \cdot i} - 2 \cdot \left\| \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) \right\| \geq \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \left\| \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \text{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j}) \right\| \geq \quad [\text{theorem: 10.85(3)}]
\end{aligned}$$

$$\begin{aligned}
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} |\operatorname{Re}(a_i \cdot z^i \cdot \overline{a_j \cdot z^j})| && \stackrel{[\text{lemma: 17.18}]}{=} \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|\operatorname{Re}(a_i \cdot \overline{a_j} \cdot z^i \cdot \overline{z^j})\| && \stackrel{[\text{lemma: 17.18(4)}]}{\geq} \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z^i \cdot \overline{z^j}\| && \geq [\text{lemma: 17.18(3)}] \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z^i\| \cdot \|\overline{z^j}\| && \stackrel{[\text{lemma: 17.18(2)}]}{=} \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z^i\| \cdot \|z^j\| && \geq [\text{lemma: 17.18(5)}] \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^i \cdot \|z\|^j && = \\
& |a_n|^2 \cdot |z|^{2 \cdot n} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && \stackrel{0 \leq |z|^{2 \cdot n} \in \mathbb{R} \wedge [\text{lemma: 17.18}]}{=} \\
& |a_n|^2 \cdot \| |z|^{2 \cdot n} \| - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && = \\
& |a_n|^2 \cdot \|z^n \cdot \overline{z^n}\| - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && \stackrel{[\text{lemma: 17.18(3)}]}{\geq} \\
& |a_n|^2 \cdot \frac{\|z^n\| \cdot \|\overline{z^n}\|}{2} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && \stackrel{[\text{lemma: 17.18(2)}]}{=} \\
& |a_n|^2 \cdot \frac{\|z^n\| \cdot \|z^n\|}{2} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && = \\
& |a_n|^2 \cdot \frac{\|z^n\|^2}{2} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && \geq [\text{lemma: 17.18(7)}] \\
& |a_n|^2 \cdot \frac{1}{2} \cdot \left(\frac{\|z\|^n}{2^n} \right)^2 - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && = \\
& |a_n|^2 \cdot \frac{\|z\|^{2 \cdot n}}{2^{2 \cdot n+1}} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j}
\end{aligned}$$

so that

$$(p \cdot \bar{p})(z) \geq |a_n|^2 \cdot \frac{\|z\|^{2 \cdot n}}{2^{2 \cdot n+1}} - 2 \cdot \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} \quad (17.44)$$

Using [lemma: 17.28] we have $A = \{(k, l) \in \{0, \dots, n\}^2 | k < l\} = \bigsqcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m$ where $B_m = \{(k, l) \in A | k + l = m\}$. So that

$$\begin{aligned}
& \sum_{(i,j) \in \{(k,l) \in \{0, \dots, n\}^2 | k < l\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && = \\
& \sum_{(i,j) \in \bigsqcup_{m \in \{1, \dots, 2 \cdot n - 1\}} B_m} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} && \stackrel{[\text{theorem: 11.44}]}{=} \\
& \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} \left(\sum_{(i,j) \in B_m} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} \right) && = \\
& \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} \left(\sum_{(i,j) \in \{(k,l) | k+l=m\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^{i+j} \right) && = \\
& \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} \left(\sum_{(i,j) \in \{(k,l) \in A | k+l=m\}} \|a_i \cdot \overline{a_j}\| \cdot \|z\|^m \right) && =
\end{aligned}$$

$$\sum_{m \in \{1, \dots, 2 \cdot n - 1\}} \left(\sum_{(i,j) \in \{(k,l) \in A \mid k+l=m\}} \|a_i \cdot \bar{a}_j\| \right) \cdot \|z\|^m = \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} b_m \cdot \|z\|^m$$

where $\forall m \in \{1, \dots, 2 \cdot n - 1\}$ $b_m = \sum_{(i,j) \in \{(k,l) \in A \mid k+l=m\}} \|a_i \cdot \bar{a}_j\|$. Substituting this in [eq: 17.44] gives

$$\begin{aligned} (p \cdot \bar{p})(z) &\geq |a_n|^2 \cdot \frac{\|z\|^{2 \cdot n}}{2^{2 \cdot n + 1}} - 2 \cdot \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} b_m \cdot \|z\|^m \\ &= |a_n|^2 \cdot \frac{\|z\|^{2 \cdot n}}{2^{2 \cdot n + 1}} + \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} (-2 \cdot b_m) \cdot \|z\|^m \\ &= 0 \cdot \|z\|^0 + |a_n|^2 \cdot \frac{\|z\|^{2 \cdot n}}{2^{2 \cdot n + 1}} + \sum_{m \in \{1, \dots, 2 \cdot n - 1\}} (-2 \cdot b_m) \cdot \|z\|^m \\ &= \sum_{i=0}^{2 \cdot n} c_i \cdot \|z\|^i \end{aligned} \quad (17.45)$$

where

$$\{c_i\}_{i \in \{0, \dots, 2 \cdot n\}} \text{ where } c_i = \begin{cases} 0 & \text{if } i = 0 \\ -2 \cdot b_i & \text{if } i \in \{1, \dots, 2 \cdot n - 1\} \\ \frac{1}{2^{2 \cdot n + 1}} \cdot |a_n|^2 & \text{if } i = 2 \cdot n \end{cases}$$

then as $a_n \neq 0$ we have that $0 < c_{2 \cdot n}$. Hence using [lemma: 17.16] it follows that

$$\lim_{x \rightarrow \infty} \sum_{i=0}^{2 \cdot n} c_i \cdot x^i = \infty$$

so there exist a $R' \in \mathbb{R}$ such that if $x \geq R'$ then $\sum_{i=0}^{2 \cdot n} c_i \cdot x^i \geq (p \cdot \bar{p})(0) + 1 > (p \cdot \bar{p})(0)$. Take $R = \min(R', 1)$ then we have using [eq: 17.45] that

$$0 < R \text{ and } \forall z \in \mathbb{C} \text{ with } \|z\| \geq R \text{ we have } (p \cdot \bar{p})(z) \geq \sum_{i=0}^{2 \cdot n} c_i \cdot \|z\|^i > (p \cdot \bar{p})(0) \quad (17.46)$$

Now as $\overline{B_{|||}(0, R)}$ is closed and bounded in $\langle \mathbb{C}, ||| \rangle$ [see theorems: 14.69, 14.82] it follows from [theorem: 14.332] that

$$\overline{B_{|||}(0, R)} \text{ is compact in } \langle \mathbb{C}, ||| \rangle \quad (17.47)$$

Now by [theorem: 17.19(5)] the function $p \cdot \bar{p}$ is continuous in $\langle \mathbb{C}, ||| \rangle$ so that, using the above, we can apply the extreme value theorem [theorem: 14.282]. Hence there exist a $z_0 \in \overline{B_{|||}(0, R)}$ so that

$$\forall z \in \overline{B_{|||}(0, R)} \text{ we have } (p \cdot \bar{p})(z) \geq (p \cdot \bar{p})(z_0) \quad (17.48)$$

As $0 \in \overline{B_{|||}(0, R)}$ it follows from the above that

$$(p \cdot \bar{p})(0) \geq (p \cdot \bar{p})(z_0) \quad (17.49)$$

If $z \in \mathbb{C}$ then we have either:

$$\|z\| \leq R. \text{ then by [eq: 17.48] } (p \cdot \bar{p})(z) \geq (p \cdot \bar{p})(z_0)$$

$$R < \|z\|. \text{ then by [eq: 17.46] } (p \cdot \bar{p})(z) \geq (p \cdot \bar{p})(0) \geq_{[\text{eq: 17.49}]} (p \cdot \bar{p})(z_0)$$

so we have

$$\forall z \in \mathbb{C} \text{ that } (p \cdot \bar{p})(z) \geq (p \cdot \bar{p})(z_0) \quad (17.50)$$

Define now

$$r: \mathbb{C} \rightarrow \mathbb{C} \text{ by } r(z) = p(z + z_0) \text{ and } r \cdot \bar{r}: \mathbb{C} \rightarrow \mathbb{C} \text{ by } (r \cdot \bar{r})(z) = r(z) \cdot \overline{r(z)} \quad (17.51)$$

Then we have

$$r(0) = p(0 + z_0) = p(z_0) \quad (17.52)$$

Further

$$\begin{aligned} (r \cdot \bar{r})(z) &= r(z) \cdot \overline{r(z)} \\ &= p(z + z_0) \cdot \overline{p(z + z_0)} \\ &= (p \cdot \bar{p})(z + z_0) \\ &\geq_{[\text{eq: 17.50}]} (p \cdot \bar{p})(z_0) \\ &= p(z_0) \cdot \overline{p(z_0)} \\ &\stackrel{=}{=} r(0) \cdot \overline{r(0)} \\ &\stackrel{\text{eq: 17.52}}{=} (r \cdot \bar{r})(0) \end{aligned}$$

in other words

$$\forall z \in \mathbb{C} \text{ we have } (r \cdot \bar{r})(z) - (r \cdot \bar{r})(0) \geq 0 \quad (17.53)$$

By [lemma: 17.25] and the definition of r it follows that r is a polynomial with $\text{ord}(r) = \text{ord}(p)$. Further using [lemma: 17.26] there exist a $k \in \{1, \dots, n\}$ and a polynomial of order $n - k$ such that

$$q(0) \neq 0 \text{ and } \forall z \in \mathbb{C} \text{ we have } r(z) = r(0) + z^k \cdot q(z) \quad (17.54)$$

Then we have

$$\begin{aligned} 0 &\stackrel{[\text{eq: 17.53}]}{\leq} (r \cdot \bar{r})(z) - (r \cdot \bar{r})(0) \\ &= r(z) \cdot \overline{r(z)} - r(0) \cdot \overline{r(0)} \\ &= (r(0) + z^k \cdot q(z)) \cdot \overline{(r(0) + z^k \cdot q(z))} - (r(0) + 0^k \cdot q(0)) \cdot \overline{(r(0) + 0^k \cdot q(0))} \\ &= (r(0) + z^k \cdot q(z)) \cdot \overline{(r(0) + z^k \cdot q(z))} - r(0) \cdot \overline{r(0)} \\ &\stackrel{[\text{theorem: 10.83}]}{=} (r(0) + z^k \cdot q(z)) \cdot \overline{(r(0) + z^k \cdot q(z))} - r(0) \cdot \overline{r(0)} \\ &= r(0) \cdot \overline{r(0)} + r(0) \cdot \overline{z^k \cdot q(z)} + z^k \cdot q(z) \cdot \overline{r(0)} + z^k \cdot q(z) \cdot \overline{z^k \cdot q(z)} - r(0) \cdot \overline{r(0)} \\ &= r(0) \cdot \overline{z^k \cdot q(z)} + z^k \cdot q(z) \cdot \overline{r(0)} + z^k \cdot q(z) \cdot \overline{z^k \cdot q(z)} \\ &= \overline{r(0) \cdot z^k \cdot q(z)} + z^k \cdot q(z) \cdot \overline{r(0)} + z^k \cdot q(z) \cdot \overline{z^k \cdot q(z)} \\ &= z^k \cdot q(z) \cdot \overline{r(0)} + z^k \cdot q(z) \cdot \overline{r(0)} + z^k \cdot q(z) \cdot \overline{z^k \cdot q(z)} \\ &\stackrel{[\text{theorem: 10.83}]}{=} 2 \cdot \text{Re}(z^k \cdot q(z) \cdot \overline{r(0)}) + z^k \cdot q(z) \cdot \overline{z^k \cdot q(z)} \\ &\stackrel{[\text{theorem: 10.85}]}{=} 2 \cdot \text{Re}(z^k \cdot q(z) \cdot \overline{r(0)}) + |z^k \cdot q(z)|^2 \end{aligned}$$

hence

$$|z^k \cdot q(z)|^2 + 2 \cdot \text{Re}(z^k \cdot q(z) \cdot \overline{r(0)}) \geq 0 \quad (17.55)$$

Given $x \in \mathbb{C}$ and $\delta \in \mathbb{R}^+$ then taken $z = \delta \cdot x$ and substituting this in the above gives

$$\begin{aligned} 0 &\leq |(\delta \cdot x)^k \cdot q(\delta \cdot x)|^2 + 2 \cdot \text{Re}((\delta \cdot x)^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \\ &\stackrel{[\text{theorems: 10.83, 10.80}]}{=} \delta^{2 \cdot k} \cdot |x^k \cdot q(\delta \cdot x)|^2 + 2 \cdot \delta^k \cdot \text{Re}(x^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \end{aligned}$$

and by dividing both sides of the above by $\delta^k > 0$ we have

$$0 \leq \delta^k \cdot |x^k \cdot q(\delta \cdot x)|^2 + 2 \cdot \text{Re}(x^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \quad (17.56)$$

Given $x \in \mathbb{C}$ define

$$f_x: \mathbb{R} \rightarrow \mathbb{R} \text{ by } f_x(\delta) = \delta^k \cdot |x^k \cdot q(\delta \cdot x)|^2 + 2 \cdot \text{Re}(x^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \quad (17.57)$$

We prove now that f_x is continuous by showing that it is the sum of functions composed of continuous functions defined in [theorem: 17.19]. So define

$$f_{x,1}: \mathbb{C} \rightarrow \mathbb{R} \text{ by } \langle \cdot \rangle \circ (x^k \cdot \cdot) \circ q \circ (x \cdot)$$

which is continuous as the composition the continuous mappings $\langle \cdot \rangle$, $(x^k \cdot)$, q and $(x \cdot)$ [see theorem: 17.19(1,3,4)]. Using the definition of $f_{x,1}$ we have:

$$f_{x,1}(\delta) = \langle \cdot \rangle((x^k \cdot)(q((x \cdot)(\delta)))) = \langle \cdot \rangle((x^k \cdot)(q(x \cdot \delta))) = \langle \cdot \rangle(x^k \cdot q(x \cdot \delta)) = |x^k \cdot q(\delta \cdot x)|^2 \quad (17.58)$$

Define

$$f_{x,2}: \mathbb{C} \rightarrow \mathbb{C} \text{ by } f_{x,2} = ((\cdot)^k) \cdot f_{x,1} \text{ so that } f_{x,2}(\delta) = (\cdot)^k(\delta) \cdot f_{x,1}(\delta) \stackrel{[\text{eq: 17.58}]}{=} \delta^2 \cdot |x^k \cdot q(\delta \cdot x)|^2 \quad (17.59)$$

which is continuous because it is the product of the continuous mappings $(\cdot)^k$ and $f_{x,1}$ [see theorem: 17.19(2,10)]. Define

$$f_{x,3}: \mathbb{C} \rightarrow \mathbb{R} \text{ by } f_{x,3} = \text{Re} \circ (x^k \cdot) \circ (\overline{r(0)} \cdot) \circ q \circ (x \cdot)$$

which is continuous as the composition of continuous functions [see theorem: 17.19(1,3,4,9)]. Further using the definition of $f_{x,3}$ we have

$$\begin{aligned} f_{x,3}(\delta) &= \text{Re}((x^k \cdot)(\overline{r(0)} \cdot)(q((x \cdot)(\delta)))) \\ &= \text{Re}((x^k \cdot)(\overline{r(0)} \cdot)(q(x \cdot \delta))) \\ &= \text{Re}((x^k \cdot)(\overline{r(0)} \cdot)(q(\delta \cdot x))) \\ &= \text{Re}((x^k \cdot)(\overline{r(0)} \cdot q(\delta \cdot x))) \\ &= \text{Re}((x^k \cdot)(q(\delta \cdot x) \cdot \overline{r(0)})) \\ &= \text{Re}(x^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \end{aligned} \quad (17.60)$$

Hence we have for $\delta \in \mathbb{R}$

$$\begin{aligned} ((f_{x,2})|_{\mathbb{R}} + 2 \cdot (f_{x,3})|_{\mathbb{R}})(\delta) &= (f_{x,2})|_{\mathbb{R}}(\delta) + 2 \cdot (f_{x,3})|_{\mathbb{R}}(\delta) \\ &= f_{x,2}(\delta) + 2 \cdot f_{x,3}(\delta) \\ &\stackrel{[\text{eqs: 17.59, 17.60}]}{=} \delta^k \cdot |x^k \cdot q(\delta \cdot x)|^2 + 2 \cdot \text{Re}(x^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \\ &\stackrel{[\text{eq: 17.57}]}{=} f_x(\delta) \end{aligned}$$

proving that

$$f_x = (f_{x,2})|_{\mathbb{R}} + (f_{x,3})|_{\mathbb{R}}$$

where by [theorem: 14.152] $(f_{x,2})|_{\mathbb{R}}$ and $(f_{x,3})|_{\mathbb{R}}$ are continuous. So f_x is the sum of two continuous function hence by [theorem: 14.166] it is continuous. To summarize

$$\forall x \in \mathbb{C} \ f_x: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } f(\delta) = \delta^k \cdot |x^k \cdot q(\delta \cdot x)|^2 + 2 \cdot \text{Re}(x^k \cdot q(\delta \cdot x) \cdot \overline{r(0)}) \text{ is continuous} \quad (17.61)$$

Combining this result with [eq: 17.56] results in

$$\forall x \in \mathbb{C} \text{ we have } \forall \delta \in \mathbb{R}^+ \text{ that } 0 \leq f_x(\delta) \quad (17.62)$$

What about $\delta = 0$? Assume that $f_x(0) < 0$ then $\varepsilon = -f_x(0) \in \mathbb{R}^+$ so by the continuity of f_x at 0 there exist a $\zeta \in \mathbb{R}^+$ such that $\forall \delta \in \mathbb{R}$ with $|\delta| = |\delta - 0| < \zeta$ $\|f_x(0) - f_x(\delta)\| < -f(0)$. So if we take $\delta = \frac{\zeta}{2}$ then we have

$$\left| f_x(0) - f_x\left(\frac{\zeta}{2}\right) \right| \stackrel{[\text{theorem: 17.18(1)}]}{=} \left\| f_x(0) - f_x\left(\frac{\zeta}{2}\right) \right\| < -f_x(0)$$

As $f_x(0) < 0 \leq f_x\left(\frac{\zeta}{2}\right)$ we have that $f_x\left(\frac{\zeta}{2}\right) - f_x(0) = \left| f_x(0) - f_x\left(\frac{\zeta}{2}\right) \right| < -f_x(0)$ so that $f_x\left(\frac{\zeta}{2}\right) < 0$ contradiction $0 \leq f_x\left(\frac{\zeta}{2}\right)$. Hence the assumption is wrong and we have for $x \in \mathbb{C}$ that

$$\begin{aligned} 0 &\leq f_x(0) \\ &= 0^2 \cdot |x^k \cdot q(0 \cdot x)|^2 + 2 \cdot \text{Re}(x^k \cdot q(0 \cdot x) \cdot \overline{r(0)}) \\ &= 2 \cdot \text{Re}(x^k \cdot q(0) \cdot \overline{r(0)}) \end{aligned}$$

or as $q(0) \cdot \overline{r(0)} \in \mathbb{C}$

$$0 \leq \text{Re}(x^k \cdot (a + i \cdot b)) \text{ where } q(0) \cdot \overline{r(0)} = a + i \cdot b \ \forall x \in \mathbb{C} \quad (17.63)$$

Now for $k \in \{1, \dots, n\}$ we have the following exclusive cases:

k is odd. Then $k = 2 \cdot m + 1$ [see theorem: 7.51]. Substituting $x = 1$ and $x = -1$ in [eq: 17.63] gives

$$0 \leq \operatorname{Re}(1^k \cdot (a + i \cdot b)) \stackrel{[\text{theorem: 10.55}]}{=} \operatorname{Re}(a + i \cdot b) = a$$

$$0 \leq \operatorname{Re}((-1)^k \cdot (a + i \cdot b)) = \operatorname{Re}((-1)^{2 \cdot m + 1} \cdot (a + i \cdot b)) \stackrel{[\text{theorem: 10.55}]}{=} \operatorname{Re}(-a - i \cdot b) = -a$$

hence $0 \leq a \wedge a \leq 0$ so that

$$a = 0$$

Now for m we have either:

m is even. Then $m = 2 \cdot l$ so that $k = 4 \cdot l + 1$. Substitute $x = i$ and $x = -i$ in [eq: 17.63] and take in account that $a = 0$ gives

$$0 \leq \operatorname{Re}(i^k \cdot i \cdot b) = \operatorname{Re}(i^{4 \cdot l + 1} \cdot i \cdot b) \stackrel{[\text{theorem: 10.81}]}{=} \operatorname{Re}(i \cdot i \cdot b) = \operatorname{Re}(-b) = -b$$

and

$$0 \leq \operatorname{Re}((-i)^k \cdot i \cdot b) = \operatorname{Re}((-1)^{2 \cdot m + 1} \cdot i^{4 \cdot l + 1} \cdot i \cdot b) \stackrel{[\text{theorem: 10.55, 10.81}]}{=} \operatorname{Re}(-i \cdot i \cdot b) = b$$

Hence $0 \leq b \wedge b \leq 0$ so that

$$b = 0$$

m is odd. Then $m = 2 \cdot l + 1$ so that $k = 2 \cdot (2 \cdot l + 1) + 1 = 4 \cdot l + 3$ hence if we substitute $x = i$ and $x = -i$ in [eq: 17.63] and take in account that $a = 0$ we get

$$0 \leq \operatorname{Re}(i^k \cdot i \cdot b) = \operatorname{Re}(i^{4 \cdot l + 3} \cdot i \cdot b) \stackrel{[\text{theorem: 10.81}]}{=} \operatorname{Re}((-i) \cdot i \cdot b) = \operatorname{Re}(b) = b$$

and

$$0 \leq \operatorname{Re}((-i)^k \cdot i \cdot b) = \operatorname{Re}((-1)^{2 \cdot m + 1} \cdot i^{4 \cdot l + 3} \cdot i \cdot b) \stackrel{[\text{theorem: 10.55, 10.81}]}{=} \operatorname{Re}((-1) \cdot (-i) \cdot i \cdot b) = \operatorname{Re}(-b) = -b$$

Hence $0 \leq b \wedge b \leq 0$ so that

$$b = 0$$

So we have in all cases that $a = b = 0$ so that $q(0) \cdot \overline{r(0)} = 0 + i \cdot 0 = 0$, as $q(0) \neq 0$ [see eq: 17.54] we conclude that

$$\overline{r(0)} = 0$$

k is even. Then $k = 2 \cdot m$ [see theorem: 7.51] and for m we have either:

m is odd. Then $m = 2 \cdot l + 1$ so that $k = 4 \cdot l + 2$. Substituting $x = i$ in [eq: 17.63] gives

$$\begin{aligned} 0 &\leq \operatorname{Re}(i^k \cdot (a + i \cdot b)) \\ &= \operatorname{Re}(i^{4 \cdot l + 2} \cdot (a + i \cdot b)) \\ &\stackrel{[\text{theorem: 10.81}]}{=} \operatorname{Re}(-(a + i \cdot b)) \\ &= \operatorname{Re}(-a - i \cdot b) \\ &= -a \end{aligned}$$

Substituting $x = 1$ in [eq: 17.63] gives

$$0 \leq \operatorname{Re}(1^k \cdot (a + i \cdot b)) = \operatorname{Re}(a + i \cdot b) = a$$

Hence $0 \leq a \wedge a \leq 0$ proving that

$$a = 0$$

Substituting $x = (1 + i)$ in [eq: 17.63] gives, using $a = 0$, that

$$\begin{aligned} 0 &\leq \operatorname{Re}((1 + i)^k \cdot (a + i \cdot b)) \\ &= \operatorname{Re}((1 + i)^{4 \cdot l + 2} \cdot i \cdot b) \\ &\stackrel{[\text{lemma: 17.20}]}{=} \operatorname{Re}(2 \cdot (-4)^l \cdot i \cdot i \cdot b) \\ &= \operatorname{Re}(-2 \cdot (-4)^l \cdot b) \\ &= -2 \cdot (-4)^l \cdot b \end{aligned}$$

and substituting $x = (1 - i)$ in [eq: 17.63] gives

$$\begin{aligned}
 0 &\leq \operatorname{Re}((1 - i)^k \cdot (a + i \cdot b)) \\
 &= \operatorname{Re}((1 - i)^{4 \cdot l + 2} \cdot i \cdot b) \\
 &\stackrel{[\text{lemma: 17.20}]}{=} \operatorname{Re}(-2 \cdot (-4)^l \cdot i \cdot i \cdot b) \\
 &= \operatorname{Re}(2 \cdot (-4)^l \cdot b) \\
 &= 2 \cdot (-4)^l \cdot b
 \end{aligned}$$

If now $(-4)^l \leq 0$ then we have $b \leq 0 \wedge 0 \leq b \Rightarrow b = 0$ and if $0 \leq (-4)^l$ then $0 \leq b \wedge b \leq 0 \Rightarrow b = 0$. So we have $a = b = 0$ proving that $q(0) \cdot \overline{r(0)} = 0 + i \cdot 0 = 0$. As $q(0) \neq 0$ [see eq: 17.54] we conclude that

$$\overline{r(0)} = 0$$

m is even. Then $m = 2 \cdot l$ so that $k = 2 \cdot m = 4 \cdot l$. Substituting $x = 1$ in [eq: 17.63] gives

$$0 \leq \operatorname{Re}(1^k \cdot (a + i \cdot b)) = \operatorname{Re}(a + i \cdot b) = a$$

proving that

$$0 \leq a \tag{17.64}$$

Take now $x = \left(1 + \frac{i}{k}\right)^2$ then as $k = 2 \cdot m$ we have by [lemma: 17.29] that

$$\operatorname{Re}(x^k) < 0 < \operatorname{Im}(x^k)$$

As $x^k \in \mathbb{C}$ we can write x^k as $x^k = c + i \cdot d$ where $c = \operatorname{Re}(x^k) \in \mathbb{R}$ and $d = \operatorname{Im}(x^k) \in \mathbb{R}$ so that

$$c < 0 < d \tag{17.65}$$

Substituting $x^k = c + i \cdot d$ in [eq: 17.63] gives

$$\begin{aligned}
 0 &\leq \operatorname{Re}(x^k \cdot (a + i \cdot b)) \\
 &= \operatorname{Re}((c + i \cdot d) \cdot (a + i \cdot b)) \\
 &= \operatorname{Re}(c \cdot a - b \cdot d + i \cdot (d \cdot a + c \cdot b)) \\
 &= c \cdot a - b \cdot d
 \end{aligned}$$

so that

$$0 \leq c \cdot a - b \cdot d \tag{17.66}$$

Take $y = \bar{x}$ then $y^k = (\bar{x})^k \stackrel{[\text{theorem: 10.83}]}{=} \overline{x^k}$ so that $y^k = c - i \cdot d$, substituting this in [eq: 17.63] gives

$$\begin{aligned}
 0 &\leq \operatorname{Re}(y^k \cdot (a + i \cdot b)) \\
 &= \operatorname{Re}((c - i \cdot d) \cdot (a + i \cdot b)) \\
 &= \operatorname{Re}(c \cdot a + b \cdot d + i \cdot (c \cdot b - d \cdot a)) \\
 &= c \cdot a + b \cdot d
 \end{aligned}$$

so that

$$0 \leq c \cdot a + b \cdot d \tag{17.67}$$

Adding [eq: 17.66] to [eq: 17.67] gives $0 \leq c \cdot a - b \cdot d + c \cdot a + b \cdot d = 2 \cdot c \cdot a$ so that $0 \leq c \cdot a$. As by [eq: 17.65] $c < 0$ it follows that $a \leq 0$ which combined with [eq: 17.64] proves that

$$a = 0$$

Substituting this in [eqs: 17.66 and 17.67] yields $0 \leq -b \cdot d \Rightarrow b \cdot d \leq 0$ and $0 \leq b \cdot d$ so that $b \cdot d = 0$, as by [eq: 17.65] $0 < d$ it follows that

$$b = 0$$

Hence $q(0) \cdot \overline{r(0)} = a + i \cdot b = 0$ and as $q(0) \neq 0$ [see eq: 17.54] it follows that

$$\overline{r(0)} = 0$$

So in all cases we have $\overline{r(0)} = 0$ so that $|r(0)|^2 = r(0) \cdot \overline{r(0)} = 0$ from which it follows that $r(0) = 0$. As by definition [see: 17.52] $r(0) = p(z_0)$ it follows that

$$p(z_0) = 0$$

and we have proved that there exist a root of the polynomial p . $\square$

Note 17.31. In the above proof we have used the extreme value theorem to find the existence of a root of p without actual providing a explicit expression to calculate the root of a polynomial. It is a well known fact that for a polynomial of a higher order then 4 there is no closed form general solution using radicals.

Theorem 17.32. Let $n \in \mathbb{N}$ and $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{R}$ with $a_n \neq 0$ then if we define

$$p: \mathbb{C} \rightarrow \mathbb{C} \text{ by } p(z) = \sum_{i=0}^n a_i \cdot z^i$$

[in other words p is a polynomial of order n with real coefficients] then for $z \in \mathbb{C}$ with $p(z) = 0$ we have that $p(\bar{z}) = 0$.

Proof. If $p(z) = 0$ then we have

$$\begin{aligned} 0 &= \overline{0} \\ &= \overline{p(z)} \\ &= \overline{\sum_{i=0}^n a_i \cdot z^i} \\ &\stackrel{[\text{theorem: 10.83}]}{=} \sum_{i=0}^n \overline{a_i} \cdot (\bar{z})^i \\ &\stackrel{[\text{theorem: 10.83}] \wedge a_i \in \mathbb{R}}{=} \sum_{i=0}^n a_i \cdot (\bar{z})^i \\ &= p(\bar{z}) \end{aligned}$$

$\square$

Theorem 17.33. Let $n \in \mathbb{N}$ and $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ with $a_n \neq 0$ then if we define

$$p: \mathbb{C} \rightarrow \mathbb{C} \text{ by } p(z) = \sum_{i=0}^n a_i \cdot z^i$$

[in other words p is a polynomial of order n] and $z_0 \in \mathbb{C}$ such that $p(z_0) = 0$ then there exist a $m \in \{1, \dots, n\}$ and a polynomial q with $\text{ord}(q) = n - m$ such that $\forall z \in \mathbb{C}$ we have $p(z) = (z - z_0)^m \cdot q(z)$ and $q(z_0) \neq 0$.

Proof. Define

$$r: \mathbb{C} \rightarrow \mathbb{C} \text{ by } r(z) = p(z + z_0) \tag{17.68}$$

then by [lemma: 17.25]

$$r \text{ is a polynomial with } \text{ord}(r) = \text{ord}(p) \in \mathbb{N} \tag{17.69}$$

Using [lemma: 17.26] there exist a $m \in \{1, \dots, n\}$ and a polynomial s such

$$s(0) \neq 0 \text{ ord}(s) = n - m \text{ and } \forall z \in \mathbb{C} \text{ we have } r(z) = r(0) + z^m \cdot s(z) \tag{17.70}$$

Then we have for $z \in \mathbb{C}$

$$\begin{aligned} p(z) &= p((z - z_0) + z_0) \\ &= r(z - z_0) \\ &\stackrel{[\text{eq: 17.70}]}{=} r(0) + (z - z_0)^m \cdot s(z - z_0) \\ &\stackrel{[\text{eq: 17.68}]}{=} p(0 + z_0) + (z - z_0)^m \cdot s(z - z_0) \\ &= p(z_0) + (z - z_0)^m \cdot s(z - z_0) \\ &= (z - z_0)^m \cdot s(z - z_0) \end{aligned} \tag{17.71}$$

Define

$$q: \mathbb{C} \rightarrow \mathbb{C} \text{ by } q(z) = s(z - z_0) = s(z + (-z_0)) \quad (17.72)$$

then by [lemma: 17.25] we have that

$$q \text{ is a polynomial and } \text{ord}(q) = \text{ord}(s) = n - m$$

Substituting this in [eq: 17.71] gives

$$p(z) = (z - z_0)^m \cdot q(z)$$

Further

$$q(z_0) = s(z_0 - z_0) = s(0) \neq 0 \quad \square$$

Theorem 17.34. *Let p be polynomial with $\text{ord}(p) \in \mathbb{N}$ then there exists a $c \in \mathbb{C} \setminus \{0\}$ and a $\{z_i\}_{i \in \{1, \dots, \text{ord}(p)\}} \subseteq \mathbb{C}$ such that*

$$\forall z \in \mathbb{C} \text{ we have } p(z) = c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i)$$

Further more

$$\{z \in \mathbb{C} | p(z) = 0\} = \{z_i | i \in \{1, \dots, \text{ord}(p)\}\}$$

In other words the distinct number of zero points of a polynomial is less than the order of the polynomial.

Proof. We prove this by induction on the order of the polynomial. So let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } p \text{ is a polynomial with } 1 \leq \text{ord}(p) \leq n \text{ then there exists a } c \in \mathbb{C} \setminus \{0\} \text{ and } \{z_i\}_{i \in \text{ord}(p)} \text{ such that } \forall z \in \mathbb{C} \ p(z) = c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i) \right\}$$

then we have:

$1 \in \mathbb{N}$. Then $\text{ord}(p) = 1$ so $p: \mathbb{C} \rightarrow \mathbb{C}$ is defined by $p(z) = a_0 + a_1 \cdot z$ where $a_1 \neq 0$. Take $z_1 = -\frac{a_0}{a_1}$ and $c = a_1$ then $c \neq 0$ and $c \cdot \prod_{i=1}^1 (z - z_i) = c \cdot (z - z_1) = a_1 \cdot \left(z + \frac{a_0}{a_1}\right) = a_1 \cdot z + a_0 = p(z)$. So we have that $1 \in S$

$n \in \mathbb{N} \rightarrow n + 1 \in \mathbb{N}$. Let p be a polynomial with $1 \leq \text{ord}(p) \leq n + 1$ then we have for $\text{ord}(p)$ either:

$1 \leq \text{ord}(p) \leq n$. As $n \in S$ there exist a $c \in \mathbb{C} \setminus \{0\}$ and $\{z_i\}_{i \in \{1, \dots, \text{ord}(p)\}}$ with $p(z) = c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i)$ so that in this case $n + 1 \in S$.

$1 \leq \text{ord}(p) = n + 1$. Let p be a polynomial with $\text{ord}(p) = n + 1 \in \mathbb{N}$. Using the fundamental theorem of algebra there exist a $y_0 \in \mathbb{C}$ such that $p(y_0) = 0$. Using the previous theorem [theorem: 17.33] there exist a $m \in \{1, \dots, n + 1\}$ and a polynomial q of order $(n + 1) - m$ such that

$$p(z) = (z - y_0)^m \cdot q(z) \text{ and } q(y_0) \neq 0 \quad (17.73)$$

for m we have now either:

$m = n + 1$. Then $\text{ord}(q) = n + 1 - m = 0$ so that there exist a $c \in \mathbb{C} \setminus \{0\}$ with

$$q: \mathbb{C} \rightarrow \mathbb{C} \text{ defined by } q(z) = c$$

Hence if we define $\{z_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{C}$ by $z_i = y_0$ then

$$\forall z \in \mathbb{C} \text{ we have } p(z) = c \cdot (z - y_0)^{n+1} = c \cdot \prod_{i=1}^{n+1} (z - y_0) = c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i)$$

proving that $n + 1 \in S$ in this case.

$m < n + 1$. Then $\text{ord}(q) = (n + 1) - m > 0$ so that

$$1 \leq \text{ord}(q) \leq n, \quad 1 \leq m \Rightarrow \text{ord}(q) = (n+1) - m \leq n$$

As $n \in S$ there exist a $c \in \mathbb{C} \setminus \{0\}$ and a $\{y_i\}_{i \in \{1, \dots, \text{ord}(q)\}} \subseteq \mathbb{C}$ such that $\forall z \in \mathbb{C}$ we have

$$q(z) = c \cdot \prod_{i=1}^{\text{ord}(q)} (z - y_i) = c \cdot \prod_{i=1}^{(n+1)-m} (x - y_i) \quad (17.74)$$

Define now $\{z_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{C}$ by $z_i = \begin{cases} y_0 & \text{if } i \in \{1, \dots, m\} \\ z_{i-m} & \text{if } i \in \{m+1, \dots, n+1\} \end{cases}$ then we have

$$\begin{aligned} p(z) &\stackrel{[\text{eq: 17.73}]}{=} (z - y_0)^m \cdot q(z) \\ &= \left(\prod_{i=1}^m (z - y_0) \right) \cdot q(z) \\ &= \left(\prod_{i=1}^m (z - z_i) \right) \cdot q(z) \\ &\stackrel{[\text{eq: 17.74}]}{=} c \cdot \left(\prod_{i=1}^m (z - z_i) \right) \cdot \left(\prod_{i=1}^{(n+1)-m} (x - y_i) \right) \\ &= c \cdot \left(\prod_{i=1}^m (z - z_i) \right) \cdot \left(\prod_{i=m+1}^{n+1} (x - y_{i-m}) \right) \\ &= c \cdot \left(\prod_{i=1}^m (z - z_i) \right) \cdot \left(\prod_{i=m+1}^{n+1} (x - z_i) \right) \\ &= c \cdot \prod_{i=1}^{n+1} (z - z_i) \\ &= c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i) \end{aligned}$$

proving that in this case $n+1 \in S$.

Mathematical induction proves that $S = \mathbb{N}$. Hence if p is a polynomial with $\text{ord}(p) \in \mathbb{N}$ then as $1 \leq \text{ord}(p) \in \mathbb{N} = S$ it follows that

$$\exists c \in \mathbb{C} \setminus \{0\} \text{ and } \{z_i\}_{i \in \{1, \dots, \text{ord}(p)\}} \subseteq \mathbb{C} \text{ such that } p(z) = c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i)$$

Further if $z \in \{z_i | i \in \{1, \dots, \text{ord}(p)\}\}$ then there exist a $k \in \{1, \dots, \text{ord}(p)\}$ such that $z = z_k$. Hence

$$\begin{aligned} p(z) &= p(z_k) \\ &= c \cdot \prod_{i=1}^{\text{ord}(p)} (z_k - z_i) \\ &= c \cdot \prod_{i \in \{1, \dots, \text{ord}(p)\}} (z_k - z_i) \\ &= c \cdot \prod_{i \in \{k\}} (z_k - z_i) \cdot \prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k\}} (z_k - z_i) \\ &= c \cdot (z_k - z_k) \cdot \prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{i_0\}} (z_k - z_i) \\ &= c \cdot 0 \cdot \prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{i_0\}} (z_k - z_i) \\ &= 0 \end{aligned}$$

so that $z \in \{z \in \mathbb{C} | p(z) = 0\}$. Hence

$$\{z_i | i \in \{1, \dots, \text{ord}(p)\}\} \subseteq \{z \in \mathbb{C} | p(z) = 0\} \quad (17.75)$$

Further if $z \in \{z \in S | p(z) = 0\}$ then $0 = c \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i)$, hence as $c \neq 0$, we have that

$$0 = \prod_{i=1}^{\text{ord}(p)} (z - z_i)$$

Assume that $\forall i \in \{1, \dots, \text{ord}(p)\} z_i \neq z$ then $\forall i \in \{1, \dots, \text{ord}(p)\} z - z_i \neq 0$, hence using [theorem: 11.53] $\prod_{i=1}^{\text{ord}(p)} (z - z_i) \neq 0$ contradicting $0 = \prod_{i=1}^{\text{ord}(p)} (z - z_i)$. So there must exist a $k \in \{1, \dots, \text{ord}(p)\}$ such that $z = z_k$. So $\{z \in S | p(z) = 0\} \subseteq \{z_i | i \in \{1, \dots, \text{ord}(p)\}\}$ which combined with [eq: 17.75] proves that

$$\{z_i | i \in \{1, \dots, \text{ord}(p)\}\} = \{z \in \mathbb{C} | p(z) = 0\} \quad \square$$

Lemma 17.35. *Let p be a polynomial then*

$$p \text{ has real coefficients} \Leftrightarrow \forall x \in \mathbb{R} \ p(x) \in \mathbb{R}$$

Proof.

$\Rightarrow$. If p has real coefficients then there exist a $\{a_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}$ with $a_n \neq 0$ such that $\forall z \in \mathbb{C}$ $p(z) = \sum_{i=0}^n a_i \cdot z^i$. Let $x \in \mathbb{R}$ then we have

$$\begin{aligned} \text{img}(p(x)) &= \text{Img}\left(\sum_{i=0}^n a_i \cdot x^i\right) \\ &\stackrel{[\text{example: 11.173}]}{=} \sum_{i=0}^n \text{Img}(a_i \cdot x^i) \\ &\stackrel{[\text{theorem: 10.80}] \wedge x^i \in \mathbb{R}}{=} \sum_{i=0}^n \text{Img}(a_i) \cdot x^i \\ &\stackrel{a_i \in \mathbb{R}}{=} \sum_{i=0}^n 0 \cdot x^i \\ &= 0 \end{aligned}$$

so that $p(x) \in \mathbb{R}$.

$\Leftarrow$. As p is a polynomial there exist a $\{a_i\}_{i \in \mathbb{N}} \subseteq \mathbb{C}$ with $a_n \neq 0$ such that $\forall z \in \mathbb{C}$ $p(z) = \sum_{i=0}^n a_i \cdot z^i$. Let $x \in \mathbb{R}$ then as $p(x) \in \mathbb{R}$ we have

$$\begin{aligned} 0 &= \text{Img}(p(x)) \\ &= \text{Img}\left(\sum_{i=0}^n a_i \cdot x^i\right) \\ &\stackrel{[\text{example: 11.173}]}{=} \sum_{i=0}^n \text{Img}(a_i \cdot x^i) \\ &\stackrel{[\text{theorem: 10.80}] \wedge x^i \in \mathbb{R}}{=} \sum_{i=0}^n \text{Img}(a_i) \cdot x^i \end{aligned}$$

so that $\forall x \in \mathbb{R}$ we have $\sum_{i=0}^n \text{Img}(a_i) \cdot x^i = 0$. As $\text{Img}(a_i) \in \mathbb{R}$ it follows from [theorem: 17.2] that $\forall i \in \{0, \dots, n\}$ we have $\text{Img}(a_i) = 0$ so that $a_i \in \mathbb{R}$. $\square$

Theorem 17.36. *Let p be a polynomial with $\text{ord}(p) \in \mathbb{N}$ and **real coefficients** then there exists a $c \in \mathbb{R} \setminus \{0\}$, $m, M \in \mathbb{N}_0$, $\{\lambda_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$, $\{b_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R}$ and $\{c_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R}$ such that $\forall i \in \{1, \dots, m\}$ we have*

$$(b_i)^2 < 4 \cdot c_i$$

and $\forall z \in \mathbb{C}$ we have

$$p(z) = c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) \right)$$

Note 17.37. If $m = 0$ then $\{1, \dots, m\} = \emptyset$ so that $\{\lambda_i\}_{i \in \emptyset}$ is the empty family [see example: 2.113] and $\prod_{i \in \emptyset} (z - \lambda_i) = 1$ [see definition: 11.32]. Likewise if $M = 0$ then $\{b_i\}_{i \in \{1, \dots, M\}}$ and $\{c_i\}_{i \in \{1, \dots, M\}}$ are empty families and $\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) = 1$

Proof. We prove this by induction so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } p \text{ is a polynomial with real coefficients and } 1 \leq \text{ord}(p) \leq n \text{ then there exists a } c \in \mathbb{R} \setminus \{0\}, m, M \in \mathbb{N}_0, \{\lambda_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}, \{b_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R} \text{ and } \{c_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R} \text{ such that } \forall i \in \{1, \dots, M\} \text{ we have } (b_i)^2 < 4 \cdot c_i \text{ and } \forall z \in \mathbb{C} \text{ that } p(z) = c \cdot \left(\prod_{i \in \{1, \dots, M\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) \right) \right\}$$

then we have:

1 $\in S$. If $1 \leq \text{ord}(p) \leq 1$ then $\text{ord}(p) = 1$ and there exist $a_0, a_1 \in \mathbb{R}$ such that $a_1 \neq 0$ and $p(z) = a_0 + a_1 \cdot z$. Take $c = a_1 \in \mathbb{R} \setminus \{0\}$, $m = 1$, $M = 0$, $\{\lambda_i\}_{i \in \{1, \dots, m\}}$ defined by $\lambda_1 = -\frac{a_0}{a_1} \in \mathbb{R}$, $\{b_i\}_{i \in \emptyset} = \{c_i\}_{i \in \emptyset} = \emptyset$ then we have

$$c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, 0\} = \emptyset} (z^2 + b_i \cdot z + c_i) \right) = a_1 \cdot \left(z + \frac{a_0}{a_1} \right) = a_0 + a_1 \cdot z = p(z)$$

Hence we must have that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let p be a polynomial with real coefficients and $1 \leq \text{ord}(p) \leq n + 1$. For $\text{ord}(p)$ we have either:

$\text{ord}(p) < n + 1$. Then $\text{ord}(p) \leq n$ so as $n \in S$ we have that there exists a $c \in \mathbb{R} \setminus \{0\}$, $m, M \in \mathbb{N}_0$, $\{\lambda_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$, $\{b_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R}$ and $\{c_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R}$ such that $\forall z \in \mathbb{C}$ we have $p(z) = c \cdot \left(\prod_{i \in \{1, \dots, M\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) \right)$. This proves that $n + 1 \in S$ in this case.

$\text{ord}(p) = n + 1$. As $1 \leq \text{ord}(p)$ $\text{ord}(p) \in \mathbb{N}$ so that by the previous theorem [theorem: 17.34] there exist $\{z_i\}_{i \in \{1, \dots, \text{ord}(p)\}} \subseteq \mathbb{C}$ and a $c' \in \mathbb{C} \setminus \{0\}$ such that

$$\forall z \in \mathbb{C} \text{ we have } p(z) = c' \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i) = c' \cdot \prod_{i \in \{1, \dots, \text{ord}(p)\}} (z - z_i) \quad (17.76)$$

and

$$\{z_i \mid i \in \{1, \dots, \text{ord}(p)\}\} = \{z \mid p(z) = 0\} \quad (17.77)$$

If $\{z_i\}_{i \in \{1, \dots, \text{ord}(p)\}} \subseteq \mathbb{R}$ then by taking $m = \text{ord}(p)$, $M = 0$ and $\{b_i\}_{i \in \emptyset} = \{c_i\}_{i \in \emptyset} = \emptyset$ we have that

$$p(z) = c' \cdot \left(\prod_{i=1}^{\text{ord}(p)} (z - z_i) \right) \cdot \left(\prod_{i \in \{1, \dots, 0\} = \emptyset} (z^2 + b_i \cdot z + c_i) \right) = c' \cdot \prod_{i=1}^{\text{ord}(p)} (z - z_i) \quad (17.78)$$

Take $x_0 = \max(\{z_i \mid i \in \{1, \dots, \text{ord}(p)\}\}) + 1 \in \mathbb{R}$ then $\forall i \in \{1, \dots, \text{ord}(p)\}$ $x_0 \neq z_i$ or $(x - x_i) \in \mathbb{R} \setminus \{0\}$ proving that $\prod_{i=1}^{\text{ord}(p)} (z - z_i) \in \mathbb{R} \setminus \{0\}$ hence $c' = \frac{p(x_0)}{\prod_{i=1}^{\text{ord}(p)} (x_0 - z_i)}$. As by [theorem: 17.35] $p(x_0) \in \mathbb{R}$ it follows that

$$c' \in \mathbb{R} \setminus \{0\}$$

Hence we have proved that $n + 1 \in S$ in the case where $\{z_i\}_{i \in \{1, \dots, \text{ord}(p)\}} \subseteq \mathbb{R}$.

So we must only prove the case where there exist a $k \in \{1, \dots, \text{ord}(p)\} = \{1, \dots, n+1\}$ such that $z_k \notin \mathbb{R}$. Now as p has real coefficients it follows from [lemma: 17.32] that $p(\overline{z_k}) = 0$, hence by [eq: 17.77] there exist a $l \in \{1, \dots, n+1\}$ such that $z_l = \overline{z_k}$. If $k = l$ then we would have that $z_k = \overline{z_k}$ so that by [theorem: 10.83] $z_k \in \mathbb{R}$ contradicting $z_k \notin \mathbb{R}$. Hence we must have that $k \neq l$ and $z_l = \overline{z_k}$. Next we have

$$\begin{aligned}
 p(z) &\stackrel{[\text{eq: 17.78}]}{=} c' \cdot \prod_{i \in \{1, \dots, \text{ord}(p)\}} (z - z_i) \\
 &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot \left(\prod_{i \in \{k, l\}} (z - z_i) \right) \\
 &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot (z - z_k) \cdot (z - z_l) \\
 &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot (z - z_k) \cdot (z - \overline{z_k}) \\
 &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot (z^2 - z \cdot \overline{z_k} - z_k \cdot z + z_k \cdot \overline{z_k}) \\
 &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot (z^2 - (z_k + \overline{z_k}) \cdot z + z_k \cdot \overline{z_k}) \\
 &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot (z^2 + d \cdot z + e)
 \end{aligned}$$

where $d = -(z_k + \overline{z_k}) = -2 \cdot \text{Re}(z_k) \in \mathbb{R}$ and $e = z_k \cdot \overline{z_k} = |z_k|^2 \in \mathbb{R}$. Hence we have

$$\begin{aligned}
 p(z) &= c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}} (z - z_i) \right) \cdot (z^2 + d \cdot z + e) \text{ where } d = -2 \cdot \text{Re}(z_k) \in \mathbb{R} \wedge \\
 e &= |z_k|^2 \in \mathbb{R}
 \end{aligned} \tag{17.79}$$

Now as $z_k \in \mathbb{C} \setminus \mathbb{R}$ we have that $0 \neq \text{img}(z_k)$ so that

$$d^2 = 4 \cdot (\text{Re}(z_k))^2 < 4 \cdot (\text{Re}(z_k))^2 + 4 \cdot (\text{Im}(z_k))^2 = 4 \cdot |z_k|^2 = 4 \cdot e$$

hence

$$d^2 < 4 \cdot e \tag{17.80}$$

As $k \neq l$ and $k, l \in \{1, \dots, \text{ord}(p)\}$ we have by [theorem: 10.92] that $\text{card}(\{1, \dots, \text{ord}(p)\} \setminus \{k, l\}) = \text{ord}(p) - 2$. Hence there exist a bijection

$$\beta: \{1, \dots, \text{ord}(p) - 2\} \rightarrow \{1, \dots, \text{ord}(p)\} \setminus \{k, l\}$$

so that by applying [theorem: 11.36] on [eq: 17.79] gives

$$p(z) = c' \cdot \left(\prod_{i \in \{1, \dots, \text{ord}(p) - 2\}} (z - z_{\beta(i)}) \right) \cdot (z^2 + d \cdot z + e) \tag{17.81}$$

For $i \in \{1, \dots, \text{ord}(p) - 2\}$ we have that

$p_i: \mathbb{C} \rightarrow \mathbb{C}$ defined by $p_{\beta(i)}(z) = z - z_{\beta(i)} = -z_{\beta(i)} + z$ is a polynomial with $\text{ord}(p_{\beta(i)}) = 1$

Define now

$$q: \mathbb{C} \rightarrow \mathbb{C} \text{ by } q(z) = c' \cdot \prod_{i \in \{1, \dots, \text{ord}(p) - 2\}} p_i(z) = c' \cdot \prod_{i=1}^{\text{ord}(p)-2} p_i(x) = c' \cdot \prod_{i=1}^{\text{ord}(p)-2} (z - z_{\beta(i)})$$

so that by [eq: 17.81] we have

$$p(z) = q(z) \cdot (z^2 + d \cdot z + e) \quad (17.82)$$

Using [theorem: 17.13] and [lemma: 17.10] it follows that

$$q \text{ is a polynomial with } \text{ord}(q) = \text{ord}(p) - 2 = n - 1 \quad (17.83)$$

Let $x \in \mathbb{R}$ then by [eq: 17.80] $0 < e - \frac{d^2}{4} \leq \frac{\left(x + \frac{d}{2}\right)^2}{0 \leq \left(x + \frac{d}{2}\right)^2} + e - \frac{d^2}{4}$ and as

$$\left(x + \frac{d}{2}\right)^2 + e - \frac{d^2}{4} = x^2 + x \cdot d + \frac{d^2}{4} + e - \frac{d^2}{4} = x^2 + d \cdot x + e$$

we have that $0 < x^2 + d \cdot x + e \in \mathbb{R}$ which, as $p(x) \in \mathbb{R}$, proves that $q(x) = \frac{p(x)}{x^2 + d \cdot x + e} \in \mathbb{R}$. Using [theorem: 17.35] it follows that q has real coefficients, hence as $\text{ord}(q) = n - 1 < n$ and $n \in S$ there exist $c \in \mathbb{R} \setminus \{0\}$, $m, M' \in \mathbb{N}_0$, $\{\lambda_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$, $\{b'_i\}_{i \in \{1, \dots, M'\}} \subseteq \mathbb{R}$ and $\{c'_i\}_{i \in \{1, \dots, M'\}} \subseteq \mathbb{R}$ such that $\forall i \in \{1, \dots, M'\} \ (b'_i)^2 < 4 \cdot c'_i$ and

$$\forall z \in \mathbb{C} \quad q(z) = c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M'\}} (z^2 + b'_i \cdot z + c'_i) \right) \quad (17.84)$$

Hence if we take $M = M' + 1$, $\{c_i\}_{i \in M} \subseteq \mathbb{R}$ by $b_i = \begin{cases} d & \text{if } i = M = M' + 1 \\ b'_i & \text{if } i \in \{1, \dots, M'\} \end{cases}$ and $\{c_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R}$ by $c_i = \begin{cases} e & \text{if } i = M = M' + 1 \\ c'_i & \text{if } i \in \{1, \dots, M'\} \end{cases}$ then we have $\forall i \in \{1, \dots, M\}$ that $b_i^2 < 4 \cdot c_i$. Further

$$\begin{aligned} & c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) \right) = \\ & c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M'\}} (z^2 + b_i \cdot z + c_i) \right) \cdot (z^2 + b_M \cdot z + c_M) = \\ & c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M'\}} (z^2 + b'_i \cdot z + c'_i) \right) \cdot (z^2 + d \cdot z + e) \stackrel{[\text{eq: 17.84}]}{=} \\ & \qquad \qquad \qquad q(z) \cdot (z^2 + d \cdot z + e) \stackrel{[\text{eq: 17.82}]}{=} \\ & \qquad \qquad \qquad p(z) \end{aligned}$$

Hence we have that

$$p(z) = c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) \right)$$

so that we have $n + 1 \in S$ in this case. □

Chapter 18

Inner Product Spaces

18.1 Definition and properties

Inner products extends the notion of length and orthogonality of vectors in Euclidean space.

Definition 18.1. A *real inner product space* is a pair $\langle X, \langle \rangle \rangle$ where X is a vector space over the field $\mathbb{R}$ together with a mapping [the *inner product*]

$$\langle \rangle: X \times X \rightarrow \mathbb{R} \text{ where } \langle \rangle(x, y) \text{ is noted as } \langle x, y \rangle$$

satisfying:

1. $\forall x, y \in X$ we have $\langle x, y \rangle = \langle y, x \rangle$
2. $\forall x, y, z \in X$ and $\forall \alpha \in \mathbb{R}$ we have $\langle x + \alpha \cdot y, z \rangle = \langle x, z \rangle + \alpha \cdot \langle y, z \rangle$
3. $\forall x \in X$ we have $0 \leq \langle x, x \rangle$
4. $\forall x \in X$ we have $\langle x, x \rangle = 0 \Rightarrow x = 0$

Theorem 18.2. Let $\langle X, \langle \rangle \rangle$ be a real inner product space and $Y \subseteq X$ a subspace of X then $\langle Y, \langle \rangle|_{Y \times Y} \rangle$ is a real inner product space.

Proof.

1. $\forall x, y \in Y$ we have $\langle x, y \rangle|_{Y \times Y} = \langle x, y \rangle = \langle y, x \rangle = \langle y, x \rangle|_{Y \times Y}$.
2. $\forall x, y, z \in Y$ and $\forall \alpha \in \mathbb{R}$ we have
$$\langle x + \alpha \cdot y, z \rangle|_{Y \times Y} = \langle x + \alpha \cdot y, z \rangle = \langle x, z \rangle + \alpha \cdot \langle y, z \rangle = \langle x, z \rangle|_{Y \times Y} + \alpha \cdot \langle y, z \rangle|_{Y \times Y}.$$
3. $\forall x \in Y$ we have $\langle x, x \rangle|_{Y \times Y} = \langle x, x \rangle \geq 0$
4. $\forall x \in Y$ we have $\langle x, x \rangle|_{Y \times Y} = \langle x, x \rangle = 0 \Rightarrow x = 0$ □

The inner product has the following properties.

Theorem 18.3. Let $\langle X, \langle \rangle \rangle$ be a real inner product space then we have

1. $\forall x, y, z \in X$ we have:
 - a. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
 - b. $\langle z, x + y \rangle = \langle z, x \rangle + \langle z, y \rangle$
2. $\forall x \in X$ we have $\langle x, 0 \rangle = \langle 0, x \rangle = 0$
3. Let $y, z \in X$ then if $\forall x \in X$ we have $\langle x, y \rangle = \langle x, z \rangle$ then $y = z$
4. Let $y, z \in X$ then if $\forall x \in X$ we have $\langle y, x \rangle = \langle z, x \rangle$ then $y = z$
5. $\forall x, y \in X$ and $\alpha \in \mathbb{R}$ then
 - a. $\langle \alpha \cdot x, y \rangle = \alpha \cdot \langle x, y \rangle$
 - b. $\langle x, \alpha \cdot y \rangle = \alpha \cdot \langle x, y \rangle$

Proof.

1. Given x, y, z we have:

$$\text{a. } \langle x + y, z \rangle = \langle x + 1 \cdot y, z \rangle = \langle x, z \rangle + 1 \cdot \langle y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$$

$$\text{b. } \langle z, x + y \rangle = \langle x + y, z \rangle \stackrel{(1.a)}{=} \langle x, z \rangle + \langle y, z \rangle = \langle z, x \rangle + \langle z, y \rangle$$

2. $\forall x \in X$ we have $\langle x, 0 \rangle = \langle 0, x \rangle = \langle 0 \cdot 0, x \rangle = 0 \cdot \langle 0, x \rangle = 0$

3. We have

$$\begin{aligned} \langle z - y, y \rangle &= \langle z - y, z \rangle \\ &= \langle z - y, (z - y) + y \rangle \\ &= \langle z - y, z - y \rangle + \langle z - y, y \rangle \end{aligned}$$

so that $\langle z - y, z - y \rangle = 0$. Hence $z - y = 0$ or $z = y$.

4. Let $z, y \in X$ and assume that $\forall x \in X$ we have $\langle y, x \rangle = \langle z, x \rangle$ then we have

$$\langle x, y \rangle = \langle y, x \rangle = \langle z, x \rangle = \langle x, z \rangle$$

so that by (3) we have $z = y$.

5. Given $x, y \in X$ and $\alpha \in \mathbb{R}$ we have:

$$\text{a. } \langle \alpha \cdot x, y \rangle = \langle 0 + \alpha \cdot x, y \rangle = \langle 0, y \rangle + \alpha \cdot \langle x, y \rangle \stackrel{(2)}{=} \alpha \cdot \langle x, y \rangle$$

$$\text{b. } \langle x, \alpha \cdot y \rangle = \langle \alpha \cdot y, x \rangle = \alpha \cdot \langle y, x \rangle = \alpha \cdot \langle x, y \rangle$$

□

Example 18.4. Let $n \in \mathbb{N}$ and define

$$\langle \rangle: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \text{ by } \langle x, y \rangle = \sum_{i=1}^n x_i \cdot y_i$$

then

$\langle \mathbb{R}, \langle \rangle \rangle$ is a real inner product space

Proof.

1. If $x, y \in \mathbb{R}^n$ then $\langle x, y \rangle = \sum_{i=1}^n x_i \cdot y_i = \sum_{i=1}^n y_i \cdot x_i = \langle y, x \rangle$

2. If $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ then we have

$$\begin{aligned} \langle x + \alpha \cdot y, z \rangle &= \sum_{i=1}^n (x_i + \alpha \cdot y_i) \cdot z_i \\ &= \sum_{i=1}^n (x_i \cdot z_i + \alpha \cdot y_i \cdot z_i) \\ &= \sum_{i=1}^n x_i \cdot z_i + \alpha \cdot \sum_{i=1}^n y_i \cdot z_i \\ &= \langle x, z \rangle + \alpha \cdot \langle y, z \rangle \end{aligned}$$

3. If $x \in X$ then we have trivially $0 \leq \sum_{i=1}^n x_i^2 = \langle x, x \rangle$.

4. Let $x \in X$ with $\langle x, x \rangle = 0$. Assume that $x \neq 0$ then there exist a $k \in \{1, \dots, n\}$ such that $x_k \neq 0$. Hence $0 < x_k^2 \leq \sum_{i=1}^n x_i^2 = \langle x, x \rangle = 0$ which leads to the contradiction $0 < 0$ so that we must have $x = 0$. □

Definition 18.5. A **complex inner product space** is a pair $\langle X, \langle \rangle \rangle$ where X is a vector space over the field $\mathbb{C}$ together with a mapping [the **inner product**]

$$\langle \rangle: X \times X \rightarrow \mathbb{C} \text{ where } \langle \rangle(x, y) \text{ is noted as } \langle x, y \rangle$$

satisfying:

1. $\forall x, y \in X$ we have $\langle x, y \rangle = \overline{\langle y, x \rangle}$

2. $\forall x, y, z \in X$ and $\forall \alpha \in \mathbb{R}$ we have $\langle x + \alpha \cdot y, z \rangle = \langle x, z \rangle + \alpha \cdot \langle y, z \rangle$
3. $\forall x \in X$ we have $0 \leq \langle x, x \rangle$
4. $\forall x \in X$ we have $\langle x, x \rangle = 0 \Rightarrow x = 0$

Note 18.6. As $\langle x, x \rangle = \overline{\langle x, x \rangle}$ it follows from [theorem: 10.83] that $\langle x, x \rangle \in \mathbb{R}$ so that (3) makes sense.

Note 18.7. In physics definition (2) is changed to

$$\forall x, y, z \in X \text{ and } \forall \alpha \in \mathbb{C} \text{ we have } \langle x, y + \alpha \cdot z \rangle = \langle x, y \rangle + \alpha \cdot \langle y, z \rangle$$

we use the mathematical convention here.

Theorem 18.8. Let $\langle X, \langle \rangle \rangle$ be a real inner product space and $Y \subseteq X$ a subspace of X then $\langle Y, \langle \rangle|_{Y \times Y} \rangle$ is a real inner product space.

Proof.

1. $\forall x, y \in Y$ we have $\langle x, y \rangle|_{Y \times Y} = \langle x, y \rangle = \langle y, x \rangle = \langle y, x \rangle|_{Y \times Y}$.
2. $\forall x, y, z \in Y$ and $\forall \alpha \in \mathbb{R}$ we have

$$\langle x + \alpha \cdot y, z \rangle|_{Y \times Y} = \langle x + \alpha \cdot y, z \rangle = \langle x, z \rangle + \alpha \cdot \langle y, z \rangle = \langle x, z \rangle|_{Y \times Y} + \alpha \cdot \langle y, z \rangle|_{Y \times Y}.$$

3. $\forall x \in Y$ we have $\langle x, x \rangle|_{Y \times Y} = \langle x, x \rangle \geq 0$
4. $\forall x \in Y$ we have $\langle x, x \rangle|_{Y \times Y} = \langle x, x \rangle = 0 \Rightarrow x = 0$ □

Example 18.9. Let $n \in \mathbb{N}$ and define

$$\langle \rangle: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C} \text{ by } \langle x, y \rangle = \sum_{i=1}^n x_i \cdot \overline{y_i}$$

then

$$\langle \mathbb{C}, \langle \rangle \rangle \text{ is a inner product space}$$

Proof.

1. If $x, y \in \mathbb{C}^n$ then

$$\begin{aligned} \langle x, y \rangle &= \sum_{i=1}^n x_i \cdot \overline{y_i} \\ &\stackrel{[\text{theorem: 10.83}]}{=} \sum_{i=1}^n \overline{\overline{x_i}} \cdot \overline{y_i} \\ &\stackrel{[\text{theorem: 10.83}]}{=} \sum_{i=1}^n \overline{\overline{x_i} \cdot y_i} \\ &\stackrel{[\text{theorem: 10.83}]}{=} \overline{\sum_{i=1}^n y_i \cdot \overline{x_i}} \\ &= \overline{\langle y, x \rangle} \end{aligned}$$

2. If $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ then we have

$$\begin{aligned} \langle x + \alpha \cdot y, z \rangle &= \sum_{i=1}^n (x_i + \alpha \cdot y_i) \cdot z_i \\ &= \sum_{i=1}^n (x_i \cdot z_i + \alpha \cdot y_i \cdot z_i) \\ &= \sum_{i=1}^n x_i \cdot z_i + \alpha \cdot \sum_{i=1}^n y_i \cdot z_i \\ &= \langle x, z \rangle + \alpha \cdot \langle y, z \rangle \end{aligned}$$

3. If $x \in X$ then we have trivially $0 \leq \sum_{i=1}^n |x_i|^2 \stackrel{[\text{theorem: 10.85}]}{=} \sum_{i=1}^n x_i \cdot \bar{x}_i = \langle x, x \rangle$.
4. Let $x \in X$ with $\langle x, x \rangle = 0$. Assume that $x \neq 0$ then there exist a $k \in \{1, \dots, n\}$ such that $x_k \neq 0$. Hence $0 < |x_k|^2 \leq \sum_{i=1}^n |x_i|^2 = \langle x, x \rangle = 0$ which leads to the contradiction $0 < 0$ so that we must have $x = 0$. $\square$

Theorem 18.10. Let $\langle X, \langle \rangle \rangle$ be a complex inner product space then we have

1. $\forall x \in X$ we have $\langle x, x \rangle \in \mathbb{R}$
2. $\forall x, y, z \in X$ we have:
 - a. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
 - b. $\langle z, x + y \rangle = \langle z, x \rangle + \langle z, y \rangle$
3. Let $y, z \in X$ then if $\forall x \in X$ we have $\langle x, y \rangle = \langle x, z \rangle$ then $y = z$
4. Let $y, z \in X$ then if $\forall x \in X$ we have $\langle y, x \rangle = \langle z, x \rangle$ then $y = z$
5. $\forall x \in X$ we have $\langle x, 0 \rangle = \langle 0, x \rangle = 0$
6. $\forall x, y \in X$ and $\alpha \in \mathbb{R}$ then
 - a. $\langle \alpha \cdot x, y \rangle = \alpha \cdot \langle x, y \rangle$
 - b. $\langle x, \alpha \cdot y \rangle = \bar{\alpha} \cdot \langle x, y \rangle$

Proof.

1. Let $x \in X$ then $\langle x, x \rangle = \overline{\langle x, x \rangle}$ so that by [theorem: 10.83] $\langle x, x \rangle \in \mathbb{R}$.
2. Let $x, y, z \in X$ then we have:
 - a. $\langle x + y, z \rangle = \langle x + 1 \cdot y, z \rangle = \langle x, z \rangle + 1 \cdot \langle y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
 - b. $\langle z, x + y \rangle = \overline{\langle x + y, z \rangle} = \overline{\langle x, z \rangle + \langle y, z \rangle} \stackrel{[\text{theorem: 10.83}]}{=} \overline{\langle x, z \rangle} + \overline{\langle y, z \rangle} = \langle z, x \rangle + \langle z, y \rangle$
3. We have

$$\begin{aligned}
 \langle z - y, y \rangle &= \langle z - y, z \rangle \\
 &= \langle z - y, (z - y) + y \rangle \\
 &= \langle z - y, z - y \rangle + \langle z - y, y \rangle
 \end{aligned}$$

so that $\langle z - y, z - y \rangle = 0$. Hence $z - y = 0$ or $z = y$.

4. Let $y, z \in X$ and assume that $\forall x \in X$ we have that $\langle y, x \rangle = \langle z, x \rangle$ then we have

$$\langle x, y \rangle = \overline{\langle y, x \rangle} = \overline{\langle z, x \rangle} = \langle x, z \rangle$$

so that by (3) $y = z$.

5. $\forall x \in X$ we have $\langle 0, x \rangle = \langle 0 \cdot 0, x \rangle = 0 \cdot \langle 0, x \rangle = 0$ and $\langle x, 0 \rangle = \overline{\langle 0, x \rangle} = \bar{0} = 0$
6. Let $x, y \in X$ and $\alpha \in \mathbb{C}$ then we have:

- a. $\langle \alpha \cdot x, y \rangle = \langle 0 + \alpha \cdot x, y \rangle = \langle 0, y \rangle + \alpha \cdot \langle x, y \rangle \stackrel{(2)}{=} \alpha \cdot \langle x, y \rangle$
- b. $\langle x, \alpha \cdot y \rangle = \overline{\langle \alpha \cdot y, x \rangle} = \overline{\alpha \cdot \langle y, x \rangle} \stackrel{[\text{theorem: 10.83}]}{=} \bar{\alpha} \cdot \overline{\langle y, x \rangle} = \bar{\alpha} \cdot \langle x, y \rangle$

$\square$

We can use mathematical induction to extend [theorems: 18.3, 18.10] to finite sums.

Theorem 18.11. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space, $n \in \mathbb{N}$, $y \in X$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq X$ then

$$\left\langle \sum_{i=1}^n x_i, y \right\rangle = \sum_{i=1}^n \langle x_i, y \rangle \quad \text{and} \quad \left\langle y, \sum_{i=1}^n x_i \right\rangle = \sum_{i=1}^n \langle y, x_i \rangle$$

Proof. The first part is proved by induction, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } y \in X, \{x_i\}_{i \in \{1, \dots, n\}} \text{ then } \left\langle \sum_{i=1}^n x_i, y \right\rangle = \sum_{i=1}^n \langle x_i, y \rangle \right\}$$

then we have:

$1 \in S$. If $y \in X$, $\{x_i\}_{i \in \{1\}} \subseteq X$ then $\langle \sum_{i=1}^1 x_i, y \rangle = \langle x_1, y \rangle = \sum_{i=1}^1 \langle x_i, y \rangle$, hence $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $y \in X$, $\{x_i\}_{i \in \{1, \dots, n+1\}} \subseteq X$ then we have

$$\begin{aligned} \left\langle \sum_{i=1}^{n+1} x_i, y \right\rangle &= \left\langle x_{n+1} + \sum_{i=1}^n x_i, y \right\rangle \\ &= \langle x_{n+1}, y \rangle + \left\langle \sum_{i=1}^n x_i, y \right\rangle \\ &\stackrel{n \in S}{=} \langle x_{n+1}, y \rangle + \sum_{i=1}^n \langle x_i, y \rangle \\ &= \sum_{i=1}^{n+1} \langle x_i, y \rangle \end{aligned}$$

which proves that $n+1 \in S$.

Hence by mathematical induction we have

$$\left\langle \sum_{i=1}^n x_i, y \right\rangle = \sum_{i=1}^n \langle x_i, y \rangle \quad (18.1)$$

For the second part we must make a distinction between real inner product spaces and complex inner product spaces.

real inner product space.

$$\left\langle y, \sum_{i=1}^n x_i \right\rangle = \left\langle \sum_{i=1}^n x_i, y \right\rangle \stackrel{[\text{eq: 18.1}]}{=} \sum_{i=1}^n \langle x_i, y \rangle = \sum_{i=1}^n \langle y, x_i \rangle$$

complex inner product space.

$$\left\langle y, \sum_{i=1}^n x_i \right\rangle = \overline{\left\langle \sum_{i=1}^n x_i, y \right\rangle} \stackrel{[\text{eq: 18.1}]}{=} \sum_{i=1}^n \overline{\langle x_i, y \rangle} \stackrel{[\text{theorem: 10.83}]}{=} \sum_{i=1}^n \overline{\langle x_i, y \rangle} = \sum_{i=1}^n \langle y, x_i \rangle \quad \square$$

Theorem 18.12. Let $\langle X, \langle \rangle \rangle$ be a **complex** inner product space, $L \in \text{Hom}(X)$ then we have

$$\begin{aligned} L &= 0 \\ \Updownarrow \\ \forall x \in X \text{ we have } \langle L(x), x \rangle &= 0 \end{aligned}$$

Proof.

$\Rightarrow$. If $L = 0$ then we have $\forall x \in X$ that $\langle L(x), x \rangle = \langle 0(x), x \rangle = \langle 0, x \rangle \stackrel{[\text{theorem: 18.10}]}{=} 0$.

$\Leftarrow$. Let $x, y \in X$ then we have:

$$\begin{aligned} \langle L(x+y), x+y \rangle - \langle L(x-y), x-y \rangle &= \\ \langle L(x) + L(y), x+y \rangle - \langle L(x) - L(y), x-y \rangle &= \\ \langle L(x), x \rangle + \langle L(x), y \rangle + \langle L(y), x+y \rangle - \langle L(x), x \rangle + \langle L(x), y \rangle - \langle L(y), x-y \rangle &= \\ 2 \cdot \langle L(x), y \rangle + \langle L(y), x+y \rangle - \langle L(y), x-y \rangle &= \\ 2 \cdot \langle L(x), y \rangle + \langle L(y), x \rangle + \langle L(y), y \rangle - \langle L(y), x \rangle + \langle L(y), y \rangle &= \\ 2 \cdot \langle L(x), y \rangle + 2 \cdot \langle L(y), y \rangle &= \end{aligned}$$

proving that

$$\langle L(x+y), x+y \rangle - \langle L(x-y), x-y \rangle = 2 \cdot \langle L(x), y \rangle + 2 \cdot \langle L(y), x \rangle$$

Further we have that

$$\begin{aligned} \langle L(x+i \cdot y), x+i \cdot y \rangle - \langle L(x-i \cdot y), x-i \cdot y \rangle &= \\ \langle L(x) + i \cdot L(y), x+i \cdot y \rangle - \langle L(x) - i \cdot L(y), x-i \cdot y \rangle &= \\ \langle L(x), x \rangle - i \cdot \langle L(x), y \rangle + \langle i \cdot L(y), x+i \cdot y \rangle - \langle L(x), x \rangle - \langle L(x), -i \cdot y \rangle + i \cdot \langle L(y), x-i \cdot y \rangle &= \\ -i \cdot \langle L(x), y \rangle + \langle i \cdot L(y), x+i \cdot y \rangle - \langle L(x), -i \cdot y \rangle + i \cdot \langle L(y), x-i \cdot y \rangle &= \\ -i \cdot \langle L(x), y \rangle + \langle i \cdot L(y), x+i \cdot y \rangle - i \cdot \langle L(x), y \rangle + i \cdot \langle L(y), x-i \cdot y \rangle &= \\ -2 \cdot i \cdot \langle L(x), y \rangle + \langle i \cdot L(y), x+i \cdot y \rangle + i \cdot \langle L(y), x-i \cdot y \rangle &= \\ -2 \cdot i \cdot \langle L(x), y \rangle + i \cdot \langle L(y), x \rangle + \langle L(y), y \rangle + i \cdot \langle L(y), x \rangle - \langle L(y), y \rangle &= \\ -2 \cdot i \cdot \langle L(x), y \rangle + 2 \cdot i \cdot \langle L(y), x \rangle & \end{aligned}$$

Hence we have that

$$\begin{aligned} \langle L(x+y), x+y \rangle - \langle L(x-y), x-y \rangle + i \cdot (\langle L(x+i \cdot y), x+i \cdot y \rangle - \langle L(x-i \cdot y), x-i \cdot y \rangle) &= \\ 2 \cdot \langle L(x), y \rangle + 2 \cdot \langle L(y), x \rangle + i \cdot (-2 \cdot i \cdot \langle L(x), y \rangle + 2 \cdot i \cdot \langle L(y), x \rangle) &= \\ 2 \cdot \langle L(x), y \rangle + 2 \cdot \langle L(y), x \rangle + 2 \cdot \langle L(x), y \rangle - 2 \cdot \langle L(y), x \rangle &= \\ 4 \cdot \langle L(x), y \rangle & \end{aligned}$$

As by the hypothesis $\langle L(x+y), x+y \rangle = 0$, $\langle L(x-y), x-y \rangle = 0$, $\langle L(x+i \cdot y), x+i \cdot y \rangle = 0$ and $\langle L(x-i \cdot y), x-i \cdot y \rangle = 0$ it follows that $\langle L(x), y \rangle = 0$. Take $y = L(x)$ then we have $\langle L(x), L(x) \rangle = 0$, hence we have $L(x) = 0$ for every $x \in X$ so that $L = 0$. $\square$

Note 18.13. The proof of the above theorem works only for the complex case, later we will prove that on a real finite dimensional inner product space the theorem is true for self-adjoint operators. [see theorem: 18.90].

For finite dimensional vector spaces over $\mathbb{R}$ [or $\mathbb{C}$] we can always construct a inner product making it in a real [or complex] inner product space.

Theorem 18.14. Let X be a finite dimensional vector space over the field $\mathbb{R}$ with a basis $\{e_1, \dots, e_n\}$ then if we define $\langle \rangle: X \times X \rightarrow \mathbb{R}$ by

$$\langle x, y \rangle = \sum_{i=1}^n x_i \cdot y_i$$

where $x = \sum_{i=1}^n x_i \cdot e_i$ and $y = \sum_{i=1}^n y_i \cdot e_i$ are the unique expansions of x, y in the basis $\{e_1, \dots, e_n\}$.

Proof.

1. If $x, y \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$ and $y = \sum_{i=1}^n y_i \cdot e_i$ then

$$\langle x, y \rangle = \sum_{i=1}^n x_i \cdot y_i = \sum_{i=1}^n y_i \cdot x_i = \langle y, x \rangle$$

2. If $a \in \mathbb{R}$, $x, y, z \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$, $y = \sum_{i=1}^n y_i \cdot e_i$ and $z = \sum_{i=1}^n z_i \cdot e_i$ then

$$x + \alpha \cdot y = \sum_{i=1}^n x_i \cdot e_i + \alpha \cdot \sum_{i=1}^n y_i \cdot e_i = \sum_{i=1}^n (x_i + \alpha \cdot y_i) \cdot e_i$$

so that

$$\begin{aligned} \langle x + \alpha \cdot y, z \rangle &= \sum_{i=1}^n (x_i + \alpha \cdot y_i) \cdot z_i \\ &= \sum_{i=1}^n (x_i \cdot z_i + \alpha \cdot y_i \cdot z_i) \\ &= \sum_{i=1}^n x_i \cdot z_i + \alpha \cdot \sum_{i=1}^n y_i \cdot z_i \\ &= \langle x, z \rangle + \alpha \cdot \langle y, z \rangle \end{aligned}$$

3. If $x \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$ then

$$0 \leq \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i \cdot x_i = \langle x, x \rangle$$

4. Let $x \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$ and $\langle x, x \rangle = 0$. Assume that $x \neq 0$ then there exist a $k \in \{1, \dots, n\}$ such that $x_k \neq 0$. Hence $0 < x_k^2 \leq \sum_{i=1}^n x_i^2 = \langle x, x \rangle = 0$ leading to the contradiction $0 < 0$, so we must have $x = 0$. $\square$

Theorem 18.15. Let X be a finite dimensional vector space over the field $\mathbb{C}$ with a basis $\{e_1, \dots, e_n\}$ then if we define $\langle \cdot \rangle: X \times X \rightarrow \mathbb{C}$ by

$$\langle x, y \rangle = \sum_{i=1}^n x_i \cdot \bar{y}_i$$

where $x = \sum_{i=1}^n x_i \cdot e_i$ and $y = \sum_{i=1}^n y_i \cdot e_i$ are the unique expansions of x, y in the basis $\{e_1, \dots, e_n\}$.

Proof.

1. If $x, y \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$ and $y = \sum_{i=1}^n y_i \cdot e_i$ then

$$\begin{aligned} \langle x, y \rangle &= \sum_{i=1}^n x_i \cdot \bar{y}_i \\ &\stackrel{[\text{theorem: 10.83}]}{=} \sum_{i=1}^n \bar{\bar{x}}_i \cdot \bar{y}_i \\ &\stackrel{[\text{theorem: 10.83}]}{=} \sum_{i=1}^n \overline{\bar{x}_i \cdot y_i} \\ &\stackrel{[\text{theorem: 10.83}]}{=} \overline{\sum_{i=1}^n y_i \cdot \bar{x}_i} \\ &= \overline{\langle y, x \rangle} \end{aligned}$$

2. If $a \in \mathbb{R}$, $x, y, z \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$, $y = \sum_{i=1}^n y_i \cdot e_i$ and $z = \sum_{i=1}^n z_i \cdot e_i$ then

$$x + \alpha \cdot y = \sum_{i=1}^n x_i \cdot e_i + \alpha \cdot \sum_{i=1}^n y_i \cdot e_i = \sum_{i=1}^n (x_i + \alpha \cdot y_i) \cdot e_i$$

so that

$$\begin{aligned} \langle x + \alpha \cdot y, z \rangle &= \sum_{i=1}^n (x_i + \alpha \cdot y_i) \cdot z_i \\ &= \sum_{i=1}^n (x_i \cdot z_i + \alpha \cdot y_i \cdot z_i) \\ &= \sum_{i=1}^n x_i \cdot z_i + \alpha \cdot \sum_{i=1}^n y_i \cdot z_i \\ &= \langle x, z \rangle + \alpha \cdot \langle y, z \rangle \end{aligned}$$

3. If $x \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$ then

$$0 \leq \sum_{i=1}^n |x_i|^2 = \sum_{i=1}^n x_i \cdot \bar{x}_i = \langle x, x \rangle$$

4. Let $x \in X$ with $x = \sum_{i=1}^n x_i \cdot e_i$ and $\langle x, x \rangle = 0$. Assume that $x \neq 0$ then there exist a $k \in \{1, \dots, n\}$ such that $x_k \neq 0$. Hence $0 < |x_k|^2 \leq \sum_{i=1}^n |x_i|^2 = \sum_{i=1}^n x_i \cdot \bar{x}_i = \langle x, x \rangle = 0$ leading to the contradiction $0 < 0$, so we must have $x = 0$. $\square$

18.2 Inner product norm

Given a inner product we can use the inner product to define a norm, the inner product norm.

Definition 18.16. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space then the **inner product norm** is defined by

$$\| \cdot \|: X \rightarrow \mathbb{R} \text{ where } \|x\| = \sqrt{\langle x, x \rangle}$$

Note 18.17. As $\langle x, x \rangle \in \mathbb{R}$ and $0 \leq \langle x, x \rangle$ we can indeed take the square root of $\langle x, x \rangle$ and $0 \leq \|x\|$.

As the name suggest the inner product norm is actually a norm., before we can prove this we must first proof a important inequality due Schwartz's.

Theorem 18.18. Let $\langle X, \langle \rangle \rangle$ be a inner product space then and $\| \cdot \|$ the inner product norm then $\forall x, y \in X$ we have

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

Further we have the following equivalence:

$$x, y \text{ are linear dependent} \Leftrightarrow |\langle x, y \rangle| = \|x\| \cdot \|y\|$$

Proof. We have the following cases to consider for $x, y \in X$:

$x = 0$. then x and y are linear dependent. Further we have

$$\begin{aligned} |\langle 0, y \rangle| &\stackrel{[\text{theorems: 18.3, 18.10}]}{=} 0 \\ &= 0 \cdot \|y\| \\ &\stackrel{[\text{theorems: 18.3, 18.10}]}{=} |\langle 0, 0 \rangle| \cdot \|y\| \\ &= \|x\| \cdot \|y\| \end{aligned}$$

$y = 0$. then x and y are linear dependent. Further we have

$$\begin{aligned} |\langle x, 0 \rangle| &\stackrel{[\text{theorems: 18.3, 18.10}]}{=} 0 \\ &= \|x\| \cdot 0 \\ &\stackrel{[\text{theorems: 18.3, 18.10}]}{=} \|x\| \cdot |\langle 0, 0 \rangle| \\ &= \|x\| \cdot \|y\| \end{aligned}$$

$x \neq 0 \wedge y \neq 0$. Then $\langle y, y \rangle \neq 0$ [otherwise by [theorems: 18.3, 18.10] we would have $y = 0$] so that we can define

$$\alpha = -\frac{\langle x, y \rangle}{\langle y, y \rangle} = -\frac{\langle x, y \rangle}{\|y\|^2}$$

We have now consider two cases:

real inner product. Then we have

$$\begin{aligned} 0 &\leq \langle x + \alpha \cdot y, x + \alpha \cdot y \rangle \\ &= \langle x, x + \alpha \cdot y \rangle + \alpha \cdot \langle y, x + \alpha \cdot y \rangle \\ &\stackrel{[\text{theorem: 18.3}]}{=} \langle x, x \rangle + \alpha \cdot \langle x, y \rangle + \alpha \cdot \langle y, x \rangle + \alpha^2 \cdot \langle y, y \rangle \\ &= \|x\|^2 + 2 \cdot \alpha \cdot \langle x, y \rangle + \alpha^2 \cdot \|y\|^2 \\ &= \|x\|^2 - 2 \cdot \frac{\langle x, y \rangle}{\|y\|^2} \cdot \langle x, y \rangle + \frac{\langle x, y \rangle^2}{\|y\|^4} \cdot \|y\|^2 \\ &= \|x\|^2 - 2 \cdot \frac{\langle x, y \rangle^2}{\|y\|^2} + \frac{\langle x, y \rangle^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2} \end{aligned}$$

hence $\langle x, y \rangle^2 \leq \|x\|^2 \cdot \|y\|^2$. As the square root is increasing [see theorem: 10.71] we have that

$$|\langle x, y \rangle| \stackrel{[\text{theorem: 10.73}]}{=} \sqrt{\langle x, y \rangle^2} \leq \sqrt{\|x\|^2 \cdot \|y\|^2} \stackrel{[\text{theorem: 10.73}]}{=} \|x\| \cdot \|y\| = \|x\| \cdot \|y\|$$

or

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

Further if x, y are linear dependent then there exist a $\beta \in \mathbb{R}$ with either:

$x = \beta \cdot y$. Then

$$|\langle x, y \rangle|^2 = \langle \beta \cdot y, y \rangle \cdot \langle x, y \rangle = \beta \cdot \langle y, y \rangle \cdot \langle x, y \rangle = \langle y, y \rangle \cdot \langle x, x \rangle = \|x\|^2 \cdot \|y\|^2$$

so that by [theorem: 10.73]

$$|\langle x, y \rangle| = \|x\| \cdot \|y\|$$

$y = \beta \cdot x$. Then

$$|\langle x, y \rangle|^2 = \langle x, \beta \cdot x \rangle \cdot \langle x, y \rangle = \beta \cdot \langle x, x \rangle \cdot \langle x, y \rangle = \langle x, x \rangle \cdot \langle y, y \rangle = \|x\|^2 \cdot \|y\|^2$$

so that by [theorem: 10.73]

$$|\langle x, y \rangle| = \|x\| \cdot \|y\|$$

So we have proved that

$$\text{If } x, y \text{ are linear dependent then } |\langle x, y \rangle| = \|x\| \cdot \|y\|$$

For the opposite implication. Assume that $|\langle x, y \rangle| = \|x\| \cdot \|y\|$ then we have

$$\begin{aligned} \langle \langle y, y \rangle \cdot x - \langle x, y \rangle \cdot y, \langle y, y \rangle \cdot x - \langle x, y \rangle \cdot y \rangle &= \\ \langle y, y \rangle^2 \cdot \langle x, x \rangle - \langle y, y \rangle \cdot \langle x, y \rangle \cdot \langle x, y \rangle - \langle x, y \rangle \cdot \langle y, y \rangle \cdot \langle y, x \rangle + \langle x, y \rangle \cdot \langle x, y \rangle \cdot \langle y, y \rangle &= \\ \langle y, y \rangle^2 \cdot \langle x, x \rangle - 2 \cdot \langle y, y \rangle \cdot \langle x, y \rangle^2 + \langle x, y \rangle^2 \cdot \langle y, y \rangle &= \\ \|y\|^4 \cdot \|x\|^2 - 2 \cdot \|y\|^2 \cdot \langle x, y \rangle^2 + \langle x, y \rangle^2 \cdot \|y\|^2 &= \\ \|y\|^4 \cdot \|x\|^2 - 2 \cdot \|y\|^2 \cdot \|x\|^2 \cdot \|y\|^2 + \|x\|^2 \cdot \|y\|^2 \cdot \|y\|^2 &= \\ \|y\|^4 \cdot \|x\|^2 - 2 \cdot \|y\|^4 \cdot \|x\|^2 + \|y\|^4 \cdot \|x\|^2 &= 0 \end{aligned}$$

so that $\langle y, y \rangle \cdot x - \langle x, y \rangle \cdot y = 0$. Hence $\|y\|^2 \cdot x = \langle x, y \rangle \cdot y$ or as $\|y\|^2 \neq 0$ that $x = \frac{\langle x, y \rangle}{\|y\|^2} \cdot y$. This proves that

$$\text{If } |\langle x, y \rangle| = \|x\| \cdot \|y\| \text{ then } x, y \text{ are linear dependent}$$

complex inner product. Then we have

$$\begin{aligned} 0 &\leq \langle x + \alpha \cdot y, x + \alpha \cdot y \rangle \\ &= \langle x, x + \alpha \cdot y \rangle + \alpha \cdot \langle y, x + \alpha \cdot y \rangle \\ &\stackrel{[\text{theorem: 18.10}]}{=} \langle x, x \rangle + \bar{\alpha} \cdot \langle x, y \rangle + \alpha \cdot \langle y, x \rangle + \alpha \cdot \bar{\alpha} \cdot \langle y, y \rangle \\ &= \|x\|^2 + \bar{\alpha} \cdot \langle x, y \rangle + \alpha \cdot \overline{\langle x, y \rangle} + \alpha \cdot \bar{\alpha} \cdot \|y\|^2 \\ &= \|x\|^2 - \frac{\overline{\langle x, y \rangle}}{\|y\|^2} \cdot \langle x, y \rangle - \frac{\langle x, y \rangle}{\|y\|^2} \cdot \overline{\langle x, y \rangle} + \frac{|\langle x, y \rangle|^2}{\|y\|^4} \cdot \|y\|^2 \\ &= \|x\|^2 - 2 \cdot \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \frac{|\langle x, y \rangle|^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{|\langle x, y \rangle|^2}{\|y\|^2} \end{aligned}$$

hence $\langle x, y \rangle^2 \leq \|x\|^2 \cdot \|y\|^2$. As the square root is increasing [see theorem: 10.71] we have that

$$|\langle x, y \rangle| \stackrel{[\text{theorem: 10.73}]}{=} \sqrt{\langle x, y \rangle^2} \leq \sqrt{\|x\|^2 \cdot \|y\|^2} \stackrel{[\text{theorem: 10.73}]}{=} \|x\| \cdot \|y\| = \|x\| \cdot \|y\|$$

or

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

Further if x, y are linear dependent then there exist a $\beta \in \mathbb{C}$ such that either $x = \beta \cdot y$. Then

$$\begin{aligned} |\langle x, y \rangle|^2 &= \langle x, y \rangle \cdot \overline{\langle x, y \rangle} \\ &= \langle \beta \cdot y, y \rangle \cdot \overline{\langle x, y \rangle} \\ &= \langle y, y \rangle \cdot \beta \cdot \overline{\langle y, x \rangle} \\ &= \langle y, y \rangle \cdot \langle x, x \rangle \\ &= \|x\|^2 \cdot \|y\|^2 \end{aligned}$$

so that by [theorem: 10.73]

$$|\langle x, y \rangle| = \|x\| \cdot \|y\|$$

$y = \beta \cdot x$. Then

$$\begin{aligned} |\langle x, y \rangle|^2 &= \langle x, y \rangle \cdot \overline{\langle x, y \rangle} \\ &= \langle x, y \rangle \cdot \langle y, x \rangle \\ &= \langle x, y \rangle \cdot \beta \cdot \langle x, x \rangle \\ &= \langle y, y \rangle \cdot \langle x, x \rangle \\ &= \|x\|^2 \cdot \|y\|^2 \end{aligned}$$

$$|\langle x, y \rangle|^2 = \|x\| \cdot \|y\|$$

so that by [theorem: 10.73]

$$|\langle x, y \rangle| = \|x\| \cdot \|y\|$$

So we have proved that

If x, y are linear dependent then $|\langle x, y \rangle| = \|x\| \cdot \|y\|$

For the opposite implication. Assume that $|\langle x, y \rangle| = \|x\| \cdot \|y\|$ then we have

$$\begin{aligned} \langle \langle y, y \rangle \cdot x - \langle x, y \rangle \cdot y, \langle y, y \rangle \cdot x - \langle x, y \rangle \cdot y \rangle &= \\ \langle y, y \rangle^2 \cdot \langle x, x \rangle - \langle y, y \rangle \cdot \overline{\langle x, y \rangle} \cdot \langle x, y \rangle - \langle x, y \rangle \cdot \langle y, y \rangle \cdot \langle y, x \rangle + \langle x, y \rangle \cdot \overline{\langle x, y \rangle} \cdot \langle y, y \rangle &= \\ \|y\|^4 \cdot \|x\|^2 - \|y\|^2 \cdot |\langle x, y \rangle|^2 - \|y\|^2 \cdot |\langle x, y \rangle|^2 + |\langle x, y \rangle|^2 \cdot \|y\|^2 &= \\ \|y\|^4 \cdot \|x\|^2 - \|y\|^2 \cdot \|x\|^2 \cdot \|y\|^2 - \|y\|^2 \cdot \|y\|^2 \cdot \|x\|^2 + \|x\|^2 \cdot \|y\|^2 \cdot \|y\|^2 &= \\ \|y\|^4 \cdot \|x\|^2 - \|y\|^4 \cdot \|x\|^2 - \|y\|^4 \cdot \|x\|^2 + \|x\|^2 \cdot \|y\|^4 &= \\ 0 \end{aligned}$$

so that $\langle y, y \rangle \cdot x - \langle x, y \rangle \cdot y = 0$. Hence $\|y\|^2 \cdot x = \langle x, y \rangle \cdot y$ or as $\|y\|^2 \neq 0$ that $x = \frac{\langle x, y \rangle}{\|y\|^2} \cdot y$. This proves that

If $|\langle x, y \rangle| = \|x\| \cdot \|y\|$ then x, y are linear dependent □

Theorem 18.19. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space then the inner product norm $\| \cdot \|$ defined above is a norm on X making $\langle X, \| \cdot \| \rangle$ a normed space.

Proof. We have for $\langle X, \langle \rangle \rangle$ two cases to consider:

real inner product. Then we have:

1. If $x \in X$ then $0 \leq \sqrt{\langle x, x \rangle} = \|x\|$ [see 10.67]
2. If $\alpha \in X$ then

$$\begin{aligned} \|\alpha \cdot x\| &= \sqrt{\langle \alpha \cdot x, \alpha \cdot x \rangle} \\ &= \sqrt{\langle \alpha \cdot x, \alpha \cdot x \rangle} \\ &= \sqrt{\alpha \cdot \alpha \cdot \langle x, x \rangle} \\ &= \sqrt{\alpha^2 \cdot \langle x, x \rangle} \\ &\stackrel{[\text{theorem: 10.74}]}{=} \sqrt{\alpha^2} \cdot \sqrt{\langle x, x \rangle} \\ &\stackrel{[\text{theorem: 10.73}]}{=} |\alpha| \cdot \sqrt{\langle x, x \rangle} \\ &= |\alpha| \cdot \|x\| \end{aligned}$$

3. If $x, y \in X$ then we have

$$\begin{aligned}
 \|x + y\|^2 &= \langle x + y, x + y \rangle \\
 &= \langle x, x + y \rangle + \langle y, x + y \rangle \\
 &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
 &= \|x\|^2 + \langle x, y \rangle + \langle x, y \rangle + \|y\|^2 \\
 &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \\
 &\leq \|x\|^2 + 2 \cdot |\langle x, y \rangle| + \|y\|^2 \\
 &\stackrel{[\text{theorem: 18.18}]}{\leq} \|x\|^2 + 2 \cdot \|x\| \cdot \|y\| + \|y\|^2 \\
 &= (\|x\| + \|y\|)^2
 \end{aligned}$$

which as the square root is strictly increasing [see theorem: 10.71] proves that

$$\|x + y\| \leq \|x\| + \|y\|$$

4. If $\|x\| = 0$ then $\langle x, x \rangle = \|x\|^2 = 0$ so that $x = 0$.

complex inner product. Then we have:

1. If $x \in X$ then $0 \leq \sqrt{\langle x, x \rangle} = \|x\|$ [see 10.67]

2. If $\alpha \in X$ then

$$\begin{aligned}
 \|\alpha \cdot x\| &= \sqrt{\langle \alpha \cdot x, \alpha \cdot x \rangle} \\
 &= \sqrt{\langle \alpha \cdot x, \alpha \cdot x \rangle} \\
 &= \sqrt{\alpha \cdot \bar{\alpha} \cdot \langle x, x \rangle} \\
 &= \sqrt{|\alpha|^2 \cdot \langle x, x \rangle} \\
 &\stackrel{[\text{theorem: 10.74}]}{=} \sqrt{|\alpha|^2} \cdot \sqrt{\langle x, x \rangle} \\
 &\stackrel{[\text{theorem: 10.73}]}{=} |\alpha| \cdot \sqrt{\langle x, x \rangle} \\
 &= |\alpha| \cdot \|x\|
 \end{aligned}$$

3. If $x, y \in X$ then we have

$$\begin{aligned}
 \|x + y\|^2 &= \langle x + y, x + y \rangle \\
 &= \langle x, x + y \rangle + \langle y, x + y \rangle \\
 &= \langle x, x \rangle + \langle x, y \rangle + \overline{\langle x, y \rangle} + \langle y, y \rangle \\
 &= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2 \\
 &\stackrel{[\text{theorem: 10.83}]}{=} \|x\|^2 + 2 \cdot \operatorname{Re}(\langle x, y \rangle) + \|y\|^2 \\
 &\stackrel{[\text{theorem: 10.85}]}{\leq} \|x\|^2 + 2 \cdot |\langle x, y \rangle| + \|y\|^2 \\
 &\stackrel{[\text{theorem: 18.18}]}{\leq} \|x\|^2 + 2 \cdot \|x\| \cdot \|y\| + \|y\|^2 \\
 &= (\|x\| + \|y\|)^2
 \end{aligned}$$

which as the square root is strictly increasing [see theorem: 10.71] proves that

$$\|x + y\| \leq \|x\| + \|y\|$$

4. If $\|x\| = 0$ then $\langle x, x \rangle = \|x\|^2 = 0$ so that $x = 0$. □

Example 18.20. Let $n \in \mathbb{N}$ then $(\mathbb{R}^n, \|\cdot\|_e)$ where $\|x\|_e = \sqrt{\sum_{i=1}^n x_i^2}$ is a normed space. $\|\cdot\|_e$ is called the Euclidean norm on $\mathbb{R}^n$.

Proof. Using [example: 18.4] $(\mathbb{R}, \langle \cdot, \cdot \rangle)$ where $\langle x, y \rangle = \sum_{i=1}^n x_i \cdot y_i$ is a real inner product space. Hence by [theorem: 18.19] $\|\cdot\|_e$ defined by $\|x\|_e = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i \cdot x_i} = \sqrt{\sum_{i=1}^n x_i^2}$ is a norm on $\mathbb{R}$. □

Example 18.21. Let $n \in \mathbb{N}$ then $\langle \mathbb{C}^n, |||_e \rangle$ where $\|x\|_e = \sqrt{\sum_{i=1}^n |x_i|^2}$ is a normed space. $|||_e$ is called the Euclidean norm on $\mathbb{C}^n$.

Proof. Using [example: 18.9] $\langle \mathbb{C}, \langle \rangle \rangle$ where $\langle x, y \rangle = \sum_{i=1}^n x_i \cdot \bar{y}_i$ is a real inner product space. Hence by [theorem: 18.19] $|||_e$ defined by $\|x\|_e = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i \cdot \bar{x}_i} = \sqrt{\sum_{i=1}^n |x_i|^2}$ is a norm on $\mathbb{C}$. $\square$

We can extend the above two examples to all finite dimensional spaces.

Example 18.22. Let $n \in \mathbb{N}$, X a finite dimensional space over $\mathbb{R}$ with $E = \{e_1, \dots, e_n\}$ a basis of X then $|||_e$, defined by $\|x\|_e = \sqrt{\sum_{i=1}^n x_i^2}$ where $x = \sum_{i=1}^n x_i \cdot e_i$, is a norm on X .

Proof. Using [theorem: 18.14] we have that $\langle \rangle$, defined by $\langle x, y \rangle = \sum_{i=1}^n x_i \cdot y_i$ where $x = \sum_{i=1}^n x_i \cdot e_i$, $y = \sum_{i=1}^n y_i \cdot e_i$, is a inner product on X . Hence by [theorem: 18.19] $|||_e$ defined by $\|x\|_e = \sqrt{\langle x, x \rangle} = \sum_{i=1}^n x_i \cdot x_i = \sum_{i=1}^n x_i^2$ is a norm on X . $\square$

Example 18.23. Let $\langle \mathbb{R}^2, |||_e \rangle$ be the real vector space of pairs of real numbers equipped with the Euclidean norm defined by $\|x\|_e = \sqrt{\sum_{i=1}^2 x_i^2} = \sqrt{x_1^2 + x_2^2}$ and $\langle \mathbb{C}, || \rangle$ be the space of complex numbers with the absolute value norm then there exist a linear isometric isomorphism between $\langle \mathbb{R}^2, |||_e \rangle$ and $\langle \mathbb{C}, || \rangle$. Hence by [theorem: 14.194] $\langle \mathbb{R}^2, |||_e \rangle$ and $\langle \mathbb{C}, || \rangle$ are homeomorphic.

Proof. Define $\varphi: \mathbb{R}^2 \rightarrow \mathbb{C}$ by $\varphi(x, y) = x + i \cdot y$ then we have:

injectivity. If $\varphi(x, y) = \varphi(x', y')$ then $x + i \cdot y = x' + i \cdot y'$ so that by [theorem: 10.77] $x = x'$ and $y = y'$, hence $(x, y) = (x', y')$.

surjectivity. If $z \in \mathbb{C}$ then $\varphi(\text{Re}(z), \text{Im}(z)) = \text{Re}(z) + i \cdot \text{Im}(z) = z$.

proving that

$$\varphi: \mathbb{R}^2 \rightarrow \mathbb{C} \text{ is a bijection}$$

Further if $(x, y), (x', y') \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$ then

$$\varphi((x, y) + \alpha \cdot (x', y')) = \varphi(x + \alpha \cdot x', y + \alpha \cdot y') = x + \alpha \cdot x' + i \cdot y + i \cdot \alpha \cdot y' = \varphi(x, y) + \alpha \cdot \varphi(x', y')$$

proving that

$$\varphi \text{ is a linear isomorphism}$$

Further for $(x, y) \in \mathbb{R}^2$

$$|\varphi(x, y)| = |x + i \cdot y| = \sqrt{x^2 + y^2} = \|x\|_e$$

proving that

$$\varphi \text{ is a isometric linear isomorphism}$$

$\square$

Theorem 18.24. Let $n \in \mathbb{N}$, X a finite dimensional space over $\mathbb{C}$ with $E = \{e_1, \dots, e_n\}$ a basis of X then $|||_e$, defined by $|||_e = \sqrt{\sum_{i=1}^n |x_i|^2}$ where $x = \sum_{i=1}^n x_i \cdot e_i$, is a norm on X .

Proof. Using [theorem: 18.15] we have that $\langle \rangle$, defined by $\langle x, y \rangle = \sum_{i=1}^n x_i \cdot \bar{y}_i$ where $x = \sum_{i=1}^n x_i \cdot e_i$, $y = \sum_{i=1}^n y_i \cdot e_i$, is a inner product on X . Hence by [theorem: 18.19] $|||_e$ defined by $\|x\|_e = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i \cdot \bar{x}_i} = \sqrt{\sum_{i=1}^n |x_i|^2}$ is a norm on X . $\square$

18.3 Orthogonality

As show in the previous section a inner product on a vector space allows us to define the concept of a length or norm of a vector. Next we introduce the concept of orthogonal vectors as a generalization of orthogonality in Euclidean spaces.

Definition 18.25. (Orthogonality) Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space and $x, y \in X$ then x, y are **orthogonal** noted as $x \perp y$ if $\langle x, y \rangle = 0$.

Definition 18.26. (Orthonormal Family) Let $\langle X, \langle \rangle \rangle$ be a inner product space with the Euclidean norm $\| \cdot \|$ then $\{e_i\}_{i \in I} \subseteq X$ is orthonormal if $\forall i, j \in I$ we have $\langle e_i, e_j \rangle = \delta_{i,j}$ where $\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

Example 18.27. Let $n \in \mathbb{N}$ then we have that

1. Given $\langle \mathbb{R}^n, \langle \rangle \rangle$ (see example: 18.4) then the canonical basis $\{e_i\}_{i \in \{1, \dots, n\}}$ defined by $(e_i)_j = \delta_{i,j}$ (see theorem: 11.167) is orthonormal.
2. Given $\langle \mathbb{C}^n, \langle \rangle \rangle$ (see example: 18.9) then the canonical basis $\{e_i\}_{i \in \{1, \dots, n\}}$ defined by $(e_i)_j = \delta_{i,j}$ (see theorem: 11.167) is orthonormal.

Proof.

1. Given $i, j \in \{1, \dots, n\}$ we have that

$$\langle e_i, e_j \rangle \stackrel{[\text{example: 18.4}]}{=} \sum_{k=1}^n (e_i)_k \cdot (e_j)_k = \sum_{k=1}^n \delta_{i,k} \cdot \delta_{j,k} = \sum_{k=1}^n \delta_{i,k} \cdot \delta_{i,j} = \delta_{i,j}$$

2. Given $i, j \in \{1, \dots, n\}$ we have that

$$\langle e_i, e_j \rangle \stackrel{[\text{example: 18.9}]}{=} \sum_{k=1}^n (e_i)_k \cdot \overline{(e_j)_k} = \sum_{k=1}^n \delta_{i,k} \cdot \delta_{j,k} = \sum_{k=1}^n \delta_{i,k} \cdot \delta_{i,j} = \delta_{i,j} \quad \square$$

Every orthonormal family is linear independent however the opposite is not true.

Theorem 18.28. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space and $\{e_i\}_{i \in I} \subseteq X$ a orthonormal family then $\{e_i\}_{i \in I}$ is linear independent.

Proof. Let W be a finite subset of I and $\{\alpha_i\}_{i \in W} \subseteq \mathbb{K}$ such that $\sum_{w \in W} \alpha_w \cdot e_w = 0$. By the definition of the generalized sum [see 11.32] there exist a bijection $\beta: \{1, \dots, n\} \rightarrow W$ such that

$$\sum_{k=1}^n \alpha_{\beta(i)} \cdot e_{\beta(i)} = \sum_{w \in W} \alpha_w \cdot e_w = 0$$

Let $w \in W$ then there exist a $k \in \{1, \dots, n\}$ such that $w = \beta(k)$. Hence

$$\begin{aligned} 0 &= \langle 0, e_{\beta(k)} \rangle \\ &= \left\langle \sum_{i=1}^n \alpha_{\beta(i)} \cdot e_{\beta(i)}, e_{\beta(k)} \right\rangle \\ &\stackrel{[\text{theorem: 18.11}]}{=} \sum_{i=1}^n \alpha_{\beta(i)} \cdot \langle e_{\beta(i)}, e_{\beta(k)} \rangle \\ &= \sum_{i=1}^n \alpha_{\beta(i)} \cdot \delta_{\beta(i), \beta(k)} \\ &\stackrel{\varphi \text{ is a bijection}}{=} \sum_{i=k \Leftrightarrow \beta(i) = \beta(k)}^n \alpha_{\beta(i)} \cdot \delta_{i,k} \\ &= \alpha_{\beta(i)} \\ &= \alpha_w \end{aligned}$$

Hence $\forall w \in W$ we have $\alpha_w = 0$ which by [theorem: 11.119] proves that $\{e_i\}_{i \in I}$ is linear independent. $\square$

Using orthonormal bases in finite dimensional vector spaces has the benefit of giving a simple representation for the matrix of a linear mapping or of the expansion of an element in its basis.

Theorem 18.29. *Let $\langle X, \langle \rangle_X \rangle$ be a finite dimensional real [or complex] inner product space with an orthonormal basis $E = \{e_1, \dots, e_n\}$ then $\forall x \in X$ we have $x = \sum_{i=1}^n \langle x, e_i \rangle \cdot e_i$*

Proof. If $x \in X$ then there exist a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ such that $x = \sum_{i=1}^n x_i \cdot e_i$. Let $k \in \{1, \dots, n\}$ then

$$\langle x, e_k \rangle = \left\langle \sum_{i=1}^n x_i \cdot e_i, e_k \right\rangle = \sum_{i=1}^n x_i \cdot \langle e_i, e_k \rangle = \sum_{i=1}^n x_i \cdot \delta_{i,k} = x_k$$

so that

$$x = \sum_{i=1}^n \langle x, e_i \rangle \cdot e_i \quad \square$$

Theorem 18.30. *Let $\langle X, \langle \rangle_X \rangle$ and $\langle Y, \langle \rangle_Y \rangle$ be two finite dimensional real [or complex] inner product spaces with $E = \{e_1, \dots, e_n\}$ an orthonormal basis for X and $F = \{f_1, \dots, f_m\}$ an orthonormal basis for Y . Then if $L: X \rightarrow Y$ is a linear mapping with matrix $\mathcal{M}(L; E, F)$ [see definition: 11.339] we have*

$$\forall i \in \{1, \dots, n\}, \forall j \in \{1, \dots, m\} \text{ we have } \mathcal{M}(L; E, F)_{j,i} = \langle L(e_i), f_j \rangle$$

Proof. We have for $i \in [1, \dots, n], \forall j \in \{1, \dots, m\}$ we have

$$\begin{aligned} \langle L(e_i), f_j \rangle &\stackrel{[\text{definition: 11.339}]}{=} \left\langle \sum_{k=1}^m \mathcal{M}(L; E, F)_{k,i} \cdot f_k, f_j \right\rangle \\ &\stackrel{[\text{theorem: 18.11}]}{=} \sum_{k=1}^m \mathcal{M}(L; E, F)_{k,i} \cdot \langle f_k, f_j \rangle \\ &= \sum_{k=1}^m \mathcal{M}(L; E, F)_{k,i} \cdot \delta_{k,j} \\ &= \mathcal{M}(L; E, F)_{j,i} \end{aligned} \quad \square$$

It turns out that for every finite dimensional real [or complex] inner product space we can construct an orthonormal basis.

Theorem 18.31. (Gram-Schmidt) *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space, $\| \cdot \|$ the inner product norm and $\{e_1, \dots, e_n\} \subseteq X$ a linear independent set where $n \in \mathbb{N}$ then $\{f_1, \dots, f_n\} \subseteq X$ recursively defined by*

$$f_i = \begin{cases} \frac{e_1}{\|e_1\|} & \text{if } i = 1 \\ \frac{e_i - \sum_{j=1}^{i-1} \langle e_i, f_j \rangle \cdot f_j}{\|e_i - \sum_{j=1}^{i-1} \langle e_i, f_j \rangle \cdot f_j\|} & \text{if } i \in \{2, \dots, n\} \end{cases}$$

is well defined [meaning no division by 0] and is an orthonormal set. Further we have that

$$\text{span}(\{e_1, \dots, e_n\}) = \text{span}(\{f_1, \dots, f_n\})$$

Proof. We will use induction to prove this, so define

$S = \{m \in \mathbb{N} \mid \text{If } m \in \{1, \dots, n\} \text{ then } \{f_1, \dots, f_m\} \text{ is well defined, orthonormal and } \text{span}(\{e_1, \dots, e_m\}) = \text{span}(\{f_1, \dots, f_m\})\}$

then we have:

$1 \in \mathcal{S}$. As $\{e_1, \dots, e_n\}$ is linear independent $e_1 \neq 0 \Rightarrow \|e_1\| \neq 0$ so that $f_1 = \frac{e_1}{\|e_1\|}$ is well defined.

If $i, j \in \{1\}$ then $i = j = 1$ and $\langle f_i, f_j \rangle = \langle f_1, f_1 \rangle = \|f_1\|^2 = \left\| \frac{e_1}{\|e_1\|} \right\|^2 = \left(\frac{\|e_1\|}{\|e_1\|} \right)^2 = 1 = \delta_{1,1} = \delta_{i,j}$ so that $\{f_1\}$ is orthonormal. Further we have $x \in \text{span}(\{e_1\})$ if there exist a $\alpha \in \mathbb{K}$ such that $x = \alpha \cdot e_1 = \alpha \cdot \|e_1\| \cdot \frac{e_1}{\|e_1\|} = (\alpha \cdot \|e_1\|) \cdot f_1$ so that $\text{span}(\{e_1\}) \subseteq \text{span}(\{f_1\})$, if $x \in \text{span}(\{f_1\})$ then there exist a $\alpha \in \mathbb{K}$ such that $x = \alpha \cdot f_1 = \frac{\alpha}{\|e_1\|} \cdot \|e_1\| \cdot f_1 = \frac{\alpha}{\|e_1\|} \cdot e_1$ proving that $x \in \text{span}(\{e_1\})$. So $\text{span}(\{e_1\}) = \text{span}(\{f_1\})$ and we have $1 \in \mathcal{S}$.

$m \in \mathcal{S} \Rightarrow m+1 \in \mathcal{S}$. As $m \in \mathcal{S}$ we have that $\{f_1, \dots, f_m\}$ is orthonormal and $\text{span}(\{e_1, \dots, e_m\}) = \text{span}(\{f_1, \dots, f_m\})$. Assume that $m+1 \in \{1, \dots, n\}$ then as $m \in \mathcal{S} \subseteq \mathbb{N}$ we have $m+1 \in \{2, \dots, n\}$. If $e_{m+1} - \sum_{i=1}^{m+1} \langle e_{m+1}, f_i \rangle \cdot f_i = 0$ then $e_{m+1} = \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i$ so that $e_{m+1} \in \text{span}(\{f_1, \dots, f_m\}) = \text{span}(\{e_1, \dots, e_m\})$ which is impossible because $\{e_1, \dots, e_m\}$ is linear independent. Hence we have that $e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i \neq 0$ so that

$$f_{m+1} = \frac{e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \text{ is well defined} \quad (18.2)$$

Further

$$\begin{aligned} \langle f_{m+1}, f_{m+1} \rangle &= \|f_{m+1}\|^2 \\ &= \left\| \frac{e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \right\|^2 \\ &= \left(\frac{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \right)^2 \\ &= 1 \end{aligned} \quad (18.3)$$

and $\forall k \in \{1, \dots, m\}$ we have

$$\begin{aligned} \langle f_{m+1}, f_k \rangle &= \\ &= \frac{1}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \cdot \left\langle e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i, f_k \right\rangle \\ &= \frac{1}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \cdot \left(\langle e_{m+1}, f_k \rangle - \left\langle \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i, f_k \right\rangle \right) \\ &= \frac{1}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \cdot \left(\langle e_{m+1}, f_k \rangle - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot \langle f_i, f_k \rangle \right) \\ &= \frac{1}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \cdot \left(\langle e_{m+1}, f_k \rangle - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot \delta_{i,k} \right) \\ &= \frac{1}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|} \cdot (\langle e_{m+1}, f_k \rangle - \langle e_{m+1}, f_k \rangle) \\ &= 0 \end{aligned} \quad (18.4)$$

So if $k, l \in \{1, \dots, m+1\}$ then we have either:

$k \wedge l \in \{1, \dots, m\}$. Then as $\{f_1, \dots, f_m\}$ is orthonormal we have that $\langle f_k, f_l \rangle = \delta_{k,l}$.

$k = m+1 \wedge l \in \{1, \dots, m\}$. Then $\langle f_k, f_l \rangle = \langle f_{m+1}, f_l \rangle \stackrel{\text{eq: 18.4}}{=} 0 = \delta_{k,l}$

$k \in \{1, \dots, m\} \wedge l = m+1$. Then $\langle f_k, f_l \rangle = \langle f_l, f_k \rangle = \langle f_{m+1}, f_k \rangle \stackrel{\text{eq: 18.4}}{=} 0 = \delta_{k,l}$

$k = m+1 = l$. Then $\langle f_k, f_l \rangle = \langle f_{m+1}, f_{m+1} \rangle \stackrel{\text{eq: 18.3}}{=} 1 = \delta_{k,l}$

proving that

$$\{f_1, \dots, f_{m+1}\} \text{ is orthonormal} \quad (18.5)$$

Further we have as $f_{m+1} = \frac{e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i}{\|e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i\|}$ that

$$e_{m+1} = \left\| e_{m+1} - \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i \right\| \cdot f_{m+1} + \sum_{i=1}^m \langle e_{m+1}, f_i \rangle \cdot f_i$$

so that $e_{m+1} \in \text{span}(\{f_1, \dots, f_{m+1}\})$. As

$$\{e_1, \dots, e_m\} \subseteq \text{span}(\{e_1, \dots, e_m\}) = \text{span}(\{f_1, \dots, f_m\}) \subseteq \text{span}(\{f_1, \dots, f_{m+1}\})$$

we have

$$\{e_1, \dots, e_{m+1}\} \subseteq \text{span}(\{e_1, \dots, e_{m+1}\}) \subseteq \text{span}(\{f_1, \dots, f_{m+1}\}) \quad (18.6)$$

To prove the opposite inequality not that by [theorem: 18.28] and the fact that $\{f_1, \dots, f_{m+1}\}$ is orthonormal it follows that $\{f_1, \dots, f_{m+1}\}$ is linear independent. Further, as by [theorem: 11.110], $\text{span}(\{f_1, \dots, f_{m+1}\})$ is a vector space, it follows that $\{f_1, \dots, f_{m+1}\}$ is a basis for the $m+1$ -dimensional vector space $\text{span}(\{f_1, \dots, f_{m+1}\})$. As $\{e_1, \dots, e_{m+1}\}$ is linear independent and by [eq: 18.6] a subset of $\text{span}(\{f_1, \dots, f_{m+1}\})$ it follow from [theorem: 11.155] that $\{e_1, \dots, e_{m+1}\}$ is a basis for $\text{span}(\{f_1, \dots, f_{m+1}\})$. Hence

$$\text{span}(\{e_1, \dots, e_{m+1}\}) = \text{span}(\{f_1, \dots, f_{m+1}\}) \quad (18.7)$$

Summarized, from [eqs: 18.2, 18.5 and 18.7] it follows that $m+1 \in S$. $\square$

Corollary 18.32. *Let $n \in \mathbb{N}$, $\langle X, \langle \rangle \rangle$ a finite dimensional real [or complex] inner product space with $\dim(X) = n$ then there exist a orthonormal basis $\{e_1, \dots, e_n\}$ on X .*

Proof. As X is finite dimensional with $\dim(X) = n$ there exist a basis $\{e'_1, \dots, e'_n\}$ on X . Hence $\{e'_1, \dots, e'_n\}$ is linear independent and $\text{span}(\{e'_1, \dots, e'_n\}) = X$. Using the Gram-Schmidt procedure [see theorem: 18.31] we can construct a orthonormal set $\{e_1, \dots, e_n\}$ [hence by [theorem: 18.28] a linear independent set] such that $\text{span}(\{e_1, \dots, e_n\}) = \text{span}(\{e'_1, \dots, e'_n\}) = X$. So we found a orthonormal basis on X . $\square$

Theorem 18.33. *Let $n \in \mathbb{N}$, $\langle X, \langle \rangle \rangle$ a finite dimensional real [or complex] inner product space with $\dim(X) = n$ and $\{e_1, \dots, e_n\}$ a orthonormal subset of X then $\{e_1, \dots, e_n\}$ is a basis on X .*

Proof. By [theorem: 18.28] $\{e_1, \dots, e_n\}$ is linear independent so that by [theorem: 11.155] $\{e_1, \dots, e_n\}$ is a basis for X . $\square$

18.3.1 Dual spaces and inner product

Recap the concept of the dual of a vector space [see definition: 11.186] which in the case of a real [or complex] inner vector spaces is defined by.

Definition 18.34. *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product vector space then the dual space, noted as X^* is defined by:*

real inner product space.

$$X^* = \text{Hom}(X, \mathbb{R})$$

complex inner product space.

$$X^* = \text{Hom}(X, \mathbb{C})$$

If we choose $\mathbb{K}$ to mean either $\mathbb{R}$ or $\mathbb{C}$ if $\langle X, \langle \rangle \rangle$ if $\langle X, \langle \rangle \rangle$ is a real or complex inner product space then we can use the simpler notation

$$X^* = \text{Hom}(X, \mathbb{K})$$

We show now that elements of X^* can be written using a inner product.

Theorem 18.35. *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product vector space and $u \in X$ then $\langle *, u \rangle$ defined by*

$$\langle *, u \rangle: X \rightarrow \mathbb{K} \text{ where } \langle *, u \rangle(y) = \langle y, u \rangle$$

is a element of X^ .*

Proof. Let $x, y \in X$ and $\alpha \in \mathbb{K}$ then

$$\langle *, u \rangle(x + \alpha \cdot y) = \langle x + \alpha \cdot y, u \rangle = \langle x, u \rangle + \alpha \cdot \langle y, u \rangle = \langle *, u \rangle(x) + \alpha \cdot \langle *, u \rangle(y)$$

proving that $\langle *, u \rangle \in \text{Hom}(X, \mathbb{K}) = X^*$ □

We have also the opposite of the above theorem.

Theorem 18.36. (Riesz Representation Theorem) *Let $n \in \mathbb{N}$, $\langle X, \langle \rangle \rangle$ a finite dimensional real [or complex] inner product vector space with $\dim(X) = n$ and $L \in X^*$ then there exist a **unique** $u \in X$ such that $L = \langle *, u \rangle$. In other words we have $\forall x \in X$ that $L(x) = \langle x, u \rangle$.*

Proof. Using [theorem: 18.32] there exist a orthonormal basis $\{e_1, \dots, e_n\}$ on X . We consider now the following cases:

real inner product space. Take $u = \sum_{i=1}^n \varphi(e_i) \cdot e_i$. Let $x \in X$ then there exist a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $x = \sum_{i=1}^n x_i \cdot e_i$

$$\begin{aligned} \langle x, u \rangle &= \left\langle \sum_{i=1}^n x_i \cdot e_i, u \right\rangle \\ &= \sum_{i=1}^n x_i \cdot \langle e_i, u \rangle \\ &= \sum_{i=1}^n x_i \cdot \left\langle e_i, \sum_{j=1}^n \varphi(e_j) \cdot e_j \right\rangle \\ &= \sum_{i=1}^n x_i \cdot \left\langle \sum_{j=1}^n \varphi(e_j) \cdot e_j, e_i \right\rangle \\ &= \sum_{i=1}^n x_i \cdot \left(\sum_{j=1}^n \varphi(e_j) \cdot \langle e_j, e_i \rangle \right) \\ &= \sum_{i=1}^n x_i \cdot \left(\sum_{j=1}^n \varphi(e_j) \cdot \delta_{i,j} \right) \\ &= \sum_{i=1}^n x_i \cdot \varphi(e_i) \\ &= \varphi \left(\sum_{i=1}^n x_i \cdot e_i \right) \\ &= \varphi(x) \end{aligned}$$

Let $v \in X$ be such that $\forall x \in X$ $\varphi(x) = \langle x, v \rangle$ then $\forall i \in \{1, \dots, n\}$ we have $\varphi(e_i)$

complex inner product space. Take $u = \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i$. Let $x \in X$ then there exist a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ such that $x = \sum_{i=1}^n x_i \cdot e_i$

$$\begin{aligned}
 \langle x, u \rangle &= \left\langle \sum_{i=1}^n x_i \cdot e_i, u \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \langle e_i, u \rangle \\
 &= \sum_{i=1}^n x_i \cdot \left\langle e_i, \sum_{j=1}^n \overline{\varphi(e_j)} \cdot e_j \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \overline{\left\langle \sum_{j=1}^n \overline{\varphi(e_j)} \cdot e_j, e_i \right\rangle} \\
 &= \sum_{i=1}^n x_i \cdot \overline{\sum_{j=1}^n \overline{\varphi(e_j)} \cdot \langle e_j, e_i \rangle} \\
 &= \sum_{i=1}^n x_i \cdot \overline{\sum_{j=1}^n \overline{\varphi(e_j)} \cdot \delta_{i,j}} \\
 &= \sum_{i=1}^n x_i \cdot \overline{\overline{\varphi(e_i)}} \\
 &= \sum_{i=1}^n x_i \cdot \varphi(e_i) \\
 &= \varphi(x)
 \end{aligned}$$

So in all cases we have found a $u \in X$ such that

$$\forall x \in X \text{ we have } \varphi(x) = \langle x, u \rangle$$

The above proves existence, as for uniqueness. Assume that there exist a $v \in X$ such that $\forall x \in X$ we have $\varphi(x) = \langle x, v \rangle$. Then, as $u - v \in X$, there exist a $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ such that $u - v = \sum_{i=1}^n y_i \cdot e_i$. Hence we have

$$\begin{aligned}
 \langle u - v, u - v \rangle &= \left\langle \sum_{i=1}^n y_i \cdot e_i, u - v \right\rangle \\
 &= \sum_{i=1}^n y_i \cdot \langle e_i, u - v \rangle \\
 &= \sum_{i=1}^n y_i \cdot (\langle e_i, u \rangle - \langle e_i, v \rangle) \\
 &= \sum_{i=1}^n y_i \cdot (\varphi(e_i) - \varphi(e_i)) \\
 &= \sum_{i=1}^n y_i \cdot 0 \\
 &= 0
 \end{aligned}$$

proving that $u - v = 0$ or $u = v$. □

Riesz Representation Theorem allows us to define the concept of the adjoint of a operator.

Theorem 18.37. Let $\langle X, \langle \rangle_X \rangle, \langle Y, \langle \rangle_Y \rangle$ be finite dimensional real [or complex] inner product spaces with $\dim(X), \dim(Y) \in \mathbb{N}$ and $L \in \text{Hom}(X, Y)$ then there exist a **unique** mapping

$$L^* \in \text{Hom}(Y, X)$$

such that for $y \in Y$ $L^*(y)$ we have

$$\forall x \in X \text{ that } \langle L(x), y \rangle_Y = \langle x, L^*(y) \rangle_X$$

Also we have

$$L^* \in \text{Hom}(Y, X)$$

The mapping L^* is called the **adjoint of L**

Proof. Given $y \in Y$ define

$$\varphi_y: X \rightarrow \mathbb{K} \text{ by } \varphi_y(x) = \langle L(x), y \rangle_Y$$

If $x_1, x_2 \in X$ and $\alpha \in \mathbb{K}$ then

$$\begin{aligned} \varphi_y(x_1 + \alpha \cdot x_2) &= \langle L(x_1 + \alpha \cdot x_2), y \rangle_Y \\ &= \langle L(x_1) + \alpha \cdot L(x_2), y \rangle_Y \\ &= \langle L(x_1), y \rangle_Y + \alpha \cdot \langle L(x_2), y \rangle_Y \\ &= \varphi_y(x_1) + \alpha \cdot \varphi_y(x_2) \end{aligned}$$

so that

$$\varphi_y \in \text{Hom}(X, \mathbb{K}) = X^*$$

Hence, using the Riesz Representation Theorem [see theorem: 18.36] there exist a **unique** $u_y \in X$ such that $\varphi_y = \langle *, u_y \rangle$, hence $\forall x \in X$ we have that

$$\langle L(x), y \rangle_Y = \varphi_y(x) = \langle *, u_y \rangle_X(x) = \langle x, u_y \rangle_X$$

So if we define

$$L^*: Y \rightarrow X \text{ by } L^*(y) = u_y$$

then $\forall y \in Y$ we have $\forall x \in X$ that $\langle L(x), y \rangle_Y = \langle x, u_y \rangle_X = \langle x, L^*(y) \rangle_X$. If $y_1, y_2 \in Y$ and $\alpha \in \mathbb{K}$. Then we have to consider either:

real inner product spaces. Then we have $\forall x \in X$

$$\varphi_{y_1 + \alpha \cdot y_2}(x) = \langle x, u_{y_1 + \alpha \cdot y_2} \rangle_X$$

but also

$$\begin{aligned} \varphi_{y_1 + \alpha \cdot y_2}(x) &= \langle L(x), y_1 + \alpha \cdot y_2 \rangle_Y \\ &= \langle L(x), y_1 \rangle_Y + \alpha \cdot \langle L(x), y_2 \rangle_Y \\ &= \varphi_{y_1}(x) + \alpha \cdot \varphi_{y_2}(x) \\ &= \langle x, u_{y_1} \rangle_X + \alpha \cdot \langle x, u_{y_2} \rangle_X \\ &= \langle x, u_{y_1} + \alpha \cdot u_{y_2} \rangle_X \end{aligned}$$

so that by **uniqueness** we must have that $u_{y_1 + \alpha \cdot y_2} = u_{y_1} + \alpha \cdot u_{y_2}$, hence we have that

$$L^*(y_1 + \alpha \cdot y_2) = u_{y_1 + \alpha \cdot y_2} = u_{y_1} + \alpha \cdot u_{y_2} = L(y_1) + \alpha \cdot L(y_2)$$

complex inner product spaces. Then we have $\forall x \in X$

$$\varphi_{y_1 + \alpha \cdot y_2}(x) = \langle x, u_{y_1 + \alpha \cdot y_2} \rangle$$

but also

$$\begin{aligned} \varphi_{y_1 + \alpha \cdot y_2}(x) &= \langle L(x), y_1 + \alpha \cdot y_2 \rangle_Y \\ &= \langle L(x), y_1 \rangle_Y + \bar{\alpha} \cdot \langle L(x), y_2 \rangle_Y \\ &= \varphi_{y_1}(x) + \bar{\alpha} \cdot \varphi_{y_2}(x) \\ &= \langle x, u_{y_1} \rangle_X + \bar{\alpha} \cdot \langle x, u_{y_2} \rangle_X \\ &= \langle x, u_{y_1} + \alpha \cdot u_{y_2} \rangle_X \end{aligned}$$

so that by **uniqueness** we must have that $u_{y_1+\alpha \cdot y_2} = u_{y_1} + \alpha \cdot y_2$, hence we have that

$$L^*(y_1 + \alpha \cdot y_2) = u_{y_1+\alpha \cdot y_2} = u_{y_1} + \alpha \cdot y_2 = L(y_1) + \alpha \cdot L(y_2)$$

Hence we have that

$$L^* \in \text{Hom}(Y, X)$$

Finally assume that there is another $K^* \in \text{Hom}(Y, X)$ such that $\forall y \in Y$ we have $\forall x \in X$ that $\langle L(x), y \rangle_Y = \langle x, L^*(y) \rangle_X$. Then given $y \in Y$ we have $\forall x \in X$ that

$$\begin{aligned} \varphi_y(x) &= \langle L(x), y \rangle_Y \\ &= \langle x, K^*(y) \rangle_X \end{aligned}$$

so that by uniqueness of u_y we must have that $K^*(y) = u_y = L^*(y)$ proving that $L^* = K^*$. $\square$

Using the above it follows that for $L \in \text{Hom}(X, Y)$ $L^* \in \text{Hom}(Y, X)$ so that logically $(L^*)^* \in \text{Hom}(X, Y)$. Actually we have even more, it turns out that $(L^*)^* = L$.

Theorem 18.38. *Let $\langle X, \langle \rangle_X \rangle$, $\langle Y, \langle \rangle_Y \rangle$ be finite dimensional real [or complex] inner product spaces with $\dim(X), \dim(Y) \in \mathbb{N}$ and $L \in \text{Hom}(X, Y)$ then $(L^*)^* = L$*

Proof. We must consider the following cases:

real inner product spaces. As $L^* \in \text{Hom}(Y, X)$ we have by [theorem: 18.37] that $(L^*)^* \in \text{Hom}(X, Y)$ is the unique mapping such that $\forall y \in X$ we have $\forall x \in Y$ that

$$\langle L^*(x), y \rangle_X = \langle x, (L^*)^*(y) \rangle_Y,$$

further we have also for $L \in \text{Hom}(X, Y)$ by [theorem: 18.37] that

$$\begin{aligned} \langle L^*(x), y \rangle_X &= \langle y, L^*(x) \rangle_X \\ &\stackrel{[\text{theorem: 18.37}]}{=} \langle L(y), x \rangle_Y \\ &= \langle x, L(y) \rangle_Y \end{aligned}$$

so that by uniqueness we must have that

$$(L^*)^* = L$$

complex inner product spaces. As $L^* \in \text{Hom}(Y, X)$ we have by [theorem: 18.37] that $(L^*)^* \in \text{Hom}(X, Y)$ is the unique mapping such that $\forall y \in X$ we have $\forall x \in Y$ that

$$\langle L^*(x), y \rangle_X = \langle x, (L^*)^*(y) \rangle_Y,$$

further we have also for $L \in \text{Hom}(X, Y)$ by [theorem: 18.37] that

$$\begin{aligned} \langle L^*(x), y \rangle_X &= \overline{\langle y, L^*(x) \rangle_X} \\ &\stackrel{[\text{theorem: 18.37}]}{=} \overline{\langle L(y), x \rangle_Y} \\ &= \langle x, L(y) \rangle_Y \end{aligned}$$

so that by uniqueness we must have that

$$(L^*)^* = L \quad \square$$

The operator of taking the adjoint of a linear mapping is also linear as the following theorem shows.

Theorem 18.39. *Let $\langle X, \langle \rangle_X \rangle$, $\langle Y, \langle \rangle_Y \rangle$ be finite dimensional real [or complex] inner product spaces with $\dim(X), \dim(Y) \in \mathbb{N}$, $L_1, L_2 \in \text{Hom}(X, Y)$ and $\alpha \in \mathbb{R}$ [or $\mathbb{C}$] then*

$$(L_1 + \alpha \cdot L_2)^* = L_1^* + \bar{\alpha} \cdot L_2^*$$

Proof. Using [theorem: 18.37] we have $\forall y \in Y$ that for $\forall x \in X$ that

$$\langle (L_1 + \alpha \cdot L_2)(x), y \rangle_Y = \langle x, (L_1 + \alpha \cdot L_2)^* \rangle$$

further we have also that

$$\begin{aligned} \langle (L_1 + \alpha \cdot L_2)(x), y \rangle_Y &= \langle L_1(x) + \alpha \cdot L_2(x), y \rangle_Y \\ &= \langle L_1(x), y \rangle_Y + \alpha \cdot \langle L_2(x), y \rangle_Y \\ &= \langle x, L_1^*(y) \rangle_X + \alpha \cdot \langle x, L_2^*(y) \rangle_X \\ &\quad \langle x, L_1^*(y) \rangle_X + \langle x, \bar{\alpha} \cdot L_2^*(y) \rangle_X \\ &= \langle x, L_1^*(y) + \bar{\alpha} \cdot L_2^*(y) \rangle_X \\ &= \langle x, (L_1^* + \bar{\alpha} \cdot L_2^*)(y) \rangle_X \end{aligned} \tag{18.8}$$

proving by uniqueness in [theorem: 18.37] that $(L_1 + \alpha \cdot L_2)^* = L_1^* + \alpha \cdot L_2^*$

Note 18.40. Step [eq: 18.8] is valid because in the real case $\alpha \in \mathbb{R} \Rightarrow \alpha = \bar{\alpha}$ and $\alpha \cdot \langle x, L_2^*(y) \rangle_X = \langle x, \bar{\alpha} \cdot L_2^*(y) \rangle_X$ and in the complex case $\alpha \cdot \langle x, L_2^*(y) \rangle_X = \langle x, \bar{\alpha} \cdot L_2^*(y) \rangle_X$. $\square$

Theorem 18.41. Let $\langle X, \langle \rangle_X \rangle$, $\langle Y, \langle \rangle_Y \rangle$ and $\langle Z, \langle \rangle_Z \rangle$ be finite dimensional real [or complex] inner product spaces with $\dim(X), \dim(Y) \in \mathbb{N}$, $\dim(Z) \in \mathbb{N}$, $L_1 \in \text{Hom}(X, Y)$, $L_2 \in \text{Hom}(Y, Z)$ then

$$(L_2 \circ L_1)^* = L_1^* \circ L_2^*$$

Proof. Using [theorem: 18.37] we have $\forall z \in Z$ that $\forall x \in X$ that

$$\langle (L_2 \circ L_1)(x), z \rangle_Z = \langle x, (L_2 \circ L_1)^*(z) \rangle$$

further we have

$$\begin{aligned} \langle (L_2 \circ L_1)(x), z \rangle_Z &= \langle L_2(L_1(x)), z \rangle_Z \\ &\stackrel{[\text{theorem: 18.38}]}{=} \langle L_1(x), L_2^*(z) \rangle_Y \\ &\stackrel{[\text{theorem: 18.38}]}{=} \langle x, L_1^*(L_2^*(z)) \rangle_X \\ &= \langle x, (L_1^* \circ L_2^*)(z) \rangle_X \end{aligned}$$

hence we have that

$$(L_2 \circ L_1)^* = L_1^* \circ L_2^* \quad \square$$

To examine the relation between the adjoint of a linear operator and its matrix we first had to extend the concept of the transpose M^T of a matrix M [see definition: 11.332].

Definition 18.42. Let $n, m \in \mathbb{N}$ then for $M \in \mathcal{M}_{n,m}(\mathbb{K})$ we have that M^H is defined by

$$\forall (i, j) \in \{1, \dots, m\} \times \{1, \dots, n\} \quad (M^H)_{i,j} = \overline{M_{j,i}}$$

Note 18.43. If $\mathbb{K} = \mathbb{R}$ then $\forall (i, j) \in \{1, \dots, m\} \times \{1, \dots, n\} \quad (M^H)_{i,j} = \overline{M_{j,i}} = M_{j,i} = (M^T)_{i,j}$ so that

$$M^T = M^H$$

Definition 18.44. (Symmetric Matrix) Let $n, m \in \mathbb{N}$ and $M \in \mathcal{M}_{n,m}(\mathbb{K})$ then M is symmetric if

$$M = M^T$$

Definition 18.45. (Hermitian Matrix) Let $n, m \in \mathbb{N}$ and $M \in \mathcal{M}_{n,m}(\mathbb{K})$ then M is Hermitian if

$$M = M^H$$

Note 18.46. If $\mathbb{K} = \mathbb{R}$ then $M^H = M^T$ so if M is Hermitian if and only if M is symmetric

Definition 18.47. (Unitary Matrix) Let $n, m \in \mathbb{N}$ and $M \in \mathcal{M}_{n,m}(\mathbb{K})$ then M is unitary if

$$M^H \cdot M = E = M \cdot M^H$$

Note 18.48. From the above definition it follows that if M is unitary then $M^{-1} = M^H$.

Note 18.49. If $\mathbb{K} = \mathbb{R}$ then $M^H = M^T$, hence if M is unitary then $M^{-1} = M^T$.

Theorem 18.50. Let $\langle X, \langle \rangle_X \rangle, \langle Y, \langle \rangle_Y \rangle$ be finite dimensional real [or complex] inner product spaces with $\dim(X) = n, \dim(Y) = m, n, m \in \mathbb{R}, E = \{e_1, \dots, e_n\}$ a orthonormal basis for $X, F = \{f_1, \dots, f_m\}$ a orthonormal basis for Y and $L \in \text{Hom}(X, Y)$ then for $L^* \in \text{Hom}(Y, X)$ we have that

$$\mathcal{M}(L^*; F, E) = \mathcal{M}(L, E; F)^H$$

See [definition: 11.339] for $\mathcal{M}(L^*; F, E)$ and $\mathcal{M}(L, E; F)$.

Proof. We must make a distinction between the real and complex case.

real inner product space. Using [theorem: 18.30] we have $\forall(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ that

$$\begin{aligned} (\mathcal{M}(L; E, F)^H)_{i,j} &= \overline{\mathcal{M}(L; E, F)_{j,i}} \\ &\stackrel{[\text{theorem: 18.30}]}{=} \overline{\langle L(e_i), f_j \rangle_Y} \\ &\stackrel{[\text{theorem: 18.37}]}{=} \overline{\langle e_i, L^*(f_j) \rangle_X} \\ &\stackrel{\langle \cdot, \cdot \rangle_X \in \mathbb{R}}{=} \langle e_i, L^*(f_j) \rangle_X \\ &= \langle L^*(f_j), e_i \rangle_X \\ &\stackrel{[\text{theorem: 18.30}]}{=} \mathcal{M}(L^*; F, E)_{i,j} \end{aligned}$$

proving that $\mathcal{M}(L^*; F, E) = \mathcal{M}(L, E; F)^H$.

complex inner product space. Using [theorem: 18.30] we have $\forall(i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}$ that

$$\begin{aligned} (\mathcal{M}(L; E, F)^H)_{i,j} &= \overline{\mathcal{M}(L; E, F)_{j,i}} \\ &\stackrel{[\text{theorem: 18.30}]}{=} \overline{\langle L(e_i), f_j \rangle_Y} \\ &\stackrel{[\text{theorem: 18.37}]}{=} \overline{\langle e_i, L^*(f_j) \rangle_X} \\ &= \langle L^*(f_j), e_i \rangle_X \\ &\stackrel{[\text{theorem: 18.30}]}{=} \mathcal{M}(L^*; F, E)_{i,j} \end{aligned}$$

proving that $\mathcal{M}(L^*; F, E) = \mathcal{M}(L, E; F)^H$. □

Corollary 18.51. Let $\langle X, \langle \rangle_X \rangle$ be a finite dimensional real inner product spaces with $\dim(X) \in \mathbb{N}$ and $L \in \text{Hom}(X) = \text{Hom}(X, X)$ then

$$\det(L) = \det(L^*)$$

Proof. Let $\{e_1, \dots, e_n\}$ be a basis in X then

$$\begin{aligned} \det(L) &\stackrel{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}(L; E, F)) \\ &\stackrel{[\text{theorem: 11.358}]}{=} \det(\mathcal{M}(L; E, F)^T) \\ &= \det(\mathcal{M}(L; E, F)^H) \\ &\stackrel{[\text{theorem: 18.51}]}{=} \det(L^*) \end{aligned}$$

□

18.4 Spectral Theorem

18.4.1 Polynomials and linear operators

We use now the fundamental theorem of algebra to prove the spectral theorem of linear algebra, that states under which conditions a linear operator can be diagonalized. First we introduce linear operators and the relation of linear operators and polynomials.

Definition 18.52. Let X be a vector space and $L: X \rightarrow X$ a mappings then L is a **linear operator on X** if $L \in \text{Hom}(X, X) = \text{Hom}(X)$.

Theorem 18.53. Let X be a finite dimensional and $L \in \text{Hom}(X)$ [L is a linear operator] then we have the following equivalences:

1. L is a isomorphism.
2. L is injective.
3. L is surjective.

Proof.

$1 \Rightarrow 2$. This is trivial as a bijection is injective and surjective.

$1 \Rightarrow 3$. This is trivial as a bijection is injective and surjective.

$2 \Rightarrow 1$. This follows from [theorem: 11.201].

$3 \Rightarrow 1$. As L is surjective it follows from [theorem: 11.198] that $\text{rank}(L) = \dim(X)$. As by [theorem: 11.204] $\dim(X) = \dim(\ker(L)) + \text{rank}(L)$ it follows that $\dim(\ker(L)) = 0$, hence using [theorem: 11.195] we have that L is injective. So L is a bijection and as L is linear it is by [theorem: 11.202] a linear isomorphism. $\square$

Definition 18.54. Let $n \in \mathbb{N}$, X a vector space and $\{L_i\}_{i \in \{1, \dots, n\}} \subseteq \text{Hom}(X)$ a finite family of linear operators then $\bigodot_{i=1}^n L_i: X \rightarrow X$ is recursively defined by

$$\left(\bigodot_{i=1}^n L_i \right) = \begin{cases} L_1 & \text{if } n = 1 \\ L_n \circ \bigodot_{i=1}^{n-1} L_i & \text{if } 1 < n \end{cases}$$

A special case of the above is the following

Definition 18.55. Let $n \in \mathbb{N}_0$, X a vector space and $L \in \text{Hom}(X)$ then L^n is recursively defined by

$$L^n = \begin{cases} \text{Id}_X & \text{if } n = 0 \\ L \circ L^{n-1} & \text{if } 0 < n \end{cases}$$

In essence we have defined the power of linear operators.

We show now that the power of operators behaves as normal power functions.

Theorem 18.56. Let $n, m \in \mathbb{N}_0$, X a vector space and $L \in \text{Hom}(X)$ then

$$L^{n+m} = L^n \circ L^m$$

Proof. This is trivially proved by induction, so define

$$S_n = \{m \in \mathbb{N}_0 \mid L^{n+m} = L^n \circ L^m\}$$

then we have:

$0 \in S_n$. As $L^{n+0} = L^n = L^n \circ \text{Id}_X = L^n \circ L^0$ proving that $0 \in S$.

$n \in S_n \Rightarrow n+1 \in S_n$. As

$$L^{n+(m+1)} = L^{(n+m)+1} = L^{(n+m)} \circ L \underset{m \in S_n}{=} (L^n \circ L^m) \circ L = L^n \circ (L^m \circ L) = L^n \circ L^{m+1}$$

proving that $n+1 \in S_n$. □

Theorem 18.57. *Let $n \in \mathbb{N}_0$, X a vector space and $L \in \text{Hom}(X)$ then $L^n \in \text{Hom}(X)$*

Proof. This is trivially proved by induction, so define

$$S = \{n \in \mathbb{N}_0 \mid L^n \in \text{Hom}(X)\}$$

then we have:

$0 \in S$. As $L^0 = \text{Id}_X \in \text{Hom}(X)$ it follows that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ we have that $L^n \in \text{Hom}(X)$, further as $L \in \text{Hom}(X)$ it follows from [theorem: 11.180] that $L^{n+1} = L^n \circ L^1 = L^n \circ (L^1 \circ L^0) = L^n \circ (L \circ \text{Id}_x) = L^n \circ L \in \text{Hom}(X)$ proving that $n+1 \in S$. □

Theorem 18.58. *Let X be a vector space, $n \in \mathbb{N}$ and $\{L_i\}_{i \in \{1, \dots, n\}} \subseteq \text{Hom}(X)$ such that $\forall i \in \ker(L_i) = \{0\}$ then*

$$\ker\left(\bigodot_{i=1}^n L_i\right) = \{0\}$$

In other words, using [theorem: 11.195], if $\forall i \in \{1, \dots, n\}$ L_i is injective then $\bigodot_{i=1}^n L_i$ is injective.

Proof. We use induction to prove this, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{L_i\}_{i \in \{1, \dots, n\}} \subseteq \text{Hom}(X) \text{ satisfies that } \forall i \in \{1, \dots, n\} \ker(L_i) = \{0\} \text{ then } \ker\left(\bigodot_{i=1}^n L_i\right) = \{0\} \right\}$$

we have:

$1 \in S$. Let $\{L_i\}_{i \in \{1\}} \subseteq \text{Hom}(X)$ with $\ker(L_1) = \{0\}$ then we have $\ker(\bigodot_{i=1}^1 L_i) = \ker(L_1) = \{0\}$ which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{L_i\}_{i \in \{1, \dots, n+1\}} \subseteq \text{Hom}(X)$ with $\forall i \in \{1, \dots, n+1\} \ker(L_i) = \{0\}$. If $x \in \ker(\bigodot_{i=1}^{n+1} L_i)$ then we have

$$\begin{aligned} 0 &= \left(\bigodot_{i=1}^{n+1} L_i\right)(x) \\ &= \left(L_{n+1} \circ \bigodot_{i=1}^n L_i\right)(x) \\ &= L_{n+1} \left(\left(\bigodot_{i=1}^n L_i\right)(x) \right) \end{aligned}$$

hence as $\ker(L_{n+1}) = \{0\}$ we must have that $(\bigodot_{i=1}^n L_i)(x) = 0$, hence as $n \in S$, it follows that $x = 0$. So $\ker(\bigodot_{i=1}^{n+1} L_i) = \{0\}$ proving that $n+1 \in S$. □

We are now ready to introduce polynomial linear operators.

Definition 18.59. *Let $n \in \mathbb{N}_0$, X a vector space over $\mathbb{C}$, $L \in \text{Hom}(X)$ and $p \in \mathcal{P}_n$ a polynomial of order n [see definition: 17.1] then $p[L]: X \rightarrow X$ is defined by*

$$p[L](x) = \sum_{i=0}^n a_i \cdot L^i(x)$$

where $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ is the unique family with $a_n \neq 0$ such that $p(z) = \sum_{i=0}^n a_i \cdot z^i$ [see definition: 17.1] and [theorem: 17.4]. Linear operators of the form $p[L]$ where p is a polynomial of order n are called **polynomial operators of order n** .

Note 18.60. Using [theorems: 11.185 and 18.57] it follows that $p[L] \in \text{Hom}(X)$.

We prove now that the product of polynomials is equal to the composition of the associated polynomial operator. This allows us to uplift properties of polynomials to polynomial operator.

Theorem 18.61. Let X be a vector space over $\mathbb{C}$, $p, q \in \mathcal{P}_n$ polynomial operators of order n and $L \in \text{Hom}(X)$ then we have

$$(p \cdot q)[L] = p[L] \circ q[L]$$

Note 18.62. By [theorem: 17.12] it is guaranteed that $p \cdot q$ is a polynomial so that $(p \cdot q)$ makes sense.

Proof. Let p be defined by $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ and q be defined by $\{b_i\}_{i \in \{0, \dots, m\}} \subseteq \mathbb{C}$. Then by [theorem: 17.12] we have that

$$p \cdot q \text{ has coefficients } \{c_i\}_{i \in \{0, \dots, n+m\}} \text{ where } c_i = \sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\}} a_k \cdot b_l \quad (18.9)$$

Next we calculate $p[L] \circ q[L]$, let $x \in X$ then we have

$$\begin{aligned} & (p[L] \circ q[L])(x) &= \\ & p[L](q[L](x)) &= \\ & \sum_{i=1}^n a_i \cdot L^i(q[L](x)) &= \\ & \sum_{i=1}^n a_i \cdot L^i \left(\sum_{j=1}^m b_j \cdot L^j(x) \right) & \stackrel{[\text{theorem: 18.57}]}{=} \\ & \sum_{i=1}^n a_i \cdot \sum_{j=1}^m b_j \cdot L^i(L^j(x)) &= \\ & \sum_{i=1}^n \left(\sum_{j=1}^m a_i \cdot b_j \cdot L^i(L^j(x)) \right) &= \\ & \sum_{i=1}^n \left(\sum_{j=1}^m a_i \cdot b_j \cdot (L^i \circ L^j)(x) \right) & \stackrel{[\text{theorem: 18.56}]}{=} \\ & \sum_{i=1}^n \left(\sum_{j=1}^m a_i \cdot b_j \cdot L^{i+j}(x) \right) & \stackrel{[\text{theorem: 11.48}]}{=} \\ & \sum_{(i,j) \in \{0, \dots, n\} \times \{0, \dots, m\}} a_i \cdot b_j \cdot L^{i+j}(x) & \stackrel{[\text{lemma: 17.11, theorem: 11.43}]}{=} \\ & \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\}} a_k \cdot b_l \cdot L^{k+l}(x) \right) &= \\ & \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\}} a_k \cdot b_l \cdot L^i(x) \right) &= \\ & \sum_{i=0}^{n+m} \left(\sum_{(k,l) \in \{(r,s) \in \{0, \dots, n\} \times \{0, \dots, m\} \mid r+s=i\}} a_k \cdot b_l \right) \cdot L^i(x) & \stackrel{[\text{eq: 18.9}]}{=} \\ & \sum_{i=0}^{n+m} c_i \cdot L^i(x) & \stackrel{[\text{eq: 18.9}]}{=} \\ & (p \cdot q)[L](x) \end{aligned}$$

which proves that

$$p[L] \circ q[L] = (p \cdot q)[L] \quad \square$$

The above can be extended, by induction.

Theorem 18.63. *Let $n \in \mathbb{N}$, X a vector space over $\mathbb{C}$, $\{p_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}$ a finite family of polynomials and $L \in \text{Hom}(X)$ then*

$$\left(\prod_{i=1}^n p_i \right)[L] = \bigodot_{i=1}^n p_i[L]$$

Proof. Define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{p_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P} \text{ then } \left(\prod_{i=1}^n p_i \right)[L] = \bigodot_{i=1}^n p_i[L] \right\}$$

then we have:

$1 \in S$. If $\{p_i\}_{i \in \{1\}} \subseteq \mathcal{P}$ then $(\prod_{i=1}^1 p_i)[L] = p_1[L] = \bigodot_{i=1}^1 p_i[L]$ proving that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. If $\{p_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{P}$ then

$$\begin{aligned} \left(\prod_{i=1}^{n+1} p_i \right)[L] &= \left(p_{n+1} \cdot \prod_{i=1}^n p_i \right)[L] \\ &\stackrel{[\text{theorem: 18.61}]}{=} p_{n+1}[L] \circ \left(\prod_{i=1}^n p_i \right)[L] \\ &\stackrel{n \in S}{=} p_{n+1}[L] \circ \bigodot_{i=1}^n p_i[L] \\ &\stackrel{\text{def}}{=} \bigodot_{i=1}^{n+1} p_i[L] \end{aligned}$$

proving that $n+1 \in S$. $\square$

Definition 18.64. *Let X be a vector space over $\mathbb{C}$ and $L \in \text{Hom}(X)$ then we define $\mathcal{P}[L]$ by*

$$\mathcal{P}[L] = \{p[L] \mid p \in \mathcal{P}\}$$

In essence $\mathcal{P}[L]$ is the set of all the polynomial operators based on the linear operator L .

Theorem 18.65. *Let X be a vector space over $\mathbb{C}$ and $L \in \text{Hom}(X)$ then $\langle \mathcal{P}[L], \circ \rangle$ is a Abelian semi group with neutral element Id_X .*

Proof. First if $x, y \in \mathcal{P}[L]$ then there exist $p, q \in \mathcal{P}$ such that $x = p[L]$ and $y = q[L]$, so we have by [theorem: 18.61] we have that $x \circ y = p[L] \circ q[L] = (p \cdot q)[L] \in \mathcal{P}[L]$. Hence

$$\circ: \mathcal{P}[L] \times \mathcal{P}[L] \rightarrow \mathcal{P}[L] \text{ defined by } \circ[x, y] = x \circ y$$

is well defined. Further we have:

associativity. This is trivial as composition of functions is associative.

neutral element. Take $\{a_i\}_{i \in \{0\}} \subseteq \mathbb{C}$ define by $a_0 = 1$ then p defined by $p(z) = \sum_{i=0}^0 a_i \cdot z^i = a_0 \cdot z^0 = a_0$ is a element of $\mathcal{P}$. Then $p[L]$ is defined by

$$p[L](x) = \sum_{i=0}^0 a_i \cdot L^i(z) = L^0(z) = \text{Id}_X(z)$$

proving that $\text{Id}_X \in \mathcal{P}[L]$. Further if $x \in \mathcal{P}[L]$ then we have that $x \circ \text{Id}_X = x = \text{Id}_X \circ x$. Hence Id_X is the neutral element in $\langle \mathcal{P}[L], \circ \rangle$.

commutativity. If $x, y \in \mathcal{P}[L]$ then there exist $p, q \in \mathcal{P}$ such that $x = \mathcal{P}[L]$ and $y = \mathcal{P}[L]$. SO we have

$$x \circ y = p[L] \circ q[L] \stackrel{[\text{theorem: 18.61}]}{=} (p \cdot q)[L] = (q \cdot p)[L] \stackrel{[\text{theorem: 18.61}]}{=} q[L] \circ p[L] \quad \square$$

The above can be used to prove that composition is stable under permutations.

Theorem 18.66. Let $n \in \mathbb{N}$, X a vector space over $\mathbb{C}$, $L \in \text{Hom}(X)$ then if $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}[L]$ and $\sigma \in P_n$ then we have that

$$\bigodot_{i=1}^n x_i = \bigodot_{i=1}^n x_{\sigma(i)}$$

Proof. By the previous theorem [theorem: 18.65] $\circ$ can be considered the sum of the Abelian group $\langle \mathcal{P}[L], \circ \rangle$ and $\bigodot_{i=1}^n x_i$ as a finite sum. The theorem follows then from [theorem: 11.31]. $\square$

The following theorem shows how we can factorize polynomial operators.

Theorem 18.67. Let X be a vector space over $\mathbb{C}$, $L \in \text{Hom}(X)$ and $p \in \mathcal{P}$ a polynomial of order $n \in \mathbb{N}$ [in other words a non constant polynomial] then there exist a $c \in \mathbb{C} \setminus \{0\}$ and a family $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ such that that

$$p[L] = c \cdot \bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X)$$

Proof. Using [theorem: 17.34] there exist a $c \in \mathbb{C} \setminus \{0\}$ and $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ such that

$$p(z) = c \cdot \prod_{i=1}^n (z - \lambda_i) = C_c(z) \cdot \prod_{i=1}^n (z - \lambda_i) \quad (18.10)$$

Given $i \in \{1, \dots, n\}$ we have for $p_i: \mathbb{C} \rightarrow \mathbb{C}$ defined by $p_i(z) = z - \lambda_i$ that $p_i(z) = z - \lambda_i = \sum_{j=0}^1 c_{i,j} \cdot z^j$ where $c_{i,0} = -\lambda_i$ and $c_{i,1} = 1 \neq 0$ so that p_i is a polynomial of order 1. Further, as for $\{a_i\}_{i \in \{0\}} \subseteq a_i$ with $a_0 = c$ we have $\sum_{i=0}^0 a_i \cdot z^i = a_0 \cdot z^0 = C_c(z)$ it follows that C_c is a polynomial of order 0. So if we define

$$\{q_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{P} \text{ by } q_i = \begin{cases} C_c & \text{if } i = n+1 \\ p_i & \text{if } i \in \{1, \dots, n\} \end{cases}$$

then we have for $z \in \mathbb{C}$ that

$$\begin{aligned} \left(\prod_{i=1}^{n+1} q_i \right)(z) &= \prod_{i=1}^{n+1} q_i(z) \\ &= q_{n+1}(z) \cdot \prod_{i=1}^n q_i(z) \\ &= C_c(z) \cdot \prod_{i=1}^n p_i(z) \\ &= c \cdot \prod_{i=1}^n (z - \lambda_i) \\ &= p(z) \end{aligned}$$

proving that

$$p = \prod_{i=1}^{n+1} q_i \quad (18.11)$$

Now if $i \in \{1, \dots, n\}$ then we have for $x \in X$ that

$$\begin{aligned}
 q_i[L](x) &= \begin{cases} \sum_{i=0}^0 a_i \cdot L^i(x) & \text{if } i = n+1 \\ \sum_{j=0}^0 c_{i,j} \cdot L^j(x) & \text{if } i \in \{1, \dots, n\} \end{cases} \\
 &= \begin{cases} c \cdot L^0(x) & \text{if } i = n+1 \\ -\lambda_i \cdot L^0(x) + 1 \cdot L^1(x) & \text{if } i \in \{1, \dots, n\} \end{cases} \\
 &= \begin{cases} c \cdot \text{Id}_X(x) & \text{if } i = n+1 \\ L(x) - \lambda_i \cdot \text{Id}_X(x) & \text{if } i \in \{1, \dots, n\} \end{cases} \\
 &= \begin{cases} (c \cdot \text{Id}_X)(x) & \text{if } i = n+1 \\ (L - \lambda_i \cdot \text{Id}_X)(x) & \text{if } i \in \{1, \dots, n\} \end{cases}
 \end{aligned}$$

so that

$$q_i[L] = \begin{cases} c \cdot \text{Id}_X & \text{if } i = n+1 \\ (L - \lambda_i \cdot \text{Id}_X) & \text{if } i \in \{1, \dots, n\} \end{cases} \quad (18.12)$$

Hence we have that

$$\begin{aligned}
 p[L] &\stackrel{[\text{theorem: 18.11}]}{=} \left(\prod_{i=1}^{n+1} q_i \right)[L] \\
 &\stackrel{[\text{theorem: 18.63}]}{=} \bigodot_{i=1}^{n+1} q_i[L] \\
 &= q_{n+1}[L] \circ \bigodot_{i=1}^n q_i[L] \\
 &\stackrel{[\text{eq: 18.12}]}{=} (c \cdot \text{Id}_X) \circ \bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X) \\
 &= c \cdot \bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X) \\
 &\quad \square
 \end{aligned}$$

18.4.2 Eigenvectors and eigenvalues.

Definition 18.68. Let X be a vector space over a field $\mathcal{F}$ and $L \in \text{Hom}(X)$ then $x \in X$ is a **eigenvector** with **eigenvalue** $\lambda \in \mathcal{F}$ if

$$x \neq 0 \text{ and } L(x) = \lambda \cdot x$$

Note 18.69. The condition $x \neq 0$ is essential because $L(0) = 0 = \lambda \cdot 0$ for all $\lambda \in \mathcal{F}$, hence if we want to have a special meaning for λ we must exclude the case $x = 0$.

It turns out that in a non trivial finite dimensional vector spaces over $\mathbb{C}$ every $L \in \text{Hom}(X)$ has eigenvectors with eigenvalues.

Theorem 18.70. Let X be finite dimensional vector space over $\mathbb{C}$ with $\dim(X) \in \mathbb{N}$ [in other words X is not trivial] then ever $L \in \text{Hom}(X)$ has at least one eigenvector.

Proof. Let $n = \dim(X)$ then as $n \neq 0$ there exist a $x \in X$ with $x \neq 0$. Consider now the set $W = \{L^0(x), \dots, L^n(x)\} \subseteq X$ of $n+1$ vectors in X . As $\dim(X) = n$ W must be linear dependent [see theorem: 11.161]. Hence there exist a family $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ such that $\exists k \in \{0, \dots, n\}$ with $a_k \neq 0$ such that

$$\sum_{i=0}^n a_i \cdot L^i(x) = 0$$

As $k \in \{i \in \{0, \dots, n\} | a_i \neq 0\}$, $m = \max(\{i \in \{0, \dots, n\} | a_i \neq 0\})$ exist.

$$0 = \sum_{i=0}^n a_i \cdot L^i(x) = \sum_{i \in \{0, \dots, m\}} a_i \cdot L^i(x) + \sum_{i \in \{m+1, \dots, n\}} a_i \cdot L^i(x) = \sum_{i=0}^m a_i \cdot L^i(x)$$

hence we have

$$\sum_{i=0}^m a_i \cdot L^i(x) = 0 \text{ and } a_m \neq 0 \quad (18.13)$$

If $m = 0$ then $0 = \sum_{i=0}^0 a_i \cdot L^i(x) = a_0 \cdot L^0(x) = a_0 \cdot \text{Id}(x) = a_0 \cdot x$ which, as $a_0 \neq 0$ leads to the contradiction $0 = x \neq 0$. Take the polynomial $p \in \mathcal{P}$ defined by $p(z) = \sum_{i=0}^m a_i \cdot z^i$ then as $m \in \mathbb{N}$ we have by [theorem: 18.67] that there exists a $c \in \mathbb{C} \setminus \{0\}$ and a family $\{\lambda_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}$ such that

$$\sum_{i=0}^n a_i \cdot L^i = p[L] = c \cdot \bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X)$$

So we have by [eq: 18.13]

$$0 = \sum_{i=0}^n a_i \cdot L^i(x) = \left(\sum_{i=0}^n a_i \cdot L^i \right)(x) = c \cdot \left(\bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X) \right)(x)$$

which as $c \neq 0$ proves that $(\bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X))(x) = 0$ which as $x \neq 0$ proves that

$$\ker \left(\bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X) \right) \neq \{0\}$$

Assume that $\forall i \in \{1, \dots, n\} \ker(L - \lambda_i \cdot \text{Id}_X) = \{0\}$ then by [theorem: 18.58] $\ker(\bigodot_{i=1}^n (L - \lambda_i \cdot \text{Id}_X)) = \{0\}$ contradicting the above. Hence there exist a $l \in \{1, \dots, n\}$ such that $\ker(L - \lambda_l \cdot \text{Id}_X) \neq \{0\}$, so there exist a $w \in X$ with $w \neq 0$ such that $(L - \lambda_l \cdot \text{Id}_X)(w) = L(w) - \lambda_l \cdot w = 0$ or $L(w) = \lambda_l \cdot w$. So we found a eigenvector w with eigenvalue λ_l . $\square$

Definition 18.71. (eigenspace) Let X be a vector space over a field $\mathcal{F}$ and $L \in \text{Hom}(X)$ with a eigenvalue λ then the **eigenspace** of λ note by $E(L, \lambda)$ is defined by

$$E(L, \lambda) = \{x \in X | L(x) = \lambda \cdot x\}$$

Note 18.72. As $L(0) = 0 = \lambda \cdot 0$ we have that $0 \in E(L, \lambda)$ although 0 is not a eigenvector.

Theorem 18.73. Let X be a vector space over a field $\mathcal{F}$ and $L \in \text{Hom}(X)$ then if λ is a eigenvalue of L then $E(L, \lambda)$ is a non trivial sub-space of X .

Proof. As λ is a eigenvalue of L there exist a $x \in X$ such that $L(x) = \lambda \cdot x$, hence

$$\emptyset \neq E(L, \lambda) = \{x \in X | L(x) = \lambda \cdot x\} \neq \{0\}$$

If $x, y \in E(L, \lambda)$ and $\alpha \in \mathcal{F}$ then $L(x + \alpha \cdot y) = L(x) + \alpha \cdot L(y) = \lambda \cdot x + \alpha \cdot \lambda \cdot y = \lambda \cdot (x + \alpha \cdot y)$ proving that $x + \alpha \cdot y \in E(L, \lambda)$ $\square$

Theorem 18.74. Let X be a vector space over a field $\mathcal{F}$ and $L \in \text{Hom}(X)$, $n \in \mathbb{N}$ and $\{e_i\}_{i \in \{1, \dots, n\}}$ a finite family of eigenvectors with $\forall i \in \{1, \dots, n\}$ a eigenvalue λ_i such that $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have $\lambda_i \neq \lambda_j$ then $\{e_i\}_{i \in \{1, \dots, n\}}$ is linear independent. In other words if $\{e_i\}_{i \in \{1, \dots, n\}}$ is a finite family of eigenvectors with distinct eigenvalues then $\{e_i\}_{i \in \{1, \dots, n\}}$ is linear independent.

Proof. Assume that $\{e_i\}_{i \in \{1, \dots, n\}}$ is linear dependent then by [theorem: 11.129] there exists a $k \in \{1, \dots, n\}$ and $\{\alpha_i\}_{i \in \{1, \dots, k-1\}} \subseteq \mathcal{F}$ such that $e_k \in \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i$. So the finite set

$$A = \left\{ k \in \{1, \dots, n\} | \exists \{\alpha_i\}_{i \in \{1, \dots, k-1\}} \subseteq \mathcal{F} \text{ such that } e_k = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i \right\}$$

is not empty. Hence $k = \min(A)$ exist and there exist a $\{\alpha_i\}_{i \in \{1, \dots, k-1\}} \subseteq F$ such that

$$x_k = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i$$

If $\{e_i\}_{i \in \{1, \dots, k-1\}}$ is again dependent then by [theorem: 11.129] again there exist a $l \in \{1, \dots, k-1\} \subseteq \{1, \dots, n\}$ and $\{\beta_i\}_{i \in \{1, \dots, l-1\}} \subseteq F$ such that $e_l = \sum_{i \in \{1, \dots, l-1\}} \beta_i \cdot e_i$, hence $l \in A$, contradicting the fact that $k = \min(A)$ [as $l \leq k-1 < k$]. So $\{e_i | i \in \{1, \dots, k-1\}\}$ must be linear independent. Hence we have that

$$e_k = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot x_i \text{ and } \{e_i | i \in \{1, \dots, k-1\}\} \text{ is linear independent}$$

Applying then L we get

$$\begin{aligned} \lambda_k \cdot e_k &= L(e_k) \\ &= L\left(\sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot e_i\right) \\ &= \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot L(e_i) \\ &= \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot \lambda_i \cdot e_i \end{aligned}$$

As also $\lambda_k \cdot e_k = \lambda_k \cdot \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot e_i = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot \lambda_k \cdot e_i$ we have

$$\begin{aligned} 0 &= \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot \lambda_i \cdot e_i - \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot \lambda_k \cdot e_i \\ &= \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot (\lambda_i - \lambda_k) \cdot e_i \end{aligned}$$

Given that $\{e_1, \dots, e_{k-1}\}$ is linear independent we have that $\forall i \in \{1, \dots, k-1\} \alpha_i \cdot (\lambda_i - \lambda_k) = 0$ which as $\lambda_k \neq \lambda_i$ [because $k \neq i$] proves that $\forall i \in \{1, \dots, k-1\} \alpha_i = 0$. Hence $e_k = \sum_{i \in \{1, \dots, k-1\}} \alpha_i \cdot e_i = 0$ contradicting then fact that $e_k \neq 0$ because e_k is a eigenvector. So the assumption wrong and

$\{e_1, \dots, e_n\}$ is linear independent. $\square$

Theorem 18.75. Let X be a finite dimensional vector space over a field $\mathcal{F}$ and $L \in \text{Hom}(X)$ then $\text{card}(\{\lambda \in \mathcal{F} | \lambda \text{ is a eigenvalue for } L\}) \leq \dim(X)$. In other words every linear operator has at most $\dim(X)$ distinct eigenvalues.

Proof. Let $\Lambda = \{\lambda \in \mathcal{F} | \lambda \text{ is a eigenvalue of } L\} = \{\lambda \in \mathcal{F} | \exists x \in X \text{ such that } L(x) = \lambda \cdot x\}$, If $m = \text{card}(\Lambda)$ then we can write Λ as

$$\Lambda = \{\lambda_1, \dots, \lambda_m\} \text{ [meaning that } \lambda: \{1, \dots, m\} \rightarrow \Lambda \text{ is a bijection]}$$

For each $i \in \{1, \dots, m\}$ we can find by definition a eigenvector $x \in X$ such that $L(x) = \lambda_i \cdot x$ allowing us to define a family $\{x_i\}_{i \in \{1, \dots, m\}} \subseteq X$ such that $\forall i \in \{1, \dots, m\} L(x_i) = \lambda_i \cdot x_i$ [see theorem: 3.108]. Let $i, j \in \{1, \dots, m\}$ with $i \neq j$ then, as λ is a bijection, $\lambda_i \neq \lambda_j$, so by the previous theorem [theorem: 18.74] $\{x_i\}_{i \in \{1, \dots, m\}}$ is linear independent. Hence by [theorem: 11.135] $\{x_i | i \in \{1, \dots, m\}\}$ is linear independent set with $\text{card}(\{x_i | i \in \{1, \dots, m\}\}) = m$. Using then [theorem: 11.161] it follows that we must have that $m \leq \dim(X)$ or

$$\text{card}(\{\lambda \in \mathcal{F} | \lambda \text{ is a eigenvalue of } L\}) \leq \dim(X) \quad \square$$

Theorem 18.76. Let X be a finite dimensional vector space over a field $\mathcal{F}$ and $L \in \text{Hom}(X)$ with eigenvalues $\{e_1, \dots, e_n\}$, $n \in \mathbb{N}$ then for $E(L, \lambda_1, \dots, \lambda_n)$ defined by

$$E(L; \lambda_1, \dots, \lambda_n) = \text{span}\left(\bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i)\right)$$

we have:

1. $E(L; \lambda_1, \dots, \lambda_n)$ is a sub-space of X
2. $\forall i \in \{1, \dots, n\}$ $E(L, \lambda_i)$ is a sub-space of $E(L; \lambda_1, \dots, \lambda_n)$
3. $E(L; \lambda_1, \dots, \lambda_n) = \bigoplus_{i \in \{1, \dots, n\}} E(L, \lambda_i)$
4. $\dim(E(L; \lambda_1, \dots, \lambda_n)) = \sum_{i=1}^n \dim(E(L, \lambda_i))$

Proof.

1. This follows from [theorem: 11.110].
2. Let $i \in I$ then by [theorem: 18.73] $E(L, \lambda_i)$ is a sub-space of X , hence $\forall x, y \in E(L, \lambda_i)$ and $\alpha \in \mathbb{F}$ we have

$$x + \alpha \cdot y \in E(L, \lambda_i) \subseteq \bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i) \subseteq \text{span}\left(\bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i)\right) = E(L; \lambda_1, \dots, n)$$

which, as also $E(L, \lambda_i) \neq \emptyset$, proves that $E(L, \lambda_i)$ is a sub-space of $E(L; \lambda_1, \dots, \lambda_n)$.

3. Let $x \in E(L; \lambda_1, \dots, \lambda_n) = \text{span}\left(\bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i)\right)$ then there exists a $m \in \mathbb{N}$, a $\{\alpha_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{F}$ and a $\{x_i\}_{i \in \{1, \dots, m\}} \in \bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i)$ such that

$$x = \sum_{i=1}^m \alpha_i \cdot x_i$$

If $i \in \{1, \dots, m\}$ then there exist a $k_i \in \{1, \dots, n\}$ such that $x_i \in E(L, \lambda_{k_i})$, then $\alpha_i \cdot x_i \in E(L, \lambda_{k_i})$. Define now

$$\{w_{i,j}\}_{j \in \{1, \dots, n\}} \subseteq E(L; \lambda_1, \dots, \lambda_n) \text{ by } w_{i,j} = \begin{cases} 0 & \text{if } j \in \{1, \dots, n\} \setminus \{k_i\} \\ \alpha_i \cdot x_i & \text{if } j = k_i \end{cases}$$

then we have

$$\sum_{j=1}^n w_{i,j} = \sum_{j \in \{1, \dots, n\} \setminus \{k_i\}} w_{i,j} + \sum_{j \in \{k_i\}} w_{i,j} = \alpha_i \cdot x_i$$

proving that

$$\forall i \in \{1, \dots, m\} \quad \alpha_i \cdot x_i \in \sum_{j \in \{1, \dots, n\}} E(L, \lambda_j)$$

As by (2) and [theorem: 12.4] $\sum_{j \in \{1, \dots, n\}} E(L, \lambda_j)$ is a vector space it follows from the above that $x = \sum_{i=1}^m \alpha_i \cdot x_i \in \sum_{j \in \{1, \dots, n\}} E(L, \lambda_j)$. Hence we have proved that

$$E(L; \lambda_1, \dots, \lambda_n) \subseteq \sum_{j \in \{1, \dots, n\}} E(L, \lambda_j)$$

If $x \in \sum_{j=1}^n E(L, \lambda_j)$ then $x = \sum_{j=1}^n v_j$ where $v_j \in E(L, \lambda_j) \subseteq \bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i)$ so that $x \in \text{span}\left(\bigcup_{i \in \{1, \dots, n\}} E(L, \lambda_i)\right) = E(L; \lambda_1, \dots, n)$, hence $\sum_{j \in \{1, \dots, n\}} E(L, \lambda_j) \subseteq E(L; \lambda_1, \dots, n)$. Combining this with the above proves that

$$E(L; \lambda_1, \dots, \lambda_n) = \sum_{j \in \{1, \dots, n\}} E(L, \lambda_j) \quad (18.14)$$

Let $k \in \{1, \dots, n\}$ and assume that $x \in E(L, \lambda_k) \cap \left(\sum_{j \in \{1, \dots, n\} \setminus \{k\}} E(L, \lambda_j)\right)$ then

$$x \in E(L, \lambda_k) \wedge x = \sum_{j \in \{1, \dots, n\} \setminus \{k\}} v_j \text{ where } v_j \in E(L, \lambda_j) \quad (18.15)$$

Define $\{w_i\}_{i \in \{1, \dots, n\}}$ by $w_i = \begin{cases} -x & \text{if } i = k \\ v_i & \text{if } i \in \{1, \dots, n\} \setminus \{k\} \end{cases} \in E(L, \lambda_i)$ then we have

$$\begin{aligned} 0 &= \sum_{i \in \{1, \dots, n\}} w_i \\ &= \sum_{i \in \{j \in \{1, \dots, n\} | w_i \neq 0\}} w_i + \sum_{i \in \{j \in \{1, \dots, n\} | w_i = 0\}} w_i \\ &= \sum_{i \in \{j \in \{1, \dots, n\} | w_i \neq 0\}} w_i \\ &= \sum_{i \in I} w_i \end{aligned} \quad (18.16)$$

where $I = \{i \in \{1, \dots, n\} | w_i \neq 0\}$. Then $\forall i \in I$ we have $w_i \neq 0 \wedge L(w_i) = \lambda_i \cdot w_i$ or w_i is a eigen vector of L . Assume that $I \neq \emptyset$ then there exist a $k \in \{1, \dots, n\}$ such that $I = \{i_1, \dots, i_k\}$ and $\forall r, s \in \{1, \dots, k\}$ with $r \neq s$ we have $i_r \neq i_s$ so that $\lambda_{i_r} \neq \lambda_{i_s}$, hence $\{w_{i_j}\}_{j \in \{1, \dots, i_k\}}$ is a family of eigenvectors with distinct eigenvalues. Applying then [theorem: 18.74] $\{w_{i_j}\}_{j \in \{1, \dots, i_k\}}$ is linear independent contradicting that we have proved that $\sum_{i \in I} w_i = 0$ [see eq: 18.16] Hence we must have that $I = \emptyset$ and $\forall i \in \{1, \dots, n\}$ $w_i = 0$, in particular $w_k = -x = 0 \Rightarrow x = 0$. So it follows that $E(L, \lambda_k) \cap (\sum_{j \in \{1, \dots, n\} \setminus \{k\}} E(L, \lambda_j)) \subseteq \{0\}$. As $E(L, \lambda_k)$ and $\sum_{j \in \{1, \dots, n\} \setminus \{k\}} E(L, \lambda_j)$ are vector spaces we have also that $0 \in E(L, \lambda_k) \cap (\sum_{j \in \{1, \dots, n\} \setminus \{k\}} E(L, \lambda_j))$. Hence we conclude that

$$E(L, \lambda_k) \cap \left(\sum_{j \in \{1, \dots, n\} \setminus \{k\}} E(L, \lambda_j) \right) = \{0\}$$

Finally using [theorem: 12.8] together with (1),(2) and [eq: 18.14] it follows that

$$E(L; \lambda_1, \dots, \lambda_n) = \bigoplus_{i \in \{1, \dots, n\}} E(L, \lambda_i)$$

4. Using (3) and [theorem: 12.15] it follows that

$$\dim(E(L; \lambda_1, \dots, \lambda_n)) = \sum_{i=1}^n \dim(E(L, \lambda_i)) \quad \square$$

18.4.3 Upper triangular and diagonal matrices

Definition 18.77. Let X be a vector space, Y a sub-space of X and $L \in \text{Hom}(X)$ then we say that Y is **invariant under L** if $L(Y) \subseteq Y$. In other words every vector in Y is mapped to another vector in Y .

Definition 18.78. Let $n \in \mathbb{N}$, $\mathcal{F}$ a field then a matrix $M \in \mathcal{M}_{n,n}(\mathcal{F})$ is a **upper triangular matrix** if $\forall i, j \in \{1, \dots, n\}$ with $i > j$ we have $M_{i,j} = 0$. In other words a matrix is a upper triangular matrix if the matrix is empty below its diagonal.

Example 18.79. The matrix $M = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ is a upper triangular matrix because

$$M_{1,2} = M_{1,3} = M_{3,2} = 0$$

The following theorem gives necessary and sufficient conditions for the matrix of a linear operator to be upper triangular.

Theorem 18.80. Let X be a finite dimensional vector space with $\dim(X) \in \mathbb{N}$ [hence X is not trivial], $E = \{e_1, \dots, e_n\}$ a basis for X and $L \in \text{Hom}(X)$ a linear operator then we have the following equivalences:

1. $\mathcal{M}(L; E, E)$ is upper triangular

2. $\forall i \in \{1, \dots, n\}$ we have $L(e_i) \in \text{span}(\{e_k | k \in \{1, \dots, i\}\})$
 3. $\forall i \in \{1, \dots, n\}$ we have that $\text{span}(\{e_k | k \in \{1, \dots, i\}\})$ is invariant under L

Proof. Let $M = \mathcal{M}(L; E, E)$ then we have:

1 $\Rightarrow$ 2. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned} L(e_i) &= \sum_{k=1}^n M_{k,i} \cdot e_k \\ &= \sum_{k \in \{1, \dots, i\}} M_{k,i} \cdot e_k + \sum_{k \in \{i+1, \dots, n\}} M_{k,i} \cdot e_k \\ &= \sum_{k \in \{1, \dots, i\}} M_{k,i} \cdot e_k + \sum_{k \in \{i+1, \dots, n\}} 0 \cdot e_k \\ &= \sum_{k \in \{1, \dots, i\}} M_{k,i} \cdot e_k \end{aligned}$$

so that $L(e_i) \in \text{span}(\{e_k | k \in \{1, \dots, i\}\})$.

2 $\Rightarrow$ 1. Let $i \in \{1, \dots, n\}$ then, as $L(e_i) \in \text{span}(\{e_k | k \in \{1, \dots, i\}\})$, there exist a $\{\alpha_k\}_{k \in \{1, \dots, i\}}$ such that $L(e_i) = \sum_{k=1}^i \alpha_k \cdot e_k$. Define

$$\{\beta_k\}_{k \in \{1, \dots, n\}} \text{ by } \beta_k = \begin{cases} \alpha_k & \text{if } k \in \{1, \dots, i\} \\ 0 & \text{if } k \in \{i+1, \dots, n\} \end{cases}$$

then we have

$$\begin{aligned} \sum_{i=1}^n M_{k,i} \cdot e_k &= L(e_i) \\ &= \sum_{k=1}^i \alpha_k \cdot e_k \\ &= \sum_{k \in \{1, \dots, i\}} \beta_k \cdot e_k + \sum_{k \in \{i+1, \dots, n\}} 0 \cdot e_k \\ &= \sum_{k \in \{1, \dots, i\}} \beta_k \cdot e_k + \sum_{k \in \{i+1, \dots, n\}} \beta_k \cdot e_k \\ &= \sum_{k=1}^n \beta_k \cdot e_k \end{aligned}$$

As $\{e_1, \dots, e_n\}$ is a basis it follows that $\forall k \in \{1, \dots, n\}$ $M_{k,i} = \beta_k$, hence $\forall k \in \{i+1, \dots, n\}$ we have $M_{k,i} = \beta_k = 0$. As $i \in \{1, \dots, n\}$ was chosen arbitrary it follows that $M_{k,i} = 0$ whenever $k > i$ proving that M is upper triangular.

2 $\Rightarrow$ 3. Given $i \in \{1, \dots, n\}$ take $x \in \text{span}(\{e_l | l \in \{1, \dots, i\}\})$ then there exist a $\{x_k\}_{k \in \{1, \dots, i\}}$ such that $x = \sum_{k=1}^i x_k \cdot e_k$, hence

$$L(x) = L\left(\sum_{k=1}^i x_k \cdot e_k\right) = \sum_{k=1}^i x_k \cdot L(e_k)$$

so that

$$L(x) \in \text{span}(\{L(e_k) | k \in \{1, \dots, i\}\}) \quad (18.17)$$

Now for every $k \in \{1, \dots, i\} \subseteq \{1, \dots, n\}$ we have by (2) that

$$L(e_k) \in \text{span}(\{e_l | l \in \{1, \dots, k\}\}) \subseteq_{[\text{theorem: 11.109}]} \text{span}(\{e_l | l \in \{1, \dots, i\}\})$$

so that $\{L(e_k) | k \in \{1, \dots, i\}\} \subseteq \text{span}(\{e_l | l \in \{1, \dots, i\}\})$ or using [theorem: 11.109] again

$$\text{span}(\{L(e_k) | k \in \{1, \dots, i\}\}) \subseteq \text{span}(\text{span}(\{e_l | l \in \{1, \dots, i\}\})) \stackrel{[\text{theorem: 11.112}]}{=} \text{span}(e_l | l \in \{1, \dots, i\})$$

which combined with [eq: 18.17] proves that

$$L(x) \subseteq \text{span}(e_l | l \in \{1, \dots, i\})$$

As $x \in \text{span}(\{e_k | k \in \{1, \dots, i\}\})$ we have from the above that

$$L(\text{span}(\{e_k | k \in \{1, \dots, i\}\})) \subseteq \text{span}(\{e_k | k \in \{1, \dots, i\}\})$$

showing that $\text{span}(\{e_k | k \in \{1, \dots, i\}\})$ is invariant under L .

3 $\Rightarrow$ 2. Let $i \in \{1, \dots, n\}$ then as $e_i \in \text{span}(\{e_l | l \in \{1, \dots, n\}\})$ and $\text{span}(\{e_l | l \in \{1, \dots, n\}\})$ is invariant under L we have that $L(e_i) \in \text{span}(\{e_l | l \in \{1, \dots, n\}\})$. $\square$

We show now that in finite dimensional non trivial vector spaces over $\mathbb{C}$ every linear operator can have a upper triangular matrix.

Theorem 18.81. *Let $n \in \mathbb{N}$, X a finite dimensional vector space over $\mathbb{C}$ with $\dim(X) = n$ [hence X is not trivial] and $L \in \text{Hom}(X)$ then there exist a basis $\{e_1, \dots, e_n\}$ for X such that $\mathcal{M}(L; E, E)$ is a upper triangular matrix.*

Proof. We prove this by induction, so define

$S = \{n \in \mathbb{N} | \text{If } X \text{ is a finite dimensional vector space over } \mathbb{C} \text{ with } m = \dim(X), 0 < m \leq n \text{ then for } L \in \text{Hom}(X) \text{ there exist a basis } E = \{e_1, \dots, e_m\} \text{ such that } \mathcal{M}(L; E, E)\}$

then we have:

1 $\in S$. If $\dim(X) = 1$ then there exist a basis $E = \{e_1\}$ of X and for $L \in \text{Hom}(X)$ the matrix $\mathcal{M}[L; E, E]$ is a 1×1 matrix which is automatically upper diagonal, hence $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let X be a finite dimensional basis with $1 < \dim(X) \leq n + 1$ and $L \in \text{Hom}(X)$. For $\dim(X)$ we have either:

$\dim(X) < n + 1$. Then $\dim(X) \leq n$ so that, as $n \in S$, there exist a basis $E = \{e_1, \dots, e_{\dim(X)}\}$ such that $\mathcal{M}(L; E, E)$ is upper triangular. Hence in this case we have that $n + 1 \in S$

$\dim(X) = n + 1$. Using [theorem: 18.70] there exist a eigenvector $x \in X$ with eigenvalue λ . So $(L - \lambda \cdot \text{Id}_X)(x) = 0$ and $x \neq 0$ hence

$$\ker(L - \lambda \cdot \text{Id}_X) \neq \{0\}$$

By [theorem: 11.193] $\ker(L - \lambda \cdot \text{Id}_X)$ is a sub-space of X , so that by [theorem: 11.159] $\ker(L - \lambda \cdot \text{Id}_X)$ is finite dimensional and together with the above we have

$$0 < \dim(\ker(L - \lambda \cdot \text{Id}_X)) \leq \dim(X)$$

Using [theorem: 11.204] we have that

$$\begin{aligned} \dim(X) &= \dim(\ker(L - \lambda \cdot \text{Id}_X)) + \text{rank}(L - \lambda \cdot \text{Id}_X) \\ &= \dim(\ker(L - \lambda \cdot \text{Id}_X)) + \dim((L - \lambda \cdot \text{Id}_X)(X)) \end{aligned}$$

proving by the above that

$$\dim((L - \lambda \cdot \text{Id}_X)(X)) < \dim(X)$$

Summarized

$U = (L - \lambda \cdot \text{Id}_X)(X)$ is a vector space [see theorem: 11.193] with $\dim(U) < \dim(X)$

If $u \in U$ then $L(u) = L(u) - \lambda \cdot u + \lambda \cdot u = (L - \lambda \cdot \text{Id}_X)(u) + \lambda \cdot u \in U$ [as $(L - \lambda \cdot \text{Id}_X)(u) \in (L - \lambda \cdot \text{Id}_X)(X) = U$ and U is a vector space], hence we have that

$$L(U) \subseteq U$$

From the above combined with the fact that the linear mapping to a subspace is linear [see theorem: 11.169] it follows that

$$L|_U \in \text{Hom}(U)$$

Hence, as $\dim(U) < \dim(X) = n + 1$ implies $\dim(U) \leq n$ and $n \in S$, we have that there exist a basis $V = \{v_1, \dots, v_{\dim(U)}\}$ for U such that $\mathcal{M}(L|_U; V, V)$ is upper triangular. Applying then the previous theorem [theorem: 18.80] it follows that

$$\forall j \in \{1, \dots, \dim(U)\} \quad L(v_j) = L|_U(v_j) \in \text{span}(\{v_k | k \in \{1, \dots, j\}\}) \quad (18.18)$$

As $\{v_1, \dots, v_{\dim(U)}\}$ is linear independent we can use [theorem: 11.157] to extend $\{v_1, \dots, v_{\dim(U)}\}$ to a basis $\{w_1, \dots, w_{n+1}\}$ where $\forall i \in \{1, \dots, \dim(U)\}$ we have $w_i = v_i$. Using the above on [eq: 18.18] results in

$$\forall j \in \{1, \dots, \dim(U)\} \quad \text{we have } L(w_j) \in \text{span}(\{w_k | k \in \{1, \dots, j\}\}) \quad (18.19)$$

Let $j \in \{\dim(U) + 1, \dots, n + 1\}$ then we have

$$L(w_j) = (L - \lambda \cdot \text{Id}_X)(w_j) + \lambda \cdot \text{Id}_X(w_j) = (L - \lambda \cdot \text{Id}_X)(w_j) + \lambda \cdot w_j \quad (18.20)$$

As

$$\begin{aligned} (L - \lambda \cdot \text{Id}_X)(w_j) &\in (L - \lambda \cdot \text{Id}_X)(X) \\ &= U \\ &= \text{span}(\{v_k | k \in \{1, \dots, \dim(U)\}\}) \\ &= \text{span}(\{w_k | k \in \{1, \dots, \dim(U)\}\}) \\ &\subseteq_{[\text{theorem: 11.109}]} \text{span}(\{w_k | k \in \{1, \dots, j\}\}) \end{aligned}$$

and $\text{span}(\{w_k | k \in \{1, \dots, j\}\})$ is a vector space [see theorem: 11.110] it follows from [eq: 18.20] that $L(w_j) \in \text{span}(\{w_k | k \in \{1, \dots, j\}\})$. Hence

$$\forall j \in \{\dim(U) + 1, \dots, n + 1\} \quad \text{we have } L(w_j) \in \text{span}(\{w_k | k \in \{1, \dots, j\}\})$$

Combining the above with [eq: 18.19] that

$$\forall j \in \{1, \dots, n + 1\} \quad \text{we have } L(w_j) \in \text{span}(\{w_k | k \in \{1, \dots, j\}\})$$

Applying then the previous theorem [theorem: 18.80] again it follows that for the basis $W = \{w_1, \dots, w_n\}$ of X we have

$$\mathcal{M}(L; W, W) \text{ is a upper triangular matrix}$$

Hence we conclude that $n + 1 \in S$. □

We can actual ensure that the basis that makes the matrix of a linear operator upper diagonal is orthonormal.

Theorem 18.82. (Schur's Theorem) *Let $n \in \mathbb{N}$, X a finite dimensional vector space over $\mathbb{C}$ with $\dim(X) = n$ [hence X is not trivial] and $L \in \text{Hom}(X)$ then there exist a orthonormal basis $\{e_1, \dots, e_n\}$ for X such that $\mathcal{M}(L; E, E)$ is a upper triangular matrix.*

Proof. Using the previous theorem [theorem: 18.81] there exist a basis $B = \{b_1, \dots, b_n\}$ of X such that

$$\mathcal{M}(L; B, B) \text{ is upper triangular}$$

By [theorem: 18.80] it follows then that B satisfied

$$\forall i \in \{1, \dots, n\} \quad \text{we have } L(\text{span}(\{b_k | k \in \{1, \dots, i\}\})) \subseteq \text{span}(\{b_k | k \in \{1, \dots, i\}\}) \quad (18.21)$$

We can now use the Gram-Schmidt procedure [see theorems: 18.31] to find a orthonormal set $E = \{e_1, \dots, e_n\}$ such that

$$\forall i \in \{1, \dots, n\} \quad \text{span}(\{e_1, \dots, e_i\}) = \text{span}(\{b_1, \dots, b_i\}) \quad (18.22)$$

As $\{e_1, \dots, e_n\}$ is orthonormal it follows from [theorem: 18.33] that

$$E = \{e_1, \dots, e_n\} \text{ is a basis of } X$$

Substituting [eq: 18.22] in [eq: 18.21] results in

$$\forall i \in \{1, \dots, n\} \quad \text{we have } L(\text{span}(\{e_k | k \in \{1, \dots, i\}\})) \subseteq \text{span}(\{e_k | k \in \{1, \dots, i\}\})$$

which by [theorem: 18.80] proves that

$$\mathcal{M}(L; E, E) \text{ is a upper diagonal matrix}$$

□

18.4.4 Diagonalization and spectral theorems

A special case of a upper triangular matrix is a diagonal matrix.

Definition 18.83. Let $n \in \mathbb{N}$, $\mathcal{F}$ a field then a matrix $M \in \mathcal{M}_{n,n}(\mathcal{F})$ is **diagonal** if $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ $M_{i,j} = 0$. In other words a matrix is diagonal if all no diagonal elements are 0.

Example 18.84. The matrix $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ is diagonal because

$$M_{2,1} = M_{3,1} = M_{1,2} = M_{3,2} = M_{1,3} = M_{2,3} = 0$$

Definition 18.85. (Self-adjoint linear operators) Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space then $L \in \text{Hom}(X) = \text{Hom}(X, X)$ is **self adjoint**

$$L = L^*$$

where L^* is defined in [theorem: 18.37] [so $L^* \in \text{Hom}(X, X)$ is the **unique** linear operator such that $\forall y \in X$ we have $\forall x \in X$ that $\langle x, L^*(y) \rangle = \langle L(x), y \rangle$]. In other words L is self-adjoint iff $\forall x, y \in X$ we have $\langle x, L(y) \rangle = \langle L(x), y \rangle$.

Example 18.86. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space then Id_X is a self-adjoint transformation.

Proof. Using [theorem: 11.183] $\text{Id}_X \in \text{Hom}(X)$, further $\forall x, y \in X$ we have

$$\langle x, \text{Id}_X(y) \rangle = \langle x, y \rangle = \langle \text{Id}_X(x), y \rangle$$

□

Theorem 18.87. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space with a basis $E = \{e_1, \dots, e_n\}$, $n \in \mathbb{N}$ and $L \in \text{Hom}(X)$ then

L is self-adjoint

$$\Updownarrow$$

$\mathcal{M}(L; E, E)$ is Hermitian

Proof.

$\Rightarrow$. This follows from

$$\begin{aligned} \mathcal{M}(L; E, E) & \stackrel{L=L^*}{=} \mathcal{M}(L^*; E, E) \\ & \stackrel{[\text{theorem: 18.50}]}{=} \mathcal{M}(L; E, E)^H \end{aligned}$$

so that $\mathcal{M}(L; E, E)$ is Hermitian.

$\Leftarrow$. Let $x \in X$ then there exist a $\{x_i\}_{i \in \{1, \dots, n\}} \in \mathbb{K}$ such that $x = \sum_{i=1}^n x_i \cdot e_i$. Let $k \in \{1, \dots, n\}$ then we have

$$\begin{aligned} (L(x))_k &= \sum_{i=1}^n \mathcal{M}(L; E, E)_{k,i} \cdot x_i \\ &= \sum_{i=1}^n (\mathcal{M}(L; E, E)^H)_{k,i} \cdot x_i \\ &\stackrel{[\text{theorem: 18.50}]}{=} \sum_{i=1}^n \mathcal{M}(L^*; E, E)_{k,i} \cdot x_i \\ &= (L^*(x))_k \end{aligned}$$

proving that $L(x) = L^*(x)$. As $x \in X$ was chosen arbitrary it follows that $L = L^*$ hence L is self-adjoint. $\square$

Corollary 18.88. *Let $\langle X, \langle \rangle \rangle$ be a real finite dimensional inner product space and $L \in \text{Hom}(X)$ a linear operator such that there exist a basis $\{e_1, \dots, e_n\}$ so that $\mathcal{M}(L; E, E)$ is diagonal then L is self-adjoint/*

Proof. As $\mathcal{M}(L; E, E)$ is real and diagonal we have $\forall i, j \in \{1, \dots, n\}$ that

$$\begin{aligned} (\mathcal{M}(L; E, E)^H)_{i,j} &= (\mathcal{M}(L; E, E)^*)_{j,i} \\ &= \mathcal{M}(L; E, E)_{j,i} \\ &= \mathcal{M}(L; E, E)_{i,i} \cdot \delta_{i,j} \\ &= \mathcal{M}(L; E, E)_{i,j} \end{aligned}$$

proving that $\mathcal{M}(L; E, E)$ is Hermitian, hence by [theorem: 18.87] L is self-adjoint. $\square$

All the eigenvalues of a self-adjoint mapping are real as is show in the next theorem.

Theorem 18.89. *Let $\langle X, \langle \rangle \rangle$ be a real [or compact] finite dimensional inner product space and $L \in \text{Hom}(X)$ then every eigenvalue of L is a real number.*

Proof. () If $\langle X, \langle \rangle \rangle$ is a real inner product space then the result is trivial, so consider the case of complex inner product space. Let $x \in X$ be a eigenvector with eigenvalue λ . Then we have $x \neq 0 \Rightarrow \|x\| > 0$ and $L(x) = \lambda \cdot x$ so that

$$\begin{aligned} \lambda &= \frac{\lambda \cdot \|x\|^2}{\|x\|^2} \\ &= \frac{\lambda \cdot \langle x, x \rangle}{\|x\|^2} \\ &= \frac{\langle \lambda \cdot x, x \rangle}{\|x\|^2} \\ &= \frac{\langle L(x), x \rangle}{\|x\|^2} \\ &= \frac{\langle x, L^*(x) \rangle}{\|x\|^2} \\ &\stackrel{L=L^*}{=} \frac{\langle x, L(x) \rangle}{\|x\|^2} \\ &= \frac{\langle x, \lambda \cdot x \rangle}{\|x\|^2} \\ &= \frac{\bar{\lambda} \cdot \langle x, x \rangle}{\|x\|^2} \\ &= \frac{\bar{\lambda} \cdot \|x\|^2}{\|x\|^2} \\ &= \bar{\lambda} \end{aligned}$$

proving that $\lambda \in \mathbb{R}$. $\square$

Theorem 18.90. *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space with $\dim(X) \in \mathbb{N}$ [so X is non trivial] and $L \in \text{Hom}(X)$ a self-adjoint operator then we have*

$$\begin{aligned} L &= 0 \\ &\Updownarrow \\ \forall x \in X \text{ we have } &\langle L(x), x \rangle = 0 \end{aligned}$$

Proof. The complex case was already prove in [theorem: 18.12] so we must only prove the real case.

$\Rightarrow$. Let $x \in X$ then $\langle L(x), x \rangle = \langle 0(x), x \rangle = \langle 0, x \rangle = 0$.

$\Leftarrow$. Let $x, y \in X$ then we have:

$$\begin{aligned}
 0 &= \langle L(x+y), x+y \rangle - \langle L(x-y), x-y \rangle \\
 &= \langle L(x) + L(y), x+y \rangle - \langle L(x) - L(y), x-y \rangle \\
 &= \langle L(x), x \rangle + \langle L(x), y \rangle + \langle L(y), x+y \rangle - \langle L(x), x \rangle + \langle L(x), y \rangle - \langle L(y), x-y \rangle \\
 &= 2 \cdot \langle L(x), y \rangle + \langle L(y), x+y \rangle - \langle L(y), x-y \rangle \\
 &= 2 \cdot \langle L(x), y \rangle + \langle L(y), x \rangle + \langle L(y), y \rangle - \langle L(y), x \rangle + \langle L(y), y \rangle \\
 &= 2 \cdot \langle L(x), y \rangle + 2 \cdot \langle L(y), x \rangle \stackrel{\text{real case}}{=} \\
 &= 2 \cdot \langle L(x), y \rangle + 2 \cdot \langle x, L(y) \rangle \\
 &= 2 \cdot \langle x, L^*(y) \rangle + 2 \cdot \langle x, L(y) \rangle \stackrel{L+L^*}{=} \\
 &= 2 \cdot \langle x, L(y) \rangle + 2 \cdot \langle x, L(y) \rangle \\
 &= 4 \cdot \langle x, L(y) \rangle
 \end{aligned}$$

so $\langle L(x), y \rangle = 0$. Take $y = L(x)$ then $\langle L(x), L(x) \rangle = 0$ so that $L(x) = 0$ proving, as $x \in X$ was chosen arbitrary, that $L = 0$. $\square$

Definition 18.91. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces then $L \in \text{Hom}(X, Y)$ is a **linear isometry** if $\forall x \in X \ \|L(x)\|_Y = \|x\|_X$. Is additional L is a bijection [so that L is a linear isomorphism] then L is called a **linear isometric isomorphism**

Theorem 18.92. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $L \in \text{Hom}(X, Y)$ a linear isometry then L is injective.

Proof. If $L(x) = L(y)$ then $L(x - y) = L(x) - L(y) = 0$ so that $\|x - y\|_X = \|L(x - y)\|_Y = \|0\|_Y = 0$ so that $x = y$. $\square$

Theorem 18.93. Let $\langle X, \|\cdot\|_X \rangle, \langle Y, \|\cdot\|_Y \rangle$ be normed spaces and $L \in \text{Hom}(X, Y)$ a linear isometric isomorphism then $L^{-1} \in \text{Hom}(Y, X)$ is a linear isometric isomorphism.

Proof. Using [theorem: 11.178] L^{-1} is a linear isomorphism [hence linear and bijective]. Let $y \in Y$ then $L^{-1}(y) \in X$ so that $\|L^{-1}(y)\|_X = \|L(L^{-1}(y))\|_Y = \|y\|_Y$. $\square$

Definition 18.94. Let $\langle X, \langle \cdot, \cdot \rangle \rangle$ be a real [or complex] finite dimensional inner product space then $U \in \text{Hom}(X)$ is **unitary** if $U^* \circ U = \text{Id}_X = U \circ U^*$.

Actually the above definition can be simplified as is shown in the following theorem.

Theorem 18.95. Let $\langle X, \langle \cdot, \cdot \rangle \rangle$ be a real [or complex] finite dimensional inner product space and $U \in \text{Hom}(X)$ then

$$U \text{ is unitary} \Leftrightarrow U^* \circ U = \text{Id}_X$$

Proof.

$\Rightarrow$. This is trivial given the definition of a unitary operator.

$\Leftarrow$. Let $x \in \ker(U)$ then $U(x) = 0$ so that

$$x = \text{Id}_X(x) = (U^* \circ U)(x) = U^*(U(x)) = U^*(0) \stackrel{U^* \in \text{hom}(X)}{=} 0$$

which proves that U is injective. Applying then [corollary: 11.202] prove that

$$U \text{ is a linear isomorphism (hence bijective)}$$

So we have that $U^{-1} = \text{Id}_X \circ U^{-1} = (U^* \circ U) \circ U^{-1} = U^* \circ (U \circ U^{-1}) = U^* \circ \text{Id}_x = U^*$ from which it follows that $U \circ U^* = U \circ U^{-1} = \text{Id}_X$. Hence U is a unitary operator by definition. $\square$

Theorem 18.96. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space and $U \in \text{Hom}(X)$ then we have:

$$\begin{aligned} U \text{ is unitary} \\ \Updownarrow \\ \forall x, y \in X \text{ we have } \langle U(x), U(y) \rangle = \langle x, y \rangle \end{aligned}$$

Proof.

$\Rightarrow$. Let $x, y \in X$ then we have

$$\begin{aligned} \langle x, y \rangle &= \langle x, \text{Id}_X(y) \rangle \\ &= \langle x, (U^* \circ U)(y) \rangle \\ &= \langle x, U^*(U(y)) \rangle \\ &= \langle U(x), U(y) \rangle \end{aligned}$$

$\Leftarrow$. Given $x, y \in X$ we have

$$\begin{aligned} \langle x, (U^* \circ U)(y) \rangle &= \langle x, U^*(U(y)) \rangle \\ &= \langle U(x), U(y) \rangle \\ &= \langle x, y \rangle \\ &= \langle x, \text{Id}_X(y) \rangle \end{aligned}$$

Hence we have given $y \in X$ for $(U^* \circ U)(y), \text{Id}_X(y)$ that $\forall x \in X \langle x, (U^* \circ U)(y) \rangle = \langle x, \text{Id}_X(y) \rangle$ so that by [theorems: 18.3, 18.10] $(U^* \circ U)(y) = \text{Id}_X(y)$ which proves that

$$U^* \circ U = \text{Id}_X \quad (18.23)$$

Applying then [theorem: 18.95] proves that U is a unitary operator. $\square$

There is a relation between a unitary operators and linear isometries.

Theorem 18.97. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space and $U \in \text{Hom}(X)$ a unitary operator then we have

1. $U^* = U^{-1}$, $U = (U^*)^{-1}$ and U^* , U^{-1} are unitary.
2. U is a linear isomorphism
3. U is a linear isometric isomorphism [in the normed space with norm $\|x\| = \sqrt{\langle x, x \rangle}$]

Proof.

1. As $U^* \circ U = \text{Id}_X = U \circ U^*$ it follows that $U^{-1} = U^*$ and $(U^*)^{-1} = U$. Further we have

$$(U^*)^* \circ U^* \underset{[\text{theorem: 18.38}]}{=} U \circ U^* = \text{Id}_X = U^* \circ U \underset{[\text{theorem: 18.38}]}{=} U^* \circ (U^*)^*$$

proving that $U^* = U^{-1}$ is unitary.

2. As $U^* \circ U = \text{Id}_X = U \circ U^*$ it follows that U is a bijection, hence as U is linear it follows that U is a linear isomorphism.

3. Using (1) we have that U is a bijection, further $\forall x \in X$ we have that

$$\|U(x)\| = \sqrt{\langle U(x), U(x) \rangle} \underset{[\text{theorem: 18.96}]}{=} \sqrt{\langle x, x \rangle} = \|x\|$$

proving that U is a linear isometric isomorphism. $\square$

Example 18.98. We have the following examples of unitary operators:

1. If $\langle X, \langle \rangle \rangle$ is a real [or complex] inner product space then:
 - a. Id_X is unitary operator
 - b. $-\text{Id}_X$ is a unitary operator

2. If $\langle X, \langle \rangle \rangle$ is a real [or complex] inner product space then $i \cdot \text{Id}_X$ is a unitary operator.

Note 18.99. In general $i \cdot \text{Id}_X$ is not a self-adjoint operator, for if $i \cdot \text{Id}_X$ was a self-adjoint operator then we would have $i \cdot \langle x, y \rangle = \langle i \cdot \text{Id}_X(x), y \rangle = \langle x, (i \cdot \text{Id}_X)^*(y) \rangle = \langle x, i \cdot \text{Id}_X(y) \rangle = -i \cdot \langle x, y \rangle$ from which it would follow that $\langle x, y \rangle = 0$ for every $x, y \in X$. This can only be true if X is the trivial vector space $\{0\}$. Hence in general it is not true that a unitary operator is self-adjoint.

Proof.

1. We have:

a. Because $\text{Id}_X \circ (\text{Id}_X)^* = (\text{Id}_X)^* \stackrel{[\text{example: 18.86}]}{=} \text{Id}_X = (\text{Id}_X)^* = (\text{Id}_X)^* \circ \text{Id}_X$ it follows that Id_X is a unitary operator.

b. Let $x, y \in X$ then we have $\langle -\text{Id}_X(x), -\text{Id}_X(y) \rangle = \langle \text{Id}_X(x), \text{Id}_X(y) \rangle = \langle x, y \rangle$ so that by [theorem: 18.96] $-\text{Id}_X$ is a unitary operator.

2. Given $x, y \in X$ we have $\langle i \cdot \text{Id}_X(x), i \cdot \text{Id}_X(y) \rangle = i \cdot (-i) \cdot \langle \text{Id}_X(x), \text{Id}_X(y) \rangle = \langle x, y \rangle$, hence, using [theorem: 18.96], $i \cdot \text{Id}_X$ is a unitary operator. $\square$

Theorem 18.100. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space and $U \in \text{Hom}(X)$ a unitary transformation then $\det(U^*) \stackrel{[\text{theorem: 18.51}]}{=} \det(U) = \pm 1$.

Proof. We have

$$\begin{aligned} 1 & \stackrel{[\text{theorem: 11.313}]}{=} \det(\text{Id}_X) \\ & = \det(U^* \circ U) \\ & \stackrel{[\text{theorem: 11.315}]}{=} \det(U^*) \cdot \det(U) \\ & \stackrel{[\text{theorem: 18.51}]}{=} \det(U) \cdot \det(U) \\ & = (\det(U))^2 \end{aligned}$$

so that $\det(U) = \pm 1$. $\square$

The following theorem shows the relation between a unitary transformation and a unitary matrix.

Theorem 18.101. Let $n \in \mathbb{N}$, $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner vector space with $\dim(X) = n$, $B = \{b_1, \dots, b_n\}$ a basis for X and $U \in \text{Hom}(X)$ then

U is a unitary transformation

$\Updownarrow$

$\mathcal{M}(U; B, B)$ is a unitary matrix [see definition: 18.47]

Proof.

$\Rightarrow$. We have

$$\begin{aligned} \mathcal{M}(U; B, B)^H \cdot \mathcal{M}(U; B, B) & \stackrel{[\text{theorem: 18.50}]}{=} \mathcal{M}(U^*; B, B) \cdot \mathcal{M}(U; B, B) \\ & \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(U^* \circ U; B, B) \\ & = \mathcal{M}(\text{Id}_X; B, B) \\ & \stackrel{[\text{theorem: 11.325}]}{=} E \\ & = \mathcal{M}(\text{Id}_X; B, B) \\ & = \mathcal{M}(U \circ U^*; B, B) \\ & \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(U; B, B) \cdot \mathcal{M}(U^*; B, B) \\ & \stackrel{[\text{theorem: 18.50}]}{=} \mathcal{M}(U; B, B) \cdot \mathcal{M}(U; B, B)^H \end{aligned}$$

which proves that $\mathcal{M}(U; B, B)$ is a unitary matrix.

$\Leftarrow$. Let $x \in X$ then there exist a $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ such that $x = \sum_{i=1}^n x_i \cdot b_i$. Then we have

$$\begin{aligned}
 ((U^* \circ U)(x))_i &= \sum_{j=1}^n \mathcal{M}(U^* \circ U; B, B)_{i,j} \cdot x_j \\
 &\stackrel{[\text{theorem: 18.50}]}{=} \sum_{j=1}^n (\mathcal{M}(U^*; B, B) \cdot \mathcal{M}(U; B, B))_{i,j} \cdot x_j \\
 &\stackrel{[\text{theorem: 18.50}]}{=} \sum_{j=1}^n (\mathcal{M}(U; B, B)^H \cdot \mathcal{M}(U; B, B))_{i,j} \cdot x_j \\
 &= \sum_{j=1}^n E_{i,j} \cdot x_j \\
 &= \sum_{j=1}^n \delta_{i,j} \cdot x_j \\
 &= x_i
 \end{aligned}$$

proving that $(U^* \circ U)(x) = x$, hence $U^* \circ U = \text{Id}_X$. $\square$

Definition 18.102. Let $n \in \mathbb{N}$, X a finite dimensional vector space with two bases $E = \{e_1, \dots, e_n\}$ and $F = \{f_1, \dots, f_n\}$ then the mapping L defined by $\forall i \in \{1, \dots, n\} L(e_i) = f_i$ [see theorem: 11.337] is called a coordinate transformation from E to F .

Theorem 18.103. Let $n \in \mathbb{N}$, X a finite dimensional vector space with bases $E = \{e_1, \dots, e_n\}$ and $F = \{f_1, \dots, f_n\}$ then for the linear mappings $U, V \in \text{Hom}(X)$ defined by $\forall i \in \{1, \dots, n\} U(e_i) = f_i$ and $V(f_i) = e_i$ [in other words U is the coordinate transformation from U to V and V is the coordinate transformation from V to U] we have:

1. $U \circ V = \text{Id}_X = V \circ U$ [or equivalently $U = V^{-1}$ and $V = U^{-1}$]
2. $\mathcal{M}(U; E, E) = \mathcal{M}(V; F, F)^{-1}$ and $\mathcal{M}(V; F, F) = \mathcal{M}(U; E, E)^{-1}$
3. $\mathcal{M}(V; E, E) = \mathcal{M}(U; E, E)^{-1}$
4. If $L \in \text{Hom}(X)$ then

$$\begin{aligned}
 \mathcal{M}(L; E, E) &= \mathcal{M}(U; E, E) \cdot \mathcal{M}(L; F, F) \cdot \mathcal{M}(U; E, E)^{-1} \\
 \mathcal{M}(L; F, F) &= \mathcal{M}(U; E, E)^{-1} \cdot \mathcal{M}(L; E, E) \cdot \mathcal{M}(U; E, E)
 \end{aligned}$$

Proof.

1. Let $x \in X$ then there exist a unique $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ and $\{x'_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ such that $x = \sum_{i=1}^n x_i \cdot e_i$ and $x = \sum_{i=1}^n x'_i \cdot f_i$. Hence we have

$$\begin{aligned}
 (U \circ V)(x) &= U(V(x)) \\
 &= U\left(V\left(\sum_{i=1}^n x'_i \cdot f_i\right)\right) \\
 &= U\left(\sum_{i=1}^n x'_i \cdot V(f_i)\right) \\
 &= U\left(\sum_{i=1}^n x'_i \cdot e_i\right) \\
 &= \sum_{i=1}^n x'_i \cdot U(e_i) \\
 &= \sum_{i=1}^n x'_i \cdot f_i \\
 &= x \\
 &= \text{Id}_X(x)
 \end{aligned}$$

proving that

$$U \circ V = \text{Id}_X$$

Further we have

$$\begin{aligned} (V \circ U)(x) &= V(U(x)) \\ &= V\left(U\left(\sum_{i=1}^n x_i \cdot e_i\right)\right) \\ &= V\left(\sum_{i=1}^n x_i \cdot U(e_i)\right) \\ &= V\left(\sum_{i=1}^n x_i \cdot f_i\right) \\ &= \sum_{i=1}^n x_i \cdot V(f_i) \\ &= \sum_{i=1}^n x_i \cdot e_i \\ &= x \\ &= \text{Id}_X(x) \end{aligned}$$

so that

$$V \circ U = \text{Id}_X$$

2. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned} \sum_{l=1}^n E_{l,i} \cdot f_l &= \sum_{l=1}^n \delta_{l,j} \cdot f_l \\ &= f_i \\ &= U(e_i) \\ &\stackrel{[\text{definition: 11.339}]}{=} \sum_{k=1}^n \mathcal{M}(U; E, E)_{k,i} \cdot e_k \\ &= \sum_{k=1}^n \mathcal{M}(U; E, E)_{k,i} \cdot V(f_k) \\ &= \sum_{k=1}^n \mathcal{M}(U; E, E)_{k,i} \cdot \left(\sum_{l=1}^n \mathcal{M}(V; F, F)_{l,k} \cdot f_l \right) \\ &= \sum_{k=1}^n \left(\sum_{l=1}^n \mathcal{M}(V; F, F)_{l,k} \cdot \mathcal{M}(U; E, E)_{k,i} \cdot f_l \right) \\ &\stackrel{[\text{theorem: 11.45}]}{=} \sum_{l=1}^n \left(\sum_{k=1}^n \mathcal{M}(V; F, F)_{l,k} \cdot \mathcal{M}(U; E, E)_{k,i} \cdot f_l \right) \\ &= \sum_{l=1}^n \left(\sum_{k=1}^n \mathcal{M}(V; F, F)_{l,k} \cdot \mathcal{M}(U; E, E)_{k,i} \right) \cdot f_l \\ &= \sum_{l=1}^n (\mathcal{M}(V; F, F) \cdot \mathcal{M}(U; E, E))_{l,i} \cdot f_l \end{aligned}$$

which as F is a basis proves that $\forall l \in \{1, \dots, n\} E_{l,i} = (\mathcal{M}(V; F, F) \cdot \mathcal{M}(U; E, E))_{l,i}$. As $i \in \{1, \dots, n\}$ was chosen arbitrary it follows that

$$\mathcal{M}(V; V, F) \cdot \mathcal{M}(U; E, E) = E \tag{18.24}$$

Further we have

$$\begin{aligned}
 \sum_{l=1}^n E_{l,i} \cdot e_l &= \sum_{l=1}^n \delta_{l,i} \cdot e_i \\
 &= e_i \\
 &= V(f_i) \\
 &\stackrel{[\text{definition: 11.339}]}{=} \sum_{l=1}^n \mathcal{M}(V; F, F)_{l,i} \cdot f_l \\
 &= \sum_{l=1}^n \mathcal{M}(V; F, F)_{l,i} \cdot U(e_l) \\
 &\stackrel{[\text{definition: 11.339}]}{=} \sum_{l=1}^n \mathcal{M}(V; F, F)_{l,i} \cdot \left(\sum_{k=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot e_k \right) \\
 &= \sum_{l=1}^n \left(\sum_{k=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot \mathcal{M}(V; F, F)_{l,i} \cdot e_k \right) \\
 &\stackrel{[\text{theorem: 11.45}]}{=} \sum_{k=1}^n \left(\sum_{l=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot \mathcal{M}(V; F, F)_{l,i} \cdot e_k \right) \\
 &= \sum_{k=1}^n \left(\sum_{l=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot \mathcal{M}(V; F, F)_{l,i} \right) \cdot e_k \\
 &= \sum_{k=1}^n (\mathcal{M}(U; E, E) \cdot \mathcal{M}(V; F, F))_{k,i} \cdot e_k
 \end{aligned}$$

which as E is a basis proves that $\forall l \in \{1, \dots, n\} \ E_{l,i} = (\mathcal{M}(U; E, E) \cdot \mathcal{M}(V; F, F))_{l,i}$. As $i \in \{1, \dots, n\}$ was chosen arbitrary it follows that $E = \mathcal{M}(U; E, E) \cdot \mathcal{M}(V; F, F)$. Combining this result with [eq: 18.24] proves that

$$\mathcal{M}(U; E, E) = \mathcal{M}(V; F, F)^{-1} \text{ and } \mathcal{M}(V; F, F) = \mathcal{M}(U; E, E)^{-1}$$

3. This follows from

$$\begin{aligned}
 \mathcal{M}(U; E, E) \cdot \mathcal{M}(V; E, E) &\stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(U \circ V; E, E) \\
 &\stackrel{(1)}{=} \mathcal{M}(\text{Id}_X; E, E) \\
 &\stackrel{[\text{theorem: 11.343}]}{=} E \\
 &\stackrel{[\text{theorem: 11.343}]}{=} \mathcal{M}(\text{Id}_X; E, E) \\
 &\stackrel{(1)}{=} \mathcal{M}(V \circ U; E, E) \\
 &\stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(V; E, E) \cdot \mathcal{M}(U; E, E)
 \end{aligned}$$

proving that

$$\mathcal{M}(V; E, E) = \mathcal{M}(U; E, E)^{-1}$$

4. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned}
 L(e_i) &= L(V(f_i)) \\
 &\stackrel{[\text{definition: 11.339}]}{=} L\left(\sum_{k=1}^n \mathcal{M}(V; F, F)_{k,i} \cdot f_k\right) \\
 &= \sum_{k=1}^n \mathcal{M}(V; F, F)_{k,i} \cdot L(f_k) \\
 &\stackrel{[\text{definition: 11.339}]}{=} \sum_{k=1}^n \mathcal{M}(V; F, F)_{k,i} \cdot \left(\sum_{l=1}^n \mathcal{M}(L; F, F)_{l,k} \cdot f_l\right)
 \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^n \left(\sum_{l=1}^n \mathcal{M}(L; F, F)_{l,k} \cdot \mathcal{M}(V; F, F)_{k,i} \cdot f_l \right) \\
&\stackrel{[\text{theorem: 11.45}]}{=} \sum_{l=1}^n \left(\sum_{k=1}^n \mathcal{M}(L; F, F)_{l,k} \cdot \mathcal{M}(V; F, F)_{k,i} \cdot f_l \right) \\
&= \sum_{l=1}^n \left(\sum_{k=1}^n \mathcal{M}(L; F, F)_{l,k} \cdot \mathcal{M}(V; F, F)_{k,i} \right) \cdot f_l \\
&= \sum_{l=1}^n (\mathcal{M}(L; F, F) \cdot \mathcal{M}(V; F, F))_{l,i} \cdot f_l \\
&= \sum_{l=1}^n (\mathcal{M}(L; F, F) \cdot \mathcal{M}(V; F, F))_{l,i} \cdot U(e_l) \\
&\stackrel{[\text{definition: 11.339}]}{=} \sum_{l=1}^n (\mathcal{M}(L; F, F) \cdot \mathcal{M}(V; F, F))_{l,i} \cdot \left(\sum_{k=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot e_k \right) \\
&= \sum_{l=1}^n \left(\sum_{k=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot (\mathcal{M}(L; F, F) \cdot \mathcal{M}(V; F, F))_{l,i} \cdot e_k \right) \\
&\stackrel{[\text{theorem: 11.45}]}{=} \sum_{k=1}^n \left(\sum_{l=1}^n \mathcal{M}(U; E, E)_{k,l} \cdot (\mathcal{M}(L; F, F) \cdot \mathcal{M}(V; F, F))_{l,i} \right) \cdot e_k \\
&= \sum_{k=1}^n (\mathcal{M}(U; E, E) \cdot (\mathcal{M}(L; F, F) \cdot \mathcal{M}(V; F, F)))_{k,i} \cdot e_k \\
&\stackrel{(2)}{=} \sum_{k=1}^n (\mathcal{M}(U; E, E) \cdot (\mathcal{M}(L; F, F) \cdot \mathcal{M}(U; E, E)^{-1}))_{k,i} \cdot e_k
\end{aligned}$$

which by [definition: 11.339] proves that

$$\mathcal{M}(L; E, E) = \mathcal{M}(U; E, E) \cdot \mathcal{M}(L; F, F) \cdot \mathcal{M}(U; E, E)^{-1} \quad (18.25)$$

Finally

$$\begin{aligned}
&\mathcal{M}(U; E, E)^{-1} \cdot \mathcal{M}(L; E, E) \cdot \mathcal{M}(U; E, E) \stackrel{[\text{eq: 18.25}]}{=} \\
&\mathcal{M}(U; E, E)^{-1} \cdot \mathcal{M}(U; E, E) \cdot \mathcal{M}(L; F, F) \cdot \mathcal{M}(U; E, E)^{-1} \cdot \mathcal{M}(U; E, E) = \\
&\mathcal{M}(L; F, F)
\end{aligned}$$

proving that

$$\mathcal{M}(L; F, F) = \mathcal{M}(U; E, E)^{-1} \cdot \mathcal{M}(L; E, E) \cdot \mathcal{M}(U; E, E)$$

□

For inner product spaces we have additionally.

Theorem 18.104. *Let $n \in \mathbb{N}$, $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner vector space with two orthonormal bases $E = \{e_1, \dots, e_n\}$, $F = \{f_1, \dots, f_n\}$ then for the linear mappings $U, V \in \text{Hom}(X)$ defined by $\forall i \in \{1, \dots, n\}$ $U(e_i) = f_i$ and $V(f_i) = e_i$ [in other words U is the coordinate transformation from U to V and V is the coordinate transformation from V to U] we have that:*

1. U is a unitary transformation.
2. $\mathcal{M}(U; E, E)^{-1} = \mathcal{M}(U; E, E)^H$ so that

$$\mathcal{M}(U; E, E) \cdot \mathcal{M}(U; E, E)^H = E = \mathcal{M}(U; E, E)^H \cdot \mathcal{M}(U; E, E)$$

proving that $\mathcal{M}(U; E, E)$ is a unitary matrix.

Proof.

1. Let $x, y \in X$ then there exists $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$, $\{x'_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$, $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{K}$ and $\{y'_i\}_{i \in \{1, \dots, n\}}$ such that $x = \sum_{i=1}^n x_i \cdot e_i = \sum_{i=1}^n x'_i \cdot f_i$ and $y = \sum_{i=1}^n y_i \cdot e_i = \sum_{i=1}^n y'_i \cdot f_i$. Then we have two case to consider:

real case.

$$\begin{aligned}
 \langle U(x), U(x) \rangle &= \left\langle U\left(\sum_{i=1}^n x_i \cdot e_i\right), U\left(\sum_{k=1}^n y_k \cdot e_k\right) \right\rangle \\
 &= \left\langle \sum_{i=1}^n x_i \cdot U(e_i), \sum_{k=1}^n y_k \cdot U(e_k) \right\rangle \\
 &= \left\langle \sum_{i=1}^n x_i \cdot f_i, \sum_{k=1}^n y_k \cdot f_k \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \left\langle f_i, \sum_{k=1}^n y_k \cdot f_k \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \sum_{k=1}^n y_k \cdot \langle f_i, f_k \rangle \\
 &\stackrel{\text{orthonormality}}{=} \sum_{i=1}^n x_i \cdot \sum_{k=1}^n y_k \cdot \delta_{i,k} \\
 &\stackrel{\text{orthonormality}}{=} \sum_{i=1}^n x_i \cdot \sum_{k=1}^n y_k \cdot \langle e_i, e_k \rangle \\
 &= \sum_{i=1}^n x_i \cdot \left\langle e_i, \sum_{k=1}^n y_k \cdot e_k \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \langle e_i, y \rangle \\
 &= \left\langle \sum_{i=1}^n x_i \cdot e_i, y \right\rangle \\
 &= \langle x, y \rangle
 \end{aligned}$$

proving by [theorem: 18.96] that U is a unitary operator.

complex case.

$$\begin{aligned}
 \langle U(x), U(x) \rangle &= \left\langle U\left(\sum_{i=1}^n x_i \cdot e_i\right), U\left(\sum_{k=1}^n y_k \cdot e_k\right) \right\rangle \\
 &= \left\langle \sum_{i=1}^n x_i \cdot U(e_i), \sum_{k=1}^n y_k \cdot U(e_k) \right\rangle \\
 &= \left\langle \sum_{i=1}^n x_i \cdot f_i, \sum_{k=1}^n y_k \cdot f_k \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \left\langle f_i, \sum_{k=1}^n y_k \cdot f_k \right\rangle \\
 &= \sum_{i=1}^n x_i \cdot \sum_{k=1}^n \bar{y}_k \cdot \langle f_i, f_k \rangle \\
 &\stackrel{\text{orthonormality}}{=} \sum_{i=1}^n x_i \cdot \sum_{k=1}^n \bar{y}_k \cdot \delta_{i,k}
 \end{aligned}$$

$$\begin{aligned}
& \stackrel{\text{orthonormality}}{=} \sum_{i=1}^n x_i \cdot \sum_{k=1}^n \overline{y_k} \cdot \langle e_i, e_k \rangle \\
& = \sum_{i=1}^n x_i \cdot \left\langle e_i, \sum_{k=1}^n y_k \cdot e_k \right\rangle \\
& = \sum_{i=1}^n x_i \cdot \langle e_i, y \rangle \\
& = \left\langle \sum_{i=1}^n x_i \cdot e_i, y \right\rangle \\
& = \langle x, y \rangle
\end{aligned}$$

2. This follows from

$$\begin{aligned}
\mathcal{M}(U; E, E)^H \cdot \mathcal{M}(U; E, E) & \stackrel{[\text{theorem: 18.50}]}{=} \mathcal{M}(U^*; E, E) \cdot \mathcal{M}(U; E, E) \\
& \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(U^* \circ U; E, E) \\
& \stackrel{(1)}{=} \mathcal{M}(\text{Id}_X; E, E) \\
& \stackrel{[\text{theorem: 11.343}]}{=} E \\
& \stackrel{[\text{theorem: 11.343}]}{=} \mathcal{M}(\text{Id}_X; E, E) \\
& \stackrel{(1)}{=} \mathcal{M}(U \circ U^*; E, E) \\
& \stackrel{[\text{theorem: 11.347}]}{=} \mathcal{M}(U; E, E) \cdot \mathcal{M}(U^*; E, E) \\
& \stackrel{[\text{theorem: 18.50}]}{=} \mathcal{M}(U; E, E) \cdot \mathcal{M}(U; E, E) \\
& \square
\end{aligned}$$

Definition 18.105. (Normal operator) Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space then $L \in \text{Hom}(X)$ is **normal** if $L \circ L^* = L^* \circ L$.

Lemma 18.106. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space then every unitary and self-adjoint operator is normal.

Proof. If L is self adjoint then $L \circ L^* = L \circ L = L^* \circ L$ proving that L is unitary. Further if L is unitary then $L \circ L^* = \text{Id}_X = L^* \circ L$ proving that L is unitary in this case. $\square$

Theorem 18.107. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space and $L \in \text{Hom}(X)$ then we have

$$\begin{aligned}
& L \text{ is normal} \\
& \Updownarrow \\
& \forall x \in X \text{ we have } \|L(x)\| = \|L^*(x)\| \text{ [using the inner product norm]}
\end{aligned}$$

Proof. First we have that

$$\begin{aligned}
(L \circ L^* - L^* \circ L)^* & \stackrel{[\text{theorem: 18.39}]}{=} (L \circ L^*)^* - (L^* \circ L)^* \\
& \stackrel{[\text{theorem: 18.41}]}{=} (L^*)^* \circ L^* - L^* \circ (L^*)^* \\
& \stackrel{[\text{theorem: 18.38}]}{=} L \circ L^* - L^* \circ L
\end{aligned}$$

proving that

$$L \circ L^* - L^* \circ L \text{ is self-adjoint} \quad (18.26)$$

Further we have

$$\begin{aligned}
L \text{ is normal} & \Leftrightarrow L \circ L^* = L^* \circ L \\
& \Leftrightarrow L \circ L^* - L^* \circ L = 0 \\
& \stackrel{[\text{theorem: 18.90}] \text{ with [eq: 18.26]}}{\Leftrightarrow} \forall x \in X \text{ we have } \langle (L \circ L^* - L^* \circ L)(x), x \rangle = 0 \\
& \Leftrightarrow \forall x \in X \text{ we have } \langle (L \circ L^*)(x), x \rangle - \langle (L^* \circ L)(x), x \rangle = 0 \\
& \Leftrightarrow \forall x \in X \text{ we have } \langle (L \circ L^*)(x), x \rangle = \langle (L^* \circ L)(x), x \rangle \\
& \Leftrightarrow \forall x \in X \text{ we have } \langle L(L^*(x)), x \rangle = \langle L^*(L(x)), x \rangle \\
& \Leftrightarrow \forall x \in X \text{ we have } \langle L^*(x), L^*(x) \rangle = \langle L(x), (L^*)^*(x) \rangle \\
& \Leftrightarrow \forall x \in X \text{ we have } \langle L^*(x), L^*(x) \rangle = \langle L(x), L(x) \rangle \\
& \Leftrightarrow \forall x \in X \text{ we have } \|L^*(x)\|^2 = \|L(x)\|^2 \\
& \stackrel{0 \leq \| \cdot \| \wedge [\text{theorem: 10.66}]}{\Leftrightarrow} \forall x \in X \text{ we have } \|L^*(x)\| = \|L(x)\|
\end{aligned}$$

proving the theorem. $\square$

The following shows the relation between diagonalization of the matrix of a linear mapping and eigenvalues.

Theorem 18.108. *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space with $\dim(X) = n \in \mathbb{N}$ and $L \in \text{Hom}(X)$ then the following are equivalent:*

1. *There exist a orthonormal basis $E = \{e_1, \dots, e_n\}$ of X such that $\mathcal{M}(L; E, E)$ is diagonal.*
 2. *X has a orthonormal basis $E = \{e_1, \dots, e_n\}$ such that $\forall i \in \{1, \dots, n\}$ e_i is a eigenvector of L .*
- Further if (1) or (2) is true the the eigenvalue of e_i is $\mathcal{M}(L; E, E)_{i,i}$.*

Proof.

- 1 $\Rightarrow$ 2.** As there exist a $E = \{e_1, \dots, e_n\}$ such that $\forall i, j \in \{1, \dots, n\}$ with $i \neq j$ we have $\mathcal{M}(L; E, E)_{i,j} = 0$. Hence given $i \in \{1, \dots, n\}$ we have

$$\begin{aligned}
L(e_i) &= \sum_{k \in \{1, \dots, n\}} \mathcal{M}(L; E, E)_{k,i} e_k \\
&= \sum_{k \in \{1, \dots, n\} \setminus k \neq i} \mathcal{M}(L; E, E)_{k,i} e_k + \sum_{k \in \{i\}} \mathcal{M}(L; E, E)_{k,i} e_k \\
&= \mathcal{M}(L; E, E)_{i,i} \cdot e_i
\end{aligned}$$

proving that e_i is a eigenvector of L with the eigenvalue $\mathcal{M}(L; E, E)_{i,i}$

- 2 $\Rightarrow$ 1.** As $\forall i \in \{1, \dots, n\}$ e_i is a eigenvector of L , hence there exist a $\lambda_i \in \mathbb{K}$ such that $L(e_i) = \lambda_i \cdot e_i$. Hence

$$\begin{aligned}
L(e_i) &= \lambda_i \cdot e_i \\
&= \sum_{k=1}^n \lambda_k \cdot \delta_{i,k} \cdot e_k
\end{aligned}$$

which proves by [definition: 11.339] that $\forall i, k \in \{1, \dots, n\}$ $\mathcal{M}(L; E, E)_{k,i} = \lambda_k \cdot \delta_{i,k}$ proving that $\mathcal{M}(L; E, E)$ is diagonal. $\square$

We prove now the Complex Spectral Theorem which proves that the matrix of a normal operator can be diagonalized or equivalent that there exist a basis of X of eigenvectors for the normal operators.

Theorem 18.109. (Complex Spectral Theorem) *Let $\langle X, \langle \rangle \rangle$ be complex finite dimensional inner product space with $\dim(X) = n \in \mathbb{N}$ and $L \in \text{Hom}(X)$ then the following are equivalent*

1. *L is normal*
2. *There exist a orthonormal basis $E = \{e_1, \dots, e_n\}$ of X such that $\mathcal{M}(L; E, E)$ is diagonal.*
3. *X has a orthonormal basis $E = \{e_1, \dots, e_n\}$ such that $\forall i \in \{1, \dots, n\}$ e_i is a eigenvector of L .*

Proof.

1 $\Rightarrow$ 2. Using Schur's theorem [see theorem: 18.82] there exist a orthonormal basis $\{e_1, \dots, e_n\}$ for X such that $M = \mathcal{M}(L; E, E)$ is a upper triangular matrix. We use now a induction argument to prove that

$$\forall m \in \{1, \dots, n\} \text{ we have } \forall i \in \{1, \dots, m\} \wedge \forall j \in \{i+1, \dots, n\} \text{ we have } M_{i,j} = 0 \quad (18.27)$$

So define

$$S_n = \{m \in \mathbb{N} | \text{If } m \leq n \text{ then } \forall i \in \{1, \dots, m\} \wedge \forall j \in \{i+1, \dots, n\} M_{i,j} = 0\}$$

then we have:

1 $\in S_n$. As M is upper triangular we have

$$L(e_1) = \sum_{k \in \{1, \dots, n\}} M_{k,1} \cdot e_k = M_{1,1} + \sum_{k \in \{2, \dots, n\}} M_{k,1} \cdot e_k = M_{1,1} + 0 = M_{1,1} \cdot e_1$$

so that $\|L(e_1)\|^2 = \langle L(e_1), L(e_1) \rangle = \langle M_{1,1} \cdot e_1, M_{1,1} \cdot e_1 \rangle = M_{1,1} \cdot \overline{M_{1,1}} \cdot \langle e_1, e_1 \rangle = |M_{1,1}|^2$ proving that

$$\|L(e_1)\|^2 = |M_{1,1}|^2 \quad (18.28)$$

Further we have

$$\begin{aligned} \|L^*(e_1)\|^2 &= \langle L^*(e_1), L^*(e_1) \rangle \\ &= \left\langle \sum_{k=1}^n \mathcal{M}(L^*; E, E)_{k,1} \cdot e_k, \sum_{l=1}^n \mathcal{M}(L^*; E, E)_{l,1} \cdot e_l \right\rangle \\ &\stackrel{[\text{theorem: 18.50}]}{=} \left\langle \sum_{k=1}^n \mathcal{M}(L; E, E)_{k,1}^H \cdot e_k, \sum_{l=1}^n \mathcal{M}(L; E, E)_{l,1}^H \cdot e_l \right\rangle \\ &= \left\langle \sum_{k=1}^n \overline{\mathcal{M}(L; E, E)_{1,k}} \cdot e_k, \sum_{l=1}^n \overline{\mathcal{M}(L; E, E)_{1,l}} \cdot e_l \right\rangle \\ &= \left\langle \sum_{k=1}^n \overline{M_{1,k}} \cdot e_k, \sum_{l=1}^n \overline{M_{1,l}} \cdot e_l \right\rangle \\ &= \sum_{k=1}^n \overline{M_{1,k}} \cdot \left\langle e_k, \sum_{l=1}^n \overline{M_{1,l}} \cdot e_l \right\rangle \\ &= \sum_{k=1}^n \overline{M_{1,k}} \cdot \left(\sum_{l=1}^n \overline{M_{1,l}} \cdot \langle e_k, e_l \rangle \right) \\ &= \sum_{k=1}^n \overline{M_{1,k}} \cdot \left(\sum_{l=1}^n M_{1,l} \cdot \langle e_k, e_l \rangle \right) \\ &= \sum_{k=1}^n \overline{M_{1,k}} \cdot \left(\sum_{l=1}^n M_{1,l} \cdot \delta_{k,l} \right) \\ &= \sum_{k=1}^n \overline{M_{1,k}} \cdot M_{1,k} \\ &= \sum_{k=1}^n |M_{1,k}|^2 \end{aligned}$$

proving that

$$\|L^*(e_1)\|^2 = \sum_{k=1}^n |M_{1,k}|^2 \quad (18.29)$$

As L is normal we have by [theorem: 18.107] we have that $\|L(e_1)\| = \|L(e_1)\|$ so that by [eqs: 18.28 and 18.29] we have

$$|M_{1,1}|^2 = \sum_{k=1}^n |M_{1,k}|^2 = |M_{1,1}|^2 + \sum_{k \in \{2, \dots, n\}} |M_{1,k}|^2$$

so that $\sum_{k \in \{2, \dots, n\}} |M_{1,k}|^2 = 0$ hence $\forall k \in \{2, \dots, n\}$ we have $M_{1,k} = 0$. In other words $\forall i \in \{1, \dots, 1\}$ we have $\forall j \in \{1+1, \dots, n\}$ we have $M_{i,j} = 0$. Hence it is proved that $1 \in S_n$.

$m \in S_n \Rightarrow m+a \in S_n$. If $m+1 \leq n$ then $m < n$ and as $m \in S$ we have that

$$\forall i \in \{1, \dots, m\} \text{ that } \forall j \in \{m+1, \dots, n\} \quad M_{i,j} = 0 \quad (18.30)$$

Further we have

$$\begin{aligned} L(e_{m+1}) &= \sum_{k \in \{1, \dots, n\}} M_{k,m+1} \cdot e_k \\ &= \sum_{k \in \{1, \dots, m+1\}} M_{k,m+1} \cdot e_k + \sum_{k \in \{m+2, \dots, n\}} M_{k,m+1} \cdot e_k \\ &\stackrel{M \text{ is upper triangular}}{=} \sum_{k \in \{1, \dots, m+1\}} M_{k,m+1} \cdot e_k + 0 \\ &= \sum_{k \in \{1, \dots, m+1\}} M_{k,m+1} \cdot e_k \\ &= M_{m+1,m+1} \cdot e_{m+1} + \sum_{k \in \{1, \dots, m\}} M_{k,m+1} \cdot e_k \\ &\stackrel{[\text{eq: 18.30}]}{=} M_{m+1,m+1} \cdot e_{m+1} + 0 \end{aligned}$$

so that

$$\|L(e_{m+1})\|^2 = \langle L(e_{m+1}), L(e_{m+1}) \rangle = \langle M_{m+1,m+1} \cdot e_{m+1}, M_{m+1,m+1} \cdot e_{m+1} \rangle = |M_{m+1,m+1}|^2$$

Hence we have

$$\|L(e_{m+1})\|^2 = |M_{m+1,m+1}|^2 \quad (18.31)$$

Further we have

$$\begin{aligned} \|L^*(e_{m+1})\|^2 &= \langle L^*(e_{m+1}), L^*(e_{m+1}) \rangle \\ &= \left\langle \sum_{k=1}^n \mathcal{M}(L^*; E, E)_{k,m+1} \cdot e_k, \sum_{l=1}^n \mathcal{M}(L^*; E, E)_{l,m+1} \cdot e_l \right\rangle \\ &\stackrel{[\text{theorem: 18.50}]}{=} \left\langle \sum_{k=1}^n \mathcal{M}(L; E, E)_{k,m+1}^H \cdot e_k, \sum_{l=1}^n \mathcal{M}(L; E, E)_{l,m+1}^H \cdot e_l \right\rangle \\ &= \left\langle \sum_{k=1}^n \overline{\mathcal{M}(L; E, E)_{m+1,k}} \cdot e_k, \sum_{l=1}^n \overline{\mathcal{M}(L; E, E)_{m+1,l}} \cdot e_l \right\rangle \\ &= \left\langle \sum_{k=1}^n \overline{M_{m+1,k}} \cdot e_k, \sum_{l=1}^n \overline{M_{m+1,l}} \cdot e_l \right\rangle \\ &= \sum_{k=1}^n \overline{M_{m+1,k}} \cdot \left\langle e_k, \sum_{l=1}^n \overline{M_{m+1,l}} \cdot e_l \right\rangle \\ &= \sum_{k=1}^n \overline{M_{m+1,k}} \cdot \sum_{l=1}^n \overline{M_{m+1,l}} \cdot \langle e_k, e_l \rangle \\ &= \sum_{k=1}^n \overline{M_{m+1,k}} \cdot \sum_{l=1}^n M_{m+1,l} \cdot \delta_{k,l} \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^n \overline{M_{m+1,k}} \cdot M_{m+1,k} \\
&= \sum_{k=1}^n |M_{m+1,k}|^2
\end{aligned}$$

proving that

$$\|L^*(e_{m+1})\|^2 = \sum_{k=1}^n |M_{m+1,k}|^2 \quad (18.32)$$

As L is normal we have by [theorem: 18.107] we have that $\|L(e_{m+1})\| = \|L^*(e_{m+1})\|$ we have by [eqs: 18.31 and 18.32] that

$$|M_{m+1,m+1}|^2 = \sum_{k=1}^n |M_{m+1,k}|^2 = |M_{m+1,m+1}|^2 + \sum_{k \in \{1, \dots, n\} \setminus \{m+1\}} |M_{m+1,k}|^2$$

so that $\sum_{k \in \{1, \dots, n\} \setminus \{m+1\}} |M_{m+1,k}|^2 = 0$, hence $\forall k \in \{1, \dots, n\} \setminus \{m+1\} \ M_{m+1,k} = 0$. Hence $\forall j \in \{(m+1)+1, \dots, n\}$ we have $M_{m+1,j} = 0$ which combined with [eq: 18.30] proves that $\forall i \in \{1, \dots, m+1\}$ that $\forall j \in \{(m+1)+1, \dots, n\} \ M_{i,j} = 0$. Hence we have proved that $m+1 \in S_n$.

Mathematical induction proves that $S_n = \mathbb{N}$ hence for

$$\forall m \in \{1, \dots, n\} \text{ we have } \forall i \in \{1, \dots, m\} \text{ that } \forall j \in \{m+1, \dots, n\} \text{ we have } M_{i,j} = 0 \quad (18.33)$$

Let $k, l \in \{1, \dots, n\}$ with $k \neq l$ then we have either:

$k < l$. As $k \in \{1, \dots, n\}$, $k \in \{1, \dots, k\}$ and $k+1 \leq l \Rightarrow l \in \{k+1, \dots, n\}$ it follows that $M_{k,l} = 0$.

$l < k$. As $l \in \{1, \dots, n\}$, $l \in \{1, \dots, l\}$ and $l+1 \leq k \Rightarrow k \in \{l+1, \dots, n\}$ it follows that $M_{k,l} = 0$.

which proves that

$$\mathcal{M}(L; E, E) = M \text{ is diagonal.}$$

2 $\Rightarrow$ 1. As there exist a basis $\{e_1, \dots, e_n\}$ such that $\mathcal{M}(L; E, E)$ is diagonal. Let $i, j \in \{1, \dots, n\}$ with $i \neq j$ then $\mathcal{M}(L^*; E, E)_{i,j} \underset{[\text{theorem: 18.50}]}{=} \overline{\mathcal{M}(L; E, E)_{j,i}^H} = \overline{\mathcal{M}(L; E, E)_{j,i}} = \overline{0} = 0$ so that $\mathcal{M}(L^*; E, E)$ is also diagonal. Let $i \in \{1, \dots, n\}$ then we have

$$\begin{aligned}
(L \circ L^*)(e_i) &= L(L^*(e_i)) \\
&= L\left(\sum_{k=1}^n \mathcal{M}(L^*; E, E)_{k,i} \cdot e_k\right) \\
&= L(\mathcal{M}(L^*; E, E)_{i,i} \cdot e_i) \\
&= \mathcal{M}(L^*; E, E) \cdot L(e_i) \\
&= \mathcal{M}(L^*; E, E) \cdot \left(\sum_{k=1}^n \mathcal{M}(L; E, E)_{k,i} \cdot e_k\right) \\
&= \mathcal{M}(L^*; E, E) \cdot \mathcal{M}(L; E, E)_{i,i} \cdot e_i \\
&= \mathcal{M}(L; E, E) \cdot \mathcal{M}(L^*; E, E)_{i,i} \cdot e_i \\
&= \mathcal{M}(L; E, E) \cdot \sum_{k=1}^n \mathcal{M}(L^*; E, E)_{k,i} \cdot e_k \\
&= \mathcal{M}(L; E, E) \cdot L^*(e_i) \\
&= L^*(\mathcal{M}(L; E, E) \cdot e_i) \\
&= L^*\left(\sum_{k=1}^n \mathcal{M}(L; E, E)_{k,i} \cdot e_k\right) \\
&= L^*(L(e_i)) \\
&= (L^* \circ L)(e_i)
\end{aligned}$$

So that $\forall i \in \{1, \dots, n\}$ we have

$$(L \circ L^*)(e_i) = (L^* \circ L)(e_i) \quad (18.34)$$

Let $x \in X$ then we have that $x = \sum_{i=1}^n x_i \cdot e_i$

$$\begin{aligned} (L \circ L^*)(x) &= (L \circ L^*)\left(\sum_{i=1}^n x_i \cdot e_i\right) \\ &= \sum_{i=1}^n x_i \cdot (L \circ L^*)(e_i) \\ &= \sum_{i=1}^n x_i \cdot (L^* \circ L)(e_i) \\ &= (L^* \circ L)\left(\sum_{i=1}^n x_i \cdot e_i\right) \\ &= (L^* \circ L)(x) \end{aligned}$$

proving that $L \circ L^* = L^* \circ L$. Hence L is normal.

2 $\Leftrightarrow$ 3. This is actually proved in [theorem: 18.108] □

Corollary 18.110. *Let $\langle X, \langle \rangle \rangle$ be a complete finite dimensional inner product spaces with $\dim(X) = n \in \mathbb{N}$, $E = \{e_1, \dots, e_n\}$ a orthonormal basis of X , $L \in \text{Hom}(X)$ a normal operator then there exist a unitary operator $U \in \text{Hom}(X)$ such that $\mathcal{M}(U^* \circ L \circ U; E, E)$ is a diagonal matrix.*

Proof. Using the Complex Spectral Theorem there exist a orthonormal basis $F = \{f_1, \dots, f_n\}$ such that $\mathcal{M}(L; F, F)$ is diagonal. By [theorem: 18.104] there exist a unitary mapping $U \in \text{Hom}(X)$ such that $\forall i \in \{1, \dots, n\}$ $U(e_i) = f_i$. Then we have $\forall i, j \in \{1, \dots, n\}$ that

$$\begin{aligned} \mathcal{M}(U^* \circ L \circ U; E, E)_{i,j} &\stackrel{[\text{theorem: 18.30}]}{=} \langle (U^* \circ L \circ U)(e_j), e_i \rangle \\ &= \langle U^*(L(U(e_j))), e_i \rangle \\ &= \langle L(U(e_j)), (U^*)^*(e_i) \rangle \\ &\stackrel{[\text{theorem: 18.38}]}{=} \langle L(U(e_j)), U(e_i) \rangle \\ &= \langle L(f_j), f_i \rangle \\ &\stackrel{[\text{theorem: 18.30}]}{=} \mathcal{M}(L; F, F)_{i,j} \end{aligned}$$

so that $\mathcal{M}(U^* \circ L \circ U; E, E) = \mathcal{M}(L; F, F)$ proving that $\mathcal{M}(U^* \circ L \circ U; E, E)$ is diagonal. □

Having proved the spectral theorem in the complex case we proceed now to prove the spectral theorem for the real case/ First we need a little lemma.

Lemma 18.111. *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space, $L \in \text{Hom}(X)$ a self-adjoint linear operator, $b, c \in \mathbb{R}$ such that $b^2 < 4 \cdot c$ then $L^2 + b \cdot L + c \cdot \text{Id}_X$ is a isomorphism.*

Proof. By [definition: 18.59] it follows that $L^2 + b \cdot L + c \cdot \text{Id}_X \in \text{Hom}(X)$ so that using [theorem: 11.202] we only have to prove that $L^2 + \beta \cdot L + c \cdot \text{Id}_X$ is injective. For every $x \in X$ with $x \neq 0$ we have using the Cauchy Schwarz inequality [see theorem: 18.18] we have

$$-b \cdot \langle L(x), x \rangle = \langle -b \cdot L(x), x \rangle \leq \| -b \cdot L(x) \| \cdot \| x \| = |b| \cdot \| L(x) \| \cdot \| x \|^2$$

or

$$-|b| \cdot \| L(x) \| \cdot \| x \| \leq b \cdot \langle L(x), x \rangle \quad (18.35)$$

Further

$$\begin{aligned}
 \langle (L^2 + b \cdot L + c \cdot \text{Id}_x)(x), x \rangle &= \langle L^2(x) + b \cdot L(x) + c \cdot x, x \rangle \\
 &= \langle L^2(x), x \rangle + b \cdot \langle L(x), x \rangle + c \cdot \langle x, x \rangle \\
 &= \langle L(x), L^*(x) \rangle + b \cdot \langle L(x), x \rangle + c \cdot \langle x, x \rangle \\
 &\stackrel{L \text{ is adjoint}}{=} \langle L(x), L(x) \rangle + b \cdot \langle L(x), x \rangle + c \cdot \langle x, x \rangle \\
 &= \|L(x)\|^2 + b \cdot \langle L(x), x \rangle + c \cdot \|x\|^2 \\
 &\geq_{[\text{eq: 18.35}]} \|L(x)\|^2 - |b| \cdot \|L(x)\| + c \cdot \|x\|^2
 \end{aligned}$$

proving

$$\|L(x)\|^2 - |b| \cdot \|L(x)\| + c \cdot \|x\|^2 \leq \langle (L^2 + b \cdot L + c \cdot \text{Id}_x)(x), x \rangle \quad (18.36)$$

Next

$$\begin{aligned}
 \left(\|L(x)\| - \frac{|b| \cdot \|x\|}{2} \right)^2 + \left(c - \frac{b^2}{4} \right) \cdot \|x\|^2 &= \\
 \|L(x)\|^2 - |b| \cdot \|L(x)\| \cdot \|x\| + \frac{|b|^2 \cdot \|x\|^2}{4} + c \cdot \|x\|^2 - \frac{b^2}{4} \cdot \|x\|^2 &\stackrel{b \in \mathbb{R} \Rightarrow \overline{b^2} = |b|^2}{=} \\
 \|L(x)\|^2 - |b| \cdot \|L(x)\| \cdot \|x\| + c \cdot \|x\|^2 &\stackrel{[\text{eq: 18.36}]}{\leq} \\
 \langle (L^2 + b \cdot L + c \cdot \text{Id}_x)(x), x \rangle &\quad (18.37)
 \end{aligned}$$

Now as $0 \leq \left(\|L(x)\| - \frac{|b| \cdot \|x\|}{2} \right)^2$ and $b^2 < 4 \cdot c \Rightarrow 0 < c - \frac{b^2}{4}$ so that as $0 < \|x\|$ we have

$$0 < \left(\|L(x)\| - \frac{|b| \cdot \|x\|}{2} \right)^2 + \left(c - \frac{b^2}{4} \right) \cdot \|x\|^2$$

which by [eq: 18.37] results in

$$0 < \langle (L^2 + b \cdot L + c \cdot \text{Id}_x)(x), x \rangle$$

so we must have that $(L^2 + b \cdot L + c \cdot \text{Id}_x)(x) \neq 0$. Hence $\ker(L^2 + b \cdot L + c \cdot \text{Id}_x) = \{0\}$ so that by [theorem: 11.195]

$$L^2 + b \cdot L + c \cdot \text{Id}_x \text{ is injective} \quad \square$$

We are now ready to prove that in the real case self-adjoint operators have a eigenvector with areal eigenvalue.

Theorem 18.112. *Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional inner product space and $L \in \text{Hom}(X)$ a self-adjoint operator then L has a eigenvector.*

Note 18.113. *Using [theorem: 18.112] all the eigenvalues are real.*

Proof. By [theorem: 18.70] every $L \in \text{Home}(X)$ in a complex finite dimensional inner product space has a eigenvector, hence this is also true for a self-adjoint operator. So we must only prove the theorem for a real finite dimensional inner product space. Let $n = \dim(X) \in \mathbb{N}$ and choose $x \in X$ with $x \neq 0$ [this exist because $n \in \mathbb{N}$ so that X is non trivial]. Consider now the set $W = \{L^0(x), \dots, L^n(x)\} \subseteq X$ of $n+1$ vectors in X . As $\dim(X) = n$ W must be linear dependent [see theorem: 11.161]. So there exist a family $\{a_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{R}$ such that

$$\{i \in \{0, \dots, n\} | a_i \neq 0\} \neq \emptyset \text{ and } \sum_{i=0}^n a_i \cdot L^i(x) = 0$$

As $\{i \in \{0, \dots, n\} | a_i \neq 0\}$ is non empty and finite $N = \max(\{i \in \{0, \dots, n\} | a_i \neq 0\})$ exist. So we have

$$\begin{aligned}
 0 &= \sum_{i \in \{0, \dots, n\}} a_i \cdot L^i(x) = \sum_{i \in \{0, \dots, N\}} a_i \cdot L^i(x) + \sum_{i \in \{0, \dots, N\}} a_i \cdot L^i(x) = \sum_{i \in \{0, \dots, N\}} a_i \cdot L^i(x) \\
 \text{or} \quad &\sum_{i=0}^N a_i \cdot L^i(x) = 0 \quad (18.38)
 \end{aligned}$$

If $N = 0$ then $0 = \sum_{i=0}^N a_i \cdot L^i(x) = a_0 \cdot L^0(x) = a_0 \cdot x$ which as $a_0 \neq 0$ proves that $x = 0$ contradicting $x \neq 0$. Hence we have

$$N \neq 0 \quad (18.39)$$

So if we define the polynomial $p \in \mathcal{P}$ by $p(z) = \sum_{i=0}^N a_i \cdot L^i(x)$ then $\text{ord}(p) = N \in \mathbb{N}$. Hence by [theorem: 17.36] there exists a $c \in \mathbb{R} \setminus \{0\}$, $m, M \in \mathbb{N}_0$, $\{b_i\}_{i \in \{1, \dots, M\}} \subseteq \mathbb{R}$ and $\{c_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $\forall i \in \{1, \dots, M\}$ we have

$$(b_i)^2 < 4 \cdot c_i$$

$$\forall z \in \mathbb{C} \text{ we have } p(z) = c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i)$$

Define now

$$\{p_i\}_{i \in \{1, \dots, m+M+1\}} \text{ by } p_i(z) = \begin{cases} c = c \cdot z^0 & \text{if } i = 1 \\ z - \lambda_{i-1} & \text{if } i \in \{2, \dots, m+1\} \\ z^2 + b_{i-(m+1)} \cdot z + c_{i-m} & \text{if } i \in \{m+2, \dots, m+M+1\} \end{cases}$$

then we have $\forall c \in \mathbb{C}$ that

$$\begin{aligned} \prod_{i \in \{1, \dots, m+M+1\}} p_i(z) &= \\ \left(\sum_{i \in \{1\}} p_i(z) \right) \cdot \left(\prod_{i \in \{2, \dots, m+1\}} p_i(z) \right) \cdot \left(\prod_{i \in \{m+2, \dots, m+M+1\}} p_i(z) \right) &= \\ c \cdot \left(\prod_{i \in \{2, \dots, m+1\}} (z - \lambda_{i-1}) \right) \cdot \left(\prod_{i \in \{m+2, \dots, m+M+1\}} (z^2 + b_{i-(m+1)} \cdot z + c_{i-(m+1)}) \right) &= \\ c \cdot \left(\prod_{i \in \{1, \dots, m\}} (z - \lambda_i) \right) \cdot \left(\prod_{i \in \{1, \dots, M\}} (z^2 + b_i \cdot z + c_i) \right) &= \\ p(z) \end{aligned}$$

Using [theorem: 18.63] we have then that

$$\sum_{i=0}^N a_i \cdot L^i \underset{[\text{definition: 18.59}]}{=} p[L] \underset{[\text{theorem: 18.63}]}{=} \bigodot_{i=1}^{n+M+1} p_i[L]$$

Now as for $x \neq 0$ we have $p[L](x) = \sum_{i=0}^N a_i \cdot L^i(x) \underset{[\text{eq: 18.38}]}{=} 0$ it follows that $\ker(\bigodot_{i=1}^{n+M+1} p_i[L]) \neq \emptyset$, hence using [theorem: 18.58] there exist a $k \in \{1, \dots, m+M+1\}$ such that $\ker(p_k[L]) \neq \emptyset$ or

$$p_k[L](y) \text{ is not injective} \quad (18.40)$$

For k we have to consider the following cases

$k = 1$. Then as p_1 is defined by $p_1(z) = c \cdot z^0$ we have $p_1[L] = c \cdot \text{Id}_X$. As $p_1[L]$ is not injective $\ker(p_1[L]) \neq \emptyset$ so that there exist a $x \neq 0$ such that $0 = p_k[L](x) = c \cdot x$ which as $c \neq 0$ leads to the contradiction $0 \neq x = 0$. Hence this case will never occur.

$k \in \{m+2, \dots, m+M+1\}$. Then as p_k is defined by $p_k(z) = z^2 + b_k \cdot z + c_k$ we have that

$$p_k[L] = L^2 + b_k \cdot L + c_k \cdot \text{Id}_X$$

which as $(b_i)^2 < 4 \cdot c_i$ proves by [lemma: 18.111] that $p_k[L]$ is a bijection hence injective contradicting [eq: 18.40]. So this case never occurs.

So the only valid case is the case where $k \in \{2, \dots, m+1\}$. Then $p_k(z) = z - \lambda_k$ and $p_k[L] = L - \lambda_k \cdot \text{Id}_X$. As $\ker(p_k[L]) \neq \emptyset$ there exist a $x \in X$ with $x \neq 0$ such that $0 = p_k[L](x) = L(x) - \lambda_k \cdot x$ or $L(x) = \lambda_k \cdot x$. Hence we found a eigenvector of L . $\square$

Definition 18.114. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space, $U \subseteq X$ then the **orthogonal complement** of U noted by $U^\perp$ is defined by

$$U^\perp = \{x \in X \mid \forall y \in U \text{ we have } \langle x, y \rangle = 0\} \subseteq X$$

In other words the orthogonal complement of a set is the collection of all elements in X that are orthogonal to every element of the given set.

Theorem 18.115. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space then we have:

1. If $U \subseteq X$ then $U^\perp$ is a sub-space of X .
2. $\{0\}^\perp = X$
3. $X^\perp = \{0\}$
4. If $U \subseteq X$ then $U \cap U^\perp \subseteq \{0\}$
5. If $U \subseteq X$ is a sub-space then $U \cap U^\perp = \{0\}$
6. If $U, W \subseteq X$ with $U \subseteq W$ then $W^\perp \subseteq U^\perp$

Proof.

1. If $x, y \in U^\perp$, $\alpha \in \mathbb{K}$ then $\forall u \in U$ we have $\langle x + \alpha \cdot y, u \rangle = \langle x, u \rangle + \alpha \cdot \langle y, u \rangle = 0 + 0 = 0$ proving that $x + \alpha \cdot y \in U^\perp$. Further as $\forall u \in U$ $\langle 0, u \rangle = 0$ it follows that $0 \in U^\perp$, hence $U^\perp \neq \emptyset$.
2. As $\forall x \in X$ we have $\langle x, 0 \rangle = 0$ it follows that $X \subseteq \{0\}^\perp \subseteq X$ so that $X = \{0\}^\perp$.
3. First $\forall x \in X$ we have $\langle 0, x \rangle = 0$ so that $\{0\} \subseteq X^\perp$, second if $y \in X^\perp$ then as $X^\perp \subseteq X$ $y \in X$, so $\langle y, y \rangle = 0$, hence $y = 0 \in \{0\}$ proving that $X^\perp \subseteq \{0\}$. Hence it follows that

$$X^\perp = \{0\}$$

4. Let $x \in U \cap U^\perp$ then $x \in U$ and as also $x \in U^\perp$ we have $\langle x, x \rangle = 0$ so that $x = 0 \in \{0\}$. Hence it follows that $U \cap U^\perp \subseteq \{0\}$.
5. As U is a sub-space it follows that $0 \in U$, as by (1) $U^\perp$ is a sub-space we have also a $0 \in U^\perp$, so $\{0\} \subseteq U \cap U^\perp$. Combining this with (4) results in $U \cap U^\perp = \{0\}$.
6. If $x \in W^\perp$ then $\forall u \in U$ we have, as $U \subseteq W$ that $u \in W$ hence $\langle x, u \rangle = 0$ proving that $x \in U^\perp$, hence $W^\perp \subseteq U^\perp$. $\square$

Theorem 18.116. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] inner product space, U a finite dimensional sub-space of X then

$$X = U \oplus U^\perp$$

Proof. We have the following cases to consider for U :

U is trivial. Then $U = \{0\}$ and $U^\perp = \{0\}^\perp \stackrel{[\text{theorem: 18.115}]}{=} X$ so that

$$U \oplus U^\perp = \{0\} \oplus X \stackrel{[\text{example: 11.213}]}{=} X$$

U is not trivial. As U is finite dimensional there exist [theorem: 18.32] an orthonormal basis $E = \{e_1, \dots, e_n\}$, $n \in \mathbb{N}$ for U . Let $x \in X$ then we have for $y = \sum_{i=1}^n \langle x, e_i \rangle \cdot e_i$ and $z = x - y$ that clearly

$$x = y + z \text{ and } y \in U$$

Further if $u \in U$ then we have by [theorem: 18.29] $u = \sum_{i=1}^n \langle u, e_i \rangle \cdot e_i$ then we have:

real case.

$$\begin{aligned} \langle z, u \rangle &= \langle x - y, u \rangle \\ &= \langle x, u \rangle - \langle y, u \rangle \\ &= \langle x, u \rangle - \left\langle \sum_{i=1}^n \langle x, e_i \rangle \cdot e_i, u \right\rangle \\ &= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \langle e_i, u \rangle \\ &= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \left\langle e_i, \sum_{k=1}^n \langle u, e_k \rangle \cdot e_k \right\rangle \end{aligned}$$

$$\begin{aligned}
&= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \sum_{k=1}^n u_k \cdot \langle e_i, e_k \rangle \\
&= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \sum_{k=1}^n u_k \cdot \delta_{i,k} \\
&= \langle x, u \rangle - \sum_{i=1}^n u_i \cdot \langle x, e_i \rangle \\
&= \langle x, u \rangle - \left\langle x, \sum_{i=1}^n u_i \cdot e_i \right\rangle \\
&= \langle x, u \rangle - \langle x, u \rangle \\
&= 0
\end{aligned}$$

complex case.

$$\begin{aligned}
\langle z, u \rangle &= \langle x - y, u \rangle \\
&= \langle x, u \rangle - \langle y, u \rangle \\
&= \langle x, u \rangle - \left\langle x, \sum_{i=1}^n \langle x, e_i \rangle \cdot e_i, u \right\rangle \\
&= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \langle e_i, u \rangle \\
&= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \left\langle e_i, \sum_{k=1}^n u_k \cdot e_k \right\rangle \\
&= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \sum_{k=1}^n \overline{u_k} \cdot \langle e_i, e_k \rangle \\
&= \langle x, u \rangle - \sum_{i=1}^n \langle x, e_i \rangle \cdot \sum_{k=1}^n \overline{u_k} \cdot \delta_{i,k} \\
&= \langle x, u \rangle - \sum_{i=1}^n \overline{u_i} \cdot \langle x, e_i \rangle \\
&= \langle x, u \rangle - \left\langle x, \sum_{i=1}^n u_i \cdot e_i \right\rangle \\
&= \langle x, u \rangle - \langle x, u \rangle \\
&= 0
\end{aligned}$$

so we have in all cases that $\langle z, u \rangle = 0$, proving that $z \in U^\perp$. Hence we have that

$$X = U + U^\perp$$

As by [theorem: 18.115] $U \cap U^\perp = \{0\}$ it follows from [theorem: 11.210] that

$$X = U \oplus U^\perp$$

□

Theorem 18.117. Let $\langle X, \langle \rangle \rangle$ be a real [or complex] finite dimensional space, $L \in \text{Hom}(X)$ a self-adjoint operator and $U \subseteq X$ a sub-space of X so that $L(U) \subseteq U$ then

1. $L(U^\perp) \subseteq U^\perp$
2. $L|_U \in \text{Hom}(U)$ is self-adjoint in $\langle U, \langle \rangle|_{U \times U} \rangle$
3. $L|_{U^\perp} \in \text{Hom}(U^\perp)$ is self-adjoint in $\langle U^\perp, \langle \rangle|_{U^\perp \times U^\perp} \rangle$

Using [theorems: 18.2, 18.8] for a proof that $\langle U, \langle \rangle|_{U \times U} \rangle$ and $\langle U^\perp, \langle \rangle|_{U^\perp \times U^\perp} \rangle$ are inner product space.

Proof.

1. Let $y \in L(U^\perp)$ then there exist a $x \in U^\perp$ such that $y = L(x)$. If $u \in U$ then $L(u) \in U$ and we have $\langle y, u \rangle = \langle L(x), u \rangle = \langle x, L^*(u) \rangle = \langle x, L(u) \rangle \underset{L(u) \subseteq U \wedge x \in U^\perp}{=} 0$, hence $y \in U^\perp$. So $L(U^\perp) \subseteq U^\perp$.

2. Let $x, y \in U$ then $L|_U(x) = L(x) \in U$ and $L|_U(y) = L(y) \in U$ so that

$$\langle L|_U(x), y \rangle_{|U \times U} = \langle L(x), y \rangle = \langle x, L(y) \rangle = \langle x, L|_U(y) \rangle_{|U \times U}$$

proving by [theorem: 18.85] that $L|_U$ is self-adjoint.

3. This follows from (2) by replacing U by $U^\perp$ and the fact that $U^\perp$ is also a subspace in X [see theorem: 18.115]. $\square$

We are now ready to prove the Spectral Theorem in the real case.

Theorem 18.118. (Real Spectral Theorem) *Let $\langle X, \langle \rangle \rangle$ be a real finite dimensional space with $\dim(X) \in \mathbb{N}$ and $L \in \text{Hom}(X)$ then the following are equivalent:*

1. L is self-adjoint
2. There exist a orthonormal base $\{e_1, \dots, e_n\}$ for X such that $\forall i \in \{1, \dots, n\}$ e_i is a eigenvector of L .
3. There exists a orthonormal basis $E = \{e_1, \dots, e_n\}$ for X such that $\mathcal{M}(L; E, E)$ is diagonal.

Proof.

1 $\Rightarrow$ 2. We prove this by induction on $\dim(X)$ so define

$S = \{n \in \mathbb{N} | \text{If } \langle X, \langle \rangle \rangle \text{ is a real inner product space with } \dim(X) = n \text{ and } L \in \text{Hom}(X) \text{ is self-adjoint then there exist a orthonormal base } \{e_1, \dots, e_n\} \text{ for } X \text{ such that } \forall i \in \{1, \dots, n\} e_i \text{ is a eigenvector of } L\}$

then we have:

1 $\in S$. If $\dim(X) = 1$ then there exist by [theorem: 18.32] a orthonormal basis $\{e_1\}$ for X . As $L(e_1) \in X$ there exist a $\lambda_1 \in \mathbb{R}$ such that $L(e_1) = \lambda_1 \cdot e_1$ proving that e_1 is a eigenvector of X . Hence $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $\langle X, \langle \rangle \rangle$ be a real finite inner product space with $\dim(X) = n + 1$ and take $L \in \text{Hom}(X)$. As L is self-adjoint there exist by [theorem: 18.112] a eigenvector x with a real eigenvalue λ , so that

$$x \neq 0, L(x) = \lambda \cdot x \text{ and } \lambda \in \mathbb{R}$$

Define $e_{n+1} = \frac{x}{\|x\|}$ then we have

$$e_{n+1} \neq 0, \|e_{n+1}\| = 1 \text{ and } L(e_{n+1}) = \frac{L(x)}{\|x\|} = \lambda \cdot \frac{x}{\|x\|} = \lambda \cdot e_{n+1}$$

Define now $U = \text{span}(\{e_{n+1}\}) = \{\alpha \cdot e_{n+1} | \alpha \in \mathbb{R}\}$ which is a vector space with $\dim(U) = 1$. Using [theorem: 18.114] we have then that

$$X = U \oplus U^\perp$$

As $U^\perp \in X$ we have by [theorem: 11.159] that $U^\perp$ is finite dimensional, so we can use [theorem: 11.212] to get $n + 1 = \dim(X) = \dim(U \oplus U^\perp) = \dim(U) + \dim(U^\perp) = 1 + \dim(U^\perp)$, hence $\dim(U^\perp) = n$. So by [theorem: 18.2] we have that

$$\langle U^\perp, \langle \rangle_{|U^\perp} \rangle \text{ is a real inner product space with } \dim(U^\perp) = n$$

Further by the previous theorem [theorem: 18.117] we have that $L|_{U^\perp}$ is a self-adjoint operator in $\langle U^\perp, \langle \rangle_{|U^\perp} \rangle$. Hence as $n \in S$ there exist a orthonormal basis $\{e_1, \dots, e_n\}$ for $U^\perp$ such that $\forall i \in \{e_1, \dots, e_n\}$ e_i is a eigenvector of $U^\perp$ with eigenvector λ_i . Consider now

$$E = \{e_1, \dots, e_n, e_{n+1}\}$$

Let $i, j \in \{1, \dots, n+1\}$ with $i \neq j$ then we have either:

$i, j \in \{1, \dots, n\}$. Then as $\{e_1, \dots, e_n\}$ is orthonormal $\langle e_i, e_j \rangle = 0$

$i = n+1 \wedge j \in \{1, \dots, n\}$. Then as $e_j \in U^\perp$ and $e_{n+1} \in U$ $\langle e_i, e_j \rangle = \langle e_{n+1}, e_j \rangle = 0$

$i \in \{1, \dots, n\} \wedge j = n+1$. Then as $e_i \in U^\perp$ and $e_{n+1} \in U$ $\langle e_i, e_j \rangle = \langle e_i, e_{n+1} \rangle = 0$

hence we have that $\{e_1, \dots, e_n, e_{n+1}\}$ is a orthonormal set. Using then [theorem: 18.33] it follows that

$$\{e_1, \dots, e_{n+1}\} \text{ is a orthonormal basis for } X$$

Further if $i \in \{1, \dots, n+1\}$ then we have either:

$i \in \{1, \dots, n\}$. Then $L(e_i) = L|_{U^\perp}(e_i) = \lambda_i \cdot e_i$ so that e_i is a eigenvector of L

$i = n+1$. Then $L(e_i) = L(e_{n+1}) = \lambda \cdot e_{n+1}$ so that e_i is a eigenvector of L

hence $\forall i \in \{1, \dots, n+1\}$ e_i is a eigenvector of L . So $n+1 \in S$.

2 $\Rightarrow$ 1. Let $\{e_1, \dots, e_n\}$ be a orthonormal basis for X such that $\forall i \in \{1, \dots, n\}$ e_i is a eigenvector of L . Then if $x, y \in X$ there exist $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$, $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $x = \sum_{i=1}^n x_i \cdot e_i$ and $y = \sum_{i=1}^n y_i \cdot e_i$. So

$$\begin{aligned} \langle L(x), y \rangle &= \left\langle L\left(\sum_{i=1}^n x_i \cdot e_i\right), \sum_{j=1}^n y_j \cdot e_j \right\rangle \\ &= \left\langle \sum_{i=1}^n x_i \cdot L(e_i), \sum_{j=1}^n y_j \cdot e_j \right\rangle \\ &= \left\langle \sum_{i=1}^n x_i \cdot \lambda_i \cdot e_i, \sum_{j=1}^n y_j \cdot e_j \right\rangle \\ &= \sum_{i=1}^n x_i \cdot \lambda_i \cdot \left\langle e_i, \sum_{j=1}^n y_j \cdot e_j \right\rangle \\ &= \sum_{i=1}^n \left(x_i \cdot \lambda_i \cdot \sum_{j=1}^n y_j \cdot \langle e_i, e_j \rangle \right) \\ &= \sum_{i=1}^n \left(x_i \cdot \lambda_i \cdot \sum_{j=1}^n y_j \cdot \delta_{i,j} \right) \\ &= \sum_{i=1}^n \lambda_i \cdot x_i \cdot y_i \\ \langle x, L(y) \rangle &= \left\langle \sum_{i=1}^n x_i \cdot e_i, L\left(\sum_{j=1}^n y_j \cdot e_j\right) \right\rangle \\ &= \sum_{i=1}^n x_i \cdot \left\langle e_i, \sum_{j=1}^n y_j \cdot L(e_j) \right\rangle \\ &= \sum_{i=1}^n x_i \cdot \left(\sum_{j=1}^n y_j \cdot \lambda_j \cdot \langle e_i, e_j \rangle \right) \\ &= \sum_{i=1}^n x_i \cdot \left(\sum_{j=1}^n y_j \cdot \lambda_j \cdot \delta_{i,j} \right) \\ &= \sum_{i=1}^n \lambda_i \cdot x_i \cdot y_i \end{aligned}$$

From the above it follows that $\langle L(x), y \rangle = \langle x, L(y) \rangle$ which as $x, y \in X$ where chosen arbitrary proves that L is self-adjoint.

2 $\Leftrightarrow$ 3. This is actually proved in [theorem: 18.108]. $\square$

Corollary 18.119. Let $\langle X, \langle \rangle \rangle$ be a real finite dimensional space with $\dim(X) \in \mathbb{N}$ with $\dim(X) = n \in \mathbb{N}$, $E = \{e_1, \dots, e_n\}$ a orthonormal basis and $L \in \text{Hom}(X)$ a self-adjoint operator then there exist a unitary operator $U \in \text{Hom}(X)$ such that $\mathcal{M}(U^* \circ L \circ U; E, E)$ is diagonal.

Proof. Using the real spectral theorem [theorem: 18.118] there exist a orthonormal basis $F = \{f_1, \dots, f_n\}$ such $\mathcal{M}(L; F, F)$ is diagonal. Take now the unitary mapping $U \in \text{Hom}(X)$ defined by $\forall i \in \{1, \dots, n\}$ $U(e_i) = f_i$ [see theorem: 18.104] then we have for $i, j \in \{1, \dots, n\}$ that

$$\begin{aligned} \mathcal{M}(U^* \circ L \circ U; E, E)_{i,j} &\stackrel{[\text{theorem: 18.30}]}{=} \langle (U^* \circ L \circ U)(e_j), e_i \rangle \\ &= \langle U^*(L(U(e_j))), e_i \rangle \\ &= \langle L(U(e_j)), (U^*)^*(e_i) \rangle \\ &\stackrel{[\text{theorem: 18.38}]}{=} \langle L(U(e_j)), U(e_i) \rangle \\ &= \langle L(f_j), f_i \rangle \\ &\stackrel{[\text{theorem: 18.30}]}{=} \mathcal{M}(L; F, F)_{i,j} \end{aligned}$$

So $\mathcal{M}(U^* \circ L \circ U; E, E) = \mathcal{M}(L; F, F)$ which, as $\mathcal{M}(L; F, F)$ is diagonal, proves that $\mathcal{M}(U^* \circ L \circ U; E, E)$ is diagonal. $\square$

Theorem 18.120. Let $n \in \mathbb{N}$ then we have:

1. If $M \in \mathcal{M}_{n,n}(\mathbb{R})$ is a symmetric matrix then there exist a unitary matrix U such that $U^T \cdot M \cdot U \stackrel{U \text{ is unitary}}{=} U^{-1} \cdot M \cdot U$ is diagonal.
2. If $M \in \mathcal{M}_{n,n}(\mathbb{C})$ is a hermitian matrix then there exist a unitary matrix U such that $U^H \cdot M \cdot U \stackrel{U \text{ is unitary}}{=} U^{-1} \cdot M \cdot U$ is diagonal.

Proof.

1. Take the inner product vector space $\langle \mathbb{R}^n, \langle \rangle \rangle$ defined in [example: 18.4] together with the canonical orthonormal basis $E = \{e_1, \dots, e + n\}$ defined by $(e_i)_j = \delta_{i,j}$ [see example: 18.27]. Define $L \in \text{Hom}(\mathbb{R})$ by $\forall i \in \{1, \dots, n\}$ $L(e_i) = \sum_{k=1}^n M_{i,k} \cdot e_k$ so that

$$\mathcal{M}(L; E, E) = M \tag{18.41}$$

As M is symmetric it is Hermitian [see definition: 18.45] we have by [theorem: 18.87] that L is a self-adjoint operator. Using the real spectral theorem [see theorem: 18.118] there exist a orthonormal basis $F = \{f_1, \dots, f_n\}$ such that $\mathcal{M}(L; F, F)$ is diagonal. Let V be the coordinate transformation from E to F then using [theorem: 18.103]

$$\mathcal{M}(L; F, F) = \mathcal{M}(V; E, E)^{-1} \cdot \mathcal{M}(L; E, E) \cdot \mathcal{M}(V; E, E)$$

so that

$$\mathcal{M}(V; E, E)^{-1} \cdot M \cdot \mathcal{M}(V; E, E) \text{ is a diagonal matrix}$$

Using [theorem: 18.104] it follows that $\mathcal{M}(V; E, E)$ is unitary. So if we take $U = \mathcal{M}(V; E, E)$ then we found a unitary matrix U such that $U^{-1} \cdot M \cdot U$ is diagonal.

2. Take the inner product vector space $\langle \mathbb{C}^n, \langle \rangle \rangle$ defined in [example: 18.9] together with the canonical orthonormal basis $E = \{e_1, \dots, e + n\}$ defined by $(e_i)_j = \delta_{i,j}$ [see example: 18.27]. Define $L \in \text{Hom}(\mathbb{R})$ by $\forall i \in \{1, \dots, n\}$ $L(e_i) = \sum_{k=1}^n M_{i,k} \cdot e_k$ so that

$$\mathcal{M}(L; E, E) = M \tag{18.42}$$

As M is Hermitian [see definition: 18.45] we have by [theorem: 18.87] that L is a self-adjoint operator hence normal by [theorem: 18.106]. Using the complex spectral theorem [see theorem: 18.109] there exist a orthonormal basis $F = \{f_1, \dots, f_n\}$ such that $\mathcal{M}(L; F, F)$ is diagonal. Let V be the coordinate transformation from E to F then using [theorem: 18.103]

$$\mathcal{M}(L; F, F) = \mathcal{M}(V; E, E)^{-1} \cdot \mathcal{M}(L; E, E) \cdot \mathcal{M}(V; E, E)$$

so that

$\mathcal{M}(V; E, E)^{-1} \cdot M \cdot \mathcal{M}(V; E, E)$ is a diagonal matrix

Using [theorem: 18.104] it follows that $\mathcal{M}(V; E, E)$ is unitary. So if we take $U = \mathcal{M}(V; E, E)$ then we found a unitary matrix U such that $U^{-1} \cdot M \cdot U$ is diagonal. $\square$

Chapter 19

The exponential function

19.1 Power series

19.1.1 Definition and converge domain

Definition 19.1. (Power Series) A *power series* is a expression of the form

$$\sum_{i=0}^{\infty} a_i \cdot z^i$$

where $\{a_i\}_{i \in \mathbb{N}_0} \subseteq \mathbb{C}$ is a sequence of complex numbers. If for a given $z \in \mathbb{C}$ $\sum_{i=0}^{\infty} a_i \cdot z^i$ converges then $\sum_{i=0}^{\infty} a_i \cdot z^i$ refers to the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n a_i \cdot z^i$.

Note 19.2. We have bit of notation overloading here the same expression $\sum_{i=0}^{\infty} a_i \cdot z^i$ stands for

1. A **power series** which is essential another notation of a family $\{a_i\}_{i \in \mathbb{N}_0} \subseteq \mathbb{C}$
2. A series if we have chosen a certain a complex number z .
3. The $\lim_{n \rightarrow \infty} \sum_{i=1}^n a_i \cdot z^i$ [which is a value in $\mathbb{C}$] if the series $\sum_{i=0}^{\infty} a_i \cdot z^i$ converges.

We use context to figure out which meaning must be attached to the expression $\sum_{i=0}^{\infty} a_i \cdot z^i$.

Given a sequence $\{a_i\}_{i \in \mathbb{N}_0} \subseteq \mathbb{C}$ then the convergence of $\sum_{i=0}^{\infty} a_i \cdot z^i$ depends on the value of z , the convergence domain captures this idea.

Definition 19.3. Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ be a power series then

1. The **convergence domain** of $\sum_{i=0}^{\infty} a_i \cdot z^i$ noted as $\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ is defined by

$$\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] = \left\{z \in \mathbb{C} \mid \sum_{i=0}^{\infty} a_i \cdot z^i \text{ converges} \right\}$$

2. The **absolute convergence domain** of $\sum_{i=0}^{\infty} a_i \cdot z^i$ noted as $\bar{\mathcal{D}}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ is defined by

$$\bar{\mathcal{D}}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] = \left\{z \in \mathbb{C} \mid \sum_{i=0}^{\infty} |a_i| \cdot |z|^i \text{ converges} \right\}$$

Example 19.4. Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ be a power series then if $z=0$ it follows that $\sum_{i=0}^{\infty} a_i \cdot z^i$ converges and

$$\sum_{i=0}^{\infty} a_i \cdot z^i = a_0$$

Proof. If $z=0$ then $\forall i \in \mathbb{N}$ we have $a_i \cdot z^i = 0$ so that for $n \in \mathbb{N}$ we have that

$$\sum_{i=0}^n a_i \cdot z^i = a_0 \cdot z^0 + \sum_{i=1}^n a_i \cdot z^i = a_0$$

so that by [example: 14.347] $\lim_{n \rightarrow \infty} \sum_{i=0}^n a_i \cdot z^i$ converges to a_0 . $\square$

Theorem 19.5. *Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ be a power series then we have*

$$0 \in \overline{\mathcal{D}} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right] \subseteq \mathcal{D} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right]$$

Proof. First if $z=0$ then $|z|=0$ and by [example: 19.4] we have that $\sum_{i=0}^{\infty} |a_i| \cdot 0^i$ converges to $|a_0|$ proving that

$$0 \in \overline{\mathcal{D}} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right]$$

Second if $z \in \overline{\mathcal{D}}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ then $\sum_{i=0}^{\infty} |a_i| \cdot |z|^i$ converges, as $\forall i \in \mathbb{N}_0$ we have that $|a_i \cdot z^i| \stackrel{[\text{theorem: 10.85}]}{=} |a_i| \cdot |z|^i$ it follows from [theorem: 14.437] that $\sum_{i=0}^{\infty} a_i \cdot z^i$ converges. Hence $z \in \mathcal{D}(\sum_{i=0}^{\infty} a_i \cdot z^i)$ so that

$$\overline{\mathcal{D}} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right] \subseteq \mathcal{D} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right] \quad \square$$

Theorem 19.6. *Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ a power series and $0 \neq z_0 \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ then $\forall \rho \in \mathbb{R}$ with $0 \leq \rho < |z_0|$ we have that $\sum_{i=0}^{\infty} |a_i| \cdot \rho^i$ converges.*

Proof. Let $0 \leq \rho < |z_0|$ then we have for ρ either:

$\rho = 0$. Then by [example: 19.4] $\sum_{i=0}^{\infty} |a_i| \cdot \rho^i$ converges.

$0 < \rho$. As $z_0 \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ we have that $\sum_{i=0}^{\infty} a_i \cdot z_0^i$ converges. Using [theorem: 14.413] it follow then that $\lim_{i \rightarrow \infty} a_i \cdot z_0^i = 0$, hence by [theorem: 14.386] $\{a_i\}_{i \in \mathbb{N}_0}$ is Cauchy and thus bounded by [theorem: 14.385]. So we have

$$\exists K \in \mathbb{R}^+ \text{ such that } \forall i \in \mathbb{N}_0 \text{ we have } |a_i \cdot z_0^i| \leq K \quad (19.1)$$

Now $\forall i \in \mathbb{N}_0$ we have $|a_i| \cdot \rho^i \stackrel{0 \neq |z_0|}{=} |a_i| \cdot |z_0|^i \cdot \left(\frac{\rho}{|z_0|}\right)^i = |a_i \cdot z_0^i| \cdot q^i$ where $q = \left|\frac{\rho}{|z_0|}\right| < 1$. Combining this with [eq: 19.1] gives

$$\forall i \in \mathbb{N}_0 \text{ we have } |a_i| \cdot \rho^i \leq K \cdot q^i \text{ where } q_i < 1 \quad (19.2)$$

Using [example: 14.412] it follows that $\sum_{i=0}^{\infty} q^i$ converges and thus by [theorem: 14.418] $\sum_{i=0}^{\infty} K \cdot q^i$ converges. Hence by dominant convergence [theorem: 14.440] and [eq: 19.2] we have that $\sum_{i=1}^{\infty} |a_i| \cdot \rho^i$. $\square$

Corollary 19.7. *Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ be a power series, $0 \neq z_0 \in \overline{\mathcal{D}}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ then*

$$B_{||}(0, |z_0|) \subseteq \overline{\mathcal{D}} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right]$$

Proof. Take $z \in B_{||}(0, |z_0|)$ then $0 \leq |z| < |z_0|$ so that by the previous theorem [theorem: 19.6] $\sum_{i=0}^{\infty} |a_i| \cdot |z|^i$ converges. Hence by definition $z \in \overline{\mathcal{D}}[\sum_{i=0}^{\infty} a_i \cdot z^i]$. $\square$

The above suggests the following definition

Definition 19.8. (Convergence Radius) *Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ a power series then the **convergence radius** of $\sum_{i=0}^{\infty} a_i \cdot z^i$ noted as $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ is defined by*

$$\mathcal{R} \left[\sum_{i=0}^{\infty} a_i \cdot z^i \right] = \begin{cases} \sup (\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[) & \text{if } \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[\text{ is bounded above} \\ \infty & \text{if } \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[\text{ is not bounded above} \end{cases}$$

Note 19.9. As by [theorem: 19.5] $0 \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ and $\mathbb{R}$ is conditional complete [see theorem: 10.18] $\sup (\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[)$ exist if $\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ is bounded above, hence the definition is valid. Further as $0 < \infty$ and $0 \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ it follows that $0 \leq \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$.

Given the above definition we have following properties of the convergence radius of a power series.

Theorem 19.10. Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ be a power series then we have for $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$

1. If $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \infty$ then

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

2.

$$\begin{aligned} \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] &= \infty \\ \Updownarrow \\ \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] &= \mathbb{C} = \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \end{aligned}$$

Proof.

1. If $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \infty$ then we have either:

$\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] = 0$. Let $z \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$, Assume that $z \neq 0$ then $0 < |z|$ so there exist a $\rho \in \mathbb{R}^+$ such that $0 < \rho < |z|$. Hence

$$\rho \in [0, \infty[\cap B_{||}(0, |z|) \subseteq_{[\text{theorem: 19.7}]} [0, \infty[\cap \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]$$

so that $0 < \rho < \sup ([0, \infty[\cap \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]) = \mathcal{R}(\sum_{i=0}^{\infty} a_i \cdot z^i) = 0$ leading to the contradiction $0 < 0$. Hence the assumption is wrong and we must have $z = 0$. So we have

$$\begin{aligned} B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) &= \emptyset \\ &\subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \\ &\subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \\ &= \{0\} \\ &= \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)} \end{aligned}$$

proving

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

$0 < \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$. Let $z \in B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i])$ then we have either:

$z = 0$. Then $z = 0 \in \overline{\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]}$.

$z \neq 0$. Then $0 < |z| < \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] = \sup (\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[)$, so there exist a $z' \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ such that $|z| < |z'| \leq \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$. Hence $z \in B_{||}(0, z') \subseteq_{[\text{corollary: 19.7}]} \overline{\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]}$.

So in all cases we have

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \quad (19.3)$$

Let $z \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$. Assume that $z \notin \overline{B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i])}$ then we have $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < |z|$, hence there exist a ρ such that $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \rho < |z|$. As $z \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ it follows from [theorem: 19.7] that $\rho \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ so that $\rho \leq \sup(\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[) = \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ contradicting $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \rho$. So the assumption is wrong and we must have that $z \in \overline{B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i])}$. Hence $\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \subseteq B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i])$ which combined with [eq: 19.3] proves that

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

2.

$\Rightarrow$. If $z \in \mathbb{C}$ then as $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] = \infty$, $\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ is not bounded above so that there exist a $r \in \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] \cap [0, \infty[$ such that $|z| < r = |r|$ proving that $z \in B_{||}(0, |r|) \subseteq_{\text{eq: 19.7}} \overline{\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]}$ proving that

$$\mathbb{C} \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \mathbb{C}$$

so that

$$\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] = \mathbb{C} = \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]}$$

$\Leftarrow$. Assume that $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \infty$. As $\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] = \mathbb{C} = \overline{\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]}$ it follows from (1) that

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

which as $\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i] = \mathbb{C} = \overline{\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]}$ proves that $\mathbb{C} \subseteq \overline{B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i])}$, so that as $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] \in \mathbb{C}$ we have the contradiction $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$. Hence the assumption is wrong and we must have that $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] = \infty$. $\square$

To simplify theorems we extend the concept of a open and closed ball to the case where the radius is ∞ .

Definition 19.11. Let $r \in [0, \infty[$ and $z \in \mathbb{C}$ then we define the open and closed ball $B_{||}(z, r)$ and $\overline{B_{||}(z, r)}$ by

$$B_{||}(z, r) = \begin{cases} \mathbb{C} & \text{if } r = \infty \\ \{y \in \mathbb{C} \mid |y - z| < r\} & \text{if } r < \infty \end{cases}$$

$$\overline{B_{||}(z, r)} = \begin{cases} \mathbb{C} & \text{if } r = \infty \\ \{y \in \mathbb{C} \mid |y - z| \leq r\} & \text{if } r < \infty \end{cases}$$

Note 19.12. If $r \in [0, \infty[$ then this reduces to the definition of a open or closed ball in $\mathbb{C}$. Note also as $\mathbb{C}$ is open and closed that $B_{||}(z, r)$ is open and by [theorem: 14.69] $\overline{B_{||}(z, r)}$ is closed.

Using this definition we can simplify [theorem: 19.10] to the following theorem.

Theorem 19.13. Let $\sum_{i=0}^{\infty} a_i \cdot z^i$ be a power series then we have

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

Proof. For $\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ we have either:

$\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] < \infty$. Then we have by [theorem: 19.10] we have that

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

$\mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i] = \infty$. Then by [theorem: 19.10] $\overline{\mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]} = \mathbb{C} = \mathcal{D}[\sum_{i=0}^{\infty} a_i \cdot z^i]$ and by definition $B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i]) = \mathbb{C} = \overline{B_{||}(0, \mathcal{R}[\sum_{i=0}^{\infty} a_i \cdot z^i])}$. So that

$$B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]} \subseteq \mathcal{D}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{i=0}^{\infty} a_i \cdot z^i\right]\right)}$$

□

Lemma 19.14. Let $\rho, \rho' \in [0, \infty]$ then if $B_{||}(0, \rho) \subseteq \overline{B_{||}(0, \rho')}$ we have $\rho \leq \rho'$.

Proof. For δ' we have the following possibilities:

$\rho' \in]0, \infty[$. If $\rho = \infty$ then $\rho' \in \mathbb{C} = B_{||}(x, \rho) \subseteq B_{||}(x, \rho)$ leading to the contradiction $\rho' < \rho'$ hence we must have that $\rho \in [0, \rho[$. Assume that $\rho' < \rho$ then there exist a δ such that $\rho' < \delta < \rho$ so that $\delta \in B_{||}(z, \rho) \subseteq \overline{B_{||}(z, \rho')}$ hence $\rho' < \delta \leq \rho'$ giving the contradiction $\rho' < \rho'$. Hence we must have in this case that $\rho \leq \rho'$.

$\rho' = 0$. Then $B_{||}(z, \rho) \subseteq \overline{B_{||}(z, \rho')} = \{0\}$. Assume that $0 < \rho$ then there exist a δ such that $0 < \delta < \rho$ then $\delta \in \{0\}$ giving the contradiction $0 < 0$. Hence we must have that $\rho = 0$.

$\rho' = \infty$. Then $\rho \leq \rho'$

□

The concept of convergence domain of a power series allows us to define the power function that maps a complex number in the convergence domain to the convergence limit of the polynomial.

Definition 19.15. Let $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ be a power series then we define

$$\sum_{n=0}^{\infty} a_n \cdot (*)^n: \mathcal{D}\left(\sum_{n=0}^{\infty} a_n \cdot z^n\right) \rightarrow \mathbb{C} \text{ by } \left(\sum_{n=0}^n a_n \cdot (*)^n\right)(z) = \sum_{n=0}^{\infty} a_n \cdot z^n$$

19.1.2 Convergence criteria

We introduce now different criteria for a series to be convergent.

Lemma 19.16. Let $n \in \mathbb{N}$, $\{z_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$, $\{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$ and define

$$\{s_i\}_{i \in \{0, \dots, n\}} \text{ by } s_i = \sum_{j=0}^i z_j$$

then we have

$$\sum_{i=0}^n \alpha_i \cdot z_i = \left(\sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1})\right) + \alpha_n \cdot s_n$$

Proof. We use induction on n for this proof, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{z_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}, \{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C} \text{ and } \{s_i\}_{i \in \{0, \dots, m\}} \text{ defined by } s_i = \sum_{j=0}^i z_j \text{ then } \right. \\ \left. \sum_{i=0}^n \alpha_i \cdot z_i = \left(\sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1})\right) + \alpha_n \cdot s_n \right\}$$

then we have:

$1 \in S$. Then we have

$$\begin{aligned}
 \left(\sum_{i=0}^0 s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + \alpha_1 \cdot s_1 &= s_0 \cdot (\alpha_0 - \alpha_1) + \alpha_1 \cdot s_1 \\
 &= \left(\sum_{i=0}^0 z_i \right) \cdot (\alpha_0 - \alpha_1) + \alpha_1 \cdot \sum_{i=0}^1 z_i \\
 &= z_0 \cdot (\alpha_0 - \alpha_1) + \alpha_1 \cdot (z_0 + z_1) \\
 &= z_0 \cdot \alpha_0 + \alpha_1 \cdot z_1 \\
 &= \sum_{i=0}^1 \alpha_i \cdot z_i
 \end{aligned}$$

so that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. We have

$$\begin{aligned}
 \sum_{i=0}^{n+1} \alpha_i \cdot z_i &= \left(\sum_{i=0}^n \alpha_i \cdot z_i \right) + \alpha_{n+1} \cdot z_{n+1} \\
 &\stackrel{n \in S}{=} \left(\sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + \alpha_n \cdot s_n + \alpha_{n+1} \cdot z_{n+1} \\
 &= \left(\sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + (\alpha_n \cdot s_n - \alpha_{n+1} \cdot s_n) + (\alpha_{n+1} \cdot z_{n+1} + \alpha_{n+1} \cdot s_n) \\
 &= \left(\sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + (\alpha_n - \alpha_{n+1}) \cdot s_n + \alpha_{n+1} \cdot (z_{n+1} + s_n) \\
 &= \left(\sum_{i=0}^n s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + \alpha_{n+1} \cdot \left(z_{n+1} + \sum_{i=0}^n z_i \right) \\
 &= \left(\sum_{i=0}^n s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + \alpha_{n+1} \cdot \sum_{i=0}^{n+1} z_i \\
 &= \left(\sum_{i=0}^n s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + \alpha_{n+1} \cdot s_{n+1}
 \end{aligned}$$

proving that $n+1 \in S$. □

Lemma 19.17. (Inequality of Abel) Let $n \in \mathbb{N}_0$, $\{z_i\}_{i \in \{0, \dots, n\}} \subseteq \mathbb{C}$, $\{\alpha_i\}_{i \in \{0, \dots, n\}} \subseteq [0, \infty[$ be such that $\forall i \in \{0, \dots, n-1\}$ we have $\alpha_{i+1} \leq \alpha_i$, $\{s_i\}_{i \in \{0, \dots, n\}}$ defined by $s_i = \sum_{j=0}^i z_j$ and $\mu \in [0, \infty[$ such that $\forall i \in \{0, \dots, n\}$ we have $|s_i| \leq \mu$ then we have for $\sigma = \sum_{i=0}^n \alpha_i \cdot z_i$ that $|\sigma| \leq \alpha_0 \cdot \mu$.

Proof. For $n \in \mathbb{N}_0$ we have two cases to consider:

$n = 0$. Then $|\sigma| = |\sum_{i=0}^0 \alpha_i \cdot z_i| = |\alpha_0 \cdot z_0| = |\alpha_0 \cdot \sum_{i=0}^0 z_i| = |\alpha_0 \cdot s_0| = |\alpha_0| \cdot |s_0| < |\alpha_0| \cdot \mu$ proving that

$$|\sigma| \leq |\alpha_0| \cdot \mu$$

$n \neq 0$. Then $n \in \mathbb{N}$ so we can use the previous lemma [lemma: 19.16] in the following

$$\begin{aligned}
 |\sigma| &= \left| \sum_{i=0}^n \alpha_i \cdot z_i \right| \\
 &\stackrel{[\text{lemma: 19.16}]}{=} \left| \left(\sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1}) \right) + \alpha_n \cdot s_n \right| \\
 &\leq \left| \sum_{i=0}^{n-1} s_i \cdot (\alpha_i - \alpha_{i+1}) \right| + |\alpha_n| \cdot |s_n|
 \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{i=0}^{n-1} |s_i| \cdot |\alpha_i - \alpha_{i+1}| + |\alpha_n| \cdot |s_n| \\
&\stackrel{\alpha_n \in [0, \infty[}{=} \sum_{i=0}^{n-1} |s_i| \cdot |\alpha_i - \alpha_{i+1}| + \alpha_n \cdot |s_n| \\
&\stackrel{\alpha_{i+1} < \alpha_i}{=} \sum_{i=0}^{n-1} |s_i| \cdot (\alpha_i - \alpha_{i+1}) + \alpha_n \cdot |s_n| \\
&\leq \sum_{i=0}^{n-1} \mu \cdot (\alpha_i - \alpha_{i+1}) + \mu \cdot \alpha_n \\
&= \mu \cdot \left(\left(\sum_{i=0}^{n-1} (\alpha_i - \alpha_{i+1}) \right) + \alpha_n \right) \\
&\stackrel{[\text{theorem: 11.10}]}{=} \mu \cdot ((\alpha_0 - \alpha_n) + \alpha_n) \\
&= \mu \cdot \alpha_0 \\
&\quad \square
\end{aligned}$$

We use now the above lemma to prove the convergence criteria of Abel Dirichlet.

Theorem 19.18. (Criteria of Abel-Dirichlet) Let $\{z_i\}_{i \in \mathbb{N}_0} \subseteq \mathbb{C}$ and $\{\varepsilon_i\}_{i \in \mathbb{N}_0} \subseteq [0, \infty[$ such that $\forall i \in \mathbb{N}$ we have $\varepsilon_{i+1} \leq \varepsilon_i$ then we have

1. If there exist a $C \in [0, \infty[$ such that $\forall i \in \mathbb{N}_0$ we have $|\sum_{j=0}^i z_j| \leq C$ and $\lim_{i \rightarrow \infty} \varepsilon_i = 0$ then $\sum_{i=0}^{\infty} \varepsilon_i \cdot z_i$ is convergent.
2. If $\sum_{i=0}^{\infty} z_i$ is convergent then $\sum_{i=1}^{\infty} \varepsilon_i \cdot z_i$ is convergent.

Proof.

1. Given $n, m \in \mathbb{N}$ with $n \leq m$ define

$$\begin{aligned}
\{x_i^{(n)}\}_{i \in \{0, \dots, m-n\}} &\subseteq \mathbb{C} \quad \text{by } x_i^{(n)} = z_{n+i} \\
\{s_i^{(n)}\}_{i \in \{0, \dots, m-n\}} &\subseteq \mathbb{C} \quad \text{by } s_i^{(n)} = \sum_{j=0}^i x_j^{(n)} \\
\{\alpha_i^{(n)}\}_{i \in \{0, \dots, m-n\}} &\quad \text{by } \alpha_i^{(n)} = \varepsilon_{i+n}
\end{aligned}$$

Then we have

$$\forall i \in \{1, \dots, m-n-1\} \text{ that } \alpha_{i+1}^{(n)} = \varepsilon_{(i+n)+1} \leq \varepsilon_{i+n} = \alpha_i^{(n)} \text{ and } \alpha_i^{(n)} = \varepsilon_{i+n} \in [0, \infty[\quad (19.4)$$

Further given $i \in \{1, \dots, m-n\}$ we have

$$s_i^{(n)} = \sum_j^i x_j^{(n)} = \sum_{j=0}^i z_{n+j} = \sum_{j=n}^{i+n} z_j = \sum_{j=0}^{i+n} z_j - \sum_{j=0}^{n-1} z_j = s_{n+i} - s_{n-1}$$

so that $|s_i^{(n)}| = |s_{n+i}| + |s_{n-1}| \leq C + C = 2 \cdot C$. Hence we have

$$\forall i \in \{1, \dots, m-n-1\} \text{ that } s_i^{(n)} = s_{n+i} - s_{n-1} \text{ and } \forall i \in \{0, \dots, m-n\} \text{ that } |s_i^{(n)}| \leq 2 \cdot C$$

Using then the inequality of Abel [see theorem: 19.17] we have that

$$\left| \sum_{i=0}^{m-n} \alpha_i \cdot x_i^{(n)} \right| \leq \alpha_0 \cdot 2 \cdot C = \varepsilon_n \cdot 2 \cdot C$$

As $\sum_{i=0}^{m-n} \alpha_i \cdot x_i^{(n)} = \sum_{i=0}^{m-n} \varepsilon_{n+i} \cdot z_{n+i} = \sum_{i=n}^m \varepsilon_i \cdot z_i$ we have, using the above, that

$$\left| \sum_{i=n}^m \varepsilon_i \cdot z_i \right| \leq \varepsilon_n \cdot 2 \cdot C \quad (19.5)$$

Take now $\varepsilon \in \mathbb{R}^+$ then as $\lim_{i \rightarrow \infty} \varepsilon_i = 0$ there exist a $N \in \mathbb{N}$ such that $\forall n \geq N$ $\varepsilon_n = |\varepsilon_n - 0| < \frac{\varepsilon}{2 \cdot C + 1}$, combining this with [eq: 19.5] gives $|\sum_{i=n}^m \varepsilon_i \cdot z_i| < \frac{2 \cdot C}{2 \cdot C + 1} \cdot \varepsilon < \varepsilon$. Hence we have proved that

$$\forall \varepsilon > 0 \text{ there exist a } N \in \mathbb{N} \text{ such that } \forall n \geq N \text{ and } m \geq n \left| \sum_{i=n}^m \varepsilon_i \cdot z_i \right| < \varepsilon$$

As $\langle \mathbb{C}, || \rangle$ is a Banach space [see corollary: 14.397] we can use [theorem: 14.414] to prove that

$$\sum_{i=0}^{\infty} \varepsilon_i \cdot z_i \text{ converges}$$

2. As $\{\varepsilon_i\}_{i \in \mathbb{N}_0}$ is a decreasing sequence and $\forall i \in \mathbb{N}_0$ $0 \leq \varepsilon_i$ it follows from [theorem: 14.376] it follows that $\alpha = \lim_{i \rightarrow \infty} \varepsilon_i$ exist. Hence by [theorem: 14.346] it follows that

$$\lim_{i \rightarrow \infty} (\varepsilon_i - \alpha) = 0 \quad (19.6)$$

Further as $\sum_{i=0}^{\infty} z_i$ converges it follows that $\lim_{n \rightarrow \infty} \sum_{i=0}^n z_i$ exist, hence by [theorem: 14.364] there exist a $C \in \mathbb{R}^+$ such that

$$\forall i \in \mathbb{N}_0 \text{ we have that } \left| \sum_{j=0}^i z_j \right| \leq C \quad (19.7)$$

Using (1) together with [eqs: 19.6, 19.7] proves that

$$\sum_{i=0}^{\infty} (\varepsilon_i - \alpha) \cdot z_i \text{ converges}$$

As by the hypothesis $\sum_{i=0}^{\infty} z_i$ converges it follows from [theorem: 14.418] that

$$\sum_{i=0}^{\infty} \alpha \cdot z_i \text{ converges}$$

Using [theorem: 14.418] again it follows that $\sum_{i=0}^{\infty} ((\varepsilon_i - \alpha) \cdot z_i + \alpha \cdot z_i)$ converges hence

$$\sum_{i=0}^{\infty} \varepsilon_i \cdot z_i \text{ converges}$$

□

Lemma 19.19. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}^+$, $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}^+$ be such that $\forall i \in \{k, \dots, \infty\}$ we have $\frac{x_{i+1}}{y_{i+1}} \leq \frac{x_i}{y_i}$ then if $\sum_{i=k}^{\infty} y_i$ converges it follows that $\sum_{i=k}^{\infty} x_i$ converges.

Proof. Take $\alpha = \frac{x_k}{y_k}$ then we have $\forall i \in \{k, \dots, \infty\}$ that $\frac{x_i}{y_i} \leq \frac{x_k}{y_k} = \alpha \Rightarrow_{0 < y_i} x_i \leq \alpha \cdot y_i$ or as $0 < x_i$ we have that $|x_i| \leq \alpha \cdot y_i$. As $\sum_{i=k}^{\infty} y_i$ is convergent it follows by [theorem: 14.418] that $\sum_{i=k}^{\infty} \alpha \cdot y_i$ is convergent. Hence using dominant convergence [see theorem: 14.440] it follows that $\sum_{i=k}^{\infty} |x_i|$ converges. Finally, as $\forall i \in \{k, \dots, \infty\}$ we have $0 < x_i \Rightarrow x_i = |x_i|$, it follows that $\sum_{i=k}^{\infty} x_i$ converges. □

Theorem 19.20. (d'Alembert) Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq]0, \infty[$ then we have

1. If $\alpha = \liminf_{i \rightarrow \infty} \frac{x_{i+1}}{x_i}$ exist and $1 < \alpha$ then $\sum_{i=k}^{\infty} x_i$ is not convergent.
2. If $\beta = \limsup_{i \rightarrow \infty} \frac{x_{i+1}}{x_i}$ exist and $\beta < 1$ then $\sum_{i=k}^{\infty} x_i$ is convergent.

Proof.

1. If $1 < \alpha$ then $1 < \sup \left(\left\{ \inf \left(\left\{ \frac{x_{i+1}}{x_i} \mid i \in \{n, \dots, \infty\} \right\} \right) \mid n \in \{k, \dots, \infty\} \right\} \right)$ hence there exist a $n \in \{k, \dots, \infty\}$ such that $1 < \inf \left(\left\{ \frac{x_{i+1}}{x_i} \mid i \in \{n, \dots, \infty\} \right\} \right)$, so $\forall i \in \{n, \dots, \infty\}$ we have $1 < \frac{x_{i+1}}{x_i}$ or $x_i < x_{i+1}$ hence

$$\forall i \in \{n+1, \dots, \infty\} \text{ we have } 0 < x_n < x_i \quad (19.8)$$

Assume now that $\sum_{i=k}^{\infty} x_i$ converges then by [theorem: 14.413] $\lim_{i \rightarrow \infty} x_i = 0$ so there exist a $N \in \{k, \dots, \infty\}$ such that $\forall i \geq N$ we have that $x_i = |x_i - 0| < x_n$. Hence for $m = \max(n+1, N)$ it follows that $x_n <_{\text{eq: 19.8}} x_i < x_n$ leading to the contradiction $x_n < x_n$. Hence $\sum_{i=k}^{\infty} x_i$ does not converges.

2. As $\beta < 1$ it follows that there exist a ρ such that $0 \leq \beta < \rho < 1$ or

$$\inf \left(\left\{ \sup \left(\left\{ \frac{x_{i+1}}{x_i} \mid i \in \{n, \dots, \infty\} \right\} \right) \mid n \in \{k, \dots, \infty\} \right\} \right) < \rho \quad (19.9)$$

hence there exist a $n \in \{k, \dots, \infty\}$ such that $\sup \left(\left\{ \frac{x_{i+1}}{x_i} \mid i \in \{n, \dots, \infty\} \right\} \right) < \rho$ hence

$$\forall i \in \{n, \dots, \infty\} \text{ we have that } \frac{x_{i+1}}{x_i} < \rho = \frac{\rho^{i+1}}{\rho^i} \quad (19.10)$$

Now as $\rho < 1$ we have by [example: 14.412] that $\sum_{i=1}^{\infty} \rho^i$ is convergent, hence, using [theorem: 14.416], $\sum_{i=n}^{\infty} \rho^i$ is convergent. Using [lemma: 19.19] together with [eqs: 19.9, 19.10] proves then that $\sum_{i=n}^{\infty} x_i$ is convergent. Finally, by [theorem: 14.416] again, we have that

$$\sum_{i=k}^{\infty} x_i \text{ converges.} \quad \square$$

19.1.3 Definition of $\exp(z)$

We will define the exponential function $\exp(x)$ as $\exp(z) = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n$ so we have to look at the convergence domain of the power series $\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n$.

Theorem 19.21. *Let $z \in \mathbb{C}$ then*

1. $\overline{\mathcal{D}}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n] = \mathbb{C} = \mathcal{D}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n]$ [in other words $\forall z \in \mathbb{C}$ we have that $\sum_{n=0}^{\infty} \frac{1}{n!} \cdot |z|^n$ and $\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n$ converges]
2. $\mathcal{R}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n] = \infty$
3. $B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n]) = \overline{\mathcal{D}}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n] = \mathcal{D}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n] = \overline{B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n])} = \mathbb{C}$
4. If $z = 0$ then $\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n = 1$

Proof.

1. Let $z \in \mathbb{C}$ then we have either:

$z = 0$. Then $|z| = 0$ and by [example: 19.4] $\sum_{n=0}^{\infty} \left| \frac{1}{n!} \right| \cdot |z|^n$ converges.

$z \neq 0$. Then we have $\forall n \in \mathbb{N}_0$ that

$$0 < \left| \frac{1}{n!} \cdot z^n \right| \text{ and } \frac{\left| \frac{1}{(n+1)!} \cdot z^{n+1} \right|}{\left| \frac{1}{n!} \cdot z^n \right|} = \frac{z}{n+1} \quad (19.11)$$

Using a consequence of the Archimedean property of the real numbers [see theorem: 10.30] there exist a $N \in \mathbb{N}_0$ such that $2 \cdot |z| < N$ so that $\forall n \in \{N, \dots, \infty\}$ we have $2 \cdot |z| < N \leq n+1$ hence $\frac{|z|}{n+1} < \frac{1}{2}$. So

$$\forall n \in \{N, \dots, \infty\} \text{ we have that } 0 < \frac{|z|}{n+1} \leq \frac{1}{2} \quad (19.12)$$

and further

$$\forall n \in \{0, \dots, \infty\} \text{ we have } 0 < \frac{|z|}{n+1} \leq \max \left(\frac{1}{2}, \max \left(\frac{|z|}{n+1} \mid n \in \{1, \dots, N-1\} \right) \right) \quad (19.13)$$

So $\forall k \in \{0, \dots, \infty\}$ the set $\left\{\frac{|z|}{n+1} | n \in \{0, \dots, \infty\}\right\}$ is bounded above hence by the conditional completeness of $\mathbb{R}$ [see theorem: 10.18] it follows that

$$\forall k \in \{0, \dots, \infty\} \quad 0 \leq \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) \quad (19.14)$$

In particular we have using [eq: 19.12] that

$$\forall k \in \{N, \dots, \infty\} \quad 0 \leq \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) \leq \frac{1}{2} \quad (19.15)$$

Using [eq: 19.14] it follows that $\left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{0, \dots, \infty\} \right\}$ is bounded below by 0 so that

$$\inf \left(\left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{0, \dots, \infty\} \right\} \right) \text{ exist}$$

Further using [eq: 19.15] it follows that $\left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{N, \dots, \infty\} \right\}$ is bounded below by 0 but also above by $\frac{1}{2}$ so that

$$\inf \left(\left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{N, \dots, \infty\} \right\} \right) \leq \frac{1}{2}$$

As $\left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{N, \dots, \infty\} \right\} \subseteq \left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{0, \dots, \infty\} \right\}$ it follows that

$$\inf \left(\left\{ \sup \left(\left\{ \frac{|z|}{n+1} | n \in \{k, \dots, \infty\} \right\} \right) | n \in \{0, \dots, \infty\} \right\} \right) \leq \frac{1}{2}$$

or

$$\limsup_{n \rightarrow \infty} \frac{|z|}{n+1} \leq \frac{1}{2} < 1$$

Using then [eq: 19.11] it follows that

$$\limsup_{n \rightarrow \infty} \frac{\left| \frac{1}{(n+1)!} \cdot z^{n+1} \right|}{\left| \frac{1}{n!} \cdot z^n \right|} < 1 \text{ and } \left\{ \frac{\left| \frac{1}{(n+1)!} \cdot z^{n+1} \right|}{\left| \frac{1}{n!} \cdot z^n \right|} \right\}_{n \in \mathbb{N}_0} \subseteq]0, \infty[$$

Applying then d'Alembert [see theorem: 19.20] proves that

$$\sum_{n=0}^{\infty} \frac{|z|^n}{n!} \text{ converges}$$

Hence $\mathbb{C} \subseteq \overline{\mathcal{D}} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] \subseteq \mathcal{D} \left(\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right) \subseteq \mathbb{C}$ or

$$\overline{\mathcal{D}} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] = \mathbb{C} = \mathcal{D} \left(\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right)$$

2. Combining (1) with [theorem: 19.10] results in $\mathcal{R} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] = \infty$.
3. As by (2) $\mathcal{R} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] = \infty$ we have by definition [see definition: 19.11] that $B_{||}(0, \mathcal{R} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right]) = \overline{B_{||}(0, \mathcal{R} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right])} = \mathbb{C}$. Hence combining this with (1) gives

$$B_{||} \left(0, \mathcal{R} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] \right) = \overline{\mathcal{D} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right]} = \mathcal{D} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] = \overline{B_{||} \left(0, \mathcal{R} \left[\sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n \right] \right)}$$

4. If $z = 0$ then by [example: 19.4] we have that $\sum_{n=0}^{\infty} \frac{z^n}{n!} = \frac{1}{0!} = 1$ □

As $\mathcal{D}\left[\sum_{n=0}^{\infty} \frac{z^n}{n!}\right] = \mathbb{C}$ we can define a function that maps every complex number z to $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ this function is the exponential function.

Definition 19.22. (Exponential function) The exponential function $\exp: \mathbb{C} \rightarrow \mathbb{C}$ is defined by

$$\exp: \mathbb{C} \rightarrow \mathbb{C} \text{ where } \exp(x) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

Note 19.23. Using the definition of the power function [see definition: 19.15] the exponential function is a example of a power function, in fact $\exp = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot (*)^n$.

19.1.4 Differentiation of the exponential function

Given a power series $\sum_{n=0}^{\infty} a_n \cdot z^n$ with $0 < \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]$ we have by [theorem: 19.13] that

$$0 \in B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right) \subseteq \mathcal{D}\left(\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right) = \text{dom}\left(\sum_{n=0}^{\infty} a_n \cdot (*)^n\right)$$

it make sense to talk about differentiation of the power function $(\sum_{n=0}^{\infty} a_n \cdot (*)^n)_{|B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])}$ at 0. If we would naively extend differentiation of a finite sum of a function then we would have

$$\begin{aligned} D\left(\sum_{n=0}^{\infty} a_n \cdot (*)^n\right) &= \sum_{n=0}^{\infty} a_n \cdot D((*)^n) \\ &= \sum_{n=0}^{\infty} a_n \cdot n \cdot (*)^{n-1} \\ &= a_0 \cdot 0 \cdot (*)^0 + \sum_{n=1}^{\infty} a_n \cdot n \cdot (*)^{n-1} \\ &= \sum_{n=1}^{\infty} a_n \cdot n \cdot (*)^{n-1} \\ &= \sum_{n=0}^{\infty} a_{n+1} \cdot (n+1) \cdot (*)^n \end{aligned}$$

which is again a power function. Of course we must ensure that $\sum_{n=0}^{\infty} a_{n+1} \cdot (n+1) \cdot z^n$ is a convergence power series and prove that we can differentiate under a infinite sum which is actually a limit of partial sums. Nevertheless $\sum_{n=0}^{\infty} a_{n+1} \cdot (n+1) \cdot z^n$ is a power series.

Definition 19.24. (Derived Power Series) Let $\sum_{n=0}^{\infty} a_n \cdot z^n$ be a power series then the **derived power series** of $\sum_{n=0}^{\infty} a_n \cdot z^n$ is the series $\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$.

We have the following usefully properties of the derived power series of a power series.

Lemma 19.25. Let $\sum_{n=0}^{\infty} a_n \cdot z^n$ be a power series then

1. $\mathcal{D}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n] \subseteq \mathcal{D}[\sum_{n=0}^{\infty} a_n \cdot z^n]$
2. $\mathcal{R}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n] = \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]$

Proof.

1. Let $z \in \mathcal{D}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n]$ then by definition $\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$ is convergent. Given $\left\{\frac{1}{n+1}\right\}_{n \in \mathbb{N}_0} \subseteq \mathbb{R}$ we have $\forall n \in \mathbb{N}_0$ that $\frac{1}{(n+1)+1} < \frac{1}{n+1}$ so by Abel Dirichlet [theorem: 19.18 (2)] we have that $\sum_{n=0}^{\infty} \frac{1}{n+1} \cdot (n+1) \cdot a_{n+1} \cdot z^n$ is convergent, hence

$$\sum_{n=0}^{\infty} a_{n+1} \cdot z^n \text{ is convergent}$$

Using [theorem: 14.418] on the above proves that $\sum_{n=0}^{\infty} z \cdot a_{n+1} \cdot z^n$ is convergent, hence

$$\sum_{n=0}^{\infty} a_{n+1} \cdot z^{n+1} \text{ is convergent}$$

Using then [theorem: 14.415] it follows that

$$\sum_{n=0}^{\infty} a_n \cdot z^n \text{ is convergent}$$

hence $z \in \mathcal{D}[\sum_{n=0}^{\infty} a_n \cdot z^n]$. So we have that

$$\mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right] \subseteq \mathcal{D}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]$$

2. Using [theorem: 19.13] we have

$$B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]\right) \subseteq \overline{\mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]} \subseteq \mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]$$

by (1) we have

$$\mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right] \subseteq \mathcal{D}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]$$

and by [theorem: 19.13] again we have

$$\mathcal{D}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right)}$$

it follows that

$$B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]\right) \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right)}$$

so that by [theorem: 19.14] we have that

$$\mathcal{R}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right] \leq \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right] \quad (19.16)$$

Let $z \in \mathbb{C}$ such that $0 < |z| < \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]$ then there exist a $\rho \in \mathbb{R}$ such that

$$0 < |z| < \rho < \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]$$

By the above $\rho \in B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]) \subseteq_{[\text{theorem: 19.13}]} \mathcal{D}[\sum_{n=0}^{\infty} a_n \cdot z^n]$ so that

$$\sum_{n=0}^{\infty} a_n \cdot \rho^n \text{ is convergent}$$

So, by [theorem: 14.413], there exist a $K \in \mathbb{R}^+$ such that

$$\forall n \in \mathbb{N}_0 \quad |a_n| \cdot \rho^n = |a_n \cdot \rho^n| \leq K \quad (19.17)$$

Now for $n \in \mathbb{N}_0$ we have that

$$\begin{aligned} |(n+1) \cdot a_{n+1} \cdot z^n| &= (n+1) \cdot |a_{n+1}| \cdot |z|^n \\ &= \frac{n+1}{\rho} \cdot \left(\frac{|z|}{\rho}\right)^n \cdot a_{n+1} \cdot \rho^{n+1} \\ &\leq_{[\text{eq: 19.17}]} K \cdot \frac{n+1}{\rho} \cdot \left(\frac{|z|}{\rho}\right)^n \\ &= K \cdot \frac{n+1}{\rho} \cdot q^n \end{aligned}$$

where $q = \frac{|z|}{\rho} < 1$. To summarize we have proved that

$$\forall n \in \mathbb{N}_0 \models |(n+1) \cdot a_{n+1} \cdot z^n| \leq K \cdot \frac{n+1}{\rho} \cdot q^n \text{ where } 0 < q < 1 \quad (19.18)$$

Now for the power series $\sum_{n=0}^{\infty} (n+1) \cdot q^n$ we have that $\forall n \in \mathbb{N}_0 \frac{((n+1)+1) \cdot q^{n+1}}{(n+1) \cdot q^n} = \frac{n+2}{n+1} \cdot q$. By [example: 14.350] $\lim_{n \rightarrow \infty} \frac{n+2}{n+1} = 1$ so that by [theorem: 14.358] $\lim_{n \rightarrow \infty} \left(\frac{n+2}{n+1} \cdot q \right) = q < 1$, using [theorem: 14.380] it follows that $\limsup_{n \rightarrow \infty} \left(\frac{n+2}{n+1} \cdot q \right)$ exist and $\limsup_{n \rightarrow \infty} \left(\frac{n+2}{n+1} \cdot q \right) < 1$. Hence we can use d'Alembert [theorem: 19.20] proving that $\sum_{n=0}^{\infty} (n+1) \cdot q^n$ converges. Using then [theorem: 14.418] together with [eq: 19.18] it follows that $\sum_{n=0}^{\infty} K \cdot \frac{n+1}{\rho} \cdot q^n$ is convergent. Using then dominant convergence [see theorem: 14.437] we conclude that $\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$ converges, hence $z \in \mathcal{D}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n]$. To summarize

$$\forall z \in \mathbb{C} \text{ with } 0 < |z| < \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right] \text{ we have that } z \in \mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]$$

As also $0 \in \mathcal{D}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n]$ it follows that

$$B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right) \subseteq \mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right] \subseteq \overline{B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]\right)}$$

So by [theorem: 19.14] it follows that $\mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n] \leq \mathcal{R}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n]$ which together with [eq: 19.16] proves that

$$\mathcal{R}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right] = \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right] \quad \square$$

Hence we have found the conditions where a derived power series converges. We prove now that the derived power series of a power series is indeed its derivative.

Theorem 19.26. *Let $\sum_{n=0}^{\infty} a_n \cdot z^n$ be a power series with $0 < \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]$ then we have*

1. $\forall z \in B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$ we have that $\sum_{n=0}^{\infty} a_n \cdot z^n$ and $\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$ converges
2. $\forall z \in B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$ the function

$$\sum_{n=0}^{\infty} a_n \cdot (*)^n : B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right) \rightarrow \mathbb{C} \text{ defined by } \left(\sum_{n=0}^{\infty} a_n \cdot (*)^n\right)(z) = \sum_{n=0}^{\infty} a_n \cdot z^n$$

has a derivative $(\sum_{n=0}^{\infty} a_n \cdot (*)^n)'_z$ at z and

$$\left(\sum_{n=0}^{\infty} a_n \cdot (*)^n\right)'_z = \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$$

Proof.

1. As $\mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n] \stackrel{=}{\underset{[\text{theorem: 19.25}]}{\mathcal{R}[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n]}}$ it follows that

$$\begin{aligned} B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right) &= B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right]\right) \subseteq \mathcal{D}\left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n\right] \\ &\subseteq \mathcal{D}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right] \end{aligned}$$

So that $\forall z \in B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$ we have that $\sum_{n=0}^{\infty} a_n \cdot z^n$ and $\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$ converges.

2. Define $f = \sum_{n=0}^{\infty} a_n \cdot (*)^n$ and $g = \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n$. Given $z \in B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$ take $\varepsilon \in \mathbb{R}^+$. As $|z| < \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]$ there exist a $\delta' \in \mathbb{R}^+$ such that $|z| < \delta' < \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]$, further, as $B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$ is open there exist a $\delta'' \in \mathbb{R}^+$ such that $B_{||}(z, \delta'') \subseteq B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$. Hence if we take $\delta = \min(\delta', \delta'')$ then we have

$$|z| < \delta < \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right] \text{ and } z \in B_{||}(z, \delta) \subseteq B_{||}\left(0, \mathcal{R}\left[\sum_{n=0}^{\infty} a_n \cdot z^n\right]\right)$$

Let $h \in \mathbb{C}$ with $0 < |h| < \min(\delta, \delta - |z|)$ then $|h+z| \leq |h| + |z| < \delta - |z| + |z| = \delta < \delta''$ so that $h+z \in B_{||}(z, \delta'') \subseteq B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n])$ or $h \in (B_{||}(0, \mathcal{R}[\sum_{n=0}^{\infty} a_n \cdot z^n]))_z$. Then we have

$$\begin{aligned} & \frac{f(z+h) - f(z)}{h} - g(z) = \\ & \frac{1}{h} \cdot (f(z+h) - g(z) - h \cdot g(z)) = \\ & \frac{1}{h} \cdot \left(\sum_{n=0}^{\infty} a_n \cdot (z+h)^n - \sum_{n=0}^{\infty} a_n \cdot z^n - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) \quad [\text{theorem: 17.24}] = \\ & \frac{1}{h} \cdot \left(\sum_{n=0}^{\infty} a_n \cdot \left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \sum_{n=0}^{\infty} a_n \cdot z^n - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\ & \frac{1}{h} \cdot \left(\sum_{n=0}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - z^n \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\ & \frac{1}{h} \cdot \left(\sum_{n=0}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - z^n \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) \quad [\text{theorem: 14.416}] = \\ & \frac{1}{h} \cdot \left(a_0 \cdot \left(\left(\sum_{i=0}^0 \binom{0}{i} \cdot h^i \cdot z^{0-i} \right) - z^0 \right) + \sum_{n=1}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \right. \right. \\ & \left. \left. z^n \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) \quad [\text{theorem: 14.416}] = \\ & \frac{1}{h} \cdot \left(a_0 \cdot \left(\left(\binom{0}{0} \cdot h^0 \cdot z^{0-0} \right) - z^0 \right) + \sum_{n=1}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - z^n \right) - \right. \\ & \left. h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\ & \frac{1}{h} \cdot \left(a_0 \cdot 0 + \sum_{n=1}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - z^n \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot \right. \\ & \left. z^n \right) = \\ & \frac{1}{h} \cdot \left(\sum_{n=1}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - z^n \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\ & \frac{1}{h} \cdot \left(\sum_{n=1}^{\infty} a_n \cdot \left(\left(\sum_{i=0}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \binom{0}{0} \cdot h^0 \cdot z^{n-0} \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot \right. \\ & \left. z^n \right) = \\ & \frac{1}{h} \cdot \left(\sum_{n=1}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) \quad [\text{theorem: 14.416}] = \end{aligned}$$

$$\begin{aligned}
& \frac{1}{h} \cdot \left(a_1 \cdot \left(\sum_{i=1}^1 \binom{1}{i} \cdot h^i \cdot z^{1-i} \right) + \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \right. \\
& \left. h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\
& \frac{1}{h} \cdot \left(a_1 \cdot \left(\binom{1}{1} \cdot h^1 \cdot z^{1-1} \right) + \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \right. \\
& \left. h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\
& \frac{1}{h} \cdot \left(a_1 \cdot h + \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - h \cdot \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) \quad [\text{theorem: 14.416}] \\
& \frac{1}{h} \cdot \left(a_1 \cdot h + \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - h \cdot (0+1) \cdot a_{0+1} \cdot z^0 - \right. \\
& \left. h \cdot \sum_{n=1}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\
& \frac{1}{h} \cdot \left(a_1 \cdot h + \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - h \cdot a_1 - h \cdot \sum_{n=1}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) = \\
& \frac{1}{h} \cdot \left(\sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - h \cdot \sum_{n=1}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right) \quad [\text{theorem: 14.415}] \\
& \frac{1}{h} \cdot \left(\sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - h \cdot \sum_{n=2}^{\infty} n \cdot a_n \cdot z^{n-1} \right) \quad [\text{theorem: 14.418}] \\
& \frac{1}{h} \cdot \left(\sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \sum_{n=2}^{\infty} n \cdot a_n \cdot h \cdot z^{n-1} \right) \quad [\text{theorem: 14.418}] \\
& \frac{1}{h} \cdot \left(\sum_{n=2}^{\infty} \left(a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - n \cdot a_n \cdot h \cdot z^{n-1} \right) \right) = \\
& \frac{1}{h} \cdot \left(\sum_{n=2}^{\infty} \left(a_n \cdot \left(\sum_{i=1}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) - \binom{n}{1} \cdot a_n \cdot h^1 \cdot z^{n-1} \right) \right) = \\
& \frac{1}{h} \cdot \left(\sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=2}^n \binom{n}{i} \cdot h^i \cdot z^{n-i} \right) \right) \quad [\text{theorem: 14.418}] \\
& \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=2}^n \binom{n}{i} \cdot h^{i-1} \cdot z^{n-i} \right)
\end{aligned}$$

To summarize the above lengthy computation we have proved that

$$\frac{f(z+h) - f(z)}{h} - g(z) = \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=2}^n \binom{n}{i} \cdot h^{i-1} \cdot z^{n-i} \right) \quad (19.19)$$

Now let $n \in \{2, \dots, \infty\}$ then we have

$$\begin{aligned}
& \left| \sum_{i=2}^n \binom{n}{i} \cdot h^{i-1} \cdot z^{n-i} \right| = \\
& |h| \cdot \left| \sum_{i=2}^n \binom{n}{i} \cdot h^{i-2} \cdot z^{n-i} \right| \leq
\end{aligned}$$

$$\begin{aligned}
& |h| \cdot \sum_{i=2}^n \binom{n}{i} \cdot |h|^{i-2} \cdot |z|^{n-i} = \\
& |h| \cdot \sum_{i=2}^n \frac{n!}{(n-i)! \cdot i!} \cdot |h|^{i-2} \cdot |z|^{n-i} \stackrel{n, i \geq 2}{=} \\
& |h| \cdot \sum_{i=2}^n \frac{n \cdot (n-1) \cdot (n-2)!}{(n-i)! \cdot i \cdot (i-1) \cdot (i-2)!} \cdot |h|^{i-2} \cdot |z|^{n-i} \leq i \geq 2 \Rightarrow i \cdot (i-1) \geq 1 \\
& |h| \cdot \sum_{i=2}^n \frac{n \cdot (n-1) \cdot (n-2)!}{(n-i)! \cdot (i-2)!} \cdot |h|^{i-2} \cdot |z|^{n-i} = \\
& |h| \cdot n \cdot (n-1) \cdot \sum_{i=2}^n \frac{(n-2)!}{(n-i)! \cdot (i-2)!} \cdot |h|^{i-2} \cdot |z|^{n-i} = \\
& |h| \cdot n \cdot (n-1) \cdot \sum_{i=2}^n \frac{(n-2)!}{((n-2)-(i-2))! \cdot (i-2)!} \cdot |h|^{i-2} \cdot |z|^{(n-2)-(i-2)} = \\
& |h| \cdot n \cdot (n-1) \cdot \sum_{i=0}^{n-2} \frac{(n-2)!}{((n-2)-i)! \cdot i!} \cdot |h|^i \cdot |z|^{(n-2)-i} \stackrel{[\text{theorem: 17.24}]}{=} \\
& |h| \cdot n \cdot (n-1) \cdot (|z| + |h|)^{n-2}
\end{aligned}$$

proving

$$\sum_{i=2}^n \binom{n}{i} \cdot |h|^{i-2} \cdot |z|^{n-i} \leq |h| \cdot n \cdot (n-1) \cdot (|z| + |h|)^{n-2} \quad (19.20)$$

As

$$\begin{aligned}
|z| + |h| & < \delta - |z| + |z| \\
& = \delta \\
& < \mathcal{R} \left[\sum_{n=0}^{\infty} a_n \cdot z^n \right] \\
& \stackrel{[\text{theorem: 19.25}]}{=} \mathcal{R} \left[\sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \right] \\
& \stackrel{[\text{theorem: 19.25}]}{=} \mathcal{R} \left[\sum_{n=0}^{\infty} ((n+1)+1) \cdot (n+1) \cdot a_{(n+1)+1} \cdot z^n \right] \\
& = \mathcal{R} \left[\sum_{n=0}^{\infty} (n+2) \cdot (n+1) \cdot a_{n+2} \cdot z^n \right]
\end{aligned}$$

so

that

$|z| + |h| \in B_{||}(\mathcal{R}[\sum_{n=0}^{\infty} (n+2) \cdot (n+1) \cdot a_{n+2} \cdot z^n]) \subseteq \overline{\mathcal{D}}[\sum_{n=0}^{\infty} (n+2) \cdot (n+1) \cdot a_{n+2} \cdot z^n]$. Hence we have that $\sum_{n=0}^{\infty} (n+2) \cdot (n+1) \cdot |a_{n+2}| \cdot (|z| + |h|)^n$ converges, using [theorem: 14.415] it follows that $\sum_{n=2}^{\infty} n \cdot (n-1) \cdot |a_n| \cdot (|z| + |h|)^{n-2}$ converges. Further using [theorem: 14.418] it follows that $\sum_{n=0}^{\infty} |h| \cdot n \cdot (n-1) \cdot |a_n| \cdot (|z| + |h|)^{n-2}$ converges. As by [eq: 19.20]

$$\sum_{i=2}^n \binom{n}{i} \cdot |h|^{i-2} \cdot |z|^{n-i} \leq |h| \cdot n \cdot (n-1) \cdot (|z| + |h|)^{n-2}$$

we can use dominant convergence [see theorem: 14.437] proving that

$$\sum_{n=2}^{\infty} \left(\sum_{i=2}^n \binom{n}{i} \cdot |h|^{i-2} \cdot |z|^{n-i} \right)$$

converges and that

$$\left| \sum_{n=2}^{\infty} \left(\sum_{i=2}^n \binom{n}{i} \cdot |h|^{i-2} \cdot |z|^{n-i} \right) \right| \leq \sum_{n=0}^{\infty} |h| \cdot n \cdot (n-1) \cdot |a_n| \cdot (|z| + |h|)^{n-2}$$

Hence we have

$$\begin{aligned} \left| \frac{f(z+h) - f(z)}{h} - g(z) \right| &\stackrel{[\text{eq: 19.19}]}{=} \left| \sum_{n=2}^{\infty} a_n \cdot \left(\sum_{i=2}^n \binom{n}{i} \cdot h^{i-1} \cdot z^{n-i} \right) \right| \\ &\leq \sum_{n=0}^{\infty} |h| \cdot n \cdot (n-1) \cdot |a_n| \cdot (|z| + |h|)^{n-2} \\ &\stackrel{[\text{theorem: 14.418}]}{=} |h| \cdot \sum_{n=0}^{\infty} |h| \cdot n \cdot (n-1) \cdot |a_n| \cdot (|z| + |h|)^{n-2} \end{aligned}$$

Take now $\delta_\varepsilon = \min \left(\delta, \frac{\varepsilon}{1 + \sum_{n=0}^{\infty} |h| \cdot n \cdot (n-1) \cdot |a_n| \cdot (|z| + |h|)^{n-2}} \right)$ then if $0 < |h| < \delta_\varepsilon$ it follows that

$$\left| \frac{f(z+h) - f(z)}{h} - g(z) \right| < \varepsilon$$

which proves that

$$f'_z \text{ exist and } f'_z = g(z)$$

Using the definition of f and g we have finally that $(\sum_{n=0}^{\infty} a_n \cdot (*)^n)'_z$ exist and

$$\left(\sum_{n=0}^{\infty} a_n \cdot (*)^n \right)'_z = \sum_{n=0}^{\infty} (n+1) \cdot a_{n+1} \cdot z^n \quad \square$$

We can use the above theorem to prove that the exponential function has a derivative at every complex number.

Theorem 19.27. $\forall z \in \mathbb{C}$ the exponential function

$$\exp: \mathbb{C} \rightarrow \mathbb{C} \text{ defined by } \exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

or

$$\exp = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot (*)^n$$

has a derivative $\exp'_z$ where

$$\exp'_z = \exp(z)$$

In other words if we define $\exp'$ by

$$\exp': \mathbb{C} \rightarrow \mathbb{C} \text{ by } \exp'(x) = \exp'_x$$

then

$$\exp' = \exp$$

Proof. Using [theorem: 19.21] it follows that $B_{||}(0, \mathcal{D}[\sum_{n=0}^{\infty} \frac{z^n}{n!}]) = \mathbb{C}$ then by the previous theorem [theorem: 19.26] we have that $\forall z \in \mathbb{C}$ $(\sum_{n=0}^{\infty} \frac{1}{n!} \cdot (*)^n)'_z$ exist and

$$\exp'_z = \left(\sum_{n=0}^{\infty} \frac{1}{n!} \cdot (*)^n \right)'_z = \sum_{n=0}^{\infty} \frac{n+1}{(n+1)!} \cdot z^n = \sum_{n=0}^{\infty} \frac{z^n}{n!} = \left(\sum_{n=0}^{\infty} \frac{(*)^n}{n!} \right)(z) = \exp(z) \quad \square$$

A direct consequence of the above is that the exponential function is continuous.

Theorem 19.28. For $\exp: \mathbb{C} \rightarrow \mathbb{C}$ we have:

1. $\exp$ is continuous
2. $\forall z \in \mathbb{C}$ we have $\forall n \in \mathbb{N}$ that $\exp$ has a n -th derivative at z and $\exp_z^{(n)} = \exp(z)$

Proof.

1. As $\forall z \in \mathbb{C}$ $\exp'_z$ exist it follows from [theorem: 16.64] that $\exp$ is differentiable at z , hence, using [theorem: 16.63], $\exp$ is continuous at x .

2. The prove is easily done by induction, so let $z \in \mathbb{C}$ and define

$$S_z = \{n \in S \mid \exp_z^{(n)} \text{ exist and } \exp_z^{(n)} = \exp(z)\}$$

then we have:

$1 \in S$. As $\exp'_z$ exist and $\exp_z^{(1)} = \exp'_z = \exp(z)$ we have that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ it follows that $\forall z \in \mathbb{C}$ $\exp_z^{(n)}$ exist and $\exp_z^{(n)} = \exp(z)$ so that the function

$$\exp^{(n)}: \mathbb{C} \rightarrow \mathbb{C} \quad \exp^{(n)}(z) = \exp_z^n$$

is equal to $\exp: \mathbb{C} \rightarrow \mathbb{C}$, hence $\exp^{(n)} = \exp$. Let $z \in \mathbb{C}$ then as $\exp'_z$ exist and $\exp'_z = \exp(z)$ it follows by definition that $\exp_z^{(n+1)}$ exist and that

$$\exp_z^{(n+1)} = (\exp^{(n)})'_z = \exp'_z = \exp(z)$$

So it follows that $n+1 \in S$. □

19.1.5 Properties of the exponential function

Next we show that the exponential behaves a lot like a power of a number. This motivates the common notation to write $\exp(z)$ as e^z . First we need some theorems about series.

Theorem 19.29. (Mertens) Let $\sum_{i=0}^{\infty} a_i$ be a absolute convergent series in $\mathbb{C}$ and $\sum_{i=0}^{\infty} b_i$ a convergent series in $\mathbb{C}$. If $\{c_i\}_{i \in \mathbb{N}_0} \subseteq \mathbb{C}$ is defined by $c_i = \sum_{j=0}^i a_j \cdot b_{i-j}$ then $\sum_{i=0}^{\infty} c_i$ converges and

$$\sum_{i=0}^{\infty} c_i = \left(\sum_{i=0}^{\infty} a_i \right) \cdot \left(\sum_{i=0}^{\infty} b_i \right)$$

Proof. Let $A = \sum_{i=0}^{\infty} a_i$ and $B = \sum_{i=0}^{\infty} b_i$. We prove now by induction that

$$\forall n \in \mathbb{N}_0 \quad \sum_{i=0}^n c_i = \sum_{i=0}^n a_i \cdot \left(\sum_{j=0}^{n-i} b_j \right) \quad (19.21)$$

Proof. Let $S = \{n \in \mathbb{N}_0 \mid \sum_{i=0}^n c_i = \sum_{i=0}^n a_i \cdot (\sum_{j=0}^{n-i} b_j)\}$ then we have:

$0 \in S$. $\sum_{i=0}^0 c_i = c_0 = \sum_{j=0}^0 a_j \cdot b_{0-j} = a_0 \cdot b_0 = \sum_{i=0}^0 a_i \cdot b_0 = \sum_{i=0}^0 a_i \cdot (\sum_{j=0}^{0-i} b_j)$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Then

$$\begin{aligned} & \sum_{i=0}^{(n+1)} a_i \cdot \left(\sum_{j=0}^{(n+1)-i} b_j \right) = \\ & a_{n+1} \cdot \sum_{j=0}^{(n+1)-(n+1)} b_j + \sum_{i=0}^n a_i \cdot \left(\sum_{j=0}^{(n+1)-i} b_j \right) = \\ & a_{n+1} \cdot b_0 + \sum_{i=0}^n a_i \cdot \left(b_{n+1-i} + \sum_{j=0}^{n-i} b_j \right) = \\ & a_{n+1} \cdot b_0 + \sum_{i=0}^n a_i \cdot b_{n+1-i} + \sum_{i=0}^n a_i \cdot \left(\sum_{j=0}^{n-i} b_j \right) \stackrel{n \in S}{=} \\ & a_{n+1} \cdot b_0 + \sum_{i=0}^n a_i \cdot b_{n+1-i} + \sum_{i=0}^n c_i = \end{aligned}$$

$$\begin{aligned}
a_{n+1} \cdot b_{(n+1)-(n+1)} + \sum_{i=0}^n a_i \cdot b_{n+1-i} + \sum_{i=0}^n c_i &= \\
\sum_{i=0}^{n+1} a_i \cdot b_{n+1-i} + \sum_{i=0}^n c_i &= \\
c_{n+1} + \sum_{i=0}^n c_i &= \\
\sum_{i=0}^{n+1} c_i
\end{aligned}$$

which proves that $n+1 \in S$.

Mathematical induction proves then [eq: 19.21]. $\square$

By [theorem: 14.416] we have for $n \in \mathbb{N}_0$ that $B = \sum_{i=0}^{\infty} b_i = \sum_{i=0}^{n-i} b_i + \sum_{i=n-i+1}^{\infty} b_i$, hence $B - \sum_{i=n-i+1}^{\infty} b_i = \sum_{i=0}^{n-i} b_i$. Substituting this in [eq: 19.21] gives that $\forall n \in \mathbb{N}_0$

$$\sum_{i=0}^n c_i = \sum_{i=0}^n a_i \cdot \left(B - \sum_{i=n-i+1}^{\infty} b_i \right) = B \cdot \sum_{i=0}^n a_i - \sum_{i=0}^n a_i \cdot \left(\sum_{j=n-i+1}^{\infty} b_j \right)$$

Hence

$$\begin{aligned}
\left| \sum_{i=0}^n c_i - A \cdot B \right| &= \left| B \cdot \sum_{i=0}^n a_i - A \cdot B - \sum_{i=0}^n a_i \cdot \left(\sum_{j=n-i+1}^{\infty} b_j \right) \right| \\
&= \left| \left(\sum_{i=0}^n a_i - A \right) \cdot B - \sum_{i=0}^n a_i \cdot \left(\sum_{j=n-i+1}^{\infty} b_j \right) \right| \\
&\leq \left| \sum_{i=0}^n a_i - A \right| + \left| \sum_{i=0}^n a_i \cdot \left(\sum_{j=n-i+1}^{\infty} b_j \right) \right| \\
&\leq \left| \sum_{i=0}^n a_i - A \right| + \sum_{i=0}^n \left| a_i \cdot \left(\sum_{j=n-i+1}^{\infty} b_j \right) \right| \\
&= \left| \sum_{i=0}^n a_i - A \right| + \sum_{i=0}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right)
\end{aligned}$$

so that

$$\left| \sum_{i=0}^n c_i - A \cdot B \right| \leq \left| \sum_{i=0}^n a_i - A \right| + \sum_{i=0}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) \quad (19.22)$$

Let $\varepsilon \in \mathbb{R}^+$ then as $\sum_{i=0}^{\infty} a_i = A$ there exist a $N_1 \in \mathbb{N}_0$ such that

$$\forall n \geq N_1 \text{ we have } \left| \sum_{i=0}^n a_i - A \right| < \frac{\varepsilon}{2} \quad (19.23)$$

Further as $\sum_{i=0}^{\infty} b_i$ converges it follow from [theorem: 14.417]

$$\lim_{n \rightarrow \infty} \sum_{i=n}^{\infty} b_i = 0 \quad (19.24)$$

Using then [theorem: 14.364] on the above proves that there exist a $M \in \mathbb{R}^+$ such that

$$\forall n \in \mathbb{N}_0 \text{ we have } \left| \sum_{i=n}^{\infty} b_i \right| \leq M \quad (19.25)$$

As $\sum_{i=0}^{\infty} a_i$ is absolute convergent $\sum_{i=0}^n |a_i|$ is convergent, given that $\lim_{n \rightarrow \infty} \sum_{i=n}^{\infty} b_i = 0$ there exist a $N_2 \in \mathbb{N}_0$ such that

$$\forall n \geq N_2 \text{ we have that } \left| \sum_{i=n}^{\infty} b_i \right| < \frac{\varepsilon}{4 \cdot (\sum_{i=1}^{\infty} |\alpha_i| + 1)} \quad (19.26)$$

Further as by [theorem: 14.417] $\lim_{n \rightarrow \infty} \sum_{i=n}^{\infty} |a_i| = 0$ there exist a $N_3 \in \mathbb{N}_0$ such that

$$\sum_{i=N_3+1}^{\infty} |a_i| < \frac{\varepsilon}{4 \cdot M} \quad (19.27)$$

Let $N_4 = \max(N_2, N_3)$ and $N = \max(N_1, 2 \cdot N_4)$ then for $n \geq N$ we have that $n \geq 2 \cdot N_4$, so that $\forall i \in \{1, \dots, N_4\}$ we have that $n - i \geq 2 \cdot N_4 - N_4 = N_4 \geq N_2$. Hence if $n \geq N$ then we have

$$\forall i \in \{1, \dots, N_4\} \text{ that } n - i + 1 \geq N_2 \quad (19.28)$$

and

$$\begin{aligned} & \sum_{i=0}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) = \\ & \sum_{i=0}^{N_4} \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) + \sum_{i=N_4+1}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) \leq_{[\text{eqs: 19.28, 19.26}]} \\ & \sum_{i=0}^{N_4} \left(|a_i| \cdot \frac{\varepsilon}{4 \cdot (\sum_{i=1}^{\infty} |\alpha_i| + 1)} \right) + \sum_{i=N_4+1}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) = \\ & \frac{\varepsilon}{4 \cdot (\sum_{i=1}^{\infty} |\alpha_i| + 1)} \cdot \sum_{i=0}^{N_4} |a_i| + \sum_{i=N_4+1}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) \leq_{[\text{theorem: 14.416}]} \\ & \frac{\varepsilon}{4 \cdot (\sum_{i=1}^{\infty} |\alpha_i| + 1)} \cdot \sum_{i=0}^{\infty} |a_i| + \sum_{i=N_4+1}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) < \\ & \frac{\varepsilon}{4} + \sum_{i=N_4+1}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) \leq_{[\text{eq: 19.25}]} \\ & \frac{\varepsilon}{4} + \sum_{i=N_4+1}^n (|a_i| \cdot M) = \\ & \frac{\varepsilon}{4} + M \cdot \sum_{i=N_4+1}^n |a_i| \leq_{[\text{theorem: 14.416}]} \\ & \frac{\varepsilon}{4} + M \cdot \sum_{i=N_4+1}^{\infty} |a_i| \leq_{[\text{eq: 19.27}]} \\ & \frac{\varepsilon}{4} + M \cdot \frac{\varepsilon}{4 \cdot M} = \\ & \frac{\varepsilon}{2} \end{aligned}$$

proving that

$$\sum_{i=0}^n \left(|a_i| \cdot \left| \sum_{j=n-i+1}^{\infty} b_j \right| \right) < \frac{\varepsilon}{2}$$

Combining the above with [eq: 19.22] and [eq: 19.23] it follows that

$$\forall n \geq N \text{ we have } \left| \sum_{i=0}^n c_i - A \cdot B \right| < \varepsilon$$

Hence

$$\sum_{i=0}^{\infty} c_i \text{ converges and } \sum_{i=0}^{\infty} c_i = A \cdot B = \left(\sum_{i=0}^{\infty} a_i \right) \cdot \left(\sum_{i=0}^{\infty} b_i \right) \quad \square$$

We apply now the above theorem to the product of power series.

Theorem 19.30. Let $\sum_{n=0}^{\infty} a_n \cdot z^n$, $\sum_{n=0}^{\infty} b_n \cdot z^n$ then if $z \in \overline{\mathcal{D}}(\sum_{n=0}^{\infty} a_n \cdot z^n) \cap \mathcal{D}(\sum_{n=0}^{\infty} b_n \cdot z^n)$ we have that $z \in \mathcal{D}(\sum_{i=0}^{\infty} c_i \cdot z^i)$ where $c_i = \sum_{j=0}^i a_j \cdot b_{i-j}$. Further

$$\sum_{i=0}^{\infty} c_i = \left(\sum_{n=0}^{\infty} a_n \cdot z^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \cdot z^n \right)$$

Proof. Let $z \in \overline{\mathcal{D}}(\sum_{n=0}^{\infty} a_n \cdot z^n) \cap \mathcal{D}(\sum_{n=0}^{\infty} b_n \cdot z^n)$ then $\sum_{n=0}^{\infty} a_n \cdot z^n$ is absolute convergent and $\sum_{n=0}^{\infty} b_n \cdot z^n$ is convergent. By the previous theorem [theorem: 19.29] it follows that

$$\sum_{i=0}^{\infty} \left(\sum_{j=0}^i (a_j \cdot z^j) \cdot (b_{i-j} \cdot z^{i-j}) \right)$$

converges and

$$\sum_{i=0}^{\infty} \left(\sum_{j=0}^i (a_j \cdot z^j) \cdot (b_{i-j} \cdot z^{i-j}) \right) = \left(\sum_{n=0}^{\infty} a_n \cdot z^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \cdot z^n \right)$$

Further we have

$$\begin{aligned} \sum_{i=0}^{\infty} \left(\sum_{j=0}^i (a_j \cdot z^j) \cdot (b_{i-j} \cdot z^{i-j}) \right) &= \sum_{i=0}^{\infty} \left(\sum_{j=0}^i a_j \cdot b_{i-j} \cdot z^i \right) \\ &= \sum_{i=0}^{\infty} \left(\sum_{j=0}^i a_j \cdot b_{i-j} \right) \cdot z^i \\ &= \sum_{i=0}^{\infty} c_i \cdot z^i \end{aligned}$$

proving that $\sum_{i=0}^{\infty} c_i \cdot z^i$ converges and $\sum_{i=0}^{\infty} c_i \cdot z^i = (\sum_{n=0}^{\infty} a_n \cdot z^n) \cdot (\sum_{n=0}^{\infty} b_n \cdot z^n)$. $\square$

Theorem 19.31. Let $x, y \in \mathbb{C}$ then $\exp(x + y) = \exp(x) \cdot \exp(y)$

Proof. Let $x, y \in \mathbb{C}$ then as $\mathbb{C} \stackrel{[\text{theorem: 19.21}]}{=} \overline{\mathcal{D}}\left[\sum_{n=0}^{\infty} \frac{z^n}{n!}\right]$ it follows that $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ and $\sum_{n=0}^{\infty} \frac{y^n}{n!}$ are absolute convergent [see definition: 14.436]. Hence we can use the above theorem [theorem: 19.30] giving that

$$\begin{aligned} \exp(x) \cdot \exp(y) &= \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} \right) \cdot \left(\sum_{n=0}^{\infty} \frac{y^n}{n!} \right) \\ &\stackrel{[\text{theorem: 19.29}]}{=} \sum_{n=0}^{\infty} \left(\sum_{i=0}^n \frac{x^i}{i!} \cdot \frac{y^{i-j}}{(i-j)!} \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \cdot \sum_{i=0}^n \left(\frac{n!}{i! \cdot (i-j)!} \cdot x^i \cdot y^{i-j} \right) \\ &\stackrel{[\text{theorem: 17.24}]}{=} \sum_{n=0}^{\infty} \frac{(x+y)^i}{n!} \\ &= \exp(x+y) \end{aligned} \quad \square$$

Theorem 19.32. *The exponential function $\exp: \mathbb{C} \rightarrow \mathbb{C}$ has the following properties:*

1. $\exp(0) = 1$
2. $\forall z \in \mathbb{C}$ we have $\exp(z) \neq 0$.
3. $\forall z \in \mathbb{C}$ we have $\exp(-z) = \frac{1}{\exp(z)}$.
4. $\forall z \in \mathbb{R}$ we have $0 < \exp(z)$ or in other words $\exp(\mathbb{R}) \subseteq \mathbb{R}^+$.
5. $\forall z \in \mathbb{R}$ with $0 < z$ we have $1 < \exp(z)$.
6. $\forall x, y \in \mathbb{R}$ with $x < y$ we have $\exp(x) < \exp(y)$ [$\exp$ is strictly increasing].
7. $\forall x, y \in \mathbb{R}$ with $x < y$ it follows that $\exp$ is convex on $[x, y]$ [see definition: 16.212].

Proof.

1. If $z = 0$ then $\exp(0) = \exp(z) = \sum_{i=0}^{\infty} \frac{1}{i!} \cdot z^i \stackrel{[\text{example: 19.4}]}{=} \frac{1}{0!} = 1$.

2. Assume that $\exists z \in \mathbb{C}$ such that $\exp(z) = 0$. Then

$$1 = \exp(0) = \exp(z + (-z)) \stackrel{[\text{theorem: 19.31}]}{=} \exp(z) \cdot \exp(-z) = 0 \cdot \exp(-z) = 0$$

leading to the contradiction $1 = 0$. So the assumption is wrong and we have $\forall z \in \mathbb{C}$ that $\exp(z) \neq 0$.

3. $1 = \exp(0) = \exp(z + (-z)) = \exp(z) \cdot \exp(-z)$ so using (2) we have $\exp(-z) = \frac{1}{\exp(z)}$.

4. If $z \in \mathbb{R}$ then we have 3 cases to consider for z :

$$z \in \mathbb{R}^+. \text{ Then } \forall i \in \mathbb{N}_0 \text{ we have } 0 < \frac{z^i}{i!} \text{ [see theorem: 10.51] so that by [theorem: 14.422]} \\ 0 < \sum_{i=0}^{\infty} \frac{z^i}{i!} = \exp(z).$$

$$z = 0. \text{ Then } 0 < 1 = \exp(0) = \exp(z).$$

$$z \in \mathbb{R}^-. \text{ Then } -z \in \mathbb{R}^+ \text{ then } 0 < \frac{1}{\exp(-z)} \stackrel{(3)}{=} \exp(z).$$

Hence in all cases we have $0 < \exp(z)$.

5. If $0 < z$ then $\forall i \in \mathbb{N}_0$ we have $0 < \frac{z^i}{i!}$ [see theorem: 10.51] so that by [theorem: 14.422] $0 < \sum_{i=1}^{\infty} \frac{z^i}{i!} = \exp(z)$. Hence $\exp(z) = \sum_{i=0}^{\infty} \frac{z^i}{i!} \stackrel{[\text{theorem: 14.416}]}{=} \frac{z^0}{0!} + \sum_{i=1}^{\infty} \frac{z^i}{i!} = 1 + \sum_{i=1}^{\infty} \frac{z^i}{i!} > 1$.

6. Let $x < y$ then $0 < y - x$ so that by (5) $1 < \exp(y - x)$, as $0 < \exp(x)$ it follows that

$$\exp(x) < \exp(x) \cdot \exp(y - x) \cdot \exp(x + y - x) = \exp(y)$$

7. Let $x \in \mathbb{R}$ then $\exp_x^{(2)} \stackrel{[\text{theorem: 19.28}]}{=} \exp(x) > 0$ [by (4) so that by [theorem: 16.214] is convex. $\square$]

We introduce now what is probably the most interesting number in mathematics [maybe with the exception of π which we will also define in this chapter] which is Euler's number.

Definition 19.33. (Euler's number) *Euler's number noted as e is defined as*

$$e = \exp(1) = \sum_{n=1}^{\infty} \frac{1}{n!}$$

Using Euler's number we see that the exponential of natural numbers behaves as a power of Euler's number.

Theorem 19.34. $\forall n \in \mathbb{N}_0$ we have

1. $e \in \mathbb{R}^+$ and $1 < e$
2. $\exp(n) = e^n$
3. $\exp(-n) = \frac{1}{e^n}$

Proof.

1. As $1 \in \mathbb{R}^+$ it follows from [theorem: 19.32 (5)] that $1 < \exp(1) = e$
2. We prove this by induction so define:

$$S = \{n \in \mathbb{N}_0 | \exp(n) = e^n\}$$

then we have:

$0 \in S$. As $\exp(0) \stackrel{[\text{theorem: 19.32}]}{=} 1 = e^0$ it follows that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $\exp(n+1) \stackrel{[\text{theorem: 19.31}]}{=} \exp(n) \cdot \exp(1) \stackrel{n \in S}{=} e^n \cdot e = e^{n+1}$ it follows that $n+1 \in S$.

3. $\exp(-n) \stackrel{[\text{theorem: 19.32}]}{=} \frac{1}{\exp(n)} \stackrel{[\text{theorem: 19.32}]}{=} \frac{1}{e^n}$ $\square$

The above theorem suggest that we could extend this to $\exp(z) = e^z$ where z is not a whole number. To be able to do this we must find a inverse of the exponential function which will be only possible on real numbers.

Theorem 19.35. Let $y \in \mathbb{R}^+$ then there exist a $x \in \mathbb{R}$ such that $y < \exp(x)$

Proof. By [theorem: 19.34] it follows that $1 < e$ so that by [theorem: 10.61] there exist a $n \in \mathbb{N}$ such that $y \leq e^n$. As by [theorem: 19.34] $e^n = \exp(n)$ it follows that $y \leq \exp(n)$ where $n \in \mathbb{N} \subseteq \mathbb{R}$. $\square$

Theorem 19.36. $\exp|_{\mathbb{R}}: \mathbb{R} \rightarrow \mathbb{R}^+$ is a bijection, $\exp(\mathbb{R}^+) =]1, \infty[$ and $\exp(\mathbb{R}^-) =]0, 1[$

Proof. First we have to prove that $\exp|_{\mathbb{R}}: \mathbb{R} \rightarrow \mathbb{R}^+$ is a bijection.

injectivity. Let $x, y \in \mathbb{R}$ such that $\exp(x) = \exp|_{\mathbb{R}}(x) = \exp|_{\mathbb{R}}(y) = \exp(y)$. Assume that $x \neq y$ then we may without loss of generality assume that $x < y$. Then by [theorem: 19.32 (6)] it follows that $\exp(x) < \exp(y)$ contradicting $\exp(x) = \exp(y)$. Hence the assumption must be wrong and we must have that $x = y$.

surjectivity. Let $y \in \mathbb{R}^+ =]0, \infty[$ then we have the following possibilities:

$y = 1$. Then $\exp(0) = 1 = y$.

$1 < y$. Then by [theorem: 19.35] there exist a $z \in \mathbb{R}$ such that $y < \exp(z)$ hence we have as $1 = \exp(0)$ that $\exp(0) < y < \exp(z)$. Further by [theorem: 19.28] on $\mathbb{R}$ hence continuous on $[0, z]$. Using the intermediate value theorem [see theorem: 14.467] there exist a $x \in [0, z]$ such that $\exp(x) = y$.

$0 < y < 1$. Then $1 < \frac{1}{y}$ so that by the previous case there exist a $z \in \mathbb{R}$ such that $\exp(z) = \frac{1}{y}$. Take $x = -z$ then $\exp(x) = \exp(-y) \stackrel{[\text{theorem: 19.32 (3)}]}{=} \frac{1}{\exp(y)} = \frac{1}{\frac{1}{y}} = y$

So in all cases we found a $x \in \mathbb{R}$ such that $\exp(x) = y$ which proves surjectivity.

For the remainder of the theorem. Let $x \in \mathbb{R}^+$ then by [theorem: 19.32(5)] we have that $1 < \exp(x)$ hence

$$\exp(\mathbb{R}^+) \subseteq]1, \infty[\tag{19.29}$$

If $x \in \mathbb{R}^-$ then $-x \in \mathbb{R}^+$ so that by [theorem: 19.32(5)] we have $1 < \exp(-x) = \frac{1}{\exp(x)}$ so that $\exp(x) < 1$. Hence we have proved that

$$\exp(\mathbb{R}^-) \subseteq]0, 1[\quad (19.30)$$

If $y \in]1, \infty[\subseteq \mathbb{R}^+$ then by surjectivity of $\exp|_{\mathbb{R}}$ there exist a $x \in \mathbb{R}$ such that $\exp(x) = y$. Assume that $x \in \mathbb{R}^-$ then by [eq: 19.30] $y = \exp(x) \in]0, 1[$ contradicting $y \in]1, \infty[$ hence we must have that $x \in \mathbb{R}_0^+$ or as $\exp(0) = 1 \neq y$ we must have $x \in \mathbb{R}^+$, hence $y \in \exp(\mathbb{R}^+)$. So $]1, \infty[\subseteq \exp(\mathbb{R}^+)$ which combined with [eq: 19.29] gives

$$\exp(\mathbb{R}) =]1, \infty[\quad (19.31)$$

If $y \in]0, 1[\subseteq \mathbb{R}^+$ then by surjectivity there exist a $x \in \mathbb{R}$ such that $\exp(x) = y$. Assume that $x \in \mathbb{R}^+$ then by [eq: 19.29] $y = \exp(x) \in]1, \infty[$ contradicting $y \in]0, 1[$. Hence we must have that $x \in \mathbb{R}_0^-$, which, as $\exp(0) = 1 \neq y$, means that $x \in \mathbb{R}^-$, hence $y \in \exp(\mathbb{R}^-)$. So $]0, 1[\subseteq \exp(\mathbb{R}^-)$ which combined with [eq: 19.30] proves

$$\exp(\mathbb{R}^-) =]0, 1[\quad \square$$

As every bijection has a inverse the function $\exp|_{\mathbb{R}}: \mathbb{R} \rightarrow \mathbb{R}^+$ has a inverse called the natural logarithm.

Definition 19.37. (Natural Logarithm) *The natural logarithm noted as $\log$ is the inverse of $\exp|_{\mathbb{R}}: \mathbb{R} \rightarrow \mathbb{R}^+$. In other words*

$$\log: \mathbb{R}^+ \rightarrow \mathbb{R} \text{ is defined by } \log = (\exp|_{\mathbb{R}})^{-1}$$

Based on the definition of the logarithm and the properties of the exponential function we have the following properties of the logarithm.

Theorem 19.38. $\log: \mathbb{R}^+ \rightarrow \mathbb{R}$ has the following properties:

1. $\forall x \in \mathbb{R}^+$ we have $\exp(\log(x)) = x$.
2. $\forall x \in \mathbb{R}$ we have $\log(\exp(x)) = x$.
3. $\log(e) = 1$.
4. $\log(1) = 0$.
5. $\forall x, y \in \mathbb{R}^+$ we have $\log(x \cdot y) = \log(x) + \log(y)$.
6. $\forall x \in \mathbb{R}^+$ we have $\log\left(\frac{1}{x}\right) = -\log(x)$.
7. $\forall x \in \mathbb{R}^+$ $\log$ has a derivative $\log'_x$ where $\log'_x = \frac{1}{x}$.
8. $\log: \mathbb{R}^+ \rightarrow \mathbb{R}$ is strictly increasing and $\forall x < 1$ we have $\log(x) < 0$.

Proof.

1. Let $x \in \mathbb{R}^+$ then $\exp(\log(x)) = \exp|_{\mathbb{R}}((\exp|_{\mathbb{R}})^{-1}(x)) = x$.
2. Let $x \in \mathbb{R}$ then $\log(\exp(x)) = (\exp|_{\mathbb{R}})^{-1}(\exp|_{\mathbb{R}}(x)) = x$.
3. $\log(e) = (\exp|_{\mathbb{R}})^{-1}(\exp(1)) = (\exp|_{\mathbb{R}})^{-1}(\exp|_{\mathbb{R}}(1)) = 1$.
4. $\log(1) = (\exp|_{\mathbb{R}})^{-1}(\exp(0)) = (\exp|_{\mathbb{R}})^{-1}(\exp|_{\mathbb{R}}(0)) = 0$.
5. Let $x, y \in \mathbb{R}^+$ then

$$\exp|_{\mathbb{R}}(\log(x) + \log(y)) = \exp(\log(x) + \log(y)) = \exp(\log(x)) \cdot \exp(\log(y)) \stackrel{(1)}{=} x \cdot y$$

$$\text{so that } \log(x) + \log(y) = (\exp|_{\mathbb{R}})^{-1}(x \cdot y) = \log(x \cdot y).$$

6. Let $x \in \mathbb{R}^+$ then by (1) $\exp|_{\mathbb{R}}(-\log(x)) = \exp(-\log(x)) \stackrel{[\text{theorem: 19.32}]}{=} \frac{1}{\exp(\log(x))} \stackrel{(1)}{=} \frac{1}{x}$ so that $-\log(x) = (\exp|_{\mathbb{R}})^{-1}\left(\frac{1}{x}\right) = \log\left(\frac{1}{x}\right)$.

7. Let $x \in \mathbb{R}^+$ and take $y = \log(x)$. As $\exp'_y = \exp(y) > 0$ and $\exp' = \exp$ is continuous we have by the inverse function theorem [see theorem: 16.351(3)] that there exist a open $U \subseteq \mathbb{R}$ and $V \subseteq \mathbb{R}^+$ with $y \in U$ and $\exp(y) \in V$ $\xRightarrow{\exp(y) = \exp(\log(x))}$ $x \in V$ such that $\exp|_U: U \rightarrow V$ has a inverse $(\exp|_U)^{-1}: V \rightarrow U$ and

$$\forall z \in U \text{ we have } ((\exp|_U)^{-1})'_{\exp(z)} = \frac{1}{\exp'_z} \quad (19.32)$$

As $\forall v \in V$ we have $(\exp|_U \circ \log|_V)(v) = \exp|_U(\log|_V(v)) = \exp(\log(v)) = v = \text{Id}_V(v)$ and $\forall u \in U$ we have $(\log|_V \circ \exp|_U)(u) = \log|_V(\exp|_U(u)) = \log(\exp(u)) = u = \text{Id}_U(u)$ so that $(\exp|_U)^{-1} = \log|_V$. Substituting this in [eq: 19.32] we have

$$\forall z \in V \text{ we have } (\log|_V)'_{\exp(z)} = \frac{1}{\exp'_z} \underset{[\text{theorem: 19.27}]}{=} \frac{1}{\exp(z)}$$

As $\log(x) = y \in U$ we have by the above that $((\log|_V)'_{\exp(\log(x))} = \frac{1}{\exp(\log(x))})$ so that

$$(\log|_V)'_x = \frac{1}{x}$$

Finally using the locality of the derivative [theorem: 16.36] it follows that $\log$ has a derivative at x and $\log'_x = \frac{1}{x}$.

8. Let $x, y \in \mathbb{R}^+$ with $x < y$. Assume that $\log(y) \leq \log(x)$ then as by [theorem: 19.32] $\exp$ is strict increasing it follows that $y = \exp(\log(y)) \leq \exp(\log(x)) = x$ contradicting $x < y$. Hence the assumption is wrong and we must have that $\log(x) < \log(y)$. Further if $x < 1$ then $\log(x) < \log(1) \underset{(4)}{=} 0$. $\square$

19.1.6 Generalized power

We are now ready to extend the n -th power of a real number where n is a natural number to the cases where n is not a natural number.

Definition 19.39. (Generalized Power) Let $a \in [0, \infty[$ and $p \in \mathbb{R}$ then we define a^p as follows

$$a^p = \begin{cases} \exp(p \cdot \log(a)) & \text{if } a \in]0, \dots, \infty[\\ 0 & \text{if } a = 0 \wedge p \neq 0 \\ 1 & \text{if } a = 0 \wedge p = 0 \end{cases}$$

Note 19.40. In this text $0^0 = 1$ because we have for $n \in \mathbb{N}_0$ defined a^n algebraic by $a^n = \begin{cases} 1 & \text{if } n = 0 \\ a \cdot a^{n-1} & \text{if } n \in \mathbb{N} \end{cases}$ [see definition: 10.48]. Many mathematical text argue that if $n \in \mathbb{N}$ then $0^n = 0 \cdot 0^{n-1} = 0$ so that 0^0 should be 0 instead or that 0^0 is undefined. As we have already used the algebraic definition $0^0 = 1$ in polynomial power series we adopt the convention that $0^0 = 1$.

Note 19.41. As the domain of the natural logarithm is $\mathbb{R}^+$ we define a^p only for $a \in \mathbb{R}_0^+$ another reason is that we will show later that $a^{\frac{1}{2}} = \sqrt{a}$ and $\sqrt{-1}$ is not defined in $\mathbb{R}$.

Having defined a^p we have the following properties of a^p

Theorem 19.42. We have the following properties for the generalized power

1. $\forall a \in [0, \infty[\ a^0 = 1$.
2. $\forall a \in]0, \infty[$ and $\forall x \in \mathbb{R}$ we have that $0 < a^x$ and $0 \leq 0^x$
3. $\forall a \in [0, \infty[$ and $\forall x \in \mathbb{R} \setminus \{0\}$ we have that $a^x = 0 \Leftrightarrow a = 0$.
4. $\forall a \in [0, \infty[$ and $\forall n \in \mathbb{N}_0$ we have $a^n = \begin{cases} a & \text{if } n = 1 \\ a \cdot a^{n-1} & \text{if } n \in \mathbb{N} \end{cases} = \underbrace{a \cdot \dots \cdot a}_n$ [see definition: 10.48].

5. $\forall x \in \mathbb{R}$ we have $1^x = 1$.

6. $\forall x \in \mathbb{R}$ we have $e^x = \exp(x)$.

7. $\forall a \in]0, \infty[$ we have $a^{-1} = \frac{1}{a}$.

Note 19.43. For $a=0$ we have $a^{-1}=0$ and $\frac{1}{a}$ is undefined so the condition $0 < a$ is required.

8. $\forall a \in]0, \infty[, \forall x \in \mathbb{R}$ we have $\frac{a^x}{a} = a^{x-1}$.

Note 19.44. For $a=0$ we have $\frac{a^n}{a}$ is undefined so the condition $0 < a$ is needed.

9. $\forall a \in]0, \infty[$ and $\forall x \in \mathbb{R}$ we have $a^{-x} = \frac{1}{a^x}$.

Note 19.45. For $a=0$ we have $\frac{1}{a^n}$ is undefined so the condition $0 < a$ is needed.

10. $\forall a \in]0, \infty[$ and $\forall x, y \in \mathbb{R}$ we have $(a^x)^y = a^{x \cdot y}$.

Note 19.46. $0 < a$ is essential because $0^1 \cdot 0^0 = 0 \cdot 1 = 0 \neq 1 = 0^0 = 0^{1 \cdot 0}$.

11. $\forall a \in]0, \infty[$ and $\forall x \in \mathbb{R}$ we have $\log(a^x) = x \cdot \log(a)$

Note 19.47. If $a=0$ then $a^1=0^1=0$ so that $\log(a^x)$ is not defined, hence $0 < a$ is a needed condition.

12. $\forall a, b \in]0, \infty[$ we have $\forall x \in \mathbb{R}$ that $(a \cdot b)^x = a^x \cdot b^x$.

Note 19.48. If $a=0$ and $b=1$ then for $p=0$ we have $(a \cdot b)^x = 0^0 = 1$ and $a^x \cdot b^x = 0^0 \cdot 1^0 = 0$ so that $a, b \in]0, \infty[$ is indeed required.

13. $\forall a, b \in]0, \infty[$ we have $\forall x \in \mathbb{R}$ that $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$.

14. $\forall x, y \in \mathbb{R}$ we have $a^{x+y} = a^x \cdot a^y$.

15. $\forall x \in]0, \infty[$ we have that $(*)^x: [0, \infty[\rightarrow \mathbb{R}$ defined by $(*)^x(y) = y^x$ is strictly increasing.

Proof.

1. For $a \in [0, \infty[$ we have either:

$a = 0$. Then $a^0 = 1$

$]0, \infty[$. Then $a^0 = \exp(0 \cdot \log(a)) = \exp(0) \stackrel{[\text{theorem: 19.32}]}{=} 1$

2. If $]0, \infty[$ then $a^x = \exp(x \cdot \log(a)) > 0$ [because [theorem: 19.32(4)]]]. Further by definition we have that $0^x = \begin{cases} 1 & \text{if } x=0 \\ 0 & \text{if } x \neq 0 \end{cases} \geq 0$.

3. Let $a^x = 0$ and assume that $a \neq 0$ then $]0, \infty[$ so that $0 = \exp(x \cdot \log(a))$. As by [theorem: 19.32 (4)] $0 < \exp(x \cdot \log(a))$ we reach the contradiction $0 > 0$. Hence we have

$$a^x = 0 \Rightarrow a = 0$$

If $a=0$ then as $x \neq 0$ we have that $a^x = 0$ so that

$$a = 0 \Rightarrow a^x = 0$$

4. To avoid confusion of notation write $a^{[n]}$ for the old definition of the power [see definition: 10.48] so $a^{[n]} = \begin{cases} 1 & \text{if } n=0 \\ a \cdot a^{n-1} & \text{if } n \in \mathbb{N} \end{cases}$. Let $n \in \mathbb{N}_0$ then for $a \in [0, \dots, \infty]$ we have either:

$a = 0$. Then for $n \in \mathbb{N}_0$ we have either:

$$n = 0. \text{ Then } a^n = a^0 = \begin{cases} \exp(n \cdot \log(a)) & \text{if } a \in]0, \infty[\\ 0 & \text{if } a=0 \wedge n \neq 0 \\ 1 & \text{if } a=0 \wedge n=0 \end{cases} = 1 = a^{[n]}.$$

$$n \neq 0. \text{ Then } a^n = 0^n = \begin{cases} \exp(n \cdot \log(a)) & \text{if } a \in]0, \infty[\\ 0 & \text{if } a = 0 \wedge n \neq 0 \\ 1 & \text{if } a = 0 \wedge n = 0 \end{cases} = 0 = a^{[n]}$$

$a \in]0, \dots, \infty[$. We use now induction to prove this, so define

$$S_a = \{n \in \mathbb{N}_0 \mid a^n = a^{[n]}\}$$

then we have:

$0 \in S$. As $a^0 \stackrel{a \in]0, \infty[}{=} \exp(0 \cdot \log(a)) = \exp(0) = 1 = a^{[0]}$, it follows that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. We have then

$$\begin{aligned} a^{n+1} & \stackrel{a \in]0, \infty[}{=} \exp((n+1) \cdot \log(a)) \\ & = \exp(n \cdot \log(a) + \log(a)) \\ & \stackrel{[\text{theorem: 19.31}]}{=} \exp(n \cdot \log(a)) \cdot \exp(\log(a)) \\ & \stackrel{n \in S}{=} a^{[n]} \cdot \exp(\log(a)) \\ & \stackrel{[\text{theorem: 19.38}]}{=} a^{[n]} \cdot a \\ & = a^{[n+1]} \end{aligned}$$

proving that $n+1 \in S$.

Mathematical induction proves then that in this case $\forall n \in \mathbb{N}_0$ we have that $a^n = a^{[n]}$.

So in all possible cases we have

$$a^n = a^{[n]} = \begin{cases} 1 & \text{if } n = 0 \\ a \cdot a^{n-1} & \text{if } n \in \mathbb{N} \end{cases}$$

5. As $1 \in]0, \dots, \infty[$ we have that

$$1^x = \exp(x \cdot \log(1)) \stackrel{[\text{theorem: 19.38}]}{=} \exp(x \cdot 0) = \exp(0) \stackrel{[\text{theorem: 19.32}]}{=} 1$$

6. As by [theorem: 19.34] $1 < e$ we have that $e \in]0, \dots, \infty[$ so that

$$e^x = \exp(x \cdot \log(e)) \stackrel{[\text{theorem: 19.38}]}{=} \exp(x \cdot 1) = \exp(x)$$

7. If $a \in]0, \dots, \infty[$ then

$$a^{-1} = \exp((-1) \cdot \log(a)) = \exp(-\log(a)) \stackrel{[\text{theorem: 19.32}]}{=} \frac{1}{\exp(\log(a))} \stackrel{[\text{theorem: 19.38}]}{=} \frac{1}{a}$$

8. If $a \in]0, \dots, \infty[$ then

$$\begin{aligned} \frac{a^x}{a} & = \frac{\exp(x \cdot \log(a))}{a} \\ & = \frac{\exp(((x-1)+1) \cdot \log(a))}{a} \\ & = \frac{\exp((x-1) \cdot \log(a) + \log(a))}{a} \\ & \stackrel{[\text{theorem: 19.32}]}{=} \frac{\exp((x-1) \cdot \log(a)) \cdot \exp(\log(a))}{a} \\ & = \frac{a^{x-1} \cdot \exp(\log(a))}{a} \\ & \stackrel{[\text{theorem: 19.38}]}{=} \frac{a^{x-1} \cdot a}{a} \\ & = a^{x-1} \end{aligned}$$

9. If $a \in]0, \infty[$ then

$$\begin{aligned} a^{-x} &= \exp(-x \cdot \log(a)) \\ &\stackrel{[theorem: 19.32]}{=} \frac{1}{\exp(x \cdot \log(a))} \\ &= \frac{1}{a^x} \end{aligned}$$

10. As $a \in]0, \infty[$ we have

$$\begin{aligned} (a^x)^y &= \exp(y \cdot \log(a^x)) \\ &= \exp(y \cdot \log(\exp(x \cdot \log(a)))) \\ &\stackrel{[theorem: 19.38]}{=} \exp(y \cdot x \cdot \log(a)) \\ &= \exp((x \cdot y) \cdot \log(a)) \\ &= a^{x \cdot y} \end{aligned}$$

11. Let $a \in]0, \infty[$ and $x \in \mathbb{R}$ then

$$\begin{aligned} \log(a^x) &= \log(\exp(x \cdot \log(a))) \\ &\stackrel{[theorem: 19.38]}{=} x \cdot \log(a) \end{aligned}$$

12. As $a, b \in]0, \infty[$ then $a \cdot b \in]0, \infty[$ so that

$$\begin{aligned} (a \cdot b)^x &= \exp(x \cdot \log(a \cdot b)) \\ &\stackrel{[theorem: 19.38]}{=} \exp(x \cdot (\log(a) + \log(b))) \\ &= \exp(x \cdot \log(a) + x \cdot \log(b)) \\ &\stackrel{[theorem: 19.31]}{=} \exp(x \cdot \log(a)) \cdot \exp(x \cdot \log(b)) \\ &= a^x \cdot b^x \end{aligned}$$

13. As $a, b \in]0, \infty[$ $\frac{a}{b}$ is defined and $\frac{a}{b} \in]0, \infty[$ so that

$$\begin{aligned} \left(\frac{a}{b}\right)^x &= \exp\left(x \cdot \log\left(\frac{a}{b}\right)\right) \\ &= \exp\left(x \cdot \log\left(a \cdot \frac{1}{b}\right)\right) \\ &\stackrel{[theorem: 19.38]}{=} \exp\left(x \cdot \left(\log(a) + \log\left(\frac{1}{b}\right)\right)\right) \\ &\stackrel{[theorem: 19.38]}{=} \exp(x \cdot (\log(a) + (-\log(b)))) \\ &= \exp(x \cdot \log(a) + (-x \cdot \log(b))) \\ &\stackrel{[theorem: 19.32]}{=} \exp(x \cdot \log(a)) \cdot \exp(-x \cdot \log(b)) \\ &\stackrel{[theorem: 19.32]}{=} \frac{\exp(x \cdot \log(a))}{\exp(x \cdot \log(b))} \\ &= \frac{a^x}{b^x} \end{aligned}$$

14. Let $a, b \in]0, \infty[$ then

$$\begin{aligned} a^{x+y} &= \exp((x+y) \cdot \log(a)) \\ &= \exp(x \cdot \log(a) + y \cdot \log(a)) \\ &\stackrel{[theorem: 19.31]}{=} \exp(x \cdot \log(a)) \cdot \exp(y \cdot \log(a)) \\ &= a^x \cdot a^y \end{aligned}$$

15. Let $x \in]0, \infty[$ then for every $y, z \in [0, \infty[$ with $y < z$ we have for y either:

$y = 0$. Then as $x \in]0, \infty[$ we have by [definition: 19.39] that $y^x = 0$. Further as $0 = y < z$ we have by [definition: 19.39] that $z^x = \exp(x \cdot \log(z))$ where by [theorem: 19.32(4)] $0 < \exp(x \cdot \log(z))$. Hence we have $y^x = 0 < \exp(x \cdot \log(z)) = z^x$ proving that

$$y^x < z^x$$

$0 < y$. Then $y, z \in]0, \infty[$ and as $x \in]0, \infty[$ we have by [definition: 19.39] that

$$y^x = \exp(x \cdot \log(y)) \wedge z^x = \exp(x \cdot \log(z))$$

As $\log$ is strictly increasing [see theorem: 19.38] it follows that $\log(y) < \log(z)$ and as $0 < \log(y), \log(z)$ and $0 < x$ we have $x \cdot \log(y) < x \cdot \log(z)$. Further, as $\exp$ is strictly increasing [see theorem: 19.32], it follows that $\exp(x \cdot \log(y)) < \exp(x \cdot \log(z))$ so that

$$y^x < z^x \quad \square$$

The above theorem can be used to prove that $\sqrt{x} = x^{\frac{1}{2}}$.

Theorem 19.49. Let $\sqrt{\cdot}: [0, \infty[\rightarrow [0, \infty[$ be defined by $\sqrt{\cdot}(x) = \sqrt{x}$ and $(\star)^{\frac{1}{2}}: [0, \infty[\rightarrow [0, \infty[$ be defined by $(\star)^{\frac{1}{2}}(x) = x^{\frac{1}{2}}$ then $\sqrt{\cdot} = (\star)^{\frac{1}{2}}$. In other words we have $\forall x \in [0, \infty[$ that $\sqrt{x} = x^{\frac{1}{2}}$.

Proof. Using [definition: 10.67] we have that $(\star)^2: [0, \infty[\rightarrow [0, \infty[$ is a bijection with inverse function $\sqrt{\cdot} = ((\star)^2)^{-1}$. Define now

$$(\star)^{\frac{1}{2}}: [0, \infty[\rightarrow [0, \infty[\text{ by } (\star)^{\frac{1}{2}}(x) = x^{\frac{1}{2}}$$

Let $x \in [0, \infty[$ then we have

$$\left((\star)^2 \circ (\star)^{\frac{1}{2}} \right)(x) = (\star)^2 \left((\star)^{\frac{1}{2}}(x) \right) = \left(x^{\frac{1}{2}} \right)^2 \stackrel{[\text{theorem: 19.42(10)}]}{=} x^{\frac{1}{2} \cdot 2} = x^1 = \stackrel{[\text{theorem: 19.42(4)}]}{=} x = \text{Id}_{[0, \infty[}(x)$$

further we have

$$\left((\star)^{\frac{1}{2}} \circ (\star)^2 \right)(x) = (\star)^{\frac{1}{2}}((\star)^2(x)) = (x^2)^{\frac{1}{2}} \stackrel{[\text{theorem: 19.42(10)}]}{=} x^{2 \cdot \frac{1}{2}} = x^1 = \stackrel{[\text{theorem: 19.42(4)}]}{=} x = \text{Id}_{[0, \infty[}(x)$$

which proves that $(\star)^2 \circ (\star)^{\frac{1}{2}} = \text{Id}_{[0, \infty[} = (\star)^{\frac{1}{2}} \circ (\star)^2$. Hence we have

$$\sqrt{\cdot} = ((\star)^2)^{-1} = (\star)^{\frac{1}{2}} \quad \square$$

Theorem 19.50. Let $x \in \mathbb{R}$ then the function

$$(\star)^x:]0, \infty[= \mathbb{R}^+ \rightarrow \mathbb{R} \text{ defined by } (\star)^x(y) = y^x$$

has a derivative $((\star)^x)'_y$ at every $y \in]0, \infty[$ with $((\star)^x)'_y = x \cdot y^{x-1}$

Proof. Let $x \in \mathbb{R}$ and define

$$\varphi: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } \varphi(y) = x \cdot y$$

then by [example: 16.182] it follows that $\forall y \in \mathbb{R}$ has a derivative at y and

$$\varphi'_y = x$$

Further if $z \in \mathbb{R}^+$ then

$$(\exp \circ \varphi \circ \log)(z) = \exp(\varphi(\log(z))) = \exp(x \cdot \log(z)) = z^x = (\star)^x(z)$$

proving that

$$(\star)^x = \exp \circ \varphi \circ \log$$

As by [theorem: 19.27] $\exp$ has a derivative at every $y \in \mathbb{R}$ with $\exp'_y = \exp(y)$ and by [theorem: 19.38] $\log$ has a derivative at every y with $\log'_y = \frac{1}{y}$ we can use the chain rule [see theorem: 16.68] twice proving that $(\star)^x = \exp \circ \varphi \circ \log$ has a derivative for every $y \in \mathbb{R}^+$ and

$$\begin{aligned}
 ((\star)^x)'_y &= (\exp \circ \varphi \circ \log)'_y \\
 &= (\exp \circ (\varphi \circ \log))'_y \\
 &= \exp'_{(\varphi \circ \log)(y)} \cdot (\varphi \circ \log)'_y \\
 &= \exp((\varphi \circ \log)(y)) \cdot (\varphi \circ \log)'_y \\
 &= \exp(\varphi(\log(y))) \cdot (\varphi \circ \log)'_y \\
 &= \exp(x \cdot \log(y)) \cdot (\varphi \circ \log)'_y \\
 &= y^x \cdot (\varphi'_{\log(y)} \cdot \log'_y) \\
 &= y^x \cdot x \cdot \frac{1}{y} \\
 &\stackrel{[\text{theorem: 19.42 (8)}]}{=} x \cdot y^{x-1}
 \end{aligned}$$

□

Corollary 19.51. *Let $x \in \mathbb{R}$ then we have $\forall n \in \mathbb{N}$ that the function*

$$(\star)^x:]0, \infty[\rightarrow \mathbb{R} \text{ defined by } (\star)^x(y) = y^x$$

has a n -times derivative $((\star)^x)^{(n)}_y$ at every $y \in]0, \infty[$ with $((\star)^x)^{(n)}_y = (\prod_{i=0}^{n-1} (x-i)) \cdot y^{x-n}$.

Note 19.52. *Using [theorem: 16.171] it follows that $\forall n \in \mathbb{N}$ and $\forall y \in]0, \infty[$ that $(\star)^x$ is n -times differentiable at y . Hence using [theorem: 16.157] $(\star)^x$ is of class C^∞ .*

Proof. Let $y \in]0, \infty[$ and define

$$S = \left\{ n \in \mathbb{N} \mid \forall y \in]0, \infty[\text{ } (\star)^x \text{ has a } n\text{-times derivative } ((\star)^x)^{(n)}_y = \left(\prod_{i=0}^{n-1} (x-i) \right) \cdot y^{x-n} \text{ at } y \right\}$$

then we have:

1 $\in S$. Using [theorem: 19.50] we have that $\forall y \in]0, \infty[$ $(\star)^x$ has a derivative $((\star)^x)'_y = x \cdot (\star)^{x-1}$ at y . Hence we have by definition that $(\star)^x$ has a 1-times derivative at y with

$$((\star)^x)^{(1)}_y = x \cdot (\star)^{x-1} = \left(\prod_{i=0}^{1-1} x \right) \cdot (\star)^{x-1}$$

which proves that $1 \in S$

$n \in S \Rightarrow n+1 \in S$. As $n \in S$ we have that $\forall y \in]0, \infty[$ $((\star)^x)^{(n)}_y$ exists with

$$((\star)^x)^{(n)}_y = \left(\prod_{i=0}^{n-1} (x-i) \right) \cdot y^{x-n}$$

So we can define

$$((\star)^x)^{(n)}:]0, \infty[\rightarrow \mathbb{R} \text{ by } ((\star)^x)^{(n)}(y) = ((\star)^x)^{(n)}_y = \left(\prod_{i=0}^{n-1} (x-i) \right) \cdot y^{x-n}$$

Using [theorems: 19.50, 16.40] on the above function it follows that $((\star)^x)^{(n)}'_y$ exist and

$$(((\star)^x)^{(n)})'_y = (x-n) \cdot \left(\prod_{i=0}^{n-1} (x-i) \right) \cdot y^{x-n-1} = \left(\prod_{i=0}^{(n+1)-1} (x-i) \right) \cdot y^{x-(n+1)}$$

so that by definition $((\star)^x)^{(n+1)}_y$ exist and $((\star)^x)^{(n+1)}_y =$

$$(((\star)^x)^{(n)})^{(n+1)}_y = (((\star)^x)^{(n)})'_y = \left(\prod_{i=0}^{(n+1)-1} (x-i) \right) \cdot y^{x-(n+1)}$$

which proves that $n + 1 \in S$. □

Corollary 19.53. We have $\forall n \in \mathbb{N}$ that $\forall n \in \mathbb{N}$ we have that the function

$$\sqrt{\cdot}:]0, \infty[\rightarrow \mathbb{R} \text{ defined by } \sqrt{\cdot}(x) = \sqrt{x}$$

has a n -times derivative where

$$(\sqrt{\cdot})_y^n = \left(\prod_{i=0}^{n-1} \left(\frac{1}{2} - 1 \right) \right) \cdot (\cdot)^{\frac{1}{2}-n}$$

Using [theorem: 16.171] it follows that $\forall n \in \mathbb{N}$ and $\forall y \in]0, \infty[$ that $\sqrt{\cdot}$ is n -times differentiable at y . Hence using [theorem: 16.157] $\sqrt{\cdot}$ is of class C^∞ .

Proof. This follows directly from [theorems: 19.49, 19.51] □

Theorem 19.54. Let $a \in]0, \infty[$ then $\forall y \in \mathbb{R}$ we have that the function

$$a^{(\cdot)}: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } a^{(\cdot)}(x) = a^x$$

has a derivative at y and $(a^{(\cdot)})'_x = \log(a) \cdot a^x$.

Proof. Define

$$\varphi: \mathbb{R} \rightarrow \mathbb{R} \text{ by } \varphi(x) = x \cdot \log(a)$$

then by [example: 16.182] it follows that $\forall x \in \mathbb{R}$ has a derivative at x and

$$\varphi'_x = \log(a)$$

Further we have for $x \in \mathbb{R}$ that $(\exp \circ \varphi)(x) = \exp(\varphi(x)) = \exp(x \cdot \log(a)) = a^x = a^{(\cdot)}(x)$ proving that

$$a^{(\cdot)} = \exp \circ \varphi$$

As by [theorem: 19.27] $\exp$ has a derivative at every $x \in \mathbb{R}$ with $\exp'_x = \exp(x)$ we can use the chain rule [see theorem: 16.68] to prove that $a^{(\cdot)}$ has a derivative at every $x \in \mathbb{R}$ and that

$$\begin{aligned} (a^{(\cdot)})'_x &= (\exp \circ \varphi)'_x \\ &= \exp'_{\varphi(x)} \cdot \varphi'_x \\ &= \exp(\varphi(x)) \cdot \varphi'_x \\ &= \exp(x \cdot \log(a)) \cdot \varphi'_x \\ &= a^x \cdot \varphi'_x \\ &= \log(a) \cdot a^x \end{aligned}$$

□

Theorem 19.55. Let $x \in]0, \infty[$ then $(\cdot)^x: [0, \infty[\rightarrow \mathbb{R}$ defined by $(\cdot)^x(y) = y^x$ is continuous on $[0, \infty[$ [where we use the sub-space topology of $\langle \mathbb{R}, \|\cdot\| \rangle$ on the $[0, \infty[$

Proof. Let $y \in [0, \infty[$ then we have either:

$y = 0$. Let $\varepsilon \in \mathbb{R}^+$ take then $\delta_1 = \log(\varepsilon) \in \mathbb{R}$ and $\delta = \exp\left(\frac{\delta_1}{x}\right)$. If $z \in B_{\|\cdot\|}(0, \delta) \cap [0, \infty[$ then we have either:

$z = 0$. Then $(\cdot)^x(z) \underset{z \neq 0}{=} 0 < \varepsilon$

$z \neq 0$. Then $0 < z < \delta$ and, as $\log$ is strictly increasing by [theorem: 19.38], we have that $\log(z) < \log(\delta)$. As $0 < x, \log(z), \log(\delta)$ it follows that $x \cdot \log(z) < x \cdot \log(\delta)$. Further, as $\exp$ is strictly increasing by [theorem: 19.32], we have that

$$\exp(x \cdot \log(z)) < \exp(x \cdot \log(\delta))$$

Now $(\cdot)^x(z) = z^x = \exp(x \cdot \log(z))$ and

$$\exp(x \cdot \log(\delta)) = \exp\left(x \cdot \log\left(\exp\left(\frac{\delta_1}{x}\right)\right)\right) = \exp\left(x \cdot \frac{\delta_1}{x}\right) = \exp(\delta_1) = \exp(\log(\varepsilon)) = \varepsilon$$

so that $(*)^x(z) < \varepsilon$.

So in all cases we have that $(*)^x(z) < \varepsilon$, as further by [theorem: 19.42 (2,3)] $0 \leq (*)^x(x)$ it follows that $|(*)^x(z)| < \varepsilon$ so that $(*)^x(z) \in B_{||}(0, \varepsilon)$. This proves that

$$(*)^x(B_{||}(0, \delta) \cap]0, \infty[) \subseteq B_{||}(0, \varepsilon) \stackrel{(*)^x(0)}{=} B_{||}((*)^x(0), \varepsilon)$$

so that $(*)^x$ is continuous at 0.

y ≠ 0. Let $\varepsilon \in \mathbb{R}^+$. Then $y \in]0, \infty[$ and, as $f:]0, \infty[\rightarrow \mathbb{R}$ defined by $f^x(z) = z^x$ has a derivative at y [see theorem: 19.50], f is continuous at y [see theorems: 16.63, 16.64]. So there exist a $\delta_1 \in \mathbb{R}^+$ such that

$$f(B_{||}(y, \delta_1) \cap]0, \infty[) \subseteq B_{||}(f(y), \varepsilon)$$

As $0 < y$ we have that $\delta = \min(\delta_1, y) \in \mathbb{R}^+$ if now $0 \in B_{||}(y, \delta)$ then $y = |y| = |y - 0| < \delta_1 \leq y$ leading to the contradiction $y < y$. Hence $0 \notin B_{||}(y, \delta)$ so that for $z \in B_{||}(y, \delta) \cap]0, \infty[$ we have $z \in B_{||}(y, \delta) \cap]0, \infty[\subseteq B_{||}(y, \delta) \cap]0, \infty[$ or

$$B_{||}(y, \delta) \cap]0, \infty[= B_{||}(y, \delta) \cap]0, \infty[$$

Hence if $z \in (*)^x(B_{||}(y, \delta) \cap]0, \infty[) = (*)^x(B_{||}(y, \delta) \cap]0, \infty[)$ then there exist a $u \in B_{||}(y, \delta) \cap]0, \infty[$ such that $z = (*)^x(u) = u^x = f(u) \in f(B_{||}(y, \delta_1) \cap]0, \infty[)$ proving that

$$(*)^x(B_{||}(y, \delta) \cap]0, \infty[) \subseteq f(B_{||}(y, \delta_1) \cap]0, \infty[) \subseteq B_{||}(f(y), \varepsilon)$$

As $y \neq 0$ then $f(y) = y^x = (*)^x(y)$ we have proved that

$$(*)^x(B_{||}(y, \delta) \cap]0, \infty[) \subseteq B_{||}((*)^x(y), \varepsilon)$$

so that $(*)^x$ is continuous at y . □

Corollary 19.56. *The function $\sqrt{*}:]0, \infty[\rightarrow \mathbb{R}$ defined by $\sqrt{*}(x) = \sqrt{x}$ is continuous on $]0, \infty[$ [where we use the sub-space topology of $\langle \mathbb{R}, || \rangle$ on the $]0, \infty[$*

Proof. This follows from [theorems: 19.49, 19.55]. □

The previous theorem is needed to prove a important inequality using generalized powers, first we have to prove a little lemma.

Lemma 19.57. *Let $p \in]1, \infty[$ and define*

$$f_p:]0, \infty[\rightarrow \mathbb{R} \text{ by } f_p(x) = (1 - x^{1-p}) + (p-1) \cdot (1-x)$$

then we have for $x \in]0, \infty[$

$$x = 1 \Leftrightarrow f_p(x) = 0$$

Proof.

$\Rightarrow$. If $x = 1$ then

$$f_p(x) = f_p(1) = (1 - 1^{1-p}) + (p-1) \cdot (1-1) \stackrel{[\text{theorem: 19.42(5)}]}{=} (1-1) + (p-1) \cdot (1-1) = 0$$

$\Leftarrow$. As

$$\begin{aligned} (-(*)^{1-p} + C_p + (1-p) \cdot \text{Id}_{]0, \infty[})(x) &= -(*)^{1-p}(x) + C_p(x) + (1-p) \cdot \text{Id}_{]0, \infty[}(x) \\ &= -x^{1-p} + p + (1-p) \cdot x \\ &= (1 - x^{1-p}) + p - 1 - (p-1) \cdot x \\ &= (1 - x^{1-p}) + (p-1) \cdot (1-x) \\ &= f_p(x) \end{aligned}$$

so that

$$f_p = -(*)^{1-p} + C_p + (1-p) \cdot \text{Id}_{]0, \infty[} \quad (19.33)$$

As $(\star)^{(1-p)}$, C_p and $\text{Id}_{]0,\infty[}$ has a derivative at every $x \in]0, \infty[$ [see examples: 19.50, 16.30, 16.31] it follows that f_p has a derivative at every $x \in]0, \infty[$ and

$$\begin{aligned}(f_p)'_x &= -((\star)^{1-p})'_x + (1-p) \cdot (\text{Id}_{]0,\infty[})'_x \\ &= -(1-p) \cdot x^{(1-p)-1} + (1-p) \\ &= (p-1) \cdot x^{-p} - (p-1) \\ &= ((p-1) \cdot (\star)^{-p} + C_{-(p-1)})(x)\end{aligned}$$

So that

$$(f_p)'_1 = (p-1) \cdot 1^{-p} - (p-1) = 0 \quad (19.34)$$

Further if we define $(f_p)':]0, \infty[\rightarrow \mathbb{R}$ by $(f_p)'(x) = (f_p)'_x$ then

$$(f_p)' = (p-1) \cdot (\star)^{-p} + C_{-(p-1)}$$

Using [examples: 19.50, 16.30] again it follows that $(f_p)'$ has a derivative at every $x \in]0, \infty[$ and

$$(f_p)''_x = (p-1) \cdot (-p) \cdot x^{-p-1} = p \cdot (1-p) \cdot x^{-p-1}$$

As $0 < x$ we have by [theorem: 19.42] that $0 < x^{-p-1}$, further as $p \in]1, \infty[$ we have $0 < p \cdot (p-1)$ so that

$$\forall x \in]0, \infty[\text{ we have } 0 < (f_p)''_x$$

Hence using [theorem: 16.211] it follows that

$$(f_p)' \text{ is strictly increasing} \quad (19.35)$$

As $(\star)^{1-p}$ is continuous [see theorem: 19.55], C_p is continuous and $\text{Id}_{]0,\infty[}$ is continuous [see examples: 14.142, 14.143] we can use [theorem: 14.165] proving that f_p is continuous,

$$f_p \text{ is continuous on }]0, \infty[\quad (19.36)$$

Let $x \in]0, \infty[$ such that $f_p(x) = 0$. Assume that $x \neq 1$ then we have either:

$x < 1$. Then by [eq: 19.36] and [theorem: 14.152] $(f_p)|_{[x,1]}$ is continuous, further $\forall z \in]x, 1[$ we have that $f_p|_{]x,1[}$ has a derivative on $]x, 1[$ and $f_p|_{[x,1]}(x) = f(x) = 0 = f(1) = f_p|_{[1,x]}(1)$ so that we can apply Rolle's theorem [theorem: 16.205] to find a $\zeta \in]x, 1[$ such that $(f_p)'_\zeta = 0$. As $(f_p)'$ is strict increasing we have

$$0 = (f_p)'_\zeta = (f_p)'(\zeta) < (f_p)'(1) = (f_p)'_1 \stackrel{[\text{eq: 19.34}]}{=} 0$$

leading to the contradiction $0 < 0$.

$1 < x$. Then by [eq: 19.36] and [theorem: 14.152] $(f_p)|_{[1,x]}$ is continuous, further $\forall z \in]1, x[$ we have that $f_p|_{]1,x[}$ has a derivative on $]1, x[$ and $f_p|_{[1,x]}(x) = f(x) = 0 = f(1) = f_p|_{[1,x]}(1)$ so that we can apply Rolle's theorem [theorem: 16.205] to find a $\zeta \in]1, x[$ such that $(f_p)'_\zeta = 0$. As $(f_p)'$ is strict increasing we have

$$0 \stackrel{[\text{eq: 19.34}]}{=} (f_p)'_1 = (f_p)'(1) < (f_p)'(\zeta) = (f_p)'_\zeta = 0$$

leading to the contradiction $0 < 0$.

As in ail cases we have a contradiction the assumption is wrong and we must have

$$x = 1 \quad \square$$

Theorem 19.58. (Young's inequality) Let $p, q \in]0, \infty[$ such that $\frac{1}{p} + \frac{1}{q} = 1$ and $a, b \in]0, \infty[= \mathbb{R}^+$ then we have:

$$1. \ a \cdot b \leq \frac{a^p}{p} + \frac{b^q}{q}$$

2. Further for equality the following are equivalent

$$a. \quad a \cdot b = \frac{a^p}{p} + \frac{b^q}{q}$$

$$b. \quad b = a^{p-1}$$

$$c. \quad a^p = b^q$$

Note 19.59. If $p \leq 1$ then $\frac{1}{p} \geq 1$ so that $\frac{1}{q} = 1 - \frac{1}{p} \leq 0$ contradicting $0 < q \Rightarrow 0 < \frac{1}{q}$ and if $q \leq 1$ then $\frac{1}{q} \geq 1$ so that $\frac{1}{p} = 1 - \frac{1}{q} \leq 0$ contradicting $0 < p \Rightarrow 0 < \frac{1}{p}$. Hence the condition $\frac{1}{p} + \frac{1}{q} = 1 \wedge p, q \in]0, 1[$ automatically includes $a, b \in]1, \infty[$.

Proof.

1. We have

$$\begin{aligned} a \cdot b &\stackrel{[\text{theorem: 19.38}]}{=} \exp(\log(a \cdot b)) \\ &\stackrel{[\text{theorem: 19.38}]}{=} \exp(\log(a) + \log(b)) \\ &= \exp\left(\frac{1}{p} \cdot p \cdot \log(a) + \frac{1}{q} \cdot q \cdot \log(b)\right) \\ &\stackrel{[\text{theorem: 19.42}]}{=} \exp\left(\frac{1}{p} \cdot \log(a^p) + \frac{1}{q} \cdot \log(b^q)\right) \end{aligned} \quad (19.37)$$

For $\log(a^p)$, $\log(b^q)$ we have either:

$\log(a^p) = \log(b^q)$. Then

$$\begin{aligned} \exp\left(\frac{1}{p} \cdot \log(a^p) + \frac{1}{q} \cdot \log(b^q)\right) &= \exp\left(\left(\frac{1}{p} + \frac{1}{q}\right) \cdot \log(a^p)\right) \\ &= \exp(\log(a^p)) \\ &= \left(\frac{1}{p} + \frac{1}{q}\right) \cdot \exp(\log(a^p)) \\ &= \frac{1}{p} \cdot \exp(\log(a^p)) + \frac{1}{q} \cdot \exp(\log(a^p)) \\ &= \frac{1}{p} \cdot \exp(\log(a^p)) + \frac{1}{q} \cdot \exp(\log(b^q)) \end{aligned}$$

$\log(a^p) < \log(b^q)$. Then as $\exp$ is convex [see theorem: 19.32] it follows from [theorem: 16.213] that

$$\exp\left(\frac{1}{p} \cdot \log(a^p) + \frac{1}{q} \cdot \log(b^q)\right) \leq \frac{1}{p} \cdot \exp(\log(a^p)) + \frac{1}{q} \cdot \exp(\log(b^q))$$

$\log(a^p) > \log(b^q)$. Then as $\exp$ is convex [see theorem: 19.32] it follows from [theorem: 16.213] that

$$\begin{aligned} \exp\left(\frac{1}{p} \cdot \log(a^p) + \frac{1}{q} \cdot \log(b^q)\right) &= \exp\left(\frac{1}{q} \cdot \log(b^q) + \frac{1}{p} \cdot \log(a^p)\right) \\ &\leq \frac{1}{q} \cdot \exp(\log(b^q)) + \frac{1}{p} \cdot \exp(\log(a^p)) \\ &= \frac{1}{p} \cdot \exp(\log(a^p)) + \frac{1}{q} \cdot \exp(\log(b^q)) \end{aligned}$$

so in all cases we have $\exp\left(\frac{1}{p} \cdot \log(a^p) + \frac{1}{q} \cdot \log(b^q)\right) \leq \frac{1}{p} \cdot \exp(\log(a^p)) + \frac{1}{q} \cdot \exp(\log(b^q))$ which combined with [eq: 19.37] gives

$$\begin{aligned} a \cdot b &\leq \frac{1}{p} \cdot \exp(\log(a^p)) + \frac{1}{q} \cdot \exp(\log(b^q)) \\ &\stackrel{[\text{eq: 19.38}]}{=} \frac{a^p}{p} + \frac{b^q}{q} \end{aligned}$$

2. First for $p, q \in]0, \infty[$ we have

$$\begin{aligned}
 \frac{1}{p} + \frac{1}{q} = 1 & \stackrel{p \neq 0}{\Leftrightarrow} p \cdot \left(\frac{1}{p} + \frac{1}{q} \right) = p \\
 & \Leftrightarrow 1 + \frac{p}{q} = p \\
 & \Leftrightarrow \frac{p}{q} = p - 1 \\
 \frac{1}{p} + \frac{1}{q} = 1 & \stackrel{q \neq 0}{\Leftrightarrow} q \cdot \left(\frac{1}{p} + \frac{1}{q} \right) = q \\
 & \Leftrightarrow \frac{q}{p} + 1 = q \\
 & \Leftrightarrow \frac{q}{p} = q - 1
 \end{aligned}$$

so that if $\frac{1}{p} + \frac{1}{q} = 1$ then

$$\frac{p}{q} = p - 1 \wedge \frac{q}{p} = q - 1 \Rightarrow (p - 1) \cdot (q - 1) = 1 \quad (19.38)$$

Now for the equivalences:

$a \Rightarrow b$. As $a \cdot b = \frac{a^p}{p} + \frac{b^q}{q}$ then we have

$$\begin{aligned}
 a \cdot b &= \frac{a^p}{p} + \frac{b^q}{q} \\
 &= \frac{q \cdot a^p + p \cdot b^q}{p \cdot q} \\
 &= \frac{q \cdot a^p + p \cdot b^q}{p + q}
 \end{aligned}$$

$\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow \frac{p+q}{p \cdot q} = 1 \Rightarrow \frac{1}{p \cdot q} = \frac{1}{p+q}$

so that [as $a \cdot b \in]0, \infty[$ we have

$$\begin{aligned}
 p + q &= \frac{q \cdot a^p + p \cdot b^q}{a \cdot b} && \Rightarrow \\
 p + q &= q \cdot \frac{a^p}{a \cdot b} + p \cdot \frac{b^q}{a \cdot b} && \stackrel{0 < a, b \wedge [\text{theorem: 19.42(8)}]}{\Rightarrow} \\
 p + q &= q \cdot \frac{a^{p-1}}{b} + p \cdot \frac{b^{q-1}}{b} && \Rightarrow \\
 p \cdot \left(1 - \frac{b^{q-1}}{b} \right) &= q \cdot \left(\frac{a^{p-1}}{b} - 1 \right) && \Rightarrow \\
 \frac{a^{p-1}}{b} - 1 &= \frac{p}{q} \cdot \left(1 - \frac{b^{q-1}}{b} \right) && [\text{eq: 19.38}] \\
 \frac{a^{p-1}}{b} - 1 &= (p - 1) \cdot \left(1 - \frac{b^{q-1}}{b} \right) && \stackrel{[\text{theorem: 19.42(9)}]}{\Rightarrow} \\
 \frac{b^{-1}}{a^{-(p-1)}} - 1 &= (p - 1) \cdot \left(1 - \frac{b^{q-1}}{b} \right) && [\text{eq: 19.38}] \\
 \frac{b^{-(q-1) \cdot (p-1)}}{a^{-(p-1)}} - 1 &= (p - 1) \cdot \left(1 - \frac{b^{q-1}}{b} \right) && = \\
 \frac{b^{(q-1) \cdot (1-p)}}{a^{(1-p)}} - 1 &= (p - 1) \cdot \left(1 - \frac{b^{q-1}}{b} \right) && \stackrel{[\text{theorem: 19.42}]}{\Rightarrow} \\
 \left(\frac{b^{q-1}}{a} \right)^{1-p} - 1 &= (p - 1) \cdot \left(1 - \frac{b^{q-1}}{b} \right) && = \\
 0 &= \left(1 - \left(\frac{b^{q-1}}{a} \right)^{1-p} + (p - 1) \cdot \left(1 - \frac{b^{q-1}}{b} \right) \right) && [\text{lemma: 19.57}] \\
 &= f_p \left(\frac{b^{q-1}}{a} \right)
 \end{aligned}$$

so that by [lemma: 19.57] we conclude that $\frac{b^{q-1}}{a} = 1$ or $a = b^{q-1}$. Hence

$$a^{p-1} = (b^{q-1})^{1-q} \stackrel{[\text{theorem: 19.42}]}{=} b^{(q-1) \cdot (p-1)} \stackrel{[\text{eq: 19.38}]}{\Rightarrow} b^1 = b$$

proving that

$$b = a^{p-1}$$

$b \Rightarrow a$. If $b = a^{p-1}$ then

$$\begin{aligned} \frac{a^p}{p} + \frac{b^q}{q} &= \frac{a^p}{p} + \frac{(a^{p-1})^q}{q} \\ &\stackrel{[\text{eq: 19.38}]}{=} \frac{a^p}{p} + \frac{\left(a^{\frac{p}{q}}\right)^q}{q} \\ &\stackrel{[\text{theorem: 19.42}]}{=} \frac{a^p}{p} + \frac{a^{\frac{p}{q} \cdot q}}{q} \\ &= \frac{a^p}{p} + \frac{a^p}{q} \\ &= \left(\frac{1}{p} + \frac{1}{q}\right) \cdot a^p \\ &= a^p \\ &= a^{(p-1)+1} \\ &\stackrel{[\text{theorem: 19.42}]}{=} a^{(p-1)} \cdot a^1 \\ &= b \cdot a^1 \\ &= a \cdot b \end{aligned}$$

$b \Rightarrow c$. If $b = a^{p-1}$ then we have

$$\begin{aligned} b^q &= (a^{p-1})^q \\ &\stackrel{[\text{eq: 19.38}]}{=} \left(a^{\frac{p}{q}}\right)^q \\ &\stackrel{[\text{theorem: 19.42}]}{=} a^{\frac{p}{q} \cdot q} \\ &= a^p \end{aligned}$$

$c \Rightarrow b$. If $a^p = b^q$ then

$$\begin{aligned} b &= b^{q \cdot \frac{1}{q}} \\ &\stackrel{[\text{theorem: 19.42}]}{=} (b^q)^{\frac{1}{q}} \\ &= (a^p)^{\frac{1}{q}} \\ &\stackrel{[\text{theorem: 19.42}]}{=} a^{\frac{p}{q}} \\ &\stackrel{[\text{eq: 19.38}]}{=} a^{p-1} \end{aligned}$$

□

19.1.7 Trigonometric functions

For the generalized powers and the natural logarithm we studied the behavior of the exponential function in the real domain. For trigonometric functions we consider the behavior of the exponential function in the complex domain.

Theorem 19.60.

1. $\forall z \in \mathbb{C}$ we have that $\overline{\exp(z)} = \exp(\bar{z})$
2. $\forall z \in \mathbb{C}$ we have that $|\exp(z)| = \exp(\operatorname{Re}(z))$
3. $\forall x + i \cdot y \in \mathbb{C}$ we have that $|\exp(z)| = \exp(x)$ and $|\exp(i \cdot y)| = \exp(0) = 1$

Proof.

1. We have

$$\begin{aligned}
 \exp(\bar{z}) &= \sum_{i=0}^n \frac{\bar{z}^i}{i!} \\
 &= \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{\bar{z}^i}{i!} \\
 &\stackrel{[\text{theorem: 10.83}]}{=} \lim_{n \rightarrow \infty} \overline{\sum_{i=0}^n \frac{z^i}{i!}} \\
 &\stackrel{[\text{theorem: 14.362}]}{=} \overline{\lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{z^i}{i!}} \\
 &= \overline{\exp(z)}
 \end{aligned}$$

2. We have

$$\begin{aligned}
 |\exp(z)|^2 &\stackrel{[\text{theorem: 10.85}]}{=} \exp(z) \cdot \overline{\exp(z)} \\
 &\stackrel{(1)}{=} \exp(z) \cdot \exp(\bar{z}) \\
 &\stackrel{[\text{theorem: 19.31}]}{=} \exp(z + \bar{z}) \\
 &\stackrel{[\text{theorem: 10.83}]}{=} \exp(2 \cdot \operatorname{Re}(z)) \\
 &= \exp(\operatorname{Re}(z) + \operatorname{Re}(z)) \\
 &\stackrel{[\text{theorem: 19.31}]}{=} \exp(\operatorname{Re}(z)) \cdot \exp(\operatorname{Re}(z)) \\
 &= \exp(\operatorname{Re}(z))^2
 \end{aligned}$$

Now as $|\exp(z)| \in \mathbb{R}^+$ and $\exp(\operatorname{Re}(z)) \in \mathbb{R}^+$ [see theorem: 19.32] it follows from [theorem: 10.67] that $\exp(z) = \exp(\operatorname{Re}(z))$.

3. If $z = x + i \cdot y$ then $|\exp(z)| \stackrel{(2)}{=} \exp(\operatorname{Re}(z)) = \exp(x)$, further

$$|\exp(i \cdot y)| \stackrel{(2)}{=} \exp(\operatorname{Re}(i \cdot y)) = \exp(0) \stackrel{[\text{theorem: 19.32}]}{=} 1 \quad \square$$

We define now the trigonometric functions in terms of the exponential functions=.

Definition 19.61. (Trigonometric functions)

1. $\cos: \mathbb{C} \rightarrow \mathbb{C}$ is defined by $\cos(z) = \frac{\exp(i \cdot z) + \exp(-i \cdot z)}{2} \stackrel{[\text{theorem: 19.42(6)}]}{=} \frac{e^{i \cdot z} + e^{-i \cdot z}}{2}$
2. $\sin: \mathbb{C} \rightarrow \mathbb{C}$ is defined by $\sin(z) = \frac{\exp(i \cdot z) - \exp(-i \cdot z)}{2 \cdot i} \stackrel{[\text{theorem: 19.42(6)}]}{=} \frac{e^{i \cdot z} - e^{-i \cdot z}}{2 \cdot i}$

Theorem 19.62. We have the following properties for the trigonometric function.

1. $\forall z \in \mathbb{C}$ we have $\cos(z) + i \cdot \sin(z) = \exp(i \cdot z) \stackrel{[\text{theorem: 19.42(6)}]}{=} e^{i \cdot z}$.
2. $\cos: \mathbb{C} \rightarrow \mathbb{C}$ and $\sin: \mathbb{C} \rightarrow \mathbb{C}$ are continuous.

3. $\cos(0) = 1$.

4. $\sin(0) = 0$.

5. $\forall z \in \mathbb{C}$ we have:

a. $\cos(-z) = \cos(z)$.

b. $\sin(-z) = -\sin(z)$.

6. $\forall z \in \mathbb{C}$ we have $\cos^2(z) + \sin^2(z) = 1$

7. $\forall x, y \in \mathbb{C}$ we have:

a. $\cos(x+y) = \cos(x) \cdot \cos(y) - \sin(x) \cdot \sin(y)$.

b. $\sin(x+y) = \sin(x) \cdot \cos(y) + \cos(x) \cdot \sin(y)$.

8. $\forall x \in \mathbb{R}$ we have:

a. $\cos(x) \in \mathbb{R}$ and $\operatorname{Re}(e^{i \cdot x}) \stackrel{[\text{theorem: 19.42(6)}]}{=} \operatorname{Re}(\exp(i \cdot x)) = \cos(x)$

b. $\sin(x) \in \mathbb{R}$ and $\operatorname{Im}(e^{i \cdot x}) \stackrel{[\text{theorem: 19.42(6)}]}{=} \operatorname{Im}(\exp(i \cdot x)) = \sin(x)$

9. $\forall x \in \mathbb{R}$ we have

$$|e^{i \cdot x}| \stackrel{[\text{theorem: 19.42(6)}]}{=} |\exp(i \cdot x)| = 1 \text{ and } |e^{-i \cdot x}| \stackrel{[\text{theorem: 19.42(6)}]}{=} |\exp(-i \cdot x)| = 1$$

10. $\forall x \in \mathbb{R}$ we have $|\cos(x)| \in [0, 1]$ and $|\sin(x)| \in [0, 1]$ or in other words as $\cos(x), \sin(x) \in \mathbb{R}$
 $-1 \leq \cos(x), \sin(x) \leq 1$

Proof.

1. Let $z \in \mathbb{C}$ then

$$\begin{aligned} \cos(z) + i \cdot \sin(z) &= \frac{\exp(i \cdot z) + \exp(-i \cdot z)}{2} + i \cdot \frac{\exp(i \cdot z) - \exp(-i \cdot z)}{2 \cdot i} \\ &= \frac{\exp(i \cdot z) + \exp(-i \cdot z) + \exp(i \cdot z) - \exp(-i \cdot z)}{2} \\ &= \exp(i \cdot z) \end{aligned}$$

2. Note that $h: \mathbb{C} \rightarrow \mathbb{C}$ defined by $f = i \cdot \operatorname{Id}_{\mathbb{C}}$ and $g: \mathbb{C} \rightarrow \mathbb{C}$ defined by $g = -i \cdot \operatorname{Id}_{\mathbb{C}}$ are continuous [see example: 14.143] and theorem: 14.165]. As $\exp$ is continuous [see theorem: 19.28] it follows from [theorem: 14.158] that $\exp \circ f$ and $\exp \circ g$ are continuous. Hence using [theorem: 14.165] $\frac{1}{2} \cdot (\exp \circ f + \exp \circ g)$ and $\frac{1}{2 \cdot i} \cdot (\exp \circ f - \exp \circ g)$ are continuous. Let $z \in \mathbb{C}$ then

$$\left(\frac{1}{2} \cdot (\exp \circ f + \exp \circ g) \right)(z) = \frac{\exp(f(z)) + \exp(g(z))}{2} = \frac{\exp(i \cdot z) + \exp(-i \cdot z)}{2} = \cos(z)$$

and

$$\left(\frac{1}{2 \cdot i} \cdot (\exp \circ f - \exp \circ g) \right)(z) = \frac{\exp(f(z)) - \exp(g(z))}{2 \cdot i} = \frac{\exp(i \cdot z) - \exp(-i \cdot z)}{2 \cdot i} = \sin(z)$$

proving that $\frac{1}{2} \cdot (\exp \circ f + \exp \circ g) = \cos$ and $\frac{1}{2 \cdot i} \cdot (\exp \circ f - \exp \circ g) = \sin$. Hence $\cos$ and $\sin$ are continuous.

3. $\cos(0) = \frac{\exp(i \cdot 0) + \exp(i \cdot 0)}{2} = \frac{\exp(0) + \exp(0)}{2} \stackrel{[\text{theorem: 19.32}]}{=} \frac{1+1}{2} = 1$.

4. $\sin(0) = \frac{\exp(i \cdot 0) - \exp(i \cdot 0)}{2 \cdot i} = \frac{\exp(0) - \exp(0)}{2 \cdot i} = 0$.

5. Let $z \in \mathbb{C}$ then we have:

a. $\cos(z) = \frac{\exp(-i \cdot z) + \exp(-(-i \cdot z))}{2} = \frac{\exp(i \cdot z) + \exp(-i \cdot z)}{2} = \cos(z)$.

$$\text{b. } \sin(z) = \frac{\exp(-i \cdot z) - \exp(-(-i \cdot z))}{2 \cdot i} = \frac{\exp(-i \cdot z) - \exp(i \cdot z)}{2 \cdot i} = -\frac{\exp(i \cdot z) - \exp(-i \cdot z)}{2 \cdot i} = -\sin(z)$$

6. Let $z \in \mathbb{C}$ then

$$\begin{aligned} \cos^2(z) + \sin^2(z) &= \frac{(\exp(i \cdot z) + \exp(-i \cdot z))^2}{4} + \frac{(\exp(i \cdot z) - \exp(-i \cdot z))^2}{-4} \\ &= \frac{1}{4} \cdot ((\exp(i \cdot z) + \exp(-i \cdot z))^2 - (\exp(i \cdot z) - \exp(-i \cdot z))^2) \\ &= \frac{1}{4} \cdot ((\exp(i \cdot z))^2 + (\exp(-i \cdot z))^2 + 2 \cdot \exp(i \cdot z) \cdot \exp(-i \cdot z) - (\exp(i \cdot z))^2 - (\exp(-i \cdot z))^2 + 2 \cdot \exp(i \cdot z) \cdot \exp(-i \cdot z)) \\ &= \frac{1}{4} \cdot 4 \cdot \exp(i \cdot z) \cdot \exp(-i \cdot z) \\ &= \exp(i \cdot z) \cdot \exp(-i \cdot z) \\ &\stackrel{[\text{theorem: 19.31}]}{=} \exp(i \cdot z - i \cdot z) \\ &= \exp(0) \\ &\stackrel{[\text{theorem: 19.32}]}{=} 1 \end{aligned}$$

7. Let $a_x = \exp(i \cdot x)$, $b_x = \exp(-i \cdot x)$, $a_y = \exp(i \cdot y)$ and $b_y = \exp(-i \cdot y)$ then we have

a.

$$\begin{aligned} \cos(x) \cdot \cos(y) - \sin(x) \cdot \sin(y) &= \\ \frac{1}{4} \cdot (a_x + b_x) \cdot (a_y + b_y) + \frac{1}{4} \cdot (a_x - b_x) \cdot (a_y - b_y) &= \\ \frac{1}{4} \cdot [(a_x \cdot a_y + a_x \cdot b_y + b_x \cdot a_y + b_x \cdot b_y) + (a_x \cdot a_y - a_x \cdot b_y - b_x \cdot a_y + b_x \cdot b_y)] &= \\ \frac{1}{4} \cdot [(a_x \cdot a_y + a_x \cdot b_y + b_x \cdot a_y + b_x \cdot b_y) + (a_x \cdot a_y - a_x \cdot b_y - b_x \cdot a_y + b_x \cdot b_y)] &= \\ \frac{1}{4} \cdot [a_x \cdot a_y + a_x \cdot b_y + b_x \cdot a_y + b_x \cdot b_y + a_x \cdot a_y - a_x \cdot b_y - b_x \cdot a_y + b_x \cdot b_y] &= \\ \frac{1}{4} \cdot (2 \cdot a_x \cdot a_y + 2 \cdot b_x \cdot b_y) &= \\ \frac{1}{2} \cdot (\exp(i \cdot x) \cdot \exp(i \cdot y) + \exp(-i \cdot x) \cdot \exp(-i \cdot y)) &= \\ \frac{1}{2} \cdot (\exp(i \cdot x + i \cdot y) + \exp(-i \cdot x - i \cdot y)) &= \\ \frac{1}{2} \cdot (\exp(i \cdot (x + y)) + \exp(-i \cdot (x + y))) &= \\ \cos(x + y) \end{aligned}$$

b.

$$\begin{aligned} \sin(x) \cdot \cos(y) + \cos(x) \cdot \sin(y) &= \\ \frac{1}{4 \cdot i} \cdot ((a_x - b_x) \cdot (a_y + b_y) + (a_x + b_x) \cdot (a_y - b_y)) &= \\ \frac{1}{4 \cdot i} \cdot (a_x \cdot a_y + a_x \cdot b_y - b_x \cdot a_y - b_x \cdot b_y + a_x \cdot a_y - a_x \cdot b_y + b_x \cdot a_y - b_x \cdot b_y) &= \\ \frac{1}{4 \cdot i} \cdot (2 \cdot a_x \cdot a_y - 2 \cdot b_x \cdot b_y) &= \\ \frac{1}{2 \cdot i} \cdot (\exp(i \cdot x) \cdot \exp(i \cdot y) - \exp(-i \cdot x) \cdot \exp(-i \cdot y)) &= \\ \frac{1}{2 \cdot i} \cdot (\exp(i \cdot x + i \cdot y) - \exp(-i \cdot x - i \cdot y)) &= \\ \frac{1}{2 \cdot i} \cdot (\exp(i \cdot (x + y)) - \exp(-i \cdot (x + y))) &= \\ \sin(x + y) \end{aligned}$$

8. Let $x \in \mathbb{R}$ then

a.

$$\begin{aligned}
 \cos(x) &= \frac{\exp(i \cdot x) + \exp(-i \cdot x)}{2} \\
 &= \frac{\exp(i \cdot x) + \exp(\overline{i \cdot x})}{2} \\
 &\stackrel{[\text{theorem: 19.60}]}{=} \frac{\exp(i \cdot x) + \overline{\exp(i \cdot x)}}{2} \\
 &\stackrel{[\text{theorem: 10.83}]}{=} \frac{2 \cdot \operatorname{Re}(\exp(i \cdot x))}{2} \\
 &= \operatorname{Re}(\exp(i \cdot x)) \in \mathbb{R}
 \end{aligned}$$

b.

$$\begin{aligned}
 \sin(x) &= \frac{\exp(i \cdot x) - \exp(-i \cdot x)}{2 \cdot i} \\
 &= \frac{\exp(i \cdot x) - \exp(\overline{i \cdot x})}{2 \cdot i} \\
 &\stackrel{[\text{theorem: 19.60}]}{=} \frac{2 \cdot i \cdot \operatorname{Im}(\exp(i \cdot x))}{2 \cdot i} \\
 &= \operatorname{Im}(\exp(i \cdot x)) \in \mathbb{R}
 \end{aligned}$$

9. Let $x \in \mathbb{R}$ then

$$|e^{i \cdot x}| \stackrel{[\text{theorem: 19.42(6)}]}{=} |\exp(i \cdot x)| \stackrel{[\text{theorem: 19.60 (3)}]}{=} 1$$

further

$$|\exp(-i \cdot x)| \stackrel{x \in \mathbb{R}}{=} |\exp(\overline{i \cdot x})| \stackrel{[\text{theorem: 19.60}]}{=} |\overline{\exp(i \cdot x)}| \stackrel{[\text{theorem: 10.85}]}{=} |\exp(i \cdot x)| = 1$$

10. Let $x \in \mathbb{R}$ then we have:

$$|\cos(x)| = \frac{1}{2} \cdot |e^{i \cdot x} + e^{-i \cdot x}| \leq \frac{1}{2} \cdot (|e^{i \cdot x}| + |e^{-i \cdot x}|) \stackrel{(\text{9})}{=} \frac{1}{2} \cdot 2 = 1$$

and

$$|\cos(x)| = \left| \frac{1}{2 \cdot i} \right| \cdot |e^{i \cdot x} - e^{-i \cdot x}| \leq \frac{1}{2} \cdot (|e^{i \cdot x}| + |e^{-i \cdot x}|) \stackrel{(\text{9})}{=} \frac{1}{2} \cdot 2 = 1$$

□

Theorem 19.63. $\forall z \in \mathbb{C}$ we have that $\cos: \mathbb{C} \rightarrow \mathbb{C}$ has a derivative at z with

$$\cos'_z = -\sin(z)$$

and $\sin: \mathbb{C} \rightarrow \mathbb{C}$ has a derivative at z with

$$\sin'_z = \cos(z)$$

Proof. As $\forall z \in \mathbb{C}$ we have that

$$\left(\frac{1}{2} \cdot (\exp \circ (i \cdot \operatorname{Id}_{\mathbb{C}}) + \exp \circ (-i \cdot \operatorname{Id}_{\mathbb{C}})) \right)(z) = \frac{1}{2} \cdot (\exp(i \cdot x) + \exp(-i \cdot x)) = \cos(x)$$

hence

$$\cos = \frac{1}{2} \cdot (\exp \circ (i \cdot \operatorname{Id}_{\mathbb{C}}) + \exp \circ (-i \cdot \operatorname{Id}_{\mathbb{C}}))$$

Likewise we have that

$$\left(\frac{1}{2 \cdot i} \cdot (\exp \circ (i \cdot \operatorname{Id}_{\mathbb{C}}) - \exp \circ (-i \cdot \operatorname{Id}_{\mathbb{C}})) \right)(z) = \frac{1}{2 \cdot i} \cdot (\exp(i \cdot x) - \exp(-i \cdot x)) = \sin(x)$$

hence

$$\sin = \frac{1}{2} \cdot (\exp \circ (i \cdot \operatorname{Id}_{\mathbb{C}}) - \exp \circ (-i \cdot \operatorname{Id}_{\mathbb{C}}))$$

Let $z \in \mathbb{C}$ then by [theorem: 19.27], [examples: 16.31], [theorem: 16.40] and the chain rule [theorem: 16.68] it follows that $\sin$ and $\cos$ has a derivative at z and

$$\begin{aligned}
 \cos'_z &= \left(\frac{1}{2} \cdot (\exp \circ (i \cdot \text{Id}_{\mathbb{C}}) + \exp \circ (-i \cdot \text{Id}_{\mathbb{C}})) \right)'_z \\
 &= \frac{1}{2} \cdot ((\exp \circ (i \cdot \text{Id}_{\mathbb{C}}))'_z + (\exp \circ (-i \cdot \text{Id}_{\mathbb{C}}))'_z) \\
 &= \frac{1}{2} \cdot (\exp'_{(i \cdot \text{Id}_{\mathbb{C}})(z)} \cdot i \cdot (\text{Id}_{\mathbb{C}})'_z + \exp'_{(-i \cdot \text{Id}_{\mathbb{C}})(z)} \cdot (-i) \cdot (\text{Id}_{\mathbb{C}})'_z) \\
 &= \frac{1}{2} \cdot (\exp'_{i \cdot z} \cdot i \cdot 1 + \exp'_{-i \cdot z} \cdot (-i) \cdot 1) \\
 &= \frac{1}{2} \cdot (i \cdot \exp(i \cdot z) - i \cdot \exp(-i \cdot z)) \\
 &= \frac{1}{2} \cdot \frac{i}{i} \cdot i \cdot (\exp(i \cdot z) - \exp(-i \cdot z)) \\
 &= -\frac{1}{2 \cdot i} \cdot (\exp(i \cdot z) - \exp(-i \cdot z)) \\
 &= \sin(z) \\
 \sin'_z &= \left(\frac{1}{2 \cdot i} \cdot (\exp \circ (i \cdot \text{Id}_{\mathbb{C}}) - \exp \circ (-i \cdot \text{Id}_{\mathbb{C}})) \right)'_z \\
 &= \frac{1}{2 \cdot i} \cdot ((\exp \circ (i \cdot \text{Id}_{\mathbb{C}}))'_z - (\exp \circ (-i \cdot \text{Id}_{\mathbb{C}}))'_z) \\
 &= \frac{1}{2 \cdot i} \cdot (\exp'_{(i \cdot \text{Id}_{\mathbb{C}})(z)} \cdot i \cdot (\text{Id}_{\mathbb{C}})'_z - \exp'_{(-i \cdot \text{Id}_{\mathbb{C}})(z)} \cdot (-i) \cdot (\text{Id}_{\mathbb{C}})'_z) \\
 &= \frac{1}{2 \cdot i} \cdot (\exp'_{i \cdot z} \cdot i \cdot 1 + \exp'_{-i \cdot z} \cdot i \cdot 1) \\
 &= \frac{1}{2 \cdot i} \cdot i \cdot (\exp(i \cdot z) + \exp(-i \cdot z)) \\
 &= \frac{1}{2} \cdot (\exp(i \cdot z) + \exp(-i \cdot z)) \\
 &= \cos(z)
 \end{aligned}$$

□

The next theorem shows that there exist a $x_0 \in \mathbb{R}$ such that $\cos(x_0) = 0$. This will be used to define the π and show that $\sin$ and $\cos$ are periodic.

Theorem 19.64. $\exists x_0 \in [0, \infty[$ such that $\cos(x_0) = 0$.

Proof. We proceed by contradiction. So assume that

$$(\text{assumption}) \quad \forall x \in [0, \infty[\quad \cos(x) \neq 0 \quad (19.39)$$

Assume that there exist $x \in [0, \infty[$ such that $\cos(x) < 0 < 1 \stackrel{[\text{theorem: 19.62}]}{=} \cos(0)$ then

$$\cos|_{[0, x]}(x) = \cos(x) \leq 0 < \cos|_{[0, x]}(0)$$

Further as $\cos: \mathbb{R} \rightarrow \mathbb{R}$ is continuous it follows from [theorem: 14.152] that $\cos|_{[0, x]}$ is continuous. So we can use the intermediate value theorem [theorem: 14.467] to find a $y \in [0, x]$ such that $\cos(y) = \cos|_{[0, 1]}(y) = 0$ which contradicts [eq: 19.39]. So $\forall x \in [0, \infty[$ we must have that $0 \leq \cos(x)$ or using [eq: 19.39] that $0 < \cos(x)$, as by [theorem: 19.63] $\sin'_x$ exist and $\sin'_x = \cos(x) > 0$ it follows from [theorem: 16.211]

$$\sin \text{ is strictly increasing on } [0, \infty[\quad (19.40)$$

As $\sin(0) \stackrel{[\text{theorem: 19.62}]}{=} 0$ we conclude that

$$\forall x \in]0, \infty[\quad \text{we have } 0 < \sin(x) \quad (19.41)$$

As $\forall x, y \in \mathbb{R}$ we have by [theorem: 19.62] that $\cos(x) < 1$ and $-\cos(y) < 1$ it follows that

$$\forall x, y \in \mathbb{R} \quad \text{we have } \cos(x) - \cos(y) < 1 + 1 = 2 \quad (19.42)$$

Let $x, y \in \mathbb{R}$ with $0 < x < y$. By [theorem: 19.63] we have that $\forall t \in \mathbb{R}$ $\cos'_t$ exist and $\cos'_t = -\sin(t)$ so that if we define

$$\cos': \mathbb{R} \rightarrow \mathbb{R} \text{ by } \cos'(t) = \cos'_t$$

it follows that

$$\cos' = -\sin \text{ which is continuous [see theorem: 19.62(2)]}$$

So using [theorems: 16.10, 16.36] it follows that

$$\cos|_{[x,y]} \text{ has a derivative at every } t \in [x, y] \text{ and } (\cos|_{[x,y]})' = \sin|_{[x,y]} \quad (19.43)$$

As $\sin$ is continuous on $\mathbb{R}$, hence by [theorem: 14.152] on $[x, y]$ the Riemann integral exist [see theorem: 15.25] and we have

$$\begin{aligned} \int_x^y \sin|_{[x,y]} &= \int_x^y -(\cos|_{[x,y]})' \\ &\stackrel{[\text{theorem: 15.28}]}{=} -\int_x^y (\cos|_{[x,y]})' \\ &\stackrel{[\text{eq: 19.43}] \text{ and } [\text{theorem: 16.216}]}{=} -(\cos|_{[x,y]}(y) - \cos|_{[x,y]}(x)) \\ &= \cos(x) - \cos(y) \\ &\stackrel{[\text{eq: 19.42}]}{\leq} 2 \end{aligned}$$

proving that

$$\int_x^y \sin|_{[x,y]} \leq 2 \quad (19.44)$$

Define now the constant function

$$C_{\sin(x)}: [x, y] \rightarrow \mathbb{R} \text{ by } C_{\sin(x)} = \sin(x)$$

Then

$$\forall t \in [x, y] \text{ we have } C_{\sin(x)}(t) = \sin(x) \stackrel{[\text{eq: 19.40}]}{\leq} \sin(t) = \sin|_{[x,y]}(t) \quad (19.45)$$

Now

$$\begin{aligned} \sin(x) \cdot (y - x) &\stackrel{[\text{theorem: 15.29}]}{=} \int_x^y C_{\sin(x)} \\ &\stackrel{[\text{eq: 19.45}] \text{ and } [\text{theorem: 15.26}]}{\leq} \int_x^y \sin|_{[x,y]} \\ &\stackrel{[\text{eq: 19.44}]}{\leq} 2 \end{aligned}$$

Hence taking in account that by [eq: 19.41] $0 < \sin(x)$ it follows that

$$\forall x, y \in \mathbb{R} \text{ with } 0 < x < y \text{ we have } y - x < \frac{2}{\sin(x)}$$

Take now $x \in \mathbb{R}^+$ and $y = \frac{2}{\sin(x)} + x + 1$ then, as $0 < \sin(x)$, we have that $0 < x < y$ so that $\frac{2}{\sin(x)} + 1 = y - x < \frac{2}{\sin(x)}$ leading to the contradiction $1 < 0$. Hence the assumption [eq: 19.39] is wrong and there exist a $x_0 \in [0, \infty[$ such that $\cos(x_0) = 0$. $\square$

Lemma 19.65. $\{x \in [0, \infty[\mid \cos(x) = 0\} = \cos^{-1}(\{0\}) \cap [0, \infty[$ has a minimum.

Proof. Using [theorem: 19.64] it follows that $\cos^{-1}(\{0\}) \cap [0, \infty[\neq \emptyset$, further $\cos^{-1}(\{0\}) \cap [0, \infty[$ is bounded below by 0, hence, as $\mathbb{R}$ is conditional complete [see theorem: 10.18], it follows that

$$x_0 = \inf(\cos^{-1}(\{0\}) \cap [0, \infty[) \text{ exist and } 0 \leq x_0$$

As $\{0\}$ is closed [see theorems: 14.242, 14.244] and $\cos$ is continuous it follows from [theorem: 14.154] that $\cos^{-1}(\{0\})$. Further by [theorems: 14.105] $[0, \infty[$ is closed so that

$$\cos^{-1}(\{0\}) \cap [0, \infty[\text{ is closed}$$

Assume now that $x_0 \notin \cos^{-1}(\{0\}) \cap [0, \infty[$ then $x_0 \in \mathbb{R} \setminus \cos^{-1}(\{0\}) \cap [0, \infty[$ a open set, so there exist a $\delta \in \mathbb{R}^+$ such that $x_0 \in]x_0 - \delta, x_0 + \delta[\subseteq \mathbb{R} \setminus \cos^{-1}(\{0\}) \cap [0, \infty[$ or

$$]x_0 - \delta, x_0 + \delta[\cap \cos^{-1}(\{0\}) \cap [0, \infty[= \emptyset \quad (19.46)$$

As $\inf(\cos^{-1}(\{0\}) \cap [0, \infty]) = x_0 < x_0 + \delta$ we have that there exist a $y \in \cos^{-1}(\{0\}) \cap [0, \infty[$ such that $x_0 - \delta < x_0 \leq y < x_0 + \delta$ so that $y \in]x_0 - \delta, x_0 + \delta[\cap \cos^{-1}(\{0\}) \cap [0, \infty[$ contradicting [eq: 19.46]. Hence we have that $x_0 \in \cos^{-1}(\{0\}) \cap [0, \infty[$ or

$$x_0 = \min(\cos^{-1}(\{0\}) \cap [0, \infty[) \quad \square$$

The above allows us to finally define π .

Definition 19.66. π is defined by

$$\pi = 2 \cdot \min(\cos^{-1}(\{0\}) \cap [0, \infty[)$$

so that

$$\frac{\pi}{2} = \min(\cos^{-1}(\{0\}) \cap [0, \infty[)$$

In other words $\frac{\pi}{2}$ is the smallest positive real number x_0 such that $\cos(x_0) = 0$.

Note 19.67. As $\cos(0) \stackrel{[theorem: 19.62]}{=} 1$ we must have that $0 < \pi$.

Next we prove that π and the trigonometric functions have the usual properties.

Theorem 19.68.

1. $\forall x \in [0, \frac{\pi}{2}[$ we have $0 < \cos(x)$ and by the above definition $\cos(\frac{\pi}{2}) = 0$.
2. $\forall x \in]0, \frac{\pi}{2}]$ we have $0 < \sin(x)$ and $\sin(\frac{\pi}{2}) = 1$.

Proof.

1. Assume that $\exists x \in [0, \frac{\pi}{2}[$ such that $\cos(x) \leq 0$. If $\cos(x) = 0$ then as $\frac{\pi}{2} = \min(\cos^{-1}(\{0\}) \cap [0, \infty[)$ we have $\frac{\pi}{2} \leq x < \frac{\pi}{2}$ a contradiction, so we must have that $\cos(x) < 0$. Then we have

$$\cos_{|[0, \frac{\pi}{2}[}(x) < 0 < 1 \stackrel{[eq: 19.62]}{=} \cos(0) = \cos_{|[0, \frac{\pi}{2}]}(0)$$

As $\cos_{|[0, \frac{\pi}{2}]}$ is continuous on $[0, \frac{\pi}{2}]$ [see theorems: 19.62(2)] we can use the intermediate value theorem [theorem: 14.467] to find a $z \in [0, x]$ such that $\cos(z) = \cos_{|[0, x]}(z) = 0$ hence $z \in \cos^{-1}(\{0\}) \cap [0, \infty[$. So we have

$$z \leq x < \frac{\pi}{2} = \min(\cos^{-1}(\{0\}) \cap [0, \infty[) \leq z$$

a contradiction. So the assumption is wrong and we must have that $\forall x \in [0, \frac{\pi}{2}[$ we have $0 < \cos(x)$.

2. As $\forall x \in [0, \frac{\pi}{2}[$ we have $\sin'_x \stackrel{[theorem: 19.63]}{=} \cos(x) > 0$ [by (1)] it follows from [theorem: 16.211] that $\sin$ is strictly increasing on $[0, \frac{\pi}{2}[$. Hence if $x \in]0, \frac{\pi}{2}[$ we have $0 < x$ so that $0 \stackrel{[theorem: 19.62]}{=} \sin(0) < \sin(x)$. So

$$\forall x \in]0, \frac{\pi}{2}[\text{ we have } 0 < \sin(x) \quad (19.47)$$

As $\forall x \in [0, \frac{\pi}{2}] = [0, \frac{\pi}{2}[\cup \{\frac{\pi}{2}\}$ we have $0 \leq \cos(x) \stackrel{[theorem: 19.63]}{=} \sin'_x$ it follows from [theorem: 16.211] $\sin$ is increasing on $[0, \frac{\pi}{2}]$ so that $\forall x \in [0, \frac{\pi}{2}]$ we have that $0 = \sin(0) \leq \sin(x)$. More specific

$$0 \leq \sin\left(\frac{\pi}{2}\right)$$

Now $1 \stackrel{[theorem: 19.62(6)]}{=} \sin^2\left(\frac{\pi}{2}\right) + \cos^2\left(\frac{\pi}{2}\right) = \sin^2\left(\frac{\pi}{2}\right) + 0^2 = \sin^2(x)$ proving that $\sin^2(x) = 1$.

Using then [definition: 10.67] together with $0 \leq \sin\left(\frac{\pi}{2}\right)$ proves that

$$\sin\left(\frac{\pi}{2}\right) = 1$$

Combining this with [eq: 19.47] gives

$$\forall x \in]0, \frac{\pi}{2}] \text{ we have } 0 < \sin(x) \text{ and } \sin\left(\frac{\pi}{2}\right) = 1$$

□

Theorem 19.69. *We have for $\sin$ and $\cos$ the following properties:*

1. $\cos(0) = 1$
2. $\cos\left(\frac{\pi}{2}\right) = 0$
3. $\sin(0) = 0$
4. $\sin\left(\frac{\pi}{2}\right) = 1$
5. $\cos(\pi) = -1$
6. $\sin(\pi) = 0$
7. $\cos(2 \cdot \pi) = 1$
8. $\sin(2 \cdot \pi) = 0$
9. $\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \sqrt{\frac{1}{2}}$
10. $\forall x \in \mathbb{C}$ we have $\cos\left(x + \frac{\pi}{2}\right) = -\sin(x)$
11. $\forall x \in \mathbb{C}$ we have $\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$
12. $\forall x \in \mathbb{C} \forall n \in \mathbb{N}_0$ we have $\cos(x \pm 2 \cdot n \cdot \pi) = \cos(x)$
13. $\forall x \in \mathbb{C} \forall n \in \mathbb{N}_0$ we have $\sin(x \pm 2 \cdot n \cdot \pi) = \sin(x)$
14. $\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ we have that $0 < \cos(x)$ and $\forall x \in]0, \pi[$ we have $0 < \sin(x)$
15. $\forall x \in]0, \pi[$ we have $\cos(x) \neq 1 \wedge \sin(x) \neq -1$ and $\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ we have $\sin(x) \neq 1 \wedge \sin(x) \neq -1$
16. $\forall x \in \mathbb{C} \forall n \in \mathbb{N}_0$ we have $e^{i \cdot (x \pm 2 \cdot n \cdot \pi)} \underset{[\text{theorem: 19.42}]}{=} \exp(i \cdot (x \pm 2 \cdot n \cdot \pi)) = \exp(i \cdot x)$

Proof.

1. This was proved in [theorem: 19.62].
2. As $\frac{\pi}{2} = \min(\cos^{-1}(\{0\}) \cap [0, \infty[)$ it follows that $\cos\left(\frac{\pi}{2}\right) = 0$.
3. This was proved in [theorem: 19.62].
4. This is proved in [theorem: 19.68].
5. $\cos(\pi) = \cos\left(\frac{\pi}{2} + \frac{\pi}{2}\right) \underset{[\text{theorem: 19.62}]}{=} \cos\left(\frac{\pi}{2}\right) \cdot \cos\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{2}\right) \cdot \sin\left(\frac{\pi}{2}\right) \underset{(2,4)}{=} 0 \cdot 0 - 1 \cdot 1 = -1$
6. $\sin(\pi) = \sin\left(\frac{\pi}{2} + \frac{\pi}{2}\right) \underset{[\text{theorem: 19.62}]}{=} \sin\left(\frac{\pi}{2}\right) \cdot \cos\left(\frac{\pi}{2}\right) - \cos\left(\frac{\pi}{2}\right) \cdot \sin\left(\frac{\pi}{2}\right) \underset{(2,4)}{=} 1 \cdot 0 - 0 \cdot 1 = 0$
7. $\cos(2 \cdot \pi) = \cos(\pi + \pi) \underset{[\text{theorem: 19.62}]}{=} \cos(\pi) \cdot \cos(\pi) - \sin(\pi) \cdot \sin(\pi) \underset{(5,6)}{=} -1 \cdot -1 = 0 \cdot 0 = 1$
8. $\sin(2 \cdot \pi) = \sin(\pi + \pi) \underset{[\text{theorem: 19.62}]}{=} \sin(\pi) \cdot \cos(\pi) - \cos(\pi) \cdot \sin(\pi) \underset{(5,6)}{=} 0 \cdot -1 - (-1) \cdot 0 = 0$
9. We have

$$\begin{aligned}
 0 &= \cos\left(\frac{\pi}{2}\right) \\
 &= \cos\left(\frac{\pi}{4} + \frac{\pi}{4}\right) \\
 &\underset{[\text{theorem: 19.62(7)}]}{=} \cos\left(\frac{\pi}{4}\right) \cdot \cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right) \cdot \sin\left(\frac{\pi}{4}\right) \\
 &= \cos^2\left(\frac{\pi}{4}\right) - \sin^2\left(\frac{\pi}{4}\right)
 \end{aligned}$$

so that $\cos^2\left(\frac{\pi}{4}\right) = \sin^2\left(\frac{\pi}{4}\right)$. As by [theorem: 19.68] $0 \leq \cos\left(\frac{\pi}{4}\right), \sin\left(\frac{\pi}{4}\right)$ it follows from [definition: 10.67] that

$$\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)$$

Next

$$1 \underset{[\text{theorem: 19.62}]}{=} \sin^2\left(\frac{\pi}{4}\right) + \cos^2\left(\frac{\pi}{4}\right) = 2 \cdot \sin^2\left(\frac{\pi}{4}\right)$$

hence $\sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2}$ which as $0 \leq \sin\left(\frac{\pi}{4}\right), \frac{1}{2}$ proves that

$$\sqrt{\frac{1}{2}} = \sin\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right)$$

10. We have $\forall z \in \mathbb{C}$ that

$$\cos\left(x + \frac{\pi}{2}\right) \stackrel{[\text{theorem: 19.62}]}{=} \cos(x) \cdot \cos\left(\frac{\pi}{2}\right) - \sin(x) \cdot \sin\left(\frac{\pi}{2}\right) \stackrel{(2,4)}{=} \cos(x) \cdot 0 - \sin(x) \cdot 1 = -\sin(x)$$

11. We have $\forall z \in \mathbb{C}$ that

$$\sin\left(x + \frac{\pi}{2}\right) \stackrel{[\text{theorem: 19.62}]}{=} \sin(x) \cdot \cos\left(\frac{\pi}{2}\right) + \cos(x) \cdot \sin\left(\frac{\pi}{2}\right) \stackrel{(2,4)}{=} \sin(x) \cdot 0 + \cos(x) \cdot 1 = \cos(x)$$

12. We prove this by induction, so let

$$S = \{n \in \mathbb{N}_0 \mid \cos(x \pm 2 \cdot n \cdot \pi) = \cos(x)\}$$

then we have:

0 $\in S$. As $\cos(x \pm 2 \cdot 0 \cdot \pi) = \cos(x \pm 0) = \cos(x)$ it follows that $0 \in S$

$n \in S \Rightarrow n + 1 \in S$. We have

$$\begin{aligned} \cos(x \pm 2 \cdot (n+1) \cdot \pi) &= \\ \cos((x \pm 2 \cdot n \cdot \pi) \pm 2 \cdot \pi) &\stackrel{[\text{theorem: 19.62}]}{=} \\ \cos(x \pm 2 \cdot n \cdot \pi) \cdot \cos(\pm \pi) - \sin(x \pm 2 \cdot n \cdot \pi) \cdot \sin(\pm \pi) &\stackrel{[\text{theorem: 19.62}]}{=} \\ \cos(x \pm 2 \cdot n \cdot \pi) \cdot \cos(\pi) - \sin(x \pm 2 \cdot n \cdot \pi) \cdot (\pm \sin(\pi)) &\stackrel{(7,8)}{=} \\ \cos(x \pm 2 \cdot n \cdot \pi) \cdot 1 - \sin(x \pm 2 \cdot n \cdot \pi) \cdot 0 &= \\ \cos(x \pm 2 \cdot n \cdot \pi) &\stackrel{n \in S}{=} \\ \cos(x) & \end{aligned}$$

proving that $n + 1 \in \mathbb{N}_0$

13. We prove this by induction, so let

$$S = \{n \in \mathbb{N}_0 \mid \sin(x \pm 2 \cdot n \cdot \pi) = \sin(x)\}$$

then we have:

0 $\in S$. As $\sin(x \pm 2 \cdot 0 \cdot \pi) = \sin(x \pm 0) = \sin(x)$ it follows that $0 \in S$

$n \in S \Rightarrow n + 1 \in S$. We have

$$\begin{aligned} \sin(x \pm 2 \cdot (n+1) \cdot \pi) &= \\ \sin((x \pm 2 \cdot n \cdot \pi) \pm 2 \cdot \pi) &\stackrel{[\text{theorem: 19.57}]}{=} \\ \sin(x \pm 2 \cdot n \cdot \pi) \cdot \cos(\pm \pi) + \cos(x \pm 2 \cdot n \cdot \pi) \cdot \sin(\pm \pi) &\stackrel{[\text{theorem: 19.57}]}{=} \\ \sin(x \pm 2 \cdot n \cdot \pi) \cdot \cos(\pi) + \cos(x \pm 2 \cdot n \cdot \pi) \cdot (\pm \sin(\pi)) &\stackrel{(7,8)}{=} \\ \sin(x \pm 2 \cdot n \cdot \pi) \cdot 1 - \cos(x \pm 2 \cdot n \cdot \pi) \cdot 0 &= \\ \sin(x \pm 2 \cdot n \cdot \pi) &\stackrel{n \in S}{=} \\ \sin(x) & \end{aligned}$$

proving that $n + 1 \in \mathbb{N}_0$

14. Using [theorem: 19.68] we have

$$\forall x \in \left[0, \frac{\pi}{2}\right[\text{ we have } 0 < \cos(x) \text{ and } \forall x \in \left]0, \frac{\pi}{2}\right] \text{ we have } 0 < \sin(x) \quad (19.48)$$

Let $x \in \left]-\frac{\pi}{2}, 0\right]$ then $x + \frac{\pi}{2} \in \left]0, \frac{\pi}{2}\right]$ so that $0 < \stackrel{[\text{eq: 19.48}]}{\sin\left(x + \frac{\pi}{2}\right)} \stackrel{(11)}{=} \cos(x)$ hence

$$\forall x \in \left]-\frac{\pi}{2}, 0\right] \text{ we have } 0 < \cos(x) \quad (19.49)$$

Let $x \in [\frac{\pi}{2}, \pi[$ then $x - \frac{\pi}{2} \in [0, \frac{\pi}{2}[$ so that $0 < \cos(x - \frac{\pi}{2}) \stackrel{(11)}{=} \sin((x - \frac{\pi}{2}) + \frac{\pi}{2}) = \sin(x)$ hence

$$\forall x \in [\frac{\pi}{2}, \pi[\text{ we have } 0 < \sin(x) \quad (19.50)$$

Combining [eqs: 19.48, 19.49 and 19.50] results in

$$\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[\text{ we have } 0 < \cos(x) \text{ and } \forall x \in]0, \pi[\text{ we have } 0 < \sin(x)$$

15. Assume that there exist a $x \in]0, \pi[$ such that $\cos(x) = 1$ or $\cos(x) = -1$. Then as by (14) $0 < \sin(x)$ we have that $0 < \sin^2(x) \stackrel{[\text{theorem: 19.62(6)}]}{=} 1 - \cos^2(x) = 1 - 1 = 0$ giving the contradiction $10 < 0$. Hence the assumption is wrong and we must have that

$$\forall x \in]0, \pi[\text{ we have } \cos(x) \neq 1 \text{ and } \cos(x) \neq -1 \quad (19.51)$$

Let $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ then $x + \frac{\pi}{2} \in]0, \pi[$ so that $-\sin(x) \stackrel{(10)}{=} \cos(x)$ which as by [eq: 19.51] $\cos(x) \neq 1$ and $\cos(x) \neq -1$ proves that $\sin(x) \neq -1$ and $\sin(x) \neq 1$. Hence

$$\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[\text{ we have } \sin(x) \neq 1 \text{ and } \sin(x) \neq -1$$

16. We have

$$\begin{aligned} \exp(i \cdot (x \pm 2 \cdot n \cdot p)) &\stackrel{[\text{theorem: 19.62}]}{=} \cos(x \pm 2 \cdot n \cdot p) + i \cdot \sin(x \pm 2 \cdot n \cdot p) \\ &\stackrel{(12,13)}{=} \cos(x) + i \cdot \sin(x) \\ &\stackrel{[\text{theorem: 19.62}]}{=} \exp(i \cdot x) \end{aligned}$$

□

We have also what is probably the most beautiful mathematical equations (because it combines the fundamental constants 0, 1, π and e).

Theorem 19.70. (Euler's equation) *We have $e^{i \cdot \pi} + 1 = 0$*

Proof. We have

$$\begin{aligned} e^{i \cdot \pi} &\stackrel{[\text{theorem: 19.42}]}{=} \exp(i \cdot \pi) \\ &\stackrel{[\text{theorem: 19.62}]}{=} \cos(\pi) + i \cdot \sin(\pi) \\ &\stackrel{[\text{theorem: 19.69}]}{=} -1 \end{aligned}$$

hence

$$e^{i \cdot \pi} + 1 = 0 \quad \square$$

Next we introduce the inverses of the functions $\cos|_{[0, \pi]}$ and $\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}$. Of course we must first prove that $\cos|_{[0, \pi]}$ and $\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}$ are indeed bijections.

Theorem 19.71.

1. *The function $\cos|_{[0, \pi]}: [0, \pi] \rightarrow [-1, 1]$ is a bijection.*
2. *The function $\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}: [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$ is a bijection.*

Proof.

1. We have

injectivity. Let $x, y \in [0, \pi]$ with $\cos(x) = \cos(y)$. Without losing generality we can take $x \leq y$ [if not interchange x and y]. Assume that $x \neq y$ then we have $x < y$ and we must look at the following cases:

$x = 0 < y = \pi$. Then we would have

$$1 = \cos(0) \stackrel{[\text{theorem: 19.69}]}{=} \cos(x) = \cos(y) \stackrel{[\text{theorem: 19.69}]}{=} 0$$

leading to the contradiction $1 \neq 0$.

$x = 0 < y < \pi$. As $y \in]0, \pi[$ it follows from [theorem: 19.69(15)] that $\cos(y) \neq 1$ which as $1 \stackrel{[theorem: 19.69]}{=} \cos(0) = \cos(x) = \cos(y)$ leads to the contradiction $1 \neq 1$.

$0 < x < y = \pi$. As $x \in]0, \pi[$ it follows from [theorem: 19.69(15)] that $\cos(x) \neq -1$ which as $-1 \stackrel{[theorem: 19.69]}{=} \cos(\pi) = \cos(x)$ leads to the contradiction $-1 \neq -1$.

$0 < x < y < \pi$. As $\forall z \in]0, \pi[$ we have $\cos'_z \stackrel{[theorem: 19.63]}{=} \sin(z) > 0$ [see theorem: 19.69] it follows from [theorem: 16.211] that $\cos$ is strictly increasing on $]0, \pi[$. Hence $\cos(x) < \cos(y)$ contradicting $\cos(x) = \cos(y)$.

As in all cases we have a contradiction the assumption is wrong and we must have that $x = y$ proving injectivity.

surjectivity. Let $y \in [-1, 1]$ then we have either:

$y = -1$. Then for $\pi \in [0, \pi]$ we have $\cos(\pi) \stackrel{[theorem: 19.69]}{=} -1 = y$.

$y = 1$. Then $0 \in [0, \pi]$ we have $\cos(0) \stackrel{[theorem: 19.69]}{=} 1 = y$.

$y \in]-1, 1[$. By [theorem: 19.62] and [theorem: 14.152] it follows that $\cos|_{[0, \pi]}$ is continuous, further we have

$$\cos|_{[0, \pi]}(\pi) = \cos(\pi) = -1 < y < 1 = \cos(0) = \cos|_{[0, \pi]}(0)$$

So we can use the intermediate value theorem [theorem: 14.467] to find a $x \in [0, \pi]$ such that $\cos(x) = y$

So in all cases we have found a $x \in [0, \pi]$ such that $\cos(x) = y$ proving surjectivity.

2. We have

injectivity. Let $x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ with $\sin(x) = \sin(y)$. Without loosing generality we can take $x \leq y$ [if not interchange x and y]. Assume that $x \neq y$ then we have $x < y$ and we must look at the following cases:

$x = -\frac{\pi}{2} < y = \frac{\pi}{2}$. Then we would have

$$\begin{aligned} -1 &\stackrel{[theorem: 19.69]}{=} -\sin\left(\frac{\pi}{2}\right) \\ &\stackrel{[theorem: 19.62]}{=} \sin\left(-\frac{\pi}{2}\right) \\ &= \sin(x) \\ &= \sin(y) \\ &= \sin\left(\frac{\pi}{2}\right) \\ &\stackrel{[theorem: 19.69]}{=} 1 \end{aligned}$$

leading to the contradiction $-1 \neq 1$.

$x = -\frac{\pi}{2} < y < \frac{\pi}{2}$. As $y \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ it follows from [theorem: 19.69(15)] that $\sin(y) \neq -1$ which as $-1 \stackrel{[theorem: 19.69]}{=} \sin(-\frac{\pi}{2}) = \sin(x) = \sin(y)$ leads to the contradiction $-1 \neq -1$.

$-\frac{\pi}{2} < x < y = \frac{\pi}{2}$. As $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ it follows from [theorem: 19.69(15)] that $\sin(x) \neq -1$ which as $-1 \stackrel{[theorem: 19.69]}{=} \sin(-\frac{\pi}{2}) = \sin(x) = \sin(y)$ leads to the contradiction $-1 \neq -1$.

$-\frac{\pi}{2} < x < y < \frac{\pi}{2}$. As $\forall z \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ we have $\sin'_z \stackrel{[theorem: 19.63]}{=} \cos(z) > 0$ [see theorem: 19.69] it follows from [theorem: 16.211] that $\sin$ is strictly increasing on $]-\frac{\pi}{2}, \frac{\pi}{2}[$. Hence $\sin(x) < \sin(y)$ contradicting $\sin(x) = \sin(y)$.

As in all cases we have a contradiction the assumption is wrong and we must have that $x = y$ proving injectivity.

surjectivity. Let $y \in [-1, 1]$ then we have either:

$y = -1$. Then for $-\frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ we have

$$\sin\left(-\frac{\pi}{2}\right) \stackrel{[\text{theorem: 19.62}]}{=} -\sin\left(\frac{\pi}{2}\right) \stackrel{[\text{theorem: 19.69}]}{=} -1 = y.$$

$y = 1$. Then for $\frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ we have $\sin\left(\frac{\pi}{2}\right) \stackrel{[\text{theorem: 19.69}]}{=} 1 = y$.

$y \in]-1, 1[$. By [theorem: 19.62] and [theorem: 14.152] it follows that $\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}$ is continuous, further we have

$$\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}\left(-\frac{\pi}{2}\right) = \sin\left(-\frac{\pi}{2}\right) = -1 < y < 1 = \sin\left(\frac{\pi}{2}\right) = \sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}\left(\frac{\pi}{2}\right)$$

So we can use the intermediate value theorem [theorem: 14.467] to find a $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\cos(x) = y$.

So in all cases we have found a $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\cos(x) = y$ proving surjectivity. $\square$

Definition 19.72. The above theorem allows us to define arccosine and arcsine as follows

arccosine: $[-1, 1] \rightarrow [0, \pi]$ is defined by $\text{arccosine} = (\cos|_{[0, \pi]})^{-1}$

and

arcsine: $[-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ is defined by $\text{arcsine} = (\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]})^{-1}$

We have then the following trivial properties for arccosine or arcsine

Theorem 19.73. We have for arccosine and arcsine

1. $\forall x \in [-1, 1]$ we have $\cos(\text{arccosine}(x)) = x$ and $\sin(\text{arcsine}(x)) = x$
2. $\forall x \in [0, \pi]$ we have $\text{arccosine}(\cos(x)) = x$
3. $\forall x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ we have $\text{arcsine}(\sin(x)) = x$
4. $\text{arccosine}(-1) = \pi$
5. $\text{arccosine}(0) = \frac{\pi}{2}$
6. $\text{arccosine}(1) = 0$
7. $\text{arcsine}(-1) = -\frac{\pi}{2}$
8. $\text{arcsine}(0) = 0$
9. $\text{arcsine}(1) = \frac{\pi}{2}$

Proof.

1. If $x \in [-1, 1]$ then

$$\cos(\text{arccosine}(x)) \stackrel{\text{arccosine}([[-1, 1]]) = [0, \pi]}{=} \cos|_{[0, \pi]}(\text{arccosine}(x)) = \cos|_{[0, \pi]}((\cos|_{[0, \pi]})^{-1}(x)) = x$$

Further

$$\begin{aligned} \sin(\text{arcsine}(x)) &\stackrel{\text{arcsine}([[-1, 1]]) = [-\frac{\pi}{2}, \frac{\pi}{2}]}{=} \sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}(\text{arcsine}(x)) \\ &= \sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}((\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]})^{-1}(x)) \\ &= x \end{aligned}$$

2. Let $x \in [0, \pi]$ then

$$\text{arccos}(\cos(x)) = \text{arccos}(\cos|_{[0, \pi]}(x)) = (\cos|_{[0, \pi]})^{-1}(\cos|_{[0, \pi]}(x)) = x$$

3. Let $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ then

$$\arcsin(\sin(x)) = \arcsin(\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}(x)) = (\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]})^{-1}(\sin|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}(x)) = x$$

4. As $\cos(\pi) = -1$ and $\pi \in [0, \pi]$ we have $\pi \stackrel{(1)}{=} \arccos(\cos(\pi)) = \arccos(-1)$.
5. As $\cos(\frac{\pi}{2}) = 0$ and $\frac{\pi}{2} \in [0, \pi]$ we have $\frac{\pi}{2} \stackrel{(1)}{=} \arccos(\cos(\frac{\pi}{2})) = \arccos(0)$.
6. As $\cos(0) = 1$ and $0 \in [0, \pi]$ we have $0 \stackrel{(1)}{=} \arccos(\cos(0)) = \arccos(1)$.
7. As $\sin(-\frac{\pi}{2}) = -\sin(\frac{\pi}{2}) = -1$ and $-\frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ we have $-\frac{\pi}{2} \stackrel{(1)}{=} \arcsin(\sin(-\frac{\pi}{2})) = \arcsin(-1)$.
8. As $\sin(0) = 0$ and $0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ we have $0 \stackrel{(1)}{=} \arcsin(\sin(0)) = \arcsin(0)$.
9. As $\sin(\frac{\pi}{2}) = 1$ and $\frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ we have $\frac{\pi}{2} \stackrel{(1)}{=} \arcsin(\sin(\frac{\pi}{2})) = \arcsin(1)$ □

Once we have defined the sinus and cosinus we can define the tangent as follows:

Definition 19.74. The tangent function noted as $\tan$ is defined by

$$\tan:]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R} \text{ where } \tan(x) = \frac{\sin(x)}{\cos(x)}$$

in other words

$$\tan = \frac{\sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}}{\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}}$$

Note 19.75. As by [theorem: 19.69(14)] $0 < \cos(x)$ for every x in $]-\frac{\pi}{2}, \frac{\pi}{2}[$ this function is well defined.

Theorem 19.76. The tangent function has the following properties:

1. $\tan(0) = 0$
2. $\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ $\tan$ has a derivative $\tan'_x$ where $\tan'_x = \frac{1}{\cos^2(x)} > 0$
3. $\tan$ is strictly increasing on $]-\frac{\pi}{2}, \frac{\pi}{2}[$
4. $\tan$ is a continuous function
5. $\tan:]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R}$ is a bijective function

Proof.

1. $\tan(0) = \frac{\sin(0)}{\cos(0)} \stackrel{[\text{theorem: 19.62}]}{=} \frac{0}{1} = 1$
2. Let $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$. Using [theorem: 19.63] and [theorem: 16.36] it follows that $\sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}$ and $\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}$ has a derivative at x with $(\sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[})'_x = \cos(x)$ and $(\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[})'_x = -\sin(x)$. As $\tan = \frac{\sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}}{\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}}$ it follows from [theorem: 16.109] that $\tan$ has a derivative at x and that

$$\begin{aligned} \tan'_x &= \left(\frac{\sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}}{\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}} \right)'_x \\ &= \frac{(\sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[})'_x \cdot \cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}(x) - (\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[})'_x \cdot \sin|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}(x)}{(\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}(x))^2} \\ &= \frac{\cos(x) \cdot \cos(x) + \sin(x) \cdot \sin(x)}{\cos^2(x)} \\ &= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} \\ &\stackrel{[\text{theorem: 19.62}]}{=} \frac{1}{\cos^2(x)} \end{aligned}$$

Further for $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ we have by [theorem: 19.69] that $0 < \cos(x)$ so that $\frac{1}{\cos^2(x)} > 0$.

3. Let $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ then, as $\tan'_x = \frac{1}{\cos^2(x)}$ and $0 < \cos(x)$ [see theorem: 19.69(14)] it follows that $0 < \tan(x)$. Hence using [theorem: 16.211] it follows that $\tan$ is strictly increasing on $]-\frac{\pi}{2}, \frac{\pi}{2}[$,
4. As $\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ $\tan'_x$ exist it follows from [theorem: 16.64] that $\tan$ is differentiable at x , hence, using [theorem: 16.63], $\tan$ is continuous at x .
5. We have for $\tan:]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R}$

injectivity. Let $x, y \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ be such that $\tan(x) = \tan(y)$. Assume that $x \neq y$ then we have either $x < y$ or $y < x$. As $\tan$ is strictly increasing by (3) we would then have that $\tan(x) < \tan(y)$ or $\tan(y) < \tan(x)$ which contradicts $\tan(x) = \tan(y)$. Hence we must have that $x = y$.

surjectivity. Let $y \in \mathbb{R}^+$. Take $z = \frac{y}{\sqrt{2}} + 1$ then $0 < \frac{1}{z} < 1$. As by [theorem: 19.71] $\cos:]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R}$ is a bijection there exist a $x' \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ such that

$$\frac{1}{z} = \cos(x')$$

Further as $\cos(0) = 1$ and $\cos(\frac{\pi}{2}) = 0$ and $\frac{1}{z} \neq 0, 1$ we must have that

$$x_1 \in]0, \frac{\pi}{2}[$$

As by [theorems: 19.63, 19.69] $\forall t \in]0, \frac{\pi}{2}[\subseteq]0, \pi[$ $\sin'_t = \cos(t) > 0$ and $\cos'_t = -\sin(t) < 0$ it follows by [theorem: 16.211] that $\sin$ is strictly increasing and $\cos$ is strictly decreasing. So if we take

$$x_2 = \max\left(\frac{\pi}{4}, x_1\right) \in]0, \frac{\pi}{2}[$$

then we have

$$\frac{1}{\sqrt{2}} \stackrel{[\text{theorem: 19.69}]}{=} \sin\left(\frac{\pi}{4}\right) \leq \sin(x_2) \text{ and } \frac{1}{z} = \cos(x_1) \geq \cos(x_2) \text{ or } z \leq \frac{1}{\cos(x_2)}$$

hence

$$\begin{aligned} \tan(x_2) &= \frac{\sin(x_2)}{\cos(x_2)} \\ &\geq \sqrt{2} \cdot \frac{1}{\cos(x_2)} \\ &\geq \sqrt{2} \cdot z \\ &>_{z=\frac{y}{\sqrt{2}}+1} \sqrt{2} \cdot \frac{y}{\sqrt{2}} \\ &= y \end{aligned}$$

So we have found a $x_2 \in]0, \frac{\pi}{2}[$ such that

$$\tan|_{[0, x_2]}(0) = \tan(0) = 0 < y < \tan(x_2) = \tan|_{[0, x_2]}(x_2)$$

By [theorem: 14.152] and (3) it follows that $\tan|_{[0, x_2]}$ is continuous. So we can use the intermediate value theorem [see theorem: 14.467] to find a $x_3 \in [0, x_2]$ such that $\tan(x_3) = \tan|_{[0, x_2]}(x) = y$. As $\tan(0) \stackrel{(1)}{=} 0 \neq y$ and $x_2 < \frac{\pi}{2}$ we have found a $x_3 \in]0, \frac{\pi}{2}[$ such that $\tan(x_3) = y$. To summarize

$$\forall y \in \mathbb{R}^+ \text{ there exist a } x \in]0, \frac{\pi}{2}[\text{ such that } \tan(x) = y \quad (19.52)$$

Now to prove surjectivity: Let $y \in \mathbb{R}$ then we have either:

$y = 0$. Then by (1) $\tan(0) = 0 = y$.

$0 < y$. Then by [eq: 19.52] there exist a $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ such that $\tan(x) = y$

$y < 0$. Then $-\frac{1}{y} \in \mathbb{R}^+$ and by [eq: 19.52] there exist a $x' \in]0, \frac{\pi}{2}[$ such that $\tan(x') = -\frac{1}{y}$. Take $x = x' - \frac{\pi}{2}$ then $x \in]-\frac{\pi}{2}, 0[\subseteq]-\frac{\pi}{2}, \frac{\pi}{2}[$ and

$$y = -\frac{1}{\tan(x')} = \frac{\cos(x')}{\sin(x')} = \frac{\cos(x + \frac{\pi}{2})}{\sin(x + \frac{\pi}{2})} \stackrel{[\text{theorem: 19.69}]}{=} -\frac{\sin(x)}{\cos(x)} = \tan(x)$$

So in all cases we there exist a $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ such that $y = \tan(x)$ proving surjectivity $\square$

As by the previous theorem $\tan:]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R}$ is a bijection it has a inverse called the arctan.

Definition 19.77. The arctangent function is defined by

$$\arctan: \mathbb{R} \rightarrow]-\frac{\pi}{2}, \frac{\pi}{2}[\text{ is defined by } \arctan = \tan^{-1}$$

which as the inverse of a bijection is itself a bijection.

Theorem 19.78. The arctangent function $\arctan: \mathbb{R} \rightarrow]-\frac{\pi}{2}, \frac{\pi}{2}[$ has the following properties:

1. $\forall x \in \mathbb{R} \tan(\arctan(x)) = x$
2. $\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[\arctan(\tan(x)) = x$
3. $\arctan(0) = 0$
4. $\forall x \in \mathbb{R} \arctan$ has a derivative at x where $\arctan'_x = \frac{1}{1+x^2}$

Proof.

1. Let $x \in \mathbb{R}$ then $\tan(\arctan(x)) = \tan(\tan^{-1}(x)) = x$
2. Let $x \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ then $\arctan(\tan(x)) = \tan^{-1}(\tan(x)) = x$
3. As $\tan(0) \stackrel{[\text{theorem: 19.76}]}{=} 0$ it follows that $\arctan(0) = \tan^{-1}(0) = 0$
4. Let $x \in \mathbb{R}$ and take $y = \arctan(x) \in]-\frac{\pi}{2}, \frac{\pi}{2}[$. We want to apply the inverse function theorem [see theorem: 16.351] so we must prove that

$$\tan':]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R} \text{ by } \tan'(x) = \tan'_x$$

is continuous and that $\tan'_y \neq 0$. Using [theorem: 19.76(2)] it follows that $\tan' = \frac{1}{(\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[})^2}$,

as $\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[}$ is continuous [see theorems: 19.62, 14.152], so by [theorem: 14.172] it follows that $\tan' = \frac{1}{(\cos|_{]-\frac{\pi}{2}, \frac{\pi}{2}[})^2}$ is continuous, further $\tan'_y \stackrel{[\text{theorem: 19.76(2)}]}{=} \frac{1}{\cos^2(y)} > 0$. By the inverse function theorem [see theorem: 16.351(3)] there exist a open $U \subseteq]-\frac{\pi}{2}, \frac{\pi}{2}[$ and $V \subseteq \mathbb{R}$ with $y \in U$ and $\tan(y) \in V$ $\xRightarrow{\tan(y) = \tan(\arctan(x)) = x} x \in V$ such that $\tan|_U: U \rightarrow V$ has a inverse $(\tan|_U)^{-1}: V \rightarrow U$ and

$$\forall z \in V \text{ we have } ((\tan|_U)^{-1})'_{\tan(z)} = \frac{1}{\tan'_z} \quad (19.53)$$

As $\forall v \in V$ we have $(\tan|_U \circ \arctan|_V)(v) = \tan|_U(\arctan|_V(v)) = \tan(\arctan(v)) = v = \text{Id}_V(v)$ and $\forall u \in U$ we have $(\arctan|_V \circ \tan|_U)(u) = \arctan|_V(\tan|_U(u)) = \arctan(\tan(u)) = u = \text{Id}_U(u)$ so that $(\tan|_U)^{-1} = \arctan|_V$. Substituting this in [eq: 19.53] we have

$$\forall z \in V \text{ we have } (\arctan|_V)'_{\tan(z)} = \frac{1}{\tan'_z} \stackrel{[\text{theorem: 19.76}]}{=} \frac{1}{\left(\frac{1}{\cos^2(z)}\right)} = \frac{1}{\left(\frac{\cos^2(z) + \sin^2(z)}{\cos^2(z)}\right)} = \frac{1}{1 + (\tan(z))^2}$$

As $\arctan(x) = y \in V$ we have by the above that $(\arctan|_V)'_{\tan(\arctan(x))} = \frac{1}{1 + (\tan(\arctan(s)))^2}$ so that

$$(\arctan|_V)'_x = \frac{1}{1+x^2}$$

Finally using the locality of the derivative [theorem: 16.36] it follows that $\arctan$ has a derivative at x and $\arctan'_x = \frac{1}{1+x^2}$. $\square$

Chapter 20

Extended real numbers

Measure Theory deals with measuring sets which is the act of assigning to a set a non negative number, the measure of the set, that somehow quantify how big this set is. This process of measuring satisfies countable additivity meaning that the measure of the countable union of sets is the sum of the measures of each set in this union. Hence we have to deal with series of non negative numbers which are not always convergent in $\mathbb{R}$ meaning that the measure of a set would not be always defined as a real number. To solve this problem we extend the set of real numbers by adding two elements, ∞ which will be bigger then every real number and $-\infty$ which will be smaller then every real number.

Definition 20.1. The set of extended real numbers $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$$

where $\infty, -\infty \notin \mathbb{R}$ and $\infty \neq -\infty$. In other words we have that

$$\overline{\mathbb{R}} = \mathbb{R} \sqcup \{\infty\} \sqcup \{-\infty\}$$

Note 20.2. Using [lemma: 10.98] there exist two elements $(\emptyset, 0)$, $(\mathbb{Q}, 0)$ such that $(\emptyset, 0) \notin \mathbb{R}$, $(\mathbb{Q}, 0) \notin \mathbb{R}$ and $(\emptyset, 0) \neq (\mathbb{Q}, 0)$. So we can choose $\infty = (\emptyset, 0)$ and $-\infty = (\mathbb{Q}, 0)$. However we never make use of the exact nature of ∞ , $-\infty$ the only thing that matters is that $\infty, -\infty \notin \mathbb{R}$ and that $\infty \neq -\infty$.

20.1 Order relation in $\overline{\mathbb{R}}$

Definition 20.3. A $x \in \overline{\mathbb{R}}$ is called a finite real number if $x \in \mathbb{R}$ so $\mathbb{R}$ is the set of finite real numbers.

We introduce now a fully order on $\overline{\mathbb{R}}$ based on the order defined on the set of real numbers. To avoid confusion with notation, $\leq$ will be the order relation on $\overline{\mathbb{R}}$ and $\leq_{\mathbb{R}}$ is the order relation on $\mathbb{R}$.

Definition 20.4. The relation $\leq \subseteq \overline{\mathbb{R}} \times \overline{\mathbb{R}}$ is defined by

$$\leq = \leq_{\mathbb{R}} \cup (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$$

where $\leq_{\mathbb{R}} \subseteq \mathbb{R} \times \mathbb{R}$ is the order relation on $\mathbb{R}$.

Notation 20.5. As usual $x \leq y$ is another notation for $(x, y) \in \leq$

Note 20.6. As $\infty \notin \mathbb{R}$, $-\infty \notin \mathbb{R}$ and $-\infty \neq \infty$ we have that

$$\begin{aligned} \leq_{\mathbb{R}} \cap (\overline{\mathbb{R}} \times \{\infty\}) &= \emptyset \\ \leq_{\mathbb{R}} \cap (\{-\infty\} \times \overline{\mathbb{R}}) &= \emptyset \\ (\overline{\mathbb{R}} \times \{\infty\}) \cap (\{-\infty\} \times \overline{\mathbb{R}}) &= \{(-\infty, \infty)\} \end{aligned}$$

Note 20.7. Using the definition we have that $\forall x \in \overline{\mathbb{R}}$ we have that $(x, \infty) \in \overline{\mathbb{R}} \times \{\infty\}$ and $(-\infty, x) \in \{-\infty\} \times \overline{\mathbb{R}}$ so that

$$\forall x \in \overline{\mathbb{R}} \text{ we have } -\infty \leq x \text{ and } x \leq \infty$$

So in essence ∞ is a number that is bigger than all numbers in $\overline{\mathbb{R}}$ and $-\infty$ is a number that is less than every number in $\overline{\mathbb{R}}$.

Note 20.8. If $x, y \in \mathbb{R}$ then if $x \leq y$ it follows that $x \leq_{\mathbb{R}} y$

Proof. As $x, y \in \mathbb{R}$, $\infty, -\infty \notin \mathbb{R}$, $(x, y) \notin (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$ so we must have that $(x, y) \in \leq_{\mathbb{R}}$ or $x \leq_{\mathbb{R}} y$. $\square$

We proof now that $(\overline{\mathbb{R}}, \leq)$ is fully ordered partial ordered set

Theorem 20.9. $(\overline{\mathbb{R}}, \leq)$ is totally ordered.

Proof. First we have to prove that $\leq$ is a partial ordered.

reflectivity. Let $x \in \overline{\mathbb{R}}$ then we have either:

$x = \infty$. Then $(x, x) = (\infty, \infty) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ proving that $x \leq x$.

$x = -\infty$. Then $(x, x) = (-\infty, -\infty) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ proving that $x \leq x$.

$x \in \mathbb{R}$. Then as $\leq_{\mathbb{R}}$ is a partial order it follows that $(x, x) \in \leq_{\mathbb{R}} \subseteq \leq$ proving that $x \leq x$.

anti-symmetry. If $x \leq y \wedge y \leq x$ then $(x, y), (y, x) \in \leq_{\mathbb{R}} \cup (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$ and we have for (x, y) the following cases:

$(x, y) \in \leq_{\mathbb{R}}$. Then as $(y, x) \in \leq_{\mathbb{R}} \cup (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$ we have either:

$(y, x) \in \leq_{\mathbb{R}}$. As $\leq_{\mathbb{R}}$ is a partial order we have that $x = y$.

$(y, x) \in \overline{\mathbb{R}} \times \{\infty\}$. Then $x \in \mathbb{R} \wedge x = \infty$ a contradiction so this cases does not occur.

$(y, x) \in \{-\infty\} \times \overline{\mathbb{R}}$. Then $y \in \mathbb{R} \wedge y = -\infty$ a contradiction so this cases does not occur.

$(x, y) \in \overline{\mathbb{R}} \times \{\infty\}$. Then as $(y, x) \in \leq_{\mathbb{R}} \cup (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$ we have either:

$(y, x) \in \leq_{\mathbb{R}}$. Then $y = \infty \wedge y \in \mathbb{R}$ a contradiction so this cases does not occur.

$(y, x) \in \overline{\mathbb{R}} \times \{\infty\}$. Then $y = \infty \wedge x = \infty$ so that $x = y$.

$(y, x) \in \{-\infty\} \times \overline{\mathbb{R}}$. Then $y = \infty \wedge y = -\infty$ a contradiction so this cases does not occur.

$(x, y) \in \{-\infty\} \times \overline{\mathbb{R}}$. Then as $(y, x) \in \leq_{\mathbb{R}} \cup (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$ we have either:

$(y, x) \in \leq_{\mathbb{R}}$. Then $x = -\infty \wedge x \in \mathbb{R}$ a contradiction so this cases does not occur.

$(y, x) \in \overline{\mathbb{R}} \times \{\infty\}$. Then $x = -\infty \wedge x = \infty$ a contradiction so this cases does not occur.

$(y, x) \in \{-\infty\} \times \overline{\mathbb{R}}$. Then $x = -\infty \wedge y = -\infty$ so that $x = y$.

Hence in all valid cases we have that $x = y$.

transitivity. If $x \leq y \wedge y \leq z$ then $(x, y), (y, z) \in \leq_{\mathbb{R}} \cup (\overline{\mathbb{R}} \times \{\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}})$ and we have for (x, y) either:

$(x, y) \in \leq_{\mathbb{R}}$. Then for (y, z) we have either:

$(y, z) \in \leq_{\mathbb{R}}$. Then $x \leq_{\mathbb{R}} y \wedge y \leq_{\mathbb{R}} z$ and as $\leq_{\mathbb{R}}$ is a partial order it follows that $x \leq_{\mathbb{R}} z$ hence $(x, z) \in \leq$ so that $x \leq z$.

$(y, z) \in \overline{\mathbb{R}} \times \{\infty\}$. As $z = \infty$ we have that $(x, z) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $x \leq z$.

$(y, z) \in \{-\infty\} \times \overline{\mathbb{R}}$. As $(x, y) \in \leq_{\mathbb{R}} \subseteq \mathbb{R} \times \mathbb{R}$ we have that $y \neq -\infty \wedge y = -\infty$ so this case can never occur.

$(x, y) \in \overline{\mathbb{R}} \times \{\infty\}$. Then for (y, z) we have either:

$(y, z) \in \leq_{\mathbb{R}}$. Then as $(y, z) \in \leq_{\mathbb{R}} \subseteq \mathbb{R} \times \mathbb{R}$ we have that $y \neq \infty \wedge y = \infty$ so this case can never occur.

$(y, z) \in \overline{\mathbb{R}} \times \{\infty\}$. As $z = \infty$ we have $(x, z) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $x \leq z$.

$(y, z) \in \{-\infty\} \times \overline{\mathbb{R}}$. As $y = \infty \wedge y = -\infty$ this case never occurs.

$(x, y) \in \{-\infty\} \times \overline{\mathbb{R}}$. Then for (y, z) we have either:

$(y, z) \in \leq_{\mathbb{R}}$. As $x = -\infty$ we have that $(x, z) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ so that $x \leq z$.

$(y, z) \in \overline{\mathbb{R}} \times \{\infty\}$. As $z = \infty$ we have that $(x, z) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $x \leq z$.

$(y, z) \in \{-\infty\} \times \overline{\mathbb{R}}$. As $x = -\infty$ we have that $(x, z) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ so that $x \leq z$.

Hence in all valid cases we have that $x \leq z$.

Next we have to prove that $\leq$ is a total order. So let $x, y \in \overline{\mathbb{R}}$ then the following cases are possible:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then as $\leq_{\mathbb{R}}$ is a total order we have $(x, y) \in \leq_{\mathbb{R}} \subseteq \leq$ so that $x \leq y$.

$x = \infty \wedge y \in \mathbb{R}$. Then $(y, x) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $y \leq x$.

$x = -\infty \wedge y \in \mathbb{R}$. Then $(x, y) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ so that $x \leq y$.

$x \in \mathbb{R} \wedge y = \infty$. Then $(x, y) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $x \leq y$.

$x = \infty \wedge y = \infty$. Then $(x, y) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $x \leq y$.

$x = -\infty \wedge y = \infty$. Then $(x, y) \in \overline{\mathbb{R}} \times \{\infty\} \subseteq \leq$ so that $x \leq y$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $(y, x) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ so that $y \leq x$.

$x = \infty \wedge y = -\infty$. Then $(y, x) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ so that $y \leq x$.

$x = -\infty \wedge y = -\infty$. Then $(y, x) \in \{-\infty\} \times \overline{\mathbb{R}} \subseteq \leq$ so that $y \leq x$. □

TODO check rest

Theorem 20.10. *Let $x \in \mathbb{R}$ then we have $-\infty < x < \infty$*

Proof. As $(x, \infty) \in \mathbb{R} \times \{\infty\} \subseteq \leq$ we have $x \leq \infty$ and $x \neq \infty$ so that $x < \infty$. Further as $(-\infty, x) \in \{-\infty\} \times \mathbb{R}$ we have that $-\infty \leq x$ and $x \neq -\infty$ so that $-\infty < x$. □

Theorem 20.11. *Let $x, y \in \overline{\mathbb{R}}$ with $x < y$ then $x \neq \infty$ and $y \neq -\infty$.*

Proof. Assume that $x = \infty$ then by definition $y \leq x \Rightarrow_{x < y} y < y$ a contradiction so we must have that $x \neq \infty$. Assume that $y = -\infty$ then by definition $y \leq x \Rightarrow_{x < y} y < y$ a contradiction so we must have that $y \neq -\infty$. □

The rational numbers are still dense in $\overline{\mathbb{R}}$ as the following theorem shows.

Theorem 20.12. *Let $x, y \in \overline{\mathbb{R}}$ with $x < y$ then $\exists q \in \mathbb{Q}$ such that $x < q < y$.*

Proof. As $x < y$ we have by the previous theorem [theorem: 20.11] $x \neq \infty \wedge y \neq -\infty$ so we must consider only the following cases for x, y :

$x \in \mathbb{R} \wedge y = \infty$. Then as $x, x+1 \in \mathbb{R}$ and $x < x+1$ we can use [theorem: 10.32] to find a $q \in \mathbb{Q}$ such that $x < q < x+1 < \infty = y$ hence $x < q < y$.

$x = -\infty \wedge y = \infty$. Then for $q = 0 \in \mathbb{Q}$ we have $x < q < y$.

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using [theorem: 10.32] there exist a $q \in \mathbb{Q}$ such that $x < q < y$.

$x = -\infty \wedge y \in \mathbb{R}$. Then $y, y-1 \in \mathbb{R}$ and $y-1 < y$ we can use [theorem: 10.32] to find a $q \in \mathbb{Q}$ such that $x = -\infty < y-1 < q < y$ hence $x < q < y$. $\square$

The primary reason for the extended real is that every non empty set has a supremum and infimum.

Theorem 20.13. *Let $\emptyset \neq A \subseteq \overline{\mathbb{R}}$ then $\sup(A)$ and $\inf(A)$ exist. Further we have:*

1. $\sup(A)$ has the following values for the possible cases of A :

$A = \{-\infty\}$. Then $\sup(A) = -\infty$

$A \neq \{-\infty\}$. Then we have the following possible sub-cases:

$\infty \in A$. Then $\sup(A) = \infty$

$\infty \notin A$. Then we have the following sub-cases:

$\exists u \in \mathbb{R} \vdash \forall x \in A$ we have $x \leq u$. Then $\sup(A) = \sup_{\mathbb{R}}(A \cap \mathbb{R}) \in \mathbb{R}$

$\forall u \in \mathbb{R} \vdash \exists x \in A$ we have $x \not\leq u$. Then $\sup(A) = \infty$

2. $\inf(A)$ has the following values for the possible cases of A :

$A = \{\infty\}$. Then $\inf(A) = \infty$

$A \neq \{\infty\}$. Then we have the following possible sub-cases:

$-\infty \in A$. Then $\inf(A) = -\infty$

$-\infty \notin A$. Then we have the following sub-cases:

$\exists l \in \mathbb{R} \vdash \forall x \in A$ we have $l \leq x$. Then $\inf(A) = \inf_{\mathbb{R}}(A \cap \mathbb{R}) \in \mathbb{R}$

$\forall l \in \mathbb{R} \vdash \exists x \in A$ we have $l \not\leq x$. Then $\inf(A) = -\infty$

Where $\sup(A)$ is the supremum of A using $\leq$, $\sup_{\mathbb{R}}(A \cap \mathbb{R})$ is the supremum of $A \cap \mathbb{R}$ using $\leq_{\mathbb{R}}$, $\inf(A)$ is the infimum of A using $\leq$ and $\inf_{\mathbb{R}}(A \cap \mathbb{R})$ is the infimum of $A \cap \mathbb{R}$ using $\leq_{\mathbb{R}0}$.

Proof. First we proof that $\sup(A)$ exist. For A we have either:

$A = \{-\infty\}$. As $-\infty \leq -\infty$ $-\infty$ is a upper bound of A . If $u \in \overline{\mathbb{R}}$ is another upper bound then, as $-\infty \leq u$ it follows that $-\infty$ is the lowest upper bound of A hence $\sup(A) = -\infty$ and

$$\sup(\{-\infty\}) = -\infty$$

$A \neq \{-\infty\}$. Then we have either:

$\infty \in A$. As $\forall x \in A$ we have $x \leq \infty$ so ∞ is clearly a upper bound for A . If u is a another upper bound for A then, as $\infty \in A$, we must have that $\infty \leq u$ so ∞ is the lowest upper bound of A . Hence $\sup(A) = \infty$ or

$$\text{If } \infty \in A \text{ then } \sup(A) = \infty$$

$\infty \notin A$. As $A \neq \emptyset$ and $A \neq \{-\infty\}$ there must be a $z \in A$ with $z \neq -\infty$, as $\infty \notin A$ it follows that $z \in \mathbb{R}$. Hence we conclude that

$$A \cap \mathbb{R} \neq \emptyset$$

Consider now for A the following cases:

$\exists u \in \mathbb{R} \vdash \forall x \in A$ we have $x \leq u$. Then $\forall x \in A \cap \mathbb{R}$ we have $x \leq u$ or, as $x, u \in \mathbb{R}$ that $x \leq_{\mathbb{R}} u$, so $\emptyset \neq A \cap \mathbb{R}$ is bounded above by u . As $\langle \mathbb{R}, \leq_{\mathbb{R}} \rangle$ is conditional complete [see theorem: 10.18] $A \cap \mathbb{R}$ has a supremum $S = \sup_{\mathbb{R}}(A \cap \mathbb{R}) \in \mathbb{R}$. Let $x \in A$, then, given that $\infty \notin A$, we have either:

$x \in A \cap \mathbb{R}$. Then $x \leq_{\mathbb{R}} S$ so that $x \leq S$

$x = -\infty$. Then $-\infty \leq S$

So in all cases we have $x \leq S$ which proves that

$$S \text{ is a upper bound of } A [\text{using } \leq]$$

Let U be another upper bound of A [using $\leq$] then we have either:

$U \in \mathbb{R}$. Then $\forall x \in A \cap \mathbb{R} \subseteq A$ we have $x \leq U$, hence, as $x, U \in \mathbb{R}$, $x \leq_{\mathbb{R}} U$ so that U is another upper bound of $A \cap \mathbb{R}$ using $\leq_{\mathbb{R}}$. So we must have that $S \leq U$.

$U = \infty$. Then trivially $S \leq U$.

$U = -\infty$. Then $\forall x \in A$ we have $-\infty \leq x \leq -\infty \Rightarrow x = -\infty$ so that $A \subseteq \{-\infty\}$ or $A \cap \mathbb{R} \subseteq \{-\infty\} \cap \mathbb{R} = \emptyset$ contradicting the fact that $A \cap \mathbb{R} = \emptyset$. So this case will never occur.

As in all valid cases $S \leq U$ it follows that S is the lowest upper bound of A or S is the supremum of A . So

$$\sup(A) = \sup_{\mathbb{R}}(A \cap \mathbb{R})$$

$\forall u \in \mathbb{R} \vdash \exists x \in A$ with $x \not\leq u$. As $\forall x \in \overline{\mathbb{R}}$ we have that $x \leq \infty$ it follows that ∞ is a upper bound of A [using $\leq$]. Let u be another upper bound of A . Assume that $\infty \not\leq u$ then by [theorem: 3.39] $u < \infty$ or $u \neq \infty$ so that u is either:

$u = -\infty$. Then $\forall x \in A$ we have $-\infty \leq x \leq u = -\infty \Rightarrow x = -\infty$ so that $A \subseteq \{-\infty\}$ or $A \cap \mathbb{R} \subseteq \{-\infty\} \cap \mathbb{R} = \emptyset$ contradicting the fact that $A \cap \mathbb{R} = \emptyset$.

$u \in \mathbb{R}$. Then $\exists x \in A$ such that $x \not\leq u$ contradicting the fact that u is a upper bound.

As in all cases we have a contradiction we must have that the assumption is wrong and that $\infty \leq u$. Hence ∞ is the lowest upper bound of A [using $\leq$] or ∞ is the supremum of A , so

$$\sup(A) = \infty$$

So in all possible cases of A we have found that $\sup(A)$ exists. Next we prove that $\inf(A)$ exist, for A we have either:

$A = \{\infty\}$. As $\infty \leq \infty$ ∞ is a lower bound of A . If $l \in \overline{\mathbb{R}}$ is another lower bound then, as $l \leq \infty$ it follows that ∞ is the greatest upper bound of A hence $\inf(A) = \infty$ and

$$\inf(\{\infty\}) = \infty$$

$A \neq \{\infty\}$. Then we have either:

$-\infty \in A$. As $\forall x \in A$ we have $-\infty \leq x$ so $-\infty$ is clearly a lower bound for A . If l is a another lower bound for A then, as $-\infty \in A$, we must have that $u \leq -\infty$ so $-\infty$ is the greatest lower bound of A . Hence $\inf(A) = -\infty$ or

$$\text{If } -\infty \in A \text{ then } \inf(A) = -\infty$$

$-\infty \notin A$. As $A \neq \emptyset$ and $A \neq \{\infty\}$ there must be a $z \in A$ with $z \neq \infty$, as $-\infty \notin A$ it follows that $z \in \mathbb{R}$. Hence we conclude that

$$A \cap \mathbb{R} \neq \emptyset$$

Consider now for A the following cases:

$\exists l \in \mathbb{R} \vdash \forall x \in A$ we have $l \leq x$. Then $\forall x \in A \cap \mathbb{R}$ we have $l \leq x$ or, as $x, l \in \mathbb{R}$ that $l \leq_{\mathbb{R}} x$, so $\emptyset \neq A \cap \mathbb{R}$ is bounded below by l . As $(\mathbb{R}, \leq_{\mathbb{R}})$ is conditional complete [see theorem: 10.18] $A \cap \mathbb{R}$ has a infimum $I = \inf_{\mathbb{R}}(A \cap \mathbb{R}) \in \mathbb{R}$. Let $x \in A$, then, given that $-\infty \notin A$, we have either:

$x \in A \cap \mathbb{R}$. Then $I \leq_{\mathbb{R}} x$ so that $I \leq x$

$x = \infty$. Then $I \leq x$

So in all cases we have $I \leq x$ which proves that

$$I \text{ is a lower bound of } A \text{ [using } \leq]$$

Let L be another lower bound of A [using $\leq$] then we have either:

$L \in \mathbb{R}$. Then $\forall x \in A \cap \mathbb{R} \subseteq A$ we have $L \leq x$, hence, as $x, L \in \mathbb{R}$, $L \leq_{\mathbb{R}} x$ so that L is another lower bound of $A \cap \mathbb{R}$ using $\leq_{\mathbb{R}}$. So we must have that $L \leq I$.

$L = \infty$. Then $\forall x \in A$ we have $\infty \leq x \leq \infty \Rightarrow x = \infty$ so that $A \subseteq \{\infty\}$ or $A \cap \mathbb{R} \subseteq \{\infty\} \cap \mathbb{R} = \emptyset$ contradicting the fact that $A \cap \mathbb{R} = \emptyset$. So this case will never occur.

$L = -\infty$. Then $L \leq I$

As in all valid cases $L \leq I$ it follows that I is the greatest lower bound of A or I is the infimum of A . So

$$\inf(A) = \inf_{\mathbb{R}}(A \cap \mathbb{R})$$

$\forall l \in \mathbb{R} \vdash \exists x \in A$ with $l \not\leq x$. As $\forall x \in \overline{\mathbb{R}}$ we have that $-\infty \leq x$ it follows that $-\infty$ is a lower bound of A [using $\leq$]. Let l be another lower bound of A . Assume that $l \not\leq -\infty$ then by [theorem: 3.39] $-\infty < l$ or $l \neq -\infty$ so that l is either:

$l = \infty$. Then $\forall x \in A$ we have $\infty = l \leq x \leq \infty \Rightarrow x = \infty$ so that $A \subseteq \{\infty\}$ or $A \cap \mathbb{R} \subseteq \{\infty\} \cap \mathbb{R} = \emptyset$ contradicting the fact that $A \cap \mathbb{R} = \emptyset$.

$l \in \mathbb{R}$. Then $\exists x \in A$ such that $l \not\leq x$ contradicting the fact that l is a lower bound.

As in all cases we have a contradiction we must have that the assumption is wrong and that $l \leq -\infty$. Hence $-\infty$ is the greatest lower bound of A [using $\leq$] or $-\infty$ is the infimum of A . So

$$\inf(A) = -\infty$$

So in all possible cases of A we have proved that $\inf(A)$ exist. □

Corollary 20.14. *If $\emptyset \neq A \subseteq \mathbb{R}$ then $\sup(A) \neq -\infty$ and $\inf(A) \neq \infty$*

Proof. If $\sup(A) = -\infty$ then $\forall x \in A$ we have $-\infty \leq x \leq -\infty$ so that $x = -\infty$, hence $A \subseteq \{-\infty\}$ so that we get $\emptyset \neq A = A \cap \mathbb{R} \subseteq \{-\infty\} \cap \mathbb{R} = \emptyset$ a contradiction. Hence

$$\sup(A) \neq -\infty$$

Likewise if $\inf(A) = \infty$ then $\forall x \in A$ we have $\infty \leq x \leq \infty$ so that $x = \infty$, hence $A \subseteq \{\infty\}$ so that we get $\emptyset \neq A = A \cap \mathbb{R} \subseteq \{\infty\} \cap \mathbb{R} = \emptyset$ a contradiction. Hence

$$\inf(A) \neq \infty$$
□

The next theorem shows the necessary and sufficient constraint for the supremum and infimum to be not finite.

Corollary 20.15. *Let $\emptyset \neq A \subseteq \overline{\mathbb{R}}$ then*

1. $\sup(A) = \infty \Leftrightarrow \infty \in A$ or $\forall u \in \mathbb{R}$ there exist a $x \in A \cap \mathbb{R}$ with $x \not\leq u$ or $u < x$.
2. $\inf(A) = -\infty \Leftrightarrow -\infty \in A$ or $\forall l \in \mathbb{R}$ there exist a $x \in A \cap \mathbb{R}$ with $l \not\leq x$ or $x < l$.
3. $\sup(A) = -\infty \Leftrightarrow A = \{-\infty\}$
4. $\inf(A) = \infty \Leftrightarrow A = \{\infty\}$

Proof.

1. From [theorem: 20.13 (1)] it follows that the only possible cases where $\sup(A) = \infty$ are that either $\infty \in A$ or $\forall u \in \mathbb{R}$ there exist a $x \in A \cap \mathbb{R}$ with $x \not\leq u$.

2. From [theorem: 20.13 (2)] it follows that the only possible cases where $\inf(A) = -\infty$ are that either $-\infty \in A$ or $\forall l \in \mathbb{R}$ there exist a $x \in A \cap \mathbb{R}$ with $l \not\leq x$.
3. From [theorem: 20.13 (1)] it follows that the only possible cases where $\sup(A) = -\infty$ is the case where $A = \{-\infty\}$.
4. From [theorem: 20.13 (2)] it follows that the only possible cases where $\inf(A) = \infty$ is the case where $A = \{\infty\}$. $\square$

20.2 Arithmetic operations on $\overline{\mathbb{R}}$

The sum operator on $\overline{\mathbb{R}}$ is an extension of the sum on $\mathbb{R}$ such that most inequalities involving sums are still satisfied.

Definition 20.16. *The sum operator is defined by*

$$(+): (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(\infty, -\infty), (-\infty, \infty)\} \rightarrow \overline{\mathbb{R}} \text{ by } (+)(x, y) \underset{\text{notation}}{=} x + y$$

where

$$x + y = \begin{cases} \infty & \text{if } x = \infty \wedge y \in \overline{\mathbb{R}} \setminus \{-\infty\} \\ \infty & \text{if } y = \infty \wedge x \in \overline{\mathbb{R}} \setminus \{-\infty\} \\ -\infty & \text{if } x = -\infty \wedge y \in \overline{\mathbb{R}} \setminus \{\infty\} \\ -\infty & \text{if } y = -\infty \wedge x \in \overline{\mathbb{R}} \setminus \{\infty\} \\ x +_{\mathbb{R}} y & \text{if } x, y \in \mathbb{R} \text{ where } +_{\mathbb{R}} \text{ is the sum defined for the real numbers} \end{cases}$$

Note 20.17. $\infty + (-\infty)$ and $-\infty + \infty$ are not defined at all.

Note 20.18. Let $x \in \overline{\mathbb{R}}$ then for $0 \in \mathbb{R} \subseteq \overline{\mathbb{R}}$ we have that $0 + x = x = x + 0$

Proof. For $x \in \overline{\mathbb{R}}$ we have either:

$x \in \mathbb{R}$. Then $x + 0 = x +_{\mathbb{R}} 0 = x$ and $0 + x = 0 +_{\mathbb{R}} x = x$.

$x = -\infty$. Then $x + 0 = -\infty + 0 = -\infty = x$ and $0 + x = 0 + (-\infty) = -\infty = x$.

$x = \infty$. Then $x + 0 = \infty + 0 = \infty = x$ and $0 + x = 0 + \infty = \infty = x$. $\square$

Definition 20.19. *The multiplication operator $\cdot$ is defined by*

$$(\cdot): \overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}} \text{ by } (\cdot)(x, y) \underset{\text{notation}}{=} x \cdot y$$

where for $(x, y) \in \overline{\mathbb{R}} \times \overline{\mathbb{R}}$ we define $x \cdot y$ as follows:

$x = 0 \vee y = 0$. Then $x \cdot y = 0$

$x < 0 \wedge y = \infty$. Then $x \cdot y = -\infty$

$0 < x \wedge y = \infty$. Then $x \cdot y = \infty$

$x < 0 \wedge y = -\infty$. Then $x \cdot y = \infty$

$0 < x \wedge y = -\infty$. Then $x \cdot y = -\infty$

$x = -\infty \wedge y \in]-\infty, 0[$. Then $x \cdot y = \infty$

$x = \infty \wedge y \in]-\infty, 0[$. Then $x \cdot y = -\infty$

$x \in \mathbb{R} \setminus \{0\} \wedge y \in]-\infty, 0[$. Then $x \cdot y = x \cdot_{\mathbb{R}} y$

$x = -\infty \wedge y \in]0, \infty[$. Then $x \cdot y = -\infty$

$x = \infty \wedge y \in]0, \infty[$. Then $x \cdot y = \infty$

$x \in \mathbb{R} \setminus \{0\} \wedge y \in]0, \infty[$. Then $x \cdot y = x \cdot_{\mathbb{R}} y$

where $x \cdot_{\mathbb{R}} y$ is the product in $\mathbb{R}$.

Note 20.20. In contrast with the sum the product in $\overline{\mathbb{R}}$ is defined $\forall x, y \in \overline{\mathbb{R}}$.

Next we define the inverse operator.

Definition 20.21. The inverse operator $(\cdot)^{-1}$ is defined by

$$(\cdot)^{-1}: \overline{\mathbb{R}} \setminus \{0\} \rightarrow \overline{\mathbb{R}} \text{ by } (\cdot)^{-1} = \begin{cases} 0 & \text{if } x = -\infty \\ 0 & \text{if } x = \infty \\ x^{-1} & \text{if } x \in \mathbb{R} \text{ where } x^{-1} \text{ is the inverse defined in } \mathbb{R} \end{cases}$$

Notation 20.22. As usually we note $(\cdot)^{-1}(x)$ as x^{-1} , $1/x$ or $\frac{1}{x}$

Note 20.23. In contrast with real numbers where for $x \in \mathbb{R} \setminus \{0\}$ we have $x \cdot \frac{1}{x} = 1 = \frac{1}{x} \cdot x$ this is not the case for $x = -\infty$ and $x = \infty$ because we have $\infty \cdot \frac{1}{\infty} = \infty \cdot 0 = 0$ and $(-\infty) \cdot \frac{1}{-\infty} = (-\infty) \cdot 0 = 0$.

Definition 20.24. The negate operator $-$ is defined by

$$(-): \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}} \text{ where } (-)(x) \underset{\text{notation}}{=} -x \underset{\text{defined}}{=} (-1) \cdot x$$

So as $(-1) \in]0, \infty[$ we have that

$$-x = \begin{cases} \infty & \text{if } x = -\infty \\ -\infty & \text{if } x = \infty \\ (-1) \cdot_{\mathbb{R}} x & \text{if } x \in \mathbb{R} \end{cases}$$

As the sum is not defined for every $x, y \in \overline{\mathbb{R}}$ $\langle \overline{\mathbb{R}}, + \rangle$ is not a group however if we restrict ourselves to non negative numbers then we have a Abelian semi group which allows us to work with generalized sums [see section: 11.1].

Definition 20.25. The set of positive extended real numbers noted as $\overline{\mathbb{R}}^+$ and the set of non negative extended real numbers noted as $\overline{\mathbb{R}}_0^+$ are defined by

$$\overline{\mathbb{R}}^+ =]0, \infty] = \{x \in \overline{\mathbb{R}} | 0 < x\}$$

and

$$\overline{\mathbb{R}}_0^+ = [0, \infty] = \{x \in \overline{\mathbb{R}} | 0 \leq x\}$$

Theorem 20.26. $\langle \overline{\mathbb{R}}_0^+, + \rangle$ is a Abelian semi-group with neutral element $0 \in \mathbb{R}$ where $+$ is the restriction to $\overline{\mathbb{R}}_0^+$ of the sum defined in [definition: 20.16].

Proof. First for $x, y \in \overline{\mathbb{R}}_0^+$ we have $0 \leq x \wedge 0 \leq y$ so that $0 \leq y \underset{[\text{note: 20.18}]}{=} 0 + y \underset{[\text{theorem: 20.29}]}{\leq} x + y$ proving that $x + y \in \overline{\mathbb{R}}_0^+$. Hence

$$+: \overline{\mathbb{R}}_0^+ \times \overline{\mathbb{R}}_0^+ \rightarrow \overline{\mathbb{R}}_0^+ \text{ defined by } +(x, y) \in x + y$$

is a well defined function. Next we need to prove the Abelian semi-group axioms:

neutral element. This was proved in [note: 20.18].

associativity. Let $x, y, z \in \overline{\mathbb{R}}_0^+$ then we have as $-\infty \notin \overline{\mathbb{R}}_0^+$ [because $-\infty < 0$] that

$$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}. \text{ Then } x + (y + z) = x +_{\mathbb{R}} (y +_{\mathbb{R}} z) = (x +_{\mathbb{R}} y) +_{\mathbb{R}} z = (x + y) + z.$$

$$x = \infty \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}. \text{ Then}$$

$$x + (y + z) = \infty + (y + z) = \infty = \infty + z = (x + \infty) + z = (x + y) + z$$

$$x \in \mathbb{R} \wedge y = \infty \wedge z \in \mathbb{R}. \text{ Then}$$

$$x + (y + y) = x + (\infty + z) = x + \infty = \infty + z = (x + \infty) + z = (x + y) + z$$

$x = \infty \wedge y = \infty \wedge z \in \mathbb{R}$. Then

$$x + (y + z) = \infty + (y + z) = \infty = \infty + z = (x + y) + z$$

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z = \infty$. Then

$$x + (y + z) = x + (y + \infty) = x + \infty = \infty = (x + y) + \infty = (x + y) + z$$

$x = \infty \wedge y \in \mathbb{R} \wedge z = \infty$. Then

$$x + (y + z) = \infty + (y + z) = \infty = (x + y) + \infty = (x + y) + z$$

$x \in \mathbb{R} \wedge y = \infty \wedge z = \infty$. Then

$$x + (y + z) = x + (\infty + \infty) = x + \infty = \infty = (x + y) + \infty = (x + y) + z$$

$x = \infty \wedge y = \infty \wedge z = \infty$. Then

$$x + (y + z) = x + (\infty + \infty) = x + \infty = \infty = (x + y) + \infty = (x + y) + z$$

So in all cases we have $x + (y + z) = (x + y) + z$.

commutativity. Let $x, y \in \overline{\mathbb{R}}_0^+$ then we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then $x + y = x +_{\mathbb{R}} y = y +_{\mathbb{R}} x = y + x$.

$x = \infty \wedge y \in \mathbb{R}$. Then $x + y = \infty + y = \infty = y + \infty = y + x$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x + y = x + \infty = \infty = \infty + x = y + x$.

$x = \infty \wedge y = \infty$. Then $x + y = \infty + \infty = y + x$.

So in all valid cases we have $x + y = y + x$. □

We have to be very careful if we combine sums and inequalities on $\overline{\mathbb{R}}$ the reasoning that works in $\mathbb{R}$ does not work always in $\overline{\mathbb{R}}$. For example if $x = 10, y = 4$ and $z = \infty$ then $x + z = y + z = \infty$ but $x \neq y$, also we have that $x + z \leq y + z$ but $x \not\leq y$. The following theorems shows what is possible.

Theorem 20.27. Let $x, y \in \overline{\mathbb{R}}$ be such that $0 \leq y$ and $(x, -y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ [so that $x - y$ is well defined] then $x - y \leq x$

Proof. As $(x, -y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ and $0 \leq y$ we have for x either:

$x = -\infty$. Then $-y \neq \infty \Rightarrow y \neq -\infty$ so for y we have either:

$y = \infty$. Then $x - y = -\infty + (-\infty) = -\infty = x$ so that $x - y \leq x$.

$y \in \mathbb{R}$. Then $x - y = -\infty + (-y) = -\infty = x$ so that $x - y \leq x$.

$x = \infty$. Then $-y \neq -\infty \Rightarrow y \neq \infty$ so for y we have either:

$y = -\infty$. Then $x - y = \infty + \infty = \infty = x$ so that $x - y \leq x$.

$y \in \mathbb{R}$. Then $x - y = \infty - y = \infty = x$ so that $x - y \leq x$.

$x \in \mathbb{R}$. Then as $0 \leq y$ we have the following possible cases for y :

$y = \infty$. Then $x - y = x + (-\infty) = -\infty \leq x$ so that $x - y \leq x$.

$y \in \mathbb{R}$. Then using the properties of the real numnners we have $x - y \leq x$. □

Proof. As either x or y are finite we have to consider two cases:

$x \in \mathbb{R}$. then for y we have either:

$y \in \mathbb{R}$. then by the properties of the real numbers we have $x - y \leq x$

$y = \infty$. then $x - y = x + (-\infty) = -\infty < x$ so that $x - y \leq x$

$y \in \mathbb{R}$. then for x we have either:

$x \in \mathbb{R}$. then by the properties of the real numbers we have $x - y \leq x$

$x = \infty$. then $x - y = x + (-y) = \infty \leq \infty = x$ giving $x - y \leq x$ □

Theorem 20.28. *If $x, y \in \overline{\mathbb{R}}$ and $z \in \mathbb{R}$ [so that $x + z$ and $y + z$ are well defined] then we have*

1. *if $x + z = y + z$ we have $x = y$*
2. *if $x + z \leq y + z$ we have $x \leq y$*
3. *if $x + z < y + z$ we have $x < y$*
4. *if $x + z = y$ we have $x = y - z \stackrel{\text{def}}{=} y + (-z)$ [where $y - z$ is well defined]*
5. *if $x + z \leq y$ we have $x \leq y - z \stackrel{\text{def}}{=} y + (-z)$ [where $y - z$ is well defined]*
6. *if $x + z < y$ we have $x < y - z \stackrel{\text{def}}{=} y + (-z)$ [where $y - z$ is well defined]*
7. *if $x \leq y + z$ we have $x + (-z) \leq y$*
8. *if $x < y + z$ we have $x + (-z) < y$*

Proof.

1. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x = y$.
 $x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x + z = y + z \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.
 $x = -\infty \wedge y \in \mathbb{R}$. Then $-\infty = x + z = y + z \in \mathbb{R}$ which as $-\infty \notin \mathbb{R}$ is not a valid case.
 $x \in \mathbb{R} \wedge y = \infty$. Then $\infty = y + z = x + z \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.
 $x = \infty \wedge y = \infty$. Then $x = y$.
 $x = -\infty \wedge y = \infty$. Then $-\infty = x + z = y + z = \infty$ which as $-\infty \neq \infty$ is not a valid case.
 $x \in \mathbb{R} \wedge y = -\infty$. Then $-\infty = y + z = x + z \in \mathbb{R}$ which as $-\infty \notin \mathbb{R}$ is not a valid case.
 $x = \infty \wedge y = -\infty$. Then $\infty = x + z = y + z = -\infty$ which as $-\infty \neq \infty$ is not a valid case.
 $x = -\infty \wedge y = -\infty$. Then $x = y$.

So in all valid cases we have $x = y$.

2. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x \leq y$.
 $x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x + z \leq y + z \in \mathbb{R}$ which as $y + z < \infty$ proves that $\infty = x + y \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.
 $x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty \leq y$.
 $x \in \mathbb{R} \wedge y = \infty$. Then $x \leq \infty = y$,
 $x = \infty \wedge y = \infty$. Then $x = \infty = y$ so that $x \leq y$.
 $x = -\infty \wedge y = \infty$. Then $x = -\infty \leq \infty = y$.
 $x \in \mathbb{R} \wedge y = -\infty$. As $x + z \in \mathbb{R}$ we have $-\infty \leq x + z \leq y + z \leq -\infty$ so that $-\infty = x + z \in \mathbb{R}$ which as $-\infty \notin \mathbb{R}$ is not a valid case.
 $x = \infty \wedge y = -\infty$. As $\infty = x + z \leq y + z = -\infty$ and $-\infty \leq \infty$ we would have $-\infty = \infty$ so this is not a valid case.
 $x = -\infty \wedge y = -\infty$. Then $x = -\infty = y$ so that $x \leq y$.

So in all valid cases we have $x \leq y$.

3. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x \leq y$.
 $x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x + z < y + z \in \mathbb{R}$ which as $y + z < \infty$ proves that $\infty = x + y \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty < y$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x < \infty = y$.

$x = \infty \wedge y = \infty$. Then $\infty = x + z < y + z = \infty$ so this case does not apply.

$x = -\infty \wedge y = \infty$. Then $x = -\infty < \infty = y$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $x + z \in \mathbb{R}$ so that $-\infty < x + z < y + z = -\infty$ so this case does not apply.

$x = \infty \wedge y = -\infty$. Then $\infty = x + z < y + z = -\infty$ so this case does not apply.

$x = -\infty \wedge y = -\infty$. Then $-\infty = x + z < y + z = -\infty$ so this case does not apply.

So in all valid cases we have $x < y$.

4. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x = y - z$.

$x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x + z = y \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.

$x = -\infty \wedge y \in \mathbb{R}$. Then $-\infty = x + z = y \in \mathbb{R}$ which as $-\infty \notin \mathbb{R}$ is not a valid case.

$x \in \mathbb{R} \wedge y = \infty$. Then $\infty = y = x + z \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.

$x = \infty \wedge y = \infty$. Then $y - z = \infty = x$.

$x = -\infty \wedge y = \infty$. Then $-\infty = x + z = y = \infty$ which as $-\infty \neq \infty$ is not a valid case.

$x \in \mathbb{R} \wedge y = -\infty$. Then $-\infty = y = x + z \in \mathbb{R}$ which as $-\infty \notin \mathbb{R}$ is not a valid case.

$x = \infty \wedge y = -\infty$. Then $\infty = x + z = y = -\infty$ which as $-\infty \neq \infty$ is not a valid case.

$x = -\infty \wedge y = -\infty$. Then $y - z = -\infty = x$ so that $x = y - z$.

So in all valid cases we have $x = y - z$.

5. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x \leq y - z$.

$x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x + z \leq y \leq \infty$ [as $y \in \mathbb{R}$] so $\infty = y \in \mathbb{R}$ which as $\infty \notin \mathbb{R}$ is not a valid case.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty < y - z$ so that $x \leq y - z$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x \leq \infty = y - z$.

$x = \infty \wedge y = \infty$. Then $y - z = \infty = x$ so that $x \leq y - z$.

$x = -\infty \wedge y = \infty$. Then $x = -\infty < \infty = y - z$ so that $x \leq y - z$.

$x \in \mathbb{R} \wedge y = -\infty$. As $x + z \in \mathbb{R}$ we have $-\infty < x + z \leq y = -\infty$ so this is not a valid case.

$x = \infty \wedge y = -\infty$. Then $\infty = x + z \leq y = -\infty < \infty$ so this is not a valid case.

$x = -\infty \wedge y = -\infty$. Then $x = -\infty = y - z$ so that $x \leq y - z$.

So in all valid cases we have $x \leq y - z$.

6. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x < y - z$.

$x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x + z < y \leq \infty$ [as $y \in \mathbb{R}$] so this case is not valid.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty < y - z$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x < \infty = y - z$ so that $x < y - z$.

$x = \infty \wedge y = \infty$. Then $\infty = x + z < y = \infty$ so this case is not valid.

$x = -\infty \wedge y = \infty$. Then $x = -\infty < \infty = y - z$ so that $x < y - z$.

$x \in \mathbb{R} \wedge y = -\infty$. As $x \in \mathbb{R}$ we have $-\infty < x + z < y = -\infty$ so this case is not valid.

$x = \infty \wedge y = -\infty$. Then $-\infty < \infty = x + z < y = -\infty$ so this case is not valid.

$x = -\infty \wedge y = -\infty$. Then $-\infty = x + z < y = -\infty$ so this case is not valid.

So in all valid cases we have $x < y - z$.

7. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x - z \leq y$.

$x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x \leq y + z < \infty$ [as $x + z \in \mathbb{R}$] so this case is not valid.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x - z = -\infty < y$ so that $x - z \leq y$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x - z < \infty = y$.

$x = \infty \wedge y = \infty$. Then $x - z = \infty = y$ so that $x - z \leq y$.

$x = -\infty \wedge y = \infty$. Then $x - z = -\infty < \infty = y$ so that $x - z \leq y$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $-\infty < x \leq y + z = -\infty$ so this case is not valid.

$x = \infty \wedge y = -\infty$. Then $-\infty < \infty = x \leq y + z = -\infty$ so this case is not valid.

$x = -\infty \wedge y = -\infty$. Then $-\infty = x \leq y + z = -\infty$ so this case is not valid.

So in all valid cases we have $x - z \leq y$.

8. For x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of the real numbers we have $x - z < y$.

$x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x < y + z < \infty$ [as $x + z \in \mathbb{R}$] so this case is not valid.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x - z = -\infty < y$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x - z < \infty$.

$x = \infty \wedge y = \infty$. Then $\infty = x < y + z = \infty$ so this case is not valid.

$x = -\infty \wedge y = \infty$. Then $x - z = -\infty < \infty = y$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $x < \infty < y + z = -\infty < \infty$ so this case is not valid.

$x = \infty \wedge y = -\infty$. Then $-\infty < \infty = x < y + z = -\infty$ so this case is not valid.

$x = -\infty \wedge y = -\infty$. Then $-\infty = x < y + z = -\infty$ so this case is not valid.

So in all valid cases we have $x - z < y$. □

Theorem 20.29. *Let*

$$(x, z) \in \overline{\mathbb{R}} \times \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

and

$$(y, z) \in \overline{\mathbb{R}} \times \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

[so that $x + z$ and $y + z$ are well defined] then we have:

1. If $x \leq y$ then $x + z \leq y + z$.
2. If $0 \leq x$ then $z \leq z + x$.
3. If $x < y$ and $z \in \mathbb{R}$ then $x + z < y + z$.

Note 20.30. The condition $z \in \mathbb{R}$ in (3) is necessary, for example if $x = 2$, $y = 3$ and $z = \infty$ then $x < y$ but $x + z = \infty = y + z$ so that $x + z \not< y + z$.

Proof.

1. We have the following cases to check for x, y, z :

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. Then using the properties of $\mathbb{R}$ we have $x + z \leq y + z$.

$x = -\infty \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. Then $x + y = -\infty < y + z$ [as $y + z \in \mathbb{R}$].

$x = \infty \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. Then $-\infty < \infty = x \leq y < \infty$ so this is not a valid case.

$x \in \mathbb{R} \wedge y = -\infty \wedge z \in \mathbb{R}$. Then $-\infty < x \leq y = -\infty$ so this is not a valid case.

$x \in -\infty \wedge y = -\infty \wedge z \in \mathbb{R}$. Then $x + z = -\infty = y + z$ so that $x + z \leq y + z$

$x \in \infty \wedge y = -\infty \wedge z \in \mathbb{R}$. Then $\infty = x \leq y = -\infty < \infty$ so this is not a valid case.

$x \in \mathbb{R} \wedge y = \infty \wedge z \in \mathbb{R}$. Then as $x + z \in \mathbb{R}$ we have $x + z < \infty = y + z$.

$x = -\infty \wedge y = \infty \wedge z \in \mathbb{R}$. Then $x + z = -\infty < \infty = y + z$,

$x = \infty \wedge y = \infty \wedge z \in \mathbb{R}$. Then $x + z = \infty = y + z$ so that $x + y \leq y + z$.

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z = -\infty$. Then $x + z = -\infty = y + z$ so that

$x = -\infty \wedge y \in \mathbb{R} \wedge z = -\infty$. Then $x + z = -\infty = y + z$ so that $x + y \leq y + z$.

$x = \infty \wedge y \in \mathbb{R} \wedge z = -\infty$. This cases is excluded because

$$(x, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x \in \mathbb{R} \wedge y = -\infty \wedge z = -\infty$. Then $-\infty < x \leq y = -\infty$ so this is not a valid case.

$x = -\infty \wedge y = -\infty \wedge z = -\infty$. Then $x + z = -\infty = y + z$ so that $x + z \leq y + z$.

$x = \infty \wedge y = -\infty \wedge z = -\infty$. This cases is excluded because

$$(x, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x \in \mathbb{R} \wedge y = \infty \wedge z = -\infty$. This cases is excluded because

$$(y, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x = -\infty \wedge y = \infty \wedge z = -\infty$. This cases is excluded because

$$(y, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x = \infty \wedge y = \infty \wedge z = -\infty$. This cases is excluded because

$$(x, z), (y, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z = \infty$. Then $x + z = \infty = y + z$ so that $x + z \leq y + z$.

$x = -\infty \wedge y \in \mathbb{R} \wedge z = \infty$. This cases is excluded because

$$(x, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x = \infty \wedge y \in \mathbb{R} \wedge z = \infty$. Then $\infty = x < y < \infty$ so this is not a valid case.

$x \in \mathbb{R} \wedge y = -\infty \wedge z = \infty$. This cases is excluded because

$$(y, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x = -\infty \wedge y = -\infty \wedge z = \infty$. This cases is excluded because

$$(x, z)(y, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x = \infty \wedge y = -\infty \wedge z = \infty$. This cases is excluded because

$$(y, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x \in \mathbb{R} \wedge y = \infty \wedge z = \infty$. Then $x + z = \infty = y + z$ so that $x + z \leq y + z$.

$x = -\infty \wedge y = \infty \wedge z = \infty$. This cases is excluded because

$$(x, z) \in \overline{\mathbb{R}} \setminus \{(-\infty, \infty), (\infty, -\infty)\}.$$

$x = \infty \wedge y = \infty \wedge z = \infty$. Then $x + z = \infty = y + z$ so that $x + y \leq y + z$.

Hence in all valid cases we have $x + y \leq y + z$.

2. As $0 \leq x$ we have $z = 0 + z \leq x + z$
(1)

3. As $x < y$ implies $x \leq y$ it follows from (1) that

$$x + z \leq y + z$$

Assume that $x + z = y + z$ then, as $z \in \mathbb{R}$, it follows from [theorem: 20.28(1)] that $x = y$ contradicting $x < y$. Hence the assumption is wrong and $x + y \neq y + z$ so that

$$x + y < y + z \quad \square$$

Theorem 20.31. Let $(x, y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ [so that $x + y$ is well defined, $z \in \overline{\mathbb{R}}$ such that $x + y < z$ then

$$(z, -y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\} \text{ [so that } x - y \text{ is well defined]}$$

and

$$x < z - y$$

Proof. If $x = \infty$ or $y = \infty$ then $x + y = \infty < z \leq \infty$ leading to the contradiction $\infty < \infty$. Hence we must have that

$$x, y \in \overline{\mathbb{R}} \setminus \{\infty\} = \mathbb{R} \cup \{-\infty\} \Rightarrow -y \in \mathbb{R} \cup \{\infty\}$$

Further If $z = -\infty$ then $-\infty \leq x + y < -\infty$ leading to the contradiction $-\infty < -\infty$ hence we have that

$$z \in \overline{\mathbb{R}} \setminus \{-\infty\} = \mathbb{R} \cup \{\infty\}$$

Hence

$$(z, -y) \in (\mathbb{R} \cup \{\infty\}) \times (\mathbb{R} \cup \{\infty\}) \subseteq (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

hence

$$z - y \text{ is well defined}$$

For z we have then the following cases to consider:

$z \in \mathbb{R}$. For x, y we have then the following possible cases:

$x = -\infty \wedge y = -\infty$. Then $x = -\infty < \infty = z + \infty = z + (-y) = z - y$ so that $x < z - y$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $x < \infty = z + \infty = z + (-y) = z - y$ so that $x < z - y$.

$x = -\infty \wedge y \in \mathbb{R}$. Then $z - y \in \mathbb{R}$ so that $x = -\infty < z - y$ so that $x < z - y$.

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then as $x + y < z$ it follows from properties of $\mathbb{R}$ that $x < z - y$.

$z \in \infty$. For x, y we have then the following possible cases:

$x = -\infty \wedge y = -\infty$. Then $x = -\infty < \infty = \infty + \infty = z + (-y) = z - y$ so that $x < z - y$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $x < \infty = \infty + \infty = z + (-y) = z - y$ so that $x < z - y$.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty < \infty = \infty - y = z - y$ so that $x < z - y$.

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then $x < \infty = \infty + (-y) = z - y$ so that $x < z - y$.

So in all cases we have that

$$x < z - y \quad \square$$

Theorem 20.32. Let $x, z \in \mathbb{R} \wedge y \in \overline{\mathbb{R}}$ [so that $y - z$ and $y - x$ are well defined] then if $x < y - z$ it follows that $z < y - x$

Proof. If $y = -\infty$ then we have as $x, z \in \mathbb{R}$ that $-\infty < x < y - z = (-\infty) + (-z) = -\infty$ which is a contradiction. Hence for y we have either:

$y = \infty$. Then as $x, z \in \mathbb{R}$ we have $z < \infty = \infty - x = y - x$ which implies that $z < y - x$.

$y \in \mathbb{R}$. Then as $x, y, z \in \mathbb{R}$ we have from $x < y - z$ by the properties of the real numbers that $z < y - x$. $\square$

Theorem 20.33. *Let $x, y \in \mathbb{R} \wedge z \in \overline{\mathbb{R}}$ [so that $y - z, y - x$ are well defined] then if $x < y - z$ it follows that $z < x - y$*

Proof. If $z = \infty$ then $x < y - \infty = -\infty$ which as $x \in \mathbb{R} \Rightarrow -\infty < x$ is a contradiction. Hence for z we have either:

$z = -\infty$. Then as $x, y \in \mathbb{R} \Rightarrow x - y \in \mathbb{R}$ we have that $z = -\infty < x - y$.

$z \in \mathbb{R}$. Then as $x, y, z \in \mathbb{R}$ we have from $x < y - z$ by the properties of the real numbers that $z < y - x$. $\square$

Theorem 20.34. *Let $x, y \in \overline{\mathbb{R}}$ with $0 < x \wedge 0 \leq y$ then $0 < x + y$*

Proof. As $0 < x$ and $0 \leq y$ then we have that $x \neq -\infty \neq y$ so that for x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then by the properties of the real numbers we have $0 < x + y$.

$x = \infty \wedge y \in \mathbb{R}$. Then $0 < \infty = x + y$.

$x \in \mathbb{R} \wedge y = \infty$. Then $0 < \infty = x + y$

$x = \infty \wedge y = \infty$. Then $0 < \infty = x + y$ $\square$

Theorem 20.35. *Let $x, y \in \overline{\mathbb{R}}$ such that $0 \leq x \wedge 0 \leq y \wedge z = x + y$ then $x \leq z$.*

Proof. As $0 \leq x$ and $0 \leq y$ then we have that $x \neq -\infty \neq y$ so that for x, y we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then by the properties of the real numbers we have $x \leq x + y = z$.

$x = \infty \wedge y \in \mathbb{R}$. Then $x = \infty = \infty + y = z$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x < \infty = x + \infty = x + y = z$.

$x = \infty \wedge y = \infty$. Then $x = \infty = x + y = z$. $\square$

Theorem 20.36. *Let $x, y \in \overline{\mathbb{R}}$ then we have:*

1. *If $0 < x \wedge 0 < y$ then $0 < x \cdot y$.*
2. *If $0 \leq x \wedge 0 \leq y$ then $0 \leq x \cdot y$.*
3. *If $x < 0 \wedge 0 < y$ then $x \cdot y < 0$.*
4. *If $0 < x \wedge y < 0$ then $x \cdot y < 0$.*
5. *If $x < 0 \wedge y < 0$ then $0 < x \cdot y$.*
6. *If $x, y \in \overline{\mathbb{R}}$ with $0 < x \leq y$ then $\frac{1}{y} \leq \frac{1}{x}$*

Proof.

1. For $0 < x \wedge 0 < y$ we have either:

$x \in \mathbb{R}^+ \wedge y \in \mathbb{R}^+$. Then by the properties of the real numbers we have $0 < x \cdot y$.

$x = \infty \wedge y \in \mathbb{R}^+$. Then $0 < \infty = \infty \cdot y = x \cdot y$.

$x \in \mathbb{R}^+ \wedge y = \infty$. Then $0 < \infty = x \cdot \infty = x \cdot y$.

$x = \infty \wedge y = \infty$. Then $0 < \infty = \infty \cdot \infty = x \cdot y$.

2. For $0 \leq x \wedge 0 \leq y$ we have either:

$x = 0 \vee y = 0$. Then $x \cdot y = 0$ so that $0 \leq x \cdot y$

$x \neq 0 \wedge y \neq 0$. Then $0 < x$ and $0 < y$ so that by (1) $0 < x \cdot y$ hence $0 \leq x \cdot y$.

3. For $x < 0 \wedge 0 < y$ we have either:

$x \in \mathbb{R}^- \wedge y \in \mathbb{R}^+$. Then by the properties of the real numbers we have $x \cdot y < 0$.

$x = -\infty \wedge y \in \mathbb{R}^+$. Then $x \cdot y = -\infty < 0$.

$x \in \mathbb{R}^- \wedge y = \infty$. Then $x \cdot y = -\infty < 0$

$x = -\infty \wedge y = \infty$. Then $x \cdot y = -\infty < 0$

4. For $0 < x \wedge y < 0$ we have either:

$x \in \mathbb{R}^+ \wedge y \in \mathbb{R}^-$. Then by the properties of the real numbers $x \cdot y < 0$.

$x = \infty \wedge y \in \mathbb{R}^-$. Then $x \cdot y = -\infty < 0$.

$x \in \mathbb{R}^+ \wedge y = -\infty$. Then $x \cdot y = -\infty < 0$.

$x = \infty \wedge y = -\infty$. Then $x \cdot y = -\infty < 0$.

5. For $x < 0 \wedge y < 0$ we have either:

$x \in \mathbb{R}^- \wedge y \in \mathbb{R}^-$. Then by the properties of the real numbers $0 < x \cdot y$.

$x = -\infty \wedge y \in \mathbb{R}^-$. Then $0 < \infty = x \cdot y$.

$x \in \mathbb{R}^- \wedge y = -\infty$. Then $0 < \infty = x \cdot y$.

$x = -\infty \wedge y = -\infty$. Then $0 < \infty = x \cdot y$.

6. If $x, y \in \overline{\mathbb{R}}$ with $0 < x \leq y$ we have for x either:

$x \in \mathbb{R}^+$. As $x \leq y$ we have for y either:

$y \in \mathbb{R}^+$. Then by the properties of the real numbers we have $\frac{1}{y} \leq \frac{1}{x}$.

$y = \infty$. Then $\frac{1}{y} = \frac{1}{\infty} = 0 < \frac{1}{x}$.

$x = \infty$. Then $y = \infty$ so that $\frac{1}{y} = \frac{1}{\infty} = \frac{1}{x}$ so that $\frac{1}{y} \leq \frac{1}{x}$. □

Corollary 20.37. *If $x \in \overline{\mathbb{R}}$ then we have:*

1. *If $0 < x$ then $-x < 0$*
2. *If $0 \leq x$ then $-x \leq 0$*
3. *If $x < 0$ then $0 < -x$*
4. *If $x \leq 0$ then $0 \leq -x$*

Proof. As $-1 < 0$ we have:

1. $-x = (-1) \cdot x \underset{[\text{theorem: 20.36}]}{<} (-1) \cdot 0 = 0$
2. $-x = (-1) \cdot x \underset{[\text{theorem: 20.36}]}{\leq} (-1) \cdot 0 = 0$
3. $-x = (-1) \cdot x \underset{[\text{theorem: 20.36}]}{>} (-1) \cdot 0 = 0$
4. $-x = (-1) \cdot x \underset{[\text{theorem: 20.36}]}{\geq} (-1) \cdot 0 = 0$ □

Although the product $x \cdot y$ is defined for every $x, y \in \overline{\mathbb{R}}$ $\langle \overline{\mathbb{R}}, \cdot \rangle$ is not a group, for example 0 has still no inverse. However $\langle \overline{\mathbb{R}}, \cdot \rangle$ is a Abelian semi group.

Theorem 20.38. *$\langle \overline{\mathbb{R}}, \cdot \rangle$ is a Abelian semi-group with neutral element 1*

Proof. We have to prove the semi-group axioms for $(\cdot): \overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}}$.

associativity. Let x, y, z then we have either:

$x = 0 \vee y = 0$. Then $x \cdot (y \cdot z) = 0 \cdot (y \cdot z) = 0 = 0 \cdot z = (x \cdot 0) \cdot z = (x \cdot y) \cdot z$

$x < 0 \wedge y = \infty$. Then for z we have either:

$z = 0$. Then $x \cdot (y \cdot z) = x \cdot (y \cdot 0) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$

$z < 0$. Then

$$x \cdot (y \cdot z) = x \cdot (-\infty) = (-\infty) \cdot (-\infty) = \infty = (-\infty) \cdot z = (x \cdot y) \cdot z$$

$0 < z$. Then

$$x \cdot (y \cdot z) = x \cdot \infty = -\infty = \infty \cdot z = (x \cdot y) \cdot z$$

$0 < x \wedge y = \infty$. Then for z we have either:

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot (y \cdot 0) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$$z < 0. \text{ Then } x \cdot (y \cdot z) = x \cdot (-\infty) = -\infty = \infty \cdot z = (x \cdot y) \cdot z$$

$$0 < z. \text{ Then } x \cdot (y \cdot z) = x \cdot \infty = \infty = \infty \cdot z = (x \cdot y) \cdot z$$

$x < 0 \wedge y = -\infty$. Then for z we have either:

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$$z < 0. \text{ Then } x \cdot (y \cdot z) = x \cdot \infty = -\infty = \infty \cdot z = (x \cdot y) \cdot z$$

$$0 < z. \text{ Then } x \cdot (y \cdot z) = x \cdot (-\infty) = \infty = \infty \cdot z = (x \cdot y) \cdot z$$

$0 < x \wedge y = -\infty$. Then for z we have either:

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$$z < 0. \text{ Then } x \cdot (y \cdot z) = x \cdot \infty = \infty = (-\infty) \cdot z = (x \cdot y) \cdot z$$

$$0 < z. \text{ Then } x \cdot (y \cdot z) = x \cdot (-\infty) = -\infty = (-\infty) \cdot z = (x \cdot y) \cdot z$$

$x = -\infty \wedge y \in]-\infty, 0[$. for z we have either:

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$$z < 0. \text{ Then by [theorem: 20.36] } 0 < y \cdot z \text{ so that}$$

$$x \cdot (y \cdot z) = -\infty = \infty \cdot z = (x \cdot y) \cdot z$$

$0 < z$. Then by [theorem: 20.36] $y \cdot z < 0$ so that

$$x \cdot (y \cdot z) = \infty = \infty \cdot z = (x \cdot y) \cdot z$$

$x = \infty \wedge y \in]-\infty, 0[$. Then for z we have either

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$$z < 0. \text{ Then by [theorem: 20.36] } 0 < y \cdot z \text{ so that}$$

$$x \cdot (y \cdot z) = \infty = (-\infty) \cdot z = (x \cdot y) \cdot z$$

$0 < z$. Then by [theorem: 20.36] $y \cdot z < 0$ so that

$$x \cdot (y \cdot z) = -\infty = (-\infty) \cdot z = (x \cdot y) \cdot z$$

$x \in \mathbb{R} \setminus \{0\} \wedge y \in]-\infty, 0[$. Then for z we have either:

$z = \infty$. Then for x we have either:

$$x < 0. \text{ Then by [theorem: 20.36] } 0 < x \cdot y \text{ so that}$$

$$x \cdot (y \cdot z) = x \cdot (-\infty) = \infty = (x \cdot y) \cdot \infty = (x \cdot y) \cdot z$$

$$0 < x. \text{ Then by [theorem: 20.36] } x \cdot y < 0 \text{ so that}$$

$$x \cdot (y \cdot z) = x \cdot (-\infty) = -\infty = (x \cdot y) \cdot \infty = (x \cdot y) \cdot z$$

$z = -\infty$. Then for x we have either:

$$x < 0. \text{ Then by [theorem: 20.36] } 0 < x \cdot y \text{ so that}$$

$$x \cdot (y \cdot z) = x \cdot \infty = -\infty = (x \cdot y) \cdot (-\infty) = (x \cdot y) \cdot z$$

$$0 < x. \text{ Then by [theorem: 20.36] } x \cdot y < 0 \text{ so that}$$

$$x \cdot (y \cdot z) = x \cdot \infty = \infty = (x \cdot y) \cdot (-\infty) = (x \cdot y) \cdot z$$

$z \in \mathbb{R}$. Then

$$x \cdot (y \cdot z) = x \cdot_{|\mathbb{R}} (y \cdot_{\mathbb{R}} z) = (x \cdot_{\mathbb{R}} y) \cdot_{\mathbb{R}} z = (x \cdot y) \cdot z$$

$x = -\infty \wedge y \in]0, \infty[$. Then for z we have either:

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$z < 0$. Then by [theorem: 20.36] $y \cdot z < 0$ so that

$$x \cdot (y \cdot z) = \infty = (-\infty) \cdot z = (x \cdot y) \cdot z$$

$0 < z$. Then by [theorem: 20.36] $0 < y \cdot z$ so that

$$x \cdot (y \cdot z) = -\infty = -\infty \cdot z = (x \cdot y) \cdot z$$

$x = \infty \wedge y \in]0, \infty[$. Then for z we have either:

$$z = 0. \text{ Then } x \cdot (y \cdot z) = x \cdot 0 = 0 = (x \cdot y) \cdot 0 = (x \cdot y) \cdot z$$

$z < 0$. Then by [theorem: 20.36] $y \cdot z < 0$ so that

$$x \cdot (y \cdot z) = -\infty = \infty \cdot z = (x \cdot y) \cdot z$$

$0 < z$. Then by [theorem: 20.36] $0 < y \cdot z$ so that

$$x \cdot (y \cdot z) = \infty = \infty \cdot z = (x \cdot y) \cdot z$$

$x \in \mathbb{R} \setminus \{0\} \wedge y \in]0, \infty[$. Then for z we have either:

$z = \infty$. Then for x we have either:

$x < 0$. Then by [theorem: 20.36] $x \cdot y < 0$ so that

$$x \cdot (y \cdot z) = x \cdot \infty = -\infty = (x \cdot y) \cdot \infty = (x \cdot y) \cdot z$$

$0 < x$. Then by [theorem: 20.36] $0 < x \cdot y$ so that

$$x \cdot (y \cdot z) = x \cdot \infty = \infty = (x \cdot y) \cdot \infty = (x \cdot y) \cdot z$$

$z = -\infty$. Then for x we have either:

$x < 0$. Then by [theorem: 20.36] $x \cdot y < 0$ so that

$$x \cdot (y \cdot z) = x \cdot (-\infty) = \infty = (x \cdot y) \cdot (-\infty) = (x \cdot y) \cdot z$$

$0 < x$. Then by [theorem: 20.36] $0 < x \cdot y$ so that

$$x \cdot (y \cdot z) = x \cdot (-\infty) = -\infty = (x \cdot y) \cdot (-\infty) = (x \cdot y) \cdot z$$

$z \in \mathbb{R}$. Then

$$x \cdot (y \cdot z) = x \cdot_{|\mathbb{R}} (y \cdot_{\mathbb{R}} z) = (x \cdot_{\mathbb{R}} y) \cdot_{\mathbb{R}} z = (x \cdot y) \cdot z$$

So in all cases we have $(x \cdot y) \cdot z = x \cdot (y \cdot z)$

neutral element. Let $x \in \overline{\mathbb{R}}$ then we have for x either:

$$x = \infty. \text{ Then } 1 \cdot \infty = \infty = \infty \cdot 1$$

$$x = -\infty. \text{ Then } 1 \cdot (-\infty) = -\infty = (-\infty) \cdot 1$$

$$x \in \mathbb{R}. \text{ Then } 1 \cdot x = 1 \cdot_{\mathbb{R}} x = x = x \cdot_{\mathbb{R}} 1 = x \cdot 1$$

commutativity. Then for $x, y \in \overline{\mathbb{R}}$ we have either

$$x = 0 \vee y = 0. \text{ Then } x \cdot y = 0 = y \cdot x$$

$x < 0 \wedge y = \infty$. Then $x \cdot y \stackrel{\text{def}}{=} -\infty$ and for x we have either:

$$x = -\infty. \text{ Then } y \cdot x \stackrel{0 < \infty = y}{=} -\infty = x \cdot y$$

$$x \in]-\infty, 0[. \text{ Then } y \cdot x = -\infty = x \cdot y.$$

$0 < x \wedge y = \infty$. Then $x \cdot y \stackrel{\text{def}}{=} \infty$ and for x we have either:

$x = \infty$. Then $y \cdot x \stackrel{0 < \infty = y}{=} \infty = x \cdot y$.

$x \in]0, \infty[$. Then $y \cdot x = \infty = x \cdot y$.

$x < 0 \wedge y = -\infty$. Then $x \cdot y \stackrel{\text{def}}{=} \infty$ and for x we have either:

$x = -\infty$. Then $y \cdot x \stackrel{y = -\infty < 0}{=} \infty = x \cdot y$.

$x \in]-\infty, 0[$. Then $y \cdot x = \infty = x \cdot y$.

$0 < x \wedge y = -\infty$. Then $x \cdot y = -\infty$ and for x we have either:

$x = \infty$. Then $y \cdot x \stackrel{y = -\infty < 0}{=} -\infty = x \cdot y$

$x \in]0, \infty[$. Then $y \cdot x = -\infty = x \cdot y$

$x = -\infty \wedge y \in]-\infty, 0[$. Then $x \cdot y = \infty \stackrel{y < 0}{=} y \cdot x$

$x = \infty \wedge y \in]-\infty, 0[$. Then $x \cdot y = -\infty \stackrel{y < 0}{=} y \cdot x$

$x \in \mathbb{R} \setminus \{0\} \wedge y \in]-\infty, 0[$. Then $x \cdot y = x \cdot_{\mathbb{R}} y = y \cdot_{\mathbb{R}} x = y \cdot x$

$x = -\infty \wedge y \in]0, \infty[$. Then $x \cdot y = -\infty \stackrel{0 < y}{=} y \cdot x$

$x = \infty \wedge y \in]0, \infty[$. Then $x \cdot y = \infty \stackrel{0 < y}{=} y \cdot x$

$x \in \mathbb{R} \setminus \{0\} \wedge y \in]0, \infty[$. Then $x \cdot y = x \cdot_{\mathbb{R}} y = y \cdot_{\mathbb{R}} x = y \cdot x$ □

Corollary 20.39. Let $x, y \in \overline{\mathbb{R}}$ then $-(x \cdot y) = (-x) \cdot y = x \cdot (-y)$

Proof. We have

$$-(x \cdot y) = (-1) \cdot (x \cdot y) \stackrel{[\text{theorem: 20.38}]}{=} ((-1) \cdot x) \cdot y = (-x) \cdot y$$

and

$$-(x \cdot y) = (-1) \cdot (x \cdot y) \stackrel{[\text{theorem: 20.38}]}{=} (-1) \cdot (y \cdot x) \stackrel{[\text{theorem: 20.38}]}{=} ((-1) \cdot y) \cdot x \stackrel{[\text{theorem: 20.38}]}{=} x \cdot (-y)$$

□

Theorem 20.40. If $x, y \in \overline{\mathbb{R}}$ with $x \leq y$ then for $\lambda \in \overline{\mathbb{R}}$ we have

1. If $0 \leq \lambda$ then $\lambda \cdot x \leq \lambda \cdot y$
2. If $\lambda \leq 0$ then $\lambda \cdot y \leq \lambda \cdot x$

Proof.

1. For $0 \leq \lambda$ we have either:

$\lambda = 0$. Then $\lambda \cdot x = 0 \cdot x = 0 = 0 \cdot y = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$\lambda \in \mathbb{R}^+$. Then for x we have either:

$x \in \mathbb{R}$. Then for y we have as $-\infty < x \leq y$ that

$y \in \mathbb{R}$. Then using [theorem: 10.14] we have

$$\lambda \cdot x = \lambda \cdot_{\mathbb{R}} x \stackrel{[\text{theorem: 10.14}]}{\leq_{\mathbb{R}}} \lambda \cdot_{\mathbb{R}} y = \lambda \cdot y$$

so that

$$\lambda \cdot x \leq \lambda \cdot y$$

$y = \infty$. Then $\lambda \cdot x \in \mathbb{R}$ so that $\lambda \cdot x \leq \infty = \lambda \cdot \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$x = -\infty$. Then for y we have either:

$y \in \mathbb{R}$. Then $\lambda \cdot y \in \mathbb{R}$ so that $\lambda \cdot x = \lambda \cdot (-\infty) = -\infty < \lambda \cdot y$ hence $\lambda \cdot x \leq \lambda \cdot y$.

$y = -\infty$. Then $\lambda \cdot x = \lambda \cdot (-\infty) = -\infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$y = \infty$. Then $\lambda \cdot x = \lambda \cdot (-\infty) = -\infty < \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$x = \infty$. Then as $x \leq y$ we have that $y = \infty$ hence $\lambda \cdot x = \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$\lambda = \infty$. Then for x we have either:

$x < 0$. Then for y we have either:

$y < 0$. Then $\lambda \cdot x = -\infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$y = 0$. Then $\lambda \cdot x = -\infty < 0 = \lambda \cdot \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$0 < y$. Then $\lambda \cdot x = -\infty < \infty = \lambda \cdot \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$x = 0$. Then as $x \leq y$ we have that y is either:

$y = 0$. Then $\lambda \cdot x = 0 = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$0 < y$. Then $\lambda \cdot x = 0 < \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$0 < x$. Then as $x \leq y$ we have that $0 < y$ so that $\lambda \cdot x = \infty = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

2. For $\lambda \leq 0$ we have either:

$\lambda = 0$. Then $\lambda \cdot x = 0 \cdot x = 0 = 0 \cdot y = \lambda \cdot y$ so that $\lambda \cdot x \leq \lambda \cdot y$.

$\lambda \in \mathbb{R}^-$. Then for x we have either:

$x \in \mathbb{R}$. Then for y we have as $-\infty < x \leq y$ that

$y \in \mathbb{R}$. Then using [theorem: 10.14] we have

$$\lambda \cdot y = \lambda \cdot_{\mathbb{R}} y \leq_{\mathbb{R}} \lambda \cdot_{\mathbb{R}} x = \lambda \cdot x$$

[theorem: 10.14]

so that

$$\lambda \cdot y \leq \lambda \cdot x$$

$y = \infty$. Then $\lambda \cdot x \in \mathbb{R}$ so that $\lambda \cdot y = -\infty < \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$x = -\infty$. Then for y we have either:

$y \in \mathbb{R}$. Then $\lambda \cdot y \in \mathbb{R}$ so that $\lambda \cdot y < \infty = \lambda \cdot x$ hence $\lambda \cdot y \leq \lambda \cdot x$.

$y = -\infty$. Then $\lambda \cdot y = \infty = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$y = \infty$. Then $\lambda \cdot y = -\infty < \infty = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$x = \infty$. Then as $x \leq y$ we have that $y = \infty$ hence $\lambda \cdot y = -\infty = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$\lambda = -\infty$. Then for x we have either:

$x < 0$. Then for y we have either:

$y < 0$. Then $\lambda \cdot y = \infty = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$y = 0$. Then $\lambda \cdot y = 0 < \infty = \lambda \cdot (-\infty) = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$0 < y$. Then $\lambda \cdot y = -\infty < \infty = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$x = 0$. Then as $x \leq y$ we have that y is either:

$y = 0$. Then $\lambda \cdot y = 0 = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$0 < y$. Then $\lambda \cdot y = -\infty < 0 = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$.

$0 < x$. Then as $x \leq y$ we have that $0 < y$ so that $\lambda \cdot y = -\infty = \lambda \cdot x$ so that $\lambda \cdot y \leq \lambda \cdot x$. □

Corollary 20.41. *If $x, y \in \overline{\mathbb{R}}$ then we have*

1. *If $-x = -y$ then $x = y$*
2. *If $x \leq y$ then $-y \leq -x$*
3. *If $x < y$ then $-y < -x$*

Proof.

1. We have

$$\begin{aligned}
 x &\stackrel{[\text{theorem: 20.38}]}{=} 1 \cdot x \\
 &= ((-1) \cdot (-1)) \cdot x \\
 &\stackrel{[\text{theorem: 20.38}]}{=} (-1) \cdot ((-1) \cdot x) \\
 &= (-1) \cdot (-x) \\
 &= (-1) \cdot (-y) \\
 &= (-1) \cdot ((-1) \cdot y) \\
 &\stackrel{[\text{theorem: 20.38}]}{=} ((-1) \cdot (-1)) \cdot y \\
 &= 1 \cdot y \\
 &= y
 \end{aligned}$$

2. Take $\lambda = -1 \leq 0$ then $-y = \lambda \cdot y \stackrel{[\text{theorem: 20.40}]}{\leq} \lambda \cdot x = -x$

3. If $x < y$ then $x \neq y$ and $x < y$, using (2) we have then that $(-y) \leq -x$ and using (1) that $-x \neq -y$ proving that $-y < -x$ $\square$

We have also a variant of the distributive law in $\overline{\mathbb{R}}$ but again we must be careful. For example $\infty \cdot (1 - 3) = \infty \cdot (-2) = -\infty$ but $\infty \cdot 1 + \infty \cdot (-3)$ is not defined because $\infty \cdot 1 = \infty$ and $\infty \cdot (-3) = -\infty$.

Theorem 20.42. Let $\alpha \in \mathbb{R}$, $(x, y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ then $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$

Proof. For (x, y) we have either:

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. then $\alpha \cdot (x + y) = \alpha \cdot_{\mathbb{R}} (x +_{\mathbb{R}} y) = \alpha \cdot_{\mathbb{R}} x +_{\mathbb{R}} \alpha \cdot_{\mathbb{R}} y = \alpha \cdot x + \alpha \cdot y$

$x \in \mathbb{R} \wedge y = \infty$. then for α we have either

$\alpha = 0$. then $\alpha \cdot (x + y) = 0 = 0 + 0 = \alpha \cdot x + \alpha \cdot y$.

$0 < \alpha$. then $\alpha \cdot (x + y) = \alpha \cdot \infty = \infty = \alpha \cdot x + \infty = \alpha \cdot x + \alpha \cdot y$

$\alpha < 0$. then $\alpha \cdot (x + y) = \alpha \cdot \infty = -\infty = \alpha \cdot x + (-\infty) = \alpha \cdot x + \alpha \cdot y$

$x \in \mathbb{R} \wedge y = -\infty$. then for α we have either

$\alpha = 0$. then $\alpha \cdot (x + y) = 0 = 0 + 0 = \alpha \cdot x + \alpha \cdot y$.

$0 < \alpha$. then $\alpha \cdot (x + y) = \alpha \cdot (-\infty) = -\infty = \alpha \cdot x + (-\infty) = \alpha \cdot x + \alpha \cdot y$

$\alpha < 0$. then $\alpha \cdot (x + y) = \alpha \cdot (-\infty) = \infty = \alpha \cdot x + \infty = \alpha \cdot x + \alpha \cdot y$

$x = \infty \wedge y \in \mathbb{R}$. then for α we have either

$\alpha = 0$. then $\alpha \cdot (x + y) = 0 = 0 + 0 = \alpha \cdot x + \alpha \cdot y$.

$0 < \alpha$. then $\alpha \cdot (x + y) = \alpha \cdot \infty = \infty = \infty + \alpha \cdot y = \alpha \cdot x + \alpha \cdot y$

$\alpha < 0$. then $\alpha \cdot (x + y) = \alpha \cdot \infty = -\infty = -\infty + \alpha \cdot y = \alpha \cdot x + \alpha \cdot y$

$x = \infty \wedge y = \infty$. then for α we have either

$\alpha = 0$. then $\alpha \cdot (x + y) = 0 = 0 + 0 = \alpha \cdot x + \alpha \cdot y$.

$0 < \alpha$. then $\alpha \cdot (x + y) = \alpha \cdot \infty = \infty = \infty + \infty = \alpha \cdot x + \alpha \cdot y$

$\alpha < 0$. then $\alpha \cdot (x + y) = \alpha \cdot \infty = -\infty = -\infty + (-\infty) = \alpha \cdot x + \alpha \cdot y$

$x = -\infty \wedge y \in \mathbb{R}$. then for α we have either

$\alpha = 0$. then $\alpha \cdot (x + y) = 0 = 0 + 0 = \alpha \cdot x + \alpha \cdot y$.

$0 < \alpha$. then $\alpha \cdot (x + y) = \alpha \cdot (-\infty) = -\infty = -\infty + \alpha \cdot y = \alpha \cdot x + \alpha \cdot y$

$\alpha < 0$. then $\alpha \cdot (x + y) = \alpha \cdot (-\infty) = \infty = \infty + \alpha \cdot y = \alpha \cdot x + \alpha \cdot y$

$x = -\infty \wedge y = -\infty$. then for α we have either

$\alpha = 0$. then $\alpha \cdot (x + y) = 0 = 0 + 0 = \alpha \cdot x + \alpha \cdot y$.

$0 < \alpha$. then $\alpha \cdot (x + y) = \alpha \cdot (-\infty) = -\infty = -\infty + (-\infty) = \alpha \cdot x + \alpha \cdot y$

$\alpha < 0$. then $\alpha \cdot (x + y) = \alpha \cdot (-\infty) = \infty = -\infty + \infty = \alpha \cdot x + \alpha \cdot y$

so in all cases we have $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$. $\square$

Theorem 20.43. *Let $x, y \in \overline{\mathbb{R}}$ such that $\forall \varepsilon \in \mathbb{R}^+$ we have $x \leq y + \varepsilon$ then $x \leq y$.*

Proof. If $\forall \varepsilon \in \mathbb{R}^+$ we have $x \leq y + \varepsilon$ then we have for x, y to verify the following cases:

$x = -\infty \wedge y = -\infty$. Then $x = y$ so that $x \leq y$

$x \in \mathbb{R} \wedge y = -\infty$. Then $-\infty < x \leq y + \varepsilon = (-\infty) + \varepsilon = -\infty$ leading to the contradiction $-\infty < -\infty$, hence this cases never materialize.

$x = \infty \wedge y = -\infty$. Then $\infty = x \leq y + \varepsilon = -\infty + \varepsilon = -\infty$ leading to the contradiction $\infty < -\infty$ so this case never occurs.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty < y$ so that $x \leq y$.

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then by [theorem: 10.31 (2)] it follows that $x \leq y$

$x = \infty \wedge y \in \mathbb{R}$. Then $\infty = x \leq y + \varepsilon < \infty$ leading to the contradiction $\infty < \infty$, hence this cases never occurs.

$x = -\infty \wedge y = \infty$. Then $x = -\infty < \infty = y$ so that $x \leq y$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x < \infty = y$ so that $x \leq y$

$x = \infty \wedge y = \infty$. Then $x = y$ so that $x \leq y$

So in all valid cases we have that $x \leq y$ $\square$

20.3 Topology on $\overline{\mathbb{R}}$

First we define the absolute value on $\overline{\mathbb{R}}$.

Definition 20.44. *Let $x \in \overline{\mathbb{R}}$ then the absolute value noted as $|x|$ is defined in analogy of the absolute value on $\mathbb{R}$ as follows*

$$|x| = \begin{cases} x & \text{if } 0 \leq x \\ -x & \text{if } x < 0 \end{cases}$$

Note 20.45. $\forall x \in \overline{\mathbb{R}}$ we have that $x \leq |x|$

Note 20.46. If $x \in \mathbb{R}$ then $|x| = |x|_{\mathbb{R}}$ where $||_{\mathbb{R}}$ is the absolute value defined in $\mathbb{R}$ [see example: 14.99].

Proof. For $x \in \overline{\mathbb{R}}$ we have either:

$0 \leq x$. Then $x = |x|$ so that $x \leq |x|$

$x < 0$. Then by [theorem: 20.37] $0 < -x = |x|$ so that $x < 0 < |x|$ or $x \leq |x|$ $\square$

The absolute value of the extended real numbers has similar properties as the absolute value on the real numbers.

Theorem 20.47. *The absolute value has the following properties:*

1. $|\infty| = \infty$
2. $|-\infty| = \infty$
3. For $x \in \overline{\mathbb{R}}$ we have $|x| = 0 \Leftrightarrow x = 0$.
4. $\forall x, y \in \overline{\mathbb{R}}$ we have $|x \cdot y| = |x| \cdot |y|$.
5. $\forall (x, y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ we have $|x + y| \leq |x| + |y|$

Proof.

1. As $0 < \infty$ it follows that $|\infty| = \infty$.
2. As $-\infty < 0$ it follows that $|\infty| = -(-\infty) = (-1) \cdot (-\infty) = \infty$
3. We have

$\Rightarrow$. Assume that $x \neq 0$ then we have either $0 < x \Rightarrow 0 \neq x = |x| \neq 0$ leading to the contradiction $0 \neq 0$ or $x < 0 \xrightarrow{[\text{theorem: 20.37}]} 0 < -x = |x| = 0$ leading to the contradiction $0 < 0$. Hence the assumption is false and we must have that $x = 0$.

$\Leftarrow$. As $0 \leq 0$ it follows that $|0| = 0$.

4. For $x, y \in \overline{\mathbb{R}}$ we have the following cases to consider:

$x = 0 \wedge y = 0$. Then $|x \cdot y| = |0| \stackrel{(3)}{=} 0$.

$x < 0 \wedge y = 0$. Then $|x \cdot y| = |0| \stackrel{(3)}{=} 0$.

$0 < x \wedge y = 0$. Then $|x \cdot y| = |0| \stackrel{(3)}{=} 0$.

$x = 0 \wedge y < 0$. Then $|x \cdot y| = |0| \stackrel{(3)}{=} 0$.

$x < 0 \wedge y < 0$. Then by [theorem: 20.36] $0 < x \cdot y$ so that $|x \cdot y| = x \cdot y = |x| \cdot |y|$.

$0 < x \wedge y < 0$. Then by [theorem: 20.36] $x \cdot y < 0$ so that

$$|x \cdot y| = -(x \cdot y) \stackrel{[\text{theorem: 20.39}]}{=} x \cdot (-y) = |x| \cdot |y|$$

$x = 0 \wedge 0 < y$. Then $|x \cdot y| = |0| \stackrel{(3)}{=} 0$.

$x < 0 \wedge 0 < y$. Then by [theorem: 20.36] $x \cdot y < 0$ so that

$$|x \cdot y| = -(x \cdot y) = (-x) \cdot y = |x| \cdot |y|$$

$0 < x \wedge 0 < y$. Then by [theorem: 20.36] $0 < x \cdot y$ so that

$$|x \cdot y| = x \cdot y = |x| \cdot |y|$$

5. For $(x, y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ we have either:

$x = -\infty \wedge y = -\infty$. Then $|x + y| = |-\infty| \stackrel{(2)}{=} \infty = \infty + \infty = |-\infty| + |-\infty| = |x| + |y|$

$x \in \mathbb{R} \wedge y = -\infty$. Then $|x + y| = |-\infty| \stackrel{(2)}{=} |x| + \infty \stackrel{(2)}{=} |x| + |-\infty| = |x| + |y|$

$x = \infty \wedge y = \infty$. Then $|x + y| = |\infty| \stackrel{(1)}{=} \infty = \infty + \infty \stackrel{(1)}{=} |\infty| + |\infty| = |x| + |y|$

$x \in \mathbb{R} \wedge y = \infty$. Then $|x + y| = |\infty| \stackrel{(1)}{=} \infty = |x| + \infty \stackrel{(1)}{=} |x| + |\infty| = |x| + |y|$

$x = \infty \wedge y \in \mathbb{R}$. Then $|x + y| = |\infty| \stackrel{(1)}{=} \infty = \infty + |y| \stackrel{(1)}{=} |\infty| + |y| = |x| + |y|$

$x = -\infty \wedge y \in \mathbb{R}$. Then $|x + y| = |-\infty| \stackrel{(2)}{=} \infty = \infty + |y| \stackrel{(2)}{=} |-\infty| + |y| = |x| + |y|$

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then $|x + y| = |x + y|_{\mathbb{R}} \leq |x|_{\mathbb{R}} + |y|_{\mathbb{R}} = |x| + |y|$ □

Unfortunately $(\overline{\mathbb{R}}, | \cdot |)$ is not a normed space because $\overline{\mathbb{R}}$ is not a vector space [remember $\infty + (-\infty)$ and $(-\infty) + \infty$ are not defined. So we can not base a topology on a norm instead we use a generating basis [see theorem: 14.43].

Definition 20.48. *The following sets are defined*

1. $\mathcal{B}_{||} = \{[a, b] \mid a, b \in \mathbb{R} \wedge a < b\}$ which is the generating basis for the natural topology $\mathcal{T}_{||}$ on $\mathbb{R}$ [see theorem 14.104].
2. $\mathcal{B}_{\infty} = \{[x, \infty] \mid x \in \mathbb{R}\}$
3. $\mathcal{B}_{-\infty} = \{[-\infty, x] \mid x \in \mathbb{R}\}$

$$4. \mathcal{B}_{\overline{\mathbb{R}}} = \mathcal{B}_{||} \cup \mathcal{B}_{\infty} \cup \mathcal{B}_{-\infty}$$

We prove now that $\mathcal{B}_{\overline{\mathbb{R}}}$ satisfies the conditions to be a generating basis [see theorem: 14.43].

Lemma 20.49. *The set $\mathcal{B}_{\overline{\mathbb{R}}}$ satisfies*

1. $\forall x \in \overline{\mathbb{R}}$ there exist a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ such that $x \in B$.
2. $\forall B_1, B_2 \in \mathcal{B}_{\overline{\mathbb{R}}}$ we have that $\forall x \in B_1 \cap B_2$ there exist a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ such that $x \in B \subseteq B_1 \cap B_2$

Proof.

1. If $x \in \overline{\mathbb{R}}$ then we have either

$x \in \mathbb{R}$. Then $x \in]x-1, x+1[\in \mathcal{B}_{\overline{\mathbb{R}}}$

$x = -\infty$. Then $x = -\infty \in [-\infty, 0[\in \mathcal{B}_{\overline{\mathbb{R}}}$

$x = \infty$. Then $x = \infty \in]0, \infty] \in \mathcal{B}_{\overline{\mathbb{R}}}$

2. If $B_1, B_2 \in \mathcal{B}_{\overline{\mathbb{R}}}$ and $x \in B_1 \cap B_2$ then we have for B_1, B_2 either:

$B_1 \in \mathcal{B}_{||} \wedge B_2 \in \mathcal{B}_{||}$. Then there exists a_1, b_1, a_2, b_2 with $B_1 =]a_1, b_1[\wedge B_2 =]a_2, b_2[$, as $x \in B_1 \cap B_2$ $a_1 < x < b_1$ and $a_2 < x < b_2$. Define $B =]a, b[$ where $a = \max(a_1, a_2)$ and $b = \min(b_1, b_2)$ then $a < x < b$ or $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $a_1, a_2 \leq a < y < b \leq b_1, b_2$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{-\infty} \wedge B_2 \in \mathcal{B}_{||}$. Then there exists $a, b, c \in \mathbb{R}$ such that $B_2 =]a, b[$ and $B_1 = [-\infty, c[$, as $x \in B_1 \cap B_2$ we have that $x < c$ and $a < x < b$. Define $B =]a, d[$ where $d = \min(c, b)$ then $a < x < d$ so that $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $a < y < d \leq c, b$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{\infty} \wedge B_2 \in \mathcal{B}_{||}$. Then there exists $a, b, c \in \mathbb{R}$ such that $B_2 =]a, b[$ and $B_1 =]c, \infty]$, as $x \in B_1 \cap B_2$ we have that $c < x$ and $a < x < b$. Define $B =]d, b[$ where $d = \max(c, a)$ then $d < x < b$ so that $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $c, a \leq d < y < b$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{||} \wedge B_2 \in \mathcal{B}_{-\infty}$. Then there exists $a, b, c \in \mathbb{R}$ such that $B_1 =]a, b[$ and $B_2 = [-\infty, c[$, as $x \in B_1 \cap B_2$ we have $a < x < b$ and $x < c$. Define $B =]a, d[$ where $d = \min(b, c)$ then $a < x < d$ so that $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $a < y < d \leq b, c$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{-\infty} \wedge B_2 \in \mathcal{B}_{-\infty}$. Then there exists $c_1, c_2 \in \mathbb{R}$ such that $B_1 = [-\infty, c_1[$ and $B_2 = [-\infty, c_2[$, as $x \in B_1 \cap B_2$ we have that $x < c_1, c_2$ hence if we define $B = [-\infty, c[$ where $c = \min(c_1, c_2) \in \mathbb{R}$ then $x < c$ so that $x \in B \in \mathcal{B}_{-\infty}$. Further if $y \in B$ then $y < c \leq c_1, c_2$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{-\infty} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{\infty} \wedge B_2 \in \mathcal{B}_{-\infty}$. Then there exists $c_1, c_2 \in \mathbb{R}$ such that $B_1 =]c_1, \infty]$ and $B_2 = [-\infty, c_2[$, as $x \in B_1 \cap B_2$ we have that $c_1 < x < c_2$. Define $B =]c_1, c_2[$ then $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $c_1 < y < c_2$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{||} \wedge B_2 \in \mathcal{B}_{\infty}$. Then there exists $a, b, c \in \mathbb{R}$ such that $B_1 =]a, b[$ and $B_2 =]c, \infty]$, as $x \in B_1 \cap B_2$ we have that $a < x < b$ and $c < x$. Define $B =]d, b[$ where $d = \max(a, c)$ then we have $d < x < b$ so that $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $a, c \leq d < y < b$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{-\infty} \wedge B_2 \in \mathcal{B}_{\infty}$. Then there exists $c_1, c_2 \in \mathbb{R}$ such that $B_1 = [-\infty, c_1[$ and $B_2 =]c_2, \infty]$, as $x \in B_1 \cap B_2$ we have $c_2 < x < c_1$. Define $B =]c_2, c_1[$ then $x \in B \in \mathcal{B}_{||}$. Further if $y \in B$ then $c_2 < y < c_1$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$.

$B_1 \in \mathcal{B}_{\infty} \wedge B_2 \in \mathcal{B}_{\infty}$. Then there exists $c_1, c_2 \in \mathbb{R}$ such that $B_1 =]c_1, \infty]$ and $B_2 =]c_2, \infty]$, as $x \in B_1 \cap B_2$ we have $c_1 < x \wedge c_2 < x$. Define $B =]c, \infty]$ where $c = \max(c_1, c_2) \in \mathbb{R}$ then $c < x$ so that $x \in B_{\infty}$. Further if $y \in B$ then $c_1, c_2 \leq c < y$ so that $y \in B_1 \cap B_2$. Hence $x \in B \subseteq B_1 \cap B_2$ where $B \in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$. $\square$

We use now $\mathcal{B}_{\overline{\mathbb{R}}}$ to generate the canonical topology on $\overline{\mathbb{R}}$.

Definition 20.50. The canonical topology on $\overline{\mathbb{R}}$ noted as $\mathcal{T}_{\overline{\mathbb{R}}}$ is defined as

$$\mathcal{T}_{\overline{\mathbb{R}}} = \{U \subseteq \overline{\mathbb{R}} \mid \forall x \in U \text{ we have that } \exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } x \in B \subseteq U\}$$

which is indeed a topology by [theorem: 14.43] and [lemma: 20.49].

It turns out that the subspace topology of $\mathcal{T}_{\overline{\mathbb{R}}}$ [see theorem: 14.8] is equal to the canonical topology $\mathcal{T}_{\mathbb{R}}$ on $\mathbb{R}$.

Theorem 20.51. Let $\mathcal{T}_{\overline{\mathbb{R}}}$ be the canonical topology on $\overline{\mathbb{R}}$ and $\mathcal{T}_{\mathbb{R}}$ the canonical topology on $\mathbb{R}$ [see theorem: 14.104] then

$$\mathcal{T}_{\mathbb{R}} \subseteq \mathcal{T}_{\overline{\mathbb{R}}}$$

and

$$\mathcal{T}_{\mathbb{R}} = \{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\} \stackrel{[\text{theorem: 14.8}]}{=} (\mathcal{T}_{\overline{\mathbb{R}}})|_{\mathbb{R}}$$

Proof. By definition

$$\mathcal{T}_{\overline{\mathbb{R}}} = \{U \subseteq \overline{\mathbb{R}} \mid \forall x \in U \text{ we have that } \exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } x \in B \subseteq U\}$$

Further by [theorem: 14.104]

$$\mathcal{T}_{\mathbb{R}} = \{U \subseteq \mathbb{R} \mid \forall x \in U \text{ we have that } \exists B \in \mathcal{B}_{\mathbb{R}} \text{ such that } x \in B \subseteq U\}$$

where

$$\mathcal{B}_{\mathbb{R}} = \{]a, b[\mid a, b \in \mathbb{R} \wedge a < b\} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$$

Let $U \in \mathcal{T}_{\mathbb{R}}$ then $\forall x \in U \exists B \in \mathcal{B}_{\mathbb{R}} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$ such that $x \in B \subseteq U$ proving that $U \in \mathcal{T}_{\overline{\mathbb{R}}}$, hence we have

$$\mathcal{T}_{\mathbb{R}} \subseteq \mathcal{T}_{\overline{\mathbb{R}}} \quad (20.1)$$

If $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ then $U \subseteq \overline{\mathbb{R}} \Rightarrow U = U \cap \mathbb{R}$ and by [eq: 20.1] $U \in \mathcal{T}_{\mathbb{R}}$ so that $U \in \{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\}$ hence

$$\mathcal{T}_{\overline{\mathbb{R}}} \subseteq \{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\} \quad (20.2)$$

If $U \in \{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\}$ then there exist a $V \in \mathcal{T}_{\overline{\mathbb{R}}}$ such that

$$U = V \cap \mathbb{R} \quad (20.3)$$

If $x \in U$ then $x \in \mathbb{R}$ and $x \in V \in \mathcal{T}_{\overline{\mathbb{R}}}$, so there exists a $B'_x \in \mathcal{B}_{\overline{\mathbb{R}}}$ such that $x \in B'_x \subseteq V$. For B'_x we have then either:

$B'_x \in \mathcal{B}_{\mathbb{R}}$. Then there exist a $a, b \in \mathbb{R}$ such that $B'_x =]a, b[\subseteq \mathbb{R}$ hence if we take $B_x = B'_x$ then

$$x \in B_x = B'_x \cap \mathbb{R} \subseteq V \cap \mathbb{R} = U$$

$B'_x \in \mathcal{B}_{-\infty}$. Then $\exists c \in \mathbb{R}$ such that $B'_x = [-\infty, c[$. Define $B_x =]x - 1, c[\subseteq \mathbb{R}$ then $x - 1 < x < c$ so that $x \in B_x \in \mathcal{B}_{\mathbb{R}}$. Further for $y \in B_x$ we have $y < c$ so that $y \in B'_x$ proving that $x \in B_x \subseteq B'_x$. Hence

$$x \in B_x = B'_x \cap \mathbb{R} \subseteq V \cap \mathbb{R} = U$$

$B'_x \in \mathcal{B}_{\infty}$. Then $\exists c \in \mathbb{R}$ such that $B'_x =]c, \infty[$. Define $B_x =]c, x + 1[\subseteq \mathbb{R}$ then $c < x < x + 1$ so that $x \in B_x \in \mathcal{B}_{\mathbb{R}}$. Further for $y \in B_x$ we have $c < y$ so that $y \in B'_x$ proving that $x \in B_x \subseteq B'_x$. Hence

$$x \in B_x = B'_x \cap \mathbb{R} \subseteq V \cap \mathbb{R} = U$$

so in all cases we found a $B_x \in \mathcal{B}_{\mathbb{R}}$ such that $x \in B_x \subseteq U$. As x was choose arbitrary it follows that $U \in \mathcal{T}_{\mathbb{R}}$. Hence $\{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\} \subseteq \mathcal{T}_{\mathbb{R}}$ which combined with [eq: 20.2] gives

$$\mathcal{T}_{\mathbb{R}} = \{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\} \quad \square$$

Theorem 20.52. *We have the following closed sets in $\mathcal{T}_{\overline{\mathbb{R}}}$*

1. $\forall x \in \mathbb{R} \ [-\infty, x] \text{ is closed.}$
2. $\forall x \in \mathbb{R} \ [x, \infty] \text{ is closed,}$
3. $\forall x \in \mathbb{R} \ \{x\} \text{ is closed.}$
4. $[-\infty, \infty] \text{ is closed.}$
5. $\{-\infty, \infty\} \text{ is closed}$
6. $\{-\infty\} \text{ is closed.}$
7. $\{\infty\} \text{ is closed.}$

Proof.

1. As

$$\begin{aligned} y \in \overline{\mathbb{R}} \setminus [-\infty, x] &\Leftrightarrow y \in \overline{\mathbb{R}} \wedge \neg(-\infty \leq y \leq x) \\ &\Leftrightarrow y \in \overline{\mathbb{R}} \wedge (y < -\infty \vee x < y) \\ &\Leftrightarrow y \in \overline{\mathbb{R}} \wedge x < y \\ &\Leftrightarrow y \in]x, \infty] \end{aligned}$$

so that $\overline{\mathbb{R}} \setminus [-\infty, x] =]x, \infty] \in \mathcal{T}_{\overline{\mathbb{R}}}$, hence $[-\infty, x]$ is closed.

2. As

$$\begin{aligned} y \in \overline{\mathbb{R}} \setminus [x, \infty] &\Leftrightarrow y \in \overline{\mathbb{R}} \wedge \neg(x \leq y \leq \infty) \\ &\Leftrightarrow y \in \overline{\mathbb{R}} \wedge (y < x \vee \infty < y) \\ &\Leftrightarrow y \in \overline{\mathbb{R}} \wedge y < x \\ &\Leftrightarrow y \in [-\infty, x[\end{aligned}$$

so that $\overline{\mathbb{R}} \setminus [x, \infty] = [-\infty, x[\in \mathcal{T}_{\overline{\mathbb{R}}}$ which proves that $[x, \infty]$ is closed.

3. As $\{x\} = [-\infty, x] \cap [x, \infty]$ it follows from (1) and (2) that $\{x\}$ is closed.
4. This is trivial as $\overline{\mathbb{R}} = [-\infty, \infty]$.
5. As $\overline{\mathbb{R}} \setminus \{-\infty, \infty\} = \mathbb{R} \in \mathcal{T}_{\overline{\mathbb{R}}} \subseteq \mathcal{T}_{\overline{\mathbb{R}}}$ it follows that $\{-\infty, \infty\}$ is closed.
6. As $\{-\infty\} = \{-\infty, \infty\} \cap [-\infty, 0]$ it follows from (2) and (5) that $\{-\infty\}$ is closed.
7. As $\{\infty\} = \{-\infty, \infty\} \cap [0, \infty]$ it follows from (1) and (5) that $\{\infty\}$ is closed. □

Theorem 20.53.

1. If $x \in \mathbb{R}$ then and $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ with $x \in U$ then $\exists \varepsilon \in \mathbb{R}^+$ such that $x \in]x - \varepsilon, x + \varepsilon[\subseteq U$.
2. If $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ with $-\infty \in U$ then $\exists \varepsilon \in \mathbb{R}^+$ such that $-\infty \in [-\infty, -\varepsilon[\subseteq U$.
3. If $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ with $\infty \in U$ then $\exists \varepsilon \in \mathbb{R}^+$ such that $\infty \in]\varepsilon, \infty] \subseteq U$.

Proof.

1. If $x \in U$ then as $\mathcal{B}_{\overline{\mathbb{R}}}$ is the generating basis for $\mathcal{T}_{\overline{\mathbb{R}}}$ there exist a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ such that $x \in B \subseteq U$. For B we have either:

$B \in \mathcal{B}_{\parallel}$. As $\mathcal{B}_{\parallel} \stackrel{[\text{theorem: 14.104}]}{=} \{]y - \varepsilon, y + \varepsilon[\mid y \in \mathbb{R} \wedge \varepsilon \in \mathbb{R}^+ \}$ there exists a $y \in \mathbb{R}$ and a $\delta \in \mathbb{R}^+$ so that $B =]y - \delta, y + \delta[$. As $x \in B$ we have $y - \delta < x < y + \delta$, define $\varepsilon = \min(x - (y - \delta), (y + \delta) - x)$ then if $z \in]x - \varepsilon, x + \varepsilon[$ we have

$$y - \delta = x - (x - (y - \delta)) \leq x - \varepsilon < z < x + \varepsilon \leq x + (y + \delta) - x = y + \delta$$

so that $x \in B$. Hence $x \in]x - \varepsilon, x + \varepsilon[\subseteq B \subseteq U$.

$B \in \mathcal{B}_{-\infty}$. Then there exist a $c \in \mathbb{R}$ such that $B = [-\infty, c[$, as $x \in \mathbb{R}$ and $x \in B$ we have $-\infty < x < c$. Take $\varepsilon = c - x \in \mathbb{R}^+$ then if $y \in]x - \varepsilon, x + \varepsilon[\subseteq \mathbb{R}$ we have $-\infty < y < x + \varepsilon = x + (c - x) = c$ so that $x \in]x - \varepsilon, x + \varepsilon[\subseteq [-\infty, c[= B \subseteq U$.

$B \in \mathcal{B}_{||}$. Then there exist a $c \in \mathbb{R}$ such that $B =]c, \infty]$, as $x \in \mathbb{R}$ and $x \in B$ we have $c < x < \infty$. Take $\varepsilon = x - c \in \mathbb{R}$ then if $y \in]x - \varepsilon, x + \varepsilon[\subseteq \mathbb{R}$ we have $c = x - (x - c) = x - \varepsilon < y < \infty$ so that $x \in]x - \varepsilon, x + \varepsilon[\subseteq]c, \infty] = B \subseteq U$.

2. As $\mathcal{B}_{\overline{\mathbb{R}}}$ is the generating basis for $\mathcal{T}_{\overline{\mathbb{R}}}$ there exists a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ such that $-\infty \in B \subseteq U$. For B we have either:

$B \in \mathcal{B}_{||}$. Then there exists $a, b \in \mathbb{R}$ such that $B =]a, b[$, as $-\infty \in B$ we have $-\infty < a < -\infty$ a contradiction. So this case does not apply.

$B \in \mathcal{B}_{-\infty}$. Then there exists a $c \in \mathbb{R}$ such that $B = [-\infty, c[$, take $\varepsilon = -\min(c, -1) \in \mathbb{R}^+$. If $y \in [-\infty, -\varepsilon[$ we have that $-\infty \leq y < -\varepsilon = \min(c, -1) \leq c$ so that $-\infty \in [-\infty, -\varepsilon[\subseteq B \subseteq U$.

$B \in \mathcal{B}_{\infty}$. Then there exists a $c \in \mathbb{R}$ such that $B =]c, \infty]$, as $-\infty \in B$ we have that $-\infty < c < -\infty$ a contradiction. So this case does not apply.

3. As $\mathcal{B}_{\overline{\mathbb{R}}}$ is the generating basis for $\mathcal{T}_{\overline{\mathbb{R}}}$ there exists a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ such that $\infty \in B \subseteq U$. For B we have either:

$B \in \mathcal{B}_{||}$. Then there exists $a, b \in \mathbb{R}$ such that $B =]a, b[$, as $\infty \in B$ we have $\infty < b < \infty$ a contradiction. So this case does not apply.

$B \in \mathcal{B}_{-\infty}$. Then there exists a $c \in \mathbb{R}$ such that $B = [-\infty, c[$, as $\infty \in B$ we have that $\infty < c < \infty$ a contradiction. So this case does not apply.

$B \in \mathcal{B}_{\infty}$. Then there exists a $c \in \mathbb{R}$ such that $B =]c, \infty]$, take $\varepsilon = \max(1, c) \in \mathbb{R}^+$. If $y \in]\varepsilon, \infty]$ then $c \leq \varepsilon < y \leq \infty$ so that $\infty \in]\varepsilon, \infty] \subseteq B \subseteq U$. $\square$

We show now that continuous functions with codomain $\mathbb{R}$ are also continuous if we use the codomain $\overline{\mathbb{R}}$.

Theorem 20.54. *Let $\langle X, \mathcal{T} \rangle$ be a topological space and let $f: X \rightarrow \mathbb{R}$ be a continuous function [using the topology $\mathcal{T}_{||}$] then $f: X \rightarrow \overline{\mathbb{R}}$ is also continuous using the topology $\mathcal{T}_{\overline{\mathbb{R}}}$.*

Proof. Let $x \in X$ and let $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ be such that $f(x) \in U$. As $f(X) \subseteq \mathbb{R}$ we have that $f(x) \in \mathbb{R}$, then by [theorem: 20.53] there exist a $\varepsilon \in \mathbb{R}^+$ such that $f(x) \in]f(x) - \varepsilon, f(x) + \varepsilon[\subseteq U$. As $f: X \rightarrow \mathbb{R}$ is continuous and $]f(x) - \varepsilon, f(x) + \varepsilon[$ is open in $\mathbb{R}$ it follows that $V = f^{-1}(]f(x) - \varepsilon, f(x) + \varepsilon[) \in \mathcal{T}$. As $f(x) \in]f(x) - \varepsilon, f(x) + \varepsilon[$ we have that $x \in V$, further

$$f(V) = f(f^{-1}(]f(x) - \varepsilon, f(x) + \varepsilon[)) \subseteq]f(x) - \varepsilon, f(x) + \varepsilon[\subseteq U$$

So using [theorem: 14.130] f is continuous at x , as $x \in X$ was chosen arbitrary it follows that $f: X \rightarrow \overline{\mathbb{R}}$ is continuous. $\square$

Theorem 20.55. *If $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ then for $x \in \mathbb{R}$ we have that*

1. $x + U = \{x + u | u \in U\} \in \mathcal{T}_{\overline{\mathbb{R}}}$ [in other words if U is open then $x + U$ is open]
2. If $x \neq 0$ then $x \cdot U = \{x \cdot u | u \in U\} \in \mathcal{T}_{\overline{\mathbb{R}}}$ [in other words if U is open then $x \cdot U$ is open]

Proof.

1. Let $y \in x + U$ then there exist a $u \in U$ such that $y = x + u$. For $u \in U$ we have either:

$u \in \mathbb{R}$. Using [theorem: 20.53] there exists a $\varepsilon \in \mathbb{R}^+$ such that

$$u \in]u - \varepsilon, u + \varepsilon[\subseteq U$$

Let $z \in]y - \varepsilon, y + \varepsilon[$ then $(x + u) - \varepsilon = y - \varepsilon < z < y + \varepsilon = (x + u) + \varepsilon$ hence $u - \varepsilon < z - x < u + \varepsilon$ or $z - x \in]u - \varepsilon, u + \varepsilon[\subseteq U$ so that $z = z + (z - x) \in x + U$, hence $y \in]y - \varepsilon, y + \varepsilon[\subseteq x + U$. As $]y - \varepsilon, y + \varepsilon[\in \mathcal{B}_{||} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$ we have by taking $B =]y - \varepsilon, y + \varepsilon[$ we have that

$$\exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } y \in B \subseteq x + U$$

$u = -\infty$. Using [theorem: 20.53] there exists a $\varepsilon \in \mathbb{R}^+$ such that

$$u = -\infty \in [-\infty, -\varepsilon[\subseteq U$$

Let $z \in [-\infty, -\varepsilon + x[$ then $-\infty \leq z < -\varepsilon + x$ so that $-\infty \leq z - x < -\varepsilon$ proving that $z - x \in [-\infty, -\varepsilon[\subseteq U$, hence $z = x + (z - x) \in x + U$ or $[-\infty, -\varepsilon + x[\subseteq x + U$. As $y = x + u = x + (-\infty) = -\infty$ it follows that $y \in [-\infty, -\varepsilon + x[\subseteq x + U$. Hence if we take $B = [-\infty, -\varepsilon + x[\in \mathcal{B}_{-\infty} \subseteq \mathcal{B}_{\mathbb{R}}$ we have that

$$\exists B \in \mathcal{B}_{\mathbb{R}} \text{ such that } y \in B \subseteq x + U$$

$u \in \infty$. Using [theorem: 20.53] there exists a $\varepsilon \in \mathbb{R}^+$ such that

$$u = \infty \in]\varepsilon, \infty] \subseteq U$$

Let $z \in]\varepsilon + x, \infty]$ then $\varepsilon + x < z \leq \infty$ so that $\varepsilon < z - x \leq \infty$ proving that $z - x \in]\varepsilon, \infty] \subseteq U$, hence $z = x + (z - x) \in x + U$ or $] \varepsilon + x, \infty] \subseteq x + U$. As $y = x + u = x + \infty = \infty$ it follows that $y \in]\varepsilon + x, \infty] \subseteq x + U$. Hence if we take $B =]\varepsilon + x, \infty] \in \mathcal{B}_{-\infty} \subseteq \mathcal{B}_{\mathbb{R}}$ we have that

$$\exists B \in \mathcal{B}_{\mathbb{R}} \text{ such that } y \in B \subseteq x + U$$

As $y \in x + U$ was chosen arbitrary and in all cases we found a $B \in \mathcal{B}_{\mathbb{R}}$ [the generating basis of $\mathcal{T}_{\mathbb{R}}$] such that $y \in B \subseteq x + U$ it follows that $x + U \in \mathcal{T}_{\mathbb{R}}$.

2. If $y \in x \cdot U$ then there exist a $u \in U$ such that $y = x \cdot u$. For $u \in U$ we have the following possibilities.

$u \in \mathbb{R}$. Using [theorem: 20.53] there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$u \in]u - \varepsilon, u + \varepsilon[\subseteq U$$

For $x \neq 0$ we have now either:

$0 < x$. Then $x \cdot \varepsilon \in \mathbb{R}^+$ so that $x \cdot u - x \cdot \varepsilon < x \cdot u < x \cdot u + x \cdot \varepsilon$. so that

$$y = x \cdot u \in]x \cdot u - x \cdot \varepsilon, x \cdot u + x \cdot \varepsilon[$$

Further if $z \in]x \cdot u - x \cdot \varepsilon, x \cdot u + x \cdot \varepsilon[$ then $x \cdot u - x \cdot \varepsilon < z < x \cdot u + x \cdot \varepsilon$ so, as $0 < x$, $u - \varepsilon < \frac{z}{x} < u + \varepsilon$. Hence $\frac{z}{x} \in]u - \varepsilon, u + \varepsilon[\subseteq U$ so that $z = x \cdot \frac{z}{x} \in x \cdot U$ proving that

$$y = x \cdot u \in]x \cdot u - x \cdot \varepsilon, x \cdot u + x \cdot \varepsilon[\subseteq x \cdot U$$

Take $B =]x \cdot u - x \cdot \varepsilon, x \cdot u + x \cdot \varepsilon[\in \mathcal{B}_{||} \subseteq \mathcal{B}_{\mathbb{R}}$ then we have that

$$\exists B \in \mathcal{B}_{\mathbb{R}} \text{ such that } y \in B \subseteq x \cdot U$$

$x < 0$. Then $-x \cdot \varepsilon \in \mathbb{R}^+$ so that $x \cdot u + x \cdot \varepsilon < x \cdot u < x \cdot u - x \cdot \varepsilon$, so that

$$y = x \cdot u \in]x \cdot u + x \cdot \varepsilon, x \cdot u - x \cdot \varepsilon[$$

Further if $z \in]x \cdot u + x \cdot \varepsilon, x \cdot u - x \cdot \varepsilon[$ then $x \cdot u + x \cdot \varepsilon < z < x \cdot u - x \cdot \varepsilon$ or, as $x < 0 \Rightarrow \frac{1}{x} < 0$ that $u - \varepsilon < \frac{z}{x} < u + \varepsilon$. Hence $\frac{z}{x} \in]u - \varepsilon, u + \varepsilon[\subseteq U$ so that $z = x \cdot \frac{z}{x} \in x \cdot U$ proving that

$$y \in]x \cdot u + x \cdot \varepsilon, x \cdot u - x \cdot \varepsilon[\subseteq x \cdot U$$

Take $B =]x \cdot u + x \cdot \varepsilon, x \cdot u - x \cdot \varepsilon[\in \mathcal{B}_{||} \subseteq \mathcal{B}_{\mathbb{R}}$ then we have that

$$\exists B \in \mathcal{B}_{\mathbb{R}} \text{ such that } y \in B \subseteq x \cdot U$$

As $y \in x \cdot U$ was chosen arbitrary and in all cases we found a $B \in \mathcal{B}_{\mathbb{R}}$ [the generating basis of $\mathcal{T}_{\mathbb{R}}$] such that $y \in B \subseteq x \cdot U$ it follows that $x \cdot U \in \mathcal{T}_{\mathbb{R}}$.

$u = -\infty$. Using [theorem: 20.53] there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$-\infty \in [-\infty, -\varepsilon[\subseteq U$$

For $x \neq 0$ we have either:

$0 < x$. then

$$y = x \cdot (-\infty) = -\infty \in [-\infty, -x \cdot \varepsilon[\text{ where } x \cdot \varepsilon \in \mathbb{R}^+$$

further if $z \in [-\infty, -x \cdot \varepsilon[$ then $-\infty \leq z < -x \cdot \varepsilon$ so that $-\infty \leq \frac{z}{x} < -\varepsilon$ proving that $\frac{z}{x} \in [-\infty, -\varepsilon[\subseteq U$ or that $z = x \cdot \frac{z}{x} \in x \cdot U$. Hence

$$y \in [-\infty, -x \cdot \varepsilon[\subset x \cdot U$$

Take $B = [-\infty, -x \cdot \varepsilon[\in \mathcal{B}_{-\infty} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$ then we have

$$\exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } y \in B \subseteq x \cdot U$$

$x < 0$. then

$$y = x \cdot (-\infty) = \infty \in]-x \cdot \varepsilon, \infty] \text{ where } -x \cdot \varepsilon \in \mathbb{R}^+$$

further if $z \in]-x \cdot \varepsilon, \infty]$ then $-x \cdot \varepsilon < z \leq \infty$ so that $-\infty = \frac{1}{x} \cdot \infty \leq \frac{z}{x} < -x \cdot \varepsilon$ proving that $\frac{z}{x} \in [-\infty, -x \cdot \varepsilon[\subseteq U$ so that $z = x \cdot \frac{z}{x} \in x \cdot U$. Hence

$$y \in]-x \cdot \varepsilon, \infty] \subset x \cdot U$$

Take $B =]-x \cdot \varepsilon, \infty] \in \mathcal{B}_{\infty} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$ then we have

$$\exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } y \in B \subseteq x \cdot U$$

As $y \in x \cdot U$ was chosen arbitrary and in all cases we found a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ [the generating basis of $\mathcal{T}_{\overline{\mathbb{R}}}$] such that $y \in B \subseteq x \cdot U$ it follows that $x \cdot U \in \mathcal{T}_{\overline{\mathbb{R}}}$.

$u = \infty$. Using [theorem: 20.53] there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$\infty \in]\varepsilon, \infty] \subseteq U$$

For $x \neq 0$ we have either:

$0 < x$. then

$$y = x \cdot \infty = \infty \in]x \cdot \varepsilon, \infty] \text{ where } x \cdot \varepsilon \in \mathbb{R}^+$$

further if $z \in]x \cdot \varepsilon, \infty]$ then $x \cdot \varepsilon < z \leq \infty$ so that $\varepsilon < \frac{z}{x} \leq \infty$, proving that $\frac{z}{x} \in]\varepsilon, \infty] \subseteq U$ or that $z = x \cdot \frac{z}{x} \in x \cdot U$. Hence

$$y \in]x \cdot \varepsilon, \infty] \subset x \cdot U$$

Take $B =]x \cdot \varepsilon, \infty] \in \mathcal{B}_{\infty} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$ then we have

$$\exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } y \in B \subseteq x \cdot U$$

$x < 0$. then

$$y = x \cdot \infty = -\infty \in [-\infty, x \cdot \varepsilon[\text{ where } -x \cdot \varepsilon \in \mathbb{R}^+$$

further if $z \in [-\infty, x \cdot \varepsilon[$ then $-\infty \leq z < x \cdot \varepsilon$ so that $\varepsilon < \frac{z}{x} \leq \infty$, proving that $\frac{z}{x} \in]\varepsilon, \infty] \subseteq U$ so that $z = x \cdot \frac{z}{x} \in x \cdot U$. Hence

$$y \in [-\infty, x \cdot \varepsilon[\subset x \cdot U$$

Take $B = [-\infty, x \cdot \varepsilon[\in \mathcal{B}_{-\infty} \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$ then we have that

$$\exists B \in \mathcal{B}_{\overline{\mathbb{R}}} \text{ such that } y \in B \subseteq x \cdot U$$

As $y \in x \cdot U$ was chosen arbitrary and in all cases we found a $B \in \mathcal{B}_{\overline{\mathbb{R}}}$ [the generating basis of $\mathcal{T}_{\overline{\mathbb{R}}}$] such that $y \in B \subseteq x \cdot U$ it follows that $x \cdot U \in \mathcal{T}_{\overline{\mathbb{R}}}$. $\square$

Theorem 20.56. $\mathfrak{T}_{\overline{\mathbb{R}}}$ is Hausdorff [see definition: 14.241].

Proof. Let $x, y \in \overline{\mathbb{R}}$ with $x \neq y$ then we have to check the following cases for x, y :

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then we have as $x \neq y$ we have either:

$x < y$. Take $\varepsilon = \frac{y-x}{2} \in \mathbb{R}^+$ then $x \in]x-\varepsilon, x+\varepsilon[\in \mathcal{T}_{\overline{\mathbb{R}}}$ and $y \in]y-\varepsilon, y+\varepsilon[\in \mathcal{T}_{\overline{\mathbb{R}}}$. Further

$$]x-\varepsilon, x+\varepsilon[=]x-\varepsilon, x + \frac{y-x}{2}[=]x-\varepsilon, \frac{x+y}{2}[$$

and

$$]y - \varepsilon, y + \varepsilon[=]y - \frac{y - x}{2}, y + \varepsilon[=]\frac{x + y}{2}, y + \varepsilon[$$

so that $]x - \varepsilon, x + \varepsilon[\cap]y - \varepsilon, y + \varepsilon[= \emptyset$.

$y < x$. Take $\varepsilon = \frac{x - y}{2} \in \mathbb{R}^+$ then $x \in]x - \varepsilon, x + \varepsilon[\in \mathcal{T}_{\overline{\mathbb{R}}}$ and $y \in]y - \varepsilon, y + \varepsilon[\in \mathcal{T}_{\overline{\mathbb{R}}}$. Further

$$]x - \varepsilon, x + \varepsilon[=]x - \frac{x - y}{2}, x + y[=]\frac{x + y}{2}, x + y[$$

and

$$]y - \varepsilon, y + \varepsilon[=]y - \varepsilon, y + \frac{x - y}{2}[=]y - \varepsilon, \frac{x + y}{2}[$$

so that $]x - \varepsilon, x + \varepsilon[\cap]y - \varepsilon, y + \varepsilon[= \emptyset$.

$x = -\infty \wedge y \in \mathbb{R}$. Then $x = -\infty \in [-\infty, y - 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$, $y \in]y - 1, y + 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$ and $[-\infty, y - 1[\cap]y - 1, y + 1[= \emptyset$.

$x = \infty \wedge y \in \mathbb{R}$. Then $x = \infty \in]y + 1, \infty] \in \mathcal{T}_{\overline{\mathbb{R}}}$, $y \in]y - 1, y + 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$ and $]y + 1, \infty] \cap]y - 1, y + 1[= \emptyset$.

$x \in \mathbb{R} \wedge y = -\infty$. Then $x \in]x - 1, x + 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$, $y = -\infty \in [-\infty, x - 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$ and $]x - 1, x + 1[\cap [-\infty, x - 1[= \emptyset$.

$x = -\infty \wedge y = -\infty$. As $x \neq y$ this is not a valid case.

$x = \infty \wedge y = -\infty$. Then $x \in]1, \infty] \in \mathcal{T}_{\overline{\mathbb{R}}}$, $y \in [-\infty, 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$ and $]1, \infty] \cap [-\infty, 1[= \emptyset$.

$x \in \mathbb{R} \wedge y = \infty$. Then $x \in]x - 1, x + 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$, $y \in]x + 1, \infty] \in \mathcal{T}_{\overline{\mathbb{R}}}$ and $]x - 1, x + 1[\cap]x + 1, \infty] = \emptyset$.

$x = -\infty \wedge y = \infty$. Then $x \in [-\infty, 1[\in \mathcal{T}_{\overline{\mathbb{R}}}$, $y \in]1, \infty] \in \mathcal{T}_{\overline{\mathbb{R}}}$ and $[-\infty, 1[\cap]1, \infty] = \emptyset$.

$x = \infty \wedge y = \infty$. As $x \neq y$ this is not a valid case □

20.4 Sequences in $\overline{\mathbb{R}}$

We will now extend the concept of a limit of sequences in $\mathbb{R}$ to the limit of sequences in $\overline{\mathbb{R}}$ in such a way that the limit of increasing/decreasing sequences always exists. So if we have a series of non negative numbers then the limit of the partial sums will always have a limit, in other words series of non negative numbers always converges. To do this we make use of the fact that in $\overline{\mathbb{R}}$ every non empty subset has a supremum and infimum and define the limit of a sequence in terms of supremum and infimum.

Theorem 20.57. *Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be a sequence of extended real numbers then we have:*

1. $\inf(\{x_i | i \in \{k, \dots, \infty\}\})$ and $\sup(\{x_i | i \in \{k, \dots, \infty\}\})$ exists
2. $\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$ exists
3. $\sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$ exists
4. $\forall m \in \{k, \dots, \infty\}$ we have

$$\sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{m, \dots, \infty\}\})$$

5. $\forall m \in \{k, \dots, \infty\}$ we have

$$\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{m, \dots, \infty\}\})$$

Proof.

1. As $\{x_i | i \in \{k, \dots, \infty\}\} \neq \emptyset$ this follows from [theorem: 20.13].
2. As $\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\} \neq \emptyset$ this follows from [theorem: 20.13].

3. As $\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\} \neq \emptyset$ this follows from [theorem: 20.13].
4. Given $l \in \{k, \dots, \infty\}$ take $S_l = \inf(\{x_i | i \in \{l, \dots, \infty\}\})$ then we have

$$\{S_l | l \in \{m, \dots, \infty\}\} \subseteq \{S_l | l \in \{k, \dots, \infty\}\}$$

so that by [theorem: 3.77]

$$\sup(\{S_l | l \in \{m, \dots, \infty\}\}) \leq \sup(\{S_l | l \in \{k, \dots, \infty\}\}) \quad (20.4)$$

For the opposite inclusion, given $l \in \{k, \dots, \infty\}$ take $n = \max(m, l)$ then $m, l \leq n$ so that

$$\{x_i | i \in \{l, \dots, \infty\}\} \subseteq \{x_i | i \in \{n, \dots, \infty\}\}$$

or using [theorem: 20.13]

$$S_n = \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \leq \inf(\{x_i | i \in \{l, \dots, \infty\}\}) = S_l$$

Hence for every $S_l \in \{S_l | l \in \{k, \dots, \infty\}\}$ we find a $S_n \in \{S_l | l \in \{m, \dots, \infty\}\}$ such that $S_n \leq S_l$ which by [theorem: 3.78] proves that $\sup(\{S_l | l \in \{k, \dots, \infty\}\}) \leq \sup(\{S_l | l \in \{m, \dots, \infty\}\})$. Combining this result with [eq: 20.4] gives

$$\sup(\{S_l | l \in \{m, \dots, \infty\}\}) = \sup(\{S_l | l \in \{k, \dots, \infty\}\})$$

or

$$\sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{m, \dots, \infty\}\})$$

5. Given $l \in \{k, \dots, \infty\}$ take $S_l = \sup(\{x_i | i \in \{l, \dots, \infty\}\})$ then we have

$$\{S_l | l \in \{m, \dots, \infty\}\} \subseteq \{S_l | l \in \{k, \dots, \infty\}\}$$

so that by [theorem: 3.77]

$$\inf(\{S_l | l \in \{k, \dots, \infty\}\}) \leq \inf(\{S_l | l \in \{m, \dots, \infty\}\}) \quad (20.5)$$

For the opposite inclusion, given $l \in \{k, \dots, \infty\}$ take $n = \max(m, l)$ then $m, l \leq n$ so that

$$\{x_i | i \in \{l, \dots, \infty\}\} \subseteq \{x_i | i \in \{n, \dots, \infty\}\}$$

or using [theorem: 20.13]

$$S_n = \sup(\{x_i | i \in \{l, \dots, \infty\}\}) \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\}) = S_l$$

Hence for every $S_l \in \{S_l | l \in \{k, \dots, \infty\}\}$ we find a $S_n \in \{S_l | l \in \{m, \dots, \infty\}\}$ such that $S_l \leq S_n$ which by [theorem: 3.78] proves that $\inf(\{S_l | l \in \{m, \dots, \infty\}\}) \leq \inf(\{S_l | l \in \{k, \dots, \infty\}\})$. Combining this result with [eq: 20.5] gives

$$\inf(\{S_l | l \in \{m, \dots, \infty\}\}) = \inf(\{S_l | l \in \{k, \dots, \infty\}\})$$

or

$$\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{m, \dots, \infty\}\}) \quad \square$$

Using the above theorem we can then define:

Definition 20.58. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be a sequence of extended real numbers then we define:

$$\liminf_{i \rightarrow \infty} x_i = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$$

$$\limsup_{i \rightarrow \infty} x_i = \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$$

Note 20.59. () The previous theorem [theorem: 20.57] ensures that the above definition is well defined for every sequence of extended real numbers. Note also that the notations $\liminf_{i \rightarrow \infty} x_i$, $\limsup_{i \rightarrow \infty} x_i$ do not mention the starting index k , which is indeed not needed because of [theorem: 20.57 (4), (5)].

Example 20.60. Let $k \in \mathbb{N}_0$, $x \in \overline{\mathbb{R}}$ then for $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ defined by $x_i = x$ we have that

$$\liminf_{i \rightarrow \infty} x_i = x = \limsup_{i \rightarrow \infty} x_i$$

Proof.

$$\begin{aligned} \liminf_{i \rightarrow \infty} x_i &= \sup (\{\inf (\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \sup (\{\inf (\{x\}) | l \in \{k, \dots, \infty\}\}) \\ &= \sup (\{x | n \in \{k, \dots, \infty\}\}) \\ &= \sup (\{x\}) \\ &= x \\ &= \inf (\{x\}) \\ &= \inf (\{x | n \in \{k, \dots, \infty\}\}) \\ &= \inf (\{\sup (\{x\}) | l \in \{k, \dots, \infty\}\}) \\ &= \inf (\{\sup (\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \limsup_{n \rightarrow \infty} x_n \end{aligned}$$

□

Theorem 20.61. Let $k \in \mathbb{N}_0$, $x \in \overline{\mathbb{R}}$ then for $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ we have $\forall n \in \{k, \dots, \infty\}$ that

$$\liminf_{i \rightarrow \infty} x_i = \liminf_{i \rightarrow \infty} x_{i+n} \text{ and } \limsup_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_{i+n}$$

Proof. Let $l \in \{k, \dots, \infty\}$ then we have

$$\begin{aligned} x \in \{x_{i+n} | i \in \{l, \dots, \infty\}\} &\Leftrightarrow x = x_{i+n} \text{ where } i \in \{l, \dots, \infty\} \\ &\Leftrightarrow x = x_i \text{ where } i \in \{l+n, \dots, \infty\} \\ &\Leftrightarrow x \in \{x_i | i \in \{l+n, \dots, \infty\}\} \end{aligned}$$

so that

$$\forall l \in \{k, \dots, \infty\} \{x_{i+n} | i \in \{l, \dots, \infty\}\} = \{x_i | i \in \{l+n, \dots, \infty\}\} \quad (20.6)$$

Let $l \in \{k, \dots, \infty\}$ and define $I_l = \inf (\{x_i | i \in \{l, \dots, \infty\}\})$ and $S_l = \sup (\{x_i | i \in \{l, \dots, \infty\}\})$ then we have

$$\begin{aligned} x \in \{I_{l+n} | l \in \{k, \dots, \infty\}\} &\Leftrightarrow x = I_{l+n} \text{ where } l \in \{k, \dots, \infty\} \\ &\Leftrightarrow x = I_l \text{ where } l \in \{k+n, \dots, \infty\} \\ &\Leftrightarrow x \in \{I_l | l \in \{k+n, \dots, \infty\}\} \\ x \in \{S_{l+n} | l \in \{k, \dots, \infty\}\} &\Leftrightarrow x = S_{l+n} \text{ where } l \in \{k, \dots, \infty\} \\ &\Leftrightarrow x = S_l \text{ where } l \in \{k+n, \dots, \infty\} \\ &\Leftrightarrow x \in \{S_l | l \in \{k+n, \dots, \infty\}\} \end{aligned}$$

so that

$$\{I_{l+n} | l \in \{k, \dots, \infty\}\} = \{I_l | l \in \{k+n, \dots, \infty\}\} \wedge \{S_{l+n} | l \in \{k, \dots, \infty\}\} = \{S_l | l \in \{k+n, \dots, \infty\}\} \quad (20.7)$$

So

$$\begin{aligned} \liminf_{i \rightarrow \infty} x_{i+n} &= \sup (\{\inf (\{x_{i+n} | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.6}]}{=} \sup (\{\inf (\{x_i | i \in \{l+n, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.7}]}{=} \sup (\{\inf (\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k+n, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.57}]}{=} \sup (\{\inf (\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \liminf_{i \rightarrow \infty} x_i \end{aligned}$$

and

$$\begin{aligned}
 \limsup_{i \rightarrow \infty} x_{i+n} &= \inf(\{\sup(\{x_{i+n} | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\stackrel{[\text{eq: 20.6}]}{=} \inf(\{\sup(\{x_i | i \in \{l+n, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\stackrel{[\text{eq: 20.7}]}{=} \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k+n, \dots, \infty\}\}) \\
 &\stackrel{[\text{theorem: 20.57}]}{=} \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &= \limsup_{i \rightarrow \infty} x_i
 \end{aligned}$$

□

Theorem 20.62. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ and $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be such that $\forall i \in \{k, \dots, \infty\} \ x_i \leq y_i$ then

1. $\liminf_{i \rightarrow \infty} x_i \leq \liminf_{i \rightarrow \infty} y_i$.
2. $\limsup_{i \rightarrow \infty} x_i \leq \limsup_{i \rightarrow \infty} y_i$.

Proof.

1. Let $l \in \{k, \dots, \infty\}$ then using [theorem: 3.78] we have that

$$\inf(\{x_i | i \in \{l, \dots, \infty\}\}) \leq \inf(\{y_i | i \in \{l, \dots, \infty\}\})$$

Applying [theorem: 3.78] again we have that

$$\begin{aligned}
 \liminf_{n \rightarrow \infty} x_n &\stackrel{\text{def}}{=} \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\leq \sup(\{\inf(\{y_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &= \liminf_{n \rightarrow \infty} y_n
 \end{aligned}$$

2. Let $l \in \{k, \dots, \infty\}$ then using [theorem: 3.78] we have that

$$\sup(\{x_i | i \in \{l, \dots, \infty\}\}) \leq \sup(\{y_i | i \in \{l, \dots, \infty\}\})$$

Applying [theorem: 3.78] again we have that

$$\begin{aligned}
 \liminf_{n \rightarrow \infty} x_n &\stackrel{\text{def}}{=} \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\leq \sup(\{\inf(\{y_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &= \liminf_{n \rightarrow \infty} y_n
 \end{aligned}$$

□

Theorem 20.63. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ and $x \in \overline{\mathbb{R}}$ then we have:

1. If $\forall i \in \{k, \dots, \infty\}$ we have $x \leq x_i$ then $x \leq \liminf_{i \rightarrow \infty} x_i$ and $x \leq \limsup_{i \rightarrow \infty} x_i$.
2. If $\forall i \in \{k, \dots, \infty\}$ we have $x_i \leq x$ then $\liminf_{i \rightarrow \infty} x_i \leq x$ and $\limsup_{i \rightarrow \infty} x_i \leq x$.

Proof.

1. Let $l \in \{k, \dots, \infty\}$ then $\forall i \in \{l, \dots, \infty\} \subseteq \{k, \dots, \infty\}$ we have $x \leq x_i$ so that

$$x \leq \inf(\{x_i | i \in \{l, \dots, \infty\}\}) \text{ and } x \leq \sup(\{x_i | i \in \{l, \dots, \infty\}\})$$

hence we have that

$$x \leq \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \liminf_{i \rightarrow \infty} x_i$$

and

$$x \leq \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \limsup_{i \rightarrow \infty} x_i$$

2. Let $l \in \{k, \dots, \infty\}$ then $\forall i \in \{l, \dots, \infty\} \subseteq \{k, \dots, \infty\}$ we have $x_i \leq x$ so that

$$\inf(\{x_i | i \in \{l, \dots, \infty\}\}) \leq x \text{ and } \sup(\{x_i | i \in \{l, \dots, \infty\}\}) \leq x$$

hence we have that

$$\liminf_{i \rightarrow \infty} x_i = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \leq x$$

and

$$\limsup_{i \rightarrow \infty} x_i = \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \leq x \quad \square$$

Lemma 20.64. *Let $\emptyset \neq S \subseteq \overline{\mathbb{R}}$ then we have:*

1. *If $y \in \mathbb{R}$ then*

- a. *$\sup(S) + y$ is well defined then for $S + y = \{s + y = s \in S\}$ we have that*

$$\sup(S + y) = \sup(S) + y$$

- b. *$\inf(S) + y$ is well defined then for $S + y = \{s + y = s \in S\}$ we have that*

$$\inf(S + y) = \inf(S) + y$$

2. *If $y \in [0, \infty]$ and $S \subseteq [0, \infty]$ we have that $\sup(S) + y$ is well defined and for $S + y = \{s + y | s \in S\}$ we have that*

$$\sup(S + y) = \sup(S) + y$$

3. *If $y \in [0, \infty]$ and $S \subseteq [0, \infty]$ we have that $\inf(S) + y$ is well defined and for $S + y = \{s + y | s \in S\}$ we have that*

$$\inf(S + y) = \inf(S) + y$$

Proof.

1. As $y \in \mathbb{R}$ we have $\forall x \in \overline{\mathbb{R}}$ that $x + y$ is well defined, $\sup(S)$ exists and $\sup(S) \in \overline{\mathbb{R}}$ it follows that $\sup(S + y)$ is well defined and $S + y$ is well defined.

- a. $\forall s \in S$ we have that $s \leq \sup(S)$ so by [theorem: 20.29] $s + y \leq \sup(S)$ so that

$$\sup(\{s + y | s \in S\}) \leq \sup(S) + y \quad (20.8)$$

Assume that $\sup(\{s + y | s \in S\}) < \sup(S) + y$ then by [theorem: 20.28]

$$\sup(\{s + y | s \in S\}) - y < \sup(S)$$

so there exist a $s \in S$ such that $\sup(\{s + y | s \in S\}) - y < s$ so that by [theorem: 20.29] $\sup(\{s + y | s \in S\}) < s + y \leq \sup(\{s + y | s \in S\})$ leading to the contradiction $\sup(\{s + y | s \in S\}) < \sup(\{s + y | s \in S\})$. Hence the assumption is wrong and we must have $\sup(S) + y \leq \sup(\{s + y | s \in S\})$. Combining this with [eq: 20.8] gives

$$\sup(S + y) = \sup(\{s + y | s \in S\}) = \sup(S) + y$$

- b. $\forall s \in S$ we have that $\inf(S) \leq s$ so by [theorem: 20.29] $\inf(S) + y \leq s + y$ so that

$$\inf(S) + y \leq \inf(\{s + y | s \in S\}) \quad (20.9)$$

Assume that $\inf(S) + y < \inf(\{s + y | s \in S\})$ then by [theorem: 20.28]

$$\inf(S) < \inf(\{s + y | s \in S\}) - y$$

so there exist a $s \in S$ such that $s < \inf(\{s + y | s \in S\}) - y$ so that by [theorem: 20.29] $\inf(\{s + y | s \in S\}) \leq s + y < \inf(\{s + y | s \in S\})$ leading to the contradiction $\inf(\{s + y | s \in S\}) < \inf(\{s + y | s \in S\})$. Hence the assumption is wrong and we must have $\inf(\{s + y | s \in S\}) \leq \inf(S) + y$. Combining this with [eq: 20.9] gives

$$\inf(S + y) = \inf(\{s + y | s \in S\}) = \inf(S) + y$$

2. For $y \in [0, \infty]$ we have either:

$y \in \mathbb{R}$. Then by (1.a) we have that $\sup(A) + y$ is well defined and

$$\sup(\{s + y | s \in S\}) = \sup(S) + y$$

$y = \infty$. Then as $S \subseteq [0, \infty]$ we have $\forall s \in S$ that $s + y = \infty$ so that $\{s + y | s \in S\} = \{\infty\}$ so that $\sup(\{s + y | s \in S\}) = \infty$. Further as $\forall s \in S$ we have $0 \leq s$ it follows that $-\infty < 0 \leq \sup(S)$ so that $\infty = \sup(S) + \infty = \sup(S) + y$ is well defined. Hence

$$\sup(\{s + y | s \in S\}) = \infty = \sup(S) + \infty = \sup(S) + y$$

3. For $y \in [-\infty, 0]$ we have either:

$y \in \mathbb{R}$. Then by (1.b) we have that $\inf(A) + y$ is well defined and

$$\inf(\{s + y | s \in S\}) = \inf(S) + y$$

$y = -\infty$. Then as $S \subseteq [-\infty, 0]$ we have $\forall s \in S$ that $s + y = -\infty$ so that $\{s + y | s \in S\} = \{-\infty\}$ so that $\inf(\{s + y | s \in S\}) = -\infty$. Further as $\forall s \in S$ we have $s \leq 0$ it follows that $\inf(S) \leq 0 < \infty$ so that $-\infty = \inf(S) + (-\infty) = \inf(S) + y$ is well defined. Hence

$$\inf(\{s + y | s \in S\}) = -\infty = \inf(S) + (-\infty) = \inf(S) + y \quad \square$$

Theorem 20.65. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$, $x \in \mathbb{R}$ [hence $\forall i \in \{k, \dots, \infty\}$ we have that $x_i + x$ is well defined] then for $\{x_i + x\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ we have

1. $\liminf_{i \rightarrow \infty} (x_i + x) = \left(\liminf_{i \rightarrow \infty} x_i \right) + x$
2. $\limsup_{i \rightarrow \infty} (x_i + x) = \left(\limsup_{i \rightarrow \infty} x_i \right) + x$

Proof.

1. We have

$$\begin{aligned} \liminf_{i \rightarrow \infty} (x_i + x) &= \sup(\{\inf(\{x_i + x | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.64}]}{=} \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) + x | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.64}]}{=} \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) + x \\ &= \left(\liminf_{i \rightarrow \infty} x_i \right) + x \end{aligned}$$

2. We have

$$\begin{aligned} \limsup_{i \rightarrow \infty} (x_i + x) &= \inf(\{\sup(\{x_i + x | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.64}]}{=} \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) + x | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.64}]}{=} \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) + x \\ &= \left(\limsup_{i \rightarrow \infty} x_i \right) + x \end{aligned} \quad \square$$

Theorem 20.66. Let $\emptyset \neq S \subseteq \overline{\mathbb{R}}$ then

1. $-\sup(S) = \inf(\{-s | s \in S\})$
2. $-\inf(S) = \sup(\{-s | s \in S\})$
3. If $\alpha \in [0, \infty[$ then $\alpha \cdot \sup(S) = \sup(\{\alpha \cdot s | s \in S\})$
4. If $\alpha \in [0, \infty[$ then $\alpha \cdot \inf(S) = \inf(\{\alpha \cdot s | s \in S\})$

Proof.

1. Let $s \in S$ then by [theorem: 20.37] $s \leq \sup(S)$ so that $-\sup(S) \leq -s$ so that

$$-\sup(S) \leq \inf(\{-s | s \in S\}) \quad (20.10)$$

Assume that $-\sup(S) < \inf(\{-s | s \in S\})$ then by [corollary: 20.41] $-\inf(\{-s | s \in S\}) < \sup(S)$ so that there exist a $s \in S$ such that $-\inf(\{-s | s \in S\}) < s$ so that by [corollary: 20.41] again we have $-s < \inf(\{-s | s \in S\})$, hence $\inf(\{-s | s \in S\}) < -s < \inf(\{-s | s \in S\})$ a contradiction. So the assumption must be false and we have $\inf(\{-s | s \in S\}) \leq -\sup(S)$, combining this with [eq: 20.10] proves that

$$-\sup(S) = \inf(\{-s | s \in S\})$$

2. Take $A = \{-s | s \in S\}$ then $S = \{-s | s \in A\}$ then

$$-\inf(S) = -\inf(\{-s | s \in A\}) \stackrel{(1)}{=} -(-\sup(A)) = \sup(\{-s | s \in S\})$$

3. For $\alpha \in [0, \infty[$ we have either:

$\alpha = 0$. Then $\{\alpha \cdot s | s \in S\} = \{0\}$ so that

$$\sup(\{\alpha \cdot s | s \in S\}) = \sup(\{0\}) = 0 = 0 \cdot \sup(S) = \alpha \cdot \sup(S)$$

proving that $\alpha \cdot \sup(S) = \sup(\{\alpha \cdot s | s \in S\})$.

$\alpha \neq 0$. Let $s \in S$ then $s \leq \sup(s)$ then, as $0 < \alpha$, we have by [theorem: 20.40] that $\alpha \cdot s \leq \alpha \cdot \sup(S)$. Hence

$$\sup(\{\alpha \cdot s | s \in S\}) \leq \alpha \cdot \sup(S) \quad (20.11)$$

Assume that $\sup(\{\alpha \cdot s | s \in S\}) < \alpha \cdot \sup(S)$ then as $0 < \alpha \in \mathbb{R}$ we have $0 < \frac{1}{\alpha}$ so that by [theorem: 20.40] $\frac{1}{\alpha} \cdot \sup(\{\alpha \cdot s | s \in S\}) < \sup(S)$. Hence there exist a $s \in S$ such that $\frac{1}{\alpha} \cdot \sup(\{\alpha \cdot s | s \in S\}) < s$ so that by [theorem: 20.40] again we have $\sup(\{\alpha \cdot s | s \in S\}) < \alpha \cdot s \leq \sup(\{\alpha \cdot s | s \in S\})$ a contradiction. Hence the assumption must be false and we have that $\alpha \cdot \sup(S) \leq \sup(\{\alpha \cdot s | s \in S\})$, combining this with [eq: 20.11] we have

$$\alpha \cdot \sup(S) = \sup(\{\alpha \cdot s | s \in S\})$$

4. For $\alpha \in [0, \infty[$ we have either:

$\alpha = 0$. Then $\{\alpha \cdot s | s \in S\} = \{0\}$ so that

$$\inf(\{\alpha \cdot s | s \in S\}) = \inf(\{0\}) = 0 = 0 \cdot \inf(S) = \alpha \cdot \inf(S)$$

proving that $\alpha \cdot \inf(S) = \inf(\{\alpha \cdot s | s \in S\})$.

$\alpha \neq 0$. Let $s \in S$ then $\inf(s) \leq s$ then, as $0 < \alpha$, we have by [theorem: 20.40] that $\alpha \cdot \inf(S) \leq \alpha \cdot s$. Hence

$$\alpha \cdot \inf(S) \leq \inf(\{\alpha \cdot s | s \in S\}) \quad (20.12)$$

Assume that $\alpha \cdot \inf(S) < \inf(\{\alpha \cdot s | s \in S\})$ then as $0 < \alpha \in \mathbb{R}$ we have $0 < \frac{1}{\alpha}$ so that by [theorem: 20.40] $\inf(S) < \frac{1}{\alpha} \cdot \inf(\{\alpha \cdot s | s \in S\})$. Hence there exist a $s \in S$ such that $s < \frac{1}{\alpha} \cdot \inf(\{\alpha \cdot s | s \in S\})$ so that by [theorem: 20.40] again we have $\inf(\{\alpha \cdot s | s \in S\}) \leq \alpha \cdot s < \inf(\{\alpha \cdot s | s \in S\})$ a contradiction. Hence the assumption must be false and we have that $\inf(\{\alpha \cdot s | s \in S\}) \leq \alpha \cdot \inf(S)$, combining this with [eq: 20.12] we have

$$\alpha \cdot \inf(S) = \inf(\{\alpha \cdot s | s \in S\}) \quad \square$$

Theorem 20.67. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ then we have:

1. $\liminf_{i \rightarrow \infty} (-x_i) = -\limsup_{i \rightarrow \infty} x_i$.

$$2. \limsup_{i \rightarrow \infty} (-x_i) = -\liminf_{i \rightarrow \infty} x_i.$$

Proof.

1.

$$\begin{aligned} \liminf_{i \rightarrow \infty} (-x_i) &= \sup(\{\inf(\{-x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.66}]}{=} \sup(\{-\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.66}]}{=} -\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= -\limsup_{i \rightarrow \infty} x_i \end{aligned}$$

2.

$$\begin{aligned} \limsup_{i \rightarrow \infty} (-x_i) &= \inf(\{\sup(\{-x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.66}]}{=} \inf(\{-\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{theorem: 20.66}]}{=} -\sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= -\liminf_{i \rightarrow \infty} x_i \end{aligned}$$

□

Lemma 20.68. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$, $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ such that $\forall i \in \{k, \dots, \infty\}$ we have [so that $x_i + y_i$ is well defined] then we have:

1. If $\forall i \in \{k, \dots, \infty\}$ $x_i \leq x_{i+1} \wedge y_i \leq y_{i+1}$ then

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) = \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

2. If $\forall i \in \{k, \dots, \infty\}$ $x_{i+1} \leq x_i \wedge y_{i+1} \leq y_i$ then

$$\inf(\{x_i | i \in \{k, \dots, \infty\}\}) + \inf(\{y_i | i \in \{k, \dots, \infty\}\}) = \inf(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

Proof.

1. As $\forall i \in \{k, \dots, \infty\}$ $x_i \leq \sup(\{x_i | i \in \{k, \dots, \infty\}\})$ and $y_i \leq \sup(\{y_i | i \in \{k, \dots, \infty\}\})$ so that by [theorem: 20.29] that

$$x_i + y_i \leq \sup(\{x_i | i \in \{k, \dots, \infty\}\}) + y_i \quad (20.13)$$

By [theorem: 20.14] and the fact that $\{x_i | i \in \{k, \dots, \infty\}\}, \{y_i | i \in \{k, \dots, \infty\}\} \neq -\infty$ so that

$$(y_i, \sup(\{x_i | i \in \{k, \dots, \infty\}\})) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

and

$$(\sup(\{y_i | i \in \{k, \dots, \infty\}\}), \sup(\{x_i | i \in \{k, \dots, \infty\}\})) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Hence by [theorem: 20.29] we have that

$$y_i + \sup(\{x_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{y_i | i \in \{k, \dots, \infty\}\}) + \sup(\{x_i | i \in \{k, \dots, \infty\}\})$$

Using the above, [eq: 20.13] and commutativity we have that

$$\begin{aligned} x_i + y_i &\leq \sup(\{y_i | i \in \{k, \dots, \infty\}\}) + \sup(\{x_i | i \in \{k, \dots, \infty\}\}) \\ \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) &\leq \sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) \end{aligned} \quad (20.14)$$

Let $m \in \{k, \dots, \infty\}$ then for every $n \in \{k, \dots, \infty\}$ we have either:

$n < m$. Then $x_m + y_n \leq x_m + y_m \leq \sup(\{x_i + y_i | i \in \{k, \dots, n\}\})$ so that

$$y_n \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) - x_m$$

$m \leq n$. Then $x_m + y_n \leq x_n + y_n \leq \sup(\{x_i + y_i | i \in \{k, \dots, n\}\})$ so that

$$y_n \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) - x_m$$

As in all cases we have $y_n \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) - x_m$ it follows that

$$\forall m \in \{k, \dots, \infty\} \sup(\{y_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) - x_m \quad (20.15)$$

As $\forall i \in \{k, \dots, \infty\} x_i \in \mathbb{R}$ we have by [theorem: 20.14] only the following possibilities for $\sup(\{y_i | i \in \{k, \dots, \infty\}\})$:

sup $(\{y_i | i \in \{k, \dots, \infty\}\}) = \infty$. Then by [eq: 20.15] $\sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) = \infty$ so that

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) = \infty = \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

hence

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

sup $(\{y_i | i \in \{k, \dots, \infty\}\}) \in \mathbb{R}$. Then by [theorems: 20.28, 20.29] and [eq: 20.15] we have

$$x_m \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) - \sup(\{y_i | i \in \{k, \dots, \infty\}\})$$

so that

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\}) - \sup(\{y_i | i \in \{k, \dots, \infty\}\})$$

which by [theorems: 20.28] proves that

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

So in all cases we have

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

which combined with [eq: 20.14] proves that

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{y_i | i \in \{k, \dots, \infty\}\}) = \sup(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

2. Consider the sequences $\{-x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$, $\{-y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ then as $\forall i \in \{k, \dots, \infty\} x_{i+1} \leq x_i \wedge y_{i+1} \leq y_i$ it follows from [theorem: 20.41] that $-x_i \leq -x_{i+1} \wedge -y_i \leq -y_{i+1}$. Hence we can use (1) to get

$$\sup(\{-x_i | i \in \{k, \dots, \infty\}\}) + \sup(\{-y_i | i \in \{k, \dots, \infty\}\}) = \sup(\{-(x_i + y_i) | i \in \{k, \dots, \infty\}\})$$

Then using [theorem: 20.66] on the above we have

$$-\inf(\{x_i | i \in \{k, \dots, \infty\}\}) - \inf(\{y_i | i \in \{k, \dots, \infty\}\}) = -\inf(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

hence

$$\inf(\{x_i | i \in \{k, \dots, \infty\}\}) + \inf(\{y_i | i \in \{k, \dots, \infty\}\}) = \inf(\{x_i + y_i | i \in \{k, \dots, \infty\}\})$$

□

Theorem 20.69. Let $k \in \mathbb{N}_0$ then $\liminf_{i \rightarrow \infty}$ and $\limsup_{i \rightarrow \infty}$ have the following properties

1. If $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ is a sequence of extended reals and $0 \leq \alpha \in \mathbb{R}$ then

$$a. \liminf_{i \rightarrow \infty}(\alpha \cdot x_i) = \alpha \cdot \liminf_{i \rightarrow \infty} x_i$$

$$b. \limsup_{i \rightarrow \infty}(\alpha \cdot x_i) = \alpha \cdot \limsup_{i \rightarrow \infty} x_i$$

2. If $\{x_i\}_{i \in \{k, \dots, n\}}, \{y_i\}_{i \in \{k, \dots, n\}} \subseteq \mathbb{R}$ are sequences of **real numbers** so that $\left(\liminf_{i \rightarrow \infty} x_i\right) + \left(\liminf_{i \rightarrow \infty} y_i\right)$ are well defined then
- $\left(\liminf_{i \rightarrow \infty} x_i\right) + \left(\liminf_{i \rightarrow \infty} y_i\right) \leq \liminf_{i \rightarrow \infty} (x_i + y_i)$
 - $\limsup_{i \rightarrow \infty} (x_i + y_i) \leq \left(\liminf_{i \rightarrow \infty} x_i\right) + \left(\liminf_{i \rightarrow \infty} y_i\right)$

Proof.

1.

$$\begin{aligned}
 \liminf_{i \rightarrow \infty} (\alpha \cdot x_i) &= \sup(\{\inf(\{\alpha \cdot x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\stackrel{[\text{theorem: 20.66}]}{=} \sup(\{\alpha \cdot \inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\stackrel{[\text{theorem: 20.66}]}{=} \alpha \cdot \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &= \alpha \cdot \liminf_{i \rightarrow \infty} x_i
 \end{aligned}$$

and

$$\begin{aligned}
 \limsup_{i \rightarrow \infty} (\alpha \cdot x_i) &= \inf(\{\sup(\{\alpha \cdot x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\stackrel{[\text{theorem: 20.66}]}{=} \inf(\{\alpha \cdot \sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &\stackrel{[\text{theorem: 20.66}]}{=} \alpha \cdot \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\
 &= \alpha \cdot \limsup_{i \rightarrow \infty} x_i
 \end{aligned}$$

2. Let $l \in \{k, \dots, \infty\}$ then we have that $\forall i \in \{l, \dots, \infty\} \inf(\{x_i | i \in \{k, \dots, \infty\}\}) \leq x_i \in \mathbb{R}$ and $\inf(\{y_i | i \in \{k, \dots, \infty\}\}) \leq y_i \in \mathbb{R}$ so that by [theorem: 20.29]

$$\inf(\{x_i | i \in \{l, \dots, \infty\}\}) + y_i \leq x_i + y_i \quad (20.16)$$

Further by [theorem: 20.14] we have that $\inf(\{x_i | i \in \{l, \dots, \infty\}\}), \inf(\{y_i | i \in \{l, \dots, \infty\}\}) \neq \infty$ so that

$$(y_i, \inf(\{x_i | i \in \{l, \dots, \infty\}\})) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

and

$$(\inf(\{y_i | i \in \{l, \dots, \infty\}\}), \inf(\{x_i | i \in \{l, \dots, \infty\}\})) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Hence using [theorem: 20.29] we have that

$$\inf(\{y_i | i \in \{l, \dots, \infty\}\}) + \inf(\{x_i | i \in \{l, \dots, \infty\}\}) \leq y_i + \inf(\{x_i | i \in \{l, \dots, \infty\}\})$$

Using the above, [eq: 20.16] and commutativity we have that

$$\inf(\{x_i | i \in \{l, \dots, \infty\}\}) + \inf(\{y_i | i \in \{l, \dots, \infty\}\}) \leq x_i + y_i$$

So that

$$\inf(\{x_i | i \in \{l, \dots, \infty\}\}) + \inf(\{y_i | i \in \{l, \dots, \infty\}\}) \leq \inf(\{x_i + y_i | i \in \{l, \dots, \infty\}\})$$

Hence

$$\begin{aligned}
 \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) + \inf(\{y_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) &\leq \\
 \sup(\{\inf(\{x_i + y_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) &= \\
 \liminf_{i \rightarrow \infty} (x_i + y_i) &\quad (20.17)
 \end{aligned}$$

Now $\forall l \in \{k, \dots, \infty\}$ we have that

$$\inf(\{x_i | i \in \{l, \dots, \infty\}\}) \leq \inf(\{x_i | i \in \{l+1, \dots, \infty\}\})$$

and

$$\inf(\{y_i | i \in \{l, \dots, \infty\}\}) \leq \inf(\{y_i | i \in \{l+1, \dots, \infty\}\})$$

so that by [theorem: 20.68]

$$\begin{aligned} & \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) + \inf(\{y_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \\ & \sup(\{\{x_i | i \in \{l+1, \dots, \infty\}\} | l \in \{k, \dots, \infty\}\}) + \sup(\{\{y_i | i \in \{l+1, \dots, \infty\}\} | l \in \{k, \dots, \infty\}\}) = \\ & \liminf_{i \rightarrow \infty} x_i + \liminf_{i \rightarrow \infty} y_i \end{aligned}$$

Substituting the above in [eq: 20.17] gives finally

$$\forall \{x_i\}_{i \in \{k, \dots, \infty\}}, \{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R} \quad \liminf_{i \rightarrow \infty} x_i + \liminf_{i \rightarrow \infty} y_i \leq \liminf_{i \rightarrow \infty} (x_i + y_i) \quad (20.18)$$

Consider the sequences $\{-x_i\}_{i \in \{k, \dots, \infty\}}, \{-y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ then by [eq: 20.18] we have that

$$\liminf_{i \rightarrow \infty} (-x_i) + \liminf_{i \rightarrow \infty} (-y_i) \leq \liminf_{i \rightarrow \infty} (-x_i + y_i)$$

Then using [theorem: 20.66] on the above gives

$$-\limsup_{i \rightarrow \infty} x_i + \left(-\limsup_{i \rightarrow \infty} y_i\right) \leq -\limsup_{i \rightarrow \infty} (x_i + y_i)$$

hence

$$\limsup_{i \rightarrow \infty} (x_i + y_i) \leq \limsup_{i \rightarrow \infty} x_i + \limsup_{i \rightarrow \infty} y_i \quad \square$$

Theorem 20.70. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ then $\liminf_{i \rightarrow \infty} x_i \leq \limsup_{i \rightarrow \infty} x_i$

Proof. Fix $n \in \{k, \dots, \infty\}$ then $\forall m \in \{k, \dots, \infty\}$ we have either:

$m \in \{k, \dots, n-1\}$. then $m < n$ so that $\{x_i | i \in \{n, \dots, \infty\}\} \subseteq \{x_i | i \in \{m, \dots, \infty\}\}$ hence

$$\inf(\{x_i | i \in \{m, \dots, \infty\}\}) \leq \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \leq x_n \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

$m \in \{n, \dots, \infty\}$. then $\inf(\{x_i | i \in \{m, \dots, \infty\}\}) \leq x_n \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$

hence we have

$$\forall m \in \{k, \dots, \infty\} \text{ we have } \inf(\{x_i | i \in \{m, \dots, \infty\}\}) \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

so we have

$$\sup(\{\inf(\{x_i | i \in \{m, \dots, \infty\}\}) | m \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

or

$$\forall n \in \{k, \dots, \infty\} \quad \liminf_{i \rightarrow \infty} x_i \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$$

so that

$$\liminf_{i \rightarrow \infty} x_i \leq \inf(\{\sup(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\}) = \limsup_{i \rightarrow \infty} x_i \quad \square$$

Remark 20.71. If we note $\leq_{\mathbb{R}}$ as the order relation on $\mathbb{R}$, $\sup_{\mathbb{R}}$ and $\inf_{\mathbb{R}}$ the supremum and infimum in $\mathbb{R}$ and $\lim_{\mathbb{R}}$ the limit in $\langle \mathbb{R}, \|\cdot\|_{\mathbb{R}} \rangle$ where $\|\cdot\|_{\mathbb{R}}$ is the absolute value on $\mathbb{R}$ then we have in [definition: 14.377] defined $\lim_{\mathbb{R}} \inf$ and $\lim_{\mathbb{R}} \sup$ defined by

1. $\lim_{\mathbb{R}} \sup_{n \rightarrow \infty} x_n$ exist if

a. $\forall n \in \{k, \dots, \infty\} \quad \sup_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\})$ exist

b. $\lim_{\mathbb{R}} \sup_{n \rightarrow \infty} \sup_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\})$ exist

If $\lim_{\mathbb{R}} \sup_{n \rightarrow \infty} x_n$ exist then $\lim_{\mathbb{R}} \sup_{n \rightarrow \infty} x_n = \lim_{\mathbb{R}} \sup_{m \rightarrow \infty} \sup_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\})$

2. $\lim_{\mathbb{R}} \inf_{n \rightarrow \infty} x_n$ exist if

a. $\forall n \in \{k, \dots, \infty\} \quad \inf_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\})$ exist

b. $\lim_{\mathbb{R}} \inf_{n \rightarrow \infty} \inf_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\})$ exist

If $\lim_{n \rightarrow \infty} \inf x_n$ exist then $\lim_{n \rightarrow \infty} \inf x_n = \lim_{m \rightarrow \infty} \inf_{n \in \{m, \dots, \infty\}} (\{x_i | i \in \{n, \dots, \infty\}\})$

Using [theorem: 14.379] the above definition is equivalent with

1.

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sup x_n \text{ exist} \\ & \Updownarrow \\ & \forall n \in \{k, \dots, \infty\} \sup_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exist} \\ & \text{and} \\ & \inf_{\mathbb{R}}(\sup_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}) \text{ exist} \end{aligned}$$

Further if $\lim_{n \rightarrow \infty} \sup x_n$ exist then

$$\lim_{n \rightarrow \infty} \sup x_n = \inf(\{\sup_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}\})$$

2.

$$\begin{aligned} & \lim_{n \rightarrow \infty} \inf x_n \text{ exist} \\ & \Updownarrow \\ & \forall n \in \{k, \dots, \infty\} \inf_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\}) \text{ exist} \\ & \text{and} \\ & \sup_{\mathbb{R}}(\inf_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\}) \text{ exist} \end{aligned}$$

Further if $\lim_{n \rightarrow \infty} \inf x_n$ exist then

$$\lim_{n \rightarrow \infty} \inf x_n = \sup_{\mathbb{R}}(\inf_{\mathbb{R}}(\{x_i | i \in \{n, \dots, \infty\}\}) | n \in \{k, \dots, \infty\})$$

Using [theorem: 14.380] we have that

$$\lim_{n \rightarrow \infty} x_n \text{ exist} \Leftrightarrow \lim_{n \rightarrow \infty} \inf x_n \text{ and } \lim_{n \rightarrow \infty} \sup x_n \text{ exists and } \lim_{n \rightarrow \infty} \inf x_n = \lim_{n \rightarrow \infty} \sup x_n$$

Further if $\lim_{n \rightarrow \infty} x_i$ exist then

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf x_i = \lim_{n \rightarrow \infty} \sup x_n$$

This suggest the following extension of a limit in $\mathbb{R}$ to the limit in $\overline{\mathbb{R}}$

Definition 20.72. Let $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ such that $\lim_{i \rightarrow \infty} \inf x_i = \lim_{i \rightarrow \infty} \sup x_i$ then we say that $\lim_{i \rightarrow \infty} x_i$ exist and we have by definition that

$$\lim_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} \inf x_i = \lim_{i \rightarrow \infty} \sup x_i$$

Note 20.73. In this chapter we use $\lim$ for the $\lim$ in $\overline{\mathbb{R}}$ which is defined using only the order relation $\leq$. As for $\mathbb{R}$ $\lim_{\mathbb{R}}$ represents the limit in $\mathbb{R}$ which is based on the absolute value $||_{\mathbb{R}}$ in $\mathbb{R}$ but can be defined by the order relation $\leq_{\mathbb{R}}$ in $\mathbb{R}$ where $\leq_{\mathbb{R}} \subseteq \leq$. Further $\sup$, $\inf$, $\liminf$, $\limsup$ are all defined in $\overline{\mathbb{R}}$ using the order relation $\leq$ on $\overline{\mathbb{R}}$ and $\sup_{\mathbb{R}}$, $\inf_{\mathbb{R}}$ are defined on $\mathbb{R}$ using the order relation $\leq_{\mathbb{R}}$ on $\mathbb{R}$ and $\lim_{\mathbb{R}} \inf$, $\lim_{\mathbb{R}} \sup$ are defined in [definition: 14.377] or [theorem: 14.379].

Example 20.74. Let $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ is defined by $x_i = x$ where $x \in \overline{\mathbb{R}}$ then $\lim_{i \rightarrow \infty} x_i = x$.

Proof. This follows from the definition and [example: 20.60]. □

Example 20.75. Let $k \in \mathbb{N}_0$ then the sequence $\{(-\infty)^i\}_{i \in \{k, \dots, \infty\}}$ has no limit

Proof. Let $l \in \{1, \dots, \infty\}$ then we have $\{(-\infty)^i | i \in \{l, \dots, \infty\}\} = \{-\infty, \infty\}$ so that

$$\{\sup(\{(-\infty)^i | i \in \{l, \dots, \infty\}\})\} = \{\infty\} \text{ and } \{\sup(\{(-\infty)^i | i \in \{l, \dots, \infty\}\})\} = \{-\infty\}$$

hence

$$\limsup_{i \rightarrow \infty} (-\infty)^i = \inf(\{\sup(\{(-\infty)^i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \infty$$

and

$$\liminf_{i \rightarrow \infty} (-\infty)^i = \sup(\{\inf(\{(-\infty)^i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = -\infty$$

so that

$$\limsup_{i \rightarrow \infty} (-\infty)^i \neq \liminf_{i \rightarrow \infty} (-\infty)^i \quad \square$$

We have the following relation between $\lim_{\mathbb{R}}$ the limit in $\mathbb{R}$ and $\lim$ the limit in $\overline{\mathbb{R}}$.

Theorem 20.76. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ a sequence of **real** numbers then we have:

1. If $\lim_{\mathbb{R}} x_i$ exists then $\liminf_{i \rightarrow \infty} x_i = \lim_{\mathbb{R}} x_i = \limsup_{i \rightarrow \infty} x_i$
2. If $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i \in \mathbb{R}$ then $\lim_{\mathbb{R}} x_i$ exists and $\liminf_{i \rightarrow \infty} x_i = \lim_{\mathbb{R}} x_i = \limsup_{i \rightarrow \infty} x_i$

Proof.

1. Let $x = \lim_{\mathbb{R}} x_i \in \mathbb{R}$. Assume that $\liminf_{i \rightarrow \infty} x_i < x$ then as $x \in \mathbb{R} \Rightarrow x < \infty$ we have either:

$\liminf_{i \rightarrow \infty} x_i = -\infty$. Then $\sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = -\infty$ so that by [theorem: 20.15] it follows that $\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\} = \{-\infty\}$ hence

$$\inf(\{x_i | i \in \{k, \dots, \infty\}\}) = -\infty \quad (20.19)$$

As $\lim_{\mathbb{R}} x_i$ exist it follows from [theorem: 14.364] that there exists a $L \in \mathbb{R}$ such that $\forall i \in \{k, \dots, \infty\} |x_i|_{\mathbb{R}} \leq L < L + 1$. So if we take $K = L + 1 \in \mathbb{R}$ then we have

$$\forall i \in \{k, \dots, \infty\} \text{ we have } -K < x_i < K$$

So $-\infty < -K < x_i \forall i \in \{k, \dots, \infty\}$ hence $-\infty < -K \leq \inf(\{x_i | i \in \{k, \dots, \infty\}\})$ contradicting [eq: 20.19].

$\liminf_{i \rightarrow \infty} x_i \in \mathbb{R}$. As $\mathbb{R} \ni \liminf_{i \rightarrow \infty} x_i < x \in \mathbb{R}$ we have that $\varepsilon = x - \liminf_{i \rightarrow \infty} x_i \in \mathbb{R}^+$. As $\lim_{\mathbb{R}} x_i = x$ there exist a $N \in \{k, \dots, \infty\}$ such that

$$\forall n \in \{N, \dots, \infty\} \text{ we have } |x_n - x|_{\mathbb{R}} <_{\mathbb{R}} \varepsilon \quad (20.20)$$

Now

$$\begin{aligned} \inf(\{x_i | i \in \{N, \dots, \infty\}\}) &\leq \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \liminf_{i \rightarrow \infty} x_i \\ &= x - \varepsilon \\ &< x - \frac{\varepsilon}{2} \end{aligned}$$

Hence by the definition of the infimum there exist a $M \in \{N, \dots, \infty\}$ such that $x_M < x - \frac{\varepsilon}{2} \xRightarrow{x_M, x \in \mathbb{R}} x_M <_{\mathbb{R}} x - \frac{\varepsilon}{2}$. Hence $|x_M - x|_{\mathbb{R}} \geq_{\mathbb{R}} x - x_M >_{\mathbb{R}} \frac{\varepsilon}{2}$ contradicting [20.20].

So in all cases we reach a contradiction proving that the assumption is wrong, hence we have that

$$x \leq \liminf_{i \rightarrow \infty} x_i \leq \limsup_{i \rightarrow \infty} x_i \quad \text{[theorem: 20.70]} \quad (20.21)$$

Assume that $x < \limsup_{i \rightarrow \infty} x_i$. Then as $-\infty < x < \limsup_{i \rightarrow \infty} x_i$ we have for $\limsup_{i \rightarrow \infty} x_i$ either:

$\limsup_{i \rightarrow \infty} x_i = \infty$. Then $\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \infty$ so that by [theorem: 20.15] it follows that $\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\} = \{\infty\}$. Hence

$$\sup(\{x_i | i \in \{k, \dots, \infty\}\}) = \infty \quad (20.22)$$

As $\lim_{\mathbb{R}} x_i$ exist it follows from [theorem: 14.364] that there exists a $L \in \mathbb{R}$ such that $\forall i \in \{k, \dots, \infty\} |x_i|_{\mathbb{R}} \leq L < L + 1$. So if we take $K = L + 1 \in \mathbb{R}$ then we have

$$\forall i \in \{k, \dots, \infty\} \text{ we have } -K < x_i < K$$

So $\sup(\{x_i | i \in \{k, \dots, \infty\}\}) \leq K < \infty$ contradiction [eq: 20.22].

$\limsup_{i \rightarrow \infty} x_i \in \mathbb{R}$. As $\mathbb{R} \ni x < \limsup_{i \rightarrow \infty} x_i \in \mathbb{R}$ we have that $\varepsilon = \limsup_{i \rightarrow \infty} x_i - x \in \mathbb{R}^+$. As $\lim_{\mathbb{R}} x_i = x$ there exist a $N \in \{k, \dots, \infty\}$ such that

$$\forall n \in \{N, \dots, \infty\} \text{ we have } |x_n - x|_{\mathbb{R}} < \varepsilon \quad (20.23)$$

Now

$$\begin{aligned} \sup(\{x_i | i \in \{N, \dots, \infty\}\}) &\geq \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \limsup_{i \rightarrow \infty} x_i \\ &= x + \varepsilon \\ &> x + \frac{\varepsilon}{2} \end{aligned}$$

So using the definition of the supremum there exist a $M \in \{N, \dots, \infty\}$ such that $x_M > x + \frac{\varepsilon}{2} \xRightarrow{x_M, x \in \mathbb{R}} x_M >_{\mathbb{R}} x + \frac{\varepsilon}{2}$. Hence $|x_M - x| \geq_{\mathbb{R}} x_M - x >_{\mathbb{R}} \frac{\varepsilon}{2}$ contradicting [eq: 20.23].

As in all cases we have a contradiction the assumption must be wrong so that

$$\limsup_{i \rightarrow \infty} x_i \leq x.$$

Combining this with [eq: 20.21] we have

$$\liminf_{i \rightarrow \infty} x_i \leq \limsup_{i \rightarrow \infty} x_i \leq x \leq \liminf_{i \rightarrow \infty} x_i \leq \limsup_{i \rightarrow \infty} x_i$$

[theorem: 20.70] [theorem: 20.70]

so that

$$\liminf_{i \rightarrow \infty} x_i = \lim_{\mathbb{R}} x_i = \limsup_{i \rightarrow \infty} x_i$$

2. Let $x = \liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i \in \mathbb{R}$. Take $\varepsilon \in \mathbb{R}^+$ then we have

$$x - \varepsilon < x = \liminf_{i \rightarrow \infty} x_i = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$$

Hence by the definition of the supremum there exist a $M_1 \in \{k, \dots, \infty\}$ such that

$$x - \varepsilon < \inf(\{x_i | i \in \{M_1, \dots, \infty\}\})$$

Hence

$$\forall n \in \{M_1, \dots, \infty\} \text{ we have } x - \varepsilon < x_n \quad (20.24)$$

Further as $x = \limsup_{i \rightarrow \infty} x_i$ we have that $\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = x < x + \varepsilon$ so that by the definition of the infimum there exist a $M_2 \in \{k, \dots, \infty\}$ such that

$$\sup(\{x_i | i \in \{M_2, \dots, \infty\}\}) < x + \varepsilon$$

Hence

$$\forall n \in \{M_2, \dots, \infty\} \text{ we have } x_n < x + \varepsilon \quad (20.25)$$

Take $M = \max(M_1, M_2)$ then we have by [eqs: 20.24, 20.25] that $\forall n \in \{M, \dots, \infty\}$ $x - \varepsilon < x_n < x + \varepsilon$ or as $x, \varepsilon \in \mathbb{R}$ that $x - \varepsilon <_{\mathbb{R}} x_n <_{\mathbb{R}} x + \varepsilon \Rightarrow |x_n - x|_{\mathbb{R}} <_{\mathbb{R}} \varepsilon$ proving that

$$\lim_{i \rightarrow \infty} x_i \text{ exists and } \lim_{i \rightarrow \infty} x_i = x = \liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i \quad \square$$

A consequence of the above theorem is that $\lim$ is an extension of $\lim_{\mathbb{R}}$ to $\overline{\mathbb{R}}$.

Corollary 20.77. *Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathbb{R}$ a sequence of **real** numbers then we have*

$$\lim_{i \rightarrow \infty} x_i \text{ exists} \Leftrightarrow \lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i \in \mathbb{R}$$

Further if $\lim_{i \rightarrow \infty} x_i$ exists or $\lim_{i \rightarrow \infty} x_i$ exists then $\lim_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i$

Proof.

$\Rightarrow$. Using [theorem: 20.76] it follows that $\liminf_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i$ so that by definition $\lim_{i \rightarrow \infty} x_i$ exists and $\mathbb{R} \ni \lim_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i$.

$\Leftarrow$. If $\lim_{i \rightarrow \infty} x_i$ exists and $\lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ we have that $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i \in \mathbb{R}$. Hence by [theorem: 20.76] $\lim_{i \rightarrow \infty} x_i$ exists and $\lim_{i \rightarrow \infty} x_i = \liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i$. $\square$

Example 20.78. We have

1. For $\{\frac{1}{i}\}_{i \in \{1, \dots, \infty\}}$ we have that $\lim_{i \rightarrow \infty} \frac{1}{i}$ exist and $\lim_{i \rightarrow \infty} \frac{1}{i} = 0$
2. For $\{\frac{1}{i}\}_{i \in \{1, \dots, \infty\}}$ we have that $\lim_{i \rightarrow \infty} \frac{(-1)^i}{i}$ exist and $\lim_{i \rightarrow \infty} \frac{(-1)^i}{i} = 0$

Proof.

1. Note that $\{\frac{1}{i}\}_{i \in \{1, \dots, \infty\}} \subseteq \mathbb{R}$ and by [example: 14.349] $\lim_{i \rightarrow \infty} \frac{1}{i} = 0$ exist. Hence by the previous corollary [corollary: 20.77] $\lim_{i \rightarrow \infty} \frac{1}{i} = 0$ exist.
2. Note that $\{\frac{1}{i}\}_{i \in \{1, \dots, \infty\}} \subseteq \mathbb{R}$ and by [example: 14.366] $\lim_{i \rightarrow \infty} \frac{(-1)^i}{i} = 0$ exist. Hence by the previous corollary [corollary: 20.77] $\lim_{i \rightarrow \infty} \frac{(-1)^i}{i} = 0$ exist. $\square$

The idea of a limit in $\mathbb{R}$ is that the sequence approaches its limit the higher the index is, we show now that the limit in $\overline{\mathbb{R}}$.

Theorem 20.79. *Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ then we have the following equivalences.*

1. $\lim_{i \rightarrow \infty} x_i = \infty \Leftrightarrow \forall C \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\}$ we have $C < x_n$
2. $\lim_{i \rightarrow \infty} x_i = -\infty \Leftrightarrow \forall C \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\}$ we have $x_n < -C$
3. If $x \in \mathbb{R}$ then $\lim_{i \rightarrow \infty} x_i = x \Leftrightarrow \forall \varepsilon \in \mathbb{R}^+ \exists N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\}$ $x - \varepsilon < x_n < x + \varepsilon$

Note 20.80. *As $x, \varepsilon \in \mathbb{R}$ we have that $-\infty < x - \varepsilon < x_n < x + \varepsilon < \infty$ so $x - \varepsilon < x_n < x + \varepsilon$ is equivalent with $x_n \in \mathbb{R}$ and $|x - x_n| < \varepsilon$.*

Proof.

1. $\Rightarrow$. If $\lim_{i \rightarrow \infty} x_i = \infty$ then $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = \infty$ so that $\sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \infty$

Hence using [theorem: 20.15] we have that either:

$\infty \in \{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}$. Then there exist a $N \in \{k, \dots, \infty\}$ such that $\inf(\{x_i | i \in \{N, \dots, \infty\}\}) = \infty$. Using [theorem: 20.15] again it follows that $\{x_i | i \in \{N, \dots, \infty\}\} = \{\infty\}$ so that $\forall n \in \{N, \dots, \infty\}$ we have $x_n = \infty$. Hence if $C \in \mathbb{R}^+$ then $\forall n \in \{N, \dots, \infty\}$ then $C < \infty = x_n$.

$\forall u \in \mathbb{R} \exists x \in \{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\} \cap \mathbb{R} \vdash u < x$. Then given $C \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $C < \inf(\{x_i | i \in \{N, \dots, \infty\}\})$ hence $\forall n \in \{N, \dots, \infty\}$ we have $C < \inf(\{x_i | i \in \{N, \dots, \infty\}\}) \leq x_n$.

so in all cases we have that

$$\forall C \in \mathbb{R}_+ \text{ there } \exists N \in \{k, \dots, \infty\} \text{ such that } \forall n \geq N \text{ we have } C < x_n$$

$\Leftarrow$. Assume that $\liminf_{i \rightarrow \infty} x_i < \infty$. Take $C = \max\left(\liminf_{i \rightarrow \infty} x_i, 1\right) \in \mathbb{R}^+$ then by the hypothesis there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n \in \{N, \dots, \infty\}$ we have $C + 1 < x_n$. Hence $C + 1$ is a lower bound of $\{x_i | i \in \{N, \dots, \infty\}\}$ so that

$$C + 1 \leq \inf(\{x_i | i \in \{N, \dots, \infty\}\}) \quad (20.26)$$

So

$$\begin{aligned} \liminf_{i \rightarrow \infty} x_i &\leq \max\left(\liminf_{i \rightarrow \infty} x_i, 1\right) \\ &= C \\ &< C + 1 \\ &\leq \inf(\{x_i | i \in \{N, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.26}]}{\leq} \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \liminf_{i \rightarrow \infty} x_i \end{aligned}$$

leading to the contradiction $\liminf_{i \rightarrow \infty} x_i < \liminf_{i \rightarrow \infty} x_i$. Hence

$$\infty \leq \liminf_{i \rightarrow \infty} x_i \stackrel{[\text{theorem: 20.70}]}{\leq} \limsup_{i \rightarrow \infty} x_i \leq \infty$$

proving that $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = \infty$ so that

$$\lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i = \infty$$

2.

$\Rightarrow$. If $\lim_{i \rightarrow \infty} x_i = -\infty$ then $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = -\infty$ so that

$$\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = -\infty$$

Hence using [theorem: 20.15] we have that either:

$-\infty \in \{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}$. Then there exist a $N \in \{k, \dots, \infty\}$ such that $\sup(\{x_i | i \in \{N, \dots, \infty\}\}) = -\infty$. Using [theorem: 20.15] again it follows that $\{x_i | i \in \{N, \dots, \infty\}\} = \{-\infty\}$ so that $\forall n \in \{N, \dots, \infty\}$ we have $x_n = -\infty$. Hence if $C \in \mathbb{R}^+$ then $\forall n \in \{N, \dots, \infty\}$ then $x_n = -\infty < -C$.

$\forall u \in \mathbb{R} \exists x \in \{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\} \cap \mathbb{R} \vdash x < u$. Then given $C \in \mathbb{R}^+$ there exist a $N \in \{k, \dots, \infty\}$ such that $\sup(\{x_i | i \in \{N, \dots, \infty\}\}) < -C$ hence $\forall n \in \{N, \dots, \infty\}$ we have $x_n < \sup(\{x_i | i \in \{N, \dots, \infty\}\}) < -C$.

so in all cases we have that

$$\forall C \in \mathbb{R}_+ \text{ there } \exists N \in \{k, \dots, \infty\} \text{ such that } \forall n \geq N \text{ we have } x_n < -C$$

$\Leftarrow$. Assume that $-\infty < \limsup_{i \rightarrow \infty} x_i$. Take $C = -\min \left(\limsup_{i \rightarrow \infty} x_i, -1 \right) \in \mathbb{R}^+$ then by the hypothesis there exists a $N \in \{k, \dots, n\}$ such that $\forall n \in \{N, \dots, \infty\}$ we have $x_n < -(C+1)$. Hence $-(C+1)$ is a upper bound for $\{x_i | i \in \{N, \dots, \infty\}\}$ so that

$$\sup(\{x_i | i \in \{N, \dots, \infty\}\}) \leq -(C+1) \quad (20.27)$$

hence

$$\begin{aligned} \limsup_{i \rightarrow \infty} x_i &\geq \min \left(\limsup_{i \rightarrow \infty} x_i, -1 \right) \\ &= -C \\ &> -(C+1) \\ &\geq \sup(\{x_i | i \in \{N, \dots, \infty\}\}) \\ &\quad \text{[eq: 20.27]} \\ &\geq \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &= \limsup_{i \rightarrow \infty} x_i \end{aligned}$$

leading to the contradiction $\limsup_{i \rightarrow \infty} x_i < \limsup_{i \rightarrow \infty} x_i$. Hence

$$-\infty \leq \liminf_{i \rightarrow \infty} x_i \leq \limsup_{i \rightarrow \infty} x_i = -\infty \quad \text{[theorem: 20.70]}$$

proving that $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = -\infty$ so that

$$\lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i = -\infty$$

3.

$\Rightarrow$. If $\lim_{i \rightarrow \infty} x_i = x \in \mathbb{R}$ exist then $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i$ so that

$$x = \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$$

and

$$x = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$$

Let $\varepsilon \in \mathbb{R}^+$ then we have $x - \varepsilon < x = \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\})$ so that there exist a $N_1 \in \{k, \dots, \infty\}$ such that

$$x - \varepsilon < \inf(\{x_i | i \in \{N_1, \dots, \infty\}\})$$

hence for $n \in \{N_1, \dots, \infty\}$ we have that

$$x - \varepsilon < \inf(\{x_i | i \in \{N_1, \dots, \infty\}\}) \leq x_n$$

Likewise, as $\inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = x < x + \varepsilon$ there exist a $N_2 \in \{k, \dots, \infty\}$ such that $\sup(\{x_i | i \in \{N, \dots, \infty\}\}) < x + \varepsilon$ hence for $n \in \{N_2, \dots, \infty\}$ we have

$$x_n \leq \sup(\{x_i | i \in \{N, \dots, \infty\}\}) < x + \varepsilon$$

Take $N = \max(N_1, N_2)$ then for $n \in \{N, \dots, \infty\}$ we have

$$x - \varepsilon < x_n < x + \varepsilon \text{ or equivalently } |x_n - x| < \varepsilon$$

$\Leftarrow$. Let $\varepsilon \in \mathbb{R}^+$ then by the hypothesis there exist a N_ε such that $\forall n \in \{N_\varepsilon, \dots, \infty\}$ we have $x - \varepsilon < x_n < x + \varepsilon$. Hence we have that $\{x_i | i \in \{N_\varepsilon, \dots, \infty\}\}$ has a lower bound $x - \varepsilon$ and a upper bound $x + \varepsilon$. Hence

$$x - \varepsilon \leq \inf(\{x_i | i \in \{N_\varepsilon, \dots, \infty\}\}) \text{ and } \sup(\{x_i | i \in \{N_\varepsilon, \dots, \infty\}\}) < x + \varepsilon$$

As

$$\inf(\{x_i | i \in \{N_\varepsilon, \dots, \infty\}\}) \leq \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) = \liminf_{i \rightarrow \infty} x_i$$

and

$$\limsup_{i \rightarrow \infty} x_i = \inf (\{\sup (\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \leq \sup (\{x_i | i \in \{N_\varepsilon, \dots, \infty\}\})$$

it follows that $x - \varepsilon < \liminf_{i \rightarrow \infty} x_i \stackrel{[\text{theorem: 20.70}]}{\leq} \limsup_{i \rightarrow \infty} x_i < x + \varepsilon$. Given that $x \in \mathbb{R}$ we have that

$$x - \varepsilon <_{\mathbb{R}} \liminf_{i \rightarrow \infty} x_i \stackrel{[\text{theorem: 20.70}]}{\leq_{\mathbb{R}}} \limsup_{i \rightarrow \infty} x_i <_{\mathbb{R}} x + \varepsilon$$

So as $\varepsilon \in \mathbb{R}^+$ was chosen arbitrary it follows from [theorem: 10.31] that

$$x \leq_{\mathbb{R}} \liminf_{i \rightarrow \infty} x_i \leq_{\mathbb{R}} \limsup_{i \rightarrow \infty} x_i \leq_{\mathbb{R}} x$$

so that $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = x$ proving that

$$\lim_{i \rightarrow \infty} x_i \text{ exists and that } \lim_{i \rightarrow \infty} x_i = x$$

□

Corollary 20.81. Let $k \in \mathbb{N}$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ with $x = \lim_{n \rightarrow \infty} x_n = x$ then $\lim_{n \rightarrow \infty} |x_n| = |x|$

Proof. We have three cases to consider for x

$x \in \mathbb{R}$. using the previous theorem [see theorem: 20.79] we have $\forall \varepsilon \in \mathbb{R}^+$ that $\exists N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \in \mathbb{R} \wedge |x - x_n| < \varepsilon \stackrel{||x| - |x_n|| \leq |x - x_n|}{\Rightarrow} x_n \in \mathbb{R} \wedge ||x| - |x_n|| < \varepsilon$ proving by the previous theorem again that $\lim_{i \rightarrow \infty} |x_i| = |x|$.

$x = \infty$. using the previous theorem [see theorem: 20.79] we have $\forall C \in \mathbb{R}^+$ that there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $C \leq x_n \leq |x_n|$ proving by the previous theorem that $\lim_{i \rightarrow \infty} x_n = \infty = |x|$.

$x = -\infty$. using the previous theorem [see theorem: 20.79] we have $\forall C \in \mathbb{R}^+$ that there exists a $N \in \{k, \dots, \infty\}$ such that $\forall n \geq N$ we have $x_n \leq -C \Rightarrow C \leq -x_n \leq |x_n|$ proving by the previous theorem that $\lim_{i \rightarrow \infty} x_n = -\infty = |x|$ □

Example 20.82. Let $\{x_n\}_{n \in \mathbb{N}}$ be defined by $x_n = n$ then $\lim_{n \rightarrow \infty} x_n$ exist and $\lim_{n \rightarrow \infty} x_n = \infty$.

Proof. Let $C \in \mathbb{R}^+$ then by [theorem: 10.30] there exist a $N \in \mathbb{N}$ such that $C < N$. Hence if $n \in \{N, \dots, \infty\}$ then $C < N < n = x_n$. So using the previous theorem [theorem: 20.79] it follows that $\lim_{n \rightarrow \infty} x_n$ exist and $\lim_{n \rightarrow \infty} x_n = \infty$. □

We can now derive the different properties of the limit in $\overline{\mathbb{R}}$ based on the properties of $\limsup_{i \rightarrow \infty}$ and $\liminf_{i \rightarrow \infty}$.

Theorem 20.83. Let $\emptyset \neq A \subseteq \overline{\mathbb{R}}$ be a non empty set of extended real numbers then we have

1. $\exists \{x_i\}_{i \in \mathbb{N}} \subseteq A$ such that $\lim_{i \rightarrow \infty} x_i$ exist and $\lim_{i \rightarrow \infty} x_i = \sup(A)$.
2. $\exists \{x_i\}_{i \in \mathbb{N}} \subseteq A$ such that $\lim_{i \rightarrow \infty} x_i$ exist and $\lim_{i \rightarrow \infty} x_i = \inf(A)$

Proof.

1. For $\sup(A)$ we have the following cases to consider:

$\sup(A) = -\infty$. Then $\forall x \in A$ we have $-\infty \leq x \leq -\infty$ which, as $A \neq \emptyset$, proves that $A = \{-\infty\}$. Define now $\{x_i\}_{i \in \mathbb{N}_0} \subseteq A = \{-\infty\}$ by $x_i = -\infty$ then by [example: 20.74] we have

$$\lim_{n \rightarrow \infty} x_n = -\infty = \sup(A)$$

sup(A) = ∞ . Then $\forall n \in \mathbb{N}$ we have that $n < \infty = \sup(A)$, hence, using the definition of the supremum, there exists a $x_n \in A$ such that $n < x_n \leq \sup(A)$. This defines a sequence $\{x_n\}_{n \in \mathbb{N}_0} \subseteq A$ such that $\forall n \in \mathbb{N}$ $n < x_n \leq \infty$. Let $C \in \mathbb{R}_+$ then by [theorem: 10.30] there exists a $N \in \mathbb{N}$ such that $C < N$, hence $\forall n \geq N$ we have $C < N \leq n < x_n$. So using [theorem: 20.79] it follows that

$$\lim_{n \rightarrow \infty} x_n = \infty = \sup(A)$$

sup(A) $\in \mathbb{R}$. Then $\forall n \in \mathbb{N}$ we have that $\sup(A) - \frac{1}{n} < \sup(A)$, hence using the definition of the supremum, there exists a $x_n \in A$ such that $\sup(A) - \frac{1}{n} < x_n \leq \sup(A)$. This defines a sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq A$ such that $\forall n \in \mathbb{N}$ $\sup(A) - \frac{1}{n} < x_n \leq \sup(A)$. Take now $\varepsilon \in \mathbb{R}^+$ then using [theorem: 10.30] there exists a $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$. Take $n \in \{N, \dots, \infty\}$ then we have

$$\sup(A) - \varepsilon < \sup(A) - \frac{1}{N} \leq \sup(A) - \frac{1}{n} < x_n \leq \sup(A) < \sup(A) + \varepsilon.$$

Which proves, using [theorem: 20.79], that

$$\lim_{n \rightarrow \infty} x_n = \sup(A)$$

2. For **inf**(A) we have the following cases to consider:

inf(A) = $-\infty$. Then $\forall n \in \mathbb{N}$ we have that $-\infty < -n$, hence, using the definition of the infimum, there exists a $x_n \in A$ such that $-\infty = \inf(A) \leq x_n < -n$. This defines a sequence $\{x_n\}_{n \in \mathbb{N}_0} \subseteq A$ such that $\forall n \in \mathbb{N}$ $-\infty \leq x_n < -n$. Let $C \in \mathbb{R}_+$ then by [theorem: 10.30] there exists a $N \in \mathbb{N}$ such that $C < N$, hence $-N < -C$. Let $n \geq N$ then $x_n < -n \leq -N < -C$. Using [theorem: 20.79] it follows that

$$\lim_{n \rightarrow \infty} x_n = -\infty = \inf(A)$$

inf(A) = ∞ . Then $\forall x \in A$ we have $\infty \leq x \leq \infty$, which, as $A \neq \emptyset$, proves that $A = \{\infty\}$. Define now $\{x_i\}_{i \in \mathbb{N}_0} \subseteq A$ by $x_i = \infty$. Then by [example: 20.74] we have that

$$\lim_{n \rightarrow \infty} x_n = \infty = \inf(A)$$

inf(A) $\in \mathbb{R}$. Then $\forall n \in \mathbb{N}_0$ we have $\inf(A) < \inf(A) + \frac{1}{n}$, hence, using the definition of the infimum, there exists a $x_n \in A$ such that $\inf(A) \leq x_n < \inf(A) + \frac{1}{n}$. So we have found a sequence $\{x_n\}_{n \in \mathbb{N}_0} \subseteq A$ such that $\forall n \in \mathbb{N}$ we have $\inf(A) \leq x_n < \inf(A) + \frac{1}{n}$. Let $\varepsilon \in \mathbb{R}^+$ then, using [theorem: 10.30] there exists a $N \in \mathbb{N}_0$ such that $\frac{1}{N} < \varepsilon$. Let $n \in \{N, \dots, \infty\}$ then we have

$$\inf(A) - \varepsilon < \inf(A) \leq x_n < \inf(A) + \frac{1}{n} \leq \inf(A) + \frac{1}{N} < \inf(A) + \varepsilon.$$

Using [theorem: 20.79] it follows that

$$\lim_{n \rightarrow \infty} x_n = \inf(A)$$

□

Theorem 20.84. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ and $n \in \mathbb{N}_0$ then if $\lim_{i \rightarrow \infty} x_i$ exists we have that $\lim_{i \rightarrow \infty} x_{i+n}$ exists and $\lim_{i \rightarrow \infty} x_{i+n} = \lim_{i \rightarrow \infty} x_i$.

Proof. As $\lim_{i \rightarrow \infty} x_i$ we have that $\liminf_{i \rightarrow \infty} x_i = \limsup_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i$. Using [theorem: 20.61] it follows that $\liminf_{i \rightarrow \infty} x_{i+n} = \limsup_{i \rightarrow \infty} x_{i+n}$ so that $\lim_{i \rightarrow \infty} x_{i+n}$ exists and $\lim_{i \rightarrow \infty} x_{i+n} = \lim_{i \rightarrow \infty} x_{i+n}$. □

Theorem 20.85. The limit in $\overline{\mathbb{R}}$ has the following properties where $k \in \mathbb{N}_0$

1. Let $x \in \overline{\mathbb{R}}$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ then we have

a. If $\forall i \in \{k, \dots, \infty\}$ $x \leq x_i$ then $x \leq \lim_{i \rightarrow \infty} x_i$

- b. If $\forall i \in \{k, \dots, \infty\} x_i \leq x$ then $\lim_{i \rightarrow \infty} x_i \leq x$
2. Let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ and $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be such that $\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i$ exist and $\forall i \in \{k, \dots, \infty\} x_i \leq y_i$ then $\lim_{i \rightarrow \infty} x_i \leq \lim_{i \rightarrow \infty} y_i$.
3. If $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ has a limit and $\alpha \in \mathbb{R}$ then $\{\alpha \cdot x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ has a limit and $\lim_{i \rightarrow \infty} (\alpha \cdot x_i) = \alpha \cdot \lim_{i \rightarrow \infty} x_i$.
4. If $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ has a limit, $x \in \overline{\mathbb{R}}$ and we have that $\forall i \in \{k, \dots, \infty\} (x_i, x) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ then

$$\lim_{i \rightarrow \infty} (x_i + x) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + x) = \lim_{i \rightarrow \infty} x_i + x$$

5. Let $\{x_i\}_{i \in \{k, \dots, n\}} \subseteq \overline{\mathbb{R}}$ and $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ such that $\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i$ exist, $\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i\right) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ and $\forall i \in \{k, \dots, \infty\}$ we have that $(x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ then

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

6. Let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ and $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be sequences with limits, $\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i\right) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ [so that $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$ is well defined] then there exist a $n \in \{k, \dots, \infty\}$ such that $\forall i \in \{n, \dots, \infty\} (x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ [hence $x_i + y_i$ is defined] and $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ has a limit with

$$\lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

7. Let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ and $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be sequences of extended reals with limits such that for $x = \lim_{i \rightarrow \infty} x_i$ and $y = \lim_{i \rightarrow \infty} y_i$ we have

$$(x, y) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(0, -\infty), (0, \infty), (-\infty, 0), (\infty, 0)\}$$

$$\text{then } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} x_i\right) \cdot \left(\lim_{i \rightarrow \infty} y_i\right)$$

8. Let $\{x_i\}_{i \in \{k, \dots, \infty\}}, \{y_i\}_{i \in \{k, \dots, \infty\}}$ be sequences of extended reals with limits then we have

$$a. \lim_{i \rightarrow \infty} (\min(x_i, y_i)) = \min\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i\right)$$

$$b. \lim_{i \rightarrow \infty} (\max(x_i, y_i)) = \max\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i\right)$$

Note 20.86. The condition in (3) that $\alpha \in \mathbb{R}$ is required. For example the sequence $\left\{\frac{(-1)^i}{i}\right\}_{i \in \{1, \dots, \infty\}}$ converges to 0 but $\left\{\infty \cdot \frac{(-1)^i}{i}\right\}_i = \{(-\infty)^i\}_{i \in \{1, \dots, \infty\}}$ does not converges [see examples: 20.75, 20.78].

Note 20.87. The extra conditions in (5) are needed, for example take $\{x_i\}_{i \in \mathbb{N}_0}$ defined by $x_i = \frac{1}{i}$ and $\{y_i\}_{i \in \mathbb{N}_0}$ defined by $y_i = i^2$ then $\lim_{i \rightarrow \infty} x_i = 0$ and $\lim_{i \rightarrow \infty} y_i = \infty$ but $\lim_{i \rightarrow \infty} (x_i \cdot y_i) = \lim_{i \rightarrow \infty} i = \infty \neq \infty \cdot 0 = 0$

Proof.

1.

- a. As $\forall i \in \{k, \dots, \infty\} x \leq x_i$ it follows from [theorem: 20.63] that $x \leq \liminf_{i \rightarrow \infty} x_i$ proving, as $\liminf_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i$, that $x \leq \lim_{i \rightarrow \infty} x_i$.

- b. As $\forall i \in \{k, \dots, \infty\} x_i \leq x$ it follows from [theorem: 20.63] that $\liminf_{i \rightarrow \infty} x_i \leq x$ proving, as $\liminf_{i \rightarrow \infty} x_i = \lim_{i \rightarrow \infty} x_i$, that $x \leq \lim_{i \rightarrow \infty} x_i$.

2. As $\forall i \in \{k, \dots, \infty\}$ $x_i \leq y_i$ we have by [theorem: 20.62] that $\liminf_{i \rightarrow \infty} x_i \leq \liminf_{i \rightarrow \infty} y_i$. Hence

$$\lim_{i \rightarrow \infty} x_i \stackrel{\text{def}}{=} \liminf_{i \rightarrow \infty} x_i \leq \liminf_{i \rightarrow \infty} y_i \stackrel{\text{def}}{=} \lim_{i \rightarrow \infty} y_i$$

3. For $\alpha \in \mathbb{R}$ we have the following cases to consider:

$0 \leq \alpha$. Then we have

$$\begin{aligned} \liminf_{i \rightarrow \infty} (\alpha \cdot x_i) &\stackrel{[\text{theorem: 20.69}]}{=} \alpha \cdot \liminf_{i \rightarrow \infty} x_i \\ &\stackrel{\text{def}}{=} \alpha \cdot \lim_{i \rightarrow \infty} x_i \\ &\stackrel{\text{def}}{=} \alpha \cdot \limsup_{i \rightarrow \infty} x_i \\ &\stackrel{[\text{theorem: 20.69}]}{=} \limsup_{i \rightarrow \infty} (\alpha \cdot x_i) \end{aligned}$$

so that $\lim_{i \rightarrow \infty} (\alpha \cdot x_i)$ exists and $\lim_{i \rightarrow \infty} (\alpha \cdot x_i) = \alpha \cdot \lim_{i \rightarrow \infty} x_i$

$\alpha < 0$. Then $0 < -\alpha$ so that

$$\begin{aligned} \liminf_{i \rightarrow \infty} (\alpha \cdot x_i) &\stackrel{[\text{theorem: 20.67}]}{=} -\limsup_{i \rightarrow \infty} ((-\alpha) \cdot x_i) \\ &\stackrel{[\text{theorem: 20.69}]}{=} -\left((-\alpha) \cdot \limsup_{i \rightarrow \infty} x_i \right) \\ &= \alpha \cdot \limsup_{i \rightarrow \infty} x_i \\ &\stackrel{\text{def}}{=} \alpha \cdot \lim_{i \rightarrow \infty} x_i \\ &\stackrel{\text{def}}{=} \alpha \cdot \liminf_{i \rightarrow \infty} x_i \\ &= -\left((-\alpha) \cdot \liminf_{i \rightarrow \infty} x_i \right) \\ &\stackrel{[\text{theorem: 20.69}]}{=} -\liminf_{i \rightarrow \infty} ((-\alpha) \cdot x_i) \\ &\stackrel{[\text{theorem: 20.67}]}{=} \limsup_{i \rightarrow \infty} (\alpha \cdot x_i) \end{aligned}$$

so that $\liminf_{i \rightarrow \infty} (\alpha \cdot x_i) = \limsup_{i \rightarrow \infty} (\alpha \cdot x_i) = \alpha \cdot \lim_{i \rightarrow \infty} x_i$. Hence

$$\lim_{i \rightarrow \infty} (\alpha \cdot x_i) \text{ exist and } \lim_{i \rightarrow \infty} (\alpha \cdot x_i) = \alpha \cdot \lim_{i \rightarrow \infty} x_i$$

4. Using [theorem: 20.65] we have that

$$\begin{aligned} \liminf_{i \rightarrow \infty} (x_i + x) &\stackrel{[\text{theorem: 20.65}]}{=} \left(\liminf_{i \rightarrow \infty} x_i \right) + x \\ &\stackrel{\text{def}}{=} \left(\lim_{i \rightarrow \infty} x_i \right) + x \\ &\stackrel{\text{def}}{=} \left(\limsup_{i \rightarrow \infty} x_i \right) + x \\ &\stackrel{[\text{theorem: 20.65}]}{=} \limsup_{i \rightarrow \infty} (x_i + x) \end{aligned}$$

Hence

$$\lim_{i \rightarrow \infty} (x_i + x) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + x) = \left(\lim_{i \rightarrow \infty} x_i \right) + x$$

5. As $\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i \right) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ we have the following cases to consider:

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Take $x = \lim_{i \rightarrow \infty} x_i$ and $y = \lim_{i \rightarrow \infty} y_i$. Let $\varepsilon \in \mathbb{R}^+$ then by [theorem: 20.79] there exist $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } |x_i - x| < \frac{\varepsilon}{2}$$

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } |y_i - y| < \frac{\varepsilon}{2}$$

take $N = \max(N_1, N_2)$ then $\forall i \in \{N, \dots, \infty\}$ we have that $x_i + y_i$ is well defined and

$$|x_i + y_i - (x + y)| \leq |x_i - x| + |y_i - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Hence using [theorem: 20.79] it follows that $\lim_{i \rightarrow \infty} (x_i + y_i)$ exist and

$$\lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Take $y = \lim_{i \rightarrow \infty} y_i$. By [theorem: 20.79] there exist a $N_1 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_1, \dots, \infty\}$ we have $|y_i - y| < 1 \Rightarrow y_i < y + 1$. Let $C \in \mathbb{R}^+$ then for $C' = \max(C + y + 1, 1) \in \mathbb{R}^+$ we have $C + y + 1 \leq C'$ so that $-C' \leq -C - y - 1$. By [theorem: 20.79] there exist a $N_2 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_2, \dots, \infty\}$ we have $x_i < -C' \leq -C - y - 1$. Take $N = \max(N_1, N_2)$ then for $i \in \{N, \dots, \infty\}$ we have

$$x_i + y_i < -C' + y_i \leq -C' + y + 1 \leq -C - y - 1 + y + 1 = -C$$

Which by [theorem: 20.79] proves that $\lim_{i \rightarrow \infty} (x_i + y_i)$ exist and

$$\lim_{i \rightarrow \infty} (x_i + y_i) = -\infty = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Take $y = \lim_{i \rightarrow \infty} y_i$. By [theorem: 20.79] there exist a $N_1 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_1, \dots, \infty\}$ we have $|y_i - y| < 1 \Rightarrow y - 1 < y_i$. Let $C \in \mathbb{R}^+$ then for $C' = \max(C - (y - 1), 1) \in \mathbb{R}^+$ we have $C - (y - 1) \leq C'$. By [theorem: 20.79] there exist a $N_2 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_2, \dots, \infty\}$ we have we have $C - (y - 1) \leq C' < x_i$. Take $N = \max(N_1, N_2)$ then for $i \in \{N, \dots, \infty\}$ we have

$$x_i + y_i > C - (y - 1) + y_i > C - (y - 1) + y - 1 = C$$

Using [theorem: 20.79] proves that $\lim_{i \rightarrow \infty} (x_i + y_i)$ exist and

$$\lim_{i \rightarrow \infty} (x_i + y_i) = \infty = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Take $x = \lim_{i \rightarrow \infty} x_i$. By [theorem: 20.79] there exist a $N_1 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_1, \dots, \infty\}$ we have $|x_i - x| < 1 \Rightarrow x_i < x + 1$. Let $C \in \mathbb{R}^+$ then for $C' = \max(C + x + 1, 1) \in \mathbb{R}^+$ we have $C + x + 1 \leq C'$ so that $-C' \leq -C - x - 1$. By [theorem: 20.79] there exist a $N_2 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_2, \dots, \infty\}$ we have $y_i < -C' \leq -C - x - 1$. Take $N = \max(N_1, N_2)$ then for $i \in \{N, \dots, \infty\}$ we have

$$x_i + y_i < x_i + (-C') \leq x + 1 + (-C') \leq x + 1 - C - x - 1 = -C$$

Which by [theorem: 20.79] proves that $\lim_{i \rightarrow \infty} (x_i + y_i)$ exist and

$$\lim_{i \rightarrow \infty} (x_i + y_i) = -\infty = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Let $C \in \mathbb{R}^+$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_1, \dots, \infty\}$ $x_i < \frac{C}{2}$ and $\forall i \in \{N_2, \dots, \infty\}$ we have $y_i < \frac{C}{2}$. Take $N = \max(N_1, N_2)$ then for $i \in \{N, \dots, \infty\}$ we have

$$x_i + y_i < \frac{C}{2} + \frac{C}{2} = C$$

Using [theorem: 20.79] it follows that

$$\lim_{i \rightarrow \infty} (x_i + y_i) = -\infty = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Take $x = \lim_{i \rightarrow \infty} x_i$. By [theorem: 20.79] there exist a $N_1 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_1, \dots, \infty\}$ we have $|x_i - x| < 1 \Rightarrow x - 1 < x_i$. Let $C \in \mathbb{R}^+$ then for $C' = \max(C - (x - 1), 1) \in \mathbb{R}^+$ we have $C - (x - 1) \leq C'$. By [theorem: 20.79] there exist a $N_2 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_2, \dots, \infty\}$ we have we have $C - (x - 1) \leq C' < y_i$. Take $N = \max(N_1, N_2)$ then for $i \in \{N, \dots, \infty\}$ we have

$$x_i + y_i > x_i + C - (x - 1) > x - 1 + C - (x - 1) = C$$

Using [theorem: 20.79] proves that $\lim_{i \rightarrow \infty} (x_i + y_i)$ exist and

$$\lim_{i \rightarrow \infty} (x_i + y_i) = \infty = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Let $C \in \mathbb{R}^+$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_1, \dots, \infty\}$ $\frac{C}{2} < x_i$ and $\forall i \in \{N_2, \dots, \infty\}$ we have $\frac{C}{2} < y_i$. Take $N = \max(N_1, N_2)$ then for $i \in \{N, \dots, \infty\}$ we have

$$x_i + y_i > \frac{C}{2} + \frac{C}{2} = C$$

Using [theorem: 20.79] it follows that

$$\lim_{i \rightarrow \infty} (x_i + y_i) = \infty = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

6. As $\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i\right) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$ we have the following cases to consider:

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Let $x = \lim_{i \rightarrow \infty} x_i$ and $y = \lim_{i \rightarrow \infty} y_i$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} |x_i - x| < 1 \Rightarrow -\infty < x - 1 < x_i < x + 1 < \infty$$

and

$$\forall i \in \{N_2, \dots, \infty\} |y_i - y| < 1 \Rightarrow -\infty < y - 1 < y_i < y + 1 < \infty$$

Hence if take $n = \max(N_1, N_2)$ then we have

$$\forall i \in \{n, \dots, \infty\} \text{ that } (x_i, y_i) \in \mathbb{R} \times \mathbb{R} \subseteq (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Let $\varepsilon \in \mathbb{R}^+$ then [theorem: 20.79] there exists $M_1, M_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{M_1, \dots, \infty\} |y_i - y| < \frac{\varepsilon}{2}$$

and

$$\forall i \in \{M_1, \dots, \infty\} |x_i - x| < \frac{\varepsilon}{2}$$

Take $N = \max(M_1, M_2, n)$ then $\forall i \in \{N, \dots, \infty\}$ we have that $x_i + y_i$ is well defined and

$$|x_i + y_i - (x + y)| \leq |x_i - x| + |y_i - y| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Hence using [theorem: 20.79] it follows that $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit

$$\lim_{i \rightarrow \infty} (x_i + y_i) = x + y = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Let $y = \lim_{i \rightarrow \infty} y_i$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } x_i < -1 < \infty$$

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } |y - 1| < 1 \Rightarrow -\infty < y - 1 < y_i < y + 1 < \infty$$

Hence if we take $n = \max(N_1, N_2)$ we have that

$$\forall i \in \{n, \dots, \infty\} \text{ that } (x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Further take $C \in \mathbb{R}^+$ and take $C' = \max(C + (y + 1), 1) \in \mathbb{R}^+$ then $C + (y + 1) \leq C'$ or $-C' \leq -C - (y + 1)$. By [theorem: 20.79] there exist a $N_3 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_3, \dots, \infty\}$ we have that $x_i < -C'$. Hence if $N = \max(n, N_3) \in \{n, \dots, \infty\}$ then we have for $i \in \{N, \dots, \infty\}$ that $x_i < -C'$ and as $y_i \in \mathbb{R}$ it follows from [theorem: 20.29] that $x_i + y_i < -C' + y_i$. Hence

$$\begin{aligned} x_i + y_i &< -C' + y_i \\ &\leq -C - (y + 1) + y_i \\ &< -C - (y + 1) + y + 1 \\ &= -C \end{aligned}$$

Using [theorem: 20.79] it follows then that $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit and $\lim_{i \rightarrow \infty} (x_i + y_i) = -\infty$. As $\lim_{i \rightarrow \infty} x_i = -\infty$ and $\lim_{i \rightarrow \infty} y_i \in \mathbb{R}$ it follows that $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i = -\infty$. So

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Let $y = \lim_{i \rightarrow \infty} y_i$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } -\infty < 1 < x_i$$

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } |y - 1| < 1 \Rightarrow -\infty < y - 1 < y_i < y + 1 < \infty$$

Hence if we take $n = \max(N_1, N_2)$ we have that

$$\forall i \in \{n, \dots, \infty\} \text{ that } (x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Further take $C \in \mathbb{R}^+$ and take $C' = \max(C - (y - 1), 1) \in \mathbb{R}^+$ then $C - (y - 1) \leq C'$. By [theorem: 20.79] there exist a $N_3 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_3, \dots, \infty\}$ we have that $C' < x_i$. Hence if $N = \max(n, N_3) \in \{n, \dots, \infty\}$ then we have for $i \in \{N, \dots, \infty\}$ that $C' < x_i$ and as $y_i \in \mathbb{R}$ it follows from [theorem: 20.29] that $C' + y_i < x_i + y_i$. Hence

$$\begin{aligned} x_i + y_i &> C' + y_i \\ &\geq C - (y - 1) + y_i \\ &> C - (y - 1) + (y - 1) \\ &= C \end{aligned}$$

Using [theorem: 20.79] it follows then that $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit and $\lim_{i \rightarrow \infty} (x_i + y_i) = \infty$. As $\lim_{i \rightarrow \infty} x_i = \infty$ and $\lim_{i \rightarrow \infty} y_i \in \mathbb{R}$ it follows that $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i = \infty$. So

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Let $x = \lim_{i \rightarrow \infty} x_i$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } y_i < -1 < \infty$$

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } |x - 1| < 1 \Rightarrow -\infty < x - 1 < x_i < x + 1 < \infty$$

Hence if we take $n = \max(N_1, N_2)$ we have that

$$\forall i \in \{n, \dots, \infty\} \text{ that } (x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Further take $C \in \mathbb{R}^+$ and take $C' = \max(C + (x + 1), 1) \in \mathbb{R}^+$ then $C + (x + 1) \leq C'$ or $-C' \leq -C - (x + 1)$. By [theorem: 20.79] there exist a $N_3 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_3, \dots, \infty\}$ we have that $y_i < -C'$. Hence if $N = \max(n, N_3) \in \{n, \dots, \infty\}$ then we have for $i \in \{N, \dots, \infty\}$ that $y_i < -C'$ and as $y_i \in \mathbb{R}$ it follows from [theorem: 20.29] that $x_i + y_i < -C' + x_i$. Hence

$$\begin{aligned} x_i + y_i &< -C' + x_i \\ &\leq -C - (x + 1) + x_i \\ &< -C - (x + 1) + x + 1 \\ &= -C \end{aligned}$$

Using [theorem: 20.79] it follows then that $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit and $\lim_{i \rightarrow \infty} (x_i + y_i) = -\infty$. As $\lim_{i \rightarrow \infty} y_i = -\infty$ and $\lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ it follows that $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i = -\infty$. So

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Using [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \ x_i < -1 < \infty \text{ and } \forall i \in \{N_2, \dots, \infty\} \ y_i < -1 < \infty$$

So if we take $n = \max(N_1, N_2)$ then for $i \in \{n, \dots, \infty\}$ we have

$$(x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Further if $C \in \mathbb{R}^+$ then there exists $N_3, N_4 \in \{n, \dots, \infty\}$ such that

$$\forall i \in \{N_3, \dots, \infty\} \ x_i < -\frac{C}{2} \text{ and } \forall i \in \{N_4, \dots, \infty\} \ y_i < -\frac{C}{2}$$

Take $N = \max(N_3, N_4) \in \{n, \dots, \infty\}$ then $\forall i \in \{N, \dots, \infty\}$ we have by [theorem: 20.29] that

$$x_i + y_i < -\frac{C}{2} + \left(-\frac{C}{2}\right) = -C$$

So that by [theorem: 20.79] $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit and $\lim_{i \rightarrow \infty} (x_i + y_i) = -\infty$, hence, as $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i = -\infty + (-\infty) = -\infty$, it follows that

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Let $x = \lim_{i \rightarrow \infty} x_i$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } -\infty < 1 < y_i$$

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } |x - 1| < 1 \Rightarrow -\infty < x - 1 < x_i < x + 1 < \infty$$

Hence if we take $n = \max(N_1, N_2)$ we have that

$$\forall i \in \{n, \dots, \infty\} \text{ that } (x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Further take $C \in \mathbb{R}^+$ and take $C' = \max(C - (x - 1), 1) \in \mathbb{R}^+$ then $C - (x - 1) \leq C'$. By [theorem: 20.79] there exist a $N_3 \in \{k, \dots, \infty\}$ such that $\forall i \in \{N_3, \dots, \infty\}$ we have that $C' < y_i$. Hence if $N = \max(n, N_3) \in \{n, \dots, \infty\}$ then we have for $i \in \{N, \dots, \infty\}$ that $C' < y_i$ and as $x_i \in \mathbb{R}$ it follows from [theorem: 20.29] that $C' + x_i < x_i + y_i$. Hence

$$\begin{aligned} x_i + y_i &> C' + x_i \\ &\geq C - (x - 1) + x_i \\ &> C - (x - 1) + (x - 1) \\ &= C \end{aligned}$$

Using [theorem: 20.79] it follows then that $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit and $\lim_{i \rightarrow \infty} (x_i + y_i) = \infty$. As $\lim_{i \rightarrow \infty} y_i = \infty$ and $\lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ it follows that $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i = \infty$. So

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Using [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \quad -\infty < 1 < x_i \text{ and } \forall i \in \{N_2, \dots, \infty\} \quad -\infty < 1 < y_i$$

So if we take $n = \max(N_1, N_2)$ then for $i \in \{n, \dots, \infty\}$ we have

$$(x_i, y_i) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(-\infty, \infty), (\infty, -\infty)\}$$

Further if $C \in \mathbb{R}^+$ then there exists $N_3, N_4 \in \{n, \dots, \infty\}$ such that

$$\forall i \in \{N_3, \dots, \infty\} \quad \frac{C}{2} < x_i \text{ and } \forall i \in \{N_4, \dots, \infty\} \quad \frac{C}{2} < y_i$$

Take $N = \max(N_3, N_4) \in \{n, \dots, \infty\}$ then $\forall i \in \{N, \dots, \infty\}$ we have by [theorem: 20.29] that

$$C = \frac{C}{2} + \frac{C}{2} < x_i + y_i$$

So that by [theorem: 20.79] $\{x_i + y_i\}_{i \in \{n, \dots, \infty\}}$ has a limit and $\lim_{i \rightarrow \infty} (x_i + y_i) = \infty$, hence, as $\lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i = \infty + \infty = \infty$, it follows that

$$\lim_{i \rightarrow \infty} (x_i + y_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i + y_i) = \lim_{i \rightarrow \infty} x_i + \lim_{i \rightarrow \infty} y_i$$

7. As $\left(\lim_{i \rightarrow \infty} x_i, \lim_{i \rightarrow \infty} y_i\right) \in (\overline{\mathbb{R}} \times \overline{\mathbb{R}}) \setminus \{(0, -\infty), (0, \infty), (-\infty, 0), (\infty, 0)\}$ we have either:

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R}$. Let $x = \lim_{i \rightarrow \infty} x_i$ and $y = \lim_{i \rightarrow \infty} y_i$. Take $\varepsilon \in \mathbb{R}^+$. Using [theorem: 20.79] there exists a $N_1 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } |x - x_i| < 1 \Rightarrow |x_i| = |x - x_i| + |x| < 1 + |x|$$

Further as $1 + |x| \in \mathbb{R}^+$ there exists a $N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_2, \dots, \infty\} \quad |x_i - x| < \frac{\varepsilon}{2 \cdot (1 + |y|)} < \infty$$

and there exist a $N_3 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_3, \dots, \infty\} \quad |y_i - y| < \frac{\varepsilon}{2 \cdot (1 + |x|)} < \infty$$

Let $N = \max(N_1, N_2, N_3)$ then we have for $i \in \{N, \dots, \infty\}$ that $-\infty < x - \varepsilon < x_i < x + \varepsilon < \infty$ and $-\infty < y - \varepsilon < y_i < y + \varepsilon < \infty$ so that $x_i, y_i \in \mathbb{R}$ hence

$$0 \leq |x_i|, |y_i|, |x|, |y| \in \mathbb{R}$$

$$\begin{aligned} |x_i \cdot y_i - x \cdot y| &= |x_i \cdot y_i - x_i \cdot y + x_i \cdot y - x \cdot y| \\ &= |x_i \cdot (y_i - y) + (x_i - x) \cdot y| \\ &\leq |x_i \cdot (y_i - y)| + |(x_i - x) \cdot y| \\ &= |x_i| \cdot |y_i - y| + |x_i - x| \cdot |y| \\ &\leq |x_i| \cdot \frac{\varepsilon}{2 \cdot (1 + |x|)} + \frac{\varepsilon}{2 \cdot (1 + |y|)} \cdot |y| \\ &< (1 + |x_i|) \cdot \frac{\varepsilon}{2 \cdot (1 + |x|)} + \frac{\varepsilon}{2 \cdot (1 + |y|)} \cdot |y| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

Using [theorem: 20.79] again it follows that

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) \text{ exists and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R} \setminus \{0\}$. Let $C \in \mathbb{R}^+$. For $y = \lim_{i \rightarrow \infty} y_i$ we have either:

$0 < y$. For $\frac{y}{2} \in \mathbb{R}^+$ we have by [theorem: 20.79] that there exist a N_1 such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } -\infty < \frac{y}{2} = y - \frac{y}{2} < y_i < y + \frac{y}{2} < \infty \quad (20.28)$$

Let $C \in \mathbb{R}^+$ then $\frac{2 \cdot (C+1)}{y} \in \mathbb{R}^+$ so by [theorem: 20.79] there exist a $N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } \frac{2 \cdot (C+1)}{y} < x_i \quad (20.29)$$

Let $N = \max(N_1, N_2)$ then $\forall i \in \{N, \dots, \infty\}$ we have as $\frac{2 \cdot (C+1)}{y} \in \mathbb{R}^+$ by [theorem: 20.40] that

$$C+1 = \frac{y}{2} \cdot \frac{2 \cdot (C+1)}{y} \underset{[\text{eq: 20.28}]}{\leq} \frac{2 \cdot (C+1)}{y} \cdot y_i$$

Further as by [eq: 20.28] $0 < y_i$ we have by [theorem: 20.40] and [eq: 20.29] that

$$\frac{2 \cdot (C+1)}{y} \cdot y_i < x_i \cdot y_i = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

So that $C < C+1 < x_i \cdot y_i \forall i \in \{N, \dots, \infty\}$, hence using [theorem: 20.79] we have that

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) \text{ exists and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \infty = \infty \cdot y = x \cdot y$$

$y < 0$. By (3) we have that $-y \in \lim_{i \rightarrow \infty} (-y_i)$. For $-\frac{y}{2} \in \mathbb{R}^+$ we have by [theorem: 20.79] that there exist a N_1 such that

$$\forall i \in \{N_1, \dots, \infty\} \text{ we have } -\infty < -\frac{y}{2} = -y - \left(-\frac{y}{2}\right) < -y_i < -y + \left(-\frac{y}{2}\right) \quad (20.30)$$

Let $C \in \mathbb{R}^+$ then $-\frac{2 \cdot (C+1)}{y} \in \mathbb{R}^+$ so by [theorem: 20.79] there exist a $N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_2, \dots, \infty\} \text{ we have } -\frac{2 \cdot (C+1)}{y} < x_i \quad (20.31)$$

Let $N = \max(N_1, N_2)$ then $\forall i \in \{N, \dots, \infty\}$ we have as $-\frac{2 \cdot (C+1)}{y} \in \mathbb{R}^+$ by [theorem: 20.40] that

$$C+1 = \left(-\frac{y}{2}\right) \cdot \left(-\frac{2 \cdot (C+1)}{y}\right) \underset{[\text{eq: 20.30}]}{<} \left(-\frac{2 \cdot (C+1)}{y}\right) \cdot (-y_i)$$

Further as by [eq: 20.30] $0 < -y_i$ so that by [theorem: 20.40] and [eq: 20.31]

$$-\frac{2 \cdot (C+1)}{y} \cdot (-y_i) < x_i \cdot (-y_i) = -(x_i \cdot y_i)$$

hence

$$C < C+1 < -(x_i \cdot y_i)$$

Hence by [theorem: 20.79] $\lim_{i \rightarrow \infty} (-(x_i \cdot y_i))$ exists and $\lim_{i \rightarrow \infty} (-(x_i \cdot y_i)) = \infty$, using (3) it follows that $\lim_{i \rightarrow \infty} (x_i \cdot y_i)$ exist and

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = -\lim_{i \rightarrow \infty} (-(x_i \cdot y_i)) = -\infty = \infty \cdot y = x \cdot y$$

or

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) \text{ exists and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = x \cdot y = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R} \setminus \{0\}$. Define $\{x'_i\}_{i \in \{k, \dots, \infty\}}$ by $x'_i = -x_i$ then $\lim_{i \rightarrow \infty} x'_i = -\lim_{i \rightarrow \infty} x_i = \infty$. Hence this reduces to the case $\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R} \setminus \{0\}$ by substituting x_i by x'_i . So we have that $\lim_{i \rightarrow \infty} (x'_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} x'_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$ so

$$\begin{aligned} \lim_{i \rightarrow \infty} (x_i \cdot y_i) &\stackrel{(3)}{=} -\lim_{i \rightarrow \infty} (-(x_i \cdot y_i)) \\ &= \lim_{i \rightarrow \infty} (x'_i \cdot y_i) \\ &= \left(\lim_{i \rightarrow \infty} x'_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right) \\ &= \left(\lim_{i \rightarrow \infty} (-x_i) \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right) \\ &= -\left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right) \end{aligned}$$

Hence

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) \text{ exists and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \setminus \{0\} \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Define $\{x'_i\}_{i \in \{k, \dots, \infty\}}$, $\{y'_i\}_{i \in \{k, \dots, \infty\}}$ by $x'_i = y_i \wedge y'_i = x_i$. Then this case reduces to the case $\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R} \setminus \{0\}$ hence

$$\lim_{i \rightarrow \infty} (x'_i \cdot y'_i) \text{ exists and } \lim_{i \rightarrow \infty} (x'_i \cdot y'_i) = \left(\lim_{i \rightarrow \infty} x'_i \right) \cdot \left(\lim_{i \rightarrow \infty} y'_i \right) = \left(\lim_{i \rightarrow \infty} y_i \right) \cdot \left(\lim_{i \rightarrow \infty} x_i \right)$$

so that

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = \lim_{i \rightarrow \infty} (y_i \cdot x_i) = \lim_{i \rightarrow \infty} (x'_i \cdot y'_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} y_i \right) \cdot \left(\lim_{i \rightarrow \infty} x_i \right)$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Let $C \in \mathbb{R}^+$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \quad 1 < x_i \text{ and } \forall i \in \{N_2, \dots, \infty\} \quad (C+1) < y_i$$

Take $N = \max(N_1, N_2)$ then we have by [theorem: 20.40] that $C+1 \leq (C+1) \cdot x_i$ and $(C+1) \cdot x_i \leq x_i \cdot y_i$ so that

$$C < C+1 \leq x_i \cdot y_i$$

Hence using [theorem: 20.79] it follows that $\lim_{i \rightarrow \infty} (x_i \cdot y_i)$ exists and

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = \infty = \infty \cdot \infty = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i = \infty$. Let $C \in \mathbb{R}^+$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \quad x_i < -1 \Rightarrow 1 < -x_i \text{ and } \forall i \in \{N_2, \dots, \infty\} \quad (C+1) < y_i$$

Let $N = \max(N_1, N_2)$ then by [theorem: 20.40] $(C+1) \leq (-x_i) \cdot (C+1)$ and $(-x_i) \cdot (C+1) \leq (-x_i) \cdot y_i = -(x_i \cdot y_i)$, hence

$$C < C+1 \leq -(x_i \cdot y_i)$$

By [theorem: 20.79] it follows that $\lim_{i \rightarrow \infty} (-(x_i \cdot y_i)) = \infty$, hence by (3) we have that

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = -\infty = -\infty \cdot \infty = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

$\lim_{i \rightarrow \infty} x_i \in \mathbb{R} \setminus \{0\} \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Define $\{x'_i\}_{i \in \{k, \dots, \infty\}}, \{y'_i\}_{i \in \{k, \dots, \infty\}}$ by $x'_i = y_i \wedge y'_i = x_i$. Then this cases reduces to the case $\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i \in \mathbb{R} \setminus \{0\}$ hence

$$\lim_{i \rightarrow \infty} (x'_i \cdot y'_i) \text{ exists and } \lim_{i \rightarrow \infty} (x'_i \cdot y'_i) = \left(\lim_{i \rightarrow \infty} x'_i \right) \cdot \left(\lim_{i \rightarrow \infty} y'_i \right) = \left(\lim_{i \rightarrow \infty} y_i \right) \cdot \left(\lim_{i \rightarrow \infty} x_i \right)$$

so that

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = \lim_{i \rightarrow \infty} (y_i \cdot x_i) = \lim_{i \rightarrow \infty} (x'_i \cdot y'_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} y_i \right) \cdot \left(\lim_{i \rightarrow \infty} x_i \right)$$

$\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Let $C \in \mathbb{R}^+$ then by [theorem: 20.79] there exists $N_1, N_2 \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N_1, \dots, \infty\} \ x_i < -1 \Rightarrow 1 < -x_i \text{ and } \forall i \in \{N_2, \dots, \infty\} \ y_i < -(C+1) \Rightarrow C+1 < -y_i$$

Take $N = \max(N_1, N_2)$ then we have by [theorem: 20.40] that $C+1 \leq (C+1) \cdot (-x_i)$ and $(C+1) \cdot (-x_i) \leq (-x_i) \cdot (-y_i) = x_i \cdot y_i$ so that

$$C < C+1 \leq x_i \cdot y_i$$

Hence using [theorem: 20.79] it follows that $\lim_{i \rightarrow \infty} (x_i \cdot y_i)$ exists and

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = \infty = (-\infty) \cdot (-\infty) = \left(\lim_{i \rightarrow \infty} x_i \right) \cdot \left(\lim_{i \rightarrow \infty} y_i \right)$$

$\lim_{i \rightarrow \infty} x_i = \infty \wedge \lim_{i \rightarrow \infty} y_i = -\infty$. Define $\{x'_i\}_{i \in \{k, \dots, \infty\}}, \{y'_i\}_{i \in \{k, \dots, \infty\}}$ by $x'_i = y_i \wedge y'_i = x_i$. Then this cases reduces to the case $\lim_{i \rightarrow \infty} x_i = -\infty \wedge \lim_{i \rightarrow \infty} y_i = \infty$ hence

$$\lim_{i \rightarrow \infty} (x'_i \cdot y'_i) \text{ exists and } \lim_{i \rightarrow \infty} (x'_i \cdot y'_i) = \left(\lim_{i \rightarrow \infty} x'_i \right) \cdot \left(\lim_{i \rightarrow \infty} y'_i \right) = \left(\lim_{i \rightarrow \infty} y_i \right) \cdot \left(\lim_{i \rightarrow \infty} x_i \right)$$

so that

$$\lim_{i \rightarrow \infty} (x_i \cdot y_i) = \lim_{i \rightarrow \infty} (y_i \cdot x_i) = \lim_{i \rightarrow \infty} (x'_i \cdot y'_i) \text{ exist and } \lim_{i \rightarrow \infty} (x_i \cdot y_i) = \left(\lim_{i \rightarrow \infty} y_i \right) \cdot \left(\lim_{i \rightarrow \infty} x_i \right)$$

□

Corollary 20.88. Let $k \in \mathbb{N}$ and $\{\{x_{i,j}\}_{j \in \{k, \dots, \infty\}}\}_{i \in \{1, \dots, m\}}$ a family of families such that $\forall i \in \{1, \dots, n\} \ \{x_{i,j}\}_{j \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ and $\lim_{j \rightarrow \infty} x_{i,j}$ exist then $\lim_{j \rightarrow \infty} (\sum_{i=1}^n x_{i,j})$ exists and

$$\lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) = \sum_{i=1}^n \left(\lim_{j \rightarrow \infty} x_{i,j} \right)$$

Proof. We prove this by induction so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{\{x_{i,j}\}_{j \in \{k, \dots, \infty\}}\}_{i \in \{1, \dots, n\}} \text{ satisfies } \forall i \in \{1, \dots, n\} \ \{x_{i,j}\}_{j \in \{k, \dots, \infty\}} \subseteq [0, \infty] \text{ and } \lim_{j \rightarrow \infty} x_{i,j} \text{ exists then } \lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) \text{ exists and } \lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) = \sum_{i=1}^n \left(\lim_{j \rightarrow \infty} x_{i,j} \right) \right\}$$

then we have:

$1 \in S$. If for $\{\{x_{i,j}\}_{j \in \{k, \dots, \infty\}}\}_{i \in \{1\}}$ we have that $\{x_{1,j}\}_{j \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ and $\lim_{j \rightarrow \infty} x_{1,j}$ exists then, as $x_{1,j} = \sum_{i=1}^1 x_{i,j}$, it follows that $\lim_{j \rightarrow \infty} (\sum_{i=1}^1 x_{i,j})$ exists and $\lim_{j \rightarrow \infty} (\sum_{i=1}^1 x_{i,j}) = \lim_{j \rightarrow \infty} x_{1,j} = \sum_{i=1}^1 \left(\lim_{j \rightarrow \infty} x_{i,j} \right)$. Hence $1 \in S$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{\{x_{i,j}\}_{j \in \{k, \dots, \infty\}}\}_{i \in \{1, \dots, n+1\}}$ be a family of families such that $\forall i \in \{1, \dots, n+1\} \{x_{1,j}\}_{j \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ and $\lim_{j \rightarrow \infty} x_{i,j}$ exist. Then as $n \in \mathcal{S}$ we have that

$$\lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) \text{ exists and } \lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) = \sum_{i=1}^n \left(\lim_{j \rightarrow \infty} x_{i,j} \right)$$

Using the previous theorem it follows then that

$$\lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} + x_{n+1,j} \right) \text{ exists and } \lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} + x_{n+1,j} \right) = \lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) + \lim_{j \rightarrow \infty} x_{n+1,j}$$

So, as

$$\sum_{i=1}^n x_{i,j} + x_{n+1,j} = \sum_{i=1}^{n+1} x_{i,j}$$

and

$$\lim_{j \rightarrow \infty} \left(\sum_{i=1}^n x_{i,j} \right) + \lim_{j \rightarrow \infty} x_{n+1,j} = \sum_{i=1}^n \left(\lim_{j \rightarrow \infty} x_{i,j} \right) + \lim_{j \rightarrow \infty} x_{n+1,j} = \sum_{i=1}^{n+1} \left(\lim_{j \rightarrow \infty} x_{i,j} \right)$$

it follows that

$$\lim_{j \rightarrow \infty} \sum_{i=1}^{n+1} x_{i,j} = \sum_{i=1}^{n+1} \left(\lim_{j \rightarrow \infty} x_{i,j} \right)$$

which proves that $n+1 \in \mathcal{S}$. □

Next we prove that if $\lim_{i \rightarrow \infty} x_i$ exists then $\lim_{i \rightarrow \infty} \frac{1}{x_i}$ exist and $\lim_{i \rightarrow \infty} \frac{1}{x_i} = \frac{1}{\lim_{i \rightarrow \infty} x_i}$. To do this we must first prove a little lemma.

Lemma 20.89. *Let $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be sequence of extended real numbers such that $\forall i \in \{k, \dots, \infty\} x_i \neq 0$ and $\lim_{i \rightarrow \infty} x_i$ exists and $0 \neq \lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ then $0 < \inf(\{|x_i| | i \in \{k, \dots, \infty\}\})$*

Proof. Take $x = \lim_{i \rightarrow \infty} x_i$ and define $\varepsilon = \frac{|x|}{2} \in \mathbb{R}^+$. As $\lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ it follows from [theorem: 20.79] that there exist a $N \in \{k, \dots, \infty\}$ such that

$$\forall i \in \{N, \dots, \infty\} \text{ we have } |x_i - x| < \varepsilon$$

Assume that $\exists i \in \{N, \dots, \infty\}$ such that $|x_i| < \varepsilon$ then we have

$$|x| = |x - x_i + x_i| \leq |x - x_i| + |x_i| < \varepsilon + \varepsilon = 2 \cdot \varepsilon = 2 \cdot \frac{|x|}{2} = |x|$$

giving the contradiction $|x| < |x|$. Hence the assumption is wrong and we must have that

$$\forall i \in \{N, \dots, \infty\} \varepsilon \leq |x_i|$$

Take $M' = \min(\{|x_i| | i \in \{k, \dots, N\}\})$ then as $\forall i \in \{k, \dots, \infty\} x_i \neq 0$ it follows that $M' \in \mathbb{R}^+$. Take $M = \min(M', \varepsilon) \in \mathbb{R}^+$ then we have for $i \in \{k, \dots, \infty\}$ either:

$i \in \{k, \dots, N\}$. Then $|x_i| \geq M' \geq M$

$i \in \{N+1, \dots, \infty\}$. Then $i \in \{N, \dots, \infty\}$ so that $|x_i| \geq \varepsilon \geq M$

hence $\forall i \in \{k, \dots, \infty\} |x_i| \geq M$ or M is a lower bound of $\{|x_i| | i \in \{k, \dots, \infty\}\}$ proving that

$$\inf(\{|x_i| | i \in \{k, \dots, \infty\}\}) \geq M > 0$$

□

Theorem 20.90. *Let $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ be sequence of extended real numbers such that $\lim_{i \rightarrow \infty} x_i$ exists and $\lim_{i \rightarrow \infty} x_i \neq 0$ then there exist a $n \in \{k, \dots, \infty\}$ such that $\forall i \in \{n, \dots, \infty\} x_i \neq 0$ and for $\left\{ \frac{1}{x_i} \right\}_{i \in \{n, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ we have that*

$$\lim_{i \rightarrow \infty} \frac{1}{x_i} \text{ exists and } \lim_{i \rightarrow \infty} \frac{1}{x_i} = \frac{1}{\lim_{i \rightarrow \infty} x_i}$$

Proof. Let $x = \lim_{i \rightarrow \infty} x_i$ then we have either:

$x = \infty$. Then by [theorem: 20.79] there exists a $n \in \{k, \dots, \infty\}$ such that $\forall i \in \{n, \dots, \infty\}$ we have $0 < 1 < x_i$ hence $\left\{\frac{1}{x_i}\right\}_{i \in \{n, \dots, \infty\}}$ is well defined. Take $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.30] there exist a $C \in \mathbb{N}$ such that $0 < \frac{1}{C} < \varepsilon$. As $\lim_{i \rightarrow \infty} x_i = \infty$ there exist a $m \in \{k, \dots, \infty\}$ such that $\forall i \in \{m, \dots, \infty\}$ $0 < C < x_i$. Take $N = \max(m, n)$ then for $i \in \{N, \dots, \infty\}$ we have by [theorem: 20.36] that $0 < \frac{1}{x_i} \leq \frac{1}{C} < \varepsilon$, as $\frac{1}{x_i} = \left|\frac{1}{x_i} - 0\right|$, that $\left|\frac{1}{x_i} - 0\right| < \varepsilon$. Hence by [theorem: 20.79] it follows that

$$\lim_{i \rightarrow \infty} \frac{1}{x_i} \text{ exist and } \lim_{i \rightarrow \infty} \frac{1}{x_i} = 0 = \frac{1}{\infty} = \frac{1}{\lim_{i \rightarrow \infty} x_i}$$

$x = -\infty$. Let $\varepsilon \in \mathbb{R}^+$. Then by [theorem: 20.79] there exists a $n \in \{k, \dots, \infty\}$ such that $\forall i \in \{n, \dots, \infty\}$ we have $x_i < -1 < 0$ hence $\left\{\frac{1}{x_i}\right\}_{i \in \{n, \dots, \infty\}}$ is well defined. Take $\varepsilon \in \mathbb{R}^+$ then by [theorem: 10.30] there exist a $C \in \mathbb{N}$ such that $0 < \frac{1}{C} < \varepsilon$. As $\lim_{i \rightarrow \infty} x_i = -\infty$ there exist a $m \in \{k, \dots, \infty\}$ such that $x_i < -C$. Take $N = \max(m, n)$ then for $i \in \{N, \dots, \infty\}$ we have that $x_i < 0 \Rightarrow 0 < -x_i$ and $x_i < -C \Rightarrow 0 < C < -x_i$. Applying [theorem: 20.36] it follows that $0 < -\frac{1}{x_i} < \frac{1}{C}$ so that $\left|\frac{1}{x_i} - 0\right| = -\frac{1}{x_i} < \frac{1}{C} = \varepsilon$. Hence by [theorem: 20.79] it follows that

$$\lim_{i \rightarrow \infty} \frac{1}{x_i} \text{ exist and } \lim_{i \rightarrow \infty} \frac{1}{x_i} = 0 = \frac{1}{-\infty} = \frac{1}{\lim_{i \rightarrow \infty} x_i}$$

$x \in \mathbb{R} \setminus \{0\}$. Let $\delta = \frac{|x|}{2}$ then as $x \neq 0$ we have that $\delta \in \mathbb{R}^+$ so that by [theorem: 20.79] there exist a $n \in \{k, \dots, \infty\}$ such that $\forall i \in \{n, \dots, \infty\}$ we have $|x - x_i| < \delta = \frac{|x|}{2}$. Assume that there exist a $i \in \{n, \dots, \infty\}$ such that $x_i = 0$ then $|x| = |x - 0| = |x - x_i| < \delta = \frac{|x|}{2}$ a contradiction. So the assumption is false and we must have that $\forall i \in \{n, \dots, \infty\}$ $|x_i| \neq 0$. By [lemma: 20.89] we have that $I = \inf(\{|x_i| \mid i \in \{n, \dots, \infty\}\}) \in \mathbb{R}^+$. Hence

$$\forall i \in \{n, \dots, \infty\} 0 < I \leq |x_i|$$

Take now $\varepsilon \in \mathbb{R}^+$ then by [theorem: 20.79] there exist a $M \in \{n, \dots, \infty\}$ such that

$$\forall i \in \{M, \dots, \infty\} |x_i - x| < \varepsilon \cdot |x| \cdot I$$

Take $M = \max(n, M)$ then $\forall i \in \{M, \dots, \infty\}$ we have $|x_i - x| < \varepsilon \cdot |x| \cdot I < \varepsilon \cdot |x| \cdot |x_i|$ so that

$$\left|\frac{1}{x_i} - \frac{1}{x}\right| = \left|\frac{x - x_i}{x_i \cdot x}\right| = \frac{|x - x_i|}{|x| \cdot |x_i|} \leq \frac{|x - x_i|}{|x| \cdot I} < \frac{\varepsilon \cdot |x| \cdot |x_i|}{|x| \cdot |x_i|} = \varepsilon$$

which by [theorem: 20.79] proves that

$$\lim_{i \rightarrow \infty} \frac{1}{x_i} \text{ exist and } \lim_{i \rightarrow \infty} \frac{1}{x_i} = \frac{1}{x} = \frac{1}{\lim_{i \rightarrow \infty} x_i}$$

□

20.5 Series of non negative sequences

We want to deal with series in $\overline{\mathbb{R}}$ however there is a complication $\langle \overline{\mathbb{R}}, + \rangle$ is not a semi-group so that a partial sum $\sum_{i=0}^n x_i$ is not always defined. However by [theorem: 20.26] $\langle \mathbb{R}_0^+, + \rangle$ is a commutative semi-group so if we limit ourselves to non negative extended real numbers then we can use everything defined in [Section: 11.1]. If we limit ourselves to non negative extended real numbers then if $\{x_i\}_{i \in \mathbb{N}_0} \subseteq \mathbb{R}_0^+$ $\{\sum_{i=0}^n x_i\}_{n \in \mathbb{N}_0}$ is a increasing sequence, as series are defined as $\lim_{n \rightarrow \infty} \sum_{i=0}^n x_i$ it makes sense to study first limits of increasing/decreasing sequences.

Lemma 20.91. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ then we have:

1. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is increasing [in other words $\forall i \in \{k, \dots, \infty\}$ we have $x_i \leq x_{i+1}$] then

$$\forall n \in \{k, \dots, \infty\} \text{ we have } \sup(\{x_i | i \in \{n, \dots, \infty\}\}) = \sup(\{x_i | i \in \{k, \dots, \infty\}\})$$
2. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is decreasing [in other words $\forall i \in \{k, \dots, \infty\}$ we have $x_{i+1} \leq x_i$] then

$$\forall n \in \{k, \dots, \infty\} \text{ we have } \inf(\{x_i | i \in \{n, \dots, \infty\}\}) = \inf(\{x_i | i \in \{k, \dots, \infty\}\})$$

Proof. Let $n \in \{k, \dots, \infty\}$ then we have:

1. As $\{x_i | i \in \{n, \dots, \infty\}\} \subseteq \{x_i | i \in \{k, \dots, \infty\}\}$ we have by [theorem: 3.77] that

$$\sup(\{x_i | i \in \{n, \dots, \infty\}\}) \leq \sup(\{x_i | i \in \{k, \dots, \infty\}\}) \quad (20.32)$$

Next if $x \in \{x_i | i \in \{k, \dots, \infty\}\}$ then there exist a $i \in \{k, \dots, \infty\}$ such that $x = x_i$, for i we have either:

$i \in \{k, \dots, n-1\}$. Then $x = x_i \leq x_n \in \{x_i | i \in \{n, \dots, \infty\}\}$

$i \in \{n, \dots, \infty\}$. Then $x = x_i \in \{x_i | i \in \{n, \dots, \infty\}\}$

so in all cases there exist a $y \in \{x_i | i \in \{n, \dots, \infty\}\}$ such that $x \leq y$. Hence by [theorem: 3.78] it follows that $\sup(\{x_i | i \in \{k, \dots, \infty\}\}) \leq \sup(\{x_i | i \in \{n, \dots, \infty\}\})$. Combining this with [eq: 20.32] proves that

$$\sup(\{x_i | i \in \{n, \dots, \infty\}\}) = \sup(\{x_i | i \in \{k, \dots, \infty\}\})$$

2. As $\{x_i | i \in \{n, \dots, \infty\}\} \subseteq \{x_i | i \in \{k, \dots, \infty\}\}$ we have by [theorem: 3.77] that

$$\inf(\{x_i | i \in \{k, \dots, \infty\}\}) \leq \inf(\{x_i | i \in \{n, \dots, \infty\}\}) \quad (20.33)$$

Next if $x \in \{x_i | i \in \{k, \dots, \infty\}\}$ then there exist a $i \in \{k, \dots, \infty\}$ such that $x = x_i$, for i we have either:

$i \in \{k, \dots, n-1\}$. Then $x = x_i \geq x_n \in \{x_i | i \in \{n, \dots, \infty\}\}$

$i \in \{n, \dots, \infty\}$. Then $x = x_i \in \{x_i | i \in \{n, \dots, \infty\}\}$

so in all cases there exist a $y \in \{x_i | i \in \{n, \dots, \infty\}\}$ such that $y \leq x$. Hence by [theorem: 3.78] it follows that $\inf(\{x_i | i \in \{k, \dots, \infty\}\}) \geq \inf(\{x_i | i \in \{n, \dots, \infty\}\})$. Combining this with [eq: 20.33] proves that

$$\inf(\{x_i | i \in \{n, \dots, \infty\}\}) = \inf(\{x_i | i \in \{k, \dots, \infty\}\}) \quad \square$$

Theorem 20.92. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ then we have:

1. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is increasing [in other words $\forall i \in \{k, \dots, \infty\}$ we have $x_i \leq x_{i+1}$] then

$$\lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i = \sup(\{x_i | i \in \{k, \dots, \infty\}\})$$
2. If $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is decreasing [in other words $\forall i \in \{k, \dots, \infty\}$ we have $x_{i+1} \leq x_i$] then

$$\lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i = \inf(\{x_i | i \in \{k, \dots, \infty\}\})$$

Proof.

1. First

$$\begin{aligned} \limsup_{i \rightarrow \infty} x_i &= \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{lemma: 20.91}]}{=} \inf(\{\sup(\{x_i | i \in \{k, \dots, \infty\}\})\}) \\ &= \sup(\{x_i | i \in \{k, \dots, \infty\}\}) \end{aligned} \quad (20.34)$$

Next as $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is increasing we have for $l \in \{k, \dots, \infty\}$ that $\forall i \in \{l, \dots, \infty\}$ that $x_l \leq x_i$ so that $\inf(\{x_i | i \in \{l, \dots, \infty\}\}) \leq x_l \leq \inf(\{x_i | i \in \{l, \dots, \infty\}\})$ proving that

$$x_l = \inf(\{x_i | i \in \{l, \dots, \infty\}\}) \quad (20.35)$$

So

$$\begin{aligned} \liminf_{i \rightarrow \infty} x_i &= \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.35}]}{=} \sup(\{x_l | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.34}]}{=} \limsup_{i \rightarrow \infty} x_i \end{aligned}$$

so that

$$\lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i = \sup(\{x_l | l \in \{k, \dots, \infty\}\})$$

2. First

$$\begin{aligned} \liminf_{i \rightarrow \infty} x_i &= \sup(\{\inf(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{lemma: 20.91}]}{=} \sup(\{\inf(\{x_i | i \in \{k, \dots, \infty\}\})\}) \\ &= \inf(\{x_i | i \in \{k, \dots, \infty\}\}) \end{aligned} \quad (20.36)$$

Next as $\{x_i\}_{i \in \{k, \dots, \infty\}}$ is decreasing we have for $l \in \{k, \dots, \infty\}$ that $\forall i \in \{l, \dots, \infty\}$ that $x_i \leq x_l$ so that $\sup(\{x_i | i \in \{l, \dots, \infty\}\}) \leq x_l \leq \sup(\{x_i | i \in \{l, \dots, \infty\}\})$ proving that

$$x_l = \sup(\{x_i | i \in \{l, \dots, \infty\}\}) \quad (20.37)$$

So

$$\begin{aligned} \limsup_{i \rightarrow \infty} x_i &= \inf(\{\sup(\{x_i | i \in \{l, \dots, \infty\}\}) | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.37}]}{=} \inf(\{x_l | l \in \{k, \dots, \infty\}\}) \\ &\stackrel{[\text{eq: 20.36}]}{=} \liminf_{i \rightarrow \infty} x_i \end{aligned}$$

so that

$$\lim_{i \rightarrow \infty} x_i \text{ exist and } \lim_{i \rightarrow \infty} x_i = \inf(\{x_l | l \in \{k, \dots, \infty\}\}) \quad \square$$

To be able to define the infinite sum of non negative extended we must prove that the partial sums are increasing. For this we extend [theorem: 11.55] to be valid for the extended real numbers.

Theorem 20.93. *We have for families of non negative numbers [members of $[0, \infty] = \mathbb{R}^+ \cup \{0\}$ that:*

1. *If $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq [0, \infty]$ a finite family of non negative extended real numbers then*

$$0 \leq \sum_{i=0}^n x_i$$

2. *If I is a finite set and $\{x_i\}_{i \in I} \subseteq [0, \infty]$ a finite family of non negative extended real numbers then*

$$0 \leq \sum_{i \in I} x_i$$

3. *If $n \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{0, \dots, n\}} \subseteq [0, \infty]$, $\{y_i\}_{i \in \{0, \dots, n\}} \subseteq [0, \infty]$ be finite families of non negative extended real numbers such that $\forall i \in \{0, \dots, n\}$ $x_i \leq y_i$ then*

$$\sum_{i=0}^n x_i \leq \sum_{i=0}^n y_i$$

4. If I is a finite set and $\{x_i\}_{i \in I} \subseteq [0, \infty]$, $\{y_i\}_{i \in I} \subseteq [0, \infty]$ be finite families of non negative extended real numbers such that $\forall i \in I$ $x_i \leq y_i$ then

$$\sum_{i=0}^n x_i \leq \sum_{i=0}^n y_i$$

5. If $n, m \in \mathbb{N}_0$ and $\{x_i\}_{i \in n+m} \subseteq [0, \dots, \infty]$ then

$$\sum_{i=0}^n x_i \leq \sum_{i=0}^{n+m} x_i$$

6. If I is a finite set, $J \subseteq I$ and $\{x_i\}_{i \in I} \subseteq [0, \infty]$ a finite family of non negative extended real numbers then

$$\sum_{i \in J} x_i \leq \sum_{i \in I} x_i$$

Proof.

1. We use induction to prove this, so define

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{For every } \{x_i\}_{i \in \{0, \dots, \infty\}} \subseteq [0, \infty] \text{ we have } 0 \leq \sum_{i=0}^n x_i \right\}$$

then we have:

$0 \in S$. For $\{x_i\}_{i \in \{0\}} \subseteq [0, \infty]$ we have $0 \leq x_0$ so that $\sum_{i=0}^0 x_i = x_0 \geq 0$ hence $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq [0, \infty]$ be a family of non negative numbers then we have $0 \leq x_{n+1}$ and, as $n \in S$, that $0 \leq \sum_{i=0}^n x_i$. Hence we have

$$0 \underset{[\text{theorem: 20.29}]}{\leq} x_{n+1} + \sum_{i=0}^n x_i = \sum_{i=0}^{n+1} x_i$$

proving that $n+1 \in S$.

2. For I we have either

$I = \emptyset$. Then by [definition: 11.32] $0 = \sum_{i \in I} x_i$.

$I \neq \emptyset$. Then by [definition: 11.32] there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that $\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)}$. As by (1) we have that $0 \leq \sum_{i=0}^{n-1} x_{\beta(i)}$ so that $0 \leq \sum_{i \in I} x_i$.

Hence in all cases we have $0 \leq \sum_{i \in I} x_i$.

3. We use induction to prove this, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{For every } \{x_i\}_{i \in \{0, \dots, \infty\}} \subseteq [0, \infty], \{y_i\}_{i \in \{0, \dots, \infty\}} \subseteq [0, \infty] \text{ with } \forall i \in \{0, \dots, n\} \right. \\ \left. x_i \leq y_i \text{ we have } \sum_{i=0}^n x_i \leq \sum_{i=0}^n y_i \right\}$$

then we have:

$0 \in S$. For $\{x_i\}_{i \in \{0\}} \subseteq [0, \infty]$, $\{y_i\}_{i \in \{0\}} \subseteq [0, \infty]$ we have that $\sum_{i=0}^0 x_i = x_0 \leq y_0 = \sum_{i=0}^0 y_i$ proving that $0 \in S$.

$n \in S \Rightarrow n+1 \in S$. As $x_{n+1} \leq y_{n+1}$ and $\sum_{i=0}^n x_i \leq \sum_{i=0}^n y_i$ we have

$$\sum_{i=0}^{n+1} x_i = x_{n+1} + \sum_{i=0}^n x_i \underset{[\text{theorem: 20.29}]}{\leq} y_{n+1} + \sum_{i=0}^n y_i = \sum_{i=0}^{n+1} y_i$$

proving that $n+1 \in S$.

4. For I we have either

$I = \emptyset$. Then by [definition: 11.32] $\sum_{i \in I} x_i = \sum_{i \in I} y_i$.

$I \neq \emptyset$. Then by [definition: 11.32] there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that $\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)}$ and $\sum_{i \in I} y_i = \sum_{i=0}^{n-1} y_{\beta(i)}$. As by (3) we have that $\sum_{i=0}^{n-1} x_{\beta(i)} \leq \sum_{i=0}^{n-1} y_{\beta(i)}$ so that $\sum_{i \in I} x_i \leq \sum_{i \in I} y_i$.

5.

$$\begin{aligned} \sum_{i=0}^n x_i &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{0, \dots, n\}} x_i \\ &\leq \sum_{i \in \{0, \dots, n\}} x_i + \sum_{i \in \{n+1, \dots, n+m\}} x_i \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in \{0, \dots, n+m\}} x_i \\ &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{i=0}^{n+m} x_i \end{aligned}$$

6.

$$\begin{aligned} \sum_{i \in J} x_i &\stackrel{(2) [\text{theorem: 20.29}]}{\leq} \sum_{i \in J} x_i + \sum_{i \in I \setminus J} x_i \\ &\stackrel{[\text{theorem: 11.43}]}{=} \sum_{i \in I} x_i \end{aligned}$$

□

Theorem 20.94. Let $k \in \mathbb{N}_0$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ then for $\{\sum_{i=k}^n x_k\}_{n \in \{k, \dots, \infty\}} \subseteq \overline{\mathbb{R}}$ we have

$$\left\{ \sum_{i=k}^n x_k \right\}_{n \in \{k, \dots, \infty\}} \subseteq [0, \infty]$$

and

$$\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \text{ exist and } \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i = \sup \left(\left\{ \sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\} \right\} \right)$$

Proof. Let $n \in \{k, \dots, \infty\}$ then we have by [theorem: 20.93] that

$$0 \leq \sum_{i=0}^{n-k} x_{i+k} \stackrel{[\text{definition: 11.12}]}{=} \sum_{i=k}^n x_i$$

and

$$\begin{aligned} \sum_{i=k}^n x_i &\stackrel{[\text{definition: 11.12}]}{=} \sum_{i=0}^{n-k} x_{i+k} \\ &\stackrel{[\text{theorem: 20.93}]}{\leq} \sum_{i=0}^{(n-k)+1} x_{i+k} \\ &= \sum_{i=0}^{(n+1)-k} x_{i+k} \\ &= \sum_{i=k}^{n+1} x_i \end{aligned}$$

proving that $\{\sum_{i=k}^n x_i\}_{n \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ is a increasing sequence. By [theorem: 20.92] it follows then that

$$\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \text{ exist and } \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i = \sup \left(\left\{ \sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\} \right\} \right) \quad \square$$

The above theorem motivates the following definition for a denumerable sums.

Definition 20.95. Let $k \in \mathbb{N}_0$ and let $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ be a family of non negative extended real numbers then $\sum_{i=k}^{\infty} x_i$ is defined by

$$\sum_{i=k}^{\infty} x_i = \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$$

Note 20.96. By [theorem: 20.94] $\lim_{n \rightarrow \infty} \sum_{i=k}^n x_i$ always exist so that $\sum_{i=k}^{\infty} x_i$ is always defined and that

$$\sum_{i=k}^{\infty} x_i = \sup \left(\left\{ \sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\} \right\} \right) \in [0, \infty]$$

Example 20.97. Let $\{x_i\}_{i \in \{1, \dots, \infty\}} \subseteq [0, \infty]$ defined by $x_i = 0$ then $\sum_{i=1}^{\infty} x_i = 0$

Proof. We have

$$\sum_{i=1}^{\infty} x_i = \sup \left(\left\{ \sum_{i=1}^n x_i \mid n \in \{1, \dots, \infty\} \right\} \right) = \sup (\{0\}) = 0$$

□

We prove now the usual properties for $\sum_{i=k}^{\infty} x_i$, first we extend [theorem: 11.8] to the set of extended real numbers:

Theorem 20.98. Let $\alpha \in \mathbb{R}$ then we have:

1. If $k \in \mathbb{N}_0$, $n \in \{k, \dots, \infty\}$ and $\{x_i\}_{i \in \{k, \dots, n\}} \subseteq [0, \infty]$ then $\sum_{i=k}^n (\alpha \cdot x_i) = \alpha \cdot \sum_{i=k}^n x_i$
2. If I is a finite set, $\{x_i\}_{i \in I} \subseteq [0, \infty]$ then $\sum_{i \in I} (\alpha \cdot x_i) = \alpha \cdot \sum_{i \in I} x_i$

Proof.

1. We use induction in the proof, so let

$$S = \left\{ n \in \mathbb{N}_0 \mid \text{For every } \{x_i\}_{i \in \{0, \dots, n\}} \subseteq [0, \infty] \text{ we have } \sum_{i=0}^n (\alpha \cdot x_i) = \alpha \cdot \sum_{i=0}^n x_i \right\}$$

then we have:

0 ∈ S. If $\{x_i\}_{i \in \{0\}} \subseteq [0, \infty]$ then $\sum_{i=0}^0 (\alpha \cdot x_i) = \alpha \cdot x_0 = \alpha \cdot \sum_{i=0}^0 x_i$ proving that $0 \in S$.

n ∈ S ⇒ n + 1 ∈ S. If $\{x_i\}_{i \in \{0, \dots, n+1\}} \subseteq [0, \infty]$ then we have

$$\begin{aligned} \sum_{i=0}^{n+1} (\alpha \cdot x_i) &= \alpha \cdot x_{n+1} + \sum_{i=0}^n (\alpha \cdot x_i) \\ &\stackrel{n \in S}{=} \alpha \cdot x_{n+1} + \alpha \cdot \sum_{i=0}^n x_i \\ &\stackrel{[\text{theorem: 20.42}]}{=} \alpha \cdot \left(x_{n+1} + \sum_{i=0}^n x_i \right) \\ &= \alpha \cdot \sum_{i=0}^{n+1} x_i \end{aligned}$$

Hence using mathematical induction we have that

$$\forall \{x_i\}_{i \in \{0, \dots, n\}} \subseteq [0, \infty], n \in \mathbb{N}_0 \text{ we have } \sum_{i=0}^n (\alpha \cdot x_i) = \alpha \cdot \sum_{i=0}^n x_i \quad (20.38)$$

Let $k \in \mathbb{N}_0, n \in \{k, \dots, \infty\}$ then we have

$$\sum_{i=k}^n (\alpha \cdot x_i) \stackrel{[\text{definition: 11.12}]}{=} \sum_{i=0}^{n-k} (\alpha \cdot x_{i+k}) \stackrel{[\text{eq: 20.38}]}{=} \alpha \cdot \sum_{i=0}^{n-k} x_{i+k} \stackrel{[\text{definition: 11.12}]}{=} \alpha \cdot \sum_{i=k}^n x_i$$

2. For I we have either

$I = \emptyset$. Then by [definition: 11.32]

$$\alpha \cdot \sum_{i \in I} x_i = \alpha \cdot \sum_{i \in \emptyset} x_i = 0 = \sum_{i \in \emptyset} (\alpha \cdot x_i) = \sum_{i \in I} (\alpha \cdot x_i)$$

$I \neq \emptyset$. Then by [definition: 11.32] there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{0, \dots, n-1\} \rightarrow I$ such that $\sum_{i \in I} (\alpha \cdot x_i) = \sum_{i=0}^{n-1} (\alpha \cdot x_{\beta(i)})$ and $\sum_{i \in I} x_i = \sum_{i=0}^{n-1} x_{\beta(i)}$. As by (1) we have $\sum_{i=0}^{n-1} (\alpha \cdot x_{\beta(i)}) = \alpha \cdot \sum_{i=0}^{n-1} x_{\beta(i)}$ it follows that $\sum_{i \in I} (\alpha \cdot x_i) = \alpha \cdot \sum_{i \in I} x_i$. $\square$

Theorem 20.99. Let $k \in \mathbb{N}_0$ then we have

1. If $\alpha \in \mathbb{R}$ and $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ then $\sum_{i=k}^{\infty} (\alpha \cdot x_i) = \alpha \cdot \sum_{i=k}^{\infty} x_i$
2. If $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$, $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ then $\sum_{i=k}^{\infty} (x_i + y_i) = \sum_{i=k}^{\infty} x_i + \sum_{i=k}^{\infty} y_i$.
3. If $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$, $\{y_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ are such that $\forall i \in \{k, \dots, \infty\}$ we have $x_i \leq y_i$ then $\sum_{i=k}^{\infty} x_i \leq \sum_{i=k}^{\infty} y_i$.

Proof.

1. We have

$$\begin{aligned} \sum_{i=k}^{\infty} (\alpha \cdot x_i) &= \lim_{n \rightarrow \infty} \sum_{i=k}^n (\alpha \cdot x_i) \\ &\stackrel{[\text{theorem: 20.98}]}{=} \lim_{n \rightarrow \infty} \left(\alpha \cdot \sum_{i=k}^n x_i \right) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \alpha \cdot \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \\ &= \alpha \cdot \sum_{i=k}^{\infty} x_i \end{aligned}$$

2. We have

$$\begin{aligned} \sum_{i=k}^{\infty} (x_i + y_i) &= \lim_{n \rightarrow \infty} \sum_{i=k}^n (x_i + y_i) \\ &\stackrel{[\text{theorem: 11.17}]}{=} \lim_{n \rightarrow \infty} \left(\sum_{i=k}^n x_i + \sum_{i=k}^n y_i \right) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} \sum_{i=k}^{\infty} x_i + \lim_{n \rightarrow \infty} \sum_{i=k}^n y_i \\ &= \sum_{i=k}^{\infty} x_i + \sum_{i=k}^{\infty} y_i \end{aligned}$$

3. Let $n \in \{k, \dots, \infty\}$ then we have by [theorem: 20.93] that $\sum_{i=k}^n x_i \leq \sum_{i=k}^n y_i$ so that

$$\sum_{i=k}^{\infty} x_i = \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \leq \lim_{n \rightarrow \infty} \sum_{i=k}^n y_i = \sum_{i=k}^{\infty} y_i \quad \square$$

Theorem 20.100. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ then for $l \in \{k, \dots, \infty\}$ we have

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^l x_i + \sum_{i=l+1}^{\infty} x_i$$

Proof. Let $n \in \{l+1, \dots, \infty\}$ then by [theorem: 11.23]

$$\sum_{i=k}^n x_i = \sum_{i=k}^l x_i + \sum_{i=l+1}^n x_i \quad (20.39)$$

Next we have

$$\begin{aligned}
 \sum_{i=k}^{\infty} x_i &= \lim_{n \rightarrow \infty} \sum_{i=k}^n x_i \\
 &\stackrel{[\text{theorem: 20.84}]}{=} \lim_{n \rightarrow \infty} \sum_{i=k}^{n+(l+1)} x_i \\
 &\stackrel{[\text{eq: 20.39}]}{=} \lim_{n \rightarrow \infty} \left(\sum_{i=k}^l x_i + \sum_{i=l+1}^{n+(l+1)} x_i \right) \\
 &\stackrel{[\text{theorem: 20.85 (4)}]}{=} \sum_{i=k}^l x_i + \lim_{n \rightarrow \infty} \sum_{i=l+1}^{n+(l+1)} x_i \\
 &\stackrel{[\text{theorem: 20.84}]}{=} \sum_{i=k}^l x_i + \lim_{n \rightarrow \infty} \sum_{i=l+1}^n x_i \\
 &= \sum_{i=k}^l x_i + \sum_{i=l+1}^{\infty} x_i
 \end{aligned}$$

□

Theorem 20.101. Let $k \in \mathbb{N}_0$, $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ then if $\sigma: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a permutation then

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{\infty} x_{\sigma(i)}$$

Proof. Let $\sigma: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ be a bijection. Let $s \in \{\sum_{i=k}^n x_{\sigma(i)} | n \in \{k, \dots, \infty\}\}$ then there exist a $n \in \{k, \dots, \infty\}$ such that

$$s = \sum_{i=k}^n x_{\sigma(i)}$$

Let $m_n = \max(\{\sigma(i) | i \in \{k, \dots, n\}\})$ then $\forall i \in \{k, \dots, n\}$ we have $\sigma(i) \leq m_n \Rightarrow \sigma(i) \in \{k, \dots, m_n\}$. Hence we have that

$$\sigma(\{k, \dots, n\}) = \{\sigma(i) | i \in \{k, \dots, n\}\} \subseteq \{k, \dots, m_n\} \quad (20.40)$$

As σ is injective it follows that

$\sigma|_{\{k, \dots, n\}}: \{k, \dots, n\} \rightarrow \sigma(\{k, \dots, n\})$ is a bijection

$$\begin{aligned}
 s &= \sum_{i=k}^n x_{\sigma(i)} \\
 &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{k, \dots, n\}} x_{\sigma(i)} \\
 &= \sum_{i \in \{k, \dots, m\}} x_{\sigma|_{\{k, \dots, n\}}(i)} \\
 &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \sigma(\{k, \dots, n\})} x_i \\
 &\stackrel{[\text{eq: 20.40}]\&[\text{theorem: 20.93}]}{\leq} \sum_{i \in \{k, \dots, m_n\}} x_i \in \left\{ \sum_{i=k}^n x_i | n \in \{k, \dots, \infty\} \right\}
 \end{aligned}$$

So $\forall s \in \{\sum_{i=k}^n x_{\sigma(i)} | n \in \{k, \dots, \infty\}\}$ we found a $\sum_{i \in \{k, \dots, m_n\}} x_i \in \{\sum_{i=k}^n x_i | n \in \{k, \dots, \infty\}\}$ such that

$$s \leq \sum_{i \in \{k, \dots, m_n\}} x_i$$

Hence using [theorem: 3.78] it follows that

$$\sum_{i=k}^{\infty} x_{\sigma(i)} = \sup \left(\left\{ \sum_{i=k}^n x_{\sigma(i)} | n \in \{k, \dots, \infty\} \right\} \right) \leq \sup \left(\left\{ \sum_{i=k}^n x_i | n \in \{k, \dots, \infty\} \right\} \right) = \sum_{i=k}^{\infty} x_i$$

So we have proved that

$$\text{If } \sigma: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\} \text{ is a bijection, } \{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty] \text{ we have } \sum_{i=k}^{\infty} x_{\sigma(i)} \leq \sum_{i=k}^{\infty} x_i \quad (20.41)$$

For the opposite inequality if $\sigma: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a bijection then

$$\sum_{i=k}^{\infty} x_{\sigma(i)} \leq \sum_{i=k}^{\infty} x_i \quad (20.42)$$

and as $\sigma^{-1}: \{k, \dots, \infty\} \rightarrow \{k, \dots, \infty\}$ is a bijection we have also

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{\infty} x_{\sigma^{-1}(\sigma(i))} \leq \sum_{i=k}^{\infty} x_{\sigma(i)}$$

which combined with [eq: 20.42] gives

$$\sum_{i=k}^{\infty} x_i = \sum_{i=k}^{\infty} x_{\sigma(i)} \quad \square$$

In analogy with [definition: 11.32] the above theorem allows us to define the sum of a denumerable set of extended non negative numbers.

Definition 20.102. Let I be a denumerable set and $\{x_n\}_{n \in I}$ then $\sum_{i \in I} x_i$ is defined by

$$\sum_{i \in I} x_i = \sum_{i=0}^{\infty} x_{\beta(i)}$$

where

$$\beta: \mathbb{N}_0 \rightarrow I \text{ is a bijection}$$

Note 20.103. As I is denumerable there exist a bijection from $\mathbb{N}_0$ to I . Further if $\gamma: \mathbb{N}_0 \rightarrow I$ is another bijection then we have that $\beta^{-1} \circ \gamma: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is a bijection so that

$$\sum_{i=0}^{\infty} x_{\beta(i)} \stackrel{[\text{theorem: 20.101}]}{=} \sum_{i=0}^{\infty} x_{\beta((\beta^{-1} \circ \gamma)(i))} = \sum_{i=0}^{\infty} x_{\gamma(i)}$$

so that this definition is independent of the chosen bijection.

The next theorem shows how we can interchange sums.

Theorem 20.104. Let $k, l \in \mathbb{N}$ then we have:

1. If $n \in \{k, \dots, \infty\}$, $m \in \{l, \dots, \infty\}$ and $\{x_{i,j}\}_{(i,j) \in \{k, \dots, n\} \times \{l, \dots, m\}} \subseteq [0, \infty]$ then

$$\sum_{i=k}^n \left(\sum_{j=l}^m x_{i,j} \right) = \sum_{j=l}^m \left(\sum_{i=k}^n x_{i,j} \right)$$

2. Let $n \in \{l, \dots, \infty\}$ and $\{x_{i,j}\}_{(i,j) \in \{k, \dots, \infty\} \times \{l, \dots, n\}} \subseteq [0, \infty]$ then

$$\sum_{i=k}^{\infty} \left(\sum_{j=l}^n x_{i,j} \right) = \sum_{j=k}^n \left(\sum_{i=l}^{\infty} x_{i,j} \right)$$

3. Let $n \in \{l, \dots, \infty\}$ and $\{x_{i,j}\}_{(i,j) \in \{l, \dots, n\} \times \{k, \dots, \infty\}} \subseteq [0, \infty]$ then

$$\sum_{i=l}^n \left(\sum_{j=k}^{\infty} x_{i,j} \right) = \sum_{j=k}^{\infty} \left(\sum_{i=l}^n x_{i,j} \right)$$

4. Let $\{x_{i,j}\}_{(i,j) \in \{k, \dots, \infty\} \times \{l, \dots, \infty\}} \subseteq [0, \infty]$ then

$$\sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) = \sum_{j=k}^{\infty} \left(\sum_{i=l}^{\infty} x_{i,j} \right)$$

Proof.

1. This was already proved in [theorem: 11.19].

2. We prove this by induction, so let

$$S = \left\{ n \in \{l, \dots, \infty\} \mid \text{If } \{x_{i,j}\}_{(i,j) \in \{k, \dots, \infty\} \times \{l, \dots, n\}} \subseteq [0, \infty] \text{ then } \sum_{i=k}^{\infty} \left(\sum_{j=l}^n x_{i,j} \right) = \sum_{j=k}^n \left(\sum_{i=l}^{\infty} x_{i,j} \right) \right\}$$

then we have:

$l \in S$. Let $\{x_{i,j}\}_{(i,j) \in \{k, \dots, \infty\} \times \{l\}} \subseteq [0, \infty]$ then we have

$$\begin{aligned} \sum_{i=k}^{\infty} \left(\sum_{j=l}^l x_{i,j} \right) &= \sum_{i=k}^{\infty} x_{i,l} \\ &= \sum_{j=l}^l \left(\sum_{i=k}^{\infty} x_{i,j} \right) \end{aligned}$$

so that $l \in S$.

$n \in S \rightarrow n+1 \in S$. Let $\{x_{i,j}\}_{(i,j) \in \{k, \dots, \infty\} \times \{l, \dots, n+1\}} \subseteq [0, \infty]$ then we have

$$\begin{aligned} \sum_{i=k}^{\infty} \left(\sum_{j=l}^{n+1} x_{i,j} \right) &= \sum_{i=k}^{\infty} \left(\sum_{j=l}^n x_{i,j} + x_{i,n+1} \right) \\ &\stackrel{[\text{theorem: 20.99}]}{=} \sum_{i=k}^{\infty} \left(\sum_{j=l}^n x_{i,j} \right) + \sum_{i=k}^{\infty} x_{i,n+1} \\ &\stackrel{n \in S}{=} \sum_{j=l}^n \left(\sum_{i=k}^{\infty} x_{i,j} \right) + \sum_{i=k}^{\infty} x_{i,n+1} \\ &= \sum_{j=l}^{n+1} \left(\sum_{i=k}^{\infty} x_{i,j} \right) \end{aligned}$$

proving that $n+1 \in S$

3. Define $\{y_{(i,j)}\}_{(i,j) \in \{k, \dots, n\} \times \{l, \dots, \infty\}}$ by $y_{(i,j)} = x_{(j,i)}$ then we have

$$\begin{aligned} \sum_{i=k}^n \left(\sum_{j=l}^{\infty} x_{i,j} \right) &= \sum_{i=k}^n \left(\sum_{j=l}^{\infty} y_{j,i} \right) \\ &\stackrel{(2)}{=} \sum_{j=l}^{\infty} \left(\sum_{i=k}^n y_{j,i} \right) \\ &= \sum_{j=l}^{\infty} \left(\sum_{i=k}^n x_{i,j} \right) \end{aligned}$$

4. Let $n \in \{k, \dots, \infty\}$ then

$$\begin{aligned} \sum_{i=k}^n \left(\sum_{j=l}^{\infty} x_{i,j} \right) &\stackrel{(3)}{=} \sum_{j=l}^{\infty} \left(\sum_{i=k}^n x_{i,j} \right) \\ &\stackrel{[\text{theorem: 20.99}]}{\leq} \sum_{j=l}^{\infty} \sup \left(\left\{ \sum_{i=k}^n x_{i,j} \mid n \in \{k, \dots, \infty\} \right\} \right) \\ &\stackrel{[\text{definition: 20.95}]}{=} \sum_{j=l}^{\infty} \left(\sum_{i=k}^{\infty} x_{i,j} \right) \end{aligned}$$

So $\sum_{j=l}^{\infty} (\sum_{i=k}^{\infty} x_{i,j})$ is a upper bound of $\{\sum_{i=k}^n (\sum_{j=l}^{\infty} x_{i,j}) \mid n \in \{k, \dots, \infty\}\}$ from which it follows that

$$\sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) = \sup \left(\sum_{i=k}^n \left(\sum_{j=l}^{\infty} x_{i,j} \right) \mid n \in \{k, \dots, \infty\} \right) \leq \sum_{j=l}^{\infty} \left(\sum_{i=k}^{\infty} x_{i,j} \right) \quad (20.43)$$

For the opposite inequality take $m \in \{l, \dots, \infty\}$ then we have

$$\begin{aligned} \sum_{j=l}^m \left(\sum_{i=k}^{\infty} x_{i,j} \right) &\stackrel{(2)}{=} \sum_{i=k}^{\infty} \left(\sum_{j=l}^m x_{i,j} \right) \\ &\stackrel{[\text{theorem: 20.99}]}{\leq} \sum_{i=k}^{\infty} \sup \left(\left\{ \sum_{j=l}^m x_{i,j} \mid m \in \{l, \dots, \infty\} \right\} \right) \\ &\stackrel{[\text{definition: 20.95}]}{=} \sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) \end{aligned}$$

So $\sum_{i=k}^{\infty} (\sum_{j=l}^{\infty} x_{i,j})$ is a upper bound of $\{\sum_{j=l}^m (\sum_{i=k}^{\infty} x_{i,j}) \mid m \in \{l, \dots, \infty\}\}$ proving that

$$\sum_{j=l}^{\infty} \left(\sum_{i=k}^{\infty} x_{i,j} \right) = \sup \left(\left\{ \sum_{j=l}^m \left(\sum_{i=k}^{\infty} x_{i,j} \right) \mid m \in \{l, \dots, \infty\} \right\} \right) \leq \sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right)$$

Combining this with [eq: 20.43] proves

$$\sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) = \sum_{j=l}^{\infty} \left(\sum_{i=k}^{\infty} x_{i,j} \right) \quad \square$$

Every finite sum of denumerable sums can be written as a denumerable sum.

Lemma 20.105. Let $k, l, m \in \mathbb{N}_0$ and $n \in \{k, \dots, \infty\}$, $\{x_{i,j}\}_{(i,j) \in \{k, \dots, n\} \times \{l, \dots, \infty\}} \subseteq [0, \infty]$ and

$$\beta: \{m, \dots, \infty\} \rightarrow \{k, \dots, n\} \times \{l, \dots, \infty\}$$

a bijection then

$$\sum_{i=m}^{\infty} x_{\sigma(i)_1, \sigma(i)_2} = \sum_{j=l}^{\infty} \left(\sum_{i=k}^n x_{i,j} \right) \stackrel{[\text{theorem: 20.104}]}{=} \sum_{i=k}^n \left(\sum_{j=l}^{\infty} x_{i,j} \right)$$

Note 20.106. By [theorem: 6.65] there exist always a bijection between $\{m, \dots, \infty\}$ and $\{k, \dots, n\} \times \{l, \dots, \infty\}$ for a given $m \in \mathbb{N}_0$.

Proof. Let $r \in \{m, \dots, \infty\}$ and define $N_r = \max(\{\beta(i)_2 | i \in \{m, \dots, r\}\}) \in \{l, \dots, \infty\}$ then if $i \in \{m, \dots, r\}$ we have that $\beta(i)_2 \in \{k, \dots, N_r\}$ so that $\beta(i) = (\beta(i)_1, \beta(i)_2) \in \{k, \dots, n\} \times \{l, \dots, N_r\}$. Hence

$$\beta(\{m, \dots, r\}) \subseteq \{k, \dots, n\} \times \{l, \dots, N_r\}$$

Further as β is a bijection

$$\beta|_{\{m, \dots, r\}}: \{m, \dots, r\} \rightarrow \beta(\{m, \dots, r\}) \text{ is a bijection}$$

Next

$$\begin{aligned} \sum_{i=m}^r x_{\beta(i)_1, \beta(i)_2} &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{m, \dots, r\}} x_{\beta(i)_1, \beta(i)_2} \\ &= \sum_{i \in \{m, \dots, r\}} x_{\sigma|_{\{m, \dots, r\}}(i)_1, \sigma|_{\{m, \dots, r\}}(i)_2} \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{(i,j) \in \beta(\{m, \dots, r\})} x_{(i,j)_1, (i,j)_2} \\ &= \sum_{(i,j) \in \beta(\{m, \dots, r\})} x_{i,j} \\ &\stackrel{[\text{theorem: 20.93}]}{\leq} \sum_{(i,j) \in \{k, \dots, n\} \times \{l, \dots, N_r\}} x_{i,j} \\ &\stackrel{[\text{theorem: 11.45}]}{=} \sum_{i \in \{k, \dots, n\}} \left(\sum_{j \in \{l, \dots, N_r\}} x_{i,j} \right) \\ &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{i=k}^n \left(\sum_{j=l}^{N_r} x_{i,j} \right) \\ &\stackrel{[\text{theorem: 20.99}]}{\leq} \sum_{i=k}^n \sup \left(\left\{ \sum_{j=l}^s x_{i,j} \mid s \in \{l, \dots, \infty\} \right\} \right) \\ &= \sum_{i=k}^n \left(\sum_{j=l}^{\infty} x_{i,j} \right) \end{aligned}$$

Hence as $r \in \{m, \dots, \infty\}$ was chosen arbitrary it follows that

$$\sum_{i=m}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sup \left(\left\{ \sum_{i=m}^r x_{\beta(i)_1, \beta(i)_2} \mid r \in \{m, \dots, \infty\} \right\} \right) \leq \sum_{i=k}^n \left(\sum_{j=l}^{\infty} x_{i,j} \right) \quad (20.44)$$

For the opposite inequality. Let $r \in \{l, \dots, \infty\}$ and take

$$M_r = \max((\beta^{-1})(\{k, \dots, n\} \times \{l, \dots, r\})) \in \{m, \dots, \infty\}$$

So $(\beta^{-1})(\{k, \dots, n\} \times \{l, \dots, r\}) \subseteq \{m, \dots, M_r\}$ hence

$$\{k, \dots, n\} \times \{l, \dots, r\} = \beta((\beta^{-1})(\{k, \dots, n\} \times \{l, \dots, r\})) \subseteq \beta(\{m, \dots, M_r\}) \quad (20.45)$$

As β is a bijection it follows that

$$\begin{aligned}
 & \beta|_{\{m, \dots, M_r\}}: \{m, \dots, M_r\} \rightarrow \beta(\{m, \dots, M_r\}) \\
 & \sum_{j=l}^r \left(\sum_{i=k}^n x_{i,j} \right) \stackrel{[\text{theorem: 20.104}]}{=} \sum_{i=k}^n \left(\sum_{j=l}^r x_{i,j} \right) \\
 & \stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{k, \dots, n\}} \left(\sum_{j \in \{l, \dots, r\}} x_{i,j} \right) \\
 & \stackrel{[\text{theorem: 11.45}]}{=} \sum_{(i,j) \in \{k, \dots, n\} \times \{l, \dots, r\}} x_{i,j} \\
 & \stackrel{[\text{theorem: 20.99}] \wedge [\text{eq: 20.45}]}{\leq} \sum_{(i,j) \in \beta(m, \dots, M_r)} x_{i,j} \\
 & = \sum_{(i,j) \in \beta(m, \dots, M_r)} x_{(i,j)_1, (i,j)_2} \\
 & \stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{m, \dots, M_r\}} x_{\beta|_{\{m, \dots, M_r\}}(i)_1, \beta|_{\{m, \dots, M_r\}}(i)_2} \\
 & = \sum_{i \in \{m, \dots, M_r\}} x_{\beta(i)_1, \beta(i)_2} \\
 & \stackrel{[\text{theorem: 11.35}]}{=} \sum_{i=m}^{M_r} x_{\beta(i)_1, \beta(i)_2} \\
 & \leq \sup \left(\left\{ \sum_{i=m}^s x_{\beta(i)_1, \beta(i)_2} \mid s \in \{m, \dots, \infty\} \right\} \right) \\
 & = \sum_{i=m}^{\infty} x_{\beta(i)_1, \beta(i)_2}
 \end{aligned}$$

Hence as $r \in \{l, \dots, \infty\}$ was chosen arbitrary we have that

$$\sum_{j=l}^{\infty} \left(\sum_{i=k}^n x_{i,j} \right) = \sup \left(\left\{ \sum_{j=l}^s \left(\sum_{i=k}^n x_{i,j} \right) \mid s \in \{l, \dots, \infty\} \right\} \right) \leq \sum_{i=m}^{\infty} x_{\beta(i)_1, \beta(i)_2}$$

which combined with [eq: 20.44] proves that

$$\sum_{i=m}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sum_{j=l}^{\infty} \left(\sum_{i=k}^n x_{i,j} \right)$$

□

Next we extend the above theorem for arbitrary finite sums of different size.

Theorem 20.107. *Let $k, l \in \mathbb{N}_0$ and $\{x_{(i,j)}\}_{(i,j) \in \bigcup_{i \in \{k, \dots, \infty\}} \{i\} \times \{n_i, \dots, m_i\}} \subseteq [0, \infty]$ [where $\forall i \in \{k, \dots, \infty\}$ we have $n_i \leq m_i$] then if*

$$\beta: \{l, \dots, \infty\} \rightarrow \bigcup_{i \in \{k, \dots, \infty\}} \{i\} \times \{n_i, \dots, m_i\}$$

is a bijection then we have

$$\sum_{i=l}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sum_{i=k}^{\infty} \left(\sum_{j=n_i}^{m_i} x_{(i,j)} \right)$$

Proof. Let $i \in \{k, \dots, \infty\}$ and define

$$\tau_i: \{n_i, \dots, m_i\} \rightarrow \{i\} \times \{n_i, \dots, m_i\} \text{ where } \tau_i(j) = (i, j) \text{ which is a bijection}$$

Proof.

injectivity. If $\tau_i(r) = \tau_i(s)$ then $(i, r) = (i, s)$ so that $r = s$.

surjectivity. If $(r, s) \in \{i\} \times \{n_i, \dots, m_i\}$ then $s \in \{n_i, \dots, m_i\}$ and $r = i$ so that $\tau_i(s) = (i, s) = (r, s)$ proving surjectivity. $\square$

Further

$$\begin{aligned} \sum_{(k,l) \in \tau_i(\{n_i, \dots, m_i\})} x_{k,l} &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{j \in \{n_i, \dots, m_i\}} x_{\tau_i(j)_1, \tau_i(j)_2} \\ &= \sum_{j \in \{n_i, \dots, m_i\}} x_{i,j} \\ &= \sum_{j=n_i}^{m_i} x_{i,j} \end{aligned} \quad (20.46)$$

Let $m \in \{l, \dots, \infty\}$ and define $N_m = \max(\beta(i)_1 | i \in \{l, \dots, m\})$. If $i \in \{l, \dots, m\}$ then, as $\beta(i) \in \bigcup_{j \in \{k, \dots, \infty\}} \{j\} \times \{n_i, \dots, m_i\}$ there exist a $j \in \{k, \dots, \infty\}$ such that $j = \beta(i)_1 \leq N_m$ and $\beta(i)_2 \in \{n_j, \dots, m_j\}$ proving that

$$\beta(\{l, \dots, m\}) \subseteq \bigcup_{i \in \{k, \dots, N_m\}} \{i\} \times \{n_i, \dots, m_i\}$$

Further if $i, j \in \{k, \dots, N_m\}$ with $i \neq j$ then $\{i\} \times \{n_i, \dots, m_i\} \cap \{j\} \times \{n_j, \dots, m_j\} = \emptyset$ hence we have

$$\beta(\{l, \dots, m\}) \subseteq \bigsqcup_{i \in \{k, \dots, N_m\}} \{i\} \times \{n_i, \dots, m_i\} \text{ a disjoint union}$$

and as β is a bijection

$$\beta_{\{l, \dots, m\}}: \{l, \dots, m\} \rightarrow \beta(\{l, \dots, m\}) \text{ is a bijection}$$

$$\begin{aligned} \sum_{i=l}^m x_{\beta(i)_1, \beta(i)_2} &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{l, \dots, m\}} x_{\beta(i)_1, \beta(i)_2} \\ &= \sum_{i \in \{l, \dots, m\}} x_{\beta_{\{l, \dots, m\}}(i)_1, \beta_{\{l, \dots, m\}}(i)_2} \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{(i,j) \in \beta(\{l, \dots, m\})} x_{(i,j)_1, (i,j)_2} \\ &\stackrel{[\text{theorem: 20.93}]}{\leq} \sum_{(i,j) \in \bigsqcup_{i \in \{k, \dots, N_m\}} \{i\} \times \{n_i, \dots, m_i\}} x_{(i,j)_1, (i,j)_2} \\ &= \sum_{(i,j) \in \bigsqcup_{i \in \{k, \dots, N_m\}} \{i\} \times \{n_i, \dots, m_i\}} x_{i,j} \\ &\stackrel{[\text{theorem: 11.44}]}{=} \sum_{r \in \{k, \dots, N_m\}} \left(\sum_{(i,j) \in \{r\} \times \{n_r, \dots, m_r\}} x_{i,j} \right) \\ &= \sum_{r \in \{k, \dots, N_m\}} \left(\sum_{j \in \{n_r, \dots, m_r\}} x_{r,j} \right) \\ &\stackrel{[\text{theorem: 11.35}]}{=} \sum_{r=k}^{N_m} \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) \\ &\stackrel{[\text{theorem: 20.93}]}{\leq} \sup \left(\left\{ \sum_{r=k}^n \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) \mid n \in \{k, \dots, \infty\} \right\} \right) \\ &= \sum_{r=k}^{\infty} \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) \end{aligned}$$

So $m \in \{l, \dots, \infty\}$ was chosen arbitrary we have that

$$\sum_{i=l}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sup \left(\left\{ \sum_{i=l}^m x_{\beta(i)_1, \beta(i)_2} \mid m \in \{l, \dots, \infty\} \right\} \right) \leq \sum_{r=k}^{\infty} \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) \quad (20.47)$$

For the opposite inequality take $m \in \{k, \dots, \infty\}$ and take

$$M_m = \max \left(\left\{ (\beta^{-1}) \left(\bigcup_{i \in \{k, \dots, m\}} \{i\} \times \{n_i, \dots, m_i\} \right) \right\} \right)$$

then $(\beta^{-1}) \left(\bigcup_{i \in \{k, \dots, m\}} \{i\} \times \{n_i, \dots, m_i\} \right) \subseteq \{l, \dots, M_m\}$ then

$$\bigcup_{i \in \{k, \dots, m\}} \{i\} \times \{n_i, \dots, m_i\} = \beta \left((\beta^{-1}) \left(\bigcup_{i \in \{k, \dots, m\}} \{i\} \times \{n_i, \dots, m_i\} \right) \right) \subseteq \beta(\{l, \dots, M_m\}) \quad (20.48)$$

Next

$$\begin{aligned} \sum_{r=k}^m \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) &\stackrel{[\text{eq: 20.46}]}{=} \sum_{r=k}^m \left(\sum_{(i,j) \in \{r\} \times \{n_r, \dots, m_r\}} x_{i,j} \right) \\ &= \sum_{r \in \{k, \dots, m\}} \left(\sum_{(i,j) \in \{r\} \times \{n_r, \dots, m_r\}} x_{i,j} \right) \\ &\stackrel{[\text{theorem: 11.44}]}{=} \sum_{(i,j) \in \bigcup_{r \in \{k, \dots, m\}} (r) \times \{n_r, \dots, m_r\}} x_{i,j} \\ &\stackrel{[\text{theorem: 20.93}] \wedge [\text{eq: 20.48}]}{\leq} \sum_{(i,j) \in \beta(\{l, \dots, M_m\})} x_{i,j} \\ &= \sum_{(i,j) \in \beta(\{l, \dots, M_m\})} x_{(i,j)_1, (i,j)_2} \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{l, \dots, M_m\}} x_{\beta(i)_1, \beta(i)_2} \\ &= \sum_{i=l}^{M_m} x_{\beta(i)_1, \beta(i)_2} \\ &\leq \sup \left(\left\{ \sum_{i=l}^n x_{\beta(i)_1, \beta(i)_2} \mid n \in \{l, \dots, \infty\} \right\} \right) \\ &= \sum_{i=l}^{\infty} x_{\beta(i)_1, \beta(i)_2} \end{aligned}$$

As $m \in \{k, \dots, \infty\}$ was chosen arbitrary it follows that

$$\sum_{r=k}^{\infty} \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) = \sup \left(\left\{ \sum_{r=k}^n \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) \mid n \in \{r, \dots, \infty\} \right\} \right) \leq \sum_{i=l}^{\infty} x_{\beta(i)_1, \beta(i)_2}$$

which combined with [eq: 20.47] proves that

$$\sum_{i=l}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sum_{r=k}^{\infty} \left(\sum_{j=n_r}^{m_r} x_{r,j} \right) \quad \square$$

Theorem 20.108. Let $k, l, n \in \mathbb{N}_0$ and $\{x_{i,j}\}_{(i,j) \in \{k, \dots, \infty\} \times \{l, \dots, \infty\}} \subseteq [0, \infty]$ and

$\beta: \{n, \dots, \infty\} \rightarrow \{k, \dots, \infty\} \times \{l, \dots, \infty\}$ a bijection

then

$$\sum_{i=0}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right)$$

Note 20.109. By [theorem: 6.65] there exist always a bijection between $\{n, \dots, \infty\}$ and $\{k, \dots, \infty\} \times \{l, \dots, \infty\}$ for a given $m \in \mathbb{N}_0$.

Proof. Let $m \in \{n, \dots, \infty\}$ and define $N_m = \max(\{\beta(i)_1 | i \in \{n, \dots, m\}\})$ and $M_m = \max(\{\beta(i)_2 | i \in \{n, \dots, m\}\})$ then we have that

$$\beta(\{n, \dots, m\}) \subseteq \{k, \dots, N_m\} \times \{l, \dots, M_m\} \quad (20.49)$$

Then we have

$$\begin{aligned} \sum_{i=n}^m x_{\beta(i)_1, \beta(i)_2} & \stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{n, \dots, m\}} x_{\beta(i)_1, \beta(i)_2} \\ & \stackrel{[\text{theorem: 11.36}]}{=} \sum_{(i,j) \in \beta(\{n, \dots, m\})} x_{(i,j)_1, (i,j)_2} \\ & = \sum_{(i,j) \in \beta(\{n, \dots, m\})} x_{i,j} \\ & \stackrel{[\text{theorem: 20.93}] \wedge [\text{eq: 20.49}]}{\leq} \sum_{(i,j) \in \{k, \dots, N_m\} \times \{l, \dots, M_m\}} x_{i,j} \\ & \stackrel{[\text{theorem: 11.45}]}{=} \sum_{i \in \{k, \dots, N_m\}} \left(\sum_{j \in \{l, \dots, M_m\}} x_{i,j} \right) \\ & \stackrel{[\text{theorem: 11.35}]}{=} \sum_{i=k}^{N_m} \left(\sum_{j=l}^{M_m} x_{i,j} \right) \\ & \stackrel{[\text{theorem: 20.93}]}{\leq} \sum_{i=k}^{N_m} \sup \left(\left\{ \sum_{j=l}^r x_{i,j} \mid r \in \{l, \dots, \infty\} \right\} \right) \\ & = \sum_{i=k}^{N_m} \left(\sum_{j=l}^{M_m} x_{i,j} \right) \\ & \leq \sup \left(\left\{ \sum_{i=k}^r \left(\sum_{j=l}^{M_m} x_{i,j} \right) \mid r \in \{k, \dots, \infty\} \right\} \right) \\ & = \sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) \end{aligned}$$

As $m \in \{n, \dots, \infty\}$ was chosen arbitrary it follows that

$$\sum_{i=n}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sup \left(\left\{ \sum_{i=n}^m x_{\beta(i)_1, \beta(i)_2} \mid m \in \{n, \dots, \infty\} \right\} \right) \leq \sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) \quad (20.50)$$

For the opposite inequality let $r \in \{k, \dots, \infty\}$, $s \in \{l, \dots, \infty\}$ and take

$$N = \max((\beta^{-1})(\{k, \dots, r\} \times \{l, \dots, s\}))$$

then $(\beta^{-1})(\{k, \dots, r\} \times \{l, \dots, s\}) \subseteq \{n, \dots, N\}$ so that

$$\{k, \dots, r\} \times \{l, \dots, s\} = \beta((\beta^{-1})(\{k, \dots, r\} \times \{l, \dots, s\})) \subseteq \beta(\{n, \dots, N\}) \quad (20.51)$$

Next

$$\begin{aligned} \sum_{i=k}^r \left(\sum_{j=l}^s x_{i,j} \right) & \stackrel{[\text{theorem: 11.35}]}{=} \sum_{i \in \{k, \dots, r\}} \left(\sum_{j \in \{l, \dots, s\}} x_{i,j} \right) \\ & \stackrel{[\text{theorem: 11.45}]}{=} \sum_{(i,j) \in \{k, \dots, r\} \times \{l, \dots, s\}} x_{i,j} \\ & \stackrel{[\text{theorem: 20.93}] \wedge [\text{eq: 20.51}]}{\leq} \sum_{(i,j) \in \beta(\{n, \dots, N\})} x_{i,j} \end{aligned}$$

$$\begin{aligned}
&= \sum_{(i,j) \in \beta(n, \dots, N)} x_{(i,j)_1, (i,j)_2} \\
&\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{n, \dots, N\}} x_{\beta(i)_1, \beta(i)_2} \\
&= \sum_{i=n}^N x_{\beta(i)_1, \beta(i)_2} \\
&\leq \sup \left(\left\{ \sum_{i=n}^m x_{\beta(i)_1, \beta(i)_2} \mid m \in \{n, \dots, \infty\} \right\} \right) \\
&= \sum_{i=n}^{\infty} x_{\beta(i)_1, \beta(i)_2}
\end{aligned}$$

Hence as $r \in \{k, \dots, \infty\}$ was chosen arbitrary we have that

$$\sum_{i=k}^{\infty} \left(\sum_{j=l}^s x_{i,j} \right) = \sup \left(\left\{ \sum_{i=k}^m \left(\sum_{j=l}^s x_{i,j} \right) \mid m \in \{k, \dots, \infty\} \right\} \right) \leq \sum_{i=n}^{\infty} x_{\beta(i)_1, \beta(i)_2} \quad (20.52)$$

Further

$$\begin{aligned}
\sum_{j=l}^s \left(\sum_{i=k}^{\infty} x_{i,j} \right) &\stackrel{[\text{theorem: 20.104}]}{=} \sum_{i=k}^{\infty} \left(\sum_{j=l}^s x_{i,j} \right) \\
&\stackrel{[\text{eq: 20.52}]}{\leq} \sum_{i=n}^{\infty} x_{\beta(i)_1, \beta(i)_2}
\end{aligned}$$

So as $s \in \{l, \dots, \infty\}$ is chosen arbitrary we have that

$$\sum_{j=l}^{\infty} \left(\sum_{i=k}^{\infty} x_{i,j} \right) = \sup \left(\left\{ \sum_{j=l}^m \left(\sum_{i=k}^{\infty} x_{i,j} \right) \mid m \in \{l, \dots, \infty\} \right\} \right) \leq \sum_{i=n}^{\infty} x_{\beta(i)_1, \beta(i)_2}$$

hence

$$\sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right) \stackrel{[\text{theorem: 20.104}]}{=} \sum_{j=l}^{\infty} \left(\sum_{i=k}^{\infty} x_{i,j} \right) \leq \sum_{i=n}^{\infty} x_{\beta(i)_1, \beta(i)_2}$$

which combined with [eq: 20.50] proves that

$$\sum_{i=0}^{\infty} x_{\beta(i)_1, \beta(i)_2} = \sum_{i=k}^{\infty} \left(\sum_{j=l}^{\infty} x_{i,j} \right)$$

□

If a series is finite then all the terms in the series are real numbers.

Lemma 20.110. *Let $k \in \mathbb{N}_0$, $n \in \{k, \dots, \infty\}$ and $\{x_i\}_{i \in \{k, \dots, n\}} \subseteq [0, \infty]$ then we have*

$$\sum_{i=k}^n x_i = \infty \Leftrightarrow \infty \in \{x_i \mid i \in \{k, \dots, n\}\}$$

Proof.

$\Rightarrow$. We use induction to prove this, so let

$$S = \left\{ n \in \{k, \dots, \infty\} \mid \text{If } \{x_i\}_{i \in \{k, \dots, n\}} \subseteq [0, \infty] \text{ satisfies } \sum_{i=k}^n x_i \text{ then } \infty \in \{x_i \mid i \in \{k, \dots, \infty\}\} \right\}$$

then we have:

$k \in S$. For $\{x_i\}_{i \in \{k\}} \subseteq [0, \infty]$ we have $x_k = \sum_{i=k}^k x_i = \infty$ hence $\infty \in \{x_i \mid i \in \{k\}\}$ proving that $k \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{k, \dots, n+1\}} \subseteq [0, \infty]$ be such that $\sum_{i=k}^{n+1} x_i = \infty$ then, as $\sum_{i=k}^{n+1} x_i = x_{n+1} + \sum_{i=k}^n x_i$, we have either:

$x_{n+1} = \infty$. Then $\infty \in \{x_i | i \in \{k, \dots, n+1\}\}$

$\sum_{i=k}^n x_i$. Then as $n \in S$ $\infty \in \{x_i | i \in \{k, \dots, n\}\} \subseteq \infty \in \{x_i | i \in \{k, \dots, n+1\}\}$

So in all possible cases we have $\infty \in \{x_i | i \in \{k, \dots, n+1\}\}$ which proves that $n+1 \in S$

$\Leftarrow$. We use also induction for this, so let

$$S = \left\{ n \in \{k, \dots, \infty\} \mid \text{If } \{x_i\}_{i \in \{k, \dots, n\}} \subseteq [0, \infty] \text{ satisfies } \infty \in \left\{ x_i \mid i \in \{k, \dots, n\} \text{ then } \sum_{i=k}^n x_i = \infty \right\} \right\}$$

the we have:

$k \in S$. If $\{x_i\}_{i \in \{k\}} \subseteq [0, \infty]$ satisfies $\infty \in \{x_i | i \in \{k, \dots, k\}\} = \{x_i\}$ then $x_k = \infty$ so that $\sum_{i=k}^k x_i = x_k = \infty$. Hence $k \in S$.

$n \in S \Rightarrow n+1 \in S$. If $\{x_i\}_{i \in \{k, \dots, n+1\}} \subseteq [0, \infty]$ satisfies $\infty \in \{x_i | i \in \{k, \dots, n+1\}\}$ then there exist a $i \in \{k, \dots, n+1\}$ such that $x_i = \infty$. For i we have either:

$i = n+1$. Then $\sum_{i=k}^{n+1} x_i = x_{n+1} + \sum_{i=k}^n x_i = \infty + \sum_{i=k}^n x_i = \infty$

$i \in \{k, \dots, n\}$. Then $\sum_{i=k}^{n+1} x_i = x_{n+1} + \sum_{i=k}^n x_i \stackrel{n \in S}{=} x_{n+1} + \infty = \infty$

So in all cases we have $\sum_{i=k}^{n+1} x_i = \infty$ proving that $n+1 \in S$. $\square$

Theorem 20.111. Let $k \in \mathbb{N}_0$ and for $\{x_i\}_{i \in \{k, \dots, \infty\}} \subseteq [0, \infty]$ we have $\sum_{i=k}^{\infty} x_i < \infty$ then $\forall i \in \{k, \dots, \infty\}$ we have $x_i < \infty$

Proof. As $\sup(\{\sum_{i=k}^n x_i \mid n \in \{k, \dots, \infty\}\}) = \sum_{i=k}^{\infty} x_i < \infty$ it follows from [theorem: 20.15] that $\forall n \in \{k, \dots, \infty\}$ we have that $\sum_{i=k}^n x_i < \infty \stackrel{[\text{lemma: 20.110}]}{=} x_n = \infty$. $\square$

Chapter 21

Measures and Integration

21.1 Preliminaries

Theorem 21.1. *Let $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}}$ be a finite family of finite sets then $\prod_{i \in \{1, \dots, n\}} A_i$ is finite and $\text{card}(\prod_{i \in \{1, \dots, n\}} A_i) = \prod_{i=1}^n \text{card}(A_i)$*

Proof. First using [theorem: 6.58] we have that

$$\prod_{i \in \{1, \dots, n\}} A_i \text{ is finite}$$

Let

$$\mathcal{S} = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \text{ is a finite family of finite sets then } \text{card}\left(\prod_{i \in \{1, \dots, n\}} A_i\right) = \prod_{i=1}^n \text{card}(A_i) \right\}$$

then we have:

$1 \in \mathcal{S}$. If $\{A_i\}_{i \in \{1\}}$ is a finite family of finite sets then by [theorem: 2.147] $\prod_{i \in \{1\}} A_i$ is bijective to A_1 so that $\text{card}(\prod_{i \in \{1\}} A_i) = \text{card}(A_1) = \prod_{i \in \{1\}} \text{card}(A_i)$ which proves that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}}$ be a finite family of sets then we have by [theorem: 6.56] that $\prod_{i \in \{1, \dots, n+1\}} A_i$ is bijective to $(\prod_{i \in \{1, \dots, n\}} A_i) \times A_{n+1}$ so that

$$\begin{aligned} \text{card}\left(\prod_{i \in \{1, \dots, n+1\}} A_i\right) &= \text{card}\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \times A_{n+1}\right) \\ &\stackrel{[\text{theorem: 10.92}]}{=} \text{card}\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot \text{card}(A_{n+1}) \\ &\stackrel{=_{n \in \mathcal{S}}}{=} \left(\prod_{i=1}^n \text{card}(A_i)\right) \cdot \text{card}(A_{n+1}) \\ &= \prod_{i=1}^{n+1} \text{card}(A_i) \end{aligned}$$

which proves that $n+1 \in \mathcal{S}$. □

Theorem 21.2. *Let A, B, C, D be sets then*

1. $(A \cdot B) \setminus (C \cdot D) = ((A \setminus C) \cdot B) \cup (A \cdot (B \setminus D))$
2. $(A \cdot B) \setminus (C \cdot D) = (A \setminus C) \cdot (B \setminus D) \cup (A \setminus C) \cdot (B \cap D) \cup (A \cap C) \cdot (B \setminus D)$ and
 - a. $((A \setminus C) \cdot (B \setminus D)) \cap ((A \setminus C) \cdot (B \cap D)) = \emptyset$
 - b. $((A \setminus C) \cdot (B \setminus D)) \cap ((A \cap C) \cdot (B \setminus D)) = \emptyset$
 - c. $((A \setminus C) \cdot (B \cap D)) \cap ((A \cap C) \cdot (B \setminus D)) = \emptyset$

Proof.

1. We have the following equivalences

$$\begin{aligned}
 (x, y) \in (A \cdot B) \setminus (C \cdot D) &\Leftrightarrow (x, y) \in A \cdot B \wedge (x, y) \notin C \cdot D \\
 &\Leftrightarrow x \in A \wedge y \in B \wedge \neg(x \in C \wedge y \in D) \\
 &\Leftrightarrow x \in A \wedge y \in B \wedge (x \notin C \vee y \notin D) \\
 &\Leftrightarrow (x \in A \wedge y \in B \wedge x \notin C) \vee (x \in A \wedge y \in B \wedge y \notin D) \\
 &\Leftrightarrow (x \in A \setminus C \wedge y \in B) \vee (x \in A \wedge y \in B \setminus D) \\
 &\Leftrightarrow (x, y) \in (A \setminus C) \cdot B \vee (x, y) \in A \cdot (B \setminus D) \\
 &\Leftrightarrow (x, y) \in (A \setminus C) \cdot B \bigcup (A \cdot (B \setminus D))
 \end{aligned}$$

which proves that

$$(A \cdot B) \setminus (C \cdot D) = ((A \setminus C) \cdot B) \bigcup (A \cdot (B \setminus D))$$

2. Next we have

$$\begin{aligned}
 &(A \cdot B) \setminus (C \cdot D) = \\
 &((A \setminus C) \cdot B) \bigcup (A \cdot (B \setminus D)) \quad [\text{theorem: 1.39}] \\
 &(((A \setminus C) \cdot ((B \setminus D) \bigcup (B \cap D))) \bigcup (((A \setminus C) \bigcup (A \cap C)) \cdot (B \setminus D))) \quad [\text{theorem: 6.89}] \\
 &\underbrace{(A \setminus C) \cdot (B \setminus D)}_I \bigcup (A \setminus C) \cdot (B \cap D) \bigcup \underbrace{(A \setminus C) \cdot (B \setminus D)}_I \bigcup (A \cap C) \cdot (B \setminus D) = \\
 &(A \setminus C) \cdot (B \setminus D) \bigcup (A \setminus C) \cdot (B \cap D) \bigcup (A \cap C) \cdot (B \setminus D)
 \end{aligned}$$

proving

$$(A \cdot B) \setminus (C \cdot D) = (A \setminus C) \cdot (B \setminus D) \bigcup (A \setminus C) \cdot (B \cap D) \bigcup (A \cap C) \cdot (B \setminus D)$$

a.

$$\begin{aligned}
 &((A \setminus C) \cdot (B \setminus D)) \cap ((A \setminus C) \cdot (B \cap D)) \quad [\text{theorem: 6.89}] \\
 &(A \setminus C) \cdot ((B \setminus D) \cap (B \cap D)) = \\
 &(A \setminus C) \cdot \emptyset \quad [\text{theorem: 6.89}] \\
 &\emptyset
 \end{aligned}$$

b.

$$\begin{aligned}
 &((A \setminus C) \cdot (B \setminus D)) \cap ((A \cap C) \cdot (B \setminus D)) \quad [\text{theorem: 6.89}] \\
 &((A \setminus C) \cap (A \cap C)) \cdot (B \setminus D) = \\
 &\emptyset \cap (B \setminus D) \quad [\text{theorem: 6.89}] \\
 &\emptyset
 \end{aligned}$$

c.

$$\begin{aligned}
 &((A \setminus C) \cdot (B \cap D)) \cap ((A \cap C) \cdot (B \setminus D)) \quad [\text{theorem: 6.89}] \\
 &(A \cdot (B \cap D)) \cap (A \cdot (B \setminus D)) \quad [\text{theorem: 6.89}] \\
 &A \cdot ((B \cap D) \cap (B \setminus D)) = \\
 &A \cdot \emptyset \quad [\text{theorem: 6.89}] \\
 &\emptyset
 \end{aligned}$$

□

Definition 21.3. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$, I a set then a family of sets $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ is **pairwise disjoint** if

$$\forall i, j \in I \text{ with } i \neq j \text{ we have } A_i \cap A_j = \emptyset$$

Notation 21.4. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$, I a set and $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ then if we write

$$\bigsqcup_{i \in I} A_i$$

we mean

$$\bigsqcup_{i \in I} A_i$$

and additional that $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ is pairwise disjoint.

Theorem 21.5. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$, I, J sets and $\{A_i\}_{i \in J} \subseteq \mathcal{A}$ be a pairwise disjoint family and $\sigma: I \rightarrow J$ a bijection then $\{A_{\sigma(i)}\}_{i \in I} \subseteq \mathcal{A}$ is pairwise disjoint and $\bigcup_{i \in I} A_{\sigma(i)} = \bigcup_{i \in J} A_i$. In other words using the above theorem we have

$$\bigsqcup_{i \in J} A_i = \bigsqcup_{i \in I} A_{\sigma(i)}$$

Proof. Using [theorem: 2.127] and the fact that a bijection is a surjection we have

$$\bigcup_{i \in I} A_{\sigma(i)} = \bigcup_{i \in J} A_i$$

Let $i, j \in I$ be such that $i \neq j$ then as σ is injective we have $\sigma(i) \neq \sigma(j)$ [if $\sigma(i) = \sigma(j)$ then $i = j$] so that $A_{\sigma(i)} \cap A_{\sigma(j)} = \emptyset$. Hence $\{A_{\sigma(i)}\}_{i \in \{1, \dots, n\}}$ is pairwise disjoint. $\square$

The following set based theorems and lemmas will be usefully later

Lemma 21.6. Let N be either ∞ or $N \in \mathbb{N}$ $\{A_i\}_{i \in \{1, \dots, N\}} \subseteq \mathcal{A}$ be a sequence or finite family of sets then for

$$\{B_i\}_{i \in \{1, \dots, N\}} \text{ defined by } B_i = \begin{cases} A_i & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, j-1\}} A_j & \text{if } i \in \{2, \dots, N\} \end{cases}$$

we have

$$\forall i \in \{1, \dots, N\} \quad B_i \subseteq A_i$$

$$\bigsqcup_{i \in \{1, \dots, N\}} B_i = \bigcup_{i \in \{1, \dots, N\}} A_i$$

and

$\forall i, j \in \{1, \dots, N\}$ with $i \neq j$ then $B_i \cap B_j = \emptyset$ [in other words $\{B_i\}_{i \in \{1, \dots, N\}}$ is pairwise disjoint]

Proof. We have

$$\forall i \in \{1, \dots, N\} \text{ that } B_i = \begin{cases} A_1 \subseteq A_i & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j \subseteq A_i & \text{if } i \in \{2, \dots, N\} \end{cases} \subseteq A_i$$

From the above it follows that

$$\bigcup_{i \in \{1, \dots, N\}} B_i \subseteq \bigcup_{i \in \{1, \dots, N\}} A_i \quad (21.1)$$

If $x \in \bigcup_{i \in \{1, \dots, N\}} A_i$ then there exist a $k \in \{1, \dots, N\}$ such that $x \in A_k$. Define $M = \{i \in \{1, \dots, k\} | x \in A_i\} \subseteq \{1, \dots, k\}$ then $k \in M$ and M is finite so that

$$m = \min(M) \text{ exist}$$

As $m \in M \subseteq \{1, \dots, k\}$ we have $x \in A_m$ and for m either:

$m = 1$. Then $x \in A_1 = B_1$ so that $x \in \bigcup_{i \in \{1, \dots, N\}} B_i$.

$1 < m$. Then for $\forall j \in \{1, \dots, m-1\}$ we have $x \notin A_j$ so that $x \notin \bigcup_{j \in \{1, \dots, m-1\}} A_j$ hence $x \in A_m \setminus \bigcup_{j \in \{1, \dots, m-1\}} A_j = B_m$. So $x \in \bigcup_{i \in \{1, \dots, N\}} B_i$.

As $x \in \bigcup_{i \in \{1, \dots, N\}} A_i$ was chosen arbitrary it follows that $\bigcup_{i \in \{1, \dots, N\}} A_i \subseteq \bigcup_{i \in \{1, \dots, N\}} B_i$ which combined with [eq: 21.1] proves that

$$\bigcup_{i \in \{1, \dots, N\}} B_i = \bigcup_{i \in \{1, \dots, N\}} A_i$$

Let $i, j \in \{1, \dots, N\}$ with $i \neq j$ then we may assume that $i < j$ [otherwise exchange i and j]. Assume that $x \in B_i \cap B_j$ then for i we have either:

$i = 1$. Then $x \in B_1 = A_1$ and $x \in B_j = A_j \setminus \bigcup_{k \in \{1, \dots, j-1\}} A_k$. So $\forall k \in \{1, \dots, j-1\}$ we have that $x \notin A_k$, hence, as $1 = i < j \Rightarrow 1 \in \{1, \dots, j-1\}$, it follows that $x \notin A_1$ contradicting $x \in B_1 = A_1$.

$1 < j$. Then $x \in B_j = A_j \setminus \bigcup_{k \in \{1, \dots, j-1\}} A_k \Rightarrow \forall k \in \{1, \dots, j-1\}$ we have $x \notin A_k$. As $i < j \Rightarrow i \in \{1, \dots, j-1\}$ it follows that $x \notin A_i$ contradicting $x \in A_i \setminus \bigcup_{k \in \{1, \dots, i-1\}} A_k \subseteq A_i$.

As in all cases we have a contradiction the assumption is wrong and we must have that

$$B_i \cap B_j = \emptyset \quad \square$$

Lemma 21.7. Let $\{A_i\}_{i \in \mathbb{N}}$ be a sequence of sets then if we define $\{B_i\}_{i \in \mathbb{N}}$ by $B_i = \bigcup_{j \in \{1, \dots, i\}} A_j$ we have

$$\bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i \text{ and } \forall i \in \mathbb{N} \text{ we have } B_i \subseteq B_{i+1}$$

Proof. First if $i \in \mathbb{N}$ and $x \in B_i$ then there exists a $j \in \{1, \dots, i\}$ such that

$$x \in A_j \subseteq \bigcup_{j \in \{1, \dots, i+1\}} A_j = B_{i+1}$$

proving that

$$\forall i \in \mathbb{N} \text{ we have } B_i \subseteq B_{i+1} \quad (21.2)$$

Second if $x \in \bigcup_{i \in \mathbb{N}_0} A_i$ then $\exists i \in \mathbb{N}$ such that $x \in A_i$. So as $i \in \{1, \dots, i\}$ we have that $x \in \bigcup_{j \in \{1, \dots, i\}} A_j = B_i$ which proves that $x \in \bigcup_{i \in \mathbb{N}} B_i$, or

$$\bigcup_{i \in \mathbb{N}} A_i \subseteq \bigcup_{i \in \mathbb{N}} B_i \quad (21.3)$$

Third if $x \in \bigcup_{i \in \mathbb{N}} B_i$ then $\exists i \in \mathbb{N}$ such that $x \in B_i = \bigcup_{j \in \{1, \dots, i\}} A_j$, so there exists a $j \in \{1, \dots, i\} \subseteq \mathbb{N}$ with $x \in A_j$, proving that $x \in \bigcup_{i \in \mathbb{N}} A_i$. Hence $\bigcup_{i \in \mathbb{N}} B_i \subseteq \bigcup_{i \in \mathbb{N}} A_i$ which combined with [eq: 21.3] gives

$$\bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i \quad (21.4)$$

The theorem is then proved by [eqs: 21.2, 21.3 and 21.4]. $\square$

Definition 21.8. Let X be a set and $\mathcal{A} \subseteq \mathcal{P}(X)$ then

1. $\mathcal{A}$ is $\cup$ -stable if $\forall A, B \in \mathcal{A}$ we have $A \cup B \in \mathcal{A}$
2. $\mathcal{A}$ is $\cap$ -stable if $\forall A, B \in \mathcal{A}$ we have $A \cap B \in \mathcal{A}$

Theorem 21.9. Let X be a set and $\mathcal{A} \subseteq \mathcal{P}(X)$ then we have:

1. $\mathcal{A}$ is $\cup$ -stable $\Leftrightarrow \forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $n \in \mathbb{N}$ we have $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$

2. $\mathcal{A}$ is $\cap$ -stable $\Leftrightarrow \forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}, n \in \mathbb{N}$ we have $\bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$

Proof. This is easily proved by induction

1.

$\Rightarrow$. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ then } \bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A} \right\}$$

then we have:

$1 \in S$. If $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A}$ then $\bigcup_{i \in \{1\}} A_i = A_1 \in \mathcal{A}$ so that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A}$ then $A_{n+1} \in \mathcal{A}$ and, as $n \in S$, $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$. Hence $\bigcup_{i \in \{1, \dots, n+1\}} A_i = A_{n+1} \cup \left(\bigcup_{i \in \{1, \dots, n\}} A_i \right) \in \mathcal{A}$ which proves that $n+1 \in S$

$\Leftarrow$. Let $A, B \in \mathcal{A}$ and define $\{A_i\}_{i \in \{1, 2\}} \subseteq \mathcal{A}$ by $A_1 = A$ and $A_2 = B$ then $A \cup B = A_1 \cup A_2 = \bigcup_{i \in \{1, 2\}} A_i \in \mathcal{A}$.

2.

$\Rightarrow$. We prove this by induction so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ then } \bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{A} \right\}$$

then we have:

$1 \in S$. If $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A}$ then $\bigcap_{i \in \{1\}} A_i = A_1 \in \mathcal{A}$ so that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A}$ then $A_{n+1} \in \mathcal{A}$ and, as $n \in S$, $\bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$. Hence $\bigcap_{i \in \{1, \dots, n+1\}} A_i = A_{n+1} \cap \left(\bigcap_{i \in \{1, \dots, n\}} A_i \right) \in \mathcal{A}$ which proves that $n+1 \in S$

$\Leftarrow$. Let $A, B \in \mathcal{A}$ and define $\{A_i\}_{i \in \{1, 2\}} \subseteq \mathcal{A}$ by $A_1 = A$ and $A_2 = B$ then $A \cap B = A_1 \cap A_2 = \bigcap_{i \in \{1, 2\}} A_i \in \mathcal{A}$. $\square$

Lemma 21.10. Let $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}}$ a finite family of sets such that $\forall i \in \{1, \dots, n\}$ there exist a family $\{B_{i,j}\}_{j \in I_i}$ such that $A_i = \bigcup_{j \in I_i} B_{i,j}$ then we have:

1. $\bigcap_{i \in \{1, \dots, n\}} A_i = \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right)$
2. If $\forall i \in \{1, \dots, n\}$ $\{B_{i,j}\}_{j \in I_i}$ is pairwise disjoint then we have that $\left\{ \bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right\}_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i}$ is pairwise disjoint. In other words

$$\bigcap_{i \in \{1, \dots, n\}} A_i = \bigsqcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right)$$

Proof.

1. We use induction to prove this, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \text{ is such that } \forall i \in \{1, \dots, n\} \exists \{B_{i,j}\}_{j \in I_i} \text{ with } A_i = \bigcup_{j \in I_i} B_{i,j} \right.$$

$$\left. \text{then } \bigcap_{i \in \{1, \dots, n\}} A_i = \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right) \right\}$$

then we have:

$1 \in \mathcal{S}$. If $\{A_i\}_{i \in \{1,1\}}$ satisfies that there exist a $\{B_{1,j}\}_{j \in I_1}$ such that $A_1 = \bigcup_{j \in I_1} B_{1,j}$ then

$$\begin{aligned} \bigcap_{i \in \{1\}} A_i &= A_1 \\ &= \bigcup_{j \in I_1} B_{1,j} \\ &= \bigcup_{j \in I_1} \left(\bigcap_{k \in \{1\}} B_{k,j} \right) \\ &= \bigcup_{(j) \in \prod_{i \in \{1\}} I_i} \left(\bigcap_{k \in \{1\}} B_{k,j} \right) \\ &= \bigcup_{\sigma \in \prod_{i \in \{1\}} I_i} \left(\bigcap_{k \in \{1\}} B_{k,\sigma_k} \right) \end{aligned}$$

proving that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}}$ is such that $\forall i \in \{1, \dots, n+1\} \exists \{B_{i,j}\}_{j \in I_i}$ with $A_i = \bigcup_{j \in I_i} B_{i,j}$. Then as $n \in \mathcal{S}$ we have that

$$\bigcap_{i \in \{1, \dots, n\}} A_i = \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \quad (21.5)$$

Hence

$$\begin{aligned} &\bigcap_{i \in \{1, \dots, n+1\}} A_i = \\ &\left(\bigcap_{i \in \{1, \dots, n\}} A_i \right) \cap A_{n+1} \stackrel{n \in \mathcal{S}}{=} \\ &\left(\bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \right) \cap A_{n+1} \stackrel{[\text{theorem: 2.138}]}{=} \\ &\bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \cap A_{n+1} \right) = \\ &\bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \cap \left(\bigcup_{j \in I_{n+1}} B_{n+1,j} \right) \right) \stackrel{[\text{theorem: 2.138}]}{=} \\ &\bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcup_{j \in I_{n+1}} \left(B_{n+1,j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \right) \right) \right) = \end{aligned}$$

So that

$$\begin{aligned} &\bigcap_{i \in \{1, \dots, n+1\}} A_i = \\ &\bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcup_{j \in I_{n+1}} \left(B_{n+1,j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \right) \right) \right) \quad (21.6) \end{aligned}$$

If $x \in \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcup_{j \in I_{n+1}} \left(B_{n+1,j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \right) \right) \right)$ then there exist a $\sigma = (\sigma_1, \dots, \sigma_n)$ such that $x \in \left(\bigcup_{j \in I_{n+1}} \left(B_{n+1,j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right) \right) \right)$, hence there exist a $j \in I_{n+1}$ such that

$$x \in B_{n+1,j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k,\sigma_k} \right)$$

Define

$$\tau \in \bigcup_{i \in \{1, \dots, n+1\}} I_i \text{ by } \tau_i = \begin{cases} \sigma_i & \text{if } i \in \{1, \dots, n\} \\ j & \text{if } i = n+1 \end{cases}$$

then $x \in B_{n+1, \tau_{n+1}} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \tau_k} \right) = \bigcap_{k \in \{1, \dots, n+1\}} B_{k, \tau_k}$. Hence

$$x \in \bigcup_{\tau \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \tau_k} \right)$$

which proves that

$$\begin{aligned} \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcup_{j \in I_{n+1}} \left(B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right) \right) \right) \right) \subseteq \\ \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \sigma_k} \right) \end{aligned} \quad (21.7)$$

If $x \in \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \sigma_k} \right)$ then there exist a $\sigma \in \prod_{i \in \{1, \dots, n+1\}} I_i$ such that $x \in \bigcap_{k \in \{1, \dots, n+1\}} B_{k, \sigma_k}$. Define $\eta \in \prod_{i \in \{1, \dots, n\}} I_i$ by $\eta_l = \sigma_l$ and $j = \sigma_{n+1} \in I_{n+1}$ then

$$x \in B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \eta_k} \right) \subseteq \bigcup_{j \in I_{n+1}} \left(B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \eta_k} \right) \right)$$

so that $x \in \bigcup_{j \in I_{n+1}} \left(B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \eta_k} \right) \right)$ or as $\eta \in \prod_{i \in \{1, \dots, n\}} I_i$ we have that $x \in \bigcup_{\eta \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\bigcup_{j \in I_{n+1}} \left(B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \eta_k} \right) \right) \right)$. Hence

$$\begin{aligned} \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \sigma_k} \right) \subseteq \\ \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcup_{j \in I_{n+1}} \left(B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right) \right) \right) \right) \end{aligned}$$

which combined with [eq: 21.7] proves that

$$\begin{aligned} \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i} \left(\left(\bigcup_{j \in I_{n+1}} \left(B_{n+1, j} \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right) \right) \right) \right) = \\ \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \sigma_k} \right) \end{aligned}$$

Combining the above with [eq: 21.6] proves that

$$\bigcap_{i \in \{1, \dots, n+1\}} A_i = \bigcup_{\sigma \in \prod_{i \in \{1, \dots, n+1\}} I_i} \left(\bigcap_{k \in \{1, \dots, n+1\}} B_{k, \sigma_k} \right)$$

Hence we have that $n+1 \in \mathcal{S}$

2. Assume that $\sigma, \tau \in \prod_{i \in \{1, \dots, n\}} I_i$ with $\sigma \neq \tau$ then there exist $i \in \{1, \dots, n\}$ such that $\sigma_i \neq \tau_i$. As $\{B_{i, j}\}_{j \in \{1, \dots, I_i\}}$ is pairwise disjoint we have that $B_{i, \sigma_i} \cap B_{i, \tau_i} = \emptyset$. Hence

$$\left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k} \right) \cap \left(\bigcap_{k \in \{1, \dots, n\}} B_{k, \tau_k} \right) \subseteq B_{i, \sigma_i} \cap B_{i, \tau_i} = \emptyset$$

proving that $\{\bigcap_{k \in \{1, \dots, n\}} B_{k, \sigma_k}\}_{\sigma \in \prod_{i \in \{1, \dots, n\}} I_i}$ is pairwise disjoint.

□

Theorem 21.11. Let $\{A_i\}_{i \in I}$ be a family of sets such that $\forall i \in I$ there exist a $\{B_{i,j}\}_{j \in I_i}$ such that $A_i = \bigcup_{j \in I_i} B_{i,j}$ then

1. $\bigcup_{i \in I} A_i = \bigcup_{i \in I} (\bigcup_{j \in I_i} B_{i,j}) = \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$
2. If additional $\{A_i\}_{i \in I}$ is pairwise disjoint and $\forall i \in I$ $\{B_{i,j}\}_{j \in I_i}$ is pairwise disjoint then

$$\{B_{\sigma_1, \sigma_2}\}_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} \text{ is pairwise disjoint}$$

so that we have

$$\bigsqcup_{i \in I} A_i = \bigsqcup_{i \in I} \left(\bigsqcup_{j \in I_i} B_{i,j} \right) = \bigsqcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$$

Proof.

1. Let $x \in \bigcup_{i \in I} A_i$ then there exist a $i \in I$ such that $x \in A_i$, hence, as $A_i = \bigcup_{j \in I_i} B_{i,j}$ there exist a $j \in I_i$ such that $x \in B_{i,j}$. As $\sigma = (i, j) \in \{i\} \times I_i \subseteq \bigcup_{i \in I} \{i\} \times I_i$ it follows $x \in B_{\sigma_1, \sigma_2}$, hence we have that $\bigcup_{i \in I} A_i \subseteq \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$. Further as for $\sigma \in \bigcup_{i \in I} \{i\} \times I_i$ we have that $B_{\sigma_1, \sigma_2} \subseteq \bigcup_{j \in I_{\sigma_1}} B_{\sigma_1, j} = A_{\sigma_1} \subseteq \bigcup_{i \in I} A_i$ it follows that $\bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2} \subseteq \bigcup_{i \in I} A_i$. So it follows that

$$\bigcup_{i \in I} A_i = \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$$

2. Let $\sigma, \tau \in \bigcup_{i \in I} \{i\} \times I_i$ with $\sigma \neq \tau$ then we have either:

$\sigma_1 = \tau_1$. Then we must have that $\sigma_2 \neq \tau_2$ so that $B_{\sigma_1, \sigma_2} \cap B_{\tau_1, \tau_2} = B_{\sigma_1, \sigma_2} \cap B_{\sigma_1, \tau_2} = \emptyset$ [because $\{B_{\sigma_1, j}\}_{j \in I_{\sigma_1}}$ is pairwise disjoint].

$\sigma_1 \neq \tau_1$. Then $B_{\sigma_1, \sigma_2} \cap B_{\tau_1, \tau_2} \subseteq (\bigcup_{j \in I_{\sigma_1}} B_{\sigma_1, j}) \cap (\bigcup_{j \in I_{\tau_1}} B_{\tau_1, j}) = A_{\sigma_1} \cap A_{\tau_1} = \emptyset$ [because $\{A_i\}_{i \in I}$ is pairwise disjoint].

Hence $\{B_{\sigma_1, \sigma_2}\}_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i}$ is pairwise disjoint. □

Corollary 21.12. Let $\{A_i\}_{i \in I}$ be a family of sets such that $\forall i \in I$ there exist a $\{B_{i,j}\}_{j \in I_i}$ such that $A_i = \bigcup_{j \in I_i} B_{i,j}$ and $\beta: J \rightarrow \bigcup_{i \in I} \{i\} \times I_i$ a bijection then

1. $\bigcup_{i \in I} A_i = \bigcup_{i \in I} (\bigcup_{j \in I_i} B_{i,j}) = \bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2}$
2. If additional $\{A_i\}_{i \in I}$ is pairwise disjoint and $\forall i \in I$ $\{B_{i,j}\}_{j \in I_i}$ is pairwise disjoint then

$$\{B_{\beta(j)_1, \beta(j)_2}\}_{j \in J} \text{ is pairwise disjoint}$$

so that

$$\bigsqcup_{i \in I} A_i = \bigsqcup_{i \in I} \left(\bigsqcup_{j \in I_i} B_{i,j} \right) = \bigsqcup_{j \in J} B_{\beta(j)_1, \beta(j)_2}$$

Proof.

1. Using [theorem: 21.11] we have that

$$\bigcup_{i \in I} A_i = \bigcup_{i \in I} \left(\bigcup_{j \in I_i} B_{i,j} \right) = \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2} \quad (21.8)$$

Let $x \in \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$ then there exist a $\sigma \in \bigcup_{i \in I} \{i\} \times I_i$ such that $x \in B_{\sigma_1, \sigma_2}$. As β is a bijection hence surjective there exist a $j \in J$ such that $\beta(j) = \sigma$ so that $x \in B_{\beta(j)_1, \beta(j)_2}$, hence $x \in \bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2}$. So

$$\bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2} \subseteq \bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2} \quad (21.9)$$

If $x \in \bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2}$ then $\exists j \in J$ such that $x \in B_{\beta(j)_1, \beta(j)_2}$ which as $\beta(j) \in \bigcup_{i \in I} \{i\} \times I_i$ proves that $x \in \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$. Hence $\bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2} \subseteq \bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2}$ which combined with [eq: 21.9] results in

$$\bigcup_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} B_{\sigma_1, \sigma_2} = \bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2} \quad (21.10)$$

Substituting the above in [eq: 21.8] proves finally that

$$\bigcup_{i \in I} A_i = \bigcup_{i \in I} \left(\bigcup_{j \in I_i} B_{i,j} \right) = \bigcup_{j \in J} B_{\beta(j)_1, \beta(j)_2}$$

2. Assume that additional $\{A_i\}_{i \in I}$ is pairwise disjoint and $\forall i \in I$ $\{B_{i,j}\}_{j \in I_i}$ is pairwise disjoint then by [theorem: 21.11]

$$\{B_{\sigma_1, \sigma_2}\}_{\sigma \in \bigcup_{i \in I} \{i\} \times I_i} \text{ pairwise disjoint} \quad (21.11)$$

Let $i, j \in J$ with $i \neq j$ then as β is injective we have $\beta(i) \neq \beta(j)$ hence we have by the above $B_{\beta(i)_1, \beta(i)_2} \cap B_{\beta(j)_1, \beta(j)_2} = \emptyset$ proving that

$$\{B_{\beta(i)_1, \beta(i)_2}\}_{i \in J} \text{ is pairwise disjoint} \quad \square$$

Theorem 21.13. Let X be a set, $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ and $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ such that

$$X = \bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i$$

then there exists a bijection $\beta: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ such that

$$X = \bigcup_{i \in \mathbb{N}} (A_{\beta(i)_1} \cap B_{\beta(i)_2})$$

Proof. As $\mathbb{N}$ is denumerable we have by [theorem: 6.63] that there exists a bijection $\beta: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Let $i \in \mathbb{N}$ then $A_i = A_i \cap X = A_i \cap \left(\bigcup_{j \in \mathbb{N}} B_j \right) = \bigcup_{j \in \mathbb{N}} (A_i \cap B_j)$ so that

$$X = \bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} \left(\bigcup_{j \in \mathbb{N}} (A_i \cap B_j) \right)$$

Let $x \in \bigcup_{i \in \mathbb{N}} (A_{\beta(i)_1} \cap B_{\beta(i)_2})$ then there exists a $i \in \mathbb{N}$ such that $x \in A_{\beta(i)_1} \cap B_{\beta(i)_2} \subseteq X$ proving that

$$\bigcup_{i \in \mathbb{N}_0} (A_{\beta(i)_1} \cap B_{\beta(i)_2}) \subseteq X \quad (21.12)$$

If $x \in X$ then $\exists i \in \mathbb{N}$ so that $x \in \bigcup_{j \in \mathbb{N}} (A_i \cap B_j)$, hence there exists a $j \in \mathbb{N}$ such that $x \in A_i \cap B_j$. As β is a bijection there exists a $k \in \mathbb{N}$ such that $\beta(k) = (i, j)$, hence

$$x \in A_{\beta(k)_1} \cap B_{\beta(k)_2} \subseteq \bigcup_{i \in \mathbb{N}} (A_{\beta(i)_1} \cap B_{\beta(i)_2})$$

proving that

$$X \subseteq \bigcup_{i \in \mathbb{N}} (A_{\beta(i)_1} \cap B_{\beta(i)_2}) \quad (21.13)$$

Finally from 21.12 and 21.13 it follows that

$$X = \bigcup_{i \in \mathbb{N}} (A_{\beta(i)_1} \cap B_{\beta(i)_2}) \quad \square$$

Definition 21.14. (Monotone set function) Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then a set function $\mu: \mathcal{A} \rightarrow [0, \infty]$ is **monotone** if $\forall A, B \in \mathcal{A}$ with $A \subseteq B$ we have $\mu(A) \leq \mu(B)$.

Definition 21.15. (Additive set function) Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then a set function $\mu: \mathcal{A} \rightarrow [0, \infty]$ is **finite additive on $\mathcal{A}$** if $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $n \in \mathbb{N}$ pairwise disjoint with $\bigsqcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$ we have

$$\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

Definition 21.16. (Sub-additive set function) Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then a set function $\mu: \mathcal{A} \rightarrow [0, \infty]$ is **sub-additive** if $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ with $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$ we have

$$\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) \leq \sum_{i=1}^n \mu(A_i)$$

Definition 21.17. (Countable additive set function) Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then a set function $\mu: \mathcal{A} \rightarrow [0, \infty]$ is **countable additive on $\mathcal{A}$** if $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ pairwise disjoint with $\bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ we have

$$\mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Note 21.18. As $\forall i \in \mathbb{N}$ $\mu(A_i) \in [0, \infty]$ it follows from [definition: 20.95] that $\sum_{i=1}^{\infty} \mu(A_i)$ is well defined.

Definition 21.19. (Countable sub-additive set function) Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then a set function $\mu: \mathcal{A} \rightarrow [0, \infty]$ is **countable sub-additive** if $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ we have

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

Note 21.20. As $\forall i \in \mathbb{N}$ $\mu(A_i) \in [0, \infty]$ it follows from [definition: 20.95] that $\sum_{i=1}^{\infty} \mu(A_i)$ is well defined.

Countable sets are either finite or denumerable so why do we talk about countable additivity instead of denumerable additivity [because $\mathbb{N}$ is a denumerable set]. However the next theorem shows that countable additivity implies finite additivity which justifies the use of the term countable additivity.

Theorem 21.21. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ such that $\emptyset \in \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a countable additive set function on $\mathcal{A}$ satisfying $\mu(\emptyset) = 0$ then μ is finite additive on $\mathcal{A}$.

Note 21.22. The requirement that $\emptyset \in \mathcal{A}$ and $\mu(\emptyset) = 0$ is essential for the proof to work.

Proof. Let $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ be a pairwise disjoint finite family such that $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ \emptyset & \text{if } i \in \{n+1, \dots, \infty\} \end{cases} \in \mathcal{A}$$

Let $x \in \bigcup_{i \in \mathbb{N}} B_i$ then there exist a $i \in \mathbb{N}$ such that $x \in B_i$. Assume that $i \in \{n+1, \dots, \infty\}$ then $x \in B_i = \emptyset$ a contradiction so we must have that $i \in \{1, \dots, n\}$. Hence $x \in B_i = A_i$ proving that $x \in \bigcup_{i \in \{1, \dots, n\}} A_i$, so

$$\bigcup_{i \in \mathbb{N}} B_i \subseteq \bigcup_{i \in \{1, \dots, n\}} A_i \quad (21.14)$$

Further if $x \in \bigcup_{i \in \{1, \dots, n\}} A_i$ then there exist a $i \in \{1, \dots, n\}$ such that $x \in A_i = B_i$ so that $x \in \bigcup_{i \in \mathbb{N}} B_i$ proving that $\bigcup_{i \in \{1, \dots, n\}} A_i \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$. Combining this with [eq: 21.14] proves that

$$\bigcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A} \quad (21.15)$$

Let $i, j \in \mathbb{N}$ with $i \neq j$ then we have the following possible cases:

$i \in \{1, \dots, n\} \wedge j \in \{1, \dots, n\}$. Then $B_i \cap B_j = A_i \cap A_j \stackrel{=}{=} \emptyset$

$i \in \{n+1, \dots, \infty\} \wedge j \in \{1, \dots, n\}$. Then $B_i \cap B_j = \emptyset \cap A_j = \emptyset$

$i \in \{1, \dots, n\} \wedge j \in \{n+1, \dots, \infty\}$. Then $B_i \cap B_j = A_i \cap \emptyset = \emptyset$

$i \in \{n+1, \dots, \infty\} \wedge j \in \{n+1, \dots, \infty\}$. Then $B_i \cap B_j = \emptyset \cap \emptyset = \emptyset$

Hence we conclude that

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ is pairwise disjoint} \quad (21.16)$$

As by [eq: 21.15] $\bigcup_{i \in \mathbb{N}} B_i \in \mathcal{A}$ we have by the above and the countable additivity of μ that

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \sum_{i=1}^{\infty} \mu(B_i) \quad (21.17)$$

Next we have

$$\begin{aligned} \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) &\stackrel{[\text{eq: 21.15}]}{=} \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\ &\stackrel{[\text{eq: 21.17}]}{=} \sum_{i=1}^{\infty} \mu(B_i) \\ &\stackrel{[\text{theorem: 20.100}]}{=} \sum_{i=1}^n \mu(B_i) + \sum_{i=n+1}^{\infty} \mu(B_i) \\ &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) \\ &\stackrel{\mu(\emptyset)=0 \wedge [\text{theorem: 20.97}]}{=} \sum_{i=1}^n \mu(A_i) \end{aligned}$$

Hence $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $n \in \mathbb{N}$ pairwise disjoint we have $\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$ proving that μ is finite additive on $\mathcal{A}$. $\square$

Theorem 21.23. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ such that $\emptyset \in \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a countable sub-additive function then μ is sub-additivity.

Proof. Let $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ be a finite family such that $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ \emptyset & \text{if } i \in \{n+1, \dots, \infty\} \end{cases} \in \mathcal{A}$$

Let $x \in \bigcup_{i \in \mathbb{N}} B_i$ then there exist a $i \in \mathbb{N}$ such that $x \in B_i$. Assume that $i \in \{n+1, \dots, \infty\}$ then $x \in B_i = \emptyset$ a contradiction so we must have that $i \in \{1, \dots, n\}$. Hence $x \in B_i = A_i$ proving that $x \in \bigcup_{i \in \{1, \dots, n\}} A_i$, so

$$\bigcup_{i \in \mathbb{N}} B_i \subseteq \bigcup_{i \in \{1, \dots, n\}} A_i \quad (21.18)$$

Further if $x \in \bigcup_{i \in \{1, \dots, n\}} A_i$ then there exist a $i \in \{1, \dots, n\}$ such that $x \in A_i = B_i$ so that $x \in \bigcup_{i \in \mathbb{N}} B_i$ proving that $\bigcup_{i \in \{1, \dots, n\}} A_i \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$. Combining this with [eq: 21.18] proves that

$$\bigcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A} \quad (21.19)$$

As $\bigcup_{i \in \mathbb{N}} B_i \in \mathcal{A}$ and μ is countable additive it follows that

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \leq \sum_{i=1}^{\infty} \mu(B_i) \quad (21.20)$$

Further

$$\begin{aligned}
 \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) & \stackrel{[\text{eq: 21.19}]}{=} \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\
 & \stackrel{[\text{eq: 21.20}]}{\leq} \sum_{i=1}^{\infty} \mu(B_i) \\
 & \stackrel{[\text{theorem: 20.100}]}{=} \sum_{i=1}^n \mu(B_i) + \sum_{i=n+1}^{\infty} \mu(B_i) \\
 & = \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) \\
 & \stackrel{\mu(\emptyset)=0 \wedge [\text{theorem: 20.97}]}{=} \sum_{i=1}^n \mu(A_i)
 \end{aligned}$$

□

21.2 Rings, algebras and σ -algebras

Definition 21.24. (Ring) Let X be a set then $\mathcal{R} \subseteq \mathcal{P}(X)$ [the set of subsets of X] is a **ring** on X if $\mathcal{R}$ satisfies:

1. $\emptyset \in \mathcal{R}$.
2. If $A, B \in \mathcal{R}$ then $A \cup B \in \mathcal{R}$ [in other words $\mathcal{R}$ is $\cup$ -stable].
3. If $A, B \in \mathcal{R}$ then $A \setminus B \in \mathcal{R}$.

Rather trivial examples of rings are the following:

Example 21.25. Let X be a set then $\{\emptyset\}$ is a ring on X

Proof.

1. As $\emptyset \in \{\emptyset\}$ we have that $\{\emptyset\} \neq \emptyset$.
2. If $A, B \in \{\emptyset\}$ then $A = B = \emptyset$ so that $A \cup B = \emptyset \in \{\emptyset\}$.
3. If $A, B \in \{\emptyset\}$ then $A = B = \emptyset$ so that $A \setminus B = \emptyset \setminus \emptyset = \emptyset \in \{\emptyset\}$. □

Example 21.26. Let X be a set then $\mathcal{P}(X)$ is a ring on X .

Proof. As $X \subseteq X$ $X \in \mathcal{P}(X)$ so that

$$\mathcal{P}(X) \neq \emptyset$$

If $A, B \in \mathcal{P}(X)$ then $A, B \subseteq X$ so that $A \cup B \subseteq X$ and $A \setminus B \subseteq X$ so that

$$A \cup B \in \mathcal{P}(X) \text{ and } A \setminus B \in \mathcal{P}(X) \quad \square$$

Theorem 21.27. Let X be a set and $\mathcal{R}$ a ring on X then we have

1. $\emptyset \in \mathcal{R}$.
2. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R}$ a finite family of sets in $\mathcal{R}$ then $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{R}$.
3. If $A, B \in \mathcal{R}$ then $A \cap B \in \mathcal{R}$ [in other words $\mathcal{R}$ is $\cap$ -stable].
4. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R}$ a finite family of sets in $\mathcal{R}$ then $\bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{R}$.

Proof.

1. As $\mathcal{R} \neq \emptyset$ there exist a $A \in \mathcal{R}$ so that $\emptyset = A \setminus A \in \mathcal{R}$

2. This follows from [theorem: 21.9].
3. If $A, B \in \mathcal{R}$ then $A \setminus B \in \mathcal{R}$ so that $A \setminus (A \setminus B) \in \mathcal{R}$ which as $A \setminus (A \setminus B) \stackrel{[\text{theorem: 1.36}]}{=} A \cap B$ proves that $A \cap B \in \mathcal{R}$.
4. This follows from [theorem: 21.9] and (3). $\square$

A algebra on X is a ring $\mathcal{R}$ where additional $X \in \mathcal{R}$.

Definition 21.28. (Algebra) Let X be a set then $\mathcal{A} \subseteq \mathcal{P}(X)$ is a **algebra** on X if $\mathcal{A}$ satisfies

1. $\mathcal{A}$ is a ring on X .
2. $X \in \mathcal{A}$.

Example 21.29. Let $X \neq \emptyset$ be a non empty set then $\{\emptyset\}$ is a ring [see example: 21.25] but not a algebra because $X \notin \{\emptyset\}$, however $\{X, \emptyset\}$ is a algebra on X .

Proof.

1. As $X, \emptyset \in \{X, \emptyset\}$ $\{X, \emptyset\} \neq \emptyset$.
2. If $A, B \in \{X, \emptyset\}$ then we have either:

$A = X \wedge B = X$. Then $A \cup B = X \in \{X, \emptyset\}$ and $A \setminus B = X \setminus X = \emptyset \in \{X, \emptyset\}$.

$A = \emptyset \wedge B = X$. Then $A \cup B = X \in \{X, \emptyset\}$ and $A \setminus B = \emptyset \setminus X = \emptyset \in \{X, \emptyset\}$.

$A = X \wedge B = \emptyset$. Then $A \cup B = X \in \{X, \emptyset\}$ and $A \setminus B = X \setminus \emptyset = X \in \{X, \emptyset\}$.

$A = \emptyset \wedge B = \emptyset$. Then $A \cup B = \emptyset \in \{X, \emptyset\}$ and $A \setminus B = \emptyset \setminus \emptyset = \emptyset \in \{X, \emptyset\}$.

so in all cases we have $A \cup B \in \{X, \emptyset\}$ and $A \setminus B \in \{X, \emptyset\}$. $\square$

Example 21.30. Let X be a set then $\mathcal{P}(X)$ is a ring on X .

Proof. Using [example: 21.26] it follows that $\mathcal{A}$ is a ring. Further as $X \subset X$ it follows that $X \in \mathcal{P}(X)$. $\square$

The following summarizes some trivial properties of a algebra.

Theorem 21.31. Let X be a set and $\mathcal{A}$ a algebra on X then we have:

1. $\emptyset \in \mathcal{A}$
2. $X \in \mathcal{A}$
3. $\forall A, B \in \mathcal{A}$ we have that $A \cup B \in \mathcal{A}$, $A \cap B \in \mathcal{A}$ and $A \setminus B \in \mathcal{A}$
4. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ a finite family of sets in $\mathcal{A}$ then $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$
5. If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ a finite family of sets in $\mathcal{A}$ then $\bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$
6. $\forall A \in \mathcal{A}$ we have that $X \setminus A \in \mathcal{A}$

Proof.

1. As $\mathcal{A}$ is a ring this follows from [theorem: 21.27].
2. This is stated in the definition of a algebra.
3. As $\mathcal{A}$ is a ring this follows from [theorem: 21.27] and the definition of a ring.
4. As $\mathcal{A}$ is a ring this follows from [theorem: 21.27].
5. As $\mathcal{A}$ is a ring this follows from [theorem: 21.27].
6. As $X \in \mathcal{A}$ and $A \in \mathcal{A}$ we have by (3) that $X \setminus A \in \mathcal{A}$. $\square$

In most books we have the following equivalent definitions of a algebra.

Theorem 21.32. *Let X be a set and $\mathcal{A} \subseteq \mathcal{P}(X)$ then the following are equivalent:*

1. $\mathcal{A}$ is a algebra on X
2. $\mathcal{A}$ satisfies:
 - a. $X \in \mathcal{A}$
 - b. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$
 - c. $\forall A, B \in \mathcal{A}$ we have $A \cap B \in \mathcal{A}$ [in other words $\mathcal{A}$ is $\cap$ -stable]
3. $\mathcal{A}$ satisfies:
 - a. $X \in \mathcal{A}$
 - b. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$
 - c. $\forall A, B \in \mathcal{A}$ we have $A \cup B \in \mathcal{A}$ [in other words $\mathcal{A}$ is $\cup$ -stable]
4. $\mathcal{A}$ satisfies:
 - a. $X \in \mathcal{A}$
 - b. $\forall A, B \in \mathcal{A}$ we have $A \setminus B \in \mathcal{A}$

Proof.

1 $\Rightarrow$ 2. If $\mathcal{A}$ is a algebra then by definition $\mathcal{A}$ is a ring and $X \in \mathcal{A}$ so that

1. $X \in \mathcal{A}$
2. $\forall A \in \mathcal{A}$ we have, as $X \in \mathcal{A}$, that $X \setminus A \in \mathcal{A}$ [by the definition of a ring].
3. $\forall A, B \in \mathcal{A}$ we have by the definition of a ring that $A \cap B \in \mathcal{A}$.

2 $\Rightarrow$ 3. We have:

1. $X \in \mathcal{A}$ by the hypothesis.
2. $X \setminus A$ by the hypothesis.
3. Let $A, B \in \mathcal{A}$ then we have that $X \setminus A \in \mathcal{A}$ and $X \setminus B \in \mathcal{A}$ so that $(X \setminus A) \cap (X \setminus B) \in \mathcal{A}$, hence

$$X \setminus ((X \setminus A) \cap (X \setminus B)) \in \mathcal{A}$$

Further we have

$$\begin{aligned} X \setminus ((X \setminus A) \cap (X \setminus B)) &\stackrel{[\text{theorem: 1.34}]}{=} (X \setminus (X \setminus A)) \cup (X \setminus (X \setminus B)) \\ &\stackrel{A, B \subseteq X \wedge [\text{theorem: 1.36}]}{=} A \cup B \end{aligned}$$

so that $A \cup B \in \mathcal{A}$.

3 $\Rightarrow$ 4. We have

1. $X \in \mathcal{A}$ by the hypothesis
2. Let $A, B \in \mathcal{A}$ then $X \setminus A \in \mathcal{A}$ so that $(X \setminus A) \cup B \in \mathcal{A}$ hence

$$X \setminus ((X \setminus A) \cup B) \in \mathcal{A}$$

Further we have

$$\begin{aligned} X \setminus ((X \setminus A) \cup B) &\stackrel{[\text{theorem: 1.34}]}{=} (X \setminus (X \setminus A)) \cap (X \setminus B) \\ &\stackrel{A \subseteq X \wedge [\text{theorem: 1.36}]}{=} A \cap (X \setminus B) \\ &\stackrel{A, B \subseteq X \wedge [\text{theorem: 1.36}]}{=} A \setminus B \end{aligned}$$

so that $A \setminus B \in \mathcal{A}$

4 $\Rightarrow$ 1. Then we have

1. $X \in \mathcal{A}$ by the hypothesis.

2. If $A, B \in \mathcal{A}$ then by the hypothesis $A \setminus B \in \mathcal{A}$ so that $A \setminus (A \setminus B) \in \mathcal{A}$, as by [theorem: 1.36] $A \setminus (A \setminus B) = A \cap B$ it follows that $A \cap B \in \mathcal{A}$ hence we have proved that

$$\forall A, B \in \mathcal{A} \text{ we have } A \cap B \in \mathcal{A}$$

Further if $A, B \in \mathcal{A}$ then as $X \in \mathcal{A}$ we have that $(X \setminus A), (X \setminus B) \in \mathcal{A}$ so that $(X \setminus A) \cap (X \setminus B) \in \mathcal{A}$, hence we have that

$$X \setminus ((X \setminus A) \cap (X \setminus B)) \in \mathcal{A}$$

Now

$$\begin{aligned} X \setminus ((X \setminus A) \cap (X \setminus B)) &\stackrel{[\text{theorem: 1.34}]}{=} (X \setminus (X \setminus A)) \cup (X \setminus (X \setminus B)) \\ &\stackrel{A, B \subseteq X \wedge [\text{theorem: 1.36}]}{=} A \cup B \end{aligned}$$

proving that

$$\forall A, B \in \mathcal{A} \text{ we have } A \cup B \in \mathcal{A}$$

So as $X \in \mathcal{A} \Rightarrow \mathcal{A} \neq \emptyset$ and $\forall A, B \in \mathcal{A}$ we have $A \cup B \in \mathcal{A}$ and $A \setminus B \in \mathcal{A}$ it follows that $\mathcal{A}$ is a algebra. $\square$

We also want to deal with countable unions of sets and for this we define a sigma algebra.

Definition 21.33. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then $\mathcal{A}$ is a sigma algebra [noted as σ -algebra] on X if

1. $X \in \mathcal{A}$.
2. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$.
3. If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ is a countable family of sets in $\mathcal{A}$ then $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$.

Note 21.34. If $A, B \in \mathcal{A}$ then if we define $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ by $A_i = \begin{cases} A & \text{if } i=1 \\ B & \text{if } i \in \{2, \dots, \infty\} \end{cases}$ then

$$A \cup B = A_1 \cup \left(\bigcup_{i \in \{2, \dots, \infty\}} A_i \right) = \bigcup_{i \in \mathbb{N}} A_i$$

so that every σ -algebra on X is a algebra. So we have

$$\sigma\text{-algebra} \xRightarrow{\text{is a}} \text{algebra} \xRightarrow{\text{is a}} \text{ring}$$

Definition 21.35. (Measurable Space) Let X be a non empty set and $\mathcal{A}$ a σ -algebra on X then the couple $\langle X, \mathcal{A} \rangle$ is called a **measurable space**.

We have the following equivalent definitions of a σ -algebra

Theorem 21.36. Let X be a set, $\mathcal{A} \subseteq \mathcal{P}(X)$ then we have the following equivalences:

1. $\mathcal{A}$ is a σ -algebra on X
2. $\mathcal{A}$ is a algebra and $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ **pairwise disjoint** we have $\bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$
3. $\mathcal{A}$ is a algebra and $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$
4. $\mathcal{A}$ satisfies:
 - a. $X \in \mathcal{A}$
 - b. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$
 - c. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$
5. $\mathcal{A}$ satisfies:
 - a. $\emptyset \in \mathcal{A}$

b. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$

c. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$

6. $\mathcal{A}$ satisfies:

a. $\emptyset \in \mathcal{A}$

b. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$

c. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{A}$

7. $\mathcal{A}$ satisfies:

a. $X \in \mathcal{A}$

b. $\forall A \in \mathcal{A}$ we have $X \setminus A \in \mathcal{A}$

c. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{A}$

Proof.

1 $\Rightarrow$ 2. As $\mathcal{A}$ is a sigma-algebra it follows that

$\mathcal{A}$ is a algebra

Further if $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ is pairwise disjoint then by the definition of a σ -algebra it follows that

$$\bigsqcup_{i \in \mathbb{N}} A_i \stackrel{\text{def}}{=} \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

2 $\Rightarrow$ 3. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ and define

$$\{B_i\}_{i \in \mathbb{N}} \text{ by } B_i = \begin{cases} A_1 & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j & \text{if } i \in \{2, \dots, \infty\} \end{cases}$$

As $\mathcal{A}$ is a algebra we have by [theorem: 21.31] that $\forall i \in \mathbb{N}$ we have $B_i \in \mathcal{A}$ so that

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$$

Further by [lemma: 21.6] $\{B_i\}_{i \in \mathbb{N}}$ is pairwise disjoint so that by (2)

$$\bigsqcup_{i \in \mathbb{N}} B_i \in \mathcal{A}$$

As further $\bigcup_{i \in \mathbb{N}} A_i \stackrel{=}{=} \bigsqcup_{i \in \mathbb{N}} B_i$ it follows that

$$\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

3 $\Rightarrow$ 4. As $\mathcal{A}$ is a algebra we have by [theorem: 21.31] that

$$X \in \mathcal{A}$$

$$\forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A}$$

Finally by (3) we have

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

4 $\Rightarrow$ 5. By (4) we have

$$\emptyset \in \mathcal{A}$$

$$\forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A}$$

and

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

As $\emptyset \in \mathcal{A}$ we have $X = X \setminus \emptyset \in \mathcal{A}$ so that we have

$$X \in \mathcal{A}$$

5 $\Rightarrow$ 6. We have by (5) that

$$\emptyset \in \mathcal{A} \text{ and } \forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A}$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$. If $i \in \mathbb{N}$ then as $A_i \in \mathcal{A}$ it follows that $X \setminus A_i \in \mathcal{A}$ further we have by (5) that $\bigcup_{i \in \mathbb{N}} (X \setminus A_i) \in \mathcal{A}$ so that

$$X \setminus \left(\bigcup_{i \in \mathbb{N}} (X \setminus A_i) \right) \in \mathcal{A}$$

As

$$\begin{aligned} X \setminus \left(\bigcup_{i \in \mathbb{N}} (X \setminus A_i) \right) &\stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{i \in \mathbb{N}} (X \setminus (X \setminus A_i)) \\ &\stackrel{A_i \subseteq X \wedge [\text{theorem: 1.36}]}{=} \bigcap_{i \in \mathbb{N}} A_i \end{aligned}$$

it follows that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcap_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

6 $\Rightarrow$ 7. By (6) we have that $\emptyset \in \mathcal{A}$ so that by (6) again we have

$$X = X \setminus \emptyset \in \mathcal{A}$$

further by (6) we have

$$\forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A} \text{ and } \forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcap_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

6 $\Rightarrow$ 1. By (7) we have

$$X \in \mathcal{A} \text{ and } \forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A}$$

Further if $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ then $\forall i \in \mathbb{N}$ we have $A_i \in \mathcal{A}$ so that $X \setminus A_i \in \mathcal{A}$. Hence we have by (6)

$$X \setminus \left(\bigcap_{i \in \mathbb{N}} (X \setminus A_i) \right) \in \mathcal{A}$$

Now

$$\begin{aligned} X \setminus \left(\bigcap_{i \in \mathbb{N}} (X \setminus A_i) \right) &\stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{i \in \mathbb{N}} (X \setminus (X \setminus A_i)) \\ &\stackrel{A_i \subseteq X \wedge [\text{theorem: 1.36}]}{=} \bigcup_{i \in \mathbb{N}} A_i \end{aligned}$$

so that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

This proves that $\mathcal{A}$ is a σ -algebra. □

Example 21.37. Let $X \neq \emptyset$ be a set then $\{\emptyset\}$ is not a σ -algebra on X [see example: 21.29] but $\{\emptyset, X\}$ is a σ -algebra on X .

Proof. Using [example: 21.29] it follows that $\{\emptyset, X\}$ is a algebra on X so that

$$X \in \mathcal{A} \text{ and } \forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A}$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \{\emptyset, X\}$ then we have two cases to consider:

$\forall i \in \mathbb{N}$ we have $A_i \neq \emptyset$. Then $\forall i \in \mathbb{N}$ $A_i = X$ so that $\bigcap_{i \in \mathbb{N}} A_i = X \in \mathcal{A}$

$\exists i \in \mathbb{N}$ with $A_i = \emptyset$. Then $\bigcap_{i \in \mathbb{N}} A_i = \emptyset \in \mathcal{A}$

so that $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{A}$. Applying then [theorem: 21.36] proves that $\{\emptyset, X\}$ is a σ -algebra on X . □

Example 21.38. Let X be a set then $\mathcal{P}(X)$ is a σ -algebra on X

Proof. Using [example: 21.30] $\mathcal{P}(X)$ is a algebra on X . Further for $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ we have $\forall i \in \mathbb{N}$ that $A_i \subseteq X$ so that $\bigcup_{i \in \mathbb{N}} A_i \subseteq X$ proving that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{P}(X)$. Applying then [theorem: 21.36] proves that $\mathcal{P}(X)$ is a σ -algebra on X . $\square$

Theorem 21.39. Let X be a set, $\mathcal{A}$ a σ -algebra on X , I a countable set, $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ a countable family of element of $\mathcal{A}$ then we have

1. $\bigcup_{i \in I} A_i \in \mathcal{A}$
2. $\bigcap_{i \in I} A_i \in \mathcal{A}$

Proof.

1. For I we have either

I is finite. Then there exist a bijection $\beta: \{1, \dots, n\} \rightarrow I$ where $n \in \mathbb{N}_0$. Let $x \in \bigcup_{i \in I} A_i$ then there exist a $i \in I$ such that $x \in A_i$. As β is a bijection hence surjective there exist a $j \in \{1, \dots, n\}$ such that $i = \beta(j)$ so that $x \in A_{\beta(j)} \subseteq \bigcup_{j \in \{1, \dots, n\}} A_{\beta(j)}$. Hence we have that

$$\bigcup_{i \in I} A_i \subseteq \bigcup_{j \in \{1, \dots, n\}} A_{\beta(j)} \quad (21.21)$$

Likewise if $x \in \bigcup_{j \in \{1, \dots, n\}} A_{\beta(j)}$ then there exist a $j \in \{1, \dots, n\}$ such that $x \in A_{\beta(j)}$ which as $\beta(j) \in I$ proves that $x \in \bigcup_{i \in I} A_i$, hence $\bigcup_{j \in \{1, \dots, n\}} A_{\beta(j)} \subseteq \bigcup_{i \in I} A_i$. Combining this with [eq: 21.21] proves that

$$\bigcup_{i \in I} A_i = \bigcup_{j \in \{1, \dots, n\}} A_{\beta(j)}$$

As $\forall j \in \{1, \dots, n\}$ $A_{\beta(j)} \in \mathcal{A}$ [because $\beta(j) \in I$] it follows, as a σ -algebra is a ring, from [theorem: 21.27] that $\bigcup_{j \in \{1, \dots, n\}} A_{\beta(j)} \in \mathcal{A}$. Hence

$$\bigcup_{i \in I} A_i \in \mathcal{A}$$

I is denumerable. Then there exist a bijection $\beta: \mathbb{N} \rightarrow I$. Let $x \in \bigcup_{i \in I} A_i$ then there exist a $i \in I$ such that $x \in A_i$. As β is a bijection hence surjective there exist a $j \in \mathbb{N}$ such that $i = \beta(j)$ so that $x \in A_{\beta(j)} \subseteq \bigcup_{j \in \mathbb{N}} A_{\beta(j)}$. Hence we have that

$$\bigcup_{i \in I} A_i \subseteq \bigcup_{j \in \mathbb{N}} A_{\beta(j)} \quad (21.22)$$

Likewise if $x \in \bigcup_{j \in \mathbb{N}} A_{\beta(j)}$ then there exist a $j \in \mathbb{N}$ such that $x \in A_{\beta(j)}$ which as $\beta(j) \in I$ proves that $x \in \bigcup_{i \in I} A_i$, hence $\bigcup_{j \in \mathbb{N}} A_{\beta(j)} \subseteq \bigcup_{i \in I} A_i$. Combining this with [eq: 21.22] proves that

$$\bigcup_{i \in I} A_i = \bigcup_{j \in \mathbb{N}} A_{\beta(j)}$$

As $\forall j \in \mathbb{N}$ $A_{\beta(j)} \in \mathcal{A}$ [because $\beta(j) \in I$] it follows, as a σ -algebra is a ring, from [theorem: 21.27] that $\bigcup_{j \in \mathbb{N}} A_{\beta(j)} \in \mathcal{A}$. Hence

$$\bigcup_{i \in I} A_i \in \mathcal{A}$$

2. Let $i \in I$ then $A_i \in \mathcal{A}$ hence $X \setminus A_i \in \mathcal{A}$ so that by (1) $\bigcup_{i \in I} (X \setminus A_i) \in \mathcal{A}$ from which it follows that

$$X \setminus \left(\bigcup_{i \in I} (X \setminus A_i) \right) \in \mathcal{A}$$

As we have

$$\begin{aligned} X \setminus \left(\bigcup_{i \in I} (X \setminus A_i) \right) &\stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{i \in I} (X \setminus (X \setminus A_i)) \\ &\stackrel{A_i \subseteq X \wedge [\text{theorem: 1.36}]}{=} \bigcap_{i \in I} A_i \end{aligned}$$

it follows that

$$\bigcap_{i \in I} A_i \in \mathcal{A} \quad \square$$

Theorem 21.40. *Let X be a set and let $\mathcal{A}$ be a algebra on X then we have:*

1. *If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ A_i \subseteq A_{i+1}$ we have $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ then $\mathcal{A}$ is a σ -algebra on X .*
2. *If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i$ we have $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{A}$ then $\mathcal{A}$ is a σ -algebra on X .*

Proof.

1. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$. Define $\{B_i\}_{i \in \mathbb{N}}$ by $B_i = \bigcup_{j \in \{1, \dots, i\}} A_j$ then as $\mathcal{A}$ is a algebra we have by [theorem: 21.31] that $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$. Further for $i \in \mathbb{N}$ we have

$$B_i = \bigcup_{j \in \{1, \dots, i\}} A_j \subseteq \bigcup_{j \in \{1, \dots, i+1\}} A_j = B_{i+1}$$

hence by the hypothesis we have

$$\bigcup_{i \in \mathbb{N}} B_i \in \mathcal{A} \quad (21.23)$$

As $\forall i \in \mathbb{N}$ we have $A_i \subseteq \bigcup_{j \in \{1, \dots, i\}} A_j = B_i$ it follows that

$$\bigcup_{i \in \mathbb{N}} A_i \subseteq \bigcup_{i \in \mathbb{N}} B_i \quad (21.24)$$

Further if $x \in \bigcup_{i \in \mathbb{N}} B_i$ then there exist a $i \in \mathbb{N}$ such that $x \in B_i = \bigcup_{j \in \{1, \dots, i\}} A_j$, hence there exist a $j \in \{1, \dots, i\}$ such that $x \in A_j \subseteq \bigcup_{i \in \mathbb{N}} A_i$. This proves that $\bigcup_{i \in \mathbb{N}} B_i \subseteq \bigcup_{i \in \mathbb{N}} A_i$ which combined with [eq: 21.24] gives us $\bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i$, hence using [eq: 21.23] we have that

$$\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

Applying then [theorem: 21.36] it follows that $\mathcal{A}$ is a σ -algebra.

2. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be such that $\forall i \in \mathbb{N}$ we have $A_i \subseteq A_{i+1}$. Let $i \in \mathbb{N}$ then as $\mathcal{A}$ is a algebra $X \setminus A_i \in \mathcal{A}$, further if $x \in X \setminus A_{i+1}$ then $x \in X$ and $x \notin A_i$ [as $A_i \subseteq A_{i+1}$] so that $x \in X \setminus A_i$. Hence $\{X \setminus A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ and $\forall i \in \mathbb{N}$ we have $X \setminus A_{i+1} \subseteq X \setminus A_i$ so by the hypothesis we have that $\bigcap_{i \in \mathbb{N}} (X \setminus A_i) \in \mathcal{A}$. As $\mathcal{A}$ is a algebra we have then that

$$X \setminus \left(\bigcap_{i \in \mathbb{N}} (X \setminus A_i) \right) \in \mathcal{A}$$

Further

$$X \setminus \left(\bigcap_{i \in \mathbb{N}} (X \setminus A_i) \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{i \in \mathbb{N}} (X \setminus (X \setminus A_i)) \stackrel{A_i \subseteq X \wedge [\text{theorem: 1.36}]}{=} \bigcup_{i \in \mathbb{N}} A_i$$

so that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$. Hence we have proved that $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ A_i \subseteq A_{i+1}$ $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ so that by (1) $\mathcal{A}$ is a σ -algebra. $\square$

We show now how we can generate a σ algebra based on a arbitrary subset of $\mathcal{P}(X)$. First we prove that for a non empty collection of σ -algebra's the intersection of this collection is again a σ -algebra.

Lemma 21.41. *Let X be a set and $\mathfrak{A}$ a non empty set of σ -algebra's on X then $\bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$ is a σ -algebra on X .*

Proof. As $\mathfrak{A} \neq \emptyset$ there exist a $\mathcal{S} \in \mathfrak{A}$ with $\mathcal{S} \subseteq \mathcal{P}(X)$ so that we have by [theorem: 1.66] that $\bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A} \subseteq \mathcal{S} \subseteq \mathcal{P}(X)$ proving that

$$\bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A} \subseteq \mathcal{P}(X)$$

As $\forall \mathcal{A} \in \mathfrak{A}$ $\mathcal{A}$ is a σ -algebra on X we have by [theorem: 21.36] that $X \in \mathcal{A}$ hence

$$X \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$$

Let $A \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$ then for $\mathcal{A} \in \mathfrak{A}$ we have $A \in \mathcal{A}$ so that $X \setminus A \in \mathcal{A}$ which as $\mathcal{A} \in \mathfrak{A}$ was chosen arbitrary proves that $X \setminus A \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$. Hence as $A \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$ was chosen arbitrary we have that

$$\forall A \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A} \text{ we have } X \setminus A \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$$

If $\{A_i\}_{i \in \mathbb{N}} \subseteq \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$ then for $\mathcal{A} \in \mathfrak{A}$ we have $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ so that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ which, as $\mathcal{A} \in \mathfrak{A}$ was chosen arbitrary, proves that $\bigcup_{i \in \mathbb{N}} A_i \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$. Hence as $\{A_i\}_{i \in \mathbb{N}} \subseteq \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$ was chosen arbitrary we have that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \bigcap_{\mathcal{A} \in \mathfrak{A}} \mathcal{A}$$

□

We use now the above theorem to show how we can generate a σ -algebra out of a non empty subset of $\mathcal{P}(X)$.

Theorem 21.42. *Let X be a set $\mathcal{A} \subseteq \mathcal{P}(X)$ then there exist a **unique** algebra $\sigma[\mathcal{A}]$ on X such that:*

1. $\mathcal{A} \subseteq \sigma[\mathcal{A}]$
2. If $\mathcal{B}$ is another σ -algebra with $\mathcal{A} \subseteq \mathcal{B}$ then $\sigma[\mathcal{A}] \subseteq \mathcal{B}$

This unique smallest σ -algebra that contains $\mathcal{A}$ is called the σ -algebra generated by $\mathcal{A}$

Proof. Define $\mathfrak{A} = \{\mathcal{T} | \mathcal{T} \text{ is a } \sigma\text{-algebra on } X \text{ with } \mathcal{A} \subseteq \mathcal{T}\}$. As $\mathcal{A} \subseteq \mathcal{P}(X)$ and $\mathcal{P}(X)$ is a σ -algebra on X [see example: 21.38] it follows that $\mathcal{P}(X) \in \mathfrak{A}$ hence $\mathfrak{A}$ is not empty. Using then previous lemma [lemma: 21.41] it follows that

$$\bigcap_{\mathcal{T} \in \mathfrak{A}} \mathcal{T} \text{ is a } \sigma\text{-algebra on } X$$

As $\forall \mathcal{T} \in \mathfrak{A}$ $\mathcal{A} \subseteq \mathcal{T}$ it follows from [theorem: 1.66] that

$$\mathcal{A} \subseteq \bigcap_{\mathcal{T} \in \mathfrak{A}} \mathcal{T}$$

Finally assume that $\mathcal{B}$ is another σ -algebra on X such that $\mathcal{A} \subseteq \mathcal{B}$ then $\mathcal{B} \in \mathfrak{A}$ so that by [theorem: 1.66]

$$\bigcap_{\mathcal{T} \in \mathfrak{A}} \mathcal{T} \subseteq \mathcal{B}$$

Taking $\sigma[\mathcal{A}] = \bigcap_{\mathcal{T} \in \mathfrak{A}} \mathcal{T}$ then proves the theorem. □

Corollary 21.43. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\sigma[\mathcal{A}] = \mathcal{A}$*

Proof. As $\mathcal{A}$ is a σ -algebra and $\mathcal{A} \subseteq \mathcal{A}$ it follows from [theorem: 21.42] that $\mathcal{A} \subseteq \sigma[\mathcal{A}] \subseteq \mathcal{A}$ which proves that $\sigma[\mathcal{A}] = \mathcal{A}$. □

We introduce now the concept of a measurable sup-space.

Theorem 21.44. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $B \subseteq X$ then $\mathcal{A}|_B$ defined by

$$\mathcal{A}|_B = \{B \cap A \mid A \in \mathcal{A}\}$$

is a σ -algebra on B called the **trace of $\mathcal{A}$ in B** . If additional $B \in \mathcal{A}$ then we have that

$$\mathcal{A}|_B \subseteq \mathcal{A}$$

and

$$\mathcal{A}|_B = \{A \in \mathcal{A} \mid A \subseteq B\}$$

Note 21.45. If $B = X$ then we have $\mathcal{A}|_X = \{X \cap A \mid A \in \mathcal{A}\} = \{A \mid A \in \mathcal{A}\} = \mathcal{A}$ so that we have $\langle X, \mathcal{A}|_X \rangle = \langle X, \mathcal{A} \rangle$.

Proof. As $X \in \mathcal{A}$ and $B = B \cap X$ it follows that

$$B \in \mathcal{A}|_B$$

Let $C \in \mathcal{A}|_B$ then there exist a $C' \in \mathcal{A}$ such that $C = B \cap C'$. As $\mathcal{A}$ is a σ -algebra we have that $X \setminus C' \in \mathcal{A}$, as further

$$\begin{aligned} B \setminus C &= B \setminus (B \cap C') \\ &\stackrel{[\text{theorem: 1.34}]}{=} (B \setminus B) \cup (B \setminus C') \\ &= B \setminus C' \\ &\stackrel{B \subseteq X}{=} (B \cap X) \setminus C' \\ &\stackrel{[\text{theorem: 1.30}]}{=} B \cap (X \setminus C') \end{aligned}$$

which proves that $B \setminus C \in \mathcal{A}|_B$. Hence

$$\forall C \in \mathcal{A}|_B \text{ we have that } B \setminus C \in \mathcal{A}|_B$$

Let $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}|_B$ then there exist a $\{C'_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that $\forall i \in \mathbb{N} \ C_i = B \cap C'_i$. As $\mathcal{A}$ is a σ -algebra we have that $\bigcup_{i \in \mathbb{N}} C'_i \in \mathcal{A}$. Now $\bigcup_{i \in \mathbb{N}} C_i = \bigcup_{i \in \mathbb{N}} (B \cap C'_i) \stackrel{[\text{theorem: 2.138}]}{=} B \cap (\bigcup_{i \in \mathbb{N}} C'_i)$ so that $\bigcup_{i \in \mathbb{N}} C_i \in \mathcal{A}|_B$. Hence we have that

$$\forall \{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}|_B \quad \bigcup_{i \in \mathbb{N}} C_i \in \mathcal{A}|_B$$

So all the conditions for a σ -algebra are satisfied so that

$$\mathcal{A}|_B \text{ is a } \sigma\text{-algebra}$$

Assume now that additional $B \in \mathcal{A}$. If $C \in \mathcal{A}|_B$ then there exist a $C' \in \mathcal{A}$ such that $C = B \cap C'$, further by [theorems: 21.36, 21.31] $B \cap C' \in \mathcal{A}$ so that $C \in \mathcal{A}$. Hence we have that

$$\mathcal{A}|_B \subseteq \mathcal{A}$$

If $C \in \mathcal{A}|_B$ then by the above $C \in \mathcal{A}$ and $C = B \cap C' \subseteq B$ so that $C \in \{A \in \mathcal{A} \mid A \subseteq B\}$. Further if $C \in \{A \in \mathcal{A} \mid A \subseteq B\}$ then $C \in \mathcal{A}$ and $C \stackrel{C \subseteq B}{=} B \cap C$ so that $C \in \mathcal{A}|_B$. Hence we have

$$\mathcal{A}|_B = \{A \in \mathcal{A} \mid A \subseteq B\} \quad \square$$

21.2.1 Borel Algebra

A special case of a σ -algebra in the context of topological spaces is the Borel algebra.

Definition 21.46. Let $\langle X, \mathcal{T} \rangle$ be a topological space then the Borel algebra on X noted as $\mathcal{B}[X, \mathcal{T}]$ is defined to be $\mathcal{B}[X, \mathcal{T}] = \sigma[\mathcal{T}]$. In other words the Borel algebra on a topological space is the σ -algebra generated by $\mathcal{T}$.

Theorem 21.47. Let $\langle X, \mathcal{T} \rangle$ be a topological space then $\mathcal{B}[X, \mathcal{T}] = \sigma[\{C \subseteq X \mid C \text{ is closed}\}]$.

Proof. If $C \in \{C \subseteq X \mid C \text{ is closed}\}$ then C is closed, hence there exist a open $U \in \mathcal{T}$ such that $C = X \setminus U$. As $U \in \mathcal{T} \subseteq \sigma[\mathcal{T}]$ it follows that $C = X \setminus U \in \sigma[\mathcal{T}]$. So $\{C \subseteq X \mid C \text{ is closed}\} \subseteq \sigma[\mathcal{T}]$ a σ -algebra, hence using [theorem: 21.42] we have

$$\sigma[\{C \subseteq X \mid C \text{ is closed}\}] \subseteq \sigma[\mathcal{T}]$$

If $U \in \mathcal{T}$ then $X \setminus U$ is closed hence $X \setminus U \in \{C \subseteq X \mid C \text{ is closed}\} \in \sigma[\{C \subseteq X \mid C \text{ is closed}\}]$, hence $U = X \setminus (X \setminus U) \in \sigma[\{C \subseteq X \mid C \text{ is closed}\}]$ proving that $\mathcal{T} \subseteq \sigma[\{C \subseteq X \mid C \text{ is closed}\}]$. So using [theorem: 21.42] we have

$$\sigma[\mathcal{T}] \subseteq \sigma[\{C \subseteq X \mid C \text{ is closed}\}]$$

This proves that

$$\mathcal{B}[X, \mathcal{T}] = \sigma[\mathcal{T}] = \sigma[\{C \subseteq X \mid C \text{ is closed}\}] \quad \square$$

Theorem 21.48. Let $\langle X, \mathcal{T}_X \rangle$ and $\langle Y, \mathcal{T}_Y \rangle$ be topological spaces and $f: X \rightarrow Y$ a continuous function then $\forall A \in \mathcal{B}[Y, \mathcal{T}_Y]$ we have that $f^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_X]$.

Proof. Let $\mathcal{A} = \{A \in \mathcal{P}(Y) \mid f^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_X]\}$ then we prove that $\mathcal{A}$ is a σ -algebra containing $\mathcal{T}_Y$. First, as $f^{-1}(Y) = X \in \mathcal{B}[X, \mathcal{T}_X]$, it follows that

$$Y \in \mathcal{A}$$

If $A \in \mathcal{A}$ then $A \subseteq Y$ and $f^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_X]$, as further

$$f^{-1}(Y \setminus A) \stackrel{[\text{theorem: 2.102}]}{=} X \setminus f^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_X]$$

it follows that $Y \setminus A \in \mathcal{B}[X, \mathcal{T}_X]$. Hence we conclude that

$$\forall A \in \mathcal{A} \text{ we have } Y \setminus A \in \mathcal{A}$$

If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ then $\forall i \in \mathbb{N}$ we have $A_i \subseteq Y$ and $f^{-1}(A_i) \in \mathcal{B}[X, \mathcal{T}_X]$, so that $\bigcup_{i \in \mathbb{N}} A_i \subseteq Y$ and $f^{-1}(\bigcup_{i \in \mathbb{N}} A_i) \stackrel{[\text{theorem: 2.144}]}{=} \bigcup_{i \in \mathbb{N}} f^{-1}(A_i) \stackrel{\mathcal{B}[X, \mathcal{T}_X] \text{ is a } \sigma\text{-algebra}}{\in} \mathcal{B}[X, \mathcal{T}_X]$ which proves that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$$

From the above it follows then from [theorems: 21.36] that

$$\mathcal{A} \text{ is a } \sigma\text{-algebra}$$

If $U \in \mathcal{T}_Y$ then $U \subseteq Y$ and, as f is continuous, that $f^{-1}(U) \in \mathcal{T}_X \subseteq \sigma(\mathcal{T}_X) = \mathcal{B}[X, \mathcal{T}_X]$ from which it follows that $U \in \mathcal{A}$ hence we have that

$$\mathcal{T}_Y \subseteq \mathcal{A}$$

Finally using [theorem: 21.42] it follows that $\mathcal{B}[Y, \mathcal{T}_Y] = \sigma[\mathcal{T}_Y] \subseteq \mathcal{A}$ from which it follows that $\forall A \in \mathcal{B}[Y, \mathcal{T}_Y]$ we have $f^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_X]$. $\square$

Corollary 21.49. Let $\langle X, \|\cdot\| \rangle$ be a normed space with the norm topology $\mathcal{T}_{\|\cdot\|}$ then

1. $\forall x \in X$ and $\forall A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$ we have

$$x + A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$$

$$\text{where } x + A = \{x + a \mid a \in A\} = \{a + x \mid a \in A\} = x + A.$$

2. $\forall \alpha \in \mathbb{K}$ and $\forall A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$ we have

$$\alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$$

$$\text{where } \alpha \cdot A = \{\alpha \cdot a \mid a \in A\}$$

Proof.

1. Take $x \in X$ then by [theorem: 14.182] $\delta_{-x}: X \rightarrow X$ defined by $\delta_{-x}(y) = -x + y$ is a continuous function. Hence by [theorem: 21.48] we have that

$$\forall A \in \mathcal{B}[X, \mathcal{T}_{||}] \text{ that } (\delta_{-x})^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_{||}] \quad (21.25)$$

Further if $y \in (\delta_{-x})^{-1}(A)$ then $-x + y = \delta_{-x}(y) \in A$ hence $y = x + (-x + y) \in x + A$ which proves that

$$(\delta_{-x})^{-1}(A) \subseteq x + A \quad (21.26)$$

If $y \in x + A$ then there exist a $a \in A$ such that $y = x + a$ so that

$$(\delta_{-x})(y) = -x + y = -x + (-x + a) = a \in A$$

from which it follows that $y \in (\delta_{-x})^{-1}(A)$ which proves that $x + A \subseteq (\delta_{-x})^{-1}(A)$. Combining this with [eq: 21.26] proves that $(\delta_{-x})^{-1}(A) = x + A$. Combining this with [eq: 21.25] proves that

$$\forall x \in X \text{ and } \forall A \in \mathcal{B}[X, \mathcal{T}_{||}] \text{ we have } x + A \in \mathcal{B}[X, \mathcal{T}_{||}]$$

2. Take $\alpha \in \mathbb{K}$ and $A \in \mathcal{B}[X, \mathcal{T}_{||}]$ then we have two cases for α to consider:

$\alpha = 0$. For A we have either:

$$A = \emptyset. \text{ Then } \alpha \cdot A = \emptyset \in \mathcal{B}[X, \mathcal{T}_{||}]$$

$A \neq \emptyset$. Then $\alpha \cdot A = \{0\}$ which is closed [see theorems: 14.242, 14.244] so that by [theorem: 21.47] $\alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{||}]$.

$\alpha \neq 0$. Using [theorem: 14.164] the function $\mu_{1/\alpha}: X \rightarrow X$ by $\mu_{1/\alpha}(x) = \frac{1}{\alpha} \cdot x$ is a continuous function. Hence by [theorem: 21.48] we have that

$$(\mu_{1/\alpha})^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_{||}] \quad (21.27)$$

If $y \in (\mu_{1/\alpha})^{-1}(A)$ then $\frac{1}{\alpha} \cdot y = (\mu_{1/\alpha})(y) \in A$ so that $y = \alpha \cdot (\frac{1}{\alpha} \cdot y) \in \alpha \cdot A$, further if $y \in \alpha \cdot A$ then there exist a $a \in A$ such that $y = \alpha \cdot a$ so that $\frac{1}{\alpha} \cdot y = \frac{1}{\alpha} \cdot (\alpha \cdot a) = a \in A$ which proves that $y \in (\mu_{1/\alpha})^{-1}(A)$. Hence $\alpha \cdot A = (\mu_{1/\alpha})^{-1}(A)$. Combining this with [eq: 21.27] proves that

$$\alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{||}]$$

In other words we have

$$\forall \alpha \in \mathbb{K} \text{ and } \forall A \in \mathcal{B}[X, \mathcal{T}_{||}] \text{ that } \alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{||}] \quad \square$$

21.2.1.1 Borel algebra on $\mathbb{R}$

A important Borel algebra is the Borel algebra on the set of real numbers.

Definition 21.50. Let $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ be the set of real numbers with the canonical topology $\mathcal{T}_{||}$ based on the absolute value norm [see theorem: 14.104] then $\mathcal{B}[\mathbb{R}]$ is the Borel algebra based on $\mathcal{T}_{||}$. In other words

$$\mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] = \sigma[\mathcal{T}_{||}]$$

We will now find equivalent definitions for $\mathcal{B}[\mathbb{R}]$ that will be useful if we define a measure. First we introduce the concept of bounded intervals on $\mathbb{R}$.

Definition 21.51. We define the following set of bounded intervals:

1. $\mathcal{I}_{||} = \{[a, b] | a, b \in \mathbb{R} \wedge [a, b] \neq \emptyset\}$ [the set of non empty closed bounded intervals]
2. $\mathcal{I}_{|} = \{[a, b[| a, b \in \mathbb{R} \wedge [a, b[\neq \emptyset\}$ [the set of non empty right open bounded intervals]

3. $\mathcal{I}_{]]} = \{]a, b[\mid a, b \in \mathbb{R} \wedge]a, b[\neq \emptyset\}$ [the set of non empty left open bounded intervals]
4. $\mathcal{I}_{][} = \{]a, b[\mid a, b \in \mathbb{R} \wedge]a, b[\neq \emptyset\}$ [the set of non empty open bounded intervals]
5. $\mathcal{I} = \mathcal{I}_{[]} \cup \mathcal{I}_{[[} \cup \mathcal{I}_{]]} \cup \mathcal{I}_{][}$

Note 21.52. If $b = a$ then $]a, b[= [a, b[=]a, b] = \emptyset$ and if $b < a$ then $[a, b] = [a, b[=]a, b] =]a, b[= \emptyset$ then we have valid intervals but they are not elements of $\mathcal{I}_{[]}$, $\mathcal{I}_{[[}$, $\mathcal{I}_{]]}$ and $\mathcal{I}_{][}$. The reason to exclude these intervals is that the boundary points of empty intervals are not well defined [for example $[1, 0[= [2, 0[= [0, 0[$ which is needed for defining the length of the interval based on the boundary points.

For case to case analysis the following lemma will be useful.

Lemma 21.53. We have that $\mathcal{I}_{[]}$, $\mathcal{I}_{[[}$, $\mathcal{I}_{]]}$ and $\mathcal{I}_{][}$ are pairwise disjoint. In other words we have

$$\begin{aligned}
 \mathcal{I}_{[]} \cap \mathcal{I}_{[[} &= \emptyset \\
 \mathcal{I}_{[]} \cap \mathcal{I}_{]]} &= \emptyset \\
 \mathcal{I}_{[]} \cap \mathcal{I}_{][} &= \emptyset \\
 \mathcal{I}_{[[} \cap \mathcal{I}_{[]}] &= \emptyset \\
 \mathcal{I}_{[[} \cap \mathcal{I}_{]]} &= \emptyset \\
 \mathcal{I}_{[[} \cap \mathcal{I}_{][} &= \emptyset \\
 \mathcal{I}_{]]} \cap \mathcal{I}_{[]} &= \emptyset \\
 \mathcal{I}_{]]} \cap \mathcal{I}_{[[} &= \emptyset \\
 \mathcal{I}_{]]} \cap \mathcal{I}_{][} &= \emptyset \\
 \mathcal{I}_{][} \cap \mathcal{I}_{[]} &= \emptyset \\
 \mathcal{I}_{][} \cap \mathcal{I}_{[[} &= \emptyset \\
 \mathcal{I}_{][} \cap \mathcal{I}_{]]} &= \emptyset
 \end{aligned}$$

so that

$$\mathcal{I} = \mathcal{I}_{[]} \sqcup \mathcal{I}_{[[} \sqcup \mathcal{I}_{]]} \sqcup \mathcal{I}_{][}$$

Proof. We have to check the following cases:

$\mathcal{I}_{[]} \cap \mathcal{I}_{[[}$. If $I \in \mathcal{I}_{[]} \cap \mathcal{I}_{[[}$ then $\emptyset \neq I = [a, b] = [c, d[$. As $[a, b] \neq \emptyset$ it follows from [theorem: 3.140] that $a \leq b$ so that $b \in [a, b] = [c, d[$ hence $c \leq b < d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c \leq b < x < d$ proving that $x \in [c, d[= [a, b]$ so that $b < x \leq b$ leading to the contradiction $b < b$. Hence we have

$$\mathcal{I}_{[]} \cap \mathcal{I}_{[[} = \emptyset \quad (21.28)$$

$\mathcal{I}_{[]} \cap \mathcal{I}_{]]}$. If $I \in \mathcal{I}_{[]} \cap \mathcal{I}_{]]}$ then $\emptyset = [a, b] =]c, d]$. As $[a, b] \neq \emptyset$ it follows from [theorem: 3.140] that $a \leq b$ so that $a \in [a, b] =]c, d]$ hence $c < a \leq d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c < x < a \leq d$ proving that $x \in]c, d] = [a, b]$ so that $a \leq x < a$ leading to the contradiction $a < a$. Hence we have

$$\mathcal{I}_{[]} \cap \mathcal{I}_{]]} = \emptyset \quad (21.29)$$

$\mathcal{I}_{[]} \cap \mathcal{I}_{][}$. If $I \in \mathcal{I}_{[]} \cap \mathcal{I}_{][}$ then $\emptyset \neq I = [a, b] =]c, d[$. As $[a, b] \neq \emptyset$ it follows from [theorem: 3.140] that $a \leq b$ so that $b \in [a, b] =]c, d[$ hence $c < b < d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c < b < x < d$ proving that $x \in]c, d[= [a, b]$ so that $b < x \leq b$ leading to the contradiction $b < b$. Hence we have

$$\mathcal{I}_{[]} \cap \mathcal{I}_{][} = \emptyset \quad (21.30)$$

$\mathcal{I}_{[[\cap \mathcal{I}_[]}$. Then we have

$$\mathcal{I}_{[[\cap \mathcal{I}_[] = \mathcal{I}_[] \cap \mathcal{I}_{[[[_{\text{eq: } \overline{21.28}}] \emptyset$$

$\mathcal{I}_{[[\cap \mathcal{I}_[]$. If $I \in \mathcal{I}_{[[\cap \mathcal{I}_[]$ then $\emptyset \neq I = [a, b[=]c, d]$. As $[a, b[\neq \emptyset$ it follows from [theorem: 3.140] that $a < b$ so that $a \in [a, b[=]c, d]$ hence $c < a \leq d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c < x < a \leq d$ proving that $x \in]c, d] = [a, b[$ so that $a \leq x < a$ leading to the contradiction $a < a$. Hence we have

$$\mathcal{I}_{[[\cap \mathcal{I}_[] = \emptyset \quad (21.31)$$

$\mathcal{I}_{[[\cap \mathcal{I}_[]$. If $I \in \mathcal{I}_{[[\cap \mathcal{I}_[]$ then $\emptyset \neq I = [a, b[=]c, d[$. As $[a, b[\neq \emptyset$ it follows from [theorem: 3.140] that $a < b$ so that $a \in [a, b[=]c, d[$ hence $c < a < d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c < x < a < d$ proving that $x \in]c, d[= [a, b[$ so that $a \leq x < a$ leading to the contradiction $a < a$. Hence we have

$$\mathcal{I}_{[[\cap \mathcal{I}_[] = \emptyset \quad (21.32)$$

$\mathcal{I}_[] \cap \mathcal{I}_[]$. Then we have

$$\mathcal{I}_[] \cap \mathcal{I}_[] = \mathcal{I}_[] \cap \mathcal{I}_[]_{[_{\text{eq: } \overline{21.29}}] \emptyset$$

$\mathcal{I}_[] \cap \mathcal{I}_{[[$. If $I \in \mathcal{I}_[] \cap \mathcal{I}_{[[$ then $I =]a, b] = [c, d[$. As $]a, b] \neq \emptyset$ we have by [theorem: 3.140] that $a < b$ so that $b \in]a, b] = [c, d[$ hence $c \leq b < d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c \leq b < x < d$ proving that $x \in [c, d[=]a, b]$ so that $b < x \leq b$ leading to the contradiction $b < b$. Hence we have

$$\mathcal{I}_[] \cap \mathcal{I}_{[[= \emptyset \quad (21.33)$$

$\mathcal{I}_[] \cap \mathcal{I}_[]$. If $I \in \mathcal{I}_[] \cap \mathcal{I}_[]$ then $I =]a, b] =]c, d[$. As $]a, b] \neq \emptyset$ we have by [theorem: 3.140] that $a < b$ so that $b \in]a, b] =]c, d[$ hence $c < b < d$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that $c < b < x < d$ proving that $x \in]c, d[=]a, b]$ so that $b < x \leq b$ leading to the contradiction $b < b$. Hence we have

$$\mathcal{I}_[] \cap \mathcal{I}_[] = \emptyset \quad (21.34)$$

$\mathcal{I}_[] \cap \mathcal{I}_[]$. Then we have

$$\mathcal{I}_[] \cap \mathcal{I}_[] = \mathcal{I}_[] \cap \mathcal{I}_[]_{[_{\text{eq: } \overline{21.30}}] \emptyset$$

$\mathcal{I}_[] \cap \mathcal{I}_{[[$. Then we have

$$\mathcal{I}_[] \cap \mathcal{I}_{[[= \mathcal{I}_{[[} \cap \mathcal{I}_[]_{[_{\text{eq: } \overline{21.32}}] \emptyset$$

$\mathcal{I}_[] \cup \mathcal{I}_[]$. Then we have

$$\mathcal{I}_[] \cup \mathcal{I}_[] = \mathcal{I}_[] \cup \mathcal{I}_[]_{[_{\text{eq: } \overline{21.34}}] \emptyset$$

□

The following allows us to switch between the different types of non empty bounded intervals.

Lemma 21.54. *Let $a, b \in \mathbb{R}$ then we have*

1. $]a, b] = \bigcap_{n \in \mathbb{N}}]a, b + \frac{1}{n}[$
2. $]a, b[= \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[$
3. $[a, b] = \bigcap_{n \in \mathbb{N}} [a - \frac{1}{n}, b]$
4. $[a, b] = \bigcap_{n \in \mathbb{N}} [a, b + \frac{1}{n}[$
5. $[a, b] = \bigcap_{n \in \mathbb{N}} [a - \frac{1}{n}, b + \frac{1}{n}[$
6. $] -\infty, a] = \bigcap_{n \in \mathbb{N}}] -\infty, a + \frac{1}{n}[$
7. $[a, \infty[= \bigcap_{n \in \mathbb{N}} [a - \frac{1}{n}, \infty[$

8. $]a, b] = \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b]$
9. $[a, b[= \bigcup_{n \in \mathbb{N}} [a, b - \frac{1}{n}]$
10. $]a, b[= \bigcup_{n \in \mathbb{N}}]a, b - \frac{1}{n}]$
11. $]a, b[= \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b[$
12. $]a, b[= \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b - \frac{1}{n}]$
13. $] -\infty, a[= \bigcup_{n \in \mathbb{N}}] -\infty, a - \frac{1}{n}]$
14. $]a, \infty[= \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, \infty[$
15. $[a, b[= [a, \infty[\setminus]b, \infty[$
16. $]a, b] =] -\infty, b] \setminus] -\infty, a]$
17. $]a, \infty[= \mathbb{R} \setminus] -\infty, a]$

Proof.

1. Let $x \in]a, b]$ then $a < x \leq b$ and as $\forall n \in \mathbb{N} \ b < b + \frac{1}{n}$ we have $a < x < b + \frac{1}{n}$ so it follows that $x \in]a, b + \frac{1}{n}[$. Hence

$$]a, b] \subseteq \bigcap_{n \in \mathbb{N}}]a, b + \frac{1}{n}[\quad (21.35)$$

If $x \in \bigcap_{n \in \mathbb{N}}]a, b + \frac{1}{n}[$ then $\forall n \in \mathbb{N} \ x \in]a, b + \frac{1}{n}[$ or $a < x < b + \frac{1}{n}$ which by [corollary: 10.31] proves that $a < x \leq b$ hence $x \in]a, b]$. So it follows that $\bigcap_{n \in \mathbb{N}}]a, b + \frac{1}{n}[\subseteq]a, b]$ which using [eq: 21.35] proves that

$$]a, b] = \bigcap_{n \in \mathbb{N}}]a, b + \frac{1}{n}[$$

2. Let $x \in [a, b[$ then $a \leq x < b$ and as $\forall n \in \mathbb{N} \ a - \frac{1}{n} < a$ it follows that $x \in]a - \frac{1}{n}, b[$. Hence

$$[a, b[\subseteq \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[\quad (21.36)$$

If $x \in \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[$ then $\forall n \in \mathbb{N} \ a - \frac{1}{n} < x < b \Rightarrow a < x + \frac{1}{n} \wedge x < b$ which by [corollary: 10.31] proves that $a \leq x < b$ hence $x \in [a, b[$. So $\bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[\subseteq [a, b[$ which using [eq: 21.36] prove that

$$[a, b[= \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[$$

3. Let $x \in [a, b]$ then $a \leq x < b$ and as $\forall n \in \mathbb{N} \ a - \frac{1}{n} < a \Rightarrow x \in]a - \frac{1}{n}, b]$. Hence

$$[a, b] \subseteq \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b] \quad (21.37)$$

If $x \in \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b]$ then $\forall n \in \mathbb{N} \ a - \frac{1}{n} < x \leq b$ or $a < x + \frac{1}{n}$ which by [corollary: 10.31] proves that $a \leq x$ giving $x \in [a, b]$. So we have $\bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b] \subseteq [a, b]$ which combined with [eq: 21.37] gives

$$[a, b] = \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b]$$

4. Let $x \in [a, b]$ then $a \leq x \leq b$ so that $\forall n \in \mathbb{N} \ a \leq x \leq b < b + \frac{1}{n}$. Which proves that

$$[a, b] \subseteq \bigcap_{n \in \mathbb{N}} [a, b + \frac{1}{n}] \quad (21.38)$$

If $x \in \bigcap_{n \in \mathbb{N}} [a, b + \frac{1}{n}[$ then $\forall n \in \mathbb{N}$ we have $a \leq x < b + \frac{1}{n}$. Using [corollary: 10.31] we have $x \leq b$ proving that $x \in [a, b]$. Hence $\bigcap_{n \in \mathbb{N}} [a, b + \frac{1}{n}[\subseteq [a, b]$ which combined with [eq: 21.38] gives

$$[a, b] = \bigcap_{n \in \mathbb{N}} \left[a, b + \frac{1}{n}[$$

5. Let $x \in [a, b]$ so that $a \leq x \leq b$ hence $\forall n \in \mathbb{N}$ we have $a - \frac{1}{n} < x < b + \frac{1}{n} \Rightarrow x \in]a - \frac{1}{n}, b + \frac{1}{n}[$. Hence

$$[a, b] \subseteq \bigcap_{n \in \mathbb{N}} \left] a - \frac{1}{n}, b + \frac{1}{n}[\quad (21.39)$$

Let $x \in \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b + \frac{1}{n}[$ then $\forall n \in \mathbb{N}$ we have $a - \frac{1}{n} < x < b + \frac{1}{n} \Rightarrow a < x + \frac{1}{n} \wedge x < b + \frac{1}{n}$. Using [corollary: 10.31] we have $a \leq x \wedge x \leq b$ or $x \in [a, b]$. Hence $\bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b + \frac{1}{n}[\subseteq [a, b]$ which together with [eq: 21.39] proves

$$[a, b] = \bigcap_{n \in \mathbb{N}} \left] a - \frac{1}{n}, b + \frac{1}{n}[$$

6. Let $x \in]-\infty, a]$ then $x \leq a$ so that $\forall n \in \mathbb{N}$ we have $x < a + \frac{1}{n}$, hence it follows that $x \in]-\infty, a + \frac{1}{n}[$. Hence

$$]-\infty, a] \subseteq \bigcap_{n \in \mathbb{N}} \left] -\infty, a + \frac{1}{n}[\quad (21.40)$$

Let $x \in \bigcap_{n \in \mathbb{N}}]-\infty, a + \frac{1}{n}[$ then $\forall n \in \mathbb{N}$ we have $x < a + \frac{1}{n}$ which by [corollary: 10.31] gives $x \leq a$ or $x \in]-\infty, a]$. Hence $\bigcap_{n \in \mathbb{N}}]-\infty, a + \frac{1}{n}[\subseteq]-\infty, a]$ which together with [eq: 21.40] proves

$$\bigcap_{n \in \mathbb{N}} \left] -\infty, a + \frac{1}{n}[=]-\infty, a]$$

7. Let $x \in [a, \infty[$ then $a \leq x$ so that $\forall n \in \mathbb{N}$ we have $a - \frac{1}{n} < x$ giving $x \in]a - \frac{1}{n}, \infty[$ proving that

$$[a, \infty[\subseteq \bigcap_{n \in \mathbb{N}} \left] a - \frac{1}{n}, \infty[\quad (21.41)$$

Let $x \in \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty[$ then $\forall n \in \mathbb{N}$ we have $a - \frac{1}{n} < x \Rightarrow a < x + \frac{1}{n}$ which by [corollary: 10.31] proves that $a \leq x \Rightarrow x \in [a, \infty[$. Hence $\bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty[\subseteq [a, \infty[$ which together with [eq: 21.41] proves that

$$[a, \infty[= \bigcap_{n \in \mathbb{N}} \left] a - \frac{1}{n}, \infty[$$

8. Let $x \in]a, b]$ then $a < x \leq b$ and using [corollary: 10.30] there exists a $n \in \mathbb{N}$ such that $a < a + \frac{1}{n} < x \leq b$ proving that $x \in [a + \frac{1}{n}, b]$. Hence

$$]a, b] \subseteq \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{n}, b \right] \quad (21.42)$$

Let $x \in \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b]$ then there exists a $n \in \mathbb{N}$ such that $x \in [a + \frac{1}{n}, b] \Rightarrow a < a + \frac{1}{n} \leq x \leq b$. Hence $x \in]a, b]$ proving that $\bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b] \subseteq]a, b]$ which by [eq: 21.42] gives

$$]a, b] = \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{n}, b \right]$$

9. Let $x \in [a, b]$ then $a \leq x < b$ and using [corollary: 10.30] there exists a $n \in \mathbb{N}$ such that $a \leq x < x + \frac{1}{n} < b \Rightarrow a \leq x \wedge x < b - \frac{1}{n}$ proving that $x \in [a, b - \frac{1}{n}]$. So

$$[a, b[\subseteq \bigcup_{n \in \mathbb{N}} \left[a, b - \frac{1}{n} \right] \quad (21.43)$$

If $x \in \bigcup_{n \in \mathbb{N}} [a, b - \frac{1}{n}]$ then $\exists n \in \mathbb{N}$ such that $x \in [a, b - \frac{1}{n}] \Rightarrow a \leq x \leq b - \frac{1}{n} < b$ proving that $x \in [a, b[$. Hence $\bigcup_{n \in \mathbb{N}} [a, b - \frac{1}{n}] \subseteq [a, b[$ which combined with 21.43 gives

$$[a, b[= \bigcup_{n \in \mathbb{N}} \left[a, b - \frac{1}{n} \right]$$

10. Let $x \in]a, b[$ then $a < x < b$ and using [corollary: 10.30] there exists a $n \in \mathbb{N}$ such that $a < x < x + \frac{1}{n} < b$ hence $a < x < b - \frac{1}{n}$ proving that $x \in]a, b - \frac{1}{n}]$. So

$$]a, b[\subseteq \bigcup_{n \in \mathbb{N}} \left] a, b - \frac{1}{n} \right] \quad (21.44)$$

If $x \in \bigcup_{n \in \mathbb{N}}]a, b - \frac{1}{n}]$ then $\exists n \in \mathbb{N}$ such that $a < x \leq b - \frac{1}{n} < b \Rightarrow x \in]a, b[$ proving that $\bigcup_{n \in \mathbb{N}}]a, b - \frac{1}{n}] \subseteq]a, b[$. Combining this with [eq: 21.44] gives

$$]a, b[= \bigcup_{n \in \mathbb{N}} \left] a, b - \frac{1}{n} \right]$$

11. Let $x \in]a, b[$ then $a < x < b$ and using [corollary: 10.30] there exists a $n \in \mathbb{N}$ such that $a < a + \frac{1}{n} < x < b$ proving that $x \in [a + \frac{1}{n}, b[$. Hence

$$]a, b[\subseteq \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{n}, b \right[\quad (21.45)$$

If $x \in \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b[$ then $\exists n \in \mathbb{N}$ such that $a < a + \frac{1}{n} < x < b \Rightarrow x \in]a, b[$ proving that $\bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b[\subseteq]a, b[$. Combining this with [eq: 21.45] gives

$$]a, b[= \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{n}, b \right[$$

12. Let $x \in]a, b[$ then $a < x < b$ and using [corollary: 10.30] there exists $n_1, n_2 \in \mathbb{N}$ such that $a < a + \frac{1}{n_1} < x < x + \frac{1}{n_2} < b \Rightarrow a + \frac{1}{n_1} < x < b - \frac{1}{n_2}$. Take $m = \max(n_1, n_2)$ then $\frac{1}{m} \leq \frac{1}{n_1}, \frac{1}{n_2}$ so that $a + \frac{1}{m} \leq a + \frac{1}{n_1} < x < x + \frac{1}{n_2} \leq b - \frac{1}{n_2} \leq b - \frac{1}{m}$ proving that $x \in [a + \frac{1}{m}, b - \frac{1}{m}]$. Hence

$$]a, b[\subseteq \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{m}, b - \frac{1}{m} \right] \quad (21.46)$$

If $x \in \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b - \frac{1}{n}]$ then $\exists n \in \mathbb{N}$ such that $x \in [a + \frac{1}{n}, b - \frac{1}{n}] \Rightarrow a < a + \frac{1}{n} \leq x < b - \frac{1}{n} < b$ proving that $a < x < b$ or $x \in]a, b[$. So $\bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b - \frac{1}{n}] \subseteq]a, b[$ which together with [eq: 21.46] gives

$$]a, b[= \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{m}, b - \frac{1}{m} \right]$$

13. Let $x \in]-\infty, a[$ then $x < a$ and using [corollary: 10.30] there exists a $n \in \mathbb{N}$ such that $x + \frac{1}{n} < a \Rightarrow x < a - \frac{1}{n} \Rightarrow x \in]-\infty, a - \frac{1}{n}]$ proving that

$$]-\infty, a[\subseteq \bigcup_{n \in \mathbb{N}} \left] -\infty, a - \frac{1}{n} \right] \quad (21.47)$$

If $x \in \bigcup_{n \in \mathbb{N}}]-\infty, a - \frac{1}{n}]$ then $\exists n \in \mathbb{N}$ such that $x \in]-\infty, a - \frac{1}{n}] \Rightarrow x \leq a - \frac{1}{n} < a$ giving $x \in]-\infty, a[$ and proving that $\bigcup_{n \in \mathbb{N}}]-\infty, a - \frac{1}{n}] \subseteq]-\infty, a[$. Using this together with [eq: 21.47] proves

$$]-\infty, a[= \bigcup_{n \in \mathbb{N}} \left] -\infty, a - \frac{1}{n} \right]$$

14. Let $x \in]a, \infty[$ then $a < x$ and using [corollary: 10.30] there exists a $n \in \mathbb{N}$ such that $a + \frac{1}{n} < x \Rightarrow x \in [a + \frac{1}{n}, \infty[$. Hence

$$]a, \infty[\subseteq \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{n}, \infty[\quad (21.48)$$

If $x \in \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, \infty[$ then $\exists n \in \mathbb{N}$ such that $x \in [a + \frac{1}{n}, \infty[\Rightarrow a < a + \frac{1}{n} \leq x$ hence $x \in]a, \infty[$. This proves that $\bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, \infty[\subseteq]a, \infty[$, combining this with [eq: 21.48] gives

$$]a, \infty[= \bigcup_{n \in \mathbb{N}+0} \left[a + \frac{1}{n}, \infty[$$

15. Let $x \in [a, b[$ then $a \leq x \wedge x < b \Rightarrow a \leq x \wedge \neg(b \leq x)$ so that $x \in [a, \infty[$ and $x \notin [b, \infty[$ proving that

$$[a, b[\subseteq [a, \infty[\setminus [b, \infty[\quad (21.49)$$

If $x \in [a, \infty[\setminus [b, \infty[$ then $a \leq x \wedge \neg(b \leq x)$ or $a \leq x$ and $x < b$ proving that $[a, \infty[\setminus [b, \infty[\subseteq [a, b[$. Combining this with [eq: 21.49] proves that

$$[a, b[= [a, \infty[\setminus [b, \infty[$$

16. Let $x \in]a, b]$ then $a < x \wedge x \leq b$ then $\neg(x \leq a) \wedge x \leq b$ so that $x \notin]-\infty, a] \wedge x \in]-\infty, b]$ proving that $x \in]-\infty, b] \setminus]-\infty, a]$. For the opposite inclusion let $x \in]-\infty, b] \setminus]-\infty, a]$ then $x \leq b$ and $\neg(x \leq a)$ proving that $x \leq b$ and $a < x$ or $x \in]a, b]$. Hence

$$]a, b] =]-\infty, b] \setminus]-\infty, a]$$

17. If $x \in]a, \infty[$ then $x \in \mathbb{R} \wedge a < x$ so that $x \not\leq a \Rightarrow x \notin]-\infty, a]$ hence $x \in \mathbb{R} \setminus]-\infty, a]$, further if $x \in \mathbb{R} \setminus]-\infty, a]$ then $x \in \mathbb{R} \wedge x \notin]-\infty, a] \Rightarrow x \not\leq a \Rightarrow a < x$ or $x \in]a, \infty[$ hence

$$]a, \infty[= \mathbb{R} \setminus]-\infty, a] \quad \square$$

We are now ready to prove alternative forms for $\mathcal{B}[\mathbb{R}]$.

Theorem 21.55. *We have the following alternative forms for $\mathcal{B}[\mathbb{R}] \stackrel{\text{definition}}{=} \sigma[\mathcal{T}_{||}]$*

$$\begin{aligned} \sigma[\{C \subseteq \mathbb{R} \mid C \text{ is closed in } \mathcal{T}_{||}\}] &\stackrel{[\text{theorem: 21.47}]}{=} \mathcal{B}[\mathbb{R}] \\ &= \sigma[\mathcal{T}_{||}] \\ &= \sigma[\mathcal{I}_{||}] \\ &= \sigma[\mathcal{I}_{|}] \\ &= \sigma[\mathcal{I}_{\perp}] \\ &= \sigma[\mathcal{I}_{\lceil}] \\ &= \sigma[\{[a, \infty[\mid a \in \mathbb{R}\}] \\ &= \sigma[\{]a, \infty[\mid a \in \mathbb{R}\}] \\ &= \sigma[\{]-\infty, a] \mid a \in \mathbb{R}\}] \\ &= \sigma[\{]-\infty, a[\mid a \in \mathbb{R}\}] \end{aligned}$$

Proof. Let $U \in \mathcal{T}_{||}$ then by [corollary: 14.110] there exist a countable family $\{]a_i, b_i[\mid i \in I \subseteq \mathcal{I}_{\lceil}\}$ such that

$$\forall i \in I \]a_i, \beta_i[\neq \emptyset \text{ and } U = \bigcup_{i \in I}]a_i, \beta_i[\quad (21.50)$$

As $\mathcal{I}_{\lceil} \subseteq \sigma[\mathcal{I}_{\lceil}]$ which is a σ -algebra it follows from [theorem: 21.39] that $U = \bigcup_{i \in I}]a_i, \beta_i[\in \sigma(\mathcal{I}_{\lceil})$. Given that $U \in \mathcal{T}_{||}$ was chosen arbitrary it follows that $\mathcal{T}_{||} \subseteq \sigma[\mathcal{I}_{\lceil}]$. Using then [theorem: 21.42] it follows that

$$\mathcal{B}[\mathbb{R}] \stackrel{\text{def}}{=} \sigma[\mathcal{T}_{||}] \subseteq \sigma[\mathcal{I}_{\lceil}] \quad (21.51)$$

If $]a, b[\in \mathcal{I}_{\downarrow}$ then by [lemma: 21.54] $]a, b[= \bigcup_{n \in \mathbb{N}}]a, b - \frac{1}{n}]$. Let $n \in \mathbb{N}$ then we have either:

$$]a, b - \frac{1}{n}] = \emptyset. \text{ Then }]a, b - \frac{1}{n}] = \emptyset \in \sigma[\mathcal{I}_{\downarrow}]$$

$$]a, b - \frac{1}{n}] \neq \emptyset. \text{ Then }]a, b - \frac{1}{n}] \in \sigma[\mathcal{I}_{\downarrow}]$$

so that $\{]a, b - \frac{1}{n}]\}_{n \in \mathbb{N}} \subseteq \sigma[\mathcal{I}_{\downarrow}]$ from which it follows by [definition: 21.33] that

$$]a, b[= \bigcup_{n \in \mathbb{N}}]a, b - \frac{1}{n}] \in \sigma[\mathcal{I}_{\downarrow}]$$

. Hence we have $\mathcal{I}_{\downarrow} \subseteq \sigma[\mathcal{I}_{\downarrow}]$ a σ -algebra on $\mathbb{R}$. Using [theorem: 21.42] it follows then that

$$\sigma[\mathcal{I}_{\downarrow}] \subseteq \sigma[\mathcal{I}_{\downarrow}] \quad (21.52)$$

If $]a, b[\in \mathcal{I}_{\uparrow}$ then by [lemma: 21.54] $]a, b[= \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b[$. For $n \in \mathbb{N}$ we have either:

$$[a + \frac{1}{n}, b[= \emptyset. \text{ Then } [a + \frac{1}{n}, b[= \emptyset \in \sigma[\mathcal{I}_{\uparrow}]$$

$$[a + \frac{1}{n}, b[\neq \emptyset. \text{ Then } [a + \frac{1}{n}, b[\in \mathcal{I}_{\uparrow} \subseteq \sigma[\mathcal{I}_{\uparrow}]$$

so that $\{[a + \frac{1}{n}, b[\}_{n \in \mathbb{N}} \subseteq \sigma[\mathcal{I}_{\uparrow}]$ from which it follows by [definition: 21.33] that

$$]a, b[= \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b[\in \sigma[\mathcal{I}_{\uparrow}]$$

Hence we have $\mathcal{I}_{\uparrow} \subseteq \sigma[\mathcal{I}_{\uparrow}]$ a σ -algebra on $\mathbb{R}$ so that by [theorem: 21.42]

$$\sigma[\mathcal{I}_{\uparrow}] \subseteq \sigma[\mathcal{I}_{\uparrow}] \quad (21.53)$$

If $[a, b] \in \mathcal{I}_{\downarrow}$ then by [lemma: 21.54] $[a, b] = \bigcap_{n \in \mathbb{N}} [a, b + \frac{1}{n}[$. For $n \in \mathbb{N}$ we have either:

$$[a, b + \frac{1}{n}[= \emptyset. \text{ Then } [a, b + \frac{1}{n}[= \emptyset \in \sigma[\mathcal{I}_{\downarrow}]$$

$$[a, b + \frac{1}{n}[\neq \emptyset. \text{ Then } [a, b + \frac{1}{n}[\in \mathcal{I}_{\downarrow} \subseteq \sigma[\mathcal{I}_{\downarrow}]$$

so that $\{[a, b + \frac{1}{n}[\}_{n \in \mathbb{N}} \subseteq \sigma[\mathcal{I}_{\downarrow}]$ from which it follows by [theorem: 21.36] that

$$[a, b] = \bigcap_{n \in \mathbb{N}} [a, b + \frac{1}{n}[\in \sigma[\mathcal{I}_{\downarrow}]$$

Hence we have $\mathcal{I}_{\downarrow} \subseteq \sigma[\mathcal{I}_{\downarrow}]$ a σ -algebra on $\mathbb{R}$ so that by [theorem: 21.42]

$$\sigma[\mathcal{I}_{\downarrow}] \subseteq \sigma[\mathcal{I}_{\downarrow}] \quad (21.54)$$

If $[a, b[\in \mathcal{I}_{\uparrow}$ then by [lemma: 21.54] $[a, b[= \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[$. As $\forall n \in \mathbb{N}$ $]a - \frac{1}{n}, b[$ is open it follows that $\{]a - \frac{1}{n}, b[\}_{n \in \mathbb{N}} \subseteq \mathcal{I}_{\uparrow}$ so that by [theorem: 21.36] $[a, b[= \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, b[\in \sigma[\mathcal{I}_{\uparrow}]$. Hence we have that $\mathcal{I}_{\uparrow} \subseteq \sigma[\mathcal{I}_{\uparrow}]$ so that

$$\sigma[\mathcal{I}_{\uparrow}] \subseteq \sigma[\mathcal{I}_{\uparrow}] \stackrel{\text{def}}{=} \mathcal{B}[\mathbb{R}] \quad (21.55)$$

Combining [eqs: 21.51, 21.52, 21.53, 21.54 and 21.55] it follows that

$$\mathcal{B}[\mathbb{R}] = \sigma[\mathcal{I}_{\uparrow}] = \sigma[\mathcal{I}_{\downarrow}] = \sigma[\mathcal{I}_{\downarrow}] = \sigma[\mathcal{I}_{\uparrow}] = \sigma[\mathcal{I}_{\downarrow}] \quad (21.56)$$

If $[a, b[\in \mathcal{I}_{\downarrow}$ then using [lemma: 21.54] we have that

$$[a, b] = [a, \infty] \setminus [b, \infty[\underset{\text{[theorem: 21.31, 21.36]}}{\in} \sigma[\{[a, \infty[\mid a \in \mathbb{R}\}]$$

proving $\mathcal{I}_{\downarrow} \subseteq \sigma[\{[a, \infty[\mid a \in \mathbb{R}\}]$ a σ -algebra. Hence using [theorem: 21.42] we have

$$\sigma[\mathcal{I}_{\downarrow}] \subseteq \sigma[\{[a, \infty[\mid a \in \mathbb{R}\}] \quad (21.57)$$

If $[a, \infty[\in \{[a, \infty[| a \in \mathbb{R}\}$ then using [lemma: 21.54] we have that $[a, \infty[= \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty[$. As $\{]a - \frac{1}{n}, \infty[\}_{n \in \mathbb{N}} \subseteq \{[a, \infty[| a \in \mathbb{R}\} \subseteq \sigma\{[a, \infty[| a \in \mathbb{R}\}$ it follows from [theorem: 21.36] that

$$[a, \infty[= \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty[\in \sigma\{[a, \infty[| a \in \mathbb{R}\}$$

Hence $\{[a, \infty[| a \in \mathbb{R}\} \subseteq \sigma\{[a, \infty[| a \in \mathbb{R}\}$ so that by [theorem: 21.42]

$$\sigma\{[a, \infty[| a \in \mathbb{R}\} \subseteq \sigma\{[a, \infty[| a \in \mathbb{R}\} \quad (21.58)$$

Let $]a, \infty[\in \{]a, \infty[| a \in \mathbb{R}\}$ then by [lemma: 21.54] $]a, \infty[= \mathbb{R} \setminus]-\infty, a]$ $\in \sigma\{]-\infty, a] | a \in \mathbb{R}\}$ so that $\{]a, \infty[| a \in \mathbb{R}\} \subseteq \sigma\{]-\infty, a] | a \in \mathbb{R}\}$. Using [theorem: 21.42] it follows that

$$\sigma\{]a, \infty[| a \in \mathbb{R}\} \subseteq \sigma\{]-\infty, a] | a \in \mathbb{R}\} \quad (21.59)$$

Let $] -\infty, a] \in \{]-\infty, a] | a \in \mathbb{R}\}$ then by [lemma: 21.54] $] -\infty, a] = \bigcap_{n \in \mathbb{N}}] -\infty, a + \frac{1}{n}[$. As $\{]-\infty, a + \frac{1}{n}[\}_{n \in \mathbb{N}} \subseteq \{]-\infty, a] | a \in \mathbb{R}\} \subseteq \sigma\{]-\infty, a] | a \in \mathbb{R}\}$ it follows from [theorem: 21.36] that $] -\infty, a] = \bigcap_{n \in \mathbb{N}}] -\infty, a + \frac{1}{n}[\in \sigma\{]-\infty, a] | a \in \mathbb{R}\}$. Hence

$$\{]-\infty, a] | a \in \mathbb{R}\} \subseteq \sigma\{]-\infty, a] | a \in \mathbb{R}\}$$

which by [theorem: 21.42] proves that

$$\sigma\{]-\infty, a] | a \in \mathbb{R}\} \subseteq \sigma\{]-\infty, a] | a \in \mathbb{R}\} \quad (21.60)$$

Let $] -\infty, a[\in \{]-\infty, a[| a \in \mathbb{R}\}$ then as $] -\infty, a[$ is open it follows that $\{]-\infty, a[| a \in \mathbb{R}\} \subseteq \mathcal{T}_{||} \subseteq \sigma[\mathcal{T}_{||}]$ and using [theorem: 21.42] it follows that

$$\sigma\{]-\infty, a[| a \in \mathbb{R}\} \subseteq \sigma[\mathcal{T}_{||}] \stackrel{[\text{eq: 21.56}]}{=} \sigma[\mathcal{I}_{||}] \quad (21.61)$$

Combining [eqs: 21.57, 21.58, 21.59, 21.60 and 21.61] gives

$$\sigma[\mathcal{I}_{||}] = \sigma\{[a, \infty[| a \in \mathbb{R}\} = \sigma\{[a, \infty[| a \in \mathbb{R}\} = \sigma\{]-\infty, a] | a \in \mathbb{R}\} = \sigma\{]-\infty, a[| a \in \mathbb{R}\}$$

Finally combining the above with [eq: 21.56] proves that

$$\begin{aligned} \mathcal{B}[\mathbb{R}] &= \sigma[\mathcal{T}_{||}] \\ &= \sigma[\mathcal{I}_{||}] \\ &= \sigma[\mathcal{I}_{||}] \\ &= \sigma[\mathcal{I}_{||}] \\ &= \sigma[\mathcal{I}_{||}] \\ &= \sigma\{[a, \infty[| a \in \mathbb{R}\} \\ &= \sigma\{]a, \infty[| a \in \mathbb{R}\} \\ &= \sigma\{]-\infty, a] | a \in \mathbb{R}\} \\ &= \sigma\{]-\infty, a[| a \in \mathbb{R}\} \end{aligned}$$

□

21.2.1.2 Borel algebra on $\overline{\mathbb{R}}$

We look now at the Borel algebra on the set of extended real numbers $\overline{\mathbb{R}}$.

Definition 21.56. Let $\langle \overline{\mathbb{R}}, \mathcal{T}_{\overline{\mathbb{R}}} \rangle$ be the topological space consisting of the extended real numbers with the canonical topology $\mathcal{T}_{\overline{\mathbb{R}}}$ [see definition: 20.50] then the Borel algebra on $\overline{\mathbb{R}}$ is noted as $\mathcal{B}[\overline{\mathbb{R}}]$ hence

$$\mathcal{B}[\overline{\mathbb{R}}] = \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$$

Just as in the real case we formulate some alternative definitions for the Borel algebra on the extended real numbers.

Lemma 21.57. *We have*

1. $\forall a \in \mathbb{R}$ we have $[a, \infty] = \bigcap_{n \in \mathbb{N}_0}]a - \frac{1}{n}, \infty]$
2. $\{\infty\} = \bigcap_{n \in \mathbb{N}_0}]n, \infty]$
3. $\{-\infty\} = \bigcap_{n \in \mathbb{N}_0} [-\infty, -n[$

Proof.

1. Let $n \in \mathbb{N}$ then if $x \in [a, \infty]$ $a \leq x \leq \infty \Rightarrow a - \frac{1}{n} < a \leq x \leq \infty$ so that $[a, \infty] \subseteq]a - \frac{1}{n}, \infty]$, hence

$$[a, \infty] \subseteq \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty] \quad (21.62)$$

If $x \in \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty]$ then $x \in]a - \frac{1}{1}, \infty] \Rightarrow a - 1 < x$. Assume that $x < a$ then $x \in \mathbb{R}$ and using [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $x + \frac{1}{n} < a$ or $x < a - \frac{1}{n}$ so that $x \notin]a - \frac{1}{n}, \infty]$ contradicting $x \in \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty]$. Hence the assumption is wrong and we have $a \leq x \leq \infty$ so that $x \in [a, \infty]$ proving that $\bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty] \subseteq [a, \infty]$, combining this with [eq: 21.62] results in

$$[a, \infty] = \bigcap_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty]$$

2. As $\forall n \in \mathbb{N}$ we have $n < \infty \leq \infty$ it follows that $\infty \in \bigcap_{n \in \mathbb{N}}]n, \infty]$ so that

$$\{\infty\} \subseteq \bigcap_{n \in \mathbb{N}}]n, \infty] \quad (21.63)$$

Let $x \in \bigcap_{n \in \mathbb{N}}]n, \infty]$. Then $x \in [1, \infty] \Rightarrow -\infty < 1 < x \leq \infty$ so that $x \neq -\infty$. Assume that $x \neq \infty$ then $x \in \mathbb{R}$ and by [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $x < n \Rightarrow x \notin]n, \infty]$ contradicting $x \in \bigcap_{n \in \mathbb{N}}]n, \infty]$, hence the assumption is wrong and we must have that $x = \infty \in \{\infty\}$. So $\bigcap_{n \in \mathbb{N}}]n, \infty] \subseteq \{\infty\}$ which combined with [eq: 21.63] proves that

$$\{\infty\} = \bigcap_{n \in \mathbb{N}}]n, \infty]$$

3. As $\forall n \in \mathbb{N}$ we have $-\infty \leq -\infty < -n$ it follows that $-\infty \in \bigcap_{n \in \mathbb{N}} [-\infty, -n[$ so that

$$\{-\infty\} \subseteq \bigcap_{n \in \mathbb{N}} [-\infty, -n[\quad (21.64)$$

Let $x \in \bigcap_{n \in \mathbb{N}} [-\infty, -n[$. Then $x \in [-\infty, -1[\Rightarrow x < -1 < \infty$ so that $x \neq \infty$. Assume that $x \neq -\infty$ then $x \in \mathbb{R}$ and by [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $-x < n \Rightarrow -n < x$ so that $x \notin [-\infty, -n[$ contradicting $x \in \bigcap_{n \in \mathbb{N}} [-\infty, -n[$, hence the assumption is wrong and we must have that $x = -\infty \in \{-\infty\}$. So $\bigcap_{n \in \mathbb{N}} [-\infty, -n[\subseteq \{-\infty\}$ which combined with [eq: 21.64] proves that

$$\{-\infty\} = \bigcap_{n \in \mathbb{N}} [-\infty, -n[\quad \square$$

Lemma 21.58. *For $\sigma[\{[a, \infty] | a \in \mathbb{R}\}]$ we have the following*

1. $\forall a \in \mathbb{R}$ we have that $[a, \infty] \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$
2. $\forall a \in \mathbb{R}$ we have that $[-\infty, a] \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$
3. $\forall a, b \in \mathbb{R}$ we have that $]a, b[\in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$
4. $\forall U \in \mathcal{T}_{||}$ we have that $U \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$
5. $\{\infty\} \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$
6. $\{-\infty\} \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$

Proof.

1. Let $a \in \mathbb{R}$ then $\{]a - \frac{1}{n}, \infty]\}_{n \in \mathbb{N}} \subseteq \{]a, \infty[|a \in \mathbb{R}\} \subseteq \sigma(\{]a, \infty[|a \in \mathbb{R}\})$, hence $\bigcup_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty] \in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$. As $[a, \infty] \stackrel{[\text{lemma: 21.57}]}{=} \bigcup_{n \in \mathbb{N}}]a - \frac{1}{n}, \infty]$ we have by [theorem: 21.36]

$$[a, \infty] \in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$$

2. Let $a \in \mathbb{R}$ then

$$\begin{aligned} x \in \overline{\mathbb{R}} \setminus [a, \infty] &\Leftrightarrow x \in [-\infty, \infty] \wedge x \notin [a, \infty] \\ &\Leftrightarrow (-\infty \leq x \wedge x \leq \infty) \wedge \neg(a \leq x \wedge x \leq \infty) \\ &\Leftrightarrow (-\infty \leq x \wedge x \leq \infty) \wedge (x < a \vee \infty < x) \\ &\stackrel{\infty < x \text{ is false}}{\Leftrightarrow} -\infty \leq x \wedge x \leq \infty \wedge x < a \\ &\Leftrightarrow -\infty \leq x \wedge x < a \\ &\Leftrightarrow x \in [-\infty, a[\end{aligned}$$

proving that $[-\infty, a[= \overline{\mathbb{R}} \setminus [a, \infty]$ which, as by (1) $[a, \infty] \in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$ and $\sigma(\{]a, \infty[|a \in \mathbb{R}\})$ is a σ -algebra, proves by [theorem: 21.36]

$$[-\infty, a[\in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$$

3. Let $a, b \in \mathbb{R}$ then we have either:

$b < a$. Then $[a, b] = \emptyset \in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$

$a \leq b$. Then

$$\begin{aligned} x \in]a, \infty[\cap [-\infty, b[&\Leftrightarrow x \in]a, \infty[\wedge x \in [-\infty, b[\\ &\Leftrightarrow a < x \wedge x \leq \infty \wedge -\infty \leq x \wedge x < b \\ &\Leftrightarrow a < x \wedge x < b \\ &\Leftrightarrow x \in]a, b[\end{aligned}$$

proving that $]a, b[=]a, \infty[\cap [-\infty, b[$. As $]a, \infty[\in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$ and by (2) $[-\infty, b[\in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$ we have [theorem] $]a, \infty[\cap [-\infty, b[\in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$. So by [theorems: 21.32, 21.36]

$$]a, b[\in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$$

4. Let U be a open set in $\mathbb{R}$ then by [corollary: 14.110] there exist a countable set I and family $\{a_i, b_i\}_{i \in I}$ such that $U = \bigcup_{i \in I}]a_i, b_i[$. By (3) it follows that $\{]a_i, b_i\}_{i \in I} \subseteq \sigma(\{]a, \infty[|a \in \mathbb{R}\})$ so that using [theorem: 21.39] $U = \bigcup_{i \in I}]a_i, b_i[\in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$.

5. Using [lemma: 21.57] we have that $\{\infty\} = \bigcap_{n \in \mathbb{N}_0}]n, \infty]$ which, as $\{]n, \infty]\}_{n \in \mathbb{N}_0} \subseteq (\{]a, \infty[|a \in \mathbb{R}\}) \subseteq \sigma(\{]a, \infty[|a \in \mathbb{R}\})$, proves by [theorem: 21.36]

$$\{\infty\} \in \sigma(\{]a, \infty[|a \in \mathbb{R}\})$$

6. Using [lemma: 21.57] we have that $\{-\infty\} = \bigcap_{n \in \mathbb{N}_0} [-\infty, -n[$ which, as by (2) $\{[-\infty, -n]\}_{n \in \mathbb{N}_0} \subseteq \sigma(\{]a, \infty[|a \in \mathbb{R}\})$, proves by [theorem: 21.36] that

$$\{-\infty\} \in \sigma(\{]a, \infty[|a \in \mathbb{R}\}) \quad \square$$

Lemma 21.59. $\mathcal{B}[\overline{\mathbb{R}}] \stackrel{\text{definition}}{=} \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] = \sigma(\{]a, \infty[|a \in \mathbb{R}\})$

Proof. Let $I \in \{]a, \infty[|a \in \mathbb{R}\}$ then there exist a $a \in \mathbb{R}$ such that $I =]a, \infty[$. By [definition: 20.48] it follows that $I \in \mathcal{B}_{\infty} \subseteq \mathcal{T}_{\overline{\mathbb{R}}}$, so that $\{]a, \infty[|a \in \mathbb{R}\} \subseteq \mathcal{T}_{\overline{\mathbb{R}}} \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$. Hence using [theorem: 21.42]

$$\sigma(\{]a, \infty[|a \in \mathbb{R}\}) \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \quad (21.65)$$

For the opposite inclusion, let $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ then by [theorem: 20.51] we have that $U \cap \mathbb{R} \in \mathcal{T}_{||}$. From $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty\} \cup \{\infty\}$ it follows that

$$U = U \cap \overline{\mathbb{R}} = (U \cap \mathbb{R}) \cup (U \cap \{-\infty\}) \cup (U \cap \{\infty\}) \quad (21.66)$$

Now as $U \cap \mathbb{R} \in \mathcal{T}_{||}$ we have by [lemma: 21.58] that

$$U \cap \mathbb{R} \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}] \quad (21.67)$$

Also $U \cap \{-\infty\}$ is either $\emptyset \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$ or $\{-\infty\} \underset{[\text{lemma: 21.58}]}{\in} \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$ so that

$$U \cap \{-\infty\} \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}] \quad (21.68)$$

Finally $U \cap \{\infty\}$ is either $\emptyset \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$ or $\{\infty\} \underset{[\text{lemma: 21.58}]}{\in} \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$ it follows that

$$U \cap \{\infty\} \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}] \quad (21.69)$$

From [eqs: 21.66, 21.67, 21.68 and 21.69] it follows from [theorems: 21.32, 21.36] that $U \in \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$. Hence $\mathcal{T}_{\overline{\mathbb{R}}} \subseteq \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$ [a sigma algebra on $\overline{\mathbb{R}}$] so that $\sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \subseteq \sigma[\{[a, \infty] | a \in \mathbb{R}\}]$, combining this with [eq: 21.65] gives

$$\sigma[\{[a, \infty] | a \in \mathbb{R}\}] = \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \quad \square$$

The relation between $\mathcal{B}[\mathbb{R}]$ and $\mathcal{B}[\overline{\mathbb{R}}]$ is expressed in the following theorem.

Lemma 21.60. $\sigma[\mathcal{T}_{\overline{\mathbb{R}}}] = \{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \underset{\mathcal{B}[\mathbb{R}] = \sigma[\mathcal{T}_{||}]}{=} \{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \sigma[\mathcal{T}_{||}]\}$

Proof. First we prove that

$$\{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \text{ is a } \sigma\text{-algebra on } \overline{\mathbb{R}} \quad (21.70)$$

Proof. As $\overline{\mathbb{R}} \cap \mathbb{R} = \mathbb{R} \in \mathcal{B}[\mathbb{R}]$ it follows that

$$\overline{\mathbb{R}} \in \{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \quad (21.71)$$

Let $U \in \{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$ then $U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]$ hence $\mathbb{R} \setminus (U \cap \mathbb{R}) \in \mathcal{B}[\mathbb{R}]$, further

$$(\overline{\mathbb{R}} \setminus U) \cap \mathbb{R} \underset{[\text{theorem: 1.36}]}{=} (\overline{\mathbb{R}} \cap \mathbb{R}) \setminus U = \mathbb{R} \setminus U \text{ and } \mathbb{R} \setminus (U \cap \mathbb{R}) \underset{[\text{eq: 1.34}]}{=} (\mathbb{R} \setminus U) \cup (\mathbb{R} \setminus \mathbb{R}) = \mathbb{R} \setminus U$$

so that $(\overline{\mathbb{R}} \setminus U) \cap \mathbb{R} = \mathbb{R} \setminus (U \cap \mathbb{R}) \in \mathcal{B}[\mathbb{R}]$, proving that $(\overline{\mathbb{R}} \setminus U) \in \{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$. In other words

$$\forall U \in \{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \text{ we have } \overline{\mathbb{R}} \setminus U \in \{U \in \mathcal{P}[\overline{\mathbb{R}}] | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \quad (21.72)$$

Let $\{U_i\}_{i \in \mathbb{N}} \subseteq \{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$ then $\forall i \in \mathbb{N} \ U_i \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]$ so that $(\bigcup_{i \in \mathbb{N}} U_i) \cap \mathbb{R} = \bigcup_{i \in \mathbb{N}} (U_i \cap \mathbb{R}) \in \mathcal{B}[\mathbb{R}]$. Hence

$$\forall \{U_i\}_{i \in \mathbb{N}} \subseteq \{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \quad \bigcup_{i \in \mathbb{N}} (U_i \cap \mathbb{R}) \in \mathcal{B}[\mathbb{R}] \quad (21.73)$$

Finally [eqs: 21.71, 21.72 and 21.73] proves [eq: 21.70]. $\square$

Take $U \in \mathcal{T}_{\overline{\mathbb{R}}}$ then by [theorem: 20.51] we have that $U \cap \mathbb{R} \in \mathcal{T}_{||} \subseteq \sigma[\mathcal{T}_{||}] \underset{\text{definition}}{=} \mathcal{B}[\mathbb{R}]$ proving that $U \in \{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$ so that

$$\mathcal{T}_{\overline{\mathbb{R}}} \subseteq \{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$$

As $\{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$ is a σ -algebra on $\overline{\mathbb{R}}$ [see eq: 21.70] it follows from [theorem: 21.42] that

$$\mathcal{B}[\overline{\mathbb{R}}] \underset{\text{definition}}{=} \sigma(\mathcal{T}_{\overline{\mathbb{R}}}) \subseteq \{U \in \mathcal{P}(\overline{\mathbb{R}}) | U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \quad (21.74)$$

For the opposite inclusion define

$$S = \{U \cap \mathbb{R} \mid U \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \} \subseteq \mathcal{P}(\mathbb{R}) \quad (21.75)$$

then as $\mathcal{T}_{\overline{\mathbb{R}}} \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ we have

$$\mathcal{T}_{\overline{\mathbb{R}}} \underset{[\text{theorem: 20.51}]}{=} \{U \cap \mathbb{R} \mid U \in \mathcal{T}_{\overline{\mathbb{R}}}\} \subseteq \{U \cap \mathbb{R} \mid U \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \} = S \quad (21.76)$$

Further as $\mathbb{R} \in \mathcal{T}_{\overline{\mathbb{R}}} \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ we have for $U \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ that $U \cap \mathbb{R} \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ proving

$$S \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \quad (21.77)$$

Next we prove that

$$S \text{ is a } \sigma\text{-algebra on } \mathbb{R} \quad (21.78)$$

Proof. As $\mathbb{R} = \mathbb{R} \cap \mathbb{R}$ and $\mathbb{R} \in \mathcal{T}_{\overline{\mathbb{R}}} \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ it follows that

$$\mathbb{R} \in S \quad (21.79)$$

If $A \in S$ then there exist a $U \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ such that $A = U \cap \mathbb{R}$ so that

$$\mathbb{R} \setminus A = \mathbb{R} \setminus (U \cap \mathbb{R}) \underset{[\text{eq: 1.34}]}{=} (\mathbb{R} \setminus U) \cup (\mathbb{R} \setminus \mathbb{R}) = \mathbb{R} \setminus U$$

which, as $\mathbb{R}, U \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \xRightarrow{[\text{eq: 21.32}]} \mathbb{R} \setminus U \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$, proves that

$$\mathbb{R} \setminus A \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$$

Hence $(\mathbb{R} \setminus A) \cap \mathbb{R} = \mathbb{R} \setminus A \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ so that $\mathbb{R} \setminus A \in S$. Summarized

$$\forall A \in S \text{ we have } \mathbb{R} \setminus A \in S \quad (21.80)$$

If $\{A_i\}_{i \in \mathbb{N}} \subseteq S$ then there exists a $\{U_i\}_{i \in \mathbb{N}} \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ such that $\forall i \in \mathbb{N}$ we have $A_i = U_i \cap \mathbb{R}$, then $\bigcup_{i \in \mathbb{N}} A_i = (\bigcup_{i \in \mathbb{N}} (U_i \cap \mathbb{R})) = (\bigcup_{i \in \mathbb{N}} U_i) \cap \mathbb{R}$ which, as $\bigcup_{i \in \mathbb{N}} U_i \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$, proves that $\bigcup_{i \in \mathbb{N}} A_i \in S$. In other words

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq S \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in S \quad (21.81)$$

Finally [eqs: 21.79, 21.80 and 21.81] proves that S is a σ -algebra on $\mathbb{R}$ thus proving [eq: 21.78]. $\square$

So we have proved that $\mathcal{T}_{\overline{\mathbb{R}}} \subseteq S$ [see eq: 21.76] and S is a σ -algebra on $\mathbb{R}$ from which it follows, using [theorem: 21.42], that

$$\mathcal{B}[\mathbb{R}] \underset{\text{definition}}{=} \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \subseteq S \underset{[\text{eq: 21.77}]}{\subseteq} \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \quad (21.82)$$

Let $A \in \{U \in \mathcal{P}[\overline{\mathbb{R}}] \mid U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}$ then $A \subseteq \overline{\mathbb{R}}$ so that $A = A \cap \overline{\mathbb{R}} = A \cap (\mathbb{R} \cup \{-\infty\}, \{\infty\})$ giving

$$A = (A \cap \mathbb{R}) \cap (A \cap \{-\infty\}) \cap (A \cap \{\infty\}) \quad (21.83)$$

where

1. $A \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}] \xRightarrow{[\text{eq: 21.82}]} A \cap \mathbb{R} \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$
2. $A \cap \{-\infty\}$ is either $\emptyset \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ or $\{-\infty\} \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ [as $-\infty$ is closed [theorem: 20.52] and closed sets are part of $\sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ by [theorem: 21.47]].
3. $A \cap \{\infty\}$ is either $\emptyset \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ or $\{\infty\} \in \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ [as ∞ is closed by [theorem: 20.52] and closed sets are part of $\sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ by [theorem: 21.47]].

From the above we conclude that $\{U \in \mathcal{P}[\overline{\mathbb{R}}] \mid U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \subseteq \sigma[\mathcal{T}_{\overline{\mathbb{R}}}]$ which combined with [eq: 21.74] proves that

$$\sigma[\mathcal{T}_{\overline{\mathbb{R}}}] = \{U \in \mathcal{P}[\overline{\mathbb{R}}] \mid U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} \quad \square$$

Summarizing the above two lemmas in a theorem we get:

Theorem 21.61. *The Borel algebra on $\overline{\mathbb{R}}$ satisfies*

$$\begin{aligned}\mathcal{B}[\overline{\mathbb{R}}] & \stackrel{\text{definition}}{=} \sigma[\mathcal{T}_{\overline{\mathbb{R}}}] \\ & \stackrel{[\text{theorem: 21.47}]}{=} \sigma[\{C \mid C \text{ is closed in } \mathcal{T}_{\overline{\mathbb{R}}}\}] \\ & = \sigma[\{[a, \infty) \mid a \in \mathbb{R}\}] \\ & = \{U \subseteq \mathcal{P}[\overline{\mathbb{R}}] \mid U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\}\end{aligned}$$

Further we have that

$$\mathcal{B}[\mathbb{R}] \subseteq \mathcal{B}[\overline{\mathbb{R}}]$$

Proof. The first part of the theorem follows from [lemmas: 21.59, 21.60]. For the last part. Let $A \in \mathcal{B}[\mathbb{R}]$ then $A \subseteq \mathbb{R} \subseteq \overline{\mathbb{R}}$ so that $A \in \mathcal{P}(\overline{\mathbb{R}})$ and $A = A \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]$ hence

$$A \in \{U \subseteq \mathcal{P}[\overline{\mathbb{R}}] \mid U \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]\} = \mathcal{B}[\overline{\mathbb{R}}].$$

which proves that $\mathcal{B}[\mathbb{R}] \subseteq \mathcal{B}[\overline{\mathbb{R}}]$. $\square$

21.2.1.3 Borel algebra on $\mathbb{R}^n$

Just as in for the real numbers and the extended real numbers we have many ways of generating the Borel algebra on $\mathbb{R}$.

Definition 21.62. *Let $n \in \mathbb{N}$ then the Borel algebra on $\mathbb{R}^n$ is noted as $\mathcal{B}[\mathbb{R}^n]$ so*

$$\mathcal{B}[\mathbb{R}^n] = \sigma[\mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n}]$$

where $\mathcal{T}_{\|\cdot\|_n}$ is the canonical topology on $\mathbb{R}^n$ [see example: 14.101]

First we extend the concept of generalized intervals to intervals on $\mathbb{R}^n$

Definition 21.63. *Let $n \in \mathbb{N}$ and $x, y \in \mathbb{R}^n$ then $x \leq_n y$ if $\forall i \in \{1, \dots, n\} x_i \leq y_i$ further we have $x <_n y$ if $\forall i \in \{1, \dots, n\} x_i < y_i$.*

Note 21.64. *If $x_i < y_i$ then $\forall i \in \{1, \dots, n\}$ we have $x_i < y_i \Rightarrow x_i \leq y_i \wedge x_i \neq y_i$ so that $x \leq y \wedge x \neq y$ which proves that*

$$x <_n y \Rightarrow x \leq_n y \wedge x \neq y$$

However the opposite is not true. for example $(1, 2) \leq_2 (2, 2)$ and $(1, 2) \neq (2, 2)$ but $(1, 2) \not<_2 (2, 2)$. So this definition is different from the normal definition of a strict inequality.

This relation makes $\langle \mathbb{R}^n, \leq_n \rangle$ a partial ordered set.

Theorem 21.65. *Let $n \in \mathbb{N}$ then $\langle \mathbb{R}^n, \leq_n \rangle$ is a partial ordered set.*

Note 21.66. $\leq_n$ is however not a total order, for example for $(1, 2), (2, 1)$ we have that $(1, 2) \not\leq_2 (2, 1)$ and $(2, 1) \not\leq_2 (1, 2)$.

Proof. We have

reflexivity. If $x \in \mathbb{R}^n$ then $\forall i \in \{1, \dots, n\}$ we have $x_i = x_i \Rightarrow x_i \leq x_i$ proving that $x \leq_n x$.

anti-symmetry. If $x, y \in \mathbb{R}^n$ satisfies $x \leq y \wedge y \leq x$ then we have $\forall i \in \mathbb{N} x_i \leq y_i$ and $y_i \leq x_i$ so that $x_i = y_i$ from which it follows that $x = y$.

transitivity. If $x \leq_n y$ and $y \leq_n z$ then we have $\forall i \in \{1, \dots, n\}$ that $x_i \leq y_i \wedge y_i \leq z_i \Rightarrow x_i \leq z_i$ from which it follows that $x \leq_n z$. $\square$

Using this order relation we can extend the concept of intervals to intervals in $\mathbb{R}^n$.

Definition 21.67. Let $n \in \mathbb{N}_0$, $a, b \in \mathbb{R}^n$ such that $a \leq_n b$ then we have

$$\begin{aligned}
[a, b] &= \{x \in \mathbb{R}^n \mid a \leq_n x \wedge x \leq_n b\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } a_i \leq x_i \wedge x_i \leq b_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in [a_i, b_i]\} \\
&= \prod_{i \in \{1, \dots, n\}} [a_i, b_i] \\
[a, b[&= \{x \in \mathbb{R}^n \mid a \leq_n x \wedge x <_n b\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } a_i \leq x_i \wedge x_i < b_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in [a_i, b_i[\} \\
&= \prod_{i < \{1, \dots, n\}} [a_i, b_i[\\
]a, b] &= \{x \in \mathbb{R}^n \mid a <_n x \wedge x \leq_n b\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } a_i < x_i \wedge x_i \leq b_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in]a_i, b_i]\} \\
&= \prod_{i \in \{1, \dots, n\}}]a_i, b_i] \\
]a, b[&= \{x \in \mathbb{R}^n \mid a <_n x \wedge x <_n b\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } a_i < x_i \wedge x_i < b_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in]a_i, b_i[\} \\
&= \prod_{i < \{1, \dots, n\}}]a_i, b_i[\\
]-\infty, a] &= \{x \in \mathbb{R}^n \mid x \leq_n a\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \leq a_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in]-\infty, a_i]\} \\
&= \prod_{i < \{1, \dots, n\}}]-\infty, a_i] \\
]-\infty, a[&= \{x \in \mathbb{R}^n \mid x <_n a\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i < a_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in]-\infty, a_i[\} \\
&= \prod_{i < \{1, \dots, n\}}]-\infty, a_i[\\
[a, \infty[&= \{x \in \mathbb{R}^n \mid a \leq_n x\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } a_i \leq x_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in [a_i, \infty[\} \\
&= \prod_{i < \{1, \dots, n\}} [a_i, \infty[\\
]a, \infty[&= \{x \in \mathbb{R}^n \mid a <_n x\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } a_i < x_i\} \\
&= \{x \in \mathbb{R}^n \mid \forall i \in \{1, \dots, n\} \text{ we have } x_i \in]a_i, \infty[\} \\
&= \prod_{i < \{1, \dots, n\}}]a_i, \infty[
\end{aligned}$$

To simply notation we have also the following definitions:

Definition 21.68. Let $n \in \mathbb{N}_0$ and $a, b \in \mathbb{R}$ then we define the following sets

1. $[a, b]^n = \{x \in \mathbb{R}^n \mid a \leq x_i \leq b\}$
2. $[a, b[^n = \{x \in \mathbb{R}^n \mid a \leq x_i < b\}$
3. $]a, b]^n = \{x \in \mathbb{R}^n \mid a < x_i \leq b\}$
4. $]a, b[^n = \{x \in \mathbb{R}^n \mid a < x_i < b\}$

5. $] -\infty, a]^n = \{x \in \mathbb{R}^n \mid x_i \leq a\}$
6. $] -\infty, a[^n = \{x \in \mathbb{R}^n \mid x_i < a\}$
7. $[a, \infty[^n = \{x \in \mathbb{R}^n \mid a \leq x_i\}$
8. $[a, \infty]^n = \{x \in \mathbb{R}^n \mid a < x_i\}$

Note 21.69. If we take $\alpha, \beta \in \mathbb{R}^n$ by $\alpha_i = a$ and $\beta_i = b$ then we have that

$$\begin{aligned}
 [a, b]^n &= [\alpha, \beta] \\
 [a, b[^n &= [\alpha, \beta[\\
]a, b]^n &=]\alpha, \beta] \\
]a, b[^n &=]\alpha, \beta[\\
]-\infty, a]^n &=]\infty, \alpha] \\
]-\infty, a[^n &=]\infty, \alpha[\\
 [a, \infty[^n &= [\alpha, \infty[\\
 [a, \infty]^n &= [\alpha, \infty]
 \end{aligned}$$

Proposition 21.70. $\forall n \in \mathbb{N}$ we have $\mathbb{R}^n = \bigcup_{i \in \mathbb{N}}]-i, i]^n = \bigcup_{i \in \mathbb{N}} [-i, i]^n$

Proof. First if $x \in \bigcup_{i \in \mathbb{N}}]-i, i[^n$ then $\exists i \in \mathbb{N}$ such that $x \in]-i, i[^n$ hence $\forall j \in \{1, \dots, n\}$ $x_j \in]-i, i[\subseteq [-i, i]$ so that $x \in [-i, i]^n$ from which it follows that $\bigcup_{i \in \mathbb{N}}]-i, i[^n \subseteq \bigcup_{i \in \mathbb{N}} [-i, i]^n$. As further $\bigcup_{i \in \mathbb{N}} [-i, i]^n \subseteq \mathbb{R}^n$ it follows that

$$\bigcup_{i \in \mathbb{N}}]-i, i[^n \subseteq \bigcup_{i \in \mathbb{N}} [-i, i]^n \subseteq \mathbb{R}^n \quad (21.84)$$

Let $x \in \mathbb{R}^n$ then $\forall j \in \{1, \dots, n\}$ $x_j \in \mathbb{R} \stackrel{[\text{corollary: 10.30}]}{=} \bigcup_{i \in \mathbb{N}}]-i, i[$ so that there exist a $i_j \in \mathbb{N}$ such that $x_j \in]-i_j, i_j[$. Take $k = \max(\{i_j \mid j \in \{1, \dots, n\}\})$ then we have $\forall j \in \{1, \dots, n\}$ that $x_j \in]-i_j, i_j[\subseteq]-k, k[$ so that $x \in]-k, k[^n$. Hence we have $\mathbb{R}^n \subseteq \bigcup_{i \in \mathbb{N}}]-i, i[^n$ which combined with [eq: 21.84] proves that

$$\mathbb{R}^n = \bigcup_{i \in \mathbb{N}}]-i, i[^n = \bigcup_{i \in \mathbb{N}} [-i, i]^n \quad \square$$

We specify now the sufficient and necessary condition for a interval in $\mathbb{R}^n$ to be empty.

Lemma 21.71. Let $n \in \mathbb{N}$ and $a, b \in \mathbb{R}^n$ then we have

1. $[a, b[= \emptyset \Leftrightarrow \exists i \in \{1, \dots, n\}$ such that $a_i \geq b_i$
2. $]a, b] = \emptyset \Leftrightarrow \exists i \in \{1, \dots, n\}$ such that $a_i \geq b_i$
3. $]a, b[= \emptyset \Leftrightarrow \exists i \in \{1, \dots, n\}$ such that $a_i \geq b_i$
4. $[a, b] = \emptyset \Leftrightarrow \exists i \in \{1, \dots, n\}$ such that $a_i > b_i$

Proof.

1. As $[a, b[= \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$ we have
 - $\Rightarrow$. If $[a, b[= \emptyset$ then by [theorem 3.111] $\exists i \in \{1, \dots, n\}$ such that $[a_i, b_i[= \emptyset$. If $a_i < b_i$ then by [theorem: 10.32] there exists a $x \in \mathbb{R}$ such that $a_i < x < b_i \Rightarrow x \in [a_i, b_i[$ contradicting $[a_i, b_i[= \emptyset$, so we must have $a_i \geq b_i$.
 - $\Leftarrow$. If there exists a $i \in \{1, \dots, n\}$ with $a_i \geq b_i$ then if $x \in [a_i, b_i[$ we have $a_i \leq x \wedge x < b_i \leq a_i$ leading to the contradiction $a_i < a_i$ so $[a_i, b_i[= \emptyset$. Hence using [theorem 3.111] $[a, b[= \emptyset$.
2.
 - $\Rightarrow$. If $]a, b] = \emptyset$ then by [theorem 3.111] $\exists i \in \{1, \dots, n\}$ such that $]a_i, b_i] = \emptyset$. If $a_i < b_i$ then by [theorem: 10.32] there exists a $x \in \mathbb{R}$ such that $a_i < x < b_i \Rightarrow x \in]a_i, b_i]$ contradicting $]a_i, b_i] = \emptyset$, so we must have $a_i \geq b_i$.

$\Leftarrow$. If there exists a $i \in \{1, \dots, n\}$ with $a_i \geq b_i$ then if $x \in]a_i, b_i]$ we have $a_i < x \wedge x \leq b_i \leq a_i$ leading to the contradiction $a_i < a_i$ so $]a_i, b_i] = \emptyset$. Hence using [theorem 3.111] $]a, b] = \emptyset$.

3.

$\Rightarrow$. If $]a, b[= \emptyset$ then by [theorem 3.111] $\exists i \in \{1, \dots, n\}$ such that $]a_i, b_i[= \emptyset$. If $a_i < b_i$ then by [theorem: 10.32] there exists a $x \in \mathbb{R}$ such that $a_i < x < b_i \Rightarrow x \in]a_i, b_i[$ contradicting $]a_i, b_i[= \emptyset$, so we must have $a_i \geq b_i$.

$\Leftarrow$. If there exists a $i \in \{1, \dots, n\}$ with $a_i \geq b_i$ then if $x \in]a_i, b_i[$ we have $a_i < x \wedge x < b_i \leq a_i$ leading to the contradiction $a_i < a_i$ so $]a_i, b_i[= \emptyset$. Hence using [theorem 3.111] $]a, b[= \emptyset$.

4.

$\Rightarrow$. If $[a, b] = \emptyset$ then by [theorem 3.111] $\exists i \in \{1, \dots, n\}$ such that $[a_i, b_i] = \emptyset$. If $a_i \leq b_i$ then $a_i \in [a_i, b_i]$ contradicting $[a_i, b_i] = \emptyset$, so we must have $a_i > b_i$.

$\Leftarrow$. If there exists a $i \in \{1, \dots, n\}$ with $a_i > b_i$ then if $x \in [a_i, b_i]$ we have $a_i \leq x \wedge x \leq b_i < a_i$ leading to the contradiction $a_i < a_i$ so $[a_i, b_i] = \emptyset$. Hence using [theorem 3.111] $[a, b] = \emptyset$. $\square$

Next we introduce the classes of intervals in $\mathbb{R}^n$ that can be used to generate the Borel algebra on $\mathbb{R}^n$.

Definition 21.72. Let $n \in \mathbb{N}$ then

$$\mathcal{R}_{\square}^n = \{[a, b] \mid a, b \in \mathbb{R}^n \wedge a \leq b\}$$

$$\mathcal{R}_{\llbracket}^n = \{[a, b[\mid a, b \in \mathbb{R}^n \wedge a < b\}$$

$$\mathcal{R}_{\rrbracket}^n = \{]a, b] \mid a, b \in \mathbb{R}^n \wedge a < b\}$$

$$\mathcal{R}_{\llbracket\rrbracket}^n = \{]a, b[\mid a, b \in \mathbb{R}^n \wedge a < b\}$$

Note 21.73. If $a < b$ then $\forall i \in \{1, \dots, n\}$ $a_i < b_i$ so that by [lemma: 21.71]

$$[a, b[,]a, b],]a, b[\neq \emptyset$$

Likewise if $a \leq b$ then $\forall i \in \{1, \dots, n\}$ $a_i \leq b_i$ so that by [lemma: 21.71]

$$[a, b] \neq \emptyset$$

Hence

$$\forall I \in \mathcal{R}_{\square}^n \text{ we have that } I \neq \emptyset$$

$$\forall I \in \mathcal{R}_{\llbracket}^n \text{ we have that } I \neq \emptyset$$

$$\forall I \in \mathcal{R}_{\rrbracket}^n \text{ we have that } I \neq \emptyset$$

$$\forall I \in \mathcal{R}_{\llbracket\rrbracket}^n \text{ we have that } I \neq \emptyset$$

Theorem 21.74. Let $n \in \mathbb{N}$ then we have

1. $\forall R \in \mathcal{R}_{\square}^n$ we have that R is closed in $\mathcal{T}_{\square}$.
2. $\forall R \in \mathcal{R}_{\llbracket}^n$ we have that R is compact in $\mathcal{T}_{\square}$.
3. $\forall R \in \mathcal{R}_{\rrbracket}^n$ we have that R is open in $\mathcal{T}_{\square}$ [in other words $R \in \mathcal{T}_{\square}$]

Proof.

1. As $R \in \mathcal{R}_{\square}^n$ there exist $a, b \in \mathbb{R}^n$ with $a \leq b$. Let $x \in \mathbb{R}^n \setminus [a, b]$ then $\exists i \in \{1, \dots, n\}$ such that $x_i \notin [a_i, b_i]$. Hence we have either:

$\mathbf{x_i < a_i}$. Take then $\varepsilon = a_i - x_i \in \mathbb{R}^+$ then $\forall y \in B_{\|\cdot\|_n}(x, \varepsilon)$ we have

$$y_i - x_i \leq |y_i - x_i| \leq \max(\{|y_i - x_i| \mid i \in \{1, \dots, n\}\}) = \|x - y\|_n < \varepsilon = a_i - x_i$$

so that $y_i < a_i$ from which it follows that $y_i \notin [a_i, b_i]$. Hence $y \in \mathbb{R}^n \setminus [a, b]$ so that

$$x \in B_{\|\cdot\|_n}(x, \varepsilon) \subseteq \mathbb{R}^n \setminus [a, b]$$

$b_i < x_i$. Take then $\varepsilon = x_i - b_i \in \mathbb{R}^+$ then $\forall y \in B_{\|\cdot\|_n}(x, \varepsilon)$ we have that

$$x_i - y_i \leq |y_i - x_i| \leq \max(\{|y_i - x_i| : i \in \{1, \dots, n\}\}) = \|y_i - x_i\|_n < \varepsilon = x_i - b_i$$

so that $b_i < y_i$ from which it follows that $y_i \notin [a_i, b_i]$. Hence $y \in \mathbb{R}^n \setminus [a, b]$ so that

$$x \in B_{\|\cdot\|_n}(x, \varepsilon) \subseteq \mathbb{R}^n \setminus [a, b]$$

So $\forall x \in \mathbb{R}^n \setminus [a, b]$ we found a $\varepsilon \in \mathbb{R}^+$ such that $x \in B_{\|\cdot\|_n}(x, \varepsilon) \subseteq \mathbb{R}^n \setminus [a, b]$, from this it follows that $\mathbb{R}^n \setminus [a, b]$ is open which proves that

$$R = [a, b] \text{ is a closed set}$$

2. As $R \in \mathcal{R}_{\|\cdot\|_n}^n$ there exists, $a, b \in \mathbb{R}^n$ with $\forall i \in \{1, \dots, n\}$ $a_i \leq b_i$ such that $R = [a, b] = \prod_{i \in \{1, \dots, n\}} [a_i, b_i]$. Using [theorem: 14.323] it follows then that R is compact.
3. Let $R \in \mathcal{R}_{\|\cdot\|_n}^n$ then there exists $a, b \in \mathbb{R}^n$ with $a < b$ such that $R =]a, b[= \prod_{i \in \{1, \dots, n\}}]a_i, b_i[$. As $\forall i \in \{1, \dots, n\}$ $]a_i, b_i[$ is open in $\mathcal{T}_{\|\cdot\|_n}$ [see theorem: 14.105] it follows from [definition: 14.45] and [theorems: 14.51, 14.98] that

$$\mathbb{R} =]a, b[\text{ is a open set in } \mathcal{T}_{\|\cdot\|_n} \quad \square$$

Later we have to define volumes [which are positive] for n -dimensional rectangles $\prod_{i \in \{1, \dots, n\}} [a_i, b_i[$ which will be defined as the product of the sides of this n -dimensional rectangle, for this we need the following lemma.

Lemma 21.75. *Let $n \in \mathbb{N}$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+$ then $\prod_{i \in \{1, \dots, n\}} x_i \in \mathbb{R}^+$*

Proof. We prove this by induction, so let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+ \text{ then } \prod_{i \in \{1, \dots, n\}} x_i \in \mathbb{R}^+ \right\}$$

then we have:

$1 \in S$. If $\{x_i\}_{i \in \{1\}} \subseteq \mathbb{R}^+$ then $\prod_{i \in \{1\}} x_i = x_1 \in \mathbb{R}^+$ which proves that $1 \in S$.

$n \in S \Rightarrow n + 1 \in S$. Let $\{x_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{R}^+$ then as $n \in S$ we have $\prod_{i=1}^n x_i \in \mathbb{R}^+$, as also $x_{n+1} \in \mathbb{R}^+$ it follows from [theorem: 10.14] that

$$0 < \left(\prod_{i \in \{1, \dots, n\}} x_i \right) \cdot x_{n+1} = \prod_{i \in \{1, \dots, n+1\}} x_i$$

so that $n + 1 \in S$ $\square$

The following lemma allows us to define a function that maps a element of $\mathcal{R}_{\|\cdot\|_n}^n$ to its endpoints. Using this we will be able to define the volume of elements of $\mathcal{R}_{\|\cdot\|_n}^n$.

Lemma 21.76. *Let $n \in \mathbb{N}$ and $I \in \mathcal{R}_{\|\cdot\|_n}^n$ then there exists a **unique** $a, b \in \mathbb{R}^n$ with $a < b$ such that $I = [a, b[$*

Proof. If $I \in \mathcal{R}_{\|\cdot\|_n}^n$ then by definition of $\mathcal{R}_{\|\cdot\|_n}^n$ there exist $a, b \in \mathbb{R}^n$ with $a < b$ such that $I = [a, b[$. Assume that there exists also $a', b' \in \mathbb{R}^n$ such that $I = [a', b'[$. Then $\prod_{i \in \{1, \dots, n\}} [a_i, b_i[= \prod_{i \in \{1, \dots, n\}} [a'_i, b'_i[$ and $\forall i \in \{1, \dots, n\}$ $[a_i, b_i[\neq \emptyset$, hence using [theorem: 3.114] we have that

$$\forall i \in \{1, \dots, n\} \text{ that } \emptyset \neq [a_i, b_i[= [a'_i, b'_i[$$

Let $i \in \{1, \dots, n\}$ then as $\emptyset \neq [a_i, b_i[= [a'_i, b'_i[$ it follows from [theorem: 3.145] that $a_i = a'_i$ and $b_i = b'_i$ so that $a = a', b = b'$ proving uniqueness. $\square$

Thanks to the above lemmas the following functions are well defined.

Definition 21.77. Let $n \in \mathbb{N}$ then we define:

1. $\text{begin}: \mathcal{R}_{||}^n \rightarrow \mathbb{R}$ by $\text{begin}(I) = a$ where $I = [a, b[$ and $a < b$
2. $\text{end}: \mathcal{R}_{||}^n \rightarrow \mathbb{R}$ by $\text{end}(I) = b$ where $I = [a, b[$ and $a < b$
3. $v^n: \mathcal{R}_{||}^n \rightarrow]0, \infty[= \mathbb{R}^+$ where $v^n(I) = \prod_{i \in \{1, \dots, n\}} (\text{begin}(I)_i - \text{end}(I)_i)$

Note 21.78. As $\text{begin}(I) = a < b = \text{end}(I)$ we have $\forall i \in \{1, \dots, n\}$

$$\text{begin}(I)_i < \text{end}(I)_i \Rightarrow \text{end}(I)_i - \text{begin}(I)_i \in [0, \infty] \text{ so that by [lemma: 21.75]}$$

$v^n(R) \in \mathbb{R}^+$ as is needed in (3).

We introduce now the set of Dyadic cubes that are a countable subset of $\mathcal{R}_{||}^n$ and will be used later to express open sets as a countable union.

Definition 21.79. (Dyadic Cubes) Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0$ then we define $\mathcal{D}_m^n$ by

$$\mathcal{D}_m^n = \left\{ \left[\frac{z}{2^m}, \frac{z+1}{2^m} \right[\mid z \in \mathbb{Z}^n \right\} \subseteq \mathcal{R}_{||}^n$$

Elements of $\mathcal{D}_m^n$ are called Dyadic cubes of order m . Further $\mathcal{D}^n$ defined by

$$\mathcal{D}^n = \bigcup_{m \in \mathbb{N}_0} \mathcal{D}_m^n \subseteq \mathcal{R}_{||}^n$$

is the set of all Dyadic cubes.

Note 21.80. $\forall i \in \{1, \dots, n\}$ we have that $\text{len}([\frac{z_i}{2^m}, \frac{z_i+1}{2^m} [) = \frac{1}{2^m}$ which proves that all sides of $[\frac{z}{2^m}, \frac{z+1}{2^m} [$ are equal, hence the name Dyadic **cube**.

Note 21.81. As $\frac{z}{2^m} < \frac{z+1}{2^m}$ we have that $\frac{z}{2^m} \in [\frac{z}{2^m}, \frac{z+1}{2^m} [$ so that Dyadic cubes are not empty.

Lemma 21.82. Let $k, l \in \mathbb{N}_0$ with $k \leq l$, $r, q \in \mathbb{Z}^n$ then for $[\frac{r}{2^l}, \frac{r+1}{2^l} [$, $[\frac{q}{2^k}, \frac{q+1}{2^k} [$ we have

$$\begin{aligned} \left[\frac{r}{2^l}, \frac{r+1}{2^l} \right[\cap \left[\frac{q}{2^k}, \frac{q+1}{2^k} \right[&\neq \emptyset \Rightarrow \left[\frac{r}{2^l}, \frac{r+1}{2^l} \right[\subseteq \left[\frac{q}{2^k}, \frac{q+1}{2^k} \right[\\ \left[\frac{r}{2^l}, \frac{r+1}{2^l} \right[\cap \left[\frac{q}{2^k}, \frac{q+1}{2^k} \right[&= \emptyset \Rightarrow \left[\frac{r}{2^l}, \frac{r+1}{2^l} \right[\not\subseteq \left[\frac{q}{2^k}, \frac{q+1}{2^k} \right[\end{aligned}$$

in other words. We have either $[\frac{r}{2^l}, \frac{r+1}{2^l} [\cap [\frac{q}{2^k}, \frac{q+1}{2^k} [= \emptyset$ or $[\frac{r}{2^l}, \frac{r+1}{2^l} [\subseteq [\frac{q}{2^k}, \frac{q+1}{2^k} [$ but not both.

Proof. Take $I = [\frac{q}{2^k}, \frac{q+1}{2^k} [$ and $J = [\frac{r}{2^l}, \frac{r+1}{2^l} [$. As $k \leq l$ we have $0 \leq l - k = n \in \mathbb{N}_0$. Next

$$\begin{aligned} J &= \left[\frac{r}{2^k}, \frac{r+1}{2^k} \right[\\ &= \left[\frac{r}{2^k} \cdot \frac{2^{l-k}}{2^{l-k}}, \frac{r+1}{2^k} \cdot \frac{2^{l-k}}{2^{l-k}} \right[\\ &= \left[\frac{r \cdot 2^{l-k}}{2^l}, \frac{r \cdot 2^{l-k} + 2^{l-k}}{2^l} \right[\\ &= \left[\frac{r \cdot 2^n}{2^l}, \frac{r \cdot 2^n + 2^n}{2^l} \right[\end{aligned} \tag{21.85}$$

Assume that $I \cap J \neq \emptyset$ so that $\exists x \in I \cap J$ with $\forall i \in \{1, \dots, n\}$

$$\frac{q_i}{2^l} \leq x_i < \frac{q_i + 1}{2^l} \wedge \frac{r_i \cdot 2^n}{2^l} \leq x_i < \frac{r_i \cdot 2^n + 2^n}{2^l} \quad (21.86)$$

If now $\frac{q_i + 1}{2^l} \leq \frac{r_i \cdot 2^n}{2^l}$ then $x_i < \frac{q_i + 1}{2^l} \leq \frac{r_i \cdot 2^n}{2^l} \leq x_i$ giving the contradiction $x_i < x_i$ so we must have

$$\frac{r_i \cdot 2^n}{2^l} < \frac{q_i + 1}{2^l} \quad (21.87)$$

Further if $\frac{r_i \cdot 2^n + 2^n}{2^l} \leq \frac{q_i}{2^l}$ then $x_i < \frac{r_i \cdot 2^n + 2^n}{2^l} \leq \frac{q_i}{2^l} \leq x_i$ giving the contradiction $x_i < x_i$ again so we must have that

$$\frac{q_i}{2^l} < \frac{r_i \cdot 2^n + 2^n}{2^l} \quad (21.88)$$

Multiplying 21.87 and 21.88 by 2^l gives $r_i \cdot 2^n < q_i + 1$ and $q_i < r_i \cdot 2^n + 2^n$. As $r_i \cdot 2^n, q_i + 1, r_i \cdot 2^n + 2^n \in \mathbb{Z}$ [as $n \in \mathbb{N}_0$] we have by [theorem: 10.28] that $r_i \cdot 2^n \leq q_i$ and $q_i + 1 \leq r_i \cdot 2^n + 2^n$. Dividing by 2^l gives

$$\forall i \in \{1, \dots, n\} \quad \frac{r_i \cdot 2^n}{2^l} \leq \frac{q_i}{2^l} \wedge \frac{q_i + 1}{2^l} \leq \frac{r_i \cdot 2^n + 2^n}{2^l} \quad (21.89)$$

Next if $x \in I$ then $\forall i \in \{1, \dots, n\} \quad \frac{q_i}{2^l} \leq x_i < \frac{q_i + 1}{2^l} \xRightarrow{21.89} \frac{r_i \cdot 2^n}{2^l} \leq x_i < \frac{r_i \cdot 2^n + 2^n}{2^l}$ so that $x \in \left[\frac{r \cdot 2^n}{2^l}, \frac{r \cdot 2^n + 2^n}{2^l} \right[$ proving using [eq: 21.85] that $x \in J$. Hence we have

$$\text{If } I \cap J \neq \emptyset \text{ then } I \subseteq J$$

Finally if $I \cap J = \emptyset$ then if $I \subseteq J$ we have $\emptyset = I \cap J = I$ a contradiction as $\frac{q}{2^l} \in I$. Hence we have either $I \cap J = \emptyset$ or $I \subseteq J$ but not both. $\square$

Theorem 21.83. *Let $n \in \mathbb{N}$ then we have*

1. $\forall m \in \mathbb{N}_0$ and $\forall R, Q \in \mathcal{D}_m^n$ with $R \neq Q$ we have $R \cap Q = \emptyset$
2. Let $k, l \in \mathbb{N}_0$ with $k \leq l$ then if $R \in \mathcal{D}_k^n$ and $Q \in \mathcal{D}_l^n$ we have that

$$\begin{aligned} R \cap Q &\neq \emptyset \Rightarrow R \subseteq Q \\ R \cap Q &= \emptyset \Rightarrow R \not\subseteq Q \end{aligned}$$

3. $\forall m \in \mathbb{N}_0$ we have that $\mathbb{R}^n = \bigsqcup_{R \in \mathcal{D}_m^n} R$ [here $\bigsqcup_{R \in \mathcal{D}_m^n} R$ is a disjoint union because of (1)].
4. $\forall m \in \mathbb{N}_0$ and $\forall R \in \mathcal{D}_m^n$ we have that $v^n(R) = \frac{1}{2^{n \cdot m}} < 1$

Proof.

1. Let $m \in \mathbb{N}_0$ and $R, Q \in \mathcal{D}_m^n$ with $R \neq Q$. Then there exist $r, z \in \mathbb{Z}^n$ such that

$$R = \left[\frac{r}{2^m}, \frac{r+1}{2^m} \right[= \prod_{i \in \{1, \dots, n\}} \left[\frac{r_i}{2^m}, \frac{r_i+1}{2^m} \right[\text{ and } Q = \left[\frac{z}{2^m}, \frac{z+1}{2^m} \right[= \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^m}, \frac{z_i+1}{2^m} \right[$$

As $R \neq Q$ there exist a $i \in \{1, \dots, n\}$ such that $\left[\frac{r_i}{2^m}, \frac{r_i+1}{2^m} \right[\neq \left[\frac{z_i}{2^m}, \frac{z_i+1}{2^m} \right[$, hence we must have that $r_i \neq z_i$. We can always assume that $r_i < z_i$ [if not exchange R and Q] so that $r_i + 1 \leq z_i$ [as $r_i, z_i \in \mathbb{Z}$ we can use [theorem: 10.28]]. Assume that $x \in \left[\frac{r_i}{2^m}, \frac{r_i+1}{2^m} \right[\cap \left[\frac{z_i}{2^m}, \frac{z_i+1}{2^m} \right[$ then $\frac{r_i}{2^m} \leq x < \frac{r_i+1}{2^m} \leq \frac{z_i}{2^m} \leq x < \frac{z_i+1}{2^m}$ leading to the contradiction $x < x$ hence we must have that $\left[\frac{r_i}{2^m}, \frac{r_i+1}{2^m} \right[\cap \left[\frac{z_i}{2^m}, \frac{z_i+1}{2^m} \right[= \emptyset$. Using [theorem: 3.111] it follows that

$$\prod_{i \in \{1, \dots, n\}} \left[\frac{r_i}{2^m}, \frac{r_i+1}{2^m} \right[\cap \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^m}, \frac{z_i+1}{2^m} \right[= \emptyset$$

or

$$R \cap Q = \emptyset$$

2. Let $k, l \in \mathbb{N}_0$ with $k \leq l$ and $R \in \mathcal{D}_l^n$, $Q \in \mathcal{D}_k^n$ then there exists $r, q \in \mathbb{Z}^n$ with $R = \left[\frac{r}{2^l}, \frac{r+1}{2^l} \right]$, $Q = \left[\frac{q}{2^k}, \frac{q+1}{2^k} \right]$. Using [lemma: 21.82] we have then that

$$\text{If } R \cap Q \neq \emptyset \text{ then } R \subseteq Q$$

$$\text{If } R \cap Q = \emptyset \text{ then } R \not\subseteq Q$$

3. Let $m \in \mathbb{N}$ and $x \in \mathbb{R}^n$ then $\forall i \in \{1, \dots, n\}$ we have $x_i \in \mathbb{R}$. Using [theorem: 10.30] there exist a $z_i \in \mathbb{Z}$ such that $z_i \leq x_i \cdot 2^m < z_i + 1$, hence $\frac{z_i}{2^m} \leq x_i < \frac{z_i + 1}{2^m}$. So if we take $z = (z_1, \dots, z_m)$ then $x \in \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^m}, \frac{z_i + 1}{2^m} \right] = \left[\frac{z}{2^m}, \frac{z+1}{2^m} \right] \in \mathcal{D}_m^n$ proving that $\mathbb{R}^n \subseteq \bigcup_{R \in \mathcal{D}_m^n} R$. As $\forall R \in \mathcal{D}_m^n$ $R \subseteq \mathbb{R}^n$ we have proved that

$$\mathbb{R}^n = \bigcup_{R \in \mathcal{D}_m^n} R$$

Finally by (1) we have that $\forall R, R' \in \mathcal{D}_m^n$ with $R \neq R'$ we have $R \cap R' = \emptyset$ so that

$$\mathbb{R}^n = \bigsqcup_{R \in \mathcal{D}_m^n} R$$

4. If $R \in \mathcal{R}_{||}^n$ then $\exists z \in \mathbb{Z}^n$ such that $R = \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^m}, \frac{z_i + 1}{2^m} \right]$ so that

$$v^n(R) = \prod_{i=1}^n \left(\frac{z_i + 1 - z_i}{2^m} \right) = \prod_{i=1}^n \frac{1}{2^m} = \frac{1}{2^{n \cdot m}} \quad \square$$

One of the reasons that we introduce the Dyadic cubes as a subset of $\mathcal{R}_{||}^n$ is that $\mathcal{D}_m^n$ is a denumerable set in contrast with $\mathcal{R}_{||}^n$ which we need giving the definition of a σ -algebra. This is proved in the next lemma.

Lemma 21.84. *Let $n \in \mathbb{N}$ then*

1. $\forall m \in \mathbb{N}_0$ $\mathcal{D}_m^n$ is denumerable.
2. $\mathcal{D}^n$ is denumerable

Proof.

1. Let $m \in \mathbb{N}_0$ then as $\mathbb{Z}$ is denumerable [see theorem: 10.5] we have by [theorem: 6.67] that $\mathbb{Z}^n$ is also denumerable hence there exist a bijection

$$\alpha: \mathbb{N} \rightarrow \mathbb{Z}^n$$

Define now $\beta: \mathbb{Z}^n \rightarrow \mathcal{D}_m^n$ by $\beta(z) = \left[\frac{z}{2^m}, \frac{z+1}{2^m} \right]$ then β is a bijection.

injectivity. If $\beta(z) = \beta(w)$ then $\left[\frac{z}{2^m}, \frac{z+1}{2^m} \right] = \left[\frac{w}{2^m}, \frac{w+1}{2^m} \right]$. Using the fact that a Dyadic cube is not empty together with [lemma: 21.76] gives $\frac{z}{2^m} = \frac{w}{2^m}$ proving that $z = w$.

surjectivity. This follows from the definition of $\mathcal{D}_m^n$.

So $\beta \circ \alpha: \mathbb{N} \rightarrow \mathcal{D}_m^n$ is a bijection between $\mathbb{N}$ and $\mathcal{D}_m^n$ which proves that $\mathcal{D}_m^n$ is denumerable.

2. Using [theorem: 6.69] it follows that $\mathcal{D}^n = \bigcup_{m \in \mathbb{N}} \mathcal{D}_m^n$ is denumerable. $\square$

We are now finally ready to prove that every non empty open set on $\mathbb{R}^n$ can be written as a denumerable union of pairwise disjoint Dyadic cubes.

Lemma 21.85. *Let $n \in \mathbb{N}$, $\mathcal{T}_{||,n}$ the canonical topology on $\mathbb{R}$ [see example: 14.101] then if $\emptyset \neq U \in \mathcal{T}_{||,n}$ there exist a **pairwise disjoint family** $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}^n$ such that $U = \bigsqcup_{i \in \mathbb{N}} D_i$.*

Proof. Let $\emptyset \neq U \in \mathcal{T}_{||,n}$ be a non empty open set in $\mathbb{R}^n$. Given $m \in \mathbb{N}_0$ define

$$\mathcal{S}_m = \{D \in \mathcal{D}_m^n \mid D \subseteq U\} \subseteq \mathcal{D}_m^n \quad (21.90)$$

and recursively

$$\mathcal{T}_m = \begin{cases} S_0 & \text{if } m = 0 \\ \{D \in \mathcal{S}_m \mid \forall i \in \{0, \dots, m-1\} \text{ we have } \forall R \in \mathcal{T}_i \text{ that } R \cap D = \emptyset\} & \text{if } m \in \mathbb{N} \end{cases} \quad (21.91)$$

Using the above definitions we have the following inclusions

$$\forall m \in \mathbb{N}_0 \text{ we have } \mathcal{T}_m \subseteq \mathcal{S}_m \subseteq \mathcal{D}_m^n \subseteq \mathcal{D}^n \quad (21.92)$$

Further we define

$$\mathcal{T} = \bigcup_{m \in \mathbb{N}_0} \mathcal{T}_m \quad (21.93)$$

If $D \in \mathcal{T}$ then there exist a $i \in \mathbb{N}_0$ such that $D \in \mathcal{T}_i \subseteq \mathcal{S}_i$ proving by [eq: 21.90] that $D \subseteq U$, hence we have

$$\bigcup_{D \in \mathcal{T}} D \subseteq U \quad (21.94)$$

Proving the opposite inclusion is a little more difficult. Let $x \in U$ then there exist a $\varepsilon \in \mathbb{R}^+$ such that

$$x \in B_{\|\cdot\|_n}(x, \varepsilon) \subseteq U$$

Using the Archimedean property of the real numbers [see theorem: 10.30] there exist $n_\varepsilon \in \mathbb{N}$ such that

$$0 < \frac{1}{n_x} < \varepsilon \quad (21.95)$$

Using [theorem: 21.83 (3)] there exist a $D_x \in \mathcal{D}_{n_x}^n$ such that $x \in D_x$. Hence there exist a $z \in \mathbb{Z}^n$ such that

$$D_x = \left[\frac{z}{2^{n_x}}, \frac{z+1}{2^{n_x}} \right] = \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^{n_x}}, \frac{z_i+1}{2^{n_x}} \right] \in \mathcal{D}_{n_x}^n \text{ such that } x \in D_x$$

from which it follows that

$$\forall i \in \{1, \dots, n\} \text{ we have } \frac{z_i}{2^{n_x}} \leq x_i < \frac{z_i+1}{2^{n_x}} \quad (21.96)$$

Let $y \in D_x$ then $y \in \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^{n_x}}, \frac{z_i+1}{2^{n_x}} \right]$ so that

$$\forall i \in \{1, \dots, n\} \text{ we have } \frac{z_i}{2^{n_x}} \leq y_i < \frac{z_i+1}{2^{n_x}} \quad (21.97)$$

Hence we have

$$\begin{aligned} y_i - x_i & \underset{[\text{eq: 21.97}]}{<} \frac{z_i+1}{2^{n_x}} - x_i \\ & \underset{[\text{eq: 21.96}]}{\leq} \frac{z_i+1}{2^{n_x}} - \frac{z_i}{2^{n_x}} \\ & = \frac{1}{2^{n_x}} \\ & \underset{[\text{eq: 21.95}]}{<} \varepsilon \\ x_i - y_i & \underset{[\text{eq: 21.96}]}{<} \frac{z_i+1}{2^{n_x}} - y_i \\ & \underset{[\text{eq: 21.97}]}{\leq} \frac{z_i+1}{2^{n_x}} - \frac{z_i}{2^{n_x}} \\ & \underset{[\text{eq: 21.95}]}{<} \varepsilon \end{aligned}$$

which proves that $\forall i \in \{1, \dots, n\} \mid y_i - x_i < \varepsilon$. Hence $\|y - x\|_n = \max(\{|y_i - x_i| \mid i \in \{1, \dots, n\}\}) < \varepsilon$ so that $y \in B_{\|\cdot\|_n}(x, \varepsilon)$. In conclusion

$$x \in D_x \subseteq B_{\|\cdot\|_n}(x, \varepsilon) \subseteq U$$

As $D_x \in \mathcal{D}_{n_x}^n$ it follows from the above and [eq: 21.90] that

$$x \in D_x \in \mathcal{S}_{n_x} \quad (21.98)$$

Define now

$$\mathcal{K}_x = \{i \in \{0, \dots, n_x\} \mid \exists D \in \mathcal{S}_i \mid x \in D\} \subseteq \{0, \dots, n_x\} \quad (21.99)$$

Then by [eq: 21.98] $n_x \in \mathcal{K}_x$ so that $\mathcal{K}_x \neq \emptyset$, as further $\mathcal{K}_x$ is finite it follows that

$$m_x = \min(\mathcal{K}_x) \text{ exist} \quad (21.100)$$

For $m_x \in \mathcal{K}_x \subseteq \{0, \dots, n_x\}$ we have now two possible cases:

$m_x = 0$. Then $\exists D \in \mathcal{S}_0$ such that $x \in D$. As by [eqs: 21.91, 21.93] $\mathcal{S}_0 = \mathcal{T}_0$ and $\mathcal{T}_0 \subseteq \mathcal{T}$ we that $x \in D \in \mathcal{T}$. In other words $x \in \bigcup_{D \in \mathcal{T}} D$.

$m_x \in \{1, \dots, n_x\}$. Then $\exists D \in \mathcal{S}_{m_x} \subseteq \mathcal{D}_{m_x}^n$ such that $x \in D$ and $\forall i \in \{0, \dots, m_x - 1\}$ we have that $\forall R \in \mathcal{S}_i$ that $x \notin R$. Let $i \in \{0, \dots, m_x - 1\}$ and assume that $\exists R \in \mathcal{S}_i \subseteq \mathcal{D}_i^n$ such that $R \cap D \neq \emptyset$ then by [theorem: 21.83 (2)] and the fact that $i \leq m_x$ it follows that $D \subseteq R$, as $x \in D$ it follows that $x \in R$ contradicting $x \notin R$ hence the assumption is wrong and we must have that $\forall R \in \mathcal{S}_i$ we have that $R \cap D = \emptyset$. As this is true $\forall i \in \{0, \dots, m_x - 1\}$ it follows from [definition: 21.91] that $D \in \mathcal{T}_{m_x} \subseteq \mathcal{T}$. In other words we have $x \in D \in \mathcal{T}$ which proves that $x \in \bigcup_{D \in \mathcal{T}} D$.

So in all cases we have that $x \in \bigcup_{D \in \mathcal{T}} D$ which as $x \in U$ was chosen arbitrary proves that $U \subseteq \bigcup_{D \in \mathcal{T}} D$. Combining this with [eq: 21.93] proves that

$$U = \bigcup_{D \in \mathcal{T}} D \quad (21.101)$$

Next we must prove that $\bigcup_{D \in \mathcal{T}} D$ is a pairwise disjoint union. Let $D, E \in \mathcal{T} = \bigcup_{m \in \mathbb{N}_0} \mathcal{T}_m$ with $D \neq E$ then there exist $k, l \in \mathbb{N}_0$ such that $D \in \mathcal{T}_k \subseteq \mathcal{D}_k^n$ and $E \in \mathcal{T}_l \subseteq \mathcal{D}_l^n$. For k, l we have the following possible cases:

$k = l$. Then using [theorem: 21.83 (1)] it follows that $D \cap E = \emptyset$.

$k < l$. Then using [eq: 21.91] that $D \cap E = \emptyset$.

$l < k$. Then using [eq: 21.91] that $D \cap E = \emptyset$.

combining the above with [eq: 21.101] gives then

$$\forall D, E \in \mathcal{T} \text{ with } D \neq E \text{ we have } D \cap E = \emptyset \quad (21.102)$$

Next we prove that $\mathcal{T}$ is denumerable, first we prove that $\mathcal{T}$ is not finite (hence by definition infinite). Assume that $\mathcal{T}$ is finite. As $U \neq \emptyset$ we must have that $\mathcal{T} \neq \emptyset$ [because $U = \bigcup_{D \in \emptyset} D = \emptyset$] so that $m = \min(\{(\text{begin}(D))_1 \mid D \in \mathcal{T}\})$ exist. Hence there exist a $R = [a, b[\in \mathcal{T}$ such that $a_1 = m$. As $a \in [a, b[= R$ and by [eq: 21.101] $R \subseteq U$ it follows that $a \in U$. Hence there exist a $\varepsilon \in \mathbb{R}^+$ such that $a \in B_{\|\cdot\|_n}(a, \varepsilon) \subseteq U$. Define e by $(e)_i = \begin{cases} 1 & i=1 \\ 0 & \text{if } i \in \{2, \dots, n\} \end{cases}$ then for $i \in \{1, \dots, n\}$ we have

$$\left| \left(a - \left(a - \frac{\varepsilon}{2} \cdot e \right) \right)_i \right| = \left| \frac{\varepsilon}{2} \cdot e_i \right| = \left| \frac{\varepsilon}{2} \right| \cdot \left| \begin{cases} 1 & \text{if } i=1 \\ 0 & \text{if } i \in \{2, \dots, n\} \end{cases} \right| \leq \frac{\varepsilon}{2} < \varepsilon$$

so that $\|a - (a - \frac{\varepsilon}{2} \cdot e)\|_n < \varepsilon$. Hence $a - \frac{\varepsilon}{2} \cdot e \in B_{\|\cdot\|_n}(a, \varepsilon) \subseteq U$ then by [eq: 21.101] there exist a $D' \in \mathcal{T}$ such that $a - \frac{\varepsilon}{2} \cdot e \in D'$. As $D' = [\text{begin}(D'), \text{end}(D')[$ it follows that $\text{begin}(D') \leq a - \frac{\varepsilon}{2} \cdot e$ so that $(\text{begin}(D'))_1 \leq (a - \frac{\varepsilon}{2} \cdot e)_1 = a_1 - \frac{\varepsilon}{2} \cdot 1 = m - \frac{\varepsilon}{2}$. As $m = \min(\{(\text{begin}(D))_1 \mid D \in \mathcal{T}\})$ $m \leq (\text{begin}(D'))_1$ which leads to the contradiction $m \leq m - \frac{\varepsilon}{2} < m$, so the assumption is wrong and we must have that

$$\mathcal{T} \text{ is not finite or } \mathcal{T} \text{ is infinite} \quad (21.103)$$

As $\mathcal{D}^n$ is denumerable [see theorem: 21.84] and $\mathcal{T} \subseteq \mathcal{D}^n$ it follows from the above and [theorem: 6.60] that

$$\mathcal{T} \text{ is denumerable}$$

So there exist a bijection $\beta: \mathbb{N} \rightarrow \mathcal{T}$. Define then $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathcal{T} \subseteq \mathcal{D}^n$ by $D_i = \beta(i)$ then we have

$$\bigcup_{i \in \mathbb{N}} D_i \stackrel{[\text{theorem: 2.127}]}{=} \bigcup_{D \in \mathcal{T}} D = U$$

Further if $i, j \in \mathbb{N}$ with $i \neq j$ then as β is a bijection, hence injective, it follows that $D_i \neq D_j$, so that using [eq: 21.102] we have $D_i \cap D_j = \emptyset$. Hence

$$U = \bigsqcup_{i \in \mathbb{N}} D_i \quad \square$$

Lemma 21.86. *Let $n \in \mathbb{N}$ and $R \in \mathcal{R}_{\parallel}^n$ with $R \neq \emptyset$ then there exist a sequence $\{U_i\}_{i \in \mathbb{N}} \subseteq \mathcal{T}_{\parallel n}$ of open sets such that $R = \bigcap_{i \in \mathbb{N}} U_i$*

Proof. As $R \in \mathcal{R}_{\parallel}^n$ there exists a $a, b \in \mathbb{R}^n$ with $a \leq b$ such that $R = [a, b[= \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$. Define then $\{U_i\}_{i \in \mathbb{N}}$ by $U_i = \prod_{j \in \{1, \dots, n\}}]a_j - \frac{1}{i}, b_j[$. As $]a_j - \frac{1}{i}, b_j[\in \mathcal{T}_{\parallel}$ [see theorem: 14.104] and $\mathcal{T}_{\parallel n}$ is the product topology based on $\mathcal{T}_{\parallel}$ [see 14.101] it follows that $\{U_i\}_{i \in \mathbb{N}} \subseteq \mathcal{T}_{\parallel n}$. Let $i \in \mathbb{N}$ and $x \in R$ then we have $\forall j \in \{1, \dots, n\}$ that $x_j \in [a_j, b_j[\Rightarrow a_j \leq x_j < b_j \Rightarrow a_j - \frac{1}{i} < a_j \leq x_j < b_j \Rightarrow x_j \in]a_j - \frac{1}{i}, b_j[$, proving using [theorem: 3.112] that $R \subseteq U_i$. Hence we have

$$R \subseteq \bigcap_{i \in \mathbb{N}} U_i \quad (21.104)$$

For the opposite inequality let $x \in \bigcap_{i \in \mathbb{N}} U_i$ then $\forall i \in \mathbb{N}$ we have $x \in U_i$ so that $\forall j \in \{1, \dots, n\}$ we have $x_j \in]a_j - \frac{1}{i}, b_j[\Rightarrow a_j - \frac{1}{i} < x_j < b_j$. Assume that $\exists j \in \{1, \dots, n\}$ such that $x_j < a_j$ then using [theorem: 10.30] there exists a $i \in \mathbb{N}$ such that $x_j < x_j + \frac{1}{i} < a_j \Rightarrow x_j < a_j - \frac{1}{i}$ contradicting the fact that $a_j - \frac{1}{i} < x_j$. Hence $\forall j \in \{1, \dots, n\}$ we have $a_j \leq x_j < b_j$ proving that $x \in R$ so that $\bigcap_{i \in \mathbb{N}} U_i \subseteq R$. This together with [eq: 21.104] proves

$$R = \bigcap_{i \in \mathbb{N}} U_i \quad \square$$

We are ready now to prove a alternative definition for the Borel algebra on $\mathbb{R}^n$.

Theorem 21.87. *Let $n \in \mathbb{N}$ then we have*

$$\sigma[\{C \subseteq \mathbb{R}^n \mid C \text{ is close in } \mathcal{T}_{\parallel n}\}] = \mathcal{B}[\mathbb{R}^n] = \sigma[\mathcal{R}_{\parallel}^n]$$

Proof. Using [theorem: 21.47] we have

$$\sigma[\{C \subseteq \mathbb{R}^n \mid C \text{ is close in } \mathcal{T}_{\parallel n}\}] \stackrel{[\text{theorem: 21.47}]}{=} \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{\parallel n}] \stackrel{\text{def}}{=} \mathcal{B}[\mathbb{R}^n]$$

so we only have to prove that $\mathcal{B}[\mathbb{R}^n, \mathcal{T}_{\parallel n}] = \sigma[\mathcal{R}_{\parallel}^n]$. Let $U \in \mathcal{T}_{\parallel n}$ then we have either:

$U = \emptyset$. Then $U = \emptyset \in \sigma[\mathcal{R}_{\parallel}^n]$.

$U \neq \emptyset$. Then using [lemma: 21.85] there exist a sequence $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}^n \subseteq \mathcal{R}_{\parallel}^n$ such that $U = \bigcup_{i \in \mathbb{N}} D_i$ which as $\sigma[\mathcal{R}_{\parallel}^n]$ is a σ -algebra proves that $U \in \sigma[\mathcal{R}_{\parallel}^n]$

Hence we have $\mathcal{T}_{\parallel n} \subseteq \sigma[\mathcal{R}_{\parallel}^n]$ which by [theorem: 21.42] gives

$$\sigma[\mathcal{T}_{\parallel n}] \subseteq \sigma[\mathcal{R}_{\parallel}^n] \quad (21.105)$$

For the opposite inclusion let $R \in \mathcal{R}_{\parallel}^n$ then by [lemma: 21.86] there exist a $\{U_i\}_{i \in \mathbb{N}} \subseteq \mathcal{T}_{\parallel n}$ such that $R = \bigcap_{i \in \mathbb{N}} U_i$. As $\sigma[\mathcal{T}_{\parallel n}]$ is a σ -algebra we have by [theorem: 21.36] that $R \in \sigma[\mathcal{T}_{\parallel n}]$. Hence $\mathcal{R}_{\parallel}^n \subseteq \sigma[\mathcal{T}_{\parallel n}]$ which by [theorem: 21.42] gives $\sigma[\mathcal{R}_{\parallel}^n] \subseteq \sigma[\mathcal{T}_{\parallel n}]$. Combining this with [eq: 21.105] gives

$$\sigma[\mathcal{R}_{\parallel}^n] = \sigma[\mathcal{T}_{\parallel n}] \stackrel{\text{def}}{=} \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{\parallel n}] = \mathcal{B}[\mathbb{R}^n] \quad \square$$

Definition 21.88. Let $n \in \mathbb{N}$ then for $i \in \{1, \dots, n\}$ and $a \in \mathbb{R}$ we define $] -\infty, a[_i \subseteq \mathbb{R}^n$ by

$$]-\infty, a[_i = \{x \in \mathbb{R}^n | x_i < a\}$$

The set $] -\infty, a[_i$ is called a **lower half open space**. The set of all lower half spaces is noted as $\mathcal{H}^n$ hence

$$\mathcal{H}^n = \{]-\infty, a[_i | i \in \{1, \dots, n\} \wedge a \in \mathbb{R}\}$$

It turns out that every lower half space is a open set.

Lemma 21.89. Let $n \in \mathbb{N}$ then $\mathcal{H}^n \subseteq \mathcal{T}_{|||n}$ [the canonical topology on $\mathbb{R}^n$]

Proof. Let $H \in \mathcal{H}^n$ then there exists a $i \in \{1, \dots, n\}$ and a $a \in \mathbb{R}$ such that $H = \{x \in \mathbb{R}^n | x_i < a\}$. Take $x \in H$ then $x_i < a$ so that $\varepsilon = a - x_i > 0$. Let $y \in B_{|||n}(x, \varepsilon)$ then

$$y_i - x_i \leq |y_i - x_i| \leq \max(\{|y_i - x_i| | i \in \{1, \dots, n\}\}) < \varepsilon = a - x_i$$

proving that $y_i < a$ so that $y \in H$. Hence $x \in B_{|||n}(x, \varepsilon) \subseteq H$ which proves that H is open. $\square$

Next we show that every half open interval can be written as a finite intersection of differences of lower half open spaces.

Lemma 21.90. Let $n \in \mathbb{N}$, $a, b \in \mathbb{R}^n$ then $[a, b[= \bigcap_{i \in \{1, \dots, n\}} (]-\infty, b_i[_i \setminus]-\infty, a_i[_i)$

Proof. Let $x \in [a, b[$ then $\forall i \in \{1, \dots, n\}$ we have $a_i \leq x_i \wedge x_i < b_i$ or $\neg(x_i < a_i) \wedge x_i < b_i$ so that $x \notin]-\infty, a_i[_i \wedge x \in]-\infty, b_i[_i \Rightarrow x \in]-\infty, b_i[_i \setminus]-\infty, a_i[_i$. Hence we have

$$[a, b[\subseteq \bigcap_{i \in \{1, \dots, n\}} (]-\infty, b_i[_i \setminus]-\infty, a_i[_i) \quad (21.106)$$

For the opposite inclusion, let $x \in \bigcap_{i \in \{1, \dots, n\}} (]-\infty, b_i[_i \setminus]-\infty, a_i[_i)$ then $\forall i \in \{1, \dots, n\}$ we have $x \in]-\infty, b_i[_i \setminus]-\infty, a_i[_i$ hence $x_i < b_i \wedge \neg(x_i < a_i) \Rightarrow x_i < b_i \wedge a_i \leq x_i$ proving that $x \in [a, b[$. So $\bigcap_{i \in \{1, \dots, n\}} (]-\infty, b_i[_i \setminus]-\infty, a_i[_i) \subseteq [a, b[$ which together with [eq: 21.106] proves that

$$[a, b[= \bigcap_{i \in \{1, \dots, n\}} (]-\infty, b_i[_i \setminus]-\infty, a_i[_i) \quad \square$$

Theorem 21.91. Let $n \in \mathbb{N}$ then $\mathcal{B}[\mathbb{R}^n] = \sigma[\mathcal{H}^n]$

Proof. If $H \in \mathcal{H}^n$ then H is open [see lemma: 21.89], so that $H \in \sigma[\mathcal{T}_{\mathbb{R}^n}] \stackrel{[\text{theorem: 21.194}]}{=} \mathcal{B}[\mathbb{R}^n]$, which proves that $\mathcal{H}^n \subseteq \mathcal{B}[\mathbb{R}^n]$, so using [theorem: 21.42]

$$\sigma[\mathcal{H}^n] \subseteq \mathcal{B}[\mathbb{R}^n] \quad (21.107)$$

Let $R \in \mathcal{R}_{||}^n$ then $\exists a, b \in \mathbb{R}^n$ such that $R = [a, b[$, using the previous lemma [see lemma: 21.90] we have then that $R = [a, b[= \bigcap_{i \in \{1, \dots, n\}} (]-\infty, b_i[_i \setminus]-\infty, a_i[_i) \stackrel{[\text{theorem: 21.36, 21.32}]}{\in} \sigma[\mathcal{H}^n]$. So $\sigma[\mathcal{R}_{||}^n] \subseteq \sigma[\mathcal{H}^n]$ which as $\sigma[\mathcal{R}] \stackrel{[\text{theorem: 21.194}]}{=} \mathcal{B}[\mathbb{R}^n]$ proves that $\mathcal{B}[\mathbb{R}^n] \subseteq \sigma[\mathcal{H}^n]$. Combining this with 21.107 gives

$$\sigma[\mathcal{H}^n] = \mathcal{B}[\mathbb{R}^n] \quad \square$$

Theorem 21.92. Let $n \in \mathbb{N}$ then $\mathcal{R}_{||}^n \cup \mathcal{R}_{||}^n \cup \mathcal{R}_{||}^n \cup \mathcal{R}_{||}^n \subseteq \mathcal{B}[\mathbb{R}^n]$

Proof. If $R \in \mathcal{R}_{||}^n \cup \mathcal{R}_{||}^n \cup \mathcal{R}_{||}^n \cup \mathcal{R}_{||}^n$ then we have either:

$R \in \mathcal{R}_{||}^n$. By [theorem: 21.74] R is a closed set so that by [theorem: 21.47] $R \in \mathcal{B}[\mathbb{R}^n]$.

$R \in \mathcal{R}_{||}^n$. By [theorem: 21.74] R is a open set so that $R \in \mathcal{B}[\mathbb{R}^n]$.

$R \in \mathcal{R}_{||}^n$. By [theorem: 21.87] it follows that $R \in \mathcal{B}[\mathbb{R}^n]$.

$R \in \mathcal{R}_{\mathbb{R}^n}^n$. Then there exists $a, b \in \mathbb{R}^n$ with $a < b$ such that $R =]a, b]$. As is already proved we have that $[a, b] \in \mathcal{B}[\mathbb{R}^n]$ and $]a, b + 1[\in \mathcal{B}[\mathbb{R}^n]$, hence we have that

$$[a, b] \cap]a, b + 1[\in \mathcal{B}[\mathbb{R}^n]$$

Now

$$\begin{aligned} x \in [a, b] \cap]a, b + 1[&\Leftrightarrow \forall i \in \{1, \dots, n\} x_i \in [a_i, b_i] \wedge x_i \in]a_i, b_i + 1[\\ &\Leftrightarrow \forall i \in \{1, \dots, n\} a_i \leq x_i \wedge x_i \leq b_i \wedge a_i < x_i \wedge x_i < b_i + 1 \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} a_i < x_i \wedge x_i \leq b_i \\ &\Leftrightarrow x \in]a, b] \end{aligned}$$

so that

$$R =]a, b] = [a, b] \cap]a, b + 1[\in \mathcal{B}[\mathbb{R}^n]$$

□

21.3 Measures

21.3.1 Measure definition and properties

We turn our attention now on associating with every set in a measure algebra a non negative number that represent the ‘size’ or ‘volume’ of this set. This leads to the study of set functions that satisfies the additive properties expected of a ‘volume’. Thanks to the extra properties of a σ -algebra we have the following equivalently statements about additivity defined in [definitions: 21.15, 21.16, 21.17 and 21.19]

Theorem 21.93. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space [see definition: 21.35] and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a set function then we have the following :*

1. μ is **finite additive** $\Leftrightarrow$ for every **pairwise disjoint** families $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ where $n \in \mathbb{N}$ we have

$$\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

2. μ is **finite sub-additive** $\Leftrightarrow$ for every $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $n \in \mathbb{N}$ we have that

$$\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) \leq \sum_{i=1}^n \mu(A_i)$$

3. μ is **countable additive** $\Leftrightarrow$ for all **pairwise disjoint** $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have that

$$\mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

4. μ is **countable sub-additive** if $\Leftrightarrow$ for all $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

Proof.

1. This follows from the fact that $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ we have, as $\mathcal{A}$ is a σ -algebra, that $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$ and the definition of additivity [see definition: 21.15].
2. This follows from the fact that $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ we have, as $\mathcal{A}$ is a σ -algebra, that $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{A}$ and the definition of sub-additivity [see definition: 21.16].

3. This follows from the fact that $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have, as $\mathcal{A}$ is a σ -algebra, that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ and the definition of countable additivity [see definition: 21.17].
4. This follows from the fact that $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have, as $\mathcal{A}$ is a σ -algebra, that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$ and the definition of countable sub-additivity [see definition: 21.19]. $\square$

Using mathematical induction it is easy to specify a necessary and sufficient condition for finite additivity and finite sub additivity.

Lemma 21.94. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a set function then we have*

1. μ is finite additive $\Leftrightarrow \forall A, B \in \mathcal{A}$ with $A \cap B = \emptyset$ we have $\mu(A \sqcup B) = \mu(A) + \mu(B)$
2. μ is finite sub additive $\Leftrightarrow \forall A, B \in \mathcal{A}$ we have $\mu(A \cup B) \leq \mu(A) + \mu(B)$

Proof.

1.

$\Rightarrow$. Let $A, B \in \mathcal{A}$ with $A \cap B = \emptyset$ then $\{A_i\}_{i \in \{1, \dots, 2\}}$ defined by $A_i = \begin{cases} A & \text{if } i = 1 \\ B & \text{if } i = 2 \end{cases}$ is pairwise disjoint and $\{A_i\}_{i \in \{1, \dots, 2\}} \subseteq \mathcal{A}$. So by assumption we have that

$$\mu(A \sqcup B) = \mu\left(\bigsqcup_{i \in \{1, \dots, 2\}} A_i\right) = \sum_{i=1}^2 \mu(A_i) = \mu(A_1) + \mu(A_2) = \mu(A) + \mu(B)$$

$\Leftarrow$. Let

$$\mathcal{S} = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ is a pairwise disjoint family then we have that } \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum \mu(A_i) \right\}$$

then we have:

$1 \in \mathcal{S}$. Let $\{A_i\}_{i \in \{1, \dots, 1\}} \subseteq \mathcal{A}$ then $\mu(\bigsqcup_{i \in \{1, \dots, 1\}} A_i) = \mu(A_1) = \sum_{i=1}^1 \mu(A_i)$ proving that $1 \in \mathcal{S}$

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A}$ be pairwise disjoint. Take $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$ and $B = A_{n+1}$, then $A \cup B = \bigcup_{i \in \{1, \dots, n+1\}} A_i$ and $A \cap B = (\bigsqcup_{i \in \{1, \dots, n\}} A_i) \cap A_{n+1} = \bigcup_{i \in \{1, \dots, n\}} (A_i \cap A_{n+1}) = \emptyset$. Hence we have

$$\begin{aligned} \mu\left(\bigcup_{i \in \{1, \dots, n+1\}} A_i\right) &= \mu(A \sqcup B) \\ &= \mu(A) + \mu(B) \\ &= \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) + \mu(A_{n+1}) \\ &\stackrel{n \in \mathcal{S}}{=} \left(\sum_{i=1}^n \mu(A_i)\right) + \mu(A_{n+1}) \\ &= \sum_{i=1}^{n+1} \mu(A_i) \end{aligned}$$

proving that $n+1 \in \mathcal{S}$.

Using mathematical induction we have that $\mathbb{N} = \mathcal{S}$. So $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint we have $\mu(\bigcup_{i \in \{1, \dots, n+1\}} A_i) = \sum_{i=1}^n \mu(A_i)$

2.

$\Rightarrow$. Let $A, B \in \mathcal{A}$ then $\{A_i\}_{i \in \{1, \dots, 2\}}$ defined by $A_i = \begin{cases} A & \text{if } i=1 \\ B & \text{if } i=2 \end{cases}$ satisfies $\{A_i\}_{i \in \{1, \dots, 2\}} \subseteq \mathcal{A}$. So by assumption we have that

$$\mu(A \cup B) = \mu\left(\bigcup_{i \in \{1, \dots, 2\}} A_i\right) \leq \sum_{i=1}^2 \mu(A_i) = \mu(A_1) + \mu(A_2) = \mu(A) + \mu(B)$$

$\Leftarrow$. Let

$$\mathcal{S} = \left\{ n \in \mathbb{N}_0 \mid \forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ we have } \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) \leq \sum_{i=1}^n \mu(A_i) \right\}$$

then:

$1 \in \mathcal{S}$. Let $\{A_i\}_{i \in \{1, \dots, 1\}} \subseteq \mathcal{A}$ then $\mu\left(\bigcup_{i \in \{1, \dots, 1\}} A_i\right) = \mu(A_1) = \sum_{i=1}^1 \mu(A_i)$ proving that $1 \in \mathcal{S}$

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A}$ pairwise disjoint. Take $A = \bigcup_{i \in \{1, \dots, n\}} A_i$ and $B = A_{n+1}$, then $A \cup B = \bigcup_{i \in \{1, \dots, n+1\}} A_i$. Hence we have

$$\begin{aligned} \mu\left(\bigcup_{i \in \{1, \dots, n+1\}} A_i\right) &= \mu(A \cup B) \\ &\leq \mu(A) + \mu(B) \\ &= \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) + \mu(A_{n+1}) \\ &\leq_{n \in \mathcal{S}} \left(\sum_{i=1}^n \mu(A_i)\right) + \mu(A_{n+1}) \\ &= \sum_{i=1}^{n+1} \mu(A_i) \end{aligned}$$

proving that $n+1 \in \mathcal{S}$.

Using mathematical induction we have that $\mathbb{N} = \mathcal{S}$. So $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ we have $\mu\left(\bigcup_{i \in \{1, \dots, n+1\}} A_i\right) \leq \sum_{i=1}^n \mu(A_i)$ $\square$

Theorem 21.95. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then if $\mu: \mathcal{A} \rightarrow [0, \infty]$ satisfies:

1. $\mu(\emptyset) = 0$
2. μ is countable additive

then μ is additive.

Proof. This was actually proved in [theorem: 21.95]. $\square$

We are ready now to define a measure on a σ -algebra which is a countable set function that associate with every set in the σ -algebra a non negative number which represents the ‘size’ or ‘volume’ of this set.

Definition 21.96. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\mu: \mathcal{A} \rightarrow [0, \infty]$ is a **measure on $\mathcal{A}$** if

1. $\mu(\emptyset) = 0$
2. For every **pairwise disjoint** $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

[in other word μ is countable additive].

A trivial example of a measure is the following.

Example 21.97. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\mu: \mathcal{A} \rightarrow [0, \infty]$ defined by $\mu(A) = 0$ is a measure on $\mathcal{A}$.

Proof. We have

1. $\mu(\emptyset) = 0$
2. If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ then $\sum_{i=1}^{\infty} \mu(A_i) = \sum_{i=1}^{\infty} 0 \stackrel{[\text{example: 20.97}]}{=} 0 = \mu(\bigcup_{i \in \mathbb{N}} A_i)$ □

Definition 21.98. (Measure Space) A measure space is a triple $\langle X, \mathcal{A}, \mu \rangle$ where X is a non empty set, $\mathcal{A}$ is a σ -algebra on X and $\mu: \mathcal{A} \rightarrow [0, \infty]$ is a measure on $\mathcal{A}$.

Theorem 21.99. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then

1. $\forall A, B \in \mathcal{A}$ with $A \subseteq B$ we have $\mu(A) \leq \mu(B)$
2. $\forall A, B \in \mathcal{A}$ with $A \subseteq B$ and $\mu(A) < \infty$ we have $\mu(B \setminus A) = \mu(B) - \mu(A)$

Proof.

1. If $A \subseteq B$ then $A \cap (B \setminus A) = \emptyset$ and $B = A \cup (B \setminus A)$ so that by [theorem: 21.95] we have

$$\mu(B) = \mu(A) + \mu(B \setminus A)$$

Further as $0 \leq \mu(B \setminus A)$ we have by [theorem: 20.35] that $\mu(A) \leq \mu(B)$.

2. If $A \subseteq B$ and $\mu(A) < \infty$ then as $0 \leq \mu(A)$ we have that $\mu(A) \in \mathbb{R}$, further by (1)

$$\mu(B) = \mu(A) + \mu(B \setminus A)$$

so that by [theorem: 20.28] $\mu(B \setminus A) = \mu(B) - \mu(A)$. □

As sequences of pairwise disjoint sets in a σ -algebra plays a important role in the definition of a measure the following theorem will be useful.

Theorem 21.100. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ then there exist a **pairwise disjoint family** $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that

$$\forall i \in \mathbb{N} \text{ we have } B_i \subseteq A_i$$

and

$$\bigcup_{i \in \mathbb{N}} A_i = \bigsqcup_{i \in \mathbb{N}} B_i$$

Proof. Using [lemma: 21.6] we have for $\{B_i\}_{i \in \mathbb{N}}$ defined by

$$B_i = \begin{cases} A_1 & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j & \text{if } i \in \{2, \dots, \infty\} \end{cases}$$

that

$$\forall i \in \mathbb{N} \text{ we have } B_i \subseteq A_i$$

$$\bigcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \mathbb{N}} A_i$$

and

$$\forall i, j \in \mathbb{N} \text{ with } i \neq j \text{ then } B_i \cap B_j = \emptyset$$

Let $i \in \mathbb{N}$ then we have either:

$i = 1$. Then $B_i = B_1 = A_1 \in \mathcal{A}$

$1 < i$. Then by [theorems: 21.36, 21.31] $B_i = A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j \in \mathcal{A}$

so that

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \quad \square$$

Theorem 21.101. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then for every $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

Proof. Using [theorem: 21.100] there exists a **pairwise disjoint** family $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that

$$\forall i \in \mathbb{N} \ B_i \subseteq A_i \text{ and } \bigsqcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \mathbb{N}} A_i \quad (21.108)$$

Using [theorem: 21.99] on the above we have

$$\forall i \in \mathbb{N} \text{ we have } \mu(B_i) \leq \mu(A_i) \quad (21.109)$$

So

$$\begin{aligned} \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) & \stackrel{[\text{eq: 21.108}]}{=} \mu\left(\bigsqcup_{i \in \mathbb{N}} B_i\right) \\ & = \sum_{i=1}^{\infty} \mu(B_i) \\ & \stackrel{[\text{eq: 21.109}] \wedge [\text{theorem: 20.99}]}{\leq} \sum_{i=1}^{\infty} \mu(A_i) \end{aligned}$$

□

Corollary 21.102. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $n \in \mathbb{N}$ then for every $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ we have

$$\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) \leq \sum_{i=1}^n \mu(A_i)$$

Proof. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ \emptyset & \text{if } i \in \mathbb{N} \setminus \{1, \dots, n\} \end{cases}$$

then we have

$$\begin{aligned} \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) & = \mu\left(\left(\bigcup_{i \in \{1, \dots, n\}} B_i\right) \cup \emptyset\right) \\ & = \mu\left(\left(\bigcup_{i \in \{1, \dots, n\}} B_i\right) \cup \left(\bigcup_{i \in \mathbb{N} \setminus \{1, \dots, n\}} B_i\right)\right) \\ & \stackrel{[\text{theorem: 2.136}]}{=} \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\ & \stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{i=1}^{\infty} \mu(B_i) \\ & \stackrel{[\text{theorem: 20.100}]}{=} \sum_{i=1}^n \mu(B_i) + \sum_{i=n+1}^{\infty} \mu(B_i) \\ & = \sum_{i=1}^n \mu(A_i) \end{aligned}$$

□

Once we have defined a measure space we can define σ -finite sets and σ -finite measures.

Definition 21.103. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $A \subseteq X$ is **σ -finite under μ** if there exist a $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ \mu(A_i) < \infty$ such that $A = \bigcup_{i \in \mathbb{N}} A_i$.

We can find two equivalent definitions for σ -finite sets.

Theorem 21.104. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $A \subseteq X$ then we have:*

A is σ -finite under $\mu \Leftrightarrow \exists \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ pairwise disjoint with $\forall i \in \mathbb{N} \mu(A_i) < \infty$ such that $A = \bigsqcup_{i \in \mathbb{N}} A_i$

Proof.

$\Rightarrow$. As A is σ -finite under μ there exist a $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that $A = \bigcup_{i \in \mathbb{N}} B_i$ with $\forall i \in \mathbb{N} \mu(A_i) < \infty$. Define now

$$\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } A_i = \begin{cases} B_i & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} B_j & \text{if } i \in \mathbb{N} \setminus \{1\} \end{cases} \quad B_i \in \mathcal{A} \text{ and } \forall i \in \mathbb{N} A_i \subseteq B_i$$

then by [lemma: 21.6] we have that

$$\{A_i\}_{i \in \mathbb{N}} \text{ is pairwise disjoint and } \bigsqcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i = A$$

Further, as $\forall i \in \mathbb{N}$ we have $A_i \subseteq B_i$, it follows from [theorem: 21.99] that $\mu(A_i) \leq \mu(B_i) < \infty$.

$\Leftarrow$. This is trivial. $\square$

Theorem 21.105. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $A \subseteq X$ then we have:*

$$\begin{aligned} & A \text{ is } \sigma\text{-finite under } \mu \\ & \Updownarrow \\ & \exists \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ with } \forall i \in \mathbb{N} \mu(A_i) \leq \mu(A_{i+1}) \wedge \mu(A_i) < \infty \text{ such that } A = \bigcup_{i \in \mathbb{N}} A_i \end{aligned}$$

Proof.

$\Rightarrow$. As A is σ -finite there exist a $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that $A = \bigcup_{i \in \mathbb{N}} B_i$. Define now

$$\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } A_i = \bigcup_{j \in \{1, \dots, i\}} B_j \underset{[\text{theorem: 21.39}]}{\in} \mathcal{A}$$

Using [lemma: 21.7] it follows that

$$\forall i \in \mathbb{N} A_i \subseteq A_{i+1} \text{ and } \bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i = A$$

Finally we have also

$$\forall i \in \mathbb{N} \text{ that } \mu(A_i) = \mu\left(\bigcup_{j \in \{1, \dots, i\}} B_j\right) \underset{[\text{corollary: 21.102}]}{\leq} \sum_{j=1}^i \mu(B_j) < \infty$$

$\Leftarrow$. This is trivial $\square$

Definition 21.106. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then a measure $\mu: \mathcal{A} \rightarrow [0, \infty]$ is a **σ -finite measure** if X is σ -finite. In other words μ is σ -finite if there exists a $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \mu(A_i) < \infty$ such that $X = \bigcup_{i \in \mathbb{N}} A_i$.*

Note 21.107. *IF μ is a **finite** measure then we have for $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ defined by $A_i = X$ that $\forall i \in \mathbb{N} \mu(A_i) = \mu(X) < \infty$ and $X = \bigcup_{i \in \mathbb{N}} A_i$ so that μ is σ -finite. In other words every finite measure is automatically σ -finite.*

We look now at new ways of defining new measures based on existing measures.

Theorem 21.108. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then we have:*

1. *If $\mu_1: \mathcal{A} \rightarrow [0, \infty]$ and $\mu_2: \mathcal{A} \rightarrow [0, \infty]$ are measures on $\mathcal{A}$ then*

$$\mu_1 + \mu_2: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } (\mu_1 + \mu_2)(A) = \mu_1(A) + \mu_2(A)$$

is a measure on $\mathcal{A}$.

2. If $\mu: \mathcal{A} \rightarrow [0, \infty]$ is a measure on $\mathcal{A}$ and $\alpha \in [0, \infty]$ then

$$\alpha \cdot \mu: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$$

is a measure on $\mathcal{A}$.

3. If $\mu: \mathcal{A} \rightarrow [0, \infty]$ is a measure and $B \in \mathcal{A}$ then

$$\mu_{[B]}: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu_{[B]}(A) = \mu(A \cap B)$$

is a measure on $\mathcal{A}$.

Note 21.109. As $A, B \in \mathcal{A}$ we have $A \cap B \in \mathcal{A}$ so that μ_B is well defined.

Proof.

1. First we have

$$(\mu_1 + \mu_2)(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0 + 0 = 0$$

Secondly if $\{A_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{A}$ is pairwise disjoint then

$$\begin{aligned} (\mu_1 + \mu_2)\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) &= \mu_1\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) + \mu_2\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) \\ &= \sum_{i=1}^{\infty} \mu_1(A_i) + \sum_{i=1}^{\infty} \mu_2(A_i) \\ &\stackrel{[\text{theorem: 20.99}]}{=} \sum_{i=1}^{\infty} (\mu_1(A_i) + \mu_2(A_i)) \\ &= \sum_{i=1}^{\infty} (\mu_1 + \mu_2)(A_i) \end{aligned}$$

proving that $\mu_1 + \mu_2$ is a measure.

2. First we have

$$(\alpha \cdot \mu)(\emptyset) = \alpha \cdot \mu(\emptyset) = \alpha \cdot 0 = 0$$

Secondly if $\{A_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{A}$ is pairwise disjoint then

$$\begin{aligned} (\alpha \cdot \mu)\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) &= \alpha \cdot \mu\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) \\ &= \alpha \cdot \sum_{i=1}^{\infty} \mu(A_i) \\ &\stackrel{[\text{theorem: 20.99}]}{=} \sum_{i=1}^{\infty} (\alpha \cdot \mu(A_i)) \\ &= \sum_{i=1}^{\infty} (\alpha \cdot \mu)(A_i) \end{aligned}$$

proving that $\alpha \cdot \mu$ is a measure.

3. First we have

$$\mu_{[B]}(\emptyset) = \mu(\emptyset \cap B) = \mu(\emptyset) = 0$$

Secondly if $\{A_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{A}$ is pairwise disjoint then

$$\mu_{[B]}\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \mu\left(\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) \cap B\right) = \mu\left(\bigsqcup_{i \in \mathbb{N}} (A_i \cap B)\right) = \sum_{i=1}^{\infty} \mu(A_i \cap B) = \sum_{i=1}^{\infty} \mu_{[B]}(A_i)$$

□

Theorem 21.110. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and let $\mathcal{B}$ a σ -algebra on X such that $\mathcal{B} \subseteq \mathcal{A}$ then $\langle X, \mathcal{B}, \mu|_{\mathcal{B}} \rangle$ is a measure space where $\mu|_{\mathcal{B}}$ is the restriction of μ to $\mathcal{B}$ defined by

$$\mu|_{\mathcal{B}}: \mathcal{B} \rightarrow [0, \infty] \text{ where } \mu|_{\mathcal{B}}(B) = \mu(B)$$

Proof. As $\mathcal{B}$ is already a σ -algebra on X we only have to proof that $\mu|_{\mathcal{B}}$ is a measure on $\mathcal{B}$.

1. As $\emptyset \in \mathcal{B} \subseteq \mathcal{A}$ we have $\mu|_{\mathcal{B}}(\emptyset) = \mu(\emptyset) = 0$
2. If $\{A_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{B} \subseteq \mathcal{A}$ is a pairwise disjoint family then

$$\mu|_{\mathcal{B}}\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) = \mu\left(\bigsqcup_{i \in \mathbb{N}_0} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) = \sum_{i=1}^{\infty} \mu|_{\mathcal{B}}(A_i) \quad \square$$

Theorem 21.111. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and define

$$\mathcal{A}|_A = \{B \in \mathcal{A} | B \subseteq A\} \subseteq \mathcal{A}$$

$$\mu|_{\mathcal{A}|_A}: \mathcal{A}|_A \rightarrow [0, \infty] \text{ by } \mu|_{\mathcal{A}|_A}(B) = \mu(B)$$

then we have that

1. $\mathcal{A}|_A = \{A \cap B | B \in \mathcal{A}\}$ [the trace of $\mathcal{A}$ in B]
2. $\mathcal{A}|_A$ is a σ -algebra on A
3. $\mu|_{\mathcal{A}|_A}$ is a measure on $\mathcal{A}|_A$ making $\langle A, \mathcal{A}|_A, \mu|_{\mathcal{A}|_A} \rangle$ a measure space

Measure spaces of the form $\langle A, \mathcal{A}|_A, \mu|_{\mathcal{A}|_A} \rangle$ are called **measure sub-spaces of** $\langle X, \mathcal{A}, \mu \rangle$

Note 21.112. If $A = X$ then $\mathcal{A}|_X = \{X \cap B | B \in \mathcal{A}\} = \mathcal{A}$ so that $\mu|_{\mathcal{A}|_X} = \mu$, in other words we have that $\langle X, \mathcal{A}|_X, \mu|_{\mathcal{A}|_X} \rangle = \langle X, \mathcal{A}, \mu \rangle$.

Proof. Using [theorem: 21.44] it follows that

$$\mathcal{A}|_A = \{B \cap A | B \in \mathcal{A}\} \text{ and } \mathcal{A}|_A \text{ is a } \sigma\text{-algebra}$$

As additional $\mathcal{A}|_B \subseteq \mathcal{A}$ it follows from the previous theorem [theorem: 21.110] that $\mu|_{\mathcal{A}|_A}$ is a measure on $\mathcal{A}|_A$. $\square$

Definition 21.113. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then μ is a **finite measure** if $\mu(X) < \infty$

We have the following equivalent definition of a finite measure.

Theorem 21.114. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space

$$\mu \text{ is a finite measure} \Leftrightarrow \forall A \in \mathcal{A} \text{ we have } \mu(A) < \infty$$

Proof.

- $$\Rightarrow. \text{ If } A \in \mathcal{A} \text{ then } A \subseteq X \text{ so that by [theorem: 21.99] } \mu(A) \leq \mu(X) < \infty.$$
- $$\Leftarrow. \text{ If } \forall A \in \mathcal{A} \mu(A) < \infty \text{ we have as } X \in \mathcal{A} \text{ that } \mu(X) < \infty. \quad \square$$

Theorem 21.115. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then

1. If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ satisfies $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}$ then

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) = \sup(\{\mu(A_i) | i \in \mathbb{N}\})$$

2. If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ satisfies $\forall i \in \mathbb{N} A_{i+1} \subseteq A_i$ and $\exists k \in \mathbb{N}$ such that $\mu(A_k) < \infty$ then

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) = \inf(\{\mu(A_i) | i \in \mathbb{N}\})$$

Proof.

1. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_1 & \text{if } i = 1 \\ A_i \setminus A_{i-1} & \text{if } i \in \{2, \dots, \infty\} \end{cases} \in \mathcal{A}$$

Let $i, j \in \mathbb{N}$ with $i \neq j$ then we may always assume that $i < j$ [otherwise exchange i and j]. Assume that $x \in B_i \cap B_j$ then we have for i either:

$i = 1$. Then $1 = i < j \Rightarrow i \leq j - 1 \wedge j \in \{2, \dots, \infty\}$ so that $x \in B_i = B_1 = A_1 \subseteq A_{j-1}$ and $x \in B_j = A_j \setminus A_{j-1}$ we have $x \in A_{j-1}$ and $x \notin A_{j-1}$ a contradiction..

$1 < i$. Then $1 < i < j \Rightarrow i \leq j - 1 \wedge j \in \{2, \dots, \infty\}$ so that $x \in A_i \subseteq A_{j-1}$ and $x \in B_j = A_j \setminus A_{j-1}$ so that $x \in A_{j-1}$ and $x \notin A_{j-1}$ a contradiction.

so the assumption must be wrong proving that $B_i \cap B_j = \emptyset$. Hence

$$\forall i, j \in \mathbb{N} \text{ we have } B_i \cap B_j = \emptyset \quad (21.110)$$

Next if $n \in \mathbb{N}$ then $\forall i \in \{1, \dots, n\}$ we have that $B_i = \begin{cases} A_1 \subseteq A_i & \text{if } i = 1 \\ A_i \setminus A_{i-1} \subseteq A_i & \text{if } i \in \{2, \dots, \infty\} \end{cases} \subseteq A_i \subseteq A_n$ so that

$$\bigcup_{i \in \{1, \dots, n\}} B_i \subseteq A_n \quad (21.111)$$

Further if $x \in A_n$ then $n \in \{i \in \{1, \dots, n\} | x \in A_i\} \subseteq \{1, \dots, n\}$ so that $m = \min(\{i \in \{1, \dots, n\} | x \in A_i\})$ exists. We have now two cases to consider for m :

$m = 1$. then $x \in A_1 = B_1 \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$

$1 < m$. then $x \notin A_{m-1}$ and $x \in A_m$ so that $x \in A_m \setminus A_{m-1} = B_m \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$

proving that $A_n \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$ which by [eq: 21.111] proves that $A_n = \bigcup_{i \in \{1, \dots, n\}} B_i$. Hence

$$\forall n \in \mathbb{N} \text{ we have } A_n = \bigsqcup_{i \in \{1, \dots, n\}} B_i \quad (21.112)$$

From the above it follows that $\forall n \in \mathbb{N} \ A_n = \bigsqcup_{i \in \{1, \dots, n\}} B_i \subseteq \bigsqcup_{i \in \mathbb{N}} B_i$ proving that $\bigcup_{n \in \mathbb{N}} A_n \subseteq \bigcup_{n \in \mathbb{N}} B_n$, further $\forall i \in \mathbb{N}$ we have $B_i = \begin{cases} A_1 \subseteq A_i & \text{if } i = 1 \\ A_i \setminus A_{i-1} \subseteq A_i & \text{if } i \in \{2, \dots, \infty\} \end{cases} \subseteq A_i \subseteq \bigcup_{i \in \mathbb{N}} A_i$ so that $\bigcup_{i \in \mathbb{N}} B_i \subseteq \bigcup_{i \in \mathbb{N}} A_i$. Hence we have

$$\bigcup_{i \in \mathbb{N}} A_i = \bigsqcup_{i \in \mathbb{N}} B_i \quad (21.113)$$

Next we have

$$\begin{aligned} \mu\left(\bigcup_{i \in \mathbb{N}_0} A_i\right) &= \mu\left(\bigsqcup_{i \in \mathbb{N}_0} B_i\right) \\ &= \sum_{i=1}^{\infty} \mu(B_i) \\ &\stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \mu(B_i) \right) \\ &\stackrel{[\text{theorem: 21.95}]}{=} \lim_{n \rightarrow \infty} \left(\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} B_i\right) \right) \\ &\stackrel{[\text{eq: 21.112}]}{=} \lim_{n \rightarrow \infty} \mu(A_n) \end{aligned}$$

So that

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

As $\forall i \in \mathbb{N}$ we have $A_i \subseteq A_{i+1} \xRightarrow{[\text{theorem: 21.99}]} \mu(A_i) \leq \mu(A_{i+1})$ so that by [theorem: 20.92]

$$\lim_{n \rightarrow \infty} \mu(A_n) = \sup \{ \mu(A_n) | n \in \mathbb{N} \}$$

2. First we consider the cases where $k=1$. Then $\forall i \in \mathbb{N}$ we have $A_i \subseteq A_1$ so that by [theorem: 21.99] $\mu(A_i) \leq \mu(A_1) < \infty$ proving that

$$\forall i \in \mathbb{N} \text{ we have } \mu(A_i) < \infty \quad (21.114)$$

Define $\{B_i\}_{i \in \mathbb{N}}$ by $B_i = A_1 \setminus A_i$. Let $i \in \mathbb{N}$ then for $x \in B_i$ we have $x \in A_1 \wedge x \notin A_i \supseteq A_{i+1}$ so that $x \in A_1 \setminus A_{i+1} = B_{i+1}$. Hence we have

$$\forall i \in \mathbb{N} \ B_i \subseteq B_{i+1} \quad (21.115)$$

So using (1) it follows that

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \lim_{i \rightarrow \infty} \mu(B_i) \quad (21.116)$$

Now

$$\begin{aligned} A_1 \setminus \left(\bigcup_{i \in \mathbb{N}} B_i\right) &\stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{i \in \mathbb{N}} (A_1 \setminus B_i) \\ &= \bigcap_{i \in \mathbb{N}} (A_1 \setminus (A_1 \setminus A_i)) \\ &\stackrel{A_i \subseteq A_1 \wedge [\text{theorem: 2.140}]}{=} \bigcap_{i \in \mathbb{N}} A_i \end{aligned} \quad (21.117)$$

As $\forall i \in \mathbb{N} \ B_i = A_1 \setminus A_i \subseteq A_1$ it follows that

$$\mu(B_i) \stackrel{[\text{theorem: 21.99}]}{\leq} \mu(A_1) < \infty \quad (21.118)$$

and $\bigcup_{i \in \mathbb{N}} B_i \subseteq A_1$ so that

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \stackrel{[\text{theorem: 21.99}]}{\leq} \mu(A_1) < \infty \quad (21.119)$$

Next

$$\begin{aligned} \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eq: 21.117}]}{=} \mu\left(A_1 \setminus \left(\bigcup_{i \in \mathbb{N}} B_i\right)\right) \\ &\stackrel{[\text{eq: 21.119}] \wedge [\text{theorem: 21.99}]}{=} \mu(A_1) - \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\ &\stackrel{[\text{eq: 21.116}]}{=} \mu(A_1) - \lim_{i \rightarrow \infty} \mu(B_i) \\ &= \mu(A_1) - \lim_{i \rightarrow \infty} \mu(A_1 \setminus A_i) \\ &\stackrel{[\text{eq: 21.114}] \wedge [\text{theorem: 21.99}]}{=} \mu(A_1) - \lim_{i \rightarrow \infty} (\mu(A_1) - \mu(A_i)) \\ &\stackrel{[\text{theorems: 20.85, 20.74}]}{=} \mu(A_1) - \left(\mu(A_1) - \lim_{i \rightarrow \infty} \mu(A_i)\right) \\ &= \lim_{i \rightarrow \infty} \mu(A_i) \end{aligned}$$

hence we have that

$$\forall \{A_i\}_{i \in \mathbb{N}} \text{ with } \mu(A_1) < \infty \text{ and } \forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i \text{ we have } \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) \quad (21.120)$$

Next we proof the general case where $1 < k$. Define $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ by $C_i = A_{i+(k-1)}$ then we have as $\mu(A_k) < \infty$ that $\mu(C_1) = \mu(A_{1+(k-1)}) = \mu(A_1) < \infty$. Further $\forall i \in \mathbb{N}$ we have that $C_{i+1} = A_{(i+(k-1))+1} \subseteq A_{i+(k-1)} = C_i$. So all conditions of [eq: 21.120] are satisfied and we have that

$$\mu\left(\bigcap_{i \in \mathbb{N}} C_i\right) = \lim_{i \rightarrow \infty} \mu(C_i)$$

As $\lim_{i \rightarrow \infty} \mu(C_i) = \lim_{i \rightarrow \infty} \mu(A_{i+(k-1)}) \stackrel{[\text{theorem: 20.84}]}{=} \lim_{i \rightarrow \infty} \mu(A_i)$ so that we have

$$\mu\left(\bigcap_{i \in \mathbb{N}} C_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) \quad (21.121)$$

Let $x \in \bigcap_{i \in \mathbb{N}} A_i$. Then $\forall i \in \mathbb{N}$ we have, as $i + (k-1) \in \mathbb{N}$ [because $1 < k$], that $x \in A_{i+(k-1)} = C_i$ hence

$$\bigcap_{i \in \mathbb{N}} A_i \subseteq \bigcap_{i \in \mathbb{N}} C_i \quad (21.122)$$

Let $x \in \bigcap_{i \in \mathbb{N}} C_i$. For $i \in \mathbb{N}$ we have either:

$i \in \{1, \dots, k-1\}$. Then $x \in C_1 = A_{1+(k-1)} = A_k \subseteq_{i \leq k-1 < k} A_i$ so that $x \in A_i$.

$i \in \{k, \dots, \infty\}$. Then $i - (k-1) \in \mathbb{N}$ so that $x \in C_{i-(k-1)} = A_{(i-(k-1))+(k-1)} = A_i$ so that $x \in A_i$

as $i \in \mathbb{N}$ was chosen arbitrary it follows that $x \in \bigcap_{i \in \mathbb{N}} A_i$, hence $\bigcap_{i \in \mathbb{N}} C_i \subseteq \bigcap_{i \in \mathbb{N}} A_i$ which combined with [eq: 21.122] proves that

$$\bigcap_{i \in \mathbb{N}} A_i = \bigcap_{i \in \mathbb{N}} C_i \quad (21.123)$$

So combining [eq: 21.121] and [eq: 21.123] it follows that

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

which proves the case $1 < k$. □

The above theorem has the following inverse which provides ways to prove if a set function is actually a measure.

Theorem 21.116. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a set function that satisfies:*

1. $\mu(\emptyset) = 0$.
2. If $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ is a **pairwise disjoint family** then

$$\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

then we have:

1. If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ that satisfies $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and $\mu(\bigcup_{i \in \mathbb{N}} A_i) = \lim_{i \rightarrow \infty} \mu(A_i)$ then μ is a measure on $\mathcal{A}$.
2. If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ that satisfies $\forall i \in \mathbb{N} A_{i+1} \subseteq A_i$ and $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and $\mu(\bigcap_{i \in \mathbb{N}} A_i) = 0$ then μ is a measure on $\mathcal{A}$.

Proof.

1. Let $\{A_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{A}$ be a pairwise disjoint family. Define

$$\{B_i\}_{i \in \mathbb{N}} \text{ by } B_i = \bigsqcup_{j \in \{1, \dots, i\}} A_j \in \mathcal{A}$$

then by [lemma: 21.7] we have that

$$\bigsqcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i \text{ and } \forall i \in \mathbb{N} \text{ we have } B_i \subseteq B_{i+1} \quad (21.124)$$

Then by the hypothesis

$$\lim_{i \rightarrow \infty} B_i \text{ exist and } \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \lim_{i \rightarrow \infty} \mu(B_i) \quad (21.125)$$

Next

$$\begin{aligned}
 \mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eq: 21.124}]}{=} \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\
 &\stackrel{[\text{eq: 21.125}]}{=} \lim_{i \rightarrow \infty} \mu(B_i) \\
 &= \lim_{i \rightarrow \infty} \mu\left(\bigsqcup_{j \in \{1, \dots, i\}} A_j\right) \\
 &\stackrel{\text{hypothesis}}{=} \lim_{i \rightarrow \infty} \left(\sum_{j=1}^i \mu(A_j)\right) \\
 &\stackrel{\text{def}}{=} \sum_{j=1}^{\infty} \mu(A_j)
 \end{aligned}$$

proving countable additivity. Hence μ is a measure on $\mathcal{A}$.

2. Let $\{A_i\}_{i \in \mathbb{N}_o} \subseteq \mathcal{A}$ be a pairwise disjoint family. Define

$$\{B_i\}_{i \in \mathbb{N}} \text{ by } B_i = \bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j \stackrel{[\text{theorem: 21.39}]}{\in} \mathcal{A}$$

Let $i \in \mathbb{N}$ then $B_{i+1} = \bigcup_{j \in \{i+2, \dots, \infty\}} A_j \subseteq \bigcup_{j \in \{i+1, \dots, \infty\}} A_j = B_i$ so that

$$\forall i \in \mathbb{N} \text{ we have } B_{i+1} \subseteq B_i$$

Let $x \in \bigcap_{i \in \mathbb{N}} B_i$ then $x \in B_1 = \bigcup_{j \in \{2, \dots, \infty\}} A_j$ so that $\exists k \in \{2, \dots, \infty\}$ such that $x \in A_k$, as $x \in \bigcap_{i \in \mathbb{N}} B_i$ we must have that $x \in B_k = \bigcup_{j \in \{k+1, \dots, \infty\}} A_j$ so that $\exists l \in \{k+1, \dots, \infty\}$ such that $x \in A_l$. Hence it follows that $x \in A_k \cap A_l \stackrel{k \neq l}{=} \emptyset$ leading to the contradiction $x \in \emptyset$. Hence we have that

$$\bigcap_{i \in \mathbb{N}} B_i = \emptyset$$

Let $i \in \mathbb{N}$ then $\bigsqcup_{j \in \mathbb{N}} A_j = (\bigsqcup_{j \in \{1, \dots, i\}} A_j) \cup (\bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j)$. Assume $x \in (\bigsqcup_{j \in \{1, \dots, i\}} A_j) \cap (\bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j)$ then there exist a $k \in \{1, \dots, i\}$ and a $l \in \{i+1, \dots, \infty\}$ such that $x \in A_k \wedge x \in A_l \Rightarrow x \in A_k \cap A_l \stackrel{k \neq l}{=} \emptyset$ leading to the contradiction $x \in \emptyset$. Hence we must have that $(\bigsqcup_{j \in \{1, \dots, i\}} A_j) \cap (\bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j) = \emptyset$. By the hypothesis we have then

$$\begin{aligned}
 \mu\left(\bigcup_{j \in \mathbb{N}} A_j\right) &= \mu\left(\left(\bigsqcup_{j \in \{1, \dots, i\}} A_j\right) \sqcup \left(\bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j\right)\right) \\
 &\stackrel{\mu \text{ is finite additive}}{=} \mu\left(\bigsqcup_{j \in \{1, \dots, i\}} A_j\right) + \mu\left(\bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j\right) \\
 &\stackrel{\mu \text{ is finite additive}}{=} \sum_{j=1}^i \mu(A_j) + \mu\left(\bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j\right) \\
 &= \sum_{j=1}^i \mu(A_j) + \mu(B_i)
 \end{aligned} \tag{21.126}$$

By [definition: 20.95]

$$\lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(A_j) \text{ exist and } \sum_{i=1}^{\infty} \mu(A_i) = \lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(A_j) \tag{21.127}$$

and by the hypothesis

$$\lim_{i \rightarrow \infty} \mu(B_i) \text{ exist and } \lim_{i \rightarrow \infty} \mu(B_i) = 0 \tag{21.128}$$

Next we have

$$\begin{aligned}
\mu\left(\bigcup_{j \in \mathbb{N}} A_j\right) & \stackrel{[\text{theorem: 20.74}]}{=} \lim_{i \rightarrow \infty} \left(\mu\left(\bigcup_{j \in \mathbb{N}} A_j\right) \right) \\
& \stackrel{[\text{eq: 21.126}]}{=} \lim_{i \rightarrow \infty} \left(\sum_{j=1}^i \mu(A_j) + \mu(B_i) \right) \\
& \stackrel{[\text{theorem: 20.85}] \wedge [\text{eqs: 21.127, 21.128}]}{=} \lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(A_j) + \lim_{i \rightarrow \infty} \mu(B_i) \\
& \stackrel{[\text{eqs: 21.127, 21.128}]}{=} \sum_{i=1}^{\infty} \mu(A_i) + 0 \\
& = \sum_{i=1}^{\infty} \mu(A_i)
\end{aligned}$$

So μ is countable additive proving that μ is a measure. $\square$

21.3.2 Carathéodory Theorem

The Carathéodory theorem provides a general method for constructing a measure. Given a set X and a non negative set function on $\mathcal{P}(X)$ the Carathéodory theorem allows us to reduce $\mathcal{P}(X)$ to a σ -algebra on X such that the restriction of the set function to this σ -algebra is a measure on this σ -algebra. The idea is that $\mathcal{P}(X)$ is in general too big to be used as a σ -algebra on which we can define a measure. As the restriction of the set function is a measure it must satisfy some extra conditions. For example every σ -algebra contains $\emptyset$ and a measure must map $\emptyset$ to 0 so the set function on $\mathcal{P}(X)$ must also map $\emptyset$ to 0. This is the idea of an outer-measure.

Definition 21.117. (Outer Measure) Let X be a set then $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ is a **outer measure** on X if

1. $\mu^*(\emptyset) = 0$
2. For $A, B \in \mathcal{P}(X)$ with $A \subseteq B$ we have $\mu^*(A) \leq \mu^*(B)$ [in other words μ^* is monotone]
3. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ we have $\mu^*(\bigcup_{i \in \mathbb{N}} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ [in other words μ^* is countable sub-additive]

It is easy to prove that an outer measure is finitely sub-additive.

Theorem 21.118. Let X be a set and $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ is an outer measure on X , $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \in \mathcal{P}(X)$ then $\mu^*(\bigcup_{i \in \{1, \dots, n\}} A_i) \leq \sum_{i=1}^n \mu^*(A_i)$

Proof. Define $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ by $C_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ \emptyset & \text{if } i \in \{n+1, \dots, \infty\} \end{cases}$ then we have

$$\begin{aligned}
\mu^*\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) & = \mu^*\left(\bigcup_{i \in \{1, \dots, n\}} C_i\right) \\
& \stackrel{\mu^* \text{ is an outer measure}}{\leq} \sum_{i=1}^{\infty} \mu^*(C_i) \\
& \stackrel{[\text{theorem: 20.100}]}{=} \sum_{i=1}^n \mu^*(C_i) + \sum_{i=n+1}^{\infty} \mu^*(C_i) \\
& = \sum_{i=1}^n \mu^*(A_i) + \sum_{i=n+1}^{\infty} 0 \\
& \stackrel{[\text{theorem: 20.97}]}{=} \sum_{i=1}^n \mu^*(A_i)
\end{aligned}$$

$\square$

We need now to reduce $\mathcal{P}(X)$ to a σ -algebra in such a way that the restriction of the outer-measure to this σ -algebra becomes a measure. This is the idea of μ^* -measurable sets.

Definition 21.119. (μ^* -measurable sets) Let X be a set and $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ a outer measure on X then $A \in \mathcal{P}(X)$ is μ^* -measurable if

$$\forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A)$$

The set of all μ^* -measurable sets in $\mathcal{P}(X)$ is noted as $\mathcal{M}[\mu^*]$, hence

$$\mathcal{M}[\mu^*] = \{A \in \mathcal{P}(X) | \forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A)\}$$

Using the properties of a outer-measure we can simplify the above definition.

Theorem 21.120. Let X be a set and $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ and $A \in \mathcal{P}(X)$ then we have

$$A \text{ is } \mu^*\text{-measurable} \Leftrightarrow \forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) \geq \mu^*(B \cap A) + \mu^*(B \setminus A)$$

Hence we have the following alternative definition for $\mathcal{M}[\mu^*]$

$$\mathcal{M}[\mu^*] = \{A \in \mathcal{P}(X) | \forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) \geq \mu^*(B \cap A) + \mu^*(B \setminus A)\}$$

Proof.

$\Rightarrow$. This is trivial as $\mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A)$ implies $\mu^*(B) \geq \mu^*(B \cap A) + \mu^*(B \setminus A)$.

$\Leftarrow$. Let $B \in \mathcal{P}(X)$ then as $B = (B \cap A) \cup (B \setminus A)$ we have by [theorem: 21.118] that $\mu^*(B) \leq \mu^*(B \cap A) + \mu^*(B \setminus A)$, combining this with the hypothesis it follows that

$$\mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A) \quad \square$$

Theorem 21.121. Let X be a set, $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ a outer measure and $A \in \mathcal{P}(X)$ then

1. If $\mu^*(A) = 0$ then $A \in \mathcal{M}[\mu^*]$.
2. If $\mu^*(X \setminus A) = 0$ then $A \in \mathcal{M}[\mu^*]$.

Proof.

1. Let $B \in \mathcal{P}(X)$ then as $B \cap A \subseteq A$ we have $\mu^*(B \cap A) \leq \mu^*(A) = 0$ so that $\mu^*(B \cap A) = 0$, further as $B \setminus A \subseteq B$ we have $\mu^*(B \setminus A) \leq \mu^*(B)$. Hence it follows that $\forall B \in \mathcal{P}(X)$ we have

$$\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \mu^*(B)$$

proving by the previous theorem [theorem: 21.120] that $A \in \mathcal{M}[\mu^*]$.

2. Let $B \in \mathcal{P}(X)$ then as $B \setminus A \subseteq X \setminus A$ we have $\mu^*(B \setminus A) \leq \mu^*(X \setminus A) = 0$ so that $\mu^*(B \setminus A) = 0$, further as $B \cap A \subseteq B$ we have $\mu^*(B \cap A) \leq \mu^*(B)$. Hence it follows that

$$\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \mu^*(B)$$

proving by the previous theorem [theorem: 21.120] that $A \in \mathcal{M}[\mu^*]$. $\square$

We prove now that $\mathcal{M}[\mu^*]$ is a σ -algebra and the restriction of μ^* on $\mathcal{M}[\mu^*]$ is a measure on $\mathcal{M}[\mu^*]$.

Theorem 21.122. (Carathéodory Theorem) Let X be a set and $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ a outer measure on X then we have:

1. $\mathcal{M}[\mu^*]$ is a σ -algebra on X .
2. $(\mu^*)|_{\mathcal{M}[\mu^*]}: \mathcal{M}[\mu^*] \rightarrow [0, \infty]$ is a measure on $\mathcal{M}[\mu^*]$.

Proof.

1. Let $B \in \mathcal{P}(X)$ then

$$\mu^*(B) = \mu^*(B \cap X) + 0 = \mu^*(B \cap X) + \mu^*(\emptyset) = \mu^*(B \cap X) + \mu^*(B \setminus X)$$

proving that

$$X \in \mathcal{M}[\mu^*] \quad (21.129)$$

If $A \in \mathcal{M}[\mu^*]$ then for $B \in \mathcal{P}(X)$ we have

$$\begin{aligned} \mu^*(B \cap (X \setminus A)) + \mu^*(B \setminus (X \setminus A)) &\stackrel{B \subseteq X \wedge [\text{theorem: 1.36 (6)}]}{=} \mu^*(B \cap (X \setminus A)) + \mu^*(B \cap A) \\ &\stackrel{B \subseteq X \wedge [\text{theorem: 1.36 (8)}]}{=} \mu^*(B \setminus A) + \mu^*(B \cap A) \\ &\stackrel{A \in \overline{\mathcal{M}}[\mu^*]}{=} \mu^*(B) \end{aligned}$$

so that $X \setminus A \in \mathcal{M}[\mu^*]$. Hence we have

$$\forall A \in \mathcal{M}[\mu^*] \text{ we have } X \setminus A \in \mathcal{M}[\mu^*] \quad (21.130)$$

Let $A, B \in \mathcal{M}[\mu^*]$ then for $C \in \mathcal{P}(X)$ we have

$$\begin{aligned} \mu^*(C \cap (A \cup B)) &\stackrel{A \in \overline{\mathcal{M}}[\mu^*]}{=} \mu^*((C \cap (A \cup B)) \cap A) + \mu^*((C \cap (A \cup B)) \setminus A) \\ &= \mu^*(C \cap ((A \cup B) \cap A)) + \mu^*(C \cap (A \cup B) \setminus A) \\ &= \mu^*(C \cap A) + \mu^*(C \cap (A \cup B) \setminus A) \\ &\stackrel{[\text{theorem: 1.30 (7)}]}{=} \mu^*(C \cap A) + \mu^*(C \cap ((A \cup B) \setminus A)) \\ &\stackrel{[\text{theorem: 1.30 (6)}]}{=} \mu^*(C \cap A) + \mu^*(C \cap ((A \setminus A) \cup (B \setminus A))) \\ &= \mu^*(C \cap A) + \mu^*(C \cap (B \setminus A)) \end{aligned}$$

proving that

$$\mu^*(C \cap (A \cup B)) = \mu^*(C \cap A) + \mu^*(C \cap (B \setminus A)) \quad (21.131)$$

So we have

$$\begin{aligned} &\mu^*(C \cap (A \cup B)) + \mu^*(C \setminus (A \cup B)) \stackrel{[\text{eq: 21.131}]}{=} \\ &\mu^*(C \cap A) + \mu^*(C \cap (B \setminus A)) + \mu^*(C \setminus (A \cup B)) \stackrel{[\text{theorem: 1.34(1)}]}{=} \\ &\mu^*(C \cap A) + \mu^*(C \cap (B \setminus A)) + \mu^*((C \setminus A) \setminus B) \stackrel{[\text{theorem: 1.30(7)}]}{=} \\ &\mu^*(C \cap A) + \mu^*((C \setminus A) \cap B) + \mu^*((C \setminus A) \setminus B) \stackrel{B \in \overline{\mathcal{M}}[\mu^*]}{=} \\ &\mu^*(C \cap A) + \mu^*(C \setminus A) \stackrel{A \in \overline{\mathcal{M}}[\mu^*]}{=} \mu^*(C) \end{aligned}$$

proving that $A \cup B$ is μ^* -measurable. Hence we have

$$\forall A, B \in \mathcal{M}[\mu^*] \text{ we have that } A \cup B \in \mathcal{M}[\mu^*] \text{ [in other words } \mathcal{M}[\mu^*] \text{ is } \cup\text{-stable}] \quad (21.132)$$

From [eqs: 21.129, 21.130 and 21.132] and [theorem: 21.32] it follows that

$$\mathcal{M}[\mu^*] \text{ is a algebra} \quad (21.133)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[\mu^*]$ **pairwise disjoint** and $B \in \mathcal{P}(X)$. We show then by mathematical induction that $\forall n \in \mathbb{N}$

$$\mu^*(B) = \sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \quad (21.134)$$

Proof. Define

$$S = \left\{ n \in \mathbb{N} \mid \mu^*(B) = \sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \right\}$$

then we have:

$1 \in \mathcal{S}$. As $A_1 \in \mathcal{M}[\mu^*]$ we have

$$\begin{aligned}\mu^*(B) &= \mu^*(B \cap A_1) + \mu^*(B \setminus A_1) \\ &= \sum_{i=1}^1 \mu^*[B \cap A_i] + \mu^*\left(B \setminus \bigsqcup_{i \in \{1\}} A_i\right)\end{aligned}$$

which proves that $1 \in \mathcal{S}$

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. If $x \in B \cap A_{n+1}$ then $x \in B$ and $x \in A_{n+1}$ then $\forall i \in \{1, \dots, n\}$ we have by pairwise disjointness $x \notin A_i$ so that $x \notin \bigsqcup_{i \in \{1, \dots, n\}} A_i$ or $x \in (B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i) \cap A_{n+1}$. Hence $B \cap A_{n+1} \subseteq (B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i) \cap A_{n+1} \subseteq B \cap A_{n+1}$. Summarized

$$\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \cap A_{n+1} = B \cap A_{n+1} \quad (21.135)$$

As $A_{n+1} \in \mathcal{M}[\mu^*]$ we have

$$\begin{aligned}& \mu^*\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \\ & \mu^*\left(\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \cap A_{n+1}\right) + \mu^*\left(\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \setminus A_{n+1}\right) \quad [\text{eq: 21.135}] \\ & \mu^*(B \cap A_{n+1}) + \mu^*\left(\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \setminus A_{n+1}\right) \quad [\text{theorem: 1.34}] \\ & \mu^*(B \cap A_{n+1}) + \mu^*\left(B \setminus \left(\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \cup A_{n+1}\right)\right) = \\ & \mu^*(B \cap A_{n+1}) + \mu^*\left(B \setminus \left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i\right)\right)\end{aligned}$$

proving that

$$\mu^*\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \mu^*(B \cap A_{n+1}) + \mu^*\left(B \setminus \left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i\right)\right) \quad (21.136)$$

Finally

$$\begin{aligned}\mu^*(B) & \stackrel{=}{=}_{n \in \mathcal{S}} \sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \\ & \stackrel{=}{=}_{[\text{eq: 21.136}]} \sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*(B \cap A_{n+1}) + \mu^*\left(B \setminus \left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i\right)\right) \\ & = \sum_{i=1}^{n+1} \mu^*(B \cap A_i) + \mu^*\left(B \setminus \left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i\right)\right)\end{aligned}$$

which proves that $n+1 \in \mathcal{S}$.

Mathematical induction proves then [eq: 21.134]. $\square$

Let $n \in \mathbb{N}$ then we have $\bigsqcup_{i \in \{1, \dots, n\}} A_i \subseteq \bigsqcup_{i \in \mathbb{N}} A_i$ so that $B \setminus \bigsqcup_{i \in \mathbb{N}} A_i \subseteq B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i$ hence $\mu^*(B \setminus \bigsqcup_{i \in \mathbb{N}} A_i) \leq \mu^*(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i)$. So

$$\begin{aligned}\sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigsqcup_{i \in \mathbb{N}} A_i\right) & \leq \sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \\ & \stackrel{=}{=}_{[\text{eq: 21.134}]} \mu^*(B) \quad (21.137)\end{aligned}$$

Next

$$\begin{aligned}
& \sum_{i=1}^{\infty} \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) = \\
& \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \mu^*(B \cap A_i) \right) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) \quad [\text{theorem: 20.85}] = \\
& \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) \right) \quad [\text{theorem: 20.85}] \wedge [\text{eq: 21.137}] \leq \\
& \lim_{n \rightarrow \infty} \mu^*(B) \quad [\text{theorem: 20.74}] = \mu^*(B)
\end{aligned}$$

proving that $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[\mu^*]$ and $B \in \mathcal{P}(X)$

$$\sum_{i=1}^{\infty} \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) \leq \mu^*(B) \quad (21.138)$$

Hence we have for $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[\mu^*]$ and $B \in \mathcal{P}(X)$ that

$$\begin{aligned}
& \mu^*\left(B \cap \left(\bigcup_{i \in \mathbb{N}} A_i\right)\right) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) = \\
& \mu^*\left(\bigcup_{i \in \mathbb{N}} (B \cap A_i)\right) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) \quad \mu^* \text{ is a outer measure} \leq \\
& \sum_{i=1}^{\infty} \mu^*(B \cap A_i) + \mu^*\left(B \setminus \bigcup_{i \in \mathbb{N}} A_i\right) \quad [\text{eq: 21.138}] \leq \mu^*(B)
\end{aligned}$$

So by [theorem: 21.120] it follows that $\bigcup_{i \in \mathbb{N}} A_i$ is μ^* -measurable. Hence we have proved that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[\mu^*] \text{ pairwise disjoint we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{M}[\mu^*] \quad (21.139)$$

As $\mathcal{M}[\mu^*]$ is a algebra [see eq: 21.133] we can use the alternative definition for a σ -algebra [see theorem: 21.36] and the above to prove that

$\mathcal{M}[\mu^*]$ is a σ -algebra

2. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[\mu^*]$ a pairwise disjoint family then for $B = \bigcup_{j \in \mathbb{N}} A_j \in \mathcal{P}(X)$ we have

$$\begin{aligned}
\sum_{i=1}^{\infty} (\mu^*)|_{\mathcal{M}[\mu^*]}(A_i) &= \sum_{i=1}^{\infty} \mu^*(A_i) \\
&= \sum_{i=1}^{\infty} \mu^*\left(\left(\bigcup_{j \in \mathbb{N}} A_j\right) \cap A_i\right) + 0 \\
&= \sum_{i=1}^{\infty} \mu^*\left(\left(\bigcup_{j \in \mathbb{N}} A_j\right) \cap A_i\right) + \mu^*\left(\left(\bigcup_{j \in \mathbb{N}} A_j\right) \setminus \left(\bigcup_{j \in \mathbb{N}} A_j\right)\right) \\
&= \sum_{i=1}^{\infty} \mu^*(B \cap A_i) + \mu^*\left(B \setminus \left(\bigcup_{j \in \mathbb{N}} A_j\right)\right) \\
&\leq \mu^*(B) \quad [\text{eq: 21.138}] \\
&= \mu^*\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\
&= (\mu^*)|_{\mathcal{M}[\mu^*]}\left(\bigcup_{i \in \mathbb{N}} A_i\right)
\end{aligned}$$

As further by the sub-additivity of μ^* we have

$$(\mu^*)_{|\mathcal{M}[\mu^*]} \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) = \mu^* \left(\bigcup_{i \in \mathbb{N}} A_i \right) \leq \sum_{i=1}^{\infty} \mu^*(A_i) = \sum_{i=1}^{\infty} (\mu^*)_{|\mathcal{M}[\mu^*]}(A_i)$$

it follows that

$$(\mu^*)_{|\mathcal{M}[\mu^*]} \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) = \sum_{i=1}^{\infty} (\mu^*)_{|\mathcal{M}[\mu^*]}(A_i)$$

which together with $(\mu^*)_{|\mathcal{M}[\mu^*]}(\emptyset) = \mu^*(\emptyset) = 0$ proves

$$(\mu^*)_{|\mathcal{M}[\mu^*]} \text{ is a measure on } \mathcal{M}[\mu^*]$$

□

21.3.3 Uniqueness of measure extensions

We will use the Carathéodory theorem to extend set functions defined on a subset $\mathcal{G} \subseteq \mathcal{P}(X)$ to a measure on $\sigma[\mathcal{G}]$. In this section we will examine under which conditions this extensions is unique. To simplify our proofs we first introduce the concept of Dynkin systems.

21.3.3.1 Dynkin systems

Definition 21.123. (Dynkin System) Let X be a set then $\mathfrak{D} \subseteq \mathcal{P}(X)$ is a **Dynkin system** on X if

1. $X \in \mathfrak{D}$
2. If $D \in \mathfrak{D}$ then $X \setminus D \in \mathfrak{D}$
3. If $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}$ is a **pairwise disjoint family** then $\bigsqcup_{i \in \mathbb{N}} D_i \in \mathfrak{D}$

Note 21.124. Using [definition: 21.33] every σ -algebra is automatically a Dynkin system however the opposite is in general not true.

Some of the properties of a Dynkin system are similar to the properties of a σ -algebra.

Theorem 21.125. Let X be a set and $\mathfrak{D}$ is a Dynkin system on X then we have:

1. $\emptyset \in \mathfrak{D}$
2. Let $n \in \mathbb{N}$ and $\{D_i\}_{i \in \{1, \dots, n\}} \subseteq \mathfrak{D}$ pairwise disjoint then $\bigsqcup_{i \in \{1, \dots, n\}} D_i \in \mathfrak{D}$

Proof.

1. As $X \in \mathfrak{D}$ we have that $\emptyset = X \setminus X \in \mathfrak{D}$
2. Define $\{E_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}$ by $E_i = \begin{cases} \emptyset & \text{if } i \in \{n+1, \dots, \infty\} \\ D_i & \text{if } i \in \{1, \dots, n\} \end{cases}$. Let $i, j \in \mathbb{N}$ with $i \neq j$ then we have either

$$i \leq n \wedge j \leq n. \text{ Then } E_i \cap E_j = D_i \cap D_j = \emptyset$$

$$n < i \wedge j \leq n. \text{ Then } E_i \cap E_j = \emptyset \cap D_j = \emptyset$$

$$i \leq n \wedge n < j. \text{ Then } E_i \cap E_j = D_i \cap \emptyset = \emptyset$$

$$n < i \wedge n < j. \text{ Then } E_i \cap E_j = \emptyset \cap \emptyset = \emptyset$$

which proves that $\{E_i\}_{i \in \mathbb{N}}$ is pairwise disjoint. So

$$\bigsqcup_{i \in \mathbb{N}} E_i \in \mathfrak{D}$$

If $x \in \bigcup_{i \in \{1, \dots, n\}} D_i$ then $\exists i \in \{1, \dots, n\}$ such that $x \in D_i = E_i$ so that $x \in \bigcup_{i \in \mathbb{N}} E_i$. If $x \in \bigcup_{i \in \mathbb{N}} E_i$ then there exist a $i \in \mathbb{N}$ such that $x \in E_i$, as $\forall i \in \{n+1, \dots, \infty\} E_i = \emptyset$ we must have that $i \in \{1, \dots, n\}$ so that $E_i = D_i$ proving that $x \in \bigcup_{i \in \{1, \dots, n\}} D_i$. Hence we have

$$\bigcup_{i \in \{1, \dots, n\}} D_i = \bigcup_{i \in \mathbb{N}} E_i \in \mathfrak{D}$$

□

Example 21.126. Let X be a set then $\mathcal{P}(X)$ is a Dynkin system on X

Proof. As $\mathcal{P}(X)$ is a σ -algebra on X [see example: 21.38] and every σ -algebra is a Dynkin system it follows that $\mathcal{P}(X)$ is a Dynkin system on X . □

Given a set of subsets of X we can, just like with σ -algebras, find the smallest Dynkin system that contains the set of subsets.

Lemma 21.127. Let X be a set and $\mathcal{X}$ a non empty set of Dynkin systems on X then $\bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$ is a Dynkin system on X .

Proof. As $\mathcal{X} \neq \emptyset$ there exist a $\mathfrak{D} \in \mathcal{X}$ with $\mathfrak{D} \subseteq \mathcal{P}(X)$ so that we have by [theorem: 1.66] that $\bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} \subseteq \mathfrak{D} \subseteq \mathcal{P}(X)$ proving that

$$\bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} \subseteq \mathcal{P}(X)$$

As $\forall \mathfrak{D} \in \mathfrak{D}$ $\mathfrak{D}$ is a Dynkin system on X we have $X \in \mathfrak{D}$ hence

$$X \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$$

Let $A \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$ then for $\mathfrak{D} \in \mathcal{X}$ we have $A \in \mathfrak{D}$ so that $X \setminus A \in \mathfrak{D}$ which as $\mathfrak{D} \in \mathcal{X}$ was chosen arbitrary proves that $X \setminus A \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$. Hence, as $A \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$ was chosen arbitrary, we have that

$$\forall A \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} \text{ we have } X \setminus A \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$$

If $\{D_i\}_{i \in \mathbb{N}} \subseteq \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$ is pairwise disjoint then for $\mathfrak{D} \in \mathcal{X}$ we have $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}$ so that $\bigcup_{i \in \mathbb{N}} D_i \in \mathfrak{D}$ which as $\mathfrak{D} \in \mathcal{X}$ was chosen arbitrary proves that $\bigcup_{i \in \mathbb{N}} D_i \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$. Hence, as $\{D_i\}_{i \in \mathbb{N}} \subseteq \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$ was chosen arbitrary, we have that

$$\text{For every pairwise disjoint family } \{D_i\}_{i \in \mathbb{N}} \subseteq \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} \text{ we have } \bigcup_{i \in \mathbb{N}} D_i \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} \quad \square$$

Theorem 21.128. Let X be a set $\mathfrak{D} \subseteq \mathcal{P}(X)$ then there exist a **unique** Dynkin system $\delta[\mathfrak{A}]$ on X that satisfies:

1. $\mathfrak{D} \subseteq \delta[\mathfrak{D}]$
2. If $\mathfrak{D}'$ is another Dynkin system with $\mathfrak{D} \subseteq \mathfrak{D}'$ then $\delta[\mathfrak{D}] \subseteq \mathfrak{D}'$
3. $\delta[\mathfrak{D}] \subseteq \sigma[\mathfrak{D}]$

This unique smallest Dynkin system that contains $\mathfrak{D}$ is called the Dynkin system generated by $\mathfrak{D}$

Proof.

1. Define $\mathcal{X} = \{\mathfrak{D} \subseteq \mathcal{P}(X) | \mathfrak{D} \text{ is a Dynkin system on } X \text{ and } \mathfrak{D} \subseteq \mathfrak{D}\}$ then as $\mathcal{P}(X)$ is a Dynkin system [see example: 21.126] and $\mathfrak{D} \subseteq \mathcal{P}(X)$ it follows that $\mathcal{X} \neq \emptyset$. Hence by [lemma: 21.127] it follows that, if we take $\delta[\mathfrak{D}] = \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$, that

$$\delta[\mathfrak{D}] \text{ is a Dynkin system on } X$$

If $D \in \mathcal{D}$ then, as $\forall \mathfrak{D} \in \mathcal{X}$ we have by definition of $\mathcal{X}$ that $\mathcal{D} \subseteq \mathfrak{D}$, it follows that $D \in \mathfrak{D}$, hence $D \in \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D}$. So that

$$\mathcal{D} \subseteq \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} = \delta[\mathcal{D}]$$

2. Assume that $\mathfrak{D}'$ is another Dynkin system on X such that $\mathcal{D} \subseteq \mathfrak{D}'$ then $\mathfrak{D}' \in \mathcal{X}$ so that by [theorem: 1.66]

$$\delta[\mathcal{D}] = \bigcap_{\mathfrak{D} \in \mathcal{X}} \mathfrak{D} \subseteq \mathfrak{D}'$$

3. As $\mathcal{D} \subseteq \sigma[\mathcal{D}]$, $\sigma[\mathcal{D}]$ is a σ -algebra hence a Dynkin system on X it follows from (2) that

$$\delta[\mathcal{D}] \subseteq \sigma[\mathcal{D}] \quad \square$$

We show now the sufficient and necessary condition for a Dynkin system to be a σ -algebra.

Theorem 21.129. *Let X be a set and $\mathfrak{D}$ a Dynkin system on X then we have*

$$\mathfrak{D} \text{ is a } \sigma\text{-algebra} \Leftrightarrow \forall A, B \in \mathfrak{D} \text{ we have } A \bigcap B \in \mathfrak{D} \text{ [in other words } \mathfrak{D} \text{ is } \cap\text{-stable}]$$

Proof.

$\Rightarrow$. Let $A, B \in \mathfrak{D}$ then, as $\mathfrak{D}$ is a σ -algebra, it follows from [theorems: 21.36, 21.32] that $A \bigcap B \in \mathfrak{D}$.

$\Leftarrow$. As $\mathfrak{D}$ is a Dynkin system and $\cap$ -stable we have already that

$$X \in \mathfrak{D}, \forall D \in \mathfrak{D} \text{ we have } X \setminus D \in \mathfrak{D} \text{ and } \forall A, B \in \mathfrak{D} \text{ we have } A \bigcap B \in \mathfrak{D} \quad (21.140)$$

Let $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}$. Define

$$\{E_i\}_{i \in \mathbb{N}} \text{ by } E_i = \begin{cases} D_1 & \text{if } i = 1 \\ D_i \bigcap \left(\bigcap_{j \in \{1, \dots, i-1\}} (X \setminus D_j) \right) & \text{if } i \in \{2, \dots, \infty\} \end{cases} \stackrel{[\text{eq: 21.140}] \wedge [\text{theorem: 21.9}]}{\in} \mathfrak{D}$$

Then $\forall i \in \mathbb{N}$ we have

$$E_i = \begin{cases} D_1 \subseteq D_i & \text{if } i = 1 \\ D_i \bigcap \left(\bigcap_{j \in \{1, \dots, i-1\}} (X \setminus D_j) \right) \subseteq D_i & \text{if } i \in \{2, \dots, \infty\} \end{cases} \subseteq D_i \quad (21.141)$$

so that

$$\bigcup_{i \in \mathbb{N}} E_i \subseteq \bigcup_{i \in \mathbb{N}} D_i \quad (21.142)$$

If $x \in \bigcup_{i \in \mathbb{N}} D_i$ then there exist a $k \in \mathbb{N}$ such that $x \in D_k$. Define

$$S = \{i \in \{1, \dots, k\} \mid x \in D_i\} \subseteq \{1, \dots, k\}$$

then $k \in S \Rightarrow S \neq \emptyset$ and S is finite so that $m = \min(S)$ exists. For m we have either:

$$\mathbf{m = 1.} \text{ Then } x \in D_1 = E_1 \subseteq \bigcup_{i \in \mathbb{N}} E_i$$

$$\mathbf{1 < m.} \text{ Then } x \in D_m \text{ and } \forall i \in \{1, \dots, m-1\} \ x \notin D_i \Rightarrow x \in X \setminus D_i \text{ so that}$$

$$x \in D_m \bigcap \left(\bigcap_{i \in \{1, \dots, m-1\}} (X \setminus D_i) \right) = E_m \subseteq \bigcup_{i \in \mathbb{N}} E_i$$

Hence $\bigcup_{i \in \mathbb{N}} D_i \subseteq \bigcup_{i \in \mathbb{N}} E_i$ and we have by [eq: 21.142] that

$$\bigcup_{i \in \mathbb{N}} E_i = \bigcup_{i \in \mathbb{N}} D_i \quad (21.143)$$

Let $i, j \in \mathbb{N}$ with $i \neq j$ then we may always assume that $i < j$ [otherwise exchange i and j]. Assume that $x \in E_i \cap E_j$ then $x \in E_i \subseteq D_i$ and $x \in E_j = D_j \cap (\bigcap_{k \in \{1, \dots, j-1\}} (X \setminus D_k)) \subseteq (X \setminus D_i) \Rightarrow x \notin D_i$ so that we reach the contradiction $x \in D_i$ and $x \notin D_i$. Hence we have

$$\forall i, j \in \mathbb{N} \text{ with } i \neq j \text{ we have } E_i \cap E_j = \emptyset$$

So by the definition of a Dynkin system it follows that

$$\bigcup_{i \in \mathbb{N}} D_i \stackrel{[\text{eq: 21.143}]}{=} \bigcup_{i \in \mathbb{N}} E_i \in \mathfrak{D}$$

Finally by the above, [eq: 21.140] and [definition: 21.33] it follows that $\mathfrak{D}$ is a σ -algebra on X . $\square$

Theorem 21.130. *If $\mathcal{D} \subseteq \mathcal{P}(\mathcal{A})$ is such that $\forall A, B \in \mathfrak{D} \ A \cap B \in \mathcal{D}$ [in other words $\mathfrak{D}$ is $\cap$ -stable] then $\delta(\mathcal{D}) = \sigma(\mathcal{D})$*

Proof. Using [theorem: 21.128] it follows that

$$\delta[\mathcal{D}] \subseteq \sigma[\mathcal{D}] \quad (21.144)$$

For the opposite inclusion. Define for every $D \in \delta[\mathcal{D}]$

$$\mathfrak{D}_D = \{E \subseteq X \mid E \cap D \in \delta[\mathcal{D}]\} \quad (21.145)$$

As $X \subseteq X$ and $X \cap D = D \in \delta[\mathcal{D}]$ it follows that

$$X \in \mathfrak{D}_D \quad (21.146)$$

Let $E \in \mathfrak{D}_D$ then $X \setminus E \subseteq X$ and

$$\begin{aligned} X \setminus ((E \cap D) \cup (X \setminus D)) &\stackrel{[\text{theorem: 1.34}]}{=} (X \setminus (E \cap D)) \cap (X \setminus (X \setminus D)) \\ &\stackrel{[\text{theorem: 1.36}] \wedge D \subseteq X}{=} (X \setminus (E \cap D)) \cap D \\ &\stackrel{[\text{theorem: 1.34}]}{=} ((X \setminus E) \cup (X \setminus D)) \cap D \\ &\stackrel{[\text{theorem: 1.29}]}{=} ((X \setminus E) \cap D) \cup ((X \setminus D) \cap D) \\ &= ((X \setminus E) \cap D) \cup \emptyset \\ &= (X \setminus E) \cap D \end{aligned}$$

Further $(E \cap D) \cap (X \setminus D) = \emptyset$ so that

$$(X \setminus E) \cap D = X \setminus ((E \cap D) \sqcup (X \setminus D)) \text{ [union is disjoint union]} \quad (21.147)$$

As $E \in \mathfrak{D}_D$ we have by [eq: 21.145] that $E \cap D \in \delta[\mathcal{D}]$, further as $\delta[\mathcal{D}]$ is a Dynkin system and $D \in \delta[\mathcal{D}]$ it follows that $X \setminus D \in \delta[\mathcal{D}]$ so that $(E \cap D) \sqcup (X \setminus D) \in \delta[\mathcal{D}]$ and finally that

$$X \setminus ((E \cap D) \sqcup (X \setminus D)) \in \delta[\mathcal{D}]$$

Hence by [eq: 21.147] it follows that

$$\forall E \in \mathfrak{D}_D \text{ that } X \setminus E \in \mathfrak{D}_D \quad (21.148)$$

Let $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}_D$ be pairwise disjoint then $\forall i, j \in \mathbb{N}$ with $i \neq j$ we have $(D_i \cap D) \cap (D_j \cap D) \subseteq D_i \cap D_j = \emptyset$, further $\forall i \in \mathbb{N}$ we have by the definition of $\mathfrak{D}_D$ that $D_i \cap D \in \delta[\mathcal{D}]$. Hence $\{D_i \cap D\}_{i \in \mathbb{N}} \subseteq \delta[\mathcal{D}]$ is pairwise disjoint, so that, as $\delta[\mathcal{D}]$ is a Dynkin system, we have

$$\left(\bigsqcup_{i \in \mathbb{N}} D_i \right) \cap D = \bigsqcup_{i \in \mathbb{N}} (D_i \cap D) \in \delta[\mathcal{D}]$$

proving that

$$\bigsqcup_{i \in \mathbb{N}} D_i \in \mathfrak{D}_D \quad (21.149)$$

From [eqs: 21.146. 21.148 and 21.149] it follows that $\mathfrak{D}_D$ is a Dynkin system on X , hence

$$\forall D \in \delta[\mathcal{D}] \text{ we have that } \mathfrak{D}_D \text{ is a Dynkin system} \quad (21.150)$$

Let $G \in \mathcal{D} \subseteq \delta[\mathcal{D}]$ then by the hypothesis we have $\forall D \in \mathcal{D}$ that $G \cap D \in \mathcal{D} \Rightarrow D \in \mathfrak{D}_G$, hence $\mathcal{D} \subseteq \mathfrak{D}_G$. As $\mathfrak{D}_G$ is a Dynkin system and $\delta[\mathcal{D}]$ is the smallest Dynkin system containing $\mathcal{D}$ it follows that $\delta[\mathcal{D}] \subseteq \mathfrak{D}_G$. Hence we have proved that

$$\forall G \in \mathcal{D} \text{ we have } \delta[\mathcal{D}] \subseteq \mathfrak{D}_G \quad (21.151)$$

Let $G \in \mathcal{D}$ and $D \in \delta[\mathcal{D}] \stackrel{[eq: 21.151]}{\subseteq} \mathfrak{D}_G$ then by the definition of $\mathfrak{D}_G$ we have $G \cap D \in \delta[\mathcal{D}]$ so that $G \in \mathfrak{D}_D$. In other words $\forall D \in \delta[\mathcal{D}]$ we have $\forall G \in \mathcal{D}$ that $G \in \mathfrak{D}_D$ proving that

$$\forall D \in \delta[\mathcal{D}] \text{ we have } \mathcal{D} \subseteq \mathfrak{D}_D$$

As $\delta[\mathcal{D}]$ is the smallest Dynkin system containing $\mathcal{D}$ and $\mathfrak{D}_D$ is a Dynkin system [see eq: 21.150] it follows from the above that

$$\forall D \in \delta[\mathcal{D}] \text{ that } \delta[\mathcal{D}] \subseteq \mathfrak{D}_D$$

Let $A, B \in \delta[\mathcal{D}]$ then by the above $\delta[\mathcal{D}] \subseteq \mathfrak{D}_B$ and, as $A \in \delta(\mathcal{D})$, $A \in \mathfrak{D}_B \stackrel{[eq: 21.145]}{\Rightarrow} A \cap B \in \delta[\mathcal{D}]$. Hence we have

$$\forall A, B \in \delta[\mathcal{D}] \text{ we have } A \cap B \in \delta[\mathcal{D}] \text{ where } \delta[\mathcal{D}] \text{ is a Dynkin system}$$

Hence by [theorem: 21.129] it follows finally that

$$\delta[\mathcal{D}] \text{ is a } \sigma\text{-algebra} \quad \square$$

We are now ready to prove that every set function on $\mathcal{G} \subseteq \mathcal{P}(X)$ has a unique extension to a measure on $\sigma[\mathcal{G}]$ if certain conditions are met.

Theorem 21.131. *Let X be a set, $\mathcal{G} \subseteq \mathcal{P}(X)$ such that*

1. $\forall A, B \in \mathcal{G}$ we have $A \cap B \in \mathcal{G}$ [in other words $\mathcal{G}$ is $\cap$ -stable]
2. $\exists \{G_i\}_{i \in \mathbb{N}} \subseteq \mathcal{G}$ such that $\forall i \in \mathbb{N}$ we have $G_i \subseteq G_{i+1}$ and $\bigcup_{i \in \mathbb{N}} G_i = X$

and μ, ν are measures on $\sigma[\mathcal{G}]$ such that

3. $\forall A \in \mathcal{G} \mu(A) = \nu(A)$
4. $\forall i \in \mathbb{N} \mu(G_i) < \infty$ [hence as $G_i \in \mathcal{G}$ we have also $\nu(G_i) < \infty$]

then we have that

$$\mu = \nu$$

Proof. Given $n \in \mathbb{N}$ define

$$\mathfrak{D}_n = \{A \in \sigma[\mathcal{G}] \mid \mu(G_n \cap A) = \nu(G_n \cap A)\} \subseteq \sigma[\mathcal{G}] \quad (21.152)$$

We prove then that $\mathfrak{D}_n$ is a Dynkin system that contains $\mathcal{G}$ for every $n \in \mathbb{N}$. Take $G \in \mathcal{G} \subseteq \sigma(\mathcal{G})$ then $G, G_n \in \mathcal{G}$ so that by (1) $G_n \cap G \in \mathcal{G}$, as μ, ν are the same on $\mathcal{G}$ we have $\mu(G_n \cap G) = \nu(G_n \cap G)$. Hence we have

$$\mathcal{G} \subseteq \mathfrak{D}_n \quad (21.153)$$

As $X \in \sigma[\mathcal{G}]$ and $\mu(G_n \cap X) = \mu(G_n) \stackrel{G_n \in \mathcal{G}}{=} \nu(G_n) = \nu(G_n \cap X)$ it follows that

$$X \in \mathfrak{D}_n \quad (21.154)$$

Let $A \in \mathfrak{D}_n$ then we have $G_n \setminus (G_n \cap A) \stackrel{[\text{theorem: 1.34}]}{=} (G_n \setminus G_n) \cup (G_n \setminus A) = \emptyset \cup (G_n \setminus A) = G_n \setminus A$ so that

$$G_n \setminus (G_n \cap A) = G_n \setminus A \quad (21.155)$$

Further $\mu(G_n \cap A) \stackrel{[\text{theorem: 21.99}]}{\leq} \mu(G_n) < \infty$ and $\nu(G_n \cap A) \stackrel{[\text{theorem: 21.99}]}{\leq} \nu(G_n) < \infty$ so that by [theorem: 21.99]

$$\mu(G_n \setminus (G_n \cap A)) = \mu(G_n) - \mu(G_n \cap A) \wedge \nu(G_n \setminus (G_n \cap A)) = \nu(G_n) - \nu(G_n \cap A) \quad (21.156)$$

hence

$$\begin{aligned} \mu(G_n \cap (X \setminus A)) &\stackrel{[\text{theorem: 1.36(8)}]}{=} \mu(G_n \setminus A) \\ &\stackrel{[\text{eq: 21.155}]}{=} \mu(G_n \setminus (G_n \cap A)) \\ &\stackrel{[\text{eq: 21.156}]}{=} \mu(G_n) - \mu(G_n \cap A) \\ &\stackrel{A \in \mathfrak{D}_n}{=} \mu(G_n) - \nu(G_n \cap A) \\ &\stackrel{G_n \in \mathcal{G}}{=} \nu(G_n) - \nu(G_n \cap A) \\ &\stackrel{[\text{eq: 21.156}]}{=} \nu(G_n \setminus (G_n \cap A)) \\ &\stackrel{[\text{eq: 21.155}]}{=} \nu(G_n \setminus A) \\ &\stackrel{[\text{theorem: 1.36(8)}]}{=} \nu(G_n \cap (X \setminus A)) \end{aligned}$$

proving that $(X \setminus A) \in \mathfrak{D}_n$, further as $A \in \mathfrak{D}_n \subseteq \sigma[\mathcal{G}]$ we have $X \setminus A \in \sigma[\mathcal{G}]$. Hence

$$\forall A \in \mathfrak{D}_n \text{ we have } X \setminus A \in \mathfrak{D}_n \quad (21.157)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}_n$ be a pairwise disjoint family. Then $\forall i \in \mathbb{N}$ with $i \neq j$ we have $(G_n \cap A_i) \cap (G_n \cap A_j) \subseteq A_i \cap A_j = \emptyset$ so that $\{G_n \cap A_i\}_{i \in \mathbb{N}}$ is pairwise disjoint. Hence, as μ, ν are measures, it follows that

$$\mu\left(\bigsqcup_{i \in \mathbb{N}} (G_n \cap A_i)\right) = \sum_{i \in \mathbb{N}} \mu(G_n \cap A_i) \stackrel{A_i \in \mathfrak{D}_n}{=} \sum_{i \in \mathbb{N}} \nu(G_n \cap A_i) = \nu\left(\bigsqcup_{i \in \mathbb{N}} (G_n \cap A_i)\right)$$

which, as $G_n \cap (\bigsqcup_{i \in \mathbb{N}} A_i) = \bigsqcup_{i \in \mathbb{N}} (G_n \cap A_i)$ and $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}_n \subseteq \sigma[\mathcal{G}]$, proves that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathfrak{D}_n \text{ we have } \bigsqcup_{i \in \mathbb{N}} A_i \in \mathfrak{D}_n \quad (21.158)$$

By [eqs: 21.154, 21.157 and 21.158] we have

$$\mathfrak{D}_n \text{ is a Dynkin system} \quad (21.159)$$

As $\forall A, B \in \mathcal{G}$ we have by (1) that $A \cap B \in \mathcal{G}$ we have by [theorem: 21.130] that

$$\delta[\mathcal{G}] = \sigma[\mathcal{G}] \quad (21.160)$$

As by [eq: 21.153] $\mathcal{G} \subseteq \mathfrak{D}_n$ and $\mathfrak{D}_n$ is a Dynkin system it follows that

$$\sigma[\mathcal{G}] \stackrel{[\text{eq: 21.160}]}{=} \delta[\mathcal{G}] \subseteq \mathfrak{D}_n \stackrel{[\text{eq: 21.152}]}{\subseteq} \sigma[\mathcal{G}]$$

Hence we have $\forall n \in \mathbb{N}$ that $\sigma[\mathcal{G}] = \mathfrak{D}_n$ so that using the definition of $\mathfrak{D}_n$ [see eq: 21.152]

$$\forall n \in \mathbb{N} \text{ we have } \forall A \in \sigma[\mathcal{G}] \text{ that } \mu(G_n \cap A) = \nu(G_n \cap A) \quad (21.161)$$

Let $A \in \sigma[\mathcal{G}]$ then $\forall i \in \mathbb{N}$ we have $G_i \cap A \subseteq G_{i+1} \cap A \in \sigma[\mathcal{G}]$ and

$$A = A \cap X = A \cap \left(\bigcup_{i \in \mathbb{N}} G_i \right) = \bigcup_{i \in \mathbb{N}} (G_i \cap A) \quad (21.162)$$

so that by [theorem: 21.115] we have that

$$\lim_{i \rightarrow \infty} \mu(G_i \cap A) \text{ exists and } \lim_{i \rightarrow \infty} \mu(G_i \cap A) = \mu\left(\bigcup_{i \in \mathbb{N}} (G_i \cap A)\right)_{[\text{eq: 21.162}]} \stackrel{=}{=} \mu(A) \quad (21.163)$$

and

$$\lim_{i \rightarrow \infty} \nu(G_i \cap A) \text{ exists and } \lim_{i \rightarrow \infty} \nu(G_i \cap A) = \mu\left(\bigcup_{i \in \mathbb{N}} (G_i \cap A)\right)_{[\text{eq: 21.162}]} \stackrel{=}{=} \nu(A) \quad (21.164)$$

Hence we have for $A \in \sigma[\mathcal{G}]$ that

$$\mu(A)_{[\text{eq: 21.163}]} \stackrel{=}{=} \lim_{i \rightarrow \infty} \mu(G_i \cap A)_{[\text{eq: 21.161}]} \stackrel{=}{=} \lim_{i \rightarrow \infty} \nu(G_i \cap A)_{[\text{eq: 21.164}]} \stackrel{=}{=} \nu(A)$$

proving that

$$\mu = \nu$$

□

21.3.4 Constructing measures and σ -algebras

21.3.4.1 Constructing outer measures

Let X be a set then given $\mathcal{G} \subseteq \mathcal{P}(X)$ we want to extend a function $\mu: \mathcal{G} \rightarrow [0, \infty]$ to a measure on a σ -algebra containing $\mathcal{G}$. To use the Carathéodory theorem for this task we have to construct a outer measure out of μ . This is the subject of this section. First we need some definitions that specifies the minimum requirements for this task.

Definition 21.132. (Proto-Ring) Let X be a set then $\mathcal{S} \subseteq \mathcal{P}(X)$ is a **proto-ring** on X if

$$\emptyset \subseteq \mathcal{S}$$

Definition 21.133. (Proto-Measure) Let X be a set then $\mathcal{S}$ a proto-ring on X then a function $\mu: \mathcal{S} \rightarrow [0, \infty]$ is a **proto-measure** on $\mathcal{S}$ if

$$\mu(\emptyset) = 0$$

Next we shown how to construct a outer-measure from a proto-measure.

Theorem 21.134. Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a proto-ring and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a proto-measure on $\mathcal{S}$ then

1.

$$\mu^*: \mathcal{P}(X) \rightarrow [0, \infty] \text{ defined by } \mu^*(A) = \begin{cases} \inf(\{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}) & \text{if } \mathcal{S}[A] \neq \emptyset \\ \infty & \text{if } \mathcal{S}[A] = \emptyset \end{cases}$$

where

$$\mathcal{S}[A] = \left\{ \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \mid A \subseteq \bigcup_{i \in \mathbb{N}} A_i \right\}$$

is a outer measure.

2. If additional we have that

$$a. \forall A, B \in \mathcal{S} \text{ that } A \cap B \in \mathcal{S}$$

$$b. \forall A, B \in \mathcal{S} \text{ with } A \subseteq B \text{ we have } \mu(A) \leq \mu(B)$$

$$c. \forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \text{ with } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{S} \text{ we have } \mu(\bigcup_{i \in \mathbb{N}} A_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

then

$$\forall A \in \mathcal{S} \text{ we have } \mu^*(A) = \mu(A)$$

Note 21.135. The outer measure defined in (1) is not necessary a extension of μ , for that we need the extra conditions 2.a, 2.b, 2.c.

Note 21.136. If $\mathcal{S}[A] \neq \emptyset$ then $\{\sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\} \neq \emptyset$ so that by [theorem: 20.13] $\inf(\{\sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\})$ exist. Further as $\forall \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]$ $0 \leq \sum_{i=1}^{\infty} \mu(A_i)$ we have that $0 \leq \inf(\{\sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\})$. This proves that μ^* is well defined.

Proof.

1. Define $\{N_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ by $N_i = \emptyset \in \mathcal{S}$ then $\emptyset \subseteq \bigcup_{i \in \mathbb{N}} N_i$ so that $\{N_i\}_{i \in \mathbb{N}} \in \mathcal{S}[\emptyset]$, hence $\mathcal{S}[\emptyset] \neq \emptyset$. So $0 = \sum_{i=1}^{\infty} \mu(\emptyset) = \sum_{i=1}^{\infty} \mu(N_i) \in \{\sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}$, hence

$$0 \leq \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A] \right\} \right) \leq 0$$

proving that $\mu^*(\emptyset) = \inf(\{\sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}) = 0$. Hence we have

$$\mu^*(\emptyset) = 0 \quad (21.165)$$

Let $A, B \in \mathcal{P}(X)$ with $A \subseteq B$. Then for $\mathcal{S}[B]$ we have either:

$\mathcal{S}[B] = \emptyset$. Then $\mu^*[A] \leq \infty = \mu^*[B]$

$\mathcal{S}[B] \neq \emptyset$. Let $\{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B]$ then $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ and $B \subseteq \bigcup_{i \in \mathbb{N}} A_i$ hence, as $A \subseteq B$, we have $A \subseteq \bigcup_{i \in \mathbb{N}} A_i$ so that $\{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]$. Hence

$$\emptyset \neq \mathcal{S}[B] \subseteq \mathcal{S}[A]$$

From the above it follows that

$$\left\{ \sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B] \right\} \subseteq \left\{ \sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A] \right\},$$

so that by [theorem: 3.77]

$$\begin{aligned} \mu^*[A] &= \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A] \right\} \right) \\ &\leq \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B] \right\} \right) \\ &= \mu^*[B] \end{aligned}$$

To summarize

$$\forall A, B \in \mathcal{P}(X) \text{ with } A \subseteq B \text{ we have } \mu^*(A) \leq \mu^*(B) \quad (21.166)$$

Let $\{T_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X)$ then we have two cases to consider:

$\exists i \in \mathbb{N}$ such that $\mu^*(T_i) = \infty$. Then $\mu^*(\bigcup_{i \in \mathbb{N}} T_i) \leq \infty = \sum_{i=1}^{\infty} \mu^*(T_i)$.

$\forall i \in \mathbb{N}$ we have $\mu^*(T_i) < \infty$. Then we must have $\forall i \in \mathbb{N}$ that $\mathcal{S}[T_i] \neq \emptyset$ so that

$$\mu^*(T_i) = \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[T_i] \right\} \right)$$

Take $\varepsilon \in \mathbb{R}^+$. Given $n \in \mathbb{N}$ we have, as $0 \leq \mu^*(T_n) < \infty$, that $\mu^*(T_n) < \mu^*(T_n) + \frac{\varepsilon}{2^n}$, so as $\mu^*(T_n) = \inf(\{\sum_{i=1}^{\infty} \mu(A_i) | \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[T_n]\})$, there exist a $\{I_{n,i}\}_{i \in \mathbb{N}} \in \mathcal{S}[T_n]$ [so that $T_n \subseteq \bigcup_{i \in \mathbb{N}} I_{n,i}$] that satisfies $\mu^*(T_n) \leq \sum_{i=1}^{\infty} \mu(I_{n,i}) < \mu^*(T_n) + \frac{\varepsilon}{2^n}$. Hence we have

$$\mu^*(T_n) \leq \sum_{i=1}^{\infty} \mu(I_{n,i}) < \mu^*(T_n) + \frac{\varepsilon}{2^n} \text{ and } T_n \subseteq \bigcup_{i \in \mathbb{N}} I_{n,i} \quad (21.167)$$

As $\mathbb{N} \times \mathbb{N}$ is denumerable [see theorem: 6.63] there exist a bijection $\beta: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Let $x \in \bigcup_{i \in \mathbb{N}} T_i$ then $\exists k \in \mathbb{N}$ such that $x \in T_k \subseteq \bigcup_{i \in \mathbb{N}} I_{k,i}$, hence $\exists l \in \mathbb{N}$ such that $x \in I_{k,l}$. As $(k, l) \in \mathbb{N} \times \mathbb{N}$ there exist a $i \in \mathbb{N}$ such that $\beta(i) = (k, l)$ so that $x \in I_{\beta(i)_1, \beta(i)_2} \subseteq \bigcup_{i \in \mathbb{N}} I_{\beta(i)_1, \beta(i)_2}$. To summarize we have proved that

$$\bigcup_{i \in \mathbb{N}} T_i \subseteq \bigcup_{i \in \mathbb{N}} I_{\beta(i)_1, \beta(i)_2} \text{ so that } \{I_{\beta(i)_1, \beta(i)_2}\}_{i \in \mathbb{N}} \in \mathcal{S} \left[\bigcup_{i \in \mathbb{N}} T_i \right] \quad (21.168)$$

Hence $\mathcal{S}[\bigcup_{i \in \mathbb{N}} T_i] \neq \emptyset$ so that

$$\begin{aligned} \mu^* \left(\bigcup_{i \in \mathbb{N}} T_i \right) &= \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S} \left[\bigcup_{i \in \mathbb{N}} T_i \right] \right\} \right) \\ &\stackrel{[\text{eq: 21.168}]}{\leq} \sum_{i=1}^{\infty} \mu(I_{\beta(i)_1, \beta(i)_2}) \\ &\stackrel{[\text{lemma: 20.108}]}{=} \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} \mu(I_{i,j}) \right) \\ &\stackrel{[\text{eq: 21.167}]}{\leq} \sum_{i=1}^{\infty} \left(\mu^*(T_i) + \frac{\varepsilon}{2^i} \right) \\ &\stackrel{[\text{theorem: 20.99}]}{=} \sum_{i=1}^{\infty} \mu^*(T_i) + \sum_{i=1}^{\infty} \frac{\varepsilon}{2^i} \\ &\stackrel{[\text{theorem: 20.99}]}{=} \sum_{i=1}^{\infty} \mu^*(T_i) + \varepsilon \cdot \sum_{i=1}^{\infty} \frac{1}{2^i} \\ &= \sum_{i=1}^{\infty} \mu^*(T_i) + \varepsilon \cdot \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2^i} \\ &\stackrel{[\text{theorem: 20.77}] \wedge [\text{example: 14.412}]}{=} \sum_{i=1}^{\infty} \mu^*(T_i) + \varepsilon \cdot \frac{1}{1 - \frac{1}{2}} \\ &= \sum_{i=1}^{\infty} \mu^*(T_i) + \varepsilon \end{aligned}$$

Hence we have

$$\forall \varepsilon \in \mathbb{R}^+ \text{ that } \mu^* \left(\bigcup_{i \in \mathbb{N}} T_i \right) \leq \sum_{i=1}^{\infty} \mu^*(T_i) + \varepsilon$$

which, as $\varepsilon \in \mathbb{R}^+$ is chosen arbitrary, proves by [theorem: 20.43] that

$$\mu^* \left(\bigcup_{i \in \mathbb{N}} T_i \right) \leq \sum_{i=1}^{\infty} \mu^*(T_i).$$

To summarize we have in all cases that

$$\forall \{T_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(X) \text{ we have } \mu^* \left(\bigcup_{i \in \mathbb{N}} T_i \right) \leq \sum_{i=1}^{\infty} \mu^*(T_i) \quad (21.169)$$

Taking in account the definition of a outer measure and [eqs: 21.165, 21.166 and 21.169] it follows that

μ^* is a outer measure on X

2. Let $A \in \mathcal{S}$ then for $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ defined by $A_i = \begin{cases} A & \text{if } i = 1 \\ \emptyset & \text{if } i \in \{2, \dots, \infty\} \end{cases} \in \mathcal{S}$ we have $A = \bigcup_{i \in \mathbb{N}} A_i$ so that $A \in S[A] \Rightarrow S[A] \neq \emptyset$ and

$$\sum_{i=1}^{\infty} \mu(A_i) \stackrel{[\text{theorem: 20.100}]}{=} \sum_{i=1}^1 \mu(A_i) + \sum_{i=2}^{\infty} \mu(A_i) = \mu(A_1) + \sum_{i=2}^{\infty} 0 = \mu(A)$$

so that $\mu(A) \in \{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in S[A]\}$. Hence

$$\mu^*(A) = \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in S[A] \right\} \right) \leq \mu(A)$$

proving that

$$\forall A \in \mathcal{S} \text{ we have } \mu^*(A) \leq \mu(A) \quad (21.170)$$

For the opposite inequality, let $A \in \mathcal{S}$ and $\{A_i\}_{i \in \mathbb{N}} \subseteq S[A]$ then $A \subseteq \bigcup_{i \in \mathbb{N}} A_i$ so that $\bigcup_{i \in \mathbb{N}} (A \cap A_i) = A \in \mathcal{S}$. By (2.a) we have $\{A \cap A_i\}_{i \in \mathbb{N}} \subseteq S$ and further

$$\mu(A) = \mu \left(\bigcup_{i \in \mathbb{N}} (A \cap A_i) \right) \stackrel{(2.c)}{\leq} \sum_{i=1}^{\infty} \mu(A \cap A_i) \stackrel{(2.b)}{\leq} \sum_{i=1}^{\infty} \mu(A_i)$$

Hence $\mu(A)$ is a lower bound of $\{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in S[A]\}$, therefore $\mu(A) \leq \mu^*(A)$. Combining this with [eq: 21.170] gives finally

$$\mu(A) = \mu^*(A) \quad \square$$

As is mentioned in [note: 21.135] if we want that the above theorem extend a proto-measure to a outer measure we need extra conditions on the proto-ring and the proto-measure. This motivates the definition of a semi-ring and a pre-measure below.

Definition 21.137. (Semi-Ring) Let X be a set then $S \subseteq \mathcal{P}(X)$ is a **semi-ring** on X if

1. $\emptyset \in \mathcal{S}$
2. $\forall A, B \in \mathcal{S}$ we have $A \cap B \in \mathcal{S}$ [in other words $\mathcal{S}$ is $\cap$ -stable]
3. $\forall A, B \in \mathcal{S}$ there exist a finite pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that

$$A \setminus B = \bigsqcup_{i \in \{1, \dots, n\}} A_i$$

Note 21.138. As by (2) $\mathcal{S}$ is $\cap$ -stable it follows from [lemma: 21.9] that $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ we have that $\bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{S}$.

Note 21.139. As $\emptyset \in \mathcal{S}$ it follows from the definition of a proto-ring [see definition: 21.132] that every semi-ring is a proto-ring.

Definition 21.140. (Pre-Measure) Let X be a set, $\mathcal{S}$ a semi-ring on X then a set function $\mu: \mathcal{S} \rightarrow [0, \infty]$ is a **pre-measure** if

1. $\mu(\emptyset) = 0$
2. For every pairwise disjoint family $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ such that $\bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{S}$ we have that

$$\mu \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) = \sum_{i=1}^{\infty} \mu(A_i)$$

In other words $\mu: \mathcal{S} \rightarrow [0, \infty]$ is countable additive on $\mathcal{A}$

Note 21.141. As $\mu(\emptyset) = 0$ it follows from the definition of a proto-measure [see definition: 21.133] that that a pre-measure on a semi-ring is a proto-measure on a proto-ring.

The above definitions do not exactly cover the extra conditions 1.a, 2.b and 2.c in [theorem: 21.134]. The next theorems and lemmas will ensure that these conditions will be satisfied by semi-rings and pre-measures.

Lemma 21.142. *Let X be a set, $\mathcal{S}$ a semi-ring on X , $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ and $A \in \mathcal{S}$ then there exist a pairwise disjoint family $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$, $m \in \mathbb{N}$ such that*

$$A \setminus \bigcup_{i \in \{1, \dots, n\}} A_i = \bigsqcup_{i \in \{1, \dots, m\}} B_i$$

Proof. We use induction to prove this. Let $A \in \mathcal{S}$ and define

$$S_A = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S} \text{ then there exist a pairwise disjoint } \{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}, m \in \mathbb{N} \right. \\ \left. \text{such that } A \setminus \bigcup_{i \in \{1, \dots, n\}} A_i = \bigsqcup_{i \in \{1, \dots, m\}} B_i \right\}$$

then we have:

1 $\in S_A$. If $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ then $A \setminus \bigcup_{i \in \{1\}} A_i = A \setminus A_1$. As $\mathcal{S}$ is a semi-ring it follows that there exist a pairwise disjoint family $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$, $m \in \mathbb{N}$ such that

$$A \setminus \bigcup_{i \in \{1\}} A_i = A \setminus A_1 = \bigsqcup_{i \in \{1, \dots, m\}} B_i$$

which proves that $1 \in S_A$.

$n \in S_A \Rightarrow n+1 \in S_A$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}$ then as $n \in S$ there exist a $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$ pairwise disjoint such that

$$A \setminus \bigcup_{i \in \{1, \dots, n\}} A_i = \bigsqcup_{i \in \{1, \dots, m\}} B_i \quad (21.171)$$

Further we have

$$\begin{aligned} A \setminus \bigcup_{i \in \{1, \dots, n+1\}} A_i &= A \setminus \left(\left(\bigcup_{i \in \{1, \dots, n\}} A_i \right) \cup A_{n+1} \right) \\ &\stackrel{[\text{theorem: 1.34}]}{=} \left(A \setminus \left(\bigcup_{i \in \{1, \dots, n\}} A_i \right) \right) \setminus A_{n+1} \\ &\stackrel{[\text{eq: 21.171}]}{=} \left(\bigsqcup_{i \in \{1, \dots, m\}} B_i \right) \setminus A_{n+1} \\ &\stackrel{[\text{theorem: 2.141}]}{=} \bigsqcup_{i \in \{1, \dots, m\}} (B_i \setminus A_{n+1}) \end{aligned} \quad (21.172)$$

If $i, j \in \{1, \dots, m\}$ with $i \neq j$ then we have $(B_i \setminus A_{n+1}) \cap (B_j \setminus A_{n+1}) \subseteq B_i \cap B_j = \emptyset$ proving that

$$\{B_i \setminus A_{n+1}\}_{i \in \{1, \dots, m\}} \text{ is pairwise disjoint} \quad (21.173)$$

Let $i \in \{1, \dots, m\}$ then as $\mathcal{S}$ is a semi-ring and $B_i, A_{n+1} \in \mathcal{S}$ there exist a pairwise disjoint $\{C_{i,j}\}_{j \in \{1, \dots, n_i\}} \subseteq \mathcal{S}$ such that

$$B_i \setminus A_{n+1} = \bigsqcup_{j \in \{1, \dots, n_i\}} C_{i,j} \quad (21.174)$$

Hence using [lemma: 21.11]

$$\bigsqcup_{i \in \{1, \dots, m\}} (B_i \setminus A_{n+1}) = \bigsqcup_{\sigma \in \bigcup_{i \in \{1, \dots, m\}} \{i\} \times \{1, \dots, n_i\}} C_{\sigma_1, \sigma_2} \quad (21.175)$$

and

$$\{C_{\sigma_1, \sigma_2}\}_{\sigma \in \bigcup_{i \in \{1, \dots, m\}} \{i\} \times \{1, \dots, n_i\}} \subseteq \mathcal{S} \text{ is pairwise disjoint} \quad (21.176)$$

As $\bigcup_{i \in \{1, \dots, m\}} \{i\} \times \{1, \dots, n_i\}$ is finite [see theorems: 6.41, 6.36] there exist a bijection such that $\beta: \{1, \dots, N\} \rightarrow \bigcup_{i \in \{1, \dots, m\}} \{i\} \times \{1, \dots, n_i\}$. Let $i, j \in \{1, \dots, N\}$ with $i \neq j$ then, as β is injective, we have $\beta(i) \neq \beta(j)$. For $\beta(i)_1, \beta(j)_1$ we have either:

$\beta(i)_1 = \beta(j)_1$. Then we must have that $\beta(i)_2 \neq \beta(j)_2$ so that

$$C_{\beta(i)_1, \beta(i)_2} \cap C_{\beta(j)_1, \beta(j)_2} = C_{\beta(i)_1, \beta(i)_2} \cap C_{\beta(i)_1, \beta(j)_2} = \emptyset$$

$\beta(i)_1 \neq \beta(j)_1$. Then we have

$$\begin{aligned} C_{\beta(i)_1, \beta(i)_2} \cap C_{\beta(j)_1, \beta(j)_2} &\subseteq \left(\bigcup_{k \in \{1, \dots, n_{\beta(i)_1}\}} C_{\beta(i)_1, k} \right) \cap \left(\bigcup_{k \in \{1, \dots, n_{\beta(j)_1}\}} C_{\beta(j)_1, k} \right) \\ &= B_{\beta(i)_1} \cap B_{\beta(j)_1} \\ &= \emptyset \end{aligned}$$

which proves that

$$\{C_{\beta(i)_1, \beta(i)_2}\}_{i \in \{1, \dots, N\}} \subseteq \mathcal{S} \text{ is pairwise disjoint} \quad (21.177)$$

Further

$$\begin{aligned} A \setminus \bigcup_{i \in \{1, \dots, n+1\}} A_i &\stackrel{[\text{eq: 21.172}]}{=} \bigcap_{i \in \{1, \dots, m\}} (B_i \setminus A_{n+1}) \\ &\stackrel{[\text{eq: 21.174}]}{=} \bigcap_{\sigma \in \bigcup_{i \in \{1, \dots, m\}} \{i\} \times \{1, \dots, n_i\}} C_{\sigma_1, \sigma_2} \\ &\stackrel{[\text{theorem: 2.127}]}{=} \bigcap_{i \in \{1, \dots, N\}} C_{\beta(i)_1, \beta(i)_2} \end{aligned}$$

which proves that $n+1 \in \mathcal{S}$. $\square$

Theorem 21.143. Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a pre-measure then $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ pairwise disjoint, $n \in \mathbb{N}$ with $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{S}$ we have

$$\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

In other words a pre-measure is additive.

Proof. As for a pre-measure we have $\mu(\emptyset) = 0$ and $\mu: \mathcal{S} \rightarrow [0, \infty]$ is countable additive it follows from [theorem: 21.95] that $\mu: \mathcal{S} \rightarrow [0, \infty]$ is additive. In other words $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ pairwise disjoint, $n \in \mathbb{N}$ with $\bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{S}$ we have

$$\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i) \quad \square$$

Theorem 21.144. Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a additive set function [see definition: 21.15] then we have:

1. $\forall A \in \mathcal{S}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ **pairwise disjoint** such that $\bigcup_{i \in \{1, \dots, n\}} A_i \subseteq A$ we have

$$\sum_{i=1}^n \mu(A_i) \leq \mu(A)$$

2. $\forall A, B \in \mathcal{S}$ with $A \subseteq B$ we have $\mu(A) \leq \mu(B)$

Proof.

1. As $A \in \mathcal{S}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ we have by [lemma: 21.142] that there exist a pairwise disjoint $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$, $m \in \mathbb{N}$ such that

$$A \setminus \bigcup_{i \in \{1, \dots, m\}} A_i = \bigcup_{i \in \{1, \dots, m\}} B_i \quad (21.178)$$

Define

$$\{D_i\}_{i \in \{1, \dots, n+m\}} \subseteq \mathcal{S} \text{ by } D_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ B_{i-n} & \text{if } i \in \{n+1, \dots, n+m\} \end{cases} \in \mathcal{S}$$

Let $i, j \in \{1, \dots, n+m\}$ with $i \neq j$ then we can always assume that $i < j$ [otherwise exchange i and j]. For i we have either:

$i \in \{1, \dots, n\}$. Then for j we have either:

$j \in \{1, \dots, n\}$. Then $D_i \cap D_j = A_i \cap A_j$ $\stackrel{\text{[eq: 21.178]}}{=} \emptyset$ $\stackrel{\text{[theorem: 1.39]}}{=}$ $\emptyset$ since $\{A_i\}_{i \in \{1, \dots, n\}}$ is pairwise disjoint

$j \in \{n+1, \dots, n+m\}$. Then

$$D_i \cap D_j = A_i \cap B_{j-n} \stackrel{\text{[eq: 21.178]}}{\subseteq} A_i \cap \left(A \setminus \bigcup_{i \in \{1, \dots, n\}} A_i \right) = \emptyset$$

$i \in \{n+1, \dots, n+m\}$. Then as $i < j$ we have $j \in \{n+1, \dots, n+m\}$ so that

$$D_i \cap D_j = B_{i-n} \cap B_{j-n} \stackrel{\text{[eq: 21.178]}}{=} \emptyset$$

hence

$$\{D_i\}_{i \in \{1, \dots, n+m\}} \subseteq \mathcal{S} \text{ is pairwise disjoint} \quad (21.179)$$

Further as $\bigsqcup_{i \in \{1, \dots, n\}} A_i \subseteq A$ we have

$$\begin{aligned} A &\stackrel{\text{[theorem: 1.39]}}{=} \left(A \setminus \bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \cup \left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \\ &\stackrel{\text{[eq: 21.178]}}{=} \left(\bigsqcup_{i \in \{1, \dots, m\}} B_i \right) \sqcup \left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \\ &= \left(\bigsqcup_{i \in \{1, \dots, m\}} D_{i+n} \right) \sqcup \left(\bigsqcup_{i \in \{1, \dots, n\}} D_i \right) \\ &\stackrel{\text{[theorem: 2.127]}}{=} \left(\bigsqcup_{i \in \{n+1, \dots, n+m\}} D_i \right) \sqcup \left(\bigsqcup_{i \in \{1, \dots, n\}} D_i \right) \\ &\stackrel{\text{[theorem: 2.136]}}{=} \bigsqcup_{i \in \{1, \dots, n+m\}} D_i \end{aligned} \quad (21.180)$$

So as $\bigsqcup_{i \in \{1, \dots, n+m\}} D_i = A \in \mathcal{S}$ we have

$$\begin{aligned} \mu(A) &\stackrel{\text{[eq: 21.180]}}{=} \mu \left(\bigsqcup_{i \in \{1, \dots, n+m\}} D_i \right) \\ &\stackrel{\text{[theorem: 2.136]}}{=} \sum_{i=1}^{n+m} \mu(D_i) \\ &= \sum_{i=1}^n \mu(D_i) + \sum_{i=n+1}^{n+m} \mu(D_i) \\ &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{n+m} \mu(D_i) \end{aligned}$$

so that by [theorem: 20.35] we have

$$\sum_{i=1}^n \mu(A_i) \leq \mu(A)$$

2. Let $A, B \in \mathcal{S}$ with $A \subseteq B$. Take then $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ by $A_i = A$ then we have $\bigsqcup_{i \in \{1\}} A_i = A \subseteq B$, hence by (1) it follows that $\mu(A) = \sum_{i=1}^1 \mu(A_i) \leq \mu(B)$ $\square$

Corollary 21.145. Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a pre-measure on $\mathcal{S}$ [see definition: 21.140] then we have:

1. $\forall A \in \mathcal{S}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ pairwise disjoint such that $\bigcup_{i \in \{1, \dots, n\}} A_i \subseteq A$ we have

$$\sum_{i=1}^n \mu(A_i) \leq \mu(A)$$

2. $\forall A, B \in \mathcal{S}$ with $A \subseteq B$ we have $\mu(A) \leq \mu(B)$

Proof. As a pre-measure is additive [see theorem: 21.143] this follows from the previous theorem [theorem: 21.144]. $\square$

Theorem 21.146. Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a pre-measure then

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \text{ with } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{S} \text{ we have } \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \leq \sum_{i=0}^{\infty} \mu(A_i)$$

In other words μ is countable sub-additive.

Proof. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ be a sequence such that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{S}$. Define

$$\{B_i\}_{i \in \mathbb{N}} \text{ by } B_i = \begin{cases} A_1 & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j & \text{if } i \in \{2, \dots, \infty\} \end{cases}$$

then by [lemma: 21.6]

$$\{B_i\}_{i \in \mathbb{N}} \text{ is pairwise disjoint and } \bigcup_{i \in \mathbb{N}} A_i = \bigsqcup_{i \in \mathbb{N}} B_i \text{ and } \forall i \in \mathbb{N} \ B_i \subseteq A_i \quad (21.181)$$

Note that the sets B_i are not elements of $\mathcal{S}$, however we will show that each B_i is a finite pairwise disjoint union of sets in $\mathcal{S}$ so that we can write $\bigcup_{i \in \mathbb{N}} A_i$ as countable pairwise disjoint union of elements of $\mathcal{S}$, allowing use to apply countable additivity. Let $i \in \mathbb{N}$ then we have either:

$i = 1$. Define $n_1 = 1$ and $\{C_{1,j}\}_{j \in \{1, \dots, n_1\}} \subseteq \mathcal{S}$ by $C_{1,1} = A_1 \in \mathcal{S}$ then $\{C_{1,j}\}_{j \in \{1, \dots, n_1\}} \subseteq \mathcal{S}$ is pairwise disjoint and $B_1 = A_1 = \bigsqcup_{j \in \{1, \dots, n_1\}} C_{1,j}$

$i \in \{2, \dots, \infty\}$. By [lemma: 21.142] there exist a pairwise disjoint family $\{C_{i,j}\}_{j \in \{1, \dots, n_i\}} \subseteq \mathcal{S}$ such that $B_i = A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j = \bigsqcup_{j \in \{1, \dots, n_i\}} C_{i,j}$.

Hence we have

$$\forall i \in \mathbb{N} \ \exists \{C_{i,j}\}_{j \in \{1, \dots, n_i\}} \subseteq \mathcal{S} \text{ pairwise disjoint such that } B_i = \bigsqcup_{j \in \{1, \dots, n_i\}} C_{i,j} \quad (21.182)$$

By [corollary: 6.80] it follows that $\bigcup_{i \in \mathbb{N}} \{i\} \times \{1, \dots, n_i\}$ is denumerable, hence there exist a bijection

$$\beta: \mathbb{N} \rightarrow \bigcup_{i \in \mathbb{N}} \{i\} \times \{1, \dots, n_i\}$$

Using the above and [eq: 21.182] allows us to use [corollary: 21.12] giving

$$\{C_{\beta(i)_1, \beta(i)_2}\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \text{ is pairwise disjoint and } \bigcup_{i \in \mathbb{N}} A_i \stackrel{[\text{eq: 21.181}]}{=} \bigsqcup_{i \in \mathbb{N}} B_i = \bigsqcup_{i \in \mathbb{N}} C_{\beta(i)_1, \beta(i)_2} \quad (21.183)$$

So we have, as $\bigsqcup_{i \in \mathbb{N}} C_{\beta(i)_1, \beta(i)_2} = \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{S}$, that

$$\begin{aligned} \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eq: 21.183}]}{=} \mu\left(\bigsqcup_{i \in \mathbb{N}} C_{\beta(i)_1, \beta(i)_2}\right) \\ &\stackrel{\mu \text{ is a pre-measure}}{=} \sum_{i=1}^{\infty} \mu(C_{\beta(i)_1, \beta(i)_2}) \\ &\stackrel{[\text{theorem: 20.107}]}{=} \sum_{i=1}^{\infty} \left(\sum_{j=1}^{n_i} \mu(C_{i,j}) \right) \end{aligned} \quad (21.184)$$

As $\forall i \in \mathbb{N} \bigcup_{j \in \{1, \dots, n_i\}} C_{i,j} \stackrel{=}{=} B_i \subseteq A_i$ it follows from [corollary: 21.145(1)] and [eq: 21.183] that $\sum_{j=1}^{n_i} \mu(C_{i,j}) \leq \mu(A_i)$, which combined with [eq: 21.184] and [theorem: 20.99] gives

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i) \quad \square$$

We can now combine the above theorems to prove a version of [theorem: 21.134] for semi-rings and pre-measures.

Theorem 21.147. *Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a pre-measure on $\mathcal{S}$ then*

$$\mu^*: \mathcal{P}(X) \rightarrow [0, \infty] \text{ defined by } \mu^*(A) = \begin{cases} \inf(\{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}) & \text{if } \mathcal{S}[A] \neq \emptyset \\ \infty & \text{if } \mathcal{S}[A] = \emptyset \end{cases}$$

where

$$\mathcal{S}[A] = \left\{ \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \mid A \subseteq \bigcup_{i \in \mathbb{N}} A_i \right\}$$

is a outer measure and

$$\forall A \in \mathcal{S} \text{ we have } \mu^*(A) = \mu(A)$$

so that

$$\mu^* \text{ is a extension of } \mu$$

Proof. As μ is a pre-measure [see definition: 21.140] we have that $\mu(\emptyset) = 0$ hence

$$\mu \text{ is a proto-measure on } \mathcal{S}$$

Using [theorems: 21.145, 21.146] it follows that

$$\begin{aligned} & \forall A, B \in \mathcal{S} \text{ with } A \subseteq B \text{ we have } \mu(A) \leq \mu(B) \\ & \forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \text{ such that } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{S} \text{ we have } \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i) \end{aligned}$$

and as $\mathcal{S}$ is a semi-ring we have

$$\forall A, B \in \mathcal{S} \text{ we have } A \cap B \in \mathcal{S}$$

So all conditions for [theorem: 21.134] are satisfies so that

$$\mu^* \text{ is a outer measure on } S \text{ and } \forall A \in \mathcal{S} \text{ we have } \mu^*(A) = \mu(A) \quad \square$$

In the above proof we use the fact that a pre-measure is countable sub-additive [see theorem 21.146] but never the countable additivity that is part of the definition of a pre-measure. So why do we define a pre-measure to be countable additive and not just countable sub-additive? The answer lies in proving the existence of a **unique extension** of a pre-measure on a semi-ring $\mathcal{S}$ to a measure on $\sigma[\mathcal{S}]$. To extend the pre-measure μ on $\mathcal{S}$ we first use the previous theorem [theorem: 21.134] to create a outer-measure μ^* that extend μ , next we use the Carathéodory theorem [theorem: 21.122] to find a measure on $\mathcal{M}[\mu^*]$. However to make sure that the constructed measure is also a measure on $\sigma[\mathcal{S}]$ we still have to prove that $\sigma[\mathcal{S}] \subseteq \mathcal{M}[\mu^*]$ and this is where we need additivity of μ . The countable additivity of a pre-measure ensures additional the additivity and countable sub-additivity of the pre-measure. Another way to look at this is the following, if μ on $\mathcal{S}$ is extended to a countable additive measure on $\sigma[\mathcal{S}]$ then we must have, as $\mathcal{S} \subseteq \sigma[\mathcal{S}]$, that μ is also countable additive.

Theorem 21.148. *Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a pre-measure on $\mathcal{S}$ and define*

$$\mu^*: \mathcal{P}(X) \rightarrow [0, \infty] \text{ by } \mu^*(A) = \begin{cases} \infty & \text{if } \mathcal{S}[A] = \emptyset \\ \inf(\{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}) & \text{if } \mathcal{S}[A] \neq \emptyset \end{cases}$$

where

$$\mathcal{S}[A] = \left\{ \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \mid A \subseteq \bigcup_{i \in \mathbb{N}} A_i \right\}$$

and

$$\mathcal{M}[\mu^*] = \{A \in \mathcal{P}(X) \mid \forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A)\}$$

then

μ^* is a outer measure

$\mathcal{M}[\mu^*]$ is a σ -algebra on X

and

$$(\mu^*)|_{\mathcal{S}} = \mu \text{ [in other words } \mu^* \text{ extends } \mu]$$

and

$$(\mu^*)|_{\mathcal{M}[\mu^*]}: \mathcal{M}[\mu^*] \rightarrow [0, \infty] \text{ is a measure on } \mathcal{M}[\mu^*]$$

and

$$\mathcal{S} \subseteq \mathcal{M}[\mu^*]$$

so that by [theorem: 21.42]

$$\sigma[\mathcal{S}] \subseteq \mathcal{M}[\mu^*]$$

and by [theorem: 21.110]

$$(\mu^*)|_{\sigma[\mathcal{S}]}: \sigma[\mathcal{S}] \rightarrow [0, \infty] \text{ is a measure on } \sigma[\mathcal{S}]$$

Hence

$$\langle X, \mathcal{M}[\mu^*], (\mu^*)|_{\mathcal{M}[\mu^*]} \rangle \text{ and } \langle X, \sigma[\mathcal{S}], (\mu^*)|_{\sigma[\mathcal{S}]} \rangle \text{ are measure spaces}$$

where

$$\forall A \in \mathcal{S} \text{ we have } \mu(A) = \mu^*(A) = (\mu^*)|_{\mathcal{M}[\mu^*]}(A) = (\mu^*)|_{\sigma[\mathcal{S}]}(A)$$

If additional there exist a $\{G_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ such that $\forall i \in \mathbb{N} \ G_i \subseteq G_{i+1} \wedge \mu(G_i) < \infty$ and $X = \bigcup_{i \in \mathbb{N}} G_i$ then if $\nu: \sigma[\mathcal{S}] \rightarrow [0, \infty]$ is a another measure on $\sigma[\mathcal{S}]$ such that $\nu|_{\mathcal{S}} = (\mu^*)|_{\mathcal{S}}$ then

$$\nu = (\mu^*)|_{\sigma[\mathcal{S}]}$$

In other words the extension of μ to a measure $(\mu^*)|_{\sigma[\mathcal{S}]}$ on $\sigma[\mathcal{S}]$ is unique.

Note 21.149. We need additivity on μ to prove that $\mathcal{S} \subseteq \mathcal{M}[\mu^*]$ which is essential for the last part of the theorem.

Proof. Using [theorem: 21.147] we have

$$\mu^*: \mathcal{P}(X) \rightarrow [0, \infty] \text{ defined by } \mu^*(A) = \begin{cases} \inf(\{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}) & \text{if } \mathcal{S}[A] \neq \emptyset \\ \infty & \text{if } \mathcal{S}[A] = \emptyset \end{cases} \quad (21.185)$$

where

$$\mathcal{S}[A] = \left\{ \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S} \mid A \subseteq \bigcup_{i \in \mathbb{N}} A_i \right\} \quad (21.186)$$

is a outer measure and

$$\forall A \in \mathcal{S} \text{ we have } \mu^*(A) = \mu(A) \text{ hence } (\mu^*)|_{\mathcal{S}} = \mu \quad (21.187)$$

As μ^* is a outer measure we can use the Carathéodory [theorem: 21.122] so that

$$\mathcal{M}[\mu^*] = \{A \in \mathcal{P}(X) \mid \forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A)\}$$

is a σ -algebra on X and

$$(\mu^*)|_{\mathcal{M}[\mu^*]}: \mathcal{M}[\mu^*] \rightarrow [0, \infty] \text{ is a measure on the } \sigma\text{-algebra } \mathcal{M}[\mu^*] \quad (21.188)$$

so that

$$\langle X, \mathcal{M}[\mu^*], (\mu^*)|_{\mathcal{M}[\mu^*]} \rangle \text{ a measure space} \quad (21.189)$$

Next we have to prove that $\mathcal{S} \subseteq \mathcal{M}[\mu^*]$. Let $R, T \in \mathcal{S}$ then as $\mathcal{S}$ is a semi-ring there exist a $m \in \mathbb{N}$, $\{S_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$ pairwise disjoint such that

$$T \setminus R = \bigsqcup_{i \in \{1, \dots, m\}} S_i \text{ and } R \cap T \in \mathcal{S} \quad (21.190)$$

Define now

$$\{D_i\}_{i \in \{1, \dots, m+1\}} \subseteq \mathcal{S} \text{ by } D_i = \begin{cases} S_i & \text{if } i \in \{1, \dots, m\} \\ R \cap T & \text{if } i = m+1 \end{cases} \in \mathcal{S}$$

Let $i, j \in \{1, \dots, m+1\}$ with $i \neq j$ then we may always assume that $i < j$ [otherwise exchange i and j], so that we have the following possibilities for j :

$j = m+1$. Then, as $i < j$, $i \in \{1, \dots, m\}$ so that

$$\begin{aligned} D_i \cap D_j &= S_i \cap (R \cap T) \\ &\subseteq \left(\bigsqcup_{i \in \{1, \dots, m\}} S_i \right) \cap (R \cap T) \\ &\stackrel{[\text{eq: 21.190}]}{=} (T \setminus R) \cap (R \cap T) \\ &\subseteq (T \setminus R) \cap R \\ &= \emptyset \end{aligned}$$

hence we have $D_i \cap D_j = \emptyset$.

$j \in \{1, \dots, m\}$. Then, as $i < j$, $i \in \{1, \dots, m\}$ so that $D_i \cap D_j = S_i \cap S_j = \emptyset$

which proves that

$$\{D_i\}_{i \in \{1, \dots, m+1\}} \subseteq \mathcal{S} \text{ is pairwise disjoint}$$

Further

$$\begin{aligned} T &= (T \setminus R) \cup (T \cap R) \\ &= \left(\bigsqcup_{i \in \{1, \dots, m\}} S_i \right) \cup (T \cap R) \\ &= \left(\bigsqcup_{i \in \{1, \dots, m\}} D_i \right) \cup D_{m+1} \\ &= \bigsqcup_{i \in \{1, \dots, m+1\}} D_i \end{aligned}$$

Hence

$$\bigsqcup_{i \in \{1, \dots, m+1\}} D_i = T \in \mathcal{S} \quad (21.191)$$

As by [theorem: 21.143] μ is additive on $\mathcal{S}$ we have

$$\begin{aligned} \mu(T) &\stackrel{[\text{eq: 21.191}]}{=} \mu\left(\bigsqcup_{i \in \{1, \dots, m+1\}} D_i\right) \\ &\stackrel{\mu \text{ is additive}}{=} \sum_{i=1}^{m+1} \mu(D_i) \\ &= \mu(D_{m+1}) + \sum_{i=1}^m \mu(D_i) \\ &= \mu(R \cap T) + \sum_{i=1}^m \mu(S_i) \end{aligned}$$

hence

$$\mu(T) = \mu(R \cap T) + \sum_{i=1}^m \mu(S_i) \quad (21.192)$$

Further as a outer measure is sub-additive [see definition: 21.117 and theorem: 21.23] we have

$$\begin{aligned}
 \mu^*(R \cap T) + \mu^*(T \setminus R) & \stackrel{[\text{eq: 21.190}]}{=} \mu^*(R \cap T) + \mu^*\left(\bigsqcup_{i \in \{1, \dots, m\}} S_i\right) \\
 & \stackrel{\text{sub-additivity of } \mu^*}{\leq} \mu^*(R \cap T) + \sum_{i=1}^m \mu^*(S_i) \\
 & \stackrel{[\text{eq: 21.187}]}{=} \mu(R \cap T) + \sum_{i=1}^m \mu(S_i) \\
 & \stackrel{[\text{eq: 21.192}]}{=} \mu(T)
 \end{aligned}$$

To summarize we have proved the following:

$$\forall R, S \in \mathcal{S} \text{ we have that } \mu^*(R \cap T) + \mu^*(T \setminus R) \leq \mu(T) \quad (21.193)$$

Let $A \in \mathcal{S}$ and take $B \in \mathcal{P}(X)$. For B we have either:

$\mathcal{S}[B] = \emptyset$. Then $\mu^*(B \setminus A) + \mu^*(B \cap A) \leq \infty = \mu^*(B)$

$\mathcal{S}[B] \neq \emptyset$. Let $\{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B]$ then we have

$$\begin{aligned}
 \sum_{i=1}^{\infty} \mu^*(A_i \cap A) + \sum_{i=1}^{\infty} \mu^*(A_i \setminus A) & \stackrel{[\text{theorem: 20.99}]}{=} \\
 \sum_{i=1}^{\infty} (\mu^*(A_i \cap A) + \mu^*(A_i \setminus A)) & \stackrel{[\text{theorem: 20.99}] \text{ and } [\text{eq: 21.193}]}{\leq} \sum_{i=1}^{\infty} \mu(A_i)
 \end{aligned} \quad (21.194)$$

As $\{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B]$ we have $B \subseteq \bigcup_{i \in \mathbb{N}} A_i$ so that

$$B \setminus A \subseteq \left(\bigcup_{i \in \mathbb{N}} A_i \right) \setminus A = \bigcup_{i \in \mathbb{N}} (A_i \setminus A) \text{ and } B \cap A \subseteq \left(\bigcup_{i \in \mathbb{N}} A_i \right) \cap A = \bigcup_{i \in \mathbb{N}} (A_i \cap A)$$

As μ^* is a outer measure hence monotone by definition [see definition: 21.117(2)] it follows from the above that

$$\mu^*(B \setminus A) \leq \mu^*\left(\bigcup_{i \in \mathbb{N}} (A_i \setminus A)\right) \text{ and } \mu^*(B \cap A) \leq \mu^*\left(\bigcup_{i \in \mathbb{N}} (A_i \cap A)\right) \quad (21.195)$$

Further using the sub-additivity of a outer measure [see definition: 21.117(3)] we have

$$\mu^*\left(\bigcup_{i \in \mathbb{N}} (A_i \setminus A)\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i \setminus A) \text{ and } \mu^*\left(\bigcup_{i \in \mathbb{N}} (A_i \cap A)\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i \cap A)$$

which combined with [eq: 21.195] gives

$$\mu^*(B \setminus A) \leq \sum_{i=1}^{\infty} \mu^*(A_i \setminus A) \text{ and } \mu^*(B \cap A) \leq \sum_{i=1}^{\infty} \mu^*(A_i \cap A)$$

so that

$$\begin{aligned}
 \mu^*(B \setminus A) + \mu^*(B \cap A) & \leq \sum_{i=1}^{\infty} \mu^*(A_i \setminus A) + \sum_{i=1}^{\infty} \mu^*(A_i \cap A) \\
 & \stackrel{[\text{eq: 21.194}]}{\leq} \sum_{i=1}^{\infty} \mu(A_i)
 \end{aligned}$$

So $\mu^*(B \setminus A) + \mu^*(B \cap A)$ is a lower bound for $\{\sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B]\}$ so that

$$\mu^*(B \setminus A) + \mu^*(B \cap A) \leq \inf \left(\left\{ \sum_{i=1}^{\infty} \mu(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[B] \right\} \right) \stackrel{[\text{eq: 21.185}]}{=} \mu^*(B)$$

So in all cases we have that $\mu^*(B \setminus A) + \mu^*(B \cap A) \leq \mu^*(B)$ which by [theorem: 21.120] means that $A \in \mathcal{M}[\mu^*]$. As $A \in \mathcal{S}$ was chosen arbitrary we conclude that

$$\mathcal{S} \subseteq \mathcal{M}[\mu^*] \quad (21.196)$$

As $\mathcal{M}[\mu^*]$ is a σ -algebra it follows from [theorem: 21.42] that

$$\mathcal{S} \subseteq \sigma[\mathcal{S}] \subseteq \mathcal{M}[\mu^*] \quad (21.197)$$

From the above, [eq: 21.188] and [theorem: 21.110] it follows that

$$(\mu^*)|_{\sigma[\mathcal{S}]} \stackrel{[theorem: 2.92 (3)]}{=} ((\mu^*)|_{\mathcal{M}[\mu^*]})|_{\sigma[\mathcal{S}]} \text{ is a measure on } \sigma[\mathcal{S}]$$

making

$$\langle X, \sigma[\mathcal{S}], (\mu^*)|_{\sigma[\mathcal{S}]} \rangle \text{ a measure space}$$

Finally assume additional that there exist a $\{G_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ such that $\forall i \in \mathbb{N} \ G_i \subseteq G_{i+1} \wedge \mu(G_i) < \infty$. Let $\nu: \sigma[\mathcal{S}] \rightarrow [0, \infty]$ be a measure such that $\forall A \in \mathcal{S}$ we have $\mu(A) = \nu(A)$ then as $(\mu^*)|_{\sigma[\mathcal{S}]}(A) = \mu^*(A) \stackrel{[eq: 21.187]}{=} \mu(A)$ it follows that $\forall A \in \mathcal{S} \ (\mu^*)|_{\sigma[\mathcal{S}]}(A) = \nu(A)$. Hence by [theorem: 21.131] it follows that

$$(\mu^*)|_{\sigma[\mathcal{S}]} = \nu$$

proving uniqueness. $\square$

21.3.4.2 Contents and pre-measures

So given a pre-measure μ on a semi-ring $\mathcal{S}$ we can extend μ to a measure on $\sigma[\mathcal{S}]$ and from there to a measure on $\mathcal{M}[\mathcal{S}]$. In the next section we show a way of proving that a additive set-function on a ring [see definition: 21.24] can be extended [if certain conditions are satisfied] to a pre-measure. First we show that every ring is actually a semi-ring.

Theorem 21.150. *Let X be a set, $\mathcal{R} \subseteq \mathcal{P}(X)$ a ring [see definition: 21.24] then $\mathcal{R}$ is a semi-ring [see definition: 21.137].*

Proof. As $\mathcal{R}$ is a ring we have by [theorem: 21.27] that

$$\emptyset \in \mathcal{R} \text{ and } \forall A, B \in \mathcal{R} \text{ we have } A \cap B \in \mathcal{R}$$

Further if $A, B \in \mathcal{R}$ then by the definition of a ring [see definition: 21.24] $A \setminus B \in \mathcal{R}$. So if we define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{R}$ by $A_1 = A \setminus B$ we have that $A \setminus B = \bigsqcup_{i \in \{1\}} A_i$. Hence we have proved that

$$\forall A, B \in \mathcal{R} \ \exists \{A_i\}_{i \in \{1, \dots, 1\}} \subseteq \mathcal{R} \text{ pairwise disjoint such that } A \setminus B = \bigsqcup_{i \in \{1, \dots, 1\}} A_i \quad \square$$

Next we define what a **content** is.

Definition 21.151. *Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring then $\mu: \mathcal{S} \rightarrow [0, \infty]$ is a **content** on $\mathcal{S}$ if*

1. $\mu(\emptyset) = 0$
2. $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ pairwise disjoint such that $\bigsqcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{S}$ we have $\mu(\bigsqcup_{i \in \{1, \dots, n\}} A_i) = \sum_{i=1}^n \mu(A_i)$.

A content is **finite** if $\forall A \in \mathcal{S} \ \mu(A) < \infty$.

Note 21.152. *If $\mathcal{S}$ is a ring then (2) can be replaced by*

$$\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}, \ n \in \mathbb{N} \text{ we have } \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

because [theorem: 21.27] ensures that $\bigsqcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{S}$.

Lemma 21.153. *Let X be a set $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring then a pre-measure $\mu: \mathcal{S} \rightarrow [0, \infty]$ is a content*

Proof. From the definition of a pre-measure [see definition: 21.140] it follows that

$$\mu(\emptyset) = 0$$

Further $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}, n \in \mathbb{N}$ pairwise disjoint such that $\bigsqcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{S}$ we have by [theorem: 21.143] that

$$\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

□

We have a simpler equivalent definition of a content on a ring.

Theorem 21.154. *Let X be a set, $\mathcal{R}$ a ring on X and $\mu: \mathcal{R} \rightarrow [0, \infty]$ a function such that $\mu(\emptyset) = 0$ [in other words μ is a proto-measure] then*

$$\mu \text{ is a content} \Leftrightarrow \forall A, B \in \mathcal{R} \text{ with } A \cap B = \emptyset \text{ we have } \mu(A \sqcup B) = \mu(A) + \mu(B)$$

Proof.

$\Rightarrow$. Let $A, B \in \mathcal{R}$ with $A \cap B = \emptyset$ and define $\{A_i\}_{i \in \{1, 2\}} \subseteq \mathcal{R}$ by $A_1 = A$ and $A_2 = B$, then $\{A_i\}_{i \in \{1, 2\}}$ is pairwise disjoint and $\mu(A \sqcup B) = \mu(\bigcup_{i \in \{1, 2\}} A_i) = \sum_{i=1}^2 \mu(A_i) = \mu(A) + \mu(B)$.

$\Leftarrow$. We prove this by induction so let

$$\mathcal{S} = \left\{ n \in \mathbb{N} \mid \text{if } \{A_i\}_{i \in \{1, \dots, n\}} \text{ is a pairwise disjoint family of sets in } \mathcal{R} \text{ then } \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i) \right\}$$

then we have

$1 \in \mathcal{S}$. If $\{A_i\}_{i \in \{1, \dots, 1\}}$ is a pairwise disjoint family of sets in $\mathcal{R}$ then

$$\mu\left(\bigsqcup_{i \in \{1, \dots, 1\}} A_i\right) = \mu(A_1) = \sum_{i=1}^1 \mu(A_i)$$

proving that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. If $\{A_i\}_{i \in \{1, \dots, n+1\}}$ is a pairwise disjoint family of sets in $\mathcal{R}$ then we have $\bigsqcup_{i \in \{1, \dots, n+1\}} A_i = (\bigsqcup_{i \in \{1, \dots, n\}} A_i) \sqcup A_{n+1}$ so that

$$\begin{aligned} \mu\left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i\right) &= \mu\left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i\right) \\ &= \mu\left(\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \sqcup A_{n+1}\right) \\ &= \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) + \mu(A_{n+1}) \\ &\stackrel{n \in \mathcal{S}}{=} \left(\sum_{i=1}^n \mu(A_i)\right) + \mu(A_{n+1}) \\ &= \sum_{i=1}^{n+1} \mu(A_i) \end{aligned}$$

proving that $n+1 \in \mathcal{S}$. □

A content has the following properties:

Theorem 21.155. *Let X be a set, $\mathcal{R}$ a ring on X and $\mu: \mathcal{R} \rightarrow [0, \infty]$ a content on $\mathcal{R}$ then we have:*

1. $\forall A, B \in \mathcal{R}$ we have $\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$
2. $\forall A, B \in \mathcal{R}$ with $A \subseteq B$ we have $\mu(A) \leq \mu(B)$ [μ is monotone].
3. $\forall A, B \in \mathcal{R}$ with $A \subseteq B$ and $\mu(A) < \infty$ we have $\mu(B \setminus A) = \mu(B) - \mu(A)$.
4. $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R}$, $n \in \mathbb{N}$ we have $\mu(\bigcup_{i \in \{1, \dots, n\}} A_i) \leq \sum_{i=1}^n \mu(A_i)$ [sub-additivity]
5. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ pairwise disjoint such that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{R}$ we have $\sum_{i=1}^{\infty} \mu(A_i) \leq \mu(\bigcup_{i \in \mathbb{N}} A_i)$.

Proof. First as $\mathcal{R}$ is a ring we have that $\forall A, B \in \mathcal{R}$ $B \setminus A \in \mathcal{R}$, further

1. As $A \cup (B \setminus A) \stackrel{[\text{theorem: 1.39}]}{=} A \cup B$ with $A \cap (B \setminus A) \stackrel{[\text{theorem: 1.39}]}{=} \emptyset$ and μ is a content we have

$$\mu(A \cup B) = \mu(A) + \mu(B \setminus A) \quad (21.198)$$

Further $B \stackrel{[\text{theorem: 1.39}]}{=} (A \cap B) \cup (B \setminus A)$ with $(A \cap B) \cap (B \setminus A) \stackrel{[\text{theorem: 1.39}]}{=} \emptyset$ so that

$$\mu(B) = \mu(A \cap B) + \mu(B \setminus A) \quad (21.199)$$

For $\mu(B \setminus A)$ we have either:

$\mu(B \setminus A) < \infty$. As μ is non negative we can sum [eq: 21.198] and [eq: 21.199] giving

$$\mu(A \cup B) + \mu(A \cap B) + \mu(B \setminus A) = \mu(B) + \mu(A) + \mu(B \setminus A)$$

and we can use [theorem: 20.28] to get

$$\mu(A \cup B) + \mu(A \cap B) = \mu(B) + \mu(A)$$

$\mu(B \setminus A) = \infty$. Then

$$\mu(A \cup B) \stackrel{[\text{eq: 21.198}]}{=} \mu(A) + \mu(B \setminus A) = \mu(A) + \infty = \infty$$

and

$$\mu(B) \stackrel{[\text{eq: 21.199}]}{=} \mu(A \cap B) + \mu(B \setminus A) = \mu(A \cap B) + \infty = \infty$$

so that

$$\mu(A \cup B) + \mu(A \cap B) = \infty + \mu(A \cap B) = \infty = \mu(A) + \infty = \mu(A) + \mu(B)$$

So in all cases we have

$$\mu(A \cup B) + \mu(A \cap B) = \mu(B) + \mu(A)$$

2. If $A, B \in \mathcal{R}$ with $A \subseteq B$ then $A \cup B = B$ so that

$$\mu(B) = \mu(A \cup B) \stackrel{[\text{eq: 21.198}]}{=} \mu(A) + \mu(B \setminus A) \quad (21.200)$$

then by [theorem: 20.35] it follows that

$$\mu(A) \leq \mu(B)$$

3. As $A \subseteq B$ we have by [eq: 21.200] that $\mu(B) = \mu(A) + \mu(B \setminus A)$ so as $\mu(A) < \infty$ we can safely add $-\mu(A)$ to both sides giving

$$\mu(B) - \mu(A) = \mu(A) + \mu(B \setminus A) - \mu(A) = \mu(B \setminus A)$$

4. Let $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R}$ and define

$$\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R} \text{ by } B_i = \begin{cases} A_1 & \text{if } i = 1 \\ A_i \setminus \bigcup_{j \in \{1, \dots, i-1\}} A_j & \text{if } i > 1 \end{cases} \in \mathcal{R} \text{ as } \mathcal{R} \text{ is a ring and } [\text{theorem: 21.27}]$$

so that

$$\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R} \text{ and } \forall i \in \{1, \dots, n\} \ B_i \subseteq A_i \stackrel{(2)}{\Rightarrow} \mu(B_i) \leq \mu(A_i) \quad (21.201)$$

Using [lemma: 21.6] we have that

$$\{B_i\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint and } \bigcup_{i \in \{1, \dots, n\}} A_i = \bigsqcup_{i \in \{1, \dots, n\}} B_i \quad (21.202)$$

Hence

$$\begin{aligned} \mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) &\stackrel{[\text{eq: 21.202}]}{=} \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} B_i\right) \\ &\stackrel{[\text{definition: 21.151}]}{=} \sum_{i=1}^n \mu(B_i) \\ &\stackrel{[\text{eq: 21.201}]}{\leq} \sum_{i=1}^n \mu(A_i) \end{aligned}$$

5. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ be a pairwise disjoint sequence of sets in $\mathcal{R}$ such that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{R}$. Let $n \in \mathbb{N}$ then we have $\bigsqcup_{i \in \{1, \dots, n\}} A_i \subseteq \bigsqcup_{i \in \mathbb{N}} A_i$ so that

$$\sum_{i=1}^n \mu(A_i) \stackrel{[\text{definition: 21.151}]}{=} \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \stackrel{(2)}{\leq} \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \quad (21.203)$$

Hence $\mu(\bigcup_{i \in \mathbb{N}} A_i)$ is an upper bound of $\{\sum_{i=1}^n \mu(A_i) | n \in \mathbb{N}\}$ so that

$$\sup\left(\left\{\sum_{i=1}^n \mu(A_i) | n \in \mathbb{N}\right\}\right) \leq \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

Hence

$$\sum_{i=1}^{\infty} \mu(A_i) \stackrel{[\text{definition: 20.95}]}{=} \sup\left(\left\{\sum_{i=1}^n \mu(A_i) | n \in \mathbb{N}\right\}\right) \leq \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

□

Similar to [theorems: 21.115, 21.116] we have the following necessary and sufficient conditions for a content to be a pre-measure.

Theorem 21.156. *Let X be a set, $\mathcal{R} \subseteq X$ a ring on X and $\mu: \mathcal{R} \rightarrow [0, \infty]$ a content then for the following statements:*

1. μ is a pre-measure
2. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ such that $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}$ and $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{R}$ we have $\mu(\bigcup_{i \in \mathbb{N}} A_i) = \lim_{i \rightarrow \infty} \mu(A_i)$.
3. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ such that $\forall i \in \mathbb{N} A_{i+1} \subseteq A_i$ and $\mu(A_i) < \infty$ and $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{R}$ we have $\mu(\bigcap_{i \in \mathbb{N}} A_i) = \lim_{i \rightarrow \infty} \mu(A_i)$.
4. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ such that $\forall i \in \mathbb{N} A_{i+1} \subseteq A_i$ and $\mu(A_i) < \infty$ and $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$ we have $\lim_{i \rightarrow \infty} \mu(A_i) = 0$.

we have $(1) \Leftrightarrow (2) \Rightarrow (3) \Leftrightarrow (4)$, Further if μ is finite then $(1) \Leftrightarrow (2) \Leftrightarrow (3) \Leftrightarrow (4)$

Proof. First as $\mathcal{R}$ is a ring we have by [theorem: 21.27] that

$$\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{R} \quad \bigcup_{i \in \{1, \dots, n\}} A_i \in \mathcal{R} \text{ and } \bigcap_{i \in \{1, \dots, n\}} A_i \in \mathcal{R} \quad (21.204)$$

(1) $\Rightarrow$ (2). Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ be such that $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}$ and $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{R}$. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R} \text{ by } B_i = \begin{cases} A_1 & \text{if } i = 1 \\ A_i \setminus A_{i-1} & \text{if } i \in \{2, \dots, \infty\} \end{cases} \in \mathcal{R} \text{ } \mathcal{R} \text{ is a ring}$$

Then by the above definition we have

$$\forall i \in \mathbb{N} \quad B_i \subseteq A_i \quad (21.205)$$

Let $i, j \in \mathbb{N}$ with $i \neq j$. We may always assume that $i < j$ [otherwise exchange i and j], then for i we have either

$i = 1$. Then $1 < j$ so that $B_1 = A_1$ and $B_j = A_j \setminus A_{j-1}$. As $1 < j \Rightarrow 1 \leq j-1$ we have $A_1 \subseteq A_{j-1}$ so that by [theorem: 1.36] $B_j = A_j \setminus A_{j-1} \subseteq A_j \setminus A_1 = A_j \setminus B_1$. Hence

$$B_i \cap B_j \subseteq B_1 \cap (A_j \setminus B_1) = \emptyset$$

$i \in \{2, \dots, \infty\}$. Then $i, j \in \{2, \dots, \infty\}$ so that $B_i = A_i \setminus A_{i-1} \wedge B_j = A_j \setminus A_{j-1}$. As $i < j \Rightarrow i \leq j-1$ we have $B_i \subseteq A_i \subseteq A_{j-1}$ so that by [theorem: 1.36] $B_j = A_j \setminus A_{j-1} \subseteq A_j \setminus B_i$. Hence

$$B_j \cap B_i \subseteq (A_j \setminus B_i) \cap B_i = \emptyset$$

So it follows that

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R} \text{ is pairwise disjoint} \quad (21.206)$$

Let $n \in \mathbb{N}$. As $\forall i \in \{1, \dots, n\}$ we have $B_i \subseteq A_i \subseteq A_n$ from which it follows that [eq: 21.205]

$$\bigsqcup_{i \in \{1, \dots, n\}} B_i \subseteq A_n \quad (21.207)$$

If $x \in A_n$ then $n \in \{i \in \{1, \dots, n\} | x \in A_i\}$ a finite set so that $j = \min(\{i \in \{1, \dots, n\} | x \in A_i\})$ exists. For j we have either:

$j = 1$. Then $x \in A_1 = B_1$ so that $x \in \bigsqcup_{i \in \{1, \dots, n\}} B_i$

$1 < j$. Then $x \in A_j$ and $x \notin A_{j-1}$ so that $x \in A_j \setminus A_{j-1} = B_j \subseteq \bigsqcup_{i \in \{1, \dots, n\}} B_i$

hence $A_n \subseteq \bigsqcup_{i \in \{1, \dots, n\}} B_i$ which combined with [eq: 21.207] we have

$$\forall n \in \mathbb{N} \text{ we have } A_n = \bigsqcup_{i \in \{1, \dots, n\}} B_i \quad (21.208)$$

Let $x \in \bigcup_{n \in \mathbb{N}} A_n$ then $\exists n \in \mathbb{N}$ so that $x \in A_n$ and by [eq: 21.208] $\exists m \in \{1, \dots, n\} \subseteq \mathbb{N}$ such that $x \in B_m$, hence $\bigcup_{n \in \mathbb{N}} A_n \subseteq \bigcup_{i \in \mathbb{N}} B_i$. As also $B_n \subseteq A_n \subseteq \bigcup_{n \in \mathbb{N}} A_n$ it follows that [eq: 21.205]

$$\bigsqcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{S} \quad (21.209)$$

As μ is a pre-measure hence countable additive we have that

$$\begin{aligned} \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) &= \mu\left(\bigsqcup_{i \in \mathbb{N}} B_i\right) \\ &= \sum_{i=1}^{\infty} \mu(B_i) \\ &\stackrel{[\text{definition: 20.95}]}{=} \lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(B_j) \\ &\stackrel{[\text{theorem: 21.143}]}{=} \lim_{i \rightarrow \infty} \mu\left(\bigcup_{j=1}^i B_j\right) \\ &\stackrel{[\text{eq: 21.208}]}{=} \lim_{i \rightarrow \infty} \mu(A_i) \end{aligned}$$

Hence we have

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

(2) $\Rightarrow$ (1). As for a content $\mu(\emptyset) = 0$ we only have to prove countable additivity. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ be a pairwise disjoint sequence of sets in $\mathcal{R}$ such that $\bigcup_{i \in \mathbb{N}} A_i$. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R} \text{ by } B_i = \bigsqcup_{j \in \{1, \dots, i\}} A_j$$

Then we have

$$\forall i \in \mathbb{N} \quad B_i = \bigsqcup_{j \in \{1, \dots, i\}} A_j \subseteq \left(\bigsqcup_{j \in \{1, \dots, i\}} A_j \right) \cup A_{i+1} = \bigsqcup_{j \in \{1, \dots, i+1\}} A_j = B_{i+1} \quad (21.210)$$

If $x \in \bigsqcup_{i \in \mathbb{N}} A_i$ then $\exists i \in \mathbb{N}$ such that $x \in A_i \subseteq \bigsqcup_{j \in \{1, \dots, i\}} A_j = B_i \subseteq \bigcup_{i \in \mathbb{N}} B_i$, hence $\bigcup_{i \in \mathbb{N}} A_i \subseteq \bigcup_{i \in \mathbb{N}} B_i$, as also $B_i = \bigcup_{j \in \{1, \dots, i\}} A_j \subseteq \bigcup_{j \in \mathbb{N}} A_j$ it follows that $\bigcup_{i \in \mathbb{N}} B_i \subseteq \bigcup_{j \in \mathbb{N}} A_j$. Hence

$$\bigcup_{i \in \mathbb{N}} B_i = \bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{R} \quad (21.211)$$

Using (2) together with [eqs: 21.210] it follows that

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \stackrel{(2)}{=} \lim_{i \rightarrow \infty} \mu(B_i) = \lim_{i \rightarrow \infty} \mu\left(\bigsqcup_{j \in \{1, \dots, i\}} A_j\right) \quad (21.212)$$

A μ is a content hence additive we have $\forall i \in \mathbb{N}$ that $\mu\left(\bigsqcup_{j \in \{1, \dots, i\}} A_j\right) = \sum_{j=1}^i \mu(A_j)$ so that by the above

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(A_j) \stackrel{[\text{definition: 20.95}]}{=} \sum_{i=1}^{\infty} \mu(A_i)$$

Finally using [eq: 21.211] it follows that $\bigcup_{i \in \mathbb{N}} B_i = \bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{R}$ so that

$$\mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

which proves countable additivity.

(2) $\Rightarrow$ (3). Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ such that $\forall i \in \mathbb{N} \quad \mu(A_i) < \infty$, $A_{i+1} \subseteq A_i$ and $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{R}$. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R} \text{ by } B_i = A_1 \setminus A_i \quad \begin{array}{l} \in \mathcal{R} \\ \mathcal{R} \text{ is a ring} \end{array}$$

Then we have

$$\forall i \in \mathbb{N} \quad B_i = A_1 \setminus A_i \stackrel{[\text{theorem: 1.36}]}{\subseteq} A_1 \setminus A_{i+1} = B_{i+1} \quad (21.213)$$

and

$$A_1 \setminus \left(\bigcap_{i \in \mathbb{N}} A_i \right) \stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{i \in \mathbb{N}} (A_1 \setminus A_i) = \bigcup_{i \in \mathbb{N}} B_i \quad (21.214)$$

As $\mathcal{R}$ is a ring and $A_1, \bigcap_{i \in \mathbb{N}} A_i \in \mathcal{R}$ we have that $\bigcup_{i \in \mathbb{N}} B_i = A_1 \setminus \left(\bigcap_{i \in \mathbb{N}} A_i \right) \in \mathcal{R}$. Hence we can apply (2)

$$\mu\left(A_1 \setminus \bigcap_{i \in \mathbb{N}} A_i\right) = \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \stackrel{(2)}{=} \lim_{i \rightarrow \infty} \mu(B_i) = \lim_{i \rightarrow \infty} \mu(A_1 \setminus A_i) \quad (21.215)$$

Let $i \in \mathbb{N}$ then $\bigcap_{i \in \mathbb{N}} A_i \subseteq A_i$ so that by [theorem: 21.155 (2)] $\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) \leq \mu(A_i) < \infty$, further we have $A_1 \setminus A_i \subseteq A_1 \Rightarrow \mu(A_1 \setminus A_i) \leq \mu(A_1) < \infty$. Hence using [theorem: 21.155(3)] we have

$$\forall i \in \mathbb{N} \text{ that } \mu(B_i) = \mu(A_1 \setminus A_i) = \mu(A_1) - \mu(A_i) \quad (21.216)$$

and

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \stackrel{[\text{eq: 21.214}]}{=} \mu\left(A_1 \setminus \bigcap_{i \in \mathbb{N}} A_i\right) = \mu(A_1) - \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) \quad (21.217)$$

So

$$\begin{aligned} \mu(A_1) - \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eqs: 21.215, 21.217}]}{=} \lim_{i \rightarrow \infty} \mu(A_1 \setminus A_i) \\ &\stackrel{[\text{eq: 21.216}]}{=} \lim_{i \rightarrow \infty} (\mu(A_1) - \mu(A_i)) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \mu(A_1) - \lim_{i \rightarrow \infty} \mu(A_i) \end{aligned}$$

As $\mu(A_1) < \infty \Rightarrow \mu(A_1) \in \mathbb{R}$ we can safely subtract $\mu(A)$ from both sides from the above equation to get $-\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = -\lim_{i \rightarrow \infty} \mu(A_i)$ from which it follows that

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

(3) $\Rightarrow$ (4). Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ be such that $\forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i$ and $\mu(A_i) < \infty$ and $\bigcap_{i \in \mathbb{N}} A_i = \emptyset \in \mathcal{S}$ then by (3) we have

$$0 = \mu(\emptyset) = \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

(4) $\Rightarrow$ (3). Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ be such that $\forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i$ and $\mu(A_i) < \infty$ and $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{R}$. Take $A = \bigcap_{i \in \mathbb{N}} A_i \in \mathcal{R}$ and define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R} \text{ by } B_i = A_i \setminus A \underset{\mathcal{R} \text{ is a ring}}{\in} \mathcal{R}$$

then we have

$$\forall i \in \mathbb{N} \text{ that } B_{i+1} = A_{i+1} \setminus A \subseteq A_i \setminus A = B_i$$

Further we have

$$\emptyset = A \setminus A = \left(\bigcap_{i \in \mathbb{N}} A_i\right) \setminus A \stackrel{[\text{theorem: 2.141}]}{=} \bigcap_{i \in \mathbb{N}} (A_i \setminus A) = \bigcap_{i \in \mathbb{N}} B_i$$

Hence using (4) we have

$$0 = \lim_{i \rightarrow \infty} \mu(B_i) = \lim_{i \rightarrow \infty} (\mu(A_i \setminus A)) \tag{21.218}$$

As $A = \bigcap_{i \in \mathbb{N}} A_i \subseteq A_i$ it follows from [theorem: 21.155(2)] that $\mu(A) \leq \mu(A_i) < \infty$, hence we have by [theorem: 21.155(3)] that

$$\forall i \in \mathbb{N} \text{ we have } \mu(A_i \setminus A) = \mu(A_i) - \mu(A) \tag{21.219}$$

Hence we have

$$\begin{aligned} 0 &\stackrel{[\text{eq: 21.218}]}{=} \lim_{i \rightarrow \infty} \mu(A_i \setminus A) \\ &\stackrel{[\text{eq: 21.219}]}{=} \lim_{i \rightarrow \infty} (\mu(A_i) - \mu(A)) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{i \rightarrow \infty} \mu(A_i) - \mu(A) \end{aligned}$$

As $\mu(A) < \infty \Rightarrow \mu(A) \in \mathbb{R}$ we have

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \mu(A) = \lim_{i \rightarrow \infty} \mu(A_i)$$

Now for the last part we have to prove that if μ is finite then (4) $\Rightarrow$ (2) [because then from (1) $\Leftrightarrow$ (2) $\Rightarrow$ (3) $\Leftrightarrow$ (4) it follows that (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) $\Leftrightarrow$ (4)].

(4) $\Rightarrow$ (2). Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ is such that $\forall i \in \mathbb{N}$ we have $A_i \subseteq A_{i+1}$ and $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{R}$. Take $A = \bigcup_{i \in \mathbb{N}} A_i$ and define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R} \text{ by } B_i = A \setminus A_i \underset{\mathcal{R} \text{ is a ring}}{\in} \mathcal{R}$$

Then we have

$$\forall i \in \mathbb{N} \text{ that } B_{i+1} = A \setminus A_{i+1} \subseteq A \setminus A_i = B_i \text{ and } \mu(B_i) \underset{\mu \text{ is finite}}{<} \infty$$

[theorem: 1.36]

Further we have

$$\emptyset = A \setminus A = A \setminus \left(\bigcup_{i \in \mathbb{N}} A_i \right) \underset{\text{[theorem: 2.141]}}{=} \bigcap_{i \in \mathbb{N}} (A \setminus A_i) = \bigcap_{i \in \mathbb{N}} B_i$$

So using (4) we have that

$$\lim_{i \rightarrow \infty} \mu(A \setminus A_i) = \lim_{i \rightarrow \infty} \mu(B_i) = 0 \quad (21.220)$$

Let $i \in \mathbb{N}$ then, as $A_i \subseteq \bigcup_{i \in \mathbb{N}} A_i = A$ and $\mu(A_i) \underset{\mu \text{ is finite}}{<} \infty$, it follows from [theorem: 21.155(3)] that $\mu(A \setminus A_i) = \mu(A) - \mu(A_i)$. Hence we have

$$\begin{aligned} 0 & \underset{\text{[eq: 21.220]}}{=} \lim_{i \rightarrow \infty} \mu(A \setminus A_i) \\ & = \lim_{i \rightarrow \infty} (\mu(A) - \mu(A_i)) \\ & \underset{\text{[theorem: 20.85]}}{=} \mu(A) - \lim_{i \rightarrow \infty} \mu(A_i) \end{aligned}$$

As μ is finite it follows from the above that

$$\mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \mu(A) = \lim_{i \rightarrow \infty} \mu(A_i) \quad \square$$

The above equivalences helps us to determine that a content on a ring is a pre-measure. If we start with a ring based on a topological space then we can make use of the properties of the topological space to help us to prove that a content is a pre-measure. For this we introduce approximation by compact classes. First we recap the definition of a compact class [see definition: 14.286].

Definition 21.157. Let X be a set then $\mathcal{C} \subseteq \mathcal{P}(X)$ is a **compact class** if $\forall \{K_i\}_{i \in \mathbb{N}} \subseteq \mathcal{C}$ with $\bigcap_{i \in \mathbb{N}} K_i = \emptyset$ there exist a $N \in \mathbb{N}$ such that $\bigcap_{i \in \{1, \dots, N\}} K_i = \emptyset$.

We have then the following theorem that can be used to determine if a content is a pre-measure.

Theorem 21.158. Let X be a set, $\mathcal{R}$ a ring on X and $\mu: X \rightarrow [0, \infty]$ a **finite** content. If there exist a compact class $\mathcal{C} \subseteq \mathcal{P}(X)$ such that $\forall A \in \mathcal{R}$ and $\forall \varepsilon \in \mathbb{R}^+$ we have that $\exists K_\varepsilon \in \mathcal{C}$ and $\exists R_\varepsilon \in \mathcal{R}$ such that

$$R_\varepsilon \subseteq K_\varepsilon \subseteq A \text{ and } \mu(A \setminus R_\varepsilon) < \varepsilon$$

then μ is a pre-measure. In other words if we can 'approximate' $\mathcal{R}$ by a complex class $\mathcal{C}$ then μ is a pre-measure.

Proof. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}$ be such that

$$\forall i \in \mathbb{N} \text{ we have } A_{i+1} \subseteq A_i \text{ and } \bigcap_{i \in \mathbb{N}} A_i = \emptyset \quad (21.221)$$

As μ is finite we have automatically

$$\forall i \in \mathbb{N} \text{ we have } \mu(A_i) < \infty \quad (21.222)$$

So if we prove that $\lim_{i \rightarrow \infty} \mu(A_i) = 0$ then it follows from [theorem: 21.156] that μ is a pre-measure.

Let $\varepsilon \in \mathbb{R}^+$ and $n \in \mathbb{N}$ then by the assumption there exist a $K_{n,\varepsilon} \in \mathcal{C}$ and a $R_{n,\varepsilon} \in \mathcal{R}$ such that $R_{n,\varepsilon} \subseteq K_{n,\varepsilon} \subseteq A_n$ and $\mu(A \setminus R_{n,\varepsilon}) < \frac{\varepsilon}{2^n}$. Hence there exist a $\{K_{n,\varepsilon}\}_{n \in \mathbb{N}} \subseteq \mathcal{C}$ and $\{R_{n,\varepsilon}\}_{n \in \mathbb{N}} \subseteq \mathcal{R}$ such that

$$\forall n \in \mathbb{N} \text{ we have } R_{n,\varepsilon} \subseteq K_{n,\varepsilon} \subseteq A_n \text{ and } \mu(A \setminus R_{n,\varepsilon}) < \frac{\varepsilon}{2^n} \quad (21.223)$$

Then we have $\bigcap_{n \in \mathbb{N}} K_{n,\varepsilon} \subseteq \bigcap_{n \in \mathbb{N}} A_n \underset{\text{[eq: 21.221]}}{=} \emptyset$ so that

$$\bigcap_{n \in \mathbb{N}} K_{n,\varepsilon} = \emptyset$$

By the definition of a compact class there exist a $N_\varepsilon \in \mathbb{N}$ such that $\bigcap_{n \in \{1, \dots, N_\varepsilon\}} K_{n, \varepsilon} = \emptyset$. As $\bigcap_{n \in \{1, \dots, N_\varepsilon\}} R_{n, \varepsilon} \subseteq \bigcap_{n \in \{1, \dots, N_\varepsilon\}} K_{n, \varepsilon}$ it follows that

$$\bigcap_{n \in \{1, \dots, N_\varepsilon\}} R_{n, \varepsilon} = \bigcap_{n \in \{1, \dots, N_\varepsilon\}} K_{n, \varepsilon} = \emptyset \quad (21.224)$$

Let $n \in \{1, \dots, N_\varepsilon\}$ then $A_{N_\varepsilon} \subseteq_{[\text{eq: 21.221}]} A_n$ so that $A_{N_\varepsilon} \subseteq \bigcap_{n \in \{1, \dots, N_\varepsilon\}} A_n \subseteq A_{N_\varepsilon}$ proving that

$$A_{N_\varepsilon} = \bigcap_{n \in \{1, \dots, N_\varepsilon\}} A_n \quad (21.225)$$

Let $x \in \bigcap_{n \in \{1, \dots, N_\varepsilon\}} A_n$. Assume that $x \notin \bigcup_{n \in \{1, \dots, N_\varepsilon\}} (A_n \setminus R_{n, \varepsilon})$ then $\forall m \in \{1, \dots, N_\varepsilon\}$ we have $x \notin A_m \setminus R_{m, \varepsilon}$, as $x \in \bigcap_{n \in \{1, \dots, N_\varepsilon\}} A_n \subseteq A_m$, it follows that $x \in R_{m, \varepsilon}$ [otherwise $x \in A_m \setminus R_{m, \varepsilon}$]. So $x \in \bigcap_{n \in \{1, \dots, N_\varepsilon\}} R_{n, \varepsilon}$ contradicting [eq: 21.224]. Hence the assumption is wrong and we must have that $x \in \bigcup_{n \in \{1, \dots, N_\varepsilon\}} (A_n \setminus R_{n, \varepsilon})$ which proves that

$$A_{N_\varepsilon} \stackrel{[\text{eq: 21.225}]}{=} \bigcap_{n \in \{1, \dots, N_\varepsilon\}} A_n \subseteq \bigcup_{n \in \{1, \dots, N_\varepsilon\}} (A_n \setminus R_{n, \varepsilon}) \underset{\mathcal{R} \text{ is a ring}}{\in} \mathcal{R} \quad (21.226)$$

Hence using the above we have

$$\begin{aligned} \mu(A_{N_\varepsilon}) &\stackrel{[\text{theorem: 21.155(2)}] \wedge [\text{eq: 21.226}]}{\leq} \mu\left(\bigcup_{n \in \{1, \dots, N_\varepsilon\}} (A_n \setminus R_{n, \varepsilon})\right) \\ &\stackrel{[\text{theorem: 21.155(5)}]}{\leq} \sum_{i=1}^{N_\varepsilon} \mu(A_n \setminus R_{n, \varepsilon}) \\ &\stackrel{[\text{eq: 21.223}]}{\leq} \sum_{i=1}^{N_\varepsilon} \frac{\varepsilon}{2^n} \\ &= \varepsilon \cdot \sum_{i=1}^N \left(\frac{1}{2}\right)^n \\ &\stackrel{[\text{lemma: 14.411}]}{=} \varepsilon \cdot \left(\frac{1}{2}\right) \cdot \frac{1 - \left(\frac{1}{2}\right)^N}{1 - \frac{1}{2}} \\ &= \varepsilon \cdot \left(1 - \frac{1}{2^N}\right) \\ &< \varepsilon \end{aligned}$$

Let $n \in \{N_\varepsilon, \dots, \infty\}$ then we have $|\mu(A_n) - 0| \underset{0 \leq \mu(A_n)}{=} \mu(A_n) \underset{[\text{eq: 21.221}]}{\leq} \mu(A_{N_\varepsilon}) < \varepsilon$. To summarize we have proved that

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a } N_\varepsilon \in \mathbb{N} \text{ such that } \forall n \in \{N_\varepsilon, \dots, \infty\} \text{ we have } |\mu(A_n) - 0| < \varepsilon$$

which proves that $\lim_{n \rightarrow \infty} \mu(A_n) = 0$. Using the fact that μ is finite and [theorem: 20.77] it follows that

$$\lim_{n \rightarrow \infty} \mu(A_i) = 0$$

So using [theorem: 21.156] it follows that μ is a pre-measure. $\square$

The above theorem is especially useful if we have a ring on a topological Hausdorff space because every set of compact sets is a compact class [see theorem: 14.287]. This is expressed in the following theorem.

Theorem 21.159. *Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space,, $\mathcal{R}$ a ring on X and $\mu: X \rightarrow [0, \infty]$ a **finite** content. If $\forall A \in \mathcal{R}$ and $\forall \varepsilon \in \mathbb{R}^+$ we have that $\exists K_\varepsilon$ compact and $\exists R_\varepsilon \in \mathcal{R}$ such that*

$$R_\varepsilon \subseteq K_\varepsilon \subseteq A \text{ and } \mu(A \setminus R_\varepsilon) < \varepsilon$$

then μ is a pre-measure.

Proof. If $\mathcal{C} = \{A \in \mathcal{P}(X) | A \text{ is compact}\}$ is the set of compact subsets of X then by [theorem: 14.287] $\mathcal{C}$ is a compact class. The proof follows then from [theorem: 21.158]. $\square$

Proving that $\mathcal{R} \subseteq \mathcal{P}(X)$ is a ring is not always possible, especially proving that $\forall A, B \in \mathcal{R}$ we have that $A \setminus B \in \mathcal{R}$. For example the difference of two half-open intervals in $\mathcal{I}_{[}$ is not always a half-open interval, hence $\mathcal{R}_{[}$ is not a ring, however it will be proved later that $\mathcal{I}_{[}$ is a semi-ring on $\mathbb{R}$. Luckily it turns out that a semi-ring with a content on it can easily be extended to a ring with a content on it. First we show how we can extend a semi-ring to a smallest ring containing the semi-ring.

Definition 21.160. Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring on X then $\mathcal{R}[\mathcal{S}]$ is defined by

$$\mathcal{R}[\mathcal{S}] = \left\{ \bigsqcup_{i \in \{1, \dots, n\}} A_i | n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S} \text{ is pairwise disjoint} \right\}$$

So $\mathcal{R}[\mathcal{S}]$ is the set of all finite pairwise disjoint unions of sets in $\mathcal{S}$.

Note 21.161. As for $A, B \in \mathcal{S}$ we have by definition of a semi-ring that $\exists \{C_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ pairwise disjoint such that $A \setminus B = \bigsqcup_{i \in \{1, \dots, n\}} C_i$ it follows that

$$\forall A, B \in \mathcal{S} \text{ we have } A \setminus B \in \mathcal{S}$$

To prove that $\mathcal{R}[\mathcal{S}]$ is a ring we have to prove that $\emptyset \in \mathcal{R}[\mathcal{S}]$ and $\forall A, B \in \mathcal{R}[\mathcal{S}]$ that $A \cup B \in \mathcal{R}[\mathcal{S}] \wedge A \setminus B \in \mathcal{R}[\mathcal{S}]$. We start with proving that $\forall A, B \in \mathcal{R}[\mathcal{S}]$ $A \cup B \in \mathcal{R}[\mathcal{S}]$ using the following two lemma's.

Lemma 21.162. Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring on X , $A \in \mathcal{S}$ and $B \in \mathcal{R}[\mathcal{S}]$ then

$$A \cup B \in \mathcal{R}[\mathcal{S}]$$

Proof. As $B \in \mathcal{R}[\mathcal{S}]$ there exist a pairwise disjoint $\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that

$$B = \bigsqcup_{i \in \{1, \dots, n\}} B_i$$

Define

$$\{C_i\}_{i \in \{1, \dots, n+1\}} \text{ by } C_i = \begin{cases} B_i \setminus A & \text{if } i \in \{1, \dots, n\} \\ A & \text{if } i = n+1 \end{cases}$$

Then if $x \in A \cup B$ we have either:

$x \in A$. Then $x \in C_{n+1} \subseteq \bigcup_{i \in \{1, \dots, n+1\}} C_i$

$x \notin A$. Then we must have that $x \in B$ so that there exist a $i \in \{1, \dots, n\}$ such that

$$x \in B_i \xrightarrow{x \notin A} x \in B_i \setminus A = C_i \subseteq \bigcup_{i \in \{1, \dots, n+1\}} C_i$$

proving that

$$A \cup B \subseteq \bigcup_{i \in \{1, \dots, n+1\}} C_i \tag{21.227}$$

Let $x \in \bigcup_{i \in \{1, \dots, n+1\}} C_i$ then there exist a $i \in \{1, \dots, n+1\}$ such that $x \in C_i$. For i we have either

$i \in \{1, \dots, n\}$. Then $x \in C_i = B_i \setminus A \subseteq B_i \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i = B \subseteq A \cup B$

$i = n+1$. Then $x \in C_{n+1} = A \subseteq A \cup B$

hence $\bigcup_{i \in \{1, \dots, n+1\}} C_i \subseteq A \cup B$. Combining this with [eq: 21.227] results in

$$A \cup B = \bigcup_{i \in \{1, \dots, n+1\}} C_i \tag{21.228}$$

Let $i, j \in \{1, \dots, n+1\}$ with $i \neq j$ then for i we have either:

$i \in \{1, \dots, n\}$. Then for j we have either:

$j \in \{1, \dots, n\}$. Then $C_i \cap C_j = (B_i \setminus A) \cap (B_j \setminus A) \subseteq B_i \cap B_j = \emptyset$

$j = n + 1$. Then $C_i \cap C_j = (B_i \setminus A) \cap A = \emptyset$

$i = n + 1$. Then as $i \neq j$ we have that $j \in \{1, \dots, n\}$ so that $C_i \cap C_j = A \cap (B_j \setminus A) = \emptyset$

hence

$$\{C_i\}_{i \in \{1, \dots, n+1\}} \text{ is pairwise disjoint so that } A \cup B = \bigsqcup_{i \in \{1, \dots, n+1\}} C_i \quad (21.229)$$

Let $i \in \{1, \dots, n+1\}$ then we have either:

$i \in \{1, \dots, n\}$. As $A, B_i \in \mathcal{S}$ and $\mathcal{S}$ is a semi-ring there exist a pairwise disjoint $\{D_{i,j}\}_{j \in \{1, \dots, n_i\}} \subseteq \mathcal{S}$ such that $C_i = B_i \setminus A = \bigsqcup_{j \in \{1, \dots, n_i\}} D_{i,j}$.

$i = n + 1$. Take $n_1 = 1$ and $\{D_{1,j}\}_{j \in \{1\}} \subseteq \mathcal{S}$ by $D_{1,1} = A$ so that $C_{n+1} = \bigsqcup_{j \in \{1, \dots, n_1\}} D_{1,j}$

Summarized we have that

$$\forall i \in \{1, \dots, n+1\} \exists \{D_{i,j}\}_{j \in \{1, \dots, n_i\}} \subseteq \mathcal{S} \text{ with } C_i = \bigsqcup_{j \in \{1, \dots, n_i\}} D_{i,j} \quad (21.230)$$

As $\bigcup_{i \in \{1, \dots, n+1\}} \{i\} \times \{1, \dots, n_i\}$ is finite by [theorems: 6.41, 6.36] there exists a bijection

$$\beta: \{1, \dots, N\} \rightarrow \bigcup_{i \in \{1, \dots, n+1\}} \{i\} \times \{1, \dots, n_i\} \quad (21.231)$$

Using [eqs: 21.229, 21.230 and 21.231] allows us to apply [theorem: 21.12] that

$$\{D_{\beta(i)_1 \beta(i)_2}\}_{i \in \{1, \dots, N\}} \subseteq \mathcal{S} \text{ is pairwise disjoint and } \bigsqcup_{i \in \{1, \dots, n+1\}} C_i = \bigsqcup_{i \in \{1, \dots, N\}} D_{\beta(i)_1, \beta(i)_2}$$

Hence using [eq: 21.229] it follows that $A \cup B = \bigsqcup_{i \in \{1, \dots, N\}} D_{\beta(i)_1, \beta(i)_2}$ which proves that

$$A \cup B \in \mathcal{S}[\mathcal{R}] \quad \square$$

We use the above lemma in a proof by induction to prove that that $\mathcal{R}[\mathcal{S}]$ is $\cup$ -stable.

Lemma 21.163. *Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring on X and $A, B \in \mathcal{R}[\mathcal{S}]$ then $A \cup B \in \mathcal{R}[\mathcal{S}]$*

Proof. Let $B \in \mathcal{R}[\mathcal{S}]$ and define

$$\mathcal{S}_B = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S} \text{ pairwise disjoint then } \left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \cup B \in \mathcal{R}[\mathcal{S}] \right\}$$

then we have:

$1 \in \mathcal{S}_B$. If $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ is pairwise disjoint then, as $A_1 \in \mathcal{S}$, we have by [lemma: 21.162] that $(\bigsqcup_{i \in \{1\}} A_i) \cup B = A_1 \cup B \in \mathcal{R}[\mathcal{S}]$ which proves that $1 \in \mathcal{S}_B$.

$n \in \mathcal{S}_B \Rightarrow n + 1 \in \mathcal{S}_B$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}$ be a pairwise disjoint family of sets in $\mathcal{S}$. As $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ is pairwise disjoint and $n \in \mathcal{S}_B$ we have that

$$\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \cup B \in \mathcal{R}[\mathcal{S}]$$

Hence, as $A_{n+1} \in \mathcal{S}$, we have by the previous lemma [lemma: 21.162] that

$$A_{n+1} \cup \left(\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \cup B \right) \in \mathcal{R}[\mathcal{S}]$$

which as $A_{n+1} \cup ((\bigsqcup_{i \in \{1, \dots, n\}} A_i) \cup B) = (\bigsqcup_{i \in \{1, \dots, n+1\}} A_i) \cup B$ proves that

$$\left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i \right) \cup B$$

so that $n + 1 \in \mathcal{S}_B$.

Mathematical induction proves then that $\mathcal{S}_B = \mathbb{N}$. Finally if $A \in \mathcal{R}[\mathcal{S}]$ then there exist a $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$, hence, as $n \in \mathbb{N} = \mathcal{S}_B$, it follows that $A \cup B = (\bigsqcup_{i \in \{1, \dots, n\}} A_i) \cup B \in \mathcal{R}[\mathcal{S}]$. $\square$

Next we prove that $\forall A, B \in \mathcal{R}[\mathcal{S}]$ that $A \setminus B \in \mathcal{R}[\mathcal{S}]$, we do this in 2 sets.

Lemma 21.164. *Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring on X , $A \in \mathcal{R}[\mathcal{S}]$ and $B \in \mathcal{S}$ then $A \setminus B \in \mathcal{R}[\mathcal{S}]$.*

Proof. We use mathematical induction in the proof. Define

$$\mathcal{S}_B = \left\{ n \in \mathbb{N} \mid \text{If } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S} \text{ pairwise disjoint then } \left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \setminus B \in \mathcal{R}[\mathcal{S}] \right\}$$

then we have:

1 $\in \mathcal{S}_B$. Let $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ pairwise disjoint then $(\bigsqcup_{i \in \{1\}} A_i) \setminus B = A_1 \setminus B \in \mathcal{R}[\mathcal{S}]$ [using the note in the definition of $\mathcal{R}[\mathcal{S}]$ see [definition: 21.160]], hence $1 \in \mathcal{S}_B$.

$n \in \mathcal{S}_B \Rightarrow n+1 \in \mathcal{S}_B$. Let $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}$ pairwise disjoint. Then we have

$$\begin{aligned} \left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i \right) \setminus B &= \left(\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \sqcup A_{n+1} \right) \setminus B \\ &= \left(\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \setminus B \right) \sqcup (A_{n+1} \setminus B) \end{aligned} \quad (21.232)$$

As $n \in \mathcal{S}_B$ we have that $(\bigsqcup_{i \in \{1, \dots, n\}} A_i) \setminus B \in \mathcal{R}[\mathcal{S}]$. Further using the fact that $A_{n+1}, B \in \mathcal{S}$ it follows from the note in [definition: 21.160] that $A_{n+1} \setminus B \in \mathcal{R}[\mathcal{S}]$. Hence using [lemma: 21.163] that $((\bigsqcup_{i \in \{1, \dots, n\}} A_i) \setminus B) \cup (A_{n+1} \setminus B) \in \mathcal{R}[\mathcal{S}]$. So by [eq: 21.232] it follows that

$$\left(\bigsqcup_{i \in \{1, \dots, n+1\}} A_i \right) \setminus B \in \mathcal{S}_B$$

which proves that $n+1 \in \mathcal{S}_B$.

By mathematical induction it follows that

$$\mathcal{S}_B = \mathbb{N}$$

Finally let $A \in \mathcal{R}[\mathcal{S}]$ then there exist a pairwise disjoint $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$. As $n \in \mathbb{N} = \mathcal{S}_B$ we have that $A \setminus B = (\bigsqcup_{i \in \{1, \dots, n\}} A_i) \setminus B \in \mathcal{R}[\mathcal{S}]$. $\square$

Lemma 21.165. *Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring on X then $\forall A, B \in \mathcal{R}[\mathcal{S}]$ we have that $A \setminus B \in \mathcal{R}[\mathcal{S}]$.*

Proof. Let $A \in \mathcal{R}[\mathcal{S}]$ and define

$$\mathcal{S}_A = \left\{ n \in \mathbb{N} \mid \text{If } \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S} \text{ is pairwise disjoint then } A \setminus \bigsqcup_{i \in \{1, \dots, n\}} B_i \in \mathcal{R}[\mathcal{S}] \right\}$$

then we have:

1 $\in \mathcal{S}_A$. Let $\{B_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ be pairwise disjoint then by the previous lemma [lemma: 21.164] we have that $A \setminus \bigsqcup_{i \in \{1\}} B_i = A \setminus B_1 \in \mathcal{R}[\mathcal{S}]$ which proves that $1 \in \mathcal{S}_A$.

$n \in \mathcal{S}_A \Rightarrow n+1 \in \mathcal{S}_A$. Let $\{B_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}$ be pairwise disjoint. Then

$$\begin{aligned} A \setminus \bigsqcup_{i \in \{1, \dots, n+1\}} B_i &= A \setminus \left(\left(\bigsqcup_{i \in \{1, \dots, n\}} B_i \right) \sqcup B_{n+1} \right) \\ &\stackrel{[\text{theorem: 1.34}]}{=} \left(A \setminus \bigsqcup_{i \in \{1, \dots, n\}} B_i \right) \setminus B_{n+1} \end{aligned}$$

As $n \in S_A$ we have that $A \setminus \bigsqcup_{i \in \{1, \dots, n\}} B_i \in \mathcal{R}[S]$ so that by [lemma: 21.164] $(A \setminus \bigsqcup_{i \in \{1, \dots, n\}} B_i) \setminus B_{n+1} \in \mathcal{R}[S]$. Hence $A \setminus \bigsqcup_{i \in \{1, \dots, n+1\}} B_i \in \mathcal{R}[S]$ which proves that $n+1 \in S_A$.

Mathematical induction proves that

$$S_A = \mathbb{N}$$

Finally if $B \in \mathcal{R}[S]$ then by definition of $\mathcal{R}[S]$ there exist a pairwise disjoint $\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that $B = \bigsqcup_{i \in \{1, \dots, n\}} B_i$. As $n \in \mathbb{N} = S_A$ $A \setminus B = A \setminus \bigsqcup_{i \in \{1, \dots, n\}} B_i \in \mathcal{R}[S]$. $\square$

We are now ready to prove that $\mathcal{R}[S]$ is a ring more specific

Theorem 21.166. *Let X be a set, $\mathcal{S} \subseteq \mathcal{P}(X)$ a semi-ring on X then $\mathcal{R}[S]$ is a ring on X such that:*

1. $\mathcal{S} \subseteq \mathcal{R}[S]$
2. If $\mathcal{T}$ is another ring on X with $\mathcal{S} \subseteq \mathcal{T}$ then $\mathcal{R}[S] \subseteq \mathcal{T}$

In other words the set of all disjoint finite unions of sets in $\mathcal{S}$ is the smallest ring covering $\mathcal{S}$.

Proof. Let $A \in \mathcal{S}$ and define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ by $A_1 = A$ then $\{A_i\}_{i \in \{1\}}$ is pairwise disjoint and $A = \bigsqcup_{i \in \{1\}} A_i$, hence $A \in \mathcal{R}[S]$ proving that

$$\mathcal{S} \subseteq \mathcal{R}[S] \quad (21.233)$$

As $\mathcal{S}$ is a semi-ring we have $\emptyset \in \mathcal{S}$ so by the above we have

$$\emptyset \in \mathcal{R}[S] \quad (21.234)$$

Let $A, B \in \mathcal{R}[S]$ then by [lemma: 21.163] and [lemma: 21.165] we have that

$$A \bigcup B \in \mathcal{R}[S] \text{ and } A \setminus B \in \mathcal{R}[S] \quad (21.235)$$

So by [eqs: 21.233, 21.234 and 21.235] it follows that

$$\mathcal{R}[S] \text{ is a ring on } X$$

It remains to prove that $\mathcal{R}[S]$ is the smallest ring containing $\mathcal{S}$. So let $\mathcal{T}$ be another ring such that $\mathcal{S} \subseteq \mathcal{T}$. If $A \in \mathcal{R}[S]$ then there exist a pairwise disjoint finite family $\{A_i\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{S} \subseteq \mathcal{T}$ such that $A = \bigsqcup_{i \in \{1, \dots, k\}} A_i$. As $\mathcal{T}$ is a ring it follows from [theorem: 21.27] that $A = \bigsqcup_{i \in \{1, \dots, k\}} A_i \in \mathcal{T}$ which proves that

$$\mathcal{R}[S] \subseteq \mathcal{T} \quad \square$$

Next we want to extend a content on a semi-ring $\mathcal{S}$ to a content on the ring $\mathcal{R}[S]$.

Theorem 21.167. *Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: \mathcal{S} \rightarrow [0, \infty]$ a content on $\mathcal{S}$ then*

1. *There exist a content $\nu: \mathcal{R}[S] \rightarrow [0, \infty]$ such that $\nu|_{\mathcal{S}} = \mu$ [in other words ν is a extension of μ]*
2. *If $\omega: \mathcal{R}[S] \rightarrow [0, \infty]$ is another content such that $\omega|_{\mathcal{S}} = \mu$ then $\omega = \nu$ [in other words the extension is unique].*
3. *If μ is finite then ν is finite.*

Proof.

1. First we proof existence. Let $A \in \mathcal{R}[S]$ then there exist a pairwise disjoint finite family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$. As ν will be a content on $\mathcal{R}[S]$ we must have that $\nu(A) = \sum_{i=1}^n \nu(A_i) \stackrel{A_i \in \mathcal{S}}{=} \sum_{i=1}^n \mu(A_i)$. So it make sense to define

$$\nu: \mathcal{R}[S] \rightarrow [0, \infty] \text{ by } \nu(A) = \sum_{i=1}^n \mu(A_i)$$

$$\text{where } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}, n \in \mathbb{N} \text{ is pairwise disjoint and } A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$$

Of course this only works if $\nu(A)$ is independent of the chosen family $\{A_i\}_{i \in \{1, \dots, n\}}$. So assume that for $A \in \mathcal{R}[\mathcal{S}]$ we have two pairwise disjoint families $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ and $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$, $m \in \mathbb{N}$ such that

$$A = \bigsqcup_{i \in \{1, \dots, n\}} A_i = \bigsqcup_{i \in \{1, \dots, m\}} B_i$$

Then we have

$$\begin{aligned} \sum_{i=1}^n \mu(A_i) & \stackrel{A_i \cap \overline{A} = A_i}{=} \sum_{i=1}^n \mu(A_i \cap A) \\ & = \sum_{i=1}^n \mu\left(A_i \cap \left(\bigsqcup_{j \in \{1, \dots, m\}} B_j\right)\right) \\ & \stackrel{[\text{theorem: 2.138}]}{=} \sum_{i=1}^n \mu\left(\bigsqcup_{j \in \{1, \dots, m\}} \mu(A_i \cap B_j)\right) \\ & \stackrel{A_i \cap \overline{B_j} \in \mathcal{S}}{=} \sum_{i=1}^n \left(\sum_{j=1}^m \mu(A_i \cap B_j)\right) \\ & \stackrel{[\text{theorem: 20.104}]}{=} \sum_{j=1}^m \left(\sum_{i=1}^n \mu(A_i \cap B_j)\right) \\ & \stackrel{A_i \cap \overline{B_j} \in \mathcal{S}}{=} \sum_{j=1}^m \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} (A_i \cap B_j)\right) \\ & \stackrel{[\text{theorem: 2.138}]}{=} \sum_{j=1}^m \mu\left(\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \cap B_j\right) \\ & = \sum_{j=1}^m \mu(A \cap B_j) \\ & = \sum_{j=1}^m \mu(B_j) \end{aligned}$$

proving that $\nu: \mathcal{R}[\mathcal{S}] \rightarrow [0, \infty]$ is well defined. Next we prove that ν is an extension of $\mathcal{S}$. Let $A \in \mathcal{S}$ then if we define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}$ by $A_1 = A$ then $\{A_i\}_{i \in \{1\}}$ is clearly disjoint and $\bigsqcup_{i \in \{1\}} A_i = A_1 = A$ so that $\nu(A) = \sum_{i=1}^1 \mu(A_i) = \mu(A_1) = \mu(A)$ proving that

$$\nu|_{\mathcal{S}} = \mu \tag{21.236}$$

Next we use [theorem: 21.154] to prove that ν is a content on $\mathcal{R}[\mathcal{S}]$. As $\emptyset \in \mathcal{S}$ we have by [eq: 21.236] that

$$\nu(\emptyset) \stackrel{\emptyset \in \mathcal{S}}{=} \mu(\emptyset) = 0 \tag{21.237}$$

Let $A, B \in \mathcal{R}[\mathcal{S}]$ such that $A \cap B = \emptyset$ then there exist pairwise disjoint families $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ and $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}$, $m \in \mathbb{N}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$, $B = \bigsqcup_{i \in \{1, \dots, m\}} B_i$. Define

$$\{C_i\}_{i \in \{1, \dots, n+m\}} \subseteq \mathcal{S} \text{ by } C_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ B_{i-n} & \text{if } i \in \{n+1, \dots, n+m\} \end{cases}$$

Let $i, j \in \{1, \dots, n+m\}$ with $i \neq j$ then we may assume that $i < j$ [otherwise exchange i and j], for i we have either:

$i \in \{1, \dots, n\}$. Then for j we have either:

$j \in \{1, \dots, n\}$. Then $C_i \cap C_j = A_i \cap A_j \stackrel{i \neq j}{=} \emptyset$.

$j \in \{n+1, \dots, n+m\}$. Then $C_i \cap C_j = A_{i-n} \cap B_{j-n} \subseteq A \cap B = \emptyset$.

$i \in \{n+1, \dots, n+m\}$. Then $j \in \{n+1, \dots, n+m\}$ $C_i \cap C_j = A_{i-n} \cap B_{j-n} \subseteq A \cap B = \emptyset$ proving that

$$\{C_i\}_{i \in \{1, \dots, n+m\}} \subseteq \mathcal{S} \text{ is pairwise disjoint} \quad (21.238)$$

Let $x \in A \cup B$ then either $x \in A = \bigcup_{i \in \{1, \dots, n\}} A_i \Rightarrow \exists i \in \{1, \dots, n\}$ with $x \in A_i = C_i$ or $x \in B = \bigcup_{i \in \{1, \dots, m\}} B_i \Rightarrow \exists i \in \{1, \dots, m\}$ with $x \in B_i = C_{i+n}$, hence $x \in \bigcup_{i \in \{1, \dots, n+m\}} C_i$, proving that

$$A \sqcup B \subseteq \bigcup_{i \in \{1, \dots, n+m\}} C_i. \quad (21.239)$$

For $i \in \{1, \dots, n+m\}$ we have either $i \in \{1, \dots, n\}$ then $C_i = A_i \subseteq A \cup B$ or $i \in \{n+1, \dots, n+m\}$ then $C_i = B_{i-n} \subseteq A \cup B$, hence in all cases we have $C_i \subseteq A \cup B$. This proves that $\bigcup_{i \in \{1, \dots, n+m\}} C_i \subseteq A \cup B$ which combines with [eq: 21.239] gives

$$A \sqcup B = \bigsqcup_{i \in \{1, \dots, n+m\}} C_i \quad (21.240)$$

Hence we have

$$\begin{aligned} \nu(A \cup B) &= \nu\left(\bigsqcup_{i \in \{1, \dots, n+m\}} C_i\right) \\ &\stackrel{\text{def}}{=} \sum_{i=1}^{n+m} \mu(C_i) \\ &\stackrel{[\text{theorem: 11.23}]}{=} \sum_{i=1}^n \mu(C_i) + \sum_{i=n+1}^{n+m} \mu(C_i) \\ &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{n+m} \mu(B_{i-n}) \\ &\stackrel{[\text{theorem: 11.12}]}{=} \sum_{i=1}^n \mu(A_i) + \sum_{i=1}^m \mu(B_i) \\ &= \nu(A) + \nu(B) \end{aligned}$$

proving that

$$\forall A, B \in \mathcal{R}[\mathcal{S}] \text{ with } A \cap B = \emptyset \text{ we have } \nu(A \sqcup B) = \nu(A) + \nu(B) \quad (21.241)$$

Using [eqs: 21.237, 21.241] we can apply [theorem: 21.154] to prove that

$$\nu: \mathcal{R}[\mathcal{S}] \rightarrow [0, \infty] \text{ is a content on } \mathcal{R}[\mathcal{S}]$$

2. Let $\omega: \mathcal{R}[\mathcal{S}] \rightarrow [0, \infty]$ be another content on $\mathcal{R}[\mathcal{S}]$ such that $\omega|_{\mathcal{S}} = \mu$. Let $A \in \mathcal{R}[\mathcal{S}]$ then there exist a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$. Hence

$$\begin{aligned} \omega(A) &= \omega\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \\ &\stackrel{\omega \text{ is a content}}{=} \sum_{i=1}^n \omega(A_i) \\ &\stackrel{\omega|_{\mathcal{S}} = \mu}{=} \sum_{i=1}^n \mu(A_i) \\ &\stackrel{\text{def of } \nu}{=} \nu(A) \end{aligned}$$

so that

$$\omega = \nu$$

3. If $A \in \mathcal{R}[\mathcal{S}]$ then there exist a finite pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$. Then $\nu(A) = \sum_{i=1}^n \mu(A_i) < \infty$ [because μ is assumed to be finite]. $\square$

We can now rephrase [theorem: 21.158] to work with semi-rings instead of rings.

Theorem 21.168. *Let X be a set, $\mathcal{S}$ a semi-ring on X and $\mu: X \rightarrow [0, \infty]$ a **finite** content. If there exist a compact class $\mathcal{C} \subseteq \mathcal{P}(X)$ such that*

1. $\forall \{K_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{C}$, $n \in \mathbb{N}$ we have $\bigcup_{i \in \{1, \dots, n\}} K_i \in \mathcal{C}$
2. $\forall A \in \mathcal{S}$ and $\forall \varepsilon \in \mathbb{R}^+$ we have that $\exists K_\varepsilon \in \mathcal{C}$ and $\exists A_\varepsilon \in \mathcal{S}$ such that

$$A_\varepsilon \subseteq K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(A_\varepsilon) < \varepsilon$$

then

μ is a pre-measure

Proof. Using [theorem: 21.167] there exist a **finite** content $\bar{\mu}: \mathcal{R}[\mathcal{S}] \rightarrow [0, \infty]$ on the ring $\mathcal{R}[\mathcal{S}]$ such that

$$(\bar{\mu})|_{\mathcal{S}} = \mu \quad (21.242)$$

Let $A \in \mathcal{R}[\mathcal{S}]$ then by definition there exist a pairwise disjoint $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$. Let $\varepsilon \in \mathbb{R}^+$ then by the hypothesis there exists $\{A_{i,\varepsilon}\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$ and $\{K_{i,\varepsilon}\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{C}$ such that

$$\forall i \in \{1, \dots, n\} \text{ we have } A_{i,\varepsilon} \subseteq K_{i,\varepsilon} \subseteq A_i \text{ and } \mu(A_i) - \mu(A_{i,\varepsilon}) < \frac{\varepsilon}{2^n} \quad (21.243)$$

Define

$$K_\varepsilon = \bigcup_{i \in \{1, \dots, n\}} K_{i,\varepsilon} \in \mathcal{C} \text{ [by (1)]} \quad (21.244)$$

and

$$A_\varepsilon = \bigcup_{i \in \{1, \dots, n\}} A_{i,\varepsilon} \in \mathcal{R}[\mathcal{S}] \text{ [because } \mathcal{S} \subseteq \mathcal{R}[\mathcal{S}] \text{ and } \mathcal{R}[\mathcal{S}] \text{ is a ring hence } \cup\text{-stable}] \quad (21.245)$$

then we have

$$A_\varepsilon = \bigcup_{i \in \{1, \dots, n\}} A_{i,\varepsilon} \subseteq \bigcup_{i \in \{1, \dots, n\}} K_{i,\varepsilon} = K_\varepsilon$$

and

$$K_\varepsilon = \bigcup_{i \in \{1, \dots, n\}} K_{i,\varepsilon} \subseteq \bigcup_{i \in \{1, \dots, n\}} A_i = A$$

so that we have

$$A_\varepsilon \in \mathcal{R}[\mathcal{S}], K_\varepsilon \in \mathcal{C} \text{ and } A_\varepsilon \subseteq K_\varepsilon \subseteq A \quad (21.246)$$

Next

$$\begin{aligned} A \setminus A_\varepsilon &= \left(\bigcup_{i \in \{1, \dots, n\}} A_i \right) \setminus \left(\bigcup_{j \in \{1, \dots, n\}} A_{j,\varepsilon} \right) \\ &\stackrel{[\text{theorem: 2.141}]}{=} \bigcup_{i \in \{1, \dots, n\}} \left(A_i \setminus \left(\bigcup_{j \in \{1, \dots, n\}} A_{j,\varepsilon} \right) \right) \\ &\stackrel{[\text{theorem: 2.140}]}{=} \bigcup_{i \in \{1, \dots, n\}} \left(\bigcap_{j \in \{1, \dots, n\}} (A_i \setminus A_{j,\varepsilon}) \right) \\ &\subseteq \bigcup_{i \in \{1, \dots, n\}} (A_i \setminus A_{i,\varepsilon}) \end{aligned}$$

Further if $i, j \in \{1, \dots, n\}$ with $i \neq j$ then $(A_i \setminus A_{i,\varepsilon}) \cap (A_j \setminus A_{j,\varepsilon}) \subseteq A_i \cap A_j = \emptyset$. To summarize we have that

$$A \setminus A_\varepsilon \subseteq \bigsqcup_{i \in \{1, \dots, n\}} (A_i \setminus A_{i,\varepsilon}) \text{ and } \{A_i \setminus A_{i,\varepsilon}\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint} \quad (21.247)$$

As $\bar{\mu}$ is a **finite** content on the ring $\mathcal{R}[\mathcal{S}]$, $A \setminus A_\varepsilon \in \mathcal{R}[\mathcal{S}]$ and $\forall i \in \{1, \dots, n\}$ $A_i \setminus A_{i,\varepsilon} \in \mathcal{R}[\mathcal{S}]$ [because $\mathcal{R}[\mathcal{S}]$ is a ring and $A, A_\varepsilon, A_i, A_{i,\varepsilon} \in \mathcal{R}[\mathcal{S}]$] we have

$$\begin{aligned}
 \bar{\mu}(A \setminus A_\varepsilon) &\stackrel{[\text{theorem: 21.155(2)}]}{\leq} \bar{\mu}\left(\bigsqcup_{i \in \{1, \dots, n\}} (A_i \setminus A_{i,\varepsilon})\right) \\
 &= \sum_{i=1}^n \bar{\mu}(A_i \setminus A_{i,\varepsilon}) \\
 &\stackrel{[\text{theorem: 21.155(3)}]}{=} \sum_{i=1}^n (\bar{\mu}(A_i) - \bar{\mu}(A_{i,\varepsilon})) \\
 \bar{\mu} \text{ extends } \mu \wedge A_i, A_{i,\varepsilon} \in \mathcal{S} &\stackrel{=}{=} \sum_{i=1}^n (\mu(A_i) - \mu(A_{i,\varepsilon})) \\
 &\stackrel{[\text{eq: 21.243}]}{\leq} \sum_{i=1}^m \frac{\varepsilon}{2^i} \\
 &= \varepsilon \cdot \sum_{i=1}^n \left(\frac{1}{2}\right)^i \\
 &\stackrel{[\text{lemma: 14.411}]}{=} \varepsilon \cdot \left(\frac{1}{2}\right) \cdot \frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} \\
 &= \varepsilon \cdot \left(1 - \left(\frac{1}{2}\right)^n\right) \\
 &< \varepsilon
 \end{aligned}$$

Hence

$$\bar{\mu}(A \setminus A_\varepsilon) < \varepsilon \quad (21.248)$$

Using [eq: 21.246] and [eq: 21.248] allows us to apply [theorem: 21.158] proving that

$$\bar{\mu} \text{ is a pre-measure on } \mathcal{R}[\mathcal{S}]$$

As μ is a content

$$\mu(\emptyset) = 0 \quad (21.249)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}$ be pairwise disjoint such that $\bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{S}$ then we have

$$\begin{aligned}
 \mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eq: 21.242}]}{=} \bar{\mu}\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) \\
 &\stackrel{\bar{\mu} \text{ is a pre-measure}}{=} \sum_{i=1}^{\infty} \bar{\mu}(A_i) \\
 &\stackrel{[\text{eq: 21.242}]}{=} \sum_{i=1}^{\infty} \mu(A_i)
 \end{aligned}$$

which together with [eq: 21.249] proves that μ is a pre-measure. $\square$

If we deal with topological spaces then we have a even simpler test to see if the extension of a content on a semi-ring $\mathcal{S}$ to the ring $\mathcal{R}[\mathcal{S}]$.

Theorem 21.169. *Let $\langle X, \mathcal{T} \rangle$ be a Hausdorff topological space, $\mathcal{S}$ a semi-ring on X and $\mu: X \rightarrow [0, \infty]$ a **finite** content. If $\forall A \in \mathcal{S}$ and $\forall \varepsilon \in \mathbb{R}^+$ there exists a compact K_ε and a $A_\varepsilon \in \mathcal{S}$ such that*

$$A_\varepsilon \subseteq K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(A_\varepsilon) < \varepsilon$$

then

$$\mu: \mathcal{R}[\mathcal{S}] \rightarrow [0, \infty] \text{ is a pre-measure on } \mathcal{S}$$

Proof. First using [theorem: 14.287] it follows that

$$\mathcal{C} = \{A \in \mathcal{P}(X) | A \text{ is compact}\} \text{ is a compact class} \quad (21.250)$$

Further:

1. $\forall \{K_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{C}$, $n \in \mathbb{N}$ we have by [theorem: 14.271] that $\bigcup_{i \in \{1, \dots, n\}} K_i$ is compact hence $\bigcup_{i \in \{1, \dots, n\}} K_i \in \mathcal{C}$.
2. $\forall A \in \mathcal{S}$ and $\forall \varepsilon \in \mathbb{R}^+$ we have by the hypothesis that $\exists K_\varepsilon \in \mathcal{C}$ and $\exists A_\varepsilon \in \mathcal{S}$ such that

$$A_\varepsilon \subseteq K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(A_\varepsilon) < \varepsilon$$

So using the previous theorem [theorem: 21.168] we have that μ is a pre-measure. $\square$

21.3.4.3 Using the Carathéodory theorem on $\mathbb{R}$

We use now the theory developed in the previous sections to construct a σ -algebra and a measure on $\mathbb{R}$. The procedure that we will follow is the following:

1. Prove that $\mathcal{I}_{[[} = \{[a, b[| a, b \in \mathbb{R} \wedge [a, b] \neq \emptyset\} \cup \{\emptyset\}$ [see theorem: 21.55] is semi-ring.
2. Prove that $\text{len}: \mathcal{I}_{[[} \cup \{\emptyset\} \rightarrow [0, \infty]$ defined by $\text{len}(I) = \begin{cases} b - a & \text{if } I = [a, b[\in \mathcal{I}_{[[} \\ 0 & \text{if } I = \emptyset \end{cases}$ [the function that calculates the length of a half-open interval] is a finite content on the semi-ring $\mathcal{I}_{[[}$.
3. Use [theorem: 21.148] a derivative of the Carathéodory Theorem to extend the pre-measure to a measure [the Lebesgue measure] on a σ -algebra [the Lebesgue algebra on $\mathbb{R}$] that contains $\sigma[\mathcal{I}_{[[}] = \mathcal{B}[\mathbb{R}]$ and whose restriction on $\mathcal{B}[\mathbb{R}]$ is also a measure that we also call the Lebesgue measure.

Definition 21.170. The set $\mathcal{S}_{[[} \subseteq \mathcal{P}(\mathbb{R})$ is defined by

$$\mathcal{S}_{[[} = \{[a, b[| a, b \in \mathbb{R}\}$$

Theorem 21.171. $\mathcal{S}_{[[} = \mathcal{I}_{[[} \cup \{\emptyset\}$ and $\{\emptyset\} \cap \mathcal{I}_{[[} = \emptyset$ in other words

$$\mathcal{S}_{[[} = \mathcal{I}_{[[} \bigsqcup \{\emptyset\}$$

where $\mathcal{I}_{[[} \stackrel{\text{definition: 21.51}}{=} \{[a, b[| a, b \in \mathbb{R} \wedge [a, b] \neq \emptyset\}$

Proof. If $I \in \mathcal{S}_{[[}$ then there exists $a, b \in \mathbb{R}$ such that $I = [a, b[$. For a, b we have either

$a < b$. Then $[a, b[\neq \emptyset$ so that $I \in \mathcal{I}_{[[} \subseteq \mathcal{I}_{[[} \cup \{\emptyset\}$

$b \leq a$. Then $[a, b[= \emptyset$ so that $I = \emptyset \in \{\emptyset\} = \mathcal{I}_{[[} \cup \{\emptyset\}$

hence we have that

$$\mathcal{S}_{[[} \subseteq \mathcal{I}_{[[} \cup \{\emptyset\} \quad (21.251)$$

If $I \in \mathcal{I}_{[[} \cup \{\emptyset\}$ then either we have:

$I \in \mathcal{I}_{[[}$. Then $\exists a, b \in \mathbb{R}$ [with $[a, b[\neq \emptyset$] such that $I = [a, b[$ so that $I \in \mathcal{S}_{[[}$.

$I \in \{\emptyset\}$. Then $I = \emptyset = [0, 0[$ so that $I \in \mathcal{S}_{[[}$

which proves that $\mathcal{I}_{[[} \cup \{\emptyset\} \subseteq \mathcal{S}_{[[}$. Combining this with [eq: 21.251] gives

$$\mathcal{S}_{[[} = \mathcal{I}_{[[} \cup \{\emptyset\}$$

Further if $I \in \mathcal{I}_{[[} \cap \{\emptyset\}$ then $I = \emptyset$ and there exists $a, b \in \mathbb{R}$ such that $I = [a, b[\neq \emptyset$ which is a contradiction. Hence

$$\mathcal{I}_{[[} \cap \{\emptyset\} = \emptyset \quad \square$$

Theorem 21.172. $\mathcal{S}_{[[}$ is semi-ring

Proof. First note that by [theorem: 21.171] that

$$\mathcal{S}_{||} = \{[a, b] \mid a, b \in \mathbb{R}\} = \mathcal{I}_{||} \bigcup \{\emptyset\}$$

Further we have:

$$1. \emptyset \in \mathcal{I}_{||} \bigcup \{\emptyset\} = \mathcal{S}_{||}$$

2. If $[a, b], [c, d] \in \mathcal{S}_{||}$ then we have either:

$$[a, b] = \emptyset \wedge [c, d] = \emptyset. \text{ Then } [a, b] \cap [c, d] = \emptyset \cap \emptyset = \emptyset \in \mathcal{S}_{||}.$$

$$[a, b] \in \mathcal{I}_{||} \wedge [c, d] = \emptyset. \text{ Then } [a, b] \cap [c, d] = [a, b] \cap \emptyset = \emptyset \in \mathcal{S}_{||}.$$

$$[a, b] = \emptyset \wedge [c, d] \in \mathcal{I}_{||}. \text{ Then } [a, b] \cap [c, d] = \emptyset \cap [c, d] = \emptyset \in \mathcal{S}_{||}.$$

$$[a, b] \in \mathcal{I}_{||} \wedge [c, d] \in \mathcal{I}_{||}. \text{ Then for } [a, b] \cap [c, d] \text{ we have either}$$

$$[a, b] \cap [c, d] = \emptyset. \text{ Then } [a, b] \cap [c, d] = \emptyset \in \mathcal{S}_{||}$$

$$[a, b] \cap [c, d] \neq \emptyset. \text{ Then for } a \text{ we have either:}$$

$$a \leq c. \text{ For } b \text{ we have either:}$$

$$b \leq d. \text{ Let } y \in [c, b] \text{ then } a \leq c \leq y < b \leq d \text{ so that } y \in [a, b] \wedge y \in [c, d] \text{ hence } y \in [a, b] \cap [c, d], \text{ proving that}$$

$$[c, b] \subseteq [a, b] \cap [c, d]$$

$$\text{Let } y \in [a, b] \cap [c, d] \text{ then } y \in [a, b] \wedge y \in [c, d] \text{ so that}$$

$$a \leq y < b \wedge c \leq y < d \Rightarrow c \leq y < b$$

$$\text{proving that } y \in [c, b] \text{ so that } [a, b] \cap [c, d] \subseteq [c, b]. \text{ Hence}$$

$$\emptyset \neq [a, b] \cap [c, d] = [c, b] \in \mathcal{I}_{||} \subseteq \mathcal{S}_{||}$$

$$d < b. \text{ Let } y \in [c, d] \text{ then } a \leq c \leq y < d < b \text{ so that } y \in [a, b] \text{ hence } [c, d] \subseteq [a, b] \text{ so that}$$

$$[a, b] \cap [c, d] = [c, d] \in \mathcal{S}_{||}$$

$$c < a. \text{ For } b \text{ we have either:}$$

$$b \leq d. \text{ Let } y \in [a, b] \text{ then } c < a \leq y < b \leq d \text{ so that } y \in [c, d] \text{ hence } [a, b] \subseteq [c, d] \text{ so that}$$

$$[a, b] \cap [c, d] = [a, b] \in \mathcal{S}_{||}$$

$$d < b. \text{ Let } y \in [a, d] \text{ then } c < a \leq y < d < b \text{ so that } y \in [a, b] \wedge y \in [c, d] \text{ hence } y \in [a, b] \cap [c, d], \text{ proving that}$$

$$[a, d] \subseteq [a, b] \cap [c, d]$$

$$\text{Let } y \in [a, b] \cap [c, d] \text{ then } y \in [a, b] \wedge y \in [c, d] \text{ so that}$$

$$a \leq y < b \wedge c \leq y < d \Rightarrow a \leq y < d$$

$$\text{proving that } y \in [a, d] \text{ or } [a, b] \cap [c, d] \subseteq [a, d]. \text{ Hence}$$

$$[a, b] \cap [c, d] = [a, d] \in \mathcal{S}_{||}$$

so in all cases we have that

$$[a, b] \cap [c, d] \in \mathcal{S}_{||}$$

3. Let $[a, b], [c, d] \in \mathcal{S}_{||}$ then we have either:

$$[a, b] = \emptyset \wedge [c, d] = \emptyset. \text{ Then } [a, b] \setminus [c, d] = \emptyset \setminus \emptyset = \emptyset = \bigcup_{i \in \{1\}} A_i \text{ where } \{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{||} \text{ is defined by } A_i = \emptyset.$$

$[a, b[\in \mathcal{I}_{[[} \wedge [c, d[= \emptyset$. Then $[a, b[\setminus [c, d[= [a, b[\setminus \emptyset = [a, b[= \bigsqcup_{i \in \{1\}} A_i$ where $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{[[}$ is defined by $A_i = [a, b[$.

$[a, b[= \emptyset \wedge [c, d[\in \mathcal{I}_{[[}$. Then $[a, b[\setminus [c, d[= \emptyset \setminus [c, d[= \emptyset = \bigsqcup_{i \in \{1\}} A_i$ where $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{[[}$ is defined by $A_i = \emptyset$.

$[a, b[\in \mathcal{I}_{[[} \wedge [c, d[\in \mathcal{I}_{[[}$. Then for $[a, b[\cap [c, d[$ we have either:

$[a, b[\cap [c, d[= \emptyset$. Then $[a, b[\setminus [c, d[= [a, b[= \bigsqcup_{i \in \{1\}} A_i$ where $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{[[}$ is defined by $A_i = [a, b[$.

$[a, b[\cap [c, d[\neq \emptyset$. As $[a, b[\neq \emptyset \neq [c, d[$ we have

$$a < b \wedge c < d \quad (21.252)$$

further there exist a x such that $a \leq x < b \wedge c \leq x < d$, hence

$$a < d \wedge c < b \quad (21.253)$$

For a, c we have either:

$a \leq c$. Then for b, d we have either:

$b \leq d$. Let $y \in [a, c[$ then $a \leq y < c \stackrel{[\text{eq: 21.253}]}{<} b$ so that $y \in [a, b[$. Assume that $y \in [c, d[$ then $c \leq y$ contradicting $y < c$ hence $y \notin [c, d[$ so that $y \in [a, b[\setminus [c, d[$. So we have

$$[a, c[\subseteq [a, b[\setminus [c, d[\quad (21.254)$$

On the other hand if $y \in [a, b[\setminus [c, d[$ we have $a \leq y < b \leq d$ and $\neg(c \leq y < d) \Rightarrow y < c \vee d \leq y$. Assume that $d \leq y$ then, as also $y < b \leq d \Rightarrow y < d$, we would have a contradiction so that we must have $d \not\leq y$ so that $y < c$. Hence we have $a \leq y < c$ which proves that $[a, b[\setminus [c, d[\subseteq [a, c[$, combining this with [eq: 21.254] proves

$$[a, b[\setminus [c, d[= [a, c[\in \mathcal{S}_{[[}$$

So if we define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{[[}$ by $A_1 = [a, c[$ we have

$$[a, b[\setminus [c, d[= \bigsqcup_{i \in \{1\}} A_i$$

$d < b$. Let $y \in [a, c[\cap [d, b[$ then $a \leq y < c \wedge d \leq y < b$, as by [eq: 21.252] $c < d$ we reach the contradicting $y < y$ so that

$$[a, c[\cap [d, b[= \emptyset \quad (21.255)$$

Let $y \in [a, c[\cup [d, b[$ then by the above we have either:

$y \in [a, c[$. Then $a \leq y < c \stackrel{[\text{eq: 21.253}]}{<} b$ so that $y \in [a, b[$. Assume that $y \in [c, d[$ then $c \leq y < d$ contradicting $y < c$ hence we have $y \notin [c, d[$ so

$$y \in [a, b[\setminus [c, d[$$

$y \in [d, b[$. Then $a \stackrel{[\text{eq: 21.253}]}{<} d \leq y < b$ so that $y \in [a, b[$. Assume that $y \in [c, d[$ then $c \leq y < d$ contradicting $d \leq y$, hence we must have $y \notin [c, d[$ or

$$y \in [a, b[\setminus [c, d[$$

as in all cases we have $y \in [a, b[\setminus [c, d[$ it follows that

$$[a, c[\cup [d, b[\subseteq [a, b[\setminus [c, d[\quad (21.256)$$

On the other hand if $y \in [a, b[\setminus [c, d[$ then

$$\begin{aligned} a \leq y < b \wedge \neg(c \leq y \wedge y < d) &= \\ a \leq y < b \wedge (y < c \vee d \leq y) &= \\ (a \leq y < b \wedge y < c) \vee (a \leq y < b \wedge d \leq y) \end{aligned}$$

so that $y \in [a, c[\vee y \in [d, b[\Rightarrow y \in [a, c[\cup [d, b[$ which proves that $[a, b[\setminus [c, d[\subseteq [a, c[\cup [d, b[$. Combining this with [eq: 21.256]

$$[a, b[\setminus [c, d[= [a, c[\cup [d, b[$$

Further if $x \in [a, c[\cap [d, b[$ then $a \leq x < c \wedge d \leq x < b$ so that $d < c$ which contradicts with [eq: 21.252]. Hence we have

$$[a, c[\cap [d, b[= \emptyset$$

Hence if we define $\{A_i\}_{i \in \{1,2\}} \subseteq \mathcal{S}_{||}$ by $A_1 = [a, c[$, $A_2 = [d, b[$ then $\{A_i\}_{i \in \{1,2\}}$ is pairwise disjoint and

$$[a, b[\setminus [c, d[= \bigsqcup_{i \in \{1,2\}} A_i$$

$c < a$. Then for b we have either:

$b \leq d$. If $y \in [a, b[$ then $c < a \leq y < b \leq d$ so that $y \in [c, d[$, hence $[a, b[\subseteq [c, d[$ so that $[a, b[\setminus [c, d[= \emptyset$. Hence if we define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{||}$ by $A_1 = \emptyset$ it follows that

$$[a, b[\setminus [c, d[= \bigsqcup_{i \in \{1\}} A_i$$

$d < b$. Let $y \in [d, b[$ then $a \leq d \leq y < b$ so that $y \in [a, b[$. [eq: 21.253]

Assume that $y \in [c, d[$ then $c \leq y < d$ contradicting $d \leq y$. Hence $y \in [a, b[\setminus [c, d[$ which proves that

$$[d, b[\subseteq [a, b[\setminus [c, d[\quad (21.257)$$

If $y \in [a, b[\setminus [c, d[$ then

$$\begin{aligned} a \leq y < b \wedge \neg(c \leq y < d) &= \\ a \leq y < b \wedge (y < c \vee d \leq y) &= \\ (a \leq y < b \wedge y < c) \vee (a \leq y < b \wedge d \leq y) &\stackrel{c < a}{=} \\ (c < a \leq y < b \wedge y < c) \vee (c < a \leq y < b \wedge d \leq y) &\stackrel{c \not\leq c}{=} \\ c < a \leq y < b \wedge d \leq y \end{aligned}$$

so that $y \in [d, b[$. Hence $[a, b[\setminus [c, d[\subseteq [d, b[$ which combined with [eq: 21.257] gives

$$[a, b[\setminus [c, d[= [d, b[$$

Define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{S}_{||}$ by $A_1 = [d, b[$ then

$$[a, b[\setminus [c, d[= \bigsqcup_{i \in \{1\}} A_i$$

So in all cases we found a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{S}_{||}$ such that

$$[a, b[\setminus [c, d[= \bigsqcup_{i \in \{1, \dots, m\}} A_i$$

□

Theorem 21.173. For $\mathcal{S}_{||} = \mathcal{I}_{||} \sqcup \{\emptyset\}$ where $\mathcal{I}_{||} = \{[a, b[\mid a, b \in \mathbb{R} \text{ such that } [a, b[\neq \emptyset\}$ we have

$$\sigma[\mathcal{S}_{||}] = \sigma[\mathcal{I}_{||}]$$

so that using [theorem: 21.55]

$$\begin{aligned} \sigma[\{C \subseteq \mathbb{R} \mid C \text{ is closed in } \mathcal{I}_{||}\}] &\stackrel{[\text{theorem: 21.47}]}{=} \mathcal{B}[\mathbb{R}] \\ &= \sigma[\mathcal{I}_{||}] \\ &= \sigma[\mathcal{I}_{|}] \\ &= \sigma[\mathcal{I}_{|}] \\ &= \sigma[\mathcal{I}_{|}] \\ &= \sigma[\mathcal{I}_{|}] \\ &= \sigma[\mathcal{I}_{|}] \\ &= \sigma[\mathcal{S}_{||}] \\ &= \sigma[\{[a, \infty[\mid a \in \mathbb{R}\}] \\ &= \sigma[\{]a, \infty[\mid a \in \mathbb{R}\}] \\ &= \sigma[\{]-\infty, a[\mid a \in \mathbb{R}\}] \\ &= \sigma[\{]-\infty, a[\mid a \in \mathbb{R}\}] \end{aligned}$$

Proof. As $\mathcal{I}_{||} \subseteq \mathcal{S}_{||} \subseteq \sigma[\mathcal{S}_{||}]$ it follows from [theorem: 21.42] and the fact that $\sigma[\mathcal{S}_{||}]$ is a σ -algebra that

$$\sigma[\mathcal{I}_{||}] \subseteq \sigma[\mathcal{S}_{||}] \quad (21.258)$$

Let $A \in \mathcal{S}_{||}$ then either $A \in \mathcal{I}_{||} \subseteq \sigma[\mathcal{I}_{||}]$ or $A = \emptyset \in \sigma[\mathcal{I}_{||}]$ which proves that $\mathcal{S}_{||} \subseteq \sigma[\mathcal{I}_{||}]$. Hence using [theorem: 21.42] and the fact that $\sigma[\mathcal{I}_{||}]$ is a σ -algebra it follows that $\sigma[\mathcal{S}_{||}] \subseteq \sigma[\mathcal{I}_{||}]$. Combining this with [eq: 21.258] proves that

$$\sigma[\mathcal{S}_{||}] = \sigma[\mathcal{I}_{||}] \quad \square$$

Definition 21.174. We define

$$\text{begin: } \mathcal{I}_{||} \rightarrow \mathbb{R} \text{ by } \text{begin}(I) = a \text{ where } I = [a, b[$$

$$\text{end: } \mathcal{I}_{||} \rightarrow \mathbb{R} \text{ by } \text{end}(I) = b \text{ where } I = [a, b[$$

Note 21.175. If $\emptyset \neq I = [a, b[= [c, d[$ then by [theorem: 3.145(7)] we have $a = c \wedge b = d$ so these functions are well defined.

Note 21.176. As $I \neq \emptyset$ we have always that $\text{begin}(I) < \text{end}(I)$

Using the above two functions we can now define the length function.

Definition 21.177. We define

$$\text{len: } \mathcal{S}_{||} \rightarrow [0, \infty] \text{ by } \text{len}(I) = \begin{cases} 0 & \text{if } I = \emptyset \\ \text{end}(I) - \text{begin}(I) & \text{if } I \in \mathcal{I}_{||} \end{cases}$$

In other words the function len calculates the length of a interval.

To simplify the proof that len is a content we need to order the half open intervals based on their boundary points. We define now a total order on $\mathcal{I}_{||}$ the set of half-open intervals.

Theorem 21.178. The relation $\preceq \subseteq \mathcal{I}_{||} \times \mathcal{I}_{||}$ defined by

$$[a, b[\preceq [c, d[\text{ if } a < c \vee (a = c \wedge b \leq d)$$

is a total order on $\mathcal{I}_{[}$ making $\langle \mathcal{I}_{[}, \preceq \rangle$ a totally ordered set.

Proof. We have

reflexivity. Let $[a, b[\in \mathcal{I}_{[}$ then as $a = a$ and $b = b \Rightarrow b \leq b$ it follows that $[a, b[\preceq [a, b[$ proving reflexivity.

anti-symmetry. Let $[a, b[, [c, d[\in \mathcal{I}_{[}$ be such that $[a, b[\preceq [c, d[$ and $[c, d[\preceq [a, b[$. As $[a, b[\preceq [c, d[$ we have either:

$a < c$. Then as $[c, d[\preceq [a, b[$ we have either $c = a$ or $c < a$ which both give a contradiction, so this cases never occurs.

$a = c$. Then from $[a, b[\preceq [c, d[$ it follows that $b \leq d$ and from $[c, d[\preceq [a, b[$ it follows that $d \leq b$. Hence we have $a = c \wedge b = d$ so that $[a, b[= [c, d[$

transitivity. Assume that $[a, b[\preceq [c, d[$ and $[c, d[\preceq [e, f[$. As $[a, b[\preceq [c, d[$ we have either:

$a < c$. Then as $[c, d[\preceq [e, f[$ we have either:

$c < e$. Then $a < e$ so that $[a, b[\preceq [e, f[$

$c = e$. Then $a < e$ so that $[a, b[\preceq [e, f[$

$a = c$. Then as $[a, b[\preceq [c, d[$ we must have that

$$b \leq d \tag{21.259}$$

As $[c, d[\preceq [e, f[$ we have either:

$c < e$. Then $a < e$ so that $[a, b[\preceq [e, f[$.

$c = e$. Then as $[c, d[\preceq [e, f[$ we have that $b \leq d \leq f$ which combined with $a = c = e$ proves that $[a, b[\preceq [e, f[$. [eq: 21.259]

So in all cases we have $[a, b[\preceq [e, f[$ proving transitivity.

totally. If $[a, b[, [c, d[$ then, as $a, b, c, d \in \mathbb{R}$ and $\langle \mathbb{R}, \leq \rangle$ is totally ordered, we have for a, c either:

$a < c$. Then $[a, b[\preceq [c, d[$

$a = c$. Then as $\langle \mathbb{R}, \leq \rangle$ is totally ordered we have for b, d either:

$b < d$. Then $[a, b[\preceq [c, d[$

$d \leq b$. Then $[c, d[\preceq [a, b[$

$c < a$. Then $[c, d[\preceq [a, b[$ □

So we can sort half open intervals which will be used in the next theorem.

Theorem 21.179. Let $n \in \mathbb{N}$, $\{[a_i, b_i[\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_{[}$ pairwise disjoint such that

$$\bigsqcup_{i \in \{1, \dots, n\}} [a_i, b_i[\in \mathcal{I}_{[}$$

then there exist a bijection $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that

$$\forall i \in \{1, \dots, n-1\} \text{ we have } b_{\sigma(i)} = a_{\sigma(i+1)}$$

and

$$\bigsqcup_{i \in \{1, \dots, n\}} [a_i, b_i[= [a_{\sigma(1)}, b_{\sigma(n)}]$$

Proof. For $n \in \mathbb{N}$ we have either:

$n = 1$. Take $\sigma = \text{Id}_{\{1\}}: \{1\} \rightarrow \{1\}$ then σ is a bijection, further as $\{1, \dots, n-1\} = \emptyset$ we have vacuously that $\forall i \in \{1, \dots, n-1\} b_{\sigma(i)} = a_{\sigma(i+1)}$, finally $\bigsqcup_{i \in \{1\}} [a_i, b_i[= [a_1, b_1[= [a_{\sigma(1)}, b_{\sigma(1)}]$.

$n \in \{2, \dots, \infty\}$. As $\langle \mathcal{I}_[, \preccurlyeq \rangle$ is totally ordered there exist by [corollary: 6.55] a bijection $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that

$$\forall i, j \in \{1, \dots, n\} \text{ with } i \leq j \text{ we have } [a_{\sigma(i)}, b_{\sigma(i)}[\preccurlyeq [a_{\sigma(j)}, b_{\sigma(j)}[\quad (21.260)$$

Let $i, j \in \{1, \dots, n\}$ with $i \leq j$ then as $[a_{\sigma(i)}, b_{\sigma(i)}[\preccurlyeq [a_{\sigma(j)}, b_{\sigma(j)}[$ we have either $a_{\sigma(i)} < a_{\sigma(j)} \Rightarrow a_{\sigma(i)} \leq a_{\sigma(j)}$ or $a_{\sigma(i)} = a_{\sigma(i+1)} \Rightarrow a_{\sigma(i)} \leq a_{\sigma(j)}$ so that

$$\forall i, j \in \{1, \dots, n\} \text{ with } i \leq j \text{ we have } a_{\sigma(i)} \leq a_{\sigma(j)} \quad (21.261)$$

Further by [theorem: 21.5] we have that

$$\{[a_{\sigma(i)}, b_{\sigma(i)}[\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_[\text{ is pairwise disjoint} \quad (21.262)$$

Assume that there exist a $k \in \{1, \dots, n-1\}$ such that $a_{\sigma(k+1)} < b_{\sigma(k)}$. Then by [eq: 21.261] $a_{\sigma(k)} \leq a_{\sigma(k+1)} < b_{\sigma(k)}$ from which it follows that $a_{\sigma(k+1)} \in [a_{\sigma(k)}, b_{\sigma(k)}[\cap [a_{\sigma(k+1)}, b_{\sigma(k+1)}[$ contradicting $[a_{\sigma(k)}, b_{\sigma(k)}[\cap [a_{\sigma(k+1)}, b_{\sigma(k+1)}[\stackrel{[eq: 21.262]}{=} \emptyset$. Hence the assumption is wrong and we conclude that

$$\forall i \in \{1, \dots, n-1\} \text{ we have } b_{\sigma(i)} \leq a_{\sigma(i+1)} \quad (21.263)$$

As $\bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[\in \mathcal{I}_[$ there exist a $c, d \in \mathbb{R}$ with $c < d$ such that

$$\bigcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[\stackrel{[theorem: 2.127]}{=} \bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[= [c, d[\quad (21.264)$$

If $i = \{1, \dots, n-1\}$ then by [eq: 21.263] we have $b_{\sigma(i)} \leq a_{\sigma(i+1)}$ and as $i \leq n-1 < n \Rightarrow i+1 \leq n$ it follows from [eq: 21.261] that $a_{\sigma(i+1)} \leq a_{\sigma(n)} < b_{\sigma(n)}$ so that $b_{\sigma(i)} < b_{\sigma(n)}$, hence

$$\forall i \in \{1, \dots, n\} \text{ we have } b_{\sigma(i)} \leq b_{\sigma(n)} \quad (21.265)$$

Let $i \in \{1, \dots, n\}$ then for $x \in [a_{\sigma(i)}, b_{\sigma(i)}[$ we have $a_{\sigma(1)} \stackrel{[eq: 21.261]}{\leq} a_{\sigma(i)} \leq x < b_{\sigma(i)} \stackrel{[eq: 21.265]}{\leq} b_{\sigma(n)}$ so that $x \in [a_{\sigma(1)}, b_{\sigma(n)}[$ hence $[a_{\sigma(i)}, b_{\sigma(i)}[\subseteq [a_{\sigma(1)}, b_{\sigma(n)}[$. So

$$[c, d[\stackrel{[eq: 21.264]}{=} \bigcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[\subseteq [a_{\sigma(1)}, b_{\sigma(n)}[\quad (21.266)$$

As $[c, d[\subseteq [a_{\sigma(1)}, b_{\sigma(n)}[$ it follows from [theorem: 3.145(9)] that

$$a_{\sigma(1)} \leq c \wedge d \leq b_{\sigma(n)} \quad (21.267)$$

As $a_{\sigma(1)} \in [a_{\sigma(1)}, b_{\sigma(1)}[\subseteq \bigcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[\stackrel{[eq: 21.264]}{=} [c, d[$ it follows that $c \leq a_{\sigma(1)}$ which combined with [eq: 21.267] proves that

$$a_{\sigma(1)} = c \quad (21.268)$$

As $a_{\sigma(n)} \in [a_{\sigma(n)}, b_{\sigma(n)}[\subseteq \bigcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[\stackrel{[eq: 21.264]}{=} [c, d[$ we have $c \leq a_{\sigma(n)} < d$. Assume that $d < b_{\sigma(n)}$ then $a_{\sigma(n)} < d < b_{\sigma(n)}$, proving that $d \in [a_{\sigma(n)}, b_{\sigma(n)}[\subseteq [c, d[$ leading to the contradiction $d < d$. Hence we must have $b_{\sigma(n)} \leq d$ which combined with [eq: 21.267] proves

$$b_{\sigma(n)} = d \quad (21.269)$$

Summarizing [eqs: 21.264, 21.268 and 21.269] we have

$$\bigsqcup_{i \in \{1, \dots, n\}} [a_i, b_i[= \bigsqcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[= [a_{\sigma(1)}, b_{\sigma(n)}[\quad (21.270)$$

Assume now that there exist a $k \in \{1, \dots, n-1\}$ such that $b_{\sigma(k)} \neq a_{\sigma(k+1)}$. If $a_{\sigma(k+1)} < b_{\sigma(k)}$ then by [eq: 21.263] $b_{\sigma(k)} \leq a_{\sigma(k+1)}$ and we reach the contradiction $a_{\sigma(k+1)} < a_{\sigma(k+1)}$ so we must have that $b_{\sigma(k)} \leq a_{\sigma(k+1)}$ which as $b_{\sigma(k)} \neq a_{\sigma(k+1)}$ proves that $b_{\sigma(k)} < a_{\sigma(k+1)}$. Hence there exist a $x \in \mathbb{R}$ such that

$$b_{\sigma(k)} < x < a_{\sigma(k+1)} \text{ so that } x \notin [a_{\sigma(k)}, b_{\sigma(k)}[\text{ and } x \notin [a_{\sigma(k+1)}, b_{\sigma(k+1)}[\quad (21.271)$$

If $i \in \{1, \dots, k-1\}$ then $b_{\sigma(i)} \underset{[\text{eq: 21.263}]}{\leq} a_{\sigma(i+1)} \underset{[\text{eq: 21.261}]}{\leq} a_{\sigma(k)} < b_{\sigma(k)} < x$ so that $x \notin [a_{\sigma(i)}, b_{\sigma(i)}[$.
 Further if $i \in \{k+1, \dots, n\}$ then $x < a_{\sigma(k+1)} \leq a_{\sigma(i)}$ so that $x \notin [a_{\sigma(i)}, b_{\sigma(i)}[$. From this we conclude that $\forall i \in \{1, \dots, n\}$ $x \notin [a_{\sigma(i)}, b_{\sigma(i)}[$ or

$$x \notin \bigcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[= [a_{\sigma(1)}, b_{\sigma(n)}[\quad (21.272)$$

As $a_{\sigma(1)} \underset{[\text{eq: 21.261}]}{\leq} a_{\sigma(k)} < b_{\sigma(k)} < x$ and $x < a_{\sigma(k+1)} \underset{[\text{eq: 21.261}]}{\leq} a_{\sigma(n)} < b_{\sigma(n)}$ so that $x \in [a_{\sigma(1)}, b_{\sigma(n)}[$ which contradicts [eq: 21.272]. Hence the assumption is false and we must have

$$\forall i \in \{1, \dots, n-1\} \text{ that } b_{\sigma(i)} = a_{\sigma(i+1)} \quad \square$$

We are now ready to prove that len is a content on the semi-ring $\mathcal{S}_{[[$.

Lemma 21.180. *The length function $\text{len}: \mathcal{S}_{[[} \rightarrow [0, \infty]$ defined by $\text{len}(I) = \begin{cases} 0 & \text{if } I = \emptyset \\ \text{end}(I) - \text{begin}(I) & \text{if } I \in \mathcal{I}_{[[} \end{cases}$ [see definition: 21.177] is a finite content on the semi-ring $\mathcal{S}_{[[$.*

Proof. First by definition we have

$$\text{len}(\emptyset) = 0 \quad (21.273)$$

Second $\forall I \in \mathcal{S}_{[[}$ we have $\text{len}(I) = \begin{cases} 0 < \infty & \text{if } I = \emptyset \\ \text{end}(I) - \text{begin}(I) < \infty & \text{if } I \in \mathcal{I}_{[[} \end{cases}$ which proves that

$$\text{len} \text{ is a finite} \quad (21.274)$$

Next we prove that len is additive on $\mathcal{I}_{[[$. Let $\{I_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_{[[}$, $n \in \mathbb{N}$ pairwise disjoint such that $\bigsqcup_{i \in \{1, \dots, n\}} I_i \in \mathcal{I}_{[[}$ then we have for $n \in \mathbb{N}$ either:

$$n = 1. \text{ Then } \text{len}(\bigsqcup_{i \in \{1, \dots, n\}} I_i) = \text{len}(\bigsqcup_{i \in \{1\}} I_i) = \text{len}(I_1) = \sum_{i=1}^1 \text{len}(I_i) = \sum_{i=1}^n \text{len}(I_i)$$

$1 < n$. Then $\forall i \in \{1, \dots, n\}$ there exists $a_i, b_i \in \mathbb{R}$ with $a_i < b_i$ such that $I_i = [a_i, b_i[$. Using [theorem: 21.179] there exist a bijection $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that

$$\bigsqcup_{i \in \{1, \dots, n\}} I_i = \bigsqcup_{i \in \{1, \dots, n\}} [a_i, b_i[= \bigsqcup_{i \in \{1, \dots, n\}} [a_{\sigma(i)}, b_{\sigma(i)}[= [a_{\sigma(1)}, b_{\sigma(n)}[\quad (21.275)$$

such that

$$\forall i \in \{1, \dots, n-1\} \text{ } b_{\sigma(i)} = a_{\sigma(i+1)} \quad (21.276)$$

Then

$$\begin{aligned} \sum_{i=1}^n \text{len}(I_i) &\underset{[\text{theorem: 11.31}]}{=} \sum_{i=1}^n \text{len}(I_{\sigma(i)}) \\ &= \sum_{i=1}^n \text{len}([a_{\sigma(i)}, b_{\sigma(i)}[) \\ &= \sum_{i=1}^n (b_{\sigma(i)} - a_{\sigma(i)}) \\ &\underset{1 < n}{=} (b_{\sigma(n)} - a_{\sigma(n)}) + \sum_{i=1}^{n-1} (b_{\sigma(i)} - a_{\sigma(i)}) \\ &\underset{[\text{eq: 21.276}]}{=} (b_{\sigma(n)} - a_{\sigma(n)}) + \sum_{i=1}^{n-1} (a_{\sigma(i+1)} - a_{\sigma(i)}) \\ &\underset{[\text{theorem: 11.10}]}{=} b_{\sigma(n)} - a_{\sigma(n)} + a_{\sigma(n)} - a_{\sigma(1)} \\ &= b_{\sigma(n)} - a_{\sigma(1)} \\ &= \text{len}([a_{\sigma(1)}, b_{\sigma(n)}[) \\ &\underset{[\text{eq: 21.275}]}{=} \text{len}\left(\bigsqcup_{i \in \{1, \dots, n\}} I_i\right) \end{aligned}$$

To summarize: $\forall \{I_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_{||}$ pairwise disjoint with $\bigsqcup_{i \in \{1, \dots, n\}} I_i \in \mathcal{I}_{||}$ we have

$$\text{len} \left(\bigsqcup_{i \in \{1, \dots, n\}} I_i \right) = \sum_{i=1}^n \text{len}(I_i) \quad (21.277)$$

Next we extend the above to $\mathcal{S}_{||}$. Let $\{I_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}_{||}$, $n \in \mathbb{N}$ such that $\bigsqcup_{i \in \{1, \dots, n\}} I_i \in \mathcal{S}_{||}$. Define $N = \{i \in \{1, \dots, n\} | I_i \neq \emptyset\}$ then for N we have either:

$N = \emptyset$. Then

$$\text{len} \left(\bigsqcup_{i \in \{1, \dots, n\}} I_i \right) = \text{len}(\emptyset) = 0 = \sum_{i=1}^n 0 = \sum_{i=1}^n \text{len}(I_i)$$

$N \neq \emptyset$. Then as $\emptyset \neq N$ is finite there exist a $m \in \{1, \dots, n\}$ and a bijection $\beta: \{1, \dots, m\} \rightarrow N$. Further we have

$$\begin{aligned} \text{len} \left(\bigsqcup_{i \in \{1, \dots, n\}} I_i \right) &\stackrel{[\text{theorem: 2.142}]}{=} \text{len} \left(\bigsqcup_{i \in N} I_i \right) \\ &\stackrel{[\text{theorem: 2.127}]}{=} \text{len} \left(\bigsqcup_{i \in \{1, \dots, m\}} I_{\sigma(i)} \right) \\ &\stackrel{[\text{eq: 21.277}]}{=} \sum_{i=1}^m \text{len}(I_{\sigma(i)}) \\ &= \sum_{i \in \{1, \dots, m\}} \text{len}(I_{\sigma(i)}) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in N} \text{len}(I_i) \\ &= \sum_{i \in N} \text{len}(I_i) + \sum_{i \in \{1, \dots, n\} \setminus N} 0 \\ &= \sum_{i \in N} \text{len}(I_i) + \sum_{i \in \{1, \dots, n\} \setminus N} \text{len}(I_i) \\ &= \sum_{i \in \{1, \dots, n\}} \text{len}(I_i) \\ &= \sum_{i=1}^n \text{len}(I_i) \end{aligned}$$

So in all cases we have that

$$\text{Let } \{I_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}_{||}, \quad n \in \mathbb{N} \quad \text{pairwise disjoint with } \bigsqcup_{i \in \{1, \dots, n\}} I_i \in \mathcal{S}_{||} \quad \text{then} \\ \text{len} \left(\bigsqcup_{i \in \{1, \dots, n\}} I_i \right) = \sum_{i=1}^n \text{len}(I_i) \quad (21.278)$$

So using [eqs: 21.273, 21.274 and 21.278] it follows that len is a finite content on the semi-ring $\mathcal{S}_{||}$. $\square$

Next we prove that len is a pre-measure on $\mathcal{S}_{||}$.

Lemma 21.181. *The function $\text{len}: \mathcal{S}_{||} \rightarrow [0, \infty]$ [see definition: 21.177] is a pre-measure on the semi-ring $\mathcal{S}_{||}$*

Proof. Consider the topological space $(\mathbb{R}, \mathcal{T}_{||})$. Let $A \in \mathcal{S}_{||}$ and $\varepsilon \in \mathbb{R}^+$ then for A we have either:

$A = \emptyset$. Take then $A_\varepsilon = \emptyset \in \mathcal{S}_{||}$ and $K_\varepsilon = \emptyset$. Then K_ε is compact by [example: 14.269], $A_\varepsilon \subseteq K_\varepsilon \subseteq A$ and $\text{len}(A) - \text{len}(A_\varepsilon) = 0 - 0 = 0 < \varepsilon$.

$A \neq \emptyset$. Then there exists $a, b \in \mathbb{R}$ with $a < b$ such that $A = [a, b]$. Define then $A_\varepsilon = [a, b - \frac{\varepsilon}{2}]$ and $K_\varepsilon = [a, b - \frac{\varepsilon}{2}]$. Then by the Heine Borel theorem [theorem: 14.279] K_ε is compact. Further $A_\varepsilon \subseteq K_\varepsilon \subseteq A$ and $\text{len}(A) - \text{len}(A_\varepsilon) = \text{len}([a, b]) - \text{len}([a, b - \frac{\varepsilon}{2}]) = b - a - (b - \frac{\varepsilon}{2} - a) = \frac{\varepsilon}{2} < \varepsilon$

Hence using [theorem: 21.169] it follows that

$$\text{len is a pre-measure on } \mathcal{S}_{[\cdot]} \quad \square$$

Next we use a derivative of the Carathéodry theorem [theorem: 21.148] to extend len to the Lebesgue measure on Lebesgue measurable sets in $\mathbb{R}$.

Theorem 21.182. *If we define*

$$\lambda^*: \mathcal{P}[\mathbb{R}] \rightarrow [0, \infty] \text{ by } \lambda^*(A) = \begin{cases} \infty & \text{if } \mathcal{S}[A] = \emptyset \\ \inf(\{\sum_{i=1}^{\infty} \text{len}(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A]\}) & \text{if } \mathcal{S}[A] \neq \emptyset \end{cases}$$

where

$$\mathcal{S}[A] = \left\{ \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}_{[\cdot]} \mid A \subseteq \bigcup_{i \in \mathbb{N}} A_i \right\}$$

then

λ^* is a outer measure

$$\mathcal{L}[\mathbb{R}] = \{A \in \mathcal{P}(\mathbb{R}) \mid \forall B \in \mathcal{P}(\mathbb{R}) \text{ we have } \lambda^*(B) = \lambda^*(B \cap A) + \lambda^*(B \setminus A)\}$$

is a σ -algebra on $\mathbb{R}$ such that

$$\sigma[\mathcal{S}_{[\cdot]}] \subseteq \mathcal{L}[\mathbb{R}]$$

and

$$\bar{\lambda} = (\lambda^*)|_{\mathcal{L}[\mathbb{R}]}: \mathcal{L}[\mathbb{R}] \rightarrow [0, \infty] \text{ is a measure on } \mathcal{L}[\mathbb{R}]$$

and

$$\lambda = (\lambda^*)|_{\sigma[\mathcal{S}_{[\cdot]}]}: \sigma[\mathcal{S}_{[\cdot]}] \rightarrow [0, \infty] \text{ is a measure on } \sigma[\mathcal{S}_{[\cdot]}]$$

and

$$(\lambda^*)|_{\mathcal{S}_{[\cdot]}} = \text{len}$$

Further if $\mu: \sigma[\mathcal{S}_{[\cdot]}] \rightarrow [0, \infty]$ is another measure such that $\mu|_{\mathcal{S}_{[\cdot]}} = \text{len}$ then

$$\mu = \lambda$$

So

$$\langle \mathbb{R}, \mathcal{L}[\mathbb{R}], \bar{\lambda} \rangle \text{ is a measure space called the Lebesgue measure space on } \mathbb{R}$$

where

$$\mathcal{L}[\mathbb{R}] \text{ is called the set of Lebesgue measurable sets}$$

and

$$\bar{\lambda} \text{ the Lebesgue measure}$$

Note 21.183. As by [theorem: 21.173] $\mathcal{B}[\mathbb{R}] = \sigma[\mathcal{S}_{[\cdot]}]$ we have also that

$$\mathcal{B}[\mathbb{R}] \subseteq \mathcal{L}[\mathbb{R}]$$

and

$$\lambda = (\bar{\lambda})|_{\mathcal{B}[\mathbb{R}]}: \mathcal{B}[\mathbb{R}] \rightarrow [0, \infty] \text{ is a measure on } \mathcal{B}[\mathbb{R}]$$

So

$$\langle \mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda \rangle \text{ is a measure space called the Borel measure space on } \mathbb{R}$$

where

$\mathcal{B}[\mathbb{R}]$ is called the set of Borel sets on $\mathbb{R}$

and

λ is called the Borel measure on $\mathbb{R}$

Proof. Using [lemma: 21.181] we have that

len is a pre-measure on the semi-ring $\mathcal{S}_{\llbracket}$

Define $\{[-i, i]\}_{i \in \mathbb{N}} \subseteq \mathcal{S}_{\llbracket}$ then we have

$$\forall i \in \mathbb{N} \text{ that } [-i, i] \subseteq [-(i+1), i+1[\text{ and } \text{len}([-i, i]) = 2 \cdot i < \infty$$

If $x \in \mathbb{R}$ then by [theorem: 10.30] there exist $n, m \in \mathbb{N}$ such that $-n < x < m$. Let $i = \max(n, m)$ then $-i \leq -n < x < m < i$ so that $x \in [-i, i[$ from which it follows that $\mathbb{R} \subseteq \bigcup_{i \in \mathbb{N}} [-i, i[$. As $\forall i \in \mathbb{N}$ $[-i, i] \subseteq \mathbb{R}$ we have $\bigcup_{i \in \mathbb{N}} [-i, i] \subseteq \mathbb{R}$. Hence we have

$$\mathbb{R} = \bigcup_{i \in \mathbb{N}} [-i, i[$$

The proof follows then from applying [theorem: 21.148]. $\square$

21.3.4.4 Product of semi-rings

If we want to use the Carathéodory theorem to create a measure on $\mathcal{B}[\mathbb{R}^n]$ we have to start with a semi-ring, as $\mathcal{B}[\mathbb{R}^n] = \sigma[\mathcal{R}_{\llbracket}^n]$ it seems sensible to prove that $\mathcal{R}_{\llbracket}^n \cup \{\emptyset\}$ is a semi-ring on $\mathbb{R}^n$. First note that $R \in \mathcal{R}_{\llbracket}^n \cup \{\emptyset\}$ can be written as $R = \prod_{i \in \{1, \dots, n\}} I_i$ where $I_i \in \mathcal{S}_{\llbracket}$. This suggests that we look at finite products of semi-rings.

Definition 21.184. Let $n \in \mathbb{N} \setminus \{1\}$, $\{X_i, \mathcal{S}_i\}_{i \in \{1, \dots, n\}}$ a finite family such that $\forall i \in \{1, \dots, n\}$ $\mathcal{S}_i \subseteq \mathcal{P}(X_i)$ then $\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ is defined by

$$\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i = \left\{ \prod_{i \in \{1, \dots, n\}} A_i \mid \forall i \in \{1, \dots, n\} A_i \in \mathcal{S}_i \right\}$$

Note 21.185. In the particular cases where $n=2$ we note $\odot_{i \in \{1, 2\}} \mathcal{S}_i$ as $\mathcal{S}_1 \odot \mathcal{S}_2$. In other words

$$\mathcal{S}_1 \odot \mathcal{S}_2 = \{A \cdot B \mid A \in \mathcal{S}_1 \wedge B \in \mathcal{S}_2\}$$

Next we will prove using mathematical induction that $\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ is a semi-ring on $\prod_{i \in \{1, \dots, n\}} X_i$ if $\forall i \in \{1, \dots, n\}$ $\mathcal{S}_i$ is a semi-ring on X_i . The next lemma will be used in the base case and the induction step.

Lemma 21.186. Let X_1, X_2 be two sets and let $\mathcal{S}_1 \subseteq \mathcal{P}(X_1)$, $\mathcal{S}_2 \subseteq \mathcal{P}(X_2)$ be semi-rings on X_1 and X_2 respectively then $\mathcal{S}_1 \odot \mathcal{S}_2$ is a semi-ring on $X_1 \cdot X_2$.

Proof. As $\mathcal{S}_1, \mathcal{S}_2$ are semi-rings we have that $\emptyset \in \mathcal{S}_1 \wedge \emptyset \in \mathcal{S}_2$ so that

$$\emptyset = \emptyset \cdot \emptyset \in \mathcal{S}_1 \odot \mathcal{S}_2 \tag{21.279}$$

Let $R, S \in \mathcal{S}_1 \odot \mathcal{S}_2$ then there exists $A, C \in \mathcal{S}_1$ and $B, D \in \mathcal{S}_2$ such that

$$R = A \cdot B \text{ and } S = C \cdot D$$

Then we have

$$\begin{aligned} R \bigcap S &= (A \cdot B) \bigcap (C \cdot D) \\ &\stackrel{[\text{theorem: 6.89}]}{=} (A \bigcap C) \cdot (B \bigcap D) \end{aligned}$$

given that $\mathcal{S}_1, \mathcal{S}_2$ are semi-rings it follows that $A \cap C \in \mathcal{S}_1 \wedge B \cap D \in \mathcal{S}_2$ so that

$$R \cap S \in \mathcal{S}_1 \odot \mathcal{S}_2 \quad (21.280)$$

Using [theorem: 21.2] we have that

$$\begin{aligned} R \setminus S &= (A \cdot B) \setminus (C \cdot D) \\ &= (A \setminus C) \cdot (B \setminus D) \cup (A \setminus C) \cdot (B \cap D) \cup (A \cap C) \cdot (B \setminus D) \end{aligned}$$

where

1. $((A \setminus C) \cdot (B \setminus D)) \cap ((A \setminus C) \cdot (B \cap D)) = \emptyset$
2. $((A \setminus C) \cdot (B \setminus D)) \cap ((A \cap C) \cdot (B \setminus D)) = \emptyset$
3. $((A \setminus C) \cdot (B \cap D)) \cap ((A \cap C) \cdot (B \setminus D)) = \emptyset$

So if we define $\{R_i\}_{i \in \{1,2,3\}}$ by

$$\begin{aligned} R_1 &= (A \setminus C) \cdot (B \setminus D) \\ R_2 &= (A \setminus C) \cdot (B \cap D) \\ R_3 &= (A \cap C) \cdot (B \setminus D) \end{aligned}$$

then we have

$$\{R_i\}_{i \in \{1,2,3\}} \text{ is pairwise disjoint and } R \setminus S = \bigsqcup_{i \in \{1,2,3\}} R_i \quad (21.281)$$

As $A, C \in \mathcal{S}_1$ a semi ring there exist a pairwise disjoint $\{A_i\}_{i \in \{1, \dots, m_1\}} \subseteq \mathcal{S}_1$ such that

$$A \setminus C = \bigsqcup_{i \in \{1, \dots, m_1\}} A_i \quad (21.282)$$

likewise as $B, D \in \mathcal{S}_1$ there exist a pairwise disjoint $\{B_i\}_{i \in \{1, \dots, m_2\}} \subseteq \mathcal{S}_2$ such that

$$B \setminus D = \bigsqcup_{i \in \{1, \dots, m_2\}} B_i \quad (21.283)$$

Now for $i \in \{1, 2, 3\}$ we have either:

$i = 1$. Then we have

$$\begin{aligned} R_1 &= (A \setminus C) \cdot (B \setminus D) \\ &\stackrel{[\text{eq: 21.282, 21.283}]}{=} \left(\bigsqcup_{i \in \{1, \dots, m_1\}} A_i \right) \cdot \left(\bigsqcup_{j \in \{1, \dots, m_2\}} B_j \right) \\ &\stackrel{[\text{theorem: 6.90}]}{=} \bigcup_{(i,j) \in \{1, \dots, m_1\} \times \{1, \dots, m_2\}} (A_i \cdot B_j) \end{aligned}$$

By [theorem: 6.41] $\{1, \dots, m_1\} \times \{1, \dots, m_2\}$ is finite so there exist a bijection

$$\sigma: \{1, \dots, N_1\} \rightarrow \{1, \dots, m_1\} \times \{1, \dots, m_2\}$$

then we have

$$R_1 = \bigcup_{(i,j) \in \{1, \dots, m_1\} \times \{1, \dots, m_2\}} (A_i \cdot B_j) \stackrel{[\text{theorem: 2.127}]}{=} \bigcup_{i \in \{1, \dots, N_1\}} (A_{\sigma(i)_1} \cdot B_{\sigma(i)_2})$$

Define

$$\{R_{1,i}\}_{i \in \{1, \dots, N_1\}} \text{ by } R_{1,i} = A_{\sigma(i)_1} \cdot B_{\sigma(i)_2}$$

then we have $\forall i \in \{1, \dots, N_1\}$, as $A_{\sigma(i)_1} \in \mathcal{S}_1$, $B_{\sigma(i)_2} \in \mathcal{S}_2$, that $R_{1,i} = A_{\sigma(i)_1} \cdot B_{\sigma(i)_2} \in \mathcal{S}_1 \odot \mathcal{S}_2$. Hence we have

$$\{R_{1,i}\}_{i \in \{1, \dots, N_1\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2 \text{ and } R_1 = \bigcup_{i \in \{1, \dots, N_1\}} R_{1,i}$$

Further if $i, j \in \{1, \dots, N_1\}$ with $i \neq j$ we have, as σ is a bijection, that $\sigma(i) \neq \sigma(j)$, for $\sigma(i)_1, \sigma(j)_1$ we have either

$\sigma(i)_1 = \sigma(j)_1$. Then $\sigma(i)_2 \neq \sigma(j)_2$ so that

$$\begin{aligned} R_{1,i} \cap R_{1,j} &= (A_{\sigma(i)_1} \cdot B_{\sigma(i)_2}) \cap (A_{\sigma(j)_1} \cdot B_{\sigma(j)_2}) \\ &= (A_{\sigma(i)_1} \cap A_{\sigma(j)_1}) \cdot (B_{\sigma(i)_2} \cap B_{\sigma(j)_2}) \\ &= (A_{\sigma(i)_1} \cap A_{\sigma(j)_1}) \cdot \emptyset \\ &= \emptyset \end{aligned}$$

$\sigma(i)_1 \neq \sigma(j)_1$. Then

$$\begin{aligned} R_{1,i} \cap R_{1,j} &= (A_{\sigma(i)_1} \cdot B_{\sigma(i)_2}) \cap (A_{\sigma(j)_1} \cdot B_{\sigma(j)_2}) \\ &= (A_{\sigma(i)_1} \cap A_{\sigma(j)_1}) \cdot (B_{\sigma(i)_2} \cap B_{\sigma(j)_2}) \\ &= \emptyset \cdot (B_{\sigma(i)_2} \cap B_{\sigma(j)_2}) \\ &= \emptyset \end{aligned}$$

which proves that

$$\{R_{1,i}\}_{i \in \{1, \dots, N_1\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2 \text{ is pairwise disjoint and } R_1 = \bigsqcup_{i \in \{1, \dots, N_1\}} R_{1,i}$$

$i = 2$. Then

$$\begin{aligned} R_2 &= (A \setminus C) \cdot (B \cap D) \\ &\stackrel{[\text{eq: 21.282}]}{=} \left(\bigsqcup_{i \in \{1, \dots, m_1\}} A_i \right) \cdot (B \cap D) \\ &\stackrel{[\text{theorem: 6.90}]}{=} \bigcup_{i \in \{1, \dots, m_1\}} (A_i \cdot (B \cap D)) \\ &= \bigcup_{i \in \{1, \dots, N_2\}} R_{2,i} \end{aligned}$$

where

$$N_2 = m_1 \text{ and } \{R_{2,i}\}_{i \in \{1, \dots, N_2\}} \text{ is defined by } R_{2,i} = A_i \cdot (B \cap D)$$

Let $i \in \{1, \dots, N_2\}$ then, as $A_i \in \mathcal{S}_1$, $B, D \in \mathcal{S}_2 \Rightarrow B \cap D \in \mathcal{S}_2$, $R_{2,i} = A_i \cdot (B \cap D) \in \mathcal{S}_1 \odot \mathcal{S}_2$ so that

$$\{R_{2,i}\}_{i \in \{1, \dots, N_2\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2$$

Further if $i, j \in \{1, \dots, N_2\}$ with $i \neq j$ then we have

$$R_{2,i} \cap R_{2,j} = (A_i \cdot (B \cap D)) \cap (A_j \cdot (B \cap D)) = (A_i \cap A_j) \cdot (B \cap D) = \emptyset \cdot (B \cap D) = \emptyset$$

So we have proved that

$$\{R_{2,i}\}_{i \in \{1, \dots, N_2\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2 \text{ is pairwise disjoint and } R_2 = \bigsqcup_{i \in \{1, \dots, N_2\}} R_{2,i}$$

$i = 3$. Then

$$\begin{aligned} R_3 &= (A \cap C) \cdot (B \setminus D) \\ &\stackrel{[\text{eq: 21.283}]}{=} (A \cap C) \cdot \left(\bigsqcup_{i \in \{1, \dots, m_2\}} B_i \right) \\ &\stackrel{[\text{theorem: 6.90}]}{=} \bigcup_{i \in \{1, \dots, m_2\}} ((A \cap C) \cdot B_i) \\ &= \bigcup_{i \in \{1, \dots, N_3\}} R_{3,i} \end{aligned}$$

where

$$N_3 = m_2 \text{ and } \{R_{3,i}\}_{i \in \{1, \dots, N_3\}} \text{ is defined by } R_{3,i} = (A \cap C) \cdot B_i$$

Let $i \in \{1, \dots, N_3\}$ then, as $B_i \in \mathcal{S}_2$, $A, C \in \mathcal{S}_1 \Rightarrow A \cap C \in \mathcal{S}_1$, $R_{3,i} = ((A \cap C) \cdot B_i) \in \mathcal{S}_1 \odot \mathcal{S}_2$ so that

$$\{R_{3,i}\}_{i \in \{1, \dots, N_3\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2$$

Further if $i, j \in \{1, \dots, N_3\}$ with $i \neq j$ then we have

$$R_{3,i} \cap R_{3,j} = ((A \cap C) \cdot B_i) \cap ((A \cap C) \cdot B_j) = (A \cap C) \cdot (B_i \cap B_j) = (A \cap C) \cdot \emptyset = \emptyset$$

So we have proved that

$$\{R_{3,i}\}_{i \in \{1, \dots, N_3\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2 \text{ is pairwise disjoint and } R_3 = \bigsqcup_{i \in \{1, \dots, N_3\}} R_{3,i}$$

To summarize the above, we have a pairwise disjoint $\{R_i\}_{i \in \{1, 2, 3\}}$ such that

$$R \setminus S \stackrel{[\text{eq: 21.281}]}{=} \bigsqcup_{i \in \{1, 2, 3\}} R_i$$

and $\forall i \in \{1, 2, 3\}$ there exist a pairwise disjoint $\{R_{i,j}\}_{j \in \{1, \dots, N_i\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2$ such

$$R_i = \bigsqcup_{j \in \{1, \dots, N_i\}} R_{i,j}$$

Further as $\bigcup_{i \in \{1, 2, 3\}} \{i\} \cdot \{1, \dots, N_i\}$ is finite by [theorems: 6.41, 6.36] there exist a bijection

$$\sigma: \{1, \dots, N\} \rightarrow \bigcup_{i \in \{1, 2, 3\}} \{i\} \cdot \{1, \dots, N_i\}$$

Hence we can apply [corollary: 21.12] to get that

$$\{R_{\sigma(i)_1, \sigma(i)_2}\}_{i \in \{1, \dots, N\}} \subseteq \mathcal{S}_1 \odot \mathcal{S}_2 \text{ is pairwise disjoint and } R \setminus S = \bigsqcup_{i \in \{1, \dots, N\}} R_{\sigma(i)_1, \sigma(i)_2} \quad (21.284)$$

Finally from [eqs: 21.279, 21.280 and the above eq: 21.284] it follows that

$$\mathcal{S}_1 \odot \mathcal{S}_2 \text{ is a semi-ring on } X_1 \cdot X_2 \quad \square$$

If we want to use the above lemma in the proof by mathematical induction that $\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ is a semi-ring we have a small problem: $\prod_{i \in \{1, \dots, n+1\}} X_i$ is not equal to $(\prod_{i \in \{1, \dots, n\}} X_i) \cdot X_{n+1}$. Luckily we can prove that $(\prod_{i \in \{1, \dots, n\}} X_i) \cdot X_{n+1}$ is bijective to $\prod_{i \in \{1, \dots, n+1\}} X_i$. So it make sense to see how a bijection can be used to generate a semi-ring based on a semi-ring.

Theorem 21.187. *Let X, Y be sets, $\beta: X \rightarrow Y$ a bijection and $\mathcal{S} \subseteq X$ a semi-ring on X then $\{\beta(A) | A \in \mathcal{S}\}$ is a semi-ring on Y .*

Proof. As $\beta(\emptyset) = \emptyset$ it follows that

$$\emptyset \in \{\beta(A) | A \in \mathcal{S}\} \quad (21.285)$$

Let $R, S \in \{\beta(A) | A \in \mathcal{S}\}$ there exist $A, B \in \mathcal{S}$ such that $R = \beta(A)$, $S = \beta(B)$. Using [theorem: 2.103(4)] and the fact that β is a bijection hence injective we have that

$$R \cap S = \beta(A) \cap \beta(B) \stackrel{[\text{theorem: 2.103(4)}]}{=} \beta(A \cap B)$$

As $\mathcal{S}$ is a semi-ring $A \cap B \in \mathcal{S}$ so that $\beta(A \cap B) \in \{\beta(A) | A \in \mathcal{S}\}$ which combined with the above proves that

$$\forall R, S \in \{\beta(A) | A \in \mathcal{S}\} \text{ we have } R \cap S \in \{\beta(A) | A \in \mathcal{S}\} \quad (21.286)$$

Using [theorem: 2.102(5)] and the fact that β is injective it follows that

$$R \setminus S = \beta(A) \setminus \beta(B) \stackrel{[\text{theorem: 2.102(5)}]}{=} \beta(A \setminus B) \quad (21.287)$$

As $A, B \in \mathcal{S}$ and $\mathcal{S}$ is a semi-ring there exist a pairwise disjoint $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}$, $n \in \mathbb{N}$ such that

$$A \setminus B = \bigsqcup_{i \in \{1, \dots, n\}} A_i \quad (21.288)$$

Let $i, j \in \{1, \dots, n\}$ with $i \neq j$ then $\beta(A_i) \cap \beta(A_j) \stackrel{[\text{theorem: 2.103(4)}]}{=} \beta(A_i \cap A_j) = \beta(\emptyset) = \emptyset$ so that

$$\beta(A_i)_{i \in \{1, \dots, n\}} \subseteq \{\beta(A) | A \in \mathcal{S}\} \text{ is pairwise disjoint} \quad (21.289)$$

Further

$$\bigsqcup_{i \in \{1, \dots, n\}} \beta(A_i) \stackrel{[\text{theorem: 2.144}]}{=} \beta\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \stackrel{[\text{eq: 21.288}]}{=} \beta(A \setminus B) \stackrel{[\text{eq: 21.287}]}{=} R \setminus S \quad (21.290)$$

Define $\{R_i\}_{i \in \{1, \dots, n\}} \subseteq \{\beta(A) | A \in \mathcal{S}\}$ by $R_i = \beta(A_i)$ then [eqs: 21.289, 21.290] can be summarized as

$$\forall R, S \in \{\beta(A) | A \in \mathcal{S}\} \exists \{R_i\}_{i \in \{1, \dots, n\}} \subseteq \{\beta(A) | A \in \mathcal{S}\} \text{ such that } R \setminus S = \bigsqcup_{i \in \{1, \dots, n\}} R_i \quad (21.291)$$

By [eqs: 21.285, 21.286 and 21.291] it follows that

$$\{\beta(A) | A \in \mathcal{S}\} \text{ is a semi-ring on } Y \quad \square$$

Next we construct a bijection between $(\prod_{i \in \{1, \dots, n\}} X_i) \cdot X_{n+1}$ and $\prod_{i \in \{1, \dots, n+1\}} X_i$.

Lemma 21.188. *Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1, \dots, n+1\}}$ a finite family such that $\forall i \in \{1, \dots, n\} \mathcal{S}_i \subseteq \mathcal{P}(X_i)$ then*

1.

$$\beta: \left(\prod_{i \in \{1, \dots, n\}} X_i \right) \cdot X_{n+1} \rightarrow \prod_{i \in \{1, \dots, n+1\}} X_i \text{ defined by } \beta(x, y)_i = \begin{cases} x_i & \text{if } i \in \{1, \dots, n\} \\ y & \text{if } i = n+1 \end{cases}$$

is a bijection

$$2. \odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i = \{\beta(B) | B \in (\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot S_{n+1}\}$$

Note 21.189. *Be careful in the notation used in (2) $\beta(B)$ is the image of the set B in the domain $(\prod_{i \in \{1, \dots, n\}} X_i) \cdot X_{n+1}$ of the bijection β . So $\odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i$ is not equal to $\beta((\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot S_{n+1})$.*

Proof.

1. We have:

injectivity. Assume that $\beta(x, y) = \beta(u, v)$ then $\forall i \in \{1, \dots, n\}$ we have

$$x_i = \beta(x, y)_i = \beta(u, v)_i = u_i$$

so that $x = u$ and $y = \beta(x, y)_{n+1} = \beta(u, v)_{n+1} = v$. So $(x, y) = (u, v)$ which proves injectivity.

surjectivity. Let $u \in \prod_{i \in \{1, \dots, n+1\}} X_i$ and define $x \in \prod_{i \in \{1, \dots, n\}} X_i$ by $x_i = u_i$ and $y \in X_{n+1}$ by $y = u_{n+1}$ then $\forall i \in \{1, \dots, n+1\}$ we have

$$\beta(x, y)_i = \begin{cases} x_i = u_i & \text{if } i \in \{1, \dots, n\} \\ u_{n+1} & \text{if } i = n+1 \end{cases} = u_i$$

proving that $\beta(x, y) = u$

2. Let $\{A_i\}_{i \in \{1, \dots, n+1\}}$ be a family of sets such that $\forall i \in \{1, \dots, n+1\} A_i \in \mathcal{S}_i \subseteq \mathcal{P}(X_i)$. Let $u \in \beta\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right)$ then $u = \beta(x, y)$ where $x \in \prod_{i \in \{1, \dots, n\}} A_i \Rightarrow \forall i \in \{1, \dots, n\} x_i \in A_i$ and $y \in A_{n+1}$. So

$$\forall i \in \{1, \dots, n+1\} \beta(x, y)_i = \begin{cases} x_i \in A_i & \text{if } i \in \{1, \dots, n\} \\ y \in A_i & \text{if } i = n+1 \end{cases} \in A_i$$

so that $u \in \prod_{i \in \{1, \dots, n+1\}} A_i$. Hence

$$\beta\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right) \subseteq \prod_{i \in \{1, \dots, n+1\}} A_i \quad (21.292)$$

If $u \in \prod_{i \in \{1, \dots, n+1\}} A_i$ then $\forall i \in \{1, \dots, n+1\}$ we have $u_i \in A_i$. Define $x \in \prod_{i \in \{1, \dots, n\}} X_i$ by $x_i = u_i \in A_i$ and $y = u_{n+1} \in A_{n+1}$ then $(x, y) \in \left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}$. further

$$\beta(x, y) = \begin{cases} x_i = u_i & \text{if } i \in \{1, \dots, n\} \\ y = u_{n+1} & \text{if } i = n+1 \end{cases} = u_i$$

proving that $u = \beta(x, y) \in \beta\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right)$. Hence

$$\prod_{i \in \{1, \dots, n+1\}} A_i \subseteq \beta\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right)$$

which combined with [eq: 21.292] proves that

$$\prod_{i \in \{1, \dots, n+1\}} A_i = \beta\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right) \quad (21.293)$$

Let now $A \in \odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i$ then $A = \prod_{i \in \{1, \dots, n+1\}} A_i$ where $\forall i \in \{1, \dots, n+1\} A_i \in \mathcal{S}_i$. Then $\prod_{i \in \{1, \dots, n\}} A_i \in \odot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ and $A_{n+1} \in \mathcal{S}_{n+1}$ so that

$$\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1} \in \left(\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i\right) \odot \mathcal{S}_{n+1}$$

Hence as

$$A = \prod_{i \in \{1, \dots, n+1\}} A_i \stackrel{[\text{eq: 21.293}]}{=} \beta\left(\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right)\right)$$

it follows that $A \in \{\beta(B) | B \in (\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot \mathcal{S}_{n+1}\}$ which proves that

$$\odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i \subseteq \left\{ \beta(B) | B \in \left(\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i\right) \odot \mathcal{S}_{n+1} \right\} \quad (21.294)$$

On the other hand if $B \in \{\beta(B) | B \in (\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot \mathcal{S}_{n+1}\}$ then there exists a $A \in (\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot \mathcal{S}_{n+1}$ such that $B = \beta(A)$. As $A \in (\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot \mathcal{S}_{n+1}$ there exist a $\{A_i\}_{i \in \{1, \dots, n\}}$ with $\forall i \in \{1, \dots, n\} A_i \in \mathcal{S}_i$ and a $A_{n+1} \in \mathcal{S}_{n+1}$ such that

$$B = \beta(A) = \beta\left(\left(\prod_{i \in \{1, \dots, n\}} A_i\right) \cdot A_{n+1}\right)$$

Using [eq: 21.293] on the above proves that that

$$B = \prod_{i \in \{1, \dots, n+1\}} A_i \in \odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i$$

hence $\{\beta(B)|B \in (\bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot S_{n+1}\} \subseteq \bigodot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i$ which combined with [eq: 21.294] proves that

$$\bigodot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i = \left\{ \beta(B) | B \in \left(\bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i \right) \odot S_{n+1} \right\}$$

□

We are now ready to use mathematical induction to prove that $\bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ is a semi-ring on $\prod_{i \in \{1, \dots, n\}} X_i$.

Theorem 21.190. *Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family such that $\forall i \in \{1, \dots, n\}$ $\mathcal{S}_i$ is a semi-ring on X_i then for $\bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ defined by*

$$\bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i = \left\{ \prod_{i \in \{1, \dots, n\}} A_i | \forall i \in \{1, \dots, n\} A_i \in \mathcal{S}_i \right\}$$

we have that

$$\bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i \text{ is a semi-ring on } \prod_{i \in \{1, \dots, n\}} X_i$$

Proof. We prove this by mathematical induction, so define

$$\mathcal{S} = \left\{ n \in \mathbb{N} | \text{If } \{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1, \dots, n\}} \text{ is a family such that } \forall i \in \{1, \dots, n\} \mathcal{S}_i \text{ is a semi-ring on } X_i \text{ then } \bigodot_{i \in \{1, \dots, n\}} \mathcal{S}_i \text{ is a semi-ring on } \prod_{i \in \{1, \dots, n\}} X_i \right\}$$

then we have:

1 $\in \mathcal{S}$. Let $\{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1\}}$ be such that $\mathcal{S}_1$ is a semi-ring on X_1 . Define

$$\beta: X_1 \rightarrow \prod_{i \in \{1\}} X_i \text{ where } \beta(x) \text{ is defined by } \beta(x)_1 = x \text{ [in other words } \beta(x) = (x)]$$

then we have:

injectivity. If $\beta(x) = \beta(y)$ then $x = \beta(x)_1 = \beta(y)_1 = y$.

surjectivity. If $y \in \prod_{i \in \{1\}} X_i$ then for $x = y_1$ we have $\forall i \in \{1\}$ that $\beta(x)_1 = x = y_1$ so that $\beta(x) = y$.

so that

β is a bijection

Let $A \in \bigodot_{i \in \{1\}} \mathcal{S}_i = \{\prod_{i \in \{1\}} A_i | \forall i \in \{1\} A_i \in \mathcal{S}_i\}$ then $A = \prod_{i \in \{1\}} A_i$ where $A_1 \in \mathcal{S}_1$. If $x \in A = \prod_{i \in \{1\}} A_i$ then $x_1 \in A_1$ so that $x = \beta(x_1) \in \beta(A_1)$ so that

$$A \subseteq \beta(A_1)$$

If $x \in \beta(A_1)$ then there exist a $a \in A_1$ such that $x = \beta(a) = (a)$ hence $x_1 = \beta(a)_1 = a \in A_1$ so that $x \in \prod_{i \in \{1\}} A_i = A$ proving that

$$\beta(A_1) \subseteq A$$

Hence $A = \beta(A_1) \in \{\beta(B) | B \in \mathcal{S}_1\}$ from which it follows that

$$\bigodot_{i \in \{1\}} \mathcal{S}_i \subseteq \{\beta(B) | B \in \mathcal{S}_1\} \tag{21.295}$$

For the opposite inclusion, let $A \in \{\beta(B) | B \in \mathcal{S}_1\}$ then there exist a $A_1 \in \mathcal{S}_1$ such that $A = \beta(A_1)$. If $x \in A = \beta(A_1)$ then there exist a $a \in A_1$ such that $x = \beta(a)$ so that $x_1 = \beta(a)_1 = a \in A_1$ hence $x \in \prod_{i \in \{1\}} A_i$ or and if $x \in \prod_{i \in \{1\}} A_i$ then $x_1 \in A_1$ so that $x = \beta(x_1) \in \beta(A_1)$, proving that $A = \prod_{i \in \{1\}} A_i \in \odot_{i \in \{1\}} \mathcal{S}_i$. Hence we have that $\{\beta(B) | B \in \mathcal{S}_1\} \subseteq \odot_{i \in \{1\}} \mathcal{S}_i$ which combined with [eq: 21.295] proves that

$$\odot_{i \in \{1\}} \mathcal{S}_i = \{\beta(B) | B \in \mathcal{S}_1\}$$

Finally as $\mathcal{S}_1$ is a semi-ring it follows from [lemma: 21.187] that $\odot_{i \in \{1\}} \mathcal{S}_i$ is a semi-ring which proves that $1 \in \mathcal{S}$.

2 $\in \mathcal{S}$. If $\{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1,2\}}$ is such that $\mathcal{S}_1$ is a semi-ring on X_1 and $\mathcal{S}_2$ is a semi-ring on X_2 then by [lemma: 21.186] $\odot_{i \in \{1,2\}} \mathcal{S}_i = \mathcal{S}_1 \odot \mathcal{S}_2$ is a semi-ring on $X_1 \cdot X_2 = \prod_{i \in \{1,2\}} X_i$. Hence $2 \in \mathcal{S}$.

n $\in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1, \dots, n+1\}}$ be a finite family such that $\forall i \in \{1, \dots, n+1\}$ we have that $\mathcal{S}_i$ is a semi-ring on X_i . As $n \in \mathcal{S}$ it follows that $\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ is a semi-ring on $\prod_{i \in \{1, \dots, n\}} X_i$. As also $\mathcal{S}_{n+1}$ is a semi-ring on X_{n+1} it follows from [lemma: 21.186] that

$$\left(\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i \right) \odot \mathcal{S}_{n+1} \text{ is a semi-ring on } \left(\prod_{i \in \{1, \dots, n\}} X_i \right) \cdot X_{n+1}$$

Using the previous lemma [lemma: 21.188] there exist a bijection

$$\beta: \left(\prod_{i \in \{1, \dots, n\}} X_i \right) \cdot X_{n+1} \rightarrow \prod_{i \in \{1, \dots, n+1\}} X_i$$

such that

$$\odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i = \left\{ \beta(B) | B \in \left(\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i \right) \odot \mathcal{S}_{n+1} \right\}$$

By [theorem: 21.187] it follows then that $\{\beta(B) | B \in (\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i) \odot \mathcal{S}_{n+1}\}$ is a semi-ring on $\prod_{i \in \{1, \dots, n+1\}} X_i$. Hence $\odot_{i \in \{1, \dots, n+1\}} \mathcal{S}_i$ is a semi-ring on $\prod_{i \in \{1, \dots, n+1\}} X_i$ proving that $n+1 \in \mathcal{S}$ $\square$

Corollary 21.191. Let $n \in \mathbb{N}$, X a set and $\mathcal{S}$ a semi-ring on X then

$$\mathcal{S}^n = \left\{ \prod_{i \in \{1, \dots, n\}} A_i | \forall i \in \{1, \dots, n\} A_i \in \mathcal{S} \right\} = \left\{ \prod_{i \in \{1, \dots, n\}} A_i | \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S} \right\}$$

is a semi-ring on X^n

Proof. Define $\{\langle X_i, \mathcal{S}_i \rangle\}_{i \in \{1, \dots, n\}}$ by $X_i = X$ and $\mathcal{S}_i = \mathcal{S}$. Then by the previous theorem [theorem: 21.190] we have that $\odot_{i \in \{1, \dots, n\}} \mathcal{S}_i$ is a semi-ring on $\prod_{i \in \{1, \dots, n\}} X_i$. Now by definition we have that

$$X^n \stackrel{\text{definition}}{=} \prod_{i \in \{1, \dots, n\}} X_i$$

and

$$\begin{aligned} \odot_{i \in \{1, \dots, n\}} \mathcal{S}_i &= \left\{ \prod_{i \in \{1, \dots, n\}} A_i | \forall i \in \{1, \dots, n\} A_i \in \mathcal{S}_i \right\} \\ &\stackrel{\mathcal{S}_i = \mathcal{S}}{=} \left\{ \prod_{i \in \{1, \dots, n\}} A_i | \forall i \in \{1, \dots, n\} A_i \in \mathcal{S} \right\} \\ &= \mathcal{S}^n \end{aligned}$$

so that $\mathcal{S}^n$ is a semi-ring on X^n . $\square$

21.3.4.5 Semi-ring on $\mathbb{R}^n$

We are now ready to prove that $\mathcal{R}_{||}^n \cup \{\emptyset\}$ is a semi-ring.

Theorem 21.192. *Let $n \in \mathbb{N}$ then we have*

$$\begin{aligned} \mathcal{R}_{||}^n &= \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \right\} \stackrel{\text{def}}{=} \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \{I_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_{||} \right\} \\ \mathcal{R}_{||}^n \cap \{\emptyset\} &= \emptyset \\ \mathcal{R}_{||}^n \cup \{\emptyset\} &= \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{S}_{||} \right\} \stackrel{\text{def}}{=} \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \{I_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}_{||} \right\} \end{aligned}$$

Proof. If $R \in \mathcal{R}_{||}^n$ then by [definition: 21.72] there exists $a, b \in \mathbb{R}^n$ with $a < b \Rightarrow a_i < b_i$ such that

$$R = [a, b[\stackrel{\text{definition: 21.67}}{=} \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$$

As $\forall i \in \{1, \dots, n\} \ a_i < b_i$ so that $[a_i, b_i[\neq \emptyset$ it follows that $[a_i, b_i[\in \mathcal{I}_{||}$, hence $R \in \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \right\}$ proving that

$$\mathcal{R}_{||}^n \subseteq \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \right\} \quad (21.296)$$

Further if $R \in \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \right\}$ then there exists a $\{I_i\}_{i \in \{1, \dots, n\}}$ with $\forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||}$ such that $R = \prod_{i \in \{1, \dots, n\}} I_i$. Hence $\forall i \in \{1, \dots, n\}$ there exists $a_i, b_i \in \mathbb{R}$ with $a_i < b_i$ such that $I_i = [a_i, b_i[$. Define $a = (a_1, \dots, a_n) \in \mathbb{R}^n$, $(b_1, \dots, b_n) \in \mathbb{R}^n$ then $a < b$ and $[a, b[\stackrel{\text{definition: 21.67}}{=} \prod_{i \in \{1, \dots, n\}} [a_i, b_i[= \prod_{i \in \{1, \dots, n\}} I_i = R$ from which it follows that $R \in \mathcal{R}_{||}^n$. So $\left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \right\} \subseteq \mathcal{R}_{||}^n$, combining this with [eq: 21.296] gives

$$\mathcal{R}_{||}^n = \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \right\} \quad (21.297)$$

Let $R \in \mathcal{R}_{||}^n$ then there exists $\{I_i\}_{i \in \{1, \dots, n\}}$ with $I_i \in \mathcal{I}_{||}$ [so that $I_i \neq \emptyset$] such that $R = \prod_{i \in \{1, \dots, n\}} I_i$. Applying now [theorem: 3.110] it follows that $R \neq \emptyset$ so that

$$\mathcal{R}_{||}^n \cap \{\emptyset\} = \emptyset \quad (21.298)$$

If $R \in \mathcal{R}_{||}^n \cup \{\emptyset\}$ then we have either:

$R \in \mathcal{R}_{||}^n$. Then by [eq: 21.297] $R = \prod_{i \in \{1, \dots, n\}} I_i$ where $\forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||} \subseteq \mathcal{S}_{||}$ so that $R \in \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{S}_{||} \right\}$.

$R = \emptyset$. Define $\{I_i\}_{i \in \{1, \dots, n\}}$ by $I_i = \emptyset \in \mathcal{S}_{||}$ then $\emptyset \stackrel{\text{theorem: 2.146}}{=} \prod_{i \in \{1, \dots, n\}} I_i$ so that $R \in \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{S}_{||} \right\}$.

hence we have that

$$\mathcal{R}_{||}^n \cup \{\emptyset\} \subseteq \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{S}_{||} \right\} \quad (21.299)$$

Let $R \in \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} \ I_i \in \mathcal{S}_{||} \right\}$ then $R = \prod_{i \in \{1, \dots, n\}} I_i$ where $\forall i \in \{1, \dots, n\} \ I_i \in \mathcal{S}_{||}$. Then we have either:

$\forall i \in \{1, \dots, n\} \ I_i \neq \emptyset$. Then by [theorem: 21.171] it follows that $\forall i \in \{1, \dots, n\} \ I_i \in \mathcal{I}_{||}$ so that $I_i = [a_i, b_i[\neq \emptyset \Rightarrow a_i < b_i$, hence for $a = (a_1, \dots, a_n), b = (b_1, \dots, b_n) \in \mathbb{R}^n$ we have $a < b$ so that

$$R = \prod_{i \in \{1, \dots, n\}} [a_i, b_i[\subseteq \mathcal{R}_{||}^n \subseteq \mathcal{R}_{||}^n \cup \{\emptyset\}$$

$\exists i \in \{1, \dots, n\}$ such that $I_i = \emptyset$. Then

$$R = \prod_{i \in \{1, \dots, n\}} I_i \stackrel{[\text{theorem: 2.146}]}{=} \emptyset \in \mathcal{R}_{||}^n \cup \{\emptyset\}$$

proving that $\{\prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} I_i \in \mathcal{S}_{||}\} \subseteq \mathcal{R}_{||}^n \cap \{\emptyset\}$. Combining this with [eq: 21.299] proves finally that

$$\mathcal{R}_{||}^n \cup \{\emptyset\} = \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} I_i \in \mathcal{S}_{||} \right\} \quad \square$$

A direct consequence of the above theorem is the following:

Theorem 21.193. *Let $n \in \mathbb{N}$ then $\mathcal{R}_{||}^n \cup \{\emptyset\}$ is a semi-ring on $\mathbb{R}^n$*

Proof. First we have

$$\mathcal{R}_{||}^n \cup \{\emptyset\} \stackrel{[\text{theorem: 21.192}]}{=} \left\{ \prod_{i \in \{1, \dots, n\}} I_i \mid \forall i \in \{1, \dots, n\} I_i \in \mathcal{S}_{||} \right\} = (\mathcal{S}_{||})^n$$

so that by [theorems: 21.172, 21.191]

$$\mathcal{R}_{||}^n \cup \{\emptyset\} \text{ is a semi-ring} \quad \square$$

Theorem 21.194. *Let $n \in \mathbb{N}$ then we have that*

$$\sigma[\{C \subseteq \mathbb{R}^n \mid C \text{ is closed in } \mathcal{T}_{|||n}\}] = \beta[\mathbb{R}^n] = \sigma[\mathcal{R}_{||}^n] = \sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}]$$

Proof. By [theorem: 21.87] we only have to prove that $\sigma[\mathcal{R}_{||}^n] = \sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}]$. As $\mathcal{R}_{||}^n \subseteq \mathcal{R}_{||}^n \cup \{\emptyset\} \subseteq \sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}]$ and $\sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}]$ is a σ -algebra it follows from [theorem: 21.42] that

$$\sigma[\mathcal{R}_{||}^n] \subseteq \sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}] \quad (21.300)$$

Let $A \in \mathcal{R}_{||}^n \cup \{\emptyset\}$ then either $A \in \mathcal{R}_{||}^n \subseteq \sigma[\mathcal{R}_{||}^n]$ or $A = \emptyset \in \sigma[\mathcal{R}_{||}^n]$ so that $\mathcal{R}_{||}^n \cup \{\emptyset\} \subseteq \sigma[\mathcal{R}_{||}^n]$, as $\sigma[\mathcal{R}_{||}^n]$ is a σ -algebra it follows from [theorem: 21.42] that $\sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}] \subseteq \sigma[\mathcal{R}_{||}^n]$. Combining this with [eq: 21.300] proves

$$\sigma[\mathcal{R}_{||}^n] = \sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}] \quad \square$$

21.3.4.6 Content on the semi-ring $\mathcal{R}_{||}^n \cup \{\emptyset\}$

Definition 21.195. (Content of a rectangle) *Let $n \in \mathbb{N}$ then we define*

$$\text{vol}^n: \mathcal{R}_{||}^n \cup \{\emptyset\} \rightarrow [0, \infty] \text{ by } \text{vol}^n(R) = \begin{cases} 0 & \text{if } R = \emptyset \\ v^n(R) & \text{if } R \neq \emptyset \end{cases}$$

where $v^n(R): \mathcal{R}_{||}^n \rightarrow [0, \infty]$ is defined by $v^n(R) = \prod_{i=1}^n (\text{begin}(R)_i - \text{end}(R)_i) \in \mathbb{R}^+$ [see definition: 21.77].

Note 21.196. *Using the definition it follows that $\text{vol}^n(R)$ is finite and*

$$\forall R \in \mathcal{R}_{||}^n \cup \{\emptyset\} \text{ we have } \text{vol}^n(R) \neq 0 \Leftrightarrow R \in \mathcal{R}_{||}^n$$

Next it remains to prove that vol^n is actual a content on $\mathcal{R}_{||}^n \cup \{\emptyset\}$, the most difficult part is proving additivity of vol^n . To illustrate how to prove this consider the following 2-dimensional case [see figure:] where

$$R = \bigsqcup_{i \in \{1, \dots, 4\}} R_i$$

then we have to prove that

$$\text{vol}^2(R) = \sum_{i=1}^4 \text{vol}^2(R_i)$$

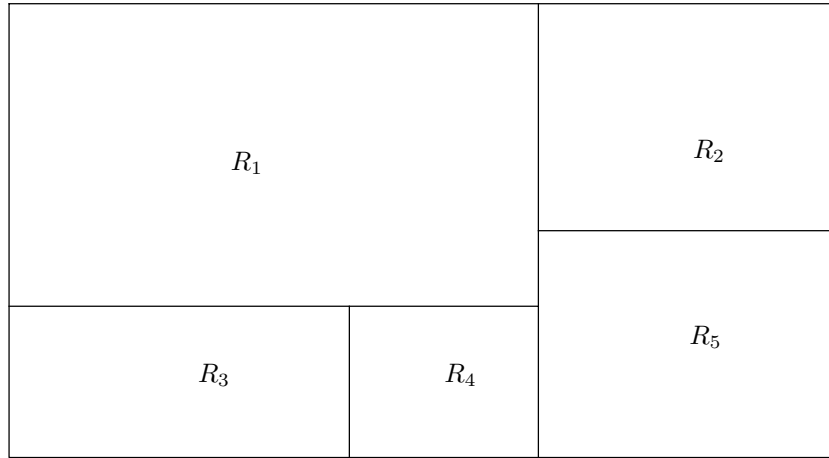


Figure 21.1.

To do this we divide R_1 , R_2 , R_3 , R_4 and R_5 in sub-cubes by partitioning their sides in sub intervals.

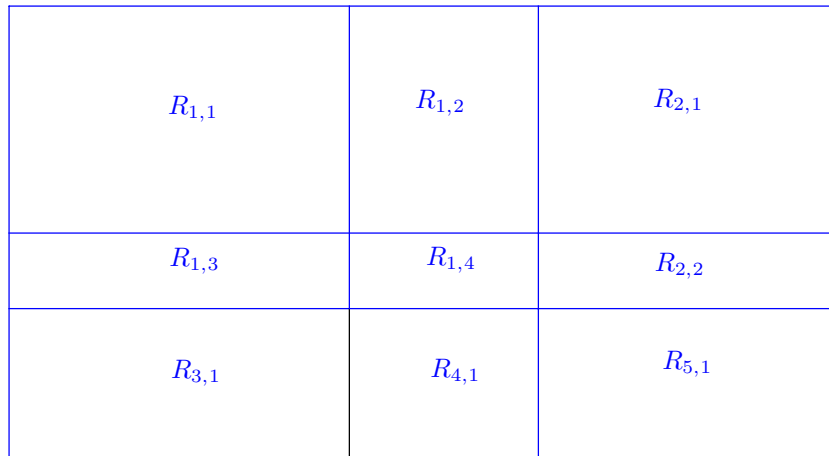


Figure 21.2.

so that

$$\begin{aligned} R_1 &= R_{1,1} + R_{1,2} + R_{1,3} + R_{1,4} \\ R_2 &= R_{2,1} + R_{2,2} \\ R_3 &= R_{3,1} \\ R_4 &= R_{4,1} \\ R_5 &= R_{5,1} \end{aligned}$$

then we prove that

$$\begin{aligned} \text{vol}^2(R) &= \text{vol}^2(R_{1,1}) + \text{vol}^2(R_{1,2}) + \text{vol}^2(R_{1,3}) + \text{vol}^2(R_{1,4}) + \text{vol}^2(R_{2,1}) + \text{vol}^2(R_{2,2}) + \text{vol}^2(R_{3,1}) + \\ &\quad \text{vol}^2(R_{4,1}) + \text{vol}^2(R_{5,1}) \end{aligned}$$

and apply associativity to get

$$\begin{aligned} \text{vol}^2(R) &= (\text{vol}^2(R_{1,1}) + \text{vol}^2(R_{1,2}) + \text{vol}^2(R_{1,3}) + \text{vol}^2(R_{1,4})) + (\text{vol}^2(R_{2,1}) + \text{vol}^2(R_{2,2})) + \\ &\quad (\text{vol}^2(R_{3,1}) + (\text{vol}^2(R_{3,1}) + (\text{vol}^2(R_{4,1}) + (\text{vol}^2(R_{5,1})))) \\ &= \text{vol}^2(R_1) + \text{vol}^2(R_2) + \text{vol}^2(R_3) + \text{vol}^2(R_4) + \text{vol}^2(R_5) \end{aligned}$$

which proves additivity. To actually turn these remarks in a proof is tedious so we split it in different lemma's. First we show that every increasing finite family of real numbers can be used to construct a pairwise disjoint union of half open intervals that is itself a half open interval. This will be used later to turn a union of half open intervals that is itself a half open interval in a disjoint union of half open intervals.

Lemma 21.197. *Let $n, m \in \mathbb{N}$ with $n \leq m$, $\{a_i\}_{i \in \{n, \dots, m\}} \subseteq \mathbb{R}$ such that $\forall i \in \{n, \dots, m-1\}$ we have $a_i \leq a_{i+1}$ then*

$$\{[a_i, a_{i+1}[\}_{i \in \{n, \dots, m-1\}} \text{ is pairwise disjoint}$$

and

$$\bigsqcup_{i \in \{n, \dots, m-1\}} [a_i, a_{i+1}[= [a_n, a_m[$$

Proof. Let $i, j \in \{n, \dots, m-1\}$ with $i \neq j$ then we may assume that $i < j$ [otherwise exchange i and j]. Assume that $x \in [a_i, a_{i+1}[\cap [a_j, a_{j+1}[$ then $a_i \leq x < a_{i+1} \leq a_j \leq x < a_{j+1}$ leading to the contradiction $x < x$, hence the assumption is wrong proving that $[a_i, a_{i+1}[\cap [a_j, a_{j+1}[= \emptyset$. So

$$\{[a_i, a_{i+1}[\}_{i \in \{n, \dots, m-1\}} \text{ is pairwise disjoint}$$

Let $x \in [a_n, a_m[$ then $a_n \leq x < a_m$ so that $n \in \{i \in \{n, \dots, m\} | a_i \leq x\}$ a finite set. Hence $M = \max(\{i \in \{n, \dots, m\} | a_i \leq x\})$ exist. As $x < a_m \Rightarrow a_m \not\leq x$ we have $M \neq m$ so that $M \in \{n, \dots, m-1\} \Rightarrow M+1 \in \{n, \dots, m\}$. As $M = \max(\{i \in \{n, \dots, m\} | a_i \leq x\})$ it follows that $a_M \leq x < a_{M+1}$ so that $x \in [a_M, a_{M+1}[\subseteq \bigsqcup_{i \in \{n, \dots, m-1\}} [a_i, a_{i+1}[$. Hence we have proved that

$$[a_n, a_m[\subseteq \bigsqcup_{i \in \{n, \dots, m-1\}} [a_i, a_{i+1}[\quad (21.301)$$

Let $x \in \bigsqcup_{i \in \{n, \dots, m-1\}} [a_i, a_{i+1}[$ then there exist a $i \in \{n, \dots, m-1\}$ such that $a_i \leq x < a_{i+1}$. As $n \leq i$ and $i \leq m-1 < m$ we have by the hypothesis that $a_n \leq a_i \wedge a_{i+1} \leq a_m$ so that $a_n \leq x < a_m$ or $x \in [a_n, a_m[$. Hence $\bigsqcup_{i \in \{n, \dots, m-1\}} [a_i, a_{i+1}[\subseteq [a_n, a_m[$ which combined with [eq: 21.301] gives

$$\bigsqcup_{i \in \{n, \dots, m-1\}} [a_i, a_{i+1}[= [a_n, a_m[\quad \square$$

Next we show that a half open interval that is a union of half open sub-intervals is the disjoint union of sub intervals that are based on the boundary points of the half open sub-intervals. For example:

$$\begin{aligned} [-4, 2[&= [-3, 2[\cup [-1, 1[\cup [-4, 0[\cup [0, 2[\\ &= [-4, -3[\cup [-3, -1[\cup [-1, 0[\cup [0, 1[\cup [1, 2[\end{aligned}$$

to do this we sort the boundary points $\{-3, 2, -1, 1, -4, 0, 2\}$ giving the sorted set

$$\{-4, -3, -1, 0, 1, 2\}$$

and use the the previous lemma to create the final list of intervals

$$[-4, -3[, [-3, -1[, [-1, 0[, [0, 1[, [1, 2[$$

Lemma 21.198. *Let $n \in \mathbb{N}$ and $\{[a_i, b_i[\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_{\mathbb{R}}$ such that $\bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[= [a, b[$ then if we take*

$$A = \{\alpha_i | i \in \{1, \dots, n\}\} \cup \{b_i | i \in \{1, \dots, n\}\}$$

there exists a $m \in \mathbb{N} \setminus \{1\}$ and a bijection $\beta: \{1, \dots, m\} \rightarrow A$ such that:

1. $\beta(1) = a \wedge \beta(m) = b$
2. $\forall i, j \in \{1, \dots, m\}$ with $i < j$ we have $\beta(i) < \beta(j)$
3. $\forall x, y \in A$ with $x < y$ we have $\beta^{-1}(x) < \beta^{-1}(y)$
4. $\forall i \in \{1, \dots, n\}$ we have $\beta^{-1}(a_i) < \beta^{-1}(b_i)$
5. $\forall i \in \{1, \dots, n\}$ $\{[\beta(j), \beta(j+1)[\}_{j \in \{\beta^{-1}(a_i), \dots, \beta^{-1}(b_i)-1\}}$ is pairwise disjoint and $[a_i, b_i[= \bigsqcup_{j \in \{\beta^{-1}(a_i), \dots, \beta^{-1}(b_i)-1\}} [\beta(j), \beta(j+1)[$

6. $\{[\beta(i), \beta(i+1)]\}_{i \in \{1, \dots, m-1\}}$ is pairwise disjoint and $\bigsqcup_{i \in \{1, \dots, m-1\}} [\beta(i), \beta(i+1)[= [a, b[$

Proof. Define $A = \{a_i | i \in \{1, \dots, n\}\} \cup \{b_i | i \in \{1, \dots, n\}\}$. As $[a_1, b_1[\neq \emptyset$ we have that $[a, b[= \bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[\neq \emptyset$ so that

$$a < b \quad (21.302)$$

Assume that $a \notin A$. As $a \in [a, b[= \bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[$ there exist a $i \in \{1, \dots, n\}$ such that $a \in [a_i, b_i[\Rightarrow a_i \leq a < b_i \xrightarrow{a \notin A} a_i < a < b_i$. Using the density of the real numbers [theorem: 10.32] there exist a $t \in \mathbb{R}$ such that $a_i < t < a < b_i \Rightarrow t \notin [a, b[\wedge t \in [a_i, b_i[\subseteq [a, b[$ a contradiction. So the assumption is wrong and we have

$$a \in A \quad (21.303)$$

Assume that $b \notin A$. Given $i \in \{1, \dots, n\}$ we have $[a_i, b_i[\subseteq \bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[\subseteq [a, b[$ so that by [theorem: 3.145] $b_i \leq b \xrightarrow{b \notin A} b_i < b$. As also $a < b$ it follows that $\max(\{a\} \cup \{b_i | i \in \{1, \dots, n\}\}) < b$. Hence using the density of the real numbers [theorem: 10.32] there exists a $t \in \mathbb{R}$ with $\max(\{a\} \cup \{b_i | i \in \{1, \dots, n\}\}) < t < b$. Hence $\forall i \in \{1, \dots, n\}$ $t \notin [a_i, b_i[\wedge a < t < b$. So $t \notin \bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[= [a, b[\wedge t \in [a, b[$ again a contradiction. Hence the assumption is wrong and we have

$$b \in A \quad (21.304)$$

As $\forall i \in \{1, \dots, n\}$ we have $[a_i, b_i[\subseteq \bigcup_{i \in \{1, \dots, n\}} [a_i, b_i[= [a, b[$ it follows from theorem: 3.145] that $a \leq a_i < b_i \leq b$ proving that

$$a = \min(A) \wedge b = \max(A)$$

Using [theorems: 6.34, 6.45] it follows that A is finite and as $a, b \in A$ $\text{card}(A) \in \mathbb{N} \setminus \{1\}$, hence using [theorem: 10.97] it follows that there exist a bijection

$$\beta: \{1, \dots, m\} \rightarrow A \text{ where } m = \text{card}(A) \text{ such that } \forall i \in \{1, \dots, m-1\} \beta(i) < \beta(i+1) \quad (21.305)$$

As $\forall i \in \{1, \dots, m\}$ we have $\beta(1) \leq \beta(i)$ and $\beta(i) \leq \beta(m)$ it follows that $\beta(1) = \min(A) = a \wedge \beta(m) = \max(A) = b$, hence

$$a = \beta(1) \wedge b = \beta(m) \quad (21.306)$$

Let $x, y \in A$ such that $x < y$. If $\beta^{-1}(y) \leq \beta^{-1}(x)$ it follows from [eq: 21.305] that $y = \beta(\beta^{-1}(y)) \leq \beta(\beta^{-1}(x)) = x$ contradicting $x < y$. Hence we have

$$\forall x, y \in A \text{ with } x < y \text{ we have } \beta^{-1}(x) < \beta^{-1}(y) \quad (21.307)$$

Let $i \in \{1, \dots, n\}$ then $a_i, b_i \in A$ and $a_i < b_i$ [as $[a_i, b_i[\in \mathcal{I}]$ so that by the above $\beta^{-1}(a_i) < \beta^{-1}(b_i)$. In other words

$$\forall i \in \{1, \dots, n\} \text{ we have } \beta^{-1}(a_i) < \beta^{-1}(b_i) \quad (21.308)$$

Let $i \in \{1, \dots, n\}$ then for $j \in \{\beta^{-1}(a_i), \dots, \beta^{-1}(b_i) - 1\}$ we have $\beta(j) < \beta(j)$ so that by [lemma: 21.197] we have

$$\{[\beta(j), \beta(j+1)]\}_{j \in \{\beta^{-1}(a_i), \dots, \beta^{-1}(b_i) - 1\}}$$
 is pairwise disjoint (21.309)

and

$$\bigsqcup_{j \in \{\beta^{-1}(a_i), \dots, \beta^{-1}(b_i) - 1\}} [\beta(j), \beta(j+1)[= [\beta(\beta^{-1}(a_i)), \beta(\beta^{-1}(b_i))][= [a_i, b_i[\quad (21.310)$$

Using [lemma: 21.197] on [eq: 21.305] gives

$$\{[\beta(i), \beta(i-1)]\}_{i \in \{1, \dots, m-1\}} \text{ is pairwise disjoint} \quad (21.311)$$

and

$$\bigsqcup_{i \in \{1, \dots, m-1\}} [\beta(i), \beta(i+1)[= [\beta(1), \beta(m)][\stackrel{[\text{eq: 21.306}]}{=} [a, b[\quad (21.312)$$

So (1) is proved by [eq: 21.306], (2) is proved by [eq: 21.305], (3) is proved by [eq: 21.307], (4) is proved by [eq: 21.308], (5) by [eqs: 21.309, 21.310] and finally (6) by [eqs: 21.311, 21.312]. $\square$

Next we show how the volume of a half open rectangle partitioned by its sides can be calculated. Giving the follow example of a partitioned rectangle R

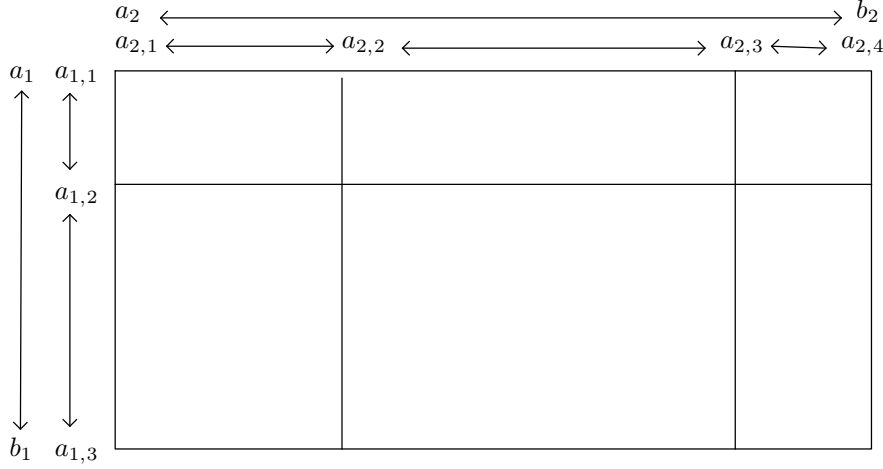


Figure 21.3.

we have then

$$\begin{aligned}
 \prod_{i \in \{1, \dots, 2\}} [a_i, b_i[&= [a_1, b_1[\cdot [a_2, b_2[\\
 &= [a_{1,1}, a_{1,2}[\cdot [a_{2,1}, a_{2,2}[\sqcup [a_{1,1}, a_{1,2}[\cdot [a_{2,2}, a_{2,3}[\sqcup [a_{1,1}, a_{1,2}[\cdot [a_{2,3}, a_{2,4}[\sqcup \\
 &\quad [a_{1,2}, a_{1,3}[\cdot [a_{2,1}, a_{2,2}[\sqcup [a_{1,2}, a_{1,3}[\cdot [a_{2,2}, a_{2,3}[\sqcup [a_{1,2}, a_{1,3}[\cdot [a_{2,3}, a_{2,4}[\\
 &= \bigsqcup_{\rho \in \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}} [a_{1,\rho_1}, a_{1,\rho_1+1}[\cdot [a_{2,\rho_2}, a_{2,\rho_2+1}[\\
 &= \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, 2\}} \{n_i, \dots, m_i-1\}} \left(\prod_{i \in \{1, \dots, 2\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right)
 \end{aligned}$$

and

$$\begin{aligned}
 \prod_{i \in \{1, \dots, 2\}} (b_i - a_i) &= (b_1 - a_1) \cdot (b_2 - a_2) \\
 &= (a_{1,2} - a_{1,1}) \cdot (a_{2,2} - a_{2,1}) + (a_{1,2} - a_{1,1}) \cdot (a_{2,3} - a_{2,2}) + (a_{1,2} - a_{1,1}) \cdot (a_{2,4} - a_{2,3}) \\
 &\quad + (a_{1,3} - a_{1,2}) \cdot (a_{2,2} - a_{2,1}) + (a_{1,3} - a_{1,2}) \cdot (a_{2,3} - a_{2,2}) + (a_{1,3} - a_{1,2}) \cdot (a_{2,4} - a_{2,3}) \\
 &= \bigsqcup_{\rho \in \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}} (a_{1,\rho_1+1} - a_{1,\rho_1}) \cdot (a_{2,\rho_2+1} - a_{2,\rho_2}) \\
 &= \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, 2\}} \{n_i, \dots, m_i-1\}} \left(\prod_{i \in \{1, \dots, 2\}} (a_{i,\rho_i+1} - a_{i,\rho_i}) \right)
 \end{aligned}$$

Lemma 21.199. Let $n \in \mathbb{N}$, $\{[a_i, b_i[\mid i \in \{1, \dots, n\}\} \subseteq \mathcal{I}_[$ such that $\forall i \in \{1, \dots, n\}$ there exists $n_i, m_i \in \mathbb{N}$ with $n_i \leq m_i$ and a $\{a_{i,j}\}_{j \in \{n_i, \dots, m_i\}} \subseteq \mathbb{R}$ such that $\forall j \in \{n_i, \dots, m_i-1\}$ $a_{i,j} \leq a_{i,j+1}$ and $a_{i,n_i} = a_i \wedge a_{i,m_i} = b_i$ then

$$\begin{aligned}
 &\left\{ \prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right\}_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i-1\}} \text{ is pairwise disjoint} \\
 &\prod_{i \in \{1, \dots, n\}} [a_i, b_i[= \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i-1\}} \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right) \\
 &\prod_{i=1}^n (b_i - a_i) = \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i-1\}} \left(\prod_{i \in \{1, \dots, n\}} (a_{i,\rho_i+1} - a_{i,\rho_i}) \right)
 \end{aligned}$$

Proof. Assume that $\exists i \in \{1, \dots, n\}$ such that $n_i = m_i$ then $a_i = a_{i,n_i} = a_{i,m_i} = b_i$ contradicting $a_i < b_i$ [because $[a_i, b_i] \neq \emptyset$], hence the assumption is false and we have

$$\forall i \in \{1, \dots, n\} \text{ we have } n_i < m_i \text{ or } n_i \leq m_i - 1 \quad (21.313)$$

Let $i \in \{1, \dots, n\}$ then, as $\forall j \in \{n_i, \dots, m_i - 1\}$ $a_{i,j} \leq a_{i,j+1}$ it follows from [lemma: 21.197] that

$$\{[a_{i,j}, a_{i,j+1}[]\}_{j \in \{n_i, \dots, m_i - 1\}} \text{ is pairwise disjoint} \quad (21.314)$$

and

$$[a_i, b_i[= [a_{i,n_i}, a_{i,m_i}[= \bigsqcup_{j \in \{n_i, \dots, m_i - 1\}} [a_{i,j}, a_{i,j+1}[\quad (21.315)$$

Let $x \in \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$ then $\forall i \in \{1, \dots, n\}$ we have $x_i \in [a_i, b_i[$, hence by the above there exist a $\rho_i \in \{n_i, \dots, m_i - 1\}$ such that $x_i \in [a_{i,\rho_i}, a_{i,\rho_i+1}[$. So for $\rho = (\rho_1, \dots, \rho_n) \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}$ it follows that $\forall i \in \{1, \dots, n\}$ $x_i \in [a_{i,\rho_i}, a_{i,\rho_i+1}[$ proving that $x \in \prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[$. Hence we have

$$\prod_{i \in \{1, \dots, n\}} [a_i, b_i[\subseteq \bigcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}} \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right) \quad (21.316)$$

If $x \in \bigcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}} \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right)$ then there exist a $\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}$ such that $x \in \prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[$. Hence $\forall i \in \{1, \dots, n\}$ we have $x_i \in [a_{i,\rho_i}, a_{i,\rho_i+1}[\subseteq [a_i, b_i[$ [eq: 21.315] proving that $x \in \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$. So it follows that

$\bigcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}} \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right) \subseteq \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$ which combined with [eq: 21.316] proves that

$$\prod_{i \in \{1, \dots, n\}} [a_i, b_i[= \bigcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}} \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right) \quad (21.317)$$

Let $\rho, \sigma \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}$ with $\rho \neq \sigma$ then there exist a $k \in \{1, \dots, n\}$ such that $\rho(k) \neq \sigma(k)$. Assume that there exist a $x \in \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right) \cap \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\sigma_i}, a_{i,\sigma_i+1}[\right)$ then $x_k \in [a_{k,\rho_k}, a_{k,\rho_k+1}[\cap [a_{k,\sigma_k}, a_{k,\sigma_k+1}[\stackrel{\rho_k \neq \sigma_k}{=} \emptyset$ [see eq: 21.314] which is a contradiction, hence the assumption is wrong and we must have that

$$\left(\prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right) \cap \left(\prod_{i \in \{1, \dots, n\}} [a_{i,\sigma_i}, a_{i,\sigma_i+1}[\right) = \emptyset.$$

So we have

$$\left\{ \prod_{i \in \{1, \dots, n\}} [a_{i,\rho_i}, a_{i,\rho_i+1}[\right\}_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i - 1\}} \text{ is pairwise disjoint} \quad (21.318)$$

Finally we have

$$\begin{aligned} \prod_{i=1}^n (b_i - a_i) &\stackrel{[\text{definition: 21.177}]}{=} \prod_{i=1}^n \text{len}([a_i, b_i[) \\ &\stackrel{[\text{eq: 21.314, 21.315}]}{=} \prod_{i=1}^n \text{len} \left(\bigsqcup_{j \in \{n_i, \dots, m_i\}} [a_{i,j}, a_{i,j+1}[\right) \\ &\stackrel{[\text{lemma: 21.180}]}{=} \prod_{i=1}^n \left(\sum_{j \in \{n_i, \dots, m_i\}} \text{len}([a_{i,j}, a_{i,j+1}[) \right) \\ &\stackrel{[\text{theorem: 11.49}]}{=} \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i\}} \left(\prod_{i \in \{1, \dots, n\}} \text{len}([a_{i,\rho_i}, a_{i,\rho_i+1}[) \right) \\ &= \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{n_i, \dots, m_i\}} \left(\prod_{i \in \{1, \dots, n\}} (a_{i,\rho_i+1} - a_{i,\rho_i}) \right) \end{aligned}$$

which proves the last part of the theorem. $\square$

The following lemma shows that if the union of half open rectangles is a half open rectangle then each side of the of the open rectangle is a union of the corresponding sides of the sub rectangles. For example in the following 2-dimensional case we have that

$$\begin{aligned}
 [a_1, b_1[&= [a_{1,1}, b_{1,1}[\cup [a_{2,1}, b_{2,1}[\cup [a_{3,1}, b_{3,1}[\cup [a_{4,1}, b_{5,1}[\cup [a_{5,1}, b_{5,1}[\\
 &= \bigcup_{i \in \{1, \dots, 5\}} [a_{i,1}, b_{i,1}[\\
 [a_2, b_2[&= [a_{1,2}, b_{1,2}[\cup [a_{2,2}, b_{2,2}[\cup [a_{3,2}, b_{3,2}[\cup [a_{4,2}, b_{4,2}[\cup [a_{5,2}, b_{5,2}[\\
 &= \bigcup_{i \in \{1, \dots, 5\}} [a_{i,2}, b_{i,2}[
 \end{aligned}$$

Figure 21.4.

Lemma 21.200. Let $n, m \in \mathbb{N}$ and let $\{\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{R}_{\mathbb{I}}^n$ such that

$$\bigcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\right) = \prod_{j \in \{1, \dots, n\}} [a_j, b_j[\in \mathcal{R}_{\mathbb{I}}^n$$

then $\forall j \in \{1, \dots, n\}$ we have

$$[a_j, b_j[= \bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[$$

Proof. Take $j \in \{1, \dots, n\}$. If $t \in \bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[$ then there exist a $i \in \{1, \dots, m\}$ such that $t \in [a_{i,j}, b_{i,j}[$. As $\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\in \mathcal{R}_{\mathbb{I}}^n$ $\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\neq \emptyset$ and it follows from [theorems: 3.110, 3.109] that the projection function $\pi_j: \prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\rightarrow [a_{i,j}, b_{i,j}[$ is a surjection hence there exists a $x \in \prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[$ such that $x_j = \pi_j(x) = t$. As

$$\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\subseteq \bigsqcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\right) = \prod_{j \in \{1, \dots, n\}} [a_j, b_j[$$

we have that $x \in \prod_{j \in \{1, \dots, n\}} [a_j, b_j[$ so that $t = \pi_j(x) = x_j \in [a_j, b_j[$. Hence we have

$$\bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[\subseteq [a_j, b_j[\quad (21.319)$$

Let $t \in [a_j, b_j[$. As $\prod_{j \in \{1, \dots, n\}} [a_j, b_j[\in \mathcal{R}_{||}^n$ $\prod_{j \in \{1, \dots, n\}} [a_j, b_j[\neq \emptyset$ so that by [theorems: 3.110, 3.109] the projection function $\pi_j: \prod_{j \in \{1, \dots, n\}} [a_j, b_j[\rightarrow [a_j, b_j[$ is a surjection, hence there exist a $x \in \prod_{j \in \{1, \dots, n\}} [a_j, b_j[= \bigsqcup_{i \in \{1, \dots, m\}} (\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[$ such that $t = \pi_j(x)$. Then there exists a $i \in \{1, \dots, m\}$ such that $x \in \prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[$. So it follows that $t = \pi_j(x) = x_j \in [a_{i,j}, b_{i,j}[\subseteq \bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[$ which proves that $[a_j, b_j[\subseteq \bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[$. Combining this with [eq: 21.319] proves that

$$[a_j, b_j[= \bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[\quad \square$$

We show now that the volume of the pairwise distinct union of half open non empty rectangles is equal to the sum of the volumes of these rectangles if this distinct union is also a half open rectangle.

Lemma 21.201. *Let $n, m \in \mathbb{N}$, $\{\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{R}_{||}^n$ pairwise disjoint such that $\bigsqcup_{i \in \{1, \dots, m\}} (\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[) = \prod_{i \in \{1, \dots, n\}} [a_i, b_i[\in \mathcal{R}_{||}^n$ then we have*

$$\prod_{i \in \{1, \dots, n\}} (b_i - a_i) = \sum_{i=1}^m \left(\prod_{j=1}^n (b_{i,j} - a_{i,j}) \right)$$

Proof. First using the previous lemma [lemma: 21.200] we have that

$$\forall j \in \{1, \dots, n\} [a_j, b_j[= \bigcup_{i \in \{1, \dots, m\}} [a_{i,j}, b_{i,j}[\quad (21.320)$$

Let $j \in \{1, \dots, n\}$. Using [lemma: 21.198] there exists a $n_j \in \mathbb{N} \setminus \{1\} \Rightarrow 1 < n_j$ and a bijection

$$\beta_j: \{1, \dots, n_j\} \rightarrow \{a_{i,j} | i \in \{1, \dots, m\}\} \cup \{b_{i,j} | i \in \{1, \dots, m\}\} \quad (21.321)$$

such that

$$\beta_j(1) = a_j \wedge \beta_j(n_j) = b_j \quad (21.322)$$

$$\forall k, l \in \{1, \dots, n_j\} \text{ with } k < l \text{ we have } \beta_j(k) < \beta_j(l) \quad (21.323)$$

$$\forall x, y \in \{a_{i,j} | i \in \{1, \dots, m\}\} \cup \{b_{i,j} | i \in \{1, \dots, m\}\} \text{ with } x < y \text{ we have } (\beta_j)^{-1}(x) < (\beta_j)^{-1}(y) \quad (21.324)$$

$$\forall i \in \{1, \dots, m\} \text{ we have } (\beta_j)^{-1}(a_{i,j}) < (\beta_j)^{-1}(b_{i,j}) \quad (21.325)$$

$$\forall i \in \{1, \dots, m\} [a_{i,j}, b_{i,j}[= \bigsqcup_{l \in \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j})-1\}} [\beta_j(l), \beta_j(l+1)[\quad (21.326)$$

$$[a_j, b_j[= \bigsqcup_{i \in \{1, \dots, n_j-1\}} [\beta_j(i), \beta_j(i+1)[\quad (21.327)$$

Because of [eqs: 21.322 21.323] we can apply [lemma: 21.199] giving

$$\prod_{j \in \{1, \dots, n\}} [a_j, b_j[= \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, n_j-1\}} \left(\prod_{j \in \{1, \dots, n\}} [\beta_j(\rho_j) - \beta_j(\rho_j+1)[\right) \quad (21.328)$$

$$\prod_{j=1}^n (b_j - a_j) = \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, n_j-1\}} \left(\prod_{j \in \{1, \dots, n\}} (\beta_j(\rho_j+1) - \beta_j(\rho_j)) \right) \quad (21.329)$$

Let $i \in \{1, \dots, m\}$. As $a_{i,j} = \beta_j((\beta_j)^{-1}(a_{i,j}))$ and $b_{i,j} = \beta_j((\beta_j)^{-1}(b_{i,j}))$ and $\forall i \in \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\}$ we have by [eq: 21.323] that $\beta_j(i) < \beta_j(i+1)$ we can use [lemma: 21.199] again to get

$$\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[= \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\}} \left(\prod_{j \in \{1, \dots, n\}} [\beta_j(\rho_j), \beta_j(\rho_j) + 1[\right) \quad (21.330)$$

$$\prod_{j \in \{1, \dots, n\}} (b_{i,j} - a_{i,j}) = \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\}} \left(\prod_{j \in \{1, \dots, n\}} (\beta_j(\rho_j + 1) - \beta_j(j)) \right) \quad (21.331)$$

To relate [eqs: 21.329] to [eq: 21.331] we prove now the following:

$$\bigsqcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\} \right) = \prod_{j \in \{1, \dots, n\}} \{1, \dots, n_j - 1\} \quad (21.332)$$

Proof. Let $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$. From [eq: 21.321] it follows that $(\beta_j)^{-1}(a_{i,j}), (\beta_j)^{-1}(b_{i,j}) \in \{1, \dots, n_j\}$, further by [eq: 21.325] we have $(\beta_j)^{-1}(a_{i,j}) \leq (\beta_j)^{-1}(b_{i,j}) - 1$ so that $\{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\} \subseteq \{1, \dots, n_j - 1\}$. Applying [theorem: 2.149] gives us then $\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\} \subseteq \prod_{j \in \{1, \dots, m\}} \{1, \dots, n_j - 1\}$, hence we have

$$\bigcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\} \right) \subseteq \prod_{j \in \{1, \dots, n\}} \{1, \dots, n_j - 1\} \quad (21.333)$$

For the opposite inclusion let $\rho \in \prod_{j \in \{1, \dots, n\}} \{1, \dots, n_j - 1\}$. Let $j \in \{1, \dots, n\}$ then $\rho_j \in \{1, \dots, n_j - 1\}$ and we have by [eq: 21.323] that $\beta_j(\rho_j) < \beta_j(\rho_j + 1)$ so that $\beta_j(\rho_j) \in [\beta_j(\rho_j), \beta_j(\rho_j)[$. Hence

$$\begin{aligned} (\beta_1(\rho_1), \dots, \beta_n(\rho_n)) &\in \prod_{j \in \{1, \dots, n\}} [\beta_j(\rho_j), \beta_j(\rho_j)[\\ &\subseteq \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, n_j - 1\}} \left(\prod_{j \in \{1, \dots, n\}} [\beta_j(\rho_j), \beta_j(\rho_j + 1)[\right) \\ &\stackrel{[\text{eq: 21.328}]}{=} \prod_{j \in \{1, \dots, n\}} [a_j, b_j[\\ &\stackrel{\text{hypothesis}}{=} \bigsqcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\right) \end{aligned}$$

Hence

$$\exists k \in \{1, \dots, m\} \text{ such that } (\beta_1(\rho_1), \dots, \beta_n(\rho_n)) \in \prod_{j \in \{1, \dots, n\}} [a_{k,j}, b_{k,j}[\quad (21.334)$$

Assume now that $\exists j \in \{1, \dots, n\}$ such that $b_{k,j} < \beta_j(\rho_j + 1)$ then $\beta_j(\rho_j) \stackrel{[\text{eq: 21.334}]}{<} b_{k,j} < \beta_j(\rho_j + 1)$ so that by [eq: 21.324] $(\beta_j)^{-1}(\beta_j(\rho_j)) < (\beta_j)^{-1}(b_{k,j}) < (\beta_j)^{-1}(\rho_j + 1)$, hence $\rho_j < (\beta_j)^{-1}(b_{k,j}) < \rho_j + 1$. As $(\beta_j)^{-1}(b_{k,j})$ is a whole number [because $(\beta_j)^{-1}(b_{k,j}) \in \{1, \dots, n_j\}$] and $\rho_j \in \{1, \dots, n_j - 1\} \Rightarrow j$ is a whole number it follows that $\rho_j + 1 \leq (\beta_j)^{-1}(b_{k,j}) < \rho_j + 1$ leading to the contradiction $\rho_j + 1 < \rho_j + 1$. Hence the assumption is wrong and we must have that

$$\forall j \in \{1, \dots, n\} \quad \beta_j(\rho_j + 1) \leq b_{k,j} \quad (21.335)$$

So given $j \in \{1, \dots, n\}$ we have $a_{k,j} \leq_{[\text{eq: 21.334}]} \beta_j(\rho_j) <_{[\text{eq: 21.323}]} \beta_j(\rho_j + 1) \leq_{[\text{eq: 21.335}]} b_{k,j}$. applying then [eq: 21.324] proves that $(\beta_j)^{-1}(a_{k,j}) \leq \rho_j < \rho_j + 1 \leq (\beta_j)^{-1}(b_{k,j})$, hence $(\beta_j)^{-1}(a_{k,j}) \leq \rho_j \leq (\beta_j)^{-1}(b_{k,j}) - 1$ proving that $\rho_j \in \{(\beta_j)^{-1}(a_{k,j}), (\beta_j)^{-1}(b_{k,j}) - 1\}$ from which it follows that $\rho \in \prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{k,j}), (\beta_j)^{-1}(b_{k,j}) - 1\}$. This prove that

$$\prod_{j \in \{1, \dots, n\}} \{1, \dots, n_j - 1\} \subseteq \bigcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\} \right)$$

which combined with [eq: 21.333] results in

$$\prod_{j \in \{1, \dots, n\}} \{1, \dots, n_j - 1\} = \bigcup_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\} \right) \quad (21.336)$$

To finish the prove for [eq: 21.332] we must prove that it is a pairwise disjoint union. So let $k, l \in \{1, \dots, m\}$ with $k \neq l$ and assume that

$$\exists r \in \prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{k,j}), \dots, (\beta_j)^{-1}(b_{k,j}) - 1\} \cap \prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{l,j}), \dots, (\beta_j)^{-1}(b_{l,j}) - 1\}$$

then $\forall j \in \{1, \dots, n\}$

$$(\beta_j)^{-1}(a_{k,j}) \leq r \leq (\beta_j)^{-1}(b_{k,j}) - 1 \wedge (\beta_j)^{-1}(a_{l,j}) \leq r \leq (\beta_j)^{-1}(b_{l,j}) - 1$$

or

$$(\beta_j)^{-1}(a_{k,j}) \leq r < (\beta_j)^{-1}(b_{k,j}) \wedge (\beta_j)^{-1}(a_{l,j}) \leq r < (\beta_j)^{-1}(b_{l,j})$$

applying then [eq: 21.323] gives

$$a_{k,j} \leq \beta_j(r) < b_{k,j} \wedge a_{l,j} \leq \beta_j(r) < b_{l,j} \text{ or } \beta_j(r) \in [a_{k,j}, b_{k,j}[\wedge \beta_j(r) \in [a_{l,j}, b_{l,j}[$$

proving that $(\beta_1(r), \dots, \beta_n(r)) \in (\prod_{i \in \{1, \dots, n\}} [a_{k,i}, b_{k,i}[) \cap (\prod_{i \in \{1, \dots, n\}} [a_{l,i}, b_{l,i}[)$ which as by the hypothesis $(\prod_{i \in \{1, \dots, n\}} [a_{k,i}, b_{k,i}[) \cap (\prod_{i \in \{1, \dots, n\}} [a_{l,i}, b_{l,i}[) = \emptyset$ is a contradiction hence we must have that

$$\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{k,j}), \dots, (\beta_j)^{-1}(b_{k,j}) - 1\} \cap \prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{l,j}), \dots, (\beta_j)^{-1}(b_{l,j}) - 1\} = \emptyset$$

the above together with [eq: 21.336] proves finally [eq: 21.332]. $\square$

Finally we are able to finish the prove of the lemma.

$$\begin{aligned} & \prod_{i=1}^n (b_i - a_i) \quad [\text{eq: 21.329}] \\ & \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, n_i - 1\}} \left(\prod_{j \in \{1, \dots, n\}} (\beta_j(\rho_j + 1) - \beta_j(\rho_j)) \right) \quad [\text{eq: 21.332}] \\ & \sum_{\rho \in \prod_{i \in \{1, \dots, m\}} (\prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\})} \left(\prod_{j \in \{1, \dots, n\}} (\beta_j(\rho_j + 1) - \right. \\ & \left. \beta_j(\rho_j)) \right) \quad [\text{theorem: 11.44}] \\ & \sum_{i \in \{1, \dots, m\}} \left(\sum_{\rho \in \prod_{j \in \{1, \dots, n\}} \{(\beta_j)^{-1}(a_{i,j}), \dots, (\beta_j)^{-1}(b_{i,j}) - 1\}} (\beta_j(\rho_j + 1) - \beta_j(\rho_j)) \right) \quad [\text{eq: 21.331}] \\ & \sum_{i \in \{1, \dots, m\}} \left(\prod_{j \in \{1, \dots, n\}} (b_{i,j} - a_{i,j}) \right) \end{aligned}$$

which proves the lemma. $\square$

Now we are finally ready to prove that $\text{vol}^n: \mathcal{R}_{\llbracket \cup \{\emptyset\}}^n \rightarrow [0, \infty]$ is a finite content.

Theorem 21.202. *Let $n \in \mathbb{N}$ then $\text{vol}^n: \mathcal{R}_{\llbracket \cup \{\emptyset\}}^n \rightarrow [0, \infty]$ [see definition: 21.195] is a finite content on the semi-ring $\mathcal{R}_{\llbracket \cup \{\emptyset\}}^n$ on $\mathbb{R}^n$ [see theorem: 21.193]*

Proof. First from the definition of vol^n [see definition: 21.195] it follows that

$$\text{vol}^n \text{ is finite} \quad (21.337)$$

and

$$\text{vol}^n(\emptyset) = 0 \quad (21.338)$$

Let $m \in \mathbb{N}$ and $\{R_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{R}_{\llbracket \cup \{\emptyset\}}^n$ be pairwise disjoint such that $\bigsqcup_{i \in \{1, \dots, m\}} R_i \in \mathcal{R}_{\llbracket \cup \{\emptyset\}}^n$ then we have two possibilities for $\bigsqcup_{i \in \{1, \dots, m\}} R_i$

$\bigsqcup_{i \in \{1, \dots, m\}} R_i = \emptyset$. Then we must have $\forall i \in \{1, \dots, m\}$ that $R_i = \emptyset$ so that

$$\text{vol}^n\left(\bigsqcup_{i \in \{1, \dots, m\}} R_i\right) = \text{vol}^n(\emptyset) = 0 = \sum_{i=1}^m 0 = \sum_{i=1}^m \text{vol}^n(\emptyset) = \sum_{i=1}^m \text{vol}^n(R_i)$$

$\bigsqcup_{i \in \{1, \dots, m\}} R_i \neq \emptyset$. Then for $R = \bigsqcup_{i \in \{1, \dots, m\}} R_i$ there exist a $\{[a_i, b_i[]_{i \in \{1, \dots, n\}} \subseteq \mathcal{I}_{\llbracket}$ such that

$$R = \prod_{i \in \{1, \dots, n\}} [a_i, b_i[\in \mathcal{R}_{\llbracket}^n \quad (21.339)$$

Then there exist a $i \in \{1, \dots, m\}$ such that $R_i \neq \emptyset$ so that $I = \{i \in \{1, \dots, m\} | R_i \neq \emptyset\} \neq \emptyset$ hence there exist a $k \in \mathbb{N}$ and a bijection

$$\beta: \{1, \dots, k\} \rightarrow I$$

Then $\forall i \in \{1, \dots, k\}$ $R_{\beta(i)} \neq \emptyset$ and $\forall i, j \in \{1, \dots, k\}$ with $i \neq j$ we have, as β is injective, that $\beta(i) \neq \beta(j)$ so that $R_{\beta(i)} \cap R_{\beta(j)} = \emptyset$. So that

$$\{R_{\beta(i)}\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{R}_{\llbracket}^n \text{ is pairwise disjoint} \quad (21.340)$$

Let $R = \bigsqcup_{i \in \{1, \dots, m\}} R_i$ then we have

$$R = \bigsqcup_{i \in \{1, \dots, m\}} R_i \stackrel{[\text{theorem: 21.142}]}{=} \bigsqcup_{i \in I} R_i \stackrel{[\text{theorem: 21.127}]}{=} \bigsqcup_{i \in \{1, \dots, k\}} R_{\beta(i)} \quad (21.341)$$

Further as $\{R_{\beta(i)}\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{R}_{\llbracket}^n$ we have $\forall i \in \{1, \dots, k\}$ that

$$R_{\beta(i)} = \prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\subseteq \mathcal{R}_{\llbracket}^n. \quad (21.342)$$

So that

$$\prod_{i \in \{1, \dots, n\}} [a_i, b_i[\stackrel{[\text{eq: 21.339}]}{=} \bigsqcup_{i \in \{1, \dots, k\}} \left(\prod_{j \in \{1, \dots, n\}} [a_{i,j}, b_{i,j}[\right) \quad (21.343)$$

Then we have

$$\begin{aligned}
 \text{vol}^n(R) &= \prod_{i=1}^n (\text{begin}(R)_i = \text{end}(R)_i) \\
 &= \prod_{i=1}^n (b_i - a_i) \\
 &\stackrel{\substack{[\text{eq: 21.343}] \\ [\text{lemma: 21.201}]}}{=} \sum_{i=1}^k \left(\prod_{j=1}^n (b_{i,j} - a_{i,j}) \right) \\
 &\stackrel{[\text{eq: 21.342}]}{=} \sum_{i=1}^k \left(\prod_{j=1}^n (\text{begin}(R_{\beta(i)})_j - \text{end}(R_{\beta(i)})_j) \right) \\
 &= \sum_{i=1}^k \text{vol}^n(R_{\beta(i)}) \\
 &= \sum_{i \in \{1, \dots, k\}} \text{vol}^n(R_{\beta(i)}) \\
 &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in I} \text{vol}^n(R_i) \\
 &= \sum_{i \in I} \text{vol}^n(R_i) + 0 \\
 &\stackrel{i \notin I = R_i = \emptyset}{=} \sum_{i \in I} \text{vol}^n(R_i) + \sum_{i \in \{1, \dots, m\} \setminus I} \text{vol}^n(R_i) \\
 &= \sum_{i \in \{1, \dots, m\}} \text{vol}^n(R_i) \\
 &= \sum_{i=1}^m \text{vol}^n(R_i)
 \end{aligned}$$

Hence

$$\text{vol}^n \left(\bigsqcup_{i \in \{1, \dots, m\}} R_i \right) = \sum_{i=1}^m \text{vol}^n(R_i)$$

which finished the proof of the theorem. $\square$

Next we use [theorem: 21.169] to prove that vol^n is actually a pre-measure on the semi-ring $\mathcal{R} \setminus \{\emptyset\}$. To do this we need the following lemma that will allow us to 'approximate' a half open rectangle by a compact set.

Lemma 21.203. *Let $n \in \mathbb{N}$, $\varepsilon \in \mathbb{R}^+$ and $\{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+$ then $\exists \{\varepsilon_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+$ with $0 < \varepsilon_i < x_i$ such that $\prod_{i=1}^n x_i < (\prod_{i=1}^n (x_i - \varepsilon_i)) + \varepsilon$*

Proof. We use induction on n . Let $\varepsilon \in \mathbb{R}^+$ and define

$$S_\varepsilon = \left\{ n \in \mathbb{N} \mid \text{If } \{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+ \text{ then } \exists \{\varepsilon_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+ \text{ with } \forall i \in \{1, \dots, n\} \ 0 < \varepsilon_i < x_i \text{ such that } \prod_{i=1}^n x_i < \left(\prod_{i=1}^n (x_i - \varepsilon_i) \right) + \varepsilon \right\}$$

then we have:

$1 \in S_\varepsilon$. If $\{x_i\}_{i \in \{1, \dots, 1\}} \subseteq \mathbb{R}^+$ then $x_1 \in \mathbb{R}^+$, define $\varepsilon_1 = \min\left(\frac{x_1}{2}, \frac{\varepsilon}{2}\right) \in \mathbb{R}^+$ then $0 < \varepsilon_1 \leq \frac{x_1}{2} < x_1$ and $\prod_{i=1}^1 x_i - \prod_{i=1}^1 (x_i - \varepsilon_i) = x_1 - (x_1 - \varepsilon_1) = \varepsilon_1 \leq \frac{\varepsilon}{2} < \varepsilon$ so that $\prod_{i=1}^1 x_i < (\prod_{i=1}^1 (x_i - \varepsilon_i)) + \varepsilon$ proving that $1 \in S_\varepsilon$.

$n \in S_\varepsilon \Rightarrow n+1 \in S_\varepsilon$. Assume that $\{x_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{R}^+$ then as $n \in S$ there exists $\{\varepsilon_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+$ with $0 < \varepsilon_i < x_i$ and $\prod_{i=1}^n x_i < (\prod_{i=1}^n (x_i - \varepsilon_i)) + \frac{\varepsilon}{2 \cdot x_{n+1}}$ so that as $0 < x_{n+1}$ we have

$$x_{n+1} \cdot \prod_{i=1}^n x_i - x_{n+1} \cdot \prod_{i=1}^n (x_i - \varepsilon_i) < \frac{\varepsilon}{2} \quad (21.344)$$

Take now $\varepsilon_{n+1} = \min \left(\frac{\varepsilon}{4 \cdot \prod_{i=1}^n x_i}, \frac{x_{n+1}}{2} \right)$ then $0 < \varepsilon_{n+1} \leq \frac{x_{n+1}}{2} < x_{n+1}$ and $\varepsilon_{n+1} \cdot (\prod_{i=1}^n x_i) \leq \frac{\varepsilon}{4} < \frac{\varepsilon}{2}$. Now

$$\begin{aligned} \prod_{i=1}^{n+1} x_i - \prod_{i=1}^{n+1} (x_i - \varepsilon_i) &= x_{n+1} \cdot \prod_{i=1}^n x_i - (x_{n+1} - \varepsilon_{n+1}) \cdot \prod_{i=1}^n (x_i - \varepsilon_i) \\ &= x_{n+1} \cdot \left(\prod_{i=1}^n x_i - \prod_{i=1}^n (x_i - \varepsilon_i) \right) + \varepsilon_{n+1} \cdot \prod_{i=1}^n (x_i - \varepsilon_i) \\ &\leq x_{n+1} \cdot \left(\prod_{i=1}^n x_i - \prod_{i=1}^n (x_i - \varepsilon_i) \right) + \frac{\varepsilon}{4} \\ &\stackrel{[\text{eq: 21.344}]}{<} \frac{\varepsilon}{2} + \frac{\varepsilon}{4} \\ &< \varepsilon \end{aligned}$$

proving that $n+1 \in S_\varepsilon$. □

Now we can prove that the volume function is a pre-measure.

Theorem 21.204. *Let $n \in \mathbb{N}$ then the function $\text{vol}^n: \mathcal{R}_{||}^n \cup \{\emptyset\} \rightarrow [0, \infty]$ is a finite pre-measure on the semi-ring $\mathcal{R}_{||}^n \cup \{\emptyset\}$.*

Proof. Consider the topological space $\langle \mathbb{R}^n, |||_n \rangle$ based on the canonical topology $|||_n$ [see theorem: 14.101]. Let $A \in \mathcal{R}_{||}^n \cup \{\emptyset\}$ then for A we have either:

$A = \emptyset$. Take then $A_\varepsilon = \emptyset \in \mathcal{R}_{||}^n \cup \{\emptyset\}$ and $K_\varepsilon = \emptyset$. Then K_ε is compact by [example: 14.269], $A_\varepsilon \subseteq K_\varepsilon \subseteq A$ and $\text{vol}^n(A) - \text{vol}^n(A_\varepsilon) = 0 - 0 = 0 < \varepsilon$.

$A \neq \emptyset$. Then there exists $a, b \in \mathbb{R}^n$ with $a < b$ such that $A = [a, b[= \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$. As $\forall i \in \{1, \dots, n\}$ $a_i < b_i \Rightarrow b_i - a_i \in \mathbb{R}^+$ we have by [lemma: 21.203] that there exists a $\{\varepsilon_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}^+$ such that

$$\forall i \in \{1, \dots, n\} \quad 0 < \varepsilon_i < b_i - a_i \text{ and } \prod_{i=1}^n (b_i - a_i) < \left(\prod_{i=1}^n ((b_i - a_i) - \varepsilon_i) \right) + \varepsilon \quad (21.345)$$

From the above it follows that $\forall i \in \{1, \dots, n\}$ $a_i < b_i - \varepsilon_i$ so that

$$A_\varepsilon = \prod_{i \in \{1, \dots, n\}} [a_i, b_i - \varepsilon_i[\in \mathcal{R}_{||}^n \subseteq \mathcal{R}_{||}^n \cup \{\emptyset\}$$

Further by [theorem: 14.323]

$$K_\varepsilon = \prod_{i \in \{1, \dots, n\}} [a, b_i - \varepsilon_i] \text{ is compact}$$

As $\forall i \in \{1, \dots, n\}$ $[a_i, b_i - \varepsilon_i[\subseteq [a_i, b_i - \varepsilon_i] \subseteq [a_i, b_i]$ it follows from [theorem: 2.149] that

$$A_\varepsilon \subseteq K_\varepsilon \subseteq A$$

Further

$$\text{vol}^n(A) - \text{vol}^n(A_\varepsilon) = \prod_{i=1}^n (b_i - a_i) - \prod_{i=1}^n ((b_i - a_i) - \varepsilon_i) \underset{[\text{eq: 21.345}]}{<} \varepsilon$$

So in all cases we found a $A_\varepsilon \in \mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}$ and a compact set K_ε such that $A_\varepsilon \subseteq K_\varepsilon \subseteq A$ and $\text{vol}^n(A) - \text{vol}^n(A_\varepsilon) < \varepsilon$. Finally using [theorem: 21.169] it follows that

$$\text{vol}^n: \mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\} \rightarrow [0, \infty] \text{ is a pre-measure on } \mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}$$

which is finite by [theorem: 21.202]. $\square$

Next we use a version of the Carathéodry theorem [theorem: 21.148] to extend vol^n to the Lebesgue measure on Lebesgue measurable sets in $\mathbb{R}^n$.

Theorem 21.205. *If we define*

$$(\lambda^n)^*: \mathcal{P}[\mathbb{R}^n] \rightarrow [0, \infty] \text{ by } (\lambda^n)^*(A) = \begin{cases} \infty & \text{if } \mathcal{S}[A] = \emptyset \\ \inf \left(\left\{ \sum_{i=1}^{\infty} \text{vol}^n(A_i) \mid \{A_i\}_{i \in \mathbb{N}} \in \mathcal{S}[A] \right\} \right) & \text{if } \mathcal{S}[A] \neq \emptyset \end{cases}$$

where

$$\mathcal{S}[A] = \left\{ \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\} \mid A \subseteq \bigcup_{i \in \mathbb{N}} A_i \right\}$$

then

$$(\lambda^n)^* \text{ is a outer measure}$$

$$\mathcal{L}[\mathbb{R}^n] = \{A \in \mathcal{P}[\mathbb{R}^n] \mid \forall B \in \mathcal{P}(\mathbb{R}^n) \text{ we have } (\lambda^n)^*(B) = (\lambda^n)^*(B \cap A) + (\lambda^n)^*(B \setminus A)\}$$

is a σ -algebra on $\mathbb{R}^n$ such that

$$\sigma[\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}] \subseteq \mathcal{L}[\mathbb{R}^n]$$

and

$$\bar{\lambda}^n = ((\lambda^n)^*)|_{\mathcal{L}[\mathbb{R}^n]}: \mathcal{L}[\mathbb{R}^n] \rightarrow [0, \infty] \text{ is a measure on } \mathcal{L}[\mathbb{R}^n]$$

and

$$\lambda^n = ((\lambda^n)^*)|_{\sigma[\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}]}: \sigma[\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}] \rightarrow [0, \infty] \text{ is a measure on } \sigma[\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}]$$

and

$$((\lambda^n)^*)|_{\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}} = \text{vol}^n$$

Further if $\mu: \sigma[\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}] \rightarrow [0, \infty]$ is another measure such that $\mu|_{\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}} = \text{vol}^n$ then

$$\mu = \lambda^n$$

We have then

$$\langle \mathbb{R}^n, \mathcal{L}[\mathbb{R}^n], \bar{\lambda}^n \rangle \text{ is a measure space called the Lebesgue measure space on } \mathbb{R}^n$$

where

$$\mathcal{L}[\mathbb{R}^n] \text{ is called the set of Lebesgue measurable sets in } \mathbb{R}^n$$

and

$$\bar{\lambda}^n \text{ is called the Lebesgue measure on } \mathbb{R}^n$$

Note 21.206. As by [theorem: 21.194] $\mathcal{B}[\mathbb{R}^n] = \sigma[\mathcal{R}_{\text{fin}}^n \setminus \{\emptyset\}]$ we have also

$$\mathcal{B}[\mathbb{R}^n] \subseteq \mathcal{L}[\mathbb{R}^n]$$

and

$$\lambda^n: \mathcal{B}[\mathbb{R}^n] \rightarrow [0, \infty] \text{ is a measure on } \mathcal{B}[\mathbb{R}^n]$$

so that

$$(\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n) \text{ is a measure space called the Borel measure space on } \mathbb{R}^n$$

and

$$\lambda^n \text{ is called the Borel measure on } \mathbb{R}^n$$

Proof. Using [theorem: 21.204] we have that

$$\text{vol}^n: \mathcal{R}_{||}^n \rightarrow [0, \infty] \text{ is a pre-measure on } \mathbb{R}^n \quad (21.346)$$

Define now $\{G_i\}_{i \in \mathbb{N}} \subseteq \mathcal{R}_{||}^n$ by $G_i = \prod_{j \in \{1, \dots, n\}} [-i, i[$. Then as $\forall i \in \mathbb{N} \quad [-i, i[\subseteq [-(i+1), i+1[$ it follows from [theorem: 2.149] that $G_i \subseteq G_{i+1}$, hence we have

$$\forall i \in \mathbb{N} \text{ we have } G_i \subseteq G_{i+1} \text{ and as } \text{vol}^n \text{ is finite that } \text{vol}^n(G_i) < \infty \quad (21.347)$$

Let $x \in \mathbb{R}^n$ then $\forall i \in \{1, \dots, n\}$ we have by [theorem: 10.30] that there exists $n_i, m_i \in \mathbb{N}$ such that $-n_i < x_i < m_i$. Take $k = \max(\{n_i | i \in \mathbb{N}\} \cup \{m_i | i \in \mathbb{N}\}) \in \mathbb{N}$ then $\forall i \in \mathbb{N}$ we have $-k < n_i < x_i < m_i < k$ so that $x \in G_k \subseteq \bigcup_{i \in \mathbb{N}} G_i \subseteq \mathbb{R}^n$. Hence we have

$$\mathbb{R}^n = \bigcup_{i \in \mathbb{N}} G_i \quad (21.348)$$

The proof follows then from [eqs: 21.346, 21.347, 21.348] and [theorem: 21.148]. $\square$

We have the following relation between the Lebesgue measure of half open rectangles and closed rectangles.

Theorem 21.207. Let $n \in \mathbb{N}$ and $a, b \in \mathbb{R}^n$ with $a \leq b$ then $\lambda^n([a, b]) = \lambda^n([a, b])$

Note 21.208. Using [theorem: 21.92] $[a, b[, [a, b] \in \mathcal{B}[\mathbb{R}^n]$ so that $\lambda^n([a, b])$ and $\lambda^n([a, b])$ are well defined.

Proof. Let $a, b \in \mathbb{R}^n$ with $a \leq b$. If $x \in [a, b]$ then $\forall m \in \mathbb{N}$ we have $\forall i \in \{1, \dots, n\}$ that $a_i \leq x_i \leq b_i < b_i + \frac{1}{m}$ so that $x \in [a, b + \frac{1}{m}[$ which proves that

$$[a, b] \subseteq \bigcap_{m \in \mathbb{N}} \left[a, b + \frac{1}{m} \right[\quad (21.349)$$

Let $x \in \bigcap_{m \in \mathbb{N}} [a, b + \frac{1}{m}[$ then $\forall m \in \mathbb{N}$ we have $x \in [a, b + \frac{1}{m}[$. Assume that $x \notin [a, b]$ then there exist a $i \in \{1, \dots, n\}$ such that $x_i < a \vee b_i < x_i$. As $x \in [a, b + \frac{1}{m}[\Rightarrow a_i \leq x_i$ we must have that $b_i < x_i$. Using [theorem: 10.30] there exist a $m_i \in \mathbb{N}$ such that $b_i < b_i + \frac{1}{m_i} < x_i$ from which it follows that $x \notin [a, b + \frac{1}{m_i}[$ which contradicts $x \in \bigcap_{m \in \mathbb{N}} [a, b + \frac{1}{m}[$, hence the assumption is false and we must have that $x \in [a, b]$. So $\bigcap_{m \in \mathbb{N}} [a, b + \frac{1}{m}[\subseteq [a, b]$ which combined with [eq: 21.349] proves that

$$[a, b] = \bigcap_{m \in \mathbb{N}} \left[a, b + \frac{1}{m} \right[\quad (21.350)$$

Let $m \in \mathbb{N}$ then we have $\lambda^n([a, b + \frac{1}{m}[) \stackrel{[\text{theorem: 21.205}]}{=} \text{vol}^n([a, b + \frac{1}{m}[) = \prod_{i=1}^n (b_i + \frac{1}{m} - a_i) < \infty$ so that

$$\forall m \in \mathbb{N} \text{ we have } \lambda^n\left(\left[a, b + \frac{1}{m}\right[\right) < \infty \quad (21.351)$$

If $x \in \left[a, b + \frac{1}{m+1}\right[$ then $\forall i \in \{1, \dots, n\}$ we have $a_i \leq x_i < b_i + \frac{1}{m+1} < b_i + \frac{1}{m}$ so that $x \in \left[a, b + \frac{1}{m}\right[$ hence we have

$$\forall m \in \mathbb{N} \text{ we have } \left[a, b + \frac{1}{m+1}\right[\subseteq \left[a, b + \frac{1}{m}\right[\quad (21.352)$$

So the conditions for [theorem: 21.115] are satisfied by [eqs: 21.351, 21.352] so that

$$\lambda^n \left(\bigcap_{m \in \mathbb{N}} \left[a, b + \frac{1}{m}\right[\right) = \lim_{m \rightarrow \infty} \lambda^n \left(\left[a, b + \frac{1}{m}\right[\right)$$

so that

$$\begin{aligned} \lambda^n([a, b]) &\stackrel{[\text{eq: 21.350}]}{=} \lambda^n \left(\bigcap_{m \in \mathbb{N}} \left[a, b + \frac{1}{m}\right[\right) \\ &= \lim_{m \rightarrow \infty} \lambda^n \left(\left[a, b + \frac{1}{m}\right[\right) \\ &= \lim_{m \rightarrow \infty} \left(\prod_{i \in \{1, \dots, n\}} \left(b_i + \frac{1}{m} - a_i \right) \right) \\ &\stackrel{[\text{theorem: 14.360}]}{=} \prod_{i \in \{1, \dots, n\}} \left(\lim_{m \rightarrow \infty} \left(b_i + \frac{1}{m} - a_i \right) \right) \\ &= \prod_{i \in \{1, \dots, n\}} (b_i - a_i) \\ &= \lambda^n([a, b]) \\ &\quad \square \end{aligned}$$

Theorem 21.209. Let $n \in \mathbb{N}$ and $a, b \in \mathbb{R}^n$ with $a \leq b$ then $\lambda^n([a, b]) = \lambda^n(]a, b[)$

Note 21.210. Using [theorem: 21.92] $[a, b[,]a, b[\in \mathcal{B}[\mathbb{R}^n]$ so that $\lambda^n([a, b])$ and $\lambda^n(]a, b[)$ are well defined.

Proof. Let $a, b \in \mathbb{R}^n$ with $a \leq b$. If $x \in]a, b[$ then $\forall i \in \{1, \dots, n\}$ we have $a_i < x_i < b_i$, using [theorem: 10.30] there exist a $m_i \in \mathbb{N}$ such that $a_i + \frac{1}{m_i} < x_i$, take $m \in \max(\{m_i | i \in \{1, \dots, n\}\})$ then $a_i + \frac{1}{m} \leq a_i + \frac{1}{m_i} < x_i < b_i \Rightarrow x_i \in [a_i + \frac{1}{m}, b_i[$. Hence $x \in \prod_{i \in \{1, \dots, n\}} [a_i + \frac{1}{m}, b_i[$ which proves that

$$]a, b[\subseteq \bigcup_{m \in \mathbb{N}} \left[a + \frac{1}{m}, b\right[$$

Let $x \in \bigcup_{m \in \mathbb{N}} [a + \frac{1}{m}, b[$ then $\exists m \in \mathbb{N}$ such that $x \in [a + \frac{1}{m}, b[$, hence $\forall i \in \{1, \dots, n\}$ we have $a_i < a_i + \frac{1}{m} \leq x_i < b_i \Rightarrow x_i \in]a_i, b_i[\Rightarrow x \in]a, b[$ we conclude that

$$]a, b[= \bigcup_{m \in \mathbb{N}} \left[a + \frac{1}{m}, b\right[\quad (21.353)$$

Further if $x \in [a + \frac{1}{m}, b[$ then $\forall i \in \{1, \dots, n\}$ $a_i + \frac{1}{m} \leq x_i < b_i \Rightarrow a_i + \frac{1}{m+1} < a_i + \frac{1}{m} \leq x_i < b_i \Rightarrow x \in [a + \frac{1}{m+1}, b[$ which proves that

$$\forall m \in \mathbb{N} \text{ we have } \left[a + \frac{1}{m}, b\right[\subseteq \left[a + \frac{1}{m+1}, b\right[\quad (21.354)$$

So the conditions for [theorem: 21.115] are satisfied by [eqs: 21.353, 21.354] which gives us that

$$\lambda^n \left(\bigcup_{m \in \mathbb{N}} \left[a + \frac{1}{m}, b\right[\right) = \lim_{m \rightarrow \infty} \lambda^n \left(\left[a + \frac{1}{m}, b\right[\right)$$

hence we have

$$\begin{aligned}
 \lambda^n([a, b]) &\stackrel{[\text{eq: 21.353}]}{=} \lambda^n\left(\bigcup_{m \in \mathbb{N}} \left[a + \frac{1}{m}, b\right]\right) \\
 &= \lim_{m \rightarrow \infty} \lambda^n\left(\left[a + \frac{1}{m}, b\right]\right) \\
 &= \lim_{m \rightarrow \infty} \left(\prod_{i \in \{1, \dots, n\}} \left(b_i + \frac{1}{m} - a_i\right)\right) \\
 &\stackrel{[\text{theorem: 14.360}]}{=} \prod_{i \in \{1, \dots, n\}} \left(\lim_{m \rightarrow \infty} \left(b_i + \frac{1}{m} - a_i\right)\right) \\
 &= \prod_{i \in \{1, \dots, n\}} (b_i - a_i) \\
 &= \lambda^n([a, b]) \\
 &\quad \square
 \end{aligned}$$

21.3.5 Completeness of measures

Definition 21.211. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $B \in \mathcal{P}(X)$ is **μ -negligible** [sometimes also called μ -null] if there exist a $A \in \mathcal{A}$ with $\mu(A) = 0$ such that $B \subseteq A$. The set of all μ -negligible sets is noted as $\mathcal{N}_\mu$ hence

$$\mathcal{N}_\mu = \{A \in \mathcal{P}(X) \mid \exists B \in \mathcal{A} \text{ with } \mu(B) = 0 \text{ such that } A \subseteq B\}$$

Lemma 21.212. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space, I a countable set [i.e. finite or denumerable] and $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ a countable family of sets such that $\forall i \in I \mu(A_i) = 0$ then $\mu(\bigcup_{i \in I} A_i) = 0$

Note 21.213. [theorem: 21.39] ensures that $\bigcup_{i \in I} A_i \in \mathcal{A}$ so that $\mu(\bigcup_{i \in I} A_i)$ is well defined.

Proof. As I is countable we have either:

I is finite. Then there exist a $n \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, n\} \rightarrow I$. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_{\beta(i)} & \text{if } i \in \{1, \dots, n\} \\ \emptyset & \text{if } i \in \mathbb{N} \setminus \{1, \dots, n\} \end{cases}$$

then we have

$$\forall i \in \mathbb{N} \text{ that } \mu(B_i) = 0$$

Let $x \in \bigcup_{i \in I} A_i$ then there exist a $i \in I$ such that $x \in A_i$. As β is a bijection there exist a $j \in \{1, \dots, n\}$ such that $i = \beta(j)$ hence $x \in A_{\beta(j)} = B_j \subseteq \bigcup_{j \in \mathbb{N}} B_j$ so that

$$\bigcup_{i \in I} A_i \subseteq \bigcup_{j \in \mathbb{N}} B_j$$

On the other hand if $x \in \bigcup_{j \in \mathbb{N}} B_j$ then $\exists j \in \mathbb{N}$ such that $x \in B_j$. We must have that $j \in \{1, \dots, n\}$ for if $j \in \mathbb{N} \setminus \{1, \dots, n\}$ then $B_j = \emptyset$, hence $x \in B_j = A_{\beta(j)} \subseteq \bigcup_{i \in I} A_i$ which proves that $\bigcup_{j \in \mathbb{N}} B_j \subseteq \bigcup_{i \in I} A_i$. Combining this result with the above gives

$$\bigcup_{i \in I} A_i = \bigcup_{i \in \mathbb{N}} B_i$$

Using the above we have

$$\begin{aligned}
 \mu\left(\bigcup_{i \in I} A_i\right) &= \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\
 &\stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{i=1}^{\infty} \mu(B_i) \\
 &= \sum_{i=1}^{\infty} 0 \\
 &= 0
 \end{aligned}$$

which proves, as $0 \leq \mu(\bigcup_{i \in I} A_i)$, that

$$\mu\left(\bigcup_{i \in I} A_i\right) = 0$$

I is denumerable. Then there exist a bijection $\beta: \mathbb{N} \rightarrow I$ then

$$\forall i \in \mathbb{N} \quad \mu(A_{\beta(i)}) = 0$$

Further

$$\begin{aligned}
 \mu\left(\bigcup_{i \in I} A_i\right) &\stackrel{[\text{eq: 2.127}]}{=} \mu\left(\bigcup_{i \in \mathbb{N}} A_{\beta(i)}\right) \\
 &\stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{i=1}^{\infty} \mu(A_{\beta(i)}) \\
 &= 0
 \end{aligned}$$

which proves, as $0 \leq \mu(\bigcup_{i \in I} A_i)$, that

$$\mu\left(\bigcup_{i \in I} A_i\right) = 0$$

□

We have the following properties for μ -negligible sets.

Theorem 21.214. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space then we have*

1. $\emptyset \in \mathcal{N}_\mu$
2. If $A \in \mathcal{N}_\mu$ then for every $B \subseteq A$ we have $B \in \mathcal{N}_\mu$
3. If I is countable, $\{A_i\}_{i \in I} \subseteq \mathcal{N}_\mu$ a countable family of μ -negligible sets then we have that
 - a. $\bigcup_{i \in I} A_i \subseteq \mathcal{N}_\mu$
 - b. If $I \neq \emptyset$ then $\bigcap_{i \in I} A_i \subseteq \mathcal{N}_\mu$

Proof.

1. This is trivial as $\emptyset \subseteq \emptyset \in \mathcal{A}$ and $\mu(\emptyset) = 0$.
2. Let $A \in \mathcal{N}_\mu$ and $B \subseteq A$. As $A \in \mathcal{N}_\mu$ there exists a $C \in \mathcal{A}$ with $\mu(C) = 0$ such that $A \subseteq C$ which as $B \subseteq A$ proves that $B \subseteq C$ so that $B \in \mathcal{N}_\mu$.
3. Assume that $\{A_i\}_{i \in I} \subseteq \mathcal{N}_\mu$ then there exists a $\{B_i\}_{i \in I} \subseteq \mathcal{A}$ such that $\forall i \in I \quad \mu(B_i) = 0$ and $A_i \subseteq B_i$. Hence $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} B_i$ and using [lemma: 21.212] we have that

$$\mu\left(\bigcup_{i \in I} B_i\right) = 0$$

which proves that

$$\bigcup_{i \in I} A_i \in \mathcal{N}_\mu$$

Finally if $I \neq \emptyset$ then there exist a $k \in I$ so that $\bigcap_{i \in I} A_i \subseteq A_k \in \mathcal{N}_\mu$ it follows from (2) that $\bigcap_{i \in \mathbb{N}} A_i \in \mathcal{N}_\mu$. □

Definition 21.215. (Complete Measures) Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then μ is **complete** if $\forall A \in \mathcal{A}$ with $\mu(A) = 0$ we have $\forall B \in \mathcal{P}(X)$ with $B \subseteq A$ that $B \in \mathcal{A}$. Every measure space with a complete measure is called a **complete measure space**.

Theorem 21.216. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then

$$\mu \text{ is complete} \Leftrightarrow \mathcal{N}_\mu \subseteq \mathcal{A}$$

Proof.

$\Rightarrow$. If $A \in \mathcal{N}_\mu$ then there exists a $B \in \mathcal{A}$ with $\mu(B) = 0$ such that $A \subseteq B$. As μ is complete $B \in \mathcal{A}$ it follows that $A \in \mathcal{A}$. Hence $\mathcal{N}_\mu \subseteq \mathcal{A}$.

$\Leftarrow$. Let $A \in \mathcal{A}$ with $\mu(A) = 0$ then for $B \in \mathcal{P}(X)$ with $B \subseteq A$ we have by definition of $\mathcal{N}_\mu$ that $B \in \mathcal{N}_\mu$. Hence, as $\mathcal{N}_\mu \subseteq \mathcal{A}$, it follows that $B \in \mathcal{A}$ which proves that μ is complete. $\square$

We prove now that every measure space constructed via the Carathéodory procedure is a complete measure space.

Theorem 21.217. Let X be a set and $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ is a **outer measure** then the measure space

$$\langle X, \mathcal{M}[\mu^*], (\mu^*)|_{\mathcal{M}[\mu^*]} \rangle \text{ [see theorem: 21.122]}$$

where

$$\mathcal{M}[\mu^*] \stackrel{\text{definition: 21.119}}{=} \{A \in \mathcal{P}(X) | \forall B \in \mathcal{P}(X) \text{ we have } \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A)\}$$

is a complete measure space.

Proof. If $A \in \mathcal{N}_{(\mu^*)|_{\mathcal{M}[\mu^*]}}$ then there exists a $B \in \mathcal{M}[\mu^*]$ with $\mu^*(B) = (\mu^*)|_{\mathcal{M}[\mu^*]}(B) = 0$ such that $A \subseteq B$. Hence by the definition of a outer measure [see definition: 21.117] we have that $\mu^*(A) \leq \mu^*(B) = 0$ which proves that $\mu^*(A) = 0$. Applying then [theorem: 21.121] proves that $A \in \mathcal{M}[\mu^*]$. Hence $\mathcal{N}_{(\mu^*)|_{\mathcal{M}[\mu^*]}} \subseteq \mathcal{M}[\mu^*]$ which by the previous theorem [theorem: 21.216] proves that $\langle X, \mathcal{M}[\mu^*], (\mu^*)|_{\mathcal{M}[\mu^*]} \rangle$ is a complete measure space. $\square$

Corollary 21.218. We have that

1. $\langle \mathbb{R}, \mathcal{L}[\mathbb{R}], \bar{\lambda} \rangle$ is a complete measure space.
2. Let $n \in \mathbb{N}$ then $\langle \mathbb{R}^n, \mathcal{L}[\mathbb{R}^n], \bar{\lambda}^n \rangle$ is a complete measure space

Proof.

1. This follows from the previous theorem [theorem: 21.217] and the definition of $\langle \mathbb{R}, \mathcal{L}[\mathbb{R}], \bar{\lambda} \rangle$ [see theorem: 21.182].
2. This follows from the previous theorem [theorem: 21.217] and the definition of $\langle \mathbb{R}^n, \mathcal{L}[\mathbb{R}^n], \bar{\lambda}^n \rangle$ [see theorem: 21.205]. $\square$

Sometimes we have to deal with arbitrary subsets of sets with measure zero, it is then desirable that there exists a extension of measure space to a bigger measure space in which these sets are elements of the bigger σ -algebra. To do this we introduce the concept of the **completion** of a measure space.

Definition 21.219. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then the **completion** of $\mathcal{A}$ noted as $\mathcal{A}_\mu$ is defined by

$$\mathcal{A}_\mu = \{A \in \mathcal{P}(X) | \exists E, F \in \mathcal{A} \text{ such that } E \subseteq A \subseteq F \text{ and } \mu(F \setminus E) = 0\}$$

elements of $\mathcal{A}_\mu$ are called μ -measurable sets.

Note 21.220. As by [theorem: 21.36] a σ -algebra is a algebra it follows from [theorem: 21.31] that $\forall E, F \in \mathcal{A} \mu \setminus F \in \mathcal{A}$ so the above is well defined.

Next we define a set function of $\bar{\mu}: \mathcal{A}_\mu \rightarrow [0, \infty]$, first we need a little lemma needed for the existence of such a function.

Lemma 21.221. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space $A \in \mathcal{P}(X)$ and $E, F \in \mathcal{A}$ such that $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = 0$ then

$$\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\} \neq \emptyset \text{ [so that } \sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\}) \text{ exists]}$$

and

$$\mu(F) = \mu(E) = \sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\})$$

Proof. As $E \in \mathcal{A}$ and $E \subseteq A$ it follows that $\mu(E) \in \{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\}$ hence

$$\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\} \neq \emptyset \quad (21.355)$$

and

$$\mu(E) \leq \sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\}) \quad (21.356)$$

As $E \subseteq A \subseteq F$ we have $E \subseteq F$ so that by [theorem: 1.39] $F = (F \setminus E) \sqcup E$ hence $\mu(F) = \mu(F \setminus E) + \mu(E) = 0 + \mu(E) = \mu(E)$ proving that

$$\mu(E) = \mu(F) \quad (21.357)$$

If $B \in \mathcal{A}$ with $B \subseteq A$ then $B \subseteq F$ so that by [theorem: 21.99] $\mu(B) \leq \mu(F) = \mu(E)$ hence

$$\sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\}) \leq \mu(E)$$

which combining with [eq: 21.356] proves that

$$\mu(E) = \sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\}) \quad \square$$

Theorem 21.222. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then if we define

$$\bar{\mu}: \mathcal{A}_\mu \rightarrow [0, \infty] \text{ by } \bar{\mu}(A) = \sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\})$$

then

$$\forall A \in \mathcal{A}_\mu \text{ we have if } E, F \in \mathcal{A} \text{ satisfies } E \subseteq A \subseteq F \text{ and } \mu(F \setminus E) = 0 \text{ then } \mu(E) = \mu(F) = \bar{\mu}(A)$$

and

$$\langle X, \mathcal{A}_\mu, \bar{\mu} \rangle \text{ is a **complete** measure space}$$

and

$$\mathcal{A} \subseteq \mathcal{A}_\mu$$

and

$$\mu = (\bar{\mu})|_{\mathcal{A}}$$

In other words the measure space $\langle X, \mathcal{A}_\mu, \bar{\mu} \rangle$ is a extension of the measure space $\langle X, \mathcal{A}, \mu \rangle$. $\langle X, \mathcal{A}_\mu, \bar{\mu} \rangle$ is called the **completion of the measure space** $\langle X, \mathcal{A}, \mu \rangle$

Note 21.223. $\bar{\mu}$ is well defined because if $A \in \mathcal{A}_\mu$ then there exists $E, F \in \mathcal{A}$ with $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = 0$ so that by [lemma: 21.221] $\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\} \neq \emptyset$.

Proof. Let $A \in \mathcal{A}_\mu$ then if $E, F \in \mathcal{A}$ is such that $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = 0$ it follows from [lemma: 21.221] that

$$\mu(E) = \mu(F) = \sup(\{\mu(B) | B \in \mathcal{A} \text{ with } B \subseteq A\}) = \bar{\mu}(A) \quad (21.358)$$

Next we prove that $\mathcal{A}_\mu$ contains $\mathcal{A}$ and is a σ -algebra. Let $A \in \mathcal{A}$ and take $E = A \in \mathcal{A}$, $F = A \in \mathcal{A}$ then $E = A = F$ or $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = \mu(A \setminus A) = \mu(\emptyset) = 0$ so that $A \in \mathcal{A}_\mu$. Hence we have

$$\mathcal{A} \subseteq \mathcal{A}_\mu \quad (21.359)$$

As $\mathcal{A}$ is a σ -algebra $\emptyset \in \mathcal{A}$ hence using the above it follows that

$$\emptyset \in \mathcal{A}_\mu \quad (21.360)$$

Let $A \in \mathcal{A}_\mu$ then there exists $E, F \in \mathcal{A}$ such that $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = 0$. As $\mathcal{A}$ is a σ -algebra we have that

$$X \setminus E, X \setminus F \in \mathcal{A} \quad (21.361)$$

further applying [theorem: 1.36] gives

$$X \setminus F \subseteq X \setminus A \subseteq X \setminus E \quad (21.362)$$

Next

$$\begin{aligned} (X \setminus E) \setminus (X \setminus F) &= (X \setminus E) \cap (X \setminus F)^c \\ &= (X \setminus E) \cap (X \cap F^c)^c \\ &= (X \setminus E) \cap (X^c \cup F) \\ &= ((X \setminus E) \cap X^c) \cup ((X \setminus E) \cap F) \\ &\stackrel{X \setminus E \subseteq X}{=} ((X \setminus E) \cap F) \\ &= (X \cap E^c \cap F) \\ &= E^c \cap F \\ &= F \setminus E \end{aligned}$$

so that

$$\mu((X \setminus E) \setminus (X \setminus F)) = \mu(F \setminus E) = 0 \quad (21.363)$$

From [eqs: 21.361, 21.362 and 21.363] it follows that $X \setminus A \in \mathcal{A}_\mu$. In other words

$$\forall A \in \mathcal{A}_\mu \text{ we have } X \setminus A \in \mathcal{A}_\mu \quad (21.364)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_\mu$ then there exists $\{E_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$, $\{F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that $\forall i \in \mathbb{N} \ E_i \subseteq A_i \subseteq F_i$ and $\mu(F_i \setminus E_i) = 0$. Then we have

$$\bigcup_{i \in \mathbb{N}} E_i \subseteq \bigcup_{i \in \mathbb{N}} A_i \subseteq \bigcup_{i \in \mathbb{N}} F_i \text{ and, as } \mathcal{A} \text{ is a } \sigma\text{-algebra, } \bigcup_{i \in \mathbb{N}} E_i, \bigcup_{i \in \mathbb{N}} F_i \in \mathcal{A} \quad (21.365)$$

If $x \in (\bigcup_{i \in \mathbb{N}} F_i) \setminus (\bigcup_{i \in \mathbb{N}} E_i)$ then $x \in \bigcup_{i \in \mathbb{N}} F_i$ and $x \notin \bigcup_{i \in \mathbb{N}} E_i$, hence there exist a $i \in \mathbb{N}$ such that $x \in F_i$ and $\forall j \in \mathbb{N} \ x \notin E_j$ in particular $x \notin E_i$, so $x \in F_i \setminus E_i$. So $(\bigcup_{i \in \mathbb{N}} F_i) \setminus (\bigcup_{i \in \mathbb{N}} E_i) \subseteq \bigcup_{i \in \mathbb{N}} (F_i \setminus E_i)$ where as $\mathcal{A}$ is a σ -algebra $F_i \setminus E_i \in \mathcal{A}$, hence

$$\mu\left(\left(\bigcup_{i \in \mathbb{N}} F_i\right) \setminus \left(\bigcup_{i \in \mathbb{N}} E_i\right)\right) \leq \mu\left(\bigcup_{i \in \mathbb{N}} (F_i \setminus E_i)\right) \stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{i=1}^{\infty} \mu(F_i \setminus E_i) = 0 \quad (21.366)$$

From [eq: 21.365] and [eq: 21.366] it follows that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}_\mu$ hence

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_\mu \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}_\mu \quad (21.367)$$

From [eqs: 21.360, 21.364 and 21.367] it follows that

$$\mathcal{A}_\mu \text{ is a } \sigma\text{-algebra} \quad (21.368)$$

Now as $\emptyset \in \mathcal{A}$, $\emptyset \subseteq \emptyset \subseteq \emptyset$ and $\mu(\emptyset \setminus \emptyset) = 0$ it follows from [eq: 21.358] that

$$\bar{\mu}(\emptyset) = \mu(\emptyset) = 0 \quad (21.369)$$

If additional $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_\mu$ is pairwise disjoint then $\forall i \in \mathbb{N} \ E_i \subseteq A_i \subseteq F_i$ and $\mu(F_i \setminus E_i) = 0$. Let $i, j \in \mathbb{N}$ with $i \neq j$ then $E_i \cap E_j \subseteq A_i \cap A_j = \emptyset$ which proves that $\{E_i\}_{i \in \mathbb{N}}$ is pairwise disjoint. So

$$\begin{aligned} \bar{\mu}\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eq: 21.365, 21.366, 21.358}]}{=} \mu\left(\bigsqcup_{i \in \mathbb{N}} E_i\right) \\ &\stackrel{\mu \text{ is a measure}}{=} \sum_{i=1}^{\infty} \mu(E_i) \\ &\stackrel{[\text{eq: 21.358}]}{=} \sum_{i=1}^{\infty} \bar{\mu}(A_i) \end{aligned}$$

The above together with [eq: 21.369] proves that

$$\bar{\mu} \text{ is a measure so that } \langle X, \mathcal{A}_\mu, \bar{\mu} \rangle \text{ is a measure space} \quad (21.370)$$

Let $A \in \mathcal{A}$ then as $A \subseteq A \subseteq A$ and $\mu(A \setminus A) = \mu(\emptyset) = 0$ we have by [eq: 21.358] that $\bar{\mu}(A) = \mu(A)$, hence

$$(\bar{\mu})|_A = \mu \quad (21.371)$$

Let $A \in \mathcal{N}_{\bar{\mu}}$ then there exist a $B \in \mathcal{A}_\mu$ with $\bar{\mu}(B) = 0$ such that $A \subseteq B$. As $B \in \mathcal{A}_\mu$ there exists $E, F \in \mathcal{A}$ with $E \subseteq B \subseteq F$ and $\mu(F \setminus E) = 0$, hence by [eq: 21.358] $\mu(F) = \bar{\mu}(B) = 0$. Next as $\emptyset \subseteq A \subseteq B \subseteq F$ and $\mu(F \setminus \emptyset) = \mu(F) = 0$ it follows that $A \in \mathcal{A}_\mu$. This proves that $\mathcal{N}_{\bar{\mu}} \subseteq \mathcal{A}_\mu$ so that by [theorem: 21.216]

$$\langle X, \mathcal{A}_\mu, \bar{\mu} \rangle \text{ is complete} \quad \square$$

21.4 Measurable Functions

21.4.1 Measurable function definition

We define now the functions that can be integrated.

Definition 21.224. (Measurable Function) Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be two measurable spaces, $A \subseteq X$ then a function

$$f: A \rightarrow Y$$

is **$\mathcal{A}, \mathcal{B}$ -measurable** if $\forall B \in \mathcal{B}$ we have that $f^{-1}(B) \in \mathcal{A}|_A$ where $\mathcal{A}|_A = \{A \cap C \mid C \in \mathcal{A}\}$ is the trace algebra of $\mathcal{A}$ defined in [theorem: 21.44]. The set of $\mathcal{A}, \mathcal{B}$ -measurable function graphs between A and Y is noted as $\mathcal{M}[A, \mathcal{A}, Y, \mathcal{B}]$ hence

$$\begin{aligned} \mathcal{M}[X, \mathcal{A}, Y, \mathcal{B}] &= \{f \in Y^A \mid f: A \rightarrow Y \text{ is } \mathcal{A}, \mathcal{B}\text{-measurable}\} \\ &= \{f \in Y^A : \forall B \in \mathcal{B} \text{ we have } f^{-1}(B) \in \mathcal{A}|_A\} \end{aligned}$$

Note 21.225. A set can have many σ -algebras defined on it, so the same function can at the same time be measurable and not measurable, depending on the σ -algebras used.

Note 21.226. If $A = X$ then by [theorem: 21.44] $\mathcal{A}|_X = \mathcal{A}$ so that $f: X \rightarrow Y$ is $\mathcal{A}, \mathcal{B}$ -measurable iff $\forall B \in \mathcal{B}$ we have $f^{-1}(B) \in \mathcal{A}$.

Lemma 21.227. Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be two measurable spaces, $A \subseteq X$, $\langle A, \mathcal{A}|_A \rangle$ the measurable space defined in [theorem: 21.44] and $f: A \rightarrow Y$ is a function then we have

$$f \text{ is } \mathcal{A}, \mathcal{B}\text{-measurable} \Leftrightarrow f \text{ is } \mathcal{A}|_A, \mathcal{B}\text{-measurable}$$

so that

$$\mathcal{M}[X, \mathcal{A}, Y, \mathcal{B}] = \mathcal{M}[A, \mathcal{A}|_A, Y, \mathcal{B}]$$

Proof.

$\Rightarrow$. Let $B \in \mathcal{B}$ then as $f: A \rightarrow Y$ is $\mathcal{A}, \mathcal{B}$ -measurable we have by definition that $f^{-1}(B) \in \mathcal{A}|_A$ which by the same definition proves that $f: A \rightarrow Y$ is $\mathcal{A}|_A, \mathcal{B}$ -measurable.

$\Leftarrow$. Let $B \in \mathcal{B}$ then as $f: A \rightarrow Y$ is $\mathcal{A}|_A, \mathcal{B}$ -measurable we have by definition that $f^{-1}(B) \in \mathcal{A}|_A$ which by the same definition proves that $f: A \rightarrow Y$ is $\mathcal{A}, \mathcal{B}$ -measurable. $\square$

Example 21.228. Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be measurable spaces, $A \subseteq X$ then $C_y: A \rightarrow Y$ defined by $C_y(x) = y$ [the constant function] is $\mathcal{A}, \mathcal{B}$ -measurable. In other words $C_y \in \mathcal{M}[A, \mathcal{A}, Y, \mathcal{B}]$.

Proof. Let $B \in \mathcal{B}$ then we have $C_y^{-1}(B) = A \in \mathcal{A}|_A$ which proves that C_y is $\mathcal{A}, \mathcal{B}$ -measurable. $\square$

In the case of generated σ -algebras there is a easier way to check if a function is measurable.

Theorem 21.229. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$, Y a set, $\mathcal{B} \subseteq \mathcal{P}(Y)$ so that $\langle Y, \sigma[\mathcal{B}] \rangle$ is a measurable space and $f: A \rightarrow Y$ a function then*

$$f \text{ is } \mathcal{A}, \sigma[\mathcal{B}]\text{-measurable} \Leftrightarrow \forall B \in \mathcal{B} \text{ we have } f^{-1}(B) \in \mathcal{A}|_A$$

Proof.

$\Rightarrow$. Let $B \in \mathcal{B}$ then as $\mathcal{B} \subseteq \sigma[\mathcal{B}]$ we have that $B \in \sigma[\mathcal{B}]$. Hence, as f is $\mathcal{A}, \sigma[\mathcal{B}]$ -measurable, it follows that $f^{-1}(B) \in \mathcal{A}|_A$.

$\Leftarrow$. First we prove that $\mathcal{H} = \{B \in \mathcal{P}(Y) | f^{-1}(B) \in \mathcal{A}|_A\}$ is a σ -algebra containing $\mathcal{B}$. Let $B \in \mathcal{B}$ then by the hypothesis $f^{-1}(B) \in \mathcal{A}|_A$ so that $B \in \mathcal{H}$, hence

$$\mathcal{B} \subseteq \mathcal{H} \quad (21.372)$$

As $f^{-1}(Y) = A \in \mathcal{A}|_A$ we have also

$$Y \in \mathcal{H} \quad (21.373)$$

Let $B \in \mathcal{H}$ then $f^{-1}(B) \in \mathcal{A}|_A$ so that $f^{-1}(Y \setminus B) \stackrel{[\text{theorem: 2.102(4)}]}{=} A \setminus f^{-1}(B) \in \mathcal{A}|_A$. From this it follows that $Y \setminus B \in \mathcal{H}$. Hence

$$\forall B \in \mathcal{H} \text{ we have that } Y \setminus B \in \mathcal{H} \quad (21.374)$$

Let $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{H}$ then $\forall i \in \mathbb{N} \ f^{-1}(B_i) \in \mathcal{A}|_A$. As $f^{-1}(\bigcup_{i \in \mathbb{N}} B_i) \stackrel{[\text{theorem: 2.144}]}{=} \bigcup_{i \in \mathbb{N}} f^{-1}(B_i)$ and $\mathcal{A}|_A$ is a σ -algebra it follows that $f^{-1}(\bigcup_{i \in \mathbb{N}} B_i) \in \mathcal{A}|_A$ so that $\bigcup_{i \in \mathbb{N}} B_i \in \mathcal{H}$. Hence

$$\forall \{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{H} \text{ we have that } \bigcup_{i \in \mathbb{N}} B_i \in \mathcal{H} \quad (21.375)$$

From [eqs: 21.372, 21.373, 21.374 and 21.375] it follows that $\mathcal{H}$ is a σ -algebra containing $\mathcal{B}$ hence, using [theorem: 21.42], it follows that $\sigma[\mathcal{B}] \subseteq \mathcal{H}$. So $\forall B \in \sigma[\mathcal{B}]$ we have $B \in \mathcal{H}$ meaning that $f^{-1}(B) \in \mathcal{A}|_A$ which proves that f is $\mathcal{A}, \sigma[\mathcal{B}]$ -measurable. $\square$

Theorem 21.230. *Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle, \langle Z, \mathcal{C} \rangle$ be measurable spaces, $A \subseteq X, B \subseteq Y, f: A \rightarrow Y$ a $\mathcal{A}, \mathcal{B}$ -measurable function and $g: B \rightarrow Z$ a $\mathcal{B}, \mathcal{C}$ -measurable function such that $f(A) \subseteq B$ then*

$$g \circ f: X \rightarrow Z$$

is a $\mathcal{A}, \mathcal{C}$ measurable function.

Proof. Take $D \in \mathcal{C}$ then, as g is $\mathcal{B}, \mathcal{C}$ measurable, $g^{-1}(D) \in \mathcal{B}|_B$. Hence there exist a $C \in \mathcal{B}$ such that $g^{-1}(D) = B \cap C$. As f is $\mathcal{A}, \mathcal{B}$ -measurable and $C \in \mathcal{B}$ we have $f^{-1}(C) \in \mathcal{A}|_A$. Now

$$\begin{aligned} (g \circ f)^{-1}(D) &\stackrel{[\text{theorem: 2.92}]}{=} f^{-1}(g^{-1}(D)) \\ &= f^{-1}(B \cap C) \\ &\stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(B) \cap f^{-1}(C) \\ &\stackrel{[\text{theorem: 2.51}]}{=} A \cap f^{-1}(C) \\ &= f^{-1}(C) \end{aligned}$$

which proves that $(g \circ f)^{-1}(D) = f^{-1}(C) \in \mathcal{A}|_A$. Hence $g \circ f$ is $\mathcal{A}, \mathcal{C}$ -measurable. $\square$

The following theorem shows that a restriction of a measurable function is measurable.

Theorem 21.231. *Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be measurable spaces, $A, B \subseteq X$ with $A \subseteq B$ and $f: B \rightarrow Y$ a $\mathcal{A}, \mathcal{B}$ -measurable function then $f|_A: A \rightarrow Y$ is a $\mathcal{A}, \mathcal{B}$ -measurable function and $f|_A: A$ In other words if $f \in \mathcal{M}[B, \mathcal{A}, Y, \mathcal{B}]$ then*

$$f|_A \in \mathcal{M}[A, \mathcal{A}, Y, \mathcal{B}] \stackrel{[\text{lemma: 21.227}]}{=} \mathcal{M}[A, \mathcal{A}|_A, Y, \mathcal{B}]$$

Note 21.232. If $B = X$ then we have that if $f \in \mathcal{M}[X, \mathcal{A}, Y, \mathcal{B}]$ then

$$f|_A \in \mathcal{M}[A, \mathcal{A}, Y, \mathcal{B}] = \mathcal{M}[A, \mathcal{A}|_A, Y, \mathcal{B}]$$

Proof. Let $D \in \mathcal{B}$ then

$$(f|_A)^{-1}(D) \stackrel{[\text{theorem: 2.92}]}{=} A \cap f^{-1}(D)$$

As f is $\mathcal{A}, \mathcal{B}$ -measurable we have that $f^{-1}(D) \in \mathcal{A}|_B$ so that there exist a $E \in \mathcal{A}$ such that $f^{-1}(D) = B \cap E$. Hence $(f|_A)^{-1}(D) = A \cap B \cap E \stackrel{A \subseteq B}{=} A \cap E \in \mathcal{A}|_A$. Hence $f|_A$ is $\mathcal{A}, \mathcal{B}$ -measurable or

$$f|_A \in \mathcal{M}[A, \mathcal{A}, Y, \mathcal{B}] \quad \square$$

Next we show the measurability of a extension of a function

Definition 21.233. Let X, Y be sets, $A \subseteq X$, $y \in Y$ and $f: A \rightarrow Y$ a function then we define

$$f_{A \uparrow yX}: X \rightarrow Y \text{ by } f_{A \uparrow yX}(x) = \begin{cases} f(x) & \text{if } x \in A \\ y & \text{if } x \in X \setminus A \end{cases}$$

Note 21.234. Let $f: X \rightarrow X$ and $x \in X$ then $f_{X \uparrow yX}(x) = \begin{cases} f(x) & \text{if } x \in X \\ y & \text{if } x \in X \setminus X = \emptyset \end{cases} = f(x)$ so that $f_{X \uparrow yX} = f$

Note 21.235. Let $x \in A$ and then $(f_{A \uparrow yX})|_A(x) \stackrel{x \in A}{=} f_{A \uparrow yX}(x) \stackrel{x \in A}{=} f(x)$ so that $(f_{A \uparrow yX})|_A = f$

Theorem 21.236. Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be measurable spaces, $A \in \mathcal{A}$, $y \in Y$ and $f: A \rightarrow Y$ a function then

$$f \text{ is } \mathcal{A}, \mathcal{B}\text{-measurable} \Leftrightarrow f_{A \uparrow yX} \text{ is } \mathcal{A}, \mathcal{B}\text{-measurable}$$

In other words we have

$$f \in \mathcal{M}[A, \mathcal{A}, Y, \mathcal{B}] \Leftrightarrow f_{A \uparrow yX} \in \mathcal{M}[X, \mathcal{A}, Y, \mathcal{B}]$$

Proof.

$\Rightarrow$. Let $B \in \mathcal{B}$ then we have either:

$y \in B$. Let $t \in (f_{A \uparrow yX})^{-1}(B)$ then we have either:

$t \in A$. Then $f(t) = (f_{A \uparrow yX})(t) \in B$ so that $t \in f^{-1}(B) \subseteq f^{-1}(B) \cup (X \setminus A)$

$t \in X \setminus A$. Then clearly $t \in X \setminus A \subseteq f^{-1}(B) \cup (X \setminus A)$

hence we have

$$(f_{A \uparrow yX})^{-1}(B) \subseteq f^{-1}(B) \cup (X \setminus A) \quad (21.376)$$

If $t \in f^{-1}(B) \cup (X \setminus A)$ then for t we have either:

$t \in A$. Then $t \in f^{-1}(B)$ so that $(f_{A \uparrow yX})(t) = f(t) \in B$ which proves that $t \in (f_{A \uparrow yX})^{-1}(B)$.

$t \in X \setminus A$. Then $f_{A \uparrow yX}(t) = y \in B$ which proves that $t \in (f_{A \uparrow yX})^{-1}(B)$

hence we have $f^{-1}(B) \cup (X \setminus A) \subseteq (f_{A \uparrow yX})^{-1}(B)$. Combining this with [eq: 21.376] proves that

$$(f_{A \uparrow yX})^{-1}(B) = f^{-1}(B) \cup (X \setminus A) \quad (21.377)$$

As f is $\mathcal{A}, \mathcal{B}$ measurable $f^{-1}(B) \in \mathcal{A}|_A$ so that $\exists E \in \mathcal{A}$ such that $f^{-1}(B) = A \cap E$. Hence

$$\begin{aligned} f^{-1}(B) \cup (X \setminus A) &= (A \cap E) \cup (X \setminus A) \\ &= ((A \cap (X \setminus A))) \cup E \cap (X \setminus A) \\ &= \emptyset \cup (E \cap (X \setminus A)) \\ &= E \cap (X \setminus A) \end{aligned}$$

As $E, A \in \mathcal{A}$ we have that $E \cap (X \setminus A) \in \mathcal{A}$ so that

$$f^{-1}(B) \cup (X \setminus A) \in \mathcal{A}$$

This together with [eq: 21.377] proves that

$$(f_{A \upharpoonright Y_X})^{-1}(B) \in \mathcal{A} = \mathcal{A}_{|X}$$

$y \in Y \setminus B$. Let $t \in (f_{A \upharpoonright Y_X})^{-1}(B)$. If $t \in X \setminus A$ then $f_{A \upharpoonright Y_X}(t) = y \in B$ contradicting $y \in Y \setminus B$, hence we must have that $t \in A$, further we have $f(t) \stackrel{t \in A}{=} f_{A \upharpoonright Y_X}(t) \in B$ so that $t \in f^{-1}(B)$. Hence $t \in A \cap f^{-1}(B)$ which proves that

$$(f_{A \upharpoonright Y_X})^{-1}(B) \subseteq A \cap f^{-1}(B) \quad (21.378)$$

Let $t \in A \cap f^{-1}(B)$ then $t \in A$ and $f(t) \in B$ so that $f_{A \upharpoonright Y_X}(t) = f(t) \in B$ which proves that $t \in (f_{A \upharpoonright Y_X})^{-1}(B)$. Hence $A \cap f^{-1}(B) \subseteq (f_{A \upharpoonright Y_X})^{-1}(B)$ which combined with [eq: 21.378] proves that

$$(f_{A \upharpoonright Y_X})^{-1}(B) = A \cap f^{-1}(B)$$

As f is $\mathcal{A}, \mathcal{B}$ -measurable $f^{-1}(B) \in \mathcal{A}_{|A}$ so there exist a $E \in \mathcal{A}$ such that $f^{-1}(B) = A \cap E$. Hence

$$(f_{A \upharpoonright Y_X})^{-1}(B) = A \cap (A \cap E) = A \cap E$$

As $A, E \in \mathcal{A}$ it follows that $A \cap E \in \mathcal{A}$ hence we conclude that

$$(f_{A \upharpoonright Y_X})^{-1}(B) \in \mathcal{A} = \mathcal{A}_{|X}$$

Hence we have proved that $\forall B \in \mathcal{B} \ (f_{A \upharpoonright Y_X})^{-1}(B) \in \mathcal{A}_{|X}$ which proves that

$$f_{A \upharpoonright Y_X} \text{ is } \mathcal{A}, \mathcal{B}\text{-measurable}$$

$\Leftarrow$. Assume that $f_{A \upharpoonright Y_X}$ is $\mathcal{A}, \mathcal{B}$ -measurable then, as $(f_{A \upharpoonright Y})_{|A} = f$ [see 21.235], it follows from [theorem: 21.231] that f is $\mathcal{A}, \mathcal{B}$ -measurable. $\square$

We have the following more specialized cases of measurable functions.

Definition 21.237. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be two topological spaces then a $\mathcal{B}[X, \mathcal{T}_X], \mathcal{B}[Y, \mathcal{T}_Y]$ -measurable $f: X \rightarrow Y$ function is called a **Borel measurable** function.

Example 21.238. A $\mathcal{B}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable function $f: A \rightarrow \overline{\mathbb{R}}$ is a Borel measurable function.

Example 21.239. A $\mathcal{B}[\mathbb{R}^n], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable function $f: A \rightarrow \overline{\mathbb{R}}$ is a Borel measurable function.

For integration we are only interested in measurable functions to $\overline{\mathbb{R}}$ or $\mathbb{R}$ so we use some simplified notations for these functions.

Definition 21.240. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$ then a

$$\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable function } f: A \rightarrow \overline{\mathbb{R}}$$

is called a **$\mathcal{A}$ -measurable** function. Further we define

$$\mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] = \{f \in \overline{\mathbb{R}}^A \mid f: A \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{A}\text{-measurable}\} = \{f \in \overline{\mathbb{R}}^A \mid f: A \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{A}, \mathcal{B}\text{-measurable}\}$$

$$\mathcal{M}[A, \mathcal{A}, \mathbb{R}] = \{f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \mid f(A) \subseteq \mathbb{R}\} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

For non negative measurable functions we have the following definitions

Definition 21.241. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$ then we define

$$\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}] = \{f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \mid f(A) \subseteq [0, \infty]\}$$

$$\mathcal{M}^+[A, \mathcal{A}, \mathbb{R}] = \{f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \mid f(A) \subseteq [0, \infty[\subseteq \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]\}$$

where $\mathcal{M}[A, \mathcal{A}, \mathbb{R}]$, $\mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ is defined in [definition: 21.240]

Note 21.242. As $[0, \infty[\subseteq \mathbb{R} \subseteq \overline{\mathbb{R}}$ and $[0, \infty[\subseteq [0, \infty] \subseteq \overline{\mathbb{R}}$ we have also

$$\mathcal{M}^+[A, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}] \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Theorem 21.243. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$ then we have:

1. If $f: A \rightarrow \mathbb{R}$ is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ -measurable then $f: A \rightarrow \overline{\mathbb{R}}$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable [in other words $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$].
2. If $f: A \rightarrow \overline{\mathbb{R}}$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable and $f(X) \subseteq \mathbb{R}$ then $f: X \rightarrow \mathbb{R}$ is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ -measurable. In other words if $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ then $f: X \rightarrow \mathbb{R}$ is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ -measurable.

Proof.

1. Let $B \in \mathcal{B}[\overline{\mathbb{R}}]$ then by [theorem: 21.61] $B \cap \mathbb{R} \in \mathcal{B}[\mathbb{R}]$. As f is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ -measurable it follows that $f^{-1}(B \cap \mathbb{R}) \in \mathcal{A}|_A$. Now

$$f^{-1}(B \cap \mathbb{R}) \stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(B) \cap f^{-1}(\mathbb{R}) = f^{-1}(B) \cap \mathbb{R} = f^{-1}(B)$$

so that $f^{-1}(B) \in \mathcal{A}|_A$. Hence f is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable.

2. As $f(X) \subseteq \mathbb{R}$ it follows that $f: A \rightarrow \mathbb{R}$ is a well defined function. Let $B \in \mathcal{B}[\mathbb{R}]$ then $B \subseteq \mathbb{R}$ so that $B \cap \mathbb{R} = B \in \mathcal{B}[\mathbb{R}]$ so that by [theorem: 21.61] $B \in \mathcal{B}[\overline{\mathbb{R}}]$. As f is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable it follows that $f^{-1}(B) \in \mathcal{A}|_A$ which proves that $f: A \rightarrow \mathbb{R}$ is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ -measurable. $\square$

Next we define Lebesgue measurable functions.

Definition 21.244. A $\mathcal{L}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable function $f: \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is called a Lebesgue measurable function and the set of $\mathcal{L}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable function is noted as $\mathcal{L}$. So

$$\mathcal{L} = \{f \in \overline{\mathbb{R}}^{\mathbb{R}} | f: \mathbb{R} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{L}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}\}$$

See [theorems: 21.182, 21.205] for the definitions of $\mathcal{L}$, $\mathcal{L}^n$.

Definition 21.245. Let $n \in \mathbb{N}$ then a $\mathcal{L}[\mathbb{R}^n], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable function $f: \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is called a Lebesgue measurable function and the set of $\mathcal{L}[\mathbb{R}^n], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable function is noted as $\mathcal{L}^n$. So

$$\mathcal{L}^n = \{f \in \overline{\mathbb{R}}^{\mathbb{R}^n} | f: \mathbb{R}^n \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{L}[\mathbb{R}^n], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}\}$$

Note that although the composition of measurable functions is measurable [see theorem: 21.230] this is not the case for Lebesgue measurable functions. For example if $f: \mathbb{R} \rightarrow \overline{\mathbb{R}}$ and $g: \mathbb{R} \rightarrow \overline{\mathbb{R}}$ are Lebesgue measurable functions such that $f(\mathbb{R}) \subseteq \mathbb{R}$ then by [theorem: 21.243] $f: \mathbb{R} \rightarrow \mathbb{R}$ is $\mathcal{L}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable and g is $\mathcal{L}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable but not $\mathcal{B}[\mathbb{R}], \mathcal{B}[\mathbb{R}]$ -measurable, so we can not apply the composition theorem. If however g is $\mathcal{B}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable then $g \circ f$ is by the previous theorem $\mathcal{L}[\mathbb{R}], \mathcal{B}[\mathbb{R}]$ -measurable hence Lebesgue measurable.

We show now how we can extend a measurable extended real valued function to a measurable function.

Definition 21.246. Let X be a set, $A \subseteq X$ and $f: A \rightarrow \overline{\mathbb{R}}$ a function then we define

$$f_{A \uparrow X}: X \rightarrow \overline{\mathbb{R}} \text{ by } f_{A \uparrow X} = f_{A \uparrow^0 X} \text{ [see definition: 21.233]}$$

in other words $f_{A \uparrow X}$ is defined by

$$\forall x \in X \text{ we have } f_{A \uparrow X}(x) = \begin{cases} f(x) & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases}$$

Note 21.247. As $\forall x \in A$ we have $(f_{A \upharpoonright X})|_A(x) = f_{A \upharpoonright X}(x) \stackrel{=}{=} f(x)$ it follows that $(f_{A \upharpoonright X})|_A = f$

Lemma 21.248. Let X be a set, $A \subseteq X$, $B \subseteq A$ and $f: A \rightarrow \overline{\mathbb{R}}$ a function then we have

$$\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X} = (\mathcal{X}_{A,B} \cdot f)_{A \upharpoonright X}$$

Here the characteristics function $\mathcal{X}_{X,B}: X \rightarrow \{0, 1\}$ is defined by $\mathcal{X}_{X,B}(x) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \in X \setminus B \end{cases}$ [see example: 2.48].

Proof. Let $x \in X$ then we have either:

$x \in B$. Then, as $B \subseteq A$ we have that $x \in A$. Hence

$$\begin{aligned} (\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X})(x) &= \mathcal{X}_{X,B}(x) \cdot f_{A \upharpoonright X}(x) \\ &\stackrel{=}{=} \mathcal{X}_{X,B}(x) \cdot f(x) \\ &= (\mathcal{X}_{X,B} \cdot f)(x) \\ &\stackrel{=}{=} (\mathcal{X}_{A,B} \cdot f)_{A \upharpoonright X}(x) \end{aligned}$$

which proves that

$$\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X} = (\mathcal{X}_{A,B} \cdot f)_{A \upharpoonright X}$$

$x \in X \setminus B$. Then we have either

$x \in A$. Then

$$\begin{aligned} (\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X})(x) &= \mathcal{X}_{X,B}(x) \cdot f_{A \upharpoonright X}(x) \\ &\stackrel{=}{=} \mathcal{X}_{X,B}(x) \cdot f(x) \\ &= (\mathcal{X}_{X,B} \cdot f)(x) \\ &\stackrel{=}{=} (\mathcal{X}_{X,B} \cdot f)_{A \upharpoonright X}(x) \end{aligned}$$

so that

$$\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X} = (\mathcal{X}_{A,B} \cdot f)_{A \upharpoonright X}$$

$x \in X \setminus A$. Then

$$\begin{aligned} (\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X})(x) &= \mathcal{X}_{X,B}(x) \cdot f_{A \upharpoonright X}(x) \\ &\stackrel{=}{=} \mathcal{X}_{X,B}(x) \cdot 0 \\ &= 0 \\ &\stackrel{=}{=} (\mathcal{X}_{X,B} \cdot f)_{A \upharpoonright X}(x) \end{aligned}$$

so that

$$\mathcal{X}_{X,B} \cdot f_{A \upharpoonright X} = (\mathcal{X}_{A,B} \cdot f)_{A \upharpoonright X} \quad \square$$

Theorem 21.249. Let X be a set, $A \subseteq X$ and $f: A \rightarrow \overline{\mathbb{R}}$ a function then

$$\mathcal{X}_{X,A} \cdot f_{A \upharpoonright X} = f_{A \upharpoonright X}$$

Proof. Let $x \in X$ then we have either:

$x \in A$. Then $(\mathcal{X}_{X,A} \cdot f_{A \upharpoonright X})(x) = \mathcal{X}_{X,A}(x) \cdot f_{A \upharpoonright X}(x) = 1 \cdot f(x) = f(x) = f_{A \upharpoonright X}(x)$.

$x \in X \setminus A$. Then $(\mathcal{X}_{X,A} \cdot f_{A \upharpoonright X})(x) = \mathcal{X}_{X,A}(x) \cdot f_{A \upharpoonright X}(x) = 0 \cdot 0 = 0 = f_{A \upharpoonright X}(x)$.

proving that in all cases $(\mathcal{X}_{X,A} \cdot f_{A \upharpoonright X})(x) = f_{A \upharpoonright X}(x)$ from which it follows that

$$\mathcal{X}_{X,A} \cdot f_{A \upharpoonright X} = f_{A \upharpoonright X} \quad \square$$

Theorem 21.250. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$ and $f: A \rightarrow \overline{\mathbb{R}}$ a function then we have

$$f_{A \upharpoonright X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \Leftrightarrow f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Note 21.251. Consider the measurable space $\langle A, \mathcal{A}|_A \rangle$ [see theorem: 21.44] then we have

$$\begin{aligned}\mathcal{M}[A, \mathcal{A}|_A] &= \{f \in \overline{\mathbb{R}}^A \mid f \text{ is } \mathcal{A}|_A\text{-measurable}\} \\ &= \{f \in \overline{\mathbb{R}}^A \mid \forall B \in \mathcal{B}[\overline{\mathbb{R}}] \ f^{-1}(B) \in \mathcal{A}|_A\} \\ &= \{f \in \overline{\mathbb{R}}^A \mid f \text{ is } \mathcal{A}\text{-measurable}\} \\ &= \mathcal{M}[A, \mathcal{A}]\end{aligned}$$

so that we also have

$$f_{A \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \Leftrightarrow f \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$$

Proof.

$\Rightarrow$. If $f_{A \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that $f_{A \uparrow X} \stackrel{\text{def}}{=} f_{A \uparrow^0 X}$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable so that by [theorem: 21.236] f is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable, hence $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

$\Leftarrow$. If $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then f is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable so that by [theorem: 21.236] $f_{A \uparrow X} = f_{A \uparrow^0 X}$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable, hence $f_{A \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$. $\square$

Most books on measure theory give a alternative definition for $\mathcal{A}$ -measurable functions based on the following definitions.

Definition 21.252. Let X be a set, $A \subseteq X$ then we define

1. If $f: A \rightarrow \overline{\mathbb{R}}, g: A \rightarrow \overline{\mathbb{R}}$ then given $x \in \overline{\mathbb{R}}$

$$\begin{aligned}\{f < x\} &\stackrel{\text{def}}{=} f^{-1}([-\infty, x]) = \{t \in A \mid f(t) < x\} \subseteq A \\ \{f \leq x\} &\stackrel{\text{def}}{=} f^{-1}([-\infty, x]) = \{t \in A \mid f(t) \leq x\} \subseteq A \\ \{f > x\} &\stackrel{\text{def}}{=} f^{-1}([x, \infty]) = \{t \in A \mid x < f(t)\} \subseteq A \\ \{f \geq x\} &\stackrel{\text{def}}{=} f^{-1}([x, \infty]) = \{t \in A \mid x \leq f(t)\} \subseteq A \\ \{f = x\} &\stackrel{\text{def}}{=} f^{-1}(\{x\}) = \{t \in A \mid x = f(t)\} \subseteq A \\ \{f \leq g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) \leq g(x)\} \subseteq A \\ \{f < g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) < g(x)\} \subseteq A \\ \{f \geq g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) \geq g(x)\} \subseteq A \\ \{f > g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) > g(x)\} \subseteq A \\ \{f = g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) = g(x)\} \subseteq A\end{aligned}$$

2. If $f: A \rightarrow \mathbb{R}, g: A \rightarrow \mathbb{R}$ then given $x \in \mathbb{R}$ we define

$$\begin{aligned}\{f < x\} &\stackrel{\text{def}}{=} f^{-1}([-\infty, x]) = \{t \in A \mid f(t) < x\} \subseteq A \\ \{f \leq x\} &\stackrel{\text{def}}{=} f^{-1}([-\infty, x]) = \{t \in A \mid f(t) \leq x\} \subseteq A \\ \{f > x\} &\stackrel{\text{def}}{=} f^{-1}([x, \infty]) = \{t \in A \mid x < f(t)\} \subseteq A \\ \{f \geq x\} &\stackrel{\text{def}}{=} f^{-1}([x, \infty]) = \{t \in A \mid x \leq f(t)\} \subseteq A \\ \{f = x\} &\stackrel{\text{def}}{=} f^{-1}(\{x\}) = \{t \in A \mid x = f(t)\} \subseteq A \\ \{f \leq g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) \leq g(x)\} \subseteq A \\ \{f < g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) < g(x)\} \subseteq A \\ \{f \geq g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) \geq g(x)\} \subseteq A \\ \{f > g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) > g(x)\} \subseteq A \\ \{f = g\} &\stackrel{\text{def}}{=} \{x \in A \mid f(x) = g(x)\} \subseteq A\end{aligned}$$

Lemma 21.253. Let X be a set, $A \subseteq X$, $f: A \rightarrow \overline{\mathbb{R}}$ and $x \in \mathbb{R}$ then

1. $\{f \geq x\} = \bigcap_{n \in \mathbb{N}} \{f > x - \frac{1}{n}\}$
2. $A \setminus \{f \geq x\} = \{f < x\}$

3. $\{f \leq x\} = \bigcap_{n \in \mathbb{N}} \left\{f < x + \frac{1}{n}\right\}$
 4. $A \setminus \{f \leq x\} = \{f > x\}$

Proof.

1. Let $n \in \mathbb{N}$ then if $t \in \{f \geq x\}$ we have that $x \leq f(t) \xRightarrow{x \in \mathbb{R}} x - \frac{1}{n} < f(t) \Rightarrow t \in \left\{x - \frac{1}{n} < f\right\}$ proving

$$\{f \geq x\} \subseteq \bigcap_{n \in \mathbb{N}} \left\{f > x - \frac{1}{n}\right\} \quad (21.379)$$

If $t \in \bigcap_{n \in \mathbb{N}} \left\{f > x - \frac{1}{n}\right\}$ then $\forall n \in \mathbb{N}$ we have $x - \frac{1}{n} < f(t)$. For $f(t)$ we have as $x - \frac{1}{n} \in \mathbb{R}$ either:

$f(t) = \infty$. Then $x \leq f(t)$ so that $t \in \{f \geq x\}$.

$f(t) \in \mathbb{R}$. Then $\forall n \in \mathbb{N}$ $x < f(t) + \frac{1}{n}$ so that by [theorem: 10.31] $x \leq f(t)$ or $t \in \{f \geq x\}$.

Hence $\bigcap_{n \in \mathbb{N}} \left\{f > x - \frac{1}{n}\right\} \subseteq \{f \geq x\}$, combining this with [eq: 21.379] proves

$$\{f \geq x\} = \bigcap_{n \in \mathbb{N}} \left\{f > x - \frac{1}{n}\right\}$$

2.

$$\begin{aligned} A \setminus \{f \geq x\} &= A \setminus f^{-1}([x, \infty]) \\ &\stackrel{[\text{theorem: 2.102(4)}]}{=} f^{-1}(\overline{\mathbb{R}} \setminus [x, \infty]) \\ &= f^{-1}([-\infty, x]) \\ &= \{f < x\} \end{aligned}$$

3. Let $n \in \mathbb{N}$ then if $t \in \{f \leq x\}$ we have that $f(t) \leq x \xRightarrow{x \in \mathbb{R}} f(t) < x + \frac{1}{n}$ or $t \in \left\{f < x + \frac{1}{n}\right\}$. So

$$\{f \leq x\} \subseteq \bigcap_{n \in \mathbb{N}} \left\{f < x + \frac{1}{n}\right\} \quad (21.380)$$

Let $t \in \bigcap_{n \in \mathbb{N}} \left\{f < x + \frac{1}{n}\right\}$ then $\forall n \in \mathbb{N}$ we have $f(t) < x + \frac{1}{n}$. As $x + \frac{1}{n} \in \mathbb{R}$ we have for $f(t)$ either:

$f(t) = -\infty$. Then $f(t) \leq x$ so that $t \in \{f \leq x\}$.

$f(t) \in \mathbb{R}$. Then $\forall n \in \mathbb{N}$ $f(t) < x + \frac{1}{n}$ so that by [theorem: 10.31] $f(t) \leq x$ so that $t \in \{f \leq x\}$.

Hence $\bigcap_{n \in \mathbb{N}} \left\{f < x + \frac{1}{n}\right\} \subseteq \{f \leq x\}$, combining this with [eq: 21.380] proves that

$$\{f \leq x\} = \bigcap_{n \in \mathbb{N}} \left\{f < x + \frac{1}{n}\right\}$$

4.

$$\begin{aligned} A \setminus \{f \leq x\} &= A \setminus f^{-1}([-\infty, x]) \\ &\stackrel{[\text{theorem: 2.102(4)}]}{=} f^{-1}(\overline{\mathbb{R}} \setminus [-\infty, x]) \\ &= f^{-1}(]x, \infty]) \\ &= \{x < f\} \end{aligned}$$

□

We are now ready to state the following equivalences for $\mathcal{A}$ -measurability [which is used in most books about measure theory].

Theorem 21.254. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$ and $f: A \rightarrow \overline{\mathbb{R}}$ a function then we have the following equivalences:*

1. f is $\mathcal{A}$ -measurable [or equivalently $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$]

2. $\forall x \in \mathbb{R}$ we have $\{f > x\} \in \mathcal{A}|_A$
3. $\forall x \in \mathbb{R}$ we have $\{f \geq x\} \in \mathcal{A}|_A$
4. $\forall x \in \mathbb{R}$ we have $\{f < x\} \in \mathcal{A}|_A$
5. $\forall x \in \mathbb{R}$ we have $\{f \leq x\} \in \mathcal{A}|_A$

Proof.

1 $\Rightarrow$ 2. Take $x \in \mathbb{R}$ then $]x, \infty] \in \sigma[\{]a, \infty[| a \in \mathbb{R}\}] \stackrel{[\text{theorem: 21.61}]}{=} \mathcal{B}[\overline{\mathbb{R}}]$ and we have, as f is $\mathcal{A}$ -measurable, that $\{f > x\} = f^{-1}(]x, \infty]) \in \mathcal{A}|_A$.

2 $\Rightarrow$ 1. Let $B \in \{]a, \infty[| a \in \mathbb{R}\}$ then $B =]x, \infty]$ where $x \in \mathbb{R}$, hence by (2)

$$f^{-1}(B) = f^{-1}(]x, \infty]) = \{f > x\} \in \mathcal{A}|_A.$$

So using [theorem: 21.229] together with $\sigma[\{]a, \infty[| a \in \mathbb{R}\}] \stackrel{[\text{theorem: 21.61}]}{=} \mathcal{B}[\overline{\mathbb{R}}]$ it follows that f is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable or in other words $\mathcal{A}$ -measurable.

2 $\Rightarrow$ 3. Let $x \in \mathbb{R}$ then using (2) we have $\forall n \in \mathbb{N}$ that $\{f > x - \frac{1}{n}\} \in \mathcal{A}|_A$, hence by [theorem: 21.36] it follows that $\bigcap_{n \in \mathbb{N}} \{f > x - \frac{1}{n}\} \in \mathcal{A}|_A$. Using [lemma: 21.253] $\{f \geq x\} = \bigcap_{n \in \mathbb{N}} \{f > x - \frac{1}{n}\}$ proving that $\{f \geq x\} \in \mathcal{A}|_A$.

3 $\Rightarrow$ 4. Let $x \in \mathbb{R}$ then $\{f < x\} \stackrel{[\text{lemma: 21.253}]}{=} A \setminus \{f \geq x\} \in \mathcal{A}|_A$ [as $\{f \geq x\} \in \mathcal{A}|_A$ by (3)].

4 $\Rightarrow$ 5. Let $x \in \mathbb{R}$ then by (4) we have $\forall n \in \mathbb{N}$ that $\{f < x + \frac{1}{n}\} \in \mathcal{A}|_A$, hence by [theorem: 21.36] it follows that $\bigcap_{n \in \mathbb{N}} \{f < x + \frac{1}{n}\} \in \mathcal{A}|_A$. Using [lemma: 21.253] $\{f \leq x\} = \bigcap_{n \in \mathbb{N}_0} \{f < x + \frac{1}{n}\}$ proving that $\{f \leq x\} \in \mathcal{A}|_A$.

5 $\Rightarrow$ 2. Let $x \in \mathbb{R}$ then $\{f > x\} \stackrel{[\text{lemma: 21.253}]}{=} A \setminus \{f \leq x\} \stackrel{(5)}{\in} \mathcal{A}|_A$. □

21.4.2 Measurable function properties

Theorem 21.255. Let $\langle X, \mathcal{T}_X \rangle, \langle Y, \mathcal{T}_Y \rangle$ be topological spaces and $f: X \rightarrow Y$ a continuous function then f is $\mathcal{B}[X, \mathcal{T}_X], \mathcal{B}[Y, \mathcal{T}_Y]$ -measurable. In other words continuous functions are Borel measurable.

Proof. If $U \in \mathcal{T}_Y$ then as f is continuous we have $f^{-1}(U) \in \mathcal{T}_X \subseteq \mathcal{B}[X, \mathcal{T}_X]$ so that by [theorem: 21.229] we have that f is $\mathcal{B}[X, \mathcal{T}_X], \mathcal{B}[Y, \mathcal{T}_Y]$ -measurable. □

Theorem 21.256. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$ and let $f: A \rightarrow \overline{\mathbb{R}}$ and $g: A \rightarrow \overline{\mathbb{R}}$ be $\mathcal{A}$ -measurable functions then we have:

1. $\{f < g\} \in \mathcal{A}|_A$
2. $\{f \leq g\} \in \mathcal{A}|_A$
3. $\{f > g\} \in \mathcal{A}|_A$
4. $\{f \geq g\} \in \mathcal{A}|_A$
5. $\{f = g\} \in \mathcal{A}|_A$
6. $\{f \neq g\} \in \mathcal{A}|_A$
7. If $x \in \overline{\mathbb{R}}$ then $\{f = x\} \in \mathcal{A}|_A$

Proof.

1. Let $t \in \{f < g\}$ then $f(t) < g(t)$. Using the density of the rational numbers in the extended real numbers [see theorem: 20.12] there exist a $q \in \mathbb{Q}$ such that $f(t) < q < g(t)$, so that $t \in \{f < q\}$ and $t \in \{g > q\}$, proving $t \in \{f < q\} \cap \{g > q\}$. Hence we have

$$\{f < g\} \subseteq \bigcup_{q \in \mathbb{Q}} (\{f < q\} \cap \{g > q\}) \quad (21.381)$$

If $t \in \bigcup_{q \in \mathbb{Q}} (\{f < q\} \cap \{g > q\})$ then there exist a $q \in \mathbb{Q}$ such that $t \in \{f < q\} \cap \{g > q\}$, hence $f(t) < q \wedge q < g(t) \Rightarrow f(t) < g(t)$ proving that $t \in \{f < g\}$. Hence $\bigcup_{q \in \mathbb{Q}} (\{f < q\} \cap \{g > q\}) \subseteq \{f < g\}$ which combined with [eq: 21.381] proves that

$$\{f < g\} = \bigcup_{q \in \mathbb{Q}} (\{f < q\} \cap \{g > q\})$$

Using [theorem: 21.254] $\forall q \in \mathbb{Q}$ $\{f < q\}, \{g > q\} \in \mathcal{A}|_A$ so that $\{f < q\} \cap \{g > q\} \in \mathcal{A}|_A$. As $\mathbb{Q}$ is denumerable [see theorem: 10.5] it follows from [theorem: 21.39] that $\bigcup_{q \in \mathbb{Q}} (\{f < q\} \cap \{g > q\}) \in \mathcal{A}|_A$ so that

$$\{f < g\} \in \mathcal{A}$$

2. We have

$$\begin{aligned} t \in (A \setminus \{g < f\}) &\Leftrightarrow t \in A \wedge \neg(g(t) < f(t)) \\ &\Leftrightarrow t \in A \wedge f(t) \leq g(t) \\ &\Leftrightarrow t \in \{f \leq g\} \end{aligned}$$

proving that $A \setminus \{g < f\} = \{f \leq g\}$. Using (1) we have $\{g < f\} \in \mathcal{A}|_A$ hence $A \setminus \{g < f\} \in \mathcal{A}|_A$ which proves that

$$\{f \leq g\} \in \mathcal{A}|_A$$

3. As $\{f > g\} = \{x \in A | f(x) > g(x)\} = \{x \in A | g(x) < f(x)\} = \{g < f\}$ this follows from (1).

4. As $\{f \geq g\} = \{x \in A | f(x) \geq g(x)\} = \{x \in A | g(x) \leq f(x)\} = \{g \leq f\}$ this follows from (2)

5. We have

$$\begin{aligned} t \in (\{f \leq g\} \setminus \{f < g\}) &\Leftrightarrow t \in \{f \leq g\} \wedge t \notin \{f < g\} \\ &\Leftrightarrow t \in A \wedge f(t) \leq g(t) \wedge \neg(f(t) < g(t)) \\ &\Leftrightarrow t \in A \wedge f(t) \leq g(t) \wedge g(t) \leq f(t) \\ &\Leftrightarrow t \in A \wedge f(t) = g(t) \\ &\Leftrightarrow t \in \{f = g\} \end{aligned}$$

proving that $\{f = g\} = \{f \leq g\} \setminus \{f > g\} \in \mathcal{A}|_A$ [using (1), (2) and [theorems: 21.32, 21.36]

6. We have

$$\begin{aligned} t \in A \setminus \{f = g\} &\Leftrightarrow t \in A \wedge \neg(f(t) = g(t)) \\ &\Leftrightarrow t \in A \wedge f(t) \neq g(t) \\ &\Leftrightarrow t \in \{f \neq g\} \end{aligned}$$

proving that $\{f \neq g\} = A \setminus \{f = g\} \in \mathcal{A}|_A$ [using (3)]

7. Let $x \in \overline{\mathbb{R}}$ then as $\{x\}$ is closed [see theorem: 20.52], we have by 21.47 that $\{x\} \in \mathcal{B}[\overline{\mathbb{R}}]$ which as f is $\mathcal{A}$ -measurable proves that $\{f = x\} = f^{-1}(\{x\}) \in \mathcal{A}|_A$. $\square$

If the sum of two measurable functions is well defined then the sum is also a measurable function.

Theorem 21.257. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A, B \subseteq X$ and $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}], g \in \mathcal{M}[B, \mathcal{A}, \overline{\mathbb{R}}]$ then for

$$f + g: D \rightarrow \overline{\mathbb{R}} \text{ defined by } (f + g)(x) = f(x) + g(x)$$

where

$$D = \{x \in A \cap B | (f(x), g(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}\}$$

we have that

$$f + g \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}]$$

Note 21.258. D is chosen such that $\forall x \in D$ $f(x) + g(x)$ is well defined.

Note 21.259. If $A = B$ and $f, g \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}]$ then $\forall x \in A \cap B = A$ we have $(f(x), g(x)) \in \mathbb{R} \times \mathbb{R} \Rightarrow (f(x), g(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ so that $D = A$ from which it follows that $f + g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$.

Note 21.260. If $A = B$ and $f, g \in \mathcal{M}^+[D, \mathcal{A}, \overline{\mathbb{R}}]$ then $\forall x \in A \cap B = A$ we have $f(x), g(x) \in [0, \infty]$ so that $(f(x), g(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$. Hence $D = A$ so that $f + g \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$.

Proof. If $t \in \{f + g < x\}$ then $t \in D \subseteq A \cap B$ and $f|_D(t) + g|_D(t) = f(t) + g(t) < x$ so that by [theorem: 20.31] $f|_D(t) < x - g|_D(t)$. Using the density of $\mathbb{Q}$ in $\overline{\mathbb{R}}$ [see theorem: 20.12] there exists a $q \in \mathbb{Q}$ such that $f|_D(t) < q < x - g|_D(t)$, hence $f|_D(t) < q$ and by [theorem: 20.33] $g|_D(t) < x - q$, which proves that $t \in \{f|_D < q\} \cap \{g|_D < x - q\} \subseteq \bigcup_{q \in \mathbb{Q}} (\{f|_D < q\} \cap \{g|_D < x - q\})$. Hence

$$\{f + g < x\} \subseteq \bigcup_{q \in \mathbb{Q}} (\{f|_D < q\} \cap \{g|_D < x - q\}) \quad (21.382)$$

If $t \in \bigcup_{q \in \mathbb{Q}} (\{f|_D < q\} \cap \{g|_D < x - q\})$ then there exist a $q \in \mathbb{Q}$ such that $t \in \{f|_D < q\} \cap \{g|_D < x - q\}$ hence $t \in D$ and $f(t) = f|_D(t) < q \wedge g(t) = g|_D(t) < x - q$. As $(f(t), g(t)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and $f(t) < q < \infty \wedge g(t) < x - q < \infty$ we have for $f(t), g(t)$ either:

$f(t) = -\infty \wedge g(t) = -\infty$. Then $f(t) + g(t) = -\infty < x$.

$f(t) \in \mathbb{R} \wedge g(t) = -\infty$. Then $f(t) + g(t) = -\infty < x$.

$f(t) = -\infty \wedge g(t) \in \mathbb{R}$. Then $f(t) + g(t) = -\infty < x$.

$f(t) \in \mathbb{R} \wedge g(t) \in \mathbb{R}$. Then as $f(t) < q \wedge g(t) < x - q$ we have $f(t) + g(t) < q + (x - q) = x$

So in all cases we have $f(t) + g(t) < x$ which proves that $t \in \{f + g < x\}$. Hence we have

$$\bigcup_{q \in \mathbb{Q}} (\{f|_D < q\} \cap \{g|_D < x - q\}) \subseteq \{f + g < x\}$$

which combined with [eq: 21.382] proves that

$$\{f + g < x\} = \bigcup_{q \in \mathbb{Q}} (\{f|_D < q\} \cap \{g|_D < x - q\})$$

Let $x \in \mathbb{Q}$ then as $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, $g \in \mathcal{M}[B, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.231] that $f|_D, g|_D \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}]$ so that by [theorem: 21.254] $\{f|_D < q\}, \{g|_D < x - q\} \in \mathcal{A}|_D$ hence we have

$$\forall q \in \mathbb{Q} \{f|_D < q\} \cap \{g|_D < x - q\} \in \mathcal{A}|_D$$

Using [theorem: 21.39] and the fact that $\mathbb{Q}$ is countable [see theorem: 10.5] it follows that $\bigcup_{q \in \mathbb{Q}} (\{f|_D < q\} \cap \{g|_D < x - q\}) \in \mathcal{A}|_D$ hence we have

$$\{f + g < x\} \in \mathcal{A}|_D$$

As $x \in \mathbb{R}$ was chosen arbitrary it follows from the above and [theorem: 21.254] that

$$f + g \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}] \quad \square$$

Next we extend the above theorem to finite sums of $\mathcal{A}$ -measurable functions using mathematical induction.

Theorem 21.261. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$, $n \in \mathbb{N}$ and

$$\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \text{ (or } \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}])$$

$$\sum_{i=1}^n f_i: A \rightarrow Y \text{ defined by } \left(\sum_{i=1}^n f_i \right)(x) = \sum_{i=1}^n f_i(x)$$

then

$$\sum_{i=1}^n f_i \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}] \text{ (or } \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}])$$

Note 21.262. The condition $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ (or $\mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$) ensures that $\sum_{i=1}^n f_i$ is well defined. In principle we can have a more specific condition to ensure that $\sum_{i=1}^n f$ is well defined but this leads to a more complex proof than just using mathematical induction.

Proof. We use induction to prove this, so define

$$\mathcal{S} = \left\{ n \in \mathbb{N} \mid \text{Let } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{R}] \text{ (or } \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]) \text{ then } \sum_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}] \text{ (or } \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]) \right\}$$

then we have:

$1 \in \mathcal{S}$. Let $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ [or $\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$] then $\forall x \in A$ $(\sum_{i=1}^1 f_i)(x) = \sum_{i=1}^1 f_i(x) = f_1(x)$ which proves that

$$\sum_{i=1}^1 f = f_1 \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}] \text{ [or } \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]]$$

hence $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ [or $\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$]. Then $\forall x \in A$ we have

$$\left(\sum_{i=1}^{n+1} f_i \right)(x) = \sum_{i=1}^{n+1} f_i(x) = f_{n+1}(x) + \sum_{i=1}^n f_i(x) = \left(f_{n+1} + \sum_{i=1}^n f_i \right)(x)$$

so that

$$\sum_{i=1}^{n+1} f_i = f_{n+1} + \sum_{i=1}^n f_i \quad (21.383)$$

As $n \in \mathcal{S}$ we have $\sum_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ [or $\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$] which, as $f_{n+1} \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ [or $\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$] proves, using [theorem: 21.257], that

$$f_{n+1} + \sum_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}] \text{ [or } \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]]$$

Combining this with [eq: 21.383] proves that $\sum_{i=1}^{n+1} f_i \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ [or $\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$] so that $n+1 \in \mathcal{S}$. $\square$

Having covered the case of sums of measurable functions we consider now the multiplication of a function by a scalar.

Theorem 21.263. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$, $\alpha \in \mathbb{R}$ and $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ where $\alpha \cdot f$ is defined pointwise by $\alpha \cdot f: A \rightarrow \overline{\mathbb{R}}$ by $(\alpha \cdot f)(x) = \alpha \cdot f(x)$.

Note 21.264. If additional $f(X) \subseteq \mathbb{R}$ then $\forall x \in X$ we have $(\alpha \cdot f)(x) = \alpha \cdot f(x) \in \mathbb{R}$ so that $(\alpha \cdot f)(X) \subseteq \mathbb{R}$, hence $\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$. In other words if $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ then $\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$.

Proof. For $\alpha \in \mathbb{R}$ we have 3 cases to consider.

$\alpha = 0$. Then $\forall x \in A$ $(\alpha \cdot f)(x) = 0 \cdot f(x) = 0$ so that $\alpha \cdot f = C_0$, hence using [theorem: 21.228] it follows that $\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

$0 < \alpha$. Let $x \in \mathbb{R}$ then

$$\begin{aligned} t \in \{\alpha \cdot f < x\} &\Leftrightarrow \alpha \cdot f(t) < x \\ &\Leftrightarrow f(t) < \frac{x}{\alpha} \\ &\Leftrightarrow t \in \left\{ f < \frac{x}{\alpha} \right\} \end{aligned}$$

proving that $\{\alpha \cdot f < x\} = \left\{ f < \frac{x}{\alpha} \right\} \in \mathcal{A}|_A$ [using theorem: 21.254]]. Using [theorem: 21.254] again we have then that $\alpha \cdot f$ is $\mathcal{A}$ -measurable.

$\alpha < 0$. Let $x \in \mathbb{R}$ then

$$\begin{aligned} t \in \{\alpha \cdot f < x\} &\Leftrightarrow \alpha \cdot f(t) < x \\ &\Leftrightarrow f(t) > \frac{x}{\alpha} \\ &\Leftrightarrow t \in \left\{f > \frac{x}{\alpha}\right\} \end{aligned}$$

proving that $\{\alpha \cdot f < x\} = \left\{f > \frac{x}{\alpha}\right\} \in \mathcal{A}|_A$ [using [theorem: 21.254]]. Using [theorem: 21.254] again we have then that $\alpha \cdot f$ is $\mathcal{A}$ -measurable.

so in all cases we have that $\alpha \cdot f$ is $\mathcal{A}$ -measurable, hence

$$\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \quad \square$$

Theorem 21.265. Let $\langle X, \mathcal{A} \rangle$ be a measurable space

1. If $A \subseteq X$, $B \subseteq X$, $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, $g \in \mathcal{M}[B, \mathcal{A}, \overline{\mathbb{R}}]$ then for

$$f - g: D \rightarrow \overline{\mathbb{R}} \text{ defined by } (f - g)(x) = f(x) - g(x)$$

where

$$D = \left\{x \in A \cap B \mid (f(x), -g(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}\right\}$$

we have that

$$f - g \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}]$$

Note 21.266. If $A = B$ and $f, g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ or $f, g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then $D = A$ so that $f - g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ or $f - g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$.

2. If $A \subseteq X$ and $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ then for

$$f^2: A \rightarrow \overline{\mathbb{R}} \text{ defined by } f^2(x) = (f(x))^2$$

we have that

$$f^2 \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}].$$

3. If $A \subseteq X$ and $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ then $f \cdot g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ where

$$f \cdot g: X \rightarrow \overline{\mathbb{R}} \text{ is defined by } (f \cdot g)(x) = f(x) \cdot g(x)$$

4. If $A \subseteq X$ and $f, g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ then we have that

$$D = \{x \in A \mid g(x) \neq 0\} \in \mathcal{A}|_A$$

and for

$$\frac{f}{g}: D \rightarrow \overline{\mathbb{R}} \text{ defined by } \frac{f}{g}(x) = \frac{f(x)}{g(x)}$$

we have that

$$\frac{f}{g} \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}]$$

Proof.

1. As $g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have by [theorem: 21.263] that

$$(-1) \cdot g \in \mathcal{M}[B, \mathcal{A}, \overline{\mathbb{R}}].$$

Further $\forall x \in X$

$$(f + (-1) \cdot g)(x) = f(x) + ((-1) \cdot g)(x) = f(x) + (-1) \cdot g(x) = f(x) - g(x) = (f - g)(x)$$

which proves that

$$f + (-1) \cdot g = f - g$$

Hence using [theorem: 21.257]

$$f - g = f + (-1) \cdot g \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}].$$

2. Let $x \in \mathbb{R}$ then we have either:

$x \leq 0$. If $t \in \{f^2 < x\}$ then we have $0 < (f(t))^2 = f^2(t) < x \leq 0$ leading to the contradiction $0 < 0$. Hence $\{f^2 < x\} = \emptyset \in \mathcal{A}|_A$.

$0 < x$. Then

$$\begin{aligned} t \in \{f^2 < x\} &\Leftrightarrow f^2(t) < x \\ &\stackrel{[\text{theorem: 10.72}]}{\Leftrightarrow} -\sqrt{x} < f(t) < \sqrt{x} \\ &\Leftrightarrow t \in \{f > -\sqrt{x}\} \wedge t \in \{f < \sqrt{x}\} \\ &\Leftrightarrow t \in \{f > -\sqrt{x}\} \cap \{f < \sqrt{x}\} \end{aligned}$$

proving that $\{f^2 < x\} = \{f > -\sqrt{x}\} \cap \{f < \sqrt{x}\}$. As $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.254] that $\{f > -\sqrt{x}\}, \{f < \sqrt{x}\} \in \mathcal{A}|_A \Rightarrow \{f > -\sqrt{x}\} \cap \{f < \sqrt{x}\} \in \mathcal{A}|_A$, hence $\{f^2 < x\} \in \mathcal{A}|_A$.

As in all cases $\{f^2 < x\} \in \mathcal{A}|_A$ we have by [theorem: 21.254] that $f^2 \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$.

3. As $f, g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ we have by [theorem: 21.257] that $f + g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$. Applying (2) it follows that $f^2, g^2, (f + g)^2 \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$. Using (1) twice we have that $((f + g)^2 - f^2) - g^2 \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ so that by [theorem: 21.263]

$$\frac{1}{2} \cdot (((f + g)^2 - f^2) - g^2) \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}] \quad (21.384)$$

Finally if $x \in A$ then

$$\begin{aligned} \left(\frac{1}{2} \cdot ((f + g)^2 - f^2 - g^2) \right)(x) &= \frac{1}{2} \cdot (((f(x) + g(x))^2 - f^2(x)) - g^2(x)) \\ &= \frac{1}{2} \cdot (f^2(x) + g^2(x) + 2 \cdot f(x) \cdot g(x) - f^2(x) - g^2(x)) \\ &= f(x) \cdot g(x) \\ &= (f \cdot g)(x) \end{aligned}$$

which proves that $f \cdot g = \frac{1}{2} \cdot (((f + g)^2 - f^2) - g^2)$ and by [eq: 21.384] that

$$f \cdot g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$$

4. Let $D = \{x \in A | g(x) \neq 0\}$ then

$$\begin{aligned} t \in D &\Leftrightarrow t \in \{x \in A | g(x) \neq 0\} \\ &\Leftrightarrow t \in A \wedge (g(t) < 0 \vee 0 < g(t)) \\ &\Leftrightarrow (t \in A \wedge g(t) < 0) \vee (t \in A \wedge 0 < g(t)) \\ &\Leftrightarrow t \in \{g < 0\} \vee t \in \{g > 0\} \\ &\Leftrightarrow t \in \{g < 0\} \cup \{g > 0\} \end{aligned}$$

so that $D = \{g < 0\} \cup \{g > 0\}$. As g is $\mathcal{A}$ -measurable we have by [theorem: 21.254] that $\{g < 0\}, \{g > 0\} \in \mathcal{A}|_A$ proving that

$$D = \{x \in A | g(x) \neq 0\} = \{g < 0\} \cup \{g > 0\} \in \mathcal{A}|_A \quad (21.385)$$

Further as $f, g \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}] = \{h \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] | f(A) \subseteq \mathbb{R}\}$ and $D \subseteq A$ it follows from [theorem: 21.231] that

$$f|_D, g|_D \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}] \quad (21.386)$$

Let $x \in \mathbb{R}$. If $t \in (\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\})$ then $t \in D$, $g(t) = g|_D(t) > 0$ and $f(t) = f|_D(t) < x \cdot g|_D(t) = x \cdot g(t)$ so that $t \in D$ and $\frac{f(t)}{g(t)} = \frac{f|_D(t)}{g|_D(t)} < x$ or $t \in \left\{\frac{f}{g} < x\right\}$. Hence we have

$$(\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}) \subseteq \left\{\frac{f}{g} < x\right\} \quad (21.387)$$

If $t \in \{g|_D < 0\} \cap \{f|_D > x \cdot g|_D\}$ then $t \in D$, $g(t) = g|_D(t) < 0$ and $f(t) = f|_D(t) > x \cdot g|_D(t) = x \cdot g(t)$ so that $t \in D$ and $\frac{f}{g}(t) = \frac{f|_D(t)}{g|_D(t)} = \frac{f(t)}{g(t)} < x$ or $t \in \left\{\frac{f}{g} < x\right\}$. Hence

$$\{g|_D < 0\} \cap \{f|_D > x \cdot g|_D\} \subseteq \left\{\frac{f}{g} < x\right\} \quad (21.388)$$

Combining [eq: 21.387, 21.388] proves that

$$(\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}) \cup (\{g|_D < 0\} \cap \{f|_D > x \cdot g|_D\}) \subseteq \left\{\frac{f}{g} < x\right\} \quad (21.389)$$

If $t \in \left\{\frac{f}{g} < x\right\}$ then $t \in D$ and $\frac{f|_D(t)}{g|_D(t)} = \frac{f(t)}{g(t)} = \frac{f}{g}(t) < x$ then for $g(t)$ we have, as $t \in D \Rightarrow g(t) \neq 0$, either:

0 < g(t). Then $t \in D$, $0 < g|_D(t)$ and $f|_D(t) < x \cdot g|_D(t)$ so that

$$t \in \{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}$$

g(t) < 0. Then $t \in D$, $g|_D(t) < 0$ and $f|_D(t) > x \cdot g|_D(t)$ so that

$$t \in \{g|_D < 0\} \cap \{f|_D(t) > x \cdot g|_D\}$$

Hence $t \in (\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}) \cup (\{g|_D < 0\} \cap \{f|_D(t) > x \cdot g|_D\})$ proving that

$$\left\{\frac{f}{g} < x\right\} \subseteq (\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}) \cup (\{g|_D < 0\} \cap \{f|_D(t) > x \cdot g|_D\})$$

which combined with [eq: 21.389] proves that

$$\left\{\frac{f}{g} < x\right\} = (\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}) \cup (\{g|_D < 0\} \cap \{f|_D(t) > x \cdot g|_D\})$$

As $f|_D, g|_D \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}]$ [see eq: 21.386] it follows from [theorem: 21.254] it follows that

$$\{g|_D > 0\}, \{g|_D < 0\} \in \mathcal{A}|_D$$

Further using [theorem: 21.263] $x \cdot g|_D \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}]$ so that by [theorem: 21.256]

$$\{f|_D < x \cdot g|_D\}, \{f|_D(t) > x \cdot g|_D\} \in \mathcal{A}|_D$$

Hence $(\{g|_D > 0\} \cap \{f|_D < x \cdot g|_D\}) \cup (\{g|_D < 0\} \cap \{f|_D(t) > x \cdot g|_D\}) \in \mathcal{A}|_D$ or $\left\{\frac{f}{g} < x\right\} \in \mathcal{A}|_D$. Finally using [theorem: 21.254] it follows that

$$\frac{f}{g} \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}] \quad \square$$

Theorem 21.267. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$ and $f, g \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then

1. $f \vee g: A \rightarrow [0, \infty]$ defined by $(f \vee g)(x) = \max(f(x), g(x))$ is a $\mathcal{A}$ -measurable function. In other words $f \vee g \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.
2. $f \wedge g: A \rightarrow [0, \infty]$ defined by $(f \wedge g)(x) = \min(f(x), g(x))$ is a $\mathcal{A}$ -measurable function. In other words $f \wedge g \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

Proof.

1. Let $x \in \mathbb{R}$ then

$$\begin{aligned} t \in \{(f \vee g) < x\} &\Leftrightarrow (f \vee g)(t) < x \\ &\Leftrightarrow \max(f(t), g(t)) < x \\ &\Leftrightarrow f(t) < x \wedge g(t) < x \\ &\Leftrightarrow t \in \{f < x\} \wedge t \in \{g < x\} \\ &\Leftrightarrow t \in \{f < x\} \cap \{g < x\} \end{aligned}$$

which proves that

$$\{(f \vee g) < x\} = \{f < x\} \cap \{g < x\}$$

Using [theorem: 21.254] and the fact that $f, g \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that $\{f < x\}, \{g < x\} \in \mathcal{A}|_A$ from which it follows, using the above, that $\{(f \vee g) < x\} \in \mathcal{A}|_A$. Applying [theorem: 21.254] again proves that $f \vee g$ is $\mathcal{A}$ -measurable.

2. Let $x \in \mathbb{R}$ then

$$\begin{aligned} t \in \{(f \wedge g) > x\} &\Leftrightarrow (f \wedge g)(t) > x \\ &\Leftrightarrow \min(f(t), g(t)) > x \\ &\Leftrightarrow f(t) > x \wedge g(t) > x \\ &\Leftrightarrow t \in \{f > x\} \wedge t \in \{g > x\} \\ &\Leftrightarrow t \in \{f > x\} \cap \{g > x\} \end{aligned}$$

which proves that

$$\{(f \vee g) > x\} = \{f > x\} \cap \{g > x\}$$

Using [theorem: 21.254] and the fact that $f, g \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that $\{f > x\}, \{g > x\} \in \mathcal{A}|_A$ from which it follows, using the above, that $\{(f \vee g) > x\} \in \mathcal{A}|_A$. Applying [theorem: 21.254] again proves that $f \vee g$ is $\mathcal{A}$ -measurable. $\square$

Using mathematical induction we can extend the above to a finite family of $\mathcal{A}$ -measurable functions.

Corollary 21.268. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$, $n \in \mathbb{N}$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then*

1. $\bigvee_{i=1}^n f_i: A \rightarrow \overline{\mathbb{R}}$ defined by $(\bigvee_{i=1}^n f_i)(x) = \max(\{f_i(x) | i \in \{1, \dots, n\}\})$ is $\mathcal{A}$ -measurable. In other words $\bigvee_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.
2. $\bigwedge_{i=1}^n f_i: A \rightarrow \overline{\mathbb{R}}$ defined by $(\bigwedge_{i=1}^n f_i)(x) = \min(\{f_i(x) | i \in \{1, \dots, n\}\})$ is $\mathcal{A}$ -measurable. In other words $\bigwedge_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

Proof. We use recursion to prove this.

1. Let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \text{ then } \bigvee_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \right\}$$

then we have:

$1 \in S$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $(\bigvee_{i=1}^1 f_i)(x) = \max(\{f_1(x)\}) = f_1(x)$ so that $\bigvee_{i=1}^1 f_i = f_1 \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, hence $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $\forall x \in A$ we have

$$\begin{aligned} \left(\bigvee_{i=1}^{n+1} f_i \right)(x) &= \max(\{f_i(x) | i \in \{1, \dots, n+1\}\}) \\ &\stackrel{[\text{corollary: 3.67}]}{=} \max(\max(\{f_i(x) | i \in \{1, \dots, n\}\}), f_{n+1}(x)) \\ &= \max\left(\left(\bigvee_{i=1}^n f_i\right)(x), f_{n+1}(x)\right) \\ &= \left(\left(\bigvee_{i=1}^n f_i\right) \vee f_{n+1}\right)(x) \end{aligned}$$

so that

$$\bigvee_{i=1}^{n+1} f_i = \left(\bigvee_{i=1}^n f_i\right) \vee f_{n+1}$$

As $n \in S$ we have that $\bigvee_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, hence, using [theorem: 21.267] it follows that $(\bigvee_{i=1}^{n+1} f_i) \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, proving $n+1 \in S$.

2. Let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \text{ then } \bigwedge_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \right\}$$

then we have:

1 $\in S$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $(\bigwedge_{i=1}^1 f_i)(x) = \min(\{f_1(x)\}) = f_1(x)$ so that $\bigwedge_{i=1}^1 f_i = f_1 \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, hence $1 \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $\forall x \in X$ we have

$$\begin{aligned} \left(\bigwedge_{i=1}^{n+1} f_i \right)(x) &= \min(\{f_i(x) \mid i \in \{1, \dots, n+1\}\}) \\ &\stackrel{[\text{corollary: 3.67}]}{=} \min(\min(\{f_i(x) \mid i \in \{1, \dots, n\}\}), f_{n+1}(x)) \\ &= \min\left(\left(\bigwedge_{i=1}^n f_i\right)(x), f_{n+1}(x)\right) \\ &= \left(\left(\bigwedge_{i=1}^n f_i\right) \wedge f_{n+1}\right)(x) \end{aligned}$$

so that

$$\bigwedge_{i=1}^{n+1} f_i = \left(\bigwedge_{i=1}^n f_i\right) \wedge f_{n+1}$$

As $n \in S$ we have that $\bigwedge_{i=1}^n f_i \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$, hence, using [theorem: 21.267] it follows that $(\bigwedge_{i=1}^{n+1} f_i) \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$, proving $n+1 \in S$. $\square$

We now look at the measurability of the $\sup$, $\inf$, $\limsup$ and $\liminf$ and limit of a sequence of measurable functions.

Theorem 21.269. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$, $k \in \mathbb{N}$ and $\{f_i\}_{i \in \{k, \dots, \infty\}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ a sequence of $\mathcal{A}$ -measurable functions then we have:

1. The function

$$\sup_{n \in \{k, \dots, \infty\}} f_n: A \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right)(x) = \sup(\{f_n(x) \mid n \in \{k, \dots, \infty\}\})$$

is $\mathcal{A}$ -measurable. In other words $\sup_{n \rightarrow \infty} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

2. The function

$$\inf_{n \in \{k, \dots, \infty\}} f_n: A \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right)(x) = \inf(\{f_n(x) \mid n \in \{k, \dots, \infty\}\})$$

is $\mathcal{A}$ -measurable. In other words $\inf_{n \in \{k, \dots, \infty\}} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

3. The function

$$\liminf_{n \rightarrow \infty} f_n: A \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\liminf_{n \rightarrow \infty} f_n \right)(x) = \liminf_{n \rightarrow \infty} f_n(x)$$

is $\mathcal{A}$ -measurable. In other words $\liminf_{n \rightarrow \infty} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

4. The function

$$\limsup_{n \rightarrow \infty} f_n: A \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\limsup_{n \rightarrow \infty} f_n \right)(x) = \limsup_{n \rightarrow \infty} f_n(x)$$

is $\mathcal{A}$ -measurable. In other words $\limsup_{n \rightarrow \infty} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$.

5. Let $D = \left\{ x \in A \mid \left(\limsup_{n \rightarrow \infty} f_n \right)(x) = \left(\liminf_{n \rightarrow \infty} f_n \right)(x) \right\}$ [this is the set where by [definition: 20.72] limit $f_n(x)$ exist so $D = \left\{ x \in X \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist} \right\}$] then

$$D \in \mathcal{A}|_A$$

and for the function

$$\lim_{n \rightarrow \infty} f_n: D \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

we have $\lim_{n \rightarrow \infty} f_n \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}]$.

6. If $\forall x \in X$ we have $\forall n \in \{k, \dots, \infty\}$ that $f_n(x) \leq f_{n+1}(x)$ then we have that

$$\lim_{n \rightarrow \infty} f_n: A \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

is well defined and

$$\lim_{n \rightarrow \infty} f_n = \sup_{n \in \{k, \dots, \infty\}} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Proof.

1. Let $x \in \mathbb{R}$. If $t \in \left\{ \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right) \leq x \right\}$ then $\sup(\{f_n(t) \mid n \in \{k, \dots, \infty\}\}) \leq x$. So $\forall n \in \{k, \dots, \infty\}$ we have $f_n(t) \leq x \Rightarrow t \in \{f_n \leq x\}$. Hence

$$\left\{ \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right) \leq x \right\} \subseteq \bigcap_{n \in \{k, \dots, \infty\}} \{f_n \leq x\} \quad (21.390)$$

If $t \in \bigcap_{n \in \{k, \dots, \infty\}} \{f_n \leq x\}$ then we have $\forall n \in \{k, \dots, \infty\}$ that $t \in \{f_n \leq x\} \Rightarrow f_n(t) \leq x$. So x is an upper bound for $\{f_n(t) \mid n \in \{k, \dots, \infty\}\}$ from which it follows that $\sup(\{f_n(t) \mid n \in \{k, \dots, \infty\}\}) \leq x$ or $t \in \left\{ \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right) \leq x \right\}$ proving that

$\bigcap_{n \in \{k, \dots, \infty\}} \{f_n \leq x\} \subseteq \left\{ \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right) \leq x \right\}$. Combining this with [eq: 21.390] gives

$$\left\{ \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right) \leq x \right\} = \bigcap_{n \in \{k, \dots, \infty\}} \{f_n \leq x\} \quad (21.391)$$

As $\forall n \in \{k, \dots, \infty\}$ f_n is $\mathcal{A}$ -measurable it follows from [theorem: 21.254] that $\{f_n \leq x\} \in \mathcal{A}|_A$. Using [theorem: 21.36] we have that $\bigcap_{n \in \{k, \dots, \infty\}} \{f_n \leq x\} \in \mathcal{A}|_A$ or, using [eq: 21.391], that $\left\{ \left(\sup_{n \in \{k, \dots, \infty\}} f_n \right) \leq x \right\} \in \mathcal{A}|_A$. Applying then [theorem: 21.254] proves that

$$\sup_{n \in \{k, \dots, \infty\}} f_n \text{ is } \mathcal{A}\text{-measurable}$$

2. Let $x \in \mathbb{R}$. If $t \in \left\{ \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right) < x \right\}$ then $\inf(\{f_n(t) \mid n \in \{k, \dots, \infty\}\}) < x$. So $\exists n \in \{k, \dots, \infty\}$ such that $f_n(t) < x \Rightarrow t \in \{f_n < x\}$. Hence

$$\left\{ \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right) < x \right\} \subseteq \bigcup_{n \in \{k, \dots, \infty\}} \{f_n < x\} \quad (21.392)$$

If $t \in \bigcup_{n \in \{k, \dots, \infty\}} \{f_n < x\}$ then $\exists n \in \{k, \dots, \infty\}$ such that $t \in \{f_n < x\} \Rightarrow f_n(t) < x$, hence $\inf(\{f_n(t) \mid n \in \{k, \dots, \infty\}\}) \leq f_n(t) < x$, proving that $t \in \left\{ \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right) < x \right\}$. So $\bigcup_{n \in \{k, \dots, \infty\}} \{f_n < x\} \subseteq \left\{ \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right) < x \right\}$ which by [eq: 21.392] gives

$$\left\{ \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right) < x \right\} = \bigcup_{n \in \{k, \dots, \infty\}} \{f_n < x\} \quad (21.393)$$

As $\forall n \in \{k, \dots, \infty\}$ f_n is $\mathcal{A}$ -measurable it follows from [theorem: 21.254] that $\{f_n < x\} \in \mathcal{A}_{|A}$, applying then [definition: 21.33] proves that $\bigcup_{n \in \{k, \dots, \infty\}} \{f_n < x\} \in \mathcal{A}_{|A}$ or using [eq: 21.393] that $\left\{ \left(\inf_{n \in \{k, \dots, \infty\}} f_n \right) < x \right\} \in \mathcal{A}_{|A}$. Using then [theorem: 21.254] it follows that

$$\inf_{n \in \{k, \dots, \infty\}} f_n \text{ is } \mathcal{A}\text{-measurable}$$

3. Let $x \in A$ then

$$\begin{aligned} \left(\liminf_{n \rightarrow \infty} f_n \right)(x) &= \liminf_{n \rightarrow \infty} f_n(x) \\ &\stackrel{[\text{definition: 20.58}]}{=} \sup \left(\left\{ \inf \{ f_i | i \in \{l, \dots, \infty\} \} \mid l \in \{k, \dots, \infty\} \right\} \right) \\ &= \sup \left(\left\{ \left(\inf_{i \in \{l, \dots, \infty\}} f_i \right)(x) \mid l \in \{k, \dots, \infty\} \right\} \right) \\ &= \sup_{l \in \{k, \dots, \infty\}} \left(\left(\inf_{i \in \{l, \dots, \infty\}} f_i \right)(x) \right) \\ &= \left(\sup_{l \in \{k, \dots, \infty\}} \left(\inf_{i \in \{l, \dots, \infty\}} f_i \right) \right)(x) \end{aligned}$$

which proves that

$$\liminf_{n \rightarrow \infty} f_n = \sup_{l \in \{k, \dots, \infty\}} \left(\inf_{i \in \{l, \dots, \infty\}} f_i \right) \quad (21.394)$$

Now $\forall l \in \{k, \dots, \infty\}$ we have by (2) that $\inf_{i \in \{l, \dots, \infty\}} f_i$ is $\mathcal{A}$ -measurable. So applying (1) it follows that $\sup_{l \in \{k, \dots, \infty\}} \left(\inf_{i \in \{l, \dots, \infty\}} f_i \right)$ is $\mathcal{A}$ -measurable. Hence using [eq: 21.394] it follows that $\liminf_{n \rightarrow \infty} f_n$ is $\mathcal{A}$ -measurable.

4. Let $x \in A$ then

$$\begin{aligned} \left(\limsup_{n \rightarrow \infty} f_n \right)(x) &= \limsup_{n \rightarrow \infty} f_n(x) \\ &\stackrel{[\text{definition: 20.58}]}{=} \inf \left(\left\{ \sup \{ f_i | i \in \{l, \dots, \infty\} \} \mid l \in \{k, \dots, \infty\} \right\} \right) \\ &= \inf \left(\left\{ \left(\sup_{i \in \{l, \dots, \infty\}} f_i \right)(x) \mid l \in \{k, \dots, \infty\} \right\} \right) \\ &= \inf_{l \in \{k, \dots, \infty\}} \left(\left(\sup_{i \in \{l, \dots, \infty\}} f_i \right)(x) \right) \\ &= \left(\inf_{l \in \{k, \dots, \infty\}} \left(\sup_{i \in \{l, \dots, \infty\}} f_i \right) \right)(x) \end{aligned}$$

which proves that

$$\limsup_{n \rightarrow \infty} f_n = \inf_{l \in \{k, \dots, \infty\}} \left(\sup_{i \in \{l, \dots, \infty\}} f_i \right) \quad (21.395)$$

Now $\forall l \in \{k, \dots, \infty\}$ we have by (1) that $\sup_{i \in \{l, \dots, \infty\}} f_i$ is $\mathcal{A}$ -measurable. So applying (2) it follows that $\inf_{l \in \{k, \dots, \infty\}} \left(\sup_{i \in \{l, \dots, \infty\}} f_i \right)$ is $\mathcal{A}$ -measurable. Hence using [eq: 21.394] it follows that $\limsup_{n \rightarrow \infty} f_n$ is $\mathcal{A}$ -measurable.

5. From (3),(4) it follows that $\liminf_{n \rightarrow \infty} f_n$ and $\limsup_{n \rightarrow \infty} f_n$ are $\mathcal{A}$ -measurable, combining this fact with [theorem: 21.256], proves that $\left\{ \limsup_{n \rightarrow \infty} f_n = \limsup_{n \rightarrow \infty} f_n \right\} \in \mathcal{A}_{|A}$. As $D = \left\{ x \in A \mid \left(\limsup_{n \rightarrow \infty} f_n \right)(x) = \left(\limsup_{n \rightarrow \infty} f_n \right)(x) \right\} = \left\{ \limsup_{n \rightarrow \infty} f_n = \limsup_{n \rightarrow \infty} f_n \right\}$ it follows that

$$D \in \mathcal{A}_{|A} \quad (21.396)$$

If $x \in D$ then by definition $\limsup_{n \rightarrow \infty} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x)$ so that by [definition: 20.72] $\lim_{n \rightarrow \infty} f_n(x)$ exist and $\lim_{n \rightarrow \infty} f_n(x) = \limsup_{n \rightarrow \infty} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x)$. This allows us to define the function

$$\lim_{n \rightarrow \infty} f_n: D \rightarrow [0, \infty] \text{ where } \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x) = \limsup_{n \rightarrow \infty} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x) \quad (21.397)$$

Let $x \in \mathbb{R}$ then

$$\begin{aligned} t \in \left\{ \left(\limsup_{n \rightarrow \infty} f_n \right)_{|D} \leq x \right\} &\Leftrightarrow t \in D \wedge \left(\limsup_{n \rightarrow \infty} f_n \right)_{|D}(t) \leq x \\ &\Leftrightarrow t \in D \wedge \left(\limsup_{n \rightarrow \infty} f_n \right)(t) \leq x \\ &\stackrel{[\text{eq: 21.397}]}{\Leftrightarrow} t \in D \wedge \left(\lim_{n \rightarrow \infty} f_n \right)(t) \leq x \\ &\Leftrightarrow t \in \left\{ \left(\lim_{n \rightarrow \infty} f_n \right) \leq x \right\} \end{aligned}$$

which proves that

$$\left\{ \left(\lim_{n \rightarrow \infty} f_n \right) \leq x \right\} = \left\{ \left(\limsup_{n \rightarrow \infty} f_n \right)_{|D} \leq x \right\}$$

As by (4) $\limsup_{n \rightarrow \infty} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ and $D \subseteq A$ it follows from [theorem: 21.231] that $\left(\limsup_{n \rightarrow \infty} f_n \right)_{|D} \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}]$. Hence using [theorem: 21.254] it follows that

$$\left\{ \left(\limsup_{n \rightarrow \infty} f_n \right)_{|D} \leq x \right\} \in \mathcal{A}_{|D}$$

so that

$$\left\{ \left(\lim_{n \rightarrow \infty} f_n \right) \leq x \right\} \in \mathcal{A}_{|D}$$

Finally using [theorem: 21.254] again it follows that

$$\lim_{n \rightarrow \infty} f_n: D \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{A}\text{-measurable}$$

6. Let $x \in X$ then as $\forall n \in \{k, \dots, \infty\}$ $f_n(x) \leq f_{n+1}(x)$ we have by [theorem: 20.92] that

$$\lim_{n \rightarrow \infty} f_n(x) \text{ exist and } \lim_{n \rightarrow \infty} f_n(x) = \sup (\{f_n(x) | n \in \{k, \dots, \infty\}\})$$

so that

$$\lim_{n \rightarrow \infty} f_n: X \rightarrow \overline{\mathbb{R}} \text{ where } \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

is a well defined function and

$$\lim_{n \rightarrow \infty} f_n = \sup_{n \in \{k, \dots, \infty\}} f_n$$

Finally by (1) we have that

$$\lim_{n \rightarrow \infty} f_n = \sup_{n \in \{k, \dots, \infty\}} f_n \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \quad \square$$

Theorem 21.270. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$ and $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ a sequence of $\mathcal{A}$ -measurable functions then for

$$f_\infty: X \rightarrow \overline{\mathbb{R}} \text{ defined by } f_\infty(x) = \begin{cases} \lim_{n \rightarrow \infty} f_n(x) & \text{if } x \in D \\ 0 & \text{if } x \in X \setminus D \end{cases}$$

where

$$D = \left\{ x \in A \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist} \right\} = \left\{ x \in A \mid \left(\limsup_{n \rightarrow \infty} f_n \right)(x) = \left(\liminf_{n \rightarrow \infty} f_n \right)(x) \right\}$$

we have that

$$f_\infty \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ or in other words } f_\infty \text{ is } \mathcal{A}\text{-measurable}$$

Proof. Using the previous theorem [theorem: 21.269] it follows that

$$D \in \mathcal{A}|_A \text{ and for the function } \lim_{n \rightarrow \infty} f_n: D \rightarrow \overline{\mathbb{R}} \text{ defined by } \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

we have $\lim_{n \rightarrow \infty} f_n \in \mathcal{M}[D, \mathcal{A}, \overline{\mathbb{R}}]$. As $D \in \mathcal{A}|_A$ there exist a $E \in \mathcal{A}$ such that $D = A \cap E$, further as $A \in \mathcal{A}$ we have also that $A \cap E \in \mathcal{A}$ from which it follows that $D \in \mathcal{A}$. Further $\forall x \in X$ we have

$$\left(\lim_{n \rightarrow \infty} f_n \right)_{D \upharpoonright X}(x) = \begin{cases} \left(\lim_{n \rightarrow \infty} f_n \right)(x) & \text{if } x \in D \\ 0 & \text{if } x \in A \setminus D \end{cases} = \begin{cases} \lim_{n \rightarrow \infty} f_n(x) & \text{if } x \in D \\ 0 & \text{if } x \in A \setminus D \end{cases} = f_\infty(x)$$

which proves that $f_\infty = \left(\lim_{n \rightarrow \infty} f_n \right)_{D \upharpoonright X}$. Hence using [theorem: 21.250]

$$f_\infty \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

□

A very useful $\mathcal{A}$ -measurable function is the Characteristic function [see also example: 2.48].

Definition 21.271. Let X be a set, $A \in \mathcal{P}(X)$ then the Characteristic function $\mathcal{X}_{X,A}$ is defined by

$$\mathcal{X}_{X,A}: X \rightarrow \overline{\mathbb{R}} \text{ where } \mathcal{X}_{X,A}(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases}$$

If X is understood from the context then we use the notation $\mathcal{X}_A$ instead of $\mathcal{X}_{X,A}$

Theorem 21.272. Let X be a set and $A \subseteq X$, $B \subseteq X$ then

1. $\mathcal{X}_{X,A} \cdot \mathcal{X}_{X,B} = \mathcal{X}_{X,A \cap B}$ [where $\mathcal{X}_{X,A} \cdot \mathcal{X}_{X,B}$ is pointwise defined].
2. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function then $(\mathcal{X}_{X,B} \cdot f)|_B = f|_B$ [where $\mathcal{X}_{X,A} \cdot f$ is pointwise defined].
3. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function then $(\mathcal{X}_{X,B} \cdot f)|_A = \mathcal{X}_{A,B} \cdot f|_A$ [where $\mathcal{X}_{X,B} \cdot f$ is pointwise defined].

Proof.

1. Let $x \in X$ then we have either:

$x \in A \cap B$. Then $x \in A \wedge x \in B$ so that

$$(\mathcal{X}_{X,A} \cdot \mathcal{X}_{X,B})(x) = \mathcal{X}_{X,A}(x) \cdot \mathcal{X}_{X,B}(x) = 1 \cdot 1 = 1 = \mathcal{X}_{X,A \cap B}(x)$$

$x \in X \setminus (A \cap B)$. Then $x \notin A \vee x \notin B$ so that

$$(\mathcal{X}_{X,A} \cdot \mathcal{X}_{X,B})(x) = \mathcal{X}_{X,A}(x) \cdot \mathcal{X}_{X,B}(x) = 0 = \mathcal{X}_{X,A \cap B}(x)$$

which proves that $\mathcal{X}_{X,A} \cdot \mathcal{X}_{X,B} = \mathcal{X}_{X,A \cap B}$

2. Let $x \in B$ then we have $(\mathcal{X}_{X,B} \cdot f)|_B(x) = (\mathcal{X}_{X,B} \cdot f)(x) = 1 \cdot f(x) = f(x) = f|_B(x)$ which proves that $(\mathcal{X}_{X,B} \cdot f)|_B = f|_B$.
3. Let $x \in A$ then we have for x either:

$x \in B$. Then

$$\begin{aligned} (\mathcal{X}_{X,B} \cdot f)|_A(x) & \stackrel{x \in A}{=} (\mathcal{X}_{X,B} \cdot f)(x) \\ & = \mathcal{X}_{X,B}(x) \cdot f(x) \\ & \stackrel{x \in A, B}{=} \mathcal{X}_{A,B}(x) \cdot f(x) \\ & \stackrel{x \in A}{=} \mathcal{X}_{A,B}(x) \cdot f|_A(x) \\ & = (\mathcal{X}_{A,B} \cdot f|_A)(x) \end{aligned}$$

$x \notin B$. Then

$$(\mathcal{X}_{X,B} \cdot f)|_A(x) = (\mathcal{X}_{X,B} \cdot f)(x) = \mathcal{X}_{X,B}(x) \cdot f(x) = 0 = \mathcal{X}_{A,B}(x) \cdot f|_A(x) = (\mathcal{X}_{A,B} \cdot f|_A)(x)$$

so that

$$(\mathcal{X}_{X,B} \cdot f)|_A = \mathcal{X}_{A,B} \cdot f|_A \quad \square$$

Theorem 21.273. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$ and $B \in \mathcal{A}$ with $B \subseteq A$ then

$$\mathcal{X}_{A,B} \in \mathcal{M}^+[A, \mathcal{A}, \mathbb{R}]$$

Note 21.274. If $A = X$ then $B \in \mathcal{A}$ implies that $B \subseteq A$ so that we have

$$\mathcal{X}_{X,A} \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

Proof. Let $x \in \mathbb{R}$ then we have three possibilities:

$x < 0$. If $t \in A$ then $\mathcal{X}_{A,B}(t) = \begin{cases} 1 & \text{if } t \in B \\ 0 & \text{if } t \in B \setminus A \end{cases} \geq 0 > x$ so that $t \in \{\mathcal{X}_{A,B} > x\}$. Hence we have $A \subseteq \{\mathcal{X}_{A,B} > x\}$. As trivially $\{\mathcal{X}_{A,B} > x\} \subseteq A$ it follows that

$$\{\mathcal{X}_{A,B} > x\} = A \in \mathcal{A}|_A$$

$0 \leq x < 1$. If $t \in \{\mathcal{X}_{A,B} > x\}$ then $t \in A$ and $\mathcal{X}_{A,B}(t) > x$. Assume that $t \in A \setminus B$ then $\mathcal{X}_{A,B}(t) = 0$ so that $0 > x \geq 0$ giving the contradiction $0 > 0$, hence the assumption must be false which proves that $t \in B$. Hence we have

$$\{\mathcal{X}_{A,B} > x\} \subseteq B \quad (21.398)$$

On the other hand if $t \in B \subseteq A$ then $t \in A$ and $\mathcal{X}_{A,B}(t) = 1 > x$ so that $t \in \{\mathcal{X}_{A,B} > x\}$ which proves that $B \subseteq \{\mathcal{X}_{A,B} > x\}$, combining this with [eq: 21.398] gives

$$\{\mathcal{X}_{A,B} > x\} = B$$

Now as $B \in \mathcal{A}$ and $B \stackrel{B \subseteq A}{=} A \cap B$ it follows that $B \in \mathcal{A}|_A$ which by the above proves that

$$\{\mathcal{X}_{A,B} > x\} \in \mathcal{A}|_A$$

$1 \leq x$. Assume $t \in \{\mathcal{X}_{A,B} > x\}$ then $\mathcal{X}_{A,B}(t) > x \geq 1$, as $\mathcal{X}_{A,B}(t) = \begin{cases} 1 & \text{if } t \in B \\ 0 & \text{if } t \in A \setminus B \end{cases} \leq 1$ we have the contradiction $1 > 1$. So the assumption is false and we must have

$$\{\mathcal{X}_{A,B} > x\} = \emptyset \in \mathcal{A}|_A$$

So in all cases we have $\{\mathcal{X}_{A,B} > x\}$ which by [theorem: 21.254] proves that

$$\mathcal{X}_{A,B} \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

As $\mathcal{X}_{A,B}(A) = \{0, 1\} \subseteq [0, \infty[$ it follows from the above that

$$\mathcal{X}_{A,B} \in \mathcal{M}^+[A, \mathcal{A}, \mathbb{R}] \quad \square$$

Lemma 21.275. Let $X, A \subseteq X$ be a set and $f: X \rightarrow \overline{\mathbb{R}}$ a function then for $B \subseteq \overline{\mathbb{R}}$ we have

1. If $0 \notin B$ then $(\mathcal{X}_{X,A} \cdot f)^{-1}(B) = A \cap f^{-1}(B)$
2. If $0 \in B$ then $(\mathcal{X}_{X,A} \cdot f)^{-1}(B) = (X \setminus A) \cup f^{-1}(B)$

Proof.

1. Let $x \in (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$ then $(\mathcal{X}_{X,A} \cdot f)(x) \in B$. Assume that $x \in X \setminus A$ then we would have $0 = \mathcal{X}_{X,A}(x) \cdot f(x) = (\mathcal{X}_{X,A} \cdot f)(x) \in B$ contradicting the hypothesis $0 \notin B$ so we must have that $x \in A$. Then $f(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = (\mathcal{X}_{X,A} \cdot f)(x) \in B$ proving that $x \in A \cap f^{-1}(B)$. Hence

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) \subseteq A \cap f^{-1}(B) \quad (21.399)$$

If $x \in A \cap f^{-1}(B)$ then $x \in A$ and $f(x) \in B$ so that $(\mathcal{X}_{X,A} \cdot f)(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = f(x) \in B$ proving that $x \in (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$. So $A \cap f^{-1}(B) \subseteq (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$ which by [eq: 21.399] proves that

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) = A \cap f^{-1}(B)$$

2. Let $x \in (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$ then $(\mathcal{X}_{X,A} \cdot f)(x) \in B$, consider the following possible cases for x :
 $x \in A$. Then $f(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = (\mathcal{X}_{X,A} \cdot f)(x) \in B$ proving that

$$x \in f^{-1}(B) \subseteq (X \setminus A) \cup f^{-1}(B)$$

$x \in X \setminus A$. Then clearly $x \in (X \setminus A) \cup f^{-1}(B)$

Hence we have

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) \subseteq (X \setminus A) \cup f^{-1}(B) \quad (21.400)$$

On the other hand if $x \in (X \setminus A) \cup f^{-1}(B)$ then we have either

$x \in X \setminus A$. Then $(\mathcal{X}_{X,A} \cdot f)(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = 0 \in B$ proving that $x \in (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$.

$x \in A$. Then $x \in f^{-1}(B)$ so that $(\mathcal{X}_{X,A} \cdot f)(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = f(x) \in B$ proving that $x \in (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$.

Hence $(X \setminus A) \cup f^{-1}(B) \subseteq (\mathcal{X}_{X,A} \cdot f)^{-1}(B)$ which by [eq: 21.400] gives

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) = (X \setminus A) \cup f^{-1}(B) \quad \square$$

Corollary 21.276. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \subseteq X$, $B \in \mathcal{A}$ with $B \subseteq A$ and $f: A \rightarrow \overline{\mathbb{R}}$ a function then we have:

1. If $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$
2. If $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ then $\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$
3. If $f \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $\mathcal{X}_{A,B} \cdot f \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$
4. If $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ then $\mathcal{X}_{A,B} \cdot f \in \mathcal{M}^+[A, \mathcal{A}, \mathbb{R}]$

Proof.

1. As $B \in \mathcal{A}$, $B \subseteq A$ we have that $B = A \cap B \in \mathcal{A}|_A$ so that

$$B \in \mathcal{A}|_A \quad (21.401)$$

Let $C \in \mathcal{B}[\overline{\mathbb{R}}]$ then we for C either:

$0 \notin C$. Then by [lemma: 21.275]

$$(\mathcal{X}_{A,B} \cdot f)^{-1}(C) = B \cap f^{-1}(C)$$

As $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ $f^{-1}(C) \in \mathcal{A}|_A$ and $B \in \mathcal{A}|_A$ [see eq: 21.401] $B \cap f^{-1}(C) \in \mathcal{A}|_A$ so that

$$(\mathcal{X}_{A,B} \cdot f)^{-1}(C) \in \mathcal{A}|_A$$

$0 \in C$. Then [lemma: 21.275]

$$(\mathcal{X}_{A,B} \cdot f)^{-1}(C) = (A \setminus B) \cup f^{-1}(C)$$

As $A \in \mathcal{A}|_A$ and by [eq: 21.401] $B \in \mathcal{A}|_A$ we have that $A \setminus B \in \mathcal{A}|_A$, further, as $f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ $f^{-1}(C) \in \mathcal{A}|_A$. Hence $(A \setminus B) \cup f^{-1}(C) \in \mathcal{A}|_A$ which proves that

$$(\mathcal{X}_{A,B} \cdot f)^{-1}(C) \in \mathcal{A}|_A$$

So in all cases we have $(\mathcal{X}_{A,B} \cdot f)^{-1}(C) \in \mathcal{A}|_A$ so that by [definition: 21.224]

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}, \mathcal{B}[\overline{\mathbb{R}}]] = \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

2. If $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ then $f(A) \subseteq \mathbb{R}$ and $\mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$. Using (1) it follows that

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Further $\forall x \in A$ we have $(\mathcal{X}_{X,A} \cdot f)(x) = \begin{cases} f(x) \in \mathbb{R} & \text{if } x \in A \\ 0 \in \mathbb{R} & \text{if } x \in X \setminus A \end{cases} \in \mathbb{R}$ so that

$$(\mathcal{X}_{A,B} \cdot f)(A) \subseteq \mathbb{R}$$

Hence we have

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$$

3. If $f \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$ then $f(A) \subseteq [0, \infty]$ and $\mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$. Using (1) it follows that

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Further $\forall x \in A$ we have $(\mathcal{X}_{X,A} \cdot f)(x) = \begin{cases} f(x) \in [0, \infty] & \text{if } x \in A \\ 0 \in [0, \infty] & \text{if } x \in X \setminus A \end{cases} \in [0, \infty]$ so that

$$(\mathcal{X}_{A,B} \cdot f)(A) \subseteq [0, \infty]$$

Hence we have

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$$

4. If $f \in \mathcal{M}^+[A, \mathcal{A}, \mathbb{R}]$ then $f(A) \subseteq [0, \infty[$ and $\mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$. Using (1) it follows that

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Further $\forall x \in A$ we have $(\mathcal{X}_{X,A} \cdot f)(x) = \begin{cases} f(x) \in [0, \infty[& \text{if } x \in A \\ 0 \in [0, \infty[& \text{if } x \in X \setminus A \end{cases} \in [0, \infty[$ so that

$$(\mathcal{X}_{A,B} \cdot f)(A) \subseteq [0, \infty[$$

Hence we have

$$\mathcal{X}_{A,B} \cdot f \in \mathcal{M}^+[A, \mathcal{A}, \mathbb{R}]$$

□

Theorem 21.277. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$ and $f: X \rightarrow \overline{\mathbb{R}}$ a function then we have

$$\mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \Leftrightarrow f|_A \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Proof.

$\Rightarrow$. Let $B \in \mathcal{B}[\overline{\mathbb{R}}]$ then as $\mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) \in \mathcal{A}|_X = \mathcal{A} \quad (21.402)$$

For B we have now two possible cases:

0 $\in B$. Then using [lemma: 21.275]

$$(X \setminus A) \bigcup f^{-1}(B) = (\mathcal{X}_{X,A} \cdot f)^{-1}(B) \underset{[\text{eq: 21.402}]}{\in} \mathcal{A} \quad (21.403)$$

Hence we have that

$$A \bigcap ((X \setminus A) \bigcup f^{-1}(B)) \in \mathcal{A}|_A \quad (21.404)$$

Further

$$\begin{aligned} A \bigcap ((X \setminus A) \bigcup f^{-1}(B)) &= (A \bigcap (X \setminus A)) \bigcup (A \bigcap f^{-1}(B)) \\ &= \emptyset \bigcup (A \bigcap f^{-1}(B)) \\ &= A \bigcap f^{-1}(B) \\ &\underset{[\text{theorem: 2.92}]}{=} (f|_A)^{-1}(B) \end{aligned}$$

so that by [eq: 21.404] we must conclude that

$$(f|_A)^{-1}(B) \in \mathcal{A}|_A$$

0 $\notin B$. Then using [lemma: 21.275] we have that

$$A \bigcap f^{-1}(B) = (\mathcal{X}_{X,A} \cdot f)^{-1}(B) \quad (21.405)$$

Hence we have by [eq: 21.402] that $A \bigcap f^{-1}(B) \in \mathcal{A}$ which as

$$A \bigcap f^{-1}(B) = A \bigcap (A \bigcap f^{-1}(B))$$

proves that

$$A \cap f^{-1}(B) \in \mathcal{A}_{|A}$$

Using [theorem: 2.92] we have that $(f|_A)^{-1}(B) = A \cap f^{-1}(B)$ so that by the above

$$(f|_A)^{-1}(B) \in \mathcal{A}_{|A}$$

As in the two possible cases we have $(f|_A)^{-1}(A) \in \mathcal{A}_{|A}$ it follows from [definition: 21.224] that

$$f|_A \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}, \mathcal{B}[\overline{\mathbb{R}}]] = \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$$

$\Leftarrow$. Let $B \in \mathcal{B}[\overline{\mathbb{R}}]$ then as $f|_A \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that

$$(f|_A)^{-1}(B) \in \mathcal{A}_{|A}$$

so that, as $(f|_A)^{-1}(B) \stackrel{[\text{theorem: 2.92}]}{=} A \cap f^{-1}(B)$,

$$A \cap f^{-1}(B) \in \mathcal{A}_{|A}$$

Hence there exist a $E \in \mathcal{A}$ such that

$$A \cap f^{-1}(B) = A \cap E$$

As by the hypothesis $A \in \mathcal{A}$ it follows that $A \cap E \in \mathcal{A}$ so that using the above we have

$$A \cap f^{-1}(B) \in \mathcal{A} \tag{21.406}$$

From $A \in \mathcal{A}$ it follows also that

$$X \setminus A \in \mathcal{A} \tag{21.407}$$

Hence using [eqs: 21.406, 21.407] we have that

$$(X \setminus A) \cup (A \cap f^{-1}(B)) \in \mathcal{A} \tag{21.408}$$

Further

$$\begin{aligned} (X \setminus A) \cup (A \cap f^{-1}(B)) &= ((X \setminus A) \cup A) \cap ((X \setminus A) \cap f^{-1}(B)) \\ &\stackrel{A \subseteq X}{=} X \cap ((X \setminus A) \cap f^{-1}(B)) \\ &= (X \setminus A) \cap f^{-1}(B) \end{aligned}$$

so that by [eq: 21.408]

$$(X \setminus A) \cap f^{-1}(B) \in \mathcal{A} \tag{21.409}$$

Now for B we have either:

$0 \in B$. Then by [lemma: 21.275]

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) = (X \setminus A) \cup f^{-1}(B)$$

so that by [eq: 21.409]

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) \in \mathcal{A} = \mathcal{A}_{|X}$$

$0 \notin B$. Then by [lemma: 21.275]

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) = A \cap f^{-1}(B)$$

so that using [eq: 21.406]

$$(\mathcal{X}_{X,A} \cdot f)^{-1}(B) \in \mathcal{A} = \mathcal{A}_{|X}$$

Hence $\forall B \in \mathcal{B}[\overline{\mathbb{R}}]$ we have $(\mathcal{X}_{X,A} \cdot f)^{-1}(B) \in \mathcal{A}_{|X}$ which by definition [see definition: 21.224] proves that

$$\mathcal{X}_{X \setminus A} \cdot f \in \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}, \mathcal{B}[\overline{\mathbb{R}}]] = \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}] \quad \square$$

21.4.3 Almost everywhere properties

Using the concept of μ -negligible sets [see definition: 21.211] we can introduce the term **almost everywhere**. A statement about elements in a set is said to be almost everywhere true if it is true for every element in the set except for a μ -negligible subset. The idea is that we can ignore the μ -negligible part because it will not influence the final result.

Definition 21.278. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$ and $P(x)$ a predicate then $P(x)$ is μ -a.e. true if $\{x | \neg P(x)\} \in \mathcal{N}_\mu$. In other words using the definition of $\mathcal{N}_\mu$ [see definition: 21.211]

$$P(x) \text{ is } \mu\text{-a.e. true or } P(x) \text{ holds } \mu\text{-a.e.}$$

if there exists a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x | \neg P(x)\} \subseteq N$$

Using the above we have the following definitions that will be used in this text.

Definition 21.279. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$ then we define:

1. If $f: A \rightarrow \overline{\mathbb{R}}$ and $g: A \rightarrow \overline{\mathbb{R}}$ are functions then $f = g$ μ -a.e. if there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in A | f(x) \neq g(x)\} \subseteq N$.
2. If $f: A \rightarrow \overline{\mathbb{R}}$ and $g: A \rightarrow \overline{\mathbb{R}}$ are functions then $f \leq g$ μ -a.e. if there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in A | g(x) < f(x)\} \subseteq N$.
3. If $f: A \rightarrow \overline{\mathbb{R}}$ and $g: A \rightarrow \overline{\mathbb{R}}$ are functions then $f < g$ μ -a.e. if there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in A | g(x) \leq f(x)\} \subseteq N$.
4. If $f: A \rightarrow \overline{\mathbb{R}}$ then $f = 0$ μ -a.e. if there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in A | f(x) \neq 0\} \subseteq N$.
5. Let $\{f_n\}_{n \in \mathbb{N}} \subseteq \overline{\mathbb{R}}^X$ be a sequence of functions and $f: A \rightarrow \overline{\mathbb{R}}$ then $\lim_{n \rightarrow \infty} f_n = f$ μ -a.e. if there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$X \setminus \left\{ x \in A \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist and } \lim_{n \rightarrow \infty} f_n(x) = f(x) \right\} \subseteq N$$

One way that almost everywhere equality is used is in masking undesirable points from the domain of a function using a characteristic function. For this the following theorem is useful.

Theorem 21.280. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$, $B \in \mathcal{N}_\mu$ [in other words there exist a $N \in \mathcal{S}$ with $\mu(N) = 0$ such that $B \subseteq N$] and $f: A \rightarrow \overline{\mathbb{R}}$ then for $\mathcal{X}_{A, A \setminus B} \cdot f: A \rightarrow \overline{\mathbb{R}}$ we have that

$$\mathcal{X}_{A, A \setminus B} \cdot f = f \text{ } \mu\text{-a.e.}$$

Proof. Let $x \in \{x \in A | (\mathcal{X}_{A, A \setminus B} \cdot f)(x) \neq f(x)\}$ then $x \in A$ and $\mathcal{X}_{A, A \setminus B}(x) \cdot f(x) \neq f(x)$. Assume that $x \notin B$ then $x \in A \setminus B$ so that $f(x) \neq \mathcal{X}_{A, A \setminus B}(x) \cdot f(x) = 1 \cdot f(x) = f(x)$ which is a contradiction. Hence the assumption is false and we must have that $x \in B$, which proves that

$$\{x \in A | (\mathcal{X}_{A, A \setminus B} \cdot f)(x) \neq f(x)\} \subseteq B \subseteq N$$

So by definition

$$\mathcal{X}_{X \setminus A} \cdot f = f \text{ } \mu\text{-a.e.}$$

□

Theorem 21.281. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$ together with functions

$$\begin{aligned} f: A &\rightarrow \overline{\mathbb{R}} \\ g: A &\rightarrow \overline{\mathbb{R}} \\ h: A &\rightarrow \overline{\mathbb{R}} \\ k: A &\rightarrow \overline{\mathbb{R}} \end{aligned}$$

then we have:

1. If $f = g$ μ -a.e. and $g = h$ μ -a.e. then $f = h$ μ -a.e.
2. If $f \leq g$ μ -a.e. and $g \leq h$ μ -a.e. then $f \leq h$ μ -a.e.
3. If $f \leq g$ μ -a.e. and $g \leq f$ μ -a.e. then $f = g$ μ -a.e.
4. If $\alpha \in \mathbb{R} \setminus \{0\}$ then if $f = g$ μ -a.e. it follows that $\alpha \cdot f = \alpha \cdot g$ μ -a.e.
5. Let

$$D = \{x \in A \mid (f(x), g(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\} \wedge (h(x), k(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}\}$$

then we have for

$$f + g: D \rightarrow \overline{\mathbb{R}} \text{ defined by } f(x) + g(x)$$

$$h + k: D \rightarrow \overline{\mathbb{R}} \text{ defined by } h(x) + g(x)$$

that

$$f + g = h + k \text{ } \mu\text{-a.e.}$$

Proof.

1. As $f = g$ μ -a.e and $g = h$ μ -a.e there exists $N_1, N_2 \in \mathcal{A}$ with $\mu(N_1) = 0 = \mu(N_2)$ such that

$$\{x \in A \mid f(x) \neq g(x)\} \subseteq N_1 \text{ and } \{x \in A \mid g(x) \neq h(x)\} \subseteq N_2$$

Let $x \in \{x \in A \mid f(x) \neq h(x)\}$ then $f(x) \neq g(x)$ or $g(x) \neq h(x)$ [otherwise $f(x) = g(x)$ and $g(x) = h(x)$ so that $f(x) = h(x)$]. Hence $x \in \{x \in A \mid f(x) \neq g(x)\} \cup \{x \in A \mid g(x) \neq h(x)\}$ which proves that

$$\{x \in A \mid f(x) \neq h(x)\} \subseteq \{x \in A \mid f(x) \neq g(x)\} \cup \{x \in A \mid g(x) \neq h(x)\} \subseteq N_1 \cup N_2$$

As $N_1 \cup N_2 \in \mathcal{A}$ and $\mu(N_1 \cup N_2) \underset{[\text{theorem: 21.102}]}{\leq} \mu(N_1) + \mu(N_2) = 0 + 0 = 0$ it follows that

$$f = h \text{ } \mu\text{-a.e.}$$

2. As $f \leq g$ μ -a.e. and $g \leq h$ μ -a.e. there exists $N_1, N_2 \in \mathcal{A}$ with $\mu(N_1) = 0 = \mu(N_2)$ such that

$$\{x \in A \mid g(x) < f(x)\} \subseteq N_1 \text{ and } \{x \in A \mid h(x) < g(x)\} \subseteq N_2$$

Let $x \in \{x \in A \mid h(x) < f(x)\}$ then $g(x) < f(x)$ or $h(x) < g(x)$ [otherwise $f(x) \leq g(x)$ and $g(x) \leq h(x)$ so that $f(x) \leq h(x)$]. Hence $x \in \{x \in A \mid g(x) < f(x)\} \cup \{x \in A \mid h(x) < g(x)\}$ which proves that

$$\{x \in A \mid h(x) < f(x)\} \subseteq \{x \in A \mid g(x) < f(x)\} \cup \{x \in A \mid h(x) < g(x)\} \subseteq N_1 \cup N_2$$

As $N_1 \cup N_2 \in \mathcal{A}$ and $\mu(N_1 \cup N_2) \underset{[\text{theorem: 21.102}]}{\leq} \mu(N_1) + \mu(N_2) = 0 + 0 = 0$ it follows that

$$f \leq h \text{ } \mu\text{-a.e.}$$

3. As $f \leq g$ μ -a.e. and $g \leq h$ μ -a.e. there exist $N_1, N_2 \in \mathcal{A}$ with $\mu(N_1) = 0 = \mu(N_2)$ such that

$$\{x \in A \mid g(x) < f(x)\} \subseteq N_1 \text{ and } \{x \in A \mid f(x) < g(x)\} \subseteq N_2$$

Let $x \in \{A \mid f(x) \neq g(x)\}$ then $f(x) < g(x)$ or $g(x) < f(x)$ so that $x \in \{x \in A \mid g(x) < f(x)\} \cup \{x \in A \mid f(x) < g(x)\}$ which proves that

$$\{A \mid f(x) \neq g(x)\} \subseteq \{x \in A \mid g(x) < f(x)\} \cup \{x \in A \mid f(x) < g(x)\} \subseteq N_1 \cup N_2$$

As $N_1 \cup N_2 \in \mathcal{A}$ and $\mu(N_1 \cup N_2) \underset{[\text{theorem: 21.102}]}{\leq} \mu(N_1) + \mu(N_2) = 0 + 0 = 0$ it follows that

$$f = h \text{ } \mu\text{-a.e.}$$

4. As $f = g$ μ -a.e. there exists a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in A \mid f(x) \neq g(x)\} \subseteq N$$

Let now $x \in \{x \in A \mid \alpha \cdot f(x) \neq \alpha \cdot g(x)\}$ then $\alpha \cdot f(x) \neq \alpha \cdot g(x) \xrightarrow{\alpha \neq 0} f(x) \neq g(x)$ proving that $\{x \in A \mid \alpha \cdot f(x) \neq \alpha \cdot g(x)\} \subseteq \{x \in A \mid f(x) \neq g(x)\} \subseteq N$. So

$$\alpha \cdot f \neq \alpha \cdot g \text{ } \mu\text{-a.e.}$$

5. As $f = h$ μ -a.e and $g = k$ μ -a.e there exists $N_1, N_2 \in \mathcal{A}$ with $\mu(N_1) = 0 = \mu(N_2)$ such that

$$\{x \in A \mid f(x) \neq h(x)\} \subseteq N_1 \text{ and } \{x \in A \mid g(x) \neq k(x)\} \subseteq N_2$$

Let $x \in \{x \in D \mid f(x) + g(x) \neq h(x) + k(x)\}$ then $x \in D$ and $f(x) \neq h(x)$ or $g(x) \neq k(x)$ [other wise $f(x) = h(x)$ and $g(x) = k(x)$ so that, as $x \in D$, $f(x) + g(x) = h(x) + k(x)$] which proves that

$$x \in \{x \in A \mid f(x) \neq h(x)\} \cup \{x \in A \mid g(x) \neq k(x)\} \subseteq N_1 \cup N_2$$

Hence

$$\{x \in D \mid f(x) + g(x) \neq h(x) + k(x)\} \subseteq N_1 \cup N_2$$

As $N_1 \cup N_2 \in \mathcal{A}$ and $\mu(N_1 \cup N_2) \underset{[\text{theorem: 21.102}]}{\leq} \mu(N_1) + \mu(N_2) = 0 + 0 = 0$ it follows that

$$f + g = h + k \text{ } \mu\text{-a.e.} \quad \square$$

Theorem 21.282. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$ and $f: A \rightarrow \overline{\mathbb{R}}$ $g: A \rightarrow \overline{\mathbb{R}}$ functions with $f(A), g(A) \subseteq \mathbb{R}$ then

$$f = g \text{ } \mu\text{-a.e.} \Leftrightarrow f - g = 0 \text{ } \mu\text{-a.e.}$$

Note 21.283. $f(A), g(A) \subseteq \mathbb{R}$ is needed to ensure that $g - f$ is well defined.

Proof. Note that

$$\begin{aligned} x \in \{x \in A \mid f(x) \neq g(x)\} &\Leftrightarrow x \in A \wedge f(x) \neq g(x) \\ &\Leftrightarrow x \in A \wedge (f - g)(x) \neq 0 \\ &\Leftrightarrow x \in \{x \in A \mid (f - g)(x) \neq 0\} \end{aligned}$$

proving that $\{x \in A \mid f(x) \neq g(x)\} = \{x \in A \mid (f - g)(x) \neq 0\}$, from which we have trivially that

$$f = g \text{ } \mu\text{-a.e.} \Leftrightarrow f - g = 0 \text{ } \mu\text{-a.e.} \quad \square$$

The following theorem shows that for a complete measure space a function that is a.e. equal to a measurable function is also measurable.

Theorem 21.284. Let $\langle X, \mathcal{A}, \mu \rangle$ be a **complete** measure space [see theorem: 21.215], $A \in \mathcal{A}$, $f: A \rightarrow \overline{\mathbb{R}}$ a $\mathcal{A}$ -measurable function and $g: A \rightarrow \overline{\mathbb{R}}$ a function so that $f = g$ μ -a.e. then g is $\mathcal{A}$ -measurable.

Note 21.285. As $X \in \mathcal{A}$ we have that if $f: X \rightarrow \overline{\mathbb{R}}$ is a $\mathcal{A}$ -measurable function and $g: X \rightarrow \overline{\mathbb{R}}$ is a function such that $f = g$ μ -a.e. then g is $\mathcal{A}$ -measurable.

Proof. As $f = g$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{t \in A \mid f(t) \neq g(t)\} \subseteq N \quad (21.410)$$

Hence

$$\forall t \in A \setminus N \text{ we have } f(t) = g(t) \quad (21.411)$$

Let $x \in \mathbb{R}$. Let $t \in \{g \leq x\} = \{t \in A \mid g(t) \leq x\} \subseteq A$ then for t we have two possibilities

$t \in N$. Then $t \in N \cap \{g \leq x\}$

$t \in A \setminus N$. Then $f(t) = g(t) \leq x$ so that $t \in (A \setminus N) \cap \{f \leq x\}$

hence we have that

$$\{g \leq x\} \subseteq (N \cap \{g \leq x\}) \cup ((A \setminus N) \cap \{f \leq x\}) \quad (21.412)$$

Let $t \in (N \cap \{g \leq x\}) \cup ((A \setminus N) \cap \{f \leq x\})$ then we have either:

$t \in N \cap \{g \leq x\}$. Then trivially $t \in \{g \leq x\}$.

$t \in (A \setminus N) \cap \{f \leq x\}$. Then as $t \in A \setminus N \subseteq A$ we have $f(t) = g(t)$ and $f(t) \leq x$ so that $g(t) \leq x$, hence $t \in \{g \leq x\}$.

So in all cases we have $t \in \{g \leq x\}$, hence $(N \cap \{g \leq x\}) \cup ((A \setminus N) \cap \{f \leq x\}) \subseteq \{g \leq x\}$. Combining this with [eq: 21.412] proves that:

$$\{g \leq x\} = (N \cap \{g \leq x\}) \cup ((A \setminus N) \cap \{f \leq x\}) \quad (21.413)$$

As $N \cap \{g \leq x\} \subseteq N$, $\mu(N) = 0$ and $\langle X, \mathcal{A}, \mu \rangle$ is a complete measure space it follows by [definition: 21.215] that $N \cap \{g \leq x\} \in \mathcal{A}$. As further $N \cap \{g \leq x\} \subseteq \{g \leq x\} \subseteq A$ we have

$$N \cap \{g \leq x\} = A \cap (N \cap \{g \leq x\}) \in \mathcal{A}|_A \quad (21.414)$$

As f is $\mathcal{A}$ -measurable $\{f \leq x\} \in \mathcal{A}|_A$ so there exist a $E \in \mathcal{A}$ such that $\{f \leq x\} = A \cap E$. Further as $A \in \mathcal{A}$ and $N \in \mathcal{A}$ $A \setminus N \in \mathcal{A}$ so that $(A \setminus N) \cap E \in \mathcal{A}$. Further

$$(A \setminus N) \cap \{f \leq x\} = (A \setminus N) \cap (A \cap E) = A \cap ((A \setminus N) \cap E)$$

from which it follows that

$$(A \setminus N) \cap \{f \leq x\} \in \mathcal{A}|_A \quad (21.415)$$

As $\mathcal{A}|_A$ is a sigma algebra it follows from [eqs: 21.414, 21.415] that

$$(N \cap \{g \leq x\}) \cup ((A \setminus N) \cap \{f \leq x\}) \in \mathcal{A}|_A$$

which combined with [eq: 21.413] proves that

$$\{g \leq x\} \in \mathcal{A}|_A$$

Finally using [theorem: 21.254] we have

$$g \text{ is } \mathcal{A}\text{-measure} \quad \square$$

Theorem 21.286. Let $\langle X, \mathcal{A}, \mu \rangle$ be a **complete measure space**, $A \in \mathcal{A}$ and $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}[A, \mathcal{A}, \overline{\mathbb{R}}]$ a sequence of $\mathcal{A}$ -measurable functions and $f: X \rightarrow \overline{\mathbb{R}}$ a function such that $\lim_{n \rightarrow \infty} f_n = f$ μ -a.e. then $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$.

Note 21.287. As $X \in \mathcal{A}$ we have for $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f: X \rightarrow \overline{\mathbb{R}}$ a function such that $\lim_{n \rightarrow \infty} f_n = f$ μ -a.e. then $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$.

Proof. Let $D = \left\{x \in A \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist}\right\}$ then we have by [theorem: 21.270] that

$$f_\infty: X \rightarrow \overline{\mathbb{R}} \text{ where } f_\infty(x) = \begin{cases} \lim_{n \rightarrow \infty} f_n(x) & \text{if } x \in D \\ 0 & \text{if } x \in X \setminus D \end{cases} \text{ is } \mathcal{A}\text{-measurable}$$

As $\lim_{n \rightarrow \infty} f_n = f$ μ -a.e. we have by [definition: 21.279(5)] there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$X \setminus \left\{x \in X \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist and } \lim_{n \rightarrow \infty} f_n(x) = f(x)\right\} \subseteq N$$

If $x \in \left\{x \in X \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist and } \lim_{n \rightarrow \infty} f_n(x) = f(x)\right\}$ then $\lim_{n \rightarrow \infty} f_n(x)$ exists and $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ so that $x \in D$ and $f(x) = \lim_{n \rightarrow \infty} f_n(x) = f_\infty(x)$ which proves that $x \in \{x \in X \mid f(x) = f_\infty(x)\}$. Hence

$$\left\{x \in X \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist and } \lim_{n \rightarrow \infty} f_n(x) = f(x)\right\} \subseteq \{x \in X \mid f(x) = f_\infty(x)\}$$

so that

$$X \setminus \{x \in X \mid f(x) = f_\infty(x)\} \subseteq X \setminus \left\{x \in X \mid \lim_{n \rightarrow \infty} f_n(x) \text{ exist and } \lim_{n \rightarrow \infty} f_n(x) = f(x)\right\} \subseteq N$$

Further

$$\begin{aligned} x \in \{x \in X \mid f(x) \neq f_\infty(x)\} &\Leftrightarrow x \in X \wedge f(x) \neq f_\infty(x) \\ &\Leftrightarrow x \in X \wedge x \notin \{x \in X \mid f(x) = f_\infty(x)\} \\ &\Leftrightarrow x \in X \setminus \{x \in X \mid f(x) = f_\infty(x)\} \end{aligned}$$

which proves that

$$\{x \in X \mid f(x) \neq f_\infty(x)\} = X \setminus \{x \in X \mid f(x) = f_\infty(x)\} \subseteq N$$

Hence it follows that $f = f_\infty$ μ -a.e.. Using then the previous theorem [theorem: 21.284] we must conclude that

f is $\mathcal{A}$ -measurable. □

21.5 The measure integral

We will build up the definition of the measure integral in faces:

1. We define a integral over measurable functions whose image is finite the so called **simple functions**.
2. Extend this integral to non negative measurable functions.
3. Extend the integral of non negative measurable function to general measurable functions.
4. Finally extend the integral to complex valued functions.

21.5.1 Integral of simple functions

Definition 21.288. Let X be a non empty set then a function $f: X \rightarrow \overline{\mathbb{R}}$ is a **simple function** if $f(X)$ is a **finite set** and $f(X) \subseteq \mathbb{R}$. The set of all simple function graphs is noted as $\mathcal{S}[X]$ so

$$\mathcal{S}[X] = \{f \in \overline{\mathbb{R}}^X \mid f(X) \subseteq \mathbb{R} \text{ and } f(X) \text{ is a finite set}\}$$

We have the following alternative definition of a simple function.

Theorem 21.289. Let X be a non empty set then for a function $f: X \rightarrow \overline{\mathbb{R}}$ with $f(X) \subseteq \mathbb{R}$ we have the following equivalences:

1. f is a simple function [in other words $f \in \mathcal{S}[X]$]
2. There exist a $n \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, n\} \rightarrow f(X)$ such that for $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$ defined by $A_i = f^{-1}(\{\beta(i)\})$ we have:
 - a. $\{A_i\}_{i \in \{1, \dots, n\}}$ is pairwise disjoint
 - b. $X = \bigsqcup_{i \in \{1, \dots, n\}} A_i$
 - c. $f = \sum_{i=1}^n \beta(i) \cdot \mathcal{X}_{A_i}$
3. There exist a $n \in \mathbb{N}$ and a pairwise disjoint $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{A_i}$$

Proof.

1 $\Rightarrow$ 2. As f is a simple function $f(X)$ is finite, further as $X \neq \emptyset$ $f(X) \neq \emptyset$, hence there exists a $n \in \mathbb{N}$ and a bijection

$$\beta: \{1, \dots, n\} \rightarrow f(X)$$

Define $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$ by $A_i = f^{-1}(\{\beta(i)\})$. Let $i, j \in \{1, \dots, n\}$ with $i \neq j$ then

$$\begin{aligned} A_i \cap A_j &= f^{-1}(\{\beta(i)\}) \cap f^{-1}(\{\beta(j)\}) \\ &\stackrel{[\text{theorem: 2.103}]}{=} f^{-1}(\{\beta(i)\} \cap \{\beta(j)\}) \\ &\stackrel{i \neq j \Rightarrow \beta(i) \neq \beta(j) \Rightarrow \{\beta(i)\} \cap \{\beta(j)\} = \emptyset}{=} f^{-1}(\emptyset) \\ &= \emptyset \end{aligned}$$

which proves that

$$\{A_i\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint} \quad (21.416)$$

Let $x \in X$ then $f(x) \in f(X)$ so, using the surjectivity of β , there exist a $i \in \{1, \dots, n\}$ such that $f(x) = \beta(i)$. Hence $x \in f^{-1}(\{\beta(i)\}) = A_i \subseteq \bigsqcup_{i \in \{1, \dots, n\}} A_i$ which proves that $X \subseteq \bigsqcup_{i \in \{1, \dots, n\}} A_i$ or, as trivially $\bigsqcup_{i \in \{1, \dots, n\}} A_i \subseteq X$, that

$$X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \quad (21.417)$$

Let $x \in X$ then by [eqs: 21.416, 21.417] there exists a $i \in \{1, \dots, n\}$ such that $x \in A_i$ and $\forall j \in \{1, \dots, n\} \setminus \{i\}$ $x \notin A_j$. Hence

$$\begin{aligned} \left(\sum_{k=1}^n f(\beta(k)) \cdot \mathcal{X}_{X, A_k} \right)(x) &= \sum_{k \in \{1, \dots, n\}} f(\beta(k)) \cdot \mathcal{X}_{X, A_k}(x) \\ &= \sum_{k \in \{i\}} f(\beta(k)) \cdot \mathcal{X}_{X, A_k}(x) + \sum_{k \in \{1, \dots, n\} \setminus \{i\}} f(\beta(k)) \cdot \mathcal{X}_{X, A_k}(x) \\ &= f(\beta(i)) \cdot \mathcal{X}_{X, A_i}(x) + \sum_{k \in \{1, \dots, n\} \setminus \{i\}} f(\beta(k)) \cdot 0 \\ &= f(\beta(i)) \\ &= f(x) \end{aligned}$$

proving that

$$f = \sum_{k=1}^n f(\beta(k)) \cdot \mathcal{X}_{X, A_k} \quad (21.418)$$

2 $\Rightarrow$ 3. By (2) there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, n\} \rightarrow f(X) \subseteq \mathbb{R}$ such that for $\{A_i\}_{i \in \{1, \dots, n\}}$ defined by $A_i = \beta^{-1}(\{\beta(i)\})$ we have

$$\{A_i\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint, } X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \text{ and } f = \sum_{i=1}^n f(\beta(i)) \cdot \mathcal{X}_{X, A_i}$$

Define $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq f(X) \subseteq \mathbb{R}$ by $f_i = f(\beta(i)) \in \mathbb{R}$ then we have by the above also

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$$

which finish the proof of (3).

3 $\Rightarrow$ 1. By (3) there exists a $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$, $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that

$$\{A_i\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint and } f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$$

Let $y \in f(X)$ then there exists a $x \in X$ such that $y = f(x)$. Using the above there exist a $i \in \{1, \dots, n\}$ such that $x \in A_i$ and $\forall j \in \{1, \dots, n\} \setminus \{i\}$ $x \notin A_j$ hence

$$\begin{aligned}
 y &= f(x) \\
 &= \left(\sum_{k=1}^n f_k \cdot \mathcal{X}_{A_k} \right)(x) \\
 &= \sum_{k \in \{1, \dots, n\}} f_k \cdot \mathcal{X}_{A_k}(x) \\
 &= \sum_{k \in \{i\}} f_k \cdot \mathcal{X}_{A_k}(x) + \sum_{k \in \{1, \dots, n\} \setminus \{i\}} f_k \cdot \mathcal{X}_{A_k}(x) \\
 &= f_i \cdot \mathcal{X}_{A_i}(x) + \sum_{k \in \{1, \dots, n\} \setminus \{i\}} f_k \cdot 0 \\
 &= f_i \\
 &\in \{f_i | i \in \{1, \dots, n\}\}
 \end{aligned}$$

from which it follows that $f(X) \subseteq \{f_i | i \in \{1, \dots, n\}\}$ a finite set. As the subset of a finite set is finite [see theorem: 6.43] it follows that $f(X)$ is a finite set, hence f is a simple function by definition. $\square$

We have the following important result for functions of the form $f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{A_i}$ like simple functions.

Theorem 21.290. *Let X be a non empty set, $n \in \mathbb{N}$, I finite, $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ pairwise disjoint and $\{f_i\}_{i \in I} \subseteq \mathbb{R}$ then for*

$$f = \sum_{i \in I} f_i \cdot \mathcal{X}_{X, A_i} : X \rightarrow \overline{\mathbb{R}}$$

we have that:

1. $\forall i \in I$ we have $\forall x \in A_i$ that $f(x) = f_i$
2. $\forall x \in X \setminus (\bigsqcup_{i \in I} A_i)$ we have $f(x) = 0$

Proof.

1. Let $i \in I$ then for $x \in A_i$ we have, as $\{A_i\}_{i \in I}$ is pairwise disjoint, that $\forall j \in I \setminus \{i\}$ $x \notin A_j$. So

$$\begin{aligned}
 f(x) &= \sum_{k \in I} f_k \cdot \mathcal{X}_{X, A_k}(x) \\
 &= \sum_{k \in \{i\}} f_k \cdot \mathcal{X}_{X, A_k}(x) + \sum_{k \in I \setminus \{i\}} f_k \cdot \mathcal{X}_{X, A_k}(x) \\
 &= f_i \cdot \mathcal{X}_{X, A_i}(x) + \sum_{k \in I \setminus \{i\}} f_k \cdot 0 \\
 &= f_i
 \end{aligned}$$

2. If $x \in X \setminus (\bigsqcup_{i \in I} A_i)$ then $\forall i \in I$ we have $x \notin A_i$ so that

$$f(x) = \sum_{i \in I} f_i \cdot \mathcal{X}_{X, A_i}(x) = \sum_{i \in I} f_i \cdot 0 = 0 \quad \square$$

The set of simple functions is stable under pointwise sum and scalar product.

Theorem 21.291. *Let X be a set then we have*

1. $\forall f \in \mathcal{S}[X]$ and $\forall \alpha \in \mathbb{R}$ we have $\alpha \cdot f \in \mathcal{S}[X]$
2. $\forall f, g \in \mathcal{S}[X]$ we have $f + g \in \mathcal{S}[X]$

Proof.

1. We have to consider two cases for α

$\alpha = 0$. Then $(\alpha \cdot f)(X) = \{0\}$ which is finite and non empty.

$\alpha \neq 0$. As f is a simple function there exists a $n \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, n\} \rightarrow f(X)$. Define now

$$\alpha_f: \{1, \dots, n\} \rightarrow (\alpha \cdot f)(X) \text{ by } \alpha_f(i) = \alpha \cdot \beta(i)$$

If $y \in (\alpha \cdot f)(X)$ then there exists a $x \in X$ such that

$$y = (\alpha \cdot f)(x) = \alpha \cdot f(x) \xrightarrow{\alpha \neq 0} f(x) = \frac{y}{\alpha}$$

As $f(x) \in f(X)$ there exists a $i \in \{1, \dots, n\}$ such that

$$\beta(i) = f(x) = \frac{y}{\alpha} \Rightarrow y = \alpha \cdot \beta(i) = \alpha_f(i)$$

proving that α_f is a surjection. So using [theorem: 6.44] we have that $(\alpha \cdot f)(X)$ is finite.

so in all cases $(\alpha \cdot f)(X)$ is finite proving that $\alpha \cdot f$ is a simple function.

2. As f and g are simple functions we have that $f(X)$ and $g(X)$ are finite. As the product of finite sets is finite [see theorem: 6.41] we have that

$$f(X) \times g(X) \text{ is finite}$$

Define now

$$s: f(X) \times g(Y) \rightarrow \{x + y | (x, y) \in f(X) \times g(Y)\} \text{ by } s(x, y) = x + y$$

Then if $z \in \{x + y | (x, y) \in f(X) \times g(Y)\}$ there exists a $x \in f(X)$ and a $y \in f(Y)$ such that $z = x + y = s(x, y)$ which proves that s is surjective. Applying then [theorem: 6.44] proves that

$$\{x + y | (x, y) \in f(X) \times g(X)\} \text{ is finite}$$

Let $y \in (f + g)(X)$ then there exists a $x \in X$ such that

$$y = f(x) + g(x) \in \{x + y | (x, y) \in f(X) \times g(X)\}.$$

Hence $(f + g)(X) \subseteq \{x + y | (x, y) \in f(X) \times g(X)\}$ which, as we have proved that $\{x + y | (x, y) \in f(X) \times g(X)\}$ is finite, proves using [theorem: 6.43] that $(f + g)(X)$ is finite.. This proves that $f + g$ is a simple function. $\square$

An example of a simple function is a function $f: \mathbb{R} \rightarrow \overline{\mathbb{R}}$ of the form

$$f = f_1 \cdot \mathcal{X}_{A_1} + f_2 \cdot \mathcal{X}_{A_2} + f_3 \cdot \mathcal{X}_{A_3} + f_4 \cdot \mathcal{X}_{A_4} \text{ where } f(X) = \{f_1, f_2, f_3, f_4\}$$

is shown in the following figure:

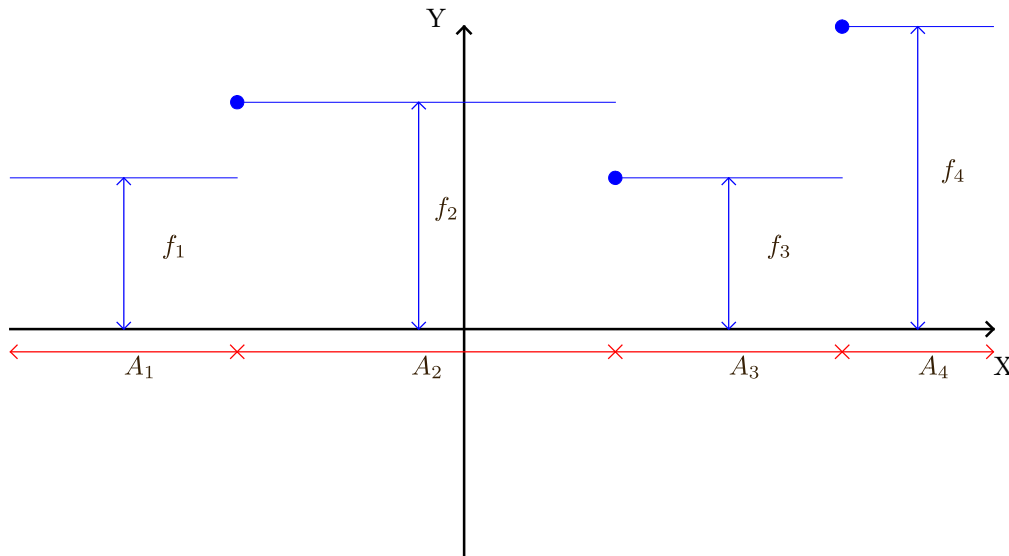


Figure 21.5.

The area below the above function and the X -axis is intuitively equal to the sum

$$f_1 \cdot \text{size}(A_1) + f_2 \cdot \text{size}(A_2) + f_3 \cdot \text{size}(A_3) + f_4 \cdot \text{size}(A_4)$$

so the integral of f which calculates this area should be of the form

$$\int f = f_1 \cdot \text{size}(A_1) + f_2 \cdot \text{size}(A_2) + f_3 \cdot \text{size}(A_3) + f_4 \cdot \text{size}(A_4)$$

hence we need a way to calculate the sizes of A_1, A_2, A_3 and A_4 , in other words A_1, A_2, A_3 and A_4 must be measurable so that we can use the measure function to calculate the sizes. The integral becomes then

$$\int f = f_1 \cdot \mu(A_1) + f_2 \cdot \mu(A_2) + f_3 \cdot \mu(A_3) + f_4 \cdot \mu(A_4)$$

For this to work we must have $A_1, A_2, A_3, A_4 \in \mathcal{A}$ and as $\mu(A_i)$ can be ∞ f_1, f_2, f_3, f_4 must be non negative. This motivates the following definition.

Definition 21.292. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then the set of $\mathcal{A}$ -measurable real valued simple functions is noted as $\mathcal{S}[X, \mathcal{A}]$. Hence

$$\mathcal{S}[X, \mathcal{A}] = \{f \in \mathcal{S}[X] \mid f: X \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{A}\text{-measurable}\} \subseteq \mathcal{S}[X]$$

Note 21.293. If $f \in \mathcal{S}[X, \mathcal{A}]$ then $f \in \mathcal{S}[X] \Rightarrow f(X) \subseteq \mathbb{R}$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that

$$\mathcal{S}[X, \mathcal{A}] \subseteq \mathcal{S}[X] \cap \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

Further if $f \in \mathcal{S}[X] \cap \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ then $f \in \mathcal{S}[X]$ and $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}(X, \mathcal{A}, \overline{\mathbb{R}})$ which proves that $\mathcal{S}[X] \cap \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{S}[X, \mathcal{A}]$. Hence we have

$$\mathcal{S}[X, \mathcal{A}] = \mathcal{S}[X] \cap \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}(X, \mathcal{A}, \overline{\mathbb{R}})$$

We look now at an alternative definition for $\mathcal{S}[X, \mathcal{A}]$. First we need a little lemma.

Lemma 21.294. Let X be a set, $n \in \mathbb{N}$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$ pairwise disjoint then for

$$\{B_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{P}(X) \text{ defined by } B_i = \begin{cases} X \setminus (\bigsqcup_{i \in \{1, \dots, n\}} A_i) & \text{if } i = n+1 \\ A_i & \text{if } i \in \{1, \dots, n\} \end{cases} \subseteq X$$

we have

$$\{B_i\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint}$$

and

$$X = \bigsqcup_{i \in \{1, \dots, n+1\}} A_i$$

Proof. If $i, j \in \{1, \dots, n+1\}$ with $i \neq j$ then we have for i, j either:

$i = n+1$. Then $j \in \{1, \dots, n\}$ so that

$$A_i \cap A_j = \left(X \setminus \left(\bigsqcup_{k \in \{1, \dots, n\}} A_k \right) \right) \cap A_j \subseteq (X \setminus A_j) \cap A_j = \emptyset.$$

$j = n+1$. Then $i \in \{1, \dots, n\}$ so that

$$A_i \cap A_j = A_i \cap \left(X \setminus \left(\bigsqcup_{k \in \{1, \dots, n\}} A_k \right) \right) \subseteq A_i \cap (X \setminus A_i) = \emptyset.$$

$i, j \in \{1, \dots, n\}$. Then as $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$ is pairwise disjoint we have that $A_i \cap A_j = \emptyset$.

which proves that

$\{B_i\}_{i \in \{1, \dots, n\}}$ is pairwise disjoint

Further

$$\begin{aligned}
 \bigsqcup_{i \in \{1, \dots, n+1\}} B_i &\stackrel{[\text{theorem: 2.136}]}{=} \left(\bigsqcup_{i \in \{1, \dots, n\}} B_i \right) \cup \left(\bigcup_{i \in \{n+1\}} B_i \right) \\
 &= \left(\bigsqcup_{i \in \{1, \dots, n\}} B_i \right) \cup B_{n+1} \\
 &= \left(\bigsqcup_{i \in \{1, \dots, n\}} B_i \right) \cup \left(X \setminus \left(\bigsqcup_{i \in \{1, \dots, n\}} B_i \right) \right) \\
 &\stackrel{=}{=} \bigsqcup_{i \in \{1, \dots, n\}} B_i \subseteq X \quad X
 \end{aligned}$$

hence

$$X = \bigsqcup_{i \in \{1, \dots, n+1\}} B_i$$

□

We have now the following equivalences for $\mathcal{S}[X, \mathcal{A}]$.

Theorem 21.295. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $f: X \rightarrow \overline{\mathbb{R}}$ a function then we have the following equivalences:*

1. $f \in \mathcal{S}[X, \mathcal{A}]$ [in other words $f \in \mathcal{S}[X] \cap \mathcal{M}[X, \mathbf{A}, \mathbb{R}]$]
2. $f \in \mathcal{S}[X]$ and $\forall y \in f(X) \ f^{-1}(\{y\}) \in \mathcal{A}$
3. There exists a $n \in \mathbb{N}$, a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ of measurable sets and a family $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ of real numbers so that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$$

Proof.

1 $\Rightarrow$ 2. As $f \in \mathcal{S}[X, \mathcal{A}] = \mathcal{S}[X] \cap \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ $f \in \mathcal{S}[X]$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$. Let $y \in f(X)$ then, as $f(X) \subseteq \mathbb{R}$ it follows that $y \in \mathbb{R}$. Applying then [theorem: 21.256] it follows that $f^{-1}(\{y\}) \stackrel{[\text{definition: 21.252}]}{=} \{f = y\} \in \mathcal{A}$.

2 $\Rightarrow$ 3. As $f \in \mathcal{S}[X]$ we can use [theorem: 21.289] to get a $n \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, n\} \rightarrow f(X)$ such that for $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{P}(X)$ defined by $A_i = f^{-1}(\beta(i))$ we have

$$f = \sum_{i=1}^n \beta(i) \cdot \mathcal{X}_{X, A_i} \text{ and } \{A_i\}_{i \in \{1, \dots, n\}} \text{ is pairwise disjoint}$$

Using (2) we have $\forall i \in \{1, \dots, n\} \ A_i = f^{-1}(\{\beta(i)\}) \in \mathcal{A}$ so that

$$\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$$

Finally, define $\{f_i\}_{i \in \{1, \dots, n\}}$ by $f_i = \beta(i) \in f(X) \stackrel{\subseteq}{\subseteq} \mathbb{R}$ then

$$\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R} \text{ and } f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{A_i}$$

3 $\Rightarrow$ 1. Define

$$\{B_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} X \setminus (\bigsqcup_{i=1, \dots, n} A_i) & \text{if } i = n+1 \\ A_i & \text{if } i \in \{1, \dots, n\} \end{cases} \in \mathcal{A}$$

$$\{g_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{R} \text{ by } g_i = \begin{cases} 0 & \text{if } i = n+1 \\ f_i & \text{if } i \in \{1, \dots, n\} \end{cases} \in \mathbb{R}$$

Then using [lemma: 21.294] we have that

$$\{g_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{R}, \{B_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A} \text{ is pairwise disjoint, } X = \bigsqcup_{i=1}^{n+1} B_i \quad (21.419)$$

Further

$$\sum_{i=1}^{n+1} g_i \cdot \chi_{X, B_i}(x) = g_{n+1} \cdot \chi_{X, B_{n+1}} + \sum_{i=1}^n g_i \cdot \chi_{X, B_i}(x) = 0 + \sum_{i=1}^n f_i \cdot \chi_{X, A_i} = f(x)$$

so that

$$f = \sum_{i=1}^{n+1} g_i \cdot \chi_{X, B_i} \quad (21.420)$$

Let $x \in X$ then there exist a $i \in \{1, \dots, n+1\}$ such that $x \in B_i$ hence $f(x) \stackrel{[\text{theorem: 21.290}]}{=} g_i \in \mathbb{R}$ which proves that

$$f(X) \subseteq \mathbb{R}$$

So by [theorem: 21.289] we have

$$f \in \mathcal{S}[X]$$

Define $\{I_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{P}(\{1, \dots, n+1\})$ by

$$I_i = \{j \in \{1, \dots, n+1\} \mid f(B_j) = \{g_i\}\} \subseteq \{1, \dots, n+1\}$$

then we prove that

$$\forall i \in \{1, \dots, n+1\} \text{ we have } f^{-1}(\{g_i\}) = \bigsqcup_{j \in I_i} B_j \stackrel{[\text{theorem: 21.39}]}{\in} \mathcal{A} \quad (21.421)$$

Proof. Let $i \in \{1, \dots, n+1\}$. Take $x \in f^{-1}(\{g_i\})$ then $f(x) = g_i$. As $x \in X = \bigcup_{k \in \{1, \dots, n\}} B_k$ there exist a $k \in \{1, \dots, n\}$ such that

$$x \in B_k$$

Using [theorem: 21.290] it follows that

$$g_i = f(x) = g_k.$$

Let $y \in B_k$ then by [theorem: 21.290] $f(y) = g_k = g_i$ which proves that

$$f(B_k) = \{g_i\}$$

so that $k \in I_i$. Hence $B_k \subseteq \bigsqcup_{j \in I_i} B_j$ which as $x \in B_k$ proves that $x \in \bigsqcup_{j \in I_i} B_j$. So we have proves that

$$f^{-1}(\{g_i\}) \subseteq \bigsqcup_{j \in I_i} B_j \quad (21.422)$$

For the opposite inclusion let $x \in \bigsqcup_{j \in I_i} B_j$ then there exist a $j \in I_i$ such that $x \in B_j$. By the definition of I_i we have $f(B_j) = \{g_i\}$ so that $f(x) = g_i$ or $x \in f^{-1}(\{g_i\})$. Hence we have that $\bigsqcup_{j \in I_i} B_j \subseteq f^{-1}(\{g_i\})$ which combined with [eq: 21.422] gives

$$f^{-1}(\{g_i\}) = \bigsqcup_{j \in I_i} B_j \quad \square$$

Let $x \in X = \bigsqcup_{i \in \{1, \dots, n+1\}} B_i$ then there exist a $i \in \{1, \dots, n+1\}$ such that $x \in B_i$. Using [theorem: 21.290] it follows that $f(x) = g_i$ so that $x \in f^{-1}(\{g_i\}) \subseteq \bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(\{g_i\})$. Hence we have $X \subseteq \bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(\{g_i\})$ and as trivially $\bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(\{g_i\}) \subseteq X$ it follows that

$$X = \bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(\{g_i\}) \quad (21.423)$$

We are now ready to prove that f is $\mathcal{A}$ -measurable. Let $B \in \mathcal{B}[\overline{\mathbb{R}}]$ then as $f^{-1}(B) \subseteq X$ we have

$$\begin{aligned}
 f^{-1}(B) &= f^{-1}(B) \cap X \\
 &\stackrel{[\text{eq: 21.423}]}{=} f^{-1}(B) \cap \left(\bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(\{g_i\}) \right) \\
 &= \bigcup_{i \in \{1, \dots, n+1\}} (f^{-1}(B) \cap f^{-1}(\{g_i\})) \\
 &\stackrel{[\text{theorem: 2.103}]}{=} \bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(B \cap \{g_i\})
 \end{aligned} \tag{21.424}$$

Let $i \in \{1, \dots, n+1\}$ then for g_i we have either:

$g_i \in B$. Then $B \cap \{g_i\} = \{g_i\}$ so that $f^{-1}(B) = f^{-1}(\{g_i\}) \in \mathcal{A}$ [see eq: 21.421]

$g_i \notin B$. Then $f^{-1}(B \cap \{g_i\}) = f^{-1}(\emptyset) = \emptyset \in \mathcal{A}$

Hence $\forall i \in \{1, \dots, n+1\}$ $f^{-1}(B \cap \{g_i\}) \in \mathcal{A}$ so that $\bigcup_{i \in \{1, \dots, n+1\}} f^{-1}(B \cap \{g_i\}) \in \mathcal{A}$ which by [eq: 21.424] proves that $f^{-1}(B) \in \mathcal{A}$. As $B \in \mathcal{B}[\overline{\mathbb{R}}]$ was chosen arbitrary it follows by definition that

f is $\mathcal{A}$ -measurable $\square$

We show now that $\mathcal{S}[X, \mathcal{A}]$ is stable under the pointwise sum and scalar product.

Theorem 21.296. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then we have*

1. $\forall f \in \mathcal{S}[X, \mathcal{A}]$ and $\forall \alpha \in \mathbb{R}$ we have $\alpha \cdot f \in \mathcal{S}[X, \mathcal{A}]$
2. $\forall f, g \in \mathcal{S}[X, \mathcal{A}]$ we have $f + g \in \mathcal{S}[X, \mathcal{A}]$

Proof.

1. As $f \in \mathcal{S}[X, \mathcal{A}]$ we have that $f \in \mathcal{S}[X]$ and f is $\mathcal{A}$ -measurable. By [theorem: 21.291] we have $\alpha \cdot f \in \mathcal{S}[X]$ and by [theorem: 21.263] that $\alpha \cdot f$ is $\mathcal{A}$ -measurable. Hence $\alpha \cdot f \in \mathcal{S}[X, \mathcal{A}]$.
2. As $f, g \in \mathcal{S}[X, \mathcal{A}]$ we have that $f, g \in \mathcal{S}[X]$ and f, g are $\mathcal{A}$ -measurable. By [theorem: 21.291] we have $f + g \in \mathcal{S}[X]$. Further using [theorem: 21.257] it follows that $f + g$ is $\mathcal{A}$ -measurable. Hence $f + g \in \mathcal{S}[X, \mathcal{A}]$. $\square$

When defining integral we are only interested in non negative measurable simple functions hence the following definition.

Definition 21.297. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then the set of non negative $\mathcal{A}$ -measurable real valued simple functions is noted as $\mathcal{S}^+[X, \mathcal{A}]$ hence*

$$\mathcal{S}^+[X, \mathcal{A}] = \{f \in \mathcal{S}[X, \mathcal{A}] \mid f(X) \subseteq [0, \infty[\}$$

Note 21.298. *As by [definition: 21.292] $\mathcal{S}[X, \mathcal{A}] \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that*

$$\mathcal{S}^+[X, \mathcal{A}] \subseteq \mathcal{S}[X, \mathcal{A}] \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Note 21.299. *If $f \in \mathcal{S}^+[X, \mathcal{A}]$ then $f \in \mathcal{S}[X]$, $f(X) \subset [0, \infty[$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ which proves that*

$$\mathcal{S}^+[X, \mathcal{A}] \subseteq \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

A trivial example of a non negative measurable simple function is the constant function.

Example 21.300. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $a \in [0, \infty[$ then $C_a \in \mathcal{S}^+[X, \mathcal{A}]$ where

$$C_a: X \rightarrow \overline{\mathbb{R}} \text{ is defined by } C_a(x) = a$$

Proof. As $C_a(X) = \{a\}$, a finite set, we have that $C_a \in \mathcal{S}[X]$ and $C_a(X) \subseteq [0, \infty[$. Let $B \in \mathcal{B}[\mathbb{R}]$ then for a we have either:

$a \in B$. Then $(C_a)^{-1}(B) = \{x \in X \mid a = C_a(x) \in B\} = X \in \mathcal{A}$

$a \notin B$. Then $(C_a)^{-1}(B) = \{x \in X \mid a = C_a(x) \in B\} = \emptyset \in \mathcal{A}$

so that C_a is $\mathcal{A}$ -measurable. Hence it follows that $C_a \in \mathcal{S}^+[X, \mathcal{A}]$. $\square$

We have a simpler alternative definition for $\mathcal{S}^+[X, \mathcal{A}]$

Theorem 21.301. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then we have the following equivalences:*

1. $f \in \mathcal{S}^+[X, \mathcal{A}]$ [in other words $f \in \mathcal{S}[X]$, $f(X) \subseteq [0, \infty[$ and f is $\mathcal{A}$ -measurable].
2. $f \in \mathcal{S}[X]$, $f(X) \subseteq [0, \infty[$ and $\forall y \in f(X) \ f^{-1}(\{y\}) \in \mathcal{A}$
3. There exists a $n \in \mathbb{N}$, a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ of measurable sets and a family $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ of real non negative numbers so that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$$

Proof.

1 $\Rightarrow$ 2. As $f \in \mathcal{S}^+[X, \mathcal{A}]$ we have that $f \in \mathcal{S}[X, \mathcal{A}]$ and $f(X) \subseteq [0, \infty[$. Using [theorem: 21.295] it follows that $f \in \mathcal{S}[X]$ and $\forall y \in f(X) \ f^{-1}(\{y\}) \in \mathcal{A}$.

2 $\Rightarrow$ 3. As $f \in \mathcal{S}[X]$ and $\forall y \in f(X) \ f^{-1}(\{y\}) \in \mathcal{A}$ it follows from [theorem: 21.295] that there exist a $n \in \mathbb{N}$, a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ and a $\{f'_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that

$$f = \sum_{i=1}^n f'_i \cdot \mathcal{X}_{X, A_i}$$

Define

$$\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[\text{ by } f_i = \max(0, f'_i)$$

Consider now

$$g = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$$

Let $x \in X$ then we have either:

$x \in \bigcup_{i \in \{1, \dots, n\}} A_i$. Then there exist a $i \in \{1, \dots, n\}$ such that $x \in A_i$. Using [theorem: 21.290] we have that

$$f'_i = f(x) \wedge f_i = g(x)$$

Further as $f'_i = f(x) \subseteq f(X) \subseteq [0, \infty[$ it follows that $f'_i = \max(f'_i, 0) = f_i$ so that $f(x) = g(x)$.

$x \in X \setminus (\bigcup_{i \in \{1, \dots, n\}} A_i)$. Then using [theorem: 21.290] we have that $f(x) = 0 = g(x)$

As in all cases we have $f(x) = g(x)$ it follows that $f = g$ hence we have that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i} \text{ where } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[\text{ and } \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ is pairwise disjoint}$$

3 $\Rightarrow$ 1. By the hypothesis we have that $f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$ where $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ is pairwise disjoint and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$. Using [theorem: 21.295] it follows that $f \in \mathcal{S}[X, \mathcal{A}]$. Let $x \in X$ then we have

$$f(x) \stackrel{[\text{theorem: 21.290}]}{=} \begin{cases} f_i \in [0, \infty[\text{ if } x \in \bigcup_{i \in \{1, \dots, n\}} A_i \\ 0 \in [0, \infty[\text{ if } x \in X \setminus \bigcup_{i \in \{1, \dots, n\}} A_i \end{cases} \in [0, \infty[$$

which proves that $f(X) \subseteq [0, \infty[$. Hence we have

$$f \in \mathcal{S}^+[X, \mathcal{A}]$$

$\square$

Example 21.302. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$, $a, b \in [0, \infty[$ then for the function

$$f: X \rightarrow \overline{\mathbb{R}} \text{ where } f(x) = \begin{cases} a & \text{if } x \in A \\ b & \text{if } x \in X \setminus A \end{cases}$$

we have that

$$f \in \mathcal{S}^+[X, \mathcal{A}]$$

Proof. As $f(X) = \{a, b\}$ a finite set it follows that $f \in \mathcal{S}[X]$ and $f(X) \subseteq [0, \infty[$. Further for a, b we have either:

$a = b$. Then $f^{-1}(\{a\}) = f^{-1}(\{b\}) = X \in \mathcal{A}$.

$a \neq b$. Then, as $\forall x \in A$ we have $f(x) = a \in \{a\}$, $A \subseteq f^{-1}(\{a\})$. If $x \in f^{-1}(\{a\})$ then $f(x) = a$ and we must have that $x \in A$ [if $x \in X \setminus A$ then $f(x) = b \neq a$] so that $f^{-1}(\{a\}) \subseteq A$. Hence we have

$$f^{-1}(\{a\}) = A \in \mathcal{A}$$

Likewise we have $\forall x \in X \setminus A$ that $f(x) = b \in \{b\}$ proving that $X \setminus A \subseteq f^{-1}(\{b\})$. If $x \in f^{-1}(\{b\})$ then $f(x) = b$ and we must have that $x \in X \setminus A$ [if $x \in A$ then $f(x) = a \neq b$] so that $f^{-1}(\{b\}) \subseteq X \setminus A$. Hence we have

$$f^{-1}(\{b\}) = X \setminus A \in \mathcal{A}$$

So in all cases we have that $f^{-1}(\{a\}), f^{-1}(\{b\}) \in \mathcal{A}$ which by [theorem: 21.301] proves that

$$f \in \mathcal{S}^+[X, \mathcal{A}]$$

□

Given a non negative measurable simple function $f = \sum_{i=1}^n \alpha_i \cdot \mathcal{X}_{X, A_i}$ we want to define the integral of f as $\int f = \sum_{i=1}^n \alpha_i \cdot \mu(A_i)$. However the same function f can have another representation $f = \sum_{i=1}^m \beta_i \cdot \mathcal{X}_{X, B_i}$, so we must show that the integral is independent of the chosen representation of a simple function.

Theorem 21.303. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{S}^+[X, \mathcal{A}]$ then if $n, m \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint, $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{A}$ pairwise disjoint, $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ and $\{b_i\}_{i \in \{1, \dots, m\}} \subseteq [0, \infty[$ satisfies

$$\sum_{i=1}^n a_i \cdot \mathcal{X}_{X, A_i} = f = \sum_{i=1}^m b_i \cdot \mathcal{X}_{X, B_i}$$

then we have

$$\sum_{i=1}^n a_i \cdot \mu(A_i) = \sum_{i=1}^m b_i \cdot \mu(B_i)$$

Note 21.304. The condition that $f \in \mathcal{S}^+[X, \mathcal{A}]$ is needed because this ensures that $\{a_i\}_{i \in \{1, \dots, n\}}$, $\{b_i\}_{i \in \{1, \dots, m\}}$ are non negative so that $\sum_{i=1}^n a_i \cdot \mu(A_i)$ and $\sum_{i=1}^m b_i \cdot \mu(B_i)$ are well defined [as $\mu(A_i), \mu(B_i)$ can be ∞].

Proof. First we remove all the terms where $a_i \cdot \mu(A_i) = 0$ and $b_i \cdot \mu(B_i) = 0$, this will not change the sum. Define $I = \{i \in \{1, \dots, n\} \mid A_i \neq \emptyset \wedge a_i \neq 0\}$ and $J = \{i \in \{1, \dots, m\} \mid B_i \neq \emptyset \wedge b_i \neq 0\}$. Then $\forall i \in \{1, \dots, n\} \setminus I$ we have $a_i \cdot \mu(A_i) \stackrel{a_i=0 \vee A_i=\emptyset}{=} 0$ and $\forall i \in \{1, \dots, m\} \setminus J$ $b_i \cdot \mu(B_i) \stackrel{b_i=0 \vee B_i=\emptyset}{=} 0$ so that

$$\begin{aligned} \left(\sum_{i=1}^n a_i \cdot \mathcal{X}_{X, A_i} \right)(x) &= \sum_{i=1}^n a_i \cdot \mathcal{X}_{X, A_i}(x) \\ &= \sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i}(x) + \sum_{i \in \{1, \dots, n\} \setminus I} a_i \cdot \mathcal{X}_{X, A_i}(x) \\ &= \sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i}(x) + 0 \\ &= \sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i}(x) \end{aligned}$$

and

$$\begin{aligned}
 \left(\sum_{i=1}^m b_i \cdot \mathcal{X}_{X, B_i} \right)(x) &= \sum_{i=1}^m b_i \cdot \mathcal{X}_{X, B_i}(x) \\
 &= \sum_{i \in J} b_i \cdot \mathcal{X}_{X, B_i}(x) + \sum_{i \in \{1, \dots, m\} \setminus J} b_i \cdot \mathcal{X}_{X, B_i}(x) \\
 &= \sum_{i \in J} b_i \cdot \mathcal{X}_{X, B_i}(x) + 0 \\
 &= \sum_{i \in J} b_i \cdot \mathcal{X}_{X, B_i}(x)
 \end{aligned}$$

proving that

$$\sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i} = f = \sum_{i \in J} b_i \cdot \mathcal{X}_{X, B_i} \quad (21.425)$$

As $\{A_i\}_{i \in \{1, \dots, n\}}$ and $\{B_i\}_{i \in \{1, \dots, m\}}$ are pairwise disjoint we have

$$\{A_i\}_{i \in I} \subseteq \mathcal{A} \text{ and } \{B_i\}_{i \in J} \subseteq \mathcal{A} \text{ are pairwise disjoint} \quad (21.426)$$

Further if $x \in \bigsqcup_{i \in I} A_i$ then there exist a $i \in I$ such that $x \in A_i$, so that by [theorem: 21.290] $f(x) = a_i \neq 0$. Assume that $x \notin \bigsqcup_{i \in J} B_i$ then $f(x) = \sum_{i \in J} b_i \cdot \mathcal{X}_{X, B_i}(x) = \sum_{i \in J} b_i \cdot 0 = 0$ contradicting $f(x) \neq 0$. Hence the assumption is wrong and we must have that $x \in \bigsqcup_{i \in J} B_i$. So

$$\bigsqcup_{i \in I} A_i \subseteq \bigsqcup_{i \in J} B_i \quad (21.427)$$

Likewise if $x \in \bigsqcup_{i \in J} B_i$ then there exist a $i \in J$ such that $x \in B_i$, so that by [theorem: 21.290] $f(x) = b_i \neq 0$. Assume that $x \notin \bigsqcup_{i \in I} A_i$ then $f(x) = \sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i}(x) = \sum_{i \in I} a_i \cdot 0 = 0$ contradicting $f(x) \neq 0$. Hence the assumption is wrong and we must have that $x \in \bigsqcup_{i \in I} A_i$. So $\bigsqcup_{i \in J} B_i \subseteq \bigsqcup_{i \in I} A_i$, which combined with [eq: 21.427] gives

$$\bigsqcup_{i \in I} A_i = \bigsqcup_{i \in J} B_i \quad (21.428)$$

Let $i \in \{1, \dots, n\}$ then, as $\{B_j\}_{j \in J}$ are pairwise disjoint, $\{A_i \cap B_j\}_{j \in J}$ are pairwise disjoint and

$$A_i = A_i \cap \left(\bigsqcup_{j \in J} B_j \right) \stackrel{[\text{eq: 21.428}]}{=} A_i \cap \left(\bigsqcup_{j \in J} B_j \right) = \bigsqcup_{j \in J} (A_i \cap B_j) \quad (21.429)$$

Let $i \in \{1, \dots, m\}$ then, as $\{A_j\}_{j \in I}$ are pairwise disjoint $\{A_j \cap B_i\}_{j \in I}$ are pairwise disjoint and

$$B_i = B_i \cap \left(\bigsqcup_{j \in I} A_j \right) \stackrel{[\text{eq: 21.428}]}{=} B_i \cap \left(\bigsqcup_{j \in I} A_j \right) = \bigsqcup_{j \in I} (A_j \cap B_i) \quad (21.430)$$

Define $K = \{(i, j) \in I \times J \mid A_i \cap B_j \neq \emptyset\}$ then if $(i, j) \in K$ we have $A_i \cap B_j \neq \emptyset$ so there exist a $x \in A_i \cap B_j$ and

$$a_i \stackrel{[\text{theorem: 21.290}]}{=} f(x) \stackrel{[\text{theorem: 21.290}]}{=} b_j$$

In other words

$$\forall (i, j) \in K \text{ we have } a_i = b_j \quad (21.431)$$

Now

$$\begin{aligned}
\sum_{i=1}^n a_i \cdot \mu(A_i) &= \sum_{i \in I} a_i \cdot \mu(A_i) + \sum_{i \in \{1, \dots, n\} \setminus I} a_i \cdot \mu(A_i) \\
&= \sum_{i \in I} a_i \cdot \mu(A_i) + \sum_{i \in \{1, \dots, n\} \setminus I} 0 \cdot \mu(A_i) \\
&= \sum_{i \in I} a_i \cdot \mu(A_i) \\
&\stackrel{[\text{eq: 21.429}]}{=} \sum_{i \in I} a_i \cdot \mu\left(\bigsqcup_{j \in J} (A_i \cap B_j)\right) \\
&= \sum_{i \in I} \left(a_i \cdot \sum_{j \in J} \mu(A_i \cap B_j)\right) \\
&= \sum_{i \in I} \left(\sum_{j \in J} a_i \cdot \mu(A_i \cap B_j)\right) \\
&\stackrel{[\text{theorem: 11.45}]}{=} \sum_{(i,j) \in I \times J} a_i \cdot \mu(A_i \cap B_j) \\
&= \sum_{(i,j) \in K} a_i \cdot \mu(A_i \cap B_j) + \sum_{(i,j) \in (I \times J) \setminus K} a_i \cdot \mu(A_i \cap B_j) \\
&= \sum_{(i,j) \in K} a_i \cdot \mu(A_i \cap B_j) + \sum_{(i,j) \in (I \times J) \setminus K} a_i \cdot 0 \\
&= \sum_{(i,j) \in K} a_i \cdot \mu(A_i \cap B_j) \\
&\stackrel{[\text{eq: 21.431}]}{=} \sum_{(i,j) \in K} b_j \cdot \mu(A_i \cap B_j) \\
&= \sum_{(i,j) \in K} b_j \cdot \mu(A_i \cap B_j) + \sum_{(i,j) \in (I \times J) \setminus K} b_j \cdot 0 \\
&= \sum_{(i,j) \in K} b_j \cdot \mu(A_i \cap B_j) + \sum_{(i,j) \in (I \times J) \setminus K} b_j \cdot \mu(A_i \cap B_j) \\
&= \sum_{(i,j) \in I \times J} b_j \cdot \mu(A_i \cap B_j) \\
&\stackrel{[\text{theorem: 11.45}]}{=} \sum_{j \in J} \left(\sum_{i \in I} (b_j \cdot \mu(A_i \cap B_j))\right) \\
&= \sum_{j \in J} \left(b_j \cdot \sum_{i \in I} \mu(A_i \cap B_j)\right) \\
&= \sum_{j \in J} b_j \cdot \mu\left(\bigsqcup_{i \in I} (A_i \cap B_j)\right) \\
&\stackrel{[\text{eq: 21.430}]}{=} \sum_{j \in J} b_j \cdot \mu(B_j) \\
&= \sum_{j \in J} b_j \cdot \mu(B_j) + \sum_{j \in \{1, \dots, m\} \setminus J} b_j \cdot \mu(B_j) \\
&= \sum_{j=1}^m b_j \cdot \mu(B_j)
\end{aligned}$$

Hence we have

$$\sum_{i=1}^n a_i \cdot \mu(A_i) = \sum_{j=1}^m b_j \cdot \mu(B_j)$$

□

Thanks to the two previous theorems [theorems: 21.301, 21.303] the following definition is well defined [existence and uniqueness].

Definition 21.305. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{S}^+[X, \mathcal{A}]$ then we define $\int^S f d\mu$ by

$$\int^S f d\mu = \sum_{i=1}^n a_i \cdot \mu(A_i) \in [0, \infty[$$

where

$$f = \sum_{i=1}^n a_i \cdot \chi_{X, A_i}, \quad n \in \mathbb{N}, \quad \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ is pairwise disjoint and } \{a_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$$

is a representation of f .

Note 21.306. For the moment we use the special symbol $\int^S f d\mu$ to differentiate the integral of a simple function from the more general integral $\int f d\mu$ that will be based on the integral of simple functions.

Example 21.307. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $a \in [0, \infty[$ and $C_a: X \rightarrow \overline{\mathbb{R}}$ defined by $C_a(x) = a$ then $C_a \in \mathcal{S}^+[X, \mathcal{A}]$ and $\int^S C_a d\mu = a \cdot \mu(X)$.

Proof. Define $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A}$ by $A_1 = X$ and $\{a_i\}_{i \in \{1\}} \subseteq [0, \infty[$ by $a_1 = a$ then $\forall x \in X$ we have $(\sum_{i=1}^1 a_i \cdot \chi_{X, A_i})(x) = \sum_{i=1}^1 a_i \cdot \chi_{X, A_i}(x) = a \cdot \chi_X(x) = a = C_a(x)$ so that $C_a = \sum_{i=1}^1 a_i \cdot \chi_{X, A_i}$. Hence by [theorem: 21.301]

$$C_a \in \mathcal{S}^+[X, \mathcal{A}]$$

and using [definition: 21.305]

$$\int^C C_a d\mu = \sum_{i=1}^1 a_i \cdot \mu(A_i) = a_1 \cdot \mu(A_1) = a \cdot \mu(X) \quad \square$$

The following lemma will be useful if we have to compare or add simple functions.

Lemma 21.308. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $f, g \in \mathcal{S}^+[X, \mathcal{A}]$ then there exist a $n \in \mathbb{N}$, a pairwise disjoint family $\{C_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ of sets in $\mathcal{A}$, a family $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ and a family $\{g_i\}_{i \in \{1, \dots, n\}}$ such that

$$X = \bigsqcup_{i \in \{1, \dots, n\}} C_i$$

$$f = \sum_{i=1}^n f_i \cdot \chi_{X, C_i} \text{ and } g = \sum_{i=1}^n g_i \cdot \chi_{X, C_i}$$

Proof. As $f, g \in \mathcal{S}^+[X, \mathcal{A}]$ we have by [theorem: 21.301] that there exists $n', m' \in \mathbb{N}$, $\{A'_i\}_{i \in \{1, \dots, n'\}} \subseteq \mathcal{A}$ pairwise disjoint, $\{B'_i\}_{i \in \{1, \dots, m'\}} \subseteq \mathcal{A}$ pairwise disjoint, $\{a'_i\}_{i \in \{1, \dots, n'\}} \subseteq [0, \infty[$ and $\{b'_i\}_{i \in \{1, \dots, m'\}} \subseteq [0, \infty[$ so that

$$f = \sum_{i=1}^{n'} a'_i \cdot \chi_{X, A'_i} \text{ and } g = \sum_{i=1}^{m'} b'_i \cdot \chi_{X, B'_i} \quad (21.432)$$

Define now

$$\{A_i\}_{i \in \{1, \dots, n'+1\}} \subseteq \mathcal{A} \text{ by } A_i = \begin{cases} X \setminus \bigcup_{i \in \{1, \dots, n'\}} A'_i & \text{if } i = n' + 1 \\ A'_i & \text{if } i \in \{1, \dots, n'\} \end{cases}$$

$$\{B_i\}_{i \in \{1, \dots, m'+1\}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} X \setminus \bigcup_{i \in \{1, \dots, m'\}} B'_i & \text{if } i = m' + 1 \\ B'_i & \text{if } i \in \{1, \dots, m'\} \end{cases}$$

$$\{a_i\}_{i \in \{1, \dots, n'+1\}} \text{ by } a_i = \begin{cases} 0 & \text{if } i = n' + 1 \\ a'_i & \text{if } i \in \{1, \dots, n'\} \end{cases}$$

$$\{b_i\}_{i \in \{1, \dots, m'+1\}} \text{ by } b_i = \begin{cases} 0 & \text{if } i = m' + 1 \\ b'_i & \text{if } i \in \{1, \dots, m'\} \end{cases}$$

then by [lemma: 21.294] $\{A_i\}_{i \in \{1, \dots, n'+1\}}$, $\{B_i\}_{i \in \{1, \dots, m'+1\}}$ are pairwise disjoint and

$$X = \bigsqcup_{i \in \{1, \dots, n'+1\}} A_i = \bigsqcup_{i \in \{1, \dots, m'+1\}} B_i \quad (21.433)$$

Hence

$$\forall i \in \{1, \dots, n'+1\} \quad A_i = A_i \cap X = A_i \cap \left(\bigsqcup_{j \in \{1, \dots, m'+1\}} B_j \right) = \bigsqcup_{j \in \{1, \dots, m'+1\}} (A_i \cap B_j) \quad (21.434)$$

and

$$\forall i \in \{1, \dots, m'+1\} \quad B_i = B_i \cap X = B_i \cap \left(\bigsqcup_{j \in \{1, \dots, n'+1\}} A_j \right) = \bigsqcup_{j \in \{1, \dots, n'+1\}} (A_j \cap B_i) \quad (21.435)$$

so that

$$X = \bigsqcup_{i=1}^{n'+1} \left(\bigsqcup_{j=1}^{m'+1} (A_i \cap B_j) \right) = \bigsqcup_{i=1}^{m'+1} \left(\bigsqcup_{j=1}^{n'+1} (A_j \cap B_i) \right) \quad (21.436)$$

Using [theorem: 6.41] $\{1, \dots, n'+1\} \times \{1, \dots, m'+1\}$ is finite and there exists a bijection

$$\beta: \{1, \dots, n\} \rightarrow \{1, \dots, n'+1\} \times \{1, \dots, m'+1\} \text{ where } n = (n'+1) \cdot (m'+1)$$

Define now

$$\begin{aligned} \{C_i\}_{i \in \{1, \dots, n\}} &\subseteq A \quad \text{by } C_i = A_{\beta(i)_1} \cap B_{\beta(i)_2} \\ \{f_i\}_{i \in \{1, \dots, n\}} &\quad \text{by } f_i = a_{\beta(i)_1} \\ \{g_i\}_{i \in \{1, \dots, n\}} &\quad \text{by } g_i = b_{\beta(i)_2} \end{aligned}$$

Let $x \in X$ then by [eq: 21.436] there exists a $i \in \{1, \dots, n'+1\}$ such that $x \in \bigsqcup_{j \in \{1, \dots, m'+1\}} (A_i \cap B_j)$, hence there exist a $j \in \{1, \dots, m'+1\}$ such that $x \in A_i \cap B_j$. As β is a bijection there exist a $k \in \{1, \dots, n\}$ such that $\beta(k) = (i, j)$ hence $x \in A_{\beta(k)_1} \cap B_{\beta(k)_2} = C_k \subseteq \bigcup_{k \in \{1, \dots, n\}} C_k$. So $X \subseteq \bigcup_{k \in \{1, \dots, n\}} C_k$ and as trivially $\bigcup_{k \in \{1, \dots, n\}} C_k \subseteq X$ it follows that

$$X = \bigcup_{i \in \{1, \dots, n\}} C_i \quad (21.437)$$

Next we prove that $\{C_i\}_{i \in \{1, \dots, n\}}$ is pairwise disjoint. So let $i, j \in \{1, \dots, n\}$ with $i \neq j$ then as β is a bijection we have that $\beta(i) \neq \beta(j)$, for $\beta(i)_1, \beta(j)_1$ we have then either:

$\beta(i)_1 = \beta(j)_1$. Then we must have that $\beta(i)_2 \neq \beta(j)_2$ so that

$$C_i \cap C_j = (A_{\beta(i)_1} \cap B_{\beta(i)_2}) \cap (A_{\beta(j)_1} \cap B_{\beta(j)_2}) \subseteq B_{\beta(i)_2} \cap B_{\beta(j)_2} = \emptyset$$

$\beta(i)_1 \neq \beta(j)_1$. Then we have that

$$C_i \cap C_j = (A_{\beta(i)_1} \cap B_{\beta(i)_2}) \cap (A_{\beta(j)_1} \cap B_{\beta(j)_2}) \subseteq A_{\beta(i)_1} \cap A_{\beta(j)_1} = \emptyset$$

which proves that

$$\{C_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ is pairwise disjoint and } X = \bigsqcup_{i \in \{1, \dots, n\}} C_i \quad (21.438)$$

Let $x \in X$ then by [eq: 21.438] there exist a $i \in \{1, \dots, n\}$ such that $x \in C_i = A_{\beta(i)_1} \cap B_{\beta(i)_2}$ hence $x \in A_{\beta(i)_1}$. For $\beta(i)_1$ we have the following possibilities:

$\beta(i)_1 = n' + 1$. Then $x \in A_{n'+1} = X \setminus (\bigsqcup_{i \in \{1, \dots, n'\}} A'_i)$ so that by [theorem: 21.290] and [eq: 21.432] $f(x) = 0$. Further using [theorem: 21.290] and [eq: 21.438] it follows also that

$$\left(\sum_{i=1}^n f_i \cdot \mathcal{X}_{C_i} \right)(x) = f_i = a_{\beta(i)_1} = a_{n'+1} = 0 = f(x)$$

which proves that $f(x) = (\sum_{i=1}^n f_i \cdot \mathcal{X}_{C_i})(x)$.

$\beta(i)_1 \in \{1, \dots, n'\}$. Then $x \in A_{\beta(i)_1} = A'_{\beta(i)_1}$ so that by [theorem: 21.290] and [eq: 21.432] $f(x) = a'_{\beta(i)_1}$. Further using [theorem: 21.290] and [eq: 21.438] we have that

$$\left(\sum_{i=1}^n f_i \cdot \mathcal{X}_{X, C_i} \right)(x) = f_i = a_{\beta(i)_1} = a'_{\beta(i)_1} = f(x)$$

so that in this cases we have also that $f(x) = (\sum_{i=1}^n f_i \cdot \mathcal{X}_{X, C_i})(x)$.

To summarize we have proved that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, C_i}$$

We use now the same technique for g . Let $x \in X$ then by [eq: 21.438] there exist a $i \in \{1, \dots, n\}$ such that $x \in C_i = A_{\beta(i)_1} \cap B_{\beta(i)_2}$ hence $x \in B_{\beta(i)_2}$. For $\beta(i)_2$ we have the following possibilities:

$\beta(i)_2 = m' + 1$. Then $x \in B_{n'+1} = X \setminus (\bigcup_{i \in \{1, \dots, n'\}} B_i)$ so that by [theorem: 21.290] and [eq: 21.432] $g(x) = 0$. Further using [theorem: 21.290] and [eq: 21.438] it follows also that

$$\left(\sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i} \right)(x) = g_i = b_{\beta(i)_2} = b_{m'+1} = 0 = g(x)$$

which proves that $g(x) = (\sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i})(x)$.

$\beta(i)_2 \in \{1, \dots, m'\}$. Then $x \in B_{\beta(i)_2} = B'_{\beta(i)_2}$ so that by [theorem: 21.290] and [eq: 21.432] $g(x) = b'_{\beta(i)_2}$. Further using [theorem: 21.290] and [eq: 21.438] we have that

$$\left(\sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i} \right)(x) = g_i = b_{\beta(i)_2} = b'_{\beta(i)_2} = g(x)$$

so that in this cases we have also that $g(x) = (\sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i})(x)$.

To summarize we have proved that

$$g = \sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i}$$

□

We prove now that the integral of non negative measurable simple functions has the usual properties of integration.

Theorem 21.309. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have*

1. If $\alpha \in [0, \infty[$ and $f \in \mathcal{S}^+[X, \mathcal{A}]$ then $\alpha \cdot f \in \mathcal{S}^+[X, \mathcal{A}]$ and $\int^S (\alpha \cdot f) d\mu = \alpha \cdot \int^S f d\mu$.
2. If $f, g \in \mathcal{S}^+[X, \mathcal{A}]$ then
 - a. $f + g \in \mathcal{S}^+[X, \mathcal{A}]$ and $\int^S (f + g) d\mu = \int^S f d\mu + \int^S g d\mu$.
 - b. If $\forall x \in X$ $f(x) \leq g(x)$ then $\int^S f d\mu \leq \int^S g d\mu$.
 - c. $f \wedge g \in \mathcal{S}^+[X, \mathcal{A}]$, $\int^S (f \wedge g) d\mu \leq \int^S f d\mu$ and $\int^S (f \wedge g) d\mu \leq \int^S g d\mu$.
 - d. $f \vee g \in \mathcal{S}^+[X, \mathcal{A}]$, $\int^S f d\mu \leq \int^S (f \vee g) d\mu$ and $\int^S g d\mu \leq \int^S (f \vee g) d\mu$.
3. If $f \in \mathcal{S}^+[X, \mathcal{A}]$ satisfies $f = 0$ μ -a.e. then $\int^S f d\mu = 0$.

Proof.

1. As $f \in \mathcal{S}^+[X, \mathcal{A}]$ there exists by [theorem: 21.301] a pairwise disjoint $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ and a $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ such that $f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$. Further we have by definition that

$$\int^S f d\mu = \sum_{i=1}^n f_i \cdot \mu(A_i)$$

Then $\{\alpha \cdot a_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ and $\forall x \in X$ we have

$$\begin{aligned} (\alpha \cdot f)(x) &= \alpha \cdot f(x) \\ &= \alpha \cdot \sum_{i=1}^n a_i \cdot \mathcal{X}_{X, A_i}(x) \\ &= \sum_{i=1}^n (\alpha \cdot a_i) \cdot \mathcal{X}_{X, A_i}(x) \\ &= \left(\sum_{i=1}^n (\alpha \cdot a_i) \cdot \mathcal{X}_{X, A_i} \right)(x) \end{aligned}$$

proving that

$$\alpha \cdot f = \sum_{i=1}^n (\alpha \cdot a_i) \cdot \mathcal{X}_{X, A_i}$$

Using [theorem: 21.301] it follows that

$$\alpha \cdot f \in \mathcal{S}^+[X, \mathcal{A}]$$

Further we have

$$\begin{aligned} \int^S \alpha \cdot f d\mu &= \sum_{i=1}^n (\alpha \cdot f_i) \cdot \mu(A_i) \\ &= \alpha \cdot \sum_{i=1}^n f_i \cdot \mu(A_i) \\ &= \alpha \cdot \int^S f d\mu \end{aligned}$$

2. As $f, g \in \mathcal{S}^+[X, \mathcal{A}]$ there exists by [lemma: 21.308] a $\{C_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint, $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ and $\{g_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ such that

$$X = \bigsqcup_{i \in \{1, \dots, n\}} A_i, \quad f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, C_i} \text{ and } g = \sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i} \quad (21.439)$$

By the definition of $\int^S$ we have also

$$\int^S f d\mu = \sum_{i=1}^n f_i \cdot \mu(C_i) \text{ and } \int^S g d\mu = \sum_{i=1}^n g_i \cdot \mu(C_i) \quad (21.440)$$

Next we have:

- a. $\{f_i + g_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ and

$$\begin{aligned} (f + g)(x) &= f(x) + g(x) \\ &= \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, C_i}(x) + \sum_{i=1}^n g_i \cdot \mathcal{X}_{X, C_i}(x) \\ &= \sum_{i=1}^n (f_i \cdot \mathcal{X}_{X, C_i}(x) + g_i \cdot \mathcal{X}_{X, C_i}(x)) \\ &= \sum_{i=1}^n (f_i + g_i) \cdot \mathcal{X}_{X, C_i}(x) \\ &= \left(\sum_{i=1}^n (f_i + g_i) \cdot \mathcal{X}_{X, C_i} \right)(x) \end{aligned}$$

which proves that

$$f + g = \sum_{i=1}^n (f_i + g_i) \cdot \mathcal{X}_{X, C_i}$$

So using [theorem: 21.301] it follows that

$$f + g \in \mathcal{S}^+[X, \mathcal{A}]$$

and using the definition of the integral

$$\begin{aligned} \int^S (f + g) d\mu &= \sum_{i=1}^n (f_i + g_i) \cdot \mu(C_i) \\ &= \sum_{i=1}^n f_i \cdot \mu(C_i) + \sum_{i=1}^n g_i \cdot \mu(C_i) \\ &\stackrel{[\text{eq: 21.440}]}{=} \int^S f d\mu + \int^S g d\mu \end{aligned}$$

b. Assume that $\forall x \in X f(x) \leq g(x)$. Define $I = \{i \in \{1, \dots, n\} \mid C_i \neq \emptyset\} \subseteq \{1, \dots, n\}$ then $\forall i \in I$ there exist a $C_i \neq \emptyset \Rightarrow \exists x \in C_i$ so that $f_i \stackrel{[\text{theorem: 21.290}]}{=} f(x) \leq g(x) \stackrel{[\text{theorem: 21.290}]}{=} g_i$. Hence we have

$$\forall i \in I f_i \leq g_i \quad (21.441)$$

Next

$$\begin{aligned} \int^S f d\mu &= \sum_{i=1}^n f_i \cdot \mu(C_i) \\ &= \sum_{i \in I} f_i \cdot \mu(C_i) + \sum_{i \in \{1, \dots, n\} \setminus I} f_i \cdot \mu(C_i) \\ &= \sum_{i \in I} f_i \cdot \mu(C_i) \\ &\leq \sum_{i \in I} g_i \cdot \mu(C_i) \\ &\stackrel{[\text{eq: 21.441}]}{=} \sum_{i \in I} g_i \cdot \mu(C_i) + \sum_{i \in \{1, \dots, n\} \setminus I} g_i \cdot \mu(C_i) \\ &= \sum_{i=1}^n g_i \cdot \mu(C_i) \\ &= \int^S g d\mu \end{aligned}$$

c. Let $x \in X$ then as $X \stackrel{[\text{eq: 21.439}]}{=} \bigsqcup_{i \in \{1, \dots, n\}} C_i$ there exist a $i \in \{1, \dots, n\}$ such that $x \in C_i$. Using [eq: 21.439] together with [theorem: 21.290] gives then $f(x) = f_i$ and $g(x) = g_i$. So

$$\begin{aligned} (f \wedge g)(x) &= \min(f(x), g(x)) \\ &= \min(f_i, g_i) \\ &\stackrel{[\text{theorem: 21.290}]}{=} \left(\sum_{i=1}^n \min(f_i, g_i) \cdot \mathcal{X}_{C_i} \right)(x) \end{aligned}$$

which proves that

$$(f \wedge g) = \sum_{i=1}^n \min(f_i, g_i) \cdot \mathcal{X}_{C_i}$$

and, as $\{\min(f_i, g_i)\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$, it follows from [theorem: 21.301] that

$$f \wedge g \in \mathcal{S}^+[X, \mathcal{A}]$$

Further as $\forall x \in X (f \wedge g)(x) = \min(f(x), g(x)) \leq f(x), g(x)$ it follows from (b) that

$$\int^S (f \wedge g) d\mu \leq \int^S f d\mu \text{ and } \int^S (f \wedge g) d\mu \leq \int^S g d\mu$$

- d. Let $x \in X$ then as $X \stackrel{[\text{eq: 21.439}]}{=} \bigsqcup_{i \in \{1, \dots, n\}} C_i$ there exist a $i \in \{1, \dots, n\}$ such that $x \in C_i$. Using [eq: 21.439] together with [theorem: 21.290] gives then $f(x) = f_i$ and $g(x) = g_i$. So

$$\begin{aligned} (f \vee g)(x) &= \max(f(x), g(x)) \\ &= \max(f_i, g_i) \\ &\stackrel{[\text{theorem: 21.290}]}{=} \left(\sum_{i=1}^n \max(f_i, g_i) \cdot \mathcal{X}_{X, C_i} \right)(x) \end{aligned}$$

which proves that

$$(f \vee g) = \sum_{i=1}^n \max(f_i, g_i) \cdot \mathcal{X}_{X, C_i}$$

and, as $\{\max(f_i, g_i)\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ it follows from [theorem: 21.301] that

$$f \vee g \in \mathcal{S}^+[X, \mathcal{A}]$$

Further as $\forall x \in X f(x), g(x) \leq \max(f(x), g(x)) = f \vee g$ it follows from (b) that

$$\int^S f d\mu \leq \int^S (f \vee g) d\mu \text{ and } \int^S g d\mu \leq \int^S (f \vee g) d\mu$$

3. As $f = 0$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in X \mid 0 \neq f(x)\} \subseteq N \quad (21.442)$$

As $f \in \mathcal{S}^+[X, \mathcal{A}]$ there exists by [theorem: 21.301] a $n \in \mathbb{N}$, a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ and a $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ such that $f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X, A_i}$. Let $i \in \{1, \dots, n\}$ then for f_i we have either:

$f_i = \mathbf{0}$. Then $f_i \cdot \mu(A_i) = 0$

$\mathbf{0} < f_i$. Then $\forall x \in A_i$ we have $f(x) \stackrel{[\text{theorem: 21.290}]}{=} a_i \neq 0$ so that $x \in \{x \in X \mid 0 \neq f(x)\}$.

Hence $A_i \subseteq \{x \in X \mid 0 \neq f(x)\} \stackrel{[\text{eq: 21.442}]}{\subseteq} N$. So $\mu(A_i) \stackrel{[\text{theorem: 21.99}]}{\leq} \mu(N) = 0$ from which

it follows that $f \cdot \mu(A_i) = 0$

So $\forall i \in \{1, \dots, n\}$ we have that $f_i \cdot \mu(A_i) = 0$ proving that

$$\int^S f d\mu = \sum_{i=1}^n f_i \cdot \mu(A_i) = 0 \quad \square$$

Using mathematical induction we have the following extension of the above theorem.

Corollary 21.310. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $n \in \mathbb{N}$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then we have that

1. $\sum_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}]$
2. $\bigwedge_{i \in \{1, \dots, n\}} f_i \in \mathcal{S}^+[X, \mathcal{A}]$ where
$$\bigwedge_{i \in \{1, \dots, n\}} f_i: X \rightarrow \overline{\mathbb{R}} \text{ is defined by } \left(\bigwedge_{i \in \{1, \dots, n\}} f_i \right)(x) = \min(\{f_i(x) \mid i \in \{1, \dots, n\}\})$$
3. $\bigvee_{i \in \{1, \dots, n\}} f_i \in \mathcal{S}^+[X, \mathcal{A}]$ where
$$\bigvee_{i \in \{1, \dots, n\}} f_i: X \rightarrow \overline{\mathbb{R}} \text{ is defined by } \left(\bigvee_{i \in \{1, \dots, n\}} f_i \right)(x) = \max(\{f_i(x) \mid i \in \{1, \dots, n\}\})$$

Proof.

1. Let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \text{ then } \sum_{i=1}^n f_i \in \mathcal{S}[X, \mathcal{A}] \right\}$$

then we have:

$\mathbf{1} \in \mathcal{S}$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then $\forall x \in X$ $(\sum_{i=1}^1 f_i)(x) = \sum_{i=1}^1 f_i(x) = f_1(x)$ so that $\sum_{i=1}^1 f_i = f_1 \in \mathcal{S}^+[X, \mathcal{A}]$ which proves that $\mathbf{1} \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. If $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then we have $\forall x \in X$

$$\begin{aligned} \left(\sum_{i=1}^{n+1} f_i \right)(x) &= \sum_{i=1}^{n+1} f_i(x) \\ &= f_{n+1}(x) + \sum_{i=1}^n f_i(x) \\ &= \left(f_{n+1} + \sum_{i=1}^n f_i \right)(x) \end{aligned}$$

As $n \in \mathcal{S}$ we have that $\sum_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}]$ which as $f_{n+1} \in \mathcal{S}^+[X, \mathcal{A}]$ proves by [theorem: 21.309] that $f_{n+1} + \sum_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}]$. Hence using the above

$$\sum_{i=1}^{n+1} f_i \in \mathcal{S}^+[X, \mathcal{A}]$$

which proves that $n+1 \in \mathcal{S}$.

2. Let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \text{ then } \bigwedge_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}] \right\}$$

then we have:

$\mathbf{1} \in \mathcal{S}$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then $(\bigwedge_{i=1}^1 f_i)(x) = \min(\{f_1(x)\}) = f_1(x)$ so that $\bigwedge_{i=1}^1 f_i = f_1 \in \mathcal{S}^+[X, \mathcal{A}]$, hence $\mathbf{1} \in S$

$n \in S \Rightarrow n+1 \in S$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then $\forall x \in A$ we have

$$\begin{aligned} \left(\bigwedge_{i=1}^{n+1} f_i \right)(x) &= \min(\{f_i(x) \mid i \in \{1, \dots, n+1\}\}) \\ &\stackrel{[\text{corollary: 3.67}]}{=} \min(\min(\{f_i(x) \mid i \in \{1, \dots, n+1\}\}), f_{n+1}(x)) \\ &= \min\left(\left(\bigwedge_{i=1}^n f_i\right)(x), f_{n+1}(x)\right) \\ &= \left(\left(\bigwedge_{i=1}^n f_i\right) \wedge f_{n+1}\right)(x) \end{aligned}$$

so that

$$\bigwedge_{i=1}^{n+1} f_i = \left(\bigwedge_{i=1}^n f_i\right) \wedge f_{n+1}$$

As $n \in S$ we have that $\bigwedge_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}]$, hence, using [theorem: 21.309] it follows that $(\bigwedge_{i=1}^{n+1} f_i) \in \mathcal{S}^+[X, \mathcal{A}]$, proving $n+1 \in S$.

3. Let

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \text{ then } \bigvee_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}] \right\}$$

then we have:

$\mathbf{1} \in \mathcal{S}$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then $(\bigvee_{i=1}^1 f_i)(x) = \max(\{f_1(x)\}) = f_1(x)$ so that $\bigvee_{i=1}^1 f_i = f_1 \in \mathcal{S}^+[X, \mathcal{A}]$, hence $\mathbf{1} \in S$

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ then $\forall x \in A$ we have

$$\begin{aligned} \left(\bigvee_{i=1}^{n+1} f_i \right)(x) &= \max(\{f_i(x) | i \in \{1, \dots, n+1\}\}) \\ &\stackrel{[\text{corollary: 3.67}]}{=} \max(\max(\{f_i(x) | i \in \{1, \dots, n+1\}\}), f_{n+1}(x)) \\ &= \max\left(\left(\bigvee_{i=1}^n f_i\right)(x), f_{n+1}(x)\right) \\ &= \left(\left(\bigvee_{i=1}^n f_i\right) \vee f_{n+1}\right)(x) \end{aligned}$$

so that

$$\bigvee_{i=1}^{n+1} f_i = \left(\bigvee_{i=1}^n f_i\right) \vee f_{n+1}$$

As $n \in \mathcal{S}$ we have that $\bigvee_{i=1}^n f_i \in \mathcal{S}^+[X, \mathcal{A}]$, hence, using [theorem: 21.309] it follows that $(\bigvee_{i=1}^{n+1} f_i) \in \mathcal{S}^+[X, \mathcal{A}]$, proving $n+1 \in \mathcal{S}$. □

The following theorem is the simple function variant of the monotone convergence theorem that is one of the central convergence theorems that sets the measure integral apart from the Riemann integral.

Theorem 21.311. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that

1. $\forall x \in X$ we have $\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x)$

Note 21.312. As $\forall x \in X$ we have $\forall k \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x)$ we have by [theorem: 20.92] that $\lim_{n \rightarrow \infty} f_n(x)$ exist, as is needed for the requirements of this theorem.

2. $\lim_{n \rightarrow \infty} f_n \in \mathcal{S}^+[X, \mathcal{A}]$ where

$$\lim_{n \rightarrow \infty} f_n: X \rightarrow \overline{\mathbb{R}} \text{ is defined by } \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

then

$$\lim_{n \rightarrow \infty} \int^S f_n d\mu \text{ exist and } \lim_{n \rightarrow \infty} \int^S f_n d\mu = \int^S \lim_{n \rightarrow \infty} f_n d\mu$$

Proof. As $\forall x \in X$ we have $\forall k \in \mathbb{N}$ that $f_k(x) \leq f_{k+1}(x)$ it follows from [theorem: 21.309] that

$$\forall k \in \mathbb{N} \text{ we have } \int^S f_k d\mu \leq \int^S f_{k+1} d\mu \quad (21.443)$$

So using [theorem: 20.92] it follows that

$$\lim_{k \rightarrow \infty} \int^S f_k d\mu \text{ exists and } \lim_{k \rightarrow \infty} \int^S f_k d\mu = \sup \left(\left\{ \int^S f_k d\mu | k \in \mathbb{N} \right\} \right) \geq 0 \quad (21.444)$$

Further as $\forall x \in X$ we have $\forall k \in \mathbb{N}$ that $f_k(x) \leq f_{k+1}(x)$ we have by [theorem: 20.92] again that

$$\lim_{k \rightarrow \infty} f_k(x) \text{ exists and } \lim_{k \rightarrow \infty} f_k(x) = \sup(\{f_k(x) | k \in \mathbb{N}\}) \quad (21.445)$$

so that

$$\forall x \in X \text{ we have } \forall k \in \mathbb{N} \text{ that } f_k(x) \leq \lim_{k \rightarrow \infty} f_k(x) \quad (21.446)$$

As $\lim_{k \rightarrow \infty} f_k \in \mathcal{S}^+[X, \mathcal{A}]$ we have that $\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$ exist and using [theorem: 21.309] it follows then that $\forall k \in \mathbb{N} \ \int^S f_k d\mu \leq \int^S \left(\lim_{i \rightarrow \infty} f_i \right) d\mu$ so that using [theorem: 20.85]

$$0 \leq \lim_{[\text{eq: 21.444}] k \rightarrow \infty} \int^S f_k d\mu \leq \int^S \left(\lim_{i \rightarrow \infty} f_i \right) d\mu \quad (21.447)$$

Next we prove the opposite inequality. As $\lim_{k \rightarrow \infty} f_k \in \mathcal{S}^+[X, \mathcal{A}]$ there exists by [theorem: 21.301] a $n \in \mathbb{N}$, a pairwise disjoint family $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ and a family $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ such that

$$\lim_{k \rightarrow \infty} f_k = \sum_{i=1}^n a_i \cdot \mathcal{X}_{X, A_i} \quad (21.448)$$

For $\{a_i\}_{i \in \{1, \dots, n\}}$ we have two possible cases:

$\forall i \in \{1, \dots, n\} \ a_i = 0$. Then $\int^{\mathcal{S}} \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \stackrel{[\text{eq: 21.448}]}{=} \sum_{i=1}^n a_i \cdot \mu(A_i) = \sum_{i=1}^n 0 \cdot \mu(A_i) = 0$ so that we have [using eq: 21.447]

$$0 \leq \lim_{i \rightarrow \infty} \int^{\mathcal{S}} f_i d\mu \stackrel{[\text{eq: 21.447}]}{\leq} \int^{\mathcal{S}} \left(\lim_{k \rightarrow \infty} f_k \right) d\mu = 0$$

which proves that in this case

$$\lim_{k \rightarrow \infty} \int^{\mathcal{S}} f_k d\mu = 0 = \int^{\mathcal{S}} \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$$

$\exists i \in \{1, \dots, n\} \ a_i \neq 0$. Then $I = \{i \in \{1, \dots, n\} | a_i \neq 0\} \neq \emptyset$ so that there exists a $m \in \mathbb{N}$ and a bijection $\beta: \{1, \dots, m\} \rightarrow I$. Let $x \in X$ then

$$\begin{aligned} \left(\lim_{k \rightarrow \infty} f_k \right)(x) &\stackrel{[\text{eq: 21.448}]}{=} \sum_{i \in \{1, \dots, n\}} a_i \cdot \mathcal{X}_{X, A_i}(x) \\ &= \sum_{i \in \{1, \dots, n\} \setminus I} a_i \cdot \mathcal{X}_{X, A_i}(x) + \sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i}(x) \\ &= \sum_{i \in I} a_i \cdot \mathcal{X}_{X, A_i}(x) \\ &\stackrel{[\text{theorem: 11.36}]}{=} \sum_{i \in \{1, \dots, m\}} a_{\beta(i)} \cdot \mathcal{X}_{X, A_{\beta(i)}}(x) \\ &= \left(\sum_{i=1}^m a_{\beta(i)} \cdot \mathcal{X}_{X, A_{\beta(i)}} \right)(x) \end{aligned}$$

proving, if we define $\{b_i\}_{i \in \{1, \dots, m\}}$ by $b_i = a_{\beta(i)}$ and $\{B_i\}_{i \in \{1, \dots, m\}}$ by $B_i = A_{\beta(i)} \in \mathcal{A}$, that

$$\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{A} \text{ and } \lim_{k \rightarrow \infty} f_k = \sum_{i=1}^m b_i \cdot \mathcal{X}_{X, B_i} \text{ and } \forall i \in \{1, \dots, m\} \ b_i \in \mathbb{R}^+ \quad (21.449)$$

Further, if $i, j \in \{1, \dots, m\}$ with $i \neq j$ then $\beta(i) \neq \beta(j)$ so that $B_i \cap B_j = A_{\beta(i)} \cap A_{\beta(j)} = \emptyset$ which proves that

$$\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{A} \text{ is pairwise disjoint} \quad (21.450)$$

Let $\varepsilon \in \mathbb{R}^+$ with $\varepsilon < 1$ then $0 < 1 - \varepsilon$. Let $i \in \{1, \dots, m\}$ and define

$$\{B_{i,j}\}_{j \in \mathbb{N}} \text{ by } B_{i,j} = \{x \in B_i | f_j(x) \geq (1 - \varepsilon) \cdot b_i\} \subseteq B_i \quad (21.451)$$

then as

$$\{x \in B_i | f_j(x) \geq (1 - \varepsilon) \cdot b_i\} = B_i \cap \{x \in X | f_j(x) \geq (1 - \varepsilon) \cdot b_i\} \stackrel{[\text{def: 21.252}]}{=} B_i \cap \{f_j \geq (1 - \varepsilon) \cdot b_i\}$$

and $B_i, \{f_j \geq (1 - \varepsilon) \cdot b_i\} \in \mathcal{A}$ [see: theorem: 21.254] it follows that

$$\forall i \in \{1, \dots, m\} \text{ we have } \{B_{i,j}\}_{j \in \mathbb{N}} \subseteq \mathcal{A} \quad (21.452)$$

If $x \in B_i$ then by [theorem: 21.290] and [eqs: 21.449, 21.450] we have $\left(\lim_{k \rightarrow \infty} f_k \right)(x) = b_i$. As by [eq: 21.449] $b_i \neq 0$ we have $(1 - \varepsilon) \cdot b_i < b_i = \left(\lim_{k \rightarrow \infty} f_k \right)(x) \stackrel{[\text{eq: 21.445}]}{=} \sup(\{f_i(x) | i \in \mathbb{N}\})$. Hence there exist a $k \in \mathbb{N}$ such that $(1 - \varepsilon) \cdot b_i < f_k(x)$ which proves that $x \in B_{i,k} \subseteq \bigcup_{j \in \mathbb{N}} B_{i,j}$. So that $B_i \subseteq \bigcup_{j \in \mathbb{N}} B_{i,j}$. As $\forall j \in \mathbb{N}$ we have $B_{i,j} \subseteq B_i \Rightarrow \bigcup_{j \in \mathbb{N}} B_{i,j} \subseteq B_i$ it follows that

$$\forall i \in \{1, \dots, m\} \ B_i = \bigcup_{j \in \mathbb{N}} B_{i,j} \quad (21.453)$$

Let $k \in \mathbb{N}$ then $\forall i, j \in \{1, \dots, m\}$ with $i \neq j$ we have $B_{i,k} \cap B_{j,k} \subseteq B_i \cap B_j = \emptyset$ which proves that

$$\forall k \in \mathbb{N} \text{ we have that } \{B_{i,k}\}_{i \in \{1, \dots, m\}} \text{ is pairwise disjoint} \quad (21.454)$$

Let $i \in \{1, \dots, m\}$ and $k \in \mathbb{N}$ then if $x \in B_{i,k}$ we have that $(1 - \varepsilon) \cdot b_i \leq f_k(x) \leq f_{k+1}(x)$, so $x \in B_{i,k+1}$, hence

$$\forall i \in \{1, \dots, m\} \text{ we have } \forall k \in \mathbb{N} \text{ that } B_{i,k} \subseteq B_{i,k+1} \quad (21.455)$$

Using now [theorem: 21.115 (1)] it follows that

$$\forall i \in \{1, \dots, m\} \mu(B_i) \underset{[\text{eq: 21.453}]}{=} \mu\left(\bigcup_{j \in \mathbb{N}} B_{i,j}\right) = \lim_{j \rightarrow \infty} \mu(B_{i,j}) = \sup(\{\mu(B_{i,j}) | j \in \mathbb{N}\}) \quad (21.456)$$

Define now

$$\{g_k\}_{k \in \mathbb{N}} \text{ by } g_k = \sum_{i=1}^m (1 - \varepsilon) \cdot b_i \cdot \mathcal{X}_{B_{i,k}} \quad (21.457)$$

then by [theorem: 21.301] together with [eqs: 21.457, 21.452, 21.454] we have that

$$\{g_k\}_{k \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \quad (21.458)$$

Let $x \in X$ and $k \in \mathbb{N}$ then for x we have either:

$x \in \bigcup_{i \in \{1, \dots, m\}} B_{i,k}$. Then there exist a $i \in \{1, \dots, m\}$ such that $x \in B_{i,k}$ so that $(1 - \varepsilon) \cdot b_i \leq f_k(x)$ which, as $g_k(x) \underset{[\text{theorem: 21.290}]}{=} (1 - \varepsilon) \cdot b_i \leq f_k(x)$, proves that

$$g_k(x) \leq f_k(x)$$

Further as $B_{i,k} \underset{[\text{eq: 21.455}]}{\subseteq} B_{i,k+1}$ we have that $x \in B_{i,k+1}$ so that

$$g_k(x) \underset{[\text{theorem: 21.290}]}{=} (1 - \varepsilon) \cdot b_{k+1} \underset{[\text{theorem: 21.290}]}{=} g_{k+1}(x)$$

proving, as equality implies inequality, that

$$g_k(x) \leq g_{k+1}(x)$$

$x \notin \bigcup_{i \in \{1, \dots, m\}} B_{i,k}$. Then

$$g_k(x) = \sum_{i=1}^m (1 - \varepsilon) \cdot b_i \cdot \mathcal{X}_{B_{i,k}}(x) = \sum_{i=1}^m (1 - \varepsilon) \cdot b_i \cdot 0 = 0$$

so that

$$g_k(x) = 0 \leq f_k(x)$$

and

$$g_k(x) = 0 \leq g_{k+1}(x)$$

So we must conclude that

$$\forall x \in X \text{ we have } \forall k \in \mathbb{N} \text{ that } g_k(x) \leq g_{k+1}(x) \text{ and } g_k(x) \leq f_k(x) \quad (21.459)$$

From the above we have using [theorem: 21.309] that $\forall k \in \mathbb{N} \int^S g_k d\mu \leq \int^S g_{k+1} d\mu$. So using [theorem: 20.92] $\lim_{k \rightarrow \infty} \int^S g_k d\mu$ exists. Lets calculate this limit. By [theorem: 20.85] and [eq: 21.456] we have that $\forall i \in \{1, \dots, m\} \lim_{k \rightarrow \infty} ((1 - \varepsilon) \cdot b_i \cdot \mu(B_{i,k}))$ exists and

$$\lim_{k \rightarrow \infty} ((1 - \varepsilon) \cdot b_i \cdot \mu(B_{i,k})) = (1 - \varepsilon) \cdot b_i \cdot \lim_{k \rightarrow \infty} \mu(B_{i,k}) \underset{[\text{eq: 21.456}]}{=} (1 - \varepsilon) \cdot b_i \cdot \mu(B_i)$$

Hence using [theorem: 20.88] on the above it follows that $\lim_{k \rightarrow \infty} (\sum_{i=1}^m (1 - \varepsilon) \cdot b_i \cdot \mu(B_{i,k}))$ exists and

$$\lim_{k \rightarrow \infty} \left(\sum_{i=1}^m (1 - \varepsilon) \cdot b_i \cdot \mu(B_{i,k}) \right) = (1 - \varepsilon) \cdot \sum_{i=1}^m b_i \cdot \mu(B_i) \quad (21.460)$$

so

$$\begin{aligned} \lim_{k \rightarrow \infty} \int^S g_k d\mu &\stackrel{[\text{eq: 21.457}]}{=} \lim_{k \rightarrow \infty} \left(\sum_{i=1}^m (1-\varepsilon) \cdot b_i \cdot \mu(B_{i,k}) \right) \\ &\stackrel{[\text{eq: 21.460}]}{=} (1-\varepsilon) \cdot \sum_{i=1}^m b_i \cdot \mu(B_i) \\ &\stackrel{[\text{eq: 21.449}]}{=} (1-\varepsilon) \cdot \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \end{aligned}$$

Hence

$$\lim_{k \rightarrow \infty} \int^S g_k d\mu = (1-\varepsilon) \cdot \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \quad (21.461)$$

By [eq: 21.459] we have $\forall x \in X$ that $\forall k \in \mathbb{N}$ $g_k(x) \leq f_k(x)$, so it follows from [theorem: 21.309] that $\int^S g_k d\mu \leq \int^S f_k d\mu$. Using then [theorem: 20.85] we have that

$$\lim_{k \rightarrow \infty} \int^S g_k d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

If we combine this with [eq: 21.461] we get

$$(1-\varepsilon) \cdot \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu \leq \infty \quad (21.462)$$

Now for $\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$ we have either:

$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu = \infty$. Then as $0 < (1-\varepsilon)$ we have $(1-\varepsilon) \cdot \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu = \infty$ so that by [eq: 21.462] $\lim_{k \rightarrow \infty} \int^S f_k d\mu = \infty = \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$ so that

$$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu < \infty$. By [eq: 21.447] we have $0 \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu \leq \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu < \infty$ so that $\lim_{k \rightarrow \infty} \int^S f_k d\mu \in \mathbb{R} \wedge \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \in \mathbb{R}$. Hence by [eq: 21.462] we have that

$$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu + \varepsilon \cdot \int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$$

As $\varepsilon \in \mathbb{R}^+$ with $\varepsilon < 1$ was chosen arbitrary it follows from [theorem: 10.31] that

$$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

which proves that in all cases

$$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

Combining this with [eq: 21.447] proves that

$$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu = \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

So in all possible cases we have

$$\int^S \left(\lim_{k \rightarrow \infty} f_k \right) d\mu = \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

□

21.5.2 Integral of non negative functions

21.5.2.1 Definition of the integral of non negative functions

We extend now the integral $\int^S$ of measurable non negative simple functions to non negative measurable functions.

Definition 21.313. Let $\langle X, \mathcal{A}, \mu \rangle$ and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then we define $\int^+ f d\mu$ by

$$\int^+ f d\mu = \sup \left(\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x) \right\} \right)$$

Note 21.314. As by [example: 21.300] $C_0 \in \mathcal{S}^+[X, \mathcal{A}]$ and $\forall x \in X$ $C_0(x) = 0 \leq f(x)$ it follows that $\{\int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x)\} \neq \emptyset$ so that

$$\sup \left(\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x) \right\} \right)$$

exists. Hence the definition is well defined.

Note 21.315. As by the definition of the integral of a non negative simple function [see definition: 21.305] $\forall g \in \mathcal{S}^+[X, \mathcal{A}]$ $0 \leq \int^S g d\mu$ it follows that

$$0 \leq \int^+ f d\mu$$

Note 21.316. In the definition of the integral $\int^+ f d\mu$ we don't use the $\mathcal{A}$ -measurability of f because $\sup \left(\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x) \right\} \right)$ is also well defined for a non negative function that is not measurable. However the requirement that f is $\mathcal{A}$ -measurable will be useful later to construct an alternative definition based on a limiting process of simple functions.

We show now that $\int^+$ is actually an extension of $\int^S$.

Theorem 21.317. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space and $f \in \mathcal{S}^+[X, \mathcal{A}]$ then $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ and

$$\int^S f d\mu = \int^+ f d\mu$$

Proof. As $f \in \mathcal{S}^+[X, \mathcal{A}]$ we have by [definition: 21.297] that f is $\mathcal{A}$ -measurable, hence $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and $f(X) \subseteq [0, \infty[$ so that

$$f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

Further, as trivially, $\forall x \in X$ $f(x) \leq f(x)$ and $f \in \mathcal{S}^+[X, \mathcal{A}]$ it follows that

$$\int^S f d\mu \in \left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x) \right\}$$

so that

$$\int^S f d\mu \leq \sup \left(\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x) \right\} \right) = \int^+ f d\mu \quad (21.463)$$

Let $g \in \mathcal{S}^+[X, \mathcal{A}]$ be such that $\forall x \in X$ we have $g(x) \leq f(x)$ then by [theorem: 21.309] we have that $\int^S g d\mu \leq \int^S f d\mu$. So it follows that $\int^S f d\mu$ is an upper bound of $\{\int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x)\}$ which proves that

$$\sup \left(\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \text{ we have } g(x) \leq f(x) \right\} \right) \leq \int^S f d\mu$$

Combining the above with [eq: 21.463] gives

$$\int^S f d\mu = \int^+ f d\mu \quad \square$$

Example 21.318. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $a \in [0, \infty]$ then $C_a \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ and

$$\int^+ C_a d\mu = a \cdot \mu(X)$$

Proof. Using [theorem: 21.300] $C_a \in \mathcal{S}^+[X, \mathcal{A}]$, hence using [theorem: 21.317] $C_a \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ and

$$\int^+ C_a d\mu = \int^S C_a d\mu \stackrel{[\text{theorem: 21.307}]}{=} a \cdot \mu(X) \quad \square$$

Example 21.319. $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space, $A \in \mathcal{A}$ and $\alpha \in [0, \infty[$ then $\alpha \cdot \mathcal{X}_{X,A} \in \mathcal{S}^+[X, \mathcal{A}] \subseteq \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ and $\int^+(\alpha \cdot \mathcal{X}_{X,A}) d\mu = \int^S (\alpha \cdot \mathcal{X}_{X,A}) d\mu = \alpha \cdot \mu(A)$.

Proof. Take $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A}$ by $A_1 = A$ and $\{a_i\}_{i \in \{1\}} = \alpha$ then

$$\forall x \in X \text{ we have } (\alpha \cdot \mathcal{X}_{X,A})(x) = \alpha \cdot \mathcal{X}_{X,A}(x) = \sum_{i=1}^1 a_i \cdot \mathcal{X}_{X,A_i}$$

proving that $\alpha \cdot \mathcal{X}_A = \sum_{i=1}^1 a_i \cdot \mathcal{X}_{A_i}$. From [theorem: 21.301] it follows then that

$$\alpha \cdot \mathcal{X}_A \in \mathcal{S}^+[X, \mathcal{A}]$$

Using the definition of $\int^S$ [see definition: 21.305] it follows that

$$\int^S (\alpha \cdot \mathcal{X}_{X,A}) d\mu = \sum_{i=1}^1 a_i \cdot \mu(A_i) = \alpha \cdot \mu(A)$$

Hence using [theorem: 21.317] $\alpha \cdot \mathcal{X}_{X,A} \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ and $\int^+(\alpha \cdot \mathcal{X}_{X,A}) d\mu = \alpha \cdot \mu(A_i)$. $\square$

21.5.2.2 Integral of non negative measurable functions on a measurable subset

Next we define the integral on a measurable subset.

Definition 21.320. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$\int_A^+ f d\mu = \int^+ (\mathcal{X}_{A,X} \cdot f) d\mu$$

Note 21.321. Using [theorem: 21.276] $\mathcal{X}_{X,A} \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ as is needed in this definition.

If $\langle X, \mathcal{A}, \mu \rangle$ is a measure space and $A \in \mathcal{A}$ then by [theorem: 21.111] we can construct the measure subspace $\langle A, \mathcal{A}|_A, \mu|_{\mathcal{A}|_A} \rangle$ where $\mathcal{A}|_A = \{B \in \mathcal{A} | B \subseteq A\} = \{A \cap B | B \in \mathcal{A}\}$ and $\mu|_{\mathcal{A}|_A}: \mathcal{A}|_A \rightarrow [0, \infty]$ is the restriction of μ to $\mathcal{A}|_A$. A natural question to ask is if there is a correlation between $\int_A^+ f d\mu$ and $\int_{\mathcal{A}|_A}^+ f d\mu|_{\mathcal{A}|_A}$. As the following theorem shows the answer is yes.

Theorem 21.322. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $f: X \rightarrow \overline{\mathbb{R}}$ a function such that $\mathcal{X}_A \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$f|_A \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$$

and

$$\int_A^+ f d\mu \stackrel{[\text{see definition: 21.320}]}{=} \int^+ (\mathcal{X}_A \cdot f) d\mu = \int^+ f|_A d\mu|_{\mathcal{A}|_A}$$

where

$$\mathcal{A}|_A = \{B \in \mathcal{A} | B \subseteq A\} \text{ see [definition: 21.44]}$$

$\mu|_{\mathcal{A}|_A}$ is the restriction of μ to $\mathcal{A}|_A$

Proof. As $\mathcal{X}_{X,A} \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.277] that

$$f|_A \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$$

Hence by definition of $\mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$ we have $\forall B \in \mathcal{B}[\overline{\mathbb{R}}]$ we have that $(f|_A)^{-1}(B) \in \mathcal{A}|_A$. Which proves that $f|_A \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$ hence we have

$$f|_A \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$$

which proves the first part of the theorem. Now using the definition of $\int^+$ [see definition: 21.313] we have that

$$\int^+ f|_A d\mu|_{\mathcal{A}|_A} = \sup \left(\left\{ \int^S g d\mu|_{\mathcal{A}|_A} | g \in \mathcal{S}^+[A, \mathcal{A}|_A] \wedge \forall x \in A \text{ we have } g(x) \leq f|_A(x) \right\} \right) \quad (21.464)$$

$$\int^+ (\mathcal{X}_A \cdot f) d\mu = \sup \left(\left\{ \int^S g d\mu | g \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } g(x) \leq (\mathcal{X}_{X,A} \cdot f)(x) \right\} \right) \quad (21.465)$$

So to prove the theorem we have to prove that the above supremum are the same. Let $I \in \left\{ \int^S g d\mu|_{\mathcal{A}|_A} | g \in \mathcal{S}^+[A, \mathcal{A}|_A] \wedge \forall x \in A \text{ we have } g(x) \leq f|_A(x) \right\}$ then there exist a $g \in \mathcal{S}^+[A, \mathcal{A}|_A]$ with $\forall x \in A$ $g(x) \leq f|_A(x) = f(x)$ such that $I = \int^S g d\mu|_{\mathcal{A}|_A}$. Using the alternate definition of simple functions [see theorem: 21.301] there exist a $n \in \mathbb{N}$, a $\{C_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}|_A = \{B \in \mathcal{A} | B \subseteq A\} \subseteq \mathcal{A}$ pairwise disjoint and $\{\alpha_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ such that

$$g = \sum_{i=1}^n \alpha_i \cdot \mathcal{X}_{A, C_i} \quad (21.466)$$

Define now

$$h = \sum_{i=1}^n \alpha_i \cdot \mathcal{X}_{X, C_i} \quad (21.467)$$

then by [theorem: 21.301] it follows that

$$h \in \mathcal{S}^+[X, \mathcal{A}] \quad (21.468)$$

Further if $x \in X$ then we have either:

$x \in \bigsqcup_{i \in \{1, \dots, n\}} C_i$. Then $\exists i \in \{1, \dots, n\}$ such that $x \in C_i \subseteq A$ so that

$$\begin{aligned} h(x) &= \left(\sum_{i=1}^n \alpha_i \cdot \mathcal{X}_{X, C_i} \right)(x) \\ &\stackrel{[\text{theorem: 21.290}]}{=} \alpha_i \\ &\stackrel{[\text{theorem: 21.290}]}{=} \left(\sum_{i=1}^n \alpha_i \cdot \mathcal{X}_{A, C_i} \right)(x) \\ &= g(x) \\ &\leq f|_A(x) \\ &= f(x) \\ &\stackrel{x \in A}{=} \mathcal{X}_{X, A}(x) \cdot f(x) \\ &= (X_{X, A} \cdot f)(x) \end{aligned}$$

$x \in X \setminus \bigsqcup_{i \in \{1, \dots, n\}} C_i$. Then by [theorem: 21.290] $h(x) = 0 \leq (\mathcal{X}_{X, A} \cdot f)(x)$ [as f is non negative].

which proves that

$$\forall x \in X \text{ we have } h(x) \leq (\mathcal{X}_{X,A} \cdot f)(x) \quad (21.469)$$

Further we have

$$\begin{aligned} I &= \int^S g d\mu|_{\mathcal{A}|_A} \\ &\stackrel{[\text{eq:21.466}] \wedge [\text{definition: 21.305}]}{=} \sum_{i=1}^n \alpha_i \cdot \mu|_{\mathcal{A}|_A}(C_i) \\ &\stackrel{C_i \in \mathcal{A}}{=} \sum_{i=1}^n \alpha_i \cdot \mu(C_i) \\ &\stackrel{[\text{eq:21.467}] \wedge [\text{definition: 21.305}]}{=} \int^S h d\mu \end{aligned}$$

The above together with [eqs: 21.468 and 21.469] proves that

$$I \in \left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } g(x) \leq (\mathcal{X}_A \cdot f)(x) \right\}.$$

So we have proved that

$$\begin{aligned} &\left\{ \int^S g d\mu|_{\mathcal{A}|_A} \mid g \in \mathcal{S}^+[A, \mathcal{A}|_A] \wedge \forall x \in A \text{ we have } g(x) \leq f|_A(x) \right\} \subseteq \\ &\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } g(x) \leq (\mathcal{X}_{X,A} \cdot f)(x) \right\} \end{aligned} \quad (21.470)$$

For the opposite inequality. Let $I \in \left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } g(x) \leq (\mathcal{X}_{X,A} \cdot f)(x) \right\}$ then there exist a $k \in \mathcal{S}^+[X, \mathcal{A}]$ with $\forall x \in X \ k(x) \leq (\mathcal{X}_{X,A} \cdot f)(x)$ such that $I = \int^S k d\mu$. Now for $x \in X$ we have either:

$$x \in A. \text{ Then } k(x) = \mathcal{X}_{X,A}(x) \cdot k(x) = (\mathcal{X}_{X,A} \cdot k)(x)$$

$$x \in X \setminus A. \text{ As } 0 \leq k(x) \leq (\mathcal{X}_{X,A} \cdot f)(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = 0 \text{ we have}$$

$$k(x) = 0 = \mathcal{X}_{X,A}(x) \cdot k(x) = (\mathcal{X}_{X,A} \cdot k)(x)$$

hence we have proved that

$$k = \mathcal{X}_{X,A} \cdot k \quad (21.471)$$

As $k \in \mathcal{S}^+[X, \mathcal{A}]$ there exist by [theorem: 21.301] a $m \in \mathbb{N}$, a $\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint and a $\{\beta_i\}_{i \in \{1, \dots, m\}} \subseteq [0, \infty[$ such that

$$k = \sum_{i=1}^m \beta_i \cdot \mathcal{X}_{X, B_i} \quad (21.472)$$

Then we have

$$k \stackrel{[\text{eq: 21.471}]}{=} \mathcal{X}_{X,A} \cdot \sum_{i=1}^m \beta_i \cdot \mathcal{X}_{X, B_i} = \sum_{i=1}^m \beta_i \cdot \mathcal{X}_{X,A} \cdot \mathcal{X}_{X, B_i} \stackrel{[\text{theorem: 21.272}]}{=} \sum_{i=1}^m \beta_i \cdot \mathcal{X}_{X, A \cap B_i} \quad (21.473)$$

Define now

$$l = \sum_{i=1}^m \beta_i \cdot \mathcal{X}_{A, A \cap B_i} \quad (21.474)$$

As $\forall i \in \{1, \dots, m\} \ B_i \in \mathcal{A} \xRightarrow{A \in \mathcal{A}} A \cap B_i \in \mathcal{A}$ and $A \cap B_i \subseteq A$ it follows that $\{A \cap B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{A}|_A$ [pairwise disjoint]. So using [theorem: 21.301] we have that

$$l \in \mathcal{S}^+[A, \mathcal{A}|_A] \quad (21.475)$$

Further if $x \in A$ then we have either:

$x \in \bigsqcup_{i \in \{1, \dots, m\}} (A \cap B_i)$. Then $\exists i \in \{1, \dots, m\}$ such that $x \in A \cap B_i \subseteq B_i$, A so that

$$\begin{aligned}
 l(x) &= \left(\sum_{i=1}^m \beta_i \cdot \mathcal{X}_{A, A \cap B_i} \right)(x) \\
 &\stackrel{[\text{theorem: 21.290}]}{=} \beta_i \\
 &\stackrel{[\text{theorem: 21.290}]}{=} \left(\sum_{i=1}^m \beta_i \cdot \mathcal{X}_{X, B_i} \right)(x) \\
 &\stackrel{[\text{eq: 21.472}]}{=} k(x) \\
 &\leq (\mathcal{X}_{X, A} \cdot f)(x) \\
 &= \mathcal{X}_{X, A}(x) \cdot f(x) \\
 &\stackrel{x \in A}{=} f(x) \\
 &\stackrel{x \in A}{=} f|_A(x)
 \end{aligned}$$

$x \notin A \setminus \bigsqcup_{i \in \{1, \dots, m\}} (A \cap B_i)$. Then by [theorem: 21.290] $l(x) = 0 \leq f|_A(x)$ [as f is non negative]

which proves that

$$\forall x \in A \text{ we have } l(x) \leq f|_A(x) \quad (21.476)$$

Further

$$\begin{aligned}
 I &= \int^S k d\mu \\
 &\stackrel{[\text{eq: 21.472}] \wedge [\text{definition: 21.305}]}{=} \sum_{i=1}^m \beta_i \cdot \mu(A \cap B_i) \\
 &\stackrel{A \cap B_i \in \mathcal{A}|_A}{=} \sum_{i=1}^m \beta_i \cdot \mu|_{\mathcal{A}|_A}(A \cap B_i) \\
 &\stackrel{[\text{eq: 21.474}] \wedge [\text{definition: 21.305}]}{=} \int^S l d\mu|_{\mathcal{A}|_A}
 \end{aligned}$$

Combining the above with [eqs: 21.475 and 21.476] proves that

$$I \in \left\{ \int^S g d\mu|_{\mathcal{A}|_A} \mid g \in \mathcal{S}^+[A, \mathcal{A}|_A] \wedge \forall x \in A \text{ we have } g(x) \leq f|_A(x) \right\}$$

hence

$$\begin{aligned}
 &\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } g(x) \leq (\mathcal{X}_A \cdot f)(x) \right\} \subseteq \\
 &\left\{ \int^S g d\mu|_{\mathcal{A}|_A} \mid g \in \mathcal{S}^+[A, \mathcal{A}|_A] \wedge \forall x \in A \text{ we have } g(x) \leq f|_A(x) \right\}
 \end{aligned}$$

which combined with [eq: 21.470] gives

$$\begin{aligned}
 &\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } g(x) \leq (\mathcal{X}_A \cdot f)(x) \right\} = \\
 &\left\{ \int^S g d\mu|_{\mathcal{A}|_A} \mid g \in \mathcal{S}^+[A, \mathcal{A}|_A] \wedge \forall x \in A \text{ we have } g(x) \leq f|_A(x) \right\}
 \end{aligned}$$

Finally substituting the above in [eqs: 21.464 and 21.465] proves that

$$\int^+ (\mathcal{X}_A \cdot f) d\mu = \int^+ f|_A d\mu|_{\mathcal{A}|_A} \quad \square$$

Corollary 21.323. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then $\forall A \in \mathcal{A}$ we have

$$f|_A \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$$

and

$$\int^+ (\mathcal{X}_{X,A} \cdot f) = \int_A^+ f d\mu = \int^+ f|_A d\mu|_{\mathcal{A}|_A}$$

Proof. Using [theorem: 21.276] and the fact that $(\mathcal{X}_{X,A} \cdot f)(X) \subseteq [0, \infty]$ that $\mathcal{X}_{X,A} \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that by [theorem: 21.322] $f|_A \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and

$$\int_A^+ f d\mu \stackrel{[\text{definition: 21.320}]}{=} \int^+ (\mathcal{X}_{X,A} \cdot f) = \int^+ f|_A d\mu|_{\mathcal{A}|_A}$$

□

The previous theorem shows how we can calculate the integral of the restriction of a function to a measurable subset. The next theorem show how we can calculate the integral of the extension of a function on a measurable subset.

Theorem 21.324. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $f \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$ then

$$f_{A \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ f d\mu|_{\mathcal{A}|_A} = \int^+ f_{A \uparrow X} d\mu$$

where

$$\mathcal{A}|_A = \{B \in \mathcal{A} | B \subseteq A\} \text{ see [definition: 21.44]}$$

$$\mu|_{\mathcal{A}|_A} \text{ is the restriction of } \mu \text{ to } \mathcal{A}|_A$$

$$f_{A \uparrow X}: X \rightarrow \overline{\mathbb{R}} \text{ is defined by } f_{A \uparrow X}(x) = \begin{cases} f(x) & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases} \text{ [see definition: 21.246]}$$

Proof. As $f \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$ we have by definition that $\forall B \in \mathcal{B}[\overline{\mathbb{R}}]$ we have $f^{-1}(B) \in \mathcal{A}|_A$ so that $f \in \mathcal{M}^+[A, \mathcal{A}, \overline{\mathbb{R}}]$. Hence by [theorem: 21.250] it follows that

$$f_{A \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

which proves the first part of the lemma. Now for the second part note that $\forall x \in X$ we have either:

$$x \in A. \text{ Then } (\mathcal{X}_{X,A} \cdot f_{A \uparrow X})(x) = \mathcal{X}_{X \setminus A}(x) \cdot f_{A \uparrow X}(x) \stackrel{x \in A}{=} 1 \cdot f_{A \uparrow X}(x) = f_{A \uparrow X}(x).$$

$$x \in X \setminus A. \text{ Then } (\mathcal{X}_{X,A} \cdot f_{A \uparrow X})(x) = \mathcal{X}_{X \setminus A}(x) \cdot f_{A \uparrow X}(x) = 0 = f_{A \uparrow X}(x).$$

which prove that

$$\mathcal{X}_{X,A} \cdot f_{A \uparrow X} = f_{A \uparrow X}$$

so that

$$\begin{aligned} \int^+ f_{A \uparrow X} d\mu &= \int^+ (\mathcal{X}_{X,A} \cdot f_{A \uparrow X}) d\mu \\ &\stackrel{[\text{theorem: 21.322}]}{=} \int^+ (f_{A \uparrow X})|_A d\mu|_{\mathcal{A}|_A} \\ &\stackrel{[\text{definition: 21.246}]}{=} \int^+ f d\mu|_{\mathcal{A}|_A} \end{aligned}$$

□

21.5.2.3 Integral of non negative measurable functions as a limit of the integral of simple functions

We show now a different definition of the integral of non negative measurable functions. This equivalent definition is based on the fact that every non negative measurable function can be written as the limit of non negative simple functions as is proved in the next theorem.

Theorem 21.325. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then there exists a sequence $\{f_k\}_{k \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that $\forall x \in X$ we have

$$\forall n \in \mathbb{N} \text{ we have } f_n(x) \leq f_{n+1}(x)$$

and

$$\lim_{n \rightarrow \infty} f_n(x) \text{ exists with } f(x) = \lim_{n \rightarrow \infty} f_n(x) = \sup(\{f_n(x) | n \in \mathbb{N}\})$$

In other words we have that

$$f = \lim_{n \rightarrow \infty} f_n$$

Proof. The strategy of the proof is, given $n \in \mathbb{N}$, to divide $f(X) \cap [0, n[$ in $n \cdot 2^n$ intervals of the form $\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right]$ and approximate f by f_n where

$$f_n(x) = \begin{cases} \frac{k-1}{2^n} & \text{if } x \in f(X) \cap [0, n[\text{ where } k \text{ satisfies } x \in f^{-1}\left(\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right]\right) \\ n & \text{if } x \in X \setminus \left(\bigcup_{k \in I_n} A_{n,k}\right) \end{cases}$$

and prove that $f = \lim_{n \rightarrow \infty} f_n$. The following illustrates this for $n=2$ and $X = \mathbb{R}$ here we have that f is defined by

$x \in$	$f^{-1}([0, 1/4])$	$\Rightarrow$	$f_2(x) = 0/4$
$x \in$	$f^{-1}([1/4, 2/4])$	$\Rightarrow$	$f_2(x) = 1/4$
$x \in$	$f^{-1}([2/4, 3/4])$	$\Rightarrow$	$f_2(x) = 2/4$
$x \in$	$f^{-1}([3/4, 4/4])$	$\Rightarrow$	$f_2(x) = 3/4$
$x \in$	$f^{-1}([4/4, 5/4])$	$\Rightarrow$	$f_2(x) = 4/4$
$x \in$	$f^{-1}([5/4, 6/4])$	$\Rightarrow$	$f_2(x) = 5/4$
$x \in$	$f^{-1}([6/4, 7/4])$	$\Rightarrow$	$f_2(x) = 6/4$
$x \in$	$f^{-1}([7/4, 8/4])$	$\Rightarrow$	$f_2(x) = 7/4$
$x \in$	$f^{-1}([8/4, \infty])$	$\Rightarrow$	$f_2(x) = 2$

so that

$$\begin{aligned} f_2 = & \mathcal{X}_{f^{-1}([0, 1/4])} \cdot \frac{0}{4} + \mathcal{X}_{f^{-1}([1/4, 2/4])} \cdot \frac{1}{4} + \mathcal{X}_{f^{-1}([2/4, 3/4])} \cdot \frac{2}{4} + \mathcal{X}_{f^{-1}([3/4, 4/4])} \cdot \frac{3}{4} + \mathcal{X}_{f^{-1}([4/4, 5/4])} \cdot \frac{4}{4} + \\ & \mathcal{X}_{f^{-1}([5/4, 6/4])} \cdot \frac{5}{4} + \mathcal{X}_{f^{-1}([6/4, 7/4])} \cdot \frac{6}{4} + \mathcal{X}_{f^{-1}([7/4, 8/4])} \cdot \frac{7}{4} + \mathcal{X}_{f^{-1}([8/4, \infty])} \cdot \frac{8}{4} \end{aligned} \quad (21.477)$$

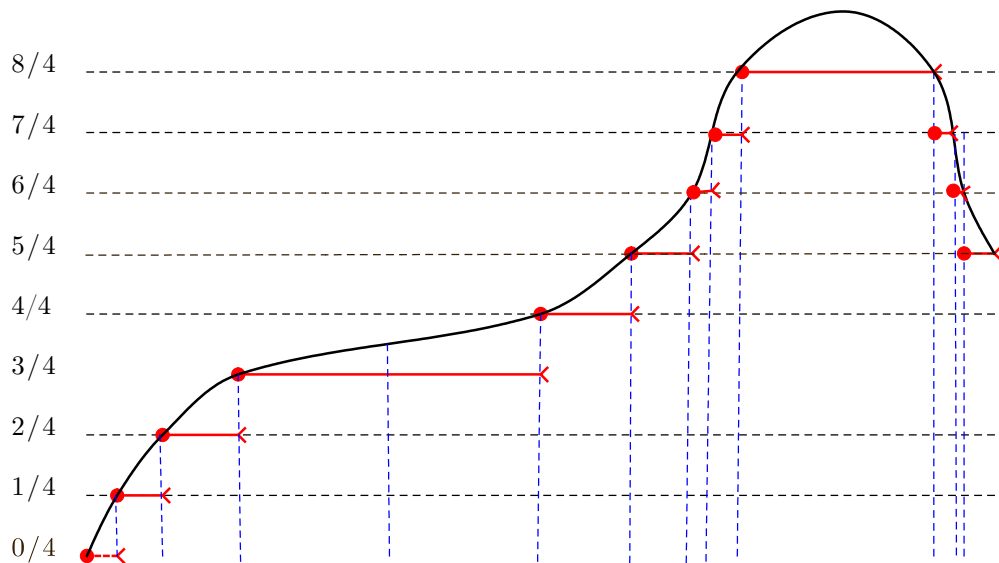


Figure 21.6. f approximated by f_2

Now for the details of the proof. Define

$$\{I_n\}_{n \in \mathbb{N}} \subseteq \mathcal{P}(\mathbb{N}) \text{ by } I_n = \{i \in \mathbb{N} \mid i \leq n \cdot 2^n\} = \{1, \dots, n \cdot 2^n\} \quad (21.478)$$

Let $n \in \mathbb{N}$. Define for $k \in I_n$

$$A_{n,k} = \left\{ x \in X \mid \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n} \right\} = f^{-1} \left(\left[\frac{k-1}{2^n}, \frac{k}{2^n} \right[\right) \quad (21.479)$$

Then we have

$$\begin{aligned} x \in A_{n,k} &\Leftrightarrow \frac{k-1}{2^n} \leq x \wedge x < \frac{k}{2^n} \\ &\Leftrightarrow x \in \left\{ f \geq \frac{k-1}{2^n} \right\} \wedge x \in \left\{ f < \frac{k}{2^n} \right\} \\ &\Leftrightarrow x \in \left\{ f \geq \frac{k-1}{2^n} \right\} \cap \left\{ f < \frac{k}{2^n} \right\} \end{aligned}$$

which proves that

$$A_{n,k} = \left\{ f \geq \frac{k-1}{2^n} \right\} \cap \left\{ f < \frac{k}{2^n} \right\} \quad (21.480)$$

As f is $\mathcal{A}$ -measurable it follows from [theorem: 21.254] that $\left\{ f \geq \frac{k-1}{2^n} \right\}, \left\{ f < \frac{k}{2^n} \right\} \in \mathcal{A}$ so that

$$\forall k \in I_n \text{ we have } A_{n,k} \in \mathcal{A} \quad (21.481)$$

As $I_n = \{1, \dots, n \cdot 2^n\}$ is finite hence countable it follows from [theorem: 21.39] that

$$\bigcup_{k \in I_n} A_{n,k} \in \mathcal{A} \quad (21.482)$$

Let $l, k \in I_n$ with $l \neq k$ then we may assume that $l < k \Rightarrow l \leq k-1$ [otherwise interchange l and k]. Assume that $x \in A_{n,k} \cap A_{n,l}$ then $\frac{l-1}{2^n} \leq f(x) < \frac{l}{2^n} \leq \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$ leading to the contradiction $f(x) < f(x)$, hence the assumption is wrong and we have that $A_{n,k} \cap A_{n,l} = \emptyset$. In other words we have proved that

$$\{A_{n,k}\}_{k \in I_n} \text{ is pairwise disjoint} \quad (21.483)$$

Given $n \in \mathbb{N}$ define

$$\begin{aligned} \{B_{n,k}\}_{k \in \{1, \dots, n \cdot 2^{n+1}\}} &\subseteq \mathcal{A} \text{ by } B_{n,k} = \begin{cases} A_{n,k} & \text{if } k \in \{1, \dots, n \cdot 2^n\} = I_n \\ X \setminus (\bigcup_{i \in I_n} A_{n,i}) & \text{if } k = n \cdot 2^n + 1 \end{cases} \underset{[\text{eq: 21.481, 21.482}]}{\in} \mathcal{A} \\ \{f_{n,k}\}_{k \in \{1, \dots, n \cdot 2^{n+1}\}} &\subseteq [0, \infty[\text{ by } f_{n,k} = \begin{cases} \frac{k-1}{2^n} & \text{if } k \in \{1, \dots, n \cdot 2^n\} = I_n \\ n & \text{if } k = n \cdot 2^n + 1 \end{cases} \end{aligned}$$

Using [lemma: 21.294] together with [eq: 21.483] it follows then that

$$\{B_{n,k}\}_{k \in \{1, \dots, n \cdot 2^{n+1}\}} \subseteq \mathcal{A} \text{ is pairwise disjoint and } \bigsqcup_{k \in \{1, \dots, n \cdot 2^{n+1}\}} B_{n,k} = X \quad (21.484)$$

Define now

$$f_n: X \rightarrow \overline{\mathbb{R}} \text{ by } f_n = \sum_{k=1}^{n \cdot 2^{n+1}} f_{n,k} \cdot \mathcal{X}_{X, B_{n,k}} \quad (21.485)$$

Using then [theorem: 21.301] together with [eq: 21.484] proves that

$$\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \quad (21.486)$$

Let $x \in X$ then we have either:

$\exists k \in I_n$ with $x \in A_{n,k}$. Then as $A_{n,k} = B_{n,k}$ we have by [theorem: 21.290] and [eqs: 21.484, 21.485] that $f_n(x) = f_{n,k} = \frac{k-1}{2^n}$.

$\forall k \in I_n$ we have $x \notin A_{n,k}$. Then $x \in X \setminus (\bigcup_{k \in I_n} A_{n,k}) = B_{n, n \cdot 2^n + 1}$ so that by [theorem: 21.290] and [eqs: 21.484, 21.485] we have $f_n(x) = f_{n, n \cdot 2^n + 1} = n$.

To summarize we have

$$\forall x \in X \quad f_n(x) = \begin{cases} \frac{k-1}{2^n} & \text{if } \exists k \in I_n \text{ with } x \in A_{n,k} \\ n & \text{if } x \in X \setminus (\bigcup_{k \in I_n} A_{n,k}) \end{cases} \quad (21.487)$$

Let $n \in \mathbb{N}$ and take $x \in f^{-1}([0, n])$ then $0 \leq f(x) < n$. Define $B_x = \{k \in I_n \mid f(x) < \frac{k}{2^n}\} \subseteq I_n$ a finite set. As $n \cdot 2^n \in I_n$ and $f(x) < n = \frac{n \cdot 2^n}{2^n}$ we have that $n \cdot 2^n \in B_x$ so that $B_x \neq \emptyset$. Hence $l = \min(B_x)$ exist. There are two possible cases for l

$l = 1$. Then $\frac{1-1}{2^n} = 0 \leq f(x) < \frac{1}{2^n}$ so that $x \in A_{n,1} \subseteq \bigcup_{k \in I_n} A_{n,k}$.

$1 < l$. Then $1 \leq l-1 < l \leq n \cdot 2^n$ so that $l-1 \in I_n$. As $l-1 < l = \min(B_x)$ we must have that $\frac{l-1}{2^n} \leq f(x) < \frac{l}{2^n}$ so that $x \in A_{n,l} \subseteq \bigcup_{k \in I_n} A_{n,k}$

So we have proved that $f^{-1}([0, n]) \subseteq \bigcup_{k \in I_n} A_{n,k}$. For the opposite inclusion, we have for $k \in I_n$ that $A_{n,k} = f^{-1}\left(\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)\right) \subseteq f^{-1}([0, 1]) \Rightarrow \bigcup_{k \in I_n} A_{n,k} \subseteq f^{-1}([0, 1])$. Summarized we have

$$\forall n \in \mathbb{N} \text{ we have } f^{-1}([0, n]) = \bigcup_{k \in I_n} A_{n,k} \quad (21.488)$$

Let $x \in X$ and $n \in \mathbb{N}$ then we have the following possible cases

$f(x) < n$. Then $x \in f^{-1}([0, n]) \stackrel{[\text{eq: 21.488}]}{=} \bigcup_{k \in I_n} A_{n,k}$ hence there exist a $k \in I_n$ [which is unique by [eq: 21.483]] such that $x \in A_{n,k}$, hence by [eq: 21.487] we have $f_n(x) = \frac{k-1}{2^n}$. As $x \in A_{n,k}$ we have also $\frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$ so that

$$f_n(x) \leq f(x) \text{ and } f_n(x) = \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$$

$n \leq f(x)$. Then $x \notin f^{-1}([0, n]) \stackrel{[\text{eq: 21.488}]}{=} \bigcup_{k \in I_n} A_{n,k}$ so that $x \in X \setminus \bigcup_{k \in I_n} A_{n,k}$ and from [eq: 21.487] we have that

$$f_n(x) = n \leq f(x)$$

Summarizing the above we have three results

$$\forall x \in X \text{ we have } f_n(x) \leq f(x) \quad (21.489)$$

$$\forall x \in X, \forall n \in \mathbb{N} \text{ with } f(x) < n \text{ there exist a } k \in I_n \text{ with } f_n(x) = \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n} \quad (21.490)$$

$$\forall x \in X, \forall n \in \mathbb{N} \text{ with } n \leq f(x) \text{ we have } f_n(x) = n \quad (21.491)$$

Next we show that $\{f_n\}_{n \in \mathbb{N}}$ is increasing. Let $x \in X$ then we have for $n \in \mathbb{N}$ either:

$f(x) < n$. Then $f(x) < n, n+1$ so using [eq: 21.490] there exists $k \in I_n, l \in I_{n+1}$ such that

$$f_n(x) = \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n} \wedge f_{n+1}(x) = \frac{l-1}{2^{n+1}} \leq f(x) < \frac{l}{2^{n+1}}$$

From this it follows that

$$\begin{aligned} \frac{k-1}{2^n} < \frac{l}{2^{n+1}} &\Rightarrow \frac{2 \cdot k - 2}{2^{n+1}} < \frac{l}{2^{n+1}} \\ &\Rightarrow 2 \cdot k - 2 < l \\ &\Rightarrow 2 \cdot k - 2 \leq l - 1 \\ &\Rightarrow \frac{2 \cdot k - 2}{2^{n+1}} \leq \frac{l - 1}{2^{n+1}} \\ &\Rightarrow \frac{k-1}{2^n} \leq \frac{l-1}{2^{n+1}} \\ &\Rightarrow f_n(x) \leq f_{n+1}(x) \end{aligned}$$

proving

$$f_n(x) \leq f_{n+1}(x)$$

$n \leq f(x) < n+1$. Then by [eq: 21.491] $f_n(x) = n$ and by [eq: 21.490] there exist a $k \in I_{n+1}$ such that $f_{n+1}(x) = \frac{k-1}{2^{n+1}} \leq f(x) < \frac{k}{2^{n+1}}$. As $n \leq f(x)$ it follows that

$$\begin{aligned} n < \frac{k}{2^{n+1}} &\Rightarrow n \cdot 2^{n+1} < k \\ &\Rightarrow n \cdot 2^{n+1} \leq k-1 \\ &\Rightarrow n \leq \frac{k-1}{2^{n+1}} \\ &\Rightarrow f_n(x) \leq f_{n+1}(x) \end{aligned}$$

proving

$$f_n(x) \leq f_{n+1}(x)$$

$n+1 \leq f(x)$. Then $n, n+1 \leq f(x)$ so that by [eq: 21.491] $f_n(x) = n < n+1 = f_{n+1}(x)$ so that we have again

$$f_n(x) \leq f_{n+1}(x)$$

So we have proved that

$$\forall x \in X \text{ we have } \forall n \in \mathbb{N} \text{ that } f_n(x) \leq f_{n+1}(x) \quad (21.492)$$

Using then [theorem: 20.92] on the above it follows that

$$\forall x \in X \lim_{n \rightarrow \infty} f_n(x) \text{ exists and } \lim_{n \rightarrow \infty} f_n(x) = \sup(\{f_n(x) | n \in \mathbb{N}\}) \quad (21.493)$$

Let $x \in X$. Using [eq: 21.489] we have that $\forall n \in \mathbb{N} \ f_n(x) \leq f(x)$ so that

$$\sup(\{f_n(x) | n \in \mathbb{N}\}) \leq f(x) \quad (21.494)$$

For the opposite inclusion. Assume that $\sup(\{f_n(x) | n \in \mathbb{N}_0\}) < f(x)$ then

$$\sup(\{f_n(x) | n \in \mathbb{N}_0\}) \in \mathbb{R}$$

and for $f(x)$ we have either:

$f(x) = \infty$. By [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $\sup(\{f_n(x) | n \in \mathbb{N}_0\}) < n < \infty = f(x)$. Using then [eq: 21.491] we have $f_n(x) = n$ so that $n = f_n(x) \leq \sup(\{f_n(x) | n \in \mathbb{N}_0\}) < n$ leading to the contradiction $n < n$.

$f(x) < \infty$. As $\sup(\{f_n(x) | n \in \mathbb{N}\}) < f(x) \Rightarrow f(x) - \sup(\{f_n(x) | n \in \mathbb{N}\}) \in \mathbb{R}^+$ there exist by [theorem: 10.64] a $n_1 \in \mathbb{N}$ such that

$$\frac{1}{2^{n_1}} < f(x) - \sup(\{f_n(x) | n \in \mathbb{N}\})$$

Further using [theorem: 10.30] there exist a $n_2 \in \mathbb{N}$ such that $f(x) < n_2$. Let $n = \max(n_1, n_2)$ then $f(x) < n$ so that by [eq: 21.490] there exist a $k \in I_n$ such that

$$f_n(x) = \frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}.$$

Then

$$f(x) - f_n(x) < \frac{k}{2^n} - \frac{k-1}{2^n} = \frac{1}{2^n} \leq \frac{1}{2^{n_1}} < f(x) - \sup(\{f_n(x) | n \in \mathbb{N}\})$$

so that $\sup(\{f_n(x) | n \in \mathbb{N}\}) < f_n(x)$ which as $f_n(x) \leq \sup(\{f_n(x) | n \in \mathbb{N}\})$ is a contradiction.

As in all cases we reach a contradiction the assumption is invalid and we must have that $f(x) \leq \sup(\{f_n(x) | n \in \mathbb{N}\})$ which combined with [eq: 21.494] proves that $f(x) = \sup(\{f_n(x) | n \in \mathbb{N}\})$, hence using [eq: 21.493] we have

$$\forall x \in X \text{ we have } \lim_{n \rightarrow \infty} f_n(x) \text{ exist and } f(x) = \lim_{n \rightarrow \infty} f_n(x) = \sup(\{f_n(x) | n \in \mathbb{N}\}) \quad \square$$

We set out now to provide an alternative definition of $\int^+ f$. As we can approximate f by a sequence of simple functions $\{f_n\}_{n \in \mathbb{N}}$ we will approximate $\int^+ f d\mu$ by $\int^S f_n d\mu$. The following illustrates this for $n=2$ and $X = \mathbb{R}$ here

$x \in$	$f^{-1}([0, 1/4])$	$\Rightarrow$	$f_2(x) = 0/4$
$x \in$	$f^{-1}([1/4, 2/4])$	$\Rightarrow$	$f_2(x) = 1/4$
$x \in$	$f^{-1}([2/4, 3/4])$	$\Rightarrow$	$f_2(x) = 2/4$
$x \in$	$f^{-1}([3/4, 4/4])$	$\Rightarrow$	$f_2(x) = 3/4$
$x \in$	$f^{-1}([4/4, 5/4])$	$\Rightarrow$	$f_2(x) = 4/4$
$x \in$	$f^{-1}([5/4, 6/4])$	$\Rightarrow$	$f_2(x) = 5/4$
$x \in$	$f^{-1}([6/4, 7/4])$	$\Rightarrow$	$f_2(x) = 6/4$
$x \in$	$f^{-1}([7/4, 8/4])$	$\Rightarrow$	$f_2(x) = 7/4$
$x \in$	$f^{-1}([8/4, \infty])$	$\Rightarrow$	$f_2(x) = 2$

so that

$$f_2 = \mathcal{X}_{f^{-1}([0, 1/4])} \cdot \frac{0}{4} + \mathcal{X}_{f^{-1}([1/4, 2/4])} \cdot \frac{1}{4} + \mathcal{X}_{f^{-1}([2/4, 3/4])} \cdot \frac{2}{4} + \mathcal{X}_{f^{-1}([3/4, 4/4])} \cdot \frac{3}{4} + \mathcal{X}_{f^{-1}([4/4, 5/4])} \cdot \frac{4}{4} + \mathcal{X}_{f^{-1}([5/4, 6/4])} \cdot \frac{5}{4} + \mathcal{X}_{f^{-1}([6/4, 7/4])} \cdot \frac{6}{4} + \mathcal{X}_{f^{-1}([7/4, 8/4])} \cdot \frac{7}{4} + \mathcal{X}_{f^{-1}([8/4, \infty])} \cdot \frac{8}{4}$$

and the simple integral becomes

$$\begin{aligned} \int^S f_2 d\mu &= \frac{0}{4} \cdot \mu\left(f^{-1}\left(\left[0, \frac{1}{4}\right]\right)\right) + \frac{1}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{1}{4}, \frac{2}{4}\right]\right)\right) + \frac{2}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{2}{4}, \frac{3}{4}\right]\right)\right) + \\ &\quad \frac{3}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{3}{4}, \frac{4}{4}\right]\right)\right) + \frac{4}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{4}{4}, \frac{5}{4}\right]\right)\right) + \frac{5}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{5}{4}, \frac{6}{4}\right]\right)\right) + \\ &\quad \frac{6}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{6}{4}, \frac{7}{4}\right]\right)\right) + \frac{7}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{7}{4}, \frac{8}{4}\right]\right)\right) + \frac{8}{4} \cdot \mu\left(f^{-1}\left(\left[\frac{8}{4}, \infty\right]\right)\right) \end{aligned}$$

which is illustrated in the below figure

Note 21.326. For the integral $\int^S f_2 d\mu$ to make sense the sets $f^{-1}\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right]$ must be measurable, which is the reason why we must require that f is measurable. The above figure suggests that the surface beneath f is approximated from below by an increasing sequence $\int^S f_n d\mu$ this will provide us with an alternative definition for $\int^+ f d\mu$.

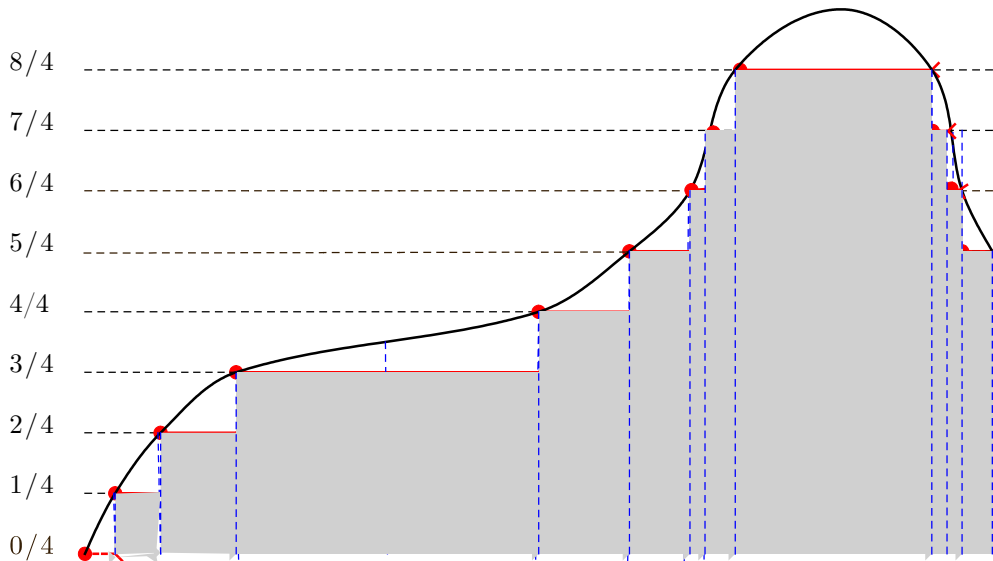


Figure 21.7. f approximated by f_2

Theorem 21.327. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $\{f_k\}_{k \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that

$$\forall x \in X \text{ we have } f_k(x) \leq f_{k+1}(x)$$

and

$$\lim_{k \rightarrow \infty} f_k \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

then

$$\lim_{k \rightarrow \infty} \int^S f_k d\mu \text{ exist}$$

and

$$\int^+ \left(\lim_{k \rightarrow \infty} f_k \right) d\mu = \sup \left(\left\{ \int^S f_k d\mu \mid k \in \mathbb{N} \right\} \right) = \lim_{k \rightarrow \infty} \int^S f_k d\mu \quad [\text{theorem: 21.317}] = \lim_{k \rightarrow \infty} \int^+ f_k d\mu$$

Note 21.328. As $\forall x \in X$ we have $\forall k \in \mathbb{N} \ f_k(x) \leq f_{k+1}(x)$ we have by [theorem: 20.92] that $\lim_{k \rightarrow \infty} f_k(x)$ exist, as is needed for the requirements of this theorem.

Proof. As $\forall x \in X$ we have $\forall k \in \mathbb{N} \ f_k(x) \leq f_{k+1}(x)$ we have by [theorem: 20.92] that

$$\forall x \in X \ \lim_{k \rightarrow \infty} f_k(x) \text{ exists and } \lim_{k \rightarrow \infty} f_k(x) = \sup \{ f_k(x) \mid k \in \mathbb{N} \} \quad (21.495)$$

so that

$$\forall x \in X \text{ we have } \forall k \in \mathbb{N} \text{ that } f_k(x) \leq \lim_{k \rightarrow \infty} f_k(x) \quad (21.496)$$

Further using [theorem: 21.309] we have that $\forall k \in \mathbb{N} \ \int^S f_k d\mu \leq \int^S f_{k+1} d\mu$ which by [theorem: 20.92] proves that

$$\lim_{k \rightarrow \infty} \int^S f_k d\mu \text{ exists and } \lim_{k \rightarrow \infty} \int^S f_k d\mu = \sup \left(\left\{ \int^S f_k d\mu \mid k \in \mathbb{N} \right\} \right) \quad (21.497)$$

So that

$$\forall k \in \mathbb{N} \text{ we have } \int^S f_k d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu \quad (21.498)$$

Let $k \in \mathbb{N}$. As by [eq: 21.496] $\forall x \in X \ f_k(x) \leq \lim_{k \rightarrow \infty} f_k(x) = \left(\lim_{k \rightarrow \infty} f_k \right)(x)$ and $f_k \in \mathcal{S}^+[X, \mathcal{A}]$ we have that $\int^S f_k d\mu \in \left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \ g(x) \leq \left(\lim_{k \rightarrow \infty} f_k \right)(x) \right\}$ so that

$$\int^S f_k d\mu \leq \sup \left(\left\{ \int^S g d\mu \mid g \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \forall x \in X \ g(x) \leq \left(\lim_{k \rightarrow \infty} f_k \right)(x) \right\} \right)$$

hence we have by [definition: 21.313] that

$$\forall k \in \mathbb{N} \ \int^S f_k d\mu \leq \int^+ \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \quad (21.499)$$

So $\int^+ \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$ is a upper bound for $\left\{ \int^S f_k d\mu \mid k \in \mathbb{N} \right\}$ from which it follows that $\sup \left(\left\{ \int^S f_k d\mu \mid k \in \mathbb{N} \right\} \right) \leq \int^+ \left(\lim_{k \rightarrow \infty} f_k \right) d\mu$ which by [eq: 21.497] proves that

$$\lim_{k \rightarrow \infty} \int^S f_k d\mu \leq \int^+ \left(\lim_{k \rightarrow \infty} f_k \right) d\mu \quad (21.500)$$

For the opposite inequality. Let $g \in \mathcal{S}^+[X, \mathcal{A}]$ be such that $\forall x \in X \ g(x) \leq \left(\lim_{k \rightarrow \infty} f_k \right)(x) = \lim_{k \rightarrow \infty} f_k(x)$. Then using [theorem: 21.309] it follows that

$$\forall k \in \mathbb{N} \ g \wedge f_k \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \int^S (g \wedge f_k) d\mu \leq \int^S f_k d\mu \quad (21.501)$$

Let $x \in X$ then given $k \in \mathbb{N}$ we have

$$(g \wedge f_k)(x) = \min(g(x), f_k(x)) \leq \min(g(x), f_{k+1}(x)) = (g \wedge f_{k+1})(x) \quad (21.502)$$

so that by [theorem: 20.92]

$$\lim_{k \rightarrow \infty} (g \wedge f_k)(x) \text{ exists and } \lim_{k \rightarrow \infty} (g \wedge f_k)(x) = \sup(\{(g \wedge f_k)(x) | k \in \mathbb{N}\}) \quad (21.503)$$

As $\forall k \in \mathbb{N} (g \wedge f_k)(x) = \min(g(x), f_k(x)) \leq g(x)$ we have that

$$\sup(\{(g \wedge f_k)(x) | k \in \mathbb{N}\}) \leq g(x) \quad (21.504)$$

As $g(x) \leq \left(\lim_{k \rightarrow \infty} f_k\right)(x)$ we have for $g(x)$ either:

$g(x) = \left(\lim_{k \rightarrow \infty} f_k\right)(x)$. Then $g(x) \stackrel{[\text{eq: 21.495}]}{=} \sup(\{f_k(x) | k \in \mathbb{N}\})$ so that $\forall k \in \mathbb{N}$ we have $f_k(x) \leq g(x)$ hence $(g \wedge f_k)(x) = \min(g(x), f_k(x)) = f_k(x)$ which proves that

$$\sup(\{(g \wedge f_k)(x) | k \in \mathbb{N}\}) = \sup(\{f_k(x) | k \in \mathbb{N}\}) \stackrel{[\text{eq: 21.495}]}{=} \left(\lim_{k \rightarrow \infty} f_k\right)(x) = g(x)$$

$g(x) < \left(\lim_{k \rightarrow \infty} f_k\right)(x)$. Then by [eq: 21.495] $g(x) < \sup(\{f_k(x) | k \in \mathbb{N}\})$ so that there exist a $k \in \mathbb{N}$ such that $g(x) < f_k(x)$ hence $(g \wedge f_k)(x) = \min(g(x), f_k(x)) = g(x)$. Hence $g(x) \in \{(g \wedge f_k)(x) | k \in \mathbb{N}\}$ so that $g(x) \leq \sup(\{(g \wedge f_k)(x) | k \in \mathbb{N}\})$ which by [eq: 21.504] proves that

$$g(x) = \sup(\{(g \wedge f_k)(x) | k \in \mathbb{N}\})$$

So in all cases $g(x) = \sup(\{(g \wedge f_k)(x) | k \in \mathbb{N}\})$ which by [eq: 21.503] proves $\lim_{k \rightarrow \infty} (g \wedge f_k)(x) = g(x)$ hence

$$\lim_{k \rightarrow \infty} (g \wedge f_k) = g \in \mathcal{S}^+[X, \mathcal{A}] \quad (21.505)$$

The above and [eq: 21.502] allows us to apply [theorem: 21.311] to get that

$$\lim_{k \rightarrow \infty} \int^S (g \wedge f_k) d\mu \text{ exists and } \lim_{k \rightarrow \infty} \int^S (g \wedge f_k) d\mu = \int^S \left(\lim_{k \rightarrow \infty} (g \wedge f_k)\right) d\mu = \int^S g d\mu \quad (21.506)$$

By [eq: 21.501] $\int^S (g \wedge f_k) d\mu \leq \int^S f_k d\mu$ so that by [theorem: 20.85] we have

$$\lim_{k \rightarrow \infty} \int^S (g \wedge f_k) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

Combining this with [eq: 21.506] gives

$$\int^S g d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

So $\lim_{k \rightarrow \infty} \int^S f_k d\mu$ is an upper bound for $\left\{ \int^S g d\mu | g \in \mathcal{S}^+[X, \mathcal{A}] \text{ with } g(x) \leq \left(\lim_{k \rightarrow \infty} f_k\right)(x) \right\}$ proving that

$$\sup\left(\left\{ \int^S g d\mu | g \in \mathcal{S}^+[X, \mathcal{A}] \text{ with } g(x) \leq \left(\lim_{k \rightarrow \infty} f_k\right)(x) \right\}\right) \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

As $\int^+ \left(\lim_{k \rightarrow \infty} f_k\right) d\mu = \sup\left(\left\{ \int^S g d\mu | g \in \mathcal{S}^+[X, \mathcal{A}] \text{ with } g(x) \leq \left(\lim_{k \rightarrow \infty} f_k\right)(x) \right\}\right)$ we have that

$$\int^+ \left(\lim_{k \rightarrow \infty} f_k\right) d\mu \leq \lim_{k \rightarrow \infty} \int^S f_k d\mu$$

Combining this with [eq: 21.500] proves that

$$\int^+ \left(\lim_{k \rightarrow \infty} f_k\right) d\mu = \lim_{k \rightarrow \infty} \int^S f_k d\mu \stackrel{[\text{eq: 21.497}]}{=} \sup\left(\left\{ \int^S f_k d\mu | k \in \mathbb{N} \right\}\right) \quad \square$$

Next we state a alternative way to define the integral $\int^+$. The benefit of this alternative definition is that it is based on a limiting process making it easier to lift properties of $\int^S$ to $\int^+$. In this theorem the measurability of f is a requirement.

Theorem 21.329. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then there exist a sequence $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that*

$$\forall x \in X \text{ we have } \forall n \in \mathbb{N} \text{ that } f_n(x) \leq f_{n+1}(x)$$

and

$$\forall x \in X \text{ we have } \lim_{n \rightarrow \infty} f_n(x) \text{ exists and } f(x) = \lim_{n \rightarrow \infty} f_n(x) = \sup(\{f_n(x) | n \in \mathbb{N}\})$$

and

$$\lim_{n \rightarrow \infty} \int^S f_n d\mu \text{ exist and } \int^+ f d\mu = \lim_{n \rightarrow \infty} \int^S f_n d\mu = \sup\left(\left\{\int^S f_n d\mu | n \in \mathbb{N}\right\}\right)$$

Proof. Using [theorem: 21.325] there exists a $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that $\forall x \in X$ we have $\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x)$ and $\lim_{n \rightarrow \infty} f_n(x) = f(x) = \sup(\{f_n(x) | n \in \mathbb{N}\})$ so that

$$\lim_{n \rightarrow \infty} f_n = f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Using this result together with the previous theorem [theorem: 21.327] proves that $\lim_{n \rightarrow \infty} \int^S f_n d\mu$ exists and

$$\int^+ f d\mu = \int^+ \left(\lim_{n \rightarrow \infty} f_n\right) d\mu = \lim_{n \rightarrow \infty} \int^S f_n d\mu \quad \square$$

21.5.2.4 Properties of the integral of non negative measurable functions

We prove now that $\int^+$ has similar properties as $\int^S$.

Theorem 21.330. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $f = 0$ μ -a.e. then $\int^+ f d\mu = 0$.*

Proof. As $f = 0$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in X | 0 < f(x)\} \underset{f(X) \subseteq [0, \infty]}{\subseteq} \{x \in X | 0 \neq f(x)\} \subseteq N$$

Let $g \in \mathcal{S}^+[X, \mathcal{A}]$ be such that $\forall x \in X \ g(x) \leq f(x)$. Then for

$$x \in \{x \in X | g(x) \neq 0\} \underset{g(X) \subseteq [0, \infty]}{\subseteq} \{x \in X | 0 < g(x)\}$$

we have $0 < g(x) \leq f(x)$ so that $x \in \{x \in X | 0 < f(x)\} \subseteq N$. Hence $\{x \in X | g(x) \neq 0\} \subseteq N$ which proves that $g = 0$ μ -a.e.. Applying then [theorem: 21.309 (5)] results in $\int^S g d\mu = 0$. So we have proved that

$$\left\{\int^S g d\mu | g \in \mathcal{S}^+[X, \mathcal{A}] \text{ with } \forall x \in X \ g(x) \leq f(x)\right\} = \{0\}$$

Hence

$$\int^+ f d\mu = \sup\left(\int g d\mu | g \in \mathcal{S}^+[X, \mathcal{A}] \text{ with } \forall x \in X \ g(x) \leq f(x)\right) = 0 \quad \square$$

Theorem 21.331. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f, g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then we have*

1. $\forall \alpha \in [0, \infty[$ we have $\alpha \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+(\alpha \cdot f) d\mu = \alpha \cdot \int^+ f d\mu$.
2. $f + g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+(f + g) d\mu = \int^+ f d\mu + \int^+ g d\mu$.
3. If $\forall x \in X$ we have $f(x) \leq g(x)$ then $\int^+ f d\mu \leq \int^+ g d\mu$
4. $\forall A \in \mathcal{A}$ we have that $\int_A^+ f d\mu \underset{\text{definition: 21.320}}{=} \int^+(\chi_{X,A} \cdot f) d\mu \leq \int^+ f d\mu$

Proof. First using [theorem: 21.329] there exists $\{f_n\}_{n \in \mathbb{N}}, \{g_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that $\forall x \in X$ we have $\forall n \in \mathbb{N}$ that $f_n(x) \leq f_{n+1}(x)$, $g_n(x) \leq g_{n+1}(x)$ and

$$f = \lim_{n \rightarrow \infty} f_n, g = \lim_{n \rightarrow \infty} g_n, \int^+ f d\mu = \lim_{n \rightarrow \infty} \int^S f_n d\mu \text{ and } \int^+ g d\mu = \lim_{n \rightarrow \infty} \int^S g_n d\mu \quad (21.507)$$

then we have:

1. Using [theorems: 21.263, 21.309] and the fact that $0 < \alpha$ it follows that

$$\alpha \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \{\alpha \cdot f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$$

Given $x \in X$ we have

$$\forall n \in \mathbb{N} \text{ we have } (\alpha \cdot f_n)(x) = \alpha \cdot f_n(x) \underset{0 < \alpha}{\leq} \alpha \cdot f_{n+1}(x) = (\alpha \cdot f_{n+1})(x)$$

and

$$\begin{aligned} (\alpha \cdot f)(x) &= \alpha \cdot f(x) \\ &\stackrel{[\text{eq: 21.507}]}{=} \alpha \cdot \lim_{n \rightarrow \infty} f_n(x) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} (\alpha \cdot f_n(x)) \\ &= \lim_{n \rightarrow \infty} (\alpha \cdot f_n)(x) \end{aligned}$$

Hence using [theorem: 21.327] we have that $\lim_{n \rightarrow \infty} \int^S (\alpha \cdot f_n) d\mu$ exists and

$$\begin{aligned} \int^+ (\alpha \cdot f) d\mu &= \lim_{n \rightarrow \infty} \int^S (\alpha \cdot f_n) d\mu \\ &\stackrel{[\text{theorem: 21.309}]}{=} \lim_{n \rightarrow \infty} \left(\alpha \cdot \int^S f_n d\mu \right) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \alpha \cdot \lim_{n \rightarrow \infty} \int^S f_n d\mu \\ &\stackrel{[\text{eq: 21.507}]}{=} \alpha \cdot \int^+ f_n d\mu \end{aligned}$$

2. Using [theorems: 21.257, 21.309] it follows that

$$f + g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \{f_n + g_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$$

Given $x \in X$ we have

$$\forall n \in \mathbb{N} \text{ we have } (f_n + g_n)(x) = f_n(x) + g_n(x) \leq f_{n+1}(x) + g_{n+1}(x) = (f_{n+1} + g_{n+1})(x)$$

and

$$\begin{aligned} (f + g)(x) &= f(x) + g(x) \\ &\stackrel{[\text{eq: 21.507}]}{=} \lim_{n \rightarrow \infty} f_n(x) + \lim_{n \rightarrow \infty} g_n(x) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} (f_n(x) + g_n(x)) \\ &= \lim_{n \rightarrow \infty} (f_n + g_n)(x) \end{aligned}$$

Hence using [theorem: 21.327] we have that $\lim_{n \rightarrow \infty} \int^S (f_n + g_n) d\mu$ exists and

$$\begin{aligned} \int^+ (f + g) d\mu &= \lim_{n \rightarrow \infty} \int^S (f_n + g_n) d\mu \\ &\stackrel{[\text{theorem: 21.309}]}{=} \lim_{n \rightarrow \infty} \left(\int^S f_n d\mu + \int^S g_n d\mu \right) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} \int^S f_n d\mu + \lim_{n \rightarrow \infty} \int^S g_n d\mu \\ &\stackrel{[\text{eq: 21.507}]}{=} \int^+ f_n d\mu + \int^+ g_n d\mu \end{aligned}$$

3. Assume that $\forall x \in X$ we have $f(x) \leq g(x)$. Let

$$y \in \left\{ \int^S h d\mu \mid h \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } h(x) \leq f(x) \right\}$$

then there exist a $h \in \mathcal{S}^+[X, \mathcal{A}]$ satisfying $\forall x \in X$ $h(x) \leq f(x)$ such that $y = \int^S h d\mu$. As $\forall x \in X$ $f(x) \leq g(x)$ we have also $h(x) \leq g(x)$ and it follows that

$$y \in \left\{ \int^S h d\mu \mid h \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } h(x) \leq g(x) \right\}$$

Hence

$$\begin{aligned} \left\{ \int^S h d\mu \mid h \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } h(x) \leq f(x) \right\} &\subseteq \\ \left\{ \int^S h d\mu \mid h \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } h(x) \leq g(x) \right\} \end{aligned}$$

So

$$\begin{aligned} \int^+ f d\mu &\stackrel{\text{def}}{=} \sup \left(\left\{ \int^S h d\mu \mid h \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } h(x) \leq f(x) \right\} \right) \\ &\stackrel{[\text{theorem: 3.77}]}{\leq} \sup \left(\left\{ \int^S h d\mu \mid h \in \mathcal{S}^+[X, \mathcal{A}] \wedge \forall x \in X \text{ we have } h(x) \leq g(x) \right\} \right) \\ &= \int^+ g d\mu \end{aligned}$$

4. Let $x \in X$ then $(\mathcal{X}_{X,A} \cdot f)(x) = \mathcal{X}_{X,A}(x) \cdot f(x) = \begin{cases} f(x) & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases} \leq f(x)$ so that by (3)

$$\int^+ (\mathcal{X}_{X,A} \cdot f) d\mu \leq \int^+ f d\mu \quad \square$$

Next we use induction to extend the last part of the previous theorem to a finite sum of measurable functions.

Theorem 21.332. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $n \in \mathbb{N}$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$\sum_{i=1}^n f_i \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ \left(\sum_{i=1}^n f_i \right) d\mu = \sum_{i=1}^n \int^+ f_i d\mu$$

Proof. We use induction to prove this, so let

$$\mathcal{S} = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ then } \sum_{i=1}^n f_i \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ \left(\sum_{i=1}^n f_i \right) d\mu = \sum_{i=1}^n \int^+ f_i d\mu \right\}$$

then we have:

$1 \in \mathcal{S}$. Let $\{f_i\}_{i \in \{1, \dots, 1\}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then $\sum_{i=1}^1 f_i = f_1 \in \mathcal{M}^+[X, \mathcal{A}]$ and

$$\int^+ \left(\sum_{i=1}^1 f_i \right) d\mu = \int^+ f_1 d\mu = \sum_{i=1}^1 \int^+ f_i d\mu$$

proving that $1 \in \mathcal{S}$.

$\mathbf{n} \in \mathcal{S} \Rightarrow \mathbf{n} + \mathbf{1} \in \mathcal{S}$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then as $n \in \mathcal{S}$ we have that $\sum_{i=1}^n f_i \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ hence

$$\sum_{i=1}^{n+1} f_i = \left(\sum_{i=1}^n f_i \right) + f_{n+1} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad [\text{using [theorem: 21.331 (2)}]$$

Further we have

$$\begin{aligned} \int^+ \left(\sum_{i=1}^{n+1} f_i \right) d\mu &= \int^+ \left(\left(\sum_{i=1}^n f_i \right) + f_{n+1} \right) d\mu \\ &\stackrel{[\text{theorem: 21.331(2)}]}{=} \int^+ \left(\sum_{i=1}^n f_i \right) d\mu + \int^+ f_{n+1} d\mu \\ &\stackrel{=}{=} \left(\sum_{i=1}^n \int^+ f_i d\mu \right) + \int^+ f_{n+1} d\mu \\ &= \sum_{i=1}^{n+1} \int^+ f_i d\mu \end{aligned}$$

proving that $n+1 \in \mathcal{S}$. □

Theorem 21.333. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and let $f, g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $f = g$ μ -a.e. then

$$\int^+ f d\mu = \int^+ g d\mu$$

Proof. Define

$$H = \{x \in X \mid f(x) \neq g(x)\} = \{f \neq g\}$$

As f, g are $\mathcal{A}$ -measurable we have by [theorem: 21.256] that $H \in \mathcal{A}$. Further as $f = g$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $H \subseteq N$ hence $\mu(H) \leq \mu(N) = 0$. Summarized we have

$$H \in \mathcal{A} \text{ and } \mu(H) = 0 \quad (21.508)$$

Define

$$h: X \rightarrow [0, \infty] \text{ by } h(x) = \begin{cases} \infty & \text{if } x \in H \\ 0 & \text{if } x \in X \setminus H \end{cases}$$

Let $x \in \mathbb{R}$ then we have either:

0 ≤ x. Then if $t \in \{h > x\}$ we have $h(t) > x \geq 0$ so that $x \in H$ [otherwise $h(t) = 0 \not> 0$] so that $\{h > x\} \subseteq H$. If $t \in H$ then $h(t) = \infty > x$ so that $H \subseteq \{h > x\}$. Summarized we have

$$\{h > x\} = H \in \mathcal{A}$$

x < 0. Then if $t \in X$ we have either $t \in H$ so that $h(t) = \infty > x \Rightarrow t \in \{h > x\}$ or $t \in X \setminus H$ giving $h(t) = 0 > x \Rightarrow t \in \{h > x\}$, hence $X \subseteq \{h > x\} \subseteq X$ proving

$$\{h > x\} = X \in \mathcal{A}$$

So $\forall x \in \mathbb{R}$ we have $\{h > x\} \in \mathcal{A}$ from which it follows, using [theorem: 21.254] and the fact that $h(X) \in [0, \infty]$, that

$$h \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.509)$$

Let $n \in \mathbb{N}$ and define

$$h_n: X \rightarrow \overline{\mathbb{R}} \text{ by } h_n = n \cdot \mathcal{X}_{X, H}$$

then by [example: 21.319] we have

$$h_n = n \cdot \mathcal{X}_{X, H} \in \mathcal{S}^+[X, \mathcal{A}] \text{ and } \int^S h_n d\mu = n \cdot \mu(H) \stackrel{[\text{eq: 21.508}]}{=} 0 \quad (21.510)$$

Further we have

$$\forall n \in \mathbb{N}, \forall x \in X \text{ that } h_n(x) = \begin{cases} n & \text{if } x \in H \\ 0 & \text{if } x \in X \setminus H \end{cases} \leq \begin{cases} n+1 & \text{if } x \in H \\ 0 & \text{if } x \in X \setminus H \end{cases} = h_{n+1}(x) \quad (21.511)$$

Let $x \in X$ then we have either:

$$\mathbf{x} \in \mathbf{H}. \text{ Then } \lim_{n \rightarrow \infty} h_n(x) = \lim_{n \rightarrow \infty} n \stackrel{[\text{theorem: 20.82}]}{=} \infty = h(x).$$

$$\mathbf{x} \in \mathbf{X} \setminus \mathbf{H}. \text{ Then } \lim_{n \rightarrow \infty} h_n(x) = \lim_{n \rightarrow \infty} 0 = 0 = h(x)$$

proving that $\lim_{n \rightarrow \infty} h_n(x) = h(x)$ so that

$$\lim_{n \rightarrow \infty} h_n = h \stackrel{[\text{eq: 21.509}]}{\in} \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

This fact together with [eqs: 21.510 and 21.511] allows us to use [theorem: 21.327] giving

$$\lim_{n \rightarrow \infty} \int^S h_n d\mu \text{ exist and } \int^+ h d\mu = \lim_{n \rightarrow \infty} \int^S h_n d\mu \stackrel{[\text{eq: 21.510}]}{=} \lim_{n \rightarrow \infty} 0 = 0 \quad (21.512)$$

Further if we look again at the possibilities of $x \in X$ we have

$$\mathbf{x} \in \mathbf{H}. \text{ Then } h(x) = \infty \text{ so that } f(x) \leq \infty = g(x) + \infty = g(x) + h(x) \text{ and } g(x) \leq \infty = f(x) + \infty = f(x) + h(x) \text{ so that } f(x) \leq g(x) + h(x) \text{ and } g(x) \leq f(x) + h(x).$$

$$\mathbf{x} \in \mathbf{X} \setminus \mathbf{H}. \text{ Then } f(x) = g(x) = g(x) + 0 = g(x) + h(x) \text{ and } g(x) = f(x) = f(x) + 0 = f(x) + h(x)$$

So in all cases we have $f(x) \leq g(x) + h(x) \wedge g(x) \leq f(x) + h(x)$. Using [theorem: 21.331] twice we have that

$$\int^+ f d\mu \leq \int^+ (f + h) d\mu = \int^+ g d\mu + \int^+ h d\mu \stackrel{[\text{eq: 21.512}]}{=} \int^+ g d\mu$$

and

$$\int^+ g d\mu \leq \int^+ (f + h) d\mu = \int^+ f d\mu + \int^+ h d\mu \stackrel{[\text{eq: 21.512}]}{=} \int^+ f d\mu$$

which proves that

$$\int^+ f d\mu = \int^+ f d\mu \quad \square$$

Next we prove that every non negative function with a finite integral is μ -a.e. finite. First we need a lemma.

Lemma 21.334. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space, $t \in \mathbb{R}^+$ and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then for*

$$A_t = \{f \geq t\} \stackrel{\text{def}}{=} \{x \in X \mid f(x) \geq t\}$$

we have

$$\mu(A_t) \leq \frac{1}{t} \cdot \int_{A_t}^+ f d\mu \leq \frac{1}{t} \cdot \int^+ f d\mu$$

Note 21.335. *Using [theorem: 21.254] $A_t \in \mathcal{A}$ as is required for the existence of $\int_{A_t}^+ f d\mu$ in this lemma.*

Proof. Consider the function $t \cdot \mathcal{X}_{X, A_t} = \sum_{i=1}^1 a_i \cdot \mathcal{X}_{X, A_i}$ where $a_1 = t \in \mathbb{R}^+$ and $A_1 = A_t$ then as $A_t = \{f \geq t\} \in \mathcal{A}$ [as f is $\mathcal{A}$ -measurable] we have that

$$t \cdot \mathcal{X}_{X, A_t} \stackrel{[\text{theorem: 21.301}]}{\in} \mathcal{S}^+[X, \mathcal{A}]$$

and

$$\int^+ (t \cdot \mathcal{X}_{X, A_t}) d\mu \stackrel{[\text{theorem: 21.317}]}{=} \int^S (t \cdot \mathcal{X}_{X, A_t}) d\mu \stackrel{\text{def}}{=} \sum_{i=1}^1 a_i \mu(A_i) = t \cdot \mu(A_t) \quad (21.513)$$

Let $x \in X$ then we have either

$$\mathbf{x} \in \mathbf{A}_t. \text{ Then } 0 < t \leq f(x) \text{ and as } \mathcal{X}_{X, A_t}(x) = 1 \text{ we have } 0 \leq t \cdot \mathcal{X}_{X, A_t}(x) \leq f(x) \cdot \mathcal{X}_{X, A_t} \leq f(x)$$

$x \in X \setminus A_t$. Then $\mathcal{X}_{X, A_t}(x) = 0$ so that trivially $0 \leq t \cdot \mathcal{X}_{X, A_t}(x) \leq f(x) \cdot \mathcal{X}_{X, A_t}(x) \leq f(x)$ hence in all cases we have $0 \leq t \cdot \mathcal{X}_{X, A_t}(x) \leq \mathcal{X}_{X, A_t} \cdot f(x) \leq f(x)$ so that by [theorem: 21.331]

$$t \cdot \mu(A_i) \stackrel{[\text{eq: 21.513}]}{=} \int^+ (t \cdot \mathcal{X}_{X, A_t}) d\mu \leq \int^+ (\mathcal{X}_{X, A_t} \cdot f) d\mu \leq \int^+ f d\mu$$

proving that

$$\mu(A_i) \leq \frac{1}{t} \cdot \int^+ f d\mu \leq \frac{1}{t} \cdot \int^+ f d\mu \quad \square$$

Theorem 21.336. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $\int^+ f d\mu < \infty$ then

$$\mu(\{x \in X : f(x) = \infty\}) = 0$$

proving that

$$f \neq \infty \text{ } \mu\text{-a.e.}$$

Note 21.337. Using [theorem: 21.256] we have that $\{x \in X | f(x) = \infty\} = \{f = \infty\} \in \mathcal{A}$ so that $\mu(\{x \in X | f(x) = \infty\})$ is well defined.

Proof. Let $n \in \mathbb{N}_0$ then $n \in \mathbb{R}^+$ so using the above lemma [lemma: 21.334] we have that

$$\mu(\{x \in X | f(x) \geq n\}) \leq \frac{1}{n} \cdot \int^+ f d\mu$$

Now if $x \in \{x \in X | f(x) = \infty\}$ we have $n \leq \infty = f(x)$ so that $x \in \{x \in X | f(x) \geq n\}$ proving that

$$\{f = \infty\} = \{x \in X | f(x) = \infty\} \subseteq \{x \in X | f(x) \geq n\}$$

As $\{f = \infty\} \in \mathcal{A}$ [see theorem: 21.256] we have

$$\mu(\{x \in X | f(x) = \infty\}) \leq \mu(\{x \in X | f(x) \geq n\}) \leq \frac{1}{n} \cdot \int^+ f d\mu$$

As n is chosen arbitrary and $0 \leq \int^+ f d\mu < \infty \Rightarrow$ we have by [theorem: 10.31] that

$$\mu(\{x \in X | f(x) = \infty\}) = 0$$

proving that

$$f \neq \infty \text{ } \mu\text{-a.e.} \quad \square$$

21.5.3 Measure integral

21.5.3.1 Definition of the measure integral

We extend now the integral $\int^+$ of non negative measurable functions to the measure integral $\int$ of measurable functions. To do this we write a measurable function f as the difference of two non negative measurable functions f^+ and f^- , so $f = f^+ - f^-$ and define $\int f d\mu$ as $\int f d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu$. Of course this only works if $\int^+ f^+ d\mu - \int^+ f^- d\mu$ is well defined, so we require that $\int^+ f^+ d\mu, \int^+ f^- d\mu$ are finite. First we define f^+, f^- .

Definition 21.338. Let X be a set and $f: X \rightarrow \overline{\mathbb{R}}$ a function then we define:

1. $f^+: X \rightarrow \overline{\mathbb{R}}$ by $f^+(x) = \max(f(x), 0) \in [0, \infty]$
2. $f^-: X \rightarrow \overline{\mathbb{R}}$ by $f^-(x) = -\min(-f(x), 0) \in [0, \infty]$
3. $|f|: X \rightarrow \overline{\mathbb{R}}$ by $|f|(x) = |f(x)|$

Note 21.339. From the definition it follows that $f^+(X) \subseteq [0, \infty], f^-(X) \subseteq [0, \infty]$ and $|f|(X) \subseteq [0, \infty]$.

Example 21.340. Let X be a set and for $y \in \mathbb{R}$ define $C_y: X \rightarrow \overline{\mathbb{R}}$ by $C_y(x) = y$ that

1. If $y \in [0, \infty]$ then $(C_y)^+ = C_y$ and $(C_y)^- = C_0$
2. If $y \in [-\infty, 0]$ then $(C_y)^+ = C_0$ and $(C_y)^- = -C_y$

Proof.

1. Let $y \in [0, \infty]$ then for $x \in X$ we have

$$(C_y)^+(x) = \max(C_y(x), 0) = \max(y, 0) = y = C_y(x)$$

$$(C_y)^-(x) = -\min(C_y(x), 0) = -\min(y, 0) = 0 = C_0(x)$$

2. Let $y \in [-\infty, 0]$ then for $x \in X$ we have

$$(C_y)^+(x) = \max(C_y(x), 0) = \max(y, 0) = 0 = C_0(x)$$

$$(C_y)^-(x) = -\min(C_y(x), 0) = -\min(y, 0) = -y = C_0(x) \quad \square$$

We have the following alternative definition for f^+ , f^- and $|f|$.

Theorem 21.341. Let X be a set and $f: X \rightarrow \overline{\mathbb{R}}$ a function then we have

1. $f^+ = f \vee C_0$
2. $f^- = -(f \wedge C_0)$
3. $|f| = f^+ + f^-$
4. $\forall x \in X \ (f^+(x), -f^-(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ so that $f^+ - f^-$ is well defined and

$$f = f^+ - f^-$$

Proof.

1. Let $x \in X$ then $f^+(x) = \max(f(x), 0) = \max(f(x), C_0(x)) = (f \vee C_0)(x)$. proving that

$$f^+ = f \vee C_0$$

2. Let $x \in X$ then $f^-(x) = -\min(f(x), 0) = -\min(f(x), C_0(x)) = -(f \wedge C_0)(x) = (-(f \wedge C_0))(x)$ proving that

$$f^- = -(f \wedge C_0)$$

3. Let $x \in X$ then for $f(x)$ we have either:

$0 \leq f(x)$. Then

$$\begin{aligned} (f^+ + f^-)(x) &= f^+(x) + f^-(x) \\ &= \max(f(x), 0) + (-\min(f(x), 0)) \\ &= f(x) + (-0) \\ &= f(x) \\ &= |f(x)| \\ &= |f|(x) \end{aligned}$$

$f(x) < 0$. Then

$$\begin{aligned} (f^+ + f^-)(x) &= f^+(x) + f^-(x) \\ &= \max(f(x), 0) + (-\min(f(x), 0)) \\ &= 0 + (-f(x)) \\ &= -f(x) \\ &= |f(x)| \\ &= |f|(x) \end{aligned}$$

which proves that

$$|f| = f^+ + f^-$$

4. Let $x \in X$ then we have the following cases for $f(x)$

$f(x) = \infty$. Then

$$f^+(x) = \max(f(x), 0) = \max(\infty, 0) = \infty = f(x)$$

and

$$f^-(x) = -\min(f(x), 0) = -\min(\infty, 0) = 0$$

so that $(f^+(x), -f^-(x)) = (\infty, 0) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and

$$(f^+ - f^-)(x) = f^+(x) - f^-(x) = f(x) - 0 = f(x)$$

$f(x) = -\infty$. Then

$$f^+(x) = \max(f(x), 0) = \max(-\infty, 0) = 0$$

and

$$f^-(x) = -\min(f(x), 0) = -\min(-\infty, 0) = \infty$$

so that $(f^+(x), -f^-(x)) = (0, -\infty) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and

$$(f^+ - f^-)(x) = f^+(x) - f^-(x) = 0 - \infty = -\infty = f(x)$$

$f(x) \in \mathbb{R}$. Then for $f(x)$ we have either:

$f(x) < 0$. Then we have

$$f^+(x) = \max(f(x), 0) = 0$$

and

$$f^-(x) = -\min(f(x), 0) = -f(x)$$

so that $(f^+(x), -f^-(x)) = (0, -f(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and

$$(f^+ - f^-)(x) = f^+(x) - f^-(x) = 0 - (-f(x)) = f(x)$$

$0 \leq f(x)$. Then we have

$$f^+(x) = \max(f(x), 0) = f(x)$$

and

$$f^-(x) = -\min(f(x), 0) = -0 = 0$$

so that $(f^+(x), -f^-(x)) = (f(x), 0) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and

$$(f^+ - f^-)(x) = f^+(x) - f^-(x) = f(x) - 0 = f(x)$$

So in all cases we have $(f^+(x), -f^-(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and $(f^+ - f^-)(x) = f(x)$. Hence

$$f = f^+ - f^- \quad \square$$

Theorem 21.342. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $f: X \rightarrow \overline{\mathbb{R}}$ then we have:

1. $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \Leftrightarrow f^+ \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f^- \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$
2. If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then $|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$

Note 21.343. If $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ then $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f(X) \subseteq \mathbb{R} \Rightarrow \forall x \in X \ f(x) \in \mathbb{R} \Rightarrow f^+(x) = \max(f(x), 0) \in \mathbb{R} \wedge f^-(x) = -\min(f(x), 0) \wedge |f(x)| \in \mathbb{R}$ so that $f^+(X), f^-(X), |f|(X) \subseteq \mathbb{R}$ hence $f^+, f^-, |f| \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$.

Proof.

1.

$\Rightarrow$. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $C_0 \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see example: 21.228] it follows from [theorems: 21.267, 21.263 and 21.341] that

$$f^+ = f \vee C_0 \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } f^- = -(f \wedge C_0) \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

which as $\forall x \in X$ $0 \leq f^+(x), f^-(x)$ proves that

$$f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

$\Leftarrow$. As $f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and by [theorem: 21.341]

$$\forall x \in X \ (f^+(x), -f^-(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\} \text{ and } f = f^+ - f^-$$

which by using [theorem: 21.265] proves that

$$f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

2. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from (1) that $f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that by [theorem: 21.257] $|f| \stackrel{[\text{theorem: 21.341}]}{=} f^+ + f^- \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$. As $\forall x \in X$ $0 \leq |f|(x)$ it follows that

$$|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

□

Theorem 21.344. *Let X be a set then we have:*

1. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function and $\alpha \in [0, \infty[$ then $(\alpha \cdot f)^+ = \alpha \cdot f^+$ and $(\alpha \cdot f)^- = \alpha \cdot f^-$
2. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function and $\alpha \in]-\infty, 0]$ then $(\alpha \cdot f)^+ = -\alpha \cdot f^-$ and $(\alpha \cdot f)^- = -\alpha \cdot f^+$
3. If $f: X \rightarrow \overline{\mathbb{R}}, g: X \rightarrow \overline{\mathbb{R}}$ are functions with $f(X) \subseteq \mathbb{R} \wedge g(X) \subseteq \mathbb{R}$ then we have $\forall x \in X$ that
 - a. $(f + g)^+(x) \leq f^+(x) + g^+(x) = (f^+ + g^+)(x)$
 - b. $(f + g)^-(x) \leq f^-(x) + g^-(x) = (f^- + g^-)(x)$
4. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function then $\forall x \in X$ we have $f^+(x) \leq |f(x)|$ and $f^-(x) \leq |f(x)|$
5. If $A \subseteq X$ and $f: X \rightarrow \overline{\mathbb{R}}$ is a function then $(f|_A)^+ = (f^+)|_A$ and $(f|_A)^- = (f^-)|_A$
6. If $A \subseteq X$ and $f: A \rightarrow \overline{\mathbb{R}}$ is a function then $(f_{A \uparrow X})^+ = (f^+)_{A \uparrow X}$ and $(f_{A \uparrow X})^- = (f^-)_{A \uparrow X}$ [see definition: 21.246]
7. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function with $f(X) \subseteq [0, \infty]$ then $f = f^+$ and $f^- = C_0$
8. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function with $f(X) \subseteq [0, \infty]$ then $(f \cdot g)^+ = f \cdot g^+$ and $(f \cdot g)^- = f \cdot g^-$
9. If $f: X \rightarrow \overline{\mathbb{R}}$ is a function, $A \subseteq X$ then $\mathcal{X}_{X,A} \cdot f$ then

$$(\mathcal{X}_{X,A} \cdot f)^+ = \mathcal{X}_{X,A} \cdot f^+ \text{ and } (\mathcal{X}_{X,A} \cdot f)^- = \mathcal{X}_{X,A} \cdot f^-$$

Proof.

1. Let $x \in X$ then we have either:

$f(x) \leq 0$. then $\alpha \cdot f(x) \leq 0$ so that

$$\begin{aligned} (\alpha \cdot f)^+(x) &= \max((\alpha \cdot f)(x), 0) \\ &= \max(\alpha \cdot f(x), 0) \\ &= 0 \\ &= \alpha \cdot 0 \\ &= \alpha \cdot \max(f(x), 0) \\ &= \alpha \cdot f^+(x) \end{aligned}$$

and

$$\begin{aligned}
 (\alpha \cdot f)^-(x) &= -\min((\alpha \cdot f)(x), 0) \\
 &= -\min(\alpha \cdot f(x), 0) \\
 &= -\alpha \cdot f(x) \\
 &= \alpha \cdot (-f(x)) \\
 &= \alpha \cdot (-\min(f(x), 0)) \\
 &= \alpha \cdot f^-(x)
 \end{aligned}$$

$0 < f(x)$. then $0 < \alpha \cdot f(x)$

$$\begin{aligned}
 (\alpha \cdot f)^+(x) &= \max((\alpha \cdot f)(x), 0) \\
 &= \max(\alpha \cdot f(x), 0) \\
 &= \alpha \cdot f(x) \\
 &= \alpha \cdot \max(f(x), 0) \\
 &= \alpha \cdot f^+(x)
 \end{aligned}$$

and

$$\begin{aligned}
 (\alpha \cdot f)^-(x) &= -\min((\alpha \cdot f)(x), 0) \\
 &= -\min(\alpha \cdot f(x), 0) \\
 &= 0 \\
 &= \alpha \cdot 0 \\
 &= \alpha \cdot (-\min(f(x), 0)) \\
 &= \alpha \cdot f^-(x)
 \end{aligned}$$

2. Let $x \in X$ then for x we have either:

$f(x) \leq 0$. then $0 \leq \alpha \cdot f(x)$

$$\begin{aligned}
 (\alpha \cdot f)^+(x) &= \max((\alpha \cdot f)(x), 0) \\
 &= \max(\alpha \cdot f(x), 0) \\
 &= \alpha \cdot f(x) \\
 &= \alpha \cdot \min(f(x), 0) \\
 &= -\alpha \cdot f^-(x)
 \end{aligned}$$

and

$$\begin{aligned}
 (\alpha \cdot f)^-(x) &= -\min((\alpha \cdot f)(x), 0) \\
 &= -\min(\alpha \cdot f(x), 0) \\
 &= 0 \\
 &= -\alpha \cdot 0 \\
 &= -\alpha \cdot \max(f(x), 0) \\
 &= -\alpha \cdot f^+(x)
 \end{aligned}$$

$0 < f(x)$. then $\alpha \cdot f(x) < 0$

$$\begin{aligned}
 (\alpha \cdot f)^+(x) &= \max((\alpha \cdot f)(x), 0) \\
 &= \max(\alpha \cdot f(x), 0) \\
 &= 0 \\
 &= \alpha \cdot 0 \\
 &= \alpha \cdot (\min(f(x), 0)) \\
 &= -\alpha \cdot f^-(x)
 \end{aligned}$$

and

$$\begin{aligned}
 (\alpha \cdot f)^-(x) &= -\min((\alpha \cdot f)(x), 0) \\
 &= -\min(\alpha \cdot f(x), 0) \\
 &= -\alpha \cdot f(x) \\
 &= -\alpha \cdot \max(f(x), 0) \\
 &= -\alpha \cdot f^+(x)
 \end{aligned}$$

3. Let $x \in X$ then

a. As $f(x) \leq \max(f(x), 0) = f^+(x)$ and $g(x) \leq \max(g(x), 0) = g^+(x)$ we have

$$\begin{aligned}
 (f+g)(x) &= f(x) + g(x) \\
 &\leq f^+(x) + g^+(x)
 \end{aligned}$$

which as $0 \leq f^+(x) + g^+(x)$ proves that $\max((f+g)(x), 0) \leq f^+(x) + g^+(x)$ so that

$$\begin{aligned}
 (f+g)^+(x) &= \max((f+g)(x), 0) \\
 &\leq f^+(x) + g^+(x)
 \end{aligned}$$

b. Then $-f^-(x) = \min(f(x), 0) \leq f(x)$ and $-g^-(x) = \min(g(x), 0) \leq g(x)$ so that

$$\begin{aligned}
 -f^-(x) - g^-(x) &\leq f(x) + g(x) \\
 &\leq (f+g)(x)
 \end{aligned}$$

which as $-f^-(x) - g^-(x) \leq 0$ proves that $-f^-(x) - g^-(x) \leq \min((f+g)(x), 0)$ so that

$$\begin{aligned}
 (f+g)^-(x) &= -\min((f+g)(x), 0) \\
 &\leq f^-(x) + g^-(x)
 \end{aligned}$$

4. Let $x \in X$ then $f(x) \leq |f(x)|$ and $0 \leq |f(x)|$ so that

$$f^+(x) = \max(f(x), 0) \leq |f(x)|$$

Further $-f(x) \leq |f(x)| \Rightarrow -|f(x)| \leq f(x)$ and $-|f(x)| \leq 0$ so that $-|f(x)| \leq \min(f(x), 0)$, hence

$$f^-(x) = -\min(f(x), 0) \leq |f(x)|$$

5. Let $x \in A \subseteq X$ then $(f|_A)^+(x) = \max(f|_A(x), 0) \underset{x \in A}{=} \max(f(x), 0) = f^+(x) \underset{x \in A}{=} (f^+)|_A(x)$ and $(f|_A)^-(x) = -\min(f|_A(x), 0) \underset{x \in A}{=} -\min(f(x), 0) = f^-(x) \underset{x \in A}{=} (f^-)|_A(x)$ proving that

$$(f|_A)^+ = (f^+)|_A \text{ and } (f|_A)^- = (f^-)|_A$$

6. Let $x \in A \subseteq X$ then we must consider two cases:

$x \in A$. Then

$$\begin{aligned}
 (f_{A \uparrow X})^+(x) &= \max((f_{A \uparrow X})(x), 0) \\
 &= \max(f(x), 0) \\
 &= f^+(x) \\
 &= (f^+)_{A \uparrow X}(x) \\
 (f_{A \uparrow X})^-(x) &= -\min((f_{A \uparrow X})(x), 0) \\
 &= -\min(f(x), 0) \\
 &= f^-(x) \\
 &= (f^-)_{A \uparrow X}(x)
 \end{aligned}$$

$x \in X \setminus A$. Then

$$\begin{aligned}
 (f_{A \uparrow X})^+(x) &= \max(f_{A \uparrow X}(x), 0) \\
 &= \max(0, 0) \\
 &= 0 \\
 &= (f^+)_{A \uparrow X}(x) \\
 (f_{A \uparrow X})^-(x) &= -\min(f_{A \uparrow X}(x), 0) \\
 &= -\min(0, 0) \\
 &= 0 \\
 &= (f^-)_{A \uparrow X}(x)
 \end{aligned}$$

Hence we have

$$(f_{A \uparrow X})^+ = (f^+)_{A \uparrow X} \text{ and } (f_{A \uparrow X})^- = (f^-)_{A \uparrow X}$$

7. Let $x \in X$ then we have, as $f(X) \subseteq [0, \infty]$, that $0 \leq f(x)$ hence

$$f^+(x) = \max(f(x), 0) = f(x) \text{ and } f^-(x) = -\min(f(x), 0) = 0 = C_0(x)$$

which proves that

$$f = f^+ \text{ and } f^- = C_0$$

8. Let $x \in X$ then for $g(x)$ we have either:

$0 \leq g(x)$. Then as $0 \leq f(x)$ we have $0 \leq f(x) \cdot g(x) = (f \cdot g)(x)$ so that

$$\begin{aligned}
 (f \cdot g)^+(x) &= \max((f \cdot g)(x), 0) \\
 &= (f \cdot g)(x) \\
 &= f(x) \cdot g(x) \\
 &= f(x) \cdot \max(g(x), 0) \\
 &= f(x) \cdot g^+(x) \\
 &= (f \cdot g^+)(x) \\
 (f \cdot g)^-(x) &= -\min((f \cdot g)(x), 0) \\
 &= 0 \\
 &= f(x) \cdot 0 \\
 &= f(x) \cdot (-\min(g(x), 0)) \\
 &= f(x) \cdot g^-(x) \\
 &= (f \cdot g^-)(x)
 \end{aligned}$$

$g(x) < 0$. Then as $0 \leq f(x)$ we have $(f \cdot g)(x) = f(x) \cdot g(x) < 0$ so that

$$\begin{aligned}
 (f \cdot g)^+(x) &= \max((f \cdot g)(x), 0) \\
 &= 0 \\
 &= f(x) \cdot 0 \\
 &= f(x) \cdot \max(g(x), 0) \\
 &= f(x) \cdot g^+(x) \\
 &= (f \cdot g^+)(x) \\
 (f \cdot g)^-(x) &= -\min((f \cdot g)(x), 0) \\
 &= -(f \cdot g)(x) \\
 &= f(x) \cdot (-g(x)) \\
 &= f(x) \cdot (-\min(g(x), 0)) \\
 &= f(x) \cdot g^-(x) \\
 &= (f \cdot g^-)(x)
 \end{aligned}$$

Hence

$$(f \cdot g)^+ = f \cdot g^+ \text{ and } (f \cdot g)^- = f \cdot g^-$$

9. As $\forall x \in X \mathcal{X}_{X,A}(x) \in \{1, 0\} \Rightarrow 0 \leq \mathcal{X}_{X,A}(x)$ it follows from (8) that

$$(\mathcal{X}_{X,A} \cdot f)^+ = \mathcal{X}_{X,A} \cdot f^+ \text{ and } (\mathcal{X}_{X,A} \cdot f)^- = \mathcal{X}_{X,A} \cdot f^- \quad \square$$

We are ready now to define the μ -measure integral and μ -integrable functions [the functions for which the μ -measure integral is defined].

Definition 21.345. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then we say that

$$\int f d\mu \text{ exist}$$

if

$$\int^+ f^+ d\mu < \infty \text{ or } \int^+ f^- d\mu < \infty$$

and in this case $\int f d\mu$ is defined by

$$\int f d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu$$

We say that

$$f \text{ is } \mu\text{-integrable if } \int^+ f^+ d\mu < \infty \text{ and } \int^+ f^- d\mu < \infty$$

The set of all measurable functions for which the measure integral exist is noted as $\overline{\mathcal{L}}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ in other words

$$\overline{\mathcal{L}}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] = \left\{ f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \int^+ f^+ d\mu < \infty \vee \int^+ f^- d\mu < \infty \right\}$$

The set of all μ -integrable function is noted as $\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ in other words

$$\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] = \left\{ f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \int^+ f^+ d\mu < \infty \wedge \int^+ f^- d\mu < \infty \right\}$$

Likewise we define

$$\overline{\mathcal{L}}[X, \mathcal{A}, \mu, \mathbb{R}] = \left\{ f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \mid \int^+ f^+ d\mu < \infty \vee \int^+ f^- d\mu < \infty \right\}$$

the set of real valued functions that have a measure integral and

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] = \left\{ f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \mid \int^+ f^+ d\mu < \infty \vee \int^+ f^- d\mu < \infty \right\}$$

the set of real valued functions that are μ -integrable.

Note 21.346. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.342] that $f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that $\int^+ f^+ d\mu$ and $\int^+ f^- d\mu$ are well defined.

Note 21.347. If f is μ -integrable then clearly $\int f d\mu$ exist however the opposite is not necessary true. For example consider $\langle X, \mathcal{A}, \mu \rangle$ be a measure space such that $\mu(X) = \infty$ then we have for $C_1: X \rightarrow \overline{\mathbb{R}}$ defined by $C_1(x) = 1$ that by [example: 21.339] $(C_1)^+ = C_1$ and $(C_1)^- = C_0$ so that by [example: 21.318] $\int^+ (C_1)^+ d\mu = 1 \cdot \mu(X) = \infty$ and $\int^+ (C_1)^- d\mu = 0 \cdot \mu(X) = 0$. Hence $\int C_1 d\mu$ exist and is equal to ∞ but C_1 is not μ -integrable.

Note 21.348. Using the above definitions we have

$$\mathcal{L}[X, \mathcal{A}, \mathbb{R}, \mu] \subseteq \mathcal{L}[X, \mathcal{A}, \overline{\mathbb{R}}, \mu] \subseteq \overline{\mathcal{L}}[X, \mathcal{A}, \overline{\mathbb{R}}, \mu]$$

and

$$\mathcal{L}[X, \mathcal{A}, \mathbb{R}, \mu] \subseteq \overline{\mathcal{L}}[X, \mathcal{A}, \mathbb{R}, \mu]$$

Example 21.349. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $C_0 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ [hence C_0 is μ -integrable] and $\int C_0 d\mu = 0$

Proof. Let $x \in X$ then we have that $(C_0)^+(x) = \max(C_0(x), 0) = 0 = C_0(x)$ and also $(C_0)^-(x) = -\min(C_0(x), 0) = 0 = C_0(x)$ so that

$$(C_0)^+ = C_0 = (C_0)^-$$

Hence we have $\int^+(C_0)^+ d\mu = \int^+ C_0 d\mu \stackrel{[\text{example: 21.330}]}{=} 0 \stackrel{[\text{example: 21.330}]}{=} \int^+ C_0 d\mu = \int^+(C_0)^- d\mu = 0$. As further $C_0 \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ [see theorem: 21.228] it follows that

$$C_0 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \int C_0 d\mu = 0 \quad \square$$

We show now a alternative definition of the measure integral and integrable functions. First we need a little lemma.

Lemma 21.350. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $\forall x \in X$ $(f_1(x), -f_2(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ [so that $f_1 - f_2$ is well defined] then we have that

$$\begin{aligned} (f_1 - f_2)^+, (f_1 - f_2)^- &\in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \\ \text{and} \\ f_1 - f_2 &\in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \\ \text{and} \\ \int^+ (f_1 - f_2)^+ &\leq \int^+ f_1 d\mu \text{ and } \int^+ (f_1 - f_2)^- d\mu \leq \int^+ f_2 d\mu \end{aligned}$$

Proof. As $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.265] that

$$f_1 - f_2 \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Using then [theorem: 21.342] it follows that

$$(f_1 - f_2)^+, (f_1 - f_2)^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Let $x \in X$ then for $(f_1 - f_2)(x)$ we have either:

$0 \leq (f_1 - f_2)(x)$. Then

$$\begin{aligned} (f_1 - f_2)^+(x) &= \max((f_1 - f_2)(x), 0) \\ &= (f_1 - f_2)(x) \\ &= f_1(x) - f_2(x) \\ &\leq f_1(x) \\ &\stackrel{0 \leq f_2(x) \wedge [\text{theorem: 20.27}]}{=} \\ (f_1 - f_2)^-(x) &= -\min((f_1 - f_2)(x), 0) \\ &= 0 \\ &\leq f_2(x) \\ &\stackrel{0 \leq f_2(x)}{=} \end{aligned}$$

$(f_1 - f_2)(x) < 0$. Then

$$\begin{aligned} (f_1 - f_2)^+(x) &= \max((f_1 - f_2)(x), 0) \\ &= 0 \\ &\stackrel{0 \leq f_1(x)}{=} f_1(x) \\ (f_1 - f_2)(x) &= -\min((f_1 - f_2)(x), 0) \\ &= -(f_1 - f_2)(x) \\ &= f_2(x) - f_1(x) \\ &\leq f_2(x) \\ &\stackrel{0 \leq f_1(x) \wedge [\text{theorem: 20.27}]}{=} \end{aligned}$$

Hence we have

$$\forall x \in X \quad (f_1 - f_2)^+(x) \leq f_1(x) \wedge (f_1 - f_2)^-(x) \leq f_2(x)$$

Using then [theorem: 21.331(3)] it follows that

$$\int^+ (f_1 - f_2)^+ d\mu \leq \int^+ f_1 d\mu \quad \text{and} \quad \int^+ (f_1 - f_2)^- d\mu \leq \int^+ f_2 d\mu$$

□

Theorem 21.351. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have for $f: X \rightarrow \overline{\mathbb{R}}$*

1. *If there exist $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ with*

a. $\forall x \in X \quad (f_1(x), -f_2(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ *[so that $f_1 - f_2$ is well defined]*

b. $\int^+ f_1 d\mu \in \mathbb{R}$ **or** $\int^- f_2 d\mu \in \mathbb{R}$

such that

$$f = f_1 - f_2$$

then

$$f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$$

2. *If there exist $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ with*

a. $\forall x \in X \quad (f_1(x), -f_2(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ *[so that $f_1 - f_2$ is well defined]*

b. $\int^+ f_1 d\mu \in \mathbb{R}$ **and** $\int^- f_2 d\mu \in \mathbb{R}$

such that

$$f = f_1 - f_2$$

then

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \quad \text{and} \quad \int f d\mu = \int^+ f_1 d\mu - \int^+ f_2 d\mu$$

Proof.

1. As $f = f_1 - f_2$ where $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$, $\forall x \in X \quad (f_1(x), -f_2(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and $\int^+ f_1 d\mu < \infty$ **or** $\int^+ f_2 d\mu < \infty$ it follows from [lemma: 21.350] that

$$f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad \text{and} \quad f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ f^+ d\mu \leq \int^+ f_1 d\mu \quad \text{and} \quad \int^+ f^- d\mu \leq \int^+ f_2 d\mu$$

Hence we have

$$\int^+ f^+ d\mu < \infty \quad \text{or} \quad \int^+ f^- d\mu < \infty$$

which by [definition: 21.345] proves that

$$f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$$

2. As $f = f_1 - f_2$ where $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$, $\forall x \in X \quad (f_1(x), -f_2(x)) \notin \{(-\infty, \infty), (\infty, -\infty)\}$ and $\int^+ f_1 d\mu < \infty$ **and** $\int^+ f_2 d\mu < \infty$ it follows from [lemma: 21.350] that

$$f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad \text{and} \quad f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ f^+ d\mu \leq \int^+ f_1 d\mu \quad \text{and} \quad \int^+ f^- d\mu \leq \int^+ f_2 d\mu$$

Hence we have

$$\int^+ f^+ d\mu < \infty \quad \text{and} \quad \int^+ f^- d\mu < \infty$$

which by [definition: 21.345] proves that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

Further, as $\int^+ f_1 d\mu < \infty$ and $\int^+ f_2 d\mu < \infty$, it follows from [theorem: 21.336] that

$$N_1 = \{x \in X \mid f_1(x) = \infty\} \in \mathcal{A} \text{ and } \mu(N_1) = 0$$

$$N_2 = \{x \in X \mid f_2(x) = \infty\} \in \mathcal{A} \text{ and } \mu(N_2) = 0$$

Take

$$N = N_1 \cup N_2 \text{ then } N \in \mathcal{A} \text{ and } X \setminus N \in \mathcal{A}$$

then we have $\mu(N) = \mu(N_1 \cup N_2) \leq \mu(N_1) + \mu(N_2) = 0 + 0$ so that

$$\mu(N) = 0$$

Let $x \in X \setminus N = (X \setminus N_1) \cap (X \setminus N_2)$ then we have $f_1(x), f_2(x) \in \mathbb{R}$ so that $f(x) = f_1(x) - f_2(x) \in \mathbb{R}$, further as $f(x) \stackrel{[\text{theorem: 21.341}]}{=} f^+(x) - f^-(x)$ we must have that $f^+(x), f^-(x) \in \mathbb{R}$. To summarize we have

$$\forall x \in X \setminus N \quad f(x), f^+(x), f^-(x), f_1(x), f_2(x) \in \mathbb{R}$$

from the above and $f^+(x) - f^-(x) \stackrel{[\text{theorem: 21.341}]}{=} f(x) = f_1(x) - f_2(x)$ it follows that

$$\forall x \in X \setminus N \text{ we have } f^+(x) + f_2(x) = f^-(x) + f_1(x) \quad (21.514)$$

so that that $\forall x \in X$ we have

$$\begin{aligned} (\mathcal{X}_{X, X \setminus N} \cdot f^+ + \mathcal{X}_{X, X \setminus N} \cdot f_2)(x) &= \begin{cases} f^+(x) + f_2(x) & \text{if } x \in X \setminus N \\ 0 & \text{if } x \in N \end{cases} \\ &\stackrel{[\text{eq: 21.514}]}{=} \begin{cases} f^-(x) + f_1(x) & \text{if } x \in X \setminus N \\ 0 & \text{if } x \in N \end{cases} \\ &= (\mathcal{X}_{X, X \setminus N} \cdot f^- + \mathcal{X}_{X, X \setminus N} \cdot f_1)(x) \end{aligned}$$

which proves that

$$\mathcal{X}_{X, X \setminus N} \cdot f^+ + \mathcal{X}_{X, X \setminus N} \cdot f_2 = \mathcal{X}_{X, N} \cdot f^- + \mathcal{X}_{X, N} \cdot f_1$$

As by [theorem: 21.276] $\mathcal{X}_{X, X \setminus N} \cdot f^+, \mathcal{X}_{X, X \setminus N} \cdot f_2, \mathcal{X}_{X, X \setminus N} \cdot f^-, \mathcal{X}_{X, X \setminus N} \cdot f_1 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have by [theorem: 21.331] that

$$\int^+ (\mathcal{X}_{X, N} \cdot f^+) d\mu + \int^+ (\mathcal{X}_{X, N} \cdot f_2) d\mu = \int^+ (\mathcal{X}_{X, N} \cdot f^-) d\mu + \int^+ (\mathcal{X}_{X, N} \cdot f_1) d\mu$$

Combining this with [theorems: 21.276, 21.280, 21.333] gives

$$\int^+ f^+ d\mu + \int^+ f_2 d\mu = \int^+ f^- d\mu + \int^+ f_1 d\mu$$

which as all the terms in the above are finite proves that

$$\int f d\mu \stackrel{\text{def}}{=} \int^+ f^+ d\mu - \int^+ f^- d\mu = \int f_1 d\mu - \int f_2 d\mu \quad \square$$

We show now that $\int$ is a extension of $\int^S$ and $\int^+$.

Theorem 21.352. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have:*

1. *If $f \in \mathcal{S}^+[X, \mathcal{A}]$ then*

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}], \quad \int f d\mu = \int^S f d\mu < \infty$$

2. If $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} \text{ and } \int f d\mu = \int^+ f d\mu$$

3. If $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+ f d\mu < \infty$ then f is μ -integrable [in other words $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$] and

$$\int f d\mu = \int^+ f d\mu$$

4. If $f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ and $f(X) \subseteq [0, \infty]$ then $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int f d\mu = \int^+ f d\mu < \infty$.

5. If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $f(X) \subseteq [0, \infty]$ then $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int f d\mu = \int^+ f d\mu < \infty$.

Note 21.353. If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ and $f(X) \subseteq [0, \infty]$ then $f(X) \subseteq [0, \infty[$ so that in this special case we have $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ and $\int f d\mu = \int^+ f d\mu < \infty$.

Proof.

1. As $f \in \mathcal{S}^+[X, \mathcal{A}]$ we have by [definition: 21.297] that $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and by [definition: 21.305]

$$\int^S f d\mu \in [0, \infty[$$

From [theorem: 21.344] it follows that

$$f^+ = f \in \mathcal{S}^+[X, \mathcal{A}] \underset{[\text{theorem: 21.317}]}{\subseteq} \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \text{ and } f^- = C_0$$

then $\int^+ f^+ d\mu \underset{[\text{theorem: 21.317}]}{=} \int^S f^+ d\mu = \int^S f d\mu \in [0, \infty[$ and $\int^+ f^- d\mu \underset{[\text{theorem: 21.318}]}{=} 0 \cdot \mu(X) = 0 \in [0, \infty[$. So we have proved that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \int f d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu = \int^S f^+ d\mu - 0 \underset{f^+ = f}{=} \int^S f d\mu < \infty$$

2. As $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have $f(X) \subset [0, \infty]$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$. Further by [theorem: 21.344]

$$f^+ = f \text{ and } f^- = C_0$$

As $\int^+ f^- d\mu = \int^+ C_0 d\mu \underset{[\text{theorem: 21.318}]}{=} 0 \cdot \mu(X) = 0 < \infty$ it follows that

$$f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} \text{ and } \int f d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu = \int^+ f d\mu - 0 = \int^+ f d\mu$$

3. As $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+ f d\mu < \infty$ we have

$$\begin{aligned} \int^+ f^+ d\mu &\underset{[\text{theorem: 21.344(7)}]}{=} \int^+ f d\mu < \infty \\ \int^+ f^- d\mu &\underset{[\text{theorem: 21.344(7)}]}{=} \int^+ C_0 d\mu = 0 < \infty \end{aligned}$$

from which it follows that f is μ -integrable and

$$\int f d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu = \int^+ f d\mu - 0 = \int^+ f d\mu$$

4. As $f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f(X) \subseteq [0, \infty]$ we have by definition that $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that by [2] $\int f d\mu = \int^+ f d\mu$.

5. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \subseteq \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ so that by (3) we have that

$$\int f d\mu = \int^+ f d\mu \underset{f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}{<} \infty$$

□

21.5.3.2 Integration on measurable sub-spaces

Definition 21.354. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $f: X \rightarrow \overline{\mathbb{R}}$ be a function then we say that

$$\int_A f d\mu \text{ exist if } \int \mathcal{X}_{X,A} \cdot f d\mu \text{ exist}$$

and in this case $\int_A f d\mu$ is defined by

$$\int_A f d\mu = \int \mathcal{X}_{X,A} d\mu$$

We say that

f is μ -integrable over A if $\mathcal{X}_{X,A}$ is μ -integrable

The set of all functions for which $\int_A f d\mu$ exist is noted as $\overline{\mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ hence

$$\begin{aligned} & \overline{\mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} = \\ & \{f \in \overline{\mathbb{R}}^X \mid \mathcal{X}_{X,A} \cdot f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}\} = \\ & \left\{ \mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu < \infty \vee \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu < \infty \right\} \quad [\text{theorem: 21.344(9)}] = \\ & \left\{ \mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \int^+ \mathcal{X}_{X,A} \cdot f^+ d\mu < \infty \vee \int^+ \mathcal{X}_{X,A} \cdot f^- d\mu < \infty \right\} \end{aligned}$$

The set of all functions that are μ -integrable on A is noted as $\mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ hence

$$\begin{aligned} & \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] = \\ & \{f \in \overline{\mathbb{R}}^X \mid \mathcal{X}_{X,A} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]\} = \\ & \left\{ \mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu < \infty \wedge \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu < \infty \right\} \quad [\text{theorem: 21.344(9)}] = \\ & \left\{ \mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \int^+ \mathcal{X}_{X,A} \cdot f^+ d\mu < \infty \wedge \int^+ \mathcal{X}_{X,A} \cdot f^- d\mu < \infty \right\} \end{aligned}$$

Note 21.355. As $\forall x \in X (\mathcal{X}_{X,X} \cdot f)(x) = \mathcal{X}_{X,X}(x) \cdot f(x) = 1 \cdot f(x) = f(x)$ it follows that

$$\mathcal{X}_{X,X} \cdot f = f$$

so that

$$\overline{\mathcal{L}_X[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} = \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \mathcal{L}_X[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] = \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

and

$$\int_X f d\mu = \int f d\mu$$

Theorem 21.356. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $A \in \mathcal{A}$ then

$$\begin{aligned} \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} & \subseteq \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \\ \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] & \subseteq \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \end{aligned}$$

Proof. Let $x \in X$ then we have either:

$x \in A$. Then

$$(\mathcal{X}_{X,A} \cdot f)^+(x) \underset{[\text{theorem: 21.344(8)}]}{=} (\mathcal{X}_{X,A} \cdot f^+)(x) = \mathcal{X}_{X,A}(x) \cdot f^+(x) = 1 \cdot f^+(x) = f^+(x)$$

$$(\mathcal{X}_{X,A} \cdot f)^-(x) \underset{[\text{theorem: 21.344(8)}]}{=} (\mathcal{X}_{X,A} \cdot f^-)(x) = \mathcal{X}_{X,A}(x) \cdot f^-(x) = 1 \cdot f^-(x) = f^-(x)$$

so that

$$(\mathcal{X}_{X,A} \cdot f)^+(x) \leq f^+(x) \wedge (\mathcal{X}_{X,A} \cdot f)^-(x) \leq f^-(x)$$

$x \in X \setminus A$. Then

$$(\mathcal{X}_{X,A} \cdot f)^+(x) \stackrel{[\text{theorem: 21.344(8)}]}{=} (\mathcal{X}_{X,A} \cdot f^+)(x) = \mathcal{X}_{X,A}(x) \cdot f^+(x) = 0 \cdot f^+(x) = 0 \leq f^+(x)$$

$$(\mathcal{X}_{X,A} \cdot f)^-(x) \stackrel{[\text{theorem: 21.344(8)}]}{=} (\mathcal{X}_{X,A} \cdot f^-)(x) = \mathcal{X}_{X,A}(x) \cdot f^-(x) = 0 \cdot f^-(x) = 0 \leq f^-(x)$$

so that

$$(\mathcal{X}_{X,A} \cdot f)^+(x) \leq f^+(x) \wedge (\mathcal{X}_{X,A} \cdot f)^-(x) \leq f^-(x)$$

hence in all cases we have

$$(\mathcal{X}_{X,A} \cdot f)^+(x) \leq f^+(x) \wedge (\mathcal{X}_{X,A} \cdot f)^-(x) \leq f^-(x)$$

If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ or $f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ then $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \stackrel{[\text{theorem: 21.276}]}{\Rightarrow} \mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and by [theorem: 21.331] we have

$$\int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu \leq \int^+ f^+ d\mu \text{ and } \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu \leq \int^+ f^- d\mu$$

If $f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ then we have $\int^+ f^+ d\mu < \infty$ or $\int^+ f^- d\mu < \infty$ so that by the above

$$\int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu < \infty \text{ or } \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu < \infty$$

which proves that $f \in \overline{\mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ so that

$$\overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]} \subseteq \overline{\mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$$

If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then we have $\int^+ f^+ d\mu < \infty$ and $\int^+ f^- d\mu < \infty$ so that by the above

$$\int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu < \infty \text{ and } \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu < \infty$$

which proves that $f \in \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ so that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \subseteq f \in \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \quad \square$$

Theorem 21.357. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $\langle A, \mathcal{A}|_A, \mu|_{\mathcal{A}|_A} \rangle$ a measure subspace of $\langle X, \mathcal{A}, \mu \rangle$ [see theorem: 21.111] then we have for $f: X \rightarrow \overline{\mathbb{R}}$

1. Then the following are equivalent

a. $f \in \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ [meaning $(\mathcal{X}_{X,A} \cdot f) \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ or equivalently f is μ -integrable over A]

b. $f|_A \in \mathcal{L}[A, \mathcal{A}|_A, \mu|_{\mathcal{A}|_A}, \overline{\mathbb{R}}]$ [meaning $f|_A$ is $\mu|_{\mathcal{A}|_A}$ -integrable]

Further in case of (a) [and thus also case (b)] we have

$$\int_A f d\mu = \int f|_A d\mu|_{\mathcal{A}|_A}$$

2. The following are equivalent

a. $f \in \mathcal{L}[A, \mathcal{A}, \mu|_{\mathcal{A}|_A}, \overline{\mathbb{R}}]$

b. $f_{A \uparrow X} \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$

Further in case of (a) [and thus also of (b)] we have

$$\int f_{A \uparrow X} d\mu = \int f d\mu|_{\mathcal{A}|_A}$$

Proof.

1.

a $\Rightarrow$ b. As $f \in \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have that $\mathcal{X}_{X,A} \cdot f$ is μ -integrable, which by definition means that

$$\mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \stackrel{[\text{theorem: 21.342}]}{\Rightarrow} (\mathcal{X}_{X,A} \cdot f)^+, (\mathcal{X}_{X,A} \cdot f)^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.515)$$

and

$$\int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu < \infty \wedge \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu < \infty \quad (21.516)$$

By [theorem: 21.344(9)] we have $(\mathcal{X}_{X,A} f)^+ = \mathcal{X}_{X,A} \cdot f^+$ and $(\mathcal{X}_{X,A} f)^- = \mathcal{X}_{X,A} \cdot f^-$ which after substituting this result in [eq: 21.515, 21.516] gives

$$\mathcal{X}_{X,A} \cdot f^+, \mathcal{X}_{X,A} \cdot f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.517)$$

$$\int^+ \mathcal{X}_{X,A} \cdot f^+ d\mu = \int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu < \infty \quad (21.518)$$

$$\int^- \mathcal{X}_{X,A} \cdot f^- d\mu = \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu < \infty \quad (21.519)$$

Using then [theorem: 21.322] on the above three equations gives

$$(f^+)_{|A}, (f^-)_{|A} \in \mathcal{M}^+[A, \mathcal{A}_{|A}, \overline{\mathbb{R}}] \quad (21.520)$$

$$\int^+ (f^+)_{|A} d\mu_{|A} = \int^+ \mathcal{X}_{X,A} \cdot f^+ d\mu < \infty \quad (21.521)$$

$$\int^+ (f^-)_{|A} d\mu_{|A} = \int^+ \mathcal{X}_{X,A} \cdot f^- d\mu < \infty \quad (21.522)$$

Using [theorem: 21.344(5)] we have that $(f^+)_{|A} = (f|_A)^+ \wedge (f^-)_{|A} = (f|_A)^-$ which after substituting this result in the previous three equations gives

$$(f|_A)^+, (f|_A)^- \in \mathcal{M}^+[A, \mathcal{A}_{|A}, \mu_{|A}] \quad (21.523)$$

$$\int^+ (f|_A)^+ d\mu_{|A} = \int^+ (f^+)_{|A} d\mu_{|A} < \infty \quad (21.524)$$

$$\int^+ (f|_A)^- d\mu_{|A} = \int^+ (f^-)_{|A} d\mu_{|A} < \infty \quad (21.525)$$

Hence we have by definition that

$$f|_A \in \mathcal{L}[A, \mathcal{A}_{|A}, \mu_{|A}, \overline{\mathbb{R}}]$$

and

$$\begin{aligned} \int f|_A d\mu_{|A} & \stackrel{\text{def}}{=} \int^+ (f|_A)^+ d\mu_{|A} - \int^+ (f|_A)^- d\mu_{|A} \\ & \stackrel{[\text{eqs: 21.524, 21.525}]}{=} \int^+ (f^+)_{|A} d\mu_{|A} - \int^+ (f^-)_{|A} d\mu_{|A} \\ & \stackrel{[\text{eqs: 21.521, 21.522}]}{=} \int^+ \mathcal{X}_{X,A} \cdot f^+ d\mu - \int^+ \mathcal{X}_{X,A} \cdot f^- d\mu \\ & \stackrel{[\text{eqs: 21.518, 21.519}]}{=} \int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu - \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu \\ & \stackrel{\text{def}}{=} \int (\mathcal{X}_{X,A} \cdot f) d\mu \\ & \stackrel{\text{def}}{=} \int_A f d\mu \end{aligned}$$

giving

$$\int_A f d\mu = \int f|_A d\mu|_{\mathcal{A}|_A}$$

b $\Rightarrow$ **a**. Because $f|_A$ is $\mu|_{\mathcal{A}|_A}$ -integrable we have by definition that

$$f|_A \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \xRightarrow{[\text{theorem: 21.342}]} (f|_A)^+, (f|_A)^- \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \quad (21.526)$$

$$\int^+ (f|_A)^+ d\mu|_{\mathcal{A}|_A} < \infty \text{ and } \int^+ (f|_A)^- d\mu|_{\mathcal{A}|_A} < \infty \quad (21.527)$$

Using [theorem: 21.344(5)] it follows that $(f|_A)^+ = (f^+)|_A$ and $(f|_A)^- = (f^-)|_A$ which after substituting this result in [eq: 21.526] gives

$$f|_A \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \text{ and } (f^+)|_A, (f^-)|_A \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$$

Using then [theorem: 21.277] on the above results in

$$\mathcal{X}_{X,A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \mathcal{X}_A \cdot f^+, \mathcal{X}_A \cdot f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.528)$$

Using [theorem: 21.322] on the above gives

$$\begin{aligned} \int^+ (\mathcal{X}_A \cdot f^+) d\mu &= \int^+ (f^+)|_A d\mu|_{\mathcal{A}|_A} \xRightarrow{[\text{theorem: 21.344(5)}]} \int^+ (f|_A)^+ d\mu|_{\mathcal{A}|_A} \stackrel{[\text{eq: 21.527}]}{<} \infty \\ \int^+ (\mathcal{X}_A \cdot f^-) d\mu &= \int^+ (f^-)|_A d\mu|_{\mathcal{A}|_A} \xRightarrow{[\text{theorem: 21.344(5)}]} \int^+ (f|_A)^- d\mu|_{\mathcal{A}|_A} \stackrel{[\text{eq: 21.527}]}{<} \infty \end{aligned}$$

Finally as by [theorem: 21.344(9)] $\mathcal{X}_A \cdot f^+ = (\mathcal{X}_A \cdot f)^+ \wedge \mathcal{X}_A \cdot f^- = (\mathcal{X}_A \cdot f)^-$ it follows from the above that

$$\int^+ (\mathcal{X}_A \cdot f)^+ d\mu < \infty \wedge \int^+ (\mathcal{X}_A \cdot f)^- d\mu < \infty$$

which together with $\mathcal{X}_A \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.528] proves $\mathcal{X}_A \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$. Hence by [definition: 21.354]

$$f \in \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

2.

a $\Rightarrow$ **b**. As f is $\mu|_{\mathcal{A}|_A}$ -integrable we have that

$$f \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \xRightarrow{[\text{theorem: 21.342}]} f^+, f^- \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \quad (21.529)$$

$$\int^+ f^+ d\mu|_{\mathcal{A}|_A} < \infty \text{ and } \int^+ f^- d\mu|_{\mathcal{A}|_A} < \infty \quad (21.530)$$

Using [theorem: 21.324] it follows that

$$(f^+)_{A \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ (f^+)_{A \uparrow X} d\mu = \int^+ f^+ d\mu|_{\mathcal{A}|_A} < \infty$$

$$(f^-)_{A \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ (f^-)_{A \uparrow X} d\mu = \int^+ f^- d\mu|_{\mathcal{A}|_A} < \infty$$

As by [theorem: 21.344(6)] $(f^+)_{A \uparrow X} = (f_{A \uparrow X})^+$ and $(f^-)_{A \uparrow X} = (f_{A \uparrow X})^-$ the above becomes:

$$(f_{A \uparrow X})^+ \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ (f_{A \uparrow X})^+ d\mu = \int^+ f^+ d\mu|_{\mathcal{A}|_A} < \infty \quad (21.531)$$

$$(f_{A \uparrow X})^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ (f_{A \uparrow X})^- d\mu = \int^+ f^- d\mu|_{\mathcal{A}|_A} < \infty \quad (21.532)$$

Using the above and the fact that $f \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \xRightarrow{[\text{theorem: 21.324}]} f_{A \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ proves that

$f_{A \uparrow X}$ is μ -integrable

Further we have that

$$\begin{aligned} \int f_{A \uparrow X} d\mu &\stackrel{\text{def}}{=} \int^+ (f_{A \uparrow X})^+ d\mu - \int^+ (f_{A \uparrow X})^- d\mu \\ &\stackrel{[\text{eqs: 21.531, 21.532}]}{=} \int^+ f^+ d\mu|_{\mathcal{A}|_A} - \int^+ f^- d\mu|_{\mathcal{A}|_A} \\ &\stackrel{\text{def}}{=} \int f d\mu|_{\mathcal{A}|_A} \end{aligned}$$

b $\Rightarrow$ a. As $f_{A \uparrow X}$ is μ -integrable we have by definition that

$$f_{A \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \xRightarrow{[\text{theorem: 21.342}]} (f_{A \uparrow X})^+, (f_{A \uparrow X})^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

$$\int^+ (f_{A \uparrow X})^+ d\mu < \infty \wedge \int^+ (f_{A \uparrow X})^- d\mu < \infty$$

Inserting $(f_{A \uparrow X})^+ \stackrel{[\text{theorem: 21.344(5)}]}{=} (f^+)_{A \uparrow X}$ and $(f_{A \uparrow X})^- \stackrel{[\text{theorem: 21.344(5)}]}{=} (f^-)_{A \uparrow X}$ in the above gives

$$f_{A \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } (f^+)_{A \uparrow X}, (f^-)_{A \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.533)$$

$$\int^+ (f^+)_{A \uparrow X} d\mu < \infty \wedge \int^+ (f^-)_{A \uparrow X} d\mu < \infty \quad (21.534)$$

Using [theorem: 21.250] on [eq: 21.533] gives

$$f \in \mathcal{M}[A, \mathcal{A}|_A, \overline{\mathbb{R}}] \text{ and } f^+, f^- \in \mathcal{M}^+[A, \mathcal{A}|_A, \overline{\mathbb{R}}]$$

Using [theorem: 21.324] on the above we have

$$\int^+ f^+ d\mu|_{\mathcal{A}|_A} = \int^+ (f^+)_{A \uparrow X} d\mu \stackrel{[\text{eq: 21.534}]}{<} \infty \wedge \int^+ f^- d\mu|_{\mathcal{A}|_A} = \int^+ (f^-)_{A \uparrow X} d\mu \stackrel{[\text{eq: 21.534}]}{<} \infty$$

From the two equations above it follows by definition that

$$f \in \mathcal{L}[A, \mathcal{A}|_A, \mu|_{\mathcal{A}|_A}, \overline{\mathbb{R}}] \quad \square$$

21.5.3.3 Properties of the measure integral

Theorem 21.358. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $f = 0$ μ -a.e. then $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $\int f d\mu = 0$

Proof. As $f = 0$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in X | f(x) \neq 0\} \subseteq N$$

Let $x \in \{x \in X | f^+(x) \neq 0\}$ then $\max(f(x), 0) \neq 0$ so that $f(x) \neq 0$ or $x \in \{x \in X | f(x) \neq 0\} \subseteq N$, hence $\{x \in X | f^+(x) \neq 0\} \subseteq N$ which proves that $f^+ = 0$ is μ -a.e.. Applying then [theorem: 21.330] proves that

$$\int^+ f^+ d\mu = 0 < \infty$$

If $x \in \{x \in X | f^-(x) \neq 0\}$ then $-\min(f(x), 0) \neq 0$ so that $f(x) \neq 0$ or $x \in \{x \in X | f(x) \neq 0\} \subseteq N$, hence $\{x \in X | f^-(x) \neq 0\} \subseteq N$ which proves that $f^- = 0$ is μ -a.e.. Applying then [theorem: 21.330] proves that

$$\int^+ f^- d\mu = 0 < \infty$$

As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+ f^+ d\mu, \int^+ f^- d\mu \subseteq [0, \infty[$ it follows that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

and

$$\int f d\mu \stackrel{\text{def}}{=} \int^+ f^+ d\mu - \int^+ f^- d\mu = 0 - 0 = 0 \quad \square$$

Theorem 21.359. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f_1, g_1, f_2, g_2 \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $f_1(X) \subseteq [0, \infty[, f_2(X) \subseteq [0, \infty[, g_1(X) \subseteq [0, \infty[, g_2(X) \subseteq [0, \infty[$ and $f_1 - f_2 = g_1 - g_2$ then

$$\int^+ f_1 d\mu - \int^+ f_2 d\mu = \int f_1 d\mu - \int f_2 d\mu = \int g_1 d\mu - \int g_2 d\mu = \int^+ g_1 d\mu - \int^+ g_2 d\mu$$

Proof. Let $x \in X$ then from $f_1 - f_2 = g_1 - g_2$ we have

$$\begin{aligned} (f_1 - f_2)(x) &= (g_1 - g_2)(x) && \Rightarrow && f_1(x) - f_2(x) &= g_1(x) - g_2(x) \\ & && \xRightarrow{f_1(x), f_2(x), g_1(x), g_2(x) \in [0, \infty[} && f_1(x) + g_2(x) &= f_2(x) + g_1(x) \\ & && \Rightarrow && (f_1 + g_2)(x) &= (f_2 + g_1)(x) \end{aligned}$$

proving that

$$f_1 + g_2 = f_2 + g_1 \quad (21.535)$$

As $f_1, f_2, g_1, g_2 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ and $f_1(X), g_1(X), f_2(X), g_2(X) \subseteq [0, \infty[$ we have

$$f_1, g_1, f_2, g_2 \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

and

$$\int f_1 d\mu \stackrel{[\text{theorem: 21.352 (3)}]}{=} \int^+ f_1 d\mu < \infty \quad (21.536)$$

$$\int f_2 d\mu \stackrel{[\text{theorem: 21.352 (3)}]}{=} \int^+ f_2 d\mu < \infty \quad (21.537)$$

$$\int g_1 d\mu \stackrel{[\text{theorem: 21.352 (3)}]}{=} \int^+ g_1 d\mu < \infty \quad (21.538)$$

$$\int g_2 d\mu \stackrel{[\text{theorem: 21.352 (3)}]}{=} \int^+ g_2 d\mu < \infty \quad (21.539)$$

Using [theorem: 21.331] we have that

$$\int^+ f_1 d\mu + \int^+ g_2 d\mu = \int^+ (f_1 + g_2) d\mu \stackrel{[\text{eq: 21.535}]}{=} \int^+ (f_2 + g_1) d\mu = \int^+ f_2 d\mu + \int^+ g_1 d\mu$$

So we have as the integrals are finite that

$$\int^+ f_1 d\mu - \int^+ f_2 d\mu = \int^+ g_1 d\mu - \int^+ g_2 d\mu$$

which combined with [eqs: 21.536, 21.230, 21.538 and 21.539] proves that

$$\int^+ f_1 d\mu - \int^+ f_2 d\mu = \int f_1 d\mu - \int f_2 d\mu = \int g_1 d\mu - \int g_2 d\mu = \int^+ g_1 d\mu - \int^+ g_2 d\mu \quad \square$$

Theorem 21.360. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have:

1. If $f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ then $\forall \alpha \in \mathbb{R} \ \alpha \cdot f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ and $\int (\alpha \cdot f) d\mu = \alpha \cdot \int f d\mu$.
2. If $f \in \mathcal{L}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then $\forall \alpha \in \mathbb{R} \ \alpha \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $\int (\alpha \cdot f) d\mu = \alpha \cdot \int f d\mu$.

3. If $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ then $f + g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ and $\int (f + g) d\mu = \int f d\mu + \int g d\mu$.

Note 21.361. As $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that $f(X), g(X) \subseteq \mathbb{R}$ so that $f + g$ is well defined.

4. If $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ satisfies $\forall x \in X f(x) \leq g(x)$ then $\int f d\mu \leq \int g d\mu$.

Proof.

1. As $f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$ we have $f \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $\int^+ f^+ d\mu < \infty$ or $\int^+ f^- d\mu < \infty$. Using [theorems: 21.342, 21.263] it follows that

$$\alpha \cdot f, (\alpha \cdot f)^+, (\alpha \cdot f)^- \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \quad (21.540)$$

Consider now the following cases for α

$\alpha = 0$. Then $\alpha \cdot f = C_0$ so that by [example: 21.349]

$$\alpha \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \subseteq \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$$

and

$$\int (\alpha \cdot f) d\mu = 0 = 0 \cdot \int f d\mu = \alpha \cdot \int f d\mu$$

$0 < \alpha$. Then we have that

$$(\alpha \cdot f)^+ \stackrel{[\text{theorem: 21.344}]}{=} \alpha \cdot f^+ \text{ and } (\alpha \cdot f)^- \stackrel{[\text{theorem: 21.344}]}{=} \alpha \cdot f^-$$

so

$$\begin{aligned} \int^+ (\alpha \cdot f)^+ d\mu &= \int^+ \alpha \cdot f^+ d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} \alpha \cdot \int^+ f^+ d\mu \end{aligned} \quad (21.541)$$

$$\begin{aligned} \int^+ (\alpha \cdot f)^- d\mu &= \int^+ \alpha \cdot f^- d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} \alpha \cdot \int^+ f^- d\mu \end{aligned} \quad (21.542)$$

from which it follows that

$$\int^+ (\alpha \cdot f)^+ d\mu < \infty \text{ or } \int^+ (\alpha \cdot f)^- d\mu < \infty \quad (21.543)$$

From [eqs: 21.540, 21.541, 21.542 and 21.543] it follows that

$$\alpha \cdot f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$$

and

$$\begin{aligned} \int (\alpha \cdot f) d\mu &\stackrel{\text{def}}{=} \int^+ (\alpha \cdot f)^+ d\mu - \int^+ (\alpha \cdot f)^- d\mu \\ &\stackrel{[\text{eq: 21.541, 21.542}]}{=} \alpha \cdot \int^+ f^+ d\mu - \alpha \cdot \int^+ f^- d\mu \\ &\stackrel{[\text{theorem: 20.42}]}{=} \alpha \cdot \left(\int^+ f^+ d\mu - \int^+ f^- d\mu \right) \\ &= \alpha \cdot \int f d\mu \end{aligned}$$

$\alpha < 0$. Then we have that $-\alpha \in [0, \infty[$ and

$$(\alpha \cdot f)^+ \stackrel{[\text{theorem: 21.344}]}{=} -\alpha \cdot f^- \text{ and } (\alpha \cdot f)^- \stackrel{[\text{theorem: 21.344}]}{=} -\alpha \cdot f^+$$

so that

$$\begin{aligned} \int^+ (\alpha \cdot f)^+ d\mu &= \int^+ ((-\alpha) \cdot f^-) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} (-\alpha) \cdot \int^+ f^- d\mu \end{aligned} \quad (21.544)$$

$$\begin{aligned} \int^+ (\alpha \cdot f)^- d\mu &= \int^+ ((-\alpha) \cdot f^+) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} (-\alpha) \cdot \int^+ f^+ d\mu \end{aligned} \quad (21.545)$$

from which it follows that

$$\int^+ (\alpha \cdot f)^+ d\mu < \infty \text{ or } \int^+ (\alpha \cdot f)^- d\mu < \infty \quad (21.546)$$

From [eqs: 21.540, 21.544, 21.545 and 21.546] it follows that

$$\alpha \cdot f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]}$$

and

$$\begin{aligned} \int (\alpha \cdot f) d\mu &= \int^+ (\alpha \cdot f)^+ d\mu - \int^+ (\alpha \cdot f)^- d\mu \\ &\stackrel{[\text{eq: 21.544, 21.546}]}{=} (-\alpha) \cdot \int^+ f^- d\mu - \left((-\alpha) \cdot \int^+ f^+ d\mu \right) \\ &\stackrel{[\text{theorem: 20.42}]}{=} \alpha \cdot \left(\int^+ f^+ d\mu - \int^+ f^- d\mu \right) \\ &= \alpha \cdot \int f d\mu \end{aligned}$$

2. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and $\int^+ f^+ d\mu < \infty$ and $\int^+ f^- d\mu < \infty$. Using [theorem : 21.342, 21.263] we have that

$$\alpha \cdot f, (\alpha \cdot f)^+, (\alpha \cdot f)^- \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \quad (21.547)$$

Consider the following cases for α

$\alpha = 0$. Then $\alpha \cdot f = C_0$ so that by [example: 21.349] that

$$\alpha \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \subseteq \overline{\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]}$$

and

$$\int (\alpha \cdot f) d\mu = 0 = 0 \cdot \int f d\mu = \alpha \cdot \int f d\mu$$

$0 < \alpha$. Then we have that

$$(\alpha \cdot f)^+ \stackrel{[\text{theorem: 21.344}]}{=} \alpha \cdot f^+ \text{ and } (\alpha \cdot f)^- \stackrel{[\text{theorem: 21.344}]}{=} \alpha \cdot f^- \quad (21.548)$$

so

$$\begin{aligned} \int^+ (\alpha \cdot f)^+ d\mu &= \int^+ (\alpha \cdot f^+) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} \alpha \cdot \int^+ f^+ d\mu < \infty \end{aligned} \quad (21.549)$$

$$\begin{aligned} \int^+ (\alpha \cdot f)^- d\mu &= \int^+ (\alpha \cdot f^-) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} \alpha \cdot \int^+ f^- d\mu < \infty \end{aligned} \quad (21.550)$$

Using [eqs: 21.547, 21.549 and 21.550] it follows that

$\alpha \cdot f$ is μ -integrable

and

$$\begin{aligned} \int (\alpha \cdot f) d\mu &= \int^+ (\alpha \cdot f)^+ d\mu - \int^+ (\alpha \cdot f)^- d\mu \\ &\stackrel{[\text{eqs: 21.549, 21.550}]}{=} \alpha \cdot \int^+ f^+ d\mu - \alpha \cdot \int^+ f^- d\mu \\ &= \alpha \cdot \left(\int^+ f^+ d\mu - \int^+ f^- d\mu \right) \\ &= \alpha \cdot \int f d\mu \end{aligned}$$

$\alpha < 0$. Then we have that $-\alpha \in [0, \infty[$ and

$$(\alpha \cdot f)^+ \stackrel{[\text{theorem: 21.344}]}{=} -\alpha \cdot f^- \text{ and } (\alpha \cdot f)^- \stackrel{[\text{theorem: 21.344}]}{=} -\alpha \cdot f^+ \quad (21.551)$$

so

$$\begin{aligned} \int^+ (\alpha \cdot f)^+ d\mu &= \int^+ (-\alpha \cdot f^-) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} (-\alpha) \cdot \int^+ f^- d\mu < \infty \end{aligned} \quad (21.552)$$

$$\begin{aligned} \int^+ (\alpha \cdot f)^- d\mu &= \int^+ (-\alpha \cdot f^+) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} (-\alpha) \cdot \int^+ f^+ d\mu < \infty \end{aligned} \quad (21.553)$$

Using [eqs: 21.547, 21.552 and 21.553] it follows that

$\alpha \cdot f$ is μ -integrable

and

$$\begin{aligned} \int (\alpha \cdot f) d\mu &= \int^+ (\alpha \cdot f)^+ d\mu - \int^+ (\alpha \cdot f)^- d\mu \\ &\stackrel{[\text{eqs: 21.552, 21.553}]}{=} (-\alpha) \cdot \int^+ f^- d\mu - (-\alpha) \cdot \int^+ f^+ d\mu \\ &= \alpha \cdot \left(\int^+ f^+ d\mu - \int^+ f^- d\mu \right) \\ &= \alpha \cdot \int f d\mu \end{aligned}$$

3. As $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ and $f(X), g(X) \subseteq \mathbb{R}$ we have that

$$f + g \text{ is well defined, } f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \text{ and } \int^+ f^+ d\mu, \int^+ f^- d\mu, \int^+ g^+ d\mu, \int^+ g^- d\mu < \infty$$

Using [theorem: 21.257] it follows that

$$f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

and by [theorem: 21.342] we have that

$$f^+, g^+, f^-, (f + g)^+, (f + g)^- \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

Further using [theorem: 21.257] we have also

$$f^+ + g^+, f^- + g^- \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

Further we have by [theorem: 21.344] that

$$\forall x \in X \quad (f + g)^+(x) \leq (f^+ + g^+)(x) \text{ and } (f + g)^-(x) \leq (f^- + g^-)(x)$$

so that we can apply [theorem: 21.331] giving

$$\begin{aligned}\int^+(f+g)^+d\mu &\leq \int^+(f^++g^+)d\mu \stackrel{[\text{theorem: 21.331}]}{=} \int^+f^+d\mu + \int^+g^+d\mu < \infty \\ \int^+(f+g)^-d\mu &\leq \int^+(f^-+g^-)d\mu \stackrel{[\text{theorem: 21.331}]}{=} \int^+f^-d\mu + \int^+g^-d\mu < \infty\end{aligned}$$

which proves that

$$f+g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

Next we have that

$$\begin{aligned}(f+g)^+ - (f+g)^- &\stackrel{[\text{theorem: 21.341}]}{=} f+g \\ &\stackrel{[\text{theorem: 21.341}]}{=} (f^+ - f^-) + (g^+ - g^-) \\ &= (f^+ + g^+) - (f^- + g^-)\end{aligned}\tag{21.554}$$

Finally

$$\begin{aligned}\int (f+g)d\mu &\stackrel{\text{def}}{=} \int^+(f+g)^+d\mu - \int^+(f+g)^-d\mu \\ &\stackrel{[\text{theorem: 21.359}], [\text{eq: 21.554}]}{=} \int^+(f^++g^+)d\mu - \int^+(f^-+g^-)d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} \int^+f^+d\mu + \int^+g^+d\mu - \left(\int^+f^-d\mu + \int^+g^-d\mu \right) \\ &= \left(\int^+f^+d\mu - \int^+f^-d\mu \right) + \left(\int^+g^+d\mu - \int^+g^-d\mu \right) \\ &= \int f d\mu + \int g d\mu\end{aligned}$$

4. First we prove the result for real functions. So let $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ then $f(X), g(X) \subseteq \mathbb{R}$ and for $-1 \in \mathbb{R}$ we have by (2) that $(-1) \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$, hence by (3) it follows that

$$f - g = f + (-1) \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]\tag{21.555}$$

As $\forall x \in X$ $f(x) \leq g(x) \Rightarrow (g-f)(x) = g(x) - f(x) \in [0, \infty[$ it follows from [theorem: 21.352] that

$$\int (g-f)d\mu = \int^+(g-f) \geq 0\tag{21.556}$$

Further we have

$$\begin{aligned}\int g d\mu - \int f d\mu &= \int g d\mu + (-1) \cdot \int f d\mu \\ &\stackrel{(2)}{=} \int g d\mu + \int ((-1) \cdot f) d\mu \\ &\stackrel{(2)}{=} \int (g + (-1) \cdot f) d\mu \\ &= \int (g - f) d\mu \\ &\stackrel{[\text{eq: 21.556}]}{\geq} 0\end{aligned}$$

which proves that

$$\int f d\mu \leq \int g d\mu$$

In other words we have proved that

$$\forall f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ we have } \int f d\mu \leq \int g d\mu\tag{21.557}$$

We extend now the result to extended real functions. So let $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then by definition we have:

$$f, g \in \mathcal{M}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.558)$$

$$\int^+ f^+ d\mu < \infty \wedge \int^+ f^- d\mu < \infty \wedge \int^+ g^+ d\mu < \infty \wedge \int^+ g^- d\mu < \infty \quad (21.559)$$

Using [theorem: 21.342] it follows that

$$f^+, f^-, g^+, g^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.560)$$

Using [theorem: 21.336] it follows that

$$\begin{aligned} N_1 &= \{x \in X \mid f^+(x) = \infty\} \in \mathcal{A} \text{ and } \mu(N_1) = 0 \\ N_2 &= \{x \in X \mid f^-(x) = \infty\} \in \mathcal{A} \text{ and } \mu(N_2) = 0 \\ N_3 &= \{x \in X \mid g^+(x) = \infty\} \in \mathcal{A} \text{ and } \mu(N_3) = 0 \\ N_4 &= \{x \in X \mid g^-(x) = \infty\} \in \mathcal{A} \text{ and } \mu(N_4) = 0 \end{aligned}$$

Take

$$N = N_1 \cup N_2 \cup N_3 \cup N_4 \text{ then } N \in \mathcal{A} \text{ and } X \setminus N \in \mathcal{A}$$

then we have $\mu(N) = \mu(N_1 \cup N_2 \cup N_3 \cup N_4) \leq \mu(N_1) + \mu(N_2) + \mu(N_3) + \mu(N_4) = 0$ so that

$$\mu(N) = 0$$

Using now [theorem: 21.276] on [eqs: 21.558] we have

$$\mathcal{X}_{X, X \setminus N} \cdot f, \mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.561)$$

so that by [theorem: 21.342]

$$(\mathcal{X}_{X, X \setminus N} \cdot f)^+, (\mathcal{X}_{X, X \setminus N} \cdot f)^-, (\mathcal{X}_{X, X \setminus N} \cdot g)^+, (\mathcal{X}_{X, X \setminus N} \cdot g)^- \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.562)$$

Let $x \in X$ then we have either:

$x \in N$. Then we have

$$\begin{aligned} (\mathcal{X}_{X \setminus N} \cdot f)(x) &= \mathcal{X}_{X \setminus N}(x) \cdot f(x) = 0 \in \mathbb{R} \\ (\mathcal{X}_{X \setminus N} \cdot g)(x) &= \mathcal{X}_{X \setminus N}(x) \cdot g(x) = 0 \in \mathbb{R} \end{aligned}$$

So that $(\mathcal{X}_{X \setminus N} \cdot f)(x) = 0 = (\mathcal{X}_{X \setminus N} \cdot g)(x)$ proving that

$$(\mathcal{X}_{X \setminus N} \cdot f)(x) \leq \mathcal{X}_{X \setminus N}(x) \cdot g(x)$$

$x \in X \setminus N$. Then we have $x \in (X \setminus N_1) \cap (X \setminus N_2) \cap (X \setminus N_3) \cap (X \setminus N_4)$ so that $f^+(x) \in \mathbb{R}$, $f^-(x) \in \mathbb{R}$, $g^+(x) \in \mathbb{R}$, $g^-(x) \in \mathbb{R}$. Further

$$\begin{aligned} (\mathcal{X}_{X, X \setminus N} \cdot f)(x) &= \mathcal{X}_{X, X \setminus N}(x) \cdot f(x) = f(x) \stackrel{[\text{theorem: 21.341}]}{=} f^+(x) - f^-(x) \in \mathbb{R} \\ (\mathcal{X}_{X, X \setminus N} \cdot g)(x) &= \mathcal{X}_{X, X \setminus N}(x) \cdot g(x) = g(x) \stackrel{[\text{theorem: 21.341}]}{=} g^+(x) - g^-(x) \in \mathbb{R} \end{aligned}$$

Further we have

$$(\mathcal{X}_{X, X \setminus N} \cdot f)(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot f(x) = f(x) \leq g(x) = \mathcal{X}_{X \setminus N}(x) \cdot g(x) = (\mathcal{X}_{X, X \setminus N} \cdot g)(x)$$

which proves that

$$(\mathcal{X}_{X, X \setminus N} \cdot f)(x) \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x)$$

To summarize we have proved that

$$(\mathcal{X}_{X, X \setminus N} \cdot f)(X) \subseteq \mathbb{R} \wedge (\mathcal{X}_{X, X \setminus N} \cdot g) \subseteq \mathbb{R} \wedge \forall x \in X \quad (21.563)$$

$$\forall x \in X \text{ we have } (\mathcal{X}_{X, X \setminus N} \cdot f)(x) \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \quad (21.564)$$

Now using [theorems: 21.280,21.333,21.344(9)] we have that

$$\int^+ (\mathcal{X}_{X,X \setminus N} \cdot f)^+ d\mu = \int^+ \mathcal{X}_{X,X \setminus N} \cdot f^+ = \int^+ f^+ d\mu < \infty \quad (21.565)$$

$$\int^+ (\mathcal{X}_{X,X \setminus N} \cdot f)^- d\mu = \int^+ \mathcal{X}_{X,X \setminus N} \cdot f^- = \int^+ f^- d\mu < \infty \quad (21.566)$$

$$\int^+ (\mathcal{X}_{X,X \setminus N} \cdot g)^+ d\mu = \int^+ \mathcal{X}_{X,X \setminus N} \cdot g^+ = \int^+ g^+ d\mu < \infty \quad (21.567)$$

$$\int^+ (\mathcal{X}_{X,X \setminus N} \cdot g)^- d\mu = \int^+ \mathcal{X}_{X,X \setminus N} \cdot g^- = \int^+ g^- d\mu < \infty \quad (21.568)$$

Combining [eqs: 21.561,21.562, 21.565,21.566,21.567,21.568] it follows that

$$\mathcal{X}_{X,X \setminus N} \cdot f, \mathcal{X}_{X,X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

and using [eqs: 21.563, 21.564] we have

$$\mathcal{X}_{X,X \setminus N} \cdot f, \mathcal{X}_{X,X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \wedge \forall x \in X \text{ we have } (\mathcal{X}_{X,X \setminus N} \cdot f)(x) \leq (\mathcal{X}_{X,X \setminus N} \cdot g)(x)$$

Using then the case for real functions [see eq: 21.557] we have

$$\int (\mathcal{X}_{X,X \setminus N} \cdot f) d\mu \leq \int (\mathcal{X}_{X,X \setminus N} \cdot g) d\mu \quad (21.569)$$

Finally

$$\begin{aligned} \int f d\mu & \stackrel{\text{def}}{=} \int^+ f^+ d\mu - \int^+ f^- d\mu \\ & \stackrel{[\text{eqs: 21.565,21.566}]}{=} \int^+ (\mathcal{X}_{X,X \setminus N} \cdot f)^+ d\mu - \int^+ (\mathcal{X}_{X,X \setminus N} \cdot f)^- d\mu \\ & \stackrel{\text{def}}{=} \int (\mathcal{X}_{X,X \setminus N} \cdot f) d\mu \\ & \stackrel{[\text{eq: 21.569}]}{\leq} \int (\mathcal{X}_{X,X \setminus N} \cdot g) d\mu \\ & \stackrel{\text{def}}{=} \int^+ (\mathcal{X}_{X,X \setminus N} \cdot g)^+ d\mu - \int^+ (\mathcal{X}_{X,X \setminus N} \cdot g)^- d\mu \\ & \stackrel{[\text{eq: 21.567, 21.568}]}{=} \int^+ g^+ d\mu - \int^+ g^- d\mu \\ & = \int g d\mu \end{aligned}$$

which proves that

$$\int f d\mu \leq \int g d\mu \quad \square$$

We have now the following alternative definitions for integrability using the absolute value

Theorem 21.362. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have:*

1. *If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then*

$$\int |f| d\mu \text{ exist}$$

2. *If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then*

$$|f| \text{ is } \mu\text{-integrable} \Leftrightarrow \int^+ |f| d\mu < \infty$$

Further if f is integrable then

$$\int |f| d\mu = \int^+ |f| d\mu$$

3. If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$f \text{ is } \mu\text{-integrable} \Leftrightarrow |f| \text{ is } \mu\text{-integrable}$$

Further if f or $|f|$ is μ -integrable then

$$\left| \int f d\mu \right| \leq \int |f| d\mu < \infty$$

so that $\int f d\mu \in \mathbb{R}$.

4. If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then $|f| \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$

5. We have

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \Leftrightarrow |f| \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

Further if $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ or $|f| \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then

$$\left| \int f d\mu \right| \leq \int |f| d\mu < \infty$$

Note 21.363. As $g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \Leftrightarrow g(X) \subseteq \mathbb{R} \wedge g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have also that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \Leftrightarrow |f| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

Proof.

1. Let $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then by [theorem: 21.342] we have $|f| \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$. Further as $|f|(X) \subseteq [0, \infty]$ it follows from [theorem: 21.344(7)] that $(|f|)^- = C_0$, so that

$$\int^+ (|f|)^- d\mu = \int^+ C_0 d\mu \stackrel{[\text{example: 21.330}]}{=} 0 < \infty$$

Hence we have by definition that

$$\int |f| d\mu \text{ exist}$$

2. Let $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then we have by [theorem: 21.342] $|f| \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$

$\Rightarrow$. As $|f|(X) \subseteq [0, \infty]$ it follows from [theorem: 21.344(7)] that $(|f|)^+ = |f|$, further as $|f|$ is μ -integrable we have also that $\int (|f|)^+ d\mu < \infty$. Hence we have that

$$\int^+ |f| d\mu < \infty$$

Further we have

$$\begin{aligned} \int |f| d\mu &= \int^+ (|f|)^+ d\mu - \int^+ (|f|)^- d\mu \\ &\stackrel{[\text{theorem: 21.344(7)}]}{=} \int^+ |f| d\mu + \int^+ C_0 d\mu \\ &\stackrel{[\text{example: 21.330}]}{=} \int^+ |f| d\mu \end{aligned}$$

or

$$\int |f| d\mu = \int^+ |f| d\mu$$

$\Leftarrow$. Assume that $\int^+ |f| d\mu < \infty$ then we have

$$\begin{aligned} \int^+ (|f|)^+ d\mu &\stackrel{[\text{theorem: 21.344(7)}]}{=} \int^+ |f| d\mu < \infty \\ \int^+ (|f|)^- d\mu &\stackrel{[\text{theorem: 21.344(7)}]}{=} \int^+ C_0 d\mu \stackrel{[\text{example: 21.330}]}{=} 0 < \infty \end{aligned}$$

so that by definition

$|f|$ is μ -integrable

3. Let $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then we have by [theorem: 21.342] $|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$

$\Rightarrow$. If f is μ -integrable then by definition we have

$$\int^+ f^+ d\mu < \infty \wedge \int^+ f^- d\mu < \infty$$

As $|f|(X) \subseteq [0, \infty]$ it follows

$$\begin{aligned} (|f|)^+ & \stackrel{[\text{theorem: 21.344(7)}]}{=} |f| \stackrel{[\text{theorem: 21.341}]}{=} f^+ + f^- \\ (|f|)^- & \stackrel{[\text{theorem: 21.344(7)}]}{=} C_0 \end{aligned}$$

Hence we have

$$\int^+ (|f|)^+ d\mu \stackrel{[\text{theorem: 21.331}]}{=} \int^+ f^+ d\mu + \int^+ f^- d\mu < \infty \quad (21.570)$$

$$\int^+ (|f|)^- d\mu = \int^+ C_0 d\mu \stackrel{[\text{example: 21.330}]}{=} 0 < \infty \quad (21.571)$$

from which it follows that

$|f|$ is μ -integrable

Further we have

$$\begin{aligned} \left| \int f d\mu \right| &= \left| \int^+ f^+ d\mu - \int^+ f^- d\mu \right| \\ &\leq \left| \int^+ f^+ d\mu \right| + \left| \int^+ f^- d\mu \right| \\ &\stackrel{0 \leq \int^+}{=} \int^+ f^+ d\mu + \int^+ f^- d\mu \\ &\stackrel{[\text{eq: 21.570}]}{=} \int^+ (|f|)^+ d\mu - 0 \\ &\stackrel{[\text{eq: 21.571}]}{=} \int^+ (|f|)^+ d\mu - \int^+ (|f|)^- d\mu \\ &= \int |f| d\mu \end{aligned}$$

or in other words

$$\left| \int f d\mu \right| \leq \int |f| d\mu$$

$\Leftarrow$. If $|f|$ is integrable then

$$\int^+ |f| d\mu \stackrel{[\text{theorem: 21.344(7)}]}{=} \int^+ (|f|)^+ d\mu + \int^+ C_0 d\mu = \int^+ (|f|)^+ d\mu < \infty$$

By [theorem: 21.344(4)] we have $\forall x \in X$ $f^+(x) \leq |f|(x) \wedge f^-(x) \leq |f|(x)$, hence using [theorem: 21.331] we have that

$$\int^+ f^+ d\mu \leq \int^+ |f| d\mu < \infty \wedge \int^+ f^- d\mu \leq \int^+ |f| d\mu < \infty$$

Hence by definition

f is μ -integrable

4. If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ then by (1) $\int |f| d\mu$ exist so that by definition

$$f \in \overline{\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}$$

5. Then we have:

$\Rightarrow$. If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then by definition $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and f is μ -integrable, using [theorem: 21.342] it follows that $|f| \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and by (3) that $|f|$ is μ -integrable. Hence we have by definition that $|f| \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$. Further by (3)

$$\left| \int f d\mu \right| \leq \int |f| d\mu$$

$\Leftarrow$. If $|f| \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then by definition $|f| \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $|f|$ is μ -integrable, using [theorem: 21.342] it follows that $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and by (3) that f is μ -integrable. Hence we have by definition that $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$. Further by (3)

$$\left| \int f d\mu \right| \leq \int |f| d\mu \quad \square$$

If a function is μ -a.e. to a μ -integrable function then the function is μ -integrable function.

Theorem 21.364. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f: X \rightarrow \overline{\mathbb{R}}$ and $g: X \rightarrow \overline{\mathbb{R}}$ two functions such that $f = g$ μ -a.e. then we have

1. If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int f d\mu$ exist then $\int g d\mu$ exist and $\int f d\mu = \int g d\mu$.
2. If $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and f is μ -integrable then

$$g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } g \text{ is } \mu\text{-integrable and } \int f d\mu = \int g d\mu$$

3. If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then

$$g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int f d\mu = \int g d\mu$$

Proof.

1. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f = g$ μ -a.e. it follows from [theorem: 21.284] that

$$g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Define

$$H = \{x \in X \mid f(x) \neq g(x)\} = \{f \neq g\}$$

As $f, g \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ it follows from [theorem: 21.256] that $H \in \mathcal{A}_{|X} = \mathcal{A}$. Further as $f = g$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $H \subseteq N \Rightarrow \mu(H) \leq \mu(N) = 0$, hence we have:

$$H \in \mathcal{A} \wedge \mu(H) = 0 \quad (21.572)$$

As $f, g \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have by [theorem: 21.342] that $f^+, f^-, g^+, g^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$. Let $x \in X$ then we have

$$\text{If } f^+(x) \neq g^+(x) \Rightarrow \max(f(x), 0) \neq \max(g(x), 0) \Rightarrow f(x) \neq g(x)$$

$$\text{If } f^-(x) \neq g^-(x) \Rightarrow -\min(f(x), 0) \neq -\min(g(x), 0) \Rightarrow f(x) \neq g(x)$$

which proves that

$$\{x \in A \mid f^+(x) \neq g^+(x)\} \subseteq H \text{ and } \{x \in A \mid f^-(x) \neq g^-(x)\} \subseteq H$$

As $\mu(H) \stackrel{[eq: 21.572]}{=} 0$ we have by definition that

$$f^+ = g^+ \text{ } \mu\text{-a.e. and } f^- = g^- \text{ } \mu\text{-a.e.}$$

Using [theorem: 21.333] together with the above gives

$$\int^+ g^+ d\mu = \int^+ f^+ d\mu < \infty \text{ or } \int^+ g^- d\mu = \int^+ f^- d\mu < \infty$$

which proves that

$$\int g d\mu \text{ exist}$$

Further

$$\int g d\mu = \int^+ g^+ d\mu - \int^+ g^- d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu = \int f d\mu$$

so that

$$\int f d\mu = \int g d\mu$$

2. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f = g$ μ -a.e. it follows from [theorem: 21.284] that

$$g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Define

$$H = \{x \in X \mid f(x) \neq g(x)\} = \{f \neq g\}$$

As $f, g \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ it follows from [theorem: 21.256] that $H \in \mathcal{A}|_X = \mathcal{A}$. Further as $f = g$ μ -a.e. there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $H \subseteq N \Rightarrow \mu(H) \leq \mu(N) = 0$, hence we have:

$$H \in \mathcal{A} \wedge \mu(H) = 0 \quad (21.573)$$

As $f, g \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have by [theorem: 21.342] that $f^+, f^-, g^+, g^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$. Let $x \in X$ then we have

$$\text{If } f^+(x) \neq g^+(x) \Rightarrow \max(f(x), 0) \neq \max(g(x), 0) \Rightarrow f(x) \neq g(x)$$

$$\text{If } f^-(x) \neq g^-(x) \Rightarrow -\min(f(x), 0) \neq -\min(g(x), 0) \Rightarrow f(x) \neq g(x)$$

which proves that

$$\{x \in A \mid f^+(x) \neq g^+(x)\} \subseteq H \text{ and } \{x \in A \mid f^-(x) \neq g^-(x)\} \subseteq H$$

As $\mu(H) \stackrel{[eq: 21.573]}{=} 0$ we have by definition that

$$f^+ = g^+ \text{ } \mu\text{-a.e. and } f^- = g^- \text{ } \mu\text{-a.e.}$$

Using [theorem: 21.333] together with the above gives

$$\int^+ g^+ d\mu = \int^+ f^+ d\mu < \infty \text{ and } \int^+ g^- d\mu = \int^+ f^- d\mu < \infty$$

which proves that

$$g \text{ is } \mu\text{-integrable}$$

Further

$$\int g d\mu = \int^+ g^+ d\mu - \int^+ g^- d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu = \int f d\mu$$

so that

$$\int f d\mu = \int g d\mu$$

3. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have by definition that $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and f is μ -integrable. Hence by (1) we have that

$$g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}], g \text{ is } \mu\text{-integrable and } \int f d\mu = \int g d\mu$$

Hence we have by definition that

$$g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int f d\mu = \int g d\mu \quad \square$$

Theorem 21.365. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $|f| \leq g$ μ -a.e. then we have:

1. If g is μ -integrable then f is μ -integrable and $\int f d\mu \leq \int g d\mu$.
2. If $g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $\int f d\mu \leq \int g d\mu$.

Proof.

1. We prove this in two stages, first the case where $\forall x \in X \ |f|(x) \leq g(x)$ and then extend this to the case where $|f| \leq g$ μ -a.e.
 - a. Assume that $\forall x \in X$ we have $|f|(x) \leq g(x)$. As $g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \Rightarrow g(X) \subseteq [0, \infty]$ and g is μ -integrable we have that

$$\int^+ g d\mu \stackrel{[\text{theorem: 21.344(7)}]}{=} \int^+ g^+ d\mu < \infty$$

As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.342] that

$$f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Let $x \in X$ then $f^+(x) \stackrel{[\text{theorem: 21.344}]}{\leq} |f(x)| \leq g(x)$ and $f^-(x) \stackrel{[\text{theorem: 21.344}]}{\leq} |f(x)| \leq g(x)$ so that by [theorem: 21.331]

$$\int^+ f^+ d\mu \leq \int^+ g d\mu < \infty \text{ and } \int^+ f^- d\mu \leq \int^+ g d\mu < \infty$$

From the above it follows by definition that

f is μ -Integrable

As $\forall x \in X \ |f(x)| \leq g(x)$ it follows using [theorem: 21.360] that

$$\int f d\mu \leq \int g d\mu$$

- b. Assume now that $|f| \leq g$ μ -a.e. Then there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in X \mid \neg(|f(x)| \leq g(x))\} \subseteq N$$

so that

$$\forall x \in X \setminus N \text{ we have } |f(x)| \leq g(x)$$

Let $x \in X$ then we have either:

$$x \in N. \text{ Then } |(\mathcal{X}_{X, X \setminus N} \cdot f)|(x) = |\mathcal{X}_{X, X \setminus N}(x) \cdot f(x)| = |0 \cdot f(x)| = 0 \leq g(x)$$

$$x \in X \setminus N. \text{ Then } |(\mathcal{X}_{X, X \setminus N} \cdot f)|(x) = |\mathcal{X}_{X, X \setminus N}(x) \cdot f(x)| = |f(x)| \leq g(x)$$

hence we have that

$$\forall x \in X \ |\mathcal{X}_{X, X \setminus N} \cdot f| \leq g(x)$$

Further as $N \in \mathcal{A} \Rightarrow X \setminus N \in \mathcal{A}$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.276] that

$$\mathcal{X}_{X, X \setminus N} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

So all the conditions for (1.a) are satisfied for $\mathcal{X}_{X, X \setminus N} \cdot f$ and we have that

$$\mathcal{X}_{X, X \setminus N} \cdot f \text{ is } \mu\text{-integrable and } \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu \leq \int g d\mu$$

Now as $\mu(N) = 0$ it follows from [theorem: 21.280] that $\mathcal{X}_{X, X \setminus N} \cdot f = f$ μ -a.e., using then [theorem: 21.364] it follows that

$$f \text{ is } \mu\text{-integrable and } \int f d\mu = \int (\mathcal{X}_{X \setminus N} \cdot f) d\mu \leq \int g d\mu$$

2. If $g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then by definition g is μ -integrable and it follows from (1) and the fact that $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ that f is μ -integrable and $\int f d\mu \leq \int g d\mu$ hence we have

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ with } \int f d\mu \leq \int g d\mu \quad \square$$

If the measure integral of a measurable function is zero then function is almost everywhere zero.

Theorem 21.366. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $\int |f| d\mu = 0$ then $f = 0$ μ -a.e..

Note 21.367. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem 21.362] that $\int |f| d\mu$ exist so the theorem makes sense.

Proof. As $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.342] that $|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$. Hence using [lemma: 21.334] we have:

$$\forall n \in \mathbb{N} \text{ that } \mu\left(\left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\}\right) \leq n \cdot \int^+ |f| d\mu = 0$$

from which it follows that

$$\forall n \in \mathbb{N} \text{ that } \mu\left(\left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\}\right) = 0 \quad (21.574)$$

Let $x \in \{x \in X \mid f(x) \neq 0\}$ then $0 < |f(x)|$ so that by [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < |f(x)| \Rightarrow x \in \{x \in X \mid |f(x)| \geq \frac{1}{n}\} \subseteq \bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| \geq \frac{1}{n}\}$. Likewise if $x \in \bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| \geq \frac{1}{n}\}$ then there exist a $n \in \mathbb{N}$ such that $x \in \{x \in X \mid |f(x)| \geq \frac{1}{n}\}$, hence $|f(x)| \geq \frac{1}{n} > 0$ so that $f(x) \neq 0$ which proves that

$$\bigcup_{n \in \mathbb{N}} \left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\} = \{x \in X \mid f(x) \neq 0\} \quad (21.575)$$

As $|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have by [theorem: 21.254] that

$$\left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\}_{n \in \mathbb{N}} \subseteq \mathcal{A} \text{ so that } \bigcup_{n \in \mathbb{N}} \left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\} \in \mathcal{A} \quad (21.576)$$

Hence

$$\begin{aligned} \mu(\{x \in X \mid f(x) \neq 0\}) & \stackrel{[\text{eq: 21.575}]}{=} \mu\left(\bigcup_{n \in \mathbb{N}} \left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\}\right) \\ & \stackrel{[\text{eq: 21.576}]] \wedge [\text{theorem: 21.101}]}{\leq} \sum_{n=1}^{\infty} \mu\left(\left\{x \in X \mid |f(x)| \geq \frac{1}{n}\right\}\right) \\ & \stackrel{[\text{eq: 21.574}]}{=} 0 \end{aligned}$$

which proves that

$$f = 0 \text{ } \mu\text{-a.e.} \quad \square$$

Theorem 21.368. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ such that $\forall A \in \mathcal{A}$ we have $0 \leq \int_A f d\mu$ then $f \geq 0$ μ -a.e.

Note 21.369. Using [theorem: 21.356] we have that $\mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \subseteq \mathcal{L}_A[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ so that $\int_A f d\mu$ is well defined].

Proof. Let $A = \{x \in X \mid 0 \not\leq f(x)\} = \{x \in X \mid f(x) < 0\} = \{f < 0\}$. Now as $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have by [theorem: 21.254] that

$$A \in \mathcal{A}|_X = \mathcal{A}$$

If $x \in X$ then we have either:

$x \in A$. Then $f(x) < 0$ and $\mathcal{X}_{X,A}(x) = 1$ so that $(\mathcal{X}_A \cdot f)(x) = f(x) < 0$ hence

$$(\mathcal{X}_{X,A} \cdot f)^+(x) = \max((\mathcal{X}_{X,A} \cdot f)(x), 0) = 0$$

$x \in X \setminus A$. Then $\mathcal{X}_{X,A}(x) = 0$ so that $(\mathcal{X}_{X,A} \cdot f)(x) = 0$ hence

$$(\mathcal{X}_{X,A} \cdot f)^+(x) = \max((\mathcal{X}_A \cdot f)(x), 0) = \max(0, 0) = 0$$

so we have proved that $(\mathcal{X}_{X,A} \cdot f)^+ = C_0$. Hence using [example: 21.318] we have that

$$\int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu = 0 \quad (21.577)$$

So we have

$$0 \leq \int_A f d\mu = \int (\mathcal{X}_{X,A} \cdot f) d\mu = \int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu - \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu = - \int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu \leq 0$$

which proves that

$$\int^+ (\mathcal{X}_{X,A} \cdot f)^- d\mu = 0 \quad (21.578)$$

Hence

$$\begin{aligned} \int^+ |\mathcal{X}_{X,A} \cdot f| d\mu &\stackrel{[\text{theorem: 21.341}]}{=} \int^+ ((\mathcal{X}_{X,A} \cdot f)^+ + (\mathcal{X}_{X,A} \cdot f)^-) d\mu \\ &\stackrel{[\text{theorem: 21.331}]}{=} \int^+ (\mathcal{X}_{X,A} \cdot f)^+ d\mu + \int (\mathcal{X}_{X,A} \cdot f)^- d\mu \\ &\stackrel{[\text{eqs: 21.577, 21.578}]}{=} 0 \end{aligned}$$

Using [theorem: 21.366(1)] on the above it follows that $\mathcal{X}_{X,A} \cdot f = 0$ μ -a.e.. Hence there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in X \mid (\mathcal{X}_{X,A} \cdot f) \neq 0\} \subseteq N$$

If $x \in \{x \in X \mid f(x) \not\geq 0\} = \{f < 0\}$ then $f(x) < 0$ and $\mathcal{X}_{X,A}(x) = 1$ proving that $(\mathcal{X}_A \cdot f)(x) < 0 \Rightarrow (\mathcal{X}_A \cdot f)(x) \neq 0$, hence $x \in \{x \in X \mid (\mathcal{X}_A \cdot f) \neq 0\} \subseteq N$. So

$$\{x \in X \mid f(x) \not\geq 0\} \subseteq N \text{ with } \mu(N) = 0$$

from which it follows that

$$f \geq 0 \text{ } \mu\text{-a.e.} \quad \square$$

Theorem 21.370. Let $\langle X, \mathcal{A}, \mu \rangle$ and $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then

$$\{x \in X \mid |f(x)| = \infty\} \in \mathcal{A} \text{ and } \mu(\{x \in X \mid |f(x)| = \infty\}) = 0$$

In other words $|f| < \infty$ μ -a.e..

Proof. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have by [theorem: 21.362] that $|f| \in \mathcal{L}[X, \mathcal{A}, \mu]$. Hence by [theorem: 21.352] we have that $|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+ |f| d\mu < \infty$. Using [theorem: 21.336] it follows then that $\{x \in X \mid |f(x)| = \infty\} \in \mathcal{A}$ and $\mu(\{x \in X \mid |f(x)| = \infty\}) = 0$. $\square$

We show now that a extended function f is μ -integrable if and only there exist a real valued μ -integrable function g that is always everywhere equal to f .

Theorem 21.371. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space

$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \Leftrightarrow f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\exists g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ with $g(X) \subseteq \mathbb{R}$ such that $f = g$ μ -a.e.

Proof. We have

$\Rightarrow$. Let $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ [so that by definition $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$] then we must create a function $g: X \rightarrow \overline{\mathbb{R}}$ with $g(X) \subseteq \mathbb{R}$ such that $f = g$ μ -a.e.. To do this define the set A of all elements of X where $f(x)$ is either $+\infty$ or $-\infty$ [or in other words $|f(x)| = \infty$].

$$A = \{x \in X \mid |f(x)| = \infty\} = \{|f| = \infty\}$$

By [theorem: 21.370] it follows that

$$A \in \mathcal{A}, X \setminus A \in \mathcal{A} \text{ and } \mu(A) = 0 \quad (21.579)$$

Define now

$$g = \mathcal{X}_{X, X \setminus A} \cdot f: X \rightarrow \overline{\mathbb{R}}$$

then using [theorem: 21.276] we have that

$$g = \mathcal{X}_{X, X \setminus A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Further if $x \in X$ then either $x \in A$ giving $g(x) \stackrel{x \notin X \setminus A}{=} 0 \in \mathbb{R}$ or $x \in X \setminus A$ then $|g(x)| = |f(x)| \neq \infty$ so that $g(x) \in \mathbb{R}$ hence

$$g(X) \subseteq \mathbb{R}$$

As $\mu(A) = 0$ it follows from [theorem: 21.280] that $\mathcal{X}_{X, X \setminus A} \cdot f = f$ μ -a.e. hence

$$g = f \text{ } \mu\text{-a.e.}$$

Using [theorem: 21.364] together with the above proves that

$$g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

$\Leftarrow$. Let $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$. As there exist a function $g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ such that $f = g$ μ -a.e. we have by [theorem: 21.364] $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$. $\square$

21.5.4 Limit theorems

One benefit that the measure integral has over the Riemann integral are the limit theorems which where actually the prime reason for the invention of the measure integral.

Theorem 21.372. (Monotone Convergence Theorem (1)) Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that $\forall x \in X$

$$\forall n \in \mathbb{N} \quad f_n(x) \leq f_{n+1}(x)$$

then we have for the function

$$f = \lim_{n \rightarrow \infty} f_n: X \rightarrow \overline{\mathbb{R}} \text{ defined by } f(x) = \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

that f is well defined and

$$f = \lim_{n \rightarrow \infty} f_n = \sup_{n \in \mathbb{N}} f_n \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

$$\lim_{n \rightarrow \infty} \int^+ f_n d\mu \text{ exists and } \int^+ f d\mu = \lim_{n \rightarrow \infty} \int^+ f_n d\mu = \sup \left(\left\{ \int^+ f_n d\mu \mid n \in \mathbb{N} \right\} \right)$$

Note 21.373. As $\forall n \in \mathbb{N} \quad f_n(x) \leq f_{n+1}(x)$ it follows from [theorem: 20.92] that $\lim_{n \rightarrow \infty} f_n(x)$ exist and $\lim_{n \rightarrow \infty} f_n(x) = \sup \{f_n(x) \mid n \in \mathbb{N}\}$

Note 21.374. We do not require that f_n is μ -integrable, however as $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.352(2)] that $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{L}[X, \mathcal{A}, \overline{\mathbb{R}}]$ from which it follows that $\int f_n d\mu$ exist and $\int f_n d\mu = \int^+ f_n d\mu$. Hence we have also

$$\lim_{n \rightarrow \infty} \int f_n d\mu \text{ exists and } \int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu = \sup \left(\left\{ \int f_n d\mu \mid n \in \mathbb{N} \right\} \right)$$

Proof. Using [theorem: 21.269(6)] it follows that

$$f = \lim_{n \rightarrow \infty} f_n: X \rightarrow \overline{\mathbb{R}} \text{ defined by } f(x) = \left(\lim_{n \rightarrow \infty} f_n \right)(x) = \lim_{n \rightarrow \infty} f_n(x)$$

is well defined and

$$f = \lim_{n \rightarrow \infty} f_n = \sup_{n \in \mathbb{N}} f_n \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.580)$$

Further as $\forall n \in \mathbb{N} \ f_n \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \Rightarrow \forall x \in X \ 0 \leq f_n(x)$ it follows from the above that

$$f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Let $x \in X$ then as $\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x)$ we have by [theorem: 21.331] that $\int^+ f_n d\mu \leq \int^+ f_{n+1} d\mu$ so that by [theorem: 20.92]

$$\lim_{n \rightarrow \infty} \int^+ f_n d\mu \text{ exist and } \lim_{n \rightarrow \infty} \int^+ f_n d\mu = \sup \left(\left\{ \int^+ f_n d\mu \mid n \in \mathbb{N} \right\} \right) \quad (21.581)$$

As $\forall n \in \mathbb{N}$ we have $f_n(x) \leq \sup(\{f_n(x) \mid n \in \mathbb{N}\}) = \lim_{n \rightarrow \infty} f_n(x) = f(x)$ we have by [theorem: 21.331] that $\int^+ f_n d\mu \leq \int^+ f d\mu$. Hence we have that $\int^+ f d\mu$ is an upper bound for $\{\int^+ f_n d\mu \mid n \in \mathbb{N}\}$ so that

$$\lim_{n \rightarrow \infty} \int^+ f_n d\mu = \sup \left(\left\{ \int^+ f_n d\mu \mid n \in \mathbb{N} \right\} \right) \leq \int^+ f d\mu \quad (21.582)$$

So it remains to prove that $\int^+ f d\mu \leq \lim_{n \rightarrow \infty} \int^+ f_n d\mu$. Let $x \in X$ and $n \in \mathbb{N}$ then as $f_n \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have by [theorem: 21.329] that there exist a $\{g_{n,i}\}_{i \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that

$$\forall i \in \mathbb{N} \text{ we have } g_{n,i}(x) \leq g_{n,i+1}(x) \quad (21.583)$$

$$f_n(x) = \lim_{i \rightarrow \infty} g_{n,i} = \sup(\{g_{n,i}(x) \mid i \in \mathbb{N}\}) \quad (21.584)$$

$$\lim_{i \rightarrow \infty} \int^S g_{n,i} d\mu \text{ exist and } \int^+ f_n d\mu = \lim_{i \rightarrow \infty} \int^S g_{n,i} d\mu \quad (21.585)$$

Define now

$$\{h_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \text{ by } h_n = \left(\bigvee_{i \in \{1, \dots, n\}} g_{i,n} \right) \Big|_{\text{[theorem: 21.310]}} \in \mathcal{S}^+[X, \mathcal{A}] \quad (21.586)$$

Let $x \in X$ and $n \in \mathbb{N}$ then $\forall i \in \{1, \dots, n\}$ we have

$$\begin{aligned} g_{i,n}(x) &\leq g_{i,n+1}(x) \\ &\leq \max(\{g_{i,n+1}(x) \mid i \in \{1, \dots, n+1\}\}) \\ &= \left(\bigvee_{i \in \{1, \dots, n+1\}} g_{i,n+1} \right)(x) \\ &= h_{n+1}(x) \end{aligned} \quad (21.587)$$

hence $h_n(x) = \left(\bigvee_{i \in \{1, \dots, n\}} g_{i,n} \right)(x) = \max(\{g_{i,n}(x) \mid i \in \{1, \dots, n\}\}) \underset{\text{[eq: 21.587]}}{\leq} h_{n+1}(x)$. So we have that

$$\forall x \in X \text{ we have } \forall n \in \mathbb{N} \text{ that } h_n(x) \leq h_{n+1}(x) \quad (21.588)$$

Further as $\forall i \in \{1, \dots, n\} \ g_{i,n}(x) \leq \sup(\{g_{i,j}(x) \mid j \in \mathbb{N}\}) \underset{\text{[eq: 21.584]}}{=} f_i(x) \underset{i \leq n \Rightarrow f_i(x) \leq f_n(x)}{\leq} f_n(x)$ we have that

$$\forall n \in \mathbb{N} \ h_n(x) = \left(\bigvee_{i \in \{1, \dots, n\}} g_{i,n} \right)(x) = \max(\{g_{i,n}(x) \mid i \in \{1, \dots, n\}\}) \leq f_n(x) \quad (21.589)$$

As $\forall n \in \mathbb{N}$ we have $f_n(x) \leq \sup(\{f_n(x) \mid n \in \mathbb{N}\}) = \lim_{n \rightarrow \infty} f_n(x) = f(x)$ it follows that

$$\forall n \in \mathbb{N} \text{ we have } h_n(x) \leq f(x) \quad (21.590)$$

Using [eq: 21.588] together with [theorem: 20.92] proves that

$$\lim_{n \rightarrow \infty} h_n(x) \text{ exist and } \lim_{n \rightarrow \infty} h_n(x) = \sup (\{h_n(x) | n \in \mathbb{N}\}) \underset{\text{[eq: 21.590]}}{\leq} f(x) \quad (21.591)$$

We prove now by contradiction that $f(x) \leq \sup (\{h_n(x) | n \in \mathbb{N}\})$. So assume that $\sup (\{h_n(x) | n \in \mathbb{N}\}) < f(x)$. Then as $f(x) = \sup (\{f_n(x) | n \in \mathbb{N}\})$ there exist a $n \in \mathbb{N}$ such that

$$\sup (\{h_n(x) | n \in \mathbb{N}\}) < f_n(x) \quad (21.592)$$

As by [eq: 21.584] $f_n(x) = \sup (\{g_{n,i}(x) | i \in \mathbb{N}\})$ there exist a $k \in \mathbb{N}$ such that

$$\sup (\{h_n(x) | n \in \mathbb{N}\}) < g_{n,k}(x) \underset{k \leq \max(h,k)}{\leq} g_{n,\max(n,k)}(x) \quad (21.593)$$

now

$$\begin{aligned} g_{n,\max(n,k)}(x) &\underset{n \in \{1, \dots, \max(n,k)\}}{\leq} \max (\{g_{i,\max(n,k)}(x) | i \in \{1, \dots, \max(n,k)\}\}) \\ &= \left(\bigvee_{i \in \{1, \dots, \max(n,k)\}} g_{i,\max(n,k)} \right)(x) \\ &= h_{\max(n,k)}(x) \end{aligned} \quad (21.594)$$

So we have

$$\begin{aligned} h_{\max(n,k)}(x) &\leq \sup (\{h_n(x) | n \in \mathbb{N}\}) \\ &\underset{\text{[eq: 21.593]}}{<} g_{n,\max(n,k)}(x) \\ &\underset{\text{[eq: 21.594]}}{\leq} h_{\max(n,k)}(x) \end{aligned}$$

which leads to the contradiction $h_{\max(n,k)}(x) < h_{\max(n,k)}(x)$. So the assumption is wrong and we have that $f(x) \leq \sup (\{h_n(x) | n \in \mathbb{N}\})$ which combined with [eq: 21.591] gives $f(x) = \sup (\{h_n(x) | n \in \mathbb{N}\})$. As $x \in X$ was chosen arbitrary it follows that

$$\forall x \in X \text{ we have } f(x) = \sup (\{h_n(x) | n \in \mathbb{N}\}) = \lim_{n \rightarrow \infty} h_n(x)$$

or

$$\lim_{n \rightarrow \infty} h_n = f \underset{\text{[eq: 21.580]}}{\in} \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.595)$$

Next

$$\begin{aligned} \int^+ f d\mu &\underset{\text{[eq: 21.595]}}{=} \int^+ \left(\lim_{n \rightarrow \infty} h_n(x) \right) d\mu \\ &\underset{\text{[theorem: 21.327] \wedge \text{[eqs: 21.588, 21.595]}}}{=} \lim_{n \rightarrow \infty} \int^S h_n d\mu \\ &\underset{\text{[theorem: 21.317]}}{=} \lim_{n \rightarrow \infty} \int^+ h_n d\mu \\ &\underset{\text{[eq: 21.589] \wedge \text{[theorems: 20.85, 21.331]}}}{\leq} \lim_{n \rightarrow \infty} \int^+ f_n d\mu \end{aligned}$$

which together with [eq: 21.582] proves that

$$\int^+ f d\mu = \lim_{n \rightarrow \infty} \int^+ f_n d\mu = \sup \left(\left\{ \int^+ f_n d\mu | n \in \mathbb{N} \right\} \right) \quad \square$$

If we weaken the requirements $\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x)$ and $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ to be hold almost everywhere we get another version of the Monotone Convergence Theorem.

Theorem 21.375. (Monotone Convergence Theorem (2)) *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$. If there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that*

$$\left\{ x \in X | \neg \left(\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x) \wedge \lim_{n \rightarrow \infty} f_n(x) = f(x) \right) \right\} \subseteq N$$

then we have

$$\int^+ f d\mu = \lim_{n \rightarrow \infty} \int^+ f_n d\mu$$

In other words if $\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x) \wedge \lim_{n \rightarrow \infty} f_n(x) = f(x)$ is μ -a.e. true then

$$\int^+ f d\mu = \lim_{n \rightarrow \infty} \int^+ f_n d\mu$$

Note 21.376. We do not require that f, f_n is μ -integrable, however as $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.352(2)] that $f \in \overline{\mathcal{L}[X, \mathcal{A}, \overline{\mathbb{R}}]}$ and $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{L}[X, \mathcal{A}, \overline{\mathbb{R}}]$ from which it follows that $\int f_n d\mu$ exist and $\int f_n d\mu = \int^+ f_n d\mu$. Hence we have also

$$\lim_{n \rightarrow \infty} \int f_n d\mu \text{ exists and } \int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$$

Proof. Let

$$A = \left\{ x \in X \mid \neg \left(\forall n \in \mathbb{N} \ f_n(x) \leq f_{n+1}(x) \wedge \lim_{n \rightarrow \infty} f_n(x) = f(x) \right) \right\}$$

then by the hypothesis there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$A \subseteq N \Rightarrow X \setminus N \subseteq X \setminus A$$

Now for $x \in X$ we have either:

$x \in N$. Then $\forall n \in \mathbb{N}$

$$(\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = 0 \cdot f_n(x) = 0 = 0 \cdot f_{n+1}(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot f_{n+1}(x) = (\mathcal{X}_{X, X \setminus N} \cdot f_{n+1})(x)$$

and

$$\begin{aligned} (\mathcal{X}_{X, X \setminus N} \cdot f)(x) &= 0 \cdot f(x) \\ &= 0 \\ &= \lim_{n \rightarrow \infty} (0 \cdot f_n(x)) \\ &= \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N}(x) \cdot f_n(x)) \\ &= \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \end{aligned}$$

$x \in X \setminus N$. Then

$$\forall n \in \mathbb{N} \ (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = 1 \cdot f_n(x) \underset{x \in X \setminus N \Rightarrow x \notin A}{\leq} f_{n+1}(x) = 1 \cdot f_{n+1}(x) = (\mathcal{X}_{X, X \setminus N} \cdot f_{n+1})(x)$$

and

$$\begin{aligned} (\mathcal{X}_{X, X \setminus N} \cdot f)(x) &= \mathcal{X}_{X, X \setminus N}(x) \cdot f(x) \\ &= f(x) \\ &\underset{x \in X \setminus N \Rightarrow x \notin A}{=} \lim_{n \rightarrow \infty} f_n(x) \\ &= \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N}(x) \cdot f_n(x)) \\ &= \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \end{aligned}$$

So we have $\forall x \in X$ that

$$\forall n \in \mathbb{N} \text{ we have } (\mathcal{X}_{X \setminus N} \cdot f_n)(x) \leq (\mathcal{X}_{X \setminus N} \cdot f_{n+1})(x) \text{ and } (\mathcal{X}_{X \setminus N} \cdot f)(x) = \lim_{n \rightarrow \infty} (\mathcal{X}_{X \setminus N} \cdot f_n)(x)$$

As $N \in \mathcal{A} \Rightarrow X \setminus N \in \mathcal{A}$ it follows from [theorem: 21.276] that $\mathcal{X}_{X \setminus N} \cdot f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\{\mathcal{X}_{X \setminus N} \cdot f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A}$. So the conditions for [theorem: 21.372] are satisfied and we have that

$$\lim_{n \rightarrow \infty} \int^+ (\mathcal{X}_{X \setminus N} \cdot f_n) d\mu \text{ exist and } \lim_{n \rightarrow \infty} \int^+ (\mathcal{X}_{X \setminus N} \cdot f_n) d\mu = \int^+ (\mathcal{X}_{X \setminus N} \cdot f) d\mu \quad (21.596)$$

Further using [theorem: 21.280] we have that

$$f = (\mathcal{X}_{X \setminus N} \cdot f) \text{ } \mu\text{-a.e. and } \forall n \in \mathbb{N} \text{ we have } f_n = (\mathcal{X}_{X \setminus N} \cdot f_n) \text{ } \mu\text{-a.e.}$$

Hence using [theorem: 21.333] we have that

$$\int^+ f d\mu \underset{[\text{theorem: 21.333}]}{=} \int^+ (\mathcal{X}_{X \setminus N} \cdot f) d\mu \underset{[\text{eq: 21.596}]}{=} \lim_{n \rightarrow \infty} \int^+ (\mathcal{X}_{X \setminus N} \cdot f_n) d\mu \underset{[\text{theorem: 21.333}]}{=} \lim_{n \rightarrow \infty} \int^+ f_n d\mu$$

proving that

$$\int^+ f d\mu = \lim_{n \rightarrow \infty} \int^+ f_n d\mu \quad \square$$

A consequence of the above theorem is the Beppo Levi's theorem.

Theorem 21.377. (Beppo Levi's Theorem) *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then we have that*

$$\sum_{i=1}^{\infty} f_i \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ \left(\sum_{i=1}^{\infty} f_i \right) d\mu = \sum_{i=1}^{\infty} \int^+ f_i d\mu$$

where

$$\sum_{i=1}^{\infty} f_i : X \rightarrow \overline{\mathbb{R}} \text{ is defined by } \left(\sum_{i=1}^{\infty} f_i \right)(x) = \sum_{i=1}^{\infty} f_i(x)$$

Note 21.378. *As $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have $\{f_i(x)\}_{i \in \mathbb{N}} \subseteq [0, \infty]$ so that by [definition: 20.95] $\sum_{i=1}^{\infty} f_i(x)$ is well defined, similar as $\{\int^+ f_i d\mu\}_{i \in \mathbb{N}} \subseteq [0, \infty]$ we have that $\sum_{i=1}^{\infty} \int^+ f_i d\mu$ is well defined.*

Note 21.379. *As $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\sum_{i=1}^{\infty} f_i \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.352(2)] that $\int (\sum_{i=1}^n f_i) d\mu$ exist, $\int (\sum_{i=1}^n f_i) d\mu = \int^+ (\sum_{i=1}^n f_i) d\mu \quad \forall n \in \mathbb{N}$ $\int f_n d\mu$ exist and $\int f_n d\mu = \int^+ f_n d\mu$. Hence we have also*

$$\int \left(\sum_{i=1}^{\infty} f_i \right) d\mu = \sum_{i=1}^{\infty} \int f_i d\mu$$

Proof. Let $n \in \mathbb{N}$ then by [theorem: 21.332]

$$\sum_{i=1}^n f_i \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ \left(\sum_{i=1}^n f_i \right) d\mu = \sum_{i=1}^n \left(\int^+ f_i d\mu \right) \quad (21.597)$$

so that

$$\left\{ \sum_{i=1}^n f_i \right\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Let $x \in X$ then by [definition: 20.95] $\lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x)$ exist and $\lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x) = \sum_{i=1}^{\infty} f_i(x) \in [0, \infty]$. Hence using [theorem: 21.269] we have that

$$\lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f_i \right) \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Using now the Monotone Convergence Theorem [theorem: 21.372] we have that

$$\lim_{n \rightarrow \infty} \int^+ \left(\sum_{i=1}^n f_i \right) d\mu \text{ exists and } \int^+ \left(\lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f_i \right) \right) d\mu = \lim_{n \rightarrow \infty} \int^+ \left(\sum_{i=1}^n f_i \right) d\mu \quad (21.598)$$

As $\left(\lim_{n \rightarrow \infty} (\sum_{i=1}^n f_i)\right)(x) = \lim_{n \rightarrow \infty} ((\sum_{i=1}^n f_i)(x)) = \lim_{n \rightarrow \infty} (\sum_{i=1}^n f_i(x)) \stackrel{\text{def}}{=} \sum_{i=1}^{\infty} f_i(x) \stackrel{\text{def}}{=} (\sum_{i=1}^{\infty} f_i)(x)$ we have

$$\lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f_i \right) = \sum_{i=1}^{\infty} f_i$$

Substituting this in [eq: 21.598] gives finally

$$\int^+ \left(\sum_{i=1}^{\infty} f_i \right) d\mu \stackrel{[\text{eq: 21.598}]}{=} \lim_{n \rightarrow \infty} \int^+ \left(\sum_{i=1}^n f_i \right) d\mu \stackrel{[\text{eq: 21.597}]}{=} \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \left(\int^+ f_i d\mu \right) \right) \stackrel{\text{def}}{=} \sum_{i=1}^{\infty} \left(\int^+ f_i d\mu \right) \quad \square$$

A application of the Beppo Levi's Theorem is a way of defining a new measure on a measure space based on a non negative measurable function.

Theorem 21.380. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then if we define*

$$\nu: \mathcal{A} \rightarrow [0, \infty] \text{ by } \nu(A) = \int_A^+ f d\mu$$

we have that ν is a measure on $\langle X, \mathcal{A} \rangle$. Further we have

$$\nu \text{ is finite} \Leftrightarrow f \text{ is } \mu\text{-integrable}$$

Proof. Let $x \in X$ then $(\mathcal{X}_{X, \emptyset} \cdot f)(x) = \mathcal{X}_{X, \emptyset}(x) \cdot f(x) = 0 \cdot f(x) = C_0$ which proves that

$$\nu(\emptyset) = \int_{\emptyset}^+ f d\mu = \int^+ (\mathcal{X}_{X, \emptyset} \cdot f) d\mu = \int^+ C_0 d\mu \stackrel{[\text{example: 21.318}]}{=} 0 \quad (21.599)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be a pairwise disjoint family. Define

$$\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ by } f_i = \mathcal{X}_{X, A_i} \cdot f \stackrel{[\text{theorem: 21.276}]}{\in} \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.600)$$

Let $A = \bigsqcup_{i \in \mathbb{N}} A_i$. Take $x \in X$ then for $f(x)$ we have the following cases to consider:

$f(x) = \infty$. then for x and $n \in \mathbb{N}$ we have either:

$x \in X \setminus A$. Then $(\mathcal{X}_{X, A} \cdot f)(x) = \mathcal{X}_{X, A}(x) \cdot f(x) = 0$ and

$$\left(\sum_{i=1}^n f_i \right)(x) = \sum_{i=1}^n (\mathcal{X}_{X, A_i}(x) \cdot f(x)) = \sum_{i=1}^n 0 = 0$$

so that

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x) = 0 = (\mathcal{X}_{X, A} \cdot f)(x)$$

$x \in A$. Then $(\mathcal{X}_{X, A} \cdot f)(x) = \mathcal{X}_{X, A}(x) \cdot f(x) = f(x) = \infty$ and

$$\left(\sum_{i=1}^n f_i \right)(x) = \sum_{i=1}^n (\mathcal{X}_{X, A_i}(x) \cdot f(x)) = \sum_{i=1}^n \infty = \infty$$

proving that

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x) = (\mathcal{X}_{X, A} \cdot f)(x)$$

$f(x) < \infty$. Then for x and $n \in \mathbb{N}$ we have either:

$x \in X \setminus A$. then $(\mathcal{X}_{X, A} \cdot f)(x) = \mathcal{X}_{X, A}(x) \cdot f(x) = 0$ and

$$\left(\sum_{i=1}^n f_i \right)(x) = \sum_{i=1}^n (\mathcal{X}_{X, A_i}(x) \cdot f(x)) = \sum_{i=1}^n 0 = 0$$

so that

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x) = 0 = (X_{X,A} \cdot f)(x)$$

$x \in A$. Let $\varepsilon > 0$, as $x \in A = \bigsqcup_{i \in \mathbb{N}} A_i$ there exists a **unique** $M \in \mathbb{N}_0$ such that $x \in A_M$ and $\forall i \in \mathbb{N} \setminus \{M\}$ $x \notin A_i$. If $n \in \{M, \dots, \infty\}$ then we have

$$\begin{aligned} & \left| (f \cdot \mathcal{X}_{X,A})(x) - \sum_{i=1}^n f_i(x) \right| = \\ & \left| f(x) \cdot \mathcal{X}_{X,A}(x) - \sum_{i=1}^n f_i(x) \right| \quad [\text{eq: 21.600}] \\ & \left| f(x) - \sum_{i=1}^n (\mathcal{X}_{X,A_i}(x) \cdot f(x)) \right| = \\ & \left| f(x) - \sum_{i \in \mathbb{N} \setminus \{M\}} (\mathcal{X}_{X,A_i}(x) \cdot f(x)) \right| = \\ & \quad |f(x) - \mathcal{X}_{X,A_N}(x) \cdot f(x)| = \\ & \quad |f(x) - f(x)| = \\ & \quad 0 < \\ & \quad \varepsilon \end{aligned}$$

proving that

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x) = (\mathcal{X}_{X,A} \cdot f)(x)$$

this proves that in all cases

$$\sum_{i=1}^{\infty} f_i(x) = (\mathcal{X}_{X,A} \cdot f)(x)$$

Applying then Beppo Levi [see 21.377] gives

$$\begin{aligned} \nu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &= \nu(A) \\ &= \int^+ (\mathcal{X}_A \cdot f) d\mu \\ &= \int^+ \left(\sum_{i=1}^{\infty} f_i\right) d\mu \\ & \stackrel{[\text{theorem: 21.377}]}{=} \sum_{i=1}^{\infty} \int^+ f_i d\mu \\ & \stackrel{[\text{eq: 21.600}]}{=} \sum_{i=1}^{\infty} \int^+ (\mathcal{X}_{A_i} \cdot f) d\mu \\ &= \sum_{i=1}^{\infty} \int_{A_i}^+ f d\mu \\ &= \sum_{i=1}^{\infty} \nu(A_i) \end{aligned}$$

which together with [eq: 21.599] proves that

$$\nu \text{ is a measure on the measurable space } \langle X, \mathcal{A} \rangle$$

For the last part of the theorem:

$\Rightarrow$. If ν is finite then as $f = \mathcal{X}_{X,X} \cdot f$ we have $\int^+ f d\mu = \int^+ (\mathcal{X}_{X,X} \cdot f) d\mu = \int_X f d\mu = \nu(X) < \infty$ which proves using [theorem: 21.352] that f is μ -integrable.

$\Leftarrow$. Let $A \in \mathcal{A}$ and $x \in X$ then $(f \cdot \mathcal{X}_{X,A})(x) = f(x) \cdot \mathcal{X}_A(x) = \begin{cases} f(x) & \text{if } x \in A \\ 0 \leq f(x) & \text{if } x \in X \setminus A \end{cases} \leq f(x)$. Hence using [theorem: 21.331] we have that $\nu(A) = \int_A^+ f d\mu = \int^+ (\mathcal{X}_A \cdot f) d\mu \leq \int^+ f d\mu < \infty$ which proves that ν is a finite measure. $\square$

Lemma 21.381. (Fatou's lemma) Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$\int^+ \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu \leq \liminf_{n \rightarrow \infty} \int^+ f_n d\mu$$

Note 21.382. Using [theorem: 21.269] we have that $\liminf_{n \rightarrow \infty} f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that $\int^+ \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu$ is well defined. Further using [theorem: 21.352] we have that $\int \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu$ with $\int \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu = \int^+ \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu$ and $\forall n \in \mathbb{N}$ $\int f_n d\mu$ exist with $\int f_n d\mu = \int^+ f_n d\mu$, hence

$$\int \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu$$

Note 21.383. We can not assume that $\lim_{n \rightarrow \infty} \int^+ f_n d\mu$ exists which is the reason $\liminf_{n \rightarrow \infty} \int^+ f_n d\mu$ is used instead.

Proof. Define

$$\{g_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ by } g_n(x) = \inf(\{f_i(x) | i \in \{n, \dots, \infty\}\}) \underset{[\text{theorem: 21.269}]}{\in} \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.601)$$

Let $x \in X$ and $n \in \mathbb{N}$ then $\{f_i(x) | i \in \{n+1, \dots, \infty\}\} \subseteq \{f_i(x) | i \in \{n, \dots, \infty\}\}$ so that by [theorem: 3.77] we have

$$\inf(\{f_i(x) | i \in \{n, \dots, \infty\}\}) \leq \inf(\{f_i(x) | i \in \{n+1, \dots, \infty\}\})$$

which as also

$$\inf(\{f_i(x) | i \in \{n, \dots, \infty\}\}) \leq f_n(x)$$

proves that

$$\forall n \in \mathbb{N} \quad g_n(x) \leq f_n(x) \text{ and } g_n(x) \leq g_{n+1}(x) \quad (21.602)$$

So we have by [theorem: 20.92] that $\lim_{n \rightarrow \infty} g_n(x)$ exist and

$$\begin{aligned} \lim_{n \rightarrow \infty} g_n(x) &\underset{[\text{theorem: 20.92}]}{=} \sup(\{g_n(x) | n \in \mathbb{N}\}) \\ &= \sup(\{\inf(\{f_i(x) | i \in \{n, \dots, \infty\}\}) | n \in \mathbb{N}\}) \\ &\underset{\text{def}}{=} \liminf_{n \rightarrow \infty} f_i(x) \end{aligned}$$

We can now apply the Monotone Convergence Theorem [see theorem: 21.372] giving

$$\begin{aligned} \int^+ \left(\liminf_{n \rightarrow \infty} f_i \right) d\mu &= \int^+ \left(\lim_{n \rightarrow \infty} g_n \right) d\mu \\ &\underset{[\text{theorem: 21.372}]}{=} \lim_{n \rightarrow \infty} \int^+ g_n d\mu \\ &\underset{\text{def}}{=} \liminf_{n \rightarrow \infty} \int^+ g_n d\mu \end{aligned} \quad (21.603)$$

Using [eq: 21.602] with [theorem: 21.331] we have $\forall n \in \mathbb{N}$ that $\int^+ g_n d\mu \leq \int^+ f_n d\mu$. So we have using [theorem: 20.62] that $\liminf_{n \rightarrow \infty} \int^+ g_n d\mu \leq \liminf_{n \rightarrow \infty} \int^+ f_n d\mu$. Substituting this in [eq: 21.603] gives finally

$$\int^+ \left(\liminf_{n \rightarrow \infty} f_i \right) d\mu \leq \liminf_{n \rightarrow \infty} \int^+ f_n d\mu \quad \square$$

Theorem 21.384. (Lebesgue's Dominated Convergence Theorem) Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ with $g(X) \subseteq [0, \infty]$, $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ such that

$$(f = \lim_{n \rightarrow \infty} f_n \text{ and } \forall n \in \mathbb{N} \text{ we have } |f_n| \leq g) \text{ holds } \mu\text{-a.e.}$$

In other words there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\left\{ x \in X \mid \neg \left(\lim_{n \rightarrow \infty} f_n(x) \text{ exists with } f(x) = \lim_{n \rightarrow \infty} f_n(x) \text{ and } \forall n \in \mathbb{N} \text{ we have } |f_n(x)| \leq g(x) \right) \right\}$$

then we have:

1. $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$
2. $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$
3. $\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$

Proof. Let $N_1 = \{x \in X \mid g(x) = \infty\}$ then it follows from [theorem: 21.370] that

$$N_1 \in \mathcal{A} \text{ and } \mu(N_1) = 0$$

By the hypothesis there exist a $N_2 \in \mathcal{A}$ with $\mu(N_2) = 0$ such that

$$\left\{ x \in X \mid \neg \left(\lim_{n \rightarrow \infty} f_n(x) \text{ exists with } f(x) = \lim_{n \rightarrow \infty} f_n(x) \text{ and } \forall n \in \mathbb{N} \text{ we have } |f_n(x)| \leq g(x) \right) \right\} \subseteq N_2$$

Take $N = N_1 \cup N_2$ then we have

$$N \in \mathcal{A} \wedge \mu(N) \leq \mu(N_1) + \mu(N_2) = 0 + 0 = 0 \quad (21.604)$$

Hence $\forall x \in X \setminus N = (X \setminus N_1) \cap (X \setminus N_2)$ we have

$$g(x) \neq \infty, \lim_{n \rightarrow \infty} f_n(x) \text{ exist with } f(x) = \lim_{n \rightarrow \infty} f_n(x) \text{ and } \forall n \in \mathbb{N} |f_n(x)| \leq g(x) \quad (21.605)$$

Using [theorem: 21.276] it follows that

$$\mathcal{X}_{X, X \setminus N} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}], \text{ and } \mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.606)$$

$$\forall n \in \mathbb{N} \mathcal{X}_{X, X \setminus N} \cdot f_n \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.607)$$

Using [theorem: 21.342] we have

$$|\mathcal{X}_{X, X \setminus N} \cdot f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \forall n \in \mathbb{N} |\mathcal{X}_{X, X \setminus N} \cdot f_n| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.608)$$

Further using [theorem: 21.280] together with the fact that $\mu(N) = 0$ we have that

$$g = \mathcal{X}_{X, \mathcal{A}} \cdot g \text{ } \mu\text{-a.e. and } f = \mathcal{X}_{X, X \setminus N} \cdot f \text{ } \mu\text{-a.e.} \quad (21.609)$$

$$\forall n \in \mathbb{N} f_n = \mathcal{X}_{X, X \setminus N} \cdot f_n \text{ } \mu\text{-a.e.} \quad (21.610)$$

Let $x \in X$ then we have either:

$x \in N$. Then we have $(\mathcal{X}_{X, X \setminus N} \cdot g)(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot g(x) = 0 \cdot g(x) = 0$ hence

$$0 \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) < \infty$$

Further $\forall n \in \mathbb{N} |\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) = |\mathcal{X}_{X, X \setminus N}(x) \cdot f_n(x)| = |0| = 0 \leq \mathcal{X}_{X, X \setminus N}(x) \cdot g(x)$ so that

$$\forall n \in \mathbb{N} \text{ we have } |\mathcal{X}_{X, X \setminus N} \cdot f_n| \leq \mathcal{X}_{X, X \setminus N}(x) \cdot g(x) < \infty$$

As $\forall n \in \mathbb{N} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot f_n(x) = 0$ and $(\mathcal{X}_{X, X \setminus N} \cdot f)(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot f(x) = 0$ it follows that

$$\lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \text{ exist and } \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = (\mathcal{X}_{X, X \setminus N} \cdot f)(x)$$

$x \in X \setminus N$. Then we have $(\mathcal{X}_{X, X \setminus N} \cdot g)(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot g(x) = g(x) \neq \infty$ so that [eq: 21.605]

$$0 \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) < \infty$$

Further $\forall n \in \mathbb{N}$ we have

$$|\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) = |\mathcal{X}_{X, X \setminus N}(x) \cdot f_n(x)| = |f_n(x)| \leq g(x) = (\mathcal{X}_{X, X \setminus N} \cdot g)(x) < \infty$$
[eq: 21.605]

As $\forall n \in \mathbb{N}$ $(\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = \mathcal{X}_{X, X \setminus N}(x) \cdot f_n(x) = f_n(x)$ and $(\mathcal{X}_{X, X \setminus N} \cdot f)(x) = f(x)$ it follows from [eq: 21.605] that

$$\lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \text{ exist and } \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = (\mathcal{X}_{X, X \setminus N} \cdot f)(x)$$

so we have that $\forall x \in X$

$$0 \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) < \infty \quad (21.611)$$

$$\forall n \in \mathbb{N} \text{ we have } |\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) < \infty \quad (21.612)$$

$$\lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \text{ exist and } \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) = (\mathcal{X}_{X, X \setminus N} \cdot f)(x) \quad (21.613)$$

Applying [corollary: 20.81] on the above [see eq: 21.613] gives

$$\forall x \in X \lim_{n \rightarrow \infty} |\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) \text{ exist and } \lim_{n \rightarrow \infty} |\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) = |\mathcal{X}_{X, X \setminus N} \cdot f|(x) \quad (21.614)$$

As $g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$, $g = \mathcal{X}_{X, X \setminus N} \cdot g$ μ -a.e. [see eq: 21.609] it follows from [theorem: 21.364] that

$$\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu = \int g d\mu \underset{g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]}{<} \infty \quad (21.615)$$

Next we prove that $\forall n \in \mathbb{N}$ $f_n \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $\int f_n d\mu = \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu$. Let $n \in \mathbb{N}$. As $\forall x \in X$ $|\mathcal{X}_{X, X \setminus N} \cdot f_n| \leq \mathcal{X}_{X, X \setminus N}(x) \cdot g(x)$ and $|\mathcal{X}_{X, X \setminus N} \cdot f_n| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$, $\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eqs: 21.606, 21.607] it follows from [theorem: 21.331]

$$\int^+ |\mathcal{X}_{X, X \setminus N} \cdot f_n| d\mu \leq \int^+ (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu \underset{[\text{eq: 21.615}][\text{theorem: 21.352}]}{=} \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu \underset{[\text{eq: 21.615}]}{<} \infty$$

As $\mathcal{X}_{X, X \setminus N} \cdot f_n \in \mathcal{M}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ [see: 21.608] it follows from the above by [theorem: 21.362] that

$$\mathcal{X}_{X, X \setminus N} \cdot f_n \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \quad (21.616)$$

As $f_n = \mathcal{X}_{X, X \setminus N} \cdot f_n$ μ -a.e. [see eq: 21.610] and $f_n \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.364] that $f_n \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ with $\int f_n d\mu = \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu$, in other words we have proved that

$$\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \forall n \in \mathbb{N} \int f_n d\mu = \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \quad (21.617)$$

Next we prove that $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$. Let $x \in X$ then we have

$$\begin{aligned} |\mathcal{X}_{X, X \setminus N} \cdot f|(x) & \underset{[\text{eq: 21.614}]}{=} \lim_{n \rightarrow \infty} |\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) \\ & \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \\ & \underset{[\text{theorem: 20.85}] \wedge [\text{eq: 21.612}]}{<} \infty \end{aligned}$$

Hence we have that

$$\forall x \in X \quad |\mathcal{X}_{X, X \setminus N} \cdot f|(x) \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) < \infty \quad (21.618)$$

Using then [theorem: 21.362] it follows that

$$\int^+ |\mathcal{X}_{X, X \setminus N} \cdot f| d\mu \leq \int^+ (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu \underset{[\text{eq: 21.615}]}{<} \infty$$

Hence as $\mathcal{X}_{X, X \setminus N} \cdot f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.606] it follows from [theorem: 21.362] that

$$\mathcal{X}_{X, X \setminus N} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \quad (21.619)$$

As $f = \mathcal{X}_{X, X \setminus N} \cdot f$ μ -a.e. [see eq: 21.609] it follows from [theorem: 21.364] that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int f d\mu = \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu \quad (21.620)$$

Next we prove that $\lim_{n \rightarrow \infty} \int f_n d\mu$ exist and that $\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$. Now by [eqs: 21.611, 21.618, 21.612] we have

$$\forall x \in X \quad (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \in \mathbb{R} \wedge (\mathcal{X}_{X, X \setminus N} \cdot f)(x) \in \mathbb{R} \wedge \forall n \in \mathbb{N} \quad (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \in \mathbb{R} \quad (21.621)$$

so that all the sums and the differences of the above are well defined. Let $x \in X$ and $n \in \mathbb{N}$ then as $|\mathcal{X}_{X, X \setminus N} \cdot f_n|(x) \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x)$ [see eq: 21.612] it follows that

$$\begin{aligned} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) &\leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \\ -(\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) &\leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \end{aligned}$$

which, using the fact that the terms are real [see eq: 21.620] gives

$$0 \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) - (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \text{ and } 0 \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) + (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x)$$

As $\mathcal{X}_{X, X \setminus N} \cdot g, \mathcal{X}_{X, X \setminus N} \cdot f_n \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eqs: 21.606, 21.607] it follows from [theorems: 21.257, 21.265] that

$$\{\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.622)$$

$$\{\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.623)$$

Using the above we have

$$\begin{aligned} (\mathcal{X}_{X, X \setminus N} \cdot g)(x) - (\mathcal{X}_{X, X \setminus N} \cdot f)(x) &\stackrel{[\text{eq: 21.613}]}{=} (\mathcal{X}_{X, X \setminus N} \cdot g)(x) - \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ &\stackrel{[\text{example: 20.74}]}{=} \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g)(x) - \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} ((\mathcal{X}_{X, X \setminus N} \cdot g)(x) - (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x)) \\ &= \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ &\stackrel{\text{def limit}}{=} \liminf_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ (\mathcal{X}_{X, X \setminus N} \cdot g)(x) + (\mathcal{X}_{X, X \setminus N} \cdot f)(x) &\stackrel{[\text{eq: 21.613}]}{=} (\mathcal{X}_{X, X \setminus N} \cdot g)(x) + \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ &\stackrel{[\text{example: 20.74}]}{=} \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g)(x) + \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} ((\mathcal{X}_{X, X \setminus N} \cdot g)(x) + (\mathcal{X}_{X, X \setminus N} \cdot f_n)(x)) \\ &= \lim_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \\ &\stackrel{\text{def limit}}{=} \liminf_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f_n)(x) \end{aligned}$$

so that

$$\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f = \liminf_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f_n) \quad (21.624)$$

$$\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f = \liminf_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f_n) \quad (21.625)$$

As $\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$, $\mathcal{X}_{X, X \setminus N} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ and $\forall n \in \mathbb{N} \quad \mathcal{X}_{X, X \setminus N} \cdot f_n \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ [see eqs: 21.615, 21.619, 21.616] it follows from [eq: 21.621] that

$$\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \mathcal{X}_{X, X \setminus N} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.626)$$

$$\forall n \in \mathbb{N} \text{ we have } \mathcal{X}_{X, X \setminus N} \cdot f_n \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.627)$$

Hence we have

$$\begin{aligned}
& \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu - \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu && [\text{theorem: 21.360}] \wedge [\text{eq: 21.626}] \\
& \int (\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f) d\mu && [\text{eq: 21.624}] \\
& \int \liminf_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu && \leq \\
& \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot g - \mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu && [\text{theorem: 21.381}] \wedge [\text{eq: 21.622}] \\
& \liminf_{n \rightarrow \infty} \left(\int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu + \left(- \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \right) \right) && [\text{theorem: 20.65}] \\
& \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu + \liminf_{n \rightarrow \infty} \left(- \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \right) && [\text{theorem: 20.67}] \\
& \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu - \limsup_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu
\end{aligned}$$

which as the measure integrals are real numbers proves that

$$\limsup_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \leq \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu \quad (21.628)$$

Like wise we have

$$\begin{aligned}
& \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu + \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu && [\text{theorem: 21.360}] \wedge [\text{eq: 21.626}] \\
& \int (\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f) d\mu && [\text{eq: 21.625}] \\
& \int \liminf_{n \rightarrow \infty} (\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu && \leq \\
& \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot g + \mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu && [\text{theorem: 21.381}] \wedge [\text{eq: 21.623}] \\
& \liminf_{n \rightarrow \infty} \left(\int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu + \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \right) && [\text{theorem: 20.65}] \\
& \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu + \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu && = \\
& \int (\mathcal{X}_{X, X \setminus N} \cdot g) d\mu + \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu
\end{aligned}$$

which as the measure integrals are real numbers proves that

$$\int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu \leq \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \quad (21.629)$$

Hence we have

$$\begin{aligned}
& \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu && \leq && \liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \\
& && [\text{eq: 21.629}] && \\
& && \leq && \limsup_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu \\
& && [\text{theorem: 20.70}] && \\
& && \leq && \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu \\
& && [\text{eq: 21.628}] &&
\end{aligned}$$

so that

$$\liminf_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu = \limsup_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu = \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu$$

hence we have that

$$\lim_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f_n) d\mu \text{ exist and } \lim_{n \rightarrow \infty} \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu = \int (\mathcal{X}_{X, X \setminus N} \cdot f) d\mu$$

Combining this with [eqs: 21.617, 21.620] it follows that

$$\lim_{n \rightarrow \infty} \int f_n d\mu \text{ exist and } \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

which together with [eq: 21.617, 21.620] finally proves the theorem. $\square$

21.5.5 Complex integration

If we want to develop a theory of integration of complex valued functions we need first to replace the measure space $\langle \mathbb{R}, \mathcal{B}[\mathbb{R}] \rangle$ with a measure space based on $\mathbb{C}$ instead of $\mathbb{R}$. Luckily we have by [examples: 14.102, 14.103, 14.123] two equivalent norms

$$|\cdot|: \mathbb{C} \rightarrow \mathbb{R} \text{ defined by } |z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$$

$$\|\cdot\|_{\mathbb{C}}: \mathbb{C} \rightarrow \mathbb{R} \text{ defined by } \|z\|_{\mathbb{C}} = \max(|\operatorname{Re}(z)|, |\operatorname{Im}(z)|)$$

that defines the canonical topology on $\mathbb{C}$ which we note as $\mathcal{T}_{\mathbb{C}}$, hence $\mathcal{T}_{\mathbb{C}} = \mathcal{T}_{|\cdot|} = \mathcal{T}_{\|\cdot\|_{\mathbb{C}}}$.

Definition 21.385. The measurable space $\langle \mathbb{C}, \mathcal{B}[\mathbb{C}] \rangle$ is defined by

$$\mathcal{B}[\mathbb{C}] = \sigma[\mathbb{C}, \mathcal{T}_{\mathbb{C}}] = \sigma[\mathbb{C}, \mathcal{T}_{|\cdot|}] = \sigma[\mathbb{C}, \mathcal{T}_{\|\cdot\|_{\mathbb{C}}}]$$

To be able to express the measurability of a complex valued function in the measurability of the real and imaginary part of the function we find an alternative definition for $\sigma[\mathbb{C}]$. In essence we must prove that

$$\mathcal{B}[\mathbb{C}] = \sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y, \delta, \rho \in \mathbb{R}\}]$$

which will allow us to use [theorem: 21.229] to prove the measurability of a complex valued function.

Definition 21.386. Let $A, B \subseteq \mathbb{R}$ then $A + i \cdot B = \{x + i \cdot y | x \in A \wedge y \in B\}$

Note 21.387. Using [theorem: 10.77] and [definition: 10.79] it follows that

$$z \in A + i \cdot B \Leftrightarrow \operatorname{Re}(z) \in A \text{ and } \operatorname{Im}(z) \in B$$

Lemma 21.388. Let U, V be open sets in $\mathcal{T}_{||}$ the canonical topology on $\mathbb{R}$ then $U + i \cdot V$ is an open set in $\mathcal{T}_{\mathbb{C}}$.

Proof. Let $z \in U + i \cdot V$ then $\operatorname{Re}(z) \in U$ and $\operatorname{Im}(z) \in V$. As U and V are open there exists a $\delta_x, \delta_y \in \mathbb{R}^+$ such that $\operatorname{Re}(z) \in B_{||}(\operatorname{Re}(z), \delta_x) \subseteq U$ and $\operatorname{Im}(z) \in B_{||}(\operatorname{Im}(z), \delta_y) \subseteq V$. Let $\delta = \min(\delta_x, \delta_y)$ then for $w \in B_{||\mathbb{C}}(z, \delta)$ we have $\|z - w\|_{\mathbb{C}} < \delta$ hence we have

$$|\operatorname{Re}(z) - \operatorname{Re}(w)| = |\operatorname{Re}(z - w)| \leq \max(|\operatorname{Re}(z - w)|, |\operatorname{Im}(z - w)|) < \delta \leq \delta_1$$

$$|\operatorname{Im}(z) - \operatorname{Im}(w)| = |\operatorname{Im}(z - w)| \leq \max(|\operatorname{Re}(z - w)|, |\operatorname{Im}(z - w)|) < \delta \leq \delta_2$$

so that $\operatorname{Re}(w) \in B_{||}(\operatorname{Re}(z), \delta_1) \subseteq U$ and $\operatorname{Im}(w) \in B_{||}(\operatorname{Im}(z), \delta_2) \subseteq V$, hence

$$w = \operatorname{Re}(w) + i \cdot \operatorname{Im}(w) \in U + i \cdot V$$

From this it follows that

$$w \in B_{||\mathbb{C}}(w, \delta) \subseteq U + i \cdot V$$

which proves that $U + i \cdot V$ is open. $\square$

Lemma 21.389. Let $U \in \mathcal{T}_{\mathbb{C}}$ be an open set in $\mathcal{T}_{\mathbb{C}}$ and $z \in U$ then there exist $q_1, q_2 \in \mathbb{Q}$, $\delta_1, \delta_2 \subseteq \mathbb{Q}^+$ such that

$$z \in B_{||}(q_1, \delta_1) + i \cdot B_{||}(q_2, \delta_2) \subseteq U$$

Proof. As U is open and $z \in U$ there exist a $\delta' \in \mathbb{R}^+$ such that

$$z \in B_{||\cdot||_{\mathbb{C}}}(z, \delta') \subseteq U \quad (21.630)$$

Take

$$\delta = \frac{\delta'}{2} \quad (21.631)$$

As $\operatorname{Re}(z) - \delta < \operatorname{Re}(z)$ we have by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] that there exist a $q_1 \in \mathbb{Q}$ such that

$$\operatorname{Re}(z) - \delta < q_1 < \operatorname{Re}(z) \quad (21.632)$$

hence we have $\operatorname{Re}(z) - q_1 < \delta$ and using [theorem: 10.32] again we find a $\delta_1 \in \mathbb{Q}$ such that

$$\operatorname{Re}(z) - q_1 < \delta_1 < \delta \quad (21.633)$$

As by [eq: 21.632] $0 < \operatorname{Re}(z) - q_1$ it follows from the above that

$$\delta_1 \in \mathbb{Q}^+$$

Further we have $q_1 - \operatorname{Re}(z) \underset{[\text{eq: 21.632}]}{<} 0 < \delta_1$ and $\operatorname{Re}(z) - q_1 \underset{[\text{eq: 21.632}]}{<} \delta_1$ which proves that $|q_1 - \operatorname{Re}(z)| < \delta_1$ or that

$$\operatorname{Re}(z) \in B_{||\cdot||}(q_1, \delta_1) \quad (21.634)$$

Likewise, as $\operatorname{Im}(z) - \delta < \operatorname{Im}(z)$ we have by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] that there exist a $q_2 \in \mathbb{Q}$ such that

$$\operatorname{Im}(z) - \delta < q_2 < \operatorname{Im}(z) \quad (21.635)$$

hence we have $\operatorname{Im}(z) - q_2 < \delta$ and using [theorem: 10.32] again we find a $\delta_2 \in \mathbb{Q}$ such that

$$\operatorname{Im}(z) - q_2 < \delta_2 < \delta \quad (21.636)$$

As by [eq: 21.635] $0 < \operatorname{Im}(z) - q_2$ it follows from the above that

$$\delta_2 \in \mathbb{Q}^+$$

Further we have $q_2 - \operatorname{Im}(z) \underset{[\text{eq: 21.635}]}{<} 0 < \delta_2$ and $\operatorname{Im}(z) - q_2 \underset{[\text{eq: 21.636}]}{<} \delta_2$ which proves that $|q_2 - \operatorname{Im}(z)| < \delta_2$ or that

$$\operatorname{Im}(z) \in B_{||\cdot||}(q_2, \delta_2) \quad (21.637)$$

Now as $\operatorname{Re}(z) \in B_{||\cdot||}(q_1, \delta_1) \wedge \operatorname{Im}(z) \in B_{||\cdot||}(q_2, \delta_2)$ [see eqs: 21.634, 21.637] it follows that

$$z \in B_{||\cdot||}(q_1, \delta_1) + i \cdot B_{||\cdot||}(q_2, \delta_2)$$

Further if $w \in B_{||\cdot||}(q_1, \delta_1) + i \cdot B_{||\cdot||}(q_2, \delta_2)$ then $\operatorname{Re}(w) \in B_{||\cdot||}(q_1, \delta_1)$ and $\operatorname{Im}(w) \in B_{||\cdot||}(q_2, \delta_2)$ so that

$$|\operatorname{Re}(z) - \operatorname{Re}(w)| \leq |\operatorname{Re}(z) - q_1| + |q_1 - \operatorname{Re}(w)| < \delta_1 + \delta_1 = 2 \cdot \delta_1 \underset{[\text{eq: 21.633}]}{<} 2 \cdot \delta \underset{[\text{eq: 21.631}]}{=} \delta'$$

and

$$|\operatorname{Im}(z) - \operatorname{Im}(w)| \leq |\operatorname{Im}(z) - q_2| + |q_2 - \operatorname{Im}(w)| < \delta_2 + \delta_2 = 2 \cdot \delta_2 \underset{[\text{eq: 21.633}]}{<} 2 \cdot \delta \underset{[\text{eq: 21.631}]}{=} \delta'$$

hence we have

$$\|z - w\|_{\mathbb{C}} = \max(|\operatorname{Re}(z - w)|, |\operatorname{Im}(z - w)|) = \max(|\operatorname{Re}(z) - \operatorname{Re}(w)|, |\operatorname{Im}(z) - \operatorname{Im}(w)|) < \delta'$$

which proves that $w \in B_{||\cdot||_{\mathbb{C}}}(z, \delta') \underset{[\text{eq: 21.630}]}{\subseteq} U$. Hence we have proved that

$$z \in B_{||\cdot||}(q_1, \delta_1) + i \cdot B_{||\cdot||}(q_2, \delta_2) \subseteq W \quad \square$$

Lemma 21.390. Let $U \in \mathcal{T}_{\mathbb{C}}$ be a open set in $\mathcal{T}_{\mathbb{C}}$ then there exist

$$\{B_{||\cdot||}(q_1, q_2) + i \cdot B_{||\cdot||}(q_3, q_4)\}_{q \in I}$$

where I is countable such that $\forall i \in I, q_1, q_3 \in \mathbb{Q}, q_2, q_4 \in \mathbb{Q}^+$ and

$$\bigcup_{q \in I} (B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)) = U$$

Proof. Define

$$I = \{q \in \mathbb{Q}^4 | (B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)) \subseteq U \wedge q_1, q_3 \in \mathbb{Q} \wedge q_2, q_4 \in \mathbb{Q}^+\} \subseteq \mathbb{Q}^4$$

As $\mathbb{Q}$ is denumerable [see theorem: 10.5(7)] it follows from [theorem: 6.67] that $\mathbb{Q}^4$ is denumerable and, as $I \subseteq \mathbb{Q}^4$, we have by [theorem: 6.60] that

I is countable

As $\forall q \in I$ we have that $B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4) \subseteq U$ it follows that

$$\bigcup_{q \in I} (B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)) \subseteq U \quad (21.638)$$

Let $z \in U$ then by [lemma: 21.389] there exists $q = (q_1, q_2, q_3, q_4) \in \mathbb{Q}^4$ such that

$$z \in B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4) \subseteq U \wedge q_2, q_4 \in \mathbb{Q}^+$$

Hence we have $z \in \bigcup_{q \in I} B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)$ or $U \subseteq \bigcup_{q \in I} B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)$ which combined with [eq: 21.638] proves that

$$\bigcup_{q \in I} (B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)) = U \quad \square$$

Theorem 21.391. Let $\mathbb{C}$ be the complex space then

$$\mathcal{B}[\mathbb{C}] = \sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}]$$

Proof. Let $A \in \{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}$ then there exist $x, y \in \mathbb{R}$ and $\delta, \rho \in \mathbb{R}^+$ such that

$$A = B_{||}(x, \delta) + i \cdot B_{||}(y, \rho)$$

As $B_{||}(x, \delta), B_{||}(y, \rho)$ are open sets in $\mathbb{R}$ it follows from [lemma: 21.388] that A is a open set in $\mathbb{C}$, hence

$$\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\} \subseteq \mathcal{T}_{\mathbb{C}} \subseteq \sigma[\mathcal{T}_{\mathbb{C}}]$$

So that by [theorem: 21.42]

$$\sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}] \subseteq \sigma[\mathcal{T}_{\mathbb{C}}] \quad (21.639)$$

On the other hand if $U \in \mathcal{T}_{\mathbb{C}}$ then by [lemma: 21.389] there exist a countable family

$$\{B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)\}_{q \in I}$$

such that

$$U = \bigcup_{q \in I} B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4)$$

As I is countable it follows from [theorem: 21.39]

$$U = \bigcup_{q \in I} B_{||}(q_1, q_2) + i \cdot B_{||}(q_3, q_4) \in \sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}]$$

which proves that

$$\mathcal{T}_{\mathbb{C}} \subseteq \sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y, \delta, \rho \in \mathbb{R}\}]$$

Applying [theorem: 21.42] again proves then that

$$\sigma[\mathcal{T}_{\mathbb{C}}] \subseteq \sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}]$$

which combined with [eq: 21.639] proves that

$$\mathcal{B}[\mathbb{C}] \stackrel{\text{def}}{=} \sigma[\mathcal{T}_{\mathbb{C}}] = \sigma[\{B_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}] \quad \square$$

21.5.5.1 Measurability of complex valued functions

Note how we define the integral for a function $f: X \rightarrow \overline{\mathbb{R}}$ based on the integral $\int^+$ of non negative functions. First we write f as $f = f^+ - f^-$ where f^+, f^- are non negative functions and define then $\int f d\mu$ by $\int f d\mu = \int^+ f^+ d\mu - \int^+ f^- d\mu$. We use the same strategy for complex valued functions $f: X \rightarrow \mathbb{C}$. Write f as $f = \text{Re}(f) + i \cdot \text{Im}(f)$ and define then $\int^{\mathbb{C}} f d\mu = \int \text{Re}(f) + i \cdot \int \text{Im}(f) d\mu$. We start with defining $\text{Re}(f)$ and $\text{Im}(f)$.

Definition 21.392. Let X be a set and $f: X \rightarrow \mathbb{C}$ then we define

$$\text{Re}(f): X \rightarrow \overline{\mathbb{R}} \text{ by } \text{Re}(f)(x) = \text{Re}(f(x))$$

$$\text{Im}(f): X \rightarrow \overline{\mathbb{R}} \text{ by } \text{Im}(f)(x) = \text{Im}(f(x))$$

Note 21.393. As $\forall x \in X \text{Re}(f)(x) = \text{Re}(f(x)) \in \mathbb{R}$ and $\text{Im}(f)(x) = \text{Im}(f(x)) \in \mathbb{R}$ it follows that

$$\text{Re}(f)(X) \subseteq \mathbb{R} \text{ and } \text{Im}(f)(X) \subseteq \mathbb{R}$$

As in the previous section many theorems are based on almost everywhere equality [typically to get rid of having to deal with infinite values or problems with the existence of limit]. So we extend the concept of μ -a.e. equality to complex valued functions.

Definition 21.394. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \subseteq X$ and $f: A \rightarrow \mathbb{C}, g: A \rightarrow \mathbb{C}$ be functions then we say that $f = g$ μ -a.e. if there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that

$$\{x \in A | f(x) \neq g(x)\} \subseteq N$$

The following theorem shows a simple way to test if two complex valued functions are almost everywhere equal.

Theorem 21.395. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $f: X \rightarrow \mathbb{C}$ and $g: X \rightarrow \mathbb{C}$ then

$$f = g \text{ } \mu\text{-a.e.} \Leftrightarrow \text{Re}(f) = \text{Re}(g) \text{ } \mu\text{-a.e. and } \text{Im}(f) = \text{Im}(g) \text{ } \mu\text{-a.e.}$$

Proof.

$\Rightarrow$. As $f = g$ μ -a.e. there exists a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in A | f(x) \neq g(x)\} \subseteq N$. Assume that $x \in \{x \in A | \text{Re}(f)(x) \neq \text{Re}(g)(x)\}$ then $\text{Re}(f(x)) \neq \text{Re}(g(x)) \Rightarrow f(x) \neq g(x)$ proving that $\{x \in A | \text{Re}(f)(x) \neq \text{Re}(g)(x)\} \subseteq \{x \in A | f(x) \neq g(x)\} \subseteq N$. So we have that

$$\text{Re}(f) = \text{Re}(g) \text{ } \mu\text{-a.e.}$$

If $x \in \{x \in A | \text{Im}(f)(x) \neq \text{Im}(g)(x)\}$ then $\text{Im}(f(x)) \neq \text{Im}(g(x)) \Rightarrow f(x) \neq g(x)$ proving that $\{x \in A | \text{Im}(f)(x) \neq \text{Im}(g)(x)\} \subseteq \{x \in A | f(x) \neq g(x)\} \subseteq N$. Hence

$$\text{Im}(f) = \text{Im}(g) \text{ } \mu\text{-a.e.}$$

$\Leftarrow$. As $\text{Re}(f) = \text{Re}(g)$ μ -a.e. and $\text{Im}(f) = \text{Im}(g)$ μ -a.e. there exists $N_1, N_2 \in \mathcal{A}$ such that $\mu(N_1) = 0 = \mu(N_2)$, $\{x \in A | \text{Re}(f)(x) \neq \text{Re}(g)(x)\} \subseteq N_1$ and $\{x \in A | \text{Im}(f)(x) \neq \text{Im}(g)(x)\} \subseteq N_2$. Let $x \in \{x \in A | f(x) \neq g(x)\}$ then $f(x) \neq g(x)$ so that either $\text{Re}(f(x)) \neq \text{Re}(g(x))$ or $\text{Im}(f(x)) \neq \text{Im}(g(x))$, hence

$$x \in \{x \in A | \text{Re}(f)(x) \neq \text{Re}(g)(x)\} \cup \{x \in A | \text{Im}(f)(x) \neq \text{Im}(g)(x)\} \subseteq N_1 \cup N_2$$

proving that $\{x \in A | f(x) \neq g(x)\} \subseteq N_1 \cup N_2$. As $\mu(N_1 \cup N_2) \leq \mu(N_1) + \mu(N_2) = 0$ it follows that

$$f = g \text{ } \mu\text{-a.e.} \quad \square$$

Once we have defined the measurable space we can define the set of all measurable functions to $\mathbb{C}$.

Definition 21.396. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$ then the set of all $\mathcal{A}, \mathcal{B}[\mathbb{C}]$ -measurable functions from A to $\mathbb{C}$ is noted as $\mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ so

$$\mathcal{M}[A, \mathcal{A}, \mathbb{C}] = \{f \in \mathbb{C}^A \mid f: A \rightarrow \mathbb{C} \text{ is } \mathcal{A}, \mathcal{B}[\mathbb{C}]\text{-measurable}\}$$

Note 21.397. Sometimes we want to state things that are true for $\mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and $\mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, in those cases we use the notation $\mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ that stands for either $\mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ or $\mathcal{M}[X, \mathcal{A}, \mathbb{C}]$.

Theorem 21.398. We have that

$$\text{Re}: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}$$

$$\text{Img}: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}$$

$$||: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}$$

or, as clearly $\text{Re}(\mathbb{C}) \subseteq \mathbb{R}$, $\text{Img}(\mathbb{C}) \subseteq \mathbb{R}$ and $||(\mathbb{C}) \subseteq [0, \infty[\subseteq \mathbb{R}$, we have that

$$\text{Re}, \text{Img} \in \mathcal{M}[\mathbb{C}, \mathcal{B}[\mathbb{C}], \mathbb{R}] \text{ and } || \in \mathcal{M}^+[\mathbb{C}, \mathcal{B}[\mathbb{C}], \mathbb{R}]$$

Proof. Using [example: 14.139] it follows that $\text{Re}: \mathbb{C} \rightarrow \mathbb{R}$ and $\text{Img}: \mathbb{C} \rightarrow \mathbb{R}$ are continuous, hence by [theorem: 21.255] it follows that $\text{Re}: \mathbb{C} \rightarrow \mathbb{R}$ and $\text{Img}: \mathbb{C} \rightarrow \mathbb{R}$ are $\mathcal{B}[\mathbb{C}], \mathcal{B}[\mathbb{R}]$ -measurable. Using further [theorem: 21.243] it follows that

$$\text{Re}: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}$$

$$\text{Img}: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}$$

As $||$ is a norm on $\mathbb{C}$ it follows from [theorem: 14.177] that $||: \mathbb{C} \rightarrow \mathbb{R}$ is continuous, hence by [theorem: 21.255] it follows that $||: \mathbb{C} \rightarrow \mathbb{R}$ is $\mathcal{B}[\mathbb{C}], \mathcal{B}[\mathbb{R}]$ -measurable. Using further [theorem: 21.243] it follows that

$$||: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable} \quad \square$$

The following equivalent definition will be crucial in proving the properties of complex valued measurable functions because it allows us to express measurability of f as the measurability of $\text{Re}(f)$ and $\text{Img}(f)$.

Theorem 21.399. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$ and $f: A \rightarrow \mathbb{C}$ a function then we have

$$f \text{ is } \mathcal{A}, \mathcal{B}[\mathbb{C}]\text{-measurable} \Leftrightarrow \text{Re}(f) \text{ is } \mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable and } \text{Img}(f) \text{ is } \mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable}$$

or in other words, as $\text{Re}(f)(X) \subseteq \mathbb{R}$, $\text{Img}(f)(X) \subseteq \mathbb{R}$,

$$f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \Leftrightarrow \text{Re}(f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \text{ and } \text{Img}(f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

Proof.

$\Rightarrow$. Let $x \in X$ then we have

$$\text{Re}(f)(x) = \text{Re}(f(x)) = (\text{Re} \circ f)(x) \text{ and } \text{Img}(f)(x) = \text{Img}(f(x)) = (\text{Img} \circ f)(x)$$

which proves that

$$\text{Re}(f) = \text{Re} \circ f \text{ and } \text{Img}(f) = \text{Img} \circ f$$

As f is $\mathcal{A}, \mathcal{B}[\mathbb{C}]$ -measurable and by [theorem: 21.398] Re, Img are $\mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable it follows from [theorem: 21.230] that $\text{Re}(f)$ and $\text{Img}(f)$ are $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable.

$\Leftarrow$. Let $A \in \{\mathcal{B}_{||}(x, \delta) + i \cdot \mathcal{B}_{||}(y, \rho) \mid x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}$ then there exist $x, y \in \mathbb{R}$ and $\delta, \rho \in \mathbb{R}^+$ such that

$$A = \mathcal{B}_{||}(x, \delta) + i \cdot \mathcal{B}_{||}(y, \rho)$$

Let $z \in f^{-1}(A)$ then as $f(z) \in A$ and

$$f(z) = \operatorname{Re}(f(z)) + i \cdot \operatorname{Im}(f(z)) = \operatorname{Re}(f)(z) + i \cdot \operatorname{Im}(f)(z)$$

it follows that $\operatorname{Re}(f)(z) \in B_{||}(x, \delta)$ and $\operatorname{Im}(f)(z) \in B_{||}(y, \rho)$. Hence $z \in (\operatorname{Re}(f))^{-1}(B_{||}(x, \delta))$ and $z \in (\operatorname{Im}(f))^{-1}(B_{||}(y, \rho))$ from which it follows that

$$f^{-1}(A) \subseteq (\operatorname{Re}(f))^{-1}(B_{||}(x, \delta)) \cap (\operatorname{Im}(f))^{-1}(B_{||}(y, \rho)) \quad (21.640)$$

Further if $z \in (\operatorname{Re}(f))^{-1}(B_{||}(x, \delta)) \cap (\operatorname{Im}(f))^{-1}(B_{||}(y, \rho))$ then

$$\operatorname{Re}(f(z)) = \operatorname{Re}(f)(z) \in B_{||}(x, \delta) \text{ and } \operatorname{Im}(f(z)) = \operatorname{Im}(f)(z) \in B_{||}(y, \rho)$$

so that

$$f(z) = \operatorname{Re}(f(z)) + i \cdot \operatorname{Im}(f(z)) \in B_{||}(x, \delta) + i \cdot B_{||}(y, \rho)$$

or $z \in f^{-1}(B_{||}(x, \delta) + i \cdot B_{||}(y, \rho)) = f^{-1}(A)$. Hence

$$(\operatorname{Re}(f))^{-1}(B_{||}(x, \delta)) \cap (\operatorname{Im}(f))^{-1}(B_{||}(y, \rho)) \subseteq f^{-1}(A)$$

Combining this with [eq: 21.640] proves that

$$f^{-1}(A) = (\operatorname{Re}(f))^{-1}(B_{||}(x, \delta)) \cap (\operatorname{Im}(f))^{-1}(B_{||}(y, \rho))$$

As $B_{||}(x, \delta)$, $B_{||}(y, \rho)$ are open sets in $\mathbb{R}$ it follows that

$$B_{||}(x, \delta), B_{||}(y, \rho) \in \mathcal{T}_{||} \subseteq_{[\text{theorem: 20.51}]} \mathcal{R}_{\mathbb{R}} \subseteq \mathcal{B}[\mathbb{R}]$$

so that, as $\operatorname{Re}(f), \operatorname{Im}(f)$ are $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ measurable, $(\operatorname{Re}(f))^{-1}(B_{||}(x, \delta)) \in \mathcal{A}$ and $(\operatorname{Im}(f))^{-1}(B_{||}(y, \rho)) \in \mathcal{A}$. Hence we have that $f^{-1}(A) \in \mathcal{A}$. To summarize we have proved that

$$\forall A \in \{\mathcal{B}_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\} \text{ we have } A \in \mathcal{A}$$

Finally, as $\mathcal{B}[\mathbb{C}] =_{[\text{theorem: 21.391}]} \sigma[\{\mathcal{B}_{||}(x, \delta) + i \cdot B_{||}(y, \rho) | x, y \in \mathbb{R} \wedge \delta, \rho \in \mathbb{R}^+\}]$, it follows from [theorem: 21.229] that

$$f \text{ is } \mathcal{A}, \mathcal{B}[\mathbb{C}]\text{-measurable} \quad \square$$

We have now similar properties for $\mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ as for $\mathcal{M}[X, \mathcal{A}]$ that can be easily proved using the previous theorem. First we define some pointwise definitions:

Definition 21.400. Let X be a set then we define

1. If $\alpha \in \mathbb{C}$ and $f: X \rightarrow \mathbb{C}$ a function then $\alpha \cdot f: X \rightarrow \mathbb{C}$ is defined by $(\alpha \cdot f)(x) = \alpha \cdot f(x)$
2. If $f: X \rightarrow \mathbb{C}$ is a function then $|f|: X \rightarrow \mathbb{R}$ is defined by $|f|(x) = |f(x)|$.
3. If $f: X \rightarrow \mathbb{C}$ and $g: X \rightarrow \mathbb{C}$ are functions then we define:
 - a. $f + g: X \rightarrow \mathbb{C}$ by $(f + g)(x) = f(x) + g(x)$
 - b. $f \cdot g: X \rightarrow \mathbb{C}$ by $(f \cdot g)(x) = f(x) \cdot g(x)$
4. If $n \in \mathbb{N}$, $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{C}^X$ then $\sum_{i=1}^n f_i: X \rightarrow \mathbb{C}$ is defined by $(\sum_{i=1}^n f_i)(x) = \sum_{i=1}^n f_i(x)$
5. If $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathbb{C}^X$ then

$$\lim_{i \rightarrow \infty} f_i: \left\{x \in X \mid \lim_{i \rightarrow \infty} f_i(x) \text{ exist}\right\} \rightarrow \mathbb{C} \text{ is defined by } \left(\lim_{i \rightarrow \infty} f_i\right)(x) = \lim_{i \rightarrow \infty} f_i(x)$$

Theorem 21.401. Let $\langle X, \mathcal{A} \rangle$, $A \in \mathcal{A}$ be a measurable space then we have

1. $\forall f, g \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ we have that $f + g \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$.
2. If $n \in \mathbb{N}$, $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ then $\sum_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$.

3. $\forall \alpha \in \mathbb{C}$ and $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ we have that $\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$.
4. $\forall f, g \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ we have $f \cdot g \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$.
5. If $A \in \mathcal{A}$ and $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ then $\chi_{X, A} \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$.
6. If $\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ and $D = \left\{x \in X \mid \lim_{i \rightarrow \infty} f_i(x) \text{ exist}\right\}$ then $D \in \mathcal{A}$ and $\lim_{i \rightarrow \infty} f_i \in \mathcal{M}[D, \mathcal{A}, \mathbb{C}]$.

Proof.

1. As $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ we have by the previous theorem [theorem: 21.399] that

$$\operatorname{Re}(f), \operatorname{Im}(f), \operatorname{Re}(g), \operatorname{Im}(g) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \quad (21.641)$$

Let $x \in X$ then

$$\begin{aligned} (f+g)(x) &= f(x) + g(x) \\ &= \operatorname{Re}(f(x)) + i \cdot \operatorname{Im}(f(x)) + \operatorname{Re}(g(x)) + i \cdot \operatorname{Im}(g(x)) \\ &= (\operatorname{Re}(f(x)) + \operatorname{Re}(g(x))) + i \cdot (\operatorname{Im}(f(x)) + \operatorname{Im}(g(x))) \\ &= (\operatorname{Re}(f) + \operatorname{Re}(g))(x) + i \cdot (\operatorname{Im}(f) + \operatorname{Im}(g))(x) \end{aligned}$$

proving that

$$\operatorname{Re}(f+g) = \operatorname{Re}(f) + \operatorname{Re}(g) \text{ and } \operatorname{Im}(f+g) = \operatorname{Im}(f) + \operatorname{Im}(g)$$

Using the above, [eq: 21.641] and [theorem: 21.257] we have that $\operatorname{Re}(f+g), \operatorname{Im}(f+g) \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$ so that by the previous theorem [see theorem: 21.399]

$$f+g \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$$

2. This is easily proved by induction. So let

$$\mathcal{S} = \left\{ n \in \mathbb{N}_0 \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{C}] \text{ then } \sum_{i=1}^n f_i \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}] \right\}$$

then we have:

1 $\in \mathcal{S}$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ then $\sum_{i=1}^1 f_i = f_1 \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ which proves that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n+1 \in \mathcal{S}$. then if $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$. As $n \in \mathcal{S}$ we have

$$\sum_{i=1}^n f_i \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$$

which, as $f_{n+1} \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, proves by (1) that

$$\sum_{i=1}^{n+1} f_i = f_{n+1} + \sum_{i=1}^n f_i \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$$

from which it follows that $n+1 \in \mathcal{S}$.

3. As $f \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ we have by the previous theorem [theorem: 21.399] that

$$\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}] \quad (21.642)$$

Let $x \in X$ then

$$\begin{aligned} (\alpha \cdot f)(x) &= \\ \alpha \cdot f(x) &= \\ (\operatorname{Re}(\alpha) + i \cdot \operatorname{Im}(\alpha)) \cdot (\operatorname{Re}(f(x)) + i \cdot \operatorname{Im}(f(x))) &= \\ (\operatorname{Re}(\alpha) \cdot \operatorname{Re}(f(x)) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f(x))) + i \cdot (\operatorname{Re}(\alpha) \cdot \operatorname{Im}(f(x)) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f(x))) &= \\ (\operatorname{Re}(\alpha) \cdot \operatorname{Re}(f) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f))(x) + i \cdot (\operatorname{Re}(\alpha) \cdot \operatorname{Im}(f) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f))(x) &= \end{aligned}$$

proving that

$$\operatorname{Re}(\alpha \cdot f) = \operatorname{Re}(\alpha) \cdot \operatorname{Re}(f) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f) \wedge \operatorname{Im}(\alpha \cdot f) = \operatorname{Re}(\alpha) \cdot \operatorname{Im}(f) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f)$$

Using the above, [eq: 21.642 and [theorems: 21.263 and 21.257] we have that

$$\operatorname{Re}(\alpha \cdot f), \operatorname{Im}(\alpha \cdot f) \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$$

which by [theorem: 21.399] proves

$$\alpha \cdot f \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$$

4. As $f, g \in \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ we have by [theorem: 21.399] that $\operatorname{Re}(f), \operatorname{Re}(g), \operatorname{Im}(f), \operatorname{Im}(g) \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$. Further for $x \in X$ we have

$$\begin{aligned} (f \cdot g)(x) &= \\ f(x) \cdot g(x) &= \\ (\operatorname{Re}(f(x)) + i \cdot \operatorname{Im}(f(x))) \cdot (\operatorname{Re}(g(x)) + i \cdot \operatorname{Im}(g(x))) &= \\ \operatorname{Re}(f(x)) \cdot \operatorname{Re}(g(x)) - \operatorname{Im}(f(x)) \cdot \operatorname{Im}(g(x)) + i \cdot (\operatorname{Re}(f(x)) \cdot \operatorname{Im}(g(x)) + \operatorname{Im}(f(x)) \cdot \operatorname{Re}(g(x))) &= \\ (\operatorname{Re}(f) \cdot \operatorname{Re}(g) - \operatorname{Im}(f) \cdot \operatorname{Im}(g))(x) + i \cdot (\operatorname{Re}(f) \cdot \operatorname{Im}(g) + \operatorname{Im}(f) \cdot \operatorname{Re}(g))(x) \end{aligned}$$

proving that

$$\begin{aligned} \operatorname{Re}(f \cdot g) &= \operatorname{Re}(f) \cdot \operatorname{Re}(g) - \operatorname{Im}(f) \cdot \operatorname{Im}(g) \\ \operatorname{Im}(f \cdot g) &= \operatorname{Re}(f) \cdot \operatorname{Im}(g) + \operatorname{Im}(f) \cdot \operatorname{Re}(g) \end{aligned}$$

Using then [theorems: 21.265, 21.257] it follows that

$$\operatorname{Re}(f \cdot g), \operatorname{Im}(f \cdot g) \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$$

which by [theorem: 21.399] proves that $f \cdot g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$.

5. As $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ we have by [theorem: 21.399] that

$$\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

Further we have for $x \in X$

$$\begin{aligned} \operatorname{Re}(\mathcal{X}_{X,A} \cdot f)(x) &= \operatorname{Re}(\mathcal{X}_{X,A}(x) \cdot f(x))_{\mathcal{X}_{X,A}(x) \in \mathbb{R}} \mathcal{X}_{X,A}(x) \cdot \operatorname{Re}(f(x)) = (\mathcal{X}_{X,A} \cdot \operatorname{Re}(f))(x) \\ \operatorname{Im}(\mathcal{X}_{X,A} \cdot f)(x) &= \operatorname{Im}(\mathcal{X}_{X,A}(x) \cdot f(x))_{\mathcal{X}_{X,A}(x) \in \mathbb{R}} \mathcal{X}_{X,A}(x) \cdot \operatorname{Im}(f(x)) = (\mathcal{X}_{X,A} \cdot \operatorname{Im}(f))(x) \end{aligned}$$

proving that

$$\operatorname{Re}(\mathcal{X}_{X,A} \cdot f) = \mathcal{X}_{X,A} \cdot \operatorname{Re}(f) \wedge \operatorname{Im}(\mathcal{X}_{X,A} \cdot f) = \mathcal{X}_{X,A} \cdot \operatorname{Im}(f)$$

so that by [theorem: 21.276]

$$\operatorname{Re}(\mathcal{X}_A \cdot f), \operatorname{Im}(\mathcal{X}_A \cdot f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

hence using [theorem: 21.399] we have that $\mathcal{X}_A \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$.

6. As $\{f_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{M}[A, \mathcal{A}, \mathbb{C}]$ we have by [theorem: 21.399] that

$$\forall i \in \mathbb{N} \operatorname{Re}(f_i), \operatorname{Im}(f_i) \in \mathcal{M}[A, \mathcal{A}, \mathbb{R}]$$

Further using [theorem: 14.361] we have that $\lim_{i \rightarrow \infty} f_i(x)$ exists if and only if $\lim_{i \rightarrow \infty} \operatorname{Re}(f_i(x))$ and $\lim_{i \rightarrow \infty} \operatorname{Im}(f_i(x))$ exists and are real. So if

$$D_1 = \left\{ x \in A \mid \lim_{i \rightarrow \infty} \operatorname{Re}(f_i(x)) \text{ exists and is real} \right\}$$

$$D_2 = \left\{ x \in A \mid \lim_{i \rightarrow \infty} \operatorname{Im}(f_i(x)) \text{ exists and is real} \right\}$$

then we have

$$D = \left\{ x \in A \mid \lim_{i \rightarrow \infty} f_i(x) \text{ exists} \right\} = D_1 \cap D_2$$

From [theorem: 21.269(6)] we have that $D_1 \in \mathcal{A}, D_2 \in \mathcal{A}$ proving that

$$D = D_1 \cap D_2 \in \mathcal{A}$$

Further using [theorem: 21.269(6)] again we have for

$$\lim_{i \rightarrow \infty} \operatorname{Re}(f_i): D_1 \rightarrow \overline{\mathbb{R}} \text{ and } \lim_{i \rightarrow \infty} \operatorname{Im}(f_i): D_2 \rightarrow \overline{\mathbb{R}}$$

that

$$\lim_{i \rightarrow \infty} \operatorname{Re}(f_i) \in \mathcal{M}[D_1, \mathcal{A}, \mathbb{R}] \text{ and } \lim_{i \rightarrow \infty} \operatorname{Im}(f_i) \in \mathcal{M}[D_2, \mathcal{A}, \mathbb{R}]$$

As $D \subseteq D_1, D_2$ we have by [theorem: 21.231] that

$$\left(\lim_{i \rightarrow \infty} \operatorname{Re}(f_i) \right)_{|D}, \left(\lim_{i \rightarrow \infty} \operatorname{Im}(f_i) \right)_{|D} \in \mathcal{M}[D, \mathcal{A}, \mathbb{R}].$$

Further $\forall x \in D$ we have by [theorem: 14.361]

$$\begin{aligned} & \left(\left(\lim_{n \rightarrow \infty} \operatorname{Re}(f_n) \right)_{|D} + i \cdot \left(\lim_{n \rightarrow \infty} \operatorname{Im}(f_n) \right)_{|D} \right)(x) = \\ & \left(\lim_{n \rightarrow \infty} \operatorname{Re}(f_n) \right)_{|D}(x) + i \cdot \left(\lim_{n \rightarrow \infty} \operatorname{Im}(f_n) \right)_{|D}(x) = \\ & \left(\lim_{n \rightarrow \infty} \operatorname{Re}(f_n) \right)(x) + i \cdot \left(\lim_{n \rightarrow \infty} \operatorname{Im}(f_n) \right)(x) \\ & \lim_{n \rightarrow \infty} \operatorname{Re}(f_n(x)) + i \cdot \lim_{n \rightarrow \infty} \operatorname{Im}(f_n(x)) \stackrel{[\text{theorem: 14.361}]}{=} \\ & \lim_{n \rightarrow \infty} (\operatorname{Re}(f_n(x)) + i \cdot \operatorname{Im}(f_n(x))) = \\ & \lim_{n \rightarrow \infty} f_n(x) \end{aligned}$$

proving that $\lim_{n \rightarrow \infty} f_n = \left(\lim_{n \rightarrow \infty} \operatorname{Re}(f_n) \right)_{|D} + i \cdot \left(\lim_{n \rightarrow \infty} \operatorname{Im}(f_n) \right)_{|D}$. Using [theorem: 21.399] we finally have that

$$\lim_{n \rightarrow \infty} f_n \in \mathcal{M}[D, \mathcal{A}, \mathbb{C}]$$

□

Theorem 21.402. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $A \in \mathcal{A}$ and $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ then

$$|f| \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

Proof. As $\forall x \in A$ we have $(|f| \circ f)(x) = |f(x)| = |f|(x)$ it follows that $|f| = |f| \circ f$. By [theorem: 21.398] $|f|: \mathbb{C} \rightarrow \overline{\mathbb{R}}$ is $\mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]$ -measurable. So using [theorem: 21.230] it follows that $|f|$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable or as $|f|(X) \subseteq [0, \infty[$ that

$$|f| \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

□

21.5.5.2 Complex Measure Integral

We can now define the integral of a complex valued function based on the integrals of the real and imaginary components of this function.

Definition 21.403. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f: X \rightarrow \mathbb{C}$ a function then

$$f \text{ is } \mu\text{-complex integrable if } \operatorname{Re}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \wedge \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

and the complex integral of f noted as $\int^{\mathbb{C}} f d\mu$ is defined by

$$\int^{\mathbb{C}} f d\mu = \int \operatorname{Re}(f) d\mu + i \cdot \int \operatorname{Im}(f) d\mu$$

The set of all complex integrable functions is noted as $\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ hence

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] = \{f \in \mathbb{C}^X \mid f: X \rightarrow \mathbb{C} \text{ is } \mu\text{-complex integrable}\}$$

Example 21.404. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $C_0 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ and $\int^{\mathbb{C}} C_0 d\mu = 0$ where $C_0: X \rightarrow \mathbb{C}$ is defined by $C_0(x)$.

Proof. Let $x \in X$ then

$$\operatorname{Re}(C_0)(x) = \operatorname{Re}(C_0(x)) = \operatorname{Re}(0) = 0 = C_0(x)$$

and

$$\operatorname{Im}(C_0)(x) = \operatorname{Im}(C_0(x)) = \operatorname{Im}(0) = 0 = C_0(x)$$

This proves that $\operatorname{Re}(C_0) = C_0 \underset{[\text{note: 21.349}]}{\in} \mathcal{L}(X, \mathcal{A}, \mu, \mathbb{R})$ and $\operatorname{Im}(C_0) = C_0 \underset{[\text{note: 21.349}]}{\in} \mathcal{L}(X, \mathcal{A}, \mu, \mathbb{R})$.

Hence we have by definition that $C_0 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$. Further we have

$$\int^{\mathbb{C}} C_0 d\mu = \int \operatorname{Re}(C_0) d\mu + i \cdot \int \operatorname{Im}(C_0) d\mu = \int C_0 d\mu + i \cdot \int C_0 d\mu \underset{[\text{note: 21.349}]}{=} 0$$

□

Theorem 21.405. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ then

$$\operatorname{Re}(\mathcal{X}_{X,A} \cdot f) = \mathcal{X}_{X,A} \cdot \operatorname{Re}(f) \text{ and } \operatorname{Im}(\mathcal{X}_{X,A} \cdot f) = \mathcal{X}_{X,A} \cdot \operatorname{Im}(f)$$

$$\mathcal{X}_{X,A} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

Proof. If $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ then $\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that by [theorem: 21.356]

$$\mathcal{X}_{X,A} \cdot \operatorname{Re}(f), \mathcal{X}_{X,A} \cdot \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}].$$

Let $x \in X$ then

$$\operatorname{Re}(\mathcal{X}_{X,A} \cdot f)(x) = \operatorname{Re}(\mathcal{X}_{X \setminus A}(x) \cdot f(x)) = \mathcal{X}_{X,A}(x) \cdot \operatorname{Re}(f)(x) = (\mathcal{X}_{X,A} \cdot \operatorname{Re}(f))(x)$$

$$\operatorname{Im}(\mathcal{X}_{X,A} \cdot f)(x) = \operatorname{Im}(\mathcal{X}_{X \setminus A}(x) \cdot f(x)) = \mathcal{X}_{X,A}(x) \cdot \operatorname{Im}(f)(x) = (\mathcal{X}_{X,A} \cdot \operatorname{Im}(f))(x)$$

which proves that

$$\operatorname{Re}(\mathcal{X}_{X,A} \cdot f) = \mathcal{X}_{X,A} \cdot \operatorname{Re}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \operatorname{Im}(\mathcal{X}_{X,A} \cdot f) = \mathcal{X}_{X,A} \cdot \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

Hence we have by definition that

$$\mathcal{X}_{X,A} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

□

The above theorem allows us to define the integral over a measurable set.

Definition 21.406. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $A \in \mathcal{A}$ and $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ then we define $\int_A^{\mathbb{C}} f d\mu$ by

$$\int_A^{\mathbb{C}} f d\mu = \int^{\mathbb{C}} (\mathcal{X}_{X,A} \cdot f) d\mu$$

Note 21.407. As

$$\begin{aligned} \int_A^{\mathbb{C}} f d\mu &= \int^{\mathbb{C}} (\mathcal{X}_{X,A} \cdot f) d\mu \\ &\stackrel{\text{def}}{=} \int \operatorname{Re}(\mathcal{X}_{X,A} \cdot f) d\mu + i \cdot \int \operatorname{Im}(\mathcal{X}_{X,A} \cdot f) d\mu \\ &\stackrel{[\text{theorem: 21.405}]}{=} \int (\mathcal{X}_{X,A} \cdot \operatorname{Re}(f)) d\mu + i \cdot \int (\mathcal{X}_{X,A} \cdot \operatorname{Im}(f)) d\mu \\ &= \int_A \operatorname{Re}(f) d\mu + i \cdot \int_A \operatorname{Im}(f) d\mu \end{aligned}$$

we have that

$$\int_A^{\mathbb{C}} f d\mu = \int_A \operatorname{Re}(f) d\mu + i \cdot \int_A \operatorname{Im}(f) d\mu$$

The following shows that the complex integral is an extension of the measure integral.

Theorem 21.408. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

and

$$\forall f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ we have that } \int^{\mathbb{C}} f d\mu = \int f d\mu$$

Proof. Let $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ then $f(X) \subseteq \mathbb{R}$. Now for the function $f: X \rightarrow \mathbb{C}$ we have as $f(X) \subseteq \mathbb{R}$ that $\forall x \in X$ $\text{Re}(f)(x) = \text{Re}(f(x)) = f(x)$ and $\text{Im}(f)(x) = \text{Im}(f(x)) = 0 = C_0(x)$. Hence we have

$$\text{Re}(f) = f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \text{Im}(f) = C_0 \underset{[\text{theorem: 21.358}]}{\in} \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.643)$$

which proves that $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$. So

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

Further

$$\int^{\mathbb{C}} f d\mu = \int \text{Re}(f) d\mu + i \cdot \int \text{Im}(f) d\mu \underset{[\text{eq: 21.643}]}{=} \int f d\mu + i \cdot \int C_0 d\mu \underset{[\text{theorem: 21.358}]}{=} \int f d\mu \quad \square$$

Theorem 21.409. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have:

1. $\forall f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that $f + g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ and

$$\int^{\mathbb{C}} (f + g) d\mu = \int^{\mathbb{C}} f d\mu + \int^{\mathbb{C}} g d\mu$$

2. $\forall \alpha \in \mathbb{C}$ and $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that $\alpha \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ and

$$\int^{\mathbb{C}} (\alpha \cdot f) d\mu = \alpha \cdot \int^{\mathbb{C}} f d\mu$$

Proof.

1. As $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ we have by definition that

$$\text{Re}(f), \text{Re}(g), \text{Im}(f), \text{Im}(g) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

Let $x \in X$ then

$$\text{Re}(f + g)(x) = \text{Re}(f(x) + g(x)) = \text{Re}(f(x)) + \text{Re}(g(x)) = (\text{Re}(f) + \text{Re}(g))(x)$$

$$\text{Im}(f + g)(x) = \text{Im}(f(x) + g(x)) = \text{Im}(f(x)) + \text{Im}(g(x)) = (\text{Im}(f) + \text{Im}(g))(x)$$

so that

$$\text{Re}(f + g) = \text{Re}(f) + \text{Re}(g) \text{ and } \text{Im}(f + g) = \text{Im}(f) + \text{Im}(g)$$

Hence using [theorem: 21.360] it follows that

$$\text{Re}(f + g), \text{Im}(f + g) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

which proves that

$$f + g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

Further

$$\begin{aligned} \int^{\mathbb{C}} (f + g) d\mu &= \int \text{Re}(f + g) d\mu + i \cdot \int \text{Im}(f + g) d\mu \\ &= \int (\text{Re}(f) + \text{Re}(g)) d\mu + i \cdot \int (\text{Im}(f) + \text{Im}(g)) d\mu \\ &\underset{[\text{theorem: 21.360}]}{=} \int \text{Re}(f) d\mu + \int \text{Re}(g) d\mu + i \cdot \int \text{Im}(f) d\mu + i \cdot \int \text{Im}(g) d\mu \\ &= \int^{\mathbb{C}} f d\mu + \int^{\mathbb{C}} g d\mu \end{aligned}$$

2. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ we have by definition that

$$\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{L}(X, \mathcal{A}, \mu, \mathbb{R}) \text{ and } \int^{\mathbb{C}} f d\mu = \int \operatorname{Re}(f) d\mu + i \cdot \int \operatorname{Im}(f) d\mu$$

Let $x \in X$ then

$$\begin{aligned} (\alpha \cdot f)(x) &= \\ \alpha \cdot f(x) &= \\ (\operatorname{Re}(\alpha) + i \cdot \operatorname{Im}(\alpha)) \cdot (\operatorname{Re}(f(x)) + i \cdot \operatorname{Im}(f(x))) &= \\ \operatorname{Re}(\alpha) \cdot \operatorname{Re}(f(x)) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f(x)) + i \cdot (\operatorname{Re}(\alpha) \cdot \operatorname{Im}(f(x)) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f(x))) &= \\ ((\operatorname{Re}(\alpha) \cdot \operatorname{Re}(f) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f)) + i \cdot (\operatorname{Re}(\alpha) \cdot \operatorname{Im}(f) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f)))(x) \end{aligned}$$

which proves that

$$\operatorname{Re}(\alpha \cdot f) = \operatorname{Re}(\alpha) \cdot \operatorname{Re}(f) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f) \wedge \operatorname{Im}(\alpha \cdot f) = \operatorname{Re}(\alpha) \cdot \operatorname{Im}(f) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f)$$

Using [theorem: 21.360] on the above we have that $\operatorname{Re}(\alpha \cdot f), \operatorname{Im}(\alpha \cdot f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that by definition

$$\alpha \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

Further

$$\begin{aligned} \int^{\mathbb{C}} (\alpha \cdot f) d\mu &= \\ \int (\operatorname{Re}(\alpha) \cdot \operatorname{Re}(f) - \operatorname{Im}(\alpha) \cdot \operatorname{Im}(f)) d\mu + i \cdot \int (\operatorname{Re}(\alpha) \cdot \operatorname{Im}(f) + \operatorname{Im}(\alpha) \cdot \operatorname{Re}(f)) d\mu &= \\ \operatorname{Re}(\alpha) \cdot \int \operatorname{Re}(f) d\mu - \operatorname{Im}(\alpha) \cdot \int \operatorname{Im}(f) d\mu + i \cdot \operatorname{Re}(\alpha) \cdot \int \operatorname{Im}(f) d\mu + & \\ i \cdot \operatorname{Im}(\alpha) \cdot \int \operatorname{Re}(f) d\mu &= \\ \operatorname{Re}(\alpha) \cdot \left(\int \operatorname{Re}(f) d\mu + i \cdot \int \operatorname{Im}(f) d\mu \right) + i \cdot \operatorname{Im}(\alpha) \cdot \left(\int \operatorname{Re}(f) d\mu + \int \operatorname{Im}(f) d\mu \right) &= \\ \operatorname{Re}(\alpha) \cdot \int^{\mathbb{C}} f d\mu + i \cdot \operatorname{Im}(\alpha) \cdot \int^{\mathbb{C}} f d\mu &= \\ (\operatorname{Re}(\alpha) + i \cdot \operatorname{Im}(\alpha)) \cdot \int^{\mathbb{C}} f d\mu &= \\ \alpha \cdot \int^{\mathbb{C}} f d\mu \end{aligned}$$

□

We have the following equivalent definition of μ -complex integrable functions:

Theorem 21.410. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f: X \rightarrow \mathbb{C}$ a complex valued function then

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] \Leftrightarrow f \in \mathcal{M}[X, \mathcal{A}, \mu, \mathbb{C}] \text{ and } |f| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

Further if $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ then we have

$$\left| \int^{\mathbb{C}} f d\mu \right| \leq \int |f| d\mu$$

Proof.

$\Rightarrow$. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that $\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that $\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$. Using then [theorem: 21.399] it follows that

$$f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \tag{21.644}$$

and using [theorem: 21.402]

$$|f| \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \tag{21.645}$$

Further as $\operatorname{Re}(f), \operatorname{Im}(g) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ we have by [theorem: 21.362] that

$$|\operatorname{Re}(f)|, |\operatorname{Im}(f)| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

so that

$$\int^+ |\operatorname{Re}(f)| d\mu \underset{[\text{theorem: 21.344(7)}]}{=} \int^+ |\operatorname{Re}(f)|^+ d\mu < \infty$$

$$\int^+ |\operatorname{Im}(f)| d\mu \underset{[\text{theorem: 21.344(7)}]}{=} \int^+ |\operatorname{Im}(f)|^+ d\mu < \infty$$

Let $x \in X$ then

$$|f|(x) = |f(x)| \underset{[\text{theorem: 10.85(8)}]}{\leq} |\operatorname{Re}(f(x))| + |\operatorname{Im}(f(x))| = (|\operatorname{Re}(f)| + |\operatorname{Im}(f)|)(x)$$

So that by [theorem: 21.331]

$$\int^+ |f| d\mu \leq \int^+ (|\operatorname{Re}(f)| + |\operatorname{Im}(f)|) d\mu \underset{[\text{theorem: 21.331}]}{=} \int^+ |\operatorname{Re}(f)| d\mu + \int^+ |\operatorname{Im}(f)| d\mu < \infty$$

which together with [eq: 21.645] proves by [theorem: 21.352] that

$$|f| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

$\Leftarrow$. As $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ we have using [theorem: 21.399] that

$$\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

Let $x \in X$ then

$$|\operatorname{Re}(f)|(x) = |\operatorname{Re}(f(x))| \underset{[\text{theorem: 10.85(3)}]}{\leq} |f(x)| = |f|(x)$$

$$|\operatorname{Im}(f)|(x) = |\operatorname{Im}(f(x))| \underset{[\text{theorem: 10.85(3)}]}{\leq} |f(x)| = |f|(x)$$

Using then [theorem: 21.365] it follows that $\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ hence we have by definition that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

For the last part define

$$w(f) = \begin{cases} 1 & \text{if } \left| \int^{\mathbb{C}} f d\mu \right| = 0 \\ \frac{\int^{\mathbb{C}} f d\mu}{\left| \int^{\mathbb{C}} f d\mu \right|} & \text{if } \left| \int^{\mathbb{C}} f d\mu \right| \neq 0 \end{cases} \in \mathbb{C} \text{ so that } |w(f)| = 1$$

then for $\int^{\mathbb{C}} f d\mu$ we have either:

$\int f^{\mathbb{C}} d\mu = 0$. Then $\left| \int f d\mu \right| = 0$ and $w(f) = 1$ so that $w(f) \cdot \left| \int^{\mathbb{C}} f d\mu \right| = 0 = \int^{\mathbb{C}} f d\mu$

$\int f^{\mathbb{C}} d\mu \neq 0$. Then $\left| \int^{\mathbb{C}} f d\mu \right| \neq 0$ and $w(f) = \frac{\int^{\mathbb{C}} f d\mu}{\left| \int^{\mathbb{C}} f d\mu \right|}$ so that $w(f) \cdot \left| \int^{\mathbb{C}} f d\mu \right| = \int^{\mathbb{C}} f d\mu$

Summarized we have

$$\int^{\mathbb{C}} f d\mu = w(f) \cdot \left| \int^{\mathbb{C}} f d\mu \right| \text{ and } |w(f)| = 1$$

Then we have

$$\begin{aligned} \left| \int^{\mathbb{C}} f d\mu \right| &= \frac{1}{w(f)} \cdot \int^{\mathbb{C}} f d\mu \\ &\underset{[\text{theorem: 21.409}]}{=} \int^{\mathbb{C}} \left(\frac{1}{w(f)} \cdot f \right) d\mu \\ &\underset{\text{def}}{=} \int \operatorname{Re} \left(\frac{1}{w(f)} \cdot f \right) d\mu + i \cdot \int \operatorname{Im} \left(\frac{1}{w(f)} \cdot f \right) d\mu \\ &\underset{f \operatorname{Re} \left(\frac{1}{w(f)} \cdot f \right) d\mu, f \operatorname{Im} \left(\frac{1}{w(f)} \cdot f \right) d\mu, \left| \int^{\mathbb{C}} f d\mu \right| \in \mathbb{R}}{=} \int \operatorname{Re} \left(\frac{1}{w(f)} \cdot f \right) d\mu \end{aligned} \quad (21.646)$$

Let $x \in X$ then using [theorem: 10.85] we have that

$$\operatorname{Re}\left(\frac{1}{w(f)} \cdot f(x)\right) \leq \left|\frac{1}{w(f)} \cdot f(x)\right| = \left|\frac{1}{w(f)}\right| \cdot |f(x)|_{|w(f)|=1} |f(x)| = |f|(x)$$

Hence using [theorem: 21.360] we have that $\int \operatorname{Re}\left(\frac{1}{w(f)} \cdot f(x)\right) d\mu \leq \int |f| d\mu$ which combined with [eq: 21.646] proves that

$$\left|\int^{\mathbb{C}} f d\mu\right| \leq \int |f| d\mu \quad \square$$

21.5.6 The spaces $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$

21.5.6.1 Definition of $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$

In this section we introduce infinite Banach spaces that play a very important role in physics and functional analyses. Be careful, in the following we use the same symbol $\|$ for the norm on $\mathbb{R}$ and on $\mathbb{C}$ [see examples: 14.99, 14.102], context will indicate which norm is used.

Definition 21.411. Let $p \in [1, \infty[$ then we define

$$\begin{aligned} \|\cdot\|^p: \mathbb{R} &\rightarrow \overline{\mathbb{R}} \text{ by } (\|\cdot\|^p)(x) = |x|^p \\ \|\cdot\|^p: \mathbb{C} &\rightarrow \overline{\mathbb{R}} \text{ by } (\|\cdot\|^p)(x) = |x|^p \end{aligned}$$

If X is a set and $f: X \rightarrow \mathbb{R}$ then we define

$$|f|^p = \|\cdot\|^p \circ f: X \rightarrow \overline{\mathbb{R}} \text{ so that } |f|^p(x) = |f(x)|^p$$

If X is a set and $f: X \rightarrow \mathbb{C}$ then we define

$$|f|^p = \|\cdot\|^p \circ f: X \rightarrow \overline{\mathbb{R}} \text{ so that } |f|^p(x) = |f(x)|^p$$

Note 21.412. Using [theorem: 19.42 (2,3)] we have that $\forall x \in [0, \infty[$ $x^p \in [0, \infty[$ so that

$$\|f\|(X) \subseteq [0, \infty[\wedge \|f\|^p(X) \subseteq [0, \infty[\wedge |f|^p(X) \subseteq [0, \infty[$$

Definition 21.413. Let $p \in [1, \infty[$ and $\langle X, \mathcal{A}, \mu \rangle$ a measures space, $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$ then we define $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ as follows

1. If $\mathbb{K} = \mathbb{R}$ then $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \mid f(X) \subseteq \mathbb{R} \wedge |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]\}$
2. If $\mathbb{K} = \mathbb{C}$ then $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \mid |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]\}$

We prove now that $\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ is the special case of $\mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$.

Theorem 21.414. Let $p \in [1, \infty[$ and $\langle X, \mathcal{A}, \mu \rangle$ a measure space then

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}] = \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$$

in other words we have

1. $\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] = \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{R}]$
2. $\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] = \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{C}]$

Proof. First as $\forall x \in X$ we have $|f|^1(x) = |f(x)|^1 \stackrel{[\text{theorem: 19.42(4)}]}{=} |f(x)| = |f|(x)$ we have that

$$|f| = |f|^1$$

1. Let $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ then $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and by [theorem: 21.362] $|f|^1 = |f| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \subseteq \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.647)$$

Further if $f \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{R}]$ then $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and $|f| = |f|^1 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$. Hence $\int^+ |f| d\mu = \int^+ |f|^1 d\mu < \infty$ so that by [theorem: 21.362] $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ which proves that $\mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{R}] \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$, combining this with [eq: 21.647] gives

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] = \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{R}]$$

2. Let $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ then we have by [theorem: 21.410] that $|f|^1 = |f| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that by definition $f \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{C}]$. Hence we have

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] \subseteq \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{C}] \quad (21.648)$$

If $f \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{C}]$ then by definition $f \in \mathcal{M}[X, \mathcal{A}, \mu, \mathbb{C}]$ and $|f| = |f|^1 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ which by [theorem: 21.410] proves that $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$, hence $\mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{C}] \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$. Combining this result with [eq: 21.648] proves that

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] = \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{C}] \quad \square$$

Example 21.415. Let $p \in [1, \infty[$ then for $C_0: X \rightarrow \mathbb{K}$ defined by $C_0(x) = 0$ we have that

$$C_0 \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

Proof. First using [theorem: 21.228] it follows that

$$C_0 \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$$

Further we have $\forall x \in X$ that $|C_0|^p(x) = |C_0(x)|^p = |0|^p \stackrel{\text{[definition: 19.39]} \wedge p \neq 0}{=} 0 = C_0(x)$ which proves that $|C_0|^p = C_0$. As by [example: 21.349] $C_0 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ it follows that

$$C_0 \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \quad \square$$

Next we want to prove that $\langle \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}], +, \cdot \rangle$ is a vector space over $\mathbb{R}$ and that $\langle \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}], +, \cdot \rangle$ is a vector space over $\mathbb{C}$. To do this we need the following three lemma's.

Lemma 21.416. Let $p \in [1, \infty[$ then we have that:

1. $\| \cdot \|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$.
2. If $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ then $|f|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$.
3. $\| \cdot \|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{C}]$.
4. If $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ then $|f|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$.

Proof.

1. Using [theorem: 14.177] $\| \cdot \|: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function [using the canonical topologies on $\mathbb{R}$] with $\|(\mathbb{R}) \subseteq [0, \infty[$. Further using [theorem: 19.55] we have that

$$(\star)^p: [0, \infty[\rightarrow \mathbb{R} \text{ defined by } (\star)^p(x) = x^p$$

is continuous. As $\forall x \in \mathbb{R}$ $\|x\|^p(x) = |x|^p = (\star)^p(|x|) = ((\star)^p \circ \|)(x)$ it follows that $\| \cdot \|^p = (\star)^p \circ \|$. Hence we have by [theorem: 14.158] that

$$\| \cdot \|^p: \mathbb{R} \rightarrow \mathbb{R} \text{ is continuous using the topologies } \mathcal{T}_{\|}, \mathcal{T}_{\|}$$

By [theorem: 21.255] we have then that $\| \cdot \|^p$ is $\mathcal{B}[\mathbb{R}], \mathcal{B}[\mathbb{R}]$ -measurable. Further using [theorem: 21.243] on the above it follows that

$$\| \cdot \|^p: \mathbb{R} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{R}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable} \quad (21.649)$$

Finally as $\| \cdot \|^p(\mathbb{R}) \subseteq [0, \infty[$ it follows that

$$\| \cdot \|^p \in \{f \in \mathcal{M}[\mathbb{R}, \mathcal{B}[\mathbb{R}], \overline{\mathbb{R}}] \mid f(X) \subseteq [0, \infty[\} \stackrel{\text{def}}{=} \mathcal{M}^+[\mathbb{R}, \mathcal{A}, \mathbb{R}]$$

2. Let $x \in X$ then $|f|^p = ||^p \circ f$. As $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ we have that f is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable and it follows from [theorem: 21.243] that $f: X \rightarrow \mathbb{R}$ is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ measurable. Hence using [theorem: 21.230] together with [eq: 21.649] proves that $|f|^p$ is $\mathcal{A}, \mathcal{B}[\mathbb{R}]$ -measurable. Finally as $|f|^p(X) \subseteq [0, \infty[$ we have that

$$|f|^p \in \{f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid |f|^p(X) \subseteq [0, \infty[\} \stackrel{\text{def}}{=} \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

3. Using [theorem: 14.177] $||: \mathbb{C} \rightarrow \mathbb{R}$ is a continuous function [using the canonical topologies on $\mathbb{C}$ and $\mathbb{R}$. Further using [theorem: 19.55] we have that

$$(*)^p: [0, \infty[\rightarrow \mathbb{R} \text{ is continuous}$$

As $\forall z \in \mathbb{C}$ we have $||^p(z) = |z|^p = (\star)^p(|z|) = ((\star)^p \circ ||)(z)$ it follows that $||^p = (\star)^p \circ ||$. Hence we have by [theorem: 14.158] that

$$||^p: \mathbb{C} \rightarrow \mathbb{R} \text{ is continuous using the topologies } \mathcal{T}_{\mathbb{C}}, \mathcal{T}_{\mathbb{R}}$$

Hence by [theorem: 21.255] $||^p$ is $\mathcal{B}[\mathbb{C}], \mathcal{B}[\mathbb{R}]$ -measurable. Using then [theorem: 21.243] it follows that

$$||^p: \mathbb{C} \rightarrow \overline{\mathbb{R}} \text{ is } \mathcal{B}[\mathbb{C}], \mathcal{B}[\overline{\mathbb{R}}]\text{-measurable} \quad (21.650)$$

Finally as $||^p(X) \subseteq [0, \infty[$ it follows that

$$||^p \in \{f \in \mathcal{M}[\mathbb{C}, \mathcal{B}[\mathbb{C}], \overline{\mathbb{R}}] \mid ||^p(\mathbb{C}) \subseteq [0, \infty[\} = \mathcal{M}^+[\mathbb{C}, \mathcal{A}, \mathbb{R}]$$

4. Let $x \in X$ then $|f|^p = ||^p \circ f$. Hence using [theorem: 21.230], [eq: 21.650] and the fact that f is $\mathcal{A}, \mathcal{B}[\mathbb{C}]$ -measurable it follows that $|f|^p$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable. Finally as $|f|^p(X) \subseteq [0, \infty[$ we have that

$$|f|^p \in \{f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \mid |f|^p(X) \subseteq [0, \infty[\} = \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad \square$$

Lemma 21.417. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $p \in [1, \infty[$, $g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ such that $\forall x \in X$ we have $|f(x)| \leq |g(x)|$ then $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$.*

Proof. As $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ we have by [lemma: 21.416] that

$$|f|^p \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$$

Let $x \in X$ then by the hypothesis $|f(x)| \leq |g(x)|$ so that by the fact that the power function is monotone [see theorem: 19.42(15)] we have $|f(x)|^p \leq |g(x)|^p$, summarized

$$\forall x \in X \text{ we have } ||f(x)|^p| = |f(x)|^p \leq |g(x)|^p$$

Further as $g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have that

$$|g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ so that } |g|^p(X) \subseteq [0, \infty[$$

Hence the conditions for [theorem: 21.365] are satisfied so that

$$|f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

which as $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ proves that

$$f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \quad \square$$

Lemma 21.418. *Let $p \in]0, \infty[$ and $z_1, z_2 \in \mathbb{K}$ then $|z_1 + z_2|^p \leq 2^p \cdot (|z_1|^p + |z_2|^p)$*

Proof. For $|z_1|, |z_2|$ we have the following possible cases:

$$|z_1| \leq |z_2|. \text{ Then}$$

$$|z_1 + z_2| \leq |z_1| + |z_2| \leq 2 \cdot |z_2|$$

So that

$$\begin{aligned}
 |z_1 + z_2|^p &\stackrel{[\text{theorem: 19.42(15)}]}{\leq} (2 \cdot |z_2|)^p \\
 &\stackrel{[\text{theorem: 19.42(12)}]}{=} 2^p \cdot |z_2|^p \\
 &\leq 2^p \cdot |z_1|^p + 2^p \cdot |z_2|^p \\
 &\leq 2^p \cdot (|z_1|^p + |z_2|^p)
 \end{aligned}$$

$|z_2| < |z_1|$. Then

$$|z_1 + z_2| \leq |z_1| + |z_2| < 2 \cdot |z_1|$$

So that

$$\begin{aligned}
 |z_1 + z_2|^p &\stackrel{[\text{theorem: 19.42(15)}]}{<} (2 \cdot |z_1|)^p \\
 &\stackrel{[\text{theorem: 19.42(12)}]}{=} 2^p \cdot |z_1|^p \\
 &\leq 2^p \cdot |z_1|^p + 2^p \cdot |z_2|^p \\
 &\leq 2^p \cdot (|z_1|^p + |z_2|^p)
 \end{aligned}$$

So in all cases we have

$$|z_1 + z_2|^p \leq 2^p \cdot (|z_1|^p + |z_2|^p) \quad \square$$

Theorem 21.419. *Let $p \in [1, \infty[$ and $\langle X, \mathcal{A}, \mu \rangle$ a measure space then we have:*

1. $\forall f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ we have $f + g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$.
2. $\forall \alpha \in \mathbb{R}$ and $\forall f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that $\alpha \cdot f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$.
3. $\forall f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$ we have $f + g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$.
4. $\forall \alpha \in \mathbb{C}$ and $\forall f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that $\alpha \cdot f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$.

Note 21.420. *If we use the convention then $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ then we can rephrase the theorem as follows*

$$\begin{aligned}
 &\forall f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } f + g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \\
 &\forall f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ and } \alpha \in \mathbb{K} \text{ we have } \alpha \cdot f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]
 \end{aligned}$$

Proof.

1. As $f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that

$$f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \text{ and } |f|^p, |g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \stackrel{[\text{theorem: 21.352(5)}]}{\Rightarrow} |f|^p, |g|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$$

From [theorem: 21.257] it follows that

$$f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \quad (21.651)$$

so that using [lemma: 21.416]

$$|f + g|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.652)$$

Let $x \in X$ then

$$|f + g|^p(x) = |f(x) + g(x)|^p \stackrel{[\text{lemma: 21.418}]}{\leq} 2^p \cdot (|f(x)|^p + |g(x)|^p) = (2^p \cdot (|f|^p + |g|^p))(x)$$

applying then [theorem: 21.331] we have

$$\int^+ |f + g|^p d\mu \leq 2^p \cdot \left(\int^+ |f|^p d\mu + \int^+ |g|^p d\mu \right) \quad (21.653)$$

As $|f|^p, |g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ we have by [theorem: 21.352(6)] that $\int^+ |f|^p d\mu < \infty$ and $\int^+ |g|^p d\mu < \infty$. So from the above [see eq: 21.652] we have that $\int^+ |f + g|^p d\mu < \infty$. Combining this with [eq: 21.652] allows us to use [theorem: 21.352] to prove that

$$|f + g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

As $f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ [see eq: 21.651] we have that

$$f + g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$$

2. As $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that

$$f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \text{ and } |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

From [theorem: 21.263] it follows then that

$$\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

Further for $x \in X$ we have

$$|\alpha \cdot f|^p(x) = |\alpha \cdot f(x)|^p = (|\alpha| \cdot |f(x)|)^p \stackrel{[\text{theorem: 19.42}]}{=} |\alpha|^p \cdot |f(x)|^p = (|\alpha|^p \cdot |f|^p)(x)$$

which proves that

$$|\alpha \cdot f|^p = |\alpha| \cdot |f|^p$$

As $|f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ it follows from [theorem: 21.360] that $|\alpha|^p \cdot |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that $|\alpha \cdot f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$, which combined with $\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ proves that

$$\alpha \cdot f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$$

3. As $f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that

$$f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \text{ and } |f|^p, |g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \stackrel{[\text{theorem: 21.352(5)}]}{\Rightarrow} |f|^p, |g|^p \in \mathcal{M}^+[X, \mathcal{A}, \mu]$$

From [theorem: 21.401] it follows that

$$f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \tag{21.654}$$

so that using [lemma: 21.416]

$$|f + g|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \tag{21.655}$$

Let $x \in X$ then

$$|f + g|^p(x) = |f(x) + g(x)|^p \stackrel{[\text{lemma: 21.418}]}{\leq} 2^p \cdot (|f(x)|^p + |g(x)|^p) = (2^p \cdot (|f|^p + |g|^p))(x)$$

applying then [theorem: 21.331] we have

$$\int^+ |f + g|^p d\mu \leq 2^p \cdot \left(\int^+ |f|^p d\mu + \int^+ |g|^p d\mu \right) \tag{21.656}$$

As $|f|^p, |g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ we have by [theorem: 21.352(6)] that $\int^+ |f|^p d\mu < \infty$ and $\int^+ |g|^p d\mu < \infty$. So from the above [see eq: 21.656] we have that $\int^+ |f + g|^p d\mu < \infty$. Combining this with [eq: 21.655] allows us to use [theorem: 21.352] to prove that $|f + g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$. As $f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ [see eq: 21.654] we have that

$$|f + g|^p \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$$

4. As $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that

$$f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \text{ and } |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

From [theorem: 21.401] it follows then that

$$\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$$

Further for $x \in X$ we have

$$|\alpha \cdot f|^p(x) = |\alpha \cdot f(x)|^p = (|\alpha| \cdot |f(x)|)^p \stackrel{[\text{theorem: 19.42}]}{=} |\alpha|^p \cdot |f(x)|^p = (|\alpha|^p \cdot |f|^p)(x)$$

which proves that

$$|\alpha \cdot f|^p = |\alpha| \cdot |f|^p$$

As $|f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ it follows from [theorem: 21.360] that $|\alpha|^p \cdot |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that $|\alpha \cdot f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$, which combined with $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ proves that

$$\alpha \cdot f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{C}]$$

□

We extend now $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ where $p \in [1, \infty[$ to the case where $p = \infty$.

Definition 21.421. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $\mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ is defined by

1. If $\mathbb{K} = \mathbb{R}$ then $\mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] | \exists M \in \mathbb{R}^+ \text{ such that } \forall x \in X \ |f(x)| \leq M\}$
2. If $\mathbb{K} = \mathbb{C}$ then $\mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] | \exists M \in \mathbb{R}^+ \text{ such that } \forall x \in X \ |f(x)| \leq M\}$

In essence $\mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ is the set of bounded measurable functions.

Theorem 21.422. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then

1. $\forall f, g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ we have $f + g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$
2. $\forall f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ and $\alpha \in \mathbb{R}$ we have that $\alpha \cdot f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$
3. $\forall f, g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$ we have $f + g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$
4. $\forall f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$ and $\alpha \in \mathbb{C}$ we have that $\alpha \cdot f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$

In other words using the notation $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ we have that

$$\begin{aligned} \forall f, g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } f + g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \\ \forall f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } \forall \alpha \in \mathbb{K} \text{ that } \alpha \cdot f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \end{aligned}$$

Proof.

1. As $f, g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, hence by [theorem: 21.257] $f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$. Further there exists $M_1, M_2 \in \mathbb{R}^+$ such that $\forall x \in X$ we have $|f(x)| \leq M_1$, $|g(x)| \leq M_2$. If we take $M = M_1 + M_2$ then $\forall x \in X$ we have

$$|(f + g)(x)| = |f(x) + g(x)| \leq |f(x)| + |g(x)| \leq M_1 + M_2 = M$$

which proves that

$$f + g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$$

2. As $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ we have that $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, hence by [theorem: 21.263] $\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$. Further there exists a $M' \in \mathbb{R}^+$ such that $\forall x \in X \ |f(x)| \leq M'$. Take $M = |\alpha| \cdot M' + 1 \in \mathbb{R}^+$ then given $x \in X$ we have

$$|(\alpha \cdot f)(x)| = |\alpha \cdot f(x)| = |\alpha| \cdot |f(x)| \leq |\alpha| \cdot M' < |\alpha| \cdot M' + 1 = M$$

So we have

$$\alpha \cdot f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$$

3. As $f, g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, hence by [theorem: 21.409] $f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, and a M_1, M_2 such that $\forall x \in X$ we have $|f(x)| \leq M_1$, $|g(x)| \leq M_2$. If $M = M_1 + M_2$ then $\forall x \in X$ we have

$$|(f + g)(x)| = |f(x) + g(x)| \leq |f(x)| + |g(x)| \leq M_1 + M_2 = M.$$

So we have that

$$f + g \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$$

4. As $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$ we have that $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, hence by [theorem: 21.409] $\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ and there exists a $M' \in \mathbb{R}^+$ such that $\forall x \in X \ |f(x)| \leq M'$. Take $M = |\alpha| \cdot M' + 1 \in \mathbb{R}^+$ then given $x \in X$ we have

$$|(\alpha \cdot f)(x)| = |\alpha \cdot f(x)| = |\alpha| \cdot |f(x)| \leq |\alpha| \cdot M' < |\alpha| \cdot M' + 1 = M.$$

So we have

$$\alpha \cdot f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{C}]$$

□

We are now ready to prove that $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ is a vector space.

Theorem 21.423. Let $p \in [1, \infty]$, $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and $\langle X, \mathcal{A}, \mu \rangle$ a measure space then if we define

$$+ : \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \times \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ where } \forall x \in X \ (f + g)(x) = f(x) + g(x)$$

$$\cdot : \mathbb{K} \times \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ where } \forall x \in X \ (\alpha \cdot f)(x) = \alpha \cdot f(x)$$

we have that

$$\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle \text{ is a vector space over the field } \mathbb{K} \text{ with neutral element } C_0$$

Proof. By [example: 21.415] $C_0 \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ so that

$$\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}] \neq \emptyset$$

As further $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \subseteq \mathbb{K}^X$ it follows from [theorem: 11.69] and [theorem: 11.61] together with [theorems: 21.419, 21.422] that

$$\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle \text{ is a vector space over the field } \mathbb{K} \text{ with neutral element } C_0 \quad \square$$

Theorem 21.424. Let $p \in [1, \infty]$ and $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then $|f| \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$

Proof. For $p \in [1, \infty]$ we have two possible cases to consider:

$p = \infty$. As $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by definition that

$$f \in \mathcal{M}[X, \mathcal{A}, \mu, \mathbb{K}] \text{ and } \exists M \in \mathbb{R}^+ \text{ such that } \forall x \in X \text{ we have } |f(x)| \leq M < \infty$$

Using [theorems: 21.342, 21.402] and $|f|(X) \subseteq \mathbb{R}$ it follows that $|f| \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, further $\forall x \in X$ we have $||f(x)|| = |f(x)| \leq M$. Hence we have by definition that

$$|f| \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$$

$p \in [1, \infty[$. As $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by definition that

$$f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \Rightarrow f(X) \subseteq \mathbb{K} \text{ and } |f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

Using [theorems: 21.342, 21.402] it follows that $|f| \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, further as $||f||^p = |f|^p$ it follows that $||f||^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$. Hence

$$|f| \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}] \quad \square$$

Next we define what later will become a pseudo norm on $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$. We start with the case where $p \in [1, \infty[$.

Definition 21.425. Let $p \in [1, \infty[$ and $\langle X, \mathcal{A}, \mu \rangle$ a measure space then we define

$$\| \cdot \|_p : \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow \mathbb{R} \text{ by } \|f\|_p = \left(\int |f|^p d\mu \right)^{\frac{1}{p}}$$

Note 21.426. As $\forall f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \ (\int |f|^p d\mu)^{\frac{1}{p}} \in [0, \infty[$ [see theorem: 21.362 and definition: 19.39] and [theorem: we have that

$$\forall f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \ \|f\|_p \in [0, \infty[$$

Theorem 21.427. Let $p \in [1, \infty[$, $\langle X, \mathcal{A}, \mu \rangle$ a measure space and $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $\|f\|_p = 0$ then we have that $|f|^p = 0$ μ -a.e. and $f = 0$ μ -a.e.

Proof. Let $x \in X$ then as $|f(x)| \in [0, \infty[$ we have by [definition: 19.39] that $|f(x)|^p \in [0, \infty[$, hence we have $\|f\|^p(x) = \|f\|^p(x) = |f|^p(x)$ so that $\|f\|^p = |f|^p$. As $\|f\|_p = 0$ we have that

$$\left(\int |f|^p d\mu \right)^{\frac{1}{p}} = 0$$

so that by [theorem: 19.42(3)]

$$\int ||f|^p|d\mu = \int |f|^p d\mu = 0$$

Using then [theorem: 21.366] it follows that $|f|^p = 0$ μ -a.e. So there exist a $N \in \mathcal{A}$ with $\mu(A) = 0$ such that $\{x \in X \mid |f(x)|^p \neq 0\} \subseteq N$. Let $x \in \{x \in X \mid |f(x)|^p \neq 0\}$ then

$$f(x) \neq 0 \Rightarrow |f(x)| \neq 0 \xrightarrow{[\text{theorem: 19.42(3)}]} |f(x)|^p \neq 0$$

so that $x \in \{x \in X \mid |f(x)|^p \neq 0\} \subseteq N$, hence $\{x \in X \mid |f(x)|^p \neq 0\} \subseteq N$ which proves that

$$f = 0 \text{ } \mu\text{-a.e.}$$

□

Next we define $|||_\infty: \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow [0, \infty]$

Definition 21.428. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then N is **locally μ -null** if $\forall A \in \mathcal{A}$ with $\mu(A) < \infty$ we have that $A \cap N \in \mathcal{N}_\mu$ [see definition: 21.211].

A trivial example is of a locally μ -null set is the empty set.

Example 21.429. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $\emptyset$ is locally μ -null.

Proof. Let $A \in \mathcal{A}$ then $A \cap \emptyset = \emptyset \in \mathcal{A}$ with $\mu(\emptyset) = 0$ proving that $\emptyset$ is locally μ -null. □

We have the following properties for μ -null sets.

Theorem 21.430. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space then

1. Every μ -negligible set is locally μ -null. In other words $\forall N \in \mathcal{N}_\mu$ we have that N is locally μ -null.
2. If B is locally μ -null and $A \subseteq B$ then A is locally μ -null.
3. If I is a countable set [finite or denumerable] and $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ a countable family of locally μ -null sets then $\bigcup_{i \in I} A_i$ is locally μ -null.

Proof.

1. Let $N \in \mathcal{N}_\mu$ then there exist a $A \in \mathcal{A}$ with $\mu(A) = 0$ such that $N \subseteq A$. Let $B \in \mathcal{A}$ with $\mu(B) < \infty$ then $N \cap B \subseteq N \subseteq A$ so that $N \cap B \in \mathcal{N}_\mu$. So we have by definition that N is locally μ -null.
2. Let $D \in \mathcal{A}$ with $\mu(D) < \infty$ then as B is locally μ -null we have that $B \cap D \in \mathcal{N}_\mu$. Hence there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $B \cap D \subseteq N$. As $A \subseteq B$ we have that $A \cap D \subseteq B \cap D \subseteq N$ which proves that $A \cap D \in \mathcal{N}_\mu$. Hence A is locally μ -null.
3. Let $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ be a countable family of locally μ -null sets. Let $A \in \mathcal{A}$ with $\mu(A) < \infty$. Given $i \in I$ we have, as A_i is locally μ -null, that $A \cap A_i \in \mathcal{N}_\mu$, so by [theorem: 21.214] we have that $\bigcup_{i \in I} (A \cap A_i) \in \mathcal{N}_\mu$. So that

$$A \cap \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (A \cap A_i) \in \mathcal{N}_\mu$$

which proves that

$$\bigcup_{i \in I} A_i \text{ is locally } \mu\text{-null}$$

□

We are now ready to define $|||_\infty$

Definition 21.431. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we define

$$|||_\infty: \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow \mathbb{R} \text{ by } \|f\|_\infty = \inf(\{M \in [0, \infty] \mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\})$$

Note 21.432. As $\{M \in [0, \infty] \mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\} \subseteq [0, \infty[$ we have that

$$\forall f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \quad \|f\|_\infty \in [0, \infty[$$

Proof. Of course for this definition to make sense the $\inf$ must exist and be finite. Let $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ then $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ and there exist a $M \in \mathbb{R}^+$ such that $\forall x \in X$ we have $|f(x)| \leq M$. From this it follows that $\{x \in X \mid |f(x)| > M\} = \emptyset$ which is a locally μ -null set [see example: 21.429]. So we have proved that

$$M \in \{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}$$

from which it follows that

$$\inf(\{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}) \text{ exist}$$

and

$$0 \leq \inf(\{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}) \leq M$$

which proves, as $M < \infty$, that

$$\inf(\{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}) \in [0, \infty[. \quad \square$$

Next we prove Holder's Inequality that essential says that for $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$ and $f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have

$$f \cdot g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$$

and

$$\int |f \cdot g| d\mu \leq \|f\|_p \cdot \|g\|_q$$

This inequality will then be used to prove the Minkowsky inequality

$$\forall p \in [1, \infty] \text{ and } f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } \|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Lemma 21.433. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space and $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ then

$$\{x \in X \mid |f(x)| > \|f\|_\infty\} \text{ is locally } \mu\text{-null}$$

Proof. Let $n \in \mathbb{N}$ then, as $\|f\|_\infty \in \mathbb{R}$, we have

$$\inf(\{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}) = \|f\|_\infty < \|f\|_\infty + \frac{1}{n}$$

Hence there exist a M_n such that $\{x \in X \mid |f(x)| > M_n\}$ is locally μ -null with

$$\|f\|_\infty \leq M_n < \|f\|_\infty + \frac{1}{n}$$

So we have by a consequence of the Axiom of Choice [theorem: 3.108] a $\{M_n\}_{n \in \mathbb{N}} \subseteq [0, \infty[$ such that

$$\forall n \in \mathbb{N} \{x \in X \mid |f(x)| > M_n\} \text{ is local } \mu\text{-null and } \|f\|_\infty \leq M_n < \|f\|_\infty + \frac{1}{n} \quad (21.657)$$

If $x \in \{x \in X \mid |f(x)| > \|f\|_\infty\}$ then $\|f\|_\infty < |f(x)|$ so that using [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $\|f\|_\infty < \|f\|_\infty + \frac{1}{n} < |f(x)|$, so $M_n < |f(x)|$ from which we deduce that $x \in \{x \in X \mid |f(x)| > M_n\} \subseteq \bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| > M_n\}$. Hence we have

$$\{x \in X \mid |f(x)| > \|f\|_\infty\} \subseteq \bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| > M_n\} \quad (21.658)$$

If $x \in \bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| > M_n\}$ then there exist a $n \in \mathbb{N}$ such that $x \in \{x \in X \mid |f(x)| > M_n\} \Rightarrow M_n < |f(x)|$, hence by [eq: 21.657] $\|f\|_\infty \leq M_n < |f(x)|$ so that $x \in \{x \in X \mid |f(x)| > \|f\|_\infty\}$ which proves that $\bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| > M_n\} \subseteq \{x \in X \mid |f(x)| > \|f\|_\infty\}$. Combining this with [eq: 21.658] proves

$$\{x \in X \mid |f(x)| > \|f\|_\infty\} = \bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| > M_n\}$$

As by [eq: 21.657] we have $\forall n \in \mathbb{N}$ that $\{x \in X \mid |f(x)| > M_n\}$ is locally μ -null, it follows from [theorem: 21.430] that $\bigcup_{n \in \mathbb{N}} \{x \in X \mid |f(x)| > M_n\}$ is locally μ -finite, hence we have that

$$\{x \in X \mid |f(x)| > \|f\|_\infty\} \text{ is locally } \mu\text{-finite} \quad \square$$

The following lemma and theorem will be crucial for the properties of $\|\cdot\|_p$.

Lemma 21.434. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ then $\{x \in X \mid f(x) \neq 0\}$ is σ -finite under μ . [see definition: 21.103].*

Proof. Let $n \in \mathbb{N}$ and define

$$A_n = \left\{x \in X \mid |f|(x) \geq \frac{1}{n}\right\} = \left\{|f| \geq \frac{1}{n}\right\}$$

As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have by [theorem: 21.362] that $|f| \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ hence $|f| \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ so that by [theorem: 21.254]

$$A_n = \left\{|f| \geq \frac{1}{n}\right\} \in \mathcal{A}$$

As $|f| \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have that

$$\int^+ |f| d\mu \underset{[\text{theorem: 21.344(7)}]}{=} \int^+ |f|^+ d\mu < \infty$$

Using [lemma: 21.334] we have further that

$$\mu(A_n) = \mu\left(\left\{|f| \geq \frac{1}{n}\right\}\right) \leq \frac{1}{\frac{1}{n}} \cdot \int^+ |f| d\mu = n \cdot \int^+ |f| d\mu < \infty$$

To summarize

$$\{A_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A} \text{ and } \forall n \in \mathbb{N} \text{ we have } \mu(A_n) < \infty \quad (21.659)$$

Let $x \in \{x \in X \mid f(x) \neq 0\}$ then $0 < |f(x)|$ so that by [theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < |f(x)|$, hence $x \in A_n$. This proves that

$$\{x \in X \mid f(x) \neq 0\} \subseteq \bigcup_{n \in \mathbb{N}} A_n \quad (21.660)$$

Further if $x \in \bigcup_{n \in \mathbb{N}} A_n$ then there exist a $n \in \mathbb{N}$ such that $x \in A_n$. Then $0 < \frac{1}{n} \leq |f(x)| \Rightarrow f(x) \neq 0$ so that $x \in \{x \in X \mid f(x) \neq 0\}$. So it follows that $\bigcup_{n \in \mathbb{N}} A_n \subseteq \{x \in X \mid f(x) \neq 0\}$ which combined with [eq: 21.660] proves that

$$\{x \in X \mid f(x) \neq 0\} = \bigcup_{n \in \mathbb{N}} A_n$$

Hence by [definition: 21.103] $\{x \in X \mid f(x) \neq 0\}$ is μ -finite under μ . $\square$

Theorem 21.435. (Holder's Inequality) *Let $p, q \in [1, \infty]$ be such that $\frac{1}{p} + \frac{1}{q} = 1$, $\langle X, \mathcal{A}, \mu \rangle$ a measure space and $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$, $g \in \mathcal{L}^q[X, \mathcal{A}, \mu, \mathbb{K}]$ then $f \cdot g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$ and*

$$\int |f \cdot g| d\mu \leq \|f\|_p \cdot \|g\|_q$$

Proof. For p, q we must consider the following cases:

$p = \infty$. Then as $1 = \frac{1}{p} + \frac{1}{q} \underset{[\text{definition: 20.21}]}{=} 0 + \frac{1}{q} = \frac{1}{q}$ we have that $q = 1$ [if $q = -\infty, \infty$ then $\frac{1}{q} = 0 \neq 1$ so we must have that $q = (1)^{-1} = 1$], so

$$g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}] \underset{[\text{theorem: 21.414}]}{=} \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.661)$$

Using then [theorem: 21.362, 21.410] it follows that

$$|g| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.662)$$

By [theorem: 21.434] it follows that $\{x \in X \mid |g(x)| \neq 0\}$ is μ -finite, so there exist a $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \mu(A_i) < \infty$ such that

$$A = \{x \in X \mid |g(x)| \neq 0\} \subseteq \bigcup_{i \in \mathbb{N}} A_i \quad (21.663)$$

Further as $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by [lemma: 21.433] that

$$B = \{x \in X \mid |f(x)| > \|f\|_\infty\} \text{ is locally } \mu\text{-null} \quad (21.664)$$

Using the definitions of A and B we have

$$\begin{aligned} A \cap B &\stackrel{[\text{eq: 21.663}]}{\subseteq} \left(\bigcup_{i \in \mathbb{N}} A_i \right) \cap B \\ &= \bigcup_{i \in \mathbb{N}} (A_i \cap B) \\ &= \bigcup_{i \in \mathbb{N}} (A_i \cap \{x \in X \mid |f(x)| > \|f\|_\infty\}) \end{aligned} \quad (21.665)$$

As $B = \{x \in X \mid |f(x)| > \|f\|_\infty\}$ is locally μ -null and $\mu(A_i) < \infty$ we have that $A_i \cap \{x \in X \mid |f(x)| > \|f\|_\infty\} \in \mathcal{N}_\mu$. So there exist a $N_i \in \mathcal{A}$ with $\mu(N_i) = 0$ such that $A_i \cap \{x \in X \mid |f(x)| > \|f\|_\infty\} \subseteq N_i$. Substituting this in [eq: 21.665] gives

$$\{x \in X \mid |g(x)| \neq 0\} \cap \{x \in X \mid |f(x)| > \|f\|_\infty\} = A \cap B \subseteq \bigcup_{i \in \mathbb{N}} N_i \quad (21.666)$$

and

$$\mu\left(\bigcup_{i \in \mathbb{N}} N_i\right) \stackrel{[\text{theorem: 21.212}]}{=} 0 \quad (21.667)$$

If $x \in \{x \in X \mid |f \cdot g|(x) > \|f\|_\infty \cdot |g(x)|\}$ then $|f(x)| \cdot |g(x)| > \|f\|_\infty \cdot |g(x)|$ and we must have that $|g(x)| \neq 0$ [otherwise we have $0 = |f(x)| \cdot |g(x)| > \|f\|_\infty \cdot |g(x)| = 0$ leading to the contradiction $0 > 0$], hence by dividing the inequality by $|g(x)|$ we have $|f(x)| > \|f\|_\infty$. So $x \in \{x \in X \mid |g(x)| \neq 0\} \cap \{x \in X \mid |f(x)| > \|f\|_\infty\}$ proving that

$$\{x \in X \mid |f \cdot g|(x) > \|f\|_\infty \cdot |g(x)|\} \subseteq \{x \in X \mid |g(x)| \neq 0\} \cap \{x \in X \mid |f(x)| > \|f\|_\infty\}$$

Using [eq: 21.666] it follows that

$$\{x \in X \mid |f \cdot g|(x) > \|f\|_\infty \cdot |g(x)|\} \subseteq \bigcup_{i \in \mathbb{N}} N_i$$

which as $\mu(\bigcup_{i \in \mathbb{N}} N_i) = 0$ [see eq: 21.667] proves that

$$|f \cdot g| \leq \|f\|_\infty \cdot |g| \text{ } \mu\text{-a.e.} \quad (21.668)$$

As $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ and $g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$ we have that $f, g \in \mathcal{M}[X, \mathcal{A}, \mu, \mathbb{K}]$, using then [theorems: 21.265, 21.401] and [lemma: 21.416] we have that

$$|f \cdot g| = |f \cdot g|^1 \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.669)$$

By [eq: 21.662] $|g| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ so that by [theorems: 21.263, 21.409] $\|f\|_\infty \cdot |g| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$. Using then [eq: 21.668] together with [theorem: 21.365] and the fact that by [eq: 21.669] $|f \cdot g|(X) \subseteq \mathbb{R}$ proves that

$$|f \cdot g|^1 = |f \cdot g| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \int |f \cdot g| d\mu \leq \int \|f\|_\infty \cdot |g| d\mu = \|f\|_\infty \cdot \int |g| d\mu \quad (21.670)$$

Hence we have by definition that

$$f \cdot g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$$

As $\|g\|_q \stackrel{=}{=} \|g\|_1 = (\int |g|^1 d\mu)^{\frac{1}{1}} = \int |g| d\mu$ it follows from [eq: 21.670] that

$$\int |f \cdot g| d\mu \leq \|f\|_\infty \cdot \|g\|_q = \|f\|_p \cdot \|g\|_q$$

$q = \infty$. This reduces to the case $p = \infty$ if we switch p with q and f with g .

$p \neq \infty \wedge q \neq \infty$. Let $x \in X$ then we have either:

$$f(x) = 0 \vee g(x) = 0. \text{ Then } |f(x) \cdot g(x)| = 0 \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}$$

$f(x) \neq 0 \wedge g(x) \neq 0$. Using Young's inequality [see 19.58] we have that

$$|f(x) \cdot g(x)| = |f(x)| \cdot |g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}$$

from which it follows that

$$\forall x \in X \quad |f(x) \cdot g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q} \quad (21.671)$$

As $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$, $g \in \mathcal{L}^q[X, \mathcal{A}, \mu, \mathbb{K}]$ it follows that $|f|^p, |g|^q \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$. Hence using [theorem: 21.360] we have that $\frac{|f|^p}{p} + \frac{|g|^q}{q} \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$. Further as $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ [see definition: 21.413] we have by [theorems: 21.265, 21.342, 21.401, 21.402] that $|f \cdot g| \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{K}]$ [so that $|f \cdot g|(X) \subseteq [0, 1] \subseteq \mathbb{R}$]. Using then [eq: 21.671] together with [theorem: 21.365] proves that

$$|f \cdot g|^1 = |f \cdot g| \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ so that by definition } f \cdot g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.672)$$

and

$$\int |f \cdot g| d\mu \leq \int \left(\frac{|f|^p}{p} + \frac{|g|^q}{q} \right) d\mu = \frac{1}{p} \cdot \int |f|^p d\mu + \frac{1}{q} \cdot \int |g|^q d\mu \quad (21.673)$$

We examine now the case where $\|f\|_p = 1 = \|g\|_q$. In this case we have that

$$\begin{aligned} 1_{[\text{theorem: 19.42}]} &= 1^p = (\|f\|_p)^p = \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} \right)^p \stackrel{[\text{theorem: 19.42}]}{=} \int |f|^p d\mu \\ 1_{[\text{theorem: 19.42}]} &= 1^q = (\|g\|_q)^q = \left(\left(\int |g|^q d\mu \right)^{\frac{1}{q}} \right)^q \stackrel{[\text{theorem: 19.42}]}{=} \int |g|^q d\mu \end{aligned}$$

Substituting this in [eq: 21.673] gives

$$\int |f \cdot g| d\mu \leq \frac{1}{p} + \frac{1}{q} = 1$$

To summarize we have

If $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$, $g \in \mathcal{L}^q[X, \mathcal{A}, \mu, \mathbb{K}]$ then $f \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ and if $\|f\|_p = 1 = \|g\|_q$ then $\int |f \cdot g| d\mu \leq 1 = \|f\|_p \cdot \|g\|_q$ (21.674)

proving the theorem in this special case. Now for the general case we have for $\|f\|_p, \|g\|_q$ the following possible cases:

$\|f\|_p \neq 0 \wedge \|g\|_q \neq 0$. Take then $f' = \frac{f}{\|f\|_p}$ and $g' = \frac{g}{\|g\|_q}$ so that by [theorem: 21.419] $f' \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$, $g' \in \mathcal{L}^q[X, \mathcal{A}, \mu, \mathbb{K}]$. Further we have $\forall x \in X$ that

$$\begin{aligned} \left| \frac{f}{\|f\|_p} \right|^p(x) &= \left(\frac{|f(x)|}{\|f\|_p} \right)^p \stackrel{[\text{theorem: 19.42}]}{=} \left(\frac{|f(x)|^p}{(\|f\|_p)^p} \right) = \left(\frac{|f|^p}{(\|f\|_p)^p} \right)(x) \\ \left| \frac{g}{\|g\|_q} \right|^q(x) &= \left(\frac{|g(x)|}{\|g\|_q} \right)^q \stackrel{[\text{theorem: 19.42}]}{=} \left(\frac{|g(x)|^q}{(\|g\|_q)^q} \right) = \left(\frac{|g|^q}{(\|g\|_q)^q} \right)(x) \end{aligned}$$

which proves that

$$\left| \frac{f}{\|f\|_p} \right|^p = \frac{|f|^p}{(\|f\|_p)^p} \wedge \left| \frac{g}{\|g\|_q} \right|^q = \frac{|g|^q}{(\|g\|_q)^q}$$

so that

$$\begin{aligned}
 \|f'\|_p &= \left(\int \left| \frac{f}{\|f\|_p} \right|^p d\mu \right)^{\frac{1}{p}} \\
 &= \left(\int \frac{|f|^p}{(\|f\|_p)^p} d\mu \right)^{\frac{1}{p}} \\
 &= \left(\left(\frac{1}{\|f\|_p} \right)^p \cdot \int |f|^p d\mu \right)^{\frac{1}{p}} \\
 &\stackrel{[\text{theorem: 19.42}]}{=} \frac{1}{\|f\|_p} \cdot \left(\int |f|^p d\mu \right)^{\frac{1}{p}} \\
 &= \frac{1}{\|f\|_p} \cdot \|f\|_p \\
 &= 1 \\
 \|g'\|_q &= \left(\int \left| \frac{g}{\|g\|_q} \right|^q d\mu \right)^{\frac{1}{q}} \\
 &= \left(\int \frac{|g|^q}{(\|g\|_q)^q} d\mu \right)^{\frac{1}{q}} \\
 &= \left(\left(\frac{1}{\|g\|_q} \right)^q \cdot \int |g|^q d\mu \right)^{\frac{1}{q}} \\
 &\stackrel{[\text{theorem: 19.42}]}{=} \frac{1}{\|g\|_q} \cdot \left(\int |g|^q d\mu \right)^{\frac{1}{q}} \\
 &= \frac{1}{\|g\|_q} \cdot \|g\|_q \\
 &= 1
 \end{aligned}$$

So we can apply [eq: 21.674] giving

$$f' \cdot g' \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}] \text{ and } \int |f' \cdot g'| d\mu \leq 1 \quad (21.675)$$

Using [theorem: 21.419] we have $f \cdot g = (\|f\|_p \cdot \|g\|_q) \cdot (f' \cdot g') \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$ and

$$\int (f \cdot g) d\mu = \|f\|_p \cdot \|g\|_q \cdot \int (f' \cdot g') d\mu \stackrel{[\text{eq: 21.675}]}{\leq} \|f\|_p \cdot \|g\|_q \cdot 1$$

in other words

$$f \cdot g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}] \text{ and } \int (f \cdot g) d\mu \leq \|f\|_p \cdot \|g\|_q$$

$\|f\|_p = 0$. Then by [theorem: 21.427] we have $f = 0$ μ -a.e so there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in X \mid f(x) \neq 0\} \subseteq N$. If $x \in \{x \in X \mid f \cdot g(x) \neq 0\}$ then $f(x) \neq 0$ [otherwise $|f \cdot g|(x) = |f(x) \cdot g(x)| = |0| = 0$] so that $x \in \{x \in X \mid f(x) \neq 0\} \subseteq N$. Hence $\{x \in X \mid f \cdot g(x) \neq 0\} \subseteq N$ which proves that $|f \cdot g| = 0$ μ -a.e.. Using [theorem: 21.358] it follows that

$$\int |f \cdot g| d\mu = 0 \leq \|f\|_p \cdot \|g\|_q$$

$\|g\|_q = 0$. Then by [theorem: 21.427] we have $g = 0$ μ -a.e so there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in X \mid g(x) \neq 0\} \subseteq N$. If $x \in \{x \in X \mid f \cdot g(x) \neq 0\}$ then $g(x) \neq 0$ [otherwise $|f \cdot g|(x) = |f(x) \cdot g(x)| = |0| = 0$] so that $x \in \{x \in X \mid g(x) \neq 0\} \subseteq N$. Hence $\{x \in X \mid f \cdot g(x) \neq 0\} \subseteq N$ which proves that $|f \cdot g| = 0$ μ -a.e.. Using [theorem: 21.358] it follows that

$$\int |f \cdot g| d\mu = 0 \leq \|f\|_p \cdot \|g\|_q$$

So in all cases we have that by [eq: 21.672]

$$f \cdot g \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}]$$

$$\int |f \cdot g| d\mu \leq \|f\|_p \cdot \|g\|_q$$

which proves the theorem for the case where $p \neq \infty \wedge q \neq \infty$. $\square$

Theorem 21.436. (Minkowski's Inequality) Let $p \in [1, \infty]$, $\langle X, \mathcal{A}, \mu \rangle$ a measure space then we have $\forall f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ that $\|f + g\|_p \leq \|f\|_p + \|g\|_p$

Note 21.437. Using [theorem: 21.423] $f + g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ so that $\|f + g\|_p$ is well defined.

Proof. As $f + g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by definition that

$$f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \text{ and } |f + g|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.676)$$

For $p \in [1, \infty]$ we have the following possible cases:

$p = \infty$. Define $F = \{x \in X \mid |f(x)| > \|f\|_\infty\}$ and $G = \{x \in X \mid |g(x)| \geq \|g\|_\infty\}$ then using [lemma: 21.433] we have that F is locally μ -null and G is locally μ -null so that by [theorem: 21.430]

$$F \cup G \text{ is locally } \mu\text{-null} \quad (21.677)$$

Let $x \in \{x \in X \mid |(f + g)(x)| > \|f\|_\infty + \|g\|_\infty\}$ then

$$\|f\|_\infty + \|g\|_\infty < |(f + g)(x)| \leq |f(x)| + |g(x)|$$

Assume that $|f(x)| \leq \|f\|_\infty \wedge |g(x)| \leq \|g\|_\infty$ then we would have $|f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty < |f(x)| + |g(x)|$ a contradiction, Hence the assumption is wrong and we must have $\|f\|_\infty < |f(x)| \vee \|g\|_\infty < |g(x)|$ so that $x \in F \cup G$. Thus it is proved that

$$\{x \in X \mid |(f + g)(x)| > \|f\|_\infty + \|g\|_\infty\} \subseteq F \cup G$$

Using the above, [eq: 21.677] together with [theorem: 21.430] proves that

$$\{x \in X \mid |(f + g)(x)| > \|f\|_\infty + \|g\|_\infty\} \text{ is locally } \mu\text{-null}$$

As by [definition: 21.431] $\|f\|_\infty + \|g\|_\infty \in [0, \infty[$ it follows that

$$\|f\|_\infty + \|g\|_\infty \in \{M \in [0, \infty[\mid \{x \in X \mid |(f + g)(x)| > M\} \text{ is locally } \mu\text{-null}\}$$

so that

$$\begin{aligned} \|f + g\|_\infty & \stackrel{[\text{definition: 21.431}]}{=} \inf(\{M \in [0, \infty[\mid \{x \in X \mid |(f + g)(x)| > M\} \text{ is locally } \mu\text{-null}\}) \\ & \leq \|f\|_\infty + \|g\|_\infty \end{aligned}$$

Hence we have proved that

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

$p = 1$. Given $x \in X$ we have that $|f + g|(x) = |f(x) + g(x)| \leq |f(x)| + |g(x)|$ so that by [theorem: 21.360] we have $\int |f + g| d\mu \leq \int |f| d\mu + \int |g| d\mu$. Hence we have

$$\begin{aligned} \|f + g\|_1 &= \left(\int |f + g|^1 d\mu \right)^{\frac{1}{1}} \\ &= \int |f + g| d\mu \\ &\leq \int |f| d\mu + \int |g| d\mu \\ &= \left(\int |f|^1 d\mu \right)^{\frac{1}{1}} + \left(\int |g|^1 d\mu \right)^{\frac{1}{1}} \\ &= \|f\|_1 + \|g\|_1 \end{aligned}$$

proving that

$$\|f + g\|_p \leq \|f\|_p$$

$p \in]1, \infty[$. Then $0 < p - 1$. Let $q = \frac{p}{p-1}$ then $\frac{1}{p} + \frac{1}{q} = \frac{1}{p} + \frac{p-1}{p} = \frac{p}{p} = 1$ and

$$(p-1) \cdot q = p \quad (21.678)$$

So it follows that $(|f + g|^{p-1})^q \stackrel{[\text{theorem: 19.42}]}{=} |f + g|^{(p-1) \cdot q} = |f + g|^p$. So using [eq: 21.676] it follows that $(|f + g|^{p-1})^q \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ so that

$$|f + g|^{p-1} \in \mathcal{L}^q[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.679)$$

Now as $f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$, $\frac{1}{p} + \frac{1}{q} = 1$ we have using the Holder's inequality [see theorem: 21.435] together with the above that

$$f \cdot |f + g|^{p-1} \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}] \wedge g \cdot |f + g|^{p-1} \in \mathcal{L}^1[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.680)$$

$$\int |f \cdot |f + g|^{p-1}| d\mu \leq \|f\|_p \cdot \| |f + g|^{p-1} \|_q \quad (21.681)$$

$$\int |g \cdot |f + g|^{p-1}| d\mu \leq \|g\|_p \cdot \| |f + g|^{p-1} \|_q \quad (21.682)$$

Further we have $\forall x \in X$ that

$$\begin{aligned} |f + g|^p(x) &= |f(x) + g(x)|^p \\ &\stackrel{[\text{theorem: 19.42}]}{=} |f(x) + g(x)| \cdot |f(x) + g(x)|^{p-1} \\ &\leq (|f(x)| + |g(x)|) \cdot |f(x) + g(x)|^{p-1} \\ &= |f(x)| \cdot |f(x) + g(x)|^{p-1} + |g(x)| \cdot |f(x) + g(x)|^{p-1} \\ &= (|f| \cdot |f + g|^{p-1} + |g| \cdot |f + g|^{p-1})(x) \end{aligned}$$

Using [theorem: 21.360] we have then

$$\begin{aligned} \int |f + g|^p d\mu &\leq \int (|f| \cdot |f + g|^{p-1} + |g| \cdot |f + g|^{p-1}) d\mu \\ &\stackrel{[\text{theorem: 21.360}]}{=} \int (|f| \cdot |f + g|^{p-1}) d\mu + \int (|g| \cdot |f + g|^{p-1}) d\mu \\ &\stackrel{[\text{eqs: 21.681, 21.682}]}{\leq} \|f\|_p \cdot \| |f + g|^{p-1} \|_q + \|g\|_p \cdot \| |f + g|^{p-1} \|_q \\ &= (\|f\|_p + \|g\|_p) \cdot \| |f + g|^{p-1} \|_q \\ &= (\|f\|_p + \|g\|_p) \cdot \left(\int (|f + g|^{p-1})^q d\mu \right)^{\frac{1}{q}} \\ &\stackrel{[\text{theorem: 19.42}]}{=} (\|f\|_p + \|g\|_p) \cdot \left(\int |f + g|^{(p-1) \cdot q} d\mu \right)^{\frac{1}{q}} \\ &\stackrel{[\text{eq: 21.678}]}{=} (\|f\|_p + \|g\|_p) \cdot \left(\int |f + g|^p d\mu \right)^{\frac{1}{q}} \end{aligned}$$

so that

$$\int |f + g|^p d\mu \leq (\|f\|_p + \|g\|_p) \cdot \left(\int |f + g|^p d\mu \right)^{\frac{1}{q}} \quad (21.683)$$

For $\int |f + g|^p d\mu$ we have either:

$\int |f + g|^p d\mu = 0$. Then we have

$$\|f + g\|_p = \left(\int |f + g|^p d\mu \right)^{\frac{1}{p}} = 0^{\frac{1}{p}} \stackrel{[\text{eq: 19.42}]}{=} 0 \leq \|f\|_p + \|g\|_p$$

so that

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

$\int |f + g|^p d\mu \neq 0$. Then by [definition: 19.39] we have $(\int |f + g|^p d\mu)^{\frac{1}{q}} \neq 0$ so that using [eq: 21.683] we have

$$\begin{aligned} \|f\|_p + \|g\|_p &\geq \frac{\int |f + g|^p d\mu}{(\int |f + g|^p d\mu)^{\frac{1}{q}}} \\ &\stackrel{[\text{eq: 19.42}]}{=} \left(\int |f + g|^p d\mu \right)^{1 - \frac{1}{q}} \\ &= \left(\int |f + g|^p d\mu \right)^{\frac{1}{p}} \\ &= \|f + g\|_p \end{aligned}$$

hence

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

So in all cases we have

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p \quad \square$$

We can use mathematical induction to extend the above to a finite sum.

Theorem 21.438. Let $p \in [1, \infty]$, $n \in \mathbb{N}$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then $\sum_{i=1}^n f_i \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and $\|\sum_{i=1}^n f_i\|_p \leq \sum_{i=1}^n \|f_i\|_p$.

Proof. This is easily proved by mathematical induction. So let

$$\mathcal{S} = \left\{ n \in \mathbb{N} \mid \text{If } \{f_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ then } \sum_{i=1}^n f_i \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ and } \left\| \sum_{i=1}^n f_i \right\|_p \leq \sum_{i=1}^n \|f_i\|_p \right\}$$

then we have:

$1 \in \mathcal{S}$. If $\{f_i\}_{i \in \{1\}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then $\sum_{i=1}^1 f_i = f_1 \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and

$$\left\| \sum_{i=1}^1 f_i \right\|_p = \|f_1\|_p = \sum_{i=1}^1 \|f_i\|_p$$

proving that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n + 1 \in \mathcal{S}$. Let $\{f_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then we have, as $n \in \mathcal{S}$, that $\sum_{i=1}^n f_i \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$. Hence using [theorem: 21.423] it follows that

$$\sum_{i=1}^{n+1} f_i = \sum_{i=1}^n f_i + f_{n+1} \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

and

$$\begin{aligned} \left\| \sum_{i=1}^{n+1} f_i \right\|_p &= \left\| \sum_{i=1}^n f_i + f_{n+1} \right\|_p \\ &\stackrel{[\text{theorem: 21.436}]}{\leq} \left\| \sum_{i=1}^n f_i \right\|_p + \|f_{n+1}\|_p \\ &\stackrel{n \in \mathcal{S}}{\leq} \sum_{i=1}^n \|f_i\|_p + \|f_{n+1}\|_p \\ &= \sum_{i=1}^{n+1} \|f_i\|_p \end{aligned}$$

proving that $n + 1 \in \mathcal{S}$. $\square$

We are now ready to prove that $\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_p \rangle$ is a pseudo normed space.

Theorem 21.439. Let $p \in [1, \dots, \infty]$ and $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_p \rangle$ is a pseudo normed vector space over $\mathbb{K}$.

Note 21.440. Using [theorem: 21.423] we have that $\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle$ is a vector space.

Note 21.441. Let $A \in \mathcal{A}$ be such that $\mu(A) = 0$ and $A \neq \emptyset$. As $A \neq \emptyset$ there exist a $a \in A$ so that $\mathcal{X}_{X,A}(a) \neq C_0(a)$ which proves that in this case $\mathcal{X}_{X,A} \neq C_0$. Further using [example: 21.319] we have $0 = \mu(A) = \int^+ \mathcal{X}_{X,A} d\mu = \int \mathcal{X}_{X,A} d\mu = (\int (\mathcal{X}_{X,A})^1 d\mu)^{\frac{1}{1}} = \|\mathcal{X}_{X,A}\|_1$. Hence it follows that in this case $\|\cdot\|_1$ is only a pseudo norm not a norm.

Proof. By [definitions: 21.425, 21.431] we have that

$$\|\cdot\|_p: \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow \mathbb{R} \text{ satisfies } \forall f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } \|f\|_p \in [0, \infty[\quad (21.684)$$

Further we have using the Minkowski's Inequality [see theorem: 21.436] that

$$\forall f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}] \quad \|f + g\|_p \leq \|f\|_p + \|g\|_p \quad (21.685)$$

Let $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ and $\alpha \in \mathbb{K}$. For $p \in [1, \infty]$ we have either:

$p \in [1, \infty[$. Then

$$\begin{aligned} \|\alpha \cdot f\|_p &= \left(\int |\alpha \cdot f|^p d\mu \right)^{\frac{1}{p}} \\ &= \left(\int (|\alpha| \cdot |f|)^p d\mu \right)^{\frac{1}{p}} \\ &\stackrel{[\text{theorem: 19.42}]}{=} \left(\int (|\alpha|^p \cdot |f|^p) d\mu \right)^{\frac{1}{p}} \\ &\stackrel{[\text{theorem: 21.360}]}{=} \left(|\alpha|^p \cdot \int |f|^p d\mu \right)^{\frac{1}{p}} \\ &\stackrel{[\text{theorem: 19.42}]}{=} (|\alpha|^p)^{\frac{1}{p}} \cdot \left(\int |f|^p d\mu \right)^{\frac{1}{p}} \\ &\stackrel{[\text{theorem: 19.42}]}{=} |\alpha| \cdot \left(\int |f|^p d\mu \right)^{\frac{1}{p}} \\ &= |\alpha| \cdot \|f\|_p \end{aligned}$$

proving that

$$\|\alpha \cdot f\|_p = |\alpha| \cdot \|f\|_p$$

$p = \infty$. For $\alpha \in \mathbb{K}$ we have either:

$\alpha = 0$. Then $\forall x \in X$ we have $|\alpha \cdot f|(x) = 0$ so that $\{x \in X \mid |\alpha \cdot f|(x) > 0\} = \emptyset$ and we have by [example: 21.429] that

$$\{x \in X \mid |\alpha \cdot f|(x) > 0\} \text{ is locally } \mu\text{-null}$$

Hence $0 \in \{M \in [0, \infty[\mid \{x \in X \mid |\alpha \cdot f|(x) > 0\} \text{ is locally } \mu\text{-null}\}$ so that

$$0 \leq \|\alpha \cdot f\|_\infty = \inf(\{M \in [0, \infty[\mid \{x \in X \mid |\alpha \cdot f|(x) > 0\} \text{ is locally } \mu\text{-null}\}) \leq 0$$

which proves that

$$\|\alpha \cdot f\|_\infty = 0 = |\alpha| \cdot \|f\|_\infty$$

$\alpha \neq 0$. Let $M \in [0, \infty[$ then we have

$$\begin{aligned} x \in \{x \in X \mid |(\alpha \cdot f)(x)| > M\} &\Leftrightarrow |\alpha \cdot f(x)| > M \\ &\stackrel{|\alpha| \neq 0}{\Leftrightarrow} |\alpha| \cdot |f(x)| > M \\ &\stackrel{|\alpha| \neq 0}{\Leftrightarrow} |f(x)| > \frac{M}{|\alpha|} \\ &\Leftrightarrow x \in \left\{x \in X \mid |f(x)| > \frac{M}{|\alpha|}\right\} \end{aligned}$$

proving that

$$\{x \in X \mid |(\alpha \cdot f)(x)| > M\} = \left\{x \in X \mid |f(x)| > \frac{M}{|\alpha|}\right\} \quad (21.686)$$

Define

$$\begin{aligned} A &= \{M \in [0, \infty \mid \{x \in X \mid |\alpha \cdot f(x)| > M\} \text{ is locally } \mu\text{-null}\} \\ B &= \{M \in [0, \infty \mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\} \end{aligned}$$

If $M \in A$ then $\{x \in X \mid |\alpha \cdot f(x)| > M\}$ is locally μ -null and using [eq: 21.686] we have that $\left\{x \in X \mid |f(x)| > \frac{M}{|\alpha|}\right\}$ is locally μ -null. Hence $M' = \frac{M}{|\alpha|} \in B$ or $M \in \{|\alpha| \cdot M' \mid M' \in B\}$ giving

$$A \subseteq \{|\alpha| \cdot M' \mid M' \in B\} \quad (21.687)$$

On the other hand, if $M \in \{|\alpha| \cdot M' \mid M' \in B\}$ then there exists a $M' \in B$ such that $M = |\alpha| \cdot M'$ or $\frac{M}{|\alpha|} = M' \in B$ hence $\left\{x \in X \mid |f(x)| > \frac{M}{|\alpha|}\right\}$ is locally μ -null. Using [eq: 21.686] we have then that $\{x \in X \mid |(\alpha \cdot f)(x)| > M\}$ is locally μ -null or $M \in A$, proving that $\{|\alpha| \cdot M' \mid M' \in B\} \subseteq A$. Combining this with [eq: 21.687] proves that

$$A = \{|\alpha| \cdot M' \mid M' \in B\} \quad (21.688)$$

As $\|\alpha \cdot f\|_\infty \stackrel{\text{def}}{=} \inf(A)$ we have by [eq: 21.688] that $\forall M \in B \quad \|\alpha \cdot f\|_\infty \leq |\alpha| \cdot M$ so that $\frac{\|\alpha \cdot f\|_\infty}{|\alpha|} \leq M$ hence $\frac{\|\alpha \cdot f\|_\infty}{|\alpha|} \leq \inf(B) = \|f\|_\infty$ or

$$\|\alpha \cdot f\|_\infty \leq |\alpha| \cdot \|f\|_\infty \quad (21.689)$$

Further if $M \in \{|\alpha| \cdot M' \mid M' \in B\}$ then $M = |\alpha| \cdot M'$ where $M' \in B$ so that $\frac{M}{|\alpha|} = M' \in B$ hence $\|f\|_\infty = \inf(B) \leq \frac{M}{|\alpha|}$ or $|\alpha| \cdot \|f\|_\infty \leq M$ so that

$$|\alpha| \cdot \|f\|_\infty \leq \inf(\{|\alpha| \cdot M' \mid M' \in B\}) \stackrel{[\text{eq: 21.688}]}{=} \inf(A) = \|\alpha \cdot f\|_\infty$$

which combined with [eq: 21.689] proves that

$$\|\alpha \cdot f\|_p = \|\alpha \cdot f\|_\infty = |\alpha| \cdot \|f\|_\infty = |\alpha| \cdot \|f\|_p \quad \square$$

It would be nice that $\|\cdot\|_p$ is actually a norm. The problem is that for $p \in [1, \infty[$ $\|f\|_p = 0$ is equivalent with $(\int |f|^p d\mu)^{\frac{1}{p}} = 0$ so that $\int |f|^p = 0$, however this means that $|f|^p = 0$ μ -a.e. not that $|f| = 0$. In essence we have to create a new vector space where the collection of all functions μ -a.e. to 0 is the new neutral element. To do this we consider first the sub space of $\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ consisted of functions f with $\|f\|_p = 0$ and use this sub space to construct the factor space [see theorem: 11.78]. This will allow us to define a norm on this factor space based on the pseudo norm.

Definition 21.442. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $p \in [1, \infty]$ then we define the **p -null set** $\mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ by

$$\mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}] = \{f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \mid \|f\|_p = 0\}$$

We have the following equivalences of p -null sets.

Theorem 21.443. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have

1. If $p \in [1, \infty[$ then

$$f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}] \Leftrightarrow f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \text{ and } f = 0 \text{ } \mu\text{-a.e.}$$

2. If $p = \infty$ then

$$f \in \mathcal{N}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \Leftrightarrow f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \text{ and } \{x \in X \mid f(x) \neq 0\} \text{ is locally } \mu\text{-null}$$

Proof.

1. First if $p \in [1, \infty[$ then

$$\begin{aligned} x \in \{x \in X \mid f(x) \neq 0\} &\Leftrightarrow f(x) \neq 0 \\ &\stackrel{\|\cdot\| \text{ is a norm}}{\Leftrightarrow} |f(x)| \neq 0 \\ &\stackrel{[\text{theorem: 19.42}]}{\Leftrightarrow} |f(x)|^p \neq 0 \\ &\Leftrightarrow x \in \{x \in X \mid |f(x)|^p \neq 0\} \end{aligned}$$

which proves that

$$\forall p \in [1, \infty[\text{ we have } \{x \in X \mid f(x) \neq 0\} = \{x \in X \mid |f(x)|^p \neq 0\} \quad (21.690)$$

$\Rightarrow$. As $f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have that $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and $\|f\|_p = 0$. Hence $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ and by [theorem: 21.427] $f = 0$ μ -a.e..

$\Leftarrow$. If $f = 0$ μ -a.e. then there exist a $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $\{x \in X \mid f(x) \neq 0\} \subseteq N$, using [eq: 21.690] we have $\{x \in X \mid |f(x)|^p \neq 0\} \subseteq N$ so that

$$|f|^p = 0 \text{ } \mu\text{-a.e.}$$

As $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ we have by the above, $f(X) \subseteq \mathbb{K} \Rightarrow |f|^p(X) \subseteq \mathbb{R}$ and [theorem: 21.358] that

$$|f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ and } \int |f|^p d\mu = 0 \Rightarrow \|f\|_p = \left(\int |f|^p d\mu \right)^{\frac{1}{p}} \stackrel{[\text{theorem: 19.42}]}{=} 0$$

so that $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ and $\|f\|_p = 0$ which proves that

$$f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

2. First we have

$$\begin{aligned} x \in \{x \in X \mid |f(x)| \neq 0\} &\Leftrightarrow |f(x)| \neq 0 \\ &\stackrel{0 \leq |f(x)|}{\Leftrightarrow} |f(x)| > 0 \\ &\Leftrightarrow \{x \in X \mid |f(x)| > 0\} \end{aligned}$$

so that

$$\{x \in X \mid |f(x)| \neq 0\} = \{x \in X \mid |f(x)| > 0\} \quad (21.691)$$

$\Rightarrow$. As $f \in \mathcal{N}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ we have that

$$f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$$

and $\|f\|_\infty = 0$. Using [lemma: 21.433] it follows that $\{x \in X \mid |f(x)| > \|f\|_\infty\}$ is locally μ -null. As $\|f\|_\infty = 0$ it follows then that $\{x \in X \mid |f(x)| > 0\}$ is locally μ -null or as $\{x \in X \mid |f(x)| \neq 0\} \stackrel{[\text{eq: 21.691}]}{=} \{x \in X \mid |f(x)| > 0\}$ it follows that

$$\{x \in X \mid |f(x)| \neq 0\} \text{ is locally } \mu\text{-null}$$

$\Leftarrow$. As $\{x \in X \mid |f(x)| \neq 0\}$ is locally μ -null we have by [eq: 21.691] $\{x \in X \mid |f(x)| > 0\}$ is locally μ -null. Hence

$$0 \in \{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}$$

so that $0 \leq \|f\|_\infty = \inf(\{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}) \leq 0$ which proves that

$$\|f\|_\infty = 0$$

As by the hypothesis $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ it follows from the above that

$$f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

□

We prove now that $\mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ is a sub-space of $\mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$.

Theorem 21.444. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $p \in [0, \infty]$ then $\langle \mathcal{N}[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle$ is a sub-space of $\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle$*

Proof. As $C_0 \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$, C_0 is the neutral element [see theorem: 21.423] and by [theorem: 21.439] $\|\cdot\|_p$ is a pseudo norm it follows that $\|C_0\|_p = 0$ hence

$$C_0 \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}] \Rightarrow \mathcal{N}[X, \mathcal{A}, \mu, \mathbb{K}] \neq \emptyset$$

If $f, g \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then $\|f\|_p = 0 = \|g\|_p$ so that $\|f + g\|_p \leq \|f\|_p + \|g\|_p = 0$ hence $\|f + g\|_p = 0$. So we have:

$$\forall f, g \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } f + g \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

Further if $\alpha \in \mathbb{K}$, $f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then $\|f\|_p = 0$ so that $\|\alpha \cdot f\|_p = |\alpha| \cdot \|f\|_p = 0$ which proves that

$$\forall f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}], \forall \alpha \in \mathbb{K} \text{ we have } \alpha \cdot f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

This concludes the proof that $\langle \mathcal{N}[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle$ is a sub-space of $\langle \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle$. $\square$

We use now [theorem: 11.78] to define $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$.

Definition 21.445. *Let $p \in [1, \infty]$ and $\langle X, \mathcal{A}, \mu \rangle$ a measure space then by [theorems: 21.444 and 11.77]*

$$\sim_p \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \times \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ where } f \sim_p g \text{ if } f - g \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

is a equivalence relation. For every $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we can then define the equivalence class $[f]_p$ by

$$[f]_p = \{g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] | g \sim_p f\} = \{g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] | f - g \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]\}$$

The set $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$ is then defined as the factor space $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] / \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ so that by [theorem: 11.78] we have

$$L^p[X, \mathcal{A}, \mu, \mathbb{K}] = \{[x]_p | x \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]\}$$

Using [theorem: 11.78] the following operators are well defined

$$+ : L^p[X, \mathcal{A}, \mu, \mathbb{K}] \times L^p[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow L^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ by } [f]_p + [g]_p = [f + g]_p$$

$$\cdot : \mathbb{K} \times L^p[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow L^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ by } \alpha \cdot [f]_p = [\alpha \cdot f]_p$$

and

$$\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], +, \cdot \rangle \text{ forms a vector space with neutral element } [C_0]_p$$

Next we show that we can define a norm on $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$.

Theorem 21.446. *Let $p \in [1, \infty]$ and $\langle X, \mathcal{A}, \mu \rangle$ a measure space then*

$$\|\cdot\|_{L^p} : L(X, \mathcal{A}, \mu, \mathbb{K}) \rightarrow \mathbb{R} \text{ defined by } \|x\|_{L^p} = \|f\| \text{ where } x = [f]_p$$

is a well defined function that defines a norm on $L(X, \mathcal{A}, \mu, \mathbb{K})$ making

$$\langle L^p(X, \mathcal{A}, \mu, \mathbb{K}), \|\cdot\|_{L^p} \rangle \text{ a normed space}$$

Proof. First we must prove that $\|\cdot\|_{L^p}$ is well defined. Let $f \in L^p(X, \mathcal{A}, \mu, \mathbb{K})$ then there exist a $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $x = [f]_p$. Assume that there exist another $g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $x = [g]_p$ then $[f]_p = [g]_p$ so that $f - g \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$. Hence we have that $\|f - g\|_p = 0$ so that

$$\|f\|_p - \|g\|_p \underset{\text{[theorem: 14.91]}}{\leq} \|f - g\|_p = 0$$

so that $\|f\|_p = \|g\|_p$. Hence it follows that

$$\|\cdot\|_{L^p} : L[X, \mathcal{A}, \mu, \mathbb{K}] \rightarrow \mathbb{R} \text{ defined by } \|x\|_{L^p} = \|f\| \text{ where } x = [f]_p \text{ is well defined} \quad (21.692)$$

Let $x \in L[X, \mathcal{A}, \mu, \mathbb{K}]$ then $x = [f]_p$ where $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$ and we have that $\|x\|_{L^p} = \|[f]_p\|_{L^p} = \|f\|_p \in [0, \infty[$ hence:

$$\forall x \in L[X, \mathcal{A}, \mu, \mathbb{K}] \text{ we have } \|x\|_p \in [0, \infty[\quad (21.693)$$

Next let $x, y \in L^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then there exist $f, g \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $x = [f]_p$ and $y = [g]_p$ then

$$\begin{aligned} \|x + y\|_{L^p} &= \|[f]_p + [g]_p\|_{L^p} \\ &= \|[f + g]_p\|_{L^p} \\ &= \|f + g\|_p \\ &\leq \|f\|_p + \|g\|_p \\ &= \|[f]_p\|_{L^p} + \|[g]_p\|_{L^p} \\ &= \|x\|_{L^p} + \|y\|_{L^p} \end{aligned}$$

So we have proved that

$$\forall x, y \in L^p[X, \mathcal{A}, \mu, \mathbb{K}] \quad \|x + y\|_{L^p} \leq \|x\|_{L^p} + \|y\|_{L^p} \quad (21.694)$$

Further if $\alpha \in \mathbb{K}$ and $x \in L^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then there exist a $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $x = [f]_p$ so that

$$\begin{aligned} \|\alpha \cdot x\|_{L^p} &= \|\alpha \cdot [f]_p\|_{L^p} \\ &= \|[\alpha \cdot f]_p\|_{L^p} \\ &= \|\alpha \cdot f\|_p \\ &= |\alpha| \cdot \|f\|_p \\ &= |\alpha| \cdot \|[f]_p\|_{L^p} \\ &= |\alpha| \cdot \|x\|_{L^p} \end{aligned}$$

hence we have

$$\forall \alpha \in \mathbb{K}, \forall x \in L^p[X, \mathcal{A}, \mu, \mathbb{K}] \quad \|\alpha \cdot x\|_{L^p} = |\alpha| \cdot \|x\|_{L^p} \quad (21.695)$$

Finally if $x \in L[X, \mathcal{A}, \mu, \mathbb{K}]$ is such that $\|x\|_{L^p} = 0$ then there exist a $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $x = [f]_p$ and $\|f\|_p = \|[f]_p\|_{L^p} = \|x\|_{L^p} = 0$, hence it follows that $f \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$. As further $C_0 \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have that $f - C_0 \in \mathcal{N}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ so that $x = [f]_p = [C_0]_p$ the neutral element in $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$. In other words we have

$$\forall x \in L^p[X, \mathcal{A}, \mu, \mathbb{K}] \text{ with } \|x\|_{L^p} = 0 \text{ we have } x = [C_0]_p \quad (21.696)$$

So by [eqs: 21.692, 21.693, 21.694, 21.695 and 21.696] it follows that $\|\cdot\|_{L^p}$ is a norm. $\square$

Next we prove that $\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^p} \rangle$ is actually a Banach space.

21.5.6.2 The Banach space $\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^p} \rangle$

We will prove now that the norm $\|\cdot\|_p$ is complete. First we need some lemmas.

Lemma 21.447. *Let $p \in [0, \infty[$ and $\{x_i\}_{i \in \mathbb{N}} \subseteq [0, \infty[$ such that $\lim_{i \rightarrow \infty} x_i$ exist and $\lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ then*

$$\begin{aligned} \lim_{i \rightarrow \infty} (x_i)^p \text{ exists and } \lim_{i \rightarrow \infty} (x_i)^p &= \left(\lim_{i \rightarrow \infty} x_i \right)^p \\ \lim_{i \rightarrow \infty} |x_i|^p \text{ exist and } \lim_{i \rightarrow \infty} |x_i|^p &= \left| \lim_{i \rightarrow \infty} x_i \right|^p \end{aligned}$$

Proof. As $[*]^p: [0, \infty[\rightarrow \mathbb{R}$ defined by $[*]^p(x) = x^p$ is continuous on $[0, \infty[$ [see theorem: 19.55] we have by [theorem: 14.371] that $\lim_{i \rightarrow \infty} (x_i)^p = \lim_{i \rightarrow \infty} (*)(x_i)$ exist and $\lim_{i \rightarrow \infty} (x_i)^p = \lim_{i \rightarrow \infty} (*)(x_i) = (*)^p\left(\lim_{i \rightarrow \infty} x_i\right) = \left(\lim_{i \rightarrow \infty} x_i\right)^p$. In other words

$$\forall \{x_i\}_{i \in \mathbb{N}} \subseteq [0, \infty[\text{ we have } \lim_{i \rightarrow \infty} (x_i)^p \text{ exists and } \lim_{i \rightarrow \infty} (x_i)^p = \left(\lim_{i \rightarrow \infty} x_i \right)^p \quad (21.697)$$

Further as $\lim_{i \rightarrow \infty} x_i$ exist and $\lim_{i \rightarrow \infty} x_i \in \mathbb{R}$ it follows from [theorem: 14.374] that for $\{|x_i|\}_{i \in \mathbb{N}} \subseteq [0, \infty[$ $\lim_{i \rightarrow \infty} |x_i|$ exist and $\lim_{i \rightarrow \infty} |x_i| = \left| \lim_{i \rightarrow \infty} x_i \right|$. Using then [eq: 21.697] it follows that

$$\lim_{i \rightarrow \infty} |x_i|^p \text{ exist and } \lim_{i \rightarrow \infty} |x_i|^p = \left(\lim_{i \rightarrow \infty} |x_i| \right)^p = \left| \lim_{i \rightarrow \infty} x_i \right|^p \quad \square$$

Lemma 21.448. Let $p \in [1, \infty]$, $\langle X, \mathcal{A}, \mu \rangle$ be measure space and $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ then we have $\|f\|_p = \| |f| \|_p$

Note 21.449. Using [theorem: 21.424] $|f| \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ as is required in this lemma.

Proof. For p we have either

$p = \infty$. then as $|f| = |(|f|)|$ we have that

$$\begin{aligned} \|f\|_\infty &= \inf(\{M \in [0, \infty[\mid \{x \in X \mid |f(x)| > M\} \text{ is locally } \mu\text{-null}\}) \\ &= \inf(\{M \in [0, \infty[\mid \{x \in X \mid |(|f|)(x)| > M\} \text{ is locally } \mu\text{-null}\}) \\ &= \|(|f|)\|_\infty \end{aligned}$$

$p \in]1, \infty[$. then as $|f| = |(|f|)|$ we have that

$$\begin{aligned} \|f\|_p &= \left(\int |f|^p \right)^{\frac{1}{p}} \\ &= \left(\int |(|f|)|^p \right)^{\frac{1}{p}} \\ &= \|(|f|)\|_p \end{aligned} \quad \square$$

Lemma 21.450. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $\langle L^\infty[X, \mathcal{A}, \mu, \mathbb{K}], \| \cdot \|_{L^\infty} \rangle$ is a Banach space

Proof. Using [theorem: 21.446] it follows that

$$\langle L^\infty[X, \mathcal{A}, \mu, \mathbb{K}], \| \cdot \|_{L^\infty} \rangle \text{ is a normed space}$$

So the only thing left to prove is that this space is complete. Let $\{f_i\}_{i \in \mathbb{N}} \subseteq L^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ be a Cauchy sequence then there exist a $\{F_i\}_{i \in \mathbb{N}} \subset \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ such that $\forall i \in \mathbb{N} \ f_i = [F_i]_p$. Let $\varepsilon \in \mathbb{R}^+$ then as $\{f_i\}_{i \in \mathbb{N}}$ is Cauchy there exist a $N_\varepsilon \in \mathbb{N}$ such that $\forall n, m \in \{N_\varepsilon, \dots, \infty\}$ we have $\|f_n - f_m\|_{L^\infty} < \varepsilon$. So $\|F_n - F_m\|_\infty = \|[F_n - F_m]_\infty\|_{L^\infty} = \|[F_n]_\infty - [F_m]_\infty\|_{L^\infty} = \|f_n - f_m\|_{L^\infty} < \varepsilon$ hence we have proved that

$$\forall \varepsilon \in \mathbb{R}^+ \ \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall n, m \in \{N_\varepsilon, \dots, \infty\} \text{ we have } \|F_n - F_m\|_\infty < \varepsilon \quad (21.698)$$

Let $n, m \in \mathbb{N}$ then as $F_n, F_m \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ we have that $F_n - F_m \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ so that by [lemma: 21.433]

$$A_{n,m} = \{x \in X \mid |F_n(x) - F_m(x)| > \|F_n - F_m\|_\infty\} \text{ is locally } \mu\text{-null} \quad (21.699)$$

Using [theorem: 21.430] it follow that

$$A = \bigcup_{(i,j) \in \mathbb{N} \times \mathbb{N}} A_{i,j} \text{ is locally } \mu\text{-null} \quad (21.700)$$

As $F_n - F_m \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by [definition: 21.421] that $F_n - F_m \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ so that by [theorem: 21.416 with $p=1$] $|F_n - F_m| \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ hence $|F_n - F_m|$ is $\mathcal{A}$ -measurable so that by [theorem: 21.254] $A_{n,m} = \{|F_n - F_m| > \|F_n - F_m\|_\infty\} \in \mathcal{A}$. As $\mathcal{A}$ is a σ -algebra it follows then that

$$\forall n, m \in \mathbb{N} \ A_{n,m} \in \mathcal{A} \wedge A = \bigcup_{(i,j) \in \mathbb{N} \times \mathbb{N}} A_{i,j} \in \mathcal{A} \wedge X \setminus A \in \mathcal{A} \quad (21.701)$$

Now

$$\begin{aligned} x \in X \setminus A &\Leftrightarrow \forall n, m \in \mathbb{N} \ x \notin A_{n,m} = \{x \in X \mid |F_n(x) - F_m(x)| > \|F_n - F_m\|_\infty\} \\ &\Leftrightarrow \forall n, m \in \mathbb{N} \ |F_n(x) - F_m(x)| \leq \|F_n - F_m\|_\infty \\ &\Leftrightarrow x \in \{x \in X \mid \forall n, m \in \mathbb{N} \text{ we have } |F_n(x) - F_m(x)| \leq \|F_n - F_m\|_\infty\} \end{aligned}$$

so that

$$X \setminus A = \{x \in X \mid \forall n, m \in \mathbb{N} \text{ we have } |F_n(x) - F_m(x)| \leq \|F_n - F_m\|_\infty\} \quad (21.702)$$

Let $x \in X \setminus A$ and take $\varepsilon \in \mathbb{R}^+$ then by [eq: 21.698] there exist a $N_\varepsilon \in \mathbb{N}$ such that $\forall n, m \in \{N_\varepsilon, \dots, \infty\}$ we have $|F_n(x) - F_m(x)| \leq \|F_n - F_m\|_\infty < \varepsilon$ which proves that

$$\forall x \in X \setminus A \quad \{F_n(x)\}_{n \in \mathbb{N}} \subseteq \mathbb{K} \text{ is a Cauchy sequence in } \langle \mathbb{K}, \|\cdot\| \rangle$$

As $\langle \mathbb{K}, \|\cdot\| \rangle$ is complete [see examples: 14.394, 14.397] it follows that

$$\forall x \in X \setminus A \quad \lim_{i \rightarrow \infty} F_i(x) \text{ exist} \quad (21.703)$$

Define

$$F: X \rightarrow \mathbb{K} \text{ by } F(x) = \begin{cases} \lim_{i \rightarrow \infty} F_i(x) & \text{if } x \in X \setminus A \\ 0 & \text{if } x \in A \end{cases} \quad (21.704)$$

As $\forall i \in \mathbb{N} \quad F_i \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ it follows that $F_i \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$, hence using [theorem: 21.276, 21.401] with [eq: 21.701] it follows that $\{\mathcal{X}_{X, X \setminus A} \cdot F_i\}_{i \in \mathbb{N}} \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ hence

$$\{\mathcal{X}_{X, X \setminus A} \cdot F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \quad (21.705)$$

Let $x \in X$ then we either:

$$x \in A. \text{ Then } \lim_{i \rightarrow \infty} (\mathcal{X}_{X, X \setminus A} \cdot f_i)(x) = \lim_{i \rightarrow \infty} 0 = 0 = F(x)$$

$$x \in X \setminus A. \text{ Then } \lim_{i \rightarrow \infty} (\mathcal{X}_{X, X \setminus A} \cdot f_i)(x) = \lim_{i \rightarrow \infty} f_i(x) = F(x)$$

hence $\forall x \in X$ we have that $\lim_{i \rightarrow \infty} (\mathcal{X}_{X, X \setminus A} \cdot f)(x)$ exist and $\lim_{i \rightarrow \infty} (\mathcal{X}_{X, X \setminus A} \cdot f)(x) = F(x)$. Using then [theorems: 21.269, 21.401] together with [eq: 21.705] proves that

$$F \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \quad (21.706)$$

Let $\varepsilon \in \mathbb{R}^+$ and take $x \in X \setminus A$. Let $n \in \{N_{\frac{\varepsilon}{2}}, \dots, \infty\}$ [see eq: 21.698 for the definition of $N_{\frac{\varepsilon}{2}}$] then by [eq: 21.704] there exist a $K_x \in \mathbb{N}$ such that $\forall m \in [K_x, \dots, \infty]$ we have $|F(x) - F_m(x)| < \frac{\varepsilon}{2}$. Take $N = \max(N_{\frac{\varepsilon}{2}}, K_x)$ then we have

$$\begin{aligned} |F(x) - F_n(x)| &= |F(x) - F_N(x) + F_N(x) - F_n(x)| \\ &\leq |F(x) - F_N(x)| + |F_N(x) - F_n(x)| \\ &< \frac{\varepsilon}{2} + |F_N(x) - F_n(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &\stackrel{[\text{eq: 21.698}]}{=} \varepsilon \end{aligned}$$

which proves, if we take $M_\varepsilon = N_{\frac{\varepsilon}{2}}$, that

$$\forall \varepsilon \in \mathbb{R}^+ \exists M_\varepsilon \in \mathbb{N} \text{ such that } \forall n \in \{M_\varepsilon, \dots, \infty\} \text{ we have } \forall x \in X \setminus A \text{ that } |F(x) - F_n(x)| < \varepsilon \quad (21.707)$$

As $1 \in \mathbb{R}^+$ we have by the above that $\exists M_1 \in \mathbb{N}$ such that $\forall x \in X \setminus A$ we have $|F(x) - F_{M_1}(x)| < 1$. As $F_{M_1} \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}]$ it follows from [definition: 21.421] that there exist a $M \in \mathbb{R}^+$ such that $\forall x \in X \quad |F_{M_1}(x)| \leq M$. Let $x \in X$ then we have either:

$$x \in X \setminus A. \text{ Then } |F(x)| = |F(x) - F_{M_1}(x) + F_{M_1}(x)| \leq |F(x) - F_{M_1}(x)| + |F_{M_1}(x)| < 1 + M.$$

$$x \in A. \text{ Then } |F(x)| = 0 \leq 1 + M$$

Hence F is bounded so that by [definition: 21.421] and [eq: 21.706] it follows that

$$F \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \text{ so that } f = [F]_\infty \in L^\infty[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.708)$$

Let $\varepsilon \in \mathbb{R}^+$. If $n \in \{M_{\varepsilon/2}, \dots, \infty\}$ then for $y \in \{x \in X \mid |F(x) - F_n(x)| > \frac{\varepsilon}{2}\}$ we have $y \in A$ [for if $y \in X \setminus A$ then by [eq: 21.707] $|F(y) - F_n(y)| < \frac{\varepsilon}{2}$]. Hence we have

$$\left\{x \in X \mid |F(x) - F_n(x)| > \frac{\varepsilon}{2}\right\} \subseteq A$$

As A is locally μ -null [see eq: 21.700] it follows from [theorem: 21.430] that

$$\left\{x \in X \mid |F(x) - F_n(x)| > \frac{\varepsilon}{2}\right\} \text{ is locally } \mu\text{-null}$$

So

$$\frac{\varepsilon}{2} \in \{M \in [0, \infty] \mid \{F(x) - F_n(x) > M\} \text{ is locally } \mu\text{-null}\}$$

Hence

$$\|F - F_n\|_\infty = \inf(\{M \in [0, \infty] \mid \{F(x) - F_n(x) > M\} \text{ is locally } \mu\text{-null}\}) \leq \frac{\varepsilon}{2} < \varepsilon$$

Hence we have proved that

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a } N \in \mathbb{N} \text{ such that } \forall n \in \{N, \dots, \infty\} \text{ we have } \|F - F_n\|_\infty < \varepsilon \quad (21.709)$$

Let $n \in \mathbb{N}$ then we have

$$\begin{aligned} \|f - f_n\|_{L^\infty} &\stackrel{[\text{eq: 21.708}]}{=} \|[F]_\infty - [F_n]_\infty\|_{L^\infty} \\ &= \|[F - F_n]_\infty\|_{L^\infty} \\ &= \|F - F_n\|_\infty \end{aligned}$$

which combined with [eq: 21.709] proves that

$$\forall \varepsilon \in \mathbb{R}^+ \exists N \in \mathbb{N} \text{ such that } \forall n \in \{N, \dots, \infty\} \text{ we have } \|f - f_n\|_{L^\infty} < \varepsilon$$

which proves that

$$\lim_{i \rightarrow \infty} f_i \text{ converges in } \langle L^\infty[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^\infty} \rangle$$

so that

$$\langle L^\infty[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^\infty} \rangle \text{ is complete} \quad \square$$

Lemma 21.451. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $p \in [1, \infty[$ then $\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^p} \rangle$ is a Banach space.*

Proof. Using [theorem: 21.446] it follows that

$$\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^p} \rangle \text{ is a normed space}$$

So it remains to prove that $\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], \|\cdot\|_{L^p} \rangle$ is complete. We use [theorem: 14.441], stating that a normed space is complete if and only if every absolute convergent series is convergent, to prove this lemma. So let $\{f_i\}_{i \in \mathbb{N}} \subseteq L^p[X, \mathcal{A}, \mu, \mathbb{K}]$ be a absolute convergent series. Then by definition

$$\sum_{i=1}^{\infty} \|f_i\|_{L^p} \text{ converges to a element in } [0, \infty[\subseteq \mathbb{R} \quad (21.710)$$

As $\{f_i\}_{i \in \mathbb{N}} \subseteq L^p[X, \mathcal{A}, \mu, \mathbb{K}]$ there exist a $\{F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ such that

$$\forall i \in \mathbb{N} \quad f_i = [F_i]_p \quad (21.711)$$

So $\forall i \in \mathbb{N}$ we have that $\|f_i\|_{L^p} = \|[F_i]_p\|_{L^p} \stackrel{\text{def}}{=} \|F_i\|_p$ which combined with [eq: 21.710] gives

$$\sum_{i=1}^{\infty} \|F_i\|_p \text{ converges to a element in } [0, \infty[\subseteq \mathbb{R} \quad (21.712)$$

As $\{F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by [theorem: 21.424] that $\{|F_i|\}_{i \in \mathbb{N}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ so that by [theorem: 21.438] we have

$$\left\{ \sum_{i=1}^n |F_i| \right\}_{n \in \mathbb{N}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.713)$$

By definition of $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ [see definition: 21.413] it follows that

$$\left\{ \left| \sum_{i=1}^n F_i \right|^p \right\}_{n \in \mathbb{N}} \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

As $\forall n \in \mathbb{N}$ we have $\forall x \in X$ that $(\sum_{i=1}^n |F_i|)(x) = \sum_{i=1}^n |F_i(x)| \geq 0$ we have also

$$\left\{ \left(\sum_{i=1}^n |F_i| \right)^p \right\}_{n \in \mathbb{N}} = \left\{ \left| \sum_{i=1}^n F_i \right|^p \right\}_{n \in \mathbb{N}}$$

or summarized

$$\left\{ \left(\sum_{i=1}^n |F_i| \right)^p \right\}_{n \in \mathbb{N}} = \left\{ \left| \sum_{i=1}^n F_i \right|^p \right\}_{n \in \mathbb{N}} \subseteq \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.714)$$

Using [theorem: 21.352] on the above we have that

$$\forall n \in \mathbb{N} \left(\sum_{i=1}^n |F_i| \right)^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \text{ and } \int \left(\sum_{i=1}^n |F_i| \right)^p d\mu = \int^+ \left(\sum_{i=1}^n |F_i| \right)^p d\mu < \infty \quad (21.715)$$

Given $x \in X$ and $n \in \mathbb{N}$ then we have $\sum_{i=1}^n |F_i(x)| \leq |F_{n+1}(x)| + \sum_{i=1}^n |F_i(x)| = \sum_{i=1}^{n+1} |F_i(x)|$ which together with the monotony of the power function [see theorem: 19.42] proves that

$$\forall x \in X \quad \forall n \in \mathbb{N} \quad \sum_{i=1}^n |F_i(x)| \leq \sum_{i=1}^{n+1} |F_i(x)| \text{ and } \left(\sum_{i=1}^n |F_i(x)| \right)^p \leq \left(\sum_{i=1}^{n+1} |F_i(x)| \right)^p \quad (21.716)$$

All the conditions for the Monotone Convergence Theorem [see theorem: 21.372] are now satisfied by [eqs: 21.715, 21.716] so that we have

$$g: X \rightarrow \overline{\mathbb{R}} \text{ by } g(x) = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n |F_i(x)| \right)^p = \sup \left(\left\{ \left(\sum_{i=1}^n |F_i(x)| \right)^p \mid n \in \mathbb{N} \right\} \right) \text{ is well defined} \quad (21.717)$$

$$g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.718)$$

$$\lim_{n \rightarrow \infty} \int^+ \left(\sum_{i=1}^n |F_i| \right)^p d\mu = \sup \left(\left\{ \int^+ \left(\sum_{i=1}^n |F_i| \right)^p d\mu \right\} \right) \quad (21.719)$$

$$\int^+ g d\mu = \lim_{n \rightarrow \infty} \int^+ \left(\sum_{i=1}^n |F_i| \right)^p d\mu \quad (21.720)$$

Next we have for $n \in \mathbb{N}$

$$\begin{aligned} \int^+ \left(\sum_{i=1}^n |F_i| \right)^p d\mu & \stackrel{[\text{eq: 21.715}]}{=} \int \left(\sum_{i=1}^n |F_i| \right)^p d\mu \\ & \stackrel{[\text{eq: 21.714}]}{=} \int \left| \sum_{i=1}^n F_i \right|^p d\mu \\ & \stackrel{[\text{definition: 21.425}]}{=} \left(\left\| \sum_{i=1}^n F_i \right\|_p \right)^p \\ & \stackrel{[\text{theorem: 19.42}], \| \cdot \|_p \text{ is a norm}}{\leq} \left(\sum_{i=1}^n \|F_i\|_p \right)^p \\ & \stackrel{[\text{lemma: 21.448}]}{=} \left(\sum_{i=1}^n \|F_i\| \right)^p \\ & \stackrel{[\text{theorem: 19.42}]}{\leq} \left(\sup \left(\left\{ \sum_{i=1}^n \|F_i\| \mid n \in \mathbb{N} \right\} \right) \right)^p \\ & \stackrel{[\text{definition: 20.95}]}{=} \left(\sum_{i=1}^{\infty} \|F_i\|_p \right)^p \end{aligned}$$

so that we have

$$\forall n \in \mathbb{N} \text{ we have } \int^+ \left(\sum_{i=1}^n |F_i| \right)^p d\mu \leq \left(\sum_{i=1}^n \|F_i\|_p \right)^p \quad (21.721)$$

Next

$$\begin{aligned} \int^+ g d\mu & \stackrel{[\text{eq: 21.720}]}{=} \lim_{n \rightarrow \infty} \int^+ \left(\sum_{i=1}^n |F_i| \right) d\mu \\ & \stackrel{[\text{eq: 21.719}]}{=} \sup \left(\left\{ \int^+ \left(\sum_{i=1}^n |F_i| \right) d\mu \mid n \in \mathbb{N} \right\} \right) \\ & \stackrel{[\text{eq: 21.721}]}{\leq} \left(\sum_{i=1}^{\infty} \|F_i\|_p \right)^p \\ & \stackrel{[\text{eq: 21.712}]}{<} \infty \end{aligned}$$

Hence as $g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.718] we have by [theorem: 21.352] that

$$g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \quad (21.722)$$

So using [theorem: 21.370] it follows that

$$N = \{x \in X \mid |g(x)| = \infty\} \in \mathcal{A} \text{ and } \mu(N) = 0 \quad (21.723)$$

The above allows us to replace g by a real valued function by multiplying it with $\mathcal{X}_{X, X \setminus N}$. As $g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.276] that

$$\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Further as $\mu(N) = 0$ it follows from [theorem: 21.280] $g = \mathcal{X}_{X, X \setminus N} \cdot g$ μ -a.e. so that, as by [eq: 21.722] $g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$, it follows from [theorem: 21.364] that

$$\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

As $\forall x \in X$ $0 \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \stackrel{g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]}{=} \mathcal{X}_{X, X \setminus N}(x) \cdot |g(x)| = \begin{cases} 0 < \infty & \text{if } x \in N \\ |g(x)| < \infty & \text{if } x \in X \setminus N \end{cases} < \infty$ it follows that

$$\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.724)$$

Let $x \in X \setminus N$ then as $\lim_{n \rightarrow \infty} (\sum_{i=1}^n |F_i(x)|)^p \stackrel{[\text{eq: 21.717}]}{=} g(x) \stackrel{[\text{eq: 21.723}]}{\neq} \infty$ it follows from [corollary: 20.77] that $\lim_{n \rightarrow \infty} (\sum_{i=1}^n |F_i(x)|)^p$ exist in $\mathbb{R}$. Using [lemma: 21.447] proves then that $\lim_{n \rightarrow \infty} ((\sum_{i=1}^n |F_i(x)|)^p)^{\frac{1}{p}}$ exist in $\mathbb{R}$ or in other words we have that

$$\forall x \in X \setminus N \sum_{i=1}^{\infty} |F_i(x)| \text{ converges in } \mathbb{R} \quad (21.725)$$

Using a consequence of dominant convergence [see corollary: 14.437] on the above for the Banach space $\langle \mathbb{C}, || \rangle$ [or $\langle \mathbb{R}, || \rangle$] proves that

$$\forall x \in X \setminus N \sum_{i=1}^{\infty} F_i(x) = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n F_i(x) \right) \in \mathbb{K} \text{ exist} \quad (21.726)$$

Define now

$$F: X \rightarrow \mathbb{K} \text{ by } F(x) = \begin{cases} \lim_{n \rightarrow \infty} (\sum_{i=1}^n F_i(x)) & \text{if } x \in X \setminus N \\ 0 & \text{if } x \in N \end{cases} \in \mathbb{K} \quad (21.727)$$

Next we prove that $F \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$. As $\{F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ we have by [definition: 21.413] that $\{F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$, using then [theorems: 21.276, 21.401], it follows that

$$\{\mathcal{X}_{X \setminus N} \cdot F_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \quad (21.728)$$

finally by [theorems: 21.261, 21.401] we have:

$$\left\{ \sum_{i=1}^n F_i \right\}_{n \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \text{ and } \left\{ \sum_{i=1}^n (\mathcal{X}_{X, X \setminus N} \cdot F_i) \right\}_{n \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \quad (21.729)$$

If $x \in X$ then

$$\lim_{n \rightarrow \infty} \left(\sum_{i=1}^n (\mathcal{X}_{X, X \setminus N} \cdot F_i)(x) \right) = \begin{cases} \lim_{n \rightarrow \infty} \sum_{i=1}^n F_i(x) & \text{if } x \in X \setminus N \\ \lim_{n \rightarrow \infty} 0 = 0 & \end{cases} \stackrel{[\text{eq: 21.727}]}{=} F(x) \quad (21.730)$$

So using [theorems: 21.269, 21.401] on the above [eqs: 21.729, 21.730] it follows that

$$F \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \text{ and by [lemma: 21.416] that } |F|^p \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \quad (21.731)$$

Let $x \in X$ and $n \in \mathbb{N}$ then we have either:

$$x \in X \setminus N.$$

$$\begin{aligned} \left| \sum_{i=1}^n F_i(x) \right|^p & \stackrel{[\text{theorem: 19.42}]}{\leq} \left(\sum_{i=1}^n |F_i(x)| \right)^p \\ & \leq \sup \left(\left\{ \left(\sum_{i=1}^n |F_i(x)| \right)^p \mid n \in \mathbb{N} \right\} \right) \\ & \stackrel{[\text{eq: 21.717}]}{=} g(x) \end{aligned}$$

so that

$$\left| \sum_{i=1}^n F_i(x) \right|^p \leq g(x) \quad (21.732)$$

As $x \in X \setminus N$ it follows from [eq: 21.726] that $\lim_{n \rightarrow \infty} (\sum_{i=1}^n F_i(x))$ exist hence by [corollary: 14.374] we have that

$$\lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right| \text{ exist and } \lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right| = \left| \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n F_i(x) \right) \right|$$

Using [lemma: 21.447] on the above it follows that

$$\lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right|^p \text{ exist}$$

and

$$\lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right|^p = \left(\lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right| \right)^p = \left| \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n F_i(x) \right) \right|^p$$

hence

$$\lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right|^p = \left| \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n F_i(x) \right) \right|^p \quad (21.733)$$

Next we have

$$\begin{aligned} |F(x)|^p & \stackrel{[\text{eq: 21.727}] \wedge x \in X \setminus N}{=} \left| \lim_{n \rightarrow \infty} \sum_{i=1}^n F_i(x) \right|^p \\ & \stackrel{[\text{eq: 21.733}]}{=} \lim_{n \rightarrow \infty} \left| \sum_{i=1}^n F_i(x) \right|^p \\ & \stackrel{[\text{theorem: 14.363}] \wedge [\text{eq: 21.732}]}{\leq} g(x) \\ & \stackrel{x \in X \setminus N}{=} \mathcal{X}_{X, X \setminus N}(x) \cdot g(x) \\ & = (\mathcal{X}_{X, X \setminus N} \cdot g)(x) \end{aligned}$$

So we have proved that in this case that

$$|F(x)|^p \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x)$$

$x \in N$. Then $|F(x)|^p \stackrel{[\text{eq: 21.727}]}{=} 0 \leq 0 \cdot g(x) = \mathcal{X}_{X, X \setminus N}(x) * g(x) = (\mathcal{X}_{X, X \setminus N} \cdot g)(x)$ hence we have in this case that

$$|F(x)|^p \leq (\mathcal{X}_{X, X \setminus N} \cdot g)(x)$$

Hence we have proved that

$$\forall x \in X \text{ we have } 0 \leq |F(x)|^p = |F(x)|^p \leq (\mathcal{X}_{X \setminus N} \cdot g)(x)$$

As $\mathcal{X}_{X, X \setminus N} \cdot g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ [see eq: 21.724], $g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.718], $|F|^p \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ [see eq: 21.731] it follows from the above by [theorem: 21.365] that

$$|F|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

As also by [eq: 21.727] $\forall x \in \mathbb{K} \ F(x) \in \mathbb{K} \Rightarrow |F(x)| \in [0, \infty[\subseteq \mathbb{R}$ we have that

$$|F|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

which as $F \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ [see eq: 21.731] proves by [definition: 21.413] that

$$F \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}] \quad (21.734)$$

Finally we prove that $\sum_{i=1}^{\infty} f_i$ converges in $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$. Let $x \in X \setminus N$ then by [eq: 21.727] $F(x) = \lim_{n \rightarrow \infty} (\sum_{i=1}^n F_i(x))$ and it follows from [theorem: 14.346] that $\lim_{n \rightarrow \infty} |F(x) - \sum_{i=1}^n F_i(x)| = 0$. Hence, using [lemma: 21.447], it follows that

$$\lim_{n \rightarrow \infty} \left| F(x) - \sum_{i=1}^n F_i(x) \right|^p = \left(\lim_{n \rightarrow \infty} \left| F(x) - \sum_{i=1}^n F_i(x) \right| \right)^p = 0^p \stackrel{p \neq 0}{=} 0$$

which, as $\mu(N) = 0$, proves that

$$\lim_{n \rightarrow \infty} \left| F - \sum_{i=1}^n F_i \right|^p = C_0 \text{ } \mu\text{-a.e. and } C_0 \stackrel{[\text{example: 21.228}]}{\in} \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.735)$$

Further as $F, \{\sum_{i=1}^n F_i\}_{n \in \mathbb{N}} \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$ [see eqs: 21.731, 21.729], it follows from [theorems: 21.265, 21.401] that $\{F - \sum_{i=1}^n F_i\}_{n \in \mathbb{N}} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$, applying then [theorem: 21.416] gives

$$\left\{ \left| F - \sum_{i=1}^n F_i \right|^p \right\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.736)$$

If $x \in X \setminus N$ then for $n \in \mathbb{N}$ we have

$$\begin{aligned} \left| F(x) - \sum_{i=1}^n F_i(x) \right| & \stackrel{[\text{eq: 21.730}]}{=} \left| \sum_{i=1}^{\infty} F_i(x) - \sum_{i=1}^n F_i(x) \right| \\ & \stackrel{[\text{theorem: 14.416}]}{=} \left| \sum_{i=n+1}^{\infty} F_i(x) \right| \\ & \stackrel{[\text{eq: 21.725}] \wedge [\text{theorem: 14.439}]}{\leq} \sum_{i=n+1}^{\infty} |F_i(x)| \\ & \leq \sum_{i=1}^{\infty} |F_i(x)| \end{aligned}$$

hence using [theorem: 19.42] $|F(x) - \sum_{i=1}^n F_i(x)|^p \leq (\sum_{i=1}^{\infty} |F_i(x)|)^p \stackrel{[\text{eq: 21.717}]}{=} g(x)$. So we have as $\mu(N) = 0$ that

$$\forall n \in \mathbb{N} \text{ we have } \left| F - \sum_{i=1}^n F_i \right| \leq g \text{ } \mu\text{-a.e.} \quad (21.737)$$

So all the conditions for the Lebesgue Dominant Convergence Theorem [see theorem: 21.384] are satisfied by [eqs: 21.722, 21.718, 21.735, 21.736, 21.737] from which it follows that

$$\lim_{n \rightarrow \infty} \int \left| F - \sum_{i=1}^n F_i \right|^p d\mu = \int \left(\lim_{n \rightarrow \infty} \left| F - \sum_{i=1}^n F_i \right|^p \right) d\mu_{[\text{eq: 21.735}]} = \int C_0 d\mu_{[\text{example: 21.349}]} = 0$$

which proves that

$$\lim_{n \rightarrow \infty} \left(\left\| F - \sum_{i=1}^n F_i \right\|_p \right)^p = 0$$

Using now [lemma: 21.447]

$$\lim_{n \rightarrow \infty} \left(\left\| F - \sum_{i=1}^n F_i \right\|_p \right) = \lim_{n \rightarrow \infty} \left(\left(\left\| F - \sum_{i=1}^n F_i \right\|_p \right)^p \right)^{1/p} = \left(\lim_{n \rightarrow \infty} \left(\left\| F - \sum_{i=1}^n F_i \right\|_p \right)^p \right)^{1/p} = 0^{1/p} = 0$$

so that

$$\lim_{n \rightarrow \infty} \left\| F - \sum_{i=1}^n F_i \right\|_p = 0 \quad (21.738)$$

Finally as $F \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{K}]$ [see eq: 21.734] we have by definition that

$$[F]_p \in L^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

Let $n \in \mathbb{N}$ then we have

$$\begin{aligned} \left\| [F]_p - \sum_{i=1}^n f_i \right\|_{L^p} &= \left\| [F]_p - \sum_{i=1}^n [F_i]_p \right\|_{L^p} \\ &= \left\| \left[F - \sum_{i=1}^n F_i \right] \right\|_{L^p} \\ &= \left\| F - \sum_{i=1}^n F_i \right\|_p \end{aligned}$$

so that by [eq: 21.738]

$$\lim_{n \rightarrow \infty} \left\| [F]_p - \sum_{i=1}^n f_i \right\|_{L^p} = 0$$

Hence using [theorem: 14.346] it follows that

$$\sum_{i=1}^{\infty} f_i \text{ converges and } \sum_{i=1}^{\infty} f_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n f_i = [F]_p \in L^p[X, \mathcal{A}, \mu, \mathbb{K}]$$

In other words we have proved that every absolute convergent series converges which by [theorem: 14.441] proves that

$$\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], |||_p \rangle \text{ is a Banach space} \quad \square$$

Summarizing the previous lemma's we have that $L^p[X, \mathcal{A}, \mu, \mathbb{K}]$ forms a Banach space using the norm $|||_{L^p}$.

Theorem 21.452. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $p \in [1, \infty]$ then $\langle L^p[X, \mathcal{A}, \mu, \mathbb{K}], |||_{L^p} \rangle$ is a Banach space.*

Proof. This follows from [lemma: 21.450] and [lemma: 21.451]. $\square$

21.5.6.3 Density

We show now that $\mathcal{S}[X, \mathcal{A}]$ is dense in $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ [see theorem: 14.117 and definition: 14.53].

Theorem 21.453. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $p \in [1, \infty[$ and $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ then $\forall \varepsilon \in \mathbb{R}^+$ there exist a $g \in \mathcal{S}[X, \mathcal{A}] \cap \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $\|f - g\|_p < \varepsilon$.*

Note 21.454. *Using [theorem: 14.117] theorem it follows that $\mathcal{S}[X, \mathcal{A}]$ is dense in $\mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$.*

Proof. If $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ then $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ so that using [theorem: 21.342] we have $f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$. Using [theorem: 21.325] there exists $\{h_n\}_{n \in \mathbb{N}}, \{g_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$ such that

$$\forall x \in X \text{ we have } \forall n \in \mathbb{N} \text{ that } h_n(x) \leq h_{n+1}(x) \wedge g_n(x) \leq g_{n+1}(x) \quad (21.739)$$

and

$$f^+ = \lim_{n \rightarrow \infty} h_n \underset{[\text{theorem: 20.92}]}{=} \sup(\{h_n | n \in \mathbb{N}\}) \wedge f^- = \lim_{n \rightarrow \infty} g_n \underset{[\text{theorem: 20.92}]}{=} \sup(\{g_n | n \in \mathbb{N}\}) \quad (21.740)$$

Define now

$$\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}[X, \mathcal{A}] \underset{[\text{definition: 21.292}]}{\subseteq} \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ by } f_n = h_n - g_n \underset{[\text{theorem: 21.296}]}{\in} \mathcal{S}[X, \mathcal{A}]$$

Given $x \in X$ we have $\forall n \in \mathbb{N}$

$$\begin{aligned} |f_n(x)| &\leq |(h_n - g_n)(x)| \\ &\leq |h_n(x)| + |g_n(x)| \\ &\underset{0 \leq h_n(x), g_n(x)}{=} h_n(x) + g_n(x) \\ &\underset{[\text{eq: 21.740}]}{\leq} f^+(x) + f^-(x) \\ &\underset{[\text{theorem: 21.341}]}{=} |f(x)| \end{aligned}$$

So we can use [lemma: 21.417] giving

$$f_n \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}] \underset{f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]}{\Rightarrow} f_n - f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.741)$$

Hence we have by definition

$$\forall n \in \mathbb{N} \quad |f_n - f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \Rightarrow |f_n - f|^p \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.742)$$

Further we have

$$\begin{aligned} |f_n(x) - f(x)| &= |h_n(x) - g_n(x) - f(x)| \\ &\underset{[\text{theorem: 21.341}]}{=} |h_n(x) - g_n(x) - (f^+(x) - f^-(x))| \\ &= |h_n(x) - f^+(x) - (g_n(x) - f^-(x))| \\ &\leq |h_n(x) - f^+(x)| + |g_n(x) - f^-(x)| \\ &\underset{[\text{eq: 21.740}]}{=} f^+(x) - h_n(x) + f^-(x) - g_n(x) \\ &\underset{0 \leq h_n(x), g_n(x)}{\leq} f^+(x) + f^-(x) \\ &\underset{[\text{theorem: 21.341}]}{=} |f(x)| \end{aligned}$$

which, as the power function is monotone [see theorem: 19.42], results in

$$\forall x \in X \text{ we have } \forall n \in \mathbb{N} \text{ that } |(f_n - f)(x)|^p \leq |f(x)|^p \quad (21.743)$$

Let $x \in X$ then as $f(x) \in \mathbb{R}$ we have that $f^+(x), f^-(x) \in \mathbb{R}$ so that by [eq: 21.740] $\lim_{n \rightarrow \infty} h_n(x) \in \mathbb{R}$ and $\lim_{n \rightarrow \infty} g_n(x) \in \mathbb{R}$ so that

$$\begin{aligned} \lim_{n \rightarrow \infty} f_n(x) &= \lim_{n \rightarrow \infty} (h_n(x) - g_n(x)) \\ &\underset{[\text{theorems: 14.358, 14.357}]}{=} \lim_{n \rightarrow \infty} h_n(x) - \lim_{n \rightarrow \infty} g_n(x) \\ &\underset{[\text{eq: 21.740}]}{=} f^+(x) - f^-(x) \\ &= f(x) \end{aligned}$$

which, using [theorem: 14.346], proves that $\lim_{n \rightarrow \infty} |f_n(x) - f(x)| = 0$. Using [lemma: 21.447] we have

$$\lim_{n \rightarrow \infty} |(f_n - f)(x)|^p = \left(\lim_{n \rightarrow \infty} (f_n - f)(x) \right)^p = 0^p \underset{p \neq 0}{=} 0 = C_0(x)$$

so that

$$\lim_{n \rightarrow \infty} |f_n - f|^p = C_0 \quad (21.744)$$

As $f \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ we have by definition that

$$|f|^p \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.745)$$

The conditions for the Lebesgue Dominant Convergence Theorem [theorem: 21.384] are now satisfied by the fact that $C_0 \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ [see example: 21.228] and [eqs: 21.742, 21.743, 21.744, 21.745]. Hence by applying this theorem we get

$$\lim_{n \rightarrow \infty} \int |f_n - f|^p d\mu = \int \left(\lim_{n \rightarrow \infty} |f_n - f|^p \right) d\mu \stackrel{[\text{eq: 21.744}]}{=} \int C_0 d\mu = 0$$

proving

$$\lim_{n \rightarrow \infty} (\|f_n - f\|_p)^p = 0$$

So given $\varepsilon \in \mathbb{R}^+$ there exist a $N \in \mathbb{N}$ such that $(\|f_N - f\|_p)^p = |(\|f_N - f\|_p)^p - 0| < \varepsilon^p$. Hence using [theorem: 19.42] we have that $\|f_N - f\|_p = ((\|f_N - f\|_p)^p)^{1/p} < (\varepsilon^p)^{1/p} = \varepsilon$. Further we have $f_N \in \mathcal{S}[X, \mathcal{A}]$ and by [eq: 21.741] $f_N \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$. Hence we have proved by taking $g = f_N$ that

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a } g \in \mathcal{S}[X, \mathcal{A}] \cap \mathcal{L}^p(X, \mathcal{A}, \mu, \mathbb{R}) \text{ such that } \|g - f\|_p < \varepsilon \quad \square$$

We have a similar theorem for $p = \infty$, first we need a little lemma.

Lemma 21.455. *Let $a, b \in \mathbb{R}$ with $a \leq b$ then given $\varepsilon \in \mathbb{R}^+$ there exists a $n \in \{2, \dots, \infty\}$ and a family $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $[a, b] \subseteq \bigcup_{i \in \{1, \dots, n-1\}} [a_i, a_{i+1}[$, $\forall i \in \{1, \dots, n-1\}$ $a_i < a_{i+1}$ and $a_{i+1} - a_i < \varepsilon$.*

Proof. Take $l = b - a$. As $\varepsilon > 0$ we have by [theorem: 10.30] that there exists a $N \in \mathbb{N}$ such that $\frac{2 \cdot l}{\varepsilon} < N$ or $l < N \cdot \frac{\varepsilon}{2}$. Define now $n = N + 1$, so that $n \in \{2, \dots, \infty\}$, and $\forall i \in \{1, \dots, n\}$ $a_i = a + (i-1) \cdot \frac{\varepsilon}{2}$. Then $\forall i \in \{1, \dots, n-1\}$ we have

$$a_i = a + (i-1) \cdot \frac{\varepsilon}{2} < a + i \cdot \frac{\varepsilon}{2} = a_{i+1}$$

and

$$a_{i+1} - a_i = a + i \cdot \frac{\varepsilon}{2} - a - (i-1) \cdot \frac{\varepsilon}{2} = \frac{\varepsilon}{2} < \varepsilon$$

Further $a_1 = a$ and $a_n = a + (n-1) \cdot \frac{\varepsilon}{2} = a + N \cdot \frac{\varepsilon}{2} > a + l = b$ so that

$$b < a_n$$

Given $x \in [a, b]$ then $a \leq x \leq b$, define $I_x = \{i \in \{1, \dots, n-1\} | a_i \leq x\} \subseteq \{1, \dots, n-1\}$. Then as $a_1 = a \leq x$ we have that the finite set I_x is not empty so that $m = \max(I_x)$ is well defined. We have now two cases to consider:

$m = n - 1$. Then $a_{n-1} \leq x$ and as $x \leq b < a_n$ it follows that $x \in [a_{n-1}, a_n[= [a_m, a_{m+1}[$

$m < n - 1$. Then $a_m \leq x$ and as $m+1 \leq n-1$ we must have $x < a_{m+1}$ proving that $x \in [a_m, a_{m+1}[$

So we conclude that $x \in \bigcup_{i \in \{1, \dots, n-1\}} [a_i, a_{i+1}[$ or that

$$[a, b] \subseteq \bigcup_{i \in \{1, \dots, n-1\}} [a_i, a_{i+1}[\quad \square$$

Theorem 21.456. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ then $\forall \varepsilon \in \mathbb{R}^+$ there exist a $g \in \mathcal{S}[X, \mathcal{A}]$ such that $\|f - g\| < \varepsilon$.*

Note 21.457. *Using [theorem: 14.117] theorem it follows that $\mathcal{S}[X, \mathcal{A}]$ is dense in $\mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$*

Proof. Let $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ and take $\varepsilon \in \mathbb{R}^+$. Using [lemma: 21.455] there exists a $n \in \{2, \dots, \infty\}$ and a family $\{a_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ with

$$[-\|f\|_\infty, \|f\|_\infty] \subseteq \bigcup_{i \in \{1, \dots, n-1\}} [a_i, a_{i+1}[\text{ and } \forall i \in \{1, \dots, n-1\} \ a_i < a_{i+1} \wedge a_{i+1} - a_i < \frac{\varepsilon}{2} \quad (21.746)$$

Let $i \in \{1, \dots, n-1\}$ then we have that $[a_i, a_{i+1}[\in \mathcal{I} \subseteq \sigma[\mathcal{I}] \stackrel{[\text{theorem: 21.55}]}{=} \mathcal{B}[\mathbb{R}] \stackrel{[\text{theorem: 21.61}]}{\subseteq} \mathcal{B}[\overline{\mathbb{R}}]$. As $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ so that $f^{-1}([a_i, a_{i+1}[) \in \mathcal{A}$. Hence we can define

$$\{A_i\}_{i \in \{1, \dots, n-1\}} \subseteq \mathcal{A} \text{ by } A_i = f^{-1}([a_i, a_{i+1}[) \quad (21.747)$$

Next

$$\begin{aligned} f^{-1}([- \|f\|_\infty, \|f\|_\infty]) &\stackrel{[\text{eq: 21.746}]}{\subseteq} f^{-1}\left(\bigcup_{i \in \{1, \dots, n-1\}} [a_i, a_{i+1}[\right) \\ &= \bigcup_{i \in \{1, \dots, n-1\}} f^{-1}([a_i, a_{i+1}[) \\ &\stackrel{[\text{eq: 21.747}]}{=} \bigcup_{i \in \{1, \dots, n-1\}} A_i \end{aligned}$$

proving that

$$f^{-1}([- \|f\|_\infty, \|f\|_\infty]) \subseteq \bigcup_{i \in \{1, \dots, n-1\}} A_i \in \mathcal{A} \quad (21.748)$$

Let $i, j \in \{1, \dots, n-1\}$ with $i \neq j$ then we may assume without loss of generality that $i < j$ [otherwise switch i and j]. Assume that $x \in A_i \cap A_j$ then $f(x) \in [a_i, a_{i+1}[\cap [a_j, a_{j+1}[$ so that $a_i \leq f(x) < a_{i+1} \leq a_j \leq f(x) < a_{j+1}$ leading to the contradiction $f(x) < f(x)$. Hence the assumption is wrong and we must have that $A_i \cap A_j = \emptyset$. So its proved that

$$\{A_i\}_{i \in \{1, \dots, n-1\}} \text{ is pairwise disjoint} \quad (21.749)$$

Using then [theorem: 21.295] we have that

$$g = \sum_{i=1}^{n-1} a_i \cdot \chi_{A_i} \in \mathcal{S}[X, \mathcal{A}] \quad (21.750)$$

If $x \in \bigcup_{i \in \{1, \dots, n-1\}} A_i$ then by [eq: 21.749] there exist a unique $i \in \{1, \dots, n-1\}$ such that $x \in A_i$ from which it follows that $f(x) \in [a_i, a_{i+1}[\Rightarrow a_i \leq f(x) < a_{i+1}$ so that $0 \leq f(x) - a_i < a_{i+1} - a_i = \frac{\varepsilon}{2}$. As $g(x) \stackrel{[\text{theorem: 21.290}]}{=} a_i$ it follows that $|f(x) - g(x)| < \frac{\varepsilon}{2}$. Hence

$$\forall x \in \bigcup_{i \in \{1, \dots, n-1\}} A_i \text{ we have } |f(x) - g(x)| < \frac{\varepsilon}{2} \quad (21.751)$$

If $x \in \{x \in X \mid |f(x) - g(x)| > \frac{\varepsilon}{2}\}$ then using the above we must have that $x \in X \setminus \bigcup_{i \in \{1, \dots, n-1\}} A_i$. Using [eq: 21.748] it follows that $x \notin f^{-1}([- \|f\|_\infty, \|f\|_\infty])$ so that $f(x) \notin [- \|f\|_\infty, \|f\|_\infty]$ hence $f(x) < - \|f\|_\infty \vee \|f\|_\infty < f(x) \Rightarrow \|f\|_\infty < -f(x) \leq |f(x)| \vee \|f\|_\infty < f(x) \leq |f(x)|$ from which we conclude that $\|f\|_\infty < |f(x)|$. So that

$$\left\{x \in X \mid |f(x) - g(x)| > \frac{\varepsilon}{2}\right\} \subseteq \{x \in X \mid |f(x)| > \|f\|_\infty\} \quad (21.752)$$

As $f \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ we have by [lemma: 21.433] that $\{x \in X \mid |f(x)| > \|f\|_\infty\}$ is locally μ -null so that by [theorem: 21.430] $\{x \in X \mid |f(x) - g(x)| > \frac{\varepsilon}{2}\}$ is locally μ -null. So we have that $\frac{\varepsilon}{2} \in \{M \in [0, \infty[\mid \{x \in X \mid |(f-g)(x)| > M\} \text{ is locally } \mu\text{-null}\}$ so that

$$\|f - g\|_\infty = \inf(\{M \in [0, \infty[\mid \{x \in X \mid |(f-g)(x)| > M\} \text{ is locally } \mu\text{-null}\}) \leq \frac{\varepsilon}{2} < \varepsilon$$

In other words

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a } g \in \mathcal{S}[X, \mathcal{A}] \text{ such that } \|f - g\|_\infty < \varepsilon \quad \square$$

The two previous theorem motivates the following definition

Definition 21.458. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $p \in [1, \infty]$ then we define

$$\mathcal{S}^p[X, \mathcal{A}, \mu, \mathbb{R}] = \begin{cases} \{[f]_\infty \mid f \in \mathcal{S}[X, \mathcal{A}] \text{ if } p = \infty\} \\ \{[f]_p \mid f \in \mathcal{S}[X, \mathcal{A}] \cap \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]\} \text{ if } p \in [1, \infty[\end{cases} \subseteq L^p[X, \mathcal{A}, \mu, \mathbb{R}]$$

Note 21.459. If $f \in \mathcal{S}[X, \mathcal{A}]$ then f is $\mathcal{A}$ -measurable and $f(X)$ is finite and non empty [as $X \neq \emptyset$], hence f is bounded above by $M = \max(\{|y| | y \in f(X)\})$ proving that $\mathcal{S}[X, \mathcal{A}] \subseteq \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$.

Using this definition we can extend the two previous theorem to the Banach space $L^p[X, \mathcal{A}, \mu, \mathbb{R}]$

Theorem 21.460. Let $\langle X, \mathcal{A}, \mu \rangle$ be measure space, $p \in [1, \infty]$ and $f \in L^p[X, \mathcal{A}, \mu, \mathbb{R}]$ then $\forall \varepsilon \in \mathbb{R}^+$ there exist a $g \in \mathcal{S}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $\|f - g\|_{L^p} < \varepsilon$

Note 21.461. Using the topological space $\langle L^p[X, \mathcal{A}, \mu, \mathbb{R}], \|\cdot\|_{L^p} \rangle$ we have by [theorem: 14.117] that $\mathcal{S}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ is dense in $L^p[X, \mathcal{A}, \mu, \mathbb{R}]$. So that by definition of dense subsets [see definition: 14.53] we have

$$\overline{\mathcal{S}^p[X, \mathcal{A}, \mu, \mathbb{R}]} = L^p[X, \mathcal{A}, \mu, \mathbb{R}]$$

Proof. Let $p \in [1, \dots, \infty]$ then we have either:

$p = \infty$. As $f \in L^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ there exists a $F \in \mathcal{L}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $f = [F]_\infty$. Using [theorem: 21.456] there exists a $G \in \mathcal{S}[X, \mathcal{A}]$ such that $\|F - G\|_\infty < \varepsilon$. Take $g = [G]_\infty \in \mathcal{S}^\infty[X, \mathcal{A}, \mu, \mathbb{R}]$ then we have

$$\|f - g\|_{L^\infty} = \|[F]_\infty - [G]_\infty\|_{L^\infty} = \|[F - G]\|_{L^\infty} = \|F - G\|_\infty < \varepsilon$$

$p \in [1, \infty[$. As $f \in L^p[X, \mathcal{A}, \mu, \mathbb{R}]$ there exists a $F \in \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $f = [F]_p$. Using [theorem: 21.453] there exists a $G \in \mathcal{S}[X, \mathcal{A}] \cap \mathcal{L}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $\|F - G\|_p < \infty$. Take $g = [G]_p \in \mathcal{S}^p[X, \mathcal{A}, \mu, \mathbb{R}]$ then we have

$$\|f - g\|_{L^p} = \|[F]_p - [G]_p\|_{L^p} = \|[F - G]\|_{L^p} = \|F - G\|_p < \varepsilon \quad \square$$

21.6 Signed and complex measures

Up to now a measure μ is defined to be a non negative real valued function. In this section we extend this to measures that are real or complex.

21.6.1 Signed measures

21.6.1.1 Definition and properties

First we must extend the concept of finite sums in $\overline{\mathbb{R}}$ and ensure that they are always well defined. The problem is that $-\infty + \infty$ and $\infty + (-\infty)$ is not defined in $\overline{\mathbb{R}}$. To avoid this we restrict ourselves to the subsets $\mathbb{R} \cup \{\infty\}$ or $\mathbb{R} \cup \{-\infty\}$.

Definition 21.462. We define

$$\mathbb{R}_\infty = \mathbb{R} \cup \{\infty\}$$

$$\mathbb{R}_{-\infty} = \mathbb{R} \cup \{-\infty\}$$

Theorem 21.463. Using the definition of $+$ in $\overline{\mathbb{R}}$ restricted to $\mathbb{R}_\infty, \mathbb{R}_{-\infty}$ we have the following

1. $\langle \mathbb{R}_\infty, + \rangle$ is a Abelian semi-group
2. $\langle \mathbb{R}_{-\infty}, + \rangle$ is a Abelian semi-group

Proof.

1. We must prove the axioms of a Abelian semi group:

associativity. Let $x, y, z \in \mathbb{R}_\infty$ then we have for $x, y, z \in \mathbb{R}_\infty$ either:

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. then using the properties of $+$ in $\mathbb{R}$ we have

$$(x + y) + z = x + (y + z)$$

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z = \infty$. Then

$$\begin{aligned}(x + y) + z &= (x + y) + \infty \\ &= \infty \\ &= x + \infty \\ &= x + (y + \infty) \\ &= x + (y + z)\end{aligned}$$

$x \in \mathbb{R} \wedge y = \infty \wedge z \in \mathbb{R}$. Then

$$\begin{aligned}(x + y) + z &= (x + \infty) + z \\ &= \infty + z \\ &= \infty \\ &= x + \infty \\ &= x + (\infty + z) \\ &= x + (y + z)\end{aligned}$$

$x \in \mathbb{R} \wedge y = \infty \wedge z = \infty$. Then

$$\begin{aligned}(x + y) + z &= (x + \infty) + \infty \\ &= \infty \\ &= x + \infty \\ &= x + (\infty + \infty) \\ &= x + (y + z)\end{aligned}$$

$x = \infty \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. Then

$$\begin{aligned}(x + y) + z &= (\infty + y) + z \\ &= \infty + z \\ &= \infty \\ &= \infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

$x = \infty \wedge y \in \mathbb{R} \wedge z = \infty$. Then

$$\begin{aligned}(x + y) + z &= (\infty + y) + \infty \\ &= \infty \\ &= \infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

$x = \infty \wedge y = \infty \wedge z \in \mathbb{R}$. Then

$$\begin{aligned}(x + y) + z &= (\infty + \infty) + z \\ &= \infty + z \\ &= \infty \\ &= \infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

$x = \infty \wedge y = \infty \wedge z = \infty$. Then

$$\begin{aligned}(x + y) + z &= (\infty + \infty) + \infty \\ &= \infty \\ &= \infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

neutral element. Then we have for $x \in \mathbb{R}_\infty$ either

$x \in \mathbb{R}$. then using the properties of $+$ in the real numbers we have

$$x + 0 = x = 0 + x$$

$x = \infty$. then

$$x + 0 = \infty + 0 = \infty = 0 + \infty = 0 + x$$

commutativity. Then we have for $x, y \in \mathbb{R}_\infty$ either

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of $+$ in $\mathbb{R}$ we have

$$x + y = y + x$$

$x \in \mathbb{R} \wedge y = \infty$. Then

$$x + y = x + \infty = \infty = \infty + x = y + x$$

$x = \infty \wedge y \in \mathbb{R}$. Then

$$x + y = \infty + y = \infty = y + \infty = y + x$$

$x = \infty \wedge y = \infty$. Then

$$x + y = \infty + \infty = \infty = \infty + \infty = y + x$$

2. We must prove the axioms of a Abelian semi group:

associativity. Let $x, y, z \in \mathbb{R}_{-\infty}$ then we have for $x, y, z \in \mathbb{R}_{-\infty}$ either:

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. Then using the properties of $+$ in $\mathbb{R}$ we have

$$(x + y) + z = x + (y + z)$$

$x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z = -\infty$. Then

$$\begin{aligned} (x + y) + z &= (x + y) + (-\infty) \\ &= -\infty \\ &= x + (-\infty) \\ &= x + (y + (-\infty)) \\ &= x + (y + z) \end{aligned}$$

$x \in \mathbb{R} \wedge y = -\infty \wedge z \in \mathbb{R}$. Then

$$\begin{aligned} (x + y) + z &= (x + (-\infty)) + z \\ &= -\infty + z \\ &= -\infty \\ &= x + (-\infty) \\ &= x + ((-\infty) + y) \\ &= x + (y + z) \end{aligned}$$

$x \in \mathbb{R} \wedge y = -\infty \wedge z = -\infty$. Then

$$\begin{aligned} (x + y) + z &= (x + (-\infty)) + (-\infty) \\ &= -\infty \\ &= x + (-\infty) \\ &= x + (-\infty + (-\infty)) \\ &= x + (y + z) \end{aligned}$$

$x = -\infty \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}$. Then

$$\begin{aligned} (x + y) + z &= (-\infty + y) + z \\ &= -\infty + z \\ &= -\infty \\ &= -\infty + (y + z) \\ &= x + (y + z) \end{aligned}$$

$x = -\infty \wedge y \in \mathbb{R} \wedge z = -\infty$. Then

$$\begin{aligned}(x + y) + z &= (-\infty + y) + (-\infty) \\ &= -\infty \\ &= -\infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

$x = -\infty \wedge y = -\infty \wedge z \in \mathbb{R}$. Then

$$\begin{aligned}(x + y) + z &= (-\infty + (-\infty)) + z \\ &= -\infty + z \\ &= -\infty \\ &= -\infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

$x = -\infty \wedge y = -\infty \wedge z = -\infty$. Then

$$\begin{aligned}(x + y) + z &= (-\infty + (-\infty)) + (-\infty) \\ &= -\infty \\ &= -\infty + (y + z) \\ &= x + (y + z)\end{aligned}$$

natural element. Then we have for $x \in \mathbb{R}_{-\infty}$ either

$x \in \mathbb{R}$. Then using the properties of $+$ in the real numbers we have

$$x + 0 = x = 0 + x$$

$x = -\infty$. Then

$$x + 0 = -\infty + 0 = -\infty = 0 + (-\infty) = 0 + x$$

commutativity. Then we have for $x, y \in \mathbb{R}_{\infty}$ either

$x \in \mathbb{R} \wedge y \in \mathbb{R}$. Then using the properties of $+$ in $\mathbb{R}$ we have

$$x + y = y + x$$

$x \in \mathbb{R} \wedge y = -\infty$. Then

$$x + y = x + (-\infty) = -\infty = -\infty + x = y + x$$

$x = -\infty \wedge y \in \mathbb{R}$. Then

$$x + y = -\infty + y = -\infty = y + (-\infty) = y + x$$

$x = -\infty \wedge y = -\infty$. Then

$$x + y = -\infty + (-\infty) = -\infty = -\infty + (-\infty) = y + x$$

□

As $\langle \mathbb{R}_{\infty}, + \rangle$ and $\langle \mathbb{R}_{-\infty}, + \rangle$ are semi groups we can now use [section: 11.1] to work with finite sums in these semi-groups.

Theorem 21.464. Let $n \in \mathbb{N}$, $\alpha \in \mathbb{R}$ then we have $\forall \{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}_{\infty}$ [or $\mathbb{R}_{-\infty}$]

1. $\forall \{x_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}_{\infty}$ we have $\alpha \cdot \sum_{i=1}^n x_i = \sum_{i=1}^n (\alpha \cdot x_i)$
2. $\forall \{x_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}_{-\infty}$ we have $\alpha \cdot \sum_{i=1}^n x_i = \sum_{i=1}^n (\alpha \cdot x_i)$

Note 21.465. $\langle \mathbb{R}_{\infty}, +, \cdot \rangle$ is not a ring so we can not use [theorem: 11.8] in this proof]

Proof. We use induction to prove this, so define

$$S = \left\{ n \in \mathbb{N} \mid \text{If } \{x_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}_{\infty} [\text{or } \mathbb{R}_{-\infty}] \text{ then } \alpha \cdot \sum_{i=1}^n x_i = \sum_{i=1}^n (\alpha \cdot x_i) \right\}$$

then we have:

$1 \in S$. Let $\{x_i\}_{i \in \{1\}} \subseteq \mathbb{R}_\infty$ [or $\mathbb{R}_{-\infty}$] then $\alpha \cdot \sum_{i=1}^1 x_i = \alpha \cdot x_1 = \sum_{i=1}^1 (\alpha \cdot x_i)$ which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Let $\{x_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathbb{R}_+$ [or $\mathbb{R}_{-\infty}$] then we have that

$$\begin{aligned} \sum_{i=1}^{n+1} (\alpha \cdot x_i) &= (\alpha \cdot x_{n+1}) + \sum_{i=1}^n (\alpha \cdot x_i) \\ &\stackrel{n \in S}{=} (\alpha \cdot x_{n+1}) + \alpha \cdot \sum_{i=1}^n x_i \\ &\stackrel{[\text{theorem: 20.42}]}{=} \alpha \cdot \left(x_{n+1} + \sum_{i=1}^n x_i \right) \\ &= \alpha \cdot \sum_{i=1}^{n+1} x_i \end{aligned}$$

which proves that $n+1 \in S$. □

Next we extend the definition of infinite sums of non negative numbers in $\overline{\mathbb{R}}$ [see definition: 20.95] to infinite sums of elements in $\mathbb{R}_\infty$ or $\mathbb{R}_-$. Unfortunately this comes at a cost, in contrast to the sum of non negative extended real numbers the infinite sum of elements in $\mathbb{R}_\infty$ or $\mathbb{R}_{-\infty}$ is not guaranteed to exist.

Definition 21.466.

1. If $\{x_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}_\infty$ then we say that $\sum_{i=1}^\infty x_i$ exist if $\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i$ exist. If this is the case then $\sum_{i=1}^\infty x_i$ is defined by

$$\sum_{i=1}^\infty x_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n x_i$$

Note 21.467. If $\{x_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}_\infty$ then we can still have that $\sum_{i=1}^\infty x_i = -\infty$.

2. If $\{x_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}_{-\infty}$ then we say that $\sum_{i=1}^\infty x_i$ exist if $\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i$ exist and $\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i \in \mathbb{R}_{-\infty}$. If this is the case then $\sum_{i=1}^\infty x_i$ is defined by

$$\sum_{i=1}^\infty x_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n x_i$$

Note 21.468. If $\{x_i\}_{i \in \mathbb{N}} \subseteq \mathbb{R}_\infty$ then we can still have that $\sum_{i=1}^\infty x_i = -\infty$.

Note 21.469. $\lim_{n \rightarrow \infty} x_i$ is defined in [definition: 20.72]

We are now ready to define a signed measure on a measurable space.

Definition 21.470. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is a **signed measure on $\mathcal{A}$** if

1. $\mu(\emptyset) = 0$
2. $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$ or $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$
3. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ pairwise disjoint we have that $\sum_{i=1}^\infty \mu(A_i)$ exist and

$$\mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^\infty \mu(A_i)$$

Note 21.471. The condition (3) implies that for every pairwise disjoint $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ we have if μ is a signed measure on $\mathcal{A}$ that:

- i. $\sum_{i=1}^{\infty} \mu(A_i)$ exist
- ii. If $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$ then $\mu(\bigsqcup_{i \in \mathbb{N}} A_i) \in \mathbb{R}_{\infty}$ so that (3) implies that $\sum_{i=1}^{\infty} \mu(A_i) \in \mathbb{R}_{\infty}$.
- iii. If $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$ then $\mu(\bigsqcup_{i \in \mathbb{N}} A_i) \in \mathbb{R}_{-\infty}$ so that (3) implies that $\sum_{i=1}^{\infty} \mu(A_i) \in \mathbb{R}_{-\infty}$.
- iv. If $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ is a bijection then $\bigsqcup_{i \in \mathbb{N}} A_i = \bigsqcup_{i \in \mathbb{N}} A_{\sigma(i)}$ so that we have automatically

$$\sum_{i=1}^{\infty} \mu(A_{\sigma(i)}) = \mu\left(\bigsqcup_{i \in \mathbb{N}} A_{\sigma(i)}\right) = \mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Definition 21.472. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then a signed measure $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is finite if

$$\forall A \in \mathcal{A} \text{ we have } \mu(A) \in \mathbb{R}$$

Given two measures where one is finite we can create a signed measure as the difference of these measures.

Theorem 21.473. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu_1: \mathcal{A} \rightarrow [0, \infty]$, $\mu_2: \mathcal{A} \rightarrow [0, \infty]$ two measures on $\mathcal{A}$ where μ_2 is finite or μ_1 is finite so that

$$\mu_1 - \mu_2: \mathcal{A} \rightarrow \overline{\mathbb{R}} \text{ defined by } (\mu_1 - \mu_2)(A)$$

is well defined then we have that

- 1. $\mu_1 - \mu_2: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is a signed measure on $\mathcal{A}$.
- 2. If $\forall A \in \mathcal{A}$ we have $\mu_2(A) \leq \mu_1(A)$ then $\mu_1 - \mu_2: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is a measure on $\mathcal{A}$.

Proof.

- 1. We have

$$(\mu_1 - \mu_2)(\emptyset) = \mu_1(\emptyset) - \mu_2(\emptyset) = 0 - 0 = 0 \quad (21.753)$$

For finiteness we have two possible cases:

μ_1 is finite. Then for $A \in \mathcal{A}$ we have either:

$$\mu_2(A) = \infty. \text{ Then } (\mu_1 - \mu_2)(A) = \mu_1(A) - \mu_2(A) = \mu_1(A) - \infty = -\infty$$

$$\mu_2(A) \in \mathbb{R}. \text{ Then } (\mu_1 - \mu_2)(A) = \mu_1(A) - \mu_2(A) \in \mathbb{R}$$

Hence we have

$$\{(\mu_1 - \mu_2)(A) | A \in \mathcal{A}\} \in \mathbb{R}_{-\infty}$$

μ_2 is finite. Then for $A \in \mathcal{A}$ we have either:

$$\mu_1(A) = \infty. \text{ Then } (\mu_1 - \mu_2)(A) = \mu_1(A) - \mu_2(A) = \infty - \mu_2(A) = \infty$$

$$\mu_1(A) \in \mathbb{R}. \text{ Then } (\mu_1 - \mu_2)(A) = \mu_1(A) - \mu_2(A) \in \mathbb{R}$$

Hence we have

$$\{(\mu_1 - \mu_2)(A) | A \in \mathcal{A}\} \in \mathbb{R}_{\infty}$$

So it follows that

$$\{(\mu_1 - \mu_2)(A) | A \in \mathcal{A}\} \in \mathbb{R}_{\infty} \text{ or } \{(\mu_1 - \mu_2)(A) | A \in \mathcal{A}\} \in \mathbb{R}_{-\infty} \quad (21.754)$$

Next we use induction to prove

$$\forall n \in \mathbb{N} \text{ we have } \forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ pairwise disjoint } \sum_{i=1}^n (\mu_1 - \mu_2)(A_i) = \sum_{i=1}^n \mu_1(A_i) - \sum_{i=1}^n \mu_2(A_i) \quad (21.755)$$

Proof. Let

$$S = \left\{ n \in \mathbb{N} \mid \forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ pairwise disjoint we have } \sum_{i=1}^n (\mu_1 - \mu_2)(A_i) = \sum_{i=1}^n \mu_1(A_i) - \sum_{i=1}^n \mu_2(A_i) \right\}$$

then we have:

$1 \in S$. Then for $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A}$ we have

$$\sum_{i=1}^1 (\mu_1 - \mu_2)(A_i) = \mu_1(A_1) - \mu_2(A_1) = \sum_{i=1}^1 \mu_1(A_i) - \sum_{i=1}^1 \mu_2(A_i)$$

which proves that $1 \in S$.

$n \in S \Rightarrow n+1 \in S$. Then for $\{A_i\}_{i \in \{1, \dots, n+1\}} \subseteq \mathcal{A}$ pairwise disjoint we have that

$$\begin{aligned} \sum_{i=1}^{n+1} (\mu_1 - \mu_2)(A_i) &= (\mu_1 - \mu_2)(A_{n+1}) + \sum_{i=1}^n (\mu_1 - \mu_2)(A_i) \\ &\stackrel{n \in S}{=} (\mu_1 - \mu_2)(A_{n+1}) + \sum_{i=1}^n \mu_1(A_i) - \sum_{i=1}^n \mu_2(A_i) \\ &= \mu_1(A_{n+1}) + \sum_{i=1}^n \mu_1(A_i) - \mu_2(A_{n+1}) - \sum_{i=1}^n \mu_2(A_i) \\ &= \sum_{i=1}^{n+1} \mu_1(A_i) - \sum_{i=1}^{n+1} \mu_2(A_i) \end{aligned}$$

which proves that $n+1 \in S$.

Mathematical induction proves [eq: 21.755] □

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then we have, as μ_1, μ_2 are signed measures, that

$$\begin{aligned} \sum_{i=1}^{\infty} \mu_1(A_i) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_1(A_i) \text{ exist and } \mu_1\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_1(A_i) \\ \sum_{i=1}^{\infty} \mu_2(A_i) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_2(A_i) \text{ exist and } \mu_2\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_2(A_i) \end{aligned}$$

Then using [theorem: 20.85(3,5)] we have that $\lim_{n \rightarrow \infty} (\sum_{i=1}^n \mu_1(A_i) - \sum_{i=1}^n \mu_2(A_i))$ exist and

$$\lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \mu_1(A_i) - \sum_{i=1}^n \mu_2(A_i) \right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_1(A_i) - \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_2(A_i) \quad (21.756)$$

Hence

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{i=1}^n (\mu_1 - \mu_2)(A_i) &\stackrel{[\text{eq: 21.755}]}{=} \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \mu_1(A_i) - \sum_{i=1}^n \mu_2(A_i) \right) \\ &\stackrel{[\text{eq: 21.756}]}{=} \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_1(A_i) - \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_2(A_i) \\ &= \mu_1\left(\bigcup_{i \in \mathbb{N}} A_i\right) - \mu_2\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\ &= (\mu_1 - \mu_2)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \end{aligned}$$

As $\sum_{i=1}^{\infty} (\mu_1 - \mu_2)(A_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n (\mu_1 - \mu_2)(A_i)$ it follows that

$$\sum_{i=1}^{\infty} (\mu_1 - \mu_2)(A_i) \text{ exist and } \sum_{i=1}^{\infty} (\mu_1 - \mu_2)(A_i) = (\mu_1 - \mu_2) \left(\bigcup_{i \in \mathbb{N}} A_i \right) \quad (21.757)$$

From [eqs: 21.753. 21.754 and 21.757] it follows that $\mu_1 - \mu_2$ is a signed measure on $\mathcal{A}$.

2. Assume that $\forall A \in \mathcal{A}$ we have $\mu_1(A) \leq \mu_2(A)$. For μ_1, μ_2 we have either:

μ_1 is finite. Then as $-\infty < \mu_1(A) \leq \mu_2(A)$ we have for $\mu_2(A)$ either:

$\mu_2(A) \in \mathbb{R}$. Then $0 \leq \mu_2(A) - \mu_1(A)$ so that $(\mu_2 - \mu_1)(A) \in [0, \infty]$

$\mu_2(A) = \infty$. Then $\mu_2(A) - \mu_1(A) = \infty - \mu_1(A) = \infty$ so that $(\mu_2 - \mu_1)(A) \in [0, \infty]$

μ_2 is finite. Then as $0 \leq \mu_1(A) \leq \mu_2(A) < \infty$ we have that $\mu_2(A) - \mu_1(A) \in [0, \infty[$ so that $(\mu_2 - \mu_1)(A) \in [0, \infty]$.

From the above we conclude that

$$\mu_2 - \mu_1: \mathcal{A} \rightarrow [0, \infty]$$

is a well defined function which together with [eqs: 21.753 and 21.757] proves that

$$\mu_2 - \mu_1: \mathcal{A} \rightarrow [0, \infty] \text{ is a measure on } \mathcal{A} \quad \square$$

Theorem 21.474. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ and $\alpha \in \mathbb{R}$ then $\alpha \cdot \mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined by $(\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$ is a signed measure.

Proof. We have

$$(\alpha \cdot \mu)(\emptyset) \underset{\mu \text{ is a signed measure}}{=} \alpha \cdot \mu(\emptyset) = \alpha \cdot 0 = 0 \quad (21.758)$$

As μ is a signed measure we have either:

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$. For $\alpha \in \mathbb{R}$ we have either:

$0 = \alpha$. Then if $x \in \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\}$ there exist a $A \in \mathcal{A}$ such that

$$x = (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A) = 0 \cdot \mu(A) = 0 \in \mathbb{R}_{\infty}$$

so that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$$

$0 < \alpha$. Then if $x \in \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\}$ there exist a $A \in \mathcal{A}$ such that $x = (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$. For $\mu(A) \in \mathbb{R}_{\infty}$ we have either:

$\mu(A) \in \mathbb{R}$. Then $x = \alpha \cdot \mu(A) \in \mathbb{R}$ so that $x \in \mathbb{R}_{\infty}$

$\mu(A) = \infty$. Then $x = \alpha \cdot \infty = \infty$ so that $x \in \mathbb{R}_{\infty}$

So that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$$

$\alpha < 0$. Then if $x \in \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\}$ there exist a $A \in \mathcal{A}$ such that $x = (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$. For $\mu(A) \in \mathbb{R}_{\infty}$ we have either:

$\mu(A) \in \mathbb{R}$. Then $x = \alpha \cdot \mu(A) \in \mathbb{R}$ so that $x \in \mathbb{R}_{-\infty}$

$\mu(A) = \infty$. Then $x = \alpha \cdot \infty = -\infty$ so that $x \in \mathbb{R}_{-\infty}$

So that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$$

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$. For $\alpha \in \mathbb{R}$ we have either:

$0 = \alpha$. Then if $x \in \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\}$ there exist a $A \in \mathcal{A}$ such that

$$x = (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A) = 0 \cdot \mu(A) = 0 \in \mathbb{R}_{\infty}$$

so that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$$

$0 < \alpha$. Then if $x \in \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\}$ there exist a $A \in \mathcal{A}$ such that $x = (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$. For $\mu(A) \in \mathbb{R}_{-\infty}$ we have either:

$\mu(A) \in \mathbb{R}$. Then $x = \alpha \cdot \mu(A) \in \mathbb{R}$ so that $x \in \mathbb{R}_{-\infty}$

$\mu(A) = -\infty$. Then $x = \alpha \cdot (-\infty) = -\infty$ so that $x \in \mathbb{R}_{-\infty}$

So that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$$

$\alpha < 0$. Then if $x \in \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\}$ there exist a $A \in \mathcal{A}$ such that $x = (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$. For $\mu(A) \in \mathbb{R}_{-\infty}$ we have either:

$\mu(A) \in \mathbb{R}$. Then $x = \alpha \cdot \mu(A) \in \mathbb{R}$ so that $x \in \mathbb{R}_{\infty}$

$\mu(A) = -\infty$. Then $x = \alpha \cdot (-\infty) = \infty$ so that $x \in \mathbb{R}_{\infty}$

So that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$$

Hence we have proved that

$$\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_+ \text{ or } \{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty} \quad (21.759)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be a pairwise disjoint family of measurable sets. Then as μ is a signed measure we have that

$$\sum_{i=1}^{\infty} \mu(A_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) \text{ exist and } \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) \quad (21.760)$$

Using [theorem: 20.85(3)] we have that

$$\lim_{n \rightarrow \infty} \left(\alpha \cdot \sum_{i=1}^n \mu(A_i) \right) \text{ exist and } \lim_{n \rightarrow \infty} \left(\alpha \cdot \sum_{i=1}^n \mu(A_i) \right) = \alpha \cdot \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) \quad (21.761)$$

As $\forall n \in \mathbb{N}$ we have $\alpha \cdot \sum_{i=1}^n \mu(A_i) \stackrel{[\text{theorem: 21.464}]}{=} \sum_{i=1}^n (\alpha \cdot \mu(A_i)) = \sum_{i=1}^n (\alpha \cdot \mu)(A_i)$ it follows that

$$\begin{aligned} \sum_{i=1}^{\infty} (\alpha \cdot \mu)(A_i) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (\alpha \cdot \mu(A_i)) \\ &= \lim_{n \rightarrow \infty} \left(\alpha \cdot \sum_{i=1}^n \mu(A_i) \right) \\ &\stackrel{[\text{eq: 21.761}]}{=} \alpha \cdot \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) \\ &\stackrel{[\text{eq: 21.760}]}{=} \alpha \cdot \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\ &= (\alpha \cdot \mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \end{aligned}$$

In other words

$$\sum_{i=1}^{\infty} (\alpha \cdot \mu)(A_i) \text{ exist and } \sum_{i=1}^{\infty} (\alpha \cdot \mu)(A_i) = (\alpha \cdot \mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \quad \square$$

Theorem 21.475. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $n \in \mathbb{N}$ and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ be a signed measure on $\mathcal{A}$ then $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint we have that

$$\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

In other words μ is finite additive.

Proof. Let $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ be pairwise disjoint. Define now

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_i & \text{if } i \in \{1, \dots, n\} \\ \emptyset & \text{if } i \in \mathbb{N} \setminus \{1, \dots, n\} \end{cases}$$

If $k, l \in \mathbb{N}$ with $k \neq l$ then we have either

$$k, l \in \{1, \dots, n\}. \text{ then } B_k \cap B_l = A_k \cap A_l = \emptyset$$

$$k \notin \{1, \dots, n\}. \text{ then } B_k \cap B_l = \emptyset \cap B_l = \emptyset$$

$$l \notin \{1, \dots, n\}. \text{ then } B_k \cap B_l = B_k \cap \emptyset = \emptyset$$

which proves that

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ is pairwise disjoint}$$

So we have by the definition of a signed measure that $\sum_{i=1}^{\infty} \mu(B_i) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(B_i)$ exists and

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \sum_{i=1}^{\infty} \mu(B_i) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(B_i) \quad (21.762)$$

Now if $n < k$ we have that

$$\begin{aligned} \sum_{i=1}^k \mu(B_i) &= \sum_{i \in \{1, \dots, k\}} \mu(B_i) \\ &\stackrel{[\text{theorem: 11.44}]}{=} \sum_{i \in \{1, \dots, n\}} \mu(B_i) + \sum_{i \in \{n+1, \dots, k\}} \mu(B_i) \\ &= \sum_{i \in \{1, \dots, n\}} \mu(B_i) + \sum_{i \in \{n+1, \dots, k\}} \mu(\emptyset) \\ &= \sum_{i \in \{1, \dots, n\}} \mu(B_i) \\ &= \sum_{i \in \{1, \dots, n\}} \mu(A_i) \\ &= \sum_{i=1}^n \mu(A_i) \end{aligned} \quad (21.763)$$

Further

$$\begin{aligned} \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(B_i) &\stackrel{[\text{theorem: 20.84}]}{=} \lim_{k \rightarrow \infty} \sum_{i=1}^{n+k+1} \mu(B_i) \\ &\stackrel{[\text{eq: 21.763}]}{=} \lim_{k \rightarrow \infty} \sum_{i=1}^n \mu(A_i) \\ &= \sum_{i=1}^{\infty} \mu(A_i) \end{aligned} \quad (21.764)$$

As

$$\bigcup_{i \in \mathbb{N}} B_i = \left(\bigcup_{i \in \{1, \dots, n\}} B_i \right) \cup \left(\bigcup_{i \in \{n+1, \dots, \infty\}} B_i \right) = \bigcup_{i \in \{1, \dots, n\}} A_i$$

we have using the above and [eqs: 21.762 and 21.764] that

$$\mu\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i) \quad \square$$

Signed measures have many of the same properties as normal measures.

Theorem 21.476. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ then we have:

1. If $A \in \mathcal{A}$ satisfies $\mu(A) \in \mathbb{R}$ then we have $\forall B \in \mathcal{A}$ with $B \subseteq A$ that $\mu(B) \in \mathbb{R}$

2. If $A, B \in \mathcal{A}$ with $B \subseteq A$ and $\mu(B) \in \mathbb{R}$ then $\mu(A \setminus B) = \mu(A) - \mu(B)$

Proof. If $B \subseteq A$ then $A = (A \setminus B) \cup B$ and $(A \setminus B) \cap B = \emptyset$ so that by [theorem: 21.475] we have

$$\mu(A) = \mu(A \setminus B) + \mu(B) \quad (21.765)$$

1. Assume that $\mu(A) \in \mathbb{R}$ and $B \subseteq A$. For μ we have either:

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$. If $\mu(B) = \infty$ then $\mu(A) \stackrel{[\text{eq: 21.765}]}{=} \mu(A \setminus B) + \infty = \infty$ contradicting $\mu(A) \in \mathbb{R}$ hence we must have that $\mu(B) \in \mathbb{R}$.

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$. If $\mu(B) = -\infty$ then $\mu(A) \stackrel{[\text{eq: 21.765}]}{=} \mu(A \setminus B) + (-\infty) = -\infty$ contradicting $\mu(A) \in \mathbb{R}$ hence we must have that $\mu(B) \in \mathbb{R}$.

2. If $B \subseteq A$ and $\mu(B) \in \mathbb{R}$ then for μ we have either:

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$. Then for $\mu(A)$ we have either:

$\mu(A) \in \mathbb{R}$. If $\mu(A \setminus B) = \infty$ then from [eq: 21.765] we have that $\mu(A) = \infty + \mu(B) = \infty$ contradicting $\mu(A) \in \mathbb{R}$. Hence we have $\mu(B), \mu(A), \mu(A \setminus B) \in \mathbb{R}$ so that from [eq: 21.765] it follows that $\mu(A \setminus B) = \mu(A) - \mu(B)$.

$\mu(A) = \infty$. If $\mu(A \setminus B) \in \mathbb{R}$ then by [eq: 21.765] $\mu(A) = \mu(A \setminus B) + \mu(B) \in \mathbb{R}$ contradicting $\mu(A) = \infty$. Hence we must have that $\mu(A \setminus B) = \infty$ so that $\mu(A \setminus B) = \infty = \infty - \mu(B) = \mu(A) - \mu(B)$.

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$. Then for $\mu(A)$ we have either:

$\mu(A) \in \mathbb{R}$. If $\mu(A \setminus B) = -\infty$ then from [eq: 21.765] we have that $\mu(A) = -\infty + \mu(B) = -\infty$ contradicting $\mu(A) \in \mathbb{R}$. Hence we have $\mu(B), \mu(A), \mu(A \setminus B) \in \mathbb{R}$ so that from [eq: 21.765] it follows that $\mu(A \setminus B) = \mu(A) - \mu(B)$.

$\mu(A) = -\infty$. If $\mu(A \setminus B) \in \mathbb{R}$ then by [eq: 21.765] $\mu(A) = \mu(A \setminus B) + \mu(B) \in \mathbb{R}$ contradicting $\mu(A) = -\infty$. Hence we must have that $\mu(A \setminus B) = -\infty$ so that $\mu(A \setminus B) = -\infty = -\infty - \mu(B) = \mu(A) - \mu(B)$. $\square$

Theorem 21.477. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and let $\mu: \mathcal{A} \rightarrow \mathbb{R}$ be a signed measure on $\langle X, \mathcal{A} \rangle$. Then we have

1. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ A_i \subseteq A_{i+1}$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

2. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i$ such that $\exists k \in \mathbb{N}$ with $\mu(A_k) \in \mathbb{R}$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

Proof.

1. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \begin{cases} A_i & \text{if } i = 1 \\ A_i \setminus A_{i-1} & \text{if } i \in \{2, \dots, \infty\} \end{cases} \in \mathcal{A}$$

Let $i, j \in \mathbb{N}$ with $i \neq j$ then we may always assume that $i < j$ [otherwise exchange i and j] then for i we have either:

$i = 1$. Then $1 < j$ so that $1 \leq j - 1$ from which it follows that $A_1 \subseteq A_{j-1} \Rightarrow A_j \setminus A_{j-1} \subseteq A_j \setminus A_1$ hence $B_i \cap B_j = A_1 \cap (A_j \setminus A_{j-1}) \subseteq A_1 \cap (A_j \setminus A_1) = \emptyset$ so that $B_i \cap B_j = \emptyset$

$i \in \{2, \dots, \infty\}$. Then as $i < j$ we have $i \leq j - 1$ so that $A_i \subseteq A_{j-1}$ from which it follows that $A_j \setminus A_{j-1} \subseteq A_j \setminus A_i$. So

$$B_i \cap B_j = (A_i \setminus A_{i-1}) \cap (A_j \setminus A_{j-1}) \subseteq (A_i \setminus A_{i-1}) \cap (A_j \setminus A_i) = \emptyset$$

or $B_i \cap B_j = \emptyset$.

Hence

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ is pairwise disjoint} \quad (21.766)$$

Let $n \in \mathbb{N}$ then $\forall i \in \{1, \dots, n\}$ we have that $B_i = \begin{cases} A_1 \subseteq A_i & \text{if } i = 1 \\ A_i \setminus A_{i-1} \subseteq A_i & \text{if } i \in \{2, \dots, \infty\} \end{cases} \subseteq A_i \subseteq A_n$ so that $\bigcup_{i \in \{1, \dots, n\}} B_i \subseteq A_n \subseteq \bigcup_{i \in \mathbb{N}} A_i$. Summarized we have

$$\forall n \in \mathbb{N} \text{ we have } \bigcup_{i \in \{1, \dots, n\}} B_i \subseteq A_n \subseteq \bigcup_{i \in \mathbb{N}} A_i \quad (21.767)$$

Let $n \in \mathbb{N}$ then for $x \in A_n$ we have that $n \in \{i \in \{1, \dots, n\} \mid x \in A_i\}$ a finite set so that $m = \min(\{i \in \{1, \dots, n\} \mid x \in A_i\})$ exist. For m we have either:

$m = 1$. Then $x \in A_1 = B_1 \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$

$1 < m$. Then $x \notin A_{m-1}$ and $x \in A_m$ so that $x \in A_m \setminus A_{m-1} = B_m \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i$

So we have that $A_n \subseteq \bigcup_{i \in \{1, \dots, n\}} B_i \subseteq \bigcup_{i \in \mathbb{N}} B_i$ which combined with [eq: 21.767] gives

$$\forall n \in \mathbb{N} \text{ we have } A_n = \bigcup_{i \in \{1, \dots, n\}} B_i \text{ and } \bigcup_{i \in \mathbb{N}} B_i = \bigcup_{i \in \mathbb{N}} A_i \quad (21.768)$$

By the definition of a signed measure we have that

$$\sum_{i=1}^{\infty} \mu(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(B_i) \text{ exist and } \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \sum_{i=1}^{\infty} \mu(B_i) \quad (21.769)$$

Finally

$$\begin{aligned} \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) &\stackrel{[\text{eq: 21.768}]}{=} \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\ &\stackrel{[\text{eq: 21.769}]}{=} \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(B_i) \\ &\stackrel{[\text{theorem: 21.475}]}{=} \lim_{n \rightarrow \infty} \mu\left(\bigcup_{i \in \{1, \dots, n\}} B_i\right) \\ &\stackrel{[\text{eq: 21.768}]}{=} \lim_{n \rightarrow \infty} \mu(A_n) \end{aligned}$$

hence

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

2. First we consider the case where $k = 1$. So let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be such that $\mu(A_1) \in \mathbb{R}$ and $\forall i \in \mathbb{N}$ we have $A_{i+1} \subseteq A_i$. Then $\forall i \in \mathbb{N}$ we have $A_i \subseteq A_1$ so that by [theorem: 21.476] we have

$$\forall i \in \mathbb{N} \text{ we have } \mu(A_i) \in \mathbb{R} \quad (21.770)$$

Define now

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = A_1 \setminus A_i \in \mathcal{A}$$

If $i \in \mathbb{N}$ then we have $B_i = A_1 \setminus A_i \subseteq A_1 \setminus A_{i+1} = B_{i+1}$ so that

$$\forall i \in \mathbb{N} \text{ we have } B_i \subseteq B_{i+1}$$

Using (1) it follows that

$$\lim_{i \rightarrow \infty} \mu(B_i) \text{ exist and } \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \lim_{i \rightarrow \infty} \mu(B_i) \quad (21.771)$$

Now

$$\begin{aligned}
 A_1 \setminus \left(\bigcup_{i \in \mathbb{N}} B_i \right) & \stackrel{[\text{theorem: 2.140}]}{=} \bigcap_{i \in \mathbb{N}} (A_1 \setminus B_i) \\
 & = \bigcap_{i \in \mathbb{N}} (A_1 \setminus (A_1 \setminus A_i)) \\
 & \stackrel{[\text{theorem: 1.36}]}{=} \bigcap_{i \in \mathbb{N}} A_i
 \end{aligned}$$

so that

$$A_1 \setminus \left(\bigcup_{i \in \mathbb{N}} B_i \right) = \bigcap_{i \in \mathbb{N}} A_i \quad (21.772)$$

As $\forall i \in \mathbb{N} \ B_i = A_1 \setminus A_i \subseteq A_1$ we have $\bigcup_{i \in \mathbb{N}} B_i \subseteq A_1$ hence as $\mu(A_1) \in \mathbb{R}$ it follows from [theorem: 21.476] that

$$\mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \in \mathbb{R}, \quad \bigcup_{i \in \mathbb{N}} B_i \subseteq A_1 \text{ and } \forall i \in \mathbb{N} \ B_i \subseteq A_1 \text{ so that } \mu(B_i) \in \mathbb{R} \quad (21.773)$$

Next

$$\begin{aligned}
 \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) & \stackrel{[\text{eq: 21.772}]}{=} \mu\left(A_1 \setminus \left(\bigcup_{i \in \mathbb{N}} B_i\right)\right) \\
 & \stackrel{[\text{eq: 21.773}] \wedge [\text{theorem: 21.476}]}{=} \mu(A_1) - \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\
 & \stackrel{[\text{eq: 21.771}]}{=} \mu(A_1) - \lim_{i \rightarrow \infty} \mu(B_i) \\
 & \stackrel{B_i = A_1 \setminus A_i, [\text{theorem: 21.476}], [\text{eq: 21.773}]}{=} \mu(A_1) - \lim_{i \rightarrow \infty} (\mu(A_1) - \mu(A_i)) \\
 & \stackrel{[\text{theorem: 20.85}]}{=} \mu(A_1) - \mu(A_1) + \lim_{i \rightarrow \infty} \mu(A_i) \\
 & = \lim_{i \rightarrow \infty} \mu(A_i)
 \end{aligned}$$

To summarize we have

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ with } \mu(A_1) \in \mathbb{R} \text{ and } \forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i \quad \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) \quad (21.774)$$

We consider now the general case where $1 \leq k$. So let $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be such that $\exists k \in \mathbb{N}$ with $\mu(C_k) \in \mathbb{R}$ and $\forall i \in \mathbb{N} \ C_{i+1} \subseteq C_i$. Define

$$\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } A_i = C_{i+(k-1)} \in \mathcal{A}$$

Then $\mu(A_1) = C_{1+(k-1)} = C_k \in \mathbb{R}$ and $\forall i \in \mathbb{N}$ we have $A_{i+1} = C_{(i+1)+(k-1)} \subseteq C_{i+(k-1)} = A_i$. So we can apply [eq: 21.774] giving

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) = \lim_{i \rightarrow \infty} \mu(C_{i+(k-1)}) \stackrel{[\text{theorem: 20.84}]}{=} \lim_{i \rightarrow \infty} \mu(C_i) \quad (21.775)$$

If $x \in \bigcap_{i \in \mathbb{N}} C_i$ then $\forall i \in \mathbb{N}$ we have, as $i + (k-1) \in \mathbb{N}$, that $x \in C_{i+(k-1)} = A_i$ proving that

$$\bigcap_{i \in \mathbb{N}} C_i \subseteq \bigcap_{i \in \mathbb{N}} A_i \quad (21.776)$$

If $x \in \bigcap_{i \in \mathbb{N}} A_i$ then $\forall i \in \mathbb{N}$ we have $x \in A_i$. Let $j \in \mathbb{N}$ then we have either:

$j \in \{k, \dots, \infty\}$. Then $j - (k-1) \in \mathbb{N}$ so that $x \in A_{j-(k-1)} = C_{j-(k-1)+(k-1)} = C_j$

$j \in \{1, \dots, k-1\}$. Then $j < k$ so that $C_k \subseteq C_j$, as $x \in A_1 = C_{1+(k-1)} = C_k$ we have $x \in C_j$.

From the above it follows that $\forall j \in \mathbb{N}$ we have $x \in C_j$ so that $x \in \bigcap_{j \in \mathbb{N}} C_j$ which proves that $\bigcap_{i \in \mathbb{N}} A_i \subseteq \bigcap_{i \in \mathbb{N}} C_i$. Combining this with [eq: 21.776] we have

$$\bigcap_{i \in \mathbb{N}} C_i = \bigcap_{i \in \mathbb{N}} A_i$$

Substituting this in [eq: 21.775] gives finally

$$\mu\left(\bigcap_{i \in \mathbb{N}} C_i\right) = \lim_{i \rightarrow \infty} \mu(C_i) \quad \square$$

For the opposite of the previous theorem we have:

Theorem 21.478. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a extended real function such that*

- a) $\mu(\emptyset) = 0$
- b) $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$ or $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$
- c) If $n \in \mathbb{N}$ and $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ is pairwise disjoint we have $\mu(\bigcup_{i \in \{1, \dots, n\}} A_i) = \sum_{i=1}^n \mu(A_i)$

then

- 1. If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

then μ is a signed measure on $\mathcal{A}$.

- 2. If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} A_{i+1} \subseteq A_i$ and $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and $\lim_{i \rightarrow \infty} \mu(A_i) = 0$ then μ is a signed measure on $\mathcal{A}$.

Proof.

- 1. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \bigsqcup_{j \in \{1, \dots, i\}} A_j \in \mathcal{A}$$

Then using [lemma: 21.7] we have that

$$\bigsqcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i \text{ and } \forall i \in \mathbb{N} B_i \subseteq B_{i+1} \quad (21.777)$$

Next

$$\begin{aligned} \mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &= \mu\left(\bigcup_{i \in \mathbb{N}} B_i\right) \\ &\stackrel{\text{assumption}}{=} \lim_{i \rightarrow \infty} \mu(B_i) \\ &= \lim_{i \rightarrow \infty} \mu\left(\bigsqcup_{j \in \{1, \dots, i\}} A_j\right) \\ &\stackrel{(c)}{=} \lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(A_j) \end{aligned}$$

which together with (a) and (b) proves that μ is a signed measure on $\mathcal{A}$.

- 2. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \bigcup_{j \in \{i, \dots, \infty\}} A_j \in \mathcal{A}$$

Let $i \in \mathbb{N}$ then $B_{i+1} = \bigcup_{j \in \{i+1, \dots, \infty\}} A_j \stackrel{[\text{theorem: 2.136}]}{\subseteq} \bigcup_{j \in \{i, \dots, \infty\}} A_j = B_i$ so that

$$\forall i \in \mathbb{N} \text{ we have } B_{i+1} \subseteq B_i \quad (21.778)$$

Assume that $x \in \bigcap_{i \in \mathbb{N}} B_i$ then $x \in B_1 = \bigcup_{j \in \mathbb{N}} A_j$ so that there exist a $j \in \mathbb{N}$ such that $x \in A_j$. Now $\forall k \in \{j+1, \dots, \infty\}$ we have as $j \neq k$ that $A_j \cap A_k = \emptyset$ so that it follows that $x \notin A_k$ so that $x \notin \bigcup_{j \in \{j+1, \dots, \infty\}} A_j = B_{j+1}$ which contradicts $x \in \bigcap_{i \in \mathbb{N}} B_i$. Hence the assumption is wrong and we must have

$$\bigcap_{i \in \mathbb{N}} B_i = \emptyset \quad (21.779)$$

Using the hypothesis we have that $\lim_{i \rightarrow \infty} \mu(B_i)$ exist and $\lim_{i \rightarrow \infty} \mu(B_i) = 0$. Using [theorem: 20.84] it follows then that

$$\lim_{i \rightarrow \infty} \mu(B_{i+1}) \text{ exist and } \lim_{i \rightarrow \infty} \mu(B_{i+1}) = 0 \quad (21.780)$$

Let $k \in \mathbb{N}$. If $x \in (\bigcup_{i \in \{1, \dots, k\}} A_i) \cap (\bigcup_{i \in \{k+1, \dots, \infty\}} A_i)$ then there exist a $i \in \{1, \dots, k\}$ and a $j \in \{k+1, \dots, \infty\}$ such that $x \in A_i \wedge x \in A_j \Rightarrow x \in A_i \cap A_j$. However as $i \in \{1, \dots, k\} \wedge j \in \{k+1, \dots, \infty\}$ we have $i \leq k < k+1 < j \Rightarrow i \neq j$ so that $A_i \cap A_j = \emptyset$ and we reach a contradiction. Hence we have

$$\forall k \in \mathbb{N} \quad \left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) \cap \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i \right) = \emptyset$$

and using [theorem: 2.136]

$$\bigcup_{i \in \mathbb{N}} A_i = \left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) \cup \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i \right)$$

By (c) [finite additivity] we have

$$\begin{aligned} \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) &= \mu\left(\left(\bigcup_{i \in \{1, \dots, k\}} A_i\right) \cup \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i\right)\right) \\ &= \mu\left(\bigcup_{i \in \{1, \dots, k\}} A_i\right) + \mu\left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i\right) \\ &= \mu\left(\bigcup_{i \in \{1, \dots, k\}} A_i\right) + \mu(B_{k+1}) \\ &\stackrel{(c)}{=} \sum_{i=1}^k \mu(A_i) + \mu(B_{k+1}) \end{aligned}$$

which proves that

$$\forall k \in \mathbb{N} \text{ we have } \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^k \mu(A_i) + \mu(B_{k+1}) \quad (21.781)$$

For μ we have now either:

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$. Then for $\mu(\bigcup_{i \in \mathbb{N}} A_i)$ we have either:

$\mu(\bigcup_{i \in \mathbb{N}} A_i) \in \mathbb{R}$. Then from [eq: 21.781] it follows that $\sum_{i=1}^k \mu(A_i), \mu(B_{k+1}) \in \mathbb{R}$ so that

$$\forall k \in \mathbb{N} \text{ we have } \left| \sum_{i=1}^k \mu(A_i) - \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \right| = |\mu(B_{k+1})|$$

Let $\varepsilon \in \mathbb{R}^+$ then as $\lim_{i \rightarrow \infty} \mu(B_{k+1}) \stackrel{[\text{eq: 21.780}]}{=} 0$ there exist a $N \in \mathbb{N}$ such that

$\forall k \in \{N, \dots, \infty\} \quad |\mu(B_{k+1})| < \varepsilon \Rightarrow \left| \sum_{i=1}^k \mu(A_i) - \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \right| < \varepsilon$. This proves that

$$\lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(A_i) \text{ exist and } \sum_{i=1}^{\infty} \mu(A_i) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(A_i) = \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

$\mu(\bigcup_{i \in \mathbb{N}} A_i) = \infty$. As $\lim_{i \rightarrow \infty} \mu(B_{k+1}) \stackrel{[\text{eq: 21.780}]}{=} 0$ there exist a $N \in \mathbb{N}$ such that $\forall n \in \{N, \dots, \infty\} |\mu(B_{n+1})| < 1$. So we have $\forall n \in \{N, \dots, \infty\}$ that $\mu(B_{n+1}) \in \mathbb{R}$ which combined with [eq: 21.781] proves that $\forall n \in \{N, \dots, \infty\} \sum_{i=1}^n \mu(A_i) = \infty$. Hence if $C \in \mathbb{R}^+$ we have that $\forall n \in \{N, \dots, \infty\} C < \sum_{i=1}^n \mu(A_i)$ which proves by [theorem: 20.79] that $\lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i)$ exist and $\lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) = \infty$. So that

$$\sum_{i=1}^{\infty} \mu(A_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) = \infty = \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$. For $\mu(\bigcup_{i \in \mathbb{N}} A_i)$ we have either:

$\mu(\bigcup_{i \in \mathbb{N}} A_i) \in \mathbb{R}$. Then from [eq: 21.781] it follows that $\sum_{i=1}^k \mu(A_i), \mu(B_{k+1}) \in \mathbb{R}$ so that

$$\forall k \in \mathbb{N} \text{ we have } \left| \sum_{i=1}^k \mu(A_i) - \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \right| = |\mu(B_{k+1})|$$

Let $\varepsilon \in \mathbb{R}^+$ then as $\lim_{i \rightarrow \infty} \mu(B_{k+1}) \stackrel{[\text{eq: 21.780}]}{=} 0$ there exist a $N \in \mathbb{N}$ such that $\forall k \in \{N, \dots, \infty\} |\mu(B_{k+1})| < \varepsilon \Rightarrow |\sum_{i=1}^k \mu(A_i) - \mu(\bigcup_{i \in \mathbb{N}} A_i)| < \varepsilon$. This proves that

$$\lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(A_i) \text{ exist and } \sum_{i=1}^{\infty} \mu(A_i) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(A_i) = \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

$\mu(\bigcup_{i \in \mathbb{N}} A_i) = -\infty$. As $\lim_{i \rightarrow \infty} \mu(B_{k+1}) \stackrel{[\text{eq: 21.780}]}{=} 0$ there exist a $N \in \mathbb{N}$ such that $\forall n \in \{N, \dots, \infty\} |\mu(B_{n+1})| < 1$. So we have $\forall n \in \{N, \dots, \infty\}$ that $\mu(B_{n+1}) \in \mathbb{R}$ which combined with [eq: 21.781] proves that $\forall n \in \{N, \dots, \infty\} \sum_{i=1}^n \mu(A_i) = -\infty$. Hence if $C \in \mathbb{R}^+$ we have that $\forall n \in \{N, \dots, \infty\} \sum_{i=1}^n \mu(A_i) < -C$ which proves by [theorem: 20.79] that $\lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i)$ exist and $\lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) = -\infty$. So that

$$\sum_{i=1}^{\infty} \mu(A_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) = -\infty = \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

So in all cases we have that

$$\sum_{i=1}^{\infty} \mu(A_i) \text{ exist and } \sum_{i=1}^{\infty} \mu(A_i) = \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

This together with (a) and (b) proves that μ is signed measure. $\square$

21.6.1.2 Hahn and Jordan decomposition

In [theorem: 21.473] we have showed that for two measures, where one of them is finite, the difference of these measures is a signed measure. In this section we want to show that every signed measure can be written as the difference of two measures.

Definition 21.479. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ then for $A \subseteq X$ we say that

1. A is a **positive set for μ** if $A \in \mathcal{A}$ and $\forall B \in \mathcal{A}$ with $B \subseteq A$ we have $0 \leq \mu(B)$.
2. A is a **negative set for μ** if $A \in \mathcal{A}$ and $\forall B \in \mathcal{A}$ with $B \subseteq A$ we have $\mu(B) \leq 0$.

Example 21.480. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ then $\emptyset$ is a positive set for μ and $\emptyset$ is a negative set for μ .

Proof. First $\emptyset \in \mathcal{A}$, second if $E \subseteq \emptyset$ then $E = \emptyset$ and $\mu(E) = \mu(\emptyset) = 0$ hence $\emptyset$ is a positive and negative set for μ . $\square$

Lemma 21.481. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ then we have for the signed measure $-\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined by $(-\mu)(A) = -\mu(A) = (-1) \cdot \mu(A)$ [see theorem: 21.474]

1. A is a positive set for $\mu \Leftrightarrow A$ is a negative set for $-\mu$
2. A is a negative set for $\mu \Leftrightarrow A$ is positive set for $-\mu$

Proof.

1.

$\Rightarrow$. If A is a positive set for μ then we have $A \in \mathcal{A}$ and for $B \in \mathcal{A}$ with $B \subseteq A$ that $0 \leq \mu(B)$, hence $(-\mu)(B) = -\mu(B) \leq 0$ which proves that A is a negative set for $-\mu$.

$\Leftarrow$. If A is a negative set for $-\mu$ then we have $A \in \mathcal{A}$ and for $B \in \mathcal{A}$ with $B \subseteq A$ that $(-\mu)(B) \leq 0$, hence $-\mu(B) \leq 0$ so that $0 \leq \mu(B)$ which proves that μ is a positive set for μ .

2.

$\Rightarrow$. If A is a negative set for μ then we have $A \in \mathcal{A}$ and for $B \in \mathcal{A}$ with $B \subseteq A$ that $\mu(B) \leq 0$, hence $(-\mu)(B) = -\mu(B) \geq 0$ which proves that A is a positive set for $-\mu$.

$\Leftarrow$. If A is a positive set for $-\mu$ then we have $A \in \mathcal{A}$ and for $B \in \mathcal{A}$ with $B \subseteq A$ that $0 \leq (-\mu)(B)$, hence $0 \leq -\mu(B)$ so that $\mu(B) \leq 0$ which proves that μ is a negative set for μ . $\square$

Lemma 21.482. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ then we have

1. If A is a positive set for μ then $\forall B \in \mathcal{A}$ with $B \subseteq A$ we have $\mu(B) \leq \mu(A)$
2. If A is a negative set for μ then $\forall B \in \mathcal{A}$ with $B \subseteq A$ we have $\mu(A) \leq \mu(B)$

Proof.

1. As A is a positive set for μ and $A, B, A \setminus B \in \mathcal{A}$ with $A \subseteq A, B \subseteq A, A \setminus B \subseteq A$ we have that

$$0 \leq \mu(A) \wedge 0 \leq \mu(B) \wedge 0 \leq \mu(A \setminus B)$$

Now as $B \subseteq A$ we have $A = (A \setminus B) \sqcup B$, hence

$$\mu(A) = \mu(A \setminus B) + \mu(B)$$

For $\mu(A)$ we have now as $0 \leq \mu(A)$ either

$\mu(A) = \infty$. then $\mu(B) \leq \mu(A)$.

$\mu(A) < \infty$. then we have by [theorem: 21.476] that $\mu(A \setminus B) \in \mathbb{R}$ and $\mu(B) \in \mathbb{R}$ so that $\mu(B) = \mu(A) - \mu(A \setminus B)$ giving $\mu(B) \leq \mu(A)$.

So in all cases we have $\mu(B) \leq \mu(A)$

2. As A is a negative set for μ and $A, B, A \setminus B \in \mathcal{A}$ with $A \subseteq A, B \subseteq A, A \setminus B \subseteq A$ we have that

$$\mu(A) \leq 0 \wedge \mu(B) \leq 0 \wedge \mu(A \setminus B) \leq 0$$

Now as $B \subseteq A$ we have $A = (A \setminus B) \sqcup B$, hence

$$\mu(A) = \mu(A \setminus B) + \mu(B)$$

For $\mu(A)$ we have now as $\mu(A) \leq 0$ either

$\mu(A) = -\infty$. then $\mu(A) \leq \mu(B)$

$-\infty < \mu(A)$. then by [theorem: 21.476] $\mu(A \setminus B) \in \mathbb{R}$ and $\mu(B) \in \mathbb{R}$ so that, as $\mu(A \setminus B) \leq 0$, that $\mu(A) = \mu(A \setminus B) + \mu(B) \leq \mu(B)$.

So in all cases we have $\mu(A) \leq \mu(B)$. $\square$

Lemma 21.483. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ be a signed measure on $\mathcal{A}$. Let $A \in \mathcal{A}$ be such that $\mu(A) \in \mathbb{R}$ then there exist a **negative set** B such that $B \subseteq A$ and $\mu(B) \leq \mu(A)$.

Proof. Consider the set $\{\mu(B)|B \in \mathcal{A} \wedge B \subseteq A\}$ then as $\mu(\emptyset) = 0$, $\emptyset \in \mathcal{A}$ and $\emptyset \subseteq A$ we have that $0 \in \{\mu(B)|B \in \mathcal{A} \wedge B \subseteq A\} \Rightarrow \{\mu(B)|B \in \mathcal{A} \wedge B \subseteq A\} \neq \emptyset$ so that

$$\delta_1 = \sup(\{\mu(B)|B \in \mathcal{A} \wedge B \subseteq A\}) \text{ exist and } 0 \leq \delta_1 \in \overline{\mathbb{R}} \quad (21.782)$$

For δ_1 we have now two possible cases:

$$\delta_1 = \infty. \text{ Then } 0 \leq \min\left(\frac{\delta_1}{2}, 1\right) = 1 < \infty = \delta_1$$

$$\delta_1 < \infty. \text{ Then } 0 \leq \min\left(\frac{\delta_1}{2}, 1\right) < \frac{\delta_1}{2} < \delta_1$$

Hence we have

$$0 \leq \min\left(\frac{\delta_1}{2}, 1\right) < \delta_1$$

so that there exist $A_1 \in \{B \in \mathcal{A} | B \subseteq A\}$ with $\min\left(\frac{\delta_1}{2}, 1\right) < \mu(A_1) \leq \delta_1$. Define now

$$\{(\rho_n, B_n)\}_{n \in \mathbb{N}}$$

recursively by:

$n = 1$. Then we define (ρ_1, B_1) by $\rho_1 = \delta_1$ and $B_1 = A_1$ so that

$$B_1 \in \mathcal{A} \wedge B_1 \subseteq A \text{ and } 0 \leq \min\left(\frac{\rho_1}{2}, 1\right) < \mu(A_1) \leq \rho_1$$

$n \in \{2, \dots, \infty\}$. Then $\emptyset \subseteq A \setminus (\bigcup_{i=1}^{n-1} B_i)$, $\emptyset \in \mathcal{A}$ and $0 = \mu(\emptyset)$. So that

$$0 \in \left\{ \mu(B) | B \in \mathcal{A} \wedge B \subseteq A \setminus \left(\bigcup_{i=1}^{n-1} B_i \right) \right\}$$

hence

$$\rho_n = \sup\left(\left\{ \mu(B) | B \in \mathcal{A} \wedge B \subseteq A \setminus \left(\bigcup_{i=1}^{n-1} B_i \right) \right\}\right) \text{ exist and } 0 \leq \rho_n$$

For ρ_n we have either:

$$\rho_n = \infty. \text{ Then } 0 \leq \min\left(\frac{\rho_n}{2}, 1\right) = 1 < \infty = \rho_n$$

$$\rho_n < \infty. \text{ Then } 0 \leq \min\left(\frac{\rho_n}{2}, 1\right) \leq \frac{\rho_n}{2} < \rho_n$$

so that

$$0 \leq \min\left(\frac{\rho_n}{2}, 1\right) < \rho_n$$

Hence there exist a $B_n \in \{B \in \mathcal{A} | B \subseteq A \setminus (\bigcup_{i \in \{1, \dots, n-1\}} B_i)\}$ such that $0 \leq \min\left(\frac{\rho_n}{2}, 1\right) < \mu(B_n) \leq \rho_n$ so that

$$B_n \in \mathcal{A} \wedge B_n \subseteq A \setminus \left(\bigcup_{i \in \{1, \dots, n-1\}} B_i \right) \text{ and } 0 \leq \min\left(\frac{\rho_n}{2}, 1\right) < \mu(B_n) \leq \rho_n$$

So we have constructed a sequence $\{(\rho_n, B_n)\}_{n \in \mathbb{N}}$ such that $\forall n \in \mathbb{N}$

$$\rho_n = \begin{cases} \sup(\{\mu(B)|B \in \mathcal{A} \wedge B \subseteq A\}) & \text{if } n = 1 \\ \sup(\{\mu(B)|B \in \mathcal{A} \wedge B \subseteq A \setminus (\bigcup_{k \in \{1, \dots, n-1\}} B_k)\}) & \text{if } n \in \{2, \dots, \infty\} \end{cases} \quad (21.783)$$

$$B_n \in \begin{cases} \{B \in \mathcal{A} | B \subseteq A\} & \text{if } n = 1 \\ \{B \in \mathcal{A} | B \subseteq A \setminus (\bigcup_{i \in \{1, \dots, n-1\}} B_i)\} & \text{if } n \in \{2, \dots, \infty\} \end{cases} \quad (21.784)$$

$$0 \leq \min\left(\frac{\rho_n}{2}, 1\right) < \mu(B_n) \leq \rho_n \quad (21.785)$$

Let $i, j \in \mathbb{N}$ with $i \neq j$ then we may always assume that $i < j$ [otherwise exchange i and j]. Then as $i \leq j - 1$ we have that $B_j \subseteq A \setminus (\bigcup_{k \in \{1, \dots, j-1\}} B_k) \subseteq A \setminus B_i$ so that $B_j \cap B_i \subseteq (A \setminus B_i) \cap B_i = \emptyset$ which proves that

$$\{B_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A} \text{ is pairwise disjoint} \quad (21.786)$$

Define

$$B_\infty = \bigsqcup_{n \in \mathbb{N}} B_n \in \mathcal{A} \text{ and } B = A \setminus B_\infty \in \mathcal{A} \text{ then } B \subseteq A \quad (21.787)$$

We will prove now that B is the required negative set. As $\forall n \in \mathbb{N}$ we have from [eq: 21.785] that

$$0 \leq \min\left(\frac{\delta_1}{2}, 1\right) < \mu(B_1) \leq \sum_{i=1}^n \mu(B_i) \leq \sup\left(\left\{\sum_{i=1}^n \mu(B_i) \mid n \in \mathbb{N}\right\}\right) = \sum_{i=1}^{\infty} \mu(B_i)$$

so that $0 < \sum_{i=1}^{\infty} \mu(B_i)$. As further μ is a signed measure we have

$$0 < \sum_{i=1}^{\infty} \mu(B_i)_{[\text{eq: 21.786}]} \mu\left(\bigsqcup_{n \in \mathbb{N}} B_n\right)_{[\text{eq: 21.787}]} \mu(B_\infty)$$

so that

$$0 < \sum_{i=1}^{\infty} \mu(B_i) = \mu(B_\infty) \quad (21.788)$$

Using [eq: 21.784] we have $\forall i \in \mathbb{N}$ that $B_i \subseteq A$ so that $B_\infty \subseteq A$ hence $A = (A \setminus B_\infty) \sqcup B_\infty = B \sqcup B_\infty$ so that by [theorem: 21.475]

$$\mu(A) = \mu(B) + \mu(B_\infty) \quad (21.789)$$

As $0 < \mu(B_\infty)$ we have by [theorem: 20.29] that $\mu(B) = \mu(B) + 0 \leq \mu(B) + \mu(B_\infty) = \mu(A)$ so that

$$B \in \mathcal{A} \text{ and } \mu(B) \leq \mu(A) \quad (21.790)$$

This proves the last part of the lemma, the only thing that still must be proved is that B is a negative set for μ .

As by [eq: 21.784] $\forall n \in \mathbb{N}$ $B_n \subseteq A$ it follows that $B_\infty = \bigcup_{n \in \mathbb{N}} B_n \subseteq A$, further by the hypothesis $\mu(A) \in \mathbb{R}$, so using [theorem: 21.476] we have $\mu(B_\infty) \in \mathbb{R}$ so that by [eq: 21.788] $\sum_{i=1}^{\infty} \mu(B_i) \in \mathbb{R}$, applying then [theorem: 14.413] proves that

$$\lim_{i \rightarrow \infty} \mu(B_i) = 0$$

Let $\varepsilon \in \mathbb{R}^+$ then, using the above, there exist a $N \in \mathbb{N}$ such that $\forall n \in \{N, \dots, \infty\}$

$$\mu(B_n)_{[\text{21.785}]} |\mu(B_n) - 0| < \min\left(\frac{\varepsilon}{2}, \frac{1}{2}\right)$$

Using [eq: 21.785] $\min\left(\frac{\rho_n}{2}, 1\right) < \mu(B_n)$ so that $\min\left(\frac{\rho_n}{2}, 1\right) < \min\left(\frac{\varepsilon}{2}, \frac{1}{2}\right)$. Assume that $1 < \frac{\rho_n}{2}$ then $\min\left(\frac{\rho_n}{2}, 1\right) = 1 < \min\left(\frac{\varepsilon}{2}, \frac{1}{2}\right) \leq \frac{1}{2}$ giving the contradiction $1 \leq \frac{1}{2}$. Hence we must have $\frac{\rho_n}{2} \leq 1$ so that $\frac{\rho_n}{2} = \min\left(\frac{\rho_n}{2}, 1\right) < \min\left(\frac{\varepsilon}{2}, \frac{1}{2}\right) \leq \frac{\varepsilon}{2}$ from which it follows that $|\rho_n - 0| = \rho_n < \varepsilon$. This proves that

$$\lim_{i \rightarrow \infty} \rho_n = 0 \quad (21.791)$$

Let $E \in \mathcal{A}$ with $E \subseteq B$. For $n \in \mathbb{N}$ we have either:

$n = 1$. Then, as $B = A \setminus B_\infty \subseteq A$, we have that $E \subseteq A$, which proves that $E \in \{B \in \mathcal{A} \mid B \subseteq A\}$.

From this it follows that

$$\mu(E) \leq \sup(\{\mu(B) \mid B \in \mathcal{A} \wedge B \subseteq A\})_{[\text{eq: 21.783}]} \rho_1$$

$n \in \{2, \dots, \infty\}$. Then as $\bigcup_{i \in \{1, \dots, n-1\}} B_i \subseteq \bigcup_{i \in \mathbb{N}} B_i = B_\infty$ we have that

$$E \subseteq B = A \setminus B_\infty \subseteq A \setminus \left(\bigcup_{i \in \{1, \dots, n-1\}} B_i\right)$$

so that $E \in \{B \in \mathcal{A} \mid B \subseteq \setminus(\bigcup_{i \in \{1, \dots, n-1\}} B_i)\}$. Hence

$$\mu(E) \leq \sup\left(\left\{\mu(B) \mid B \in \mathcal{A} \wedge B \subseteq \setminus\left(\bigcup_{i \in \{1, \dots, n-1\}} B_i\right)\right\}\right) = \rho_n$$

So we have proved that

$$\forall n \in \mathbb{N} \text{ we have } \mu(E) \leq \rho_n \quad (21.792)$$

Assume that $0 < \mu(E)$ then as $\lim_{n \rightarrow \infty} \rho_n \stackrel{[eq: 21.791]}{=} 0$ there exist a $N \in \mathbb{N}$ such that $\rho_n = |\rho_n - 0| < \mu(E)$ contradicting [eq: 21.792]. Hence we must have that $\mu(E) \leq 0$. So the assumption is false and we must have that $\mu(E) \leq 0$. In other words we have proved that

$$\forall E \in \mathcal{A} \text{ with } E \subseteq B \text{ we have } \mu(E) \leq 0$$

which proves that

$$B \text{ is a negative set for } \mu$$

The above combined with [eq: 21.790] proves the lemma. $\square$

Lemma 21.484. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$. If A and B are negative sets for μ such that $A \cap B = \emptyset$ then $A \cup B$ is a negative set for μ .*

Proof. Let $C \in \mathcal{A}$ be such that $C \subseteq A \cup B$. Then $C = C \cap (A \cup B) = (C \cap A) \cup (C \cap B)$ and $(C \cap A) \cap (C \cap B) = C \cap (A \cap B) = \emptyset$. So

$$\mu(C) = \mu(C \cap A) + \mu(C \cap B)$$

As A, B are negative sets for μ , $C \cap A, C \cap B \in \mathcal{A}$ and $C \cap A \subseteq A$, $C \cap B \subseteq B$, we have $\mu(C \cap A) \leq 0$ and $\mu(C \cap B) \leq 0$. So we have that

$$\mu(C) = \mu(C \cap A) + \mu(C \cap B) \leq 0$$

Hence $A \cup B$ is a negative set. $\square$

Definition 21.485. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$. Then (P, N) is a **Hahn Decomposition of μ** if*

$$\begin{aligned} X &= P \cup N \text{ and } P \cap N = \emptyset \\ P &\text{ is a positive set for } \mu \\ N &\text{ is a negative set for } \mu \end{aligned}$$

It is possible that there are two Hahn Decomposition of a measure but then there is a relation between the positive and negative sets.

Theorem 21.486. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ and $(P_1, N_1), (P_2, N_2)$ be two Hahn decomposition's of μ then $\mu(P_1 \cap N_2) = 0 = \mu(P_2 \cap N_1)$.*

Proof. As $P_1 \cap N_2 \in \mathcal{A}$, $P_1 \cap N_2 \subseteq P_1, P_1 \cap N_2 \subseteq N_2$, P_1 a positive set for μ and N_2 a negative set for μ , we have $0 \leq \mu(P_1 \cap N_2) \leq 0$ proving that

$$\mu(P_1 \cap N_2) = 0$$

Likewise, as $P_2 \cap N_1 \in \mathcal{A}$, $P_2 \cap N_1 \subseteq P_2, P_2 \cap N_1 \subseteq N_1$, P_2 a positive set for μ and N_1 a negative set for μ , we have $0 \leq \mu(P_2 \cap N_1) \leq 0$ proving that

$$\mu(P_2 \cap N_1) = 0 \quad \square$$

Theorem 21.487. (Hahn Decomposition Theorem) *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\langle X, \mathcal{A} \rangle$. Then there exists a Hahn decomposition of μ . In other words there exists subsets $P, N \subseteq X$ such that*

$$\begin{aligned} X &= P \cup N \text{ and } P \cap N = \emptyset \\ P &\text{ is a positive set for } \mu \\ N &\text{ is a negative set for } \mu \end{aligned}$$

Proof. For μ we have two cases to consider:

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$. As $\emptyset$ is a negative set for μ [see example: 21.480] and $\mu(\emptyset) = 0$ it follows that $0 \in \{\mu(A) | A \text{ is a negative set for } \mu\}$ so that

$$L = \inf(\{\mu(A) | A \text{ is a negative set for } \mu\}) \text{ exist and } L \leq 0 \quad (21.793)$$

Using [theorem: 20.83] there exist a sequence $\{x_i\}_{i \in \mathbb{N}} \subseteq \{\mu(A) | A \text{ is a negative set for } \mu\}$ such that $\lim_{n \rightarrow \infty} x_n$ exist and $\lim_{n \rightarrow \infty} x_n = L$. Hence there exist a sequence $\{A_n\}_{n \in \mathbb{N}}$ of negative sets for μ such that $\forall n \in \mathbb{N}$ we have $\mu(A_n) = x_n$ so that

$$\lim_{n \rightarrow \infty} \mu(A_n) \text{ exist and } \lim_{n \rightarrow \infty} \mu(A_n) = L \quad (21.794)$$

By the definition of a negative set we have $\{A_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A}$ so that

$$N = \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A} \quad (21.795)$$

We prove now that N is a negative set for μ . Using [theorem: 21.100] there exist a sequence $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ is pairwise disjoint, } N = \bigcup_{i \in \mathbb{N}} B_i \text{ and } \forall i \in \mathbb{N} \text{ we have } B_i \subseteq A_i \quad (21.796)$$

Let $E \in \mathcal{A}$ with $E \subseteq N$ then

$$\begin{aligned} \mu(E) &= \mu(E \cap N) \\ &= \mu\left(E \cap \left(\bigcup_{i \in \mathbb{N}} B_i\right)\right) \\ &= \mu\left(\bigcup_{i \in \mathbb{N}} (E \cap B_i)\right) \\ &= \sum_{i=1}^{\infty} \mu(E \cap B_i) \end{aligned} \quad (21.797)$$

Given $i \in \mathbb{N}$ we have by [eq: 21.796] $E \cap B_i \in \mathcal{A}$ and $E \cap B_i \subseteq B_i \subseteq A_i$. As A_i is a negative set for μ it follows that $\mu(E \cap B_i) \leq 0$. Hence

$$\mu(E) \underset{[\text{eq: 21.797}]}{=} \sum_{i=1}^{\infty} \mu(E \cap B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(E \cap B_i) \underset{[\text{eq: 20.85}]}{\leq} 0$$

So $\forall E \in \mathcal{A}$ with $E \subseteq N$ we have $\mu(E) \leq 0$ which proves that

$$N \text{ is a negative set for } \mu \quad (21.798)$$

From the above it follows that $\mu(N) \in \{\mu(A) | A \text{ is a negative set for } \mu\}$ so that

$$L = \inf(\{\mu(A) | A \text{ is a negative set for } \mu\}) \leq \mu(N) \quad (21.799)$$

Further given $n \in \mathbb{N}$ $A_n \in \mathcal{A}$, $A_n \subseteq \bigcup_{i \in \mathbb{N}} A_i = N$ and N is a negative set. So it follows from [theorem: 21.482] $\mu(N) \leq \mu(A_n)$. Using [theorem: 20.85] it follows then that $\mu(N) \leq \lim_{n \rightarrow \infty} \mu(A_n) \underset{[\text{eq: 21.794}]}{=} L$ which combined with [eq: 21.799]

$$L = \mu(N) \quad (21.800)$$

As $L \underset{[\text{eq: 21.793}]}{\leq} 0$ we have from the above that $\mu(N) \leq 0 < \infty$ so that $\mu(N) \neq \infty$, further as

$\mu(N) \underset{N \in \mathcal{A}}{\in} \{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_+$ we have that $\mu(N) \neq -\infty$. Hence we have

$$\mu(N) \in \mathbb{R} \quad (21.801)$$

Let $P = X \setminus N$ then we prove that P is a positive set for μ . As $N \in \mathcal{A}$ we have that $P \in \mathcal{A}$. Assume that there exist a $C \in \mathcal{A}$ with $C \subseteq P$ such that $\mu(C) < 0$. As $\mu(C) \in \{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$ we have $\mu(C) \neq -\infty$ so that $-\infty < \mu(C) < 0$, hence $\mu(C) \in \mathbb{R}$. Using then [lemma: 21.483] it follows that there exist a negative set B for μ such that

$$B \subseteq C \text{ and } \mu(B) \leq \mu(C) < 0 \quad (21.802)$$

Also we have $N \cap B \subseteq N \cap C \subseteq N \cap P = N \cap (X \setminus N) = \emptyset$ so that

$$N \cap B = \emptyset \quad (21.803)$$

As N, B are negative sets for μ we have by the above and [lemma: 21.484] that

$$N \cup B \text{ is a negative set for } \mu \quad (21.804)$$

Now as $\mu(N) \in \mathbb{R}$ and $\mu(B) < 0$ [see eq: 21.801, 21.802] we have that $\mu(N) + \mu(B) < \mu(N) = L$. So

$$\begin{aligned} \mu(N \cup B) &\stackrel{N \cap B = \emptyset}{=} \mu(N) + \mu(B) \\ &< L \\ &\stackrel{[\text{eq: 21.799}]}{=} \inf(\{\mu(A) | A \text{ is a negative set for } \mu\}) \\ &\stackrel{[\text{eq: 21.804}]}{\leq} \mu(N \cup B) \end{aligned}$$

leading to the contradiction $\mu(N \cup B) < \mu(N \cup B)$. So the assumption is wrong and we must have that $\forall C \in \mathcal{A}$ with $C \subseteq P$ we have $0 \leq \mu(C)$ which proves that

$$P \text{ is a positive set for } \mu \quad (21.805)$$

To summarize

$$N \text{ is a negative set for } \mu, P \text{ is a positive set for } \mu, \emptyset = P \cap N \text{ and } X = P \cup N$$

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$. Using [lemma: 21.481] we have that $\nu = (-\mu): \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is a signed measure on $\mathcal{A}$. Further if $x \in \{\nu(A) | A \in \mathcal{A}\}$ then there exist a $A \in \mathcal{A}$ such that $x = \nu(A) = -\mu(A)$. As we have either $\mu(A) \in \mathbb{R} \Rightarrow -\mu(A) \in \mathbb{R}$ or $\mu(A) = -\infty \Rightarrow -\mu(A) = \infty$ it follows that $\{\nu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_\infty$. Using then the previous case [see eq: 21.805] for ν there exist a positive set N for ν and a negative set P for ν such that $P \cap N = \emptyset$ and $X = P \cup N$. Using [lemma: 21.481] it follows that P is a positive set for $-\nu = -(-\mu) = \mu$ and N is a negative set for $-\nu = -(-\mu) = \mu$. $\square$

The Hahn Decomposition allows us to write every signed measure as the difference of two measures.

Theorem 21.488. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure on $\mathcal{A}$ and (P, N) a Hahn decomposition of μ then we have:*

$$\begin{aligned} \forall A \in \mathcal{A} \quad \mu(A \cap P) &\in [0, \infty] \wedge -\mu(A \cap N) \in [0, \infty] \\ \mu^+: \mathcal{A} &\rightarrow [0, \infty] \text{ defined by } \mu^+(A) = \mu(A \cap P) \text{ is a measure on } \mathcal{A} \\ \mu^-: \mathcal{A} &\rightarrow [0, \infty] \text{ defined by } \mu^-(A) = -\mu(A \cap N) \text{ is a measure on } \mathcal{A} \\ \mu^+ \text{ or } \mu^- &\text{ is finite} \\ \forall A \in \mathcal{A} \text{ we have } \mu(A) &= \mu^+(A) - \mu^-(A) \end{aligned}$$

Further we have that

$$\forall A \in \mathcal{A} \quad \mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \text{ and } \mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\})$$

Proof. Let $A \in \mathcal{A}$. As P is a positive set for μ and N is a negative set for μ we have that $N, P \in \mathcal{A}$ so that $A \cap P, A \cap N \in \mathcal{A}$. As also $A \cap P \subseteq P, A \cap N \subseteq N$ it follows that $0 \leq \mu(A \cap P)$ and $\mu(A \cap N) \leq 0$ so that

$$\forall A \in \mathcal{A} \text{ we have } \mu(A \cap P) \in [0, \infty] \wedge -\mu(A \cap N) \in [0, \infty] \quad (21.806)$$

Further we have

$$\mu^+(\emptyset) = \mu(\emptyset \cap P) = \mu(\emptyset) = 0 \text{ and } \mu^-(\emptyset) = -\mu(\emptyset \cap N) = -\mu(\emptyset) = 0$$

and for every $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint that

$$\begin{aligned} \mu^+\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &= \mu\left(\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) \cap P\right) \\ &= \mu\left(\bigsqcup_{i \in \mathbb{N}} (A_i \cap P)\right) \\ &= \sum_{i=1}^{\infty} \mu(A_i \cap P) \\ &= \sum_{i=1}^{\infty} \mu^+(A_i) \\ \mu^-\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &= -\mu\left(\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) \cap N\right) \\ &= -\mu\left(\bigsqcup_{i \in \mathbb{N}} (A_i \cap N)\right) \\ &= (-1) \cdot \sum_{i=1}^{\infty} \mu(A_i \cap N) \\ &= (-1) \cdot \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i \cap N) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \lim_{n \rightarrow \infty} \sum_{i=1}^n (-\mu(A_i \cap N)) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu^-(A_i) \\ &= \sum_{i=1}^{\infty} \mu^-(A_i) \end{aligned}$$

Which proves that

$$\mu^+: \mathcal{A} \rightarrow [0, \infty] \text{ and } \mu^-: \mathcal{A} \rightarrow [0, \infty] \text{ are measures} \quad (21.807)$$

For μ we have either:

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$. Suppose that there exist a $A \in \mathcal{A}$ such that $\mu^-(A) = \infty$ then $\infty = -\mu(A \cap N) \Rightarrow \mu(A \cap N) = -\infty$ contradicting $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{\infty}$. Hence we have that $\forall A \in \mathcal{A} \mu^-(A) \in [0, \infty[$, in other words μ^- is finite.

$\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$. Suppose that there exist a $A \in \mathcal{A}$ such that $\mu^+(A) = \infty$ then $\mu(A \cap N) = \infty$ contradicting $\{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}_{-\infty}$. Hence we have that $\forall A \in \mathcal{A} \mu^+(A) \in [0, \infty[$, in other words μ^+ is finite.

Hence we have that

$$\mu^+ \text{ is finite or } \mu^- \text{ is finite} \quad (21.808)$$

Let $A \in \mathcal{A}$ then as $X = P \sqcup N$ and $P \cap N = \emptyset$ we have

$$\begin{aligned} \mu(A) &= \mu(A \cap X) \\ &= \mu(A \cap (P \sqcup N)) \\ &= \mu((A \cap P) \sqcup (A \cap N)) \\ &\stackrel{P \cap N = \emptyset}{=} \mu(A \cap P) + \mu(A \cap N) \\ &= \mu^+(A) - \mu^-(A) \end{aligned}$$

Hence

$$\forall A \in \mathcal{A} \text{ we have } \mu(A) = \mu^+(A) - \mu^-(A) \quad (21.809)$$

So the only thing left to prove is that we can define μ^+ and μ^- in terms of the supremum. Let $A \in \mathcal{A}$ then if $B \in \mathcal{A}$ with $B \subseteq A$ we have

$$\begin{aligned} \mu(B) & \underset{[\text{eq: 21.809}]}{=} \mu^+(B) - \mu^-(B) \underset{[\text{eq: 21.808}] \wedge [\text{theorem: 20.27}]}{\leq} \mu^+(B) \underset{[\text{theorem: 21.99}]}{\leq} \mu^+(A) \\ \text{so that} \quad & \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \leq \mu^+(A) \end{aligned} \quad (21.810)$$

Further, as $A \cap P \in \mathcal{A}$, $A \cap P \subseteq A$ and $\mu^+(A) = \mu(A \cap P)$, we have $\mu^+(A) \in \{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}$ so that $\mu^+(A) \leq \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\})$. Combining this with [eq: 21.810] proves that $\mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\})$. Hence

$$\forall A \in \mathcal{A} \text{ we have } \mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \quad (21.811)$$

Similar, let $A \in \mathcal{A}$ then if $B \in \mathcal{A}$ with $B \subseteq A$ we have

$$\begin{aligned} -\mu(B) & \underset{[\text{eq: 21.809}]}{=} \mu^-(B) - \mu^+(B) \underset{[\text{eq: 21.808}] \wedge [\text{theorem: 20.27}]}{\leq} \mu^-(B) \underset{[\text{theorem: 21.99}]}{\leq} \mu^-(A) \\ \text{so that} \quad & \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \leq \mu^-(A) \end{aligned} \quad (21.812)$$

Further as $A \cap N \in \mathcal{A}$, $A \cap N \subseteq A$ and $\mu^-(A) = -\mu(N \cap A)$, we have that $\mu^-(A) \in \{-\mu(B) | B \in \mathcal{A} \text{ and } B \subseteq A\}$ so that $\mu^-(A) \leq \sup(\{-\mu(B) | B \in \mathcal{A} \text{ and } B \subseteq A\})$. Combining this with [eq: 21.812] proves that $\mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\})$. Hence

$$\forall A \in \mathcal{A} \text{ we have } \mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \quad (21.813) \quad \square$$

Theorem 21.489. (Jordan Decomposition) Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ a signed measure then

$\forall A \in \mathcal{A}$ we have $\sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \subseteq [0, \infty]$ and $\sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \subseteq [0, \infty]$

$\mu^+: \mathcal{A} \rightarrow [0, \infty]$ defined by $\mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\})$ is a measure on $\mathcal{A}$

$\mu^-: \mathcal{A} \rightarrow [0, \infty]$ defined by $\mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\})$ is a measure on $\mathcal{A}$

μ^+ or μ^- are finite

If μ is finite then μ^+ and μ^- are finite

$$\mu = \mu^+ - \mu^-$$

Further there exist a Hahn Decomposition (P, N) of μ such that

$$\forall A \in \mathcal{A} \text{ we have } \mu^+(A) = \mu(A \cap P) \text{ and } \mu^-(A) = -\mu(A \cap N)$$

The function

$$|\mu| = \mu^+ + \mu^-: \mathcal{A} \rightarrow [0, \infty] \text{ is a measure and } \forall A \in \mathcal{A} \text{ we have } |\mu(A)| \leq |\mu|(A)$$

For every Hahn Decomposition (P', N') of μ we have that

$$\forall A \in \mathcal{A} \mu^+(A) = \mu(A \cap P') \text{ and } \mu^-(A) = -\mu(A \cap N')$$

Further if there exists a measure ν on $\langle X, \mathcal{A} \rangle$ such that $\forall A \in \mathcal{A} |\mu(A)| \leq \nu(A)$ then $\forall A \in \mathcal{A}$ we have that $|\mu|(A) \leq \nu(A)$. In other words $|\mu|$ is the smallest measure such that $\forall A \in \mathcal{A} |\mu(A)| < \nu(A)$. The pair (μ^+, μ^-) is called a **Jordan Decomposition** of μ . μ^+ is called the **positive part of μ** and μ^- is called the **negative part of μ** . The function $|\mu|$ is called the **variation of μ** . the **total variation** of the signed measure μ noted as $\|\mu\|$ is defined to be

$$\|\mu\| = |\mu|(X) = \mu^+(X) + \mu^-(X)$$

Proof. Using the Hahn Decomposition Theorem [see theorem: 21.487] there exist a Hahn Decomposition (P, N) of μ . Using the previous theorem [theorem: 21.488] we have that

$$\forall A \in \mathcal{A} \text{ we have } \mu(A \cap P) \in [0, \infty] \wedge -\mu(A \cap N) \in [0, \infty] \quad (21.814)$$

$$\mu^+: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu^+(A) = \mu(A \cap P) \text{ is a measure on } \mathcal{A} \quad (21.815)$$

$$\mu^-: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu^-(A) = -\mu(A \cap N) \text{ is a measure on } \mathcal{A} \quad (21.816)$$

$$\mu^+ \text{ or } \mu^- \text{ is finite} \quad (21.817)$$

$$\forall A \in \mathcal{A} \text{ we have } \mu(A) = \mu^+(A) - \mu^-(A) \quad (21.818)$$

$$\forall A \in \mathcal{A} \mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \text{ and } \mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \quad (21.819)$$

Applying [eq: 21.819] on [eqs: 21.814, 21.815, 21.816, 21.817 and 21.818] we have that

$$\forall A \in \mathcal{A} \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \in [0, \infty] \text{ and } \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) \in [0, \infty]$$

$$\mu^+: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) = \mu(A \cap P) \text{ is a measure on } \mathcal{A}$$

$$\mu^-: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) = -\mu(A \cap N) \text{ is a measure on } \mathcal{A}$$

$$\mu^+ \text{ or } \mu^- \text{ is finite}$$

$$\forall A \in \mathcal{A} \text{ we have } \mu(A) = \mu^+(A) - \mu^-(A)$$

As μ^+, μ^- are measures on $\mathcal{A}$ we have by [theorem: 21.108] that

$$|\mu| \stackrel{\text{def}}{=} \mu^+ + \mu^-: \mathcal{A} \rightarrow [0, \infty] \text{ is a measure on } \mathcal{A}$$

If μ is finite then $\forall A \in \mathcal{A}$ we have $\mu^+(A) = \mu(A \cap P) \in \mathbb{R} \wedge \mu^-(A) = \mu(A \cap N) \in \mathbb{R}$ so that we have

$$\text{If } \mu \text{ is finite then } \mu^+ \text{ and } \mu^- \text{ are finite}$$

Let $A \in \mathcal{A}$ then

$$|\mu(A)| = |\mu^+(A) - \mu^-(A)| \leq |\mu^+(A)| + |\mu^-(A)| \stackrel{0 \leq \mu^+(A), \mu^-(A)}{=} \mu^+(A) + \mu^-(A) = |\mu|(A)$$

which proves that

$$\forall A \in \mathcal{A} \text{ we have } |\mu(A)| \leq |\mu|(A)$$

Further if (P', N') is any Hahn Decomposition of μ we have by [theorem: 21.488] that

$$\mu^+(A) = \sup(\{\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) = \mu(A \cap P')$$

$$\mu^-(A) = \sup(\{-\mu(B) | B \in \mathcal{A} \wedge B \subseteq A\}) = \mu(A \cap N')$$

Let ν be a measure on $\langle X, \mathcal{A} \rangle$ such that $\forall A \in \mathcal{A}$ we have $|\mu(A)| < \nu(A)$. If $A \in \mathcal{A}$ then we have

$$\begin{aligned} \mu^+(A) &\stackrel{[\text{eq: 21.815}]}{=} \mu(A \cap P) \\ &\stackrel{[\text{eq: 21.814}]}{=} |\mu(A \cap P)| \\ &\leq \nu(A \cap P) \\ \mu^-(A) &\stackrel{[\text{eq: 21.816}]}{=} -\mu(A \cap N) \\ &\stackrel{[\text{eq: 21.814}]}{=} |\mu(A \cap N)| \\ &\leq \nu(A \cap N) \end{aligned}$$

So that

$$\begin{aligned}
 |\mu|(A) &= \mu^+(A) + \mu^-(A) \\
 &\leq \nu(A \cap P) + \nu(A \cap N) \\
 &\stackrel{(A \cap P) \cap (A \cap N) = \emptyset}{=} \nu((A \cap P) \sqcup (A \cap N)) \\
 &= \nu(A \cap (P \sqcup N)) \\
 &= \nu(A \cap X) \\
 &= \nu(A)
 \end{aligned}$$

which proves that

$$\forall A \in \mathcal{A} \text{ we have } |\mu|(A) \leq \nu(A)$$

□

21.6.2 Complex Measure

21.6.2.1 Definition and properties

Just as we defined signed measures it is possible to define complex measures.

Definition 21.490. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then a function $\mu: \mathcal{A} \rightarrow \mathbb{C}$ is a **complex measure** if

1. $\mu(\emptyset) = 0$
2. $\forall \{A_i\}_{i \in \mathbb{N}_0} \subseteq \mathcal{A}$ pairwise disjoint we have

$$\sum_{i=1}^{\infty} \mu(A_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) \text{ exists [using the norm } |z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}]$$

and

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Note 21.491. As $\forall A \in \mathcal{A}$ we have that $\mu(A) \in \mathbb{C}$ it follows that $\operatorname{Im}(\mu(A)), \operatorname{Re}(\mu(A)) \in \mathbb{R}$, or the real and imaginary parts are finite.

We can express a complex measure as the sum of two signed measures.

Theorem 21.492. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: X \rightarrow \mathbb{C}$ a complex measure then

$\operatorname{Re}(\mu): X \rightarrow \overline{\mathbb{R}}$ defined by $\operatorname{Re}(\mu)(A) = \operatorname{Re}(\mu(A))$ is a finite signed measure on $\mathcal{A}$

$\operatorname{Im}(\mu): X \rightarrow \overline{\mathbb{R}}$ defined by $\operatorname{Im}(\mu)(A) = \operatorname{Im}(\mu(A))$ is a finite signed measure on $\mathcal{A}$

Note 21.493. As $\forall A \in \mathcal{A}$ we have

$$\mu(A) = \operatorname{Re}(\mu(A)) + i \cdot \operatorname{Im}(\mu(A)) = \operatorname{Re}(\mu)(A) + \operatorname{Im}(\mu)(A) = (\operatorname{Re}(\mu) + i \cdot \operatorname{Im}(\mu))(A)$$

we have that

$$\mu = \operatorname{Re}(\mu) + i \cdot \operatorname{Im}(\mu)$$

Proof. First we have that

$$\operatorname{Re}(\mu)(\emptyset) = \operatorname{Re}(\mu(\emptyset)) = \operatorname{Re}(0) = 0 \text{ and } \operatorname{Im}(\mu)(\emptyset) = \operatorname{Im}(\mu(\emptyset)) = \operatorname{Im}(0) = 0 \quad (21.820)$$

$$\forall A \in \mathcal{A} \operatorname{Re}(\mu)(A) = \operatorname{Re}(\mu(A)) \in \mathbb{R} \text{ and } \operatorname{Im}(\mu)(A) = \operatorname{Im}(\mu(A)) \in \mathbb{R} \quad (21.821)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then $\sum_{i=1}^{\infty} \mu(A_i)$ exist and $\sum_{i=1}^{\infty} \mu(A_i) = \mu(\bigcup_{i \in \mathbb{N}} A_i)$. Using [theorem: 14.419] it follows that

$$\sum_{i \rightarrow \infty} \operatorname{Re}(\mu(A_i)) \text{ exist with } \sum_{i \rightarrow \infty} \operatorname{Re}(\mu(A_i)) = \operatorname{Re}\left(\sum_{i=1}^{\infty} \mu(A_i)\right) \quad (21.822)$$

$$\sum_{i \rightarrow \infty} \operatorname{Im}(\mu(A_i)) \text{ exist with } \sum_{i \rightarrow \infty} \operatorname{Im}(\mu(A_i)) = \operatorname{Im}\left(\sum_{i=1}^{\infty} \mu(A_i)\right) \quad (21.823)$$

Hence we have, as $\forall i \in \mathbb{N} \operatorname{Re}(\mu)(A_i) = \operatorname{Re}(\mu(A_i))$ and $\operatorname{Im}(\mu)(A_i) = \operatorname{Im}(\mu(A_i))$, that

$$\begin{aligned} \sum_{i=1}^{\infty} \operatorname{Re}(\mu)(A_i) &\text{ exist and } \sum_{i=1}^{\infty} \operatorname{Im}(\mu)(A_i) \text{ exist} \\ \sum_{i=1}^{\infty} \operatorname{Re}(\mu)(A_i) &= \operatorname{Re}\left(\sum_{i=1}^{\infty} \mu(A_i)\right) \stackrel{[\text{eq: 21.822}]}{=} \operatorname{Re}\left(\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)\right) = \operatorname{Re}(\mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\ \sum_{i=1}^{\infty} \operatorname{Im}(\mu)(A_i) &= \operatorname{Im}\left(\sum_{i=1}^{\infty} \mu(A_i)\right) \stackrel{[\text{eq: 21.823}]}{=} \operatorname{Im}\left(\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)\right) = \operatorname{Im}(\mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \end{aligned}$$

The above together with [eqs: 21.820, 21.821] proves that

$\operatorname{Re}(\mu)$ and $\operatorname{Im}(\mu)$ are finite signed measures □

Complex measures are evidently finite additive

Theorem 21.494. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $n \in \mathbb{N}$ and $\mu: \mathcal{A} \rightarrow \mathbb{C}$ a complex measure on $\mathcal{A}$ then*

1. $\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint we have that

$$\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \mu(A_i)$$

2. $\forall A, B \in \mathcal{A}$ with $B \subseteq A$ we have $\mu(A \setminus B) = \mu(A) - \mu(B)$

In other words μ is finite additive.

Proof.

1. Using the previous theorem [theorem: 21.492] we have that $\operatorname{Re}(\mu)$ and $\operatorname{Im}(\mu)$ are signed measures, hence by [theorem: 21.475] we have that

$$\operatorname{Re}(\mu)\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \operatorname{Re}(\mu)(A_i) \text{ and } \operatorname{Im}(\mu)\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n \operatorname{Im}(\mu)(A_i)$$

So that

$$\begin{aligned} \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) &= \operatorname{Re}\left(\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right)\right) + i \cdot \operatorname{Im}\left(\mu\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right)\right) \\ &= \operatorname{Re}(\mu)\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) + i \cdot \operatorname{Im}(\mu)\left(\bigsqcup_{i \in \{1, \dots, n\}} A_i\right) \\ &= \sum_{i=1}^n \operatorname{Re}(\mu)(A_i) + i \cdot \sum_{i=1}^n \operatorname{Im}(\mu)(A_i) \\ &= \sum_{i=1}^n (\operatorname{Re}(\mu)(A_i) + i \cdot \operatorname{Im}(\mu)(A_i)) \\ &= \sum_{i=1}^n (\operatorname{Re}(\mu(A_i)) + i \cdot \operatorname{Im}(\mu(A_i))) \\ &= \sum_{i=1}^n \mu(A_i) \end{aligned}$$

2. As $B \subseteq A$ we have $A = (A \setminus B) \cup B$ with $(A \setminus B) \cap B = \emptyset$ so that $\mu(A) \stackrel{(1)}{=} \mu(A \setminus B) + \mu(B)$ so that $\mu(A \setminus B) = \mu(A) - \mu(B)$. $\square$

We have now similar theorems as [theorems: 21.477, 21.478] for complex measures.

Theorem 21.495. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: \mathcal{A} \rightarrow \mathbb{C}$ a complex measure on $\mathcal{A}$ then we have:*

1. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N}_0 \ A_i \subseteq A_{i+1}$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and

$$\mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

2. $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N}_0 \ A_{i+1} \subseteq A_i$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and

$$\mu\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i)$$

Proof.

1. As $\text{Re}(\mu)$, $\text{Im}(\mu)$ are **finite signed** measures [see theorem: 21.492] it follows from [theorem: 21.477] that

$$\lim_{i \rightarrow \infty} \text{Re}(\mu)(A_i) \text{ exist and } \lim_{i \rightarrow \infty} \text{Im}(\mu)(A_i) \text{ exist}$$

and

$$\lim_{i \rightarrow \infty} \text{Re}(\mu)(A_i) = \text{Re}(\mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \text{ and } \lim_{i \rightarrow \infty} \text{Im}(\mu)(A_i) = \text{Im}(\mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

Using [theorem: 14.361] on the above we have then that

$$\lim_{i \rightarrow \infty} \mu(A_i) \stackrel{[\text{theorem: 21.492}]}{=} \lim_{i \rightarrow \infty} (\text{Re}(\mu)(A_i) + i \cdot \text{Im}(\mu)(A_i)) \text{ exist}$$

and

$$\begin{aligned} \lim_{i \rightarrow \infty} \mu(A_i) &\stackrel{[\text{theorem: 21.492}]}{=} \lim_{i \rightarrow \infty} (\text{Re}(\mu)(A_i) + i \cdot \text{Im}(\mu)(A_i)) \\ &\stackrel{[\text{theorem: 14.361}]}{=} \lim_{i \rightarrow \infty} \text{Re}(\mu)(A_i) + i \cdot \lim_{i \rightarrow \infty} \text{Im}(\mu)(A_i) \\ &= \text{Re}(\mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) + i \cdot \text{Im}(\mu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\ &= \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \end{aligned}$$

Summarized

$$\lim_{i \rightarrow \infty} \mu(A_i) \text{ exist and } \lim_{i \rightarrow \infty} \mu(A_i) = \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

2. As $\text{Re}(\mu)$, $\text{Im}(\mu)$ are **finite signed** measures [see theorem: 21.492] it follows from [theorem: 21.477] that

$$\lim_{i \rightarrow \infty} \text{Re}(\mu)(A_i) \text{ exist and } \lim_{i \rightarrow \infty} \text{Im}(\mu)(A_i) \text{ exist}$$

and

$$\lim_{i \rightarrow \infty} \text{Re}(\mu)(A_i) = \text{Re}(\mu)\left(\bigcap_{i \in \mathbb{N}} A_i\right) \text{ and } \lim_{i \rightarrow \infty} \text{Im}(\mu)(A_i) = \text{Im}(\mu)\left(\bigcap_{i \in \mathbb{N}} A_i\right)$$

Using [theorem: 14.361] on the above we have then that

$$\lim_{i \rightarrow \infty} \mu(A_i) \stackrel{[\text{theorem: 21.492}]}{=} \lim_{i \rightarrow \infty} (\text{Re}(\mu)(A_i) + i \cdot \text{Im}(\mu)(A_i)) \text{ exist}$$

and

$$\begin{aligned}
 \lim_{i \rightarrow \infty} \mu(A_i) &\stackrel{[\text{theorem: 21.492}]}{=} \lim_{i \rightarrow \infty} (\operatorname{Re}(\mu)(A_i) + i \cdot \operatorname{Im}(\mu)(A_i)) \\
 &\stackrel{[\text{theorem: 14.361}]}{=} \lim_{i \rightarrow \infty} \operatorname{Re}(\mu)(A_i) + i \cdot \lim_{i \rightarrow \infty} \operatorname{Im}(\mu)(A_i) \\
 &= \operatorname{Re}(\mu) \left(\bigcap_{i \in \mathbb{N}} A_i \right) + i \cdot \operatorname{Im} \left(\bigcap_{i \in \mathbb{N}} A_i \right) \\
 &= \mu \left(\bigcap_{i \in \mathbb{N}} A_i \right)
 \end{aligned}$$

Summarized

$$\lim_{i \rightarrow \infty} \mu(A_i) \text{ exist and } \lim_{i \rightarrow \infty} \mu(A_i) = \mu \left(\bigcap_{i \in \mathbb{N}} A_i \right) \quad \square$$

Theorem 21.496. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and let $\mu: X \rightarrow \mathbb{C}$ be a complex function such that

- a) $\mu(\emptyset) = 0$
 b) For all finite pairwise disjoint measurable sets $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $n \in \mathbb{N}$ we have $\mu(\bigsqcup_{i \in \{1, \dots, n\}} A_i) = \sum_{i=1}^n \mu(A_i)$

then we have

1. If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and $\mu(\bigcup_{i \in \mathbb{N}_0} A_i) = \lim_{n \rightarrow \infty} \mu(A_n)$ then μ is a complex measure on $\mathcal{A}$.
2. If $\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} A_{i+1} \subseteq A_i$ and $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$ we have that $\lim_{i \rightarrow \infty} \mu(A_i)$ exists and $\lim_{i \rightarrow \infty} \mu(A_i) = 0$ then μ is a complex measure on $\mathcal{A}$.

Proof.

1. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be a pairwise disjoint sequence of measurable sets. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_n = \bigsqcup_{i \in \{1, \dots, n\}} A_i$$

Then using [lemma: 21.7] we have that

$$\bigcup_{i \in \mathbb{N}} B_i = \bigsqcup_{i \in \mathbb{N}} A_i \text{ and } \forall i \in \mathbb{N} \text{ we have } B_i \subseteq B_{i+1}$$

Hence we have

$$\begin{aligned}
 \mu \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) &= \mu \left(\bigcup_{i \in \mathbb{N}} B_i \right) \\
 &\stackrel{\text{hypothesis}}{=} \lim_{i \rightarrow \infty} \mu(B_i) \text{ [where the limit exist]} \\
 &= \lim_{i \rightarrow \infty} \mu \left(\bigsqcup_{j \in \{1, \dots, i\}} A_j \right) \\
 &\stackrel{(a)}{=} \lim_{i \rightarrow \infty} \sum_{j=1}^i \mu(A_j) \\
 &= \sum_{j=1}^{\infty} \mu(A_j)
 \end{aligned}$$

which together with (a) proves that μ is a complex measure.

2. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be a pairwise disjoint sequence of measurable sets. Define

$$\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_i = \bigsqcup_{j \in \{i, \dots, \infty\}} A_j$$

Then $\forall i \in \mathbb{N}$ we have $B_{i+1} = \bigsqcup_{j \in \{i+1, \dots, \infty\}} A_j \stackrel{[\text{theorem: 2.136}]}{\subseteq} \bigsqcup_{j \in \{i, \dots, \infty\}} A_j = B_i$ proving that

$$\forall i \in \mathbb{N} \text{ we have } B_{i+1} \subseteq B_i \quad (21.824)$$

Assume that $x \in \bigcap_{i \in \mathbb{N}} B_i$ then $x \in B_1$ so there exist a $k \in \{1, \dots, \infty\}$ such that $x \in A_k$. From $x \in \bigcap_{i \in \mathbb{N}} B_i$ we have $x \in B_{k+1}$ so there exist a $j \in \{k+1, \dots, \infty\}$ such that $x \in A_j$ hence $x \in A_k \cap A_j \stackrel{k \neq j}{=} \emptyset$ giving the contradiction $x \in \emptyset$. So the assumption is false and we must have

$$\bigcap_{i \in \mathbb{N}} B_i = \emptyset$$

Using the hypothesis it follows that $\lim_{i \rightarrow \infty} \mu(B_i)$ exist and $\lim_{i \rightarrow \infty} \mu(B_i) = 0$. Using then [theorem: 14.341] it follows that

$$\lim_{i \rightarrow \infty} \mu(B_{i+1}) = 0 \quad (21.825)$$

Let $k \in \mathbb{N}$. If $x \in \left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) \cap \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i \right)$ then $\exists i \in \{1, \dots, k\}$ and a $j \in \{k+1, \dots, \infty\}$ such that $x \in A_i \cap A_j$. As $i \leq k < k+1 \leq j$ we have that $A_i \cap A_j = \emptyset$ contradicting $x \in A_i \cap A_j$ so we have

$$\left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) \cap \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i \right) = \emptyset$$

and

$$\bigcup_{i \in \mathbb{N}} A_i = \left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) \cup \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i \right)$$

So that

$$\begin{aligned} \mu \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) &\stackrel{(b)}{=} \mu \left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) + \mu \left(\bigcup_{i \in \{k+1, \dots, \infty\}} A_i \right) \\ &= \mu \left(\bigcup_{i \in \{1, \dots, k\}} A_i \right) + \mu(B_{k+1}) \\ &\stackrel{(b)}{=} \sum_{i=1}^k \mu(A_i) + \mu(B_{k+1}) \end{aligned}$$

Hence we have that

$$\mu \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) = \sum_{i=1}^k \mu(A_i) + \mu(B_{k+1})$$

so that

$$\forall k \in \mathbb{N} \text{ we have } \left| \mu \left(\bigcup_{i \in \mathbb{N}} A_i \right) - \sum_{i=1}^k \mu(A_i) \right| = |\mu(B_{k+1})| \quad (21.826)$$

Let $\varepsilon \in \mathbb{R}^+$. Then by [eq: 21.825] there exist a $N \in \mathbb{N}$ such that $\forall k \in \{N, \dots, \infty\}$

$$|\mu(B_{k+1})| < \varepsilon \stackrel{[\text{eq: 21.826}]}{\Rightarrow} \left| \mu \left(\bigcup_{i \in \mathbb{N}} A_i \right) - \sum_{i=1}^k \mu(A_i) \right| < \varepsilon$$

which proves that

$$\sum_{i=1}^{\infty} \mu(A_i) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(A_i) \text{ exist and } \sum_{i=1}^{\infty} \mu(A_i) = \mu \left(\bigcup_{i \in \mathbb{N}} A_i \right)$$

This together with (a) proves that

$$\mu \text{ is a complex measure}$$

□

21.6.2.2 Jordan decomposition of a complex measure

As every complex measure μ can be written as $\mu = \text{Re}(\mu) + i \cdot \text{Im}(\mu)$, we can do a Jordan Decomposition of the real and imaginary part of μ . This lead to the following definition.

Theorem 21.497. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and $\mu: X \rightarrow \mathbb{C}$ a complex measure then

$\mu_r^+: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined by $\mu_r^+(A) = \sup \{ \operatorname{Re}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a finite measure on $\mathcal{A}$

$\mu_r^-: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined by $\mu_r^-(A) = \sup \{ -\operatorname{Re}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a finite measure on $\mathcal{A}$

$\mu_c^+: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined by $\mu_c^+(A) = \sup \{ \operatorname{Im}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a finite measure on $\mathcal{A}$

$\mu_c^-: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ defined by $\mu_c^-(A) = \sup \{ -\operatorname{Im}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a finite measure on $\mathcal{A}$

and

$$\mu = \mu_r^+ - \mu_r^- + i \cdot (\mu_c^+ - \mu_c^-)$$

The quadruple is called the *Jordan Decomposition* of μ

Proof. Using [theorem: 21.492] it follows that $\operatorname{Re}(\mu)$ and $\operatorname{Im}(\mu)$ are finite signed measures such that

$$\mu = \operatorname{Re}(\mu) + i \cdot \operatorname{Im}(\mu)$$

Using [theorem: 21.489] we have that

$(\operatorname{Re}(\mu))^+: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ by $(\operatorname{Re}(\mu))^+(A) = \sup \{ \operatorname{Re}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a measure on $\mathcal{A}$

$(\operatorname{Re}(\mu))^-: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ by $(\operatorname{Re}(\mu))^-(A) = \sup \{ -\operatorname{Re}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a measure on $\mathcal{A}$

$(\operatorname{Im}(\mu))^+: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ by $(\operatorname{Im}(\mu))^+(A) = \sup \{ \operatorname{Im}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a measure on $\mathcal{A}$

$(\operatorname{Im}(\mu))^-: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ by $(\operatorname{Im}(\mu))^-(A) = \sup \{ -\operatorname{Im}(\mu)(B) \mid B \in \mathcal{A} \wedge B \subseteq A \}$ is a measure on $\mathcal{A}$

and

$$\operatorname{Re}(\mu) = (\operatorname{Re}(\mu))^+ - (\operatorname{Re}(\mu))^- \text{ and } \operatorname{Im}(\mu) = (\operatorname{Im}(\mu))^+ - (\operatorname{Im}(\mu))^-$$

As $\operatorname{Re}(\mu), \operatorname{Im}(\mu)$ are finite we have by [theorem: 21.489] that

$$(\operatorname{Re}(\mu))^+, (\operatorname{Re}(\mu))^-, (\operatorname{Im}(\mu))^+, (\operatorname{Im}(\mu))^- \text{ are finite measures}$$

Further

$$\mu = (\operatorname{Re}(\mu))^+ + (\operatorname{Re}(\mu))^- + i \cdot ((\operatorname{Im}(\mu))^+ - (\operatorname{Im}(\mu))^-)$$

The theorem is then proved by taking $\mu_r^+ = (\operatorname{Re}(\mu))^+$, $\mu_r^- = (\operatorname{Re}(\mu))^-$, $\mu_c^+ = (\operatorname{Im}(\mu))^+$ and $\mu_c^- = (\operatorname{Im}(\mu))^-$. $\square$

Definition 21.498. (Variation of a Complex Measure) Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu: \mathcal{A} \rightarrow \mathbb{C}$ a complex measure then the **variation** of μ noted by $|\mu|$ is defined by

$$|\mu|: \mathcal{A} \rightarrow [0, \infty] \text{ where } |\mu|(A) = \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right)$$

In the above definition we use $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$ to indicate that $\{A_i\}_{i \in \{1, \dots, n\}}$ is pairwise disjoint and $A = \bigcup_{i \in \{1, \dots, n\}} A_i$. The **total variation** of μ noted by $\|\mu\|$ is defined by

$$\|\mu\| = |\mu|(X)$$

in other words

$$\|\mu\| = \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right)$$

Note 21.499. As $A = \bigsqcup_{i \in \{1\}} A_i$ where $A_1 = A$ we have that

$$\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \neq \emptyset$$

so that $\sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right)$ is well defined.

Theorem 21.500. Let $\langle X, \mathcal{A} \rangle$ and $\mu: \mathcal{A} \rightarrow \mathbb{C}$ a complex measure then the variation

$$|\mu|: \mathcal{A} \rightarrow [0, \infty]$$

is a finite measure. Further we have that

$$\forall A \in \mathcal{A} \text{ we have that } |\mu(A)| \leq |\mu|(A)$$

Proof. First we prove that $|\mu|$ is finite. Use [theorem: 21.497] to get a Jordan decomposition of **finite measures** $(\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-)$ of μ . Let $A \in \mathcal{A}$ and take $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint with $A = \bigsqcup_{i=1}^n A_i$ then we have

$$\begin{aligned} \sum_{i=1}^n |\mu(A_i)| &= \\ \sum_{i=1}^n |\mu_r^+(A_i) + \mu_r^-(A_i) + i \cdot (\mu_c^+(A_i) - \mu_c^-(A_i))| &\leq \\ \sum_{i=1}^n (|\mu_r^+(A_i)| + |\mu_r^-(A_i)| + |\mu_c^+(A_i)| + |\mu_c^-(A_i)|) &= \\ \sum_{i=1}^n |\mu_r^+(A_i)| + \sum_{i=1}^n |\mu_r^-(A_i)| + \sum_{i=1}^n |\mu_c^+(A_i)| + \sum_{i=1}^n |\mu_c^-(A_i)| & \text{measures are non negative} \\ \sum_{i=1}^n \mu_r^+(A_i) + \sum_{i=1}^n \mu_r^-(A_i) + \sum_{i=1}^n \mu_c^+(A_i) + \sum_{i=1}^n \mu_c^-(A_i) &= \\ \mu_r^+\left(\bigsqcup_{i=1}^n A_i\right) + \mu_r^-\left(\bigsqcup_{i=1}^n A_i\right) + \mu_c^+\left(\bigsqcup_{i=1}^n A_i\right) + \mu_c^-\left(\bigsqcup_{i=1}^n A_i\right) &= \\ \mu_r^+(A) + \mu_r^-(A) + \mu_c^+(A) + \mu_c^-(A) & \end{aligned}$$

which proves that $\mu_r^+(A) + \mu_r^-(A) + \mu_c^+(A) + \mu_c^-(A)$ is a real upper bound of

$$\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\}$$

So that by [theorems: 10.18] we have that $\forall A \in \mathcal{A}$

$$\begin{aligned} |\mu|(A) &= \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) \\ &\leq \mu_r^+(A) + \mu_r^-(A) + \mu_c^+(A) + \mu_c^-(A) \in \mathbb{R} \end{aligned}$$

which proves that

$$\forall A \in \mathcal{A} \text{ we have } |\mu|(A) \in \mathbb{R} \text{ and } |\mu|(A) \leq \mu_r^+(A) + \mu_r^-(A) + \mu_c^+(A) + \mu_c^-(A) \in \mathbb{R} \quad (21.827)$$

Next we use [theorem: 21.116] to prove that $|\mu|$ is a measure. Let $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ be a pairwise disjoint sequence of measurable sets such that $\emptyset = \bigsqcup_{i \in \mathbb{N}} A_i$ then we have $\forall i \in \mathbb{N}$ that $A_i = \emptyset \Rightarrow |\mu(A_i)| = |\emptyset| = 0$. Hence $\sum_{i=1}^n |\mu(A_i)| = 0$ so that

$$|\mu|(\emptyset) = \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge \emptyset = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) = \sup(\{0\}) = 0$$

proving that

$$|\mu|(\emptyset) = 0 \quad (21.828)$$

Let $A, B \in \mathcal{A}$ with $A \cap B = \emptyset$. Let $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ be pairwise disjoint such that $A \cup B = \bigsqcup_{i \in \{1, \dots, n\}} A_i$. Then given $i \in \{1, \dots, n\}$ we have $A_i = A_i \cap (A \cup B) = (A_i \cap A) \sqcup (A_i \cap B)$ so that $|\mu(A_i)| \stackrel{[\text{theorem: 21.494}]}{=} |\mu(A_i \cap A) + \mu(A_i \cap B)| \leq |\mu(A_i \cap A)| + |\mu(A_i \cap B)|$. Hence we have that

$$\begin{aligned} \sum_{i=1}^n |\mu(A_i)| &\leq \sum_{i=1}^n (|\mu(A_i \cap A)| + |\mu(A_i \cap B)|) \\ &= \sum_{i=1}^n |\mu(A_i \cap A)| + \sum_{i=1}^n |\mu(A_i \cap B)| \end{aligned} \quad (21.829)$$

Further

$$A = A \cap (A \cup B) = A \cap \left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) = \bigsqcup_{i \in \{1, \dots, n\}} (A_i \cap A)$$

and

$$B = B \cap (A \cup B) = B \cap \left(\bigsqcup_{i \in \{1, \dots, n\}} A_i \right) = \bigsqcup_{i \in \{1, \dots, n\}} (A_i \cap B)$$

Hence

$$\begin{aligned} \sum_{i=1}^n |\mu(A_i \cap A)| &\leq \sup \left(\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i=1}^n B_i \right\} \right) \\ &= |\mu|(A) \\ \sum_{i=1}^n |\mu(A_i \cap B)| &\leq \sup \left(\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge B = \bigsqcup_{i=1}^n B_i \right\} \right) \\ &= |\mu|(B) \end{aligned}$$

which combined with [eq: 21.829] proves that

$$\sum_{i=1}^n |\mu(A_i)| \leq |\mu|(A) + |\mu|(B)$$

Hence $|\mu|(A) + |\mu|(B)$ is an upper bound for

$$\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A \sqcup B = \bigsqcup_{i=1}^n B_i \right\}$$

so that

$$\begin{aligned} |\mu|(A \sqcup B) &= \sup \left(\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A \sqcup B = \bigsqcup_{i=1}^n B_i \right\} \right) \\ &\leq |\mu|(A) + |\mu|(B) \end{aligned}$$

which proves that

$$|\mu|(A \sqcup B) \leq |\mu|(A) + |\mu|(B) \quad (21.830)$$

For the opposite inequality. Let $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $\{B_i\}_{i \in \{1, \dots, m\}} \subseteq \mathcal{A}$ be pairwise disjoint such that $A = \bigsqcup_{i=1}^n A_i$, $B = \bigsqcup_{i=1}^m B_i$. Define

$$\{C_i\}_{i \in \{1, \dots, n+m\}} \text{ by } \begin{cases} C_i = A_i & \text{if } i \in \{1, \dots, n\} \\ C_i = B_{i-n} & \text{if } i \in \{n+1, \dots, n+m\} \end{cases}$$

then

$$\begin{aligned} x \in A \cup B &\Leftrightarrow x \in A \vee x \in B \\ &\Leftrightarrow (\exists i \in \{1, \dots, n\} x \in A_i) \vee (\exists j \in \{1, \dots, m\} x \in B_j) \\ &\Leftrightarrow (\exists i \in \{1, \dots, n\} x \in C_i) \vee (\exists j \in \{1, \dots, m\} x \in C_{j+n}) \\ &\Leftrightarrow (\exists i \in \{1, \dots, n\} x \in C_i) \vee (\exists j \in \{n+1, \dots, n+m\} x \in C_j) \\ &\Leftrightarrow \exists i \in \{1, \dots, n+m\} x \in C_i \\ &\Leftrightarrow \bigcup_{i \in \{1, \dots, n+m\}} C_i \end{aligned}$$

proving that

$$A \cup B = \bigcup_{i=1}^{n+m} C_i \quad (21.831)$$

Further If $i, j \in \{1, \dots, n+m\}$ with $i \neq j$ then we have either

$$i \in \{1, \dots, n\} \wedge j \in \{1, \dots, n\}. \text{ Then } C_i \cap C_j = A_i \cap A_j = \emptyset$$

$$i \in \{1, \dots, n\} \wedge j \in \{n+1, \dots, n+m\}. \text{ Then } C_i \cap C_j = A_i \cap B_{j-n} \subseteq A \cap B = \emptyset$$

$$i \in \{n+1, \dots, n+m\} \wedge j \in \{1, \dots, n\}. \text{ Then } C_i \cap C_j = B_{i-n} \cap A_j \subseteq B \cap A = \emptyset$$

$$i \in \{n+1, \dots, n+m\} \wedge j \in \{n+1, \dots, n+m\}. \text{ Then } C_i \cap C_j = B_{i-n} \cap B_{j-n} = \emptyset$$

hence we have that

$$\{C_i\}_{i \in \{1, \dots, n+m\}} \subseteq \mathcal{A} \text{ is pairwise disjoint} \quad (21.832)$$

So

$$\begin{aligned} \sum_{i=1}^n |\mu(A_i)| + \sum_{i=1}^m |\mu(B_i)| &= \sum_{i=1}^n |\mu(C_i)| + \sum_{i=1}^m |\mu(C_{i+n})| \\ &= \sum_{i=1}^n |\mu(C_i)| + \sum_{i=n+1}^{n+m} |\mu(C_i)| \\ &= \sum_{i=1}^{n+m} |\mu(C_i)| \\ &\leq \sup \left(\left\{ \sum_{i=1}^n |\mu(C_i)| \mid n \in \mathbb{N} \wedge \{C_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge \bigcup_{i=1}^n C_i = A \cup B \right\} \right) \\ &= |\mu|(A \sqcup B) \end{aligned} \quad (21.833)$$

Then as $\sum_{i=1}^m |\mu(B_i)| \in \mathbb{R}$ we have that $\sum_{i=1}^n |\mu(A_i)| \leq |\mu|(A \sqcup B) - \sum_{i=1}^m |\mu(B_i)|$ so that $|\mu|(A \sqcup B) - \sum_{i=1}^m |\mu(B_i)|$ is an upper bound for

$$\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigcup_{i=1}^n B_i \right\}$$

so that

$$|\mu|(A) = \sup \left(\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigcup_{i=1}^n B_i \right\} \right) \leq |\mu|(A \sqcup B) - \sum_{i=1}^m |\mu(B_i)|$$

As by [eq: 21.827] $|\mu|(A) \in \mathbb{R}$, $|\mu|(A \sqcup B) \in \mathbb{R}$, $\sum_{i=1}^m |\mu(B_i)| \in \mathbb{R}$ we have that

$$\sum_{i=1}^m |\mu(B_i)| \leq |\mu|(A \sqcup B) - |\mu|(A)$$

Hence $|\mu|(A \sqcup B) - |\mu|(A)$ is an upper bound for

$$\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge B = \bigcup_{i=1}^n B_i \right\}$$

so that

$$|\mu|(B) = \sup \left(\left\{ \sum_{i=1}^n |\mu(B_i)| \mid n \in \mathbb{N} \wedge \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge B = \bigcup_{i=1}^n B_i \right\} \right) \leq |\mu|(A \sqcup B) - |\mu|(A)$$

which proves that $|\mu|(A) + |\mu|(B) \leq |\mu|(A \sqcup B)$. Combining this with [eq: 21.830] proves that $|\mu|(A \sqcup B) = |\mu|(A) + |\mu|(B)$. Using then [theorem: 21.9] we have that

$$\forall \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ pairwise disjoint we have } |\mu|\left(\bigcup_{i \in \{1, \dots, n\}} A_i\right) = \sum_{i=1}^n |\mu|(A_i) \quad (21.834)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i$ such that $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$. Then as $\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-$ are finite measures we have by [theorem: 21.115]

$$\lim_{i \rightarrow \infty} \mu_r^+(A_i) = \mu_r^+\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \mu_r^+(\emptyset) = 0$$

$$\lim_{i \rightarrow \infty} \mu_r^-(A_i) = \mu_r^-\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \mu_r^-(\emptyset) = 0$$

$$\lim_{i \rightarrow \infty} \mu_c^+(A_i) = \mu_c^+\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \mu_c^+(\emptyset) = 0$$

$$\lim_{i \rightarrow \infty} \mu_c^-(A_i) = \mu_c^-\left(\bigcap_{i \in \mathbb{N}} A_i\right) = \mu_c^-(\emptyset) = 0$$

Let $\varepsilon \in \mathbb{R}^+$ then there exists $N_r^+ \in \mathbb{N}, N_r^- \in \mathbb{N}, N_c^+ \in \mathbb{N}, N_c^- \in \mathbb{N}$ such that

$$\forall n \in \{N_r^+, \dots, \infty\} \ \mu_r^+(A_i) = |\mu_r^+(A_i) - 0| < \frac{\varepsilon}{4}$$

$$\forall n \in \{N_r^-, \dots, \infty\} \ \mu_r^-(A_i) = |\mu_r^-(A_i) - 0| < \frac{\varepsilon}{4}$$

$$\forall n \in \{N_c^+, \dots, \infty\} \ \mu_c^+(A_i) = |\mu_c^+(A_i) - 0| < \frac{\varepsilon}{4}$$

$$\forall n \in \{N_c^-, \dots, \infty\} \ \mu_c^-(A_i) = |\mu_c^-(A_i) - 0| < \frac{\varepsilon}{4}$$

Take $N = \max(N_r^+, N_r^-, N_c^+, N_c^-)$ then we have $\forall i \in \{N, \dots, \infty\}$

$$||\mu|(A_i) - 0| = |\mu|(A_i) \leq_{[\text{eq: 21.827}]} \mu_r^+(A_i) + \mu_r^-(A_i) + \mu_c^+(A_i) + \mu_c^-(A_i) < \varepsilon$$

which proves that $\lim_{i \rightarrow \infty} |\mu|(A_i) = 0$. Hence we have proved that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \text{ with } \forall i \in \mathbb{N} \ A_{i+1} \subseteq A_i \text{ and } \bigcap_{i \in \mathbb{N}} A_i = \emptyset \text{ we have } \lim_{i \rightarrow \infty} \mu(A_i) = 0 \quad (21.835)$$

So the conditions for [theorem: 21.116] are satisfied by [eqs: 21.825, 21.834 and 21.835] proves that $|\mu|$ is a measure on $\mathcal{A}$. This together with [eq: 21.827] proves that

$$|\mu|: \mathcal{A} \rightarrow [0, \infty] \text{ is a finite measure on } \mathcal{A}$$

Finally let $A \in \mathcal{A}$. Take

$$\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A} \text{ by } A_1 = A$$

then we have that $\{A_i\}_{i \in \{1\}} \subseteq \mathcal{A}$ is pairwise disjoint and $A = \bigsqcup_{i \in \{1\}} A_i$ so that

$$|\mu(A)| = |\mu(A_1)| = \sum_{i=1}^1 |\mu(A_i)| \in \left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N}_0 \in \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\}$$

so that $|\mu(A)| \leq \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N}_0 \in \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) = |\mu|(A)$. Hence we have

$$\forall A \in \mathcal{A} \text{ we have } |\mu(A)| \leq |\mu|(A) \quad \square$$

21.6.3 Banach space of signed (complex) measures

We show now that the collection of finite signed measures [or complex measures] is a Banach space.

Definition 21.501. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then we define $M(X, \mathcal{A}, \mathbb{K})$ where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$ by

$$M(X, \mathcal{A}, \mathbb{R}) = \{\mu \in \mathbb{R}^{\mathcal{A}} \mid \mu: \mathcal{A} \rightarrow \overline{\mathbb{R}} \text{ is a **finite** signed measure on } \mathcal{A}\}$$

$$M(X, \mathcal{A}, \mathbb{C}) = \{\mu \in \mathbb{C}^{\mathcal{A}} \mid \mu: \mathcal{A} \rightarrow \mathbb{C} \text{ is a complex measure on } \mathcal{A}\}$$

Theorem 21.502. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then

$$\forall \mu \in M(X, \mathcal{A}, \mathbb{K}) \text{ we have } \forall A \in \mathcal{A} \text{ that } |\mu|(A) \in [0, \infty[$$

More specific we have for the total variation $|||$ that

$$||\mu|| \underset{\text{definitions: } 21.489, 21.498}{=} |\mu|(X) \in [0, \infty[$$

Proof. We must consider the following cases for $M(X, \mathcal{A}, \mathbb{K})$:

$\mathbb{K} = \mathbb{R}$. If $\mu \in M[X, \mathcal{A}, \mathbb{K}] = M[X, \mathcal{A}, \mathbb{R}]$ then μ is a finite signed measure. Using [theorem: 21.489] we have that $\mu = \mu^+ - \mu^-$ where μ^+, μ^- are finite measures on $\mathcal{A}$ and $|\mu| = \mu^+ + \mu^-$. Hence we have

$$\forall A \in \mathcal{A} \text{ we have } |\mu|(A) = \mu^+(A) + \mu^-(A) \in [0, \infty[.$$

$\mathbb{K} = \mathbb{C}$. If $\mu \in M[X, \mathcal{A}, \mathbb{K}] = M[X, \mathcal{A}, \mathbb{C}]$ then μ is a complex measure. Hence using [theorem: 21.500] $|\mu|$ is a finite measure so that

$$\forall A \in \mathcal{A} \text{ we have } |\mu|(A) \in \mathbb{R} \quad \square$$

We construct now on $M[X, \mathcal{A}, \mathbb{K}]$ a addition operator $+$ and a scalar product operator $\cdot$ making $\langle M[X, \mathcal{A}, \mathbb{K}] \rangle$ a vector space.

Theorem 21.503. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\langle M[X, \mathcal{A}, \mathbb{K}], +, \cdot \rangle$ is a vector space over $\mathbb{K}$ where

$$+: M[X, \mathcal{A}, \mathbb{K}] \times M[X, \mathcal{A}, \mathbb{K}] \rightarrow M[X, \mathcal{A}, \mathbb{K}] \text{ is defined by } (\mu + \nu)(A) = \mu(A) + \nu(A)$$

$$\cdot: \mathbb{K} \times M[X, \mathcal{A}, \mathbb{K}] \rightarrow M[X, \mathcal{A}, \mathbb{K}] \text{ is defined by } (\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$$

$$C_0 \in M[X, \mathcal{A}, \mathbb{K}] \text{ defined by } C_0(A) = 0 \text{ is the neutral element}$$

Proof. We have for $\mathbb{K}$ either:

$\mathbb{K} = \mathbb{R}$. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then

$$\begin{aligned} C_0\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &= 0 = \sum_{i=1}^{\infty} 0 = \sum_{i=1}^{\infty} C_0(A_i) \\ \{C_0(A) \mid A \in \mathcal{A}\} &= \{0\} \subseteq \mathbb{R} \subseteq \mathbb{R}_{\infty} \\ C_0(\emptyset) &= 0 \end{aligned}$$

so that $C_0: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is a finite signed measure. Hence

$$C_0 \in M(X, \mathcal{A}, \mathbb{R}) \Rightarrow M(X, \mathcal{A}, \mathbb{R}) \neq \emptyset \quad (21.836)$$

Let $\mu, \nu \in M[X, \mathcal{A}, \mathbb{R}]$. Then

$$(\mu + \nu)(\emptyset) = \mu(\emptyset) + \nu(\emptyset) = 0 + 0 = 0$$

$$\{(\mu + \nu)(A) \mid A \in \mathcal{A}\} = \{\mu(A) + \nu(A) \mid A \in \mathcal{A}\} \subseteq \mathbb{R} \subseteq \mathbb{R}_{\infty}$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then we have that

$$\begin{aligned} \sum_{i=1}^{\infty} \mu(A_i) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(A_i) \text{ exist and } \sum_{i=1}^{\infty} \mu(A_i) = \mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) \in \mathbb{R} \\ \sum_{i=1}^{\infty} \nu(A_i) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \nu(A_i) \text{ exist and } \sum_{i=1}^{\infty} \nu(A_i) = \nu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) \in \mathbb{R} \end{aligned}$$

Using then the fact that μ, ν are finite measures it follows from [theorem: 14.418] that

$$\sum_{i=1}^{\infty} (\mu(A_i) + \nu(A_i))$$

exist and

$$\begin{aligned} \sum_{i=1}^{\infty} (\mu + \nu)(A_i) &= \sum_{i=1}^{\infty} (\mu(A_i) + \nu(A_i)) \\ &= \sum_{i=1}^{\infty} \mu(A_i) + \sum_{i=1}^{\infty} \nu(A_i) \\ &= \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) + \nu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\ &= (\mu + \nu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \end{aligned}$$

which proves that $\mu + \nu$ is a finite signed measure. Hence

$$\forall \mu, \nu \in M[X, \mathcal{A}, \mathbb{R}] \text{ we have that } \mu + \nu \in M[X, \mathcal{A}, \mu] \quad (21.837)$$

Let $\alpha \in \mathbb{R}$, $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ then by [theorem: 21.474] $\alpha \cdot \mu$ is a signed measure, as also $\{(\alpha \cdot \mu)(A) | A \in \mathcal{A}\} = \{\alpha \cdot \mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R}$ it follows that

$$\forall \mu \in M(X, \mathcal{A}, \mathbb{R}) \text{ we have } \forall \alpha \in \mathbb{R} \text{ that } \alpha \cdot \mu \in M(X, \mathcal{A}, \mathbb{R}) \quad (21.838)$$

Using [eqs: 21.836, 21.837 and 21.838] together with [theorems: 11.69, 11.61] it follows that

$$\langle M(X, \mathcal{A}, \mathbb{R}), +, \cdot \rangle \text{ is a vector space over } \mathbb{R}$$

$\mathbb{K} = \mathbb{C}$. Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then

$$\begin{aligned} C_0\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) &= 0 = \sum_{i=1}^{\infty} 0 = \sum_{i=1}^{\infty} C_0(A_i) \\ C_0(\emptyset) &= 0 \end{aligned}$$

so that $C_0: \mathcal{A} \rightarrow \overline{\mathbb{R}}$ is a complex measure. Hence

$$C_0 \in M(X, \mathcal{A}, \mathbb{C}) \Rightarrow M(X, \mathcal{A}, \mathbb{C}) \neq \emptyset \quad (21.839)$$

Let $\mu, \nu \in M(X, \mathcal{A}, \mathbb{C})$ then we have

$$(\mu + \nu)(\emptyset) = \mu(\emptyset) + \nu(\emptyset) = 0 + 0 = 0 \quad (21.840)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then $\sum_{i=1}^{\infty} \mu(A_i)$, $\sum_{i=1}^{\infty} \nu(A_i)$ exists and $\sum_{i=1}^{\infty} \mu(A_i) = \mu(\bigcup_{i \in \mathbb{N}} A_i)$, $\sum_{i=1}^{\infty} \nu(A_i) = \nu(\bigcup_{i \in \mathbb{N}} A_i)$. By [theorem: 14.418] we have that $\sum_{i=1}^{\infty} (\mu(A_i) + \nu(A_i))$ exist and

$$\begin{aligned} \sum_{i=1}^{\infty} (\mu + \nu)(A_i) &= \sum_{i=1}^{\infty} (\mu(A_i) + \nu(A_i)) \\ &= \sum_{i=1}^{\infty} \mu(A_i) + \sum_{i=1}^{\infty} \nu(A_i) \\ &= \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) + \nu\left(\bigcup_{i \in \mathbb{N}} A_i\right) \\ &= (\mu + \nu)\left(\bigcup_{i \in \mathbb{N}} A_i\right) \end{aligned}$$

which proves that $\mu + \nu$ is a complex measure. Hence

$$\forall \mu, \nu \in M[X, \mathcal{A}, \mathbb{R}] \text{ we have that } \mu + \nu \in M[X, \mathcal{A}, \mu] \quad (21.841)$$

Let $\mu \in M(X, \mathcal{A}, \mathbb{C})$ and $\alpha \in \mathbb{C}$ then we have

$$(\alpha \cdot \nu)(\emptyset) = \alpha \cdot \nu(\emptyset) = \alpha \cdot 0 = 0$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ be pairwise disjoint then $\sum_{i=1}^{\infty} \mu(A_i)$ exist and $\sum_{i=1}^{\infty} \mu(A_i) = \mu(\bigcup_{i \in \mathbb{N}} A_i)$ so that by [theorem: 14.418] $\sum_{i=1}^{\infty} \alpha \cdot \mu(A_i)$ exist and

$$\sum_{i=1}^{\infty} (\alpha \cdot \mu)(A_i) = \sum_{i=1}^{\infty} \alpha \cdot \mu(A_i) = \alpha \cdot \sum_{i=1}^{\infty} \mu(A_i) = \alpha \cdot \mu\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = (\alpha \cdot \mu)\left(\bigsqcup_{i \in \mathbb{N}} A_i\right)$$

which proves that $\alpha \cdot \mu \in M[X, \mathcal{A}, \mathbb{C}]$. Hence

$$\forall u \in M(X, \mathcal{A}, \mathbb{C}) \text{ we have } \forall \alpha \in \mathbb{C} \text{ we have } \alpha \cdot \mu \in M[X, \mathcal{A}, \mathbb{C}] \quad (21.842)$$

Using [eqs: 21.840, 21.841 and 21.842] together with [theorems: 11.69, 11.61] it follows that

$$\langle M[X, \mathcal{A}, \mathbb{C}], +, \cdot \rangle \text{ is a vector space over } \mathbb{C} \quad \square$$

Next we prove that $\langle M[X, \mathcal{A}, \mathbb{K}], ||| \rangle$ is a normed space.

Theorem 21.504. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\langle M[X, \mathcal{A}, \mathbb{R}], ||| \rangle$ is a normed space. Here $|||$ is defined [see definition: 21.489] to be $\|\mu\| = |\mu|(X)$.*

Proof. Let $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ and let (μ^+, μ^-) be the Jordan Decomposition of μ and $((\alpha \cdot \mu)^+, (\alpha \cdot \mu)^-)$ be the Jordan Decomposition of $\alpha \cdot \mu$ [see theorem: 21.489 for the existence]. Let further (P, N) be a Hahn Decomposition of μ [see theorem: 21.487]. For $\alpha \in \mathbb{R}$ we have either:

$0 \leq \alpha$. Let $A \in \mathcal{A}$ with $A \subseteq P$ [a positive set for μ] then we have $0 \leq \mu(A)$ so that $0 \leq \alpha \cdot \mu(A) = (\alpha \cdot \mu)(A)$ which proves by [definition: 21.479] that P is a positive set for $\alpha \cdot \mu$. Let $A \in \mathcal{A}$ with $A \subseteq N$ [a negative set for μ] then we have $\mu(A) \leq 0$ so that $(\alpha \cdot \mu)(A) = \alpha \cdot \mu(A) \leq 0$ which proves that N is a negative set for $\alpha \cdot \mu$. Hence we have that

(P, N) is a Hahn Decomposition of $\alpha \cdot \mu$

Then we have

$$\begin{aligned} \|\alpha \cdot \mu\| &= |\alpha \cdot \mu|(X) \\ &\stackrel{\text{def}}{=} (\alpha \cdot \mu)^+(X) + (\alpha \cdot \mu)^-(X) \\ &\stackrel{[\text{theorem: 21.489}]}{=} (\alpha \cdot \mu)(X \cap P) - (\alpha \cdot \mu)(X \cap N) \\ &= \alpha \cdot \mu(X \cap P) - \alpha \cdot \mu(X \cap N) \\ &= \alpha \cdot (\mu(X \cap P) - \mu(X \cap N)) \\ &= \alpha \cdot (\mu^+(X) + \mu^-(X)) \\ &= \alpha \cdot |\mu|(X) \\ &= \alpha \cdot \|\mu\| \\ &\stackrel{0 \leq \alpha}{=} |\alpha| \cdot \|\mu\| \end{aligned}$$

$\alpha < 0$. Then if $A \in \mathcal{A}$ with $A \subseteq P$ [a positive set for μ] then $0 \leq \mu(A)$ so that $(\alpha \cdot \mu)(A) = \alpha \cdot \mu(A) \leq 0$ which proves that P is a negative set for $\alpha \cdot \mu$. Likewise if $A \in \mathcal{A}$ with $A \subseteq N$ [a negative set for μ] then $\mu(A) \leq 0$ so that $0 \leq \alpha \cdot \mu(A) = (\alpha \cdot \mu)(A)$ which proves that N is a positive set for $\alpha \cdot \mu$. Hence

(N, P) is a Hahn Decomposition of $\alpha \cdot \mu$

Then we have

$$\begin{aligned}
 \|\alpha \cdot \mu\| &= |\alpha \cdot \mu|(X) \\
 &\stackrel{\text{def}}{=} (\alpha \cdot \mu)^+(X) + (\alpha \cdot \mu)^-(X) \\
 &\stackrel{[\text{theorem: 21.489}]}{=} (\alpha \cdot \mu)(X \cap N) - (\alpha \cdot \mu)(X \cap P) \\
 &= \alpha \cdot \mu(X \cap N) - \alpha \cdot \mu(X \cap P) \\
 &= -\alpha \cdot (\mu(X \cap P) - \mu(X \cap N)) \\
 &\stackrel{\alpha \leq 0}{=} |\alpha| \cdot (\mu(X \cap P) - \mu(X \cap N)) \\
 &= |\alpha| \cdot (\mu^+(X) + \mu^-(X)) \\
 &= |\alpha| \cdot \|\mu\|
 \end{aligned}$$

So we have proved that

$$\forall \mu \in M[X, \mathcal{A}, \mathbb{R}] \text{ we have } \forall \alpha \in \mathbb{R} \text{ that } \|\alpha \cdot \mu\| = |\alpha| \cdot \|\mu\| \quad (21.843)$$

Let $\mu_1, \mu_2 \in M[X, \mathcal{A}, \mathbb{R}]$ then μ_1, μ_2 are finite and $\mu_1 + \mu_2 \in M[X, \mathcal{A}, \mathbb{R}]$. By the Hahn Decomposition Theorem [see theorem: 21.487] there exists Hahn decomposition's $(P_1, N_1), (P_2, N_2), (P_{12}, N_{12})$ of $\mu_1, \mu_2, \mu_1 + \mu_2$ respectively. Then we have

$$\begin{aligned}
 \mu_1(P_{12}) &\stackrel{[\text{theorem: 21.489}]}{=} (\mu_1)^+(P_{12}) - (\mu_1)^-(P_{12}) \\
 &= \mu_1(P_{12} \cap P_1) + \mu_1(P_{12} \cap N_1) \\
 &\leq \mu_1(P_{12} \cap P_1) \\
 &\stackrel{P_{12} \cap N_1 \subseteq N_1 \Rightarrow \mu_1(P_{12} \cap N_1) \leq 0 \wedge \mu_1 \text{ is finite}}{=} \\
 &\stackrel{[\text{lemma: 21.482}]}{\leq} \mu_1(P_1) \quad (21.844)
 \end{aligned}$$

$$\begin{aligned}
 \mu_1(N_{12}) &\stackrel{[\text{theorem: 21.489}]}{=} (\mu_1)^+(N_{12}) - (\mu_1)^-(N_{12}) \\
 &= \mu_1(N_{12} \cap P_1) + \mu_1(N_{12} \cap N_1) \\
 &\geq \mu_1(N_{12} \cap N_1) \\
 &\stackrel{N_{12} \cap P_1 \subseteq P_1 \Rightarrow \mu_1(N_{12} \cap P_1) \geq 0 \wedge \mu_1 \text{ is finite}}{=} \\
 &\stackrel{[\text{lemma: 21.482}]}{\geq} \mu_1(N_1) \quad (21.845)
 \end{aligned}$$

$$\begin{aligned}
 \mu_2(P_{12}) &\stackrel{[\text{theorem: 21.489}]}{=} (\mu_2)^+(P_{12}) - (\mu_2)^-(P_{12}) \\
 &= \mu_2(P_{12} \cap P_2) + \mu_2(P_{12} \cap N_2) \\
 &\leq \mu_2(P_{12} \cap P_2) \\
 &\stackrel{P_{12} \cap N_2 \subseteq N_2 \Rightarrow \mu_2(P_{12} \cap N_2) \leq 0 \wedge \mu_2 \text{ is finite}}{=} \\
 &\stackrel{[\text{lemma: 21.482}]}{\leq} \mu_2(P_2) \quad (21.846)
 \end{aligned}$$

$$\begin{aligned}
 \mu_2(N_{12}) &\stackrel{[\text{theorem: 21.489}]}{=} (\mu_2)^+(N_{12}) - (\mu_2)^-(N_{12}) \\
 &= \mu_2(N_{12} \cap P_2) + \mu_2(N_{12} \cap N_2) \\
 &\geq \mu_2(N_{12} \cap N_2) \\
 &\stackrel{N_{12} \cap P_2 \subseteq P_2 \Rightarrow \mu_2(N_{12} \cap P_2) \geq 0 \wedge \mu_2 \text{ is finite}}{=} \\
 &\stackrel{[\text{lemma: 21.482}]}{\geq} \mu_2(N_2) \quad (21.847)
 \end{aligned}$$

Now

$$\begin{aligned}
\|\mu_1 + \mu_2\| &= |\mu_1 + \mu_2|(X) \\
&= (\mu_1 + \mu_2)^+(X) + (\mu_1 + \mu_2)^-(X) \\
&= (\mu_1 + \mu_2)(X \cap P_{12}) - (\mu_1 + \mu_2)(X \cap N_{12}) \\
&= (\mu_1 + \mu_2)(P_{12}) - (\mu_1 + \mu_2)(N_{12}) \\
&= \mu_1(P_{12}) + \mu_2(P_{12}) - \mu_1(N_{12}) - \mu_2(N_{12}) \\
&\leq \mu_1(P_{12}) + \mu_2(P_{12}) - \mu_1(N_1) - \mu_2(N_2) \\
&\quad \text{[eqs: 21.845, 21.847]} \\
&\leq \mu_1(P_1) - \mu_1(N_1) + \mu_2(P_2) - \mu_2(N_2) \\
&\quad \text{[eqs: 21.844, 21.846]} \\
&= \mu_1(X \cap P_1) - \mu_1(X \cap N_1) + \mu_2(X \cap P_2) - \mu_2(X \cap N_2) \\
&= (\mu_1)^+(X) + (\mu_1)^-(X) + (\mu_2)^+(X) + (\mu_2)^-(X) \\
&= |\mu_1|(X) + |\mu_2|(X) \\
&= \|\mu_1\| + \|\mu_2\|
\end{aligned}$$

Hence we have proved the triangular inequality

$$\forall \mu_1, \mu_2 \in M[X, \mathcal{A}, \mathbb{R}] \text{ we have } \|\mu_1 + \mu_2\| \leq \|\mu_1\| + \|\mu_2\| \quad (21.848)$$

Let $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ be such that $\|\mu\| = 0$ then we have $\forall A \in \mathcal{A}$ that

$$|\mu(A)| \underset{\text{[theorem: 21.489]}}{\leq} |\mu|(A) \underset{\text{[theorem: 21.99] } \wedge |\mu| \text{ is a measure}}{\leq} |\mu|(X) = \|\mu\| = 0$$

so that $\forall A \in \mathcal{A} \mu(A) = 0 = C_0(A)$ which proves that

$$\text{If } \|\mu\| = 0 \text{ then } \mu = C_0 \text{ the neutral element in } M[X, \mathcal{A}, \mathbb{R}] \quad (21.849)$$

As $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ μ is a **finite** signed measure so that by [theorem: 21.489] $|\mu|$ is a finite measure. Hence $\|\mu\| = |\mu|(X) \in [0, \infty[$, this together with [eqs: 21.843, 21.848 and 21.849] proves that

$$\|\cdot\| \text{ is a norm on } M[X, \mathcal{A}, \mathbb{R}] \quad \square$$

Next we prove the same for the vector space $\langle M[X, \mathcal{A}, \mathbb{C}], +, \cdot \rangle$

Theorem 21.505. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\langle M[X, \mathcal{A}, \mathbb{C}], \|\cdot\| \rangle$ is a normed space. Here $\|\cdot\|$ is defined by*

$$\|u\| = |\mu|(X) = \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right)$$

[see definition: 21.498].

Proof. Let $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ then by [theorem: 21.500] it follows that $|\mu|$ is a finite measure, hence we have that $\|\mu\| = |\mu|(X) \in [0, \infty[$. In other words

$$\forall \mu \in M[X, \mathcal{A}, \mathbb{C}] \text{ we have } \|\mu\| \in [0, \infty[\quad (21.850)$$

Let $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ and $\alpha \in \mathbb{C}$ then by [theorem: 21.500] $|\mu|$ is a finite measure so that

$$\left| \sup \left(\left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) \right) \right| \in \mathbb{R} \text{ exist} \quad (21.851)$$

Further we have

$$\begin{aligned}
 & \frac{\|\alpha \cdot \mu\|}{|\alpha \cdot \mu|(X)} \stackrel{[\text{definition: 21.498}]}{=} \\
 & \sup \left(\left\{ \sum_{i=1}^n |(\alpha \cdot \mu)(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) = \\
 & \sup \left(\left\{ |\alpha| \cdot \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) \stackrel{[\text{theorem: 10.22}] \wedge [\text{eq: 21.851}]}{=} \\
 & |\alpha| \cdot \sup \left(\left\{ \sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) = \\
 & \frac{|\alpha| \cdot \|\mu\|}{|\alpha| \cdot \|\mu\|} =
 \end{aligned}$$

which proves that

$$\forall \mu \in M[X, \mathcal{A}, \mathbb{C}] \text{ we have } \forall \alpha \in \mathbb{C} \text{ that } \|\alpha \cdot \mu\| = |\alpha| \cdot \|\mu\| \quad (21.852)$$

Let $\mu_1, \mu_2 \in M[X, \mathcal{A}, \mathbb{C}]$ and

$$x \in \left\{ \sum_{i=1}^n |(\mu_1 + \mu_2)(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\}$$

then $\exists \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint with $X = \bigsqcup_{i \in \{1, \dots, n\}} A_i$ such that

$$\begin{aligned}
 x &= \sum_{i=1}^n |(\mu_1 + \mu_2)(A_i)| \\
 &= \sum_{i=1}^n |\mu_1(A_i) + \mu_2(A_i)| \\
 &\leq \sum_{i=1}^n (|\mu_1(A_i)| + |\mu_2(A_i)|) \\
 &= \sum_{i=1}^n |\mu_1(A_i)| + \sum_{i=1}^n |\mu_2(A_i)| \\
 &\leq \sup \left(\left\{ \sum_{i=1}^n |\mu_1(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) + \\
 &\quad \sup \left(\left\{ \sum_{i=1}^n |\mu_2(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) \\
 &= \|\mu_1\| + \|\mu_2\|
 \end{aligned}$$

Hence $\|\mu_1\| + \|\mu_2\|$ is an upper bound for the set

$$\left\{ \sum_{i=1}^n |(\mu_1 + \mu_2)(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\}$$

so that

$$\begin{aligned}
 \|\mu_1 + \mu_2\| &= \sup \left(\left\{ \sum_{i=1}^n |(\mu_1 + \mu_2)(A_i)| \mid n \in \mathbb{N} \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right\} \right) \leq \\
 &\|\mu_1\| + \|\mu_2\|
 \end{aligned}$$

which proves that

$$\forall \mu_1, \mu_2 \in M[X, \mathcal{A}, \mathbb{C}] \text{ we have } \|\mu_1 + \mu_2\| \leq \|\mu_1\| + \|\mu_2\| \quad (21.853)$$

Further if $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ is such that $\|\mu\| = 0$ then we have for $A \in \mathcal{A}$ that

$$|\mu(A)| \underset{[\text{theorem: 21.500}]}{\leq} |\mu|(A) \underset{[\text{theorem: 21.99}] \wedge |\mu| \text{ is a measure}}{\leq} |\mu|(X) = \|\mu\| = 0$$

so that $\forall A \in \mathcal{A} \mu(A) = 0 = C_0(A)$ which proves that

$$\text{If } \|\mu\| = 0 \text{ then } \mu = C_0 \text{ the neutral element in } M[X, \mathcal{A}, \mathbb{R}] \quad (21.854)$$

From [eqs: 21.850, 21.852, 21.853 and 21.854] it follows that

$$\|\cdot\| \text{ is a norm on } M(X, \mathcal{A}, \mathbb{C}) \quad \square$$

Next we prove that $\langle M[X, \mathcal{A}, \mathbb{K}], \|\cdot\| \rangle$ is not only a normed space but also a Banach space.

Theorem 21.506. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then $\langle M[X, \mathcal{A}, \mathbb{K}], \|\cdot\| \rangle$ is a Banach space*

Proof. Using [theorems: 21.504, 21.505] it follows that

$$\langle M[X, \mathcal{A}, \mathbb{K}], \|\cdot\| \rangle \text{ is a normed space}$$

So to prove that $\langle M[X, \mathcal{A}, \mathbb{K}], \|\cdot\| \rangle$ is a Banach space we must prove that every Cauchy sequence in $M[X, \mathcal{A}, \mathbb{K}]$ is a convergent series in $M[X, \mathcal{A}, \mathbb{K}]$. So let $\{\mu_i\}_{i \in \mathbb{N}} \subseteq M[X, \mathcal{A}, \mathbb{K}]$ be a Cauchy sequence then we have that

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a } N_\varepsilon \in \mathbb{N} \text{ with } \forall n, m \in \{N_\varepsilon, \dots, \infty\} \quad |\mu_n - \mu_m|(X) \underset{\text{def}}{=} \|\mu_n - \mu_m\| < \varepsilon \quad (21.855)$$

Let $\varepsilon \in \mathbb{R}^+$ then we have, as $\forall n, m \in \{N_\varepsilon, \dots, \infty\}$ $|\mu_n - \mu_m|$ is a finite measure [see theorems: 21.503, 21.489, 21.500], that $\forall A \in \mathcal{A}$

$$\begin{aligned} |\mu_n(A) - \mu_m(A)| &= |(\mu_n - \mu_m)(A)| \\ &\underset{[\text{theorems: 21.489, 21.500}]}{\leq} |\mu_n - \mu_m|(A) \\ &\underset{[\text{theorem: 21.99}]}{\leq} |\mu_n - \mu_m|(X) \\ &= \|\mu_n - \mu_m\| \\ &\underset{[\text{eq: 21.855}]}{<} \varepsilon \end{aligned}$$

In other words

$$\forall \varepsilon > 0 \quad \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall n, m \in \{N_\varepsilon, \dots, \infty\} \text{ and } \forall A \in \mathcal{A} \text{ we have } |\mu_n(A) - \mu_m(A)| < \varepsilon \quad (21.856)$$

Hence we have that $\forall A \in \mathcal{A} \quad \{\mu_n(A)\}_{n \in \mathbb{N}} \subseteq \mathbb{K}$ is a Cauchy sequence in $\langle \mathbb{K}, \|\cdot\| \rangle$. As by [examples: 14.394, 14.397] $\langle \mathbb{K}, \|\cdot\| \rangle$ is a Banach space it follows that

$$\forall A \in \mathcal{A} \quad \lim_{n \rightarrow \infty} \mu_n(A) \in \mathbb{K} \text{ exist}$$

Hence we can define the following function

$$\mu: \mathcal{A} \rightarrow \mathbb{K} \text{ where } \mu(A) = \lim_{n \rightarrow \infty} \mu_n(A) \in \mathbb{K} \quad (21.857)$$

From which it follows that

$$\text{If } \mathbb{K} = \mathbb{R} \text{ then } \{\mu(A) | A \in \mathcal{A}\} \subseteq \mathbb{R} \subseteq \mathbb{R}_\infty \quad (21.858)$$

Take $\varepsilon \in \mathbb{R}^+$ and let $n \in \{N_\varepsilon, \dots, \infty\}$ then given a arbitrary $A \in \mathcal{A}$ we have as $\mu(A) = \lim_{n \rightarrow \infty} \mu_n(A)$ that there exist a M_ε such that $\forall k \in \{M_\varepsilon, \dots, \infty\}$ we have $|\mu_k(A) - \mu(A)| < \frac{\varepsilon}{2}$. Let $M = \max(N_\varepsilon, M_\varepsilon)$ then we have

$$\begin{aligned} |\mu_n(A) - \mu(A)| &= |\mu_n(A) - \mu_M(A) + \mu_M(A) - \mu(A)| \\ &\leq |\mu_n(A) - \mu_M(A)| + |\mu_M(A) - \mu(A)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

which proves, if we take $K_\varepsilon = N_{\frac{\varepsilon}{2}}$, that

$$\forall \varepsilon \in \mathbb{R}^+ \exists K_\varepsilon \in \mathbb{N} \text{ such that } \forall n \in \{K_\varepsilon, \dots, \infty\} \text{ we have } \forall A \in \mathcal{A} \quad |\mu_n(A) - \mu(A)| < \varepsilon \quad (21.859)$$

Now we prove that $\mu \in M[X, \mathcal{A}, \mathbb{K}]$. In this proof we will use [theorems: 21.478, 21.496], so our first task is to prove that the conditions for these theorems are satisfied. First we have

$$\mu(\emptyset) = \lim_{n \rightarrow \infty} \mu_n(\emptyset) = \lim_{n \rightarrow \infty} 0 = 0 \quad (21.860)$$

Let $\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$, $n \in \mathbb{N}$ be a finite pairwise disjoint family of measurable sets. Let $\varepsilon \in \mathbb{R}^+$ then $\forall i \in \{1, \dots, n\}$ we have, as $\mu(A_i) \underset{[\text{eq: 21.857}]}{=} \lim_{k \rightarrow \infty} \mu_k(A_i)$, that there exist a $N_i \in \mathbb{N}$ such that $\forall k \in \{N_i, \dots, \infty\}$ we have $|\mu_k(B_i) - \mu(B_i)| < \frac{\varepsilon}{n}$. Take $N = \max(\{N_i | i \in \{1, \dots, n\}\})$ then we have $\forall k \in \{N, \dots, \infty\}$ that

$$\begin{aligned} \left| \sum_{i=1}^n \mu_k(B_i) - \sum_{i=1}^n \mu(B_i) \right| &= \left| \sum_{i=1}^n (\mu_k(B_i) - \mu(B_i)) \right| \\ &\leq \sum_{i=1}^n |\mu_k(B_i) - \mu(B_i)| \\ &\underset{N_i \leq N}{\leq} \sum_{i=1}^n \frac{\varepsilon}{n} \\ &= \varepsilon \end{aligned}$$

so that

$$\lim_{k \rightarrow \infty} \sum_{i=1}^n \mu_k(B_i) = \sum_{i=1}^n \mu(B_i) \quad (21.861)$$

Further we have

$$\begin{aligned} \mu\left(\bigsqcup_{i \in \{1, \dots, n\}} B_i\right) &\underset{[\text{eq: 21.857}]}{=} \lim_{k \rightarrow \infty} \mu_k\left(\bigsqcup_{i \in \{1, \dots, n\}} B_i\right) \\ &\underset{\mu_k \in M[\overline{X}, \mathcal{A}, \mathbb{K}]}{=} \lim_{k \rightarrow \infty} \sum_{i=1}^n \mu_k(B_i) \\ &\underset{[\text{eq: 21.861}]}{=} \sum_{i=1}^n \mu(B_i) \end{aligned}$$

proving finite additivity. To summarize

$$\forall \{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \text{ pairwise disjoint we have that } \mu\left(\bigsqcup_{i=1}^n B_i\right) = \sum_{i=1}^n \mu(B_i) \quad (21.862)$$

Let now $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ with $\forall i \in \mathbb{N} \quad A_{i+1} \subseteq A_i$ such that $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$. For $\mathbb{K}$ we have then two slightly different cases to consider:

$\mathbb{K} = \mathbb{R}$. Let $i \in \mathbb{N}$ then $\mu_i \in M(X, \mathcal{A}, \mathbb{R}) \Rightarrow \mu_i$ is a finite signed measure, hence using [theorem: 21.477] it follows that $\lim_{k \rightarrow \infty} \mu_i(A_k) = \mu_i(\bigcap_{k \in \mathbb{N}} A_k) = \mu_i(\emptyset) = 0$.

$\mathbb{K} = \mathbb{C}$. Let $i \in \mathbb{N}$ then $\mu_i \in M[X, \mathcal{A}, \mathbb{C}] \Rightarrow \mu_i$ is a complex measure so that using [theorem: 21.495] we have that $\lim_{k \rightarrow \infty} \mu_i(A_k) = \mu_i(\bigcap_{k \in \mathbb{N}} A_k) = \mu_i(\emptyset) = 0$.

So in all cases we have

$$\forall i \in \mathbb{N} \text{ that } \lim_{k \rightarrow \infty} \mu_i(A_k) = 0 \quad (21.863)$$

Let $\varepsilon \in \mathbb{R}^+$ then by [eq: 21.859] there exist a $N \in \mathbb{N}$ such that $\forall A \in \mathcal{A} \quad |\mu_N(A) - \mu(A)| < \frac{\varepsilon}{2}$. From [eq: 21.863] it follows that there exist a $K \in \mathbb{N}$ such that $\forall n \in \{K, \dots, \infty\}$ we have $|\mu_N(A_n)| < \frac{\varepsilon}{2}$. Let $n \in \{K, \dots, \infty\}$ then we have

$$|\mu(A_n)| \leq |\mu(A_n) - \mu_N(A_n) + \mu_N(A_n)| \leq |\mu(A_n) - \mu_N(A_n)| + |\mu_N(A_n)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

which proves that $\lim_{n \rightarrow \infty} \mu(A_n) = 0$. To summarize we have proved that

$$\forall \{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{A} \text{ with } A_{k+1} \subseteq A_k \text{ such that } \bigcap_{k \in \mathbb{N}} A_k = \emptyset \text{ we have } \lim_{k \rightarrow \infty} \mu(A_k) = 0 \quad (21.864)$$

Again we look at the two cases for $\mathbb{K}$

$\mathbb{K} = \mathbb{R}$. Then [eqs: 21.860, 21.858, 21.862 and 21.864] allows us to use [them: 21.478] to prove that μ is a signed measure. As by [eq: 21.858] μ is finite it follows that

$$\mu \in M[X, \mathcal{A}, \mathbb{R}]$$

$\mathbb{K} = \mathbb{C}$. Then [eqs: 21.860, 21.862 and 21.864] allows us to use [theorem: 21.496] to prove that μ is a complex measure. Hence

$$\mu \in M[X, \mathcal{A}, \mathbb{C}]$$

So we have proved that for $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$

$$\mu \in M[X, \mathcal{A}, \mathbb{K}] \quad (21.865)$$

Next we must prove that $\lim_{n \rightarrow \infty} \mu_n$ converges using the norm $\|\cdot\|$. Consider for $\mathbb{K}$ either:

$\mathbb{K} = \mathbb{R}$. Let $\varepsilon \in \mathbb{R}^+$ then for $n \in \{K_{\frac{\varepsilon}{4}}, \dots, \infty\}$ we have by [eq: 21.859] that $\forall A \in \mathcal{A}$

$$(\mu_n - \mu)(A) = \mu_n(A) - \mu(A) \leq |\mu_n(A) - \mu(A)| < \frac{\varepsilon}{4}$$

$$-(\mu_n - \mu)(A) = -(\mu_n(A) - \mu(A)) \leq |\mu_n(A) - \mu(A)| < \frac{\varepsilon}{4}$$

so that $\frac{\varepsilon}{4}$ is a upper bound for

$$\{(\mu_n - \mu)(A) | A \in \mathcal{A} \wedge A \subseteq X\} \text{ and } \{-(\mu_n - \mu)(A) | A \in \mathcal{A} \wedge A \subseteq X\}$$

Hence we have that

$$(\mu_n - \mu)^+(X) \stackrel{[\text{theorem: 21.489}]}{=} \sup \{(\mu_n - \mu)(A) | A \in \mathcal{A} \wedge A \subseteq X\} < \frac{\varepsilon}{4}$$

$$(\mu_n - \mu)^-(X) \stackrel{[\text{theorem: 21.489}]}{=} \sup \{-(\mu_n - \mu)(A) | A \in \mathcal{A} \wedge A \subseteq X\} < \frac{\varepsilon}{4}$$

so that

$$\|\mu_n - \mu\| = (\mu_n - \mu)^+(X) + (\mu_n - \mu)^-(X) = \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2} < \varepsilon$$

which proves that

$$\mu = \lim_{n \rightarrow \infty} \mu_n$$

$\mathbb{K} = \mathbb{C}$. Let $\varepsilon \in \mathbb{R}^+$. As $\{\mu_n\}_{n \in \mathbb{N}}$ is Cauchy there exist a $N \in \mathbb{N}$ such that $\forall n, m \in \{N, \dots, \infty\}$ we have $\|\mu_n - \mu_m\| < \frac{\varepsilon}{4}$. Hence using [definition: 21.498] we have that

$$\sup \left(\left\{ \sum_{i=1}^k |(\mu_n - \mu_m)(A_i)| \mid \{A_i\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, k\}} A_i \right\} \right) < \frac{\varepsilon}{4} \quad (21.866)$$

Take $\{A_i\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{A}$ pairwise disjoint such that $X = \bigsqcup_{i \in \{1, \dots, k\}} A_i$. Using [eq: 21.859] there exist a $K \in \mathbb{N}$ such that $\forall l \in \{K, \dots, \infty\}$ we have $\forall i \in \{1, \dots, k\}$ that $|(\mu_l - \mu)(A_i)| < \frac{\varepsilon}{4 \cdot k}$. So that

$$\sum_{i=1}^k |(\mu_l - \mu)(A_i)| < \sum_{i=1}^k \frac{\varepsilon}{4 \cdot k} = \frac{\varepsilon}{4} \quad (21.867)$$

Take now $n \in \{N, \dots, \infty\}$ and $M = \max(N, K)$ then we have

$$\begin{aligned}
 \sum_{i=1}^k |(\mu_n - \mu)(A_i)| &= \\
 \sum_{i=1}^k |\mu_n(A_i) - \mu(A_i)| &= \\
 \sum_{i=1}^k |\mu_n(A_i) - \mu_M(A_i) + \mu_M(A_i) - \mu(A_i)| &\leq \\
 \sum_{i=1}^k |\mu_n(A_i) - \mu_M(A_i)| + \sum_{i=1}^k |\mu_M(A_i) - \mu(A_i)| &\stackrel{[eq: 21.867]}{<} \\
 \sum_{i=1}^k |\mu_n(A_i) - \mu_M(A_i)| + \frac{\varepsilon}{4} &\leq \\
 \sup \left(\left(\sum_{i=1}^k |(\mu_n - \mu_M)(A_i)| \mid \{A_i\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, k\}} A_i \right) + \frac{\varepsilon}{4} \right) &\stackrel{[eq: 21.866]}{<} \\
 \frac{\varepsilon}{4} + \frac{\varepsilon}{4} &= \\
 \frac{\varepsilon}{2}
 \end{aligned}$$

So $\frac{\varepsilon}{2}$ is an upper bound for $\left\{ \sum_{i=1}^k |(\mu_n - \mu)(A_i)| \mid \{A_i\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, k\}} A_i \right\}$ which proves that

$$\|\mu_n - \mu\|_{\text{def}} = \sup \left(\sum_{i=1}^k |(\mu_n - \mu)(A_i)| \mid \{A_i\}_{i \in \{1, \dots, k\}} \subseteq \mathcal{A} \wedge X = \bigsqcup_{i \in \{1, \dots, k\}} A_i \right) \leq \frac{\varepsilon}{2} < \varepsilon$$

Hence we have that

$$\mu = \lim_{n \rightarrow \infty} \mu_n$$

In conclusion we have proved that for every Cauchy sequence $\{\mu_n\}_{n \in \mathbb{N}} \subseteq M[X, \mathcal{A}, \mathbb{K}]$ $\mu = \lim_{n \rightarrow \infty} \mu_k$ exist in $\langle M[X, \mathcal{A}, \mathbb{K}], ||| \rangle$ or in other words that

$$\langle M[X, \mathcal{A}, \mathbb{K}], ||| \rangle \text{ is a Banach space}$$

□

21.6.3.1 Integrals based on signed or complex measures.

As a signed measure μ can be written as $\mu = \mu^+ - \mu^-$ where (μ^+, μ^-) is the Jordan decomposition of μ it make sense to define $\int f d\mu$ as $\int f d\mu = \int f d\mu^+ - \int f d\mu^-$. Likewise as a complex measure can be written as $\mu = \mu_r^+ - \mu_r^- + i \cdot (\mu_c^+ - \mu_c^-)$ where $(\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-)$ is the Jordan decomposition of μ it make sense to define $\int f d\mu$ as $\int f d\mu = \int f d\mu_r^+ - \int f d\mu_r^- + i \cdot (\int f d\mu_c^+ - \int f d\mu_c^-)$ but this works only if the integrals on the left side are finite. Hence as with the definition of μ -integrable functions we need an extra condition on the measurable functions. This is provided in the following definition.

Definition 21.507. Let $\langle X, \mathcal{A} \rangle$ be a measurable space then the set of **bounded measurable functions** $B[X, \mathcal{A}, \mathbb{K}]$ is defined by

$$B[X, \mathcal{A}, \mathbb{K}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{K}] \mid \exists M \in [0, \infty[\text{ such that } \forall x \in X \text{ we have } |f(x)| \leq M\} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{K}]$$

more specific we have

$$B[X, \mathcal{A}, \mathbb{R}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}] \mid \exists M \in [0, \infty[\text{ such that } \forall x \in X \text{ we have } |f(x)| \leq M\} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

$$B[X, \mathcal{A}, \mathbb{C}] = \{f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \mid \exists M \in [0, \infty[\text{ such that } \forall x \in X \text{ we have } |f(x)| \leq M\} \subseteq \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$$

We show now that that $B[X, \mathcal{A}, \mathbb{K}]$ is actually a normed vector space.

Theorem 21.508. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space then we have*

1. $\langle B[X, \mathcal{A}, \mathbb{R}], +, \cdot \rangle$ is a vector space [where $+$ and $\cdot$ are pointwise defined] and C_0 defined by $C_0(x) = 0$ is the neutral element.
2. $\langle B[X, \mathcal{A}, \mathbb{C}], +, \cdot \rangle$ is a vector space [where $+$ and $\cdot$ are pointwise defined] and C_0 defined by $C_0(x) = 0$ is the neutral element.
3. If we define

$$\|f\|_\infty: B[X, \mathcal{A}, \mathbb{K}] \rightarrow [0, \infty[\text{ by } \|f\|_\infty = \sup(\{|f(x)| | x \in X\})$$

then $\langle B[X, \mathcal{A}, \mathbb{K}], \|\cdot\|_\infty \rangle$ is a normed space.

Proof.

1. As $C_0 \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ [see example: 21.228] and $\forall x \in X \ |C_0(x)| = |0| \leq 0$ it follows that

$$C_0 \in B[X, \mathcal{A}, \mathbb{R}] \Rightarrow \emptyset \neq B[X, \mathcal{A}, \mathbb{R}]$$

Let $f, g \in B[X, \mathcal{A}, \mathbb{R}]$ then $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and there exist a $M_1, M_2 \in [0, \infty[$ such that $\forall x \in X$ we have $|f(x)| \leq M_1$ and $|g(x)| \leq M_2$. Using [theorem: 21.257] it follows that $f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, further we have $\forall x \in X$ that $|(f + g)(x)| \leq |f(x)| + |g(x)| \leq M_1 + M_2$ which proves that $f + g \in B[X, \mathcal{A}, \mathbb{R}]$. Hence we have

$$\forall f, g \in B[X, \mathcal{A}, \mathbb{R}] \ f + g \in B[X, \mathcal{A}, \mathbb{R}]$$

If $\alpha \in \mathbb{R}$ then by [theorem: 21.263] we have $\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, further $\forall x \in X$ we have $|(\alpha \cdot f)(x)| = |\alpha| \cdot |f(x)| \leq |\alpha| \cdot M_1 \leq |\alpha| \cdot M_1 + 1$ which proves that $\alpha \cdot f \in B[X, \mathcal{A}, \mathbb{R}]$. Hence we have

$$\forall f \in B[X, \mathcal{A}, \mathbb{R}] \text{ we have } \alpha \cdot f \in B[X, \mathcal{A}, \mathbb{R}]$$

Using now [theorem: 11.69] and [theorem: 11.61] it follows that

$$\langle B[X, \mathcal{A}, \mathbb{R}], +, \cdot \rangle \text{ is a vector space}$$

2. As $C_0 \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ [see example: 21.228] and $\forall x \in X \ |C_0(x)| = 0 \leq 0$ it follows that

$$C_0 \in B[X, \mathcal{A}, \mathbb{C}] \Rightarrow \emptyset \neq B[X, \mathcal{A}, \mathbb{C}]$$

Let $f, g \in B[X, \mathcal{A}, \mathbb{C}]$ then $f, g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$ and there exist a $M_1, M_2 \in [0, \infty[$ such that $\forall x \in X$ we have $|f(x)| \leq M_1$ and $|g(x)| \leq M_2$. Using [theorem: 21.401] it follows that $f + g \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, further we have $\forall x \in X$ that $|(f + g)(x)| \leq |f(x)| + |g(x)| \leq M_1 + M_2$ which proves that $f + g \in B[X, \mathcal{A}, \mathbb{C}]$. Hence we have

$$\forall f, g \in B[X, \mathcal{A}, \mathbb{C}] \ f + g \in B[X, \mathcal{A}, \mathbb{C}]$$

If $\alpha \in \mathbb{R}$ then by [theorem: 21.401] we have $\alpha \cdot f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}]$, further $\forall x \in X$ we have $|(\alpha \cdot f)(x)| = |\alpha| \cdot |f(x)| \leq |\alpha| \cdot M_1 \leq |\alpha| \cdot M_1 + 1$ which proves that $\alpha \cdot f \in B[X, \mathcal{A}, \mathbb{C}]$. Hence we have

$$\forall f \in B[X, \mathcal{A}, \mathbb{C}] \text{ we have } \alpha \cdot f \in B[X, \mathcal{A}, \mathbb{C}]$$

Using [theorem: 11.69] and [theorem: 11.61] it follows that

$$\langle B[X, \mathcal{A}, \mathbb{C}], +, \cdot \rangle \text{ is a vector space}$$

3. Let $f \in B[X, \mathcal{A}, \mathbb{K}]$ then $\exists M \in [0, \infty[$ such that $\forall x \in X$ we have $|f(x)| \leq M$ so that M is a upper bound of $\{|f(x)| | x \in X\}$ which is not empty as $X \neq \emptyset$ by the definition of a measurable space [see definition: 21.35]. As $\mathbb{R}$ is conditional complete [see theorem: 10.18] it follows that

$$\sup(\{|f(x)| | x \in X\}) \in \mathbb{R} \text{ exist}$$

So the function

$$\|\cdot\|_\infty: B[X, \mathcal{A}, \mathbb{K}] \rightarrow [0, \infty[\text{ where } \|f\|_\infty = \sup(\{|f(x)| | x \in X\}) \quad (21.868)$$

is well defined. Let $f, g \in B[X, \mathcal{A}, \mathbb{K}]$ then we have then we have $\forall x \in X$ that

$$|(f+g)(x)| \leq |f(x)| + |g(x)| \leq \sup(|f(x)| | x \in X) + \sup(|g(x)| | x \in X) \leq \|f\|_\infty + \|g\|_\infty$$

so that $\|f\|_\infty + \|g\|_\infty$ is an upper bound of $\{|(f+g)(x)| | x \in X\}$. Hence

$$\|f+g\|_\infty = \sup(|(f+g)(x)| | x \in X) \leq \|f\|_\infty + \|g\|_\infty$$

which proves that

$$\forall f, g \in B[X, \mathcal{A}, \mathbb{K}] \text{ we have } \|f+g\|_\infty \leq \|f\|_\infty + \|g\|_\infty \quad (21.869)$$

Let $\alpha \in \mathbb{R}$ and $f \in M[X, \mathcal{A}, \mathbb{K}]$ then

$$\begin{aligned} \|\alpha \cdot f\|_\infty &= \sup(|(\alpha \cdot f)(x)| | x \in X) \\ &= \sup(|\alpha| \cdot |f(x)| | x \in X) \\ &\stackrel{[\text{theorem: 10.22}]}{=} |\alpha| \cdot \sup(|f(x)| | x \in X) \\ &= |\alpha| \cdot \|f\|_\infty \end{aligned}$$

hence we have

$$\forall \alpha \in \mathbb{R}, \forall f \in M[X, \mathcal{A}, \mathbb{K}] \text{ we have } \|\alpha \cdot f\|_\infty = |\alpha| \cdot \|f\|_\infty \quad (21.870)$$

If $f \in M[X, \mathcal{A}, \mathbb{K}]$ is such that $\|f\|_\infty = 0$ then we have $\forall x \in X$

$$0 \leq |f(x)| \leq \sup(|f(x)| | x \in X) = \|f\|_\infty = 0$$

proving that $\forall x \in X \quad |f(x)| = 0 = C_0(x)$. Hence

$$\forall f \in B[X, \mathcal{A}, \mathbb{K}] \text{ with } \|f\|_\infty = 0 \text{ we have that } f = C_0 \quad (21.871)$$

From [eqs: 21.868, 21.869, 21.870 and 21.871] it follows by definition that $\|\cdot\|_\infty$ is a norm on $B[X, \mathcal{A}, \mathbb{K}]$. $\square$

Next we show that all bounded measurable functions are integrable.

Theorem 21.509. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, μ a finite measure and $f \in B(X, \mathcal{A}, \mathbb{K})$ then*

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{K}]$$

In other words if $f \in B[X, \mathcal{A}, \mathbb{K}]$ then $\int f d\mu$ exist.

Proof. For $\mathbb{K}$ we have either

$\mathbb{K} = \mathbb{R}$. Then $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and there exist a $M \in \mathbb{R}$ such that $\forall x \in X$ we have $|f(x)| \leq M$. By [theorem: 21.342] $f^+, f^- \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$, further we have by [theorem: 21.344] that $\forall x \in X$ $f_+(x) \leq |f(x)| \leq M = C_M(x)$. So

$$\begin{aligned} \int^+ f^+ d\mu &\stackrel{[\text{theorem: 21.331}]}{\leq} \int^+ C_M d\mu \stackrel{[\text{example: 21.318}]}{\leq} M \cdot \mu(A) \stackrel{\mu \text{ is finite}}{<} \infty \\ \int^+ f^- d\mu &\stackrel{[\text{theorem: 21.331}]}{\leq} \int^+ C_M d\mu \stackrel{[\text{example: 21.318}]}{\leq} M \cdot \mu(A) \stackrel{\mu \text{ is finite}}{<} \infty \end{aligned}$$

Hence by [definition: 21.345] we have that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

$\mathbb{K} = \mathbb{C}$. Then $f \in \mathcal{M}[X, \mathcal{A}, \mathbb{C}] \stackrel{[\text{theorem: 21.399}]}{\Rightarrow} \text{Re}(f), \text{Im}(f) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$ and there exist a $M \in \mathbb{R}$ such that $\forall x \in X$ we have $|f(x)| \leq M$. Then we have

$$\forall x \in X \text{ we have } |\text{Re}(f)(x)| = |\text{Re}(f(x))| \stackrel{[\text{theorem: 10.85}]}{\leq} |f(x)| \leq M$$

$$\forall x \in X \text{ we have } |\text{Im}(f)(x)| = |\text{Im}(f(x))| \stackrel{[\text{theorem: 10.85}]}{\leq} |f(x)| \leq M$$

Using then the case $\mathbb{K} = \mathbb{R}$ on the above proves that $\operatorname{Re}(f), \operatorname{Im}(f) \in \mathcal{L}[X, \mathcal{A}, \mathbb{R}]$. Hence it follows from [definition: 21.403] that $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$. $\square$

Now we are ready to define integration using signed and complex measures:

Definition 21.510. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ and $f \in B(X, \mathcal{A}, \mathbb{R})$ then we define $\int f d\mu$ as

$$\int f d\mu = \int f d\mu^+ - \int f d\mu^- \text{ where } (\mu^+, \mu^-) \text{ is the Jordon Decomposition of } \mu$$

Note 21.511. As $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ we have by [theorem: 21.489] that μ^+, μ^- are finite so that by [theorem: 21.509] $f \in \mathcal{L}[X, \mathcal{A}, \mu^+, \mathbb{R}] \wedge f \in \mathcal{L}[X, \mathcal{A}, \mu^-, \mathbb{R}]$. Hence $\int f d\mu^+ - \int f d\mu^-$ is well defined.

Definition 21.512. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ and $f \in B(X, \mathcal{A}, \mathbb{C})$ then we define $\int^{\mathbb{C}} f d\mu$ as

$$\int^{\mathbb{C}} f d\mu = \int^{\mathbb{C}} f d\mu^+ - \int^{\mathbb{C}} f d\mu^- \text{ where } (\mu^+, \mu^-) \text{ is the Jordon Decomposition of } \mu$$

Note 21.513. As $\mu \in M[X, \mathcal{A}, \mathbb{R}]$ we have by [theorem: 21.489] that μ^+, μ^- are finite measures so that by [theorem: 21.509] $f \in \mathcal{L}[X, \mathcal{A}, \mu^+, \mathbb{C}] \wedge f \in \mathcal{L}[X, \mathcal{A}, \mu^-, \mathbb{C}]$. Hence $\int^{\mathbb{C}} f d\mu^+ - \int^{\mathbb{C}} f d\mu^-$ is well defined.

Definition 21.514. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ and $f \in B(X, \mathcal{A}, \mathbb{R})$ then we define $\int f d\mu$ as

$$\int f d\mu = \int f d\mu_r^+ - \int f d\mu_r^- + i \cdot \left(\int f d\mu_c^+ - \int f d\mu_c^- \right)$$

where $(\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-)$ is the Jordon Decomposition of μ .

Note 21.515. As $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ we have by [theorem: 21.497] that $\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-$ are finite measures so that by [theorem: 21.509] $f \in \mathcal{L}[X, \mathcal{A}, \mu_r^+, \mathbb{R}] \wedge f \in \mathcal{L}[X, \mathcal{A}, \mu_r^-, \mathbb{R}]$. Hence $\int f d\mu_r^+, \int f d\mu_r^-, \int f d\mu_c^+, \int f d\mu_c^-$ are well defined.

Definition 21.516. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ and $f \in B(X, \mathcal{A}, \mathbb{C})$ then we define $\int^{\mathbb{C}} f d\mu$ as

$$\int^{\mathbb{C}} f d\mu = \int^{\mathbb{C}} f d\mu_r^+ - \int^{\mathbb{C}} f d\mu_r^- + i \cdot \left(\int^{\mathbb{C}} f d\mu_c^+ - \int^{\mathbb{C}} f d\mu_c^- \right)$$

where $(\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-)$ is the Jordon Decomposition of μ .

Note 21.517. As $\mu \in M[X, \mathcal{A}, \mathbb{C}]$ we have by [theorem: 21.497] that $\mu_r^+, \mu_r^-, \mu_c^+, \mu_c^-$ are finite measures so that by [theorem: 21.509] $f \in \mathcal{L}[X, \mathcal{A}, \mu_r^+, \mathbb{R}] \wedge f \in \mathcal{L}[X, \mathcal{A}, \mu_r^-, \mathbb{R}]$. Hence $\int f d\mu_r^+, \int f d\mu_r^-, \int f d\mu_c^+, \int f d\mu_c^-$ are well defined.

21.6.3.2 Radon-Nikodym Theorem

In this section we prove how a measure can be expressed as the integral of a 'density' function using another measure. In essence if μ, ν are two measures on a measurable space $\langle X, \mathcal{A} \rangle$ then if certain conditions are met we have

$$\forall A \in \mathcal{A} \quad \nu(A) = \int_A f d\mu$$

We will prove the Radon-Nikodym Theorem for the following cases:

- finite measures
- σ -finite measures

- signed measures
- complex measures

First we examine the conditions needed for the Radon-Nikodym to be valid.

Definition 21.518. Let $\langle X, \mathcal{A} \rangle$ be a measurable space and μ, ν measures on $\mathcal{A}$ then ν is **absolute continuous with respect to μ** noted as $\nu \ll \mu$ if

$$\forall A \in \mathcal{A} \text{ we have if } \mu(A) = 0 \text{ then } \nu(A) = 0$$

Example 21.519. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then $\mu \ll \mu$

Proof. As $\forall A \in \mathcal{A}$ we have $\mu(A) = 0 \Rightarrow \mu(A) = 0$ it follows trivially that $\mu \ll \mu$ $\square$

Example 21.520. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ then

$$\nu: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \nu(A) = \int^+ (\mathcal{X}_A \cdot f) d\mu$$

is a measure on $\mathcal{A}$ and

$$\nu \ll \mu$$

Proof. Using [theorem: 21.380] it follows that $\nu: \mathcal{A} \rightarrow [0, \infty]$ is a measure on $\mathcal{A}$. Let $A \in \mathcal{A}$ be such that $\mu(A) = 0$ then if $x \in \{x \in X | (f \cdot \mathcal{X}_A)(x) \neq 0\}$ we must have that $x \in A$ [for if $x \in X \setminus A$ then $(\mathcal{X}_A \cdot f) = \mathcal{X}_A(x) \cdot f(x) = 0$] which proves that $\{x \in X | (f \cdot \mathcal{X}_A)(x) \neq 0\} \subseteq A \in \mathcal{A}$ with $\mu(A) = 0$. Hence $\mathcal{X}_A \cdot f = 0$ μ -a.e.. Using [theorem: 21.330] it follows that $\nu(A) = \int^+ (\mathcal{X}_A \cdot f) d\mu = 0$. $\square$

The following theorem explains why we use the name **absolute continuity** for $\nu \ll \mu$ and shows the similarity with absolute continuity on functions based on the ε - δ definition. To prove this theorem we first need a little lemma.

Lemma 21.521. Let $n \in \mathbb{N}$ then $\sum_{k=n}^{\infty} \frac{1}{2^k} = \frac{1}{2^{n-1}}$

Proof. For $n \in \mathbb{N}$ we have either:

$$n = 1. \text{ Then } \sum_{k=n}^{\infty} \frac{1}{2^k} = \sum_{k=1}^{\infty} \frac{1}{2^k} \text{ [example: 14.412]} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1 = \frac{1}{2^{1-1}} = \frac{1}{2^{n-1}}$$

$n \in \mathbb{N} \setminus \{1\}$. Then

$$\begin{aligned} 1 & \text{ [example: 14.412]} = \sum_{k=1}^{\infty} \frac{1}{2^k} \\ & \text{ [theorem: 14.416]} = \sum_{k=1}^{n-1} \frac{1}{2^k} + \sum_{k=n}^{\infty} \frac{1}{2^k} \\ & \text{ [lemma: 14.411]} = \frac{1}{2} \cdot \left(\frac{1 - \frac{1}{2^{n-1}}}{1 - \frac{1}{2}} \right) + \sum_{k=n}^{\infty} \frac{1}{2^k} \\ & = 1 - \frac{1}{2^{n-1}} + \sum_{k=n}^{\infty} \frac{1}{2^k} \end{aligned}$$

which proves that $\sum_{k=n}^{\infty} \frac{1}{2^k} = \frac{1}{2^{n-1}}$. $\square$

Theorem 21.522. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and let ν be a **finite measure** on $\mathcal{A}$ then we have

$$\begin{aligned} \nu & \ll \mu \\ \Updownarrow \\ \forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall A \in \mathcal{A} \text{ with } \mu(A) < \delta \text{ we have } \nu(A) < \varepsilon \end{aligned}$$

Proof.

$\Rightarrow$. We prove this by contradiction. So assume that there exist a $\varepsilon \in \mathbb{R}^+$ such that $\forall \delta \in \mathbb{R}^+$ there exist $A \in \mathcal{A}$ with $\mu(A) < \delta$ such that $\varepsilon \leq \nu(A)$. More specific $\forall k \in \mathbb{N}$ there exist a $A_k \in \mathcal{A}$ with $\mu(A_k) < \frac{1}{2^k}$ such that $\varepsilon \leq \nu(A_k)$. Define now

$$\{B_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A} \text{ by } B_n = \bigcup_{k \in \{n, \dots, \infty\}} A_k \text{ and } B = \bigcap_{k \in \mathbb{N}} B_k \in \mathcal{A}$$

Then we have $\forall n \in \mathbb{N}$ that

$$B_{n+1} = \bigcup_{k \in \{n+1, \dots, \infty\}} A_k \subseteq \bigcup_{k \in \{n, \dots, \infty\}} A_k = B_n$$

Further we have

$$\mu(B_1) = \mu\left(\bigcup_{k \in \mathbb{N}} A_k\right) \underset{[\text{theorem: 21.101}]}{\leq} \sum_{k=1}^{\infty} \mu(A_k) \underset{[\text{theorem: 20.99}]}{\leq} \sum_{k=1}^{\infty} \frac{1}{2^k} \underset{[\text{example: 14.412}]}{=} 1 < \infty$$

Hence we can use [theorem: 21.115] to get

$$\mu(B) = \mu\left(\bigcap_{k \in \mathbb{N}} B_k\right) = \lim_{k \rightarrow \infty} \mu(B_k) = \inf(\{\mu(B_i) | i \in \mathbb{N}\}) \quad (21.872)$$

Further, as ν is finite, we have by [theorem: 21.115] that

$$\nu(B) = \nu\left(\bigcap_{k \in \mathbb{N}} B_k\right) = \lim_{k \rightarrow \infty} \nu(B_k) = \inf(\{\nu(B_i) | i \in \mathbb{N}\}) \quad (21.873)$$

Also for $n \in \mathbb{N}$ we have

$$\mu(B_n) = \mu\left(\bigcup_{k \in \{n, \dots, \infty\}} A_k\right) \underset{[\text{theorem: 21.101}]}{\leq} \sum_{k=n}^{\infty} \mu(A_k) \leq \sum_{k=n}^{\infty} \frac{1}{2^k} \underset{[\text{lemma: 21.521}]}{\leq} \frac{1}{2^{n-1}}$$

so that by [example: 14.348] and [theorem: 14.355] $\lim_{k \rightarrow \infty} \mu(B_k) = 0$ which combined with [eq: 21.872] gives $\mu(B) = 0$. As $\nu \ll \mu$ we must have also

$$\nu(B) = 0 \quad (21.874)$$

Now $\forall n \in \mathbb{N}$ we have $A_n \subseteq \bigcup_{k \in \{n, \dots, \infty\}} A_k = B_n$ so that $\varepsilon \leq \nu(A_n) \leq \nu(B_n)$. Hence ε is a lower bound of $\{\nu(B_n) | n \in \mathbb{N}\}$ from which it follows that $0 < \varepsilon \leq \inf(\{\nu(B_n) | n \in \mathbb{N}\}) \underset{[\text{eq: 21.873}]}{=} \nu(B)$ which contradicts [eq: 21.874]. So the assumption is wrong and we must have that

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta \in \mathbb{R}^+ \text{ such that } \forall A \in \mathcal{A} \text{ with } \mu(A) < \delta \text{ we have } \nu(A) < \varepsilon$$

$\Leftarrow$. Let $A \in \mathcal{A}$ be such that $\mu(A) = 0$ and assume that $0 < \nu(A) \in \mathbb{R}^+$. Then by the hypothesis there exists a $\delta > 0$ such if $\mu(A) < \delta$ then $\nu(A) < \nu(A)$ a contradiction. Hence the assumption is wrong and we must have $\nu(A) = 0$ which proves that

$$\nu \ll \mu \quad \square$$

We are now ready to prove the first version of the Radon-Nikodym theorem. This theorem states that given a measurable space $\langle X, \mathcal{A} \rangle$ and two σ -finite measures μ, ν on $\mathcal{A}$ such that $\nu \ll \mu$ then there exist a μ -a.e. unique non negative measurable function h such that

$$\nu(A) = \int_A^+ h d\mu = \int_A^+ (\chi_A \cdot h) d\mu.$$

We do this in different steps.

1. First we prove that if ν is finite and $\forall A \in \mathcal{A} \int_A^+ h d\mu = \nu(A) = \int_A^+ f d\mu$ then $f = h$ μ -a.e.
2. We extended (1) to the cases where ν is σ -finite
3. Next we prove that there exist a h such that $\forall A \in \mathcal{A} \nu(A) = \int_A^+ f d\mu$

Lemma 21.523. *Let X be a set and $f: X \rightarrow \mathbb{R}$, $h: X \rightarrow \mathbb{R}$ be two functions then we have*

1. $(f - h)^+ = \mathcal{X}_{X,F} \cdot (f - h)$ where $F = \{h < f\} \stackrel{\text{def}}{=} \{x \in X \mid h(x) < f(x)\}$
2. $(f - h)^- = \mathcal{X}_{X,G} \cdot (h - f)$ where $G = \{f < h\} \stackrel{\text{def}}{=} \{x \in X \mid f(x) < h(x)\}$

Proof.

1. Let $x \in X$ then we have either:

$x \in F$. then $(f - h)(x) = f(x) - h(x) > 0$ so that

$$\begin{aligned} (f - h)^+(x) &= \max((f - h)(x), 0) \\ &= (f - h)(x) \\ &= \mathcal{X}_{X,F}(x) \cdot (f - h)(x) \\ &= (\mathcal{X}_{X,F} \cdot (f - h))(x) \end{aligned}$$

$x \in X \setminus F$. then $(f - h)(x) = f(x) - h(x) \leq 0$ so that

$$(f - h)^+(x) = \max((f - h)(x), 0) = 0 = \mathcal{X}_{X,F}(x) \cdot (f - h)(x) = (\mathcal{X}_{X,F} \cdot (f - h))(x)$$

proving that

$$(f - h)^+ = \mathcal{X}_{X,F} \cdot (f - h)$$

2. Let $x \in X$ then we have either:

$x \in G$. then $(f - h)(x) = f(x) - h(x) < 0$ so that

$$\begin{aligned} (f - h)^-(x) &= -\min((f - h)(x), 0) \\ &= h(x) - f(x) \\ &= \mathcal{X}_{X,G}(x) \cdot (h - f)(x) \\ &= (\mathcal{X}_{X,G} \cdot (h - f))(x) \end{aligned}$$

$x \in X \setminus G$. then $(f - h)(x) = f(x) - h(x) \geq 0$ so that

$$(f - h)^-(x) = -\min((f - h)(x), 0) = 0 = \mathcal{X}_{X,G}(x) \cdot (h - f)(x) = (\mathcal{X}_{X,G} \cdot (h - f))(x)$$

proving that

$$(f - h)^- = \mathcal{X}_{X,G} \cdot (h - f)$$

□

Lemma 21.524. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, μ a measure on $\mathcal{A}$, ν a **finite** measure on $\mathcal{A}$ and $h, f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ such that*

$$\forall A \in \mathcal{A} \quad \int_A^+ f d\mu = \nu(A) = \int_A^+ h d\mu$$

then

$$f = h \text{ } \mu\text{-a.e.}$$

Here $\int_A^+ f d\mu \stackrel{\text{definition: 21.320}}{=} \int^+(\mathcal{X}_A \cdot f) d\mu$ and $\int_A^+ h d\mu \stackrel{\text{definition: 21.320}}{=} \int^+(\mathcal{X}_A \cdot h) d\mu$

Proof. Let $A \in \mathcal{A}$ then as ν is finite we have $0 \leq \nu(A) < \infty$ so that

$$0 \leq \int^+(\mathcal{X}_{X,A} \cdot h) d\mu = \int_A^+ h d\mu = \int^+(\mathcal{X}_{X,A} \cdot f) d\mu = \int_A^+ f d\mu = \nu(A) < \infty$$

hence, using [theorem: 21.352] it follows that

$$\mathcal{X}_{X,A} \cdot h \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \wedge \mathcal{X}_{X,A} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

and

$$\int^+ \mathcal{X}_{X,A} \cdot h d\mu = \int \mathcal{X}_{X,A} \cdot h d\mu = \int^+ \mathcal{X}_{X,A} \cdot f d\mu = \int \mathcal{X}_{X,A} \cdot f d\mu$$

Using the linearity of $\int$ [see theorem: 21.360] it follows that

$$\mathcal{X}_{X,A} \cdot (f - h) = \mathcal{X}_{X,A} \cdot f - \mathcal{X}_{X,A} \cdot h \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

and

$$\int \mathcal{X}_{X,A} \cdot (f - h) d\mu = \int \mathcal{X}_{X,A} \cdot f d\mu - \int \mathcal{X}_{X,A} \cdot h d\mu = 0$$

or

$$\forall A \in \mathcal{A} \text{ we have } \int \mathcal{X}_{X,A} \cdot (f - h) d\mu = 0 \quad (21.875)$$

Define now $F = \{h < f\}$ and $G = \{f < h\}$ then $F, G \in \mathcal{A}$ [see theorem: 21.256] so that using [lemma: 21.523]

$$(f - h)^+ = \mathcal{X}_{X,F} \cdot (f - h) \text{ and } (f - h)^- = \mathcal{X}_{X,G} \cdot (h - f) = -\mathcal{X}_{X,G} \cdot (f - h) \quad (21.876)$$

Then

$$\begin{aligned} \int (f - h)^+ d\mu &\stackrel{[\text{eq: 21.876}]}{=} \int (\mathcal{X}_{X,F} \cdot (f - h)) d\mu \stackrel{[\text{eq: 21.875}]}{=} 0 \\ \int (f - h)^- d\mu &\stackrel{[\text{eq: 21.876}]}{=} - \int (\mathcal{X}_{X,G} \cdot (f - h)) d\mu \stackrel{[\text{eq: 21.875}]}{=} 0 \end{aligned}$$

so that

$$\begin{aligned} \int |f - h| d\mu &\stackrel{[\text{theorem: 21.341}]}{=} \int ((f - h)^+ + (f - h)^-) d\mu \\ &\stackrel{[\text{theorem: 21.360}]}{=} \int (f - h)^+ d\mu + \int (f - h)^- d\mu \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Using now [theorem: 21.366] it follows that $f - h = 0$ μ -a.e. or using [theorem: 21.282] it follows that

$$f = h \text{ } \mu\text{-a.e.} \quad \square$$

Next we extend the previous lemma to the cases where ν is a σ -finite measure [see definition: 21.106].

Lemma 21.525. *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, μ a measure on $\mathcal{A}$, ν a σ -finite measure on $\mathcal{A}$ and $f, h \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ such that*

$$\forall A \in \mathcal{A} \text{ we have } \int_A^+ h d\mu = \nu(A) = \int_A^+ f d\mu$$

then we have that

$$h = f \text{ } \mu\text{-a.e.}$$

Proof. As ν is σ -finite we have by [definition: 21.106] and [theorem: 21.100] that there exist a $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ pairwise disjoint such that

$$X = \bigsqcup_{i \in \mathbb{N}} A_i \text{ and } \forall i \in \mathbb{N} \text{ we have } \nu(A_i) < \infty \quad (21.877)$$

Let $i \in \mathbb{N}$ then and take the measure spaces [see theorem: 21.111]

$$\langle A_i, \mathcal{A}|_{A_i}, \nu|_{\mathcal{A}|_{A_i}} \rangle \quad \langle A_i, \mathcal{A}|_{A_i}, \mu|_{\mathcal{A}|_{A_i}} \rangle$$

where

$$\mathcal{A}|_{A_i} = \{B \in \mathcal{A} | B \subseteq A_i\} \text{ and } \nu|_{\mathcal{A}|_{A_i}}, \mu|_{\mathcal{A}|_{A_i}} \text{ are the restrictions of } \nu, \mu \text{ to } \mathcal{A}|_{A_i}$$

Then for $C \in \mathcal{A}|_{A_i}$ we have $C \in \mathcal{A}$ and $C \subseteq A_i$ so that $\nu|_{\mathcal{A}|_{A_i}}(C) = \nu(C) \leq \nu(A_i) < \infty$ which proves that

$$\nu|_{\mathcal{A}|_{A_i}} \text{ is a finite measure} \quad (21.878)$$

Using [theorem: 21.231] and the fact that $f, h \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that

$$f|_{A_i}, h|_{A_i} \in \mathcal{M}[A_i, \mathcal{A}|_{A_i}, \overline{\mathbb{R}}] \quad (21.879)$$

Let $C \in \mathcal{A}_{|A_i}$ then $C \in \mathcal{A}$ and $C \subseteq A_i$ then using [theorem: 21.276] that

$$\mathcal{X}_{A_i, C} \cdot f_{|A_i}, \mathcal{X}_{A_i, C} \cdot h_{|A_i} \in \mathcal{M}[A_i, \mathcal{A}_{|A_i}, \overline{\mathbb{R}}] \text{ [see definition: 21.271]}$$

so that the following is well defined

$$\begin{aligned} \int_C^+ h_{|A_i} d\mu_{|A_i} &\stackrel{\text{def}}{=} \int^+ (\mathcal{X}_{A_i, C} \cdot h_{|A_i}) d\mu_{|A_i} \\ &\stackrel{\text{[theorem: 21.272]}}{=} \int^+ (\mathcal{X}_{X, C} \cdot h)_{|A_i} d\mu_{|A_i} \\ &\stackrel{\text{[theorem: 21.322]}}{=} \int^+ (\mathcal{X}_{X, A_i} \cdot (\mathcal{X}_{X, C} \cdot h)) d\mu \\ &\stackrel{\text{[theorem: 21.272]}}{=} \int^+ (\mathcal{X}_{X, A_i \cap C} \cdot h) d\mu \\ &\stackrel{C \subseteq A_i}{=} \int^+ (\mathcal{X}_{X, C} \cdot h) d\mu \\ &\stackrel{\text{def}}{=} \int_C^+ h d\mu \\ &= \nu(C) \\ &\stackrel{C \in \mathcal{A}_{|A_i}}{=} \nu_{|A_i}(C) \\ \int_C^+ f_{|A_i} d\mu_{|A_i} &\stackrel{\text{def}}{=} \int^+ (\mathcal{X}_{A_i, C} \cdot f_{|A_i}) d\mu_{|A_i} \\ &\stackrel{\text{[theorem: 21.272]}}{=} \int^+ (\mathcal{X}_{X, C} \cdot f)_{|A_i} d\mu_{|A_i} \\ &\stackrel{\text{[theorem: 21.322]}}{=} \int^+ (\mathcal{X}_{X, A_i} \cdot (\mathcal{X}_{X, C} \cdot f)) d\mu \\ &\stackrel{\text{[theorem: 21.272]}}{=} \int^+ (\mathcal{X}_{X, A_i \cap C} \cdot f) d\mu \\ &\stackrel{C \subseteq A_i}{=} \int^+ (\mathcal{X}_{X, C} \cdot f) d\mu \\ &\stackrel{\text{def}}{=} \int_C^+ f d\mu \\ &= \nu(C) \\ &\stackrel{C \in \mathcal{A}_{|A_i}}{=} \nu_{|A_i}(C) \end{aligned}$$

which proves that

$$\forall C \in \mathcal{A}_{|A_i} \text{ we have } \int_C^+ h_{|A_i} d\mu_{|A_i} = \nu_{|A_i}(C) = \int_C^+ f_{|A_i} d\mu_{|A_i} \quad (21.880)$$

The above together with [eqs: 21.878, 21.879] allows us to apply the previous lemma [see lemma: 21.524] giving $h_{|A_i} = f_{|A_i} \mu_{|A_i}$ -a.e. hence as $i \in \mathbb{N}$ was chosen arbitrary we have

$$\forall i \in \mathbb{N} \text{ we have } h_{|A_i} = f_{|A_i} \mu_{|A_i} \text{-a.e.} \quad (21.881)$$

Using the definition of $\mu_{|A_i}$ -a.e. the above becomes

$$\forall i \in \mathbb{N} \exists N_i \in \mathcal{A}_{|A_i} \subseteq \mathcal{A} \text{ such that } \{x \in A_i | h_{|A_i}(x) \neq f_{|A_i}(x)\} \subseteq N_i \text{ and } \mu_{|A_i}(N_i) = 0 \quad (21.882)$$

Let now $x \in \{x \in X | h(x) \neq f(x)\}$ then as $X = \bigsqcup_{i \in \mathbb{N}} A_i$ there exist a $i \in \mathbb{N}$ such that $x \in A_i$ and $h_{|A_i}(x) = h(x) \neq f(x) = f_{|A_i}(x)$ so that by [eq: 21.882] $x \in N_i \subseteq \bigcup_{i \in \mathbb{N}} N_i$. Hence we have

$$\{x \in X | h(x) \neq f(x)\} \subseteq \bigcup_{i \in \mathbb{N}} N_i$$

which as also

$$\mu\left(\bigcup_{i \in \mathbb{N}_0} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = \sum_{i=1}^{\infty} \mu|_{\mathcal{A}_{A_i}}(N_i) = \sum_{i=1}^{\infty} 0 = 0$$

proves that

$$h = f \text{ } \mu\text{-a.e.}$$

□

Next we prove the existence part of the Radon-Nikodym theorem for finite measures.

Lemma 21.526. (Radon-Nikodym Theorem (finite case)) *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and let μ and ν be **finite** measures on $\mathcal{A}$ such that $\nu \ll \mu$ [in other words ν is absolute convergent in relation to μ] then there exist a $h \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ [hence h is finite] such that*

$$\forall A \in \mathcal{A} \text{ we have } \nu(A) = \int_A^+ h d\mu \stackrel{\text{def}}{=} \int_A^+ (\mathcal{X}_{X,A} \cdot h) d\mu$$

Proof. Define

$$\mathcal{F} = \left\{ f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \mid \forall A \in \mathcal{A} \text{ we have } \int_A^+ f d\mu \stackrel{\text{def}}{=} \int_A^+ (\mathcal{X}_{X,A} \cdot f) d\mu \leq \nu(A) \right\}$$

Take $f_1, f_2 \in \mathcal{F}$ then using [theorem: 21.267] it follows that

$$f_1 \vee f_2 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.883)$$

Let $A \in \mathcal{A}$ and define

$$A_1 = \{x \in A \mid f_1(x) > f_2(x)\} = A \cap \{x \in X \mid f_1(x) > f_2(x)\} = A \cap \{f_1 > f_2\} \stackrel{[\text{theorem: 21.256}]}{\in} \mathcal{A}$$

$$A_2 = \{x \in A \mid f_1(x) \leq f_2(x)\} = A \cap \{x \in X \mid f_1(x) \leq f_2(x)\} = A \cap \{f_1 \leq f_2\} \stackrel{[\text{theorem: 21.256}]}{\in} \mathcal{A}$$

Then clearly we have

$$A = A_1 \cup A_2 \text{ and } A_1 \cap A_2 = \emptyset$$

Let $x \in X$ then we have either:

$x \in \mathbf{X} \setminus \mathbf{A}$. Then $x \notin A \Rightarrow x \notin A_1, A_2$ so that

$$\begin{aligned} (\mathcal{X}_{X,A} \cdot (f_1 \vee f_2))(x) &= \mathcal{X}_{X,A}(x) \cdot \max(f_1(x), f_2(x)) \\ &= 0 \cdot \max(f_1(x), f_2(x)) \\ &= 0 \\ &= 0 \cdot f_1(x) + 0 \cdot f_2(x) \\ &= \mathcal{X}_{X,A_1}(x) \cdot f_1(x) + \mathcal{X}_{X,A_2}(x) \cdot f_2(x) \\ &= (\mathcal{X}_{X,A_1} \cdot f_1 + \mathcal{X}_{X,A_2} \cdot f_2)(x) \end{aligned}$$

$x \in \mathbf{A}_1$. Then $f_1(x) > f_2(x)$ and $x \notin A_2$ so that $\max(f_1(x), f_2(x)) = f_1(x)$ hence

$$\begin{aligned} (\mathcal{X}_{X,A} \cdot (f_1 \vee f_2))(x) &= \mathcal{X}_{X,A}(x) \cdot \max(f_1(x), f_2(x)) \\ &= f_1(x) \\ &= \mathcal{X}_{X,A_1} \cdot f_1(x) + 0 \cdot f_2(x) \\ &= \mathcal{X}_{X,A_1}(x) \cdot f_1(x) + \mathcal{X}_{X,A_2}(x) \cdot f_2(x) \\ &= (\mathcal{X}_{X,A_1} \cdot f_1 + \mathcal{X}_{X,A_2} \cdot f_2)(x) \end{aligned}$$

$x \in \mathbf{A}_2$. Then $f_1(x) \leq f_2(x)$ so that $\max(f_1(x), f_2(x)) = f_2(x)$ hence

$$\begin{aligned} (\mathcal{X}_{X,A} \cdot (f_1 \vee f_2))(x) &= \mathcal{X}_{X,A}(x) \cdot \max(f_1(x), f_2(x)) \\ &= f_2(x) \\ &= 0 \cdot f_1(x) + \mathcal{X}_{X,A_2}(x) \cdot f_2(x) \\ &= \mathcal{X}_{X,A_1}(x) \cdot f_1(x) + \mathcal{X}_{X,A_2}(x) \cdot f_2(x) \\ &= (\mathcal{X}_{X,A_1} \cdot f_1 + \mathcal{X}_{X,A_2} \cdot f_2)(x) \end{aligned}$$

which proves that

$$\mathcal{X}_{X \setminus A} \cdot (f_1 \vee f_2) = \mathcal{X}_{X, A_1} \cdot f_1 + \mathcal{X}_{X, A_2} \cdot f_2$$

As $f_1, f_2 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorems: 21.276, 21.257, 21.267] that

$$\begin{aligned} \int_A^+ (f_1 \vee f_2) d\mu &\stackrel{\text{def}}{=} \int^+ (\mathcal{X}_{X, A} \cdot (f_1 \vee f_2)) d\mu \\ &= \int^+ (\mathcal{X}_{X, A_1} \cdot f_1 + \mathcal{X}_{X, A_2} \cdot f_2) d\mu \\ &= \int^+ (\mathcal{X}_{X, A_1} \cdot f_1) d\mu + \int^+ (\mathcal{X}_{X, A_2} \cdot f_2) d\mu \\ &= \int_{A_1}^+ f_1 d\mu + \int_{A_2}^+ f_2 d\mu \\ &\leq \nu(A_1) + \nu(A_2) \\ &\stackrel{\nu \text{ is a measure}}{=} \nu(A_1 \cup A_2) \\ &= \nu(A) \end{aligned}$$

which proves that $f_1 \vee f_2 \in \mathcal{F}$. To summarize

$$\forall f_1, f_2 \in \mathcal{F} \text{ we have } f_1 \vee f_2 \in \mathcal{F} \quad (21.884)$$

By [example: 21.318] $C_0 \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\forall A \in \mathcal{A}$

$$\int_A^+ C_0 d\mu = \int^+ (\mathcal{X}_{X, A} \cdot C_0) d\mu = \int^+ C_0 d\mu = 0 \leq \nu(A) \quad \nu \text{ is a measure}$$

which proves that $C_0 \in \mathcal{F}$ so that

$$\mathcal{F} \neq \emptyset \quad (21.885)$$

From the above it follows that $\{\int^+ f d\mu \mid f \in \mathcal{F}\} \neq \emptyset$ so that by [theorem: 20.83] there exist a $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{F}$ such that

$$\lim_{n \rightarrow \infty} \int^+ f_n d\mu = \sup \left(\left\{ \int^+ f d\mu \mid f \in \mathcal{F} \right\} \right) \text{ exist [it could be } \infty] \quad (21.886)$$

Define $\{g_n\}_{n \in \mathbb{N}}$ recursively by

$$\{g_n\}_{n \in \mathbb{N}} \text{ by } g_n = \begin{cases} f_1 & \text{if } n = 1 \\ g_n = g_{n-1} \vee f_n & \text{if } n \in \{2, \dots, \infty\} \end{cases}$$

then we prove by induction that

$$\forall n \in \mathbb{N} \quad g_n \in \mathcal{F} \text{ and } \forall x \in X \quad 0 \leq f_n(x) \leq g_n(x) \leq g_{n+1}(x) \quad (21.887)$$

Proof. Take

$$\mathcal{S} = \{n \in \mathbb{N} \mid g_n \in \mathcal{F} \wedge \forall x \in X \text{ we have } 0 \leq f_n(x) \leq g_n(x) \wedge g_n(x) \leq g_{n+1}(x)\}$$

then we have

$$1 \in \mathcal{S}. \text{ if } n = 1 \text{ Then } g_1 = f_1 \in \mathcal{F} \text{ and}$$

$$\forall x \in X \quad 0 \leq f_1(x) = g_1(x) \leq \max(g_1(x), f_2(x)) = (g_1 \vee f_2)(x) = g_2(x)$$

proving that $1 \in \mathcal{S}$.

$n \in \mathcal{S} \Rightarrow n + 1 \in \mathcal{S}$. Then $g_n \in \mathcal{F}$ [as $n \in \mathcal{S}$] and using [eq: 21.884] we have that

$$g_{n+1} = g_n \vee f_{n+1} \in \mathcal{F}$$

Further $\forall x \in X$ we have $0 \leq f_n(x) \leq g_n(x)$ and

$$f_{n+1}(x) \leq \max(g_n(x), f_{n+1}(x)) = (g_n \vee f_{n+1})(x) = g_{n+1}(x)$$

and

$$g_{n+1}(x) \leq \max(g_{n+1}(x), f_{(n+1)+1}(x)) = (g_{n+1} \vee f_{(n+1)+1})(x) = g_{(n+1)+1}(x).$$

which proves that $n+1 \in \mathcal{S}$.

Mathematical induction finishes then the proof of [eq: 21.887]. $\square$

Now, as $\forall n \in \mathbb{N}$ $g_n \in \mathcal{F}$, we have that $\int^+ g_n d\mu \leq \sup(\{\int^+ f d\mu | f \in \mathcal{F}\})$, further from [eq: 21.887] it follows using [theorem: 21.331] that $\int^+ g_n d\mu \leq \int^+ g_{n+1} d\mu$, hence $\{\int^+ g_n d\mu\}_{n \in \mathbb{N}}$ is increasing. So using [theorem: 20.92]

$$\lim_{n \rightarrow \infty} \int^+ g_n d\mu = \sup\left(\left\{\int^+ g_n d\mu | n \in \mathbb{N}\right\}\right) \leq \sup\left(\left\{\int^+ f d\mu | f \in \mathcal{F}\right\}\right) \quad (21.888)$$

Applying [theorem: 21.331] on [eq: 21.887] gives $\forall n \in \mathbb{N}$ that $\int^+ f_n d\mu \leq \int^+ g_n d\mu$. Using then [theorem: 20.85] and [eq: 21.886] it follows that $\sup(\{\int^+ f d\mu | f \in \mathcal{F}\}) = \lim_{n \rightarrow \infty} \int^+ f_n d\mu \leq \lim_{n \rightarrow \infty} \int^+ g_n d\mu$. Combining this with [eq: 21.888] proves that

$$\lim_{n \rightarrow \infty} \int^+ g_n d\mu = \sup\left(\left\{\int^+ f d\mu | f \in \mathcal{F}\right\}\right) \quad (21.889)$$

As $\forall x \in X$, $\forall n \in \mathbb{N}$ $g_n(x) \leq g_{n+1}(x)$ [see eq: 21.887] we have by [theorem: 20.92] that $\lim_{n \rightarrow \infty} g_n(x)$ exist and $\lim_{n \rightarrow \infty} g_n(x) = \sup(\{g_n(x) | n \in \mathbb{N}\})$. Using [theorem: 21.269] it follows that the function defined by

$$g = \lim_{n \rightarrow \infty} g_n: X \rightarrow \overline{\mathbb{R}} \text{ where } g(x) = \lim_{n \rightarrow \infty} g_n(x) = \sup(\{g_n(x) | n \in \mathbb{N}\}) \underset{\text{[eq: 21.887]}}{\geq} 0 \quad (21.890)$$

exist and

$$g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.891)$$

So we can use the Monotone Convergence Theorem [see theorem: 21.372] to get

$$\int^+ g d\mu = \int^+ \left(\lim_{n \rightarrow \infty} g_n\right) d\mu = \lim_{n \rightarrow \infty} \int^+ g_n d\mu \underset{\text{[eq: 21.889]}}{=} \sup\left(\left\{\int^+ f d\mu | f \in \mathcal{F}\right\}\right) \quad (21.892)$$

Let $A \in \mathcal{A}$ then as $\{g_n\}_{n \in \mathbb{N}} \subseteq \mathcal{F} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [corollary: 21.276] that

$$\{\mathcal{X}_{X,A} \cdot g_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

Further

$$\forall x \in X \quad \forall n \in \mathbb{N} \quad (\mathcal{X}_{X,A} \cdot g_n)(x) = \begin{cases} g_n(x) \leq g_{n+1}(x) & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases} \leq (\mathcal{X}_{X,A} \cdot g_{n+1})(x)$$

Also for $x \in X$ we have either:

$$x \in A. \text{ Then } \lim_{n \rightarrow \infty} (\mathcal{X}_{X,A} \cdot g_n)(x) = \lim_{n \rightarrow \infty} g_n(x) = g(x) = (\mathcal{X}_{X,A} \cdot g)(x)$$

$$x \in X \setminus A. \text{ Then } \lim_{n \rightarrow \infty} (\mathcal{X}_{X,A} \cdot g_n)(x) = \lim_{n \rightarrow \infty} 0 = 0 = (\mathcal{X}_{X,A} \cdot g)(x)$$

proving that

$$\mathcal{X}_{X,A} \cdot g = \lim_{n \rightarrow \infty} (\mathcal{X}_{X,A} \cdot g_n)$$

Hence all conditions for the Monotone Convergence Theorem [see theorem: 21.372] are satisfied so that

$$\int_A^+ g d\mu \underset{\text{def}}{=} \int^+ (\mathcal{X}_{X,A} \cdot g) d\mu = \int^+ \left(\lim_{n \rightarrow \infty} (\mathcal{X}_{X,A} \cdot g_n)\right) d\mu = \lim_{n \rightarrow \infty} \int^+ (\mathcal{X}_{X,A} \cdot g_n) d\mu \underset{\text{def}}{=} \lim_{n \rightarrow \infty} \int_A^+ g_n d\mu$$

As $\forall n \in \mathbb{N}$ $g_n \underset{\text{[eq: 21.887]}}{\in} \mathcal{F}$ we have by the definition of $\mathcal{F}$ that $\int_A^+ g_n d\mu \leq \nu(A)$, hence using [theorem: 20.85] we have that $\lim_{n \rightarrow \infty} \int_A^+ g_n d\mu \leq \nu(A)$. To summarize we have

$$\forall A \in \mathcal{A} \text{ we have } \int_A^+ g d\mu = \lim_{n \rightarrow \infty} \int_A^+ g_n d\mu \leq \nu(A) \underset{\nu \text{ is finite}}{<} \infty \quad (21.893)$$

In the special case of $A = X \in \mathcal{A}$ we have using the above that

$$\int^+ g d\mu = \int^+ (\mathcal{X}_{X,X} \cdot g) d\mu = \int_X^+ g d\mu < \infty \quad (21.894)$$

Using [eq: 21.891, 21.893] it follows from the definition of $\mathcal{F}$ that

$$g = \lim_{n \rightarrow \infty} g_n \in \mathcal{F} \quad (21.895)$$

As by [eq: 21.891] $g \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.380] that

$$\rho: A \rightarrow [0, \infty] \text{ defined by } \rho(A) = \int_A^+ g d\mu \text{ is a measure on } \mathcal{A} \quad (21.896)$$

As $\forall A \in \mathcal{A} \rho(A) = \int_A^+ g d\mu \leq \nu(A) < \infty$ it follows that ρ is a finite measure, as also ν is finite [theorem: 21.473] applies, so that

$$\nu_0 = \nu - \rho: \mathcal{A} \rightarrow [0, \infty] \text{ where } 0 \leq \nu_0(A) = \nu(A) - \rho(A) = \nu(A) - \int_A^+ g d\mu \text{ is a finite measure} \quad (21.897)$$

We prove now by contradiction that $\forall A \in \mathcal{A} \nu_0(A) \leq 0$. Assume that $\exists A_0 \in \mathcal{A}$ such that $0 < \nu_0(A_0)$. As $A_0 \subseteq X \Rightarrow 0 < \nu_0(A_0) \leq \nu_0(X) \Rightarrow \nu_0(X) \in \mathbb{R}^+$, further as μ is finite $\mu(X) + 1 \in \mathbb{R}^+$, hence $\varepsilon = \frac{\nu_0(X)}{2 \cdot (\mu(X) + 1)} \in \mathbb{R}^+$. For ε we have then $\varepsilon \cdot \mu(X) = \frac{\nu_0(X) \cdot \mu(X)}{2 \cdot (\mu(X) + 1)} < \frac{\nu_0(X)}{2} < \nu_0(X)$. To summarize we found a $\varepsilon \in \mathbb{R}^+$ such that

$$\varepsilon \cdot \mu(X) < \nu_0(X) \quad (21.898)$$

As ν_0 is a finite measure [see eq: 21.897] and $\varepsilon \cdot \mu$ is a measure [see theorem: 21.108] it follows from [theorem: 21.473] that

$$\nu_0 - \varepsilon \cdot \mu \text{ is a signed measure on } \mathcal{A} \quad (21.899)$$

Let (P, N) be a Hahn decomposition of $\nu_0 - \varepsilon \cdot \mu$ [see theorem: 21.487] then for $A \in \mathcal{A}$ we have $A \cap P \in \mathcal{A}$ and $A \cap P \subseteq P$ so that $0 \leq (\nu_0 - \varepsilon \cdot \mu)(A \cap P) = \nu_0(A \cap P) - \varepsilon \cdot \mu(A \cap P)$. As μ, ν_0 are finite measures it follows that

$$\forall A \in \mathcal{A} \text{ we have } \varepsilon \cdot \mu(A \cap P) \leq \nu_0(A \cap P) \quad (21.900)$$

Now for $A \in \mathcal{A}$ we have, taking in account that $\int_A^+ g d\mu, \nu_0(A)$ are finite [see eqs: 21.894, 21.897], that

$$\begin{aligned} \nu(A) & \stackrel{[\text{eq: 21.897}]}{=} \left(\int_A^+ g d\mu \right) + \nu_0(A) \\ & \stackrel{\nu_0 \text{ is a measure}}{\geq} \left(\int_A^+ g d\mu \right) + \nu_0(A \cap P) \\ & \stackrel{[\text{eq: 21.900}]}{\geq} \left(\int_A^+ g d\mu \right) + \varepsilon \cdot \mu(A \cap P) \\ & \stackrel{[\text{example: 21.319}]}{=} \int_A^+ g d\mu + \int^+ (\varepsilon \cdot \mathcal{X}_{X, A \cap P}) d\mu \\ & \stackrel{[\text{theorem: 21.272}]}{=} \int^+ (\mathcal{X}_{X, A} \cdot g) d\mu + \int^+ (\mathcal{X}_{X, A} \cdot (\varepsilon \cdot \mathcal{X}_{X, P})) d\mu \\ & \stackrel{[\text{theorem: 21.331}]}{=} \int^+ (\mathcal{X}_{X, A} \cdot (g + \varepsilon \cdot \mathcal{X}_{X, P})) d\mu \\ & = \int_A^+ (g + \varepsilon \cdot \mathcal{X}_{X, P}) d\mu \end{aligned}$$

which using the definition of $\mathcal{F}$ and the fact that $g + \varepsilon \cdot \mathcal{X}_{X, P} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.891, theorems: 21.272, 21.331], proves that

$$g + \varepsilon \cdot \mathcal{X}_{X, P} \in \mathcal{F} \quad (21.901)$$

If $\mu(P) = 0$ then as $\nu \ll \mu$ we must have $\nu(P) = 0$ so that by [eq: 21.897]

$$0 \leq \nu_0(P) = \nu(P) - \rho(P) = -\rho(P) \underset{[\text{eq: 21.896}]}{\leq} 0$$

so that $\mu(P) = 0 = \nu_0(P)$, hence:

$$\begin{aligned} \nu_0(X) - \varepsilon \cdot \mu(X) & \underset{X = \overline{P} \sqcup N}{=} \nu_0(P) + \nu_0(N) - \varepsilon \cdot \mu(P) - \varepsilon \cdot \mu(N) \\ & = \nu_0(N) - \varepsilon \cdot \mu(N) \\ & = (\nu_0 - \varepsilon \cdot \mu)(N) \\ & \leq 0 \end{aligned}$$

(P, N is a Hahn decomposition of $\nu_0 - \varepsilon \cdot \mu$)

which contradicts [eq: 21.898]. Hence we must have that $0 < \mu(P)$ from which it follows that

$$\begin{aligned} \int^+ (g + \varepsilon \cdot \mathcal{X}_{X,P}) d\mu & \underset{[\text{theorem: 21.331}]}{=} \int^+ g d\mu + \int^+ (\varepsilon \cdot \mathcal{X}_{X,P}) d\mu \\ & \underset{[\text{example: 21.319}]}{=} \int^+ g d\mu + \varepsilon \cdot \mu(P) \end{aligned}$$

As $\int^+ g d\mu < \infty$ [see eq: 21.894] and $0 < \varepsilon \cdot \mu(P) \underset{\mu \text{ is finite}}{<} \infty$ it follows that

$$\int^+ g d\mu < \int^+ (g + \varepsilon \cdot \mathcal{X}_{X,P}) d\mu$$

However

$$\int^+ (g + \varepsilon \cdot \mathcal{X}_{X,P}) d\mu \underset{[\text{eq: 21.901}]}{\leq} \sup \left(\left\{ \int^+ f | f \in \mathcal{F} \right\} \right) \underset{[\text{eq: 21.892}]}{=} \int^+ g d\mu$$

which leads to the contradiction $\int^+ g d\mu < \int^+ g d\mu$. Hence the assumption $\exists A_0 \in \mathcal{A}$ such that $0 < \nu_0(A_0)$ is false and we must have $\forall A \in \mathcal{A}$ that $\nu_0(A) \leq 0$ which combined with [eq: 21.897] proves that

$$\forall A \in \mathcal{A} \text{ we have } \nu_0(A) = 0$$

Hence $\forall A \in \mathcal{A}$ we have $\nu(A) - \int_A^+ g d\mu \underset{[\text{eq: 21.897}]}{=} \nu_0(A) = 0$ so that

$$\forall A \in \mathcal{A} \text{ we have } \int^+ (\mathcal{X}_{X,A} \cdot g) d\mu = \int_A^+ g d\mu = \nu(A) \underset{\nu \text{ is finite}}{<} \infty \quad (21.902)$$

By [eq: 21.894] $\int^+ g d\mu < \infty$ so that by [theorem: 21.336] we have

$$M = \{x \in X | g(x) = \infty\} \in \mathcal{A} \text{ and } \mu(M) = 0$$

Define now $h = \mathcal{X}_{X, X \setminus M} \cdot g$ then we have $\forall x \in X$ either:

$$\mathbf{x} \in \mathbf{M}. \text{ Then } (\mathcal{X}_{X, X \setminus M} \cdot g)(x) = 0 \cdot g(x) = 0 \in \mathbb{R}$$

$$\mathbf{X} \in \mathbf{X} \setminus \mathbf{M}. \text{ Then } (\mathcal{X}_{X, X \setminus M} \cdot g)(x) = g(x) \underset{x \in M}{\in} \mathbb{R}$$

So $(\mathcal{X}_{X, X \setminus M} \cdot g)(X) \subseteq \mathbb{R}$ and as $g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \underset{[\text{theorem: 21.276}]}{\Rightarrow} \mathcal{X}_{X, X \setminus M} \cdot g \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows that

$$h = \mathcal{X}_{X, X \setminus M} \cdot g \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.903)$$

Let $A \in \mathcal{A}$. As $\mu(M) = 0$ we have by [theorem: 21.280] that $\mathcal{X}_{X, X \setminus M} \cdot (\mathcal{X}_{X,A} \cdot g) = \mathcal{X}_{X,A} \cdot g$ μ -a.e. so that by [theorem: 21.333] we have

$$\int^+ \mathcal{X}_{X, X \setminus M} \cdot (\mathcal{X}_{X,A} \cdot g) d\mu = \int^+ (\mathcal{X}_{X,A} \cdot g) d\mu \underset{[\text{eq: 21.902}]}{=} \nu(A) \quad (21.904)$$

Further we have

$$\int_A^+ h d\mu = \int^+ (\mathcal{X}_{X,A} \cdot h) d\mu = \int^+ (\mathcal{X}_{X,A} \cdot (\mathcal{X}_{X, X \setminus M} \cdot g)) d\mu = \int^+ (\mathcal{X}_{X, X \setminus M} \cdot (\mathcal{X}_{X,A} \cdot g)) d\mu$$

which combined with [eq: 21.904] proves that

$$\forall A \in \mathcal{A} \quad \nu(A) = \int_A^+ h d\mu$$

the above together with [eq: 21.903] finishes the theorem. $\square$

Next step is to extend the above theorem to σ -finite measures.

Theorem 21.527. (Radon-Nikodym Theorem) *Let $\langle X, \mathcal{A} \rangle$ be a measurable space and let μ, ν be σ -finite measures on $\mathcal{A}$. If $\nu \ll \mu$ [in other words ν is absolute convergent in relation to μ] then there exist a $h \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ [so that h is finite] such that*

$$\forall A \in \mathcal{A} \quad \nu(A) = \int_A^+ h d\mu \stackrel{\text{def}}{=} \int_A^+ (\mathcal{X}_{X,A} \cdot h) d\mu$$

Further if there is another $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ such that $\forall A \in \mathcal{A} \quad \nu(A) = \int_A^+ f d\mu$ then

$$f = h \quad \mu\text{-a.e.}$$

Proof. As μ, ν are σ -finite measures there exist by [theorem: 21.104] two pairwise disjoint families $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$, $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that $\forall i \in \mathbb{N} \quad \mu(A_i), \nu(B_i) \in \mathbb{R}$ and $\bigsqcup_{i \in \mathbb{N}} A_i = X = \bigsqcup_{i \in \mathbb{N}} B_i$. Using [theorem: 21.13] there exist a bijection $\beta: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ such that

$$X = \bigcup_{i \in \mathbb{N}} (A_{\beta(i)_1} \cap B_{\beta(i)_2}) \quad \text{where} \quad \{A_{\beta(i)_1} \cap B_{\beta(i)_2}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$$

Further if $i, j \in \mathbb{N}$ with $i \neq j$ then as β is a bijection $\beta(i) \neq \beta(j)$. For $\beta(i)_1, \beta(j)_1$ we have either $\beta(i)_1 \neq \beta(j)_1$. Then we have

$$(A_{\beta(i)_1} \cap B_{\beta(i)_2}) \cap (A_{\beta(j)_1} \cap B_{\beta(j)_2}) \subseteq A_{\beta(i)_1} \cap A_{\beta(j)_1} = \emptyset$$

$\beta(i)_1 = \beta(j)_1$. Then we must have $\beta(i)_2 \neq \beta(j)_2$ so that

$$(A_{\beta(i)_1} \cap B_{\beta(i)_2}) \cap (A_{\beta(j)_1} \cap B_{\beta(j)_2}) \subseteq B_{\beta(i)_2} \cap B_{\beta(j)_2} = \emptyset$$

Further $\forall i \in \mathbb{N}$ we have $\mu(A_{\beta(i)_1} \cap B_{\beta(i)_2}) \leq \mu(A_{\beta(i)_1}) < \infty$ and $\nu(A_{\beta(i)_1} \cap B_{\beta(i)_2}) \leq \nu(B_{\beta(i)_2}) < \infty$. So if we define

$$\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \quad \text{by} \quad C_i = A_{\beta(i)_1} \cap B_{\beta(i)_2}$$

then we have

$$\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \quad \text{is pairwise disjoint, } X = \bigsqcup_{i \in \mathbb{N}} C_i \quad \text{and} \quad \forall i \in \mathbb{N} \quad \mu(C_i) < \infty \wedge \nu(C_i) < \infty \quad (21.905)$$

Let $i \in \mathbb{N}$. Consider the measurable space $\langle C_i, \mathcal{A}|_{C_i} \rangle$ with measures $\mu|_{\mathcal{A}|_{C_i}}, \nu|_{\mathcal{A}|_{C_i}}$ [see theorem: 21.111]. Then if $A \in \mathcal{A}|_{C_i}$ we have $A \in \mathcal{A} \wedge A \subseteq C_i$ and $\mu|_{\mathcal{A}|_{C_i}}(A) = \mu(A) \leq \mu(C_i) < \infty$ and $\nu|_{\mathcal{A}|_{C_i}}(A) = \nu(A) \leq \nu(C_i) < \infty$ hence we have that

$$\forall i \in \mathbb{N} \quad \text{we have } \mu|_{\mathcal{A}|_{C_i}} \text{ and } \nu|_{\mathcal{A}|_{C_i}} \text{ are finite measures on } \mathcal{A}|_{C_i}$$

If $A \in \mathcal{A}|_{C_i}$ such that $\mu|_{C_i}(A) = 0$ then $\mu(A) = 0$, as $\nu \ll \mu$ it follows that $\nu(A) = 0$, so that $\nu|_{\mathcal{A}|_{C_i}}(A) = \nu(A) = 0$. Hence we have that

$$\forall i \in \mathbb{N} \quad \text{we have } \nu|_{\mathcal{A}|_{C_i}} \ll \mu|_{\mathcal{A}|_{C_i}}$$

So we can apply the finite version of the Radon-Nikodym Theorem [theorem: 21.526] to get

$$\forall i \in \mathbb{N} \quad \exists h_i \in \mathcal{M}^+[C_i, \mathcal{A}|_{C_i}, \mathbb{R}] \quad \text{such that} \quad \forall A \in \mathcal{A}|_{C_i} \quad \nu|_{\mathcal{A}|_{C_i}}(A) = \int_A^+ h_i \cdot d\mu|_{\mathcal{A}|_{C_i}} \quad (21.906)$$

Let $i \in \mathbb{N}$. As $h_i \in \mathcal{M}[C_i, \mathcal{A}|_{C_i}, \mathbb{R}]$ we have, using [theorem: 21.324], that $(h_i)_{C_i \uparrow X} \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$. As further $\forall x \in X \quad (h_i)_{C_i \uparrow X}(x) = \begin{cases} h_i(x) \in [0, \infty[& \text{if } x \in C_i \\ 0 \in [0, \infty[& \text{if } x \in X \setminus C_i \end{cases}$ it follows that

$$\{(h_i)_{C_i \uparrow X}\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.907)$$

Let $A \in \mathcal{A}_{|C_i}$ then $A \in \mathcal{A} \wedge A \subseteq C_i$ and by [theorem: 21.276] $\mathcal{X}_{C_i, A} \cdot h_i \in \mathcal{M}^+[C_i, \mathcal{A}_{|C_i}, \mathbb{R}]$. Applying then [theorem: 21.324] results in

$$(\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

$$\int^+ (\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X} d\mu \stackrel{[\text{theorem: 21.324}]}{=} \int^+ (\mathcal{X}_{C_i, A} \cdot h_i) d\mu_{|A|C_i}$$

Using [theorem: 21.248] we have that $\mathcal{X}_{X, A} \cdot (h_i)_{|C_i \uparrow X} = (\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X}$ so that

$$\int^+ \mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X} d\mu = \int^+ (\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X} d\mu = \int^+ (\mathcal{X}_{C_i, A} \cdot h_i) d\mu_{|A|C_i}$$

or

$$\int_A^+ (h_i)_{C_i \uparrow X} d\mu = \int_A^+ h_i d\mu_{|A|C_i}$$

Further for $x \in X$ we have $((\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X})(x) = \begin{cases} (\mathcal{X}_{C_i, A} \cdot h_i)(x) \in \mathbb{R} & \text{if } x \in C_i \\ 0 \in \mathbb{R} & \text{if } x \notin C_i \end{cases}$ so that $(\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X} \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$. To summarize we have that

$$\{(\mathcal{X}_{C_i, A} \cdot h_i)_{C_i \uparrow X}\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \text{ and } \forall i \in \mathbb{N}, \forall A \in \mathcal{A}_{|C_i} \int_A^+ (h_i)_{C_i \uparrow X} d\mu = \int_A^+ h_i d\mu_{|A|C_i} \quad (21.908)$$

Given $x \in X = \bigsqcup_{i \in \mathbb{N}} C_i$ there exist a **unique** $i \in \mathbb{N}$ such that $x \in C_i$ which allows us to define the following function:

$$h: X \rightarrow \overline{\mathbb{R}} \text{ by } h(x) = h_i(x) \text{ where } i \text{ is such that } x \in C_i$$

Then as $\forall i \in \mathbb{N} \ h_i \in \mathcal{M}[C_i, \mathcal{A}_{|C_i}, \mathbb{R}] \Rightarrow h_i(X) \subseteq [0, \infty[$ it follows that

$$h(X) \subseteq [0, \infty[\quad (21.909)$$

Let $A \in \mathcal{A}$ and $x \in X$. Let $\varepsilon \in \mathbb{R}^+$ then as $x \in X = \bigsqcup_{i \in \mathbb{N}} C_i$ there exist a $N \in \mathbb{N}$ such that $x \in C_N$ so that $h(x) = h_N(x)$. Hence $(h_N)_{C_N \uparrow X}(x) \stackrel{x \in C_N}{=} h_N(x)$ and $\forall i \in \mathbb{N} \setminus \{N\}$ we have $x \notin C_i$ so that $(h_i)_{C_i \uparrow X}(x) = 0$. Let $n \in \{N, \dots, \infty\}$ then we have for $x \in C_N$ either:

$x \in A$. Then

$$\begin{aligned} \left| (\mathcal{X}_{X, A} \cdot h)(x) - \sum_{i=1}^n (\mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X})(x) \right| &= \left| h(x) - \sum_{i=1}^n (h_i)_{C_i \uparrow X}(x) \right| \\ &\stackrel{x \in C_N}{=} \left| h_N(x) - \sum_{i=1}^n (h_i)_{C_i \uparrow X}(x) \right| \\ &= |h_N(x) - h_N(x)| \\ &= 0 \\ &< \varepsilon \end{aligned}$$

$x \in X \setminus A$. Then

$$\begin{aligned} \left| (\mathcal{X}_{X, A} \cdot h)(x) - \sum_{i=1}^n (\mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X})(x) \right| &= 0 - \sum_{i=1}^n 0 \\ &= 0 \\ &< \varepsilon \end{aligned}$$

which proves that $\sum_{i=1}^{\infty} (\mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X})(x)$ exist and $\sum_{i=1}^{\infty} (\mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X})(x) = (\mathcal{X}_{X, A} \cdot h)(x)$. Hence we have that

$$\forall A \in \mathcal{A} \text{ we have } \sum_{i=1}^{\infty} (\mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X}) = \mathcal{X}_{X, A} \cdot h \quad (21.910)$$

Further using [theorem: 21.276] on [eq: 21.907] it follows that

$$\{\mathcal{X}_{X, A} \cdot (h_i)_{C_i \uparrow X}\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.911)$$

So the conditions for the Beppo Levi's Theorem [see theorem: 21.377] from which it follows that

$$\mathcal{X}_{A,X} \cdot h = \sum_{i=1}^{\infty} (\mathcal{X}_{X,A} \cdot (h_i)_{C_i \uparrow X}) \in \mathcal{M}^+[X, \mathcal{A}, \bar{R}] \quad (21.912)$$

and

$$\int^+ (\mathcal{X}_{X,A} \cdot h) d\mu = \int^+ \left(\sum_{i=1}^{\infty} (\mathcal{X}_{X,A} \cdot (h_i)_{C_i \uparrow X}) \right) d\mu = \sum_{i=1}^{\infty} \int^+ (\mathcal{X}_{X,A} \cdot (h_i)_{C_i \uparrow X}) d\mu \quad (21.913)$$

As $X \in \mathcal{A}$ and $\mathcal{X}_{X,X} \cdot h = h$ it follows from [eq: 21.912] and $h(X) \in [0, \infty[$ [see eq: 21.909] that

$$h \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}] \quad (21.914)$$

Further we have for $A \in \mathcal{A}$ that $A = A \cap X = A \cap (\bigsqcup_{i \in \mathbb{N}} C_i) = \bigsqcup_{i \in \mathbb{N}} (A \cap C_i)$ where $\forall i \in \mathbb{N} A \cap C_i \in \mathcal{A} \wedge A \cap C_i \subseteq C_i$ so that $A \cap C_i \in \mathcal{A}_{|C_i}$. Hence

$$\begin{aligned} \nu(A) &= \nu\left(\bigsqcup_{i \in \mathbb{N}} (A \cap C_i)\right) \\ &= \sum_{i=1}^{\infty} \nu(A \cap C_i) \\ &\stackrel{A \cap C_i \in \mathcal{A}_{|C_i}}{=} \sum_{i=1}^{\infty} \nu_{|A \cap C_i}(A \cap C_i) \\ &\stackrel{[\text{eq: 21.906}]}{=} \sum_{i=1}^{\infty} \left(\int_{A \cap C_i}^+ h_i d\mu_{|A \cap C_i} \right) \\ &\stackrel{[\text{eq: 21.908}]}{=} \sum_{i=1}^{\infty} \left(\int_{A \cap C_i}^+ (h_i)_{C_i \uparrow X} d\mu \right) \\ &\stackrel{\text{def}}{=} \sum_{i=1}^{\infty} \left(\int^+ (\mathcal{X}_{X,A \cap C_i} \cdot (h_i)_{C_i \uparrow X}) d\mu \right) \\ &\stackrel{[\text{theorem: 21.272}]}{=} \sum_{i=1}^{\infty} \left(\int^+ (\mathcal{X}_{X,A} \cdot (\mathcal{X}_{X,C_i} \cdot (h_i)_{C_i \uparrow X})) d\mu \right) \\ &\stackrel{[\text{definition: 21.249}]}{=} \sum_{i=1}^{\infty} \left(\int^+ (\mathcal{X}_{X,A} \cdot (h_i)_{C_i \uparrow X}) d\mu \right) \\ &\stackrel{[\text{eq: 21.913}]}{=} \int^+ (\mathcal{X}_{X,A} \cdot h) d\mu \\ &\stackrel{\text{def}}{=} \int_A^+ h d\mu \end{aligned}$$

Hence we have found a $h \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ [see eq: 21.914] such that $\forall A \in \mathcal{A}$ we have $\nu(A) = \int_A^+ h d\mu$ which proves existence. Now for uniqueness, assume that there exist a $f \in \mathcal{M}^+(X, \mathcal{A}, \mathbb{R})$ such that $\forall A \in \mathcal{A} \nu(A) = \int_A^+ f d\mu$ then by [lemma: 21.525] it follows that

$$f = h \text{ } \mu\text{-a.e.} \quad \square$$

We extend now the Radon-Nikodym theorem to signed and complex measures. First we extend the concept of absolute continuity.

Definition 21.528. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we define:

1. If ν is a signed measure on $\mathcal{A}$ then ν is **absolute continuous with respect to μ noted as $\nu \ll \mu$** if $|\nu| \ll \mu$ where $|\nu|$ is the variation of the signed measure ν [see theorem: 21.489].
2. If ν is a complex measure on $\mathcal{A}$ then ν is **absolute continuous with respect to μ noted as $\nu \ll \mu$** if $|\nu| \ll \mu$ where $|\nu|$ is the variation of the complex measure ν [see theorem: 21.498].

We have the following equivalent definition for absolute contiguity on signed (complex) measures.

Theorem 21.529. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space then we have the following equivalences:*

1. *If ν is a signed measure on $\mathcal{A}$ with Jordan decomposition (ν^+, ν^-) [see theorem: 21.489] then we have*

$$\nu \ll \mu \Leftrightarrow \nu^+ \ll \mu \wedge \nu^- \ll \mu$$

2. *If ν is a complex measure on $\mathcal{A}$ with Jordan decomposition $(\nu_r^+, \nu_r^-, \nu_c^+, \nu_c^-)$ [see theorem: 21.497] then we have*

$$\nu \ll \mu \Leftrightarrow \nu_r^+ \ll \mu \wedge \nu_r^- \ll \mu \wedge \nu_c^+ \ll \mu \wedge \nu_c^- \ll \mu$$

Proof.

1. Let ν^+, ν^- be the Jordan decomposition of ν then by [theorem: 21.489] ν^+, ν^- are measures on $\mathcal{A}$ and $|\nu| = \nu^+ + \nu^-$.

$\Rightarrow$. Let $A \in \mathcal{A}$ with $\mu(A) = 0$ then as $\nu \ll \mu$ we have by definition that $|\nu| \ll \mu$ so that $|\nu|(A) = 0$, hence $\nu^+(A) + \nu^-(A) = (\nu^+ + \nu^-)(A) = 0$. As $0 \leq \nu^+(A), \nu^-(A)$ [because ν^+, ν^- are measures] we have

$$0 \leq \nu^+(A), \nu^-(A) \leq \nu^+(A) + \nu^-(A) = 0$$

so that $\nu^+(A) = 0 = \nu^-(A)$ which proves that $\nu^+ \ll \mu \wedge \nu^- \ll \mu$.

$\Leftarrow$. Let $A \in \mathcal{A}$ with $\mu(A) = 0$ then as $\nu^+ \ll \mu \wedge \nu^- \ll \mu$ we have that $\nu^+(A) = 0 = \nu^-(A)$ so that $|\nu|(A) = \nu^+(A) + \nu^-(A) = 0$ which proves that $|\nu| \ll \mu$ so that by definition $\nu \ll \mu$.

2. Let $\nu_r^+, \nu_r^-, \nu_c^+, \nu_c^-$ be the Jordan decomposition of ν then by [theorem: 21.497] we have that $\nu_r^+, \nu_r^-, \nu_c^+, \nu_c^-$ are finite measures and $\nu = \nu_r^+ - \nu_r^- + i \cdot (\nu_c^+ - \nu_c^-)$.

$\Rightarrow$. Let $A \in \mathcal{A}$ such that $\mu(A) = 0$. As $\nu \ll \mu$ it follows that $|\nu|(A) = 0$, so, as $|\nu(A)| \leq |\nu|(A)$ [see theorem: 21.500], it follows that $\nu(A) = 0$. So we must have that

$$\nu_r^+(A) + \nu_r^-(A) = 0 = \nu_c^+(A) + \nu_c^-(A)$$

hence

$$0 \leq \nu_r^+(A), \nu_r^-(A) \leq \nu_r^+(A) + \nu_r^-(A) = 0 \text{ and } 0 \leq \nu_c^+(A), \nu_c^-(A) \leq \nu_c^+(A) + \nu_c^-(A) = 0$$

which proves that

$$\nu_r^+(A) = 0 \wedge \nu_r^-(A) = 0 \wedge \nu_c^+(A) = 0 \wedge \nu_c^-(A) = 0$$

So we have

$$\nu_r^+ \ll \mu \wedge \nu_r^- \ll \mu \wedge \nu_c^+ \ll \mu \wedge \nu_c^- \ll \mu$$

$\Leftarrow$. Let $A \in \mathcal{A}$ such that $\mu(A) = 0$. As $\nu_r^+ \ll \mu \wedge \nu_r^- \ll \mu \wedge \nu_c^+ \ll \mu \wedge \nu_c^- \ll \mu$ we have that

$$\nu_r^+(A) = \nu_r^-(A) = \nu_c^+(A) = \nu_c^-(A) = 0$$

Take $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ pairwise disjoint such that $A = \bigsqcup_{i \in \{1, \dots, n\}} A_i$ then we have $\forall i \in \mathbb{N} \ \nu_r^+(A_i) \leq \nu_r^+(A) = 0, \ \nu_r^-(A_i) \leq \nu_r^-(A) = 0, \ \nu_c^+(A_i) \leq \nu_c^+(A) = 0$ and $\nu_c^-(A_i) \leq \nu_c^-(A) = 0$, so that $\nu_r^+(A_i) = \nu_r^-(A_i) = \nu_c^+(A_i) = \nu_c^-(A_i) = 0$. Hence

$$\begin{aligned} \sum_{i=1}^n |\nu(A_i)| &= \sum_{i=1}^n |\nu_r^+(A_i) - \nu_r^-(A_i) + i \cdot \nu_c^+(A_i) - i \cdot \nu_c^-(A_i)| \\ &\leq \sum_{i=1}^n (|\nu_r^+(A_i)| + |\nu_r^-(A_i)| + |\nu_c^+(A_i)| + |\nu_c^-(A_i)|) \\ &= \sum_{i=1}^n 0 \\ &= 0 \end{aligned}$$

so that

$$\begin{aligned} |\nu|(A) &= \sup \left(\sum_{i=1}^n |\mu(A_i)| \mid n \in \mathbb{N}_0 \wedge \{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A} \wedge A = \bigsqcup_{i \in \{1, \dots, n\}} A_i \right) \\ &= \sup(\{0\}) \\ &= 0 \end{aligned}$$

proving that

$$\nu \ll \mu$$

□

To prove the Radon-Nikodym theorem for signed or complex measures, we first prove the uniqueness condition.

Lemma 21.530. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f, g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ satisfies*

$$\forall A \in \mathcal{A} \quad \int_A f d\mu = \int_A g d\mu$$

then

$$f = g \text{ } \mu\text{-a.e.}$$

Proof. Let $A \in \mathcal{A}$ then we have by [theorems: 21.356, 21.360]

$$\begin{aligned} \int (\mathcal{X}_{X,A} \cdot (f - g)) d\mu &= \int (\mathcal{X}_{X,A} \cdot f - \mathcal{X}_{X,A} \cdot g) d\mu \\ &= \int (\mathcal{X}_{X,A} \cdot f) d\mu - \int (\mathcal{X}_{X,A} \cdot g) d\mu \\ &= \nu(A) - \nu(A) \\ &= 0 \end{aligned}$$

so that

$$\forall A \in \mathcal{A} \text{ we have } \int (\mathcal{X}_{X,A} \cdot (f - g)) d\mu = 0 \quad (21.915)$$

Define now $F = \{f > g\}$ and $G = \{f < g\}$ then $\{f < g\}, \{g < f\} \in \mathcal{A}$ [see theorem: 21.256] and we have by [theorem: 21.523] that

$$(f - g)^+ = \mathcal{X}_F \cdot (f - g) \text{ and } (f - g)^- = \mathcal{X}_G \cdot (g - f) = -\mathcal{X}_G \cdot (f - g) \quad (21.916)$$

Next we have

$$\int |(f - g)^+| d\mu = \int (f - g)^+ d\mu = \int \mathcal{X}_F \cdot (f - g) d\mu \stackrel{[\text{eq: 21.915}]}{=} 0$$

and

$$\int |(f - g)^-| d\mu = \int (f - g)^- d\mu = - \int \mathcal{X}_G \cdot (f - g) d\mu \stackrel{[\text{eq: 21.915}]}{=} 0$$

Using [theorem: 21.366] we have then that $(f - h)^+ = 0$ μ -a.e. and $(f - h)^- = 0$ μ -a.e.. As by [theorem: 21.341] $(f - h) = (f - h)^+ - (f - h)^-$ it follows from [theorem: 21.281] that $(f - h) = 0$ μ -a.e.. Hence by [theorem: 21.282]

$$f = h \text{ } \mu\text{-a.e.}$$

which concludes the lemma. □

We have a similar lemma for complex measures.

Lemma 21.531. *Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space and $f, g \in \mathcal{L}[X, \mathcal{A}, \mathbb{C}]$ such that*

$$\forall A \in \mathcal{A} \text{ we have } \int_A f d\mu = \int_A g d\mu$$

then

$$f = g \text{ } \mu\text{-a.e.}$$

Proof. Let $A \in \mathcal{A}$ then

$$\begin{aligned} \int \operatorname{Re}(\mathcal{X}_A \cdot (f - g)) d\mu + i \cdot \int \operatorname{Im}(\mathcal{X}_A \cdot (f - g)) d\mu &\stackrel{\text{def}}{=} \int^{\mathbb{C}} \mathcal{X}_A \cdot (f - g) d\mu \\ &\stackrel{[\text{theorem: } 21.409]}{=} \int^{\mathbb{C}} \mathcal{X}_A \cdot f d\mu - \int^{\mathbb{C}} \mathcal{X}_A \cdot g d\mu \\ &\stackrel{\text{def}}{=} \int_A^{\mathbb{C}} f d\mu - \int_A^{\mathbb{C}} g d\mu \\ &= \nu(A) - \nu(A) \\ &= 0 \end{aligned}$$

hence

$$\int \operatorname{Re}(\mathcal{X}_A \cdot (f - g)) d\mu = 0 \wedge \int \operatorname{Im}(\mathcal{X}_A \cdot (f - g)) d\mu = 0 \quad (21.917)$$

As $\mathcal{X}_A$ is real it follows from the above and [theorem: 21.405] that

$$\forall A \in \mathcal{A} \text{ we have } \int \mathcal{X}_A \cdot (\operatorname{Re}(f) - \operatorname{Re}(g)) d\mu = 0 \wedge \int \mathcal{X}_A \cdot (\operatorname{Im}(f) - \operatorname{Im}(g)) d\mu = 0 \quad (21.918)$$

Define

$$F_r = \{\operatorname{Re}(f) > \operatorname{Re}(g)\}, G_r = \{\operatorname{Re}(f) < \operatorname{Re}(g)\}, F_c = \{\operatorname{Im}(f) > \operatorname{Im}(g)\}, G_c = \{\operatorname{Im}(f) < \operatorname{Im}(g)\}$$

Using [definition: 21.403] and [definition: 21.345] we have

$$\operatorname{Re}(f), \operatorname{Re}(g), \operatorname{Im}(f), \operatorname{Im}(g) \in \mathcal{M}[X, \mathcal{A}, \mathbb{R}]$$

and it follows by [theorem: 21.256] that

$$F_r, G_r, F_c, G_c \in \mathcal{A} \quad (21.919)$$

Next we have

$$\begin{aligned} \int |(\operatorname{Re}(f) - \operatorname{Re}(g))^+| d\mu &= \int (\operatorname{Re}(f) - \operatorname{Re}(g))^+ d\mu \\ &\stackrel{[\text{theorem: } 21.523]}{=} \int \mathcal{X}_{F_r} \cdot (\operatorname{Re}(f) - \operatorname{Re}(g)) d\mu \\ &\stackrel{[\text{eqs: } 21.918, 21.919]}{=} 0 \\ \int |(\operatorname{Re}(f) - \operatorname{Re}(g))^-| d\mu &= \int (\operatorname{Re}(f) - \operatorname{Re}(g))^- d\mu \\ &\stackrel{[\text{theorem: } 21.523]}{=} \int \mathcal{X}_{G_r} \cdot (\operatorname{Re}(f) - \operatorname{Re}(g)) d\mu \\ &\stackrel{[\text{eqs: } 21.918, 21.919]}{=} 0 \\ \int |(\operatorname{Im}(f) - \operatorname{Im}(g))^+| d\mu &= \int (\operatorname{Im}(f) - \operatorname{Im}(g))^+ d\mu \\ &\stackrel{[\text{theorem: } 21.523]}{=} \int \mathcal{X}_{F_c} \cdot (\operatorname{Im}(f) - \operatorname{Im}(g)) d\mu \\ &\stackrel{[\text{eqs: } 21.918, 21.919]}{=} 0 \\ \int |(\operatorname{Im}(f) - \operatorname{Im}(g))^-| d\mu &= \int (\operatorname{Im}(f) - \operatorname{Im}(g))^- d\mu \\ &\stackrel{[\text{theorem: } 21.523]}{=} \int \mathcal{X}_{G_c} \cdot (\operatorname{Im}(f) - \operatorname{Im}(g)) d\mu \\ &\stackrel{[\text{eqs: } 21.918, 21.919]}{=} 0 \end{aligned}$$

Using [theorem: 21.366] on the above gives

$$\begin{aligned} (\operatorname{Re}(f) - \operatorname{Re}(g))^+ &= 0 \quad \mu\text{-a.e.} \\ (\operatorname{Re}(f) - \operatorname{Re}(g))^- &= 0 \quad \mu\text{-a.e.} \\ (\operatorname{Im}(f) - \operatorname{Im}(g))^+ &= 0 \quad \mu\text{-a.e.} \\ (\operatorname{Im}(f) - \operatorname{Im}(g))^- &= 0 \quad \mu\text{-a.e.} \end{aligned}$$

By [theorem: 21.341]

$$\operatorname{Re}(f) - \operatorname{Re}(g) = (\operatorname{Re}(f) - \operatorname{Re}(g))^+ - (\operatorname{Re}(f) - \operatorname{Re}(g))^-$$

and

$$\operatorname{Im}(f) - \operatorname{Im}(g) = (\operatorname{Im}(f) - \operatorname{Im}(g))^+ - (\operatorname{Im}(f) - \operatorname{Im}(g))^-$$

so it follows from [theorem: 21.281] that $\operatorname{Re}(f) - \operatorname{Re}(g) = 0$ μ -a.e and $\operatorname{Im}(f) - \operatorname{Im}(g) = 0$ μ -a.e.. Applying [theorem: 21.282] proves that gives $\operatorname{Re}(f) = \operatorname{Re}(g)$ μ -a.e. and $\operatorname{Im}(f) = \operatorname{Im}(g)$ μ -a.e. Finally using [theorem: 21.395] again proves that

$$f = g \quad \mu\text{-a.e.} \quad \square$$

We are now finally ready to prove the Radon-Nikodym Theorem for signed and complex measures.

Theorem 21.532. (Radon-Nikodym Theorem (Signed/Complex)) *Let $\langle X, \mathcal{A} \rangle$ be a measurable space, μ a σ -finite measure on $\langle X, \mathcal{A} \rangle$ then we have*

1. *If ν is a **finite signed** measure on $\langle X, \mathcal{A} \rangle$ so that $\nu \ll \mu$ then there exists a $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ such that*
 - a. $\forall A \in \mathcal{A}$ we have $\nu(A) = \int_A f d\mu$
 - b. *If $g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $\forall A \in \mathcal{A}$ $\nu(A) = \int_A g d\mu$ then $f = g$ μ -a.e.*
2. *If ν is a **complex** measure on $\langle X, \mathcal{A} \rangle$ so that $\nu \ll \mu$ then there exists a $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ such that*
 - a. $\forall A \in \mathcal{A}$ we have $\nu(A) = \int_A f d\mu$
 - b. *If $g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ such that $\forall A \in \mathcal{A}$ $\nu(A) = \int_A g d\mu$ then $f = g$ μ -a.e.*

Proof.

1. As ν is a signed measure there exist a Jordan Decomposition [see theorem: 21.489] (ν^+, ν^-) of ν , so that $\nu = \nu^+ - \nu^-$. As ν is finite we have also that ν^-, ν^+ are finite measures on $\mathcal{A}$ and as finite measures are automatically σ -finite [see definition: 21.106] ν^+, ν^- are σ -finite. Using the Radon-Nikodym Theorem [see 21.527] there exists $f_+, f_- \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ such that

$$\forall A \in \mathcal{A} \quad \int_A (\mathcal{X}_{X,A} \cdot f_+) d\mu = \int_A^+ f_+ d\mu = \nu^+(A) < \infty \wedge \int_A^+ (\mathcal{X}_{X,A} \cdot f_-) d\mu \int_A^+ f_- d\mu = \nu^-(A) < \infty$$

Using [theorems: 21.276, 21.352] on the above gives

$$\forall A \in \mathcal{A} \text{ we have } \mathcal{X}_{X,A} \cdot f_+, \mathcal{X}_{X,A} \cdot f_- \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \Big|_{\text{take } \bar{A}=X} \Rightarrow f_+, f_- \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

and

$$\forall A \in \mathcal{A} \text{ we have } \int_A f_+ d\mu = \int \mathcal{X}_{X,A} \cdot f_+ d\mu = \nu^+(A) \wedge \int_A f_- d\mu = \int \mathcal{X}_{X,A} \cdot f_- d\mu = \nu^-(A)$$

Take $f = f_+ - f_-$ so that $\mathcal{X}_{X,A} \cdot f = \mathcal{X}_{X,A} \cdot f_+ - \mathcal{X}_{X,A} \cdot f_-$ then by [theorem: 21.360] we have

$$\forall A \in \mathcal{A} \quad \mathcal{X}_{X,A} \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad \text{and} \quad f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$$

and

$$\forall A \in \mathcal{A} \quad \int_A f d\mu = \int \mathcal{X}_{X,A} \cdot f d\mu = \int \mathcal{X}_{X,A} \cdot f_+ d\mu - \int \mathcal{X}_{X,A} \cdot f_- d\mu = \nu^+(A) - \nu^-(A) = \nu(A)$$

Summarizing

$$\exists f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \text{ such that } \forall A \in \mathcal{A} \quad \int_A f d\mu = \nu(A)$$

Finally if there exists another $g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $\forall A \in \mathcal{A} \int_A g d\mu = \nu(A)$ then using [lemma: 21.530] it follows that

$$f = g \text{ } \mu\text{-a.e.}$$

2. As ν is a complex measure there exists a Jordan Decomposition [see theorem: 21.497] $(\nu_r^+, \nu_r^-, \nu_c^+, \nu_c^-)$ of ν so that $\nu = (\nu_r^+ - \nu_r^-) + i \cdot (\nu_c^+ - \nu_c^-)$ where $\nu_r^+, \nu_r^-, \nu_c^+, \nu_c^-$ are finite measures on $\mathcal{A}$ [hence σ -finite on $\mathcal{A}$]. Using the Radon-Nikodym Theorem [see 21.527] there exists $f_+^r, f_-^r, f_+^c, f_-^c \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ such that

$$\begin{aligned} \forall A \in \mathcal{A} \text{ we have } \int_A^+ \mathcal{X}_{X,A} \cdot f_+^r d\mu &= \int_A^+ f_+^r d\mu = \nu_r^+(A) < \infty \\ \int_A^+ \mathcal{X}_{X,A} \cdot f_-^r d\mu &= \int_A^+ f_-^r d\mu = \nu_r^-(A) < \infty \\ \int_A^+ \mathcal{X}_{X,A} \cdot f_+^c d\mu &= \int_A^+ f_+^c d\mu = \nu_c^+(A) < \infty \\ \int_A^+ \mathcal{X}_{X,A} \cdot f_-^c d\mu &= \int_A^+ f_-^c d\mu = \nu_c^-(A) < \infty \end{aligned}$$

Using [theorem: 21.352] on the above gives

$$\forall A \in \mathcal{A} \text{ we have } \mathcal{X}_{X,A} \cdot f_+^r, \mathcal{X}_{X,A} \cdot f_-^r, \mathcal{X}_{X,A} \cdot f_+^c, \mathcal{X}_{X,A} \cdot f_-^c \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.920)$$

and taking $A = X$ in the above

$$f_+^r, f_-^r, f_+^c, f_-^c \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \quad (21.921)$$

further

$$\begin{aligned} \forall A \in \mathcal{A} \text{ we have } \int_A f_+^r d\mu &= \int \mathcal{X}_{X,A} \cdot f_+^r d\mu = \nu_r^+(A) \\ \int_A f_-^r d\mu &= \int \mathcal{X}_{X,A} \cdot f_-^r d\mu = \nu_r^-(A) \\ \int_A f_+^c d\mu &= \int \mathcal{X}_{X,A} \cdot f_+^c d\mu = \nu_c^+(A) \\ \int_A f_-^c d\mu &= \int \mathcal{X}_{X,A} \cdot f_-^c d\mu = \nu_c^-(A) \end{aligned} \quad (21.922)$$

Take $f = f_+^r - f_-^r + i \cdot (f_+^c - f_-^c)$ then by combining [theorem: 21.360] and [eqs: 21.920, 21.921] we have for $A \in \mathcal{A}$

$$\begin{aligned} \operatorname{Re}(f) &= f_+^r - f_-^r \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \\ \operatorname{Re}(\mathcal{X}_A \cdot f) &= \mathcal{X}_A \cdot f_+^r - \mathcal{X}_A \cdot f_-^r \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \\ \operatorname{Im}(f) &= f_+^c - f_-^c \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \\ \operatorname{Im}(\mathcal{X}_A \cdot f) &= \mathcal{X}_A \cdot f_+^c - \mathcal{X}_A \cdot f_-^c \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}] \end{aligned}$$

which by definition of the integral of complex functions proves that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] \wedge \mathcal{X}_A \cdot f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

and

$$\begin{aligned} \int_A f d\mu &= \int \mathcal{X}_A \cdot f d\mu \\ &= \int \operatorname{Re}(\mathcal{X}_A \cdot f) d\mu + i \cdot \int \operatorname{Im}(\mathcal{X}_A \cdot f) d\mu \\ &= \int (\mathcal{X}_A \cdot f_+^r - \mathcal{X}_A \cdot f_-^r) d\mu + i \cdot \int (\mathcal{X}_A \cdot f_+^c - \mathcal{X}_A \cdot f_-^c) d\mu \\ &= \int \mathcal{X}_A \cdot f_+^r d\mu - \int \mathcal{X}_A \cdot f_-^r d\mu + i \cdot \left(\int \mathcal{X}_A \cdot f_+^c d\mu - \int \mathcal{X}_A \cdot f_-^c d\mu \right) \\ &= \nu_r^+(A) - \nu_r^-(A) + i \cdot (\nu_c^+ - \nu_c^-) \\ &= \nu(A) \end{aligned}$$

Summarizing we have

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}] \text{ and } \forall A \in \mathcal{A} \text{ we have } \int f d\mu = \nu(A)$$

Finally if there exists another $g \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ such that $\forall A \in \mathcal{A}$ we have $\int_A g d\mu = \nu(A)$ then using [lemma: 21.531] we have that

$$f = g \text{ } \mu\text{-a.e.} \quad \square$$

For the Riemann Integral on a Banach space [see definition: 15.20] we have the Fundamental Theorem of Analysis [see theorem: 16.216]. This states that if a function $F: [a, b] \rightarrow X$ [where X is a Banach space] has a derivative F' on $[a, b]$ then $\int_a^b F' = F(b) - F(a)$. If we take the usual notation of $F' = \frac{dF}{dx}$ for the derivative and $\int_a^b f dx$ the Riemann integral this becomes

$$l([F(a), F(b)]) = F(b) - F(a) = \int_a^b \frac{dF}{dx} dx$$

where $l([F(a), F(b)])$ is the measure of the interval $[F(a), F(b)]$. Compare this with the Radon-Nikodym theorem which says that for measures ν, μ with $\nu \ll \mu$ there exist a integrable function h such that

$$\nu(A) = \int_A h d\mu$$

Hence it seems logical to consider f as a kind of derivative and write f as follows $f = \frac{d\nu}{d\mu}$ so that we have

$$\nu(A) = \int_A \frac{d\nu}{d\mu} d\mu$$

This motivates the definition of the Radon-Nikodym derivative.

Definition 21.533. Let $\langle X, \mathcal{A} \rangle$ be a measurable space, μ a σ -finite measure on $\mathcal{A}$ then we have the following definitions:

1. If ν is a σ -finite measure on $\mathcal{A}$ then a Radon-Nikodym derivative of ν with respect to μ is a $f \in \mathcal{M}^+[X, \mathcal{A}, \mathbb{R}]$ such that $\forall A \in \mathcal{A}$ we have

$$\nu(A) = \int_A^+ f d\mu$$

2. If ν is a finite signed measure on $\mathcal{A}$ then a Radon-Nikodym derivative of ν with respect to μ is a $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{R}]$ such that $\forall A \in \mathcal{A}$ we have

$$\nu(A) = \int_A f d\mu$$

3. If ν is a complex measure on $\mathcal{A}$ then a Radon-Nikodym derivative of ν with respect to μ is a $f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$ such that $\forall A \in \mathcal{A}$ we have

$$\nu(A) = \int_A^{\mathbb{C}} f d\mu$$

Note 21.534. Using [theorems: 21.527, 21.532] it follows that if $\nu \ll \mu$ then there exist a Radon-Nikodym derivative of ν with respect to μ .

Note 21.535. A Radon-Nikodym derivative of ν with respect to μ is noted by $\frac{d\nu}{d\mu}$. Be aware that this notation is misleading as it suggest that there exist only one Radon-Nikodym while [theorems: 21.527, 21.532] ensure existence but not uniqueness. The best way to think of this notation is that $f = \frac{d\nu}{d\mu}$ is the same as saying that f is a Radon-Nikodym derivative so that

$$\forall A \in \mathcal{A} \text{ we have } \nu(A) = \begin{cases} \int_A^+ f d\mu & \text{if } \nu \text{ is a } \sigma\text{-finite measure} \\ \int_A f d\mu & \text{if } \nu \text{ is a } \sigma\text{-finite measure} \\ \int_A^{\mathbb{C}} f d\mu & \text{if } \nu \text{ is a complex measure} \end{cases}$$

Using this interpretation we have that if $f = \frac{d\nu}{d\mu}$ and $g = \frac{d\nu}{d\mu}$ then using [lemmas: 21.525, 21.530 and 21.531] we have that $f = g$ μ -a.e.

21.7 Product measures

In this section we examine how given a finite family of measurable spaces we can define a σ -algebra and measure on the product of these spaces. The prime objective is to show how a integral on the product of these spaces can be calculated based on the integrals on the measurable spaces.

21.7.1 Product of σ -algebras

Given a finite family $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in \{1, \dots, n\}}$ of measurable spaces we show how a σ -algebra can be defined on $\prod_{i \in \{1, \dots, n\}} X_i$. First we show how given a family $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in I}$ of measurable spaces, a set X and a family of mappings $\{f_i: X \rightarrow X_i\}_{i \in I}$ we can define on X a σ -algebra such that these mappings are measurable.

Definition 21.536. Let X, Y be sets, $f: X \rightarrow Y$ a function and $\mathcal{A} \subseteq \mathcal{P}(Y)$ then

$$f^{-1}[\mathcal{A}] = \{f^{-1}(A) | A \in \mathcal{A}\} \subseteq \mathcal{P}(X)$$

In other words $f^{-1}[\mathcal{A}]$ is the set of pre-images by f of sets in $\mathcal{A}$

Definition 21.537. Let I be a set, $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in I}$ a family of measurable sets, X a set and $\{f_i: X \rightarrow X_i\}_{i \in I}$ a family of functions then we define the σ -algebra $\sigma[f_i | i \in I]$ on X by

$$\sigma[f_i | i \in I] = \sigma \left[\bigcup_{i \in I} f_i^{-1}[\mathcal{A}_i] \right]$$

The benefit of the above σ -algebra is that it is the smallest σ -algebra on X such that every f_i is measurable.

Theorem 21.538. Let I be a set, $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in I}$ a family of measurable sets, X a non empty set and $\{f_i: X \rightarrow X_i\}_{i \in I}$ a family of functions then we have for the σ -algebra $\sigma[f_i | i \in I]$ on X that

$$\forall i \in I \ f_i: X \rightarrow X_i \text{ is } \sigma[f_i | i \in I], \mathcal{A}_i\text{-measurable}$$

Further if $\mathcal{A}$ is another σ -algebra on X such that $\forall i \in I \ f_i: X \rightarrow X_i$ is $\mathcal{A}, \mathcal{A}_i$ -measurable then

$$\sigma[f_i | i \in I] \subseteq \mathcal{A}$$

Proof. Let $i \in I$ then $\forall A \in \mathcal{A}_i$ we have that

$$f_i^{-1}(A) \in \{f^{-1}(A) | A \in \mathcal{A}_i\} = f^{-1}[\mathcal{A}_i] \subseteq \bigcup_{i \in I} f_i^{-1}[\mathcal{A}_i] \subseteq \sigma \left[\bigcup_{i \in I} f_i^{-1}[\mathcal{A}_i] \right] \stackrel{\text{def}}{=} \sigma[f_i | i \in I]$$

which proves that $f_i = \sigma[f_i | i \in I], \mathcal{A}_i$ -measurable. Hence

$$\forall i \in I \ f_i: X \rightarrow X_i \text{ is } \sigma[f_i | i \in I], \mathcal{A}_i\text{-measurable}$$

Let $\mathcal{A}$ be another σ -algebra on X such that $\forall i \in I \ f_i: X \rightarrow X_i$ is $\mathcal{A}, \mathcal{A}_i$ -measurable. Let $A \in \bigcup_{i \in I} f_i^{-1}[\mathcal{A}_i]$ then there exist a $i \in I$ such that $A \in f_i^{-1}[\mathcal{A}_i] = \{f_i^{-1}(A) | A \in \mathcal{A}_i\}$ then $\exists A' \in \mathcal{A}_i$ such that $A = f_i^{-1}(A')$. As f_i is $\mathcal{A}, \mathcal{A}_i$ -measurable it follows that $f_i^{-1}(A') \in \mathcal{A}$ so that $A \in \mathcal{A}$. Hence we have that $\bigcup_{i \in I} f_i^{-1}[\mathcal{A}_i] \subseteq \mathcal{A}$ so that

$$\sigma[f_i | i \in I] = \sigma \left[\bigcup_{i \in I} f_i^{-1}[\mathcal{A}_i] \right] \subseteq \mathcal{A} \quad \square$$

We are now ready to define the product of σ -algebras.

Definition 21.539. Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of measurable spaces then the product σ -algebra on $\prod_{i \in \{1, \dots, n\}} X_i$ noted as $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$ is defined by

$$\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n = \sigma[\pi_i; i \in \{1, \dots, n\}]$$

where $\forall i \in \{1, \dots, n\}$ $\pi_i: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow X_i$ is the i -the projection map defined by $\pi_i(x_1, \dots, x_n) = x_i$ [see definition: 2.156]. The measurable space

$$\left\langle \prod_{i \in \{1, \dots, n\}} X_i, \mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n \right\rangle$$

is called the **product** of the measurable spaces $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in \{1, \dots, n\}}$.

Convention 21.540. If $\forall i \in \{1, \dots, n\}$ $\mathcal{A}_i = \mathcal{A}$ then

$$\underbrace{\mathcal{A} \otimes \dots \otimes \mathcal{A}}_n \underset{\text{notation}}{=} \mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$$

A immediate consequence of [theorem: 21.538] it the following:

Theorem 21.541. Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of measurable spaces then

$$\forall i \in \{1, \dots, n\} \quad \pi_i: \prod_{i \in \{1, \dots, n\}} X_i \rightarrow X_i \text{ is } \mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n, \mathcal{A}_i\text{-measurable.}$$

Further if $\mathcal{A}$ is another σ -algebra on $\prod_{i \in \{1, \dots, n\}} X_i$ such that $\forall i \in \{1, \dots, n\}$ π_i is $\mathcal{A}, \mathcal{A}_i$ -measurable then

$$\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n \subseteq \mathcal{A}$$

Proof. This follows directly from [definition: 21.539] and [theorem: 21.538]. $\square$

Theorem 21.542. Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of measurable spaces such that $\forall i \in \{1, \dots, n\}$ $\mathcal{A}_i = \sigma[\mathcal{E}_i]$ and there exist a $\{E_{i,k}\}_{k \in \mathbb{N}} \subseteq \mathcal{E}_i$ such that $\forall k \in \mathbb{N}$ $E_{i,k} \subseteq E_{i,k+1}$ and $X_i = \bigcup_{k \in \mathbb{N}} E_{i,k}$ then we have that

$$\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n = \sigma \left[\left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \quad E_i \in \mathcal{E}_i \right\} \right]$$

Proof. First we prove that for any σ -algebra on $\prod_{i \in \{1, \dots, n\}} X_i$ we have that

$$\forall i \in \{1, \dots, n\} \quad \pi_i \text{ is } \mathcal{A}, \mathcal{A}_i\text{-measurable} \Leftrightarrow \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \quad E_i \in \mathcal{E}_i \right\} \subseteq \mathcal{A} \quad (21.923)$$

Proof. We have

$\Rightarrow$. Let $E \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \quad E_i \in \mathcal{E}_i \right\}$ then $E = \prod_{i \in \{1, \dots, n\}} E_i$ where $\forall i \in \{1, \dots, n\}$ $E_i \in \mathcal{E}_i$. Now

$$\begin{aligned} (x_1, \dots, x_n) \in E &= \prod_{i \in \{1, \dots, n\}} E_i && \Leftrightarrow && \forall i \in \{1, \dots, n\} \quad x_i \in E_i \\ &&& \Leftrightarrow && \forall i \in \{1, \dots, n\} \quad \pi_i(x_1, \dots, x_n) \in E_i \\ &&& \Leftrightarrow && \forall i \in \{1, \dots, n\} \quad (x_1, \dots, x_n) \in \pi_i^{-1}(E_i) \\ &&& \Leftrightarrow && (x_1, \dots, x_n) \in \bigcap_{i \in \{1, \dots, n\}} \pi_i^{-1}(E_i) \end{aligned}$$

which proves that

$$E = \bigcap_{i \in \{1, \dots, n\}} \pi_i^{-1}(E_i)$$

As $\forall i \in \{1, \dots, n\}$ $E_i \in \mathcal{E}_i \subseteq \mathcal{A}_i$ and π_i is $\mathcal{A}, \mathcal{A}_i$ -measurable we have that $\pi_i^{-1}(E_i) \in \mathcal{A}$. So it follows from the above and [theorems: 21.36] that $E \in \mathcal{A}$. Hence we have proved that

$$\left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{A}_i \right\} \subseteq \mathcal{A}$$

$\Leftarrow$. Fix $j \in \{1, \dots, n\}$ and $E \in \mathcal{A}_j$. Define

$$\{F_k\}_{k \in \mathbb{N}} \text{ by } F_k = \prod_{i \in \{1, \dots, n\}} F_{k,i} \text{ where } F_{k,i} = \begin{cases} E \in \mathcal{E}_j & \text{if } i = j \\ E_{i,k} \in \mathcal{E}_i & \text{if } i \in \{1, \dots, n\} \setminus \{j\} \end{cases}$$

Then we have as $\forall k \in \mathbb{N}$ $F_k = \prod_{i \in \{1, \dots, n\}} F_{k,i}$ and $\forall i \in \{1, \dots, n\}$ $F_{k,i} \in \mathcal{E}_i$ that $F_k \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{E}_i \right\} \subseteq \mathcal{A}$, hence

$$\{F_k\}_{k \in \mathbb{N}} \subseteq \mathcal{A} \quad (21.924)$$

Further we have $\forall k \in \mathbb{N}$ that $\forall i \in \{1, \dots, n\}$ $F_{k,i} = \begin{cases} E = F_{k+1,i} & \text{if } i = j \\ E_{i,k} \subseteq E_{i,k+1} & \text{if } i \in \{1, \dots, n\} \setminus \{j\} \end{cases} \subseteq F_{k+1,i}$ so that by [theorem: 2.149] $F_k = \prod_{i \in \{1, \dots, n\}} F_{k,i} \subseteq \prod_{i \in \{1, \dots, n\}} F_{k+1,i}$. To summarize

$$\forall k \in \mathbb{N} \quad F_k \subseteq F_{k+1} \quad (21.925)$$

Define

$$G = \prod_{i \in \{1, \dots, n\}} G_i \text{ where } G_i = \begin{cases} E & \text{if } i = j \\ X_i & \text{if } i \in \{1, \dots, n\} \setminus \{j\} \end{cases}$$

Let $k \in \mathbb{N}$ then $\forall i \in \{1, \dots, n\}$ $F_{k,i} = \begin{cases} E = G_i & \text{if } i = j \\ E_{i,k} \subseteq X_i = G_i & \text{if } i \in \{1, \dots, n\} \setminus \{j\} \end{cases} \subseteq G_i$ so that by [theorem: 2.149] we have $F_k \subseteq G$ from which it follows that

$$\bigcup_{k \in \mathbb{N}} F_k \subseteq G \quad (21.926)$$

Let $(x_1, \dots, x_n) \in G$. Take $i \in \{1, \dots, n\}$ then we have either:

$i = j$. Then $x_i \in G_i = E = F_{1,i}$ so if we take $k_i = 1$ we have $x_i \in F_{k_i,i}$

$i \in \{1, \dots, n\} \setminus \{j\}$. Then as $x_i \in G_i = X_i = \bigcup_{k \in \mathbb{N}} E_{i,k}$ there exist a $k_i \in \mathbb{N}$ such that $x_i \in E_{i,k_i} = F_{k_i,i}$

Hence if we take $k = \max(\{k_i \mid i \in \{1, \dots, n\}\})$ then $\forall i \in \{1, \dots, n\}$ we have $x_i \in F_{k_i,i} \subseteq F_{k,i}$ so that $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} F_{k,i} = F_k \subseteq \bigcup_{k \in \mathbb{N}} F_k$. Hence $G \subseteq \bigcup_{k \in \mathbb{N}} F_k$ which combined with [eq: 21.926] proves that

$$G = \bigcup_{k \in \mathbb{N}} F_k$$

As by [eq: 21.924] $\{F_k\}_{k \in \mathbb{N}} \subseteq \mathcal{A}$ it follows from the above that

$$G \in \mathcal{A}$$

Now if $(x_1, \dots, x_n) \in G$ then $\pi_j(x_1, \dots, x_n) = x_j \in G_j = E$ so that $(x_1, \dots, x_n) \in \pi_j^{-1}(E)$. Likewise if $(x_1, \dots, x_n) \in \pi_j^{-1}(E) \subseteq \prod_{i \in \{1, \dots, n\}} X_i$ then $\forall i \in \{1, \dots, n\} \setminus \{j\}$ we have $x_i \in X_i = G_i$ and for $i = j$ $x_i \in E = G_i$ so that $(x_1, \dots, x_n) \in G$. Hence we must conclude that $\pi_j^{-1}(E) = G \in \mathcal{A}$ so that

$$\forall j \in \{1, \dots, n\} \text{ we have } \forall E \in \mathcal{A}_j \text{ that } \pi_j^{-1}(E) \in \mathcal{A}$$

In other words

$$\forall j \in \{1, \dots, n\} \quad \pi_j \text{ is } \mathcal{A}, \mathcal{A}_j\text{-measurable} \quad \square$$

As $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$ is a σ -algebra on $\prod_{i \in \{1, \dots, n\}} X_i$ and by [theorem: 21.541] $\forall i \in \{1, \dots, n\}$ π_i is $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n, \mathcal{A}_i$ -measurable it follow from [eq: 21.923] that

$$\left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{E}_i \right\} \subseteq \mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$$

hence

$$\sigma\left[\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{E}_i\right\}\right] \subseteq \mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n \quad (21.927)$$

Take $\mathcal{B} = \sigma\left[\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{E}_i\right\}\right]$ then as

$$\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{E}_i\right\} \subseteq \mathcal{B}$$

and $\mathcal{B}$ is a σ -algebra it follows from [eq: 21.923] that $\forall i \in \{1, \dots, n\}$ π_i is $\mathcal{B}$, $\mathcal{A}_i$ -measurable. Hence using [theorem: 21.538] and [definition: 21.539] it follows that $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n \subseteq \mathcal{B}$ or

$$\sigma\left[\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{E}_i\right\}\right] \subseteq \mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$$

which combined with [eq: 21.927] proves that

$$\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n = \sigma\left[\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{E}_i\right\}\right] \quad \square$$

Corollary 21.543. Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{A}_i \rangle\}_{i \in \{1, \dots, n\}}$ be a finite family of measurable spaces then we have that

$$\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n = \sigma\left[\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{A}_i\right\}\right]$$

Proof. Let $i \in \{1, \dots, n\}$ then by [theorem: 21.43] we have that $\mathcal{A}_i = \sigma[\mathcal{A}_i]$. Further if we define $\{E_{i,k}\}_{k \in \mathbb{N}} \subseteq \mathcal{A}_i$ by $E_{i,k} = X_i \in \mathcal{A}_i$ then we have $\forall k \in \mathbb{N}$ we have $E_{i,k} \subseteq E_{i,k+1}$ and $X_i = \bigcup_{k \in \mathbb{N}} E_{i,k}$. So all the conditions of the previous theorem [theorem: 21.542] are satisfied which proves

$$\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n = \sigma\left[\left\{\prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} \ E_i \in \mathcal{A}_i\right\}\right] \quad \square$$

A application of [theorem: 21.542] is that we can write the Borel algebra $\mathcal{B}[\mathbb{R}^n]$ on $\mathbb{R}^n$ as a product of the Borel algebra's $\mathcal{B}[\mathbb{R}]$.

Theorem 21.544. Let $n \in \mathbb{N}$ then

$$\mathcal{B}[\mathbb{R}^n] = \underbrace{\mathcal{B}[\mathbb{R}] \otimes \dots \otimes \mathcal{B}[\mathbb{R}]}_n$$

Proof. Define $\{X_i, \mathcal{A}_i\}_{i \in \{1, \dots, n\}}$ by $X_i = \mathbb{R}$ and $\mathcal{A}_i = \mathcal{B}[\mathbb{R}]$ then

$$\mathbb{R}^n \stackrel{\text{definition}}{=} \prod_{i \in \{1, \dots, n\}} X_i$$

Further using [theorem: 21.55] we have that

$$\forall i \in \{1, \dots, n\} \ \mathcal{A}_i = \mathcal{B}[\mathbb{R}] = \sigma[\mathcal{I}_{[]}]$$

Further for $\{[-k, k]\}_{k \in \mathbb{N}} \subseteq \mathcal{I}_{[]}$ we have trivially

$$\forall n \in \mathbb{N} \ [-n, n[\subseteq [-(n+1), n+1[$$

Further if $x \in \mathbb{R}$ then by [theorem: 10.30] there exist a $k \in \mathbb{N}$ such that $|x| < k$ so that $-x < k \wedge x < k$ hence $-k < x < k \Rightarrow x \in]-k, k[\subseteq [-k, k[\subseteq \bigcup_{k \in \mathbb{N}} [-n, n[$. Hence $\mathbb{R} \subseteq \bigcup_{k \in \mathbb{N}} [-n, n[$ which as $\bigcup_{k \in \mathbb{N}} [-k, k[\subseteq \mathbb{R}$ proves that

$$\mathbb{R} = \bigcup_{k \in \mathbb{N}} [-k, k[$$

So all the conditions for [theorem: 21.542] are satisfied and we have

$$\underbrace{\mathcal{B}[\mathbb{R}] \otimes \cdots \otimes \mathcal{B}[\mathbb{R}]}_n = \sigma \left[\left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\} \right] \quad (21.928)$$

Now if $E \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\}$ then $E = \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$ where $a_i, b_i \in \mathbb{R}$ and $a_i < b_i$ [see definition: 21.51] so that by [definitions: 21.67, 21.72] $E \subseteq \mathcal{R}_{[[}^n$. Hence

$$\left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\} \subseteq \mathcal{R}_{[[}^n \quad (21.929)$$

For the opposite inclusion let $E \in \mathcal{R}_{[[}^n$ then by [definitions: 21.67, 21.72] $E = \prod_{i \in \{1, \dots, n\}} [a_i, b_i[$ where $a_i, b_i \in \mathbb{R}$ and $a_i < b_i$ so that by [see definition: 21.51] we have that $E \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\}$. Hence $\mathcal{R}_{[[}^n \subseteq \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\}$ which combined with [eq: 21.929] proves that

$$\mathcal{R}_{[[}^n = \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\}$$

Now

$$\mathcal{B}[\mathbb{R}^n] \underset{[\text{theorem: 21.194}]}{=} \sigma[\mathcal{R}_{[[}^n] = \sigma \left[\left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{I}_{[[} \right\} \right]$$

and combining this with [eq: 21.928] proves finally that

$$\mathcal{B}[\mathbb{R}^n] = \underbrace{\mathcal{B}[\mathbb{R}] \otimes \cdots \otimes \mathcal{B}[\mathbb{R}]}_n \quad \square$$

21.7.2 Product measures

Theorem 21.545. Let $n \in \{2, \dots, \infty\}$, $\{\langle X_i, \mathcal{A}_i, \mu_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of measure spaces such that $\forall i \in \{1, \dots, n\}$ we have that $\exists \mathcal{E}_i \subseteq \mathcal{P}(X)$ such that

1. $\mathcal{A}_i = \sigma[\mathcal{E}_i]$
2. $\forall A, B \in \mathcal{E}_i$ we have $A \cap B \in \mathcal{E}_i$ [in other words $\mathcal{E}_i$ is $\cap$ -stable]
3. $\exists \{E_{i,k}\}_{k \in \mathbb{N}} \subseteq \mathcal{E}_i$ such that $\forall k \in \mathbb{N}$ we have $E_{i,k} \subseteq E_{i,k+1}$, $\mu_i(E_{i,k}) < \infty$ and $X_i = \bigcup_{k \in \mathbb{N}} E_{i,k}$

then if μ, ν are two measures on $\mathcal{A}_1 \otimes \cdots \otimes \mathcal{A}_n$ such that

$$\forall \prod_{i \in \{1, \dots, n\}} E_i \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{E}_i \right\}$$

we have

$$\mu \left(\prod_{i \in \{1, \dots, n\}} E_i \right) = \prod_{i \in \{1, \dots, n\}} \mu_i(E_i) = \nu \left(\prod_{i \in \{1, \dots, n\}} E_i \right) \quad (21.930)$$

then

$$\mu = \nu$$

Proof. Let $\mathcal{E} = \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{E}_i \right\}$ then by [theorem: 21.542] we have that

$$\mathcal{A}_1 \otimes \cdots \otimes \mathcal{A}_n = \sigma[\mathcal{E}]$$

Let $E, F \in \mathcal{E}$ then $E = \prod_{i \in \{1, \dots, n\}} E_i$ and $F = \prod_{i \in \{1, \dots, n\}} F_i$ where $\forall i \in \{1, \dots, n\} E_i, F_i \in \mathcal{E}_i$ so that

$$E \cap F = \left(\prod_{i \in \{1, \dots, n\}} E_i \right) \cap \left(\prod_{i \in \{1, \dots, n\}} F_i \right) \underset{[\text{theorem: 2.150}]}{=} \prod_{i \in \{1, \dots, n\}} (E_i \cap F_i)$$

As $\forall i \in \{1, \dots, n\}$ $\mathcal{E}_i$ is $\cap$ -stable $E_i \cap F_i \in \mathcal{E}_i$ it follows that $\prod_{i \in \{1, \dots, n\}} (E_i \cap F_i) \in \mathcal{E}$ hence $E \cap F \in \mathcal{E}$. In other word

$$\mathcal{E} \text{ is } \cap\text{-stable} \quad (21.931)$$

Further if we define

$$\{F_k\}_{k \in \mathbb{N}} \subseteq \mathcal{E} \text{ by } F_k = \prod_{i \in \{1, \dots, n\}} E_{i,k} \in \mathcal{E} \quad (21.932)$$

then we have by [theorem: 2.149] and the fact that $\forall i \in \{1, \dots, n\} \forall k \in \mathbb{N} E_{i,k} \subseteq E_{i,k+1}$ we have that

$$\forall k \in \mathbb{N} F_k \subseteq F_{k+1} \quad (21.933)$$

Let $(x_1, \dots, x_n) \in \prod_{i \in \{1, \dots, n\}} X_i$ then $\forall i \in \{1, \dots, n\}$ we have that $x_i \in X_i = \bigcup_{k \in \mathbb{N}} E_{i,k} \Rightarrow \exists k_i \in \mathbb{N}$ such that $x_i \in E_{i,k_i}$. Take $m = \max(\{k_i | i \in \{1, \dots, n\}\})$ then we have $\forall i \in \{1, \dots, n\} x_i \in E_{i,k_i} \subseteq E_{i,m}$ so that $(x_1, \dots, x_n) \in F_m \subseteq \bigcup_{k \in \mathbb{N}} F_k$. Hence $\prod_{i \in \{1, \dots, n\}} X_i \subseteq \bigcup_{k \in \mathbb{N}} F_k$. Further $\forall k \in \mathbb{N}$ we have that $F_k = \prod_{i \in \{1, \dots, n\}} E_{i,k} \subseteq \prod_{i \in \{1, \dots, n\}} X_i$ so that

$$\prod_{i \in \{1, \dots, n\}} X_i = \bigcup_{k \in \mathbb{N}} F_k \quad (21.934)$$

Further we have

$$\begin{aligned} \mu(F_k) &\stackrel{[\text{eq: 21.932}]}{=} \mu\left(\prod_{i \in \{1, \dots, n\}} E_{i,k}\right) \\ &\stackrel{[\text{eq: 21.930}]}{=} \prod_{i \in \{1, \dots, n\}} \mu_i(E_{i,k}) \\ &\stackrel{[\text{eq: 21.930}]}{=} \nu\left(\prod_{i \in \{1, \dots, n\}} E_{i,k}\right) \\ &\stackrel{[\text{eq: 21.932}]}{=} \nu(F_k) \end{aligned}$$

in other words

$$\forall k \in \mathbb{N} \mu(F_k) = \prod_{i \in \{1, \dots, n\}} \mu_i(E_{i,k}) = \nu(F_k) \quad (21.935)$$

As $\forall i \in \{1, \dots, n\}$ we have $\forall k \in \mathbb{N}$ that $\mu_i(E_{i,k})$ is finite we have from the above that

$$\forall k \in \mathbb{N} \mu(F_k) < \infty \quad (21.936)$$

The conditions for [theorem: 21.131] are fulfilled by [eqs: 21.931, 21.932, 21.933, 21.934, 21.935 and 21.936], so using this theorem it follows that

$$\mu = \nu \quad \square$$

Corollary 21.546. Let $n \in \mathbb{N}$, $\{\langle X_i, \mathcal{A}_i, \mu_i \rangle\}_{i \in \{1, \dots, n\}}$ a finite family of measure spaces such that $\forall i \in \{1, \dots, n\}$ μ_i is σ -finite and μ, ν are two measures on $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$ such that

$$\forall \prod_{i \in \{1, \dots, n\}} E_i \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{A}_i \right\}$$

we have

$$\mu\left(\prod_{i \in \{1, \dots, n\}} E_i\right) = \prod_{i \in \{1, \dots, n\}} \mu_i(E_i) = \nu\left(\prod_{i \in \{1, \dots, n\}} E_i\right)$$

Proof. For $i \in \{1, \dots, n\}$ take $\varepsilon_i = \mathcal{A}_i$ then

$$\sigma[\varepsilon_i] = \sigma[\mathcal{A}_i] = \mathcal{A}_i$$

and trivially we have

$$\forall A, B \in \mathcal{E}_i = \mathcal{A}_i \text{ we have } A \cap B \in \mathcal{A}_i = \mathcal{E}_i \text{ [in other words } \mathcal{E}_i \text{ is } \cap\text{-stable}]$$

As $\forall i \in \{1, \dots, n\}$ μ_i is σ -finite [theorem: 21.105] we have that $\forall i \in \{1, \dots, n\}$ there exist a

$$\{E_{i,k}\}_{k \in \mathbb{N}} \subseteq \mathcal{A}_i = \mathcal{E}_i \text{ with } \forall k \in \mathbb{N} \text{ we have } E_{i,k} \subseteq E_{i,k+1} \wedge \mu_i(E_{i,k}) < \infty \text{ and } X_i = \bigcup_{k \in \mathbb{N}} E_{i,k}$$

By the hypothesis we have

$$\begin{aligned} & \forall \prod_{i \in \{1, \dots, n\}} E_i \in \left\{ \prod_{i \in \{1, \dots, n\}} E_i \mid \forall i \in \{1, \dots, n\} E_i \in \mathcal{A}_i = \mathcal{E}_i \right\} \\ \text{that} \quad & \mu \left(\prod_{i \in \{1, \dots, n\}} E_i \right) = \prod_{i \in \{1, \dots, n\}} \mu_i(E_i) = \nu \left(\prod_{i \in \{1, \dots, n\}} E_i \right) \end{aligned}$$

Hence all the conditions for [theorem: 21.545] are satisfied so that we have

$$\mu = \nu \quad \square$$

This theorem does not prove existence of a measure on $\mathcal{A}_1 \otimes \dots \otimes \mathcal{A}_n$ satisfying [eq: 21.930] only that if such measure exist then it is unique. The remaining part of this section is to prove that such a measure exist. We first prove existence for $n=2$. First we introduce the idea of a section of a set.

Definition 21.547. Let $\{X_i\}_{i \in \{1,2\}}$ be a family of two sets let $A \subseteq X_1 \cdot X_2$ then we define

$$\forall x \in X_1 \quad A_{[1,x]} = \{y \in X_2 \mid (x, y) \in A\} \subseteq X_2$$

$$\forall y \in X_2 \quad A_{[2,y]} = \{x \in X_1 \mid (x, y) \in A\} \subseteq X_1$$

We call $A_{[1,x]}$ the 1, x -section of A and $A_{[2,y]}$ the 2, y -section of A

Note 21.548. $X_1 \cdot X_2$ is a short notation for $\prod_{i \in \{1,2\}} X_i$

Lemma 21.549. Let $\{X_i\}_{i \in \{1,2\}}$ be a family of two sets then we have:

1. $\forall x \in X_1 \quad (X_1 \cdot X_2)_{[1,x]} = X_2$
2. $\forall y \in X_2 \quad (X_1 \cdot X_2)_{[2,y]} = X_1$
3. $\forall x \in X_1$ that $\emptyset_{[1,x]} = \emptyset$ and $\forall y \in X_2$ we have $\emptyset_{[2,y]} = \emptyset$
4. $\forall x \in X_1, \forall A, B \subseteq X_1 \cdot X_2$ we have $(A \setminus B)_{[1,x]} = A_{[1,x]} \setminus B_{[1,x]}$
5. $\forall y \in X_2, \forall A, B \subseteq X_1 \cdot X_2$ we have $(A \setminus B)_{[2,y]} = A_{[2,y]} \setminus B_{[2,y]}$
6. If $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X_1 \cdot X_2)$, $x \in X_1$ then $(\bigcup_{i \in I} A_i)_{[1,x]} = \bigcup_{i \in I} (A_i)_{[1,x]}$
7. If $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X_1 \cdot X_2)$, $y \in X_2$ then $(\bigcup_{i \in I} A_i)_{[2,y]} = \bigcup_{i \in I} (A_i)_{[2,y]}$
8. Let $A_1 \subseteq X_1, A_2 \subseteq X_2$ then we have

$$a. \forall x \in X_1 \text{ we have } (A_1 \times A_2)_{[1,x]} = \begin{cases} A_2 & \text{if } x \in A_1 \\ \emptyset & \text{if } x \in X_1 \setminus A_1 \end{cases}$$

$$b. \forall y \in X_2 \text{ we have } (A_1 \times A_2)_{[2,y]} = \begin{cases} A_1 & \text{if } y \in A_2 \\ \emptyset & \text{if } y \in X_2 \setminus A_2 \end{cases}$$

9. Let $A, B \subseteq X_1 \cdot X_2$ and $x \in X_1$ then $(A \cap B)_{[1,x]} = A_{[1,x]} \cap B_{[1,x]}$
10. Let $A, B \subseteq X_1 \cdot X_2$ and $y \in X_2$ then $(A \cap B)_{[2,y]} = A_{[2,y]} \cap B_{[2,y]}$

Proof.

1. Let $x \in X_1$. If $y \in X_2$ then $(x, y) \in X_1 \cdot X_2$ so that $y \in \{y \in X_2 \mid (x, y) \in X_1 \cdot X_2\} = (X_1 \cdot X_2)_{[1,x]}$. Hence $X_2 \subseteq (X_1 \cdot X_2)_{[1,x]}$ and as by definition $(X_1 \cdot X_2)_{[1,x]} \subseteq X_2$ it follows that

$$(X_1 \cdot X_2)_{[1,x]} = X_2$$

2. Let $y \in X_2$. If $x \in X_1$ then $(x, y) \in X_1 \cdot X_2$ so that $x \in \{x \in X_1 \mid (x, y) \in X_1 \cdot X_2\} = (X_1 \cdot X_2)_{[2,y]}$. Hence $X_1 \subseteq (X_1 \cdot X_2)_{[2,y]}$ and as by definition $(X_1 \cdot X_2)_{[2,y]} \subseteq X_1$ it follows that

$$(X_1 \cdot X_2)_{[2,y]} = X_1$$

3. Let $x \in X_1$ then $\emptyset_{[1,x]} = \{y \in X_2 | (x, y) \in \emptyset\} = \emptyset$. Let $y \in X_2$ then $\emptyset_{[2,y]} = \{x \in X_1 | (x, y) \in \emptyset\} = \emptyset$.
4. Let $x \in X_1$ and $A, B \subseteq X_1 \cdot X_2$. Then we have

$$\begin{aligned}
 y \in (A \setminus B)_{[1,x]} &\Leftrightarrow y \in \{y \in X_2 | (x, y) \in A \setminus B\} \\
 &\Leftrightarrow y \in X_2 \wedge (x, y) \in A \setminus B \\
 &\Leftrightarrow y \in X_2 \wedge (x, y) \in A \wedge (x, y) \notin B \\
 &\Leftrightarrow y \in \{y \in X_2 | (x, y) \in A\} \wedge y \notin \{y \in X_2 | (x, y) \in B\} \\
 &\Leftrightarrow y \in A_{[1,x]} \wedge y \notin B_{[1,x]} \\
 &\Leftrightarrow y \in A_{[1,x]} \setminus B_{[1,x]}
 \end{aligned}$$

so that

$$(A \setminus B)_{[1,x]} = A_{[1,x]} \setminus B_{[1,x]}$$

5. Let $y \in X_2$ and $B \subseteq X_1 \cdot X_2$. Then we have

$$\begin{aligned}
 x \in (A \setminus B)_{[2,y]} &\Leftrightarrow x \in \{x \in X_1 | (x, y) \in A \setminus B\} \\
 &\Leftrightarrow x \in X_1 \wedge (x, y) \in A \setminus B \\
 &\Leftrightarrow x \in X_1 \wedge (x, y) \in A \wedge (x, y) \notin B \\
 &\Leftrightarrow x \in \{x \in X_1 | (x, y) \in A\} \wedge x \notin \{x \in X_1 | (x, y) \in B\} \\
 &\Leftrightarrow x \in A_{[2,y]} \wedge x \notin B_{[2,y]} \\
 &\Leftrightarrow x \in A_{[2,y]} \setminus B_{[2,y]}
 \end{aligned}$$

so that

$$(A \setminus B)_{[2,y]} = A_{[2,y]} \setminus B_{[2,y]}$$

6. Let $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X_1 \cdot X_2)$ and $x \in X_1$ then we have

$$\begin{aligned}
 y \in \left(\bigcup_{i \in I} A_i \right)_{[1,x]} &\Leftrightarrow y \in X_2 \wedge (x, y) \in \bigcup_{i \in I} A_i \\
 &\Leftrightarrow y \in X_2 \wedge \exists i \in I \text{ such that } (x, y) \in A_i \\
 &\Leftrightarrow \exists i \in I \text{ such that } y \in \{y \in X_2 | (x, y) \in A_i\} \\
 &\Leftrightarrow \exists i \in I \text{ such that } y \in (A_i)_{[1,x]} \\
 &\Leftrightarrow y \in \bigcup_{i \in I} (A_i)_{[1,x]}
 \end{aligned}$$

so that

$$\left(\bigcup_{i \in I} A_i \right)_{[1,x]} = \bigcup_{i \in I} (A_i)_{[1,x]}$$

7. Let $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X_1 \cdot X_2)$ and $y \in X_2$ then we have

$$\begin{aligned}
 x \in \left(\bigcup_{i \in I} A_i \right)_{[2,y]} &\Leftrightarrow x \in X_1 \wedge (x, y) \in \bigcup_{i \in I} A_i \\
 &\Leftrightarrow x \in X_1 \wedge \exists i \in I \text{ such that } (x, y) \in A_i \\
 &\Leftrightarrow \exists i \in I \text{ such that } x \in \{x \in X_1 | (x, y) \in A_i\} \\
 &\Leftrightarrow \exists i \in I \text{ such that } x \in (A_i)_{[2,y]} \\
 &\Leftrightarrow x \in \bigcup_{i \in I} (A_i)_{[2,y]}
 \end{aligned}$$

so that

$$\left(\bigcup_{i \in I} A_i \right)_{[2,y]} = \bigcup_{i \in I} (A_i)_{[2,y]}$$

8. Let $A_1 \subseteq X_1$, $A_2 \subseteq X_2$ then we have:

a. Let $x \in X_1$ then we have either:

$x \in A_1$. Then

$$\begin{aligned} y \in (A_1 \cdot A_2)_{[1,x]} &\Leftrightarrow y \in X_2 \wedge (x, y) \in A_1 \cdot A_2 \\ &\Leftrightarrow_{x \in A_1 \wedge A_2 \subseteq X_2} y \in A_2 \end{aligned}$$

which proves that

$$(A_1 \cdot A_2)_{[1,x]} = A_2$$

$x \in X_1 \setminus A_1$. Then if $y \in (A_1 \cdot A_2)_{[1,x]}$ we have that $(x, y) \in A_1 \cdot A_2 \Rightarrow x \in A_1$ which contradicts with $x \in X_1 \setminus A_1$. Hence we must have that

$$(A_1 \cdot A_2)_{[1,x]} = \emptyset$$

b. Let $y \in X_2$ then we have either:

$y \in A_2$. Then

$$\begin{aligned} x \in (A_1 \cdot A_2)_{[2,y]} &\Leftrightarrow x \in X_1 \wedge (x, y) \in A_1 \cdot A_2 \\ &\Leftrightarrow_{y \in A_2 \wedge A_1 \subseteq X_1} x \in A_1 \end{aligned}$$

which proves that

$$(A_1 \cdot A_2)_{[2,y]} = A_1$$

$y \in X_2 \setminus A_2$. Then if $x \in (A_1 \cdot A_2)_{[2,y]}$ we have that $(x, y) \in A_1 \cdot A_2 \Rightarrow y \in A_2$ which contradicts with $y \in X_2 \setminus A_2$. Hence we must have that

$$(A_1 \cdot A_2)_{[2,y]} = \emptyset$$

9. Let $x \in X_1$, $A, B \subseteq X_1 \cdot X_2$ then we have

$$\begin{aligned} y \in (A \cap B)_{[1,x]} &\Leftrightarrow y \in \{y \in X_2 \mid (x, y) \in A \cap B\} \\ &\Leftrightarrow y \in X_2 \wedge (x, y) \in A \cap B \\ &\Leftrightarrow y \in X_2 \wedge (x, y) \in A \wedge (x, y) \in B \\ &\Leftrightarrow y \in \{y \in X_2 \mid (x, y) \in A\} \wedge y \in \{y \in X_2 \mid (x, y) \in B\} \\ &\Leftrightarrow y \in A_{[1,x]} \wedge y \in B_{[1,x]} \\ &\Leftrightarrow y \in A_{[1,x]} \cap B_{[1,x]} \end{aligned}$$

which proves that

$$(A \cap B)_{[1,x]} = A_{[1,x]} \cap B_{[1,x]}$$

10. Let $y \in X_2$, $A, B \subseteq X_1 \cdot X_2$ then we have

$$\begin{aligned} x \in (A \cap B)_{[2,y]} &\Leftrightarrow x \in \{x \in X_1 \mid (x, y) \in A \cap B\} \\ &\Leftrightarrow x \in X_1 \wedge (x, y) \in A \cap B \\ &\Leftrightarrow x \in X_1 \wedge (x, y) \in A \wedge (x, y) \in B \\ &\Leftrightarrow x \in \{x \in X_1 \mid (x, y) \in A\} \wedge x \in \{x \in X_1 \mid (x, y) \in B\} \\ &\Leftrightarrow x \in A_{[2,y]} \wedge x \in B_{[2,y]} \\ &\Leftrightarrow x \in A_{[2,y]} \cap B_{[2,y]} \end{aligned}$$

which proves that

$$(A \cap B)_{[2,y]} = A_{[2,y]} \cap B_{[2,y]}$$

□

Lemma 21.550. Let $\langle X_1, \mathcal{A}_1 \rangle$, $\langle X_2, \mathcal{A}_2 \rangle$ be two measurable spaces and $A \in \mathcal{A}_1 \otimes \mathcal{A}_2$ then we have

1. $\forall x \in X_1 \ A_{[1,x]} \in \mathcal{A}_2$

2. $\forall y \in X_2 \ A_{[2,y]} \in \mathcal{A}_1$

Note 21.551. As $\mathcal{A}_1 \otimes \mathcal{A}_2$ is a σ -algebra on $X_1 \cdot X_2 = \prod_{i \in \{1,2\}} X_i$ we have that $A \subseteq X_1 \cdot X_2$ as is required for this lemma to make sense.

Proof.

1. Let $x \in X_1$ and define

$$\mathcal{B} = \{A \subseteq \mathcal{P}(X_1 \cdot X_2) \mid A_{[1,x]} \in \mathcal{A}_2\} \quad (21.937)$$

Using [lemma: 21.549] we have that $(X_1 \cdot X_2)_{[1,x]} = X_2 \in \mathcal{A}_2$ which proves that

$$X_1 \cdot X_2 \in \mathcal{B} \quad (21.938)$$

Further if $B \in \mathcal{B}$ then $B_{[1,x]} \in \mathcal{A}_2$ and using [lemma: 21.549] we have

$$\begin{aligned} ((X_1 \cdot X_2) \setminus B)_{[1,x]} &\stackrel{[\text{lemma: 21.549(4)}]}{=} (X_1 \cdot X_2)_{[1,x]} \setminus B_{[1,x]} \\ &\stackrel{[\text{lemma: 21.549(1)}]}{=} X_2 \setminus B_{[1,x]} \\ &\in \mathcal{A}_2 \\ &\quad B_{[1,x]} \in \mathcal{A}_2 \end{aligned}$$

Hence

$$\forall B \in \mathcal{B} \text{ we have that } (X_1 \cdot X_2) \setminus B \in \mathcal{B} \quad (21.939)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}$ then $\{(A_i)_{[1,x]}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_2$ so that $\bigcup_{i \in \mathbb{N}} (A_i)_{[1,x]} \in \mathcal{A}_2$. As by [lemma: 21.549] $(\bigcup_{i \in \mathbb{N}} A_i)_{[1,x]} = \bigcup_{i \in \mathbb{N}} (A_i)_{[1,x]}$ it follows that $(\bigcup_{i \in \mathbb{N}} A_i)_{[1,x]} \in \mathcal{A}_2$ hence $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{B}$. So

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{B} \quad (21.940)$$

Combining [eqs: 21.938, 21.939, 21.940] proves that

$$\mathcal{B} \text{ is a } \sigma\text{-algebra on } X_1 \cdot X_2 \quad (21.941)$$

Let $E_1 \cdot E_2 \in \{\prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} \ E_i \in \mathcal{A}_i\}$ then $E_1 \in \mathcal{A}_1 \wedge E_2 \in \mathcal{A}_2$. Using [lemma: 21.549] we have then that $(E_1 \cdot E_2)_{[1,x]} = \begin{cases} E_2 \in \mathcal{A}_2 & \text{if } x \in E_1 \\ \emptyset \in \mathcal{A}_2 & \text{if } x \in X_1 \setminus E_1 \end{cases} \in \mathcal{A}_2$ so that $E_1 \cdot E_2 \in \mathcal{B}$. So we have proved that

$$\left\{ \prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} \ E_i \in \mathcal{A}_i \right\} \subseteq \mathcal{B}$$

As by [corollary: 21.543] $\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma[\{\prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} \ E_i \in \mathcal{A}_i\}]$ we have using the above, [eq: 21.941] and [theorem: 21.42] that

$$\mathcal{A}_1 \otimes \mathcal{A}_2 \subseteq \mathcal{B}$$

Hence $\forall A \in \mathcal{A}_1 \otimes \mathcal{A}_2$ we have $A \in \mathcal{B}$ so that

$$A_{[1,x]} \in \mathcal{A}_2$$

2. Let $y \in X_2$ and define

$$\mathcal{B} = \{A \subseteq \mathcal{P}(X_1 \cdot X_2) \mid A_{[2,y]} \in \mathcal{A}_1\} \quad (21.942)$$

Using [lemma: 21.549] we have that $(X_1 \cdot X_2)_{[2,y]} = X_1 \in \mathcal{A}_1$ which proves that

$$X_1 \cdot X_2 \in \mathcal{B} \quad (21.943)$$

Further if $B \in \mathcal{B}$ then $B_{[2,y]} \in \mathcal{A}_1$ and using [lemma: 21.549] we have

$$\begin{aligned} ((X_1 \cdot X_2) \setminus B)_{[2,y]} &\stackrel{[\text{lemma: 21.549(5)}]}{=} (X_1 \cdot X_2)_{[2,y]} \setminus B_{[2,y]} \\ &\stackrel{[\text{lemma: 21.549(2)}]}{=} X_1 \setminus B_{[2,y]} \\ &\in \mathcal{A}_1 \\ &\quad B_{[2,y]} \in \mathcal{A}_1 \end{aligned}$$

Hence

$$\forall B \in \mathcal{B} \text{ we have that } (X_1 \cdot X_2) \setminus B \in \mathcal{B} \quad (21.944)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}$ then $\{(A_i)_{[2,y]}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1$ so that $\bigcup_{i \in \mathbb{N}} (A_i)_{[2,y]} \in \mathcal{A}_1$. As by [lemma: 21.549] $(\bigcup_{i \in \mathbb{N}} A_i)_{[2,y]} = \bigcup_{i \in \mathbb{N}} (A_i)_{[2,y]}$ it follows that $(\bigcup_{i \in \mathbb{N}} A_i)_{[2,y]} \in \mathcal{A}_1$ hence $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{B}$. So

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B} \text{ we have } \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{B} \quad (21.945)$$

Combining [eqs: 21.943, 21.944, 21.945] proves that

$$\mathcal{B} \text{ is a } \sigma\text{-algebra on } X_1 \cdot X_2 \quad (21.946)$$

Let $E_1 \cdot E_2 \in \left\{ \prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} E_i \in \mathcal{A}_i \right\}$ then $E_1 \in \mathcal{A}_1 \wedge E_2 \in \mathcal{A}_2$. Using [lemma: 21.549] we have then that $(E_1 \cdot E_2)_{[2,y]} = \begin{cases} E_1 \in \mathcal{A}_1 & \text{if } x \in E_2 \\ \emptyset \in \mathcal{A}_1 & \text{if } x \in X_1 \setminus E_2 \end{cases} \in \mathcal{A}_1$ so that $E_1 \cdot E_2 \in \mathcal{B}$. So we have proved that

$$\left\{ \prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} E_i \in \mathcal{A}_i \right\} \subseteq \mathcal{B}$$

As by [corollary: 21.543] $\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma\left[\left\{ \prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} E_i \in \mathcal{A}_i \right\}\right]$ we have using the above, [eq: 21.946] and [theorem: 21.42] that

$$\mathcal{A}_1 \otimes \mathcal{A}_2 \subseteq \mathcal{B}$$

Hence $\forall A \in \mathcal{A}_1 \otimes \mathcal{A}_2$ we have $A \in \mathcal{B}$ so that

$$A_{[2,y]} \in \mathcal{A}_1 \quad \square$$

Lemma 21.552. Let $\langle X_1, \mathcal{A}_1, \mu_1 \rangle, \langle X_2, \mathcal{A}_2, \mu_2 \rangle$ be two measure spaces where μ_1, μ_2 are finite measures then we have $\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ that

1. For $f_{1,Q}: X_1 \rightarrow \overline{\mathbb{R}}$ defined by $f_{1,Q}(x) = \mu_2(Q_{[1,x]})$ we have that $f_{1,Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$
2. For $f_{2,Q}: X_2 \rightarrow \overline{\mathbb{R}}$ defined by $f_{2,Q}(x) = \mu_1(Q_{[2,x]})$ we have that $f_{2,Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$

Note 21.553. As μ_1, μ_2 are finite measures it follows that $f_{1,Q}(X_1) \subseteq [0, \infty[\wedge f_{2,Q}(X_2) \subseteq [0, \infty[$

Proof. We use the concept of Dynkin systems to prove this theorem.

1. Let $x \in X_1$ then we have $f_{1,X_1 \cdot X_2}(x) = \mu_2((X_1 \cdot X_2)_{[1,x]}) \stackrel{[\text{lemma: 21.549(1)}]}{=} \mu_2(X_2) \underset{\mu_2 \text{ is finite}}{\in} \mathbb{R}$ so that

$$f_{1,X_1 \cdot X_2} = C_{\mu_2(X_2)} \quad (21.947)$$

which by [example: 21.228] proves that

$$f_{1,X_1 \cdot X_2} \in \mathcal{M}[X_1, \mathcal{A}_1, \mathbb{R}] \quad (21.948)$$

Let $A, B \in \mathcal{A}_1 \otimes \mathcal{A}_2$ then for $x \in X_1$ we have

$$\begin{aligned} f_{1,A \setminus B}(x) &= \mu_2((A \setminus B)_{[1,x]}) \\ &\stackrel{[\text{lemma: 21.549(4)}]}{=} \mu_2(A_{[1,x]} \setminus B_{[1,x]}) \\ &\stackrel{[\text{theorem: 21.99}] \wedge \mu_2 \text{ is finite}}{=} \mu_2(A_{[1,x]}) - \mu_2(B_{[1,x]}) \\ &= f_{1,A}(x) - f_{1,B}(x) \\ &= (f_{1,A} - f_{1,B})(x) \end{aligned}$$

which proves that

$$\forall A, B \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } f_{1,A \setminus B} = f_{1,A} - f_{1,B} \quad (21.949)$$

In the special cases where $A = X_1 \cdot X_2$ the above becomes

$$\forall A \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ then } f_{1,(X_1 \cdot X_2) \setminus A} = f_{1,X_1 \cdot X_2} - f_{1,A} \quad (21.950)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2$ be pairwise disjoint. Let $x \in X_1$ and $i, j \in \mathbb{N}$ with $i \neq j$ then we have $(A_i)_{[1,x]} \cap (A_j)_{[1,x]} = \emptyset$ [lemma: 21.549(9)] $(A_i \cap A_j)_{[1,x]} = \emptyset$ [lemma: 21.549(3)] $\emptyset$. Hence

$$\forall x \in X_1 \text{ we have } \{(A_i)_{[1,x]}\}_{i \in \mathbb{N}} \text{ is pairwise disjoint} \quad (21.951)$$

So for $x \in X_1$ we have

$$\begin{aligned} f_{1, \bigsqcup_{i \in \mathbb{N}} A_i}(x) &= \mu_2 \left(\left(\bigsqcup_{i \in \mathbb{N}} A_i \right)_{[1,x]} \right) \\ &\stackrel{[\text{lemma: 21.549}]}{=} \mu_2 \left(\bigsqcup_{i \in \mathbb{N}} (A_i)_{[1,x]} \right) \\ &\stackrel{[\text{eq: 21.951}] \wedge \mu_2 \text{ is a measure}}{=} \sum_{i=1}^{\infty} \mu_2((A_i)_{[1,x]}) \\ &= \sum_{i=1}^{\infty} f_{1, A_i}(x) \\ &= \left(\sum_{i=1}^{\infty} f_{1, A_i} \right)(x) \end{aligned}$$

which proves that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ pairwise disjoint we have } f_{1, \bigsqcup_{i \in \mathbb{N}} A_i} = \sum_{i=1}^{\infty} f_{1, A_i} \quad (21.952)$$

Define now

$$\mathcal{D} = \{D \in \mathcal{A}_1 \otimes \mathcal{A}_2 \mid f_{1,D} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]\}$$

Using [eq: 21.948] it follows that

$$X_1 \cdot X_2 \in \mathcal{D} \quad (21.953)$$

If $A \in \mathcal{D}$ then $f_{1,A} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$, further by [eq: 21.948] $f_{1, X_1 \cdot X_2} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$ so that by [theorem: 21.265] $f_{1, (X_1 \cdot X_2) \setminus A} \stackrel{[\text{eq: 21.950}]}{=} f_{1, X_1 \cdot X_2} - f_{1,A} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$. Hence

$$\forall A \in \mathcal{D} \text{ we have that } (X_1 \cdot X_2) \setminus A \in \mathcal{D} \quad (21.954)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}$ be pairwise disjoint then $\forall i \in \mathbb{N}$ we have that $f_{1, A_i} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$, as $\forall x \in X_1$ $f_{1, A_i}(x) = \mu_2(A_{i, [1,x]}) \in [0, \infty[$ it follows that $f_{1, A_i} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$. Using Beppo's Theorem [see theorem: 21.377] it follows that $\sum_{i=1}^{\infty} f_{1, A_i} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$, hence, as $f_{1, \bigsqcup_{i \in \mathbb{N}} A_i} \stackrel{[\text{eq: 21.952}]}{=} \sum_{i=1}^{\infty} f_{1, A_i}$, it follows that $\bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{D}$. To summarize

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D} \text{ pairwise disjoint we have } \bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{D} \quad (21.955)$$

By [eqs: 21.953, 21.954, 21.955] it follows by [definition: 21.123] that

$$\mathcal{D} \text{ is a Dynkin system} \quad (21.956)$$

Take

$$\mathcal{E} = \left\{ \prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} \ E_i \in \mathcal{A}_i \right\} \stackrel{[\text{theorem: 21.542}]}{\subseteq} \mathcal{A}_1 \otimes \mathcal{A}_2$$

Let $E = E_1 \cdot E_2 \in \mathcal{E}$ then $\forall x \in X_1$ we have either:

$x \in E_1$. Then by [lemma: 21.549] we have that $(E_1 \cdot E_2)_{[1,x]} = E_2$ so that

$$f_{1, E_1 \cdot E_2}(x) = \mu_2((E_1 \cdot E_2)_{[1,x]}) = \mu_2(E_2) = \mu_2(E_2) \cdot \mathcal{X}_{X_1, E_1}(x)$$

$x \in X_1 \setminus E_1$. Then by [lemma: 21.549] we have that $(E_1 \cdot E_2)_{[1,x]} = \emptyset$ so that

$$f_{1,E_1 \cdot E_2}(x) = \mu_2((E_1 \cdot E_2)_{[1,x]}) = \mu_2(\emptyset) = 0 = \mu_2(E_2) \cdot \mathcal{X}_{X_1, E_1}(x)$$

hence we have that

$$f_{1,E_1 \cdot E_2} = \mu_2(E_2) \cdot \mathcal{X}_{X_1, E_1}$$

As $E_1 \in \mathcal{A}_1$ it follows from [theorems: 21.273, 21.263] that

$$f_{1,E_1 \cdot E_2} = \mu_2(E_2) \cdot \mathcal{X}_{X_1, E_1} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$$

which proves that $E = E_1 \cdot E_2 \in \mathcal{D}$. Hence we have proved that

$$\mathcal{E} \subseteq \mathcal{D} \quad (21.957)$$

If $E = E_1 \cdot E_2, F = F_1 \cdot F_2 \in \mathcal{E}$ then we have that $E_1, F_1 \in \mathcal{A}_1$ and $E_2, F_2 \in \mathcal{A}_2$ then we have

$$E \bigcap F \stackrel{[\text{theorem: 2.150}]}{=} (E_1 \bigcap F_1) \cdot (E_2 \bigcap F_2)$$

which, as $E_1 \bigcap F_1 \in \mathcal{A}_1, E_2 \bigcap F_2 \in \mathcal{A}_2$, proves that $E \bigcap F \in \mathcal{E}$. Hence we have

$$\forall E, F \in \mathcal{E} \text{ we have } E \bigcap F \in \mathcal{E} \quad (21.958)$$

Using then [theorem: 21.130] on [eq: 21.958] gives

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma \left[\prod_{i \in \{1,2\}} E_i \mid \forall i \in \{1,2\} E_i \in \mathcal{A}_i \right] = \sigma[\mathcal{E}] = \delta[\mathcal{E}]$$

As $\mathcal{E} \subseteq \mathcal{D}$ [see eq: 21.957] and $\mathcal{D}$ is a Dynkin system [see eq: 21.956] we have by [theorem: 21.128(2)] that $\delta[\mathcal{E}] \subseteq \mathcal{D}$ so that $\mathcal{A}_1 \otimes \mathcal{A}_2 \subseteq \mathcal{D}$. Hence we have that

$$\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } Q \in \mathcal{D} \text{ so that } f_{1,Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$$

2. Let $x \in X_2$ then we have $f_{2,X_1 \cdot X_2}(x) = \mu_1((X_1 \cdot X_2)_{[2,x]}) \stackrel{[\text{lemma: 21.549(1)}]}{=} \mu_1(X_1) \in \mathbb{R}$ so that

$$f_{2,X_1 \cdot X_2} = C_{\mu_1(X_1)} \quad (21.959)$$

which by [example: 21.228] proves that

$$f_{2,X_1 \cdot X_2} \in \mathcal{M}[X_2, \mathcal{A}_2, \mathbb{R}] \quad (21.960)$$

Let $A, B \in \mathcal{A}_1 \otimes \mathcal{A}_2$ then for $x \in X_2$ we have

$$\begin{aligned} f_{2,A \setminus B}(x) &= \mu_1((A \setminus B)_{[2,x]}) \\ &\stackrel{[\text{lemma: 21.549(5)}]}{=} \mu_1(A_{[2,x]} \setminus B_{[2,x]}) \\ &\stackrel{[\text{theorem: 21.99}] \wedge \mu_2 \text{ is finite}}{=} \mu_1(A_{[2,x]}) - \mu_1(B_{[2,x]}) \\ &= f_{2,A}(x) - f_{2,B}(x) \\ &= (f_{2,A} - f_{2,B})(x) \end{aligned}$$

which proves that

$$\forall A, B \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } f_{2,A \setminus B} = f_{2,A} - f_{2,B} \quad (21.961)$$

In the special cases where $A = X_1 \cdot X_2$ the above becomes

$$\forall A \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ then } f_{2,(X_1 \cdot X_2) \setminus A} = f_{2,X_1 \cdot X_2} - f_{2,A} \quad (21.962)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2$ be pairwise disjoint. Let $x \in X_2$ and $i, j \in \mathbb{N}$ with $i \neq j$ then we have $(A_i)_{[2,x]} \bigcap (A_j)_{[2,x]} \stackrel{[\text{lemma: 21.549(10)}]}{=} (A_i \bigcap A_j)_{[2,x]} = \emptyset_{[2,x]} \stackrel{[\text{lemma: 21.549}]}{=} \emptyset$. Hence

$$\forall x \in X_2 \text{ we have } \{(A_i)_{[2,x]}\}_{i \in \mathbb{N}} \text{ is pairwise disjoint} \quad (21.963)$$

So for $x \in X_2$ we have

$$\begin{aligned}
 f_{2, \bigsqcup_{i \in \mathbb{N}} A_i}(x) &= \mu_1 \left(\left(\bigsqcup_{i \in \mathbb{N}} A_i \right)_{[2, x]} \right) \\
 &\stackrel{[\text{lemma: } 21.549(7)]}{=} \mu_1 \left(\bigsqcup_{i \in \mathbb{N}} (A_i)_{[2, x]} \right) \\
 &\stackrel{[\text{eq: } 21.963] \wedge \mu_1 \text{ is a measure}}{=} \sum_{i=1}^{\infty} \mu_1((A_i)_{[2, x]}) \\
 &= \sum_{i=1}^{\infty} f_{2, A_i}(x) \\
 &= \left(\sum_{i=1}^{\infty} f_{2, A_i} \right)(x)
 \end{aligned}$$

which proves that

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ pairwise disjoint we have } f_{2, \bigsqcup_{i \in \mathbb{N}} A_i} = \sum_{i=1}^{\infty} f_{2, A_i} \quad (21.964)$$

Define now

$$\mathcal{D} = \{D \in \mathcal{A}_1 \otimes \mathcal{A}_2 \mid f_{2, D} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]\}$$

Using [eq: 21.960] it follows that

$$X_1 \cdot X_2 \in \mathcal{D} \quad (21.965)$$

If $A \in \mathcal{D}$ then $f_{2, A} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$, further by [eq: 21.960] $f_{2, X_1 \cdot X_2} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$ so that by [theorem: 21.265] $f_{2, (X_1 \cdot X_2) \setminus A} \stackrel{[\text{eq: } 21.962]}{=} f_{2, X_1 \cdot X_2} - f_{2, A} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$. Hence

$$\forall A \in \mathcal{D} \text{ we have that } (X_1 \cdot X_2) \setminus A \in \mathcal{D} \quad (21.966)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}$ be pairwise disjoint then $\forall i \in \mathbb{N}$ we have that $f_{2, A_i} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$, as $\forall x \in X_2$ $f_{2, A_i}(x) = \mu_1(A_{[2, x]}) \in [0, \infty[$ it follows that $f_{2, A_i} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$. Using Beppo's Theorem [theorem: 21.377] it follows that $\sum_{i=1}^{\infty} f_{2, A_i} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$, hence, as $f_{2, \bigsqcup_{i \in \mathbb{N}} A_i} \stackrel{[\text{eq: } 21.964]}{=} \sum_{i=1}^{\infty} f_{2, A_i}$, it follows that $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{D}$. To summarize

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D} \text{ pairwise disjoint we have } \bigsqcup_{i \in \mathbb{N}} A_i \in \mathcal{D} \quad (21.967)$$

By [eqs: 21.965, 21.966, 21.967] it follows by [definition: 21.123] that

$$\mathcal{D} \text{ is a Dynkin system} \quad (21.968)$$

Take

$$\mathcal{E} = \left\{ \prod_{i \in \{1, 2\}} E_i \mid \forall i \in \{1, 2\} E_i \in \mathcal{A}_i \right\} \stackrel{[\text{theorem: } 21.542]}{\subseteq} \mathcal{A}_1 \otimes \mathcal{A}_2$$

Let $E = E_1 \cdot E_2 \in \mathcal{E}$ then $\forall x \in X_2$ we have either:

$x \in E_2$. Then by [lemma: 21.549(8)] we have that $(E_1 \cdot E_2)_{[2, x]} = E_1$ so that

$$f_{2, E_1 \cdot E_2}(x) = \mu_1((E_1 \cdot E_2)_{[2, x]}) = \mu_1(E_1) = \mu_1(E_1) \cdot \mathcal{X}_{X_2, E_2}(x)$$

$x \in X_2 \setminus E_2$. Then [lemma: 21.549(8)] we have that $(E_1 \cdot E_2)_{[2, x]} = \emptyset$ so that

$$f_{2, E_1 \cdot E_2}(x) = \mu_1((E_1 \cdot E_2)_{[2, x]}) = \mu_1(\emptyset) = 0 = \mu_1(E_1) \cdot \mathcal{X}_{X_2, E_2}(x)$$

hence we have that

$$f_{2, E_1 \cdot E_2} = \mu_1(E_1) \cdot \mathcal{X}_{X_2, E_2}$$

As $E_1 \in \mathcal{A}_1$ it follows from [theorems: 21.273, 21.263] that

$$f_{2, E_1 \cdot E_2} = \mu_1(E_1) \cdot \mathcal{X}_{X_2, E_2} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$$

which proves that $E = E_1 \cdot E_2 \in \mathcal{D}$. Hence we have proved that

$$\mathcal{E} \subseteq \mathcal{D} \quad (21.969)$$

If $E = E_1 \cdot E_2, F = F_1 \cdot F_2 \in \mathcal{E}$ then we have that $E_1, F_1 \in \mathcal{A}_1$ and $E_2, F_2 \in \mathcal{A}_2$ then we have

$$E \cap F \stackrel{[\text{theorem: 2.150}]}{=} (E_1 \cap F_1) \cdot (E_2 \cap F_2)$$

which, as $E_1 \cap F_1 \in \mathcal{A}_1, E_2 \cap F_2 \in \mathcal{A}_2$, proves that $E \cap F \in \mathcal{E}$. Hence

$$\forall E, F \in \mathcal{E} \text{ we have } E \cap F \in \mathcal{E} \quad (21.970)$$

Using then [theorem: 21.130] on [eq: 21.970] gives

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma \left[\prod_{i \in \{1, 2\}} E_i \mid \forall i \in \{1, 2\} E_i \in \mathcal{A}_i \right] = \sigma[\mathcal{E}] = \delta[\mathcal{E}]$$

As $\mathcal{E} \subseteq \mathcal{D}$ [see eq: 21.969] and $\mathcal{D}$ is a Dynkin system [see eq: 21.968] we have by [theorem: 21.128(2)] that $\delta[\mathcal{E}] \subseteq \mathcal{D}$ so that $\mathcal{A}_1 \otimes \mathcal{A}_2 \subseteq \mathcal{D}$. Hence we have that

$$\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } Q \in \mathcal{D} \text{ so that } f_{2, Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad \square$$

We extend now the previous lemma to the case of σ -finite measures.

Lemma 21.554. *Let $\langle X_1, \mathcal{A}_1, \mu_1 \rangle, \langle X_2, \mathcal{A}_2, \mu_2 \rangle$ be two measure spaces where μ_1, μ_2 are σ -finite measures then we have $\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ that*

1. *For $f_{1, Q}: X_1 \rightarrow \overline{\mathbb{R}}$ defined by $f_{1, Q}(x) = \mu_2(Q_{[1, x]})$ we have that $f_{1, Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$*
2. *For $f_{2, Q}: X_2 \rightarrow \overline{\mathbb{R}}$ defined by $f_{2, Q}(x) = \mu_1(Q_{[2, x]})$ we have that $f_{2, Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$*

Proof.

1. As μ_2 is σ -finite it follows from [definition: 21.106] and [theorem: 21.105] that there exist a sequence $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_2$ such that $\forall i \in \mathbb{N} A_i \subseteq A_{i+1}, \mu_2(A_i) < \infty$ and $X_2 = \bigcup_{i \in \mathbb{N}} A_i$. Let $i \in \mathbb{N}$ and define the measure [see theorem: 21.108(3)]

$$\mu_{i, 2}: \mathcal{A}_2 \rightarrow [0, \infty] \text{ by } \mu_{i, 2}(A) = \mu_2(A_i \cap A) \leq \mu_2(A_i) < \infty \quad (21.971)$$

then $\mu_{i, 2}$ is a finite measure and we can apply the previous lemma [lemma: 21.552] which proves that $\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ we have for the function

$$f_{i, 1, Q}: X_1 \rightarrow \overline{\mathbb{R}} \text{ defined by } f_{i, 1, Q}(x) = \mu_{i, 2}(Q_{[1, x]}) \stackrel{[\text{eq: 21.971}]}{=} \mu_2(A_i \cap Q_{[1, x]}) \quad (21.972)$$

that

$$f_{i, 1, Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \quad (21.973)$$

Let $Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ and take $x \in X_1$ then we have for $\{A_i \cap Q_{[1, x]}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_2$ that $\forall i \in \mathbb{N} A_i \cap Q_{[1, x]} \subseteq A_{i+1} \cap Q_{[1, x]}$ and $Q_{[1, x]} = X_2 \cap Q_{[1, x]} = (\bigcup_{i \in \mathbb{N}} A_i) \cap Q_{[1, x]} = \bigcup_{i \in \mathbb{N}} (A_i \cap Q_{[1, x]})$. Using then [theorem: 21.115] it follows that

$$\mu_2(Q_{[1, x]}) = \sup(\{\mu_2(A_i \cap Q_{[1, x]}) \mid i \in \mathbb{N}\}) \stackrel{[\text{eq: 21.972}]}{=} \sup(\{f_{i, 1, Q}(x) \mid i \in \mathbb{N}\}) \quad (21.974)$$

Define now

$$f_{1, Q}: X_1 \rightarrow \overline{\mathbb{R}} \text{ by } f_{1, Q}(x) = \mu_2(Q_{[1, x]})$$

then we have by [eq: 21.974] that $f_{1, Q}(x) = \sup(\{f_{i, 1, Q}(x) \mid i \in \mathbb{N}\})$, hence

$$f_{1, Q} = \sup(\{f_{i, 1, Q} \mid i \in \mathbb{N}\})$$

As $\forall i \in \mathbb{N} \ f_{i,1,Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$ [see eq: 21.973] it follows, using [theorem: 21.269], that

$$f_{1,Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$$

2. As μ_1 is σ -finite it follows from [definition: 21.106] and [theorem: 21.105] that there exist a sequence $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1$ such that $\forall i \in \mathbb{N} \ A_i \subseteq A_{i+1}$, $\mu_1(A_i) < \infty$ and $X_1 = \bigcup_{i \in \mathbb{N}} A_i$. Let $i \in \mathbb{N}$ and define the measure [see theorem: 21.108(2)]

$$\mu_{i,1}: \mathcal{A}_1 \rightarrow [0, \infty] \text{ by } \mu_{i,1}(A) = \mu_1(A_i \cap A) \leq \mu_1(A_i) < \infty \quad (21.975)$$

then μ_2 is a finite measure and we can apply the previous lemma [lemma: 21.552] which proves that $\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ we have for the function

$$f_{i,2,Q}: X_2 \rightarrow \overline{\mathbb{R}} \text{ defined by } f_{i,2,Q}(x) = \mu_{i,1}(Q_{[2,x]}) \underset{[\text{eq: 21.975}]}{=} \mu_1(A_i \cap Q_{[2,x]}) \quad (21.976)$$

that

$$f_{i,2,Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.977)$$

Let $Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ and take $x \in X_2$ then we have for $\{A_i \cap Q_{[2,x]}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1$ that $\forall i \in \mathbb{N} \ A_i \cap Q_{[2,x]} \subseteq A_{i+1} \cap Q_{[2,x]}$ and $Q_{[2,x]} = X_1 \cap Q_{[2,x]} = (\bigcup_{i \in \mathbb{N}} A_i) \cap Q_{[2,x]} = \bigcup_{i \in \mathbb{N}} (A_i \cap Q_{[2,x]})$. Using then [theorem: 21.115] it follows that

$$\mu_1(Q_{[2,x]}) = \sup(\{\mu_1(A_i \cap Q_{[2,x]}) | i \in \mathbb{N}\}) \underset{[\text{eq: 21.976}]}{=} \sup(\{f_{i,2,Q}(x) | i \in \mathbb{N}\}) \quad (21.978)$$

Define now

$$f_{2,Q}: X_2 \rightarrow \overline{\mathbb{R}} \text{ by } f_{2,Q}(x) = \mu_1(Q_{[2,x]})$$

then we have by [eq: 21.978] that $f_{2,Q}(x) = \sup(\{f_{i,2,Q}(x) | i \in \mathbb{N}\})$, hence

$$f_{2,Q} = \sup(\{f_{i,2,Q} | i \in \mathbb{N}\})$$

As $\forall i \in \mathbb{N} \ f_{i,2,Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$ [see eq: 21.977] it follows using [theorem: 21.269] that

$$f_{2,Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad \square$$

We introduce now a notation that will simplify and clarify the rest of this section, by using the expression defining a function in the integral notation.

Convention 21.555. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measurable space then we use the following simplifying conventions

1. The notation

$$\int^+ E(x) \mu(dx)$$

is equivalent with saying that for the function defined by

$$f: X \rightarrow \overline{\mathbb{R}} \text{ where } f(x) = E(x)$$

we have that

$$f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ E(x) \mu(dx) = \int f d\mu$$

2. The notation

$$\int E(x) \mu(dx)$$

is equivalent with saying that for the function defined by

$$f: X \rightarrow \overline{\mathbb{R}} \text{ where } f(x) = E(x)$$

we have that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

and

$$\int E(x) \mu(dx) = \int f d\mu$$

3. The notation

$$\int^{\mathbb{C}} E(x) \mu(dx)$$

is equivalent with saying that for the function defined by

$$f: X \rightarrow \mathbb{C} \text{ where } f(x) = E(x)$$

we have that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \mathbb{C}]$$

and

$$\int^{\mathbb{C}} E(x) \mu(dx) = \int^{\mathbb{C}} f d\mu$$

Theorem 21.556. Let $\langle X_1, \mathcal{A}_1, \mu_1 \rangle, \langle X_2, \mathcal{A}_2, \mu_2 \rangle$ be two measure spaces where μ_1, μ_2 are **σ -finite** then we have for

$$\mu_1 \otimes \mu_2: \mathcal{A}_1 \otimes \mathcal{A}_2 \rightarrow [0, \infty] \text{ defined by } (\mu_1 \otimes \mu_2)(Q) = \int^+ \mu_2(Q_{[1,x]}) \mu_1(dx)$$

that

$$\mu_1 \otimes \mu_2 \text{ is } \sigma\text{-finite measure on } \mathcal{A}_1 \otimes \mathcal{A}_2$$

and

$$\forall A_1 \in \mathcal{A}_1, \forall A_2 \in \mathcal{A}_2 \text{ we have } (\mu_1 \otimes \mu_2)(A_1 \cdot A_2) = \mu_1(A_1) \cdot \mu_2(A_2)$$

and

$$\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } (\mu_1 \otimes \mu_2)(Q) = \int^+ \mu_2(Q_{[1,x]}) \mu_1(dx) = \int^+ \mu_1(Q_{[2,x]}) \mu_2(dx)$$

Further if there exist another measure

$$\pi: \mathcal{A}_1 \otimes \mathcal{A}_2 \rightarrow [0, \infty]$$

such that

$$\forall A_1 \in \mathcal{A}_1, \forall A_2 \in \mathcal{A}_2 \text{ we have } \pi(A_1 \cdot A_2) = \mu_1(A_1) \cdot \mu_2(A_2)$$

then

$$\pi = \mu_1 \otimes \mu_2$$

The measure $\mu_1 \otimes \mu_2$ on $\mathcal{A}_1 \otimes \mathcal{A}_2$ is called the product measure of μ_1 and μ_2 .

Note 21.557. Let $A_1 \in \mathcal{A}_1, A_2 \in \mathcal{A}_2$ then by [theorem: 21.542] $A_1 \cdot A_2 \in \mathcal{A}_1 \otimes \mathcal{A}_2$ so that $(\mu_1 \otimes \mu_2)(A_1 \cdot A_2)$ is well defined.

Note 21.558. We use the previous convention [see convention: 21.555] here, so

$$\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } (\mu_1 \otimes \mu_2)(Q) = \int^+ \mu_2(Q_{[1,x]}) \mu_1(dx) = \int^+ \mu_1(Q_{[2,x]}) \mu_2(dx)$$

in the above is equivalent with the following. For the functions

$$f_{1,Q}: X_1 \rightarrow \overline{\mathbb{R}} \text{ defined by } f_{1,Q}(x) = \mu_2(Q_{[1,x]})$$

$$f_{2,Q}: X_2 \rightarrow \overline{\mathbb{R}} \text{ defined by } f_{2,Q}(x) = \mu_1(Q_{[2,x]})$$

we have that

$$f_{1,Q} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge f_{2,Q} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$$

and

$$(\mu_1 \otimes \mu_2)(A) = \int^+ f_{1,Q} d\mu_1 = \int^+ f_{2,Q} d\mu_2$$

Proof. Let $Q \in \mathcal{A}_1 \otimes \mathcal{A}_2$ and define

$$f_{1,Q}: X_1 \rightarrow \overline{\mathbb{R}} \text{ by } f_{1,Q}(x) = \mu_2(Q_{[1,x]})$$

$$f_{2,Q}: X_2 \rightarrow \overline{\mathbb{R}} \text{ by } f_{2,Q}(x) = \mu_1(Q_{[2,x]})$$

then by [lemma: 21.554] we have that

$$f_{1,Q} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge f_{2,Q} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.979)$$

As $\forall x \in X_1$ $f_{1,Q}(x) = \mu_2(Q_{[1,x]}) \in [0, \infty]$ and $\forall x \in X_2$ $f_{2,Q}(x) = \mu_1(Q_{[2,x]}) \in [0, \infty]$ it follows from the above that

$$f_{1,Q} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge f_{2,Q} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.980)$$

So we can define the function

$$(\mu_1 \otimes \mu_2): \mathcal{A}_1 \otimes \mathcal{A}_2 \rightarrow [0, \infty] \text{ by } (\mathcal{A}_1 \otimes \mathcal{A}_2)(Q) = \int^+ f_{1,Q} d\mu_1 \stackrel{\text{notation}}{=} \int^+ \mu_2(Q_{[1,x]}) \mu_1(dx) \quad (21.981)$$

Let $x \in X_1$ then $f_{1,\emptyset}(x) = \mu_2(\emptyset_{[1,x]}) \stackrel{[\text{theorem: 21.549}]}{=} \mu_2(\emptyset) = 0$ which proves that $f_{1,\emptyset} = C_0$. Applying then [theorem: 21.318] we have that $\int^+ f_{1,\emptyset} d\mu_1 = \int^+ C_0 d\mu_1 = 0 \cdot \mu_1(X_1) = 0$ which proves that

$$(\mu_1 \otimes \mu_2)(\emptyset) = 0 \quad (21.982)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2$ be pairwise disjoint and take $x \in X_1$. Let $i, j \in \mathbb{N}$ with $i \neq j$ then

$$(A_i)_{[1,x]} \cap (A_j)_{[1,x]} \stackrel{[\text{theorem: 21.549(9)}]}{=} (A_i \cap A_j)_{[1,x]} = \emptyset_{[1,x]} \stackrel{[\text{theorem: 21.549(3)}]}{=} \emptyset$$

which proves that

$$\{(A_i)_{[1,x]}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_2 \text{ is pairwise disjoint} \quad (21.983)$$

$$\begin{aligned} f_{1, \bigsqcup_{i \in \mathbb{N}} A_i}(x) &= \mu_2\left(\left(\bigsqcup_{i \in \mathbb{N}} A_i\right)_{[1,x]}\right) \\ &\stackrel{[\text{theorem: 21.549(6)}]}{=} \mu_2\left(\bigsqcup_{i \in \mathbb{N}} (A_i)_{[1,x]}\right) \\ &\stackrel{[\text{eq: 21.983}]}{=} \sum_{i=1}^{\infty} \mu_2((A_i)_{[1,x]}) \\ &= \sum_{i=1}^{\infty} f_{1,A_i}(x) \end{aligned}$$

so that

$$f_{1, \bigsqcup_{i \in \mathbb{N}} A_i} = \sum_{i=1}^{\infty} f_{1,A_i}$$

Using the Beppo's Levi theorem [see theorem: 21.377] we have that

$$\sum_{i=1}^{\infty} f_{1,A_i} \in \mathcal{M}^+(X_1, \mathcal{A}_1, \overline{\mathbb{R}})$$

and

$$(\mu_1 \otimes \mu_2)\left(\bigsqcup_{i \in \mathbb{N}} A_i\right) = \int^+ f_{1, \bigsqcup_{i \in \mathbb{N}} A_i} d\mu_1 = \int^+ \left(\sum_{i=1}^{\infty} f_{1,A_i}\right) d\mu_1 = \sum_{i=1}^{\infty} \int^+ f_{1,A_i} d\mu_1 = \sum_{i=1}^{\infty} (\mu_1 \otimes \mu_2)(A_i)$$

Hence we have

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ pairwise disjoint then } (\mu_1 \otimes \mu_2) \left(\bigcup_{i \in \mathbb{N}} A_i \right) = \sum_{i=1}^{\infty} (\mu_1 \otimes \mu_2)(A_i)$$

The above together with [eq: 21.982] proves that $\mu_1 \otimes \mu_2$ is a measure on $\mathcal{A}_1 \otimes \mathcal{A}_2$. To summarize

$$(\mu_1 \otimes \mu_2): \mathcal{A}_1 \otimes \mathcal{A}_2 \rightarrow [0, \infty] \text{ defined by } (\mu_1 \otimes \mu_2)(Q) = \int^+ f_{1,Q} d\mu_1 \text{ is a measure on } \mathcal{A}_1 \otimes \mathcal{A}_2 \quad (21.984)$$

Let $A_1 \in \mathcal{A}_1$, $A_2 \in \mathcal{A}_2$ then we have for $x \in X_1$ that either:

$$x \in \mathbf{A}_1. \text{ Then } f_{1,A_1 \cdot A_2}(x) = \mu_2([A_1 \cdot A_2]_{[1,x]}) \stackrel{[\text{theorem: 21.549(8)}]}{=} \mu_2(A_2) = \mu_2(A_2) \cdot \mathcal{X}_{X_1, A_1}(x)$$

$$x \in X_1 \setminus \mathbf{A}_1. \text{ Then } f_{1,A_1 \cdot A_2}(x) = \mu_2([A_1 \cdot A_2]_{[1,x]}) \stackrel{[\text{theorem: 21.549(8)}]}{=} \mu_2(\emptyset) = \mu_2(A_2) \cdot \mathcal{X}_{X_1, A_1}(x)$$

which proves that

$$f_{1,A_1 \cdot A_2} = \mu_2(A_2) \cdot \mathcal{X}_{X_1, A_1}$$

Then

$$\begin{aligned} (\mu_1 \otimes \mu_2)(A_1 \cdot A_2) &= \int^+ f_{1,A_1 \cdot A_2} d\mu_1 \\ &= \int^+ (\mu_2(A_2) \cdot \mathcal{X}_{X_1, A_1}) d\mu_1 \\ &\stackrel{[\text{example: 21.319}]}{=} \mu_2(A_2) \cdot \mu_1(A_1) \\ &= \mu_1(A_1) \cdot \mu_2(A_2) \end{aligned}$$

To summarize we have

$$\forall A_1 \in \mathcal{A}_1, \forall A_2 \in \mathcal{A}_2 \text{ we have } (\mu_1 \otimes \mu_2)(A_1 \cdot A_2) = \mu_1(A_1) \cdot \mu_2(A_2) \quad (21.985)$$

Now if π is another measure on $\mathcal{A}_1 \otimes \mathcal{A}_2$ such that $\forall A_1 \in \mathcal{A}_1, \forall A_2 \in \mathcal{A}_2$ we have $\pi(A_1 \cdot A_2) = \pi_1(A_1) \cdot \pi(A_2)$ then by [corollary: 21.546]

$$\pi = \mu_1 \otimes \mu_2 \quad (21.986)$$

As μ_1, μ_2 are σ -finite there exist by [theorem: 21.105] there exist $\forall i \in \{1, 2\}$ a

$$\{E_{i,k}\}_{k \in \mathbb{N}} \subseteq \mathcal{A}_i \text{ with } \forall k \in \mathbb{N} \ E_{i,k} \subseteq E_{i,k+1} \wedge \mu(E_{i,k}) < \infty \text{ and } X_i = \bigcup_{k \in \mathbb{N}} E_{i,k} \quad (21.987)$$

Let $x = (x_1, x_2) \in X_1 \cdot X_2$ then $x_1 \in X_1 \wedge x_2 \in X_2$ so that by the above there exist $k, l \in \mathbb{N}$ such that $x_1 \in E_{1,k} \wedge x_2 \in E_{2,l}$ hence $x = (x_1, x_2) \in E_{1, \max(k,l)} \cdot E_{2, \max(k,l)} \subseteq \bigcup_{k \in \mathbb{N}} (E_{1,k} \cdot E_{2,k})$. As also $\forall k \in \mathbb{N} \ E_{1,k} \cdot E_{2,k} \subseteq X_1 \cdot X_2$ it follows that

$$\bigcup_{k \in \mathbb{N}} E_{1,k} \cdot E_{2,k} = X_1 \cdot X_2 \quad (21.988)$$

Further $\forall k \in \mathbb{N}$ we have by [theorem: 21.542] that $E_{1,k} \cdot E_{2,k} \in \mathcal{A}_1 \otimes \mathcal{A}_2$ and by [eqs: 21.985, 21.987] that $(\mu_1 \otimes \mu_2)(E_{1,k} \cdot E_{2,k}) = \mu_1(E_{1,k}) \cdot \mu_2(E_{2,k}) < \infty$, hence, taking in account [eq: 21.988], we have

$$\mu_1 \otimes \mu_2 \text{ is } \sigma\text{-finite}$$

Finally the only thing left to prove is that $(\mu_1 \otimes \mu_2)(Q) = \int^+ \mu_1(Q_{[2,x]}) \mu_2(dx)$. As $\forall Q \in \mathcal{A}$ $f_{2,Q} \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.980] we can define

$$\pi: \mathcal{A}_1 \otimes \mathcal{A}_2 \rightarrow [0, \infty] \text{ by } \pi(Q) = \int^+ f_{2,Q} d\mu_2 \stackrel{\text{notation}}{=} \int^+ \mu_1(Q_{[2,x]}) \mu_2(dx)$$

Let $x \in X_2$ then $f_{2,\emptyset}(x) = \mu_1(\emptyset_{[2,x]}) \stackrel{[\text{theorem: 21.549(3)}]}{=} \mu_1(\emptyset) = 0$ which proves that $f_{2,\emptyset} = C_0$. Applying then [theorem: 21.318] we have that $\int^+ f_{2,\emptyset} d\mu_2 = \int^+ C_0 d\mu_2 = 0 \cdot \mu_2(X_2) = 0$ which proves that

$$\pi(\emptyset) = 0 \quad (21.989)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2$ be pairwise disjoint and take $x \in X_2$. Let $i, j \in \mathbb{N}$ with $i \neq j$ then

$$(A_i)_{[2,x]} \cap (A_j)_{[2,x]} \stackrel{[\text{theorem: 21.549(10)}]}{=} (A_i \cap A_j)_{[2,x]} = \emptyset_{[2,x]} \stackrel{[\text{theorem: 21.549(3)}]}{=} \emptyset$$

which proves that

$$\{(A_i)_{[2,x]}\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \text{ is pairwise disjoint} \quad (21.990)$$

Hence

$$\begin{aligned} f_{2, \sqcup_{i \in \mathbb{N}} A_i}(x) &= \mu_1 \left(\left(\sqcup_{i \in \mathbb{N}} A_i \right)_{[2,x]} \right) \\ &\stackrel{[\text{theorem: 21.549(7)}]}{=} \mu_1 \left(\sqcup_{i \in \mathbb{N}} (A_i)_{[2,x]} \right) \\ &\stackrel{[\text{eq: 21.990}]}{=} \sum_{i=1}^{\infty} \mu_2((A_i)_{[2,x]}) \\ &= \sum_{i=1}^{\infty} f_{2, A_i}(x) \end{aligned}$$

so that

$$f_{2, \sqcup_{i \in \mathbb{N}} A_i} = \sum_{i=1}^{\infty} f_{2, A_i}$$

Using the Beppo's Levi theorem [see theorem: 21.377] we have that

$$\sum_{i=1}^{\infty} f_{2, A_i} \in \mathcal{M}^+(X_2, \mathcal{A}_2, \overline{\mathbb{R}})$$

and

$$\pi \left(\sqcup_{i \in \mathbb{N}} A_i \right) = \int^+ f_{2, \sqcup_{i \in \mathbb{N}} A_i} d\mu_2 = \int^+ \left(\sum_{i=1}^{\infty} f_{2, A_i} \right) d\mu_2 = \sum_{i=1}^{\infty} \int^+ f_{2, A_i} d\mu_2 = \sum_{i=1}^{\infty} \pi(A_i)$$

Hence we have

$$\forall \{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ pairwise disjoint then } \pi \left(\sqcup_{i \in \mathbb{N}} A_i \right) = \sum_{i=1}^{\infty} \pi(A_i)$$

The above together with [eq: 21.990] proves that π is a measure on $\mathcal{A}_1 \otimes \mathcal{A}_2$. To summarize

$$\pi: \mathcal{A}_1 \otimes \mathcal{A}_2 \rightarrow [0, \infty] \text{ defined by } (\mathcal{A}_1 \otimes \mathcal{A}_2)(Q) = \int^+ f_{2, Q} d\mu_2 \text{ is a measure on } \mathcal{A}_1 \otimes \mathcal{A}_2 \quad (21.991)$$

Let $A_1 \in \mathcal{A}_1$, $A_2 \in \mathcal{A}_2$ then we have for $x \in X_2$ that either:

$$\mathbf{x} \in \mathbf{A}_2. \text{ Then } f_{2, A_1 \cdot A_2}(x) = \mu_1([A_1 \cdot A_2]_{[2,x]}) \stackrel{[\text{theorem: 21.549(8)}]}{=} \mu_1(A_1) = \mu_1(A_1) \cdot \mathcal{X}_{X_2, A_2}(x)$$

$$\mathbf{x} \in \mathbf{X}_2 \setminus \mathbf{A}_2. \text{ Then } f_{2, A_1 \cdot A_2}(x) = \mu_1([A_1 \cdot A_2]_{[2,x]}) \stackrel{[\text{theorem: 21.549(8)}]}{=} \mu_1(\emptyset) = \mu_1(A_1) \cdot \mathcal{X}_{X_2, A_2}(x)$$

which proves that

$$f_{2, A_1 \cdot A_2} = \mu_1(A_1) \cdot \mathcal{X}_{X_2, A_2}$$

Then

$$\begin{aligned} \pi(A_1 \cdot A_2) &= \int^+ f_{2, A_1 \cdot A_2} d\mu_2 \\ &= \int^+ (\mu_1(A_1) \cdot \mathcal{X}_{X_1, A_1}) d\mu_2 \\ &\stackrel{[\text{example: 21.319}]}{=} \mu_1(A_1) \cdot \mu_2(A_2) \end{aligned}$$

To summarize we have

$$\forall A_1 \in \mathcal{A}_1, \forall A_2 \in \mathcal{A}_2 \text{ we have } \pi(A_1 \cdot A_2) = \mu_1(A_1) \cdot \mu_2(A_2) \quad (21.992)$$

Hence using [eqs: 21.991, 21.992 and 21.986] we have that $\pi = \mu_1 \otimes \mu_2$ or using the definition of π that

$$\forall Q \in \mathcal{A}_1 \otimes \mathcal{A}_2 \text{ we have } (\mu_1 \otimes \mu_2) = \int^+ f_{2, A_1 \cdot A_2} d\mu_2 \underset{\text{notation}}{=} \int^+ \mu_1(Q_{[2, x]}) \mu_2(dx) \quad \square$$

Having defined the product of measures it is time to consider integrals using this product measure. We don this in steps, starting with the integral of simple functions, using then the fact that non negative functions can be written as the limit of simple functions to consider the integral of non negative measurable functions to finish at integrable functions. First we define sections of functions.

Definition 21.559. Let X_1, X_2, Y be sets and $f: X_1 \cdot X_2 \rightarrow Y$ a function then we define:

1. $\forall x \in X_1$ $f(x, \cdot): X_2 \rightarrow Y$ where $f(x, \cdot)(y) = f(x, y)$
2. $\forall y \in X_2$ $f(\cdot, y): X_1 \rightarrow Y$ where $f(\cdot, y)(x) = f(x, y)$

We call $f(\cdot, x)$ and $f(x, \cdot)$ the sections of f .

Applying the above definition on characteristic function we have.

Lemma 21.560. Let X_1, X_2, Y be sets then we have:

1. $\forall Q \subseteq X_1 \cdot X_2$ we have $\forall x \in X_1$ that $(\mathcal{X}_{X_1 \cdot X_2, Q})(x, \cdot) = \mathcal{X}_{X_2, Q_{[1, x]}}$
2. $\forall Q \subseteq X_1 \cdot X_2$ we have $\forall y \in X_2$ that $(\mathcal{X}_{X_1 \cdot X_2, Q})(\cdot, y) = \mathcal{X}_{X_1, Q_{[2, y]}}$

Proof.

1. Let $y \in X_2$ then we have either:

$y \in Q_{[1, x]}$. Then $y \in \{y \in X_2 | (x, y) \in Q\}$ so that $(x, y) \in Q$ hence we have that

$$(\mathcal{X}_{X_1 \cdot X_2, Q})(x, \cdot)(y) = \mathcal{X}_{X_1 \cdot X_2, Q}(x, y) = 1 = \mathcal{X}_{X_2, Q_{[1, x]}}(y)$$

$y \in X_2 \setminus Q_{[1, x]}$. Then $y \notin \{y \in X_2 | (x, y) \in Q\}$ so that $(x, y) \notin Q$ hence

$$(\mathcal{X}_{X_1 \cdot X_2, Q})(x, \cdot)(y) = \mathcal{X}_{X_1 \cdot X_2, Q}(x, y) = 0 = \mathcal{X}_{X_2, Q_{[1, x]}}(y)$$

which proves that $(\mathcal{X}_{X_1 \cdot X_2, Q})(x, \cdot) = \mathcal{X}_{X_2, Q_{[1, x]}}$.

2. Let $x \in X_1$ then we have either:

$x \in Q_{[2, y]}$. Then $x \in \{x \in X_1 | (x, y) \in Q\}$ so that $(x, y) \in Q$ hence we have that

$$(\mathcal{X}_{X_1 \cdot X_2, Q})(\cdot, y)(x) = \mathcal{X}_{X_1 \cdot X_2, Q}(x, y) = 1 = \mathcal{X}_{X_1, Q_{[2, y]}}(x)$$

$x \in X_1 \setminus Q_{[2, y]}$. Then $x \notin \{x \in X_1 | (x, y) \in Q\}$ so that $(x, y) \notin Q$ hence

$$(\mathcal{X}_{X_1 \cdot X_2, Q})(\cdot, y)(x) = \mathcal{X}_{X_1 \cdot X_2, Q}(x, y) = 0 = \mathcal{X}_{X_1, Q_{[2, y]}}(x)$$

which proves that $(\mathcal{X}_{X_1 \cdot X_2, Q})(\cdot, y) = \mathcal{X}_{X_1, Q_{[2, y]}}$. $\square$

If we want to integrate the sections of a function we have to show that sections are measurable which is indeed the case as the following lemma shows.

Lemma 21.561. Let $\langle X_1, \mathcal{A}_1 \rangle, \langle X_2, \mathcal{A}_2 \rangle, \langle Y, \mathcal{B} \rangle$ be measurable spaces and

$$f: X_1 \cdot X_2 \rightarrow Y$$

a $\mathcal{A}_1 \otimes \mathcal{A}_2, \mathcal{B}$ -measurable function then:

1. $\forall x \in X_1$ $f(x, \cdot): X_2 \rightarrow Y$ is a $\mathcal{A}_2, \mathcal{B}$ -measurable function.
2. $\forall y \in X_2$ $f(\cdot, y): X_1 \rightarrow Y$ is a $\mathcal{A}_1, \mathcal{B}$ -measurable function.

Note 21.562. In particular if $\langle Y, \mathcal{B} \rangle = \langle \overline{\mathbb{R}}, \mathcal{B}[\overline{\mathbb{R}}] \rangle$ then we have for $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$ that

$\forall x \in X_1$ we have $f(x, \cdot) \in \mathcal{M}[X_2, \mathcal{A}_2, \mathbb{R}]$ and $\forall y \in X_2$ we have $f(\cdot, y) \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$

Proof.

1. Let $x \in X_1$ and $B \in \mathcal{B}$ then we have

$$\begin{aligned} y \in (f(x, \cdot))^{-1}(B) &\Leftrightarrow y \in X_2 \wedge f(x, \cdot)(y) \in B \\ &\Leftrightarrow y \in X_2 \wedge f(x, y) \in B \\ &\Leftrightarrow y \in X_2 \wedge (x, y) \in f^{-1}(B) \\ &\Leftrightarrow y \in \{y \in X_2 \mid (x, y) \in f^{-1}(B)\} \\ &\Leftrightarrow y \in (f^{-1}(B))_{[1, x]} \end{aligned}$$

which proves that

$$(f(x, \cdot))^{-1}(B) = (f^{-1}(B))_{[1, x]}$$

As f is $\mathcal{A}_1 \otimes \mathcal{A}_2, \mathcal{B}$ -measurable it follows that $f^{-1}(B) \in \mathcal{A}_1 \otimes \mathcal{A}_2$ so that by [lemma: 21.550] it follows that $(f^{-1}(B))_{[1, x]} \in \mathcal{A}_2$. Hence we have by definition that

$f(x, \cdot)$ is $\mathcal{A}_2, \mathcal{B}$ -measurable

2. Let $y \in X_2$ and $B \in \mathcal{B}$ then we have

$$\begin{aligned} x \in (f(\cdot, y))^{-1}(B) &\Leftrightarrow x \in X_1 \wedge f(\cdot, y)(x) \in B \\ &\Leftrightarrow x \in X_1 \wedge f(x, y) \in B \\ &\Leftrightarrow x \in X_1 \wedge (x, y) \in f^{-1}(B) \\ &\Leftrightarrow x \in \{x \in X_1 \mid (x, y) \in f^{-1}(B)\} \\ &\Leftrightarrow x \in (f^{-1}(B))_{[2, y]} \end{aligned}$$

which proves that

$$(f(\cdot, y))^{-1}(B) = (f^{-1}(B))_{[2, y]}$$

As f is $\mathcal{A}_1 \otimes \mathcal{A}_2, \mathcal{B}$ -measurable it follows that $f^{-1}(B) \in \mathcal{A}_1 \otimes \mathcal{A}_2$ so that by [lemma: 21.550] it follows that $(f^{-1}(B))_{[2, y]} \in \mathcal{A}_1$. Hence we have by definition that

$f(\cdot, y)$ is $\mathcal{A}_2, \mathcal{B}$ -measurable □

Lemma 21.563. Let $\langle X_1, \mathcal{A}_1, \mu_1 \rangle, \langle X_2, \mathcal{A}_2, \mu_2 \rangle$ be measurable spaces such that μ_1, μ_2 are σ -finite and

$$f \in \mathcal{S}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2] \text{ [see definition: 21.297]}$$

so that by [theorem: 21.317]

$$f \in \mathcal{M}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}]$$

then

$$\forall x \in X_1 \text{ we have } f(x, \cdot) \in \mathcal{S}^+[X_2, \mathcal{A}_2] \text{ and } \forall y \in X_2 \text{ we have } f(\cdot, y) \in \mathcal{S}^+[X_1, \mathcal{A}_1]$$

where

$$f(\cdot, y): X_1 \rightarrow \overline{\mathbb{R}} \text{ is defined by } f(\cdot, y)(x) = f(x, y)$$

$$f(x, \cdot): X_2 \rightarrow \overline{\mathbb{R}} \text{ is defined by } f(x, \cdot)(y) = f(x, y)$$

and for the functions

$$I_{f,1}: X_1 \rightarrow \overline{\mathbb{R}} \text{ defined by } I_{f,1}(x) = \int^S f(x, \cdot) d\mu_2 \underset{\text{[theorem: 21.317]}}{=} \int^+ f(x, \cdot) d\mu_2$$

$$I_{f,2}: X_2 \rightarrow \overline{\mathbb{R}} \text{ defined by } I_{f,2}(x) = \int^S f(\cdot, x) d\mu_1 \underset{\text{[theorem: 21.317]}}{=} \int^+ f(\cdot, x) d\mu_1$$

we have

$$I_{f,1} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge I_{f,2} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$$

and

$$\int^+ f d(\mu_1 \otimes \mu_2) = \int^+ I_{f,1} d\mu_1 = \int^+ I_{f,2} d\mu_2$$

Note 21.564. If we use our convention [see convention: 21.555] this becomes

$$\int^+ f d(\mu_1 \otimes \mu_2) = \int^+ \left(\int^+ f(x, y) \mu(dy) \right) \mu(dx) = \int^+ \left(\int^+ f(x, y) \mu(dx) \right) \mu(dy)$$

Proof. As $f \in \mathcal{S}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2]$ there exist by [definition: 21.305] a $n \in \mathbb{N}$, $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ and $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}_1 \otimes \mathcal{A}_2$ such that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1 \cdot X_2, A_i}$$

and

$$\int^S f d(\mu_1 \otimes \mu_2) = \sum_{i=1}^n f_i \cdot (\mu_1 \otimes \mu_2)(A_i) \quad (21.993)$$

Let $x \in X_1$ then we have $\forall y \in X_2$ that

$$\begin{aligned} f(x, \cdot)(y) &= f(x, y) \\ &= \left(\sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1 \cdot X_2, A_i} \right)(x, y) \\ &= \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1 \cdot X_2, A_i}(x, y) \\ &= \sum_{i=1}^n f_i \cdot (\mathcal{X}_{X_1 \cdot X_2, A_i})(x, \cdot)(y) \\ &\stackrel{[\text{lemma: 21.560}]}{=} \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_2, (A_i)_{[1, x]}}(y) \\ &= \left(\sum_{i=1}^n f_i \cdot \mathcal{X}_{X_2, (A_i)_{[1, x]}} \right)(y) \end{aligned}$$

Likewise if $y \in X_2$ then we have $\forall x \in X_1$ that

$$\begin{aligned} f(\cdot, y)(x) &= f(x, y) \\ &= \left(\sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1 \cdot X_2, A_i} \right)(x, y) \\ &= \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1 \cdot X_2, A_i}(x, y) \\ &= \sum_{i=1}^n f_i \cdot (\mathcal{X}_{X_1 \cdot X_2, A_i})(\cdot, y)(x) \\ &\stackrel{[\text{lemma: 21.560}]}{=} \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1, (A_i)_{[2, y]}}(x) \\ &= \left(\sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1, (A_i)_{[2, y]}} \right)(x) \end{aligned}$$

Hence we have that

$$\forall x \in X_1 \text{ we have that } f(x, \cdot) = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_2, (A_i)_{[1, x]}} \quad (21.994)$$

$$\forall y \in X_2 \text{ we have that } f(\cdot, y) = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X_1, (A_i)_{[2, y]}} \quad (21.995)$$

Using [lemma: 21.550] we have $\forall i \in \{1, \dots, n\}$ that $\forall x \in X_1 (A_i)_{[1, x]} \in \mathcal{A}_2$ and $\forall y \in X_2 (A_i)_{[2, y]} \in \mathcal{A}_1$ which proves by [theorem: 21.301] that

$$\forall x \in X_1 \text{ we have } f(x, \cdot) \in \mathcal{S}^+[X_2, \mathcal{A}_2] \text{ and } \forall y \in X_2 \text{ we have } f(\cdot, y) \in \mathcal{S}^+[X_1, \mathcal{A}_1] \quad (21.996)$$

Using the definition of the integral of simple functions [see definition: 21.305] we have then that

$$\forall x \in X_1 \text{ we have that } I_{f,1}(x) = \int^S f(x, \cdot) d\mu_2 = \sum_{i=1}^n f_i \cdot \mu_2((A_i)_{[1, x]}) \quad (21.997)$$

$$\forall y \in X_2 \text{ we have that } I_{f,2}(y) = \int^S f(\cdot, y) d\mu_1 = \sum_{i=1}^n f_i \cdot \mu_1((A_i)_{[2, y]}) \quad (21.998)$$

Define

$$f_{1,A_i}: X_1 \rightarrow \overline{\mathbb{R}} \text{ where } f_{1,A_i}(x) = \mu_2((A_i)_{[1, x]}) \quad (21.999)$$

$$f_{2,A_i}: X_2 \rightarrow \overline{\mathbb{R}} \text{ where } f_{2,A_i}(y) = \mu_1((A_i)_{[2, y]}) \quad (21.1000)$$

then we have

$$I_{f,1} = \sum_{i=1}^n f_i \cdot f_{1,A_i} \text{ and } I_{f,2} = \sum_{i=1}^n f_i \cdot f_{2,A_i} \quad (21.1001)$$

Using [theorem: 21.556] we have that $\forall i \in \{1, \dots, n\}$

$$f_{1,A_i} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge f_{2,A_i} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1002)$$

and

$$(\mu_1 \otimes \mu_2)(A_i) = \int^+ f_{1,A_i} d\mu_1 = \int^+ f_{2,A_i} d\mu_2 \quad (21.1003)$$

Using then [theorems: 21.331, 21.332] on the above equations [eqs: 21.1001, 21.1002, 21.1003] that

$$I_{f,1} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge I_{f,2} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$$

and

$$\begin{aligned} \int^+ I_{f,1} d\mu_1 &= \sum_{i=1}^n f_i \cdot \int^+ f_{1,A_i} d\mu_1 \stackrel{\text{eq: 21.1003}}{=} \sum_{i=1}^n f_i \cdot (\mu_1 \otimes \mu_2)(A_i) \\ \int^+ I_{f,2} d\mu_2 &= \sum_{i=1}^n f_i \cdot \int^+ f_{2,A_i} d\mu_2 \stackrel{\text{eq: 21.1003}}{=} \sum_{i=1}^n f_i \cdot (\mu_1 \otimes \mu_2)(A_i) \end{aligned}$$

Combining the above with [eq: 21.993] we have that

$$\int^S f d(\mu_1 \otimes \mu_2) = \int^+ I_{f,1} d\mu_1 = \int^+ I_{f,2} d\mu_2$$

Finally using [theorem: 21.317] it follows that

$$f \in \mathcal{M}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}] \text{ and } \int^+ f d(\mu_1 \otimes \mu_2) = \int^+ I_{f,1} d\mu_1 = \int^+ I_{f,2} d\mu_2 \quad \square$$

Extending the above lemma to non negative measurable functions leads to Tonelli's Theorem.

Theorem 21.565. (Tonelli) *Let $\langle X_1, \mathcal{A}_1, \mu_1 \rangle$, $\langle X_2, \mathcal{A}_2, \mu_2 \rangle$ be measurable spaces such that μ_1, μ_2 are σ -finite and*

$$f: X_1 \cdot X_2 \rightarrow \overline{\mathbb{R}}$$

a function such that $f \in \mathcal{M}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}]$. Then we have for the functions

$$I_{f,1}: X_1 \rightarrow \overline{\mathbb{R}} \text{ defined by } I_{f,1}(x) = \int^+ f(x, \cdot) d\mu_2$$

$$I_{f,2}: X_2 \rightarrow \overline{\mathbb{R}} \text{ defined by } I_{f,2}(x) = \int^+ f(\cdot, x) d\mu_1$$

where

$$f(\cdot, y): X_1 \rightarrow \overline{\mathbb{R}} \text{ is defined by } f(\cdot, y)(x) = f(x, y)$$

$$f(x, \cdot): X_2 \rightarrow \overline{\mathbb{R}} \text{ is defined by } f(x, \cdot)(y) = f(x, y)$$

that

$$I_{f,1} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \wedge I_{f,2} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$$

and

$$\int^+ f d(\mu_1 \otimes \mu_2) = \int^+ I_{f,1} d\mu_1 = \int^+ I_{f,2} d\mu_2$$

Note 21.566. Using the convention [convention: 21.555] the above is equivalent with saying that

$$\int^+ f d(\mu_1 \otimes \mu_2) = \int^+ \left(\int^+ f(x, y) \mu(dy) \right) \mu(dx) = \int^+ \left(\int^+ f(x, y) \mu(dx) \right) \mu(dy)$$

Note 21.567. The condition that μ_1, μ_2 are σ -finite is necessary to ensure that $\mu_1 \otimes \mu_2$ is a measure on $\mathcal{A}_1 \otimes \mathcal{A}_2$ [see theorem: 21.556]

Note 21.568. By [lemma: 21.561] we have that $\forall x \in X_1 \quad f(x, \cdot) \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$ and $\forall y \in X_2$ that $f(\cdot, y) \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$ so that $\int^+ f(x, \cdot) d\mu_2$ and $\int^+ f(\cdot, y) d\mu_1$ are well defined.

Proof. As $f \in \mathcal{M}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}]$ there exist by [theorem: 21.329] a sequence

$$\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{S}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2] \quad (21.1004)$$

such that

$$\forall (x, y) \in X_1 \cdot X_2 \text{ we have } \forall i \in \mathbb{N} \text{ that } f_i(x, y) \leq f_{i+1}(x, y) \quad (21.1005)$$

$$\forall (x, y) \in X_1 \cdot X_2 \text{ we have } \lim_{i \rightarrow \infty} f_i(x, y) \text{ exist and } f(x, y) = \lim_{i \rightarrow \infty} f_i(x, y) \quad (21.1006)$$

$$\lim_{n \rightarrow \infty} \int^S f_i d(\mu_1 \otimes \mu_2) \text{ exist and } \int^+ f d(\mu_1 \otimes \mu_2) = \lim_{i \rightarrow \infty} \int^S f_i d(\mu_1 \otimes \mu_2) \quad (21.1007)$$

As $\int^+$ is a extension of $\int^S$ [see theorem: 21.317] we have that

$$\{f_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1008)$$

and

$$\lim_{i \rightarrow \infty} \int^+ f_i d(\mu_1 \otimes \mu_2) \text{ exist and } \int^+ f d(\mu_1 \otimes \mu_2) = \lim_{i \rightarrow \infty} \int^+ f_i d(\mu_1 \otimes \mu_2) \quad (21.1009)$$

Let $i \in \mathbb{N}$ then by the previous lemma [see lemma: 21.563] we have that $\forall x \in X_1$ and $\forall y \in X_2$

$$f_i(x, \cdot) \in \mathcal{S}^+[X_1, \mathcal{A}_1] \subseteq \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ and } f_i(\cdot, y) \in \mathcal{S}^+[X_2, \mathcal{A}_2] \subseteq \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1010)$$

and for the functions

$$I_{i,1}: X_1 \rightarrow \overline{\mathbb{R}} \text{ defined by } I_{i,1}(x) = \int^+ f_i(x, \cdot) d\mu_2 \quad (21.1011)$$

$$I_{i,2}: X_2 \rightarrow \overline{\mathbb{R}} \text{ defined by } I_{i,2}(y) = \int^+ f_i(\cdot, y) d\mu_1 \quad (21.1012)$$

that

$$\forall i \in \mathbb{N} \quad I_{i,1} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ and } I_{i,2} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1013)$$

and

$$\forall i \in \mathbb{N} \quad \int^+ f_i d(\mu_1 \otimes \mu_2) = \int^+ I_{i,1} d\mu_1 = \int^+ I_{i,2} d\mu_2 \quad (21.1014)$$

Further

$$\forall x \in X_1, \forall y \in X_2, \forall i \in \mathbb{N} \text{ we have } f_i(x, \cdot)(y) = f_i(x, y) \leq f_{i+1}(x, y) = f_{i+1}(x, \cdot)(y) \quad (21.1015)$$

$$\forall y \in X_2, \forall x \in X_1, \forall i \in \mathbb{N} \text{ we have } f_i(\cdot, y)(x) = f_i(x, y) \leq f_{i+1}(x, y) = f_{i+1}(\cdot, y)(x) \quad (21.1016)$$

Using now [theorem: 21.331] on the above proves that $\forall i \in \mathbb{N}$

$$\forall x \in X_1 \quad \int^+ f_i(x, \cdot) d\mu_2 \leq \int^+ f_{i+1}(x, \cdot) d\mu_2 \text{ and } \forall y \in X_2 \quad \int^+ f_i(\cdot, y) d\mu_1 \leq \int^+ f_{i+1}(\cdot, y) d\mu_1$$

from which it follows that

$$\forall x \in X_1, \forall i \in \mathbb{N} \quad I_{i,1}(x) \leq I_{i+1,1}(x) \text{ and } \forall y \in X_2, \forall i \in \mathbb{N} \quad I_{i,2}(y) \leq I_{i+1,2}(y) \quad (21.1017)$$

Thanks to [eqs: 21.1010, 21.1015, 21.1016] we can apply the Monotone Convergence Theorem [see theorem: 21.372] to prove that $\forall x \in X_1$ and $\forall y \in X_2$ the following functions are well defined

$$\forall x \in X_1 \quad \lim_{i \rightarrow \infty} f_i(x, \cdot) \text{ and that } \forall y \in X_2 \quad \lim_{i \rightarrow \infty} f_i(\cdot, y)$$

and

$$\lim_{i \rightarrow \infty} f_i(x, \cdot) \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \wedge \lim_{i \rightarrow \infty} f_i(\cdot, y) \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$$

and $\lim_{i \rightarrow \infty} \int^+ f_i(x, \cdot) d\mu_2, \lim_{i \rightarrow \infty} \int^+ f_i(\cdot, y) d\mu_1$ exist with

$$\lim_{i \rightarrow \infty} \int^+ f_i(x, \cdot) d\mu_2 = \int^+ \lim_{i \rightarrow \infty} f_i(x, \cdot) d\mu_2 \text{ and } \lim_{i \rightarrow \infty} \int^+ f_i(\cdot, y) d\mu_1 = \int^+ \lim_{i \rightarrow \infty} f_i(\cdot, y) d\mu_1 \quad (21.1018)$$

As we have that

$$\begin{aligned} \left(\lim_{i \rightarrow \infty} f_i(x, \cdot) \right)(y) &= \lim_{\text{def } i \rightarrow \infty} (f_i(x, \cdot)(y)) = \lim_{i \rightarrow \infty} f_i(x, y) \underset{[\text{eq: 21.1006}]}{=} f(x, y) = f(x, \cdot)(y) \\ \left(\lim_{i \rightarrow \infty} f_i(\cdot, y) \right)(x) &= \lim_{\text{def } i \rightarrow \infty} (f_i(\cdot, y)(x)) = \lim_{i \rightarrow \infty} f_i(x, y) \underset{[\text{eq: 21.1006}]}{=} f(x, y) = f(\cdot, y)(x) \end{aligned}$$

proving that

$$\forall x \in X_1 \text{ we have } f(x, \cdot) = \lim_{i \rightarrow \infty} f_i(x, \cdot) \text{ and } \forall y \in X_2 \text{ we have } f(\cdot, y) = \lim_{i \rightarrow \infty} f_i(\cdot, y) \quad (21.1019)$$

which combined with [eq: 21.1018] proves that

$$\forall x \in X_1 \quad \int^+ f(x, \cdot) d\mu_2 = \lim_{i \rightarrow \infty} \int^+ f_i(x, \cdot) d\mu_2 \wedge \forall y \in X_2 \quad \int^+ f(\cdot, y) d\mu_1 = \lim_{i \rightarrow \infty} \int^+ f_i(\cdot, y) d\mu_1 \quad (21.1020)$$

Substituting [eqs: 21.1011, 21.1012] in the above gives

$$\forall x \in X_1 \quad \int^+ f(x, \cdot) d\mu_2 = \lim_{i \rightarrow \infty} I_{i,1}(x) \text{ and } \forall y \in X_2 \quad \int^+ f(\cdot, y) d\mu_1 = \lim_{i \rightarrow \infty} I_{i,2}(y) \quad (21.1021)$$

Using [eqs: 21.1013, 21.1017] we can use the Monone Convergence Theorem [see theorem: 21.372] again to get that

$$\lim_{i \rightarrow \infty} I_{i,1} \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ and } \lim_{i \rightarrow \infty} I_{i,2} \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \subseteq \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1022)$$

$$\lim_{i \rightarrow \infty} \int^+ I_{i,1} d\mu_1, \lim_{i \rightarrow \infty} \int^+ I_{i,2} d\mu_2 \text{ exist}$$

with

$$\int^+ \left(\lim_{i \rightarrow \infty} I_{i,1} \right) d\mu_1 = \lim_{i \rightarrow \infty} \int^+ I_{i,1} d\mu_1 \text{ and } \int^+ \left(\lim_{i \rightarrow \infty} I_{i,2} \right) d\mu_2 = \lim_{i \rightarrow \infty} \int^+ I_{i,2} d\mu_2 \quad (21.1023)$$

Define now

$$I_{f,1}: X_1 \rightarrow \overline{\mathbb{R}} \text{ by } I_{f,1}(x) = \int^+ f(x, \cdot) d\mu_2$$

$$I_{f,2}: X_1 \rightarrow \overline{\mathbb{R}} \text{ by } I_{f,2}(y) = \int^+ f(\cdot, y) d\mu_1$$

then we have by [eq: 21.1021] that $\forall x \in X_1$ $I_{f,1}(x) = \lim_{i \rightarrow \infty} I_{i,1}(x)$ and $\forall y \in X_2$ $I_{f,2}(y) = \lim_{i \rightarrow \infty} I_{i,2}(y)$ from which it follows that

$$I_{f,1} = \lim_{i \rightarrow \infty} I_{i,1} \text{ and } I_{f,2} = \lim_{i \rightarrow \infty} I_{i,2}$$

Substituting the above in [eqs: 21.1022, 21.1023] proves that

$$I_{f,1} \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ and } I_{f,2} \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$$

and

$$\begin{aligned} \int^+ I_{f,1} d\mu_1 & \stackrel{[\text{eq: 21.1023}]}{=} \lim_{i \rightarrow \infty} \int^+ I_{i,1} d\mu_1 \\ & \stackrel{[\text{eq: 21.1014}]}{=} \lim_{i \rightarrow \infty} \int^+ f_i d(\mu_1 \otimes \mu_2) \\ & \stackrel{[\text{eq: 21.1009}]}{=} \int^+ f d(\mu_1 \otimes \mu_2) \\ \int^+ I_{f,2} d\mu_2 & \stackrel{[\text{eq: 21.1023}]}{=} \lim_{i \rightarrow \infty} \int^+ I_{i,2} d\mu_2 \\ & \stackrel{[\text{eq: 21.1014}]}{=} \lim_{i \rightarrow \infty} \int^+ f_i d(\mu_1 \otimes \mu_2) \\ & \stackrel{[\text{eq: 21.1009}]}{=} \int^+ f d(\mu_1 \otimes \mu_2) \end{aligned}$$

which proves finally that

$$\int^+ f d(\mu_1 \otimes \mu_2) = \int^+ I_{f,1} d\mu_1 = \int^+ I_{f,2} d\mu_2$$

□

Next we extend Tonelli's theorem to integrable functions. As f is integrable if $f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ and $\int^+ f^+ d\mu < \infty$, $\int^+ f^- d\mu < \infty$ it makes sense to examing $(f(x, \cdot))^+, (f(\cdot, y))^+, (f(x, \cdot))^-$ and $(f(\cdot, y))^-$ which we do in the next lemma.

Lemma 21.569. *Let X_1, Y_1 be sets and $f: X_1 \cdot X_2 \rightarrow \overline{\mathbb{R}}$ be a function then we have*

$$\forall x \in X_1 \quad (f^+)(x, \cdot) = (f(x, \cdot))^+ \text{ and } (f^-)(x, \cdot) = (f(x, \cdot))^-$$

$$\forall y \in X_2 \quad (f^+)(\cdot, y) = (f(\cdot, y))^+ \text{ and } (f^-)(\cdot, y) = (f(\cdot, y))^-$$

Proof. Let $x \in X_1$ then we have $\forall y \in X_2$ that

$$(f^+)(x, \cdot)(y) = f^+(x, y) = \max(f(x, y), 0) = \max(f(x, \cdot)(y), 0) = (f(x, \cdot))^+(y)$$

$$(f^-)(x, \cdot)(y) = f^-(x, y) = -\min(f(x, y), 0) = -\min(f(x, \cdot)(y), 0) = (f(x, \cdot))^- (y)$$

hence we have

$$\forall x \in X_1 \quad (f^+)(x, \cdot) = (f(x, \cdot))^+ \text{ and } (f^-)(x, \cdot) = (f(x, \cdot))^-$$

Likewise, let $y \in X_2$ then we have $\forall x \in X_1$ that

$$(f^+)(\cdot, y)(x) = f^+(x, y) = \max(f(x, y), 0) = \max(f(\cdot, y)(x), 0) = (f(\cdot, y))^+(x)$$

$$(f^-)(\cdot, y)(x) = f^-(x, y) = -\min(f(x, y), 0) = -\min(f(\cdot, y)(x), 0) = (f(\cdot, y))^- (x)$$

hence we have

$$\forall y \in X_2 \ (f^+)(\cdot, y) = (f(\cdot, y))^+ \text{ and } (f^-)(\cdot, y) = (f(\cdot, y))^- \quad \square$$

We are now ready to prove the extension of Tonalli's theorem to the general cases.

Theorem 21.570. (Fubini's Theorem) *Let $\langle X_1, \mathcal{A}_1, \mu_1 \rangle, \langle X_2, \mathcal{A}_2, \mu_2 \rangle$ be measurable spaces where μ_1, μ_2 are σ -finite and $f \in \mathcal{L}[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mu_1 \otimes \mu_2]$ then we have*

$$D_1 = \{x \in X_1 \mid f(x, \cdot) \in \mathcal{L}[X_2, \mathcal{A}_2, \mu_2]\} \in \mathcal{A}_1 \text{ and } \mu_1(X_1 \setminus D_1) = 0$$

$$D_2 = \{x \in X_2 \mid f(\cdot, x) \in \mathcal{L}[X_1, \mathcal{A}_1, \mu_1]\} \in \mathcal{A}_2 \text{ and } \mu_2(X_2 \setminus D_2) = 0$$

$$I_{f,1} \in \mathcal{L}[X_1, \mathcal{A}_1, \mu_1] \text{ where } I_{f,1}: X_1 \rightarrow \overline{\mathbb{R}} \text{ is defined by } I_{f,1}(x) = \begin{cases} \int f(x, \cdot) d\mu_2 & \text{if } x \in D_1 \\ 0 & \text{if } x \in X_1 \setminus D_1 \end{cases}$$

$$I_{f,2} \in \mathcal{L}[X_2, \mathcal{A}_2, \mu_2] \text{ where } I_{f,2}: X_2 \rightarrow \overline{\mathbb{R}} \text{ is defined by } I_{f,2}(x) = \begin{cases} \int f(\cdot, x) d\mu_1 & \text{if } x \in D_2 \\ 0 & \text{if } x \in X_2 \setminus D_2 \end{cases}$$

$$\int I_{f,1} d\mu_1 = \int I_{f,2} d\mu_2 = \int f d(\mu_1 \otimes \mu_2)$$

Note 21.571. *As $\forall x \in X_1 \setminus D_1$ we have $x \notin D_1 \Rightarrow f(x, \cdot) \notin \mathcal{L}[X_2, \mathcal{A}_2, \mu_2, \overline{\mathbb{R}}]$ and $\mu_1(X_1 \setminus D_1) = 0$ we have that*

$$f(x, \cdot) \text{ is } \mu_1\text{-a.e } \mu_2\text{-integrable}$$

Likewise, as $\forall x \in X_2 \setminus D_2$ we have $x \notin D_2 \Rightarrow f(\cdot, x) \notin \mathcal{L}[X_1, \mathcal{A}_1, \mu_1, \overline{\mathbb{R}}]$ and $\mu_2(X_2 \setminus D_2) = 0$ we have that

$$f(\cdot, x) \text{ is } \mu_2\text{-a.e } \mu_1\text{-integrable}$$

Proof. As $f \in \mathcal{L}[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mu_1 \otimes \mu_2]$ we have by [definition: 21.345] that

$$f \in \mathcal{M}[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}] \wedge f^+, f^- \in \mathcal{M}^+[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1024)$$

$$\int^+ f^+ d(\mu_1 \otimes \mu_2) < \infty \text{ and } \int^+ f^- d(\mu_1 \otimes \mu_2) < \infty \quad (21.1025)$$

$$\int f d(\mu_1 \otimes \mu_2) = \int^+ f^+ d(\mu_1 \otimes \mu_2) - \int^+ f^- d(\mu_1 \otimes \mu_2) \quad (21.1026)$$

Using Tonelli's Theorem [theorem: 21.565] on [eq: 21.1024] proves that

$$\forall x \in X_1 \ (f^+)(x, \cdot) \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \text{ and } \forall x \in X_2 \ (f^-)(\cdot, x) \in \mathcal{M}^+(X_1, \mathcal{A}_1, \overline{\mathbb{R}}) \quad (21.1027)$$

$$I_{f,1}^+ \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ where } I_{f,1}^+: X_1 \rightarrow \overline{\mathbb{R}} \text{ is defined by } I_{f,1}^+(x) = \int^+ (f^+)(x, \cdot) d\mu_2 \quad (21.1028)$$

$$I_{f,2}^+ \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \text{ where } I_{f,2}^+: X_2 \rightarrow \overline{\mathbb{R}} \text{ is defined by } I_{f,2}^+(x) = \int^+ (f^+)(\cdot, x) d\mu_1 \quad (21.1029)$$

$$I_{f,1}^- \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ where } I_{f,1}^-: X_1 \rightarrow \overline{\mathbb{R}} \text{ is defined by } I_{f,1}^-(x) = \int^+ (f^-)(x, \cdot) d\mu_2 \quad (21.1030)$$

$$I_{f,2}^- \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \text{ where } I_{f,2}^-: X_2 \rightarrow \overline{\mathbb{R}} \text{ is defined by } I_{f,2}^-(x) = \int^+ (f^-)(\cdot, x) d\mu_1 \quad (21.1031)$$

$$\int^+ I_{f,1}^+ d\mu_1 = \int^+ I_{f,2}^+ d\mu_2 = \int^+ f^+ d(\mu_1 \otimes \mu_2) \underset{[\text{eq: 21.1025}]}{<} \infty \quad (21.1032)$$

$$\int^+ I_{f,1}^- d\mu_1 = \int^+ I_{f,2}^- d\mu_2 = \int^+ f^- d(\mu_1 \otimes \mu_2) \underset{[\text{eq: 21.1025}]}{<} \infty \quad (21.1033)$$

Using [theorem: 21.336] on [eqs: 21.1028, 21.1029, 21.1030, 21.1031] and [eqs: 21.1032, 21.1033] results in

$$N_{f,1}^+ = \{x \in X_1 \mid I_{f,1}^+(x) \neq \infty\} \in \mathcal{A}_1 \wedge \mu(X_1 \setminus N_{f,1}^+) = 0 \quad (21.1034)$$

$$N_{f,2}^+ = \{x \in X_2 \mid I_{f,2}^+(x) \neq \infty\} \in \mathcal{A}_2 \wedge \mu(X_2 \setminus N_{f,2}^+) = 0 \quad (21.1035)$$

$$N_{f,1}^- = \{x \in X_1 \mid I_{f,1}^-(x) \neq \infty\} \in \mathcal{A}_1 \wedge \mu(X_1 \setminus N_{f,1}^-) = 0 \quad (21.1036)$$

$$N_{f,2}^- = \{x \in X_2 \mid I_{f,2}^-(x) \neq \infty\} \in \mathcal{A}_2 \wedge \mu(X_2 \setminus N_{f,2}^-) = 0 \quad (21.1037)$$

As $f \in \mathcal{M}[X_1 \cdot X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \overline{\mathbb{R}}]$ [see eq: 21.1024] it follows from [lemma: 21.561] that

$$\forall x \in X_1 \ f(x, \cdot) \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \text{ and } \forall x \in X_2 \ f(\cdot, x) \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \quad (21.1038)$$

Now we have the following equivalences

$$\begin{aligned} x \in D_1 &= \{x \in X_1 \mid f(x, \cdot) \in \mathcal{L}[X_2, \mathcal{A}_2, \mu_2, \overline{\mathbb{R}}]\} \\ &\stackrel{\updownarrow}{=} \\ &\quad f(x, \cdot) \in \mathcal{M}[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \text{ by [eq: 21.1038]} \\ &\int^+ (f(x, \cdot))^+ d\mu_2 < \infty \wedge \int^+ (f(x, \cdot))^- d\mu_2 < \infty \\ &\stackrel{\updownarrow}{=} \\ &\quad \text{[lemma: 21.569]} \\ &\int^+ (f^+)(x, \cdot) d\mu_2 < \infty \wedge \int^+ (f^-)(x, \cdot) d\mu_2 < \infty \\ &\stackrel{\updownarrow}{=} \\ &\quad \text{[eqs: 21.1028, 21.1030]} \\ &I_{f,1}^+(x) < \infty \wedge I_{f,1}^-(x) < \infty \\ &\stackrel{\updownarrow}{=} \\ &\quad \text{[eqs: 21.1034, 21.1036]} \\ &x \in N_{f,1}^+ \bigcap N_{f,1}^- \end{aligned}$$

and

$$\begin{aligned} x \in D_2 &= \{x \in X_2 \mid f(\cdot, x) \in \mathcal{L}[X_1, \mathcal{A}_1, \mu_1, \overline{\mathbb{R}}]\} \\ &\stackrel{\updownarrow}{=} \\ &\quad f(\cdot, x) \in \mathcal{M}[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \text{ by [eq: 21.1038]} \\ &\int^+ (f(\cdot, x))^+ d\mu_1 < \infty \wedge \int^+ (f(\cdot, x))^- d\mu_1 < \infty \\ &\stackrel{\updownarrow}{=} \\ &\quad \text{[lemma: 21.569]} \\ &\int^+ (f^+)(\cdot, x) d\mu_1 < \infty \wedge \int^+ (f^-)(\cdot, x) d\mu_1 < \infty \\ &\stackrel{\updownarrow}{=} \\ &\quad \text{[eqs: 21.1029, 21.1031]} \\ &I_{f,2}^+(x) < \infty \wedge I_{f,2}^-(x) < \infty \\ &\stackrel{\updownarrow}{=} \\ &\quad \text{[eqs: 21.1035, 21.1037]} \\ &x \in N_{f,2}^+ \bigcap N_{f,2}^- \end{aligned}$$

To summarize the above we have

$$D_1 = N_{f,1}^+ \bigcap N_{f,1}^- \text{ and } D_2 = N_{f,2}^+ \bigcap N_{f,2}^- \quad (21.1039)$$

Using [eqs: 21.1034, 21.1035, 21.1036, 21.1037] it follows that

$$X_1 \setminus (N_{f,1}^+ \bigcap N_{f,1}^-) \in \mathcal{A}_1 \text{ and } X_2 \setminus (N_{f,2}^+ \bigcap N_{f,2}^-) \in \mathcal{A}_2$$

Further

$$\mu_1(X_1 \setminus (N_{f,1}^+ \cap N_{f,1}^-)) = \mu_1((X_1 \setminus N_{f,1}^+) \cup (X_1 \setminus N_{f,1}^-)) \leq \mu_1(X_1 \setminus N_{f,1}^+) + \mu_1(X_1 \setminus N_{f,1}^-) = 0$$

$$\mu_2(X_2 \setminus (N_{f,2}^+ \cap N_{f,2}^-)) = \mu_2((X_2 \setminus N_{f,2}^+) \cup (X_2 \setminus N_{f,2}^-)) \leq \mu_2(X_2 \setminus N_{f,2}^+) + \mu_2(X_2 \setminus N_{f,2}^-) = 0$$

which combined with [eq: 21.1039] proves that

$$D_1 = \{x \in X_1 \mid f(x, \cdot) \in \mathcal{L}[X_2, \mathcal{A}_2, \mu_2, \overline{\mathbb{R}}]\} \in \mathcal{A}_1 \text{ and } \mu_1(X_1 \setminus D_1) = 0 \quad (21.1040)$$

$$D_2 = \{x \in X_2 \mid f(\cdot, x) \in \mathcal{L}[X_1, \mathcal{A}_1, \mu_1, \overline{\mathbb{R}}]\} \in \mathcal{A}_2 \text{ and } \mu_2(X_2 \setminus D_2) = 0 \quad (21.1041)$$

The above allows us to define

$$I_{f,1}: X_1 \rightarrow \overline{\mathbb{R}} \text{ by } I_{f,1}(x) = \begin{cases} \int f(x, \cdot) d\mu_2 & \text{if } x \in D_1 \\ 0 & \text{if } x \in X_1 \setminus D_1 \end{cases} \quad (21.1042)$$

$$I_{f,2}: X_2 \rightarrow \overline{\mathbb{R}} \text{ by } I_{f,2}(x) = \begin{cases} \int f(\cdot, x) d\mu_1 & \text{if } x \in D_2 \\ 0 & \text{if } x \in X_2 \setminus D_2 \end{cases} \quad (21.1043)$$

Take

$$N_1 = X_1 \setminus D_1 \in \mathcal{A}_1 \wedge N_2 = X_2 \setminus D_2 \in \mathcal{A}_2 \text{ so that } X_1 \setminus N_1 = D_1 \wedge X_2 \setminus N_2 = D_2 \quad (21.1044)$$

As by [eqs: 21.1028, 21.1030] and [eqs: 21.1029, 21.1031] $I_{f,1}^+, I_{f,1}^- \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}]$ and $I_{f,2}^+, I_{f,2}^- \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}]$ it follows from [theorem: 21.276] that

$$\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^+, \mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^- \in \mathcal{M}^+[X_1, \mathcal{A}_1, \overline{\mathbb{R}}] \quad (21.1045)$$

$$\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^+, \mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^- \in \mathcal{M}^+[X_2, \mathcal{A}_2, \overline{\mathbb{R}}] \quad (21.1046)$$

Further we have as $\mu_1(N_1) = 0 = \mu_2(N_2)$ that by [theorems: 21.280, 21.333]

$$\int^+ (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^+) d\mu_1 = \int^+ I_{f,1}^+ d\mu_1 \underset{[\text{eq: 21.1032}]}{<} \infty \quad (21.1047)$$

$$\int^+ (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^-) d\mu_1 = \int^+ I_{f,1}^- d\mu_1 \underset{[\text{eq: 21.1033}]}{<} \infty \quad (21.1048)$$

$$\int^+ (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^+) d\mu_2 = \int^+ I_{f,2}^+ d\mu_2 \underset{[\text{eq: 21.1032}]}{<} \infty \quad (21.1049)$$

$$\int^+ (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^-) d\mu_2 = \int^+ I_{f,2}^- d\mu_2 \underset{[\text{eq: 21.1033}]}{<} \infty \quad (21.1050)$$

Further we have for $x \in X_1$ either:

$x \in N_1$. Then $x \in X_1 \setminus D_1$ so that

$$I_{f,1}(x) \underset{[\text{eq: 21.1042}]}{=} 0 = 0 - 0 = (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^+ - \mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^-)(x)$$

$x \in X_1 \setminus N_1$. Then $x \in D_1$ so that

$$\begin{aligned} I_{f,1}(x) &\underset{[\text{eq: 21.1042}]}{=} \int f(x, \cdot) d\mu_2 \\ &= \int^+ (f(x, \cdot))^+ d\mu_2 - \int^+ (f(x, \cdot))^- d\mu_2 \\ &\underset{[\text{lemma: 21.569}]}{=} \int^+ (f^+)(x, \cdot) d\mu_2 - \int^+ (f^-)(x, \cdot) d\mu_2 \\ &\underset{[\text{eqs: 21.1028, 21.1030}]}{=} I_{f,1}^+(x) - I_{f,1}^-(x) \\ &= (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^+ - \mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^-)(x) \end{aligned}$$

Likewise if $x \in X_1$ we have either:

$x \in N_2$. Then $x \in X_2 \setminus D_2$ so that

$$I_{f,2}(x) \underset{[\text{eq: 21.1043}]}{=} 0 = 0 - 0 = (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot f_{f,2}^+ - \mathcal{X}_{X_2, X_2 \setminus N_2} \cdot f_{f,2}^-)(x)$$

$x \in X_2 \setminus N_2$. Then $x \in D_2$ so that

$$\begin{aligned} I_{f,2}(x) &\underset{[\text{eq: 21.1043}]}{=} \int f(., x) d\mu_1 \\ &= \int^+ (f(., x))^+ d\mu_1 - \int^+ (f(., x))^- d\mu_1 \\ &\underset{[\text{lemma: 21.569}]}{=} \int^+ (f^+)(., x) d\mu_1 - \int^+ (f^-)(., x) d\mu_1 \\ &\underset{[\text{eqs: 21.1029, 21.1031}]}{=} I_{f,2}^+(x) - I_{f,2}^-(x) \\ &= (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot f_{f,2}^+ - \mathcal{X}_{X_2, X_2 \setminus N_2} \cdot f_{f,2}^-)(x) \end{aligned}$$

From the above it follows that

$$I_{f,1} = \mathcal{X}_{X_1, X_1 \setminus N_1} \cdot f_{f,1}^+ - \mathcal{X}_{X_1, X_1 \setminus N_1} \cdot f_{f,1}^- \text{ and } I_{f,2} = \mathcal{X}_{X_2, X_2 \setminus N_2} \cdot f_{f,2}^+ - \mathcal{X}_{X_2, X_2 \setminus N_2} \cdot f_{f,2}^- \quad (21.1051)$$

Taking in account [eqs: 21.1045, 21.1046, 21.1047, 21.1048, 21.1049, 21.1050 and 21.1051] we can use [theorem: 21.351(2)] to get

$$I_{f,1} \in \mathcal{L}[X_1, \mathcal{A}_1, \mu_1, \overline{\mathbb{R}}] \text{ and } \int I_{f,1} d\mu_1 = \int^+ (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^+) d\mu_1 - \int^+ (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^-) d\mu_1 \quad (21.1052)$$

$$I_{f,2} \in \mathcal{L}[X_2, \mathcal{A}_2, \mu_2, \overline{\mathbb{R}}] \text{ and } \int I_{f,2} d\mu_2 = \int^+ (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^+) d\mu_2 - \int^+ (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^-) d\mu_2 \quad (21.1053)$$

Now

$$\begin{aligned} &\int^+ (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^+) d\mu_1 - \int^+ (\mathcal{X}_{X_1, X_1 \setminus N_1} \cdot I_{f,1}^-) d\mu_1 \underset{[\text{eq: 21.1047, 21.1048}]}{=} \\ &\quad \int^+ I_{f,1}^+ d\mu_1 - \int^+ I_{f,1}^- d\mu_1 \underset{[\text{eq: 21.1032, 21.1033}]}{=} \\ &\quad \int^+ f^+ d(\mu_1 \otimes \mu_2) - \int^+ f^- d(\mu_1 \otimes \mu_2) \underset{[\text{eq: 21.1026}]}{=} \\ &\quad \int f d(\mu_1 \otimes \mu_2) \\ &\int^+ (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^+) d\mu_2 - \int^+ (\mathcal{X}_{X_2, X_2 \setminus N_2} \cdot I_{f,2}^-) d\mu_2 \underset{[\text{eq: 21.1049, 21.1050}]}{=} \\ &\quad \int^+ I_{f,2}^+ d\mu_2 - \int^+ I_{f,2}^- d\mu_2 \underset{[\text{eq: 21.1032, 21.1033}]}{=} \\ &\quad \int^+ f^+ d(\mu_1 \otimes \mu_2) - \int^+ f^- d(\mu_1 \otimes \mu_2) \underset{[\text{eq: 21.1026}]}{=} \\ &\quad \int f d(\mu_1 \otimes \mu_2) \end{aligned}$$

which combined with [eqs: 21.1052, 21.1053] gives finally:

$$\int I_{f,1} d\mu_1 = \int I_{f,2} d\mu_2 = \int f d(\mu_1 \otimes \mu_2)$$

□

21.8 Change of variables

In this section we see how measures and measure integral behaves under a change of variables. This study can be divided in three subsections:

Borel measures and regularity. Here we study the relation between topology and measure theory.

Pushforward or image measures. Given two measurable spaces $\langle X, \mathcal{A} \rangle$ and $\langle Y, \mathcal{B} \rangle$ and a measurable function $T: X \rightarrow Y$ (representing the change of variable) we define given a measure μ on $\mathcal{A}$ the pushforward [or image] measure $\mu[T^{-1}]$ on $\mathcal{B}$ by $\mu[T^{-1}](A) = \mu(T^{-1}(A))$. In the special cases of λ and λ^n we use the theory of Borel measures and regularity we formulate expressions that relate $\lambda[T^{-1}]$, $\lambda^n[T^{-1}]$ and λ , λ^n .

Integrals and change of variables. Here we proof the general change of variable formula

$$\int (f \circ T) d\mu = \int f d(\mu[T^{-1}])$$

and look at more specific cases for the Borel measures λ and λ^n and change of variables.

21.8.1 Regularity and Radon measures

21.8.1.1 Definition of Regular and Radon measures

First we define Borel measures and Borel spaces.

Definition 21.572. Let $\langle X, \mathcal{T} \rangle$ be a topological spaces then a measure $\mu: \mathcal{B}[X, \mathcal{T}] \rightarrow [0, \infty]$ where $\mathcal{B}[X, \mathcal{T}] = \sigma[\mathcal{T}]$ is the Borel Algebra on X [see definition: 21.46] is called a **Borel measure**. The measure space $\langle X, \mathcal{B}[X, \mathcal{T}], \mu \rangle$ is called a **Borel space**.

We have the following concrete examples of Borel spaces for $\mathbb{R}$ and $\mathbb{R}^n$.

Example 21.573. $\langle \mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda \rangle$ and $\langle \mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n \rangle$ are Borel spaces.

Proof. This follows from [theorems: 21.182, 21.205]. □

For finite Borel measures on normed spaces we have that the measure of a Borel set is only depended on the measure of the closed sets contained in the Borel set or the open sets containing the Borel set. To prove this we need first two lemma's.

Lemma 21.574. Let $\langle X, \mathcal{T} \rangle$ be a topological space and $\mu: \mathcal{B}[X, \mathcal{T}] \rightarrow [0, \infty]$ a **finite** measure on the Borel algebra $\mathcal{B}[X, \mathcal{T}] = \sigma[\mathcal{T}]$ and $A \in \mathcal{B}[X, \mathcal{T}]$ then we have

$$\mu(A) = \sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$$

$$\Updownarrow$$

$$\forall \varepsilon \in \mathbb{R}^+ \exists U \text{ open}, \exists F \text{ closed with } F \subseteq A \subseteq U \text{ and } \mu(U \setminus F) < \varepsilon$$

Note 21.575. Using [theorem: 21.47] it follows that $\mathcal{B}[X, \mathcal{T}]$ contains all open en closed sets so that the above equivalences make sense. Further as $\emptyset, X$ are open and closed we have that $\forall A \in \mathcal{B}[X, \mathcal{T}] \emptyset \in \{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}$ and $X \in \{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$ so that

$$\sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}) \text{ and } \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$$

are well defined.

Proof.

$\Rightarrow$. Let $A \in \mathcal{B}[X, \mathcal{T}]$ and take $\varepsilon \in \mathbb{R}^+$. As $\mu(A)$ is finite we have

$$\mu(A) - \frac{\varepsilon}{2} < \mu(A) = \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\})$$

so that there exist a F closed with $F \subseteq A$ such that

$$\mu(A) - \frac{\varepsilon}{2} < \mu(F) \Rightarrow -\mu(F) < -\mu(A) + \frac{\varepsilon}{2}$$

Likewise we have

$$\inf(\{\mu(U) \mid A \subseteq U \wedge U \text{ is open}\}) < \mu(A) + \frac{\varepsilon}{2}$$

so that there exist a U open with $A \subseteq U$ such that

$$\mu(U) < \mu(A) + \frac{\varepsilon}{2}$$

Hence we have that

$$F \subseteq A \subseteq U$$

Further as μ is finite we have

$$\mu(U \setminus F) \underset{[\text{theorem: 21.99}]}{=} \mu(U) - \mu(F) < \mu(A) + \frac{\varepsilon}{2} - \mu(A) + \frac{\varepsilon}{2} = \varepsilon$$

hence

$$\mu(U \setminus F) < \varepsilon$$

$\Leftarrow$. Let $A \in \mathcal{B}[X, \mathcal{T}]$. Let F be a closed set such that $F \subseteq A$ then using [theorem: 21.99] we have that $\mu(F) \leq \mu(A)$. Hence $\mu(A)$ is an upper bound of $\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}$ so that

$$\sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}) \leq \mu(A) \quad (21.1054)$$

Assume that $\sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}) < \mu(A)$ then

$$\varepsilon = \mu(A) - \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}) \in \mathbb{R}^+$$

By the hypothesis there exist a U open and a F closed such that

$$F \subseteq A \subseteq U \text{ and } \mu(U \setminus F) < \varepsilon$$

As $A \setminus F \subseteq U \setminus F$ we have by [theorem: 21.99] that $\mu(A) - \mu(F) = \mu(A \setminus F) \leq \mu(U \setminus F) < \varepsilon$ so that

$$\begin{aligned} \mu(A) &< \mu(F) + \varepsilon \\ &\leq \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}) + \varepsilon \\ &= \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}) + \mu(A) - \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\}) \\ &= \mu(A) \end{aligned}$$

which is a contradiction, hence the assumption is wrong and we must have that $\mu(A) \leq \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\})$. Combining this with [eq: 21.1054] gives

$$\mu(A) = \sup(\{\mu(F) \mid F \subseteq A \wedge F \text{ is closed}\})$$

Let U be an open set such that $A \subseteq U$ then using [theorem: 21.99] we have $\mu(A) \leq \mu(U)$. Hence $\mu(A)$ is a lower bound for $\{\mu(U) \mid A \subseteq U \wedge U \text{ is open}\}$ so that

$$\mu(A) \leq \inf(\{\mu(U) \mid A \subseteq U \wedge U \text{ is open}\}) \quad (21.1055)$$

Assume that $\mu(A) < \inf(\{\mu(U) \mid A \subseteq U \wedge U \text{ is open}\})$ then

$$\varepsilon = \inf(\{\mu(U) \mid A \subseteq U \wedge U \text{ is open}\}) - \mu(A) \in \mathbb{R}^+$$

So by the hypothesis there exist a U open and a F closed such that

$$F \subseteq A \subseteq U \text{ and } \mu(U \setminus F) < \varepsilon$$

As $F \subseteq A$ we have $U \setminus A \subseteq U \setminus F$ so that by [theorem: 21.99]

$$\mu(U) - \mu(A) = \mu(U \setminus A) \leq \mu(U \setminus F) < \varepsilon$$

so that

$$\begin{aligned} \mu(A) &> \mu(U) - \varepsilon \\ &\geq \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - \varepsilon \\ &= \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - (\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - \mu(A)) \\ &= \mu(A) \end{aligned}$$

which is a contradiction, hence the assumption is wrong and we must have that $\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \leq \mu(A)$. Combining this with [eq: 21.1055] proves that

$$\mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \quad \square$$

Lemma 21.576. *Let $\langle X, \|\cdot\| \rangle$ be a normed space with the norm topology $\mathcal{T}_{\|\cdot\|}$ and $\mu: \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}] \rightarrow [0, \infty]$ a finite Borel measure on the Borel algebra and $A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$ then $\forall \varepsilon \in \mathbb{R}^+$ there exist a open set U and a closed set F with $F \subseteq A \subseteq U$ such that $\mu(U \setminus F) < \varepsilon$.*

Proof. First note that by [theorem: 21.47] every open or closed set used in this proof is a element of $\mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$. Consider the set

$$\mathcal{A} = \{A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}] | \forall \varepsilon \in \mathbb{R}^+ \exists U \text{ open, } \exists F \text{ closed with } F \subseteq A \subseteq U \text{ such that } \mu(U \setminus F) < \varepsilon\} \quad (21.1056)$$

We prove now that $\mathcal{A}$ is a σ -algebra on X . As X is open and closed, $X \subseteq X \subseteq X$ and $\mu(X \setminus X) = \mu(\emptyset) = 0 < \varepsilon \forall \varepsilon \in \mathbb{R}^+$ it follows that

$$X \in \mathcal{A} \quad (21.1057)$$

Let $A \in \mathcal{A}$ then given $\varepsilon \in \mathbb{R}^+$ there exists a open set U , a closed set F such that $F \subseteq A \subseteq U$ and $\mu(U \setminus F) < \varepsilon$. Take $F' = X \setminus U$ and $U' = X \setminus F$ then U' is a open set and F' is a closed set, further as $F \subseteq A \subseteq U$ we have $F' = X \setminus U \subseteq X \setminus A \subseteq X \setminus F = U'$, also

$$\mu(U' \setminus F') = \mu((X \setminus F) \setminus (X \setminus U)) \stackrel{[\text{theorem: 1.36(11)}]}{=} \mu(U \setminus F) < \varepsilon$$

Hence we conclude that $X \setminus A \in \mathcal{A}$ so that

$$\forall A \in \mathcal{A} \text{ we have } X \setminus A \in \mathcal{A} \quad (21.1058)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ and take $\varepsilon \in \mathbb{R}^+$. Given $n \in \mathbb{N}$ we have as $A_n \in \mathcal{A}$ that there exists a open set F_n and a open set U_n such that

$$F_n \subseteq A_n \subseteq U_n \text{ and } \mu(U_n \setminus F_n) < \frac{\varepsilon}{2^{n+1}} \quad (21.1059)$$

Take $U = \bigcup_{n \in \mathbb{N}} U_n$ then U is a open set and $U \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$. Further take $S = \bigcup_{n \in \mathbb{N}} F_n$ [which is in general not closed] then we have that $S \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$ so that

$$U \setminus S \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$$

From the definition of U, S and [eq: 21.1059] we have

$$S \subseteq \bigcup_{n \in \mathbb{N}} A_n \subseteq U \quad (21.1060)$$

Further

$$\begin{aligned}
 \mu(U \setminus S) &= \mu\left(\left(\bigcup_{n \in \mathbb{N}} U_n\right) \setminus S\right) \\
 &= \mu\left(\bigcup_{n \in \mathbb{N}} (U_n \setminus S)\right) \\
 &\stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{n=1}^{\infty} \mu(U_n \setminus S) \\
 &\stackrel{F_n \subseteq S \Rightarrow U_n \setminus S \subseteq U \setminus F_n}{\leq} \sum_{n=1}^{\infty} \mu(U_n \setminus F_n) \\
 &= \sum_{n=1}^{\infty} \frac{\varepsilon}{2^{n+1}} \\
 &= \frac{\varepsilon}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{2^n} \\
 &\stackrel{[\text{example: 14.412}]}{=} \frac{\varepsilon}{2}
 \end{aligned}$$

which proves that

$$\mu(U \setminus S) < \frac{\varepsilon}{2} \quad (21.1061)$$

If S would be closed then we are done but unfortunately this is not always true. To solve this we find a closed set that approximate S from below. Given $n \in \mathbb{N}$ consider the closed set $S_n = \bigcup_{i \in \{1, \dots, n\}} F_n$ then by [lemma: 21.7] we have that

$$S = \bigcup_{n \in \mathbb{N}} S_n \text{ and } \forall n \in \mathbb{N} \ S_n \leq S_{n+1} \text{ and as } \{F_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have also } S_n \in \mathcal{A}$$

Using then [theorem: 21.115] on the above we have

$$\mu(S) = \lim_{n \rightarrow \infty} \mu(S_n) = \sup(\{\mu(S_n) | n \in \mathbb{N}\})$$

As μ is finite we have by [theorem: 20.79] that $\exists N \in \mathbb{N}$ such that $\forall n \in \{N, \dots, \infty\} \ |\mu(S) - \mu(S_n)| < \frac{\varepsilon}{2}$, as $\mu(S_N) \leq \sup(\{\mu(S_n) | n \in \mathbb{N}\}) = \mu(S)$ we have that

$$S_N \in \mathcal{A}, \ S_N \subseteq S \text{ and } \mu(S \setminus S_N) \stackrel{\mu \text{ is finite} \wedge [\text{theorem: 21.99}]}{=} \mu(S) - \mu(S_N) < \frac{\varepsilon}{2} \quad (21.1062)$$

Now

$$\begin{aligned}
 U \setminus S_N &\stackrel{S_N \subseteq U}{=} ((U \setminus S) \bigsqcup S) \setminus S_N \\
 &= ((U \setminus S) \setminus S_N) \bigsqcup (S \setminus S_N) \\
 &\stackrel{[\text{theorem: 1.34}]}{=} (U \setminus (S \cup S_N)) \bigsqcup (S \setminus S_N) \\
 &\stackrel{S_N \subseteq S}{=} (U \setminus S_N) \bigsqcup (S \setminus S_N)
 \end{aligned}$$

so that

$$\mu(U \setminus S_N) = \mu(U \setminus S_N) + \mu(S \setminus S_N) \stackrel{[\text{eq: 21.1061, 21.1062}]}{<} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon$$

Further using [eq: 21.1061] we have

$$S_N \subseteq S \subseteq \bigcup_{n \in \mathbb{N}} A_n \subseteq U$$

which proves that $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A}$. Hence

$$\forall \{A_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A} \text{ we have } \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A} \quad (21.1063)$$

From [eqs: 21.1057, 21.1058 and 21.1063] it follows that

$$\mathcal{A} \text{ is a } \sigma\text{-algebra} \quad (21.1064)$$

The next logical step is to prove that $\mathcal{T} \subseteq \mathcal{A}$. Let $U \in \mathcal{T}_{|||}$ then we have two possible cases:

$X \setminus U = \emptyset$. Then $U = X \in \mathcal{A}$

$X \setminus U \neq \emptyset$. Consider the distance function [see definition: 14.183]

$$\delta_{X \setminus U}: X \rightarrow \mathbb{R} \text{ where } \delta_{X \setminus U}(x) = \inf(\{\|x - y\| \mid y \in X \setminus U\})$$

Define now

$$\{F_n\}_{n \in \mathbb{N}} \text{ by } F_n = \delta_{X \setminus U}^{-1}\left(\left[\frac{1}{n}, \infty\right]\right)$$

Let $n \in \mathbb{N}$. Using the fact that $\left[\frac{1}{n}, \infty\right]$ is closed in $\mathcal{T}_{|||}$ [see theorem: 14.105], $\delta_{X \setminus U}$ is continuous [see theorem: 14.185] it follows from [theorem: 14.154] that F_n is a closed set. Further if $x \in F_n$ then $\delta_{X \setminus U}(x) \in \left[\frac{1}{n}, \infty\right]$ so that $\frac{1}{n+1} < \frac{1}{n} \leq \delta_{X \setminus U}(x) \Rightarrow \delta_{X \setminus U}(x) \in \left[\frac{1}{n+1}, \infty\right]$ or $x \in F_{n+1}$. To summarize

$$\forall n \in \mathbb{N} \text{ we have } F_n \text{ is closed and } F_n \subseteq F_{n+1} \quad (21.1065)$$

Let $x \in F_n$ then $0 < \frac{1}{n} \leq \delta_{X \setminus U}(x)$. Assume that $x \in X \setminus U$ then $0 \in \{\|x - y\| \mid y \in X \setminus U\}$ so that $\delta_{X \setminus U}(x) = \inf(\{\|x - y\| \mid y \in X \setminus U\}) \leq 0$ contradicting $0 < \delta_{X \setminus U}(x)$, hence the assumption is false and we must have that $x \notin X \setminus U$ which proves that $x \in U$. So we conclude that

$$\forall n \in \mathbb{N} \ F_n \subseteq U \text{ from which it follows that } \bigcup_{n \in \mathbb{N}} F_n \subseteq U \quad (21.1066)$$

If now $x \in U$ then there exist a $\delta \in \mathbb{R}^+$ such that $B_{|||}(x, \delta) \subseteq U$, hence $\forall y \in X \setminus U$ we have that $y \notin B_{|||}(x, \delta) \Rightarrow \delta \leq \|x - y\|$ so that δ is a lower bound for $\{\|x - y\| \mid y \in X \setminus U\}$. In other words

$$0 < \delta \leq \delta_{X \setminus U}(x) = \inf(\{\|x - y\| \mid y \in X \setminus U\})$$

Using the Archimedean property of the real numbers [see theorem: 10.30] there exist a $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \delta \leq \delta_{X \setminus U}(x)$, hence $x \in \delta_{X \setminus U}^{-1}\left(\left[\frac{1}{n}, \infty\right]\right) = F_n \subseteq \bigcup_{n \in \mathbb{N}} F_n$, this proves that $U \subseteq \bigcup_{n \in \mathbb{N}} F_n$ which combined with [eq: 21.1066] results in

$$U = \bigcup_{n \in \mathbb{N}} F_n \quad (21.1067)$$

Consider now $\{U \setminus F_n\}_{n \in \mathbb{N}} \subseteq \mathcal{B}[X, \mathcal{T}_{|||}]$ then we have

$$\forall n \in \mathbb{N} \text{ we have } U \setminus F_{n+1} \stackrel{[\text{eq: 21.1065}]}{\subseteq} U \setminus F_n$$

and

$$\bigcap_{n \in \mathbb{N}} (U \setminus F_n) = U \setminus \left(\bigcup_{n \in \mathbb{N}} F_n\right) \stackrel{[\text{eq: 21.1067}]}{=} U \setminus U = \emptyset$$

So the conditions for [theorem: 21.115] are satisfied [note that μ is finite] from which it follows that

$$\lim_{n \rightarrow \infty} \mu(U \setminus F_n) \text{ exist and } 0 = \mu(\emptyset) = \lim_{n \rightarrow \infty} \mu(U \setminus F_n)$$

Hence we have

$$\forall \varepsilon \in \mathbb{R}^+ \ \exists N \in \mathbb{N} \text{ such that } \mu(U \setminus F_N) < \varepsilon$$

As also $F_N \subseteq \bigcup_{n \in \mathbb{N}} F_n = U \subseteq U$ and F_N is closed [see eq: 21.1065] it follows that

$$U \in \mathcal{A}$$

So in all cases we have $U \in \mathcal{A}$. In other words we have proved that

$$\mathcal{T}_{|||} \subseteq \mathcal{A}$$

As we have already proved that $\mathcal{A}$ is a σ -algebra on X [see eq: 21.1064] it follows from [theorem: 21.42] that $\mathcal{B}[X, \mathcal{T}_{|||}] = \sigma[\mathcal{T}_{|||}] \subseteq \mathcal{A}$. Hence if $A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ then $A \in \mathcal{A}$ so that

$\forall \varepsilon$ there exists a closed set F and a open set U such that $F \subseteq A \subseteq U$ and $\mu(U \setminus F) < \varepsilon$ $\square$

The above two lemmas can be summarized in the following theorem.

Theorem 21.577. *Let $\langle X, ||| \rangle$ be a normed space with topology $\mathcal{T}_{|||}$ and $\mu: \mathcal{B}[X, \mathcal{T}_{|||}] \rightarrow [0, \infty]$ a finite measure then $\forall A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ we have*

$$\mu(A) = \sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$$

Proof. This follows directly from [lemma: 21.574] and [lemma: 21.576]. $\square$

Unfortunately we can not use the above corollary on the Borel measures λ and λ^n because these measures are not finite. For this we introduce the concept of locally finite measures, compact finite measures, regular measures and Radon measures.

Definition 21.578. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\mathcal{A}$ a σ -algebra on X with $\mathcal{B}[X, \mathcal{T}] \subseteq \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a measure then we define:*

1. μ is **locally finite** if $\forall x \in X$ there exist a open set U such that $x \in U$ and $\mu(U) < \infty$
2. μ is **external regular** if $\forall A \in \mathcal{A}$ we have that $\mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$
3. μ is **internal regular** if $\forall A \in \mathcal{A}$ we have that $\mu(A) = \sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\})$

Note 21.579. *As $X, \emptyset$ are open and closed and $\forall A \in \mathcal{A}$ we have $\emptyset \subseteq A \subseteq X$ we have that $\emptyset \in \{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}$ and $X \in \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ it follows that*

$\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ exist and $\sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\})$ exist

Definition 21.580. *Let $\langle X, \mathcal{T} \rangle$ be a **Hausdorff** topological space, $\mathcal{A}$ a σ -algebra on X with $\mathcal{B}[X, \mathcal{T}] \subseteq \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a measure then we define:*

1. μ is **compact finite** if $\mu(K) < \infty$ for every compact set.
2. μ is **compact internal regular** if $\forall A \in \mathcal{A}$ we have

$$\mu(A) = \sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\})$$

3. μ is **regular** if μ is **external regular** and **compact internal regular**.
4. μ is a **Radon measure** if μ is **locally finite** and **regular**.

Note 21.581. *The Hausdorff condition is essential as it ensures by [theorem: 14.276] that a compact set K is closed, hence a element of $\mathcal{A}$ so that $\mu(K)$ makes sense. Further as $\emptyset$ is compact [see example: 14.269] and $\forall A \in \mathcal{A}$ $\emptyset \subseteq A$ we have that $\emptyset \in \{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}$ so that $\sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\})$ is well defined.*

Theorem 21.582. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\mathcal{A}$ a σ -algebra on X with $\mathcal{B}[X, \mathcal{T}] \subseteq \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a measure, $\alpha \in [0, \infty[$ then we have for the measure $\alpha \cdot \mu$ [see theorem: 21.108] that:*

1. If $\mathcal{T}$ is **Hausdorff** and μ is compact finite then $\alpha \cdot \mu$ is compact finite.
2. If $\mathcal{T}$ is **Hausdorff** and μ is compact internal regular then $\alpha \cdot \mu$ is compact internal regular.
3. If μ is locally finite then $\alpha \cdot \mu$ is locally finite.
4. If μ is external regular then $\alpha \cdot \mu$ is external regular.
5. If μ is internal regular then $\alpha \cdot \mu$ is internal regular.
6. If $\mathcal{T}$ is **Hausdorff** and μ is regular then $\alpha \cdot \mu$ is regular.
7. If $\mathcal{T}$ is **Hausdorff** and μ is a Radon measure then $\alpha \cdot \mu$ is a Radon measure.

Proof.

1. As μ is compact finite we have for every compact set K that $\mu(K) < \infty$ so that $(\alpha \cdot \mu)(K) = \alpha \cdot \mu(K) < \infty$ from which it follows that $\alpha \cdot \mu$ is compact finite.
2. Let $A \in \mathcal{A}$ then as μ is compact internal regular we have that

$$\mu(A) = \sup(\{\mu(K) | C \subseteq A \wedge K \text{ is compact}\}).$$

Hence we have

$$\begin{aligned} (\alpha \cdot \mu)(A) &= \alpha \cdot \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) \\ &\stackrel{[\text{theorem: 10.22}]}{=} \sup(\{\alpha \cdot \mu(K) | K \subseteq A \wedge K \text{ is compact}\}) \\ &= \sup(\{(\alpha \cdot \mu)(K) | C \subseteq A \wedge K \text{ is compact}\}) \end{aligned}$$

3. If $x \in X$ then as μ is locally finite there exist a open set U such that $x \in U$ and $\mu(U) < \infty$ so that $(\alpha \cdot \mu)(U) = \alpha \cdot \mu(U) < \infty$ which proves that $\alpha \cdot \mu$ is locally finite.
4. Let $A \in \mathcal{A}$ then as μ is external regular we have that $\mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$. Hence we have

$$\begin{aligned} (\alpha \cdot \mu)(A) &= \alpha \cdot \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \\ &\stackrel{[\text{theorem: 10.22}]}{=} \inf(\{\alpha \cdot \mu(U) | A \subseteq U \wedge U \text{ is open}\}) \\ &= \inf(\{(\alpha \cdot \mu)(U) | A \subseteq U \wedge U \text{ is open}\}) \end{aligned}$$

5. Let $A \in \mathcal{A}$ then as μ is internal regular we have that $\mu(A) = \sup(\{\mu(F) | C \subseteq A \wedge F \text{ is closed}\})$. Hence we have

$$\begin{aligned} (\alpha \cdot \mu)(A) &= \alpha \cdot \sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}) \\ &\stackrel{[\text{theorem: 10.22}]}{=} \sup(\{\alpha \cdot \mu(F) | F \subseteq A \wedge F \text{ is closed}\}) \\ &= \sup(\{(\alpha \cdot \mu)(F) | F \subseteq A \wedge F \text{ is closed}\}) \end{aligned}$$

6. This follows from (2) and (4).

7. This follows from (3) and (6). □

The regularity of a measure tells us how a measure can be approximated by open, closed or compact sets which shows a relation between the measure and the topology. This can be used to prove uniqueness of measures.

Theorem 21.583. Let $\langle X, \mathcal{T} \rangle$ be a topological space, $\mathcal{A}$ a σ -algebra on X with $\mathcal{B}[X, \mathcal{T}] \subseteq \mathcal{A}$ then we have that:

1. If $\mu: \mathcal{A} \rightarrow [0, \infty]$ and $\nu: \mathcal{A} \rightarrow [0, \infty]$ are external regular measures such that for every open set U we have $\mu(U) = \nu(U)$ then $\mu = \nu$.
2. If $\mu: \mathcal{A} \rightarrow [0, \infty]$ and $\nu: \mathcal{A} \rightarrow [0, \infty]$ are internal regular measures such that for every closed set F we have $\mu(F) = \nu(F)$ then $\mu = \nu$.
3. If additional $\mathcal{T}$ is **Hausdorff** then we have: If $\mu: \mathcal{A} \rightarrow [0, \infty]$ and $\nu: \mathcal{A} \rightarrow [0, \infty]$ are compact internal regular measures such that for every compact set K we have $\mu(K) = \nu(K)$ then $\mu = \nu$.
4. If additional $\mathcal{T}$ is **Hausdorff** and $\mu: \mathcal{A} \rightarrow [0, \infty]$ and $\nu: \mathcal{A} \rightarrow [0, \infty]$ are Radon measures then we have:
 - a. If for every open set U $\mu(U) = \nu(U)$ then $\mu = \nu$.
 - b. If $\mathcal{T}$ is Hausdorff and for every compact set K we have that $\mu(K) = \nu(K)$ then $\mu = \nu$.

Proof. The proof is trivial

1. Let $A \in \mathcal{A}$ then we have as μ, ν are external regular we have

$$\mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) = \inf(\{\nu(U) | A \subseteq U \wedge U \text{ is open}\}) = \nu(A)$$

which proves that $\mu = \nu$.

2. Let $A \in \mathcal{A}$ then we have as μ, ν are internal regular we have

$$\mu(A) = \sup(\{\mu(F) | F \subseteq A \wedge F \text{ is closed}\}) = \sup(\{\nu(F) | F \subseteq A \wedge F \text{ is closed}\}) = \nu(A)$$

which proves that $\mu = \nu$.

3. Let $A \in \mathcal{A}$ then we have as μ, ν are compact internal regular we have

$$\mu(A) = \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) = \sup(\{\nu(K) | K \subseteq A \wedge K \text{ is compact}\}) = \nu(A)$$

which proves that $\mu = \nu$.

4. As by [definition: 21.580] a Radon measure is external regular and compact internal regular this follows from (1) and (3). □

Theorem 21.584. Let $\langle X, \mathcal{T} \rangle$ be a **Hausdorff** topological space, $\mathcal{A}$ a σ -algebra on X with $\mathcal{B}[X, \mathcal{T}] \subseteq \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a local finite measure then μ is compact finite.

Note 21.585. If $\mathcal{T}$ is a metric space or normed space then by [theorem: 14.242] $\mathcal{T}$ is Hausdorff so that this theorem is also valid for metric and normed spaces.

Proof. Let K be a compact set then $\forall x \in K$ there exist by local finiteness a open set U_x with $x \in U_x$ and $\mu(U_x) < \infty$, hence $K \subseteq \bigcup_{x \in K} U_x$. As K is compact there exists a finite set $\{x_1, \dots, x_n\} \subseteq K$ such that $K \subseteq \bigcup_{i \in \{1, \dots, n\}} U_{x_i}$. Hence

$$\mu(K) \leq \mu\left(\bigcup_{i \in \{1, \dots, n\}} U_{x_i}\right) \underset{[\text{theorem: 21.102}]}{\leq} \sum_{i \in \{1, \dots, n\}} \mu(U_{x_i}) < \infty$$

which proves that $\mu(K)$ is finite. □

Theorem 21.586. Let $\mathcal{A}$ be a σ -algebra on $\mathbb{R}$ with $\mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{\mathbb{R}}] \subseteq \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a compact finite measure then we have $\forall I \in \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \{\emptyset\}$ [see definition: 21.51] that

$$\mu(I) < \infty$$

and

$$\mu_{[I]}: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu_{[I]}(A) = \mu(A \cap I) \text{ is a finite measure}$$

Proof. Let $I \in \mathcal{I}_{\mathbb{R}}$ then $I = [a, b]$ where $a, b \in \mathbb{R} \wedge a \leq b$, using the Heine-Borel theorem [theorem: 14.279] $[a, b]$ is a compact set. As μ is compact finite it follows then that $\mu(I) < \infty$ or in other words

$$\forall I \in \mathcal{I}_{\mathbb{R}} \text{ we have } \mu(I) < \infty$$

Let $I \in \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \{\emptyset\}$ then either $I = \emptyset \Rightarrow \mu(I) = 0 < \infty$ or there exist $a, b \in \mathbb{R}$ with $a < b$ such that $I = [a, b[\subseteq [a, b] \vee I =]a, b] \subseteq [a, b] \vee I =]a, b[\subseteq [a, b] \subseteq [a, b]$ so that $\mu(I) \leq \mu([a, b]) < \infty$. Hence we have

$$\forall I \in \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \{\emptyset\} \text{ that } \mu(I) < \infty$$

Let $I \in \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \mathcal{I}_{\mathbb{R}} \cup \{\emptyset\}$ then we have then by [theorem: 21.108(3)] that $\mu_{[I]}$ is a measure on $\mathcal{A}$. Finally as $\forall A \in \mathcal{A}$ we have $\mu_{[I]}(A) = \mu(A \cap I) \leq \mu(I) < \infty$ it follows that

$$\mu_{[I]} \text{ is a finite measure on } \mathcal{A} \quad \square$$

Theorem 21.587. Let $\mathcal{A}$ be a σ -algebra on $\mathbb{R}^n$ such that $\mathcal{B}[\mathbb{R}^n] = \mathcal{B}[\mathbb{R}^n, ||\cdot||_n] \subseteq \mathcal{A}$ and $\mu: \mathcal{A} \rightarrow [0, \infty]$ a compact finite measure then we have $\forall R \in \mathcal{R}_{\mathbb{R}^n} \cup \mathcal{R}_{\mathbb{R}^n} \cup \mathcal{R}_{\mathbb{R}^n} \cup \mathcal{R}_{\mathbb{R}^n} \cup \{\emptyset\}$ that

$$\mu(R) < \infty$$

and

$\mu_{[R]}: \mathcal{A} \rightarrow [0, \infty]$ defined by $\mu_{[R]}(A) = \mu(A \cap R)$ is a finite measure

Proof. Let $R \in \mathcal{R}_{\text{II}}^n$ then by [theorem: 21.74] R is a compact set. As μ is compact finite it follows then that $\mu(R) < \infty$ or in other words

$$\forall R \in \mathcal{R}_{\text{II}}^n \text{ we have } \mu(R) < \infty$$

Let $R \in \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \{\emptyset\}$ then either $R = \emptyset \Rightarrow \mu(R) = 0 < \infty$ or there exist $a, b \in \mathbb{R}^n$ with $a < b$ such that $R = [a, b[\subseteq [a, b] \vee]a, b] \subseteq [a, b] \vee]a, b[\subseteq [a, b]$ so that $\mu(R) \leq \mu([a, b]) < \infty$. Hence we have

$$\forall R \in \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \{\emptyset\} \text{ that } \mu(R) < \infty$$

Let $I \in \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \mathcal{R}_{\text{II}}^n \cup \{\emptyset\} \subseteq \mathcal{B}[\mathbb{R}^n]$ then by [theorem: 21.108(3)]

$$\mu_{[R]}: \mathcal{A} \rightarrow [0, \infty] \text{ defined by } \mu_{[R]}(A) = \mu(A \cap R)$$

is a measure on $\mathbb{R}$. Finally as $\forall A \in \mathcal{A}$ we have $\mu_{[R]}(A) = \mu(A \cap R) \leq \mu(R) < \infty$ it follows that

$$\mu_{[R]} \text{ is a finite measure on } \mathcal{A} \quad \square$$

21.8.1.2 $\lambda, \bar{\lambda}, \bar{\lambda}^n$ and λ^n are Radon measures

We set out now to prove that the Borel and Lebesgue measures on $\mathbb{R}$ and $\mathbb{R}^n$ are Radon measures, for this we need the following two lemma's that state that compact finite measures on $\mathcal{B}[\mathbb{R}]$ and $\mathcal{B}[\mathbb{R}^n]$ are regular.

Lemma 21.588. Let $\mu: \mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{\text{II}}] \rightarrow [0, \infty]$ a **compact finite measure** then μ is regular.

Proof. First we prove external regularity. Let $A \in \mathcal{B}[\mathbb{R}]$ then we have the following possible cases to consider:

$\mu(A) = \infty$. Let $x \in \{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$ then $x = \mu(U)$ where U is open and $A \subseteq U$. Using the monotonicity of μ we have $\mu(A) \leq \mu(U) = x$ from which it follows that $\mu(A)$ is a lower bound of $\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$. Hence $\infty = \mu(A) \leq \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \leq \infty$ so that

$$\mu(A) = \infty = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$$

$\mu(A) < \infty$. If $x \in \{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$ then $x = \mu(U)$ where U is a open set with $A \subseteq U$. By the monotonicity of μ [see theorem: 21.99] it follows that $\mu(A) \leq \mu(U) = x$ so that $\mu(A)$ is a lower bound of $\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$. Hence we have that

$$\mu(A) \leq \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \quad (21.1068)$$

Assume that $\mu(A) < \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$. Let $\varepsilon \in \mathbb{R}^+$ and define $\{Q_i\}_{i \in \mathbb{N}} \subseteq \mathcal{I}_{\text{II}} \subseteq \mathcal{T}_{\text{II}}$ by $Q_i =]-i, i[$ then by [theorem: 21.586] we have for $i \in \mathbb{N}$ that

$$\mu_{[Q_i]}: \mathcal{B}[\mathbb{R}] \rightarrow [0, \infty] \text{ defined by } \mu_{[Q_i]}(A) = \mu(A \cap Q_i) \text{ is a finite measure}$$

Hence using [theorem: 21.576] we have that there exists a open U_i and a closed set F_i such that

$$F_i \subseteq A \subseteq U_i \text{ and } \mu_{[Q_i]}(U_i \setminus F_i) < \frac{\varepsilon}{2^{i+1}}$$

As $F_i \subseteq A$ we have $U_i \setminus A \subseteq U_i \setminus F_i$ so that $\mu_{[Q_i]}(U_i \setminus A) \leq \mu_{[Q_i]}(U_i \setminus F_i) < \frac{\varepsilon}{2^{i+1}}$. To summarize we have

$$\forall i \in \mathbb{N} \text{ there exist a open } U_i \text{ such that } A \subseteq U_i \text{ and } \mu_{[Q_i]}(U_i \setminus A) < \frac{\varepsilon}{2^{i+1}} \quad (21.1069)$$

Define now $U = \bigcup_{i \in \mathbb{N}} (U_i \cap Q_i)$ which is open [as $U_i, Q_i =]-i, i[$ are open]. Let $x \in A \subseteq X$ then as $\bigcup_{i \in \mathbb{N}} Q_i = \bigcup_{i \in \mathbb{N}}]-i, i[\stackrel{[\text{corollary: 10.30}]}{=} \mathbb{R}$ there exist a $i \in \mathbb{N}$ such that $x \in Q_i$, as further $A \subseteq U_i$ by [eq: 21.1069], it follows that $x \in U_i \cap Q_i \subseteq U$, hence it follows that $A \subseteq U$. Further

$$\begin{aligned}
 \mu(U \setminus A) &= \mu\left(\left(\bigcup_{i \in \mathbb{N}} (U_i \cap Q_i)\right) \setminus A\right) \\
 &= \mu\left(\bigcup_{i \in \mathbb{N}} ((U_i \cap Q_i) \setminus A)\right) \\
 &\stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{i=1}^{\infty} \mu((U_i \cap Q_i) \setminus A) \\
 &\stackrel{[\text{theorem: 1.36}]}{=} \sum_{i=1}^{\infty} \mu((U_i \setminus A) \cap Q_i) \\
 &= \sum_{i=1}^{\infty} \mu_{[Q_i]}(U_i \setminus A) \\
 &\stackrel{[\text{theorem: 14.437}] \wedge [\text{eq: 21.1069}]}{\leq} \sum_{i=1}^{\infty} \frac{\varepsilon}{2^{i+1}} \\
 &= \frac{\varepsilon}{2} \cdot \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\stackrel{[\text{example: 14.412}]}{=} \frac{\varepsilon}{2} \\
 &< \varepsilon
 \end{aligned}$$

Summarized we have

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a open set } U \text{ with } A \subseteq U \text{ and } \mu(U \setminus A) < \varepsilon \quad (21.1070)$$

From the above it follows, by using $\varepsilon = 1$, that there exist a V open such that $A \subseteq V$ and $\mu(V) - \mu(A) \stackrel{[\text{theorem: 21.99}]}{=} \mu(V \setminus A) < 1$ so that $\mu(V) < \infty$. Hence

$$\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \leq \mu(V) < \infty$$

which proves that $\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ is finite. As we have assumed that $\mu(A) < \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ we have that

$$\varepsilon = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - \mu(A) \in \mathbb{R}^+$$

So using [eq: 21.1070] there exist a open set U with $A \subseteq U$ such that $\mu(U \setminus A) < \varepsilon$. As $U \stackrel{A \subseteq U}{=} (U \setminus A) \sqcup A$, we have

$$\begin{aligned}
 \mu(U) &= \mu(U \setminus A) + \mu(A) \\
 &< \varepsilon + \mu(A) \\
 &= \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - \mu(A) + \mu(A) \\
 &= \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \\
 &\leq \mu(U) \\
 &\quad A \subseteq U \text{ open}
 \end{aligned}$$

leading to the contradiction $\mu(U) < \mu(U)$ from which it follows that the assumption $\mu(A) < \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ is false. Hence $\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \leq \mu(A)$ which combined with [eq: 21.1068] proves that

$$\mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$$

Hence we have proved that $\forall A \in \mathcal{B}[\mathbb{R}] \mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ which by definition means that

$$\mu \text{ is external regular} \quad (21.1071)$$

Next we have to prove compact internal regularity. Let $A \in \mathcal{B}[\mathbb{R}]$ then we have two possible cases:

$\mu(A) < \infty$. Define $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}[\mathbb{R}]$ by $C_i = [-i, i]$ and consider the measures $\mu_{[C_i]}: \mathcal{B}[\mathbb{R}] \rightarrow [0, \infty]$ where $\mu_{[C_i]}(B) = \mu(B \cap C_i)$ then from [theorem: 21.586] it follows that $\mu_{[C_i]}$ are finite. Let $\varepsilon \in \mathbb{R}^+$ then by [lemma: 21.576] there exist a open U_i and a closed set F_i such that $F_i \subseteq A \subseteq U_i$ and $\mu_{[C_i]}(U_i \setminus F_i) < \frac{\varepsilon}{2} \xrightarrow{A \setminus F_i \subseteq U_i \setminus F_i} \mu_{[C_i]}(A \setminus F_i) < \frac{\varepsilon}{2}$. To summarize we have

$$\forall i \in \mathbb{N} \text{ there exist a closed set } F_i \text{ such that } F_i \subseteq A \text{ and } \mu_{[C_i]}(A \setminus F_i) < \frac{\varepsilon}{2} \quad (21.1072)$$

For $\{C_i\}_{i \in \mathbb{N}}$ we have that $\forall i \in \mathbb{N} \ C_i = [-i, i] \subseteq [-(i+1), i+1] = C_{i+1}$ and $\mathbb{R} = \bigcup_{i \in \mathbb{N}} [-i, i] \subseteq \bigcup_{i \in \mathbb{N}} [-i, i] \subseteq \mathbb{R}$ so that $\mathbb{R} = \bigcup_{i \in \mathbb{N}} C_i$. Hence for $\{A \cap C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}[\mathbb{R}]$ we have that $\forall i \in \mathbb{N} \ A \cap C_i \subseteq A \cap C_{i+1}$ and $A = A \cap \mathbb{R} = A \cap (\bigcup_{i \in \mathbb{N}} C_i) = \bigcup_{i \in \mathbb{N}} (A \cap C_i)$. Hence by [theorem: 21.115] we have that

$$\mu(A) = \lim_{i \rightarrow \infty} \mu(A \cap C_i)$$

As $\mu(A) < \infty$ it follows from [theorem: 20.79] that

$$\exists N_\varepsilon \in \mathbb{N} \text{ such that } \mu(A) - \mu(A \cap C_{N_\varepsilon}) \underset{A \cap C_{N_\varepsilon} \subseteq A}{=} |\mu(A) - \mu(A \cap C_{N_\varepsilon})| < \frac{\varepsilon}{2} \quad (21.1073)$$

Now

$$\begin{aligned} \mu(A) - \mu(F_{N_\varepsilon} \cap C_{N_\varepsilon}) &\underset{\mu_{[C_{N_\varepsilon}]} \text{ is finite} \wedge [\text{eq: 21.1073}]}{<} \mu(A \cap C_{N_\varepsilon}) + \frac{\varepsilon}{2} - \mu(F_{N_\varepsilon} \cap C_{N_\varepsilon}) \\ &= \mu_{[C_{N_\varepsilon}]}(A) - \mu_{[C_{N_\varepsilon}]}(F_{N_\varepsilon}) + \frac{\varepsilon}{2} \\ &\underset{\mu_{[C_{N_\varepsilon}]} \text{ is finite}}{=} \mu_{[C_{N_\varepsilon}]}(A \setminus F_{N_\varepsilon}) + \frac{\varepsilon}{2} \\ &\underset{[\text{eq: 21.1072}]}{<} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

hence

$$\mu(A) - \mu(F_{N_\varepsilon} \cap C_{N_\varepsilon}) < \varepsilon \quad (21.1074)$$

As $F_{N_\varepsilon} \cap C_{N_\varepsilon} = F_{N_\varepsilon} \cap [-N_\varepsilon, N_\varepsilon]$ is a closed set and $F_{N_\varepsilon} \cap C_{N_\varepsilon} \subseteq [-N_\varepsilon, N_\varepsilon]$, which is compact by Heine Borell [see theorem: 14.279], it follows from [theorem: 14.277] that $F_{N_\varepsilon} \cap C_{N_\varepsilon}$ is compact. So if we take $K_\varepsilon = F_{N_\varepsilon} \cap C_{N_\varepsilon} \subseteq F_{N_\varepsilon} \underset{[\text{eq: 21.1072}]}{\subseteq} A$ then we have that

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a compact set } K_\varepsilon \text{ such that } K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(K_\varepsilon) < \varepsilon \quad (21.1075)$$

Assume now that $\sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) < \mu(A)$ then

$$\varepsilon = \mu(A) - \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) < \mu(A) \in \mathbb{R}^+$$

so that by [eq: 21.1075] there exist a compact K_ε such that

$$K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(K_\varepsilon) < \mu(A) - \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\})$$

from which it follows that

$$\sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) < \mu(K_\varepsilon) \underset{K_\varepsilon \subseteq A}{\leq} \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\})$$

which is a contradiction. Hence the assumption is wrong and we must have

$$\mu(A) \leq \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) \quad (21.1076)$$

If $x \in \{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}$ then there exist a compact set $K \subseteq A$ such that $x = \mu(K) \leq \mu(A)$. Hence $\mu(A)$ is a upper bound of $\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}$ which proves that $\sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) \leq \mu(A)$. Combing this with [eq: 21.1076] gives us $\mu(A) = \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\})$. Hence we have proved that

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ with } \mu(A) < \infty \text{ we have } \mu(A) = \sup(\{\mu(K) | K \subseteq A \wedge K \text{ is compact}\}) \quad (21.1077)$$

$\mu(A) = \infty$. Define $\{A_i\}_{i \in \mathbb{N}}$ by $A_i = A \cap]-i, i[$ then we have that

$$\forall i \in \mathbb{N} \ A_i \subseteq A_{i+1} \text{ and } A = A \cap \mathbb{R} \underset{[\text{corollary: 10.30}]}{=} A \cap \left(\bigcup_{i \in \mathbb{N}}]-i, i[\right) = \bigcup_{i \in \mathbb{N}} A_i$$

Hence using [theorem: 21.115] we have that

$$\infty = \mu(A) = \lim_{i \rightarrow \infty} \mu(A_i) \quad (21.1078)$$

As $\forall i \in \mathbb{N}$ we have $\mu(A_i) \underset{A_i = A \cap]-i, i[\subseteq]-i, i[}{\leq} \mu(]-i, i[) \underset{[\text{theorem: 21.586}]}{<} \infty$ we can use [eq: 21.1077] to get

$$\forall i \in \mathbb{N} \text{ we have } \mu(A_i) = \sup \{ \mu(K) \mid K \subseteq A_i \wedge K \text{ is compact} \} \quad (21.1079)$$

As $\forall i \in \mathbb{N} \ \mu(A_i) - 1 \underset{\mu(A_i) < \infty}{<} \mu(A_i) = \sup \{ \mu(K) \mid K \subseteq A_i \wedge K \text{ is compact} \}$ we have by the definition of the supremum that

$$\forall i \in \mathbb{N} \text{ there exist a compact set } K_i \text{ such that } K_i \subseteq A_i \text{ and } \mu(A_i) - 1 < \mu(K_i) \quad (21.1080)$$

Assume now that $\sup \{ \mu(K) \mid K \subseteq A \wedge K \text{ is compact} \} < \infty$. As $\lim_{i \rightarrow \infty} \mu(A_i) \underset{[\text{eq: 21.1078}]}{=} \infty$ we have by [theorem: 20.79] that there exist a $N \in \mathbb{N}$ such that $\sup \{ \mu(K) \mid K \subseteq A \wedge K \text{ is compact} \} + 1 < \mu(A_N)$, as also $\mu(A_N) < \infty$ we have that

$$\sup \{ \mu(K) \mid K \subseteq A \wedge K \text{ is compact} \} < \mu(A_N) - 1$$

Using [eq: 21.1080] there exist a compact K_N such that $K_N \subseteq A_N$ and $\mu(A_N) - 1 < \mu(K_N)$ so that

$$\mu(K_N) \underset{K_N \subseteq A \wedge K_N \text{ is compact}}{\leq} \sup \{ \mu(K) \mid K \subseteq A \wedge K \text{ is compact} \} < \mu(A_N) - 1 < \mu(K_N)$$

leading to the contradiction $\mu(K_N) < \mu(K_N)$. Hence the assumption is false and we must have that

$$\sup \{ \mu(K) \mid K \subseteq A \wedge K \text{ is compact} \} = \infty = \mu(A)$$

So we have

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ we have } \mu(A) = \sup \{ \mu(K) \mid K \subseteq A \wedge K \text{ is compact} \}$$

which proves that

$$\mu \text{ is compact internal regular} \quad (21.1081)$$

As μ is external regular and compact internal regular [see eqs: 21.1071, 21.1081] it follows by [definition: 21.580] that

$$\mu \text{ is regular} \quad \square$$

We have a similar theorem for $\mathcal{B}[\mathbb{R}^n]$.

Lemma 21.589. *Let $n \in \mathbb{N}$ and $\mu: \mathcal{B}[\mathbb{R}^n] = \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n}] \rightarrow [0, \infty]$ a compact finite measure then μ is regular.*

Proof. The proof is similar to the proof of [lemma: 21.588]. First we prove external regularity. Let $A \in \mathcal{B}[\mathbb{R}^n]$ then we have the following possible cases to consider:

$\mu(A) = \infty$. If $x \in \{ \mu(U) \mid A \subseteq U \wedge U \text{ is open} \}$ then $x = \mu(A)$ where $A \subseteq U$, hence $\mu(A) \leq \mu(U) = x$. So it follows that $\mu(A)$ is a lower bound of $\{ \mu(U) \mid A \subseteq U \wedge U \text{ is open} \}$ from which it follows that $\infty = \mu(A) \leq \inf \{ \mu(U) \mid A \subseteq U \wedge U \text{ is open} \} \leq \infty$, hence

$$\mu(A) = \sup \{ \mu(U) \mid A \subseteq U \wedge U \text{ is open} \}$$

$\mu(A) < \infty$. If $x \in \{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$ then $x = \mu(U)$ where U is a open set with $A \subseteq U$. So it follows that $\mu(A) \leq \mu(U) = x$ which proves that $\mu(A)$ is a lower bound of $\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}$. Hence we have that

$$\mu(A) \leq \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \quad (21.1082)$$

Assume that $\mu(A) < \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$. Let $\varepsilon \in \mathbb{R}^+$ and define $\{Q_i\}_{i \in \mathbb{N}} \subseteq \mathcal{I} \subseteq \mathcal{T}_{||}$ by $Q_i =]-i, i[^n \in \mathcal{R}_{||}^n$ then by [theorem: 21.587] we have for $i \in \mathbb{N}$ that

$$\mu_{[Q_i]}: \mathcal{B}[\mathbb{R}^n] \rightarrow [0, \infty] \text{ defined by } \mu_{[Q_i]}(A) = \mu(A \cap Q_i) \text{ is a finite measure}$$

Hence using [lemma: 21.576] we have that there exists a open U_i and a closed set F_i such that

$$F_i \subseteq A \subseteq U_i \text{ and } \mu_{[Q_i]}(U_i \setminus F_i) < \frac{\varepsilon}{2^{i+1}}$$

As $F_i \subseteq A$ we have $U_i \setminus A \subseteq U_i \setminus F_i$ it follows from the monotonicity of μ that $\mu_{[Q_i]}(U_i \setminus A) \leq \mu_{[Q_i]}(U_i \setminus F_i) < \frac{\varepsilon}{2^i}$. To summarize we have

$$\forall i \in \mathbb{N} \text{ there exist a open } U_i \text{ such that } A \subseteq U_i \text{ and } \mu_{[Q_i]}(U_i \setminus A) < \frac{\varepsilon}{2^{i+1}} \quad (21.1083)$$

Define now $U = \bigcup_{i \in \mathbb{N}} (U_i \cap Q_i)$ which is open as $U_i, Q_i =]-i, i[^n$ are open [see theorem: 21.74]. Let $x \in A \subseteq \mathbb{R}^n$ then as $\bigcup_{i \in \mathbb{N}} Q_i = \bigcup_{i \in \mathbb{N}}]-i, i[^n \stackrel{[\text{proposition: 21.70}]}{=} \mathbb{R}^n$ there exist a $i \in \mathbb{N}$ such that $x \in Q_i$, as further $A \subseteq U_i$ by [eq: 21.1083], it follows that $x \in U_i \cap Q_i \subseteq U$, hence it follows that $A \subseteq U$. Further

$$\begin{aligned} \mu(U \setminus A) &= \mu\left(\left(\bigcup_{i \in \mathbb{N}} (U_i \cap Q_i)\right) \setminus A\right) \\ &= \mu\left(\bigcup_{i \in \mathbb{N}} ((U_i \cap Q_i) \setminus A)\right) \\ &\stackrel{[\text{theorem: 21.101}]}{\leq} \sum_{i=1}^{\infty} \mu((U_i \cap Q_i) \setminus A) \\ &\stackrel{[\text{theorem: 1.36}]}{=} \sum_{i=1}^{\infty} \mu((U_i \setminus A) \cap Q_i) \\ &= \sum_{i=1}^{\infty} \mu_{[Q_i]}(U_i \setminus A) \\ &\stackrel{[\text{theorem: 14.437}] \wedge [\text{eq: 21.1083}]}{\leq} \sum_{i=1}^{\infty} \frac{\varepsilon}{2^{i+1}} \\ &= \frac{\varepsilon}{2} \cdot \sum_{i=1}^{\infty} \frac{1}{2^i} \\ &\stackrel{[\text{example: 14.412}]}{=} \frac{\varepsilon}{2} \\ &< \varepsilon \end{aligned}$$

Summarized we have

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a open set } U \text{ with } A \subseteq U \text{ and } \mu(U \setminus A) < \varepsilon \quad (21.1084)$$

From the above it follows, by using $\varepsilon = 1$, that there exist a V open such that $A \subseteq V$ and $\mu(V) - \mu(A) \stackrel{[\text{theorem: 21.99}] \wedge \mu(A) < \infty}{=} \mu(V \setminus A) < 1$ so that $\mu(V) < \infty$. Hence

$$\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \leq \mu(V) < \infty$$

which proves that $\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ is finite. As we have assumed that $\mu(A) < \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ it follows that

$$\varepsilon = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - \mu(A) \in \mathbb{R}^+$$

So using [eq: 21.1084] there exist a open set U with $A \subseteq U$ such that $\mu(U \setminus A) < \varepsilon$. As $U \stackrel{A \subseteq U}{=} (U \setminus A) \sqcup A$, we have

$$\begin{aligned} \mu(U) &= \mu(U \setminus A) + \mu(A) \\ &< \varepsilon + \mu(A) \\ &= \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) - \mu(A) + \mu(A) \\ &= \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \\ &\leq \mu(U) \\ &\quad A \subseteq U \text{ open} \end{aligned}$$

leading to the contradiction $\mu(U) < \mu(U)$ from which it follows that the assumption $\mu(A) < \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ is false. Hence $\inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\}) \leq \mu(A)$ which combined with [eq: 21.1082] proves that

$$\mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$$

So we have proved that $\forall A \in \mathcal{B}[\mathbb{R}^n] \mu(A) = \inf(\{\mu(U) | A \subseteq U \wedge U \text{ is open}\})$ which by definition means that

$$\mu \text{ is external regular} \quad (21.1085)$$

Next we have to prove compact internal regularity. Let $A \in \mathcal{B}[\mathbb{R}^n]$ then we have two possible cases:

$\mu(A) < \infty$. Define $\{C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}[\mathbb{R}^n]$ by $C_i = [-i, i]^n$ and consider the measures $\mu_{[C_i]}: \mathcal{B}[\mathbb{R}^n] \rightarrow [0, \infty]$ where $\mu_{[C_i]}(B) = \mu(B \cap C_i)$ then from [theorem: 21.586] it follows that the measures $\mu_{[C_i]}$ are finite. Let $\varepsilon \in \mathbb{R}^+$ then by [lemma: 21.576] there exist a open U_i and a closed set F_i such that $F_i \subseteq A \subseteq U_i$ and $\mu_{[C_i]}(U_i \setminus F_i) < \frac{\varepsilon}{2} \stackrel{A \setminus F_i \subseteq U_i \setminus F_i}{\Rightarrow} \mu_{[C_i]}(A \setminus F_i) < \frac{\varepsilon}{2}$. To summarize we have

$$\forall i \in \mathbb{N} \text{ there exist a closed set } F_i \text{ such that } F_i \subseteq A \text{ and } \mu_{[C_i]}(A \setminus F_i) < \frac{\varepsilon}{2} \quad (21.1086)$$

As for $\{C_i\}_{i \in \mathbb{N}}$ we have that $\forall i \in \mathbb{N} \ C_i = [-i, i]^n \subseteq [-(i+1), i+1]^n = C_{i+1}$ and $\mathbb{R}^n \stackrel{=}{=} \bigcup_{i \in \mathbb{N}} [-i, i]^n = \bigcup_{i \in \mathbb{N}} C_i$. Hence for $\{A \cap C_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}[\mathbb{R}^n]$ we have that $\forall i \in \mathbb{N} \ A \cap C_i \subseteq A \cap C_{i+1}$ and $A = A \cap \mathbb{R}^n = A \cap (\bigcup_{i \in \mathbb{N}} C_i) = \bigcup_{i \in \mathbb{N}} (A \cap C_i)$. So using [theorem: 21.115] we have that

$$\mu(A) = \lim_{i \rightarrow \infty} \mu(A \cap C_i)$$

As $\mu(A) < \infty$ it follows from [theorem: 20.79] that

$$\exists N_\varepsilon \in \mathbb{N} \text{ such that } \mu(A) - \mu(A \cap C_{N_\varepsilon}) \stackrel{A \cap C_{N_\varepsilon} \subseteq A}{=} |\mu(A) - \mu(A \cap C_{N_\varepsilon})| < \frac{\varepsilon}{2} \quad (21.1087)$$

Now

$$\begin{aligned} \mu(A) - \mu(F_{N_\varepsilon} \cap C_{N_\varepsilon}) &\stackrel{[eq: 21.1087]}{<} \mu(A \cap C_{N_\varepsilon}) + \frac{\varepsilon}{2} - \mu(F_{N_\varepsilon} \cap C_{N_\varepsilon}) \\ &= \mu_{[C_{N_\varepsilon}]}(A) - \mu_{[C_{N_\varepsilon}]}(F_{N_\varepsilon}) + \frac{\varepsilon}{2} \\ &\stackrel{\mu_{[C_{N_\varepsilon}]} \text{ is finite} \wedge [theorem: 21.99]}{=} \mu_{[C_{N_\varepsilon}]}(A \setminus F_{N_\varepsilon}) + \frac{\varepsilon}{2} \\ &\stackrel{[eq: 21.1086]}{<} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

hence

$$\mu(A) - \mu(F_{N_\varepsilon} \cap C_{N_\varepsilon}) < \varepsilon \quad (21.1088)$$

As $F_{N_\varepsilon} \cap C_{N_\varepsilon} = F_{N_\varepsilon} \cap [-N_\varepsilon, N_\varepsilon]^n$ is a closed set and $F_{N_\varepsilon} \cap C_{N_\varepsilon} \subseteq [-N_\varepsilon, N_\varepsilon]^n$, which is compact by Heine Borell [see theorem: 14.324], it follows from [theorem: 14.277] that $F_{N_\varepsilon} \cap C_{N_\varepsilon}$ is compact. So if we take $K_\varepsilon = F_{N_\varepsilon} \cap C_{N_\varepsilon} \subseteq F_{N_\varepsilon} \subseteq A$ then we have that [eq: 21.1086]

$$\forall \varepsilon \in \mathbb{R}^+ \text{ there exist a compact set } K_\varepsilon \text{ such that } K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(K_\varepsilon) < \varepsilon \quad (21.1089)$$

Assume now that $\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) < \mu(A)$ then

$$\varepsilon = \mu(A) - \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) < \mu(A) \in \mathbb{R}^+$$

so that by [eq: 21.1089] there exist a compact K_ε such that

$$K_\varepsilon \subseteq A \text{ and } \mu(A) - \mu(K_\varepsilon) < \mu(A) - \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\})$$

from which it follows that

$$\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) < \mu(K_\varepsilon) \leq \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\})$$

which is a contradiction. Hence the assumption is wrong and we must have

$$\mu(A) \leq \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) \quad (21.1090)$$

If $x \in \{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}$ then there exist a compact set $K \subseteq A$ such that $x = \mu(K) \leq \mu(A)$. Hence $\mu(A)$ is a upper bound of $\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}$ which proves that $\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) \leq \mu(A)$. Combing this with [eq: 21.1090] gives us $\mu(A) = \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\})$. Hence we have proved that

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ with } \mu(A) < \infty \text{ we have } \mu(A) = \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) \quad (21.1091)$$

$\mu(A) = \infty$. Define $\{A_i\}_{i \in \mathbb{N}}$ by $A_i = A \cap]-i, i[^n$ then we have that

$$\forall i \in \mathbb{N} \ A_i \subseteq A_{i+1} \text{ and } A = A \cap \mathbb{R}^n \underset{[\text{proposition: 21.70}]}{=} A \cap \left(\bigcup_{i \in \mathbb{N}}]-i, i[^n \right) = \bigcup_{i \in \mathbb{N}} A_i$$

Hence using [theorem: 21.115] we have that

$$\infty = \mu(A) = \lim_{i \rightarrow \infty} \mu(A_i) \quad (21.1092)$$

As $\forall i \in \mathbb{N}$ we have $\mu(A_i) \leq \mu(]-i, i[^n) \underset{[\text{theorem: 21.586}]}{<} \infty$ we can use [eq: 21.1091] to get

$$\forall i \in \mathbb{N} \text{ we have } \mu(A_i) = \sup(\{\mu(K)|K \subseteq A_i \wedge K \text{ is compact}\}) \quad (21.1093)$$

As $\forall i \in \mathbb{N} \ \mu(A_i) - 1 < \mu(A_i) = \sup(\{\mu(K)|K \subseteq A_i \wedge K \text{ is compact}\})$ we have by the definition of the supremum that

$$\forall i \in \mathbb{N} \text{ there exist a compact set } K_i \text{ such that } K_i \subseteq A_i \text{ and } \mu(A_i) - 1 < \mu(K_i) \quad (21.1094)$$

Assume now that $\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) < \infty$ then we have from $\lim_{i \rightarrow \infty} \mu(A_i) \underset{[\text{eq: 21.1092}]}{=} \infty$ by [theorem: 20.79] that there exist a $N \in \mathbb{N}$ such that $\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) + 1 < \mu(A_N)$, as also $\mu(A_N) < \infty$ we have that

$$\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) < \mu(A_N) - 1$$

Using [eq: 21.1094] there exist a K_N such that $K_N \subseteq A_N$ and $\mu(A_N) - 1 < \mu(K_N)$ so that

$$\mu(K_N) \leq \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) < \mu(A_N) - 1 < \mu(K_N)$$

leading to the contradiction $\mu(K_N) < \mu(K_N)$. Hence the assumption is false and we must have that

$$\sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\}) = \infty = \mu(A)$$

So we have

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ we have } \mu(A) = \sup(\{\mu(K)|K \subseteq A \wedge K \text{ is compact}\})$$

which proves that

$$\mu \text{ is compact internal regular} \quad (21.1095)$$

As μ is external regular and compact internal regular [see eqs: 21.1085, 21.1095] it follows by [definition: 21.580] that

$$\mu \text{ is regular} \quad \square$$

We show now that the Borel measures λ, λ^n are locally finite and thus compact finite.

Lemma 21.590.

1. The Lebesgue measure $\bar{\lambda}$ in $\langle \mathbb{R}, \mathcal{L}[\mathbb{R}], \bar{\lambda} \rangle$ [see theorem: 21.182] is locally finite.
2. The Borel measure λ in $\langle \mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda \rangle$ [see theorem: 21.182] is locally finite.
3. For $n \in \mathbb{N}$ the Lebesgue measure $\bar{\lambda}^n$ in $\langle \mathbb{R}^n, \mathcal{L}[\mathbb{R}^n], \bar{\lambda}^n \rangle$ [see theorem: 21.205] is locally finite.
4. For $n \in \mathbb{N}$ the Borel measure λ^n in $\langle \mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n \rangle$ [see theorem: 21.205] is locally finite.

Note 21.591. As by [theorem: 21.584] locally finite Borel measures are compact finite and $\mathcal{T}_{||}, \mathcal{T}_{|||}^n$ are Hausdorff [see theorem: 14.242] it follows that

1. The Lebesgue measures $\bar{\lambda}$ in $\langle \mathbb{R}, \mathcal{L}[\mathbb{R}], \bar{\lambda} \rangle$ is compact finite.
2. The Borel measure λ in $\langle \mathbb{R}, \mathcal{L}[\mathbb{R}], \lambda \rangle$ is compact finite.
3. For $n \in \mathbb{N}$ the Lebesgue measure $\bar{\lambda}^n$ in $\langle \mathbb{R}^n, \mathcal{L}[\mathbb{R}^n], \bar{\lambda}^n \rangle$ is compact finite.
4. For $n \in \mathbb{N}$ the Borel measure λ^n in $\langle \mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n \rangle$ is compact finite.

Proof.

1. Using theorem: 21.182] we have that

$$\mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] = \mathcal{B}[\mathbb{R}] \subseteq \mathcal{L}[\mathbb{R}]$$

Let $x \in \mathbb{R}$ then $x \in]x-1, x+1[$ where $]x-1, x+1[$ is open. Further

$$\begin{aligned} \bar{\lambda}(]x-1, x+1[) &\leq \bar{\lambda}([x-1, x+1]) \\ &\stackrel{[\text{theorem: 21.99}]}{=} \text{len}([x-1, x+1]) \\ &\stackrel{[\text{theorem: 21.182}]}{=} 2 \\ &< \infty \end{aligned}$$

so that by [definition: 21.578]

$\bar{\lambda}$ is locally finite

2. First we have

$$\mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] = \mathcal{B}[\mathbb{R}] \Rightarrow \mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] \subseteq \mathcal{B}[\mathbb{R}]$$

Let $x \in \mathbb{R}$ then $x \in]x-1, x+1[$ where $]x-1, x+1[$ is open. Further

$$\begin{aligned} \lambda(]x-1, x+1[) &\stackrel{[\text{theorem: 21.182}]}{=} \bar{\lambda}(]x-1, x+1[) \\ &\leq \bar{\lambda}([x-1, x+1]) \\ &\stackrel{[\text{theorem: 21.99}]}{=} \text{len}([x-1, x+1]) \\ &\stackrel{[\text{theorem: 21.182}]}{=} 2 \\ &< \infty \end{aligned}$$

so that by [definition: 21.578]

λ is locally finite

3. First we have [see theorem: 21.205] that

$$\mathcal{B}[\mathbb{R}^n, \mathcal{T}_{|||}^n] = \mathcal{B}[\mathbb{R}^n] \subseteq \mathcal{L}[\mathbb{R}^n]$$

Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ then $x \in \prod_{i \in \{1, \dots, n\}}]x_i - 1, x_i + 1[$ a open set. Further

$$\begin{aligned} \bar{\lambda}^n \left(\prod_{i \in \{1, \dots, n\}}]x_i - 1, x_i + 1[\right) &\stackrel{[\text{theorem: 21.99}]}{\leq} \bar{\lambda}^n \left(\prod_{i \in \{1, \dots, n\}} [x_i - 1, x_i + 1[\right) \\ &\stackrel{[\text{theorem: 21.205}]}{=} \text{vol}^n \left(\prod_{i \in \{1, \dots, n\}} [x_i - 1, x_i + 1[\right) \\ &= \prod_{i=1}^n 2 \\ &= 2^n \\ &< \infty \end{aligned}$$

so that by [defintion: 21.578]

λ is locally finite

4. First we have

$$\mathcal{B}[\mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n}] = \mathcal{B}[\mathbb{R}^n] \Rightarrow \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n}] \subseteq \mathcal{B}[\mathbb{R}^n]$$

Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ then $x \in \prod_{i \in \{1, \dots, n\}}]x_i - 1, x_i + 1[$ a open set. Further

$$\begin{aligned} \lambda^n \left(\prod_{i \in \{1, \dots, n\}}]x_i - 1, x_i + 1[\right) &\stackrel{[\text{theorem: 21.205}]}{=} \bar{\lambda}^n \left(\prod_{i \in \{1, \dots, n\}}]x_i - 1, x_i + 1[\right) \\ &\stackrel{[\text{theorem: 21.99}]}{\leq} \bar{\lambda}^n \left(\prod_{i \in \{1, \dots, n\}} [x_i - 1, x_i + 1[\right) \\ &\stackrel{[\text{theorem: 21.205}]}{=} \text{vol}^n \left(\prod_{i \in \{1, \dots, n\}} [x_i - 1, x_i + 1[\right) \\ &= \prod_{i=1}^n 2 \\ &= 2^n \\ &< \infty \end{aligned}$$

so that by [defintion: 21.578]

λ is locally finite

□

Now we are ready to prove that $\lambda, \bar{\lambda}$ and λ^n are Radon measures.

Theorem 21.592.

1. The Borel measure λ in $\langle \mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda \rangle$ is a Radon measure.
2. Let $n \in \mathbb{N}$ then the Borel measure λ^n in $\langle \mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n \rangle$ is a Radon measure.

Proof. First note that $\mathcal{T}_{\|\cdot\|}$ and $\mathcal{T}_{\|\cdot\|_n}$ are Hausdorff [see theorem: 14.251]

1. Using [lemma: 21.590] λ is locally finite and compact finite, hence by [lemma: 21.588] λ is regular. So by [definition: 21.580] that λ is a Radon measure.
2. Using [lemma: 21.590] λ^n is locally finite and locally compact, hence by [lemma: 21.589] λ^n is regular. So by [definition: 21.580] that λ^n is a Radon measure. □

21.8.1.3 Translation invariance and Radon Measures

Theorem 21.593. Let $\langle X, \|\cdot\| \rangle$ be a normed space with the norm topology $\mathcal{T}_{\|\cdot\|}$ then

1. $\forall x \in X$ and $\forall A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$ we have

$$x + A \in \mathcal{B}[X, \mathcal{T}_{\|\cdot\|}]$$

where $x + A = \{x + a | a \in A\} = \{a + x | a \in A\} = x + A$.

2. $\forall \alpha \in \mathbb{K}$ and $\forall A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ we have

$$\alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{|||}]$$

where $\alpha \cdot A = \{\alpha \cdot a | a \in A\}$

Proof.

1. Take $x \in X$ then by [theorem: 14.182] $\delta_{-x}: X \rightarrow X$ defined by $\delta_{-x}(y) = -x + y$ is a continuous function. Hence by [theorem: 21.48] we have that

$$\forall A \in \mathcal{B}[X, \mathcal{T}_{|||}] \text{ that } \delta_{-x}^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_{|||}] \quad (21.1096)$$

Further

$$\begin{aligned} y \in \delta_{-x}^{-1}(A) &\Leftrightarrow \delta_{-x}(y) \in A \\ &\Leftrightarrow -x + y \in A \\ &\Leftrightarrow y = x + (-x + y) \text{ where } (-x + y) \in A \\ &\Leftrightarrow y = x + a \text{ where } a \in A \\ &\Leftrightarrow y \in x + A \end{aligned}$$

which proves that $\delta_{-x}^{-1}(A) = x + A$. Combining this with [eq: 21.1096] proves that

$$\forall x \in X \text{ and } \forall A \in \mathcal{B}[X, \mathcal{T}_{|||}] \text{ we have } x + A \in \mathcal{B}[X, \mathcal{T}_{|||}]$$

2. Take $\alpha \in \mathbb{K}$ and $A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ then we have two cases for α to consider:

$\alpha = 0$. For A we have either:

$A = \emptyset$. Then $\alpha \cdot A = \emptyset \in \mathcal{B}[X, \mathcal{T}_{|||}]$.

$A \neq \emptyset$. Then $\alpha \cdot A = \{0\}$ which is closed [see theorems: 14.242, 14.244] so that by [theorem: 21.47] $\alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{|||}]$.

$\alpha \neq 0$. Using [theorem: 14.164] the function $\mu_{1 \setminus \alpha}: X \rightarrow X$ by $\mu_{1 \setminus \alpha}(x) = \frac{1}{\alpha} \cdot x$ is a continuous function. Hence by [theorem: 21.48] we have that

$$\mu_{1 \setminus \alpha}^{-1}(A) \in \mathcal{B}[X, \mathcal{T}_{|||}] \quad (21.1097)$$

Further

$$\begin{aligned} y \in \mu_{1 \setminus \alpha}^{-1}(A) &\Leftrightarrow \mu_{1 \setminus \alpha}(y) \in A \\ &\Leftrightarrow \frac{1}{\alpha} \cdot y \in A \\ &\Leftrightarrow y = \alpha \cdot \left(\frac{1}{\alpha} \cdot y \right) \text{ where } \frac{1}{\alpha} \cdot y \in A \\ &\Leftrightarrow y = \alpha \cdot a \text{ where } a \in A \\ &\Leftrightarrow y \in \alpha \cdot A \end{aligned}$$

which proves that $\mu_{1 \setminus \alpha}^{-1}(A) = \frac{1}{\alpha} \cdot A$. Combining this with [eq: 21.1097] proves that

$$\alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{|||}]$$

In other words we have

$$\forall \alpha \in \mathbb{K} \text{ and } \forall A \in \mathcal{B}[X, \mathcal{T}_{|||}] \text{ that } \alpha \cdot A \in \mathcal{B}[X, \mathcal{T}_{|||}] \quad \square$$

Definition 21.594. Let $\langle X, ||| \rangle$ be a normed space with the normed topology $\mathcal{T}_{|||}$ then a measure $\mu: \mathcal{B}[X, \mathcal{T}_{|||}] \rightarrow [0, \infty]$ is **translation invariant** if $\forall x \in X$ and $\forall A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ we have that

$$\mu(A + x) = \mu(A) \text{ where } A + x = \{a + x | a \in A\} = \{x + a | a \in A\} = x + A$$

Note 21.595. Using [corollary: 21.49] we have that $\forall x \in X$ we have $\forall A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ that $x + A \in \mathcal{B}[X, \mathcal{T}_{|||}]$ so that $\mu(x + A)$ makes sense.

We prove now that λ and λ^n are translation invariant, first we need a little lemma:

Lemma 21.596. *Let X be a vector space, $x \in X$, I a set and $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$ then we have:*

1. *If $\{A_i\}_{i \in I}$ is pairwise disjoint then $\{A_i + x\}_{i \in I}$ is pairwise disjoint.*
2. $(\bigcup_{i \in I} A_i) + x = \bigcup_{i \in I} (A_i + x)$

Proof.

1. Let $(i, j) \in I$ with $i \neq j$. Assume that $(A_i + x) \cap (A_j + x) \neq \emptyset$ then there exist a $y \in (A_i + x) \cap (A_j + x)$ from which it follows that there exists $a_i \in A_i$ and $a_j \in A_j$ such that $a_i + x = y = a_j + x$, hence $a_i = a_j$ which leads to the contradiction $a_i \in A_i \cap A_j = \emptyset$. So that the assumption is false and we must conclude that $(A_i + x) \cap (A_j + x) = \emptyset$.
2. We have

$$\begin{aligned} y \in \left(\bigcup_{i \in I} A_i \right) + x &\Leftrightarrow \exists a \in \bigcup_{i \in I} A_i \text{ such that } y = a + x \\ &\Leftrightarrow \exists i \in I \text{ such that } a \in A_i \text{ and } y = a + x \\ &\Leftrightarrow \exists i \in I \text{ such that } y \in A_i + x \\ &\Leftrightarrow y \in \bigcup_{i \in I} (A_i + x) \end{aligned}$$

which proves that

$$\left(\bigcup_{i \in I} A_i \right) + x = \bigcup_{i \in I} (A_i + x) \quad \square$$

Theorem 21.597. *The Borel measure λ in the measure space $\langle \mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda \rangle$ is translation invariant.*

Proof. Let $x \in \mathbb{R}$ and define

$$\lambda_x: \sigma[\mathcal{S}_{[]} = \mathcal{B}[\mathbb{R}] \rightarrow [0, \infty] \text{ by } \lambda_x(A) = \lambda(A + x)$$

where λ is the Borel measure on $\sigma[\mathcal{S}_{[]} = \mathcal{B}[\mathbb{R}]$ [see theorem: 21.182]. Then we have that

$$\lambda_x(\emptyset) = \lambda(\emptyset + x) = \lambda(\emptyset) = 0 \quad (21.1098)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}[\mathbb{R}]$ be pairwise disjoint then by [lemma: 21.596] we have that

$$\{A_i + x\}_{i \in \mathbb{N}} \subseteq \mathcal{B}[\mathbb{R}] \text{ is pairwise disjoint} \quad (21.1099)$$

and

$$\left(\bigsqcup_{i \in \mathbb{N}} A_i \right) + x = \bigsqcup_{i \in \mathbb{N}} (A_i + x) \quad (21.1100)$$

Hence we have

$$\begin{aligned} \lambda_x \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) &= \lambda \left(\left(\bigsqcup_{i \in \mathbb{N}} A_i \right) + x \right) \\ &\stackrel{[\text{eq: 21.1100}]}{=} \lambda \left(\bigsqcup_{i \in \mathbb{N}} (A_i + x) \right) \\ &\stackrel{[\text{eq: 21.1099}] \wedge \lambda \text{ is a measure}}{=} \sum_{i=1}^{\infty} \lambda(A_i + x) \\ &= \sum_{i=1}^{\infty} \lambda_x(A_i) \end{aligned}$$

which together with [eq: 21.1098] proves that

$$\lambda_x: \sigma[\mathcal{S}_{[]} = \mathcal{B}[\mathbb{R}] \rightarrow [0, \infty] \text{ is a measure}$$

Let $A \in \mathcal{S}_{[]} = \mathcal{I}_{[]} \cup \{\emptyset\}$ then we have for A either:

$$A = \emptyset. \text{ Then } \lambda_x(A) = \lambda_x(\emptyset) = \text{len}(\emptyset) = \text{len}(A).$$

$A \in \mathcal{I}_{\mathbb{I}}$. Then there exists $a, b \in \mathbb{R}$ with $a < b$ such that $A = [a, b[$. Hence

$$\begin{aligned}
 y \in A + x &\Leftrightarrow y \in [a, b[+ x \\
 &\Leftrightarrow \exists y' \in [a, b[\text{ with } y = y' + x \\
 &\Leftrightarrow y - x \in [a, b[\\
 &\Leftrightarrow a \leq y - x \leq b \\
 &\Leftrightarrow a + x \leq y \leq b + x \\
 &\Leftrightarrow y \in [a + x, b + x[
 \end{aligned}$$

which proves that

$$A + x = [a + x, b + x[\quad (21.1101)$$

Next

$$\begin{aligned}
 \lambda_x(A) &= \lambda(A + x) \\
 &\stackrel{[\text{eq: 21.1101}]}{=} \lambda([a + x, b + x[) \\
 &\stackrel{[\text{theorem: 21.182}]}{=} \text{len}([a + x, b + x[) \\
 &= b + x - (a + x) \\
 &= b - a \\
 &= \text{len}([a, b[)
 \end{aligned}$$

so that $\lambda_x(A) = \text{len}(A)$.

So we have proved that $\lambda_x|_{\mathcal{S}_{\mathbb{I}}} = \text{len}$ which by [theorem: 21.182] proves that $\lambda_x = \lambda$ or in other words we have

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ we have } \lambda(A + x) = \lambda_x(A) = \lambda(A) \quad \square$$

Theorem 21.598. *Let $n \in \mathbb{N}$, then the Borel measure λ^n in $\langle \mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n \rangle$ is translation invariant.*

Proof. Let $x \in \mathbb{R}^n$ and define

$$\lambda_x^n: \sigma[\mathcal{R}_{\mathbb{I}}^n \cup \{\emptyset\}] = \mathcal{B}[\mathbb{R}^n] \rightarrow [0, \infty] \text{ by } \lambda_x^n(A) = \lambda^n(A + x)$$

where λ^n is the Borel measure defined in [theorem: 21.205]. Then we have that

$$\lambda_x^n(\emptyset) = \lambda^n(\emptyset + x) = \lambda^n(\emptyset) = 0 \quad (21.1102)$$

Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \sigma[\mathcal{R}_{\mathbb{I}}^n \cup \{\emptyset\}] = \mathcal{B}[\mathbb{R}^n]$ pairwise disjoint then by [lemma: 21.596]

$$\{A_i\}_{i \in \mathbb{N}} \text{ is pairwise disjoint} \quad (21.1103)$$

and

$$\left(\bigsqcup_{i \in \mathbb{N}} A_i \right) + x = \bigsqcup_{i \in \mathbb{N}} (A_i + x) \quad (21.1104)$$

Hence we have that

$$\begin{aligned}
 \lambda_x^n \left(\bigsqcup_{i \in \mathbb{N}} A_i \right) &= \lambda^n \left(\left(\bigsqcup_{i \in \mathbb{N}} A_i \right) + x \right) \\
 &\stackrel{[\text{eq: 21.1104}]}{=} \lambda^n \left(\bigsqcup_{i \in \mathbb{N}} (A_i + x) \right) \\
 &\stackrel{[\text{eq: 21.1103}] \wedge \lambda^n \text{ is a measure}}{=} \sum_{i=1}^{\infty} \lambda^n(A_i + x) \\
 &= \sum_{i=1}^{\infty} \lambda_x^n(A_i)
 \end{aligned}$$

which together with [eq: 21.1102] proves that

$$\lambda_x^n: \sigma[\mathcal{R}_{\mathbb{I}}^n \cup \{\emptyset\}] = \mathcal{B}[\mathbb{R}^n] \text{ is a measure on } \mathcal{R}_{\mathbb{I}}^n \cup \{\emptyset\}$$

Let $A \in \mathcal{R}_{||}^n \cup \{\emptyset\}$ then we have either:

$A = \emptyset$. Then $\lambda_x^n(A) = \lambda_x^n(\emptyset) = \lambda^n(\emptyset + x) = \lambda^n(\emptyset) = 0 = \text{vol}^n(\emptyset)$

$A = \mathcal{R}_x^n$. Then there exist $a, b \in \mathbb{R}^n$ with $a < b$ such that $A = [a, b[$. Hence we have

$$\begin{aligned} y \in A + x &\Leftrightarrow y \in [a, b[+ x \\ &\Leftrightarrow \exists t \in [a, b[\text{ such that } y = t + x \\ &\Leftrightarrow \exists t \in \mathbb{R}^n \text{ such that } \forall i \in \{1, \dots, n\} \ a_i \leq t_i < b_i \text{ and } y_i = t_i + x_i \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } a_i \leq y_i - x_i < b_i \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } a_i + x_i \leq y_i < b_i + x_i \\ &\Leftrightarrow y \in [a + x, b + x[\end{aligned}$$

proving that

$$A + x = [a + x, b + x[\quad (21.1105)$$

Next

$$\begin{aligned} \lambda_x^n(A) &= \lambda^n(A + x) \\ &\stackrel{[\text{eq: 21.1105}]}{=} \lambda^n([a + x, b + x[) \\ &\stackrel{[\text{theorem: 21.205}]}{=} \text{vol}^n([a + x, b + x[) \\ &= \prod_{i=1}^n (b_i + x_i - (a_i + x_i)) \\ &= \prod_{i=1}^n (b_i - a_i) \\ &= \text{vol}^n([a, b[) \\ &= \text{vol}^n(A) \end{aligned}$$

so that $\lambda_x^n(A) = \text{vol}^n(A)$.

So we have proved that $(\lambda_x^n)_{|\mathcal{R}_{||}^n \cup \{\emptyset\}} = \text{vol}^n$ from which it follows by [theorem: 21.205] that $\lambda_x^n = \lambda^n$.

Hence if $A \in \mathcal{B}[\mathbb{R}^n] = \sigma[\mathcal{R}_{||}^n \cup \{\emptyset\}]$ then we have

$$\lambda^n(A + x) = \lambda_x^n(A) = \lambda^n(A) \quad \square$$

We prove now a surprising relation between all translation invariant Radon measures on $\mathbb{R}$ and the Borel measure λ . First we introduce Dyadic intervals [similar to Dyadic cubes in [definition 21.79]] and prove that every open set in $\mathcal{T}_{||}$ can be written as the countable disjoint union of Dyadic intervals.

Definition 21.599. (Dyadic interval) Let $n \in \mathbb{N}_0$, $z \in \mathbb{Z}$ then $\left[\frac{z}{2^n}, \frac{z+1}{2^n}\right[$ is called a Dyadic interval of order n . The set of Dyadic intervals of order n is noted as $\mathcal{D}_n$ hence

$$\mathcal{D}_n = \left\{ \left[\frac{z}{2^n}, \frac{z+1}{2^n} \right[\mid z \in \mathbb{Z} \right\}$$

and the set of all Dyadic intervals is noted as $\mathcal{D}$ hence

$$\mathcal{D} = \bigcup_{n \in \mathbb{N}} \mathcal{D}_n$$

Lemma 21.600. Let $n, k \in \mathbb{N}_0$ with $k \leq n$ and $v, z \in \mathbb{Z}$ then we have

$$\begin{aligned} \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right[\cap \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right[\neq \emptyset &\Rightarrow \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right[\subseteq \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right[\\ \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right[\cap \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right[= \emptyset &\Rightarrow \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right[\not\subseteq \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right[\end{aligned}$$

in other words we have either $\left[\frac{v}{2^n}, \frac{v+1}{2^n} \right[\cap \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right[= \emptyset$ or $\left[\frac{v}{2^n}, \frac{v+1}{2^n} \right[\subseteq \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right[$ but not both.

Proof. Take $I = [\frac{v}{2^n}, \frac{v+1}{2^n}]$ and $J = [\frac{z}{2^k}, \frac{z+1}{2^k}]$. As $k \leq n$ we have $0 \leq l = n - k$ and

$$\begin{aligned} J &= \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right] \\ &= \left[\frac{z \cdot 2^{n-k}}{2^k \cdot 2^{n-k}}, \frac{(z+1) \cdot 2^{n-k}}{2^k \cdot 2^{n-k}} \right] \\ &= \left[\frac{z \cdot 2^{n-k}}{2^k \cdot 2^{n-k}}, \frac{z \cdot 2^{n-k} + 2^{n-k}}{2^k \cdot 2^{n-k}} \right] \\ &= \left[\frac{z \cdot 2^{n-k}}{2^n}, \frac{z \cdot 2^{n-k} + 2^{n-k}}{2^n} \right] \\ &= \left[\frac{z \cdot 2^l}{2^n}, \frac{z \cdot 2^l + 2^l}{2^n} \right] \end{aligned}$$

which proves that

$$J = \left[\frac{z \cdot 2^l}{2^n}, \frac{z \cdot 2^l + 2^l}{2^n} \right] \quad (21.1106)$$

Let $I \cap J \neq \emptyset$. If $\frac{v+1}{2^n} \leq \frac{z \cdot 2^l}{2^n}$ then $I \cap J = \emptyset$ and if $\frac{z \cdot 2^l + 2^l}{2^n} \leq \frac{v}{2^n}$ we have also $I \cap J = \emptyset$, hence we must have

$$\frac{z \cdot 2^l}{2^n} < \frac{v+1}{2^n} \wedge \frac{v}{2^n} < \frac{z \cdot 2^l + 2^l}{2^n}$$

after multiplying both sides in the above by 2^n we have

$$z \cdot 2^l < v+1 \wedge v < z \cdot 2^l + 2^l \quad (21.1107)$$

Assume that $\frac{v}{2^n} < \frac{z \cdot 2^l}{2^n}$ then $v < z \cdot 2^l$ $\xrightarrow{0 \leq l \Rightarrow 2^l, v, z \in \mathbb{Z}}$ $v+1 \leq z \cdot 2^l$ $\xrightarrow{[\text{eq: 21.1107}]}$ $v+1 < z \cdot 2^l + 2^l$ leading to the contradiction $v+1 < v+1$, so we must have that

$$\frac{z \cdot 2^l}{2^n} \leq \frac{v}{2^n} \quad (21.1108)$$

Assume that $\frac{z \cdot 2^l + 2^l}{2^n} < \frac{v+1}{2^n}$ then $z \cdot 2^l + 2^l < v+1$ $\xrightarrow{0 \leq l \Rightarrow z \cdot 2^l, 2^l, v \in \mathbb{Z}}$ $z \cdot 2^l + 2^l \leq v$ $\xrightarrow{[\text{eq: 21.1107}]}$ $z \cdot 2^l + 2^l < z \cdot 2^l + 2^l$ which leads to the contradiction $z \cdot 2^l + 2^l < z \cdot 2^l + 2^l$, so we must have that

$$\frac{v+1}{2^n} \leq \frac{z \cdot 2^l + 2^l}{2^n} \quad (21.1109)$$

Hence $I = [\frac{v}{2^n}, \frac{v+1}{2^n}] \xrightarrow{[\text{eqs: 21.1108, 21.1109}]} \subseteq [\frac{z \cdot 2^l}{2^n}, \frac{z \cdot 2^l + 2^l}{2^n}] \xrightarrow{[\text{eq: 21.1124}]} J$. So we have proved that

$$\left[\frac{v}{2^n}, \frac{v+1}{2^n} \right] \cap \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right] \neq \emptyset \Rightarrow \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right] \subseteq \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right]$$

Let $[\frac{v}{2^n}, \frac{v+1}{2^n}] \cap [\frac{z}{2^k}, \frac{z+1}{2^k}] = \emptyset$. Assume that $[\frac{v}{2^n}, \frac{v+1}{2^n}] \subseteq [\frac{z}{2^k}, \frac{z+1}{2^k}]$ then

$$\frac{v}{2^n} \in \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right] = \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right] \cap \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right] = \emptyset$$

which lead to the contradiction $\frac{v}{2^n} \in \emptyset$, hence we must have that $[\frac{v}{2^n}, \frac{v+1}{2^n}] \not\subseteq [\frac{z}{2^k}, \frac{z+1}{2^k}]$. So we have

$$\left[\frac{v}{2^n}, \frac{v+1}{2^n} \right] \cap \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right] = \emptyset \Rightarrow \left[\frac{v}{2^n}, \frac{v+1}{2^n} \right] \not\subseteq \left[\frac{z}{2^k}, \frac{z+1}{2^k} \right] \quad \square$$

Lemma 21.601. *Dyadic intervals have the following properties:*

1. If $n, k \in \mathbb{N}_0$ with $k \leq n$ then we have $\forall I \in \mathcal{D}_n, \forall J \in \mathcal{D}_k$ that

$$I \cap J \neq \emptyset \Rightarrow I \subseteq J$$

$$I \cap J = \emptyset \Rightarrow I \not\subseteq J$$

2. $\forall n \in \mathbb{N}_0$ and $\forall I, J \in \mathcal{D}_n$ with $I \neq J$ we have $I \cap J = \emptyset$
3. $\forall n \in \mathbb{N}_0$ we have $\mathbb{R} = \bigsqcup_{I \in \mathcal{D}_n} I$ [by (2) pairwise disjoint union]
4. $\forall n \in \mathbb{N}_0$ and $\forall I \in \mathcal{D}_n$ we have $\text{len}(I) = 2^{-n}$
5. $\mathcal{D}$ is denumerable

Proof.

1. This follows directly from [lemma: 21.600].
2. Let $I \neq J$ and assume that $I \cap J \neq \emptyset$ as $n \leq n$ it follows from (1) that that $I \subseteq J \wedge J \subseteq I$ leading to the contradiction $I = J \wedge I \neq J$ hence the assumption is wrong and we must have that $I \cap J = \emptyset$.
3. Let $x \in \mathbb{R}$ then for $x \cdot 2^n$ we have by [theorem: 10.30] that there exist a $z \in \mathbb{Z}$ such that $z \leq x \cdot 2^n < z + 1$ from which it follows that $\frac{z}{2^n} \leq x < \frac{z+1}{2^n}$. Hence $x \in [\frac{z}{2^n}, \frac{z+1}{2^n}[\in \mathcal{D}_n$ which proves that $\mathbb{R} \subseteq \bigcup_{I \in \mathcal{D}_n} I$. As trivially $\bigcup_{I \in \mathcal{D}_n} I \subseteq \mathbb{R}$ we conclude that

$$\mathbb{R} = \bigcup_{I \in \mathcal{D}_n} I$$

4. If $I \in \mathcal{D}_n$ then there exist a $z \in \mathbb{Z}$ such that $I = [\frac{z}{2^n}, \frac{z+1}{2^n}[$ so that

$$\text{len}(I) = \text{len}\left(\left[\frac{z}{2^n}, \frac{z+1}{2^n}\right]\right) = \frac{z+1}{2^n} - \frac{z}{2^n} = \frac{1}{2^n} = 2^{-n}$$

5. Given $n \in \mathbb{N}$ define $\beta: \mathbb{Z} \rightarrow \mathcal{D}_n$ by $\beta(z) = [\frac{z}{2^n}, \frac{z+1}{2^n}[$ then β is a bijection.

injectivity. If $\beta(x) = \beta(y)$ then $[\frac{x}{2^n}, \frac{x+1}{2^n}[= [\frac{y}{2^n}, \frac{y+1}{2^n}[$ which by [theorem: 3.145] proves that $\frac{x}{2^n} = \frac{y}{2^n}$ from which it follows that $x = y$.

surjectivity. If $I \in \mathcal{D}_n$ then there exist a $z \in \mathbb{Z}$ such that $I = [\frac{z}{2^n}, \frac{z+1}{2^n}[= \beta(z)$.

As $\mathbb{Z}$ is denumerable [see theorem: 10.5] and β is a bijection it follows that $\forall n \in \mathbb{N}$ $\mathcal{D}_n$ is denumerable. Finally by [theorem: 6.69] it follows that $\mathcal{D} = \bigcup_{n \in \mathbb{N}_0} \mathcal{D}_n$ is denumerable. $\square$

Next we prove that every non empty open set can be written as a disjoint countable union of Dyadic intervals.

Theorem 21.602. Let $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ be the topological spaces based on the canonical norm $||$ on $\mathbb{R}$ then $\forall U \in \mathcal{T}_{||}$ with $U \neq \emptyset$ we have that $\exists \{I_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}$ pairwise disjoint such that

$$U = \bigsqcup_{i \in \mathbb{N}} I_i$$

Proof. Let U be a non empty open set in $\mathbb{R}$. Define now

$$\forall n \in \mathbb{N}_0 \quad S_n = \{I \in \mathcal{D}_n | I \subseteq U\} \subseteq \mathcal{D}_n \quad (21.1110)$$

$$\forall n \in \mathbb{N}_0 \quad \mathcal{I}_n = \begin{cases} S_0 & \text{if } n = 0 \\ \{I \in S_n | \forall i \in \{1, \dots, n-1\} \text{ we have } \forall J \in \mathcal{I}_i \quad I \not\subseteq J\} & \text{if } n \in \mathbb{N} \end{cases} \subseteq S_n \subseteq \mathcal{D}_n \quad (21.1111)$$

and

$$\mathcal{I} = \bigcup_{i \in \mathbb{N}_0} \mathcal{I}_i \subseteq \bigcup_{i \in \mathbb{N}_0} \mathcal{D}_n = \mathcal{D} \quad (21.1112)$$

Next we prove that $U = \bigcup_{i \in \mathbb{N}} I_i$. Let $x \in U$ then as U is open there exist a $\varepsilon_x \in \mathbb{R}^+$ such that

$$x \in]x - \varepsilon_x, x + \varepsilon_x[\subseteq U \quad (21.1113)$$

Using [theorem: 10.30] there exist a $n_x \in \mathbb{N}$ such that

$$0 < \frac{1}{n_x} < \varepsilon_x \quad (21.1114)$$

using [lemma: 21.601(3)] there exist a $I_x \in \mathcal{D}_{n_x}$ so that $x \in I_x$. Hence there exist a $z \in \mathbb{Z}$ such that

$$x \in I_x = \left[\frac{z}{2^{n_x}}, \frac{z+1}{2^{n_x}} \right] \Rightarrow \frac{z}{2^{n_x}} \leq x < \frac{z+1}{2^{n_x}} \quad (21.1115)$$

If $t \in \left[\frac{z}{2^{n_x}}, \frac{z+1}{2^{n_x}} \right] = I_x$ then $\frac{z}{2^{n_x}} \leq t < \frac{z+1}{2^{n_x}}$ so that $x - t < \frac{z+1}{2^{n_x}} - \frac{z}{2^{n_x}} = \frac{1}{2^{n_x}} \underset{[\text{eq: 21.1114}]}{<} \varepsilon_x$ so that $x - \varepsilon_x < t$. Further $t - x < \frac{z+1}{2^{n_x}} - \frac{z}{2^{n_x}} = \frac{1}{2^{n_x}} \underset{[\text{eq: 21.1114}]}{<} \varepsilon_x$ so that $t < x + \varepsilon_x$. Hence $t \in]x - \varepsilon_x, x + \varepsilon_x[\underset{[\text{eq: 21.1113}]}{\subseteq} U$ which proves that

$$x \in I_x \subseteq U \quad (21.1116)$$

Hence by definition [eq: 21.1110] we have that

$$I_x \in S_{n_x} \quad (21.1117)$$

Define

$$\mathcal{K}_x = \{i \in \{0, \dots, n_x\} \mid \exists I \in S_i \text{ such that } x \in I \subseteq U\} \text{ a finite set}$$

then by [eqs: 21.1116, 21.1117] it follows $n_x \in \mathcal{K}_x$ which proves that $\mathcal{K}_x \neq \emptyset$. Hence

$$m_x = \min(\mathcal{K}_x) \text{ exist}$$

For m_x we have either:

$m_x = 0$. Then $\exists I \in S_0$ such that $x \in I$ which as $S_0 \underset{[\text{eq: 21.1111}]}{=} \mathcal{I}_0 \subseteq \bigcup_{i \in \mathbb{N}_0} \mathcal{I}_i = \mathcal{I}$ proves that $x \in \bigcup_{I \in \mathcal{I}} I$.

$0 < m_x$. Then $\exists I \in S_{m_x}$ such that $x \in I \subseteq U$ and $\forall i \in \{1, \dots, m_x - 1\}$ we have $\forall J \in S_i$ that $\neg(x \in J \wedge J \subseteq U) \Rightarrow x \notin J \vee J \not\subseteq U$. As $J \in S_i$ we have by [eq: 21.1110] that $J \subseteq U$, hence we must have that $x \notin J$ which as $x \in I$ proves that $I \not\subseteq J$. Hence by [eq: 21.1111] we have that $I \in \mathcal{I}_{m_x} \subseteq \bigcup_{i \in \mathbb{N}_0} \mathcal{I}_i = \mathcal{I}$ which proves that $x \in \bigcup_{I \in \mathcal{I}} I$.

so in all cases we have $x \in \bigcup_{I \in \mathcal{I}} I$. As we have chosen $x \in U$ arbitrary it follows that

$$U \subseteq \bigcup_{I \in \mathcal{I}} I \quad (21.1118)$$

Let $x \in \bigcup_{I \in \mathcal{I}} I$ then there exist a $I \in \mathcal{I}$ such that $x \in I$. As $\mathcal{I} = \bigcup_{i \in \mathbb{N}_0} \mathcal{I}_i$ there exist a $n \in \mathbb{N}_0$ such that $I \in \mathcal{I}_n$ which as $\mathcal{I}_n \subseteq \mathcal{S}_n$ means that $I \subseteq U$ so that $x \in U$. Hence we have $\bigcup_{I \in \mathcal{I}} I \subseteq U$ which combined with [eq: 21.1118] proves that

$$U = \bigcup_{I \in \mathcal{I}} I \quad (21.1119)$$

Next we prove that the above union is pairwise disjoint. Let $I, J \in \mathcal{I}$ with $I \neq J$ then as $\mathcal{I} = \bigcup_{i \in \mathbb{N}_0} \mathcal{I}_i$ there exists a $n, m \in \mathbb{N}_0$ such that $I \in \mathcal{I}_n$ and $J \in \mathcal{I}_m$. For n, m we have either:

$n = m$. Then $I, J \in \mathcal{I}_n \subseteq \mathcal{D}_n$ so that by [lemma: 21.601] $I \cap J = \emptyset$

$n < m$. Then $n \in \{0, \dots, m - 1\}$ and by [eq: 21.1111] we have that $I \not\subseteq J$ hence as $I \in \mathcal{I}_n \subseteq \mathcal{D}_n$, $J \in \mathcal{I}_m \subseteq \mathcal{D}_m$ and $n < m$ it follows by [lemma: 21.601] that $I \cap J = \emptyset$.

$m < n$. Then $m \in \{0, \dots, n - 1\}$ and by [eq: 21.1111] we have that $J \not\subseteq I$ hence as $I \in \mathcal{I}_n \subseteq \mathcal{D}_n$, $J \in \mathcal{I}_m \subseteq \mathcal{D}_m$ and $m < n$ it follows by [lemma: 21.601] that $I \cap J = \emptyset$.

This proves

$$\forall I, J \in \mathcal{I} \text{ with } I \neq J \text{ we have } I \cap J = \emptyset \text{ and } U = \bigsqcup_{I \in \mathcal{I}} I \quad (21.1120)$$

Next we prove that $\mathcal{I}$ is denumerable. By [lemma: 21.601] $\mathcal{D}$ is denumerable which, as $\mathcal{I} \underset{[\text{eq: 21.1112}]}{\subseteq} \mathcal{D}$, proves by [theorem: 6.60] that $\mathcal{I}$ is either finite or denumerable. Assume that $\mathcal{I}$ is finite. As $U \neq \emptyset$ we have that $\bigsqcup_{I \in \mathcal{I}} I \neq \emptyset$ and it follows that $\mathcal{I} \neq \emptyset$. Use [theorem: 3.145] to define the function

$$\alpha: \mathcal{D} \rightarrow \mathbb{R} \text{ by } \alpha\left(\left[\frac{z}{2^n}, \frac{z+1}{2^n}\right]\right) = \frac{z}{2^n}$$

then $\alpha|_{\mathcal{I}}: \mathcal{I} \rightarrow \alpha(\mathcal{I})$ is a surjection. As $\mathcal{I}$ is finite it follows from [theorem: 6.44] that $\alpha(\mathcal{I})$ is finite, further, as $\mathcal{I} \neq \emptyset$, $\alpha(\mathcal{I}) \neq \emptyset$. Hence

$$m = \min(\alpha(\mathcal{I})) \text{ exist} \quad (21.1121)$$

Further if $I \in \mathcal{I}$ then $I = [\frac{z}{2^n}, \frac{z+1}{2^n}[$ so that $\alpha(I) = \alpha([\frac{z}{2^n}, \frac{z+1}{2^n}[) = \frac{z}{2^n} \in [\frac{z}{2^n}, \frac{z+1}{2^n}[= I \subseteq U$ which proves that $\alpha(\mathcal{I}) \subseteq U$. As $m \in \alpha(\mathcal{I})$ it follows that

$$m \in U \quad (21.1122)$$

As U is open there exist a $\varepsilon \in \mathbb{R}^+$ such that $m \in]m - \varepsilon, m + \varepsilon[\subseteq U$. Using [theorem: 10.32] there exist a $x \in \mathbb{R}$ such that

$$m - \varepsilon < x < m < m + \varepsilon \quad (21.1123)$$

From the above it follows that $x \in]m - \varepsilon, m + \varepsilon[\subseteq U$, as $U \stackrel{[\text{eq: 21.1120}]}{=} \bigcup_{I \in \mathcal{I}} I$ there exist $I = [\frac{z}{2^n}, \frac{z+1}{2^n}[\in \mathcal{I}$ such that $x \in I = [\frac{z}{2^n}, \frac{z+1}{2^n}[$ from which it follows that $\alpha(I) = \frac{z}{2^n} \leq x \stackrel{[\text{eq: 21.1123}]}{<} m$. As $I \in \mathcal{I}$ we have that $m \stackrel{[\text{eq: 21.1121}]}{=} \min(\alpha(\mathcal{I})) \leq \alpha(I)$ so that we reach the contradiction $m < m$. Hence the assumption is wrong and $\mathcal{I}$ is not finite. This proves that $\mathcal{I}$ is denumerable from which it follows that there exist a bijection $\sigma: \mathbb{N} \rightarrow \mathcal{I}$. Define $\{I_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}$ by $I_i = \sigma(I)$ then we have using [theorem: 21.5] together with [eq: 21.1120] that

$$\{I_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D} \text{ is pairwise disjoint and } U = \bigsqcup_{i \in \mathbb{N}} I_i \quad \square$$

We are ready now to prove that every translation invariant Radon measure μ is essential equal to the Borel measure λ multiplied by a constant.

Theorem 21.603. *Let $\mu: \mathcal{B}[\mathbb{R}] \rightarrow [0, \infty]$ be a translation invariant Radon measure then*

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ we have } \mu(A) = \mu([0, 1]) \cdot \lambda(A)$$

or in other words we have that

$$\mu = \mu([0, 1]) \cdot \lambda$$

Proof. First we calculate $\mu([0, 1])$ in terms of Dyadic intervals. Given $N \in \mathbb{N}$ consider the family $\{a_i\}_{i \in \{1, \dots, 2^N+1\}}$ by $a_i = \frac{i-1}{2^N}$ then we have $a_1 = 0 \wedge a_{2^N+1} = 1$ and $\forall i \in \{1, \dots, 2^N\}$ we have $a_i = \frac{i-1}{2^N} < \frac{i}{2^N} = a_{i+1}$. Then from [lemma: 21.197] it follows that

$$\forall N \in \mathbb{N} \text{ we have } \left\{ \left[\frac{i-1}{2^N}, \frac{i}{2^N} \right] \right\}_{i \in \{1, \dots, 2^N\}} = \{[a_i, a_{i+1}[\}_{i \in \{1, \dots, 2^N\}} \text{ is pairwise disjoint} \quad (21.1124)$$

and

$$\forall N \in \mathbb{N} \text{ we have } [0, 1[= \bigsqcup_{i \in \{1, \dots, 2^N\}} [a_i, a_{i+1}[= \bigsqcup_{i \in \{1, \dots, 2^N\}} \left[\frac{i-1}{2^N}, \frac{i}{2^N} \right[\quad (21.1125)$$

Let $i \in \{1, \dots, 2^N\}$ then we have

$$\begin{aligned} y \in \left[\frac{i-1}{2^N}, \frac{i}{2^N} \right[&\Leftrightarrow \frac{i-1}{2^N} \leq y < \frac{i}{2^N} \\ &\Leftrightarrow 0 \leq y - \frac{i-1}{2^N} < \frac{1}{2^N} \\ &\Leftrightarrow y - \frac{i-1}{2^N} \in \left[0, \frac{1}{2^N} \right[\\ &\Leftrightarrow y \in \left[0, \frac{1}{2^N} \right[+ \frac{i-1}{2^N} \end{aligned}$$

so that

$$\left[\frac{i-1}{2^N}, \frac{i}{2^N} \right[= \left[0, \frac{1}{2^N} \right[+ \frac{i-1}{2^N} \quad (21.1126)$$

Next

$$\begin{aligned}
 \mu([0, 1]) & \stackrel{[\text{eq: 21.1125}]}{=} \mu\left(\bigsqcup_{i \in \{1, \dots, 2^N\}} \left[\frac{i-1}{2^N}, \frac{i}{2^N}\right]\right) \\
 & \stackrel{[\text{eq: 21.1124}]}{=} \sum_{i=1}^{2^N} \mu\left(\left[\frac{i-1}{2^N}, \frac{i}{2^N}\right]\right) \\
 & \stackrel{[\text{eq: 21.1126}]}{=} \sum_{i=1}^{2^N} \mu\left(\left[0, \frac{1}{2^N}\right] + \frac{i-1}{2^N}\right) \\
 & \stackrel{\mu \text{ is translation invariant}}{=} \sum_{i=1}^{2^N} \mu\left(\left[0, \frac{1}{2^N}\right]\right) \\
 & = 2^N \cdot \mu\left(\left[0, \frac{1}{2^N}\right]\right) \\
 \forall N \in \mathbb{N} \text{ we have } \mu([0, 1]) & = 2^N \cdot \mu\left(\left[0, \frac{1}{2^N}\right]\right) \tag{21.1127}
 \end{aligned}$$

Consider now Dyadic intervals [see definition: 21.599]. Let $I \in \mathcal{D}$ then there exist a $N \in \mathbb{N}$ and a $z \in \mathbb{Z}$ such that $I = \left[\frac{z}{2^N}, \frac{z+1}{2^N}\right]$. Now

$$\begin{aligned}
 y \in \left[\frac{z}{2^N}, \frac{z+1}{2^N}\right] & \Leftrightarrow \frac{z}{2^N} \leq y < \frac{z+1}{2^N} \\
 & \Leftrightarrow 0 \leq y - \frac{z}{2^N} < \frac{z+1}{2^N} - \frac{z}{2^N} \\
 & \Leftrightarrow 0 \leq y - \frac{z}{2^N} < \frac{1}{2^N} \\
 & \Leftrightarrow y - \frac{z}{2^N} \in \left[0, \frac{1}{2^N}\right] \\
 & \Leftrightarrow y \in \left[0, \frac{1}{2^N}\right] + \frac{z}{2^N}
 \end{aligned}$$

so that

$$\begin{aligned}
 \mu(I) & = \mu\left(\left[\frac{z}{2^N}, \frac{z+1}{2^N}\right]\right) \\
 & = \mu\left(\left[0, \frac{1}{2^N}\right] + \frac{z}{2^N}\right) \\
 & \stackrel{\mu \text{ is translation invariant}}{=} \mu\left(\left[0, \frac{1}{2^N}\right]\right) \\
 & \stackrel{[\text{eq: 21.1127}]}{=} \frac{\mu([0, 1])}{2^N} \\
 \lambda(I) & = \lambda\left(\left[\frac{z}{2^N}, \frac{z+1}{2^N}\right]\right) \\
 & \stackrel{[\text{theorem: 21.182}]}{=} \text{len}\left(\left[\frac{z}{2^N}, \frac{z+1}{2^N}\right]\right) \\
 & = \frac{z+1}{2^N} - \frac{z}{2^N} \\
 & = \frac{1}{2^N}
 \end{aligned}$$

Hence we have

$$\forall I \in \mathcal{D} \text{ we have } \mu(I) = \mu([0, 1]) \cdot \lambda(I) \tag{21.1128}$$

We extend now the above relation to open sets. Let $U \in \mathcal{T}_{||}$ be a open set then we have either:

$$U = \emptyset. \text{ Then } \mu(U) = \mu(\emptyset) = 0 = \mu([0, 1]) \cdot 0 = \mu([0, 1]) \cdot \lambda(\emptyset) = \mu([0, 1]) \cdot \lambda(U).$$

$U \neq \emptyset$. Using [theorem: 21.602] there exist a pairwise disjoint $\{I_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}$ such that

$$U = \bigsqcup_{i \in \mathbb{N}} I_i$$

Hence

$$\begin{aligned} \mu(U) &= \mu\left(\bigsqcup_{i \in \mathbb{N}} I_i\right) \\ &= \sum_{i=1}^{\infty} \mu(I_i) \\ &\stackrel{[\text{eq: 21.1128}]}{=} \sum_{i=1}^{\infty} (\mu([0, 1[) \cdot \lambda(I_i)) \\ &= \mu([0, 1[) \cdot \sum_{i=1}^{\infty} \lambda(I_i) \\ &= \mu([0, 1[) \cdot \lambda\left(\bigsqcup_{i \in \mathbb{N}} I_i\right) \\ &= \mu([0, 1[) \cdot \lambda(I) \end{aligned}$$

which proves that

$$\forall U \in \mathcal{T}_{||} \text{ we have } \mu(U) = \mu([0, 1[) \cdot \lambda(U)$$

The above, the fact that μ is a Radon measure and by [theorems: 21.582, 21.592] $\mu([0, 1[) \cdot \lambda$ is a Radon measure we have using [theorem: 21.583] that

$$\mu = \mu([0, 1[) \cdot \lambda \quad \square$$

We have a similar result and proof for Radon measures on $\mathbb{R}^n$ and the Borel measure λ^n .

Theorem 21.604. *Let $n \in \mathbb{N}$, $\mu: \mathcal{B}[\mathbb{R}^n] \rightarrow [0, \infty]$ a translation invariant Radon measure then*

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \mu(A) = \mu([0, 1[^n) \cdot \lambda^n(A)$$

or in other words we have that

$$\mu = \mu([0, 1[^n) \cdot \lambda^n$$

Proof. First we calculate $\mu([0, 1[^n)$ using Dyadic cubes. Let $N \in \mathbb{N}$ and consider $\forall i \in \{1, \dots, n\}$ the family $\{a_{i,j}\}_{j \in \{1, \dots, 2^N+1\}}$ defined by $a_{i,j} = \frac{j-1}{2^N}$ then we have $a_{i,1} = 0$, $a_{i,2^N+1} = 1$ and $\forall j \in \{1, \dots, 2^N\}$ we have $a_{i,j} = \frac{j-1}{2^N} < \frac{j}{2^N} = a_{i,j+1}$. Then from [lemma: 21.197] it follows that $\forall i \in \{1, \dots, n\}$

$$\forall N \in \mathbb{N} \left\{ \left[\frac{j-1}{2^N}, \frac{j}{2^N} \right] \right\}_{j \in \{1, \dots, 2^N\}} = \{[a_{i,j}, a_{i,j+1}[)_{i \in \{1, \dots, 2^N\}} \text{ is pairwise disjoint} \quad (21.1129)$$

and

$$\forall i \in \{1, \dots, n\}, \forall N \in \mathbb{N} \text{ we have } [0, 1[= \bigsqcup_{j \in \{1, \dots, 2^N\}} [a_{i,j}, a_{i,j+1}[= \bigsqcup_{j \in \{1, \dots, 2^N\}} \left[\frac{j-1}{2^N}, \frac{j}{2^N} \right[\quad (21.1130)$$

So we have by [lemma: 21.199] that

$$\begin{aligned} [0, 1[^n &= \prod_{i \in \{1, \dots, n\}} [0, 1[\\ &\stackrel{[\text{lemma: 21.199}]}{=} \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}} \left(\prod_{j \in \{1, \dots, n\}} [a_{j,\rho_j}, a_{j,\rho_j+1}[\right) \\ &= \bigsqcup_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}} \left(\prod_{j \in \{1, \dots, n\}} \left[\frac{\rho_j-1}{2^N}, \frac{\rho_j}{2^N} \right[\right) \end{aligned}$$

so that

$$\mu([0, 1]^n) = \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}} \mu \left(\prod_{j \in \{1, \dots, n\}} \left[\frac{\rho_j - 1}{2^N}, \frac{\rho_j}{2^N} \right] \right) \quad (21.1131)$$

Now given $\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}$ we have that

$$\begin{aligned} x \in \prod_{j \in \{1, \dots, n\}} \left[\frac{\rho_j - 1}{2^N}, \frac{\rho_j}{2^N} \right] &\Leftrightarrow \forall j \in \{1, \dots, n\} x_j \in \left[\frac{\rho_j - 1}{2^N}, \frac{\rho_j}{2^N} \right] \\ &\Leftrightarrow \forall j \in \{1, \dots, n\} \frac{\rho_j - 1}{2^N} \leq x_j < \frac{\rho_j}{2^N} \\ &\Leftrightarrow \forall j \in \{1, \dots, n\} 0 \leq x_j - \frac{\rho_j - 1}{2^N} < \frac{\rho_j}{2^N} - \frac{\rho_j - 1}{2^N} \\ &\Leftrightarrow \forall j \in \{1, \dots, n\} 0 \leq x_j - \frac{\rho_j - 1}{2^N} < \frac{1}{2^N} \\ &\Leftrightarrow \forall j \in \{1, \dots, n\} x_j - \frac{\rho_j - 1}{2^N} \in \left[0, \frac{1}{2^N} \right] \\ &\Leftrightarrow \left(x - \frac{\rho - 1}{2^N} \right) \in \left[0, \frac{1}{2^N} \right]^n \\ &\Leftrightarrow x \in \left[0, \frac{1}{2^N} \right]^n + \frac{\rho - 1}{2^N} \end{aligned} \quad (21.1132)$$

so that

$$\begin{aligned} \mu([0, 1]^n) &\stackrel{[\text{eq: 21.1131}]}{=} \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}} \mu \left(\prod_{j \in \{1, \dots, n\}} \left[\frac{\rho_j - 1}{2^N}, \frac{\rho_j}{2^N} \right] \right) \\ &\stackrel{[\text{eq: 21.1132}]}{=} \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}} \mu \left(\left[0, \frac{1}{2^N} \right]^n + \frac{\rho - 1}{2^N} \right) \\ &\stackrel{\mu \text{ is translation independent}}{=} \sum_{\rho \in \prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\}} \mu \left(\left[0, \frac{1}{2^N} \right]^n \right) \\ &\stackrel{\#(\prod_{i \in \{1, \dots, n\}} \{1, \dots, 2^N\})_{[21.1]}}{=} 2^{n \cdot N} \cdot \mu \left(\left[0, \frac{1}{2^N} \right]^n \right) \end{aligned}$$

so that

$$\forall N \in \mathbb{N} \text{ we have } \mu([0, 1]^n) = 2^{n \cdot N} \cdot \mu \left(\left[0, \frac{1}{2^N} \right]^n \right) \quad (21.1133)$$

Let now $R \in \mathcal{D}^n$ be a Dyadic cube [see definition: 21.79] then there exist a $N \in \mathbb{N}$ and a $z \in \mathbb{Z}^n$ such that $D = \left[\frac{z}{2^N}, \frac{z+1}{2^N} \right] = \prod_{i \in \{1, \dots, n\}} \left[\frac{z_i}{2^N}, \frac{z_i+1}{2^N} \right]$. Then we have

$$\begin{aligned} x \in D &\Leftrightarrow x \in \left[\frac{z}{2^N}, \frac{z+1}{2^N} \right] \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } x_i \in \left[\frac{z_i}{2^N}, \frac{z_i+1}{2^N} \right] \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } \frac{z_i}{2^N} \leq x_i < \frac{z_i+1}{2^N} \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } 0 \leq x_i - \frac{z_i}{2^N} < \frac{z_i+1}{2^N} - \frac{z_i}{2^N} \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } 0 \leq x_i - \frac{z_i}{2^N} < \frac{1}{2^N} \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \text{ we have } x_i - \frac{z_i}{2^N} \in \left[0, \frac{1}{2^N} \right] \end{aligned}$$

$$\begin{aligned} &\Leftrightarrow x - \frac{z}{2^N} \in \left[0, \frac{1}{2^N}\right]^n \\ &\Leftrightarrow x \in \left[0, \frac{1}{2^N}\right]^n + \frac{z}{2^N} \end{aligned}$$

so we have that

$$D = \left[0, \frac{1}{2^N}\right]^n + \frac{z}{2^N} \quad (21.1134)$$

Hence we have

$$\begin{aligned} \mu(D) &\stackrel{[\text{eq: 21.1134}]}{=} \mu\left(\left[0, \frac{1}{2^N}\right]^n + \frac{z}{2^N}\right) \\ &\stackrel{\mu \text{ is translation invariant}}{=} \mu\left(\left[0, \frac{1}{2^N}\right]^n\right) \\ &\stackrel{[\text{eq: 21.1133}]}{=} \frac{\mu([0, 1]^n)}{2^{n \cdot N}} \\ \lambda^n(D) &= \lambda^n\left(\left[\frac{z}{2^N}, \frac{z+1}{2^N}\right]^n\right) \\ &\stackrel{[\text{theorem: 21.205}]}{=} \text{vol}^n\left(\left[\frac{z}{2^N}, \frac{z+1}{2^N}\right]^n\right) \\ &= \prod_{i=1}^n \left(\frac{z_i+1}{2^N} - \frac{z_i}{2^N}\right) \\ &= \left(\frac{1}{2^N}\right)^n \\ &= \frac{1}{2^{n \cdot N}} \end{aligned}$$

so that

$$\forall D \in \mathcal{D} \text{ we have } \mu(D) = \mu([0, 1]^n) \cdot \lambda^n(D) \quad (21.1135)$$

Let now $U \in \mathcal{T}_{|||n}$ then we have either:

$U = \emptyset$. Then $\mu(U) = \mu(\emptyset) = 0 = \mu([0, 1]^n) \cdot 0 = \mu([0, 1]^n) \cdot \lambda^n(\emptyset) = \mu([0, 1]^n) \cdot \lambda^n(U)$

$U \neq \emptyset$. Then by [theorem: 21.602] there exist a pairwise disjoint $\{D_i\}_{i \in \mathbb{N}} \subseteq \mathcal{D}$ such that

$$U = \bigsqcup_{i \in \mathbb{N}} D_i$$

Then we have

$$\begin{aligned} \mu(U) &= \mu\left(\bigsqcup_{i \in \mathbb{N}} D_i\right) \\ &= \sum_{I=1}^{\infty} \mu(D_i) \\ &\stackrel{[\text{eq: 21.1135}]}{=} \sum_{I=1}^{\infty} (\mu([0, 1]^n) \cdot \lambda^n(D_i)) \\ &= \mu([0, 1]^n) \cdot \sum_{I=1}^{\infty} \lambda^n(D_i) \\ &= \mu([0, 1]^n) \cdot \lambda^n\left(\bigsqcup_{i \in \mathbb{N}} D_i\right) \\ &= \mu([0, 1]^n) \cdot \lambda^n(U) \end{aligned}$$

which proves that

$$\forall U \in \mathcal{T}_{|||n} \text{ we have } \mu(U) = (\mu([0, 1]^n) \cdot \lambda^n)(U)$$

The above, the fact that μ is a Radon measure and by [theorems: 21.582, 21.592] $\mu([0, 1]^n) \cdot \lambda^n$ is a Radon measure we have using [theorem: 21.583] that

$$\mu = \mu([0, 1]^n) \cdot \lambda^n \quad \square$$

21.8.2 Pushforward or image measures

Theorem 21.605. Let $\langle X, \mathcal{A} \rangle$ and $\langle Y, \mathcal{B} \rangle$ be two measurable spaces, μ a measure on $\mathcal{A}$ and

$$T: X \rightarrow Y \text{ a } \mathcal{A}, \mathcal{B}\text{-measurable function}$$

then the function

$$\mu[T^{-1}]: \mathcal{B} \rightarrow [0, \infty] \text{ defined by } \mu[T^{-1}](A) = \mu(T^{-1}(A))$$

is a measure on $\mathcal{B}$. The measure $\mu[T^{-1}]$ is called the **pushforward of μ by T** .

Note 21.606. As T is $\mathcal{A}, \mathcal{B}$ -measurable we have that $\forall A \in \mathcal{B} \ T^{-1}(A) \in \mathcal{A}$ so that $\mu(T^{-1}(A))$ is well defined.

Proof. First note that

$$\mu[T^{-1}](\emptyset) = \mu(T^{-1}(\emptyset)) = \mu(\emptyset) \underset{\mu \text{ is a measure}}{=} 0 \quad (21.1136)$$

Let $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{B}$ be pairwise disjoint. Let $i, j \in \mathbb{N}$ with $i \neq j$ then

$$T^{-1}(B_i) \cap T^{-1}(B_j) \underset{[\text{theorem: 2.103}]}{=} T^{-1}(B_i \cap B_j) = T^{-1}(\emptyset) = \emptyset$$

which proves that

$$\{T^{-1}(B_i)\}_{i \in \mathbb{N}} \text{ is pairwise disjoint}$$

Further as T is $\mathcal{A}, \mathcal{B}$ -measurable we have that

$$\{T^{-1}(B_i)\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$$

So we have

$$\begin{aligned} \mu[T^{-1}]\left(\bigsqcup_{i \in \mathbb{N}} B_i\right) &= \mu\left(T^{-1}\left(\bigsqcup_{i \in \mathbb{N}} B_i\right)\right) \\ &\underset{[\text{theorem: 2.144}]}{=} \mu\left(\bigsqcup_{i \in \mathbb{N}} T^{-1}(B_i)\right) \\ &\underset{\mu \text{ is a measure}}{=} \sum_{i=1}^{\infty} \mu(T^{-1}(B_i)) \\ &= \sum_{i=1}^{\infty} \mu[T^{-1}](B_i) \end{aligned}$$

Hence it follows that

$$\mu[T^{-1}] \text{ is a measure on } \mathcal{B} \quad \square$$

Theorem 21.607. We have

1. Let $T: \mathbb{R} \rightarrow \mathbb{R}$ be a linear isomorphism and $\mu: \mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] \rightarrow [0, \infty]$ a Radon measure then

$$\mu[T^{-1}]: \mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] \rightarrow [0, \infty] \text{ defined by } \mu[T^{-1}](A) = \mu(T^{-1}(A))$$

is a well defined Radon measure.

2. Let $n \in \mathbb{N}$, $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear isomorphism and $\mu: \mathcal{B}[\mathbb{R}^n] = \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{||n}] \rightarrow [0, \infty]$ a Radon measure then

$$\mu[T^{-1}]: \mathcal{B}[\mathbb{R}^n] = \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{||n}] \rightarrow [0, \infty] \text{ defined by } \mu[T^{-1}](A) = \mu(T^{-1}(A))$$

is a well defined Radon measure.

Proof.

1. As $T: \mathbb{R} \rightarrow \mathbb{R}$ is a linear isomorphism it follows from [theorem: 14.205] that T is a homeomorphism. Hence

T is open and continuous

so that by [theorem: 21.255] T is $\mathcal{B}[\mathbb{R}, \mathcal{T}_{||}]$, $\mathcal{B}[\mathbb{R}, \mathcal{T}_{||}]$ -measurable. Hence using the previous theorem [theorem: 21.605] we have that

$$\mu[T^{-1}]: \mathcal{B}[\mathbb{R}^n] = \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{||n}] \rightarrow [0, \infty] \text{ defined by } \mu[T^{-1}](A) = \mu(T^{-1}(A))$$

is a well defined measure. Next we prove that $\mu[T^{-1}]$ is locally finite. As μ is a Radon measure it follows from [definition: 21.580] that μ is locally finite. Let $y \in \mathbb{R}$ then as T is a bijection there exist a $x \in \mathbb{R}$ such that $y = T(x)$. As μ is locally finite there exist a open set with $x \in U$ such that $\mu(U) < \infty$. As T is open $V = T(U)$ is a open set with $y = T(x) \in V$. Hence we have

$$\mu[T^{-1}](V) = \mu(T^{-1}(V)) = \mu(T^{-1}(T(U))) \stackrel{[\text{theorem: 2.57}]}{=} \mu(U) < \infty$$

which proves that

$\mu[T^{-1}]$ is locally finite

Using [theorem: 21.584] it follows then that $\mu[T]$ is compact finite and finally by [theorem: 21.588] that $\mu[T]$ is a Radon measure.

2. As $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear isomorphism it follows from [theorem: 14.207] that T is a homeomorphism. Hence

T is open and continuous

so that by [theorem: 21.255] T is $\mathcal{B}[\mathbb{R}^n, \mathcal{T}_{||n}]$, $\mathcal{B}[\mathbb{R}^n, \mathcal{T}_{||n}]$ -measurable. Hence using the previous theorem [theorem: 21.605] we have that

$$\mu[T^{-1}]: \mathcal{B}[\mathbb{R}^n] = \mathcal{B}[\mathbb{R}^n, \mathcal{T}_{||n}] \rightarrow [0, \infty] \text{ defined by } \mu[T^{-1}](A) = \mu(T^{-1}(A))$$

is a well defined measure. Next we prove that $\mu[T^{-1}]$ is locally finite. As μ is a Radon measure it follows from [definition: 21.580] that μ is locally finite. Let $y \in \mathbb{R}^n$ then as T is a bijection there exist a $x \in \mathbb{R}^n$ such that $y = T(x)$. As μ is locally finite there exist a open set with $x \in U$ such that $\mu(U) < \infty$. As T is open $V = T(U)$ is a open set with $y = T(x) \in V$. Hence we have

$$\mu[T^{-1}](V) = \mu(T^{-1}(V)) = \mu(T^{-1}(T(U))) \stackrel{[\text{theorem: 2.57}]}{=} \mu(U) < \infty$$

which proves that

$\mu[T^{-1}]$ is locally finite

Using [theorem: 21.584] it follows then that $\mu[T]$ is compact finite and finally by [theorem: 21.589] that $\mu[T]$ is a Radon measure. □

We examine now how the Borel measures λ , λ^n behaves under a linear isomorphism.

Theorem 21.608. *Let $T: \mathbb{R} \rightarrow \mathbb{R}$ be a linear isomorphism then we have $\forall A \in \mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{||}]$ that*

$$\lambda(T^{-1}(A)) = \lambda(T^{-1}([0, 1])) \cdot \lambda(A)$$

Note 21.609. *By [theorem: 14.205] T is continuous so that by [theorem: 21.255] T is $T^{-1}(A)$, $T^{-1}([0, 1]) \in \mathcal{B}[\mathbb{R}]$ so that $\lambda(T^{-1}(A))$, $\lambda(T^{-1}([0, 1]))$ is well defined.*

Proof. As λ is a Radon measure [see theorem: 21.592] it follows from the previous theorem [theorem: 21.607] that

$$\lambda[T^{-1}]: \mathcal{B}[\mathbb{R}] = \mathcal{B}[\mathbb{R}, \mathcal{T}_{||}] \rightarrow [0, \infty] \text{ is a well defined Radon measure}$$

Next we proof that $\lambda[T^{-1}]$ is translation invariant. As $L: \mathbb{R} \rightarrow \mathbb{R}$ is a linear isomorphism we have from [theorem: 11.178] that

$$K = L^{-1}: \mathbb{R} \rightarrow \mathbb{R} \text{ is a linear isomorphism}$$

Hence we have for $x \in \mathbb{R}$ that

$$\begin{aligned} y \in (T^{-1})(A+x) &\Leftrightarrow y \in K(A+x) \\ &\Leftrightarrow y = K(a+x) \text{ where } a \in A \\ &\Leftrightarrow y = K(a) + K(x) \text{ where } a \in A \\ &\Leftrightarrow y \in K(A) + K(x) \\ &\Leftrightarrow y \in (T^{-1})(A) + (K)(x) \end{aligned}$$

so that

$$T^{-1}(A+x) \underset{[\text{theorem: 2.70}]}{=} (T^{-1})(A+x) = (T^{-1})(A) + K(x) \underset{[\text{theorem: 2.70}]}{=} T^{-1}(A) + K(x)$$

Hence we have

$$\lambda[T^{-1}](A+x) = \lambda(T^{-1}(A)) = \lambda(T^{-1}(A) + K(x)) \underset{[\text{theorem: 21.597}]}{=} \lambda(T^{-1}(A)) = \lambda[T^{-1}](A)$$

which proves that

$$\lambda[T^{-1}] \text{ is translation invariant}$$

Finally using [theorem: 21.603] we have that

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ we have that } \lambda(T^{-1}(A)) = \lambda[T^{-1}](A) = \lambda(T^{-1}([0, 1])) \cdot \lambda(A) \quad \square$$

Next we calculate the value of $\lambda(T^{-1}([0, 1]))$

Theorem 21.610. *Let $T: \mathbb{R} \rightarrow \mathbb{R}$ be a linear isomorphism then we have $\forall A \in \mathcal{B}[\mathbb{R}]$ that*

$$T(1) \neq 0 \text{ and } \lambda[T^{-1}](A) = \lambda(T^{-1}(A)) = \frac{1}{|T(1)|} \cdot \lambda(A)$$

Proof. By [example: 11.179] it follows that

$$T(1) \neq 0 \text{ and } T = T(1) \cdot \text{Id}_{\mathbb{R}}$$

For $T(1)$ we have either:

$0 < T(1)$. Then we have:

$$\begin{aligned} y \in T^{-1}([0, 1]) &\Leftrightarrow T(y) \in [0, 1] \\ &\Leftrightarrow 0 \leq T(y) < 1 \\ &\Leftrightarrow 0 \leq T(1) \cdot y < 1 \\ &\Leftrightarrow 0 \leq y < \frac{1}{T(1)} \\ &\Leftrightarrow y \in \left[0, \frac{1}{T(1)}\right] \end{aligned}$$

so that

$$T^{-1}([0, 1]) = \left[0, \frac{1}{T(1)}\right]$$

Hence

$$\lambda(T^{-1}([0, 1])) = \lambda\left(\left[0, \frac{1}{T(1)}\right]\right) = \text{len}\left(\left[0, \frac{1}{T(1)}\right]\right) = \frac{1}{|T(1)|}$$

in other words

$$\text{If } T \text{ is a isomorphism with } 0 < T(1) \text{ then } \lambda(T^{-1}([0, 1])) = \frac{1}{|T(1)|} \quad (21.1137)$$

$T(1) < 0$. First note that $-\text{Id}$ is a linear isomorphism and

$$\begin{aligned} y \in (-\text{Id})^{-1}((-\text{Id})^{-1}([0, 1])) &\Leftrightarrow (-\text{Id})(y) \in (-\text{Id})^{-1}([0, 1]) \\ &\Leftrightarrow -y \in (-\text{Id})^{-1}([0, 1]) \\ &\Leftrightarrow -(-y) \in [0, 1[\\ &\Leftrightarrow y \in [0, 1[\end{aligned}$$

which proves that

$$(-\text{Id})^{-1}((-\text{Id})^{-1}([0, 1])) = [0, 1[$$

hence we have

$$\begin{aligned} 1 &= \text{len}([0, 1]) \\ &= \lambda([0, 1]) \\ &= \lambda((-\text{Id})^{-1}((-\text{Id})^{-1}([0, 1]))) \\ &\stackrel{[\text{theorem: 21.608}]}{=} \lambda((-\text{Id})^{-1}([0, 1])) \cdot \lambda((-\text{Id})^{-1}([0, 1])) \\ &= (\lambda((-\text{Id})^{-1}([0, 1])))^2 \end{aligned}$$

As λ is a measure so that $0 \leq \lambda((-\text{Id})^{-1}([0, 1]))$ it follows from the above that

$$\lambda((-\text{Id})^{-1}([0, 1])) = 1 \quad (21.1138)$$

Take $L = (-\text{Id}) \circ T$ then L is a isomorphism and $L(1) = (-\text{Id})(T(1)) = -T(1) > 0$. Hence using [eq: 21.1137] we have that

$$\lambda(L([0, 1])) = \frac{1}{|L(1)|} = \frac{1}{|-T(1)|} = \frac{1}{|T(1)|} \quad (21.1139)$$

Further

$$\begin{aligned} \lambda(L^{-1}([0, 1])) &= \lambda((-(-\text{Id}) \circ T)^{-1}([0, 1])) \\ &= \lambda(T^{-1}((-\text{Id})^{-1}([0, 1]))) \\ &\stackrel{[\text{theorem: 21.608}]}{=} \lambda(T^{-1}([0, 1])) \cdot \lambda((-\text{Id})^{-1}([0, 1])) \\ &\stackrel{[\text{eq: 21.1138}]}{=} \lambda(T^{-1}([0, 1])) \cdot 1 \\ &= \lambda(T^{-1}([0, 1])) \end{aligned}$$

which combined with [eq: 21.1139] proves that

$$\lambda(T^{-1}([0, 1])) = \frac{1}{|T(1)|}$$

So in all cases we have

$$\lambda(T^{-1}([0, 1])) = \frac{1}{|T(1)|} \quad (21.1140)$$

Hence we have

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ that } \lambda(T^{-1}(A)) \stackrel{[\text{theorem: 21.608}]}{=} \lambda(T^{-1}([0, 1])) \cdot \lambda(A) \stackrel{[\text{eq: 21.1140}]}{=} \frac{1}{|T(1)|} \cdot \lambda(A) \quad \square$$

We have now a similar result for λ^n .

Theorem 21.611. *Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear isomorphism then we have $\forall A \in \mathcal{B}[\mathbb{R}^n]$ we have*

$$\lambda^n[T^{-1}](A) = \lambda^n(T^{-1}(A)) = \lambda^n(T^{-1}([0, 1]^n)) \cdot \lambda^n(A)$$

Proof. As λ is a Radon measure [see theorem: 21.592] it follows from [theorem: 21.607] that

$\lambda^n[T^{-1}]: \mathcal{B}[\mathbb{R}^n] \rightarrow [0, \infty]$ defined by $\lambda^n[T^{-1}](A) = \lambda(T^{-1}(A))$ is a well defined Radon measure

Next we prove that $\lambda^n[T^{-1}]$ is translation invariant. As T is a linear isomorphism it follows from [theorem: 11.178] that

$$K = L^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ is a linear isomorphism}$$

Hence we have for $x \in \mathbb{R}^n$ that

$$\begin{aligned} y \in (T^{-1})(A+x) &\Leftrightarrow y \in K(A+x) \\ &\Leftrightarrow y = K(a+x) \text{ where } a \in A \\ &\Leftrightarrow y = K(a) + K(x) \text{ where } a \in A \\ &\Leftrightarrow y \in K(A) + K(x) \\ &\Leftrightarrow y \in (T^{-1})(A) + (K)(x) \end{aligned}$$

so that

$$T^{-1}(A+x) \underset{[\text{theorem: 2.70}]}{=} (T^{-1})(A+x) = (T^{-1})(A) + K(x) \underset{[\text{theorem: 2.70}]}{=} T^{-1}(A) + K(x)$$

Hence we have

$$\lambda^n[T^{-1}](A+x) = \lambda^n(T^{-1}(A+x)) = \lambda^n(T^{-1}(A) + K(x)) \underset{[\text{theorem: 21.598}]}{=} \lambda^n(T^{-1}(A)) = \lambda[T^{-1}](A)$$

which proves that

$$\lambda^n[T^{-1}] \text{ is translation invariant}$$

Finally using [theorem: 21.603] we have that

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have that } \lambda^n(T^{-1}(A)) = \lambda^n[T^{-1}](A) = \lambda^n(T^{-1}([0, 1])) \cdot \lambda^n(A) \quad \square$$

Next we calculate the value of $\lambda^n(T^{-1}([0, 1]))$. This is a little bit complicated so we spread the calculation over several lemma's.

Lemma 21.612. *Let $n \in \mathbb{N}$, $E = \{e_1, \dots, e_n\}$ the canonical basis in $\mathbb{R}^n$ [see example: 11.167] and $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ a linear isomorphism we have that the matrix $\mathcal{M}(T; E, E)$ is diagonal [so by [definition: 11.356] $\mathcal{M}(T; E, E)_{i,j} = \lambda_{i,j} \cdot \mathcal{M}(T; E, E)_{i,i}$ where $\mathcal{M}(T; E, E)_{i,i} \in [0, \infty[$ then we have*

$$\det(T) \neq 0$$

and

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \lambda^n[T^{-1}](A) = \lambda^n(T^{-1}(A)) = \frac{1}{\det(T)} \cdot \lambda^n(A)$$

Note 21.613. $\mathcal{M}(T; E, E)$ is defined [see definition: 11.339] to be the unique matrix so that

$$\forall i \in \{1, \dots, n\} \quad T(e_i) = \sum_{j=1}^n \mathcal{M}(T; E, E)_{j,i} \cdot e_j$$

Further using [theorem: 11.359] the determinant of a linear transformation is related to the determinant of the matrix of the transformation by

$$\det(T) = \det(\mathcal{M}(T; E, E))$$

Proof. Using the previous theorem [theorem: 21.611] we have that

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \lambda^n[T^{-1}](A) = \lambda^n(T^{-1}(A)) = \lambda^n(T^{-1}([0, 1^n])) \cdot \lambda^n(A) \quad (21.1141)$$

As $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear isomorphism we have by [theorem: 11.323] that

$$\det(T) \neq 0 \quad (21.1142)$$

As $\mathcal{M}[T; E, E]$ is diagonal we have

$$\det(T) \underset{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}(T; E, E)) \underset{[\text{theorem: 11.357}]}{=} \prod_{i=1}^n \mathcal{M}(T; E, E)_{i,i} \quad (21.1143)$$

As also $\forall i \in \{1, \dots, n\}$ $\mathcal{M}(T; E, E)_{i,i} \in [0, \infty[$ it follows from the above that

$$\forall i \in \{1, \dots, n\} \quad 0 < \mathcal{M}(T; E, E)_{i,i} \quad (21.1144)$$

Define

$$M \in \mathbb{R}^n \text{ by } M_i = \mathcal{M}(T; E, E)_{i,i} \quad \forall i \in \{1, \dots, n\}$$

then we have $\forall x \in \mathbb{R}^n$ that

$$T(x)_i \stackrel{[\text{example: 11.342}]}{=} \sum_{j=1}^n \mathcal{M}(T; E, E)_{j,i} x_j = \sum_{j=1}^n \delta_{i,j} \cdot \mathcal{M}(T; E, E)_{i,i} x_j = \mathcal{M}(T; E, E)_{i,i} \cdot x_i = x_i \cdot M_i$$

so that

$$\forall x \in \mathbb{R}^n \text{ we have } \forall i \in \{1, \dots, n\} \quad T(x)_i = M_i \cdot x_i \quad (21.1145)$$

Hence

$$\begin{aligned} x \in T^{-1}([0, 1]^n) &\Leftrightarrow T(x) \in [0, 1]^n \\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \quad T(x)_i \in [0, 1[\\ &\stackrel{[\text{eq: 21.1145}]}{\Leftrightarrow} \forall i \in \{1, \dots, n\} \quad x_i \cdot M_i \in [0, 1[\\ &\Leftrightarrow \forall i \in \{1, \dots, n\} \quad 0 \leq x_i \cdot M_i < 1 \\ &\stackrel{[\text{eq: 21.1144}]}{\Leftrightarrow} \forall i \in \{1, \dots, n\} \quad 0 \leq x_i < \frac{1}{M_i} \\ &\Leftrightarrow x \in \prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{M_i} \right[\end{aligned}$$

which proves that

$$T^{-1}([0, 1]^n) = \prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{M_i} \right[$$

Hence we have

$$\begin{aligned} \lambda^n[T^{-1}([0, 1]^n)] &= \lambda^n(T^{-1}([0, 1]^n)) \\ &= \lambda^n\left(\prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{M_i} \right[\right) \\ &= \text{vol}^n\left(\prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{M_i} \right[\right) \\ &= \prod_{i=1}^n \frac{1}{M_i} \\ &= \frac{1}{\prod_{i=1}^n M_i} \\ &= \frac{1}{\prod_{i=1}^n \mathcal{M}[T; E, E]_{i,i}} \\ &\stackrel{[\text{eq: 21.1143}]}{=} \det(T) \end{aligned}$$

Substituting this result in [eq: 21.1141]

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \lambda^n[T^{-1}(A)] = \lambda^n(T^{-1}(A)) = \frac{1}{\det(T)} \cdot \lambda^n(A) \quad \square$$

Next examine the behavior of λ^n under a Unitary Transformation [see definition: 18.94].

Lemma 21.614. *Let $n \in \mathbb{N}$, $U: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a **unitary** transformation [using the canonical inner product defined in [example: 18.4]] then we have that*

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ that } \lambda^n[U^{-1}(A)] = \lambda^n(U^{-1}(A)) = \lambda^n(A)$$

Note 21.615. Using [theorem: 18.97] U is a linear isomorphism so that by [theorem: 21.607] $\lambda^n[U^{-1}]$ is well defined.

Proof. First note that by [theorem: 14.329] all norms on $\mathbb{R}^n$ are equivalent, so if $\|\cdot\|_n$ is the canonical norm on $\mathbb{R}$ [see example: 14.101] and $\|\cdot\|_e$ is the Euclidean norm on $\mathbb{R}^n$ [see definition: 18.20] then

$$\mathcal{T}_{\|\cdot\|_n} = \mathcal{T}_e \quad (21.1146)$$

As by [theorem: 18.97] U is a linear isomorphism it follows from [theorem: 21.611] we have that

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \lambda^n[U^{-1}](A) = \lambda^n(U^{-1}(A)) = \lambda^n(U^{-1}([0, 1]^n)) \cdot \lambda^n(A) \quad (21.1147)$$

Take the open ball $B_{\|\cdot\|_e}(0, 1) = \{x \in \mathbb{R}^n \mid \|x\|_e < 1\}$ then $B_{\|\cdot\|_e}(0, 1) \in \mathcal{T}_{\|\cdot\|_e} \stackrel{[eq: 21.1147]}{=} \mathcal{T}_{\|\cdot\|_n} \subseteq \mathcal{B}[\mathbb{R}^n]$ then we have by the above that

$$\lambda^n(U^{-1}(B_{\|\cdot\|_e}(0, 1))) = \lambda^n(U^{-1}([0, 1]^n)) \cdot \lambda^n(B_{\|\cdot\|_e}(0, 1)) \quad (21.1148)$$

By [theorem: 18.97] U is a linear isometry so that

$$\forall x \in \mathbb{R}^n \text{ we have } \|U(x)\|_e = \|x\|_e$$

Hence we have

$$\begin{aligned} x \in B_{\|\cdot\|_e}(0, 1) &\Leftrightarrow \|x\|_e < 1 \\ &\stackrel{[eq: 21.1148]}{\Leftrightarrow} \|U(x)\|_e < 1 \\ &\Leftrightarrow U(x) \in B_{\|\cdot\|_e}(0, 1) \\ &\Leftrightarrow x \in U^{-1}(B_{\|\cdot\|_e}(0, 1)) \end{aligned}$$

so that $B_{\|\cdot\|_e}(0, 1) = U^{-1}(B_{\|\cdot\|_e}(0, 1))$, substituting this result in [eq: 21.1147] gives then

$$\lambda^n(B_{\|\cdot\|_e}(0, 1)) = \lambda^n(U^{-1}([0, 1]^n)) \cdot \lambda^n(B_{\|\cdot\|_e}(0, 1)) \quad (21.1149)$$

Let $x \in B_{\|\cdot\|_e}(0, 1)$ then we have $\|x\|_e < 1$. Let $i \in \{1, \dots, n\}$ then we have

$$-x_i, x_i \stackrel{[theorem: 10.73]}{\leq} |x_i| = \sqrt{x_i^2} \stackrel{[theorem: 10.71]}{\leq} \sqrt{\sum_{i=1}^n x_i^2} = \|x\|_e < 1$$

so that $-1 < x_i < 1$ or $x_i \in [-1, 1[$. Hence we have proved that $x \in [-1, 1]^n$ so that $B_{\|\cdot\|_e}(0, 1) \subseteq [-1, 1]^n$ from which it follows that $\lambda^n(B_{\|\cdot\|_e}(0, 1)) \leq \lambda^n([-1, 1]^n) \stackrel{[theorem: 21.205]}{=} \text{vol}^n([-1, 1]^n) = 2^n < \infty$. In other words

$$\lambda^n(B_{\|\cdot\|_e}(0, 1)) < \infty \quad (21.1150)$$

Further if $x \in \prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{\sqrt{n}}\right[$ then $\forall i \in \{1, \dots, n\}$ we have $0 < x_i < \frac{1}{\sqrt{n}}$ so that

$$\|x\|_e = \sqrt{\sum_{i=1}^n x_i^2} \stackrel{[theorem: 10.71]}{\leq} \sqrt{\sum_{i=1}^n \frac{1}{n}} = \sqrt{n \cdot \frac{1}{n}} = 1$$

which prove that $\prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{\sqrt{n}}\right[\subseteq B_{\|\cdot\|_e}(0, 1)$ so that

$$0 < n \cdot \frac{1}{\sqrt{n}} = \text{vol}^n\left(\prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{\sqrt{n}}\right[\right) = \lambda^n\left(\prod_{i \in \{1, \dots, n\}} \left[0, \frac{1}{\sqrt{n}}\right[\right) \leq \lambda^n(B_{\|\cdot\|_e}(0, 1))$$

Hence

$$0 < \lambda^n(B_{\|\cdot\|_e}(0, 1)) \quad (21.1151)$$

The above together with [eq: 21.1150] allows us to multiply both sides of [eq: 21.1149] by $\frac{1}{\lambda^n(B_{\|\cdot\|_e}(0, 1))}$ giving

$$\lambda^n(U^{-1}([0, 1]^n)) = 1$$

Substituting this in [eq: 21.1147] proves finally that

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \lambda^n[U^{-1}](A) = \lambda^n(U^{-1}(A)) = \lambda^n(A) \quad \square$$

We show now that every linear isomorphism $L: \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be written as

$$T = S \circ D \circ T$$

where S and T are unitary operators and D is a operator with a diagonal matrix [using the canonical basis of $\mathbb{R}^n$].

Lemma 21.616. *Let $n \in \mathbb{N}$, $E = \{e_1, \dots, e_n\}$ the canonical basis of $\mathbb{R}^n$ [see example: 11.167] and $L: \mathbb{R}^n \rightarrow \mathbb{R}^n$ a linear isomorphism then there exist two **unitary** operators $S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a linear isomorphism $D: \mathbb{R}^n \rightarrow \mathbb{R}^n$ where $\mathcal{M}[D; E, E]$ with*

$$\forall i, j \in \{1, \dots, n\} \mathcal{M}(D; E, E)_{i,j} = \mathcal{M}(D; E, E)_{i,i} \cdot \delta_{i,j} > 0$$

such that

$$L = S \circ D \circ T$$

Proof. Consider the adjoint L^* of L [see theorem: 18.37] then we have

$$(L^* \circ L)^* \underset{[\text{theorem: 18.41}]}{=} L^* \circ (L^*)^* \underset{[\text{theorem: 18.38}]}{=} L^* \circ L$$

which proves that $L^* \circ L$ is self adjoint. Using [theorem: 18.119] there exist unitary operator U such that

$$\mathcal{M}(U^* \circ (L^* \circ L) \circ U; E, E) \text{ is diagonal}$$

Take

$$D' = U^* \circ (L^* \circ L) \circ U \quad (21.1152)$$

then we have that

$$\forall i, j \in \{1, \dots, n\} \text{ we have } \mathcal{M}(D'; E, E)_{i,j} = \lambda'_i \cdot \delta_{i,j} \text{ where } \lambda'_i = \mathcal{M}(D'; E, E)_{i,i} \quad (21.1153)$$

so that $\forall i \in \{1, \dots, n\}$ we have

$$D'(e_i) = \sum_{j=1}^n \mathcal{M}(D'; E, E)_{j,i} \cdot e_j = \sum_{j=1}^n \lambda'_j \cdot \delta_{j,i} \cdot e_j = \lambda'_i \cdot e_i$$

which proves that

$$\forall i \in \{1, \dots, n\} \text{ we have } D'(e_i) = \lambda'_i \cdot e_i \quad (21.1154)$$

As L is a linear isomorphism and U is a linear isomorphism [see theorem: 18.97] we have that $L \circ U$ is a linear isomorphism [see theorem: 11.182] so that by [theorem: 11.195] $\ker(L \circ U) = \{0\}$. Hence we have

$$\forall i \in \{1, \dots, n\} \text{ we have that } L(U(e_i)) = (L \circ U)(e_i) \underset{e_i \neq 0}{\neq} 0 \quad (21.1155)$$

Next we have $\forall i \in \{1, \dots, n\}$ that

$$\begin{aligned} \lambda'_i & \underset{[\text{eq: 21.1153}]}{=} \mathcal{M}(D'; E, E)_{i,i} \\ & \underset{[\text{theorem: 18.30}]}{=} \langle D'(e_i), e_i \rangle \\ & \underset{[\text{eq: 21.1152}]}{=} \langle (U^* \circ (L^* \circ L) \circ U)(e_i), e_i \rangle \\ & = \langle U^*(L^*(L(U(e_i)))) , e_i \rangle \\ & \underset{[\text{theorems: 18.37}]}{=} \langle L^*(L(U(e_i))), U(e_i) \rangle \\ & \underset{[\text{theorems: 18.37}]}{=} \langle L(U(e_i)), L(U(e_i)) \rangle \\ & = (\|L(U(e_i))\|_e)^2 \\ & \underset{[\text{eq: 21.1154}]}{>} 0 \end{aligned}$$

hence we have

$$\forall i \in \{1, \dots, n\} \text{ we have } 0 < \lambda'_i = \mathcal{M}(D'; E, E)_{i,i} \quad (21.1156)$$

Using the above and [theorem: 11.337] we can define $D, D'' \in \text{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ such that

$$\forall i \in \{1, \dots, n\} \quad D(e_i) = \sqrt{\lambda'_i} \cdot e_i \text{ and } D''(e_i) = \frac{1}{\sqrt{\lambda'_i}} \cdot e_i \quad (21.1157)$$

Then we have $\forall i \in \{1, \dots, n\}$ that

$$\begin{aligned} (D \circ D'')(e_i) &= D(D''(e_i)) \\ &\stackrel{[\text{eq: 21.1157}]}{=} D\left(\frac{1}{\sqrt{\lambda'_i}} \cdot e_i\right) \\ &= \frac{1}{\sqrt{\lambda'_i}} \cdot D(e_i) \\ &\stackrel{[\text{eq: 21.1157}]}{=} \left(\frac{1}{\sqrt{\lambda'_i}} \cdot \sqrt{\lambda'_i}\right) \cdot e_i \\ &= e_i \\ &= \text{Id}_{\mathbb{R}^n}(e_i) \\ &= \left(\sqrt{\lambda'_i} \cdot \frac{1}{\sqrt{\lambda'_i}}\right) \cdot e_i \\ &= \sqrt{\lambda'_i} \cdot \left(\frac{1}{\sqrt{\lambda'_i}} \cdot e_i\right) \\ &\stackrel{[\text{eq: 21.1157}]}{=} \sqrt{\lambda'_i} \cdot D''(e_i) \\ &= D''(\sqrt{\lambda'_i} \cdot e_i) \\ &\stackrel{[\text{eq: 21.1157}]}{=} D''(D(e_i)) \\ &= (D'' \circ D)(e_i) \end{aligned}$$

so that [theorem: 11.336] we have

$$D \circ D'' = \text{Id}_{\mathbb{R}^n} = D'' \circ D$$

which proves that

$$D, D^{-1} \text{ are linear isomorphisms and } D^{-1} = D'' \quad (21.1158)$$

Further we have $\forall i, j \in \{1, \dots, n\}$ that

$$\begin{aligned} \mathcal{M}(D^{-1}; E, E)_{i,j} &\stackrel{[\text{theorem: 18.30}]}{=} \langle D^{-1}(e_j), e_i \rangle \\ &\stackrel{[\text{eq: 21.1158}]}{=} \langle D''(e_j), e_i \rangle \\ &\stackrel{[\text{eq: 21.1157}]}{=} \left\langle 1 / \sqrt{\lambda'_j} \cdot e_j, e_i \right\rangle \\ &= 1 / \sqrt{\lambda'_j} \cdot \delta_{i,j} \end{aligned}$$

so that

$$\forall i, j \in \{1, \dots, n\} \text{ we have } \mathcal{M}(D^{-1}; E, E)_{i,j} = \frac{1}{\sqrt{\lambda'_j}} \cdot \delta_{i,j} \quad (21.1159)$$

Now $\forall i \in \{1, \dots, n\}$ we have

$$\begin{aligned}
 ((L \circ U \circ D^{-1})^* \circ (L \circ U \circ D^{-1}))(e_i) & \stackrel{[\text{theorem: 18.41}]}{=} ((D^{-1})^* \circ U^* \circ L^* \circ L \circ U \circ D^{-1})(e_i) \\
 & \stackrel{[\text{theorem: 18.88}]}{=} (D^{-1} \circ (U^* \circ L^* \circ L \circ U) \circ D^{-1})(e_i) \\
 & \stackrel{[\text{eq: 21.1158}]}{=} (D'' \circ (U^* \circ L^* \circ L \circ U) \circ D'')(e_i) \\
 & \stackrel{[\text{eq: 21.1152}]}{=} (D'' \circ D' \circ D'')(e_i) \\
 & = D''(D'(D''(e_i))) \\
 & \stackrel{[\text{eq: 21.1157}]}{=} D''\left(D'\left(\frac{1}{\sqrt{\lambda'_i}} \cdot e_i\right)\right) \\
 & = \frac{1}{\sqrt{\lambda'_i}} \cdot D''(D'(e_i)) \\
 & \stackrel{[\text{eq: 21.1154}]}{=} \frac{1}{\sqrt{\lambda'_i}} \cdot D''(\lambda'_i \cdot e_i) \\
 & = \frac{\lambda'_i}{\sqrt{\lambda'_i}} \cdot D''(e_i) \\
 & \stackrel{[\text{eq: 21.1157}]}{=} \left(\frac{\lambda'_i}{\sqrt{\lambda'_i}} \cdot \frac{2}{\sqrt{\lambda'_i}}\right) \cdot e_i \\
 & = e_i \\
 & = \text{Id}_{\mathbb{R}^n}(e_i)
 \end{aligned}$$

which proves by [theorem: 11.336] that

$$(L \circ U \circ D^{-1})^* \circ (L \circ U \circ D^{-1}) = \text{Id}_{\mathbb{R}^n}$$

Using then [theorem: 18.95] it follows that $L \circ U \circ D^{-1}$ is unitary, further as U is unitary we have by [theorem: 18.97] that U^{-1} is unitary. Hence if we take $S = L \circ U \circ D^{-1}$ and $T = U^{-1}$ then we have that

$$S, T \text{ are unitary operators} \quad (21.1160)$$

Now

$$\begin{aligned}
 S \circ D \circ T &= (L \circ U \circ D^{-1}) \circ D \circ (U^{-1}) \\
 &= L \circ U \circ (D^{-1} \circ D) \circ U^{-1} \\
 &= L \circ (U \circ U^{-1}) \\
 &= L
 \end{aligned}$$

which proves that

$$L = S \circ D \circ T \quad (21.1161)$$

Further we have that $\forall i, j \in \{1, \dots, n\}$

$$\mathcal{M}[D; E, E]_{i,j} \stackrel{[\text{theorem: 18.30}]}{=} \langle D(e_j), e_i \rangle = \langle \sqrt{\lambda'_i} \cdot e_j, e_i \rangle = \sqrt{\lambda'_i} \cdot \delta_{j,i} > 0$$

which proves that

$$\forall i, j \in \{1, \dots, n\} \quad \mathcal{M}[D; E, E]_{i,j} = \mathcal{M}[D; E, E]_{i,i} \cdot \delta_{i,j} > 0$$

The above together with [eqs: 21.1160 and 21.1161] prove the lemma. $\square$

We are now ready to proof that for linear isomorphisms L we have $\lambda^n[L^{-1}](A) = \frac{1}{|\det(L)|} \cdot \lambda^n(A)$.

Theorem 21.617. *Let $n \in \mathbb{N}$, $L: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear isomorphism then $\forall A \in \mathcal{B}[\mathbb{R}^n]$ we have that*

$$\lambda^n[L^{-1}](A) = \lambda^n(L^{-1}(A)) = \frac{1}{|\det(L)|} \cdot \lambda^n(A)$$

Proof. Let $E = \{e_1, \dots, e_n\}$ be the canonical basis for $\mathbb{R}^n$ then by [lemma:21.616] there exist a linear isomorphism D where

$$\forall i, j \in \{1, \dots, n\} \text{ we have } \mathcal{M}(D; E, E)_{i,j} = \mathcal{M}(D; E, E)_{i,i} \cdot \lambda_{i,j} > 0 \quad (21.1162)$$

is diagonal and two unitary transformations S, T such that

$$L = S \circ D \circ T$$

Hence we have

$$\begin{aligned} \lambda^n[L^{-1}]([0, 1]^n) &= \lambda^n(L^{-1}([0, 1]^n)) \\ &= \lambda^n((S \circ D \circ T)^{-1}([0, 1]^n)) \\ &\stackrel{[\text{theorem: 2.23}]}{=} \lambda^n(T^{-1}(D^{-1}(S^{-1}([0, 1]^n)))) \\ &\stackrel{[\text{lemma: 21.614}]}{=} \lambda^n(D^{-1}(S^{-1}([0, 1]^n))) \\ &\stackrel{[\text{lemma: 21.612}]}{=} \frac{1}{\det(D)} \cdot \lambda^n(S^{-1}([0, 1]^n)) \\ &\stackrel{[\text{lemma: 21.614}]}{=} \frac{1}{\det(D)} \cdot \lambda^n([0, 1]^n) \\ &= \frac{1}{\det(D)} \cdot \text{vol}^n([0, 1]^n) \\ &= \frac{1}{\det(D)} \cdot \prod_{i=1}^n (1 - 0) \\ &= \frac{1}{\det(D)} \end{aligned}$$

which proves that

$$\lambda^n(L^{-1}([0, 1]^n)) = \frac{1}{\det(D)} \quad (21.1163)$$

Now

$$\begin{aligned} \det(L) &= \det(S \circ D \circ T) \\ &\stackrel{[\text{theorem: 11.315}]}{=} \det(S) \cdot \det(D) \cdot \det(T) \\ &\stackrel{[\text{theorem: 18.100}]}{=} (\pm 1) \cdot \det(D) \cdot (\pm 1) \\ &= \pm \det(D) \end{aligned}$$

further

$$\det(D) \stackrel{[\text{theorem: 11.359}]}{=} \det(\mathcal{M}(D; E, E)) \stackrel{[\text{theorem: 11.357}]}{=} \prod_{i=1}^n \mathcal{M}(D; E, E)_{i,i} \stackrel{[\text{eq: 21.1162}]}{>} 0$$

so that $\det(D) = |\det(D)| = |\pm \det(D)| = |\det(L)|$ which combined with [eq: 21.1163] proves

$$\lambda^n(L^{-1}([0, 1]^n)) = \frac{1}{|\det(L)|}$$

Finally we have $\forall A \in \mathcal{B}[\mathbb{R}^n]$ that

$$\lambda^n[L^{-1}](A) = \lambda^n(L^{-1}(A)) \stackrel{[\text{lemma: 21.611}]}{=} \lambda^n(L^{-1}([0, 1]^n)) \cdot \lambda^n(A) = \frac{1}{|\det(L)|} \cdot \lambda^n(A)$$

or

$$\lambda^n[L^{-1}](A) = \lambda^n(L^{-1}(A)) = \frac{1}{|\det(L)|} \cdot \lambda^n(A) \quad \square$$

21.8.3 Integral and change of variables

We prove now the general change of variable formula for measure integrals

$$\int (f \circ T) d\mu = \int f d(\mu[T^{-1}])$$

We do this in three steps.

1. First we prove that for $f \in \mathcal{S}^+[X, \mathcal{A}]$

$$\int^S (f \circ T) d\mu = \int^S f d(\mu[T^{-1}])$$

2. Next we use [theorem: 21.329] and the Monotone Convergence theorem [see theorem: 21.372] to prove that for $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have

$$\int^+ (f \circ T) d\mu = \int^+ f d\mu[T^{-1}]$$

3. Finally we prove the result for $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$.

Lemma 21.618. *Let X, Y be sets, $B \subseteq Y$ and $T: X \rightarrow Y$ a function then*

$$\mathcal{X}_{Y,B} \circ T = \mathcal{X}_{X,T^{-1}(B)}$$

Proof. Let $x \in X$ then we have for x either:

$x \in T^{-1}(B)$. Then $T(x) \in B$ so that $\mathcal{X}_{X,T^{-1}(B)}(x) = 1 = \mathcal{X}_{Y,B}(T(x))$.

$x \in X \setminus T^{-1}(B)$. Then $T(x) \notin B \Rightarrow T(x) \in Y \setminus B$ so that $\mathcal{X}_{X,T^{-1}(B)}(x) = 0 = \mathcal{X}_{Y,B}(T(x))$.

which proves that $\forall x \in X$ we have $\mathcal{X}_{X,T^{-1}(B)}(x) = \mathcal{X}_{Y,B}(T(x)) = (\mathcal{X}_{Y,B} \circ T)(x)$ from which it follows that

$$\mathcal{X}_{X,T^{-1}(B)} = \mathcal{X}_{Y,B} \circ T \quad \square$$

Lemma 21.619. *Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be two measurable spaces, μ a measure on $\mathcal{B}$, $T: X \rightarrow Y$ a $\mathcal{A}, \mathcal{B}$ -measurable function and $f \in \mathcal{S}^+[Y, \mathcal{B}]$ then we have that*

$$f \circ T \in \mathcal{S}^+[X, \mathcal{A}]$$

and

$$\int^S (f \circ T) d\mu = \int^S f d\mu[T^{-1}]$$

Proof. As $f \in \mathcal{S}^+[Y, \mathcal{B}]$ we have by [theorem: 21.301] that there exists a $n \in \mathbb{N}$, $\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{B}$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ so that

$$f = \sum_{i=1}^n f_i \cdot \mathcal{X}_{Y,B_i} \quad (21.1164)$$

Let $x \in X$ then we have

$$\begin{aligned} (f \circ T)(x) &= f(T(x)) \\ &= \sum_{i=1}^n f_i \cdot \mathcal{X}_{Y,B_i}(T(x)) \\ &= \sum_{i=1}^n f_i \cdot (\mathcal{X}_{Y,B_i} \circ T)(x) \\ &\stackrel{[\text{lemma: 21.618}]}{=} \sum_{i=1}^n f_i \cdot \mathcal{X}_{X,T^{-1}(B_i)}(x) \end{aligned}$$

which proves that

$$f \circ T = \sum_{i=1}^n f_i \cdot \mathcal{X}_{X,T^{-1}(B_i)} \quad (21.1165)$$

Further as T is $\mathcal{A}, \mathcal{B}$ -measurable and $\{B_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{B}$ it follows that $\{T^{-1}(B_i)\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ which together with the above proves by [theorem: 21.301] that

$$f \circ T \in \mathcal{S}^+[X, \mathcal{A}]$$

Finally

$$\begin{aligned}
 \int^S (f \circ T) d\mu & \stackrel{[\text{definition: 21.305}] \wedge [\text{eq: 21.1165}]}{=} \sum_{i=1}^n f_i \cdot \mu(T^{-1}(B_i)) \\
 & \stackrel{[\text{theorem: 21.605}]}{=} \sum_{i=1}^n f_i \cdot \mu[T^{-1}](B_i) \\
 & \stackrel{[\text{definition: 21.305}] \wedge [\text{eq: 21.1164}]}{=} \int^S f d\mu[T^{-1}]
 \end{aligned}$$

□

Lemma 21.620. Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be two measurable spaces, μ a measure on $\mathcal{A}$, $T: X \rightarrow Y$ a $\mathcal{A}, \mathcal{B}$ -measurable function and $f \in \mathcal{M}^+[Y, \mathcal{B}, \overline{\mathbb{R}}]$ then

$$f \circ T \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ (f \circ T) d\mu = \int^+ f d\mu[T^{-1}]$$

Proof. As $f \in \mathcal{M}^+[Y, \mathcal{B}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.329] that there exist a sequence

$$\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[Y, \mathcal{B}] \quad (21.1166)$$

such that

$$\forall y \in Y \text{ we have } \forall n \in \mathbb{N} \ f_n(y) \leq f_{n+1}(y) \text{ and } \lim_{n \rightarrow \infty} f_n(y) \text{ exist with } f(y) = \lim_{n \rightarrow \infty} f_n(y) \quad (21.1167)$$

and using the measure $\mu[T^{-1}]$ on $\mathcal{B}$ we have that

$$\lim_{n \rightarrow \infty} \int^S f_n d\mu[T^{-1}] \text{ exists and } \int^+ f d\mu[T^{-1}] = \lim_{n \rightarrow \infty} \int^S f_n d\mu[T^{-1}]$$

Using [theorem: 21.317] the above becomes

$$\lim_{n \rightarrow \infty} \int^+ f_n d\mu[T^{-1}] \text{ exists and } \int^+ f d\mu[T^{-1}] = \lim_{n \rightarrow \infty} \int^+ f_n d\mu[T^{-1}] \quad (21.1168)$$

Further using [lemma: 21.619] on [eq: 21.1166] we have that

$$\{f_n \circ T\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}] \text{ so that by [theorem: 21.317] } \{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}^+[Y, \mathcal{B}, \overline{\mathbb{R}}] \quad (21.1169)$$

and

$$\forall n \in \mathbb{N} \ \int^S (f_n \circ T) d\mu = \int^S f_n d\mu[T^{-1}] \stackrel{[\text{theorem: 21.317}]}{\Rightarrow} \int^+ (f_n \circ T) d\mu = \int^+ f_n d\mu[T^{-1}] \quad (21.1170)$$

Now we have

$$\forall x \in X \text{ that } (f_n \circ T)(x) = f_n(T(x)) \underset{[\text{eq: 21.1167}]}{\leq} f_{n+1}(T(x)) = (f_{n+1} \circ T)(x) \quad (21.1171)$$

and, as $\forall x \in X \ T(x) \in Y$, that by [eq: 21.1167],

$$\forall x \in X \ (f \circ T)(x) = f(T(x)) = \lim_{n \rightarrow \infty} f_n(T(x)) = \lim_{n \rightarrow \infty} (f_n \circ T)(x) \text{ exist} \quad (21.1172)$$

Thanks to [eqs: 21.1169, 21.1171, 21.1172] we can apply the Monotone Convergence Theorem [see theorem: 21.372] that

$$f \circ T \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ (f \circ T) d\mu = \lim_{n \rightarrow \infty} \int^+ (f_n \circ T) d\mu \stackrel{[\text{eq: 21.1170}]}{=} \lim_{n \rightarrow \infty} \int^+ f_n d\mu[T^{-1}] \stackrel{[\text{eq: 21.1168}]}{=} \int^+ f d\mu[T^{-1}]$$

so that

$$\int^+ (f \circ T) d\mu = \int^+ f d\mu[T^{-1}] \quad \square$$

Lemma 21.621. *Let X, Y be sets, $T: X \rightarrow Y$ a function and $f: Y \rightarrow \overline{\mathbb{R}}$ then*

$$(f \circ T)^+ = f^+ \circ T \text{ and } (f \circ T)^- = f^- \circ T$$

Proof. Let $x \in X$ then we have

$$(f^+ \circ T)(x) = f^+(T(x)) = \max(f(T(x)), 0) = \max((f \circ T)(x), 0) = (f \circ T)^+(x)$$

$$(f^- \circ T)(x) = f^-(T(x)) = -\min(f(T(x)), 0) = -\min((f \circ T)(x), 0) = (f \circ T)^-(x)$$

which proves that $(f \circ T)^+ = f^+ \circ T$ and $(f \circ T)^- = f^- \circ T$. $\square$

Theorem 21.622. (Change of Variables (1)) *Let $\langle X, \mathcal{A} \rangle, \langle Y, \mathcal{B} \rangle$ be measurable spaces, μ a measure on $\mathcal{B}$, $T: X \rightarrow Y$ a $\mathcal{A}, \mathcal{B}$ -measurable function and $f \in \mathcal{L}[X, \mathcal{A}, \mu[T^{-1}], \overline{\mathbb{R}}]$ then we have*

$$f \circ T \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

and

$$\int (f \circ T) d\mu = \int f d\mu[T^{-1}]$$

Proof. As $f \in \mathcal{L}[X, \mathcal{A}, \mu[T^{-1}], \overline{\mathbb{R}}]$ we have by [definition: 21.345] that

$$f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}] \xRightarrow{[\text{theorem: 21.342}]} f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \quad (21.1173)$$

and

$$\int^+ f^+ d\mu[T^{-1}] < \infty \text{ and } \int^+ f^- d\mu[T^{-1}] < \infty \quad (21.1174)$$

As T is $\mathcal{A}, \mathcal{B}$ -measurable and f is $\mathcal{B}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable [because $f \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$] it follows from [theorem: 21.230] that $f \circ T$ is $\mathcal{A}, \mathcal{B}[\overline{\mathbb{R}}]$ -measurable. Hence we have that

$$f \circ T \in \mathcal{M}[X, \mathcal{A}, \overline{\mathbb{R}}]$$

As $f^+, f^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ [see eq: 21.1173] it follows from [lemma: 21.620] that

$$f^+ \circ T, f^- \circ T \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$$

and

$$\int^+ (f^+ \circ T) d\mu = \int^+ f^+ d\mu[T^{-1}] \underset{[\text{eq: 21.1174}]}{<} \infty \text{ and } \int^+ (f^- \circ T) d\mu = \int^+ f^- d\mu[T^{-1}] \underset{[\text{eq: 21.1174}]}{<} \infty$$

As by [lemma: 21.621] $(f \circ T)^+ = f^+ \circ T$ and $(f \circ T)^- = f^- \circ T$ it follows from the above that

$$(f \circ T)^+, (f \circ T)^- \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}] \text{ and } \int^+ (f \circ T)^+ d\mu < \infty \wedge \int^+ (f \circ T)^- d\mu < \infty$$

which by definition means that

$$f \circ T \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

and

$$\begin{aligned} \int (f \circ T) d\mu &\stackrel{\text{def}}{=} \int^+ (f \circ T)^+ d\mu - \int^+ (f \circ T)^- d\mu \\ &= \int^+ (f^+ \circ T) d\mu - \int^+ (f^- \circ T) d\mu \\ &= \int^+ f^+ d\mu[T^{-1}] - \int^+ f^- d\mu[T^{-1}] \\ &\stackrel{\text{def}}{=} \int f d\mu[T^{-1}] \end{aligned}$$

Hence

$$\int (f \circ T) d\mu = \int f d\mu[T^{-1}] \quad \square$$

Integrating over a measurable set gives the following variation of the above theorem.

Theorem 21.623. (Change of Variables (2)) *Let $\langle X, \mathcal{A} \rangle$, $\langle Y, \mathcal{B} \rangle$ be measurable spaces, $B \in \mathcal{B}$, $T: X \rightarrow Y$ a $\mathcal{A}, \mathcal{B}$ -measurable and $f \in \mathcal{L}_B[Y, \mathcal{B}, \mu[T^{-1}], \overline{\mathbb{R}}]$ then we have*

$$\begin{aligned} T^{-1}(B) &\in \mathcal{A} \\ f \circ T &\in \mathcal{L}_{T^{-1}(B)}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \\ \int_{T^{-1}(B)} (f \circ T) d\mu &= \int_B f d\mu[T^{-1}] \end{aligned}$$

Proof. As T is $\mathcal{A}, \mathcal{B}$ -measurable we have that

$$T^{-1}(B) \in \mathcal{A}$$

Further as $f \in \mathcal{L}_B[Y, \mathcal{B}, \mu[T^{-1}], \overline{\mathbb{R}}]$ we have by [definition: 21.354] that

$$\mathcal{X}_{Y, B} \cdot f \in \mathcal{L}[Y, \mathcal{B}, \mu[T^{-1}], \overline{\mathbb{R}}] \text{ and } \int_B f d\mu[T^{-1}] = \int (\mathcal{X}_{Y, B} \cdot f) d\mu \quad (21.1175)$$

Using now [theorem: 21.622] we have that

$$(\mathcal{X}_{Y, B} \cdot f) \circ T \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int ((\mathcal{X}_{Y, B} \cdot f) \circ T) d\mu = \int (\mathcal{X}_{Y, B} \cdot f) d\mu[T^{-1}] \quad (21.1176)$$

Now we have $\forall x \in X$ that

$$\begin{aligned} ((\mathcal{X}_{Y, B} \cdot f) \circ T)(x) &= (\mathcal{X}_{Y, B} \cdot f)(T(x)) \\ &= \mathcal{X}_{Y, B}(T(x)) \cdot f(T(x)) \\ &= (\mathcal{X}_{Y, B} \circ T)(x) \cdot (f \circ T)(x) \\ &\stackrel{[\text{lemma: 21.618}]}{=} \mathcal{X}_{X, T^{-1}(B)}(x) \cdot (f \circ T)(x) \\ &= (\mathcal{X}_{X, T^{-1}(B)} \cdot (f \circ T))(x) \end{aligned}$$

so that

$$(\mathcal{X}_{Y, B} \cdot f) \circ T = \mathcal{X}_{X, T^{-1}(B)} \cdot (f \circ T)$$

Substituting the above in [eq: 21.1176] proves that

$$\mathcal{X}_{X, T^{-1}(B)} \cdot (f \circ T) \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int (\mathcal{X}_{X, T^{-1}(B)} \cdot (f \circ T)) d\mu = \int (\mathcal{X}_{Y, B} \cdot f) d\mu[T^{-1}]$$

so that by [definition: 21.354]

$$f \circ T \in \mathcal{L}_{T^{-1}(B)}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}] \text{ and } \int_{T^{-1}(B)} (f \circ T) d\mu = \int_B f d\mu[T^{-1}] \quad \square$$

In the special case where T is a linear isomorphism and the measures λ , λ^n we have the following relations [see theorems: 21.610, 21.617]

$$\forall A \in \mathcal{B}[\mathbb{R}] \text{ we have } \lambda[T^{-1}](A) = \frac{1}{|T(1)|} \cdot \lambda(A)$$

$$\forall A \in \mathcal{B}[\mathbb{R}^n] \text{ we have } \lambda^n[T^{-1}](A) = \frac{1}{|\det(L)|} \cdot \lambda^n(A)$$

so that

$$\lambda[T^{-1}] = \frac{1}{|T(1)|} \cdot \lambda \text{ and } \lambda^n[T^{-1}] = \frac{1}{|\det(L)|} \cdot \lambda^n$$

This suggests that we examine how the integral behaves when we multiply a measure by a non negative scalar. In essence we prove that for $\alpha \in [0, \infty[$

$$\int f d(\alpha \cdot \mu) = \alpha \cdot \int f d\mu$$

Again we do this in three stages

1. First we prove that for $f \in \mathcal{S}^+[X, \mathcal{A}]$

$$\int^S f d(\alpha \cdot \mu) = \alpha \cdot \int^S f d\mu$$

2. Next we use [theorem: 21.329] and the Monotone Convergence theorem [see theorem: 21.372] to prove that for $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ we have

$$\int^+ f d(\alpha \cdot \mu) = \alpha \cdot \int^+ f d\mu$$

3. Finally we prove the result for $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$

$$\int f d(\alpha \cdot \mu) = \alpha \cdot \int f d\mu$$

Lemma 21.624. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $\alpha \in [0, \infty[$ and $f \in \mathcal{S}^+[X, \mathcal{A}]$ then we have

$$\int^S f d(\alpha \cdot \mu) = \alpha \cdot \int^S f d\mu$$

Note 21.625. Using [theorem: 21.108] $\alpha \cdot \mu: \mathcal{A} \rightarrow [0, \infty[$ defined by $(\alpha \cdot \mu)(A) = \alpha \cdot \mu(A)$ is a measure on $\mathcal{A}$.

Proof. As $f \in \mathcal{S}^+[Y, \mathcal{A}]$ we have by [theorem: 21.301] that there exists a $n \in \mathbb{N}$, $\{A_i\}_{i \in \{1, \dots, n\}} \subseteq \mathcal{A}$ and $\{f_i\}_{i \in \{1, \dots, n\}} \subseteq [0, \infty[$ so that

$$f = \sum_{i=1}^n f_i \cdot \chi_{Y, A_i}$$

Hence we have

$$\int^S f d(\alpha \cdot \mu) = \sum_{i=1}^n f_i \cdot (\alpha \cdot \mu)(A_i) = \sum_{i=1}^n f_i \cdot \alpha \cdot \mu(A_i) = \alpha \cdot \sum_{i=1}^n f_i \cdot \mu(A_i) = \alpha \cdot \int^S f d\mu \quad \square$$

Lemma 21.626. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $\alpha \in [0, \infty[$ and $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ then

$$\int^+ f d(\alpha \cdot \mu) = \alpha \cdot \int^+ f d\mu$$

Proof. As $f \in \mathcal{M}^+[X, \mathcal{A}, \overline{\mathbb{R}}]$ it follows from [theorem: 21.329] that there exist a sequence

$$\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}^+[X, \mathcal{A}]$$

such that

$$\forall y \in Y \text{ we have } \forall n \in \mathbb{N} \ f_n(y) \leq f_{n+1}(y) \text{ and } \lim_{n \rightarrow \infty} f_n(y) \text{ exist with } f(y) = \lim_{n \rightarrow \infty} f_n(y)$$

and for the measure $\alpha \cdot \mu$

$$\lim_{n \rightarrow \infty} \int^S f_n d(\alpha \cdot \mu) \text{ exists and } \int^+ f d(\alpha \cdot \mu) = \lim_{n \rightarrow \infty} \int^S f_n d(\alpha \cdot \mu) \quad (21.1177)$$

and for the measure μ

$$\lim_{n \rightarrow \infty} \int^S f_n d\mu \text{ exists and } \int^+ f d\mu = \lim_{n \rightarrow \infty} \int^S f_n d\mu \quad (21.1178)$$

Let $n \in \mathbb{N}$ then by [lemma: 21.624] we have

$$\int^S f_n d(\alpha \cdot \mu) = \alpha \cdot \int^S f_n d\mu \quad (21.1179)$$

hence we have

$$\begin{aligned} \int^+ f d(\alpha \cdot \mu) &\stackrel{[\text{eq: 21.1177}]}{=} \lim_{n \rightarrow \infty} \int^S f_n d(\alpha \cdot \mu) \\ &\stackrel{[\text{eq: 21.1179}]}{=} \lim_{n \rightarrow \infty} \left(\alpha \cdot \int^S f_n d\mu \right) \\ &\stackrel{[\text{theorem: 20.85}]}{=} \alpha \cdot \lim_{n \rightarrow \infty} \int^S f_n d\mu \\ &\stackrel{[\text{eq: 21.1178}]}{=} \alpha \cdot \int^+ f d\mu \end{aligned}$$

Summarized we have

$$\int^+ f d(\alpha \cdot \mu) = \alpha \cdot \int^+ f d\mu \quad \square$$

Theorem 21.627. Let $\langle X, \mathcal{A}, \mu \rangle$ be a measure space, $\alpha \in [0, \infty[$ then

$$f \in \mathcal{L}[X, \mathcal{A}, \alpha \cdot \mu, \overline{\mathbb{R}}] \Leftrightarrow f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

Further we have if $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ or $f \in \mathcal{L}[X, \mathcal{A}, \alpha \cdot \mu, \overline{\mathbb{R}}]$ then

$$\int^+ f d(\alpha \cdot \mu) = \alpha \cdot \int^+ f d\mu$$

Proof.

$\Rightarrow$. As $f \in \mathcal{L}[X, \mathcal{A}, \alpha \cdot \mu, \overline{\mathbb{R}}]$ we have by [definition: 21.345] that

$$f \in \mathcal{M}[X, \mathcal{A}, \alpha \cdot \mu, \overline{\mathbb{R}}]$$

$$\int^+ f^+ d(\alpha \cdot \mu) < \infty \text{ and } \int^+ f^- d(\alpha \cdot \mu) < \infty$$

$$\int f d(\alpha \cdot \mu) = \int^+ f^+ d(\alpha \cdot \mu) - \int^+ f^- d(\alpha \cdot \mu) < \infty$$

Using [lemma: 21.626] we have

$$\int^+ f^+ d(\alpha \cdot \mu) = \alpha \cdot \int^+ f^+ d\mu \text{ and } \int^+ f^- d(\alpha \cdot \mu) = \alpha \cdot \int^+ f^- d\mu$$

so that as $\alpha \in [0, \infty[$

$$\int^+ f^+ d\mu < \infty \text{ and } \int^+ f^- d\mu < \infty$$

which proves that

$$f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$$

Further we have

$$\alpha \cdot \int f d\mu = \alpha \cdot \int^+ f^+ d\mu - \alpha \cdot \int^+ f^- d\mu = \int^+ f^+ d(\alpha \cdot \mu) - \int^+ f^- d(\alpha \cdot \mu) = \int f d(\alpha \cdot \mu)$$

so that

$$\int f d(\alpha \cdot \mu) = \alpha \cdot \int f d\mu$$

$\Leftarrow$. As $f \in \mathcal{L}[X, \mathcal{A}, \mu, \overline{\mathbb{R}}]$ we have by [definition: 21.345] that

$$\begin{aligned} f &\in \mathcal{M}[X, \mathcal{A}, \alpha \cdot \mu, \overline{\mathbb{R}}] \\ \int^+ f^+ d\mu &< \infty \text{ and } \int^+ f^- d\mu < \infty \\ \int f d\mu &= \int^+ f^+ d\mu - \int^+ f^- d\mu \end{aligned}$$

Using [lemma: 21.626] we have

$$\int^+ f^+ d(\alpha \cdot \mu) = \alpha \cdot \int^+ f^+ d\mu \text{ and } \int^+ f^- d(\alpha \cdot \mu) = \alpha \cdot \int^+ f^- d\mu$$

so that as $\alpha \in [0, \infty[$

$$\int^+ f^+ d(\alpha \cdot \mu) < \infty \text{ and } \int^+ f^- d(\alpha \cdot \mu) < \infty$$

which proves that

$$f \in \mathcal{L}[X, \mathcal{A}, \alpha \cdot \mu, \overline{\mathbb{R}}]$$

Further we have

$$\alpha \cdot \int f d\mu = \alpha \cdot \int^+ f^+ d\mu - \alpha \cdot \int^+ f^- d\mu = \int^+ f^+ d(\alpha \cdot \mu) - \int^+ f^- d(\alpha \cdot \mu) = \int f d(\alpha \cdot \mu) \quad \square$$

Theorem 21.628. *We have*

1. Let $T: \mathbb{R} \rightarrow \mathbb{R}$ be a linear isomorphism and $f \in \mathcal{L}[\mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda, \overline{\mathbb{R}}]$ then $f \circ T \in \mathcal{L}[\mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda, \overline{\mathbb{R}}]$ and

$$\int (f \circ T) d\lambda = \frac{1}{|L(1)|} \cdot \int f d\lambda$$

2. Let $n \in \mathbb{N}$, $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear isomorphism and $f \in \mathcal{L}[\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n, \overline{\mathbb{R}}]$ then $f \circ T \in \mathcal{L}[\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n, \overline{\mathbb{R}}]$ and

$$\int (f \circ T) d\lambda = \frac{1}{|L(1)|} \cdot \int f d\lambda^n$$

Proof.

1. Using [theorem: 21.610] we have that $\forall A \in \mathcal{B}[\mathbb{R}] \lambda[T^{-1}](A) = \frac{1}{|T(1)|} \cdot \lambda(A)$ so that

$$\lambda[T^{-1}] = \frac{1}{|T(1)|} \cdot \lambda \quad (21.1180)$$

As $f \in \mathcal{L}[\mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda, \overline{\mathbb{R}}]$ we have by [theorem: 21.627]

$$f \in \mathcal{L}\left[\mathbb{R}, \mathcal{B}[\mathbb{R}], \frac{1}{|T(1)|} \cdot \lambda, \overline{\mathbb{R}}\right]_{[\text{eq: 21.1180}]} \mathcal{L}[\mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda[T^{-1}], \overline{\mathbb{R}}]$$

and

$$\int f d\lambda_{[L^{-1}]}_{[\text{eq: 21.1180}]} \int f d\left(\frac{1}{|T(1)|} \cdot \lambda\right)_{[\text{theorem: 21.627}]} \frac{1}{|T(1)|} \cdot \int f d\lambda$$

Using now [theorem: 21.622] on the above we have that

$$f \in \mathcal{L}[\mathbb{R}, \mathcal{B}[\mathbb{R}], \lambda, \overline{\mathbb{R}}]$$

and

$$\int (f \circ T) d\lambda = \frac{1}{|T(1)|} \cdot \int f d\lambda$$

2. Using [theorem: 21.617] we have that $\forall A \in \mathcal{B}[\mathbb{R}^n] \lambda^n[T^{-1}](A) = \frac{1}{|\det(T)|} \cdot \lambda^n(A)$ so that

$$\lambda^n[T^{-1}] = \frac{1}{|\det(T)|} \cdot \lambda^n \quad (21.1181)$$

As $f \in \mathcal{L}[\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n, \overline{\mathbb{R}}]$ we have by [theorem: 21.627]

$$f \in \mathcal{L}^n \left[\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \frac{1}{|\det(T)|} \cdot \lambda^n, \overline{\mathbb{R}} \right]_{\text{[eq: 21.1181]}} \stackrel{=}{=} \mathcal{L}[\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda^n[T^{-1}], \overline{\mathbb{R}}]$$

and

$$\int f d\lambda^n[L^{-1}]_{\text{[eq: 21.1181]}} \stackrel{=}{=} \int f d\left(\frac{1}{|\det(T)|} \cdot \lambda^n \right)_{\text{[theorem: 21.627]}} \stackrel{=}{=} \frac{1}{|\det(T)|} \cdot \int f d\lambda$$

Using now [theorem: 21.622] on the above we have that

$$f \in \mathcal{L}[\mathbb{R}^n, \mathcal{B}[\mathbb{R}^n], \lambda, \overline{\mathbb{R}}]$$

and

$$\int (f \circ T) d\lambda = \frac{1}{|\det(T)|} \cdot \int f d\lambda$$

□

Chapter 22

Affine Spaces

22.1 Definition and properties

René Descartes showed that classical euclidean space [which is a collection of points, lines, planes and concepts like parallel lines and orthogonality] can be identified with $\mathbb{R}^3$ using a coordinate system. However identification does not mean that they are the same, for example in geometric space there is no concept of adding points and there is no special point why $\mathbb{R}^3$ has a canonical vector space structure and has a special point $(0, 0, 0)$. Further the identification is dependent on the coordinate axes and origin that has been choosen. Affine spaces turns out to be a better model for the euclidian space by separating the space of points from the vector space structure.

Definition 22.1. Let $n \in \mathbb{N}$ then a **real affine space of dimension n** is a triple $\langle \mathbb{A}^n, V, \ominus \rangle$ where $\mathbb{A}^n$ is a non empty set called the **set of points** and $\langle V, +, \cdot \rangle$ is a n -dimensional vector space over the field $\mathbb{R}$ called the **space of displacements** or the **space of free vectors** together with a operator

$$\ominus: \mathbb{A}^n \times \mathbb{A}^n \rightarrow V$$

satisfying:

1. $\forall P, Q, R \in \mathbb{A}^n$ we have $(P \ominus Q) + (Q \ominus R) = P \ominus R$
2. $\forall P \in \mathbb{A}^n, \forall \vec{v} \in V$ there exist a **unique** $Q \in \mathbb{A}^n$ such that $\vec{v} = Q \ominus P$. In other words if $Q \ominus P = \vec{v} = Q' \ominus P$ then $Q = Q'$.

Notation 22.2. We use the follloing notation conventions:

1. The notation $\mathbb{A}^n$ does **not** mean that $\mathbb{A}^n$ is the n -times product of a set $\mathbb{A}$ but is a notation to indicate that the dimension of the space of displacements is n .
2. Elements of $\mathbb{A}^n$ are called points and are noted as upper case symbols. Examples are P, Q, R, P_1, P_2 , etc.
3. Elements of V are called vectors and are noted with a lower case letter beneath a arrow. Examples are $\vec{v}, \vec{u}, \vec{v}_1$ etc.
4. If $(P, \vec{v}) \in \mathbb{A}^n \times V$ then the **unique** point $Q \in \mathbb{A}^n$ such that $\vec{v} = Q \ominus P$ is noted as $P \oplus \vec{v}$. Using this notation we have that for $(P, \vec{v}) \in \mathbb{A}^n \times V$

$$P \oplus \vec{v} \in \mathbb{A}^n \text{ and } (P \oplus \vec{v}) \ominus P = \vec{v}$$

and

$$\text{If } Q \in \mathbb{A}^n \text{ such that } \vec{v} = Q \ominus P \text{ then } Q = P \oplus \vec{v}$$

5. The operators $+, \cdot, -$ are always the traditional operators in the vector space $\langle V, +, \cdot \rangle$ and the field $\langle \mathbb{R}, +, \cdot \rangle$.

The following figure illustrates the definition of an affine space $\langle \mathbb{A}^2, V \rangle$

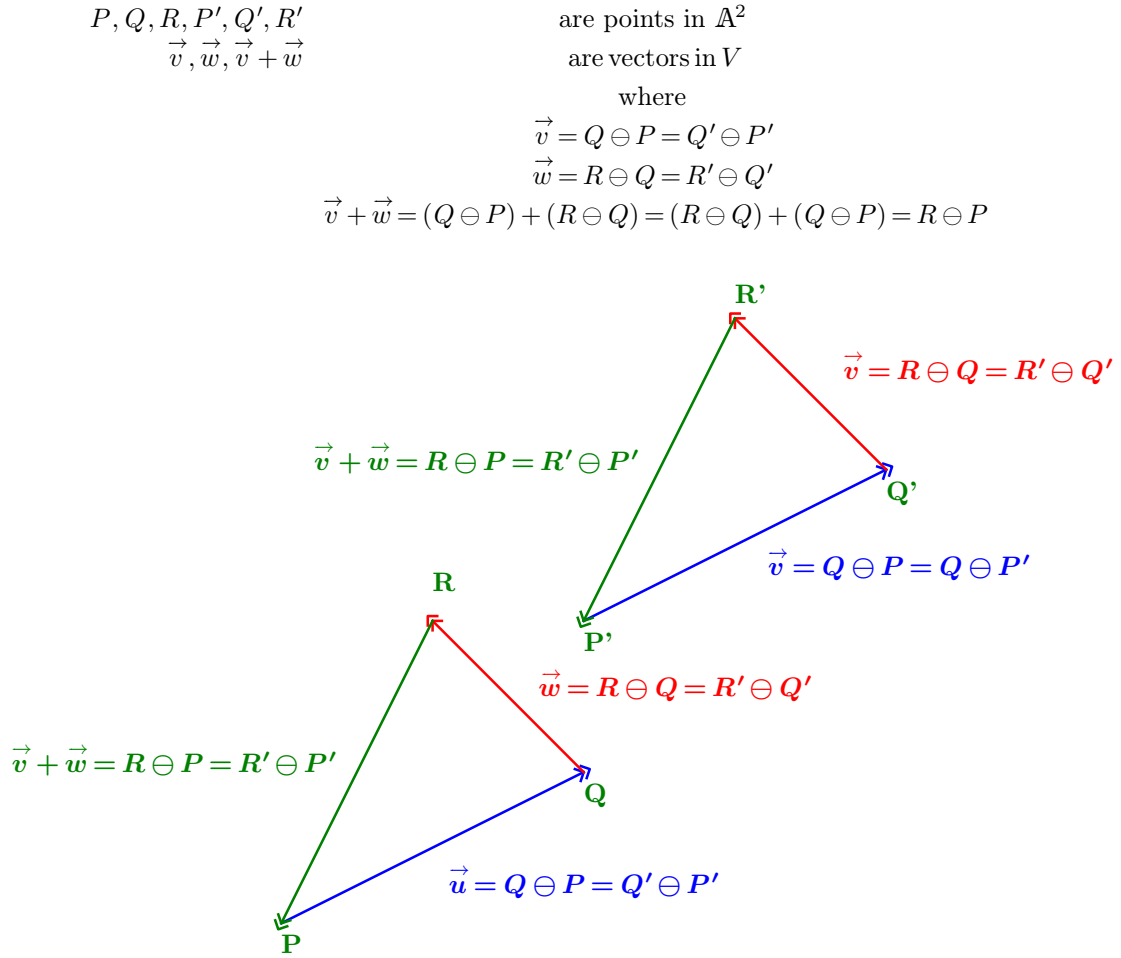


Figure 22.1.

A trivial example of an affine space is the following

Example 22.3. Let $n \in \mathbb{N}$ and take $\mathbb{A}^n = \mathbb{R}^n$, $\langle V, +, \cdot \rangle = \langle \mathbb{R}^n, +, \cdot \rangle$ and

$$\ominus: \mathbb{A}^n \times \mathbb{A}^n = \mathbb{R}^n \times \mathbb{R}^n \rightarrow V = \mathbb{R}^n \text{ where } Q \ominus P = Q - P$$

then $\langle \mathbb{A}^n, V, \ominus \rangle$ is an affine space. In other words $\langle \mathbb{R}^n, \mathbb{R}^n, - \rangle$ is an affine space which is the canonical example of an affine space.

Proof. Using [theorem: 11.72] $\langle \mathbb{R}, +, \cdot \rangle$ is a vector space. Let $P, Q, R \in \mathbb{R}^n$ then $P = (p_1, \dots, p_n)$, $Q = (q_1, \dots, q_n)$ and $R = (r_1, \dots, r_n)$ then

$$\begin{aligned}
 (P \ominus Q) + (Q \ominus R) &= (P - Q) + (Q - R) \\
 &= ((p_1, \dots, p_n) - (q_1, \dots, q_n)) + ((q_1, \dots, q_n) - (r_1, \dots, r_n)) \\
 &= (p_1 - q_1, \dots, p_n - q_n) + (q_1 - r_1, \dots, q_n - r_n) \\
 &= (p_1 - q_1 + q_1 - r_1, \dots, p_n - q_n + q_n - r_n) \\
 &= (p_1 - r_1, \dots, p_n - r_n) \\
 &= (p_1, \dots, p_n) - (r_1, \dots, r_n) \\
 &= (P \ominus R)
 \end{aligned}$$

Hence

$$\forall P, Q, R \in \mathbb{R}^n \text{ we have } (P \ominus Q) + (Q \ominus R) = P \ominus R$$

Further if $P = (p_1, \dots, p_n) \in \mathbb{R}^n$ and $\vec{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$ then if we take $Q = (p_1 + v_1, \dots, p_n + v_n)$ we have

$$Q \ominus P = (p_1 + v_1, \dots, p_n + v_n) - (p_1, \dots, p_n) = (p_1 + v_1 - p_1, \dots, p_n + v_n - p_n) = (v_1, \dots, v_n) = \vec{v}$$

Hence

$$\forall P \in \mathbb{R}^n \vec{v} \in \mathbb{R}^n \text{ there exists a } Q \in \mathbb{R}^n \text{ such that } Q \ominus P = \vec{v}$$

Further if $Q' = (q'_1, \dots, q'_n) \in \mathbb{R}^n$ such that $Q' \ominus P = \vec{v}$ then we have that

$$\forall i \in \{1, \dots, n\} \quad v_i = q'_i - p_i \Rightarrow q'_i = p_i + v_i = q_i$$

so that $Q' = Q$. □

Theorem 22.4. Let $n \in \mathbb{N}$ and $n \in \mathbb{N}$ and $\langle \mathbb{A}^n, V, \ominus \rangle$ a real affine space of dimension n then $\forall P, Q$ we have

$$P \oplus (Q \ominus P) = P$$

Proof. Let $P, Q \in \mathbb{A}^n$. For $\vec{v} = P \ominus Q \in V$ we have by [definition: 22.1] that there exist a **unique** R such that $R \ominus Q = \vec{v}$. By [definition: 22.1] $R = Q \oplus \vec{v}$ so that $(Q \oplus \vec{v}) \ominus Q = \vec{v} = P \ominus Q$ so that by uniqueness we have $P = Q \oplus \vec{v} = Q \oplus (P \ominus Q)$. □

The following theorem shows that $P \ominus Q = P' \ominus Q'$ implies that $P' \ominus P = Q' \ominus Q$ as is illustrated in the followin figure.

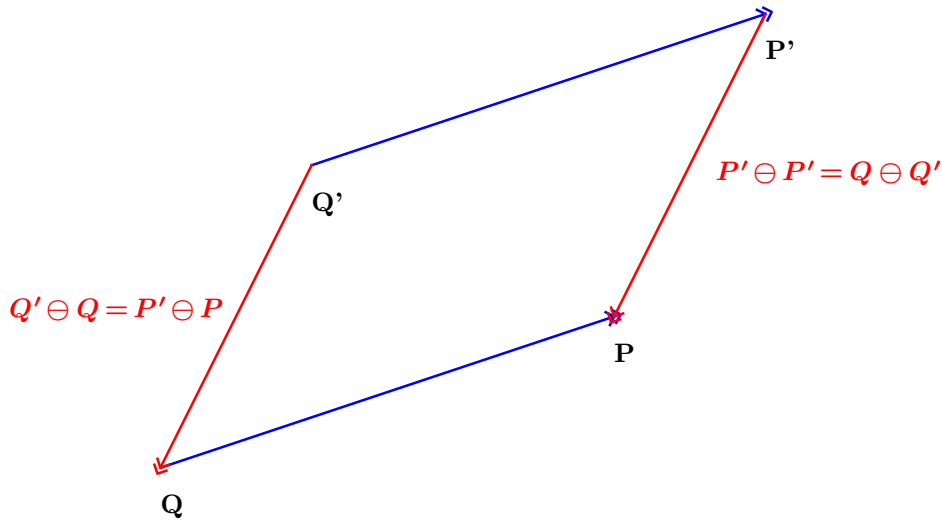


Figure 22.2.

Theorem 22.5. Let $n \in \mathbb{N}$ and $\langle \mathbb{A}^n, V, \ominus \rangle$ a real affine space of dimension n then $\forall P, P', Q, Q' \in \mathbb{A}^n$ such that $P \ominus Q = P' \ominus Q'$ then we have $P \ominus P' = Q \ominus Q'$.

Proof. As $P \ominus Q = P' \ominus Q'$ we have

$$\begin{aligned} (P \ominus Q) + (Q \ominus P') &= (P' \ominus Q') + (Q \ominus P') \stackrel{\text{def}}{\Rightarrow} (P \ominus P') = (P' \ominus Q') + (Q \ominus P') \\ &\Rightarrow (P \ominus P') = (Q \ominus P') + (P' \ominus Q') \\ &\Rightarrow P \ominus P' = Q \ominus Q' \end{aligned}$$

□

Theorem 22.6. Let $n \in \mathbb{N}$ and $\langle \mathbb{A}^n, V, \ominus \rangle$ a real affine space of dimension n then $\forall P \in \mathbb{A}^n$ we have that $P \ominus P = \vec{0}$ where $\vec{0}$ is the neutral element in V .

Proof. Let $P \in \mathbb{A}^n$ then $\forall \vec{v} \in V$ we have by [definition: 22.1, notation (4)] that

$$(P \oplus \vec{v}) \ominus P = \vec{v}$$

so that

$$(P \ominus P) + \vec{v} = (P \ominus P) + ((P \oplus \vec{v}) \ominus P) = ((P \oplus \vec{v}) \ominus P) + (P \ominus P) = (P \oplus \vec{v}) \ominus P = \vec{v}$$

which proves that $P \ominus P = \vec{0}$. □

Theorem 22.7. Let $n \in \mathbb{N}$, $\langle \mathbb{A}^n, V, \ominus \rangle$ a real affine space of dimension n and $P, Q \in \mathbb{A}^n$ then $(P \ominus Q) = -(Q \ominus P)$ where $-(Q \ominus P)$ is the additive inverse of the vector $Q \ominus P$.

Proof. As $(P \ominus Q) + (Q \ominus P) = P \ominus P \stackrel{[\text{theorem: 22.6}]}{=} \vec{0}$ it follows that $(P \ominus Q) = -(Q \ominus P)$. □

Theorem 22.8. Let $n \in \mathbb{N}$, $\langle \mathbb{A}^n, V, \ominus \rangle$ a real affine space of dimension n , $P \in \mathbb{A}^n$ and $\vec{v}, \vec{u} \in V$ then we have

$$(P \oplus \vec{v}) \oplus \vec{u} = P \oplus (\vec{v} + \vec{u})$$

Proof. As $\vec{u} = ((P \oplus \vec{v}) \oplus \vec{u}) \ominus (P \oplus \vec{v})$ and $\vec{v} = (P \oplus \vec{v}) \ominus P$ we have that

$$\vec{v} + \vec{u} = \vec{u} + \vec{v} = (((P \oplus \vec{v}) \oplus \vec{u}) \ominus (P \oplus \vec{v})) + ((P \oplus \vec{v}) \ominus P) = ((P \oplus \vec{v}) \oplus \vec{u}) \ominus P$$

as also $(P \oplus (\vec{v} + \vec{u})) \ominus P = \vec{v} + \vec{u}$ we have by uniqueness that

$$(P \oplus \vec{v}) \oplus \vec{u} = P \oplus (\vec{v} + \vec{u}) \quad \square$$

Theorem 22.9. Let $n \in \mathbb{N}$, $\langle \mathbb{A}^n, V, \ominus \rangle$ a real affine space of dimension n , $P, Q \in \mathbb{A}^n$ and $\vec{v} \in V$ then

$$P \ominus Q = (P \oplus \vec{v}) \ominus (Q \oplus \vec{v})$$

Proof. We have

$$\begin{aligned} P \ominus Q &= (P \ominus Q) + \vec{0} \\ &= (P \ominus Q) + (-\vec{v} + \vec{v}) \\ &= (P \ominus Q) + (-((Q \oplus \vec{v}) \ominus Q) + ((P \oplus \vec{v}) \ominus P)) \\ &\stackrel{[\text{theorem: 22.7}]}{=} (P \ominus Q) + ((Q \ominus (Q \oplus \vec{v})) + ((P \oplus \vec{v}) \ominus P)) \\ &\stackrel{V \text{ is a vector space}}{=} ((P \ominus Q) + (Q \ominus (Q \oplus \vec{v}))) + ((P \oplus \vec{v}) \ominus P) \\ &\stackrel{\text{affine space definition}}{=} (P \ominus (Q \oplus \vec{v})) + ((P \oplus \vec{v}) \ominus P) \\ &\stackrel{V \text{ is a vector space}}{=} ((P \oplus \vec{v}) \ominus P) + (P \ominus (Q \oplus \vec{v})) \\ &\stackrel{\text{affine space definition}}{=} (P \oplus \vec{v}) - (Q \oplus \vec{v}) \end{aligned}$$

□

We define now the concept of a Coordinate System which essential associate with points in $\mathbb{A}^n$ a tuple of coordinates.

Definition 22.10. (Local Coordinate System) Let $\langle \mathbb{A}^n, V, \ominus \rangle$ be a real affine space of dimension n then a **local coordinate system** on $\langle \mathbb{A}^n, V, \ominus \rangle$ is a mapping

$$f: U \rightarrow \mathbb{R}^n$$

where $U \subseteq \mathbb{A}^n$, $f(U)$ is a open set in $\langle \mathbb{R}^n, \|\cdot\|_n \rangle$ and f is injective. Let $P \in \mathbb{A}$ then $f(P)$ is of the form $f(P) = (x_1, \dots, x_n)$ we call $(x_1, \dots, x_n)$ the coordinates of P using the coordinate system f . If $U = \mathbb{A}^n$ then f is called a **global coordinate system** on $\langle \mathbb{A}^n, V, \ominus \rangle$.

We prove now that every real affine space of dimension n has a **global coordinate system**.

Theorem 22.11. Let $n \in \mathbb{N}$, $\langle \mathbb{A}^n, V, \ominus \rangle$ be a real affine space of dimension n , $O \in \mathbb{A}^n$ the set of points and $E = \{\vec{e}_1, \dots, \vec{e}_n\}$ a basis for V the set of displacements then the function

$$f: \mathbb{A}^n \rightarrow \mathbb{R}^n \text{ defined by } f(P) = (x_1, \dots, x_n) \text{ where } (P \ominus O) = \sum_{i=1}^n y_i \cdot \vec{e}_i$$

is a **bijection** that is a **global coordinate system** on $\langle \mathbb{A}^n, V, \ominus \rangle$. We call this **global coordinate system** a **Cartesian coordinate system with origin O and axes $\{e_1, \dots, e_n\}$** .

Note 22.12. Although this theorem ensures that every real affine space of dimension n has at least one global coordinate system not every coordinate system should be a Cartesian coordinate system. These coordinate systems that are not Cartesian are called **curvilinear coordinate systems**.

Proof. Let $P \in \mathbb{A}^n$ then $P \ominus O \in V$ so there exist by [theorem: 11.143] a **unique** $\{y_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $(P \ominus O) = \sum_{i=1}^n x_i \cdot \vec{e}_i$. Hence the function

$$f: \mathbb{A}^n \rightarrow \mathbb{R}^n \text{ defined by } f(P) = (x_1, \dots, x_n) \text{ where } (P \ominus O) = \sum_{i=1}^n x_i \cdot \vec{e}_i$$

is well defined. We prove now that

f is bijective

injectivity. If $f(P_1) = f(P_2) = (x_1, \dots, x_n)$ then

$$(P_1 \ominus O) = \sum_{i=1}^n x_i \cdot \vec{e}_i = (P_2 \ominus O)$$

and by the uniqueness in [definition: 22.1] it follows that $P_1 = P_2$.

surjectivity. If $(y_1, \dots, y_n) \in \mathbb{R}^n$ then for $P = O \oplus (\sum_{i=1}^n y_i \cdot \vec{e}_i)$ we have

$$P \ominus O = \left(O \oplus \left(\sum_{i=1}^n y_i \cdot \vec{e}_i \right) \right) \ominus O \stackrel{[\text{theorem: 22.4}]}{=} \sum_{i=1}^n y_i \cdot \vec{e}_i$$

so that $f(P) = (y_1, \dots, y_n)$ which proves surjectivity.

Because of surjectivity we have that $f(\mathbb{A}^n) = \mathbb{R}^n$ a open set in $\langle \mathbb{R}^n, \|\cdot\|_n \rangle$. So by [definition: 22.10] f is a global coordinate system on $\langle \mathbb{A}^n, V, \ominus \rangle$. $\square$

We examine now how Cartesian coordinate system transforms under a change of basis.

Theorem 22.13. Let $n \in \mathbb{N}$, $\langle \mathbb{A}^n, V, \ominus \rangle$ be a real affine space of dimension n and $E = \{\vec{e}_1, \dots, \vec{e}_n\}$, $E' = \{\vec{e}'_1, \dots, \vec{e}'_n\}$ are two bases of V . By [theorem: 11.353] there exist then a **invertible** matrix $M \in \mathcal{M}_{n,n}(\mathbb{R})$ such that

$$\forall i \in \{1, \dots, n\} \vec{e}_i = \sum_{j=1}^n M_{j,i} \cdot \vec{e}'_j \text{ and } \vec{e}'_i = \sum_{j=1}^n (M^{-1})_{j,i} \cdot \vec{e}_j$$

Let

$f: \mathbb{A}^n \rightarrow \mathbb{R}^n$ a Cartesian coordinate system with origin $O \in \mathbb{A}^n$ and axes E

$g: \mathbb{A}^n \rightarrow \mathbb{R}^n$ a Cartesian coordinate system with origin $O' \in \mathbb{A}^n$ and axes E'

then we have:

1. For $g \circ f^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $f \circ g^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ we have $\forall i \in \{1, \dots, n\}$

$$\forall (x_1, \dots, x_n) \quad ((g \circ f^{-1})(x_1, \dots, x_n))_i = \sum_{j=1}^n M_{i,j} \cdot (x_j + b_j)$$

$$\forall (x'_1, \dots, x'_n) \quad ((f \circ g^{-1})(x'_1, \dots, x'_n))_i = -b_i + \sum_{j=1}^n (M^{-1})_{i,j} \cdot x'_j$$

where $(O \ominus Q') = \sum_{i=1}^n b_i \cdot \vec{e}_i$.

2. Let $P \in \mathbb{A}^n$ then if $(x_1, \dots, x_n)$ are the coordinates of P using f and $(x'_1, \dots, x'_n)$ are the coordinates of P using g then we have

$$\forall i \in \{1, \dots, n\} \text{ that } x'_i = \sum_{j=1}^n M_{i,j} \cdot (x_j + b_j)$$

$$\forall i \in \{1, \dots, n\} \text{ that } x_i = -b_i + \sum_{j=1}^n (M^{-1})_{i,j} \cdot x'_j$$

where $O \ominus Q' = \sum_{i=1}^n b_i \cdot \vec{e}_i$

Note 22.14. Using [theorem: 22.11] we have that f, g are bijections so that f^{-1} and g^{-1} are well defined. Further as $O \ominus O' \in V$ and $\{\vec{e}_1, \dots, \vec{e}_n\}$ is a basis for V there exist a $\{b_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that $O \ominus Q' = \sum_{i=1}^n b_i \cdot \vec{e}_i$.

Proof.

1. Let $(x_1, \dots, x_n) \in \mathbb{R}$ then for $P = O \oplus (\sum_{i=1}^n x_i \cdot \vec{e}_i)$ we have that

$$P \ominus O = \left(O \oplus \left(\sum_{i=1}^n x_i \cdot \vec{e}_i \right) \right) \ominus O \stackrel{[\text{theorem: 22.4}]}{=} \sum_{i=1}^n x_i \cdot \vec{e}_i \quad (22.1)$$

so that $f(P) = (x_1, \dots, x_n)$ hence

$$P = f^{-1}(x_1, \dots, x_n) \quad (22.2)$$

Now

$$\begin{aligned} P \ominus O' &= (P \ominus O) + (O \ominus O') \\ &\stackrel{[\text{eq: 22.1}]}{=} \sum_{i=1}^n x_i \cdot \vec{e}_i + \sum_{i=1}^n b_i \cdot \vec{e}_i \\ &= \sum_{i=1}^n (x_i + b_i) \cdot \vec{e}_i \\ &= \sum_{i=1}^n (x_i + b_i) \cdot \left(\sum_{j=1}^n M_{j,i} \cdot \vec{e}'_j \right) \\ &= \sum_{i=1}^n \left(\sum_{j=1}^n ((x_i + b_i) \cdot M_{j,i} \cdot \vec{e}'_j) \right) \\ &\stackrel{[\text{theorem: 11.45}]}{=} \sum_{j=1}^n \left(\sum_{i=1}^n ((x_i + b_i) \cdot M_{j,i} \cdot \vec{e}'_j) \right) \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n ((x_i + b_i) \cdot M_{j,i}) \right) \cdot \vec{e}'_j \\ &= \sum_{i=1}^n \left(\sum_{j=1}^n M_{i,j} \cdot (x_j + b_j) \right) \cdot \vec{e}'_i \end{aligned}$$

hence we have that

$$\begin{aligned}
 (g \circ f^{-1})(x_1, \dots, x_n) &= (g(f^{-1}(x_1, \dots, x_n))) \\
 &\stackrel{[\text{eq: 22.2}]}{=} g(P) \\
 &= \left(\sum_{j=1}^n M_{1,j} \cdot (x_j + b_j), \dots, \sum_{j=1}^n M_{n,j} \cdot (x_j + b_j) \right)
 \end{aligned}$$

so that

$$\forall i \in \{1, \dots, n\} \quad ((g \circ f^{-1})(x_1, \dots, x_n))_i = \sum_{j=1}^n M_{i,j} \cdot (x_j + b_j) \quad (22.3)$$

Let $(x'_1, \dots, x'_n) \in \mathbb{R}^n$ then for $P = O' \oplus (\sum_{i=1}^n x'_i \cdot \vec{e}_i')$ we have that

$$P \ominus O' = \left(O' \oplus \left(\sum_{i=1}^n x'_i \cdot \vec{e}_i' \right) \right) \ominus O' \stackrel{[\text{theorem: 22.4}]}{=} \sum_{i=1}^n x'_i \cdot \vec{e}_i' \quad (22.4)$$

so that $g(P) = (x'_1, \dots, x'_n)$ hence

$$P = g^{-1}(x'_1, \dots, x'_n) \quad (22.5)$$

Now we have

$$\begin{aligned}
 P \ominus O &= (P \ominus O') + (O' \ominus O) \\
 &\stackrel{[\text{theorem: 22.7}]}{=} (P \ominus O') - (O \ominus O') \\
 &\stackrel{[\text{eq: 22.4}]}{=} \sum_{i=1}^n x'_i \cdot \vec{e}_i' - \sum_{i=1}^n b_i \cdot \vec{e}_i \\
 &= \sum_{i=1}^n x'_i \cdot \left(\sum_{j=1}^n (M^{-1})_{j,i} \cdot \vec{e}_j \right) - \sum_{i=1}^n b_i \cdot \vec{e}_i \\
 &= \sum_{i=1}^n \left(\sum_{j=1}^n x'_i \cdot ((M^{-1})_{j,i} \cdot \vec{e}_j) \right) - \sum_{i=1}^n b_i \cdot \vec{e}_i \\
 &\stackrel{[\text{theorem: 11.45}]}{=} \sum_{j=1}^n \left(\sum_{i=1}^n x'_i \cdot ((M^{-1})_{j,i} \cdot \vec{e}_j) \right) - \sum_{i=1}^n b_i \cdot \vec{e}_i \\
 &= \sum_{j=1}^n \left(\sum_{i=1}^n x'_i \cdot ((M^{-1})_{j,i} \cdot \vec{e}_j) \right) - \sum_{j=1}^n b_j \cdot \vec{e}_j \\
 &= \sum_{j=1}^n \left(\sum_{i=1}^n x'_i \cdot (M^{-1})_{j,i} \right) \cdot \vec{e}_j - \sum_{j=1}^n b_j \cdot \vec{e}_j \\
 &= \sum_{j=1}^n \left(-b_j + \sum_{i=1}^n x'_i \cdot (M^{-1})_{j,i} \right) \cdot \vec{e}_j \\
 &= \sum_{i=1}^n \left(-b_i + \sum_{j=1}^n x'_j \cdot (M^{-1})_{i,j} \right) \cdot \vec{e}_i
 \end{aligned}$$

hence we have

$$\begin{aligned}
 (f \circ g^{-1})(x'_1, \dots, x'_n) &= f(g^{-1}(x'_1, \dots, x'_n)) \\
 &\stackrel{[\text{eq: 22.5}]}{=} f(P) \\
 &= \left(-b_1 + \sum_{j=1}^n x'_j \cdot (M^{-1})_{1,j}, \dots, -b_n + \sum_{j=1}^n x'_j \cdot (M^{-1})_{n,j} \right)
 \end{aligned}$$

from which it follows that

$$\forall i \in \{1, \dots, n\} \text{ that } ((f \circ g^{-1})(x'_1, \dots, x'_n))_i = -b_i + \sum_{j=1}^n (M^{-1})_{i,j} \cdot x'_j \quad (22.6)$$

2. Let $P \in \mathbb{A}^n$, $(x_1, \dots, x_n) \in \mathbb{R}^n$ the coordinates of P using the coordinate system f and $(x'_1, \dots, x'_n) \in \mathbb{R}^n$ the coordinates of P using the coordinate system g . Then

$$f(P) = (x_1, \dots, x_n) \text{ and } g(P) = (x'_1, \dots, x'_n) \Rightarrow f^{-1}(x_1, \dots, x_n) = P = g^{-1}(x'_1, \dots, x'_n)$$

so that $\forall i \in \{1, \dots, n\}$

$$\begin{aligned} x'_i &= (g(P))_i \\ &= (g(f^{-1}(x_1, \dots, x_n)))_i \\ &= ((g \circ f^{-1})(x_1, \dots, x_n))_i \\ &\stackrel{[\text{eq: 22.3}]}{=} \sum_{j=1}^n M_{i,j} \cdot (x_j + b_j) \\ x_i &= (f(P))_i \\ &= (f(g^{-1}(x'_1, \dots, x'_n)))_i \\ &= ((f \circ g^{-1})(x'_1, \dots, x'_n))_i \\ &\stackrel{[\text{eq: 22.6}]}{=} -b_i + \sum_{j=1}^n (M^{-1})_{i,j} \cdot x'_j \end{aligned}$$

Summarized we have

$$\forall i \in \{1, \dots, n\} \text{ that } x'_i = \sum_{j=1}^n M_{i,j} \cdot (x_j + b_j) \text{ and } x_i = -b_i + \sum_{j=1}^n (M^{-1})_{i,j} \cdot x'_j \quad \square$$

Next we introduce the concept of transformations that preserves the affine structure of affine spaces. First we need a little lemma.

Lemma 22.15. *Let $n_1, n_2 \in \mathbb{N}$, $\langle \mathbb{A}_1^{n_1}, V_1, \ominus_1 \rangle$, $\langle \mathbb{A}_1^{n_2}, V_2, \ominus_2 \rangle$ be two real affine spaces and $\psi: \mathbb{A}^{n_1} \rightarrow \mathbb{A}^{n_2}$ a function such that $\forall P, Q \in \mathbb{A}^{n_1}$ and $\forall \vec{v} \in V_1$ we have that*

$$\psi(P \oplus_1 \vec{v}) \ominus_2 \psi(Q \oplus_1 \vec{v}) = \psi(P) \ominus_2 \psi(Q)$$

[in other words ψ is invariant under displacement by a vector] then the function

$$d\psi: V_1 \rightarrow V_2 \text{ defined by } d\psi(\vec{v}) = \psi(P) \ominus_2 \psi(Q) \text{ where } \vec{v} = P \ominus_1 Q$$

is a well defined function.

Proof. First we prove existence. As $\mathbb{A}_1^{n_1} \neq \emptyset$ there exist a $Q \in \mathbb{A}_1^{n_1}$ then by [definition: 22.1] there exist a unique P such that $\vec{v} = P \ominus_1 Q$. Next we have to prove that $d\psi(\vec{v})$ is independent of the choice of P and Q . Assume that there exist also a $P', Q' \in \mathbb{A}_1^{n_1}$ such that $\vec{v} = P' \ominus_1 Q'$. Then we have $P \ominus_1 Q = \vec{v} = P' \ominus_1 Q'$ so that $P = Q \oplus_1 \vec{v}$ and $P' = Q' \oplus_1 \vec{v}$ and

$$\begin{aligned} \psi(P) \ominus_2 \psi(Q) &\stackrel{[\text{definition: 22.1}]}{=} (\psi(P) \ominus_2 \psi(P')) + (\psi(P') \ominus_2 \psi(Q)) \\ &= (\psi(Q \oplus_1 \vec{v}) \ominus_2 \psi(Q' \oplus_1 \vec{v})) + (\psi(P') \ominus_2 \psi(Q)) \\ &\stackrel{(1)}{=} (\psi(Q) \ominus_2 \psi(Q')) + (\psi(P') \ominus_2 \psi(Q)) \\ &= (\psi(P') \ominus_2 \psi(Q)) + (\psi(Q) \ominus_2 \psi(Q')) \\ &\stackrel{[\text{definition: 22.1}]}{=} \psi(P') \ominus_2 \psi(Q') \end{aligned}$$

which proves $\psi(P) \ominus_2 \psi(Q) = \psi(P') \ominus_2 \psi(Q')$ so that $d\psi$ is well defined. $\square$

Definition 22.16. Let $n_1, n_2 \in \mathbb{N}$, $\langle \mathbb{A}_1^{n_1}, V_1, \ominus_1 \rangle$, $\langle \mathbb{A}_2^{n_2}, V_2, \ominus_2 \rangle$ be two real affine spaces then a map

$$\psi: \mathbb{A}_1^{n_1} \rightarrow \mathbb{A}_2^{n_2}$$

is called a **affine transformation** if

1. $\forall P, Q \in \mathbb{A}_1^{n_1}$ and $\forall \vec{v} \in V_1$ $\psi(P \oplus_1 \vec{v}) \ominus_2 \psi(Q \oplus_1 \vec{v}) = \psi(P) \ominus_2 \psi(Q)$. In other words ψ is invariant under displacements by a vector.
2. The map $d\psi$ defined by

$$d\psi: V_1 \rightarrow V_2 \text{ defined by } d\psi(\vec{v}) = \psi(P) \ominus_2 \psi(Q) \text{ where } \vec{v} = P \ominus_1 Q$$

is a linear map.

Note 22.17. As in principle there could be another two points P', Q' such that $P' \ominus Q' = P \ominus Q$ we must prove that $d\psi(\vec{v})$ is independent of the choice of these two points P, Q . This is done in the previous lemma [see lemma: 22.15].

In the special case where the dimensions of the affine spaces are the same we can define a isomorphisme of affine spaces.

Definition 22.18. Let $n \in \mathbb{N}$, $\langle \mathbb{A}_1^n, V_1, \ominus_1 \rangle$, $\langle \mathbb{A}_2^n, V_2, \ominus_2 \rangle$ be two real affine spaces of dimension n then a function

$$\psi: V_1 \rightarrow V_2$$

is a **affine isomorphism** if ψ is a affine mapping and a bijection,

Theorem 22.19. Let $n \in \mathbb{N}$, $\langle \mathbb{A}_1^n, V_1, \ominus_1 \rangle$, $\langle \mathbb{A}_2^n, V_2, \ominus_2 \rangle$ be two real affine spaces of dimension n and $\psi: V_1 \rightarrow V_2$ a affine isomorphism then $\psi^{-1}: V_2 \rightarrow V_1$ is a affine isomorphism and $d\psi: V_1 \rightarrow V_2$ is a linear isomorphism.

Proof. First we prove that

$$d\psi \text{ is injective} \tag{22.7}$$

Proof. Using [theorem: 11.171] we have that

$$0 \in \ker(d\psi) \tag{22.8}$$

Let $\vec{v} \in \ker(d\psi)$ then $d\psi(\vec{v}) = \vec{0}$, by definition of $d\psi$ there exist a $P, Q \in V_1$ such that $\vec{v} = P \ominus_1 Q$ and $\psi(P) \ominus_2 \psi(Q) = d\psi(\vec{v}) = \vec{0}$. Hence as also $\psi(Q) \ominus_2 \psi(Q) = \vec{0}$ we have by [definition: 22.1] that $\psi(P) = \psi(Q) \xRightarrow[\psi \text{ is bijective}]{\text{def}} P = Q$. So $\vec{v} = P \ominus_1 P$ which proves by [theorem: 22.6] that $\vec{v} = \vec{0}$. Hence $\ker(d\psi) \subseteq \{0\}$ which combine with [eq: 22.8] proves that

$$\ker(d\psi) = \vec{0}$$

Finally using the linearity of $d\psi$, the above and [theorem: 11.195] it follows that $d\psi$ is injective. $\square$

Let $P, Q \in \mathbb{A}_2^n$ and $\vec{v} \in V_2$ then as ψ is a bijection there exist $P', Q' \in \mathbb{A}_1^n$ such that $P = \psi(P')$ and $Q = \psi(Q')$ or $P' = \psi^{-1}(P)$, $Q' = \psi^{-1}(Q)$. As ψ is a affine mapping we have that

$$\forall \vec{u} \in V_1 \quad \psi(P' \oplus_1 \vec{u}) \ominus_2 \psi(Q' \oplus_1 \vec{u}) = \psi(P') \ominus_2 \psi(Q') \tag{22.9}$$

Take $\vec{u} \in V_1$ by $\vec{u} = \psi^{-1}(P \oplus_2 \vec{v}) \ominus_1 P'$ then we have

$$d\psi(\vec{u}) \stackrel{\text{def}}{=} \psi(\psi^{-1}(P \oplus_2 \vec{v})) \ominus_2 \psi(P') = (P \oplus_2 \vec{v}) \ominus_2 P = \vec{v} \tag{22.10}$$

Take $\vec{w} \in V_1$ by $\vec{w} = \psi^{-1}(Q \oplus_2 \vec{v}) \ominus_1 Q'$ then we have

$$d\psi(\vec{w}) = \psi(\psi^{-1}(Q \oplus_2 \vec{v})) \ominus_2 \psi(Q') = (Q \oplus_2 \vec{v}) \ominus_2 Q = \vec{v} \quad (22.11)$$

As $d\psi$ is injective [see eq: 22.7] it follows from [eqs: 22.10 and 22.11] that

$$\vec{u} = \vec{w} \quad (22.12)$$

Using the definitions of $\vec{u}, \vec{w}$ on the above gives

$$\psi^{-1}(P \oplus_2 \vec{v}) \ominus_1 P' = (\psi^{-1}(Q \oplus_2 \vec{v}) \ominus_1 Q')$$

so that by [theorem: 22.5]

$$\psi^{-1}(P \oplus_2 \vec{v}) \ominus_1 \psi^{-1}(Q \oplus_2 \vec{v}) = P' \ominus_1 Q' = \psi^{-1}(P) \ominus_1 \psi^{-1}(Q)$$

To summarize we have

$$\forall P, Q \in \mathbb{A}_2^n \text{ and } \vec{v} \in V_2 \text{ we have } \psi^{-1}(P \oplus_2 \vec{v}) \ominus_1 \psi^{-1}(Q \oplus_2 \vec{v}) = \psi^{-1}(P) \ominus_1 \psi^{-1}(Q) \quad (22.13)$$

The above allows us to use [lemma: 22.15] to define the function

$$d\psi^{-1}: V_2 \rightarrow V_1 \text{ where } d\psi^{-1}(\vec{v}) = \psi^{-1}(P) \ominus_1 \psi^{-1}(Q) \text{ where } \vec{v} = P \ominus_2 Q \quad (22.14)$$

Next we prove that $d\psi^{-1}$ is the inverse of $d\psi$. Let $\vec{v} \in V_1$ then $d\psi(\vec{v}) = \psi(P) \ominus_2 \psi(Q)$ where $\vec{v} = P \ominus_1 Q$ so that by [eq: 22.14] $d\psi^{-1}(d\psi(\vec{v})) = \psi^{-1}(\psi(P)) \ominus_1 \psi^{-1}(\psi(Q)) = P \ominus_1 Q = \vec{v}$. Hence

$$d\psi^{-1} \circ d\psi = \text{Id}_{V_1}$$

Likewise if $\vec{v} \in V_2$ then $d\psi^{-1}(\vec{v}) = \psi^{-1}(P) \ominus_1 \psi^{-1}(Q)$ where $\vec{v} = P \ominus_2 Q$. Then we have by definition $d\psi(d\psi^{-1}(\vec{v})) = \psi(\psi^{-1}(P)) \ominus_2 \psi(\psi^{-1}(Q)) = P \ominus_2 Q = \vec{v}$ which proves that

$$d\psi \circ d\psi^{-1} = \text{Id}_{V_2}$$

Hence by [theorem: 2.73] $d\psi$ is a bijection and as $d\psi$ is linear it follows that

$$d\psi \text{ is a linear isomorphism}$$

From the above and [theorem: 11.178] it follows that

$$d\psi^{-1} \text{ is also a linear isomorphism}$$

Combing the above with [eq: 22.13] proves that

$$\psi^{-1} \text{ is an affine isomorphism} \quad \square$$

A usefull criteria to determine if a affine mapping between two affine spaces of the same finite dimension is a isomorphism is stated in the following theorem.

Theorem 22.20. *Let $n \in \mathbb{N}$, $\langle \mathbb{A}_1^n, V_1, \ominus_1 \rangle$, $\langle \mathbb{A}_2^n, V_2, \ominus_2 \rangle$ be two real affine spaces of dimension n , $\psi: \mathbb{A}_1^n \rightarrow \mathbb{A}_2^n$ a affine mapping and $O \in \mathbb{A}_1^n$ then we have*

$$\forall P \in \mathbb{A}_1^n \text{ we have } \psi(P) = \psi(O) \oplus_2 d\psi(P \ominus_1 O)$$

Further we have

$$\psi \text{ is injective} \Leftrightarrow \psi \text{ is surjective}$$

In other words if ψ is either injective or surjective then ψ is a affine isomorphism.

Proof. Let $O \in \mathbb{A}_1^n$ then for every $P \in \mathbb{A}_1^n$ we have for $\vec{v} = P \ominus_1 O \in V_1$ by the definition of $d\psi$ that $d\psi(\vec{v}) = d\psi(P \ominus_1 O) = \psi(P) \ominus_2 \psi(O)$. Hence we have by [definition: 22.1] that

$$\psi(P) = \psi(O) \oplus_2 d\psi(\vec{v}) = \psi(O) \oplus_2 d\psi(P \ominus_1 O)$$

So we have proved that

$$\forall P \in \mathbb{A}_1^n \text{ we have } \psi(P) = \psi(O) \oplus_2 d\psi(P \ominus_1 O) \quad (22.15)$$

Next we prove that

$$\psi \text{ is injective} \Leftrightarrow d\psi \text{ is injective} \quad (22.16)$$

Proof. As $\mathbb{A}_1^n \neq \emptyset$ there exist a $O \in \mathbb{A}_1^n$

$\Rightarrow$. Let $\vec{v}_1, \vec{v}_2 \in V_1$ be such that $d\psi(\vec{v}_1) = d\psi(\vec{v}_2)$. By [definition: 22.1] there exists P_1, P_2 such that $\vec{v}_1 = P_1 \ominus_1 O$ and $\vec{v}_2 = P_2 \ominus_1 O$ then we have

$$\psi(P_1) \ominus_2 \psi(O) = d\psi(\vec{v}_1) = d\psi(\vec{v}_2) = \psi(P_2) \ominus_2 \psi(O)$$

so that by [theorem: 22.5]

$$\psi(P_1) \ominus_2 \psi(P_2) = \psi(O) \ominus_2 \psi(O) \stackrel{[\text{theorem: 22.6}]}{=} \vec{0} \stackrel{[\text{theorem: 22.6}]}{=} \psi(P_2) \ominus_2 \psi(P_2)$$

which by uniqueness in [definition: 22.1] proves that $\psi(P_1) = \psi(P_2)$. As ψ is injective we have that $P_1 = P_2$ so that $\vec{v}_1 = P_1 \ominus_1 O = P_2 \ominus_1 O = \vec{v}_2$ which proves that $\vec{v}_1 = \vec{v}_2$. Hence $d\psi$ is injective.

$\Leftarrow$. Let $P_1, P_2 \in \mathbb{A}_1^n$ be such that $\psi(P_1) = \psi(P_2)$ then we have by [eq: 22.15] that

$$\psi(O) \oplus_2 d\psi(P_1 \ominus_1 O) = \psi(P_1) = \psi(P_2) = \psi(O) \oplus_2 d\psi(P_2 \ominus_1 O) \quad (22.17)$$

so that by [theorem: 22.1] we have

$$\begin{aligned} d\psi(P_1 \ominus_1 O) &\stackrel{[\text{theorem: 22.1}]}{=} (\psi(O) \oplus_2 d\psi(P_1 \ominus_1 O)) \ominus_2 \psi(O) \\ &\stackrel{[\text{eq: 22.17}]}{=} (\psi(O) \oplus_2 d\psi(P_2 \ominus_1 O)) \ominus_2 \psi(O) \\ &\stackrel{[\text{theorem: 22.1}]}{=} d\psi(P_2 \ominus_1 O) \end{aligned}$$

which as $d\psi$ is injective proved that $P_1 \ominus_1 O = P_2 \ominus_1 O$. Using the uniqueness in [definition: 22.1] it follows that $P_1 = P_2$. Hence ψ is injective. $\square$

Next we prove that

$$\psi \text{ is surjective} \Leftrightarrow d\psi \text{ is surjective} \quad (22.18)$$

Proof. As $\mathbb{A}_1^n \neq \emptyset$ there exist a $O' \in \mathbb{A}_1^n$. Take $O = \psi(O') \in \mathbb{A}_2^n$

$\Rightarrow$. Let $\vec{v} \in V_2$ then by [definition: 22.1] there exist a $P \in V_2$ such that $\vec{v} = P \ominus_2 O$. As ψ is surjective there exist a $P' \in \mathbb{A}_1^n$ such that $P = \psi(P')$. Then we have for $\vec{\mu} = P' \ominus_1 O' \in V_1$ that $d\psi(\vec{\mu}) = d\psi(P' \ominus_1 O') = \psi(P') \ominus_2 \psi(O') = P \ominus_2 O = \vec{v}$ which proves that $d\psi$ is surjective.

$\Leftarrow$. Let $P \in \mathbb{A}_2^n$ then for $\vec{v} = P \ominus_2 O$ we have as $d\psi$ is surjective there exist a $\vec{\mu} \in V_1$ such that

$$\vec{v} = d\psi(\vec{\mu}) \quad (22.19)$$

Using [definition: 22.1] there exist a $P' \in \mathbb{A}_1^n$ such that $\vec{\mu} = P' \ominus_1 O'$ then we have by [eq: 22.15] that

$$\begin{aligned} \psi(P') &\stackrel{[\text{eq: 22.15}]}{=} \psi(O') \oplus_2 d\psi(P' \ominus_1 O') \\ &= \psi(O') \oplus_2 d\psi(\vec{\mu}) \\ &\stackrel{[\text{eq: 22.19}]}{=} O \oplus_2 \vec{v} \\ &= O \oplus_2 (P \ominus_2 O) \\ &\stackrel{[\text{theorem: 22.4}]}{=} P \end{aligned}$$

which proves surjectivity of ψ $\square$

Finally we prove that

$$\psi \text{ is injective} \Leftrightarrow \psi \text{ is surjective}$$

Proof.

$\Rightarrow$. As ψ is injective it follows from [eq: 22.16] that $d\psi: V_1 \rightarrow V_2$ is injective. As also $\dim(V_1) = \dim(V_2) = n$ it follows from [theorem: 11.201] that $d\psi$ is surjective. Hence by [eq: 22.18] we have that ψ is surjective.

$\Leftarrow$. As ψ is surjective it follows from [eq: 22.18] that $d\psi: V_1 \rightarrow V_2$ is surjective. As also $\dim(V_1) = \dim(V_2) = n$ it follows from [theorem: 11.205] that $d\psi$ is injective. Hence by [eq: 22.16] that ψ is injective. $\square$ $\square$

We can express a affine transformation using Cartesian coordinates as follows.

Theorem 22.21. *Let $n, m \in \mathbb{N}$, $\langle \mathbb{A}_1^n, V_1, \ominus_1 \rangle$, $\langle \mathbb{A}_2^m, V_2, \ominus_2 \rangle$ be two real affine spaces then we have:*

1. *Let $\psi: \mathbb{A}_1^n \rightarrow \mathbb{A}_2^m$ be a affine mapping then for*

$f_1: \mathbb{A}_1^n \rightarrow \mathbb{R}^n$ a Cartesian coordinate system with origin O_1 and axes $E = \{\vec{e}_1, \dots, \vec{e}_n\}$

$f_2: \mathbb{A}_2^m \rightarrow \mathbb{R}^m$ a Cartesian coordinate system with origin O_2 and axes $F = \{\vec{f}_1, \dots, \vec{f}_m\}$

we have that

a. *For $f_2 \circ \psi \circ f_1^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ there exist a $M \in \mathcal{M}_{m,n}(\mathbb{R})$ and a $\{c_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$ such that*

$$\forall i \in \{1, \dots, m\} \text{ we have } ((f_2 \circ \psi \circ f_1^{-1})(x_1, \dots, x_n))_i = c_i + \sum_{j=1}^n M_{i,j} \cdot x_j$$

b. *If $P \in \mathbb{A}_1^n$ has coordinates $(x_1, \dots, x_n)$ using the coordinate system f_1 and $\psi(P) \in \mathbb{A}_2^m$ has coordinates $(y_1, \dots, y_m)$ using the coordinate system f_2 then*

$$\forall i \in \{1, \dots, m\} \text{ we have } y_i = c_i + \sum_{j=1}^n M_{i,j} \cdot x_j$$

2. *Let $\psi: \mathbb{A}_1^n \rightarrow \mathbb{A}_2^m$ be a mapping such that there exist a $M \in \mathcal{M}_{m,n}(\mathbb{R})$, a $\{c_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$ and 2 coordinate systems*

$f_1: \mathbb{A}_1^n \rightarrow \mathbb{R}^n$ a Cartesian coordinate system with origin O_1 and axes $E = \{\vec{e}_1, \dots, \vec{e}_n\}$

$f_2: \mathbb{A}_2^m \rightarrow \mathbb{R}^m$ a Cartesian coordinate system with origin O_2 and axes $E' = \{\vec{f}_1, \dots, \vec{f}_m\}$

such that

$$\forall i \in \{1, \dots, m\} \text{ we have } ((f_2 \circ \psi \circ f_1^{-1})(x_1, \dots, x_n))_i = c_i + \sum_{j=1}^n M_{i,j} \cdot x_j$$

then ψ is a affine mapping.

Proof.

1. We have:

a. As $\psi: \mathbb{A}_1^n \rightarrow \mathbb{A}_2^m$ is a affine mapping we have that

$$\forall P, Q \in \mathbb{A}_1^n \text{ and } \forall \vec{v} \in V_1 \quad \varphi(P \oplus_1 \vec{v}) \ominus_2 \psi(Q \oplus_1 \vec{v}) = \psi(P) \ominus_2 \psi(Q) \quad (22.20)$$

and

$$d\psi: V_1 \rightarrow V_2 \text{ by } d\psi(\vec{v}) = \psi(P) \ominus_2 \psi(Q) \text{ where } \vec{v} = P \ominus_1 Q \text{ is a linear mapping} \quad (22.21)$$

As $d\psi$ is a linear mapping and E is a basis for V_1 and F is a basis for V_2 it follows from [theorem: 11.339] that there exist a $M \in \mathcal{M}_{m,n}[\mathbb{R}]$ such that

$$\forall i \in \{1, \dots, n\} \text{ that } d\psi(\vec{e}_i) = \sum_{j=1}^m M_{j,i} \cdot \vec{f}_j \quad (22.22)$$

Let $(x_1, \dots, x_n) \in \mathbb{R}^n$ then for $P = f_1^{-1}(x_1, \dots, x_n)$ we have, as $f_1(P) = (x_1, \dots, x_n)$ and f_1 is a Cartesian coordinate system with origin O_1 and axes E , that

$$P \ominus_1 O_1 = \sum_{i=1}^n x_i \cdot \vec{e}_i \quad (22.23)$$

As $\psi(O_1) \ominus_2 O_2 \in V_2$ and F is a basis for V_2 there exist a $\{c_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$ such that

$$\psi(O_1) \ominus_2 O_2 = \sum_{j=1}^m c_j \cdot \vec{f}_j \quad (22.24)$$

Hence

$$\begin{aligned} \psi(P) \ominus_2 O_2 &= (\psi(P) \ominus_2 \psi(O_1)) + (\psi(O_1) \ominus_2 O_2) \\ &\stackrel{[\text{eq: } 22.24]}{=} (\psi(P) \ominus_2 \psi(O_1)) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &= d\psi(P \ominus_1 O_1) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &\stackrel{[\text{eq: } 22.23]}{=} d\psi\left(\sum_{i=1}^n x_i \cdot \vec{e}_i\right) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &= \sum_{i=1}^n x_i \cdot d\psi(\vec{e}_i) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &\stackrel{[\text{eq: } 22.22]}{=} \sum_{i=1}^n x_i \cdot \left(\sum_{j=1}^m M_{j,i} \cdot \vec{f}_j\right) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &= \sum_{i=1}^n \left(\sum_{j=1}^m x_i \cdot (M_{j,i} \cdot \vec{f}_j)\right) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &\stackrel{[\text{theorem: } 11.45]}{=} \sum_{j=1}^m \left(\sum_{i=1}^n x_i \cdot (M_{j,i} \cdot \vec{f}_j)\right) + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &= \sum_{j=1}^m \left(\sum_{i=1}^n x_i \cdot (M_{j,i})\right) \cdot \vec{f}_j + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &= \sum_{j=1}^m \left(\sum_{i=1}^n M_{j,i} \cdot x_i\right) \cdot \vec{f}_j + \sum_{j=1}^m c_j \cdot \vec{f}_j \\ &= \sum_{j=1}^m \left(c_j + \sum_{i=1}^n M_{j,i} \cdot x_i\right) \cdot \vec{f}_j \end{aligned}$$

As f_2 is a Cartesian coordinate system with origin O_2 and axes F it follows that

$$f_2(\psi(P)) = \left(c_1 + \sum_{j=1}^n M_{1,j} \cdot x_j, \dots, c_m + \sum_{j=1}^n M_{m,j} \cdot x_j\right)$$

which as $(f_2 \circ \psi \circ f_1^{-1})(x_1, \dots, x_n) = f_2(\psi(f_1^{-1}(x_1, \dots, x_n))) = f_2(\psi(P))$ proves that

$$\forall i \in \{1, \dots, m\} \text{ we have } ((f_2 \circ \psi \circ f_1^{-1})(x_1, \dots, x_n))_i = c_i + \sum_{j=1}^n M_{i,j} \cdot x_j \quad (22.25)$$

b. If P has coordinates $(x_1, \dots, x_n)$ using f_1 and $\psi(P)$ has coordinates $(y_1, \dots, y_m)$ using f_2 then we have $P = f_1^{-1}(x_1, \dots, x_n)$ and $f_2(\psi(P)) = (y_1, \dots, y_m)$ so that

$$(f_2 \circ \psi \circ f_1^{-1})(x_1, \dots, x_n) = f_2(\psi(f_1^{-1}(x_1, \dots, x_n))) = f_2(\psi(P)) = (y_1, \dots, y_m)$$

Using now [eq: 22.25] on the above we have

$$\forall i \in \{1, \dots, m\} \text{ we have } y_i = c_i + \sum_{j=1}^n M_{i,j} \cdot x_j$$

2. Let $\psi: \mathbb{A}_1^n \rightarrow \mathbb{A}_2^m$ be a mapping such that there exist a $M \in \mathcal{M}_{m,n}(\mathbb{R})$, a $\{c_i\}_{i \in \{1, \dots, m\}} \subseteq \mathbb{R}$ and 2 coordinate systems

$f_1: \mathbb{A}_1^n \rightarrow \mathbb{R}^n$ a Cartesian coordinate system with origin O_1 and axes $E = \{\vec{e}_1, \dots, \vec{e}_n\}$

$f_2: \mathbb{A}_2^m \rightarrow \mathbb{R}^m$ a Cartesian coordinate system with origin O_2 and axes $F = \{\vec{f}_1, \dots, \vec{f}_m\}$

such that

$$\forall i \in \{1, \dots, m\} \text{ we have } ((f_2 \circ \psi \circ f_1^{-1})(x_1, \dots, x_n))_i = c_i + \sum_{j=1}^n M_{i,j} \cdot x_j \quad (22.26)$$

Let $P, Q \in \mathbb{A}_1^n$ and $\vec{v} \in V_1$. As E is a basis for V_1 there exist a $\{v_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that

$$\vec{v} = \sum_{i=1}^n v_i \cdot \vec{e}_i \quad (22.27)$$

As $P \ominus_1 O_1, Q \ominus_1 O_1 \in V_1$ and E is a basis we have that $\exists \{p_i\}_{i \in \{1, \dots, n\}}, \{q_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that

$$P \ominus_1 O_1 = \sum_{i=1}^n p_i \cdot \vec{e}_i \wedge Q \ominus_1 O_1 = \sum_{i=1}^n q_i \cdot \vec{e}_i \quad (22.28)$$

so that

$$f_1(P) = (p_1, \dots, p_n) \wedge f_1(Q) = (q_1, \dots, q_n) \quad (22.29)$$

or

$$P = f_1^{-1}(p_1, \dots, p_n) \wedge Q = f_1^{-1}(q_1, \dots, q_n) \quad (22.30)$$

Hence $\forall i \in \{1, \dots, m\}$ we have

$$\begin{aligned} (f_2(\psi(P)))_i &\stackrel{[\text{eq: 22.30}]}{=} (f_2(\psi(f_1^{-1}(p_1, \dots, p_n))))_i \\ &= ((f_2 \circ \psi \circ f_1^{-1})(p_1, \dots, p_n))_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot p_j \\ (f_2(\psi(Q)))_i &\stackrel{[\text{eq: 22.30}]}{=} (f_2(\psi(f_1^{-1}(q_1, \dots, q_n))))_i \\ &= ((f_2 \circ \psi \circ f_1^{-1})(q_1, \dots, q_n))_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot q_j \end{aligned}$$

As f_2 is a Cartesian coordinate system with origin O_2 and axes F we have from the above that

$$\psi(P) \ominus_2 O_2 = \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot p_j \right) \cdot \vec{f}_i \quad (22.31)$$

$$\psi(Q) \ominus_2 O_2 = \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot q_j \right) \cdot \vec{f}_i \quad (22.32)$$

so that

$$\begin{aligned} \psi(P) \ominus_2 \psi(Q) &\stackrel{[\text{definition: 22.1}]}{=} (\psi(P) \ominus_2 O_2) + (O_2 \ominus_2 \psi(Q)) \\ &\stackrel{[\text{theorem: 22.7}]}{=} (\psi(P) \ominus_2 O_2) - (\psi(Q) \ominus_2 O_2) \\ &\stackrel{[\text{eq:22.31, 22.32}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot p_j \right) \cdot \vec{f}_i - \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot q_j \right) \cdot \vec{f}_i \\ &= \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot p_j - \left(c_i + \sum_{j=1}^n M_{i,j} \cdot q_j \right) \right) \cdot \vec{f}_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot p_j - \sum_{j=1}^n M_{i,j} \cdot q_j \right) \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot (p_j - q_j) \right) \cdot \vec{f}_i \end{aligned}$$

which proves that

$$\psi(P) \ominus_2 \psi(Q) = \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot (p_j - q_j) \right) \cdot \vec{f}_i \quad (22.33)$$

Now

$$\begin{aligned} (P \oplus_1 \vec{v}) \ominus_1 O_1 &\stackrel{[\text{definition: 22.1}]}{=} ((P \oplus_1 \vec{v}) \ominus_1 P) + (P \ominus_1 O_1) \\ &\stackrel{[\text{theorem: 22.1}] \wedge [\text{eq:22.28}]}{=} \vec{v} + \sum_{i=1}^n p_i \cdot \vec{e}_i \\ &\stackrel{[\text{eq: 22.27}]}{=} \sum_{i=1}^n v_i \cdot \vec{e}_i + \sum_{i=1}^n p_i \cdot \vec{e}_i \\ &= \sum_{i=1}^n (v_i + p_i) \cdot \vec{e}_i \\ (Q \oplus_1 \vec{v}) \ominus_1 O_1 &\stackrel{[\text{definition: 22.1}]}{=} ((Q \oplus_1 \vec{v}) \ominus_1 Q) + (Q \ominus_1 O_1) \\ &\stackrel{[\text{theorem: 22.1}] \wedge [\text{eq:22.28}]}{=} \vec{v} + \sum_{i=1}^n q_i \cdot \vec{e}_i \\ &\stackrel{[\text{eq: 22.27}]}{=} \sum_{i=1}^n v_i \cdot \vec{e}_i + \sum_{i=1}^n q_i \cdot \vec{e}_i \\ &= \sum_{i=1}^n (v_i + q_i) \cdot \vec{e}_i \end{aligned}$$

So that

$$f_1(P \oplus_1 \vec{v}) = (v_1 + p_1, \dots, v_n + p_n) \text{ and } f_1(Q \oplus_1 \vec{v}) = (v_1 + q_1, \dots, v_n + q_n) \quad (22.34)$$

hence

$$P \oplus_1 \vec{v} = f_1^{-1}(v_1 + p_1, \dots, v_n + p_n) \text{ and } Q \oplus_1 \vec{v} = f_1^{-1}(v_1 + q_1, \dots, v_n + q_n) \quad (22.35)$$

Next we have $\forall i \in \{1, \dots, m\}$

$$\begin{aligned}
 (f_2(\psi(P \oplus_1 \vec{v})))_i &\stackrel{[\text{eq: 22.35}]}{=} (f_2(\psi(f_1^{-1}(v_1 + p_1, \dots, v_n + p_n))))_i \\
 &= ((f_2 \circ \psi \circ f_1^{-1})(v_1 + p_1, \dots, v_n + p_n))_i \\
 &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + p_j) \\
 (f_2(\psi(Q \oplus_1 \vec{v})))_i &\stackrel{[\text{eq: 22.35}]}{=} (f_2(\psi(f_1^{-1}(v_1 + q_1, \dots, v_n + q_n))))_i \\
 &= ((f_2 \circ \psi \circ f_1^{-1})(v_1 + q_1, \dots, v_n + q_n))_i \\
 &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + q_j)
 \end{aligned}$$

As f_2 is a Cartesian system with origin O_2 and axes F it follows from the above that

$$\psi(P \oplus \vec{v}) \ominus_2 O_2 = \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + p_j) \right) \cdot \vec{f}_i \quad (22.36)$$

$$\psi(Q \oplus \vec{v}) \ominus_2 O_2 = \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + q_j) \right) \cdot \vec{f}_i \quad (22.37)$$

so that

$$\begin{aligned}
 &\psi(P \oplus_1 \vec{v}) \ominus_2 \psi(Q \oplus_1 \vec{v}) \stackrel{[\text{definition: 22.1}]}{=} \\
 &(\psi(P \oplus_1 \vec{v}) \ominus_2 O_2) + (O_2 \ominus_2 \psi(Q \oplus_1 \vec{v})) \stackrel{[\text{theorem: 22.7}]}{=} \\
 &(\psi(P \oplus_1 \vec{v}) \ominus_2 O_2) - (\psi(Q \oplus_1 \vec{v}) \ominus_2 O_2) \stackrel{[\text{eqs: 22.36, 22.37}]}{=} \\
 &\sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + p_j) \right) \cdot \vec{f}_i - \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + q_j) \right) \cdot \vec{f}_i = \\
 &\sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + p_j) - \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (v_j + q_j) \right) \right) \cdot \vec{f}_i = \\
 &\sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot (v_j + p_j) - \sum_{j=1}^n M_{i,j} \cdot (v_j + q_j) \right) \cdot \vec{f}_i = \\
 &\sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot (v_j + p_j - (v_j + q_j)) \right) \cdot \vec{f}_i = \\
 &\sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot (p_j - q_j) \right) \cdot \vec{f}_i \stackrel{[\text{eq: 22.33}]}{=} \\
 &\psi(P) \ominus_2 \psi(Q)
 \end{aligned}$$

Hence we have that

$$\forall P, Q \in \mathbb{A}_1^n \text{ and } \vec{v} \in V_1 \text{ we have } \psi(P \oplus_1 \vec{v}) \ominus_2 \psi(Q \oplus_1 \vec{v}) = \psi(P) \ominus_2 \psi(Q) \quad (22.38)$$

Let $\vec{u}, \vec{v} \in V_1$ and $\alpha \in \mathbb{R}$ then by [definition: 22.1] there exists $P, Q, R, S \in \mathbb{A}_1^n$ such that

$$\vec{u} = P \ominus_1 O_1 \wedge \vec{v} = Q \ominus_1 O_1 \wedge R \ominus_1 O_1 = \alpha \cdot \vec{v} \wedge S \ominus_1 O_1 = \vec{u} + \alpha \cdot \vec{v} \quad (22.39)$$

hence as E is a basis for V_1 there exist $\{p_i\}_{i \in \{1, \dots, n\}}, \{q_i\}_{i \in \{1, \dots, n\}} \subseteq \mathbb{R}$ such that

$$P \ominus_1 O_1 = \vec{u} = \sum_{i=1}^n p_i \cdot \vec{e}_i \wedge Q \ominus_1 O_1 = \vec{v} = \sum_{i=1}^n q_i \cdot \vec{e}_i \wedge R \ominus_1 O_1 = \alpha \cdot \vec{v} = \sum_{i=1}^n (\alpha \cdot q_i) \cdot \vec{e}_i$$

so that

$$S \ominus_1 O_1 = \vec{u} + \alpha \cdot \vec{v} = \sum_{i=1}^n p_i \cdot \vec{e}_i + \sum_{i=1}^n (\alpha \cdot q_i) \cdot \vec{e}_i = \sum_{i=1}^n (p_i + \alpha \cdot q_i) \cdot \vec{e}_i$$

Further

$$O_1 \ominus_1 O_1 \stackrel{[\text{theorem: 22.6}]}{=} \vec{0} = \sum_{i=1}^n 0 \cdot \vec{e}_i$$

As f_1 is a Cartesain coordinate system with origin O_1 we have that

$$\begin{aligned} f_1(P) &= (p_1, \dots, p_n) \\ f_1(Q) &= (q_1, \dots, q_n) \\ f_1(R) &= (\alpha \cdot q_1, \dots, \alpha \cdot q_n) \\ f_1(S) &= (p_1 + \alpha \cdot q_1, \dots, p_n + \alpha \cdot q_n) \\ f_1(O_1) &= \left(\underbrace{0, \dots, 0}_n \right) \end{aligned}$$

So that

$$\begin{aligned} P &= f_1^{-1}(p_1, \dots, p_n) \\ Q &= f_1^{-1}(q_1, \dots, q_n) \\ R &= f_1^{-1}(\alpha \cdot q_1, \dots, \alpha \cdot q_n) \\ S &= f_1^{-1}(p_1 + \alpha \cdot q_1, \dots, p_n + \alpha \cdot q_n) \\ O_1 &= f_1^{-1}\left(\underbrace{0, \dots, 0}_n\right) \end{aligned}$$

Hence

$$\begin{aligned} (f_2(\psi(P)))_i &= (f_2(\psi(f_1^{-1}(p_1, \dots, p_n))))_i \\ &= ((f_2 \circ \psi \circ f_1^{-1})(p_1, \dots, p_n))_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot p_j \end{aligned} \tag{22.40}$$

$$\begin{aligned} (f_2(\psi(Q)))_i &= (f_2(\psi(f_1^{-1}(q_1, \dots, q_n))))_i \\ &= ((f_2 \circ \psi \circ f_1^{-1})(q_1, \dots, q_n))_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot q_j \end{aligned} \tag{22.41}$$

$$\begin{aligned} (f_2(\psi(R)))_i &= (f_2(\psi(f_1^{-1}(\alpha \cdot q_1, \dots, \alpha \cdot q_n))))_i \\ &= ((f_2 \circ \psi \circ f_1^{-1})(\alpha \cdot q_1, \dots, \alpha \cdot q_n))_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot (\alpha \cdot q_j) \end{aligned} \tag{22.42}$$

$$\begin{aligned} (f_2(\psi(S)))_i &= (f_2(\psi(f_1^{-1}(p_1 + \alpha \cdot q_1, \dots, p_n + \alpha \cdot q_n))))_i \\ &= ((f_2 \circ \psi \circ f_1^{-1})(p_1 + \alpha \cdot q_1, \dots, p_n + \alpha \cdot q_n))_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot (p_j + \alpha \cdot q_j) \end{aligned} \tag{22.43}$$

$$\begin{aligned} (f_2(\psi(O_1)))_i &= \left(f_2 \left(\psi \left(f_1^{-1} \left(\underbrace{0, \dots, 0}_n \right) \right) \right) \right)_i \\ &= \left((f_2 \circ \psi \circ f_1^{-1}) \left(\underbrace{0, \dots, 0}_n \right) \right)_i \\ &\stackrel{[\text{eq: 22.26}]}{=} c_i + \sum_{j=1}^n M_{i,j} \cdot 0 \\ &= c_i \end{aligned} \tag{22.44}$$

As f_2 is a Cartesian coordinate system with origin O_2 with axes F we have that

$$\psi(P) \ominus_2 O_2 \stackrel{[\text{eq: 22.40}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot p_j \right) \cdot \vec{f}_i \quad (22.45)$$

$$\psi(Q) \ominus_2 O_2 \stackrel{[\text{eq: 22.41}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot q_j \right) \cdot \vec{f}_i \quad (22.46)$$

$$\psi(R) \ominus_2 O_2 \stackrel{[\text{eq: 22.42}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (\alpha \cdot q_j) \right) \cdot \vec{f}_i \quad (22.47)$$

$$\psi(S) \ominus_2 O_2 \stackrel{[\text{eq: 22.43}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (p_j + \alpha \cdot q_j) \right) \cdot \vec{f}_i \quad (22.48)$$

$$\psi(O_1) \ominus_2 O_2 \stackrel{[\text{eq: 22.44}]}{=} \sum_{i=1}^m c_i \cdot \vec{f}_i \quad (22.49)$$

So that

$$\begin{aligned} d\psi(\vec{u} + \alpha \cdot \vec{v}) &\stackrel{[\text{eq: 22.39}]}{=} d\psi(S \ominus_1 O_1) \\ &= \psi(S) \ominus_2 \psi(O_1) \\ &\stackrel{[\text{definition: 22.1}]}{=} (\psi(S) \ominus_2 O_2) + (O_2 \ominus_2 \psi(O_1)) \\ &\stackrel{[\text{theorem: 22.7}]}{=} (\psi(S) \ominus_2 O_2) - (\psi(O_1) \ominus_2 O_2) \\ &\stackrel{[\text{eqs: 22.48, 22.49}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (p_j + \alpha \cdot q_j) \right) \cdot \vec{f}_i - \sum_{i=1}^m c_i \cdot \vec{f}_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot p_j \right) \cdot \vec{f}_i + \alpha \cdot \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot q_j \right) \cdot \vec{f}_i \quad (22.50) \end{aligned}$$

$$\begin{aligned} d\psi(\vec{u}) &\stackrel{[\text{eq: 22.39}]}{=} d\psi(P \ominus_1 O_1) \\ &= \psi(P) \ominus_2 \psi(O_1) \\ &\stackrel{[\text{definition: 22.1}]}{=} (\psi(P) \ominus_2 O_2) + (O_2 \ominus_2 \psi(O_1)) \\ &\stackrel{[\text{theorem: 22.7}]}{=} (\psi(P) \ominus_2 O_2) - (\psi(O_1) \ominus_2 O_2) \\ &\stackrel{[\text{eqs: 22.45, 22.49}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot p_j \right) \cdot \vec{f}_i - \sum_{i=1}^m c_i \cdot \vec{f}_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot p_j \right) \cdot \vec{f}_i \quad (22.51) \end{aligned}$$

$$\begin{aligned} d\psi(\alpha \cdot \vec{v}) &\stackrel{[\text{eq: 22.39}]}{=} \psi(R \ominus_1 O_1) \\ &= \psi(R) \ominus_2 \psi(O_1) \\ &\stackrel{[\text{definition: 22.1}]}{=} (\psi(R) \ominus_2 O_2) + (O_2 \ominus_2 \psi(O_1)) \\ &\stackrel{[\text{theorem: 22.7}]}{=} (\psi(R) \ominus_2 O_2) - (\psi(O_1) \ominus_2 O_2) \\ &\stackrel{[\text{eqs: 22.47, 22.49}]}{=} \sum_{i=1}^m \left(c_i + \sum_{j=1}^n M_{i,j} \cdot (\alpha \cdot q_j) \right) \cdot \vec{f}_i - \sum_{i=1}^m c_i \cdot \vec{f}_i \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot (\alpha \cdot q_j) \right) \cdot \vec{f}_i \\
&= \alpha \cdot \sum_{i=1}^m \left(\sum_{j=1}^n M_{i,j} \cdot q_j \right) \cdot \vec{f}_i
\end{aligned} \tag{22.52}$$

Using then [eqs: 22.50, 22.48 and 22.52] it follows that

$$d\psi(\vec{u} + \alpha \cdot \vec{v}) = d\psi(\vec{u}) + \alpha \cdot d\psi(\vec{v})$$

hence

$$d\psi: V_1 \rightarrow V_2 \text{ is linear}$$

The above together with [eq:22.38] proves that ψ is a affine mapping. $\square$

We define now the concept of a line in a affine space.

Definition 22.22. Let $n \in \mathbb{N}$, $\langle \mathbb{A}^n, V, \oplus \rangle$ be a real affine space of dimension n , $O \in \mathbb{A}^n$, $\vec{v} \in V$ then a **line l with origin O and tangent vector $\vec{v}$** is the map

$$l: \mathbb{R} \rightarrow \mathbb{A}^n \text{ defined by } l(t) = O \oplus (t \cdot \vec{v})$$

If $a, b \in \mathbb{R}$ with $a < b$ then the maps

$$\begin{aligned}
l|_{[a,b]}: [a, b] &\rightarrow \mathbb{A}^n \\
l|_{]a,b]}:]a, b] &\rightarrow \mathbb{A}^n \\
l|_{[a,b[}: [a, b[&\rightarrow \mathbb{A}^n \\
l|_{]a,b[}:]a, b[&\rightarrow \mathbb{A}^n
\end{aligned}$$

are called **segments of the line l** .

The importance of affine mappings is that they map lines to lines.

TODO

Theorem 22.23. Let $n, m \in \mathbb{N}$, $\langle \mathbb{A}_1^n, V_1, \oplus_1 \rangle, \langle \mathbb{A}_2^m, V_2, \oplus_2 \rangle$ be a real affine spaces and $\psi: \mathbb{A}_1^n \rightarrow \mathbb{A}_2^m$ a affine mapping then if l is a line with origin O and tangent vector $\vec{v}$ then $\psi \circ l$ is a line with origin $\psi(O)$ and tangent vector $d\psi(\vec{v})$.

Proof. As l is a line with origin O and tangent vector $\vec{v}$ we have that

$$l: \mathbb{R} \rightarrow \mathbb{A}_1^n \text{ is defined by } l(t) = O \oplus_1 (t \cdot \vec{v})$$

Let $t \in \mathbb{R}$ then by [definition: 22.1] $t \cdot \vec{v} = (O \oplus_1 (t \cdot \vec{v})) \ominus_1 O = l(t) \ominus_1 O$ so that

$$t \cdot d\psi(\vec{v}) \underset{d\psi \text{ is linear}}{=} d\psi(t \cdot \vec{v}) = \psi(l(t)) \ominus_2 \psi(O) = (\psi \circ l(t)) \ominus_2 \psi(O)$$

So we have that

$$\psi(O) \oplus_2 (t \cdot d\psi(\vec{v})) = \psi(O) \oplus_2 (\psi \circ l(t)) \ominus_2 \psi(O) \underset{[\text{theorem: 22.4}]}{=} (\psi \circ l)(t)$$

so we have that

$$\psi \circ l: \mathbb{R} \rightarrow \mathbb{A}_2^m \text{ is defined by } (\psi \circ l)(t) = \psi(O) \oplus_2 (t \cdot d\psi(\vec{v}))$$

which proves that $\psi \circ l$ is a line with origin $\psi(O)$ and tangent vector $d\psi(\vec{v})$. $\square$

22.2 Euclidean space

To be able to define distances, angles, areas, we need the additional structure of an inner product on the set of displacements of an affine space. This leads to the definition of a Euclidean space.

Definition 22.24. (Euclidean Space) Let $n \in \mathbb{N}$ then $\langle \mathbb{E}^n, V, \langle \rangle \rangle$ is a **real Euclidean Space of dimension n** if $\langle \mathbb{E}^n, V \rangle$ is a real affine space of dimension n and $\langle V, \langle \rangle \rangle$ is a real inner product space [see definition: 18.1]

Definition 22.25. Let $n \in \mathbb{N}$ $\langle \mathbb{E}^n, V, \langle \rangle \rangle$ be a real Euclidean Space of dimension n then a Cartesian coordinations system with origin $O \in \mathbb{E}^n$ and axes $E = \{\vec{e}_1, \dots, \vec{e}_n\}$ is **orthonormal** if E is a **orthonormal basis** [see definition: 18.26].

The inner product on the space of displacements allows us to define a metric on a Euclidean space.

Chapter 23

Differential Manifolds

23.1 Differentiable Manifold

23.1.1 Topological Manifold

This section is about the study of differentiable manifolds, which are essential spaces that locally look like the Banach space $\langle \mathbb{R}^n, \|\cdot\|_n \rangle$ [see theorem: 14.395]. This allows us to introduce the concept of differentiability on a differentiable manifold. We start with the definition of a topological manifold. Here the topology that we choose for $\mathbb{R}^n$ becomes important, we use the topology generated by the canonical norm $\|\cdot\|_n$ [see example: 14.101] because of its properties. As all norms on $\mathbb{R}^n$ are equivalent [meaning the generated topology is the same] [see theorem: 14.325] we use the following notation convention.

Notation 23.1. Let $n \in \mathbb{N}$ then the topology $\mathcal{T}_{\mathbb{R}^n}$ on $\mathbb{R}^n$ is defined by $\mathcal{T}_{\mathbb{R}^n} = \mathcal{T}_{\|\cdot\|_n}$. Hence if $\|\cdot\|$ is any norm on $\mathbb{R}^n$ then by [theorem: 14.325] $\mathcal{T}_{\|\cdot\|} = \mathcal{T}_{\mathbb{R}^n}$.

We define now a topological manifold which is a topological space that is locally homeomorph with $\langle \mathbb{R}^n, \mathcal{T}_{\mathbb{R}^n} \rangle$.

Definition 23.2. (Topological Manifold) A topological manifold $\langle M, \mathcal{T} \rangle$ of dimension $n \in \mathbb{N}$ is a second countable Hausdorff topological space $\langle M, \mathcal{T} \rangle$ [see definitions: 14.241, 14.257] such that $\forall m \in M$ there exist a open set $U \in \mathcal{T}_{\mathbb{R}^n}$ with $m \in U$ that is homeomorph with a open set $V \in \mathcal{T}_{\mathbb{R}^n}$ [using the subspace topologies $\mathcal{T}|_U$ and $(\mathcal{T}_{\mathbb{R}^n})|_U$].

The above definition suggest the following two definitions.

Definition 23.3. (Coordinate Chart) Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold then a **coordinate chart on $\langle M, \mathcal{T} \rangle$** is a pair (U, φ) where $U \subseteq M$ is a open set in M and $\varphi: U \rightarrow \mathbb{R}^n$ a function such that $\varphi(U) \in \mathcal{T}_{\mathbb{R}^n}$ and $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism [using the topologies $\mathcal{T}|_U$ and $(\mathcal{T}_{\mathbb{R}^n})|_{\varphi(U)}$] between the open sets U and $\varphi(U)$.

Note 23.4. As $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism, hence injective, we have $\forall x, y \in U$ with $\varphi(x) = \varphi(y)$ that $x = y$ which proves that $\varphi: U \rightarrow \mathbb{R}^n$ is injective.

Lemma 23.5. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and (U, φ) a coordinate chart on $\langle M, \mathcal{T} \rangle$ then for every open V in M we have that $\varphi(U \cap V)$ is a open set in $\mathbb{R}^n$.

Note 23.6. In particular if $V \subseteq U$ then $\varphi(V) = \varphi(V \cap U)$ is a open set in $\mathbb{R}^n$.

Proof. As V is open it follows that $U \cap V \in \mathcal{T}|_U$. As $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism, hence open, it follows that $\varphi(U \cap V) \in (\mathcal{T}_{\mathbb{R}^n})|_{\varphi(U)}$. Using the fact that $\varphi(U) \in \mathcal{T}_{\mathbb{R}^n}$ we have by [theorem: 14.11] that $(\mathcal{T}_{\mathbb{R}^n})|_{\varphi(U)} \subseteq \mathcal{T}_{\mathbb{R}^n}$. Hence we conclude that $\varphi(U \cap V) \in \mathcal{T}_{\mathbb{R}^n}$ or that $\varphi(U \cap V)$ is open in $\mathbb{R}^n$. $\square$

Theorem 23.7. Let $n \in \mathbb{N}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, (U, φ) a chart on $\langle M, \mathcal{T} \rangle$ and $V \in \mathcal{T}$ with $V \subseteq U$ then $(V, \varphi|_V)$ is a chart on $\langle M, \mathcal{T} \rangle$

Proof. As (U, φ) is a chart we have that

$$U \in \mathcal{T}, \varphi(U) \in \mathcal{T}_{\mathbb{R}^n} \text{ and } \varphi: U \rightarrow \varphi(U) \text{ is a homeomorphism}$$

Using [theorem: 14.193] it follows then that

$$\varphi|_V: V \rightarrow \varphi(V) = \varphi|_V(V) \text{ is a homeomorphism}$$

As $V = U \cap V$ it follows from [lemma: 23.5] also that

$$\varphi|_V(V) \stackrel{[\text{theorem: 2.92(3)}]}{=} \varphi(V) \in \mathcal{T}_{\mathbb{R}^n}$$

To summarize we have that $V \in \mathcal{T}$, $\varphi|_V(V) \in \mathcal{T}_{\mathbb{R}^n}$ and $\varphi|_V: V \rightarrow \varphi|_V(V)$ is a homeomorphism so that by definition $(V, \varphi|_V)$ is a chart on $\langle M, \mathcal{T} \rangle$. $\square$

Definition 23.8. (Coordinate Domain) Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold a open set $U \in \mathcal{T}$ is a **coordinate domain** if there exist a homeomorphism [using the topologies $\mathcal{T}|_U$ and $(\mathcal{T}_{\mathbb{R}^n})|_{\varphi(U)}$] $\varphi: U \rightarrow \varphi(U)$ where $\varphi(M)$ is open in $\mathcal{T}$

Definition 23.9. (Coordinate Ball or Cube) Let $n \in \mathbb{N}$, $\langle M, \mathcal{T} \rangle$ a topological space then a open set $U \in \mathcal{T}$ is a **coordinate n -ball** iff there exist a homeomorphism $\varphi: U \rightarrow B_{\|\cdot\|_e}(x, \delta)$ where $x \in \mathbb{R}^n$, $\delta \in \mathbb{R}^+$ and $\|\cdot\|_e$ is the Euclidean norm [see definition: 18.20]. A open set $U \in \mathcal{T}$ is a **coordinate n -cube** if there exist a homeomorphism [using the topologies $\mathcal{T}|_U$ and $(\mathcal{T}_{\mathbb{R}^n})|_{\varphi(U)}$] $\varphi: U \rightarrow B_{\|\cdot\|_n}(x, \delta)$ where $x \in \mathbb{R}^n$, $\delta \in \mathbb{R}^+$ and $\|\cdot\|_n$ is the maximum norm [see example: 14.101].

Theorem 23.10. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold then $\forall m \in M$ there exist a coordinate chart (U, φ) with $m \in U$.

Proof. Let $m \in M$ then by definition there exist a open set $U \subseteq M$ with $m \in U$ and a open set $V \subseteq \mathbb{R}^n$ such that U is homeomorphic to V . Hence there exist a homeomorphism $\varphi: U \rightarrow V$. As φ is bijective it follows that $\varphi(U) = V$. The pair $(U, \varphi) \in \mathcal{T} \times \{\varphi \in \mathbb{R}^U | U \in \mathcal{T}\}$ is then the required coordinate chart. $\square$

Definition 23.11. (Atlas) Given a n -dimensional topological manifold $\langle M, \mathcal{T} \rangle$ then $\mathcal{A} \subseteq \{(U, \varphi) | U \in \mathcal{T} \wedge \varphi \in (\mathbb{R}^n)^U\}$ is a **atlas on $\langle M, \mathcal{T} \rangle$** iff

1. $\forall (U, \varphi) \in \mathcal{A}$ we have that (U, φ) is a coordinate chart on $\langle M, \mathcal{T} \rangle$
2. $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U$

Theorem 23.12. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ be a n -dimensional topological manifold then there exist at least one atlas on $\langle M, \mathcal{T} \rangle$.

Proof. Let $m \in M$ then by [theorem: 23.10] there exist a $(U_m, \varphi_m) \in \mathcal{T} \times (\mathbb{R}^n)^U$ such that (U, φ) is a coordinate chart. Hence if we define $\mathcal{A} = \{(U_m, \varphi_m) | m \in M\}$ then $\mathcal{A}$ is the required atlas. $\square$

A trivial example of a n -dimensional topological manifold is a second countable Hausdorff topological space that is homeomorphic to a open subset of $\mathbb{R}^n$.

Example 23.13. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a second countable Hausdorff topological space such that there exists a homeomorphism $\varphi: M \rightarrow V$ where $V \in \mathcal{T}_{\mathbb{R}^n}$ then $\langle M, \mathcal{T} \rangle$ is a n -dimensional topological space with atlas $\{(M, \varphi)\}$.

Proof. As $M \in \mathcal{T}$, $\varphi(M) = V \in \mathcal{T}_{\mathbb{R}^n}$ it follows that (M, φ) is a chart on $\langle M, \varphi \rangle$. Further $\forall p \in M$ we have trivially that $p \in M$ which proves that

$$\langle M, \mathcal{T} \rangle \text{ is a } n\text{-dimensional topological manifold with atlas } \{(M, \varphi)\} \quad \square$$

Example 23.14. $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ is a 1-dimensional topological manifold and $\{(\mathbb{R}, \varphi)\}$ is a atlas on $\langle \mathbb{R}, \mathcal{T}_{||} \rangle$ where

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}^1 \text{ is defined by } \varphi(x) = (x) \text{ [here } (x) \text{ is a tuple of one element where } (x)_1 = x]$$

Proof. First using [theorem: 14.242]

$$\langle \mathbb{R}, \mathcal{T}_{||} \rangle \text{ is Hausdorff}$$

and using [example: 14.259]

$$\langle \mathbb{R}, \mathcal{T}_{||} \rangle \text{ is second countable}$$

Define

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}^1 \text{ by } \varphi(x) = (x) \text{ [here } (x) \text{ is a tuple of one element where } (x)_1 = x]$$

then by [example: 14.189] φ is a homeomorphism. Using then [theorem: 23.13] completes the proof. $\square$

Example 23.15. Let $n \in \mathbb{N}$ then $\langle \mathbb{R}^n, \mathcal{T}_{\mathbb{R}^n} \rangle$ is a n -dimensional topological manifold with atlas $\{(\mathbb{R}^n, \text{Id}_{\mathbb{R}^n})\}$.

Proof. By [example: 14.259] and [theorem: 14.242] we have that

$$\langle \mathbb{R}^n, \mathcal{T}_{\|\cdot\|_n} \rangle \text{ is second countable and Hausdorff}$$

Further by [example: 14.188] $\text{Id}_{\mathbb{R}^n}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a homeomorphism between the open sets $\mathbb{R}^n$ and $\mathbb{R}^n$. Using [example: 23.13] finishes the proof. $\square$

Every open subset of a n -dimensional topological manifold is a n -dimensional topological manifold.

Example 23.16. Let $n \in \mathbb{N}$, $\langle M, \mathcal{T} \rangle$ be a n -dimensional topological manifold and $U \in \mathcal{T}$ a open subset of M then $\langle U, \mathcal{T}_U \rangle$ is a n -dimensional topological manifold. Further if $\langle V, \varphi \rangle$ is a coordinate chart on $\langle M, \mathcal{T} \rangle$ then $\langle V \cap U, \varphi|_{V \cap U} \rangle$ is a coordinate chart on $\langle U, \mathcal{T}_U \rangle$.

Proof. Using [theorem: 14.261] it follows that

$$\langle U, \mathcal{T}_U \rangle \text{ is Hausdorff and Second Countable} \quad (23.1)$$

Further if $u \in U$ then, as $\langle M, \mathcal{T} \rangle$ is a topological manifold, there exists a open set $V \in \mathcal{T}$ with $u \in V$, and a homeomorphism $\psi: V \rightarrow \psi(V)$. As $u \in U$ and $u \in V$ it follows

$$u \in V \cap U \in \mathcal{T}_U \quad (23.2)$$

Using [lemma: 23.5] it follows that

$$\psi(V \cap U) \in \mathcal{T}_{\mathbb{R}^n} \quad (23.3)$$

Further by [theorem: 14.193] we have as $\psi: V \rightarrow \psi(V)$ is a homeomorphism and $V \cap U \subseteq V$ that

$$\psi|_{V \cap U}: V \cap U \rightarrow \psi(V \cap U) \text{ is a homeomorphism using } (\mathcal{T}_V)|_{V \cap U}, ((\mathcal{T}_{\mathbb{R}^n})|_{\psi(V)})|_{\psi(V \cap U)}$$

As $\varphi(V \cap U) \subseteq \varphi(V)$ it follows from [theorem: 14.10] that

$$((\mathcal{T}_{\mathbb{R}^n})|_{\psi(V)})|_{\varphi(V \cap U)} = (\mathcal{T}_{\mathbb{R}^n})|_{\varphi(V \cap U)}$$

Further as $V \cap U \subseteq U$ and $V \cap U \subseteq V$ it follows from [theorem: 14.10] that

$$(\mathcal{T}_U)|_{V \cap U} = \mathcal{T}_V|_{V \cap U} = (\mathcal{T}_V)|_{V \cap U}$$

Hence

$$\varphi|_{V \cap U}: V \cap U \rightarrow \varphi(V \cap U) \text{ is a homeomorphism using } (\mathcal{T}_U)|_{V \cap U}, (\mathcal{T}_{\mathbb{R}^n})|_{\varphi(V \cap U)} \quad (23.4)$$

Using the fact that $u \in U$ has been chosen arbitrary together with [eqs: 23.1, 23.2, 23.3, 23.4] proves by [definition: 23.2] that

$$\langle U, \mathcal{T}_U \rangle \text{ is a topological manifold} \quad \square$$

A less trivial example is the n -dimensional unit sphere S^n embedded in $\mathbb{R}^{n+1}$. For this example we need the following lemma.

Lemma 23.17. Let $n \in \mathbb{N}$ then

$$\begin{aligned} \text{and } \left\{ x \in \mathbb{R}^n \mid \sum_{i \in \{1, \dots, n\}} x_i^2 < 1 \right\} & \stackrel{[\text{example: 18.20}]}{=} \{ x \in \mathbb{R}^n \mid \|x\|_e^2 < 1 \} \text{ is a open set in } \mathbb{R}^n \\ \zeta: \left\{ x \in \mathbb{R}^n \mid \sum_{i=1}^n x_i^2 < 1 \right\} & \rightarrow \mathbb{R} \text{ defined by } \zeta(x) = \sqrt{1 - \sum_{i \in \{1, \dots, n\}} x_i^2} \end{aligned}$$

is of class C^∞ . In particular it follows that ζ is a continuous function.

Proof. Let $x \in \{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}$ then $\|x\|_e^2 < 1$ so that by the monotonicity of $\sqrt{\cdot}$ [see theorem: 10.72] $\|x\|_e < 1$. Let $\delta_x = 1 - \|x\|_e \in \mathbb{R}^+$ then we have for $y \in \mathbb{B}_{\|\cdot\|_e}(x, \delta_x)$ that

$$\|y\|_e = \|y - x + x\|_e \leq \|y - x\|_e + \|x\|_e < \delta_x + \|x\|_e = 1 - \|x\|_e + \|x\|_e = 1$$

so that

$$\|y\|_e < 1$$

For $\|y\|_e$ we have either

$\|y\|_e = 0$. Then $\|y\|_e^2 = 0 < 1$

$0 < \|y\|_e$. Then we have by [theorem:10.14(15)] that $\|y\|_e^2 < 1^2 = 1$

hence we have in all cases that $\|y\|_e^2 < 1$ or $y \in \{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}$. So it follows that $B_{\|\cdot\|_e}(x, \delta_x) \subseteq \{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}$ which, as this is valid for every $x \in \{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}$, proves that

$$\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\} \in \mathcal{T}_{\|\cdot\|_e} = \mathcal{T}_{\mathbb{R}^n}$$

Let $i \in \{1, \dots, n\}$ then by [theorem: 16.185] $\pi_i: \mathbb{R}^n \rightarrow \mathbb{R}$ is of class C^∞ so that by [theorem: 16.201] $(\pi_i)^2$ is also of class C^∞ . Further using [theorem: 16.178] and [example: 16.179] it follows that

$$C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \text{ is of class } C^\infty$$

Applying then [corollary: 16.156] it follows that

$$\left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)_{|\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}} \text{ is of class } C^\infty$$

Let $x \in \{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}$ then $\sum_{i \in \{1, \dots, n\}} x_i^2 = \|x\|_e^2 < 1$ so that

$$\begin{aligned} \left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)_{|\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}}(x) &= 1 - \sum_{i \in \{1, \dots, n\}} (\pi_i(x))^2 \\ &= 1 - \sum_{i \in \{1, \dots, n\}} x_i^2 \\ &\in]0, \infty[\end{aligned}$$

hence

$$\left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)_{|\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}} (\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}) \subseteq]0, \infty[$$

As by [corollary: 19.53] $\sqrt{\cdot}:]0, \infty[\rightarrow \mathbb{R}$ is of class C^∞ it follows from the chain rule [see theorem: 16.197] that

$$\sqrt{\cdot} \circ \left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)_{|\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}} \text{ is of class } C^\infty$$

Finally as $\forall x \in \{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}$

$$\begin{aligned} \left(\sqrt{\cdot} \circ \left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)_{|\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}} \right)(x) &= \\ \sqrt{\left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)_{|\{x \in \mathbb{R}^n \mid \|x\|_e^2 < 1\}}(x)} &= \\ \sqrt{\left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)(x)} &= \\ \sqrt{1 - \sum_{i \in \{1, \dots, n\}} x_i^2} &= \\ \zeta(x) \end{aligned}$$

it follows that

$$\zeta = \sqrt{\cdot} \circ \left(C_1 - \sum_{i \in \{1, \dots, n\}} (\pi_i)^2 \right)$$

which proves that

$$\zeta \text{ is of class } C^\infty$$

□

Example 23.18. Let $n \in \mathbb{N}$, $S^n = \left\{x \in \mathbb{R}^{n+1} \mid \sqrt{\sum_{i=1}^{n+1} x_i^2} = \|x\|_e = 1\right\}$ the unit sphere in $\mathbb{R}^{n+1}$ then

$$\langle S^n, (\mathcal{T}_{\mathbb{R}^n})|_{S^n} \rangle \text{ is a } n\text{-dimensional topological manifold}$$

Further if we define $\forall i \in \{1, \dots, n+1\}$

$$V_i^+ = \{x \in \mathbb{R}^{n+1} \mid 0 < x_i\} \text{ and } V_i^- = \{x \in \mathbb{R}^{n+1} \mid x_i < 0\}$$

$$\psi_i^+ : S^n \cap V_i^+ \rightarrow \left\{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1\right\} \text{ where } (\psi_i^+(x))_{j \in \{1, \dots, n\}} \stackrel{\text{def}}{=} \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ x_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases}$$

$$\psi_i^- : S^n \cap V_i^- \rightarrow \left\{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1\right\} \text{ where } (\psi_i^-(x))_{j \in \{1, \dots, n\}} \stackrel{\text{def}}{=} \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ x_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases}$$

then

$$\psi_i^+, \psi_i^- \text{ are homeomorphisms}$$

with inverses defined by

$$(\psi_i^+)^{-1} : \left\{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1\right\} \rightarrow S^n \cap V_i^+ \text{ where } ((\psi_i^+)^{-1})_j = \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ \sqrt{1 - \sum_{k=1}^n x_k^2} & \text{if } j = i \\ x_{j-1} & \text{if } j \in \{i+1, \dots, n+1\} \end{cases}$$

$$(\psi_i^-)^{-1} : \left\{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1\right\} \rightarrow S^n \cap V_i^- \text{ where } ((\psi_i^-)^{-1})_j = \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ -\sqrt{1 - \sum_{k=1}^n x_k^2} & \text{if } j = i \\ x_{j-1} & \text{if } j \in \{i+1, \dots, n+1\} \end{cases}$$

so that $\{(U_i, \beta_i) \mid i \in \{1, \dots, 2 \cdot n + 2\}\}$ where

$$U_i = \begin{cases} S^n \cap V_i^+ & \text{if } i \in \{1, \dots, n+1\} \\ S^n \cap V_{i-(n+1)}^- & \text{if } i \in \{n+2, \dots, 2 \cdot n + 2\} \end{cases} \stackrel{\text{eq: 23.8}}{\in} (\mathcal{T}_{\|\cdot\|_{n+1}})|_{S^n}$$

and

$$\beta_i = \begin{cases} \psi_i^+ & \text{if } i \in \{1, \dots, n+1\} \\ \psi_{i-(n+1)}^- & \text{if } i \in \{n+2, \dots, 2 \cdot n + 2\} \end{cases}$$

is a atlas on $\langle S^n, (\mathcal{T}_{\|\cdot\|_{n+1}})|_{S^n} \rangle$.

Proof. By [example: 14.259] and [theorem: 14.242] we have that

$$\langle \mathbb{R}^{n+1}, \mathcal{T}_{\mathbb{R}^{n+1}} \rangle \text{ is second countable and Hausdorff}$$

Applying now [theorem: 14.261] it follows that

$$\langle S^n, (\mathcal{T}_{\mathbb{R}^{n+1}})|_{S^n} \rangle \text{ is second countable and Hausdorff} \quad (23.5)$$

Given $i \in \{1, \dots, n+1\}$ define

$$V_i^+ = \{x \in \mathbb{R}^{n+1} \mid 0 < x_i\} \text{ and } V_i^- = \{x \in \mathbb{R}^{n+1} \mid x_i < 0\}$$

We prove now that V_i^+, V_i^- are open sets in $\mathbb{R}^{n+1}$. Let $x \in V_i^+$ then $0 < x_i$. Define $\delta = \frac{x_i}{2} \in \mathbb{R}^+$ then we have for $y \in B_{\|\cdot\|_{n+1}}(x, \delta)$ that

$$x_i - y_i \leq |x_i - y_i| \leq \max(\{|x_i - y_i| \mid i \in \{1, \dots, n+1\}\}) = \|x - y\|_{n+1} < \delta = \frac{x_i}{2}$$

so that $x_i - y_i < \frac{x_i}{2} \Rightarrow 0 < \frac{x_i}{2} < y_i$ from which it follows that $y \in V_i^+$. Hence we have $x \in B_{\|\cdot\|_{n+1}}(x, \delta) \subseteq V_i^+$ which proves that

$$V_i^+ \in \mathcal{T}_{\|\cdot\|_{n+1}} \text{ or } V_i^+ \text{ is open in } \mathbb{R}^{n+1} \quad (23.6)$$

Likewise, let $x \in V_i^-$ then $x_i < 0$. Define $\delta = \frac{-x_i}{2} \in \mathbb{R}^+$ then we have for $y \in B_{\|\cdot\|_{n+1}}(x, \delta)$ that

$$y_i - x_i \leq |x_i - y_i| \leq \max(\{|x_i - y_i| : i \in \{1, \dots, n+1\}\}) = \|x - y\|_{n+1} < \delta = \frac{-x_i}{2}$$

so that $y_i - x_i < \frac{-x_i}{2} \Rightarrow y_i < \frac{x_i}{2} < 0$ from which it follows that $y \in V_i^-$. Hence we have that $x \in B_{\|\cdot\|_{n+1}}(x, \delta) \subseteq V_i^-$ which proves that

$$V_i^- \in \mathcal{T}_{\mathbb{R}^{n+1}} \text{ or } V_i^- \text{ is open in } \mathbb{R}^{n+1} \quad (23.7)$$

Using now the definition of a subspace topology [see definition: 14.8] on [eqs: 23.6, 23.7] proves that

$$S^n \cap V_i^+ \in (\mathcal{T}_{\mathbb{R}^{n+1}})_{|S^n} \text{ and } S^n \cap V_i^- \in (\mathcal{T}_{\mathbb{R}^{n+1}})_{|S^n} \quad (23.8)$$

Define now

$$\psi_i: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n \text{ by } (\psi_i(x_1, \dots, x_{n+1}))_j \stackrel{\text{def}}{=} \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ x_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases}$$

Then we have $\forall x, y \in \mathbb{R}^{n+1}, \alpha \in \mathbb{R}$ that $\forall j \in \{1, \dots, n\}$

$$\begin{aligned} (\psi_i(x + \alpha \cdot y))_j &= \begin{cases} (x + \alpha \cdot y)_j & \text{if } j \in \{1, \dots, i-1\} \\ (x + \alpha \cdot y)_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases} \\ &= \begin{cases} x_j + \alpha \cdot y_j & \text{if } j \in \{1, \dots, i-1\} \\ x_{j+1} + \alpha \cdot y_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases} \\ &= \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ x_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases} + \alpha \cdot \begin{cases} y_j & \text{if } j \in \{1, \dots, i-1\} \\ y_{j+1} & \text{if } j \in \{i, \dots, n\} \end{cases} \\ &= (\psi_i(x) + \alpha \cdot \psi_i(y))_j \end{aligned}$$

so that

$$\psi_i \in \text{Hom}(\mathbb{R}^{n+1}, \mathbb{R}^n)$$

Further we have $\forall x \in \mathbb{R}^{n+1}$ that

$$\begin{aligned} \|\psi_i(x)\|_n &= \sup(\{|\psi_i(x)_j| : j \in \{1, \dots, n\}\}) \\ &= \sup\left(\left\{\begin{array}{l} |x_j| \text{ if } j \in \{1, \dots, i-1\} \\ |x_{j+1}| \text{ if } j \in \{i, \dots, n\} \end{array} \mid j \in \{1, \dots, n\}\right\}\right) \\ &= \sup(\{|x_1|, \dots, |x_{i-1}|, |x_i|, \dots, |x_{n+1}|\}) \\ &\leq \sup(\{|x_i| : i \in \{1, \dots, n+1\}\}) \\ &= \|x\|_{n+1} \end{aligned}$$

so that by [theorem: 14.201]

$$\psi_i \text{ is a continuous mapping} \quad (23.9)$$

Next we examine the images of $S^n \cap V_i^+, S^n \cap V_i^-$ by ψ_i . Let $x \in S^n \cap V_i^+$ [or $x \in S^n \cap V_i^-$] then $0 < x_i \Rightarrow 0 < |x_i|^2$ [or $x_i < 0 \Rightarrow 0 < |x_i|^2$] so that

$$\sum_{j \in \{1, \dots, n+1\} \setminus \{i\}} x_j^2 < \sum_{j \in \{1, \dots, n+1\}} x_j^2$$

As $\sqrt{\cdot}$ is monotone [see theorem: 10.71] we have that

$$\begin{aligned} \sqrt{\sum_{j \in \{1, \dots, n\}} (\psi_i(x)_j)^2} &= \sqrt{\sum_{j \in \{1, \dots, n\} \setminus \{i\}} x_j^2} \\ &< \sqrt{\sum_{j \in \{1, \dots, n+1\}} x_j^2} \\ &\stackrel{x \in S^n}{=} 1 \end{aligned}$$

So that

$$\sum_{j \in \{1, \dots, n\}} (\psi_i(x)_j)^2 < 1^2 = 1$$

from which it follows that

$$\psi_i(S^n \cap V_i^+) \subseteq \left\{ x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1 \right\}$$

and

$$\psi_i(S^n \cap V_i^-) \subseteq \left\{ x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1 \right\}$$

The above allows us to define the restriction mappings

$$\begin{aligned} \psi_i^+ &= (\psi_i)|_{S^n \cap V_i^+}: S^n \cap V_i^+ \rightarrow \left\{ x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1 \right\} \\ \psi_i^- &= (\psi_i)|_{S^n \cap V_i^-}: S^n \cap V_i^- \rightarrow \left\{ x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1 \right\} \end{aligned}$$

As ψ_i is continuous it follows from [theorems: 14.152, 14.145] that

$$\psi_i^+, \psi_i^- \text{ are continuous [using the subspace topologies]} \quad (23.10)$$

We find now the inverse of ψ^+ , ψ^- . Let $x \in \{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} |x_j|^2 < 1\}$ then we have for

$$\begin{aligned} \varphi_i^+(x) \text{ defined by } \varphi_i^+(x)_j &= \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ \sqrt{1 - \sum_{k=1}^n x_k^2} & \text{if } j = i \\ x_{j-1} & \text{if } j \in \{i+1, \dots, n+1\} \end{cases} \\ \varphi_i^-(x) \text{ defined by } \varphi_i^-(x)_j &= \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ -\sqrt{1 - \sum_{k=1}^n x_k^2} & \text{if } j = i \\ x_{j-1} & \text{if } j \in \{i+1, \dots, n+1\} \end{cases} \end{aligned}$$

that

$$\varphi_i^+(x) \in V_i^+ \text{ and } \varphi_i^-(x) \in V_i^- \quad (23.11)$$

Further we have

$$\begin{aligned} \sqrt{\sum_{k=1}^{n+1} ((\varphi_i^+(x))_k)^2} &= \sqrt{\sum_{k=1}^{i-1} ((\varphi_i^+(x))_k)^2 + \sum_{k=i}^i ((\varphi_i^+(x))_k)^2 + \sum_{k=i+1}^{n+1} ((\varphi_i^+(x))_k)^2} \\ &= \sqrt{\sum_{k=1}^{i-1} x_k^2 + \left(\sqrt{1 - \sum_{k=1}^n x_k^2} \right)^2 + \sum_{k=i+1}^{n+1} x_{k-1}^2} \\ &= \sqrt{\sum_{k=1}^{i-1} x_k^2 + \left(1 - \sum_{k=1}^n x_k^2 \right) + \sum_{k=i}^n x_k^2} \\ &= \sqrt{\sum_{k=1}^n x_k^2 + \left(1 - \sum_{k=1}^n x_k^2 \right)} \\ &= \sqrt{1} \\ &= 1 \\ \sqrt{\sum_{k=1}^{n+1} (\varphi_i^-(x)_k)^2} &= \sqrt{\sum_{k=1}^{i-1} ((\varphi_i^-(x))_k)^2 + \sum_{k=i}^i ((\varphi_i^-(x))_k)^2 + \sum_{k=i+1}^{n+1} ((\varphi_i^-(x))_k)^2} \end{aligned}$$

$$\begin{aligned}
&= \sqrt{\sum_{k=1}^{i-1} x_i^2 + \left(-\sqrt{1 - \sum_{k=1}^n x_k^2}\right)^2 + \sum_{k=i+1}^{n+1} x_{k-1}^2} \\
&= \sqrt{\sum_{k=1}^{i-1} x_i^2 + \left(1 - \sum_{k=1}^n x_k^2\right) + \sum_{k=i}^n x_k^2} \\
&= \sqrt{\sum_{k=1}^n x_k^2 + \left(1 - \sum_{k=1}^n x_k^2\right)} \\
&= \sqrt{1} \\
&= 1
\end{aligned}$$

so that $\varphi_i^+(x) \in S^n$ and $\varphi_i^-(x) \in S^n$. Combining this with [eq: 23.11] proves that

$$\forall x \in \left\{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 < 1\right\} \text{ we have } \varphi_i^+(x) \in S^n \cap V_i^+ \text{ and } \varphi_i^-(x) \in S^n \cap V_i^-$$

The above allows us to define the following functions

$$\varphi_i^+ : \left\{x \in \mathbb{R}^n \mid \sum_{j=1}^n x_j^2 < 1\right\} \rightarrow S^n \cap V_i^+ \text{ where } \varphi_i^+(x)_j = \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ \sqrt{1 - \sum_{k=1}^n x_k^2} & \text{if } j = i \\ x_{j-1} & \text{if } j \in \{i+1, \dots, n+1\} \end{cases} \quad (23.12)$$

$$\varphi_i^- : \left\{x \in \mathbb{R}^n \mid \sum_{j=1}^n x_j^2 < 1\right\} \rightarrow S^n \cap V_i^- \text{ where } \varphi_i^-(x)_j = \begin{cases} x_j & \text{if } j \in \{1, \dots, i-1\} \\ -\sqrt{1 - \sum_{k=1}^n x_k^2} & \text{if } j = i \\ x_{j-1} & \text{if } j \in \{i+1, \dots, n+1\} \end{cases} \quad (23.13)$$

Next we prove that φ_i^+ is the inverse of ψ_i^+ and φ_i^- is the inverse of ψ_i^- . Let $x \in \{x \in \mathbb{R}^n \mid \sum_{j=1}^n x_j^2 < 1\}$ then we have for $k \in \{1, \dots, n\}$

$$\begin{aligned}
((\psi_i^+ \circ \varphi_i^+)(x))_k &= (\psi_i^+(\varphi_i^+(x)))_k \\
&= \begin{cases} (\varphi_i^+(x))_k & \text{if } k \in \{1, \dots, i-1\} \\ (\varphi_i^+(x))_{k+1} & \text{if } k \in \{i, \dots, n\} \end{cases} \\
&\stackrel{k \in \{i, \dots, n\} \Rightarrow k+1 \in \{i+1, \dots, n+1\}}{=} \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ x_{(k+1)-1} & \text{if } k \in \{i, \dots, n\} \end{cases} \\
&= \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ x_k & \text{if } k \in \{i, \dots, n\} \end{cases} \\
&= x_k \\
((\psi_i^- \circ \varphi_i^-)(x))_k &= (\psi_i^-(\varphi_i^-(x)))_k \\
&= \begin{cases} (\varphi_i^-(x))_k & \text{if } k \in \{1, \dots, i-1\} \\ (\varphi_i^-(x))_{k+1} & \text{if } k \in \{i, \dots, n\} \end{cases} \\
&\stackrel{k \in \{i, \dots, n\} \Rightarrow k+1 \in \{i+1, \dots, n+1\}}{=} \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ x_{(k+1)-1} & \text{if } k \in \{i, \dots, n\} \end{cases} \\
&= \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ x_k & \text{if } k \in \{i, \dots, n\} \end{cases} \\
&= x_k
\end{aligned}$$

Hence we have that

$$\psi_i^+ \circ \varphi_i^+ = \text{Id}_{\{x \in \mathbb{R}^n \mid \sum_{j=1}^n x_j^2 < 1\}} = \psi_i^- \circ \varphi_i^- \quad (23.14)$$

Let $x \in S^n$ then we have $\sqrt{\sum_{k=1}^{n+1} x_k^2} = 1$ so that $\sum_{k=1}^n x_k^2 = 1$ hence

$$x_i^2 = 1 - \sum_{k \in \{1, \dots, n\} \setminus \{i\}} x_k^2$$

If additional $x \in V_i^+$ then $0 < x_i$ so that $x_i = \sqrt{1 - \sum_{k \in \{1, \dots, n\} \setminus \{i\}} x_k^2}$ and if $x \in V_i^-$ then $x_i < 0$ so that $x_i = -\sqrt{1 - \sum_{k \in \{1, \dots, n\} \setminus \{i\}} x_k^2}$. Summarized we have

$$\forall x \in S^n \cap V_i^+ \quad x_i = \sqrt{1 - \sum_{k \in \{1, \dots, n\} \setminus \{i\}} x_k^2} \wedge \forall x \in S^n \cap V_i^- \quad x_i = -\sqrt{1 - \sum_{k \in \{1, \dots, n\} \setminus \{i\}} x_k^2} \quad (23.15)$$

Let $x \in S^n \cap V_i^n$ then we have for $k \in \{1, \dots, n+1\}$ that

$$\begin{aligned} ((\varphi_i^+ \circ \psi_i^+)(x))_k &= (\varphi_i^+(\psi_i^+(x)))_k \\ &= \begin{cases} (\psi_i^+(x))_k & \text{if } k \in \{1, \dots, i-1\} \\ \sqrt{1 - \sum_{k=1}^n (\psi_k^+(x))^2} & \text{if } k=i \\ (\psi_i^+(x))_{k-1} & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &= \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ \sqrt{1 - \sum_{k=1}^n (\psi_k^+(x))^2} & \text{if } k=i \\ x_{(k-1)+1} & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &= \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ \sqrt{1 - \sum_{k=1}^n (\psi_k^+(x))^2} & \text{if } k=i \\ x_k & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &\stackrel{[\text{eq: } 23.15]}{=} x_k \end{aligned}$$

Likewise we have for $x \in S^n \cap V_i^-$ and $k \in \{1, \dots, n+1\}$ that

$$\begin{aligned} ((\varphi_i^- \circ \psi_i^-)(x))_k &= (\varphi_i^-(\psi_i^-(x)))_k \\ &= \begin{cases} (\psi_i^-(x))_k & \text{if } k \in \{1, \dots, i-1\} \\ -\sqrt{1 - \sum_{k=1}^n (\psi_k^-(x))^2} & \text{if } k=i \\ (\psi_i^-(x))_{k-1} & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &= \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ -\sqrt{1 - \sum_{k=1}^n (\psi_k^-(x))^2} & \text{if } k=i \\ x_{(k-1)+1} & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &= \begin{cases} x_k & \text{if } k \in \{1, \dots, i-1\} \\ -\sqrt{1 - \sum_{k=1}^n (\psi_k^-(x))^2} & \text{if } k=i \\ x_k & \text{if } k \in \{i+1, \dots, n\} \end{cases} \\ &\stackrel{[\text{eq: } 23.15]}{=} x_k \end{aligned}$$

The above proves that

$$\varphi_i^+ \circ \psi_i^+ = \text{Id}_{S^n \cap V_i^+} \text{ and } \varphi_i^- \circ \psi_i^- = \text{Id}_{S^n \cap V_i^-}$$

which combined with [eq: 23.14] proves that

$$\psi_i^+ \text{ and } \psi_i^- \text{ are bijections with } (\psi_i^+)^{-1} = \varphi_i^+ \text{ and } (\psi_i^-)^{-1} = \varphi_i^- \quad (23.16)$$

Next we prove that φ_i^+ and φ_i^- are continuous. Let $k \in \{1, \dots, n+1\}$ then we have for k either

$k = i$. By [lemma: 23.17] it follows that

$$\zeta: \left\{ x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2 \right\} \rightarrow \mathbb{R} \text{ defined by } \zeta(x) = \sqrt{1 - \sum_{j \in \{1, \dots, n\}} x_j^2}$$

is continuous. Further we have $\forall x \in \{x \in \mathbb{R}^n \mid \sum_{j \in \{1, \dots, n\}} x_j^2\}$

$$(\pi_k \circ \varphi_i^+)(x) = \pi_k(\varphi_i^+(x)) \stackrel{[\text{eq: 23.12}]}{=} \sqrt{1 - \sum_{j \in \{1, \dots, n\}} x_j^2} = \zeta(x)$$

$$(\pi_k \circ \varphi_i^-)(x) = \pi_k(\varphi_i^-(x)) \stackrel{[\text{eq: 23.13}]}{=} -\sqrt{1 - \sum_{j \in \{1, \dots, n\}} x_j^2} = -\zeta(x)$$

which proves that $\pi_k \circ \varphi_i^+ = \zeta$ and $\pi_k \circ \varphi_i^- = -\zeta$ from which it follows that

$$\pi_k \circ \varphi_i^+ \text{ and } \pi_k \circ \varphi_i^- \text{ are continuous}$$

$k \in \{1, \dots, i-1\}$. Then by [eq: 23.12, 23.13] we have that $\pi_k \circ \varphi_i^+ = \pi_k$ and $\pi_k \circ \varphi_i^- = \pi_k$ so that by [theorem: 14.160]

$$\pi_k \circ \varphi_i^+ \text{ and } \pi_k \circ \varphi_i^- \text{ are continuous}$$

$k \in \{i+1, \dots, n+1\}$. Then by [eq: 23.12, 23.13] we have that $\pi_k \circ \varphi_i^+ = \pi_{k-1}$ and $\pi_k \circ \varphi_i^- = \pi_{k-1}$ so that by [theorem: 14.160]

$$\pi_k \circ \varphi_i^+ \text{ and } \pi_k \circ \varphi_i^- \text{ are continuous}$$

Hence $\forall k \in \{1, \dots, n+1\}$ we have that $\pi_k \circ \varphi_i^+$ and $\pi_k \circ \varphi_i^-$ are continuous, so that, using [theorem: 14.161], it follows that

$$\varphi_i^+ \text{ and } \varphi_i^- \text{ are continuous} \quad (23.17)$$

Combining [eqs: 23.10, 23.16, 23.17] it follows that

$$\forall i \in \{1, \dots, n+1\} \text{ } \psi_i^+ \text{ and } \psi_i^- \text{ are homeomorphisms} \quad (23.18)$$

Define now $\{(U_i, \beta_i)\}_{i \in \{1, \dots, 2 \cdot n+2\}}$ by [see eq: 23.8]

$$U_i = \begin{cases} S^n \cap V_i^+ & \text{if } i \in \{1, \dots, n+1\} \\ S^n \cap V_{i-(n+1)}^- & \text{if } i \in \{n+2, \dots, 2 \cdot n+2\} \end{cases} \stackrel{[\text{eq: 23.8}]}{\in} (\mathcal{T}_{\|\cdot\|_{n+1}})_{|S^n}$$

and

$$\beta_i = \begin{cases} \psi_i^+ & \text{if } i \in \{1, \dots, n+1\} \\ \psi_{i-(n+1)}^- & \text{if } i \in \{n+2, \dots, 2 \cdot n+2\} \end{cases} \text{ which is a homeomorphism}$$

Finally if $x \in S^n$ then we have that $1 = \sum_{i=1}^{n+1} x_i^2$ so that $\exists k \in \{1, \dots, n+1\}$ such that $x_k \neq 0$. For x_k we have either:

$$\mathbf{0} < \mathbf{x}_k. \text{ Then } x \in \{x \in \mathbb{R}^{n+1} \mid 0 < x_k\} = V_k^+ = U_k \subseteq \bigcup_{i \in \{1, \dots, 2 \cdot n+2\}} U_i$$

$$\mathbf{x}_k < \mathbf{0}. \text{ Then } x \in \{x \in \mathbb{R}^{n+1} \mid x_k < 0\} = V_k^- = U_{k+(n+1)} \subseteq \bigcup_{i \in \{1, \dots, 2 \cdot n+2\}} U_i$$

Hence we have that $S^n \subseteq \bigcup_{i \in \{1, \dots, 2 \cdot n + 2\}} U_i$. As $\forall i \in \{1, \dots, 2 \cdot n + 2\}$

$$U_i = \begin{cases} S^n \cap V_i^+ & \text{if } i \in \{1, \dots, n+1\} \\ S^n \cap V_{i-(n+1)}^- & \text{if } i \in \{n+2, \dots, 2 \cdot n + 2\} \end{cases} \subseteq S^n$$

we conclude that

$$S^n = \bigcup_{i \in \{1, \dots, 2 \cdot n + 2\}} U_i$$

Hence we have by definition that

$$\langle S^n, (\mathcal{T}_{\|\cdot\|_n})|_{S^n} \rangle \text{ is a } n\text{-dimensional topological manifold}$$

where

$$\{(U_i, \beta_i)\}_{i \in \{1, \dots, 2 \cdot n + 2\}} \text{ is an atlas on } S^n$$

□

23.1.2 Topological properties of a Topological Manifold

As a topological manifold is locally like $\langle \mathbb{R}^n, \mathcal{T}_{\mathbb{R}^n} \rangle$ it follows that some of the topological properties of $\mathbb{R}^n$ are translated in topological properties of the topological manifold. In proving these properties it will turn out that the requirements for a topological manifold to be Hausdorff and second countable are necessary. In this section we will cover topological properties like compactness, connectness and paracompactness. First remember that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$ using the norm $\|\cdot\|_e$ as the next lemma shows.

Lemma 23.19. *Let $n \in \mathbb{N}$ then $\forall x \in \mathbb{R}^n$ and $\forall \varepsilon \in \mathbb{R}^+$ there exist a $q \in \mathbb{Q}^n$ such that $\|x - q\|_e < \varepsilon$.*

Proof. Let $x \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}^+$. Let $i \in \{1, \dots, n\}$ then as $x_i - \frac{\varepsilon}{\sqrt{n}} < x_i + \frac{\varepsilon}{\sqrt{n}}$ there exist by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] a $q_i \in \mathbb{Q}$ such that $x_i - \frac{\varepsilon}{\sqrt{n}} < q_i < x_i + \frac{\varepsilon}{\sqrt{n}}$. Hence we have $x_i - q_i < \frac{\varepsilon}{\sqrt{n}} \wedge q_i - x_i < \frac{\varepsilon}{\sqrt{n}}$ so that

$$0 \leq |x_i - q_i| < \frac{\varepsilon}{\sqrt{n}}$$

Hence we have

$$|x_i - q_i|^2 \underset{0 \leq |x_i - q_i| < \frac{\varepsilon}{\sqrt{n}}}{\leq} \frac{\varepsilon}{\sqrt{n}} \cdot |x_i - q_i| \underset{0 < \frac{\varepsilon}{2 \cdot \sqrt{n}}}{<} \left(\frac{\varepsilon}{\sqrt{n}} \right)^2 = \frac{\varepsilon^2}{n}$$

so that for $q = (q_1, \dots, q_n) \in \mathbb{Q}^n$ we have

$$\begin{aligned} \sum_{i=1}^n (x_i - q_i)^2 &= \sum_{i=1}^n |x_i - q_i|^2 \\ &< \sum_{i=1}^n \frac{\varepsilon^2}{n} \\ &= \varepsilon^2 \end{aligned}$$

As $\sqrt{\cdot}$ is strictly increasing [see theorem: 10.71] it follows from the above that

$$\|x - q\|_e = \sqrt{\sum_{i=1}^n (x_i - q_i)^2} < \sqrt{\varepsilon^2} = \varepsilon$$

□

Lemma 23.20. *Let $n \in \mathbb{N}$ and $U \in \mathcal{T}_{\mathbb{R}^n}$ a open set then*

$$\mathcal{B} = \{B_{\|\cdot\|_e}(x, r) \mid (x, r) \in \mathbb{Q}^n \times \mathbb{Q} \text{ such that } 0 < r \wedge B_{\|\cdot\|_e}(x, 2 \cdot r) \subseteq U\}$$

is a countable basis of $(\mathcal{T}_{\mathbb{R}^n})|_U$.

Proof. Let $I = \{(q, r) \in \mathbb{Q}^n \times \mathbb{Q} \mid 0 < r \wedge B_{\|\cdot\|_e}(q, 2 \cdot r) \subseteq U\} \subseteq \mathbb{Q}^n \times \mathbb{Q}$ then by [theorems: 10.5, 6.81, 6.71] I is countable. So that by [corollary: 6.74]

$$\mathcal{B} = \{B_{\|\cdot\|_e}(q, r) \mid (q, r) \in \mathbb{Q}^n \times \mathbb{Q} \text{ such that } 0 < r \wedge B_{\|\cdot\|_e}(q, 2 \cdot r) \subseteq U\} \text{ is countable} \quad (23.19)$$

Using [theorem: 14.11] we have, as U is open, that

$$(\mathcal{T}_{\mathbb{R}^n})|_U \subseteq \mathcal{T}_{\mathbb{R}^n}$$

Let $V \in (\mathcal{T}_{\mathbb{R}^n})|_U$ then by the above it follows that $V \in \mathcal{T}_{\mathbb{R}^n} = \mathcal{T}_{|||e}$. So if we take $x \in V$ there exist a $\delta \in \mathbb{R}^+$ such that

$$x \in B_{|||e}(x, \delta) \subseteq V$$

As $0 < \delta$ there exist by the density of $\mathbb{Q}$ in $\mathbb{R}$ [see theorem: 10.32] a $r' \in \mathbb{Q}$ such that $0 < r' < \delta$ so that

$$x \in B_{|||e}(x, r') \subseteq B_{|||e}(x, \delta) \subseteq V \quad (23.20)$$

Using the density of $\mathbb{Q}^n$ in $\mathbb{R}^n$ [see lemma: 23.19] there exist a $q \in \mathbb{Q}^n$ such that $\|x - q\|_e < \frac{r'}{4}$. Consider now $B_{|||e}\left(q, \frac{r'}{4}\right)$ then we have, as $\|x - q\|_e < \frac{r'}{4}$, that

$$x \in B_{|||e}\left(q, \frac{r'}{4}\right) \quad (23.21)$$

Further if $y \in B_{|||e}\left(q, \frac{r'}{2}\right)$ then $\|y - q\|_e < \frac{r'}{2}$ so that

$$\|y - x\|_e = \|y - q + q - x\|_e \leq \|y - q\|_e + \|q - x\|_e < \frac{r'}{2} + \frac{r'}{4} < \frac{r'}{2} + \frac{r'}{2} < r'$$

which proves that $B_{|||e}\left(q, \frac{r'}{2}\right) \subseteq B_{|||e}(x, r')$. Summarized we have that

$$x \underset{[\text{eq: 23.21}]}{\in} B_{|||e}\left(q, \frac{r'}{4}\right) \subseteq B_{|||e}\left(q, \frac{r'}{2}\right) \subseteq B_{|||e}(x, r') \underset{[\text{eq: 23.20}]}{\subseteq} B_{|||e}(x, \delta) \subseteq V \quad (23.22)$$

Take $r = \frac{r'}{4}$ then $0 < r \in \mathbb{Q}$, $q \in \mathbb{Q}^n$ and $B_{|||e}(q, 2 \cdot r) = B_{|||e}\left(q, \frac{r'}{2}\right) \underset{[\text{eq: 23.22}]}{\subseteq} V \subseteq U$ so that

$$B_{|||e}(q, r) \in \mathcal{B}$$

Further we have also $x \in B_{|||e}\left(q, \frac{r'}{4}\right) = B_{|||e}(q, r) = B_{|||e}\left(q, \frac{r'}{4}\right) \underset{[\text{eq: 23.22}]}{\subseteq} V$ which as $V \in (\mathcal{T}_{|||e})|_U$ and $x \in V$ was choosen arbitrary proves by [theorem: 14.38] that

$$\mathcal{B} \text{ is a basis for } (\mathcal{T}_{\mathbb{R}^n})|_U \quad \square$$

Lemma 23.21. *Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and (U, φ) a coordinate chart on $\langle M, \mathcal{T} \rangle$ then there exist a countable basis $\mathcal{B} \subseteq \mathcal{T}|_U$ of $\mathcal{T}|_U$ such that $\forall B \in \mathcal{B}$ B is a pre-compact [using the topology $\mathcal{T}$] coordinate ball.*

Note 23.22. *Remember that by [definition: 14.265] $A \subseteq M$ is pre-compact if $\bar{A}$ is compact.*

Proof. As (U, φ) is a coordinate n -chart on $\langle M, \mathcal{T} \rangle$ we have that $U \in \mathcal{T}$, $\varphi(U) \in \mathcal{T}_{\mathbb{R}^n} = \mathcal{T}_{|||e}$ and $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism using the topologies $\mathcal{T}|_U$ and $(\mathcal{T}_{|||e})|_{\varphi(U)}$. Using the previous lemma [see lemma: 23.20] we have that

$$\mathcal{A} = \{B_{|||e}(x, r) | (x, r) \in \mathbb{Q}^n \times \mathbb{Q} \text{ such that } 0 < r \wedge B_{|||e}(x, 2 \cdot r) \subseteq U\}$$

is a countable basis for $(\mathcal{T}_{\mathbb{R}^n})|_{\varphi(U)} = (\mathcal{T}_{|||e})|_{\varphi(U)}$. Let $B \in \mathcal{A}$ then there exist a $q \in \mathbb{Q}^n$, $0 < r \in \mathbb{Q}$ such that $B = B_{|||e}(q, r)$. Further $B = B_{|||e}(q, r) \subseteq \overline{B_{|||e}(q, r)} \subseteq B_{|||e}(q, 2 \cdot r) \subseteq U$. Hence we have that

$$\bar{B} \subseteq \overline{B_{|||e}(q, r)} \underset{[\text{theorems: 14.26, 14.69}]}{=} \overline{B_{|||e}(q, r)} \subseteq B_{|||e}(q, 2 \cdot r) \subseteq U$$

As $\overline{B_{|||e}(q, r)}$ is compact using the topology $\mathcal{T}_{|||e}$ [see theorem: 14.327] it follows from [theorem: 14.277] and the fact that $\bar{B}$ is closed that $\bar{B}$ is compact in $\mathcal{T}_{|||e}$. Hence using [theorem: 14.272] it follows that

$$\bar{B} \text{ is compact using the topology } (\mathcal{T}_{|||e})|_{\varphi(U)}$$

As $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism $\varphi^{-1}: \varphi(U) \rightarrow U$ is continuous, hence it follows from [theorem: 14.275] that

$$\varphi^{-1}(\overline{B}) \stackrel{[\text{theorem: 2.70}]}{=} (\varphi^{-1}(\overline{B})) \text{ is compact using the topology } \mathcal{T}_U$$

Further using [theorem: 14.191] we have that

$$\varphi^{-1}(\overline{B}) = \overline{\varphi^{-1}(B)}$$

Hence we have by [definition: 14.265] that $\varphi^{-1}(B)$ is precompact using the topology $\mathcal{T}_U$. Using now [theorem: 14.278] it follows that

$$\varphi^{-1}(B) \text{ is precompact using the topology } \mathcal{T}$$

Using [theorems: 14.10, 14.193] we have that

$$\varphi|_{\varphi^{-1}(B)}: \varphi^{-1}(B) \rightarrow \varphi(\varphi^{-1}(B)) \stackrel{\varphi \text{ is bijective}}{=} B = B_{\|\cdot\|_e}(q, r)$$

is a homeomorphism using the topologies $\mathcal{T}_{|\varphi^{-1}(B)|}$, $(\mathcal{T}_{\|\cdot\|_e})|_B$ so that by [definition: 23.9]

$$\varphi^{-1}(B) \text{ is a coordinate } n\text{-ball}$$

Further as $\mathcal{A}$ is countable we have by [theorem: 6.74] that

$$\mathcal{B} = \{\varphi^{-1}(B) | B \in \mathcal{B}\} \text{ is countable}$$

To summarize we have that

$$\mathcal{B} = \{\varphi^{-1}(B) | B \in \mathcal{B}\} \text{ is a countable set of precompact [using the topology } \mathcal{T] \text{ coordinate balls}$$

It remains to prove that $\mathcal{B}$ is a basis for $\mathcal{T}_U$. So let $V \in \mathcal{T}_U$ be an open set then, as φ is a homeomorphism hence open, $\varphi(V) \in (\mathcal{T}_{\|\cdot\|_e})|_{\varphi(U)}$. As $\mathcal{A}$ is a basis for $(\mathcal{T}_{\|\cdot\|_e})|_{\varphi(U)}$ there exists a $\{B_i\}_{i \in I} \subseteq \mathcal{B}$ such that $\varphi(V) = \bigcup_{i \in I} B_i$. Hence we have

$$V \stackrel{[\text{theorem: 2.57}]}{=} \varphi^{-1}(\varphi(V)) = \varphi^{-1}\left(\bigcup_{i \in I} B_i\right) \stackrel{[\text{theorem: 2.144}]}{=} \bigcup_{i \in I} \varphi^{-1}(B_i)$$

which as $\{\varphi^{-1}(B_i)\}_{i \in I} \subseteq \mathcal{B}$ proves that

$$\mathcal{B} \text{ is a basis for } \mathcal{T}_U$$

□

Theorem 23.23. *Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold then $\mathcal{T}$ has a countable basis of precompact coordinate balls.*

Proof. As $\langle M, \mathcal{T} \rangle$ is a topological manifold $\mathcal{T}$ is a Hausdorff second countable topology and there exist $\forall x \in M$ a coordinate chart $\langle \varphi_x, U_x \rangle$ with $x \in U_x \in \mathcal{T}$. Hence we have that

$$M = \bigcup_{x \in M} U_x$$

As $\mathcal{T}$ is second countable it follows from [theorem: 14.262] and the above that there exist a countable $\mathcal{C} \subseteq M$ such that

$$M = \bigcup_{x \in \mathcal{C}} U_x$$

By the previous lemma [lemma: 23.21] we have $\forall x \in \mathcal{C}$ a countable basis $\mathcal{B}_x$ [in the topology $\mathcal{T}_{|U_x|}$] of precompact [using the topology $\mathcal{T}$] coordinate balls. Take

$$\mathcal{B} = \bigcup_{x \in \mathcal{C}} \mathcal{B}_x$$

Then using [theorem: 6.78] it follows that

$$\mathcal{B} \text{ is countable}$$

Further $\forall B \in \mathcal{B}$ we have that $\exists x \in \mathcal{C}$ such that $B \in \mathcal{B}_x$. As $\mathcal{B}_x$ is a basis for $\mathcal{T}_{|U_x|}$ we have $\mathcal{B}_x \subseteq \mathcal{T}_{|U_x|}$ which, as $\mathcal{T}_{|U_x|} \subseteq \mathcal{T}$ [using [theorem: 14.11] and $U_x \in \mathcal{T}$], proves that $B \in \mathcal{T}$. Hence we have

$$\mathcal{B} \subseteq \mathcal{T}$$

Finally let $V \in \mathcal{T}$ be an open set then if $y \in V$ there exist a $x \in \mathcal{C}$ such that $y \in U_x$. As $y \in V \cap U_x$ a open set in the subspace topology $\mathcal{T}|_{U_x}$ and $\mathcal{B}_x$ is a basis for $\mathcal{T}|_{U_x}$ there exist a $B \in \mathcal{B}_x \subseteq \bigcup_{x \in \mathcal{C}} \mathcal{B}_x$ such that $y \in B \subseteq V \cap U_x \subseteq V$. To summarize $\forall V \in \mathcal{T}, \forall y \in V$ there exist a $B \in \mathcal{B} \subseteq \mathcal{T}$ such that $y \in B \subseteq V$ which using [theorem: 14.38] proves that

$\mathcal{B}$ is a basis for $\mathcal{T}$

□

We use the above theorem to prove the connection properties of a topological manifold.

Theorem 23.24. *Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ be a **non empty** n -dimensional topological manifold then we have that*

1. $\langle M, \mathcal{T} \rangle$ is locally path connected [see definition: 14.481].
2. M is connected $\Leftrightarrow M$ is locally connected.
3. The components of M are the same as its path components. More specifically $\forall x \in X$ we have that $\sim[x] = \sim_p[x]$. See [definitions: 14.468, 14.478].
4. Every component of M is open.
5. If C is a component of M then $\langle C, \mathcal{T}|_C \rangle$ is a topological manifold.
6. The collection $X/\sim$ of components of X is countable.

Proof.

1. Using [theorem: 23.21] there exist a basis $\mathcal{B} \subseteq \mathcal{T}$ such that $\forall B \in \mathcal{B}$ B is a coordinate ball. Hence if $B \in \mathcal{B}$ we have by [definition: 23.9] that there exist a $\delta \in \mathbb{R}^+, x \in \mathbb{R}^n$ and a homeomorphism $\varphi: B \rightarrow B_{|||e}(x, \delta)$. As $B_{|||e}(x, y)$ is convex [see theorem: 16.221] it follows from [theorem: 14.474] that $B_{|||e}(x, \delta)$ is path connected. As $\varphi^{-1}: B_{|||e}(x, \delta) \rightarrow B$ is continuous it follows from [theorem: 14.475] that B is path connected. So we have proved that $\mathcal{T}$ has a basis of path connected sets wch by [definition: 14.481] proves that

$\langle M, \mathcal{T} \rangle$ is locally path connected

2. As M is non empty and by (1) locally path connected this follows from [theorem: 14.482(4)].
3. As M is locally path connected this follows from [theorem: 14.482(2)].
4. As M is locally path connected this follows from [theorem: 14.482(3)].
5. If C is a component then by (4) C is open so that by [example: 23.16] $\langle C, \mathcal{T}|_C \rangle$ is a topological manifold.
6. Using [definition: 14.468]

$$M = \bigsqcup_{C \in M/\sim} C \text{ [union of pairwise disjoint set]}$$

where $M/\sim$ is the set of components. As by (4) $\forall C \in M/\sim$ C is open and $\langle M, \mathcal{T} \rangle$ is second countable [see definition: 23.2] it follows from [theorem: 14.262] that there exist a countable $\mathcal{A} \subseteq M/\sim$ such that

$$M = \bigsqcup_{B \in \mathcal{A}} B$$

Let $C \in M/\sim$ then by [definition: 14.468] there exist a $x \in M$ such that $C = \sim[x]$ from which it follows that $x \in C$. Assume that $C \notin \mathcal{A}$ then $\forall B \in \mathcal{A} \subseteq M/\sim$ we have that $B \neq C$ which as $M/\sim$ is pairwise disjoint means that $B \cap C = \emptyset$ so that $x \notin B$. Hence $x \notin \bigcup_{B \in \mathcal{A}} B = M$ contradicting $x \in M$. So the assumption $C \notin \mathcal{A}$ is false and we must have that $C \in \mathcal{A}$. Hence we have $M/\sim \subseteq \mathcal{A} \subseteq M/\sim$ so that $M/\sim = \mathcal{A}$ which as $\mathcal{A}$ is countable proves that

$M/\sim$ is countable

□

Theorem 23.25. *Every topological manifold $\langle M, \mathcal{T} \rangle$ is locally compact [see definition: 14.289]*

Proof. As $\mathcal{T}$ is Hausdorff and by [theorem: 23.21] $\langle M, \mathcal{T} \rangle$ has pre-compact base, it follows from [theorem: 14.290] that $\langle M, \mathcal{T} \rangle$ is locally compact. □

Next we define the concept of a paracompact spaces.

Definition 23.26. (refinement) Let $\mathcal{U}$ be a collection of sets then a collection of sets $\mathcal{V}$ is a **refinement** of $\mathcal{U}$ if $\forall V \in \mathcal{V}$ there exist a $U \in \mathcal{U}$ such that $V \subseteq U$.

Definition 23.27. (locally finite) Let $\langle X, \mathcal{T} \rangle$ be a topological space then a collection $\mathcal{U} \subseteq \mathcal{P}(X)$ of subsets of X is called **locally finite** if $\forall x \in X$ there exist a $V \in \mathcal{T}$ with $x \in V$ such that $\{U \in \mathcal{U} | U \cap V \neq \emptyset\}$ is finite.

Definition 23.28. (Paracompact Spaces) A topological space $\langle X, \mathcal{T} \rangle$ is paracompact if $\forall \mathcal{U} \subseteq \mathcal{T}$ with $X = \bigcup_{U \in \mathcal{U}} U$ there exist a locally finite refinement $\mathcal{V} \subseteq \mathcal{T}$ of $\mathcal{U}$ such that $X = \bigcup_{V \in \mathcal{V}} V$. In other words every open cover of X admits a open locally finite covering refinement.

Theorem 23.29. (Topological Manifolds are Paracompact) Let $\langle M, \mathcal{T} \rangle$ be a topological manifold, let $\mathcal{U} \subseteq \mathcal{T}$ be such that $M = \bigcup_{U \in \mathcal{U}} U$ and $\mathcal{B}$ a basis for $\mathcal{T}$ then there exist a countable locally finite open refinement $\mathcal{V}$ of $\mathcal{U}$ with $\mathcal{U} \subseteq \mathcal{B}$.

Note 23.30. As by [example: 14.35] $\mathcal{T}$ is a basis of itself it follows from the previous definition [see definition: 23.28] that $\langle M, \mathcal{T} \rangle$ is paracompact.

Proof. As $\langle M, \mathcal{T} \rangle$ is second countable, Hausdorff and locally compact [see theorem: 23.25] it follows from [theorem: 14.292] that there exist a exhausting of X by compact sets $\{C_i\}_{i \in \mathbb{N}}$. So using the definition of exhausting [see definition: 14.291] we have that

$$M = \bigcup_{i \in \mathbb{N}} C_i \text{ and } \forall i \in \mathbb{N} \ C_i \text{ is compact and } C_i \subseteq (C_{i+1})^\circ \quad (23.23)$$

As $\forall i \in \mathbb{N} \ C_i \subseteq (C_{i+1})^\circ \subseteq C_{i+1}$ it follows by [theorem: 14.14] $(C_i)^\circ \subseteq (C_{i+1})^0$. Hence we have that

$$\forall i, j \in \mathbb{N} \text{ with } i \leq j \text{ we have } C_i \subseteq C_j \text{ and } (C_i)^\circ \subseteq (C_j)^\circ \quad (23.24)$$

Define now

$$\{V_i\}_{i \in \mathbb{N}_0} \text{ by } V_i = \begin{cases} C_1 & \text{if } i = 0 \\ C_{i+1} \setminus (C_i)^\circ & \end{cases}$$

Let $i \in \mathbb{N}_0$ then we have either:

$i = 0$. Then $V_i = C_1$ a compact set.

$i \in \mathbb{N}$. Then as $\langle M, \mathcal{T} \rangle$ is Hausdorff and C_i is compact it follows from [theorem: 14.276] that C_i is closed. As $(C_i)^\circ$ is open it follows from [theorem: 14.22] that $C_{i+1} \setminus (C_i)^\circ$ is closed, hence by [theorem: 14.277] $V_i = C_{i+1} \setminus (C_i)^\circ$ is compact.

Hence

$$\forall i \in \mathbb{N}_0 \ V_i \text{ is compact} \quad (23.25)$$

Let $x \in M$ then, as $M = \bigcup_{i \in \mathbb{N}} C_i$, there exist a $n \in \mathbb{N}$ such that $x \in C_n$. Hence $\{i \in \{1, \dots, n\} | x \in C_i\} \neq \emptyset$ so that by [theorem: 6.49] $m = \min(\{i \in \{1, \dots, n\} | x \in C_i\})$ exist. For $m \in \{1, \dots, n\}$ we have either:

$m = 1$. Then $x \in C_1 = V_0 \subseteq \bigcup_{i \in \mathbb{N}_0} V_i$

$i \in \{2, \dots, n\}$. Then $x \notin C_{m-1} \Rightarrow x \notin (C_{m-1})^\circ$ and $x \in C_m$ so that

$$x \in C_m \setminus (C_{m-1})^\circ = V_{m-1} \subseteq \bigcup_{i \in \mathbb{N}_0} V_i$$

so in all cases we have $x \in \bigcup_{i \in \mathbb{N}_0} V_i$. Hence, as $\forall i \in \mathbb{N}_0, V_i \subseteq M$ it follows that

$$M = \bigcup_{i \in \mathbb{N}_0} V_i \quad (23.26)$$

Define now

$$\{W_i\}_{i \in \mathbb{N}_0} \text{ by } W_i = \begin{cases} (C_2)^\circ & \text{if } i = 0 \\ (C_3)^\circ & \text{if } i = 1 \\ (C_{i+2})^\circ \setminus C_{i-1} & \text{if } i \in \{2, \dots, \infty\} \end{cases}$$

Let $i \in \mathbb{N}_0$ then we have either:

$i = 0$. Then $W_i = (C_2)^\circ \in \mathcal{T}$ and $V_i = C_1 \underset{[\text{eq: 23.23}]}{\subseteq} (C_2)^\circ = W_i$.

$i = 1$. Then $W_i = (C_3)^\circ \in \mathcal{T}$ and $V_i = V_1 = C_2 \setminus (C_1)^\circ \subseteq C_2 \underset{[\text{eq: 23.23}]}{\subseteq} (C_3)^\circ = W_i$

$i \in \{2, \dots, \infty\}$. As $\mathcal{T}$ is Hausdorff and C_{i-1} is compact it follows from [theorem: 14.276] that C_{i-1} is closed so that by [theorem: 14.22] $W_i = (C_{i+2})^\circ \setminus C_{i-1} \in \mathcal{T}$. Further we have

$$V_i = C_{i+1} \setminus (C_i)^\circ \underset{[\text{eq: 23.23}]}{\subseteq} C_{i+1} \setminus C_{i-1} \underset{[\text{eq: 23.23}]}{\subseteq} (C_{i+2})^\circ \setminus C_{i-1} = W_i$$

To summarize the above we have

$$\forall i \in \mathbb{N}_0 \ W_i \in \mathcal{T} \text{ and } V_i \subseteq W_i \quad (23.27)$$

Let $i, j \in \mathbb{N}_0$ and consider the following two special cases for $j \in \mathbb{N}_0$

$0 \leq j < i - 2$. Now for j we have either:

$j = 0$. Then $2 < i \Rightarrow 2 \leq i - 1$ so that $W_j = W_0 = (C_2)^\circ \subseteq C_2 \underset{[\text{eq: 23.24}]}{\subseteq} C_{i-1}$ and $W_i = (C_{i+2})^\circ \setminus C_{i-1}$ so that $W_j \cap W_i = \emptyset$.

$j = 1$. Then $3 < i \Rightarrow 3 \leq i - 1$ so that $W_j = W_1 = (C_3)^\circ \subseteq C_3 \underset{[\text{eq: 23.24}]}{\subseteq} C_{i-1}$ and $W_i = (C_{i+2})^\circ \setminus C_{i-1}$ so that $W_j \cap W_i = \emptyset$.

$j \in \{2, \dots, \infty\}$. Then $j < i - 2 \Rightarrow j + 2 < i - 1$ so that

$$W_j = (C_{j+2})^\circ \setminus C_{j-1} \subseteq (C_{j+2})^\circ \subseteq C_{j+2} \underset{[\text{eq: 23.24}]}{\subseteq} C_{i-1}$$

and $W_i = (C_{i+2})^\circ \setminus C_{i-1}$ from which it follows that $W_j \cap W_i = \emptyset$.

$i + 2 < j$. Then for i we have either:

$i = 0$. Then $2 < j \Rightarrow 2 \leq j - 1$ so that $W_i = W_0 = (C_2)^\circ \subseteq C_2 \underset{[\text{eq: 23.24}]}{\subseteq} C_{j-1}$ and $W_j = (C_{j+2})^\circ \setminus C_{j-1}$ so that $W_j \cap W_i = \emptyset$.

$i = 1$. Then $3 < j \Rightarrow 3 \leq j - 1$ so that $W_i = W_1 = (C_3)^\circ \subseteq C_3 \underset{[\text{eq: 23.24}]}{\subseteq} C_{j-1}$ and $W_j = (C_{j+2})^\circ \setminus C_{j-1}$ so that $W_j \cap W_i = \emptyset$.

$i \in \{2, \dots, \infty\}$. Then $i < j - 2 \Rightarrow i + 2 \leq j - 1$ so that

$$W_i = (C_{i+2})^\circ \setminus C_{i-1} \subseteq (C_{i+2})^\circ \subseteq C_{i+2} \underset{[\text{eq: 23.24}]}{\subseteq} C_{j-1}$$

and $W_j = (C_{j+2})^\circ \setminus C_{j-1}$ so that $W_j \cap W_i = \emptyset$.

Hence we have that

$$\forall i, j \in \mathbb{N}_0 \text{ we have } W_i \cap W_j = \emptyset \text{ except maybe for } j \in \{i - 2, \dots, i + 2\} \text{ a finite set} \quad (23.28)$$

Let $i \in \mathbb{N}_0$ and $x \in V_i \underset{[\text{eq: 23.27}]}{\subseteq} W_i \in \mathcal{T}$ then, as by the hypothesis, $M = \bigcup_{U \in \mathcal{U}} U$ there exist a $U_x \in \mathcal{U} \subseteq \mathcal{T}$ such that $x \in U_x$, hence $x \in U_x \cap W_i \in \mathcal{T}$. As $\mathcal{B}$ is a basis for $\mathcal{T}$ there exist a $B_{x,i} \in \mathcal{B}$ such that $x \in B_{x,i} \subseteq U_x \cap W_i$. Define

$$\mathcal{B}_i = \{B_{x,i} | x \in V_i\} \subseteq \mathcal{B} \text{ where } x \in B_{x,i} \subseteq U_x \cap W_i \quad (23.29)$$

Let $x \in V_i$ then as $x \in B_{x,i} \in \mathcal{B}_i$ it follows that $V_i \subseteq \bigcup_{B \in \mathcal{B}_i} B$ which as V_i is compact [see eq: 23.25] means that there exist a finite $\mathcal{V}_i \subseteq \mathcal{B}_i$ such that

$$V_i \subseteq \bigcup_{B \in \mathcal{V}_i} B \text{ where } \mathcal{V}_i \subseteq \mathcal{B}_i \subseteq \mathcal{B} \quad (23.30)$$

By [theorem: 6.78] it follows that

$$\mathcal{V} = \bigcup_{i \in \mathbb{N}_0} \mathcal{V}_i \text{ is countable and } \mathcal{V} \subseteq \mathcal{B} \quad (23.31)$$

Let $x \in M$ then by [eq: 23.26] there exist a $i \in \mathbb{N}_0$ such that $x \in V_i$, hence by [eq: 23.30] there exist a $B \in \mathcal{V}_i \subseteq \bigcup_{i \in \mathbb{N}_0} \mathcal{V}_i = \mathcal{V}$ such that $x \in B$. Hence $x \in \bigcup_{B \in \mathcal{V}} B$ which proves that $M \subseteq \bigcup_{B \in \mathcal{V}} B$. As $\mathcal{V} \subseteq \mathcal{B} \subseteq \mathcal{T} \subseteq \mathcal{P}(M)$ we have also $\bigcup_{B \in \mathcal{V}} B \subseteq M$. So we have

$$M = \bigcup_{B \in \mathcal{V}} B \quad (23.32)$$

Let $B \in \mathcal{V}$ then there exist a $i \in \mathbb{N}_0$ such that $B \in \mathcal{V}_i \subseteq \mathcal{B}_i$ hence there exist by [eq: 23.29] a $x \in V_i$ such that

$$B = B_{x,i} \subseteq U_x \bigcap W_i \subseteq U_x \in \mathcal{U}$$

which by [definition: 23.26] proves that

$$\mathcal{V} \text{ is a refinement of } \mathcal{U} \quad (23.33)$$

Let $x \in M$ then by [eq: 23.32] there exist a $B \in \mathcal{V}$ such that $x \in B$. As $B \in \mathcal{V}$ there exist by [eq: 23.31] a $i \in \mathbb{N}_0$ so that $B \in \mathcal{V}_i \subseteq \mathcal{B}_i$. By [eq: 23.29] it follows then that $x \in B = B_{x,i} \subseteq U_x \bigcap W_i \subseteq W_i$. Hence we have that

$$x \in W_i \in \mathcal{T}$$

Let $B \in \{B \in \mathcal{V} | B \bigcap W_i \neq \emptyset\}$ then by [eqs: 23.30, 23.31] there exist a $j \in \mathbb{N}_0$ such that $B \in \mathcal{V}_j \subseteq \mathcal{B}_j$. Hence by [eq: 23.29] there exist a $y \in V_j$ such that $B = B_{y,j} \subseteq U_y \bigcap W_j \subseteq W_j$. As $B \bigcap W_i \neq \emptyset$ we have that $W_i \bigcap W_j \neq \emptyset$. Using [eq: 23.28] it follows then that $B \in \{B_j | j \in \{i-2, \dots, i+2\}\}$. Hence $\{B \in \mathcal{V} | B \bigcap W_i \neq \emptyset\} \subseteq \{B_j | j \in \{i-2, \dots, i+2\}\}$ which is finite so that

$$\{B \in \mathcal{V} | B \bigcap W_i \neq \emptyset\} \text{ is finite}$$

Using [definition: 23.27] and the fact that $x \in M$ has been chosen arbitrary it follows that

$$\mathcal{V} \text{ is locally finite.} \quad (23.34)$$

To summarize [eqs: 23.31, 23.33 and 23.34] we have constructed a countable locally finite refinement $\mathcal{V}$ of $\mathcal{U}$ where $\mathcal{V}$ covers M and $\mathcal{V} \subseteq \mathcal{B}$. $\square$

23.1.3 Differential structure on a topological manifold

As we have to deal with homeomorphisms between subsets the following theorem and convention will be usefull.

Theorem 23.31. Let $\langle X, \mathcal{T}_X \rangle$ and $\langle Y, \mathcal{T}_Y \rangle$ be topologies and $B \subseteq A \subseteq X$, $C \subseteq Y$ and

$$f: A \rightarrow C \text{ a homeomorphism using the topologies } (\mathcal{T}_X)|_A \text{ and } (\mathcal{T}_Y)|_C$$

then

$$f|_B: B \rightarrow f(B) \text{ is a homeomorphism using the topologies } (\mathcal{T}_X)|_B \text{ and } (\mathcal{T}_Y)|_{f(B)}$$

Proof. Using [theorem: 14.193] it follows that

$$f|_B: B \rightarrow f(B) \text{ is a homeomorphism using the topologies } ((\mathcal{T}_X)|_A)|_B \text{ and } ((\mathcal{T}_Y)|_C)|_{f(B)}$$

As $B \subseteq A$ we have $f(B) \subseteq f(A) = C$ hence using [theorem: 14.10] it follows that $((\mathcal{T}_X)|_A)|_B = (\mathcal{T}_X)|_B$ and $((\mathcal{T}_Y)|_C)|_{f(B)} = (\mathcal{T}_Y)|_{f(B)}$ from which it follows that

$$f|_B: B \rightarrow f(B) \text{ is a homeomorphism using the topologies } (\mathcal{T}_X)|_B \text{ and } (\mathcal{T}_Y)|_{f(B)} \quad \square$$

The above theorem allows us to use the following convention that greatly simplify notation.

Convention 23.32. Let $\langle X, \mathcal{T}_X \rangle$ and $\langle Y, \mathcal{T}_Y \rangle$ be topologies and $U \subseteq X$ and $V \subseteq Y$ then to avoid to say each time that $f: U \rightarrow V$ is a homeomorphism using the topologies $(\mathcal{T}_X)|_U$ and $(\mathcal{T}_Y)|_V$ we simplify the notation by just saying that $f: U \rightarrow V$ is a homeomorphism. So saying that

$$f: U \rightarrow V \text{ is a homeomorphism}$$

actually means that

$$f: U \rightarrow V \text{ is a homeomorphism using the topologies } (\mathcal{T}_X)|_U \text{ and } (\mathcal{T}_Y)|_V$$

Likewise saying that

$$U \text{ is homeomorphic with } V$$

actually means that

$$U \text{ is homeomorphic with } V \text{ using the topologies } (\mathcal{T}_X)|_U \text{ and } (\mathcal{T}_Y)|_V$$

We sketch now how the fact that a topological manifold is locally homeomorphic with a open set of $\mathbb{R}^n$ can be used to define differentiability for functions whose domain is in the manifold. Let $\langle M, \mathcal{T} \rangle$ be a n -dimensional topological manifold, $f: M \rightarrow \mathbb{R}$ a function and $m \in M$ then there exist a coordinate chart (U, φ) such that $m \in U$. As $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism and $\varphi(U)$ is open we can consider the function $f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R}$ and see if this function is of class C^m . If so we can say that f is of class C^m at m . However if we use this as a definition then C^m differentiability at m is strongly dependent on the choice of the coordinate chart, after all there could be another chart (V, ψ) with $x \in V$ so that $f \circ \psi^{-1}: \psi(U) \rightarrow \mathbb{R}$ is not of class C^m . To solve this we must require that $f \circ \psi^{-1}: \psi(U) \rightarrow \mathbb{R}$ is also of class C^m . By [corollary: 16.156] it followss that $(f \circ \varphi^{-1})|_{\varphi(U \cap V)}$ is of class C^m on $\varphi(U \cap V)$ and that $(f \circ \psi^{-1})|_{\psi(U \cap V)}$ must be of class C^m on $\psi(U \cap V)$. If we require that

$$\varphi \circ \psi^{-1}: \psi(U \cap V) \rightarrow \varphi(U \cap V) \text{ is of class } C^m$$

then the chain rule ensures that $(f \circ \varphi^{-1})|_{\varphi(U \cap V)} \circ (\varphi \circ \psi^{-1})|_{\psi(U \cap V)}$ is of class C^m . As

$$(f \circ \psi^{-1})|_{\psi(U \cap V)} = (f \circ \varphi^{-1})|_{\varphi(U \cap V)} \circ (\varphi \circ \psi^{-1})|_{\psi(U \cap V)}$$

the fact that $(f \circ \varphi^{-1})|_{\varphi(U \cap V)}$ is of class C^m automatically ensures that $(f \circ \psi^{-1})|_{\psi(U \cap V)}$ is of class C^m . The above discussion motivates the concept of C^m -compatible charts and C^m atlases.

Definition 23.33. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and (φ, U) , (ψ, V) coordinate charts such that $U \cap V \neq \emptyset$ then the function $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is called a **transition map from φ to ψ** or a **coordinate transformation**.

Note 23.34. As by [definition: 23.3] $\varphi: U \rightarrow \mathbb{R}^n$ and $\psi: V \rightarrow \mathbb{R}^n$ are injective functions [being homeomorphisms] it follows from [theorem: 2.101] that $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is a well defined bijection.

Theorem 23.35. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and (φ, U) , (ψ, V) coordinate charts such that $U \cap V \neq \emptyset$ then $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is a homeomorphism.

Proof. As by [definition: 23.3] $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism it follows from [theorem: 14.190] that $\varphi^{-1}: \varphi(U) \rightarrow U$ that is a homeomorphism. Using then [theorem: 23.31] it follows that

$$(\varphi^{-1})|_{\varphi(U \cap V)}: \varphi(U \cap V) \rightarrow \varphi^{-1}(\varphi(U \cap V)) \stackrel{[\text{theorem: 2.57}]}{=} U \cap V \text{ is a homeomorphism}$$

As $\psi: V \rightarrow \psi(V)$ is a homeomorphism it follows from [theorem: 23.31] that

$$\psi|_{U \cap V}: U \cap V \rightarrow \psi(U \cap V) \text{ is a homeomorphism}$$

Using [theorem: 14.192] it follows that

$$\psi|_{U \cap V} \circ (\varphi^{-1})|_{\varphi(U \cap V)}: \varphi(U \cap V) \rightarrow \psi(U \cap V) \text{ is a homeomorphism}$$

As further by [theorem: 2.101] $\psi \circ \varphi^{-1} = \psi|_{U \cap V} \circ (\varphi^{-1})|_{\varphi(U \cap V)}$ it follows that

$$\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V) \text{ is a homeomorphism} \quad \square$$

Definition 23.36. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a topological manifold then two coordinate charts (U, φ) and (V, ψ) are **C^m compatible** if $U \cap V = \emptyset$ or $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is a diffeomorphism of class C^m [see definition: 16.331].

Note 23.37. As for (U, φ) we have that $\varphi(U \cap U) = \varphi(U)$ and $\varphi \circ \varphi^{-1} = \text{Id}_{\varphi(U)}$ it follows from [example: 16.335] that $\varphi \circ \varphi^{-1}: \varphi(U) \rightarrow \varphi(U)$ is differentiable of class C^∞ . Hence (U, φ) is C^m compatible with (U, φ) proving that the relation **C^m compatible** is **reflexive**. Using [theorem: 16.332] and [theorem: 2.77] it follows that

$$\begin{aligned} \psi \circ \varphi^{-1}: \varphi(U \cap V) &\rightarrow \psi(U \cap V) \text{ is a diffeomorphism of class } C^m \\ \Downarrow \\ \varphi \circ \psi^{-1}: \psi(U \cap V) &\rightarrow \varphi(U \cap V) \text{ is a diffeomorphism of class } C^m \end{aligned}$$

so that the relation **C^m compatible** is **symmetric**. Be aware however that this relation is not transitive.

The relation **C^m compatible** is invariant under a restriction.
TODO check this again

Theorem 23.38. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a topological manifold, (U, φ) , (V, ψ) two charts that are C^m compatible and $W \in \mathcal{T}$ with $W \subseteq U$ then $(W, \varphi|_W)$ is a chart that is C^m compatible with (V, ψ)

Proof. Using [theorem: 23.7] it follows that

$$(W, \varphi|_W) \text{ is a coordinate chart on } \langle M, \mathcal{T} \rangle$$

Now for $W \cap V$ we have either:

$W \cap V = \emptyset$. Then by definition $(W, \varphi|_W)$ is C^m compatible with (V, ψ)

$W \cap V \neq \emptyset$. Then as $W \subseteq U$ it follows that $U \cap V \neq \emptyset$ hence

$$\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V) \text{ is a diffeomorphism of class } C^m$$

As $(W, \varphi|_W)$ is a coordinate chart on $\langle M, \mathcal{T} \rangle$ and $V \in \mathcal{T}$ it follows from [lemma: 23.5] that $\varphi|_W(V \cap W) \in \mathcal{T}_{\mathbb{R}^n}$. Further using [theorem: 2.92(3)] we have, as $V \cap W \subseteq W$ that $\varphi|_W(V \cap W) = \varphi(V \cap W)$. Hence we have that

$$\varphi(V \cap W) = \varphi|_W(V \cap W) \in \mathcal{T}_{\mathbb{R}^n}$$

So using [theorem: 16.336] it follows that

$$(\psi \circ \varphi^{-1})|_{\varphi(V \cap W)} \text{ is a diffeomorphism of class } C^m \quad (23.35)$$

Further using the homeomorphisms defined by the coordinate charts $(W, \varphi|_W)$ and (V, ψ)

$$\psi: V \rightarrow \psi(V) \text{ and } \varphi|_W: W \rightarrow \varphi|_W(W)$$

we have the coordinate transformation

$$\psi \circ (\varphi|_W)^{-1}: \varphi(V \cap W) \rightarrow \psi(V \cap W)$$

Then $\forall x \in \varphi(V \cap W)$ we have that

$$\begin{aligned} (\psi \circ (\varphi|_W)^{-1})(x) &= \psi((\varphi|_W)^{-1}(x)) \\ &= \psi(\varphi^{-1}(x)) \\ &= (\psi \circ \varphi^{-1})(x) \\ &\stackrel{x \in \varphi(V \cap W)}{=} (\psi \circ \varphi^{-1})|_{\varphi(V \cap W)}(x) \end{aligned}$$

which proves that

$$\psi \circ (\varphi|_W)^{-1} = (\psi \circ \varphi^{-1})|_{\varphi(V \cap W)}$$

Using then [eq: 23.35] we conclude that

$$\psi \circ (\varphi|_W)^{-1}: \varphi(V \cap W) \rightarrow \psi(V \cap W) \text{ is a diffeomorphism of class } C^m$$

from which it follows that $(W, \varphi|_W)$ and (V, ψ) are C^m compatible. $\square$

Definition 23.39. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold then

$$\mathcal{A} \subseteq \{(U, \varphi) | U \in \mathcal{T} \wedge \varphi \in (\mathbb{R}^n)^U\}$$

is a C^m -atlas on $\langle M, \mathcal{T} \rangle$ iff

1. $\forall (U, \varphi) \in \mathcal{A}$ (U, φ) is a coordinate chart on $\langle M, \mathcal{T} \rangle$.
2. $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U$.
3. $\forall (U, \varphi), (V, \psi) \in \mathcal{A}$ we have that (U, φ) and (V, ψ) are C^m compatible.

The set of all C^m -atlases on $\langle M, \mathcal{T} \rangle$ is noted as $\Delta^m[M, \mathcal{T}]$ in other words

$$\Delta^m[M, \mathcal{T}] = \{\mathcal{A} \subseteq \{(U, \varphi) | U \in \mathcal{T} \wedge \varphi \in (\mathbb{R}^n)^U\} | \mathcal{A} \text{ is a } C^m\text{-atlas}\}$$

Definition 23.40. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, $\mathcal{A}$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ then a coordinate chart (U, φ) on $\langle M, \mathcal{T} \rangle$ is C^r compatible with $\mathcal{A}$ iff

$$\forall (V, \psi) \in \mathcal{A} \text{ } (V, \psi) \text{ is } C^m \text{ compatible with } (U, \varphi)$$

The following lemma shows that we can extend a given C^m atlas $\mathcal{A}$ with a coordinate chart on $\langle M, \mathcal{T} \rangle$ that is C^m compatible with $\mathcal{A}$.

Lemma 23.41. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, $\mathcal{A} \in \Delta^m[M, \mathcal{T}]$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ and (U, φ) a coordinate chart that is C^m compatible with $\mathcal{A}$ then we have

$$\mathcal{A} \cup \{(U, \varphi)\} \text{ is a } C^m \text{ atlas on } \langle M, \mathcal{T} \rangle$$

Proof. Let $(V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}$ then we have either:

$(V, \psi) = (U, \varphi)$. Then as (U, φ) is a coordinate chart (V, ψ) is a coordinate chart.

$(V, \psi) \in \mathcal{A}$. Then as $\mathcal{A}$ is a C^m atlas on $\langle M, \mathcal{T} \rangle$ (V, ψ) is a coordinate chart

so that

$$\forall (V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\} \text{ } (V, \psi) \text{ is a coordinate chart} \quad (23.36)$$

As $\mathcal{A}$ is a C^m atlas we have that $M = \bigcup_{(V, \psi) \in \mathcal{A}} V \subseteq \bigcup_{(V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}} V \stackrel{\mathcal{T} \subseteq \mathcal{P}(M)}{\subseteq} M$ so that

$$M = \bigcup_{(V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}} V \quad (23.37)$$

Let $(V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}$, $(W, \gamma) \in \mathcal{A} \cup \{(U, \varphi)\}$ then we have either:

$(V, \psi) = (U, \varphi) \wedge (W, \gamma) = (U, \varphi)$. Then by reflectivity [see note: 23.37] it follows that $(V, \psi) = (U, \varphi)$ and $(W, \gamma) = (U, \varphi)$ are C^m compatible.

$(V, \psi) \neq (U, \varphi) \wedge (W, \gamma) = (U, \varphi)$. Then $(V, \psi) \in \mathcal{A}$ so that by the hypotheses $(W, \gamma) = (U, \varphi)$ and (V, ψ) are C^m compatible so that by symmetry [see note: 23.37] (V, ψ) and (W, γ) are C^m compatible.

$(V, \psi) = (U, \varphi) \wedge (W, \gamma) \neq (U, \varphi)$. Then $(W, \gamma) \in \mathcal{A}$ so that by the hypothesis $(V, \psi) = (U, \varphi)$ and (W, γ) are C^m compatible.

$(V, \psi) \neq (U, \varphi) \wedge (W, \gamma) \neq (U, \varphi)$. Then $(V, \psi) \in \mathcal{A}$ and $(W, \gamma) \in \mathcal{A}$ which, as $\mathcal{A}$ is a C^m atlas, proves that (V, ψ) and (W, γ) are C^m compatible.

To summarize we have that

$$\forall (V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}, \forall (W, \gamma) \in \mathcal{A} \cup \{(U, \varphi)\} \text{ } (V, \psi) \text{ and } (W, \gamma) \text{ are } C^m \text{ compatible} \quad (23.38)$$

Using then [definition: 23.39] together with [eq: 23.36, 23.37 and 23.38] proves that

$$\mathcal{A} \cup \{(U, \varphi)\} \text{ is a } C^m \text{ differentiable atlas on } \langle M, \mathcal{T} \rangle \quad \square$$

We are now ready to define differentiability for mappings from a manifold to the space of real numbers.

Definition 23.42. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimension topological manifold and $\mathcal{A}$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ then $f: M \rightarrow \mathbb{R}$ is of class C^m using $\mathcal{A}$ iff $\forall p \in M$ there $\exists (U, \varphi) \in \mathcal{A}$ such that

$$p \in U \text{ and } f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R} \text{ is of class } C^m$$

Lemma 23.43. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, $\mathcal{A}$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ and $f: M \rightarrow \mathbb{R}$ a real valued function that is of class C^m using $\mathcal{A}$ then for every coordinate chart (V, ψ) on $\langle M, \mathcal{T} \rangle$ that is C^m compatible with $\mathcal{A}$ we have that

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m$$

Proof. Let (V, ψ) be a coordinate chart on $\langle M, \mathcal{T} \rangle$ that is C^m compatible with $\mathcal{A}$ and consider the function

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R}$$

Let $x \in \psi(V)$ then as ψ is bijective there exist a $p \in V$ such that $x = \psi(p)$. As $f: M \rightarrow \mathbb{R}$ is of class C^m using $\mathcal{A}$ there exist a $(U, \varphi) \in \mathcal{A}$ with $p \in U$ such that

$$f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R} \text{ is of class } C^m \quad (23.39)$$

As $p \in U$ and $p \in V$ we have that $p \in U \cap V$ so that

$$x = \psi(p) \in \psi(U \cap V) \quad (23.40)$$

As $V \in \mathcal{T}$ we have, as $\varphi: U \rightarrow \varphi(U)$ is a homeomorphism, by [lemma: 23.5] that

$$\varphi(U \cap V) \in \mathcal{T}_{\mathbb{R}^n} \quad (23.41)$$

Hence using [theorem: 16.156] it follows from [eqs: 23.39, 23.41] that

$$(f \circ \varphi^{-1})|_{\varphi(U \cap V)}: \varphi(U \cap V) \rightarrow \mathbb{R} \text{ is of class } C^m$$

Further as (U, φ) and (V, ψ) are C^m compatible [because (V, ψ) is C^m compatible with $\mathcal{A}$ and $(U, \varphi) \in \mathcal{A}$] we have that $\varphi \circ \psi^{-1}: \psi(U \cap V) \rightarrow \varphi(U \cap V)$ is a diffeomorphism of class C^m hence

$$\varphi \circ \psi^{-1}: \psi(U \cap V) \rightarrow \mathbb{R} \text{ is of class } C^m \text{ with } (\varphi \circ \psi^{-1})(\psi(U \cap V)) = \varphi(U \cap V)$$

Using now the chain rule [see theorem: 16.197] it follows that

$$(f \circ \varphi^{-1})|_{\varphi(U \cap V)} \circ (\varphi \circ \psi^{-1}): \psi(U \cap V) \rightarrow \mathbb{R} \text{ is of class } C^m \quad (23.42)$$

As $\forall y \in \psi(U \cap V)$ we have

$$\begin{aligned} ((f \circ \varphi^{-1})|_{\varphi(U \cap V)} \circ (\varphi \circ \psi^{-1}))(y) &= (f \circ \varphi^{-1})|_{\varphi(U \cap V)}((\varphi \circ \psi^{-1})(y)) \\ &= (f \circ \varphi^{-1})(\varphi(\psi^{-1}(y))) \\ &= f(\varphi^{-1}(\varphi(\psi^{-1}(y)))) \\ &= f(\psi^{-1}(y)) \\ &= (f \circ \psi)^{-1}(y) \\ &= (f \circ \psi)|_{\psi(U \cap V)}(y) \end{aligned}$$

from which it follows that $(f \circ \psi)|_{\psi(U \cap V)} = (f \circ \varphi^{-1})|_{\varphi(U \cap V)} \circ (\varphi \circ \psi^{-1})$. Hence we have by [eq: 23.42] that

$$(f \circ \psi)|_{\psi(U \cap V)}: \psi(U \cap V) \rightarrow \mathbb{R} \text{ is of class } C^m \quad (23.43)$$

As $U \in \mathcal{T}$ we have that $U \cap V \in \mathcal{T}_V$ which, as ψ is a homeomorphism, proves by [lemma: 23.5] that

$$\psi(U \cap V) \subseteq \psi(V) \text{ and } \psi(U \cap V) \in \mathcal{T}_{\mathbb{R}^n} \quad (23.44)$$

To summarize [eqs: 23.40, 23.44, 23.44] we have $\forall x \in \psi(V)$ that $\psi(U \cap V) \in \mathcal{T}_{\mathbb{R}^n}$, $x \in \psi(U \cap V) \subseteq \psi(V)$ and $(f \circ \psi)|_{\psi(U \cap V)}: \psi(U \cap V) \rightarrow \mathbb{R}$ is of class C^m so that by [theorem: 16.159]

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m \quad \square$$

We can now use the previous lemma to prove that C^m compatibility using a C^m atlas is essentially independent of the choice of a chart.

Theorem 23.44. *Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, $\mathcal{A}$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ and $f: M \rightarrow \mathbb{R}$ a real valued function then we have the following equivalence*

$$\begin{aligned} f: M \rightarrow \mathbb{R} \text{ is of class } C^m \text{ using } \mathcal{A} \\ \Updownarrow \\ \forall (V, \psi) \in \mathcal{A} \text{ we have that } f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m \end{aligned}$$

Proof.

$\Rightarrow$. Let $(V, \psi) \in \mathcal{A}$ then by [definitions: 23.39, 23.40] (V, ψ) is C^m compatible with $\mathcal{A}$, hence using [lemma: 23.43] it follows that

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m$$

$\Leftarrow$. Let $p \in M$ then as $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U$ there exist a $(U, \varphi) \in \mathcal{A}$ such that $p \in U$ and by the hypothesis we have also that $f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R}$ is of class C^m . As $p \in M$ has been chosen arbitrary it follows that $f: M \rightarrow \mathbb{R}$ is of class C^m . $\square$

The next lemma shows that a atlas $\mathcal{A}$ can be extended to a atlas $\mathcal{B}$ without changing the set of functions between a manifold and $\mathbb{R}$ that are of class C^m .

Lemma 23.45. *Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, $\mathcal{A}$ a C^m atlas on $\langle M, \mathcal{T} \rangle$, (U, φ) a coordinate chart on $\langle M, \mathcal{T} \rangle$ that C^m compatible with $\mathcal{A}$ [so that by [lemma: 23.41] $\mathcal{A} \cup \{(U, \varphi)\}$ is a C^m -atlas] and $f: M \rightarrow \mathbb{R}$ a function then*

$$\begin{aligned} f \text{ is of class } C^m \text{ using } \mathcal{A} \\ \Updownarrow \\ f \text{ is of class } C^m \text{ using } \mathcal{A} \cup \{(U, \varphi)\} \end{aligned}$$

Proof.

$\Rightarrow$. Let $(V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}$ then we have either:

$(V, \psi) = (U, \varphi)$. Then (V, ψ) is C^m compatible with $\mathcal{A}$ so that by [lemma: 23.43]

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m$$

$(V, \psi) \in \mathcal{A}$. Then by [theorem: 23.44]

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m$$

This proves that $\forall (V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}$ $f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R}$ is of class C^m so that by [theorem: 23.44]

$$f \text{ is of class } C^m \text{ using } \mathcal{A} \cup \{(U, \varphi)\}$$

$\Leftarrow$. Using [theorem: 23.44] it follows that $\forall (V, \psi) \in \mathcal{A} \cup \{(U, \varphi)\}$ $f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R}$ is of class C^m . Hence we have $\forall (V, \psi) \in \mathcal{A}$ that $f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R}$ is of class C^m which by [theorem: 23.44] proves that

$$f \text{ is of class } C^m \text{ using } \mathcal{A} \quad \square$$

The above lemma shows that many atlases produces the same set of functions from a manifold to $\mathbb{R}$ that are of class C^m . To avoid this extra complication we introduce the concept of a maximal atlas.

Definition 23.46. (maximal atlas) Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold then a C^m atlas $\mathcal{A}$ on $\langle M, \mathcal{T} \rangle$ is **maximal** if for every coordinate chart (U, φ) on $\langle M, \mathcal{T} \rangle$ that is C^m compatible with $\mathcal{A}$ we have that $(U, \varphi) \in \mathcal{A}$.

One benefit of a maximal atlas is that it contains all the sub coordinate charts of its charts as is shown in the next theorem.

Theorem 23.47. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold, $\mathcal{A}$ a maximal C^m atlas on $\langle M, \mathcal{T} \rangle$ and $(U, \varphi) \in \mathcal{A}$ then $\forall W$ open in M with $W \subseteq U$ we have that $(W, \varphi|_W) \in \mathcal{A}$.

Note 23.48. In particular if V is a open set in M then $U \cap V$ is open and $U \cap V \subseteq U$ so that $(V \cap U, \varphi|_{V \cap U}) \in \mathcal{A}$

Proof. As $(U, \varphi) \in \mathcal{A}$ we have $\forall (V, \psi) \in \mathcal{A}$ that (U, φ) is C^m compatible with (V, ψ) , or using [theorem: 23.7] $(W, \varphi|_W)$ is C^m compatible with (V, ψ) . As $\mathcal{A}$ is maximal it follows then that $(W, \varphi|_W) \in \mathcal{A}$. $\square$

Given a C^m atlas on a topological manifold we can create a unique maximal atlas that contains this atlas. To do this we define an equivalence relation between atlases and construct the maximal atlas from an equivalence class.

Definition 23.49. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold then two C^m -atlases $\mathcal{A} \in \Delta^m[M, \mathcal{T}]$ and $\mathcal{B} \in \Delta^m[M, \mathcal{T}]$ on $\langle M, \mathcal{T} \rangle$ are **C^m compatible** iff $\mathcal{A} \cup \mathcal{B}$ is a C^m -atlas.

We have then the following equivalent definition of compatible atlases.

Theorem 23.50. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$ $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and $\mathcal{A} \in \Delta^m[M, \mathcal{T}]$, $\mathcal{B} \in \Delta^m[M, \mathcal{T}]$ two C^m -atlases on $\langle M, \mathcal{T} \rangle$ then we have

$\mathcal{A}$ and $\mathcal{B}$ are C^m compatible

$\Updownarrow$

$\forall (U, \varphi) \in \mathcal{A}$ and $\forall (V, \psi) \in \mathcal{B}$ we have that (U, φ) and (V, ψ) are C^m compatible

Proof.

$\Rightarrow$. As $\mathcal{A}, \mathcal{B}$ are C^m compatible $\mathcal{A} \cup \mathcal{B}$ is a C^m -compatible atlas hence we have $\forall (U, \varphi) \in \mathcal{A} \subseteq \mathcal{A} \cup \mathcal{B}$ and $\forall (V, \psi) \in \mathcal{B} \subseteq \mathcal{A} \cup \mathcal{B}$ that (U, φ) and (V, ψ) are C^m compatible.

$\Leftarrow$. We have to prove that $\mathcal{A} \cup \mathcal{B}$ is a C^m -compatible atlas. Let $m \in M$ then as $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U$ there exist a $(U, \varphi) \in \mathcal{A} \subseteq \mathcal{A} \cup \mathcal{B}$ so that $m \in U \subseteq \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{B}} U$ which proves that $M \subseteq \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{B}} U \subseteq \bigcup_{U \in \mathcal{T}} U \subseteq M$ so that

$$M = \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{B}} U$$

If $(U, \varphi) \in \mathcal{A} \cup \mathcal{B}$ we have either $(U, \varphi) \in \mathcal{A}$ so that (U, φ) is a coordinate chart on $\langle M, \mathcal{T} \rangle$ or $(U, \varphi) \in \mathcal{B}$ so that (U, φ) is a coordinate chart on $\langle M, \mathcal{T} \rangle$, in other words

$$\forall (U, \varphi) \in \mathcal{A} \cup \mathcal{B} \quad (U, \varphi) \text{ is a coordinate chart on } \langle M, \mathcal{T} \rangle$$

Further if $(U, \varphi), (V, \psi) \in \mathcal{A} \cup \mathcal{B}$ then we have either:

$(U, \varphi) \in \mathcal{A} \wedge (V, \psi) \in \mathcal{B}$. Then by the hypothesis (U, φ) and (V, ψ) are C^m compatible.

$(U, \varphi) \notin \mathcal{A} \wedge (V, \psi) \in \mathcal{B}$. Then $(U, \varphi), (V, \psi) \in \mathcal{B}$ and we have, as $\mathcal{B}$ is a C^m -atlas, that (U, φ) and (V, ψ) are C^m compatible.

$(U, \varphi) \in \mathcal{A} \wedge (V, \psi) \notin \mathcal{B}$. Then $(U, \varphi), (V, \psi) \in \mathcal{A}$ and we have, as $\mathcal{A}$ is a C^m -atlas, that (U, φ) and (V, ψ) are C^m compatible.

So

$\forall (U, \varphi) \in \mathcal{A} \cup \mathcal{B}, \forall (V, \psi) \in \mathcal{A} \cup \mathcal{B}$ we have that (U, φ) and (V, ψ) is C^m compatible

Hence it follows that

$$\mathcal{A} \cup \mathcal{B} \text{ is a } C^m\text{-atlas} \quad \square$$

We show now that the relation **is C^m compatible** between atlases is a equivalence relation.

Lemma 23.51. *Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$ $\langle M, \mathcal{T} \rangle$ be a n -dimensional topological manifold then $\approx_m \subseteq \Delta^m[M, \mathcal{T}] \times \Delta^m[M, \mathcal{T}]$ defined by $\mathcal{A} \approx_m \mathcal{B}$ iff $\mathcal{A}$ is **C^m compatible** with $\mathcal{B}$ is a equivalence relation.*

Proof.

reflectivity. Let $\mathcal{A}$ be a C^m -atlas then, as $\mathcal{A} \cup \mathcal{A} = \mathcal{A}$, $\mathcal{A} \cup \mathcal{A}$ is a C^m -atlas which proves that $\mathcal{A} \approx_m \mathcal{A}$.

symetricity. If $\mathcal{A} \approx_m \mathcal{B}$ then $\mathcal{A} \cup \mathcal{B}$ is a C^m -atlas from which it follows that $\mathcal{B} \cup \mathcal{A}$ is a C^m -atlas proving that $\mathcal{B} \approx_m \mathcal{A}$.

transitivity. If $\mathcal{A} \approx_m \mathcal{B}$ and $\mathcal{B} \approx_m \mathcal{C}$ then $\mathcal{A} \cup \mathcal{B}$ is a C^m -atlas and $\mathcal{B} \cup \mathcal{C}$ is a C^m -atlas. Let $m \in M$ then as $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U$ there exist a $(U, \varphi) \in \mathcal{A} \subseteq \mathcal{A} \cup \mathcal{C}$ with $m \in U$, hence $m \in U \subseteq \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{C}} U$ which proves that $M \subseteq \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{C}} U \subseteq \bigcup_{U \in \mathcal{T} \subseteq \mathcal{P}(M)} M$ so that

$$M = \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{C}} U \quad (23.45)$$

Further, if $(U, \varphi) \in \mathcal{A} \cup \mathcal{C}$ then either $(U, \varphi) \in \mathcal{A}$ so that (U, φ) is a coordinate chart on $\langle M, \mathcal{T} \rangle$ or $(U, \varphi) \in \mathcal{C}$ so that (U, φ) is a coordinate chart on $\langle M, \mathcal{T} \rangle$. Hence we have that

$$\forall (U, \varphi) \in \mathcal{A} \cup \mathcal{C} \quad (U, \varphi) \text{ is a coordinate chart on } \langle M, \mathcal{T} \rangle \quad (23.46)$$

Let $(U, \varphi), (V, \psi) \in \mathcal{A} \cup \mathcal{C}$ then we have either:

$(U, \varphi) \in \mathcal{A} \wedge (V, \psi) \in \mathcal{C}$. Then for $U \cap V$ we have either:

$U \cap V = \emptyset$. Then (U, φ) and (V, ψ) are C^m compatible.

$U \cap V \neq \emptyset$. Let $x \in \psi(U \cap V)$ then there exist a $p \in U \cap V$ such that $\psi(p) = x$. As $\mathcal{B}$ is a C^m atlas there exist a $(W, \gamma) \in \mathcal{B}$ such that $p \in W$, hence $p \in U \cap V \cap W$ from which it follows that $U \cap W \neq \emptyset \neq W \cap V$. As $\mathcal{A} \approx_m \mathcal{B}$ and $\mathcal{B} \approx_m \mathcal{C}$ it follows that $\mathcal{A} \cup \mathcal{B}$ and $\mathcal{B} \cup \mathcal{C}$ are C^m atlases so that

$$\varphi \circ \gamma^{-1}: \gamma(U \cap W) \rightarrow \varphi(U \cap W) \text{ is a diffeomorphism of class } C^m \quad (23.47)$$

$$\gamma \circ \psi^{-1}: \psi(W \cap V) \rightarrow \gamma(W \cap V) \text{ is a diffeomorphism of class } C^m \quad (23.48)$$

Take $Z = U \cap V \cap W$ then

$$x = \psi(p) \in \psi(Z) \subseteq \psi(W \cap V) \text{ and } \gamma(Z) \subseteq \gamma(W \cap V) \quad (23.49)$$

As U, V, W are open it follows that $Z = U \cap V \cap W$ is a open set in M , hence as $Z \subseteq U, V, W$ it follows from [lemma: 23.5] that

$$\gamma(Z) \in \mathcal{T}_{\mathbb{R}^n}, \varphi(Z) \in \mathcal{T}_{\mathbb{R}^n}, \psi(Z) \in \mathcal{T}_{\mathbb{R}^n} \quad (23.50)$$

Using [theorems: 2.23, 2.57] it it follows that

$$(\varphi \circ \gamma^{-1})(\gamma(Z)) = \varphi((\gamma^{-1})(\gamma(Z))) = \varphi(Z) \quad (23.51)$$

and

$$(\gamma \circ \psi^{-1})(\psi(Z)) = \gamma((\psi^{-1})(\psi(Z))) = \gamma(Z) \quad (23.52)$$

So using [theorem: 16.336] on the above [see eqs: 23.47, 23.48, 23.49, 23.50, 23.51, 23.52] it follows that

$$(\varphi \circ \gamma^{-1})|_{\gamma(Z)}: \gamma(Z) \rightarrow \varphi(Z) \text{ is a diffeomorphism of class } C^m$$

$$(\gamma \circ \psi^{-1})|_{\psi(Z)}: \psi(Z) \rightarrow \gamma(Z) \text{ is a diffeomorphism of class } C^m$$

Hence using the chain rule [see theorem: 16.339] on the above it follows that

$$(\varphi \circ \gamma^{-1})|_{\gamma(Z)} \circ (\gamma \circ \psi^{-1})|_{\psi(Z)} \text{ is a diffeomorphism of class } C^m \quad (23.53)$$

Further we have for $x \in \psi(Z)$ that

$$\begin{aligned} ((\varphi \circ \gamma^{-1})|_{\gamma(Z)} \circ (\gamma \circ \psi^{-1})|_{\psi(Z)})(x) &= (\varphi \circ \gamma^{-1})|_{\gamma(Z)}((\gamma \circ \psi^{-1})|_{\psi(Z)}(x)) \\ &= (\varphi \circ \gamma^{-1})((\gamma \circ \psi^{-1})(x)) \\ &= \varphi(\gamma^{-1}(\gamma(\psi^{-1}(x)))) \\ &= \varphi(\psi^{-1}(x)) \\ &= (\varphi \circ \psi^{-1})(x) \\ &\stackrel{x \in \psi(Z)}{=} (\varphi \circ \psi^{-1})|_{\psi(Z)}(x) \end{aligned}$$

from which it follows that

$$(\varphi \circ \psi^{-1})|_{\psi(Z)} = (\varphi \circ \gamma^{-1})|_{\gamma(Z)} \circ (\gamma \circ \psi^{-1})|_{\psi(Z)}$$

combining this with [eqs: 23.53, 23.50] proves that

$$\psi(Z), \varphi(Z) \in \mathcal{T}_{\mathbb{R}^n} \text{ and } (\varphi \circ \psi^{-1})|_{\psi(Z)}: \psi(Z) \rightarrow \varphi(Z) \text{ is a diffeomorphism of class } C^m$$

As this is true for every $x \in \psi(Z) \subseteq \psi(V)$ it follows from [theorem: 16.337] that

$$\varphi \circ \psi^{-1} \text{ is of class } C^m$$

which proves that (U, φ) and (V, ψ) are C^m compatible.

$(U, \varphi) \notin \mathcal{A} \wedge (V, \psi) \in \mathcal{C}$. Then $(U, \varphi), (V, \psi) \in \mathcal{C}$ so that (U, φ) and (V, ψ) are C^m compatible.

$(U, \varphi) \in \mathcal{A} \wedge (V, \psi) \notin \mathcal{C}$. Then $(U, \varphi), (V, \psi) \in \mathcal{A}$ so that (U, φ) and (V, ψ) are C^m compatible.

So in all cases we have that (U, φ) and (V, ψ) are C^m compatible which, as this is true for all $(U, \varphi), (V, \psi)$ in $\mathcal{A} \cup \mathcal{C}$, proves that $\mathcal{A} \cup \mathcal{C}$ is a C^m atlas on M . Hence

$$\mathcal{A} \approx_m \mathcal{C}$$

proving transitivity. □

Next we define a C^m differentiable structure on a topological manifold and use this to define a maximal C^m atlas

Definition 23.52. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and $\mathcal{A} \in \Delta^m[M, \mathcal{T}]$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ then a C^m differentiable structure determined by $\mathcal{A}$ noted as $\mathcal{M}[\mathcal{A}]$ is defined to be

$$\mathcal{M}[\mathcal{A}] = \bigcup_{\mathcal{B} \in \approx_m[\mathcal{A}]} \mathcal{B}$$

where

$$\approx_m[\mathcal{A}] \in \Delta^m[M, \mathcal{T}] / \approx_m$$

is the equivalence class defined by [see definitions: 3.9, 3.16 and lemma: 23.51]

$$\approx_m[\mathcal{A}] \stackrel{\text{def}}{=} \{\mathcal{B} \in \Delta^m[M, T] \mid \mathcal{A} \approx_m \mathcal{B}\} = \{\mathcal{B} \in \Delta^m[M, T] \mid \mathcal{A} \text{ is } C^m \text{ compatible with } \mathcal{B}\}$$

Theorem 23.53. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, T \rangle$ a n -dimensional topological manifold and $\mathcal{A} \in \Delta^m[M, T]$ a C^m atlas on $\langle M, T \rangle$ then we have:

1. $\mathcal{M}[\mathcal{A}]$ is a maximal C^m atlas on $\langle M, T \rangle$ such that $\mathcal{A} \subseteq \mathcal{M}[\mathcal{A}]$.
2. If $\mathcal{B}$ is a maximal C^m atlas on $\langle M, T \rangle$ such that $\mathcal{A} \subseteq \mathcal{B}$ then $\mathcal{B} = \mathcal{M}[\mathcal{A}]$. In other words $\mathcal{M}[\mathcal{A}]$ is the unique maximal C^m atlas that contains $\mathcal{A}$.
3. If $\mathcal{B} \in \Delta^m[M, T]$ then

$$\begin{aligned} \mathcal{M}[\mathcal{A}] &= \mathcal{M}[\mathcal{B}] \\ &\Updownarrow \\ \mathcal{A} \cup \mathcal{B} &\text{ is a } C^m \text{ atlas on } \langle M, T \rangle \\ &\Updownarrow \\ &\text{[definition: 23.49]} \\ \mathcal{A} &\approx_m \mathcal{B} \end{aligned}$$

Proof.

1. Let $(U, \varphi) \in \mathcal{M}[\mathcal{A}] = \bigcup_{\mathcal{B} \in \approx_m[\mathcal{A}]} \mathcal{B}$ then there exist a $\mathcal{B} \in \approx_m[\mathcal{A}]$ such that $(U, \varphi) \in \mathcal{B}$. As $\mathcal{B}$ is a C^m atlas it follows that (U, φ) is a coordinate chart. Hence we have that

$$\forall (U, \varphi) \in \mathcal{M}[\mathcal{A}] \quad (U, \varphi) \text{ is a coordinate chart} \quad (23.54)$$

Further, as $\mathcal{A} \in \approx_m[\mathcal{A}]$, it follows that

$$\mathcal{A} \subseteq \bigcup_{\mathcal{B} \in \approx_m[\mathcal{A}]} \mathcal{B} = \mathcal{M}[\mathcal{A}] \quad (23.55)$$

so that

$$\bigcup_{(U, \varphi) \in \mathcal{A}} U \subseteq \bigcup_{(U, \varphi) \in \mathcal{M}[\mathcal{A}]} U \subseteq \bigcup_{T \subseteq \overline{P}(M)} U \subseteq M$$

Given that $\mathcal{A}$ is a C^m atlas on $\langle M, T \rangle$ it follows that $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U$ which combined with the above proves that

$$M = \bigcup_{(U, \varphi) \in \mathcal{M}[\mathcal{A}]} U \quad (23.56)$$

Let $(U, \varphi) \in \mathcal{M}[\mathcal{A}]$, $(V, \psi) \in \mathcal{M}[\mathcal{A}]$ then there exist a $\mathcal{B} \in \approx_m[\mathcal{A}]$ and a $\mathcal{R} \in \approx_m[\mathcal{A}]$ such that $(U, \varphi) \in \mathcal{B}$ and $(V, \psi) \in \mathcal{R}$ which proves that

$$(U, \varphi) \in \mathcal{B} \cup \mathcal{R} \text{ and } (V, \psi) \in \mathcal{B} \cup \mathcal{R} \quad (23.57)$$

From $\mathcal{B} \in \approx_m[\mathcal{A}]$ and a $\mathcal{R} \in \approx_m[\mathcal{A}]$ it follows that $\mathcal{B} \approx_m \mathcal{A} \wedge \mathcal{A} \approx_m \mathcal{R}$ so that by transitivity $\mathcal{B} \approx_m \mathcal{R}$ which by [definition: 23.49] means that $\mathcal{B} \cup \mathcal{R}$ is a C^m atlas on $\langle M, T \rangle$. Combining this fact with [eq: 23.57] proves that (U, φ) and (V, ψ) are C^m compatible. To summarize we have proved that

$$\forall (U, \varphi) \in \mathcal{M}[\mathcal{A}], \forall (V, \psi) \in \mathcal{M}[\mathcal{A}] \text{ we have } (U, \varphi) \text{ and } (V, \psi) \text{ are } C^m \text{ compatible} \quad (23.58)$$

From [eq: 23.55, 23.54, 23.56 and 23.58] it follows that

$$\mathcal{M}[\mathcal{A}] \text{ is a } C^m \text{ atlas on } \langle M, T \rangle \text{ and } \mathcal{A} \subseteq \mathcal{M}[\mathcal{A}]$$

Next we have to prove that $\mathcal{M}[\mathcal{A}]$ is a maximal atlas using [definition: 23.46]. Let (U, φ) be a coordinate chart such that $\forall (V, \psi) \in \mathcal{M}[\mathcal{A}]$ we have (U, φ) and (V, ψ) are C^m compatible. Then, as $\mathcal{A} \subseteq \mathcal{M}[\mathcal{A}]$, we have $\forall (V, \psi) \in \mathcal{A}$ that (U, φ) and (V, ψ) are C^m compatible so that (U, φ) is C^m compatible with $\mathcal{A}$. Using now [lemma: 23.41] it follows that

$$\mathcal{A} \cup \{(U, \varphi)\} \text{ is a } C^m \text{ atlas on } \langle M, T \rangle$$

As $\mathcal{A} \cup (\mathcal{A} \cup \{(U, \varphi)\}) = \mathcal{A} \cup \{(U, \varphi)\}$ it follows from the above that $\mathcal{A} \approx_m (\mathcal{A} \cup \{(U, \varphi)\})$ so that $\mathcal{A} \cup \{(U, \varphi)\} \in \approx_m[\mathcal{A}]$ from which it follows that

$$\mathcal{A} \cup \{(U, \varphi)\} \subseteq \bigcup_{\mathcal{B} \in \approx_m} \mathcal{B} = \mathcal{M}[\mathcal{A}]$$

so that $(U, \varphi) \in \mathcal{M}[\mathcal{A}]$. Hence by [definition: 23.46] it follows that

$$\mathcal{M}[\mathcal{A}] \text{ is a maximal } C^m \text{ atlas on } \langle M, \mathcal{T} \rangle$$

2. Assume that $\mathcal{B}$ is another maximal C^m atlas on $\langle M, \mathcal{T} \rangle$ such that $\mathcal{A} \subseteq \mathcal{B}$. Then $\mathcal{A} \cup \mathcal{B} = \mathcal{B}$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ so that $\mathcal{A} \approx_m \mathcal{B}$ from which it follows that $\mathcal{B} \in \approx_m[\mathcal{A}]$. Hence $\mathcal{B} \subseteq \bigcup_{\mathcal{R} \in \approx_m[\mathcal{A}]} \mathcal{R} = \mathcal{M}[\mathcal{A}]$ which proves that

$$\mathcal{B} \subseteq \mathcal{M}[\mathcal{A}] \quad (23.59)$$

Let $(U, \varphi) \in \mathcal{M}[\mathcal{A}]$ then, as by (1) $\mathcal{M}[\mathcal{A}]$ is a C^m atlas, we have $\forall (V, \psi) \in \mathcal{M}[\mathcal{A}]$ that (U, φ) is C^m compatible with (V, ψ) . As $\mathcal{B} \subseteq \mathcal{M}[\mathcal{A}]$ we have also that $\forall (V, \psi) \in \mathcal{B}$ (U, φ) is C^m compatible with (V, ψ) [as $\mathcal{M}[\mathcal{A}]$ is a C^m atlas], hence as $\mathcal{B}$ is maximal it follows that $(U, \varphi) \in \mathcal{B}$. This proves that $\mathcal{M}[\mathcal{A}] \subseteq \mathcal{B}$ which combined with [eq: 23.59] proves that

$$\mathcal{B} = \mathcal{M}[\mathcal{A}]$$

3. Let $\mathcal{A}, \mathcal{B}$ be two C^m atlases on $\langle M, \mathcal{T} \rangle$

$\Rightarrow$. As $\mathcal{A}$ and $\mathcal{B}$ are atlases, hence collections of coordinate charts, it follows that

$$\forall (U, \varphi) \in \mathcal{A} \cup \mathcal{B} \quad (U, \varphi) \text{ is a coordinate chart} \quad (23.60)$$

Further we have, as $M = \bigcup_{(U, \varphi) \in \mathcal{A}} U \subseteq \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{B}} U \subseteq \bigcup_{\mathcal{T} \in \mathcal{P}(M)} U$, that

$$M = \bigcup_{(U, \varphi) \in \mathcal{A} \cup \mathcal{B}} U \quad (23.61)$$

By (1) we have that $\mathcal{A} \subseteq \mathcal{M}[\mathcal{A}]$ and $\mathcal{B} \subseteq \mathcal{M}[\mathcal{B}]$ which as by the hypothesis $\mathcal{M}[\mathcal{A}] = \mathcal{M}[\mathcal{B}]$ proves that

$$\mathcal{A} \cup \mathcal{B} \subseteq \mathcal{M}[\mathcal{B}]$$

Let $(U, \varphi) \in \mathcal{A} \cup \mathcal{B}$ and $(V, \psi) \in \mathcal{A} \cup \mathcal{B}$ then it follows from the above that $(U, \varphi) \in \mathcal{M}[\mathcal{B}]$ and $(V, \psi) \in \mathcal{M}[\mathcal{B}]$ which, as $\mathcal{M}[\mathcal{B}]$ is a C^m atlas, proves that (U, φ) is C^m compatible to (V, ψ) . Hence

$$\forall (U, \varphi) \in \mathcal{A} \cup \mathcal{B}, \forall (V, \psi) \in \mathcal{A} \cup \mathcal{B} \text{ we have } (U, \varphi) \text{ is } C^m \text{ compatible with } (V, \psi)$$

Combining the above with [eqs: 23.60 and 23.61] proves that

$$\mathcal{A} \cup \mathcal{B} \text{ is a } C^m \text{ atlas on } \mathcal{M}[\mathcal{B}]$$

$\Leftarrow$. Assume that $\mathcal{A} \cup \mathcal{B}$ is a C^m atlas on $\langle M, \mathcal{T} \rangle$ then by definition $\mathcal{A} \approx_m \mathcal{B}$ so that by [theorem: 3.11] $\approx_m[\mathcal{A}] = \approx_m[\mathcal{B}]$. Hence we have

$$\mathcal{M}[\mathcal{A}] = \bigcup_{\mathcal{R} \in \approx_m[\mathcal{A}]} \mathcal{R} = \bigcup_{\mathcal{R} \in \approx_m[\mathcal{B}]} \mathcal{R} = \mathcal{M}[\mathcal{B}] \quad \square$$

Theorem 23.54. Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$, $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold and $\mathcal{A} \in \Delta^m[M, \mathcal{T}]$ a C^m atlas on $\langle M, \mathcal{T} \rangle$ and $f: M \rightarrow \mathbb{R}$ a function then we have the following equivalences

$$\begin{aligned} f \text{ is of class } C^m \text{ using } \mathcal{A} \\ \Updownarrow \\ f \text{ is of class } C^m \text{ using } \mathcal{M}[\mathcal{A}] \end{aligned}$$

Proof.

$\Rightarrow$. Let $(V, \psi) \in \mathcal{M}[\mathcal{A}]$ then, as $\mathcal{M}[\mathcal{A}]$ is a C^m atlas with $\mathcal{A} \subseteq \mathcal{M}[\mathcal{A}]$ [see theorem: 23.53] it follows that

$$(V, \psi) \text{ is } C^m \text{ compatible with } \mathcal{A}$$

As f is of class C^m using $\mathcal{A}$ it follows from [lemma: 23.43] that

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R}^n \text{ is of class } C^m$$

As this is true for every $(V, \psi) \in \mathcal{M}[\mathcal{A}]$ it follows from [theorem: 23.44] that

$$f \text{ is of class } C^m \text{ using } \mathcal{M}[\mathcal{A}]$$

$\Leftarrow$. Let $(V, \psi) \in \mathcal{A}$ then as $\mathcal{A} \subseteq \mathcal{M}[\mathcal{A}]$ [see theorem: 23.53] it follows that $(V, \psi) \in \mathcal{M}[\mathcal{A}]$. As f is of class C^m using $\mathcal{M}[\mathcal{A}]$ it follows from [theorem: 23.44] that

$$f \circ \psi^{-1}: \psi(V) \rightarrow \mathbb{R} \text{ is of class } C^m$$

As this is true for every $(V, \psi) \in \mathcal{A}$ it follows from [theorem: 23.44] again that

$$f \text{ is of class } C^m \text{ using } \mathcal{A}$$

□

We are now ready to define a C^m differentiable manifold.

Definition 23.55. (C^m differentiable manifold) Let $n \in \mathbb{N}$, $m \in \mathbb{N}_0 \cup \{\infty\}$ then $\mathcal{M}^n \langle M, \mathcal{T}, \mathcal{A} \rangle$ is a n -dimensional C^m differentiable manifold iff

1. $\langle M, \mathcal{T} \rangle$ is a n -dimensional topological space
2. $\mathcal{A}$ is a maximal C^m atlas on $\langle M, \mathcal{T} \rangle$ called the C^m structure on M

Note 23.56. If we want to construct a C^m differentiable manifold we start usually with a n -dimensional topological manifold $\langle M, \mathcal{T} \rangle$, construct a C^m atlas $\mathcal{A}$ based on the coordinate charts of the topological manifold allowing us to use [theorem: 23.53 (1)] define then the n -dimensional C^m differentiable manifold

$$\mathcal{M}^n \langle M, \mathcal{T}, \mathcal{M}[\mathcal{A}] \rangle$$

Let's look now at some examples of differentiable manifolds.

Example 23.57. Let $n \in \mathbb{N}$ and $\langle M, \mathcal{T} \rangle$ a n -dimensional topological manifold with a atlas $\mathcal{A} = \{(U, \varphi)\}$ then $\mathcal{M}^n \langle M, \mathcal{T}, \mathcal{M}[\{(U, \varphi)\}] \rangle$ is a n -dimensional C^∞ differentiable manifold.

Note 23.58. If $\langle M, \mathcal{T} \rangle$ is a second countable Hausdorff topological space such that there exist a homeomorphism $\varphi: M \rightarrow V$ where $V \in \mathcal{T}_{\mathbb{R}^n}$ then by [example: 23.13] $\{(U, \varphi)\}$ is a atlas on $\langle M, \mathcal{T} \rangle$ so that $\mathcal{M} \langle M, \mathcal{T}, \mathcal{M}[\{(U, \varphi)\}] \rangle$ is a n -dimensional topological manifold.

Proof. Take $(V, \psi), (W, \gamma) \in \mathcal{A}$ then $V = W = U$ and $\psi = \gamma = \varphi$ so that the function $\gamma \circ \psi^{-1}: \psi(V \cap W) \rightarrow \gamma(V \cap W)$ is equal to $\varphi \circ \varphi^{-1}: \varphi(U) \rightarrow \varphi(U)$ or, as $\varphi \circ \varphi^{-1} = \text{Id}_{\varphi(U)}$, to the function $\text{Id}_{\varphi(U)}: \varphi(U) \rightarrow \varphi(U)$. As $\varphi(U)$ is open [because $\{(U, \varphi)\}$ is a atlas on $\langle M, \mathcal{T} \rangle$] it follows from [example: 16.335] that $\text{Id}_{\varphi(U)}$ is a diffeomorphism of class C^∞ which proves that $\{(U, \varphi)\}$ is a C^∞ atlas on $\langle M, \mathcal{T} \rangle$. Using [theorem: 23.53] it follows that $\mathcal{M}[\{(U, \varphi)\}]$ is a maximal atlas so that

$$\mathcal{M}^n \langle M, \mathcal{T}, \mathcal{M}[\{(U, \varphi)\}] \rangle \text{ is a } n\text{-dimensional } C^m \text{ differentiable manifold}$$

□

Example 23.59. $\mathcal{M}^1 \langle \mathbb{R}, \mathcal{T}_||, \mathcal{M}[\{(\mathbb{R}, \varphi)\}] \rangle$ is a 1-dimensional C^∞ differentiable manifold where

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}^1 \text{ by } \varphi(x) = (x) \text{ [here } (x) \text{ is a tuple of one element where } (x)_1 = x]$$

Proof. This follows from [examples: 23.14 and 23.57].

□

Example 23.60. Let $n \in \mathbb{N}$ then $\mathcal{M}^n\langle \mathbb{R}^n, \mathcal{T}_{\mathbb{R}^n}, \{(\text{Id}_{\mathbb{R}^n})\} \rangle$ is a n -dimensional C^∞ differentiable manifold.

Proof. This follows from [exercises: 23.15 and 23.57]. $\square$

Every open subset of a n -dimensional C^m differentiable manifold is a n -dimensional C^m differentiable manifold.

Example 23.61. Let $\mathcal{M}^n\langle M, \mathcal{T}, \mathcal{A} \rangle$ be a n -dimensional C^m differentiable manifold and $U \in \mathcal{T}$ a open set in M then $\mathcal{M}^n\langle U, \mathcal{T}|_U, \mathcal{A}|_U \rangle$ is a n -dimensional C^m differentiable manifold where $\mathcal{A}|_U = \{(U \cap V, \psi|_{U \cap V}) | (V, \psi) \in \mathcal{A}\}$.

Proof. Using [example: 23.16] it follows that

$$\langle U, \mathcal{T}|_U \rangle \text{ is a } n\text{-dimensional topological manifold}$$

and

$$\mathcal{A}|_U \text{ is an atlas on } \langle U, \mathcal{T}|_U \rangle$$

So we only have to prove that $\mathcal{A}|_U$ is a C^m -atlas. Let $\langle V, \psi \rangle, \langle W, \gamma \rangle \in \mathcal{A}|_U$ then there exists $\langle V', \psi' \rangle, \langle W', \gamma' \rangle \in \mathcal{A}$ such that

$$V = U \cap V', W = U \cap W', \psi = (\psi')|_{U \cap V'} = (\psi')|_V \text{ and } \gamma = (\gamma')|_{U \cap W'} = (\gamma')|_W$$

As $\mathcal{A}$ is a C^m atlas on M it follows that

$$\gamma' \circ (\psi')^{-1}: \psi'(V' \cap W') \rightarrow \gamma'(V' \cap W') \text{ is a } C^m \text{ diffeomorphism}$$

As $\mathcal{A}|_U$ is an atlas it follows that $W \in \mathcal{T}|_U$ so that $V \cap W \in (\mathcal{T}|_U)|_V$. As further $\psi: V \rightarrow \psi(V)$ is a homeomorphism [as $\mathcal{A}|_U$ is an atlas on $\langle U, \mathcal{T}|_U \rangle$] it follows that

$$\psi(V \cap W) \in (\mathcal{T}_{\mathbb{R}^n})_{\psi(V)} \underset{\psi(V) \in \mathcal{T}_{\mathbb{R}^n} \wedge [\text{theorem: 14.11}]}{\subseteq} \mathcal{T}_{\mathbb{R}^n}.$$

In other words

$$\psi(V \cap W) \text{ is open in } \mathcal{T}_{\mathbb{R}^n}$$

Hence using [theorem: 16.336] it follows that

$$(\gamma' \circ (\psi')^{-1})|_{\psi(V \cap W)}: \psi(V \cap W) \rightarrow (\gamma' \circ (\psi')^{-1})(\psi(V \cap W)) \text{ is a } C^m \text{ diffeomorphism} \quad (23.62)$$

As $V \cap W = (U \cap V') \cap (U \cap W') \subseteq V' \cap W' \subseteq V', W'$ it follows from [theorem: 2.92] that

$$\psi(V \cap W) = (\psi')|_V(V \cap W) \underset{[\text{theorem: 2.92}]}{=} \psi'(V \cap W) \quad (23.63)$$

$$\gamma(V \cap W) = (\gamma')|_W(V \cap W) \underset{[\text{theorem: 2.92}]}{=} \gamma'(V \cap W) \quad (23.64)$$

Further we have

$$\begin{aligned} (\gamma' \circ (\psi')^{-1})|_{\psi(V \cap W)} &\underset{[\text{eq: 23.63}]}{=} (\gamma' \circ (\psi')^{-1})|_{\psi'(V \cap W)} \\ &\underset{[\text{theorem: 2.96}]}{=} \gamma'|_{(\psi')^{-1}(\psi'(V \cap W))} \circ ((\psi')^{-1})|_{\psi'(V \cap W)} \\ &= (\gamma')|_{V \cap W} \circ ((\psi')^{-1})|_{\psi'(V \cap W)} \\ &\underset{[\text{theorem: 2.100}]}{=} (\gamma')|_{V \cap W} \circ ((\psi')|_{V \cap W})^{-1} \\ &\underset{[\text{theorem: 2.92}]}{=} ((\gamma')|_V)|_{V \cap W} \circ (((\psi')|_V)|_{V \cap W})^{-1} \\ &= \gamma|_{V \cap W} \circ (\psi|_{V \cap W})^{-1} \\ &\underset{[\text{theorem: 2.101}]}{=} \gamma \circ \psi^{-1} \end{aligned}$$

which proves that

$$(\gamma' \circ (\psi')^{-1})|_{\psi(V \cap W)} = \gamma \circ \psi^{-1} \quad (23.65)$$

Also

$$\begin{aligned}
 (\gamma' \circ (\psi')^{-1})(\psi(V \cap W)) &= (\gamma' \circ (\psi')^{-1})(\psi'(V \cap W)) \\
 &= \gamma'((\psi')^{-1}(\psi'(V \cap W))) \\
 &= \gamma'(V \cap W) \\
 &\stackrel{[\text{eq: 23.64}]}{=} \gamma(V \cap W)
 \end{aligned}$$

proving that

$$(\gamma' \circ (\psi')^{-1})(\psi(V \cap W)) = \gamma(V \cap W) \quad (23.66)$$

Substituting [eqs: 23.65, 23.66] in [eq: 23.62] it follows that

$$\gamma \circ \psi^{-1}: \psi(V \cap W) \rightarrow \gamma(V \cap W) \text{ is diffeomorphism of class } C^m$$

As this is true for every $(V, \psi), (W, \gamma) \in \mathcal{A}_U$ it follows that

$$\mathcal{A}_U \text{ is a } C^m \text{ atlas on } \langle U, \mathcal{T}_U \rangle$$

Using [theorem: 23.53] it follows that $\mathcal{M}[\mathcal{A}_U]$ is a maximal atlas so that

$$\mathcal{M}^n \langle U, \mathcal{T}_U, \mathcal{M}[\mathcal{A}_U] \rangle \text{ is a } n\text{-dimensional } C^m \text{ differentiable manifold} \quad \square$$

23.2 Differential mappings between manifolds

We extend now the definition of a C^r function between a manifold and the set of the real numbers to a C^m mapping between manifolds.

Definition 23.62. Let $n, m \in \mathbb{N}$, $r \in \mathbb{N}_0 \cup \{\infty\}$, $\mathcal{M} \langle {}^n M, \mathcal{T}_M, \mathcal{A}_M \rangle$ a n -dimensional C^r differentiable manifold, $\mathcal{M}^m \langle N, \mathcal{T}_N, \mathcal{A}_N \rangle$ a m -dimensional C^r differentiable manifold and $f: M \rightarrow N$ a function then f is a C^r mapping iff $\forall p \in M$ there $\exists (U, \varphi) \in \mathcal{A}_M$ with $p \in U$ and there $\exists (V, \psi) \in \mathcal{A}_N$ with $f(p) \in V$ and $f(U) \subseteq V$ such that

$$\psi \circ f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

Note 23.63. As $\varphi^{-1}(\varphi(U)) = U \subseteq M$ it follows from [theorem: 2.30] that $f \circ \varphi^{-1}: \varphi(U) \rightarrow N$ is a well defined function. Further as $(f \circ \varphi^{-1})(\varphi(U)) = f(\varphi^{-1}(\varphi(U))) = f(U) \subseteq V$ it follows from [theorem: 2.30] that $\psi \circ f \circ \varphi^{-1} = \psi \circ (f \circ \varphi^{-1}): \varphi(U) \rightarrow \mathbb{R}^m$ is a well defined function.

Note 23.64. As $\varphi(U) \in \mathcal{T}_{\mathbb{R}^n}$ the concept of class C^r is feasible.

Every C^r mapping between differentiable manifolds is a continuous mapping

Theorem 23.65. Let $n, m \in \mathbb{N}$, $r \in \mathbb{N}_0 \cup \{\infty\}$, $\mathcal{M} \langle {}^n M, \mathcal{T}_M, \mathcal{A}_M \rangle$ a n -dimensional C^r differentiable manifold, $\mathcal{M}^m \langle N, \mathcal{T}_N, \mathcal{A}_N \rangle$ a m -dimensional C^r differentiable manifold and $f: M \rightarrow N$ a C^r mapping then f is continuous.

Proof. Let $p \in M$ then there exists a $(\varphi, U) \in \mathcal{A}_M$ with $p \in U$ and a $(\psi, V) \in \mathcal{A}_N$ with $f(p) \in V$ and $f(U) \subseteq V$ such that

$$\psi \circ f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

hence by [theorem: 16.158] $\psi \circ f \circ \varphi^{-1}$ is of class C^0 or

$$\psi \circ f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R}^m \text{ is continuous}$$

As $\varphi: U \rightarrow \varphi(U)$ and $\psi^{-1}: \psi(V) \rightarrow \psi(V)$ are homeomorphism it follows that $\varphi: U \rightarrow \varphi(U)$ is continuous and $\psi: V \rightarrow \psi(V)$ is continuous. Further

$$(\psi \circ f \circ \varphi^{-1})(\varphi(U)) = \psi(f(\varphi^{-1}(\varphi(U)))) = \psi(f(U)) \subseteq \psi(V)$$

so that by [theorem: 14.158]

$$\psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi: U \rightarrow V \text{ is continuous}$$

As $\forall x \in U$ we have that

$$\begin{aligned} (\psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi)(x) &= \psi^{-1}(\psi(f(\varphi^{-1}(\varphi(x)))) \\ &= \psi^{-1}(\psi(f(x))) \\ &= f(x) \\ &\stackrel{x \in U}{=} f|_U(x) \end{aligned}$$

it follows that $f|_U = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi$ so that $f|_U: U \rightarrow V$ is continuous. As $p \in U$ it follows that $f|_U$ is continuous at p . Hence as U is open it follows from [theorem: 14.149] that f is continuous at p . As this is true for every $p \in M$ it follows that

$$f: M \rightarrow N \text{ is continuous} \quad \square$$

We have a equivalent definition of a C^r mapping between manifolds.

Theorem 23.66. *Let $n, m \in \mathbb{N}$, $r \in \mathbb{N}_0 \cup \{\infty\}$, $\mathcal{M}\langle nM, \mathcal{T}_M, \mathcal{A}_M \rangle$ a n -dimensional C^r differentiable manifold, $\mathcal{M}^m\langle N, \mathcal{T}_N, \mathcal{A}_N \rangle$ a m -dimensional C^r differentiable manifold and $f: M \rightarrow N$ a function then we have that the following are equivalent:*

1. f is a C^r mapping.
2. $\forall p \in M \exists (U, \varphi) \in \mathcal{A}_M$ with $p \in U$, $\exists (V, \psi) \in \mathcal{A}_N$ with $f(p) \in V$ such that $U \cap f^{-1}(V)$ is open in M and $\psi \circ f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m$ is of class C^r .
3. f is continuous and there exist a C^r atlas $\mathcal{B}_M \subseteq \mathcal{A}_M$ and a C^r atlas $\mathcal{B}_N \subseteq \mathcal{A}_N$ such that $\forall (U, \varphi) \in \mathcal{B}_M$ and $\forall (V, \psi) \in \mathcal{B}_N$ we have that

$$\psi \circ f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

Note 23.67. *As $f^{-1}(V)$ is open $U \cap f^{-1}(V)$ is automatical satisfied.*

Note 23.68. *As $\varphi^{-1}(\varphi(U \cap f^{-1}(V))) = U \cap f^{-1}(V) \subseteq M$ it follows from [theorem: 2.30] that $f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \rightarrow N$ is a well defined function. Further as*

$$(f \circ \varphi^{-1})(\varphi(U \cap f^{-1}(V))) = f(\varphi^{-1}(\varphi(\varphi(U \cap f^{-1}(V)))) = f(U \cap f^{-1}(V)) \subseteq f(f^{-1}(V)) \subseteq V$$

it follows from [theorem: 2.30] that $\psi \circ f \circ \varphi^{-1} = \psi \circ (f \circ \varphi^{-1}): \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m$ is a well defined function. Further the concept of class C^r in (2) and (3) is feasible because $U \cap f^{-1}$ is open in M .

Proof.

1 $\Rightarrow$ 2. Let $p \in M$ then as f is a C^r mapping there exist a $(U, \varphi) \in \mathcal{A}_M$ with $p \in M$ and a $(V, \psi) \in \mathcal{A}_N$ with $f(p) \in V$ such that $f(U) \subseteq V$ and

$$\psi \circ f \circ \varphi^{-1}: \varphi(U) \rightarrow \mathbb{R}^m \text{ is of class } C^m \quad (23.67)$$

By [theorem: 23.65] f is a continuous mapping so that, as $V \in \mathcal{T}_N$ [because (V, ψ) is a chart], it follows that $f^{-1}(V) \in \mathcal{T}_M$. Hence, as $U \in \mathcal{T}_M$ it follows that

$$U \cap f^{-1}(V) \text{ is open in } M \quad (23.68)$$

So using [lemma: 23.5] it follows that $\varphi(U \cap f^{-1}(V))$ is a open set in $\mathbb{R}^n$. Hence using [coollary: 16.156], it follows that

$$(\psi \circ f \circ \varphi^{-1})|_{\varphi(U \cap f^{-1}(V))}: \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m \text{ is of class } C^m$$

As $\varphi^{-1}(\varphi(U \cap f^{-1}(V))) = U \cap f^{-1}(V) \subseteq M$ it follows from [theorem: 2.30] that $f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \rightarrow N$ is a well defined function. Further as

$$\begin{aligned} (f \circ \varphi^{-1})(\varphi(U \cap f^{-1}(V))) &= f(\varphi^{-1}(\varphi(\varphi(U \cap f^{-1}(V)))) \\ &= f(U \cap f^{-1}(V)) \subseteq f(f^{-1}(V)) \subseteq V \end{aligned}$$

it follows from [theorem: 2.30] again that

$$\psi \circ f \circ \varphi^{-1} = \psi \circ (f \circ \varphi^{-1}): \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m$$

is a well defined function. Hence, using [theorem: 2.93] it follows that

$$(\psi \circ f \circ \varphi^{-1})|_{\varphi(U \cap f^{-1}(V))} = \psi \circ f \circ \varphi^{-1}$$

so that

$$(\psi \circ f \circ \varphi^{-1}): \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m \text{ is of class } C^m$$

2 $\Rightarrow$ 1. Assume that $\forall p \in M$ there exist a $(U, \varphi) \in \mathcal{A}_M$ with $p \in U$ and a $(V, \psi) \in \mathcal{A}_N$ with $f(p) \in V$ such that $U \cap f^{-1}(V)$ is open in M and

$$\psi \circ f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m \text{ is of class } C^r \quad (23.69)$$

As $f(p) \in V$ $p \in f^{-1}(V)$ so that

$$p \in U \cap f^{-1}(V) \wedge f(p) \in V \quad (23.70)$$

Further we have that $f(U \cap f^{-1}(V)) \subseteq f(f^{-1}(V)) \subseteq V$ so that

$$f(U \cap f^{-1}(V)) \subseteq V \quad (23.71)$$

As $(U, f) \in \mathcal{A}_M$ [a maximum C^r atlas] and $U \cap f^{-1}(V)$ is open it follows from [theorem: 23.47] that

$$(U \cap f^{-1}(V), \varphi|_{U \cap f^{-1}(V)}) \in \mathcal{A}_M \quad (23.72)$$

As $(\varphi|_{U \cap f^{-1}(V)})^{-1}(\varphi|_{U \cap f^{-1}(V)}(U \cap f^{-1}(V))) = U \cap f^{-1}(V) \subseteq M$ it follows from [theorem: 2.30] that $f \circ (\varphi|_{U \cap f^{-1}(V)})^{-1}: \varphi|_{U \cap f^{-1}(V)}(U \cap f^{-1}(V)) \rightarrow N$ is a well defined function. Further as

$$\begin{aligned} f \circ (\varphi|_{U \cap f^{-1}(V)})^{-1}(\varphi|_{U \cap f^{-1}(V)}(U \cap f^{-1}(V))) &= \\ f((\varphi|_{U \cap f^{-1}(V)})^{-1}(\varphi|_{U \cap f^{-1}(V)}(U \cap f^{-1}(V)))) &= \\ f(U \cap f^{-1}(V)) &\subseteq \\ f(f^{-1}(V)) &\subseteq V \end{aligned}$$

it follows from [theorem: 2.30] that

$$\psi \circ f \circ (\varphi|_{U \cap f^{-1}(V)})^{-1}: \varphi|_{U \cap f^{-1}(V)}(U \cap f^{-1}(V)) \stackrel{[\text{theorem: 2.92}]}{=} \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m$$

is a well defined function. Further $\forall x \in \varphi(U \cap f^{-1}(V))$ we have

$$\begin{aligned} \psi \circ f \circ (\varphi|_{U \cap f^{-1}(V)})^{-1}(x) &= \psi(f((\varphi|_{U \cap f^{-1}(V)})^{-1}(x))) \\ &\stackrel{[\text{theorem: 2.95}]}{=} \psi(f((\varphi^{-1})_{\varphi(U \cap f^{-1}(V))}(x))) \\ &= \psi(f(\varphi^{-1}(x))) \\ &= (\psi \circ f \circ \varphi^{-1})(x) \end{aligned}$$

which proves that $\psi \circ f \circ (\varphi|_{U \cap f^{-1}(V)})^{-1} = \psi \circ f \circ \varphi^{-1}$. So using [eq: 23.69] it follows that

$$\psi \circ f \circ (\varphi|_{U \cap f^{-1}(V)})^{-1}: \varphi|_{U \cap f^{-1}(V)}(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m \text{ is of class } C^r \quad (23.73)$$

Let $W = U \cap f^{-1}(V)$ and $\gamma = \varphi|_{U \cap f^{-1}(V)}$ then we have by [eqs: 23.70] that we found $(W, \gamma) \in \mathcal{A}_M, (V, \psi) \in \mathcal{A}_N$ such that

$$p \in W \wedge f(p) \in V \wedge f(W) \subseteq V \wedge \psi \circ f \circ \gamma^{-1}: \gamma(W) \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

2 $\Rightarrow$ 3. By $2 \Rightarrow 1$ we have that f is a C^r mapping so that by [theorem: 23.65]

$$f \text{ is continuous}$$

Let $\mathcal{B}_M = \mathcal{A}_M$ and $\mathcal{B}_N = \mathcal{A}_N$ and take $(U, \varphi) \in \mathcal{B}_M, (V, \psi) \in \mathcal{B}_N$ and consider the function

$$\psi \circ f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \rightarrow \mathbb{R}^m$$

Let $x \in \varphi(U \cap f^{-1}(V))$ then there exist a $p \in U \cap f^{-1}(V)$ such that $x = \varphi(p)$ so that

$$x = \varphi(p) \wedge f(p) \in V$$

Using (2) there exist a $(U', \varphi') \in \mathcal{A}_M$ and a $(V', \psi') \in \mathcal{A}_N$ such that

$$p \in U' \wedge f(p) \in V' \wedge U' \cap f^{-1}(V') \in \mathcal{T}_M$$

and

$$\psi' \circ f \circ (\varphi')^{-1}: \varphi'(U' \cap f^{-1}(V')) \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

As $(U, \varphi) \in \mathcal{B}_M \subseteq \mathcal{A}_M, (V, \psi) \in \mathcal{B}_N \subseteq \mathcal{A}_N$ we have as $(U', \varphi') \in \mathcal{A}_M, (V', \psi') \in \mathcal{A}_N$ that (U, φ) is C^r compatible with (U', φ') and (V, ψ) is C^r so that, as $p \in U \cap U' \Rightarrow U \cap U' \neq \emptyset$, we have

$$\varphi' \circ (\varphi)^{-1}: \varphi(U \cap U') \rightarrow \varphi'(U \cap U') \text{ is a } C^r \text{ diffeomorphism}$$

$$\psi \circ (\psi')^{-1}: \psi'(V \cap V') \rightarrow \psi(V \cap V') \text{ is a } C^r \text{ diffeomorphism}$$

so that

$$\varphi' \circ \varphi^{-1}: \varphi(U \cap U') \rightarrow \mathbb{R}^n \text{ is of class } C^r$$

$$\psi \circ (\psi')^{-1}: \psi'(V \cap V') \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

As $V, V' \in \mathcal{T}_N$ are open and f is continuous $f^{-1}(V \cap V') \in \mathcal{T}_M$ so that $U \cap U' \cap f^{-1}(V \cap V') \in \mathcal{T}_M$ so that by [lemma: 23.5]

$$\varphi(U \cap U' \cap f^{-1}(V \cap V')) \in \mathcal{T}_{\mathbb{R}^n}$$

Hence if we define $W = U \cap U' \cap f^{-1}(V \cap V')$ then

$$\varphi(W) \in \mathcal{T}_{\mathbb{R}^n} \text{ and } W \subseteq U \cap U'$$

So using [corollary: 16.156] it follows that

$$(\varphi' \circ \varphi^{-1})|_{\varphi(W)}: \varphi(W) \rightarrow \mathbb{R}^n \text{ is of class } C^r$$

Further we have

$$\begin{aligned} (\varphi' \circ \varphi^{-1})|_{\varphi(W)}(\varphi(W)) &= \varphi'((\varphi^{-1})|_{\varphi(W)}(\varphi(W))) \\ &= \varphi'(W) \\ &= \varphi'(U \cap U' \cap f^{-1}(V \cap V')) \\ &\subseteq \varphi'(U \cap f^{-1}(V)) \end{aligned}$$

Using then the chain rule [see theorem: 16.197] it follows that

$$(\psi' \circ f \circ (\varphi')^{-1}) \circ ((\varphi' \circ \varphi^{-1})|_{\varphi(W)}): \varphi(W) \rightarrow \mathbb{R}^m \text{ is of class } C^r$$

3 $\Rightarrow$ 2.

□

Chapter 24

TODO

24.1 Sets.tm

1. Spell check started/Finished
2. Definition of totally (fully ordered) set is changed from preorder + total ordering to partial order + total order. Check must be done to ensure that this does not introduce inconsistencies.
3. Check has started

24.2 NaturalNumbers.tm

1. Spell check started/Finished

24.3 Numbers.tm

1. Spell check started/Finished

24.4 LinearAlgebra.tm

1. Spell check started/Finished
2. Rewrite this to use $\mathbb{R}^2 = \mathbb{R} \cdot \mathbb{R}$

24.5 DirectSum.tm

1. Spell check started/Finished

24.6 TensorProduct.tm

1. Spell check started/Finished

24.7 Topology.tm

1. Spell check started/Finished

24.8 IntegrationBanach.tm

1. Spell check started/Finished

24.9 Differentiation.tm

1. Spell check started/Finished
2. Change definition of C^k to C^k at a point Started/Finished
3. Add n-times derivatives on $[a,b]$ and taylor expansion

24.10 FundamentalTheorem.OfAlgebra.tm

1. Spell check started/Finished
2. Check for errors started/Finished

24.11 InnerProductSpaces.tm

1. Spell check started/Finished

24.12 Exponential.tm

1. Spell check started/Finished
2. Check for errors Started

24.13 Measure.tm

1. Rewrite started/
2. Check Extended Real Numbers started
3. Base et must be non empty [Check this]

24.14 Manifold.tm

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